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# SKYSCRAPER PROJECT

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version: November 12, 2024

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# Preface

**What is this?** : At first, this is a note containing subtle or important materials I encountered while studying. I started this project in the fall of my third undergraduate year (October 2019) in Peking University, noticing that I have a poor memory and consistently forget what I thought I have already learned. So it came to me that I can compile all the proofs of theorems I cannot recall that is hard and subtle yet appearing over and over again. But gradually it turns out I want to make it as comprehensive as possible. A more accurate description of this note now is: this is a more self-contained introduction to modern (post-Grothendieck) pure mathematics that I care about.

I constantly add stuff to this note, and I regularly put them online. You can find the newest version released at [https://math.mit.edu/~hao\\_peng/skyscraper.pdf](https://math.mit.edu/~hao_peng/skyscraper.pdf). Don't be surprised to find many sections containing only titles, and the symbol “?” means “further work required here”. If you find errors or have suggestions, please feel free to email me.

**Constitutions:** The following are principles of the structure of this note, but the current version is far from it. They serve as ultimate goals.

- This note should be self-contained.
- Notations should be consistent throughout the whole note.
- Propositions should be put in the (sub)section of the most advanced term appeared in the statement, or as a corollary.
- Logical order is not necessary, but vicious circles are intolerable. Although it should also be properly ordered logically in the sense that if every notion appearing is referred to its definition, the ordering has the least number of reverse cross-references (i.e. referring to definitions after it) under admissible permutation, where admissible permutation means a permutation that preserves the tree structure of this note (i.e. the tree of chapter-section-subsection- $\dots$  ordering).
- Theories should be stated at the most generality. There's no need to give proofs for the special case but clarify the deduction from the general case, unless it is needed in the proof of the general case, then state it as a lemma.
- When facing multiple proofs, only the most elegant and essential proof should be recorded.
- References should be traced to the original author and his specific paper.
- Each part should contain less than 10 chapters. Each chapter should contain less than 20 sections. Each section should contain less than 30 pages.

**Writing Styles:** The following are writing standards of this note. References are [CMOS, Poo] and [How to write mathematics badly, by J. P. Serre]. Notice some rules of [Poo] remain to be discussed.

- Use less words and more math symbols and equations, and use plain English, so that even those who don't speak English can understand without much effort.
- Use proper cases for names of people.

- A proposition is a proven mathematical sentence.
- A theorem is a proposition of notable importance.
- A lemma is a proposition whose importance is derived from the theorem or proposition it aims to prove.
- A corollary is a proposition whose proof readily follows from its corresponding proposition or theorem.
- A conjecture is a mathematical sentence whose proof(or disproof/proof of independence) is unknown yet.
- A conjectural corollary is a mathematical sentence which is implied by the corresponding conjecture.
- A theorem & conjectural corollary is a proven mathematical sentence which can be implied by the corresponding conjecture.
- The naming of lemmas/propositions/theorems/conjectures by multiple authors follows the lexicographical order, except for historical followup works. Authors of the same paper is connected by “-” symbol, and authors of different papers are separated by a “/” symbol.
- A remark is a comment, and the book can be read independent of the remarks.(much like remarks in any programming language).

**Tips:** This is hardly a *readable* note. I use it as a dictionary. It only contains materials that I’m interested in and many proofs are still missing. Hopefully I can complete them later. The main reason why I have to latex all these materials together is that I need tons of cross-references. So I believe it’s the best way to read this book digitally, and it’s good to know how to go forwards and backwards between hyper-references on your computer. For example, on a MacOS system, the default shortcuts are  + ,  +  for Preview and  + ,  +  for Foxit Reader. Foxit Reader is stable when handling a large file.

**Acknowledgement:** Sincere thanks to Yi Tian(田翊) for answering my questions when I was learning algebraic geometry and  $p$ -adic geometry in year 2020 when I was in Peking University. His help is fundamental. Thanks to Zhiyu Zhang(张志宇) for giving me directions on mathematical study and advices on mathematical life in year 2022 when I was in MIT.

There is already a great online book [Sta] maintained by de Jong that covers considerably much of the Algebraic Geometry part of this note. I haven’t finish reading it but I reordered the materials that I learnt and kept track of it in my own way. I sometimes used different paths to optimize the proof. The idea to compile this note was inspired by [Sta]. One different feature of this note compared to [Sta] is that I want to make every proof as optimal and as general as possible(See Constitutions above). Any suggestion to optimize proofs or generalize statements is strongly encouraged.

The writing style of this note is imitating and influenced by various literature, including beautiful writings of J. S. Milne, especially [Mil17, Mil17d, Mil08], monographs of J. Lurie, especially [Lur09, Lur11], and also de Jong’s magnificent [Sta].



In arranging the contents of this note, I consulted the AMS classification (newest version from <https://mathscinet.ams.org/mathscinet/publications-search>).

**Copyright Issue:** It should be made clear that I took proofs from many different places, so only a tiny fraction of this note should be considered originated from me. I am currently too busy being a graduate student, so many references are still missing. Tell me if you think I should cite you or somebody's work. Nevertheless, I hope this note can contribute to my study and help anyone who read it. But it comes with no warranty, please use at your own risk.

**Addendum March 25 2023:** Recently voices are increasing asserting that it is impossible for anyone be sure of every detail of the propositions he used, and everyone has to write assertions he hasn't checked. I don't agree with this. I started to get in touch with V. Voevodsky's idea of machine-assisted proofs and type-theory. And I noticed this book becoming more like programming code inadvertently. Also I'm keeping an eye on Lean-formalized mathematics. With the releasing of ChatGPT4 last week, the future is both exalting and challenging.

**Addendum March 27 2023:** Today I watched Buzzard's ICM2022 talk, and this is the first time I heard of the Lean project. I am astonished to learn that the Lean project and this book share similar goals. Also, what I am writing is in fact similar to type theory based rather than set theory based. And my notes are more like Mizar. But they also differ in several ways: This book put more priority in succinctness, and proofs are only written to the degree that I can check back. I don't need absolute check of rigor. But inspired by those, I made some changes to the note to make it more like a 'pseudocode'.



*“And they said, Go to, let us build us a city and a tower, whose top may reach unto heaven;  
and let us make us a name, lest we be scattered abroad upon the face of the whole earth.”  
—Genesis 11:1-9*

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# Part I

## Foundations



# 1 | Logics

## 1.1 Underlying Logics for this book

*“The question for the ultimate foundations and the ultimate meaning of mathematics remains open; we do not know in which direction it will find its final solution nor even whether a final objective answer can be expected at all. “Mathematizing” may well be a creative activity of man, like language or music, of primary originality, whose historical decisions defy complete objective rationalization.”*

–H. Weyl

### 1 Philosophical Issue

References are <https://us.metamath.org/mpeuni/mmset.html>.

**Remark(1.1.1.1).** This section is not mathematics. All other sections are mathematics, and logical issues will not occur(except possibly in comments). ┘

## 2 Logical Foundations

We stick to the formalism point of view that mathematics is a game of symbols with no intrinsic meaning. And this book is a possible way the game is played, and record what we have done with mathematics in a digital form. Our point of view is most similar to that of Metamath:<https://us.metamath.org/mpeuni/mmset.html>.

The game has a rule to label strings of symbols(given below2), and our task is to find more and more strings of symbols labeled “LEGAL”, and for a given string of symbols, determine if we can label it “LEGAL”.

As humans, we observe interesting patterns in these “meaningless” symbol strings as they evolve from the axioms, and we attach meaning to them. One example is the set of natural numbers, whose properties match those we observe when we count everyday objects. “2” is a set, more precisely an abbreviation of a string of symbols. So “2” can never be the two apple sitting in front of you, it just behave like those two apples, because we deliberately designed it to be.

### Primitive Notions:

In essence, this book has no meaning at all, unless you attach meaning to it. And you can never truly understand what is a “set” by studying set theory. This is unavoidable, by the existence of “primitive notions”.

The primitive notion we choose is the notion of “**symbols**”. Alphabets in any human language is a symbol. And “(”, “)”, “ $\implies$ ”, “ $\wedge$ ”, “ $\emptyset$ ”, “ $\neg$ ”, “ $=$ ”, “ $\in$ ” and “ $\forall$ ” are all symbols, called **connectives**.

**Rules(Axioms):**

There are three families of rules to label strings of symbols.

- **Propositional calculus axioms with equality:**

- (Constant variables:) “const:  $\emptyset$ ” can be labeled “LEGAL”.
- (Inference Rules:) If  $\varphi, \psi$  are strings of symbols, and
  - \* “wff:  $\varphi$ ” and “wff:  $\psi$ ” are labeled “LEGAL”.
  - \* “ $\varphi$ ” and “ $\varphi \rightarrow \psi$ ” are labeled “LEGAL”.

Then  $\psi$  can also be labeled “LEGAL”.

- (Axiom for  $\wedge$ :) If  $\varphi, \psi$  are strings of symbols, then

$$\left( (\varphi \wedge \psi) \rightarrow \neg(\varphi \rightarrow \neg\psi) \right)$$

and

$$\left( \neg(\varphi \rightarrow \neg\psi) \rightarrow (\varphi \wedge \psi) \right)$$

can be labeled “LEGAL”.

- (Well-defined formulas:)
  - \* If  $x$  is a string of symbols, then

$$\text{wff: (const: } x \text{)}$$

and

$$\text{wff: (var: } x \text{)}$$

and

$$\text{wff: (wff: } x \text{)}$$

can be labeled “LEGAL”.

- \* If  $x, y$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{var: } y) \right) \rightarrow \left( (\text{wff: } x = y) \wedge (\text{wff: } x \in y) \right) \right)$$

can be labeled “LEGAL”.

- \* If  $\varphi, \psi$  are strings of symbols, then

$$\left( (\text{wff: } \varphi) \rightarrow (\text{wff: } (\neg\varphi)) \right)$$

and

$$\left( \left( (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \right) \rightarrow (\text{wff: } (\varphi \implies \psi)) \right)$$

can be labeled “LEGAL”.

- (Axioms of simplification:) If  $\varphi, \psi$  are strings of symbols, then

$$\left( \left( (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \right) \rightarrow (\varphi \implies (\psi \implies \varphi)) \right)$$

can be labeled “LEGAL”.



- (Axiom Frege:)If  $\varphi, \psi, \chi$  are strings of symbols, then

$$\left( \left( (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \wedge (\text{wff: } \eta) \right) \rightarrow \left( (\varphi \implies (\psi \implies \eta)) \implies ((\varphi \implies \psi) \implies (\varphi \implies \eta)) \right) \right)$$

can be labeled “LEGAL”.

- (Axiom transpose:)If  $\varphi, \psi$  are strings of symbols, then

$$\left( \left( (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \right) \rightarrow \left( (\neg\varphi \implies \neg\psi) \implies (\psi \implies \varphi) \right) \right)$$

can also be labeled “LEGAL”.

- (Axiom modus ponens:)If  $\varphi, \psi$  are strings of symbols, then

$$\left( \left( (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \wedge \varphi \wedge (\varphi \rightarrow \psi) \right) \rightarrow \psi \right)$$

can be labeled “LEGAL”.

- (Axiom of equality:)If  $x, y, z$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{var: } y) \wedge (\text{var: } z) \right) \rightarrow \left( (x = y) \implies ((x = z) \implies (y = z)) \right) \right)$$

can be labeled “LEGAL”.

- (Axiom of Substitution:)

- **Predicate calculus axioms with equality:**

- If  $x, \varphi, \psi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \right) \rightarrow (\text{wff: } ((\forall x)\varphi)) \right)$$

can be labeled “LEGAL”.

- (Axiom of generalization:)If  $x, \varphi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{wff: } \varphi) \right) \rightarrow \left( \varphi \implies ((\forall x)\varphi) \right) \right)$$

can be labeled “LEGAL”.

- (Axiom of quantified implication:)If  $x, \varphi, \psi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{wff: } \varphi) \wedge (\text{wff: } \psi) \right) \rightarrow \left( ((\forall x)(\varphi \rightarrow \psi)) \rightarrow ((\forall x)\varphi \rightarrow (\forall x)\psi) \right) \right)$$

can be labeled “LEGAL”.

- (Axiom of distinctness:)

- \* If  $x, y$  are strings of symbols, then

$$\left( (\text{var: } x) \rightarrow (\text{wff: } (\text{distin: } (x, y))) \right)$$

can be labeled “LEGAL”.

- \* If  $x, y, z$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, y)) \wedge (\text{distin: } (x, z)) \right) \rightarrow (\text{distin: } (x, (y = z))) \right)$$

and

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, y)) \wedge (\text{distin: } (x, z)) \right) \rightarrow (\text{distin: } (x, (y \in z))) \right)$$

can be labeled “LEGAL”.

- \* If  $x, y$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, y)) \right) \rightarrow (\text{distin: } (x, (\text{const: } y))) \right)$$

and

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, y)) \right) \rightarrow (\text{distin: } (x, (\text{var: } y))) \right)$$

and

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, y)) \right) \rightarrow (\text{distin: } (x, (\text{wff: } y))) \right)$$

can be labeled “LEGAL”.

- \* If  $x, \varphi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, \varphi)) \right) \rightarrow (\text{distin: } (x, (\neg \varphi))) \right)$$

can be labeled “LEGAL”.

- \* If  $x, \varphi, \psi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{distin: } (x, \varphi)) \wedge (\text{distin: } (x, \psi)) \right) \rightarrow (\text{distin: } (x, (\varphi \rightarrow \psi))) \right)$$

can be labeled “LEGAL”.

- \* If  $x, y, \varphi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{dist: } (x, y)) \wedge (\text{distin: } (x, \varphi)) \right) \rightarrow (\text{distin: } (x, ((\forall y)\varphi))) \right)$$

can be labeled “LEGAL”.

- \* If  $x, \varphi$  are strings of symbols, then and  $\varphi$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{wff: } \varphi) \wedge (\text{distin: } (x, \varphi)) \right) \rightarrow (\varphi \rightarrow ((\forall x)\varphi)) \right)$$

can be labeled “LEGAL”.

- **Set theory axioms(TG):**

- (Axiom of Existence:) If  $x, y$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{var: } y) \right) \rightarrow \left( \neg(\forall x)(\neg(x = y)) \right) \right)$$

can be labeled “LEGAL”.

- (Axiom of left equality:) If  $x, y, z$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{var: } y) \wedge (\text{var: } z) \right) \rightarrow \left( (x = y) \implies ((x \in z) \implies (y \in z)) \right) \right)$$

can be labeled “LEGAL”.

- (Axiom of right equality:) If  $x, y, z$  are strings of symbols, then

$$\left( \left( (\text{var: } x) \wedge (\text{var: } y) \wedge (\text{var: } z) \right) \rightarrow \left( (x = y) \implies ((z \in x) \implies (z \in y)) \right) \right)$$

can be labeled “LEGAL”.

- (Tarski-Grothendieck Axiom:)

You may noticed the strings of symbols that we have labeled “LEGAL” have something in common, i.e. you can see they are ‘well-formed formulas’, a notion which I didn’t define.

### Abbreviations and definitions:

When doing mathematics, an **abbreviation** is just a new axiom. For example, the sentence

$$“(\varphi \iff \psi)” \text{ is an abbreviation for } “((\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi))”$$

is just suggesting adding the axiom

$$\left( \left( (\varphi \iff \psi) \rightarrow ((\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)) \right) \wedge \left( ((\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)) \rightarrow (\varphi \iff \psi) \right) \right).$$

A **definition** is an assertion of a variable followed by an abbreviation, used to save space. All theorems using this definition can be reduced to wffs, where the defined new string disappears.

For example, the sentence “the power set of a set  $X$  is defined to be the set of all subsets of  $X$ ” is just suggesting adding the axioms

$$\text{var: (the power set of } X)$$

and

$$\mathcal{M}(\text{the power set of } X)$$

and

$$(y \in (\text{the power set of } X)) \iff (y \subset X).$$

Notice adding definition may make the axioms inconsistent. Thus whenever we use definitions, we need to make sure that definitions names are different.

Whenever we substitute abbreviates, we will use  $\overset{\text{subst}}{\iff}$ .

### 3 Other Choice of Foundations

Instead of using the foundations given by ZFC+TG, we can also change it to either

- A family of type theory axioms with equality as the only primitive notions, as in HOL Light, or
- A family of calculus of inductive constructions axioms, as in Coq or Lean.

I haven't try these, but may change the setup in the future.

### 4 Proofs

**Def. (1.1.4.1).** A mathematical **proof** of a proposition is a chain of logical deductions leading to the proposition from a base set of axioms. ┘

A Sample Proof:

## 2 | Mathematical Logic and Computer Science

*“It is with logic that one proves; it is with intuition that one invents.”*

–H. Poincaré

### 2.1 Mathematical Logic

References are [Logic for mathematicians, Hamilton], [Axiomatic Set Theory] and [Mathematical Logic, Mendelson]. We mostly follow Mendelson’s approach.

#### Notation(2.1.0.1).

- Beginning this section, we will be talking doing mathematics obeying the rules stated in [Logical Foundations](#). Our task is to write and only write strings of symbols and label them “LEGAL”. Our writing style is most similar to that of Mizar<http://mizar.org>.
- You may see definitions like propositional calculus/predicate calculus which already appeared in the former section, REMEMBER: this has nothing to do with them, and only can be seen as a formal representation of them. The things in the last section belongs to real world, and the things in this section and all the following belongs to this game called “mathematics”(If you are drawing papers on a paper, what you draw are not real papers). Of course you can argue truth can be learned about the real world by doing this “game”, but that depends on you attaching meaning to it.(This point has already been illustrated in the last section.)
- Don’t worry, you can still prove that the axiom of choice is independent with the ZF system by giving a model, essentially the same way proving that the Euclid’s fifth axiom is independent of the other four(in the frame of propositional calculus axioms and predicate calculus axioms, of course, everything is happening in the “game”).

┘

### 1 Peano’s Postulates for Natural Numbers

After settling [Philosophical Issue](#), I can start expressing now. In fact, I already started when I am explaining philosophical issue to you.

#### the Primordial Systems

**Def.(2.1.1.1)[the Primordial Languages].** The primordial language  $\mathcal{L}_0$  consists of

- **free variable symbols**  $a, b, c, d, m, n$ .
- **bound variable symbols**  $x, y, z, w$ .

- **constant variable symbol** 0.
- **function symbols**  $S, +, \cdot$ ,
- **predicate symbols**  $=$ .
- **punctuations symbols** “(”, “...”, “)”.
- **connective symbols**  $\neg$  and  $\implies$ .
- **quantifier symbol**  $\forall$ .

⌋

**Def. (2.1.1.2)[the (Blank)Peano Arithmetic system].** The (blank)Peano arithmetic system PA consists of the following data:

- the primordial language  $\mathcal{L}$ (2.1.1.1).
- A procedure called a **grammar** to specify which things are called **well-formed formulas**(wffs):
  - There is a procedure to specify which things are called **terms**:
    - \* 0 and free variable symbols are terms.
    - \* If  $t, s$  are terms, then  $S(t), (t + s), t \cdot s$  are terms.
  - There is a procedure to specify which things are called **atomic formulas**: If  $t, s$  are terms, then  $t = s$  is an atomic formula.
  - The procedure to determine wffs:
    - \* Atomic formulas are wffs.
    - \* if  $\varphi, \psi$  are wffs and  $x$  is a bound variable not appearing in  $\varphi$ , then  $\neg\varphi, (\varphi \implies \psi), (\forall x)\varphi(x/a)$  are wffs, where  $\varphi(x/a)$  is obtained from  $\varphi$  by replacing each occurrence of some free variable  $a$  by a bound variable  $x$  that doesn't appear in  $\varphi$ .
- A procedure to specify which well-formed formulas are called **axioms**: The following are axioms:
  - Logical Axioms: If  $\varphi, \psi, \eta$  are wffs,
    - \*  $(\varphi \implies (\psi \implies \varphi))$ .
    - \*  $((\varphi \implies (\psi \implies \eta)) \implies ((\varphi \implies \psi) \implies (\varphi \implies \eta)))$ .
    - \*  $((\neg\varphi \implies \neg\psi) \implies (\psi \implies \varphi))$ .
    - \*  $(\varphi \implies (\psi \implies \varphi))$ .
    - \*  $(\varphi \implies (\psi \implies \eta)) \implies ((\varphi \implies \psi) \implies (\varphi \implies \eta))$ .
    - \*  $((\neg\varphi \implies \neg\psi) \implies (\psi \implies \varphi))$ .
    - \*  $((\forall x)\varphi \implies \varphi(a/x))$  where  $\varphi(a/x)$  is obtained from  $\varphi$  by replacing each occurrence of the bound variable  $x$  in  $\varphi(x)$  by the free variable  $a$ .
    - \*  $((\forall x)(\varphi \implies \psi) \implies (\varphi \implies (\forall x)\psi))$ , if the free variable  $a$  in  $\varphi$  which are quantifying with  $x$  doesn't appear.
    - \* (Modus Ponens or MP)  $\varphi \wedge (\varphi \implies \psi) \rightarrow \psi$ .
    - \* (Generalizations)  $\varphi \rightarrow (\forall x)\varphi(x/a)$ , where  $x$  is a bound variable symbol that doesn't appear in  $\varphi$ , and  $\varphi(a/x)$  is obtained from  $\varphi$  by replacing each occurrence of some free variable  $a$  by  $x$ .
  - Non-Logical Axioms: If  $a, b, c$  are any free variables,  $x, y, z$  are any bound variables and  $\varphi$  a wff., then

- \*  $(a = b) \implies ((a = c) \implies (b = c)).$
- \*  $(a = b) \implies (S(a) = S(b)).$
- \*  $\neg(0 = S(a)).$
- \*  $(S(a) = S(b)) \implies (a = b).$
- \*  $(a + 0) = a.$
- \*  $a + S(b) = S(a + b).$
- \*  $a \cdot 0 = 0.$
- \*  $a \cdot S(b) = a \cdot b + b.$
- \* (Principle of mathematical induction)  $\varphi(0/a) \implies ((\forall x)(\varphi(x/a) \implies \varphi(S(x)/a)) \implies (\forall x)\varphi(x)).$

terms in the Peano arithmetic system will be called **natural numbers**.  $\lrcorner$

**Remark (2.1.1.3) [Metaphysical Issue].**

- From now any definition/proposition/rem we make should be regarded as wffs in a formal sufficiently high order system containing the blank PA system(2.1.1.2) and the class system to be defined in(2.1.3.2) and we can quantify over any other systems. In other words, I assume will can comprehend everything in this book. Such a language is called a **metasystem** for it. The symbols in this metalanguage is called a **metavariable**.
- I assume it has an axiom which is a stronger form of the mathematical induction(2.1.1.2)(i.e. Induction can apply to properties of classes).
- If you don't agree with the axioms and assumptions I made so far, you should close this file and leave right now.  $\lrcorner$

**Remark (2.1.1.4) [Logical Issue].** Pedantically, by our construction of our mathematical system, in order for propositions to be defined in a lower order system, we should have defined symbols exclusively each time we enlarge the system, and never say something like “let  $n$  be a natural number,  $n + 1 > 0$ ”, because this is not a wff. In stead, we should write  $(\forall n)(n + 1 > 0)$ , because this is a wff. in Peano's system PA, and  $n$  is born to be representing a natural number, i.e. a symbol in the system PA.

But in practice, it is annoying to do so, so we will be more tolerant on this kind of usage. Just remember always this kind of sentence can be formalized.  $\lrcorner$

## 2 Languages

**Def. (2.1.2.1) [Languages].** A language  $\mathcal{L}$  is a countable set of symbols. There may be subsets of symbols such as free symbols, bound symbols, function symbols, logical symbols, etc..  $\lrcorner$

**Remark (2.1.2.2) [Formal System].** A formal system  $\mathcal{L}$  consists of the following data:

- A language  $L$ (2.1.2.1), or equivalently, a set of symbols.
- A set of **well-formed formulas** or wffs.
- A set of **axioms**.  $\lrcorner$

**Remark(2.1.2.3) [Class System as a Metasystem].** Notice for any language or system, the class system is a metasystem of it. So we can formalize talking about everything about this system in the class system, like consistency, completeness...

Given a formal system  $L$ , we will typically use metavariables  $a, b, c, x, y, z$  whose domain is the collection of symbols in  $L$ , and use metavariables  $\varphi, \psi, \eta$  to describe wffs in  $L$ .  $\perp$

**Meta Thm. (2.1.2.4).** The primordial language  $\mathcal{L}_0$ (2.1.1.1) is a language(2.1.2.1). In fact a first order language to be defined in(2.1.5.1). The class system  $\mathcal{L}_C$  is a formal system(2.1.2.2). (i.e. they are sets).  $\perp$

**Remark (2.1.2.5).** Until now I settled all the logical issues, but remember I made several assumptions that is metaphysical and cannot be settled by logic or math:

- There is a representation of will called the primordial language(2.1.1.1),
  - There is a “talking system” that is a representation of will that can describe all other systems can enables the presence of this book(2.1.1.3).
- $\perp$

### 3 Class System(ZF)

**Def. (2.1.3.1).** In fact this theory is a theory that lies between ZF and NBG. I think it is better to just change everything to NBG?  $\perp$

**Def. (2.1.3.2) [Class System].** The **class system** ZF is a formal system(2.1.2.2) with

- Language: A first order language(2.1.5.1) with
  - Free variable symbols:
    - \*  $0, a_0$  are free variable symbols.
    - \* For any natural number  $n$ , if  $a_n$  is a free variable symbol, then  $a_{n+1}$  is also a free variable symbol.
  - Bound variable symbols:
    - \*  $x(0)$  is a bound variable symbol.
    - \* For any natural number  $n$ , if  $x_n$  is a bound variable symbol, then  $x_{n+1}$  is also a bound variable symbol.
  - Predicate symbol:  $\in, =$ .
  - Punctuation symbols: “(”, “)”, “...”, “,” and “{”, “}”, “|”, “:”.
  - Connective symbols:  $\neg$  and  $\rightarrow$ .
  - Quantifier symbol:  $\forall$ .
- Grammar:
  - A procedure to specify wffs:
    - \* If  $a, b$  are free variable, then  $a \in b$  is a wff..
    - \* If  $\varphi, \psi$  are wffs,  $a, b$  are free variables and  $x$  is a bound variable, then

$$(a \in \{x|\psi(x/a)\}), \quad (\{x|\psi(x/a)\} \in b), \quad (\{x|\varphi(x/a)\} \in \{x|\psi(x/b)\})$$

are wffs, where  $\psi(x/a)$ (resp.  $\varphi(x/b)$ ) is obtained from  $\psi$ (resp.  $\varphi$ ) by replacing a free variable  $a$ (resp.  $b$ ) by a bound variable  $x$  that doesn't appear in  $\psi$ (resp.  $\varphi$ ).



- \* If  $\varphi, \psi$  are wffs, then  $\neg\varphi$  and  $\varphi \rightarrow \psi$  are wffs.
- \* If  $\varphi$  is a wff. and  $x$  is a bound variable, then  $(\forall x)\varphi(x/a)$  is a wff.
- A procedure to specify terms:
  - \* Variables are terms.
  - \* If  $\varphi$  is a wff., then  $\{x|\varphi\}$  is a term, called a **class**.
- Axioms:
  - Logical Axioms: If  $\varphi, \psi, \eta$  are wffs,
    - \*  $(\varphi \implies (\psi \implies \varphi))$ .
    - \*  $((\varphi \implies (\psi \implies \eta)) \implies ((\varphi \implies \psi) \implies (\varphi \implies \eta)))$ .
    - \*  $((\neg\varphi \implies \neg\psi) \implies (\psi \implies \varphi))$ .
    - \*  $((\forall x)\varphi \implies \varphi(a/x))$  where  $\varphi(a/x)$  is obtained from  $\varphi$  by replacing each occurrence of the bound variable  $x$  in  $\varphi(x)$  by the free variable  $a$ .
    - \*  $((\forall x)(\varphi \implies \psi) \implies (\varphi \implies (\forall x)\psi))$ , if the free variable  $a$  in  $\varphi$  which are quantifying with  $x$  doesn't appear.
    - \* (Modus Ponens or MP)  $\varphi \wedge (\varphi \implies \psi) \rightarrow \psi$ .
    - \* (Generalizations)  $\varphi \rightarrow (\forall x)\varphi(x/a)$ , where  $x$  is a bound variable symbol that doesn't appear in  $\varphi$ , and  $\varphi(a/x)$  is obtained from  $\varphi$  by replacing each occurrence of some free variable  $a$  by  $x$ .
  - Non-logical axioms: If  $\varphi, \psi$  are wffs, then
    - C1:  $(a \in \{x|\varphi\}) \iff \varphi(a/x)$ , where  $\varphi(a/x)$  is  $\varphi$  with all occurrence of  $x$  replaced by  $a$ .
    - C2:  $(\{x|\varphi(x)\} \in a) \iff ((\exists y)((y \in a) \wedge (\forall z)((z \in y) \iff \varphi(z/x))))$ , where  $\varphi(z/x)$  is  $\varphi$  with all occurrence of  $x$  replaced by  $z$ .
    - C3:  $(\{x|\varphi(x)\} \in \{x|\psi\}) \iff ((\exists y)(\psi(y/x) \wedge (\forall z)((z \in y) \iff \varphi(z/x))))$ , where  $\varphi(z/x)$  is  $\varphi$  with all occurrence of  $x$  replaced by  $z$  and  $\psi(y/x)$  is  $\psi$  with all occurrence of  $x$  replaced by  $y$ .
    - C4: (Equality)  $(a = b) \iff (\forall x)((x \in a) \iff (x \in b))$ .
    - C5: (Axiom of Extensionality)(2.1.3.3).
    - C6: (Axiom of Pairing)(2.1.3.20).
    - C7: (Axiom of Union)(2.1.3.26).
    - C8: (Axiom of Power Set)(2.1.3.30).
    - C9: (Axiom Schema of Replacement)(2.1.3.31).
    - C10: (Axiom of Regularity/Foundation)(2.1.3.40).
    - C11: (Axiom of Infinity)?.

┘

### Equalities

**Axiom (2.1.3.3)[Axiom of Extensionality].**  $((a = b) \wedge (a \in c)) \rightarrow (b \in c)$ .

┘

**Meta Thm.(2.1.3.4)[Reduce Class to Sets].** For any wff  $\varphi$  in the class system, there exists a wff  $\varphi^*$  in the set system s.t.  $\varphi$  is deducible from  $\varphi^*$  and  $\varphi^*$  is deducible from  $\varphi$ . (2.1.4.1)?  $\lrcorner$

*Proof:* Use induction. ?  $\square$

**Remark(2.1.3.5).** This tells us that the extension from set system to class system is conservative. ?  $\lrcorner$

**Meta Thm.(2.1.3.6).** The set system ? is a subsystem(2.1.4.4) of the class system(2.1.3.2). Moreover, every term in set system is a term in the class system.  $\lrcorner$

**Def.(2.1.3.7).**

- If  $\varphi$  is a wffs, define  $\{x : \varphi\} \triangleq \{x|\varphi\}$ .
- If  $A, B$  are classes, then define  $A \notin B \triangleq \neg(A \in B)$ .
- If  $A$  is a class and  $\varphi$  is a wff., define  $(\forall x \in a)\varphi \triangleq (\forall x)(x \in a)(\wedge \varphi)$

$\lrcorner$

**Def.(2.1.3.8)[Equality between Classes].** If  $A, B$  are classes, then

$$A = B \triangleq (\forall x)((x \in A) \iff (x \in B)), \quad A \neq B \triangleq \neg(A = B)$$

$\lrcorner$

**Meta Thm.(2.1.3.9).** If  $A, B, C$  are classes, then

- $(A = B) \iff (\exists x)((x = A) \wedge (x \in B))$ .
- $(A = A)$ .
- $(A = B) \rightarrow (B = A)$ .
- $(A = B) \wedge (B = C) \rightarrow (A = C)$ .

$\lrcorner$

*Proof:* Cf.[Axiomatic Set Theory]P13.  $\square$

**Meta Thm.(2.1.3.10).**  $(a = b) \rightarrow (\varphi(a/x) \iff \varphi(b/x))$ , where  $\varphi(a/x)$  is constructed from  $\varphi$  by replacing every free occurrence of  $x$  by  $a$ .  $\lrcorner$

*Proof:* Cf.[Axiomatic Set Theory, P8]. The proof used induction on the number of logical symbols in  $\varphi$ . ?  $\square$

**Thm.(2.1.3.11)[Every Set is a Class].**  $a = \{x|x \in a\}$ .  $\lrcorner$

*Proof:* This follows from axiom (C2) and the fact  $(\forall x)((x \in a) \iff (x \in a))$ .  $\square$

**Def.(2.1.3.12)[Set Predicate].** If  $A$  is a class,  $\mathcal{M}(A) \triangleq (\exists x)(x = A)$ . This predicate formally indicate that “ $A$  is a set”.  $\lrcorner$

**Prop.(2.1.3.13).**  $\mathcal{M}(a)$ .  $\lrcorner$

*Proof:*  $a = a$ .  $\square$

**Meta Thm.(2.1.3.14).** If  $A$  is a class, then  $(A \in \{x|\varphi(x)\}) \iff (\mathcal{M}(A) \wedge \varphi(A))$ .  $\lrcorner$

*Proof:* Use induction on  $A$ . ?  $\square$

**Thm. (2.1.3.15) [Russell's Paradox].** Define  $Ru \triangleq \{x | x \notin x\}$ . Then  $\neg \mathcal{M}(Ru)$ . ┘

*Proof:* By considering the wff.  $\varphi : x \notin x$ , it follows from (2.1.3.14) that

$$(\mathcal{M}(Ru) \wedge (Ru \notin Ru)) \rightarrow (Ru \in \{x | x \notin x\}) \xrightarrow{\text{subst}} (Ru \in Ru)$$

so

$$\mathcal{M}(Ru) \rightarrow ((Ru \notin Ru) \rightarrow (Ru \in Ru)).$$

so

$$\mathcal{M}(Ru) \rightarrow (Ru \in Ru).$$

Then by axiom (C3),

$$(Ru \in Ru) \rightarrow ((\exists y)(\psi(y/x) \wedge (\forall z)((z \in y) \iff z \notin z)))$$

which is false. So  $\neg \mathcal{M}(Ru)$ . □

**Def. (2.1.3.16) [Definable Sets].** If  $\varphi$  is a wff., then we say that  $\{x | \varphi(x)\}$  is a **definable set** if  $\mathcal{M}(\{x | \varphi(x)\})$ . ┘

### Properties of Classes

**Def. (2.1.3.17) [Pairs and Ordered Pairs].**

- $\{a, b\} \triangleq \{x | (x = a) \vee (x = b)\}$ .
  - $\{a\} \triangleq \{a, a\}$ .
  - $(a, b) \triangleq \{x | (x = \{a\}) \vee (x = \{a, b\})\}$ .
  - $a_n \triangleq (a, n)$ . (notice  $n$  is a natural number).
- ┘

**Meta Def. (2.1.3.18) [MultiPairs].** For any free variable  $a_0$  and a natural number  $n$ , define a term  $\{a_1, \dots, a_n\}$ . For any bound variable  $x_0$ , define a bound term  $\{x_1, \dots, x_n\}$ . ┘

**Prop. (2.1.3.19).**

- $(c \in \{a, b\}) \iff ((c = a) \vee (c = b))$ .
  - $(c \in \{a\}) \iff (c = a)$ .
  - $(c \in (a, b)) \iff ((c = \{a\}) \vee (c = \{a, b\}))$ .
  - $(\{a\} = \{b\}) \iff (a = b)$ .
  - $(\{a\} = \{b, c\}) \iff ((a = b) \wedge (b = c))$ .
  - $((a, b) = (c, d)) \iff ((a = c) \wedge (b = d))$ .
  - $((\forall x)((a \in x) \rightarrow (b \in x))) \rightarrow (a = b)$ .
- ┘

**Axiom (2.1.3.20) [Axiom of Pairing].**  $\mathcal{M}(\{a, b\})$ . ┘

**Cor. (2.1.3.21).**  $\mathcal{M}((a, b))$ . ┘

**Meta Thm. (2.1.3.22) [Objects of Classes are Sets].** For classes  $A, B$ ,  $(A \in B) \rightarrow \mathcal{M}(A)$ . ┘

**Def. (2.1.3.23) [Unions].** For classes  $A, B$ , define

- $A \cup B \triangleq \{x | (x \in A) \vee (x \in B)\}.$
- $A \cap B \triangleq \{x | (x \in A) \wedge (x \in B)\}.$
- For a class  $A$ , define  $\cup(A) \triangleq \{x | (\exists y)((x \in y) \wedge (y \in A))\}.$

┘

**Prop. (2.1.3.24).**  $(a \cup b) = \cup(\{a, b\}).$

┘

*Proof:* Cf.[Axiomatic Set Theory, P16].

□

**Prop. (2.1.3.25).**

- $(a \in (b \cup \{b\})) \iff (a \in b) \vee (a = b).$
- $A \cup B = B \cup A.$
- $A \cap B = B \cap A.$
- $(A \cup B) \cup C = A \cup (B \cup C).$
- $(A \cap B) \cap C = A \cap (B \cap C).$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$

┘

*Proof:*

□

**Axiom (2.1.3.26) [Axioms of Union].**  $\mathcal{M}(\cup(a)).$

┘

**Cor. (2.1.3.27).**  $\mathcal{M}(a \cup b).$

┘

*Proof:*

□

**Def. (2.1.3.28) [Subclasses].** For classes  $A, B$ , define

- $A \subset B \triangleq (\forall x)((x \in A) \rightarrow (x \in B)).$
- $A \subsetneq B \triangleq ((A \subset B) \wedge (A \neq B)).$

┘

**Def. (2.1.3.29) [Power Sets].**  $\mathcal{P}(a) \triangleq \{x | x \subset a\}$ , called the **power set** of  $a$ .

┘

**Axiom (2.1.3.30) [Axiom of Power Set].**  $\mathcal{M}(\mathcal{P}(a)).$

┘

**Axiom (2.1.3.31) [Axiom Schema of Replacement].** For each wffs  $\varphi$ , there is an axiom

$$((\forall x)(\forall y)(\forall z)((\varphi(x/a, y/b) \wedge \varphi(x/a, z/b)) \rightarrow (y = z)) \rightarrow \mathcal{M}(\{y | (\exists x \in a)\varphi(x/a, y/b)\})).$$

┘

**Meta Cor. (2.1.3.32).** For any class  $A$ ,  $\mathcal{M}(a \cap A).$

┘

*Proof:* Cf.[Axiomatic Set Theory, P20].

□

**Def. (2.1.3.33) [Setminus].** For classes  $A, B$ , define  $A \setminus B \triangleq \{x | (x \in A) \wedge (x \notin B)\}.$

┘

**Meta Thm. (2.1.3.34).** If  $A$  is a class, then  $\mathcal{M}(a \setminus A).$

┘

*Proof:*  $\mathcal{M}(a \setminus A) \xrightarrow{\text{subst}} \mathcal{M}(\{x \in a \mid x \notin A\})$  is a theorem by (2.1.3.32). □

**Def. (2.1.3.35) [Empty Class].**  $0 \triangleq \{x \mid x \notin x\}$ . ┘

**Prop. (2.1.3.36).**  $a \setminus a = 0$ . ┘

*Proof:*  $a \setminus a \xrightarrow{\text{subst}} \{x \mid (x \in a) \wedge (x \notin a)\} = \{x \mid x \notin x\}$  ? □

**Cor. (2.1.3.37) [Empty Set].**  $\mathcal{M}(0)$ . ┘

**Prop. (2.1.3.38).**

- $(\forall x)(x \notin 0)$ .
  - $(a \neq 0) \iff (\exists x)(x \in a)$ .
- ┘

*Proof:* 1:  $(\forall x)(x = x)$ .  
2: By axiom (Equality),

$$(a \neq 0) \iff (\exists x)((x \in 0) \wedge (x \notin a)) \vee ((x \in a) \wedge (x \notin 0)).$$

Then it follows from item1 that  $(a \neq 0) \iff (\exists x)(x \in a)$ . □

**Meta Def. (2.1.3.39).** We define  $\{a_1 \in a_2 \in \dots \in a_n\}$  as in (2.1.4.3). ? ┘

**Axiom (2.1.3.40) [Axiom of Regularity (Foundation)].**  $(a \neq 0) \rightarrow (\exists x \in a)((x \cap a) = 0)$ . ┘

**Cor. (2.1.3.41).**  $(a \notin a)$ . ┘

*Proof:* If  $(a \in a)$ , then  $\{a\} \neq 0$  by (2.1.3.38), and unwinding the definition ?,

$$(\forall x)((x \in \{a\}) \rightarrow (x \cap \{a\}) \neq 0),$$

contradicting axiom of regularity (2.1.3.40). □

**Meta Cor. (2.1.3.42).** For any natural number  $n > 0$ ,  $\neg(a_1 \in a_2 \in \dots \in a_n \in a_1)$ . ┘

*Proof:* If  $(a_1 \in a_2 \in \dots \in a_n \in a_1)$ , then  $\{a_1, \dots, a_n\} \neq 0$  by (2.1.3.38), and

$$(\forall x)((x \in \{a_1, \dots, a_n\}) \rightarrow (x \cap \{a_1, \dots, a_n\}) \neq 0),$$

contradicting axiom of regularity (2.1.3.40). □

**Prop. (2.1.3.43) [von. Neumann Universe].** Define  $\mathcal{V} \triangleq \{x \mid x = x\}$ , then  $\neg \mathcal{M}(\mathcal{V})$ . And in fact  $\mathcal{V} = \text{Ru}$  (2.1.3.15). ┘

*Proof:* Since  $\mathcal{V} = \mathcal{V}$ , if  $\mathcal{M}(V)$ , then  $V \in V$ , by (2.1.3.14), then this together with  $\mathcal{M}(V)$  contradicts (2.1.3.41) after unwinding definition ?. (Notice that metathm (2.1.3.14) is a wff. when restricted to  $V$ ).

For the last assertion, ? □

**Meta Thm. (2.1.3.44).** For classes  $A, B$ ,

- $0 \subset A$ .

- $A \subset \mathcal{V}$ .
- $(\forall x)(x \notin A) \rightarrow (A = 0)$ .
- $(A \subset a) \rightarrow \mathcal{M}(A)$ .
- $\mathcal{M}(A) \rightarrow \mathcal{M}(A \cap B)$ .
- $A \notin A$ .

┘

**Meta Thm. (2.1.3.45)[Strong Regularity].** The axiom of regularity(2.1.3.40) implies a stronger form of regularity: For any class  $A$ ,

$$(A \neq 0) \rightarrow (\exists x \in A)((x \cap a) = 0).$$

┘

*Proof:* Cf.[Axiomatic Set Theory]P80. □

**Meta Cor. (2.1.3.46).** For any class  $A$ ,

$$(\forall x)((x \subset A) \rightarrow (x \in A)) \rightarrow (A = \mathcal{V})$$

(In words, if every set  $a$  has a property(a wff.  $\varphi$ ) whenever every element of  $a$  has this property, then every set has this property.) ┘

*Proof:* If  $(\forall x)((x \subset A) \rightarrow (x \in A))$ , denote  $B = V \setminus A$ , if  $B \neq 0$ , then by strong regularity(2.1.3.45),

$$(\exists a)((a \in B) \wedge (a \cap B) = 0).$$

So

$$(\forall y)((y \in a) \rightarrow (y \notin B)).$$

But  $y \in V$ , so

$$(\forall y)((y \in a) \rightarrow (y \notin A)).$$

And then  $a \subset A$ , and the hypothesis on  $A$  implies  $a \in A$ , contradicting the fact  $a \in B$ . So  $A = V$ , by(2.1.3.44). □

**Meta Cor. (2.1.3.47)[Inclusion Induction].** There are no infinite descending  $\in$ -chains of sets. ┘

*Proof:* The property  $\mathcal{P}$  of not having infinite descending  $\in$ -chain is a wff., so  $A = \{x | \mathcal{P}(x)\}$  is a class. And it can be shown  $(a \subset A) \rightarrow (a \in A)$ . Thus by(2.1.3.46),  $A = Ru$ , so every set has no infinite descending  $\in$ -chains. □

## 4 Formal Propositional Calculus

**Meta Def. (2.1.4.1)[Deductions].** Let  $L$  be a formal system(2.1.2.2), a **proof** is a finite sequence of wffs s.t. every wffs appearing is either an axiom or a wedge of two wffs before it.

A **theorem** in  $L$  is a wffs that is deducible from an axiom  $\eta$  of  $L$ . A proof(resp. theorem/defi) in the metasystem is called a **metaproof**(resp. **metatheorem**/defi). ┘

**Meta Def. (2.1.4.2)[Multiple Deduction].** Let  $L$  be a formal system and  $\psi$  a wff. in  $L$ , then for any natural number  $n > 0$  and any  $\{\eta_1, \dots, \eta_n\}$  (2.1.3.18), then introduce metaterms

$$\{\eta_1\} \vdash_L \psi, \quad \{\eta_1, \eta_2\} \vdash_L \psi, \quad \{\eta_1, \dots, \eta_n\} \vdash_L \psi$$

and metaaxioms:

$$\{\eta_1\} \vdash_L \psi \triangleq \vdash_L (\eta_1 \rightarrow \psi).$$

$$\{\eta_1, \dots, \eta_{n+1}\} \vdash_L \psi \triangleq \{\eta_1, \dots, \eta_n\} \vdash_L (\eta_{n+1} \rightarrow \psi)$$

It reads:  $\psi$  is **deducible form**  $\eta_1$  **to**  $\eta_n$ . ┘

**Remark (2.1.4.3).** This definition of multiple deduction shows that we have the ability to handle multiple definition involving "...". But due to limit of space, we will not do so again, and leave it to the future to formalize it. ? ┘

**Meta Def. (2.1.4.4)[Subsystem].** A formal system  $L$  is called a **subsystem** of another formal system  $L'$  or  $L'$  is an **extension system** of  $L$  if:

- There is a procedure to assign a wff. in  $L'$  for each wff. in  $L$  such that
    - Every proof in  $L$  is assigned to a proof in  $L'$ .
    - Every axiom in  $L$  is a theorem in  $L'$ .
- ┘

### Propositional Calculus

**Def. (2.1.4.5)[Formal System of Propositional Calculus].** The formal system (2.1.2.2) of **propositional calculus**  $\mathcal{L}$  is defined by the following:

- Symbols:
  - A procedure to specify which things are **variable symbols**.
  - **punctuation symbols** “(”, “)”.
  - **connective symbols**:  $\neg, \implies$ .
- Grammar:
  - Symbols are wffs.
  - If  $\varphi, \psi$  are wffs, then  $\neg\varphi$  and  $(\varphi \implies \psi)$  are wffs.
- Axioms: If  $\varphi, \psi, \eta$  are wffs, then the following are axioms:
  - L1:  $(\varphi \implies (\psi \implies \varphi))$ .
  - L2:  $((\varphi \implies (\psi \implies \eta)) \implies ((\varphi \implies \psi) \implies (\varphi \implies \eta)))$ .
  - L3:  $((\neg\varphi \implies \neg\psi) \implies (\psi \implies \varphi))$ .
  - L4: (Modus Ponens or MP) If  $\varphi, \psi$  are wffs, then  $\varphi \wedge (\varphi \rightarrow \psi) \rightarrow \psi$ .

From now on, this system is the foundation of the book, and every language follows will contain  $\mathcal{L}$ .

┘

**Def. (2.1.4.6)[Logical Abbreviations].** For wffs  $\varphi, \psi$ ,

- $(\varphi \vee \psi) \triangleq (\neg\varphi \implies \psi)$ ,
- $(\varphi \wedge \psi) \triangleq \neg(\varphi \implies \neg\psi)$ ,

- $(\varphi \iff \psi) \triangleq \neg((\varphi \implies \psi) \implies \neg(\psi \implies \varphi))$ ,
- $[ \triangleq \{$ ,
- $] \triangleq \}$ .

┘

**Meta Thm. (2.1.4.7).** let  $L$  be a language and  $\varphi, \psi, \eta$  be wffs in  $L$

- Axioms of  $L$  are theorems in  $L$ .
- $\{(\varphi \implies \psi), (\psi \implies \eta)\} \vdash_L (\varphi \implies \eta)$ .
- $\vdash_L (\neg\psi \implies (\psi \implies \varphi))$ .
- $\vdash_L ((\neg\varphi \implies \varphi) \implies \varphi)$ .

┘

*Proof:* Cf.[Hamilton, Chap2].

□

**Meta Thm. (2.1.4.8)[Deduction Theorem].** Let  $\varphi, \psi$  be wffs in  $L$ ,

- If  $\Gamma \cup \{\varphi\} \vdash_L \psi$ , then  $\Gamma \vdash_L (\varphi \implies \psi)$ .
- If  $\Gamma \vdash_L (\varphi \implies \psi)$ , then  $\Gamma \cup \{\varphi\} \vdash_L \psi$ .

┘

*Proof:* Cf.[Hamilton, P32].?

□

### Adequacy Theorem for $\mathcal{L}$

## 5 First Order Systems

### Formal Predicate Calculus

**Def. (2.1.5.1)[First Order Languages].** A first order language  $\mathcal{L}$  is a language(2.1.2.1) with

- a countable set of **bound variable symbols**.
- a countable set of **free variable symbols**.
- a countable set of **constant symbols**.
- a countable set of **predicate(relation) symbols**.
- a countable set of **functions symbols**.
- the **punctuations symbols** “(”, “)” and “,”.
- the **connective symbols**  $\neg$  and  $\implies$ .
- the **quantifier symbol**  $\forall$ .

┘

**Def. (2.1.5.2)[Formal System of Predicate Calculus].** Given a first order language  $L$ (2.1.5.1), we can define a formal system(2.1.2.2) of **predicate calculus**  $\mathcal{K}_L$  as follows:

- Symbols: symbols in  $L$ .
- Grammar:
  - There is a countable set of **terms**:
    - \* Constant symbols and free variable symbols are terms.



- \* For any natural number  $n$ , If  $f^n$  is a function symbol, and for any  $m < n$ ,  $t_m$  is a term, then  $f(t_1, \dots, t_{n_f})$  is a term.
- There is a set of **atomic formulas**: If  $R^k$  is a predicate symbol and  $t_1, \dots, t_k$  are terms, then  $R^k(t_1, \dots, t_k)$  are atomic formulas.
- There is a countable set of wffs:
  - \* Atomic formulas are wffs.
  - \* if  $\varphi, \psi$  are wffs and  $x$  is a bound variable not appearing in  $\varphi$ , then  $\neg\varphi, (\varphi \implies \psi), (\forall x)\varphi(x/a)$  are wffs, where  $\varphi(x/a)$  is obtained from  $\varphi$  by replacing each occurrence of some free variable  $a$  by a bound variable  $x$  that doesn't appear in  $\varphi$ .
- Axioms: If  $\varphi, \psi, \eta$  are wffs in  $\mathcal{K}_L$  and  $x$  is a variable, then There the following countably many axioms:
  - K1:  $(\varphi \implies (\psi \implies \varphi))$ .
  - K2:  $((\varphi \implies (\psi \implies \eta)) \implies ((\varphi \implies \psi) \implies (\varphi \implies \eta)))$ .
  - K3:  $((\neg\varphi \implies \neg\psi) \implies (\psi \implies \varphi))$ .
  - K4:  $((\forall x)\varphi \implies \varphi(a/x))$  where  $\varphi(a/x)$  is obtained from  $\varphi$  by replacing each occurrence of the bound variable  $x$  in  $\varphi(x)$  by the free variable  $a$ .
  - K5:  $((\forall x)(\varphi \implies \psi) \implies (\varphi \implies (\forall x)\psi))$ , if the free variable  $a$  in  $\varphi$  which are quantifying with  $x$  doesn't appear.
  - K6: (Modus Ponens or MP)  $\varphi \wedge (\varphi \implies \psi) \rightarrow \psi$ .
  - K7: (Generalizations)  $\varphi \rightarrow (\forall x)\varphi(x/a)$ , where  $x$  is a bound variable symbol that doesn't appear in  $\varphi$ , and  $\varphi(a/x)$  is obtained from  $\varphi$  by replacing each occurrence of some free variable  $a$  by  $x$ .

┘

**Meta Def. (2.1.5.3)[Sentences and Theories].** A **sentence** is a formula without free variables.

A **theory** is a set of sentences in a language  $\mathcal{L}$ .

┘

**Meta Thm. (2.1.5.4)[Tautologies are Theorems].** if  $\varphi$  is a wff. in a language  $L$  and it is a tautology<sup>?</sup>, then  $\varphi$  is a theorem(2.1.4.1) of  $\mathcal{K}_L$ (2.1.5.2).

┘

*Proof:* Cf.[Hamilton]P72.

□

**Meta Thm. (2.1.5.5)[Soundness Theorem for  $\mathcal{K}_L$ ].** If  $\varphi$  is a wff. in a language  $L$  and  $\vdash_{\mathcal{K}_L} \varphi$ , then  $\varphi$  is locally valid<sup>?</sup>,

┘

*Proof:* Cf.[Hamilton]P74.

□

**Meta Cor. (2.1.5.6)[Consistence for  $\mathcal{K}_L$ ].**  $\mathcal{K}_L$  is consistent.

┘

*Proof:* Cf.[Hamilton]P74.

□

**Meta Cor. (2.1.5.7).** For any wffs  $\varphi, \psi, \eta$  of  $L$ ,

$$\{(\varphi \implies \psi), (\psi \implies \eta)\} \vdash_{\mathcal{K}_L} (\varphi \implies \eta)$$

┘

*Proof:* Cf.[Hamilton]P76.

□

### Adequacy Theorem for $\mathcal{K}_L$

Def.(2.1.5.8). ┘

### Satisfaction and Truth

See[Hamilton].

## 6 Models and Examples of Systems

Def.(2.1.6.1) [(Mathematical) Languages]. A mathematical language  $\mathcal{L}$  is a first order language(2.1.5.1) given by the following data:

- A set  $\mathcal{F}$  of **function symbols**.
- A set  $\mathcal{R}$  of **relations symbols**.
- A set  $\mathcal{C}$  of **constant symbols**.

Function symbols, relations symbols, constant symbols consists of all symbols in  $\mathcal{L}$ . For simplicity, we call a mathematical language simply a language. ┘

Example(2.1.6.2) [(Mathematical) Languages].

- $\mathcal{L}_0$  is the basic language of logics.
- $\mathcal{L}_r = (+, \cdot)$  is defined to be the language of rings.
- $\mathcal{L}_{or} = (+, \cdot, <)$  is defined to be the language of ordered rings.
- $\mathcal{L}_{\omega_1, \omega}(Q)$  is defined to be the language which is an extension of  $\mathcal{L}_0$  in which we can form countable conjunctions and add a quantifier  $Q$  : there exists uncountably many. ┘

Def.(2.1.6.3) [Theories].

- $ACF$  is defined to be the theory of alg.closed fields, w.r.t  $\mathcal{L}_r$ .
- $ACF_p$  is defined to be the theory of alg.closed fields of char  $p$ .
- $DAG$  is defined to be the theory of non-trivial torsion-free divisible Abelian groups, w.r.t.  $\mathcal{L}_r$ .
- $DLO$  is the theory of dense linear orders without endpoints.
- $ODAG$  is the theory of nontrivial divisible Abelian groups.
- $RCF$  is the theory of real closed fields w.r.t  $\mathcal{L}_{or}$ . ┘

## 2.2 Set Theory

*“No one shall expel us from the paradise which Cantor has created for us.”*

–D. Hilbert

References are [Jec03], [Model Theory Marker], [Axiomatic Set Theory], [Hamilton],

### Notation(2.2.0.1).

- Use notations from [Mathematical Logic](#).
- We build theories based on the class system(2.1.3.2).
- As in(2.1.1.3), all propositions are regarded as theorems in a system that is an extension of all other systems defined in this book, in particular it is an extension of the blank PA system(2.1.1.2) and the class system  $\mathcal{L}_C$ (2.1.3.2).
- We will rebuild PA system and NBG/ZFC set theory inside primordial ZFC.(This new ZFC should be regarded as a representation of the primordial ZFC inside itself), so the logic section will no longer be cited other than this section. ?

┘

### 1 Von.Neumann-Bernays-Gödel Set Theory(NBG)

**Def.(2.2.1.1)[Inductive Sets].** A set  $I$  is called **inductive** if  $0 = \emptyset \in I$  and if  $n \in I$ , then  $n + 1 \in I$ , where  $n + 1 = S(n)$  the successor. ┘

**Axiom(2.2.1.2)[Axiom of infinity].** An inductive set(2.2.1.1) exists. ┘

**Def.(2.2.1.3)[Set of Natural Numbers].** The **set of natural numbers**  $\mathbb{N}$  is defined to be

$$\mathbb{N} = \{x \in I_0 | x \in I \text{ for all inductive set } I\},$$

where  $I_0$  is an inductive set given by(2.2.1.2). Elements of  $\mathbb{N}$  are called **natural numbers**. ┘

**Cor.(2.2.1.4)[Inductive Principle].** If  $P(x)$  is a property that  $P(0)$ , and  $P(n)$  implies  $P(n+1)$ , then  $P(n)$  for each natural number  $n$ . ┘

*Proof:* By definition  $B = \{n \in \mathbb{N} | P(n)\}$  is an inductive set, so  $\mathbb{N} \subset B$ . □

**Prop.(2.2.1.5).**  $\mathbb{N}$ (2.2.1.3) is a linearly ordered set. ┘

*Proof:* Cf.[Set Theory Jech P43]. □

**Cor.(2.2.1.6)[Inductive Principle Second Version].** If  $P(x)$  is a property that  $P(0)$ , and  $P(k)$  holds for all  $k < n$  implies  $P(n)$ , then  $P(n)$  for each natural number  $n$ . ┘

*Proof:* Use induction principle(2.2.1.4) for the property  $Q(n) : P(k)$  for all  $k < n$ . Then  $Q(n)$  implies  $Q(n+1)$ . □

## 2 Relations and Functions

**Def. (2.2.2.1) [Products].** For classes  $A, B$ ,  $A \times B$  is the class

$$\{x | (\exists y \in A)(\exists z \in B)(x = (y, z))\},$$

called the **product of  $A$  and  $B$** . ┘

**Prop. (2.2.2.2).**  $\mathcal{M}(a \times b)$ . ┘

*Proof:* Cf. [Axiomatic Set theory]P23. □

**Def. (2.2.2.3).** Inductively define ? class terms

- $A^1 \triangleq A$ ,
- $A^{n+1} \triangleq A^n \times A$ ,
- $A^{-1} \triangleq \{(x, y) | (y, x) \in A\}$ .

And define ┘

**Def. (2.2.2.4) [Relations and Functions].** Define the following predicates:

- $\text{Rel}(R) \triangleq R \subset A^2$ .
  -
- ┘

**Def. (2.2.2.5) [Bijections].** ┘

**Def. (2.2.2.6) [Finite and Infinite Sets].** For  $n \in \mathbb{N}$  and a set  $X$ , we use the sentence “ $X$  has  $n$  elements” to mean there is a bijection from  $n$  to  $X$ . We use the sentence “ $X$  is a **finite set**” to mean  $X$  has  $n$  elements for some  $n \in \mathbb{N}$ , and say it is an **infinite set** otherwise. ┘

## 3 Orderings

**Def. (2.2.3.1) [Ordering].** A **partial ordering** on a set  $A$  is a relation (2.2.2.4)  $<$  on  $A$ , or equivalently a subset  $C \subset A \times A$  that

- For no  $x \in A$ ,  $x < x$  holds.
- If  $x < y$  and  $y < z$ , then  $x < z$ .

It is called a **total ordering** if moreover it satisfies

- For every  $x, y \in A$  that  $x \neq y$ , either  $x < y$  or  $y < x$ .

A **poset** is defined to be a partially ordered set. ┘

**Def. (2.2.3.2) [Reverse Orderings].** The **reverse ordering**  $A^{\text{op}}$  of an ordered set  $A$  is the same set  $A$  with the ordering reversed. ┘

**Def. (2.2.3.3) [Cofinality].** The **cofinality** of or a poset (2.2.3.1)  $\alpha$  is the smallest cardinality  $\delta$  of a cofinal subset of  $\alpha$ . ┘

**Def. (2.2.3.4) [ $\kappa$ -Filtered Poset].** For a cardinal  $\kappa$ , a poset is called  **$\kappa$ -filtered** if for any subset unbounded from above has cardinality  $\geq \kappa$ . ┘

**Def. (2.2.3.5) [Directed Sets].** A **directed set** is a poset that 3-filtered and non-empty. ┘

**Def. (2.2.3.6).** In a poset  $P$ , two element  $p, q$  are called **compatible** if there is an  $r \in P$  that  $r < p, r < q$ .  $\lrcorner$

**Def. (2.2.3.7) [ $\kappa$ -Chain Condition].** For a cardinal  $\kappa$ , a poset  $P$  is said to satisfy the  **$\kappa$ -chain condition** if for any subset  $A \subset P$  that elements of  $A$  are pairwise incompatible(2.2.3.6), then  $|A| \leq \kappa$ .  $\lrcorner$

### Total Ordering

**Def. (2.2.3.8) [Well-Orderings].** A **linear ordering** is defined to be a total ordering.

A linear ordering is called a **well-ordering** if every nonempty subset has a minimal element.  $\lrcorner$

**Def. (2.2.3.9) [Lexicographical Ordering].** If given a family of linearly ordered set  $A_i$  indexed by a well-ordered set  $I$ , then there is a linear ordering on  $\prod_I A_i$ , where  $(f_i) < (g_i)$  iff  $(f_i) \neq (g_i)$  and for the minimal  $i_0$ (well-ordering used) that  $f_{i_0} \neq g_{i_0}$ ,  $f_{i_0} < g_{i_0}$ . It is called the **lexicographical ordering**.  $\lrcorner$

**Def. (2.2.3.10).** An ordered set  $X$  is called **dense** iff for each  $a < b$ , there is a  $x$  that  $a < x < b$ .  $\lrcorner$

**Def. (2.2.3.11) [Least Upper Bound].** An ordered set  $A$  is said to have the **least upper bound property** if any subset  $A_0 \subset A$  bounded above has a least upper bound. It is said to satisfy the **greatest lower bound property** if  $A^{op}$  satisfies the least upper bound property.  $\lrcorner$

**Prop. (2.2.3.12) [Cantor].** Any two ordered set that is countable, dense and has no endpoints are isomorphic. In particular, any of these is isomorphic to the set of rational numbers  $\mathbb{Q}$ .  $\lrcorner$

*Proof:* We will built the isomorphism by extending partial embeddings. Let  $a_0, \dots, a_n, \dots$  be an ordering of  $A$ ,  $b_0, \dots, b_n, \dots$  be an ordering of  $B$ , and we can alternatively extend mapping on  $a_n$  and  $b_n$ , as  $A, B$  are complete without endpoints. Then we get an isomorphism of  $A$  and  $B$ .  $\square$

**Cor. (2.2.3.13).** Any countable linearly ordered set can be mapped isomorphically into  $\mathbb{Q}$ .  $\lrcorner$

**Def. (2.2.3.14).** A **initial segment** of an ordered set  $W$  is the ordered set  $W[a] = \{x \in W | x < a\}$ .  $\lrcorner$

**Lemma (2.2.3.15).** If  $W$  is a well-ordered set, then any increasing function  $f : W \rightarrow W$  satisfies  $f(x) \geq x$ .  $\lrcorner$

*Proof:* If the set  $\{x | f(x) < x\}$  is not empty, then it has a minimal element  $a$ , then  $f(f(a)) < f(a)$ , contradiction.  $\square$

**Cor. (2.2.3.16).** A well-ordered set cannot be isomorphic to an initial segment of itself, and an automorphism of a well-ordered set must be identity.  $\lrcorner$

*Proof:* Use the above lemma(2.2.3.15), if it is isomorphic to  $W[a]$ , then  $f(a) < a$ , contradiction. For any automorphism,  $f(x) \geq x, f^{-1}(x) \geq x$ , so  $f(x) = x$ .  $\square$

**Prop. (2.2.3.17) [Comparison of Well-Orderings].** A cut of a well-ordered set is well-ordered. And for any two well-ordered sets  $W_1, W_2$ , either they are isomorphic, or one of them is isomorphic to a initial segment of another.  $\lrcorner$

*Proof:* The three cases are mutually exclusive by (2.2.3.16), So it suffices to show one of them holds.

Define a set  $f = \{(x, y) \in W \times W \mid W_1[x] \cong W_2[y]\}$ . (2.2.3.16) shows  $f$  is injective and monotone in both coordinates. Now we want to prove that if the domain of  $f$  is not all  $W_1$ , then it is an initial segment, and the image is all  $W_2$ , this will finish the proof.

It is clearly an initial segment  $W_1[a]$  because it is well-ordered and if  $h : W_1[x] \cong W_2[y]$  and  $x' < x$ , then  $h : W_1[x'] \cong W_2[h(y)]$ . If the image is not all of  $W_2$ , then similarly the image of  $f$  is an initial segment of  $W_2$ ,  $= W_2[b]$ . But this means  $W_1[a] \cong W_2[b]$ , so  $a, b$  is also in the domain(image), which is a contradiction.  $\square$

### Well-Quasi-Orderings

**Def. (2.2.3.18) [Well-quasi-orderings].** A **well-quasi-ordering** on a set  $A$  is a partial ordering  $\leq$  on  $A$  satisfying the following: For any infinite sequence  $(x_i) \in A^{\mathbb{N}}$ , there exists  $i < j \in \mathbb{N}$  s.t.  $x_i \leq x_j$ .  $\perp$

**Prop. (2.2.3.19).** A partial ordering on a set  $A$  is a well-quasi-ordering if  $X$  contains neither an infinite incomparable subset or an infinite decreasing chain  $x_0 > x_1 > \dots$

Moreover, for a well-quasi-ordering, in any infinite sequence, there exists an infinite ascending chain  $x_{i_0} \leq x_{i_1} \leq \dots$  where  $i_0 < i_1 \dots$   $\perp$

*Proof:* This follows from the infinite Ramsey theorem (37.4.0.2).  $\square$

**Prop. (2.2.3.20).** If  $X$  is a quasi-well-ordered set, then we can define a quasi-well-ordering on the set  $[X]^{\leq \aleph_0}$  of finite subsets of  $X$ : for any  $A, B \in [X]^{\leq \aleph_0}$ , define  $A \leq B$  if there is an injection  $f : A \rightarrow B$  s.t.  $a \leq f(a)$  for any  $a \in A$ . Then this is a quasi-well-ordering on  $[X]^{\leq \aleph_0}$ .  $\perp$

*Proof:* Suppose  $[X]^{\leq \aleph_0}$  is not quasi-well-ordered, and call the counterexamples bad sequences. Then we can choose a bad sequence  $(A_n) \in ([X]^{\leq \aleph_0})^{\mathbb{N}}$  s.t. for any  $n \in \mathbb{N}$ ,  $A_n$  is the subset with minimum cardinality s.t.  $(A_0, A_1, \dots, A_n)$  is the start of a bad sequence. Clearly  $A_n \neq \emptyset$  for any  $n \in \mathbb{N}$ .

Choose an element  $a_i$  from each  $A_i$ , and denote  $B_i = A_i \setminus \{a_i\}$ . By hypothesis, the sequence  $(a_0, a_1, \dots)$  contains a infinite ascending sequence  $a_{n_0} \leq a_{n_1} \leq \dots$ . By the choice of  $A_i$ , the sequence

$$A_0, A_1, \dots, A_{n_0-1}, B_{n_0}, B_{n_1}, \dots$$

is not a bad sequence. Clearly such an ascending pair can only be  $B_{n_i} \leq B_{n_j}$  for some  $i < j$ . But extending any injection  $f : B_{n_i} \rightarrow B_{n_j}$  to an injection  $\bar{f} : A_{n_i} \rightarrow A_{n_j}$  by  $\bar{f}(a_{n_i}) = a_{n_j}$ , we get that  $A_{n_i} \leq A_{n_j}$ , contradiction.  $\square$

### Complete Linear Ordering

**Def. (2.2.3.21) [Complete Ordering].** A **cut** of an ordered set  $X$  consists of two disjoint nonempty subsets  $A \cup B = X$  that  $a < b$  for any  $a \in A, b \in B$ .

It is called a **Dedekind cut** if  $A$  doesn't has a maximal element. It is called a **gap** if  $A$  doesn't have a maximal element and  $B$  doesn't have a minimal element.

An ordered set is called **complete** if there are no gaps.  $\perp$

**Prop. (2.2.3.22).** Any complete ordered set  $R$  has the least upper bound property and greatest lower bound property.  $\perp$

*Proof:* Consider the cut  $A = \{x | x < a \text{ for some element in } T\}$ ,  $B = \mathbb{R} - A$ , if  $T$  is bounded above,  $B$  is not empty, so this is truly a cut, and  $A$  doesn't have a maximal element, because if  $x < a \in \mathbb{R}$ , then  $x < \frac{x+a}{2} < a$ . So by completeness of  $\mathbb{R}$ ,  $B$  has a minimal element, that is, the supremum of  $A$  exists. Similarly for the case  $A$  bounded from below.  $\square$

**Prop. (2.2.3.23)[Completion of Ordering].** There is an obvious ordering on the set  $C$  of all Dedekind cuts of  $X$ , and  $X$  embeds into  $C$  by  $b \mapsto \{x | x < b\} \cup \{x | x \geq b\}$ .

$C$  is complete and has no endpoints,  $P$  is dense in  $C$ , which is called a **completion** of  $P$ .  $\lrcorner$

*Proof:* Cf.[Set Theory Jech P88].  $\square$

**Prop. (2.2.3.24) [Real Numbers].**  $\mathbb{Q}$  has a unique completion ordering  $\mathbb{R}$ , called the **set of real numbers**.  $\lrcorner$

*Proof:*  $\mathbb{R}$  is a dense linear ordering without endpoints, so by (2.2.3.12) if it is countable then it is isomorphic to  $\mathbb{Q}$ , but this is not possible because  $\mathbb{Q}$  is not complete.  $\square$

## 4 Ordinals(Cantor)

**Def. (2.2.4.1)[Ordinal Numbers].** A set is called **transitive** iff each element of  $T$  is a subset of  $T$ . A set  $\alpha$  is called a **ordinal number** iff  $\alpha$  is transitive and well-ordered by inclusion.  $\lrcorner$

**Prop. (2.2.4.2).** If  $\alpha$  is an ordinal, then  $S(\alpha) = \alpha \cup \{\alpha\}$  is also an ordinal, obviously. Thus any natural number is an ordinal by definition.

An ordinal is called a **successor ordinal** iff  $\alpha = S(\beta)$  for some  $\beta$ , and a **limit ordinal** otherwise.

$\lrcorner$

**Lemma (2.2.4.3).**

1. If  $\alpha$  is an ordinal, then  $\alpha \notin \alpha$ .
2. Every element of an ordinal is an ordinal.
3. If ordinals  $\alpha \subsetneq \beta$ , then  $\alpha \in \beta$ . That is, for ordinals,  $\subsetneq$  is the same as  $\in$ .

$\lrcorner$

*Proof:*

1. If  $\alpha \in \alpha$ , then contradiction to the fact  $\in$  is a ordering (2.2.4.1).
2. To show  $x \in \alpha$  is transitive, it suffices to show that if  $u \in v \in x$ , then  $u \in x$ , because then  $v$  is a subset of  $x$ . But this follows from the fact  $\in$  is an ordering. And because  $x \subset \alpha$ , the inclusion of  $x$  is the restriction of inclusion in  $\alpha$ , so it is a well-ordering.
3. Consider  $\beta - \alpha$ , it has a minimal element  $\gamma$ . Notice  $\gamma \subset \alpha$ , because otherwise there is an element of  $\beta - \alpha$  smaller than  $\gamma$ , by definition (2.2.4.1).  
Now we show  $\gamma = \alpha$ , then it will follow that  $\alpha \in \beta$ . For this, if  $\delta \in \alpha$  and  $\delta \notin \gamma$ , then  $\gamma \in \delta$  or  $\gamma = \delta$ . But then this implies that  $\delta \in \alpha$  because  $\alpha$  is an ordinal, contradicting the fact  $\gamma \in \beta - \alpha$ .

$\square$

**Prop. (2.2.4.4)[Ordinal is Well-Ordered].** Define the ordering of ordinal by  $\alpha < \beta$  iff  $\alpha \in \beta$ . The ordering of ordinals is a total ordering and is a well-ordering.  $\lrcorner$

*Proof:* If  $\alpha\beta \in \gamma$ , then  $\alpha \in \gamma$  because  $\gamma$  is transitive. If  $\alpha \in \beta \in \alpha$ , then  $\alpha \in \alpha$ , contradicting (2.2.4.3).

Given any two ordinals,  $\alpha \cap \beta$  is also an ordinal by definition. If  $\alpha \cap \beta = \beta$  or  $\alpha$ , then  $\alpha \subset \beta$ , hence  $\alpha \in \beta$  by (2.2.4.3). If  $\alpha \cap \beta \subsetneq \alpha$  and  $\alpha \cap \beta \subsetneq \beta$ , then  $\alpha \cap \beta \in \alpha \cap \beta$ , contradiction.

Well-ordering: Given a set of ordinals, take  $\alpha \in A$  and consider the set  $\alpha \cap A$ . If  $\alpha \cap A = \emptyset$ , then  $\alpha$  is minimal in  $A$ , because otherwise some  $\beta \in \alpha \cap A$ . If  $\alpha \cap A \neq \emptyset$ , then it has a minimal element  $\beta$  in the inclusion because  $\alpha$  is an ordinal. Then  $\beta$  is the minimal element of  $A$ .  $\square$

**Cor. (2.2.4.5)[Supremum Ordinal].** Any set of ordinals has a supremum ordinal, it is just  $\cup_{\alpha \in X} \alpha$ .  $\lrcorner$

*Proof:* Firstly  $\cup_{\alpha \in X} \alpha$  is transitive and it is well-ordered (for each subset  $A \subset X$ , choose an  $\alpha \in X$  that  $\alpha \cap A \neq \emptyset$ , then the minimal element of  $\alpha \cap A$  is just the minimal element of  $A$ .) so it is an ordinal.

Now if  $\alpha \in X$ , then  $\alpha \subset \cup X$ , so  $\alpha \leq \cup X$  by (2.2.4.3). And if  $\alpha \in \gamma$  for some ordinal  $\gamma$ , then  $\cup X \subset \gamma$ . So  $\cup X$  is truly the supremum.  $\square$

**Cor. (2.2.4.6).** For any set  $X$  of ordinals, there is an ordinal  $\alpha$  that is not in  $X$ , just choose  $S(\cup X)$ .  $\lrcorner$

**Prop. (2.2.4.7).** Every well-ordered set is isomorphic to a unique ordinal.

So we can regard an ordinal as an equivalence class of isomorphic well-ordered sets.  $\lrcorner$

*Proof:* Cf.[Set Theory Jech P111].  $\square$

**Cor. (2.2.4.8)[Cardinal as Initial Ordinal].** The axiom of choice together with (2.2.4.4) asserts that every cardinal has a unique smallest ordinal, called the **initial ordinal**. So we can identify cardinal number  $\alpha$  as an ordinal that is the initial ordinal  $\omega_\alpha$  of  $\alpha$ . Anyway, cardinal number is fewer than ordinal numbers.

The first infinite cardinal number (or the first initial ordinal) is denoted by  $\omega$  or  $\aleph_0$ .  $\lrcorner$

**Prop. (2.2.4.9)[Transfinite Induction/Recursion].** If a property defined for the set of ordinals satisfies:

1.  $P(0)$ .
2.  $P(\alpha + 1)$  if  $P(\alpha)$ .
3.  $P(\lambda)$  if  $P(\beta)$  for all  $\beta < \lambda$ .

then  $P$  is true for all ordinals.

Transfinite recursion:  $\lrcorner$

*Proof:*  $\square$

### Ordinal Arithmetic

Cf.[Set Theory Jech Chap5.5].

**Def. (2.2.4.10).** We use infinite recursion to define **addition of ordinals** as

- $\beta + 0 = \beta$
- $\beta + (\alpha + 1) = (\beta + \alpha) + 1$ , where  $\alpha + 1$  is the successor of  $\alpha$ .
- $\beta + \alpha = \sup\{\beta + \gamma \mid \gamma < \alpha\}$  for a limit ordinal  $\alpha$ .

The **multiplication of ordinals** and **exponentiation of ordinals** are defined similarly.  $\lrcorner$



**Remark(2.2.4.11) [Cardinal and Ordinal Arithmetics].** Note that the ordinal arithmetics may be smaller than the ordinal sum of the corresponding initial ordinal(2.2.4.8), because operations of initial ordinals may not be initial, the deeper reason is that the cardinal case, we can rearrange the order to get a smaller ordinal.  $\lrcorner$

**Prop. (2.2.4.12).** The addition and multiplication of ordinals are of the order type of  $\alpha \amalg \beta$  in adjunction order and  $\alpha \times \beta$  in lexicographical order respectively, Cf.[Set Theory Jech P120,122]  $\lrcorner$

### Cantor Normal Form

**Prop. (2.2.4.13) [Cantor Normal Form].** Any ordinal  $\alpha$  can be expressed uniquely as the form  $\alpha = \sum_{i < n} \omega^{\beta_i}$ , where  $\beta_0 \geq \beta_1 \geq \dots \beta_{n-1}$  are ordinals.  $\lrcorner$

*Proof:* Cf.[Jech Set Theory P124].  $\square$

**Prop. (2.2.4.14) [Goodstein Sequence].** A **weak Goodstein sequence** is a sequence that  $m_2$  is any positive integer,  $m_{k+1}$  is  $m_k$  written in  $k$ -basis and replacing the base by  $k + 1$ , and then minus 1.

The **Goodstein sequence** is a sequence that  $m_2$  is any positive integer,  $m_{k+1}$  is  $m_k$  written in  $k$ -basis and even the exponents in  $k$ -basis and replacing the base by  $k + 1$ , and then minus 1.

Then for each Goodstein sequence and weak Goodstein sequence, it reaches 1 in a finite number of times.  $\lrcorner$

*Proof:* Let  $m_k = \sum k^{a_i} b_i$ , then let the ordinal  $\alpha_k = \sum \omega^{a_i} b_i$ . Then it is clear that  $\alpha_2 > \alpha_3 > \dots$ . But if the weak Goodstein sequence doesn't terminate, we constructed a descending sequence of ordinals that doesn't terminate, contradiction(choose a minimal element).

Similarly for Goodstein sequences, just replace every base  $k$  by  $\omega$ .  $\square$

## 5 The Axiom of Choice

**Def. (2.2.5.1) [Choice Functions].** Let  $S$  be a system of sets, a function  $g$  defined on  $S$  is called a **choice function** iff  $g(X) \in X$  for each  $X \in S$ .  $\lrcorner$

**Axiom (2.2.5.2) [the Axiom of Choice, Zermelo1904].** Any system of sets has a choice function.  $\lrcorner$

**Thm. (2.2.5.3) [Zermelo1904].** The following are equivalent:

1. the axiom of choice.
2. (the well-ordering principle) Every set can be well-ordered.
3. (Zorn's lemma) If every chain in a partially ordered sets has an upper bound, then the partially ordered set has a maximal element.

$\lrcorner$

*Proof:*  $2 \rightarrow 1$ : If  $A$  is well-ordered, then  $P(A)$  clearly has a choice function, that is the minimal element of a set.

$1 \rightarrow 2$ : Use transfinite recursion, Cf.[Set Theory Jech P137].

$2 \rightarrow 3, 3 \rightarrow 2$ : Cf.[Set Theory Jech P142].  $\square$

**Prop. (2.2.5.4).** Every infinite set  $X$  has a countable subset, if the axiom of choice holds.  $\lrcorner$

*Proof:* Choose a well-ordering of it(2.2.5.3), then it is an infinite ordinal. Then the initial segment of the first ordinal  $X[\omega]$  is a countable subset.  $\square$

**Prop. (2.2.5.5).** For every infinite set  $S$ , there exists a unique aleph  $\aleph_\alpha$  that  $|S| = \aleph_\alpha$ .  $\lrcorner$

*Proof:* choose a well-ordering of  $S$  (2.2.5.3), then it is an infinite ordinal, and it has the same cardinality as an initial cardinal by (2.2.4.8), thus the result.  $\square$

**Cor. (2.2.5.6).** For any sets  $A$  and  $B$ , either  $\#A \leq \#B$  or  $\#B \leq \#A$ .  $\lrcorner$

*Proof:* Because the ordinal is totally ordered (2.2.4.4).  $\square$

## 6 Cardinals

**Thm. (2.2.6.1) [Cantor-Schröder-Bernstein].** If there is an injection from  $A$  to  $B$  and an injection from  $B$  to  $A$ , then there is a bijection from  $A$  to  $B$ . Thus the ordering of the cardinal is well-defined.

So we can denote  $\#A \leq \#B$  iff there is an injection from  $A$  to  $B$ . Then if  $\#A \leq \#B$  and  $\#B \leq \#A$ , then  $\#A = \#B$ . In particular, this is a well-defined order relation.  $\lrcorner$

*Proof:* It  $f : A \rightarrow B$ ,  $g : B \rightarrow A$  be injection, then use lemma (2.2.6.2) for  $g \circ f(A) \subset g(B) \subset A$ .  $\square$

**Lemma (2.2.6.2).** If  $A_1 \subset B \subset A$  with  $\#A = \#A_1$ , then  $\#A = \#B$ .  $\lrcorner$

*Proof:* Let  $f$  be a bijection from  $A$  to  $A_1$ . Define inductively  $A_{n+1} = f(A_n)$ ,  $B_{n+1} = f(B_n)$ . Then  $A_{n+1} \subset B_n \subset A_n$ . Let  $C_n = A_n - B_n$ ,  $C = \cup C_n$ , then  $f(C_n) = C_{n+1}$ , so  $f(C) = \cup_{i>0} C_i$ .

Now define  $g : A \rightarrow B = f(x)$  on  $C$  and  $x$  on  $A \setminus C$ , then it is a bijection from  $A$  to  $B$ .  $\square$

**Def. (2.2.6.3) [Cardinal Numbers].** A **cardinal number** is an equivalence class of sets, where equivalence is given by bijections. it is used to describe the ‘size’ of a set. We use the sentence  $\kappa \in \mathcal{Card}$  to mean that  $\kappa$  is a cardinal.

It is by the axiom of choice that any two cardinal number can be compared. ?

Denote  $\aleph_0 = \#\mathbb{N}$  (2.2.1.3).  $\lrcorner$

### Peano Arithmetics

#### Countable and Uncountable Sets

**Def. (2.2.6.4) [Countable Sets].** A set is called **countable** iff it has cardinality  $\aleph_0$  (2.2.6.3). It is called a **finite set** if it has the cardinality of  $n$  for some natural number  $n \in \mathbb{N}$ . It is called an **uncountable set** iff it is not countable or finite. It is called an **at most countable set** if it is finite or countable.  $\lrcorner$

**Prop. (2.2.6.5).** The subset or image of an at most countable set is at most countable.  $\lrcorner$

*Proof:*  $\square$

**Prop. (2.2.6.6).** The product of two at most countable sets is at most countable. (Use diagonal enumerating).  $\lrcorner$

*Proof:*  $\square$

**Prop. (2.2.6.7).** A countable union of almost countable subsets is almost countable.  $\lrcorner$

*Proof:* It suffices to prove the countable case, the rest follows from (2.2.6.5). For this, choose an enumerating  $a_n(k)$  for each  $A_n$ , the  $\cup A_n$  is the image of  $\mathbb{N} \times \mathbb{N} : (n, k) \mapsto a_n(k)$ . Then it is countable by (2.2.6.6).  $\square$

**Prop. (2.2.6.8).** The set of finite sequences and hence the set of finite subsets of a countable set is countable.  $\lrcorner$

*Proof:* The desired set equals  $\cup_k A^k$ , which is countable by (2.2.6.6) and (2.2.6.7).  $\square$

## 7 Cardinal Arithmetics

**Def. (2.2.7.1).** The **sum**, **multiplication** and **exponentiation** of two ordinal is the cardinality of the set  $A \coprod B$ ,  $A \times B$  or  $A^B$  respectively.

It is easily verified to be associative and commutative, just as usual operations.  $\lrcorner$

**Prop. (2.2.7.2).**  $\aleph_0 \times \aleph_0 = \aleph_0$  by (2.2.6.6). And  $\kappa \times \kappa = \kappa$  for any infinite cardinal, if one uses the axiom of choice by (2.2.8.3).  $\lrcorner$

**Prop. (2.2.7.3).** The image of a set  $X$  has cardinals no more than  $X$ , if axiom of choice holds.  $\lrcorner$

*Proof:* Use axiom of choice to choose an element from each inverse image  $f^{-1}(\{x\})$ , then it is an injection from  $f(X)$  to  $X$ .  $\square$

**Prop. (2.2.7.4) [Cantor].**  $\#P(X) = 2^{\#X}$ , and  $\#X < \#P(X)$ .  $\lrcorner$

*Proof:* The first is obvious, for the second, the function  $x \rightarrow \{x\}$  is an injection of  $X$  into  $P(X)$ . And there are no mapping from  $X$  onto  $P(X)$ , because if  $f$  is one, then consider  $S = \{x | x \notin f(x)\}$ , then  $S$  is not in the range of  $f$ , because if  $f(z) = S$ , then  $z \in S$  iff  $z \notin S$ , contradiction.  $\square$

**Prop. (2.2.7.5) [Cardinality Arithmetic of  $\aleph_0$ ].** For cardinality arithmetics involving  $\aleph_0$ , Cf.[Set Theory Jech P98].  $\lrcorner$

*Proof:*  $\square$

**Prop. (2.2.7.6) [Cardinality of  $\mathbb{R}$ ].**  $\#\mathbb{R} = \#P(\mathbb{N}) = 2^{\aleph_0}$ . In particular, by (2.2.7.4),  $\mathbb{R}$  is uncountable.  $\lrcorner$

*Proof:* The first equality is by (2.2.7.4). Now by the construction of  $\mathbb{R}$ , it can be embedded into  $P(\mathbb{N})$ , so  $\#\mathbb{R} \leq \#P(\mathbb{Q}) = \#P(\mathbb{N})$ . Conversely,  $|2^{\mathbb{N}}| \leq |\mathbb{R}|$  by decimal representation, so they are equal by bernstein (2.2.6.1).  $\square$

**Prop. (2.2.7.7).** if  $\#B = 2^{\aleph_0}$  and  $\#A \leq \aleph_0$ , then  $|B \setminus A| = 2^{\aleph_0}$ . In fact,  $|B - A| = |B|$  for any  $\#A < \#B$ , if one uses the axiom of choice.  $\lrcorner$

*Proof:* By (2.2.7.5), we can assume  $B = \mathbb{R} \times \mathbb{R}$ , then project  $A$  onto the coordinate axis, then  $\pi(A)$  has cardinality  $\leq \aleph_0$ , so there is a  $x_0 \notin \pi(A)$ , so  $x_0 \times \mathbb{R} \subset B - A$ , so  $|B - A| = 2^{\aleph_0}$ .

For the general case, ?  $\square$

**Conj. (2.2.7.8) [The Continuum Hypothesis, Cantor1878].** There is no cardinal  $\kappa$  that  $\aleph_0 < \kappa < 2^{\aleph_0}$ .

Notice  $2^{\aleph_0} \geq \aleph_1$  by Cantor's theorem (2.2.7.4), and this hypothesis is equivalent to  $2^{\aleph_0} = \aleph_1$ .  $\lrcorner$

*Proof:*  $\square$

For Infinite operation of Cardinal Arithmetics, Cf.[Jec03]Chap9.

## 8 Alephs

**Prop. (2.2.8.1).** For any set  $A$ , there is a least ordinal that is not equipotent to any subset of  $A$ , called the **Hartogs number** of  $A$ . This is clearly an initial ordinal.  $\lrcorner$

*Proof:* By axiom schema of replacement, any well-ordered subsets of  $A$  is equipotent to an ordinal, and also by axiom schema of replacement, there is a set  $H$  that for any well-ordering of subsets of  $A$  in  $P(A \times A)$ , this ordered sets is equipotent to a  $\alpha \in H$ . Then use (2.2.4.6) to find a minimal ordinal that is not equipotent to any subset of  $A$ . In fact, this is just  $h(A) = \{\alpha \in H \mid \alpha \text{ equipotent to some subset of } A\}$ .  $\square$

**Def. (2.2.8.2)[Aleph].** The **alephs** for ordinal numbers are defined recursively:  $\aleph_0 = \omega$ ,  $\aleph_{\alpha+1} = h(\aleph_\alpha)$ , and  $\aleph_\alpha = \sup\{\aleph_\beta \mid \beta < \alpha\}$  for a limit ordinal  $\alpha$ . By definition  $\alpha_\alpha < \aleph_\beta$  when  $\alpha < \beta$ .

Then  $\aleph_\alpha$  are all infinite initial ordinal numbers, and any infinite ordinal number is of the form  $\aleph_\alpha$  for some ordinal  $\alpha$ . So natural numbers together with alephs are just all the cardinal numbers.

Notice: to avoid confusion, when do arithmetic of ordinal numbers,  $\aleph_\alpha$  is written as  $\omega_\alpha$ .  $\lrcorner$

*Proof:* Use transfinite induction on  $\alpha$ . The only nontrivial case is when  $\alpha$  is a limit ordinal, where if  $\gamma < \aleph_\alpha$  and  $|\gamma| = |\aleph_\alpha|$ , then there is a  $\beta < \alpha$  that  $\gamma \leq \aleph_\beta$  by definition, so  $|\aleph_\alpha| < |\gamma| \leq |\aleph_\beta| < |\aleph_\alpha|$  as  $\aleph_\beta$  is an initial ordinal.

To prove that any infinite initial ordinal is an aleph, first notice that  $\alpha < \aleph_\alpha$  by a simple transfinite induction. So we may use transfinite induction on the following assertion for  $\alpha$ : if  $\Omega < \aleph_\alpha$ , then there is a  $\gamma < \alpha$  that  $\Omega = \aleph_\gamma$ . For this,  $\alpha = 0$  is trivially true, if  $\alpha = \beta + 1$ , then  $\Omega < h(\aleph_\alpha)$  implies that  $|\Omega| < |\aleph_\alpha|$  by definition. Because  $\Omega$  is initial,  $\Omega = \aleph_\beta$  or  $\Omega < \aleph_\beta$ , so by induction hypothesis it is true. If  $\alpha$  is a limit ordinal, then  $\Omega < \omega_\beta$  for some  $\beta < \alpha$ , so also by induction hypothesis it is true.  $\square$

### Aleph Arithmetics

**Prop. (2.2.8.3).**  $\aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha$ .  $\lrcorner$

*Proof:* Cf.[Set Theory Jech P134].  $\square$

**Cor. (2.2.8.4).**  $\aleph_\alpha \cdot \aleph_\beta = \aleph_\beta$  for  $\alpha \leq \beta$ , and  $n \cdot \aleph_\beta = \aleph_\beta$ .  
So  $\aleph_\alpha + \aleph_\beta = \aleph_\beta$  for  $\alpha \leq \beta$ , and  $n + \aleph_\beta = \aleph_\beta$ .  $\lrcorner$

## 9 Natural Numbers and Real Numbers

**Prop. (2.2.9.1).**  $\mathbb{R}$  is the unique ordered field in which every non-empty bounded set has a least upper bound.  $\lrcorner$

*Proof:*  $\square$

**Prop. (2.2.9.2).**  $(\mathbb{N}, <)$  (2.2.1.5) is a well-ordered set.  $\lrcorner$

*Proof:*  $\square$

**Example (2.2.9.3)[Examples of Countable Sets].**

- $\mathbb{Z}$ .
- $\mathbb{Q}$ .

$\lrcorner$

### Arithmetic of Real numbers

**Prop. (2.2.9.4).** The set of real numbers  $\mathbb{R}$ (2.2.3.24) can be endowed with a field structure, making it an ordered field.  $\lrcorner$

*Proof:* Cf.[Set Theory Jech P175].  $\square$

**Prop. (2.2.9.5).**  $\mathbb{R}$  satisfies the least upper bound hypothesis.  $\lrcorner$

*Proof:*  $\square$

## 10 Filters and Ultrafilters

**Def. (2.2.10.1) [Filter].** For a poset  $P$ , a **filter** on  $P$  is a subset  $F$  that

- If  $p < q$  and  $p \in F$ , then  $q \in F$ .
- If  $p, q \in F$ , then there is an  $r \in F$  that  $r < p, r < q$ .

$\lrcorner$

**Def. (2.2.10.2) [Filter on Sets].** Let  $S$  be a non-empty set, a **filter** on  $S$  is a filter  $F$  on  $\mathcal{P}(S)$  that  $\emptyset \notin F$ .

an **ideal** on  $S$  is a collection  $F$  of subsets of  $S$  that:

- $\emptyset \in F$  and  $S \notin F$ .
- If  $X, Y \in F$ , then  $X \cup Y \in F$ .
- If  $X \in F, X \supset Y$ , then  $Y \in F$ .

An ideal is just the dual(complement) of a filter.  $\lrcorner$

**Def. (2.2.10.3) [Finite intersection property].** A family of subsets of a set is said to have the **finite intersection property** if any finite collection of elements of this family is non-empty.  $\lrcorner$

**Lemma (2.2.10.4).** let  $G$  be a collection of subsets of  $S$  that has the finite intersection property(2.2.10.3), then there is a smallest filter  $F$  that  $G \subset F$ . It is just the collection of subsets of  $S$  that contain some finite intersection set of elements of  $G$ .  $\lrcorner$

**Def. (2.2.10.5) [Ultrafilter].** An **ultrafilter** is a filter  $F$  that for every subset  $X$ ,  $X \in F$  iff  $S - X \notin F$ . A **prime ideal** is an ideal that for every subset  $X$ ,  $X \in F$  iff  $S - X \notin F$ .

A ultrafilter is equivalent to a maximal filter. And it is equivalent to a  $\{0,1\}$ -valued finitely additive measure on  $S$ .  $\lrcorner$

*Proof:* If  $F$  is an ultrafilter, then it is maximal, because any larger filter will have some  $X, S - X$ , thus has  $\emptyset$ , contradiction.

Conversely, if  $F$  is maximal filter but not ultra, then there is a  $X$  that  $X \notin F, S - X \notin F$ . Let  $G = F \cup \{X\}$ , then any finite intersection of elements of  $G$  is not empty:  $X_1 \cap \dots \cap X_n \cap X \neq \emptyset$  otherwise  $S - X \in F$ . So there is a filter containing  $G$  by(2.2.10.4), contradiction.  $\square$

**Prop. (2.2.10.6) [Pushforward of Filters].** If  $\mathcal{F}$  is a(n) (ultra)filter on  $X$  and  $f : X \rightarrow Y$  is a function, then  $f_*(\mathcal{F}) = \{A \subset Y | f^{-1}(A) \in \mathcal{F}\}$  is a(n) (ultra)filter on  $Y$ , called the **pushforward filter** of  $\mathcal{F}$ .  $\lrcorner$

**Prop. (2.2.10.7).** For an ultrafilter  $\mathcal{F}$  on  $X$ , if  $U_i \notin \mathcal{F}$ , then  $\sum_{i=1}^n U_i \notin \mathcal{F}$ .  $\lrcorner$

*Proof:* As  $X - U_i \in \mathcal{F}$ , there intersection are in  $\mathcal{F}$ , so its complement is not in  $\mathcal{F}$ .  $\square$

**Prop. (2.2.10.8).** Any filter can be extended to an ultrafilter(maximal filter), if the axiom of choice is used.  $\lrcorner$

*Proof:* Use Zorn's lemma(2.2.5.3). It suffices to prove that a union of a chain of filters is a filter, which is trivial.  $\square$

**Cor. (2.2.10.9).** Non-principal ultrafilter exists on any infinite set. And in fact, any non-principal ultrafilter contains all the cofinite sets.

For any non-principle ultrafilter, it cannot contains a single pt  $\{x\}$ , so it contains every cofinite set.  $\lrcorner$

*Proof:* Consider any ultrafilter containing the filter of cofinite sets of  $S$ , then it is non-principal.  $\square$

**Def. (2.2.10.10) [ $\kappa$ -Completeness].** Let  $\kappa$  be an uncountable cardinal, then a field  $F$  on a set  $S$  is called  $\kappa$ -**complete** if for every cardinal  $\lambda < \kappa$ , if  $X_\alpha \in F$  for every  $\alpha < \lambda$ , then  $\bigcap_{\alpha < \lambda} X_\alpha \in F$ .

An  $\aleph_1$ -complete filter is also called a  $\sigma$ -complete filter.  $\lrcorner$

## Closed unbounded and Stationary Set

### Silver's Theorem

#### 11 Models

Cf.[Axiomatic Set Theory]P12.

**Def. (2.2.11.1).**  $\lrcorner$

### Absoluteness

#### 12 Large Cardinals

**Def. (2.2.12.1) [Abstract Measures].** Let  $S$  be a non-empty set, then an **abstract measure** on  $S$  is a non-trivial probabilistic measure  $\mu$  on the measurable space  $(S, \mathcal{P}(S))$  that  $\mu(\{a\}) = 0$  for any  $a \in S$ .  $\lrcorner$

**Def. (2.2.12.2) [Regular Cardinal].** A cardinal is called **regular** if it is not a sum of  $\lambda$  cardinals  $\kappa_i$  that  $\lambda < \kappa$  and  $\kappa_i < \kappa$ .  $\lrcorner$

**Def. (2.2.12.3) [Strong Limit].** A cardinal  $\kappa$  is called a **strong limit** if  $2^\lambda < \kappa$  for any  $\lambda < \kappa$ (2.2.12.3).  $\lrcorner$

**Def. (2.2.12.4) [Strongly Inaccessible Cardinal].** A cardinal  $\kappa$  is called **strongly inaccessible(SI)** if it is regular(2.2.12.2) and is a strongly limit.  $\lrcorner$

**Def. (2.2.12.5) [Weakly Inaccessible Cardinal].** A cardinal  $\kappa$  is called **weakly inaccessible** if it is regular and is a limit cardinal.  $\lrcorner$

**Prop. (2.2.12.6) [Measure and CH].** If there exists an abstract measure on  $2^{\aleph_0}$ , then the Continuum Hypothesis(2.2.7.8) fails.  $\lrcorner$

*Proof:* Cf.[Jec03]P242.  $\square$

**Prop. (2.2.12.7).** If there is a measure on a set  $S$ , then some cardinal  $\kappa \leq |S|$  is weakly inaccessible.  $\lrcorner$

*Proof:* Cf.[Jec03]P243. □

**Prop. (2.2.12.8).** Let  $\mu$  be a  $\{0, 1\}$ -valued measure on  $S$ , then  $U = \{X \subset S \mid \mu(X) = 1\}$  is a non-principal  $\sigma$ -complete ultrafilter of  $S$  ┘

**Prop. (2.2.12.9) [Stanislaw-Ulam Dichotomy].** If there exists a measure on some set, then either there exists a  $\{0, 1\}$ -valued measure on some set, or there exists a measure on  $2^{\aleph_0}$ . ┘

*Proof:* Cf.[Jec03]P245. □

**Def. (2.2.12.10) [Measurable Cardinals].** A **measurable cardinal** is an uncountable cardinal  $\kappa$  on which there exists a non-principal  $\kappa$ -complete ultrafilter. ┘

**Prop. (2.2.12.11).** A measurable cardinal is strongly inaccessible. ┘

*Proof:* Cf.[Jec03]P247. □

### 13 Gödel Model

**Thm. (2.2.13.1) [Gödel Incompleteness Theorem].** For any set of axioms for elementary number theory, there will always be statements which can be formulated in an elementary way, and which are decidable, but which can not be deduced using only elementary methods. ┘

*Proof:* □

**Remark (2.2.13.2).** This theorem gives a reason why many theorems, for examples the Fermat's last theorem, must be proven using analytic methods or geometric methods. ┘

### 14 Silver Machine

### 15 Forcing

**Def. (2.2.15.1) [Dense Subset].** In a poset  $P$ , a subset  $D \subset P$  is called **dense** if for any  $p \in P$ , there is a  $q \in D$  that  $q < p$ . If  $\mathcal{D}$  is a collection of dense subsets of  $P$ , a filter  $G \subset P$  is called  **$\mathcal{D}$ -generic** if  $D \cap G \neq \emptyset$  for all  $D \in \mathcal{D}$ . ┘

**Prop. (2.2.15.2).** If  $\mathcal{D}$  is a countable collection of dense subsets of  $P$ , then there is a  $\mathcal{D}$ -generic filter  $G$ . ┘

*Proof:* Let  $\mathcal{D} = \{D_1, \dots, D_n, \dots\}$ . Choose  $p_0 \in P$ , and consecutively choose  $p_n \leq p_{n-1}$  that  $p_n \in D_n$ , and define  $G = \{q \mid q \geq p_n \text{ for some } n\}$ . □

**Axiom (2.2.15.3) [Martin's Axiom].** If  $P$  is a partially ordered set satisfying the countable chain condition (2.2.3.7), and  $\mathcal{D}$  is a collection of dense subsets of  $P$  with  $|\mathcal{D}| < 2^{\aleph_0}$ , then there is a  $\mathcal{D}$ -generic filter on  $P$ . ┘

### 16 Determinacy

### 17 Stationary Set

## 2.3 Homotopy Type Theory

Constructive type theory tries to solve the following question: If any foundation of mathematics (a formal system) are doomed to be inconsistent, can a formal system still give us useful information in spite of being inconsistent? If we are not allowed to prove by contradiction or by negation of negation, then it is possible to have a formal system that may not be consistent but more amenable to computer verifications.



## 2.4 Model Theory

References are [Model Theory Marker]. The exercises of [Model Theory Marker] are important.

**Notation (2.4.0.1).**

- Use notations defined in [Set Theory](#).
- All propositions from now are wffs in class theory (or with choice or large cardinal), so there are no metatheorems.

┘

### 1 Set Models

#### Basics

**Def. (2.4.1.1) [Structures].** Given a mathematical language  $\mathcal{L}$  (2.1.6.1), an  $\mathcal{L}$ -**structure** on a set  $M$  is an assignment for each constant symbol  $c$  an element  $c^M \in M$ , for each function symbol  $f$  of arity  $n$  a function  $f^M : M^n \rightarrow M$ , and for each relation symbol  $R$  of arity  $m$  a subset  $R^M \subset M^m$ . These  $c^M, f^M, R^M$  are called **interpretations** of  $\mathcal{L}$ .

And there is a natural definition of morphisms of  $\mathcal{L}$ -structures, and an injective morphism of  $\mathcal{L}$ -structures is called an **embedding** or a **structure extension**. ┘

**Def. (2.4.1.2) [Satisfaction].** Let  $\varphi$  be a formula with free variables  $\bar{v} = (v_{i_1}, \dots, v_{i_m})$ , then we inductively define  $\mathcal{M} \models \varphi(\bar{a})$  as follows:

- If  $\varphi$  is  $t_1 = t_2$  where  $t_1, t_2$  are terms, then  $\mathcal{M} \models \varphi(\bar{a})$  if  $\sqcup_1^{\mathcal{M}}(\bar{a}) = t_2^{\mathcal{M}}(\bar{a})$ .
- If  $\varphi$  is  $R(t_1, \dots, t_n)$ , then  $\mathcal{M} \models \varphi(\bar{a})$  if  $(t_1^{\mathcal{M}}(\bar{a}), \dots, t_n^{\mathcal{M}}(\bar{a})) \in R^{\mathcal{M}}$ .
- If  $\varphi$  is  $\neg\psi$ , then  $\mathcal{M} \models \varphi(\bar{a})$  if  $\mathcal{M} \not\models \psi(\bar{a})$ .
- If  $\varphi$  is  $\psi \wedge \theta$ , then  $\mathcal{M} \models \varphi(\bar{a})$  if  $\mathcal{M} \models \psi(\bar{a})$  and  $\mathcal{M} \models \theta(\bar{a})$ .
- If  $\varphi$  is  $\exists v_j \psi(\bar{v}, v_j)$ , then  $\mathcal{M} \models \varphi(\bar{a})$  if there is a  $b \in M$  that  $\mathcal{M} \models \psi(\bar{a}, b)$ .

and we say  $M$  **satisfies**  $\varphi(\bar{a})$ , or  $\varphi(\bar{a})$  is true in  $M$ . Notice if there is no free variables,  $\varphi(\bar{a})$  just writes  $\varphi$ . ┘

**Def. (2.4.1.3) [Mathematical  $\mathcal{L}$ -Theory].** Let  $\mathcal{L}$  be a mathematical language, then a (mathematical)  $\mathcal{L}$ -**theory** is a formal system that is a consistent extension of  $\mathcal{K}_{\mathcal{L}}$  (2.1.5.2). ┘

**Def. (2.4.1.4) [Models].** If  $T$  is an  $\mathcal{L}$ -theory and  $\mathcal{M}$  a set with  $\mathcal{L}$ -structure satisfying all theorems  $\varphi \in T$ , then  $\mathcal{M}$  is called a **model** of  $T$ , and writes  $M \models T$ .

A theory  $T$  is called **satisfiable** iff there is a model  $\mathcal{M}$  for  $T$ .

A set of  $\mathcal{L}$ -structures  $\mathcal{K}$  is called an **elementary class** iff there is an  $\mathcal{L}$ -theory  $T$  that  $\mathcal{K} = \{\mathcal{M} \mid \mathcal{M} \models T\}$ .

Given a  $\mathcal{L}$ -structure on  $M$ , the **theory of  $M$**  is the set of all sentences true in  $M$ .

Two  $\mathcal{L}$ -structures  $\mathcal{M}$  and  $\mathcal{N}$  are called **elementary equivalent**, denoted by  $\mathcal{M} \equiv \mathcal{N}$ , if for all  $\mathcal{L}$ -sentences  $\varphi$ ,  $\mathcal{M} \models \varphi \iff \mathcal{N} \models \varphi$ . ┘

**Prop. (2.4.1.5).** If  $T$  is a mathematical  $\mathcal{L}$ -theory given by axioms, and every axiom of  $M$  is a set with  $\mathcal{L}$ -structure satisfying every axiom, then  $\mathcal{M}$  is a model of  $T$ . ┘

*Proof:* This follows from induction on the length of the proof of a theorem. ┘

**Prop. (2.4.1.6).** A mathematical  $\mathcal{L}$ -theory is consistent iff it has a model.  $\lrcorner$

*Proof:* Cf.[Hamilton]P91+98. ?  $\square$

**Prop. (2.4.1.7) [Quantifier-Free Formulae].** Suppose  $\mathcal{M}$  is a substructure of  $\mathcal{N}$  and  $\bar{a}$  is a tuple in  $\mathcal{M}$ . If  $\varphi(\bar{v})$  is a quantifier-free formula, then  $\mathcal{M} \models \varphi(\bar{a})$  iff  $\mathcal{N} \models \varphi(\bar{a})$ .  $\lrcorner$

*Proof:* Cf.[Model Theory P11].  $\square$

**Prop. (2.4.1.8).** If  $\mathcal{L}$ -structures  $\mathcal{M} \cong \mathcal{N}$ , then  $\mathcal{M}$  is elementarily equivalent to  $\mathcal{N}$ .  $\lrcorner$

*Proof:* This seemingly trivial proposition still needs proof, and the proof uses induction, just as that of (2.4.1.7).  $\square$

**Def. (2.4.1.9) [Logical Consequence].** Let  $T$  be an  $\mathcal{L}$ -theory and  $\varphi$  an  $\mathcal{L}$ -sentence, then  $\varphi$  is called a **logical consequence** of  $T$ , writes  $T \models \varphi$ , if for any  $\mathcal{L}$ -structure  $\mathcal{M}$  that  $\mathcal{M} \models T$ ,  $\mathcal{M} \models \varphi$ .  $\lrcorner$

### Definable Sets and Interpretability

**Def. (2.4.1.10) [Definable Sets].** Let  $\mathcal{M}$  be an  $\mathcal{L}$ -structure (2.4.1.1), a subset  $X \subset \mathcal{M}^n$  is called a **definable set** iff there is an  $\mathcal{L}$ -formula  $\varphi(v_1, \dots, v_n, w_1, \dots, w_m)$  and a tuple  $\bar{b} \in \mathcal{M}^m$  that  $X = \{\bar{a} \in \mathcal{M}^n : \mathcal{M} \models \varphi(\bar{a}, \bar{b})\}$ . Moreover if  $A \subset \mathcal{M}$ ,  $X \subset \mathcal{M}^n$  is called an **A-definable set** iff  $y_i \in A$ .  $\lrcorner$

**Prop. (2.4.1.11) [Examples of Definable Sets].** The definability of some sets are often nontrivial, using many number theories. For example, Cf.[Marker P20].  $\lrcorner$

**Prop. (2.4.1.12).** There is an inductive characterization of definable sets, Cf.[Marker P22].  $\lrcorner$

**Prop. (2.4.1.13).** If  $\mathcal{M}$  is an  $\mathcal{L}$ -structure, If  $X \subset \mathcal{M}^n$  is A-definable, then every  $\mathcal{L}$ -automorphism of  $\mathcal{M}$  that fixes A pointwise will fix X setwise.  $\lrcorner$

*Proof:* For an automorphism  $\tau$  of  $\mathcal{M}$ ,  $\mathcal{M} \models \varphi(\bar{b}, \bar{a})$  iff  $\mathcal{M} \models \varphi(\tau(\bar{b}), \tau(\bar{a})) = \varphi(\tau(\bar{b}), \bar{a})$ .  $\square$

**Cor. (2.4.1.14).**  $\mathbb{R}$  is not definable in  $\mathbb{C}$ .  $\lrcorner$

*Proof:* If  $\mathbb{R}$  is definable, it is definable over a finite set  $A \subset \mathbb{C}$ , Let  $r, s$  be algebraically independent over  $\mathbb{A}$  and  $r \in \mathbb{R}, s \notin \mathbb{R}$ . This can be done, otherwise  $\mathbb{C}$  or  $\mathbb{R}$  is finite transcendental over  $\mathbb{Q}$ , then  $|\mathbb{C}| = |\mathbb{Q}|$  (3.2.7.3), which is impossible by (2.2.3.24). Then there is an automorphism  $\sigma$  of  $\mathbb{C}$  fixing A that  $\sigma(r) = s$ , so  $\mathbb{R}$  is not definable by (2.4.1.13).  $\square$

**Def. (2.4.1.15) [Definably Interpretability].** An  $\mathcal{L}_0$ -structure  $\mathcal{N}$  is called **definably interpreted** in an  $\mathcal{L}$ -structure  $\mathcal{M}$  if there is a definable set  $X \subset \mathcal{M}^n$  that we can interpret the symbols of  $\mathcal{L}_0$  as definable subsets and functions of X and the resulting  $\mathcal{L}_0$ -structure is isomorphic to  $\mathcal{N}$ .

The usual example is that the group structure of  $GL_2(K)$  is definably interpreted in the ring structure of a field.  $\lrcorner$

**Def. (2.4.1.16) [Interpretability and Quotient Construction].** An  $\mathcal{L}_0$ -structure  $\mathcal{N}$  is called **interpretable** in an  $\mathcal{L}$ -structure  $\mathcal{M}$  iff there is a definable set  $X \in \mathcal{M}^n$  and a definable equivalence relation  $E$  on X, that we can interpret the symbols of  $\mathcal{L}_0$  as definable subsets and functions of  $X/E$  and the resulting  $\mathcal{L}_0$ -structure is isomorphic to  $\mathcal{N}$ .

The usual example is that the set structure of a projective space is interpretable in the ring structure of a field.  $\lrcorner$

**Prop. (2.4.1.17).** Any structure for a countable language can be interpreted in a graph.  $\lrcorner$

*Proof:*  $\square$

## 2 Basic Techniques

**Def. (2.4.2.1) [Definitions for Theories].** A theory  $T$  is said to have the **witness property** iff whenever  $\varphi(v)$  is an  $\mathcal{L}$ -formula with one free variable  $v$ , there is a constant symbol  $c$  that  $T \models (\exists v \varphi(v) \rightarrow \varphi(c))$ .

A theory  $T$  is called **maximal** iff for any  $\mathcal{L}$ -sentence, either  $\varphi \in T$  or  $\neg\varphi \in T$ .

A theory  $T$  is called **complete** iff for any  $\mathcal{L}$ -sentence, either  $T \models \varphi$  or  $T \models \neg\varphi$ .  $\lrcorner$

**Def. (2.4.2.2) [Consistent Theory].** A theory  $T$  is called **inconsistent** if there is a sentence  $\varphi$  that  $T \vdash \varphi \wedge \neg\varphi$ , otherwise it is called **consistent**.  $\lrcorner$

**Def. (2.4.2.3) [Recursiveness and Decidability].** A language  $\mathcal{L}$  is called **recursive** iff there is an algorithm that decides whether a sequence of symbols is an  $\mathcal{L}$ -formula.

An  $\mathcal{L}$ -theory  $T$  is called **recursive** iff there is an algorithm that decides whether a given  $\mathcal{L}$ -sentence is in  $T$ .

An  $\mathcal{L}$ -theory  $T$  is called **decidable** iff there is an algorithm that decides whether a given  $\varphi$  satisfies  $T \models \varphi$ .  $\lrcorner$

**Prop. (2.4.2.4).** If  $\mathcal{L}$  is a recursive language and  $T$  is a recursive  $\mathcal{L}$ -structure, then  $\{\varphi | T \vdash \varphi\}$  is recursively enumerable.  $\lrcorner$

*Proof:* There is a computable listing  $\sigma_1, \dots, \sigma_n \dots$  of all the finite sequences of  $\mathcal{L}$ -formulas, because  $\mathcal{L}$  is recursive. Then we can check at each stage iff  $\sigma_i$  is a proof of  $\varphi$ . This involves checking if each formula is in  $T$  (checkable because  $T$  is recursive) or it is a simple consequences of formulae before it, and finally check the last formula is  $\varphi$ . If  $\sigma_i$  is a proof of  $\varphi$ , then halt, otherwise go on to check  $\sigma_{i+1}$ .  $\square$

**Prop. (2.4.2.5).** The halting computation set is not computable.  $\lrcorner$

*Proof:* Cf. [Mathematical Logic Shoenfield] ?.  $\square$

**Cor. (2.4.2.6).** The full theory  $Th(\mathbb{N})$  of the ring structure of  $\mathbb{N}$  is undecidable.  $\lrcorner$

*Proof:* If such an algorithm exists, then we can use it to compute whether the sentence

$$\varphi(e, x) = \exists s T(\underbrace{1 + \dots + 1}_{e\text{-times}}, \underbrace{1 + \dots + 1}_{x\text{-times}}, s)$$

is computable. Then this will contradict the fact that halting computation set is not computable (2.4.2.5).  $\square$

**Prop. (2.4.2.7) [Gödel's Completeness Theorem].** Let  $T$  be an  $\mathcal{L}$ -theory and  $\varphi$  is an  $\mathcal{L}$ -sentence, then  $T \models \varphi$  iff  $T \vdash \varphi$ .  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (2.4.2.8) [Consistent and Satisfiable].** A theory  $T$  is consistent iff it is satisfiable.  $\lrcorner$

*Proof:* If  $T$  is satisfiable, then it is clearly consistent, and if  $T$  is not satisfiable, then there are no models for  $T$ , so  $T \models \varphi \wedge \neg\varphi$  by definition, so  $T \vdash \varphi \wedge \neg\varphi$  by Gödel's completeness theorem.  $\square$

**Cor. (2.4.2.9) [Lemma on Constants].** Suppose  $T \vdash \varphi(\bar{c})$  and  $\bar{c}$  is a tuple of constants not appearing in  $T$ , then  $T \vdash \forall x \varphi(x)$ .  $\lrcorner$

*Proof:* We use the Gödel's completeness theorem (2.4.2.7), and notice this theorem is obviously true for  $\vdash$  replaced by  $\models$ . Notice we can add the constants to  $\mathcal{L}$ , thus  $\varphi(\bar{c})$  has no free variables.  $\square$

### Ultraproducts of Theories

**Def. (2.4.2.10) [Ultraproducts of Theories].** If  $M_i, i \in I$  is a collection of  $\mathcal{L}$ -structures and  $\mathcal{F}$  is an ultrafilter on  $I$ , then the **ultraproduct**  $\prod_I M_i / \mathcal{F}$  of  $M_i$  is a  $\mathcal{L}$ -structure defined as:

- The underlying set  $M = \prod M_i / \sim$ , where  $(a_i) \sim (b_i)$  iff  $\{i \in I \mid a_i = b_i\} \in \mathcal{F}$ , which is an equivalent relation.
- If  $c$  is a constant symbol, then  $c^M = (c^{M_i})$ .
- If  $f$  is a function symbol, then  $f^M([a_{i1}], \dots, [a_{in}]) = [f^{M_i}(a_{i1}, \dots, a_{in})]$ .
- If  $R$  is a relation symbol, then  $([a_{i1}], \dots, [a_{in}]) \in R^M$  iff  $\{i \in I \mid (a_{i1}, \dots, a_{in}) \in R^{M_i}\} \in \mathcal{F}$ .

┘

**Prop. (2.4.2.11) [Los Theorem].** Let  $M_i, i \in I$  be  $\mathcal{L}$ -structures and  $\mathcal{F}$  be an ultrafilter on  $I$ . Let  $\varphi(\bar{x})$  be a first-order logic formula in the free variables  $\bar{x}$ , and let  $\overline{[a_i]}$  be a tuple of elements from  $\prod_I M_i / \mathcal{F}$ , then

$$\prod_I M_i / \mathcal{F} \models \varphi(\overline{[a_i]}) \iff \{i \in I \mid M_i \models \varphi(\bar{a}_i)\} \in \mathcal{F}.$$

┘

*Proof:* The proof is by induction, which is routine, Cf.[Model Theory for Algebra and Algebraic Geometry, P22]. □

**Cor. (2.4.2.12).** An ultrapower of models for a theory  $T$  is also a model for  $T$ . ┘

**Cor. (2.4.2.13) [Non-Standard Model for  $\text{Th}(\mathbb{R})$ ].** Consider  $\mathbb{R}$  in the language  $\mathcal{L}_r$ , where  $\mathcal{L}_r$  is the language of rings, (i.e., the ring structure), let  $\mathcal{F}$  be a non-principle ultrafilter on  $\mathbb{N}$  (2.2.10.9), and consider the ultrapower  $\mathcal{R} = \prod_{i \in \mathbb{N}} \mathbb{R} / \mathcal{F}$ , which is called an **ultrapower** of  $\mathbb{R}$ .

Notice each factor satisfies  $\text{Th}(\mathbb{R})$ , so  $\mathcal{R}$  also satisfies  $\text{Th}(\mathbb{R})$ . ┘

**Cor. (2.4.2.14).** Using ultrapower construction, we can find a field of characteristic 0 that has exactly one algebraic extension in each degree. ┘

*Proof:* Just use the field model  $\mathbb{F}_p$  for all  $p$  and construct their ultrapower w.r.t. a non-principal ultrafilter. □

**Prop. (2.4.2.15) [Keisler-Shelah Theorem].** Two  $\mathcal{L}$ -structures  $\mathcal{M}$  and  $\mathcal{N}$  are elementary equivalent iff there is an index set  $I$  and an ultrafilter  $\mathcal{F}$  on  $I$  that  $\prod_I \mathcal{M} / \mathcal{F} \cong \prod_I \mathcal{N} / \mathcal{F}$ . ┘

*Proof:* Cf.[C. C. Chang and H. J. Keisler, Model Theory 6.1.15]. □

### Compactness Theorem and Henkin Construction

**Lemma (2.4.2.16).** Suppose  $T$  is a maximal and finitely satisfiable  $\mathcal{L}$ -theory with the witness property, then  $T$  is satisfiable. In fact,  $T$  has  $\kappa$  constant symbols, then there is a model  $\mathcal{M} \models T$  that  $|\mathcal{M}| \leq \kappa$ . ┘

*Proof:* Cf.[Marker, P35]. □

**Lemma (2.4.2.17).** Let  $\mathcal{L}$  be a finitely satisfiable  $\mathcal{L}$ -theory, then there is a language  $\mathcal{L}^* \subset \mathcal{L}$  and a  $T \subset T^*$  a finitely satisfiable  $\mathcal{L}^*$ -theory that any  $\mathcal{L}^*$ -theory extending  $T^*$  has the witness property. And we can choose  $|\mathcal{L}^*| = |\mathcal{L}| + \aleph_0$ . ┘

*Proof:* We first show there is a language  $\mathcal{L} \subset \mathcal{L}_1$  and a finitely satisfiable  $\mathcal{L}_1$ -theory  $T \subset T_1$  that witnesses all  $\mathcal{L}$ -formulae: We add a constant symbol  $c_\varphi$  for each  $\mathcal{L}$ -formula  $\varphi$ , and add sentences  $(\exists v\varphi(v)) \rightarrow \varphi(c_\varphi)$  to  $T$ , then this  $T_1$  is finitely satisfiable, because for any finite subset  $\Delta$  of  $T_1$ , only f.m. constant symbols appear, and we have a model  $\mathcal{M}$  for  $\Delta \cap T$ , thus we can interpret  $c_\varphi$  as the element  $a$  that  $\mathcal{M} \models \varphi(a)$ , if  $\mathcal{M} \models \exists v\varphi(v)$ . Now we can inductively define  $L_n, T_n$ , and consider their union, then this satisfies the desired conditions. And the cardinality is also clear.  $\square$

**Lemma (2.4.2.18).** If  $T$  is a finitely satisfiable  $\mathcal{L}$ -theory, then there is a maximal (2.4.2.1) finitely satisfiable  $\mathcal{L}$ -theory  $\mathcal{L} \subset \mathcal{L}'$ .  $\lrcorner$

*Proof:* it is easy to construct a maximal satisfiable  $\mathcal{L}$ -theory, and it is truly maximal in sense of (2.4.2.1): for any sentence  $\varphi$ , if  $T \cup \varphi$  is not finitely satisfiable, then there is a finite set  $\Delta \subset T$  that  $\Delta \models \neg\varphi$ . Then we claim  $T \cup \neg$  is finitely satisfiable: for another finite set  $\Sigma$ , because  $\Delta \cup \Sigma$  is finitely satisfiable and  $\Delta \models \neg\varphi$ ,  $\Sigma \cup \neg\varphi$  is satisfiable.  $\square$

**Prop. (2.4.2.19) [Strong Compactness Theorem].** If  $T$  is a finitely satisfiable  $\mathcal{L}$ -theory and  $\kappa$  is an infinite cardinal that  $\kappa > |\mathcal{L}|$ , then there is a model of  $T$  of cardinality at most  $\kappa$ .  $\lrcorner$

*Proof:* By (2.4.2.17), we get a desired language  $\mathcal{L}^*$  of cardinality  $\leq \kappa$ , and a theory  $T^*$ , and by (2.4.2.18), we can assume  $T^*$  is maximal and finitely satisfiable, and it has the witness property. Then (2.4.2.16) says there is a model of cardinality  $\leq \kappa$ .  $\square$

**Cor. (2.4.2.20) [Compactness Theorem].** A theory  $T$  is satisfiable iff every finite subset of  $T$  is satisfiable.  $\lrcorner$

**Remark (2.4.2.21).** Notice the compactness theorem is also a consequence of the completeness theorem (2.4.2.7): if  $T$  is not satisfiable, then it is not consistent by (2.4.2.8). Let  $\sigma$  be a proof of a contradiction in  $T$ , then  $\sigma$  consists of f.m. sentences in  $T$ , which consists of a finite unsatisfiable subset  $T_0$  of  $T$ .

It is clear provable using ultrafilters: If there is a family of structures  $\{M_\Delta\}$  indexed by the collection of all finite subsets of  $T$ , with  $M_\Delta \models \Delta$  for all  $\Delta \in I$  where  $I$  is the set of all finite subsets of  $T$ .

Then we want to find an ultrafilter  $\mathcal{F}$  on  $I$  that for all  $\varphi \in T$ ,  $\{\Delta \mid M_\Delta \models \varphi\} \in \mathcal{F}$ , then we can use Leo's theorem (2.4.2.11) to show that  $\prod_I M_\Delta / \mathcal{F}$  is a model for all  $\varphi \in T$ . Now in fact we make pick a ultrafilter over the filter generated by all the  $A_\varphi = \{\Delta \mid \varphi \in \Delta\}$ , because  $M_\Delta \models \Delta$ . In fact, this is the case because  $A_\varphi$  has the finite intersection property trivially.  $\lrcorner$

**Cor. (2.4.2.22).** If  $T \models \varphi$ , then  $\Delta \models \varphi$  for some finite  $\Delta \subset T$ .  $\lrcorner$

*Proof:* If not, then  $\Delta \cup \{\neg\varphi\}$  is satisfiable for all finite  $\Delta \subset T$ , so  $T \cup \{\neg\varphi\}$  is finitely satisfiable, thus satisfiable by compactness theorem, but this cannot be true because  $T \models \varphi$ .  $\square$

**Cor. (2.4.2.23) [Torsion Elements].** Let  $\mathcal{L}$  be a language containing  $\{\cdot, e\}$ , the language of groups, and  $T$  is a theory extending the theory of groups, let  $\varphi(v)$  be an  $\mathcal{L}$ -formula. If for any  $n$  there is a  $G_n \models T$  and  $G_n$  has an element of finite order greater than  $n$ , then there is an  $\mathcal{L}$ -structure  $G$  that  $G \models T$  and  $G$  has an element of infinite order.

In particular, there is no formula that defines the torsion elements in any models for  $T$ .  $\lrcorner$

*Proof:* Consider a new language  $\mathcal{L}^* = \mathcal{L} \cup \{c\}$ , and  $T^*$  an  $\mathcal{L}^*$ -theory that

$$T^* = T \cup \{\varphi(c)\} \cup \{\neg(\underbrace{c \cdots c}_{n\text{-times}} = e)\},$$

then the theory  $T^*$  is finitely satisfiable by hypothesis, so  $T^*$  is satisfiable by compactness theorem.  $\square$

**Cor. (2.4.2.24) [Element Larger than All Natural Number].** Consider  $\mathcal{L}$  the language of ordered rings, let  $\text{Th}(\mathbb{N})$  be the theory of  $\mathbb{N}$ , then there is an  $\mathcal{L}$ -structure  $\mathcal{M}$  that  $\mathcal{M} \models \text{Th}(\mathbb{N})$  and  $\mathcal{M}$  has an element that is larger than every natural number.  $\lrcorner$

*Proof:* The same proof as that of (2.4.2.23), but use

$$T^* = \text{Th}(\mathbb{N}) \cup \left\{ \underbrace{1 + \dots + 1}_{n\text{-times}} < c \right\}$$

$\square$

**Prop. (2.4.2.25) [Model of a Given Cardinality].** If  $T$  is an  $\mathcal{L}$ -theory with infinite models, then if  $\kappa$  is an infinite cardinal that  $\kappa \geq |\mathcal{L}|$ , then there is a model of  $T$  of cardinality  $\kappa$ .  $\lrcorner$

*Proof:* Let  $\mathcal{L}^* = \mathcal{L} \cup \{c_\alpha : \alpha < \kappa\}$ , where  $c_\alpha$  are pairwise different new constants, and  $T^*$  be the  $\mathcal{L}^*$  theory  $T \cup \{c_\alpha \neq c_\beta, \alpha < \beta < \kappa\}$ , then any model for  $T^*$  must have cardinality  $\geq \kappa$ . But then we use strong compactness theorem (2.4.2.19), it suffices to show  $T^*$  is finitely satisfiable: for any finite  $\Delta \subset T^*$ , there are only f.m. new constant symbols, thus we can use the infinite model  $\mathcal{M}$  to interpret the constant symbols randomly, thus we are done.  $\square$

### Complete Theories

**Def. (2.4.2.26) [Categorical Theories].** Let  $\kappa$  be an infinite cardinal and  $T$  is a theory with models of size  $\kappa$ .  $T$  is called  $\kappa$ -categorical iff any two models of  $T$  of cardinality  $\kappa$  are isomorphic.  $\lrcorner$

**Prop. (2.4.2.27).** The theory of torsion-free divisible Abelian groups is  $\kappa$ -categorical for all  $\kappa > \aleph_0$ .  $\lrcorner$

*Proof:* A torsion-free Abelian group is just a vector space over  $\mathbb{Q}$ . Thus the conclusion is trivial, just notice for a cardinal  $\kappa > \aleph_0$ , a vector space of dimension  $\kappa$  has cardinality  $\kappa$ .  $\square$

**Prop. (2.4.2.28) [Vaught's Test].** Let  $T$  be a satisfiable theory with no finite models, if  $T$  is  $\kappa$ -categorical for some  $\kappa \geq |\mathcal{L}|$ , then  $T$  is complete.  $\lrcorner$

*Proof:* If there is a sentence  $\varphi$  that  $T \not\models \varphi$  and  $T \not\models \neg\varphi$ , so  $T_0 = T \cup \{\varphi\}$  and  $T_1 = T \cup \{\neg\varphi\}$  is satisfiable. They both have infinite models by hypothesis, so by (2.4.2.25) there are models of cardinal  $\kappa$  for  $T_0$  and  $T_1$ , but they cannot be isomorphic, contradiction. So  $T$  is complete.  $\square$

**Prop. (2.4.2.29) [Recursive Complete Satisfiable is Decidable].** If  $T$  is a recursive complete satisfiable theory in a recursive language  $\mathcal{L}$ , then  $T$  is decidable.  $\lrcorner$

*Proof:* Because  $T$  is satisfiable, The set of all  $\varphi$  that  $\mathcal{M} \models \varphi$  and the set of all  $\varphi$  that  $\mathcal{M} \models \neg\varphi$  are disjoint, and their sum is the set of all sentences by completeness. By Gödel's completeness theorem, this is equivalent to  $\mathcal{M} \vdash \varphi$  or  $\mathcal{M} \vdash \neg\varphi$ . Then by (2.4.2.4), they are both enumerable, so it is decidable by definition.  $\square$

**Up and Down(of Cardinality)**

**Def. (2.4.2.30) [Elementary Embedding].** An  $\mathcal{L}$ -structure embedding  $i : \mathcal{M} \rightarrow \mathcal{N}$  is called an **elementary embedding**, denoted by  $\mathcal{M} \prec \mathcal{N}$ , if for any  $\mathcal{L}$ -formula  $\varphi$ ,

$$\mathcal{M} \models \varphi(\bar{a}) \iff \mathcal{N} \models \varphi(i(\bar{a})).$$

Notice that one way implication is sufficient, because we can use negation.

Isomorphisms are elementary embeddings by (2.4.1.8).  $\lrcorner$

**Def. (2.4.2.31) [Diagrams].** Let  $\mathcal{M}$  be an  $\mathcal{L}$ -structure, then we can add to  $\mathcal{L}$  constant symbols for each element of  $M$ , and the class  $Diag(\mathcal{M})$  of **atomic diagrams** of  $\mathcal{M}$  is sentences of the form  $\varphi(m_1, \dots, m_n)$ , where  $\varphi$  is an atomic  $\mathcal{L}$ -formula or the negation of an atomic  $\mathcal{L}$ -formula, and  $\mathcal{M} \models \varphi(m_1, \dots, m_n)$ . And the class  $Diag_{el}(\mathcal{M})$  of **elementary diagrams** of  $\mathcal{M}$  is sentences of the form  $\varphi(m_1, \dots, m_n)$  where  $\varphi$  is an  $\mathcal{L}$ -formula that  $\mathcal{M} \models \varphi(m_1, \dots, m_n)$ .  $\lrcorner$

**Prop. (2.4.2.32) [Diagrams and Embedding].** The diagram and elementary diagram is very important in constructing embeddings of theories: Let  $\mathcal{N}$  be an  $\mathcal{L}_M$ -structure, then:

- If  $\mathcal{N} \models Diag(\mathcal{M})$ , there is an  $\mathcal{L}$ -embedding  $\mathcal{M} \subset \mathcal{N}$ .
- If  $\mathcal{N} \models Diag_{el}(\mathcal{M})$ , there is an elementary  $\mathcal{L}$ -embedding  $\mathcal{M} \prec \mathcal{N}$ .

$\lrcorner$

**Prop. (2.4.2.33) [Upward Löwenheim-Skolem Theorem].** Let  $\mathcal{M}$  be an infinite  $\mathcal{L}$ -structure and  $\kappa$  be an infinite cardinal that  $\kappa \geq |\mathcal{M}| + |\mathcal{L}|$ , then there is an  $\mathcal{L}$ -structure  $\mathcal{N}$  of cardinality  $\kappa$  that  $\mathcal{M}$  embeds into  $\mathcal{N}$ .  $\lrcorner$

*Proof:*  $Diag_{el}(\mathcal{M})$  is clearly satisfiable, so by (2.4.2.25), there is an  $\mathcal{L}^*$ -model  $\mathcal{N}$  of cardinality  $\kappa$  that  $\mathcal{N} \models Diag_{el}(\mathcal{M})$ . Then clearly  $\mathcal{M} \prec \mathcal{N}$ .  $\square$

**Prop. (2.4.2.34) [Tarski-Vaught Test].** A substructure  $\mathcal{M}$  of a structure  $\mathcal{N}$  is an elementary substructure iff for any formula  $\varphi(v, \bar{w})$  and  $\bar{a} \in M$ , if there is  $b \in N$  that  $\mathcal{N} \models \varphi(b, \bar{a})$ , then there is a  $c \in M$  that  $\mathcal{N} \models \varphi(c, \bar{a})$ .  $\lrcorner$

*Proof:* Cf. [Marker P45]. ?  $\square$

**Prop. (2.4.2.35) [(Downward) Löwenheim-Skolem Theorem].** Suppose  $\mathcal{M}$  is an  $\mathcal{L}$ -structure and  $X \subset \mathcal{M}$ , then there is a elementary submodel  $\mathcal{N}$  that  $X \subset \mathcal{N}$  and  $|\mathcal{N}| \leq |X| + |\mathcal{L}| + \aleph_0$ .  $\lrcorner$

*Proof:* Cf. [Marker P46].  $\square$

**Def. (2.4.2.36) [Universal Sentences].** A **universal sentence** is a sentence of the form  $\forall v \varphi(v)$ , where  $\varphi$  is quantifier-free.

For a theory  $T$ , denote by  $T_{\forall}$  the set of all of the universal sentences  $\varphi$  that  $T \models \varphi$ .

A  $\mathcal{L}$ -theory  $T$  is said to have a **universal axiomatization** iff there is a set of universal  $\mathcal{L}$ -sentences  $\Gamma$  that  $\mathcal{M} \models T$  iff  $\mathcal{M} \models \Gamma$ .  $\lrcorner$

**Prop. (2.4.2.37) [Universal Axiomatization].** An  $\mathcal{L}$ -theory  $T$  has a universal axiomatization iff for any  $\mathcal{N} \subset \mathcal{M}$ , if  $\mathcal{M} \models T$ , then  $\mathcal{N} \models T$ .  $\lrcorner$

*Proof:* Cf. [Marker P47].  $\square$



**Prop. (2.4.2.38)[Universal Consequences].** If  $T$  is an  $\mathcal{L}$ -theory, then  $\mathcal{A} \models T_{\forall}$  iff there is an  $\mathcal{M} \models T$  that  $\mathcal{A} \subset \mathcal{M}$ .  $\lrcorner$

*Proof:* If  $\mathcal{A} \subset \mathcal{M}$ , then  $\mathcal{A} \models T_{\forall}$ , by (2.4.2.37). Conversely, if  $\mathcal{A} \models T_{\forall}$ , then consider  $Diag(\mathcal{A})$ , then it suffices to show  $Diag(\mathcal{A})$  is satisfiable. If it is not, then by compactness theorem (2.4.2.20) there is a finite set that is not satisfiable, and the part coming from diagrams of  $\mathcal{M}$  can be expressed by a single formula  $\chi(\bar{a})$ , where  $\chi$  is quantifier-free,  $\bar{a} \in \mathcal{M}$ ,  $\mathcal{M} \models \chi(\bar{a})$ , and we are saying that  $T \cup \{\chi(\bar{a})\}$  is not satisfiable, then  $T \models \neg\chi(\bar{a})$ .

Now  $T \models \forall \bar{x} \neg\varphi(\bar{x})$  as  $\mathcal{L}$ -theory because  $\bar{a}$  can be designated arbitrarily, but this sentence is in  $T_{\forall}$ , thus  $\mathcal{M} \models \forall \bar{x} \neg\varphi(\bar{x})$ , in particular,  $\mathcal{M} \models \neg\varphi(\bar{a})$ , contradiction.  $\square$

**Prop. (2.4.2.39)[Elementary Chain].** If  $\mathcal{M}_i$  is a chain of  $\mathcal{L}$ -structures that  $\mathcal{M}_i \prec \mathcal{M}_j$  for any  $i < j$ , then we can define their union  $\mathcal{M} = \cup \mathcal{M}_i$ . Then  $\mathcal{M}$  is an elementary extension of each  $\mathcal{M}_i$ .  $\lrcorner$

*Proof:* Use induction on formulas to show that

$$\mathcal{M}_i \models \varphi(\bar{a}_i) \iff \mathcal{M} \models \varphi(\bar{a}_i),$$

for all  $\mathcal{L}$ -formulas  $\varphi$ . This is true for quantifier-free formula by (2.4.1.7), and if this is true for  $\varphi, \psi$ , then this is true for  $\neg\varphi$  and  $\varphi \wedge \psi$ . For the sentence  $\varphi = \exists v \psi(v, \bar{w})$ , if  $\mathcal{M}_i \models \varphi(b, \bar{a})$  for some  $b$ , then so does  $\mathcal{M}$ . Conversely, if  $\mathcal{M} \models \varphi(b, \bar{a})$  for some  $b$ , then  $b \in \mathcal{M}_j$  for some  $j$ , then  $\mathcal{M}_j \models \varphi$ , by the condition,  $\mathcal{M}_i \models \varphi$  also.  $\square$

### Back and Forth Argument

Cf.[Marker Chap2.4]. Some deep ideas are involved.

**Prop. (2.4.2.40)[Cantor].** The theory  $DLO(2.1.6.3)$  is  $\aleph_0$ -categorical and complete.  $\lrcorner$

*Proof:* It is  $\aleph_0$ -categorical by (2.2.3.12), and it is complete, by Vaught's test (2.4.2.28), as it has no finite models.  $\square$

## 3 Quantifier Elimination

**Def. (2.4.3.1)[Quantifier Elimination].** A theory is said to have **quantifier elimination** if for each formula  $\varphi(\bar{v})$ , there is a quantifier free  $\psi$  that  $T \models \forall \bar{v}(\varphi(\bar{v}) \leftrightarrow \psi(\bar{v}))$ .  $\lrcorner$

**Lemma (2.4.3.2).** Suppose  $\mathcal{L}$  contains a constant symbol  $c$ ,  $T$  is an  $\mathcal{L}$ -theory, and  $\varphi(\bar{a})$  is an  $\mathcal{L}$ -formula, then the following are equivalent:

- There is a quantifier-free  $\mathcal{L}$ -formula  $\psi(\bar{v})$  that  $T \models \forall \bar{v}(\varphi(\bar{v}) \leftrightarrow \psi(\bar{v}))$ .
- If  $\mathcal{M}, \mathcal{N}$  are models of  $T$ , and  $\mathcal{A}$  is an  $\mathcal{L}$ -structure that  $\mathcal{A} \subset \mathcal{M} \cap \mathcal{N}$ , then for all  $\bar{a} \in \mathcal{A}$ ,  $\mathcal{N} \models \varphi(\bar{a}) \iff \mathcal{M} \models \varphi(\bar{a})$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 2$  is clear, because a quantifier-free formula  $\psi(\bar{a})$  is preserved under substructure by (2.4.1.7).

$2 \rightarrow 1$ , Cf.[Marker, P74]. ?  $\square$

**Lemma (2.4.3.3).** Let  $T$  be an  $\mathcal{L}$ -theory that for any quantifier-free  $\mathcal{L}$ -formula  $\theta(\bar{v}, w)$ , there is a quantifier-free formula  $\psi(\bar{v})$  that  $T \models \forall \bar{v}(\exists w \theta(\bar{v}, w) \leftrightarrow \psi(\bar{v}))$ , then  $T$  has quantifier elimination.  $\lrcorner$



*Proof:* We want to show for each formula  $\varphi(\bar{a})$ , there is a quantifier-free formula  $\psi(\bar{a})$  that  $T \models \forall \bar{v}(\varphi(\bar{a}) \leftrightarrow \psi(\bar{v}))$ , and we use induction on the complexity of  $\varphi$ .

If  $\varphi$  is quantifier-free, this is trivial. If  $\varphi = \neg\theta_0$  or  $\varphi = \theta_0 \wedge \theta_1$ , then we are easily done. If  $\psi(\bar{v}) = \exists w\theta(\bar{v}, w)$ , and  $T \models \forall \bar{v}(\theta(\bar{v}, w) \leftrightarrow \psi_0(\bar{v}, w))$ , then  $T \models \forall \bar{v}(\varphi(\bar{v}) \leftrightarrow \exists w\psi_0(\bar{v}, w))$ . But the assumption shows there is a quantifier-free  $\psi$  that  $T \models \forall \bar{v}(\exists w\psi_0(\bar{v}, w) \leftrightarrow \psi(\bar{v}))$ , and then  $T \models \forall \bar{v}(\varphi(\bar{v}) \leftrightarrow \psi(\bar{v}))$ .  $\square$

**Prop. (2.4.3.4) [Criterion for Quantifier Elimination].** Combining (2.4.3.2) and (2.4.3.3), we get: If  $T$  is an  $\mathcal{L}$ -theory that for all quantifier-free formula  $\varphi(\bar{v}, w)$  and models  $\mathcal{M}, \mathcal{N} \models T$ , and  $\mathcal{A} \subset M \cap N, \bar{a} \in A$ , and if there is  $b \in M$  that  $\mathcal{M} \models \varphi(\bar{a}, b)$  will imply there is  $c \in \mathcal{N}$  that  $\mathcal{N} \models \varphi(\bar{a}, c)$ , then  $T$  is quantifier-free.  $\lrcorner$

**Def. (2.4.3.5) [Algebraically Prime Models].** A theory  $T$  is said to have **algebraically prime models** if for any  $\mathcal{A} \models T_\forall$  (2.4.2.36), there is a  $\mathcal{M} \models T$  and an embedding  $i : \mathcal{A} \subset \mathcal{M}$  that for any other  $\mathcal{N} \models T$ , any embedding  $j : \mathcal{A} \rightarrow \mathcal{N}$  factors through  $i$ .  $\lrcorner$

**Def. (2.4.3.6) [Simply Closed Model].** If  $\mathcal{M}, \mathcal{N}$  are models of  $T$ , and  $\mathcal{M} \subset \mathcal{N}$ , then  $\mathcal{M}$  is called **simply closed** in  $\mathcal{N}$ , denoted by  $\mathcal{M} \prec_s \mathcal{N}$ , iff for any quantifier-free formula  $\varphi(\bar{a}, w)$  and  $\bar{a} \in \mathcal{M}^n$ , if  $\mathcal{N} \models \exists w\varphi(\bar{a}, w)$ , then so does  $\mathcal{M}$ .  $\lrcorner$

**Prop. (2.4.3.7) [Quantifier Elimination Test].** If  $T$  is an  $\mathcal{L}$ -theory that has algebraically prime models, and any inclusion of models for  $T$  is simply closed, then  $T$  has quantifier elimination.  $\lrcorner$

*Proof:* This is an immediate consequence of (2.4.3.4).  $\square$

**Def. (2.4.3.8) [Model-Complete].** An  $\mathcal{L}$ -theory  $T$  is called **model-complete** if for any two models  $\mathcal{M}, \mathcal{N}$  for  $T$ ,  $\mathcal{M} \prec \mathcal{N}$  (2.4.2.30) whenever  $\mathcal{M} \subset \mathcal{N}$ . A complete theory is clearly model complete.  $\lrcorner$

**Prop. (2.4.3.9).** let  $T$  be an  $\mathcal{L}$ -theory. If  $T$  has quantifier elimination, then  $T$  is model-complete.  $\lrcorner$

*Proof:* If  $\mathcal{M} \subset \mathcal{N}$ , let  $\varphi(\bar{a})$  be an  $\mathcal{L}$ -formula, and  $\bar{a} \in \mathcal{M}$ , then there is a quantifier-free formula  $\psi(\bar{v})$  that  $T \models \forall \bar{v}(\varphi(\bar{v}) \leftrightarrow \psi(\bar{v}))$ . By (2.4.1.7), the formula  $\psi(\bar{a})$  passes between  $\mathcal{M}$  and  $\mathcal{N}$ , so

$$\mathcal{M} \models \varphi(\bar{a}) \iff \mathcal{M} \models \psi(\bar{a}) \iff \mathcal{N} \models \psi(\bar{a}) \iff \mathcal{N} \models \varphi(\bar{a}).$$

$\square$

**Prop. (2.4.3.10) [Minimal Model and Completeness].** If  $T$  is model-complete and there is a minimal model  $\mathcal{M}_0$  that  $\mathcal{M}_0$  embeds into every model of  $T$ , then  $T$  is complete.  $\lrcorner$

*Proof:* Clearly any model is elementary equivalent to  $\mathcal{M}_0$ , thus clearly  $T$  is complete.  $\square$

**Prop. (2.4.3.11) [Eliminating Algorithm].** Let  $T$  be a decidable theory with quantifier elimination, then there is an algorithm to find the elimination  $\psi$  of a given formula  $\varphi$ .  $\lrcorner$

*Proof:* We just need to find a quantifier-free  $\psi$  that  $T \models \forall \bar{v}(\varphi(\bar{v}) \leftrightarrow \psi(\bar{v}))$ . This an effective search because  $T$  is decidable, and we can eventually find  $\psi$  because  $T$  has quantifier elimination.  $\square$

### Examples of Quantifier Elimination

**Prop. (2.4.3.12).** The theory  $DLO(2.1.6.3)$  has quantifier elimination. ┘

*Proof:* Cf.[Marker P72]. □

**Lemma (2.4.3.13).**  $DAG(2.1.6.3)$  has algebraically prime models. ┘

*Proof:* This is just the alg.closure of the quotient field of an integral domain. □

**Prop. (2.4.3.14).**  $DAG$  has quantifier elimination. ┘

*Proof:* We use(2.4.3.7) By(2.4.3.13),  $DAG$  has algebraically prime models, thus it suffices to show any inclusion of models for  $DAG(2.1.6.3)$  is simply closed: For any torsion-free divisible Abelian groups  $H \subset G$ ,  $\varphi(\bar{a}, w)$  quantifier-free,  $\bar{a} \in H, b \in G$  that  $G \models \varphi(\bar{a}, w)$ . Then  $\varphi$  is a disjunction of conjunctions of atomic or negated atomic formulas, and we need to prove  $H \models \varphi(\bar{a}, w)$ .

So we may assume  $\varphi$  is conjunction of atomic formulas and negated atomic formulas:

$$\varphi(\bar{a}, w) \leftrightarrow \bigwedge (g_i + m_i w = 0) \wedge \bigwedge (h_i + m'_i w \neq 0).$$

If any  $m_i \neq 0$ , then  $b \in H$ . If all  $m_i = 0$ , then there are only f.m. constraints, and there is  $b' \in H$  that  $H \models \varphi(\bar{a}, b')$  because  $H$  is infinite. □

**Cor. (2.4.3.15).** The theory  $DAG$  is complete, by(2.4.3.10), as  $(\mathbb{Q}, +, 0)$  embeds in every model of  $DAG$ , and(2.4.3.14)(2.4.3.9). ┘

**Prop. (2.4.3.16).** The theory  $ODAG$  is a complete theory with quantifier elimination. In particular, any ordered divisible Abelian group is elementarily equivalent to  $(\mathbb{Q}, +, <)$ , by completeness. ┘

*Proof:* Cf.[Marker, P80]. □

**Prop. (2.4.3.17)[Presburger Arithmetic].** Presburger arithmetic is a complete decidable theory with quantifier elimination in the language  $\mathcal{L}^*$ . ┘

*Proof:* Cf.[Marker, P84] ?. □

### Minimal Structures and Minimal Theories

**Def. (2.4.3.18) [Minimal Structures].** An  $\mathcal{L}$ -structure  $\mathcal{M}$  is called a **minimal structure** if any definable set  $X \subset M$  is either finite or cofinite. ┘

**Def. (2.4.3.19) [Quasi-Minimal Structures].** An  $\mathcal{L}$ -structure  $\mathcal{M}$  is called a **quasi-minimal structure** if  $\mathcal{M}$  is uncountable, and for any definable set  $X \subset \mathcal{M}$  is either countable or uncountable. And it is called a **strongly quasi-minimal structure** if for any finite subset  $A \subset \mathcal{M}$ , each  $\text{Aut}_A(\mathcal{M})$ -invariant subset of  $\mathcal{M}$  is either countable or cocountable. ┘

**Def. (2.4.3.20) [Strongly-(Quasi-)Minimal Theories].** A **strongly-(quasi-)minimal theory** is a theory  $T$  s.t. any model  $\mathcal{M} \models T$  is (quasi-)minimal. ┘

**Prop. (2.4.3.21).**  $DAG(2.1.6.3)$  is a strongly-minimal theory. ┘

*Proof:* Cf.[Marker P78]. □

## 4 Alg.Closed Fields

**Lemma (2.4.4.1).**  $ACF_p$  (2.1.6.3) is  $\kappa$ -categorical for all uncountable cardinal  $\kappa$ . ┘

*Proof:* Because two alg.closed field of the same transcendental degree over the base field is isomorphic (3.2.7.4). The conclusion follows as an alg.closed field of transcendence degree  $\kappa$  has cardinality  $\kappa + \aleph_0$ . □

**Prop. (2.4.4.2).** The theory  $ACF_p$  is complete, by (2.4.4.1) and (2.4.2.28). ┘

**Cor. (2.4.4.3).**  $ACF_p$  is decidable, in particular,  $\text{Th}(\mathbb{C})$ , the first-order theory of the fields of complex numbers, is decidable. ┘

*Proof:*  $ACF_p$  is complete by (2.4.4.1) and (2.4.2.28), it is clearly recursive, and it is clearly satisfiable, so use (2.4.2.29). □

**Prop. (2.4.4.4) [First order Lefschetz Principle].** Let  $\varphi$  be a sentence in the language of rings, the following are equivalent:

1.  $\varphi$  is true in  $\mathbb{C}$ .
  2.  $\varphi$  is true in every(some) alg.closed field of char 0.
  3. For  $p$  large/there exists arbitrary large  $p$ ,  $\varphi$  is true in any alg.closed field of char  $p$ .
- ┘

*Proof:* 1, 2 are equivalent because  $ACF_p$  is complete (2.4.4.2) and use Gödel's completeness theorem. If 2 is true, then  $ACT_0 \models \varphi$ , so by (2.4.2.22), then some  $\Delta \models \varphi$ . So for  $p$  sufficiently large,  $ACF_p \models \Delta$ , so  $ACF_p \models \varphi$  for  $p$  large.

If  $ACF_0 \not\models \varphi$ , then  $ACF_0 \models \neg\varphi$  by completeness (2.4.4.2), so by above argument,  $ACF_p \models \neg\varphi$  for  $p$  large, contradiction. □

**Cor. (2.4.4.5) [Ax].** Any injective polynomial map from  $\mathbb{C}^n$  to  $\mathbb{C}^n$  is surjective. ┘

*Proof:* By Lefschetz principle (2.4.4.4), it suffices to show this for  $(\mathbb{F}_p)^{alg}$  for  $p$  large. If this is not true, then choose the coefficients of the coordinate map of  $f$ , and the coordinates of an element that is not in the image, then the subfield  $k$  they generated is algebraic over  $\mathbb{F}_p$ , so it is finite, and clearly  $f$  is also surjective on  $k^n$ , contradiction. □

**Cor. (2.4.4.6).** Let  $K \subset L$  be alg.closed fields, and  $V, W$  be varieties defined over  $K$ , and  $f$  is a polynomial isomorphism of  $V$  and  $W$  over  $L$ , then there is a polynomial isomorphism of  $V$  and  $W$  over  $K$ . ┘

*Proof:* Let  $f$  have degree  $d$ . Then we can write a formula  $\Psi$  saying that there is an embedding of  $K$  into  $\mathcal{M}$  and there is a polynomial bijection of  $V$  and  $W$  over  $\mathcal{M}$ , then  $L \models \Psi$ , and because  $ACF$  is complete (2.4.4.2),  $K \models \Psi$ , too. Thus there is also an isomorphism over  $K$ . □

**Prop. (2.4.4.7).**  $ACF_{\forall}$  is the theory of integral domains. ┘

*Proof:* Clearly a ring is a subring of an alg.closed field iff it is an integral domain, so the result follows from (2.4.2.38). □

**Prop. (2.4.4.8).**  $ACF$  has qualifier elimination. ┘

*Proof:* Use (2.4.3.5), clearly it has algebraically prime models (2.4.4.7), and we need to check simply closedness.

For this, if  $K \subset L$ , notice that a quantifier-free formula  $\varphi$  is just a conjunction of some polynomial functions and negation of polynomial functions, with their coefficients in  $K$ . If there are some polynomial, then the solution  $b$  of  $\varphi$  in  $L$  is algebraic over  $K$ , thus in  $K$  because  $K$  is alg.closed. Now if it is just negations of polynomials, then clearly  $\varphi$  is true in  $K$  for some  $c \in K$ , because  $K$  is alg.closed thus infinite.  $\square$

**Cor. (2.4.4.9).**  $ACF$  is model-complete, by (2.4.3.9).  $\lrcorner$

**Lemma (2.4.4.10).** Let  $K$  be a field, then the subsets of  $K^n$  defined by atomic formulas are exactly Zariski closed subsets. And a subset of  $K^n$  that is quantifier-definable iff it is a Boolean combination of Zariski closed subsets (constructible). (Clear).  $\lrcorner$

**Prop. (2.4.4.11).**  $ACF$  is strongly minimal.  $\lrcorner$

*Proof:* Because by quantifier elimination (2.4.4.8), every definable set is a finite Boolean combination of sets of the form  $V(p) = 0$ , where  $p \in K[X]$ , but  $V(p)$  is either finite or all of  $K$ .  $\square$

### Elimination of Imaginaries in Alg.Closed Fields

**Def. (2.4.4.12) [Algebraicity].** Let  $\mathcal{M}$  be an  $\mathcal{L}$ -structure and  $A \subset M$ , then for  $b \in M$ ,  $b \in acl(A)$  if there is a formula  $\varphi(x, \bar{y})$  and  $\bar{a} \in A$  that  $\mathcal{M} \models \varphi(b, \bar{a})$ , and  $\{x \in M \mid \mathcal{M} \models \varphi(x, \bar{a})\}$  is finite.  $\lrcorner$

**Def. (2.4.4.13) [Algebraicity over Equivalence Class].** Let  $\mathcal{M}$  be an  $\mathcal{L}$ -structure and  $E$  a definable equivalence relation on  $M^n$ , then we say a element  $c \in M$  is algebraic over  $\bar{a}/E, b_1, \dots, b_m$  if there is a formula  $\varphi(c, \bar{y}, \bar{z})$  that

- $\mathcal{M} \models \varphi(c, \bar{a}, \bar{b})$ .
- If  $\bar{a}E\bar{a}'$ , then  $\mathcal{M} \models \varphi(c, \bar{a}, \bar{b}) \leftrightarrow \varphi(c, \bar{a}', \bar{b})$ .
- $\{x \mid \mathcal{M} \models \varphi(x, \bar{a}, \bar{b})\}$  is finite.

And a sequence  $\bar{c}$  is called algebraic over  $a/E, b_1, \dots, b_m$  iff each coordinate is.  $\lrcorner$

**Lemma (2.4.4.14).** Suppose  $\bar{c}$  is algebraic over  $\bar{a}/E, \bar{d}, \bar{b}$ , and  $\bar{b}$  is algebraic over  $\bar{a}/E, \bar{d}$ , then  $\bar{c}$  is also algebraic over  $\bar{a}/E, \bar{d}$ .  $\lrcorner$

*Proof:* Cf. [Marker, P92].  $\square$

**Lemma (2.4.4.15).** Suppose  $K$  is an alg.closed field and  $E$  is a definable equivalence relation on  $K^n$ , and  $\psi(\bar{x}, \bar{y}, \bar{d})$  defines  $E$ . If  $\bar{a} \in K^n$ , then there is a  $\bar{c} \in K^n$  algebraic over  $\bar{a}/E, \bar{d}$  s.t.  $\bar{c}E\bar{a}$ .  $\lrcorner$

*Proof:* Cf. [Marker, P92]. ?  $\square$

**Prop. (2.4.4.16) [Elimination of Imaginaries].** Let  $K$  be an alg.closed field,  $A \subset K$ , and  $E$  is an  $A$ -definable equivalence relation on  $K^n$ , then for some  $l$  there is an  $A$ -definable function  $f : K^n \rightarrow K^l$  that  $\bar{x}E\bar{y}$  iff  $f(\bar{x}) = f(\bar{y})$ .  $\lrcorner$

*Proof:* Cf. [Marker, P92].  $\square$

## 5 Real-Closed Fields

**Prop. (2.4.5.1).** The class of real-closed fields is an elementary class(2.4.1.4) of  $\mathcal{L}_r$ -structures.  $\lrcorner$

*Proof:* The real-closed fields are axiomatized by:

- Axioms for fields.
- For each  $n \geq 1$ , the sentence  $\forall x_1 \dots \forall x_n (x_1^2 + \dots + x_n^2 + 1 \neq 0)$ .
- $\forall x \exists y (y^2 = x \vee y^2 + x = 0)$ .
- For each  $n \geq 0$ , the sentence  $\forall x_1 \dots \forall x_n \exists y (y^{2n+1} + \sum_{i=0}^{2n} x_i y^i = 0)$ .

These truly axiomatize real-closed fields, by(3.2.10.7).  $\square$

**Prop. (2.4.5.2).** Let  $F$  be a real-closed field, then the definable sets in  $F^n$  w.r.t the  $\mathcal{L}_{or}$  is also definable w.r.t.  $\mathcal{L}_r$ .  $\lrcorner$

*Proof:* Because we can replace all instances  $t_i < t_j$  by  $\exists v (v \neq 0 \wedge t_i + v^2 = t_j)$ .  $\square$

**Prop. (2.4.5.3).**  $RCF_{\forall}$  is the theory of ordered integral domains(easy).  $\lrcorner$

**Cor. (2.4.5.4).**  $RCF$  has algebraically prime models.  $\lrcorner$

*Proof:* Let  $D$  be an ordered domain, and let  $R$  be a real closure of the fraction field of  $D$  compatible with the ordering of  $D$ . Let  $F$  be another real-closed field extension of  $D$ , then let  $K$  be the algebraic closure of the fraction field of  $D$  in  $F$ , then  $K$  is a real-closed field by(3.2.10.7), then by(3.2.10.12) there is an isomorphism of  $R \cong K$  extending  $D \rightarrow K$ , thus embeds  $R$  into  $F$  extending  $D \rightarrow F$ .  $\square$

**Prop. (2.4.5.5).**  $RCF$  has quantifier elimination (w.r.t.  $\mathcal{L}_{or}$ ).  $\lrcorner$

*Proof:* Use(2.4.3.5), it has algebraically prime models by(2.4.5.4), and we need to check simply closedness: Let  $F \subset K$  be RCFs,  $\varphi(v, \bar{w})$  be a quantifier-free formula and let  $\bar{a} \in F, b \in K$  that  $K \models \varphi(b, \bar{a})$ . Then there are polynomials that

$$\varphi(v, \bar{a}) \leftrightarrow (\bigwedge p_i(v) = 0 \wedge \bigwedge q_i(v) > 0).$$

If any  $p_i \neq 0$ , then  $b$  is algebraic over  $F$ , but then  $b \in F$  because  $F$  is real-closed. So we may assume  $\varphi(v, \bar{a}) \leftrightarrow \bigwedge q_i(v) > 0$ . Then because  $q_i(b) > 0$  and  $q_i$  has f.m. zeros, by intermediate property(3.2.10.9), we can find a nbhd of  $b$  that is interval with endpoints in  $F$  that  $q_i > 0$ , and because  $F$  is dense(char0), there is an element  $b'$  that  $F \models \varphi(b', \bar{a})$ .  $\square$

**Cor. (2.4.5.6).**  $RCF$  is complete and decidable. Thus it is just the theory of  $(\mathbb{R}, +, \cdot, <)$ , and it is decidable.  $\lrcorner$

*Proof:* It is model-complete by(2.4.3.9), and it has a minimal model that is the field  $\mathbb{R}_{alg}$  of algebraic numbers in  $\mathbb{R}$ , then it is complete by(2.4.3.10). It is decidable by(2.4.2.29).  $\square$

**Prop. (2.4.5.7)[RCF is o-Minimal].** The theory  $RCF$ (2.1.6.3) is o-minimal.  $\lrcorner$

*Proof:* Because  $RCF$  has quantifier elimination, every definable set of a real-closed field  $R$  is semialgebraic, thus it is clearly a disjoint union of intervals and points.  $\square$

### Semi-algebraic Sets

**Def. (2.4.5.8)[Semi-algebraic Sets].** Let  $F$  be an ordered field, then  $X \subset F^m$  is called **semi-algebraic** if there it is a Boolean combination of sets of the form  $\{\underline{X} \in F^m \mid p(\underline{X}) > 0\}$ , where  $p(\underline{X}) \in F[\underline{X}]$ .  $\lrcorner$

**Thm. (2.4.5.9) [Tarski-Seidenberg].** If  $F \in \mathbf{Field}$  is real-closed, the semi-algebraic sets are closed under projection.  $\lrcorner$

*Proof:* This is because quantifier elimination(2.4.5.5) implies semialgebraic sets are just definable sets, and the projection of a definable set is also definable.  $\square$

**Cor. (2.4.5.10).** The composite of two semialgebraic functions are semialgebraic.  $\lrcorner$

**Prop. (2.4.5.11) [Closure of Semialgebraic Sets].** Let  $F$  be a real-closed field and  $A \subset F^n$  be semialgebraic, then the closure of  $A$  in the Euclidean topology is also semialgebraic.  $\lrcorner$

*Proof:* The closure of  $A$  is defined by

$$\{\bar{x} \mid \forall \varepsilon > 0, \exists \bar{y} \in A, d(\bar{x}, \bar{y}) < \varepsilon\}.$$

Thus it is definable, and thus semialgebraic by quantifier elimination(2.4.5.5).  $\square$

**Prop. (2.4.5.12).** Let  $F$  be a real-closed field and  $X \subset F^n$  be closed and bounded,  $f$  be a continuous semialgebraic function, then  $f(X)$  is closed and bounded.  $\lrcorner$

*Proof:* If  $F$  is  $\mathbb{R}$ , then this follows from(4.4.5.2) that  $X$  is compact hence the image is also compact hence closed and bounded. As we can formulate a sentence asserting the conclusion. Thus the theorem follows as  $RCF$  is complete(2.4.5.6).  $\square$

**Lemma (2.4.5.13).** For  $R$  a real-closed field and  $f : R \rightarrow R$  semialgebraic, then for any interval  $U$  there is an element  $x$  that  $f$  is continuous at  $x$ .  $\lrcorner$

*Proof:* It suffices to prove for  $R = \mathbb{R}$  by completeness of  $RCF$ . If there is an interval that  $f$  has finite range, then the inverse image of some point  $b$  is infinite, but also definable, so by  $o$ -minimality contains an interval that  $f$  is constant, hence continuous.

Otherwise, we inductively choose a chain of intervals:  $V_0 = U$ , and the image of  $V_n$  is definable thus contains an interval of length at most  $1/n$ , by  $o$ -minimality and our assumption. Now for the same reason the inverse image of this interval contains an interval, called  $V_{n+1}$ , s.t.  $\overline{V_{n+1}} \subset V_n$ , then  $\cap V_n = \cap \overline{V_n} \neq \emptyset$  because  $\mathbb{R}$  is locally compact. For any  $x \in V$ , it is easy to see  $f$  is continuous at  $x$ .  $\square$

**Prop. (2.4.5.14).** For  $R$  a real-closed field and  $f : R \rightarrow R$  semialgebraic, then  $f$  is discontinuous only at f.m. points.  $\lrcorner$

*Proof:* The discontinuous points of  $f$  is definable by

$$D = \{x \mid F \models \exists \varepsilon > 0 \forall \delta > 0 \exists y \mid x - y \mid < \delta \wedge \mid f(x) - f(y) \mid > \varepsilon\},$$

thus it is a finite union of intervals and points, by  $o$ -minimality(2.4.5.7), but it must be f.m. points, by(2.4.5.13).  $\square$

**Def. (2.4.5.15).** We can naturally define the notion of **definably connected** and **definably arcwise connected** sets in  $R^n$ .  $\lrcorner$

**Def. (2.4.5.16) [Cells].** For an real-closed field  $F$ , we inductively define  $n$ -cells:

- $X \subset F^n$  is a 0-cell if it is a single point.
- If  $X \subset F^m$  is an  $n$ -cell and  $f : X \rightarrow F$  is a continuous definable function, then  $Y = \{(\bar{x}, f(\bar{x})) | \bar{x} \in X\}$  is an  $n$ -cell.
- If  $X \subset F^m$  is an  $n$ -cell and  $f, g$  are either continuous functions from  $X$  to  $F$  or constants  $\pm\infty$ , then  $Y = \{(\bar{x}, y) | \bar{x} \in X \wedge f(\bar{x}) < y < g(\bar{x})\}$  is an  $n + 1$ -cell.

⌋

**Prop. (2.4.5.17) [Uniform Bounding].** Let  $X \subset F^{m+1}$  be semialgebraic, then there is a natural number  $N$  that if  $\bar{a} \in F^n$  and  $X_{\bar{a}} = \{y | (\bar{a}, y) \in X\}$  is finite, then  $|X_{\bar{a}}| < N$ .

⌋

*Proof:* First notice  $X_{\bar{a}}$  is definable, thus by  $\mathcal{o}$ -minimality,  $\{\bar{a} | |X_{\bar{a}}| < \infty\}$  is definable, thus we may assume these are all of  $X$ .

Now let  $\Gamma$  be the theory

$$RCF + Diag(F) + \{\exists y_1, y_2, \dots, y_m [\bigwedge_{i < j} y_i \neq y_j \wedge \bigwedge y_i \in X_{\bar{c}}]\}$$

where  $m \in \mathbb{N}$ , and  $c_1, \dots, c_n$  are new constants. (Notice  $y_i \in X_{\bar{c}}$  is a sentence).

Then  $\Gamma$  is not satisfiable, because otherwise there is an embedding  $F \subset K$ , and by model completeness  $F \prec K$ , thus there is no  $\bar{a}$  that  $X_{\bar{a}}$  is infinite, because it is definable, contradiction.

Then by compactness theorem, there is some f.m. sentences of  $\Gamma$  that is unsatisfiable, thus the conclusion follows. □

**Prop. (2.4.5.18) [Cell Decomposition].** Let  $F$  be a real-closed field, then any semialgebraic set  $X \subset F^m$  is a finite disjoint union of cells. ⌋

*Proof:* Cf. [Marker, P103]. □

## 2.5 O-Minimal Structures

References are [O-Minimal Structures] and [Tame Topology and O-minimal Structures].

**Notation(2.5.0.1).**

- Use notations defined in [Model Theory](#).
- Fix a real-closed field  $R$ .
- Fix an o-minimal structure on  $R$ (2.5.0.2).

┘

**Def.(2.5.0.2) [O-Minimal Structures on Real Closed Fields].** A structure expanding the real-closed field  $R$  is a collection  $\mathcal{S} = (\mathcal{S}^n)$ , where  $\mathcal{S}^n$  is a family of subsets of  $R^n$ , that

- Algebraic subsets are in  $\mathcal{S}$ .
- Each  $\mathcal{S}^n$  is a Boolean subalgebra of the powerset of  $R^n$ .
- $\mathcal{S}$  is stable under Cartesian product and projection.

And  $\mathcal{S}^n$  are called the **definable subsets** of  $R^n$ . And this structure is called **o-minimal** if moreover each subset in  $\mathcal{S}^1$  is a finite unions of intervals and points.

┘

**Def.(2.5.0.3) [ $r$ -parameterizations].** Let  $X \subset R^n$  be an o-minimally definable set and  $r \in \mathbb{N}$ , then an  **$r$ -parametrization** of  $X$  is a set  $S$  of maps  $(0, 1)^{\dim X} \rightarrow X$  s.t.  $\cup_{\varphi \in S} \text{Im}(\varphi) = X$  and each  $\varphi$  is  $C^r$ -continuous, and  $\varphi^{(\alpha)}$  is strongly-bounded for any  $\alpha \in \mathbb{N}^{\dim X}$  with  $|\alpha| \leq r$ .

Moreover, if  $F : X \rightarrow R^m$  is a definable function and  $S$  is an  $r$ -parametrization of  $X$ , then  $S$  is called an  **$r$ -parametrization** of  $F$  if for  $F \circ S$  is an  $r$ -parametrization of  $F(X)$ .

┘

**Thm.(2.5.0.4) [Gromov Reparameterization theorem].** For any strongly-bounded o-minimally definable subset  $X \subset R^n$  and  $r \in \mathbb{N}$ , there exists an  $r$ -parametrization of  $X$ (2.5.0.3).

And if  $F$  is a strongly-bounded o-minimal definable function on  $X$ , then there exists an  $r$ -parametrization of  $F$ (2.5.0.3).

┘

*Proof:* Cf.[P-W06]P599?.

□

**Cor.(2.5.0.5).** Let  $n, m, r \in \mathbb{Z}_+$  and  $X \subset (0, 1)^n \times R^m$  be a definable family, then there exists  $N \in \mathbb{Z}_+$  and for each  $y \in R^m$  a set  $S_y$  of  $N$  definable functions mapping  $(0, 1)^{\dim X_y}$  to  $X_y$ , s.t.

- $\cup_{\varphi \in S_y} \text{Im}(\varphi) = X_y$ ,
- $|\varphi^{(\alpha)}(x)| \leq 1$  for each  $\varphi \in S_y, \alpha \in \mathbb{N}^{\dim X_y}$  with  $|\alpha| \leq r$  and  $x \in (0, 1)^{\dim X_y}$ .

┘

*Proof:* Cf.[P-W06]P609?.

□

### 1 Theory of Exponential Functions

References are [Zil05] and [Mar06].

**Def.(2.5.1.1) [Exponential Maps].** For  $R \in \mathcal{CRing}$ , an **exponential map** on  $R$  is defined to be a non-trivial group homomorphism  $\exp : R^+ \rightarrow R^\times$ . A ring/field with an exponential map is called an **exponential ring/field**.

┘

**Prop.(2.5.1.2).** An exponential field necessarily has characteristic 0.

┘



**Def. (2.5.1.3) [Exponential Polynomial Rings].** For any exponential ring  $(A, \exp)$  and indeterminants  $X_1, \dots, X_n$ , we can define the **exponential polynomial ring**  $A[X_1, \dots, X_n]_{\exp}$ , and it satisfies an obvious universal property. ? ┘

*Proof:* □

**Thm. (2.5.1.4) [van den Dries/Henson-Rubel].** The natural embedding  $\mathbb{C}[X_1, \dots, X_n]_{\exp} \rightarrow \mathcal{O}(\mathbb{C}^n)$  is injective. ┘

*Proof:* Cf. [Exponential Rings, Exponential Polynomials and Exponential Functions]. □

**Cor. (2.5.1.5).** If  $P \in \mathbb{C}[X, Y] \setminus (\mathbb{C}[X] \cup \mathbb{C}[Y])$  is irreducible, then  $f(z) = P(z, e^z)$  has infinitely many zeros in  $\mathbb{C}$ . ┘

*Proof:* If  $f(z)$  has only f.m. zeros, then by Hadamard's factorization,  $f(z) = e^{az}q(z)$  for some  $q(X) \in \mathbb{C}[X]$ . Then it follows from (2.5.1.4) that

$$P(z, e^z) = e^{az}q(z)$$

is a formal identity. Then  $a \in \mathbb{N}$  and  $P(X, Y) = Y^a q(X)$ , which contradicts the hypothesis. □

**Def. (2.5.1.6) [Exponential Algebraic Closures].** Let  $(L, \exp)$  be an exponential field, and  $A \subset L$ . Suppose  $K \subset L$  is the exponential subfield generated by  $A$ , then the **exponential algebraic closure**  $\text{expcl}(A)$  is defined to be the set of all  $a \in L$  s.t. there exists  $n \in \mathbb{Z}_+$ ,  $f_1, \dots, f_n \in K[X_1, \dots, X_n]_{\exp}$ , and  $(b_1, \dots, b_n) \in L^n$  s.t.

$$f_i(b_1, \dots, b_n) = 0, \forall i, \quad \text{Jac}\left(\frac{f_1, \dots, f_n}{b_1, \dots, b_n}\right) \neq 0, \quad b_1 = a.$$

┘

**Prop. (2.5.1.7).** If  $A \subset \mathbb{C}_{\exp}$  is a finite subset, then  $\text{expcl}(A)$  is countable. ┘

*Proof:* □

**Def. (2.5.1.8) [Ordered Exponential Fields].** Define  $\mathbb{R}_{\exp} = (\mathbb{R}, +, \cdot, \exp, <, 0, 1)$ , and  $\mathbb{C}_{\exp} = (\mathbb{C}, +, \cdot, \exp, <, 0, 1)$  to be structures of ordered exponential fields. ┘

**Thm. (2.5.1.9) [Wilkie].**  $\mathbb{R}_{\exp}$  (2.5.1.8) is o-minimal. ┘

*Proof:* ? □

**Prop. (2.5.1.10).**  $\mathbb{Z}$  is definable in  $\mathbb{C}_{\exp}$ . ┘

*Proof:*  $\mathbb{Z} = \{x \in \mathbb{C} \mid \forall y (\exp(y) = 1 \rightarrow \exp(xy) = 1)\}$ . □

**Conj. (2.5.1.11).**  $\mathbb{C}_{\exp}$  is strongly quasi-minimal. ┘

*Proof:* □

**Conj. Cor. (2.5.1.12).**  $\mathbb{R}$  is not definable in  $\mathbb{C}_{\exp}$ . ┘

**Conj. Cor. (2.5.1.13).**  $\# \text{Aut}(\mathbb{C}_{\exp}) = 2^{2^{\aleph_0}}$ . ┘

*Proof:* □

**Conj. (2.5.1.14) [Schanuel's Conjecture over the Kernel].** Let  $(K, \exp)$  be an exponential field, then the **Schanuel's conjecture over the kernel** says: For any  $n \in \mathbb{Z}_+$  and  $\alpha_1, \dots, \alpha_n \in K$ ,

$$\text{tr.deg}_{\mathbb{Q}(\ker \exp)} \mathbb{Q}(\alpha_1, \dots, \alpha_n, \exp(\alpha_1), \dots, \exp(\alpha_n), \ker \exp) \geq \dim_{\mathbb{Q}} \left( (\ker \exp) \otimes \mathbb{Q} + \sum \mathbb{Q}\alpha_i \right) / (\ker \exp) \otimes \mathbb{Q}.$$

Notice the second term just equals  $\text{rank}(\exp(\alpha_1)^{\mathbb{Z}} \cdot \dots \cdot \exp(\alpha_n)^{\mathbb{Z}})$ . ┘

*Proof:* □

**Prop. (2.5.1.15).** The Scaneul's conjecture is weak than the Schanuel's conjecture(19.9.1.2). But on  $\mathbb{C}_{\exp}$ , they are equivalent. ? ┘

*Proof:* □

**Conj. (2.5.1.16) [Converse Schanuel's Conjecture].** Let  $K \in \text{Field}$  is uncountable with an exponential function  $\exp$  s.t.  $\ker(\exp) \cong \mathbb{Z}$ . Suppose that Schanuel's conjecture holds for  $K$ (19.9.1.2), then there exists an embedding  $(K, +, \cdot, \exp, 0, 1) \hookrightarrow \mathbb{C}_{\exp}$ . ┘

**Def. (2.5.1.17) [Strongly Exponentially-Closed Fields with Exponentials, Zilber].** Zilber showed that there is an  $\mathcal{L}_{\omega_1, \omega}(Q)$ -theory  $C_{\exp}$  of **strongly exponentially-closed fields with exponentials**. ┘

*Proof:* Cf.[Zil05]. □

**Thm. (2.5.1.18) [Zilber].**  $C_{\exp}$  is strongly quasi-minimal(2.4.3.20).

For any uncountable cardinal  $\kappa$ , there exists a unique model for  $C_{\exp}$  of cardinality  $\kappa$ .

If  $\mathcal{M} \models C_{\exp}$  and  $A \subset \mathcal{M}$  is finite, then for any  $a, b \in \text{cl}(A)$ , there exists an automorphism of  $\mathcal{M}$  taking  $a$  to  $b$ . ┘

*Proof:* □

**Conj. (2.5.1.19) [Zilber].**  $\mathbb{C}_{\exp}$  is strongly exponentially-closed. ┘

*Proof:* □

**Prop. (2.5.1.20).** Suppose Schanuel's conjecture is true(19.9.1.2), then for  $P \in \mathbb{C}[X, Y] \setminus (\mathbb{C}[X] \cup \mathbb{C}[Y])$  is irreducible,  $f(z) = P(z, e^z)$  has infinitely many alg.independent zeros in  $\mathbb{C}$ . ┘

*Proof:* Cf.[Mar06]. □

## 2 O-Minimal Structures on $\mathbb{R}^n$

References are [Dri98].

### (Globally)Subanalytic Sets

**Thm. (2.5.2.1).** (Globally)Subanalytic sets form an o-minimal structure. ┘

*Proof:* ? □

**Thm. (2.5.2.2).**  $\mathbb{R}_{\text{an}, \exp}$  is o-minimal. ┘

*Proof:* ? □

### Cell Decomposition Theorem

**Def. (2.5.2.3) [Cells].** Defined in [Tame Topology and O-minimal Structures]P50. ? ┘

**Thm. (2.5.2.4) [Cell Decomposition Theorem, Knight-Pillay-Steinhorn[KPS86]].** Each o-minimally definable set  $A \subset \mathbb{R}^n$  has a finite partition  $A = C_1 \cup \dots \cup C_n$  of cells. And if  $f : A \rightarrow \mathbb{R}^n$  is a definable map, the cells can also be chosen s.t. all restrictions  $f|_{C_i}$  are continuous. ┘

*Proof:* Cf.[Tame Topology and O-minimal Structures]P52. ? □

**Prop. (2.5.2.5) [Real O-minimal f.m. Connected Components].** An o-minimal definable set  $X \subset \mathbb{R}^n$  has only f.m. components. In particular, a semi-algebraic set  $X \subset \mathbb{R}^n$  has only f.m. connected components. ┘

*Proof:* ? □

### Definable Invariants

**Def. (2.5.2.6) [Dimensions].** The **dimension** of a non-empty o-minimal definable subset  $X \subset \mathbb{R}^m$  is defined to be

$$\dim X = \max\{i_1 + \dots + i_m \mid X \text{ contains an } (i_1, \dots, i_m)\text{-cell}\}.$$

┘

**Prop. (2.5.2.7).**

- If  $X \subset Y \subset \mathbb{R}^m$  are o-minimally definable, then  $\dim X \leq \dim Y \leq m$ .
- If  $X \subset \mathbb{R}^m, Y \subset \mathbb{R}^n$  are o-minimally definable and there is a definable bijection between  $X$  and  $Y$ , then  $\dim X = \dim Y$ .
- If  $X, Y \subset \mathbb{R}^m$  are o-minimally definable, then  $\dim(X \cup Y) = \max(\dim X, \dim Y)$ .

┘

*Proof:* Cf.[Dri98]P64. ? □

**Prop. (2.5.2.8).** Let  $S \subset \mathbb{R}^m$  be a non-empty definable set, then  $\dim \partial S \leq \dim S$ . In particular,  $\dim \bar{S} = \dim S$ . ┘

*Proof:* Cf.[Dri98]P67. ? □

## 2.6 Elementary Number Theory

### 1 Primes

**Def. (2.6.1.1) [Prime Numbers].**  $p \in \mathbb{Z}_+$  is called a (rational)**prime number** if for any  $d \in \mathbb{Z}_+$ ,  $d|p$  implies  $d = 1$  or  $d = p$ . The set of prime numbers is denoted by  $\mathbf{P}$ .  $\lrcorner$

**Lemma (2.6.1.2) [Infinitely Many Primes].**  $\#\mathbf{Prime} = \infty$ .  $\lrcorner$

*Proof:* Suppose  $\#\mathbf{Prime} < \infty$ , let  $N = \prod_{p \in \mathbf{Prime}} p$ . Then  $N+1$  has no prime factors, contradiction.  $\square$

**Prop. (2.6.1.3) [Multiplicative Groups of Remainders].** For  $p \in \mathbf{Prime}$  and  $n \in \mathbb{Z}_+$ , the  $p$ -adic logarithm map defines an isomorphism

$$(\mathbb{Z}/(p))^* \cong \begin{cases} \mathbb{Z}/(2) \times \mathbb{Z}/(2^{n-2}) & p = 2, n \geq 2 \\ \mathbb{Z}/(p^{n-1}(p-1)) & \text{otherwise} \end{cases}.$$

$\lrcorner$

*Proof:* This follows from (19.1.5.16). For the first case, notice that there is a split exact sequence

$$1 \rightarrow (1 + 4\mathbb{Z})/(1 + 2^n\mathbb{Z}) \rightarrow (\mathbb{Z}/(2^n))^* \rightarrow (\mathbb{Z}/(4))^* \rightarrow 1.$$

$\square$

**Def. (2.6.1.4) [Mersenne-Primes].** If  $p \in \mathbf{Prime}$  and  $M_p = 2^p - 1$  is also a prime, then  $M_p$  is called a **Mersenne prime**.  $\lrcorner$

**Def. (2.6.1.5) [Perfect Numbers].** A **perfect number** is a number  $n \in \mathbb{Z}_+$  s.t.

$$\sigma_1(n) = 2n \text{ (2.6.1.23).}$$

Moreover, an **abundant number** is a number  $n \in \mathbb{Z}_+$  s.t.  $\sigma(n) > 2n$ , and a **deficient number** is a number  $n \in \mathbb{Z}_+$  s.t.  $\sigma(n) < 2n$ .  $\lrcorner$

**Thm. (2.6.1.6) [Even Perfect Numbers, Euclid-Euler].**  $n \in \mathbb{Z}_+$  is an even perfect number iff there exists  $p \in \mathbf{Prime}$  s.t.  $n = 2^{p-1}(2^p - 1)$  and  $2^p - 1 \in \mathbf{Prime}$ .  $\lrcorner$

*Proof:* If  $p \in \mathbf{Prime}$  s.t.  $n = 2^{p-1}(2^p - 1)$  and  $2^p - 1 \in \mathbf{Prime}$ , then it is clear that  $n = 2^{p-1}(2^p - 1)$  is a perfect number. Conversely, suppose  $n = 2^k x$  is even and perfect, where  $2 \nmid x$ . Then

$$2^{k+1}x = \sigma(2^k x) = (2^{k+1} - 1)\sigma(x).$$

Denote  $y = x/(2^{k+1} - 1) \in \mathbb{Z}_+$ , then

$$2^{k+1}y = \sigma(x) \geq x + y = 2^{k+1}y.$$

Thus  $x$  has only two divisors:  $x$  and  $y$ . So  $y = 1$ , and  $x = 2^{k+1} - 1$  is a prime. This forces  $k+1$  to be a prime, and  $x$  being a Mersenne prime.  $\square$

**Conj. (2.6.1.7).** There are no odd perfect numbers.  $\lrcorner$

*Proof:*

□

**Thm& Conj.Cor.(2.6.1.8) [Dickson-Nielsen-Pollack].** For any  $k \in \mathbb{Z}_+$ ,

1. If  $n$  is an odd perfect number, then  $\nu(n) \geq 9$ .
2. If  $n$  is an odd perfect number with  $\nu(n) \leq k$ , then  $n < 2^{4^k}$ .
3. For  $X \in \mathbb{Z}_+$ , the number of odd perfect numbers  $n \leq X$  with  $\nu(n) \leq k$  is bounded by  $(\log x)^k$ .
4. The number of odd perfect numbers  $N$  with  $\nu(n) \leq k$  is bounded by  $4^{k^2}$ .

┘

*Proof:* 1: Cf.[Nielsen, Odd perfect numbers have at least nine distinct prime factors].

2: Cf.[Nielsen, An upper bound for odd perfect numbers].

3: Let  $N$  be an odd perfect number with  $9 \leq \nu(N) \leq k$ , let  $p_0$  be the minimal prime divisor of  $N$ , and  $p_0^{e_0} \parallel N$ . Denote  $B = p_0^{e_0}$  and  $N = BA$ . Then

$$B < \sigma(B) = \frac{2A}{\sigma A} B \leq 2B,$$

with equality on the right iff  $A = 1$ . Suppose  $A \neq 1$ , then  $\sigma(B) \nmid 2B$ . So there is a least  $p_1 \in \mathbf{P}$  s.t.  $p_1 \mid \sigma(B)$ ,  $p_1 \nmid 2B$ . So  $p_1 \mid A$ . Suppose  $p_1^{e_1} \parallel A$ , and let  $A' = A/p_1^{e_1}$ ,  $B' = Bp_1^{e_1}$ , then the same reasoning can apply to  $A'$  and  $B'$ . Finally, we get

$$A = p_1^{e_1} p_2^{e_2} \dots p_l^{e_l},$$

and  $8 \leq l = \nu(N) - 1 \leq k - 1$ . Notice

$$\frac{1}{2} = \frac{N}{\sigma(N)} \geq \prod_{p \mid N} \left(1 - \frac{1}{p}\right) \geq 1 - \sum_{p \mid N} \frac{1}{p} \geq 1 - \frac{k}{p_0},$$

so  $p_0 \leq 2k$ . Thus there are  $\leq k - 1$  possibilities of  $p_0$ . And there are  $\leq \log x / \log 3$  possibilities of  $e_0$ . Notice that if  $B$  is given, then  $A$  is totally determined by the sequence  $e_1, \dots, e_n$ . Thus there are at most

$$\frac{(k-1)(k-4)}{(\log 3)(\log 5)^{k-1}} (\log x)^k$$

possibilities of  $N = AB$ . This coefficient is decreasing for  $k \geq 9$ , and it is smaller than 1.

4 follows from item2 and item3.

□

**Thm& Conj.Cor.(2.6.1.9) [Euler].** If  $N$  is an odd perfect prime, then there exists  $p \in \mathbf{Prime}$  s.t.  $N = p^\gamma m^2$ , where  $p \equiv \gamma \equiv 1 \pmod{4}$ .

┘

*Proof:*

□

**Thm.(2.6.1.10) [Power Lifting].** If  $p \in \mathbf{P}$ ,  $a, b \in \mathbb{Z} \setminus (p)$ , and  $v_p(a - b) = k \geq 1$ , then  $v_p(a^{p^n} - b^{p^n}) = k + n$  for  $n \in \mathbb{Z}_+$ , except for  $p = 2$  and  $k = 1$ , in which case  $v_p(a^{p^n} - b^{p^n}) = k + n + 1$  for  $n \in \mathbb{Z}_+$ .

┘

*Proof:* Use induction on  $n$ : If  $a^{p^n} = b^{p^n} + p^{n+k}c$  for some  $c$  s.t.  $p \nmid c$ , then

$$a^{p^{n+1}} = b^{p^{n+1}} + p^{n+k+1}b^{p^n(p-1)}c \dots + p^{(n+k)p}c^p,$$

and the assertion follows.

□

**Prop. (2.6.1.11).** For  $n \geq 2$ , the multiplicative group of  $(\mathbb{Z}/(2^n))^* \cong \mathbb{Z}/(2) \times \mathbb{Z}/(2^{n-2})$ .  $\lrcorner$

*Proof:* it suffices to show  $3^{2^{n-3}} \not\equiv \pm 1 \pmod{2^n}$ , and  $3^{2^{n-2}} \equiv 1 \pmod{2^n}$ .  $\square$

**Prop. (2.6.1.12) [*b*-adic Decompositions and Irreducibility].** If  $b > 2$  and  $p$  is a prime, consider the  $b$ -adic expansion  $p = \sum a_n b^n$ , then the polynomial

$$\sum a_n X^n$$

is irreducible over  $\mathbb{Z}$ .  $\lrcorner$

*Proof:* If  $p(x) = h(x)r(x)$ , use the lemma?? to show that  $r(b)$  and  $h(b)$  cannot be 1, thus  $h(b)r(b) = p$  cannot happen.  $\square$

**Thm. (2.6.1.13) [Zsigmondy].** If  $a, n \in \mathbb{Z}_{\geq 2}$ , then there exists a prime divisor of  $a^n - 1$  not dividing  $a^j - 1$  for any  $0 < j < n$ , except in the following cases:

- $n = 2, a = 2^s - 1, s \geq 2$ .
- $n = 6, a = 2$ .

$\lrcorner$

*Proof:* By (3.3.2.8), it suffices to find a prime ideal of  $\Psi_n(a)$  not dividing  $n$ . If  $n = 2$ , then any prime divisor of  $a + 1$  not dividing  $a - 1$  suffices, as long as  $a$  is not of the form  $2^s - 1$ .

For  $n \geq 3$ , if all prime divisors of  $\Psi_n(a)$  divides  $n$ , we prove that  $\Psi_n(a) \in \mathbf{P}$ : Take  $p | \Psi_n(a)$ , and let  $k = \text{ord}(a, \mathbb{F}_p^\times) | (p - 1) < n$ , then  $p | \Psi_k(a)$ , and  $n/k = p^t$  for some  $t \in \mathbb{Z}_+$ . Then if  $p \neq 2$ , by power-lifting (2.6.1.10),

$$v_p(a^n - 1) = v_p(a^{n/p} - 1) + 1,$$

so  $v_p(\Psi_n(a)) = 1$ . And if  $p = 2$ , then clearly  $k = 1$ , and  $n \in 2^{\mathbb{Z}_+}$ , and  $\Psi_n(a) = a^{n/2} + 1 \equiv 2 \pmod{4}$  as  $n \neq 2$ , so it is still true that  $v_2(\Psi_n(a)) = 1$ .

And if  $\ell \in \mathbf{P} \setminus \{p\}$  is another prime dividing  $\Psi_n(a)$ , then  $n/p^t = k | (p - 1)$ ,  $n/\ell^{t'} = k' | \ell - 1$ , so  $p \leq \ell - 1 < \ell \leq p - 1$ , contradiction. So  $\Psi_n(a) = p$  is a prime.

Then suppose  $n = p^k r$  where  $(r, p) = 1$ , we prove that  $p > (b^{p-2}(b - 1))^{\varphi(r)}$  where  $b = a^{q^{k-1}}$ : By (3.3.2.5),

$$\Psi_n(a) = \Psi_r(b^p) / \Psi_r(b) \geq \left( \frac{b^p - 1}{b + 1} \right)^{\varphi(r)} \geq \left( \frac{b^{p-2}(b^2 - 1)}{b + 1} \right)^{\varphi(r)} = (b^{p-2}(b - 1))^{\varphi(r)}.$$

If  $p \geq 5$ , then we have  $b^{p-2} > p$ , so only possible cases are  $p = 2$  or  $3$ . The case  $p = 2$  is not possible, because then  $a = 2$ , and  $2 = \Psi_n(2) \equiv 1 \pmod{2}$ . So  $p = 3, a = 2, k = 1$  and  $r = 1$  or  $2$ . But  $\Psi_n(2) \neq 3$ , contradiction.  $\square$

## Integers

**Def. (2.6.1.14) [Numerical Polynomials].** A function  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  is called a **numerical polynomial** if it there exists  $a_1, \dots, a_r \in \mathbb{Z}$  that

$$f(n) = \sum_{i=0}^r a_i \binom{n}{i}.$$

for  $n$  sufficiently large.  $\lrcorner$

**Prop. (2.6.1.15).** Suppose that  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  is defined for  $n$  sufficiently large and  $g(n) = f(n) - f(n - 1)$  is a numerical polynomial, then  $f$  is a numerical polynomial.  $\lrcorner$

*Proof:* Suppose  $f(n) - f(n-1) = \sum_{i=0}^r a_i \binom{n}{i}$  for all  $n > 0$ . If we set  $g(n) = f(n) - \sum_{i=0}^r a_i \binom{n+1}{i+1}$ , then  $g(n) - g(n-1) = 0$  for  $n$  sufficiently large, so it is eventually constant, which is equal to  $a_{-1}$ . Then  $f(n) = \sum_{i=0}^r a_i \binom{n+1}{i+1} + a_{-1}$  is a numerical polynomial.  $\square$

**Cor. (2.6.1.16).** Any polynomial function with coefficients in  $\mathbb{Q}$  maps any integer sufficiently large into  $\mathbb{Z}$  iff it is a numerical function.  $\lrcorner$

**Prop. (2.6.1.17).** Let  $b, k \in \mathbb{Z}_+, n \in \mathbb{Z}$ , then  $v_p(n(n-b) \dots (n-(k-1)b)) \geq v_p(k!)$  for any  $p \nmid b$ .  $\lrcorner$

*Proof:* This is because  $v_p(n(n-b) \dots (n-kb)) \geq [k/p] + [k/p^2] + \dots \geq v_p(k!)$ .  $\square$

**Cor. (2.6.1.18).** Let  $m \in \mathbb{Z}_+$  and  $a \in \mathbb{Q}^*$  with denominator  $b$ , then for  $N$  large,  $b^N \binom{a}{m} \in \mathbb{Z}$ .  $\lrcorner$

### Useful Functions

**Def. (2.6.1.19) [Multiplicative and Additive Functions].** A function  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  is called a **multiplicative function** if  $f(mn) = f(m)f(n)$  for any  $m, n \in \mathbb{Z}_+$  s.t.  $(m, n) = 1$ . And it is called an **additive function** if  $f(mn) = f(m) + f(n)$  for any  $m, n \in \mathbb{Z}_+$  s.t.  $(m, n) = 1$ .  $\lrcorner$

**Def. (2.6.1.20) [Greatest Common Divisors].** For  $S \subset \mathbb{Z}$ , the **greatest common divisor**  $\gcd(S) \in \mathbb{Z}_+ \cup \{\infty\}$  denote the supremum of integers  $m \in \mathbb{Z}_+$  s.t.  $m|s$  for each  $s \in S$ .

For  $S \subset \mathbb{Z}^\times$ , the **least common multiple**  $\text{lcm}(S) \in \mathbb{Z}_+ \cup \{\infty\}$  denote the supremum of integers  $m \in \mathbb{Z}_+$  s.t.  $s|m$  for each  $s \in S$ .  $\lrcorner$

**Prop. (2.6.1.21).** For  $n \in \mathbb{Z}_{\geq 7}$ ,

- For any  $1 \leq m \leq n$ ,  $m \binom{n}{m} \mid \text{lcm}(\{1, 2, \dots, n\})$ .
- $\text{lcm}(\{1, 2, \dots, n\}) \geq 2^n$ .

$\lrcorner$

*Proof:* 1: By (2.6.1.33), for any  $p \in \text{Prime}$ , if  $v_p(m) = k$ , then  $k + v_p(\binom{n}{m}) \leq v_p(\text{lcm}(\{1, 2, \dots, n\}))$ . Thus the assertion follows.

2: By item1,  $n \binom{2n}{n} \mid \text{lcm}(\{1, 2, \dots, 2n\})$ , and  $(2n+1) \binom{2n}{n} = (n+1) \binom{2n+1}{n+1} \mid \text{lcm}(\{1, 2, \dots, 2n+1\})$ . So

$$n(2n+1) \binom{2n}{n} \mid \text{lcm}(\{1, 2, \dots, 2n+1\}),$$

and  $\text{lcm}(\{1, 2, \dots, 2n+1\}) \geq n4^n$ . So for  $n \geq 2$ ,  $\text{lcm}(\{1, 2, \dots, 2n+1\})$ , and for  $n \geq 4$ ,  $\text{lcm}(\{1, 2, \dots, 2n+2\}) \geq \text{lcm}(\{1, 2, \dots, 2n+1\}) \geq 2^{2n+2}$ . The case  $n \in \{7, 8\}$  can be verified directly.  $\square$

**Def. (2.6.1.22) [Prime Factors and the Liouville-Function].** Define functions:

$$\Omega : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+ : n \mapsto \sum_{p \in \text{Prime}} v_p(n), \quad \omega : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+ : n \mapsto \#\{p \in \text{Prime} : p|n\}$$

and also  $\nu(n) = (-1)^{\Omega(n)}$ , called the **Liouville-function**.  $\lrcorner$

**Def. (2.6.1.23) [ $\sigma_s$ -Function].** For  $s \in \mathbb{R}$ , the function  $\sigma_s$  is defined to be

$$\sigma_s : \mathbb{Z}_+ \rightarrow \mathbb{R}_+ : n \mapsto \sum_{d \in \mathbb{Z}_+, d|n} d^s.$$

And also denote  $\sigma(n) = \sigma_1(n)$ .

And for  $s = 1$ , denote  $\tau(n) = \sigma_0(n)$ , called the  **$\tau$ -function**.  $\lrcorner$

**Def. (2.6.1.24) [Euler's  $\phi$ -Function].** The **Euler's  $\phi$ -function** is defined to be

$$\phi : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+ : n \mapsto \#(\mathbb{Z}/(n))^* = n \prod_{p \in S(n)} \left(1 - \frac{1}{p}\right).$$

┘

*Proof:* For the explicit formula, use the fact  $\phi = \mu * n$  (2.6.2.10) is multiplicative and

$$\phi(p^k) = \mu(1)p^k + \mu(p)p^{k-1} = p^k(1 - p^{-1}).$$

for any  $p \in \text{Prime}$ ,  $k \in \mathbb{Z}_+$ . □

**Def. (2.6.1.25) [Möbius-Function].** The **Möbius-function** is defined to be the multiplicative function

$$\mu : \mathbb{Z}_+ \rightarrow \{0, 1, -1\} : n \mapsto \begin{cases} (-1)^k & n = \prod_{i=1}^k p_i, p_i \neq p_j \in \text{Prime} \\ 0 & \text{otherwise} \end{cases}.$$

┘

**Prop. (2.6.1.26) [Möbius Identities].** For  $n \in \mathbb{Z}_+$ ,

- If  $m \in \mathbb{Z}_+$  is prime to  $n$ , then  $\mu(mn) = \mu(m)\mu(n)$  (2.6.1.25).

•

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases}.$$

- For  $p \in \text{Prime}$ ,

$$\sum_{d|n, p \nmid d} \mu(d) = \begin{cases} 1 & n \in p^{\mathbb{Z}_+} \\ 0 & \text{otherwise} \end{cases}.$$

┘

*Proof:* 1: This is clear.

2: By item1, it suffices to prove for  $d = p^k$  for some  $p \in \text{Prime}$ , then this is clear.

3: By item1, it suffices to prove for  $d = p^k$  for some  $p \in \text{Prime}$ , then this is clear. □

**Prop. (2.6.1.27) [Kronecker Symbol].** Define the **Kronecker symbol** as a function

$$\mathbb{Z} \times \mathbb{Z} \rightarrow \{0, -1, 1\} : (a, m) \mapsto \left(\frac{a}{m}\right)$$

satisfying the following:

- $\left(\frac{a}{m}\right)$  is totally multiplicative in  $a$  and in  $m$ .
- $\left(\frac{a}{1}\right) = 1$  for any  $a \in \mathbb{Z}$ .
- $\left(\frac{a}{0}\right) = \begin{cases} 1 & , a \in \{\pm 1\} \\ 0 & , \text{otherwise} \end{cases}.$
- $\left(\frac{a}{-1}\right) = \begin{cases} -1 & a < 0 \\ 1 & a \geq 0 \end{cases}.$



- $\left(\frac{a}{2}\right) = \begin{cases} 0 & a \in 2\mathbb{Z} \\ (-1)^{\frac{a^2-1}{8}} & a \notin 2\mathbb{Z} \end{cases}.$
- For  $p \in \text{Prime}_{\geq 3}$ ,  $\left(\frac{a}{p}\right) = \begin{cases} 0 & , a \in (p) \\ 1 & , a \notin (p), a^{\frac{p-1}{2}} \equiv 1 \pmod{p} \\ -1 & , a \notin (p), a^{\frac{p-1}{2}} \equiv -1 \pmod{p} \end{cases}.$

┘

**Cor. (2.6.1.28).** If  $m \in 2\mathbb{Z}_+ + 1$ , then

- $\left(\frac{-1}{m}\right) = (-1)^{\frac{m-1}{2}}.$
- $\left(\frac{2}{m}\right) = (-1)^{\frac{m^2-1}{8}}.$
- $\left(\frac{n}{m}\right) = (-1)^{(n-1)(m-1)/4} \left(\frac{m}{n}\right)$  for any  $n \in \mathbb{Z}.$
- If  $D \equiv 0, 1 \pmod{4}$  and  $n \in 2\mathbb{Z}_+ + 1$  s.t.  $n \equiv m \pmod{D}$ , then  $\left(\frac{D}{n}\right) = \left(\frac{D}{m}\right).$

┘

**Lemma (2.6.1.29).**  $\sum_{x=0}^{p-1} x^k \equiv \begin{cases} -1 \pmod{p} & , (p-1)|k \\ 0 \pmod{p} & , \text{otherwise} \end{cases}.$

┘

*Proof:* Choose a  $a$  that  $a^k - 1$  is not divisible by  $p$ , this is doable iff  $k$  is not divisible by  $p-1$ , then it is clear.  $\square$

**Def. (2.6.1.30) [Integral Part].** For any  $\alpha \in \mathbb{R}$ , let  $[\alpha]$  be the maximal integer  $n$  that  $n \leq \alpha$ , called the **integral part of  $\alpha$**  and  $\{\alpha\} = \alpha - [\alpha]$ , called the **fractional part of  $\alpha$** .

Also denote  $\llbracket \alpha \rrbracket = [\alpha + \frac{1}{2}]$  to be the integer nearest to  $\alpha$  and  $\{\{\alpha\}\} = |\alpha - \llbracket \alpha \rrbracket|$  to be the distance from  $\alpha$  to the nearest integer.  $\lrcorner$

**Def. (2.6.1.31) [von.Mangoldt Function].** The **von.Mangoldt function** is defined to be

$$\Lambda : \mathbb{Z}_+ \rightarrow \mathbb{R} : \Lambda(n) = \begin{cases} \log p & , p \in \text{Prime}, n \in p^{\mathbb{Z}_+} \\ 0 & , \text{otherwise} \end{cases}.$$

┘

**Def. (2.6.1.32) [Chebyshev Functions].** For  $X \in \mathbb{R}_+$ , define the **Chebyshev function of first kind**

$$\vartheta(X) \triangleq \sum_{p \in \text{Prime}, p \leq X} \log p$$

and the **Chebyshev function of second kind**

$$\psi(X) \triangleq \sum_{n \in \mathbb{Z}_+, n \leq X} \Lambda(n) \text{ (2.6.1.31)}.$$

Notice  $\psi(X) \geq \vartheta(X).$   $\lrcorner$

### Congruences of Binomial Coefficients

References are <http://www.cecm.sfu.ca/organics/papers/granville/paper/binomial/html/binomial.html>.

**Prop. (2.6.1.33) [ $p$ -Power in Factorials].** For  $n \in \mathbb{Z}_+$ ,  $v_p(n!) = \frac{n-c(n)}{p-1}$ , where  $c(n)$  is the sum of the presentation of  $n$  in the  $p$ -adic base. In particular,

$$\frac{n-p+1}{p} \leq v_p(n!) \leq \frac{n-1}{p-1}.$$

┘

**Cor. (2.6.1.34) [Kummer].**  $v_p(\binom{a+b}{a})$  equals the number of carries when adding  $a$  and  $b$  in base  $p$ . ┘

**Prop. (2.6.1.35) [Wilson].** For  $p \in \text{Prime}$ ,  $(p-1)! \equiv -1 \pmod{p}$ . ┘

*Proof:* Consider the two polynomials

$$g(X) = \prod_{k=1}^{p-1} (X-k), \quad h(X) = X^{p-1} - 1,$$

then

$$f(X) = g(X) - h(X)$$

are identically zero on  $\mathbb{F}_q$  but has degree  $\leq p-2$ , so it must be identically 0. Thus the assertion follows.  $\square$

**Cor. (2.6.1.36).** For any  $n \in \mathbb{Z}$ ,

$$\binom{np-1}{p-1} \equiv 1 \pmod{p}.$$

And

$$\binom{2p-1}{p-1} \equiv 1 \pmod{p^r}$$

where  $r = \min(3, p-1)$ . ┘

*Proof:*  $\square$

**Prop. (2.6.1.37) [Lucas].** For  $m, r \in \mathbb{Z}$  s.t.  $n = m + r$ , let  $n_i, m_i, r_i$  be their  $i$ -th term in base- $p$ , then if  $k = v_p(\binom{n}{m})$ , then

$$\frac{(-1)^k}{p^k} \binom{n}{m} \equiv \prod_k \frac{n_k!}{m_k! r_k!} \pmod{p}.$$

┘

*Proof:* Denote  $(n!)_p = \prod_{1 \leq m \leq n, p \nmid m} m$ , then by some argument, it suffices to prove that

$$(-1)^{\lfloor n/p \rfloor} (n!)_p \equiv n_0! \pmod{p}.$$

And this follows easily from Wilson's theorem (2.6.1.35).  $\square$

**Prop. (2.6.1.38) [Glaisher].** If  $p \in \text{Prime}$ ,  $1 \leq j, k \leq p-1$  and  $n \in \mathbb{Z}$  s.t.  $n \equiv k \pmod{p-1}$ , then

$$\sum_{1 \leq m \leq n, m \equiv j \pmod{p-1}} \binom{n}{m} \equiv \binom{k}{j} \pmod{p}.$$

┘

*Proof:* Notice if the  $i$ -th term of  $n, m$  in the base  $p$  are  $n_i, m_i$ , then  $p \nmid \binom{n}{m}$  iff  $m_i < n_i$  for each  $i$ . And then

$$\sum_{1 \leq m \leq n-1, m \equiv j \pmod{p-1}} \binom{n}{m} = \sum_{(m_0, \dots, m_d)} \binom{n_0}{m_0} \cdots \binom{n_d}{m_d}.$$

But the RHS equals the sum of coefficients of  $X^j, X^{j+(p-1)}, X^{j+2(p-1)}, \dots$  in  $(X+1)^{n_0}(X+1)^{n_1} \cdots (X+1)^{n_d} = (X+1)^{\sum n_i}$ , which is also equal to

$$\sum_{1 \leq m \leq \sum n_i, m \equiv j \pmod{p-1}} \binom{\sum n_i}{m}$$

And also  $n \equiv \sum n_i \pmod{p-1}$ , so we can use induction on  $n$ , and notice that the assertion is trivial for  $n \leq p-1$ . □

**Cor. (2.6.1.39) [Hermite].** If  $n \in \mathbb{Z}_+, p \in \text{Prime}$ , then

$$\sum_{1 \leq m \leq n-1, (p-1) \mid m} \binom{n}{m} \equiv 0 \pmod{p}$$

┘

**Prop. (2.6.1.40) [Carlitz].** If  $n \in \mathbb{Z}_+, p \in \text{Prime}, r \in \mathbb{Z}_+$  and  $p^{r-1} \mid n$ , then

$$p + (p-1) \sum_{1 \leq m \leq n-1, (p-1) \mid m} \binom{n}{m} \equiv 0 \pmod{p^r}.$$

┘

*Proof:* □

**Prop. (2.6.1.41) [Morley].** For  $p \in \text{Prime}_{\geq 5}$ ,

$$(-1)^{\frac{p-1}{2}} \binom{p-1}{(p-1)/2} \equiv 4^{p-1} \pmod{p^3}.$$

┘

*Proof:* □

## 2 $L$ -Series

**Def. (2.6.2.1) [ $L$ -Series].** For  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$ , the  $L$ -**function** of  $f$  is defined to be a formal sum

$$L(f; s) = \sum_{n \in \mathbb{Z}_+} \frac{f(n)}{n^s}.$$

In particular, for  $f, g, h : \mathbb{Z}_+ \rightarrow \mathbb{C}$ , the equation

$$L(h; s) = L(f; s)L(g; s)$$

just means

$$h(n) = \sum_{d|n, d \in \mathbb{Z}_+} f(d)g\left(\frac{n}{d}\right)$$

for any  $n \in \mathbb{Z}_+$ . In this case, we denote  $h = f * g$ , called the **Dirichlet convolution** of  $f$  and  $g$ .  $\lrcorner$

**Def. (2.6.2.2) [Riemann's Zeta Function].** The **Riemann's zeta function** is defined to be the formal sum

$$\zeta(s) = L(\mathbf{1}; s) = \sum_{n \in \mathbb{Z}_+} \frac{1}{n^s} \text{ (2.6.2.1)} = \prod_{p \in \text{Prime}} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \dots\right).$$

$\lrcorner$

**Prop. (2.6.2.3).**

$$\zeta^{-1}(s) = \sum_{n \in \mathbb{Z}_+} \frac{\mu(n)}{n^s}.$$

i.e.  $\zeta * \mu = \delta$ .  $\lrcorner$

**Cor. (2.6.2.4) [Möbius-Inversion].** For any  $f, g : \mathbb{Z}_+ \rightarrow \mathbb{C}$ , the following are equivalent:

$$g(n) = \sum_{d|n} f(d), \forall n \in \mathbb{Z}_+.$$

$$f(n) = \sum_{d|n} \mu(d)g\left(\frac{n}{d}\right), \forall n \in \mathbb{Z}_+.$$

$\lrcorner$

*Proof:* The first equation is equivalent to  $L(g; s) = \zeta(s)L(f; s)$ , while the second equation is equivalent to  $L(f; s) = \zeta^{-1}(s)L(g; s)$ .  $\square$

**Cor. (2.6.2.5) [Another Möbius-inversion].** For any finite sequence  $(a_n)_{n \in \mathbb{Z}_+}, (b_m)_{m \in \mathbb{Z}_+}$  supported on square-free numbers, the following two statements are equivalent:

$$b_d = \mu(d) \sum_{d|m} a_m, \forall d \in \mathbb{Z}_+.$$

$$a_d = \mu(d) \sum_{d|m} b_m, \forall d \in \mathbb{Z}_+.$$

$\lrcorner$

*Proof:* Direct calculation.  $\square$

**Prop. (2.6.2.6).** Because  $\log : \mathbb{Z}_+ \rightarrow \mathbb{R}$  is totally additive, we have

$$\log \cdot (f * g) = (\log \cdot f) * g + f * (\log \cdot g)$$

for any  $f, g : \mathbb{Z}_+ \rightarrow \mathbb{C}$ . ┘

**Prop. (2.6.2.7).** Because  $\text{id} : \mathbb{Z}_+ \rightarrow \mathbb{R}$  is totally multiplicative, we have

$$\frac{f(n)}{n} * \frac{g(n)}{n} = \frac{(f * g)(n)}{n}.$$

for any  $f, g : \mathbb{Z}_+ \rightarrow \mathbb{C}$ . ┘

### L-Series of Elementary Functions

**Prop. (2.6.2.8).** For  $r \in \mathbb{R}$ ,

$$\zeta(s)\zeta(s-r) = \sum_{n \in \mathbb{Z}_+} \sigma_r(n)n^{-s}.$$

In particular,  $\zeta^2(s) = \sum_{n \in \mathbb{Z}_+} \sigma_0(n)n^{-s}$ . ┘

**Prop. (2.6.2.9) [Rankin-Selberg  $\zeta$ -Series].** For  $\alpha, \beta \in \mathbb{R}_+$ ,

$$\zeta(s)\zeta(s-\alpha)\zeta(s-\beta)\zeta(s-\alpha-\beta)\zeta^{-1}(2s-\alpha-\beta) = \sum_{n \in \mathbb{Z}_+} \sigma_\alpha(n)\sigma_\beta(n)n^{-s}.$$

In particular,

$$\zeta^4(s)\zeta^{-1}(2s) = \sum_{n \in \mathbb{Z}_+} \sigma_0(n)^2 n^{-s} = \prod_{p \in \text{Prime}} \left(1 + \frac{4}{p^s} + \frac{9}{p^{2s}} + \dots\right).$$

Dividing by  $\zeta(s)$ , we get

$$\zeta^3(s)\zeta^{-1}(2s) = \sum_{n \in \mathbb{Z}_+} \sigma_0(n^2)n^{-s} = \prod_{p \in \text{Prime}} \left(1 + \frac{3}{p^s} + \frac{5}{p^{2s}} + \dots\right).$$

Multiplying by  $\zeta(2s)$ , we get

$$\zeta^3(s) = \sum_{n \in \mathbb{Z}_+} \left( \sum_{n/m \in (\mathbb{Z}_+)^2} \sigma_0(m^2) \right) n^{-s}.$$

Dividing by  $\zeta(s)$  again, we get

$$\zeta^2(s)\zeta^{-2}(2s) = \sum_{n \in \mathbb{Z}_+} 2^{\omega(n)} n^{-s} \text{ (2.6.1.22)} = \prod_{p \in \text{Prime}} \left(1 + \frac{2}{p^s} + \frac{2}{p^{2s}} + \dots\right).$$

Dividing by  $\zeta(s)$  again we get

$$\zeta(s)\zeta^{-1}(2s) = \sum_{n \in \mathbb{Z}_+} |\mu(n)| n^{-s} = \prod_{p \in \text{Prime}} \left(1 + \frac{1}{p^s}\right).$$

Taking inverse, we get

$$\zeta^{-1}(s)\zeta(2s) = \sum_{n \in \mathbb{Z}_+} (-1)^{\Omega(n)} n^{-s}.$$

┘

**Prop. (2.6.2.10)** [Euler's  $\phi$ -Function].

$$\frac{\zeta(s-1)}{\zeta(s)} = \sum_{n \in \mathbb{Z}_+} \phi(n) = \prod_{p \in \text{Prime}} \left( 1 + \frac{p-1}{p^s} + \frac{p(p-1)}{p^{2s}} + \frac{p^2(p-1)}{p^{3s}} + \dots \right).$$

In other words,  $\phi = \mu * \text{id}$ . ┘

*Proof:* This follows from the fact  $n = \mathbf{1} * \phi$  and Möbius inversion (2.6.2.4). □

**Prop. (2.6.2.11).** For  $s \in \mathbb{C}$  s.t.  $\text{Re}(s) > 1$ , we have

$$-\zeta'(s) = \sum_{n \in \mathbb{Z}_+} \frac{\log n}{n^s}.$$

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \in \mathbb{Z}_+} \Lambda(n) n^{-s} \text{ (2.6.1.31)}.$$

Equivalently,  $\Lambda = \mu * \log$ , and  $\log = \mathbf{1} * \Lambda$ , by (2.6.2.9) and (2.6.2.4). ┘

*Proof:* Both sides are analytic for  $\text{Re}(s)$  large, and when  $s$  is large,

$$-\frac{\zeta'(s)}{\zeta(s)} = -\text{dlog } \zeta(s) = \sum_{p \in \text{Prime}} \text{dlog}(1 - p^{-s}) = \sum_{n \in \mathbb{Z}_+} \Lambda(n) n^{-s}.$$

□

**Cor. (2.6.2.12).**

$$\Lambda - \mathbf{1} = \mu * (L - \sigma_0).$$

┘

**Def. (2.6.2.13)** [Generalized  $\mu$ -Functions and  $\Lambda$ -Functions]. For  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$ , define  $\mu_f, \Lambda_f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  to be

$$L(f; s)^{-1} = \sum_{n \in \mathbb{Z}_+} \mu_f(n) n^{-s}, \quad -\frac{L'(f; s)}{L(f; s)} = \sum_{n \in \mathbb{Z}_+} \Lambda_f(n) n^{-s}.$$

Notice that when  $f$  is completely multiplicative, then

$$\mu_f = \mu f, \quad \Lambda_f = \Lambda f.$$

┘

### 3 Maximal Orders

**Thm. (2.6.3.1)** [Maximal Order for  $\tau(n)$ ]. For  $n \in \mathbb{Z}_+$ ,

$$\overline{\lim}_n \frac{\log \tau(n)}{\log 2 \cdot \log n / \log_{(2)} n} = 1.$$

In particular, there exists  $c \in \mathbb{R}_+$  s.t.

$$\tau(n) \leq n^{\frac{c}{\log_{(2)} n}}.$$

┘

*Proof:* Cf.[Ten15]P113.?

□

**Cor. (2.6.3.2)**[Maximal Order for  $\omega(n)$ ]. For  $n \in \mathbb{Z}_+$ ,

$$\overline{\lim}_n \frac{\omega(n)}{\log n / \log_{(2)} n} = 1, \quad \overline{\lim}_n \frac{\Omega(n)}{\log n / \log 2} = 1$$

┘

*Proof:* The first equation follows from the inequality  $2^{\omega(n)} \leq \sigma_0(n)$  and (2.6.3.2) (with the same example approaching the suplim).

The second equation follows trivially from the inequality  $2^{\Omega(n)} \leq n$ .

□

**Thm. (2.6.3.3)**[Maximal and Minimum Order for  $\phi(n)$ ]. For  $n \in \mathbb{Z}_+$ ,

$$\overline{\lim}_n \frac{\phi(n)}{n} = 1.$$

$$\lim_n \frac{\phi(n)}{e^{-\gamma_0} n / \log_{(2)} n} = 1.$$

┘

*Proof:* Cf.[Ten15]P115.?

□

## 4 Continued Fractions

**Def. (2.6.4.1)**[Continued Fractions]. A **finite continued fraction**  $[a_1, \dots, a_n]$  is an abbreviation of

$$\frac{1}{a_1 + \frac{1}{a_2 + \dots \frac{1}{a_n}}}, \quad a_i \in \mathbb{Z}_+.$$

A **continued fraction**  $[a_1, a_2, \dots]$  is an abbreviation of

$$\frac{1}{a_1 + \frac{1}{a_2 + \dots}}, \quad a_i \in \mathbb{Z}_+.$$

┘

**Def. (2.6.4.2)**[Gauss Transformation]. The **Gauss transformation** is the function  $\varphi : \mathbb{R} \rightarrow [0, 1]$  :

$$\varphi(x) = \begin{cases} 1/x - [1/x] & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

┘

**Prop. (2.6.4.3)**. For any sequence of positive numbers  $\{a_1, \dots, a_n, \dots\}$ ,  $[a_1, \dots, a_n]$  converges. In particular,  $\lim_{n \rightarrow \infty} [\varphi(x), \varphi^2(x), \dots, \varphi^n(x)] = \{x\}$  for  $x \in \mathbb{R}$ .

┘

*Proof:* We may assume the sequence is infinite. Then we notice for any  $a \in \mathbb{Z}_+$  and  $t$  varying in an interval in  $(0, 1]$  of length  $l$ , the value of  $\frac{1}{a+t}$  is in an interval of length  $l/a(a+l) \leq l/(l+1)$ . So clearly the length of possible values of  $[a_1, \dots, a_n, t]$  for  $t$  in an interval of length  $l$  is in an interval of length  $l_n$  that  $\lim_{n \rightarrow \infty} l_n = 0$ .

Then we conclude  $[a_1, \dots, a_n]$  is a Cauchy sequence for  $n$ , thus it converges.

□

**Prop. (2.6.4.4).**  $x \in \mathbb{R}$  is a rational number iff  $\varphi^m(x) = 0$  for some  $m > 0$ . ┘

*Proof:* If  $x$  is rational then this is Euclid division. If  $\varphi^m(x) = 0$ , then notice  $x = [x] + [\varphi(x), \varphi^2(x), \dots, \varphi^{m-1}(x)]$  is rational. □

**Prop. (2.6.4.5).**  $x \in \mathbb{R}$  has eventually periodic continued fractions  $[\varphi(x), \varphi^2(x), \dots, \varphi^n(x), \dots]$  iff  $[\mathbb{Q}(x) : \mathbb{Q}] \leq 2$ . ┘

*Proof:* If the fraction is finite, then  $\mathbb{Q}(x) = \mathbb{Q}$ , by (2.6.4.4). If the fraction has periodic part  $[a_1, \dots, a_n]$ , let  $t = [a_1, \dots, a_n, a_1, \dots, a_n, \dots]$ , then

$$\frac{at + b}{ct + d} = t$$

for some  $a, b, c, d \in \mathbb{Z}$ , so  $[\mathbb{Q}[t] : \mathbb{Q}] = 2$ , and hence  $[\mathbb{Q}(x) : \mathbb{Q}] = 2$ .

For the converse, ? . □



## 2.7 Complexity Theory

References are [Computational Complexity, Papadimitriou], and [Logics for Mathematicians, Hamilton].

### 1 Problems and Algorithms

**Def. (2.7.1.1) [Decision Problems].** For  $I \in \text{Set}$ , a **decision problem** on  $I$  is a subset  $X \subset I$ . The elements of  $X$  are called its **instances**. The goal is to determine the set  $X$ .  $\perp$

**Def. (2.7.1.2) [Optimization Problems].** For  $I \in \text{Set}$ , a **optimization problem** is a subset  $X \subset I$  and a cost function  $f : I \rightarrow \mathbb{R}$ . The goal is to seek the maximum of  $f$  on  $X$ .  $\perp$

**Remark (2.7.1.3) [Cobham-Edmonds Thesis].** computational problems can be feasibly computed on some computational device only if they can be computed in polynomial time.  $\perp$

### Examples of Problems

**Example (2.7.1.4) [Graph Reachability].** The input is a directed graph  $G = (V, E)$  and two nodes  $s, t \in V$ . The problem is to find if  $s$  is connected to  $t$ , and also find the shortest path from  $s$  to  $t$ .

This problem can be solved by maintaining a set  $S \subset V$ , where  $S = \{s\}$  at first, and at each step, choose a vertex  $i$  in  $S$  that stayed the longest time, delete it, and add all vertices  $j$  s.t.  $(i, j) \in E$ . Proceed until  $t \in S$ , or  $S$  becomes empty. In the first case,  $s$  is connected to  $t$ , and we find a shortest path to  $t$ . While in the second case,  $s$  is disconnected to  $t$ .

Then this problem is in  $\text{TIME}(O(n^2))$ .  $\perp$

**Example (2.7.1.5) [Maximum Flow Problems].** The input is a directed graph  $G = (V, E)$  with a source  $s \in V$  with no incoming edges and a sink  $t \in V$  with no out-coming edges, and each edge is labeled with a positive integer.

A **flow** in  $G$  from  $s$  to  $t$ , which means an assignment of non-negative integers  $f(e)$  to each edge  $e$  s.t. for each vertex  $v$  except  $s$  and  $t$ , the sum of values of edges with source  $v$  equals the sum of values of edges with target  $v$ .

Then the problem is to find a flow with maximum sum of values at  $s$ .

This problem can be solved by at first finding a flow by finding the shortest path from  $s$  to  $t$ , then at each step, we draw the difference graph, and try to find a shortest path from  $s$  to  $t$ . This is called a augmentation flow.

If the edge on a path with the lest capacity is called a **bottleneck**, then it can be proved that there are at most  $\#V^2$  iterations:

**Lemma (2.7.1.6).**

- The distance from  $s$  to any vertex doesn't decrease from one augmentation graph to the next graph.
- If an edge  $e = (i, j)$  is a bottleneck in one augmentation, then the distance of  $i$  from  $s$  must increase by at least 2 before  $-e$  becomes a bottleneck again.

$\perp$

*Proof:* 1: If the distance from  $s$  to some vertex decreases. Let  $v$  be a vertex s.t. the distance after augmentation is the shortest from  $s$ . Let  $s \rightarrow \dots \rightarrow i \rightarrow \dots \rightarrow v$  be a shortest path, then  $(v, i)$  must be in

the shortest flow. Then the distance from  $s$  to  $i$  must also decreased, and that distance is smaller than the distance to  $v$ , contradiction.

2: This follows from 1, because the shortest path from  $s$  to  $j$  cannot decrease, the new path includes  $-e = (j, i)$  instead of going to  $i$  directly, means that the distance from  $s$  to  $i$  must increased by at least 2.  $\square$

Then by (2.7.1.4), the problem is in  $\text{TIME}(O(n^4))$ .  $\lrcorner$

**Example (2.7.1.7) [Bipartite Matching].** The input is a bipartite graph  $G = (U, V, E)$  where  $\#U = \#V = n$  and  $E \subset U \times V$ , the problem is to determine whether there is a matching between  $U$  and  $V$  via  $E$ .

By adding two points  $s, t$  with  $s$  pointing to each vertex in  $U$  and each vertex in  $V$  pointing to  $t$ , we can reduce the problem to finding a flow from  $s$  to  $t$  of capacity  $n$ . Thus this decision problem is in  $\text{TIME}(O(n^4))$ , by (2.7.1.5).  $\lrcorner$

**Example (2.7.1.8) [Salesman Problem].** The salesman problem is clearly in NP.  $\lrcorner$

## 2 Turing Machines

*“It is possible to invent a single machine which can be used to compute any computable sequence.”*  
—A. Turing

**Def. (2.7.2.1) [Turing Machines].** A **Turing machine** is a tuple

$$M = (K, \Sigma, \Delta, s, \Pi, \triangleright, h, y, n, \leftarrow, \rightarrow, -),$$

where

- $h, y, n, \leftarrow, \rightarrow, -$  are disjoint singletons, called the **halting state**, **accepting state**, **rejecting state**, **left curser**, **right curser**, **stay curser** resp.
- $K$  is a finite set disjoint from  $\{h, y, n, \leftarrow, \rightarrow, -\}$ . elements of  $K \cup \{h, y, n\}$  are called **states**.
- $s \in K$  is a state, called the **initial state**.
- $\Sigma$  is a finite set disjoint from  $K \cup \{h, y, n, \leftarrow, \rightarrow, -\}$ , called the **alphabet of  $M$** , and elements of  $\Sigma$  are called **symbols**.
- $\Pi, \triangleright \in \Sigma$  are two symbols, called the **blank symbol** and **first symbol** resp..
- $\Delta \subset K \times \Sigma \times (K \cup \{h, y, n\}) \times \Sigma \times \{\leftarrow, \rightarrow, -\}$ , called the **transition relation of  $M$** .  $\delta$  must satisfy: for any states  $p, q$ , symbol  $\sigma$  and  $m \in \{\leftarrow, \rightarrow, -\}$  s.t.  $(p, \triangleright, q, \sigma, m) \in \Delta$ , we have  $\sigma = \triangleright$  and  $m = \rightarrow$ .

For normalization, we may assume that

$$\Sigma = \{1, 2, \dots, \#\Sigma\}, \quad K = \{\#\Sigma + 1, \dots, \#\Sigma + \#K\},$$

$$s = \#\Sigma + 1, \quad (h, y, n, \leftarrow, \rightarrow, -) = (\#\Sigma + \#K + 1, \dots, \#\Sigma + \#K + 6).$$

where each integer is encoded by a binary string.

$M$  is called a **deterministic Turing machine** if  $\Delta$  is a function  $\Delta : K \times \Sigma \rightarrow (K \cup \{h, y, n\}) \times \Sigma \times \{\leftarrow, \rightarrow, -\}$ , and it is called a **non-deterministic Turing machine** otherwise.

Similarly, we can define **multi-string Turing machines**.  $\lrcorner$

**Prop.(2.7.2.2) [Computing with a non-deterministic Turing Machine].** Given a non-deterministic Turing machine and a finite string  $x \in \text{String}(\Sigma \setminus \Pi)$  called the **input of the Turing machine**, a **computation procedure** is the following:

- Write  $\triangleright$  on the left of  $x$ .
- Put a cursor on  $\triangleright$ .
- set  $\text{STATE} = s$ .
- For  $i \in \mathbb{Z}_+$ , in the  $i$ -th step, if  $\text{STATE} = p \in K$  and the cursor is pointing on  $\rho$ , choose  $(q, \sigma, m) \in (K \cup \{h, y, n\}) \times \Sigma \times \{\leftarrow, \rightarrow, -\}$  s.t.  $(p, \rho, q, \sigma, m) \in \Delta$ , then replace the symbol pointed by the cursor by  $\sigma$ , replace  $\text{STATE} = q$ , and move the cursor according to  $m$ .
- If  $\text{STATE} \in \{h, y, n\}$ , stop.

Notice the cursor will never fall left to the leftmost  $\triangleright$  we wrote. But it may go to the right of  $x$ , in which case we assume it reads  $\Pi$ . So the string may become longer.

The **output of the Turing machine**  $M(x)$  is defined to be

- ‘yes’ if the machine stops with  $\text{STATE} = y$ , in which case we say this machine **accepts the input** in this procedure,
- ‘no’ if the machine stops with  $\text{STATE} = n$ , in which case we say this machine **rejects the input** in this procedure,
- the longest string following the leftmost  $\triangleright$  and ending with a symbol in  $\Sigma \setminus \Pi$ , if the machine stops with  $\text{STATE} = h$ . In which case we say this machine **halted at the input** in this procedure.
- ‘ $\nearrow$ ’ if the machine never stops in this procedure.

┘

**Def.(2.7.2.3) [Monte-Carlo Turing Machines].**

┘

**Def.(2.7.2.4) [Las-Vegas Turing Machines].**

┘

### Languages and Computation

**Def.(2.7.2.5) [Languages].** Given a set of symbols  $\Sigma$  (containing  $\Pi$ ), A **language on  $\Sigma$**  is a set of strings of symbols  $\mathcal{L} \subset \text{String}(\Sigma \setminus \Pi)$ .

┘

**Def.(2.7.2.6) [Recursive languages].** Given a set of symbols  $\Sigma$  containing  $\Pi$  and a language on  $\Sigma$ , a deterministic Turing machine  $M$  with alphabet  $\Sigma$  is said to **decide the language  $\mathcal{L}$**  if for any  $x \in \text{String}(\Sigma \setminus \Pi)$ ,  $M(x) = y$  if  $x \in \mathcal{L}$  and  $M(x) = n$  if  $x \notin \mathcal{L}$ .

$\mathcal{L}$  is called a **recursive**(or **computable/decidable**) language if it is decided by some deterministic Turing machine on  $\Sigma$ .

┘

**Def.(2.7.2.7) [Recursively Enumerable languages].** Given a set of symbols  $\Sigma$  containing  $\Pi$  and a language on  $\Sigma$ , a deterministic Turing machine  $M$  with alphabet  $\Sigma$  is said to **accept the language  $\mathcal{L}$**  if for any  $x \in \text{String}(\Sigma \setminus \Pi)$ ,  $M(x) = y$  if  $x \in \mathcal{L}$  and  $M(x) = \nearrow$  if  $x \notin \mathcal{L}$ .

$\mathcal{L}$  is called a **recursively enumerable**(or **computably enumerable/semidecidable/partially decidable/provable/Turing-recognizable**) language if it is accepted by some deterministic Turing machine on  $\Sigma$ .

The class of recursively enumerable languages is denoted by  $RE$ , and the class of complements of enumerable languages is denoted by  $\text{co-}RE$ .

┘

**Cor. (2.7.2.8).** A recursive language is recursively denumerable.  $\lrcorner$

**Def. (2.7.2.9) [Recursive Functions].** Given a set of symbols  $\Sigma$  containing  $\Pi$  and a function  $f : \text{String}(\Sigma \setminus \Pi) \rightarrow \text{String}(\Sigma)$ , a deterministic Turing machine  $M$  with alphabet  $\Sigma$  is said to **compute**  $f$  if for any  $x \in \text{String}(\Sigma \setminus \Pi)$ ,  $M(x) = f(x)$ .

$f$  is called a **recursively function** language if it is decided by some deterministic Turing machine on  $\Sigma$ .  $\lrcorner$

**Def. (2.7.2.10) [Non-deterministic Computations].** Given a set of symbols  $\Sigma$  containing  $\Pi$  and a language on  $\Sigma$ , a non-deterministic Turing machine  $M$  with alphabet  $\Sigma$  is said to **decide the language**  $\mathcal{L}$  if for any  $x \in \text{String}(\Sigma \setminus \Pi)$ ,  $M(x) = y$  iff  $x \in \mathcal{L}$ .  $\lrcorner$

### 3 Computability

**Remark (2.7.3.1) [Church-Turing Thesis].** Church proposed the assertion that every set of  $n$ -tuples of positive integers that is intuitively recursive is in fact Turing recursive.

“intuitively recursive” is not a mathematical statement, so this assertion has no mathematical proof or disproof. It is empirically right, and usually believed to be right.

The guiding role of this thesis played to mathematics should be compared to the role law of conservation of energy played to physics. Namely, once the Turing undecidability of a set has been proved, one need not spend one’s time seeking a universal method for recognizing the elements of that set.  $\lrcorner$

**Def. (2.7.3.2) [Universal Turing Machine].** There is a **universal Turing machine**  $U$  that with the input of a pair  $(M, x)$  where  $M$  is a Turing machine (2.7.2.1) and  $x$  is a input for  $M$ , the output of  $U$  is  $U(M, x) = M(x)$ .  $\lrcorner$

*Proof:* Cf. [Computational Complexity, P57].  $\square$

### 4 P and NP

*“We all know P is different than NP. We just don’t know how to prove it.”*

*–J. Hartmanis*

#### Complexity Classes

**Def. (2.7.4.1) [Complexity Classes].** Given a multi-string Turing machine model, a mode of computation, and a bound  $f : \mathbb{N} \rightarrow \mathbb{N}$  on the resources, a **complexity class** is defined as the set of all languages (2.7.2.5) decided by some multi-string Turing machine  $M$  operating in this mode s.t. for any input  $x$ , each computation procedure of  $M$  expends at most  $f(|x|)$  units of the specified resource. For example,

- $\text{TIME}(f)$ : decided by a deterministic Turing machine in  $f$  times.
- $\text{SPACE}(f)$ : decided by a deterministic Turing machine in  $f$  spaces.
- $\text{P} = \bigcup_{j \in \mathbb{Z}_+} \text{TIME}(n^j)$ : decided by a deterministic Turing machine in polynomial times.
- $\text{PSPACE} = \bigcup_{j \in \mathbb{Z}_+} \text{SPACE}(n^j)$ : decided by a deterministic Turing machine in polynomial spaces.
- $\text{EXP} = \bigcup_{j \in \mathbb{Z}_+} \text{TIME}(2^{n^j})$ : decided by a deterministic Turing machine in exponential times.

- $L = \text{SPACE}(\log n)$ : decided by a deterministic Turing machine in  $\log$  spaces.
- $\text{NTIME}(f)$ : decided by a non-deterministic Turing machine in  $f$  times.
- $\text{NP} = \bigcup_{j \in \mathbb{Z}_+} \text{NTIME}(n^j)$ : decided by a non-deterministic Turing machine in polynomial time.
- $\text{co-NP}$ : The complement is in  $\text{NP}$ .
- $\text{NSPACE}(f)$ : decided by a non-deterministic Turing machine in  $f$  spaces.
- $\text{NPSPACE} = \bigcup_{j \in \mathbb{Z}_+} \text{NSPACE}(n^j)$ : decided by a non-deterministic Turing machine in polynomial spaces.
- $\text{NL} = \text{NSPACE}(\log n)$ : decided by a non-deterministic Turing machine in  $\log$  spaces.
- $\text{BP}(f)$ : decided by a two-sided Monte-Carlo Turing machine in  $f$  times.
- $\text{BPP} = \bigcup_{j \in \mathbb{Z}_+} \text{BP}(n^j)$ : decided by a two-sided Monte-Carlo Turing machine in polynomial times.
- $\text{R}(f)$ : decided by a one-sided Monte-Carlo Turing machine in  $f$  times.
- $\text{RP} = \bigcup_{j \in \mathbb{Z}_+} \text{R}(n^j)$ : decided by a one-sided Monte-Carlo Turing machine in polynomial times.
- $\text{co-RP}$ : The complements is in  $\text{RP}$ .
- $\text{ZPP} = \text{RP} \cap \text{co-RP}$ : decided by a Las-Vegas Turing machine in polynomial expected time.

┘

**Def. (2.7.4.2) [Proper Complexity Functions].** Cf.[Computational Complexity]P140.

┘

**Def. (2.7.4.3) [Hard and Complete Problems].** For a computational complexity class  $\mathcal{C}$ , a decision problem  $X \subset I$  is called  **$\mathcal{C}$ -hard** if every problem in  $\mathcal{C}$  can be reduced to it. And it is called  **$\mathcal{C}$ -complete** if it is  $\mathcal{C}$ -hard and it is in  $\mathcal{C}$ .

┘

**Prop. (2.7.4.4).** Suppose a (deterministic or nondeterministic) Turing machine  $M$  decides a language  $\mathcal{L}$  within time/space  $f(n)$ , where  $f$  is a proper complexity function(2.7.4.2), then there is a precise Turing machine  $M'$ , which decides the same language in time/space  $O(f(n))$ .

┘

*Proof:* Cf.[Computational Complexity]P141.

□

**Prop. (2.7.4.5) [Multi-String and Quadratic Time].** Given a multi-string deterministic Turing machine  $N$  and a language  $\mathcal{L}$  decided by  $N$  in time  $f(n)$ , there is a single-string deterministic Turing machine  $M$  that determines  $\mathcal{L}$  in time  $O(f(n)^2)$ .

┘

*Proof:* Cf.[Computational Complexity]P30.

□

**Prop. (2.7.4.6) [Non-Deterministic Machines and Exponential Time].** Given a non-deterministic Turing machine  $N$ , there exists  $C = C(N) \in \mathbb{R}_{>1}$  s.t. if a language  $\mathcal{L}$  is decided by  $N$  in time  $f(n)$ , then it is decided by a 3-string deterministic Turing machine  $M$  in time  $O(C^{f(n)})$ . In particular,

$$\text{NTIME}(f(n)) \subset \bigcup_{C \in \mathbb{R}_{>1}} \text{TIME}(C^{f(n)}).$$

┘

*Proof:* Cf.[Computational Complexity]P47.

□

**Prop. (2.7.4.7).**  $\text{RP} \subset \text{NP} \subset \text{PSPACE}$ .

┘

*Proof:* □

**Prop. (2.7.4.8).**  $P \subset BPP$ . ┘

*Proof:* □

**Conj. (2.7.4.9).** What is the relation between  $BPP$  and  $NP$ ? ┘

*Proof:* □

**Conj. (2.7.4.10)**[Cook's Hypothesis].  $P \neq NP$ . ┘

*Proof:* □

**5** Inside  $P$

**6** Beyond  $NP$

**7** Algebraic Complexity Theory

## **2.8 Computer Systems**

References are [Computer Systems, a Programmer's Perspective].

- 1 Program Structure and Execution**
- 2 Running Programs on a System**
- 3 Interaction and Communication between Programs**
- 4 Error Handling**

## 2.9 Theoretical Computer Science



## 2.10 Computer Algebra

References are [Computer Algebra].

### 1 Euclid

#### Basic Algorithms

#### Euclidean Algorithms

#### Modular Algorithms

#### Resultants and gcd

### 2 Newton

#### Multiplications

**Algo. (2.10.2.1) [Schönhage–Strassen Modular Exponentiation Algorithm].** There is an algorithm to calculate  $x^r \bmod n$  for  $l$ -bits numbers  $x, r, n$  in  $O(l^2 \log l \log \log l)$  times.  $\lrcorner$

*Proof:*  $\square$

**Algo. (2.10.2.2) [Multiplication Algorithm, Harvey-van der Hoeven].** There is an algorithm that computes the product of two  $n$ -bit integers in  $O(n \log n)$  times.  $\lrcorner$

*Proof:*  $\square$

#### Fourier Transforms

**Def. (2.10.2.3) [Discrete Fourier Transform].** The **inverse Fourier transform** is defined to be:

$$(x_0, \dots, x_{n-1}) \mapsto (y_0, \dots, y_{n-1}), \quad y_k = \frac{1}{N} \sum_{j=0}^{N-1} \omega_N^{jk} x_j.$$

This is in fact represented by a van der Waerden matrix of  $\omega_N$  times  $\frac{1}{N}$ .  $\lrcorner$

**Prop. (2.10.2.4) [Fast Fourier Transform].** There is a **fast Fourier transform algorithm** that can calculate the Fourier transform in  $O(N \log N)$  time.  $\lrcorner$

*Proof:*  $\square$

#### Newton Iterations

#### Polynomial Evaluations and Interpolations

#### Linear Algebras

### 3 Gauss

#### Factorizing Polynomials over Finite Fields

#### Short Vectors in Lattices

### 4 Fermat

See also the section on computational number theory.

Primality Testing

Factorizing Integers

**5 Hilbert**

Gröbner Basis

Symbolic Integration

## 2.11 Linux

### Notation(2.11.0.1).

- Assume we are using the newest version of MacOS.

└

### 1 Terminal

**Def.(2.11.1.1).** • pwd(print work directory): Show current directory.

- ls(list): List names of all files in the current directory.
- cd: Change directory.
  - cd +“folder\_name”: Going to the folder in the current directory.
  - cd .. : Going to the parent folder.

└

### 2 Git

#### Basic Syntaxes

**Def.(2.11.2.1)[Systems].** • Git version: Show current git version.

- Git config: configuration.
  - Git config –global user.name + “user\_name”: change user name of current operations.
  - Git config –global user.email + “user\_email”: change user email of current operations.

└

**Def.(2.11.2.2)[Start a Working Area].**

- Git clone: Clone a repository into a new directory.
- Git init: Create an empty Git repository or reinitialize an existing one.

└

**Def.(2.11.2.3)[Work on the Current Change].**

- Git add + “file\_name”: Add file contents to the index. Git add .: add all file in the current directory to the index.
- Git mv: Move or rename a file, a directory, or a symlink.
- Git restore: Restore working tree files.
- Git rm: Remove files from the working tree and from the index.

└

**Def.(2.11.2.4)[Examine the History and State].**

- Git bisect: Use binary search to find the commit that introduced a bug.
- Git diff +“commit\_id”: Show changes between commits, commit and working tree, etc.
- Git grep: Print lines matching a pattern.
- Git log: Show commit logs.

- Git show: Show various types of objects.
- Git status: Show the working tree status.

└

**Def. (2.11.2.5)[Grow, Mark and Tweak Your Common History].**

- Git branch: List, create, or delete branches.
  - Git branch -a: List all branches.
  - Git branch + “**branch\_name**”: Create a branch with the given name and current changes, while stay on the current branch.
  - Git branch -M + “**branch\_name**”: Create a branch with the given name and current changes, and make it the main branch.
- Git checkout+ “**branch\_name**”: Change to the given branch.
- Git commit: Record changes to the repository. Each commit with have a unique commit id, and marked with date and author’s name and email.
  - Git commit -m “comments”: commit and add a comment to this commit.
- Git merge + “**branch\_name**”: Join this branch to the current branch(when these two branches are the same).
- Git rebase: Reapply commits on top of another base tip.
- Git reset + “**commit\_id**”: Reset current HEAD to the specified state.
- Git switch: Switch branches.
- Git tag: Create, list, delete or verify a tag object signed with GPG.

└

**Def. (2.11.2.6)[Collaborate].**

- Git fetch: Download objects and refs from another repository.
- Git pull: Update remote refs along with associated objects.
- Git push -u + “**branch\_name**”: upload branches.

└

## 2.12 C++

*“There are only two kinds of languages: the ones people complain about and the ones nobody uses.”*  
*–B. Stroustrup*

References are [C++ Primer].

### 1 Data Structures

**Def. (2.12.1.1) [Data].** What is a datum.

A **data structure** is a logical grouping of data and operations on that data. ┘

### 2 Computer System Basics

**Def. (2.12.2.1) [Buffer].** A **buffer** in a computer is a region of storage used to hold data. IO facilities often store input (or output) in a buffer and read or write the buffer independently from actions in the program. Output buffers can be explicitly flushed to force the buffer to be written. By default, reading `tt cin` flushes `tt cout`; `tt cout` is also flushed when the program ends normally. ┘

**Def. (2.12.2.2) [Console Windows].** ┘

**Def. (2.12.2.3) [Operands].** ┘

**Def. (2.12.2.4) [IO Devices].** ┘

### 3 Basic Notions

**Fact (2.12.3.1) [C++ Language].** C++ is a high-level, general-purpose programming language created by Danish computer scientist Bjarne Stroustrup.

Like other programming languages, C++ provides a common set of features, for examples

- Build-in types as integers types, characters, etc. (2.12.4.1),
- Variables (2.12.4.2), which let us give names to the objects we use.
- Expressions and statements (2.12.6.1) to manipulate values of these types.
- Control structures, such as `if` or `while`, that allow us to conditionally or repeatedly execute a set of actions.
- Functions (2.12.7.1) that let us define callable units of computation.

And it distinguishes from other programming languages in the feature of

- Classes (2.12.8.1), which let programmers define their own types that include operations as well as data, which are as easy to use as the built-in types. ┘

**Fact (2.12.3.2) [Philosophy of C++].** Throughout C++’s life, its development and evolution has been guided by a set of principles:

1. It must be driven by actual problems and its features should be immediately useful in real world programs.
2. Every feature should be implementable (with a reasonably obvious way to do so).
3. Programmers should be free to pick their own programming style, and that style should be fully supported by C++.

4. Allowing a useful feature is more important than preventing every possible misuse of C++.
5. It should provide facilities for organising programs into separate, well-defined parts, and provide facilities for combining separately developed parts.
6. No implicit violations of the type system (but allow explicit violations; that is, those explicitly requested by the programmer).
7. User-created types need to have the same support and performance as built-in types.
8. Unused features should not negatively impact created executables (e.g. in lower performance).
9. There should be no language beneath C++ (except assembly language).
10. C++ should work alongside other existing programming languages, rather than fostering its own separate and incompatible programming environment.
11. If the programmer's intent is unknown, allow the programmer to specify it by providing manual control.

┘

**Def. (2.12.3.3) [C++ Programs].** A **C++ program** is a combination of codes that can be used to solve real world problems. It is usually stored in computer files, referred to as the **source files**. ┘

**Fact (2.12.3.4) [How to Compile a C++ Program].** How to compile a program depends on the ongoing operating system and compiler. Many PC-based compilers are run from an integrated development environment (IDE) that bundles the compiler with build and analysis tools. ?

On a command-line interface, a program is usually compiled in a console window (2.12.2.2).

```
| 1 $ CC prog1.cc
```

where tt \$ is the system prompt. ┘

**Def. (2.12.3.5) [Headers].** A **header** in the C++ language is a mechanism whereby the definitions of a class or other names are made available to multiple programs. A program uses a header through a tt `#include` directive, cf. (2.12.9.1)(2.12.8.1). ┘

**Def. (2.12.3.6) [Comments].** A **comment** in a C++ program is a fragment of codes that helps whoever reads this code but ignored by the compiler. There are two kinds of comments: single-lined and multi-lined: comments start with a “//” symbol and ends with a newline:

```
| 1 // any comments you want to write.
```

and everything to the right of the slashes on the current line is ignored by the compiler, so it can contain anything but a newline.

Another kind of comments are multi-lined, which starts with a “/\*” symbol and ends with a “\*/” symbol:

```
| 1 /* any comments
| 2
| 3 that you want to write.
| 4 */
```

and everything that falls between the “/\*” and “\*/” is ignored by the compiler, so it can contain anything but a “\*/” symbol.

In particular, two fragment of comments cannot intersect each other. ┘

## 4 Variables and Types

**Def. (2.12.4.1) [Data Types].** A **type of a datum**(2.12.1.1) is an indication of the contents of this datum and also the operations that can be applied to this datum. Types are fundamental to any program: They tell us what our data mean and what operations we can perform on those data. ┘

**Def. (2.12.4.2) [Variables].** A **variable** in the C++ language is an object with name and types. A variable is defined by the following statement:

```
|1 variable_type variable_name;
```

When defining a variable, we can **initialize** it at the same time, which means giving it a value at the same time that it is created. For example,

```
|1 int v1=0;
```

┘

## 5 Operators

**Def. (2.12.5.1) [Operators].** An **operator** is a C++ language specifies an action and return an object. There are some common operators:

- The **dot operator** “tt .”, which can be used to access a member function(2.12.8.1)(on the right-hand side) of an variable(2.12.4.2) of class type(2.12.8.1)(on the left-hand side), and return this member function. For example:

```
|1 variable_name.member_function
```

- The **call operator** “tt (”, “tt )”, which can be used to call a function(2.12.7.1)(on the left-hand side), and return this function. For example,

```
|1 main()
```

or

```
|1 variable_name.member_function()
```

A pair of parenthesis operator can enclose a (possibly empty) list of arguments(2.12.7.1) used to call this function.

- The scope operator “tt ::” is used to access names(on the right-hand side) in a namespace(on the left-hand side), and returns this name. For example,

```
|1 std::cout
```

┘

## 6 Statements

**Def. (2.12.6.1) [Statements].** A statement in a C++ program(2.12.3.3) that specifies an action to take place when the program is executed. Statements can contain other statements within themselves. ┘

## 7 Functions

**Def. (2.12.7.1) [Functions].** Every C++ program contains some **functions**, which is a unit of computation that can be called by the parenthesis operator(2.12.5.1) to be used, and generate a single output from several inputs. It consists of

- A **return type**, which is the type of the value returned by a function(2.12.4.1).
- A **function name**, which is the name by which a function is known, and the function can be called using the name.
- A **parameter list**, which is a list that specifies what **arguments**(i.e. values passed to this function) can be used to call the function.
- A **function body**, which is a sequence of zero or more statements(2.12.6.1) enclosed in curly braces, that defines the actions performed by a function.

Every function is displayed as follows:

```
1 return_type function_name(parameter_list){
2     function_body
3 }
```

where the function body consists of a block of statements(2.12.6.1), each marked with a symbol “;”. The block is delimited by a pair of curly braces “{” and “}”. And the last statement must be a **return statement**, which determines the function, and also can be used to return a value to the function’s caller. If it returns a value, the value must be compatible with the return type of this function. ┘

**Def. (2.12.7.2)[Main Function].** A **main function** in a C++ program(2.12.3.3) is a function(2.12.7.1) called by the operating system to execute this program. Each program must have one and only one function named main. The main function must have integer return type. The empty main function which return a value 0 to the system:

```
1 int main(){
2     return 0;
3 }
```

The returning value 0 is usually an indication of a successes of running this function, and the returning value -1 is by default treated as a failure indicator. ┘

## 8 Classes

**Def. (2.12.8.1)[Classes].** A **class** in the C++ language is a facility for defining our own data structures together with associated operations, called its **member functions**. How these data are stored or computed is a question?.

A **class type** is a data type(2.12.4.1) defined by a class(2.12.8.1). A class determines all the operations that can be used on objects of this class type.

A class can be called by the directive `tt #include`:

```
1 #include "class_header";
```

Notice that we use the double quote “” symbols when the class is not defined by the standard library, cf.(2.12.9.1). ┘



## Copy Control

## Overloaded Operations and Conversions

## Object-Oriented Programming

## Templates and Generic Programming

## 9 Standard Library

**Def. (2.12.9.1)[Standard Library].** The **standard library** is a collection of types and functions that every C++ compiler must support. It has several sub-libraries, for example,

- The IO library.

Each library has a header(2.12.3.5), and they can be called by the directive `tt #include`:

```
|1 #include <library_header>;
```

Notice this statement must be written in a single line, and outside any functions(2.12.7.1). Usually, these statements are placed at the begging of the program.

Each type defined in the standard library is called a **library type**. ┘

**Def. (2.12.9.2) [Namespaces].** A **namespace** is a mechanism for putting names defined by a library(2.12.9.1) into a single place. Namespaces help avoid inadvertent name clashes. Namespaces allow us to avoid inadvertent collisions between the names we define and uses of those same names inside a library. The names defined by the standard library are in the namespace `tt std`. ┘

## Input and Output

**Def. (2.12.9.3)[Streams].** A **stream** in C++ language is a sequence of characters read from or written to an IO device(2.12.2.4). ┘

**Fact (2.12.9.4)[Iostream Library].** The **iostream** is a header that provides the library types(2.12.9.1) for stream-oriented(2.12.9.3) input and output. It contains two types, the **istream** providing stream-oriented input, and the **ostream** providing stream-oriented output. ┘

**Def. (2.12.9.5) [Standard Input/Output].** A **standard input/output** is an input/output stream(2.12.9.3) usually associated with the window in which the program executes.

A **standard error** is an output stream used for error reporting and warning. Ordinarily, the standard output and the standard error are tied to the window in which the program is executed. ┘

**Fact (2.12.9.6)[Some IO-Type Objects].**

- `tt cin` is an object of type `istream` used to read from the standard input.
- `tt cout` is an object of type `ostream` used to write to the standard output.
- `tt cerr` is an object of type `ostream` tied to the standard error, which often writes to the same device as the standard output. By default, writes to `tt cerr` are not buffered(2.12.2.1). Usually used for error messages or other output that is not part of the normal logic of the program.
- `tt clog` is an object of type `ostream` tied to the standard error. By default, writes to `clog` are buffered(2.12.2.1). `tt clog` is usually used to report information about program execution to a log file.
- A **manipulator** is an object, such as `tt std::endl`, that when read or written “manipulates” the stream itself.

┘

**Def. (2.12.9.7) [Some IO Operators].**

- The **output operator** “<<” writes the right-hand side operand to the output stream indicated by the left-hand operand, and returns the left-hand side. For example, `cout << "hi"` writes `tt hi` to the standard output(2.12.9.5). standard output.
- The **input operator** reads from the input stream specified by the left-hand side operand to the right-hand operand, and returns the left-hand side. For example, `cin >> i` reads the next value on the standard input(2.12.9.5) into `i`.

┘

### Sequential Containers

### Generic Algorithms

### Associative Containers

### Dynamical Memory

### Specialized Library Facilities

## 10 Tools for Large Programs

## 11 Specialized Tools and Techniques

## 2.13 Python

### Notation(2.13.0.1).

- Use basic logics and syntax as in C++.

┘

### 1 Basic Syntax

**Def.(2.13.1.1)[Comments].** Use # to comment out all texts on the right of the same line.

┘

**Remark(2.13.1.2)[Indentation].** In Python, indentation is important, as there are no “;” needed to mark the end of a enviroment.

┘

**Def.(2.13.1.3)[Strings].** Use 'any string' or ''any string'' to define strings.

Use

```
| 1  '''
| 2  any
| 3  string
| 4  '''
```

to define string in multiple lines.

┘

**Def.(2.13.1.4)[Print Function].** The print function prints out strings on the screen. It by default ends with \n to generate a new line. For example

```
| 1  print('Hello World')
```

prints out Hello World on the screen, and generate a new line.

To avoid generating a new line, we can use

```
| 1  print('Hello World', end='')
```

to end with nothing, or

```
| 1  print('Hello World', end=' ')
```

to end with a blank space.

You can also print several strings together, for example,

```
| 1  length = 5
| 2  breadth = 2
| 3  print('Area is', 2 * ( length + breadth ))
```

prints out “Area is 14” in your screen. Notice that you don’t need to manually add a blanket after the string “Area is”.

┘

**Def.(2.13.1.5)[Input Function].** The input function takes user’s input as return, e.g.:

```
| 1  something = input("Enter text: ")
```

will print out “Enter test:” to your screen. And when you enter some string and press “return”, the input function will assign your entered string as value of “something”.

┘

**Def.(2.13.1.6)[Open Functions].** The open function can be used to edit files, e.g.

```
| 1  open('poem.txt', 'w')
```

opens the file “poem.txt” in the writing mode.(if this file doesn’t exist, then it creates a file). And

```
| 1  open('poem.txt', 'r')
```

opens the file “poem.txt” in the reading mode. ┘

**Def.(2.13.1.7)[Control-Flows].** The control-flows of Python is similar to C++. For example,

- If syntax:

```

1 number = 23
2 guess = int(input('Enter an integer : '))\ref{inpvlusidnifenis}
3 if guess == number:
4     print('Congratulations, you guessed it.')
5     print('(but you do not win any prizes!)')
6 elif guess < number:
7     print('No, it is a little higher than that')
8     print('No, it is a little lower than that')
```

(2.13.1.5).

- While syntax:

```

1 number = 23
2 running = True
3 while running:
4     guess = int(input('Enter an integer : '))
5     if guess == number:
6         print('Congratulations, you guessed it.')
7         running = False
8     elif guess < number:
9         print('No, it is a little higher than that.')
10    else:
11        print('No, it is a little lower than that.')
12 else:
13     print('The while loop is over.')
14 print('Done')
```

- For syntax: which is similar to foreach syntax in C++:

```

1 for i in range(1, 5):
2     print(i)
3 else:
4     print('The for loop is over')
```

- Break syntax: get out of the control-flow loop immediately.

```

1 while True:
2     s = input('Enter something : ')
3     if s == 'quit':
4         break
5     print('Length of the string is', len(s))
6     print('Done')
```

- Continue syntax: Jump to the start of the next control-flow loop immediately.

```

1 while True:
2     s = input('Enter something : ')
3     if s == 'quit':
4         break
5     if len(s) < 3:
6         print('Too small')
7     continue
8     print('Input is of sufficient length')
```

But notice there are no “switch” syntax. ┘

## Functions

**Def.(2.13.1.8)[Defining Functions].** Functions can be defined using the syntax “def” following the name of this function and a pair of bracket embracing the parameters of this function, e.g.:

```
1 def say_hello():
2     print('hello world')

1 def print_max(a, b):
2     if a > b:
3         print(a, 'is maximum')
4     elif a == b:
5         print(a, 'is equal to', b)
6     else:
7         print(b, 'is maximum')
```

You can also define default parameter values. But notice parameters with default values must be put after those without default values, e.g.:

```
1 def say(message, times=1):
2     print(message * times)
3 say('Hello')
4 say('World', 5)
```

will print out

```
Hello
WorldWorldWorldWorldWorld
to your screen.
```

└

**Remark(2.13.1.9)[Local Parameters].** When defining a function, your operation to variables will not influence any variable with the same name outside this function definition process. For example:

```
1 x = 50
2 def func(x):
3     print('x is', x)
4     x=2
5     print('Changed local x to', x)
6 func(x)
7     print('x is still', x)
```

will print out

```
x is 50
Changed local x to 2
x is still 50
```

to your screen. If you want to operate on the variable globally, you need to use the syntax `global`,

e.g.:

```
1 x = 50
2 y=10
3 def func():
4     global x, y
5     print('x is', x)
6     x=2
7 func()
8     print(x)
```

will print “2” to your screen.

└

**Def.(2.13.1.10)[Using Functions].** When using functions, we need to provide parameter values, unless it has default setting(2.13.1.8). When providing parameters, we can either provide directly according to their position(called position arguments), or provide by calling their names(called keyword arguments), e.g.:

```

1 def func(a, b=5, c=10):
2     print('a is', a, 'and b is', b, 'and c is', c)
3 func(3, 7)
4 func(25, c=24)
5 func(c=50, a=100)

```

┘

**Def. (2.13.1.11)[Return Syntax].** In defining a function, the `return` syntax stops the function. Every function has an implicit `return None` line in the end, unless you write your own return code. You can also return a value to the function, making this function a string, e.g.

```

1 def maximum(x, y):
2     if x > y:
3         return x
4     elif x == y:
5         return 'The numbers are equal'
6     else:
7         return y
8 print(maximum(2, 3))

```

will print out 3 to your screen.

┘

**Def. (2.13.1.12)[Variable Parameters].** You can define functions with a variable number of parameters, e.g.:

```

1 def total(a=5, *numbers, **phonebook):
2     print('a', a)
3     for single_item in numbers:
4         print('single_item', single_item)
5     for first_part, second_part in phonebook.items():
6         print(first_part, second_part)
7 print(total(10, 1, 2, 3, Jack=1123, John=2231, Inge=1560))

```

will print out

```

a 10
single_item 1
single_item 2
single_item 3
Inge 1560
John 2231
Jack 1123
None

```

to you screen. Where `*param` makes all the position arguments(2.13.1.10) from now on together as a tuple, and `**param` makes all the keyword arguments(2.13.1.10) from now on together as a dictionary?.

┘

**Def. (2.13.1.13)[Documentation Strings].** You can use a function to get a document: e.g.

```

1 def print_max(x, y):
2     x = int(x)
3     y = int(y)
4     if x > y:
5         print(x, 'is maximum')
6     else:
7         print(y, 'is maximum')
8 print_max(3, 5)
9 print(print_max.__doc__)

```

will print out  
 5 is maximum  
 to your screen.

### Modules and Packages

**Def. (2.13.1.14) [Python Modules].** Every Python code file with extension name “.py” is a Python module. e.g. we can create a new file named “mymodule.py” with codes:

```
|1 def say_hi():
|2     print('Hi, this is mymodule speaking.')
|3     __version__ = '0.1'
```

where the “\_\_” makes the function private.

**Def. (2.13.1.15) [Import Modules].** You can import modules using `import` syntax, e.g. with the module defined in (2.13.1.14), we can write in another file:

```
|1 import mymodule
|2
|3 mymodule.say_hi()
|4 print('Version', mymodule.__version__)
```

Or you can import functions directly, e.g.:

```
|1 from math import sqrt
|2 print("Square root of 16 is", sqrt(16))
```

**Def. (2.13.1.16) [dir Functions].** The `dir()` function(directory) can return the name list defined by the parameter object. If the parameter is empty, then it returns the name list of current module. e.g. if you run the file

```
|1 import sys
```

and input “`dir()`”, then it will print out

```
['PS1', 'REPLHooks', '__annotations__', '__builtins__', '__doc__', '__loader__', '__name__',
to your screen.
```

**Def. (2.13.1.17) [Python Packages].** A **Python package** is a folder containing Python modules (2.13.1.14) and a special file named “`__init__.py`”.

### Data Structures

**Def. (2.13.1.18) [Data Structures in Python].** There are 4 default data structures in Python:

- **List:** List is a mutable data structure used to store a set of ordered items. Use brackets to define a list, e.g.

```
|1 shoplist = ['apple', 'mango', 'carrot', 'banana']
```

define a list named “shoplist” with 4 items.

For a list, there are several operations to it, for example, for the shoplist defined above, we can do the following operation:

```
—
|1 a = shoplist[0]
```

which is “apple”,

```

—
|1  shoplist.append('rice')
    which add an item “rice” to the shoplist.

```

```

—
|1  shoplist.sort()
    sorts the items in shoplist in alphabet order.
—

```

- Tuple: A tuple is a immutable data structure used to store items together. Use parenthesis to define a tuple, e.g.:

```

|1  zoo = ('python', 'elephant', 'penguin')
|2  singleton = (2, )

```

define a tuple named “zoo” with 3 items and a tuple named “singleton” with a single item.

- Dictionary: A dictionary is a way to assign values to keywords, e.g.

```

|1  ab = {
|2      'Swaroop': 'swaroop@swaroopch.com',
|3      'Larry': 'larry@wall.org',
|4      'Matsumoto': 'matz@ruby-lang.org',
|5      'Spammer': 'spammer@hotmail.com'
|6  }
|7  for name, address in ab.items():
|8  print('Contact {} at {}'.format(name, address))

```

will print out

Contact Swaroop at swaroop@swaroopch.com

Contact Matsumoto at matz@ruby-lang.org

Contact Larry at larry@wall.org

to your screen. And

```

|1  ab['Guido'] = 'guido@python.org'

```

will add a pair

'Guido': 'guido@python.org'

to the dictionary.

- Set:

└

**Def. (2.13.1.19) [Format Function].** The format function can construct character strings from other informations, e.g.:

```

|1  age = 20
|2  name = 'Swaroop'
|3
|4  print('{0} was {1} years old when he wrote this book'.format(name, age))
|5  print('Why is {0} playing with that python?'.format(name))

```

will print out

Swaroop was 20 years old when he wrote this book

Why is Swaroop playing with that python?

to your screen.

└



## 2 Python Standard Library

References are <https://docs.python.org/3/library/>.

**Def. (2.13.2.1) [Python Packages].** Python has many open-sourced libraries and packages, you can find them in <https://pypi.org> and download them via the package installer `pip`! (Cf. <https://pip.pypa.io/en/latest/>) in the terminal. ┘

**Def. (2.13.2.2) [How to Ask Questions].** Cf. <http://catb.org/~esr/faqs/smart-questions.html>. ┘

### Sys Module



## 3 | Algebras

### 3.1 Group Theory

*“La théorie des groupes est, pour ainsi dire, la mathématique entière, dépouillée de sa matière et réduite à une forme pure.”*

–H. Poincaré

Main References are [Finite Groups, Issac], [Lan02] and [代数学引论, 丁石孙].

**Notation(3.1.0.1).**

- Let  $G \in \mathcal{G}rp$ .
- Use notations on sets defined in [Set Theory](#).

┘

#### 1 Magmas

*“Il est peu de notions en mathématiques qui soient plus primitives que celle de loi de composition.”*

–N. Bourbaki

**Def.(3.1.1.1)[Binary Operators].** A **binary operator** on a set  $X$  is a map  $\circ : X \times X \rightarrow X$ . ┘

**Def.(3.1.1.2)[Unital Operator].** A **unital binary operator** on a set  $X$  is a map  $\circ : X \times X \rightarrow X$  that has a left and right identity element  $1_\circ$ . ┘

**Prop.(3.1.1.3)[Associative Operators].** An **associative operator** on a set  $X$  is an operator  $\circ : X \times X \rightarrow X$  s.t.  $(x \circ y) \circ z = x \circ (y \circ z)$  for any  $x, y, z \in X$ . ┘

**Def.(3.1.1.4)[Magma].** A **magma** is a set  $X \in \mathcal{S}et$  with a binary operator  $\circ : X \times X \rightarrow X$ . A **unital magma** is a magma that the operator is unital. It is called **Abelian** if  $x \circ y = y \circ x$  for any  $x, y \in X$ . ┘

**Prop.(3.1.1.5)[Eckmann-Hilton argument].** If  $\circ$  and  $\otimes$  are two unital binary operators on a set that commute with each other:  $(a \otimes b) \circ (c \otimes d) = (a \circ c) \otimes (b \circ d)$ , then they are equal and in fact commutative and associative. ┘

*Proof:* Firstly the units coincide, because

$$1_\circ = 1_\circ \circ 1_\circ = (1_\otimes \otimes 1_\circ) \circ (1_\circ \otimes 1_\otimes) = (1_\otimes \circ 1_\circ) \otimes (1_\circ \circ 1_\otimes) = 1_\otimes \otimes 1_\otimes = 1_\otimes.$$

Next

$$a \circ b = (1 \otimes a) \circ (b \otimes 1) = (1 \circ b) \otimes (a \circ 1) = b \otimes a = (b \circ 1) \otimes (1 \circ a) = (b \otimes 1) \circ (1 \otimes a) = b \circ a.$$

Thus  $\circ$  and  $\otimes$  coincide and are commutative. Finally for associativity:

$$(a \otimes b) \otimes c = (a \otimes b) \otimes (1 \otimes c) = (a \otimes 1) \otimes (b \otimes c) = a \otimes (b \otimes c).$$

□

**Def. (3.1.1.6) [Monoid].** A **monoid** is an associative unital magma  $(X, \circ)$  (3.1.1.3)(3.1.1.2)(3.1.1.4). The category of monoids are denoted by  $\mathcal{Mon}$ . ┘

## 2 Groups

**Def. (3.1.2.1) [Groups, Cayley1854].** ?, The category of Groups is denoted by  $\mathsf{Grp}$ . ┘

**Def. (3.1.2.2) [Generators].** Let  $G \in \mathsf{Grp}$  and  $S \subset G$ , then  $S$  is said to be a **set of generator** for  $G$  if the only subgroup of  $G$  containing  $S$  is  $G$ . ┘

**Def. (3.1.2.3) [Normal Subgroups].** A **normal subgroup** of a group  $G$  is a subgroup  $N$  that if  $x \in G$ , then  $x^{-1}Nx = N$ . ┘

**Def. (3.1.2.4) [Simple Groups].** A **simple group** is a group that has no normal subgroups. ┘

**Def. (3.1.2.5) [Finitely Generated Group].** A group  $G$  is called finitely generated if there is a finite subset  $S$  that the only subgroup containing  $S$  is  $G$  itself. ┘

**Def. (3.1.2.6) [Product Subgroups].** Let  $G$  be a group and  $N, H$  be subgroups such that  $N$  is normal in  $G$ , then  $NH = \{x \in G | x = nh, n \in N, h \in H\}$  is a subgroup of  $G$ , called the **product subgroup**. ┘

**Def. (3.1.2.7) [Coset].** Let  $G$  be a group and  $H$  a subgroup. Consider the equivalence relation on  $G$  s.t.  $x \sim y$  iff  $x = yh$  for some  $h \in H$ , then the equivalence classes is denoted by  $G/H$ , called the **right coset** of  $H$  in  $G$ . And  $G$  acts on this set by left translation.

Similarly we can define **left coset**  $H \backslash G$  of  $H$  in  $G$ .

Moreover, if  $H$  is normal in  $G$ , then this set is a group with structure given by  $\tau H \cdot \sigma H = \tau \sigma H$ , called the **quotient group** structure. It satisfies the universal property that any group homomorphism  $\varphi : G \rightarrow G'$  s.t.  $\varphi(H) = e$  factors through  $G/H$  uniquely. ┘

**Prop. (3.1.2.8) [Fundamental Isomorphisms].** Let  $G$  be a group and  $N, H$  be subgroups such that  $N$  is normal in  $G$ , then

- there is a natural isomorphism  $G/NH \cong (G/N)/(H/H \cap N)$  as sets, and if  $H$  is also normal, this is an isomorphism of groups.
  - there is a natural isomorphism of groups:  $H/N \cap H \cong NH/N$ .
- ┘

*Proof:* 1:

2: Consider the natural isomorphism  $N \mapsto NH/H : n \mapsto nH$ , then it is a group homomorphism, and the kernel is  $N \cap H$ , thus we are done. ┘

**Cor. (3.1.2.9).** if  $H_1, H_2$  are subgroups of a group  $G$  that has finite indexes, then  $H_1 \cap H_2$  also has finite index in  $G$ . ┘

*Proof:* By fundamental isomorphism (3.1.2.8),  $H_1/H_1 \cap H_2 \cong H_1 H_2 / H_2 \subset G/H_2$ , so  $H_1 \cap H_2$  has finite index in  $H_1$ , so by transitivity of indexes,  $H_1 \cap H_2$  has finite index in  $G$ . ┘

**Def. (3.1.2.10)[Index of Subgroup].** The **index of a subgroup**  $H$  in a group  $G$  is defined to be the number of the left coset  $G/H$ , if it is finite. Now if  $H$  has finite index in  $G$ , then  $|G/H| = |H \backslash G|$ .  $\lrcorner$

*Proof:* Because for any system of representative  $a_i$  for the left coset  $G/H$ ,  $a_i^{-1}$  is a representative for the right coset  $H \backslash G$ , and vice versa.  $\square$

**Prop. (3.1.2.11).** If a finite group  $G$  has an automorphism  $\alpha$  that  $\alpha^2 = \text{id}$  and  $\alpha$  has no fixed point other than  $e$ , then  $G$  is an Abelian group of odd order.  $\lrcorner$

*Proof:*  $G$  is clearly of odd order. Consider the map  $g \mapsto \alpha(g)g^{-1}$ , then it is injective, hence it is also surjective, and consider  $\alpha(\alpha(g)g^{-1}) = g\alpha(g)^{-1} = (\alpha(g)g^{-1})^{-1}$ , thus  $\alpha(h) = h^{-1}$  for all  $h \in G$ , thus clearly  $G$  is Abelian.  $\square$

**Prop. (3.1.2.12).** If  $H \leq G \in \mathcal{A}b^{\text{fin}}$  and  $H \neq G$ , then  $G \neq \cup g^{-1}Hg$ .  $\lrcorner$

*Proof:* There are at most  $|G/H|$  different summands in the right hand side, so it doesn't have enough elements.  $\square$

**Prop. (3.1.2.13).** If  $G$  is a f.g. group (3.1.2.5) and  $H$  is a group of finite index in  $G$ , then  $H$  is f.g.  $\lrcorner$

*Proof:* Suppose  $G$  has generators  $g_i$ , we may add their inverses to it, and let  $Ht_1, \dots, Ht_m$  are all the right cosets with  $t_1 = 1$ , then there are  $h_{ij}$  that  $t_i g_j = h_{ij} t_{k_{ij}}$ , then we claim  $H$  is generated by  $h_{ij}$ .

For this, consider any  $h = \prod g_{i_r}$ , then  $g_1 = h_{1i_r} t_{k_{1i_r}}$ , and we can do this from left to right. Now  $h = \prod h_{i_s j_s} t_{o_s}$ , then  $t_o$  must be 1, and we are done.  $\square$

**Prop. (3.1.2.14).** Let  $\varphi : G \rightarrow H$  be a map of sets between groups s.t.  $\# \text{Im}(\varphi) = \infty$ , and for each  $g \in G$ , there is a set  $S(g) \in H$  that  $\varphi(h)\varphi(g) = \varphi(hg)$  for any  $h$  s.t.  $\varphi(h) \notin S(g)$ , then  $\varphi$  is a homomorphism.  $\lrcorner$

*Proof:* For  $g, h \in G$ , take  $f$  s.t.  $\varphi(f) \notin S(g) \cup \varphi(g)^{-1}S(h) \cup \{\varphi(gh)\}$ . Then

$$\varphi(f)\varphi(g)\varphi(h) = \varphi(fg)\varphi(h) = \varphi(fgh) = \varphi(f)\varphi(gh),$$

thus  $\varphi(g)\varphi(h) = \varphi(gh)$ .  $\square$

### General Linear Groups

**Notation (3.1.2.15)[Group-Theoretic Notations for  $\text{GL}(n)$ ].** When  $G = \text{GL}(n)$ ,

- Let  $B$  be the Borel subgroup of upper triangular matrices.
- Let  $\text{Unip}(n)$  is the subgroup of upper triangle unipotent matrices.
- Let  $M_n$  is the **mirabolic subgroup** of  $B$  with  $a_{n,n} = 1$ , which is isomorphic to  $\text{GL}(n-1) \ltimes \mathbb{A}^n$ .
- Let  $T$  the group of diagonal matrices.  $T_1$  is the subgroup of  $T$  that  $a_{nn} = 1$ .
- Denote  $w^0$  the matrix  $= \sum_{i=1}^n e_{i,n-i}$ .
- If  $n = 2$ , denote  $w_0 = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$ ,  $w_1 = \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}$ ,

$$t(y) = \begin{bmatrix} y & \\ & y^{-1} \end{bmatrix}, \quad n(z) = \begin{bmatrix} 1 & z \\ & 1 \end{bmatrix} \quad \forall y \in F^*, z \in F.$$

following (3.1.5.9).

$\lrcorner$

### 3 Abelian Groups

**Remark (3.1.3.1).** An Abelian group is the same as a module over  $\mathbb{Z}$ . Thus the theory of commutative algebra applies in this case. The category of Abelian groups is denoted by  $\mathcal{Ab}$ . The category of finite Abelian groups are denoted by  $\mathcal{Ab}^{\text{fin}}$ .  $\lrcorner$

**Prop. (3.1.3.2) [Abelianization].** There is a functor from the category of Abelian semi-groups to the category of Abelian groups that is left adjoint to the forgetful functor, called the **Abelianization**.  $\lrcorner$

*Proof:* Define  $A'$  to be the quotient

$$\bigoplus_{(a,b) \in A^2} \mathbb{Z} \rightarrow \bigoplus_{x \in A} \mathbb{Z} \rightarrow A' \rightarrow 0,$$

where  $1_{(a,b)}$  is mapped to  $1_a + 1_b - 1_{a+b}$ . Then it can be seen this satisfies the universal property and it is functorial in  $A$ .  $\square$

**Prop. (3.1.3.3) [Classifying F.g. Abelian Groups].** Any finitely generated Abelian group is of the form

$$\mathbb{Z}^r \bigoplus_{p_i \text{ primes}} \bigoplus_{j \leq n_i} \mathbb{Z}/(p_i^{a_{i,j}})$$

*Proof:* As  $\mathbb{Z}$  is PID, the classifying theorem follows immediately from (3.2.4.21):  $\square$

**Prop. (3.1.3.4) [Baer-Specker Group].** The group  $\prod^{\mathbb{N}} \mathbb{Z}$  is called the **Baer-Specker group**. Then the natural homomorphism

$$\text{Hom}(\prod^{\mathbb{N}} \mathbb{Z}, \mathbb{Z}) \rightarrow \prod^{\mathbb{N}} \mathbb{Z}$$

is injective with image  $\bigoplus^{\mathbb{N}} \mathbb{Z}$ . In particular, by countability argument,  $\prod^{\mathbb{N}} \mathbb{Z}$  is not a free group (5.3.1.5). Moreover, any infinite direct product of  $\mathbb{Z}$  is not free, because otherwise the subgroup  $\prod^{\mathbb{N}} \mathbb{Z}$  would be free.  $\lrcorner$

*Proof:* The natural homomorphism is the composite  $?$

$$\text{Hom}(\prod^{\mathbb{N}} \mathbb{Z}, \mathbb{Z}) \xrightarrow{\varepsilon^*} \text{Hom}(\#^{\mathbb{N}} \mathbb{Z}, \mathbb{Z}) \xrightarrow{\eta} \prod^{\mathbb{N}} \mathbb{Z}.$$

Thus the assertion follows from (3.1.5.6) and (3.1.5.5).  $\square$

**Remark (3.1.3.5).** This can be proven in other ways, like using condensed mathematics.  $?$   $\lrcorner$

#### Finite Abelian Groups

**Prop. (3.1.3.6) [Characterizing Finite Cyclic Groups].** Let  $G \in \mathcal{Ab}^{\text{fin}}$  that  $\#\{x \in G \mid x^d = 1\} \leq d$  for any  $d \geq 1$ , then  $G$  is cyclic.  $\lrcorner$

*Proof:* Consider the subset  $G_d$  of elements of order  $d$ , if it is non-empty, choose  $y \in G_d$ , then  $\#\langle y \rangle = d$ , thus  $\langle y \rangle = \{x \in G \mid x^d = 1\}$ , which means  $\#G_d \leq \varphi(d)$ . Then  $|G| = \sum_{d \mid n} \#G_d \leq \sum_{d \mid n} \varphi(d) = n$ . Thus  $G_n \neq \emptyset$ , which means  $G$  is cyclic.  $\square$

**Cor. (3.1.3.7).** For  $k \in \text{Field}$ , any finite subgroup of  $k^\times$  is cyclic.  $\lrcorner$

**Cor. (3.1.3.8) [Primitive roots modulo  $q$ ].** For  $p \in \mathbf{P}$  and  $q \in p^{\mathbb{Z}_+}$ ,  $\mathbb{F}_q^\times$  is cyclic, i.e.  $\mathbb{F}_q^\times \cong \langle \sigma \rangle$ . Any such generator  $\sigma$  is called a **primitive root modulo  $p$** .  $\lrcorner$

## 4 Automorphism Groups

**Def. (3.1.4.1) [Automorphisms Groups].** Let  $G \in \mathcal{G}rp$ , the set of automorphism of  $G$  is a group, denoted by  $\text{Aut}(G)$ .

Then for any  $g \in G$ , there is an automorphism  $C_g \in \text{Aut}(G) = \text{Hom}(G, G) : x \mapsto gxg^{-1}$ . And the mapping  $G \rightarrow \text{Aut}(G) : g \mapsto C_g$  is a group homomorphism. All automorphisms of  $G$  of the form are called **inner automorphisms** of  $G$ . The group of inner automorphisms of  $G$  is denoted by  $\text{Inn}(G)$ . Automorphisms that are not inner are called **outer automorphisms**.  $\lrcorner$

**Prop. (3.1.4.2).** Any group of order  $> 2$  have at least 2 automorphisms.  $\lrcorner$

*Proof:* Assume the contrary, consider its inner automorphism, then it is Abelian, and then multiplying by  $p$  for  $p$  large prime is not identity, Then  $\#G|(p-1)$  for such  $G$ . Now it is clear  $|G| = 2$ , because otherwise we can choose  $p \equiv 2 \pmod{\#G}$ .  $\square$

**Def. (3.1.4.3) [Automorphic Complete Groups].** An **automorphic-complete group** is a group whose automorphisms are all inner.  $\lrcorner$

**Prop. (3.1.4.4).**  $S_n$  is the automorphism group of  $A_n$  for  $n = 5$  or  $n \geq 7$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (3.1.4.5).** If  $G$  is a non-Abelian simple group, then  $\text{Aut}(G)$  is a complete group.  $\lrcorner$

**Prop. (3.1.4.6).**  $S_n$  are complete groups except for  $S_6$ .  $\lrcorner$

*Proof:*  $S_n$  is the automorphism group of  $A_n$  for  $n = 5$  or  $n \geq 7$  by (3.1.4.4), thus it is complete by (3.1.4.5).  $\square$

**Prop. (3.1.4.7) [Wielandt].** If  $G$  is a finite group with trivial center, then the sequence

$$G < \text{Aut}(G) < \text{Aut}(\text{Aut}(G)) < \dots$$

must terminate in finite steps.  $\lrcorner$

*Proof:*  $\square$

## 5 Free Groups and Presentations

**Def. (3.1.5.1) [Free Groups].** There is a **free group** functor  $F : \text{Set} \rightarrow \mathcal{G}rp$  that is left adjoint to the forgetful functor.

The image of the set  $[n-1]$  is denoted by  $F_n$ .  $\lrcorner$

*Proof:* For the existence of  $F$ , Cf. [Lan02]P66.  $\square$

**Prop. (3.1.5.2) [Nielsen-Schreier].** A subgroup  $H$  of a free group  $G$  is a free group. Moreover, a subgroup  $H$  of finite index  $m$  in  $F_n$  is isomorphic to  $F_{1+m(n-1)}$ .  $\lrcorner$

*Proof:* A free group is the fundamental group of a graph  $X$  which is a wedge sum of circles, and there is a covering  $X_H \rightarrow X$  the  $\pi_1(X_H) = H$  by (4.15.1.28). And if  $H$  is of index  $m$  in a free group  $G$ ,  $X_H \rightarrow X$  has degree  $m$  by (37.2.1.26). Now (37.2.1.26) shows  $H = \pi_1(X_H)$  is a free group, and the final assertion follows by comparing two ways of counting Euler number  $\chi$ .  $\square$

**Prop. (3.1.5.3).** Use the same method as in (3.1.5.2), we can determine all the subgroups of index 2 in  $F_2$ . ┘

*Proof:* ? □

**Def. (3.1.5.4) [Earring Group].** For any  $n \in \mathbb{Z}_+$ , there are group homomorphisms  $F_n \rightarrow F_{n-1}$  corresponding to the map of sets

$$[n-1] \rightarrow F_{n-1} : i \mapsto \begin{cases} [i] & , i \leq n-2 \\ 1 & , i = n-1 \end{cases}.$$

The **Earring group**  $\#^{\mathbb{N}}\mathbb{Z}$  is defined to be the subgroup of

$$\varprojlim_{n \in \mathbb{N}} F_n$$

consisting of elements  $(w_0, w_1, \dots, w_n, \dots)$  s.t. if each  $w_n$  is a reduced word, then for any  $k \in \mathbb{Z}_+$ , the number of  $[k]^{\pm}$  appearing in  $w_n$  stabilizes. ┘

**Prop. (3.1.5.5).** The natural homomorphism

$$\eta : \text{Hom}(\#^{\mathbb{N}}\mathbb{Z}, \mathbb{Z}) \rightarrow \prod^{\mathbb{N}} \mathbb{Z}$$

is injective with image  $\oplus^{\mathbb{N}}\mathbb{Z}$ . ┘

*Proof:* Cf. <https://wildtopology.com/2014/05/09/the-hawaiian-earring-group-is-not-free-part-i/>. □

**Prop. (3.1.5.6).** There is a natural group homomorphism  $\varepsilon : \#^{\mathbb{N}}\mathbb{Z} \rightarrow \prod^{\mathbb{N}} \mathbb{Z}$  that is surjective. ┘

*Proof:* The  $k$ -th coordinate of this map is given by omitting all the elements that is not  $[k]$ . It is clearly surjective. □

**Prop. (3.1.5.7) [Smith].**  $\#^{\mathbb{N}}\mathbb{Z}$  is not a free group. ┘

*Proof:* Cf. <https://wildtopology.com/2014/05/09/the-hawaiian-earring-group-is-not-free-part-i/>. □

### Presentations

**Def. (3.1.5.8) [Defining Sets].** For  $G \in \text{Grp}$ , a **defining set** of  $G$  is a subset  $S \subset G$  s.t.  $G$  is generated by  $S$  subject to the relations of the form  $ab = c$  where  $a, b, c \in S$ . ┘

### List of Presentations of Important Groups

**Prop. (3.1.5.9).** For  $K \in \text{Field}$ ,  $\text{SL}(2; K)$  has a representation as:

$$\langle t(y), n(z), w_1 \rangle, \quad y \in K^{\times}, z \in F$$

quotient the relations:

$$t(y_1)t(y_2) = t(y_1y_2), \quad n(z_1)n(z_2) = n(z_1 + z_2), \quad t(y)n(z)t(y)^{-1} = n(y^2z), \quad w_1t(y)w_1 = t(-y^{-1}).$$



$$w_1 n(z) w_1 = t(-z^{-1}) n(-z) w_1 n(-z^{-1}), z \in F^*,$$

And the isomorphism is given by  $\Phi$ :

$$t(y) \mapsto \begin{bmatrix} y & \\ & y^{-1} \end{bmatrix}, \quad n(z) \mapsto \begin{bmatrix} 1 & z \\ & 1 \end{bmatrix}, \quad w_1 \mapsto \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}.$$

┘

*Proof:* The map vanishes on the relations is direct calculation. An inverse of  $\Phi$  is constructed by

$$\Psi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} n(a/c) t(-c^{-1}) w_1 n(d/c) & c \neq 0 \\ t(a) n(b/a) & c = 0 \end{cases}$$

The verification of the inverse is verified by direct calculation. □

**Prop. (3.1.5.10) [Metaplectic Modular Group].** In the metaplectic modular group  $\text{Mp}(2, \mathbb{Z})$  (26.5.1.1), define

$$\tilde{S} = \left( \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}, -i\sqrt{\tau} \right), \quad \tilde{T} = \left( \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \mathbb{1} \right), \quad \tilde{R} = \tilde{S}^2 = (-I, -i).$$

Then  $\text{Mp}(2, \mathbb{Z})$  is generated by  $\tilde{S}, \tilde{T}$  with relations

$$\tilde{S}^8 = (\tilde{S}\tilde{T})^3 = 1.$$

And the center  $C(\text{Mp}(2, \mathbb{Z})) = \langle \tilde{S}^2 \rangle \cong \mathbb{Z}/(4)$ , with  $\text{Mp}(2, \mathbb{Z})/\{1, \tilde{S}^4\} = \text{SL}(2, \mathbb{Z})$ . ┘

*Proof:* ? □

**Cor. (3.1.5.11).**  $\text{SL}(2, \mathbb{Z})$  is generated by  $S = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}, T = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$  with relations

$$S^4 = 1, \quad (ST)^3 = S^2$$

.

┘

*Proof:* [Serre, Trees, P81]. □

**Cor. (3.1.5.12) [Modular Group].**  $\text{PSL}_2(\mathbb{Z}) = \text{SL}(2, \mathbb{Z})/\{\pm 1\}$  is generated by  $S = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}, T =$

$\begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$  with relations

$$S^2 = 1, \quad (ST)^3 = 1.$$

┘

## 6 Sylow Theory

**Prop. (3.1.6.1) [Class Equation].** For a finite group  $G$ , if  $G_x = C((x))$ , then

$$|G| = |C(G)| + \sum |G|/|G_x|$$

where the summation is over non-trivial conjugate classes of  $G$ .  $\lrcorner$

*Proof:* Consider the left action of  $G$  on itself, and calculate elements.  $\square$

**Cor. (3.1.6.2) [ $p$ -Groups has non-trivial center].** if  $G$  is a  $p$ -group, then  $G$  has a non-trivial center.  $\lrcorner$

*Proof:* Consider the conjugation action of  $G$  on itself.  $\square$

**Cor. (3.1.6.3).** If  $p \nmid \#G$ , then  $G$  has an element of order  $p$ .  $\lrcorner$

*Proof:* Follows from Sylow theory and any  $p$ -group has a non-trivial center.  $\square$

**Lemma (3.1.6.4).** For any  $p$ -group  $G$  acting on a finite set  $X$ ,  $|X| \equiv |X^G| \pmod{p}$ . (trivial).  $\lrcorner$

**Prop. (3.1.6.5) [Sylow Theorem, Sylow1872].** For a finite group of order  $|G| = p^k m$ .

- There is a Sylow  $p$ -group.
  - For a Sylow  $p$ -subgroup, any  $p$ -subgroup is contained in a conjugate of  $P$ . In particular, any two Sylow  $p$ -subgroups are conjugate.
  - the number of Sylow  $p$ -groups  $n_p$  satisfies:  $n_p | m, n_p \equiv 1 \pmod{p}$ .
- $\lrcorner$

*Proof:* 1: Use induction, let  $Z = C(G)$ , if  $p \nmid |Z|$ , then  $Z$  contains a cyclic group of order  $p$ . Choose a  $p$ -Sylow subgroup of  $G/C$ , then its inverse image in  $G$  is a  $p$ -Sylow subgroup. If  $Z$  is prime to  $p$ , consider the conjugate action of  $G$  on  $G - Z$ , then some conjugacy class has order prime to  $p$ , by (3.1.6.4), then the stablizer  $H$  of this class satisfies  $[G : H]$  is prime to  $p$ . Thus  $H$  contains a  $p$ -Sylow subgroup by induction.

2: If  $Q$  is a  $p$ -subgroup, then  $Q$  acts on  $G/P$  by left translation, so it has a fixed element by (3.1.6.4),  $QxP = xP$  for some  $x$ , thus  $Q \subset xPx^{-1}$ .

3:  $n_p | m$  by considering the conjugate action of  $P$  on the set of conjugates of  $P$ , then as in the proof of item2,  $P$  is the only fixed element, so  $n_p \equiv 1 \pmod{p}$  by (3.1.6.4).  $\square$

**Lemma (3.1.6.6).** If  $G$  has a sylow subgroup  $H$  that  $|G/H|!$  is not divisible by  $|G|$ , then  $G$  is not simple.  $\lrcorner$

*Proof:* Consider the conjugate action of  $G$  on the conjugacy classes of  $H$ , then it is a group homomorphism of  $G$  into a subgroup of  $S_{|G/H|}$ , but the hypothesis shows that it is not injective, thus the kernel is non-trivial normal.  $\square$

**Prop. (3.1.6.7) [Frattini Argument].** If  $G$  is a finite group,  $N$  is normal in  $G$  and  $P$  is a Sylow subgroup of  $N$ , then  $N.N_G(P) = G$ .  $\lrcorner$

*Proof:* For any element  $g \in G$ , consider  $g^{-1}Pg \subset N$  is a Sylow subgroup of  $N$ , thus by Sylow theorem (3.1.6.5), there is a  $n \in N$  that  $g^{-1}Pg = n^{-1}Pn$ , thus  $gn^{-1} \in N_G(P)$ , thus  $g \in NN_G(P)$ .  $\square$

## 7 Split Extension

**Prop. (3.1.7.1) [Cyclic Central Extension Split].** If there is an exact sequence  $0 \rightarrow Z \rightarrow G \rightarrow C \rightarrow 0$  where  $Z \subset C(G)$  and  $C$  is cyclic, then  $G$  is Abelian.  $\lrcorner$

*Proof:* This is because we can choose an inverse image of a generator of  $C$ .  $\square$

**Prop. (3.1.7.2) [Schur-Zassenhaus].** An exact sequence of finite groups  $0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 0$  must split when  $|A|$  and  $|G|$  are relatively prime.  $\lrcorner$

*Proof:*  $\square$

**Prop. (3.1.7.3).** Let  $\alpha, \beta : G \rightarrow \text{Aut}(H)$  be two actions of  $G$  on  $H$ , then their semiproduct sequences

$$1 \rightarrow H \rightarrow G \ltimes H \rightarrow G \rightarrow 1$$

are isomorphic iff  $\alpha, \beta$  are equivalent modulo  $\text{Inn}(H)$ .  $\lrcorner$

## 8 Subnormality

**Def. (3.1.8.1) [Subnormal Series].** A **subnormal series** of  $G$  is a descending chain of groups:

$$G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_r = \{e\}$$

that  $G_{k+1}$  is normal in  $G_k$ . It is called a **composite series** iff each  $G_k/G_{k+1}$  is simple.

A subnormal series is called a **normal series** if each  $G_i$  is normal in  $G$ .  $\lrcorner$

**Lemma (3.1.8.2) [Butterfly Lemma].** Let  $H_1, H_2 \leq G, N_1 \trianglelefteq H_1, N_2 \trianglelefteq H_2$ , then there is a canonical isomorphism of groups:

$$N_1(H_1 \cap H_2)/N_1(H_1 \cap N_2) \cong N_2(H_2 \cap H_1)/N_2(H_2 \cap N_1).$$

$\lrcorner$

*Proof:* Cf. [Lan02] P20.  $\square$

**Prop. (3.1.8.3) [Schreier].** Any two subnormal series of a group  $G$  have a common refinement.  $\lrcorner$

*Proof:*  $\square$

**Cor. (3.1.8.4) [Jordan-Hölder].** For any two composition series  $G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_r = \{e\}$ ,  $G = G'_0 \triangleright G'_1 \triangleright \dots \triangleright G'_{r'} = \{e\}$ , there  $r = r'$ , and there exists a permutation  $\sigma \in S_{r-1}$  s.t.  $G_i/G_{i+1} \cong G'_{\sigma(i)}/G'_{\sigma(i)+1}$ .  $\lrcorner$

**Def. (3.1.8.5) [Central Series].** A **central series** of  $G$  is an ascending chain of subgroups:

$$1 = Z_0 < Z_1 < \dots < Z_r = G$$

s.t. each  $Z_{k+1}/Z_k$  is in the center of  $G/Z_k$ .  $\lrcorner$

**Def. (3.1.8.6) [Solvable and Nilpotent Groups].**  $G$  is called a **A solvable group** if it has a finite subnormal series (3.1.8.1)  $\{G_i\}$  s.t. each  $G_i/G_{i+1}$  is Abelian.

$G$  is called a **nilpotent group** if it has a finite central series (3.1.8.5).  $\lrcorner$

**Def. (3.1.8.7) [Solvable Index].** If  $G \in \text{Grp}^{\text{fin}}$  is solvable, then the **solvable index** of  $G$  is the integer  $m \in \mathbb{N}$  s.t. there is a subnormal sequence

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \dots \trianglelefteq G$$

that each quotient  $G_{k+1}/G_k$  is a cyclic group of prime order.  $\lrcorner$

**Def. (3.1.8.8) [Supersolvable Groups].**  $G$  is called a **supersolvable group** iff it has a finite normal series (3.1.8.1)  $\{G_i\}$  s.t.  $G_i/G_{i+1}$  is cyclic. Any supersolvable group is solvable.  $\lrcorner$

**Prop. (3.1.8.9).**

- Subgroups and quotients of a supersolvable group is supersolvable.
- Any finite nilpotent group is supersolvable.  $\lrcorner$

**Prop. (3.1.8.10).** If  $G$  is a non-Abelian supersolvable group, then there is a normal Abelian subgroup of  $G$  not contained in the center of  $G$ .  $\lrcorner$

*Proof:* Let  $G = G_1 \triangleright G_2 \triangleright \dots \triangleright G_r = 1$  be a normal series of  $G$ , then there is a maximal  $i$  s.t.  $G_{i+1}$  is contained in the center of  $G$  but  $G_i$  is not. Then  $G_i$  is Abelian and normal in  $G$ .  $\square$

**Lemma (3.1.8.11) [ $p$ -Groups are Nilpotent].** Any finite  $p$ -group is nilpotent, thus supersolvable (3.1.8.9).  $\lrcorner$

*Proof:* Using induction by (3.1.6.2), we see it has a central series, thus nilpotent (3.1.8.6).  $\square$

**Prop. (3.1.8.12) [Characterizing Nilpotent Finite Groups].** If  $\#G < \infty$ , then the following are equivalent:

- $G$  is nilpotent.
- $N_G(H) > H$  for every proper subgroup  $H < G$ .
- Every maximal subgroup of  $G$  is normal.
- Every Sylow subgroup of  $G$  is normal.
- $G$  is a direct product of its non-trivial Sylow subgroups.  $\lrcorner$

*Proof:*  $1 \rightarrow 2$ : Choose a central series  $Z_n$ , let  $Z_n \subset H$  and  $Z_{n+1} \not\subset H$ , then  $[Z_{n+1}, H] \subset [Z_{n+1}, G] \subset Z_n \subset H$ , thus  $Z_{n+1} \subset N_G(H)$ .

$2 \rightarrow 3, 4 \rightarrow 5$ : trivial.

$3 \rightarrow 4$  For any  $p$ -Sylow subgroup  $G$ , if  $N_G(P)$  is proper subgroup, then it is contained in some maximal subgroup  $M$ , and  $M$  is normal, thus by Frattini argument (3.1.6.7),  $G = N_G(P)M = M$ , contradiction.

$5 \rightarrow 1$ : By lemma (3.1.8.11).  $\square$

### Minimal Normal Subgroups

**Prop. (3.1.8.13) [Minimal Normal Subgroup].** The minimal normal subgroup  $N$  of a finite group  $G$  is a direct product of simple groups  $L^n$ .  $\lrcorner$

*Proof:* Let  $N_1$  be a maximal normal subgroup of  $N$ , then  $N/N_1$  is simple, and let  $N_i$  be the conjugates of  $N_1$  in  $G$ , then they are all maximal normal subgroup of  $N$ . the simple groups  $N/N_i$  are mutually isomorphic, and  $\cap N_i = 1$  by the minimality of  $N$ .

Now we use induction to prove  $N/N_1 \cap \dots \cap N_i$  is isomorphic to a product of  $N/N_1$ , which will finish the proof.

Now assume  $N_1 \cap \dots \cap N_{i-1} \not\subseteq N_i$ , then  $(N_1 \cap \dots \cap N_{i-1})N_i = N$ , and notice

$$N/N_1 \cap \dots \cap N_i \cong N_1 \cap \dots \cap N_{i-1}/N_1 \cap \dots \cap N_i \times N_i/N_1 \cap \dots \cap N_i \cong N/N_i \times N/N_1 \cap \dots \cap N_{i-1}.$$

□

**Prop. (3.1.8.14).** If a finite group  $|G| = \prod p_i$ , where  $p_i$  are distinct primes that  $\prod p_i$  and  $\prod (p_i - 1)$  are coprime, then  $G$  is cyclic. ┘

*Proof:* We prove all the Sylow groups are normal. Choose the maximal Sylow group  $A_n$ , then it is normal by Sylow theorem, and other Sylow groups act by conjugation is trivial (consider the center (3.1.6.2), then the center of the quotient, and so on), hence  $A_n$  is in the center. Now consider the quotient, by induction it is cyclic, hence this is a central extension of a cyclic group, hence  $G$  is Abelian (3.1.7.1), hence cyclic. □

**Prop. (3.1.8.15).** If  $\#G < \infty$  and  $p$  is the minimal prime factor of  $\#G$ , then all subgroups  $N$  of  $G$  of index  $p$  are normal. ┘

*Proof:* Consider the left action of  $G$  on  $G/H$ , then the kernel is  $\cap a^{-1}Ha$ , which is the maximal normal subgroup contained in  $H$ . Now this is group homomorphism of  $G$  into  $S_p$ , thus it has kernel at least  $|G|/p$ , so the kernel equals  $H$ , showing  $H$  is normal. □

**Prop. (3.1.8.16) [Burnside's Theorem].** If  $p, q$  are primes, then any finite groups of order  $p^a q^b$  is solvable. ┘

*Proof:* Cf. [Serre Linear representations of finite groups, P65]. □

**Prop. (3.1.8.17) [Thompson].** A finite group is not solvable iff there exist non-trivial elements  $x, y, z$  of coprime orders  $a, b, c$  that  $xy = z$ . ┘

**Prop. (3.1.8.18) [Feit-Thompson].** All finite groups that has odd order is solvable. ┘

*Proof:* □

## 9 Commutators

**Def. (3.1.9.1) [Notation].**

- $[a, b] = a^{-1}b^{-1}ab$ .
- $x^y = y^{-1}xy$ .

┘

**Prop. (3.1.9.2) [Commutator relations].** ┘

**Def. (3.1.9.3) [Metabelian Groups].** A metabelian group is a group  $G$  that  $G'$  is Abelian. ┘

**Prop. (3.1.9.4).** If  $G = AB$  where  $A, B$  are Abelian, then  $[G, G] = [A, B]$  and  $G$  is metabelian. ┘

*Proof:* The first one is easy to verify, the second because if we let  $b^{a_1} = a_2 b_2$ ,  $a^{b_1} = b_3 a_3$ , then

$$[a, b]^{a_1 b_1} = [a, b^{a_1}]^{b_1} = [a, b_2]^{b_1} = [a^{b_1}, b_2] = [a_3, b_2]$$

and similarly,  $[a, b]^{b_1 a_1} = [a_3, b_2]$ , so we have  $[a, b]$  commutes with  $[b_1^{-1}, a_1^{-1}]$ , which shows  $[A, B]$  is Abelian.  $\square$

**Prop. (3.1.9.5).** If  $f : G_1 \rightarrow G_2 \in \mathcal{G}rp$  s.t.  $G_2$  is nilpotent. Then  $f$  is surjective iff  $f^{ab} : G_1^{ab} \rightarrow G_2^{ab}$  is surjective.  $\lrcorner$

*Proof:* Define inductively  $\gamma_1(G_i) = G_i$ ,  $\gamma_{n+1}(G_i) = [\gamma_n(G_i), G_i]$ , we prove by induction that  $G_1/\gamma_j(G_1) \rightarrow G_2/\gamma_j(G_2)$  is surjective for any  $j \in \mathbb{Z}$ : The case  $j \leq 2$  follows from the hypothesis. Then it suffices to show that

$$\gamma_i(G_1)/\gamma_{i+1}(G_1) \rightarrow \gamma_i(G_2)/\gamma_{i+1}(G_2)$$

is surjective. It suffices to show that any element  $[a, b]$  where  $a \in \gamma_{i-1}(G_2)$ ,  $b \in G_2$  is contained in the image. But by induction hypothesis, there exists  $a' \in \gamma_{i-1}(G_1)$  and  $b' \in G_1$  s.t.  $f(a') = am$ ,  $f(b') = bn$  for some  $m, n \in \gamma_i(G_2)$ . Then

$$f([a', b']) = [am, bn] = [m, b]^a [a, b][am, n]^b \equiv [a, b] \pmod{\gamma_{i+1}(G_2)}.$$

$\square$

### Transfer

**Def. (3.1.9.6) [Transfer Maps].** For  $H \leq G \in \mathcal{G}rp$  s.t.  $[G : H] < \infty$ , choose a set of left coset representatives  $\{x_1, \dots, x_n\}$  of  $H$ , then for any  $g \in G$  and  $i \in [n]_+$ , there exists  $h_i \in H$  s.t.  $gx_i = x_j h_i(g)$ . Then the map

$$G \mapsto H^{ab} : g \mapsto \prod_{i \in [n]_+} h_i(g)$$

is a group homomorphism, and induces a map  $\text{Ver} : G^{ab} \rightarrow H^{ab}$ , called the **transfer map**.  $\lrcorner$

**Prop. (3.1.9.7).** If  $G$  is a met-Abelian finite group, then the transfer of  $\text{Ver} : G \rightarrow G'$  is trivial map.  $\lrcorner$

## 10 Permutation Groups

**Def. (3.1.10.1) [Permutation Groups].** For  $n \in \mathbb{Z}_+$ , denote  $\text{Sym}_n = \text{Aut}([n]_+)$ . In particular,  $\#\text{Sym}_n = n!$ .

For  $\pi \in \text{Sym}_n$ , construct a graph on  $[n]_+$  s.t.  $i$  is connected to  $\pi(i)$ , then this graph is 2-regular (may have loops) thus consists of cycles

$$(x_1, \pi(x_1), \pi^{\circ 2}(x_1), \dots, \pi^{(k_i-1)}(x_1)), \dots, (x_r, \pi(x_r), \pi^{\circ 2}(x_r), \dots, \pi^{(k_r-1)}(x_r)),$$

then we denote

$$\pi = (x_1 \pi(x_1) \pi^{\circ 2}(x_1) \dots \pi^{(k_i-1)}(x_1)) \dots (x_r \pi(x_r) \pi^{\circ 2}(x_r) \dots \pi^{(k_r-1)}(x_r)).$$

For example, if  $\pi \in \text{Sym}_5$  and  $\pi(1) = 2, \pi(2) = 3, \pi(3) = 1, \pi(4) = 5, \pi(5) = 4$ , then  $\pi$  is denoted by

$$\pi = (123)(45).$$

Sometimes the cycle of length 1 (i.e. fixed points of  $\pi$ ) can be omitted.  $\lrcorner$

**Prop. (3.1.10.2).** For  $n \in \mathbb{Z}_+$  and  $\sigma, \pi \in \text{Sym}_n$ , if

$$\pi = (i_1 i_2 \dots i_l) \dots (i_m i_{m+1} \dots i_n),$$

then

$$\sigma \pi \sigma^{-1} = (\sigma(i_1) \sigma(i_2) \dots \sigma(i_l)) \dots (\sigma(i_m) \sigma(i_{m+1}) \dots \sigma(i_n)).$$

┘

**Lemma (3.1.10.3).** If  $n \geq 3$ , then any proper normal subgroup of  $A_n$  has index divisible by 3. ┘

*Proof:* Otherwise consider  $n = [G : H]$ , then every  $p$ -power is in  $H$ . But then an element  $c$  of order 3 is in  $H$ , because  $c = c^{3k+1} = (c^{-1})^{3k+2}$  for any  $k$ . But  $A_n$  is generated by 3-Cycles.  $\square$

**Lemma (3.1.10.4).**  $A_5$  is simple. ┘

*Proof:* By (3.1.10.3), any proper normal subgroup  $H$  has order dividing 20.  $H$  cannot contain a 5-cycle, because a 5-cycle has  $\square$

**Prop. (3.1.10.5).**  $\text{Alt}_n$  is simple for  $n \geq 5$ . ┘

*Proof:* Cf. [代数学引论 P66].  $\square$

## 11 Classification of Finite Groups

**Thm. (3.1.11.1) [Classification of Finite Simple Groups].** Every finite simple group is one of the following:

- $\mathbb{Z}/(p)$ .
- $\text{Alt}_n$ .
- 16 infinite family of finite groups of Lie type.
- 26 exceptional groups, called **sporadic simple groups**.

┘

*Proof:* A full list is in <https://brauer.maths.qmul.ac.uk/Atlas/v3/> ?  $\square$

**Prop. (3.1.11.2) [Monster Group].** The largest sporadic simple group is called the **Monster group**  $\mathbb{M}$ .

$$\#\mathbb{M} = 2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71.$$

┘

*Proof:*  $\square$

**Prop. (3.1.11.3) [Happy Family].** There are 20 sporadic groups appearing as a subquotient of  $\mathbb{M}$ , and they form a set called the **happy family**. The remaining 6 groups are known as the **pariah groups**, including the **Lyons group**, **Janko groups**  $J_1, J_3, J_4$ , **Rudvalis group** and the **O’Nan group**  $\text{ON}$ . ┘

**Prop. (3.1.11.4) [Baby Monster Group].** The second largest sporadic simple group is called the **baby Monster group**  $\mathbb{B}$ . ┘

**Def. (3.1.11.5)[O’Nan Group].** ┘

**Prop. (3.1.11.6).** The O’Nan group has size

$$\#\text{ON} = 2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31.$$

┘

**Prop. (3.1.11.7)[Classification of Small Groups].**

1. A group  $G$  of prime order  $p$  or order  $p^2$  is Abelian.
2. A group  $G$  of order  $p^a q^b$  that it has  $q^b$   $p$ -Sylow subgroups, then its  $q$ -Sylow subgroup is normal thus it is not simple.
3. A non-Abelian group  $G$  of order 6 is isomorphic to  $\text{Sym}_3$ .
4. Any non-Abelian group of order 8 is isomorphic to  $D_4$  or quadratic numbers  $Q$ .
5. A group of order smaller than 60 is solvable.
6. Any simple group  $G$  of order 60 is isomorphic to  $\text{Alt}_5$ .
7. A group of order 148 is not simple, by (3.1.6.6) applied to the 37-Sylow subgroup.
8. A group of order 150 is not simple, by (3.1.6.6) applied to the 5-Sylow subgroup.

┘

*Proof:*

1. Because  $G$  has non-trivial center  $Z$  by (3.1.6.2), if  $Z = G$ , then it is Abelian, otherwise the  $\#Z = p$ , and the quotient  $G/Z$  is cyclic, thus  $G$  is Abelian by (3.1.7.1).
2. Calculating elements.
3. Consider its normal 3-Sylow group, then the quotient is cyclic thus  $G$  is semi-product which must by  $S_3$  when non-Abelian.
- 4.
- 5.
6. Consider  $G$  has 6 5-Sylow groups, thus there are 24 elements of order 5.  
 $G$  has 4 or 10 3-Sylow subgroups, if it have 4 3-Sylow subgroups, then the normalizer contains a 5-Sylow subgroup, so we have a subgroup of order 15, which must by  $\mathbb{Z}/15$ , so it contains a normal 5-Sylow subgroup, which shows there are at most  $60/15 = 4$  5-Sylow subgroups, contradiction.  
 So we have 10 3-Sylow subgroups, which shows there are at most 15 elements of order 2 or 4.  
 So we have 3 or 5 2-Sylow subgroups. If it is 3, then we can do the same as that for 3-Sylow to construct a 20-order group and reach contradiction.  
 So now it have 5 2-Sylow subgroups, and then we consider the conjugate action on Sylow subgroups, which is transitive, so it has trivial kernel, and  $G \hookrightarrow S_5$ . Now  $G = [G, G] \subset [S_5, S_5] = A_5$ .

□

**Prop. (3.1.11.8).** There is a group that is group that  $a^3 = 1$  for any  $a \in G$ , but is not Abelian. It is the uni-upper-triangular matrices in  $\text{Mat}(3; \mathbb{F}_3)$ . ┘



## 12 Profinite Groups

References are [Neukirch Cohomology of Number Fields], [Serre Galois Cohomology] [Profinite Groups Zaleskii] and [Shatz, Profinite Groups, Arithmetic and Geometry].

**Def. (3.1.12.1)[Profinite Groups].** A **profinite group** is a topological group that is an inverse limit of finite discrete groups. The category of profinite groups is denoted by  $\mathcal{ProfGrp}$ . And the subcategory of profinite Abelian groups is denoted by  $\mathcal{ProfAb}$ .  $\lrcorner$

**Lemma (3.1.12.2).** For a compact totally disconnected group  $G$ , any nbhd  $U$  of  $e$  contains a normal open subgroup.  $\lrcorner$

*Proof:*  $U$  contains a precompact nbhd of  $e$ , then by (4.12.1.24),  $U$  contains an open subgroup  $V$ , so by (4.12.1.6), there is a nbhd  $V'$  of  $e$  that  $xV'x^{-1} \subset V$  for all  $x \in G$ , this says  $\cap x^{-1}Vx$  is open, so it is an open normal subgroup.  $\square$

**Prop. (3.1.12.3)[Profinite Compact and Totally Disconnected].** A profinite group  $G$  is the same thing as a totally disconnected, compact Hausdorff topological group. In particular,  $G \cong \varprojlim G/U$  where  $U$  runs over all open normal subgroups of  $G$ .  $\lrcorner$

*Proof:* One way is because  $\lim G_i$  is a closed subgroup of  $\prod G_i$  which by Tychonoff's theorem is compact.

Conversely, by (3.1.12.2),  $G$  has a basis of  $e$  consisting of normal open subgroups, and by (4.12.1.23), the intersection of open normal subgroups is  $\{e\}$ . For any open normal subgroup  $N$  of  $G$ ,  $G/N$  is compact discrete hence finite, the map  $G \rightarrow \varprojlim G/N$  is continuous and has dense image, but  $G$  is compact and the right is Hausdorff, so the image is closed, hence it is surjective. It is injective because  $\cap N = \{e\}$ . Hence  $G \cong \varprojlim G/N$ .  $\square$

**Cor. (3.1.12.4).** A closed subgroup of a profinite group is profinite, and a quotient group is profinite.

A direct product of profinite groups are profinite, and so the inverse limit profinite groups are profinite, as it is a closed subgroup of a direct product.  $\lrcorner$

*Proof:* The closed subgroup is totally disconnected by (4.4.1.27).

To show the quotient group is totally disconnected, by (4.12.1.23), it suffice to prove  $H$  is intersection of compact open nbhds in  $G/H$ . If  $x \notin H$ , then there is an open subgroup  $U$  disjoint from  $xH$  by (4.12.1.7), so it is closed hence compact. So  $UH$  is a compact nbhd of  $H$  in  $G/H$  that doesn't contains  $xH$ , hence the result.  $\square$

**Cor. (3.1.12.5).** A closed subgroup of a profinite group is a intersection of open normal subgroups of  $G$  containing it, as  $G/H$  is profinite and as in the proof of (3.1.12.3),  $H$  is the intersection of open normal subgroups of  $G/H$ .  $\lrcorner$

**Prop. (3.1.12.6).** For  $G \in \mathcal{ProfGrp}$ , if  $U_\alpha$  is the system of open normal subgroups of  $G$ , and  $H$  is a closed subgroup of  $G$ , then there is a natural isomorphism of topological groups

$$H \cong \varprojlim H/H \cap U_\alpha,$$

and if  $H$  is normal, then there is a natural isomorphism of topological groups

$$G/H \cong \varprojlim G/HU_\alpha.$$

$\lrcorner$

*Proof:* Cf.[Central Simple Algebras]P113.?

□

**Prop. (3.1.12.7).** Any infinite profinite groups is uncountable.

┘

*Proof:* [Profinite Groups Zalesskii]Prop2.3.1.

□

**Prop. (3.1.12.8).**  $\text{ProfAb}$  is Pontryagin dual to the category of torsion abelian group. (not that hard to verify).

┘

*Proof:*

□

### Pro- $p$ -Groups

**Def. (3.1.12.9) [Surnatural Number].** To consider indexes of closed subgroups of a profinite group, the notion of surnatural numbers are needed. A **surnatural number** is a formal product  $\prod_p p^{n_p}$ ,  $n_p \in \mathbb{N} \cup \{0, \infty\}$ .

For a closed subgroup  $H$  of a profinite group  $G$ ,  $[G : H]$  is defined to be the least common multiple of  $[G/U : H/H \cap U]$  where  $U$  goes over all open normal subgroups of  $G$ . This also equals the least common multiple of  $[G : V]$  for  $V$  open containing  $H$  (because for any such  $V$ , there is an open normal subgroup  $U$  that  $HU \subset V$  (4.12.1.6)).

┘

**Prop. (3.1.12.10).** The index is compatible with composition and quotient:  $[G : K] = [G : H][H : K]$  and  $[G : H] = [G/K : H/K]$  for  $K$  closed normal in  $G$ .

$[G : H]$  is finite iff  $H$  is open. For a decreasing family of closed subgroups  $H_i$  of  $G$ ,  $[G : \cap H_i]$  equals the least common multiple of  $[G : H_i]$ .

┘

*Proof:*  $[G/U : K/K \cap U] = [G/U : H/H \cap U][H/H \cap U : K/K \cap U][G : H][H : K]$ , giving one way of inequality. For the converse, Cf.[Etale Cohomology Fulei P150]. The quotient case is trivial.

If  $[G : H]$  is finite, then For the final assertion, notice for a open subgroup  $V$ ,  $G - V$  is compact, so  $\cap H_i \subset V$  iff  $\cap H_i \subset V$  for some  $i$ .

□

**Def. (3.1.12.11) [Pro- $p$ -Group].** A profinite group  $G$  is called a **pro- $p$ -group** iff  $[G : 1]$  is a power of  $p$ .

Given a profinite group, a closed subgroup  $H$  is called **Sylow pro- $p$ -subgroup** of  $G$  if  $H$  is pro- $p$  and  $[G : H]$  is prime to  $p$ .

┘

*Proof:*

□

**Prop. (3.1.12.12).** A profinite group  $G$  is a pro- $p$ -group iff it is an inverse limit of finite  $p$ -groups. In particular, any finite quotient of a pro- $p$ -group is a  $p$ -group.

┘

*Proof:*

□

**Prop. (3.1.12.13) [Sylow Pro- $p$ -Group Exists].**

┘

*Proof:*

□

**Prop. (3.1.12.14).** Any pro- $p$  subgroup  $H$  of  $G$  is contained in a Sylow  $p$ -subgroup of  $G$ , and any two Sylow  $p$ -subgroups are conjugate. And a surjective morphism of profinite groups maps a pro- $p$  group to a pro- $p$  group.

┘

*Proof:* For any open normal subgroup  $U$  of  $G$ , let  $I_U$  be the sets of all Sylow groups of  $G/U$  containing  $H/H \cap U$ , then the map  $G/V \rightarrow G/U$  maps  $I_V$  to  $I_U$ , and  $I_U$  is finite nonempty by Sylow theory. So the inverse limit of  $I_U$  is nonempty, and let  $(P_U)$  be such an element, and  $P = \varprojlim_U P_U$ , then  $P$  is a pro- $p$  subgroup of  $G$ , and  $[G : P]$  equals the least common multiple of  $[G/U : P_U]$ , which is prime to  $p$ , so it is a Sylow  $p$ -group. Similarly, for two Sylow- $p$  subgroup, we consider  $A_U$  the set of all  $x \in G/U$  that  $x^{-1}(PU/U)x = P'U/U$ , then there is a inverse element  $x$ , and  $x^{-1}Px = P'$ .

If  $G' = G/N$ , then  $[G/N : PN/N] = [G : PN][G : P]$  is prime to  $p$ , and  $[PN/N : 1] = [P : P \cap N][P : 1]$  is a power of  $p$ , so  $PN/N$  is Sylow- $p$  in  $G'$ .  $\square$

**Prop. (3.1.12.15).** For a pro- $p$  group  $G$ , any nonzero simple  $p$ -torsion  $G$ -module is isomorphic to  $\mathbb{Z}/p\mathbb{Z}$  with trivial  $G$ -action.  $\lrcorner$

*Proof:* The action of  $G$  on  $A$  factors through a finite quotient group which is a  $p$ -group, by (24.1.3.7),  $A^G \neq 0$ , so  $A = A^G$ , then  $A$  must be  $\mathbb{Z}/(p)$ .  $\square$

## 3.2 Algebras

*“En un mot les calculs sont impraticables.”*

–E. Galois

References are [Lan02, Rom07] and [Finite Groups, Issac].

This section differs from sections on Commutative Algebras because it contains more basic, but maybe non-commutative properties. This section differs from [Group Theory](#) because it concerns objects with more structure than a group.

**Notation(3.2.0.1).**

- Use notations on groups defined in [Group Theory](#).

┘

### 1 Rings

#### Basics

**Def.(3.2.1.1)[Rings].** A **ring** is an Abelian group  $(R, +) \in \mathcal{Ab}$  together with a multiplication map

$$\times : R \times R \rightarrow R$$

s.t.

- $(R, \times)$  is a monoid.
- For any  $x, y, z \in R$ ,

$$x(y + z) = xy + xz, \quad z(x + y) = zx + zy$$

A **commutative ring** is a ring that the multiplication is commutative.

- The category of non-associative rings is denoted by  $\mathbf{naAlg}$ .
- The category of associative rings is denoted by  $\mathbf{Alg}$ .
- The category of unital associative rings is denoted by  $\mathbf{Ring}$ .
- The category of commutative rings is denoted by  $\mathbf{CAlg}$ .
- The category of commutative unital rings is denoted by  $\mathbf{CRing}$ .

┘

**Def.(3.2.1.2).**  $R \in \mathbf{Ring}$  is called

- a **simple ring** iff it has no non-trivial 2-sided ideals.
- a **prime ring** iff for any two ideals  $A, B \in \mathbf{Ring}$ ,  $AB = 0$  implies  $A = 0$  or  $B = 0$ .
- a **domain** or **integral ring** if for any  $a, b \in R$ ,  $ab = 0$  implies  $a = 0$  or  $b = 0$ .
- a **reduced ring** iff it has no non-zero nilpotent element.
- a **Dedekind-finite ring** iff for any  $a, b \in R$ ,  $ab = 1 \Rightarrow ba = 1$ .

┘

**Def.(3.2.1.3)[Fields].** A **field** is a commutative ring that every non-zero element has an inverse. The category of fields is denoted by  $\mathbf{Field}$ .

┘

**Prop. (3.2.1.4) [Characteristic].** For  $R \neq 0 \in \text{Ring}$ , there exists at most one  $p \in \mathbf{P}$  s.t.  $p \cdot 1 = 0 \in k$ .  
If such a  $p$  exists, denote  $\text{char } k = p$ , otherwise denote  $\text{char } k = 0$ .  $\lrcorner$

**Prop. (3.2.1.5).** If  $1 - ab$  is left(resp. right) invertible in a unital ring  $R$ , then so is  $1 - ba$ , and

$$(1 - ba)^{-1} = 1 + b(1 - ab)^{-1}a.$$

 $\lrcorner$ 

*Proof:* Direct Calculation.  $\square$

**Prop. (3.2.1.6) [Kaplansky].** If  $a$  has a right inverse by no left inverse in a ring, then  $a$  has infinitely many right inverses.  $\lrcorner$

*Proof:* If  $ab = 1$ , then in fact  $b + (1 - ba)a^n$  are right inverses for  $a$  for any  $n \geq 0$ , and they are distinct, because if  $b + (1 - ba)a^n = b + (1 - ba)a^m$ , then  $(1 - ba)a^n = (1 - ba)a^m$ . And by multiplying  $b$  on the right, we get  $(1 - ba)a^{n-m} = 1 - ba$ , so  $((1 - ba)a^{n-m-1} + b)a = 1$ .  $\square$

### Others

**Prop. (3.2.1.7).** If  $R \in \text{Ring}$ ,  $\#R = p^2$ , then  $R$  is commutative.  $\lrcorner$

*Proof:* Consider the center of  $R$ , it is non-trivial because  $\textcolor{red}{?}$ .  $\square$

**Prop. (3.2.1.8).** Let  $A, A'$  be  $k$ -algebras and  $B, B'$  subalgebras of  $A, A'$  with centralizers  $C, C'$ , then the centralizer of  $B \otimes_k B' \subset A \otimes_k A'$  is  $C \otimes_k C'$ .  $\lrcorner$

*Proof:* It suffices to show that  $C \otimes_k A' \cap A \otimes_k C' = C \otimes_k C'$ , which is clear because they are flat over  $k$ .  $\square$

**Def. (3.2.1.9) [Division Algebras].** A **skew field**(or **division algebra**) is an algebra that every non-zero element is invertible (but may not be unital or associative). A unital division algebra is called a **division ring**.  $\lrcorner$

**Def. (3.2.1.10) [Involutions].** For  $R \in \text{Ring}$ , an **involution** on  $R$  is a bijection  $*$  :  $R \rightarrow R$  s.t.  $(ab)^* = b^*a^*$  and  $*^{\circ 2} = \text{id}$ . It is called an **involution of first kind** if it fixes elements in the center of  $B$ . It is called an **involution of second kind** otherwise.  $\lrcorner$

## 2 Gröbner Basis

Cf.[Algebraic Aspects of Cryptography]P70.

## 3 Commutative Rings

### Bézout Domain

**Def. (3.2.3.1) [Bézout Domains].** A **Bézout domain** is an integral domain that any sum of two principal domains is also principal.  $\lrcorner$

**Prop. (3.2.3.2).** The localization of a Bézout domain is Bézout. A local ring is Bézout iff it is a valuation ring by (12.2.2.8).  $\lrcorner$

### UFDs

**Def. (3.2.3.3) [Unique Factorization Domains].** A non-zero element  $x$  in a domain  $R$  is called **irreducible element** iff for any  $y, z \in R$  that  $x = yz$ , either  $y$  or  $z$  is a unit.

A non-zero element  $x$  in a domain  $R$  is called a **prime element** iff  $(x)$  is a prime ideal. Every prime is irreducible.

A domain  $R$  is called a **UFD** or **unique factorization domain** iff every non-zero element  $x \in R$  has a factorization into irreducibles, unique up to units.  $\lrcorner$

**Def. (3.2.3.4) [Relatively Prime Elements].** A set of elements  $(x_1, \dots, x_n)$  in a domain  $R$  is called **relatively prime** if there are no irreducible elements that divides each one of them.  $\lrcorner$

**Prop. (3.2.3.5).** if  $R$  is Noetherian domain, then each element has a decomposition into irreducible.  $\lrcorner$

*Proof:* Trace the decomposition inductively, if it doesn't stop, then it contradicts with Noetherian hypothesis.  $\square$

**Prop. (3.2.3.6).** An integral domain  $R$  is a UFD(3.2.3.3) iff each element  $x$  factors into irreducible elements(3.2.3.3), and every irreducible element is a prime element(3.2.3.3). Also this is equivalent to every non-zero element factors into product of prime elements.  $\lrcorner$

*Proof:* If  $R$  is a UFD, then if  $x$  is irreducible, if  $ab \in (x)$ ,  $ab = xc$ , then  $x$  is one irreducible in the decomposition of  $a$  and  $b$ , by UFD, so  $a \in (x)$  or  $b \in (x)$ , and  $(x)$  is a prime ideal.

Conversely, if there are two decompositions  $\prod a_i = \prod b_j$ , then some  $b_j \in (a_i)$  by primeness of  $(a_i)$ , so  $b_j = a_i u$ , so  $u$  must be units, so by induction, this two decompositions are the same.

If every element factors into prime elements, then an irreducible element is a prime because it factors as a product of primes, and notice every prime is irreducible(3.2.3.3).  $\square$

**Prop. (3.2.3.7) [Gauss Lemma].** If  $R \in \mathcal{CRing}$  is a UFD, then

- The product of two primitive elements in  $R[X]$  is primitive.
- $p(X) \in R[X]$  is irreducible iff it is primitive and irreducible in  $\text{Frac}(R)[X]$ .

$\lrcorner$

*Proof:* ?  $\square$

**Cor. (3.2.3.8).** If  $R \in \mathcal{CRing}$  is a UFD, then  $R[X]$  is a UFD.  $\lrcorner$

*Proof:*  $\text{Frac}(R)[X]$  is a Euclidean domain thus a PID, so any  $p(X) \in R[X]$ ,  $p(X)$  factors into a product of irreducible polynomials in  $\text{Frac}(R)[X]$  up to units:  $p(X) = \prod_i p_i(X)$ . By multiplying elements in  $R^\times$ , we may assume that  $c_i p_i(X)$  are primitive elements contained in  $R[X]$ . Thus  $\prod_i (c_i p_i(X))$  is also a primitive element in  $R[X]$ , thus it must equal  $p(X)$  up to a unit in  $R$ . So we proved that  $p(X)$  factors into a product of irreducible elements in  $R[X]$ . And clearly the factorization is unique up to units.  $\square$

**Prop. (3.2.3.9) [Kaplansky].** An integral domain  $R$  is UFD iff every non-zero prime ideal contains a non-zero principle prime ideal. In particular, any prime of height 1 is principal.  $\lrcorner$

*Proof:* If  $R$  is a UFD, let  $\mathfrak{p}$  be a non-zero prime ideal, choose  $a \neq 0 \in \mathfrak{p}$ , and write  $a = \pi_1 \dots \pi_n$  as a product of irreducibles. Then  $\pi_i \in \mathfrak{p}$  for some  $i$ , so  $\mathfrak{p}$  contains the prime ideal  $(\pi_i)$ (3.2.3.6).

Conversely, let  $S$  be the set of all finite products of prime ideals of  $R$ , then  $S$  is a multiplicative set of  $R$ . For any non-zero element  $a \in R$ , if  $(a) \cap S = \emptyset$ , then there is a prime ideal  $P$  containing  $(a)$  and

is maximal among those avoiding elements of  $S$ . Then  $P$  contains a non-zero prime  $\pi$ , contradiction. So  $(a) \cap S \neq \emptyset$ . Let  $b \in R$  that  $ab = \pi_1 \dots \pi_n$ , then we show  $a \in S$  by induction on  $n$ .

If  $n = 1$ , then  $a$  is a unit or a prime, so we are done. For general  $n$ , if  $\pi_k | b$  for some  $k$ , then  $b = \pi_k c$ , and  $ac = \pi_1 \dots \pi_{k-1} \pi_{k+1} \dots \pi_n$ , then  $a \in S$  by induction hypothesis. Otherwise  $\pi_k | a$  for all  $k$ , thus  $a = \pi_1 \dots \pi_n c$  and  $1 = bc$ , thus  $c$  is a unit and  $a \in S$ .

Thus we proved that every non-zero element is a product of prime elements, so  $R$  is a UFD by (3.2.3.6).  $\square$

**Prop. (3.2.3.10).** If  $k \in \mathbf{Field}$ , then the ring of power series  $k[[X]]$  is a UFD.  $\lrcorner$

*Proof:* Cf.[Algebra, Lang P209].  $\square$

**Remark (3.2.3.11).** WARNING: if  $A$  is a UFD,  $A[[X]]$  may not be a UFD. Cf.[Matsumura, Commutative Ring Theory, P165].  $\lrcorner$

**Prop. (3.2.3.12) [Nagata].** If  $A$  is a Noetherian domain and  $x \in A$  is a prime element s.t.  $A[\frac{1}{x}]$  is a UFD, then  $A$  is a UFD.  $\lrcorner$

*Proof:*  $A[\frac{1}{x}]$  is normal by (5.3.5.2), and  $A$  is also normal: If  $a \in \text{Frac}(A)$  is integral over  $A$ , then  $a = r/x^m$  for some  $m \in \mathbb{Z}, r \in A \setminus (x)$ . If  $a \notin A$ , then  $r \in x$ , contradiction.

Then we can use (9.1.1.25) to show that  $\text{Cl}(\text{Spec } A) = 0$ , and then  $A$  is a UFD by (9.1.4.6).  $\square$

**Prop. (3.2.3.13).** For a field  $k$  of characteristic  $\neq 2$  and  $n \geq m \geq 5$ ,  $A = k[x_1, \dots, x_n]/(-x_1^2 + x_2^2 + \dots + x_m^2)$  is a UFD.  $\lrcorner$

*Proof:*  $A \cong k[x_1, \dots, x_n]/(x_1 x_2 + x_3^2 + \dots + x_m^2)$ , and  $(x_2)$  is a prime ideal of  $A$  s.t.  $A[\frac{1}{x}]$  is a UFD by (3.2.3.8), so  $A$  is a UFD by Nagata's lemma (3.2.3.12).  $\square$

**Prop. (3.2.3.14) [Quadratic Extension of UFDs].** If  $A \in \mathcal{CA}lg$  is an integral domain and  $Z^2 - gZ + f$  doesn't have a root in  $\text{Frac}(A)$ , then  $A[Z]/(Z^2 - gZ - f)$  is integral and normal.  $\lrcorner$

*Proof:* If  $(a + bZ)(c + dZ) = 0$ , then by hypothesis  $ad + bc + bdg = 0, ac + fbd = 0$ . Then  $b(fd^2 - c^2 - cdg) = 0$ , so  $b = 0$  or  $c = d$  by hypothesis. If  $(c, d) \neq (0, 0)$ , then  $ac = ad = 0$ , so  $a = b = 0$ .  $\square$

### PIDs

**Def. (3.2.3.15) [Euclidean Domain].**  $\lrcorner$

**Prop. (3.2.3.16) [Euclidean Domains are PID].** Euclidean domains (3.2.3.15) are PIDs.  $\lrcorner$

*Proof:*  $\square$

**Example (3.2.3.17).**

- $\mathbb{Z}$  is a PID.
- For  $k \in \mathbf{Field}$ ,  $k[X]$  is a PID.

$\lrcorner$

**Prop. (3.2.3.18) [PID Structures].** In a PID,

- An element  $t$  is irreducible iff  $(t)$  is maximal.
- A PID is UFD hence Noetherian.

- An element  $t$  is irreducible iff it is a prime.
- 

┘

*Proof:* 1:  
 2: By (3.2.3.9).  
 3: By item2 and (3.2.3.5).

□

## 4 Modules

**Def. (3.2.4.1) [Modules].** For  $R \in \mathcal{M}\text{on}$  (3.1.1.6), a (left)  $R$ -modules is defined to be ?. The category of  $R$ -modules is denoted by  $\text{Mod}_R$ .

Left modules are preferred. There are several proposition that is written in favor of right modules, they should be rectified. ┘

**Def. (3.2.4.2) [Finite Modules].** For  $R \in \mathcal{M}\text{on}$ , a **finite  $R$ -module** is an  $R$ -module  $M$  that there is a surjection  $R^n \rightarrow M$  for some  $n \geq 0$ . ┘

**Prop. (3.2.4.3).** Let  $R \in \mathcal{C}\text{Ring}$  and  $M$  be an  $R$ -module generated by  $n-1$  elements, then any  $R$ -module map  $R^n \rightarrow M$  has a nonzero kernel. ┘

*Proof:* Choose a surjection  $R^{n-1} \rightarrow M$ , then the map  $R^n \rightarrow M$  can be extended to a map  $R^n \rightarrow R^{n-1}$ . It suffices to assume  $M = R^{n-1}$ . This map is represented by a matrix. If some entry  $a_{ij} = a$  is not nilpotent, then we can localize  $R$  to  $R_a$  that  $a$  is a unit. We can assume  $a_{11} = a$ , and apply elementary row and column transformation to make  $A = \text{diag}(1, B)$ , then we finish by induction. Now if all  $a_{ij}$  are nilpotent, then  $I = (a_{ij})$  is nilpotent, and if  $m$  is the maximal integer that  $I^m \neq 0$ , then  $(I^m)^{\oplus n}$  is contained in the kernel of this morphism. □

**Cor. (3.2.4.4) [Rank of Free Modules].** If  $M$  is a free module over a nonzero commutative ring  $R$ , then any basis of  $M$  is of the same cardinality, called the **rank** of  $M$ , and any spanning subset of  $M$  has greater cardinalities. In particular,  $R^m \neq R^n$  as  $R$ -modules. ┘

**Prop. (3.2.4.5) [Fitting Lemma].** For  $R \in \mathcal{C}\text{Ring}$ ,  $M \in \text{Mod}_R$  and  $T \in \text{End}_R(M)$ , if we denote  $p$  the minimal integer that  $R(T^p) = R(T^{p+1})$  and  $q$  the minimal integer that  $N(T^q) = N(T^{q+1})$ . Then

- The morphisms are stable afterward.
- If there is a  $m, n$  that  $R(T^m) \oplus N(T^n) = X$  for a  $R$ -module endomorphism  $T \in \text{End}(M)$ , then  $p, q < \infty$  and they are equal. Moreover, if we know  $p, q < \infty$ , then we have  $R(T^p) \oplus N(T^q) = M$ . ┘

*Proof:* We notice that

$$T^i : N(T^{i+j})/N(T^i) \rightarrow R(T^i) \cap N(T^j), \quad T^i : M/(R(T^j) + N(T^i)) \rightarrow R(T^i)/R(T^{i+j})$$

are isomorphisms. Thus  $R(T^m) \oplus N(T^n) = X$  shows  $q \leq m$  and  $p \leq n$ , thus we have  $R(T^p) \oplus N(T^q) = M$ , which implies  $p \geq q$  and  $q \geq p$ . thus the result. The rest also follows easily from these isomorphisms. □

**Prop. (3.2.4.6) [Nakayama].** If  $M$  is a finite  $A$ -module, and  $I \subset A$  is an ideal that  $IM = M$ , then there is a  $a \in 1 + I$  that  $aM = 0$ .

In particular, if  $I \subset \text{rad}(A)$ , then  $a$  is a unit (5.2.6.2), so  $M = 0$ . ┘



*Proof:* Because  $M = IM$ , choose a set of generators  $\{x_i\}$  of  $M$ , then  $x_i = \sum a_{ij}x_j$ , where  $a_{ij} \in I$ . Then if the matrix  $M = (\delta_{ij} - a_{ij})$ , then  $Mx_i = 0$ . So taking the adjoint matrix, then  $\det(M)x_i = 0$ . Notice  $\det(M)$  is a morphism. But the determinant must be element like  $1 + k, k \in I$ , so we are done.  $\square$

**Cor. (3.2.4.7).** If  $M$  is a finite  $A$ -module,  $N$  a submodule and  $M = \text{rad}(A)M + N$ , then  $M = N$ .  $\lrcorner$

**Cor. (3.2.4.8).** If a finite  $R$ -module  $M$  satisfies  $M \otimes_R k(p) = 0$ , then there is a  $f \notin p$  that  $M_f = 0$ .  $\lrcorner$

*Proof:* Because  $M_p = 0$ , and the support of  $M$  is closed(finiteness used).  $\square$

**Cor. (3.2.4.9).** If an endomorphism  $\varphi$  of a finite module  $M$  over  $R$  is surjective, then it is injective.  $\lrcorner$

*Proof:* This endomorphism makes  $M$  a finite module over  $R[X]$  by letting  $X$  acts by  $\varphi$ . So the hypothesis shows  $IM = M$  where  $I = (x) \subset R[x]$ . Then Nakayama(3.2.4.6) shows there are some  $(1 + f(X)X)M = 0$ . Thus for any  $m \in M$  that  $X(m) = 0$ ,  $m = 0$ , so  $\varphi$  is injective.  $\square$

**Prop. (3.2.4.10).** If  $A \in \text{Ring}/\mathbb{C}$  and  $\alpha \in A$  is not nilpotent, then there exists a simple  $A$ -module  $M$  that  $\alpha|_M \neq 0$ .  $\lrcorner$

*Proof:* First we claim that there is some  $\lambda \neq 0 \in \mathbb{C}$  that  $a - \lambda$  is not invertible in  $A$ , this is nearly the same as the proof of(3.6.1.2), noticing that  $a$  is not nilpotent. Now we can take  $M = A/(a - \lambda)A$ , then  $a1 = \lambda \neq 0$ .  $\square$

**Prop. (3.2.4.11)[Jordan-Horder].** Cf.(3.1.8.4).  $\lrcorner$

### Tensor Module and Hom Module

**Def. (3.2.4.12)[Hom Module].** If  $B$  is an  $R$ - $S$ -bimodule and  $C$  is a  $T$ - $S$ -bimodule, then  $\text{Hom}_S(B, C)$  is naturally a  $T - R$ -module given by the action  $(tfr)(b) = tf(rb)$ .

Dually, if  $B$  is an  $S - R$ -bimodule and  $C$  is a  $S - T$ -bimodule, then  $\text{Hom}_S(B, C)$  is naturally an  $R - T$ -bimodule given by the action  $(rft)(b) = f(br)t$ .  $\lrcorner$

**Def. (3.2.4.13)[Tensor Product].** Given a ring  $R$ , a  $T$ - $R$ -bimodule  $A$  and an  $R$ - $S$ -bimodule  $B$ , their **tensor product** is a  $T$ - $S$ -bimodule defined by universal properties:  $A \times B \rightarrow A \otimes_R B$  is a  $T$ - $S$ -bimodule map, and any  $T$ - $S$ -bimodule map  $A \times B \rightarrow C$  to a  $T$ - $S$ -bimodule  $C$  factors uniquely through  $A \otimes_R B \rightarrow C$ .

The tensor product can be constructed as:

There is an adjoint:

$$- \otimes_R B : {}_T \text{Mod}_R \xleftrightarrow{\quad} {}_T \text{Mod}_S : \text{Hom}_S(B, -) .$$

In particular, tensoring commutes with colimits.

Similarly, there is an adjoint:

$$A \otimes_R - : {}_R \text{Mod}_S \xleftrightarrow{\quad} {}_T \text{Mod}_R : \text{Hom}_T(A, -) .$$

$\lrcorner$

*Proof:* We need to give an isomorphism

$$\tau : \text{Hom}_S(A \otimes_R B, C) \cong \text{Hom}_R(A, \text{Hom}_S(B, C)).$$

Given  $f \in \text{Hom}_S(A \otimes_R B, C)$ , we define

$$\tau(f) : A \rightarrow \text{Hom}_S(B, C) : (\tau(f)a)(b) = f(a \otimes b).$$

and conversely, for  $g \in \text{Hom}_R(A, \text{Hom}_S(B, C))$ ,

$$\tau^{-1}(g)(a \otimes b) = (g(a))(b).$$

The verifications is routine and the isomorphism for left modules is dual.  $\square$

**Prop. (3.2.4.14) [(Co)Induced Modules].** Given a ring homomorphism  $S \rightarrow R$ , then  $R$  is a  $S - R$ -bimodule as well as a  $R - S$ -bimodule, then we define:

- $f^* : {}_R\text{Mod} \rightarrow {}_S\text{Mod} : f^*M = {}_S M = \text{Hom}_R(R, M) = R \otimes_R M$  (3.2.4.12)(3.2.4.13), the **restriction**.
- $f_! : {}_S\text{Mod} \rightarrow {}_R\text{Mod} : f_!M = R \otimes_S M$  is the **induced module**, it is left adjoint to restriction, by (3.2.4.13).
- $f_* : {}_S\text{Mod} \rightarrow {}_R\text{Mod} : f_*M = \text{Hom}_S(R, M)$  is the **coinduced module**, it is right adjoint to restriction, by (3.2.4.13).

Dually for left modules, we define:

- $f^* : \text{Mod}_R \rightarrow \text{Mod}_S : f^*M = M_S = \text{Hom}_R(R, M) = M \otimes_R R$  (3.2.4.12)(3.2.4.13), the **restriction**.
- $f_! : \text{Mod}_S \rightarrow \text{Mod}_R : f_!M = M \otimes_S R$  is the **induced module**, it is left adjoint to restriction, by (3.2.4.13).
- $f_* : \text{Mod}_S \rightarrow \text{Mod}_R : f_*M = \text{Hom}_S(R, M)$  is the **coinduced module**, it is right adjoint to restriction, by (3.2.4.13).  $\lrcorner$

**Def. (3.2.4.15) [Algebras].** For  $R \in \mathcal{C}\text{Ring}$ , a(n) (commutative/unital/associative)**algebra** over  $R$  is a (commutative/unital/associative)magma object in the monoidal category  $(\text{Mod}_R, \otimes)$ .

- The category of  $R$ -algebras is denoted by  $\mathcal{A}lg_R$ .
- The category of commutative  $R$ -algebras is denoted by  $\mathcal{C}Alg_R$ .
- The category of commutative unital associative algebra over  $R$  is denoted by  $\mathcal{C}\text{Ring}_R$ .

Notice  $\mathcal{C}\text{Ring}_R \cong \mathcal{C}\text{Ring}/R$ .  $\lrcorner$

### Torsion-Free Modules

**Def. (3.2.4.16) [Torsion-Free Modules].** Let  $R$  be a ring, an  $R$ -module  $M$  is called **torsion-free** iff there are no non-zero divisor  $x \in R, 0 \neq f \in M$  that  $xf = 0$ .  $\lrcorner$

**Prop. (3.2.4.17).** If  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3$  and  $M_1, M_3$  are torsion-free, then  $M_2$  is torsion-free. Torsion-free is a stalk-wise property (5.1.4.2).  $\lrcorner$

*Proof:* Trivial.  $\square$

**Prop. (3.2.4.18).** Let  $M$  be a finite  $R$ -module, then  $M$  is torsion-free if it is a submodule of a finite free module.  $\lrcorner$

*Proof:* One direction is trivial, for the other, if  $M$  is torsion-free, then  $M \subset M \otimes_R K$ , and  $M \otimes_R K$  is a finite  $K$ -vector space, with basis  $e_i$ . Now let  $x_i$  be a basis of  $M$ , let  $x_i = \sum a_{ij}/b_{ij}e_j$ , then let  $e = \prod_{ij} b_{ij}$ , then  $M \subset Re_1/b \oplus \dots \oplus Re_n/b$ .  $\square$

**Prop. (3.2.4.19).** If  $M, N$  are  $R$ -modules that  $M$  is torsion-free, then  $\text{Hom}(N, M)$  is torsion free.  $\lrcorner$

*Proof:* Choose a surjection  $\bigoplus_I R \rightarrow N \rightarrow 0$ , then  $\text{Hom}(N, M) \hookrightarrow \prod_I M$  is torsion free.  $\square$

### Modules over PIDs or Bézout Domains

**Prop. (3.2.4.20)[Modules over a Bézout domain].** Let  $R$  be a Bézout domain, then

- any finite submodule of a free module over  $R$  is finite free.
- any f.p.  $R$ -module  $M$  is a summand of a free  $R$ -module and  $M_{\text{tor}}$ , where  $M_{\text{tor}} = \bigoplus R/(f_i)$  where  $f_i \in R^*$ .

$\lrcorner$

*Proof:* Cf. [Sta]0ASU. ?  $\square$

**Prop. (3.2.4.21)[Classification of Modules over PID].** Let  $R \in \mathcal{CRing}$  be a PID, then:

1. Every submodule of a free module over  $R$  is free of smaller rank. In particular, a projective module over  $R$  is free.
2. Every finite torsion-free module over  $R$  is free.
3. Every finite module over  $R$  has a primary decomposition  $M = \bigoplus_i R/(q_i)$ , where  $(q_i)$  is primary ideals.

So projective  $\iff$  free  $\iff$  torsion-free (when f.g.).  $\lrcorner$

*Proof:* 1: Choose a well ordering on the basis of  $F$ , let  $F_i$  is the submodule generated by  $e_j, j \leq i$ . Then  $\text{pr}_i(P \cap F_i) \subset R$  is an ideal of the form  $(a_i)$ , thus choose  $u_i \in P$  that  $\text{pr}_i(u_i) = a_i$ . Then  $u_i$  is a basis for  $P$ : they are linearly independent, because for any finite linear combination that are 0, the maximal coordinate are 0. It also spans  $P$ , because we can choose an element in  $P - \{u_i\}$  whose maximal nonzero coordinate  $\alpha$  is minimal among them, by well-orderedness. But we can subtract a multiple of  $u_\alpha$ , thus producing a smaller element, contradiction.

2: If it is finite torsion-free, then it is a submodule of a finite free module (3.2.4.18), so it is free by item 1.

3: Follows immediately from (5.2.5.34) and (3.2.3.18).  $\square$

**Prop. (3.2.4.22)[Primary Cyclic Decomposition].** There is a primary cyclic decomposition theorem for a torsion module  $M$  over a PID  $R$ . Thus the multisets of elementary divisors of  $M$  is a complete set of invariants for  $M$ .  $\lrcorner$

*Proof:* Cf. [Advanced Linear Algebra P153].  $\square$

**Cor. (3.2.4.23)[Invariant Factor Decomposition].** By reordering the cyclic decomposition, we can get the **invariant factor decomposition** of  $M$ , there are scalars  $d_m | d_{m-1} | \dots | d_1$  that are called the **invariant factors** of  $M$ .  $\lrcorner$

*Proof:* Cf. [Advanced Linear Algebra P157].  $\square$

**Cor. (3.2.4.24) [Elementary Factor Theorem].** Let  $F$  be a free module over a PID  $R$ , and let  $M$  be a f.g. submodule  $\neq 0$ , then there exists a basis  $\mathcal{B}$  of  $F$ , elements  $e_1, \dots, e_m$  in this basis, and non-zero elements  $a_1, \dots, a_m \in R$  that

- the elements  $a_1 e_1, \dots, a_m e_m$  forms a basis of  $M$ .
- $a_i | a_{i+1}$ .

And these  $a_i$  are uniquely determined up to units.  $\lrcorner$

### Transfinite Direct Sum Dévissage of Modules

**Def. (3.2.4.25) [Direct Sum Dévissage of Modules].** Let  $M$  be a module over a ring  $R$ , then a **direct sum dévissage** is a family of submodules  $M_\alpha$  indexed by an ordinal  $S$  such that

- $M_0 = 0$ .
- if  $\alpha + 1 \in S$ , then  $M_\alpha$  is a direct sum of  $M_{\alpha+1}$ .
- if  $\alpha$  is a limit ordinal, then  $M_\alpha = \bigcup_{\beta < \alpha} M_\beta$ .
- $\bigcup_{\alpha \in S} M_\alpha = M$ .

If moreover, for any  $\alpha \in S, \alpha + 1 \in S$ ,  $M_{\alpha+1}/M_\alpha$  is countably generated, then  $M_\alpha$  is called a **Kaplansky dévissage** of  $M$ .  $\lrcorner$

**Prop. (3.2.4.26).** Let  $M_\alpha$  be a direct sum dévissage of  $M$ , then  $M \cong \bigoplus_{\alpha \in S, \alpha+1 \in S} M_{\alpha+1}/M_\alpha$ .  $\lrcorner$

*Proof:* Cf. [Sta]058V.  $\square$

**Cor. (3.2.4.27).**  $M$  is a direct sum of countably generated modules iff  $M$  admits a Kaplansky dévissage.  $\lrcorner$

**Prop. (3.2.4.28).** Suppose  $M$  is a direct sum of countably generated modules and  $P$  is a direct sum of  $M$ , then  $P$  is also a direct sum of countably generated modules.  $\lrcorner$

*Proof:* Cf. [Sta]058X.  $\square$

**Prop. (3.2.4.29) [Direct Summand Criterion of Free Modules].** Let  $M$  be a countably generated  $R$ -module that for any direct summand  $N$  of  $M$  and an element  $x \in N$ ,  $x$  is contained in a free direct summand of  $N$ , then  $M$  is free.  $\lrcorner$

*Proof:* Let  $x_1, x_2, \dots$  be a countable set of generators for  $M$ , then we can use inductions to find free submodules  $F_1, F_2, \dots$  of  $M$  s.t.  $\bigoplus_{i=1}^n F_i$  are direct summands of  $M$  and contains  $x_1, \dots, x_n$  for any  $n$ , thus  $M = \bigoplus F_i$  is free.  $\square$

### Quadratic Forms on Modules

**Def. (3.2.4.30) [Quadratic Forms].** For  $A \in \mathcal{C}\text{Ring}$  and  $M, N \in \text{Mod}_A$ , a function  $Q : M \rightarrow N$  is called a **quadratic form** if

- $Q(am) = a^2 d(m)$  for any  $a \in A$  and  $m \in M$ .
- The function  $B_Q : M \otimes_A M \rightarrow N : (\alpha, \beta) \mapsto Q(\alpha + \beta) - Q(\alpha) - Q(\beta)$  is bilinear.

It is called **positive semi-definite form** if moreover  $N = \mathbb{R}$  as a module over  $A$  and  $Q(m) \geq 0$  for any  $m \in M$ , and it is called a **positive-definite form** if moreover  $Q(m) = 0 \iff m = 0$ .  $\lrcorner$

**Prop. (3.2.4.31) [Cauchy-Schwartz].** Let  $A \in \mathcal{A}b$  and  $d$  a positive semi-definite quadratic form over  $A$ , then

$$|d(\alpha - \beta) - d(\alpha) - d(\beta)| \leq 2\sqrt{d(\alpha)d(\beta)}.$$

┘

*Proof:* Consider the bilinear form  $B_d$ , then

$$0 \leq d(m\alpha - n\beta) = m^2d(\alpha) + n^2d(\beta) - 2mnB_d(\alpha, \beta).$$

This is true for any  $m, n \in \mathbb{Z}$ , thus the discriminant  $B_d(\alpha, \beta)^2 \leq d(\alpha)d(\beta)$ . □

**Prop. (3.2.4.32).** If  $M, N \in \mathcal{A}b$  and  $d$  is a quadratic form on  $M \times N$  s.t.  $d(0 \times N) = d(M \times 0) = 0$ , then  $d$  is bilinear in both  $M$  and  $N$ . ┘

*Proof:* The bilinear form associated to  $d$  satisfies  $B_d((a, 0), (0, b)) = d(a, b)$ , thus it is clear  $d$  is bilinear in  $M$  and  $N$ . □

**Def. (3.2.4.33) [Polarization].** Let  $\Gamma \in \mathcal{A}b$  and  $h : \Gamma \rightarrow \mathbb{R}$  is a function, then the  $r$ -th **polarization function**  $P_r(h) : \Gamma^r \rightarrow \mathbb{R}$  is defined to be

$$P_r(h)(x_1, \dots, x_r) = \frac{1}{r!} \sum_{I \subset \{1, \dots, r\}} (-1)^{r-\#I} \left( \sum_{i \in I} x_i \right).$$

┘

**Lemma (3.2.4.34).**

- Let  $\tau_a(h)(x) = h(a + x) - h(x)$ , then  $\tau_a\tau_b = \tau_{a+b} - \tau_a - \tau_b$ .
- $P_r(h)(x_1, \dots, x_r) = \tau_{x_1}\tau_{x_2} \dots \tau_{x_r}(h)(0) = \tau_{x_2} \dots \tau_{x_r}(h)(x_1)$ .
- If  $A_r \in \text{Sym}^r \Gamma^\vee$ , let  $\Delta A_r : \Gamma : \mathbb{R} : \Delta A_r(x) = \frac{1}{r!} A_r(x, \dots, x)$ , then  $\tau_a \Delta A_r(x) = \sum_{1 \leq s < r} \Delta A_s$ , where  $A_s \in \text{Sym}^s \Gamma^\vee$ , and  $A_{r-1}(x_1, \dots, x_{r-1}) = A_r(x_1, \dots, x_{r-1}, a)$ .
- If  $A_r \in \text{Sym}^r \Gamma^\vee$ ,  $P_r(\Delta A_r)(x_1, \dots, x_n) = A_r(x_1, \dots, x_r)$  and  $P_s(\Delta A_r) = 0$  for  $s > r$ .

┘

**Prop. (3.2.4.35).** There is a natural isomorphism of additive groups

$$\bigoplus_{j=0}^{r-1} \text{Sym}^j \Gamma^\vee \cong \{h : \Gamma \rightarrow \mathbb{R} | P_r(h) = 0\} : (A_0, \dots, A_{r-1}) \mapsto \sum_{i=0}^{r-1} \Delta A_i.$$

┘

*Proof:* This is an injective map by (3.2.4.34), and for any  $h$  that  $P_r(h) = 0$ , By (3.2.4.34) item1, this means  $A_{r-1} = P_{r-1}(h) \in \text{Sym}^{r-1} \Gamma^\vee$ . Let  $h' = h - \Delta A_{r-1}$ , then (3.2.4.34) item4 shows  $P_{r-1}(h') = 0$ , thus we can use induction to show  $h' = \sum_{i=1}^{r-2} \Delta A_i$ , thus  $h = \sum_{i=1}^{r-1} \Delta A_i$ . □

**Prop. (3.2.4.36).** Let  $\Gamma \in \mathcal{A}b$ , for functions  $h_1, h_2 : \Gamma \rightarrow \mathbb{R}$ , let  $h_1 = h_2 + O(1)$  denote the fact that  $|h_1 - h_2|$  is bounded on  $\Gamma$ . Then there is a natural isomorphism of additive groups:

$$\bigoplus_{j=0}^{r-1} \text{Sym}^j \Gamma^\vee \cong \{h : \Gamma \rightarrow \mathbb{R} | P_r(h) = O(1)\} / O(1) : (A_0, \dots, A_{r-1}) \mapsto h(x) = A_0 + A_1(x) + \dots + A_{r-1}(x, \dots, x).$$

┘

*Proof:* The map is injective by (3.2.4.35). To show surjectivity:

By hypothesis and (3.2.4.34), there exists  $C > 0$  s.t.

$$|P_{r-1}h(x_0 + x_1, x_2, \dots, x_{r-1}) - P_{r-1}h(x_0, x_2, \dots, x_{r-1}) - P_{r-1}h(x_1, x_2, \dots, x_{r-1})| \leq C.$$

So

$$|P_{r-1}h(2^N x_1, x_2, \dots, x_{r-1}) - 2P_{r-1}h(2^{N-1} x_1, x_2, \dots, x_{r-1})| \leq C$$

for any  $N$ , and by iterating,

$$|P_{r-1}h(2^N x_1, 2^N x_2, \dots, x_{r-1}) - 2^{r-1} P_{r-1}h(2^{N-1} x_1, 2^{N-1} x_2, \dots, 2^{N-1} x_{r-1})| \leq (2^{r-1} - 1)C.$$

$$\left| \frac{P_{r-1}h(2^N x_1, 2^N x_2, \dots, x_{r-1})}{2^{N(r-1)}} - \frac{P_{r-1}h(2^{N-1} x_1, 2^{N-1} x_2, \dots, 2^{N-1} x_{r-1})}{2^{(N-1)(r-1)}} \right| \leq \frac{2^{r-1} - 1}{2^{N(r-1)}} C.$$

Thus

$$A_{r-1}(x_1, \dots, x_{r-1}) = \lim_{N \rightarrow \infty} \frac{P_{r-1}h(2^N x_1, 2^N x_2, \dots, x_{r-1})}{2^{N(r-1)}}$$

exists, and it is clear that  $A_{r-1} \in \text{Sym}^{r-1} \Gamma^\vee$  and  $A_{r-1}(x_1, \dots, x_{r-1}) - P_{r-1}h(x_1, \dots, x_{r-1}) \leq 2^{r-1}C$ .

Let  $h' = h - \Delta A_{r-1}$ , then  $P_{r-1}h' = P_{r-1}h - A_{r-1} = O(1)$ , thus we can use induction to show  $h' = \sum_{i=1}^{r-2} \Delta A_i + O(1)$ , thus  $h = \sum_{i=1}^{r-1} \Delta A_i + O(1)$ .  $\square$

**Cor. (3.2.4.37).** Let  $\Gamma \in \mathcal{A}b$  and  $h : \Gamma \rightarrow \mathbb{R}$  be a function on an Abelian group  $\Gamma$  that

$$h\left(\sum_{i=1}^3 x_i\right) - \sum_{1 \leq i < j \leq 3} h(x_i + x_j) + \sum_{i=1}^3 h(x_i) = O(1),$$

then there exists a unique symmetric bilinear pairing  $b$  on  $\Gamma$  and  $l$  a homomorphism  $\Gamma \rightarrow \mathbb{R}$ , that

$$h(x) = \frac{1}{2}b(x, x) + l(x) + O(1).$$

┘

**Prop. (3.2.4.38).** Let  $M \in \mathcal{A}b$  and  $b : M \times M \rightarrow \mathbb{R}$  is a bilinear form on  $M$  that  $\text{rad}(B) = 0$ , so  $b$  defines a bilinear form on  $B_{\overline{R}} : M \otimes_{\mathbb{Z}} \mathbb{R} \rightarrow \mathbb{R}$ . Then  $b_{\mathbb{R}}$  is positive-definite iff for any f.g. subgroup  $M'$  of  $M$  and  $C > 0$ ,  $\{x \in M' | b(x, x) \leq C\}$  is finite.  $\square$

*Proof:* We may assume  $M$  is f.g.. As  $M/M_{\text{tor}}$  is torsion-free, it is a lattice in  $M_{\mathbb{R}}$  (19.1.5.39). Assume  $b_{\mathbb{R}}$  is positive-definite, then it defines a topology on  $M_{\mathbb{R}}$ , then  $\{x \in M' | b_{\mathbb{R}}(x, x) \leq C\}$  is finite because it is a compact discrete set.

Conversely, if  $\{x \in M' | b(x, x) \leq C\}$  is finite and  $b_{\mathbb{R}}$  is not positive-definite, then it is at least semi-positive definite. If  $b_{\mathbb{R}}(y, y) = 0$ , then by Cauchy-Schwartz inequality (3.2.4.31),  $y$  is in the radical of  $b_{\mathbb{R}}$ . By hypothesis  $y \notin M_{\mathbb{Q}}$ .

Now choose a basis  $e_1, \dots, e_n$  of  $M$ , for any  $n > 1$ , there is a  $y_n \in M$  s.t.  $y_n - ny \in S = \{\sum \alpha_i e_i | 0 \leq \alpha \leq 1\}$ , and

$$b_{\mathbb{R}}(y_n, y_n) = b_{\mathbb{R}}(y_n - ny, y_n - ny)$$

is bounded by the maximal value of  $b_{\mathbb{R}}$  on  $S$ . These  $y_n$  are all different because  $y \notin M_{\mathbb{Q}}$ , thus contradicting the hypothesis.  $\square$

**Prop. (3.2.4.39) [Descent].** Let  $M \in \mathcal{Ab}$  s.t.  $\#M/(n) < \infty$  for some  $n \geq 2$  and there is a positive semi-definite symmetric bilinear form  $B$  on  $M$  s.t.  $\{x \in M \mid B(x, x) \leq C\}$  is finite for any  $C > 0$ , then  $M$  is f.g..  $\lrcorner$

*Proof:* Choose a set of generators  $\{x_i\}$  for  $M/(n)$ . Now there is a constant  $C$  that whenever  $(x, x) \geq C$ ,

$$(x - x_i, x - x_i) < 2(x, x), \forall i,$$

because by Cauchy-Schwartz inequality (3.2.4.31),  $(x - x_i, x - x_i)$  is similar to  $(x, x)$  when  $(x, x)$  is large.

Now let  $\Gamma = \{x_1, \dots, x_s\} \cup \{x \in \Gamma \mid (x, x) < C\}$ . Then  $\Gamma$  is finite by hypothesis. We prove  $\Gamma$  generates  $M$ : Consider the infimum  $C_0$  of  $(x, x)$  that  $x$  is not generated by  $M$ , then there is a  $x$  that  $C_0 \leq (x, x) < 2C_0$ . Obviously  $C_0 \geq C$ . Let  $x = x_i + ny$  for some  $x_i \in \Gamma, y \in M$ , then

$$(y, y) = \frac{1}{n^2}(x - x_i, x - x_i) < \frac{2}{n^2}(x, x) \leq \frac{1}{2}(x, x) < C_0.$$

Thus by minimality of  $C_0$ ,  $y \in \Gamma$ , thus  $x \in \Gamma$ .  $\square$

**Prop. (3.2.4.40) [Lagrangian Decomposition].** Let  $A \in \mathcal{Ab}$  be f.g. and  $\Gamma : A \times A \rightarrow \mathbb{Q}/\mathbb{Z}$  is a bilinear alternating non-degenerate pairing, then a **Lagrangian decomposition** of  $A$  is an Abelian subgroup  $B \subset A$  s.t.  $\Gamma|_B = 0$ , called an **isotropic subgroup** of  $A$ , such that  $A \cong B \oplus B^\vee$ , and  $\Gamma$  is given by

$$\Gamma((x, \chi), (y, \psi)) = \chi(y)\psi(x)^{-1}.$$

Then  $A$  admits a Lagrangian decomposition. In particular, if  $\#A < \infty$ , then  $\#A \in (\mathbb{Z}_+)^2$ .  $\lrcorner$

*Proof:* Because  $A$  is f.g., take  $x, y$  in  $A$  s.t.  $\Gamma(x, y)$  has the maximal denominator, then it is easy to see that for any  $z \in A$ ,  $z - ax - by \in \text{span}(x, y)^\perp$  for some  $a, b \in \mathbb{Z}$ , so  $A = \text{span}(a, b) \oplus \text{span}(a, b)^\perp$ , and we can use induction.  $\square$

## 5 Fields

### Finite Fields

## 6 Field Extensions

**Def. (3.2.6.1) [Composition of Fields Extensions].**  $\lrcorner$

**Def. (3.2.6.2) [Distinguished Class of Extensions].** A family  $\mathcal{L}$  of field extensions are called a **distinguished class** iff

- It is closed under base change.
- $E/F/K \in \mathcal{L}$  iff  $F/K \in \mathcal{L}$  and  $E/F \in \mathcal{L}$ .

$\lrcorner$

### Algebraic Extensions

**Def. (3.2.6.3) [Algebraic Extensions].** An **algebraic extension** is a field extension  $L/K$  s.t. for any element  $\alpha \in L$ , there exists a nonzero polynomial  $f(X) \in K[X]$  s.t.  $f(\alpha) = 0$ .  $\lrcorner$

**Prop. (3.2.6.4).** The family of finite extensions form a distinguished class.

The family of algebraic extensions form a distinguished class.

The family of f.g. extensions form a distinguished class.  $\lrcorner$

*Proof:* Finite case is trivial. For the alg. extensions, for  $k \subset F \subset E$ , for any  $\alpha \in E$ ,  $\alpha$  satisfies a polynomial function with f.m coefficients in  $F$ , the coefficients form a subfield  $F_0$  of  $F$  which is finite over  $k$ , so  $k \subset F_0 \subset F_0(\alpha)$  is a finite tower, so it is finite, hence algebraic. The base change is easy to check.

For f.g. extensions, it suffice to check composition: ? □

**Prop. (3.2.6.5).** For an alg. extension  $k \subset E$ , any injective field map  $E \rightarrow E$  over  $k$  is an automorphism. (This is because it induce a permutation of any  $\alpha$  with its conjugates in  $E$ , so it is surjective). ⌋

**Lemma (3.2.6.6).** Let  $f \in k[X]$  be a polynomial of degree  $\geq 1$ , then there is a field  $K$  that  $f$  has a root in  $K$ . Hence for any finite set of polynomials, there is a field  $K$  that all of them have roots in  $K$ . ⌋

*Proof:* Cf. [Algebra Lang P231]. □

**Lemma (3.2.6.7).** For any  $k \in \mathbf{Field}$ , there exists  $K \in \mathbf{Field}$  s.t.  $k \subset K, \overline{K} = K$ . ⌋

*Proof:* Firstly, we construct a field that every polynomial in  $k[X]$  of degree  $\geq 1$  has a root. Consider the polynomial ring  $k[X_f]$ , where there is a indeterminant  $X_f$  for each  $f \in k[X]$  of deg  $\geq 1$ . Then the ideal generated by  $f(X_f)$  is not a unit ideal, which can be seen by constructing a finite field extension that  $f_i$  all have a root in it (3.2.6.6).

So if  $\mathfrak{m}$  is a maximal ideal containing all  $f(X_f)$ , then the quotient field is a field that all  $f$  have a root ( $X_f$ ).

So now if we construct inductively like this, and consider their union, then it is clearly a field and any polynomial of degree  $\geq 1$  have a root in it. □

**Prop. (3.2.6.8) [Algebraic Closure].** Assume the axiom of choice, for any  $K \in \mathbf{Field}$ , there exists uniquely an algebraic field extension  $K/k$  s.t.  $K$  is alg. closed, up to isomorphism over  $k$ . Any such field  $K$  is called an **algebraic closure** of  $k$ , denoted by  $\overline{k}$ . ⌋

*Proof:* Let  $E$  be a field that is alg. closed and contains  $k$  by (3.2.6.7). Let  $k^a$  be the union of subextensions that are algebraic over  $k$ .  $k^a$  is a field, by (3.2.6.4), and  $k^a$  is alg. closed, because if  $f(X)$  is a polynomial of degree  $\geq 1$  in  $k^a[X]$ , then it has a root  $\alpha \in E$ , and  $\alpha$  is algebraic over  $k^a$ , so  $\alpha \in k^a$ .

Why unique? □

**Prop. (3.2.6.9) [Finite Algebraic Extensions].** Let  $F/L/K \in \mathbf{Field}$ , there is a finite extension  $F'/K$  s.t.  $F = LF'$ . ⌋

*Proof:* Take a generator  $x_i$  of  $F/L$ , then  $x_i$  are algebraic over  $L$ , thus there are polynomials  $f_i \in L[X]$  s.t.  $f_i(x_i) = 0$ . Let  $F'$  be the fields over  $K$  generated by all the coefficients of  $f_i$ s, then  $F = LF'$ . □

**Prop. (3.2.6.10).** If  $E/F$  is an algebraic field extension, then for any  $R$  that  $E \subset R \subset F$ ,  $R$  is a field. ⌋

*Proof:* If  $\alpha \in R$ , then  $\alpha$  is algebraic over  $E$ , so there is a relation  $\alpha^n + \dots + a_0 = 0$ , so  $\alpha^{-1} = -a_0^{-1}(a_1 + \dots + \alpha^{n-1}) \in R$ . □



### Normal & Separable Extensions

**Def. (3.2.6.11) [Normal Extensions].** A field extension  $K/k$  in  $\bar{k}$  is called **normal extension** iff it satisfied the following equivalent conditions:

1. Any embedding of  $K$  into  $\bar{k}$  induce an automorphism on  $K$ .
2.  $K$  is the splitting field of a family of polynomials in  $k[X]$ .
3. Every irreducible polynomial in  $k[X]$  that has a root in  $K$  splits in  $K$ .

┘

*Proof:* Cf.[Algebra Lang P237].

□

**Prop. (3.2.6.12).** Normal extensions are stable under base change and composition, by the first definition of (3.2.6.11). ┘

**Def. (3.2.6.13) [Normal Closure].** For any field extension  $F/K$ , there is a field extension  $E/F$  that  $E/K$  is normal, called the **normal closure** of  $F/K$ , denoted by  $\tilde{F}/K$ . It is the composite of conjugates of  $F/K$ . ┘

**Def. (3.2.6.14) [Separable Degree].** Define the **separable degree**  $[E : k]_s$  of an extension  $E/k$  as the cardinality of embedding of  $E$  into  $\bar{k}$ . Separable degree commutes with composition, and when  $E/k$  is finite,  $[E : k]_s \leq [E : k]$ . ┘

**Def. (3.2.6.15) [Separable Polynomials].** A finite extension is called a **separable extension** iff  $[E : k]_s = [E : k]$ , an algebraic number  $\alpha$  over  $k$  is called **separable over  $K$**  iff  $k(\alpha)/k$  is separable. A polynomial  $f \in k[X]$  is called a **separable polynomial** iff it has no multiple roots in  $\bar{k}$ . ┘

**Def. (3.2.6.16) [Separable Extensions].** An algebraic extension  $E/k$  is called a **separable extension** iff it satisfies the following equivalent conditions:

- Every f.g. subfield is separable over  $k$ , (this is compatible because subfield of a finite separable extension is separable, by the compatibility of separable degree).
- Every element of  $E$  is separable.
- It is generated by a family of separable elements.

┘

*Proof:* If  $E/k$  is separable and  $k \subset k(\alpha) \subset E$ , then by (3.2.6.15),  $k(\alpha)$  is separable. And if it is generated by a family of separable elements  $\{\alpha_i\}$ , then any f.g. subfield can be f.g. by elements  $\{\alpha_i\}$ . Now it is a tower of separable extensions, hence separable by the compatibility of separable degree. □

**Prop. (3.2.6.17).** Separable extensions form a distinguished class. ┘

*Proof:* Cf.[Algebra Lang P241].

□

**Prop. (3.2.6.18) [Primitive Element Theorem].** A finite extension  $E/k \in \mathbf{Field}$  is primitive iff there are only f.m. middle fields. And if  $E/k$  is separable, this is satisfied. ┘

*Proof:* If  $k$  is finite, this is simple. Assume  $k$  infinite, for any two elements  $\alpha, \beta$ , consider  $k(\alpha + c_i\beta)$ , if there is only finitely many middle fields, there exists two that are equal, so  $k(\alpha, \beta) = k(\gamma)$ . Proceeding inductively,  $E$  is primitive.

Conversely, if  $k(\alpha) = E$ , every middle field corresponds to a divisor of the irreducible polynomial of  $\alpha$ . This map is injective, because for any  $g_F$ , degree of  $\alpha$  over  $F$  is the same over the degree over the coefficient field of  $g_F$ , so it must be equal to  $F$ .

If  $E/k$  is separable, Let

$$P(X) = \prod_{i \neq j} (\sigma_i \alpha + X \sigma_i \beta - \sigma_j \alpha - X \sigma_j \beta)$$

for different embedding  $\sigma_i, \sigma_j$  of  $E(\alpha, \beta)$  into  $\bar{k}$ . Then it is not identically zero, thus there exists  $c$  that  $\sigma_i(\alpha + c\beta)$  is all distinct, thus generate  $K(\alpha, \beta)$ .  $\square$

### Inseparable Extensions

**Def. (3.2.6.19) [Perfect Fields].**  $K \in \mathbf{Field}$  is called a **perfect field** if  $\text{char } k = 0$  or  $k^{\text{char } k} = k$ .  $\lrcorner$

**Prop. (3.2.6.20) [Algebraic Extensions of Perfect Fields are Perfect].** Algebraic extensions of perfect fields (3.2.6.19) are perfect fields.  $\lrcorner$

**Prop. (3.2.6.21).** Let  $k \in \mathbf{Field}$  and  $f(X) \in k[X]$  irreducible. If  $\text{char } k = 0$ , then  $f$  is separable, and if  $\text{char } k = p \in \mathbf{Prime}$ , then all the roots of  $f$  in  $\bar{k}$  have the same multiplicity and  $[k(\alpha) : k] = p^n [k(\alpha) : k]$  for some  $n \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* All roots have the same multiplicity because there are Galois actions. If the multiplicity is not 1, the derivative  $f'$  is zero, otherwise  $f$  is not irreducible. Then  $f(X) = g(X^p)$ . We can choose  $f(X) = h(X^{p^n})$  with  $h$  separable, then  $[k(\alpha) : k(\alpha^{p^n})] = p^n$ , thus the result.  $\square$

**Def. (3.2.6.22) [Inseparable Degrees].** The **inseparable degree**  $[E : k]_i$  is defined as the quotient  $[E : k] / [E : k]_s$ . An algebraic element  $\alpha$  is called **purely inseparable** over  $k$  iff there is a  $n$  that  $\alpha^{p^n} \in k$ .  $\lrcorner$

**Def. (3.2.6.23) [Purely Inseparable Extensions].** An extension is called **purely inseparable** if it satisfies the following equivalent conditions:

- $[E : k]_s = 1$ .
- Every element  $\alpha$  of  $E$  is purely inseparable over  $k$ .
- For every  $\alpha \in E$ , the irreducible equation of  $\alpha$  over  $k$  is of type  $X^{p^n} - a$ .
- It is generated by a family of purely inseparable elements.

$\lrcorner$

*Proof:* Cf. [Algebra Lang P249].  $\square$

**Cor. (3.2.6.24).**  $K \in \mathbf{Field}^p$  is perfect (3.2.6.19) iff there are no purely inseparable extensions of it.  $\lrcorner$

**Cor. (3.2.6.25) [Perfect Closure].** For any  $k \in \mathbf{Field}^p$ , there is a unique purely inseparable field extension  $k^{\text{perf}}/k$  that  $k^{\text{perf}}$  is perfect, called the **perfect closure** of  $k$ . It is generated by adding all the  $p^n$ -th roots to  $k$ .  $\lrcorner$

**Prop. (3.2.6.26).** Purely inseparable extensions form a distinguished class.  $\lrcorner$

*Proof:* Cf. [Algebra Lang P250].  $\square$

**Prop. (3.2.6.27).** If  $E/k \in \mathbf{Field}$  is algebraic and  $E_0$  be the maximal separable extension contained in  $E$ , then  $E/E_0$  is purely inseparable. And if  $E/k$  is normal, then  $E_0/k$  is normal, too.  $\lrcorner$

*Proof:* By the proof of (3.2.6.21), any  $\alpha$  has a  $p^n$  that  $\alpha^{p^n}$  that  $\alpha^{p^n}$  is separable, hence it is purely inseparable over  $E_0$  by (3.2.6.23).  $E_0/k$  is normal because any  $\sigma$  maps  $E$  to itself, and  $E_0$  to  $\sigma(E_0) \in E$  separable, hence  $\sigma(E_0) \subset E_0$ .  $\square$

### Trace and Norm

**Prop. (3.2.6.28) [Trace and Norm].** Let  $K \subset_{\text{fin}} L \in \mathbf{Field}$  and  $\Sigma = \text{Hom}(L, \overline{K})$ , then define the **trace map**  $\text{tr}_{L/K} : L \rightarrow K : x \mapsto [L : K]_i \sum_{\sigma \in \Sigma} \sigma(x)$  and the **norm map**  $\text{Nm}_{L/K} : L \rightarrow K : x \mapsto (\prod_{\sigma \in \Sigma} \sigma(x))^{[L:K]_i}$ .  $\lrcorner$

**Prop. (3.2.6.29).** For  $L/K \in \mathbf{Field}$ ,

- the norm induces a multiplicative homomorphism  $L^\times \rightarrow K^\times$ , and the trace induces an additive homomorphism  $L \rightarrow K$ .
- if  $E/F/K$  are field extensions, then  $\text{Nm}_{E/K} = \text{Nm}_{F/K} \circ \text{Nm}_{E/F}$ , and  $\text{tr}_{E/K} = \text{tr}_{F/K} \circ \text{tr}_{E/F}$ .
- if  $L = K(\alpha)$  and  $F = \text{Irr}(\alpha, k; X) = X^n - a_{n-1}X^{n-1} + \dots + (-1)^n a_0$ , then

$$\text{Nm}_{L/K}(\alpha) = a_0, \quad \text{tr}_{L/K}(\alpha) = a_{n-1}.$$

$\lrcorner$

*Proof:* Cf. [Lan02]P285.  $\square$

**Prop. (3.2.6.30) [Calculating Dual Basis].** Let  $L = K(\alpha)$  be a finite separable extension, let  $f(X) = \text{Irr}(\alpha, K; X)$ , and

$$\frac{f(X)}{(X - \alpha)} = \beta_0 + \beta_1 X + \dots + \beta_{n-1} X^{n-1},$$

then the dual basis of  $(1, \alpha, \dots, \alpha^{n-1})$  w.r.t. the trace form (3.2.6.32) is

$$\frac{\beta_0}{f'(\alpha)}, \dots, \frac{\beta_{n-1}}{f'(\alpha)}.$$

$\lrcorner$

*Proof:* Denote the roots of  $f$  be  $\alpha_i$ , then they are pairwise different, and

$$\sum_i \frac{f(X)}{X - \alpha_i} \frac{\alpha_i^r}{f'(\alpha_i)} = X^r$$

for all  $r$  by Lagrange interpolation (3.3.1.2). But this is equivalent to

$$\text{tr}\left(\frac{\alpha^r \beta_j}{f'(\alpha)}\right) = \delta_{ij}$$

$\square$

**Prop. (3.2.6.31).** For  $K \subset L \in \mathbf{Field}$  and  $x \in L$ ,  $x \in L$  acts on  $L$  via multiplication  $T_x$ . Then

$$\det(T_x) = N_{L/K}(x), \quad \text{tr}(T_x) = \text{tr}_{L/K}(x).$$

$\lrcorner$

*Proof:* Let  $F = K(x) \subset L$ , and  $p(X)$  be the minimal monic polynomial of  $x$ , then  $\text{char}(T_x|_X) = \text{char}(T_x|_K; X)^{[L:K(x)]}$ , thus we are done by (3.2.6.29).  $\square$

**Prop. (3.2.6.32) [Trace Form].** For  $L \subset_{\text{fin}} K \in \text{Field}$ , consider the pairing

$$\text{Tr}_{L/K} : L \times L \rightarrow K : (x, y) \mapsto \text{tr}(xy),$$

called the **trace form**. Then the following are equivalent:

1.  $L/K$  is separable.
2.  $\text{tr}_{L/K} \neq 0$ .
3.  $\text{Tr}_{L/K}$  is non-degenerate.

┘

*Proof:* 2, 3 are equivalent by a minute's thought. For  $1 \iff 2$ :

If  $L/K$  is inseparable, then by (3.2.6.29), if we consider  $L/K'/K$  that  $L/K'$  is purely inseparable of degree  $p$ , then it is generated by some equation  $x^p - a$ , and the root is  $\alpha$ . Then  $L = K'(\alpha)$  and the minimal polynomial of  $\alpha^i$ ,  $0 \leq i \leq p-1$  is  $x^p - a^i$ , so  $\text{tr}(\alpha^i) = 0$ , and  $\text{tr}_{L/K} = 0$ .

If  $L/K$  is separable,  $\text{tr}_{L/K} \neq 0$  by the linear independence of characters (3.2.8.4). □

**Def. (3.2.6.33) [Discriminant of a Basis].** Let  $\alpha_i$  be a basis of a separable extension  $L/K$ , then the **discriminant of the basis**  $d(\alpha_1, \dots, \alpha_n)$  is defined to be  $(\det(\sigma_i(\alpha_j))_{i,j})^2$ . Clearly,  $d(\alpha_1, \dots, \alpha_n)$  is invariant under the Galois action of  $L/K$ , thus it is an element of  $K$ . ┘

**Prop. (3.2.6.34).** Notice  $\text{tr}_{L/K}(\alpha_i \alpha_j) = \sum_k \sigma_k(\alpha_i) \sigma_k(\alpha_j)$ , thus  $(\text{tr}_{L/K}(\alpha_i \alpha_j))$  is the product of the matrices  $(\sigma_k \alpha_i)^t$  and  $(\sigma_k \alpha_j)$ , thus

$$d(\alpha_1, \dots, \alpha_n) = \det(\text{tr}_{L/K}(\alpha_i \alpha_j)).$$

So the discriminant of  $\{\alpha_1, \dots, \alpha_n\}$  is the Gram matrix of the trace norm w.r.t. this basis (3.5.11.3).

In particular, by (3.2.6.32),

$$d(\alpha_1, \dots, \alpha_n) \neq 0.$$

┘

**Prop. (3.2.6.35).** If  $L = K(\theta)$  is a separable field extension of degree  $d$ , and  $\theta_1 = \theta, \dots, \theta_n$  are the conjugates of  $\theta$ , then

$$d(1, \theta, \dots, \theta^{d-1}) = (\det \text{Van}(\theta_1, \dots, \theta_d))^2 = \prod_{i < j} (\theta_i - \theta_j)^2.$$

┘

## 7 Transcendental Extensions

**Def. (3.2.7.1) [Transcendental Basis].** Let  $k \subset K \in \text{Field}$ , a **transcendental base** of  $K/k$  is an algebraically independent subset of  $K$  that any element of  $K$  is algebraic over it.

Given any algebraically independent set  $S \subset T$  a set over which  $K$  is algebraic,  $S$  can be extended to a transcendental base contained in  $T$ , by Zorn's lemma. In particular, transcendental basis exist.

┘

**Prop. (3.2.7.2).** Any two transcendental basis have the same cardinality, called the **transcendental degree** of  $K/k$ , denoted by  $\text{tr.deg}(K/k)$ . ┘

*Proof:* If  $K/k$  has a finite transcendental basis, then let  $X = \{x_1, \dots, x_m\}$  transcendental base of minimal number,  $S = \{w_1, \dots, w_n\}$  an algebraically independent set. If  $n > m$ , we proceed by changing one element in  $X$  a time using induction and prove that  $K$  is algebraic over  $\{w_1, \dots, w_r, x_{r+1}, \dots, x_m\}$ , contradiction.

Because  $w_{r+1}$  is algebraic over  $\{w_1, \dots, w_r, x_{r+1}, \dots, x_m\}$ , we have a minimal polynomial

$$f = \sum g_j(w_{r+1}, w_1, \dots, w_r, x_{r+2}, \dots, x_m)x_{r+1}^j$$

s.t.  $f(w_{r+1}, w_1, \dots, x_m) = 0$  (after possibly renumbering  $x_i$ , this  $x$  must exist because  $S$  is itself algebraically independent). So  $x_r$  is algebraic over  $\{w_1, \dots, w_{r+1}, x_{r+2}, \dots, x_m\}$ , hence  $K$  is independent over it, too.

If  $K/k$  has an infinite basis  $B$ , let  $B'$  be another basis, then for any  $\alpha \in B^*$ , there is a finite set  $B_\alpha \subset B$  that  $\alpha$  is algebraic over  $k(B_\alpha)$ , because algebraic equation involves f.m. generators. Then we define  $B^* = \cup_{\alpha \in B'} B_\alpha$ , then it has cardinality smaller than  $B'$ . But  $B^* = B$ , because for any  $\beta \in B$ ,  $\beta$  is algebraic over  $B'$  which is algebraic over  $k(B^*)$ , thus  $\beta$  is algebraic over  $k(B^*)$ , thus  $\beta \in B^*$ .  $\square$

**Prop. (3.2.7.3).** If  $K$  is of finite transcendental degree over  $k$ , then  $\#K = \#k$ .  $\lrcorner$

*Proof:* We find a purely transcendental  $L/k$  that  $K/L$  is algebraic, then the element of  $L$  are all polynomials of finite indeterminants of elements of  $k$ , so  $|L| = |k|$  by (2.2.8.4), and similarly  $|K| = |L|$ .  $\square$

**Prop. (3.2.7.4).** For  $k \in \mathbf{Field}$ , any two alg.closed field extension  $K_1/k, K_2/k$  of the same transcendental degree over  $k$  are isomorphic over  $k$ .  $\lrcorner$

*Proof:* We define a bijection of the transcendental basis, and then extend it to an isomorphism of fields, by (3.2.6.8).  $\square$

**Cor. (3.2.7.5).** Any two alg.closed field of the same characteristic and cardinality are isomorphic.  $\lrcorner$

*Proof:* It suffices to show that their transcendental basis over the base field are of the same cardinality?  $\square$

**Cor. (3.2.7.6) [Extending Automorphisms].**

- Let  $K/k \in \mathbf{Field}$  and  $K = \overline{K}$ , then any automorphism of  $k$  extends to an automorphism of  $K$ .
- Let  $K/k \in \mathbf{Field}$  and  $\#K \leq \aleph_0, \#K > \aleph_0$ , then any homomorphism  $\alpha : k \rightarrow K$  extends to an automorphism of  $K$ .

$\lrcorner$

*Proof:* 1 follows from (3.2.7.5).

2 follows from the fact that  $\text{tr.deg}(K/k) = \text{tr.deg}(K/\alpha(k))$ .  $\square$

**Prop. (3.2.7.7) [Transitivity of Transcendence Degrees].** Let  $k \subset E \subset K \in \mathbf{Field}$ , then  $\text{tr.deg}(K/k) = \text{tr.deg}(K/E) + \text{tr.deg}(E/k)$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (3.2.7.8) [Lüroth].** The automorphism group of  $K(T)$  is  $\text{PGL}(2; K)$ .  $\lrcorner$

*Proof:* Consider  $\theta = \sigma(x) = \frac{f(x)}{g(x)}$ , then  $x$  is algebraic over  $K(\theta) : \theta g(x) - f(x) = 0$ . Now  $x$  is transcendental over  $K$ , thus  $\theta$  is transcendental over  $K$  as well. Now the minimal polynomial of  $x$  over  $K(\theta)$  is just  $\theta g(x) - f(x)$ , because it is irreducible, as it is linear over  $\theta$ . But  $K(x) = K(\theta)$ , thus the polynomial must have degree 1, so  $f(x), g(x)$  is of degree 1. Now the rest is clear.  $\square$

**Prop. (3.2.7.9) [Lüroth].** Any subfield  $K \not\subset L \subset K(T)$  is of the form  $K(u)$  where  $u \in K(X)$  is transcendental over  $K$ .  $\lrcorner$

*Proof:* It is clearly transcendental over  $K$ , so this follows from (7.13.1.18) and Riemann-Hurewitz (7.13.1.33), notice that the field extension is separable by (5.3.9.4): The non-singular complete curve corresponding to  $L$  has genus 0, and it has a rational point, so isomorphic to  $\mathbb{P}_k^1$  (7.13.8.2).  $\square$

### Linearly Disjoint Extensions

**Def. (3.2.7.10) [Linearly Disjoint Extensions].** Let  $k \subset K \in \mathbf{Field}$ ,  $L \subset \Omega = \overline{\Omega} \in \mathbf{Field}$ . Then  $K$  is said to be **linearly disjoint** from  $L$  over  $k$  if any finite subset of  $K$  that is linearly independent over  $k$  remains linearly independent over  $L$ .

When  $[K : k], [L : k] < \infty$ , this is equivalent to the fact that  $[KL : k] = [K : k] \cdot [L : k]$ .  $\lrcorner$

**Prop. (3.2.7.11).** Situation as in (3.2.7.10), if  $K$  is linearly disjoint from  $L$ , then  $L$  is linearly disjoint from  $K$ .  $\lrcorner$

*Proof:* Let  $y_1, \dots, y_n$  be elements of  $L$  that is linearly independent over  $k$ . Suppose  $x_1, \dots, x_n \in K$  are not all zero s.t.  $\sum x_i y_i = 0$ , then we may assume  $x_1, \dots, x_r$  are linearly independent and  $x_{r+1}, \dots, x_n$  are linear combinations of  $\{x_1, \dots, x_r\}$ . Suppose  $x_k = \sum_{i=1}^r a_{ki} x_i$ , then

$$\sum_{i=1}^r \left( y_i + \sum_{k=r+1}^n a_{ki} y_k \right) x_i = 0.$$

But then by the hypothesis,  $y_i + \sum_{k=r+1}^n a_{ki} y_k = 0$ , contradiction.  $\square$

**Prop. (3.2.7.12).** Let  $k \subset \Omega \in \mathbf{Field}$  and  $K, E \subset L \in \mathbf{Field}$  be subfields of  $\Omega$  containing  $k$ , then  $K, L$  are linearly disjoint over  $k$  iff  $K, E$  are linearly disjoint over  $k$  and  $KE, L$  are linearly disjoint over  $E$ .  $\lrcorner$

*Proof:* Cf. [Lan02] P361. ?  $\square$

**Def. (3.2.7.13) [Algebraically Disjoint Extensions].** Situation as in (3.2.7.10),  $K$  is said to be **algebraically disjoint** from  $L$  over  $k$  if any finite subset of  $K$  that is algebraically independent over  $k$  remains algebraically independent over  $L$ .

When  $\text{tr.deg}(K/k), \text{tr.deg}(L/k) < \infty$ , this is equivalent to the fact that  $\text{tr.deg}(KL/k) = \text{tr.deg}(K/k) + \text{tr.deg}(L/k)$ .  $\lrcorner$

**Prop. (3.2.7.14).** Situation as in (3.2.7.10), if  $K$  is algebraically disjoint from  $L$ , then  $L$  is algebraically disjoint from  $K$ .  $\lrcorner$

*Proof:* This follows from transitivity property of transcendence degrees (3.2.7.7).  $\square$

**Prop. (3.2.7.15).** Situation as in (3.2.7.10), if  $K, L$  are linearly disjoint over  $k$ , then they are algebraically disjoint over  $k$ .  $\lrcorner$

*Proof:* Let  $x_1, \dots, x_n$  be elements of  $K$  that are algebraically independent over  $k$ , suppose there are f.m.  $y_\alpha \in L$  s.t.  $\sum_\alpha y_\alpha \underline{x}^\alpha = 0$ , then  $\{\underline{x}^\alpha\}_\alpha$  are linearly dependent over  $L$ . But they are linearly independent over  $k$ , contradicting the hypothesis that  $K, L$  are linearly disjoint over  $k$ .  $\square$

## 8 Galois Theory

**Prop. (3.2.8.1) [Fixed Fields].** If  $K \subset L \in \mathbf{Field}$  and  $L = \overline{L}$ , then  $L^{\mathrm{Aut}(L/K)} = K^{\mathrm{perf}}$ .  $\lrcorner$

*Proof:*  $K^{\mathrm{perf}} \subset L^{\mathrm{Aut}(L/K)}$  is trivial, to show the other inclusion, it suffices to show that for any  $x \in L \setminus K^{\mathrm{perf}}$ , there exists  $\sigma \in \mathrm{Aut}(L/K)$  s.t.  $\sigma(x) \neq x$ . If such an  $x$  is transcendental over  $K$ , then there is an automorphism of  $K(x)$  that maps  $x$  to  $-x$ , and it extends to an automorphism of  $L$  by (3.2.7.6). And if  $x$  is integral over  $K$  but not purely inseparable over  $K$ , then there exists a conjugate  $x'$  of  $x$  s.t.  $x' \neq x$ . Then there is an automorphism of  $K(x)$  that maps  $x$  to  $x'$ , and it extends to an automorphism of  $L$ .  $\square$

**Def. (3.2.8.2) [Galois Extensions].** A **Galois field extension** is a field extension that is both normal (3.2.6.11) and (algebraic)separable (3.2.6.14). We use  $L/K \in \mathbf{GalField}$  to denote that:  $L/K \in \mathbf{Field}$  and  $L/K$  is a Galois field extension. And we use  $L/K \in \mathbf{GalField}^{\mathrm{fin}}$  to denote that:  $L \subset K \in \mathbf{Field}$  and  $L/K$  is a finite Galois field extension.  $\lrcorner$

**Def. (3.2.8.3) [Galois Closures].** Let  $F/K \in \mathbf{Field}$  be separable, then the normal closure  $E/K$  is also separable, called the **Galois closure** of  $F/K$ .

In particular, the maximal separable extension  $k^{\mathrm{sep}}/k$ , called the **separable closure** of  $k$ , is Galois over  $k$ , and  $\mathrm{Gal}(k^{\mathrm{sep}}/k)$  is denoted by  $\mathrm{Gal}_k$ .  $\lrcorner$

**Prop. (3.2.8.4) [Linear Independence of characters, Dedekind/Artin].** Let  $L \in \mathbf{Field}$ ,  $G \in \mathcal{M}\mathrm{on}$  and  $\chi_i : G \rightarrow L$  be distinct multiplicative maps, then  $\chi_i$  are linearly independent over  $L$ .  $\lrcorner$

*Proof:* Let  $\sum_{i=1}^n a_i \chi_i = 0$  and  $a_i \neq 0$  for any  $i$ , we use induction to derive contradictions: for all  $i$ .  $n = 1$  is trivial, and for general  $n$ , assume  $a_1, a_2 \neq 0$ , then the two equations  $\chi_1(h) \neq \chi_2(h)$ , then  $\sum_{i=1}^n a_i \chi_i(hg) = 0$  and  $\sum_{i=1}^n a_i \chi_i(g) = 0$  gives us a equation with smaller  $n$ , thus we are done.  $\square$

**Cor. (3.2.8.5).** If  $\alpha_i$  are different elements in  $K$  and there are element  $a_i$  that  $\sum a_i \alpha_i^v = 0$  for every  $v \geq 0$ , then  $a_i = 0$  for all  $n$ . (Seen as characters from  $\mathbb{Z}_{\geq 0} \rightarrow K$ ).  $\lrcorner$

**Thm. (3.2.8.6) [Algebraic Independence of Automorphisms, Artin].** Let  $K \in \mathbf{Field}$ , and  $G = \{\sigma_1, \dots, \sigma_n\}$  be a finite subgroup of  $\mathrm{Aut}(K)$ , then  $\{\sigma_i\}$  are algebraically independent over  $K$ .  $\lrcorner$

*Proof:* Cf. [Lang, P311]. ?  $\square$

**Thm. (3.2.8.7) [Galois Main Theorem, Artin].** Let  $K \in \mathbf{Field}$  and  $G \leq \mathrm{Aut}(K)$  be a finite subgroup. Then  $K/K^G \in \mathbf{GalField}$  and  $\mathrm{Gal}(K/K^G) \cong G$ .  $\lrcorner$

*Proof:* For every element  $x$ , set  $\{\sigma_1 x, \dots, \sigma_r x\}$  be distinct conjugates, then  $f(X) = \prod_i^r (X - \sigma_i x)$  shows that  $K$  is separable and normal over  $K^G$ . And primitive element theorem shows that  $[K : K^G] \leq |G|$ , so it must equals  $G$ .  $\square$

**Prop. (3.2.8.8).** If  $L/K \in \mathbf{GalField}^{\mathrm{fin}}$ , then there is an isomorphism of  $L$ -vector spaces:

$$L \otimes_K L \cong L \times L \times \dots \times L : (a, b) \mapsto (ab, a\sigma_1(b), \dots, a\sigma_{n-1}(b))$$

where  $\sigma_i$  are Galois elements, and  $L$  acts on the left on  $L \otimes_K L$ .  $\lrcorner$

*Proof:* Choose a primitive element  $x$  and its minimal polynomial  $f(x)$ , then  $L \cong K[X]/(f)$ , and  $L \otimes_K L \cong L[X]/(f)$ , but  $f$  decomposes completely in  $L[X]$ , thus by Chinese remainder theorem (5.1.1.8), the given map is an isomorphism of rings.  $\square$



**Prop. (3.2.8.9)[Normal Basis Theorem].** For  $L/K \in \mathbf{GalField}^{\text{fin}}$ , normal basis exists.  $\lrcorner$

*Proof:* Finite case: The Galois group is cyclic, and the linear independent of characters shows that the minimal polynomial of  $\sigma$  is  $n$ -dimensional thus equals  $X^n - 1$ . Regard  $L$  as a  $K[X]$  module thus by (3.2.4.21) is a direct sum of modules of the form  $K[X]/(f(x))$ ,  $f(x)|X^n - 1$  and the minimal polynomial for the action of  $X$  is  $X^n - 1$ . So it must be isomorphic to  $K(X)/(X^n - 1)$ .

Infinite Case: Let

$$f(\{X_\sigma\}) = \det(t_{\sigma_i, \sigma_j}) \in \mathbb{Z}[\{X_\sigma\}], \quad t_{\sigma, \tau} = X_{\sigma^{-1}\tau}$$

then  $f \neq 0$  by substituting 1 for  $X_{\text{id}}$  and 0 otherwise. So it won't vanish for all  $x \in L$  if we substitute  $X_\sigma = \sigma(x)$  because  $\{(\sigma(x))_\sigma\}$  are algebraically independent (3.2.8.6). Thus there exists  $w \in L$  s.t.

$$\det(\sigma^{-1}\tau(w)) \neq 0.$$

Now if

$$\sum a_\tau \tau(w) = 0, \quad a_\sigma \in K,$$

act by  $\sigma$  for all  $\sigma$ , we get  $[\sigma^{-1}\tau(w)]_{\tau, \sigma} [a_\sigma]_\sigma = 0$ , thus  $a_\sigma = 0$  for each  $\sigma \in \text{Gal}(L/K)$ . So  $\{\tau(w)\}$  is a  $K$ -basis of  $L$ .  $\square$

**Prop. (3.2.8.10)[Finite Fields].**  $\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \mathbb{Z}/(n)$ . and is generated by the **Frobenius automorphism**  $\varphi : x \mapsto x^p$ .  $\lrcorner$

*Proof:*  $\square$

### Infinite Galois theory

References are [Hid04]P31 and [Sta].

**Def. (3.2.8.11)[Krull Topology].** For  $L/K \in \mathbf{Field}$ , we can define a topology on  $\text{Aut}(L/K)$  as follows: A fundamental system of nbhd of  $\text{id} \in \text{Aut}(L/K)$  is given by  $\text{Aut}(L/M)$ , where  $M/K$  is f.g. This is called the **Krull topology**.  $\lrcorner$

**Thm. (3.2.8.12)[Infinite Galois Theorem, Jacobson].** For  $L/K \in \mathbf{Field}$ , the set of intermediate fields  $M$  s.t.  $L/M \in \mathbf{GalField}$  is in bijection with the set of compact subsets of  $\text{Aut}(L/M)$ .

Moreover, this bijection induces a bijection between the set of intermediate fields  $M$  f.g. over  $K$  s.t.  $L/M \in \mathbf{GalField}$  with the set of compact open subsets of  $\text{Aut}(L/M)$ . If there is a compact open subset of  $\text{Aut}(L/M)$ , then  $\text{Aut}(L/M)$  is locally compact.  $\lrcorner$

*Proof:* ? Cf. [Hid04]P35 or [Sta]0BML.

The highlight is that  $G(L/L^H) = H$  for a closed subgp  $H$  of  $\text{Gal}(L/K)$ . If  $\sigma$  fixes  $L^H$  but is not in  $H$ , because for every finite field  $M$ ,  $H \cdot G(L/M)$  corresponds to  $M/(M \cap L^H)$ , so  $\sigma G(L/M) \cap H \neq \emptyset$ . So  $\sigma$  is in the closure of  $H$  thus in  $H$ .  $\square$

**Prop. (3.2.8.13)[Infinite Fixed Fields].** Suppose  $C \in \mathbf{Field}$  s.t.  $C = \overline{C}$ , and  $U \leq \text{Aut}(C)$  with  $V \leq U$  a subgroup of finite index, then  $[C^V : C^U] < \infty$ . Moreover, if  $V \trianglelefteq U$ , then  $[C^V : C^U] \leq [U : V]$ .  $\lrcorner$

*Proof:* Define  $W = \bigcap_{g \in U} gVg^{-1} \subset V$ , then  $W$  is a subgroup of  $V$  that is normal in  $U$ . There is a natural map  $U/V \rightarrow \text{Aut}(C^U/C^V)$ , and the fixed field of the image is just  $C^V$ . By (3.2.8.7), this implies that  $C^U/C^V \in \mathbf{GalField}^{\text{fin}}$  and  $U/V \rightarrow \text{Aut}(C^U/C^V)$  is surjective. Thus

$$[C^V : C^U] \leq [C^W : C^U] \leq [U : W].$$

$\square$



**Prop. (3.2.8.14)[Linearly Disjoint Galois Extensions].** Situation as in (3.2.7.10), if  $K, L$  are linearly independent over  $k$ , then  $K \cap L = k$ , and the converse is true if  $K/k, L/k \in \mathbf{GalField}$ .  $\square$

*Proof:* For the converse, it suffices to prove for  $K/k, L/k \in \mathbf{GalField}^{\text{fin}}$ . Let  $H_L = \text{Gal}(KL/K)$  and  $H_K = \text{Gal}(KL/K)$ , we have  $H_L \cap H_K = e$  and  $H_L H_K = H$ , so we must have  $[KL : k] = [K : k] \cdot [L : k]$ .  $\square$

**Prop. (3.2.8.15)[Kummer Theory].** Let  $n \in \mathbb{Z}_+, K \in \mathbf{Field}$  s.t.  $n \in K^\times$  and  $\mu_n \subset K$ . A **Kummer extension**  $L/K$  of exponent  $n$  is a Galois extension that the Galois group is Abelian of exponent  $n$ .

Then there is an inclusion preserving isomorphism between the lattice of Kummer extensions  $L$  over  $K$  and the lattice of subgroups of  $K^\times$  containing  $(K^\times)^n$ :

$$L \mapsto \Delta = (L^\times)^n \cap K^\times, \quad \Delta \mapsto K(\sqrt[n]{\Delta}).$$

And  $\Delta/(K^\times)^n$  is isomorphic to  $\text{Gal}(L/K)^\vee = \text{Hom}(\text{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$ .  $\square$

*Proof:* The Galois cohomology of the Kummer sequence  $1 \rightarrow \mu_n \rightarrow L^\times \xrightarrow{n} (L^\times)^n \rightarrow 0$  says

$$1 \rightarrow K^\times \xrightarrow{n} (L^\times)^n \cap K^\times \xrightarrow{\delta} H^1(\text{Gal}(L/K), \mu_n) \rightarrow H^1(\text{Gal}(L/K), L^\times) = 1 \text{ (9.8.3.2)}.$$

And  $\text{Gal}(L/K)$  acts trivially on  $\mu_n \subset K^*$ , so

$$H^1(\text{Gal}(L/K), \mu_n) = \text{Hom}(\text{Gal}(L/K), \mathbb{Q}/\mathbb{Z}),$$

and

$$\delta : a \mapsto \chi_a(\sigma) = \sigma(\sqrt[n]{a})/\sqrt[n]{a} \in \mu_n.$$

Now let  $L/K \in \mathbf{GalField}$  be the maximal Kummer extension of exponent  $n$ , then  $(L^\times)^n = K^\times$ . So the assertion follows from Galois theory on applied to  $L/K$ .  $\square$

**Thm. (3.2.8.16)[Artin-Schreier Theory].** Let  $K \in p\text{-Field}$  containing  $\mathbb{F}_q$ , where  $q = p^k$ , an **Artin-Schreier extension**  $L/K$  of exponent  $q$  is a Galois extension that the Galois group is Abelian of exponent  $q$ .

Then there is an inclusion preserving isomorphism between the lattice of Artin-Schreier extensions  $L$  over  $K$  of exponent  $q$  and the lattice of subgroups of  $W_{p,k}^+(K)$  containing  $(W_{p,k}^+(\text{Frob}_p^k) - \text{id})(W_{p,k}^+(K))$ :

$$L \mapsto \Delta = (W_{p,k}^+(\text{Frob}_p^k) - \text{id})(W_{p,k}^+(L)) \cap W_{p,k}^+(K),$$

$$\Delta \mapsto K(\{b_{i1}, \dots, b_{ik} | b_i = (b_{i1}, \dots, b_{ik}), W_{p,k}^+(\text{Frob}_p^k)(b) - b \in \Delta\}).$$

And  $\Delta/(W_{p,k}^+(\text{Frob}_p^k) - \text{id})(W_{p,k}^+(K))$  is isomorphic to  $\text{Gal}(L/K)^\vee = \text{Hom}(\text{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$ .  $\square$

*Proof:* The Galois cohomology of this exact sequence

$$0 \rightarrow W_{p,k}^+(\mathbb{F}_q) \rightarrow W_{p,k}^+(L) \xrightarrow{W_{p,k}^+(\text{Frob}_p^k) - \text{id}} (W_{p,k}^+(\text{Frob}_p^k) - \text{id})(W_{p,k}^+(L)) \rightarrow 0,$$

says

$$1 \rightarrow W_{p,k}^+(K) \xrightarrow{W_{p,k}^+(\text{Frob}_p^k) - \text{id}} (W_{p,k}^+(\text{Frob}_p^k) - \text{id})(W_{p,k}^+(L)) \cap W_{p,k}^+(K)$$

$$\xrightarrow{\delta} H^1(\text{Gal}(L/K), W_{p,k}^+(\mathbb{F}_q)) \rightarrow H^1(\text{Gal}(L/K), W_{p,k}^+(L)) = 0 \text{ (9.8.3.3)}$$

And  $\text{Gal}(L/K)$  acts trivially on  $W_{p,k}^+(\mathbb{F}_q)$ , so by (5.5.3.14),

$$H^1(\text{Gal}(L/K), W_{p,k}^+(\mathbb{F}_q)) = \text{Hom}(\text{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$$

and

$$\delta : a \mapsto \chi_a(\sigma) = \sigma(b) - b \in W_{p,k}^+(\mathbb{F}_q), \quad W_{p,k}^+(\text{Frob}_p^k)(b) - b = a.$$

Next we prove for any  $b = (b_1, \dots, b_k)$ ,  $W_{p,k}^+(\text{Frob}_p^k)(b) - b \in W_{p,k}^+(K)$ ,  $K(\{b_1, \dots, b_k\})/K$  is Abelian of exponent  $q$ : Let  $K_i = K(\{b_1, \dots, b_i\})$ . By (5.5.2.7),

$$(b_1^q, \dots, b_k^q) - (b_1, \dots, b_k) \in W_{p,k}^+(K)$$

implies  $b_i^q - b_i \in K_{i-1}$ . Thus  $K_i/K_{i-1}$  is an Artin-Schreier extension of exponent  $p$ . As the Galois action is always of the form  $\sigma(s) - s \in \mathbb{F}_q$  for any  $\sigma \in \text{Gal}(K_i/K)$  and  $s \in K_k$ , we see  $K_k/K$  is also Artin-Schreier of exponent  $q = p^k$ .

Now let  $L/K$  be the maximal Artin-Schreier extension of exponent  $q$ , then  $(W_{p,k}^+(\text{Frob}_p^k) - \text{id})(W_{p,k}^+(L)) = W_{p,k}^+(K)$ . So the assertion follows from Galois theory on applied to  $L/K$ .  $\square$

**Cor. (3.2.8.17) [Artin-Schreier of Exponent  $p$ ].** Let  $p \in \text{Prime}$ ,  $K \in \text{Field}^p$ , then there is an inclusion preserving isomorphism between the lattice of Artin-Schreier extensions  $L$  over  $K$  of exponent  $p$  and the lattice of subgroups of  $K^+$  containing  $\{x^p - x | x \in K\}$ :

$$L \mapsto \Delta = \{x^p - x | x \in L\} \cap K,$$

$$\Delta \mapsto K(\{b | b^p - b \in \Delta\}).$$

And  $\Delta / \{x^p - x | x \in K\}$  is isomorphic to  $\text{Gal}(L/K)^\vee = \text{Hom}(\text{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$ .  $\lrcorner$

### Applications

**Thm. (3.2.8.18) [Unsolvability of the Quintic by Radicals, Abel1826-Galois1832].**  $\lrcorner$

*Proof:*  $\square$

## 9 Ordered Rings

**Def. (3.2.9.1) [Ordered Abelian Groups].** An **ordered Abelian group** is an Abelian group  $\Gamma \in \mathcal{Ab}$  together with a subset  $P \subset \Gamma$  s.t.

- $R = P \coprod \{0\} \coprod (-P)$  is a disjoint union.
- If  $x, y \in P$ , then  $x + y \in P$ .

Elements in  $P$  are called **positive elements**.  $\lrcorner$

**Def. (3.2.9.2) [Ordered Rings].** An **ordered ring** is a commutative ring  $R \in \mathcal{CRing}$  together with a subset  $P \subset R$  s.t.

- $R = P \coprod \{0\} \coprod (-P)$  is a disjoint union.
- If  $x, y \in P$ , then  $x + y, xy \in P$ .

Elements in  $P$  are called **positive elements**.

An **ordered field** is an ordered ring that is also a field. An **orderable ring** (resp. orderable field) is a ring (resp. field) that can be given an ordered structure.  $\lrcorner$

**Prop. (3.2.9.3).** An orderable field  $K$  (3.2.9.2) satisfies  $\text{char } K = 0$ , because  $0 \notin P \cup (-P)$ .

For any orderable field  $K$  and  $x \in K$ , we have  $x^2 \in P$ .  $\lrcorner$

**Def. (3.2.9.4) [Convex Subgroups].** Let  $\Gamma$  be an ordered Abelian group (3.2.9.1), then a **convex subgroup** of  $A$  is a subgroup  $\Delta$  that if  $a < b < c$  and  $a, c \in \Delta$ , then  $b \in \Delta$ . Notice this is in fact equivalent to if  $0 < c \in \Delta$ , then  $0 < b < c$  are also in  $\Delta$ .  $\lrcorner$

**Prop. (3.2.9.5) [Height].** Let  $\Gamma$  be an ordered Abelian group (3.2.9.1), then the set of all convex subgroups of  $\Gamma$  is well-ordered, and its ordinal is called the **height** of  $\Gamma$ .  $\lrcorner$

*Proof:* If  $\Delta_1, \Delta_2$  don't contain each other, let  $a \in \Delta_1 - \Delta_2$  and  $b \in \Delta_2 - \Delta_1$ , then changing  $\pm a, \pm b$ , we may assume  $0 < a < b$ , so  $a \in \Delta_2$ , contradiction.  $\square$

**Prop. (3.2.9.6) [Height 1 Case].** Let  $\Gamma$  be an ordered Abelian group (3.2.9.1), then the following are equivalent:

1.  $\text{ht}(\Gamma) = 1$ .
2. for all  $a, b \in \Gamma$  that  $a > 0$  and  $b \geq 0$ , there is an integer  $n$  that  $b \leq na$ .
3. there exists an ordered injection  $\Gamma \hookrightarrow (\mathbb{R}, <)$ .

$\lrcorner$

*Proof:*  $3 \rightarrow 1$  is easy.

$1 \rightarrow 2$ : Consider the convex subgroup generated by  $a$ , then it is  $\Delta$  by height condition, so  $b$  must be in it, i.e.  $b \leq na$  for some  $n$ .

$2 \rightarrow 3$ : Choose an  $a > 0$ , let the injection  $\varphi$  given by  $\varphi(b) = \sup\{\frac{n}{k} | na \leq kb\}$  for  $b > 0$  and extends to negative elements.

It is easily verified that  $\varphi(c) + \varphi(b) \leq \varphi(c + b)$ , and if  $\varphi(c) + \varphi(b) < \varphi(c + b)$ , choose a rational approximation of them, and multiply to get integers, then if  $k(c + b) \leq \varphi(c + b)ka > \varphi(b)a + \varphi(c)a + a$ , then either  $kc \geq \varphi(c)ka + a$  or  $kb \geq \varphi(b)ka + a$ , contradiction.

So this map is truly a morphism of ordered Abelian groups, and it is injective because if  $b > 0$ , then by 2, there must be an  $n$  that  $a \leq nb$ , so  $\varphi(b) \geq 1/n$ .  $\square$

## 10 Real Fields

**Def. (3.2.10.1) [Real Fields].** A field  $K$  is called **real** if  $-1$  is not a sum of squares in  $K$ . A field  $K$  is called **real closed** iff it is real, and any alg. extension that is real must be itself. An ordered field is clearly a real field by (3.2.9.3), the converse is in fact true, by (3.2.10.11). In particular, a real field is of characteristic 0.  $\lrcorner$

**Prop. (3.2.10.2).** If  $K$  is real,  $a \in K$ ,  $a$  and  $-a$  cannot both be sum of squares. If  $-a$  is not a sum of squares in  $K$ , then  $K(\sqrt{a})$  is real. Hence either  $K(\sqrt{a})$  or  $K(\sqrt{-a})$  is real.  $\lrcorner$

*Proof:* Suppose  $K(\sqrt{a})$  is not real. If  $a$  is a square, then  $K(\sqrt{a}) = K$  is real. So  $a$  is not a square,

$$-1 = \sum (b_i + c_i \sqrt{a})^2 = \sum (b_i^2 + ac_i^2 + 2b_i c_i \sqrt{a})$$

Since  $\sqrt{a} \notin K$ ,  $-1 = \sum (b_i^2 + ac_i^2)$ , so

$$-a = \frac{1 + \sum b_i^2}{\sum c_i^2}$$

This implies that  $-a$  is a sum of squares.  $\square$

**Prop. (3.2.10.3).** If the minimal polynomial  $f$  of an  $\alpha$  algebraic over a real field  $K$  is of odd degree, then  $K(\alpha)$  is real.  $\lrcorner$

*Proof:* If  $K(a)$  is not real, then  $-1 = \sum g_i(X)^2 + h(X)f(X)$ , where  $g_i$  has degree smaller than  $n$ . This can happen if  $h(X)$  has degree odd and  $\leq n-2$ . Then if  $\beta$  is a root of  $h$ , then  $K(\beta)$  is also not real. So the proof is finished if we use induction.  $\square$

**Def. (3.2.10.4) [Real Closure Exists].** For any real field  $K$ , there exists a **real closure**  $K^a$  of  $K$ . That is, it is real closed and algebraic over  $K$ .  $\lrcorner$

*Proof:* This is an easy consequence of Zorn's lemma.  $\square$

**Cor. (3.2.10.5) [Real Closed Fields Unique Ordering].** There exists a unique ordering on a real closed field  $R$ . The elements  $> 0$  are just the squares in  $R$ . Now every real closed field is assumed to have this ordering tacitly. In particular, any real closed field has  $\text{char} 0$ , so does any real field.  $\lrcorner$

*Proof:* The set of finite sum of squares in  $R$  is closed under addition and multiplication, and all of them are squares, by (3.2.10.2) and maximality of  $R$ . Also by (3.2.10.2) either  $a$  is a square or  $-a$  is a square, but not simultaneously. So it is truly an order on  $R$ .  $\square$

**Prop. (3.2.10.6) [Fundamental Theorem of Algebra].** For  $R \in \mathbf{Field}$ ,  $R$  is real closed iff  $R \neq R[\sqrt{-1}]$  and  $\overline{R} = R(\sqrt{-1})$ .  $\lrcorner$

*Proof:* One direction is trivial, the other follows from the lemma below (3.2.10.7), it satisfies the condition by (3.2.10.2) and maximality.  $\square$

**Lemma (3.2.10.7) [Equivalent Definition of Real Closed Fields].** If  $R$  is a real field that: for all  $a \in R$ ,  $\sqrt{a} \in R$  or  $\sqrt{-a} \in R$ , and any polynomial of odd degree has a root in  $R$ , then  $K = R(\sqrt{-1})$  is alg.closed.  $\lrcorner$

*Proof:* For any order of  $R$ , the first condition in fact says that any  $a > 0$  in  $R$  is a square. Now  $\frac{a+\sqrt{a^2+b^2}}{2}$  is non-negative, so there is a  $c^2 = \frac{a+\sqrt{a^2+b^2}}{2}$ , that is  $(c + \frac{b}{2c}i)^2 = a + bi$ , so  $K$  has all squares.

As  $R$  is of  $\text{char} 0$  (3.2.10.5) (3.2.9.3), so it suffices to show any Galois extension  $L/K$  is trivial. Let  $G = G(L/R)$ , and  $H$  be its 2-Sylow subgroup, then  $G = H$  by condition. Now if  $G_1 = G(L/K)$ , then  $G_1$  is nontrivial, because otherwise there is a subgroup of index 2, then its fixed field is a square extension of  $K$ , which is impossible by what we have proved. So  $G = G_1$ , that is  $L = K$ .  $\square$

**Cor. (3.2.10.8) [Complex Numbers is Alg.Closed, Gauss1799].**  $\mathbb{C} = \mathbb{R}[i]$  is alg.closed.  $\lrcorner$

**Prop. (3.2.10.9) [Intermediate Property].** An ordered field is real closed iff it has the intermediate property.  $\lrcorner$

*Proof:* If  $R$  is real closed, as  $R[i]$  is alg.closed (3.2.10.6),  $f$  can be decomposed into factors of degree 1 or 2. For a factor  $X^2 + \alpha X + \beta$ ,  $4\beta > \alpha^2$ , otherwise it has a root hence not irreducible. So the change of sign is because of a linear factor, the rest is easy.

Conversely, if it has the intermediate property, then for  $a > 0$ , consider  $p(X) = X^2 - a$ , then  $p(0) < 0, p(a+1) > 0$ , so  $p$  has a root, that is,  $a$  is a square. For a polynomial of odd degree, for  $M$  large enough,  $f(M) > 0, f(-M) < 0$ , so  $f$  has a root. So by (3.2.10.7)  $R$  is real closed.  $\square$

**Prop. (3.2.10.10) [Artin-Schreier].** If  $K \in \mathbf{Field}$ ,  $K = K^{\text{sep}}$ , and  $F \subset_{\text{fin}} K$ , then  $F = K$  or  $F$  is real closed and  $K = F(\sqrt{-1})$ .  $\lrcorner$

*Proof:* Cf. [Lan02] P299.  $\square$

### Real Fields and Order

**Prop. (3.2.10.11) [Real Field and Order].** If  $R$  is a real field, then it is orderable, in fact, if  $-a$  is not a sum of squares in  $F$ , then there is an ordering that  $a > 0$ . So a real field is equivalent to an orderable field.  $\lrcorner$

*Proof:* By (3.2.10.2),  $F(\sqrt{a})$  is real, so it has a real closure (3.2.10.4) and has the induced order (3.2.10.5), and  $a > 0$  because it is a square (3.2.9.3).  $\square$

**Prop. (3.2.10.12) [Existence and Uniqueness of Real Closure].** For any ordered field  $F$ , there is a unique real closure  $R$  of  $F$  that every positive element of  $F$  is a square in  $R$ , thus the ordering is compatible.  $\lrcorner$

*Proof:* The existence is by adding all the square roots of elements  $> 0$  to  $F$ , the resulting field is real closed because of (3.2.10.2) and the fact a union of real fields (3.2.10.2) is real.

The uniqueness: because an ordered field is of char 0 (3.2.9.3), so the primitive element theorem (3.2.6.18) applies that each finite subextension of  $R_0$  is of the form  $F(\alpha)$ , where  $\alpha$  is a root of an irreducible separable polynomial  $f$ . Then the roots of  $f$  are different so can be ordered  $\alpha_1 < \dots < \alpha_n$ . Similarly,  $f$  has the same number of different roots in  $R_1$   $\beta_1 < \dots < \beta_n$  by (3.2.10.14), so there is a map  $h : \alpha_i \rightarrow \beta_i$ , and it is the unique map that a ordered map from  $F(\alpha)$  to  $R_1$  extending id on  $F$  can be. it is this uniqueness that makes us able to use Zorn's lemma to show that there is a maximal ordered map, must be a map from  $R_0$  to  $R_1$ , which is an isomorphism, by primitive element theorem again.  $\square$

**Prop. (3.2.10.13) [Sturm's Algorithm].** Cf. [Model Theory Marker P327].  $\lrcorner$

**Cor. (3.2.10.14).** If  $F$  is an ordered field and  $R_0, R_1$  be two real closure of  $F$  that is compatible with the ordering, then any irreducible polynomial has the same number of roots in  $R_0$  and  $R_1$ .  $\lrcorner$

*Proof:* Cf. [Model Theory Marker P328].  $\square$

**Prop. (3.2.10.15) [Hilbert's 17th Problem].** If  $f$  is a positive semidefinite rational function over a real closed field  $F$ , then  $f$  is a sum of squares of rational functions.  $\lrcorner$

*Proof:* Let  $f(X_1, \dots, X_n)$  be a positive semidefinite rational function, if  $f$  is not a sum of squares of rational functions, then by (3.2.10.11), there is an ordering on  $F(\overline{X})$  that  $f < 0$ . Let  $R$  be a real closure of  $F(\overline{X})$ , then  $R \models \exists \bar{v} F(\bar{v}) < 0$ , as  $F(\overline{X}) < 0$ . But  $RCR$  is complete, by (2.4.5.6), thus  $F \models \exists \bar{v} F(\bar{v}) < 0$  also, contradiction.  $\square$

### 3.3 Polynomials

*“You probably think that one knows everything about polynomials.”*

–S. Lang

#### 1 Basics

**Prop. (3.3.1.1)[Descartes’s Rule of Sign].** Let  $p(x) = a_0x^{b_0} + a_1x^{b_1} + \dots + a_nx^{b_n}$  be a real polynomial with nonzero  $a_i$ , where  $A_0 < B_1 < \dots < b_n$ , then the number of positive roots of  $p(x)$  has the same parity with the number of consecutive changes of signs of  $(a_k)_{k=0,\dots,n}$ .  $\square$

*Proof:* Lemma: when  $a_0a_n > 0$ , the number of positive roots are even and when  $a_0a_n < 0$ , it is odd. This is seen by consider  $p(0)$  and  $p(\infty)$ .

Then we consider the derivative  $p'$  and use induction. Denote the number of changing sign by  $v(p)$  and the number of positive roots by  $z(p)$ , then if  $z_0a_1 > 0$ , then  $v(p) = v(p')$  and  $z(p) \equiv z(p') \pmod{2}$ . Then we have  $z(p) \equiv v(p) \pmod{2}$  and middle value theorem shows that  $z(p') \leq z(p) - 1$ , hence by induction and parity argument, we have  $v(p) \geq z(p)$ .

If  $a_0a_1 < 0$ , then the same method shows that  $v(p) = v(p') + 1 \geq z(p') + 1 \geq z(p)$  and the have the same parity by the lemma.  $\square$

**Prop. (3.3.1.2)[Lagrange Interpolation].** If  $K \in \mathbf{Field}$  and  $a_i$  are  $n + 1$  elements of  $K$ ,  $b_i$  are  $n + 1$  elements of  $K$ , then there is a unique polynomial  $f$  of degree no greater than  $n$  that  $f(a_i) = b_i$ .  $\square$

*Proof:* The polynomial in search is

$$f(X) = \sum \prod_{j \neq i} b_i \frac{X - a_j}{a_i - a_j}.$$

It is a polynomial of degree smaller than  $n + 1$ , and it satisfies the hypothesis. And clearly there is at most one such polynomial, otherwise their difference has  $n + 1$  zeros.  $\square$

**Cor. (3.3.1.3).** If  $f(X) = a_nX^n + \dots + a_0$ , then for any  $n + 1$  different integers  $a_0, \dots, a_n$ , there exists some  $|f(a_i)| \geq \frac{n!}{2^n} |a_n|$ .  $\square$

*Proof:* Use Lagrange interpolation and consider the leading coefficient.  $\square$

**Prop. (3.3.1.4).** If a degree  $n$  polynomial  $p$  satisfies  $p(n) = 2^n$  for  $n = 0, 1, \dots, n$ , then  $p(n + 1) = 2^{n+1} - 1$ .  $\square$

*Proof:* The polynomial in search is  $p(x) = \sum_{k=0}^n \binom{x}{k}$ .  $\square$

#### Irreducibility

**Prop. (3.3.1.5).** If  $p \in \mathbf{Prime}$  and  $f(x) = a_nx^n + \dots + a_1x + p \in \mathbb{Z}[X]$  satisfies  $\sum |a_i| < p$ , then  $f$  is irreducible in  $\mathbb{Z}[X]$ .  $\square$

*Proof:* Firstly, all of its roots has norm bigger than 1, because otherwise  $p = |\sum a_k x^k| \leq \sum |a_k| < p$ , contradiction. So if now  $f = gh$ , then  $g, h$  all have roots with norm greater than 1, in particular their constant coefficients have norm greater than 1, which is a contradiction because  $p$  is a prime.  $\square$

**Prop. (3.3.1.6) [Irreducible Reciprocal Polynomials].** Let  $K \subset \mathbb{R} \in \mathbf{Field}$  and  $f(X) \in K[X]^\times$  be an irreducible polynomial with a root  $\alpha \in \mathbb{C}$  s.t.  $|\alpha| = 1$  but  $\alpha \neq 1$ , then  $f(X)$  is reciprocal, i.e.  $f(X) = X^{\deg f} f(1/X)$ .  $\lrcorner$

*Proof:* If  $\alpha = -1$ , then  $f(X) = X + 1$  is reciprocal. If  $\alpha \neq -1$ , then  $\alpha^{-1} = \bar{\alpha} \neq \alpha$  is also a root of  $f(X)$ . But  $\alpha^{-1}$  is also a root of  $X^{\deg f} f(1/X)$  which has the same degree as  $f(X)$ , thus  $f(X) = X^{\deg f} f(1/X)$  because  $f(X)$  is irreducible.  $\square$

### Resultants

**Def. (3.3.1.7) [Resultants].** Let  $R \in \mathcal{CRing}$ , the **resultant**  $\text{res}(A, B)$  of two polynomials  $A, B \in R[X]$  of degree  $d, e$  respectively is the determinant of the linear map

$$W_e \times W_d \rightarrow W_{d+e} : (X, Y) \mapsto AX + BY,$$

where  $W_t$  is the free module of polynomials of degree  $< t$ .  $\lrcorner$

**Prop. (3.3.1.8).** The resultant can be seen as the determinant of the matrix with values the coefficient of  $A$  or  $B$  in different places, multiplying  $X$ 's with different degree and add to the last row, we can get  $A \cdot X$ 's and  $B \cdot X$ 's, so:  $\text{res}(A, B) = AC + BD$  for some  $C, D$ .

Now if  $R \subset S$  and  $A, B$  has common roots in  $S$ , then  $\text{res}(A, B) = 0$ .  $\lrcorner$

**Cor. (3.3.1.9).** Resultant is stable under Euclidean division, so it can be seen as a suitable division remainder of the two polynomial.  $\lrcorner$

**Prop. (3.3.1.10) [Resultants Detects Coprime Polynomials].** Let  $R \subset L \in \mathcal{CRing}$  and  $L \in \mathbf{Field}$ . If  $A, B \in R[X]$  decompose into linear factors in  $L$ , let  $t_i$  be roots of  $A$  and  $u_j$  be roots of  $B$ , then

$$\text{res}(A, B) = v_0^d w_0^e \prod_{i=1}^d \prod_{j=0}^e (t_i - u_j)$$

In particular,  $\text{res}(A, B) \neq 0$  iff  $A, B \in L[X]$  are coprime.  $\lrcorner$

*Proof:* See the resultant as polynomials of the roots of  $A$  and  $B$ , then we proved that if they has the same root, then  $\text{res} = 0$ , so it is divisible by  $(t_i - u_j)$  for all  $i, j$ . Then notice the RHS is homogenous of degree  $d$  in  $u_j$  and homogenous of degree  $e$  in  $t_i$ , so does  $\text{res}$ . So they are equal.  $\square$

**Prop. (3.3.1.11).** Let  $k \in \mathbf{Field}$  and  $P \in k[X]$ , then  $x \in k$  is a double root of  $P$  iff  $\gcd(P, P')(x) = 0$ .  $\lrcorner$

*Proof:*  $\square$

## 2 Cyclotomic Polynomials

Cf. [Cyclotomic Polynomials in Olympiad Number Theory].

**Def. (3.3.2.1) [Cyclotomic Polynomials].** For  $n \in \mathbb{Z}_+$ , define the **cyclotomic polynomial**

$$\Psi_n(X) = \prod_{a \in (\mathbb{Z}/(n))^*} (X - e^{2\pi i \frac{a}{n}}) \in \mathbb{C}[X].$$

Then in fact  $\Psi_n(X) \in \mathbb{Z}[X]$ .

Sometimes we also consider the homogenized version  $\Phi_n(X, Y)$  of  $\Psi_n(X)$ .  $\lrcorner$

*Proof:* Its coefficients are algebraic integers, and it is invariant under action of  $\text{Gal}_{\mathbb{Q}}$ , so its coefficients are in  $\mathbb{Q} \cap \mathcal{O}_{\overline{\mathbb{Q}}} = \mathbb{Z}$ .  $\square$

**Prop. (3.3.2.2) [Cyclotomic Polynomials are Irreducible].** For any  $n \in \mathbb{Z}_+$ , the cyclotomic polynomial  $\Psi_n(x)$  is irreducible over  $\mathbb{Z}$ .  $\lrcorner$

*Proof:* It suffices to show that for any irreducible factor  $f|\Psi_n(x)$ , if  $\xi$  is a root of  $f$  and  $(p, n) = 1$ , then  $\xi^p$  is also a root of  $f$ . Cf. [Lan02].  $\textcolor{red}{?}$   $\square$

**Prop. (3.3.2.3).** For  $a, n \in \mathbf{P}$  and  $(a, n) = 1$ ,  $\Psi_n(X^p) = \prod_{d|a} \Psi_{dn}(X)$ .  $\lrcorner$

*Proof:* This follows from counting.  $\square$

**Cor. (3.3.2.4).** For  $n \in \mathbb{Z}_+$ ,  $X^n - 1 = \prod_{d|n} \Psi_d(X)$ . Thus by Möbius inversion,  $\Psi_n(x) = \prod_{d|n} (X^d - 1)^{\mu(n/d)}$ .  $\lrcorner$

**Prop. (3.3.2.5).** For  $n \in \mathbb{Z}_+$  and  $p \in \mathbf{P}$ ,

$$\Psi_{np}(X) = \begin{cases} \Psi_n(X^p) & , p|n \\ \frac{\Psi_n(X^p)}{\Psi_n(X)} & , p \nmid n \end{cases}.$$

$\lrcorner$

*Proof:* These follows from counting.  $\square$

**Cor. (3.3.2.6).** If  $n \in \mathbb{Z}_{\geq 3}$  is odd, then  $\Psi_{2n}(X) = \Psi_n(-X)$ .  $\lrcorner$

*Proof:* It suffices to show that  $\Psi_{2n}(X^2) = \Psi_n(X)\Psi_n(-X)$ . And this is because  $\zeta \mapsto \zeta^2$  is an automorphism between primitive  $n$ -th roots, and  $\varphi(n)$  is even.  $\square$

**Prop. (3.3.2.7).** For  $m \neq n \in \mathbb{Z}_+$ ,  $\Psi_m(X)$  and  $\Psi_n(X)$  are coprime in  $\mathbb{Q}(X)$ . And if  $m, n \in \mathbb{Z}_+ \setminus (p)$ , then  $\Psi_m(X)$  and  $\Psi_n(X)$  are also coprime in  $\mathbb{F}_p[X]$ .  $\lrcorner$

*Proof:* It suffices to show that  $X^{mn} - 1$  doesn't have multiple roots. By (3.3.1.11), it suffices to notice that

$$\gcd(X^{mn} - 1, mnX^{mn-1}) = 1,$$

which is also true in  $\mathbb{F}_p[X]$  if  $m, n \in \mathbb{Z}_+ \setminus (p)$ .  $\square$

**Cor. (3.3.2.8).** For  $p \in \mathbf{P}$  and  $n \in \mathbb{Z}_+ \setminus (p)$ ,  $a \in \mathbb{Z}$ ,

$$p|\Psi_n(a) \iff p \nmid a \quad \& \quad \text{ord}(a, \mathbb{F}_p^\times) = n.$$

$\lrcorner$

**Cor. (3.3.2.9).** For  $p \in \mathbf{P}$  and  $n \in \mathbb{Z}_+ \setminus (p)$ ,  $p|\Psi_n(a)$  for some  $a \in \mathbb{Z}$  iff  $p \equiv 1 \pmod{n}$ .  $\lrcorner$

**Prop. (3.3.2.10).** If  $m \geq n \in \mathbb{Z}_+$ ,  $h \in \mathbb{Z}$ , and  $A = \gcd(\Psi_n(h), \Psi_m(h)) \neq 1$ , then there exists  $p \in \mathbf{P}$  s.t.  $A \in p^{\mathbb{Z}_+}$ , and also  $\frac{m}{n} \in p^{\mathbb{Z}_+}$ .  $\lrcorner$

*Proof:* Suppose  $p|A$ , and suppose  $m = p^a c, n = p^b d, p \nmid cd$ , then by (3.3.2.5),

$$\Psi_m(h) \equiv (\Psi_c(h))^{p^a} \pmod{p}, \quad \Psi_n(h) \equiv (\Psi_d(h))^{p^b} \pmod{p}.$$

Then it follows from (3.3.2.7) that  $c = d$ . Thus  $\frac{m}{n} \in p^{\mathbb{Z}_+}$ . The assertions follow easily from this.  $\square$



**Prop. (3.3.2.11).** For  $n \in \mathbb{Z}_+$ ,

$$\Psi_n(1) = \begin{cases} p & , n \in p^{\mathbb{Z}_+}, p \in \mathbf{P} \\ 1 & , \text{otherwise} \end{cases}$$

┘

*Proof:* This follows easily from induction and the fact

$$X^{n-1} + \dots + X + 1 = \prod_{d|n, d \neq 1} \Psi_d(X).$$

□

**Prop. (3.3.2.12) [Lower Bounds on Cyclotomic Polynomials].** there exists  $c \in \mathbb{R}_+$  s.t. for any  $n \in \mathbb{Z}_+$ , and  $a > b \in \mathbb{Z}_+$ ,

$$|\Phi_n(a, b)| \geq e^{cn}.$$

┘

*Proof:*  $\Phi_n(a, b) = \prod (a - \zeta b)$ , and  $|a - \zeta b| > 1$ . moreover, notice that for  $n$  large, nearly half of the primitive roots of unity  $\zeta$  will have negative real parts (Cf. [The Distribution of Totatives, Lehmer]), and for such  $\zeta$ ,  $|a - \zeta b| \geq \sqrt{5}$ . □

### 3 Stable Polynomials

**Def. (3.3.3.1) [Stable Polynomials].** For  $n \in \mathbb{Z}_+$ , a **real-stable polynomial**  $f \in \mathbb{C}[z_1, \dots, z_n]$  is either a zero polynomial or it satisfies that

$$f(z_1, \dots, z_n) \neq 0$$

for any  $z_1, \dots, z_n \in \mathcal{H}$ . ┘

**Cor. (3.3.3.2).** A real stable univariate polynomial is real-rooted. ┘

**Prop. (3.3.3.3) [Wagner].** If  $n \in \mathbb{Z}_+$  and  $f \in \mathbb{R}[z_1, \dots, z_n]$  is real stable, then for any  $c \in \mathbb{R}$ ,  $f(z_1, \dots, z_{n-1}, c)$  is also real stable. ┘

*Proof:* Cf. [Multivariate stable polynomials: theory and applications] Lemma 2.4. ? □

**Prop. (3.3.3.4) [Borcea-Brändén].** Let  $m, n \in \mathbb{Z}_+$  and  $A_1, \dots, A_m \in \text{Pos}_{\geq 0}(n; \mathbb{C})$ , then

$$f(z_1, \dots, z_n) = \det\left(\sum z_i A_i\right)$$

is a real stable polynomial (3.3.3.1). ┘

*Proof:* ? Cf. [Applications of stable polynomials to mixed determinants: Johnson's conjectures, unimodality, and symmetrized Fischer products] Prop 2.4.. □

**Prop. (3.3.3.5) [Borcea-Brändén].** Let  $n \in \mathbb{Z}_+$  and  $T : \mathbb{R}[z_1, \dots, z_n] \rightarrow \mathbb{R}[z_1, \dots, z_n]$  be a differential operator of the form

$$T = \sum_{\alpha, \beta \in \mathbb{N}^n} c_{\alpha, \beta} z^\alpha \partial^\beta,$$

and define

$$f_T(\mathbf{z}, \mathbf{w}) = \sum_{\alpha, \beta \in \mathbb{N}^n} c_{\alpha, \beta} z^\alpha w^\beta.$$

Then  $T$  preserves real-stable polynomials (3.3.3.1) iff  $F_T(\mathbf{z}, -\mathbf{w})$  is real-stable. ┘

*Proof:* ? Cf.[Multivariate Polya-Schur classification problems in the Weyl algebra]Thm1.3.  $\square$

**Cor. (3.3.3.6).** For any  $a, b \in \mathbb{R}_{\geq 0}$ , the operators  $T = 1 + a\partial_x$  or  $T = 1 + a\partial_x + b\partial_y$  preserves real-stable polynomials.  $\lrcorner$

### Mixed Characteristic Polynomials

**Def. (3.3.3.7) [Mixed Characteristic Polynomials].** For  $m, d \in \mathbb{Z}_+$  and  $A_1, \dots, A_m \in \text{Mat}(n; \mathbb{C})$ , the **mixed characteristic polynomial** of  $A_1, \dots, A_m$  is defined to be

$$\mu[A_1, \dots, A_m](X) = \left[ \prod_i (1 - \partial_{z_i}) \det \left( X\mathbf{1} + \sum_i z_i A_i \right) \right] \Big|_{z_1=\dots=z_m=0}.$$

It is a monic polynomial.  $\lrcorner$

**Thm. (3.3.3.8) [Mixed Characteristic Polynomials are Real-Rooted].** For  $m, d \in \mathbb{Z}_+$  and  $A_1, \dots, A_m \in \text{Pos}_{\geq 0}(n; \mathbb{C})$ , the mixed characteristic polynomial  $\mu[A_1, \dots, A_m]$  is monic and real-rooted.  $\lrcorner$

*Proof:* It follows from (3.3.3.4)(3.3.3.6) and (3.3.3.3) that  $\mu[A_1, \dots, A_m]$  is real stable. Then it is real-rooted by (3.3.3.2).  $\square$

**Thm. (3.3.3.9) [Marcus-Spielman-Srivastava].** Let  $m, d \in \mathbb{Z}_+$  and  $v_1, \dots, v_m$  be independent random column vectors in  $\mathbb{C}^d$  with finite support, and let  $A_i = \mathbf{E} v_i v_i^*$ , then

$$\mathbf{E} \det \left( X\mathbf{1} - \sum_i v_i v_i^* \right) = \mu[A_1, \dots, A_m](X) \quad (3.3.3.7).$$

$\lrcorner$

*Proof:* Using induction on  $m$ , it suffices to show that for any  $A \in \text{Mat}(d; \mathbb{C})$ ,

$$\mathbf{E} \det(A - vv^*) = \left[ (1 - \partial_t) \det(A + t \mathbf{E} vv^*) \right] \Big|_{t=0}.$$

If  $A$  is invertible, then by (3.5.9.12), the LHS equals

$$\mathbf{E} \det(A) (1 - v^* A^{-1} v) = \det(A) \mathbf{E} (1 - \text{tr}(A v^* v)) = \det(A) - \det(A) \text{tr}(A^{-1} \mathbf{E} vv^*).$$

And this equals the RHS by (3.5.9.25).

The case that  $A$  being non-invertible follows from continuity.  $\square$

**Cor. (3.3.3.10).** Let  $m, n \in \mathbb{Z}_+$  and  $u_1, \dots, u_m, v_1, \dots, v_m \in \mathbb{C}^n$ ,  $D \in \text{Pos}(n; \mathbb{C})$ , and  $p_1, \dots, p_m \in [0, 1]$ , then the univariate polynomial

$$P(X) = \sum_{S \subset [m]_+} \prod_{i \in S} p_i \prod_{i \notin S} (1 - p_i) \det \left( X\mathbf{1} + D + \sum_{i \in S} u_i u_i^* + \sum_{i \notin S} v_i v_i^* \right)$$

is real-rooted.  $\lrcorner$

*Proof:* This follows from (3.3.3.9) with binomial distributions, and also the fact that  $D$  is a sum of rank-1 matrices.  $\square$

### Interlacing Polynomials

**Def. (3.3.3.11) [Interlacing Sequences].** For  $n \in \mathbb{Z}_+$  and  $A = \{\alpha_1 \leq \alpha_2 \leq \dots \alpha_n\} \in \mathbb{R}$  and  $B = \{\beta_1 \leq \beta_2 \leq \dots \leq \beta_{n+1}\} \in \mathbb{R}$ ,  $A$  is said to interlace  $B$  iff

$$\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2 \leq \dots \leq \alpha_n \leq \beta_{n+1}.$$

┘

**Def. (3.3.3.12) [Interlacing Polynomials].** For two real-rooted monic polynomials  $f(X), g(X) \in \mathbb{R}[X]$  s.t.  $\deg(f) = \deg(g) + 1$ ,  $g$  is said to **interlace the polynomial**  $f$  iff the roots of  $g$  interlaces the roots of  $f$  (3.3.3.11). ┘

**Prop. (3.3.3.13).** Let  $k \in \mathbb{Z}_+$  and  $f_1, \dots, f_k$  be real-rooted polynomials of the same degree, then they have a common interlacing polynomial (3.3.3.12) iff for any  $t \in \mathbb{R}$ ,  $|\# \text{Zero}(f) \cap [t, \infty) - \# \text{Zero}(g) \cap [t, \infty)| \leq 1$ . ┘

**Thm. (3.3.3.14) [Chudnovsky-Seymour].** Let  $k \in \mathbb{Z}_+$  and  $f_1, \dots, f_k \in \mathbb{R}[X]$  be monic real-rooted polynomials of the same degree, then the following are equivalent:

- $\sum_i \lambda_i f_i$  is real-rooted for any  $\lambda_1, \dots, \lambda_k \in \mathbb{R}_{\geq 0}$ .
- $f_1, \dots, f_k$  have a common interlacing polynomial.
- Any two  $f_i, f_j$  have a common interlacing polynomial.
- $f_i + c f_j$  is real-rooted for any  $c \in \mathbb{R}_+$ .

┘

*Proof:*  $1 \rightarrow 4$  is trivial.

$4 \rightarrow 3$ : By (3.3.3.13), it suffices to show that for any  $t \in \mathbb{R}$ ,  $|\# \text{Zero}(f_i) \cap [t, \infty) - \# \text{Zero}(f_j) \cap [t, \infty)| \leq 1$  for any  $i, j$ . We may assume that  $f_i, f_j$  have no common zeros. Suppose for the contrary that there exists a maximal  $t \in \mathbb{R}$  s.t.  $\# \text{Zero}(f_i) \cap [t, \infty) - \# \text{Zero}(f_j) \cap [t, \infty) \geq 2$ . then  $f_i(t) = 0, f_j(t) \neq 0$ . Take  $\varepsilon \in \mathbb{R}_+$  small s.t.  $f_j, f_i \neq 0$  on  $[t - \varepsilon, t)$ . Now for any  $\lambda \in [0, 1]$ , consider the number of zeros of  $F_\lambda = \lambda f_1 + (1 - \lambda) f_2$  on the region  $\{x + iy \in \mathbb{C} | x > t - \varepsilon\}$ , then it is a constant function of  $\lambda$ . But  $F_0 = f_1, F_1 = f_2$ , contradicting the assumption. So we proved that  $|\# \text{Zero}(f_i) \cap [t, \infty) - \# \text{Zero}(f_j) \cap [t, \infty)| \leq 1$  for any  $i, j$ .

$3 \rightarrow 2$ : This follows easily from (3.3.3.13).

$2 \rightarrow 1$ : This follows easily from the intermediate property. ┘

**Cor. (3.3.3.15) [Derivatives of Interlacing Polynomials].** Let  $k \in \mathbb{Z}_+$  and  $f_1, \dots, f_k \in \mathbb{R}[X]$  be monic real-rooted polynomials of the same degree with a common interlacing polynomial, then so does  $\{f_1^\lambda, \dots, f_k^\lambda\}$ . ┘

*Proof:* This follows from the fact that the derivative of a real-rooted polynomial is also real-rooted. ┘

**Def. (3.3.3.16) [Interlacing Families].** Let  $m, n \in \mathbb{Z}_+, S_1, \dots, S_m \in \text{finSet}$ , and  $\{f_{s_1, \dots, s_m}\}$  be real-rooted degree  $n$ -polynomials with positive leading coefficients indexed over  $S_1 \times \dots \times S_m$ . For any  $k \in [m - 1]_+$  and  $s_1 \in S_1, \dots, s_k \in S_k$ , define

$$f_{s_1, \dots, s_k} = \sum_{s_{k+1} \in S_{k+1}, \dots, s_m \in S_m} f_{s_1, \dots, s_k, s_{k+1}, \dots, s_m},$$

and

$$f_{\emptyset} = \sum_{s_1 \in S_1, \dots, s_m \in S_m} f_{s_1, \dots, s_m}.$$

Then  $\{f_{s_1, \dots, s_m}\}$  is called an **interlacing family of polynomials** if for any  $k \in [m-1]_+$  and  $s_1 \in S_1, \dots, s_k \in S_k$ , the family of polynomials

$$\{f_{s_1, \dots, s_k, t}\}_{t \in S_{k+1}}$$

has a common interlacing.  $\lrcorner$

**Prop. (3.3.3.17).** Let  $k \in \mathbb{Z}_+$ ,  $f_1, \dots, f_k \in \mathbb{R}[X]$  be real-rooted polynomials with positive leading coefficients such that  $\deg(f_1) = \dots = \deg(f_k)$ , and having a common interlacing polynomial (3.3.3.12), then the polynomial

$$f_{\emptyset} = \sum_i f_i.$$

is also real-rooted, and there exists  $i \in [k]_+$  s.t. the roots of  $f_i$  is bounded by the largest root of  $f_{\emptyset}$ .  $\lrcorner$

*Proof:*  $f_{\emptyset}$  is real-rooted by (3.3.3.14).

Let  $g$  interlaces each  $f_i$ , and  $\alpha_{n-1}$  the largest root of  $g$ , then  $f_i(\alpha_{n-1}) \leq 0$  for any  $i$ . So  $f_{\emptyset}(\alpha_{n-1}) \leq 0$ . But  $f_{\emptyset}$  has positive leading coefficients, so the largest root  $\beta_n$  of  $f_{\emptyset}$  satisfies  $\beta_n \geq \alpha_{n-1}$ .

But now  $f_i(\beta_n) \geq 0$  for some  $i$ . Then this  $f_i$  has a root in  $[\alpha_{n-1}, \beta_n]$ , which is its largest root. So the roots of  $f_i$  is bounded by the largest root of  $f_{\emptyset}$ .  $\square$

**Cor. (3.3.3.18).** Let  $m, n \in \mathbb{Z}_+$ ,  $S_1, \dots, S_m \in \text{finSet}$ , and  $\{f_{s_1, \dots, s_m}\}$  be an interlacing family of real polynomials, then there exists  $s_i \in S_i$  s.t. the roots of  $f_{s_1, \dots, s_m}$  is bounded by the largest root of  $f_{\emptyset}$ .  $\lrcorner$

*Proof:* This follows easily by inductively using (3.3.3.17).  $\square$

**Thm. (3.3.3.19) [Interlacing Families from Independent Random Vectors].** Let  $m, d \in \mathbb{Z}_+$  and  $v_1, \dots, v_m$  be independent random column vectors in  $\mathbb{C}^d$  with finite support, and let  $v_i$  take values  $w_{i1}, \dots, w_{il_i}$  with probabilities  $p_{i1}, \dots, p_{il_i}$  resp.. For  $j_i \in [l_i]_+$ , define

$$q_{j_1, \dots, j_m}(X) = \prod_i p_{ij_i} \det \left( X \mathbf{1} - \sum_i w_{ij_i} w_{ij_i}^* \right) \in \mathbb{C}[X],$$

then these polynomials form an interlacing family (3.3.3.16).  $\lrcorner$

*Proof:* Notations as in (3.3.3.16), it suffices to show that for any  $k \in [m]_+$  and  $j_i \in [l_i]_+$ , the family

$$\{q_{j_1, \dots, j_k, t}\}_{t \in [l_{k+1}]_+}$$

has a common interlacing. By (3.3.3.14), it suffices to show that for any  $\lambda_1, \dots, \lambda_{l_{k+1}} \in [0, 1]$  s.t.  $\sum_j \lambda_j = 1$ ,

$$\sum_{j=1}^{l_{k+1}} \lambda_j q_{j_1, \dots, j_k, t}$$

is real-rooted. But then this follows from (3.3.3.8) with a suitable choice of the probability distribution.  $\square$

## 4 Invariant Theory

**Prop. (3.3.4.1) [Elementary Symmetric Polynomial].** For  $n$  indeterminants  $x_i$ , define the **elementary polynomials**

$$\sigma_k = \sum_{1 \leq i_1 < \dots < i_k \leq n} x_{i_1} x_{i_2} \dots x_{i_k} \in \mathbb{Z}[x_1, \dots, x_n]$$

where for  $k > n$  this expression means 0. Then any symmetric polynomial is a polynomial of the fundamental symmetric polynomials.  $\lrcorner$

*Proof:* Set the first coordinate to 0, then the rest is a polynomial of the fundamental symmetric polynomials by induction,  $f(0, x) = h(\sigma_1, \dots, \sigma_n)$ , then consider  $f - h$ , with  $h$  the same expression but  $x_n$  is included, we get it is a symmetric polynomial, and it is divisible by  $x_1$ , thus also divisible by  $\prod x_i$ , thus divide it by  $\prod x_i$  and use induction, we get  $f$  is a polynomial of elementary symmetric polynomials.  $\square$

**Prop. (3.3.4.2) [Newton Identities].** For  $n$  indeterminants  $x_i$ , define

$$s_k = \begin{cases} \sum x_i^k & , k \geq 0 \\ 0 & , k < 0 \end{cases} \in \mathbb{Z}[x_1, \dots, x_n],$$

then there are **Newton Identities**:

$$s_k - \sigma_1 s_{k-1} + \dots + (-1)^n \sigma_n s_{k-n} = 0.$$

$\lrcorner$

*Proof:* The case of  $k \geq n$  is simple. Now if  $k < n$ , then we prove by induction on  $n$ : If  $n$  is already proven, then the term is 0 if one of them is 0, but this implies that this equation is divisible by  $\prod_{i=1}^{n+1} x_i$ , but it has degree  $k \leq n$ , so it must be 0.  $\square$

**Prop. (3.3.4.3) [Chern Polynomials].** By (3.3.4.2) and induction there are polynomials  $P_k \in \mathbb{Z}[x_1, \dots, x_k]$  s.t.  $P_k(\sigma_1, \dots, \sigma_k) = s_k, k \geq 0$ , called the **Chern polynomials**. Then they satisfy:

$$\log(1 + c_1 + c_2 + \dots + c_n + \dots) = \sum_{p \geq 1} (-1)^{p-1} \frac{P_p}{p} \in \mathbb{Q}[[c_1, \dots, c_n, \dots]].$$

In particular, we can easily calculate via homogeneity that

$$P_1 = c_1, \quad P_2 = c_1^2 - 2c_2, \quad P_3 = c_1^3 - 3c_1c_2 + 3c_3, \quad P_4 = c_1^4 - 4c_1^2c_2 + 4c_1c_3 + 2c_2^2 - 4c_4.$$

$\lrcorner$

*Proof:* We can define elementary power series

$$\sigma_k = \sum_{1 \leq i_1 < \dots < i_k} x_{i_1} x_{i_2} \dots x_{i_k} \in \mathbb{Z}[[x_1, \dots, x_n, \dots]],$$

and also

$$s_k = \begin{cases} \sum x_i^k & , k \geq 0 \\ 0 & , k < 0 \end{cases} \in \mathbb{Z}[[x_1, \dots, x_n, \dots]].$$

Then the Newton identities also holds by taking limits. There is an injection

$$\mathbb{Q}[[c_1, \dots, c_n, \dots]] \rightarrow \mathbb{Q}[[x_1, \dots, x_n, \dots]] : c_i \mapsto \sigma_i,$$

and the LHS is mapped to

$$\log\left(\prod_i (1 + x_i)\right) = \prod_i \log(1 + x_i) = \sum_{p \geq 1} (-1)^p \frac{s_p}{p}$$

is the image of the RHS.  $\square$

**Prop. (3.3.4.4) [Todd Polynomials].** There are **Todd polynomials**  $Q_p \in \mathbb{Q}[\sigma_1, \dots, \sigma_p], p \geq 1$  homogenous of degree  $p$  s.t.

$$\text{Todd}(x_1, \dots, x_n, \dots) = \prod_{i=1}^n \frac{x_i}{1 - e^{-x_i}} = 1 + \sum_{p \geq 1} Q_p \in \mathbb{Q}[[x_1, x_2, \dots]]$$

for any  $n \in \mathbb{Z}_+$ , and

$$Q_1 = \frac{1}{2}c_1, \quad Q_2 = \frac{1}{12}(c_1^2 + c_2), \quad Q_3 = \frac{1}{24}c_1c_2, \quad Q_4 = \frac{1}{720}(-c_1^4 + 4c_1^2c_2 + 3c_2^2 + c_1c_3 - c_4).$$

┘

*Proof:* ?  $\square$

**Prop. (3.3.4.5) [Conjugate Invariant Polynomials].** Any polynomial on the entries of matrixes  $M_n(k)$  that is invariant under conjugation is generated by coefficients of  $\det(\lambda I + X)$  and can also be generated by  $\text{tr}(X^k)$ .  $\square$

*Proof:* We notice that the matrixes having disjoint eigenvalues is dense in  $M_n(k)$ , thus the restriction of the polynomial on these matrixes is a symmetric polynomial(3.3.4.1) thus identical to a polynomial described above. Hence they are equal.  $\square$

**Prop. (3.3.4.6).** For any polynomial on the entries of matrixes  $M_n(k)$  that  $f(BA) = f(A)$  for  $B \in O(n)$ , there is a polynomial  $F$  that  $f(A) = F(A^*A)$ . Cf.[Heat Equation and the Index Theorem Atiyah P323].  $\square$

**Prop. (3.3.4.7) [Weyl].** Any linear map  $f$  from  $(\mathbb{R}^m)^{\otimes n}$  to  $R$  that is  $O(m, \mathbb{R})$ -equivariant is a linear combinations of maps of the form:

$$v_1 \otimes v_2 \otimes \dots \otimes v_n \mapsto \langle v_{i_1}, v_{i_2} \rangle \langle v_{i_3}, v_{i_4} \rangle \dots \langle v_{n-1}, v_n \rangle.$$

Where  $i_1, \dots, i_n$  is a permutation of  $1, 2, \dots, n$  when  $n$  is even and when  $n$  is odd,  $f$  must be 0.  $\square$

*Proof:* Cf.[Heat Equation and the Index Theorem].  $\square$

## 3.4 Orthogonal Polynomials

References are [Special Functions].

### 1 Chebyshev Polynomials

**Def. (3.4.1.1) [Chebyshev Polynomials].** For any  $n \in \mathbb{N}$ , the **Chebyshev polynomial**  $S_n(T)$  is defined to be the expression of

$$\alpha^n + \alpha^{n-2} + \dots + \alpha^{2-n} + \alpha^{-n}$$

as a polynomial of  $T = \alpha + \alpha^{-1}$ . For example,

$$S_0(T) = 1, \quad S_1(T) = T, \quad S_2(T) = T^2 - 1.$$

$$S_3(T) = T^3 - 2T, \quad S_4(T) = T^4 - 3T^2 + 1, \quad S_5(T) = T^5 - 4T^3 + 3T.$$

┘

**Prop. (3.4.1.2).** For any  $n \in \mathbb{N}$ ,  $S_n(T)$  is monic of degree  $n$ . In particular,  $\{S_n(T)\}$  form a basis for  $\mathbb{Z}[T]$  over  $\mathbb{Z}$ . ┘

**Prop. (3.4.1.3).** If  $m, n \in \mathbb{N}$  with  $m \neq n$ , then

$$\int_{-1}^1 S_m(T) S_n(T) \left( \frac{2}{\pi} \sqrt{1-T^2} \right) dt = 0.$$

┘

*Proof:*

□

### 3.5 Linear Algebra

*“It is my experience that proofs involving matrices can be shortened by 50% if one throws the matrices out.”*

–E. Artin

References are [H-K71, Mey00, Rom07], [线性代数 谢启鸿], [Matrix Analysis, Horn-Johnson] and [Determinant, 高等代数 notes, 安金鹏].

**Def. (3.5.0.1) [Notations].**

- Let  $k \in \text{Field}$ .
- Let  $R \in \mathbb{C}\text{Ring}$ .

┘

In this section,  $\text{Vect}_k$  is studied, without considering any topology on  $k$  or  $V$ . More generally, the category of f.g. modules over a commutative unital ring  $R$  is studied.

#### 1 Basics

**Def. (3.5.1.1) [Vector Spaces].** A **vector space** over  $k$  is a module over  $k$  (3.2.4.1). ┘

**Def. (3.5.1.2) [Linear Operators].** A **linear operator** on a  $R$ -module  $V$  is an element of the endomorphism ring  $\text{End}_R(V)$  ┘

**Def. (3.5.1.3) [Basis].** Let  $R$  be a field, then the sets  $S$  that are linearly independent over  $R$  has maximal objects by Zorn’s lemma, and such a maximal object must span  $M$ , called a **basis** of  $M$ . ┘

**Prop. (3.5.1.4) [Dimension].** All basis of a linear  $k$ -vector space  $V$  have the same cardinality, this cardinality is called the **dimension(vector space)** or **rank(vector space)**  $\dim_k(V)$  or  $\text{rank}_k(V)$  of  $V$ . This follows immediately from (3.2.4.4). ┘

**Prop. (3.5.1.5) [Canonical way of Writing a Basis].** After so many years, I still find it confusing to write a basis and observing change of basis, so I will write it here:

A vector should always be written vertically, and so a basis should be  $\vec{e} = (e_1, \dots, e_n)$  (horizontal), and a vector with basis  $\vec{a}$  (vertical) is in fact  $\vec{e} \vec{a}$ .

A change of basis should be written  $\vec{e}' = ea$ , with  $a \in GL_n$ , and then if an operator has matrix  $A$  w.r.t. the basis  $\vec{e}$ , it then map in the basis  $\vec{e}'$   $v = \vec{e}' x = \vec{e} ax \mapsto \vec{e} Aax = \vec{e}' a^{-1} Aax$ , so it has matrix  $a^{-1} Aa$  w.r.t the basis  $\vec{e}'$ . ┘

**Prop. (3.5.1.6) [Extension of Basis].** ┘

*Proof:* ┘

**Prop. (3.5.1.7).** Let  $F$  be a subfield of  $k$  and  $U \in \text{Vect}/k$  with a  $F$ -subspace  $U'$ . Then if every finite  $F$ -linearly independent subset of  $U'$  is independent over  $k$ , then  $\dim_F(U') \leq \dim_K(U)$ . ┘

*Proof:* If the converse is true, there is a  $F$ -basis  $u'_j$  of  $U'$ , then some of  $u'_j$  is  $K$ -linearly dependent, contradiction. ┘

**Def. (3.5.1.8) [Flags].** For  $V \in \text{Vect}/k$ , a **flag** in  $V$  is a nested sequence

$$V_1 \subsetneq \dots \subsetneq V_n$$

of subspaces of  $V$ . ┘



### Matrices

**Def. (3.5.1.9) [Matrices].** Let  $R$  be a commutative ring, the set of **matrices** of size  $m \times n$  over  $R$  is an (non-commutative) algebra  $\text{Mat}(m \times n; R)$  over  $R$  whose underlying module is  $R^{\oplus n^2} = \bigoplus_{1 \leq i \leq m, 1 \leq j \leq n} e_{ij}R$ , and the algebra structure is given by

$$e_{ij}e_{ik} = e_{ik}.$$

An element  $A = \sum_{1 \leq i \leq m, 1 \leq j \leq n} a_{ij}e_{ij} \in M_{m \times n}(R)$  is denoted by  $A = (a_{ij})$ . ┘

**Def. (3.5.1.10) [Common Matrices].** For  $n \in \mathbb{Z}_+$ , let

$$\mathbf{1} = \mathbf{1}_n \stackrel{\Delta}{=} \text{diag}(1, 1, \dots, 1) \in \text{Mat}(n; \mathbb{Z}),$$

and

$$\mathbf{J} = \mathbf{J}_n \stackrel{\Delta}{=} \sum_{1 \leq i, j \leq n} e_{ij}.$$

**Prop. (3.5.1.11).**  $A, B$  are two  $n \times n$ -matrices, if  $1 - AB$  is invertible, then so does  $1 - BA$ , and ┘

$$(1 - BA)^{-1} = 1 + B(1 - AB)^{-1}A.$$

*Proof:* Immediate from (3.2.1.5) or (3.5.9.9). □

**Cor. (3.5.1.12).**  $AB$  and  $BA$  has the same characteristic polynomials. ┘

**Prop. (3.5.1.13).** For a ring  $R$ , there is an isomorphism of rings

$$\text{Mat}(n; R)^{\text{op}} \cong \text{Mat}(n; R^{\text{op}}).$$

The isomorphism is given by  $A \mapsto A^t$ . ┘

**Def. (3.5.1.14) [Notation for Matrix Groups].** Let  $R$  be a ring, denote

- $\text{GL}(n; R)$  be the subgroup of  $\text{End}_R(\bigoplus R^{\oplus n})$  consisting of invertible matrices.
- $\text{SL}(n; R)$  be the subgroup of  $\text{GL}(n, R)$  consisting of matrices of determinant 1 (3.5.9.1).
- $\text{Unip}(n; R)$  be the subgroup of  $\text{GL}(n, R)$  consisting of upper-triangular matrices.
- $P_n(R)$  be the subgroup of  $\text{GL}(n, R)$  consisting of matrices  $A = (a_{ij})$  that  $a_{ni} = 0$  for  $i < n$ .
- $Q_n(R)$  be the subgroup of  $\text{GL}(n, R)$  consisting of matrices  $A = (a_{ij})$  that  $a_{ni} = 0$  for  $i < n$  and  $a_{nn} = 1$ . ┘

**Prop. (3.5.1.15).** Let  $K$  be a topological field, there is a decomposition of spaces

$$U_n(K) \setminus \text{GL}(n; K) \cong GL(1; K) \times U_2(K) \setminus \text{GL}(2; K) \times \dots \times Q_n(K) \setminus GL(n, K)$$

where  $\text{GL}(n-1; K)$  embeds  $\text{GL}(n, K)$  obviously. ┘

**Remark (3.5.1.16).** ? This is wrong. ┘

*Proof:* By induction, it suffices to show

$$U_n(K) \setminus \text{GL}(n, K) \times U_{n-1}(K) \setminus \text{GL}(n-1, K) \times Q_n(K) \times GL(n, K).$$

Consider the map  $\text{GL}(n-1, K) \times \text{GL}(n, K) \rightarrow U_n(K) \setminus \text{GL}(n, K) : (x, y) \mapsto xy$ , then  $x_1y_1 = x_2y_2$  iff  $x_1y_1 = ux_2y_2$  for some  $u \in U_n(K)$ , iff  $y_1y_2^{-1} = x_1^{-1}ux_2$ . This is possible iff  $y_1y_2^{-1} \in Q_n$ , and for all  $(y_1, y_2)$  that  $y_1y_2^{-1} \in Q_n$ ,  $x_1y_1 = ux_2y_2$  iff  $x_1 = ux_2y_2y_1^{-1}$ . □

## 2 Rank

**Prop. (3.5.2.1) [Rank Nullity Theorem].** For  $T : V \rightarrow W \in \mathcal{V}ect_k$ , then  $\text{rank}(T) + \dim \text{Nul}(T) = \dim V$ .

┘

*Proof:* This follows from the exact sequence  $0 \rightarrow \ker(T) \rightarrow V \rightarrow \text{Im}(T) \rightarrow 0$ . □

**Prop. (3.5.2.2) [Row Rank equals Column Rank].** The row rank of a  $m \times n$  matrix  $A$  is the same as the column rank. ┘

*Proof:* Let  $A$  be the matrix of a linear map  $T : V \rightarrow W$ , then column rank equals the range of  $T$ , and the row rank equals the range of  $T^t$ , so they are equal by (3.5.3.7). □

**Prop. (3.5.2.3) [Sylvester's Inequality].** For  $U$  a  $m \times n$  matrix and  $V$  a  $n \times k$  matrix,

$$\text{Rank}(UV) \geq \text{Rank}(U) + \text{Rank}(V) - n$$

┘

*Proof:* This comes from  $\dim \ker fg \leq \dim \ker f + \dim \ker g$ , which is because  $\ker fg = g^{-1}(\ker f)$ . □

**Prop. (3.5.2.4) [Finite Field General Linear Group].** Over finite field  $\mathbb{F}_{p^k}$ ,  $\# \text{GL}(n; \mathbb{F}_{p^k}) = (p^n - 1)(p^n - p) \dots (p^n - p^{n-1})$ . ┘

*Proof:* This is because choose the rows are equivalent to choosing a basis for  $V = \bigoplus_{i=1}^n \mathbb{F}_{p^k}$ , and when choosing  $n$ -th row, it suffices to avoid an element in the span of the first  $n - 1$  rows. □

**Prop. (3.5.2.5) [Wedge and Rank].** Let  $T : V \rightarrow V'$  be a linear map, then  $\wedge^m(T) = 0$  iff  $\text{rank}(T) < m$ . ┘

*Proof:* If  $\text{rank}(T) < m$ , then  $\text{Im}(\wedge^m(T)) \subset \wedge^m \text{Im}(T) = 0$ . Conversely, if  $\text{rank}(T) \geq m$ , choose  $\{Te_1, \dots, Te_m\}$  linearly independent, then  $T(e_1 \wedge \dots \wedge e_m) = Te_1 \wedge \dots \wedge Te_m \neq 0$ . □

## 3 Dual spaces

**Def. (3.5.3.1) [Dual Spaces].** For  $V \in \mathcal{V}ect_k$ , the set of linear functionals on  $V$  with value in  $k$  form another vector space, which is denoted by  $V^*$ , called the **dual space** of  $V$ . ┘

**Def. (3.5.3.2) [Annihilator].** Let  $V$  be a vector space and  $W \subset V$  a subspace, then the **annihilator**  $W^\perp$  is the subspace of  $V^*$  consisting of linear functionals  $l$  on  $V$  s.t.  $l(W) = 0$ . ┘

**Def. (3.5.3.3) [Dual Basis].** Let  $V \in \mathcal{V}ect_k$  and  $(e_1, \dots, e_n)$  is a basis of  $V$ , then there exists unique elements  $\alpha_1, \dots, \alpha_n$  of  $V^*$  s.t.  $(e_i, \alpha_j) = \delta_{ij}$ . Any such elements  $\alpha_1, \dots, \alpha_n$  constitute a basis for  $V^*$ , and is called the **dual basis** of  $(e_1, \dots, e_n)$ . ┘

**Prop. (3.5.3.4) [Dimension of the Annihilator].** Let  $W \subset V$  be f.d. vector spaces, then  $\dim W + \dim W^\perp = \dim V$ . ┘

*Proof:* Let  $\{e_1, \dots, e_k\}$  be a basis of  $W$  and extend it to a basis  $\{e_1, \dots, e_k, e_{k+1}, \dots, e_n\}$  of  $V$  by (3.5.1.6). Consider the dual space  $e_1^*, \dots, e_n^*$ , then the annihilator of  $W$  is just  $\text{span}\{e_{k+1}^*, \dots, e_n^*\}$ . Thus  $k + (n - k) = n = \dim V$ . □

**Def. (3.5.3.5) [Transpose].** Let  $T : V \rightarrow W$  be a linear map of f.d. vector spaces, then  $T$  induces a map  $T^t : W^* \rightarrow V^*$  given by  $T^t(l) = l \circ T$ . This is a linear map, called the **transpose** of  $T$ .  $\lrcorner$

**Prop. (3.5.3.6) [Adjoint Map and Transpose].** Let  $T : V \rightarrow W$  be a linear map of f.d. vector spaces, such that w.r.t. a basis  $\{e_1, \dots, e_n\}$  of  $V$  and  $\{f_1, \dots, f_m\}$  of  $W$ ,  $T$  is given by matrix  $A = (a_{ij})$ . Then w.r.t. the dual basis  $\{f_1^*, \dots, f_m^*\}$  of  $W^*$ ,  $\{e_1^*, \dots, e_n^*\}$  of  $V^*$ , the transpose map (3.5.3.5)  $T^t : W^* \rightarrow V^*$  is given by the a matrix  $A^t = (b_{ij})$ , where  $b_{ij} = a_{ji}$ .  $A^t$  is called the **transpose matrix** of  $A$ .  $\lrcorner$

*Proof:*  $\langle e_j, T^t(f_i^*) \rangle = \langle Te_j, f_i^* \rangle = a_{ji}$ , thus  $b_{ij} = a_{ji}$ .  $\square$

**Prop. (3.5.3.7) [Transpose Range Nullity Duality].** Let  $f : V \rightarrow V'$  be a linear map between f.d. vector spaces over a field  $k$ , then

- $\text{rank}(T^t) = \text{rank}(T)$ .
- The range of  $T^t$  is the annihilator of the null space of  $T$ .

$\lrcorner$

*Proof:* 1: Notice that  $T^t(g) = 0$  iff  $\langle g, T(v) \rangle = 0$  for any  $v \in V$ , thus the null space of  $T^t$  is just the annihilator of the range of  $T$ , which equals  $\dim V' - \text{rank}(T)$  by (3.5.3.4). But this space has dimension  $\dim \ker(T^t) = \dim V' - \text{rank}(T^t)$ , so  $\text{rank}(T) = \text{rank}(T^t)$ .

2: This follows from the fact that  $(T^t)^t = T$  and the argument above.  $\square$

**Cor. (3.5.3.8).** A linear map between f.d. vector spaces is surjective iff its transpose is injective.  $\lrcorner$

**Prop. (3.5.3.9) [Infinite Dual space].** If  $\dim_K V = \infty$ , then  $\dim_K V < \dim_K V^*$ .  $\lrcorner$

*Proof:* Notice  $\text{Hom}(\oplus_{i \in I} K e_i, K) = \prod K e_i^*$ .

We prove first that if  $|K|$  is at most countable, then  $|V| = |I|$ . Notice the set  $S_n(I)$  of all  $n$ -element subsets of  $I$  is of the same cardinality of  $I$  (2.2.8.3). And the finite sums of  $K$  and  $e_i$  can be seen as a subset of  $S_n(I) \times K^n$ , so it has the same cardinality of  $I$ .

Now we can prove if  $\#K < \aleph_0$ , then  $\dim V < \dim V^*$ . This is because  $V^*$  equals the functions from  $V$  to  $K$ , which is bigger than the functions from  $V$  to  $\{0, 1\}$ , which is the power set of  $V$ , so having cardinality  $2^{|V|}$  which is bigger than  $|V|$ , by Cantor theorem (2.2.7.4).

Now generally,  $K$  is not countable, but it has a base field  $F$ , which is countable, so we consider the  $F$ -vector space  $W = \oplus_{i \in I} F e_i$ , then  $\dim_F W = \dim_K V$ , and  $\dim_F W < \dim_F W^*$ . If we can show  $\dim_F W^* \leq \dim_K V^*$ , then we are done.

For this, first consider the natural  $F$ -linear mapping  $W^* \rightarrow V^*$ , which is clearly an imbedding. Now we want to use (3.5.1.7), so we check the conditions, for  $F$ -linearly independent  $\varphi_1, \dots, \varphi_n$ , if  $\sum c_i \varphi_i = 0$ ,  $c_i \in K$ , then if we can find  $w_k \in W$  that  $\varphi_i(w_j) = \delta_{ij}$ , then this is a contradiction. But this is true, by a simple argument, using the  $F$ -linearity of  $F$ .  $\square$

## 4 Rational Forms and Jordan Forms

**Prop. (3.5.4.1) [Elementary and Invariant Factors].** A linear operator in  $L(V)$  is equivalent to a  $k[X]$ -module structure on  $V$ , and two operators are similar iff the module structure are isomorphic.

As  $k[X]$  is a PID, the elementary factors, invariant factors, cyclic and elementary decomposition theorems (3.2.4.22) can be applied to the case.  $\lrcorner$

*Proof:* Cf. [Advanced Linear Algebra P168].  $\square$

**Cor. (3.5.4.2) [Jordan Forms].**

- For any  $g \in \text{Mat}(n; k)$ , if all the eigenvalues of  $g$  are in  $k$  (e.g.  $k = \bar{k}$ ), then  $g$  is similar to a matrix of blocks  $\lambda_i \mathbf{1} + N$ ,  $Nx_i = x_{i-1}$ , called the **Jordan form** of the matrix.
- Any  $g \in \text{Mat}(n; \mathbb{R})$  is similar to a matrix of blocks of the above form together with  $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$  on the diagonal and  $I_{2 \times 2}$  on the lower side.

┘

*Proof:* 1: Because  $k = \bar{k}$ , the elementary factors are all of the form  $(x - c_i)^{m_{ij}}$ . Now in the basis

$$\{(T - c_i)^{m_i-1}v, (T - c_i)^{m_i-2}v, \dots, v\},$$

the matrix is just the Jordan form.

1: Over  $\mathbb{R}$ , the elementary factors are all of the form  $(x - c_i)^{m_{ij}}$  and  $((x - a)^2 + b^2)^{m_{ij}}$ . Then complexify it and consider a cyclic vector  $v$ , for  $(T - (a + bi)I)$ , let  $v_{n+1} = (T - (a + bi)I)v_n$ , and let  $v_n = X_n + iY_n$ , then it can be verified that  $T$  is of the Jordan form given in the basis  $X_i, Y_i$ .  $\square$

**Def. (3.5.4.3) [Companion matrix].** The **companion matrix** for a monic polynomial  $p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in k[X]$  is the matrix

$$T_{p(x)} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \ddots & & \vdots \\ \vdots & \vdots & \ddots & 0 & \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix}$$

┘

**Prop. (3.5.4.4) [Companion Matrix is Nonderogatory].** An operator is cyclic iff it is similar to a companion matrix.

A companion matrix is nonderogatory (3.5.9.28). In fact, the minimal polynomial and the maximal polynomial of the companion matrix of  $p(X)$  are both  $p(X)$ .  $\square$

*Proof:* The operator of the companion matrix is an operator  $A$  with a basis  $\{v, Av, A^2v, \dots, A^{n-1}v\}$  and  $A^n v = -\sum_{i < n} a_i A^i v$ , which is just equivalent to the fact the action of  $A$  is cyclic.

The determinant of  $T_{p(X)}$  equals  $p(X)$  by (3.5.4.14), and for the minimal polynomial, in the basis of  $\{v, Av, A^2v, \dots, A^{n-1}v\}$ , clearly for a polynomial  $f(X)$  of degree  $m < n$ ,  $f(A)v \neq 0$ , and  $p(A)v = \sum_{i < n} a_i A^i v + A^n v = 0$ . So the minimal polynomial of  $A$  is  $p(X)$ .  $\square$

**Prop. (3.5.4.5) [Rational Canonical Form].** Every matrix is similar to a sum of companion matrices (3.5.4.3) corresponding to its elementary divisors.  $\square$

**Prop. (3.5.4.6) [Invariant Factor Form].** Every matrix is similar to a sum of companion matrices, corresponding to its invariant factors.  $\square$

**Def. (3.5.4.7) [Nilpotent and Unipotent Matrices].** A **nilpotent matrix** is a matrix  $A \in \text{Mat}(n; k)$  s.t.  $A^n = \mathbf{1}$ . And it is called **maximally nilpotent** if the minimal polynomial of  $A$  is  $T^n = 1$ .

A **unipotent matrix** is a matrix  $A \in \text{Mat}(n; k)$  s.t.  $A - \mathbf{1}$  is nilpotent. And a **maximally unipotent matrix** is a matrix  $A \in \text{Mat}(n; k)$  s.t.  $A - \mathbf{1}$  is maximally nilpotent.  $\square$

### Computing the Invariant Factors

**Def. (3.5.4.8) [Elementary Row Operation].** An **elementary row operation** for a matrix  $M$  over an algebra  $A$  is one of the following:

- Multiplying one row of  $M$  by a non-zero scalar in  $A$ .
- plus the  $r$ -th row by  $c$ -times the  $s$ -th row, where  $c$  is invariant  $A$ .

And an **elementary matrix** is a matrix obtained by the identity matrix by an elementary row operation.

Two matrix is called **row equivalent** iff they can be connected by f.m. elementary row operations, and this is equivalent to  $M = PN$ , where  $P$  is a product of f.m. elementary matrices, because left multiplication by an elementary matrix is equivalent to an elementary row operation.

Similarly we can define elementary column operations. ┘

**Lemma (3.5.4.9).** The elementary row operation change the determinant only by an invariant element in  $A$ . (Clear). ┘

**Prop. (3.5.4.10).** Let  $P$  be a matrix with entries in  $F[X]$ , then the following are equivalent:

- $P$  is invertible.
  - The determinant is a nonzero scalar.
  - $P$  is row equivalent to identity matrix.
  - $P$  is a product of elementary matrices.
- ┘

*Proof:* The only hard part is  $2 \rightarrow 3$ : this is because it has determinant in  $F^*$ , thus the greatest common divisor of the first column is a scalar, thus we can use row operation to make it  $(1, 0, \dots, 0)^t$ , and then we can continue to make it a upper triangular matrix with 1 in the diagonal, and also kill the upper half part. So it is row equivalent to the identity matrix. □

**Cor. (3.5.4.11).** Let  $M, N$  be two matrices with entries in  $F[X]$ , then they are equivalent iff  $N = PM$  for some  $P$  that has determinant in  $F$ . ┘

**Def. (3.5.4.12).** We call  $M, N$  **equivalent** iff  $M, N$  are connected by a sequence of elementary row operations and column operations. This is equivalent to  $M = PNQ$  for  $P, Q$  invertible matrices in  $M_n(F[X])$ . ┘

**Prop. (3.5.4.13) [Normal Form of Companion Matrix].** For a monic polynomial  $p(X)$ , consider its companion matrix  $T_p$ , then the matrix  $xI - T_p \in M_n(F[X])$  is equivalent to  $\text{diag}(p(X), 1, \dots, 1)$ . ┘

*Proof:* Clear, if one reduces the  $x$  in the diagonal from the bottom row to the top row one by one. □

**Cor. (3.5.4.14).** For a companion matrix  $A$  of  $p$ ,  $\det(xI - A) = p$ . ┘

*Proof:* This is from (3.5.4.9), and the fact both the side are monic polynomials. □

**Def. (3.5.4.15) [Smith Normal Form].**  $A \in \text{Mat}(n; k[X])$  is called a **Smith normal form** if it is diagonal and diagonal entries  $f_i \in F[X]$  is monic and satisfies  $f_k$  divides  $f_{k+1}$ . ┘

**Prop. (3.5.4.16).** Any matrix  $M$  with entries in  $F[X]$  is equivalent to a unique Smith normal form. ┘

*Proof:* This is immediate from (16.5.5.5) applied to the PID  $F[X]$  (3.2.3.17).  $\square$

**Cor. (3.5.4.17)[Computing Invariant Factors].** The diagonal entries of the Smith form of the matrix  $xI - M \in M_n(F[X])$  are just the invariant factors of  $M$ .  $\lrcorner$

*Proof:* This is because of the uniqueness of Smith form (3.5.4.16) and the invariant factor (3.5.4.6) and (3.5.4.13).  $\square$

### Applications

**Prop. (3.5.4.18).** For any two matrices  $A, B \in M_{n \times n}(K)$ ,  $(AB)^n$  and  $(BA)^n$  are similar.  $\lrcorner$

*Proof:* It suffices to show that they have the same elementary factors. Notice that for any irreducible polynomial  $p$ , if  $p \neq x$ , then if  $p^k(AB)v = 0$ , then  $p^k(BA)Bv = 0$ , if  $p^k(BA)v = 0$ , then  $p^k(AB)Av = 0$ . Thus there are maps  $B : N(p^k(AB)) \rightarrow N(p^k(BA))$  and  $A : N(p^k(BA)) \rightarrow N(p^k(AB))$ . Now their composition are both injective, thus they have the same dimension.

And for  $p = x$ , these two both have nullity as the multiplicity of 0 in the charpoly of  $AB, BA$  (3.5.9.10), thus the same. So they have the same elementary factors, thus similar.  $\square$

**Prop. (3.5.4.19)[Matrix Similar to Transpose].** Any matrix is similar to its transpose.  $\lrcorner$

*Proof:* This is because the invariant factors can be computed using the greatest common divisors of minors by (3.5.4.16) and (3.5.4.17), and they are clearly invariant under conjugation.  $\square$

## 5 Tensor Algebras

**Def. (3.5.5.1) [Tensor Algebras].** The tensor product and tensor algebras of modules are defined in (3.2.4.13) and (5.1.1.23). In this subsection, we focus on tensor algebras of vector spaces.  $\lrcorner$

**Def. (3.5.5.2) [Symmetrized Tensors].** If  $k \in \text{Field}^0$  and  $V \in \text{Vect}_k$ , for any  $n \in \mathbb{Z}_+$ , we can define a multilinear map

$$V^n \rightarrow T^n(V) : (v_1, \dots, v_n) \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(n)}.$$

and descends to a linear map  $\sigma : S^n(V) \rightarrow T^n(V)$ , called the **symmetrizer map**. Elements in  $\text{Im}(\sigma) = \tilde{S}^n(V)$  is called **symmetrized tensors**.

Then  $\sigma^2 = \sigma$ , and  $\ker(\sigma) = I \cap T^n$ , where  $I = \ker(T(V) \rightarrow \text{Sym}(V))$ . In particular,

$$T^n(V) = \tilde{S}^n(V) \oplus (T^n(V) \cap I).$$

$\lrcorner$

*Proof:*  $\sigma^2 = \sigma$  is easy. So  $V = \text{Im}(\sigma) \oplus \ker(\sigma)$ . Now  $T^n(V) \cap I \subset \ker(\sigma)$ , and because  $\text{Im}(\sigma) \rightarrow S^n(V)$  is surjective,  $T^n(V) \cap I = \ker(\sigma)$ .  $\square$

**Def. (3.5.5.3)[Anti-Symmetrized Tensors].** Let  $k \in \text{Field}^0$  and  $V \in \text{Vect}_k$ , for any  $n \in \mathbb{N}$ , we define a multilinear map

$$V^n \rightarrow T^n(V) : (v_1, \dots, v_n) \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(n)}.$$

and descends to a linear map  $\sigma : \wedge^n(V) \rightarrow T^n(V)$ , called the **anti-symmetrizer map**. Elements in  $\text{Im}(\tau) = \tilde{\wedge}^n(V)$  is called **anti-symmetrized tensors**.

Then  $\tau^2 = \tau$ , and  $\ker(\tau) = I \cap T^n$ , where  $I = \ker(T(V) \rightarrow \wedge(V))$ . In particular,

$$T^n(V) = \tilde{\wedge}^n(V) \oplus (T^n(V) \cap I).$$

┘

*Proof:*  $\sigma^2 = \sigma$  is easy. So  $V = \text{Im}(\sigma) \oplus \ker(\sigma)$ . Now  $T^n(V) \cap I \subset \ker(\sigma)$ , and because  $\text{Im}(\sigma) \rightarrow S^n(V)$  is surjective,  $T^n(V) \cap I = \ker(\sigma)$ .  $\square$

**Prop. (3.5.5.4).** Let  $k \in \text{Field}$ ,  $A \in \text{Mat}(k)$ , then the eigenvalues of  $\text{Sym}^n A$  and  $\wedge^n A$  are all known.  $\lrcorner$

*Proof:* Because we can pass to  $\bar{k}$ , and over there  $A$  is upper-triangularizable.  $\square$

## 6 Supervector Spaces(Berezin)

**Def. (3.5.6.1)[Supervector Spaces].** For  $k \in \text{Field}^0$ , a **supervector space** over  $k$  is a  $\mathbb{Z}/(2)$ -graded vector spaces over  $k$ , i.e.  $V = V_0 \oplus V_1$ . If  $\dim V_0 = n$  and  $\dim V_1 = m$ , then  $V$  is denoted by  $k^{n|m}$ .  $\lrcorner$

**Prop. (3.5.6.2)[Tensors of Supervector Spaces].** There is a natural notion of tensors of supervector spaces, and  $S_2$  acts on  $V \otimes V$  by

$$\sigma(v \otimes w) = (-1)^{ij} w \otimes v, \quad v \in V_i, w \in V_j.$$

And then we can define the **symmetric algebra**  $S(V) = \bigoplus_{i \in \mathbb{N}} S^i(V)$  and the **exterior algebra**  $\wedge(V) = \bigoplus_{i \in \mathbb{N}} \wedge^i(V)$  s.t.

$$S(V) = S(V_0) \otimes \wedge(V_1), \quad \wedge(V) = \wedge(V_0) \otimes S(V_1).$$

┘

## 7 Spectral Theory

References are [\[Mey00\]](#)Chap7.

**Def. (3.5.7.1)[Eigenvalues and Eigenvectors].** For  $A \in \text{Mat}(n; k)$ , an **eigenvalue** of  $A$  is a number  $\lambda \in \bar{k}$  s.t there exists  $v \in (\bar{k})^n$  s.t.

$$Av = \lambda v.$$

And such a vector is called a **eigenvector** for  $A$  of eigenvalue  $\lambda$ . More generally, a **generalized eigenvalue** is a vector  $v \in (\bar{k})^n$  s.t.

$$(A - \lambda \mathbf{1})^k v = 0$$

for some  $k \in \mathbb{Z}_+$ .  $\lrcorner$

### Indices

**Def. (3.5.7.2) [Indices].** For a singular matrix  $A \in \text{Mat}(n; k)$ , there exists  $m \in \mathbb{Z}_+$  s.t.

$$k^m = \text{Im}(A^m) \oplus \ker(A^m).$$

The smallest such  $m$  is called the **index of nilpotency** of  $A$ , denoted by  $\text{ind}(A)$ . For  $A \in \text{GL}(n; k)$ , define  $\text{ind}(A) = 0$ . ┘

**Prop. (3.5.7.3) [Indices of Nilpotent Matrices].** If  $A \in \text{Mat}(n; k)$  is nilpotent, then  $\text{ind}(A)$  is the smallest number  $m$  s.t.  $A^m = 0$ . ┘

*Proof:* Cf.[Matrix Analysis and Applied Linear Algebra]P396. □

**Def. (3.5.7.4) [Indices].** For  $A \in \text{Mat}(n; k)$  and an eigenvalue  $\lambda$ , the **index of eigenvalue** of  $\lambda$  is defined to be  $\text{ind}(\lambda) = \text{ind}(A - \lambda \mathbf{1})$  (3.5.7.2). ┘

### Difference Equations, Limits, and Summability

**Prop. (3.5.7.5) [Cesáro Summability].** For  $A \in \text{Mat}(n; \mathbb{C})$ , then  $A$  is Cesáro summable, i.e.

$$G = \lim_{k \rightarrow \infty} \frac{\mathbf{1} + P + \dots + P^{k-1}}{k}$$

exists, iff  $\rho(A) < 1$ , or  $\rho(A) = 1$  with the unit eigenvalues all being semisimple.

And in this case,  $G$  is the projection onto  $\ker(\mathbf{1} - A)$  along  $\text{Im}(\mathbf{1} - A)$ .

And if  $A$  is convergent to  $G$ , then  $A$  is summable to  $G$ . ┘

*Proof:* Cf.[Matrix Analysis and Applied Linear Algebra]P633. ? □

## 8 Location and Perturbation of Eigenvalues

References are [Matrix Analysis, Horn-Johnson]Chap6.

## 9 Homogenous Invariants

### Determinants

References are [advanced determinant calculus].

**Def. (3.5.9.1) [Determinant].** For  $T \in \text{End}(V)$ , as  $\dim \wedge^n V^* = 1$ , the **determinant**  $\det(T) \in R$  is an identity  $\det(T) \in R$  determined by the identity  $\wedge^n(T^t) = \det T \cdot \text{id}_{\wedge^n V^*}$ . That is:  $L(T\alpha_1, \dots, T\alpha_n) = \det T \cdot L(\alpha_1, \dots, \alpha_n)$ . And the determinant of a matrix is defined by the linear operator it associates in a canonical basis. ┘

**Prop. (3.5.9.2) [Properties of Determinants].**

1.  $\det(\text{id}_V) = 1$ .
2.  $\det(UV) = \det U \cdot \det V$ .
3.  $T$  is invertible iff  $\det T$  is invertible, in which case  $\det(T^{-1}) = (\det T)^{-1}$ .
4. If  $\{\alpha_1, \dots, \alpha_n\}$  is a basis for  $V$ , and  $f_i$  is its dual basis, then  $\det T = f_1 \wedge \dots \wedge f_n(T\alpha_1, \dots, T\alpha_n)$ . ┘



*Proof:* All these are not hard. □

**Cor. (3.5.9.3).**  $\det(P^{-1}AP) = \det(A)$ . ┘

**Prop. (3.5.9.4).**  $\det T = \det T^t$ . ┘

*Proof:* Use (3.5.9.2), if  $\{\alpha_1, \dots, \alpha_n\}$  is a basis for  $V$ , and  $f_i$  is its dual basis, then

$$\det T^t = \alpha_1 \wedge \dots \wedge \alpha_n(T^t f_1, \dots, T^t f_n) = f_1 \wedge \dots \wedge f_n(T\alpha_1, \dots, T\alpha_n) = \det T$$

□

**Prop. (3.5.9.5) [Expansion of Determinants].** If  $A_i$  be the  $i$ -th column of  $A$ , then

$$\det A = f_1 \wedge \dots \wedge f_n(A\varepsilon_1, \dots, A\varepsilon_n) = f_1 \wedge \dots \wedge f_n(A_1, \dots, A_n) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{1 \leq i \leq n} A_{\sigma(i)i}.$$

┘

**Prop. (3.5.9.6).** For a matrix, the determinant satisfies the following properties:

1. adding a multiple of a column/row to another column/row, the determinant doesn't change.
2. Multiplying a row or a column with a scalar, then the determinant multiplies with this scalar.
3. Changing two rows or two columns makes the determinant multiply by  $-1$ .

┘

*Proof:* All this follows from 4 of (3.5.9.2). Notice the last one follows from the first two. □

**Prop. (3.5.9.7) [Laplacian Expansion Formula].** Cf. [Determinant 安金鹏 P15]. ┘

**Prop. (3.5.9.8) [Binet's Formula].** Let  $F/E$  be a Hermitian pair (3.5.11.6),  $A \in \text{Mat}(n \times m; F)$ , then

$$\det(A^*A) = \sum_{I \in \{0, \dots, n\}, \#I=m} |\det(A_I)|^2.$$

┘

*Proof:* Let  $L : F^m \rightarrow F^n$  be the corresponding map, then  $\wedge^n(L^*) \circ \wedge^n(L) = \wedge^n(L^* \circ L)$ . In the canonical basis, the matrices for  $\wedge^n(L^*)$  (resp.  $\wedge^n(L)$ ) have only one column (resp. one row) with entries  $\det(A_I)$  by definition (3.5.9.1), thus the assertion follows by (3.5.9.1) again. □

**Prop. (3.5.9.9) [Sylvester's Determinant Identity].** If  $A$  and  $B$  are matrices of sizes  $m \times n$  and  $n \times m$ , then

$$\det(\mathbf{1}_m + AB) = \det(\mathbf{1}_n + BA)$$

┘

*Proof:* Notice

$$\begin{aligned} \begin{bmatrix} 1 & A \\ B & 1 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ B & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 - BA \end{bmatrix} \begin{bmatrix} 1 & A \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & A \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 - BA & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ B & 1 \end{bmatrix} \end{aligned}$$

and use (3.5.9.2). □

**Cor. (3.5.9.10).** Multiplying by  $T$ , we see that the characteristic polynomial of  $AB$  and  $BA$  are the same.  $\lrcorner$

**Remark (3.5.9.11).** There is another proof in case  $m = n$ : It suffices to show

$$\det(\mathbf{1} + (A + X\mathbf{1})(B + X\mathbf{1})) = \det(\mathbf{1} + (B + X\mathbf{1})(A + X\mathbf{1})).$$

But notice  $A + X\mathbf{1}$  and  $B + X\mathbf{1}$  are invertible in  $\text{Mat}(n; K(X))$ , thus

$$\det(\mathbf{1} + (A + x\mathbf{1})(B + x\mathbf{1})) = \det((A + x\mathbf{1})((A + x\mathbf{1})^{-1} + (B + x\mathbf{1}))) = \det(\mathbf{1} + (B + x\mathbf{1})(A + x\mathbf{1})).$$

$\lrcorner$

**Cor. (3.5.9.12).** Let  $n \in \mathbb{Z}_+$ ,  $A \in \text{GL}(n; \mathbb{C})$  and  $u, v \in \mathbb{C}^n$ , then

$$\det(A + uv^t) = \det(A) \cdot (1 + v^t A^{-1} u).$$

$\lrcorner$

**Prop. (3.5.9.13) [Schur Complements].** If  $A$  is invertible,

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(A) \det(D - CA^{-1}B).$$

The matrix  $D - CA^{-1}B$  is called the **Schur complement** of  $A$  in  $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ .

Similarly, if  $D$  is invertible,

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(D) \det(A - BD^{-1}C).$$

The matrix  $A - BD^{-1}C$  is called the **Schur complement** of  $D$  in  $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ .  $\lrcorner$

*Proof:*

$$\det \begin{bmatrix} A & B \\ C & D - CA^{-1}B \end{bmatrix} = \det(A) \det(D - CA^{-1}B).$$

$\square$

### Applications

**Prop. (3.5.9.14).** Adjunction matrix, Cf. [Determinant 安金鹏 P16].  $\lrcorner$

*Proof:*  $\square$

**Cor. (3.5.9.15).** If  $AB = 1$ , then  $BA = 1$ .  $\lrcorner$

**Prop. (3.5.9.16) [Cramer's Rule].** Cf. [Determinant 安金鹏 P16].  $\lrcorner$

**Prop. (3.5.9.17) [Hadamard's Inequality].** For any  $A \in \text{Mat}(n; \mathbb{R})$ ,

$$|\det(A)| \leq \prod_j \left( \sum_i |a_{ij}|^2 \right)^{1/2},$$

with equality iff  $A \in O(n)$ . ┘

*Proof:* If  $\det A = 0$ , this is trivially true. And if  $A$  is non-singular, then we have a  $QR$ -decomposition  $A = QR$  (16.5.5.4), where  $Q$  is orthogonal and  $R$  is upper-triangular. Then  $|\det(A)| = |\det(R)|$ , and left multiplication by  $Q$  doesn't change the norm of column vectors of  $A$ , so it suffices to prove for  $A = R$ . And this is trivial. □

**Prop. (3.5.9.18) [Vandermonde Matrix].** For  $R \in \mathcal{CRing}$  and  $x_1, \dots, x_n \in R$ , the  $n \times n$  **Vandermonde matrix** is defined to be  $\text{Van}(x_1, \dots, x_n) \triangleq (x_i^{j-1})_{i,j}$ . Then it has determinant  $\prod_{i < j} (x_i - x_j)$ , which is invertible when  $x_i$  are pairwise different. ┘

*Proof:* Eliminate the first row by adding columns. □

**Prop. (3.5.9.19) [Pfaffian].** There is a **Pfaffian polynomial**  $\text{Pf}(M)$  s.t.  $\det M = \text{Pf}(M)^2$  for a skew-symmetric matrix. This is because a skew symmetric is equal to  $A^t \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}^k A$  for  $A$  an orthogonal matrix (3.5.11.9), so it has determinant  $(\det A)^2$  and  $A$  and depends polynomially on the entries of  $M$ . ┘

**Cor. (3.5.9.20).**

$$\text{Pf}(A^t M A) = \det A \cdot \text{Pf}(M).$$

Because we only need to consider the sign and it is determined by letting  $A = \mathbf{1}$ . ┘

**Prop. (3.5.9.21) [Determinants of Hermitian Pairs].** ┘

### Characteristic Polynomials

**Def. (3.5.9.22) [Characteristic Polynomials].** For  $k \in \text{Field}$ ,  $V \in \text{Vect}_k$  and  $T \in \text{End}(V)$ , the **characteristic polynomial** of  $T$  is defined to be

$$P(X; T) = \det(X\mathbf{1} - T|V) \in \mathbb{Z}[X] \text{ (3.5.9.1)},$$

where  $X\mathbf{1} - T$  is regarded as an element in  $\text{End}(V \otimes k(X))$ . ┘

### Traces

**Def. (3.5.9.23) [Trace].** For a  $n \times n$ -matrix  $A$ , define its **trace**  $\text{tr}(A)$  to be the minus of the coefficient of  $x^{n-1}$  in  $\det(x\mathbf{1} - A)$ . It is clear that  $\text{tr}(A) = \sum_{i=1}^n a_{ii}$ , and by (3.5.9.3) that traces are invariant under conjugacy. ┘

**Prop. (3.5.9.24).**  $\text{tr}(AB) = \text{tr}(BA)$ . ┘

*Proof:* This is because  $\det(x - AB) = \det(x - BA)$  by Sylvester determinant identity (3.5.9.9). □

**Prop. (3.5.9.25).** If  $A, B \in \text{Mat}(n; \mathbb{C})$  and  $A$  is invertible, then

$$\frac{\partial}{\partial t} \det(A + tB) = \det A \text{tr}(A^{-1}B).$$

┘

**Prop. (3.5.9.26) [Trace Formula].** For  $V \in \mathcal{V}ect/k, F \in \text{End}(V), d \in \mathbb{Z}_+$ ,

$$t \frac{\partial}{\partial t} \log \det(1 - FT^d)^{-1} = \sum_{i \geq 1} d \text{tr}(F^i|V) T^i \in 1 + Tk[[T]].$$

In particular, if  $\text{char } k = 0$ ,

$$\det(1 - FT^d|V)^{-1} = \exp \left( \sum_{i \geq 1} \text{tr}(F^i|V) \frac{T^{di}}{i} \right) \in 1 + Tk[[T]].$$

┘

*Proof:* Pass to  $\bar{k}$  and let  $k_1, \dots, k_r$  be the eigenvalues of  $F$ . □

### Minimal Polynomials

**Def. (3.5.9.27) [Minimal Polynomials].** The **minimal polynomial** of a matrix  $A$  is the polynomial  $p$  of minimal degree that  $p(A) = 0$ . It is equivalent to the maximal invariant factor of  $A$ , by (3.5.4.1).

┘

**Def. (3.5.9.28) [Non-derogatory Operator].** An operator is called **nonderogatory** iff it has only one invariant factor. ┘

**Prop. (3.5.9.29) [Generalized Cayley-Hamilton].** The characteristic polynomial of  $A$  is the product of the elementary divisors of  $A$ , thus the characteristic polynomial and minimal polynomial (3.5.9.27) of  $A$  have the same set of irreducible factors, but may not with the same multiplicity.

In particular, the characteristic polynomial is divisible the minimal polynomial. ┘

*Proof:* Because charpoly and minipoly are both invariant under similarity, assume  $A$  is in rational form (3.5.4.5), so the result follows from (3.5.4.4). □

**Prop. (3.5.9.30).** For  $k \in \text{Field}$  and  $A, C \in \text{Mat}(n; k)$ , the linear map  $\text{Mat}(n; k) \rightarrow \text{Mat}(n; k) : X \rightarrow AX - XC$  is an isomorphism iff the minimal polynomials of  $A$  and  $C$  has no common factor. ┘

*Proof:* Notice if  $AX = XC$ , then we have  $P(A)X = XP(C)$  for every polynomial  $P$ , in particular for the minimal polynomials of  $A$  and  $C$ , thus  $P(C)$  is non-invertible and  $A, C$  has a characteristic value in common. Conversely, if they have a characteristic value, then we upper triangularize  $A$  to see clearly that there is a  $X$  that  $AX = XC$  ( $X$  has only the first row). □

### Characteristic Polynomial Images

In this subsubsection, we consider the set of characteristic polynomials of elements in a subgroup of  $\text{GL}(n; k)$  as a whole.

## 10 Diagonalization and Triangulation

**Prop. (3.5.10.1).** If a linear map has matrix form  $T$  in a basis  $(X_i)$  and there is another basis  $(Y_i)$  that  $(Y_i) = (X_i)P$ , then it has matrix form  $PTP^{-1}$  in the basis  $(Y_i)$ . In particular, if  $T$  can be diagonalized, with eigenvectors  $(X_i)$ , then  $T = (X_i)D(X_i)^{-1}$ .  $\lrcorner$

**Prop. (3.5.10.2) [Relation with Minimal Polynomial].** For  $n \in \mathbb{Z}_+$ ,

- $A \in \text{GL}(n; k)$  is upper-triangulable iff its minimal polynomial is a product of linear factors.
- $A \in \text{GL}(n; k)$  is diagonalizable iff its minimal polynomial is a product of linear factors with no multiple roots.

$\lrcorner$

*Proof:* 1: If it is upper-triangulable, its minimal polynomial is a product of factors because its characteristic polynomial does (3.5.9.29). Conversely, we can find an eigenvector for  $A$ , then we quotient this vector and use induction.

2: If it is diagonalizable, then the minimal polynomial is clearly polynomials. Conversely, its elementary factors are all linear factors, thus its Jordan form is just diagonal (3.5.4.2).  $\square$

**Cor. (3.5.10.3) [Upper Triangulation over Alg. Closed Field].** If  $k \in \text{Field}, k = \bar{k}$ , then any  $A \in \text{Mat}(n; k)$  is upper-triangulable over  $k$ . Dually, it is lower-triangulable.  $\lrcorner$

*Proof:* It suffices to find an eigenvector of  $A$  and then use induction. This is clear, as the characteristic polynomial of  $A$  has a root in  $k$ .  $\square$

**Prop. (3.5.10.4) [Simultaneously Triangulation].** If  $A_i \subset \text{Mat}(n; k)$  is a commuting family of upper-triangulable matrices, then they are simultaneously triangulable.  $\lrcorner$

*Proof:* As in the proof of (3.5.10.3), by induction, it suffices to show there is a common eigenvector. Now assume there are f.m. matrices in  $\mathcal{F}$ , we induct on the number of matrices to show there is an eigenvector. Let  $\lambda$  be an eigenvalue of  $A_1$  because it is upper-triangulable, then  $N(A_1 - \lambda I)$  is invariant under  $\mathcal{F}$ , and all the matrices are upper-triangulable on  $N(A_1 - \lambda I)$ , which is seen by intersecting the flag with  $N(A_1 - \lambda I)$ . So by induction, there is a common eigenvector for  $\mathcal{F}$ .  $\square$

**Prop. (3.5.10.5) [Simultaneously Diagonalizable].** If  $A_i$  is a commuting family of diagonalizable  $n \times n$ -matrices, then they are simultaneously diagonalizable.

If  $A_i$  is a commuting family of real symmetric matrixes, then they are simultaneously orthogonally diagonalizable.  $\lrcorner$

*Proof:* We may assume there are f.m. matrices and use induction. Consider the diagonal decomposition  $V_i$  of  $V$  for  $A_1$ , then each  $V_i$  is invariant under  $\mathcal{F}$ . Notice then each  $A_i$  is diagonalizable on  $V_i$ , thus by induction,  $\mathcal{F}$  is simultaneously diagonalizable on each  $V_i$ , then  $\mathcal{F}$  is simultaneously diagonalizable.

For the second, induction on the numbers of matrices. If some matrix is  $cI$ , then clear, if some are not  $cI$ , then choose its eigenvalue decomposition, we conclude by induction hypothesis.  $\square$

**Prop. (3.5.10.6) [Invariance of Field Extension].** Let  $A \in \text{Mat}(n; k)$ , and  $E$  be the subfield generated by the entries of  $A$ , then the invariant factors of  $A$  are polynomials over  $E$ . In particular, two matrix are similar over the smallest field that they are defined.  $\lrcorner$

*Proof:* Clear, because we know how an irreducible polynomial over  $E$  factors through  $\bar{E}$ .  $\square$

### Jordan-Decompositions

**Prop. (3.5.10.7) [Jordan-Decompositions].** If  $k^{\text{sep}} = k$  and  $A \in \text{Mat}(n; k)$ , then there exists unique matrices  $A_s, A_n$  that  $A_s$  is semisimple,  $A_n$  is nilpotent,  $A = A_s + A_n$ , and  $A_s, A_n$  are polynomials of  $A$ , in particular,  $A_s$  commutes with  $A_n$ . This is called the **Jordan-decomposition** of  $A$  in  $\text{Mat}(n; k)$ .

Moreover, if  $A$  is invertible, then there exists a unique pair of matrices  $(A_s, A_u)$  in  $\text{Mat}(n; k)$  s.t.  $A_s$  is semisimple,  $A_u$  is unipotent,  $A = A_s \cdot A_u$ , and  $A_s, A_u$  are polynomials of  $A$ , in particular,  $A_s$  commutes with  $A_u$ . This is called the **Jordan-decomposition** of  $A$  in  $\text{GL}(n; k)$ .  $\square$

*Proof:* If  $k$  is alg.closed, then use (3.5.10.3). And Lagrange interpolation shows  $A_s, A_n$  are polynomials in  $A$ . For the uniqueness, notice that because they are polynomials in  $A$ , if there are two sets of Jordan decompositions  $A = A_s + A_n = A'_s + A'_n$ , then  $(A_s - A'_s) = (A'_n - A_n)$ , but the sum of commuting semisimple/nilpotent matrices is semisimple/nilpotent, so it is semisimple and nilpotent, so it can only be 0, so  $A_s = A'_s$ .

In general, because  $k$  is perfect, we can take a Galois field extension  $k'/k$  containing all eigenvalues of  $A$ . Then  $A = A_s + A_n$  where  $A_s, A_n$  has entries in  $k'$ . But then taking Galois action and using the uniqueness,  $A_s, A_n$  has entries in  $k$ .

Finally, by the Galois action again,  $A_s, A_n$  are polynomials in  $A$ .

If  $A$  is invertible, then  $A_s$  is also invertible, seen by base change to alg.closure. Then  $A = A_s(1 + A_s^{-1}A_n)$ , where  $1 + A_s^{-1}A_n$  is unipotent.  $\square$

**Prop. (3.5.10.8).** If  $T$  is a diagonalizable operator on a subspace  $V$ , then for any invariant subspace  $V'$ ,  $T|_{V'}$  is also diagonalizable.  $\square$

*Proof:* Use the eigenvalue decomposition  $V = \oplus_i V_{\lambda_i}$ , then  $V' = \oplus_i (V_{\lambda_i} \cap V')$ , which is because if  $\sum v_i \in V'$ , where  $v_i$  are in different eigenspaces, then each  $v_i \in V'$ .  $\square$

**Prop. (3.5.10.9) [Jordan-Decomposition over Finite Fields].** Let  $k \in \text{Field}^{\text{fin}}$  and  $p = \text{char } k, q = \#k, n \in \mathbb{Z}_+$ , then  $x \in \text{GL}(n; k)$  has  $p$ -power order iff  $x$  is unipotent, and  $x$  is order prime to  $p$  iff  $x$  is semisimple.  $\square$

*Proof:* If  $x$  is unipotent, then  $(x - 1)^{q^m} = x^{q^m} - 1 = 0$  for  $m$  large. Conversely, if  $x^{q^m} = 1$  for some  $m$ , then  $x$  is unipotent.

if  $x$  has order prime to  $p$ , then  $x$  is semisimple by (3.5.10.2). Conversely, if  $x$  is semisimple, then over a finite base change  $x$  is diagonal, thus  $x^{q^m-1} = 1$ , and the order of  $x$  is prime to  $p$ .  $\square$

## 11 Sesqui-linear Forms

**Def. (3.5.11.1) [Sesqui-linear Forms].** For an involutive algebra  $(R, *) \in \text{Alg}^{\text{asso}}$  and a free left  $R$ -module  $V$ , a **sesquilinear form** on  $V$  is a map  $\varphi : V \times V \rightarrow R$  s.t.

$$\varphi(ax, by) = a^* \varphi(x, y) b, \text{ where } a, b \in R, x, y \in V.$$

A sesquilinear form  $\varphi$  on  $V$  is called a **Hermitian form** (resp. **skew-Hermitian form**) if  $\varphi(x, y) = \varphi(y, x)^*$  (resp.  $\varphi(x, y) = -\varphi(y, x)^*$ ).

When  $*$  = id, it is also called a **bilinear form**/symmetric form/alternating form resp.  $\square$

**Def. (3.5.11.2) [Hermitian Matrices].** Situation as in (3.5.11.6), then a **Hermitian matrix** w.r.t.  $E/F$  is a matrix  $A \in \text{Mat}(n; E)$  s.t.  $A^t = cA$ , which corresponds to an  $E$ -linear map  $L : E^n \rightarrow E^n$  s.t.  $L^* = L$  (3.5.11.7).  $\square$

**Prop. (3.5.11.3) [Gram-Matrices].** Let  $n \in \mathbb{Z}_+$  and  $V \in \text{Vect}_k$  with a basis  $\{e_1, \dots, e_n\}$ , then a  $n \times n$  symmetric(Hermitian) matrix (3.5.11.2)  $M$  defines a bilinear(sesqui-linear) form  $B$  on  $V$  by  $(y, x) \mapsto x^t M y (x^t M \bar{y})$ . Conversely, for any bilinear(sesqui-linear) form  $B$  on  $V$ , let  $M = (a_{ij})$  where  $a_{ij} = (e_i, e_j)$ , then  $M$  is symmetric(Hermitian) matrix, called the **Gram matrix** of  $B$ .

so we will interchange freely between a symmetric(Hermitian) matrix and a bilinear(sesqui-linear) form on  $V$ .  $\lrcorner$

**Prop. (3.5.11.4) [Casimir Elements].** Let  $(R, *) \in \text{Alg}^{\text{asso}}$  and  $B$  is a non-degenerate sesquilinear form on  $R$ . Choose a basis  $\{x_i\}$  of  $V$  and a dual basis  $y_i$ , then  $\sum x_i \otimes y_i \in T(V)$  is independent of  $x_i$  chosen.  $\lrcorner$

*Proof:* Let  $V^\vee$  be the dual of the vector space  $V$ . There is an isomorphism  $V \otimes V^\vee \cong \text{End}(V)$  given by mapping  $(v, f)$  to the operator  $v' \mapsto f(v')v$ . The non-degenerate bilinear form induces naturally an isomorphism  $\beta : V \cong V^\vee : v \mapsto (\cdot, v)$ . Then under the isomorphisms

$$\text{End}(V) \cong V \otimes V' \cong V \otimes V,$$

where the second isomorphism is  $(\text{id}_V, \beta^{-1})$ . The identity map  $\text{id}_V \in \text{End}(V)$  is sent to  $\sum x_i \otimes y_i \in T(V)$ , so  $\sum x_i \otimes y_i$  is independent of the basis chosen.  $\square$

### Quadratic Spaces and Orthogonal Matrices

For symmetric bilinear forms and more about quadratic forms, see [Quadratic Forms](#).

### Hermitian Spaces

**Notation (3.5.11.5).**

- Let  $D \in \text{Az}_k$  with an involution  $*$ .

$\lrcorner$

**Def. (3.5.11.6) [Hermitian Space].** A **Hermitian space**(resp. **skew-Hermitian space**) over  $D$  is a finite free  $D$ -module  $V$  with a Hermitian form(resp. skew-Hermitian form)  $(-, -) : V \times V \rightarrow E(3.5.11.1)$  that is non-degenerate.

When  $*$  = id, these are also called **symmetric spaces** and **symplectic spaces**.  $\lrcorner$

**Def. (3.5.11.7) [Hermitian Transpose].** Let  $L : V \rightarrow W$  be a map between Hermitian spaces over  $D$ , then there exists a unique **Hermitian transpose**  $L^* : W^* \rightarrow V^*$  which is a  $F$ -anti-linear map that satisfies

$$\langle w', Lu \rangle = \langle L^* w', u \rangle, \text{ where } u \in V, w' \in W^*.$$

As the pairing on  $V$  is non-degenerate(3.5.11.6), this map is uniquely determined.

If  $V = W$ , then in coordinate forms, it can be verified that  $[L^*] = \overline{[L]}^t$ .  $\lrcorner$

### Complex Hermitian Matrices and Real Symmetric Matrices

**Prop. (3.5.11.8) [Hermitian Eigenvalues are Real].** Any eigenvalue of a Hermitian(e.g., real symmetric) matrix  $M$  is real.  $\lrcorner$

*Proof:* Consider the bilinear form defined by the matrix  $M$ , then if  $x$  is an eigenvector with eigenvalue  $\lambda$ , then  $\lambda(x, x) = (Hx, x) = (x, Hx) = \bar{\lambda}(x, x)$ , so if  $\lambda$  is not real,  $x = 0$ .  $\square$

**Thm. (3.5.11.9) [Principal Axis Theorem].** A symmetric matrix  $A$  is orthogonally diagonalizable. Similarly, a Hermitian matrix is unitarily diagonalizable.  $\lrcorner$

*Proof:* Firstly, we can find an eigenvector of  $A$ : Only the real case needs proof, and this is because any eigenvalue of  $A$  is real (3.5.11.8).

Let  $v$  be an eigenvector of  $A$  of length 1, then the orthogonal complement of  $v$  is preserved by  $A$ , so we can use induction to find an orthonormal basis consisting of eigenvectors of  $A$ , then these together with  $v$  forms an orthonormal basis consisting of eigenvectors of  $A$ .  $\square$

**Prop. (3.5.11.10) [Complex Normal operators].** More generally, a normal operator over  $\mathbb{C}$  is unitarily diagonalizable using resolution of identity (12.9.3.3) because the spectrum are discrete thus the point projection is orthogonal.  $\lrcorner$

**Prop. (3.5.11.11) [Gram-Schmidt].** Suppose  $\text{char } k \neq 2$ , then any symmetric matrix over  $k$  is congruent to a diagonal matrix.  $\lrcorner$

*Proof:* Any symmetric matrix defines a bilinear form on  $V$ . If  $B$  is not identically 0, then there is a  $x$  that  $x^t B x \neq 0$ , by polarization identity. Then  $W = \{Kx\}$  is non-degenerate, so we have  $W \oplus W^\perp = V$  by (19.5.1.8). And by induction, we are done.  $\square$

**Prop. (3.5.11.12) [Antonne-Takagi].** For any complex symmetric matrix  $A$ , there is unitary matrix  $U$  that  $UAU^t$  is a real diagonal matrix with non-negative entries.  $\lrcorner$

*Proof:* Consider  $B = A^*A$  is Hermitian and positive-semi-definite, thus there is a unitary matrix  $V$  that  $V^*BV$  is diagonal with non-negative real entries by (3.5.11.9). Now  $C = V^tAV$  is complex symmetric with  $C^*C$  real diagonal. If we let  $C = X + iY$ , then  $XY = YX$ . So by (3.5.10.5), there is a real orthogonal matrix  $W$  that  $WXW^t$  and  $WYW^t$  are diagonal. Now set  $U = WV^t$ , which is unitary,  $UAU^t$  is complex diagonal. And easily we can modify the diagonal entries to be non-negative.  $\square$

### Real Spectral Theory

**Notation (3.5.11.13).**

- Use notations on spectral theory defined in [Spectral Theory on Hilbert Spaces](#).  $\lrcorner$

**Prop. (3.5.11.14).** For a normal operator  $N$  on a real inner product space,  $N\alpha = 0$  iff  $N^t\alpha = 0$ . In particular,  $N$  and  $N^t$  has the same number of eigenvalues and dimension of eigenspaces.  $\lrcorner$

*Proof:* This is because

$$(N\alpha, N\alpha) = (N^t N\alpha, \alpha) = (NN^t\alpha, \alpha) = (N^t\alpha, N^t\alpha).$$

$\square$

**Cor. (3.5.11.15).** For a normal operator  $N$  on a real inner product space,  $\text{Im}(N) = \ker(N)^\perp$ .  $\lrcorner$

*Proof:*  $\text{Im}(N) = \ker(N^*) = \ker(N)$ .  $\square$

**Cor. (3.5.11.16).** For a normal operator  $N$  on a real inner product space, if  $N^2\alpha = 0$ , then  $N\alpha = 0$ .  $\lrcorner$

*Proof:* This is because  $N\alpha \in \ker(N) \cap \text{Im}(N) = \emptyset$ .  $\square$



**Lemma (3.5.11.17).** If  $f, g$  are relatively prime polynomials and  $T$  is a normal operator,  $f(T)\alpha = 0$  and  $g(T)\beta = 0$ , then  $\alpha$  is orthogonal to  $\beta$ .  $\lrcorner$

*Proof:* Choose polynomials  $a, b$  that  $af + bg = 1$ , then  $\alpha = b(T)g(T)\alpha$ , and

$$(\alpha, \beta) = (b(T)g(T)\alpha, \beta) = (b(T)\alpha, g(T)^*\beta)$$

Notice  $g(T)$  is also normal, and  $g(T)\beta = 0$ , thus by (3.5.11.14)  $g(T)^*\beta = 0$ . Thus  $\alpha, \beta$  are orthogonal.  $\square$

**Lemma (3.5.11.18).** Let  $V$  be a real inner product space and  $S$  an operator that  $S^2 = -1$ . Suppose  $\alpha \in V$  and  $S\alpha = -\beta$ , then

$$S^*\alpha = \beta, \quad S^*\beta = -\alpha,$$

$\alpha, \beta$  are orthogonal, and  $\|\alpha\| = \|\beta\|$ .  $\lrcorner$

*Proof:* Because

$$0 = \|S\alpha + \beta\|^2 + \|S\beta - \alpha\|^2 = \|S\alpha\|^2 + \|\beta\|^2 + 2(S\alpha, \beta) + \|S\beta\|^2 + \|\alpha\|^2 - 2(S\beta, \alpha),$$

and  $S$  is normal, we get

$$0 = \|S^*\alpha\|^2 + \|\beta\|^2 - 2(S^*\alpha, \beta) + \|S^*\beta\|^2 + \|\alpha\|^2 + 2(S^*\beta, \alpha) = \|S^*\alpha - \beta\|^2 + \|S^*\beta + \alpha\|^2,$$

which gives the desired equation. And also

$$(\alpha, \beta) = (-S^*\beta, \beta) = -(\beta, S\beta) = -(\beta, \alpha)$$

which implies  $(\alpha, \beta) = 0$ .  $\square$

**Prop. (3.5.11.19) [Real Normal Operators].** Let  $A$  be a normal matrix, then  $A$  is orthogonally congruent to matrixes of the form  $\text{diag}(B_1, \dots, B_n)$ , where  $B_i$  are  $1 \times 1$  or  $2 \times 2$  matrixes of the form

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}. \quad \lrcorner$$

*Proof:* Firstly the minimal polynomial of  $A$  is a product of different irreducible polynomials  $p = p_1 \dots p_k$ , by (3.5.11.16). Let  $f_i = p/p_i$ , then  $f_1, \dots, f_k$  are relatively prime, so there are polynomials  $g_i$  that  $1 = \sum f_i g_i$ . Then for any  $v \in V$ ,  $v = \sum f_i(T)g_i(T)v$ , and  $f_i(T)g_i(T)v$  is annihilated by  $p_i(T)$ . Let  $W_i = \ker(p_i(T))$ , then  $V = \sum W_i$ , and  $W_i$  is orthogonal to  $W_j$  by (3.5.11.17).

The restriction of  $T$  on  $W_i$  has minimal polynomial  $p_i$ . If  $p_i$  has degree 1, then  $T$  is a scalar on  $W_i$ . If  $p_i$  has degree 2, then  $p_i = (x - a_i)^2 + b_i^2$  for some  $a_i, b_i \in \mathbb{R}, b_i \neq 0$ . Then we choose a maximal  $k$  that there exists subspaces  $V_j \in W_i, j \leq k$  that

- $\dim V_j = 2$ .
- $V_j$  are pairwise orthogonal.
- $V_j$  is invariant under  $T, T^*$ , and  $T|_{V_i}$  is orthogonal congruent to  $\begin{bmatrix} a_i & b_i \\ -b_i & a_i \end{bmatrix}$ .

Then denote  $W = \bigoplus_{j=1}^k V_j$ , we prove that  $W = V$ : Suppose not, then  $W^\perp \neq 0$  is invariant under  $T$  and  $T^*$ . Denote  $S = b^{-1}(T - a)$ , then  $S^2 + 1 = 0$ . Let  $\alpha \neq 0 \in W^\perp$ ,  $\beta = -S\alpha$ , then  $S\beta = \alpha$ ,  $\alpha, \beta \in W^\perp$ . Since  $T = aI + bS$ , we have

$$T\alpha = a\alpha - b\beta, \quad T\beta = b\alpha + a\beta.$$

Now lemma (3.5.11.18) implies  $\{\alpha, \beta\}$  is invariant under  $S, S^*$  thus also  $T, T^*$ , so this gives another  $V_{k+1}$ , contradiction.  $\square$

**Cor. (3.5.11.20) [Unitary Equivalence of Normal Operators].** Let  $T, T'$  be real(complex) normal matrices, then  $T$  is orthogonally(unitarily) congruent to  $T'$  iff  $T$  and  $T'$  have the same characteristic polynomial.  $\lrcorner$

*Proof:* This follows from (3.5.11.19)(or (3.5.11.10)).  $\square$

**Cor. (3.5.11.21) [Complex Structure Form].** A real normal matrix  $J$  s.t.  $J^2 + 1 = 0$  is orthogonal congruent to  $\langle \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rangle_n$ .

In particular, as such  $J$  is equivalent to a complex structure on  $\mathbb{R}^n$ , so the set of complex structures is bijective to  $O(n)/U(\frac{n}{2})$  thus can be endowed with a structure of a homogenous space.  $\lrcorner$

*Proof:* This is because  $J$  and  $\langle \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rangle_n$  are both normal and they have the same characteristic polynomial, by (3.5.9.29).  $\square$

**Def. (3.5.11.22) [Cayley-transformation].** If  $\text{char } k \neq 2$  and  $P \in \text{Mat}(n; k)$  that has no eigenvalue  $-1$ , there is a **Cayley transformation**  $A = \frac{1-P}{1+P}$ ,  $1+A$  is invertible, and  $P = \frac{1-A}{1+A}$ .  
Then  $A$  is skew-symmetric iff  $P$  is orthogonal.  $\lrcorner$

*Proof:* If  $Av = -v$ , then  $v - Pv = -v - Pv$ , so  $2v = 0$ , so  $v = 0$ . So  $A + 1$  is invertible.

If  $P$  is orthogonal, then  $A^t = \frac{1-P^t}{1+P^t} = \frac{1-P^{-1}}{1+P^{-1}} = -A$ . Conversely, if  $A^t = -A$ , then  $P^t = \frac{1+A}{1-A} = P^{-1}$ .  $\square$

**Prop. (3.5.11.23).** If  $\text{char } k \neq 2$  and  $P$  is an orthogonal matrix of odd dimension, then  $\det P$  is an eigenvalue of  $P$ .  $\lrcorner$

*Proof:* multiplying by  $-1$ , we can assume  $\det P = 1$ . Then the action of  $P$  on  $S^{n-1}$  has a fixed point by Lefschetz trace formula. Thus  $1$  is an eigenvalue of  $P$ . Contradiction.  $\square$

### Symplectic Spaces and Symplectic Matrices

**Def. (3.5.11.24) [Lagrangian Subspaces].** For a f.d. symplectic space  $V$  over  $k$ , a **Lagrangian subspace** of  $V$  is defined to be a subspace  $W \subset V$  s.t.  $W^\perp = W$ . In particular,  $\dim W = \frac{1}{2} \dim V$ .  $\lrcorner$

**Prop. (3.5.11.25).** For  $V \in \text{Vect}^{\text{fin}}_k$ , there is a natural symplectic space structure on  $V \otimes V^*$ , defined by

$$\langle (x, \xi), (x', \xi') \rangle = \langle x', \xi \rangle - \langle x, \xi' \rangle.$$

$\lrcorner$

**Prop. (3.5.11.26) [Basis for Symplectic Spaces].** Suppose  $\text{char } k \neq 2$ , then for any f.d. symplectic vector space  $V$  over  $k$ , there exists a basis  $\{x_1, \dots, x_r, y_1, \dots, y_r, z_1, \dots, z_k\}$  s.t.  $(x_i, y_i) = -(y_i, x_i) = 1$ , and the inner product of other pairs in this basis vanish.

In particular, if  $V$  is non-degenerate, then  $\dim V \in 2\mathbb{Z}_+$ .  $\lrcorner$

*Proof:* Use induction on  $\dim V$ . If  $\dim V = 0$ , this is clear. For  $\dim V > 0$ , if there exists a pair  $(x, y) \neq 0$ , then we can take  $(x, y) = 1$ , and we can look at the complement  $\{x, y\}^\perp \subset V$  and use induction.  $\square$

**Prop. (3.5.11.27).** Given a bilinear form over  $k$ , the relation of orthogonality is symmetric iff the space is symmetric or symplectic.  $\lrcorner$

*Proof:* Let  $w = B(x, z)y - B(x, y)z$ , then  $B(x, w) = 0$ , hence we have  $B(w, x) = 0$ , that is

$$B(x, z)B(y, x) - B(x, y)B(z, x) = 0.$$

Let  $z = x$ , then  $B(x, x)[B(x, y) - B(y, x)] = 0$ .

If some  $B(u, v) \neq B(v, u)$  and  $B(w, w) \neq 0$ , then  $B(u, u) = B(v, v) = 0, B(w, v) = B(v, w), B(w, u) = B(u, w)$ , Let  $x = u$  or  $v$  we get  $B(w, v) = B(v, w) = 0 = B(w, u) = B(u, w)$ . Now  $B(u, w + v) \neq B(w + v, u)$ , hence  $B(w + v, w + v) = 0 = B(w, w)$ , contradiction.  $\square$

**Prop. (3.5.11.28) [Symplectic Group Determinant 1].**  $\text{Sp}(2n) \subset \text{SL}(2n)$ .  $\lrcorner$

*Proof:* A symplectic matrix preserves the symplectic structure thus the symplectic form  $\omega$ , hence preserves  $\omega^n$  which is a volume form, so it has determinant 1 by definition (3.5.9.1).  $\square$

### Complex Structures

**Def. (3.5.11.29) [Complex Pairs].** Let  $F \in \text{Field}$  and  $E$  an étale  $F$ -algebra of degree 2. Thus  $E/F$  is a Galois extension of degree 2 with involution  $c \in \text{Gal}(E/F)$  or  $E = F \oplus F$  with involution  $c(x, y) = (\overline{x}, \overline{y}) = (y, x)$ . Such a pair  $E/F$  is called a **complex pair**. We sometimes write  $c(x)$  as  $\overline{x}$ .  $\lrcorner$

**Prop. (3.5.11.30) [Complex Structures].** Let  $g \in \mathbb{Z}_+$  and  $V \in \text{Vect}^{2g}/\mathbb{R}$ , then the following are equivalent:

- A complex structure on  $V$ .
- An endomorphism  $J \in \text{End}_{\mathbb{R}}(V)$  s.t.  $J^2 = -\text{id}$ .
- A representation  $\rho : U^1 \rightarrow \text{End}_{\mathbb{R}}(V)$  s.t.  $V \otimes_{\mathbb{R}} \mathbb{C}$  has weights  $\pm 1$ , each with multiplicity  $g$ .

$\lrcorner$

*Proof:* 1  $\rightarrow$  2: Take  $J$  to be multiplication by  $i$ .

2  $\rightarrow$  1: The complex structure is given by  $(a + bi)x = ax + bJx$ .

2  $\rightarrow$  3: The representation is given by  $\rho(\theta) \mapsto \cos \theta + \sin \theta J$ . Then  $V_{\mathbb{C}} = \ker(J - i) \oplus \ker(J + i)$ , and conjugation interchanges these two spaces.

3  $\rightarrow$  2: Given such a representation,  $J = \rho(i)$  satisfies  $J^2 = -1$ .  $\square$

**Prop. (3.5.11.31) [Hermitian Forms on Complex Vector Spaces].** Given a complex vector space  $(V, J)/\mathbb{C}$  (3.5.11.30) and a Hermitian form  $V \times V \rightarrow \mathbb{C}$ , if we write  $(u, v) = \varphi(u, v) - i\psi(u, v)$ , then

- $\varphi$  is symmetric and  $\varphi(Ju, Jv) = \varphi(u, v)$ .
- $\psi$  is alternating and  $\psi(Ju, Jv) = \psi(u, v)$ .
- $\psi(u, v) = -\varphi(u, Jv), \varphi(u, v) = \psi(u, Jv)$ .

Conversely, if  $\varphi$  satisfies these conditions, then

$$(u, v) = \varphi(u, v) + i\varphi(u, Jv).$$

is a Hermitian form. Also  $(-, -)$  is positive or non-singular iff  $\varphi$  is.  $\lrcorner$

over  $\mathbb{H}$

**Def. (3.5.11.32) [Real Quaternion Algebra].** The **real quaternion algebra**  $\mathbb{H}$  is the space  $\mathbb{R}\{1, i, j, k\}$  subjects to the relations

$$ij = -ji = k, jk = -kj = i, ki = -ik = j, i^2 = j^2 = k^2 = -1.$$

In fact, by (19.5.5.3), any quaternion algebra over  $\mathbb{R}$  is isomorphic to  $\mathbb{H}$ .  $\lrcorner$

**Prop. (3.5.11.33) [Module of Quaternion Algebras].** There is an involution on  $\mathbb{H}$  s.t. if  $x = a + bi + cj + dk$ , then  $\bar{x} = a - bi - cj - dk$ . Then we define a the modulus of  $x \in \mathbb{H}$  as

$$|x|^2 = x\bar{x} = \bar{x}x = a^2 + b^2 + c^2 + d^2.$$

Then every non-zero element of  $\mathbb{H}$  is invertible. In particular, it is a skew field.  $\lrcorner$

**Prop. (3.5.11.34).**  $\mathbb{H}$  is isomorphic to subalgebra of matrices in  $\text{Mat}(2; \mathbb{C})$  consisting of elements of the form

$$\left\{ \begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} \mid \alpha, \beta \in \mathbb{C} \right\}$$

**Prop. (3.5.11.35).** The center of  $\mathbb{H}$  is  $\mathbb{R}$ , by (19.6.1.8).  $\lrcorner$

**Prop. (3.5.11.36) [Invertible Quaternion Matrices].** Denote  $\text{GL}(n; \mathbb{H})$  the set of invertible matrices in  $\text{Mat}(n; \mathbb{H})$ . If  $A \in \text{Mat}(n; \mathbb{H})$ ,  $A$  acts on  $\mathbb{H}^n$ , and determines a complex matrix  $A'$  in  $\text{Mat}(2n; \mathbb{C})$ .

- If  $A, B \in \text{Mat}(n; \mathbb{H})$  satisfies  $AB = 1$ , then  $BA = 1$ .
- $A \in M = \text{GL}(n; \mathbb{H})$  iff  $A' \in \text{GL}(2n; \mathbb{C})$ . In particular if we define the determinant of  $A \in \text{Mat}(n; \mathbb{H})$  as the determinant of  $A'$ , then  $\det(A) \neq 0$  iff  $A$  is invertible.  $\lrcorner$

*Proof:* 1:  $(AB)' = A'B'$ , thus  $\text{Mat}(n; \mathbb{H})$  is a subalgebra of  $\text{Mat}(2n; \mathbb{C})$ . Thus  $B'A' = 1$  by (3.5.9.15).

2: If  $A'$  is invertible, then  $A$  is a bijection, thus there are vectors  $v_1, \dots, v_n$  that  $Av_i = e_i$ . This means

$$(v_1, \dots, v_n)A = 1.$$

Thus by item1  $A$  is invertible.  $\square$

**Prop. (3.5.11.37) [Quaternion Hermitian Forms].**

- Every non-degenerate Hermitian quaternion form is of the form

$$(x, y) = \bar{x}_1 y_1 + \dots + \bar{x}_p y_p - \overline{x_{p+1}} y_{p+1} - \dots - \bar{x}_n y_n$$

in some basis, where  $p$  is uniquely determined by the form.

- Every non-degenerate skew-Hermitian quaternion form is of the form

$$(x, y) = \bar{x}_1 j y_1 + \dots + \bar{x}_n j y_n$$

in some basis.  $\lrcorner$

*Proof:* Firstly, there are some  $v$  that  $(v, v) \neq 0$ : If  $(v, v) = 0$  for all  $v$ , then  $(x, y) + (y, x) = 0$  for all  $x, y$ . If it is skew-Hermitian, this means  $(x, y)$  is real, which is impossible, unless  $(x, y) = 0$ . If it is Hermitian, this means  $(x, y)$  is imaginary, but  $(x, yi), (x, yj), (x, yk)$  are all imaginary, thus  $(x, y) = 0$ .

Then choose this  $v$ , and take the orthogonal complement of  $v$ , then by induction we can find  $v_1, \dots, v_n$  that is mutually orthogonal.

If it is Hermitian, then  $(v_i, v_i) \in \mathbb{R}$ , thus we can find some  $t_i \in \mathbb{R}$  that  $(tv_i, tv_i) = \pm 1$ .  $2p$  is the multiplicity of the eigenvalue 1 of the eigenspace of the matrix corresponding to the form, so  $p$  is uniquely defined.

If it is skew-Hermitian, then  $(v_i, v_i)$  is imaginary, thus by (16.6.1.7), there are  $u_i \in \mathbb{H}$  that  $(u_i v_i, u_i v_i) = \bar{u}_i (v_i, v_i) u_i = j$ .  $\square$

**Cor. (3.5.11.38).**

- Every non-degenerate Hermitian quaternion form is of the form

$$(x, y) = B_1(x, y) + jB_2(x, y)$$

in some basis, where  $B_1(x, y)$  is the usual canonical Hermitian form of signature  $(2p, 2q)$ , and  $B_2$  is the usual canonical skew-symmetric bilinear form. Also  $B_2(x, y) = B_1(xj, y)$ .

- Every non-degenerate skew-Hermitian quaternion form is of the form

$$(x, y) = B_1(x, y) + jB_2(x, y)$$

in some basis, where  $B_1(x, y)$  is the usual canonical skew-Hermitian form that  $iB_1$  is Hermitian of signature  $(n, -n)$ , and  $B_2$  is the usual canonical symmetric bilinear form. Also  $B_2(x, y) = B_1(xj, y)$ . in some basis.  $\lrcorner$

## 12 Positivity (Inner Spaces)

**Def. (3.5.12.1) [Inner Spaces].** A **real inner space** is a f.d. real quadratic space that  $B(v, v) \in \mathbb{R}_+$  for  $v \neq 0$ . It is necessarily non-degenerate.

A **complex inner space** is a f.d. Hermitian space  $V$  w.r.t.  $\mathbb{C}/R(3.5.11.6)$  s.t.  $B(v, v) \in \mathbb{R}_+$  for  $v \neq 0$ .  $\lrcorner$

**Prop. (3.5.12.2).** An inner metric on a vector space will induce an inner metric on the dual space, that is, asserting the dual basis of an orthonormal basis to be orthonormal. On an arbitrary basis, the matrix on the dual basis is written as  $A^{-1}$ . because we can write  $A = P^t P$ , and the dual basis transformation is like  $(P^t)^{-1}$ , so the metric matrix is  $A^{-1}$ .  $\lrcorner$

**Prop. (3.5.12.3) [Positivity and Principal Minors].** A matrix is positive symmetric (Hermitian) iff it is symmetric, and all its upper principle minors has positive determinants.

A positive symmetric (Hermitian) matrix is equivalent to a real (complex) inner space.  $\lrcorner$

*Proof:* Cf. [Hoffman P328].  $\square$

**Prop. (3.5.12.4) [Positive Eigenvalues are Positive Real].** The spectrum of a Hermitian matrix (e.g. positive symmetric matrix)  $A$  is positive and real.  $\lrcorner$

*Proof:* Let  $x$  be an eigenvector of  $A$  with eigenvalue  $\lambda$ , then  $\lambda$  is real by (3.5.11.8), and  $0 < (x, x) = x^* A x = \lambda x^* x$ , so  $\lambda > 0$ .  $\square$

**Prop. (3.5.12.5) [Positivity of Traces].** Let  $A, B \in \text{Pos}_{\geq 0}(n; \mathbb{R})$ ,  $\text{tr}(A) \geq 0$ , and  $\text{tr}(AB) \geq 0$ .  $\lrcorner$

*Proof:*  $\text{tr}(A) \geq 0$  because we can write  $A = P^{-1}DP$  where  $D$  is diagonal and  $P$  is orthogonal. Then the entries of  $D$  are non-negative, and  $\text{tr}(A) = \text{tr}(D) \geq 0$ .

Also, write  $A = P^t P$ , then  $\text{tr}(AB) = \text{tr}(PBP^t) \geq 0$ , because  $PBP^t$  is also a semi-positive matrix.

□

### 13 Matrix Analysis

**Thm. (3.5.13.1).** Let  $K \in \text{Pos}_{\geq 0}(n; \mathbb{C})$ , and the eigenvalues of  $K$  be  $\kappa_1 \geq \kappa_2 \geq \dots \geq \kappa_n \geq 0$  (3.5.12.4), then for any  $x \in \mathbb{C}^n$ ,

$$\kappa_n \|x\|^2 \leq x^t K x \leq \kappa_1 \|x\|^2.$$

$\lrcorner$

*Proof:* By [Principal Axis Theorem\(3.5.11.9\)](#), we can assume that  $K = U^{-1} \text{diag}(\kappa_1, \dots, \kappa_n) U$ , where  $U \in U(n)$ . Then

$$\kappa_n \|x\|^2 \leq \kappa_n \|Ux\|^2 \leq x^t K x = x^t U^t \text{diag}(\kappa_1, \dots, \kappa_n) U x \leq \kappa_1 \|Ux\|^2 = \kappa_1 \|x\|^2$$

□

**Thm. (3.5.13.2) [Weyl].** Let  $\varphi : [0, \infty) \rightarrow \mathbb{R}$  be a non-decreasing function s.t.  $\varphi(0) = 0$ , and  $\varphi(e^\xi)$  is a convex function of  $\xi$ . Suppose  $A \in \text{Mat}(n; \mathbb{C})$ , and  $K = A^* A$ , and let the eigenvalues of  $A$  be  $\alpha_1, \dots, \alpha_n$  with  $\lambda_i = |\alpha_i|^2$  satisfying  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ , and the eigenvalues of  $K$  be  $\kappa_1 \geq \kappa_2 \geq \dots \geq \kappa_n \in \mathbb{R}_+$  (3.5.12.4).

Then for any  $1 \leq m \leq n$ ,

$$\varphi(\lambda_1) + \dots + \varphi(\lambda_m) \leq \varphi(\kappa_1) + \dots + \varphi(\kappa_m).$$

In particular, for  $s \in \mathbb{R}_+$ ,

$$\lambda_1^s + \dots + \lambda_m^s \leq \kappa_1^s + \dots + \kappa_m^s.$$

$\lrcorner$

*Proof:* Notice first that by (12.3.12.4), it suffices to show that

$$\prod_{i=1}^m \lambda_i \leq \prod_{i=1}^m \kappa_i, \forall 1 \leq m \leq n.$$

Then notice  $\prod_{i=1}^m \lambda_i$  is the largest module of eigenvalues of  $\wedge^m A$ , and  $\prod_{i=1}^m \kappa_i$  is the largest eigenvalue of  $(\text{Sym}^m A)^* \text{Sym} A$  by (3.5.5.4), so it reduces to show that  $\lambda_1 \leq \kappa_1$ . For this, notice that if  $Ax = \alpha_1 x$ , then  $x^t K x = \lambda_1 \|x\|^2$ . But  $x^t K x \leq \kappa_1 \|x\|^2$  (3.5.13.1), so we are done.  $\square$

**Cor. (3.5.13.3).** If  $A \in \text{GL}(n, \mathbb{C})$  is invertible, then we can apply the theorem to  $A^{-1}$  to show that: For any non-increasing function  $\varphi : [0, \text{infy}) \rightarrow \mathbb{R}$  s.t.  $\lim_{x \rightarrow \infty} \varphi(x) = 0$ , and  $\varphi(e^\xi)$  is a convex function of  $\xi$ , and any  $1 \leq m \leq n$ ,

$$\varphi(\lambda_1) + \dots + \varphi(\lambda_m) \leq \varphi(\kappa_1) + \dots + \varphi(\kappa_m).$$

In particular, for  $s \in \mathbb{R}_+$ ,

$$\frac{1}{\lambda_1^s} + \dots + \frac{1}{\lambda_m^s} \leq \frac{1}{\kappa_1^s} + \dots + \frac{1}{\kappa_m^s}.$$

$\lrcorner$

### Positive Solutions

**Prop. (3.5.13.4) [Farkas' Lemma].** For  $A \in \text{Mat}(n; \mathbb{R})$  and  $b \in \mathbb{R}^n$ , exactly one of the following equation has a solution:

- $AX = b, X \geq 0$ .
- $Y^t A \leq 0, Y^t b > 0$ .

┘

*Proof:* First notice if both have a solution, then  $0 \geq Y^t AX > 0$ , contradiction. The rest follows from the Hahn-Banach separation theorem.  $\square$

**Cor. (3.5.13.5) [Gordan's Theorem].** exactly one of the following has a solution:

$$\begin{cases} AX > 0 \\ Y^t A = 0, Y \geq 0, Y \neq 0 \end{cases}$$

┘

*Proof:* If both have a solution, then  $0 = Y^t AX > 0$ , contradiction. If the first has no solution, then  $A'x = e, z \geq 0$ , where  $A' = [A, -A, -I]$  has no solution, by Farkas' lemma, there is a solution of  $Y^t A' \leq 0$  and  $Y^t b = 0$ . Which shows that  $Y^t A = 0$  and  $Y \neq 0$ .  $\square$

**Cor. (3.5.13.6).** For any subspace in  $\mathbb{R}^m$ , either it has an intersection with the open first quadrant, or its orthogonal complement has an intersection with the closed first quadrant minus 0. (Regard it has the image of a  $AX$ ).  $\square$

## 14 Perron-Frobenius Theory

References are [Mey00]Chap8 and [Matrix Analysis, Horn-Johnson]Chap8.

**Prop. (3.5.14.1) [Positive Eigenvalue].** For  $A \in \text{Mat}(n; \mathbb{R}_+)$ , then

1.  $\rho(A) \neq 0$ .
2.  $\rho(A) \in \sigma(A)$ ,
3. If  $x \in \mathbb{R}^n$  s.t.  $Ax = \rho(A)x$ , then  $A \text{abs}(x) = \rho(A) \text{abs}(x)$  and  $x \in (\mathbb{R}_{\geq 0})^n$ . In particular,  $A$  has a  $\rho(A)$ -eigenvector with positive entries.
4.  $\rho(A)$  is the only eigenvalue of  $A$  on the radius circle.

┘

*Proof:* Cf. [Matrix Analysis and Applied Linear algebra]P663. ?

- 1: If  $\rho(A) = 0$ , then  $A$  is nilpotent, so  $A^n = 0$ , which is impossible.
- 2:
- 3:
- 4:

 $\square$ 

**Prop. (3.5.14.2) [Multiplicity One and the Perron Vector].** For  $A \in \text{Mat}(n; \mathbb{R}_+)$ ,

- $\rho(A)$  has multiplicity 1.
- $A$  has a unique eigenvector  $p \in (\mathbb{R}_+)^n$  with  $\|p\| = 1$ , called the **Perron vector** of  $A$ .
- There are no other eigenvector of  $A$  with non-negative entries that is not a positive multiple of  $p$ .

$\rho(A)$  is also called the **Perron root** of  $A$ . ┘

*Proof:* We may assume that  $\rho(A) = 1$ .

1: We first show that  $\text{ind}(1) = 1$ . Cf.[Matrix Analysis and Applied Linear algebra]P664. ?

Then if  $\text{mul}(\rho(A)) = m > 1$ , then there are  $m$   $\rho(A)$ -eigenvectors for  $A$ . But for two different eigenvalues, we can use linear combination to get an eigenvector of  $A$  that has some entry 0, which contradicts(3.5.14.1).

2: This follows from(3.5.14.1).

3: Suppose  $(\lambda, v)$  is an eigenpair for  $A$  with  $v \in (\mathbb{R}_{\geq 0})^n$ , and  $u$  be the Perron vector of  $A^t$ . Then  $u^t v > 0$ , and

$$\rho(A)u^t v = u^t A v = \lambda u^t v.$$

So  $\lambda = \rho(A)$ , and the rest follows. □

**Prop. (3.5.14.3) [Collatz-Wielandt Formula].** For  $A \in \text{Mat}(n; \mathbb{R}_+)$ , the Perron root  $\rho(A)$  is given by

$$\rho(A) = \max_{\mathbf{x} \in (\mathbb{R}_{\geq 0})^n \setminus 0} \left( \min_{1 \leq i \leq n, x_i \neq 0} \frac{(A\mathbf{x})_i}{x_i} \right).$$

┘

*Proof:* For  $\mathbf{x} \in (\mathbb{R}_{\geq 0})^n \setminus 0$ , let  $\xi = \min_{1 \leq i \leq n, x_i \neq 0} \frac{(A\mathbf{x})_i}{x_i}$ , then  $0 \leq \xi \mathbf{x} \leq A\mathbf{x}$ . Let  $\mathbf{q}$  be the Perron vector for  $A^t$ , then  $\mathbf{q}^t \mathbf{x} > 0$ , and

$$\xi \mathbf{q}^t \mathbf{x} \leq \mathbf{q}^t A \mathbf{x} = \rho(A) \mathbf{q}^t \mathbf{x},$$

which implies that  $\xi \leq \rho(A)$ . □

### Non-negative Entried Irreducible Matrices

**Def. (3.5.14.4) [Irreducible Matrices].** ┘

**Def. (3.5.14.5) [Primitive Matrices].** An irreducible matrix  $A \in \text{Mat}(n; \mathbb{R}_{\geq 0})$  is called a **primitive matrix** if there are only one eigenvalue on the radius circle of  $A$ .

The number of eigenvalues on the radius circle is called the **index of primitivity** of  $A$ . ┘

**Prop. (3.5.14.6) [Characterizing Primitive Matrices].** An irreducible matrix  $A \in \text{Mat}(n; \mathbb{R}_{\geq 0})$  is primitive iff  $\lim_k (\frac{A}{\rho(A)})^k$  exists. In which case,

$$\lim_k \left( \frac{A}{\rho(A)} \right)^k = G = \frac{\mathbf{p} \mathbf{q}^t}{\mathbf{q}^t \mathbf{p}}$$

where  $\mathbf{p}, \mathbf{q}$  are Perron vectors for  $A$  and  $A^t$  resp. and  $G$  is the projection onto  $\ker(A - \rho(A)\mathbf{1})$  along  $\text{Im}(A - \rho(A)\mathbf{1})$ . ┘

*Proof:* Cf.[Matrix Analysis and Applied Linear algebra]P674. ? □

**Prop. (3.5.14.7).** If irreducible  $A \in \text{Mat}(n; \mathbb{R}_{\geq 0})$  has imprimitive index  $h$  with eigenvalues  $\{\lambda_1, \dots, \lambda_h\}$  on the radius circle, then  $\{\lambda_i\} = \mu_n$  and  $\text{mul}(\lambda_i) = 1$  for any  $i$ . ┘

*Proof:* Cf.[Matrix Analysis and Applied Linear algebra]P676. ? □

**Prop. (3.5.14.8) [Frobenius Test for Primitivity].** An irreducible matrix  $A \in \text{Mat}(n; \mathbb{R}_{\geq 0})$  is primitive, then  $A$  is primitive iff  $A^m \in \text{Mat}(n; \mathbb{R}_+)$  for some  $m \in \mathbb{Z}_+$ . ┘

*Proof:* Cf.[Matrix Analysis and Applied Linear algebra]P678. ? □



### Stochastic Matrices and Markov Chains

**Def. (3.5.14.9) [Stochastic Matrices].** A **stochastic matrix** is a matrix  $P \in \text{Mat}(n; \mathbb{R}_{\geq 0})$  s.t. each column sum of  $P$  is 1.  $\lrcorner$

**Def. (3.5.14.10) [Unit Eigenvalues].** A **unit eigenvalue** of a stochastic matrix (3.5.14.9) is a complex eigenvalue of unit length.  $\lrcorner$

**Prop. (3.5.14.11).** If  $P$  is a stochastic matrix, then  $1 \leq \rho(P) \leq \|P\|_{\infty} = 1$ , so  $\rho(P) = 1$ .  $\lrcorner$

**Prop. (3.5.14.12) [Canonical Forms for Stochastic Matrices].** If  $P \in \text{Mat}(n; \mathbb{R}_{\geq 0})$  is a stochastic matrix, then there is a permutation matrix  $Q$  s.t.

$$Q^{-1}PQ = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1r} & P_{1,r+1} & P_{1,r+2} & \dots & P_{1m} \\ 0 & P_{22} & \dots & P_{2r} & P_{2,r+1} & P_{2,r+2} & \dots & P_{2m} \\ \vdots & 0 & \ddots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & P_{rr} & P_{r,r+1} & P_{r,r+2} & \dots & P_{rm} \\ 0 & 0 & \dots & 0 & P_{r+1,r+1} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & P_{r+2,r+2} & \dots & 0 \\ \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & P_{m,m} \end{bmatrix}$$

where

- $P_{11}, \dots, P_{rr}$  are either irreducible or  $[0]_{1 \times 1}$ ,
- $P_{r+1,r+1}, \dots, P_{m,m}$  are irreducible,
- For  $1 \leq i \leq r$ , not all  $P_{i,i+j}$  are 0 for  $1 \leq j \leq m-i$ .

$\lrcorner$

*Proof:*

$\square$

**Prop. (3.5.14.13).** If  $P$  is a stochastic matrix, then every unit eigenvalue of  $P$  is semisimple, and is a root of unity.  $\lrcorner$

*Proof:* By using the canonical form (3.5.14.12), it suffices to show that  $\rho(P_{ii}) < 1$  for any  $1 \leq i \leq r$ : This is true if  $P_{ii} = 0$ , and if  $P_{ii}$  is irreducible, by the hypothesis we have  $P_{ii}\mathbf{e} < \mathbf{e}$ . So  $\rho(P_{ii}) < 1$ , otherwise we will get a contradiction by left multiplying a left Perron vector (3.5.14.2).  $\square$

**Cor. (3.5.14.14) [Stochastic Matrices are Cesàro Summable].** If  $P$  is a stochastic matrix, then  $P$  is Cesàro summable, i.e.

$$G = \lim_{k \rightarrow \infty} \frac{\mathbf{1} + P + \dots + P^{k-1}}{k}$$

exists, and  $G$  is the projection onto  $\ker(\mathbf{1} - P)$  along  $\text{Im}(\mathbf{1} - P)$ .  $\lrcorner$

*Proof:* This follows from (3.5.7.5).  $\square$

## 15 Others

**Prop. (3.5.15.1).** Let  $H^* = \bigoplus_{i=0}^{2d} H^i$  be a graded algebra over  $k$  s.t.  $\dim H^i < \infty$  for each  $i$ . Assume that for each  $0 \leq i \leq 2d$ , there is a perfect pairing

$$H^i \times H^{2d-i} \rightarrow K$$

induced from an isomorphism  $\text{tr} : H^{2d} \cong K$ , Let  $\varphi$  be a ring endomorphism of  $H^*$  that satisfies

- $\varphi(H^i) \subset H^i$ ,
- $\varphi_{2d} = \text{id}$

Then each  $\varphi_i$  is invertible, and  $\varphi_i^{-1} = \varphi_{2d-i}^t$ , where  $\varphi_{2d-i}^t$  is the transpose of  $\varphi_{2d-i}$  w.r.t. the perfect pairing  $H^i \times H^{2d-i} \rightarrow K$ . ┘

*Proof:* Let  $a \in H^i$  be any element, if  $a \neq 0$ , then because the pairing  $H^i \times H^{2d-i} \rightarrow K$  is perfect, there exists some  $b \in H^{2d-i}$  s.t.  $ab \neq 0$ . Thus  $\varphi_i(a)\varphi_{2d-i}(b) \neq 0$ , thus  $\varphi_i$  is injective. Now  $\text{tr}(\varphi_i^{-1}(a)b) = \text{tr}(\varphi_{2d}(\varphi_i^{-1}(a)b)) = \text{tr}(a\varphi_{2d-i}(b))$ , thus  $\varphi_i^{-1} = \varphi_{2d-i}^t$ . □

### 3.6 More on (Non-Commutative)Algebras

References are [Noncommutative Rings, T.Y.Lam], [Frobenius Algebras, 1], [[?Lan05](#)]Chap17 and [[Sta](#)]Chap11.

**Notation(3.6.0.1).**

- Use notations on algebras defined in [Algebras](#).
- Use notations on linear algebras defined in [Linear Algebra](#).
- Let  $R \in \text{Ring}$ .

┘

#### 1 Division Rings

**Prop.(3.6.1.1)[Wedderburn].** Any finite division ring  $D$  is a field. ┘

*Proof:* Use the class equation for the invertible elements of  $D$ , if it is not isomorphic, consider the center  $Z(D)$  of  $D$ , let  $|Z(D)| = z$ , then it is a field, and any other centralizer can be seen as a vector space over  $Z(D)$ , let it of dimension  $k$ , then  $z^n - 1 = z - 1 + \sum \frac{z^n - 1}{z^{k_i} - 1}$ . But then let  $\Psi_n$  be the cyclotomic polynomial of degree  $n$ , then  $\Psi_n(z)$  divides  $z - 1$ . But this is not true, as it is bigger.  $\square$

**Prop.(3.6.1.2).** Let  $k \in \text{Field}$ ,  $k = \bar{k}$ , then any division ring  $A$  over  $k$  with dimension  $< \#k$  is isomorphic to  $k$ .

In particular, a division ring  $A$  of at most countable dimension over  $\mathbb{C}$  is isomorphic to  $\mathbb{C}$ . ┘

*Proof:* Notice it suffices to find an eigenvalue of  $L_\varphi : A \rightarrow A$  for each  $\varphi \in A$ . But  $\{(\varphi - a)^{-1}\}$  is a uncountable set of elements of  $A$ , so some  $\sum a_i(\varphi - c_i)^{-1} = 0$ , spanning the expression, we see  $\prod_k(\varphi - \mu_k) = 0$ , so some  $\varphi - \mu_k = 0$ , contradiction.  $\square$

**Prop.(3.6.1.3)[Hua Equation].** In a division ring  $D$ , if  $a, b \neq 0$  and  $ab \neq 1$ , then

$$a - (a^{-1} + (b^{-1} - a)^{-1})^{-1} = aba$$

┘

*Proof:* suffices to show

$$1 = (1 - ab)a(a^{-1} + (b^{-1} - a)^{-1}) = (1 - ab)(1 + a(b^{-1} - a)^{-1}).$$

But this is equal to

$$(1 - ab)(1 - (b^{-1} - a)(b^{-1} - a)^{-1} + b^{-1}(b^{-1} - a)^{-1}) = (1 - ab)(1 - ab)^{-1} = 1.$$

$\square$

**Prop.(3.6.1.4)[Cartan-Brauer-Hua].** Let  $K \subset D$  be division rings s.t.  $x^{-1}Kx \in K$  for any  $x \in D^\times$ , then either  $K \subset Z(D)$  or  $K = D$ . ┘

*Proof:* Cf.[Lam, P211].  $\square$

**Remark(3.6.1.5).** For more about division rings, see [Finite Semi-simple Algebras](#) and [Brauer-Grothendieck Groups](#). ┘

## 2 Semisimplicity

**Def. (3.6.2.1).** For  $R \in \text{Ring}$ , a **simple  $R$ -module** is a  $E \in \text{Mod}_R$  that no submodules other than 0 and  $E$ . It is called **faithful** iff there is no nonzero element  $a \in R$  that  $ax = 0$  for any  $x \in E$ .

It is called **non-degenerate  $R$ -module** if  $RE = E$ .  $\lrcorner$

**Prop. (3.6.2.2) [Shur's lemma].** For  $R \in \text{Ring}$  and  $E$  a simple  $R$ -module,  $\text{End}_R(E)$  is a division ring, this is because the kernel and image are all 0 or  $E$ .  $\lrcorner$

**Cor. (3.6.2.3) [Uniqueness of Decomposition of Modules].** If an  $R$ -module  $E$  can be written as a finite direct sum of simple  $R$ -modules in multiple ways, then the multiplicity of the irreducible modules appearing in it is uniquely determined.  $\lrcorner$

*Proof:* Cf. [Lang, Algebra, P643]  $\square$

**Def. (3.6.2.4) [Semisimple Modules].** For  $R \in \text{Ring}$ , a **semisimple  $R$ -module** is a  $E \in \text{Mod}_R$  iff it satisfies the following equivalent conditions:

- It is a sum of simple modules.
- It is a direct sum of simple modules.
- Any submodule  $F$  of  $E$  has a complement in  $E$ .

$\lrcorner$

*Proof:*  $3 \rightarrow 2$ : By Zorn's lemma, it suffices to show any non-zero semisimple module contains a simple submodule: Take a  $m \neq 0 \in M$ , then we may assume  $M = Rm$ , and by Zorn's lemma there is a maximal submodule  $N$  that  $m \notin N$ , and let  $N \oplus N' = M$ , then we show  $N'$  is simple. because any submodule  $N''$  satisfies  $m \in N \oplus N''$  thus  $N'' = N'$ .

$2 \rightarrow 1$  is immediate, it suffices to show  $1 \rightarrow 3$ : for any submodule  $N \subset M$ , consider all the simple modules that intersect  $N$  trivially, denote their sum by  $V$ , I claim  $N \oplus V = M$ , otherwise, let  $S$  be a simple submodule that contained in  $N + V$ , then  $S \cap (N + V) = 0$ , so  $N \cap (S + V) = 0$ , contradicting the maximality.  $\square$

**Cor. (3.6.2.5).** For  $R \in \text{Ring}$ , any submodule and quotient module of a semisimple  $R$ -module is semisimple.  $\lrcorner$

*Proof:* The quotient is clearly a sum of simple modules, and for a submodule, its submodule has a complement.  $\square$

### Density Theorems

**Def. (3.6.2.6).** If  $E$  be a semisimple  $R$ -module, let  $R' = \text{End}_R(E)$ , then  $E$  is also a  $R'$ -module, where the action is given by  $(\varphi, x) \mapsto \varphi(x)$ . Then any element of  $R$  defines an element of  $\text{End}_{R'}(E)$  by left multiplication. If we called  $\text{End}_{R'}(E)$  the **bicommutant** of  $E$  over  $R$ , then  $\lrcorner$

**Lemma (3.6.2.7).** For a module  $E$  over  $R$ , let  $\text{End}_{R'}(E)$  be the bicommutant. If  $f \in \text{End}_{R'}(E)$  and  $x \in E$ , then there is an element  $\alpha \in R$  that  $\alpha x = f(x)$ .  $\lrcorner$

*Proof:* Since  $E$  is semisimple, write  $E = Rx \oplus F$ , and let  $\pi$  be the projection unto  $Rx$ , then  $\pi \in \text{End}_R(E)$ , and  $f(x) = f(\pi(x)) = \pi f(x) \in Rx$ .  $\square$

**Prop. (3.6.2.8) [Jacobson Density Theorem].** Let  $E$  be semisimple over  $R$  and let  $R' = \text{End}_R(E)$ . If  $f \in \text{End}_{R'}(E)$  and  $x_1, \dots, x_n \in E$ , then there is an element  $\alpha \in R$  that  $\alpha x_i = f(x_i)$  for all  $i$ .

In particular, if  $E$  is f.g. over  $R'$ , the natural map  $R \rightarrow \text{End}_{R'}(E)$  is surjective.  $\lrcorner$

*Proof:* Consider  $f^n \in \text{End}_R(E^n)$ , then it is just a  $n \times n$  matrix with entries in  $R' = \text{End}_R(E)$ . Consider the lemma(3.6.2.7) shows there is an  $\alpha \in R$  that

$$(\alpha x_1, \dots, \alpha x_n) = (f(x_1), \dots, f(x_n))$$

which is the result.  $\square$

**Cor.(3.6.2.9) [Wedderburn Theorem].** Let  $R \in \mathcal{R}$ ing and  $E$  is a faithful simple  $R$ -module. Let  $D = \text{End}_R(E)$ . If  $E$  is of f.g. over  $D$ , then  $R = \text{End}_D(E)$ .  $\lrcorner$

*Proof:* The density theorem(3.6.2.8) and the fact  $E$  is f.g. over  $D$  shows that  $R \rightarrow \text{End}_k(E)$  is surjective, and also injective because it is faithful, thus  $R = \text{End}_k(E)$ .  $\square$

**Cor.(3.6.2.10)[Burnside's Theorem].** Let  $E$  is an  $R$ -module that is f.d over an alg.closed field  $k$  and  $R$  is a subalgebra of  $\text{End}_k(E)$ . If  $E$  is simple as an  $R$ -module, then  $R = \text{End}_k(E)$ .  $\lrcorner$

*Proof:* It follows from Shur's lemma(3.6.2.2) and(3.6.1.2)  $R' = \text{End}_R(E)$  is just  $k$ , so we can use Wedderburn's theorem(3.6.2.9).  $\square$

**Cor.(3.6.2.11)[F.d. Simple Module over Commutative Algebra].** If  $R$  is commutative and  $E$  is a simple  $R$ -module of f.d. over an alg.closed field  $k$ , then  $E$  is of 1-dimensional.  $\lrcorner$

*Proof:* The theorem shows the image of  $R$  in  $\text{End}_k(E)$  is all of  $\text{End}_k(E)$ , but Shur's lemma(3.6.2.2) and(3.6.1.2) that the image of  $R$  consists of scalars, thus  $\dim_k E = 1$ .  $\square$

**Prop.(3.6.2.12) [Projection Operators].** Let  $k$  be a field and  $R$  is a  $k$ -algebra. If  $V_1, \dots, V_n$  are pairwise non-isomorphic simple  $R$ -modules of f.d. over  $k$ , then there exists elements  $e_i$  in  $R$  that acts as identity on  $V_i$  and 0 on other  $V_j$ .  $\lrcorner$

*Proof:* This is an immediate consequence of Jacobson density theorem(3.6.2.8) applied to the projection operator on  $\oplus_i V_i$ .  $\square$

**Prop.(3.6.2.13)[Characters Determine F.D. Representations(Bourbaki)].** Let  $R$  be an algebra over a field  $k \in \mathbf{Field}^0$ , and  $E_1, E_2$  be two semisimple  $R$ -module of f.d. over  $k$  s.t. they have the same character  $\chi_1 = \chi_2$ , then the  $R$ -modules  $E_1, E_2$  are isomorphic.  $\lrcorner$

*Proof:*  $E, F$  are isomorphic to direct sums of simple  $R$ -modules, so it suffices to show the multiplicities  $m, n$  of any simple module  $V$  is the same. We can find an element  $e$  that is identity on  $E$  and 0 on other simple modules  $V_i$  by(3.6.2.12), thus the trace of  $e$  on  $E, F$  are  $m \dim_k(V), n \dim_k(V)$  respectively, thus  $m = n$  because  $k$  is of char0.  $\square$

**Def.(3.6.2.14)[Matrix Coefficients].** Let  $R$  be an algebra over a field  $k \in \mathbf{Field}^0$  and  $M \in \text{Mod}_R$  is of f.d. over  $k$ , then a **matrix coefficient** of  $M$  is a function  $c : R \rightarrow k$  of the form  $c(r) = L(rx)$ , where  $L$  is a linear functional on  $M$  and  $x \in M$ .  $\lrcorner$

**Cor.(3.6.2.15).** If  $R$  is an algebra over a field  $k$  and two simple  $R$ -modules that is f.d. over  $k$  have a nonzero matrix coefficient in common, then they are isomorphic.  $\lrcorner$

*Proof:* Let  $c(r) = L_1(rx_1) = L_2(rx_2)$ . If they are non-isomorphic, then we can find a  $e \in R$  that is identity on  $M_1$  and 0 on  $M_2$  by(3.6.2.12). Let  $c(u) \neq 0$ , then

$$c(ue) = L_1(ue x_1) = L_1(ux_1) = c(u) \neq 0, \quad c(ue) = L_2(ue x_2) = L_2(0) = 0$$

contradiction.  $\square$

**Prop. (3.6.2.16) [Simple Modules of Tensor Product].** Let  $A, B$  be algebras over a field  $\Omega$ ,  $R = A \otimes B$ , then:

- If  $P$  be a simple  $R$ -module of f.d. over  $\Omega$ , then there is a simple  $A$ -module  $M$  and a simple  $B$ -module  $N$  that  $P$  is isomorphic to a quotient of  $M \otimes N$ . Also the isomorphism classes of  $M, N$  are uniquely determined.
- If  $\Omega$  is alg.closed and  $M, N$  are simple modules of  $A, B$  of f.d. over  $\Omega$ , then  $M \otimes N$  is a simple module over  $A \otimes B$ .

┘

*Proof:* 1: because  $P$  is f.d., it contains a simple  $A$ -module  $M$ . Let  $N_1 = \text{Hom}_A(M, P)$ , then it is a  $B$ -module as  $A, B$  commutes in  $A \otimes B$ . Then we can define a map  $\lambda : M \otimes N_1 \rightarrow P$ , which is a  $A \otimes B$ -module morphism. Now  $N_1$  is also of f.d., so it contains a simple  $B$ -module  $N$ , and  $\lambda$  is clearly non-zero on  $M \otimes N$ , thus its image is all of  $P$ , as  $P$  is simple.

For the uniqueness, let  $d = \dim N$ , then  $P$  is isomorphic to  $k \leq d$  copies of  $M$  as an  $A$ -module, so the isomorphism class of  $M$  is determined, so does that of  $N$ .

2: Consider the map of  $A, B$  to  $\text{End}_\Omega(M), \text{End}_\Omega(N)$  are surjective by (3.6.2.10). Then it suffices to show that  $M \otimes N$  are simple over  $\text{End}_\Omega(M) \otimes \text{End}_\Omega(N)$ , but this is just  $\text{End}_\Omega(M \otimes N)$ , over which  $M \otimes N$  is clearly simple.  $\square$

### Semisimple Rings and Simple Rings

**Def. (3.6.2.17) [Semisimple Ring].** A (left) **semisimple ring** is a ring  $R$  s.t. it is a (left) semisimple module over itself (3.6.2.4). In (3.6.2.23), we will see left-semisimple is equivalent to right-semisimple.

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**Prop. (3.6.2.18).** Any semisimple ring is left Artinian and Noetherian.  $\square$

*Proof:* Consider the decomposition  $R = \oplus_\alpha \mathfrak{A}_\alpha$ , where  $\mathfrak{A}_\alpha$  are simple left  $R$ -ideals. Then by considering the decomposition of  $1 \in R$ , clearly this is a finite sum. The rest is easy.  $\square$

**Prop. (3.6.2.19).** If  $R$  is a semisimple ring, then any  $R$ -module is semisimple.  $\square$

*Proof:* Any  $R$ -module is a quotient of a free  $R$ -module, thus semisimple, by (3.6.2.5).  $\square$

**Prop. (3.6.2.20).**  $R$  is semisimple iff all  $R$ -modules are projective.  $\square$

*Proof:* If  $R$  is left-semisimple, for any  $R$ -module  $P$  and an exact sequence

$$0 \rightarrow N \rightarrow M \rightarrow P \rightarrow 0$$

by (3.6.2.19), there is a complement of  $N$  in  $M$ , thus this sequence splits and  $P$  is projective. The converse is also clear that any submodule of any  $R$ -module has a complement.  $\square$

**Lemma (3.6.2.21).** Let  $D$  be a division ring and  $R = \text{Mat}(n; D)$ , then

- $R$  is simple, left semisimple and left Noetherian.
- $R$  has a unique left simple module  $V$ , and  $R$  acts faithfully on  $V$ , with  $R \cong nV$ .
- $\text{End}_R(V) \cong D$ .

┘

*Proof:* Cf. [Lam, P31].  $\square$

**Prop. (3.6.2.22) [Wedderburn-Artin].** Any left semisimple ring is of the form  $R \cong \text{Mat}(n_1; D_1) \times \dots \times \text{Mat}(n_r; D_r)$ , and  $D_k$  are uniquely determined division rings. There are exactly  $r$  different left simple  $R$ -modules and  $D_i$  are uniquely determined.  $\lrcorner$

*Proof:* Consider the decomposition

$$R \cong n_1 V_1 \oplus \dots \oplus n_r V_r$$

where  $V_i$  are non-isomorphic simple left  $R$ -modules. Then we can use Schur's lemma (3.6.2.2) and (3.6.2.21) to calculate the endomorphism ring, so

$$R \cong \text{End}(n_1 V) \times \dots \times \text{End}(n_r V_r) \cong \text{Mat}(n_1, D_1) \times \dots \times \text{Mat}(n_r, D_r).$$

For the uniqueness, we use (3.6.2.21), which shows  $D_i$  and  $V_i$  both can be recovered.  $\square$

**Cor. (3.6.2.23).**  $R \in \text{Ring}$  is left semisimple iff it is right semisimple. ((3.5.1.13) is used).  $\lrcorner$

**Cor. (3.6.2.24).** A semisimple commutative ring is a finite direct product of fields.  $\lrcorner$

**Def. (3.6.2.25) [Semisimple Categories].** Let  $k \in \text{Field}$  and  $\mathcal{A} \in \text{AbCat}/k$  that  $\text{End}(X)$  are all finite  $k$ -modules for  $X \in \mathcal{A}$ , then  $\mathcal{A}$  is semisimple iff  $\text{End}(X)$  is a semisimple  $k$ -algebra for any  $X$ .  $\lrcorner$

*Proof:* If  $\mathcal{A}$  is semisimple, then any  $X \in \mathcal{A}$  is a direct sum of simple objects, so  $\text{End}(X)$  is semisimple.

Conversely, if  $\text{End}(X)$  is semisimple thus a product of matrix algebras over division algebras, so  $X$  can be indecomposable only if  $\text{End}(X)$  is a division algebra. Now if  $f : M \rightarrow N$  is a morphism of indecomposable objects, if there is a map  $g : N \rightarrow M$  that  $g \circ f \neq 0$ , then  $g \circ f$  is an automorphism of  $M$ , and  $(g \circ f)^{-1} \circ g$  is a right inverse to  $f$ , so  $N$  is a direct sum of  $M$ , and thus  $f$  is an isomorphism because  $M$  is indecomposable.

Now it suffices to show any indecomposable object is simple. If  $M$  is an indecomposable object properly contained in another indecomposable object, then

$$\begin{bmatrix} 0 & 0 \\ \text{Hom}(M, N) & 0 \end{bmatrix} \subset \begin{bmatrix} \text{End}(M) & \text{Hom}(N, N) \\ \text{Hom}(M, N) & \text{End}(N) \end{bmatrix} = \text{End}(M \oplus N)$$

is a two sided nilpotent nonzero ideal, contradicting the fact  $\text{End}(M \oplus N)$  is semisimple.  $\square$

**Prop. (3.6.2.26) [Semisimplicity and Base Change].** If  $A \in \text{Alg}_k$  and  $A \otimes_k K$  is semisimple for some field extension  $K/k$ , then  $A$  is semisimple. Conversely, if  $A \in \text{Alg}_k$  is semisimple, then  $A \otimes_k K$  is semisimple for every separable field extension  $K/k$ .  $\lrcorner$

*Proof:* Cf. [Milne, Lie Algebras, Lie Groups and Algebraic Groups] P48.  $\square$

**Prop. (3.6.2.27) [Characters Determine F.D. Representations].** If  $R$  is a semisimple ring over a field  $k \in \text{Field}^0$ , then its f.d. representations are determined by their characters, by (3.6.2.13).  $\lrcorner$

**Prop. (3.6.2.28) [Simple Artinian Algebras].** For a simple ring  $R$  (3.2.1.2), the following are equivalent:

- $R$  is left semisimple.
- $R$  is left Artinian.
- $R$  has a minimal left ideal.

- $R \cong \text{Mat}(n; D)$  for some division ring  $D$ .

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*Proof:* The equivalence of 1, 4 is by Wedderburn theorem (3.6.2.22) and (3.6.2.21).  $2 \rightarrow 3$  is easy, and for  $1 \rightarrow 2$ ,  $R$  is a finite direct sum of minimal ideals by Wedderburn theorem (3.6.2.22), so it is Artinian.

For  $3 \rightarrow 1$ , consider all the ideals in  $R$  that is isomorphic to the minimal ideal  $\mathfrak{A}$ , then it is also a right ideal of  $R$ , so equals  $R$ , hence  $R$  is semisimple.  $\square$

**Cor. (3.6.2.29).** And finite simple  $k$ -algebra  $A$  is of the form  $\text{Mat}(n; D)$  where  $D$  is a finite division ring over  $k$ . This is because  $A$  is clearly left Artinian.  $\square$

**Prop. (3.6.2.30) [Double Centralizer Property].** Let  $R$  be a simple ring and  $\mathfrak{A}$  a nonzero left ideal. Let  $D = \text{End}_R(\mathfrak{A})$ , then the natural map  $f : R \rightarrow \text{End}_D(\mathfrak{A})$  is an isomorphism.  $\square$

*Proof:* since  $R$  is simple,  $f$  is injective. To show it is surjective, let  $E = \text{End}_D(\mathfrak{A})$ , then for any  $a, r \in \mathfrak{A}$  and  $h \in E$ , we have  $h(ra) = h(r)a$ , thus

$$(h \cdot f(r))a = h(ra) = h(r)a = f(h(r))a$$

which shows  $f(R)$  is a left ideal in  $E$ . And because  $\mathfrak{A}R = R$ , we have  $f(R) = f(\mathfrak{A})f(R)$ , then

$$Ef(R) = Ef(\mathfrak{A})f(R) \subset f(\mathfrak{A})f(R) = f(R).$$

which shows  $f(R)$  is a left ideal in  $E$ , and it contains 1, so  $f(R) = E$ .  $\square$

### 3 Jacobson Radicals

**Def. (3.6.3.1) [Jacobson Radical].** For  $R \in \text{Ring}$ , the **Jacobson radical** is defined to be the intersection of all maximal left ideals in  $R$ .

$R$  is called **Jacobson-semisimple** if  $\text{rad } R = 0$ .

$R$  is called **semi-local** if  $\overline{R} = R/\text{rad } R$  is semisimple (3.6.2.17).

$R$  is called **semi-primary** if  $R$  is semi-local and  $\text{rad } R$  is nilpotent.  $\square$

**Prop. (3.6.3.2) [Characterizing Jacobson Radicals].** For  $R \in \text{Ring}$  and  $y \in R$ , the following are equivalent:

- $y \in \text{rad } R$ .
- $1 - xy$  is left-invertible for any  $x \in R$ .
- $yM = 0$  for any simple left  $R$ -module  $M$ .
- $1 - xyz$  is invertible for any  $x, z \in R$ .

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*Proof:*  $1 \rightarrow 2$ : If  $1 - xy$  is not left invertible, then it is contained in a maximal left ideal  $\mathfrak{m}$ , but  $y \in \mathfrak{m}$ , then  $1 \in \mathfrak{m}$ , contradiction.

$2 \rightarrow 3$ : If  $ym \neq 0$ , then we must have  $Rym = M$ , so  $xym = m$  for some  $x \in R$ , which shows  $(1 - xy)m = 0$ , but then  $m = 0$ .

$3 \rightarrow 1$ : Consider the simple left  $R$ -module  $R/\mathfrak{m}$  for any maximal left ideal  $\mathfrak{m}$ .

$4 \rightarrow 2$  is trivial, for  $1 + 2 + 3 \rightarrow 4$ : By item 3 we know  $\text{rad } R$  is an ideal, so  $yz \in \text{rad } R$  and there is a  $u$  that  $u(1 - xyz) = 1$ . But then  $u = 1 + u(xyz)$  is also left-invertible, so  $u$  is invertible and  $1 - xyz$  is invertible.  $\square$



**Cor. (3.6.3.3)[rad( $R$ ) is 2-Sided].**  $\text{rad } R$  is the largest ideal  $\mathfrak{A}$  of  $R$  that  $1+\mathfrak{A}$  are all units. In particular,  $\text{rad}(R)$  is a 2-sided ideal of  $R$ , and it is also the intersection of maximal right ideals of  $R$ .  $\lrcorner$

**Def. (3.6.3.4) [Locally Nilpotent Subsets].** A subset of  $R \in \mathcal{C}\text{Ring}$  is called **locally nilpotent** iff every element of it is nilpotent.  $\lrcorner$

**Lemma (3.6.3.5).** if a left or right ideal  $\mathfrak{A} \subset R$  is locally nilpotent, then  $\mathfrak{A} \subset \text{rad } R$ .  $\lrcorner$

*Proof:* Suppose  $y \in \mathfrak{A}$ , then  $xy \in \mathfrak{A}$  is nilpotent. So  $1 - xy$  has an inverse, for any  $x$ . Then  $y \in \text{rad } R$ , by (3.6.3.2).  $\square$

**Prop. (3.6.3.6) [Artinian Radical Nilpotent].** In a left Artinian ring,  $\text{rad } R$  is the largest nilpotent left ideal, and it is also the largest nilpotent right ideal.  $\lrcorner$

*Proof:* By the above lemma, it suffices to show  $J = \text{rad } R$  is nilpotent. By Artinian property, the descending chain  $J \supset J^2 \supset J^3 \supset \dots$  is stabilizing, so there exists  $k$  s.t.  $J^k = J^{k+1} = I$ . Then  $I = 0$ , because otherwise we can choose a minimal non-zero left ideal  $\mathfrak{A}$  s.t.  $I\mathfrak{A} = 0$ . Then if  $a \neq 0 \in \mathfrak{A}$ ,

$$I(Ia) = I^2a = Ia = 0,$$

so  $\mathfrak{A} = Ia$ , and  $a = ya$  for some  $y \in J$ , so  $(1 - y)a = 0$ . But  $1 - y$  is a unit by (3.6.3.2), contradiction.  $\square$

**Cor. (3.6.3.7).** In a left Artinian ring, any 1-sided locally nilpotent ideal is nilpotent. By left Artinian, there is a  $k$  that  $(\text{rad } R)^k = (\text{rad } R)^{k+1} = I$ . Now if  $I \neq 0$ , then we can choose a minimal left ideal  $\mathfrak{A}$  that  $I\mathfrak{A} \neq 0$  by Artinian property. Now there is an  $a \in \mathfrak{A}$  that  $Ia \neq 0$ , so  $I(Ia) = Ia \neq 0$ , so  $Ia = \mathfrak{A}$  and  $a = ya$  for some  $y \in I$ . But  $1 - y$  is invertible, so  $a = 0$ , contradiction.  $\lrcorner$

**Prop. (3.6.3.8) [Semisimplicity and Jacobson Semisimplicity].** For  $R \in \text{Ring}$ , the following are equivalent:

- $R$  is semisimple.
- $R$  is Jacobson-semisimple and left Artinian.
- $R$  is Jacobson semisimple and satisfies DCC on principal left ideals.

$\lrcorner$

*Proof:*  $1 \rightarrow 2$ :  $R$  is left Artinian by (3.6.2.18), and consider  $R = \text{rad } R \oplus \mathfrak{B}$ , then  $\mathfrak{B}$  is contained in a maximal left ideal  $\mathfrak{m}$ , which cannot contain  $\text{rad } R$ , unless  $\text{rad } R = 0$ .

$3 \rightarrow 1$ : 3 implies that any left ideal  $\mathfrak{A}$  contains a minimal left ideal  $I$  (the minimal principal one), and every minimal left ideal  $I$  is a direct summand of  ${}_R R$  (by choosing the maximal left ideal  $\mathfrak{m}$  not containing  $I$ , because  $I \oplus \mathfrak{m} = R$ ).

Then we can deduce 1: If  $R$  is not semisimple, then take a minimal left ideal  $\mathfrak{B}_1$ , then  $R = \mathfrak{B}_1 \oplus \mathfrak{A}_1$ , and  $\mathfrak{A}_1 \neq 0$  otherwise  $R$  is semisimple, and we can choose a minimal left ideal  $\mathfrak{B}_2 \subset \mathfrak{A}_1$ , then  $\mathfrak{A}_1 = \mathfrak{B}_2 \oplus \mathfrak{A}_2$ . Continuing this way, we get a chain of left ideals  $\mathfrak{A}_1 \supset \mathfrak{A}_2 \supset \dots$ , and they are both principal because they are direct summands of  ${}_R R$ , contradicting item 3.  $\square$

**Prop. (3.6.3.9) [Hokins-Levitzki Theorem].** Let  $R$  be a semi-primary ring (3.6.3.1), then for any  $R$ -module  $M$ , the following are equivalent:

- $M$  is Noetherian.
- $M$  is Artinian.

- $M$  has a composition series.

In particular, a ring is left Artinian iff it is left Noetherian and semi-primary.  $\lrcorner$

*Proof:* It suffices to prove if  $M$  is Noetherian or Artinian,  $M$  has a composition series. Denote  $J = \text{rad } R$ , then  $J^n$  for some  $n > 0$ . We consider

$$0 \subset J^{n-1}M \subset \dots \subset JM \subset M,$$

and the quotient  $J^{k-1}M/J^kM$  is Artinian or Noetherian over  $\overline{R} = R/\text{rad } R$  which is semisimple, so it is a direct sum of simple modules, and the sum is finite, so there is a composition series.

The last assertion: A left Artinian ring is semi-primary, by (3.6.3.6) and (3.6.3.8). So the assertion follows from the equivalence of item 1 and 2.  $\square$

**Lemma (3.6.3.10).** Let  $x \in \text{rad}(R)$  where  $R$  is a  $k$ -algebra, then  $x$  is algebraic over  $k$  iff  $x$  is nilpotent.  $\lrcorner$

*Proof:* One direction is trivial. For the other, if  $x^r + a_1x^{r+1} + \dots + a_nx^{r+n} = 0$ , then because

$$1 + a_1x^1 + \dots + a_nx^n$$

is invertible, we have  $x^r = 0$ .  $\square$

**Prop. (3.6.3.11) [Amitsur].** Suppose  $k \in \mathbf{Field}$  and  $R \in \mathbf{Ring}/k$  s.t.  $\dim_k R < \#k$ , then  $\text{rad } R$  is the largest locally nilpotent ideal of  $R$ .  $\lrcorner$

*Proof:* If  $|k| < \infty$ , then  $R$  is Artinian, so  $\text{rad } R$  is nilpotent by (3.6.3.6), and it is the largest by (3.6.3.6) again. Suppose now  $k$  is infinite. By the lemma above, it suffices to show every  $r \in \text{rad } R$  is algebraic over  $k$ . Notice that  $a - r$  is invertible for  $a \in k^*$ , and  $\{(a - r)^{-1}\}$  cannot be  $k$ -linearly independent because  $\dim_k R < |k|$ , so there is a dependence relation

$$\sum_{i=1}^n b_i(a_i - r)^{-1} = 0.$$

Hence  $r$  is algebraic over  $k$ .  $\square$

**Prop. (3.6.3.12) [Amitsur].** Let  $R \in \mathbf{Ring}$  and  $S = R[T]$ . Let  $J = \text{rad } S$  and  $N = R \cap J$ , then  $N$  is a locally nilpotent ideal in  $R$ , and  $J = N[T]$ . In particular, if  $R$  is Jacobson semisimple, then  $S$  is also Jacobson semisimple (3.6.3.5).  $\lrcorner$

*Proof:* Cf. [Lam, P71].  $\square$

**Lemma (3.6.3.13).** Let  $R \in \mathbf{Ring}/k$  and  $K/k$  is a separable algebraic field extension, then if  $R$  is Jacobson semisimple, so is  $R \otimes_k K$ .  $\lrcorner$

*Proof:* Cf. [Lam, P76].  $\square$

**Prop. (3.6.3.14) [Jacobson Radical Under Base Change of Fields].** Let  $R \in \mathbf{Ring}/k$  and  $K/k$  a separable algebraic extension, then  $\text{rad}(R \otimes_k K) = (\text{rad } R) \otimes_k K$ .  $\lrcorner$

*Proof:* Cf. [Lam, P76].  $\square$

## 4 Finite Semi-simple Algebras

References are [Sta]Chap 11.

**Notation(3.6.4.1).**

- Let  $k \in \mathbf{Field}$ .

┘

**Def.(3.6.4.2) [Separable Semi-simple Algebras].** A **separable semisimple  $k$ -algebra** is a  $k$ -algebra  $A \in \mathbf{Ring}/k$  s.t.  $Z(A)$  is étale over  $k$ . Every separable semisimple  $k$ -algebra is a product of separable simple  $k$ -algebras, which are of the form  $\mathbf{Mat}(n, K)$  for some  $K/k \in \mathbf{Field}$  separable. ┘

**Prop.(3.6.4.3).** If  $A$  is a separable semisimple  $k$ -algebra, then  $A \otimes_k k'$  is a separable semisimple  $k'$ -algebra for any  $k'/k \in \mathbf{Field}$ . ┘

**Def.(3.6.4.4) [Azumaya Algebras].** A **central  $k$ -algebra** is a  $k$ -algebra  $A \in \mathbf{Ring}_k$  s.t.  $Z(A) = k$ . ┘

**Def.(3.6.4.5) [Azumaya Algebras].** An **Azumaya algebra** over  $k$  is defined to be a finite central simple  $k$ -algebra(3.2.1.2). The category of Azumaya algebras over  $k$  is denoted by  $\mathbf{Az}_k$ . ┘

**Lemma(3.6.4.6).** Let  $D/k$  be a central division ring and  $A \in \mathbf{Ring}/k$ , then any two-sided ideal  $I$  of  $A \otimes_k D$  is of the form  $J \otimes_k D$  for some two-sided ideal  $J$  of  $A$ . In particular, if  $A$  is simple, then  $A \otimes_k D$  is also simple. ┘

*Proof:* Cf.[Sta]074C or [Frobenius Algebras, 1]P371.?

□

**Prop.(3.6.4.7) [Tensor Product of Simple Algebras].** if  $A, A'$  are two simple  $k$ -algebras that  $A$  is finite central over  $k$ , then  $A \otimes_k A'$  is also simple. ┘

*Proof:* Let  $A'$  be finite central over  $k$ , then by(3.6.2.28),  $A' \cong \mathbf{Mat}(n; D)$  for some finite central division algebra over  $k$ . Then

$$A \otimes_k A' \cong \mathbf{Mat}(n; A \otimes_k D),$$

which is simple by(3.6.4.6) and(3.6.4.10). ┘

□

**Cor.(3.6.4.8).** The tensor product of two Azumaya  $k$ -algebras is an Azumaya  $k$ -algebra. ┘

*Proof:* Combine the proposition with(3.2.1.8). ┘

□

**Cor.(3.6.4.9) [Base Change].** If  $A \in \mathbf{Ring}/k$ , then for any field extension  $k'/k$ ,  $A \otimes_k k' \in \mathbf{Az}_{k'}$  iff  $A \in \mathbf{Az}_k$ . ┘

*Proof:* Combine(3.2.1.8) with(3.6.4.7). ┘

□

**Prop.(3.6.4.10) [Matrix Algebras].** Let  $R \in \mathbf{Ring}$  and  $n \in \mathbb{Z}_+$ , then

- The functors  $M \mapsto M^{\oplus n}, N \mapsto e_{11}n$  defines an equivalence of categories between  $\mathbf{Mod}_R$  and  $\mathbf{Mod}_{\mathbf{Mat}(n; R)}$ .
- Any two-sided ideal of  $\mathbf{Mat}(n; R)$  is of the form  $\mathbf{Mat}(n; I)$  for some two-sided ideal  $I$  of  $R$ .
- Then center of  $\mathbf{Mat}(n, R)$  is the equal to the center of  $R$ .

┘

*Proof:* 3 is easy. ┘

□

**Thm. (3.6.4.11) [Skolem-Noether].** Let  $A \in \text{Az}_k$  and  $B$  be a simple  $k$ -algebra, and  $f, g : B \rightarrow A \in \text{Ring}_k$ . Then there exists  $x \in A^*$  s.t.  $f(a) = x^{-1}g(a)x$  for any  $a \in A$ .  $\lrcorner$

*Proof:* Choose a simple  $A$ -module  $M$ , then  $L = \text{End}_A(M)$  is a division ring, and  $M$  has two  $B \otimes_k L^{\text{op}}$  structures induced by  $f$  and  $g$ . The  $k$ -algebra  $B \otimes_k L$  is simple by (3.6.4.7), and also  $B$  is finite simple because there is a  $k$ -homomorphism  $B \rightarrow A$ , so  $B \otimes_k L$  is finite simple. Thus the two  $B \otimes_k L$ -structures on  $M$  are isomorphic, which means there is a  $\varphi : M \rightarrow M$  intertwining these two structures. But  $\varphi$  commutes with  $L$ , meaning that  $\varphi$  is just multiplying by some  $x \in A$  by Jacobson density theorem (3.6.2.8), so  $x$  is what we want.  $\square$

**Cor. (3.6.4.12).** If  $A \in \text{Az}_k$ , then any automorphism of  $A$  is inner.  $\lrcorner$

### Splitting Fields

**Prop. (3.6.4.13) [Centralizer Theorem].** Let  $A \in \text{Az}_k$ , and  $B$  a simple sub- $k$ -algebra of  $A$ , then

- $C = C_A(B)$  is also simple.
- $\dim_k A = \dim_k B \cdot \dim_k C$ .
- $C_A(C) = B$ .

$\lrcorner$

*Proof:* Cf. [Sta]074T. ?  $\square$

**Cor. (3.6.4.14).** Let  $B \subset A \in \text{Az}_k$ , then  $C = C_A(B)$  is also in  $\text{Az}_k$ , and  $A \cong B \otimes_k C$ .

In particular, if a division ring  $D$  is of f.d. over its center  $k$ , then  $D \in \text{Az}_k$ , and  $\dim_k D$  is a square.  $\lrcorner$

*Proof:*  $\dim_k A = \dim_k(B \otimes_k C)$  by (3.6.4.13), and  $B \otimes_k C$  is simple by (3.6.4.7), so the natural map  $B \otimes_k C \rightarrow A$  is an isomorphism, and the center of  $C$  is  $k$  by (3.6.4.7).  $\square$

**Prop. (3.6.4.15).** For  $A \in \text{Az}_k$ , If  $K \subset A$  is a sub- $k$ -field, then the following are equivalent:

- $\dim_k A = [K : k]^2$ .
- $C_A(K) = K$ .
- $K$  is a maximal commutative subring of  $A$ .

$\lrcorner$

*Proof:* 1, 2 are equivalent by (3.6.4.13),  $2 \rightarrow 3$  is trivial, and  $3 \rightarrow 2$  because if  $C_A(K) \neq K$ , take any  $a \in C_A(K)$ , we get a larger commutative subring  $K(a)$  of  $A$ .  $\square$

**Cor. (3.6.4.16) [Central Simple Algebras has Square Dimensions].** If  $D \in \text{Az}_k$  is a central division algebra, then  $\dim_k D$  is a square, and every maximal subfield  $K$  of  $D$  satisfies  $\dim_k D = [K : k] = \sqrt{\dim_k D}$ .  $\lrcorner$

*Proof:* This is because any commutative subring of  $D$  is a field.  $\square$

**Def. (3.6.4.17) [Splitting Fields].** For  $k \in \text{Field}$  and  $B$  a finite semisimple  $k$ -algebra, then  $B$  is said to **split** if  $B$  is isomorphic to a product of matrix algebras over  $k$ . A field extension  $K/k$  is called a **splitting field** of  $B$  if  $B \otimes_k K$  is split.  $\lrcorner$

**Prop. (3.6.4.18) [Maximal Subfields are Splitting].** Let  $k \subset_{\text{fin}} k' \in \mathbf{Field}$ ,  $A \in \mathbf{Az}_k$ , then  $k'$  is a splitting field of  $A$  iff there exists a  $B \in \mathbf{Az}_k$  similar to  $A$  (11.2.1.5) s.t.  $k' \subset B$  and  $\dim_k B = [k' : k]^2$ .

In particular, if  $D \in \mathbf{Az}_k$  is a central division ring, any maximal subfield of  $D$  is a splitting field of  $D$ , by (3.6.4.15),  $\lrcorner$

*Proof:* Cf. [Sta]074Z. ?  $\square$

**Cor. (3.6.4.19).** If  $D \in \mathbf{Az}_k$  is a central division algebra with  $\dim_k D = d^2$ ,  $d \in \mathbb{Z}_+$  by (3.6.4.16), then for any splitting field  $k'$  for  $D$ , we have  $d \mid [k' : k]$ .  $\lrcorner$

*Proof:* By (3.6.4.18), there exists  $B \in \mathbf{Az}_k$  similar to  $D$  s.t.  $\dim_k B = [k' : k]^2$ . Then it follows from (11.2.2.2) that  $B \cong \text{Mat}(n, D)$  for some  $n \in \mathbb{Z}_+$ . Then  $n^2 d^2 = [k' : k]^2$ , so  $d \mid [k' : k]$ .  $\square$

**Prop. (3.6.4.20) [Galois Splitting Fields, Noether-Köthe].** If  $D \in \mathbf{Az}_k$ , then there exists a maximal subfield  $K \subset D$  s.t.  $K/k$  is finite separable. In particular, by (11.2.2.2) (3.6.4.18), any central simple algebra has a finite Galois splitting field.  $\lrcorner$

*Proof:* Notice it suffices to prove that: if  $D \neq k$ , there exists  $\alpha \in D \setminus k$  s.t.  $k(\alpha)/k$  is separable. This is because if we find such  $\alpha$ , then  $D' = C_D(k(\alpha)) \in \mathbf{Az}_{k(\alpha)}$ , by (3.6.4.13), so we can use induction on  $D'/k(\alpha)$ .

We may assume  $k$  is non-perfect of characteristic  $p \in \mathbf{Prime}$ , thus an infinite field. Suppose we cannot find such an element, then every element satisfies an equation of the form  $T^{p^r} - a = 0$ , where  $a \in k$ . And because  $\dim_k D < \infty$ , there exists an  $r$  s.t.  $x^{p^r} \in k$  for any  $x \in D$ . Then we can use the fact  $\#k = \infty$  to show that  $x^{p^r} \in \bar{k}$  for any  $x \in D \otimes_k \bar{k}$ . But then every  $p^r$ -th power of  $D \otimes_k \bar{k} \cong \text{Mat}(n, \bar{k})$  is central, which is not true, as  $n > 1$  and we can take  $e_{11}$ .  $\square$

**Cor. (3.6.4.21).** For  $A \in \mathbf{Ring}/k$ ,  $A \in \mathbf{Az}_k$  iff  $A \otimes_k k^{\text{sep}} \cong \text{Mat}(n, k^{\text{sep}})$  for some  $n \in \mathbb{Z}_+$ , by (3.6.4.9). In particular, if  $A \in \mathbf{Az}_k$ , then  $\dim_k A \in (\mathbb{Z}_+)^2$ .  $\lrcorner$

**Def. (3.6.4.22) [Reduced Degrees].** If  $B \in \mathbf{Ring}_{\text{fin,ss}}/k$  and  $B = \prod_i B_i$  where  $B_i$  are simple  $k$ -algebras with center  $k_i$ , define the **reduced degree** of  $B$  over  $k$  to be

$$[B : k]_{\text{red}} = \sum_i [B_i : k_i]^{1/2} [k_i : k],$$

which is an integer by (3.6.4.16). And for any field extension  $k'/k$ ,

$$[B_{k'} : k']_{\text{red}} = [B : k]_{\text{red}}.$$

$\lrcorner$

*Proof:* To show it is invariant, pass to the algebraic closure, then  $B = \prod_j \text{Mat}(n_j, \bar{k})$ , and  $[B : k]_{\text{red}} = \sum n_j$ .  $\square$

**Def. (3.6.4.23) [Reduced Norms].** Let  $B \in \mathbf{Az}_k$ . If  $x \in B$ , let  $P(T)$  be the characteristic polynomial of  $R_x : B \rightarrow B$ . Then by passing to finite Galois splitting fields (3.6.4.20), we get  $P(T) = Q(T)^n$  where  $\dim_k B = n^2$ . Then because there are Galois actions, we see  $Q(T) \in k[T]$ . This  $Q(T)$  is called the **reduced characteristic** of  $x$ . We can also routinely define **reduced norms** and **reduced trace**, denoted by  $\text{Nmrd}_{B/k}$  and  $\text{trrd}_{B/k}$ .

Then this reduced characteristic is just the usual one when  $B \cong \text{Mat}(n; k)$ . And when  $K$  is a splitting field of  $B$  containing  $x$ , then this equals the characteristic polynomial of  $x$  in  $K/k$ .

More generally, if  $k \subset_{\text{fin}} K \in \mathbf{Field}$  and  $B \in \mathbf{Az}_K$ , then we can define  $\text{Nmrd}_{B/k} = \text{Nm}_{K/k} \circ \text{Nmrd}_{B/K}$  and  $\text{trrd}_{B/k} = \text{tr}_{K/k} \circ \text{trrd}_{B/K}$ .  $\lrcorner$

*Proof:* For the last assertion, notice  $B = K \oplus Kx_1 \oplus \dots \oplus Kx_n$  as modules over  $K$ .  $\square$

**Prop. (3.6.4.24).** If  $B$  is a finite semisimple  $k$ -algebra, any maximal étale  $k$ -subalgebra of  $B$  has rank  $[B : k]_{\text{red}}$  over  $k$ .  $\lrcorner$

*Proof:* This follows from (3.6.4.16).  $\square$

**Prop. (3.6.4.25).** If  $B$  is a finite semisimple  $k$ -algebra, then for any faithful  $B$ -module  $M$ ,

$$\dim_k M \geq [B : k]_{\text{red}}.$$

And the equality holds iff  $B$  is split.  $\lrcorner$

*Proof:* This is clear from (3.6.2.21).  $\square$

**Prop. (3.6.4.26).** For  $A \in \text{Az}_k$ ,  $A \otimes_k A^{\text{op}} \cong \text{Mat}(n; k)$ , where  $n = \dim_k A$ .  $\lrcorner$

*Proof:* There is a map  $A \otimes_k A^{\text{op}} \rightarrow \text{End}_k(A) : (a \otimes a') \mapsto (x \mapsto axa')$ . By (3.6.4.8),  $A \otimes_k A^{\text{op}}$  is simple, thus this is an injective map, but both sides have the same dimension, thus this is an isomorphism.  $\square$

### Finite Semisimple $k$ -Algebras with Involutions

See [Book on Involutions].

**Prop. (3.6.4.27) [Decomposition of Semisimple Algebras with an involution].** Let  $(B, *)$  be a finite semisimple  $k$ -algebra with an involution, if  $k = \bar{k}$  and  $\text{char } k = 0$ , then it decomposes as products of pairs of the following types:

- (A) :  $\text{Mat}(n; k) \times \text{Mat}(n; k)$ ,  $(a, b)^* = (b^t, a^t)$ .
- (C) :  $\text{Mat}(n; k)$ ,  $b^* = b^t$ .
- (BD) :  $\text{Mat}(2n; k)$ ,  $b^* = Jb^tJ^{-1}$ , where  $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ .

$\lrcorner$

*Proof:* Let  $B = B_1 \times \dots \times B_r$  be the decomposition into products of simple  $k$ -algebras, where  $B_i^*$  are the minimal two-sided ideals of  $B$ . Applying  $*$ ,  $B = B_1^* \times \dots \times B_r^*$  so  $B_i^*$  is a permutation of  $B_i$ , so  $B$  is a product of algebras either simple or product of two simple algebras that  $*$  interchanges them.

If  $B$  is simple, then  $B \cong M_n(k)$  as  $k$  is alg.closed, so  $b^* = ub^tu^{-1}$  for some  $u \in M_n(k)$  by Skolem-Noether (3.6.4.11). Then  $b = b^{**} = (u^tu^{-1})^{-1}bu^tu^{-1}$ , so  $u^tu^{-1}$  is in the center, denote it by  $c$ , then  $u^t = cu$ ,  $u = c^2u$ , so  $c = \pm 1$ , and  $u$  is symmetric or skew-symmetric. Up to a congruence, we see the situation is (C) or (BD).

The other case is also easy.  $\square$

**Prop. (3.6.4.28).** Let  $A$  be a finite simple algebra over  $k$  with center  $K$ , and let  $\dagger$  be an involution on  $A$ .

- If  $\dagger$  is of the first kind, then every involution on  $A$  of the first kind are of the form  $x \rightarrow ax^\dagger a^{-1}$  for some  $a \in A^*$  with  $a^\dagger = \pm a$ .
- If  $\dagger$  is of the second kind, then every involution on  $A$  of the second kind agreeing with  $\dagger$  on  $K$  are of the form  $x \rightarrow ax^\dagger a^{-1}$  for some  $a \in A^*$  with  $a^\dagger = a$ .

┘

*Proof:* If  $*$  is an involution satisfying the conditions as above,  $* \circ \dagger$  is an automorphism of  $A$  preserving  $K$ . So by Skolem-Noether, we see  $x^* = ax^\dagger a^{-1}$  for some  $a \in A^\times$ . And because  $*$  is an involution, we have  $a^\dagger a^{-1} \in K$ . Assume  $a^\dagger = ca$  where  $c \in k^\times$ , then  $c^\dagger c = 1$ . If  $\dagger$  is of the first kind, this means that  $c = \pm 1$ . And if  $\dagger$  is of the second kind, this implies that  $c = d^\dagger d^{-1}$  for some  $d \in K^\times$  by [Mil17]P78<sup>?</sup>. Thus we can replace  $a$  by  $a/d$  to achieve  $a^\dagger = a$ .  $\square$

**Prop. (3.6.4.29) [Adjoint Involutions].** Let  $D \in \text{Az}_k$  with an involution  $*$  and  $V \in \text{Vect}^{\text{fin}}/D$ . If  $\varphi$  is a (skew-)Hermitian form over  $D$ , then there is an involution  $*_V$  on  $\text{End}_D(V)$  extending  $*$  s.t. for each  $\alpha \in \text{End}_D(V)$ ,

$$(\alpha^{*_\varphi} x, y) = (x, \alpha y), \text{ where } x, y \in V,$$

called the **adjoint involution** associated to  $\varphi$ . And if  $\varphi, \varphi'$  are two such sesquilinear forms, then  $*_\varphi \cong *_{\varphi'}$  iff  $\varphi' = c\varphi$  for some  $c \in k^\times$ .

Every involution on  $\text{End}_D(V)$  extending  $*$  appears in this way.  $\square$

*Proof:* To show that every involution on  $\text{End}_D(V)$  extending  $*$  is of this form, take the standard Hermitian sesquilinear form on  $V$  and consider its adjoint involution, then the assertion follows from (3.6.4.28).  $\square$

**Cor. (3.6.4.30) [Types of Involutions of First Kind].** If  $\text{char } k \neq 2$ ,  $D \in \text{Az}_k$  and  $*$  is an involution of first kind on  $D$ , then  $(D, *) \otimes_k k^{\text{sep}} \cong (\text{End}(V), *_\varphi)$  for some  $V \in \text{Vect}^{\text{fin}}/k^{\text{sep}}$  and a non-degenerate symmetric or alternating form  $\varphi$  on  $V$ . In the symmetric case,  $*$  is called an **involution of orthogonal type**, and in the skew-symmetric case it is called an **involution of symplectic type**. Notice the symplectic type happens only when  $\dim_k D$  is even.  $\square$

**Prop. (3.6.4.31) [(Skew)Hermitian Spaces and Involutions of the First Kind].** If  $\text{char } k \neq 2$ ,  $D \in \text{Az}_k$  and  $*$  is an involution on  $D$  of the first kind (3.2.1.10), taking adjoint form (3.6.4.29) induces bijections between

- $k^\times$ -homogeneity classes of Hermitian spaces of rank  $n$  over  $D$ , and involutions on  $\text{Mat}(n; D)$  extending  $*$  of the same type as  $*$  (3.6.4.30).
- $k^\times$ -homogeneity classes of skew-Hermitian spaces of rank  $n$  over  $D$ , and involutions on  $\text{Mat}(n; D)$  extending  $*$  of the opposite type as  $*$  (3.6.4.30).

┘

*Proof:* Extends  $*$  to an involution of first type on  $\text{Mat}(n; D)$  in the natural way  $((a_{ij})^* = (a_{ij}^*)^t)$ , which is the adjoint involution of the standard Hermitian form, and it is of the same type as  $*$ . If  $\varphi'$  be another sesquilinear form and let  $*_{\varphi'}$  be the adjoint involution, then  $*' = *_{\varphi'} = \text{Inn}(a) \circ *$  for some  $a \in \text{GL}(n; D)$  with  $a^* = \pm a$  by (3.6.4.28). Notice the sesquilinear form

$$\varphi''(x, y) = \varphi(a^{-1}x, y)$$

satisfies  $*'' = *_{\varphi''} = \text{Inn}(a) \circ *$ , we have  $\varphi''$  is a scalar multiple of  $\varphi'$  by (3.6.4.29). It is clear that this  $\varphi'$  is Hermitian iff  $a^* = a$  and skew-Hermitian iff  $a^* = -a$ . And it is clear that  $\text{Inn}(a) \circ *$  is of the same type as  $*$  iff  $a^* = a$  and is of the opposite type as  $*$  iff  $a^* = -a$ .  $\square$

**Prop. (3.6.4.32) [Transpose Involutions].** For any  $A \in \text{Alg}/k$ , the map

$$*_{\text{trans}} : A \times_k A^{\text{op}} \rightarrow A \times_k A^{\text{op}} : (a, b) \mapsto (b, a)$$

is an involution of second kind in  $A \otimes_k A^{\text{op}}$ .  $\square$



**Prop. (3.6.4.33) [Involutions Simple of Unitary Type].** For any separable semisimple algebra  $A/k$  with an involution  $*$  of second kind s.t.  $k = \{a \in A \mid a^* = a\}$ .

- The center  $K$  of  $A$  is an étale  $k$ -algebra of degree 2.
- Let  $L/k \in \mathbf{Field}$  s.t.  $K \otimes_k L \cong K_1 \times K_2 \cong L \times L$  and let  $A_1 = A \otimes_K K_1$ , then  $A_1$  is a central simple algebra over  $K_1$ , and there is an isomorphism of involutive algebras

$$(D \otimes_k L, *) = (D_1 \times D_1^{\text{op}}, *) \cong (D_1 \times D_1^{\text{op}}, *_{\text{trans}}) : a \otimes c \mapsto (a \otimes c, a^* \otimes c) \quad (3.6.4.32).$$

- $(D, *) \otimes_k k^{\text{sep}} \cong (\text{End}(V) \times \text{End}(V)^{\text{op}}, *_{\text{trans}}) \cong (\text{End}(V) \times \text{End}(V), *'_{\text{trans}})$ , where  $(a, b)^{*'_{\text{trans}}} = (b^t, a^t)$ .

Such  $A$  is said to be **simple of unitary type**. In this case,  $A$  is either a simple algebra or a product of two central simple algebras over  $k$ .  $\lrcorner$

*Proof:* Why is it an isomorphism?

Only need to check the isomorphism to the transpose involutions, but this follows from the fact that  $A_1$  is simple.  $\square$

**Prop. (3.6.4.34) [Hermitian Spaces and Involutions of Second Kind].** If  $K/k$  is a separable quadratic extension with  $\text{Gal}(K/k) = \{1, \mathbf{c}\}$ , taking adjoint form (3.6.4.29) induces bijections between  $K^\times$ -homogeneity classes of Hermitian spaces of rank  $n$  over  $K$ , and involutions on  $\text{Mat}(n; K)$  extending  $\mathbf{c}$  (3.6.4.30).  $\lrcorner$

*Proof:* The proof is the same as that of (3.6.4.31).  $\square$

## 5 Idempotent Algebras

**Lemma (3.6.5.1).** Let  $R \in \mathbf{Ring}$  and  $e, f$  be idempotents of  $R$  that  $ef = fe = e$ , then  $f = e + e'$ , where  $e'$  is an idempotent and  $M[f] = M[e] \oplus M[e']$ .

Moreover, if  $R$  is an algebra over an alg.closed field  $k$  and  $M[e]$  is a simple  $R[e]$ -module of f.d. over  $k$ , then  $\dim \text{Hom}_{R[e]}(M[e], M[f]) = 1$ .  $\lrcorner$

*Proof:* Let  $e' = f - e$ , then it is easily verified to be an idempotent, and  $ee' = e'e = 0$ , thus  $M[f] = M[e] \oplus M[e']$  is clear.

For the second, because  $R[e]$  acts by 0 in  $R[e']$ ,  $\text{Hom}_{R[e]}(M[e], M[f]) = \text{Hom}_{R[e]}(M[e], M[e])$  has dimension 1, by Shur's lemma (3.6.2.2) and (3.6.1.2).  $\square$

**Def. (3.6.5.2) [Idempotent Algebra].** For  $k \in \mathbf{Field}$ , **idempotent algebra** over  $k$  is an algebra  $\mathcal{H} \in \mathbf{Alg}_k^{\text{asso}}$  together with a set  $\mathcal{E}$  of idempotents that if  $e_1, e_2 \in \mathcal{E}$ , then there exists  $e_0 \in \mathcal{E}$  that

$$e_0 e_1 = e_1 e_0 = e_1, e_0 e_2 = e_2 e_0 = e_2,$$

and also for any  $\varphi \in \mathcal{H}$ , there exists  $e \in \mathcal{E}$  that  $e\varphi = \varphi e = \varphi$ .

We can define a partial order on  $\mathcal{E}$ :  $e < f$  iff  $ef = fe = f$ , then this order is cofinal.

If  $e$  is an idempotent, denote  $\mathcal{H}[e] = e\mathcal{H}e$ , which is a subring of  $\mathcal{H}$  with unit  $e$ . Also if  $M$  is an  $\mathcal{H}$ -module, we denote  $M[e]$  the  $R[e]$ -module  $eM$ .  $\lrcorner$

**Prop. (3.6.5.3).** If  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$  is an exact sequence of  $\mathcal{H}$ -modules,  $e \in \mathcal{E}$ , then  $0 \rightarrow M_1[e] \rightarrow M_2[e] \rightarrow M_3[e] \rightarrow 0$  is also exact.  $\lrcorner$

*Proof:* Because tensoring an idempotent is exact.  $\square$



**Def. (3.6.5.4) [Smooth Representations of Idempotent Algebras].** Let  $(\mathcal{H}, \mathcal{E})$  be an idempotent algebra, a **smooth  $\mathcal{H}$ -module**  $M$  is an  $\mathcal{H}$ -module  $M = \cup_{e \in \mathcal{E}} M[e]$ , and an **admissible  $\mathcal{H}$ -module** is a smooth  $\mathcal{H}$ -module that each  $M[e]$  is of f.d. over  $k$ .

A smooth  $\mathcal{H}$ -module is clearly non-degenerate.  $\lrcorner$

**Def. (3.6.5.5) [Contragradient Module].** If  $M$  be a smooth  $\mathcal{H}$ -module, then we can define a **contragradient  $\bar{M}$**  of  $M$ , which is the smooth  $\mathcal{H}$ -module consisting of smooth vectors in  $M^*$ , where the action is defined as  $((\varphi)\lambda)(m) = \lambda(\iota(\varphi)m)$ . Notice if  $M$  is admissible, then so does  $\bar{M}$ , because an element in  $\bar{M}[e]$  is determined by its restriction on  $M[\iota(e)]$ , so  $\bar{M}[e]$  is of f.d. if  $M[\iota(e)]$  does.  $\lrcorner$

**Prop. (3.6.5.6) [Simpleness Checked on Idempotents].** Let  $M$  be a non-zero module over an idempotent algebra  $(\mathcal{H}, \mathcal{E})$  over  $\mathbb{C}$ ,  $\mathcal{E}^0$  be a cofinal subset of  $\mathcal{E}$ , then  $M$  is a simple  $\mathcal{H}$ -module iff  $M[e]$  is simple  $\mathcal{H}[e]$ -modules for any  $e \in \mathcal{E}^0$ .  $\lrcorner$

*Proof:* If there exists a proper  $\mathcal{H}$ -submodule  $M_1$  of  $M$ , let  $M/M_1 = M_2$ , then  $M[e]/M_1[e] = M_2[e]$  for all  $e$  by (3.6.5.3), Thus we can choose  $e$  s.t.  $M_1[e] \neq 0, M_2[e] \neq 0$ , thus  $M[e]$  is reducible. Conversely, If  $W_0 \subset M[e_K]$  is a proper non-zero  $\mathcal{H}[e_K]$ -submodule, then  $[\pi(\mathcal{H})W_0][e] = W_0$ , thus  $\pi(\mathcal{H})W_0$  is a proper submodule of  $M$ .  $\square$

**Prop. (3.6.5.7) [Isomorphism Checked on Idempotents].** Let  $V_1, V_2$  be two simple admissible modules over an idempotent algebra  $(\mathcal{H}, \mathcal{E})$ , if  $e \in E$  that  $V_1[e] \cong V_2[e] \neq 0$ , then  $V_1 \cong V_2$ .  $\lrcorner$

*Proof:* Choose an isomorphism  $j : V_1[e] \rightarrow V_2[e]$ , then the subspace  $V' = \{(x, jx)\} \subset V_1[e] \oplus V_2[e] \subset V_1 \oplus V_2$  is a  $\mathcal{H}[e]$ -submodule. Let  $V = \pi(\mathcal{H})V'$ , then  $V[e] = V'$ , so  $V$  is not contained or contains any of  $V_1$  or  $V_2$ , thus  $V \cap V_1 = V \cap V_2 = 0$  as  $V_i$  are irreducible. Thus the projections  $V \rightarrow V_1, V \rightarrow V_2$  are isomorphisms as  $V_i$  are irreducible, and  $V_1 \cong V_2$ .  $\square$

**Prop. (3.6.5.8) [Extension Theorem].** Let  $(\mathcal{H}, \mathcal{E})$  be an idempotent algebra and  $e \in E$ . If  $V_e$  is a simple  $\mathcal{H}[e]$ -module, then there is a simple  $\mathcal{H}$ -module  $V$  that  $V[e] = V_e$ .  $\lrcorner$

*Proof:* Let  $V_e \cong \mathcal{H}[e]/I$ , where  $I$  is a left ideal of  $V_e$ . Let  $E_1, E_2$  be the module generated by  $I$  and  $\mathcal{H}[e]$  in  $\mathcal{H}$  by left action of  $\mathcal{H}$ , then  $E_1[e] = I$  and  $E_2[e] = \mathcal{H}[e]$ , thus  $(E_3 = E_1/E_2)[e] \cong V_e$  by (3.6.5.3). Now if  $E'$  is a submodule of  $E_3$ , then either  $E'[e] = 0$  thus  $(E_3/E')[e] \cong L$ , or  $E'[e] = L$ , thus  $E' = E_3$  by (3.6.5.7). Thus if we choose the proper submodule  $E'$  that  $E_3/E'$  is irreducible by (24.1.2.10), then  $E_3/E'[e] = L$ .  $\square$

**Prop. (3.6.5.9).** If  $\mathcal{H}$  is an idempotent algebra, then  $\mathcal{M}(\mathcal{H})$  has enough projectives.  $\lrcorner$

*Proof:* For any idempotent  $e \in \mathcal{H}$ , consider the module  $\mathcal{H}e$ , then it is projective, because  $\text{Hom}_{\mathcal{H}}(\mathcal{H}e, X) = eX$ , thus it is clearly exact. Now any  $m \in V$  has an idempotent  $e \in \mathcal{H}$  that  $ev = v$  (use definition). Thus by taking the direct sum, we are done.  $\square$

### Spherical Idempotents

**Def. (3.6.5.10) [Spherical Idempotents].** Let  $(\mathcal{H}, \mathcal{E})$  be an idempotent algebra over a field  $\Omega$ . Then an idempotent  $e^0 \in \mathcal{E}$  is called **spherical idempotent** if there exists an anti-involution  $\iota : \mathcal{H} \rightarrow \mathcal{H}$  that  $\iota(x) = x$  for any  $x \in \mathcal{H}[e^0]$ . Notice this implies  $\mathcal{H}[e^0]$  is commutative, as  $xy = \iota(xy) = yx$ .  $\lrcorner$

**Def. (3.6.5.11) [Spherical Vectors].** Let  $e^0$  be a spherical idempotent of the idempotent algebra  $(\mathcal{H}, \mathcal{E})$  and  $\iota$  is the corresponding involution. If  $M$  is an admissible  $\mathcal{H}$ -module, then a **spherical(unramified) module** is a  $\mathcal{H}$ -module  $M$  that  $M[e^0] \neq 0$ , and elements in  $M[e^0]$  are called **spherical vectors**.

Then if  $M$  is simple and spherical, then  $M$  has at most one spherical vector up to scalar, and  $\widehat{M}$  is also spherical. Thus the space of spherical vectors is fixed by  $\mathcal{H}[e^0]$ , and the action of  $\mathcal{H}[e^0]$  on that defines a **spherical character**.  $\lrcorner$

*Proof:* In fact,  $M[e^0]$  is a simple  $\mathcal{H}[e^0]$ -module (3.6.5.6), and is of f.d., and  $\mathcal{H}[e^0]$  is commutative (3.6.5.10), so (3.6.2.11) shows it is of dimension 1. so if  $m^0 \neq 0 \in M[e^0]$ , then we can define a  $\widehat{m}^0 \neq 0 \in \widehat{M}[e^0]$  as:

$$e^0 m = \widehat{m}^0(m) \cdot m^0.$$

□

**Prop. (3.6.5.12) [Isomorphism Checked on Spherical Idempotents].** Let  $(\mathcal{H}, \mathcal{E})$  be an idempotent algebra over a field  $\Omega$  with a spherical idempotent  $e^0$  (3.6.5.11). If  $M, N$  are simple admissible spherical  $\mathcal{H}$ -modules that  $M[e^0] \cong N[e^0]$  over  $\mathcal{H}[e^0]$ , then  $M \cong N$  as  $\mathcal{H}$ -modules.  $\lrcorner$

*Proof:* This follows from (3.6.5.7).  $\square$

### Restricted Tensor Product

**Def. (3.6.5.13) [Restricted Tensor Product].** Give Vector spaces  $V_v$  indexed by a set  $\Sigma$  and elements  $x_v^0 \in V_v$  be given for a.e.  $v$ , we can define the **restricted tensor product**  $\bigotimes' V_v$  as the direct limit

$$\lim_{S \subset \Sigma \text{ finite}} \bigotimes_{v \in S} V_v.$$

It can be thought of as the vector spaces spanned by all symbols  $\bigotimes_v x_v$  where  $x_v = x_v^0$  for a.e.  $v$ .

Notice if  $V_v$  are idempotent algebras and  $x_v^0 \in V_v$  are idempotents for a.e.  $v$ , then  $\prod' V_v$  also has a natural idempotent algebra structure.  $\lrcorner$

**Def. (3.6.5.14) [Tensor Product Module of Idempotent Algebras].** Given a set of idempotent algebras  $(\mathcal{H}_v, \mathcal{E}_v)$  and specify an idempotent  $e_v^0$  for a.e.  $v$ . Let  $M_v$  be  $\mathcal{H}_v$ -modules over  $k$  for all  $v$ , and assume  $M_v[e_v^0]$  be of dimension 1 a.e.  $v$ , and specify a.e. a non-zero element  $m_v^0 \in M_v[e_v^0]$

Then we can define tensor product (3.6.5.13)  $\prod' \mathcal{H}_v$  which is an idempotent algebra and  $\prod' M_v$ . Then we can define an **restricted tensor module** structure of  $M_v$  over  $\prod' \mathcal{H}_v$  by the action

$$(\bigotimes_v \varphi_v) x_v = \bigotimes_v (\varphi_v) x_v.$$

□

**Lemma (3.6.5.15) [Simple Module of Tensor Product].** Let  $(\mathcal{H}_1, \mathcal{E}_1), (\mathcal{H}_2, \mathcal{E}_2)$  be idempotent algebras over an alg.closed field  $\Omega$ , and let  $(\mathcal{H}, \mathcal{E})$  be their tensor product. If  $M_1, M_2$  are simple admissible modules over  $\mathcal{H}_1, \mathcal{H}_2$ , respectively, then  $M_1 \otimes M_2$  are simple admissible modules over  $\mathcal{H}_1 \otimes \mathcal{H}_2$ , and any simple admissible module over  $\mathcal{H}$  comes uniquely from a pair  $(M_1, M_2)$  like this.  $\lrcorner$

*Proof:* If  $M_1, M_2$  are simple admissible over  $\mathcal{H}_1, \mathcal{H}_2$  respectively, then if  $e_1 \in \mathcal{E}_1, e_2 \in \mathcal{E}_2$ , then  $(M_1 \otimes M_2)[e_1 \otimes e_2] = M_1[e_1] \otimes M_2[e_2]$  is simple and of f.d. by (3.6.5.6) and (3.6.2.16), so  $M_1 \otimes M_2$  is simple by (3.6.5.6).

Now if  $M$  is simple admissible over  $\mathcal{H}_1 \otimes \mathcal{H}_2$ , then we find an  $e_1^0 \otimes e_2^0 \in \mathcal{E}$  that  $M[e_1^0 \otimes e_2^0] \neq 0$ . Let  $\mathcal{E}_i^0 = \{e_i \in \mathcal{E}_i | e_i < e_i^0\}$ , then  $\mathcal{E}_i^0$  is cofinal in  $\mathcal{E}_i$ . Then for any  $e_i \in \mathcal{E}_i^0$ ,  $M[e_1 \otimes e_2]$  is non-zero thus simple, thus it is of the form

$$M[e_1 \otimes e_2] = M_1(e_1, e_2) \otimes M_2(e_1, e_2)$$

where  $M_i(e_1, e_2)$  are simple  $\mathcal{H}[e_i]$ -modules by (3.6.2.16).

Now we show that  $M_2(e_1, e_2)$  depends only on  $e_1$ : it suffices to show  $M_2(e_1, e_2) = M_2(f_1, e_2)$  for  $f_1 < e_1$ . For this, notice that for any idempotents  $g_1, g_2$ ,

$$M[g_1 \otimes g_2] = M_1(g_1, g_2) \otimes M_2(g_1, g_2)$$

is a finite direct sum of simple modules  $M_2(g_1, g_2)$  as an  $\mathcal{H}_2[g_2]$ -module. Notice that by (3.6.5.1),  $M[e_1 \otimes e_2] = M[f_1 \otimes e_2] \oplus M[e' \otimes e_2]$  for  $e' = f_1 - e_1$ , so by (3.6.2.3),  $M_2(f_1, e_2) = M_2(e_1, e_2)$ . Similarly we know  $M_1(e_1, e_2)$  only depends on  $e_1$ .

Next we have:

$$\dim_k \operatorname{Hom}_{H_1[e_1]}(M_1[e_1], M_1[f_1]) \geq 1, \quad f_1 \leq e_1.$$

and similar for  $\mathcal{H}_2$ . For this, it suffices to prove that  $\dim_k \operatorname{Hom}_{H_1[e_1]}(M[e_1 \otimes e_2], M[f_1 \otimes e_2]) \geq 1$ , by what we just said, but this is by the decomposition

$$M[e_1 \otimes e_2] = M[f_1 \otimes e_2] \oplus M[e' \otimes e_2]$$

above. But

$$\dim_k \operatorname{Hom}_{H_1[e_1] \otimes H_2[e_2]}(M_1(e_1) \otimes M_2(e_2), M_1(f_1) \otimes M_2(f_2)) = 1, \quad f_1 \leq e_1, f_2 \leq e_2$$

by (3.6.5.1), so the  $\geq$  above should change to  $=$ .

Now we know the homomorphism is of dimension 1, we want to choose a family of maps that is compatible for  $g_1 \leq f_1 \leq e_1 \leq e_1^0$ . For this, we can choose the maps  $\lambda(e_1, e_1^0)$  arbitrarily, then choose  $\lambda(e_1, f_1)$  to be compatible with  $\lambda(e_1, e_1^0)$  and  $\lambda(f_1, e_1^0)$ , then these are compatible choices. Hence we can define a direct limit

$$M_1 = \varinjlim_{(\mathcal{E}^0)^{op}} M(e_1).$$

It is easy to see  $M_1(e_1) \rightarrow M_1$  are all injective, and  $M_1(e_1) = M[e_1]$ . Similarly, we can define an  $M_2$  that  $M_2[e_2] = M_2(e_2)$ .

Finally, we have

$$M[e_1 \otimes e_2] = M_1[e_1] \otimes M_2[e_2] = (M_1 \otimes M_2)[e_1 \otimes e_2]$$

for all  $e_1 \in \mathcal{E}_1^0, e_2 \in \mathcal{E}_2^0$ , thus  $M \cong M_1 \otimes M_2$  by (3.6.5.7).

As for the uniqueness of  $M_1, M_2$ , notice the decomposition of  $M[e_1 \otimes e_2]$  is unique by (3.6.2.16), so  $M_i[e_i]$  is uniquely determined, thus by (3.6.5.7)  $M_i$  are determined.  $\square$

**Prop. (3.6.5.16)[Flath Theorem].** let  $(\mathcal{H}_v, \mathcal{E}_v)$  be an indexed family of idempotent  $k$ -algebras, and for a.e.  $v$  let  $e_v^0 \in \mathcal{E}_v$  be a spherical idempotent. Let  $(\mathcal{H}, \mathcal{E})$  be the restricted tensor product of  $\mathcal{H}_c$  w.r.t.  $e_v^0$ , which is an idempotent algebra. For each  $v \in \Omega$ , there is a simple  $\mathcal{H}_v$ -module  $M_v$  and for a.e.  $v$  we specify a non-zero spherical vector. Let  $\otimes_v M_v$  be the tensor product module, then it is a simple admissible  $\mathcal{H}$ -module, and every simple admissible module is of this type, with  $M_v$  uniquely determined.  $\square$

*Proof:* Firstly the tensor product is simple and admissible: For any idempotent  $e = \otimes e_v \in \mathcal{E}$ , there is a finite set  $S$  that if  $v \notin S$ , then  $e_v = e_v^0$ , hence for these  $v$   $\dim M_v[e_v] = 1$ . Then

$$M[e] = \bigotimes_{v \in S} M_v[e_v].$$

which is simple of f.d. by (3.6.5.15), so  $M$  is simple admissible by (3.6.5.6).

Conversely, for any simple  $\mathcal{H}$ -module  $M$ , we need to show it is a tensor product. If there are only f.m. indices, then this follows from (3.6.5.15).

Suppose first that  $e_v^0$  is defined and spherical for all  $v$ , and  $e = \otimes_v e_v^0$ , and  $M[e] \neq 0$ . Then  $\dim M[e] = 1$  (3.6.5.11). Let  $m$  be a spherical vector, then there is a ring homomorphism  $\gamma : \mathcal{H}[e] \rightarrow k$  defined by  $hm = \gamma(h)m$ . Because  $\mathcal{H}[e] = \otimes'_v \mathcal{H}_v[e_v^0]$ , we have

$$\gamma(\otimes_v h_v) = \prod_v \gamma_v(h_v).$$

Now if we decompose  $\mathcal{H} = \mathcal{H}_v \otimes \mathcal{H}'_v$ , then by (3.6.5.15), there exists simple admissible module  $M_v$  over  $\mathcal{H}_v$ ,  $M'_v$  over  $\mathcal{H}'_v$  respectively, that  $M = M_v \otimes M'_v$ , thus  $M[e] = M_v[e_v^0] \otimes M'_v[e'_v]$ . Now consider  $M_v$  for all  $v$ , and  $N = \otimes'_v M_v$  w.r.t.  $m_v$ , then it is simple admissible  $\mathcal{H}$ -module, and we have  $N[e] = \otimes'_v M_v[e_v^0] \cong M[e]$  as  $\mathcal{H}[e]$ -modules of dimension 1, which is because they are both simple and have the same character

$$\gamma(\otimes_v h_v) = \prod_v \gamma_v(h_v).$$

Hence  $M \cong N$ , by (3.6.5.12).

Now the general case follows from the two situations above: choose  $e \in \mathcal{E}$  that  $M[e] \neq 0$ , then let  $S$  be large that for  $v \notin S$ ,  $e_v = e_v^0$ . Then we decompose  $\mathcal{H}$  as  $\mathcal{H} = \bigotimes_{v \in S} \mathcal{H}_v \otimes (\bigotimes_{v \notin S} \mathcal{H}_v)$ , then

$$M = \bigotimes_{v \in S} M_v \otimes M' = \bigotimes_{v \in S} M_v \otimes \left( \bigotimes_{v \notin S} M_v \right) = \otimes_v M_v.$$

□

## 6 Rings and Categories

References are <https://stacks.math.columbia.edu/tag/0GNG>.

**Prop. (3.6.6.1).** Let  $A, B \in \mathbf{Ring}$  and  $F : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_B$  a functor, then the following are equivalent:

- $F$  is isomorphic to the functor  $K \otimes_A M$  for some  $K \in \mathbf{Mod}_{(B,A)}$ .
- $F$  is right exact and commutes with all direct sums.
- $F$  commutes with all colimits.
- $F$  has a right adjoint  $G$ .

┘

*Proof:* Cf. [Sta]0GNR.

□

**Prop. (3.6.6.2).** Let  $R \in \mathbf{Ring}$  and  $A, B \in \mathbf{Ring}_R$ , then there is an equivalence of categories:

$$\mathbf{Func}(\mathbf{Mod}_A, \mathbf{Mod}_B) \cong \mathbf{Mod}_{A \otimes_R B}.$$

┘

*Proof:* Cf. [Sta]0GNR.

□

### Quotient by Equivalence Relations

**Prop. (3.6.6.3) [Quotients by Equivalence Relations].** Let  $u_0, u_1 : A_0 \rightarrow A_1$  be an equivalence in the dual category of  $\mathcal{CRing}_{R_0}$  (4.1.1.24). If  $u_0$  is locally free of constant rank  $r$ , then a quotient  $u : A \rightarrow A_0$  exists, and  $u$  is locally free of constant rank  $r$ .  $\lrcorner$

*Proof:* Cf. [Mil17b]P592.  $\square$

### Morita Theory

References are [Morita Equivalence], [Frobenius Algebras, 1]Chap2 and [Fuller Rings and Categories of Modules].

**Def. (3.6.6.4).** Two ring  $R, S$  are called **Morita equivalent** if the category of  $\text{mod-}R$  is equivalent to the category of  $\text{mod-}S$ .  $\lrcorner$

**Prop. (3.6.6.5).** For an Abelian category  $\mathcal{A}$  satisfying AB3 (i.e arbitrary sum exists), An object  $P$  of  $\mathcal{A}$  is a **progenerator** if the functor  $h' : X \mapsto \text{Hom}_{\mathcal{A}}(P, X)$  is exact and and strict:  $h'(X) = 0$  implies  $X = 0$ . Then  $h'$  determines an equivalence from  $\mathcal{A}$  to  $\text{mod-}R$ , where  $R = \text{Hom}_{\mathcal{A}}(P, P)$ .

Similarly, if  $\mathcal{A}$  is an Abelian Noetherian category and  $P$  is a progenerator, then  $R$  is Noetherian and  $\mathcal{A}$  is equivalent to the category of finitely generated  $R$ -categories.  $\lrcorner$

*Proof:* Essentially surjective: construct using direct limit and cokernel.

Notice that  $h'(X) \cong h'(X') \rightarrow X \cong X'$  by strictness and A4 axiom. So let  $X = \text{Coker}(P^{\oplus I}, P^{\oplus J})$ ,

$$\begin{aligned} \text{Hom}(h'(X), h'(Y)) &= \text{Hom}(\text{Coker}(h'(P^{\oplus J}), h'(P^{\oplus I})), h'(Y)) \\ &= \ker(\text{Hom}(h'(P^{\oplus J}), h'(Y)) \rightarrow \text{Hom}(h'(P^{\oplus I}), h'(Y))) \\ &= \ker(h'(Y^{\coprod I}) \rightarrow h'(Y^{\coprod J})) \\ &= \text{Hom}(X, Y) \end{aligned}$$

$\square$

**Prop. (3.6.6.6).** In the case when  $\mathcal{A}$  is the category  $\text{mod-}R$ ,  $P$  is a generator  $\iff h' : X \mapsto \text{Hom}_R(P, X)$  is faithful  $\iff$  every  $M$  is a quotient of direct sums of  $P$ . And a **progenerator** is a f.g. projective generator.  $\lrcorner$

**Prop. (3.6.6.7).** Let  $P$  be a  $(A, B)$ -bimodule, iff  $P$  is a progenerator as a right  $B$  module, then it is a progenerator as a left  $A$  module.  $\lrcorner$

**Prop. (3.6.6.8).** Let  $P$  be a progenerator as a  $\lrcorner$

**Prop. (3.6.6.9) [Morita].** The following are equivalent:

- categories  $A\text{-mod}$  and  $B\text{-mod}$  are equivalent.
- categories  $\text{mod-}A$  and  $\text{mod-}B$  are equivalent.
- There exist a finitely generated progenerator  $P$  of  $\text{mod-}A$  that  $B \cong \text{End}_A P$ .

$\lrcorner$

*Proof:*  $2 \rightarrow 3$ :  $A$  is a progenerator in  $\text{mod-}A$ , thus when  $A \sim B$ ,  $F : \text{mod-}A \rightarrow \text{mod-}B$ ,  $A \cong \text{End}_A A = \text{End}_B F(A)$ , and  $F(A)$  is a left  $A$  module as well as a progenerator of  $B$ . Thus there is a  $(A, B)$ -bimodule  $P$  that  $A \cong \text{End}_B P$ , and a  $(B, A)$ -bimodule  $Q$  that  $B \cong \text{End}_A Q$ .  $\square$

**Prop. (3.6.6.10).** There can be defined another Morita invariance that  $R \sim S$  iff there are  $(R, S)$ -bimodule  $P$  and  $(S, R)$ -bimodule  $Q$  that  $P \otimes_S Q \cong R$  as a  $(R, R)$ -bimodule and  $Q \otimes_R P \cong S$  as a  $(S, S)$ -bimodule. This will immediately generate equivalence between  $R\text{-mod}$  and  $S\text{-mod}$  as well as equivalence between  $\text{mod-}R$  and  $\text{mod-}S$  by tensoring. And  $P$  and  $Q$  are projective modules respectively, because equivalence is a kind of adjoint.  $\lrcorner$

**Prop. (3.6.6.11).** Let  $D$  be a division ring over  $k$  of finite degree and  $A = M_r(D)$ . Let  $S = D^r$  with  $A$  acting by left multiplication and  $D$  acting by right multiplication, then there  $S$  is a simple  $A$ -module, and every  $A$ -module is a direct sum of copies of  $S$ . This means that  $S \otimes_D -$  induces an equivalence from  $\text{Mod}_D$  to  $\text{Mod}_A$ .  $\lrcorner$

**Prop. (3.6.6.12) [Properties Preserved under Morita Invariance].** Cf. [Rings and Categories of Modules P54].  $\lrcorner$

## 7 Auslander-Reiten Theory

References are [Frobenius Algebras, 1].

## 8 Frobenius-Algebras

**Notation (3.6.8.1).**

- Let  $k \in \text{Field}$ .

$\lrcorner$

**Def. (3.6.8.2) [Frobenius-Algebras].** A **Frobenius algebra** over  $k$  is a finite  $k$ -algebra s.t. the left regular representation of  $A$  and the right regular representation of  $A$  are equivalent. A Frobenius algebra  $A$  is called a **symmetric algebra** if the left regular representation of  $A$  and the right regular representation of  $A$  are conjugate by a symmetric invertible matrix.  $\lrcorner$

**Thm. (3.6.8.3) [Frobenius].** A finite  $k$ -algebra  $A$  is a Frobenius algebra (resp. symmetric algebra) iff there exists an invertible (resp. symmetric invertible) parastrophic matrix  $P(\xi)$ .  $\lrcorner$

*Proof:* Cf. [Frobenius Algebras, 1]P334.  $\square$

**Def. (3.6.8.4) [Associative Bilinear form].** For a finite  $k$ -algebra  $A$ , an **associative bilinear form** on  $A$  is a  $k$ -bilinear map  $(-, -) : A \times A \rightarrow k$  s.t.  $(a, bc) = (ab, c)$  for any  $a, b, c \in A$ .  $\lrcorner$

**Thm. (3.6.8.5) [First Brauer–Nesbitt–Nakayama Theorem].** For a finite  $k$ -algebra  $A$ , the followings are equivalent:

1.  $A$  is a Frobenius algebra (3.6.8.2).
2. There exists a non-degenerate associative bilinear form on  $A$ .
3. There exists a  $k$ -linear map  $\varphi : A \rightarrow k$  s.t.  $\ker(\varphi)$  contains no non-zero right ideals of  $A$ .
4. There exists a  $k$ -linear map  $\varphi' : A \rightarrow k$  s.t.  $\ker(\varphi')$  contains no non-zero left ideals of  $A$ .

In particular,  $A$  is a Frobenius algebra iff  $A^{\text{op}}$  is a Frobenius algebra.  $\lrcorner$

*Proof:* Cf. [Frobenius Algebras, 1]P336.

The equivalence between item2, 3, 4 are clear, by defining  $\varphi(ab) = (a, b)$ .  $\square$

**Thm. (3.6.8.6) [Second Brauer–Nesbitt–Nakayama Theorem].** For a finite  $k$ -algebra  $A$ , the followings are equivalent:

1.  $A$  is a symmetric algebra(3.6.8.2).
2. There exists a non-degenerate associative symmetric bilinear form on  $A$ .
3. There exists a  $k$ -linear map  $\varphi : A \rightarrow k$  s.t.  $\varphi(ab) = \varphi(ba)$  for all  $a, b \in A$  and  $\ker(\varphi)$  contains no two-sided ideals of  $A$ .
4. There exists an isomorphism  $A \cong D(A)$  of  $A$ -bimodules. ?

In particular,  $A$  is a symmetric algebra iff  $A^{\text{op}}$  is a symmetric algebra.  $\lrcorner$

*Proof:* Cf.[Frobenius Algebras, 1]P340.  $\square$

**Cor.(3.6.8.7) [Product of Frobenius Algebras].** Let  $A_1, \dots, A_r$  be finite  $k$ -algebras, then  $A$  is a Frobenius algebra(resp. symmetric algebra) iff each  $A_i$  is.  $\lrcorner$

*Proof:* This follows from non-degenerate associative bilinear form criterions in(3.6.8.5) and(3.6.8.6).  $\square$

**Example(3.6.8.8) [Frobenius Algebras].**

- For  $n \in \mathbb{Z}_+$ ,  $\text{Mat}(n; k)$  is a symmetric algebra with the non-degenerate associative symmetric  $k$ -bilinear form  $\text{Mat}(n; k) \times \text{Mat}(n; k) \rightarrow k : (A, B) \mapsto \text{tr}(AB)$ .
- For  $G \in \text{Grp}^{\text{fin}}$ ,  $k[G]$  is a symmetric algebra with the non-degenerate associative symmetric  $k$ -bilinear form  $k[G] \times k[G] \rightarrow k : (\sum \lambda_g g, \sum \mu_g g) = \sum_{g \in G} \lambda_g \mu_{g^{-1}}$ .
- For  $\lambda \in k^\times$ , define  $A_\lambda = k\{X, Y\}/(X^2, Y^2, XY - \lambda YX)$ , where  $k\{X, Y\}$  is the non-commutative polynomial ring of 2 variables. Then  $A$  is a 4-dimensional local  $k$ -algebra. Let  $\varphi : A_\lambda \rightarrow k$  be the  $k$ -linear map s.t.  $\varphi(1) = \varphi(x) = \varphi(y) = 0, \varphi(xy) = 1$ . Then it is easy to verify that the kernel of  $\varphi$  contains no non-zero right ideals of  $A_\lambda$ , thus  $A_\lambda$  is a Frobenius algebra by(3.6.8.5). If  $\lambda = 1$ ,  $A_\lambda$  is a symmetric algebra. And if  $\lambda \neq 1$ , then  $A$  is not a symmetric algebra, because for any  $k$ -linear map  $\varphi' : A \rightarrow k$  s.t.  $\varphi'(ab) = \varphi'(ba)$ , we have  $\varphi'(xy) = \lambda \varphi'(yx) = \lambda \varphi'(xy)$ , which implies  $\varphi'(xy) = 0$ , so  $k.xy$  is a non-zero right ideal in the kernel of  $\varphi'$ .  $\lrcorner$

**Prop.(3.6.8.9) [Semisimple Algebras are Symmetric Algebras].** Any finite semisimple algebra  $A \in \text{Ring}/k$  is a symmetric algebra(3.6.8.2).  $\lrcorner$

*Proof:* It follows from Wedderburn-Artin(3.6.2.22) and(3.6.8.7) that it suffices to prove that  $\text{Mat}(n; D)$  is a symmetric algebra for any  $n \in \mathbb{Z}_+$  and any finite division ring  $D$  over  $k$ . Cf.[Frobenius Algebras, 1]P374 ?  $\square$

**Def.(3.6.8.10) [Nakayama Automorphisms].** If  $A$  is a Frobenius algebra over  $k$ (3.6.8.2) and let  $(-, -) : A \times A \rightarrow k$  be a non-degenerate associative  $k$ -bilinear form, then there exists a unique automorphism  $\nu$  of  $A$  s.t.

$$(\nu(a), b) = (b, a), \text{ where } a, b \in A,$$

called the **Nakayama automorphism** associated to  $(-, -)$ . And for any invertible element  $c \in A^*$ ,  $\nu'(a) = \nu(cac^{-1})$  is the Nakayama automorphism associated to the non-degenerate associative  $k$ -bilinear form  $(a, b)' = (a, bc)$ .  $\lrcorner$

*Proof:* Cf.[Frobenius Algebras, 1]P345.  $\square$

**Prop.(3.6.8.11).** If  $A$  is a Frobenius algebra over  $k$ (3.6.8.2) and  $(-, -), (-, -)' : A \times A \rightarrow k$  are two non-degenerate associative  $k$ -bilinear forms, then there exists  $c \in A^*$  s.t.  $(a, b)' = (a, bc)$  for any  $a, b \in A$ . In particular, any two Nakayama automorphisms of  $A$  are conjugate(3.6.8.10).  $\lrcorner$

*Proof:* Cf.[Frobenius Algebras, 1]P346.  $\square$

## 9 Non-Associative Algebras

**Def.(3.6.9.1)[Alternative Algebras].** An **alternative algebra** is an algebra  $A \in \text{naAlg}$  s.t. for any  $x, y \in A$ ,

$$x(xy) = x^2y, \quad (yx)x = yx^2.$$

┘

**Prop.(3.6.9.2)[Properties of Alternative Algebras].** For any alternative algebra  $A$  and  $x, y, z \in A$ ,

- (Associator) Define the associator  $\text{Asso}(x, y, z) = x(yz) - (xy)z$ . Then the associator is anti-symmetric:

$$\text{Asso}(x, y, z) = -\text{Asso}(x, z, y) = -\text{Asso}(z, y, x) = -\text{Asso}(y, x, z).$$

- (Flexible Law)  $(xy)x = x(yz)$ .
- If  $xy = -yx$ , then  $x(yz) = -y(xz)$ ,  $(zx)y = -(zy)x$ .
- (Moufang Identity)  $(zx)(yz) = z(xy)z$ .
- (Powers) If we define inductively  $x^0 = 1$ ,  $x^{n+1} = x^n x$  for  $n \in \mathbb{Z}_+$ , then we have  $x^n x^m = x^{n+m}$  for any  $m, n \in \mathbb{N}$ .

┘

*Proof:* 1: The alternativity says  $\text{Asso}(x, x, y) = \text{Asso}(x, y, y) = 0$ . So

$$0 = \text{Asso}(x+y, x+y, z) = \text{Asso}(x, x, z) + \text{Asso}(x, y, z) + \text{Asso}(y, x, z) + \text{Asso}(y, y, z) = \text{Asso}(x, y, z) + \text{Asso}(y, x, z).$$

2: This is because  $\text{Asso}(x, y, x) = -\text{Asso}(y, x, x) = 0$ .

3:  $0 = \text{Asso}(x, y, z) + \text{Asso}(y, x, z) + (xy + yx)z = x(yz) + y(xz)$ .

4:

$$\begin{aligned} (zx)(yz) - ((zx)y)z &= \text{Asso}(zx, y, z) = \text{Asso}(y, z, zx) = y(z^2x) - (yz)(zx) \\ &= y(z^2x) - \text{Asso}(yz, z, x) - (yz^2)x = \text{Asso}(y, z^2, x) - \text{Asso}(yz, z, x) \\ &= \text{Asso}(y, z^2, x) - \text{Asso}(x, yz, z) = \text{Asso}(y, z^2, x) - x(y^2z) + (x(yz))z \\ &= \text{Asso}(y, z^2, x) + \text{Asso}(x, y, z)z = \text{Asso}(x, y, z^2) = \text{Asso}(x, y, z)z \end{aligned}$$

So  $(zx)(yz) = \text{Asso}(x, y, z)z + ((zx)y)z = z(xy)z$ .

5: It can be shown by flexibility and induction that  $x.x^m = x^{m+1}$  for any  $m \in \mathbb{N}$ . Then we can assume  $m \in \mathbb{Z}_+$  and use induction on  $n$  and the Moufang identity to show that

$$x^{n+1}.x^m = (x.x^n)(x^{m-1}x) = x(x^n.x^{m-1})x = x.x^{n+m-1}.x = x^{m+n+1}.$$

□

## 10 Miscellaneous

**Lemma(3.6.10.1).** Let  $R$  be a (possibly non-commutative) unital ring of characteristic 0 with the following properties:

- $R$  has no zero-divisors,
- $\text{rank}_{\mathbb{Z}} R \leq 4$ ,



- $R$  has an involution  $*$  :  $R \rightarrow R^{\text{op}}$ ,
- For any  $\alpha \in R$ ,  $\alpha\alpha^* \in \mathbb{N}$ , and  $\alpha\alpha^* = 0$  iff  $\alpha = 0$ .

Then  $R$  is isomorphic to exactly one of the following:

- $\mathbb{Z}$ .
- a  $\mathbb{Z}$ -order in an imaginary quadratic extension over  $\mathbb{Q}$ .
- a  $\mathbb{Z}$ -order in a definite quaternion algebra over  $\mathbb{Q}$ .

And the third case won't happen in characteristic 0, by (21.3.1.26).  $\lrcorner$

*Proof:* It suffices to show  $K = R \otimes \mathbb{Q}$  is either  $\mathbb{Q}$ , an imaginary quadratic field or a definite quaternion algebra. For  $\alpha \in K$ , denote  $\text{Nm}(\alpha) = \alpha\alpha^*$ ,  $\text{tr}(\alpha) = \alpha + \alpha^*$ , then  $\text{tr}\alpha = 1 + \text{Nm}(\alpha) - \text{Nm}(1 - \alpha) \in \mathbb{Q}$ .

Assume  $K \neq \mathbb{Q}$ , let  $\alpha \in K \setminus \mathbb{Q}$ . We may replace  $\alpha$  by  $\alpha - \frac{1}{2}\text{tr}(\alpha)$  to assume  $\text{tr}(\alpha) = 0$ , then  $\alpha^2 = -\text{Nm}(\alpha) < 0 \in \mathbb{Z}$ . So  $\mathbb{Q}(\alpha)$  is an imaginary quadratic field.

Assume  $K \neq \mathbb{Q}(\alpha)$ , let  $\beta \in K \setminus \mathbb{Q}(\alpha)$ . We may replace  $\beta$  by  $\beta - \frac{1}{2}\text{tr}\beta - \frac{\text{tr}(\alpha\beta)}{\alpha^2}\alpha$  to assume  $\text{tr}(\beta) = \text{tr}(\alpha\beta) = 0$ . Hence  $\beta^2 = -N\beta < 0 \in \mathbb{Z}$ , and  $\alpha\beta = -\beta\alpha$ . So  $\mathbb{Q}(\alpha, \beta)$  is a definite quadratic extension of  $\mathbb{Q}$ . Then  $K = \mathbb{Q}(\alpha, \beta)$  because they are has rank 4 over  $\mathbb{Z}$ .  $\square$

### 3.7 Lie Algebras, I

Basic references are [Ser87, ?Kna96, Car05, Mil13, ?Et121], [Lie Algebras and Lie Groups Serre].

**Notation (3.7.0.1).**

- Let  $k \in \text{Field}^0$ .
- Let  $\mathfrak{g} \in \text{LieAlg}_k$  (3.7.1.1).

┘

#### 1 Basics

**Def. (3.7.1.1) [Lie Algebras].** A **Lie algebra**  $L$  over  $k$  is an non-associative algebra over  $k$  with a bilinear **Lie-bracket** operation that satisfies:

$$[x, x] = 0, \quad [x[yz]] = [[xy]z] + [y[xz]] \text{ (Jacobi Identities)}.$$

The category of Lie-algebras over  $k$  is denoted by  $\text{LieAlg}_k$ .

It is easily deduced that  $[xy] = -[yx]$ .

Denote  $\text{adx}(y) = [xy]$ , then  $\text{ad}(x)$  are all derivatives of  $L$ .

An element  $x \in \mathfrak{g}$  is called a **nilpotent element** (resp. **semisimple element**) if  $\text{ad}(x)$  is nilpotent (resp. semisimple). ┘

**Prop. (3.7.1.2) [Associative Algebra].** Any  $A \in \text{Alg}_k^{\text{asso}}$  can be given naturally a Lie algebra structure by defining  $[xy] = xy - yx$ . In this way, we get a natural functor  $\text{Alg}_k^{\text{asso}} \rightarrow \text{LieAlg}_k : A \mapsto [A]$ . ┘

**Prop. (3.7.1.3) [Derivatives form a Lie Algebra].** For  $A \in \text{Alg}_k^{\text{asso}}$ , the set of derivatives  $\text{Der}_k(A) = \text{Der}_k(A, A)$  is a Lie algebra under the associative bracket. ┘

*Proof:*

$$\begin{aligned} [D_1, D_2](ab) &= D_1(D_2(a)b + aD_2(b)) - D_2(D_1(a)b + aD_1(b)) \\ &= D_1D_2(a)b + D_2(a)D_1(b) + D_1(a)D_2(b) + aD_1D_2(b) \\ &\quad - (D_2D_1(a)b + D_1(a)D_2(b) + D_2(a)D_1(b) + aD_2D_1(b)) \\ &= [D_1, D_2](a)b + a[D_1, D_2](b) \end{aligned}$$

□

**Prop. (3.7.1.4) [Base Change of Fields].** Let  $\mathfrak{a}$  be a subalgebra of a Lie algebra  $\mathfrak{g}$  over  $k$ , and  $k'/k$  a field extension, then  $\mathfrak{a}_{k'}$  is a subalgebra of  $\mathfrak{g}_{k'}$ , and

$$N_{\mathfrak{g}_{k'}}(\mathfrak{a}_{k'}) = N_{\mathfrak{g}}(\mathfrak{a})_{k'}.$$

$$c_{\mathfrak{g}_{k'}}(\mathfrak{a}_{k'}) = c_{\mathfrak{g}}(\mathfrak{a})_{k'}.$$

┘

*Proof:* This is because the normalizer and centralizer is defined by linear equations with coefficients in  $k$ , thus the vector space is defined over  $k$ . □

**Prop. (3.7.1.5).** Let  $D$  be a derivative of a  $k$ -algebra  $A$  that is nilpotent, then  $e^D$  is an automorphism of  $A$  (as an algebra). ┘

*Proof:* Routine calculation.  $\square$

**Def. (3.7.1.6) [Semidirect Product of Lie Algebras].** Let  $\mathfrak{g}, \mathfrak{h} \in \text{LieAlg}_k$  and  $\tau : \mathfrak{g} \rightarrow \text{Der}(\mathfrak{h})$ , then we can define a **semi-direct product Lie algebra**  $\mathfrak{g} \ltimes \mathfrak{h}$  that is isomorphic to  $\mathfrak{g} \oplus \mathfrak{h}$  as vector spaces, and  $\mathfrak{g}, \mathfrak{h}$  are subalgebras of  $\mathfrak{g} \ltimes \mathfrak{h}$ , with  $[g, h] = \tau(g)(h)$ . It can be shown that this is truly a Lie algebra.  $\lrcorner$

**Def. (3.7.1.7) [Elementary Automorphisms].** Let  $\mathfrak{g}$  be a Lie algebra, a **special automorphism** is an automorphism of  $\mathfrak{g}$  of the form  $e^{\text{ad}_{\mathfrak{g}} x}$ , where  $x$  is in the nilpotent radical (3.7.1.34).

The group of **elementary automorphisms** is the subgroup of the automorphism group of  $\mathfrak{g}$  generated by the automorphisms of the form  $e^{\text{ad}_{\mathfrak{g}}(x)}$  where  $\text{ad}(x)$  is nilpotent.  $\lrcorner$

**Def. (3.7.1.8) [Ideals].** A subspace  $\mathfrak{a} \subset \mathfrak{g}$  is called an **ideal** of  $\mathfrak{g}$  iff  $[\mathfrak{g}, \mathfrak{a}] \subset \mathfrak{a}$ . If  $I$  is an ideal of  $L$ , then  $L/I$  can be made into a Lie algebra by defining  $[I + x, I + y] = I + [xy]$ .  $\lrcorner$

**Def. (3.7.1.9) [Center].** The **center** of a Lie algebra  $\mathfrak{g}$  is the elements  $a$  that  $\text{ada} = 0$ . It is an ideal.  $\lrcorner$

**Def. (3.7.1.10) [Simple Lie Algebras].** A Lie algebra  $\mathfrak{g}$  is called **simple** if it is not 1-dimensional and it has no nontrivial ideal.  $\lrcorner$

**Def. (3.7.1.11) [Lie Algebra of Affine Maps].** For  $V \in \text{Vect}^{\text{fin}}/k$ , if we regard  $V$  as a commutative algebra, then  $\text{Der}_k(V) = \mathfrak{gl}_V$ . Then  $V \rtimes \mathfrak{gl}_V$  is a Lie algebra, denoted by  $\mathfrak{af}(V)$ .

Let  $V' = V \oplus k$ , and let  $\mathfrak{h} = \{w \in \mathfrak{gl}_{V'} \mid w(V') \subset V\}$ , which is a Lie subalgebra of  $\mathfrak{gl}_{V'}$ . If we define

$$\eta : \mathfrak{h} \rightarrow \mathfrak{gl}_V : \eta(w) = w|_V, \quad \zeta : \mathfrak{h} \rightarrow V : \zeta(w) = w(0, 1),$$

then  $(\eta, \zeta)$  defines a Lie algebra homomorphism from  $\mathfrak{h}$  to  $\mathfrak{af}(V)$ . This map is bijective, with the inverse given by sending  $(v, f) \in \mathfrak{af}(V)$  to the morphism

$$(v', c) \mapsto (f(v') + cv, 0).$$

$\lrcorner$

**Lemma (3.7.1.12).** Let  $\lambda, \mu \in k$  and  $x, y, z \in \mathfrak{g}$ , then we have

$$(\text{ad}(x) - \lambda - \mu)^m [y, z] = \sum_{i=1}^m \binom{m}{i} [(\text{ad}(x) - \lambda)^i y, (\text{ad}(x) - \mu)^{m-i} z].$$

$\lrcorner$

*Proof:*

$$\begin{aligned} (\text{ad}(x) - \lambda - \mu)^m [y, z] &= \sum_{p+q+r+s=m} (-1)^{r+s} \frac{m!}{p!q!r!s!} \lambda^r \mu^s [(\text{ad} x)^p(y), (\text{ad} x)^q(z)] \\ &= \sum_{k+l=m} \frac{m!}{k!l!} [(\text{ad}(x) - \lambda)^k y, (\text{ad}(x) - \mu)^l z]. \end{aligned}$$

$\square$

**Def. (3.7.1.13) [Killing Form].** A bilinear form  $B$  on  $\mathfrak{g}$  is called **invariant** if  $B([x, y], z) + B(x, [y, z]) = 0$ .

The **Killing form** on a Lie algebra  $\mathfrak{g}$  of f.d. is the invariant symmetric bilinear form defined by  $B(x, y) = \text{tr}(\text{adx} \circ \text{ady})$ .

if  $\mathfrak{a}$  is an ideal of a Lie algebra  $\mathfrak{g}$ , then the Killing form on  $\mathfrak{a}$  is that of the Killing form on  $\mathfrak{a}$  as a Lie algebra. This is linear algebra.  $\lrcorner$

**Prop. (3.7.1.14).** Any invariant symmetric bilinear form on a simple Lie algebra  $\mathfrak{g}$  is a multiple of the Killing form.  $\lrcorner$

**Prop. (3.7.1.15).** A subalgebra  $\mathfrak{h}$  of a Lie algebra  $\mathfrak{g}$  is commutative if it consists of semisimple elements.  $\lrcorner$

*Proof:* For an element  $x \in \mathfrak{h}$ , we need to show that  $\text{ad}_{\mathfrak{h}}(x) = 0$ . If it is not, because  $\text{ad}_{\mathfrak{h}}(x)$  is semisimple by (3.5.10.8), there is a nonzero eigenvector  $y$  at least after a base change, so  $[x, y] = cy, y \neq 0 \in \mathfrak{h}$ . So  $\text{ad}(y)(x) = -cy$ , and  $\text{ad}(y)^2(x) = 0$ , so  $\text{ad}(y)$  is non-semisimple on the subspace  $\{x, y\}$ , which means  $\text{ad}(y)$  is non-semisimple on  $\mathfrak{g}$ , by (3.5.10.8) again.  $\square$

**Lemma (3.7.1.16).** If  $\mathfrak{g} \subset \mathfrak{gl}_n$  is a Lie subalgebra, and  $a \in \mathfrak{g}$  is a nilpotent matrix, then  $\text{ad}(a)$  is also nilpotent.  $\lrcorner$

*Proof:* This is because  $\text{ad}(a) = l(a) - r(a)$ , where  $l(a)$  is left multiplication and  $r(a)$  is right multiplication. The left and right multiplication commutes, so it is clear  $\text{ad}(a)^{2n} = 0$  if  $\text{ad}(a)^n = 0$ .  $\square$

### Nilpotent and Solvable Lie Algebras

**Def. (3.7.1.17)[Nilpotent and Solvable Lie Algebras].** For  $\mathfrak{g} \in \text{LieAlg}_k$ , the **lower central series** of  $\mathfrak{g}$  is the descending sequence of ideals of  $\mathfrak{g}$  defined inductively by  $C^1\mathfrak{g} = \mathfrak{g}$  and  $C^m\mathfrak{g} = [\mathfrak{g}, C^{m-1}\mathfrak{g}]$ .

Let  $\mathfrak{g}$  be a Lie algebra, the **derived series** of  $\mathfrak{g}$  is the descending sequence of ideals of  $\mathfrak{g}$  defined inductively by  $D^1\mathfrak{g} = \mathfrak{g}$  and  $D^n\mathfrak{g} = [D^{n-1}\mathfrak{g}, D^{n-1}\mathfrak{g}]$ .

A Lie algebra is called **nilpotent Lie algebra** if there is an  $n$  that  $C^n\mathfrak{g} = 0$ . This is equivalent to  $\text{adx}_1\text{adx}_2 \dots \text{adx}_n = 0$  for any  $n$  element  $x - 1, \dots, x_n$ . It is called **solvable** if  $D^n = 0$  for some  $n$ . It is clear that  $D^n \subset C^n$ , so nilpotent Lie algebra is solvable.  $\lrcorner$

**Prop. (3.7.1.18).** The lower central series satisfies:  $[C^n\mathfrak{g}, C^m\mathfrak{g}] \subset C^{m+n}\mathfrak{g}$ .

The operation of taking derived series or lower central series commutes with base change of fields.  $\lrcorner$

*Proof:* Prove by induction on  $n$ :  $n = 0, 1$  is trivial, and if the assertion is true for  $n \geq k$ , then for  $n = k + 1$ ,  $[C^n\mathfrak{g}, C^m\mathfrak{g}] \subset [\mathfrak{g}, [C^{n-1}\mathfrak{g}, C^m\mathfrak{g}]] + [C^{n-1}\mathfrak{g}, C^{m+1}\mathfrak{g}] \subset C^{m+n}\mathfrak{g}$ .  $\square$

**Cor. (3.7.1.19).** Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$ , and  $k'/k$  is a field extension, then  $\mathfrak{g}$  is solvable/nilpotent iff  $\mathfrak{g} \otimes_k k'$  is solvable/nilpotent.  $\lrcorner$

**Prop. (3.7.1.20).** If  $\mathfrak{g}$  is a nilpotent Lie algebra, then for any subalgebra  $\mathfrak{h} \subsetneq \mathfrak{g}$ ,  $N_{\mathfrak{g}}(\mathfrak{h}) \neq \mathfrak{h}$ .  $\lrcorner$

*Proof:* Because  $\mathfrak{g}^n = 0$  for some  $n$ , take  $n$  to be the maximal one that  $\mathfrak{h} \not\subset \mathfrak{g}^n$ , the  $[\mathfrak{g}^n, \mathfrak{h}] \subset \mathfrak{g}^{n+1} \subset \mathfrak{h}$ , so  $\mathfrak{g}^n \subset N_{\mathfrak{g}}(\mathfrak{h})$ , so  $N_{\mathfrak{g}}(\mathfrak{h}) \neq \mathfrak{h}$ .  $\square$

**Prop. (3.7.1.21).** Sub-Lie-algebras, quotient Lie-algebras and extension Lie-algebras of solvable Lie-algebras are solvable.  $\lrcorner$

*Proof:* Let  $\mathfrak{h} \subset \mathfrak{g}$ , then  $D^n(\mathfrak{h}) \subset D^n(\mathfrak{g})$ , so if  $\mathfrak{g}$  is solvable, then so is  $\mathfrak{h}$ . Also the quotient is clearly solvable. For an extension of Lie algebras

$$0 \rightarrow \mathfrak{h} \rightarrow \mathfrak{g} \rightarrow \mathfrak{r} \rightarrow 0,$$

if  $\mathfrak{h}, \mathfrak{r}$  are both solvable, let  $D^m(\mathfrak{h}) = 0, D^n(\mathfrak{r}) = 0$ , then the image of  $D^n(\mathfrak{g})$  is 0 in  $\mathfrak{r}$ , so  $D^n(\mathfrak{g}) \subset \mathfrak{h}$ , so  $D^{m+n}(\mathfrak{g}) = 0$ .  $\square$

**Cor. (3.7.1.22) [Radicals].** If  $\mathfrak{a}, \mathfrak{b}$  are solvable ideals of a Lie algebra  $\mathfrak{g}$ , then the ideal  $\mathfrak{a} + \mathfrak{b}$  is also solvable, because  $0 \rightarrow \mathfrak{a} \rightarrow \mathfrak{a} + \mathfrak{b} \rightarrow \mathfrak{b}/\mathfrak{a} \cap \mathfrak{b}$ .

Let  $\mathfrak{r} \subset \mathfrak{g}$  be the sum of all solvable ideals of  $\mathfrak{g}$ , called the **radical(Lie-algebra)**, denoted by  $\text{rad}(\mathfrak{g})$ . When  $\mathfrak{g}$  is of f.d., this is the maximal solvable ideal.  $\lrcorner$

**Def. (3.7.1.23) [Semisimple Lie-Algebras].**  $\mathfrak{g} \in \text{LieAlg}_k$  is called a **semisimple Lie-algebra** if  $\text{rad}(\mathfrak{g}) = 0$  (3.7.1.22), or equivalently  $\mathfrak{g}$  has no solvable ideals or no commutative ideals. Notice  $\mathfrak{g}/\text{rad}(\mathfrak{g})$  is semisimple, by (3.7.1.22).  $\lrcorner$

**Prop. (3.7.1.24) [Lie].** Let  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  be a solvable lie algebra over an alg.closed field  $k$  of char 0, then  $\mathfrak{g}$  is upper triangulable. Equivalently, there exists a vector  $v \in V$  which is a common eigenvector for all  $X \in \mathfrak{g}$ , and moreover equivalently, any irreducible representation of  $\mathfrak{g}$  is 1-dimensional.  $\lrcorner$

*Proof:* Idea is to prove by induction on dimension of  $\mathfrak{g}$ .

Produce a codimension 1 ideal  $\mathfrak{h}$  of  $\mathfrak{g}$ . Let  $\mathfrak{g}$  be generated (as a vector space) by  $\mathfrak{h}$  and  $Y$ . Being a subalgebra of solvable algebra  $\mathfrak{g}$ ,  $\mathfrak{h}$  is itself a solvable lie algebra. Apply induction step on  $\mathfrak{h}$  and choose  $v \in V$  such that  $v$  is an eigenvector for all  $X \in \mathfrak{h}$ .

The idea is to consider set  $W$  all common eigenvectors of elements of  $\mathfrak{h}$  and produce an eigenvector of  $Y$  from this  $W$ . Let

$$W = \{v \in V | X(v) = \lambda(X)v \ \forall X \in \mathfrak{h} \text{ for a fixed } \lambda(X) \in \mathbb{C}\}.$$

Suppose  $W$  is an invariant subspace of  $Y$ , we then have restriction map  $Y : W \rightarrow W$ . As we are in complex vector space (algebraically closed) there exists an eigenvector for  $Y$  in  $W$  say  $w_0$ . Thus,  $w_0$  is common eigenvector for all elements of  $\mathfrak{g}$ .

It remains to show that  $W$  is an invariant subspace of  $Y$  i.e.,  $Y(w) \in W$  for all  $w \in W$  i.e., given  $X \in \mathfrak{h}$ , we need to have  $X(Y(w)) = \lambda(X)Y(w)$ .

Let  $w \in W$ , we have

$$\begin{aligned} X(Y(w)) &= Y(X(w)) + [X, Y](w) \\ &= Y(\lambda(X)w) + \lambda([X, Y])w \\ &= \lambda(X)Y(w) + \lambda([X, Y])w \end{aligned}$$

This is almost the same as what we want but with an extra term  $\lambda([X, Y])w$ . Suppose we prove  $\lambda([X, Y]) = 0$  for all  $X \in \mathfrak{h}$  then we are done.

Then considers subspace  $U$  spanned by elements  $\{w, Y(w), Y^2(w), \dots\}$  and then says that  $U$  is invariant subspace of each element of  $\mathfrak{h}$  and (assuming  $n$  is the smallest integer such that  $Y^{n+1}w$  is in the subspace generated by  $\{w, Y(w), \dots, Y^n(w)\}$ ) representation of an element  $Z$  of  $\mathfrak{h}$  with the basis  $\{w, Y(w), \dots, Y^n(w)\}$  is an upper triangular matrix with  $\lambda(Z)$  in the diagonal. So,  $\text{tr}(Z) = n\lambda(Z)$ .

So,  $\text{tr}([X, Y]) = n\lambda([X, Y])$ . As  $[X, Y] = XY - YX$ , we have  $\text{tr}([X, Y]) = \text{tr}(XY) - \text{tr}(YX) = 0$ . Thus,  $\lambda([X, Y]) = 0$  and we are done.  $\square$

**Cor. (3.7.1.25).** If  $\mathfrak{g}$  is a solvable algebra over an alg.closed field  $k$  of char 0, then all irreducible representations of  $\mathfrak{g}$  is of dimension 1.  $\lrcorner$

**Cor. (3.7.1.26).**  $\mathfrak{g}$  is a solvable algebra iff  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent.  $\lrcorner$

*Proof:* If  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent, then clearly  $\mathfrak{g}$  is solvable. Conversely, if  $\mathfrak{g}$  is solvable, we need to prove  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent. For this, we can assume  $k$  is alg.closed, and then  $\text{ad}(\mathfrak{g}) \subset \mathfrak{b}_{\mathfrak{g}}$  for some basis, thus  $\text{ad}([\mathfrak{g}, \mathfrak{g}]) \subset \mathfrak{n}_{\mathfrak{g}}$  is nilpotent, and the kernel of  $\text{ad}$  is an Abelian subalgebra, so  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent.  $\square$

**Cor. (3.7.1.27).** If  $\mathfrak{g}$  is a Lie subalgebra of  $\mathfrak{gl}_n(k)$  where  $k$  is an alg.closed field of char 0, then

$$\mathfrak{g} \text{ is solvable} \iff \operatorname{tr}(xy) = 0, \forall x \in \mathfrak{g}, y \in [\mathfrak{g}, \mathfrak{g}].$$

┘

*Proof:* Firstly if  $\mathfrak{g}$  is solvable, then by Lie's theorem, we can assume  $\mathfrak{g} \in \mathfrak{b}_V$  the upper-triangular matrices, so  $[\mathfrak{g}, \mathfrak{g}] \in \mathfrak{n}_V$  is nilpotent, and so  $xy \in \mathfrak{n}_V$  is also nilpotent, and  $\operatorname{tr}(xy) = 0$ .

Conversely, if  $\operatorname{tr}(xy) = 0$  for all  $x \in \mathfrak{g}, y \in [\mathfrak{g}, \mathfrak{g}]$ , we only need to prove  $[\mathfrak{g}, \mathfrak{g}]$  is solvable, so we may change  $\mathfrak{g}$  to  $[\mathfrak{g}, \mathfrak{g}]$  and assume  $\operatorname{tr}(xy) = 0$  for all  $x, y \in \mathfrak{g}$ .

Now to show  $\mathfrak{g}$  is solvable, it suffices to show  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent, or by Engel's theorem(3.7.1.29) all  $x \in [\mathfrak{g}, \mathfrak{g}]$  defines a nilpotent endomorphism on  $V$ . Choose a basis that  $x$  is upper-triangular by(3.5.10.3), and let  $x_s$  be the semisimple part of  $x$ , then it suffices to show  $x_s = 0$ , or equivalently  $\operatorname{tr}(\overline{x_s}x) = 0$ . To show this, notice  $x \in [\mathfrak{g}, \mathfrak{g}]$ , so it suffices to show  $\operatorname{tr}(\overline{x_s}[y, z]) = 0$  for any  $y, z$ . But this equals  $\operatorname{tr}([\overline{x_s}, y], z)$ . Finally, this is 0 because of the hypothesis and the fact  $\overline{x_s}$  is a polynomial in  $x$ (3.5.10.7) so  $[\overline{x_s}, y] \in \mathfrak{g}$ .  $\square$

**Cor. (3.7.1.28) [Cartan's Criteria for Solvability].**  $\mathfrak{g} \in \operatorname{LieAlg}_k$  is solvable if  $\kappa_{\mathfrak{g}}(x, y)$  for any  $x \in \mathfrak{g}, y \in [\mathfrak{g}, \mathfrak{g}]$ , where  $\kappa$  is the Killing form.  $\square$

*Proof:* By(3.7.1.19), it suffices to show for  $k$  alg.closed. Because the kernel of the adjoint map is the center of  $\mathfrak{g}$ , so  $\mathfrak{g}$  is solvable iff  $\operatorname{ad}(\mathfrak{g})$  is solvable.  $\square$

**Prop. (3.7.1.29) [Engel].** If  $(V, \rho)$  is a representation of a Lie algebra  $\mathfrak{g}$  s.t.  $\rho(x)$  is nilpotent for all  $x \in \mathfrak{g}$ , then there is a basis that  $\rho(\mathfrak{g})$  is contained in  $\mathfrak{n}_V$ , in particular  $\mathfrak{g}$  is nilpotent.  $\square$

*Proof:* It suffices to show if a sub-Lie algebra of  $\mathfrak{gl}_V$  consists of nilpotent elements, then there is a common 0-eigenvector. Use Induction, choose a maximal subalgebra  $K$  of  $L$ , then notice the normalizer of  $K$  in  $L$  is strictly containing  $K$ , because we can let  $K$  acts by adjoint on  $L/K$ , and notice  $\operatorname{ad}x = \lambda_x - \rho_x$  is nilpotent for  $x$  a nilpotent matrix, so by induction hypothesis there is an  $x \in L$  that  $[x, K] \subset K$ . But  $K$  is maximal, so it must be of codimension 1, and  $L = K + Fz$ . The 0-eigenvectors for  $K$  is a nonzero subspace by induction hypothesis. Now this space is invariant under  $z$ : for any  $h \in K$ ,

$$h(z(v)) = [h, z](v) + zh(v) = 0.$$

So now a 0-eigenvector for  $z$  in this space will suffice.  $\square$

**Cor. (3.7.1.30).** If all elements of  $L$  are ad-nilpotent(i.e.  $\operatorname{ad}x = 0$ ), then  $L$  is nilpotent. Equivalently, elements of  $L$  has a common 0-eigenvector.  $\square$

*Proof:* Consider  $\operatorname{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}_{\mathfrak{g}}$ , then the image is nilpotent by Engel's theorem(3.7.1.29), and the kernel of  $\operatorname{ad}$  is the center of  $\mathfrak{g}$ , so  $\mathfrak{g}$  is also nilpotent.  $\square$

**Cor. (3.7.1.31).** Let  $\mathfrak{a}$  be an ideal of a Lie algebra  $\mathfrak{g}$ . Then  $\mathfrak{a}$  is nilpotent if  $\operatorname{ad}_{\mathfrak{g}}(a)$  is nilpotent for any  $a \in \mathfrak{a}$ .  $\square$

*Proof:* If  $\operatorname{ad}_{\mathfrak{g}}(a)$  is nilpotent for any  $a \in \mathfrak{a}$ , then also  $\operatorname{ad}_{\mathfrak{a}}(a)$  is nilpotent, so  $\mathfrak{a}$  is nilpotent by Engel's theorem. Conversely, if  $\mathfrak{a}$  is nilpotent, then  $\operatorname{ad}_{\mathfrak{a}}(a)$  is nilpotent for any  $a \in \mathfrak{a}$ . And  $\operatorname{ad}(a)(\mathfrak{g}) \subset \mathfrak{a}$ , so  $\operatorname{ad}_{\mathfrak{g}}(a)$  is nilpotent.  $\square$

**Cor. (3.7.1.32).** The sum of two nilpotent ideals of  $\mathfrak{g}$  is nilpotent.  $\square$

*Proof:* We need to show for any  $a \in \mathfrak{a}, b \in \mathfrak{b}$ ,  $\text{ad}_{\mathfrak{g}}(a + b)$  is nilpotent. For this, we need to factor  $\mathfrak{g}$  as Jordan sequence  $0 \subset \mathfrak{g}_1 \subset \mathfrak{g}_2 \subset \dots \subset \mathfrak{g}_n = \mathfrak{g}$  over itself via the adjoint representation, then I claim that  $\text{ad}(a)(\mathfrak{g}_k) \subset \mathfrak{g}_{k-1}$ : Because  $V = \mathfrak{g}_k/\mathfrak{g}_{k-1}$  is simple, let  $V' \subset V$  consists of vectors  $v$  that  $a(v) = 0$  for any  $a \in \mathfrak{a}$ , then it is non-empty by Engel's theorem, and also it is invariant under action of  $\mathfrak{g}$ :  $a(gv) = [a, g](v) + g(av) = 0$ . So it is all of  $V$ .

Then, we have  $\text{ad}(a + b)(\mathfrak{g}_k) \subset \mathfrak{g}_{k-1}$ , thus  $\text{ad}(a + b)$  is nilpotent.  $\square$

**Cor. (3.7.1.33) [Maximal Nilpotent Ideal].** For any Lie algebra  $\mathfrak{g}$ , there exists a maximal nilpotent ideal, denoted by  $\mathfrak{n}$ .  $\lrcorner$

**Def. (3.7.1.34) [Nilpotent Radical].** The **nilpotent radical**  $\mathfrak{s} = \mathfrak{s}(\mathfrak{g})$  of a Lie algebra is the intersection of the kernels of simple representations of  $\mathfrak{g}$ .

$\mathfrak{s}$  is nilpotent for any f.d. representation of  $\mathfrak{g}$ , in particular the adjoint representation of  $\mathfrak{g}$ . Thus it is nilpotent, by (3.7.1.31).  $\lrcorner$

**Lemma (3.7.1.35).** Let  $\mathfrak{g} \subset \mathfrak{gl}_V$  be a subalgebra, and let  $\mathfrak{a}$  a commutative ideal of  $\mathfrak{g}$ . If  $V$  is simple as a  $\mathfrak{g}$ -module, then  $[\mathfrak{g}, \mathfrak{g}] \cap \mathfrak{a} = 0$ .  $\lrcorner$

*Proof:* Cf. [Mil13]P58.  $\square$

**Prop. (3.7.1.36).** Let  $\mathfrak{g}$  be a Lie algebra, and  $\mathfrak{r}$  its radical,  $\mathfrak{s}$  its nilpotent radical, then

$$\mathfrak{s} = D(\mathfrak{g}) \cap \mathfrak{r} = [\mathfrak{g}, \mathfrak{r}].$$

In particular,  $[\mathfrak{g}, \mathfrak{r}]$  is nilpotent (3.7.1.34).  $\lrcorner$

*Proof:* To show  $D(\mathfrak{g}) \cap \mathfrak{r} \subset \mathfrak{s}$ , we need to show that  $\rho(D(\mathfrak{g}) \cap \mathfrak{r}) = 0$  for any simple representation  $\rho$ . Because  $\mathfrak{r}$  is solvable, let  $r$  be the smallest integer that  $\rho(D^{r+1}(\mathfrak{r})) = 0$ , then  $\mathfrak{a} = \rho(D^r(\mathfrak{r}))$  is a commutative ideal of  $\rho(\mathfrak{g})$ . Hence by (3.7.1.35)  $D(\rho(\mathfrak{g})) \cap \mathfrak{a} = 0$ , so  $\rho(D(\mathfrak{g}) \cap D^r(\mathfrak{r})) = 0$ . Now if  $r > 0$ , then  $\rho(D^r(\mathfrak{r})) = 0$ , contradicting the minimality of  $r$ , so  $\rho(D(\mathfrak{g}) \cap \mathfrak{r}) = 0$ .

To show  $\mathfrak{s} \subset [\mathfrak{g}, \mathfrak{r}]$ , let  $\mathfrak{q} = \mathfrak{g}/[\mathfrak{g}, \mathfrak{r}]$ , and  $f$  the quotient map. Then because the kernel is solvable,  $f(\mathfrak{r})$  is the radical of  $\mathfrak{q}$  (3.7.1.22), but it is also contained in the center of  $\mathfrak{q}$ , so  $\mathfrak{q}$  is reductive, and thus has a faithful semisimple representation (3.7.4.4), then the kernel of this representation is just  $[\mathfrak{g}, \mathfrak{r}]$ , showing that  $\mathfrak{s} \subset [\mathfrak{g}, \mathfrak{r}]$ .  $\square$

**Def. (3.7.1.37) [Levi Subalgebras].** Let  $\mathfrak{g}$  be a Lie algebra and  $\mathfrak{r}$  its radical, then a Lie subalgebra  $\mathfrak{s}$  is called a **Levi subalgebra** if  $\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{s}$ .  $\lrcorner$

**Prop. (3.7.1.38) [Levi-Malcev].** Every Lie algebra over a field  $k$  of char 0 has a Levi subalgebra, and any two Levi subalgebras of  $\mathfrak{g}$  are conjugate by a special automorphism of  $\mathfrak{g}$  (3.7.1.7).  $\lrcorner$

*Proof:* If  $\mathfrak{g}$  is reductive, then Levi subalgebra exists uniquely, by (3.7.4.2) and (3.7.4.3).

if  $\mathfrak{r}$  is a minimal ideal of  $\mathfrak{g}$ , then  $[\mathfrak{g}, \mathfrak{r}] = \mathfrak{r}$ , and  $[\mathfrak{r}, \mathfrak{r}] = 0 = Z(\mathfrak{g})$ . Consider the adjoint action of  $\mathfrak{g}$  on  $\text{End}_k(\mathfrak{g})$ . Also consider the subspaces  $V, W$  of  $\text{End}_k(\mathfrak{g})$ , where  $V$  is the subspace of maps from  $\mathfrak{g}$  to  $\mathfrak{r}$  that restriction to  $\mathfrak{r}$  is a constant multiple of identity, and  $W$  is the subspaces of  $W$  consisting of maps vanishing on  $\mathfrak{r}$ . Both of  $V, W$  are invariant under action of  $\mathfrak{g}$ .

Let  $\varphi : \mathfrak{r} \rightarrow \mathfrak{g}$  be the adjoint action, which is injective and has image  $P \subset W$ . Also  $\mathfrak{P}$  is invariant under action of  $\mathfrak{g}$  (because  $\mathfrak{r}$  is an ideal).

For  $x \in \mathfrak{r}, y \in \mathfrak{g}, \alpha \in V$ ,  $(x\alpha)(y) = [x, \alpha(y)] - \alpha([x, y]) = -\lambda(\alpha)[x, y]$ , so  $x\alpha = \text{ad}(\lambda(\alpha)x)$ , which means elements of  $\mathfrak{r}$  map  $V$  into  $P$ . Thus now  $\mathfrak{r}$  acts trivially on  $V/P$ , and  $W/P$  is invariant under

the action of  $\mathfrak{g}/\mathfrak{r}$ , which is a semisimple Lie algebra. Thus by Weyl's theorem(24.4.1.2), there exists a  $\mathfrak{g}$ -stable line  $L$  that  $V/P = W/P \oplus L$ . But  $\mathfrak{g}$  acts trivially on  $L$  by(24.4.1.1).

Let  $\alpha_0$  generates  $L$  and normalized that  $\lambda(\alpha) = 1$ , then  $\mathfrak{g}\alpha_0 \in P$ , We consider the map  $\mathfrak{g} \xrightarrow{g \mapsto g\alpha_0} P \xrightarrow{\varphi^{-1}} \mathfrak{r}$ , whose restriction to  $\mathfrak{r}$  is the identity map, so its kernel is a Levi subgroup for  $\mathfrak{r}$ .

Still in this case, let  $\mathfrak{s}'$  be a second Levi subgroup for  $\mathfrak{r}$ . For each  $x \in \mathfrak{s}'$ , there is a unique  $h(x) \in \mathfrak{r}$  that  $x + h(x) \in \mathfrak{s}$ . Hence this  $h$  satisfies  $h([x, y]) = [h(x), y] + [x, h(y)]$ . But then by(24.4.1.3),  $h(x) = [a, x]$  for some  $a \in \mathfrak{r}$ . Then  $1 + \text{ad}(a)$  maps  $\mathfrak{s}$  to  $\mathfrak{s}'$ , and  $\text{ad}(a)^2 = 0$  because  $\mathfrak{r}$  is commutative. And  $\mathfrak{r} = [\mathfrak{r}, \mathfrak{g}]$ , so  $a$  is in the nilpotent radical of  $\mathfrak{g}$ , thus  $\mathfrak{s}, \mathfrak{s}'$  are conjugate by a special automorphism.

For the general case, we can use induction on the dimension of  $\mathfrak{g}$ . After the first two cases, we can assume that  $[\mathfrak{g}, \mathfrak{r}] \neq 0$  and  $\mathfrak{r}$  contains a proper non-trivial ideal. As  $[\mathfrak{g}, \mathfrak{r}]$  is nilpotent(3.7.1.36), its center is non-zero. So we can choose a maximal ideal  $\mathfrak{m}$  contained in the center of  $[\mathfrak{g}, \mathfrak{r}]$ , and  $\mathfrak{m} \neq \mathfrak{r}$ . Now  $\mathfrak{g}/\mathfrak{m}$  has radical  $\mathfrak{r}/\mathfrak{m}$  because  $\mathfrak{m}$  is solvable, so we can apply induction to find a Levi subgroup  $\mathfrak{h}'$  for  $\mathfrak{g}/\mathfrak{m}$  in  $\mathfrak{g}/\mathfrak{m}$ , and let  $\mathfrak{h}''$  be its preimage in  $\mathfrak{g}$ . Then  $\mathfrak{h}''$  has radical  $\mathfrak{m}$ , and thus by induction there is a Levi subgroup  $\mathfrak{h}$  for  $\mathfrak{m}$  in  $\mathfrak{h}''$ , and this  $\mathfrak{h}$  is a Levi subgroup for  $\mathfrak{r}$  in  $\mathfrak{g}$ . Similarly any two such Levi subgroups are conjugate by a special automorphism.

This has another cohomological proof in [Etingof, Section48].?

□

## 2 Semisimple Lie-Algebras

**Prop. (3.7.2.1)[Cartan-Killing Criteria for Semisimplicity].** A f.d. Lie algebra  $\mathfrak{g}$  is semisimple iff its Killing form(3.7.1.13) is non-degenerate. ┘

*Proof:* If  $\mathfrak{g}$  is semisimple, then the adjoint representation is faithful, thus by(3.7.9.6) the Killing form is non-degenerate. Conversely, if the Killing form is non-degenerate and  $\mathfrak{a}$  is a commutative ideal of  $\mathfrak{g}$  and  $a \in \mathfrak{a}, g \in \mathfrak{g}$ , then  $(\text{ada} \circ \text{ad}g)^2 = 0$ , so  $\text{ada} \circ \text{ad}g$  is nilpotent and has trace 0. so  $a \in \mathfrak{g}^\perp$ , which is 0 because the Killing form is non-degenerate. □

**Cor. (3.7.2.2).** Let  $\mathfrak{a}$  be a semisimple ideal of a Lie algebra  $\mathfrak{g}$ , then the orthogonal space  $\mathfrak{a}'$  w.r.t the Killing form is an ideal and is a complement for  $\mathfrak{a}$  in  $\mathfrak{g}$ , and  $\mathfrak{g} \cong \mathfrak{a} \times \mathfrak{a}'$ . ┘

*Proof:*  $\mathfrak{a}^\perp$  is an ideal because the Killing form is invariant. The Killing form of  $\mathfrak{a}$  is the restriction of the Killing form of  $\mathfrak{g}$ (3.7.1.13), so  $\mathfrak{a} \cap \mathfrak{a}' = 0$  because  $\kappa_{\mathfrak{a}}$  is non-degenerate. □

**Cor. (3.7.2.3).** A Lie algebra is semisimple iff it is isomorphic to a product of simple algebras  $\mathfrak{g} = \mathfrak{a}_1 \times \cdots \times \mathfrak{a}_r$ , and these  $\mathfrak{a}_i$  are all its minimal ideals, (Not only up to isomorphism). ┘

*Proof:* The Killing form of  $\mathfrak{a} \cap \mathfrak{a}^\perp$  is 0, thus it is solvable, by(3.7.1.28), and then 0, so we can continue the decomposition.

For any minimal nonzero ideal  $\mathfrak{a} \subset \mathfrak{g}$ , then  $[\mathfrak{g}, \mathfrak{a}]$  is an ideal contained in  $\mathfrak{a}$ . which is nonzero because  $\mathfrak{g}$  has trivial center. Then

$$\mathfrak{a} = [\mathfrak{g}, \mathfrak{a}] = \oplus_i [\mathfrak{a}, \mathfrak{a}_i]$$

so  $\mathfrak{a} \subset [\mathfrak{a}, \mathfrak{a}_i] \subset \mathfrak{a}_i$ , and then  $\mathfrak{a} = \mathfrak{a}_i$  by simplicity. □

**Cor. (3.7.2.4).** If  $\mathfrak{g}$  is semisimple, then  $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ . ┘

**Cor. (3.7.2.5).** Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$  and  $k'/k$  is a field extension, then  $\mathfrak{g}$  is semisimple iff  $\mathfrak{g} \otimes_k k'$  is semisimple. ┘

**Prop. (3.7.2.6)[Examples of Semisimple Lie Algebras].**



- The subalgebra  $\mathfrak{sl}(V)$  of all elements of  $\text{End}(V)$  of trace 0 is semisimple. ┘

**Prop. (3.7.2.7).** If  $L$  is semisimple, then every derivative of  $L$  is inner. ┘

*Proof:* This is a special case of (24.4.1.3) applied to the adjoint representation. □

**Prop. (3.7.2.8).** If  $\mathfrak{g} \in \text{LieAlg}_k$  is a semisimple sub-Lie-algebra of  $\mathfrak{gl}_n(k)$ , then it contains the semisimple and nilpotent parts of each of its elements under the Jordan decomposition (3.5.10.7). ┘

*Proof:* We may assume  $k$  is alg.closed, because the Jordan decomposition is invariant of the field that contains  $\mathfrak{g}$ , and an element is contained in a vector space can be checked after base change to a larger field. For any subspace  $W \subset V$ , let  $\mathfrak{g}_W = \{\alpha \in \mathfrak{gl}_V \mid \alpha(W) \subset W, \text{tr}(\alpha|_W) = 0\}$ , then if  $\mathfrak{g}W \subset W$ , then  $\mathfrak{g} \subset \mathfrak{g}_W$ , because every element of  $\mathfrak{g}$  is a sum of brackets by (3.7.2.4), thus have zero trace. Now consider

$$\mathfrak{g}' = \mathfrak{n}_{\mathfrak{gl}_V}(\mathfrak{g}) \bigcap_{\mathfrak{g}W \subset W} \mathfrak{g}_W.$$

If  $x \in \mathfrak{g}'$ , then so does  $x_s$  and  $x_n$ , because they are polynomials in  $x$  without constant terms, and  $\text{ad}(x)_s = \text{ad}(x_s)$ ,  $\text{ad}(x)_n = \text{ad}(x_n)$ .

So it suffices to show that  $\mathfrak{g} = \mathfrak{g}'$ . We claim that  $\mathfrak{g}' = \mathfrak{g}$ : As  $\mathfrak{g}$  is a semisimple ideal of  $\mathfrak{g}'$ , by (3.7.2.2), we have a decomposition

$$\mathfrak{g}' = \mathfrak{g} \oplus \mathfrak{g}^\perp.$$

Let  $\alpha \in \mathfrak{g}^\perp$  and  $W$  a simple  $\mathfrak{g}$ -module of  $V$ , then  $\alpha$  acts on  $W$  as a scalar, which must be 0 because  $\alpha \in \mathfrak{gl}_W$  and  $k$  has char 0. As  $W$  is a sum of simple  $\mathfrak{g}$ -modules by Weyl's theorem (24.4.1.2), we get the desired conclusion. □

**Prop. (3.7.2.9) [Abstract Jordan-Decomposition].** A **semisimple/nilpotent element**  $x$  in a Lie algebra  $\mathfrak{g}$  is an element that  $\rho(x)$  is semisimple/nilpotent for any representation  $(V, \rho)$  of  $\mathfrak{g}$ . And  $x = x_s + x_n$  is called a **Jordan decomposition** iff  $\rho(x) = \rho(x_s) + \rho(x_n)$  is a Jordan-decomposition (3.5.10.7) for any representation  $\rho$  of  $\mathfrak{g}$ .

Every element of a semisimple Lie algebra  $\mathfrak{g}$  over a field of characteristic 0 has a unique Jordan decomposition, and  $x = x_s + x_n$  is a Jordan decomposition if  $\rho(x) = \rho(x_s) + \rho(x_n)$  is a Jordan decomposition for one faithful representation  $\rho$  of  $\mathfrak{g}$ . In particular, this holds for the adjoint representation  $\text{ad}$ . ┘

*Proof:* Let  $x \in \mathfrak{g}$  and  $(V, \rho)$  a faithful representation of  $\mathfrak{g}$  (for example the adjoint representation), then there is at most one  $x = x_s + x_n$  that  $\rho(x) = \rho(x_s) + \rho(x_n)$  is the Jordan decomposition, which proves the uniqueness.

Now for any  $x \in \mathfrak{g}$ , as (3.7.2.8) shows, there do exist these two elements that  $\rho(x) = \rho(x_s) + \rho(x_n)$ . But then it can be checked directly  $\text{ad}(x) = \text{ad}(x_s) + \text{ad}(x_n)$  is the Jordan decomposition of  $\text{ad}(x)$  as an endomorphism of  $\mathfrak{g}$ , by (3.7.1.16). As the adjoint representation is faithful, this shows the Jordan decomposition is independent of the faithful representation chosen.

Now every representation is a subrepresentation of a faithful representation, so we can prove the existence. □

### Simple Lie Algebras

References are [Car05]Chap8 and [Kna96], more data about simple Lie algebras can be found in [Kna96]P508.

**Def. (3.7.2.10) [Lie Algebras of Type  $A_n$ :  $\mathfrak{sl}(n+1, \mathbb{C})$ ].** ┘

**Def. (3.7.2.11) [Lie Algebras of Type  $A_1$ ].**  $A_1$  is also called  $\mathfrak{sl}_2(\mathbb{C})$ . It has a basis  $f, h, e$  with

$$[he] = 2e, \quad [hf] = -2f, \quad [ef] = h.$$

In matrix form,

$$h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad e = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad f = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

It can also be realized by

$$H = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad R = \frac{1}{2} \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}, \quad L = \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & -1 \end{bmatrix}.$$

These two representation differ by a conjugation by the Cayley transformation  $C = -\frac{1+i}{2} \begin{bmatrix} i & 1 \\ i & -1 \end{bmatrix}$ .

It is simple by ?. The subalgebra  $\mathfrak{b}$  generated by  $X$  and  $H$  is solvable, called the canonical Borel subalgebra of  $\mathfrak{sl}_2$ . ┘

*Proof:* □

**Prop. (3.7.2.12) [Lie Algebras of Type  $B_n$ :  $\mathfrak{so}(2n+1, \mathbb{C})$ ].** ┘

**Prop. (3.7.2.13) [Lie Algebras of Type  $C_n$ :  $\mathfrak{sp}(n, \mathbb{C})$ ].** ┘

**Prop. (3.7.2.14) [Lie Algebras of Type  $D_n$ :  $\mathfrak{so}(2n, \mathbb{C})$ ].** ┘

**Prop. (3.7.2.15) [Lie Algebras of Type  $G_2$ ].** ┘

**Prop. (3.7.2.16) [Lie Algebras of Type  $F_4$ ].** ┘

**Prop. (3.7.2.17) [Lie Algebras of Type  $E_8$ ].** ┘

**Prop. (3.7.2.18) [Classification of F.D. Complex Simple Lie Algebras].** ┘

**Prop. (3.7.2.19) [Group Automorphisms of Simple Lie Algebras].** Cf.[Carter, P184]. ┘

**Prop. (3.7.2.20).**  $\mathfrak{sl}_2(k)$  is simple if  $k$  has characteristic  $\neq 2$ . ┘

**Def. (3.7.2.21) [Exponent].** Let  $\mathfrak{g}$  be simple,

$$e = \sum e_i, \quad h = 2\rho^\vee = \sum_i (2\rho^\vee, \omega_i) h_i, \quad f = \sum_i (2\rho^\vee, \omega_i) f_i,$$

then by definition,  $[h, e] = 2e$ ,  $[e, f] = h$ , and

$$[h, f] = \sum_i \sum_j (2\rho^\vee, \omega_i) \alpha_i^\vee(\alpha_j) (2\rho^\vee, \omega_j) f_j$$

$$\begin{aligned}
&= \sum_j [(\sum_i (2\rho^\vee, \omega_i) \alpha_i^\vee, \alpha_j)] (2\rho^\vee, \omega_j) f_j \\
&= \sum_j (2\rho^\vee, \alpha_j) (2\rho^\vee, \omega_j) f_j \\
&= -2f
\end{aligned}$$

So  $\{h, e, f\}$  is a  $\mathfrak{sl}_2$ -tuple, called the **principal  $\mathfrak{sl}_2$ -subalgebra** of  $\mathfrak{g}$ . The highest weights of subrepresentation of the adjoint action of this subalgebra on  $\mathfrak{g}$  is called the **exponents of  $\mathfrak{g}$** , counted with multiplicity.

Equivalently, as  $h|_{\mathfrak{g}^\alpha} = \text{ht}(\alpha)\text{id}$ , if  $r_m$  is the number of roots of  $\mathfrak{g}$  of height  $m$ , the exponents of  $\mathfrak{g}$  is the numbers  $m$  that  $r_m > r_{m+1}$ .  $\lrcorner$

**Cor. (3.7.2.22).** While  $r_m = 0$  for  $m$  large and  $r_1 = r(\mathfrak{g})$ , there are  $r$  exponents of  $\mathfrak{g}$ . Since the roots of height 2 are all  $\alpha_i + \alpha_j$  where  $i, j$  are connected by an edge in the Dynkin diagram, thus  $r_2 = r - 1$ , thus  $1 = m_1 \leq \dots \leq m_r$ ,  $\sum_{i=1}^r = |R^+|$ . And  $\mathfrak{g} \cong \oplus_{r=1}^r L_{2m_i+1}$  as a representation of the principal  $\mathfrak{sl}_2$ -subalgebra.  $\lrcorner$

**Prop. (3.7.2.23)[Exponents of Simple Lie Algebras].** The data of exponents of simple Lie algebras:

- $A_n : 1, 2, \dots, n$ .
- $B_n : 1, 3, \dots, 2n - 1$ .
- $C_n : 1, 3, \dots, 2n - 1$ .
- $D_n : 1, 3, \dots, 2n - 3$  and  $n$ .
- $G_2 : 1, 5$ .
- $F_4 : 1, 5, 7, 11$ .
- $E_8 : 1, 7, 11, 13, 17, 19, 23, 29$ .
- $E_7 : 1, 5, 7, 9, 11, 13, 17$ .
- $E_6 : 1, 4, 5, 7, 8, 11$ .

$\lrcorner$

*Proof:*

$\square$

### 3 Cartan Subalgebras

Lie algebras in this subsection are assumed to be of f.d..

**Def. (3.7.3.1)[Cartan Subalgebras].** A **Cartan subalgebra** of a Lie algebra  $\mathfrak{g}$  is a nilpotent subalgebra  $\mathfrak{h}$  that equals to its own normalizer in  $\mathfrak{g}$ .  $\lrcorner$

**Remark (3.7.3.2).** As a proper subalgebra of a nilpotent algebra is never its own normalizer (3.7.1.20), a Cartan subalgebra is a maximal nilpotent subalgebra, but a maximal nilpotent subalgebra may not be a Cartan subalgebra.

If  $k'/k$  is a field extension, then  $\mathfrak{h}$  is a Cartan subalgebra of  $\mathfrak{g}$  iff  $\mathfrak{h}_{k'}$  is a Cartan subalgebra of  $\mathfrak{g}_{k'}$ . This is because being nilpotent and the normalizer is also compatible with base change (3.7.1.4).  $\lrcorner$

**Prop. (3.7.3.3)[Diagonal Cartan Algebra].** Let  $\mathfrak{g} \subset \mathfrak{gl}_V$  be a subalgebra containing a diagonal matrix  $a = \text{diag}(a_1, \dots, a_n)$  with distinct  $a_i$ , and let  $\mathfrak{h}$  be the subspace of all diagonal matrices in  $\mathfrak{g}$ , then  $\mathfrak{h}$  is a Cartan subalgebra of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Firstly  $\mathfrak{h}$  is Abelian, and if  $b = \sum b_{ij}e_{ij} \in N_{\mathfrak{g}}(\mathfrak{h})$ , then  $[a, b] \in \mathfrak{h}$ . But

$$[a, b] = \sum_{ij} (a_{ii} - a_{jj})b_{ij}e_{ij}$$

is in  $\mathfrak{h}$  iff  $\mathfrak{b}$  is diagonal, or  $b \in \mathfrak{h}$ , so  $\mathfrak{h} = N_{\mathfrak{g}}(\mathfrak{h})$  and  $\mathfrak{h}$  is a Cartan subalgebra.  $\square$

**Def. (3.7.3.4) [Regular Elements].** Let  $\mathfrak{g}$  be a f.d. Lie algebra, for any  $x \in \mathfrak{g}$ , let  $P_x(T)$  be the characteristic polynomial of  $\text{ad}(x)$ :

$$P_x(T) = \det(T - \text{ad}(x)) = T^m + a_{m-1}(x)T^{m-1} + \dots + a_0(x).$$

Then the **rank of  $\mathfrak{g}$**  is the minimal  $n$  that  $n(x) \neq 0$  for some  $x \in \mathfrak{g}$ . A **regular element** is an element  $x \in \mathfrak{g}$  that  $a_n(x) \neq 0$ .  $\lrcorner$

**Prop. (3.7.3.5) [Regular Elements and Cartan Subalgebras].** For any regular element  $x \in \mathfrak{g}$ , the nilspace  $\mathfrak{g}_x^0$  is a Cartan subalgebra of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Let

$$U_1 = \{y \in \mathfrak{g}_x^0 \mid \text{ad}_{\mathfrak{g}}(y)|_{\mathfrak{g}_x^0} \text{ is not nilpotent}\},$$

$$U_2 = \{y \in \mathfrak{g}_x^0 \mid \text{ad}_{\mathfrak{g}}(y)|_{(\mathfrak{g}/\mathfrak{g}_x^0)} \text{ is invertible}\}.$$

They are both Zariski open subsets of  $\mathfrak{g}_x^0$ . According to Engel's theorem (3.7.1.29), to show  $\mathfrak{g}$  is nilpotent, it suffices to show that  $U_1$  is empty.  $U_2$  is non-empty because it contains  $x$ , so if  $U_1$  is non-empty,  $U_1 \cap U_2$  is non-empty and there is a  $y \in U_1 \cap U_2$ . For this  $y$ ,  $n(y) < \dim \mathfrak{g}_x^0 = n(x)$ , contradicting the regularity of  $x$ .

It remains to show that  $\mathfrak{g}_x^0$  is its own normalizer. If  $z$  normalizes  $\mathfrak{g}_x^0$ , then  $[z, x] \in \mathfrak{g}_x^0$ , which means  $(\text{ad}(x))^n[z, x] = 0$ , so  $\text{ad}(x)^{n+1}(z) = 0$ , thus  $z \in \mathfrak{g}_x^0$ .  $\square$

**Cor. (3.7.3.6) [Cartan Subalgebras Exist].** Let  $\mathfrak{g}$  be a Lie algebra over an infinity field  $k$  contains some Cartan subalgebra, and when  $k$  is alg.closed, all Cartan subalgebras come from some regular element, by (3.7.3.13).  $\lrcorner$

*Proof:* Regular elements exist because  $k$  is infinite.  $\square$

**Cor. (3.7.3.7).** Every Lie algebra over an infinite field is a sum of Cartan subalgebras.  $\lrcorner$

*Proof:* This is because the sum of Cartan subalgebras is a vector space thus Zariski closed but it contains all regular elements, which is a Zariski open subset.  $\square$

**Cor. (3.7.3.8).** Let  $\mathfrak{a}$  be a subalgebra of a Lie algebra  $\mathfrak{g}$  that  $\text{ad}_{\mathfrak{g}}(a)$  is semisimple for any  $a \in \mathfrak{a}$ , then  $\mathfrak{a}$  is contained in a Cartan subalgebra of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Cf. [Mil13]P81. ?  $\square$

**Prop. (3.7.3.9).** Let  $\mathfrak{h}$  be a Cartan subalgebra of  $\mathfrak{g}$  over an alg.closed field  $k$ . Consider the generalized eigenvalue decomposition (3.7.3.12), if  $x \in \mathfrak{g}^\alpha$ , then  $\text{ad}(x)(\mathfrak{g}^\beta) \in \mathfrak{g}^{\alpha+\beta}$  (3.7.1.12), and thus  $\text{ad}(x)$  is nilpotent. Let  $E(\mathfrak{h})$  be the subgroup of the group of elementary automorphisms (3.7.1.7) of  $\mathfrak{g}$  generated by the set of all the automorphisms  $e^{\text{ad}_{\mathfrak{g}}(x)}$ , where  $x \in \mathfrak{g}^\alpha$  for some  $\alpha \in \mathfrak{h}^* \setminus 0$ .

Now let  $\mathfrak{h}, \mathfrak{h}'$  be two Cartan subalgebras of  $\mathfrak{g}$ , then there exists  $u \in E(\mathfrak{h}), u' \in E(\mathfrak{h}')$  that  $u(\mathfrak{h}) = u'(\mathfrak{h}')$ .  $\lrcorner$

*Proof:* Number the elements of  $\mathfrak{h}^* \setminus 0$  as  $\alpha_1, \dots, \alpha_n$ , and consider the map

$$f : \mathfrak{g}^{\alpha_1} \times \dots \times \mathfrak{g}^{\alpha_n} \times \mathfrak{h} \rightarrow \mathfrak{g} : (x_1, \dots, x_n, h) \mapsto e^{\text{ad}(x_1)} \dots e^{\text{ad}(x_n)} h.$$

Given a  $h_0 \in \mathfrak{h}$ , it can be shown that

$$(df)|_{(0, \dots, 0, h_0)} : (x_1, \dots, x_n, h) \mapsto h + \sum_i [x_i, h_0].$$

Thus if we choose a regular  $h_0 \in \mathfrak{h}$ , then  $(df)|_{(0, \dots, 0, h_0)}$  is surjective from  $\mathfrak{g}$  to  $\mathfrak{g}$ . Thus  $E(\mathfrak{h})\mathfrak{h}_r$  contains a dense open subset of  $\mathfrak{g}$ . Similarly,  $E(\mathfrak{h}')\mathfrak{h}'_r$  contains a dense open subset of  $\mathfrak{g}$ . So their intersection is not empty, i.e.  $u(h) = u'(h')$  for some  $u, u', h, h'$ . Now

$$u(\mathfrak{h}) = u(\mathfrak{g}_h^0) = \mathfrak{g}_{u(h)}^0 = \mathfrak{g}_{u'(h')}^0 = u'(\mathfrak{g}_{h'}^0) = u'(\mathfrak{h}')$$

□

**Cor. (3.7.3.10).** All Cartan subalgebras in a Lie algebra have the same dimension, which is the rank of  $\mathfrak{g}$ .

*Proof:* Because we can take a base change to an alg.closed field, under which a Cartan subalgebra is also a Cartan subalgebra by (3.7.3.2), then they have the same rank. □

**Cor. (3.7.3.11) [Cartan Subalgebras are Conjugate].** Any two Cartan subalgebras of a f.d. Lie algebra over an alg.closed field  $k$  are conjugate by an elementary automorphism.

### Cartan Subalgebra of Semisimple Lie Algebras

**Lemma (3.7.3.12) [Decomposition w.r.t. a Cartan Subalgebra].** Let  $\mathfrak{h}$  be a Cartan subalgebra of a semisimple Lie algebra  $\mathfrak{g}$ , and assume that

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \mathfrak{h}^\vee \setminus 0} \mathfrak{g}^\alpha.$$

where  $R = R(\mathfrak{g}, \mathfrak{h})$  is called the **root system associated to  $(\mathfrak{g}, \mathfrak{h})$** . This is true, for example, when  $k = \bar{k}$ , by (3.7.9.10). (See (3.7.3.21) for when this decomposition is possible). □

**Cor. (3.7.3.13).** If  $k = \bar{k}$ , the set  $\mathfrak{h}_r$  of regular elements  $h$  in  $\mathfrak{h}$  that  $\mathfrak{g}_h^0 = \mathfrak{h}$  is open and dense in  $\mathfrak{h}$  in the Zariski topology. □

*Proof:* The condition is equivalent to  $\prod_{\alpha \in \mathfrak{h}^\vee \setminus 0} \alpha(h) \neq 0$ , which is an open condition. □

**Lemma (3.7.3.14).** In the decomposition above, if  $\alpha + \beta \neq 0$ , then  $\mathfrak{g}^\alpha$  and  $\mathfrak{g}^\beta$  is orthogonal w.r.t. the Killing form. □

*Proof:*  $\text{ad}(x)\text{ad}(y)\mathfrak{g}^\gamma \subset \mathfrak{g}^{\alpha+\beta+\gamma}$ , so if  $\alpha + \beta \neq 0$ , then  $\text{ad}(x)\text{ad}(y)$  is nilpotent, thus  $\kappa(x, y) = 0$ . □

**Prop. (3.7.3.15) [Cartan Subalgebras of Semisimple Lie Algebras].** Let  $\mathfrak{h}$  be a Cartan subalgebra of a semisimple Lie algebra  $\mathfrak{g}$ , then

- Every element of  $\mathfrak{h}$  is semisimple. In particular,  $\mathfrak{h}$  is commutative.
- The centralizer of  $\mathfrak{h}$  in  $\mathfrak{g}$  is  $\mathfrak{h}$ .
- The restriction of the Killing form to  $\mathfrak{h}$  is non-degenerate.

┘

*Proof:* By (3.7.3.2), it suffices to prove this after  $k$  is replaced by its alg.closure, so the generalized eigenvalue decomposition (3.7.3.12) holds. We prove 3 first: by (3.7.3.14),  $\mathfrak{h}$  is orthogonal to all  $[\mathfrak{h}, x]$  for any  $x$ . But if  $x \in \mathfrak{g}^\alpha$ , we can see that  $[\mathfrak{h}, x] = \mathfrak{g}^\alpha$ , so  $\mathfrak{h}$  is orthogonal to all  $\bigoplus \mathfrak{g}^\alpha$ , so  $\kappa$  must be non-degenerate on  $\mathfrak{h}$ .

Because  $\mathfrak{g}$  has trivial center, the adjoint representation realizes  $\mathfrak{h}$  as a subalgebra of  $\mathfrak{gl}_{\mathfrak{g}}$ , and Lie's theorem (3.7.1.24) shows there is a basis that  $\mathfrak{h} \subset \mathfrak{b}_{\mathfrak{g}}$ , hence  $\text{ad}([\mathfrak{h}, \mathfrak{h}]) \subset \mathfrak{n}_{\mathfrak{g}}$ , and so  $\text{tr}(\mathfrak{h}, [\mathfrak{h}, \mathfrak{h}]) = 0$ . As  $\kappa$  is non-degenerate on  $\mathfrak{h}$ ,  $[\mathfrak{h}, \mathfrak{h}] = 0$ , thus  $\mathfrak{h}$  is commutative. Now  $\mathfrak{h} \subset c_{\mathfrak{g}}(\mathfrak{h}) \subset N_{\mathfrak{g}}(\mathfrak{h})$ , thus  $\mathfrak{h} = c_{\mathfrak{g}}(\mathfrak{h})$ .

If  $x \in \mathfrak{h}$ , and  $x = x_s + x_n$  is the Jordan decomposition (3.7.2.9), then  $\text{ad}(x_n)$  are polynomials of  $\text{ad}(x)$  thus lies in  $\mathfrak{h}$ . Now  $\text{ad}(x_n)$  commutes with all  $\text{ad}(y)$  for  $y \in \mathfrak{h}$ , thus  $\text{ad}(y)\text{ad}(x_n)$  is nilpotent, thus  $\kappa(y, x_n) = 0$ . Thus  $x_n = 0$  as  $\kappa$  is non-degenerate on  $\mathfrak{h}$ .  $\square$

**Cor. (3.7.3.16) [Cartan Subalgebra Maximal Abelian].** The Cartan subalgebras of a semisimple Lie algebra are the maximal subalgebras consisting of semisimple elements (3.7.2.9), and they are maximal Abelian subalgebras.

WARNING: maximal Abelian subalgebra may not be Cartan subalgebras, as they may contain non-semisimple elements.  $\perp$

*Proof:* A subalgebra consisting of semisimple elements is contained in a Cartan subalgebra, by (3.7.3.8).

Conversely, if  $\mathfrak{h} \subset \mathfrak{h}'$  and  $\mathfrak{h}$  is a Cartan subalgebra and  $\mathfrak{h}'$  consists of semisimple elements, then by (3.7.1.15),  $\mathfrak{h}'$  is commutative, and thus  $\mathfrak{h}' \subset c_{\mathfrak{g}}(\mathfrak{h})$ , so  $\mathfrak{h} = \mathfrak{h}'$ .

Cartan subalgebras are Abelian by (3.7.3.15), and they are maximal Abelian because they are self-centralizing.  $\square$

**Cor. (3.7.3.17).** Every regular element is semisimple, because it is contained in a Cartan subalgebra by (3.7.3.5).  $\perp$

### Split Semisimple Lie Algebras

**Def. (3.7.3.18) [Split Semisimple Lie Algebras].** A **split Cartan subalgebra** of a semisimple Lie algebra  $\mathfrak{g}$  over a field  $k$  is a Cartan subalgebra that all the eigenvalues of the linear maps  $\text{ad}(h)$  lies in  $k$  for all  $h \in \mathfrak{h}$ . A **split semisimple Lie algebra** is a pair  $(\mathfrak{g}, \mathfrak{h})$  where  $\mathfrak{g}$  is semisimple and  $\mathfrak{h}$  is a split Cartan subalgebra.  $\perp$

**Remark (3.7.3.19).** For example, the diagonal matrices in  $\mathfrak{sl}(n)$  is a splitting Cartan subalgebra over any field.

$\mathfrak{sl}(2; \mathbb{R})$  has a non-split Cartan subalgebra  $\left\{ \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix} \mid a \in \mathbb{R} \right\}$ .  $\perp$

**Prop. (3.7.3.20).** Let  $\alpha$  be a root of the split semisimple Lie algebra  $(\mathfrak{g}, \mathfrak{h})$ , then

- The subspaces  $\mathfrak{g}^\alpha$  and  $\mathfrak{h}_\alpha = [\mathfrak{g}^\alpha, \mathfrak{g}^{-\alpha}]$  are both 1-dimensional.
- There is a unique element  $h_\alpha \in \mathfrak{h}_\alpha$  such that  $\alpha(h_\alpha) = 2$ , and  $(h_\alpha, h_\alpha) \neq 0$ .
- For each nonzero  $x_\alpha \in \mathfrak{g}^\alpha$ , there is a  $y_\alpha \in \mathfrak{g}^{-\alpha}$  such that

$$[x_\alpha, y_\alpha] = h_\alpha, \quad [h_\alpha, x_\alpha] = 2x_\alpha, \quad [h_\alpha, y_\alpha] = -2y_\alpha.$$

i.e.  $\mathfrak{s}_\alpha = \{x_\alpha, y_\alpha, h_\alpha\} \cong \mathfrak{sl}_2$ .

- $k\alpha$  is a root iff  $k = 0, \pm 1$ .

┘

*Proof:* Define  $\mathfrak{h}_\alpha = [\mathfrak{g}^\alpha, \mathfrak{g}^{-\alpha}] \subset \mathfrak{h}$ . Because the Killing form is non-degenerate on  $\mathfrak{h}$ , we can define for each  $\alpha \in R$  a unique element  $h^\alpha \in \mathfrak{h}$  that  $\alpha(h) = \kappa(h, h^\alpha)$  for all  $h \in \mathfrak{h}$ . Then  $\mathfrak{h}_\alpha$  is the subspace spanned by  $h^\alpha$ : This is because for  $x \in \mathfrak{g}^\alpha, y \in \mathfrak{g}^{-\alpha}$ ,

$$\kappa(h, [x, y]) = \kappa([h, x], y) = \alpha(x)\kappa(x, y)$$

so  $[x, y] = \kappa(x, y)h^\alpha$ . Combine this with the fact  $\kappa(\mathfrak{g}^\alpha, \mathfrak{g}^{-\alpha}) \neq 0$ , we get the fact  $\mathfrak{h}_\alpha = kh^\alpha$  is 1-dimensional.

Next, there is a unique element  $h_\alpha \in \mathfrak{h}_\alpha$  that  $\alpha(h_\alpha) = 2$ . For this, it suffices to show that  $\alpha$  doesn't vanish on  $\mathfrak{h}_\alpha$ : Otherwise let  $x \in \mathfrak{g}^\alpha$  and  $y \in \mathfrak{g}^{-\alpha}$  that  $[x, y] = h \neq 0$ , then  $[h, x] = \alpha(h)x = 0 = [h, y]$ . So  $\{x, y, h\}$  spans a solvable subalgebra  $\mathfrak{a}$  of  $\mathfrak{g}$ . As  $h \in [\mathfrak{a}, \mathfrak{a}]$ . By Lie's theorem,  $\rho(h)$  is nilpotent for any representation  $\rho$  of  $\mathfrak{a}$ . But  $h$  is in the Cartan subalgebra so  $\text{ad}_{\mathfrak{g}}(h)$  is semisimple(3.7.3.15), so  $h = 0$ , contradiction.

If  $(h_\alpha, h_\alpha) = 0$ , let  $h_\alpha = [x_\alpha, y_\alpha]$  for  $x_\alpha \in \mathfrak{g}^\alpha, y_\alpha \in \mathfrak{g}^{-\alpha}$ , then  $\{x_\alpha, y_\alpha, h_\alpha\}$  is solvable, so by Lie's theorem, there is a basis of  $\mathfrak{g}$  that the adjoint action is upper-triangular(when pass to the alg.closure). But then  $\text{ad}(h_\alpha)$  is nilpotent, but it is also semisimple, so  $h_\alpha = 0$ , contradiction.

Because  $x_\alpha \neq 0$ , there exists a unique  $y_\alpha \in \mathfrak{g}^{-\alpha}$  that  $[x_\alpha, y_\alpha] = h_\alpha$ . Now  $[h_\alpha, x_\alpha] = \alpha(h_\alpha)x_\alpha = 2x_\alpha, [h_\alpha, y_\alpha] = -\alpha(h_\alpha)y_\alpha = -2y_\alpha$ .

Finally,  $\mathfrak{h}_\alpha \oplus \bigoplus_{k \neq 0 \in \mathbb{Z}} \mathfrak{g}^{k\alpha}$  is a subrepresentation of  $x_\alpha, y_\alpha, h_\alpha \cong \mathfrak{sl}_2$ , so if  $\mathfrak{g}^{2\alpha} \neq 0$ , then  $\text{ad}(x_\alpha)$  induces an isomorphism  $\mathfrak{g}^\alpha \cong \mathfrak{g}^{2\alpha}$  by the representation theory of  $\mathfrak{sl}_2$ (24.4.1.11). But  $\mathfrak{g}^\alpha$  is generated by  $x_\alpha$ , contradiction. So  $\mathfrak{g}^{k\alpha} \neq 0$  only for  $k = 0, \pm 1$ .  $\square$

**Prop. (3.7.3.21) [Root Decompositions].** If  $(\mathfrak{g}, \mathfrak{h})$  is a split semisimple Lie algebra, then  $\text{ad}(\mathfrak{h})$  is a commuting family of semisimple endomorphisms with eigenvalues in  $k$ (3.7.3.15). Thus the decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}^\alpha$$

holds as in(3.7.3.12). And it is in fact an eigenvalue decomposition, not only generalized eigenvalue decomposition, by(3.7.3.20).

Then  $\Phi$  is a reduced root system(3.9.5.2) in  $\mathfrak{h}^\vee$ , with  $s_\alpha(\beta) = \beta - \beta(h_\alpha)\alpha$ . In particular,  $\alpha^\vee = h_\alpha$ .

┘

*Proof:* Firstly  $R$  spans  $\mathfrak{h}^\vee$ : if  $h \in \mathfrak{h}$  lies in the center of all  $\alpha \in \Phi$ , then  $[h, \mathfrak{g}^\alpha] = 0$  for all  $\alpha \in R$ , and as  $[h, \mathfrak{h}] = 0$ , this means  $h$  is in the center of  $\mathfrak{g}$ , which is trivial, so  $h = 0$ . So  $R$  spans  $\mathfrak{h}^\vee$ .

We need to prove that if  $\alpha, \beta \in \Phi$ , then  $\beta(h_\alpha) \in \mathbb{Z}$  and  $\beta - \beta(h_\alpha)\alpha \in \Phi$ . For this, regard  $\mathfrak{g}$  as a  $\mathfrak{s}_\alpha$ -module(3.7.3.20) under the adjoint action, then the assertion follows from the representation theory of  $\mathfrak{sl}(2)$ (24.4.1.11):  $h_\alpha$  acts on  $\mathfrak{g}^\beta$  by  $\beta(h_\alpha)$ , and  $y_\alpha^n$  induces an isomorphism  $\mathfrak{g}^\beta \cong \mathfrak{g}^{\beta - n\alpha}$ .

Finally,  $R$  is reduced by(3.7.3.20).  $\square$

**Prop. (3.7.3.22) [Splitting Cartan Subalgebras are Conjugate].** The group of elementary automorphisms of  $\mathfrak{g}$ (3.7.1.7) acts transitively on the set of pairs  $(\mathfrak{b}, \mathfrak{h})$  consisting of a Borel subalgebra(3.7.4.7) and a splitting Cartan subalgebra of  $\mathfrak{g}$ .  $\square$

*Proof:* Cf.[Mil13]P98.  $\square$

**Prop. (3.7.3.23) [Jacobson-Morozov].** Let  $\mathfrak{g}$  be a semisimple Lie algebra and  $e \in \mathfrak{g}$  is nilpotent, then there exists a homomorphism  $\mathfrak{sl}_2 \rightarrow \mathfrak{g}$  mapping  $X$  to  $e$ .  $\square$

*Proof:* Cf.<https://people.math.harvard.edu/~ana/part1.pdf>.  $\square$

### Recovering Split Semisimple Lie Algebras from Dynkin Diagrams

References are [Car05]Chap7.

**Def. (3.7.3.24) [Cartan Matrix].** Let  $(\mathfrak{g}, \mathfrak{h})$  be a split semisimple Lie algebra with root system  $\Phi$ , with notations in (3.7.3.20), define the Cartan matrix of  $\mathfrak{g}$  to be the Cartan matrix  $A$  of  $\Phi$ ,  $A = (a_{ij})$ , where  $a_{ij} = \alpha_j(h_{\alpha_i})$ , where  $\{\alpha_1, \dots, \alpha_n\}$  is a base  $S$  of  $\Phi$ .  $\lrcorner$

**Prop. (3.7.3.25) [Serre Relations].** Let  $(\mathfrak{g}, \mathfrak{h})$  be a split semisimple Lie algebra with root system  $R$  and a base  $S$ , and Cartan matrix  $A = (a_{ij})$ . Denote  $\mathfrak{n}_{\pm} = \bigoplus_{\alpha \in R^{\pm}} \mathfrak{g}^{\alpha}$ , then  $\mathfrak{g} = \mathfrak{n}_{-} \oplus \mathfrak{h} \oplus \mathfrak{n}_{+}$ . Take  $e_i \in \mathfrak{g}^{\alpha_i}$ ,  $f_i \in \mathfrak{g}^{-\alpha_i}$  s.t.  $\{e_i, f_i, h_i = [e_i, f_i]\} = \mathfrak{s}_i$  is a  $\mathfrak{sl}_2$ -triple (3.7.3.20). Then

- $e_i, f_i, h_i$  generate  $\mathfrak{g}$ .
- (Serre Relations):

$$[h_i, h_j] = 0, \quad [h_i, e_j] = a_{ij}, \quad [h_i, f_j] = -a_{ij}f_j, \quad [e_i, f_j] = \delta_{ij}h_i,$$

$$(\text{ad}(e_i))^{1-a_{ij}}e_j = 0, \quad (\text{ad}(f_i))^{1-a_{ij}}f_j = 0, \quad i \neq j$$

$\lrcorner$

*Proof:* 1: Because  $h_i$  generate  $\mathfrak{h}$ , it suffices to show  $e_i$  generate  $\mathfrak{n}_{+}$ , then dually  $f_i$  generate  $\mathfrak{n}_{-}$ . We prove any  $\mathfrak{g}^{\alpha}$  is in the span by induction on the height of  $\alpha$ . If  $\alpha = \sum n_i \alpha_i$ , where  $n_i > 0$ , then  $0 < (\alpha, \alpha) = \sum n_i (\alpha, \alpha_i)$ , thus  $(\alpha, \alpha_i) > 0$  for some  $i$ , thus by (3.9.5.23)  $\alpha - \alpha_i$  is a root, and the representation theory of  $\mathfrak{s}_i$  shows  $e_i$  induces an isomorphism  $\mathfrak{g}^{\alpha - \alpha_i} \cong \mathfrak{g}^{\alpha}$ . So we are done.

2: Only the last two assertions need a proof. The irreducible  $\mathfrak{s}_i$  representation generated by  $e_j$  satisfies  $[f_i, e_j] = 0$ ,  $[h_i, e_j] = a_{ij}$ , thus this submodule is isomorphic to  $W_{a_{ij}}$ , and  $(\text{ad}(e_i))^{1-a_{ij}}e_j = 0$ . The  $f_i$  case is similar.  $\square$

**Prop. (3.7.3.26) [Criterion of Semisimplicity].** Let  $\mathfrak{g} \in \text{LieAlg}_k$  and  $\mathfrak{h}$  a commutative Lie subalgebra. If

- there is a decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}^{\alpha},$$

where  $\Phi \in \mathfrak{h}^{\vee}$  is the finite set of  $\alpha \in \mathfrak{h}^{\vee} \setminus \{0\}$  that  $\mathfrak{g}^{\alpha} \neq 0$ , and  $\dim \mathfrak{g}^{\alpha} = 1$  for all  $\alpha \in \Phi$ .

- $\Phi$  generates  $\mathfrak{h}^{\vee}$ .
- If  $\alpha \in \Phi$ , then  $-\alpha \in \Phi$ , and  $[[\mathfrak{g}^{\alpha}, \mathfrak{g}^{-\alpha}], \mathfrak{g}^{\alpha}] \neq 0$ .

Then  $\mathfrak{g}$  is semisimple and  $\mathfrak{h}$  is a split Cartan subalgebra of  $\mathfrak{g}$  with root system  $\Phi$ .  $\lrcorner$

*Proof:* If  $I$  is a commutative ideal of  $\mathfrak{g}$ , by action of  $\mathfrak{h}$ , we can assume that  $\mathfrak{g}^{\alpha} \subset I$  for some  $\alpha$ . If  $\alpha \neq 0$ , then  $\mathfrak{g}^{\alpha}$  and  $[\mathfrak{g}^{-\alpha}, \mathfrak{g}^{\alpha}] \subset I$ , so  $[[\mathfrak{g}^{\alpha}, \mathfrak{g}^{-\alpha}], \mathfrak{g}^{\alpha}] = 0$ , contradiction. If  $I \subset \mathfrak{h}$ , then by the hypothesis some  $\alpha(I) \neq 0$ , so  $\mathfrak{g}^{\alpha} = [I, \mathfrak{g}^{\alpha}] \subset \mathfrak{g}^{\alpha}$ , contradiction. So  $I = 0$ .

$\mathfrak{h}$  consists of semisimple elements and it is its own centralizer, so it is a Cartan subalgebra by (3.7.3.16). It is clearly split.  $\square$

**Cor. (3.7.3.27) [Criterion of Simplicity].** Let  $(\mathfrak{g}, \mathfrak{h})$  be a split semisimple Lie algebra. A decomposition  $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$  of Lie algebras defines a decomposition  $(\mathfrak{g}, \mathfrak{h}) = (\mathfrak{g}_1, \mathfrak{h}_1) \oplus (\mathfrak{g}_2, \mathfrak{h}_2)$ , and hence a decomposition of the root system  $\Phi(\mathfrak{g}, \mathfrak{h})$ .

In particular, if the root system  $\Phi(\mathfrak{g}, \mathfrak{h})$  is indecomposable, then  $\mathfrak{g}$  is simple.  $\lrcorner$



*Proof:* Let

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}^\alpha, \quad \mathfrak{g}_1 = \mathfrak{h}_1 \oplus \bigoplus_{\alpha \in \Phi_1} \mathfrak{g}_1^\alpha, \quad \mathfrak{g}_2 = \mathfrak{h}_2 \oplus \bigoplus_{\alpha \in \Phi_2} \mathfrak{g}_2^\alpha$$

be the eigenvalue decomposition of  $\mathfrak{g}, \mathfrak{g}_1, \mathfrak{g}_2$  w.r.t. the adjoint action of  $\mathfrak{h}$ , then  $\mathfrak{h} = \mathfrak{h}_1 \oplus \mathfrak{h}_2$ , and  $\Phi = \Phi_1 \amalg \Phi_2$ .  $\square$

**Prop. (3.7.3.28)[Serre Presentations].** Let  $\mathfrak{g}(R)$  be the Lie algebra generated by  $e_i, f_i, h_i$  with defining relations as in (3.7.3.25), then

- The Lie subalgebra  $\mathfrak{n}_+$  generated by  $e_i$  has  $(\text{ad}(e_i))^{1-a_{ij}}(e_j) = 0$  as defining relations. The Lie subalgebra  $\mathfrak{n}_-$  generated by  $f_i$  has  $(\text{ad}(f_i))^{1-a_{ij}}(f_j) = 0$  as defining relations. And  $h_i$  are linearly independent.
- $\mathfrak{g}(R)$  is a sum of f.d. modules over every  $\mathfrak{sl}_2$ -triple  $\mathfrak{s}_i$ .
- $\mathfrak{g}(R)$  is of f.d..
- $\mathfrak{g}(R)$  is semisimple and has root system  $R$ .
- $\mathfrak{g}(R_1 \oplus R_2) \cong \mathfrak{g}(R_1) \oplus \mathfrak{g}(R_2)$ . In particular, by (3.7.3.27),  $\mathfrak{g}(R)$  is simple iff  $R$  is indecomposable.  $\dashv$

*Proof:* It suffices to prove for indecomposable root systems. Consider  $\widetilde{\mathfrak{g}(R)}$  the Lie algebra generated by elements  $e_i, f_i, h_i$  with the defining relations in (3.7.3.25) without the final two Serre relations, then it is  $\mathbb{Z}$ -graded, with  $\deg(e_i) = 1, \deg(f_i) = -1, \deg(h_i) = 0$ . Thus we have a decomposition

$$\widetilde{\mathfrak{g}(R)} = \widetilde{\mathfrak{n}_+} \oplus \widetilde{\mathfrak{h}} \oplus \widetilde{\mathfrak{n}_-}$$

by degree, and clearly  $\widetilde{\mathfrak{n}_+}$  is generated by  $e_i$ ,  $\widetilde{\mathfrak{h}}$  is generated by  $h_i$  and  $\widetilde{\mathfrak{n}_-}$  is generated by  $f_i$ .

Now I claim that  $\widetilde{\mathfrak{n}_+}$  is a free Lie algebra on generators  $e_i$ ,  $\widetilde{\mathfrak{n}_-}$  is a free Lie algebra on generators  $f_i$  and  $h_i$  are linearly independent. It suffices to prove for  $e_i$ , and  $f_i$  is true with the dual polarization. For this, let  $\mathfrak{h}'$  be a vector space with basis  $h'_i$ , and consider the Lie algebra  $\mathfrak{a} = FL_r \rtimes \mathfrak{h}'$ , where  $FL_r$  is the free Lie algebra generated by  $f'_i$ , and  $\mathfrak{h}'$  acts on  $FL_r$  by  $[h'_i, f'_j] = -a_{ij}f'_j$ , then  $U = U(\mathfrak{a}) = k\langle f'_1, \dots, f'_n \rangle \rtimes k[h'_1, \dots, h'_n]$ , and there is an action of  $\widetilde{\mathfrak{g}(R)}$  on  $U$  that is defined on the generators as follows: if  $w = f'_{j_1} \dots f'_{j_s}$  is a word in  $f'_i$  of weight  $\alpha$  and  $P \in k[h'_1, \dots, h'_n]$ , then

$$h_i(w \otimes P) = w \otimes (h'_i - \alpha(h_i))P$$

$$f_i(w \otimes P) = fw \otimes P$$

$$e_i(f'_{j_1} \dots f'_{j_s} \otimes P) = \sum_{k|j_k=i} f'_{j_1} \dots \widehat{f'_{j_k}} \dots f'_{j_s} (h'_i - (\alpha_{j_{k+1}} + \dots + \alpha_{j_s})(h'_i))P.$$

It can be shown that this is truly a representation, and then it induces a map  $\widetilde{\mathfrak{g}(R)} \rightarrow U : x \mapsto x(1)$ , and this maps Lie polynomials of  $f_i$  in  $\widetilde{\mathfrak{n}_+}$  to Lie polynomials of  $f'_i$ , and  $h_i$  to  $h'_i$ , so the assertions are true.

1: Now consider the elements  $S_{ij}^+ = (\text{ade}_i)^{1-a_{ij}}e_j \in \widetilde{\mathfrak{n}_+}$  and  $S_{ij}^- = (\text{ad}f_i)^{1-a_{ij}}f_j \in \widetilde{\mathfrak{n}_-}$ . Then  $[f_k, S_{ij}^+] = 0$ : If  $k \neq i, j$ , then this is true, and if  $k = j$ , then  $[f_j[e_ie_j]] = [e_i[f_je_j]] = [h_je_i] = a_{ji}e_i$ , thus  $[f_j(\text{ade}_i)^r(e_j)] = 0$  for  $r \geq 2$ . If  $a_{ij} \geq -1$ , then this is true, and if  $a_{ij} = 0$ , then  $a_{ji} = 0$  too, so the assertion is also true. If  $k = i$ , then we can prove by induction that  $[f_i(\text{ade}_i)^r(e_j)] = -r(a_{ij} + r - 1)(\text{ade}_i)^{r-1}(e_j)$ . Thus  $[f_i(\text{ade}_i)^{1-a_{ij}}(e_j)] = 0$ , too.

So by induction we see that the ideals  $I^\pm \in \widetilde{\mathfrak{n}}_\pm$  generated by such  $S_{ij}^\pm$  is ideals  $I_\pm$  of  $\widetilde{g(R)}$ . Then the ideal generated by the the Serre relations is the graded ideal  $I = I_+ \oplus I_-$ , which implies the assertion.

2: The Serre relations shows that  $e_j$  generates the representation  $W_{-a_{ij}}$  of  $\mathfrak{s}_i$  for  $j \geq i$ , and so does  $f_j$ . Also  $e_i, f_i, h_i$  generate  $W_2$ , and  $h_j$  generates  $W_0$  or  $W_0 \oplus W_2$ , so  $\mathfrak{g}(R)$  is a sum of f.d. modules over  $\mathfrak{s}_i$ , as the module generated by  $[a, b]$  is a subquotient of  $V \otimes W$  the modules generated by  $\{a\}$  and  $\{b\}$ .

3:  $\mathfrak{g}(R) = \bigoplus_{\alpha \in Q} \mathfrak{g}_\alpha$ , where  $\mathfrak{g}_\alpha$  is the subspace of  $\mathfrak{g}(R)$  of weight  $\alpha$ . Then  $\mathfrak{g}_\alpha \neq 0$  only if  $\alpha \in Q_+$  or  $Q_-$ , and each  $\mathfrak{g}_\alpha$  is of f.d.. Now we show that  $\mathfrak{g}_R \neq 0$  only if  $\alpha \in R \cup \{0\}$ , which will suffice. We prove by induction on the height of  $\alpha$ : the height 1 case is trivial, and if  $\alpha = ke_i$ , then the statement is clear as  $\mathfrak{g}^{k\alpha_i} = 0$  for  $k \geq 2$ , because  $\mathfrak{n}_+$  is generated by  $e_i$ . If it is not of the form  $\alpha = ke_i$ ,  $(\alpha, \alpha_i) > 0$  for some  $i$ , so by the representation theory of  $\mathfrak{s}_i \cong \mathfrak{sl}_2$ ,  $\mathfrak{g}^{s_i\alpha} \neq 0$ , where  $s_i\alpha = \alpha - \alpha_i^\vee(\alpha)\alpha_i \notin Q_-$ , thus  $s_i\alpha \in Q_+$ , and then by induction hypothesis  $s_i\alpha \in R$ , so  $\alpha \in R$ .

4:  $\widetilde{\mathfrak{g}(R)}^\alpha$  is 1-dimensional for any  $\alpha$ , so this is also true for  $\mathfrak{g}(R)^\alpha$ . Then  $\mathfrak{g}(R)$  is semisimple with root system  $R$  by (3.7.3.26).  $\square$

**Cor. (3.7.3.29) [Classification of Split Simple Lie Algebras].** By Simple Lie Algebras, split simple Lie algebras over  $k$  are in bijection with Dynkin diagrams  $A_l, l \geq 1, B_l, l \geq 2, C_l, l \geq 3, D_l, l \geq 4$  and  $E_6, E_7, E_8, F_4, G_2$ , by (3.9.6.5).  $\lrcorner$

### Notations for Split Semisimple Lie algebras

**Def. (3.7.3.30) [Notations for Split Semisimple Lie Algebras].** Let  $(\mathfrak{g}, \mathfrak{h})$  be a split semisimple Lie algebra, then

- Its reduced root system is denoted by  $\Phi$  (3.7.3.21).
- Choose a base  $\Delta = \{\alpha_1, \dots, \alpha_r\} \subset \Phi$ , and denote the positive (resp. negative) roots by  $\Phi^+$  (resp.  $\Phi^-$ ).
- Notation for root system is the same as in Notations for a Root System.
- $\mathfrak{g}$  has a weight decomposition  $\mathfrak{g} = \mathfrak{h} \oplus_{\alpha \in \Phi} \mathfrak{g}^\alpha$ .
- $x_\alpha \in \mathfrak{g}^\alpha, y_\alpha \in \mathfrak{g}^{-\alpha}$ , where  $\alpha \in \Phi^+$ .
- $\mathfrak{s}_i = \{x_i, y_i, h_i\}$  is a  $\mathfrak{sl}_2$ -triple, where  $x_i \in \mathfrak{g}^{\alpha_i}, y_i \in \mathfrak{g}^{-\alpha_i}$  (3.7.3.20).
- $\alpha_i^\vee = h_i$  (3.7.3.21).

$\lrcorner$

## 4 Reductive Lie Algebra

**Def. (3.7.4.1) [Reductive Lie Algebra].** A Lie algebra  $\mathfrak{g}$  is called **reductive** if  $\text{rad}(\mathfrak{g}) = Z(\mathfrak{g})$ , or equivalently  $Z(\mathfrak{g}) \subset \text{rad}(\mathfrak{g})$ .  $\lrcorner$

**Prop. (3.7.4.2).** The following conditions on a Lie algebra  $\mathfrak{g}$  are equivalent:

- $\mathfrak{g}$  is reductive.
- The adjoint representation of  $\mathfrak{g}$  is semisimple.
- $\mathfrak{g}$  is a product of a commutative Lie algebra  $\mathfrak{c}$  and a semisimple Lie algebra  $\mathfrak{b}$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2$ : The adjoint representation factors through the center of  $\mathfrak{g}$ , which is also the radical of  $\mathfrak{g}$ , so it is a representation of  $\mathfrak{g}/\text{rad}(\mathfrak{g})$ , which is semisimple(3.7.1.23), so Weyl's theorem(24.4.1.2) shows the adjoint representation is semisimple.

$2 \rightarrow 3$ : If the adjoint representation is semisimple, then  $\mathfrak{g}$  decomposes as a sum of minimal nonzero ideals  $\mathfrak{a}_i$  of  $\mathfrak{g}$ , and then  $\mathfrak{g}$  is a product of these  $\mathfrak{a}_i$ . Let  $\mathfrak{c}$  be the product of the one-dimensional ideals, then  $\mathfrak{c}$  is in the center thus commutative, and  $\mathfrak{b}$  the product of the remaining ideals, then  $\mathfrak{b}$  is semisimple because it has no solvable ideals.

$3 \rightarrow 1$  is trivial.  $\square$

**Cor. (3.7.4.3).** The decomposition of  $\mathfrak{g}$  into a product of commutative Lie algebra and a semisimple Lie algebra is unique: in fact  $\mathfrak{c}$  is the center of  $\mathfrak{g}$ , and  $\mathfrak{b} = [\mathfrak{g}, \mathfrak{g}]$ , by(3.7.2.4).  $\lrcorner$

**Prop. (3.7.4.4).** A Lie algebra  $\mathfrak{g}$  is reductive iff it has a faithful semisimple representation iff it has a trivial nilpotent radical(3.7.1.34).  $\lrcorner$

*Proof:* If  $\mathfrak{g}$  has a faithful semisimple representation, then the nilpotent radical  $\mathfrak{s} = 0$ , thus by(3.7.1.36),  $\mathfrak{r}$  is in the center of  $\mathfrak{g}$ , thus  $\mathfrak{g}$  is reductive.

Conversely, if  $\mathfrak{g}$  is reductive, then we need to show  $\mathfrak{g}$  has a faithful semisimple representation: For this, we can take the tensor product of the trivial representation of the commutative part and the adjoint representation of the semisimple part(3.7.2.3).

The last assertion is clear from(3.7.1.36).  $\square$

**Cor. (3.7.4.5) [Trace Form Criterion for Reductiveness].** If the trace form  $B_\rho$ (3.7.9.5) is non-degenerate for some representation  $(\rho, V)$  of  $\mathfrak{g}$ , then  $\mathfrak{g}$  is reductive.  $\lrcorner$

*Proof:* If  $x \in \mathfrak{s}$ , then  $\rho(x) = 0$ , thus  $B_\rho(x, y) = 0$  for any  $y$ , thus  $x = 0$ . So  $\mathfrak{s} = 0$ , and  $\mathfrak{g}$  is reductive.  $\square$

**Cor. (3.7.4.6) [Classical Lie Algebras are Reductive].** All classical Lie groups over  $\mathbb{R}$  or  $\mathbb{C}$  are reductive.  $\lrcorner$

*Proof:* Apply(3.7.4.5) their standard representations.  $\square$

**Def. (3.7.4.7) [Borel Subalgebras].** Let  $(\mathfrak{g}, \mathfrak{h})$  be a split reductive Lie algebra with a base  $\Phi$ , and consider the corresponding triangular decomposition  $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ (3.7.3.21). Denote  $\mathfrak{b}_+ = \mathfrak{h} \oplus \mathfrak{n}_+$ , and call any Lie subalgebra of  $\mathfrak{g}$  conjugate to  $\mathfrak{b}_+$  a **borel subalgebra** of  $\mathfrak{g}$ . The definition is independent of the choice of the Cartan subalgebra  $\mathfrak{h}$ , by(3.7.3.22).  $\lrcorner$

## 5 Compact Lie Algebras

References are [李群讲义, 项武义] and [?Kna96].

**Def. (3.7.5.1)[Compact Lie Algebras].** A **compact Lie algebra** is Lie algebra that is the Lie algebra of a compact Lie group.  $\lrcorner$

**Prop. (3.7.5.2)[Killing Form of Compact Lie Algebras].** The Killing form of a compact Lie algebra  $\mathfrak{g}$  is negatively semi-definite, with the kernel the center of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Choose an invariant inner product on  $\mathfrak{g}$  w.r.t. the adjoint representation of  $G$  by(24.7.3.2). Take derivative w.r.t the equation  $(\text{Ad}(g)Y, \text{Ad}(g)Z) = (Y, Z)$ , we get by(16.5.1.12),

$$(\text{ad}(X)Y, Z) + (Y, \text{ad}(X)Z) = 0.$$

so  $\text{ad}(X)$  is skew-symmetric w.r.t. this inner product, thus the eigenvalues are all purely imaginary. Then  $B(x, x) = \text{tr}(\text{ad}(x)\text{ad}(x)) \leq 0$ .  $\square$

**Cor. (3.7.5.3) [Compact Lie Algebra is Reductive].** A compact Lie algebra  $\mathfrak{g}$  is reductive.  $\lrcorner$

*Proof:* Because of the invariant inner form, any ideal  $\mathfrak{a}$  of  $\mathfrak{g}$  has a complement  $\mathfrak{a}^\perp$ , thus the adjoint representation of  $\mathfrak{g}$  is semisimple, thus  $\mathfrak{g}$  is reductive (3.7.4.2).  $\square$

**Cor. (3.7.5.4) [Compact Lie algebra Elements Semisimple].** For a compact Lie algebra  $\mathfrak{g}$ , every element is semisimple, and the eigenvalues of any adjoint operator  $\text{ad}(x)$  is purely imaginary.  $\lrcorner$

*Proof:* Because the Killing form is negative definite, thus its negation is an inner product on  $\mathfrak{g}$ , and  $\text{ad}(x)$  acts by skew-Hermitian matrices, thus has purely imaginary eigenvalues.  $\square$

**Prop. (3.7.5.5).** If  $\mathfrak{g}$  is reductive and the Killing form is negative definite on  $[\mathfrak{g}, \mathfrak{g}]$ , then  $\mathfrak{g}$  is compact.  $\lrcorner$

*Proof:* For the commutative part we can take the torus  $(S^1)^n$ , so it suffices to prove the semisimple case: for this, we consider  $\text{Int}(\mathfrak{g})^0 \subset GL(\mathfrak{g})$ , it is contained in  $O(\mathfrak{g})$  with the inner product defined by the negation of the Killing form, thus it is compact. And the Lie algebra of it is  $\text{Der}(\mathfrak{g}) = \text{ad}(\mathfrak{g})$  (3.7.2.7).  $\square$

**Prop. (3.7.5.6) [Representation of Compact Lie Algebras].** Let  $\mathfrak{g}$  be a compact Lie algebra that is the Lie algebra of a simply-connected compact Lie group, then  $\text{Rep}(\mathfrak{g})$  is semisimple.  $\lrcorner$

*Proof:*  $\square$

## 6 Singular element

### Introduction

Singular element in  $\mathfrak{g}$  is a linear space and is defined by some homogenous ideal in  $S(\mathfrak{g})$ .

The paper [Singular element] of Kostant tells in fact it is defined by some  $r$ -homogenous functions  $M^r$  in  $S(\mathfrak{g})$ , and further describes the properties of this ideal such as the  $G$ -module decomposition and as span of determinant minors.

### Preliminary

Let complex simple Lie algebra  $\mathfrak{g} = \text{Lie } G, n = l + 2r$ . The non-degenerate Killing form  $\mathcal{B} \triangleq (x, y)$  on  $\mathfrak{g}$  generate a nonsingular pair on  $S(\mathfrak{g})$  and  $\wedge(\mathfrak{g})$  by

$$(x_1 \cdots x_k, y_1 \cdots y_k) = \sum_{\sigma \in \Sigma_k} (x_1, y_{\sigma(1)}) \cdots (x_k, y_{\sigma(k)})$$

$$(x_1 \wedge \cdots \wedge x_k, y_1 \wedge \cdots \wedge y_k) = \sum_{\sigma \in \Sigma_k} sg(\sigma) (x_1, y_{\sigma(1)}) \cdots (x_k, y_{\sigma(k)})$$

So  $\mathfrak{g} \longleftrightarrow \mathfrak{g}', S(\mathfrak{g}) \longleftrightarrow S(\mathfrak{g}') \longleftrightarrow$  polynomial functions on  $\mathfrak{g}$ ; and  $S(\mathfrak{g})$  and  $\wedge(\mathfrak{g})$  are  $\mathfrak{g}$  thus  $G$  modules extending the adjoint representation.

recall that  $\delta$  and  $\partial$  are called  $\mathcal{B}$ -dual if  $(\delta x, y) = (x, \partial y)$ . Set antiderivation  $-d$   $\mathcal{B}$ -dual to the operator

$$\partial(x_1 \wedge \cdots \wedge x_p) = \sum_{i < j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \hat{x}_i \wedge \cdots \hat{x}_j \wedge \cdots \wedge x_p$$

on  $\wedge(\mathfrak{g})$  and antiderivation  $\iota(u)$   $\mathcal{B}$ -dual to the operator  $\epsilon(u)v = u \wedge v$  on  $\wedge(\mathfrak{g})$ .

Element  $v$  of  $S(\mathfrak{g})$  are called **invariant** iff  $gv = v, \forall g \in G$  and element  $u$  of  $S(\mathfrak{g})$  are called **harmonic** iff  $(u, v) = 0, \forall v$  invariant and no constant term.

Denote by  $J, H$  respectively the graded subspace of invariant and harmonic elements, then:

**Prop. (3.7.6.1)**[Separation of Variables (in [Kos63])].  $S(\mathfrak{g}) \cong J \otimes H$ . ┘

### the Ideal of Sing $\mathfrak{g}$

In the projection  $\tau : T(\mathfrak{g}) \longrightarrow U(\mathfrak{g})$ , PBW theorem asserts that  $S(\mathfrak{g}) \longrightarrow U(\mathfrak{g})$  is an isomorphism. Denote:

$$\Gamma = \tau|_{S(\mathfrak{g})}^{-1} \circ \tau$$

$\Gamma$  is a  $G$ -map (as a consequence of the next prop).

Denote by  $\Gamma_{2r,2}$  the subgroup of permutation that preserves the set of unordered pairs  $\{(1,2), (3,4), \dots, (2r-1, 2r)\}$  and let  $\Pi_r$  be a left coset representative of  $\Gamma_{2r,2}$  in  $\Gamma_{2r}$  that  $sg(\Pi_r) = 1$

In [Amitsur-Levitski], Kostant proved:

**Prop. (3.7.6.2)**[in [Kos81]].

$$\begin{aligned} \Gamma(\wedge^{2k}(\mathfrak{g})) &= R^k \in S^k(\mathfrak{g}) \\ \Gamma(x_1 \wedge \dots \wedge x_k) &\longrightarrow \sum_{v \in \Pi_k} [x_{v(1)}, x_{v(2)}] \cdots [x_{v(2k-1)}, x_{v(2k)}] \end{aligned}$$

┘

**Prop. (3.7.6.3)**[in [Kos81]].

$$M = R^r \in H^r$$

so it consists of harmonic functions. ┘

Let  $w \in \wedge^2 \mathfrak{g}$  of rank  $k$  standardized as  $v_1 \wedge v_2 + \dots + v_{2k-1} \wedge v_{2k}$ . Let

$$\text{Rad} w = \{y \in \mathfrak{g} | \iota(y)w = 0\} = \{y \in \mathfrak{g} | (w, \epsilon(y)z) = 0, \forall z\}$$

then  $w$  of rank  $2k \iff w^k \neq 0 \ \& \ w^{k+1} = 0 \iff \dim \text{Rad} w = n - 2k$ .

**Lemma (3.7.6.4).**

$$\iota(y)dx = [y, x]$$

Thus

$$\text{Rad} dx = \mathfrak{g}^x, \text{ Sing } \mathfrak{g} = \{x \in \mathfrak{g} | (dx)^r = 0\}.$$

┘

*Proof:*  $(\iota(y)dx, z) = (dx, y \wedge z) = (x, -[y, z]) = ([y, x], z)$  □

So in order to find the module  $M$ , it's the best to find the dual of

$$\gamma : S(\mathfrak{g}) \longrightarrow \wedge^{\text{even}} \mathfrak{g} : x \longrightarrow -dx$$

Luckily:

**Prop. (3.7.6.5)**[in [Kos81]].  $\gamma$  is  $\mathcal{B}$  dual to  $\Gamma$ , in particular,

$$(\Gamma(\zeta), x) = \frac{(-1)^r}{r!} (\zeta, (dx)^r) \quad (\forall \zeta \in \wedge^{2r} \mathfrak{g} \text{ and } x \in \mathfrak{g})$$

┘

So:  $f(x) = 0, \forall f \in M \iff x \in \text{Singg}$ .

**Cor. (3.7.6.6).** Let  $\mathfrak{a}$  be a CSA of  $\mathfrak{g}$ ,  $\Delta_+(\mathfrak{a})$  be the positive roots, then

$$f|_{\mathfrak{a}} = C_f \cdot \prod_{\beta \in \Delta_+(\mathfrak{a})} \beta \quad (\forall f \in M)$$

┘

*Proof:* This is because that an element in a CSA is singular iff it commutes with an element outside this CSA, and taking root decomposition, this is equivalent to annihilated by a root, and by counting degree, the cor follows.  $\square$

By propositions of [[Kos59]] a **regular nilpotent** element  $e$  is **uniquely** in a nilpotent radical  $\mathfrak{n}$  of a Borel subalgebra and that  $\mathfrak{g}^e \cap [\mathfrak{n}, \mathfrak{n}] = (\text{Singg}) \cap \mathfrak{g}^e$ . So there is a linear function  $\xi$  on  $\mathfrak{g}^e$  that  $\ker \xi = (\text{Singg}) \cap \mathfrak{g}^e$ . Thus:

**Cor. (3.7.6.7).**  $f|_{\mathfrak{g}^e} = C_f \cdot \xi^r \quad (\forall f \in M)$ .

┘

*Proof:* By counting degree, the same reason as before.  $\square$

Now we think of a natural question: Can singular elements be defined by functions of even lower degree? The answer is NO.

**Prop. (3.7.6.8).** Assume  $0 \neq f$  homogenous vanishes on  $\text{Singg}$ , then  $\deg f \geq r$

┘

*Proof:* By the last cor, if  $f$  has degree less than  $r$  then  $f$  vanishes on any CSA, but semisimple regular element, thus CSAs are Zariski dense in  $\mathfrak{g}$  (this is because semisimple elements are defined by a polynomial), so  $f = 0$ .  $\square$

Thus we have established that  $\text{Singg}$  is an algebraic set defined by a set of harmonic  $r$ -homogenous functions on  $\mathfrak{g}$  and not by functions of degree lower than  $r$ .

Next we offer a different formation of  $M$ .

### $M$ as minors of determinants

for a  $\mathcal{B}$  dual basis  $y_i, w_j$ , define a derivation

$$d_W(f \otimes u) = \sum_i^n \partial_{y_i} f \otimes \epsilon(w_i)u \quad \text{on } S(\mathfrak{g}) \otimes \wedge \mathfrak{g}.$$

Here  $\partial_{\sum a_i x_i}$  is defined as  $\sum a_i \frac{\partial}{\partial x_i}$  for a standard basis  $x_i$  of  $\mathfrak{g}$ . It's easy to verify that  $d_W$  is well defined and is a  $G$ -map (Take a different basis  $Aw_i$  and  $Bz_i$ , then  $AB^t = I$ , substitute into the formula of  $d_W$ , it doesn't change).

**Chevalley Thm** tells us  $J$  is a polynomial ring  $\mathbb{C}[p_1, \dots, p_l]$ , where  $p_i$  are homogenous polynomials of fixed degree  $d_i$  and  $\sum_{j=1}^l (d_j - 1) = r$ . So:

$$d_W p_1(x) \wedge \dots \wedge d_W p_l(x) = \sum_{1 \leq i_1 < \dots < i_l \leq n} \phi(y_{i_1}, \dots, y_{i_l})(x) w_{i_1} \wedge \dots \wedge w_{i_l}$$

Where  $\phi(y_{i_1}, \dots, y_{i_l}) = \det \partial_{y_i} p_j$  is homogenous of degree  $r$ . (counting degree).

To see this, notice that  $f \otimes u$  acts as a function from  $\mathfrak{g}$  to  $\wedge \mathfrak{g}$ :  $f \otimes u(x) = f(x)u$ . So:

$$d_W p_j(x) = \sum_{i=1}^n \partial_{z_i} p_j(x) w_i.$$

**Prop. (3.7.6.9).** for any CSA  $\mathfrak{h}$  of  $\mathfrak{g}$  and a basis  $\{v_i\}$  of  $\mathfrak{h}, \forall x \in \mathfrak{h}$ ,

$$d_W p_1(x) \wedge \cdots \wedge d_W p_l(x) = \kappa \cdot \prod_{\phi \in \Delta_+} \phi(y) v_1 \wedge \cdots \wedge v_l$$

┘

**Lemma (3.7.6.10)** [in [Kos63]].  $\{d_W p_1(x), \dots, d_W p_l(x)\}$  is linearly independant iff  $x \in \text{Regg}$ . ┘

*Proof:* Notice that  $d_W p_j$  is a  $\mathfrak{g}$ -map,  $\text{ady} \cdot d_W p_j = d_W p_j([y, x])$  so  $d_W p_j(x)$  commutes with  $\mathfrak{g}^x$ ; so  $\in \mathfrak{g}^x$ . Then the lm tells us when  $y$  is regular,  $d_W p_j(y)$  forms a basis of  $\mathfrak{g}^y$ . Considering in  $\mathfrak{g}^y$ ,  $x$  is regular iff  $\prod_{\phi \in \Delta_+} \phi(x) \neq 0$ , the prop follows.  $\square$

Next we give an explicit expression for  $\gamma_r$ .

It can be verified (taking a  $z_i$  basis) that  $dx = \frac{1}{2} \sum_{i=1}^n w_i \wedge [z_i, x]$ .

Now  $x \in \mathfrak{h}$ ,

$$dx = \sum_{\phi \in \Delta_+} \phi(x) e_\phi \wedge f_{-\phi}$$

(just take the basis  $w_i$  and  $z_i$  as a standard basis of  $\mathfrak{g}$  consisting of  $\{h_i, \dots, h_l, e_\phi, f_\phi\}$ )

So

$$\gamma_r(x^r) = r!(-1)^r \prod_{\phi \in \Delta_+} \phi(x) e_\phi \wedge f_{-\phi}$$

Let  $\mu = i^r v_1 \wedge \cdots \wedge v_l \wedge \prod_{\phi \in \Delta_+} e_\phi \wedge f_{-\phi}$  then  $(\mu, \mu) = 1$ .

Denote  $v^* = \iota(v)\mu$  for  $v \in \wedge \mathfrak{g}$ , then

$$(v_1 \wedge \cdots \wedge v_l)^* = i^r \prod_{\phi \in \Delta_+} e_\phi \wedge f_{-\phi} = C_o \gamma_r(x^r)$$

(notice that  $\iota(u)\iota(v) = \iota(v \wedge u)$  and use lm 1)

**Prop. (3.7.6.11).**  $(d_W p_1(x) \wedge \cdots \wedge d_W p_l(x))^* = \kappa_o \gamma_r(\frac{x^r}{r!}) \neq 0$ . ┘

*Proof:* For  $y \in \mathfrak{h}$  regular, this follows from previous calculations, and notice both side are  $G$ -maps, and semisimple regular elements are Zariski open, conclusion follows.  $\square$

**Lemma (3.7.6.12).**  $(s, t) = (s^*, t^*)$ , so that  $-^*$  is a  $\mathcal{B}$ -isomorphism. ┘

**Prop. (3.7.6.13).** Let  $\{w_1, \dots, w_{2r}\}$  be linearly independent and  $\{u_1, \dots, u_l\}$  be a basis of  $\{w_1, \dots, w_{2r}\}^\perp$ , then

$$\Gamma(w_1 \wedge \cdots \wedge w_{2r}) = \kappa_1 \det \partial_{u_i} p_j \neq 0$$

Thus,  $M$  is the span of all the minors  $\det \partial_{u_i} p_j$ . ┘

*Proof:* By the preceding props,

$$\begin{aligned} \det \partial_{u_i} p_j &= \phi(u_1, \dots, u_l)(x) \\ &= (d_W p_1(x) \wedge \cdots \wedge d_W p_l(x), u_1 \wedge \cdots \wedge u_l) \\ &= ((d_W p_1(x) \wedge \cdots \wedge d_W p_l(x))^*, (u_1 \wedge \cdots \wedge u_l)^*) \\ &= \kappa_o \kappa_2 \left( \phi_r \left( \frac{x^r}{r!} \right), w_1 \wedge \cdots \wedge w_{2r} \right) \\ &= \kappa^{-1} \Gamma(w_1 \wedge \cdots \wedge w_{2r})(x). \end{aligned}$$

$\square$

### $G$ -module structure of $M$

Now we show the  $G$ -module structure of  $M$ .

Let  $\theta$  be the derivative that  $\theta(x)(y) = [x, y]$  on  $\mathfrak{g}$ .  $\text{Cas} = \sum_{i=1}^n \theta(z_i)\theta(w_i)$ .

It's in fact just the action of the Casimir element in center of  $U(\mathfrak{g})$ . Let  $m_l$  and  $M_l$  be the maximal eigenvalue and eigenspace of  $\text{Cas}$ .

For a commutative Lie subalgebra  $\mathfrak{c}$  of rank  $l$ , denote by  $[\mathfrak{c}]$  the line it defines on  $\wedge^l \mathfrak{g}$ . The span of these  $[\mathfrak{c}]$  is denoted  $A_l$ . Notice that  $[g^y] \subset A_l$  for a regular  $y$ , and  $A_l$  is a  $G$ -submodule.

**Prop. (3.7.6.14)** [in [Kos65]].

$$A_l = M_l; \quad m_l = l.$$

┘

An ideal in a Borel subalgebra of  $\mathfrak{g}$  is necessarily spanned by root vectors and a prop of [[K-W09]] says any ideal of  $\dim l$  is (denoted by  $\mathcal{I}$ ) in fact abelian.

A prop in [[Kos65]] asserts that for two different ideals  $\Phi_1, \Phi_2$ , sum of their weight vectors  $\langle \Phi \rangle$  is distinct.

So  $G[\Phi_i]$  is an irreducible  $G$ -module  $V_\Phi$  with highest weight  $\langle \Phi \rangle$  and  $V_\Phi$  are inequivalent  $G$ -modules (because an irreducible representation have only one highest vector).

**Prop. (3.7.6.15)** [in [Kos65]].

$$M_l = \oplus_{\Phi \in \mathcal{I}} V_\Phi.$$

┘

Now denote  $M_{2r}$  image of  $M_l$  under the isomorphism  $u \rightarrow u^*$ , then

**Prop. (3.7.6.16)** [in [K-W09]].  $M_l$  is the span of  $G \cdot [g^x]$  for  $x$  regular.

┘

but by precious prop,

$$[g^x] = \mathbb{C} d_W p_1(x) \wedge \dots \wedge p_l(x).$$

thus  $M_{2r}$  is the span of  $G \cdot (\gamma_r(\frac{x^r}{r!}))$ ,  $x$  regular.

**Prop. (3.7.6.17)** [Final].  $\Gamma|_{M_{2r}} : M_{2r} \rightarrow M$  is an isomorphism and  $M \cong M_{2r} \cong M_l = A_l$  as  $G$ -module.

So  $M$  is a multiplicity one module with  $|\mathcal{I}|$  irreducible components.

┘

*Proof:* Notice that  $\Gamma(\zeta)(x) = (\zeta, \gamma_r(\frac{x^r}{r!}))$  and  $M_{2r}$  is the span of  $G \cdot (\gamma_r(\frac{x^r}{r!}))$ , the first part follows, and the rest is a recapitulation of previous props.  $\square$

## 7 Real Lie Algebra

**Prop. (3.7.7.1)** [Passage from Real to Complex]. If  $\mathfrak{g}_0$  is a Lie algebra over  $\mathbb{R}$  and  $\mathfrak{g} = \mathfrak{g}_0 \otimes \mathbb{C}$  its complexification, then  $\mathfrak{g}_0$  is Abelian/nilpotent/solvable/semisimple iff  $\mathfrak{g}$  does.

┘

**Def. (3.7.7.2).** A **compact real form** is a real subalgebra  $\mathfrak{l}$  of  $\mathfrak{g}$  s.t.  $\mathfrak{g}$  is the complexification of  $\mathfrak{l}$  and  $\mathfrak{l}$  is the lie algebra of a compact simply-connected Lie group.

┘

**Prop. (3.7.7.3).** A real Lie algebra is compact iff there exists an invariant inner product iff the Killing form is negative definite.

┘



*Proof:* One direction is easy, just use the average method to find a  $G$ -invariant inner product and then take derivative. For the other direction, the identity shows that a complement of an ideal is an ideal so  $\mathfrak{g}$  is decomposed into simple lie groups and reduce to the case that  $\mathfrak{g}$  is simple. The ideal is to show that  $\mathfrak{g} \cong \text{ad}(\mathfrak{g})$  is the whole outer derivative group  $\partial(\mathfrak{g})$  (the following lm). So  $\mathfrak{g}$  equals to the identity component of  $\text{Aut}(\mathfrak{g})$  which is a closed subgroup thus closed but it is also a subgroup of the compact group  $O(\mathfrak{g})$  thus it is compact.  $\square$

**Lemma (3.7.7.4).** If a real semisimple Lie algebra  $X$  has an invariant inner product, then every outer derivative is inner. (In fact, this is true by Cartan Criterion for semisimplicity (3.7.2.7)).  $\lrcorner$

*Proof:* since  $\text{ad}(X)$  is skew-symmetric, it's diagonalizable and its eigenvalue is pure imaginary, so the Killing form of  $X$  is negative definite. Now choose the complement  $\mathfrak{a}$  of  $\text{ad}(X)$  in  $\partial(X)$ , then  $\mathfrak{a} \cap X = 0$ . Thus for  $D \in \mathfrak{a}$ ,  $\text{ad}(D(g)) = [D, \text{ad}(g)] = 0$  for all  $g$  in  $X$ , so  $D = 0$ , thus  $\text{ad}(X) = \partial(X)$ .  $\square$

**Prop. (3.7.7.5).** 1. The complexification of the Lie algebra of a connected compact Lie group is reductive.  
2. A complex Lie algebra is semisimple iff it is isomorphic to the complexification of the Lie algebra of a simply-connected compact Lie group. i.e. every complex semisimple Lie algebra has a compact real form.  $\lrcorner$

*Proof:* 1: Because a connected compact Lie group is completely reducible so the does the Lie algebra and so does the complexification. So it is reductive by (3.7.4.1)4.

2: Cf.[Varadarajan Lie Groups Lie algebras and Their Representations]. The idea is to find a real form whose corresponding simply-connected group is compact.  $\square$

**Prop. (3.7.7.6).** If  $\mathfrak{g}$  is the Lie algebra of a matrix Lie group  $G$ , then:

1. every Cartan subalgebra comes from a maximal commutative subalgebra of a compact real form and any two Cartan subalgebras are conjugate under the Ad-action of  $G$ .
2. any two compact real form is conjugate under the Ad-action of  $G$ .
3. any two maximal commutative subalgebra of a compact real form is conjugate under the Ad-action of the corresponding compact compact subgroup.  $\lrcorner$

**Prop. (3.7.7.7).** A real Lie algebra is semisimple iff its complexification is semisimple. Cf.[Varadarajan].  $\lrcorner$

**Cor. (3.7.7.8).** The real Lie algebra of a compact simply-connected group is semisimple.  $\lrcorner$

Note: For the classification of real semisimple Lie algebras, Cf.[李群讲义项武义 §6]

**Prop. (3.7.7.9).** If a complex representation of a Lie group admits an invariant bilinear form, then it is non-degenerate and unique. In fact, this is equivalent to a  $G$ -map from  $V$  to  $V^*$ . Thus there is unique invariant inner product in a compact real form by the preceding proposition.  $\lrcorner$

## 8 Universal Constructions

**Notation (3.7.8.1).**

- In this subsection  $k$  can be a field of any characteristics.
- Let  $\mathfrak{g} \in \text{LieAlg}_k$ .

┘

**Def. (3.7.8.2)[Universal Enveloping Algebra].** The **universal enveloping algebra** of a Lie algebra  $\mathfrak{g}$  is defined to be

$$U(\mathfrak{g}) = T(\mathfrak{g})/J, \quad T(\mathfrak{g}) = \bigoplus_{n=0}^{\infty} \mathfrak{g}^{\otimes n}$$

which is a graded algebra  $T(\mathfrak{g})$  quotients the ideal  $J = (\{x \otimes y - y \otimes x - [xy]\})$ .

There is a natural linear map  $\sigma : \mathfrak{g} \rightarrow U(\mathfrak{g})$ .

┘

**Prop. (3.7.8.3).** The universal enveloping algebra  $U : \mathfrak{g} \mapsto U(\mathfrak{g})$  defines a functor  $\text{LieAlg} \rightarrow \text{Alg}^{\text{asso}}$  that is left adjoint to the canonical functor  $\text{Alg}_k^{\text{asso}} \rightarrow \text{LieAlg}_k$  (3.7.1.2).

┘

*Proof:* For any associative algebra  $A$  and a morphism of Lie algebras  $\mathfrak{g} \rightarrow [A]$ , there is easily seen a morphism  $U(\mathfrak{g}) \rightarrow A$ , and it is unique.  $\square$

**Cor. (3.7.8.4)[Representations as Modules].** A representation of  $\mathfrak{g}$  (3.7.11.4) is the same as a representation of  $U(\mathfrak{g})$ .

┘

**Prop. (3.7.8.5)[Poincaré-Birkhoff-Witt].** Let  $\mathfrak{g}$  be a Lie algebra, define a filtration on  $U(\mathfrak{g})$  by assigning  $F_n U(\mathfrak{g}) =$  the image of  $\bigoplus_{i=0}^n \mathfrak{g}^{\otimes i}$  in  $U(\mathfrak{g})$ . We have  $[F_i U(\mathfrak{g}), F_j U(\mathfrak{g})] \subset F_{i+j-1} U(\mathfrak{g})$ , thus  $gr U(\mathfrak{g})$  has a graded commutative ring structure. Thus there is an algebra homomorphism  $S(\mathfrak{g}) \rightarrow gr U(\mathfrak{g})$ , and this homomorphism is an isomorphism.

If  $\mathfrak{g}$  has a basis  $\{x_i\}$ ,  $i \in I$  and  $<$  is an order on  $I$ , then  $U(\mathfrak{g})$  has a basis consisting of elements  $\{x_{i_1}^{r_1} \cdots x_{i_k}^{r_k}\}$  where  $i_1 < i_2 < \cdots < i_r$ .

┘

*Proof:* We assign any monomial  $a_{i_1} \cdots a_{i_n}$  in  $a_i$ s a pair  $(k, N)$  where  $k$  is the number of factors in a monomial and  $N$  is the number of inversions (meaning the number of pairs  $1 \leq r, s \leq n$  that  $i_r > i_s$ ), pairs  $(k, N)$  are lexicographical ordered. Let  $T^{(k, N)}$  be the space of  $T(\mathfrak{g})$  generated by monomials of index  $(k, N)$ ,  $T^k = \bigcup_{N=1}^{\infty} T^{(k, N)}$ , and  $\mathcal{U}^{(k, N)}, \mathcal{U}^k$  the space of  $T(\mathfrak{g})$  the image of  $T^{(k, N)}, T^k$  in  $\mathcal{U}(\mathfrak{g})$ . We will use induction on  $(k, N)$ . Notice for any  $k$ ,  $T^k = T^{(k, N)}$  for  $N$  large.

To show that the monomials in  $(\star)$  generates  $U(\mathfrak{g})$ , if we have a monomial  $a_{i_1} a_{i_2} \cdots a_{i_s} a_{i_{s+1}} \cdots a_{i_k}$  that  $i_s > i_{s+1}$ , then

$$a_{i_1} a_{i_2} \cdots a_{i_s} a_{i_{s+1}} \cdots a_{i_k} = a_{i_1} a_{i_2} \cdots (a_{i_{s+1}} a_{i_s} + [a_{i_s}, a_{i_{s+1}}]) \cdots a_{i_k},$$

which is in  $\bigcup_{(k', N') < (k, N)} \mathcal{U}^{(k', N')}$ . So we can use induction on  $(k, N)$  to show that any element of  $\mathcal{U}(\mathfrak{g})$  is in  $\bigcup_{k=1}^{\infty} T^{(k, 0)}$ .

From now on, we write  $a_{i_1} a_{i_2} \cdots a_{i_n}$  as  $a_{i_1} a_{i_2} \cdots a_{i_n}$  for simplicity.

To show that the monomials are linearly independent, we first show that there is a linear map

$$\theta : T(\mathfrak{g}) \rightarrow R = \mathbb{C}[z_i]_{i \in I}$$

satisfying the following conditions:

$$\theta(a_{i_1} \cdots a_{i_n}) = z_{i_1} \cdots z_{i_n}, \text{ if } i_1 \leq i_2 \leq \cdots \leq i_n, \quad (\star\star)$$

$$\begin{aligned} & \theta(a_1 \dots a_{i_k} a_{i_{k+1}} \dots a_{i_n}) \\ &= \theta(a_1 \dots a_{i_{k+1}} a_{i_k} \dots a_{i_n}) + \theta(a_1 \dots [a_{i_k} a_{i_{k+1}}] \dots a_{i_n}). \end{aligned} \quad (\star \star \star)$$

We construct this map by construction on  $\cup_{(k', N') \leq (k, N)} U^{(k', N')}$  and use induction on  $(kN)$ . For  $k = 0$ , let  $\theta(1) = 1$ . If  $\theta$  is defined for any monomials of index  $(k, N)$  that  $k < n$ , define  $\theta$  on  $T^{n, 0}$  by  $\theta(a_{i_1} \dots a_{i_n}) = z_{i_1} \dots z_{i_n}$ , then it satisfies  $(\star \star)$ .

And if  $\theta$  is already defined for any  $T^{n, k}$  that  $k < i$ , suppose that the monomial  $a_{i_1} \dots a_{i_n}$  has index  $(n, i)$ , then there is a smallest  $k$  that  $a_{i_1} \dots a_{i_{k+1}} a_{i_k} \dots a_{i_n}$  has index  $i - 1$ . Then we define

$$\theta(a_{i_1} \dots a_{i_n}) = \theta(a_{i_1} \dots a_{i_{k+1}} a_{i_k} \dots a_{i_n}) + \theta(a_{i_1} \dots [a_{i_k}, a_{i_{k+1}}] \dots a_{i_n}).$$

Now we need to check that this definition satisfies  $(\star \star \star)$ :

If there is another  $k'$  that  $i_{k'} > i_{k'+1}$ , then  $k < k'$ . Suppose first that  $k + 1 < k'$ , let  $a_{i_k} = a$ ,  $a_{i_{k+1}} = b$ ,  $a_{i_{k'}} = c$ ,  $a_{i_{k'+1}} = d$ , then

$$\begin{aligned} \theta(\dots ab \dots cd \dots) &= \theta(\dots ba \dots cd \dots) + \theta(\dots [a, b] \dots cd \dots) \\ &= \theta(\dots ba \dots dc \dots) + \theta(\dots ba \dots [cd] \dots) + \theta(\dots [ab] \dots dc \dots) + \theta(\dots [ab] \dots [cd] \dots) \\ &= \theta(\dots ab \dots dc \dots) + \theta(\dots ab \dots [cd] \dots) \end{aligned}$$

where the terms except the first one are all in  $\cup_{(k', N') < (k, N)} T^{(k', N')}$  so the equalities come from induction hypothesis. So it satisfies  $(\star \star \star)$ .

Suppose next that  $k' = k + 1$ , let  $a_{i_k} = a$ ,  $a_{i_{k+1}} = b$ ,  $a_{i_{k+2}} = c$ , then

$$\begin{aligned} \theta(\dots abc \dots) &= \theta(\dots bac \dots) + \theta(\dots [ab]c \dots) \\ &= \theta(\dots bca \dots) + \theta(\dots b[ac] \dots) + \theta(\dots c[ab] \dots) + \theta(\dots [[ab]c] \dots) \\ &= \theta(\dots cba \dots) + \theta(\dots [bc]a \dots) + \theta(\dots b[ac] \dots) + \theta(\dots c[ab] \dots) + \theta(\dots [[ac]b] \dots) + \theta(\dots [a[bc]] \dots) \\ &= \theta(\dots cab \dots) + \theta(\dots [ac]b \dots) + \theta(\dots a[bc] \dots) \\ &= \theta(\dots acb \dots) + \theta(\dots a[bc] \dots) \end{aligned}$$

where the terms except the first one are all in  $\cup_{(k', N') < (k, N)} T^{(k', N')}$  so the equalities come from induction hypothesis, and in the third equality we used the Jacobi identity. So it satisfies  $(\star \star \star)$ .

Now all elements in  $J$  is a linear combination of elements of the form

$$a_1 \dots a_{i_k} a_{i_{k+1}} \dots a_{i_n} - a_1 \dots a_{i_{k+1}} a_{i_k} \dots a_{i_n} - a_1 \dots [a_{i_k} a_{i_{k+1}}] \dots a_{i_n},$$

so the map  $\theta$  factors through  $T(\mathfrak{g}) \rightarrow U(\mathfrak{g})$  to a map  $\bar{\theta} : U(\mathfrak{g}) \rightarrow R$ , and the elements  $a_{i_1} a_{i_2} \dots a_{i_k}, i_1 \leq i_2 \leq \dots \leq i_k$  are mapped to  $z_{i_1} \dots z_{i_n}$ , which are linearly independent in  $R$ , so the elements  $a_{i_1} a_{i_2} \dots a_{i_k}, i_1 \leq i_2 \leq \dots \leq i_k$  are also linearly independent in  $U(\mathfrak{g})$ .  $\square$

**Cor. (3.7.8.6).** The map  $\mathfrak{g} \rightarrow U(\mathfrak{g})$  is injective.  $\lrcorner$

**Cor. (3.7.8.7).**  $U(\mathfrak{g})$  has no zero-divisors.  $\lrcorner$

*Proof:* We can use the identities  $x \otimes y - y \otimes x = [xy]$  to make any element in their right representations under the PBW prop(3.7.8.5), so it is clear that the product of two nonzero elements cannot be 0.  $\square$

**Cor. (3.7.8.8).** If  $\mathfrak{h} \subset \mathfrak{g}$ , then the subalgebra of  $U(\mathfrak{g})$  generated by  $\mathfrak{h}$  is isomorphic to  $U(\mathfrak{h})$ .  $\lrcorner$

*Proof:* There is a natural map  $U(\mathfrak{h}) \rightarrow U(\mathfrak{g})$ , and the image is just the subgroup generated by  $\mathfrak{h}$ . The PBW theorem shows this map is injective.  $\square$

**Cor. (3.7.8.9).** If  $\mathfrak{g} = \mathfrak{a} \times \mathfrak{b}$ , then  $U(\mathfrak{g}) = U(\mathfrak{a}) \otimes U(\mathfrak{b})$ .  $\lrcorner$

**Def. (3.7.8.10) [Coproduct of  $U(\mathfrak{g})$ ].** Let  $\mathfrak{g}$  be a Lie algebra, there is a coproduct  $\Delta$  on  $T(\mathfrak{g})$  defined by  $\Delta(g) = g \otimes 1 + 1 \otimes g$ . This coproduct descends to a coproduct  $\Delta : U(\mathfrak{g}) \rightarrow U(\mathfrak{g}) \otimes U(\mathfrak{g})$ .  $\lrcorner$

*Proof:* It suffices to check that  $\Delta(J) \subset J \otimes T(\mathfrak{g}) + T(\mathfrak{g}) \otimes J$ , and this is because

$$\begin{aligned} \Delta(x \otimes y - y \otimes x - [xy]) &= (x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y) - (y \otimes 1 + 1 \otimes y)(x \otimes 1 + 1 \otimes x) - ([xy] \otimes 1 + 1 \otimes [xy]) \\ &= (x \otimes y - y \otimes x - [xy]) \otimes 1 + 1 \otimes (x \otimes y - y \otimes x - [xy]) \end{aligned}$$

$\square$

**Prop. (3.7.8.11) [ $\mathfrak{g}$ -Module Structure].**  $T(\mathfrak{g})$  is a  $\mathfrak{g}$ -module by adjoint action, and this action descends to a  $\mathfrak{g}$ -module structure on  $U(\mathfrak{g})$ .

In fact, this action is inner:  $\text{ad}(g)(z) = gz - zg$  for  $z \in U(\mathfrak{g})$ . In particular,

$$Z(U(\mathfrak{g})) = U(\mathfrak{g})^{\mathfrak{g}}.$$

$\lrcorner$

*Proof:* It suffices to show that  $\mathfrak{g}J \subset J$  and  $\mathfrak{g}I \subset I$ :

$$\begin{aligned} y(a \otimes b - b \otimes a - [ab]) &= [ya] \otimes b + a \otimes [yb] - [yb] \otimes a - b \otimes [ya] - [y[ab]] \\ &= [ya] \otimes b - b \otimes [ya] - [[ya]b] + a \otimes [yb] - [yb] \otimes a - [a[yb]] \end{aligned}$$

$\square$

**Prop. (3.7.8.12) [Transpose].** There is an anti-automorphism  $u \mapsto u^t$  of  $U(\mathfrak{g})$  that  $X^t = -X$  for  $X \in \mathfrak{g}$ .  $\lrcorner$

*Proof:* We first extend this  $t$  to an automorphism of  $T(\mathfrak{g})$ , then we compose with the obvious anti-automorphism  $T(\mathfrak{g}) \rightarrow T(\mathfrak{g})$ . Then we check that this map descends to  $U(\mathfrak{g})$ :  $(X \otimes Y - Y \otimes X - [X, Y])^t = Y \otimes X - X \otimes Y - [Y, X] \in J$ , so  $J^t \in J$ .  $\square$

**Prop. (3.7.8.13) [Graded Algebra of  $U(\mathfrak{g})$ ].** If we let  $S(\mathfrak{g}) = T(\mathfrak{g})/(x \otimes y - y \otimes x)$  be the universal symmetric algebra of  $\mathfrak{g}$ , then it is a graded algebra. There is a filtered structure on  $U(\mathfrak{g})$  given by  $U_i = \{\text{submodule generated by } a_1 a_2 \dots a_j, j \leq i\}$ , then the associated graded algebra of  $U(\mathfrak{g})$  is isomorphic to  $S(\mathfrak{g})$  by PBW theorem.  $\lrcorner$

**Cor. (3.7.8.14).** If  $W$  is a subspace of  $T^n(L)$  that is sent isomorphically onto  $S^n(L)$ , then the image of  $W$  is a complement of  $U_n(L)$  complementary to  $U_{n-1}(L)$ .  $\lrcorner$

**Cor. (3.7.8.15) [Symmetrization Map].** If  $\text{char } k = 0$ , the symmetrization map  $\sigma : S(\mathfrak{g}) \rightarrow T(\mathfrak{g}) \rightarrow U(\mathfrak{g})$  (3.5.5.2) is an isomorphism of  $\mathfrak{g}$ -modules that

$$U^n(\mathfrak{g}) = \sigma(S^n(\mathfrak{g})) \oplus U^{n-1}(\mathfrak{g}).$$

Moreover, by (3.7.8.11), this isomorphism induces an isomorphism

$$S(\mathfrak{g})^{\mathfrak{g}} \cong Z(U(\mathfrak{g})).$$

$\lrcorner$

*Proof:* It is clearly an isomorphism of vector spaces by (3.7.8.14). It suffices to show the map is compatible with  $\mathfrak{g}$ -actions: Because

$$\begin{aligned}\sigma(y_1 \dots y_n) &= \frac{1}{n!} \sum_{\sigma \in \Sigma_n} y_{\sigma(1)} \dots y_{\sigma(n)}, \\ \sigma(g(y_1 \dots y_n)) &= \sigma([gy_1] \dots y_n + \dots + y_1 \dots [gy_n]) \\ &= \frac{1}{n!} \sum_{\sigma \in \Sigma_n} ([gy_{\sigma(1)}] \dots y_{\sigma(n)} + \dots + y_{\sigma(1)} \dots [gy_{\sigma(n)}]) \\ &= g\left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} y_{\sigma(1)} \dots y_{\sigma(n)}\right)\end{aligned}$$

□

**Prop. (3.7.8.16).** If  $\text{char } k = 0$ , then the set of primitive elements (3.11.2.4) are just  $\mathfrak{g}$ . ┘

*Proof:* If  $f$  is primitive, then the leading term  $f_0$  of  $f$  is also primitive in  $\text{gr}U(\mathfrak{g}) \cong S(\mathfrak{g})$ . Now consider  $S(\mathfrak{g}) \xrightarrow{\Delta} S(\mathfrak{g}) \otimes S(\mathfrak{g}) \xrightarrow{\mu} S(\mathfrak{g})$ , then if  $f$  is of degree  $n$ , then  $2^n f_0 = 2f_0$ , which means  $n = 1$ . So  $f = c + f_0$ , and  $c = 0$ . □

**Prop. (3.7.8.17)** [ $U(\mathfrak{g})$  is Noetherian]. For a f.d. Lie algebra  $\mathfrak{g}$ ,  $U(\mathfrak{g})$  is left Noetherian. ┘

*Proof:* This is because the graded structure on  $U(\mathfrak{g})$  satisfies  $\text{gr}U(\mathfrak{g}) \cong S(\mathfrak{g}) \cong \mathbb{C}[X_1, \dots, X_n]$  is Noetherian, so  $U(\mathfrak{g})$  is Noetherian itself. □

### Free Lie Algebras

**Def. (3.7.8.18)** [Free Lie Algebras]. For  $X \in \text{Set}$ , we can define the **free Lie algebra**  $FL(X)$  to be the intersection of Lie subalgebras in  $[F(X)]$  containing  $\sigma(X)$ , where  $F(X)$  is the free algebra generated by  $X$ .

Then the free Lie algebra  $FL : X \mapsto FL(X)$  defines a functor  $\text{Set} \rightarrow \text{LieAlg}$  that is left adjoint to the forgetful functor. ┘

*Proof:* We need to show that for any Lie algebra  $L$  and a map of sets  $\theta : X \rightarrow L$ , there is a unique  $\varphi$  completing the upper left triangular diagram:

$$\begin{array}{ccccc} X & \xrightarrow{i} & FL(X) & \longrightarrow & F(X) \\ \downarrow \theta & \swarrow \varphi & & \searrow \bar{\varphi} & \downarrow \bar{\theta} \\ L & \xrightarrow{\sigma} & U(L) & & \end{array}$$

Notice  $\bar{\varphi}^{-1}(\sigma(L))$  is a Lie algebra containing  $X$  thus containing  $FL(X)$ , so it induces a  $\varphi$ .

And for the uniqueness, if there are two  $\varphi_1, \varphi_2$ , then the element that they coincide is a Lie algebra containing  $X$ , thus containing  $FL(X)$ , so  $\varphi_1 = \varphi_2$ . □

**Cor. (3.7.8.19).**  $U(FL(X)) \cong FX$  for any set  $X$ . ┘

*Proof:* Because  $U \circ FL$  and  $F$  are both left adjoint to the forgetful functor  $\text{Alg}^{\text{asso}} \rightarrow \text{Set}$ . □

### Invariant Polynomials

#### $Z(U(\mathfrak{g}))$

**Prop. (3.7.8.20) [Kostant].**  $S\mathfrak{g}$  is a free  $(S\mathfrak{g})^{\mathfrak{g}}$ -module, and for any  $V \in \text{Irr}^{\text{fd}}(\mathfrak{g})$ ,  $\text{Hom}_{\mathfrak{g}}(V, S\mathfrak{g})$  is a free  $(S\mathfrak{g})^{\mathfrak{g}}$ -module of rank  $\dim V[0]$ .  $\lrcorner$

*Proof:* Cf.[Rep of Lie Algebras, Etingof]P42.  $\square$

**Cor. (3.7.8.21) [Kostant].** When  $\text{char } k = 0$ ,  $U(\mathfrak{g})$  is a free  $Z(U(\mathfrak{g}))$ -module, and for any  $V \in \text{Irr}^{\text{fd}}(\mathfrak{g})$ ,  $\text{Hom}_{\mathfrak{g}}(V, U(\mathfrak{g}))$  is a free  $Z(U(\mathfrak{g}))$ -module of rank  $\dim V[0]$ .  $\lrcorner$

*Proof:* This follows from (3.7.8.15).  $\square$

**Thm. (3.7.8.22) [Harish-Chandra Isomorphism].** If  $(\mathfrak{g}, \mathfrak{h})$  is a split semisimple Lie algebra, then there is an isomorphism

$$\text{H-C} : Z(U(\mathfrak{g})) \cong \mathbb{C}[\mathfrak{h}^*]^{W^\bullet}.$$

(It just maps  $z \in Z(L)$  to its pure  $H$  part and transform every indeterminants  $h_i$  to  $h_i - 1$ . e.g.  $z = h^2 + 2h + 1 + 4fe \in Z(\mathfrak{sl}_2)$  is mapped to  $h^2$  in  $S(H)$ .)  $\lrcorner$

*Proof:* Cf.[Etingof, Rep of Lie Groups]P48.  $\square$

**Cor. (3.7.8.23) [Chevalley].** If  $(\mathfrak{g}, \mathfrak{h})$  is a split semisimple Lie algebra, then  $Z(U(\mathfrak{g}))$  is isomorphic to  $k[X_1, \dots, X_{\text{rank } \mathfrak{g}}]$ . In particular,  $Z(U(\mathfrak{sl}_2)) = \mathbb{C}[\frac{1}{2}h^2 + ef + fe]$  (3.7.8.24).  $\lrcorner$

*Proof:* Cf.[Carter]P236.  $\color{red}?$

Because there is a commutative diagram of isomorphisms of algebras:

$$\begin{array}{ccc} S(L)^G & \xrightarrow{\alpha} & P(L)^G \\ \downarrow \eta & & \downarrow \varphi \\ S(H)^W & \xrightarrow{\beta} & P(H)^W \end{array}$$

Where  $P$  is the polynomial ring  $\cong S(L^*)$ , the horizontal is Killing isomorphisms and vertical is the restriction maps. Cf.[Carter prop 13.32]  $\color{red}?$ .

The Harish-Chandra isomorphism (3.7.8.22) gives an isomorphism of algebras  $Z(\mathfrak{g}) \rightarrow S(\mathfrak{h})^W$   $\square$

**Def. (3.7.8.24) [Casimir Element].** If  $\mathfrak{g}$  is semisimple, by (3.7.2.1) the Killing form is non-degenerate, thus we choose a basis  $x_i$  of  $L$  and a dual basis  $y_i$ , then  $c = \sum x_i y_i$  is independent of  $x_i$  chosen by (3.5.11.4), and is called the **Casimir element** of  $U(\mathfrak{g})$ .  $\lrcorner$

**Prop. (3.7.8.25).** If  $\mathfrak{g}$  is semisimple, then the Casimir element lies in  $Z(U(\mathfrak{g}))$ .  $\lrcorner$

*Proof:*  $\square$

### Miscellaneous

**Prop. (3.7.8.26) [Grading on  $U(\mathfrak{sl}_2(\mathbb{C}))$ ].** Let  $H, R, L$  be a basis of  $\mathfrak{sl}_2(\mathbb{C})$  (3.7.2.11), if we define a grading as  $\deg R = 1, \deg H = 0, \deg L = -1$ , then this is descends to a grading on  $U(\mathfrak{g})$ , and the degree 0 part is the ring  $\mathcal{R} = \mathbb{C}[\Delta, H]$ . Also, there is a decomposition:

$$U(\mathfrak{g}) = \bigoplus_{i \geq 0} L^i \mathcal{R} \oplus \bigoplus_{i > 0} R^i \mathcal{R}.$$

$\lrcorner$

## 9 Representations

**Def. (3.7.9.1) [Representations].** A representation of a Lie algebra  $\mathfrak{g}$  over a vector space  $V$  is a Lie homomorphism  $\mathfrak{g} \rightarrow \mathfrak{gl}_V$ . The category of Representations of Lie algebras is denoted by  $\mathcal{Rep}(\mathfrak{g})$ .  $\lrcorner$

**Def. (3.7.9.2) [Tensor Product of Representations].** Let  $(V_1, \pi_1), (V_2, \pi_2)$  be two representations of a Lie algebra  $\mathfrak{g}$ , then  $(V_1 \otimes V_2, \pi_1 \otimes \pi_2)$  is a representation of  $\mathfrak{g}$  given by

$$(\pi_1 \otimes \pi_2)(g)(v_1 \otimes v_2) = \pi_1(g)v_1 \otimes v_2 + v_1 \otimes \pi_2(g)v_2.$$

$\lrcorner$

**Def. (3.7.9.3) [Representation on Tensor Algebras].** If  $(V, \rho)$  is a representation of  $\mathfrak{g}$ , then  $\mathfrak{g}$  acts on  $T(V)$  via (3.7.9.2). Also, the ideals  $I, J$  are invariant under action of  $\mathfrak{g}$ , thus the representation extends to  $\text{Sym}(V)$  and  $\wedge(V)$ . Also it preserves degree, thus it induces representations on  $\text{Sym}^k(V)$  and  $\wedge^k(V)$ .  $\lrcorner$

*Proof:*

$$g(a \otimes a) = g(a) \otimes a + a \otimes g(a) = (a + g(a)) \otimes (a + g(a)) - a \otimes a - g(a) \otimes g(a)$$

and

$$g(a \otimes b - b \otimes a) = g(a) \otimes b + a \otimes g(b) - g(b) \otimes a - b \otimes g(a) = g(a) \otimes b - b \otimes g(a) + a \otimes g(b) - g(b) \otimes a$$

$\square$

**Def. (3.7.9.4) [Dual Representation].** If  $(\varphi, V)$  is a representation of  $\mathfrak{g}$ , we define the **dual representation**  $(\varphi^*, V^\vee)$  as

$$(\varphi^*(g)(v^*), v) = (v^*, \varphi(g)v).$$

$\lrcorner$

**Def. (3.7.9.5) [Trace Form].** The **trace form** of a representation  $(V, \rho)$  of a Lie algebra  $\mathfrak{g}$  is an invariant symmetric form  $\beta_\rho$  defined by  $(x, y) \mapsto \text{tr}(\rho(x) \circ \rho(y))$ .  $\lrcorner$

*Proof:* It is invariant because

$$\begin{aligned} \text{tr}(\rho([x, y]) \circ \rho(z)) &= \text{tr}(\rho(x)\rho(y)\rho(z)) - \text{tr}(\rho(y)\rho(x)\rho(z)) \\ &= \text{tr}(\rho(x)\rho(y)\rho(z)) - \text{tr}(\rho(x)\rho(z)\rho(y)) \\ &= \text{tr}(\rho(x)\rho([y, z])). \end{aligned}$$

$\square$

**Prop. (3.7.9.6).** If  $\rho$  is a faithful representation of  $\mathfrak{g}$  and  $\mathfrak{g}$  is semisimple, then  $\beta_\rho$  is non-degenerate.  $\lrcorner$

*Proof:* The Cartan's criteria (3.7.1.27) shows  $\mathfrak{g}^\perp$  is a solvable sub-Lie algebra, so it must be 0 as  $\mathfrak{g}$  is semisimple (3.7.1.23).  $\square$

**Prop. (3.7.9.7).** Let  $L$  be a simple lie algebra, then any two non-degenerate symmetric invariant bilinear forms on  $L$  is proportional. Because any of this form corresponds to a  $L$ -morphism from  $L$  to  $L^*$ . In particular, when  $L \subset \mathfrak{gl}_n$ , the usual trace is proportional to the Killing form.  $\lrcorner$

**Remark (3.7.9.8) [Rep( $\mathfrak{g}$ )].** For  $\mathfrak{g} \in \text{Lie}_k$ , let  $\text{Rep}^{\text{fd}}(\mathfrak{g})$  denote the category of f.d. representations of  $\mathfrak{g}$ .  
 $\lrcorner$

**Prop. (3.7.9.9) [Schur's Lemma].** Let  $\dim \mathfrak{g} < \infty$  and  $M \in \text{Irr}(\mathfrak{g})$ , then  $\dim M \leq \aleph_0$ . In particular, Shur's lemma holds by (24.1.1.10). In particular,  $Z(U(\mathfrak{g}))$  acts by a character  $\chi : Z(U(\mathfrak{g})) \rightarrow k$ , called the **infinitesimal character** of  $M$ .  
 $\lrcorner$

*Proof:* It is of countable dimensional because  $\dim U(\mathfrak{g})$  is countable.  $\square$

**Prop. (3.7.9.10) [Generalized Eigenspace Decomposition for Nilpotent Lie Algebras].** If  $k = \bar{k}$ ,  $\mathfrak{g}$  is a nilpotent algebra and  $(V, \rho) \in \text{Rep}(\mathfrak{g})$ , then there is a generalized eigenspace decomposition

$$V = \bigoplus_{\lambda \in \mathfrak{g}^*} V^\lambda,$$

where  $V^\lambda$  are the generalized eigenspaces, and they are stable under action of  $\mathfrak{g}$ .  
 $\lrcorner$

*Proof:* We use an induction argument:

If each  $a \in \mathfrak{h}$  has only one eigenvalue, then  $V$  is the generalized eigenspace  $V_\lambda$  for some function  $\lambda$  on  $\mathfrak{h}$ . Then it suffices to show that  $\lambda$  is linear. But this is because by Lie's theorem elements of  $\mathfrak{h}$  has a common eigenvector.

If for some  $a_0$ ,  $\text{ad}(a_0)$  has two eigenvalues. Now  $\mathfrak{h}$  is nilpotent, so  $\mathfrak{h} \subset \mathfrak{h}_0^{a_0}$ , and hence  $\pi(h)V_\lambda^{a_0} \subset V_\lambda^{a_0}$  for any  $\lambda$  by (3.7.1.12).

As  $k$  is alg.closed,  $V$  can be written as a sum of generalized eigenspaces of  $a_0$ , and each generalized eigenspace is a subrepresentation of  $\mathfrak{h}$ , thus we can use induction.  $\square$

**Def. (3.7.9.11) [Casimir Operator].** Let  $\mathfrak{g}$  be a semisimple Lie algebra of dimension  $n$ , and  $\beta : \mathfrak{g} \times \mathfrak{g} \rightarrow k$  a non-degenerate invariant bilinear form on  $\mathfrak{g}$ . Let  $e_i$  be a basis of  $\mathfrak{g}$  and  $e'_i$  be the dual basis under  $\beta$ , then  $c = \sum e_i e'_i \in U(\mathfrak{g})$  is independent of the basis, and lies in the center of  $U(\mathfrak{g})$ .

Now the trace form  $\beta_V$  for a faithful representation  $\mathfrak{g} \rightarrow \mathfrak{gl}_V$  of  $\mathfrak{g}$  is non-degenerate and invariant (3.7.9.6), then the corresponding elements  $c_\rho$  is called the **Casimir element** of  $(V, \rho)$ , and the action  $c_V$  of  $c_\rho$  on  $V$  is called the **Casimir operator** of  $(V, \rho)$ .

The Casimir operator  $c_V$  is a  $\mathfrak{g}$ -module homomorphism, and has trace  $n$ .  
 $\lrcorner$

*Proof:* The independence of basis is by (3.5.11.4). To show it is in the center of  $\mathcal{U}(\mathfrak{g})$ , Cf. [Mil13].P50?.

Casimir operator  $c_V$  is a  $\mathfrak{g}$ -module homomorphism follows from the fact  $c_\rho$  is in the center of  $U(\mathfrak{g})$ , and its trace is

$$\text{tr}(c_V) = \sum_i \text{tr}(e_i e'_i) = \sum_i (\beta_V(e_i, e'_i)) = n$$

$\square$

**Def. (3.7.9.12) [Unimodular Lie Algebras].** A f.d Lie algebra  $\mathfrak{g}$  is called **unimodular** if  $\wedge \text{ad}$  is a trivial representation of  $\mathfrak{g}$ .  
 $\lrcorner$

**Prop. (3.7.9.13).**

- If  $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ , then  $\mathfrak{g}$  is unimodular.
- If  $\mathfrak{g}$  is nilpotent, then  $\mathfrak{g}$  is unimodular.
- If  $\mathfrak{g}_1, \mathfrak{g}_2$  is nilpotent, then  $\mathfrak{g}_1 \oplus \mathfrak{g}_2$  is unimodular.



- If  $\mathfrak{g}$  is reductive, then  $\mathfrak{g}$  is unimodular. ┘

**Lemma (3.7.9.14) [Zassenhaus].** Let  $\mathfrak{g}$  be a Lie algebra and  $\mathfrak{g}'$  an ideal of  $\mathfrak{g}$ . A representation  $\rho'$  of  $\mathfrak{g}'$  extends to a representation  $\rho$  of  $\mathfrak{g}$  that  $\mathfrak{n}_{\rho'}(\mathfrak{g}') \subset \mathfrak{n}_{\rho}(\mathfrak{g})$  if there exists a Lie subalgebra  $\mathfrak{h}$  of  $\mathfrak{g}$  that  $\mathfrak{g} = \mathfrak{g}' \oplus \mathfrak{h}$  and  $[\mathfrak{h}, \mathfrak{g}'] \subset \mathfrak{n}_{\rho'}(\mathfrak{g}')$ . If moreover  $\text{ad}_{\mathfrak{g}}(x)|_{\mathfrak{g}'}$  is nilpotent for all  $x \in \mathfrak{h}$ , then  $\rho$  can be chosen that  $\mathfrak{h} \subset \mathfrak{n}_{\rho}(\mathfrak{g})$ . ┘

*Proof:* Cf. [Mil13]P65. □

**Prop. (3.7.9.15) [Nilpotent Representation Extension].** Let  $\mathfrak{g}$  be a Lie algebra,  $\mathfrak{a}$  a nilpotent ideal of  $\mathfrak{g}$ , and  $\rho$  a representation of  $\mathfrak{a}$  that  $\rho(x)$  is nilpotent for all  $x \in \mathfrak{a}$ . Then  $\rho$  extends to a representation  $\rho'$  of  $\mathfrak{g}$  that  $\rho'(x)$  is nilpotent for all  $x \in \mathfrak{n}$  the largest nilpotent ideal of  $\mathfrak{g}$ . ┘

*Proof:* □

**Thm. (3.7.9.16) [Ado].** Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$  of char 0, then there exists a faithful representation  $\rho$  of  $\mathfrak{g}$  that  $\rho(\mathfrak{n})$  consists of nilpotent endomorphisms, where  $\mathfrak{n}$  is the largest nilpotent radical. If  $\mathfrak{g}$  is of f.d., then this representation can be chosen to be of f.d.. In particular, any finite dimensional Lie algebra can be embedded in some  $\mathfrak{gl}(n, k)$ . ┘

*Proof:* This is true for any commutative Lie algebras, for example we can use tensor products of  $c \mapsto \begin{bmatrix} 0 & c \\ 0 & 0 \end{bmatrix}$ . Choose a faithful representation of the center  $\mathfrak{c}$  of  $\mathfrak{g}$  that every element is mapped to a nilpotent endomorphism, and then extend it to a representation  $\rho_1$  of  $\mathfrak{g}$  by (3.7.9.15). Let  $\rho_2$  be the adjoint representation of  $\mathfrak{g}$ , and  $\rho = \rho_1 \oplus \rho_2$ . Then  $\ker(\rho) = \ker(\rho_1) \cap \ker(\rho_2) = \ker(\rho_1) \cap \mathfrak{c} = 0$ , so this is faithful. And it sends an element in  $\mathfrak{n}$  to a nilpotent endomorphism because each  $\rho_1$  and  $\rho_2$  do. □

**Remark (3.7.9.17).** In fact, this is true for Lie algebras over a field of char  $p$ , too, Cf. [JACOBSON, N. 1962. Lie algebras.] Chap 6.3. ┘

### Semisimple Representations

For representations of a split semisimple Lie algebra, See [Representations of Semisimple Lie Algebras](#).

## 10 Amitsur-Levitski

### Preliminary

Notice that in this paper, Kostant considers **reductive** lie groups. But in the range of this paper, the abelian part makes no contribution in the alternative part because they commutes with all elements. So We well just consider a **semisimple** Lie algebra in order to get a non-degenerate Killing form.

**Prop. (3.7.10.1).**

$$\Gamma(\wedge^{2k}(\mathfrak{g})) = R^k \in S^k(\mathfrak{g})$$

$$\Gamma(x_1 \wedge \cdots \wedge x_k) = \sum_{v \in \Pi_k} [x_{v(1)}, x_{v(2)}] \cdots [x_{v(2k-1)}, x_{v(2k)}]$$

┘

*Proof:* The proof is in fact simple, just notice that for every  $v \in \Pi_k$  a representative of the subgroup  $\Sigma_{2k,2}$  permuting the unordered pairs  $\{(1,2), (3,4), \dots, (2k-1, 2k)\}$ , the element in  $v\Sigma_{2k,2}$  in fact combine in pairs to  $[x_{v(2i-1)}, x_{v(2i)}]$  and together the  $k!$  permutation of them compose a  $[x_{v(1)}, x_{v(2)}] \cdots [x_{v(2k-1)}, x_{v(2k)}]$ .  $\square$

Later he finds the dual of  $\Gamma$ , that is:

**Prop. (3.7.10.2).**  $\gamma$  is  $\mathcal{B}$  dual to  $\Gamma$ ,

$$(\Gamma(\zeta), y_1 \cdots y_r) = (-1)^r (\zeta, dy_1 \wedge \cdots \wedge dy_r) \quad \forall \zeta \in \wedge^{2r} \mathfrak{g} \text{ and } y_i \in \mathfrak{g}.$$

In particular,

$$\Gamma(\zeta)(x) = \frac{(-1)^r}{r!} (\zeta, (dx)^r) \quad \forall \zeta \in \wedge^{2r} \mathfrak{g} \text{ and } x \in \mathfrak{g}.$$

┘

For the proof just notice that  $(-dw, x_i \wedge x_j) = (w_i, [x_i, x_j])$  and

$$(x_1 \otimes \cdots \otimes x_r, y_1 \otimes \cdots \otimes y_r) = \sum_{\sigma} (x_1, y_{\sigma(1)}) \cdots (x_r, y_{\sigma(r)})$$

So  $\dim$  of  $R^k$  equals the  $\dim$  of image of  $\gamma_r$ , that is, spanned by  $(dx)^k$  (because they are dual).

We say that a representation of  $\mathfrak{g}$  satisfies  $m$ -fold standard identity if the alternating sum of any  $m$  elements of image of  $\mathfrak{g}$  is 0. Obviously, this is equivalent to:

$$\tau(R^k(\mathfrak{g})) \subset \ker \pi_V$$

Now let  $o(\mathfrak{g})$  be the maximum rank of  $dw, w \in \mathfrak{g}$ , then by the discussion of the first paper, when  $\mathfrak{g}$  is semisimple,  $o(\mathfrak{g}) = r$ . So the  $2r$ -identity is satisfied by any representation of  $\mathfrak{g}$ .

Furthermore a prop of [Harish-Chandra] assert for any nonzero element  $u \in U(\mathfrak{g})$ , there is a representation such that  $\pi(U) \neq 0$ . So this is a sharp bound for general representations.

But one might naturally ask: Can we find the specific bound for a particular representation of a specific  $\mathfrak{g}$ ? The answer is YES.

**Prop. (3.7.10.3).**  $\gamma$  vanishes on the ideal  $J'_+ \cdot S(\mathfrak{g})$ .  $\square$

*Proof:* The proof comes from the observation  $\pi$  is a  $G$ -map and by **Cartan-Koszul** theory, invariant elements in  $\wedge \mathfrak{g}$  are naturally isomorphic to the cohomology of  $\mathfrak{g}$  and  $\gamma(w) = -dw$  is clearly exact, Thus  $\gamma(w_1 w_2 \dots w_i) = (-1)^i dw_1 \wedge \cdots \wedge dw_i$  is exact too. So  $\gamma(w) = 0$ .  $\square$

**Cor. (3.7.10.4).**  $M = R^r \in H^r$ , so it consists of harmonic functions.  $\square$

*Proof:*  $(u, \Gamma(y)) = (\gamma(u), y) = 0 \quad \forall u$  invariant, by Thm; so  $R^k = \text{Image } \Gamma$  is harmonic.  $\square$

### Generalized Amitsur-Levitski

Let  $E^k \subset U(\mathfrak{g})$  be spanned by  $y^k, y$  nilpotent in  $\mathfrak{g}$ , and  $Z$  the center.

in [Kos97] Kostant proved that the PBW isomorphism  $\delta : S(\mathfrak{g}) \longrightarrow U(\mathfrak{g})$  induces  $\delta(J) = Z, \delta(H) = E$ .

so  $\tau : T(\mathfrak{g}) \longrightarrow U(\mathfrak{g})$  induces

$$\tau(A^{2k}(\mathfrak{g})) = \delta(R^k(\mathfrak{g})) \subset E^k.$$

Define  $\epsilon(\pi)$  the minimum integer  $k$  that  $\pi(y)^k = 0, \forall y$  nilpotent in  $\mathfrak{g}$ . Then clearly:

**Prop. (3.7.10.5)[Generalized Amitsur-Levitski].**  $\pi$  satisfies the  $2\epsilon(\pi)$ -fold standard identity.  $\lrcorner$

**Prop. (3.7.10.6).** If  $\pi$  satisfies the  $m$ -identity, it satisfies the  $m+1$ -identity (By taking a summation on a fixed first element  $\sigma(1)$ ).  $\lrcorner$

**Prop. (3.7.10.7).**

- Let  $\pi$  be the natural representation of  $\mathfrak{gl}_n$  on  $\mathbb{C}^n$  then  $\epsilon(\pi) = n$ .
- If  $n$  even and  $\pi$  the natural representation of skew-symmetric matrixes on  $\mathbb{C}^n$  then  $\epsilon(\pi) = n - 1$ .

From this one derives the **classical Amitsur-Levitski prop** that  $GL_n(\mathbb{C})$  satisfies the  $n$ -fold standard identity.  $\lrcorner$

*Proof:* 1: the **abstract Jordan decomposition** which assures  $x$  is nilpotent in  $\mathfrak{gl}_n$  if  $\pi(x)$  is nilpotent.

2: comes from the **Lacobson-Morosov** Thm that any nilpotent element of  $\mathfrak{g}$  is contained in a  $\mathfrak{sl}_2$ -triple. Thus we only need to show that  $W$  is reducible considered as this  $\mathfrak{sl}_2$ -triple-module.

But then an irreducible representation of  $\mathfrak{sl}_2$  preserves a non-degenerate bilinear form it must be odd dimensional cause a non-degenerate bilinear form is equivalent to a  $\mathfrak{g}$ -map from  $V$  to  $V^*$ .

And there can be constructed an anti-symmetric form defined on the  $\mathfrak{sl}_2$ -representation on  $\text{Sym}_{2k}[x, y]$  by (3.7.10.9), so there can't exist symmetric  $\mathfrak{g}$ -invariant form. So this representation must be reducible.  $\square$

**Cor. (3.7.10.8)[Classical Amitsur-Levitski].** By (3.7.10.5),

$$[[x_1, x_2, \dots, x_n]] = \sum_{\sigma} \text{sgn}(\sigma) x_{\sigma(1)} x_{\sigma(2)} \cdots x_{\sigma(k)} = 0 \quad \forall x_i \in \mathfrak{gl}_n$$

called  $n$ -fold **standard identity**.

$\lrcorner$

**Prop. (3.7.10.9).** For a construction of the anti-symmetric form, notice

$$\pi(g)f(x_1, x_2) = f(g_{11}x_1 + g_{12}x_2, g_{21}x_1 + g_{22}x_2).$$

Set

$$v_k = \binom{m}{k} x_1^{m-k} x_2^k, \quad \Omega(v_k, v_{m-k}) = (-1)^k \binom{m}{k}. \quad \Omega(v_k, v_p) = 0, \quad k + p \neq m.$$

One verifies:

$$\Omega(g \cdot u, g \cdot v) = (\det g)^m \Omega(u, v) \quad \forall g \in GL(2, \mathbb{C})$$

So when  $m = n - 1$  is odd, this is a symplectic form preserved by  $\mathfrak{sl}_2$ .  $\lrcorner$

### A computable Formula

Finally, Kostant gave a computable formula for determining  $\epsilon(\pi)$ . Clearly we just need to consider irreducible representation.

Let  $\pi_\lambda$  be the irreducible representation of highest weight  $\lambda$ , then the dual representation  $\pi_{\lambda'}$  has highest weight the negative of the lowest weight of  $\pi_\lambda$ , that is,  $-w_o(\gamma)$ .

But then  $\lambda + \lambda'$  is a sum of simple positive roots.  $\lambda + \lambda' = \sum_{i=1}^n m_i \alpha_i$ . Put  $\epsilon(\lambda) = 1 + \sum_{i=1}^n m_i$ . then:

**Prop. (3.7.10.10).**  $\epsilon(\pi_\lambda) = \epsilon(\lambda)$ .  $\lrcorner$

*Proof:* Just choose a  $\mathfrak{sl}_2$ -triple  $\{e, x, f\}$  with  $\alpha(x) = 2 \forall \alpha$  simple root. Then  $\lambda(x)$  and  $-\lambda'(x)$  are respectively the maximal and minimal eigenvalues of  $\pi(x)$ .  $(\lambda + \lambda')(x) = 2(\epsilon(\lambda) - 1)$ . Thus  $f$  has nilpotent degree  $\epsilon(\lambda)$ . And any nilpotent element action increases the eigenvalue of a eigenvector of  $x$  by at least 2, the prop follows.  $\square$

### Further Work

(cf. [Pro76]) Another different proof of the Amitsur-Levitski theorem is given by Kostant using techniques related to **trace identities**. It turns out that method sheds more light. Later is studied the polynomial of matrices invariant under the conjugation action.

Artin conjectured that all the invariants is polynomials of the Trace polynomial  $Tr(A_1 A_2 \cdots A_n)$  (Proved)

And further, the relations among these invariant all turned out to be consequences of the prop of Hamilton-Cayley. All this is made into the **Invariant Theory**.

#### Prop. (3.7.10.11) [Interesting results].

1. If an algebra over a field of characteristic 0 satisfies the identity  $X^n = 0$ , then it satisfies all the identities of  $n \times n$  matrices.
2. The space of multilinear identities of degree  $m$  of  $n \times n$  matrices can be described completely in terms of Young diagrams.

┘

## 11 Lie $p$ -Algebras

**Remark (3.7.11.1).** In this subsection, let  $k$  be a field of characteristic  $p$ .

┘

**Def. (3.7.11.2) [Lie  $p$ -Algebras].** Let  $x_0, x_1$  be elements of a Lie algebra over  $k$  of characteristic  $p$ , then for  $0 < r < p$ , let  $s_r(x_0, x_1)$  denote  $\frac{1}{r}$  times coefficient of  $t^{r-1}$  in the expression of  $\text{ad}_{tx+y}^{p-1}(x)$ .

Then a **Lie  $p$ -algebra** is a Lie algebra  $\mathfrak{g}$  over  $k$  equipped with a map  $x \mapsto x^{[p]} : \mathfrak{g} \rightarrow \mathfrak{g}$  such that

- $(cx)^{[p]} = c^p x^{[p]}$  where  $c \in k$ .
- $\text{ad}(x^{[p]}) = (\text{ad}(x))^p$ .
- $(x + y)^{[p]} = x^{[p]} + y^{[p]} + \sum_{r=1}^{p-1} s_r(x, y)$ .

┘

**Example (3.7.11.3).** If  $L$  is an Abelian Lie algebra, then it can be regarded as a Lie  $p$ -algebra by assigning  $x^{[p]} = 0$ .

If  $A$  is an associative algebra, then it can be given a Lie  $p$ -algebra structure by assigning  $x^{[p]} = x^p$ .

┘

*Proof:* To show  $A$  is a Lie  $p$ -algebra, we check  $\text{ad}(x)^p(y) = (l_x - r_x)^p(y) = l_x^p - r_x^p(y) = \text{ad}(x^p)(y)$ . For the third formula, notice

$$\text{ad}(tx + y)^{p-1}(x) = \sum_{i=0}^{p-1} (-i)^i \binom{p-1}{i} (tx + y)^{p-1-i} x (tx + y)^i$$

Notice that  $(-i)^i \binom{p-1}{i} \equiv 1 \pmod{p}$ , so  $s_r(x, y)$  is equivalent to the sum of words of  $x, y$  with  $r$   $x$ s. So the third formula clearly holds.  $\square$

**Def. (3.7.11.4) [Representations].** A **representation of a Lie  $p$ -algebra**  $\mathfrak{g}$  over a  $k$ -vector space  $V$  is a homomorphism  $\mathfrak{g} \rightarrow \mathfrak{gl}_V$  of Lie  $p$ -algebras.

┘

**Def. (3.7.11.5) [Universal Enveloping  $p$ -Algebra].** The **universal enveloping  $p$ -algebra** of a Lie  $p$ -algebra  $\mathfrak{g}$  is defined to be

$$U^{[p]}(\mathfrak{g}) = T(\mathfrak{g})/J^{[p]}, \quad T(\mathfrak{g}) = \bigoplus_{n=0}^{\infty} \mathfrak{g}^{\otimes n}$$

which is a graded algebra  $T(\mathfrak{g})$  quotients the ideal  $J = (\{x \otimes y - y \otimes x - [xy], x^{\otimes p} - x^{[p]}\})$ .

There is a natural linear map  $\sigma : \mathfrak{g} \rightarrow U^{[p]}(\mathfrak{g})$ . ┘

**Prop. (3.7.11.6).** The universal enveloping  $p$ -algebra  $U^{[p]} : \mathfrak{g} \mapsto U^{[p]}(\mathfrak{g})$  defines a functor  $LiepAlg \rightarrow AssAlg$  that is left adjoint to the canonical functor  $AssAlg_k \rightarrow LiepAlg_k$  (3.7.11.3). ┘

*Proof:* For any associative algebra  $A$  and a morphism of Lie  $p$ -algebras  $\mathfrak{g} \rightarrow [A]$ , there is easily seen a morphism  $U^{[p]}(\mathfrak{g}) \rightarrow A$ , and it is unique. □

**Cor. (3.7.11.7) [Representation as Modules].** A representation of a Lie  $p$ -algebra  $\mathfrak{g}$  (3.7.11.4) is the same as a representation of  $U^{[p]}(\mathfrak{g})$ . ┘

**Prop. (3.7.11.8).** Let  $e_i$  be a  $k$ -vector space basis of  $\mathfrak{g}$ , then the monomials

$$\{e_{i_1}^{n_{i_1}} e_{i_2}^{n_{i_2}} \dots e_{i_r}^{n_{i_r}} \mid i_1 < i_2 < \dots < i_r, 0 < n_{i_k} < p\}$$

for different  $r$  form a basis of  $U^{[p]}(\mathfrak{g})$ . ┘

*Proof:* This is a consequence of PBW theorem (3.7.8.5). □

**Cor. (3.7.11.9).** If  $\dim_k \mathfrak{g} < \infty$ , then so does  $U^{[p]}(\mathfrak{g})$ , and the map  $i : \mathfrak{g} \rightarrow U^{[p]}(\mathfrak{g})$  is injective. ┘

### 3.8 Infinite Dimensional Lie Algebras

References are [Car05].

#### 1 Kac-Moody Algebras

The point about Kac-Moody algebras is that they share many of the properties enjoyed by semi-simple Lie algebras.

#### Representations of Symmetrizable Kac-Moody Algebras

##### Kac's Character Formula

#### Representations of Affine Kac-Moody algebras

#### 2 Borcherds Algebras

References are [Generalized Kac-Moody algebras, Borcherds].

**Def. (3.8.2.1) [Borcherds Algebras].** A **Borcherds algebra** or a generalized Kac-Moody algebra is a graded real Lie algebra  $L = \bigoplus_{n \in \mathbb{Z}} L_n$  s.t.

- $\dim L_n < \infty$  for  $n \neq 0$ .
- $L$  is diagonalizable w.r.t.  $L_0$ .
- $L$  possesses an involution  $\omega$  which is  $-\text{id}$  on  $L_0/(L_0 \cap Z(L))$  and  $\omega(L_m) = L_{-m}$ .
- $L$  possesses an invariant bilinear form  $(-, -)$  s.t.  $(L_m, L_n) = 0$  if  $m + n \neq 0$ .
- the invariant bilinear form is invariant under  $\omega$ , and the bilinear form  $\langle x, y \rangle = (x, \omega(y))$  is negative definite on  $L_m$  for any  $m \neq 0$ .
- $L_0 \subset [L, L]$ .

┘

*Proof:* Cf.[Generalized Kac-Moody algebras, Borcherds].

□

**Cor. (3.8.2.2).** For a generalized Kac-Moody algebra,

- $[L_0, L_0] = 0$ .

┘

**Def. (3.8.2.3) [Universal Borcherds Algebras].** Let  $I \subset N$  and  $\mathfrak{A} = (a_{ij}) \in \text{Mat}(I \times I; \mathbb{R})$  s.t.

- $a_{ij} < 0$  for  $i \neq j$ .
- If  $a_{ii} > 0$ , then  $2a_{ij}/a_{ii} \in \mathbb{Z}$  for any  $j$ .

┘

**Prop. (3.8.2.4).** Let  $L$  be a universal Borcherds algebra and  $I \in \text{Ideal}(L)$  is contained in  $Z(L)$ . Then  $L/I$  retains the structure of a Borcherds algebra.

┘

*Proof:* Cf.[Car05]P522.

□

**Thm. (3.8.2.5) [Classifying Borcherds Algebras].** Let  $L$  be a Borcherds algebra, then there exists a unique universal Borcherds algebra  $L_u$  (3.8.2.3) and some graded homomorphism  $f : L_u \rightarrow L$  (n.n. unique) s.t.

- $\ker(f) \subset Z(L_u)$ .
- $\text{Im}(f)$  is an ideal of  $L$ .
- $L$  is a semi-direct product of  $\text{Im}(f)$  with an Abelian Lie algebra of outer derivations lying in  $L_0$  and preserving all subspaces  $\mathbb{R}\bar{e}_i$  and  $\mathbb{R}\bar{f}_i$ .

Moreover,  $f$  respects the involutions and the bilinear forms.  $\lrcorner$

*Proof:*  $\square$

### Representations of Borchers Algebras

**Thm. (3.8.2.6) [Borchers' Character Formula].** Let  $L$  be a Borchers algebra,  $\lambda \in P^+$  a dominant integral weight and  $L_\lambda$  the corresponding irreducible highest weight representation, then

$$\text{ch } L_\lambda = \frac{\sum_{w \in W} \varepsilon(w) w \left( \sum_{\Psi} (-1)^{\#\Psi} e^{\lambda + \rho - \Psi} \right)}{e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})^{m_\alpha}},$$

where  $m_\alpha = \dim L[\alpha]$ , and  $\Psi$  runs through all finite subsets of mutually orthogonal imaginary fundamental roots, and  $\rho \in Q \otimes \mathbb{R}$  s.t.

$$\langle \rho, \alpha_i \rangle = \frac{1}{2} \langle \alpha_i, \alpha_i \rangle$$

for any real fundamental root  $\alpha_i$ .

The denominator  $\Delta_L = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})^{m_\alpha}$  is called the **Borchers' denominator** of  $L$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (3.8.2.7) [Borchers' Denominator Formula].**

$$\Delta_L = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})^{m_\alpha} = \sum_{w \in W} w \left( e^\rho \sum_{\Psi} (-1)^{\#\Psi} e^{-\sum \Phi} \right).$$

$\lrcorner$

*Proof:*  $\square$

### the Monstrous Lie Algebra

**Def. (3.8.2.8) [The Monstrous Lie Algebra].** Let  $\mathfrak{A}$  be a  $\mathbb{N} \times \mathbb{N}$  block matrix with blocks of rows and columns parametrized by  $\mathbb{N}$  s.t.

- The  $(i, j)$ -th block  $B_{i,j} \in \text{Mat}(c_j \times c_j; \mathbb{Z})$ , where  $c(i)$  is the coefficient of  $q^i$  in  $j(z)$  (26.3.4.22) if  $i \in \mathbb{Z}_+$  and  $c_0 = 1$ .
- $B_{0,0} = 2$ .
- For  $n \in \mathbb{Z}_+$ , each entry in  $B_{0,n}$  and  $B_{n,0}$  is  $-(n-1)$ .
- For  $m, n \in \mathbb{Z}_+$ , each entry in  $B_{m,n}$  is  $-(m+n)$ .

Then the **Monstrous Lie algebra** is defined to be the Borchers algebra  $\mathfrak{M} = L(\mathfrak{A})/Z(L(\mathfrak{A}))$  (3.8.2.3)(3.8.2.4).  $\lrcorner$

**Thm. (3.8.2.9) [Borcherds' Monstrous Denominator Identity].**

$$j(p) - j(q) = p^{-1} \prod_{m \in \mathbb{Z}_+, n \in \mathbb{Z}} (1 - p^m q^n)^{c_{mn}}.$$

┘

*Proof:* Cf. [Car05]P539. ?

□

### Kac-Weyl Character and Denominator Formulae

#### 3 Vertex Operator Algebras

#### 4 Moonshine Conjectures

*“In the broad light of day mathematicians check their equations and their proofs, leaving no stone unturned in their search for rigour. But, at night, under the full moon, they dream, they float among the stars and wonder at the miracle of the heavens. They are inspired. Without dreams there is no art, no mathematics, no life.”*

–M. Atiyah

References are [Bor98], [Work of Borcherds], [Vertex Operator Algebras and the Monster], [Infinite-Dimensional Lie Algebras, Kac], [Monstrous Moonshine: The first twenty-five years], [Monstrous moonshine and monstrous Lie superalgebras, Borcherds]

**Thm. (3.8.4.1) [Monstrous Moonshine].** There is a naturally defined graded infinite dimensional module, called the **monstrous module**  $V = \bigoplus_{n \in \mathbb{Z}} V_n$  of the monster group  $\mathbb{M}$  s.t. for any  $g \in \mathbb{M}$ , the McKay-Thompson series

$$T_g(\tau) = \sum_{n \geq -1} \text{tr}(g|V_n) e^{2\pi i n \tau}$$

is a Hauptmodul for a discrete subgroup  $\Gamma \subset \text{SL}(2; \mathbb{Z})$  of genus 0 and period  $r$  at the cusp. ┘

*Proof:* Cf. [Borcherds, 1992]. ?

The proof is sketched as follows:

- [Frenkel-Lepowsky-Meurman] constructed a candidate of the monstrous module  $V$ , and it has an algebraic structure making it a vertex algebra.
- Use the vertex algebra structure on  $V$  and the Goddard-Thorn no-ghost theorem in [Goddard and C. B. Thorn, Compatibility of the dual Pomeron with unitarity and the absence of ghosts in the dual resonance model] to construct a Lie algebra acted on by the monster group, called the **monstrous Lie algebra**.
- Show that the monster Lie algebra is a generalized Kac-Moody algebra (3.8.2.1), then use the (twisted) Weyl-Kac denominator formula to show that  $T_g(\tau)$  is a “completely replicable function”.
- Completely replicable functions were modular functions of Hauptmoduls for genus 0 groups.

□



**Thm. (3.8.4.2) [Duncan-Mertens-Ono, 2017].** There exists an infinite dimensional module

$$V = \bigoplus_{n \in \mathbb{Z}_+, n \equiv 0, 3 \pmod{4}} V_n$$

of the O’Nan group (3.1.11.5)  $\mathbb{ON}$  s.t. for any  $g \in \mathbb{ON}$ , the Mckay-Thompson series

$$F_g(\tau) = q^{-4} + 2 + \sum_{n>0} \text{tr}(g|V_n) e^{2\pi i n \tau}$$

is a meromorphic modular form of weight  $\frac{3}{2}$  on  $\Gamma_0(4N)$ , where  $N$  is the order of  $g$ . ┘

*Proof:* □

### 3.9 Reflection Groups and Coxeter Groups

References are [Ser87, Hum90, Kna96, Mil17, K-P23] and [Etingof, Lie Algebras].

#### Notation(3.9.0.1).

- Let  $V \in \text{Vect}^{\text{fin}}/\mathbb{R}$ .
- We usually use  $W$  to denote a Weyl-group, use  $\Phi$  to denote a root system, use  $\Psi$  to denote an affine root system, use  $\alpha$  to denote a root, use  $\psi$  to denote an affine root.

┘

#### 1 Coxeter-Systems

**Def.(3.9.1.1) [Coxeter-Systems].** A **Coxeter-system** is a pair  $(W, S)$  where  $W \in \text{Grp}$ ,  $S \subset W$  s.t.  $\#S < \infty$  and  $W$  has a representation of the form

$$W = \{S | (st)^{n(s,t)} = 1\},$$

where  $S \subset W$  and  $n(s, s) = 1, n(s, t) \in \mathbb{Z}_{\geq 2} \cup \{\infty\}$  for any  $s, t \in S$ , where  $n(s, t) = \infty$  means no relation between  $s$  and  $t$ . The pair  $(W, S)$  is called a **Coxeter-system**.

The rank of the Coxeter-system is defined to be  $\#S$ .

┘

**Def.(3.9.1.2) [Coxeter-Graphs].** For a Coxeter-system  $(W, S)$ (3.9.1.1), the **Coxeter-graph** is the simple graph  $G(W, S) \in \text{Graph}$  s.t.  $V(G(W, S)) = S$ , and  $s \neq t \in S$  are connected by an edge iff  $n(s, t) \geq 3$ .

$(W, S)$  is called an **irreducible Coxeter-system** if  $G(W, S)$  is connected.

┘

**Def.(3.9.1.3) [Length Functions].** For a Coxeter system  $(W, S)$ (3.9.1.1), define the length function  $\text{length} : W \rightarrow \mathbb{N}$  s.t. for any  $w \in W$ ,  $\text{length}(w)$  is the smallest number  $m \in \mathbb{N}$  s.t.  $w$  can be written as a product of  $m$  elements in  $S$ . A way to write  $w \in W$  as a product of  $\text{length}(w)$ -many elements in  $S$  is called a **reduced expression** of  $w$ .

┘

**Thm.(3.9.1.4) [Representations of Coxeter-Groups].** For a Coxeter-groups  $(W, S)$ (3.9.1.1), let  $V[S] = k\{v_s\}_{s \in S}$ , and define a symmetric bilinear form  $B$  on  $V$  by

$$B(s, s) = 1, \quad B(s, t) = -\cos \frac{\pi}{n(s, t)}, \quad \text{where } s, t \in S.$$

Define reflections

$$\sigma_s : V[S] \rightarrow V[S] : \sigma_s(\lambda) = \lambda - 2B(v_s, \lambda)v_s.$$

It is easy to verify that  $\sigma_s$  preserves  $B$ . Then there is a unique map  $\rho : W \rightarrow \text{O}(V, B)$  sending  $s \in S$  to  $\sigma_s$ , called the **canonical representation of Coxeter-groups**.

┘

*Proof:* We verify that for any  $s \neq t \in S$ , the order of  $\sigma_s \sigma_t$  is exactly  $n(s, t)$ : Let  $W_{s,t}$  be the subspace of  $V$  generated by  $s, t$ , and let  $W_{s,t}^\perp$  be the orthogonal complement w.r.t.  $B$ , then clearly  $\sigma_s, \sigma_t$  preserves  $W_{s,t}$  and fixes elements of  $W_{s,t}^\perp$ .

And on  $W_{s,t}$ , if  $n(s, t) = \infty$ , then it can be verified that  $(\sigma_s \sigma_t)^k v_s = 2k(v_s + v_t) + v_s$ , so  $\sigma_s \sigma_t$  has infinite order, and if  $n(s, t) < \infty$ , then  $B$  is positive-definite on  $W_{s,t}$  and  $v_s, v_t$  has angle  $\pi - \pi/n(s, t)$ . So  $\sigma_s \sigma_t$  is just rotation by angle  $2\pi/n(s, t)$ , which has order  $m$ .

Then we get a map  $\rho : W \rightarrow \text{GL}(V[S])$ . It suffices to show that  $\rho$  is injective:

□

**Thm. (3.9.1.5).** For a Coxeter-groups  $(W, S)$  (3.9.1.1) and  $w \in W, s \in S$ , if  $\text{length}(ws) > \text{length}(w)$ , then  $w(v_s) > 0$ . And if  $\text{length}(ws) < \text{length}(w)$ , then  $w(v_s) < 0$ .  $\lrcorner$

*Proof:* Cf. [Hum90]P111. ?  $\square$

**Cor. (3.9.1.6) [Canonical Representations are Faithful].** For a Coxeter-groups  $(W, S)$  (3.9.1.1), the canonical representation (3.9.1.4) is faithful.  $\lrcorner$

**Prop. (3.9.1.7) [Canonical Representations of Irreducible Coxeter-Groups].** Situation as in (3.9.1.4), if  $(W, S)$  is irreducible (3.9.1.2), then:

- Any properly-contained  $W$ -invariant subspace of  $V[S]$  is contained in the radical  $\text{rad}(V)$ .
  - If  $B$  is degenerate, then  $V$  fails to be completely reducible as a  $W$ -representation.
  - If  $B$  is non-degenerate, then  $V$  is irreducible as a  $W$ -representation.
  - $\text{End}_W(V) = \mathbb{R}$ .
- $\lrcorner$

*Proof:* Cf. [Hum90]P131 ?  $\square$

**Prop. (3.9.1.8) [Conjugations between  $S$ ].** Let  $(W, S)$  be a Coxeter-system, then  $s, s' \in S$  are conjugate in  $W$  iff there exists a sequence  $s_0 = s, s_1, \dots, s_n = s'$  s.t.  $n(s_i, s_{i+1}) \in 2\mathbb{Z} + 1$  for each  $i \in [n-1]$ .  $\lrcorner$

*Proof:* Cf. [Hum90]P110.  $\square$

## 2 Finite Reflection Groups

### Reflections

**Def. (3.9.2.1) [Reflections].** A **reflection** on  $V$  is a linear transformation  $s \in \text{End}(V)$  that has  $\dim V - 1$  eigenvalues 1 and one eigenvalue  $-1$ .  $\lrcorner$

**Prop. (3.9.2.2).** Let  $\alpha \in V$  and  $\alpha^\vee \in V^\vee$  that  $\langle \alpha, \alpha^\vee \rangle = 2$ , then

$$s_\alpha : x \mapsto x - \langle x, \alpha^\vee \rangle \alpha$$

is a reflection, and every reflection reversing the vector  $\alpha$  is of this form.  $\lrcorner$

*Proof:* Clearly  $s_\alpha$  is a reflection, and if  $s$  is any reflection, then  $\alpha^\vee$  is the composition of the quotient map  $V \rightarrow V/H$  with the map  $V/H \rightarrow \mathbb{F}$  sending  $\alpha + H$  to 1.  $\square$

**Lemma (3.9.2.3).** Let  $R \subset V$  be a finite set s.t.  $\text{span}\{R\} = V$ , then for any  $\alpha \in V$ , there exists at most one reflection  $s_\alpha$  reversing  $\alpha$  s.t.  $s_\alpha(R) \subset R$ .  $\lrcorner$

*Proof:* Let  $s, s'$  be two such reflections, then  $t = ss'$  is an automorphism that is identity on both  $\mathbb{F}\alpha$  and  $V/\mathbb{F}\alpha$ . So  $(t-1)^2 = 0$  on  $V$ . So the minimal polynomial of  $T$  divides  $(T-1)^2$ . Also because  $R$  is finite there exists an  $m$  that  $t^m = 1$  on  $R$  thus on  $V$ , so  $t = 1$  as the greatest divisor of  $(T-1)^2$  and  $T^m - 1$  is  $T - 1$ .  $\square$

**Lemma (3.9.2.4).** Let  $V$  be an inner product space, then for any vector  $\alpha$ , there exists a unique reflection  $s_\alpha$  that respects the inner product, which is

$$s_\alpha(x) = x - 2 \frac{(x, \alpha)}{(\alpha, \alpha)} \alpha.$$

┘

**Def. (3.9.2.5) [Reflection Groups].**  $W \leq \text{GL}(n; V)$  is called a **reflection group** if it is generated by reflections (3.9.2.1).

In particular, Coxeter-groups are reflection groups, by (3.9.1.4). ┘

**Prop. (3.9.2.6) [Finite Coxeter-Groups and Finite Reflection Groups].** For a Coxeter group  $(W, S)$  (3.9.1.1), the following are equivalent:

- $\#W < \infty$ .
- The canonical bilinear form  $B$  as in the canonical representation (3.9.1.4) is positive-definite.
- $W$  is a finite reflection group (3.9.2.5).

┘

*Proof:* 1  $\iff$  3 follows from (3.9.1.4) (3.9.1.6).

2  $\rightarrow$  3: If  $B$  is positive-definite, then  $W \leq \text{O}(B)$  is a discrete subgroup, thus finite.

1  $\rightarrow$  2: We may assume that  $(W, S)$  is irreducible, then  $B$  is non-degenerate by (3.9.1.7). But then  $V$  is an irreducible  $W$ -module, so there are up to a scalar only one  $W$ -invariant inner form on  $V$ . As  $W$  is finite we can choose a positive-definite invariant form on  $V$ , so  $B$  must also be positive-definite, as  $B(s, s) = 1 > 0$ .  $\square$

**Thm. (3.9.2.7) [Characterizing Reflection Groups, Shephard-Todd-Chevalley].** For  $\Gamma \leq \text{GL}(n; \mathbb{R})$ ,  $\Gamma$  is a finite reflection group (3.9.2.5) iff  $k[X_1, \dots, X_n]^\Gamma$  is a polynomial ring. And in this case,

- There are homogenous alg.ind polynomials  $p_1, \dots, p_r$ ,  $d_i = \deg(p_i)$ ,  $d_1 \leq d_2 \leq \dots \leq d_r$  s.t.  $k[X_1, \dots, X_n]^\Gamma = k[p_1, \dots, p_r]$ .

•

$$\frac{1}{\#\Gamma} \sum_{\gamma \in \Gamma} \frac{\text{tr}(\gamma)}{\det(1 - \gamma T)} = \prod_{i=1}^r \frac{1}{1 - T^{d_i}} \in k(T).$$

- $(d_1, \dots, d_r)$  are determined by  $\Gamma$ .
- $\#\Gamma = \prod_i d_i$ , and  $\Gamma$  contains  $\sum_i (d_i - 1)$  reflections.
- If  $\Gamma$  is an irreducible subgroup of  $\text{GL}(n; k)$ , then it is an abstractly irreducible group, and  $Z(\Gamma)$  is cyclic of order  $p \cdot \gcd(d_1, \dots, d_r)$ .

┘

*Proof:*  $\square$

## Classifying Finite Reflection Groups

### Weyl-Groups

**Def. (3.9.2.8) [Crystallographic Groups].** A **crystallographic group** is a subgroup of  $\text{GL}(V)$  that fixes some lattice of  $V$ . ┘

**Prop. (3.9.2.9).** If  $(W, S)$  is a Coxeter-system and  $W \leq \text{GL}(V[S])$  is crystallographic(3.9.2.8), then for any  $\alpha, \beta \in \Delta$ ?,  $n(\alpha, \beta) \in \{2, 3, 4, 6\}$ (3.9.2.6).  $\lrcorner$

*Proof:* Choose a  $W$ -invariant inner form on  $V$ , let  $W_{\alpha, \beta} = \mathbb{R}\{\alpha, \beta\}$  and  $W_{\alpha, \beta}^\perp$  be the orthogonal complement of  $W_{\alpha, \beta}$ , and  $s_\alpha s_\beta$  stabilizes  $W_{\alpha, \beta}$  and fixes every element of  $W_{\alpha, \beta}^\perp$ . Thus the trace of  $s_\alpha s_\beta$  is  $\dim V - 2 + 2 \cos \frac{2\pi}{n(\alpha, \beta)}$ . But this must be an integer, so  $n(\alpha, \beta) \in \{2, 3, 4, 6\}$ .  $\square$

**Def. (3.9.2.10) [Weyl-Groups].** A (finite) **Weyl-group** is a finite crystallographic reflection group(3.9.2.8)(3.9.2.5).  $\lrcorner$

**Prop. (3.9.2.11).**  $\lrcorner$

**Prop. (3.9.2.12) [Crystallographic Weyl-Groups].** Cf. [Hum90]P135.  $\lrcorner$

**Def. (3.9.2.13) [Hyperbolic Weyl-Groups].**  $\lrcorner$

### 3 Invariants of Finite Reflection Groups

### 4 Representations of Reflection Groups

**Thm. (3.9.4.1) [Characters of Weyl-Groups].** All finite  $\mathbb{C}$ -representations of a finite Weyl-group(3.9.2.10) is defined over  $\mathbb{Q}$ .  $\lrcorner$

*Proof:* Cf. [Hum90]P181.  $\square$

## 5 Root Systems

**Def. (3.9.5.1) [Root Systems].** A finite subset  $\Phi \subset V$  is called a **root system** if

- $\Phi$  spans  $V$  and doesn't contain 0.
  - For each  $\alpha \in \Phi$ , there exists a reflection(3.9.2.1)  $s_\alpha = \text{id} - \alpha^\vee \otimes \alpha$  for some  $\alpha^\vee \in V^\vee$  s.t.  $\langle \alpha^\vee, \alpha \rangle = 2$  s.t.  $s_\alpha$  reverses vector  $\alpha$  and  $s_\alpha(\Phi) \subset \Phi$ . Notice  $s_\alpha$  is unique by(3.9.2.3).
  - For any  $\alpha, \beta \in \Phi$ ,  $s_\alpha(\beta) - \beta$  is a multiple of  $\alpha$ , or equivalently  $\alpha^\vee(\beta) \in \mathbb{Z}$ .
- Elements of  $\Phi$  are called the **roots** of  $\Phi$ , and  $\{\alpha^\vee\}_{\alpha \in \Phi}$  is called the set of **coroots** of  $\Phi$ .  $\text{rank } V$  is called the **rank of the root system**.

The subgroup  $W(\Phi) \leq \text{GL}(V)$  generated by all  $s_\alpha$  is called the **Weyl group** of  $\Phi$ . The  $W(\Phi)$  is a finite group, as a subgroup of the group of permutations of  $\Phi$ .

Use  $(V, \Phi, W) \in \text{RootSys}$  to denote that  $\Phi$  is a root system of  $V$  with Weyl group  $W$ .

A root system is called **irreducible** iff it cannot be written as a direct sum of two root systems.

$\lrcorner$

**Def. (3.9.5.2) [Reduced Root System].** For  $(V, \Phi) \in \text{RootSys}$ (3.9.5.1), if  $\alpha, \beta \in \Phi$  are multiples of each other i.e.  $\beta = c\alpha$  for some  $c \in k^\times$ , then  $2\langle \beta, \alpha^\vee \rangle = 2c \in \mathbb{Z}$ , also  $2c^{-1} \in \mathbb{Z}$ , so  $c \in \{-1, -1/2, 1/2, 1\}$ .

A **reduced root system** is a root system  $\Phi$  s.t.  $\Phi \cap 2\Phi = \emptyset$ .  $\lrcorner$

**Prop. (3.9.5.3).** Let  $(V, \Phi, W) \in \text{RootSys}$  and  $\alpha, \beta \in \Phi$  s.t.  $\alpha^\vee = \beta^\vee$ , then  $\alpha = \beta$ .  $\lrcorner$

*Proof:* If  $\alpha = \beta$ , then  $s_\alpha(\beta) = \beta - 2\alpha$  and  $s_\beta(\alpha) = \alpha - 2\beta$ . Thus  $(s_\beta s_\alpha)^n \alpha = \alpha + 2n(\beta - \alpha) \in \Phi$ , forcing  $\alpha = \beta$ .  $\square$

**Prop. (3.9.5.4).** A root system  $\Phi$ (3.9.5.1) is irreducible iff the action of  $W(\phi)$  on  $V$  is irreducible.  $\lrcorner$

*Proof:* ? □

**Prop. (3.9.5.5) [Invariant Quadratic Form].** For  $(\Phi, V) \in \text{RootSys}(3.9.5.1)$ , there is a positive bilinear form in  $V$  that is invariant under the action of the Weyl group  $W(\Phi)$ .

Notice that for such a bilinear form, the reflection must be of the form given in (3.9.2.4), or in other words,  $\alpha^\vee(\beta) = (\frac{2\alpha}{(\alpha, \alpha)}, \beta)$ . □

*Proof:* This follows entirely because the Weyl group of  $R$  is finite, as we can take the average of any positive bilinear form under the action of reflections. □

**Def. (3.9.5.6) [Dual Root Systems].** Let  $(V, \Phi) \in \text{RootSys}$ , the set  $\Phi^\vee$  of dual vectors  $\alpha^\vee$  for  $\alpha \in \Phi$  is also a root system in  $V^\vee$ , called the **dual root system** of  $\Phi$ . And  $W(\Phi)^{\text{op}} \cong W(\Phi)$ .

Moreover,  $\Phi^{\vee\vee} \cong \Phi$ . □

*Proof:* Take an invariant quadratic form on  $V$  (3.9.5.5), then it gives an isomorphism  $V \rightarrow V^\vee$ . Then  $\alpha^\vee$  corresponds to  $\frac{2\alpha}{(\alpha, \alpha)}$  under this isomorphism, and there obviously generates  $V$ . If  $\alpha^\vee \in R^\vee$ , we can take the corresponding reflection to be  $s_{\alpha^\vee} = 1 - \alpha \otimes \alpha^\vee$ . Then

$$\begin{aligned} s_{\alpha^\vee}(\beta^\vee)(x) &= (\beta^\vee - \beta^\vee(\alpha)\alpha^\vee)(x) \\ &= \left(x, \frac{2\beta}{(\beta, \beta)}\right) - \frac{(\alpha, \beta)}{(\beta, \beta)} \left(x, \frac{2\alpha}{(\alpha, \alpha)}\right) \\ &= \left(x, \frac{2(\beta - \alpha(\alpha, \beta)/(\alpha, \alpha))}{(\beta, \beta)}\right) \\ &= \left(x, \frac{2s_\alpha(\beta)}{(\beta, \beta)}\right) = (s_\alpha(\beta))^\vee(x). \end{aligned}$$

So  $s_{\alpha^\vee}(\beta^\vee) = (s_\alpha(\beta))^\vee \in R^\vee$ . In this way, we see  $\alpha^{\vee\vee} = \alpha$ , and  $s_{\alpha^\vee}(\beta^\vee) - \beta^\vee = \beta^\vee(\alpha)\alpha^\vee$  is an integral multiple of  $\alpha^\vee$ . Also it can be seen easily from the formula above that the Weyl group is the opposite of the Weyl group of  $\Phi$ . □

**Prop. (3.9.5.7) [Rational Root Systems].** Suppose  $(\Phi, V) \in \text{RootSys}$  and  $V_0$  is the  $\mathbb{Q}$ -vector space spanned by  $\Phi$ , then  $(\Phi, V_0)$  is also a root system over  $\mathbb{Q}$  with  $V_0 \otimes_{\mathbb{Q}} k \cong V$ .

So from now on we focus on the case  $k = \mathbb{Q}$ . □

*Proof:* The non-trivial part is that the isomorphism  $V_0 \otimes_{\mathbb{Q}} \mathbb{F} \cong V$ . The natural map  $i : V_0 \otimes_{\mathbb{Q}} \mathbb{F} \rightarrow V$  is surjective because  $R$  generates  $V$ , and consider its transpose

$$i^* : V^* \rightarrow V_0^* \otimes_{\mathbb{Q}} \mathbb{F},$$

then we can see that this maps  $\alpha^\vee$  to  $\alpha_0^\vee$ , and  $R_0^\vee$  is a root system in  $V_0^*$  (3.9.5.6), thus  $\alpha_0^\vee$  generates  $V_0^\vee$ , and  $i^*$  is surjective, showing  $i$  is injective. □

**Def. (3.9.5.8) [Weyl-Chambers].** For  $(V, \Phi) \in \text{RootSys}/\mathbb{Q}$ , a **Weyl-chamber** is a connected component of  $V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in \Phi} H'_\alpha$ , where  $H'_\alpha = \{t \in V^* \mid \langle t, \alpha \rangle = 0\}$ . □

**Def. (3.9.5.9) [Bases of Root systems].** A **base of root system** is a subset  $\Delta \subset \Phi$  s.t. every elements of  $\Phi$  can be written uniquely as a linear integral combination of elements of  $\Delta$  with the same sign. Use  $(\Phi, \Delta) \in \text{RootSys}$  to denote that  $\Phi \in \text{RootSys}$  is a root system with a chosen base  $\Delta$ .

If  $\Delta$  is a base of  $\Phi$ , then we let  $\Phi^+$  denote the subset of  $\Phi$  that is non-negative integral combinations of elements of  $\Delta$ , called the **positive roots** of  $\Phi$ , and  $\Phi^- \subset \Phi$  is the subset of non-positive integral combinations of elements of  $\Delta$ , called the **negative roots** of  $\Phi$ .  $\Phi = \Phi^+ \amalg \Phi^-$ . □

**Prop. (3.9.5.10) [Bases exist].** Suppose  $(V, \Phi) \in \text{RootSys}$  and  $t \in V^\vee$  is contained in a Weyl chamber (3.9.5.8). Let  $\Delta_t^+$  (resp.  $\Delta_t^-$ ) be the set of all  $\alpha \in \Phi$  that  $t(\alpha) > 0$  (resp.  $< 0$ ), then  $\Phi = \Delta_t^+ \cup \Delta_t^-$ .

An element of  $\Delta_t^+$  is called **irreducible** if it cannot be written as the sum of two elements in  $\Delta_t^+$ . Let  $\Delta_t$  be the set of irreducible elements of  $\Delta_t^+$ , then  $\Delta_t$  is a base of  $\Phi$ .

In particular,  $(V, \Phi)$  contains a base. And if  $\Delta$  is a base and  $t \in V^\vee$  s.t.  $t(\Delta) > 0$ , then  $\Delta = \Delta_t^+$ . In this way, the set of bases of  $\Phi$  are in bijection with the set of Weyl chambers of  $\Phi$ , where the Weyl-chamber  $C_+(\Phi)$  corresponding to  $\Delta$  is called the **positive Weyl-chamber** w.r.t.  $\Delta$ .  $\lrcorner$

*Proof:* Cf. [Ser87]P38.  $\square$

**Cor. (3.9.5.11).** Suppose  $(V, \Phi) \in \text{RootSys}$  and  $t \in V^\vee$  is contained in a Weyl chamber (3.9.5.8). If  $\Delta$  is a basis of  $V$  contained in  $\Delta_t^+$  that attains the minimal value of  $t$  in  $\Delta_t^+$ , then  $\Delta = \Delta_t^+$  is a base of  $\Phi$ .  $\lrcorner$

**Prop. (3.9.5.12).** For  $(\Phi, V, \Delta) \in \text{RootSys}$ , every positive root  $\beta$  can be written as

$$\beta = \alpha_1 + \dots + \alpha_k$$

s.t.  $\alpha_i \in \Delta$  and  $\sum_{i=1}^r \alpha_i \in \Phi$  for all  $i \in [k]_+$ .  $\lrcorner$

*Proof:* Cf. [Ser87]P40.  $\square$

**Prop. (3.9.5.13) [Half Sum of Positive Roots].** For  $(\Phi, \Delta) \in \text{RootSys}_{\text{red}}$  and  $\alpha \in \Delta$ ,

$$s_\alpha(\Phi^+ \setminus \{\alpha\}) = \Phi^+ \setminus \{\alpha\}.$$

In particular if  $\rho = \frac{1}{2}(\sum_{\alpha \in \Phi^+} \alpha)$ , then  $s_\alpha \rho = \rho - \alpha$ . In particular,  $w(\rho) - \rho \in Q(\Phi)$  for any  $w \in W$ .  $\lrcorner$

*Proof:* If  $\beta \in \Phi^+$ , then  $s_\alpha(\beta) = \beta - c\alpha$  for some  $c > 0$ . Thus  $s_\alpha(\beta) \in \Phi^-$  iff  $\beta = \alpha$ .  $\square$

**Prop. (3.9.5.14) [Base and Dual System].** For  $(\Phi, \Delta) \in \text{RootSys}$ ,  $\Delta^\vee$  is a base of  $\Phi^\vee$ .  $\lrcorner$

*Proof:* By the isomorphism between  $V$  and  $V^\vee$ , it suffices to show the vectors  $\{\alpha^\vee\}_{\alpha \in \Delta}$  is a base for  $\Phi^\vee = \{\alpha^\vee\}_{\alpha \in \Phi}$ . But it is clear if  $S$  is a base corresponding to a vector  $t \in V^*$ , then  $\{\alpha^\vee, \alpha \in S\}$  is the extremal vectors corresponding to  $t$  too, so it is a base by (3.9.5.10).  $\square$

**Def. (3.9.5.15) [Parabolic Subsets].** For  $(\Phi, \Delta) \in \text{RootSys}$ , a **parabolic subset** of  $\Phi$  is a subset  $X \subset \Phi$  s.t. it is closed under addition, and  $X \cup (-X) = \Phi$ .  $\lrcorner$

### Property of the Weyl Groups

**Prop. (3.9.5.16) [Actions of Weyl Groups].** Let  $(\Phi, \Delta, W) \in \text{RootSys}_{\text{red}}$ , then  $W$  acts on  $\Phi$  thus also on  $V$  because  $\Phi$  is a basis of  $V$ , and thus also on  $V^*$  and on the set of bases and Weyl chambers of  $\Phi$ . Then:

1. For any  $t \in V^*$ , there exists  $w \in W$  that  $(w(t), \alpha) \geq 0$  for all  $\alpha \in \Delta$ .
2.  $W$  acts transitively on the set of bases of  $\Phi$ .
3. For each  $\beta \in \Phi$ , there exists some  $w \in W$  s.t.  $w(\beta) \in \Delta$ .
4. The group  $W$  is generated by  $\{s_\alpha\}_{\alpha \in \Delta}$  with relations  $\{(s_\alpha s_\beta)^{n(\alpha, \beta)} = 1\}$ . In particular, it is a Coxeter group (3.9.1.1).

5.  $W$  acts simply transitively on the set of bases of  $\Phi$ , thus also on the set of Weyl chambers(3.9.5.8).  $\lrcorner$

*Proof:* We can in fact prove this for  $W_S$  the group generated by  $s_\alpha$  where  $\alpha \in S$ .

1: Let  $\rho$  be defined in(3.9.5.13), and choose an element  $w \in W_S$  that  $w(t)(\rho)$  is maximal, then  $w(t)(\rho) \geq w(t)s_\alpha(\rho) = w(t)(\rho) - w(t)(\alpha)$ . So  $(w(t), \alpha) \geq 0$ .

2: Let  $S'$  be a base and  $t' \in V^\vee$  that  $t'(S') > 0$ (3.9.5.10), Also by 1 we can find  $w \in W$  that  $w(t)(\alpha) \geq 0$  for all  $\alpha \in S$ . And in fact  $w(t)(\alpha) > 0$  for all  $\alpha \in S$ . So  $S = S_t, S' = S_{w(t)}$ . Thus  $w$  sends  $S'$  to  $S$ .

4: Finally we prove that  $W_S = W$ : because for any  $\beta \in \Phi$ , there exists  $w \in W_S$  that  $w(\beta) \in S$ , so  $s_\beta = w^{-1}s_{w(\beta)}w \in W_S$ . To prove that the relations are full relations?.

5: If  $w \in W$  satisfies  $w(C) = C$ , we may assume  $C$  is the fundamental Weyl chamber, then  $w(S) = S$ , so by(3.9.5.18),  $l(w) = 0$ , so  $w = \text{id}$ .  $\square$

**Def. (3.9.5.17) [Length].** Let  $w \in W$ , define the **length of roots** of  $w$  to be the minimal number  $m$  that  $w$  can be written as a product of  $m$  simple reflections  $s_i$ .  $\lrcorner$

**Prop. (3.9.5.18) [Longest Element  $w_0$ ].** For  $(\Phi, \Delta, W) \in \text{RootSys}$  and  $w \in W$ ,

- Let  $n(w)$  be the number of elements  $\alpha$  in  $\Phi^+$  that  $w(\alpha) \in \Phi^-$ , then  $n(w) = l(w)$ .
- There is a unique element  $w_0 \in W$  with length  $\#\Phi^+$ . In particular,  $w_0 = -1$  if  $-1 \in W$ .
- $w_0(\Phi^+) = \Phi^-$ , and  $w_0^2 = \text{id}$ .

$\lrcorner$

*Proof:* 1: Cf.[Carter, P63]?.

2: By(3.9.5.16) there is a unique element  $w_0 \in W$  mapping  $\Delta$  to  $-\Delta$ , and it has maximal length  $\#\Phi^+$  by item1.

3:  $w_0^2(\Phi^+) = \Phi^+$ , thus  $l(w_0^2) = 0$  and  $w_0^2 = \text{id}$ .  $\square$

**Def. (3.9.5.19) [Height of Weights].** For  $\alpha = \sum n_i \alpha_i \in \Phi$ , then the **height of roots** of  $\alpha$  is denoted by  $\text{ht}(\alpha) = \sum n_i$ .  $\lrcorner$

**Prop. (3.9.5.20) [Highest Roots].** For  $(V, \Phi, \Delta) \in \text{RootSys}$  irreducible, there exists a root  $\tilde{\alpha} = \sum_{\alpha \in \Delta} n_\alpha \alpha$  that for any other root  $\sum_{\alpha \in \Delta} m_\alpha \alpha$ ,  $n_\alpha \geq m_\alpha$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (3.9.5.21) [Kostant's Partition Function].** For  $(V, \Phi, \Delta) \in \text{RootSys}$ , the **Kostant's partition function**  $\mathfrak{P}$  is a function on  $Q(\Phi)$  that  $\mathfrak{P}(\alpha)$  equals the number of ways to write  $\alpha$  as an unordered sum of positive roots in  $\Phi^+$ .  $\lrcorner$

### Cartan Matrix and Dynkin Diagrams

**Prop. (3.9.5.22) [Cartan-Numbers and Cartan-Matrices].** Let  $(V, \Phi) \in \text{RootSys}$  and  $\alpha, \beta$  be two non-propositional roots in a root system  $\Phi$ , then we can put  $n(\alpha, \beta) = \alpha^\vee(\beta) = 2(\alpha, \beta)/(\alpha, \alpha)$ . Then  $n(\alpha, \beta)$  are integers(3.9.5.1), called the **Cartan-numbers**, and  $n(\alpha, \beta)n(\beta, \alpha) = 4 \cos^2 \varphi_{\alpha, \beta}$ , where  $\varphi_{\alpha, \beta}$  is the angle between these two vectors. Then because  $4 \cos^2 \varphi_{\alpha, \beta}$  is an integer, it can only take values 0, 1, 2, 3. Then there are only 7 possibilities of the angles between  $\alpha, \beta$ , and if  $\alpha, \beta$  are not orthogonal, their ration of lengths are determined also by the angle.

The **Cartan-matrix** is the matrix  $(n(\alpha, \beta))_{\alpha, \beta \in S}$ (3.9.5.22).  $\lrcorner$



**Prop. (3.9.5.23) [String of Roots].** If  $\alpha, \beta$  are not proportional and  $n(\beta, \alpha) > 0$ , then  $\alpha - \beta$  is a root. In particular, for  $\alpha, \beta \in S$  where  $S$  is a base of  $R$ ,  $n(\alpha, \beta) \leq 0$ .  $\lrcorner$

*Proof:* (3.9.5.22) shows  $n(\beta, \alpha) = 1$  or  $n(\alpha, \beta) = 1$ . Now  $\alpha - \beta = s_\beta(\alpha)$  or  $-s_\alpha(\beta)$  is a root of  $R$ .  $\square$

**Prop. (3.9.5.24) [Cartan Matrix Determine Reduced Root Systems].** The Cartan matrix depends only on  $(V, \Phi)$  and not on  $\Delta$ , and reduced root systems are determined by its Cartan matrix up to isomorphisms.  $\lrcorner$

*Proof:* The first assertion follows from (3.9.5.16) as every two bases are conjugate. The Cartan matrix determines  $R$  because it determined the inner product between roots in a basis of  $V$ .  $\square$

**Prop. (3.9.5.25).** Let  $E$  be the group of automorphisms of  $S$  that leaves that Cartan matrix invariant, then it can be identified with the set of automorphisms of  $R$  that leave the base  $S$  invariant. Then the group  $\text{Aut}(R)$  is isomorphic to the semi-product  $E \ltimes W$ .  $\lrcorner$

*Proof:* Let  $W$  is generated by  $s_\alpha$  so invariant under  $\text{Aut}(R)$ . Now if  $u \in \text{Aut}(R)$ , then  $u(S)$  is a base of  $R$ , so there exists some  $w \in W$  that  $w(u(S)) = S$ , thus  $u \in EW$ . Also if  $w \in W \cap E$ , then  $\square$

**Def. (3.9.5.26) [Coxeter graphs].** A **coxeter graph** is a finite graph that each pair of distinct vertices are connected by 0, 1, 2 or 3 vertices.  $\lrcorner$

**Def. (3.9.5.27) [Coxeter Graph associated to Root Systems].** Let  $(V, \Phi, \Delta) \in \text{RootSys}$ , then the associated **coxeter graph** is a graph whose nodes are indexed by the elements of  $\Delta$ , and two distinct nodes  $\alpha, \beta$  are connected by  $a(\beta, \alpha)a(\alpha, \beta) = 4 \cos^2 \varphi_{\alpha, \beta}$  edges (3.9.5.22). This is independent of the choice of  $\Delta$ , by (3.9.5.24).  $\lrcorner$

**Prop. (3.9.5.28).** For  $(V, \Phi, \Delta) \in \text{RootSys}$ ,  $\Phi$  is simple iff the Coxeter graph (3.9.5.26) is connected.  $\lrcorner$

*Proof:* By the formula in (3.9.2.4),  $R$  is decomposable iff  $R = R_1 \amalg R_2$  where  $R_1, R_2$  are orthogonal to each other. Then this is equivalent to  $\varphi_{\alpha, \beta} = \pi/2$  for any  $\alpha \in R_1, \beta \in R_2$ .  $\square$

**Def. (3.9.5.29) [Dynkin Diagrams].** The coxeter graphs (3.9.5.27) cannot determine root systems up to isomorphisms, because it cannot distinguish between  $n(\alpha, \beta), n(\beta, \alpha)$ . So there are **Dynkin diagrams** which is constructed from the coxeter diagram by adding a vector from the longer vector to the shorter vector when  $n(\alpha, \beta)n(\beta, \alpha) = 2$  or 3. In particular, Dynkin diagrams determine reduced root systems, by (3.9.5.24).

A **simply-laced Dynkin diagram** is a Dynkin diagram without multiple edges. A **simply-laced root system** is a root system whose associated Dynkin diagram is simply-laced.  $\lrcorner$

### Weight Lattices

**Def. (3.9.5.30) [Weight Lattices].** For  $(V, \Phi) \in \text{RootSys}$ , define the **root lattice** as the  $\mathbb{Z}$ -lattice  $Q(\Phi)$  generated by  $\Phi$ , and also the **weight lattice**  $P(\Phi) = \{x \in V \mid \alpha^\vee(x) \in \mathbb{Z}, \forall \alpha \in \Phi\}$ .  $P(\Phi)$  has a generator by **fundamental weights**  $\omega_i$  s.t.  $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$ .  $Q(\Phi) \subset P(\Phi)$ , and  $[P(\Phi) : Q(\Phi)]$  is finite because they are both complete lattices.  $\lrcorner$

**Def. (3.9.5.31) [Root Diagrams].** A **root diagram** is a triple  $(V, \Phi, X)$  where  $(V, \Phi) \in \text{RootSys}$  is a root system and  $Q(\Phi) \subset X \subset P(\Phi)$  (3.9.5.30).  $\lrcorner$

**Prop. (3.9.5.32).**  $\rho = \sum_i \omega_i$  (3.9.5.13) (3.9.5.30).  $\lrcorner$

*Proof:* It follows from (3.9.5.13) that  $s_\alpha \rho = \rho - \alpha$ , so  $\langle \rho, \alpha_i^\vee \rangle = 1$ , thus  $\rho = \sum_i \omega_i$  by definition (3.9.5.13).  $\square$

### Dominant and Minuscale Weights

**Def. (3.9.5.33) [Dominant Weights].** For  $(V, \Phi, \Delta) \in \text{RootSys}$ ,

- A **dominant integral weight** is a weight  $\omega \in P(\Phi)$  s.t.  $\langle \omega, \alpha^\vee \rangle \in \mathbb{N}$  for any  $\alpha \in \Phi^+$ . The set of dominant weights is denoted by  $P^+(\Phi)$ .
- $Q^+(\Phi) \subset Q(\Phi)$  is the set of positive combinations of positive roots (3.9.5.9).

┘

**Prop. (3.9.5.34) [Partial-Ordering of Weights].** For  $(V, \Phi, \Delta) \in \text{RootSys}$  and  $x, x' \in V^\vee$ , the following are equivalent:

- $x$  lies in the convex hull of the finite set  $W(G, T).x'$ .
- If  $\tilde{x}, \tilde{x}' \in C_+(\Phi)$  s.t.  $W(G, T)\tilde{x} = W(G, T)x, W(G, T)\tilde{x}' = W(G, T)x'$ , then  $\tilde{x}' - x \in \mathbb{R}_{\geq 0}\Phi^\vee$ .
- If  $\tilde{x}' \in C_+(\Phi)$  s.t.  $W(G, T)\tilde{x}' = W(G, T)x'$ , then  $\tilde{x}' - wx \in \mathbb{R}_{\geq 0}\Phi^\vee$  for any  $w \in W(G, T)$ .

And we denote  $x \prec x'$  if these conditions hold.

┘

*Proof:* Cf. [R-R96] P165. ?

□

**Def. (3.9.5.35) [Minuscale Weights].** A dominant integral weight (3.9.5.33)  $\omega$  for  $(\Phi, \Delta)$  is called a **minuscale weight** if  $\langle \omega, \beta^\vee \rangle \leq 1$  for any positive coroot  $\beta^\vee$ .

┘

**Prop. (3.9.5.36) [Minuscale and Fundamental Weights].** Let  $\theta^\vee = \sum m_i \alpha_i^\vee$  be the maximal coroot, where  $m_i$  are positive integers, and  $m_i = \langle \omega_i, \theta^\vee \rangle$ .

Then any non-zero minuscale weight (3.9.5.35) is fundamental (3.9.5.30), and a fundamental weight  $\omega_i$  is minuscale iff  $m_i = 1$ .

┘

*Proof:* If  $\omega = \sum a_i \omega_i$  is minuscale, the definition (3.9.5.35) requires  $\sum a_i = 1$ , and any other coroot is of the form  $\beta^\vee = \sum n_i \alpha_i^\vee$ , where  $n_i \leq m_i$ , so  $n_i = \langle \omega_i, \beta^\vee \rangle \leq 1$ .

□

**Lemma (3.9.5.37).** If  $\omega \in Q(\Phi)$  and  $|\langle \omega, \beta^\vee \rangle| \leq 1$  for any coroot  $\beta$ , then  $\omega = 0$ .

┘

*Proof:* If not, choose a counterexample  $\omega = \sum m_i \alpha_i$  with  $\sum |m_i| > 0$  minimum, then  $(\omega, \omega) = \sum m_i (\omega, \alpha_i^\vee) > 0$ , so changing  $\omega$  to  $-\omega$  if necessary, we can assume  $m_i > 0$ ,  $(\omega, \alpha_i^\vee) > 0$  for some  $i$ , so  $(\omega, \alpha_i^\vee) = 1$ , and  $s_i \omega = \omega - \alpha_i$  is another counterexample with smaller  $|m_i|$ , contradiction.

□

**Prop. (3.9.5.38).** The following are equivalent for a dominant integral weight  $\omega$  of a root system:

- $\omega$  is minuscale (3.9.5.35).
- If  $\lambda \in P^+(\Phi)$  and  $\omega - \lambda \in Q^+(\Phi)$ , then  $\lambda = \omega$ .

┘

*Proof:* Cf. [Etingof, P141] ?

□

**Prop. (3.9.5.39) [Number of Minuscale Weights].** Every coset in  $P(\Phi)/Q(\Phi)$  contains a unique minuscale weight (3.9.5.35). This gives a bijection between the  $P(\Phi)/Q(\Phi)$  and the set of minuscale weights. In particular, the number of minuscale weights equals the determinant of the Cartan matrix.

┘

*Proof:* For any  $a \in P(\Phi)$ , let  $C = a + Q(\Phi)$  be a coset, let  $\omega \in C \cap P^+(\Phi)$  be an element with minimum  $(\omega, \rho)$ , then for any dominant weight  $\lambda < \omega \in C$ ,  $(\omega - \lambda, \rho) \geq 0$ , so  $\lambda = \omega$ . Thus  $\omega$  is minuscale (3.9.5.38).

Conversely, if  $\omega_1 \neq \omega_2 \in C$  are minuscale, then by (3.9.5.37), there is a positive coroot  $\beta^\vee$  that  $|(\omega_1 - \omega_2, \beta)| > 2$ . But then as  $\omega_1, \omega_2$  are both dominant,  $|(\omega_1 - \omega_2, \beta)| \leq 1$ , contradiction.

□

### Root Data

**Def. (3.9.5.40) [Root Data].** A **root datum** is a triple  $(X, \Phi, \vee)$  where  $X$  is a free  $\mathbb{Z}$ -module,  $\Phi \subset X$  is a finite set, and  $\vee : \Phi \rightarrow X^\vee = \text{Hom}(X, \mathbb{Z})$  is a map satisfying the following:

- $\langle \alpha, \alpha^\vee \rangle = 2$  for any  $\alpha \in \Phi$ .
- $s_\alpha(\Phi) = \Phi$  for any  $\alpha \in \Phi$ , where  $s_\alpha = \text{id} - \alpha^\vee \otimes \alpha : X \rightarrow X$ .
- The subgroup  $W \subset \text{GL}(X)$  generated by  $\{s_\alpha\}_{\alpha \in \Phi}$  is finite, called the **Weyl group** of  $(X, \Phi, \vee)$ . Use  $(X, \Phi, \vee) \in \text{RootDat}$  to denote that  $(X, \Phi, \vee)$  is a root datum.  $\square$

**Prop. (3.9.5.41) [Root Systems Associated to Root Data].** For  $(X, \Phi, \vee) \in \text{RootDat}$  (3.9.5.40), let  $V$  be the  $\mathbb{Q}$ -subspace of  $X_\mathbb{Q}$  generated by  $\Phi$ , then  $(V, \Phi) \in \text{RootSys}$  (3.9.5.1), and  $W(X, \Phi, \vee) \cong W(V, \Phi)$ .

Thus we use the notations for root systems on  $(X, \Phi, \vee)$ . In particular, the **rank of the root datum** is  $\text{rank}_\mathbb{Z} X$  and the **semisimple rank of the root datum** is  $\text{rank } V$ .  $\square$

*Proof:* Any  $\alpha \in \Phi$  is non-zero because  $\langle \alpha, \alpha^\vee \rangle = 2$ . And  $\alpha^\vee(\beta) \in \mathbb{Z}$  for  $\alpha, \beta \in \Phi$  because  $\alpha^\vee \in X^\vee$ .  $W(X, \Phi, \vee) \cong W(V, \Phi)$  follows from (3.9.5.3).  $\square$

**Prop. (3.9.5.42) [Uniqueness of  $\vee$ ].** If  $X$  is a free  $\mathbb{Z}$ -module and  $\Phi \subset X, \Phi^\vee \in X^\vee$  are finite sets, then there exists at most one  $\vee : \Phi \rightarrow X^\vee$  making it into a root datum (3.9.5.40), and this  $\vee$  is injective.

Thus we can also use  $(X, \Phi, \Phi^\vee)$  to denote that  $X$  is a free  $\mathbb{Z}$ -module and  $\Phi \subset X, \Phi^\vee \in X^\vee$  are finite sets and there exists a unique map  $\vee : \Phi \rightarrow \Phi^\vee$  making  $(X, \Phi, \vee)$  a root datum.  $\square$

*Proof:* We show that for  $(X, \Phi, \vee) \in \text{RootDat}$ ,  $\vee$  is determined by  $(X, \Phi, \Phi^\vee)$ : Let  $(V, \Phi)$  be the associated root system, then for any  $\alpha \in \vee$ , the image of  $\alpha$  in  $V^*$  is determined by  $\alpha$  by (3.9.2.3). But then also  $\alpha^\vee$  is determined, because there are no two elements  $\alpha^\vee, \beta^\vee \in \Phi^\vee$  with the same image in  $V^*$ , by (3.9.5.3).  $\square$

**Def. (3.9.5.43) [Semisimple Root Data].** A **semisimple root datum** is a root datum  $(X, \Phi, \vee) \in \text{RootDat}$  s.t.  $Q(\Phi)$  has finite index in  $X$ .  $\square$

**Prop. (3.9.5.44) [Semisimple Root Data and Rational Root Diagrams].** For  $(X, \Phi, \vee) \in \text{RootDat}$  (3.9.5.43),  $(X_\mathbb{Q}, \Phi, X)$  is a root diagram (3.9.5.31). Conversely, for any root diagram  $(V, \Phi, X)$ ,  $(X, \Phi, \vee)$  is a root datum.

Thus a semisimple root datum is equivalent to a rational root diagram.  $\square$

*Proof:*  $Q(\Phi) \subset X$  because  $\Phi \subset X$ ,  $X \subset P(\Phi)$  because  $\Phi^\vee \subset X^\vee$  (3.9.5.30).

Conversely, if  $(V, \Phi, X)$  is a root diagram, then for any  $\alpha \in \Phi$ , we have  $\alpha^\vee \in P(\Phi)^\vee \subset X^\vee$  by definition of  $P(\Phi)$  (3.9.5.30), and the first two conditions of (3.9.5.40) are satisfied by definition of root system (3.9.5.1), and the subgroup of  $\text{GL}(X)$  generated by  $\{s_\alpha\}_{\alpha \in \Phi}$  is finite because it is a subgroup of  $\text{Aut}(\Phi)$ , which is finite, thus  $(X, \Phi, \vee)$  is a root datum.  $\square$

**Prop. (3.9.5.45) [Transpose of  $s_\alpha$ ].** For  $(X, \Phi, \Phi^\vee) \in \text{RootDat}$  (3.9.5.42) and for any  $\alpha \in \Phi, x \in X, y \in X^\vee$ ,

$$\langle s_\alpha(x), y \rangle = \langle x, s_{\alpha^\vee}(y) \rangle,$$

where  $s_{\alpha^\vee} = \text{id} - \alpha \otimes \alpha^\vee$ . And we have  $s_{\alpha^\vee}(\Phi^\vee) = \Phi^\vee$ . More explicitly,  $s_{\alpha^\vee}(\beta^\vee) = (s_\alpha(\beta))^\vee$ . In particular,  $W(\Phi) \cong W(\Phi^\vee)^{\text{op}}$ .  $\square$

*Proof:*  $\langle s_\alpha(x), y \rangle = \langle x - \langle x, \alpha^\vee \rangle \alpha, y \rangle = \langle x, y \rangle - \langle x, \alpha^\vee \rangle \langle \alpha, y \rangle$  and similarly for  $\langle x, s_{\alpha^\vee}(y) \rangle$ .

To show that  $s_{\alpha^\vee}(\beta^\vee) = (s_\alpha(\beta))^\vee$ , Cf. [Mil17]P619?  $\square$

**Prop. (3.9.5.46) [Equivalent Definition of Root Data].** If  $X, X'$  are free  $\mathbb{Z}$ -modules with a perfect pairing  $\langle -, - \rangle : X \times X' \rightarrow \mathbb{Z}$ , and  $\Phi \subset X, \Phi' \subset X'$  are finite sets with a bijection  $\Phi \cong \Phi' : \alpha \mapsto \alpha'$  s.t.

- $\langle \alpha, \alpha' \rangle = 2$  for any  $\alpha \in \Phi$ .
- $s_\alpha(\Phi) = \Phi, s_{\alpha'}(\Phi') = \Phi'$  for  $\alpha \in \Phi$  and  $\alpha' \in \Phi'$ , where  $s_\alpha(x) = x - \langle x, \alpha' \rangle \alpha, s_{\alpha'}(x') = x' - \langle \alpha, x' \rangle \alpha'$ .

denote

$$Q(\Phi) = \mathbb{Z}\Phi \subset X, \quad Q(\Phi') = \mathbb{Z}\Phi' \subset X', \quad V = Q(\Phi)_{\mathbb{R}}, \quad V' = Q(\Phi')_{\mathbb{R}}, \quad X_0 = Q(\Phi')^{\perp},$$

and define  $p : X \rightarrow X' : x \mapsto \sum_{\alpha \in \Phi} \langle x, \alpha' \rangle \alpha'$ . Then:

1. For any  $\alpha \in \Phi, x \in X, y \in X'$ , we have  $\langle s_\alpha(x), y \rangle = \langle x, s_{-\alpha}(y) \rangle$ .
2.  $\langle x, p(x) \rangle = \sum_{\alpha \in \Phi} \langle x, \alpha' \rangle^2 \geq 0$ , with strict inequality if  $x \in \Phi$ .
3. For  $\alpha \in \Phi$ , we have  $2p(\alpha) = \langle \alpha, p(\alpha) \rangle \alpha'$ .
4.  $p$  induces an isomorphism  $V \cong V'$ .
5.  $Q(\Phi) \cap X_0 = 0$  and  $Q(\Phi) + X_0$  has finite index in  $X$ .
6.  $\langle -, - \rangle : V \times V' \rightarrow \mathbb{Q}$  is non-degenerate.
7.  $(X, \Phi, \Phi') \in \text{RootDat}$  (3.9.5.42)

Conversely, every  $(X, \Phi, \Phi') \in \text{RootDat}$  arises in this way.  $\lrcorner$

*Proof:* 1, 2 are trivial.

3: Cf. [Mil17]P616?.

4: By item2 and item3,  $p(Q(\Phi))$  has finite index in  $Q(\Phi')$ , which implies that  $p \otimes \mathbb{Q}$  is surjective. Thus  $\dim V \geq \dim V'$ . But by duality, we also has  $\dim V' \geq \dim V$ , and thus  $p$  is an isomorphism.

5: Firstly  $\ker(p) = X_0$  because  $\langle x, p(x) \rangle = \sum_{\alpha \in \Phi} \langle x, \alpha' \rangle^2$ , thus  $p \otimes \mathbb{Q}$  has kernel  $(X_0)_{\mathbb{Q}}$  and maps  $V$  isomorphically onto  $V'$  by item2, which implies that  $V \oplus (X_0)_{\mathbb{Q}} = X_{\mathbb{Q}}$ . And the assertions follow.

6: If  $x \in Q(\Phi)$  s.t.  $\langle x, V' \rangle = 0$ , then  $x \in Q(\Phi) \cap X_0 = 0$  by item5. As  $\dim V = \dim V'$  by item1, this implies that  $\langle -, - \rangle : V \times V' \rightarrow \mathbb{Q}$  is non-degenerate.

7: We need to show that the subgroup  $W \subset \text{GL}(X)$  generated by  $\{s_\alpha\}_{\alpha \in \Phi}$  is finite. For this, it suffices to show that  $W$  acts faithfully on  $\Phi^\vee$ : If  $w \in W$  s.t.  $w(\alpha) = \alpha$  for all  $\alpha \in \Phi^\vee$ , then  $\langle w(x) - x, \alpha \rangle = 0$  for any  $x \in X$  by (3.9.5.45). But  $w(x) - x \in Q(\Phi)$ , thus  $w(x) = x$  by item4. Thus  $w = \text{id}$ .

Conversely, for  $(X, \Phi, \Phi') \in \text{RootDat}$ , take  $X' = X^\vee$  and  $\Phi' = \Phi^\vee \subset X^\vee$ , then the above conditions are satisfied by (3.9.5.45).  $\square$

**Cor. (3.9.5.47).** For  $(X, \Phi, \Phi^\vee) \in \text{RootDat}$ , the Weyl group  $W(\Phi) \subset \text{GL}(X)$  generated by  $\{s_\alpha\}_{\alpha \in \Phi}$  acts faithfully on  $\Phi$ . In particular,  $\#W(\Phi) < \infty$  (this also follows from (3.9.5.41)).  $\lrcorner$

*Proof:* If  $w(\alpha) = \alpha$  for all  $\alpha \in \Phi^\vee$ , then for all  $x \in X$ , we have

$$\langle w(x) - x, \alpha \rangle = \langle w(x), \alpha \rangle - \langle x, \alpha \rangle = \langle x, w^{-1}\alpha \rangle - \langle x, \alpha \rangle = 0.$$

But it is clear by induction that  $w(x) - x \in Q(\Phi)$ , so by the non-negativity of the pairing, we get  $w(x) = x$  for any  $x \in X$ .  $\square$

**Cor. (3.9.5.48) [Dual Root Data].** For  $(X, \Phi, \Phi^\vee) \in \text{RootDat}$ (3.9.5.42),  $(X^\vee, \Phi^\vee, \Phi)$  is also a root datum, called the **dual root datum** of  $(X, \Phi, \Phi^\vee)$ .

Then the root system associated to  $(X^\vee, \Phi^\vee, \Phi)$ (3.9.5.41) is dual to the root system associated to  $(X, \Phi, \Phi^\vee)$ (3.9.5.6). In particular, they have the same Weyl group.  $\lrcorner$

*Proof:* This is because the conditions in(3.9.5.46) are self-dual. The rest are easy.  $\square$

**Prop. (3.9.5.49) [Classifying Root Data].** For  $(X, \Phi, \Phi^\vee) \in \text{RootDat}$ , using notation as in(3.9.5.46), the map  $\text{pr} : X \rightarrow X' = X/X_0$  is injective on  $Q(\Phi)$  and induces an isomorphism  $V \cong X'_Q$ . In particular,  $(X'_Q, \Phi' = \text{pr}(\Phi))$  is isomorphic to the root system associated to  $(X, \Phi, \Phi^\vee)$ (3.9.5.41). Then we have a map  $Y = X/Q(\Phi) \rightarrow P(\Phi')/Q(\Phi')$ (3.9.5.44) with kernel  $X_0$ .

Conversely, for any  $(V, \Phi) \in \text{RootSys}$  and a f.g.  $\mathbb{Z}$ -module  $Y$  together with a map  $\varphi : Y \rightarrow P(\Phi)/Q(\Phi)$  with torsion-free kernel, define  $X(\varphi)$  to be the kernel of the map

$$P(\Phi) \oplus Y \rightarrow P(\Phi)/Q(\Phi),$$

then  $X(\varphi)$  is a finite free  $\mathbb{Z}$ -module containing  $Q(\Phi) \oplus 0$ . Then  $(X(\varphi), \Phi)$  is a root datum with associated root system  $(V, R)$ .  $\lrcorner$

*Proof:* Why is it a root datum? We can use(3.9.5.46).  $\square$

### Isogenies of Root Data

References are [Mil17].

**Def. (3.9.5.50) [Isogenies of Root Data].** For  $(X, \Phi, \Phi^\vee), (X', \Phi', \Phi'^\vee) \in \text{RootDat}$ , an injective homomorphism  $f : X' \rightarrow X$  with finite cokernel is called an **isogeny of root data** if there is a bijection  $\Phi \cong \Phi' : \alpha \mapsto \alpha'$  and a map  $q : \Phi \rightarrow \max(\text{char } k, 1)^\mathbb{N}$  s.t.

$$f(\alpha') = q(\alpha)\alpha, \quad f^\vee(\alpha^\vee) = q(\alpha)\alpha'^\vee.$$

And  $f$  is called an **central isogeny of root data** if  $q(\alpha) = 1$  for all  $\alpha \in \Phi$ .  $\lrcorner$

**Example (3.9.5.51) [Frobenius Isogenies].** For  $(X, \Phi, \Phi^\vee) \in \text{RootSys}$  and  $q \in \max(\text{char } k, 1)^\mathbb{N}$ , the map  $x \mapsto qx : X \rightarrow X$  is an isogeny(3.9.5.50), called the **Frobenius isogenies**.  $\lrcorner$

**Prop. (3.9.5.52).** An isogeny of root data(3.9.5.50) induces a bijection of the set of bases.  $\lrcorner$

## 6 Simple Root Systems

**Notation (3.9.6.1).**

- Let  $(V, \Phi, \Delta) \in \text{RootSys}$  be simple.

$\lrcorner$

**Lemma (3.9.6.2) [Listing of Irreducible Root Systems].** The coxeter graphs  $\Gamma$  associated to irreducible root systems(3.9.5.27) are exactly the graphs  $A_n(n \geq 1), B_n(n \geq 2), D_n(n \geq 4), E_6, E_7, E_8, F_4, G_2$ . (graphs are given in [Etingod, Lie algebras]P109.)  $\lrcorner$

*Proof:* The existence of these types are given by (3.9.6.3).

For the converse, notice first that for a graph with vertices  $v_i$  and  $v_i, v_j$  are connected with  $A_{ij}$  edges, the quadratic form

$$Q(x_1, \dots, x_n) = \sum_{i=1}^n x_i^2 - \sum_{1 \leq i < j \leq n} \sqrt{A_{ij}} x_i x_j = \left\| \sum x_i v_i \right\|^2$$

is positive definite.

Next we show some constraints on the graph:

- Any subgraph of the coxeter graph, it can be shown that the corresponding quadratic form is also positive definite, in particular non-degenerate.
- : It is a tree: If it contains a circle, consider the subgraph of the circle, then the corresponding quadratic form vanishes on  $(1, 1, \dots, 1)$ , contradiction.
- It doesn't contains subgraphs listed in P77 of [Car05].

Now if  $\Gamma$  contains a triple edge, then it must be  $G_2$ , otherwise it must contain a  $\widehat{G}_2$ . If  $\Gamma$  contains no triple edges, then it contains at most one double edge, otherwise it contains some  $\widehat{C}_l$ ,  $l \geq 2$ . If it contains only one double edge, then it contains no branch points, otherwise it contains some  $\widehat{B}_l$ ,  $l \geq 3$ , so it is a chain with an edge added. If the double edge is in one end, then it is  $B_l$  for some  $l \geq 3$ , otherwise it must be  $F_4$ , in order not to contain  $\widehat{F}_4$ .

Now we assume  $\Gamma$  contains only simple edges. If it contains no branch point, then it is  $A_l$  for some  $l \geq 1$ . But  $\Gamma$  contains at most one branch point, and only 3 edges, otherwise it contains some  $\widehat{D}_l$ ,  $l \geq 4$ . If  $\Gamma$  has only one branch point, then the graph is a center with three branches. One of the branch must be a single vertex, otherwise it contains a  $\widehat{E}_6$ . Then a second branch must have  $\leq 2$  vertices, otherwise it contains a  $\widehat{E}_7$ . If the second branch has a single vertex, then  $\Gamma \cong D_l$  for some  $l \geq 4$ . If the second branch has two vertices, then the third branch has  $\leq 4$  vertices, otherwise it contains some  $\widehat{E}_8$ . So  $\Gamma \cong E_6, E_7$  or  $E_8$ .  $\square$

**Prop. (3.9.6.3) [Classifying Irreducible Root Systems].** Let  $e_n$  be the standard basis of  $\mathbb{R}^n$  with the standard bilinear form, and let  $L_n$  be the subgroup generated by  $e_n$ . Then

- $A_n$ : Let  $V$  be the hyperplane of  $\mathbb{R}^{n+1}$  orthogonal to the vector  $e_1 + \dots + e_{n+1}$ , and  $R$  be the subset of  $L_{n+1} \cap V$  consisting of vectors of length  $\sqrt{2}$ . Then  $(R, V)$  is a root system, and the Weyl group is the permutation group  $S_n$  of  $e_1, \dots, e_{n+1}$ . The polarization with  $t(e_i) = n+1-i$  gives a base consisting of  $\{e_i - e_{i+1}, i = 1, \dots, n\}$  by (3.9.5.11).
- $B_n$ : Let  $V = \mathbb{R}^n$  and  $R$  be the subset of  $L_n$  consisting of vectors of length 1 or  $\sqrt{2}$ . Then  $(R, V)$  is a root system, and the Weyl group is the permutation and sign changes of the vectors  $e_i$ , isomorphic to  $\mathbb{Z}_2^n \rtimes S_n$ . The polarization with  $t(e_i) = n+1-i$  gives a base consisting of  $\{e_1 - e_2, \dots, e_{n-1} - e_n, e_n\}$  by (3.9.5.11).
- $C_n$ : Let  $C_n$  be the dual system of  $B_n$  (3.9.5.6), which by the invariant form isomorphic to the set of  $\mathbb{R}^n$  consisting of vectors  $\pm e_i \pm e_j, \pm 2e_i$ . It has the same Weyl group as  $B_n$ . It has a base consisting of  $\{e_1 - e_2, \dots, e_{n-1} - e_n, 2e_n\}$  by (3.9.5.11).
- $D_n$ : Let  $V = \mathbb{R}^n$  and  $R$  be the set of all vectors of  $L_n$  of length  $\sqrt{2}$ . The Weyl group consists of permutations and sign changes of an even number of the vectors  $e_i$ . The polarization with  $t(e_i) = n-i$  gives a base  $\Delta = \{e_1 - e_2, \dots, e_{n-1} - e_n, e_{n-1} + e_n\}$ .
- $G_2$ : Let  $V = \mathbb{R}[\omega]$  and  $R$  be the subset of  $\mathbb{Z}[\omega]$  of norm 1 or 3. The Weyl group is isomorphic to the dihedral group. It clearly has a base consisting of  $\{1, \omega - 1\}$ .

- $F_4$ : Let  $V = \mathbb{R}^4$ , and let  $R$  be the set of vectors  $\{\pm e_i, \pm e_i \pm e_j, \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\}$ . It can be shown this is a root system. The polarization s.t.

$$t(e_1) = 8, t(e_2) = 3, t(e_3) = 2, t(e_4) = 1,$$

gives a base consisting of  $\{e_2 - e_3, e_3 - e_4, e_4, \frac{1}{2}(e_1 - e_2 - e_3 - e_4)\}$ .

- $E_8$ : Let  $V = \mathbb{R}^8$  and

$$\Phi_{E_8} = \{\pm e_i \pm e_j, i \neq j\} \cup \left\{ \frac{1}{2} \left( \sum_{i=1}^8 \pm e_i \right) \text{ with even number of minus signs} \right\}.$$

The polarization with

$$t(e_1) = 23, t(e_2) = 6, t(e_3) = 5, \dots, t(e_8) = 0$$

gives a base consisting of  $\{e_2 - e_3, \dots, e_7 - e_8, e_7 + e_8, \frac{1}{2}(e_1 - e_2 - \dots - e_7 + e_8)\}$ .

- $E_7$ :  $E_7$  is the intersection of  $E_8$  with the hypersurface  $\sum x_i = 0$ , so

$$\Phi_{E_7} = \{\pm(e_i - e_j), i \neq j\} \cup \left\{ \frac{1}{2} \left( \sum_{i=1}^8 \pm e_i \right) \text{ with 4 minus signs} \right\}.$$

The polarization with

$$t(e_1) = 18, t(e_2) = 7, t(e_3) = 6, \dots, t(e_8) = 1$$

gives a base  $\Delta = \{e_2 - e_3, \dots, e_7 - e_8, \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 + e_6 + e_7 + e_8)\}$ .

- $E_6$ :  $E_6$  is the intersection of  $E_7$  with the hypersurface  $x_7 + x_8 = 0$ .

$$\Phi_{E_6} = \{e_i - e_j | i \neq j \leq 6\} \cup \{\pm(e_7 - e_8)\} \cup \left\{ \frac{1}{2}(\pm e_1 \pm e_2 \pm \dots \pm e_7) \pm \frac{1}{2}(e_7 - e_8) \text{ with 3 minus signs before } v_i, i \leq 6 \right\}.$$

The polarization  $t$  s.t.

$$t(e_1) = 11, t(e_2) = 4, t(e_3) = 3, t(e_4) = 2, t(e_5) = 1, t(e_6) = 0, t(e_7) = 4, t(e_8) = 3$$

gives a base  $\Delta = \{e_2 - e_3, \dots, e_5 - e_6, e_7 - e_8, \frac{1}{2}(e_1 - e_2 - e_3 - e_4 + e_5 + e_6 - e_7 + e_8)\}$ .  
 $A_n^\vee \cong A_n, B_n^\vee \cong C_n, D_n^\vee \cong D_n, G_2^\vee \cong G_2, F_4^\vee \cong F_4, E_8^\vee \cong E_8, E_7^\vee \cong E_7, E_6^\vee \cong E_6.$  ┐

**Remark (3.9.6.4).** The polarizations  $t$  are intimately related to the averaged coroot  $\rho^\vee$  (3.9.6.9). ┐

**Thm. (3.9.6.5) [Classifying Dynkin Diagrams].** The Dynkin diagrams arising from irreducible root systems are exactly the diagrams  $A_n (n \geq 1), B_n (n \geq 2), C_n (n \geq 3), D_n (n \geq 4), E_6, E_7, E_8, F_4, G_2$ . (graphs are given in [Etingod, Lie algebras]P109.) ┐

*Proof:* This follows easily from (3.9.6.2) and (3.9.6.3). ┐

**Prop. (3.9.6.6) [Non-Reduced Root Systems].** For any  $n \geq 1$ , there is exactly one non-reduced irreducible root system of rank  $n$ , which is  $BC_n$ , the union of  $B_n$  and  $C_n$  in (3.9.6.3). ┐

*Proof:* Cf. [Kna96]P139. ┐

**Prop. (3.9.6.7) [Weight Lattices of Irreducible Root Systems].** Notation as in (3.9.6.3),



- $A_n$ :  $\alpha_i^\vee = \alpha_i$  for  $i \leq n$ , so  $\omega_i = (\frac{n+1-i}{n+1}, \dots, \frac{n+1-i}{n+1}, \frac{-i}{n+1}, \dots, \frac{-i}{n+1})$  ( $i$  terms) for  $1 \leq i \leq n$ . Thus

$$P = \{(x_i) \in \mathbb{R}^n | x_i - x_j \in \mathbb{Z}, \sum x_i = 0\}, \quad Q = \{(x_i) \in \mathbb{Z}^n | \sum x_i = 0\}, \quad P/Q \cong \mathbb{Z}/(n+1).$$

- $B_n$ :  $\alpha_i^\vee = \alpha_i$  for  $i < n$  and  $\alpha_n^\vee = 2e_n$ , so  $\omega_i = (1, \dots, 1, 0, \dots, 0)$  ( $i$  ones) for  $1 \leq i < n$ , and  $\omega_n = (\frac{1}{2}, \dots, \frac{1}{2})$ . Thus

$$P = \mathbb{Z}^n, \quad Q = \mathbb{Z}^n \cup [(\frac{1}{2}, \dots, \frac{1}{2}) + \mathbb{Z}^n], \quad P/Q \cong \mathbb{Z}/(2).$$

- $C_n$ :  $\alpha_i^\vee = \alpha_i$  for  $i < n$  and  $\alpha_n^\vee = e_n$ , so  $\omega_i = (1, \dots, 1, 0, \dots, 0)$  ( $i$  ones) for  $1 \leq i \leq n$ . Then  $P = \{(x_i) \in \mathbb{Z}^n | \sum x_i \in 2\mathbb{Z}\}$  and  $Q = \mathbb{Z}^n$ ,  $P/Q \cong \mathbb{Z}/(2)$ .
- $D_n$ :  $\alpha_i^\vee = \alpha_i$  for  $i \leq n$ , so  $\omega_i = (1, \dots, 1, 0, \dots, 0)$  ( $i$  ones) for  $1 \leq i \leq n-2$ ,  $\omega_{n-1} = (\frac{1}{2}, \dots, \frac{1}{2})$ ,  $\omega_n = (\frac{1}{2}, \dots, \frac{1}{2}, -\frac{1}{2})$ . So  $P/Q \cong (\mathbb{Z}/2\mathbb{Z})^2$  for  $n$  even and  $\mathbb{Z}/4\mathbb{Z}$  for  $n$  odd.
- $G_2$ : It is clear  $\omega_1 = \omega + 1$ ,  $\omega_2 = 2\omega + 1$ . So  $P = Q = \mathbb{Z}[\omega]$ .
- $F_4$ :  $\omega_1 = (1, 1, 0, 0)$ ,  $\omega_2 = (2, 1, 1, 0)$ ,  $\omega_3 = (\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ ,  $\omega_4 = (1, 0, 0, 0)$ . So  $P = Q = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) + \mathbb{Z}^4$ .
- $E_8$ :  $\omega_1 = (1, 1, 0, \dots, 0)$ , So  $P = Q = \{(x_1, \dots, x_8) | x_i \in \mathbb{Z} \text{ or } x_i \in \frac{1}{2} + \mathbb{Z}, \sum x_i \in 2\mathbb{Z}\}$ .
- $E_7$ :  $P/Q \cong \mathbb{Z}/(2)$ .
- $E_6$ :  $P/Q \cong \mathbb{Z}/(3)$ .

┘

*Proof:*

□

**Prop. (3.9.6.8)[Maximal Roots].** For  $A_n$ , the maximal root is  $\theta = e_1 - e_{n+1} = \alpha_1 + \dots + \alpha_n = \omega_1 + \omega_n$ . For  $C_n$ , the maximal root is  $\theta = 2e_1 = 2\alpha_1 + \dots + 2\alpha_{n-1} + \alpha_n = 2\omega_1$ .

But for any other irreducible root system  $\Phi$ , the maximal root  $\theta$  is a fundamental root:

- $B_n$ :  $\theta = e_1 + e_2 = \alpha_1 + 2\alpha_2 + \dots + 2\alpha_n = \omega_2$ .
- $D_n$ :  $\theta = e_1 + e_2 = \alpha_1 + 2\alpha_2 + \dots + 2\alpha_{n-2} + \alpha_{n-1} + \alpha_n = \omega_2$ .
- $G_2$ :  $\theta = 2\omega + 1 = 3\alpha_1 + 2\alpha_2 = \omega_2$ .
- $F_4$ :  $\theta = e_1 + e_2 = 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 = \omega_1$ .
- $E_8$ :  $\theta = e_1 + e_2 = 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 5\alpha_4 + 6\alpha_5 + 4e_6 + 3e_7 + 2e_8 = \omega_1$ .
- $E_7$ :  $\theta = e_1 - e_8 = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + 2\alpha_7 = \omega_6$ .
- $E_6$ :  $\theta = \varepsilon_1 - \varepsilon_6 = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4 + \alpha_5 + 2\alpha_6 = \omega_4$ .

┘

**Prop. (3.9.6.9) [ $\rho$  and  $\rho^\vee$ ].** We write  $\rho_\Phi^\vee \triangleq \rho_{\Phi^\vee}$ . Under the identification of  $V$  and  $V^*$ , If  $(R, S)$  is simply laced, then  $R \cong R^\vee$  via  $\alpha \mapsto \alpha^\vee$ , so by (3.9.6.7),

- $A_n$ :  $\rho_{A_n} = \rho_{A_n}^\vee = (\frac{n}{2}, \frac{n-2}{2}, \dots, -\frac{n}{2}) = \sum_i \frac{i(n+1-i)}{2} \alpha_i$ .
- $B_n$ :  $\rho_{B_n} = \rho_{B_n}^\vee = (\frac{2n-1}{2}, \frac{2n-3}{2}, \dots, \frac{3}{2}, \frac{1}{2}) = \sum_i \frac{i(2n-i)}{2} \alpha_i$ .
- $C_n$ :  $\rho_{C_n} = \rho_{C_n}^\vee = (n, n-1, \dots, 1) = \sum_{i \leq n-1} \frac{i(2n+1-i)}{2} \alpha_i + \frac{n(n+1)}{4} \alpha_n$ .
- $D_n$ :  $\rho_{D_n} = \rho_{D_n}^\vee = (n-1, n-2, \dots, 0) = \sum_{i \leq n-2} \frac{i(2n-1-i)}{2} \alpha_i + \frac{1}{2} \alpha_{n-1} + \frac{n(n-1)}{4} \alpha_n$ .



- $G_2 : \rho_{G_2} = 3\omega + 2 = 5\alpha + 3\beta, \rho_{G_2}^\vee = \frac{10}{3}\omega + \frac{8}{3} = 6\alpha + \frac{10}{3}\beta.$
- $F_4 : \rho_{F_4} = (\frac{11}{2}, \frac{5}{2}, \frac{3}{2}, \frac{1}{2}), \rho_{F_4}^\vee = (8, 3, 2, 1).$
- $E_8 : \rho_{E_8} = \rho_{E_8}^\vee = (23, 6, 5, 4, 3, 2, 1, 0).$
- $E_7 : \rho_{E_7} = \rho_{E_7}^\vee = (18, 7, 6, \dots, 1) - (\frac{23}{4}, \dots, \frac{23}{4}).$
- $E_6 : \rho_{E_6} = \rho_{E_6}^\vee = (\frac{15}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}, -\frac{5}{2}, -\frac{7}{2}, \frac{1}{2}, -\frac{1}{2}).$

┘

**Def. (3.9.6.10) [Coxeter Numbers].** If  $(\Phi, \Delta)$  is an irreducible, define the **Coxeter number** to be  $h_\Phi = (\theta, \rho^\vee) + 1 = \text{ht}(\theta) + 1$ , and the **dual Coxeter number**  $h_\Phi^\vee = (\rho, (\theta)^\vee) + 1$ . Clearly if  $(\Phi, S)$  is simply-laced (3.9.5.29), then  $\Phi \cong \Phi^\vee$  via  $\alpha \mapsto \alpha^\vee$ , so if  $\theta = \sum m_i \alpha_i$ , then  $\theta^\vee = \sum m_i \alpha_i^\vee$ , so  $h_R^\vee = h_R = \sum m_i + 1$ . ┘

**Prop. (3.9.6.11) [Listing of Coxeter Numbers].** We can determine the Coxeter numbers and dual coxeter numbers of irreducible root systems:

- $h_{A_n} = h_{A_n}^\vee = n + 1.$
- $h_{B_n} = 2n$ , and  $\theta^\vee = e_1 + e_2 = \alpha_1^\vee + 2\alpha_2^\vee + \dots + \alpha_n^\vee$ , so  $h_{B_n}^\vee = 2n - 1.$
- $h_{C_n} = 2n$ , and  $\theta^\vee = e_1 = \alpha_1^\vee + \dots + \alpha_n^\vee$ , so  $h_{C_n}^\vee = n + 1.$
- $h_{D_n} = h_{D_n}^\vee = 2n - 2.$
- $h_{G_2} = 6$ , and  $\theta^\vee = \frac{2}{3}(2\omega + 3) = \alpha_1^\vee + 2\alpha_2^\vee$ , so  $h_{G_2}^\vee = 4.$
- $h_{F_4} = 12$ , and  $\theta^\vee = e_1 + e_2 = 2\alpha_1^\vee + 3\alpha_2^\vee + 2\alpha_3^\vee + \alpha_4^\vee$ , so  $h_{F_4}^\vee = 9.$
- $h_{E_8} = h_{E_8}^\vee = 30.$
- $h_{E_7} = h_{E_7}^\vee = 18.$
- $h_{E_6} = h_{E_6}^\vee = 12.$

┘

**Prop. (3.9.6.12) [Listing of Minuscale Weights].** By (3.9.5.39) and (3.9.6.3)(3.9.6.7), we can determine minuscule weights when  $(\Phi, \Delta)$  is irreducible.

- $A_n$ :  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/(n+1)$ , so it has  $n+1$  minuscule weight, which are all the fundamental weights  $\omega_i$  and 0.
- $B_n$ :  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/(2)$ , so there is exactly one non-zero minuscule weight, which is  $\omega_n = (\frac{1}{2}, \dots, \frac{1}{2})$ , by (3.9.5.38).
- $C_n$ :  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/(2)$ , so there is exactly one non-zero minuscule weight, which is  $\omega_n = (1, 0, \dots, 0)$ , by (3.9.5.38).
- $D_n$ :  $\#P(\Phi)/Q(\Phi) = 4$ , so there are exactly 3 non-zero minuscule weights, which are  $\omega_1, \omega_{n-1}$  and  $\omega_n$ , with corresponding fundamental representations of dimension  $2n, 2^{n-1}, 2^{n-1}$ . This is because the maximal coroot  $\beta^\vee = \alpha_1^\vee + 2\alpha_2^\vee + \dots + 2\alpha_{n-2}^\vee + \alpha_{n-1}^\vee + \alpha_n^\vee$ .
- $G_2$ :  $\#P(\Phi)/Q(\Phi) = 1$ , so there are no non-zero minuscule weights.
- $F_4$ : There are no non-zero minuscule weights.
- $E_8$ : There are no non-zero minuscule weights.
- $E_7$ :  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/2\mathbb{Z}$ , so there is exactly one non-zero minuscule weight.

- $E_6$ :  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/3\mathbb{Z}$ , so there are exactly two non-zero minuscule weights.  $\lrcorner$

**Prop. (3.9.6.13) [Longest Element  $w_0$ ].** Notice  $-w_0$  (3.9.6.13) is an automorphism of  $(\Phi, \Delta)$  that permutes fundamental weights and simple roots, so it induces an automorphism of the Dynkin diagram of  $\Phi$ . So if the Dynkin diagram of  $\mathfrak{g}$  has no non-trivial automorphism,  $w_0 = -1$ . This is case for  $\Phi = A_1, B_n, C_n, G_2, F_4, E_7$  and  $E_8$ .

For  $D_n$ , if  $n$  is even, then  $-1 \in W$  so  $w_0 = -1$ . If  $n$  is odd, then  $w_0(e_i) = -e_i$  for  $i < n$  and  $w_0(e_n) = e_n$ . Notice also each  $s_i$  acts trivially on  $P(\Phi)/Q(\Phi)$ , so  $-w_0$  acts by  $-1$  on  $P(\Phi)/Q(\Phi)$ .

For  $A_n, n \geq 2$ ,  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/(n+1)$ , thus  $-w_0$  is the flip of the chain.

Likewise for  $E_6$ ,  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/3\mathbb{Z}$ , thus  $-w_0$  flips the two minuscule weights.

But for  $D_n$ ,  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$  for  $n$  even and  $P(\Phi)/Q(\Phi) \cong \mathbb{Z}/4\mathbb{Z}$  for  $n$  odd, so  $-w_0$  preserves the minuscule weights or flip  $w_{n-1}$  and  $w_n$ .  $\lrcorner$

## 7 Notations for a Root System

**Def. (3.9.7.1) [Notations for a Root System].** For a root system  $(\Phi, \Delta)$ ,

- $\Phi^+$  (resp.  $\Phi^-$ ) is the positive (resp. negative) roots w.r.t.  $\Delta$ .
- $\Delta = \{\alpha_1, \dots, \alpha_r\}$ .
- $s_i$  is the reflection w.r.t.  $\alpha_i$ .
- $\theta$  is the maximal root.
- $\theta^\vee$  is the maximal coroot (maximal root of the dual root system). (It may not be dual to  $\theta$ )
- $\rho = \frac{1}{2} \sum_i \alpha_i = \sum \omega_i$ .
- $\rho^\vee$  is  $\rho$  of the dual system  $\Phi^\vee$ .
- The height of weights is defined as in (3.9.5.19).
- $W$  is the Weyl group (3.9.5.1).
- $w_0 \in W$  is the unique element with length  $\#\Phi^+$  (3.9.5.18).
- $n_{ij}$  is the Cartan number  $n(\alpha_i, \alpha_j)$  (3.9.5.22).
- $Q(\Phi)$  is the root lattice (3.9.5.30).
- $Q^+(\Phi)$  is the set of positive combinations of roots (3.9.5.33).
- $P(\Phi)$  is the weight lattice, generated by the fundamental weights  $\omega_i$  (3.9.5.30).
- $P^+(\Phi) \subset P(\Phi)$  is the set of dominant integral weights (3.9.5.33).
- $\mathfrak{P}$  is the Kostant partition function on  $Q(\Phi)$  (3.9.5.21).  $\lrcorner$

## 8 Affine Root Systems

References are [B-T72, Tit74, K-P23].

**Notation (3.9.8.1) [Affine Spaces].**

- Let  $k = \mathbb{R}$ .
- Let  $V \in \mathcal{V}\text{ect}^{\text{fin}}/k$ .

- For an affine  $V$ -torsor  $A$ , denote  $A^*$  the space of affine maps  $A \rightarrow k$ .
- There is a natural exact sequence  $0 \rightarrow k \rightarrow A^* \rightarrow V^* \rightarrow 0$ , where  $A^* \rightarrow V^* : \psi \mapsto \dot{\psi}$  is the derivative map.

┘

**Def. (3.9.8.2) [Affine Root Systems].** For an affine  $V$ -torsor  $A$ , an **affine root system** on  $A$  is a subset  $\Psi \subset A^*$  s.t.

- $\Psi$  consists of non-constant functions, spans  $A^*$ , and its image  $\Phi$  under  $A^* \rightarrow V^*$  is finite.
- For any  $\psi \in \Psi$ , there exists  $\dot{\psi}^\vee \in V$  s.t.  $\langle \psi^\vee, \dot{\psi} \rangle = 2$ , and the reflection  $s_\psi$  on  $A^*$  (3.9.2.1) defined by

$$s_\psi(x) = x - \langle \dot{\psi}^\vee, x \rangle \psi, \text{ where } x \in A^*$$

preserves  $\Psi$ . Such a  $\dot{\psi}^\vee$  is unique by (3.9.2.3). Notice  $s_\psi|_{k^*} = \text{id}_{k^*}$ .

- For any  $\psi, \eta \in \Psi$  we have  $\langle \dot{\eta}^\vee, \dot{\psi} \rangle \in \mathbb{Z}$ .
- For any  $a \in \Phi$ , the set  $\{\psi \in \Psi | \dot{\psi} = a\}$  is discrete.

Elements of  $\Psi$  are called the **affine roots** of  $\Psi$ , and  $\{\dot{\psi}^\vee\}_{\psi \in \Psi}$  is called the set of **affine coroots** of  $\Psi$ .

$\text{rank } V$  is called the **rank of the affine root system**.

The subgroup  $W(\Psi) \leq \text{GL}(A^*)$  generated by  $\{s_\psi\}_{\psi \in \Psi}$  is called the **affine Weyl group** of  $\Psi$ .

Use  $(V, A, \Psi, \Phi, W) \in \text{AffRootSys}$  to denote that  $\Psi$  is an affine root system on the affine  $V$ -torsor  $A$ .

┘

**Cor. (3.9.8.3) [Weyl Group  $W(\Psi)$ ].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}$  (3.9.8.2) and each root  $\psi \in \Psi$ , denote

$$s_\psi(x) = x - \psi(x)\dot{\psi}^\vee, \text{ where } x \in A,$$

then the image of  $s_\psi$  under the map  $\text{Aff}(A)^{\text{op}} \subset \text{GL}(A^*) : T(f)(x) = f(Tx)$  is just  $s_\psi \in \text{GL}(A^*)$  defined in (3.9.8.2). Thus  $W(\Psi)$  acts on  $A$  on the right, and taking derivative induces a map

$$W(\Psi)^{\text{op}} \rightarrow W(\Phi)^{\text{op}}$$

which dualizes to  $W(\Psi) \rightarrow W(\Phi)$ .

┘

**Def. (3.9.8.4) [Extended Weyl Groups].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}$  (3.9.8.2), the **extended Weyl-group**  $W(\Psi)^{\text{ext}}$  is the group of affine automorphisms of  $A$  preserving  $\Psi$ , and has derivative in  $W(\Phi)$ . Clearly  $W(\Psi) \leq W(\Psi)^{\text{ext}}$ .

┘

**Prop. (3.9.8.5).** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}$  (3.9.8.2),

- $(V^*, \Phi, W(\Phi)) \in \text{RootSys}$  (3.9.5.1), and for each  $\psi \in \Psi$ ,  $\dot{\psi}^\vee$  is the coroot of  $\psi \in V^*$ .
- For any  $\psi, \eta \in \Psi$ , if  $\eta = r\psi + c$  for some  $r, c \in k$ , then  $r \in \{\pm\frac{1}{2}, \pm 1, \pm 2\}$  and  $\dot{\psi}^\vee = r\dot{\eta}^\vee$ .
- The natural map  $W(\Psi) \rightarrow W(\Phi)$  (3.9.8.3) is surjective.

$(V^*, \Phi, W(\Phi))$  is called the **derived root system** of  $\Psi$ .

┘

*Proof:* 1: The axioms of a root system (3.9.5.1) is easy to check.

2: This follows from (3.9.5.2).

3: Trivial.

□

**Def. (3.9.8.6) [Reduced Affine Root Systems].** A **reduced affine root system** is an affine root system  $\Psi$  (3.9.8.2) s.t.  $\Psi \cap 2\Psi = \emptyset$ .  $\lrcorner$

**Prop. (3.9.8.7).** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ ,  $W(\Psi)$  acts properly on  $A$  (3.9.8.3) (4.12.1.11).  $\lrcorner$

*Proof:* Because  $\#W(\Phi) < \infty$  (3.9.5.1), it suffices to prove that the kernel of  $W(\Psi) \rightarrow W(\Phi)$ , which is the subgroup of translations in  $W(\Psi)$ , acts properly on  $A$  (4.12.1.15) characteriszlprooperancrihausroe. Then acting properly means that the subgroup of translations is not dense in  $V$ , which is guaranteed by definition (3.9.8.2).  $\square$

**Prop. (3.9.8.8) [Affine Root Systems are Infinite].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ ,

- $W(\Psi)$  has no fixed point on  $A$ .
- $\#\Psi = \infty$ .
- $\#W(\Psi) = \infty$ .

$\lrcorner$

*Proof:* 1: If  $x \in A$  is fixed by  $W(\Psi)$ , then by (3.9.8.7),  $\psi(x) = 0$  for all  $\psi \in \Psi$ , which means that  $x = 0$  because  $\Psi$  spans  $A^*$ .

2: If  $\#W(\Psi) < \infty$ , then the center of mass of any orbit of the action of  $W(\Psi)$  on  $A$  is fixed by  $W(\Psi)$ , contradicting item 1.

3: By item 2, the kernel of the map  $W(\Psi) \rightarrow W(\Phi)$  is non-trivial, thus has a translation. But  $W(\Psi)$  preserves  $\Psi$ , so  $\Psi$  is infinite.  $\square$

**Prop. (3.9.8.9) [Another Characterization of Affine Root Systems].** For an affine  $V$ -torsor  $A$ , a subset  $\Psi \in A^*$  is an affine root system (3.9.8.2) iff

- $\Phi = \dot{\Psi} \subset V^*$  is a root system on  $V^*$  (3.9.8.20).
- For any  $\psi \in \Psi$ , define the affine endomorphism  $s_\psi$  of  $A$  that fixes the hyperplane  $H_\psi$  and with derivative  $s_{\dot{\psi}}$ , then  $s_\psi$  preserves  $\Psi$ .
- For any  $\alpha \in \Phi$ , the set  $\{\psi \in \Psi | \dot{\psi} = \alpha\}$  contains at least two elements and is discrete in  $A^*$ .

$\lrcorner$

*Proof:* If  $\Psi$  is an affine root system, then these properties are satisfied by (3.9.8.5) (3.9.8.8). Conversely, if these properties are satisfied, then  $\Psi$  doesn't contain constant functions, and by item 3 it spans  $A^*$ . And the other axioms for affine root systems follows easily.  $\square$

**Def. (3.9.8.10) [Irreducible Affine Root Systems].** An **irreducible affine root system** is an affine root system  $(V, A, \Psi, \Phi, W(\Psi))$  (3.9.8.2) s.t.  $\Phi$  is an irreducible root system (3.9.5.1).  $\lrcorner$

**Def. (3.9.8.11) [Simplicial Structure on  $A$ ].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$  and  $\psi \in \Psi$ ,  $H_\psi = \{a \in A | \psi(a) = 0\}$  is a sub affine space of  $A$ , called the **affine root hyperplane** associated to  $\psi$ , whose derivative  $\dot{H}_\psi$  is equal to  $H_{\dot{\psi}} = \{v \in V | \dot{\psi}(v) = 0\}$ .

Then the set  $\{H_\psi\}_{\psi \in \Psi}$  is locally finite, and connected components of

$$A \setminus \bigcup_{\psi \in \Psi} H_\psi$$

are called **affine chambers** of this affine root system. And connected components of

$$\bigcup_{\psi \in \Psi} H_\psi \setminus \bigcup_{\psi_1 \neq \psi_2 \in \Psi} H_{\psi_1} \cap H_{\psi_2}$$

are facets of codimension 1 of this affine root system. Continuing this way, we get a polysimplicial structure on  $A$ , and if  $\Psi$  is irreducible(3.9.8.10), this is a simplicial structure.  $\lrcorner$

*Proof:* To show that the set  $\{H_\psi\}$  is locally finite, Cf.[K-P23]P17. For the last assertion, Cf.[Bourbaki]Chapter 5.3.9.Proposition 8.  $\square$

**Prop.(3.9.8.12) [Weyl-Group Action on  $A$ ].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ , with the simplicial structure on  $A$ (3.9.8.11),

1.  $W(\Psi)$  acts as polysimplicial automorphisms on  $A$ .
2. If  $w \in W(\Psi)$  stabilizes a simplex  $\sigma \in A$ , then  $w$  fixes vertices of  $\sigma$ . And this happens iff  $w$  is a product of reflections along hyperplanes containing  $\sigma$ .
3. If  $\sigma, \tau$  are simplexes of a same chamber and  $w\sigma = \tau$  for  $w \in W(\Psi)$ , then  $\sigma = \tau$ .
4.  $W(\Psi)$  acts simply-transitively on the set of chambers of  $A$ .

$\lrcorner$

*Proof:* Cf.[Bourbaki]Chapter5.3.3,Prop1, Thm2.  $\square$

**Cor.(3.9.8.13).** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ ,  $W(\Phi)$  is normal in  $W(\Phi)^{\text{ext}}$ (3.9.8.4), and for any chamber  $C \subset A$ , if  $\Xi_C$  is the stablizer of  $C$  in  $W(\Phi)^{\text{ext}}$ , then  $W(\Phi)^{\text{ext}} = W(\Phi) \ltimes \Xi_C$ .  $\lrcorner$

**Def.(3.9.8.14) [Simple Affine Roots].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ , if  $C \subset A$  is a chamber(3.9.8.11), define  $\Psi(C)^+$ (resp.  $\Psi(C)^-$ ) to be the set of those  $\psi \in \Psi$  s.t.  $\psi(C) \geq 0$ (resp.  $\psi(C) \leq 0$ ).

And the set  $\Psi(C)^0$  of indivisible  $\psi \in \Psi$  s.t.  $\psi(C) \in \mathbb{R}_+$  and  $H_\psi$  is a face of  $C$  is called an **affine basis** of  $\Psi$ , and elements of  $\psi(C)^0$  are called **simple affine roots** w.r.t.  $C$ .

If  $S$  is the set of reflections along the elements in  $\Psi(C)^0$ , then  $(W(\Psi), S)$  is a Coxeter-system(3.9.1.1).  $\lrcorner$

*Proof:* Cf.[Bourbaki]Chapter5.3.2, Thm1.1.  $\square$

**Def.(3.9.8.15) [Direct Sum of Affine Root Systems].** For  $(V_1, A_1, \Psi_1), (V_2, A_2, \Psi_2) \in \text{AffRootSys}(3.9.8.2)$ , the natural map  $\oplus : A_1^* \oplus A_2^* \rightarrow (A_1 \oplus A_2)^*$  transports  $\Psi_1 \cup \Psi_2$  bijectively to a subset of  $(A_1 \oplus A_2)^*$  because  $\Psi_i$  contains no constant functions. If we define  $d(\psi_1 \oplus \psi_2) = \dot{\psi}_1 + \dot{\psi}_2$ , then it can be verified that  $\Psi_1 \oplus \Psi_2$  is an affine root system on  $A_1 \oplus A_2$ , called the **direct sum of affine root systems**.  $\lrcorner$

**Prop.(3.9.8.16) [Decomposing into Irreducible Affine Root Systems].** Any affine root system  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$  is naturally a direct sum of irreducible affine root systems(3.9.8.10)(3.9.8.15). In particular,  $\Psi$  is irreducible iff it is not a direct sum of two other affine root systems(3.9.8.15).  $\lrcorner$

*Proof:* Cf.[K-P23]P22.  $\square$

**Prop.(3.9.8.17) [Simple Affine Roots as a Basis].** Situation as in(3.9.8.11), if  $C \subset A$  is an affine chamber and  $\Delta = \Psi(C)^0$  is the set of simple affine roots associated to  $C$ (3.9.8.14), then

- $W(\Psi)$  is generated by  $\{s_\psi\}_{\psi \in \Delta}$ .
- $W(\Psi)$  acts simply-transitively on the set of basis of  $\Psi$ .
- $\Psi^{\text{short}} = W(\Psi) \cdot \Delta$ .

- Any  $\psi \in \Psi(C)^+$  (3.9.8.14) is a non-negative integral linear combination of elements of  $\Delta$ . In particular,  $\Delta$  spans  $A^*$ .
- For any  $\psi \neq \eta \in \Delta$ , we have  $\dot{\psi}(\dot{\eta}^\vee) \leq 0$ .

If moreover  $\Psi$  is irreducible (3.9.8.10), then

- $\Delta$  is a basis of  $A^*$ .
- The vertices  $\{x_0, \dots, x_l\}$  of  $C$  are in bijection with  $\Delta$ , s.t.  $\psi_i(x_j) \neq 0$  iff  $i = j$ .

┘

*Proof:* 1: Cf. [Bourbaki]Chap5. 3.1. Lemma2?

2: This follows from (3.9.8.12).

3: This follows from item2.

4: Let  $\Lambda$  be the lattice in  $A^*$  generated by  $\Delta$ , then it is stable under action of each  $s_\psi$  for  $\psi \in \Delta$ . Thus it is stable under action of  $W(\Psi)$ , and item3 implies that  $\Lambda$  is just the lattice generated by  $\Psi$ . Thus any  $\psi \in \Psi(C)^+$  is an integral combination of elements of  $\Delta$ . To show the coefficients are non-negative, notice we can decompose  $\Psi$  into a direct sum of irreducible affine root systems (3.9.8.16), and on each irreducible affine root system, we can evaluate at the vertices of  $C$  and apply item2 below.

5: Cf. [Bourbaki]Chap5. 3.4. Prop3?

1, 2: This is clear because in this case  $C$  is a simplex in  $A$ .

□

**Lemma (3.9.8.18).** Situation as in (3.9.8.11), if  $C \subset A$  is an affine chamber and  $\psi \in \Psi(C)^0$ , then  $s_\psi$  permutes the elements of  $\Psi(C)^+$  that are not proportional to  $\alpha$ .

┘

*Proof:* We may reduce to the case when  $\Psi$  is irreducible. Then for any  $\xi \in \Psi(C)^0$  we have  $s_\psi(\xi) = \xi - \dot{\xi}(\dot{\psi}^\vee)\alpha$ . But any element in  $\Psi(C)^+$  (resp.  $\Psi(C)^-$ ) is uniquely a non-negative combination of elements in  $\Psi(C)^0$ , so the assertion follows.

□

**Prop. (3.9.8.19) [Geometric Meaning of Length Functions for  $W(\Psi)$ ].** Situation as in (3.9.8.14), if length is the length function of the Coxeter-system  $(W(\Psi), S)$ , then

- If  $w \in W(\Psi)$  and  $\psi \in \Psi(C)^0$ , then  $\text{length}(s_\alpha w) > \text{length}(w)$  iff  $w^{-1}\psi \in \Psi(C)^+$ .
- For any  $w \in W(\Psi)$ ,  $\text{length}(w) = \#(w.\Psi(C)^{\text{short},+} \cap \Psi(C)^{\text{short},-})$ .
- If  $w = s_1 \dots s_q$  is a reduced expression where  $s_i = s_{\alpha_i}$  for  $\psi_i \in \Psi(C)^0$ , then

$$w.\Psi(C)^{\text{short},+} \cap \Psi(C)^{\text{short},-} = \{s_q \dots s_{i+1}(\psi_i) | i \in [q]_+\}.$$

┘

*Proof:* 1: By [Bourbaki]Thm5.3.2.1.(2)?,  $\text{length}(sw) > \text{length}(w)$  iff the chambers  $C$  and  $w(C)$  are on the same side of  $H_\psi \subset A$ , or equivalently,  $\alpha(w.C) \in \mathbb{R}_{\geq 0}$ .

2: Use induction on  $\text{length}(w)$ . If  $\text{length}(w) = 1$ , this follows from (3.9.8.18). For  $\text{length}(w) > 1$ , write  $w = s_\psi w'$  where  $\psi \in \Psi(C)^0$ , then

$$\begin{aligned} \#(w.\Psi(C)^{\text{short},+} \cap \Psi(C)^{\text{short},-}) &= \#(w'.\Psi(C)^{\text{short},+} \cap s_\psi.\Psi(C)^{\text{short},-}) \\ &= \#(w'.\Psi(C)^{\text{short},+} \cap (\{\psi\} \cup \Psi(C)^{\text{short},-} \setminus \{-\psi\})) \end{aligned}$$

And notice that  $\psi \in w'.\Psi(C)^{\text{short},+}$  by item1.

3: It follows from item1 that the set  $\{s_q \dots s_{i+1}(\psi_i) | i \in [q]_+\}$  is contained in  $\Psi(C)^+$  and mapped by  $w$  into  $\Psi(C)^-$ . And the equality follows from item2.

□

**Def. (3.9.8.20)** [Associated Affine Root Systems]. For  $(V, \Phi) \in \text{RootSys}(3.9.5.1)$ , take  $A = V^*$  and define

$$\Psi_\Phi = \{\alpha + n | \alpha \in \Phi, n \in I_\alpha\} \subset A^*,$$

where  $I_\alpha = \begin{cases} \mathbb{Z} & , \alpha \in \Phi^{\text{short}} \\ 2\mathbb{Z} + 1 & , \alpha \in \Phi^{\text{long}} \end{cases}$ , then  $(V^*, A, \Psi)$  is a reduced affine root system with derivative  $(V, \Phi)(3.9.8.2)$ , by (3.9.8.9).  $\perp$

**Def. (3.9.8.21)** [Dual Affine Root Systems]. Cf. [K-P23]P26.  $\perp$

**Prop. (3.9.8.22)** [Sliced Root Systems]. For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$  and  $x \in A$ , define  $\Psi_x = \{\psi \in \Psi | \psi(x) = 0\}$ , then

1.  $\Psi_x$  is a root system of the subspace of  $A_x^*$  that it generates.
2.  $d : \Psi_x \rightarrow \Phi$  is injective, making  $\Psi_x$  a sub-root system of  $\Phi$  (but may not be closed).
3. The subset  $\{\psi^\vee | \psi \in \Psi_x\} \subset \Psi^\vee(3.9.8.21)$  is identified with  $\Psi_x^\vee$ .
4. Let  $W(\Psi)_x$  be the stablizer of  $x$  in  $W(\Psi)$ , then the action of  $W(\Psi)_x$  on  $\Psi_x$  identifies  $W(\Psi)_x$  with  $W(\Psi_x)$ .
5. If  $C$  is a chamber of  $A$  containing  $x$ , then  $\Psi_x(C)^0 = \Psi(C)^0 \cap \Psi_x(3.9.8.14)$  is a basis for  $\Psi_x$ , with corresponding set of positive roots  $\Psi_x(C)^+$ .
6.  $C \mapsto \Psi_x(C)^+ = \Psi_x \cap \Psi(C)^+$  is a bijection between the set of chambers containing  $x$  with the set of Weyl-chambers in  $\Psi_x$ .
7. More generally,  $\sigma' \mapsto \Psi_x(\sigma')^+ = \{\psi \in \Psi_x | \psi(\sigma') \geq 0\}$  is an inclusion-preserving bijection between the set of simplexes containing  $x$  to the set of parabolic subsets of  $\Psi_x(3.9.5.15)$ .  $\perp$

*Proof:* 2 is easy. For 1,  $\Psi_x$  doesn't contain 0 because  $\Psi$  doesn't contain constants, and the dual element of  $\psi \in \Psi_x$  is just  $\psi^\vee = \dot{\psi}^\vee \in V \cong (A_x^*)^\vee$ , noticing that  $s_\psi$  preserves  $A_x^*$ .

3: is easy.

4 follows from (3.9.8.12) item 2.

5: All elements of  $\Psi_x(C)^0$  are short. By (3.9.8.17) each  $\psi \in \Psi_x(C)^+$  can be written as a non-negative integral combination of elements of  $\Psi(C)^0$ , but each element of  $\Psi(C)^+$  is non-negative at  $x$ , so this combination only contain those  $\psi_i$  s.t.  $\psi(x) = 0$ , i.e. those  $\psi_i \in \Psi_x(C)^0$ . To show that elements of  $\Psi_x(C)^0$  are linearly independent, write  $C = C_1 \times \dots \times C_n$  as direct sum of irreducible chambers (3.9.8.15), and  $x = (x_1, \dots, x_n)$ , then a linear relation between elements of  $\Psi_x(C)^0$  will induce an affine relation between elements of  $\Psi_{i,x_i}(C_i)^0$ . But elements of  $\Psi_{i,x_i}(C_i)^0$  all vanish at  $x_i$ , so any affine relation is in fact a linear relation, and thus trivial by (3.9.8.17).

6: Any other basis of  $\Psi_x$  is obtained from  $\Psi_x(C)^0$  by action of  $W(\Psi_x) = W(\Psi)_x$  by item 4, and  $W(\Psi_x)$  acts on  $C$  gives all chambers containing  $C$  by (3.9.8.12).

7: It is clear that  $\Psi_x(\sigma')$  is a parabolic subset of  $\Psi_x$ . Conversely, Cf. [K-P23]P28.  $\textcolor{red}{?}$   $\square$

**Def. (3.9.8.23)** [Special Points]. For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ , a point  $x \in A$  is called

- A **special point** if for each  $\psi \in \Psi$ , there exists  $\psi' \in \Psi_x(3.9.8.22)$  s.t.  $H_\psi$  and  $H_{\psi'}$  are parallel.
- A **extra special point** if  $\Phi_x$  contains a basis of  $\Phi$ .

$\perp$

**Prop. (3.9.8.24)** [Properties of Special Points]. Situation as in (3.9.8.22),



1. special points are vertices of  $A$ .
2.  $d : W(\Psi_x) \rightarrow W(\Phi)$  is injective, and it is surjective iff  $x$  is a special point(3.9.8.23).
3. Any chamber  $C$  contains an extra special point.
4. If  $x \in A$  is extra special, then  $d : \Psi_x^{\text{short}} \rightarrow \Phi^{\text{short}}$  is a bijection.
5. Extra special points are special.
6. If  $\Phi$  is reduced, then special points are extra special.
7. If  $\Psi$  is irreducible, then for any basis  $\Delta$ , there exists  $\psi_0 \in \Delta$  s.t.  $d(\Delta \setminus \psi_0)$  is a basis for  $\Phi$ .

┘

*Proof:* 1: If  $x$  is special, then  $\Psi_x$  contains  $l$ -many independent elements, so  $x$  is a vertex.

2: It is clearly injective by(3.9.8.22). For the second assertion, Cf.[Bourbaki]5.3.10.9.?

3: Take a basis  $\alpha_1, \dots, \alpha_l$  of  $\Phi$ , and take  $\psi_i \in \Psi$  s.t.  $\dot{\psi}_i = \alpha_i$  and  $H_{\psi_i}$  is a wall of  $C$ , then the hyperplanes  $H_{\psi_i}$  intersect at a point, which is extra special.

4: This follows from(3.9.5.16).

5: This follows from item4.

6: If  $\Phi$  is reduced and  $x$  is special, then  $\Psi_x \rightarrow \Phi$  is a bijection.

7: In the irreducible case vertices of  $C$  are in bijection with  $\Psi(C)^0$ (3.9.8.17), so if  $x \in C$  is an extra special vertex(exists by item3), then the elements of  $\Psi(C)^0$  except the one corresponding to  $x$  is a basis of  $\Psi_x$ , by(3.9.8.22) item5.  $\square$

**Prop.(3.9.8.25)[Closed Symmetric Subsets].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ , if  $\Phi' \subset \Phi$  is a closed symmetric subset, and  $W \subset V$  is the subspace annihilated by  $\Phi'$ , then  $\Psi' = \{\psi \in \Psi | \dot{\psi} \in \Phi'\}$  is an affine root system in  $(A/W)^*$ , and is reduced if  $\Psi$  is.  $\square$

*Proof:* Cf.[K-P23]P46.  $\square$

**Def.(3.9.8.26)[Affine Dynkin Diagrams].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ ,  $C \subset A$  an affine chamber(3.9.8.11) and  $\Psi(C)^0$  the corresponding set of simple affine roots(3.9.8.14), then the **affine Dynkin diagram**  $D(\Psi)$  is given as follows:

- Its nodes are elements of  $\Psi(C)^0$ .
- Two nodes  $\psi, \eta \in \Psi(C)^0$  are joined by bonds if  $\dot{\psi}$  or  $\dot{\eta}$  are short, in which case they are joined by  $f(\psi, \eta) = \dot{\psi}(\dot{\eta}^\vee) \cdot \dot{\eta}(\dot{\psi}^\vee)$ -many bonds.
- If  $\dot{\psi}(\dot{\eta}^\vee) \neq \dot{\eta}(\dot{\psi}^\vee) = 1$ , place an arrow on the bonds between  $\psi$  and  $\eta$ , pointing at  $\eta$ .

The isomorphism class of  $D(\Psi)$  is independent of  $C$  chosen, by(3.9.8.12).  $\square$

**Prop.(3.9.8.27) [Affine Dynkin Diagram of Dual Systems].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ ,  $D(\Psi^\vee)$ (3.9.8.26) is obtained from  $D(\Psi)$  by inverting all arrows.  $\square$

*Proof:* ?  $\square$

**Prop.(3.9.8.28) [Dynkin Diagram of Sliced Root Systems].** For  $(V, A, \Psi, \Phi, W(\Psi)) \in \text{AffRootSys}(3.9.8.2)$ , if  $C \subset A$  is an affine chamber and  $\sigma \subset C$  is a facet of  $C$ , then the Dynkin diagram of  $\Psi_\sigma$ (3.9.8.22) is obtained from  $D(\Psi)$  by removing all nodes corresponding to vertices of  $\sigma$ (3.9.8.17).  $\square$

**Prop.(3.9.8.29)[Action of  $W(\Psi)^{\text{ext}}/W(\Psi)$  on  $D(\Psi)$ ].** Cf.[K-P23]P44. ?  $\square$



### Classifying Affine Root Systems

**Prop. (3.9.8.30) [Category of Affine Spaces].** Let

- $\mathcal{C} \in \text{Cat}$  be the category of pairs  $(V, A)$  where  $V \in \text{Vect}^{\text{fin}}/k$  and  $A$  an affine  $V$ -torsor, where the morphisms are affine maps.
- $\mathcal{D} \in \text{Cat}$  be the category of pairs  $(W, \lambda)$  where  $W \in \text{Vect}^{\text{fin}}/k$  and  $\lambda$  a linear functional on  $W$ .

Then the functors

$$F : \mathcal{C} \rightarrow \mathcal{D} : (V, A) \mapsto ((A^*)^\vee, \mathbf{1}_A), f \mapsto (f^*)^\vee,$$

$$G : \mathcal{D} \rightarrow \mathcal{C} : (W, \lambda) \mapsto (\lambda^{-1}(0), \lambda^{-1}(1)), f \mapsto f|_{\lambda^{-1}(1)}$$

are mutually inverse equivalence of categories.  $\lrcorner$

*Proof:* Both natural isomorphisms  $1 \rightarrow G \circ F$  and  $1 \rightarrow F \circ G$  are easy to construct.  $\square$

**Def. (3.9.8.31) [Isomorphisms of Affine Root Systems].** For  $(V_1, A_1, \Psi_1), (V_2, A_2, \Psi_2) \in \text{AffRootSys}$ ,

- A **strong isomorphism of affine root systems** between  $\Psi_1$  and  $\Psi_2$  is an isomorphism  $f : A_1 \rightarrow A_2$  of affine spaces s.t.  $f^* : A_1^* \rightarrow A_2^*$  induces a bijection  $\Psi_1 \cong \Psi_2$  and satisfies: For any  $\eta \in A_1^*$ , if  $\psi = f^*(\eta)$ , then  $\text{df}(\dot{\psi}^\vee) = \dot{\eta}^\vee$ .
- An **isomorphism of affine root system** between  $\Psi_1$  and  $\Psi_2$  is a bijective map  $\varphi : \Psi_1 \cong \Psi_2$  s.t. if  $r, s \in \mathbb{Z}, \psi, \eta \in \Psi_1$  s.t.  $s\psi + r\eta \in \Psi_1$ , then  $\varphi(s\psi + r\eta) = s\varphi(\psi) + r\varphi(\eta)$ .

$\lrcorner$

**Prop. (3.9.8.32) [ $c(\Psi)$ ].** For  $(V, A, \Psi) \in \text{AffRootSys}$  irreducible, and  $C \subset A$  a chamber with corresponding set of simple affine roots  $\Delta$ , then

- There exists a unique collection of positive integers  $(n_\psi)_{\psi \in \Delta}$  without common denominators s.t.  $c(\Psi) = \sum_{\psi \in \Delta} n_\psi \psi$  is a constant function on  $A$ .
- A special vertex  $x$  of  $C$  is extra special iff  $n_\psi = 1$  where  $\psi \in \Delta$  corresponds to  $x$  (3.9.8.17).
- $c(\Psi)$  is independent of  $C$  chosen.

$\lrcorner$

*Proof:* Cf. [K-P23]P31.  $\square$

**Prop. (3.9.8.33).** For  $(V_1, A_1, \Psi_1), (V_2, A_2, \Psi_2) \in \text{AffRootSys}$  irreducible, any isomorphism  $\varphi : \Psi_1 \cong \Psi_2$  (3.9.8.31) extends uniquely to a vector space isomorphism  $f^* : A_1^* \rightarrow A_2^*$  that sends constant functions to constant function, hence descending to a vector space isomorphism  $f^* : V_1^* \cong V_2^*$ . Then:

- For any  $\alpha_1 \in A_1^*, \eta_1 \in \Psi_1$ , we have  $\langle \dot{\alpha}_1, \dot{\eta}_1^\vee \rangle = \langle f^*(\alpha_1), f^*(\eta_1)^\vee \rangle$ .
- There exists  $\epsilon_\varphi \in \{\pm 1\}$  s.t.  $f^*(c(\Psi_1)\mathbf{1}_{A_1}) = \epsilon_\varphi c(\Psi_2)\mathbf{1}_{A_2}$  (3.9.8.32).
- If  $c(\Psi_1) = c(\Psi_2)$  and  $\epsilon_\varphi = 1$ , then  $\varphi$  is a strong isomorphism of affine root systems (3.9.8.31) ( $\dot{\eta}$  is the shorter roots).

$\lrcorner$

*Proof:* Cf. [K-P23]P34.  $\square$

**Prop. (3.9.8.34) [Isomorphisms and Strong Isomorphisms].** For  $(V, A, \Psi), (V', A', \Psi') \in \text{AffRootSys}$ , if  $\varphi : \Psi \cong \Psi'$  is an isomorphism of affine root system (3.9.8.31), then there exists an affine root system  $\Psi''$  on  $A$  s.t.

- If  $\Psi = \Psi_1 \oplus \dots \Psi_n$ ,  $A = A_1 \oplus \dots A_n$  is the decomposition of  $\Psi$  into irreducible affine root systems (3.9.8.16), then  $\Psi'' = r_1 \Psi_1 \oplus \dots r_n \Psi_n$  for some  $r_1, \dots, r_n \in \mathbb{R}^\times$ .
- The composition of  $\varphi : \Psi \cong \Psi'$  with the obvious bijection  $\Psi'' \cong \Psi$  is a strong isomorphism of affine root systems (3.9.8.31).

┘

*Proof:*  $\varphi : \Psi_i \rightarrow \varphi(\Psi_i)$  is an isomorphism of affine root systems for each  $i$ . If  $f_i^* : A_i^* \rightarrow (A'_i)^*$  is the extension of  $\varphi|_{\Psi_i}$ , then we can choose  $r_i \in \mathbb{R}^\times$  s.t.  $f_i^*(c(\Psi_1)\mathbf{1}_{A_1}) = r_i c(\varphi(\Psi_i))\mathbf{1}_{A'_i}$  (3.9.8.32), then the composition of  $\varphi : \Psi \cong \Psi'$  with the obvious bijection  $\Psi'' \cong \Psi$  is a strong isomorphism of affine root systems by (3.9.8.33).  $\square$

**Thm. (3.9.8.35)[Classifying Irreducible Reduced Affine Root Systems].** Any irreducible reduced affine root system is isomorphic to an affine root system  $\Psi_\Phi$  associated to a root system  $\Phi$  (3.9.8.20), or its dual.  $\square$

*Proof:* Cf. [K-P23]P39.  $\square$

**Cor. (3.9.8.36)[Types of Irreducible Reduced Affine Root Systems].** Isomorphism classes of an irreducible reduced affine root system is determined by its affine Dynkin diagram (3.9.8.26), and they are in bijection with the diagrams in [K-P23]P42.  $\square$

*Proof:* If two affine root systems  $\Psi, \Psi'$  has the same affine Dynkin diagram w.r.t. chambers  $C, C'$  resp., then there is a bijection  $\varphi : \Psi(C)^0 \cong \Psi'(C')^0$ , and by (3.9.8.28) and the root system case, we have  $\Psi_S \cong \Psi'_{\varphi(S)}$  for any subset  $S \subset \Psi(C)^0$ . These isomorphism glue together into an isomorphism  $\Psi \cong \Psi'$  because  $\Psi(C)^0$  is a basis of  $A^*$  (3.9.8.17).  $\square$

The classification follows from (3.9.8.35).  $\square$

**Prop. (3.9.8.37)[Types of Irreducible Non-Reduced Affine Root Systems].** Any irreducible non-reduced affine root system  $\Psi$  is determined by the Dynkin diagram of  $\Psi^{\text{short}}$  and the affine Dynkin diagram of  $\Psi^{\text{long}}$ , and the possible situations are recorded in [K-P23]P41.  $\square$

*Proof:*  $\Phi = d(\Psi)$  is non-reduced and irreducible, so it is of type  $BC_n$ . Because  $\Phi^{\text{short}} \subset d(\Psi^{\text{short}}) \subset \Phi$ ,  $\Phi^{\text{long}} \subset d(\Psi^{\text{long}}) \subset \Phi$ , we see that  $\Psi^{\text{short}}$  is of type  $B_n, C_n^\vee$  or  $BC_n$ , and  $\Psi^{\text{long}}$  is of type  $B_n^\vee, C_n$  or  $BC_n$ . But the Weyl-groups of  $\Psi^{\text{short}}$  and  $\Psi^{\text{long}}$  agree, so?  $\square$

## 3.10 Algebraic K-Theory

References are [Algebraic K-Theory of Fields, Suslin, in Proc. ICM 1986, P222-244], [Algebraic K-Theory, Olivier Isely] and [G-S17], [nLab]. [Totaro, Milnor K-theory is the Simplest Part of Algebraic K-Theory, K-theory 6, 177-189, 1992].

### 1 Milnor K-Groups of Fields

**Def. (3.10.1.1) [ $K_0$ ].** The Grothendieck group  $K_0(A)$  for a ring  $A$  is the free group generated by f.g. projective module over  $A$  modulo exact sequences. Then we have  $P \sim Q$  iff  $P \oplus A^n \cong Q \oplus A^n$  for some  $n$ . This is a functor  $\mathcal{CAlg} \rightarrow \mathbf{Ab}$ .  $\lrcorner$

**Def. (3.10.1.2) [Milnor K-Groups].** For  $k \in \mathbf{Field}$ , define the  $n$ -th **Milnor K-group** to be the groups

$$K_0^{\text{Mil}}(k) = \mathbb{Z}, \quad K_1^{\text{Mil}}(k) = k^\times, \quad K_n^{\text{Mil}}(k) = (k^\times)^{\otimes n} / \{a_1 \otimes \dots \otimes a_n \mid \exists 1 \leq i < j \leq n, a_i + a_j = 1\}, n \geq 2.$$

The elements in  $K_n^{\text{Mil}}(k)$  are called **symbols**, and the class of  $a_1 \otimes \dots \otimes a_m$  is denoted by  $\{a_1, \dots, a_n\}$ .  $\lrcorner$

**Prop. (3.10.1.3) [Total Milnor K-Groups].** For  $m, n \in \mathbb{N}$ , the tensor product  $(k^\times)^{\otimes m} \times (k^\times)^{\otimes n} \rightarrow (k^\times)^{\otimes(m+n)}$  induces a surjective map

$$K_m^{\text{Mil}}(k) \otimes K_n^{\text{Mil}}(k) \rightarrow K_{m+n}^{\text{Mil}}(k),$$

which induces a graded group structure on

$$K_*^{\text{Mil}}(k) = \bigoplus_{n \in \mathbb{N}} K_n^{\text{Mil}}(k).$$

Then  $K_*^{\text{Mil}}(k)$  is graded commutative.  $\lrcorner$

*Proof:* Firstly we show that  $\{x, -x\} = 0$ :

$$\{x, -x\} + \{x, -(1-x)x^{-1}\} = \{x, 1-x\} = 0,$$

so

$$\{x, -x\} = -\{x, 1-x^{-1}\} = \{x^{-1}, 1-x^{-1}\} = 0.$$

Thus for any  $x, y \in k^\times$ ,

$$0 = \{xy, -xy\} = \{x, y\} + \{x, -x\} + \{y, -y\} + \{y, x\}.$$

The general cases follow by induction.  $\square$

**Prop. (3.10.1.4) [Finite Fields].** If  $k \in \mathbf{Field}^{\text{fin}}$ , then for any  $n \geq 2$ ,  $K_n^{\text{Mil}}(k) = 0$ .  $\lrcorner$

*Proof:* By (3.10.1.3), it suffices to show that  $K_2^{\text{Mil}}(k) = 0$ . And if  $\omega$  is a generator of the cyclic group  $k^\times$ , then it suffices to show that  $\{\omega, \omega\} = \{\omega, -1\} = 0$ . If  $\#k = 2^m$ , then

$$0 = \{1, \omega\} = \{\omega^{2^m-1}, \omega\} = (2^m - 1)\{\omega, \omega\},$$

thus we are done. And if  $\text{char } k \neq 2$ , then we can find two non-squares in  $k^\times$  s.t.  $a + b = 1$ . Then

$$0 = \{a, b\} = \{\omega^l, \omega^k\} = kl\{\omega, \omega\},$$

so  $\{\omega, \omega\} = 0$ .  $\square$

### Tame Symbols and Specialization Maps

**Def. (3.10.1.5) [Tame symbols and Specialization Maps].** Let  $(R, K, \kappa)$  be a DVR, then for any  $n \in \mathbb{Z}_+$ , there exists a unique homomorphism

$$\partial^{\text{Mil}} : K_n^{\text{Mil}}(K) \rightarrow K_{n-1}^{\text{Mil}}(\kappa)$$

called the **tame symbol map**, s.t. for any uniformizer  $\varpi$  and units  $u_1, \dots, u_{n-1} \in R^*$ ,

$$\partial^{\text{Mil}}(\{\varpi, u_1, \dots, u_{n-1}\}) = \{\bar{u}_1, \dots, \bar{u}_{n-1}\}.$$

Moreover, for any uniformizer  $\varpi$  and  $n \in \mathbb{Z}_+$ , there exists a **specialization map**

$$s_{\varpi}^{\text{Mil}} : K_n^{\text{Mil}}(K) \rightarrow K_n^{\text{Mil}}(\kappa)$$

s.t.

$$s_{\varpi}^{\text{Mil}}(\{\varpi^{i_1} u_1, \dots, \varpi^{i_n} u_n\}) = \{\bar{u}_1, \dots, \bar{u}_n\}$$

for any  $u_1, \dots, u_n \in R^*$ . ┘

*Proof:* Cf.[Central Simple Algebras]P217. ? □

**Cor. (3.10.1.6).**  $\partial^{\text{Mil}}(\{a, b\}) = (-1)^{v(a)v(b)} \overline{a^{-v(b)} b^{v(a)}}.$  ┘

*Proof:* □

**Thm. (3.10.1.7) [Berrick-Keating].** Let  $R, S \in \mathcal{R}\text{ing}$  and  $U \in \text{Mod}_{R-S}$ , let  $T = \begin{bmatrix} R & U \\ & S \end{bmatrix} \in \mathcal{R}\text{ing}$ , then the map

$$\pi_i : K_i(T) \rightarrow K_i(T) \oplus K_i(S)$$

is an isomorphism for any  $i \in \mathbb{Z}$ . ┘

*Proof:* [The K-Theory of Triangular Matrix Rings, Berrick-Keating, in Applications of algebraic K-theory to algebraic geometry and number theory, 1986]. □

## 2 Bloch-Kato Conjecture

**Lemma (3.10.2.1) [Hilbert's Theorem90 for  $K_2$ ].** Let  $K/k$  be a cyclic Galois field extension with a generator  $\sigma \in \text{Gal}(K/k)$ , then the complex

$$K_2^{\text{Mil}}(K) \xrightarrow{\sigma-1} K_2^{\text{Mil}}(K) \xrightarrow{\text{Nm}_{K/k}} K_2^{\text{Mil}}(k)$$

is exact. ┘

*Proof:* Cf.[Central Simple Algebras]P276. ? □

**Prop. (3.10.2.2) [Galois Symbols].** Let  $k \in \mathcal{F}\text{ield}$ ,  $m \in \mathbb{Z} \cap k^\times$ ,  $n \in \mathbb{Z}_+$ , the boundary map

$$\partial k^\times \rightarrow H^1(k, \mu_m)$$

induces a map

$$\partial^n : (k^\times)^{\otimes n} \rightarrow (H^1(k, \mu_m))^{\otimes n} \xrightarrow{\cup} H^n(k, \mu_m^{\otimes n}),$$

and this map factors through  $K_n^{\text{Mil}}(k)$  and gives a **Galois symbol map**

$$h_{k,m}^n : K_n^{\text{Mil}}(k) \rightarrow H^n(k, \mu_m^{\otimes n}).$$

┘

*Proof:* To show this map factors through  $K_n^{\text{Mil}}(k)$ , it suffices to show for  $n = 2$  and  $\partial^2(a \otimes (1-a)) = 0$ . Take an irreducible factorization

$$x^m - a = \prod_l f_l \in k[x],$$

and let  $\alpha_l$  be a root of  $f_l$  in  $k_{\text{sep}}$ ,  $K_l = k(\alpha_l)$ , then

$$(1-a) = \prod_l \text{Nm}_{K_l/k}(1-\alpha_l).$$

and

$$\partial^2(a \otimes (1-a)) = \sum_l \partial^2(a \otimes \text{Nm}_{K_l/k}(1-\alpha_l)).$$

But

$$\begin{aligned} \partial^2(a \otimes \text{Nm}_{K_l/k}(1-\alpha_l)) &= \partial(a) \cup \partial(\text{Nm}_{K_l/k}(1-\alpha_l)) = \partial(a) \cup \text{cor}_k^{K_l}(1-\alpha_l) \\ &= \text{cor}(\text{res}_k^{K_l}(\partial(a)) \cup (1-\alpha_l)) = \text{cor}(\partial_{K_l}(a) \cup (1-\alpha_l)) \end{aligned}$$

But by definition  $a \in (K_l^\times)^m$ , so  $\partial_{K_l}(a) = 0$ , by (9.8.3.9). Thus  $\partial^2(a \otimes (1-a)) = 0$ .  $\square$

**Thm. (3.10.2.3) [Bloch-Kato Conjecture, Voevodsky-Rost/Merkurjev - Suslin].** Situation as in (3.10.2.2), the Galois symbol (3.10.2.2) induces an isomorphism

$$K_n^{\text{Mil}}(k)/mK_n^{\text{Mil}}(k) \xrightarrow{h_{k,m}^n, \cong} H^n(k, \mu_m^{\otimes n}).$$

┘

*Proof:* The  $n = 1$  case follows from (9.8.3.9).

For the  $n = 2$  case ?  $\square$

### 3.11 Hopf-Algebras

References are [Dri86], [A Brief Introduction To Quantum Groups, Etingof-Semenyakin]. [A Guide to Quantum Groups].

#### 1 Poisson-Algebras

**Def. (3.11.1.1)[Poisson Algebras].** For  $R \in \mathcal{CRing}$ , a **Poisson algebra** is a triple  $(A, \cdot, \{-, -\})$  where  $\cdot, \{-, -\} : A \times A \rightarrow A$  are  $R$ -bilinear operators, s.t.

- $(A, \cdot) \in \mathcal{Alg}_k$ .
- $(A, \{-, -\}) \in \mathcal{LieAlg}_k$ .
- For any  $x, y, z \in A$ ,  $\{x, yz\} = \{x, y\}z + y\{x, z\}$ .

┘

**Cor. (3.11.1.2).** For any  $R \in \mathcal{CRing}$  and a Poisson algebra  $A$  over  $R$ , if  $a \in A$ , then  $\{a, 1\} = 0$ .

┘

**Prop. (3.11.1.3)[Commutator].** For any  $R \in \mathcal{CRing}$  and  $A \in \mathcal{Alg}_R$ , the commutator  $[x, y] = xy - yx$  defines a Poisson algebra structure on  $A$ .

┘

**Prop. (3.11.1.4)[Tensor Products of Commutative Poisson Algebras].** Let  $A, B$  be two commutative Poisson  $k$ -algebras, then there is a canonical Poisson structure on  $A \otimes_k B$ :

$$\{a \otimes a', b \otimes b'\} = \{a, a'\} \otimes bb' + aa' \otimes \{b, b'\}.$$

┘

**Prop. (3.11.1.5).** If  $R \in \mathcal{CRing}$  and  $A$  is a Poisson algebra over  $R$ , then for any  $a, b, c, d, e \in A$ ,

$$\{a, b\}e[c, d] = [a, b]e\{c, d\}.$$

┘

*Proof:* We first show that  $\{a, b\}[c, d] = [a, b]\{c, d\}$  using the two ways of calculating  $[ac, bd]$ , and then we expand the expression

$$\{a, b\}[ec, d] = [a, b]\{ec, d\}$$

to get  $\{a, b\}e[c, d] = [a, b]e\{c, d\}$ . □

#### 2 Coalgebras and Bialgebras

**Def. (3.11.2.1)[Coalgebras].** For  $R \in \mathcal{CRing}$ , a **coalgebra** is a co-monoid object in the category  $\mathcal{Mod}_R$ .

Usually for a coalgebra, its comultiplication is denoted by  $\Delta$ , and counit denoted by  $\varepsilon$ . ┘

**Remark (3.11.2.2) [Yoneda Interpretation].** A coalgebra  $(C, \Delta, \varepsilon)$  needs to satisfy the following relations:

$$\Delta^{(2)} \triangleq (\text{id} \otimes \Delta) \circ \Delta = (\Delta \otimes \text{id}) \circ \Delta,$$

$$\text{mul.} \circ (\text{id} \otimes \varepsilon) \circ \Delta = \text{mul.} \circ (\varepsilon \otimes \text{id}) \circ \Delta = \text{id}.$$

We do not need to verify all the relations defining a coalgebra  $C$ , whenever we have a functorial monoidal structure on all the set  $\text{Hom}_R(H, T)$ , we immediately recover the maps

- (Comultiplication):  $\Delta : C \rightarrow C \otimes_R C$  (or denoted by  $\mu$ ) as  $i_1 \cdot i_2$  in  $\text{Hom}_R(C, C \otimes C)$ ,

- (Counit):  $\varepsilon : C \rightarrow R$  as 1 in  $\text{Hom}_R(C, R) = C^\vee$ .

by Yoneda lemma.

$C^\vee$  is an monoid by definition, and  $C$  is called **cocommutative** iff  $C^\vee$  is commutative.  $\lrcorner$

**Prop. (3.11.2.3) [Tensor Products of Co-Algebras].** Let  $R \in \mathcal{CRing}$  and  $C, D$  be co-algebras over  $R$ , then  $C \otimes D$  is a co-algebras with

$$\Delta_{C \otimes D} = \sigma_{23} \circ (\Delta_C \otimes \Delta_D), \quad \varepsilon_{C \otimes D} = \varepsilon_C \otimes \varepsilon_D.$$

$\lrcorner$

**Def. (3.11.2.4) [Primitive Elements].** Let  $R \in \mathcal{CRing}$  and  $H$  be a coalgebra over  $R$ , then an element  $x \in H$  is called a **primitive element** if  $\mu(x) = 1 \otimes x + x \otimes 1$ . It is called **group-like element** if  $\Delta(x) = x \otimes x$ .  $\lrcorner$

**Def. (3.11.2.5) [Sweedler Notations].** For a coalgebra  $A$  and  $a \in A$ , if  $\Delta(a) = \sum_i a_{1i} \otimes a_{2i}$ , then we can write

$$\Delta(a) = \sum_{\text{Swe}} a_1 \otimes a_2.$$

And similarly, by the first defining relation of (3.11.2.2), we can also write

$$(\text{id} \otimes \Delta) \circ \Delta(a) = (\Delta \otimes \text{id}) \circ \Delta(a) = \sum_{\text{Swe}} a_1 \otimes a_2 \otimes a_3.$$

Then the second defining relation just writes

$$a = \sum_{\text{Swe}} a_1 \varepsilon(a_2) = \sum_{\text{Swe}} \varepsilon(a_1) a_2.$$

$\lrcorner$

**Prop. (3.11.2.6).** For any coalgebra  $(A, \Delta, \varepsilon)$  and  $h \in A$ , by defining relations,

$$h = \sum_{\text{Swe}} \varepsilon(h_1) h_2, \quad \varepsilon(h) = \sum_{\text{Swe}} \varepsilon(h_1) \varepsilon(h_2).$$

$\lrcorner$

**Def. (3.11.2.7) [Co-Derivations].** For a co-algebra  $(C, \Delta, \varepsilon)$ , a **co-derivation** is a map  $d : C \rightarrow C$  s.t. for any  $c \in C$ ,

$$\Delta(d(c)) = d(c_1) \otimes c_2 + c_1 \otimes d(c_2).$$

$\lrcorner$

### Co-Poisson Co-Algebras

**Def. (3.11.2.8) [Lie Co-Algebras].** For  $k \in \text{Field}$ , **Lie co-algebra** over  $k$  is a module  $C \in \text{Vect}_k$  together with a  $k$ -linear map  $q : C \rightarrow C \otimes C$  s.t.

$$(1 + t_2) \circ q = 0$$

$$(1 + t_3 + t_3^2) \circ (q \otimes \text{id}) \circ q = 0.$$

where  $t_3 : C^3 \rightarrow C^3$  is given by  $t_3(v_1 \otimes v_2 \otimes v_3) = v_3 \otimes v_1 \otimes v_2$ .  $\lrcorner$

**Def. (3.11.2.9) [Sweedler Notations].** For  $k \in \mathbf{Field}$  and a Lie co-algebra  $C$  over  $k$  and  $a \in C$ , if  $q(a) = \sum_i a_{1i} \otimes a_{2i}$ , we can write

$$q(a) = \sum_{\text{Swe}} a_{(1)} \otimes a_{(2)}.$$

And similarly, we can also write

$$(q \otimes \text{id}) \circ q(a) = \sum_{\text{Swe}} a_{(1)} \otimes a_{(2)} \otimes a_{(3)}.$$

Then the defining relation of Lie co-algebras writes

$$\begin{aligned} \sum_{\text{Swe}} a_{(1)} \otimes a_{(2)} &= - \sum_{\text{Swe}} a_{(2)} \otimes a_{(1)}, \\ \sum_{\text{Swe}} a_{(1)} \otimes a_{(2)} \otimes a_{(3)} + \sum_{\text{Swe}} a_{(3)} \otimes a_{(1)} \otimes a_{(2)} + \sum_{\text{Swe}} a_{(2)} \otimes a_{(3)} \otimes a_{(1)} &= 0. \end{aligned}$$

┘

**Def. (3.11.2.10) [Co-Poisson Co-Algebras].** For  $k \in \mathcal{CRing}$ , a **co-Poisson co-algebra** is a co-algebra  $(C, \Delta, \varepsilon)$  over  $R$  together with a  $R$ -linear map  $q : C \rightarrow C \otimes C$  s.t.

- $(C, q)$  is a Lie co-algebra (3.11.2.8).
- (Co-Leibniz rule)

$$(\Delta \otimes \text{id}) \circ q = (\text{id} \otimes q) \circ \Delta + \sigma_{23}(q \otimes \text{id}) \circ \Delta.$$

Equivalently,

$$\sum_{\text{Swe}} c_{(1)1} \otimes c_{(1)2} \otimes c_{(2)} = \sum_{\text{Swe}} c_1 \otimes c_{2(1)} \otimes c_{2(2)} + \sum_{\text{Swe}} c_{1(1)} \otimes c_2 \otimes c_{1(2)}.$$

┘

**Cor. (3.11.2.11).** For any co-Poisson co-algebra  $(C, \Delta, \varepsilon, q)$ ,

$$(\text{id} \otimes \Delta) \circ q = (q \otimes \text{id}) \circ \Delta + \sigma_{12}(\text{id} \otimes q) \circ \Delta.$$

┘

**Prop. (3.11.2.12) [Poisson-Algebras and Co-Poisson Co-algebras].** Let  $R \in \mathcal{CRing}$  and  $(C, \Delta, \varepsilon)$  be a co-algebra over  $R$  with an  $R$ -linear map  $q : C \rightarrow C \otimes C$ , then  $(C, q)$  is a co-Poisson structure iff  $(C^*, q^*)$  is a Poisson algebra.

And if  $\sigma : C \rightarrow D$  is a linear map between co-Poisson co-algebras over  $R$ , then  $\sigma$  is a homomorphism of co-Poisson co-algebras iff  $\sigma^* : D^* \rightarrow C^*$  is a homomorphism of Poisson algebras. ┘

**Prop. (3.11.2.13) [Co-Commutator].** For any  $k \in \mathbf{Field}$  and a co-algebra  $(C, \Delta, \varepsilon)$  over  $k$ , the co-commutator  $\Delta' = \Delta - \sigma_{12} \circ \Delta$  defines a co-Poisson co-algebra structure on  $C$ . ┘

**Prop. (3.11.2.14) [Tensor Products of Co-Commutative Co-Poisson Co-Algebras].** Let  $R \in \mathcal{CRing}$  and  $C, D$  be co-commutative co-Poisson co-algebras over  $R$ , then  $C \otimes D$  is also a co-commutative co-Poisson co-algebras with the co-algebra structure given by (3.11.2.3) and

$$q_{C \otimes D} = \sigma_{23} \circ (q_C \otimes \Delta_D + \Delta_C \otimes q_D),$$

i.e.,

$$q_{C \otimes D}(a \otimes b) = (a_{(1)} \otimes b_1) \otimes (a_{(2)} \otimes b_2) + (a_1 \otimes b_{(1)}) \otimes (a_2 \otimes b_{(2)}).$$

┘



*Proof:* ? □

**Prop. (3.11.2.15).** There are no co-Poisson algebra structures on a group co-algebra  $k[\Gamma]$ . ? ┘

*Proof:* □

**Prop. (3.11.2.16).** If  $C$  is a co-commutative co-algebra and  $d_1, d_2$  are two co-derivatives of  $C$  s.t.  $d_1 d_2 = d_2 d_1$ , then the map

$$q : C \rightarrow C \otimes C : q(c) = d_1(c_1) \otimes d_2(c_2) - d_2(c_1) \otimes d_1(c_2)$$

defines a co-Poisson co-algebra on  $C$ . ┘

**Prop. (3.11.2.17).** For any co-commutative co-algebra  $(C, \Delta, \varepsilon, q)$ ,

$$(q \otimes \Delta') \circ \Delta = (\Delta' \otimes q) \circ \Delta.$$

And also

$$(\Delta' \otimes \text{id} \otimes q) \circ \Delta^{(2)} = (q \otimes \text{id} \otimes \Delta') \circ \Delta^{(2)}.$$

┘

*Proof:* These follow from (3.11.2.12) and (3.11.1.5). □

**Prop. (3.11.2.18).** For any co-Poisson co-algebra  $(C, \Delta, \varepsilon, q)$  and  $f \in C^*$ ,  $(f \otimes \text{id}) \circ q : C \rightarrow C$  and  $(\text{id} \otimes f) \circ q : C \rightarrow C$  are co-derivatives of  $C$ . ┘

*Proof:* □

**Prop. (3.11.2.19).** For any co-Poisson co-algebra  $(C, \Delta, \varepsilon, q)$ ,  $(\varepsilon \otimes \text{id}) \circ q = (\text{id} \otimes \varepsilon) \circ q = 0$ , i.e.

$$\sum_{\text{Swe}} \varepsilon(c_{(1)}) c_{(2)} = \sum_{\text{Swe}} c_{(1)} \varepsilon(c_{(2)}) = 0.$$

┘

*Proof:* □

### Bialgebras

**Def. (3.11.2.20) [Bialgebras].** For  $R \in \mathcal{C}\text{Ring}$ , a **bialgebra**  $A$  is a unital  $R$ -algebra  $A$  with algebra homomorphisms  $\Delta, \varepsilon : A \rightarrow A \times A$  s.t.  $(A, \Delta, \varepsilon)$  is a co-algebra over  $R$ . Equivalently,  $\Delta$  and  $\varepsilon$  need to satisfy properties as listed in <https://en.wikipedia.org/wiki/Bialgebra>.

WARNING:  $A$  cannot be defined as the co-unital-monoid algebra in  $\text{Ring}/R$ , because tensor products are not coproducts for non-commutative rings, Cf. <https://math.stackexchange.com/questions/984151/proof-that-the-tensor-product-is-the-coproduct-in-the-category-of-r-algebras>.

┘

**Cor. (3.11.2.21).** For any bialgebra  $(A, \cdot, \Delta, \varepsilon)$ ,  $\Delta(1) = 1$ . ┘

### Co-modules

**Def. (3.11.2.22) [Co-modules].** Let  $A$  be a coalgebra over  $k \in \mathbf{Field}$ , then a right **co-module** is a  $k$ -vector space  $V$  together with a  $k$ -linear map  $\rho : V \rightarrow V \otimes A$  that satisfies

$$(\mathrm{id}_V \otimes \mu) \circ \rho = (\rho \otimes \mathrm{id}_A) \circ \rho, \quad (\mathrm{id}_V \otimes \varepsilon) \circ \rho = \mathrm{id}_V.$$

The map  $\rho$  is called the **co-action**, and a  $k$ -subspace  $W \subset V$  that  $\rho(W) \subset W \otimes A$  is called a **sub-comodule** of  $V$ .  $\lrcorner$

**Def. (3.11.2.23) [Tensor Product of Co-modules].**  $\lrcorner$

### 3 Hopf Algebras

**Def. (3.11.3.1) [Hopf Algebra].** For  $R \in \mathbf{CRing}$ , a **Hopf algebra** over a  $R$  is an  $R$ -bi-algebra  $(A, \mathrm{mul.}, \eta, \Delta, \varepsilon)$  together with an  $R$ -linear isomorphism  $S : A \rightarrow A$  (the antipode) that satisfies:

$$\mathrm{mul.} \circ (\mathrm{id} \otimes S) \circ \Delta = \mathrm{mul.} \circ (S \otimes \mathrm{id}) \circ \Delta = \eta \circ \varepsilon : A \rightarrow A,$$

or equivalently,

$$\sum_{\mathrm{Swe}} a_1 S(a_2) = \sum_{\mathrm{Swe}} S(a_1) a_2 = \varepsilon(a), \quad a \in A.$$

If  $S^2 = \mathrm{id}_A$ , then  $A$  is called an **involutive Hopf algebra**.

Similarly we can define **topological Hopf algebras** or **graded Hopf algebras**.  $\lrcorner$

**Remark (3.11.3.2).** The motivating example is  $A = \mathcal{O}(G)$  where  $G \in \mathbf{Grp}$ .  $\lrcorner$

**Prop. (3.11.3.3).** For a Hopf algebra  $A$ ,  $S$  is an anti-homomorphism both for the algebra structure and coalgebra structure.

And if  $A$  is commutative or co-commutative, then  $A$  is involutive.  $\lrcorner$

*Proof:* Cf. <https://ncatlab.org/nlab/show/Hopf+algebra>, and [Abe (1980), Theorem 2.1.4].  $\square$

**Prop. (3.11.3.4).** For any  $(A, \mathrm{mul.}, \eta, \Delta, \varepsilon, S) \in \mathbf{Hopf}$ ,  $S$  is determined by the bi-algebra structure of  $A$ .  $\lrcorner$

*Proof:*  $\square$

**Example (3.11.3.5) [Group Algebras].** For  $\Gamma \in \mathbf{Grp}$ , the group algebra  $R[\Gamma]$  with the coalgebra structure

$$\Delta : R[\Gamma] \rightarrow R[\Gamma] \times R[\Gamma] : g \mapsto g \otimes g, \quad \varepsilon : R[\Gamma] \rightarrow R : \sum a_g g \mapsto \sum a_g$$

and the antipode

$$S : R[\Gamma] \rightarrow R[\Gamma] : g \mapsto g^{-1}$$

is a Hopf algebra.  $\lrcorner$

**Prop. (3.11.3.6) [Universal Enveloping Algebras].**  $\lrcorner$

**Def. (3.11.3.7) [Hopf Ideal and Quotient Hopf Algebra].** Let  $A$  be a Hopf algebra, then a quotient Hopf algebra is a quotient  $A/I$  that has a Hopf algebra structure compatible with that of  $A$ . In another words, there are commutative diagrams

$$\begin{array}{ccccc} A & \xrightarrow{\Delta} & A \otimes_R A & & A & \xrightarrow{\varepsilon} & R & & A & \xrightarrow{S} & A \\ \downarrow & & \downarrow & & \downarrow & \nearrow & & & \downarrow & & \downarrow \\ A/I & \xrightarrow{\bar{\Delta}} & A/I \otimes_R A/I & & A/I & & & & A/I & \xrightarrow{\bar{S}} & A/I \end{array}$$

In particular, quotient Hopf algebras correspond to ideals of  $A$  s.t.

$$\Delta(I) \subset A \otimes I + I \otimes A, \quad \varepsilon(I) = 0, \quad S(I) \subset I,$$

called **Hopf ideals** of  $A$ . ┘

**Remark (3.11.3.8) [Examples].** If  $\Gamma' \triangleleft \Gamma \in \mathcal{G}rp$ , then  $R[\Gamma]$  has a quotient Hopf algebra  $R[\Gamma/\Gamma']$ . In particular,  $R[t]/(t^n - 1)$  is a quotient Hopf algebra of  $R[t, t^{-1}]$ . ┘

**Def. (3.11.3.9) [Dual Hopf Algebras].** Let  $H$  be a Hopf algebra, matrix coefficients.  $H^0$ . ? ┘

**Prop. (3.11.3.10).**  $\mathcal{R}ep(H) \cong \mathcal{M}od_{H^0}$ . ┘

### Topologists' Hopf Algebras

**Def. (3.11.3.11) [Topologist's Hopf Algebra].** A **topologist's Hopf algebra** over a commutative ring  $R$  is a unital magma object in the dual category of graded  $R$ -algebras with 0-degree term  $R$ , given by maps  $\Delta : A \rightarrow A \otimes A$ , and  $\varepsilon : A \rightarrow R$  the canonical projection map. So in particular,  $\Delta$  is of the form

$$\Delta(\alpha) = 1 \otimes \alpha + \alpha \otimes 1 + \sum_i \alpha'_i \otimes \alpha''_i, |\alpha'_i| > 0, |\alpha''_i| > 0$$

**Prop. (3.11.3.12).** The tensor product of two topologists' Hopf algebra is a topologists' Hopf algebra. ┘

**Prop. (3.11.3.13).** Let  $\mathbb{F}$  be a field, then  $\mathbb{F}[\alpha]/(\alpha^n)$ , where  $\alpha$  is placed at even dimension or  $\mathbb{F}$  has characteristic 2, is a topologists' Hopf algebra iff  $\mathbb{F}$  has positive characteristic  $p$  and  $n$  is a power of  $p$ . ┘

*Proof:* By definition, it is easy to see that  $\alpha$  is primitive, thus

$$\Delta(\alpha^n) = 0 = \sum_{0 < i < n} \binom{n}{i} \alpha^i.$$

Which then implies that  $n$  is a  $p$ -power. □

**Prop. (3.11.3.14) [Hopf-Borel].** Let  $A$  be a topologists' Hopf algebra over a perfect field  $K$ , and  $A$  is of f.d. in each degree, then:

- If  $K$  has characteristic 0,  $A$  is isomorphic as an algebra to the tensor product of an exterior product of odd-dimensional generators and a polynomial ring of even-dimensional generators.

- If  $K$  has characteristic  $p$ , then  $A$  is isomorphic as an algebra to the tensor product of algebras of the following types:
  - $K[\alpha]$ , where  $\alpha$  is even-dimensional if  $p \neq 2$ .
  - $\wedge_K[\alpha]$  where  $\alpha$  is odd-dimensional.
  - $K[\alpha]/(\alpha^{p^i})$ , where  $\alpha$  is even-dimensional if  $p \neq 2$ .

┘

*Proof:* We only prove the 0-characteristic case, Cf. [Hat02]P285. ?

□

#### 4 Commutative Hopf Algebras

**Prop. (3.11.4.1) [Commutative Hopf Algebra].** For  $R \in \mathcal{CRing}$ , a commutative Hopf algebra over  $R$  is equivalent to a cogroup object in  $\mathcal{CAlg}_R$ . A homomorphism of Hopf algebras is a morphism of algebras that represents a natural transformation of functors from  $\mathcal{CAlg}_R$  to  $\mathcal{Grp}$ . ┘

*Proof:* The critical point is to look at the definition of Hopf algebra. In this case,  $S$  is a homomorphism of algebras (3.11.3.3), and notice tensor product is just the product in the dual category, and  $m : A \otimes A \rightarrow A$  is the diagonal in the dual category, thus a cogroup object is a map  $\Delta : A \rightarrow A \otimes A$  together with a map  $\text{inv} : A \rightarrow A$  that

$$m \circ (\text{id} \otimes \text{inv}) \circ \Delta = m \circ (\text{inv} \otimes \text{id}) \circ \Delta \text{id} = \text{id} : A \rightarrow A,$$

which is exactly the definition of the Hopf algebra (3.11.3.1). □

**Cor. (3.11.4.2) [Yoneda Interpretation].** We do not need to verify all the relations defining a Hopf algebra  $H$ , whenever we have a functorial commutative group structure on all the set  $\text{Hom}_R(H, T)$ , we immediately recover the maps:

- (Comultiplication):  $\mu : H \rightarrow H \otimes_R H$  as  $i_1 \cdot i_2$  in  $\text{Hom}_R(H, H \otimes H)$ ,
- (Antipode):  $\iota : H \rightarrow H$  as  $\text{inv}$  in  $\text{Hom}_R(H, H)$ ,
- (Counit):  $\varepsilon : H \rightarrow R$  as  $1$  in  $\text{Hom}_R(H, R) = H^\vee$ .

by Yoneda lemma.

For convenience, we denote the structure maps  $\eta_H : R \rightarrow H, \mu : H \times_R H \rightarrow H, (-)^{-1} : H \rightarrow H$ .

┘

**Prop. (3.11.4.3) [Group Algebras].** Let  $\Gamma$  be a commutative group, then the group algebra  $R[\Gamma]$  (3.11.3.5) represents the group functor that maps a commutative  $R$ -algebra  $S$  to the commutative group  $\text{Hom}_{\mathcal{Grp}}(\Gamma, S^*)$ . ┘

**Cor. (3.11.4.4) [Multiplicative Groups].**  $G_{m,R} = R[t, t^{-1}]$  is a Hopf algebra that represents the group functor of multiplicative groups on  $\mathcal{CAlg}_R$ . ┘

**Cor. (3.11.4.5) [Roots of Unity  $\mu_{n,R}$ ].** Let  $\Gamma \cong \mathbb{Z}/n\mathbb{Z}$ , then  $R[\Gamma] \cong R[t]/(t^n - 1)$  is a Hopf algebra that represents the group functor of multiplicative groups of  $n$ -th roots of unity on  $\mathcal{CAlg}_R$ . ┘

**Prop. (3.11.4.6) [Additive Groups].**  $R[t]$  can be given a Hopf algebra structure that represents the group functor that maps a commutative  $R$ -algebra  $S$  to the additive group  $S^+$ . ┘

**Def. (3.11.4.7)**  $[G_{(a,b),R}]$ . Given elements  $a, b \in R$  that  $ab = 2$ , for any commutative  $R$ -algebra  $S$ , the group  $G_{(a,b),R}(S)$  of elements  $x$  of  $S$  that  $x^2 + ax = 0$  is a group under the mapping  $(m, n) \mapsto m + n + bmn$ . Notice the inverse of  $m$  is  $m$  itself. Then  $R[t]/(t^2 + at)$  can be given a Hopf algebra that represents the functor  $S \mapsto G_{(a,b),R}(S)$ .  $\lrcorner$

**Def. (3.11.4.8)**  $[V_a]$ . Let  $V$  be a vector space over  $k$ , then  $\text{Sym}(V^\vee)$  can be given a Hopf algebra structure representation the functor  $V_a : R \mapsto R \otimes_k V \cong \text{Hom}_k(V^\vee, R)$ .  $\lrcorner$

**Def. (3.11.4.9)** **[Locally Constant Functions]**. Let  $\Gamma$  be a group, then  $\underline{\Gamma}_R = \prod_{\gamma \in \Gamma} R$  represents the group functor that maps an  $R$ -algebra  $T$  to the group of locally constant functions on  $\text{Spec } R$  with value in  $\Gamma$ .  $\lrcorner$

**Remark (3.11.4.10)**. To illustrate the philosophy of (10.1.1.2), we figure out the Hopf structure of the local constant functions: a map  $\prod_{\gamma \in \Gamma} R \rightarrow T$  is equivalent to a set of idempotents  $e_\gamma$  of  $T$  that  $\sum e_\gamma = 1$ . This is equivalent to a locally constant function on  $\text{Spec } T$  that takes value  $\gamma$  on  $V(e_\gamma)$ . Then the product takes values  $\gamma\delta$  on  $V(e_\gamma \otimes e_\delta) \subset \text{Spec } T \otimes T$ , or equivalently takes values  $\gamma$  on  $V(\sum_{gg'=\gamma} e_g \otimes e_{g'})$ , so  $\Delta(e_\gamma) = \sum_{gg'=\gamma} e_g \otimes e_{g'}$ .  $\lrcorner$

**Lemma (3.11.4.11)** **[Modulo  $\ker(\varepsilon)$ ]**. For a Hopf algebra  $A$  over  $R$ , the comultiplication and counit are determined by  $\ker \varepsilon$ :

- $R \oplus \ker \varepsilon \rightarrow A : (a, b) \mapsto a + b$  is an isomorphism of  $R$ -modules.
- $\mu(a) \equiv -\varepsilon(a) + a \otimes 1 + 1 \otimes a \pmod{\ker \varepsilon \otimes_R \ker \varepsilon}$ .
- $\iota(a) \equiv -a \pmod{(\ker \varepsilon)^2}$  for  $a \in \ker \varepsilon$ .

$\lrcorner$

*Proof:* 1: this is because the counit  $0 \rightarrow \ker \varepsilon \rightarrow A \rightarrow R \rightarrow 0$  has an inverse by the  $R$ -algebra map  $R \rightarrow S$ .

2: item1 allows us to write

$$A \otimes_R A = R \oplus (\ker \varepsilon \otimes_R R) \oplus (R \otimes_R \ker \varepsilon) \oplus (\ker \varepsilon \otimes \ker \varepsilon)$$

so for  $a \in A$ ,

$$\mu(a) = b + c \otimes 1 + 1 \otimes d + z$$

where  $b \in R, c, d \in \ker \varepsilon, z \in \ker \varepsilon \otimes_R \ker \varepsilon$ . Then  $a = (\varepsilon \otimes \text{id}_A)(b + c \otimes 1 + 1 \otimes d + z) = b + d$ , and also  $a = b + c$ . Applying  $\varepsilon$  shows  $b = \varepsilon(a)$ , and thus

$$\mu(a) = \varepsilon(a) + (a - \varepsilon(a)) \otimes 1 + 1 \otimes (a - \varepsilon(a)) + z = -\varepsilon(a) + a \otimes 1 + 1 \otimes a + z.$$

3: Let  $\iota(a) = b + c$  where  $b \in R, c \in \ker \varepsilon$ , then

$$\varepsilon(a) = \text{mul.}(\iota \otimes \text{id})(-\varepsilon(a) + a \otimes 1 + 1 \otimes a + z) = -\varepsilon(a) + \iota(a) + a + \text{mul.}(\iota \otimes \text{id})(z)$$

so for  $a \in \ker \varepsilon$ ,  $\iota(a) \equiv -a \pmod{(\ker \varepsilon)^2}$ , as  $\text{mul.}(\iota \otimes \text{id})(z) \in (\ker \varepsilon)^2$ , because  $\iota$  commutes with  $\varepsilon$ .  $\square$

**Prop. (3.11.4.12)** **[Hopf Ideals of Additive Groups]**. Let  $f = \sum_{i=0}^d a_i t^i \in R[t]$  be a monic polynomial  $\neq t$ , then  $(f)$  is a Hopf ideal of  $R[t]$  iff  $R$  is of char  $p > 0$  and the derivative  $f' = 0$ .  $\lrcorner$

*Proof:* We check conditions in (3.11.3.7), the first says  $\sum a_i (t \otimes 1 + 1 \otimes t)^i$  vanishes in  $R[t]/(f) \otimes_R R[t]/(f)$ . But  $f$  is monic, so this is equivalent to  $a_i \binom{i}{j} = 0$  for any  $0 < j < i \leq d$ . But we have  $\gcd_{0 < j < i} \binom{i}{j} = p$  if  $i = p^r$  for some  $r \geq 1$  and 1 otherwise, so  $a_d = 0$  unless  $d$  is a power of  $p$ , and  $p = 0 \in R$  because  $a_d = 1$ . In particular,  $f = \sum_{i=0}^k b_i t^{p^i}$ , and it automatically satisfies the other two conditions.  $\square$

**Cor. (3.11.4.13)**  $[\alpha_{p^r, R}]$ . Let  $R$  be a commutative ring of char  $p > 0$ , then the quotient Hopf algebra  $R[t]/(t^{p^r})$  corresponding to the Hopf ideal  $(t^{p^r})$  is denoted by  $\alpha_{p^r, R}$ .

WARNING:  $\alpha_{p^r, R}$  is isomorphic to  $\mu_{p^r, R}$  as  $R$ -algebras, but they are not isomorphic as Hopf algebras.  $\lrcorner$

**Prop. (3.11.4.14) [Irreducibility of Hopf Algebras]**. For  $k \in \mathbf{Field}$ , the following Hopf algebra contains no proper Hopf ideals:

- $\mathbb{G}_{a, k}$  if char  $k = 0$ .
- $\alpha_{p, k}$  if char  $k = p > 0$ .

$\lrcorner$

*Proof:* Any Hopf ideal of  $k[t]$  is principal, thus this proposition follows from (3.11.4.12).  $\square$

**Def. (3.11.4.15) [Cokernel Hopf Algebra]**. Let  $A \rightarrow B$  be a homomorphism of Hopf Algebras over  $R$ , then the **cokernel Hopf algebra** is the algebra  $B \otimes_A R = B/B \otimes_A \ker(\varepsilon_A)$ , which represents the functor of kernels of  $\mathrm{Hom}_R(B, T) \rightarrow \mathrm{Hom}_R(A, T)$ , thus is a Hopf algebra.  $\lrcorner$

**Lemma (3.11.4.16)**. A Hopf algebra over a field  $k$  is direct limit of Hopf algebra of f.t. over  $k$ .  $\lrcorner$

**Def. (3.11.4.17) [Group-Like Elements]**. Let  $A$  be a Hopf algebra, then a **group-like element**  $a \in A$  is an invertible element that satisfies  $\mu(a) = a \otimes a \in A \otimes A$ .

If  $a$  is a group-like element in a Hopf algebra  $A$ , then  $a = (\varepsilon, \mathrm{id})\mu(a) = \varepsilon(a)a$ , so  $\varepsilon(a) = 1$ .  $\lrcorner$

**Prop. (3.11.4.18) [Group-Like Elements are Linearly Independent]**. Let  $A$  be a Hopf algebra over a field  $k$ , then the set of group-like elements are linearly independent over  $k$ .  $\lrcorner$

*Proof:* If  $e = \sum a_i e_i$  that  $e, e_i$  are all group-like elements, then

$$\mu(e) = e \otimes e = \sum c_i c_j e_i \otimes e_j = \sum c_i \mu(e_i) = \sum c_i e_i \otimes e_i$$

so  $c_i^2 = c_i$  and  $c_i c_j = 0$ . Now also notice  $1 = \varepsilon(e) = \sum c_i \varepsilon(e_i) = \sum c_i$  (3.11.4.17), contradiction.  $\square$

**Cor. (3.11.4.19)**. The set of group-like elements in the Hopf algebra  $k[\Gamma]$  (3.11.4.3) are just the set  $\Gamma \subset k[\Gamma]$ .  $\lrcorner$

**Prop. (3.11.4.20) [Cartier Theorem]**. For  $k \in \mathbf{Field}^0$ , any Hopf algebra  $A$  over  $k$  is reduced.  $\lrcorner$

*Proof:* We can base change to the algebraic closure  $\bar{k}$  of  $k$  and assume  $k$  is alg.closed. Because reducedness is stalkwise and Hilbert's Nullstellensatz, it suffices to show  $A_s$  is reduced at each  $s \in G(k)$ . The translation by  $g \in G(k)$  acts transitively on  $G(k)$ , so it suffices to show it vanishes at the kernel of the counit map  $\ker(\varepsilon)$ . Cf. [Jakob, Stix].

We may assume by taking direct limits that  $A$  is f.g. over  $k$ . Let  $\mathfrak{m}$  be the kernel of the counit  $\varepsilon : A \rightarrow k$ , then  $\mathfrak{m}/\mathfrak{m}^2$  is a f.g.  $k$ -vector space with a basis lifting to  $x_1, \dots, x_r \in \mathfrak{m}$ .

We prove that  $\mathrm{gr}_{\mathfrak{m}}(A)$  is a polynomial ring in  $x_i$ , Cf. [Shatz].  $\square$

**Prop. (3.11.4.21) [Faithfully Flatness]**. If  $A \subset B$  are f.g. Hopf algebras over a field  $k$ , then  $B$  is f.f. over  $A$ .  $\lrcorner$

*Proof:* Cf. [Milne, P73] **?**.  $\square$

**Cor. (3.11.4.22)**. If  $A \subset B$  are Hopf algebras that  $B$  is an integral domain, and let  $K, L$  be their fraction field, then  $B \cap L = A$ . In particular, if  $K = L$ , then  $A = B$ .  $\lrcorner$

*Proof:* Since  $A \rightarrow B$  is f.f. by (3.11.4.21),  $cB \cap A = cA$  for any  $c \in A$  by (3.11.4.21). Thus if  $a/c \in B$  for  $a, c \in A$ , then  $a \in cB \cap A = cA$ , so  $a/c \in A$ .  $\square$

## 5 Clifford Algebras

References are [L-W89] and [Book of Involutions].

**Def. (3.11.5.1) [Clifford Algebra].** Let  $R \in \mathcal{CRing}$  and  $(L, Q) \in \text{QuadSpa}/R$ , the **Clifford algebra** is defined to be  $\text{Cl}(L) = T(L)/\langle v \otimes v - Q(v) \rangle$ . Because the quotient ideal is even, there is a  $\mathbb{Z}/(2)$ -grading on  $\text{Cl}(L)$ :  $\text{Cl}(L) = \text{Cl}^+(L) \oplus \text{Cl}^-(L)$ .  $\lrcorner$

**Def. (3.11.5.2) [Involutions].** By functoriality, there is a map  $\text{O}(L, Q) \rightarrow \text{Aut}(\text{Cl}(L))$ .

The involution  $-\text{Id} \in \text{O}(L, Q)$  induces an involution on  $\text{Cl}(L) : (v_1 v_2 \dots v_n)' \triangleq (-v_n)(-v_{n-1}) \dots (-v_1)$ . This involution preserves the  $\mathbb{Z}/(2)$ -grading.  $\lrcorner$

**Def. (3.11.5.3) [Transpose].** There is a anti-automorphism  $(-)^t : \text{Cl}(L) \rightarrow \text{Cl}(L)$  called the **Clifford transpose** that

$$(v_1 \otimes \dots \otimes v_r)^t = v_r \otimes \dots \otimes v_1, \quad \forall v_1, \dots, v_r \in L.$$

And define the **norm map**:  $Q(x) = x^t x : \text{Cl}(L) \rightarrow \text{Cl}(L)$ , which extends the norm  $Q$  on  $L$ .  $\lrcorner$

**Prop. (3.11.5.4).** Let  $R \in \mathcal{CRing}/\mathbb{Q}$  and  $(L, Q)$  be a quadratic space over  $R$ , then there is a canonical isomorphism of  $R$ -modules

$$\Lambda^* L \cong \text{Cl}(L)$$

compatible with the filtered structure. In particular, we can speak of  $\Lambda^k L \subset \text{Cl}(L, Q)$ .  $\lrcorner$

*Proof:* The map

$$f : L \otimes \dots \otimes L \rightarrow \text{Cl}(L) : v_1 \otimes \dots \otimes v_r \rightarrow \frac{1}{r!} \sum_{\sigma \in \text{Sym}_r} \text{sgn}(\sigma) v_{\sigma(1)} \dots v_{\sigma(r)}$$

descends to a map  $\Lambda^r L \rightarrow \text{Cl}(L)$ .  $\square$

**Prop. (3.11.5.5).** Let  $L = L_1 \oplus L_2$  be an orthogonal decomposition of quadratic spaces over  $R$ , then there is a natural isomorphism of Clifford algebras

$$\text{Cl}(L) \cong \text{Cl}(L_1) \hat{\otimes} \text{Cl}(L_2),$$

where the  $\hat{\otimes}$  is the  $\mathbb{Z}/(2)$ -graded tensor product.  $\lrcorner$

**Def. (3.11.5.6) [Spin Groups].** There are subgroups of  $\text{Cl}^+(L)$ :

$$\text{Spin}(L) = \{g \in \text{Cl}^+(L) : gg' = 1, gLg^{-1} = L\}$$

$$\text{GSpin}(L) = \{g \in \text{Cl}^+(L) : gg' = \lambda(g) \in R^*, gLg^{-1} = L\}.$$

called the **spin group** and **GSpin group** of  $L$ .  $\text{Spin}(L)$  and  $\text{GSpin}(L)$  acts on  $L$  via conjugation.  $\lrcorner$

**Prop. (3.11.5.7).** There is a diagram with exact rows and columns:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mu_2 & \longrightarrow & \text{Spin}(L) & \longrightarrow & \text{SO}(L) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \parallel \\ 1 & \longrightarrow & R^* & \longrightarrow & \text{GSpin}(L) & \longrightarrow & \text{SO}(L) \longrightarrow 1 \\ & & \downarrow^2 & & \downarrow_\lambda & & \\ & & R^* & \xlongequal{\quad} & R^* & & \end{array}$$

*Proof:*  $\square$

### Real Clifford Algebras

**Prop. (3.11.5.8) [Real Clifford Algebras].** Let  $\text{Cl}_{r,s}(\mathbb{R})$  denote the real Clifford algebra of the quadratic space  $V \cong \mathbb{R}^{r+s}$  or signature  $r - s$ , and also  $\text{Cl}_n(\mathbb{R}) \triangleq \text{Cl}_{n,0}(\mathbb{R})$ ,  $\text{Cl}_n(\mathbb{R})^* = \text{Cl}_{0,n}(\mathbb{R})$ . Then clearly  $\text{Cl}_{r,s}(\mathbb{R}) \cong \text{Cl}_1(\mathbb{R})^{\widehat{\otimes} r} \widehat{\otimes} \text{Cl}_1^*(\mathbb{R})^{\widehat{\otimes} s}$ , and

$$\text{Cl}_1(\mathbb{R}) \cong \mathbb{C}, \quad \text{Cl}_1^*(\mathbb{R}) \cong \mathbb{R} \oplus \mathbb{R}, \quad \text{Cl}_2(\mathbb{R}) \cong \mathbb{H} \subset \text{Mat}(2; \mathbb{C}), \quad \text{Cl}_2^*(\mathbb{R}) \cong \text{Mat}(2; \mathbb{R}).$$

$$\text{And dim}_{\mathbb{R}} \text{Cl}_{r+s}(\mathbb{R}) = 2^{r+s}. \quad \lrcorner$$

**Def. (3.11.5.9) [Volume Elements and Chirality Elements].** Notation as in (3.11.5.8) and take an orientation of  $\mathbb{R}^{r+s}$ , then for any orientation-compatible orthonormal basis  $v_1, \dots, v_{r+s}$ , we can consider the element  $v_1 v_2 \dots v_{r+s} \in \text{Cl}_{r,s}(\mathbb{R})$ . Then it only depends on the orientation, called the **Clifford volume form**. And

$$\omega^2 = (-1)^{\frac{n(n+1)}{2}+s}, \quad v\omega = (-1)^{n-1}\omega v, \quad \forall v \in \mathbb{R}^{r+s}.$$

In particular, if  $2 \nmid r+s$ , then  $\omega \in Z(\text{Cl}_{r,s}(\mathbb{R}))$ , and if  $2 \mid r+s$ , then  $\varphi\omega = \omega\varphi'$  for any  $\varphi \in \text{Cl}_{r,s}(\mathbb{R})$ .  $\lrcorner$

**Prop. (3.11.5.10) [Volume Decompositions].** Suppose the Clifford volume form (3.11.5.9)  $\omega^2 = 1$ , and denote

$$\pi^+ = \frac{1}{2}(1 + \omega), \quad \pi^- = \frac{1}{2}(1 - \omega),$$

then  $\pi^{\pm}$  are idempotents, with  $\pi^+ + \pi^- = 1$ . And

- If  $2 \nmid r+s$ , then there is a decomposition

$$\text{Cl}_{r,s}(\mathbb{R}) = \text{Cl}_{r,s}^+(\mathbb{R}) \oplus \text{Cl}_{r,s}^-(\mathbb{R}), \quad \text{Cl}_{r,s}^{\pm}(\mathbb{R}) = \text{Cl}_{r,s}(\mathbb{R})\pi^{\pm}.$$

$$\text{and } \alpha(\text{Cl}_{r,s}^{\pm}(\mathbb{R})) = \text{Cl}_{r,s}^{\mp}(\mathbb{R}). \quad \lrcorner$$

**Prop. (3.11.5.11).** For any  $r, s \in \mathbb{N}$ , there is an isomorphism  $\text{Cl}_{r,s}(\mathbb{R}) \cong \text{Cl}_{r+1,s}^0(\mathbb{R})$ .  $\lrcorner$

*Proof:* Take the usual basis  $e_1, \dots, e_{r+1}, e_{r+2}, \dots, e_r$  of  $\mathbb{R}^{r+1+s}$  and define a map

$$f : \mathbb{R}^{r+s} \rightarrow \text{Cl}_{r+1,s}^0(\mathbb{R}) : e_i \mapsto e_{r+1}e_i,$$

then for any  $x = \sum_{i \neq r+1} x_i e_i$ , we have

$$f(x)^2 = \sum_{i,j} x_i x_j e_{r+1} e_{r+1} e_i e_{r+1} e_j = \sum_{i,j} x_i x_j e_i e_j = x \cdot x = -q(x),$$

so  $f$  extends to an algebra homomorphism  $\text{Cl}_{r,s}(\mathbb{R}) \rightarrow \text{Cl}_{r+1,s}^0(\mathbb{R})$ . This is an isomorphism by checking on a linear basis.  $\square$

**Prop. (3.11.5.12).** Let  $L : \text{Cl}_n(\mathbb{R}) \rightarrow \text{Cl}_n(\mathbb{R})$  be the linear map

$$L(\varphi) = - \sum_j e_j \varphi e_j$$

where  $e_j$  is an orthonormal basis of  $\mathbb{R}^n$ . Set  $\tilde{L} = \alpha \circ L$ , then the eigenspaces of  $\tilde{L}$  are the canonical images of  $\Lambda^p(\mathbb{R}^n) \subset \text{Cl}_n(\mathbb{R})$ . In fact,

$$\tilde{L}_{\Lambda^p} = (n - 2p) \text{Id}. \quad \lrcorner$$



*Proof:* In fact,

$$\begin{aligned}
 L(e_1 \dots e_p) &= - \sum_{j=1}^p e_j e_1 e_2 \dots e_p e_j - \sum_{j=p+1}^n e_j e_1 \dots e_p e_j \\
 &= \sum_{j=1}^p (-1)^{p-1} p e_1 e_2 \dots e_p - \sum_{j=p+1}^n (-1)^p e_1 \dots e_p \\
 &= (-1)^{n-2p} e_1 \dots e_p.
 \end{aligned}$$

□

**Prop.(3.11.5.13) [Clifford Multiplication and Interior Products].** Under the canonical vector space isomorphism  $f : \text{Cl}_n(\mathbb{R}) \cong \Lambda^*(\mathbb{R}^n)$ ,

$$f(v\varphi) = v \wedge f(\varphi) - v \lrcorner f(\varphi).$$

┘

*Proof:* Cf.[L-W89]P25.

□

**Prop.(3.11.5.14).**

$$\text{Cl}_{n+2,0} \cong \text{Cl}_{0,n} \otimes \text{Cl}_{2,0}, \quad \text{Cl}_{0,n+2} \cong \text{Cl}_{n,0} \otimes \text{Cl}_{0,2}.$$

by the mapping  $e_i \rightarrow e_i \otimes e'_1 e'_2$ ,  $e_{n+j} \rightarrow 1 \otimes e'_j$ . So we have

$$\text{Cl}_{n+8,0} \cong \text{Cl}_n \otimes \mathbb{R}(16), \quad \text{Cl}_{n+2,0} = \text{Cl}_{n+2,0} \otimes \mathbb{C} = \text{Cl}_{n,0} \otimes \mathbb{C}(2).$$

because  $\mathbb{H} \otimes \mathbb{C} = \mathbb{C}(2)$ , and

$$\begin{bmatrix}
 n & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
 \text{Cl}_{n,0} & \mathbb{R} & \mathbb{C} & \mathbb{H} & \mathbb{H} \oplus \mathbb{H} & \mathbb{H}(2) & \mathbb{C}(4) & \mathbb{R}(8) & \mathbb{R}(8) \oplus \mathbb{R}(8) \\
 \text{Cl}_{n,0} & \mathbb{C} & \mathbb{C} \oplus \mathbb{C} & & & & & & 
 \end{bmatrix}.$$

┘

## 6 Quantization

Cf.[Dri85, Dri86].

Quantizing is somewhat like replacing commutative algebras by noncommutative ones. One way to try to construct non-classical examples of quantum groups is to look for deformations, in the category of Hopf algebras, of classical algebras of functions on a group.

### (Co-)Poisson Hopf Algebras

**Def.(3.11.6.1) [Poisson-Hopf Algebras].** A **Poisson Hopf algebra** is a Hopf algebra  $(A, \text{mul.}, \eta, \Delta, \varepsilon, S)$  with a Poisson structure  $\{-, -\}$  s.t.

$$\Delta(\{a, b\}) = \sum_{\text{Swe}} \{a_1, b_1\} \otimes a_2 b_2 + \sum_{\text{Swe}} a_1 b_1 \otimes \{a_2, b_2\}.$$

Notice if  $A$  is commutative, then this just means  $\Delta : A \rightarrow A \otimes A$  is a homomorphism of Poisson structures, by(3.11.1.4).

(The motivating example is  $A = \mathcal{O}(G)$  where  $G$  is a Poisson-Lie group.)

┘

**Cor. (3.11.6.2).** For any Poisson-Hopf algebra  $(A, \text{mul.}, \eta, \Delta, \varepsilon, S, \{-, -\})$  and  $a, b \in A$ ,

$$\sum_{\text{Swe}} \{a, S(b_1)\} b_2 = \sum_{\text{Swe}} S(b_1) \{b_2, a\}.$$

┘

*Proof:* It follows from the Leibniz rule that

$$\sum_{\text{Swe}} \{a, S(b_1)\} b_2 + \sum_{\text{Swe}} S(b_1) \{a, b_2\} = \sum_{\text{Swe}} \{a, S(b_1) b_2\} \stackrel{(3.11.3.1)}{=} \{a, \varepsilon(b)\} \stackrel{(3.11.1.2)}{=} 0.$$

□

**Prop. (3.11.6.3).** If  $k \in \text{Field}$ ,  $\text{char } k \neq 2$ , and  $\mathfrak{g} \in \text{LieAlg}_k$  is non-Abelian, then there are no non-trivial Poisson Hopf algebra structures on  $U(\mathfrak{g})$ . ┘

*Proof:* Cf.[Co-Poisson structures on polynomial Hopf algebras]P816. □

**Prop. (3.11.6.4).** For a Poisson Hopf algebra  $(H, \mu, \eta, \Delta, \varepsilon, S, \{-, -\})$ , the counit  $\varepsilon : H \rightarrow R$  is a Poisson algebra homomorphism:  $\varepsilon(\{a, b\}) = 0$  for any  $a, b \in H$ . And if  $H$  is commutative, then  $S : H \rightarrow H$  is a Poisson algebra anti-homomorphism. ┘

*Proof:* For  $\varepsilon$ , because of the defining relation for coalgebras, for any  $a, b \in A$ ,

$$\varepsilon(\{a, b\}) = \varepsilon(\text{mul.} \circ (\varepsilon \otimes \text{id}) \circ \Delta(\{a, b\})) = \sum_{\text{Swe}} \varepsilon(\{a_1, b_1\}) \varepsilon(a_2 b_2) + \sum_{\text{Swe}} \varepsilon(a_1 b_1) \varepsilon(\{a_2, b_2\}),$$

but also by (3.11.2.6),

$$\varepsilon(\{a, b\}) = \varepsilon(\{\sum_{\text{Swe}} \varepsilon(a_1) a_2, \sum_{\text{Swe}} \varepsilon(b_1) b_2\}) = \sum_{\text{Swe}} \varepsilon(a_1 b_1) \{a_2, b_2\}.$$

similarly,

$$\varepsilon(\{a, b\}) = \sum_{\text{Swe}} \{a_1, b_1\} \varepsilon(a_2 b_2).$$

Thus  $\varepsilon(\{a, b\}) = 2\varepsilon(\{a, b\})$ , so  $\varepsilon(\{a, b\}) = 0$ .

For  $S$ , Firstly for any  $a, b \in A$ ,

$$\{a_1, b_1\} S(a_2 b_2) + a_1 b_1 S(\{a_2, b_2\}) = 0,$$

by (3.11.6.1) and the fact  $\varepsilon(\{a, b\}) = 0$ . Then by (3.11.6.2) (3.11.3.1) and (3.11.3.3),

$$\begin{aligned} \{S(b), S(a)\} &= \sum_{\text{Swe}} \{S(b_1), S(a)\} b_2 S(b_3) \\ (3.11.6.2) &= - \sum_{\text{Swe}} S(b_1) \{b_2, S(a)\} S(b_3) \\ &= - \sum_{\text{Swe}} S(b_1) \{b_2, S(a_1)\} a_2 S(a_3) S(b_3) \\ (3.11.6.2) &= \sum_{\text{Swe}} S(b_1) S(a_1) \{b_2, a_2\} S(a_3) S(b_3) \\ &= - \sum_{\text{Swe}} S(b_1) S(a_1) b_2 a_2 S(\{b_3, a_3\}) \\ (3.11.2.5) &= S(\{a, b\}). \end{aligned}$$

□

**Def. (3.11.6.5) [Co-Poisson Hopf Algebras].** A **co-Poisson Hopf algebra** is a Hopf algebra  $(H, \text{mul.}, \eta, \Delta, \varepsilon, S)$  equipped with a co-Poisson co-algebra structure  $q : H \rightarrow H \otimes H$  (3.11.2.10) s.t. for any  $a, b \in H$ ,

$$q(ab) = q(a)\Delta(b) + \Delta(a)q(b).$$

Notice if  $H$  is co-commutative, then this just means that  $\text{mul.} : H \otimes H \rightarrow H$  is a co-Poisson co-algebra homomorphism, by (3.11.2.14).  $\lrcorner$

**Prop. (3.11.6.6) [Poisson-Hopf Algebras and Co-Poisson Hopf-algebras].** Let  $R \in \mathcal{CRing}$  and  $(H, \text{mul.}, \eta, \Delta, \varepsilon, S)$  be a Hopf algebra over  $R$  with an  $R$ -linear map  $q : H \rightarrow H \otimes H$ , then  $(H, q)$  is a co-Poisson structure iff  $(H^*, q^*)$  is a Poisson-Hopf algebra.

And if  $\sigma : H_1 \rightarrow H_2$  is a linear map between co-Poisson Hopf algebras over  $R$ , then  $\sigma$  is a homomorphism of co-Poisson Hopf algebras iff  $\sigma^* : D^* \rightarrow C^*$  is a homomorphism of Poisson-Hopf algebras.  $\lrcorner$

**Prop. (3.11.6.7).** For a co-Poisson Hopf algebra  $(H, \mu, \eta, \Delta, \varepsilon, S, \{-, -\})$ , the unit  $\eta : R \rightarrow H$  is a co-Poisson co-algebra homomorphism:  $q(1) = 0$ . And if  $H$  is co-commutative, then  $S : H \rightarrow H$  is a co-Poisson co-algebra anti-homomorphism.  $\lrcorner$

*Proof:* This follows from (3.11.6.6) and (3.11.6.4).  $\square$

**Prop. (3.11.6.8) [Co-Poisson Hopf Structures on  $U(\mathfrak{g})$ ].** For  $k \in \mathbf{Field}^0$  and  $\mathfrak{g} \in \mathbf{LieAlg}_k$ , co-Poisson Hopf structures on  $U(\mathfrak{g})$  are equivalent to Lie bi-algebra structures on  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Cf. [A guide to quantum groups]P178.  $\square$

### Quantization of (Co-)Poisson Hopf Algebras

**Def. (3.11.6.9) [Deformations of Hopf Algebras].** For  $k \in \mathbf{Field}$  and a Hopf algebra  $(A, \text{mul.}, \eta, \Delta, \varepsilon, S)$  over  $k$ , a **deformation of Hopf algebra** is a topological Hopf algebra  $(A_{\hbar}, \text{mul.}, \eta_{\hbar}, \Delta_{\hbar}, \varepsilon_{\hbar}, S_{\hbar})$  over  $k[[\hbar]]$  s.t.

- $A_{\hbar} \cong A[[\hbar]]$  as  $k[[\hbar]]$ -modules.
- $\mu_{\hbar} \equiv \mu(\text{mod } \hbar), \Delta_{\hbar} \equiv \Delta(\text{mod } \hbar)$ .

Two deformations  $A_{\hbar}, A'_{\hbar}$  are called equivalent if there is an isomorphism of Hopf algebras  $f : A_{\hbar} \cong A'_{\hbar}$  over  $k[[\hbar]]$  s.t.  $f \equiv 1(\text{mod } \hbar)$ .  $\lrcorner$

**Def. (3.11.6.10) [Infinitesimal Deformations of Hopf Algebras].** Cf. [A guide to Hopf algebras]P172.  $\lrcorner$

**Def. (3.11.6.11) [Quantization of Poisson-Hopf Algebras].** For  $k \in \mathbf{Field}$  and a commutative Poisson algebra  $(A, \cdot, \{-, -\})$  over  $k$ , a **quantization of Poisson algebra**  $A_{\hbar}$  is a topological associative algebra  $(A_{\hbar}, \cdot_{\hbar}) \in \mathcal{Alg}_{\mathbb{C}[[\hbar]]}$  s.t.

- $A_{\hbar} \cong A[[\hbar]]$  as  $k[[\hbar]]$ -modules.

$$\cdot_{\hbar} = \cdot(\text{mod } \hbar).$$

$$\{\bar{a}, \bar{b}\} \equiv \frac{ab - ba}{\hbar}(\text{mod } \hbar), \quad a, b \in A_{\hbar}.$$

And if  $A$  is a Poisson-Hopf algebra (3.11.6.1), then a **quantization of Poisson-Hopf algebra**  $A_{\hbar}$  is a Hopf algebra deformation of  $A$  over  $k[[\hbar]]$  (3.11.6.9) that is also a quantization of the Poisson structure.  $\lrcorner$

**Def. (3.11.6.12) [Quantization of Poisson-Lie Groups].** if  $(G, \{-, -\})$  is a Poisson-Lie algebra, then a **quantization of Poisson-Lie group**  $\mathcal{F}_h(G)$  is a quantization of the Poisson-Hopf algebra  $\mathcal{O}(G)$  (3.11.6.11). And  $(G, \{-, -\})$  is called the **classical limit** of  $\mathcal{F}_h(G)$ .  $\lrcorner$

**Prop. (3.11.6.13).** Let  $k \in \text{Field}$ ,  $\text{char } k \neq 2$ , and  $\mathfrak{g} \in \text{LieAlg}_k$  is non-Abelian, then there are no non-trivial Poisson Hopf structures on  $U(\mathfrak{g})$ .  $\lrcorner$

*Proof:* ?  $\square$

## Quantized Universal Enveloping Algebras

### 7 Lie Bi-algebras

**Def. (3.11.7.1) [Lie bi-algebras].** For  $k \in \text{Field}$ , a **Lie bi-algebra** is a Lie algebra  $\mathfrak{g} \in \text{LieAlg}_k$  together with a skew-symmetric linear map  $\delta : \mathfrak{g} \rightarrow \mathfrak{g} \otimes \mathfrak{g}$  called the **co-commutator** s.t.

- $\delta^* : \mathfrak{g}^* \otimes \mathfrak{g}^* \rightarrow \mathfrak{g}^*$  is a Lie algebra structure on  $\mathfrak{g}^*$ .
- $\delta$  is a 1-cocycle on  $\mathfrak{g}$  with values in  $\mathfrak{g} \otimes \mathfrak{g}$ , i.e.

$$\delta[X, Y] = (\text{ad}_X \otimes \text{id} + \text{id} \otimes \text{ad}_X)\delta(Y) + (\text{ad}_Y \otimes \text{id} + \text{id} \otimes \text{ad}_Y)\delta(X).$$

$\lrcorner$

### 8 Yang-Baxter Equations

#### Set-Theoretical Solutions

**Def. (3.11.8.1) [Quantum Yang-Baxter Equations].** For  $S \in \text{Set}$ , a map  $R : S \times S \rightarrow S \times S$  is said to satisfy the (set-theoretic) **quantum Yang-Baxter equation** if  $R$  is a bijection and

$$R_{12} \circ R_{13} \circ R_{23} = R_{23} \circ R_{13} \circ R_{12} : S \times S \times S \rightarrow S \times S \times S.$$

$\lrcorner$

**Thm. (3.11.8.2).** Suppose  $G \in \text{Grp}$ , and  $\xi, \eta$  are left and right actions of  $G$  on itself satisfying the following condition:

$$uv = (\xi(u)v)(u\eta(v)), \quad \forall u, v \in G,$$

then

$$R : G \times G \rightarrow G \times G : (u, v) \mapsto (u\eta(v), \xi(u)v)$$

is a solution of the set-theoretic Yang-Baxter equation (3.11.8.1) on  $G$ .  $\lrcorner$

*Proof:* Cf. [ON THE SET-THEORETICAL YANG-BAXTER EQUATION].  $\square$

**Thm. (3.11.8.3) [Universal Braid relations].** Let  $\sigma : S \times S \rightarrow S \times S$  be a non-degenerate solution of the braid relation, and let  $G(S, \sigma)$  be the group generated by  $S$  subject to the relation  $uv \sim yx$  whenever  $\sigma(u, v) = (y, x)$ , with a natural map  $i : S \rightarrow G(S, \sigma)$ . Then there is a unique braiding operator  $\sigma^G$  on  $G(S, \sigma)$  commuting with  $i$ , and it is the initial object in the category of such groups with a braiding operator.  $\lrcorner$

*Proof:* Cf. [ON THE SET-THEORETICAL YANG-BAXTER EQUATION]P16.  $\square$

### 9 Quiver Hecke Algebra

Cf. [Bru13].

## 4 | Categories and Algebraic Topology

### 4.1 Categories

*“Mathematicians do not study objects, but relations among objects; they are indifferent to the replacement of objects by others as long the relations don’t change. Matter is not important, only form interests them.”*

–H. Poincaré

References are [Mac98], [Bor94], [Coend Calculus, Fosco Loregian].

**Notation (4.1.0.1).**

- Let  $\mathcal{C} \in \text{Cat}$  (4.1.4.2).

┘

#### 1 Basics

**Def. (4.1.1.1) [Categories, Eilenberg-Mac.Lane1945].** A category  $\mathcal{C}$  consists of the following data:

- A class  $\text{Ob}(\mathcal{C})$  of **objects of  $\mathcal{C}$** .
- For any  $x, y \in \text{Ob}(\mathcal{C})$  a class  $\text{Mor}(x, y)$  of **morphisms from  $x$  to  $y$** .
- For any  $(x, y, z) \in \text{Ob}(\mathcal{C})$ , a map of classes  $\circ : \text{Mor}(y, z) \times \text{Mor}(x, y) \rightarrow \text{Mor}(x, z)$ , called the **composition law** of  $\mathcal{C}$ .

that satisfies:

- (Identity) For any  $x \in \text{Ob}(\mathcal{C})$ , there exists an element  $\text{id}_x \in \text{Mor}(x, x)$  s.t.  $\text{id}_x \circ \varphi = \varphi, \psi \circ \text{id}_x = \psi$  whenever these composition makes sense. It is clear such an element is unique.
- (Associativity)  $(\varphi \circ \psi) \circ \chi = \varphi \circ (\psi \circ \chi)$  whenever these compositions make sense.

┘

**Remark (4.1.1.2).** We will sometimes say  $x \in \mathcal{C}$  to mean that  $x$  is an element of  $\text{Ob}(\mathcal{C})$  and  $f : x \rightarrow y \in \mathcal{C}$  or  $f$  is a morphism in  $\mathcal{C}$  to mean:  $x, y \in \text{Ob}(\mathcal{C})$  and  $f \in \text{Mor}(x, y)$ . Hopefully this won’t make any confusions.

┘

**Def. (4.1.1.3) [Dual Categories].** Let  $\mathcal{C}$  be a category, then the **dual category** of  $\mathcal{C}$  is a category  $\mathcal{C}^{\text{op}}$  with

- $\text{Ob}(\mathcal{C}^{\text{op}}) = \text{Ob}(\mathcal{C})$ .
- For  $x, y \in \text{Ob}(\mathcal{C}^{\text{op}}) = \text{Ob}(\mathcal{C})$ ,  $\text{Mor}_{\mathcal{C}^{\text{op}}}(x, y) = \text{Mor}_{\mathcal{C}}(y, x)$ .
- The composition law of  $\mathcal{C}^{\text{op}}$  is induced from that of  $\mathcal{C}$ .

┘

**Def. (4.1.1.4)[Isomorphisms].** Let  $\mathcal{C}$  be a category, an **isomorphism** in  $\mathcal{C}$  is a morphism  $\varphi \in \text{Mor}(x, y)$  in  $\mathcal{C}$  s.t. there exists a morphism  $\psi \in \text{Mor}(y, x)$  s.t.  $\varphi \circ \psi = \text{id}_y$  and  $\psi \circ \varphi = \text{id}_x$ . it is clear that such a  $\psi$  is unique, and if it exists, it is called the **inverse morphism** of  $\varphi$ .

For  $x, y \in \mathcal{C}$ , if there exists an isomorphism in  $\text{Mor}(x, y)$ ,  $x, y$  are said to be **equivalent objects**.

┘

**Def. (4.1.1.5) [Functors].** Let  $\mathcal{C}, \mathcal{C}'$  be categories, a **functor**  $F$  from  $\mathcal{C} \rightarrow \mathcal{C}'$  consists of the following data:

- a morphism of sets  $\text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{C}') : x \mapsto F(x)$ .
- for any  $x, y \in \text{Ob}(\mathcal{C})$ , a morphism of sets  $\text{Mor}(x, y) \rightarrow \text{Mor}(F(x), F(y)) : f \mapsto F(f)$ .

that satisfies:

- $F(\text{id}_x) = \text{id}_{F(x)}$ .
- $F(\varphi \circ \psi) = F(\varphi) \circ F(\psi)$  whenever  $\varphi \circ \psi$  is defined.

A **contravariant functor** from  $\mathcal{C}$  to  $\mathcal{C}'$  is defined to be a functor from  $\mathcal{C}^{\text{op}}$  to  $\mathcal{C}'$ .

┘

**Def. (4.1.1.6) [Identity Functor  $\text{id}_{\mathcal{C}}$ ].** For any  $\mathcal{C} \in \text{Cat}$ , there is an identity functor  $\text{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ .

┘

**Def. (4.1.1.7) [Natural Transformations].** Let  $\mathcal{C}, \mathcal{C}'$  be categories and  $F, G : \mathcal{C} \rightarrow \mathcal{C}'$  be functors between categories, then a **natural transformation**  $\eta$  from  $F$  to  $G$  consists of an element  $\eta_x$  for each  $x \in \text{Ob}(\mathcal{C})$  s.t. for any  $f : x \rightarrow y \in \mathcal{C}$ , the following diagram is commutative:

$$\begin{array}{ccc} F(x) & \xrightarrow{F(f)} & F(y) \\ \downarrow \eta_x & & \downarrow \eta_y \\ G(x) & \xrightarrow{G(f)} & G(y) \end{array} \cdot$$

Then for any two categories  $\mathcal{C}, \mathcal{C}'$ , there is a category  $\text{Fun}(\mathcal{C}, \mathcal{C}')$  consisting of the set of functors from  $\mathcal{C}$  to  $\mathcal{C}'$ , and natural transformations as morphisms between functors.

┘

**Def. (4.1.1.8) [Final and Initial Objects].** An object  $Z$  of a category is called a **final object** if  $\# \text{Hom}(X, Z) = 1$  for any object  $X$ . It is called **weakly final** if  $\text{Hom}(X, Z) \neq \emptyset$  for every object  $X$ .

Dually an object is called **(weakly)initial** if it is (weakly)initial as an object of  $\mathcal{C}^{\text{op}}$ .

┘

**Def. (4.1.1.9) [Filtered Categories].** A **filtered category** is a category  $\mathcal{I} \in \text{Cat}$  that:

- It is nonempty.
- for any  $a, b \in \mathcal{I}$ , there is some  $c \in \mathcal{I}$  with morphisms  $a \rightarrow c, b \rightarrow c$
- for any two morphisms  $a, b : x \rightarrow y$ , there is a morphism  $c : y \rightarrow z$  that  $c \circ a = c \circ b$ .

┘

**Def. (4.1.1.10) [Fully Faithful Functors].** A functor of category  $F : \mathcal{C} \rightarrow \mathcal{D}$  is called:

- a **full/faithful functor** if for any  $x, y \in \text{Ob}(\mathcal{C})$ ,  $F : \text{Hom}(X, Y) \rightarrow \text{Hom}(F(x), F(y))$  is surjective/injective.
- an **essentially surjective functor** if any object of  $\mathcal{D}$  is equivalent to some  $f(x)$ , where  $x \in \mathcal{C}$ .
- an **equivalence** if there is a functor  $G : \mathcal{D} \rightarrow \mathcal{C}$  that  $F \circ G \cong \text{id}_{\mathcal{D}}$  and  $G \circ F \cong \text{id}_{\mathcal{C}}$ .

┘

**Prop. (4.1.1.11) [Category Equivalences].** A Functor  $\mathcal{C} \rightarrow \mathcal{D}$  is an equivalence if and only if it's fully faithful and essentially surjective.

┘

*Proof:* There exist an object  $G(X) \in \mathcal{C}$  and an isomorphism  $\xi_X : FG(X) \rightarrow X$  for every  $X \in \mathcal{D}$ . Because  $F$  is fully faithful, there exists a unique morphism  $G(f) : G(X) \rightarrow G(Y)$  such that  $F(G(f)) = \xi_Y^{-1} \circ f \circ \xi_X$  for every morphism  $f : X \rightarrow Y$  in  $\mathcal{D}$ . Thus we obtain a functor  $G : \mathcal{C} \rightarrow \mathcal{D}$  as well as a natural isomorphism  $\xi : F \circ G \rightarrow \text{Id}_{\mathcal{D}}$ . Moreover, the isomorphism  $\xi_{F(Z)} : FGF(Z) \rightarrow F(Z)$  decides an isomorphism  $\eta_Z : GF(Z) \rightarrow Z$  for every  $Z \in \mathcal{C}$ . This yields a natural isomorphism  $\eta : G \circ F \rightarrow \text{Id}_{\mathcal{C}}$ .  $\square$

**Def. (4.1.1.12) [Setoids].** A **setoid** is a category that is equivalent to a category that has only the identity morphisms.  $\lrcorner$

**Def. (4.1.1.13) [Groupoids].** A **groupoid** is a category that all morphisms are isomorphisms. The full subcategory of  $\text{Cat}$  consisting of groupoids is denoted by  $\text{Grpd}$ .  $\lrcorner$

**Prop. (4.1.1.14) [Equivalence Relations].** An **equivalence relation** is a groupoid that  $\# \text{Mor}_{\mathcal{C}}(x, y) \leq 1$  for any  $x, y \in \mathcal{C}$ .  $\lrcorner$

**Prop. (4.1.1.15).** A category is equivalent to a setoid iff it is an equivalence relation.  $\lrcorner$

*Proof:*

$\square$

**Def. (4.1.1.16) [Subcategories].** Let  $\mathcal{C}$  be a category, a **subcategory**  $\mathcal{C}'$  of  $\mathcal{C}$  is a category together with a functor  $F : \mathcal{C}' \rightarrow \mathcal{C}$  s.t.  $F : \text{Ob}(\mathcal{C}') \rightarrow \text{Ob}(\mathcal{C})$  is injective and  $F$  is faithful. And it is called a **full subcategory** if  $F$  is fully faithful. It is called a **strictly-full subcategory** if for any  $y \in \mathcal{C}$ ,  $y$  is equivalent to some  $F(x)$ ,  $x \in \text{Ob}(\mathcal{C}')$  iff  $y \in F(\mathcal{C}')$ .  $\lrcorner$

**Def. (4.1.1.17) [Comma Categories].** For a category  $\mathcal{C}$  and an object  $S$ , the **comma category**  $\mathcal{C}/S$  is defined to be the category of arrows  $T \rightarrow S$  with the arrows being compatible arrows over  $S$ .  $\lrcorner$

**Def. (4.1.1.18) [Category of Arrows].** For a category  $\mathcal{C}$ , the category of arrows  $\text{Arr}(\mathcal{C})$  is a category whose objects are arrows in  $\mathcal{C}$  and a morphism  $f \rightarrow g$  is a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{h} & B \\ \downarrow f & & \downarrow g \\ A' & \xrightarrow{k} & B' \end{array}$$

$\lrcorner$

**Def. (4.1.1.19) [Twisted Arrow Category].** For a category  $\mathcal{C}$ , the category of twisted arrows  $TW(\mathcal{C})$  is a category whose objects are arrows in  $\mathcal{C}$  and a morphism  $f \rightarrow g$  is a diagram

$$\begin{array}{ccc} A & \xleftarrow{h} & B \\ \downarrow f & & \downarrow g \\ A' & \xrightarrow{k} & B' \end{array}$$

$\lrcorner$

**Def. (4.1.1.20) [Epimorphisms and Monomorphisms].** An **epimorphism** in a category is a morphism  $X \rightarrow Y$  that the map  $\text{Map}(Y, Z) \rightarrow \text{Map}(X, Z)$  induced by composition is injective. Dually, an **monomorphism** is an epimorphism in the dual category.

A **balanced category** is a category s.t. every morphism that is both epi and mono is an isomorphism.  $\lrcorner$

**Def. (4.1.1.21) [Connected Objects].** A **connected object** in  $\mathcal{C}$  is an object  $X \in \text{Ob}(\mathcal{C})$  s.t. it is not initial (4.1.1.8), and for any monomorphism  $Y \rightarrow X$  (4.1.1.20), either  $Y$  is initial or  $Y \rightarrow X$  is an isomorphism.  $\lrcorner$

**Def. (4.1.1.22) [Projective and Injective].** A **projective object**  $X$  in a category is an object that for any epimorphism (4.1.1.20)  $Y \rightarrow Z$ ,  $\text{Hom}(X, Y) \rightarrow \text{Hom}(X, Z)$  is surjective. Dually, an **injective object**  $X$  is a projective object in the dual category.  $\lrcorner$

**Def. (4.1.1.23) [Retract].** A morphism  $f$  in a category  $\mathcal{C}$  is called a **retract** of  $g$  if there are morphisms  $F, G : f \rightarrow g$  in  $\text{Mor}(\mathcal{C})$  (4.1.1.18) that  $G \circ F = \text{id}_f$ .  $\lrcorner$

**Def. (4.1.1.24) [Equivalence relations].** For  $\mathcal{C} \in \text{Cat}$  has finite products,  $X_0 \in \mathcal{C}$ , an **equivalence relation on  $X_0$**  is a tuple  $(X_1, u)$ , where  $X_1 \in \mathcal{PSh}(\mathcal{C})$  and  $u \in \text{Hom}(X_1, X_0 \times X_0)$  s.t for any  $X \in \mathcal{C}$ , the map

$$\tilde{X}_1(T) \xrightarrow{u} \tilde{X}_0(T) \times \tilde{X}_0(T)$$

is a bijection of  $\tilde{X}_1(T)$  onto the graph of an equivalence relation on  $\tilde{X}_0(T)$ . Similarly, we can define **pre-relations on  $X_0$** , **relations on  $X_0$**  and **pre-equivalence relations on  $X_0$** .  $\lrcorner$

**Prop. (4.1.1.25) [Involutions].** Situation as in (4.1.1.24), then there is a morphism  $\sigma : F_0 \rightarrow F_0$ ,  $\sigma^2 = \text{id}$ ,  $u_0 \circ \sigma = u_1$ ,  $u_1 \circ \sigma = u_0$ , by the symmetry in the definition of equivalence relations.  $\lrcorner$

**Def. (4.1.1.26) [Categorical Quotients].** Situation as in (4.1.1.24), for an equivalence relation  $u : X_1 \rightarrow X_0 \times X_0$ , a morphism  $v : X_0 \rightarrow X$  is called a categorical quotient if

- $u$  factors through  $u : X_1 \rightarrow X_0 \times_X X_0$ , and  $u : X_1 \rightarrow X_0 \times_X X_0$  is an isomorphism.
- For any  $T \in \mathcal{C}$ , there is a cokernel sequence

$$\text{Mor}(X, T) \rightarrow \text{Mor}(X_0, T) \rightrightarrows \text{Mor}(X_1, T)$$

is exact.  $\lrcorner$

### Representable Functors

**Def. (4.1.1.27) [Presheaves].** A **presheaf** on  $\mathcal{C}$  is a contravariant functor from  $\mathcal{C}$  to  $\text{Set}$ . The category of presheaves on  $\mathcal{C}$  is denoted by  $\mathcal{PSh}(\mathcal{C})$ .  $\lrcorner$

**Prop. (4.1.1.28) [Representability Criterion].** Let  $\mathcal{C}$  be a complete category,  $F : \mathcal{C} \rightarrow \text{Set}$  be a functor. Assume that  $F$  commutes with small limits, and the category  $\mathcal{I}$  of pairs  $(x, f)$  where  $x \in \mathcal{C}$ ,  $f \in F(x)$  has a cofinal family of objects indexed by a set  $I$ , then  $F$  is representable, i.e. there is an object  $x$  that  $F(y) = \text{Mor}_{\mathcal{C}}(x, y)$ , functorial in  $y$ .  $\lrcorner$

*Proof:* Because  $\mathcal{C}$  has small limits, let  $\mathcal{I}'$  be the full subcategory of  $\mathcal{I}$  generated by  $(x_i, f_i)$ , set  $x = \varprojlim_{(x_i, f_i) \in \mathcal{I}'} x_i$ . As  $F$  commutes with limits,  $F(x) = \varprojlim_{(x_i, f_i) \in \mathcal{I}'} F(x_i)$ . Hence there is a universal element  $f \in F(x)$  that maps to  $f_i$  under  $F(x \rightarrow x_i)$ .  $f$  induces a natural transformation  $\xi : \text{Mor}_{\mathcal{C}}(x, -) \rightarrow F(-)$ .

The assumption shows  $\xi$  is surjective. Now let  $x' \rightarrow x$  be the equalizer of all maps  $\varphi : x \rightarrow x$  that  $F(\varphi)f = f$ , then there is a  $f' \in F(x')$  mapping to  $f$ . then the transformation  $\xi'$  defined by  $f'$  is also surjective. Now we also want to show it is injective: if  $a, b \in \text{Mor}_{\mathcal{C}}(x', y)$  mapsto the same element, then we consider the equalizer  $e' : x'' \rightarrow x'$  of  $a, b$ , then the assumption and the fact  $F$  commutes with equalizer shows there is a  $f'' \in F(x'')$  mapping to  $f'$ .



By universality consider a morphism  $\psi : x \rightarrow x''$  that  $F(\psi)f = f''$ , then  $e \circ e' \circ \psi$  is a morphism  $x \rightarrow x$  that fixes  $f$ , thus by construction  $ee'\psi e = e$ , so  $e'\psi e = \text{id}$ , because  $e$  is a monomorphism. Then  $e'$  is an epimorphism, thus  $a = b$ .  $\square$

### Adjunctions

**Def. (4.1.1.29) [Adjunction Pairs].** A pair of functors  $(f, g)$  where  $f : \mathcal{C} \rightarrow \mathcal{D}, g : \mathcal{D} \rightarrow \mathcal{C} \in \text{Cat}$  are called an **adjunction pair** iff there is natural isomorphism of functors:

$$\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \text{Set} : \text{Hom}(f(-), -) \cong \text{Hom}(-, g(-)).$$

Equivalently this means that for any  $X_1 \rightarrow X_2 \in \mathcal{C}, Y_1 \rightarrow Y_2 \in \mathcal{D}$ ,

$$\begin{array}{ccc} f(X_1) & \longrightarrow & Y_1 \\ \downarrow & & \downarrow \\ f(X_2) & \longrightarrow & Y_2 \end{array}$$

is commutative iff its corresponding map

$$\begin{array}{ccc} X_1 & \longrightarrow & g(Y_1) \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & g(Y_2) \end{array}$$

is commutative.

Denoted an adjunction pair by  $(f, g) : \mathcal{C} \rightleftarrows \mathcal{D}$ .  $\lrcorner$

**Prop. (4.1.1.30) [Units and Counits].** If  $f, g$  are adjoints, then there are natural transformations  $u : \text{id} \rightarrow gf$ , and  $v : fg \rightarrow \text{id}$ , called the **unit/counit maps**. They satisfies  $f \xrightarrow{u} f g f \xrightarrow{v} f$  is  $\text{id}$ , and  $g \xrightarrow{u} g f g \xrightarrow{v} g$  is  $\text{id}$ .

Conversely, if there are natural morphisms  $u, v$  satisfying these two identities, then  $(f, g)$  is an adjunction, by

$$\text{Hom}(fX, Y) \rightarrow \text{Hom}(gfX, gY) \rightarrow \text{Hom}(X, gY) \rightarrow \text{Hom}(fX, fgY) \rightarrow \text{Hom}(fX, Y).$$

$\lrcorner$

*Proof:*

$\square$

**Lemma (4.1.1.31).** Let  $(f, g)$  be an adjunction pair, then

- If  $g \circ f$  is fully faithful, then  $f$  is fully faithful.
- If  $f \circ g$  is fully faithful, then  $g$  is fully faithful.

$\lrcorner$

*Proof:* Cf. [Sta]0FWV.  $\square$

**Prop. (4.1.1.32).** Let  $(f, g)$  be an adjunction pair, then

- $f$  is fully faithful iff  $u : \text{id} \rightarrow gf$  is an isomorphism.

- $g$  is fully faithful iff  $v : fg \rightarrow \text{id}$  is an isomorphism.

┘

*Proof:* 1: If  $\text{id} \cong gf$ , then  $gf$  is fully faithful, so  $f$  is fully faithful by (4.1.1.31). If  $f$  is fully faithful, then for any  $X, Y$ ,

$$\text{Hom}(X, Y) \xrightarrow{u} H(X, gfY) \cong H(fX, fY) \cong H(X, Y)$$

is a canonical isomorphism, so  $u$  is an isomorphism.

2 is dual to 1.

□

**Cor. (4.1.1.33) [Units and Equivalences].** Let  $(F, G)$  be an adjunction pair, then the following are equivalent:

- $F, G$  are both fully faithful.
- the unit and counit are both isomorphisms.
- $F, G$  defines an equivalence of categories.

┘

*Proof:*  $1 \rightarrow 2 \rightarrow 3$  follow from (4.1.1.32). And  $3 \rightarrow 1$  is clear.

□

**Prop. (4.1.1.34) [Adjunction Preserves (Co)Limits].** A right adjoint functor is left exact and it preserves injectives if its left adjoint is exact.

A left adjoint functor is right exact and it preserves projectives if its right adjoint is exact. ┘

**Prop. (4.1.1.35) [Adjoint Functor Theorem].** Let  $G : \mathcal{C} \rightarrow \mathcal{D} \in \mathbf{Cat}$ , assume  $\mathcal{C}$  is complete and  $G$  commutes with small limits. Assume for every  $y \in \mathcal{D}$ , the category of pairs  $(x, f)$  where  $x \in \mathcal{C}$  and  $f \in \text{Mor}_{\mathcal{D}}(y, G(x))$  has a cofinal family of objects indexed by a set  $I$ , then  $G$  has a left adjoint.

Similarly the dual statement holds. ┘

*Proof:* The assumption shows that for any  $y \in \mathcal{D}$ , the functor  $x \mapsto \text{Mor}_{\mathcal{D}}(y, G(x))$  satisfies the condition of (4.1.1.28), thus it is representable by an object denoted by  $F(y)$ . By Yoneda's lemma,  $F$  underlies a functor, and this functor is a left adjoint of  $G$ . □

**Prop. (4.1.1.36) [Groupoidification].** There is a functor  $\mathbf{Grpd} : \mathbf{Cat} \rightarrow \mathbf{Grpd}$  that is left adjoint to the inclusion functor, called the **groupoidification functor**. ┘

*Proof:*

□

**Prop. (4.1.1.37) [Examples of Adjunction pairs].**

- The valuation at  $k$ -th coordinate is left adjoint to the functor  $k_*(A)(i) = \prod_{\text{Hom } i, k} A$  and is exact. So  $k_*$  preserves injectives.
- The sheaf  $\Gamma$  functor is right adjoint to the constant sheaf functor over arbitrary site.
- The inclusion functor is right adjoint to the shification functor over arbitrary site.
- The forgetful functor is right adjoint to the Shification functor, and shification is exact, so it preserves injectives.
- The stalk functor is left adjoint to the skyscraper sheaf operator.

┘

### Limits and Colimits

**Prop. (4.1.1.38) [Products and Equalizers Implies Limits].** If a category admits arbitrary (resp. finite) products and equalizers, then it admits arbitrary (resp. finite) limits. Dually a category that admits coproducts and coequalizers admits all colimits.  $\lrcorner$

*Proof:* The limits over a category  $\mathcal{C}$  is an equalizer of products over the category of arrows in  $\mathcal{C}$ .  $\square$

**Def. (4.1.1.39) [Exact Functors].** A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is called a **left exact functor** if it maps finite limits to finite limits. It is called a **right exact functor** if it maps finite colimits to finite colimits. It is called an **exact functor** if it is both left exact and right exact.  $\lrcorner$

**Def. (4.1.1.40) [(Co)Complete Category].** A category is called **(co)complete** if it has all small (co)limits, i.e. (co)limits over small categories (4.1.2.1).  $\lrcorner$

**Def. (4.1.1.41) [Inverse Systems].** An **inverse system** in  $\mathcal{C}$  is a diagram  $\mathbb{Z}_+^{\text{op}} \rightarrow \mathcal{C}$ , where  $\mathbb{Z}_+$  is the category of non-negative integers with a unique morphism  $n \rightarrow m$  iff  $n \leq m$ .  $\lrcorner$

**Prop. (4.1.1.42) [Equivalent Inverse System].** Two inverse systems  $\{A_n\}, \{B_n\}$  are called equivalent if there are two non-decreasing unbounded maps  $\alpha, \beta : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$  and maps  $\alpha : A_n \rightarrow B_{\alpha(n)}, \beta : B_n \rightarrow A_{\beta(n)}$  that is compatible with the transition maps and for any  $n$ , there is a  $m$  large that  $A_m \rightarrow A_{\beta\alpha(m)} \rightarrow A_n$  coincide with the transition map  $A_m \rightarrow A_n$ , and similar for  $B_n$ . And similarly for colimits.

The limit for two inverse systems are the same, and similarly for colimits.  $\lrcorner$

**Prop. (4.1.1.43) [Filtered Colimit of Sets].** Filtered colimits in  $\mathbf{Set}$  commutes with finite limits.  $\lrcorner$

*Proof:* Cf. [Sta]002W.  $\square$

**Prop. (4.1.1.44) [Cofiltered Limits of Sets].** A cofiltered limit of nonempty sets is nonempty.  $\lrcorner$

*Proof:* Cf. [Sta]086J.  $\square$

**Def. (4.1.1.45) [Mittag-Leffler].** If  $(A_i, \varphi_{ji})$  is a directed inverse system of sets over  $I$ , then it is said to satisfy the **Mittag-Leffler condition** if  $\varphi_{ji}(A_j) \in A_i$  stabilizes. This is clearly true if  $\varphi_{ji}$  is surjective for any  $i, j$ .  $\lrcorner$

**Prop. (4.1.1.46).** If  $(A_n)$  where  $n \in \mathbb{Z}$  is a Mittag-Leffler inverse system of nonempty sets, then  $\lim A_i$  is also nonempty.  $\lrcorner$

*Proof:* Let  $A'_j = \cap_{i \geq j} \varphi_{ji} A_j$ , then  $(A'_j)$  is a filtered system that the transition maps are all surjective, and clearly  $\lim A_j = \lim A'_j$ , so it is nonempty by (4.1.1.44).  $\square$

**Prop. (4.1.1.47).**  $\mathbf{Cat}$  is complete and cocomplete.  $\lrcorner$

*Proof:*  $\square$

### Fiber Product

**Prop. (4.1.1.48).** For a category  $\mathcal{C}$ , the following are equivalent:

- It has arbitrary limits.
- it has arbitrary products and equalizer.

- it has arbitrary products and fibered products.

┘

*Proof:*  $1 \rightarrow 2, 1 \rightarrow 3$  is trivial.  $3 \rightarrow 2$ : The equalizer for  $f, g : X \rightarrow Y$  can be constructed as the base change of  $Y \rightarrow Y \otimes Y$  along  $(f, g) : X \rightarrow Y \times Y$ .  $2 \rightarrow 1$ : for any diagram  $F : I \rightarrow C$ , the fibered pullback can be constructed as the equalizer of two morphisms:

$$s, t : \prod_{i \in \text{Ob}(I)} F(i) \rightarrow \prod_{f:j \rightarrow k \in \text{Mor}(I)} F(k)$$

where  $\pi_{(f:j \rightarrow k)} s = \pi_k$ , and  $\pi_{(f:j \rightarrow k)} t = (Ff)\pi_j$ .

□

**Prop. (4.1.1.49) [Diagonal Base Change].** The diagonal commutes with base change:

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{\Delta} & (X \times_Y Z) \times_Z (X \times_Y Z) \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta} & X \times_Y X \end{array}$$

┘

*Proof:*

□

**Prop. (4.1.1.50).**  $(X \times_E Y) \times_S (Z \times_F W) = (X \times_S Z) \times_{E \times_S F} (Y \times_S W)$ .

┘

*Proof:*

□

**Prop. (4.1.1.51).** For  $f : X \rightarrow T$  and  $g : Y \rightarrow T$  and  $h : T \rightarrow S$ ,  $X \times_T Y = T \times_{T \times_S T} (X \times_S Y)$ . In particular,  $X \times_T Y \rightarrow X \times_S Y$  is a base change of  $T \rightarrow T \times_S T$ .

┘

*Proof:* For any object  $U$ ,

$$\text{Hom}(U, X \times_T Y) = \{s : U \rightarrow X, t : U \rightarrow Y \mid f \circ s = g \circ t\},$$

$$\text{Hom}(U, T \times_{T \times_S T} (X \times_S Y)) = \{r : U \rightarrow T, s : U \rightarrow X, t : U \rightarrow Y \mid h \circ f \circ s = h \circ g \circ t, r = g \circ t, r = f \circ s\},$$

so they are functorially isomorphic. Then by Yoneda's lemma (4.1.3.11) we have the desired isomorphism.

□

**Prop. (4.1.1.52).** The diagonal map  $X \rightarrow X \times_Y X$  is an isomorphism iff  $X \rightarrow Y$  is monomorphism. (Because this is equivalent to  $\text{pr}_1 = \text{pr}_2$ ).

┘

**Def. (4.1.1.53) [Mapping graph].** Let  $f : X \rightarrow Y$  be a morphism in a category with fiber products and finite products, then the **mapping graph**  $\Gamma_f$  of  $f$  is defined to be the pullback

$$\begin{array}{ccc} \Gamma_f & \longrightarrow & X \times Y \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\Delta} & Y \times Y \end{array}.$$

it can be seen that  $\Gamma_f$  is isomorphic to  $X$ .

┘

### Localizations

**Def. (4.1.1.54) [Localized Categories].** Let  $\mathcal{C} \in \text{Cat}$  and  $S \subset \text{Mor}(\mathcal{C})$ , then a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is called a **Localized category** of  $\mathcal{C}$  w.r.t  $S$  if it maps morphisms in  $S$  to isomorphisms, and any other functor with this property factors uniquely through  $F$ .  $\perp$

**Def. (4.1.1.55) [Localizing Systems].** For  $\mathcal{C} \in \text{Cat}$ , a class of morphisms  $S$  in  $\mathcal{C}$  is called a left(resp. right) **localizing system** if:

(LS1):  $S$  is closed under composition and contains all the identities.

(LS2): for every  $s \in S$  and  $f$  with the same source(resp. target), there exists  $t \in S, g \in \text{Mor}(\mathcal{C})$  s.t.  $t \circ f = g \circ s$ (resp.  $f \circ t = s \circ g$ ).

(LS3): For  $f, g \in \text{Mor}(\mathcal{C})$  with the same source and target, the existence of a  $t \in S$  s.t.  $ft = gt$  implies(resp. is implied by) the existence of a  $s \in S$  s.t.  $sf = sg$ .

It is called **localizing system** if it is both left localizing and right localizing.  $\perp$

**Def. (4.1.1.56) [Saturated Localizing Systems].** Let  $S$  be a localizing system, then it is called a **saturated localizing system** is moreover it satisfies

- if  $f, g, h \in S$  and  $fg, gh \in S$ , then  $g \in S$ .

$\perp$

**Def. (4.1.1.57) [Gabriel-Zisman Localization].** If  $\mathcal{C} \in \text{Cat}$  and  $S \subset \text{Mor}(\mathcal{C})$ , we construct a category  $S^{-1}\mathcal{C}$  as follows:

- $\text{Ob}(S^{-1}\mathcal{C}) = \text{Ob}(\mathcal{C})$ .
- For  $X, Y \in \text{Ob}(S^{-1}\mathcal{C})$ ,  $\text{Mor}(X, Y) = \{(f, s) | f : X \rightarrow Y', s : Y' \rightarrow Y, s \in S\} / \sim$ , where the equivalence relation is generated by the following relations:  $(f_1, s_1), (f_2, s_2) \in \text{Mor}_{S^{-1}\mathcal{C}}(X, Y)$  are equivalent if there exists some commutative diagram in  $\mathcal{C}$ :

$$\begin{array}{ccccc} X & \xrightarrow{f_1} & Y' & \xleftarrow{s_1} & Y \\ & \searrow f_2 & \downarrow u & \swarrow s_2 & \\ & & Y'' & & \end{array} .$$

the pair  $(f, s)$  is also denoted by  $s^{-1}f$ .

- The composition of two morphisms  $(f : X \rightarrow Y', s : Y' \rightarrow Y)$  and  $(g : Y \rightarrow Z', t : Z' \rightarrow Z)$  is defined to be the equivalence class of  $(hf : X \rightarrow Z'', ut : Z' \rightarrow Z'')$ , where  $h$  and  $u \in S$  are chosen to fit in the commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{g} & Z \\ \downarrow s & & \downarrow u \\ Y' & \xrightarrow{h} & Z'' \end{array} .$$

- The identity elements  $\text{id}_X \in \text{Mor}(S^{-1}\mathcal{C})$  is defined to be the equivalence class of  $(\text{id}_X, \text{id}_X)$ .

Then these definitions are well-defined and makes  $S^{-1}\mathcal{C}$  a category.

And dually we can define  $S^{-1}\mathcal{C}$  for a right localizing system.  $\perp$

*Proof:* Cf. [Sta]04VG,  $\square$

**Cor. (4.1.1.58) [Comparing Morphisms in  $S^{-1}\mathcal{C}$ ].** Suppose  $\mathcal{C} \in \mathcal{Cat}$  and  $S \subset \text{Mor}(\mathcal{C})$ , then given any finite set of morphism  $g_i : x \rightarrow y$  in  $S^{-1}\mathcal{C}$  with the same target, we can find their representatives  $(f_i, s)$  with common  $s \in S$ .

And two morphisms  $(f_1 : X \rightarrow Y', s : Y \rightarrow Y')$  and  $(f_2 : X \rightarrow Y', s : Y \rightarrow Y')$  are equivalent iff there exists  $a : Y' \rightarrow Y''$  s.t.  $a \circ s \in S$  and  $a \circ f_1 = a \circ f_2$ .  $\lrcorner$

*Proof:* These follow from LS1 and LS2 in the definition (4.1.1.55).  $\square$

**Cor. (4.1.1.59) [ $S^{-1}\mathcal{C}$  is the Localized Category].** If  $\mathcal{C} \in \mathcal{Cat}$  and  $S \subset \text{Mor}(\mathcal{C})$  is a left localizing system, then the rule

$$X \mapsto X, (f : X \rightarrow Y) \mapsto \text{id}_Y^{-1} f$$

defines a functor  $Q : \mathcal{C} \rightarrow S^{-1}\mathcal{C}$  that represents  $S^{-1}\mathcal{C}$  as a localizing category of  $\mathcal{C}$  w.r.t  $S$  (4.1.1.54). And  $Q$  preserves finite colimits.

The dual statement is true for a right localizing system  $S$ .  $\lrcorner$

*Proof:* It is clearly a functor, and for any  $s \in S$ ,  $Q(s)$  is an isomorphism with inverse  $(\text{id}_Y, s)$ . To show that it preserves finite colimits, notice

$$\begin{aligned} \text{Mor}_{S^{-1}\mathcal{C}}(\text{colim}_i X_i, Y) &= \text{colim}_{s:Y \rightarrow Y' \in S} \text{Mor}_{\mathcal{C}}(\text{colim}_i X_i, Y') \\ &= \text{colim}_{s:Y \rightarrow Y' \in S} \lim_i \text{Mor}_{\mathcal{C}}(X_i, Y') \\ (4.1.1.43) \quad &= \lim_i \text{colim}_{s:Y \rightarrow Y' \in S} \text{Mor}_{\mathcal{C}}(X_i, Y') \\ &= \lim_i \text{Mor}_{S^{-1}\mathcal{C}}(X_i, Y) \end{aligned}$$

To show it is the localized category, suppose  $G : \mathcal{C} \rightarrow \mathcal{D}$  is a functor s.t.  $G(s)$  are all isomorphisms for  $s \in S$ , then we can define  $H : S^{-1}\mathcal{C} \rightarrow \mathcal{D} : H(x) = x$ , and  $H((f, s)) = G(s)^{-1} \circ G(f)$ . Then it can be checked that this is well-defined and  $G = H \circ Q$ .  $\square$

**Cor. (4.1.1.60).** If  $\mathcal{C} \in \mathcal{Cat}$  and  $S \subset \text{Mor}(\mathcal{C})$  is a localizing system, then the left localizing category and the right localizing category is canonically isomorphic, by the universal property (4.1.1.54).  $\lrcorner$

**Prop. (4.1.1.61) [Saturation].** If  $\mathcal{C} \in \mathcal{Cat}$  and  $S \subset \text{Mor}(\mathcal{C})$  is a localizing system, then the morphisms in  $\mathcal{C}$  that is mapped to an isomorphism in  $S^{-1}\mathcal{C}$  is the smallest saturated localizing system containing  $S$ , called the **saturation of  $S$** . Moreover, it consists of morphisms  $f \in \text{Mor}(\mathcal{C})$  s.t. there exists  $g, h \in S$  s.t.  $gf, gh \in S$ .  $\lrcorner$

*Proof:* Cf. [Sta]05Q9.  $\square$

**Prop. (4.1.1.62) [Full Subcategories of Localized Categories].** If  $S$  is a left localizing system in a category  $\mathcal{C}$ ,  $\mathcal{C}'$  is a full subcategory of  $\mathcal{C}$  and  $S'$  is a left localizing system of  $\mathcal{C}'$  that  $S' \subset S$ . If for any  $s : X \rightarrow Y \in S$  and  $X \in \mathcal{C}'$ , there is a morphism  $t : Y \rightarrow Z$  that  $Z \in \mathcal{C}'$  and  $t \circ s \in S'$ , then the natural functor  $(S')^{-1}\mathcal{C}' \rightarrow S^{-1}\mathcal{C}$  is fully faithful. And the dual is true for right localizing systems.  $\lrcorner$

*Proof:* This is not hard from the Gabriel-Zisman localization description (4.1.1.57).  $\square$

**Def. (4.1.1.63) [Reflective Localizations].** A **reflective localization** is an adjunction  $(L, R) : \mathcal{C} \rightleftarrows \mathcal{D}$  s.t.  $R$  is fully faithful.  $\lrcorner$

**Prop. (4.1.1.64) [Reflective Localizations as Localizations].** Let  $(L, R) : \mathcal{C} \rightleftarrows \mathcal{D}$  be a reflective localization(4.1.1.63), then  $\mathcal{D}$  is equivalent to the localization  $\mathcal{C}[S^{-1}]$ , where  $S$  is the class of morphisms of  $\mathcal{C}$  that are sent to isomorphisms by  $L$   $\lrcorner$

*Proof:* Cf.[Bor94]P190.  $\square$

**Def. (4.1.1.65) [ $S$ -Local Equivalences].** Let  $\mathcal{C} \in \mathbf{Cat}$  and  $S$  a class of morphisms in  $\mathcal{C}$ .

- $c \in \mathcal{C}$  is called  **$S$ -local** if  $\mathrm{Hom}(c_2, c) \rightarrow \mathrm{Hom}(c_1, c)$  is a bijection for any  $c_1 \rightarrow c_2 \in S$ .
- $f : c_1 \rightarrow c_2 \in \mathcal{C}$  is called an  **$S$ -local equivalence** if for any  $S$ -local object  $c \in \mathcal{C}$ ,  $\mathrm{Hom}(c_2, c) \rightarrow \mathrm{Hom}(c_1, c)$  is a bijection.

$\lrcorner$

**Prop. (4.1.1.66).** Let  $(L, R) : \mathcal{C} \rightleftarrows \mathcal{D}$  is a reflective localization, and let  $S$  be the class of morphisms in  $\mathcal{C}$  that is mapped sent to isomorphisms by  $L$ , then

- The essential image of  $R$  consists precisely of  $S$ -local objects.
- The  $S$ -local

$\lrcorner$

### Group Objects

**Def. (4.1.1.67) [Group Object].** In a category  $\mathcal{C} \in \mathbf{Cat}$  with finite products and a final object  $e$ , a(n) **(Abelian)group object** is an object  $G$  that  $h_G$  is a functor from  $\mathcal{C}$  to  $\mathbf{Grp}$ (resp.  $\mathbf{Ab}$ ). And a homomorphism of group objects is a natural transformation as a functor  $\mathcal{C} \rightarrow \mathbf{Grp}$ .

This is in fact equivalent to a morphism  $m_G : G \times G \rightarrow G$  and  $i_G : G \rightarrow G$ ,  $e_G : e \rightarrow G$  that satisfy the desired commuting diagrams.  $\lrcorner$

**Def. (4.1.1.68) [Group Action].** A **(left)action** of a group object  $G$  on an object  $X$  is a map of presheaves  $h_G \times h_X \rightarrow h_X$  that for any  $U$ ,  $h_G(U) \times h_X(U) \rightarrow h_X(U)$  is a group action. This is equivalent to a morphism  $\mu : G \times X \rightarrow X$  that satisfies the desired commuting diagrams.  $\lrcorner$

**Prop. (4.1.1.69).** Let  $\mu : G \times X \rightarrow X$  be an action of a group object  $G$  on an object  $X$ , there is a commutative diagram

$$\begin{array}{ccc} G \times X & \xrightarrow{(g,x) \mapsto (g,gx)} & G \times X \\ \downarrow (g,x) \mapsto gx & & \downarrow \pi_2 \\ X & \xrightarrow{id} & X \end{array}$$

and the horizontal maps are isomorphisms.  $\lrcorner$

**Cor. (4.1.1.70).** Considering the action of  $G$  on itself, we get

$$\begin{array}{ccc} G \times G & \xrightarrow{(g,h) \mapsto (g,gh)} & G \times G \\ \downarrow m & & \downarrow \pi_2 \\ G & \xrightarrow{id} & G \end{array}$$

and the horizontal maps are isomorphisms.  $\lrcorner$

**Prop. (4.1.1.71).** A unital magma object  $G$  in the category of groups is an Abelian group.  $\lrcorner$

*Proof:* A unital magma object structure endows  $G$  with maps  $(\tilde{m}, \tilde{e})$ , and  $\tilde{m}(ab, cd) = \tilde{m}(a, c)\tilde{m}(b, d)$ , because  $\tilde{m}$  is a morphism of groups. So we can use Eckmann-Hilton argument (3.1.1.5) to show the category multiplication is the same as the group multiplication. So the commutativity of  $\tilde{m}$  with the inverse  $i$  implies that it is Abelian.  $\square$

**Prop. (4.1.1.72).** A right-lax monoidal functor (4.1.5.12) between Cartesian monoidal structures maps a unital magma object to a unital magma object.  $\lrcorner$

**Def. (4.1.1.73) [Categorical Quotient].** Let  $C$  be a category with finite products,  $G$  be a group object in  $C$ ,  $G \times X \rightarrow X$  is a left action (4.1.1.68), then a morphism  $X \rightarrow Y$  is called the **categorical quotient** of  $X$  iff  $Y$  is the coequalizer of  $G \times X \xrightarrow[\text{pr}_2]{\rho} X$ . It is called the **universal categorical quotient** of  $X$  iff its product with  $S$  is the categorical quotient for each element  $S \in C$ , in the category  $C_{/S}$ .  $\lrcorner$

## 2 Presentable Categories

Main references are [Locally presentable and accessible categories].

**Def. (4.1.2.1) [Small Categories].** Given a regular cardinal  $\kappa$  (2.2.12.2),  $S \in \text{Set}$  is called  $\kappa$ -**small** if it has cardinality smaller than  $\kappa$ .  $\mathcal{C} \in \text{Cat}$  is called  $\kappa$ -**small** if  $\text{Ob}(\mathcal{C})$  is  $\kappa$ -small, and the set of all morphisms of  $\mathcal{C}$  is  $\kappa$ -small.

Through out the whole book, we will fix a strongly inaccessible cardinal  $\kappa$  and call a set or category **small** if it is  $\kappa$ -small. And a category is called **essentially small** if it is equivalent to a small category.  $\lrcorner$

**Prop. (4.1.2.2).** Any small cocomplete category is a poset.  $\lrcorner$

*Proof:* [MacLane]P114.  $\square$

**Def. (4.1.2.3) [Compact Objects].** For a regular cardinal  $\kappa$ . Let  $\mathcal{C} \in \text{Cat}$  be a category that admits small colimits, and  $J$  be a  $\kappa$ -filtered poset and a diagram  $\{Y_\alpha\}$  indexed by  $J$ , then for  $X \in \mathcal{C}$ , there is a natural map

$$\varinjlim \text{Hom}(X, Y_j) \rightarrow \text{Hom}(X, \varinjlim Y_j).$$

$X$  is called  $\kappa$ -**compact** if this is an isomorphism for any  $\kappa$ -filtered diagram  $J$ .  $X$  is called **small** if it is  $\kappa$ -compact for some small (4.1.2.1) regular cardinal  $\kappa$ .  $\lrcorner$

**Def. (4.1.2.4) [ $\kappa$ -Accessible Categories].** For a regular cardinal  $\kappa$ , a  $\kappa$ -**accessible category** is a locally small category (4.1.2.1)  $n\text{Cat}$  that satisfies:

- $\mathcal{C}$  admits all  $\kappa$ -filtered colimits.
  - $\mathcal{C}$  is generated by a  $\kappa$ -small set  $S$  consisting of  $\kappa$ -compact objects of  $\mathcal{C}$  under  $\kappa$ -filtered colimits.
- $\mathcal{C}$  is called **accessible** if it is  $\kappa$ -accessible for some small regular cardinal  $\kappa$ . A  $\kappa$ -**accessible functor** is a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  between  $\kappa$ -cocomplete categories that preserves  $\kappa$ -filtered colimits. And a functor is called a **accessible functor** if it is  $\kappa$ -accessible for some small regular cardinal  $\kappa$ .  $\lrcorner$

**Def. (4.1.2.5) [Locally Presentable Categories].** A **locally presentable category** is a category that is both cocomplete and accessible (4.1.2.4).  $\lrcorner$

**Prop. (4.1.2.6).** Any locally presentable category is complete.  $\lrcorner$



*Proof:*

□

**Example (4.1.2.7) [Locally Presentable Categories].**

- $\mathbf{Set}$  is locally presentable.
- If  $\mathcal{C}$  is a small category, then  $\mathcal{PSh}^{\mathbf{Set}}(\mathcal{C})$  is locally presentable.
- For  $R \in \mathcal{C}\mathbf{Ring}$ ,  $\mathbf{Mod}_R$  and  $\mathbf{Ch}(R)$  are locally presentable.
- If  $T : \mathcal{C} \rightarrow \mathcal{C}$  is an accessible monad on a locally presentable category, then the category of  $T$ -algebras is locally presentable.
- Every Grothendieck topos is locally presentable.
- If  $\mathcal{M}$  is a locally presentable symmetric monoidal category, then  $\mathbf{Cat}_{\mathcal{M}}$  is locally presentable.
- $\mathbf{Top}$  is not locally presentable, but  $\mathcal{CG}$  is locally presentable.

┘

*Proof:*

□

**Cor. (4.1.2.8).**  $\mathbf{Cat}$  is locally presentable.

┘

**Prop. (4.1.2.9) [Adjoint Functor Theorem].** Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor between locally presentable categories, then

- $F$  is a left adjoint iff it preserves small colimits.
- $F$  is a right adjoint iff it is accessible and preserves small limits.

┘

*Proof:*

□

**Def. (4.1.2.10) [Accessible Reflective Localizations].** An **accessible reflective localization** is a reflective localization (4.1.1.63)  $(L, R) : \mathcal{C} \rightleftarrows \mathcal{D}$  s.t.  $R$  is accessible (4.1.2.4).

┘

**Prop. (4.1.2.11) [Classifying Locally Presentable Categories].** A category is locally presentable iff it is equivalent to an accessible reflective localization (4.1.2.10) of  $\mathcal{PSh}^{\mathbf{Set}}(\mathcal{A})$  for some small category  $\mathcal{A}$ .

┘

*Proof:* Cf. [Locally presentable and accessible categories].

□

### 3 Ends and Coends

**Def. (4.1.3.1) [Dinatural Transformation].** Given categories  $\mathcal{C}, \mathcal{D}$  and functor  $P, Q : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{D}$ , then a **dinatural transformation**  $\alpha : P \rightarrow Q$  is a family of arrows  $\alpha_C : P(C, C) \rightarrow Q(C, C)$  where  $C \in \mathcal{C}$  that for any arrow  $C \rightarrow C' \in \mathcal{C}$ , there is a commutative diagram

$$\begin{array}{ccccc} P(C', C) & \longrightarrow & P(C, C) & \xrightarrow{\alpha_C} & Q(C, C) \\ \downarrow & & & & \downarrow \\ P(C', C') & \xrightarrow{\alpha_{C'}} & Q(C', C') & \longrightarrow & Q(C, C') \end{array}$$

┘

**Def. (4.1.3.2) [Wedge and Cowedge].** Let  $P : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{D}$  be a functor, then a **wedge** for  $P$  is a binatural functor  $\Delta_D \rightarrow P$ , where  $D$  is an object of  $\mathcal{C}$ . Similarly a **cowedge** is a binatural functor  $P \rightarrow \Delta_D$ .  $\lrcorner$

**Def. (4.1.3.3) [End and Coend].** For a functor  $P : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{D}$ , the wedges and cowedges of  $P$  form categories, and we define **end** of  $P$  is just a terminal wedge, denoted by  $\int_C F(C, C)$ , and the **coend** of  $P$  is the initial cowedge, denoted by  $\int^C F(C, C)$ .  $\lrcorner$

Ends and coends are functorial w.r.t natural transformations.  $\lrcorner$

**Prop. (4.1.3.4) [Ends as Colimits].** There is a morphism

$$F \mapsto \bar{F} : \text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(Tw(\mathcal{C}), \mathcal{D}) \quad (4.1.1.19)$$

where we maps  $f : C \rightarrow C'$  to  $F(C, C')$ . Then it can be checked that

$$\int_C F(C, C) = \lim_{Tw(\mathcal{C})} \bar{F}, \quad \int^C F(C, C) = \text{colim}_{Tw(\mathcal{C})} \bar{F}.$$

**Cor. (4.1.3.5).** A functor that preserves (co)limits preserves (co)ends.  $\lrcorner$

**Cor. (4.1.3.6).** For an object  $D \in \mathcal{D}$ , we have isomorphisms:

$$\begin{aligned} \text{Hom}(\int^C F(C, C), D) &\cong \int_C \text{Hom}(F(C, C), D), \\ \text{Hom}(D, \int_C F(C, C)) &\cong \int_C \text{Hom}(D, F(C, C)). \end{aligned}$$

**Prop. (4.1.3.7) [Fubini].** Cf.[Coend Calculus, P20].  $\lrcorner$

**Prop. (4.1.3.8) [Natural Transformation as Ends].** Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be functors, then the set of natural transformations is an coend.  $\lrcorner$

$$\text{Map}(F, G) \cong \int_C \text{Hom}_{\mathcal{D}}(FC, GC)$$

### Kan Extension

Cf.[All Concepts are Kan Extensions].

**Def. (4.1.3.9) [Kan Extensions].** Given functors  $F : \mathcal{C} \rightarrow \mathcal{E}$  and  $K : \mathcal{C} \rightarrow \mathcal{D}$ , a **left Kan extension** of  $F$  along  $K$  is a functor  $\text{Lan}_K F : \mathcal{D} \rightarrow \mathcal{E}$  together with a natural transformation  $\eta : F \rightarrow \text{Lan}_K F \circ K$  that for any other pair  $(G : \mathcal{D} \rightarrow \mathcal{E}, \gamma : F \rightarrow G \circ K)$ ,  $\gamma$  factors uniquely through  $\eta$ .  $\lrcorner$

Dually, a **right Kan extension** of  $F$  over  $K$  is equivalent to a left Kan extension of  $F^{\text{op}}$  over  $K^{\text{op}}$ .  $\lrcorner$

**Prop. (4.1.3.10) [(Co)Limits as Kan Extensions].** If  $\mathcal{D}$  is the final category 1, then the left Kan extension is just the colimit of the diagram defined by  $F$ , and the right Kan extension is just the limit of the diagram defined by  $F$ .  $\lrcorner$

**Prop. (4.1.3.11)[Yoneda's Lemma].** For any  $X \in \mathcal{C}$ ,

$$\mathbf{y}_X : \mathcal{C} \rightarrow \mathbf{Set} : Y \mapsto \mathbf{Hom}(Y, X)$$

is a presheaf on  $\mathcal{C}$ , and  $\mathbf{Hom}(h_X, \mathcal{F}) \cong \mathcal{F}(X)$  for any  $\mathcal{F} \in \mathbf{PSh}(\mathcal{C})$ .

So  $X \rightarrow h_X$  is a fully faithful embedding  $\mathbf{y} : \mathcal{C} \rightarrow \mathbf{PSh}^{\mathbf{Set}}(\mathcal{C})$ . In particular, if a  $X \rightarrow Y$  induces isomorphism  $\mathbf{Hom}(W, X) \rightarrow \mathbf{Hom}(W, Y)$  for every  $W$ , then  $X \cong Y$ .

So we can regard  $\mathcal{C}$  as a fully faithful subcategory of  $\mathbf{PSh}^{\mathbf{Set}}(\mathcal{C})$ .  $\lrcorner$

*Proof:* The map  $\mathbf{Hom}(h_X, \mathcal{F}) \rightarrow \mathcal{F}(X)$  maps a  $u$  to  $u(X)(\text{id}_X)$ . And the inverse map is defined to be  $x \in \mathcal{F}(X) \mapsto (s \in \mathbf{Hom}(Y, X) \mapsto s^*(x) \in \mathcal{F}(Y)) \in \mathbf{Hom}(h_X, \mathcal{F})$ .  $\square$

**Cor. (4.1.3.12).** A **universal object** for a presheaf  $\mathcal{F}$  is a pair  $(X, \zeta)$  that  $\zeta \in \mathcal{F}(X)$  with the property that for any  $U$  and a  $\sigma \in \mathcal{F}(U)$ , there is a unique arrow  $U \rightarrow X$  that  $Ff(\zeta) = \sigma$ .

In fact, a universal object is equivalent to an isomorphism  $h_X \cong \mathcal{F}$ .  $\lrcorner$

**Prop. (4.1.3.13)[Presheaves as Colimits of Presentable Presheaves].** For  $\mathcal{C} \in \mathbf{Cat}$ , any presheaf of sets on  $\mathcal{C}$  is a colimit of presentable presheaves on  $\mathcal{C}$ . More precisely, there is an isomorphism

$$\mathcal{F} \cong \varinjlim_{h_X \rightarrow \mathcal{F}} h_X.$$

From this we see that any functor  $\mathbf{PSh}^{\mathbf{Set}}(\mathcal{C}) \rightarrow \mathcal{D}$  compatible with colimits is determined by its restriction on  $\mathcal{C}$ .  $\lrcorner$

*Proof:* For any presheaf  $\mathcal{G}$ , there is a morphism  $\mathbf{Hom}(\mathcal{F}, \mathcal{G}) \rightarrow \mathbf{Hom}(\varinjlim_{h_X \rightarrow \mathcal{F}} h_X, \mathcal{G})$ , i.e. a set of sections  $f_s \in \mathcal{G}(X)$  for every  $\mathbf{y}_X \xrightarrow{s} \mathcal{F}$ , that if  $t \circ u = s$ , then  $u^*(f_t) = f_s$ . Conversely, by Yoneda lemma, this just says that there is a morphism of presheaves  $\mathcal{F} \rightarrow \mathcal{G} : F(X) \rightarrow G(X) : s \mapsto f_s$ .  $\square$

**Cor. (4.1.3.14)[Yoneda Extensions].** For any  $\mathcal{C} \in \mathbf{Cat}$  and a cocomplete category  $\mathcal{D}$ , then any functor  $Q : \mathcal{C} \rightarrow \mathcal{D}$  extends to a functor  $|\cdot|_Q : \mathbf{PSh}^{\mathbf{Set}}(\mathcal{C}) \rightarrow \mathcal{D}$  s.t.

$$\begin{array}{ccc} \mathcal{D} & \xrightarrow{Q} & \mathcal{D} \\ \downarrow \mathbf{y} & \nearrow |\cdot|_Q & \\ \mathbf{PSh}^{\mathbf{Set}}(\mathcal{C}) & & \end{array}$$

is commutative. Moreover, there is a functor

$$\text{Sing}_Q : \mathcal{C} \rightarrow \mathbf{PSh}^{\mathbf{Set}}(\mathcal{C}) : C \mapsto (X \mapsto \mathbf{Hom}_{\mathcal{D}}(QX, C)).$$

And there is an adjunction

$$|\cdot|_Q : \mathbf{PSh}^{\mathbf{Set}}(\mathcal{C}) \rightleftarrows \mathcal{D} : \text{Sing}_Q.$$

Moreover, this assignment  $Q \mapsto (|\cdot|_Q, \text{Sing}_Q)$  induces an equivalence of categories:

$$\mathbf{Func}(\mathcal{C}, \mathcal{D}) \cong \mathbf{Adj}(\mathbf{PSh}^{\mathbf{Set}}(\mathcal{C}), \mathcal{D}).$$

$\lrcorner$

*Proof:* Define  $|\cdot|_Q$  by colimit as in (4.1.3.13), then  $\text{?}$   $\square$

**Cor. (4.1.3.15).** Any contravariant functor  $Q : \mathcal{C} \rightarrow \mathbf{Set}$  that take colimits to limits is representable.  $\lrcorner$

*Proof:* Use  $\text{Sing}_Q$  as in the last proof,  $Q$  is representable by  $\text{Sing}_Q(\text{pt})$ .  $\square$

**Def. (4.1.3.16)[J-Free Diagram].** For a cocomplete category  $\mathcal{C}$  and a small category  $\mathcal{J}$ , let  $\mathcal{J}^\delta$  be the subcategory of  $\mathcal{J}$  that has the same objects but only the identity morphisms, then an  $\mathcal{J}$ -diagram in  $\mathcal{C}$  is called **J-free** iff it is a left Kan extension of some diagram  $\mathcal{J}^\delta \rightarrow \mathcal{C}$ .  $\lrcorner$

## 4 $n$ -Categories

**Remark (4.1.4.1) [2-Categories].** For  $n \in \mathbb{N}$ ,  $n$ -categories are defined as in (4.7.1.28).  $\lrcorner$

**Prop. (4.1.4.2) [Cat].** There is a 2-category  $\mathbf{Cat}$  consisting of small categories and functors and natural transformations.  $\lrcorner$

**Def. (4.1.4.3) [(2,1)-Categories].** A **(2,1)-category** is a 2-category that all the 2-morphisms (corresponding to a 2-simplex) are isomorphisms.  $\lrcorner$

**Prop. (4.1.4.4) [Final Objects].** An object in a (2,1)-category is a final object (4.7.3.2) iff for any  $y$  there is a morphism  $y \rightarrow x$ , and any two morphisms  $y \rightarrow x$  are isomorphic by a unique 2-morphism.  $\lrcorner$

**Lemma (4.1.4.5) [2-Commutative diagrams].** Let  $\mathcal{C}$  be a 2-category, and  $g : y \rightarrow z, f : x \rightarrow z$  are arrows in  $\mathcal{C}$ , then the diagrams in  $\mathcal{C}$ :

$$\begin{array}{ccc} w & \xrightarrow{a} & x \\ \downarrow b & & \downarrow f \\ y & \xrightarrow{g} & z \end{array}$$

together with a 2-morphism from  $gb$  to  $fa$ , naturally form a 2-category. A diagrams with invertible 2-morphisms are called **2-commutative diagram** in  $\mathcal{C}$ .  $\lrcorner$

**Def. (4.1.4.6) [2-Fibered Products].** Let  $\mathcal{C}$  be a 2-category, and  $g : y \rightarrow z, f : x \rightarrow z$  are arrows in  $\mathcal{C}$ , a **2-fibered product** of  $f, g$  is a final object in the (2,1)-category of 2-commutative diagrams as defined in (4.1.4.5), and it is denoted by  $x \times_z y$ .  $\lrcorner$

## 5 Monoidal Categories

Main references are [Tensor Categories, Etingof], [Lur09].

**Def. (4.1.5.1) [Monoidal Categories].** A **monoidal category** is a category  $\mathcal{C}$  with a functor  $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  and a unit object  $1$  together with isomorphisms

$$\eta_{A,B,C} : (A \otimes B) \otimes C \rightarrow A \otimes (B \otimes C), \quad \alpha_A : A \otimes \mathbb{1} \rightarrow A, \quad \beta_A : \mathbb{1} \otimes A \rightarrow A$$

s.t.

- $\eta_{A,B,C}, \alpha_A, \beta_B$  are functorial in each coordinate.
- (MacLane Pentagon) The following diagram is commutative:

$$\begin{array}{ccc} & ((A \otimes B) \otimes C) \otimes D & \\ \eta_{A,B,C} \otimes \text{id}_D \swarrow & & \searrow \eta_{A \otimes B, C, D} \\ (A \otimes (B \otimes C)) \otimes D & & (A \otimes B) \otimes (C \otimes D) \\ \downarrow \eta_{A, B \otimes C, D} & & \downarrow \eta_{A, B, C \otimes D} \\ A \otimes ((B \otimes C) \otimes D) & \xrightarrow{\text{id}_A \otimes \eta_{B, C, D}} & A \otimes (B \otimes (C \otimes D)) \end{array}$$

- The following diagram is commutative:

$$\begin{array}{ccc} (A \otimes \mathbb{1}) \otimes B & \xrightarrow{\eta_{A, \mathbb{1}, B}} & A \otimes (\mathbb{1} \otimes B) \\ & \searrow \alpha_A \otimes \text{id}_B & \swarrow \text{id}_A \otimes \beta_B \\ & A \otimes B & \end{array}$$

┘

**Def. (4.1.5.2) [Monoidal Objects].** Let  $(\mathcal{C}, \otimes)$  be a monoidal category, then a **monoidal object** in  $\mathcal{C}$  is an object  $A \in \mathcal{C}$  s.t. there are morphisms  $\mu : A \otimes A \rightarrow A, \varepsilon : \mathbf{1} \rightarrow A$  s.t.

$$\mu(\mu \otimes \text{id}_A) = \mu(\text{id}_A \otimes \mu) \circ \eta_{A,A,A}, \quad \mu(\text{id}_A \otimes \varepsilon) = \alpha_A, \quad \mu(\varepsilon \otimes \text{id}_A) = \beta_A.$$

The full subcategory of monoidal objects in  $\mathcal{C}$  is denoted by  $\mathcal{M}\text{on}(\mathcal{C})$ .

┘

**Def. (4.1.5.3) [Dual and Opposed Category].** Let  $\mathcal{C}$  be a monoidal category, then the **opposed category**  $\mathcal{C}^{opp}$  is the same category as  $\mathcal{C}$  with the tensoring switched, which is also a monoidal category.

The dual category  $\mathcal{C}^{op}$  with the same tensoring is also a monoidal category.

┘

**Def. (4.1.5.4) [Strict Monoidal Categories].** A **strict monoidal category** is a monoidal category that the isomorphisms above (4.1.5.1) are all identities.

┘

**Prop. (4.1.5.5) [Mac Lane Coherence].** In a monoidal category, any two morphisms between two bracketings of a product  $X_1 \otimes \dots \otimes X_n$  constructed using the isomorphisms  $\eta_{A,B,C}$  are equal. Also if some  $X_i$  are 1, then we can also use  $\alpha_A$  and  $\beta_A$ .

┘

**Prop. (4.1.5.6) [Cartesian Monoidal Categories].** For a category with a final object and finite products, the product makes  $\mathcal{C}$  a symmetric monoidal category, called the **Cartesian monoidal structure**.

┘

**Prop. (4.1.5.7).** For any category  $\mathcal{C}$ , the category  $[\mathcal{C}, \mathcal{C}]$  of endofunctors has a natural monoidal structure.

┘

**Def. (4.1.5.8) [Closedness].** A monoidal category  $(\mathcal{C}, \otimes)$  is called **left-closed** if for every  $A \in \mathcal{C}$ , the functor  $N \mapsto A \otimes N$  has a right adjoint  $Y \mapsto {}^A Y$  (or denoted by  $\text{Hom}(A, Y)$  when  $\mathcal{C}$  is symmetric). Dually it is called **right-closed** if  $\mathcal{C}^{opp}$  is left-closed. It is called **closed** if it is both left-closed and right-closed.

For a Cartesian monoidal category  $\mathcal{C}$ ,  $\mathcal{C}$  is called **Cartesian-closed** if it is closed for the Cartesian monoidal structure.

┘

**Prop. (4.1.5.9).**  $\text{Cat}$  is Cartesian closed, and for any  $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \text{Cat}$ , there is an equivalence of categories

$$\text{Func}(\mathcal{A}, \text{Func}(\mathcal{B}, \mathcal{C})) \cong \text{Func}(\mathcal{A} \times \mathcal{B}, \mathcal{C}).$$

┘

*Proof:*

□

**Def. (4.1.5.10) [Reflexive Objects].** Let  $(\mathcal{C}, \otimes)$  be a symmetric monoidal category that is closed, then we denote  $\text{Hom}(Y, \mathbf{1})$  by  $Y^\vee$ , then there is a natural map  $Y \rightarrow (Y^\vee)^\vee$ . If such a map is reflexive, then  $Y$  is called a **reflexive object**.

┘

**Prop. (4.1.5.11).** If  $\mathcal{C}$  is closed, then

$$\varinjlim {}^A B \cong \varprojlim ({}^A B) \quad {}^A (\varinjlim B_i) \cong \varprojlim ({}^A B_i)$$

┘

*Proof:* Because  $\mathcal{C}$  is closed, left and right tensor  $A \otimes -$  and  $- \otimes A$  are both left adjoints thus commutes with colimits. Now  $B \mapsto {}^A B$  is a right adjoint, thus it commutes with limits. And for any  $C \in \mathcal{C}$ ,

$$\begin{aligned} \text{Hom}(C, \varinjlim {}^A B) &\cong \text{Hom}(C \otimes (\varinjlim A_i), B) \cong \text{Hom}(\varinjlim C \otimes A_i, B) \\ &\cong \varprojlim \text{Hom}(C \otimes A_i, B) \cong \varprojlim \text{Hom}(C, {}^{A_i} B) \cong \text{Hom}(C, \varprojlim ({}^{A_i} B)) \end{aligned}$$

so  $\varinjlim {}^A B \cong \varprojlim ({}^{A_i} B)$  by Yoneda's lemma(4.1.3.11).  $\square$

**Def.(4.1.5.12) [Monoidal Functor].** Let  $(\mathcal{C}, \otimes), (\mathcal{D}, \otimes)$  be monoidal categories, then a **right-lax monoidal functor** from  $\mathcal{C}$  to  $\mathcal{D}$  is a functor  $G$  together with morphisms  $\gamma_{A,B} : G(A) \otimes G(B) \rightarrow G(A \otimes B)$  for any  $A, B \in \mathcal{C}$  and a morphism  $e : \mathbb{1}_{\mathcal{D}} \rightarrow G(\mathbb{1}_{\mathcal{C}})$  s.t.

- $\gamma_{A,B}$  is functorial in each coordinate.
- (Hexagon Diagram) The following diagram is commutative:

$$\begin{array}{ccc} (G(A) \otimes G(B)) \otimes G(C) & \xrightarrow{\eta_{G(A), G(B), G(C)}} & G(A) \otimes (G(B) \otimes G(C)) \\ \downarrow \gamma_{A,B} & & \downarrow \gamma_{B,C} \\ G(A \otimes B) \otimes G(C) & & G(A) \otimes G(B \otimes C) \\ \downarrow \gamma_{A \otimes B, C} & \xrightarrow{G(\eta_{A,B,C})} & \downarrow \gamma_{A, B \otimes C} \\ G((A \otimes B) \otimes C) & \longrightarrow & G(A \otimes (B \otimes C)) \end{array}$$

- The following diagrams are commutative:

$$\begin{array}{ccccc} G(A) \otimes \mathbb{1}_{\mathcal{D}} & \xrightarrow{\text{id}_{G(A)} \otimes e} & G(A) \otimes G(\text{id}_{\mathcal{C}}) & \xrightarrow{\gamma_{A, \mathbb{1}_{\mathcal{C}}}} & G(A \otimes \mathbb{1}_{\mathcal{C}}) \\ & \searrow \alpha_{G(A)} & & \swarrow G(\alpha_A) & \\ & & G(A) & & \\ \\ \mathbb{1}_{\mathcal{D}} \otimes G(B) & \xrightarrow{e \otimes \text{id}_{G(B)}} & G(\text{id}_{\mathcal{C}}) \otimes G(B) & \xrightarrow{\gamma_{\mathbb{1}_{\mathcal{C}}, B}} & G(B \otimes \mathbb{1}_{\mathcal{C}}) \\ & \searrow \alpha_{G(B)} & & \swarrow G(\beta_B) & \\ & & G(B) & & \end{array}$$

Moreover, it is called a **monoidal functor** if  $\gamma_{A,B}$  and  $e$  are all isomorphisms.

A **monoidal natural transformation** between right-lax monoidal functors is a natural transformation that commutes with the maps  $\gamma_{A,B}$  and  $e$ .  $\lrcorner$

**Def.(4.1.5.13) [Equivalence of Monoidal Categories].** An equivalence of monoidal categories is a map that is an equivalence of categories as well as a monoidal functor.  $\lrcorner$

**Prop.(4.1.5.14) [Mac Lane Strictness].** Any monoidal category is equivalent to a strict monoidal category.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.1.5.15) [Examples].** For a monoidal category  $(\mathcal{C}, \otimes)$ , the functor  $X \mapsto \text{Hom}(\mathbb{1}, X)$  is a right-lax monoidal functor from  $\mathcal{C} \rightarrow \text{Set}$  (4.1.5.6).

The morphism  $\pi_0 : A \mapsto \pi_0(A)$  is a monoidal functor from the category of topological spaces  $\mathcal{Top}$  to the category  $\text{Set}$  because it commutes with products.  $\lrcorner$

*Proof:*  $\square$

### Symmetric Monoidal Categories

**Def. (4.1.5.16) [Symmetric Monoidal Category].** A **symmetric monoidal category** is a monoidal category  $(\mathcal{C}, \otimes)$  together with a natural transformation  $\psi$  between  $\otimes$  and  $\otimes \circ \iota : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  that  $\psi^2 = \text{id}$  and the following commutative hexagon diagram is commutative:

A **symmetric monoidal functor** between symmetric monoidal categories are required to commute with the braiding  $\psi$ .  $\lrcorner$

### Rigid Monoidal Categories

**Def. (4.1.5.17) [Duals].** Let  $\mathcal{C}$  be a monoidal category and  $V \in \mathcal{C}$ , a **left dual** of  $V$  is an element  $V^*$  together with maps

$$ev_V : V^* \otimes V \rightarrow \mathbb{1}, \quad coev_V : \mathbb{1} \mapsto V \otimes V^*$$

that satisfies

$$V \rightarrow V \otimes V^* \otimes V \rightarrow V, \quad V^* \rightarrow V^* \otimes V \otimes V^* \rightarrow V^*$$

are identities.

Similarly we can define right dual. if  $Y$  is a left/right dual to  $X$ , then  $X$  is right/left dual to  $Y$ .  $\lrcorner$

**Def. (4.1.5.18) [Dual Morphisms].** Let  $f; X \rightarrow Y$  be a morphism and  $X^*, Y^*$  the left duals of  $X$  and  $Y$ , then there is a natural **left dual map**:  $f^* : Y^* \rightarrow X^*$  given by

$$Y^* \rightarrow Y^* \otimes X \otimes X^* \rightarrow Y^* \otimes Y \otimes X^* \rightarrow X^*.$$

$\lrcorner$

**Prop. (4.1.5.19) [Monoidal Functor Preserves Duals].** Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a monoidal functor (4.1.5.12) between monoidal categories,  $X \in \mathcal{C}$  is an object with left dual  $X^*$ . Then  $F(X^*)$  is a left dual of  $F(X)$  with evaluation and coevaluation maps

$$ev_{F(X)} : F(X^*) \otimes F(X) \rightarrow F(X^* \otimes X) \rightarrow F(\mathbb{1}_{\mathcal{C}}) \rightarrow \mathbb{1}_{\mathcal{D}},$$

$$coev_{F(X)} : \mathbb{1}_{\mathcal{D}} \rightarrow F(\mathbb{1}_{\mathcal{C}}) \rightarrow F(X \otimes X^*) \rightarrow F(X) \otimes F(X^*).$$

and similarly for right duals.  $\lrcorner$

**Prop. (4.1.5.20) [Adjointness].** Let  $\mathcal{C}$  be a monoidal category and  $V \in \mathcal{C}$  with left dual  $V^*$ , then there are natural adjunction maps

$$\text{Hom}(U \otimes V, W) \cong \text{Hom}(U, W \otimes V^*), \quad \text{Hom}(V^* \otimes U, W) \cong \text{Hom}(U, V \otimes W).$$

$\lrcorner$

*Proof:* The first adjunction map is given by  $f \mapsto (f \otimes \text{id}_{V^*}) \circ (\text{id}_U \otimes coev_V)$ , and the inverse given by  $g \mapsto (W \otimes ev_V) \circ (g \otimes \text{id}_V)$ . The verification and the second one:  $\square$

**Cor. (4.1.5.21).** In particular, we can use Yoneda lemma to show the left/right adjoints are unique if they exist.  $\lrcorner$

**Invertible Objects and Grothendieck Categories**

**Def. (4.1.5.22) [Invertible Objects].** An **invertible object** in a monoidal category is a rigid object  $L$  that the evaluation maps and coevaluation maps are all isomorphisms. ? ⌋



## 4.2 Categories II

References are [Bor94, Mac98], [Coend Calculus, Fosco Loregian].

**Notation (4.2.0.1).**

- Let  $\mathcal{C} \in \mathbf{Cat}$ .

┘

### 1 Tensor Categories

**Notation (4.2.1.1).**

- Let  $k \in \mathbf{Field}$ .

┘

**Def. (4.2.1.2) [Tensor Categories].** A **tensor category** over a field  $k$  is an Artinian Abelian category over  $k$  together with a strict monoidal structure that is bilinear and satisfies  $\mathrm{End}(\mathbf{1}) = k$ .

A **tensor functor** is a functor between tensor categories that is additive and monoidal.

┘

**Prop. (4.2.1.3) [End(1)].** If  $\mathcal{C}$  is a rigid tensor category, then  $\mathrm{End}(\mathbf{1})$  is a ring that acts on objects  $X \in \mathcal{C}$  via  $X \cong \mathbf{1} \otimes X$ . The action of  $\mathrm{End}(\mathbf{1})$  commutes with  $\mathrm{End}(X)$ , in particular,  $\mathrm{End}(\mathbf{1})$  is commutative and  $\mathcal{C}$  is  $\mathrm{End}(\mathbf{1})$ -linear.

┘

### Rigid Tensor Categories

**Def. (4.2.1.4) [Rigid Tensor Categories].** Let  $\mathcal{C}$  be a monoidal category, an element  $X \in \mathcal{C}$  is called **rigid object** if it has left and right duals. A **rigid tensor category** is a tensor category (4.2.1.2) s.t. every object is rigid. In particular, by (4.1.5.20), a rigid tensor category is closed.

┘

**Prop. (4.2.1.5).** If  $\mathcal{C}$  is a rigid tensor category, then the functor

Equivalence of  $\mathcal{C}^{\mathrm{op}}$  and  $\mathcal{C}^{\mathrm{opp}}$ . ?

┘

**Cor. (4.2.1.6).** Any natural transformation of monoidal functors between rigid tensor categories is an isomorphism.

┘

*Proof:* Cf. [Milne, Tannakian Categories, P13]. ?

□

**Prop. (4.2.1.7) [Trace and Rank].** Let  $\mathcal{C}$  be a symmetric rigid tensor category, we can define a trace morphism

$$\mathrm{tr}_X : \mathrm{End}(X) \rightarrow \mathrm{End}(\mathbf{1}) : f \mapsto \mathbf{1} \rightarrow X \otimes X^* \xrightarrow{f \otimes \mathrm{id}} X \otimes X^* \rightarrow X^* \otimes X \rightarrow \mathbf{1}$$

And the **dimension** of  $X$  is defined to be  $\mathrm{tr}_X(\mathrm{id}_X) \in \mathrm{End}(\mathbf{1})$ .

┘

**Prop. (4.2.1.8) [Abelian Rigid Tensor Categories Exact].** If  $\mathcal{C}$  is an Abelian rigid tensor category, then  $\otimes$  commutes with inverse limits and direct limits in each variable.

┘

*Proof:* It commutes with direct limits because it is left adjoint to the Hom functor, and it commutes with inverse limits by considering the opposite category (4.2.1.5).

□

**Prop. (4.2.1.9).**  $\dim(X \otimes Y) = \dim X \cdot \dim Y$ . If there is an exact sequence  $0 \rightarrow X \rightarrow Z \rightarrow Y \rightarrow 0$ , then  $\dim(Z) = \dim(X) + \dim(Y)$ .

┘

**Prop. (4.2.1.10)[Decompositions].** Let  $(\mathcal{C}, \otimes)$  be a rigid Abelian tensor category and if  $U$  is a subobject of  $1$ , then  $1 = U \oplus U^\perp$  where  $U^\perp = \ker(1 \rightarrow U^\vee)$ . Consequently,  $1$  is a simple object iff  $\text{End}(1)$  is a field. And any rigid tensor category can be decomposed as rigid Abelian tensor categories  $\mathcal{C}_I$  with  $\text{End}(1_i)$  being fields.  $\lrcorner$

*Proof:* Let  $V = \text{Coker}(U \rightarrow 1)$ , by tensoring  $0 \rightarrow U \rightarrow 1 \rightarrow V \rightarrow 0$  with  $U \hookrightarrow 1$ , we get exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U & \longrightarrow & 1 & \longrightarrow & V & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & \nearrow 0 & \uparrow & & \\ 0 & \longrightarrow & U \otimes U & \longrightarrow & U & \longrightarrow & V \otimes U & \longrightarrow & 0 \end{array}$$

thus  $V \otimes U = 0$  and  $U \otimes U = U$  via  $1 \otimes 1 \cong 1$ .

For the rest, Cf.[Tannakian Categories, Milne, P14].  $\square$

**Def. (4.2.1.11)[Associative Algebra in a Symmetric Tensor Category].**  $\lrcorner$

### Tannakian Categories

**Def. (4.2.1.12)[Fiber Functor].** Let  $\mathcal{C}$  be a  $k$ -linear tensor category, then a **fiber functor** on  $\mathcal{C}$  with values in a  $k$ -algebra  $R$  is a  $k$ -linear exact faithful tensor functor  $\eta : \mathcal{C} \rightarrow \text{Mod}_R$  that takes values in the subcategory  $\mathbf{Proj}_R$ .  $\lrcorner$

**Def. (4.2.1.13)[Tannakian Category].** A **Tannakian category** is a symmetric rigid tensor category (4.2.1.4)  $\mathcal{C}$  that  $\text{End}(1) = k$  together with a fiber functor (4.2.1.12) with values in some  $R \in \mathcal{C}\text{Ring}_k$ .  $\lrcorner$

**Def. (4.2.1.14)[Neutral Tannakian Categories].** A **neutral Tannakian category** is a Tannakian category that the fiber functor is valued in  $k$ . By (10.3.2.6), such a category is equivalent to  $\text{Rep}_k(G)$  for some  $G \in \text{GrpSch}^{\text{aff}}/k$ , and such  $G$  is uniquely defined up to inner automorphisms.  $\lrcorner$

**Remark (4.2.1.15).** In particular, a Tannakian category can be thought of as an abstract version of the category of representations of an affine group scheme that has no distinguished “forgetful” functor, just as a vector space is an abstract version of  $k^n$  that has no distinguished basis.  $\lrcorner$

## 2 Enriched Categories

**Def. (4.2.2.1)[Enriched Categories].** Given a monoidal category  $(\mathcal{C}, \otimes)$ , an **enriched category**  $\mathcal{D}$  over  $\mathcal{C}$  consists of the following data:

- a collection of objects.
- For objects  $X, Y \in \mathcal{D}$ , a mapping object  $\text{Map}_{\mathcal{D}}(X, Y) \in \mathcal{C}$ .
- For objects  $X, Y, Z \in \mathcal{D}$ , a composite map  $\text{Map}_{\mathcal{D}}(X, Y) \otimes_{\mathcal{C}} \text{Map}_{\mathcal{D}}(Y, Z) \rightarrow \text{Map}_{\mathcal{D}}(X, Z)$  that is associative.
- For every  $X \in \mathcal{D}$ , a morphism  $1 \mapsto \text{Map}_{\mathcal{D}}(X, X)$  that satisfies the commutative diagrams of the identity morphism.

$\lrcorner$

**Def. (4.2.2.2)[Category of Enriched Categories].** We can naturally define morphisms and natural transformations of categories enriched over a monoidal category  $\mathcal{C}$ . The resulting category is denoted by  $\text{Cat}_{\mathcal{C}}$ .  $\lrcorner$

**Prop. (4.2.2.3)[Completeness and Cocompleteness].** Let  $\mathcal{C}$  be a complete and cocomplete symmetric monoidal closed category, then  $\text{Cat}_{\mathcal{C}}$  is also complete and cocomplete.  $\lrcorner$

*Proof:* Cf. [H. Wolff. V-cat and V-graph]. ?  $\square$

**Prop. (4.2.2.4) [Transfer of Enriched Structure].** Given a right-lax monoidal functor between monoidal categories  $F : \mathcal{C} \rightarrow \mathcal{C}'$  and a category  $\mathcal{D}$  enriched over  $\mathcal{C}$ , we may obtain a category  $F(\mathcal{D})$  enriched over  $\mathcal{C}'$  by asserting  $\text{Map}_{F(\mathcal{D})}(X, Y) = F(\text{Map}(X, Y))$ . It is an enriched category just by the definition of right-lax monoidal functors.  $\lrcorner$

**Prop. (4.2.2.5)[Underlying Category].** For a category enriched  $\mathcal{D}$  over  $\mathcal{C}$ , by (4.1.5.15), we can transfer the structure via  $\mathcal{C} \rightarrow \text{Set} : X \mapsto \text{Hom}(1, X)$ , and the resulting category is called the underlying category of  $\mathcal{D}$ .  $\lrcorner$

**Prop. (4.2.2.6).**

- A category enriched in  $\text{Set}$  is just a usual category.
- A right-closed monoidal category is enriched over itself if we define  $\text{Map}(X, Y) = Y^X$ . (Check).  $\lrcorner$

**Def. (4.2.2.7) [Tensor Category].** Let  $\mathcal{C}$  be a right-closed monoidal category and  $\mathcal{D}$  a category enriched over  $\mathcal{C}$ , then  $\mathcal{D}$  is called **tensor over  $\mathcal{C}$**  if for any  $C \in \mathcal{C}$  and  $X \in \mathcal{D}$ , there is an isomorphism of functors

$$\eta : \text{Map}_{\mathcal{D}}(X, -)^C \cong \text{Map}_{\mathcal{D}}(X \otimes C, -).$$

for some element  $X \otimes C \in \mathcal{D}$ .

In particular, this implies  $\text{Hom}_{\mathcal{C}}(C, \text{Map}(X, Y)) \cong \text{Hom}_{\mathcal{D}}(X \otimes C, Y)$ , thus  $X \otimes C$  is determined up to a unique isomorphism, and the map  $(X, C) \mapsto X \otimes C$  defines a functor  $\mathcal{D} \times \mathcal{C} \rightarrow \mathcal{D}$  that there are natural morphisms

$$X \otimes (C \otimes D) \cong (X \otimes C) \otimes D.$$

Dually, if  $\mathcal{C}$  is left-closed and there is an object  ${}^C X$  that there is an isomorphism of functors

$${}^C \text{Map}(-, X) \cong \text{Map}(-, {}^C X)$$

then  $\mathcal{D}$  is called **cotensor over  $\mathcal{C}$** .  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.2.2.8).** If  $\mathcal{C}$  is a right-closed monoidal category, then it is naturally tensor over itself, as defined in (4.2.2.6).  $\lrcorner$

### Lifting Property and Small Object Argument

**Def. (4.2.2.9)[lifting Properties].** Let  $\mathcal{C}$  be a category and  $p : A \rightarrow B, q : X \rightarrow Y$  be morphisms, then  $p$  is said to have **left lifting property** w.r.t  $q$  and  $q$  is said to have **right lifting property** w.r.t  $p$

if given any diagram 
$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow p & \nearrow & \downarrow q \\ B & \longrightarrow & Y \end{array}$$
, there is a dotted arrow completing the diagram.

For a set  $A$  of morphisms of  $\mathcal{C}$ , let  $l(A)$  denote the morphisms that have left lifting property w.r.t  $A$  and  $r(A)$  the morphisms that have right lifting property w.r.t  $A$ .  $\lrcorner$

**Def. (4.2.2.10) [Weakly Saturated Class].** Let  $\mathcal{C}$  be a category with all small colimits, then a class of morphisms of  $\mathcal{C}$  is called **weakly saturated** if it satisfies:

- Closed under pushout.
- Closed under transfinite composition: Let  $\alpha$  be an ordinal and  $\{D_\beta\}_{\beta < \alpha}$  be a system of objects in  $\mathcal{C}_{C/}$  indexed by  $\alpha$ . For  $\beta < \alpha$ , let  $D_{<\beta}$  be the colimit of system  $\{D_\gamma\}_{\gamma < \beta}$  in  $\mathcal{C}_{C/}$ , then if each  $D_{<\beta} \rightarrow D_\beta$  is in  $S$ , then  $C \rightarrow D_{<\alpha}$  is in  $S$ .
- Closed under Retraction: In the category of morphisms of  $\mathcal{C}$ , if there is a morphism  $F : f \rightarrow g, G : g \rightarrow f$  that  $G \circ F = \text{id}$ , and  $g \in S$ , then  $f \in S$ .

┘

**Cor. (4.2.2.11).** The second condition implies all isomorphisms are in  $S$ , and  $S$  is closed under composition.

┘

**Prop. (4.2.2.12) [Lifting and Retraction].** For a diagram  $\begin{array}{ccc} X & \xlongequal{\quad} & X \\ \downarrow i & \nearrow s & \downarrow p \\ Z & \xrightarrow{f} & Y \end{array}$  represents  $p$  as a retraction of

$u$ , as the diagram  $\begin{array}{ccccc} X & \xrightarrow{i} & Z & \xrightarrow{s} & X \\ \downarrow p & & \downarrow f & & \downarrow p \\ Y & \xlongequal{\quad} & Y & \xlongequal{\quad} & Y \end{array}$  shows.

Dually a diagram  $\begin{array}{ccc} X & \xrightarrow{i} & Y \\ \downarrow q & \nearrow s & \downarrow f \\ Z & \xlongequal{\quad} & Z \end{array}$  represents  $q$  as a retraction of  $i$ .

┘

**Prop. (4.2.2.13) [Small Object Argument].** Let  $\mathcal{C}$  be a presentable category and  $A_0 = \{\varphi_i : C_i \rightarrow D_i\}$  be a small collection of morphisms in  $\mathcal{C}$ , then there is a morphism  $T : \mathcal{C}^{[1]} \rightarrow \mathcal{C}^{[2]}$  taking morphisms  $f : X \rightarrow Z$  in  $\mathcal{C}$  to the diagram

$$\begin{array}{ccc} & Y & \\ f' \nearrow & & \searrow f'' \\ X & \xrightarrow{f} & Z \end{array}$$

That  $f'$  belongs to the weakly saturated class generated by  $A_0$  and  $f'' \in r(A_0)$ .

Moreover, if  $\kappa$  is a regular cardinal that each  $C_i, D_i$  is  $\kappa$ -compact, then  $T$  commutes with  $\kappa$ -filtered colimits.

┘

*Proof:* Cf.[HTT, P788].

□

**Lemma (4.2.2.14).**  $l(A)$  is weakly saturated for any set of morphisms  $A$  (Clear).

┘

**Cor. (4.2.2.15) [Generated Weakly Saturated Class].** For any presentable category  $\mathcal{C}$  and  $A$  a set of morphisms in  $\mathcal{C}$ ,  $l(r(A))$  is the smallest weakly saturated class of morphisms containing  $A$ .

┘

*Proof:* One direction of inclusion is by (4.2.2.14), for the other, if  $f : X \rightarrow Z \in l(r(A))$ , then there is a factorization  $X \xrightarrow{f'} Y \xrightarrow{f''} Z$ , where  $f' \in \overline{A}$  and  $f'' \in r(A)$ , thus  $f \in l(f'')$ , thus  $f$  is retraction of  $f'$ :

$$\begin{array}{ccc} X & \xrightarrow{f'} & Y \\ \downarrow f & \nearrow g & \downarrow f'' \\ Z & \xrightarrow{\quad} & Z \end{array} \Rightarrow \begin{array}{ccccc} X & \longrightarrow & X & \longrightarrow & X \\ \downarrow f & & \downarrow f' & & \downarrow f \\ Z & \xrightarrow{g} & Y & \xrightarrow{f''} & Z \end{array}$$

Thus  $f \in \overline{A}$ .

□

### Trees

Cf.[HTT, Appendix]

## 3 Fibered Categories

### Categories of Categories

**Def. (4.2.3.1) [2-Category of Categories over Categories].** There is a 2-category of categories over  $\mathcal{C}$ , denoted by  $\text{Cat}(\mathcal{C})$ , where the 1-morphisms are morphisms of categories over  $\mathcal{C}$  and the 2-morphisms are base-preserving natural transformations.

Two categories over  $\mathcal{C}$  are called **equivalent** if they are equivalent in this 2-category.  $\lrcorner$

**Prop. (4.2.3.2) [2-Fibered Products in the Categories of Categories].** 2-fibered products exists in  $\text{Cat}(\mathcal{C})$ .  $\lrcorner$

*Proof:* For  $F : \mathcal{A} \rightarrow \mathcal{C}, G : \mathcal{B} \rightarrow \mathcal{C} \in \text{Cat}(\mathcal{C})$ , we can define  $\mathcal{A} \times_{\mathcal{C}} \mathcal{B} \in \text{Cat}(\mathcal{C})$  as follows:

- Objects are the triples  $(A, B, f)$  where  $A \in \mathcal{A}, B \in \mathcal{B}$  and  $f : F(A) \rightarrow G(B)$  is an isomorphism in  $\mathcal{C}$ .
- Morphisms from  $(A, B, f)$  to  $(A', B', f')$  are pairs  $(a, b)$  where  $a : A \rightarrow A', b : B \rightarrow B'$  s.t. the diagram

$$\begin{array}{ccc} F(A) & \xrightarrow{f} & G(B) \\ \downarrow F(a) & & \downarrow G(b) \\ F(A') & \xrightarrow{f'} & G(B') \end{array}$$

is commutative.

$\mathcal{A} \times_{\mathcal{C}} \mathcal{B}$  is a category both over  $\mathcal{A}$  and over  $\mathcal{B}$ , and it fits into a 2-fiber products diagram

$$\begin{array}{ccc} \mathcal{A} \times_{\mathcal{C}} \mathcal{B} & \longrightarrow & \mathcal{B} \\ \downarrow & & \downarrow \\ \mathcal{A} & \longrightarrow & \mathcal{C} \end{array}$$

where the invertible 2-morphism is given by  $\psi_{(\mathcal{A}, \mathcal{B}, f)} = f : F(A) \rightarrow G(B)$ .

The verification that this defines a final object in the 2-category of 2-commutative diagrams is in [Sta]02X9.  $\square$

**Cor. (4.2.3.3).** There is an equivalence of fibre categories:

$$(\mathcal{X} \times_{\mathcal{S}} \mathcal{Y})_U \cong \mathcal{X}_U \times_{\mathcal{S}_U} \mathcal{Y}_U.$$

In particular,  $\mathcal{A} \times_{\mathcal{C}} \mathcal{B}$  is a groupoid iff  $\mathcal{A}, \mathcal{B}$  are all groupoids.  $\lrcorner$

**Prop. (4.2.3.4).** Let  $\mathcal{A} \rightarrow \mathcal{C}, \mathcal{B} \rightarrow \mathcal{C}, \mathcal{C} \rightarrow \mathcal{D}$  be functors between categories, then there is a 2-fiber product diagram

$$\begin{array}{ccc} \mathcal{A} \times_{\mathcal{C}} \mathcal{B} & \longrightarrow & \mathcal{A} \times_{\mathcal{D}} \mathcal{B} \\ \downarrow & & \downarrow \\ \mathcal{C} & \xrightarrow{\Delta_{\mathcal{C}/\mathcal{D}}} & \mathcal{C} \times_{\mathcal{D}} \mathcal{C} \end{array}$$

$\lrcorner$

*Proof:*

$\square$

### Fibered Categories

**Def. (4.2.3.5) [(Co)Cartesian-Arrows].** Let  $\text{pr} : \mathcal{F} \rightarrow \mathcal{C}$  be a morphism, then a **Cartesian-arrow** is an arrow  $\varphi : C' \rightarrow C$  in  $\mathcal{F}$  s.t. for any  $C'' \in \text{Ob}(\mathcal{F})$ , the map

$$\text{Hom}(C'', C') \rightarrow \text{Hom}(C'', C) \times_{\text{Hom}(\text{pr}(C''), \text{pr}(C))} \text{Hom}(\text{pr}(C''), \text{pr}(C'))$$

is a bijection. A **coCartesian arrow** is an arrow that corresponds to a Cartesian diagram in  $\mathcal{F}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ .  $\lrcorner$

**Prop. (4.2.3.6).**

- If  $f$  is Cartesian, then  $f \circ g$  is Cartesian iff  $g$  is Cartesian.
- An arrow in  $\mathcal{F}$  whose image is an isomorphism is Cartesian iff it is itself an isomorphism.

$\lrcorner$

*Proof:* Easy.  $\square$

**Def. (4.2.3.7) [Quasi-Fibration].** A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is called a **quasi-fibration** if for any  $X \in \mathcal{C}$  and an isomorphism  $f : F(X) \cong Y$ , there is an isomorphism  $\bar{f} : X \rightarrow \bar{Y}$  mapping to  $f$ .  $\lrcorner$

**Def. (4.2.3.8) [2-Category of Fibered Categories].** A **fibered category** over  $\mathcal{C}$  is a category over  $\mathcal{C}$   $\text{pr} : \mathcal{F} \rightarrow \mathcal{C}$  s.t. for any  $\eta \in \mathcal{F}$  and an arrow  $f : U \rightarrow \text{pr}(\eta)$ , there is a Cartesian arrow (4.2.3.5)  $\xi \rightarrow \eta$  in  $\mathcal{F}$  lifting  $f$ . A **cofibered category** over  $\mathcal{C}$  is a category  $p : \mathcal{F} \rightarrow \mathcal{C}$  that the dual category  $\text{pr}^{\text{op}} : \mathcal{F}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$  is a fibered category.

A morphism of fibered categories over  $\mathcal{C}$  is a morphism of categories over  $\mathcal{C}$  that maps Cartesian morphisms to Cartesian morphisms. A 2-morphism of fibered categories is the same as a 2-morphism of categories over categories.  $\lrcorner$

**Def. (4.2.3.9) [Split Fibered Categories].** A **split fibered category** is a fibered category  $\mathcal{F} \rightarrow \mathcal{C}$  that comes from a functor  $\mathcal{C}^{\text{op}} \rightarrow \text{Cat}$ .  $\lrcorner$

**Def. (4.2.3.10) [Cleavage].** A **cleavage** of a fibered category  $\pi : \mathcal{F} \rightarrow \mathcal{C}$  is a choice of Cartesian arrow  $f^*$  lifting  $f$  for any  $f \in \text{Arr}(\mathcal{C})$ .  $\lrcorner$

**Lemma (4.2.3.11).** Let  $F : \mathcal{F} \rightarrow \mathcal{G} \in \text{FibCat}(\mathcal{C})$  s.t.  $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$  are fully faithful for any  $U \in \mathcal{C}$ , then  $F$  is fully faithful.  $\lrcorner$

*Proof:* To show  $F$  is fully faithful, it suffices to show for objects  $X, Y$  lying over  $U, V$ ,  $F$  induces a bijection of morphisms from  $x$  to  $y$  lying over a fixed  $f : U \rightarrow V$ . Choose a Cartesian morphism  $f^*y \rightarrow y$  in  $\mathcal{S}_1$  lying over  $f$ , then this induces a bijection between morphisms from  $x$  to  $y$  lying over  $f$  and  $\text{Hom}_{\mathcal{S}_{1,U}}(x, f^*y)$ . Similarly, because  $F$  preserves Cartesian morphisms, we get a bijection between morphisms  $F(x) \rightarrow F(y)$  lying over  $f$  and  $\text{Hom}_{\mathcal{S}_{2,U}}(F(x), F(f^*y))$ . Then the desired bijection follows from the hypothesis.  $\square$

**Prop. (4.2.3.12) [Equivalence of Fibered Categories].** For  $F : \mathcal{F} \rightarrow \mathcal{G} \in \text{FibCat}(\mathcal{C})$ ,  $F$  is an equivalence iff the restriction  $F_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  is an equivalence of categories for any object  $U$  of  $\mathcal{C}$ .  $\lrcorner$

*Proof:* One direction is trivial: for the other, the proof is similar to the fact essentially surjective+fully faithful implies equivalence.

Because  $F(U)$  are equivalences, for any object  $\xi$  of  $\mathcal{G}$  over  $U$ , pick an object  $G\xi$  in  $\mathcal{F}(U)$  together with an isomorphism  $\alpha_\xi : \xi \cong F(G\xi)$ . And for any morphism  $\xi \rightarrow \eta$ , by (4.2.3.11), there is a unique arrow  $G\varphi : G\xi \rightarrow G\eta$  that  $F(G\varphi) = \alpha_\eta \circ \varphi \circ \alpha_\xi^{-1}$ .

Thus clearly there is a 2-isomorphism  $\text{id}_{\mathcal{G}} \cong F \circ G$ . It remains to construct an 2-isomorphism  $\text{id}_{\mathcal{F}} \cong G \circ F$ : For any object  $\xi'$  over  $U$ , since  $F(U)$  is fully faithful, there is a unique isomorphism  $\beta_{\xi'} : \xi' \cong G \circ F(\xi)$  in  $\mathcal{F}(U)$  that  $F\beta_{\xi'} = \alpha_{F\xi'}$ . Then this is easily checked to be an 2-isomorphism  $\beta : \text{id}_{\mathcal{F}} \cong F \circ G$ .  $\square$

**Prop. (4.2.3.13)[2-Fiber Products of Fibered Categories].** 2-fiber products exists in the category of fibered categories, and it coincides with that defined in (4.2.3.2).  $\lrcorner$

*Proof:* it suffices to show for fibered categories  $\mathcal{X}, \mathcal{Y}$  over  $\mathcal{C}$ ,  $\mathcal{X} \times_{\mathcal{C}} \mathcal{Y}$  is also fibered over  $\mathcal{C}$ . Let  $(x, y, \varphi)$  be an object of  $\mathcal{X} \times_{\mathcal{C}} \mathcal{Y}$  mapping to  $U \in \mathcal{C}$  and  $f : V \rightarrow U$  is a morphism in  $\mathcal{C}$ , choose Cartesian morphisms  $a : f^*x \rightarrow x$ ,  $b : f^*y \rightarrow y$  lying over  $f$ , then  $F(a)$  and  $G(b)$  are Cartesian. Since  $\varphi : F(x) \rightarrow G(y)$  is an isomorphism, by the property of Cartesian morphisms, there exists a unique isomorphism  $f^*\varphi : F(f^*x) \rightarrow G(f^*y) \in \mathcal{S}_V$  that  $G(b) \circ f^*\varphi = \varphi \circ F(a)$ . In other words,  $(F(a), F(b)) : (V, f^*x, f^*y, f^*\varphi) \rightarrow (U, x, y, \varphi)$  is a morphism in  $\mathcal{X} \times_{\mathcal{S}} \mathcal{Y}$ .

The verification that this morphism is Cartesian is omitted.  $\square$

**Lemma (4.2.3.14).** Let  $\mathcal{S} \rightarrow \mathcal{C}$  be a fibered category that factors through  $\mathcal{C}/U$  where  $U \in \mathcal{C}$ , then  $\mathcal{S} \rightarrow \mathcal{C}/U$  is also a fibered category.  $\lrcorner$

*Proof:* Cf. [Sta]02XR.  $\square$

**Def. (4.2.3.15)[ $G$ -Equivariant Object].** Let  $G : \mathcal{C}^{\text{op}} \rightarrow (\text{Grp})$  be a group functor and  $p_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{C}$  a fibered category, and  $X$  an object of  $\mathcal{C}$  with an action of  $G$ . A  **$G$ -equivariant object** of  $\mathcal{F}(X)$  is an object  $\rho$  of  $\mathcal{F}(X)$  that there is an action of  $G \circ p_{\mathcal{F}}$  on  $\rho$  and for any object  $U$  and  $\xi \in \mathcal{F}(U)$ , the function  $p_{\mathcal{F}} : \text{Hom}_{\mathcal{F}}(\xi, \rho) \rightarrow \text{Hom}_{\mathcal{C}}(U, X)$  is  $G(U)$ -equivariant.

The category  $\mathcal{F}^G(X)$  of  $G$ -equivariant objects of  $\mathcal{F}(X)$  consisting of  $G \circ p_{\mathcal{F}}$ -equivariant morphism of  $G$ -equivariant objects.  $\lrcorner$

**Prop. (4.2.3.16).** Let  $\pi_2$  is the projection  $G \otimes X \rightarrow X$ ,  $\rho$  be an object of  $\mathcal{F}(X)$ , then a  $G$ -equivariant structure on  $\rho$  is the same as a Cartesian arrow  $\beta : \pi_2^*\rho \rightarrow \rho$  that  $p_{\mathcal{F}}\beta = \alpha$ , and satisfies the desired commutative diagram corresponding to  $(gh)x = g(hx)$ . And a morphism of  $G$ -equivariant objects just corresponds to a morphism of pairs  $(\rho, \beta)$ .  $\lrcorner$

*Proof:* Cf. [Vistoli, P68].  $\square$

**Def. (4.2.3.17)[Presheaf of Arrows].** Let  $\mathcal{F}$  be a fibered category over  $\mathcal{C}$ , and  $\xi, \eta \in \mathcal{F}(S)$ , then we can define a quasi-functor  $\text{Hom}_{\mathcal{S}}(\xi, \eta) \rightarrow (\mathcal{C}/S)$ , where

$$\text{Hom}_{\mathcal{S}}(\xi, \eta)(U/S) = \{(\xi_1 \rightarrow \xi, \eta_1 \rightarrow \eta, \varphi)\}$$

where  $\xi_1 \rightarrow \xi, \eta_1 \rightarrow \eta$  are Cartesian arrows over  $U \rightarrow S$ , and  $\varphi : \xi_1 \rightarrow \eta_1 \in \mathcal{F}(U)$ . The arrows in  $\text{Hom}_{\mathcal{S}}(\xi, \eta)$  are uniquely defined by the property of Cartesian arrows. Then it is a quasi-functor, by (4.2.3.28).

Then it is equivalence to a presheaf  $\underline{\text{Hom}}_{\mathcal{S}}(\xi, \eta)$ , by (4.2.3.29). Equivalently, this presheaf can be defined by designating a choice of Cartesian arrows.  $\lrcorner$

**Prop. (4.2.3.18)[Splitting a Fibered Category].** Let  $\mathcal{F} \rightarrow \mathcal{C}$  be a fibered category, then there exists a canonically defined split fibered category  $\tilde{\mathcal{F}} \rightarrow \mathcal{C}$  with a canonical equivalence of fibered categories  $\tilde{\mathcal{F}} \rightarrow \mathcal{F}$  over  $\mathcal{C}$ .  $\lrcorner$

*Proof:* There is a functor  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{Cat} : U \mapsto \text{Hom}(h_U, \mathcal{F})$ , with corresponds to a split fibered category  $\tilde{\mathcal{F}}$ . There is an obvious morphism  $\tilde{\mathcal{F}} \rightarrow \mathcal{F}$ , sending an object  $\varphi : h_U \rightarrow \mathcal{F}$  to  $\varphi(\text{id}_U) \in \mathcal{F}(U)$ . And for any  $f : U \rightarrow V \in \mathcal{C}$  and a  $\varphi : h_U \rightarrow \mathcal{F}$ , we send  $f^*\varphi \rightarrow \varphi \in \tilde{\mathcal{F}}$  to  $\varphi(f : V/U \rightarrow U/U) \in \mathcal{F}$ . Then we get a canonical map of fibered categories  $\tilde{\mathcal{F}} \rightarrow \mathcal{F}$  over  $\mathcal{C}$ . It is an equivalence of categories by (4.2.3.12) and (4.2.3.31).  $\square$

### Categories Fibered in Groupoids, Sets and Equivalence Relations

**Def. (4.2.3.19)[Category Fibered in Groupoids].** A category (co)fibered in groupoids/sets/equivalence relations over  $\mathcal{C}$  is a category  $\mathcal{F}$  (co)fibered over  $\mathcal{C}$  (4.2.3.8) s.t.  $\mathcal{F}(U)$  is a groupoid/setoids/equivalence relations (4.1.1.14) for any  $U \in \mathcal{C}$ . Also we call a category fibered in equivalence relations over  $\mathcal{C}$  a **quasi-functor**.  $\lrcorner$

**Prop. (4.2.3.20)[Characterization of Category Fibered in Groupoids].** Let  $\mathcal{F}$  be a category over  $\mathcal{C}$ , then  $\mathcal{F}$  is fibered over groupoids over  $\mathcal{C}$  iff

- Every morphism in  $\mathcal{F}$  is Cartesian.
- Given any  $\eta \in \mathcal{F}, U \in \mathcal{C}$  and a morphism  $f : U \rightarrow p_{\mathcal{F}}(\eta)$ , there is an arrow  $\varphi : \xi \rightarrow \eta \in \mathcal{F}$  mapping to  $f$ .

And dually for cofibered categories.  $\lrcorner$

*Proof:* If these two holds, then  $\mathcal{F}$  is clearly fibered over  $\mathcal{C}$ , and for any arrow  $f : \xi \rightarrow \eta$  in  $\mathcal{F}(U)$ , there is a morphism  $g : \eta \rightarrow \xi$  that is right inverse to  $f$ , then clearly it is the inverse.

Conversely, if  $\mathcal{F}$  is fibered over  $\mathcal{C}$ , it suffices to check1: for any arrow  $f$ , the image in  $\mathcal{C}$  can be lifted to a Cartesian diagram in  $\mathcal{F}$ , and differs  $f$  by an isomorphism, thus  $f$  is also Cartesian by (4.2.3.6).  $\square$

**Cor. (4.2.3.21).** if  $\mathcal{A}$  is a category fibered in groupoids over  $\mathcal{B}$  and  $\mathcal{B}$  is a category fibered in groupoids over  $\mathcal{C}$ , then  $\mathcal{A}$  is a category fibered in groupoids over  $\mathcal{C}$ .  $\lrcorner$

**Prop. (4.2.3.22)[Associated Category Fibered in Groupoids].** Let  $\mathcal{F} \rightarrow \mathcal{C}$  be a fibered category, then the **associated category fibered in groupoids**  $\mathcal{F}_{\text{cart}}$  is the category obtained by deleting all the non-Cartesian arrows. Then  $\mathcal{F}_{\text{cart}}$  is a category fibered in groupoids over  $\mathcal{C}$ .  $\lrcorner$

*Proof:* Firstly  $\mathcal{F}_{\text{cart}}$  is a category by (4.2.3.6), and it is a category fibered in groupoids by (4.2.3.20).  $\square$

**Def. (4.2.3.23)[Rigid Fibered Categories].** A **rigid fibered category** is a fibered category whose associated category fibered groupoids is fibered in setoids. Equivalently, there are no isomorphisms above any  $\text{id}_U$ .  $\lrcorner$

**Prop. (4.2.3.24)[2-Fibered Products of Categories Fibered in Groupoids].** The 2-fibered products of categories fibered in groupoids over  $\mathcal{C}$  is a category fibered in groupoids over  $\mathcal{C}$ .  $\lrcorner$

*Proof:* The 2-fibered products exist by (4.2.3.13), and using (4.2.3.3), it is fibered in groupoids because the 2-fibered products of groupoids is a groupoid by (4.2.3.3).  $\square$

**Prop. (4.2.3.25)[Characterization of Categories Fibered in Setoids].** Let  $\mathcal{F}$  be a category over  $\mathcal{C}$ , then  $\mathcal{F}$  is fibered in setoids over  $\mathcal{C}$  iff for any object  $\eta$  of  $\mathcal{F}$  and an arrow  $f : U \rightarrow p_{\mathcal{F}}\eta \in \mathcal{C}$ , there is a unique arrow  $\xi \rightarrow \eta$  mapping to  $f$ .  $\lrcorner$



*Proof:* Let  $\mathcal{F}$  be a category fibered in sets, then pick a Cartesian arrow  $\tilde{f}$  over  $f$ , then any other lifting factors through this lifting by the property of Cartesian, then it is identity, because  $\mathcal{F}(U)$  is a setoid.

Conversely, if the hypothesis holds, then clearly  $\mathcal{F}(U)$  is a setoid, and the fibered category condition holds, because of the uniqueness.  $\square$

**Cor. (4.2.3.26) [Presheaves and Categories Fibered in Setoids].** Let  $\mathcal{C}$  be a category, then categories fibered in setoids over  $\mathcal{C}$  are exactly those equivalent to a presheaf over  $\mathcal{C}$ .  $\lrcorner$

**Cor. (4.2.3.27).** In particular, for any object  $X \in \mathcal{C}$ , the presheaf  $h_X$  determines a category fibered in sets, which is just the comma category  $\mathcal{C}/X \rightarrow \mathcal{C}$ .  $\lrcorner$

**Prop. (4.2.3.28) [Characterization of Quasi-Functors].** A category  $\mathcal{F}$  over  $\mathcal{C}$  is a quasi-functor iff:

- Given any object  $\eta \in \mathcal{F}$  and an arrow  $f : U \rightarrow p_{\mathcal{F}}\eta$ , there is a lifting  $\xi \rightarrow \eta$  mapping to  $f$ . And given any two such extensions, there is a morphism  $\xi' \rightarrow \xi$  commuting them.
- Given any two objects  $\xi, \eta \in \mathcal{F}$  and an arrow  $f : p_{\mathcal{F}}\xi \rightarrow p_{\mathcal{F}}\eta$ , there is at most one arrow  $\tilde{f} : \xi \rightarrow \eta$  lifting  $f$ .

$\lrcorner$

*Proof:*

$\square$

**Prop. (4.2.3.29).** A fibered category over  $\mathcal{C}$  is a quasi-functor iff it is equivalent to a presheaf.  $\lrcorner$

*Proof:* If it is equivalent to a functor, then  $\mathcal{F}(U)$  is equivalent to a setoid, thus it is an equivalent relation, by (4.1.1.15). Conversely, if  $\mathcal{F}$  is a quasi-functor, then it is fibered in groupoids, thus by (4.2.3.20) every morphism is Cartesian, and if we denote  $\Phi(U)$  the equivalence classes of  $\mathcal{F}(U)$ , then a morphism  $U \rightarrow V$  will induce a morphism  $\Phi(U) \rightarrow \Phi(V)$  by the property of Cartesian arrows. Then clearly this defines a presheaf that is equivalent to  $\mathcal{F}$ .  $\square$

### Representability

**Def. (4.2.3.30) [Representable Fibered Category].** A fibered category is called **representable** if it is equivalent to the fibered category  $\mathcal{C}/X$  defined in (4.2.3.27) for some  $X \in \mathcal{C}$ .  $\lrcorner$

**Prop. (4.2.3.31) [2-Categorical Yoneda Lemma].** Let  $\mathcal{F}$  be a fibered category over  $\mathcal{C}$  and  $X \in \mathcal{C}$ , there is an equivalence of categories:

$$\mathrm{Hom}_{\mathcal{C}}((\mathcal{C}/X), \mathcal{F}) \cong \mathcal{F}(X) : \varphi \mapsto \varphi(\mathrm{id}_X)$$

$\lrcorner$

*Proof:* To show this functor is essentially surjective, choose a choice of pullbacks of  $\mathcal{F}$ , for any  $\xi \in \mathcal{F}(X)$ , we define a  $F : \mathcal{C}/X \rightarrow \mathcal{F}$  that maps a  $\varphi : U \rightarrow X$  to  $\varphi^*\xi$ , and to any morphism in  $\mathcal{C}/X$  an arrow in  $\mathcal{F}$  induced by Cartesian property.

To show it is fully faithful, notice a natural transformation of  $\varphi, \psi \in \mathrm{Hom}_{\mathcal{C}}((\mathcal{C}/X), \mathcal{F})$  is determined by their value on  $\mathrm{id}_X$ , and any map  $\varphi(\mathrm{id}_X) \rightarrow \psi(\mathrm{id}_X)$  induces a natural transformation, by Cartesian properties.  $\square$

**Cor. (4.2.3.32) [Characterization of Representability].** Translating the 2-Yoneda lemma and (4.2.3.12), we see that a fibered category  $\mathcal{F}$  is representable by  $X \in \mathcal{C}$  iff  $\mathcal{F}$  is fibered in groupoids, and there is an object  $\xi \in \mathcal{F}(X)$  that for any object  $\rho \in \mathcal{F}$ , there is a unique arrow  $\rho \rightarrow \xi$ .  $\lrcorner$

**Cor. (4.2.3.33).** If  $\mathcal{X}, \mathcal{Y}$  are fibered categories over  $\mathcal{C}$  representable by  $U, V$  resp., then there is an isomorphism of sets

$$\mathrm{Hom}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})/2\text{-isomorphisms} \cong \mathrm{Hom}_{\mathcal{C}}(U, V)$$

⌋

*Proof:* By Yoneda lemma there is an equivalence  $\mathrm{Hom}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y}) \cong \mathrm{Hom}_{\mathcal{C}}(U, V)$ , and then  $\mathrm{Hom}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$  is an equivalence relation by (4.1.1.15), so the isomorphism is clear.  $\square$

**Def. (4.2.3.34)[Representable 1-Morphisms].** Let  $\mathcal{C}$  be a category and  $F : \mathcal{X} \rightarrow \mathcal{Y}$  be a morphism of categories fibered over  $\mathcal{C}$ , then  $F$  is called **representable** if for any  $U \in \mathcal{C}$  and a morphism  $\mathcal{C}/U \rightarrow \mathcal{Y}$ , the fibered category  $(\mathcal{C}/U) \times_{\mathcal{Y}} \mathcal{X} \rightarrow \mathcal{C}/U$  is representable (4.2.3.30) (Notice it is a fibered category by (4.2.3.13) and (4.2.3.14)).  $\lrcorner$

**Prop. (4.2.3.35)[Diagonal and Representability].** Let  $\mathcal{S}$  be a category fibered in groupoids over  $\mathcal{C}$ . Assume  $\mathcal{C}$  has fibered product, then the following are equivalent:

- $\Delta_{\mathcal{S}} : \mathcal{S} \rightarrow \mathcal{S} \times_{\mathcal{C}} \mathcal{S}$  is representable.
- For every  $U \in \mathcal{C}$ , any  $G : \mathcal{C}/U \rightarrow \mathcal{S}$  is representable.

⌋

*Proof:* Cf. [Sta]02YA.

The key to this proposition is the fibered product diagram (4.2.3.13)(4.1.1.49)

$$\begin{array}{ccc} X \times_{\mathcal{F}} Y & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow \Delta \\ X \times_S Y & \xrightarrow{f \times g} & \mathcal{F} \times_S \mathcal{F} \end{array}$$

which still holds in the 2-commutative sense.

So if  $\Delta_{\mathcal{F}}$  is schematic, then  $X \times_{\mathcal{F}} Y$  is a scheme, so  $X \rightarrow \mathcal{F}$  is schematic, for any scheme  $X$ . Conversely, consider the fibered products

$$\begin{array}{ccccc} \mathcal{F} \times_{\mathcal{F} \times_S \mathcal{F}} X & \longrightarrow & X \times_{\mathcal{F}} X & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow & & \downarrow \Delta \\ X & \xrightarrow{\Delta} & X \times_S X & \xrightarrow{f \times g} & \mathcal{F} \times_S \mathcal{F} \end{array}$$

induced by  $h = f \times g : X \mapsto \mathcal{F} \times_S \mathcal{F}$ . So in order to prove  $\Delta_{\mathcal{F}}$  is schematic, it suffices to prove  $\mathcal{F} \times_{\mathcal{F} \times_S \mathcal{F}} X$  is a scheme, and for this, it suffices to prove  $X \times_{\mathcal{F}} X$  is a scheme. But  $X \times_{\mathcal{F}} X \rightarrow X$  is a pullback of  $X \rightarrow \mathcal{F}$ , so it is a scheme.  $\square$

## 4.3 Categorical Logic

References are [Categorical Logic Notes, Jacob Lurie], [Coend Calculus, Fosco Loregian], [Harder-Narasimhan Filtrations, Huayi Chen], [Harder-Narasimhan Theory, Jonathan Pottharst], [Coend Calculus], and [Sta].

### 1 Monads and Categories

**Def. (4.3.1.1) [Monad].** Let  $\mathcal{C}$  be a category, a **monad** on  $\mathcal{C}$  is an endofunctor  $\mathcal{C} \rightarrow \mathcal{C}$  together with two natural morphisms:

- (multiplication)  $\mu : T \circ T \rightarrow T$ .
- (unit)  $\text{id}_{\mathcal{C}} \rightarrow T$ .

that satisfies associativity and unit diagrams.  $\lrcorner$

**Def. (4.3.1.2) [Algebras over Monads].** An **algebra over a monad**  $T$  is an object  $X$  together with a morphism  $\alpha : TX \rightarrow X$  that satisfies the diagrams for an algebra.  $\lrcorner$

### 2 Group Formation

The goal of this subsection is to give a formation that encompass both the group theory and algebraic group theory.

**Def. (4.3.2.1) [Group Formations].** A **group formation** is a category  $\mathcal{C} \in \mathbf{Cat}$  with a class of arrows  $\mathcal{S} \in \mathcal{C}$  consisting of monomorphisms, satisfying the following axioms:

- $G$  is complete and cocomplete.
- $G$  has a zero object  $e$ .
- $\mathcal{S}$  is stable under pullbacks.
- If  $N \rightarrow H \rightarrow G \in \mathcal{S}$  and  $H \rightarrow G$  is a monomorphism, then  $N \rightarrow H \in \mathcal{S}$ .
- Let  $N \rightarrow G \in \mathcal{S}$ , then  $N$  is called a **normal subobject** of  $G$ , and  $N = \ker(G \rightarrow \text{Coker}(N \rightarrow G))$ .
- For  $G \in \mathcal{C}$ , co-products exist in  $\mathcal{C}/G$ , and if  $N \rightarrow G, H \rightarrow G \in \mathcal{S}$ , then the co-product is also in  $\mathcal{S}$ , denoted by  $NH \rightarrow G$ .
- Let  $N \rightarrow G \in \mathcal{S}$  and  $H \rightarrow G$  a monomorphism, then
  - there is a natural isomorphism:  $H/N \cap H \cong HN/N$ .
  - If  $N \rightarrow G \in \mathcal{S}$ , then there is a natural isomorphism  $G/NH \cong (G/N)/(H/H \cap N)$ .

$\lrcorner$

**Prop. (4.3.2.2).**  $\lrcorner$

### 3 Filtrations of a Category

**Def. (4.3.3.1) [Filtrations in a Category].** Let  $\mathcal{C}$  be a category with an initial object, a **filtration** of an object  $X$  in  $\mathcal{C}$  is a family  $\mathcal{F} = (X_t)_{t \in \mathbb{R}}$  of subobjects of  $X$  indexed by  $\mathbb{R}$  that satisfies:

- (decreasing property) if  $s \leq t$ , then  $X_s \rightarrow X$  factors through  $X_t$ .
- (separation property) for sufficiently small  $t$ ,  $X_t$  is the initial object.

- (exhaustiveness) for sufficiently large  $s$ ,  $X_s = X$ .
- (right locally constant property) for any  $t \in \mathbb{R}$ , there exists  $\delta > 0$  that for any  $s \in [t, t + \delta)$ , the morphism  $X_t \rightarrow X_s$  are isomorphisms.
- (finite jump) the jump set of  $\mathcal{F}$  is finite.

Naturally, we define morphisms of filtrations.  $\lrcorner$

**Def. (4.3.3.2) [Pullback and Pushforward Filtrations].** Suppose fibered products exist in  $\mathcal{C}$ , for any morphism  $f : X \rightarrow Y$  and a filtration  $\mathcal{G} = (Y_t) \rightarrow Y$ , then the family  $f^*\mathcal{G} = (Y_t \otimes_Y X) \rightarrow X$  is a filtration on  $X$ , called the **pullback filtration**.

If  $f : X \rightarrow Y$  and  $\mathcal{F} = (X_t) \rightarrow X$  is a filtration on  $X$ , then if there is a filtration  $f_*\mathcal{F}$  on  $Y$  and a morphism of filtrations  $\mathcal{F} \rightarrow f_*\mathcal{F}$  compatible with  $f$ , then  $f_*\mathcal{F}$  is called the **pushforward filtrations**.  $\lrcorner$

## 4 Galois Categories

References are [Gro71, B-S15].

**Prop. (4.3.4.1) [Profinite Automorphism Groups].** For  $\mathcal{C} \in \mathbf{Cat}$  and  $F : \mathcal{C} \rightarrow \mathbf{Set}$ , the map

$$\mathrm{Aut}(F) \rightarrow \prod_{X \in \mathcal{C}} \mathrm{Aut}(F(X))$$

identifies  $\mathrm{Aut}(F)$  with a closed subset of  $\prod_{X \in \mathcal{C}} \mathrm{Aut}(F(X))$ . In particular, if  $\#F(X) < \infty$  for any  $X \in \mathcal{C}$ , then  $\mathrm{Aut}(F) \in \mathbf{ProfGrp}$ .  $\lrcorner$

*Proof:* Cf. [Sta]0BMR.  $\square$

**Prop. (4.3.4.2).** For  $G \in \mathbf{TopGrp}$ , consider  $\mathcal{C} = G\text{-finSet}$  and  $F : \mathcal{C} \rightarrow \mathbf{finSet}$  the forgetful functor, then  $\mathrm{Aut}(F)$  with the profinite topology defined in (4.3.4.1) is isomorphic to the profinite completion of  $G$ .  $\lrcorner$

*Proof:* Cf. [Sta]0BMU.  $\square$

**Prop. (4.3.4.3).** If  $G \in \mathbf{TopGrp}$  and  $F : G\text{-finSet} \rightarrow \mathbf{finSet}$  is an exact functor, then  $F$  is isomorphic to the forgetful functor.  $\lrcorner$

*Proof:* Cf. [Sta]0BMV.  $\square$

**Def. (4.3.4.4) [Galois-Categories].** A **Galois-category** is a pair  $(\mathcal{C}, F)$  where  $\mathcal{C} \in \mathbf{Cat}$  and  $F : \mathcal{C} \rightarrow \mathbf{finSet} \in \mathbf{Cat}$  s.t.

- $\mathcal{C}$  has finite limits and colimits,
- Each object of  $\mathcal{C}$  is a finite co-product of connected objects (4.1.1.21).
- $F$  reflects isomorphisms and is exact.

$\lrcorner$

**Prop. (4.3.4.5) [Properties of Galois-Categories].** If  $(\mathcal{C}, F)$  is a Galois-category (4.3.4.4) and  $X \rightarrow Y \in \mathbf{Cat}$ , then

- $F$  is faithful.
- $X \rightarrow Y$  is a monomorphism iff  $F(X) \rightarrow F(Y)$  is injective.

- $A \in \mathcal{C}$  is initial iff  $F(A) = \emptyset$ .
- $Z \in \mathcal{C}$  is final iff  $F(Z) = \text{pt.}$
- If  $X, Y$  are both connected objects(4.1.1.21), then  $X \rightarrow Y$  is an epimorphism.
- If  $X$  is a connected object and  $a, b : X \rightarrow Y \in \mathcal{C}$ , then  $a = b$  iff  $F(a)$  and  $F(b)$  agree on one element of  $F(X)$ .
- If  $X = \coprod_{i \in [n]_+} X_i$  and  $Y = \coprod_{j \in [m]_+} Y_j$  for  $m, n \in \mathbb{N}$  where  $X_i, Y_j$  are connected objects, then there exists a map  $\alpha : [n]_+ \rightarrow [m]_+$  s.t.  $X \rightarrow Y$  comes from a collection of maps  $X_i \rightarrow Y_{\alpha(i)}$ .

┘

**Def. (4.3.4.6) [Galois-Objects].** If  $(\mathcal{C}, F)$  is a Galois-category(4.3.4.4) and  $X \in \mathcal{C}$  is a connected object(4.1.1.21), then  $\# \text{Aut}(X) \leq \# F(X)$  by(4.3.4.5). We say  $X$  is a Galois-object if  $\# \text{Aut}(X) = \# F(X)$ , or equivalently  $\text{Aut}(X)$  acts transitively on  $F(X)$ .

┘

**Prop. (4.3.4.7) [Connected Objects Dominated by Galois-Objects].** If  $(\mathcal{C}, F)$  is a Galois-category(4.3.4.4) and  $X \in \mathcal{C}$  is a connected object(4.1.1.21), then there exists a Galois-object  $Y \in \mathcal{C}$ (4.3.4.6) with  $\text{Mor}(Y, X) \neq \emptyset$ .

┘

*Proof:* Cf.[Sta]0BN2.?

□

**Prop. (4.3.4.8).** If  $(\mathcal{C}, F)$  is a Galois-category(4.3.4.4) and  $X \in \mathcal{C}$  is a connected object(4.1.1.21), then the action of  $\text{Aut}(F)$  on  $F(X)$  is transitive.

┘

*Proof:* Cf.[Sta]0BN3.

□

**Prop. (4.3.4.9).** If  $(\mathcal{C}, F)$  is a Galois-category(4.3.4.4) and  $G = \text{Aut}(F)$  with the profinite topology defined as in(4.3.4.1), the functor  $F : \mathcal{C} \rightarrow G\text{-finSet}$  is an equivalence of categories.

┘

*Proof:* Cf.[Sta]0BN4.

□

**Prop. (4.3.4.10) [Exact Functors and Homomorphism of Automorphism Groups].** If  $(\mathcal{C}, F), (\mathcal{C}', F')$  are Galois-categories(4.3.4.4) and  $H : \mathcal{C} \rightarrow \mathcal{C}'$  is a category, then there exists an isomorphism  $t : F' \circ H \rightarrow F$ . The choice of  $t$  determines a continuous homomorphism  $h : \text{Aut}(F') \rightarrow \text{Aut}(F)$  and a 2-commutative diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{H} & \mathcal{C}' \\ \downarrow & & \downarrow \\ \text{Aut}(F)\text{-finSet} & \xrightarrow{h} & \text{Aut}(F')\text{-finSet} \end{array}$$

And the map  $h$  is independent of  $t$  up to  $G$ -inner automorphisms.

Conversely, given a continuous homomorphism  $h : \text{Aut}(F') \rightarrow \text{Aut}(F)$ , there exists an exact functor  $H : \mathcal{C} \rightarrow \mathcal{C}'$  an isomorphism  $t$  recovering  $h$  as above.

┘

*Proof:* By(4.3.4.9), we may assume that  $\mathcal{C} = \text{Aut}(F)\text{-finSet}$  and  $\mathcal{C}' = \text{Aut}(F')\text{-finSet}$ . Then the existence of  $t$  follows from(4.3.4.3). The uniqueness of  $t$  up to  $G$ -inner automorphism is obvious. The final statement is also trivial.

□

## 5 Harder-Narasimhan Formalism

References are [Pot20].

**Def. (4.3.5.1) [Harder-Narasimhan Formalism].** A **Harder-Narasimhan formalism** consists of

- An exact category  $\mathcal{C}$  (4.8.2.1).
- A function  $\deg : \text{Ob}(\mathcal{C}) \rightarrow \mathbb{Z}$  that is additive w.r.t. short exact sequences.
- An exact faithful **generic fiber functor** to an Abelian category  $F : \mathcal{C} \rightarrow \mathcal{A}$  that induces for each object  $F : \mathcal{E} \in \mathcal{C}$  a bijection

$$\{\text{strict objects of } \mathcal{E}\} \cong \{\text{subobjects of } F(\mathcal{E})\}$$

where a **strict subobject** is an object that can be prolonged to an exact sequence.

- An additive function  $\text{rank} : \mathcal{A} \rightarrow \mathbb{N}$  on  $\mathcal{A}$  that  $\text{rank}(\mathcal{L}) = 0 \iff \mathcal{L} = 0$ , and its composition with  $F$  is also called rank.
- If  $u : \mathcal{E} \rightarrow \mathcal{E}'$  is a morphism in  $\mathcal{C}$  that  $F(u)$  is an isomorphism, then  $\deg(\mathcal{E}) \leq \deg(\mathcal{E}')$  with equality iff  $u$  is an isomorphism.

┘

**Cor. (4.3.5.2).**

- We are free to choose the "kernel" for  $u$  that  $F(u)$  is surjection.
- The subobjects of subobjects are subobjects, by axiom3.

┘

**Def. (4.3.5.3) [Saturation].** Let  $X'$  be a subobject of  $X$ , then we let  $\widetilde{X'}$  denote the strict subobject of  $X$  corresponding to the subobject  $F(X')$  of  $F(X)$ , called the **saturation** of  $X'$ . The saturation satisfies:

- 
- 
- 

┘

*Proof:* Cf. [Pot20]P3. □

**Prop. (4.3.5.4).** Every morphism  $f : X \rightarrow Y$  has a kernel and a image in  $\mathcal{C}$ , and  $0 \rightarrow \ker f \rightarrow X \rightarrow \text{Im } f \rightarrow 0$  is an exact sequence. ┘

*Proof:* Cf. [Pot20]P3. □

**Prop. (4.3.5.5) [HN-Formalism on the Category of Filtered Vector Spaces].** If  $L/K$  is a field extension, there is a category  $\text{Vect Fil}_{L/K}$  consisting of  $(V, \text{Fil}^\bullet)$  where  $V \in \text{Vect}_K$  and  $\text{Fil}^\bullet$  is a finite filtration on  $V \otimes_K L$ . It is an exact category by declaring exact sequences be those induce exact sequences on the gradeds.

The generic fiber functor is  $\text{Vect Fil}_{L/K} \rightarrow \text{Vect}_K : (V, \text{Fil}^\bullet) \mapsto V$ , and rank is as usual, the **Hodge-Tate degree** is defined to be

$$t_{\text{H-T}}((V, \text{Fil}^\bullet)) = \sum i \dim_L \text{gr}^i(V \otimes_K L).$$

This is a HN-filtration. ┘

*Proof:* The axioms can be directly checked, noticing that a subfiltration  $W_n$  of a filtration  $V_n$  is a strict object iff  $W_k = W_n \cap V_k$ .  $\square$

**Def. (4.3.5.6) [Slopes].** In a HN-formalism, the **slope** is defined to be  $\text{slope}(E) = \frac{\deg(E)}{\text{rank}(E)}$ .

$\mathcal{E}$  is called **semistable of slope**  $\lambda$  iff  $\text{slope}(\mathcal{E}) = \lambda$ , and  $\text{slope}(\mathcal{E}') \leq \lambda$  for any nonzero strict subobject  $\mathcal{E}' \subset \mathcal{E}$ .  $\lrcorner$

**Prop. (4.3.5.7).** If  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$  be a short exact sequence in  $\mathcal{C}$ , then:

- If two of them have the same slope, then so does the third.
- If two of them have different slope, then we know the ordering of these slopes.

$\lrcorner$

*Proof:* Just notice that the degree and rank are all additive functions.  $\square$

**Prop. (4.3.5.8).** If  $\mathcal{E}$  is semistable of slope  $\lambda$ , then for any morphism  $u : \mathcal{E} \rightarrow \mathcal{E}''$  that  $F(u)$  is surjective,  $\mu(\mathcal{E}'') \geq \lambda$ .  $\lrcorner$

*Proof:* Take the kernel of  $F(u)$  in  $\mathcal{A}$ , which corresponds to a strict object  $\mathcal{E}'$  of  $\mathcal{E}$ , and  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$  is exact, so we can use (4.3.5.19).  $\square$

**Cor. (4.3.5.9).** If  $\mathcal{E}, \mathcal{F}$  are semistable of slopes  $\lambda > \mu$ , then  $\text{Hom}_{\mathcal{C}}(\mathcal{E}, \mathcal{F}) = 0$ .  $\lrcorner$

*Proof:* Notice  $F$  is faithful.  $\square$

**Prop. (4.3.5.10) [Semistable Objects].** If  $f : \mathcal{E} \rightarrow \mathcal{F}$  be a map of vector bundles of the same slope  $\lambda$ , then  $\ker(f)$  and  $\text{Coker}(f)$  are all semistable vector bundles of slope  $\lambda$ , and if  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$  is exact and  $\mathcal{E}', \mathcal{E}''$  are semistable of slope  $\lambda$ , then so does  $\mathcal{E}$ .  $\lrcorner$

*Proof:* Use  $F(f)$  to find the "coimage"  $A$  and the "image"  $B$  of  $f$ , then there is a map from  $F(A)$  to  $F(B)$  which is an isomorphism, but they have the same degree and rank, thus  $A \cong B$  by the last axiom. And the image must have slope  $\lambda$ . Then  $\ker(f), \text{Coker}(f)$  all can be defined, and they have the same slope  $\lambda$  by (4.3.5.19).

$\ker(f)$  is semistable because strict subobjects of  $\ker(f)$  are also strict subobjects of  $\mathcal{E}$  (4.3.5.2). And for  $\text{Coker}(f)$ , if it is not semistable, choose  $\overline{\mathcal{F}'} \subset \text{Coker}(f)$  that has slope  $> \lambda$ , let  $\mathcal{F}'$  be the inverse image, then  $0 \rightarrow \text{Im}(f) \rightarrow \mathcal{F}' \rightarrow \overline{\mathcal{F}'} \rightarrow 0$ , then by (4.3.5.19)  $\text{slope}(\mathcal{F}') > \lambda$ , contradicting the semi-stability of  $\mathcal{F}$ .

For the extension,  $\text{slope}(\mathcal{E}) = \lambda$  by (4.3.5.19), and for a strict subobject  $\mathcal{F}$  of  $\mathcal{E}$ , then we can find  $F', F''$  be strict objects of  $\mathcal{E}', \mathcal{E}''$  respectively that there is an exact sequence  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ ?, which shows  $\text{slope}(\mathcal{F}) \leq \lambda$ , so  $\mathcal{E}$  is semistable.  $\square$

### Harder-Narasimhan Filtration

**Lemma (4.3.5.11) [Final Subobjects of Maximal Slope].** Let  $X$  be an object of  $\mathcal{C}$  and  $X', X''$  its subobjects of maximal slope, then  $X' + X''$  and  $X' \cap X''$  are also of maximal slope.  $\lrcorner$

*Proof:* Cf. [Pot20] P8.  $\square$

**Def. (4.3.5.12).** Given an object  $X$  of  $\mathcal{C}$ , consider the following condition on a nonzero subobject  $X'$  of  $X$ :

For all subobjects  $X''$  of  $X$  properly containing  $X'$ ,  $\mu(X'') < \mu(X')$ .  $\lrcorner$

**Def. (4.3.5.13) [SCSS].** Let  $X'$  be a subobject of  $X$ , then the following conditions are equivalent:

- $X'$  satisfies condition (4.3.5.12) and is semistable.
- $X'$  satisfies condition (4.3.5.12) and is of maximal slope.
- $X'$  is the final object of  $X$  of maximal slope.

If  $X'$  satisfies these equivalent conditions and  $X' \neq X$ , then  $X$  is called a **strongly contradicting semi-stability** or SCSS of  $X$ .  $\lrcorner$

*Proof:* Cf. [Pot20]P8.  $\square$

**Prop. (4.3.5.14).** Every object  $X$  of  $\mathcal{C}$  admits a SCSS.  $\lrcorner$

**Def. (4.3.5.15) [Harder-Narasimhan Filtration].** Let  $\mathcal{E} \in \mathcal{C}$ , a chain of strict objects  $0 \subset \mathcal{E}_0 \subsetneq \mathcal{E}_1 \subsetneq \dots \subsetneq \mathcal{E}_m = \mathcal{E}$  is called a **Harder-Narasimhan filtration** iff each quotient  $\mathcal{E}_i/\mathcal{E}_{i-1}$  is semistable of slope  $\lambda_i$  and  $\lambda_1 > \lambda_2 > \dots > \lambda_m$ .  $\lrcorner$

**Prop. (4.3.5.16) [Faltings].** Every object  $\mathcal{E} \in \mathcal{C}$  has a unique functorial Harder-Narasimhan filtration.  $\lrcorner$

*Proof:* For uniqueness: if there are two filtrations, it suffices to show that  $\mathcal{E}'_1 = \mathcal{E}_1$ , because notice by (4.3.5.2)  $\mathcal{E}_i$  is a strict subobject of  $\mathcal{E}_j$  for any  $i < j$ , so we finish by induction on the length of the filtration and considering  $\mathcal{E}/\mathcal{E}_1$ .

For this, firstly  $\lambda_1 = \lambda'_1$ , suppose the contrary and  $\lambda_1 > \lambda'_1$ , then  $\lambda_1 > \lambda'_i$  for each  $i$ , so  $\text{Hom}(\mathcal{E}_1, \mathcal{E}'_i/\mathcal{E}'_{i-1}) = 0$  for each  $i$  by (4.3.5.32), so by induction  $\text{Hom}(\mathcal{E}_1, \mathcal{E}) = 0$ , contradiction.

Next by the same reason as in the proof above,  $\mathcal{E}_1 \hookrightarrow \mathcal{E}$  has image in  $\mathcal{E}'_1$ , and the reverse is true for  $\mathcal{E}'_1$ , so  $\mathcal{E}_1 \cong \mathcal{E}'_1$  in  $\mathcal{E}$ .

For existence: Use induction on  $\text{rank}(\mathcal{E})$ . If  $\mathcal{E}$  is semistable, then we finish. Otherwise, there is a strict subobject  $\mathcal{F}$  and  $0 \rightarrow \mathcal{F} \rightarrow \mathcal{E} \rightarrow \mathcal{G} \rightarrow 0$  that  $\text{slope}(\mathcal{F}) > \text{slope}(\mathcal{E})$ , so  $\text{rank}(\mathcal{F}), \text{rank}(\mathcal{G}) < \text{rank}(\mathcal{E})$ . Now by induction  $\mathcal{F}$  and  $\mathcal{G}$  has HN-filtration, thus by argument as above, we see that  $\mathcal{E}$  cannot have strict subobject with slope bigger than slopes appearing in the HN-filtration of  $\mathcal{F}, \mathcal{G}$ . So if we choose a strict subobject of  $\mathcal{E}_1$  of maximal rank among the strict subobjects of maximal slope, we claim the subobjects of  $\mathcal{E}/\mathcal{E}_1$  all have slopes smaller than  $\text{slope}(\mathcal{E}_1)$ : if some  $\text{slope}(\mathcal{G}) \geq \text{slope}(\mathcal{E}_1)$ , consider its inverse image  $\mathcal{G}'$ , then  $0 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{G}' \rightarrow \mathcal{G} \rightarrow 0$ , thus  $\text{slope}(\mathcal{G}') \geq \text{slope}(\mathcal{E}_1)$  and has bigger rank, contradiction. So we can use induction on  $\mathcal{E}/\mathcal{E}_1$ .  $\square$

## HN-Polygons

### Harder-Narasimhan Categories

This subsection is unnecessary. Main references are [Harder-Narasimhan Categories]

**Def. (4.3.5.17) [Harder-Narasimhan Categories].** A **Harder-Narasimhan category** consists of a geometric exact category  $(\mathcal{C}, \mathcal{E}, A)$  (4.8.2.11) consists of

1. A function  $\deg : \text{Ob}(\mathcal{C}_A) \rightarrow \mathbb{R}$  that is additive w.r.t short exact sequences in  $\mathcal{E}_A$ .
2. A function  $\text{rank} : \text{Ob}(\mathcal{C}) \rightarrow \mathbb{N}$  on  $\mathcal{A}$  that is additive w.r.t short exact sequences in  $\mathcal{E}$ ,  $\text{rank}(X) = 0 \iff X = 0$ .

The **slope** of a nonzero object  $X$  is defined to be  $\mu(X) = \deg(X)/\text{rank } X$ . And  $\mathcal{X}$  is called **semistable of slope**  $\lambda$  iff  $\mu(X) = \lambda$ , and  $\mu(X') \leq \lambda$  for any nonzero geometric subobject  $X' \subset X$ .

And the category satisfies the following axiom:



- **NH:** For any nonzero geometric object  $X$ , there exists a geometric subobject  $X_{des} \subset X$  that

$$\mu(X_{des}) = \sup\{\mu(Y) | Y \text{ is a non-zero geometric subobject of } X\}$$

and moreover, any nonzero geometric subobject  $Z$  of  $X$  that  $\mu(Z) = \mu(X_{des})$  is a geometric subobject of  $X_{des}$ .

Notice that  $X_{des}$  is semistable and unique up to isomorphisms, called the **destablization** of  $X$ .  $\lrcorner$

**Cor. (4.3.5.18).** A geometric object of rank 1 is semistable.  $\lrcorner$

*Proof:* For any nonzero geometric subobject  $X' \subset X$ ,  $\text{rank } X' = \text{rank } X = 1$ , so  $\text{rank } X/X' = 0$  hence  $X/X' = 0$  and  $X' \cong X$ . Then clearly  $\mu(X') = \mu(X)$  and  $X$  is semistable.  $\square$

**Cor. (4.3.5.19).** If  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$  be a short exact sequence in  $\mathcal{E}_A$ , then:

- If two of them have the same slope, then so does the third.
- If two of them have different slope, then we know the ordering of these slopes.

$\lrcorner$

*Proof:* Just notice that the degree and rank are all additive functions.  $\square$

**Prop. (4.3.5.20) [Abelian Categories as Harder-Narasimhan Categories].** Let  $(\mathcal{C}, \mathcal{E}, A)$  be a geometric exact category (4.8.2.11) with functions  $\deg$  and  $\text{rank}$  s.t.  $\mathcal{C} \in \mathcal{AbCat}$  and  $\mathcal{E}$  is the set of short exact sequences, then  $(\mathcal{C}, \mathcal{E}, A, \deg, \text{rank})$  is a Harder-Narasimhan category (4.3.5.17).  $\lrcorner$

*Proof:* We need to check HN: induct on  $\text{rank } X$ : The condition is clear when  $X$  is semistable, in particular when  $\text{rank } X = 1$  (4.3.5.18), and when  $X$  is not semistable, let  $Y$  be a geometric subobject of  $X$  that  $\mu(X') > \mu(X)$  and  $\text{rank } X'$  is maximal, then by induction hypothesis, there is a destablization  $Y_{des}$ , and we want to show  $X'_{des}$  is just  $X_{des}$ : Let  $Y$  be a nonzero geometric subobject of  $X$ . If  $Y$  is a geometric subobject of  $X'$ , then  $\mu(Y) \leq \mu(X'_{des})$ . If  $Y$  is not a geometric subobject of  $X'$ , then  $Y + X'$  is greater than  $X'$ , and  $\text{rank}(Y + X') > \text{rank } X'$ , so by maximality,  $\mu(Y + X') \leq \mu(X) < \mu(X')$ . Moreover, there is an exact sequence

$$0 \rightarrow Y \cap X' \rightarrow Y \oplus X' \rightarrow Y + X' \rightarrow 0$$

so

$$\begin{aligned} \deg Y &= \deg(Y \cap X') + \deg(Y + X') - \deg(X') \\ &< \mu(X'_{des}) \text{rank}(Y \cap X') + \mu(X')(\text{rank}(Y + X') - \text{rank}(X')) \\ &\leq \mu(X'_{des})(\text{rank}(Y \cap X') + \text{rank}(Y + X') - \text{rank}(X')) \\ &= \mu(X'_{des}) \text{rank}(Y). \end{aligned}$$

In particular, if  $\mu(Y) = \mu(X'_{des})$ , then  $Y$  must be a geometric subobject of  $X'$  hence a geometric subobject of  $X'_{des}$ . so HN holds for  $X$ .  $\square$

**Def. (4.3.5.21) [Harder-Narasimhan Filtration].** Let  $\mathcal{X}$  be a nonzero geometric object, a chain of admissible monomorphisms  $0 = X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_m = X$  is called a **Harder-Narasimhan filtration** of  $X$  iff each quotient  $\mathcal{E}_i/\mathcal{E}_{i-1}$  are all semistable of slope  $\lambda_i$  and  $\lambda_1 > \lambda_2 > \dots > \lambda_m$ , called the slopes associated to  $X$ .  $\lrcorner$

**Prop. (4.3.5.22) [Harder-Narasimhan Filtrations Exist].** Let  $(\mathcal{C}, \mathcal{E}, A, \deg, \text{rank})$  be a Harder-Narasimhan category (4.3.5.17), then the Harder-Narasimhan filtration exists for any nonzero geometric object  $X$ .  $\lrcorner$

*Proof:* We induct on the rank of  $X$ : if  $X$  is semistable, this is clear, so we are done in the case  $\text{rank } X = 1$  by (4.3.5.18). We choose  $X_1 = X_{des}$ , then  $X_{des}$  is semistable and  $X' = X/X_{des} \neq 0$ . Now  $\text{rank } X' < \text{rank } X$ , we can apply induction hypothesis to obtain a HN-filtration  $0 = X'_1 \xrightarrow{f'_1} X'_2 \rightarrow \dots \rightarrow X'_{n-1} \xrightarrow{f'_{n-1}} X'_n = X'$ . Now let  $X_i = X \times_{X'} X'_i$  (exists by Ex6(4.8.2.1)), then  $X_1 = X_{des}$ . Since  $X \rightarrow X'$  is an admissible epimorphism,  $X_i \rightarrow X'_i$  are also admissible epimorphisms, and there are Cartesian diagrams

$$\begin{array}{ccc} X_i & \xrightarrow{f_i} & X_{i+1} \\ \downarrow \pi_i & & \downarrow \pi_{i+1} \\ X'_i & \xrightarrow{f'_i} & X'_{i+1} \end{array}$$

and  $f_i$  are monomorphisms, so  $f_i$  is the kernel of  $X_{i+1} \rightarrow X'_{i+1}/X'_i$ , which is an admissible epimorphism, so  $f_i$  is admissible. There are natural isomorphisms  $\varphi_i : X_{i+1}/X_i \rightarrow X'_{i+1}/X'_i$  (Cf. [Harder-Narasimhan Filtrations]), and axiom A6(4.8.2.11) shows  $\varphi$  is compatible with the geometric structure, so  $X_{i+1}/X_i$  are all semistable, and notice

$$\mu(X_2/X_1) = \frac{\text{rank}(X_2)\mu(X_2) - \text{rank}(X_1)\mu(X_1)}{\text{rank}(X_2) - \text{rank}(X_1)} < \mu(X_1)$$

so  $0 \rightarrow X_1 \rightarrow X_2 \rightarrow \dots \rightarrow X_m = X$  is a HN-filtration for  $X$ .  $\square$

**Prop. (4.3.5.23) [HN-Formalism for Filtrations in an Abelian Category].** Let  $\mathcal{C}$  be an Abelian category and  $\mathcal{E}$  the set of all short exact sequences in  $\mathcal{C}$ ,  $A(X)$  is the set of isomorphism classes of filtrations on  $X$ , then  $(\mathcal{C}, \mathcal{E}, A)$  is a geometric exact category, by (4.8.2.15), given any additive rank function on  $\mathcal{C}$ , and define a degree function for any filtration  $\mathcal{F} = (X_\lambda)$  as

$$\deg(\mathcal{F}) = \int_{\mathbb{R}} \lambda d \text{rank } X_\lambda,$$

then  $\deg$  is additive w.r.t short exact sequences of filtrations, and  $(\mathcal{C}, \mathcal{E}, A, \deg, \text{rank})$  is a Harder-Narasimhan filtration.

Then a filtration is semistable iff it has only one jump. Then the HN-filtration of a filtration is just the jump set ordered decreasingly.  $\lrcorner$

**Prop. (4.3.5.24) [HN Formalism for Vector Spaces with Two Norms].** The vector spaces with two norms is a Harder-Narasimhan category, Cf. [Harder-Narasimhan Filtrations, P9].  $\lrcorner$

**Prop. (4.3.5.25) [HN-Formalism for Torsion-Free Sheaves].** The category of torsion-free sheaves on a geometrically normal projective variety of dimension  $d \geq 1$  over a field  $K$  is a Harder-Narasimhan category. Cf. [Harder-Narasimhan Filtrations, P10].  $\lrcorner$

**Prop. (4.3.5.26) [HH-Formalism for Hermitian Adelic Bundle].** Let  $K$  be a number field, then the category of Hermitian adelic bundle over  $K$  is a Harder-Narasimhan category. Cf. [Harder-Narasimhan Filtrations, P10].  $\lrcorner$

**Prop. (4.3.5.27) [HN Formalism for Filtered Vector Spaces Field].** If  $L/K \in \text{Field}$ , there is a category  $\text{FilVect}_{L/K}$  consisting of  $(V, \text{Fil})$  where  $V \in \text{Vect}/K$  and  $\text{Fil}$  is a (finite) filtration of vectors spaces over  $L$  on  $V \otimes_K L$  (4.3.3.1). It is a geometric exact category by (4.8.2.15) and a Harder-Narasimhan category by (4.3.5.20) where the rank is as usual, and the degree is defined to be

$$\deg((V, \text{Fil})) = \int_{\mathbb{R}} \lambda d \text{rank } V_\lambda \quad (4.3.5.23)$$

$\lrcorner$

### Slope Inequalities and Functoriality

**Def. (4.3.5.28) [Additional Conditions].** In this subsection, we assume the Harder-Narasimhan filtration satisfies the following axiom that is  $\lrcorner$

**Def. (4.3.5.29) [Slope Inequality Axioms].** To show the functoriality of the Harder-Narasimhan filtration, we need the following axiom:

- (SI): If  $X_1, X_2$  are two semistable geometric objects that  $\mu(X_1) > \mu(X_2)$ , then there are no non-zero morphism from  $X_1$  to  $X_2$  compatible with the geometric structures (4.8.2.12).  $\lrcorner$

**Prop. (4.3.5.30).** If SI holds, then  $\lrcorner$

**Prop. (4.3.5.31).** If  $\mathcal{E}$  is semistable of slope  $\lambda$ , then for any morphism  $u : \mathcal{E} \rightarrow \mathcal{E}''$  that  $F(u)$  is an isomorphism,  $\text{slope}(\mathcal{E}'') \geq \lambda$ .  $\lrcorner$

*Proof:* Take the kernel of  $F(u)$  in  $\mathcal{A}$ , which corresponds to a strict object  $\mathcal{E}'$  of  $\mathcal{E}$ , and  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$  is exact, so we can use (4.3.5.19).  $\square$

**Cor. (4.3.5.32).** If  $\mathcal{E}, \mathcal{F}$  are semistable of slopes  $\lambda > \mu$ , then  $\text{Hom}_{\mathcal{C}}(\mathcal{E}, \mathcal{F}) = 0$ .  $\lrcorner$

**Prop. (4.3.5.33) [Semistable Vector Bundles Form a Weak Serre Subcategory].** If  $f : \mathcal{E} \rightarrow \mathcal{F}$  be a map of vector bundles of the same slope  $\lambda$ , then  $\ker(f)$  and  $\text{Coker}(f)$  are all semistable vector bundles of slope  $\lambda$ , and if  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$  is exact and  $\mathcal{E}', \mathcal{E}''$  are semistable of slope  $\lambda$ , then so does  $\mathcal{E}$ .  $\lrcorner$

*Proof:* Use  $F(f)$  to find the "coimage"  $A$  and the "image"  $B$  of  $f$ , then there is a map from  $F(A)$  to  $F(B)$  which is an isomorphism, but they have the same degree and rank, thus  $A \cong B$  by the last axiom. And the image must have slope  $\lambda$ . Then  $\ker(f), \text{Coker}(f)$  all can be defined, and they have the same slope  $\lambda$  by (4.3.5.19).

$\ker(f)$  is semistable because strict subobjects of  $\ker(f)$  are also strict subobjects of  $\mathcal{E}$  (4.3.5.2). And for  $\text{Coker}(f)$ , if it is not semistable, choose  $\overline{\mathcal{F}'} \subset \text{Coker}(f)$  that has slope  $> \lambda$ , let  $\mathcal{F}'$  be the inverse image, then  $0 \rightarrow \text{Im}(f) \rightarrow \mathcal{F}' \rightarrow \overline{\mathcal{F}'} \rightarrow 0$ , then by (4.3.5.19)  $\text{slope}(\mathcal{F}') > \lambda$ , contradicting the semi-stability of  $\mathcal{F}$ .

For the extension,  $\text{slope}(\mathcal{E}) = \lambda$  by (4.3.5.19), and for a strict subobject  $\mathcal{F}$  of  $\mathcal{E}$ , then we can find  $F', F''$  be strict objects of  $\mathcal{E}', \mathcal{E}''$  respectively that there is an exact sequence  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ ?, which shows  $\text{slope}(\mathcal{F}) \leq \lambda$ , so  $\mathcal{E}$  is semistable.  $\square$

## 4.4 Topology I

References are [Mun00, Jec03, Mor19].

### Notation (4.4.0.1).

- Use notations on set theory defined in [Set Theory](#).
- Let  $X \in \mathcal{Top}$  (4.4.1.1).
- I usually use  $U \subset X$  to denote an open subset, and  $Z \subset X$  to denote a closed subset.

┘

### 1 Basics

**Def. (4.4.1.1) [Topological Spaces and Continuous Functions].** For  $X \in \mathcal{Set}$ , a **topology** on  $X$  is a subset  $\mathcal{T} \subset \mathcal{P}(X)$  s.t.

- $\emptyset \in \mathcal{T}, X \in \mathcal{T}$ .
- If  $\{U_\alpha\}_{\alpha \in I} \subset \mathcal{T}$ , then  $\cup_{\alpha \in I} U_\alpha \in \mathcal{T}$ .
- If  $U, V \in \mathcal{T}$ , then  $U \cap V \in \mathcal{T}$ .

A **topological space** is a pair  $(X, \mathcal{T})$  where  $X \in \mathcal{Set}$  and  $\mathcal{T} \subset \mathcal{P}(X)$  is a topology on  $X$ . Any element of  $\mathcal{T}$  is called an **open subset** of  $X$ .

For  $X, Y \in \mathcal{Top}$ , a map  $f : X \rightarrow Y$  is called a **continuous map** if for any open subset  $U \subset Y$ ,  $f^{-1}(U) \subset X$  is also open.

The category of topological spaces is denoted by  $\mathcal{Top}$ .

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**Prop. (4.4.1.2) [Kuratowski Closure Operators, Kuratowski [Kur66]].** For  $X \in \mathcal{Set}$ , a **Kuratowski closure operator** is a map  $\mathcal{P}(X) \rightarrow \mathcal{P}(X) : A \rightarrow A^*$  s.t.

- $\emptyset^* = \emptyset$ ,
- $A \subset A^*$  for any  $A \in \mathcal{P}(X)$ .
- $(A^*)^* = A^*$  for any  $A \in \mathcal{P}(X)$ .
- $(A \cup B)^* = A^* \cup B^*$  for any  $A, B \in \mathcal{P}(X)$ .

Then Given a Kuratowski operator on  $X$ ,  $\mathcal{T} = \{U \in \mathcal{P}(X) \mid (U^c)^* = U^c\}$  is a topology on  $X$  (4.4.1.1), and for any  $A \in \mathcal{P}(X)$ ,  $A^*$  is the closure of  $A$  in this topology.

┘

*Proof:* Clearly  $\emptyset, X$  are closed in this topology. If  $A, B \in \mathcal{P}(X)$  s.t.  $A^* = A, B^* = B$ , then  $(A \cup B)^* = A^* \cup B^* = A \cup B$  by hypothesis. If  $\{U_\alpha\}_{\alpha \in I} \subset \mathcal{P}(X)$  s.t.  $U_\alpha^* = U_\alpha$  for each  $\alpha \in I$ , then  $(\cap_\alpha U_\alpha)^* \subset U_\alpha^* = U_\alpha$  by item 4, thus  $(\cap_\alpha U_\alpha)^* \subset \cap_\alpha U_\alpha$ . But then we must have  $(\cap_\alpha U_\alpha)^* = \cap_\alpha U_\alpha$  by item 2.  $\square$

**Def. (4.4.1.3) [Limits].** For  $X \in \mathcal{Top}$ , and  $(x_n)_{n \in \mathbb{Z}} \subset X$ ,  $y \in X$  is called a **limit of sequence** if for any open subset  $U \subset X$ , there exists some  $N_U \in \mathbb{Z}_+$  s.t. for any  $N > N_U$ ,  $x_N \in U$ .

Let  $S \subset X$  be a subspace, then  $x \in X$  is called a **limit point** of  $S \subset X$  if for any open subset  $U \subset X$ ,  $(U \cap S) \setminus \{x\} \neq \emptyset$ .

$x \in X$  is called an **isolated point** of  $X$  if  $x$  is not a limit point of  $X \subset X$ . Equivalently,  $\{x\} \subset X$  is open. A closed subspace of  $X$  without isolated points is called a **perfect subspace**.

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**Def. (4.4.1.4) [Semi-continuous Functions].** A function from  $X \rightarrow [-\infty, \infty]$  is called **upper semi-continuous** iff  $f^{-1}([-\infty, a))$  are all open. It is called **lower semi-continuous** iff  $f^{-1}((a, \infty])$  are all open.

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**Def. (4.4.1.5) [Separable Topological Spaces].** A **separable topological space** is a topological space (4.4.1.1) it has a countable dense subset.  $\lrcorner$

**Def. (4.4.1.6) [Product Topology].** Arbitrary product exists in  $\mathcal{Top}$ , and it is constructed as follows: for a family of topology spaces  $(X_i)_I$ , the product  $\prod_I X_i$  is the set-theoretic product endowed with the topology generated by the basis  $\pi_i^{-1}(U_i)$  for  $U_i$  open in  $X_i$ .  $\lrcorner$

**Prop. (4.4.1.7) [Limits and Colimits].** Arbitrary limits and colimits exist in the category of topological spaces. The limits is given as a subspace of the product topology, and the colimits  $X = \text{colim } X_i$  is given a topology that  $U \subset X$  is open iff  $U \cap X_i$  is open for each  $i$ .  $\lrcorner$

**Prop. (4.4.1.8) [Pullback Space].** Let  $E \rightarrow X$  and  $f : X' \rightarrow X$  be maps of spaces, then there is a pullback map  $f^*E = E \times_X X' \rightarrow X'$ , called the **pullback space**.  $\lrcorner$

**Def. (4.4.1.9) [Quotient Topology].** Let  $f : X \rightarrow Y$  be a surjective map of spaces, and  $X$  has a topology, then we can define a **quotient topology** on  $Y$  that  $U \subset Y$  is open iff  $f^{-1}(U)$  is open in  $X$ . It has the universal property that any continuous map  $X \rightarrow Z$  that factors through  $f$  set-theoretically factors through  $f$  as a continuous map.

Such a map is called a **quotient map**.  $\lrcorner$

**Prop. (4.4.1.10).** A surjective open map  $\text{pr}$  is a quotient map.  $\lrcorner$

*Proof:* It is clearly that a subset  $U$  is open iff  $\text{pr}^{-1}(U)$  is open.  $\square$

**Def. (4.4.1.11) [Glueing Space].** Let  $A \subset X \in \mathcal{Top}$  and  $f : A \rightarrow Y \in \mathcal{Top}$ , then we have the glueing space  $Y \coprod_f X$ .  $\lrcorner$

**Def. (4.4.1.12) [Mapping Cylinder].** Let  $f : X \rightarrow Y$  be a map, then we define the **mapping cylinder**  $M(f) = Y \coprod_f X \times I$ , where  $X \times \{0\} \subset X \times I$  mapsto  $Y$  by  $f$ .  $\lrcorner$

**Def. (4.4.1.13) [Cone].** For  $X \in \mathcal{Top}$ , define the **cone over**  $X$  to be the space  $C(X) = (X \times I)/(X \times \{1\})$ . Notice that  $C(\emptyset) = \text{pt}$ .  $\lrcorner$

**Def. (4.4.1.14) [Mapping Cone].** Let  $f : X \rightarrow Y$  be a map, then we define the **mapping cone**  $C(f) = M(f)/X \times \{1\}$ .  $\lrcorner$

**Lemma (4.4.1.15).** If  $f : X \rightarrow Y$  is a surjective continuous map that  $f(E) \neq Y$  for any proper closed subspace of  $X$ , then for any  $U \subset X$  open,  $f(U) \subset \overline{Y \setminus f(X \setminus U)}$ .  $\lrcorner$

*Proof:* Take  $y \in f(U)$  and a nbhd  $V$  of  $y$  in  $Y$ , we show that  $V$  intersect  $Y \setminus f(X \setminus U)$ :  $W = U \cap f^{-1}(V)$  is nonempty, thus  $f(X \setminus W) \neq Y$ , take  $y' \in Y \setminus f(X \setminus W)$ , then it is clear  $y' \in Y \setminus f(X \setminus U)$ , and  $y' \in V$ .  $\square$

**Def. (4.4.1.16) [Locally Closed Subset].** A subset  $Z$  of  $X$  is called **locally closed** if for any  $z \in Z$ , there is a nbhd  $U$  of  $z$  in  $X$  that  $U \cap Z$  is closed in  $Z$ . Equivalently, a locally closed subset is the intersection of an open subset with a closed subset.  $\lrcorner$

*Proof:* An intersection of an open subset and a closed subset is clearly locally closed. Conversely, if  $Z$  is locally closed, we choose for each  $z \in Z$  a nbhd  $U_z$  that  $U_z \cap Z$  is closed in  $U_z$ , then we can show  $Z = \overline{Z} \cap \bigcup_z U_z$ . This is because if  $x \in \overline{Z} \cap \bigcup_z U_z$ , then  $x \in U_z$  for some  $z$ , and also  $x \in \overline{Z}$ , thus  $x$  is in  $Z$ .  $\square$

### Filter Langrange

**Def. (4.4.1.17) [Convergence and Filter].** For a filter  $\mathcal{F}$  on a topological space,  $\mathcal{F}$  **converges** to a point  $y$  iff any open set containing  $y$  is in  $\mathcal{F}$ .

If  $X$  is a set and  $Y$  is a topological space and  $X \rightarrow Y$  is a function, then  $y \in Y$  is a  **$\mathcal{F}$ -limit of  $f$**  if  $f_*\mathcal{F}$  converges to  $y$ .  $\lrcorner$

**Prop. (4.4.1.18) [Ultrafilter Convergence Theorem].** Let  $X, Y$  be a topological space, then:

1.  $Y$  is compact iff any ultrafilter on  $Y$  has a limit point.
2.  $Y$  is Hausdorff iff any ultrafilter on  $Y$  has at most one limit point.
3. a function  $f : X \rightarrow Y$  is continuous iff for any filter on  $X$  converging to  $x$ , the filter  $f_*(\mathcal{F})$  converges to  $y$ .

$\lrcorner$

*Proof:*

1. If  $Y$  is compact but every point is not a limit point, then for any  $x$ , there is an open set  $U_x$  that  $U_x \notin \mathcal{F}$ , but then f.m. of them covers  $Y$ , which is in  $\mathcal{F}$ , so one of them must be in  $\mathcal{F}$  by (2.2.10.7), contradiction.  
Conversely, if  $\cup_I U_i = X$  but no finite union of them cover  $X$ , then  $X - U_i$  satisfies the finite intersection property, so there is an ultrafilter containing all  $X - U_i$  by (2.2.10.4) and (2.2.10.5). Then clearly any point  $x$  is not a limit point of  $\mathcal{F}$ .
2. If  $Y$  is Hausdorff and  $x, y$  are both limit point of a filter  $\mathcal{F}$ , then there are two non-intersecting nbhd of them in  $\mathcal{F}$ , so its intersection  $\emptyset \in \mathcal{F}$ , contradiction.  
Conversely, if  $x, y$  are two point that their nbhds both intersect, then their nbhds together satisfies the finite intersection property, so there is an ultrafilter containing all of them, by (2.2.10.4) and (2.2.10.5), thus converging to both  $x$  and  $y$ .
3. This is an easy consequence considering the filter of all the nbhd containing  $x$ .

$\square$

### Connected Components

**Def. (4.4.1.19) [Connectedness].**  $X \in \mathcal{T}\text{op}$  is called **connected** if it satisfies: for any open subset  $U, V$  of  $X$  that  $U \cup V = X$  and  $U \cap V = \emptyset$ , either  $U = \emptyset$  or  $V = \emptyset$ .

$X$  is called **locally connected** if there is a basis of  $X$  consisting of connected open subsets of  $X$ .  $\lrcorner$

**Def. (4.4.1.20) [Path-Connectedness].** A space  $X$  is called **path-connected** if any two points of  $X$  can be connected by an arc.

$X$  is called **locally path-connected** if there is a basis of  $X$  consisting of path-connected open subsets of  $X$ .  $\lrcorner$

**Prop. (4.4.1.21).** If  $X$  is an ordered set with the least upper bound property and satisfies: for any  $x < y \in X$ , there exists  $z \in X$  s.t.  $x < z < y$ . Then  $X$  is connected, so are intervals and rays in  $X$ .  $\lrcorner$

*Proof:* Let  $Y$  be an interval or ray in  $X$ . Suppose for contradiction  $Y = A \amalg B$ , where  $A, B$  are open and non-empty in  $Y$ . Choose  $a \in A, b \in B$ , then  $[a, b] \subset Y$ . Let  $A_0 = A \cap [a, b]$  and  $B_0 = B \cap [a, b]$ , and  $c = \sup A_0 \in [a, b]$ .

Then  $c \notin B_0$ : If  $c \in B_0$ , then  $c \neq a$ , and because  $B_0$  is open, there exists some  $a \leq d < c$  s.t.  $(d, c] \subset B_0$ . So  $d$  is a smaller upper bound for  $A_0$ , contradiction.

And also  $c \notin A_0$ : If  $c \in A_0$ , then  $c \neq b$ , and because  $A_0$  is open, there exists some  $c < e \leq b$  s.t.  $[c, e) \subset A_0$ . Then by hypothesis, there exists  $z \in (c, e) \subset A_0$ , contradicting the fact  $c$  is an upper bound for  $A_0$ .

Then we derived a contradiction that  $c \notin Y$ , which proves  $Y$  is connected.  $\square$

**Def. (4.4.1.22) [Connected Components].** In a topological space  $X$ , if  $x \in X$ , the **connected component** of  $x$  is the maximal connected subspace of  $X$  containing  $x$ . The **path-connected component** of  $x$  is the maximal path-connected subspace of  $X$  containing  $x$   $\lrcorner$

**Prop. (4.4.1.23).**

- Any connected component of  $X$  is closed.
- If  $X$  is locally connected, then any connected component of  $X$  is clopen.
- If  $X$  is locally path-connected, then any path-connected component of  $X$  is open, and any connected component is also open.  $\lrcorner$

**Prop. (4.4.1.24) [Clopen Subsets and Connected Components].** Let  $X$  be a normal topological space and  $x \in X$ , then the connected component of  $X$  containing  $x$  is the intersection of clopen subsets containing  $x$ , denoted by  $A$ .  $\lrcorner$

*Proof:* Assume  $A$  splits into two components  $B, D$ . Since  $A$  is closed,  $B$  and  $D$  are both closed, because  $X$  is normal there are disjoint open neighborhoods  $U$  and  $V$  around  $B$  and  $D$ , respectively. The open sets  $U$  and  $V$  cover the intersection of all clopen neighborhoods of  $A$ , so cause  $X$  is compact, there must exist a finite number of clopen sets around  $A$ , say  $A_1, \dots, A_n$  such that  $U \cup V$  covers  $K = \bigcap_{i=1}^n A_i$ .

Note that  $K$  is clopen. We can assume that  $x \in U$ . It is not difficult to see that  $K \cap U$  is clopen and does not contain all of  $A$ , contradicting the definition of  $A$ .  $\square$

**Cor. (4.4.1.25).** For a compact Hausdorff topological space  $X$  and a point  $x \in X$ , the connected component of  $X$  containing  $x$  is the intersection of all compact open neighborhoods of  $x$ , because  $X$  is normal (4.4.7.6).  $\lrcorner$

**Def. (4.4.1.26) [Totally Disconnected Space].** A space is called **totally disconnected** iff any connected subset of  $X$  contains only one point.  $\lrcorner$

**Prop. (4.4.1.27).** A subspace of a totally disconnected space is totally disconnected, because totally disconnected is equivalent to the only connected subsets are pt sets.  $\lrcorner$

**Def. (4.4.1.28) [Cantor-Sets, Brouwer].** A **Cantor-set** is a nonempty, perfect, compact, metrizable and zero dimensional topological space. Equivalently, it is a topological space homeomorphic to the classical ternary Cantor-set.  $\lrcorner$

*Proof:*  $\square$



### Extremally Disconnected Space

**Def. (4.4.1.29) [Extremally Disconnected Space].** A topological space  $S$  is called **extremally disconnected** if the closure of any open subset of  $X$  is open.  $\lrcorner$

**Prop. (4.4.1.30).** Let  $X$  be an extremally disconnected space, If  $U, V$  are disjoint open subsets of  $X$ , then  $\overline{U}, \overline{V}$   $\lrcorner$

*Proof:* Because  $V \cap \overline{U} = \emptyset$ , thus similarly  $\overline{V} \cap \overline{U} = \emptyset$ .  $\square$

**Lemma (4.4.1.31).** Let  $f : X \rightarrow Y$  be a continuous surjective map of compact Hausdorff spaces that  $Y$  is extremally disconnected and  $f(Z) \neq Y$  for any proper closed subspace of  $X$ , then  $f$  is a homeomorphism.  $\lrcorner$

*Proof:* By (4.4.2.11) it suffices to show that  $f$  is injective. Suppose  $f(x) = f(x') = y$ , then choose disjoint nbhd  $U, U'$  of  $x, x'$ , and  $T = f(X \setminus U), T' = f(X \setminus U')$  closed in  $Y$ , then  $Y = T \cup T'$  and  $y$  is contained in the closure of  $Y \setminus T$  and  $Y \setminus T'$  by (4.4.1.15), but this contradicts (4.4.1.30).  $\square$

**Prop. (4.4.1.32) [Projective Spaces].** Compact Hausdorff extremally disconnected spaces are exactly the projective objects in the category of compact Hausdorff spaces.  $\lrcorner$

*Proof:* Assume  $X$  is projective, let  $U \subset X$  be open, and the complement by  $Z$ , then consider the surjection  $\overline{U} \amalg Z \rightarrow X$ , and let  $\sigma$  be the projection, then  $\sigma(U) \subset \overline{U}$ , thus  $\sigma^{-1}(\overline{U}) = \overline{U}$ , and it is open.

Conversely, if  $X$  is extremally disconnected, then by (4.4.2.13), there is a compact subset  $E \subset Y$  that  $f(E) = X$  and  $f(E') \neq X$  for all closed subspace  $E' \subset E$ . Then (4.4.1.31) says  $f|_E$  is a homeomorphism, and the inverse of it gives a desired section.  $\square$

**Prop. (4.4.1.33) [Gleason].** In an extremally disconnected space  $X$ , a convergent sequence is eventually constant. In particular,  $\mathbb{Z}_p$  is a profinite group that is not extremally disconnected.  $\lrcorner$

*Proof:* Cf. [Projective Topological Spaces]  $\square$

## 2 Compactness

**Def. (4.4.2.1) [Quasi-Compact Space].** A topological space is called **compact** or **quasi-compact** iff any open covering of it has a finite sub-covering. A subspace of a topological space is called **precompact** if its closure is compact.  $\lrcorner$

**Def. (4.4.2.2) [Quasi-Compact Morphism].** A map of topological spaces is called a **quasi-compact morphism** if the inverse image of any quasi-compact open subset is quasi-compact open.  $\lrcorner$

**Prop. (4.4.2.3) [Alexander Subbase Theorem].** A topological space is compact iff the closed subsets has the finite intersection property (2.2.10.3). In fact, it suffices to show that the family of complements of a subbasis of open sets has the finite intersection property.  $\lrcorner$

*Proof:* Cf. [Sta]08ZP.  $\square$

**Prop. (4.4.2.4).** Let  $X$  be a totally ordered set with the least upper bound property, then each closed interval of  $X$  is compact in the order topology. In particular, this applies to a complete totally ordered set  $X$  (2.2.3.22).  $\lrcorner$



*Proof:* Let  $a < b$  and  $\mathcal{U}$  a covering of  $[a, b]$ . We first show that for any  $x \in [a, b]$ , there exists some  $x < y \leq b$  that  $[x, y]$  can be covered by at most two elements of  $\mathcal{U}$ : If  $x$  has an immediate successor, then take  $y$  to be it. If  $x$  has no immediate successor, choose an element  $U$  of  $\mathcal{U}$  containing  $x$ , then  $U$  contains some  $[x, c)$ . Choose  $y \in [x, c)$ , then  $[x, y]$  is covered by a single element of  $\mathcal{U}$ .

Now let  $C$  be the set of points  $y \in (a, b]$  that  $[a, y]$  can be covered by f.m. elements of  $\mathcal{U}$ . We showed before this set is non-empty. Let  $c$  be the least upper bound of  $C$ , then  $a < c \leq b$ .

Next, we show  $c \in C$ : take an element  $U$  of  $\mathcal{U}$  containing  $c$ , then  $U$  contains some  $(d, c]$ . There must be some element of  $C$  lying in the interval  $(d, c]$ , otherwise  $d$  is an upper bound of  $C$ . Then  $[a, y]$  is covered by f.m. elements of  $\mathcal{U}$ , so  $c \in C$ .

Finally, we show  $c = b$ , but this is because otherwise we can find  $c < y \leq b$  that  $[c, y]$  is covered by 2 elements, thus  $y \in C$ , so  $y \leq c$ , contradiction.  $\square$

**Prop. (4.4.2.5) [Tychonoff].** An arbitrary direct product of compact topological spaces is compact.  $\lrcorner$

*Proof:* We prove the finite intersection property. If  $A$  is a family of subsets that any finite intersection of closure of them is nonempty, then consider a maximal family  $\mathfrak{D}$  of subsets containing  $A$  that any finite intersection of closures of them is nonempty, it exists by Zorn's lemma. Consider the projection of  $\mathfrak{D}$  onto a coordinate, then by Hypothesis, it has an intersection  $x_\alpha$ . Now we want to show  $x = (x_\alpha)$  belongs to each  $D \in \mathfrak{D}$ .

If  $U_\beta$  is any subbasis element containing  $x$ , then  $U_\beta$  intersect each  $\mathfrak{D}$  because  $x_\beta \in \pi_\beta(D)$ , so it is in  $\mathfrak{D}$ , by maximality of  $\mathfrak{D}$ . So the finite intersections are also in  $\mathfrak{D}$ , so all local basis of  $x$  are in  $\mathfrak{D}$ . This means that local basis intersect each element of  $\mathfrak{D}$ , that is, all closure of elements in  $\mathfrak{D}$  contains  $x$ .  $\square$

**Def. (4.4.2.6) [Sequentially Compact].** A subset  $A$  in a space  $X$  is called **sequentially compact** iff any sequence of points in  $A$  has a convergent subsequence in  $X$ . It is called **self sequentially compact** if it is sequentially compact in itself.  $\lrcorner$

**Prop. (4.4.2.7).**  $f : X \rightarrow Y$ ,  $X$  is compact and  $Y$  is Hausdorff, then for a descending chain  $Y_i$  of closed subsets of  $X$ ,

$$f\left(\bigcap_n Y_n\right) = \bigcap_n f(Y_n).$$

$\lrcorner$

*Proof:* The left side is compact, so if  $x \notin f(\bigcap_n Y_n)$ , there is a closed subsets  $x \in T$  that  $T \cap f(\bigcap_n Y_n) = \emptyset$ , so  $f^{-1}(T) \cap Y_n = \emptyset$ , so  $f^{-1}(T) \cap Y_n = \emptyset$  for some  $n$ , hence  $x \notin f(Y_n)$ .  $\square$

**Prop. (4.4.2.8) [Fixed Point Theorem].** If  $X$  is a compact metric space  $M$ ,  $T$  is a continuous map  $X \rightarrow X$  that  $d(x, y) < d(Tx, Ty)$ , then  $T$  has a unique fixed point in  $X$ .  $\lrcorner$

*Proof:* The uniqueness is obvious, for the existence, first notice  $T$  is obviously continuous, so consider  $d(x, Tx)$ , this is a continuous function on  $M$ , so it contains a minimum value, if it not 0, then  $d(Tx, T^2x) < d(x, Tx)$ , which is a contradiction.  $\square$

**Def. (4.4.2.9) [Proper Map].** A **proper map** is a continuous map s.t. the inverse image of compact subsets are compact.  $\lrcorner$

**Prop. (4.4.2.10).** If  $X$  is compact and  $Y$  is Hausdorff, then a continuous map  $f : X \rightarrow Y$  is proper.  $\lrcorner$

**Cor. (4.4.2.11).** A continuous bijective map from a compact space to a Hausdorff space is a homeomorphism.  $\lrcorner$

**Prop. (4.4.2.12) [Proper Continuous Maps Closed].** Let  $f : X \rightarrow Y$  be a proper continuous map with  $Y$  locally-compact Hausdorff, then  $f$  is a closed map.  $\lrcorner$

*Proof:* Let  $K \subset X$  closed, and  $y \in \overline{f(K)}$ . Choose precompact open nbhd  $U$  of  $y$ , then  $f^{-1}(\overline{U})$  is compact in  $X$ , so  $f(K \cap f^{-1}(\overline{U})) = f(K) \cap \overline{U}$  is compact and thus closed in  $Y$ . Then  $y \in f(K)$ , and  $f(K)$  is closed.  $\square$

**Lemma (4.4.2.13).** Let  $f : X \rightarrow Y$  be a continuous map of compact Hausdorff spaces, then there exists a smallest closed subset  $E$  of  $X$  that  $f(E) = Y$ .  $\lrcorner$

*Proof:* Use Zorn's lemma, noticing that the intersection of a chain of possible  $E_i$ s also maps to  $Y$ , by the intersection property (4.4.2.3).  $\square$

### Stone-Čech Compactification

**Def. (4.4.2.14) [Stone-Čech Compactification].** The **Stone-Čech Compactification**  $\beta$  is defined to be a functor from the category of sets to the category of compact Hausdorff space that is left adjoint to the forgetful functor.

The construction of  $\beta(X)$  is as follows:  $\beta X$  = the set of all ultrafilters on  $X$ , and the topology is generate by  $U_A = \{\mathcal{F} | A \in \mathcal{F}\}$  as a basis of clopen subsets. For a map  $f : X \rightarrow Y$ , the map  $\beta X \rightarrow \beta Y$  is given by  $f_*$ .  $\lrcorner$

*Proof:* First  $\beta X$  is a compact Hausdorff space: it is compact because if there are sets  $A_i$  that any ultrafilter contains at least one of them, then f.m. of them must cover  $X$ , otherwise  $X - A_i$  satisfies the finite intersection property thus is contained in some ultrafilter, by (2.2.10.4) and (2.2.10.5), contradiction, Then by (2.2.10.7) shows that any ultrafilter contains one of them. It is Hausdorff because for any two different ultrafilter, there must be an  $A$  that  $A \in \mathcal{F}_1$  and  $X - A \in \mathcal{F}_2$ .  $f_*$  is continuous because  $f^{-1}(U_A) = U_{f^{-1}(A)}$ .

Now for any map  $f : X \rightarrow Y$  where  $Y$  is a topological space, map an ultrafilter  $\mathcal{F}$  to the unique limit point of  $f_*\mathcal{F}$  in  $Y$  (existence and uniqueness by (4.4.1.18)). This map is continuous from  $\beta X$  to  $Y$  because for any open set  $V \subset Y$ ,  $U_{f^{-1}(V)}$  is mapped into  $V$ . And for any  $\beta X \rightarrow Y$  continuous, consider  $X \rightarrow Y$  which maps  $x$  to the image of the principle ultrafilter  $\mathcal{F}_x$  in  $Y$ .

This two map are mutually converses to each other, first for a  $f : X \rightarrow Y$ ,  $X \rightarrow \beta X \rightarrow Y$  is  $f$  itself, because the pushout of the principle ultrafilter clearly converges to  $f(x)$ . And for a  $\beta X \rightarrow Y$ , if  $\mathcal{F}$  doesn't map to  $\lim f_*\mathcal{F}$  but mapped to some  $t$ , then by definition, there is a nbhd  $U$  of  $t$  that  $f^{-1}(U) \notin \mathcal{F}$ , but by continuity, there is a  $\mathcal{F} \in U_B$  mapped into  $U$ . But then  $B \in f^{-1}(U)$ , otherwise if  $x \in B - f^{-1}(U)$ , then  $\mathcal{F}_x$  is mapped to  $U$ , contradiction, so  $f^{-1}(U)$  containing  $B$  is also in  $\mathcal{F}$ , contradiction.  $\square$

**Lemma (4.4.2.15).** In fact, the spaces in the image of the Stone-Čech compactification are all profinite spaces.  $\lrcorner$

*Proof:* As shown before, for any two different ultrafilter, there must be an  $A$  that  $A \in \mathcal{F}_1$  and  $X - A \in \mathcal{F}_2$ ,  $U_A$  is open and closed.  $\square$

**Prop. (4.4.2.16) [Stone Representation Theorem].** The Stone-Čech compactification  $\beta$  gives an equivalent of categories from the category of Boolean algebras to the category of profinite spaces.  $\lrcorner$

*Proof:*  $\beta B$  is a profinite space by lemma (4.4.2.15), and  $B$  can be recovered from  $\beta B$  as the Boolean algebra of all clopen subsets of  $\beta B$ , because  $\beta B$  is compact. This is a inverse isomorphism because ?  $\square$

**Prop. (4.4.2.17).** The Stone-Čech compactification of a set  $\beta S_0$  is extremally disconnected.  $\lrcorner$

*Proof:* We check condition (4.4.1.32): For any surjection  $S' \rightarrow \beta S_0$ , we may take a lift  $S_0 \rightarrow S$  arbitrarily, then by definition there is a morphism  $\beta S_0 \rightarrow S'$  extending this morphism. It is a section by the universal property.  $\square$

### Locally-Compact Space

**Def. (4.4.2.18) [Locally-Compact Space].** A Hausdorff space is called **locally-compact** if for any point  $x$ , there is a compact subset containing a nbhd of  $x$ .  $\lrcorner$

**Prop. (4.4.2.19).** If  $X$  is Hausdorff, then  $X$  is locally-compact iff for any  $x \in X$  and  $x \in U \subset X$  open, there exists a precompact nbhd  $V$  that  $x \in V \subset \bar{V} \subset U$ .  $\lrcorner$

*Proof:* One direction is trivial. For the other, for any  $x \in X$  and a nbhd  $U$  of  $x$ , let  $U_0$  be a precompact nbhd of  $X$ , then  $\bar{U}_0 \setminus U$  is closed thus compact and disjoint from  $x$ . Because  $X$  is Hausdorff, we can find a nbhd  $V'$  of  $x$  that  $\bar{V}'$  is disjoint from  $\bar{U}_0 \setminus U$ . Now let  $V = V' \cap U_0$ , then  $\bar{V} \subset \bar{U}_0$  is closed thus compact, and  $\bar{V} \subset \bar{V}' \cap \bar{U}_0 \subset U$ .  $\square$

**Cor. (4.4.2.20).** Open subsets and closed subsets of a locally-compact Hausdorff space  $X$  is locally-compact Hausdorff.  $\lrcorner$

*Proof:* If  $A \subset X$  is closed and  $x \in A$ , then there exists a precompact nbhd  $U$  of  $x \in X$ , then  $A \cap \bar{U}$  is closed in  $\bar{U}$  thus compact, and contains the nbhd  $U \cap A$  of  $x \in A$ , so  $A$  is locally-compact.

If  $A \subset X$  is open,  $x \in A$ , let  $x \in U \subset A$ , and  $U$  open in  $X$ , then we can apply (4.4.2.19) to find a precompact nbhd  $V$  that  $x \in V \subset \bar{V} \subset U$ , so  $A$  is locally-compact.  $\square$

**Prop. (4.4.2.21) [One-Point Compactification].** Let  $X$  be a space, then  $X$  is locally-compact Hausdorff iff there exists a compact Hausdorff space  $Y$  containing  $X$  s.t.  $Y \setminus X$  is a single point. Moreover, in this case,  $Y$  is unique up to homeomorphisms, called the **one-point compactification** of  $X$ .  $\lrcorner$

*Proof:* Cf. [Mun00]P183.  $\square$

**Cor. (4.4.2.22).** A space  $X$  is homeomorphic to an open subset of a compact open subspace iff it is locally-compact Hausdorff, by (4.4.2.21) and (4.4.2.20).  $\lrcorner$

**Prop. (4.4.2.23).** A locally-compact second countable Hausdorff space  $X$  has a countable basis consisting of precompact open subsets. In particular,  $X$  is  $\sigma$ -compact.  $\lrcorner$

*Proof:* Choose a countable basis  $\{U_n\}$  of  $X$ . For any  $x \in X$  and any nbhd  $U$  of  $x$ , there exists a precompact  $U_x \subset U$  containing  $x$  by (4.4.2.19). Then for some  $n(x)$ ,  $U_{n(x)} \subset U_x$  containing  $x$ , so  $U_{n(x)}$  is precompact. Thus the set of  $U_n$  that is precompact forms a countable basis of  $X$ .  $\square$

**Prop. (4.4.2.24).** If  $f : X \rightarrow Y$  is a quotient map and  $Z$  is locally-compact, then  $X \times Z \rightarrow Y \times Z$  is also a quotient map.  $\lrcorner$

*Proof:* Consider  $W$  as  $Y \times Z$  in the quotient map topology, then  $X \times Z \rightarrow W$  is continuous, which means  $X \rightarrow \text{Map}(Z, W)$  is continuous. But then  $Y \rightarrow \text{Map}(Z, W)$  is continuous because  $Y$  is a quotient map. Applying (4.4.3.7), we see  $Y \times Z \rightarrow W$  is continuous.  $W \rightarrow Y \times Z$  is continuous by quotient map hypothesis, so  $Y \times Z \cong W$ .  $\square$

### Compactly Generated Spaces

**Def. (4.4.2.25) [Compactly Generated Spaces].** A **compactly generated space** is a space that is the colimit of its compact Hausdorff subspaces. The category of compactly generated spaces is denoted by  $\mathcal{CG}$ .  $\lrcorner$

**Prop. (4.4.2.26).** For  $X \in \mathcal{CG}$  and  $Y \in \mathcal{Top}$ , a set-theoretical map  $f : X \rightarrow Y$  is continuous iff for any  $K \in \mathcal{Top}$  compact Hausdorff and a map  $i : K \rightarrow X$ , the composite map  $K \rightarrow Y$  is continuous.  $\lrcorner$

**Prop. (4.4.2.27) [Compact Generating Functor].** There is a **compact generating functor**  $(-)_c : \mathcal{Top} \rightarrow \mathcal{CG}$  right adjoint to the inclusion functor.  $\lrcorner$

*Proof:*  $k$  is constructed by  $X \mapsto \varinjlim_{K \subset X, K \text{ compact}} K$ . Notice  $X_c = X$  set-theoretically. It is easy to verify that if  $Y \in \mathcal{CG}$  and  $Y \rightarrow X$  is continuous, then the set-theoretical map  $Y \rightarrow X_c$  is also continuous. So  $(-)_c$  is right adjoint to the inclusion functor.  $\square$

**Cor. (4.4.2.28).**  $\mathcal{CG}$  is both complete and cocomplete, and the colimits coincide with colimits in  $\mathcal{Top}$ .  $\lrcorner$

**Cor. (4.4.2.29).**  $\mathcal{CG}$  is Cartesian closed (4.1.5.8).  $\lrcorner$

**Cor. (4.4.2.30).** The right adjoint to the Cartesian product is the compactification of the mapping space functor (4.4.3.1), by (4.4.3.6) and (4.4.2.27).  $\lrcorner$

**Prop. (4.4.2.31).** Locally compact Hausdorff topological spaces are compactly generated.  $\lrcorner$

*Proof:* For  $X \in \mathcal{Top}$ , if  $Z \subset X$  satisfies  $Z \cap K$  is closed for any compact Hausdorff subset  $K \subset X$ , for any  $x \in X$ , there exists a precompact nbhd  $U$  of  $x \in X$ , so  $Z \cap \overline{U}$  is closed, and  $Z \cap U$  is closed in  $U$ . Thus  $Z$  is closed in  $X$ .  $\square$

**Prop. (4.4.2.32).** Quotients and closed subspaces of a compactly generated space are compactly generated. Colimits of compactly generated spaces are compactly generated.  $\lrcorner$

*Proof:* These follow from (4.4.2.25).  $\square$

**Prop. (4.4.2.33).** If  $X \in \mathcal{CG}$  and  $Y$  is locally-compact Hausdorff, then  $X \times Y \in \mathcal{CG}$ .  $\lrcorner$

*Proof:* Notice by (4.4.3.7), a map  $X \times Y \rightarrow Z$  is continuous iff  $C \times Y \rightarrow Z$  is continuous for any compact Hausdorff subset  $C \subset X$ . The assertion is equivalent to  $X \times Y \rightarrow (X \times Y)_c$  is continuous, which is then equivalent to  $C \times Y \rightarrow (X \times Y)_c$  continuous for any compact subset  $C \subset X$ . But  $Y$  is compactly generated, and  $C$  is locally-compact, thus it suffices to show  $C \times C' \rightarrow (X \times Y)_c$  is continuous for any compact subset  $C' \subset Y$ . Then this is true because  $C \times C'$  is compact.  $\square$

## 3 Mapping Spaces

### Compact-Open Topology on Mapping Spaces

**Def. (4.4.3.1) [Compact-Open Topology].** The **compact-open topology** on a function space  $\text{Map}(X, Y)$  is a topology generated by the sets  $(K, U) = \{f : f(K) \subset U\}$ , where  $K$  is compact in  $X$  and  $U$  is open in  $Y$ .  $\lrcorner$

**Prop. (4.4.3.2).** For  $X' \rightarrow X, Y \rightarrow Y' \in \mathcal{Top}$ , the induced map  $\text{Map}(X, Y) \rightarrow \text{Map}(X', Y')$  is continuous. In particular, if  $X \cong X'$  and  $Y \cong Y'$ , then  $\text{Map}(X, Y) \cong \text{Map}(X', Y')$ .  $\lrcorner$

**Prop. (4.4.3.3).** If  $Y$  is compact and  $X$  a metric space, then the compact open topology on  $\text{Map}(Y, X)$  coincides with the uniform topology on functions.  $\lrcorner$

*Proof:* If  $f \in (K, U)$ , then  $f(K) \subset U$ , then there is a  $\varepsilon > 0$  that  $B(K, \varepsilon) \subset U$ , thus  $B(f, \varepsilon) \subset (K, U)$ .

For any  $f \in \text{Map}(Y, X)$ ,  $f(Y)$  is compact thus there are f.m. open balls  $B(f(y_i), \frac{\varepsilon}{3})$  that covers  $f(Y)$ . Now if  $K_i = \overline{f^{-1}(B(f(y_i), \frac{\varepsilon}{3}))}$ , then  $\cup K_i = Y$ , and  $f \in \cap_i (K_i, B(f(y_i), \frac{\varepsilon}{2}))$ . Also  $\cap_i (K_i, B(f(y_i), \frac{\varepsilon}{2})) \subset B(f, \varepsilon)$ , because for any  $g \in \cap_i (K_i, B(f(y_i), \frac{\varepsilon}{2}))$  and  $y \in K_i$ ,  $|f(y) - g(y)| \leq |f(y) - f(y_i)| + |f(y_i) - g(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ .  $\square$

**Def. (4.4.3.4)[Relative Maps].** Let  $A \subset X$  and  $B \subset Y$ , we denote by  $\text{Map}((X, A), (Y, B))$  the subspace of  $\text{Map}(X, Y)$  consisting of continuous functions  $f : X \rightarrow Y$  that map  $A$  into  $B$ .  $\lrcorner$

**Prop. (4.4.3.5)[Composition is Continuous].** For  $X, Y, Z \in \mathcal{T}\text{op}$ , if  $Y \in \text{LCH}$ , then the composition map

$$\text{Map}(Y, Z) \times \text{Map}(X, Y) \rightarrow \text{Map}(X, Z)$$

is continuous. In particular,  $\text{Map}(Y, Z) \times Y \rightarrow Z$  is continuous.  $\lrcorner$

*Proof:* For any compact  $K \subset X$ , open  $U \subset Y$  and  $g \circ f \in (K, U)$ ,  $f(K) \subset g^{-1}(U) \subset Y$ . Because  $Y$  is locally-compact, there is an precompact open subset  $V$  that  $f(K) \subset V \subset \overline{V} \subset g^{-1}(U)$ , then  $(\overline{V}, U) \times (K, V)$  maps into  $(K, U)$ .  $\square$

**Lemma (4.4.3.6)[Subbasis].** Let  $X$  be Hausdorff,  $W_\alpha$  be a subbasis of  $Y$ , then  $(K, W_\alpha)$  where  $K \subset X$  compact is a subbasis of  $\text{Map}(X, Y)$ .  $\lrcorner$

*Proof:* If  $f \in (K, U) \subset \text{Map}(X, Y)$ , let  $U = \bigcup_\beta U_\beta$ , where  $U_\beta = \bigcap_{j=1}^{k(\beta)} W_{\beta,j}$ . Now  $K \subset \bigcup_\beta f^{-1}(U_\beta)$ , because  $K$  is compact Hausdorff thus the partition of unity (4.4.7.8) gives us f.m. compact subsets  $K_1, \dots, K_n$  of  $K$  that  $K_i \subset f^{-1}(U_{\beta_i})$  for some  $\beta$ . Then

$$f \in \bigcap_{i=1}^n \bigcap_{j=1}^{k(\beta_i)} (K_i, W_{\beta_i,j}) = \bigcap_{i=1}^n (K_i, U_{\beta_i}) \subset (K, \bigcup_{i=1}^n U_{\beta_i}) \subset (K, U).$$

$\square$

**Prop. (4.4.3.7)[Adjointness of Mapping Space].** For  $Y \in \text{LCH}$ , then

$$\text{Map}(X \times Y, Z) \cong \text{Map}(X, \text{Map}(Y, Z))$$

as sets. And if moreover  $X$  is Hausdorff, then it is a homeomorphism.

In particular, the compact-open topology on  $\text{Map}(Y, Z)$  is the coarsest topology s.t.

$$\text{Map}(Y, Z) \times Y \rightarrow Z$$

is continuous.  $\lrcorner$

*Proof:* Let  $\varphi \in \text{Map}(X \times Y, Z)$ , for any  $x \in X$ , take  $\varphi(x) : Y \rightarrow Z : \varphi(x)(y) = \varphi(x, y)$ , then  $\varphi(x)$  is continuous, and  $\tilde{\varphi} : X \mapsto \text{Map}(Y, Z) : x \mapsto \varphi(x)$  is continuous: for any compact  $K \subset Y$  and open  $U \subset Z$ ,  $\tilde{\varphi}^{-1}((K, U)) = \{x \in X \mid \varphi(x \times K) \subset U\}$ , which is open.

Conversely, for any  $\psi : X \rightarrow \text{Map}(Y, Z)$ , let  $\tilde{\psi} : X \times Y \rightarrow Z$  be given by  $\tilde{\psi}(x, y) = \psi(x)(y)$ . Then  $\tilde{\psi}$  is continuous: For any open  $U \subset Z$ , if  $(x, y) \in \tilde{\psi}^{-1}(U)$ , then  $\psi(x)y \subset U$ . Because  $\psi(x)$  is continuous, there is a nbhd  $W$  of  $y \in Y$  that  $\varphi(x)(W) \subset U$ . Then because  $Y$  is locally-compact

Hausdorff, there is a precompact nbhd  $V$  of  $y \in Y$  that  $y \in V \subset \bar{V} \subset W$ . Then  $\psi^{-1}((\bar{V}, U))$  is a nbhd of  $x \in X$ , and  $(x, y) \in (\psi^{-1}((\bar{V}, U)), V) \subset \tilde{\psi}^{-1}(U)$ , thus  $\tilde{\psi}^{-1}(U)$  is open, and  $\tilde{\psi}$  is continuous.

Now let  $F : \varphi \mapsto \tilde{\varphi}$  and  $G : \psi \mapsto \tilde{\psi}$ , then we show  $F, G$  are both continuous: By (4.4.3.6),  $(K, (L, U))$  is a subbasis of  $\text{Map}(X, \text{Map}(Y, Z))$ .  $F$  is continuous because  $F((K \times L, U)) = (K, (L, U))$ .  $G$  is continuous because  $X \times Y$  is Hausdorff and for any  $J$  compact in  $X \times Y$ ,  $J \times J \xrightarrow{(\text{pr}_1, \text{pr}_2)} X \times Y$  is continuous, so for any  $U \subset Z$  open, there exists  $J_1 \subset X, J_2 \subset Y$  compact s.t.  $G((J_1, (J_2, U))) \subset (J, U)$ .  $\square$

**Cor. (4.4.3.8).** If  $Y$  is locally-compact and Hausdorff,  $A \subset X, B \subset Y, C \subset Z$ , then

$$\text{Map}((X \times Y, X \times B \cup A \times Y), (Z, C)) \cong \text{Map}((X, A), \text{Map}((Y, B), (Z, C)))$$

as sets, and if moreover  $X$  is Hausdorff, then it is a homeomorphism.  $\lrcorner$

**Def. (4.4.3.9) [Admissible Topology].** For  $X, Y \in \mathcal{T}\text{op}$ , an **admissible topology** on  $\text{Map}(X, Y)$  is a topology that makes the evaluation map  $e : \text{Map}(X, Y) \times X \rightarrow Y$  continuous.  $\lrcorner$

**Prop. (4.4.3.10).** The compact-open topology on  $\text{Map}(X, Y)$  is coarser than admissible topology on it (4.4.3.9). And if  $X$  is locally-compact Hausdorff, then it is admissible.  $\lrcorner$

*Proof:* It suffices to show any  $(K, U)$  is open in  $\text{Map}(X, Y)$ . Let  $f \in (K, U)$ , then for any  $x \in K$ ,  $e(f, x) \in U$ , thus there is a nbhd  $U_x$  of  $f$  and a nbhd  $W_x$  of  $x$  that  $e(U_x \times W_x) \subset U$ . Now because  $K$  is compact, we can choose a nbhd  $V$  of  $f$  that  $e(V \times K) \subset U$ . Thus  $f \in V \subset (K, U)$ , and  $(K, U)$  is open in  $\text{Map}(X, Y)$ . If  $Y$  is locally-compact Hausdorff, then  $\text{Map}(Y, Z) \times Y \rightarrow Z$  continuous by applying (4.4.3.5) with  $X = \text{pt}$ .  $\square$

### Mapping Spaces in $\mathcal{CG}$

**Def. (4.4.3.11) [Mapping Spaces].** For  $X, Y \in \mathcal{CG}$ , the mapping space  $\text{Map}_c(X, Y)$  is the compact generation (4.4.2.27) of the space of continuous functions from  $X$  to  $Y$  with the topology generated by the sets  $(K, U) = \{f : f(i(K)) \subset U\}$ , where  $K$  is a compact Hausdorff space,  $i : K \rightarrow X$  and  $U$  is open in  $Y$ .  $\lrcorner$

**Prop. (4.4.3.12).** For  $X' \rightarrow X, Y \rightarrow Y' \in \mathcal{CG}$ , the induced map  $\text{Map}_c(X, Y) \rightarrow \text{Map}_c(X', Y')$  is continuous. In particular, if  $X \cong X'$  and  $Y \cong Y'$ , then  $\text{Map}_c(X, Y) \cong \text{Map}_c(X', Y')$ .  $\lrcorner$

**Lemma (4.4.3.13) [Evaluations are Continuous].** For  $X, Y \in \mathcal{CG}$ , the composition map

$$\text{Map}_c(Y, X) \times_c Y \rightarrow X$$

is continuous.  $\lrcorner$

*Proof:* It suffices to prove that for any compact Hausdorff space  $K, F$  mapping into  $\text{Map}(Y, X), Y$ ,  $K \times F \rightarrow X$  is continuous: For  $(f_0, y) \in \text{Map}_c(Y, X) \times_c Y$  and  $f_0(y) \in U$ , as  $f_0$  is continuous, there exists a nbhd  $N$  of  $y \in Y$  s.t.  $f_0(\overline{N \cap F}) \subset U$ , and then  $(f_0, y) \in (K \cap (\overline{N \cap F}, U), N \cap F)$  is mapped into  $U$ .  $\square$

**Prop. (4.4.3.14) [Adjointness of Mapping Space].** For  $X, Y, Z \in \mathcal{CG}$ , there is a natural homeomorphism

$$\text{Map}_c(X \times_c Y, Z) \cong \text{Map}_c(X, \text{Map}_c(Y, Z))$$

$\lrcorner$

*Proof:* Let  $\varphi \in \text{Map}_c(X \times_c Y, Z)$ , for any  $x \in X$ , take  $\varphi(x) : Y \rightarrow Z : \varphi(x)(y) = \varphi(x, y)$ , then  $\varphi(x)$  is continuous, and  $\tilde{\varphi} : X \mapsto \text{Map}(Y, Z) : x \mapsto \varphi(x)$  is continuous: if  $\varphi(x) \in (K, U)$ ,  $\tilde{\varphi}^{-1}((K, U)) = \{x \in X | \varphi(x \times i(K)) \subset U\}$ , which is open.

To prove this construction is continuous, notice we have already seen that if  $f : X \times_c Y \rightarrow Z$  is continuous then  $f : Y \rightarrow \text{Map}_c(X, Z)$  is continuous. Apply this to

$$\text{Map}_c(Y \times_c X, Z) \times_c Y \times_c X \rightarrow Z$$

we see that

$$X \times_c \text{Map}_c(Y \times_c X, Z) \rightarrow \text{Map}_c(Y, Z)$$

is continuous, and

$$\text{Map}_c(Y \times_c X, Z) \rightarrow \text{Map}_c(X, \text{Map}_c(Y, Z))$$

is continuous. Similarly, as

$$\text{Map}_c(X, \text{Map}_c(X, Z)) \times_c Y \times_c X \rightarrow Z$$

is continuous, we get

$$\text{Map}_c(X, \text{Map}_c(Y, Z)) \rightarrow \text{Map}_c(X \times_c Y, Z)$$

is continuous, and it is clearly the inverse to the construction above.  $\square$

**Cor. (4.4.3.15).** For  $X, Y, Z \in \mathcal{CG}$ ,  $A \subset X, B \subset Y, C \subset Z$ , then there is a natural homeomorphism

$$\text{Map}((X \times Y, X \times B \cup A \times Y), (Z, C)) \cong \text{Map}((X, A), \text{Map}((Y, B), (Z, C)))$$

$\lrcorner$

**Cor. (4.4.3.16).** products in  $\mathcal{CG}$  commute with colimits.  $\lrcorner$

**Cor. (4.4.3.17) [Compositions are Continuous].** For  $X, Y, Z \in \mathcal{CG}$ , the composition map

$$\text{Map}_c(Y, Z) \times_c \text{Map}_c(X, Y) \rightarrow \text{Map}_c(X, Z)$$

is continuous.  $\lrcorner$

*Proof:* It suffices to show that

$$\text{Map}_c(Y, Z) \times_c \text{Map}_c(X, Y) \times_c X \rightarrow Z$$

is continuous, and this follows from (4.4.3.13).  $\square$

### Construction of Spaces

**Def. (4.4.3.18) [Path Spaces].** For  $f : A \rightarrow B \in \mathcal{T}\text{op}$ , let  $E_f$  be the subspace of  $A \times \text{Map}(I, B)$  consisting of pairs  $(a, \gamma)$  that  $\gamma(0) = a$ .

The fibers of  $E_f \rightarrow B$  are called **homotopy fibers** of  $f$ .  $\lrcorner$

**Def. (4.4.3.19) [ $n$ -Loop Space].** The  $n$ -loop space of  $X$  is defined to be  $\Omega^n(X, x_0) = M(I^n, \partial I^n; X, x_0)$ . Then by (4.4.3.8) we have

$$\Omega(\Omega^n(X, x_0), \tilde{x}_0) \cong \Omega^{n+1}(X, x_0).$$

$\lrcorner$

**Prop. (4.4.3.20) [Loop spaces are Homotopy Fibers].** Let  $f : x_0 \rightarrow B$  be a point, then  $E_f$  is the **path space**  $PB$  of all paths starting from  $x_0$ , and the homotopy fiber over  $x_0$  is just the loop space  $\Omega(X, x_0)$ .  $PB$  is contractible, with the contraction given by  $H : PB \times I \rightarrow PB : \gamma_t(x) = \gamma(tx)$ .  $\lrcorner$



## 4 Profinite Spaces

**Def. (4.4.4.1) [Profinite Spaces].** A topological space is called a **profinite space** if it is a cofiltered limit of finite discrete topological spaces. The category of profinite spaces is denoted by  $\mathcal{P}\text{rof}$ .

A profinite space is the same thing as a totally disconnected, compact Hausdorff topological space. Thus a closed subspace of a profinite space is profinite.  $\lrcorner$

*Proof:* The profinite spaces are clearly totally disconnected, compact Hausdorff (by Tychonoff).

Conversely, if it is totally disconnected and compact Hausdorff, let  $\mathcal{I}$  be the set of clopen decompositions  $X = \coprod_I U_i$  of  $X$ , then for each  $I \subset \mathcal{I}$ , there is a map  $X \rightarrow I$ , and there is a partial order on the decompositions of  $X$ . We show that the map  $X \rightarrow \lim_{I \subset \mathcal{I}} I$  is a homeomorphism. It is injective by (4.4.1.26)(4.4.1.24)(4.4.7.6). It is surjective by compactness of  $X$ , and it is clearly open, thus homeomorphism by (4.4.2.11).  $\square$

**Cor. (4.4.4.2).** A cofiltered limit of profinite spaces is profinite.  $\lrcorner$

**Prop. (4.4.4.3).** Any open covering of a profinite space has a clopen disjoint subcover.  $\lrcorner$

*Proof:* By (4.4.4.1), we may assume that  $X = \lim_{i \in I} X_i$ , where  $X_i$  is finite. Let  $f_i$  be the projection, as the limit is filtered, a fundamental family of nbhds of a point  $(x_i)_{i \in I}$  is  $\{f_j(x_j)\}_{j \in I}$ . Then for each covering, we may assume it is finite  $X = \cup_{i \in I} f_i^{-1}(x_i)$ , choose a  $j > i$  for each  $i$ , as  $I$  is cofiltered, then  $X = \coprod_{x \in X_j} f_j^{-1}(x)$  satisfies the desired property.  $\square$

**Prop. (4.4.4.4).** If  $X$  is quasi-compact and any connected component of  $X$  is the intersection of clopen sets containing it (e.g.  $X$  is normal (4.4.1.24)), then  $\pi_0(X)$  is a profinite space.  $\lrcorner$

*Proof:*  $\pi_0(X)$  is an image of  $X$ , so it is quasi-compact, also it is clearly totally disconnected. To show it is Hausdorff, let  $C, D$  be disjoint connected components of  $X$ , then  $C = \cap U_\alpha$ , where  $U_\alpha$  are clopen. Since  $C \cap D = \emptyset$ ,  $U_\alpha \cap D = \emptyset$  for some  $\alpha$ . and then the image of  $U_\alpha$  separates  $C$  and  $D$  in  $\pi_0(X)$ .  $\square$

### Locally Profinite Space

**Def. (4.4.4.5) [Locally Profinite Space].** A **locally profinite space** is a totally disconnected, locally-compact Hausdorff topological space. The category of locally profinite space is denoted by  $\text{loc}\mathcal{P}\text{rofSpa}$ .  $\lrcorner$

**Prop. (4.4.4.6).** A locally closed subsets of a locally profinite space is locally profinite. And compact subsets are profinite.  $\lrcorner$

*Proof:* Closed subsets are clearly locally profinite, for the open subsets, it is also locally-compact.  $\square$

**Cor. (4.4.4.7).** Any open covering of a compact subsets of a locally profinite space has an clopen disjoint subcover, by (4.4.4.3).  $\lrcorner$

**Prop. (4.4.4.8).** The set of all compact open subsets form a basis of the topology of  $G$ .  $\lrcorner$

**Prop. (4.4.4.9).** If  $X \in \text{loc}\mathcal{P}\text{rofSpa}$  and  $K$  is a compact subspace of  $X$ . Let  $K \subset \cup U_\alpha$  be an open covering, then there exist f.m. disjoint open compact subsets  $V_i \subset X$  that each  $V_i \cap K \subset U_\alpha$  for some  $\alpha$ , and  $K \subset \cup V_i$ .  $\lrcorner$

*Proof:* Because  $K$  is profinite (4.4.4.6), (4.4.4.3) shows there is a finite disjoint compact open subcover  $W_i$  of  $U_\alpha$ , then  $W_i = V_i \cap K$  for  $V_i$  compact, and using compactness of  $W_i$ , we can assume  $V_i$  is compact. Finally to make  $V_i$  disjoint, we can let  $V_i = V_i \setminus (\cup_{k=1}^{i-1} V_k)$ .  $\square$



### Structure Sheaves and Distributions

References are [Representations of the Group  $\mathrm{GL}(n; F)$  over Local Fields, Bernstein-Zelevinsky] and [Representation of  $p$ -adic Groups, Bernstein].

**Notation (4.4.4.10).**

- Use notations on sheaf theory defined in ?.

┘

**Def. (4.4.4.11) [Structure Sheaves and Distributions].** For  $X \in \mathrm{locProfSpa}$ , the structure sheaf  $C_X^\infty$  on  $X$  is defined to be the locally constant sheaf  $\underline{\mathbb{Z}}$  on  $X$ .

The space  $C_c^\infty(X)$  of **test functions** (or **smooth functions**) on  $X$  is the set of locally constant continuous functions with compact supports.

For  $W \in \mathcal{Ab}$ , the space of  $(W$ -valued) **distributions (profinite)** on  $X$  with values in  $W$  is defined to be  $\mathrm{Dist}(X, W) \triangleq \mathrm{Hom}(C_c^\infty(X), W)$ . And denote  $\mathrm{Dist}(X) = \mathrm{Dist}(X, \mathbb{C})$ .

More generally, if  $\mathcal{F}$  is a  $C_X^\infty$ -sheaf, then the space of **distributions on  $\mathcal{F}$**  on  $X$  valued in  $W$  is defined to be  $\mathrm{Dist}(X, \mathcal{F}; W) \triangleq \mathrm{Hom}(\Gamma_c(X, \mathcal{F}), W)$ . ┘

**Prop. (4.4.4.12) [Distributions, Bernstein-Zelevinsky[B-Z76]].** If  $X \in \mathrm{locProfSpa}$ ,  $\mathcal{F} \in \mathrm{Mod}(C_X^\infty)$ ,  $W \in \mathcal{Ab}$ , then for any open subset  $U \subset X$  and  $Z = X \setminus U$ , there is an exact sequence:

$$0 \rightarrow \Gamma_c(U, \mathcal{F}; W) \rightarrow \Gamma_c(X, \mathcal{F}; W) \rightarrow \Gamma_c(Z, \mathcal{F}; W) \rightarrow 0.$$

thus also an exact sequence:

$$0 \rightarrow \mathrm{Dist}(Z, \mathcal{F}; W) \rightarrow \mathrm{Dist}(X, \mathcal{F}; W) \rightarrow \mathrm{Dist}(U, \mathcal{F}; W) \rightarrow 0.$$

In particular, there is an exact sequence

$$0 \rightarrow \mathrm{Dist}(Z) \rightarrow \mathrm{Dist}(X) \rightarrow \mathrm{Dist}(U) \rightarrow 0.$$

┘

*Proof:* The left exactness is clear. To show the right surjectivity, let  $f \in \Gamma_c(Z, \mathcal{F})$ , then for each  $x \in \mathrm{Supp}(f)$ , there is a compact open nbhd  $U_x$  that  $f$  restricts to an element of  $\Gamma(U_x, \mathcal{F})$ . Then these  $U_x$  cover  $\mathrm{Supp}(f)$ , then by (4.4.4.7), there is a disjoint finite cover  $\{U_i\}$  of  $\mathrm{Supp}(f)$  that  $f$  are induced by  $f_i \in \Gamma(U_i, \mathcal{F})$ . Now we can replace  $U_i$  by  $U_i \setminus \bigcup_{j < i} U_j$  to let  $U_i$  be disjoint, then  $f$  extends to an element of  $\bigcup U_i$ .  $\square$

**Prop. (4.4.4.13).** If  $X, Y$  are both locally profinite, then

$$C_c^\infty(X \times Y) = C_c^\infty(X) \otimes C_c^\infty(Y)$$

┘

*Proof:* Because the subspaces of the form  $U \times V$  for  $U, V$  open form a subbasis of  $X \times Y$ , so any compact open subset of  $X \times Y$  is a disjoint union of sets of the form  $U \times V$ .  $\square$

**Def. (4.4.4.14) [Cosmooth  $C_c^\infty(X)$ -Modules].** A **cosmooth  $C_c^\infty(X)$ -module** is a  $C_c^\infty(X)$ -module  $M$  that for any  $m \in M$ , there exists some compact open  $U \subset X$  that  $m = \chi_U m$ . ┘

**Prop. (4.4.4.15) [Cosmooth  $C_c^\infty(X)$ -Module and  $C^\infty(X)$ -Sheaves].** Let  $\mathcal{F}$  be a  $C_X^\infty$ -sheaf on  $X$ , then the space  $\Gamma_c(X, \mathcal{F})$  is a cosmooth  $C_c^\infty(X)$ -module (4.4.4.14), and this defines an equivalence of categories between the category of non-degenerate  $C_c^\infty(X)$ -modules and  $C^\infty(X)$ -sheaves.

Notice that in this case, a  $C_c^\infty(X)$ -module  $M$  being non-degenerate is equivalent to: for any  $m \in M$ , there is a compact open subset  $U$  that  $\chi_U m = m$ .  $\lrcorner$

*Proof:* For any non-degenerate  $C_c^\infty(X)$ -module  $M$ , define a sheaf  $\mathcal{F}_M$  of the compatible stalks in  $\prod_{x \in X} M/M(x)$ , where

$$M(x) = \{m \in M \mid \chi_U m = 0 \text{ for some } x \in U\}.$$

Then we show these two functors are inverse to each other: One direction is clear, as the sheaf  $\mathcal{F}$  is just the sheaf of compatible stalks in  $\prod_{x \in X} \mathcal{F}_x$ , and it is easy to see  $\mathcal{F}_x \cong M/M(x)$ . For the other direction, any element in  $\Gamma_c(X, \mathcal{F}_M)$  is induced from  $f_i \in \Gamma(U_i, \mathcal{F}_M)$ , where  $U_i$  are pairwise disjoint compact open subsets, by (4.4.4.7), and it is easy to see  $\Gamma(U_i, \mathcal{F}_M) = \chi_U M$ . Thus  $\Gamma_c(X, \mathcal{F}_M) = C_c^\infty(X)M$ , and the non-degeneracy of  $M$  shows  $\Gamma_c(X, \mathcal{F}_M) \cong M$ .  $\square$

**Remark (4.4.4.16).** Notice that  $M(x)$  can be equivalently defined to be the space spanned by the elements

$$\{gm \mid g \in C_c^\infty(X), g(x) = 0, m \in M\}.$$

$\lrcorner$

## 5 Real Numbers

References are [Mun00].

### Topology

**Prop. (4.4.5.1) [ℝ is Connected].** ℝ is connected, and so are intervals and rays in ℝ.  $\lrcorner$

*Proof:* This follows from (4.4.1.21), as the hypothesis is satisfied by (2.2.9.1) and the fact for any  $a < b \in \mathbb{R}$ ,  $a < \frac{a+b}{2} < b$ .  $\square$

**Prop. (4.4.5.2).**  $\mathbb{R}^n$  satisfies the Heine-Borel property (4.4.8.7), i.e. bounded closed sets are compact.  $\lrcorner$

*Proof:* It suffices to consider the square metric. If  $A$  is bounded, then  $A$  is in  $[-N, N]^n$  for some number  $N$ .  $[-N, N]^n$  is compact by (4.4.2.4) and (4.4.2.5).  $\square$

**Prop. (4.4.5.3).** Let  $I = [a, b] \subset \mathbb{R}$  be a closed interval and  $F : I \rightarrow \mathbb{R}$  a continuous function s.t.  $I \subset F(I)$ , then there exists  $x \in I$  s.t.  $F(x) = x$ .  $\lrcorner$

*Proof:* Let  $F(c) = a, F(d) = b$ , then  $c \geq a$  and  $d \leq b$ . So  $F(c) \leq c, F(d) \geq d$ . And then the fact that  $\mathbb{R}$  is connected implies that there exists  $x \in I$  s.t.  $F(x) = x$ .  $\square$

### Borel Set

**Def. (4.4.5.4).** Let  $U$  be an ultrafilter on a set  $I$  and  $\{a_I\}$  be a bounded sequence of real numbers. Then a number  $a$  is called the  $U$ -limit of  $\{a_I\}$  is for every  $\varepsilon > 0, \{i \in I \mid |a_i - a| < \varepsilon\} \in U$ .

There is at most one limit, because  $\{i \in I \mid |a_i - a| < \varepsilon\}, \{i \in I \mid |a_i - b| < \varepsilon\}$  will be disjoint hence cannot both be in  $U$ .  $\lrcorner$

**Prop. (4.4.5.5) [Generalized Limit].** Let  $U$  be an ultrafilter on  $\mathbb{N}$ , then for any bounded sequence of real numbers  $\{a_n\}$ ,  $\lim_U a_n$  exists. i.e. There is a functional from  $l^\infty$  to  $\mathbb{R}$ .

And if  $\{a_n\}$  has a limit pt  $a$  in the usual sense, then  $\lim_U a_n = a$  for any non-principal ultrafilter  $U$ , because any  $\{i \in I \mid |a_i - a| < \varepsilon\}$  is cofinite hence in  $U$  (2.2.10.9).  $\square$

*Proof:* Let  $A_x = \{n \mid a_n < x\}$ . Then  $A_x$  is monotone, then we can choose  $c = \sup\{x \mid A_x \notin U\}$  (2.2.3.22). And it is easily verified that  $c = \lim_U a_n$ .  $\square$

**Cor. (4.4.5.6) [Density Measure].** There exists a measure  $m$  on  $\mathbb{N}$  that  $m(A) = d(A)$  for each set  $A \subset \mathbb{N}$  that has a density  $d(A)$ .  $\square$

*Proof:* Let  $U$  be a non-principal ultrafilter on  $\mathbb{N}$  (2.2.10.9), let  $m(A) = \lim_U \frac{A(n)}{n}$ . It is clearly additive and monotone. And it equals the density by (4.4.5.5)  $\square$

## 6 Separation Axioms

**Prop. (4.4.6.1).** Any Quasi-compact  $T_0$  space  $X$  contains a closed point.  $\square$

*Proof:* Consider the family of non-empty closed subsets of  $X$ , there is a minimal element by quasi-compactness. Choose a minimal element  $T$ , and let  $x \in T$ , then  $\overline{\{x\}} = T$ . Then  $x$  is closed, otherwise there is some  $x' \neq x \in \overline{\{x\}}$ , and  $\{x'\} \neq \{x\}$  because  $X$  is  $T_0$ .  $\square$

### Hausdorff

**Prop. (4.4.6.2).** If  $X$  is a Hausdorff space and  $S_1 \supset S_2 \supset \dots$  is a nested chain of compact connected subsets of  $X$ , then  $S = \bigcap S_i$  is also compact and connected.  $\square$

*Proof:*  $S$  is clearly compact. Suppose it is disconnected, then  $S = A_1 \cup A_2$  is a disjoint union of compact open subsets. Then we can find two open disjoint subset  $U_1, U_2$  of  $X$  s.t.  $A_i = U_i \cap S$ . Then by the compactness of  $S_1$  and (4.4.2.3), it can be shown that  $S_n \in U_1 \cup U_2$  for  $n$  large, so  $S_n$  is disconnected, contradiction.  $\square$

### Hausdorffization

Cf.[the Hausdorff Quotient].

### Regular

### Completely Regular

### Normal (T4)

**Prop. (4.4.6.3) [Urysohn lemma].** Let  $X$  be normal,  $A$  and  $B$  two closed subset of  $X$ , then there exists a continuous map from  $X$  to  $[0, 1]$  that maps  $A$  to 0 and  $B$  to 1.  $\square$

*Proof:* Use the countability of rational numbers to construct a family of  $U_q$  s.t.

$$p < q \Rightarrow \bar{U}_p \subset U_q$$

Then choose  $f(x) = \inf\{p \in \mathbb{Q} \mid x \in U_p\}$ , then this  $f$  meets the requirement.  $\square$

**Cor. (4.4.6.4) [Tietze extension].** If  $X$  is normal and  $Y$  is a closed subspace, then any continuous function  $f$  on  $Y$  can be extended to a continuous function on  $X$ .  $\square$

*Proof:*  $\square$

## 7 Paracompactness

**Def. (4.4.7.1) [Paracompactness].**  $X \in \mathcal{Top}$  is called **paracompact** if any open covering  $\mathcal{U}$  of  $X$  has a locally finite open refinement covering.  $\lrcorner$

**Prop. (4.4.7.2) [Characterization of Paracompactness].** If  $X$  is regular, then TFAE:

1. Each open cover of  $X$  has an open locally finite refinement.
2. Each open cover of  $X$  has a locally finite refinement.
3. Each open cover of  $X$  has a closed locally finite refinement.
4. Each open cover of  $X$  is even. i.e. for any cover, there is an open nbhd  $V$  of diagonal of  $X \times X$  such that  $\forall x, V[x] = \{y | (x, y) \in V\}$  refines the cover.
5. Each open cover of  $X$  has an open  $\sigma$ -discrete refinement.
6. Each open cover of  $X$  has an open  $\sigma$ -locally finite refinement.

If this is satisfied, then  $X$  is called **paracompact**.  $\lrcorner$

*Proof:*  $6 \rightarrow 2$ : Just minus every open set the part of open sets that appeared in families that ordered before it.  $2 + 4 \rightarrow 1$ : Use the lemma below, we can transform the cover  $\mathcal{A}$  into  $V[\mathcal{A}] \cap U_A$  which is an open locally finite cover

Cf. [General Topology Kelley] and [Mun00]P254.  $\square$

**Lemma (4.4.7.3).** If  $X$  satisfies 4, let  $U$  be a nbhd of diagonal of  $X \times X$ , then there exists a symmetric nbhd of diagonal s.t.  $V \circ V \subset U$ , where  $U \circ V = \{(x, z) | (x, y) \in U, (y, z) \in V, \exists y\}$ .  $\lrcorner$

*Proof:*  $\forall x$  in  $X$ , there is a nbhd s.t.  $W[x] \times W[x] \subset U$ , this is an open cover, so there is a nbhd  $R$  of diagonal s.t.  $R[x]$  refines it. Hence  $R[x] \times R[x] \subset U$ . Let  $V = R \cap R^{-1}$ ,  $V \circ V$  is the union of sets  $V[x] \times V[x]$ , so  $V \circ V \subset U$ .  $\square$

**Lemma (4.4.7.4).** In the preceding proposition, if  $X$  satisfies 4, Let  $\mathcal{A}$  be a locally finite (resp. discrete i.e. intersect only one) family of subsets of  $X$ , then use the last lemma, there is a nbhd  $V$  of diagonal of  $X \times X$  such that  $V[\mathcal{A}] = \{y | (x, y) \in V, \exists x \in A\}$  is locally finite (resp. discrete).  $\lrcorner$

*Proof:* Choose for every pt a nbhd satisfy the property, then it is an open cover. Choose a diagonal nbhd  $U$  for the property 4, then choose coordinate symmetric nbhd  $V$  of diagonal s.t.  $V \circ V \subset U$ . If  $V[x]$  intersect  $V[A]$ , then  $V \circ V[x]$  intersect  $A$ . Done.  $\square$

**Prop. (4.4.7.5).** A locally-compact second-countable Hausdorff space  $X$  is paracompact.  $\lrcorner$

*Proof:* Let  $\mathcal{U}$  be a covering of  $X$ , because  $X$  is second countable and locally-compact, by (4.4.2.23), we may assume  $\mathcal{U}$  is a countable covering and consisting of precompact subsets. Moreover, we can change  $U_n$  to  $U'_n = \cup_{i=1}^n U_i$ , because if  $\{B_\alpha\}$  is a locally finite refinement of  $\{\cup_{i=1}^n U_i\}$ , then  $\{B_\alpha \cap U_\alpha\}$  is a locally finite refinement of  $\{U_i\}$ . So  $U_n \subset U_{n+1}$ , and because  $\overline{U}_n$  is compact, we may assume  $\overline{U}_n \subset U_{n+1}$ .

Now let  $K_n = \overline{U}_{n+1} \setminus U_n$  (where  $U_0 = \emptyset$ ), then  $K_n$  are all compact, and  $W_n = U_{n+1} \setminus \overline{U}_{n-2}$  is an open subset containing  $K_n$ . So there are f.m. open subsets of  $W_n$  that cover  $K_n$ . There open subsets is then a refinement covering of  $\mathcal{U}$ , and they are locally finite, because any  $x \in X$  is contained in  $U_n$  for some  $n$ .  $\square$

**Prop. (4.4.7.6) [Paracompact Spaces are Normal].** A Hausdorff paracompact space  $X$  is normal. In particular, a compact Hausdorff space is normal.  $\lrcorner$

*Proof:* Firstly  $X$  is regular: Let  $x \in X$  and  $B$  a closed subset disjoint from  $x$ , then because  $X$  is Hausdorff, there is a covering of  $B$  by open subsets  $V_\alpha$  that  $x \notin \overline{V}_\alpha$ . Now consider the open covering  $\{X \setminus B, V_\alpha\}$  of  $X$ , then there is a locally finite refinement  $\{B_\beta\}$ . Those  $B_\alpha$  that intersect  $B$  is then a locally finite covering of  $B$ . Let  $U$  be the union of these open subsets, then  $B \subset U$  and  $x \notin \overline{U}$ , because of the locally finiteness.

To prove  $X$  is normal, we do the same but  $x$  replaced by a closed subset disjoint from  $B$  and use regularity.  $\square$

**Lemma (4.4.7.7)[Shrinking Lemma].** Let  $X$  be a paracompact Hausdorff space and  $\{U_\alpha\}_{\alpha \in I}$  an open covering of  $X$ , then there is an open covering  $\{V_\alpha\}$  of  $X$  that  $\overline{V}_\alpha \subset U_\alpha$  for any  $\alpha$ .  $\lrcorner$

*Proof:* Let  $\mathcal{A}$  be the family of open subsets  $A$  of  $X$  that  $\overline{A} \subset U_\alpha$  for some  $\alpha$ , then because  $X$  is normal (4.4.7.6),  $\mathcal{A}$  is a covering of  $X$ . Then we find a locally finite open covering  $\mathcal{B}$  of  $\mathcal{A}$ , and let the covering map be  $B_\beta \subset U_{f(\beta)}$ , then we can get a covering  $\mathcal{V}$  of  $X$  indexed by  $I$  that  $V_\alpha = \cup_{f(\beta)=\alpha} B_\beta$ . This is also locally finite, and  $\overline{V}_\alpha \subset U_\alpha$  by the locally finiteness of  $\mathcal{B}$ .  $\square$

**Thm. (4.4.7.8)[Partition of unity].** In a paracompact Hausdorff space, given any open cover  $\{U_\alpha\}$ , there exists a partition of unity  $\{\rho_\alpha\}$  that  $\text{Supp } \rho_\alpha \subset U_\alpha$ , and  $\{\text{Supp}(\rho_\alpha)\}$  is locally finite. Moreover, if  $X$  is locally-compact (4.4.2.18), we can assume each  $\text{Supp}(\rho_\alpha)$  is compact.  $\lrcorner$

*Proof:* Using shrinking lemma (4.4.7.7) twice, we can find locally finite open coverings  $\{W_\alpha\}, \{V_\alpha\}$  that  $\overline{W}_\alpha \subset V_\alpha$ ,  $\overline{V}_\alpha \subset U_\alpha$ . Because  $X$  is normal, we can find functions  $\psi_\alpha$  on  $X$  that  $\psi(W_\alpha) = 1, \psi_\alpha(X \setminus V_\alpha) = 0$ . Then  $\text{Supp}(\psi) \subset \overline{V}_\alpha \subset U_\alpha$ , so  $\{\text{Supp}(\psi_\alpha)\}$  is locally finite. So we can define  $\Phi(x) = \sum_\alpha \psi_\alpha(x)$ .  $\Phi(x) > 0$  for any  $x$  because  $\{W_\alpha\}$  is a covering of  $X$ . Finally, we define  $\rho_\alpha = \psi_\alpha / \Phi$ , then this is a partition of unity dominated by  $\{U_\alpha\}$ .

If  $X$  is locally-compact,  $\{U_\alpha\}$  has a refinement covering given by precompact open subsets by (4.4.2.19).  $\square$

## 8 Metric Spaces

**Def. (4.4.8.1)[Metrics].** For  $X \in \text{Set}$ , a **metric** on  $X$  is a map  $\text{dist} : B \times B \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$  s.t.

- $\text{dist}(x, y) = 0 \iff x = y$  for all  $x, y \in B$ .
- $\text{dist}(x, y) = \text{dist}(y, x)$  for all  $x, y \in B$ .
- $\text{dist}(x, z) + \text{dist}(z, y) \leq \text{dist}(x, y)$  for all  $x, y, z \in B$ . (where we assume that  $x + \infty = \infty$  for any  $x \in \mathbb{R}_{\geq 0}$  and  $\infty + \infty = \infty$ ).

$\lrcorner$

**Def. (4.4.8.2)[Metric Balls].** Let  $X \in \text{Set}$  and  $\text{dist}$  a metric on  $X$  (4.4.8.1). For  $E \subset X, \delta \in \mathbb{R}_+$ , define the **metric balls**

$$\mathbb{D}(E, \delta) = \{y \in X : \text{dist}(x, y) < \delta \exists x \in E\}, \quad \overline{\mathbb{D}}(E, \delta) = \{y \in X : \text{dist}(x, y) \leq \delta \exists x \in E\}.$$

$\lrcorner$

**Def. (4.4.8.3)[Metric Spaces].** A **metric space** is defined to be a pair  $(X, \text{dist})$  where  $X \in \text{Set}$  and  $\text{dist} : X \times X \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$  is a metric on  $X$  (4.4.8.1).

Then  $X$  has a topological structure given by  $d$  by defining the set of open balls  $\{\overline{\mathbb{D}}(x, \delta)\}_{x \in X, \delta \in \mathbb{R}_+}$  as a topological basis.

And a **metrizable space** is a topological space whose topology is induced by a metric as above. The category of metric spaces is the subcategory of  $\text{Cat}$  consisting of metric spaces, denoted by  $\text{MetSpa}$ .  $\lrcorner$

**Example (4.4.8.4) [Non-Metrizable Spaces].** An uncountable product of  $\mathbb{R}$  is not a metrizable space (4.4.8.3).  $\lrcorner$

*Proof:* Cf. [Mun00]P133.  $\square$

**Def. (4.4.8.5) [Bounded Subsets].** For  $(X, \text{dist}) \in \text{MetSpa}$  (4.4.8.3), a subset  $E \subset X$  is called a **bounded subset** if there exists  $x \in X$  and  $r \in \mathbb{R}_+$  s.t.  $E \subset \mathbb{D}(B, r)$  (4.4.8.2).  $\lrcorner$

**Prop. (4.4.8.6) [Metric Spaces are Paracompact].** Any metric space is paracompact (4.4.7.1).  $\lrcorner$

*Proof:* Cf. [Mun00]P257.  $\square$

**Def. (4.4.8.7) [Heine-Borel Property].**  $X \in \text{MetSpa}$  (4.4.8.3) is said to have the **Heine-Borel property** if every closed and bounded subset (4.4.8.5) of  $X$  is compact.  $\lrcorner$

**Prop. (4.4.8.8) [Bounded Actions on Metric Spaces].** Let  $(X, \text{dist}) \in \text{MetSpa}$  and  $A$  is a subset of isometries of  $X$ . If  $\{g(x)\}_{g \in A}$  is bounded for some  $x \in X$ , then this holds for any  $x \in X$ . If this happens,  $A$  is said to be a set of **bounded isometric actions** on  $X$ .  $\lrcorner$

*Proof:* For any  $x, y \in X$  and  $g \in A$ , we have

$$\text{dist}(g(y), y) \leq \text{dist}(g(y), g(x)) + \text{dist}(g(x), x) + \text{dist}(x, y) = 2\text{dist}(x, y) + \text{dist}(g(x), x).$$

$\square$

### Geodesics Metric Spaces

References are [Metric spaces of non-negative curvature].

**Def. (4.4.8.9) [Length of Curve Segments].** For  $(X, \text{dist}) \in \text{MetSpa}$  (4.4.8.3) and a curve segment  $\gamma : [0, 1] \rightarrow X$ , the **length of  $\gamma$**  is defined to be

$$\text{length}(\gamma) = \sup_{n \in \mathbb{Z}_+} \sup_{0=t_0 < t_1 < \dots < t_n} \sum_{i \in [n]_+} \text{dist}(\gamma(t_i), \gamma(t_{i-1})) \in \mathbb{R}_{\geq 0} \cup \{\infty\}.$$

$\gamma$  is called a **rectifiable curve segment** if  $\text{length}(\gamma) < \infty$ . And  $\gamma$  is called a **geodesic curve segment** if it satisfies that for all  $0 \leq t_1 < t_2 \leq 1$ , we have

$$\text{length}(\gamma|_{[t_1, t_2]}) = \text{dist}(\gamma(t_1), \gamma(t_2)) = \text{length}(\gamma)|t_1 - t_2|.$$

$\lrcorner$

**Def. (4.4.8.10) [Geodesic Metric Spaces].** A **geodesic metric space** is a metric space  $X \in \text{MetSpa}$  (4.4.8.3) s.t. every two points  $x, y \in X$  is the two ends of a geodesic curve segment (4.4.8.9). And a **uniquely-geodesic metric space** is a geodesic metric space s.t. any two geodesic curve segments connecting  $x$  and  $y$  has the same image.

For a uniquely-geodesic metric space and  $x, y \in X$ , denote  $[x, y]$  the image of the geodesic from  $x$  to  $y$ .  $\lrcorner$

**Def. (4.4.8.11) [Length Spaces].** For a metric space  $(X, \text{dist})$  (4.4.8.3), define

$$\text{dist}_{\text{length}} : X \times X \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\} : (x, y) \mapsto \sup_{\gamma \text{ a curve connecting } x, y} \text{length}(\gamma),$$

then  $(X, \text{dist}_{\text{length}})$  is a metric space (4.4.8.3), and  $(X, \text{dist})$  is called a **length space** if  $\text{dist}_{\text{length}} = \text{dist}$ .  $\lrcorner$

**Prop. (4.4.8.12)[Geodesic Metric Spaces are Length Spaces].** Any geodesic metric space(4.4.8.10) is a length space(4.4.8.11).

For any  $(X, \text{dist}) \in \text{MetSpa}$ ,  $(X, \text{dist}_{\text{length}})$  is a length space.  $\lrcorner$

**Prop. (4.4.8.13)[Negatively-Curved Metric Spaces].** A non-positively-curved metric space is a metric space  $(X, \text{dist}) \in \text{MetSpa}$  s.t. for any  $x, y \in X$ , there exists  $m \in X$  s.t. for any  $z \in X$  we have

$$\text{dist}(x, z)^2 + \text{dist}(z, y)^2 \geq 2\text{dist}(m, z)^2 + \frac{1}{2}\text{dist}(x, y)^2.$$

$\lrcorner$

**Prop. (4.4.8.14) [Negatively-Curved Metric Spaces are Uniquely-Geodesic].** Any non-positively-curved geodesic metric space  $X \in \text{MetSpa}$ (4.4.8.10)(4.4.8.13) is uniquely-geodesic(4.4.8.10), and for any  $x, y \in X$ , the midpoint of  $[x, y]$ (4.4.8.10) is the unique point  $m$  in(4.4.8.13) that can make the statement true(4.4.8.13).  $\lrcorner$

*Proof:* For  $x, y \in X$ , if  $m$  is the point that makes the statement happen, then for midpoint  $m'$  of any geodesic containing  $x, y$ , plug in  $z = m'$ , we get  $\text{dist}(m, m') = 0$ , thus  $m = m'$ .  $\square$

**Def. (4.4.8.15)[Convex Subsets].** For a uniquely-geodesic metric space  $(X, \text{dist}) \in \text{MetSpa}$ (4.4.8.10),

- A **convex subset** is a subset  $M \subset X$  s.t.  $[x, y] \subset M$  for any  $x, y \in M$ .
- The **radius(convex subset)** for a convex subset  $M \subset X$  is defined to be  $r(M) = \inf_{x \in M} \text{diam}(x, M)$ .
- A **barycenter** of a convex subset  $M$  is a point  $x \in M$  s.t.  $\text{diam}(x, M) = r(M)$ .

Notice that intersections of convex subsets are convex subsets.  $\lrcorner$

**Prop. (4.4.8.16).** For a non-positively-curved metric space  $(X, \text{dist}) \in \text{MetSpa}$ (4.4.8.13), then for any  $z \in X$  and  $r \in \mathbb{R}_+$ , the closed metric ball  $\overline{\mathbb{D}}(z, r)$ (4.4.8.2) is a convex subset of  $X$ (4.4.8.15).  $\lrcorner$

*Proof:* For any  $x, y \in \overline{\mathbb{D}}(z, r)$ , let  $m$  be the midpoint of  $[x, y]$ , then  $m \in \overline{\mathbb{D}}(z, r)$  by(4.4.8.13)(4.4.8.14). Then we can also take the midpoint of  $[m, x]$  and  $[m, y]$ , thus showing that there is a dense subset of  $[x, y]$  contained in  $\overline{\mathbb{D}}(z, r)$ . Hence  $[x, y] \subset \overline{\mathbb{D}}(z, r)$  as  $\overline{\mathbb{D}}(z, r)$  is closed.  $\square$

**Cor. (4.4.8.17)[Convex Hulls].** For a non-positively-curved metric space  $(X, \text{dist}) \in \text{MetSpa}$ (4.4.8.13) and a subset  $M \subset X$ , the **convex hull** of  $M$  is defined to be the smallest closed convex subsets of  $X$  containing  $M$ , denoted by  $\text{Conv}(M)$ .

If  $M$  is bounded in  $X$ , then so is  $\text{Conv}(M)$ , by(4.4.8.16).  $\lrcorner$

### Convergence in Metric Spaces

**Prop. (4.4.8.18) [Continuity for Metric Spaces].** For  $(X, \text{dist}_X), (Y, \text{dist}_Y) \in \text{MetSpa}$ , a map  $f : X \rightarrow Y$  is continuous iff for any  $x \in X$  and any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\delta \in \mathbb{R}_+$  s.t. for any  $y \in X$ ,

$$\text{dist}_X(x, y) < \delta \implies \text{dist}_Y(f(x), f(y)) < \varepsilon.$$

$\lrcorner$

*Proof:* Cf.[Munkres, P129].  $\square$

**Prop. (4.4.8.19) [Sequential-Continuity].** For  $(X, \text{dist}_X), (Y, \text{dist}_Y) \in \text{MetSpa}$ , a map  $f : X \rightarrow Y \in \text{Set}$  is continuous iff for any convergent sequence  $\{x_n\} \rightarrow x \in X$ , the sequence  $\{f(x_n)\}$  converges to  $f(x)$ .  $\lrcorner$



*Proof:* Cf.[Munkres, P130]. □

**Def. (4.4.8.20) [Uniform Convergence].** For  $X \in \mathcal{Top}$  and  $(Y, \text{dist}_Y) \in \mathcal{MetSpa}$ , a sequence of maps  $\{f_n\} : X \rightarrow Y$  is said to **converge uniformly** to a function  $f : X \rightarrow Y$  if for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $N \in \mathbb{Z}_+$  s.t. for any  $n \geq N$  and  $x \in X$ , we have

$$\text{dist}_Y(f_n(x), f(x)) < \varepsilon.$$

┘

**Prop. (4.4.8.21) [Uniform Limit Theorem].** Let  $X \in \mathcal{Top}$ ,  $(Y, \text{dist}_Y) \in \mathcal{MetSpa}$  and  $\{f_n\} : X \rightarrow Y$  is a sequence of continuous maps that converges uniformly to a map  $f : X \rightarrow Y$ , then  $f$  is also continuous. ┘

*Proof:* Cf.[Munkres, P132]. ? □

### Complete Metric Spaces

**Def. (4.4.8.22) [Cauchy Sequences].** A **Cauchy sequence** in a metric space  $X \in \mathcal{MetSpa}$  (4.4.8.3) is a sequence  $(x_n)_{n \in \mathbb{Z}_+} \in X^{\mathbb{Z}_+}$  s.t. for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $N \in \mathbb{Z}_+$  s.t.  $d(x_m, x_n) < \varepsilon$  for any  $m, n \in \mathbb{Z}_{\geq N}$ . ┘

**Def. (4.4.8.23) [Complete Metric Spaces].** A **complete metric space** is a metric space  $X$  (4.4.8.3) s.t. every Cauchy sequence (4.4.8.22) has a limit in  $X$ . ┘

**Def. (4.4.8.24) [Totally-Bounded Subsets].** For  $(X, \text{dist}) \in \mathcal{MetSpa}$ , a subset  $E \subset X$  is called a **totally bounded subset** iff for every  $\varepsilon \in \mathbb{R}_+$ , there exists a finite set  $F \subset X$  s.t.  $E \subset B(F, \varepsilon)$ . This definition is compatible with that in the case of a topological vector space (12.7.1.11) when it is an  $F$ -space. ┘

**Prop. (4.4.8.25).** The closure of a totally bounded set in a metric space is totally bounded. ┘

*Proof:* For each  $\varepsilon > 0$ , choose a finite set  $F$  that  $E \subset B(F, \varepsilon/2)$ , then  $\overline{E} \subset B(F, \varepsilon)$ . □

**Prop. (4.4.8.26) [Totally Bounded Metric Spaces are Separable].** A totally-bounded metric space  $X \in \mathcal{MetSpa}$  (4.4.8.24) is separable. ┘

*Proof:*  $\cup N_n$  is dense and countable in  $X$ , where  $N_n$  is a finite  $1/n$ -net of  $X$ . □

**Prop. (4.4.8.27) [Sequence lemma, Hausdorff].** Let  $X$  be a complete metric space, then:

1. A sequentially compact (4.4.2.6) subset  $M$  is totally bounded and the converse is true if  $X$  is complete.
2. A subset  $M$  is compact iff it is self-sequentially compact iff it is closed and sequentially compact.
3. A subset  $M$  is precompact iff it is sequentially compact (4.4.2.6).

┘

*Proof:* 1: If  $M$  is not totally bounded, then for some  $\varepsilon > 0$ , we can choose consecutively a sequence of points  $x_i$  that  $d(x_i, x_j) \geq \varepsilon$ , this cannot have a convergent subsequence in  $X$ .

Conversely, if  $M$  is totally bounded, choose a  $1/k$ -net for each  $k$ , then for any sequence in  $M$ , there is a  $y_i$  that some infinite subsequence  $\{x_n^{(1)}\} \subset B(y_1, 1)$ , and consecutively find infinite subsequences  $\{x_n^{(m)}\} \subset B(y_k, 1/k)$ , then finally choose the diagonal, then it is a Cauchy sequence.



2: If it is compact, given a sequence, if no point is a convergent point, then each point has a nbhd that contains at most one point of the sequence. Then by compactness, there are at most f.m. points, contradiction. A compact set must have the convergent point in itself because it is closed as  $M$  is Hausdorff.

Conversely, if it is self-sequentially compact, then it is totally bounded by 1. so if  $M$  is not compact, then for each  $n$  it has  $1/n$ -net  $N_n$ , then there is at least one  $x_n$  that  $B(x_n, 1/n)$  cannot be covered by f.m. of the covering, The sequence  $\{x_n\}$  has a subsequence  $\{x_{n_k}\}$  that is convergent to  $x$ . But  $x \in M$  is in some open cover, so  $B(x_{n_k}, 1/n_k)$  is contained in some open cover, contradiction.

That closed and sequentially compact is equivalent to self sequentially compact is obvious.

3: If it is precompact, then it is sequentially compact by 2, conversely, if  $x_i$  is a sequence in  $\overline{M}$ , then choose  $|y_n - x_n| \leq 1/n$ , so some sequence  $y_{n_k}$  is convergent to  $y_0 \in \overline{M}$ , so  $x_{n_k}$  also converges to  $y_0$ . So  $\overline{M}$  is self-sequentially compact, so it is compact by 2.  $\square$

**Cor. (4.4.8.28) [Arzela-Ascoli].** For  $X \in \text{CHaus}$ ,  $F \subset C(X)$  is a sequentially compact(precompact, by(4.4.8.27)) subset iff it is uniformly bounded and equicontinuous.  $\lrcorner$

*Proof:* As  $C(M)$  is complete metric space, sequentially compact is equivalent to totally bounded. If it is totally bounded, then it is clearly uniformly bounded, and for every  $\varepsilon > 0$ , find a  $\varepsilon/3$ -net for  $F$ , which means f.m. functions in  $F$  that any other function is  $\varepsilon/3$ -close to one of them. So they are equicontinuous.

Conversely, if it is uniformly bounded and equicontinuous, for every  $\varepsilon > 0$ , find a finite covering of  $M$  that for any two points  $x, y$  in one cover of them,  $|f(x) - f(y)| < \varepsilon/3$  for all  $f \in F$ . Then choose for each covering a point  $x_i$ , consider  $f : F \rightarrow \mathbb{C}^n : \varphi \mapsto (\varphi(x_1), \dots, \varphi(x_n))$ , then the image is bounded, hence precompact by(4.4.5.2), so it is totally bounded by(4.4.8.27). So we can choose a  $\varepsilon/3$ -net  $\varphi_k$  for  $x_i$  simultaneously, and it is by  $\varepsilon/3$  argument that these  $\varphi_k$  is a  $\varepsilon$ -net for  $F$ .  $\square$

**Prop. (4.4.8.29)[Fixed point theorem for Contractions].** If  $X$  is a complete metric space(4.4.8.23) and  $f : X \rightarrow X$  satisfies  $\text{dist}(f(x), f(y)) \leq \lambda \text{dist}(x, y)$  for some  $0 \leq \lambda < 1$ , then  $f$  has a unique fixed point in  $X$ . If  $X$  is moreover compact, then any  $f$  satisfying  $\text{dist}(f(x), f(y)) < \text{dist}(x, y)$  for all  $x, y \in X$  will have a unique fixed point.  $\lrcorner$

*Proof:*  $x + f(x) + f^2(x) + \dots$  is the fixed point. And uniqueness is easy. For the compact case, notice the image  $\text{Im } f^n$  is a descending chain, it must stable to some  $T$ . If  $x, y \in Y$  attains the diameter of  $Y$ , and let  $x = f(X), y = f(Y)$ , where  $X, Y \in T$ , then  $\text{dist}(x, y) < \text{dist}(X, Y) \leq \text{dist}(x, y)$ , contradiction.  $\square$

**Prop. (4.4.8.30)[Dilations are Closed].** If  $X, Y \in \text{MetSpa}$  s.t.  $X$  is complete, then if  $f : X \rightarrow Y$  is continuous dilation, i.e.  $d(f(x_1), f(x_2)) \geq d(x_1, x_2)$ , then  $f(X)$  is closed.

So a continuous dilation map on a complete metric space is a closed map.  $\lrcorner$

*Proof:* If  $y \in \overline{f(X)}$ , then because  $Y$  is metric, there are  $x_n$  that  $y = \lim f(x_n)$ . Thus  $\{f(x_n)\}$  is Cauchy in  $Y$ , and  $\{x_n\}$  is Cauchy too. So there is a  $x = \lim x_n$ , and clearly  $f(x) = y$ .  $\square$

**Thm. (4.4.8.31)[Bruhat-Tits Fixed-Point Lemma, Bruhat-Tits[B-T72]].** Let  $(X, \text{dist}) \in \text{MetSpa}$  be a complete non-positively-curved space(4.4.8.23)(4.4.8.13),

- Any non-empty closed bounded(4.4.8.5) and convex(4.4.8.15) subset  $M \subset X$  has a unique barycenter(4.4.8.15).
- If  $M \subset X$  is a non-empty bounded subset and  $G \in \text{Grp}$  acts on  $X$  by isometries and preserves  $M$ , then  $G$  has a fixed point in  $X$ . And  $M$  is closed, this fixed point can be taken in  $M$ .

┘

*Proof:* 1: Let  $r = \inf_{x \in M} \text{diam}(x, M)$ . Firstly we prove the following: If  $\varepsilon \in \mathbb{R}_+$  and  $x, y \in M$  s.t.  $\text{diam}(x, M) < r + \varepsilon$  and  $\text{diam}(y, M) < r + \varepsilon$ , then  $\text{dist}(x, y) < 16r\varepsilon$ : Let  $m$  be the midpoint of  $[x, y]$ , which is contained in  $M$  because  $M$  is convex. Thus there exists  $z \in M$  s.t.  $\text{dist}(m, z) > r - \varepsilon$ . Then the assertion follows directly from the inequality in (4.4.8.13).

Then the uniqueness follow immediately, and for the existence, choose a sequence  $\{x_i\}_{i \in \mathbb{Z}_+} \in M$  s.t.  $\text{diam}(x_i, M) \rightarrow r$ , then it follows from the claim above that  $x_i$  is a Cauchy-sequence and converges to some  $x$  which is in  $M$  because  $M$  is closed. And then  $\text{diam}(x, M) = r$  by continuity, so  $x$  is a barycenter of  $M$ .

2: The barycenter of the convex hull  $\text{Conv}(M)$  of  $M$  (4.4.8.17) is fixed by  $G$ , by item1. □

**Remark (4.4.8.32).** This is the analogue of the Cartan fixed-point theorem (Cf. [Eberlein, Geometry of nonpositively curved manifolds] Prop 1.4.6.). ┘

**Cor. (4.4.8.33) [Bounded Group of Isometries has Fixed Points].** Let  $(X, \text{dist}) \in \text{MetSpa}$  be a non-empty complete non-positively-curved space (4.4.8.23) (4.4.8.13). If  $G \in \text{Grp}$  is a bounded group of isometric actions (4.4.8.8) on  $X$ , then  $G$  has a fixed point. ┘

*Proof:* For any  $x \in X$ ,  $M = \{g(x)\}_{g \in G}$  is a subset bounded under the action of  $G$ , then we can use (4.4.8.31). □

**Prop. (4.4.8.34) [Closest-Point Map].** Let  $(X, \text{dist}) \in \text{MetSpa}$  be a complete non-positively-curved space (4.4.8.23) (4.4.8.13) and  $Y \subset X$  a closed convex subset, then

- For any  $x \in X$ , there exists a unique point  $\text{pr}_Y(x) \in Y$  closest to  $x$ .
  - $\text{pr}_Y : X \rightarrow Y$  is a continuous function.
  - $\text{pr}_Y$  is equivariant w.r.t. any isometry of  $X$  preserving  $Y$ .
- ┘

*Proof:* We first prove the following claim: For any  $x \in X$  and  $\varepsilon \in \mathbb{R}_+$ , if  $y_1, y_2 \in Y$  s.t.  $\text{dist}(x, y_1) < \text{dist}(x, Y) + \varepsilon$ ,  $\text{dist}(x, y_2) < \text{dist}(x, Y) + \varepsilon$ , then we have  $\text{dist}(y_1, y_2)^2 < 16r\varepsilon$ : Let  $m$  be the midpoint of  $[x, y]$ , which is contained in  $M$  because  $M$  is convex. Thus there exists  $z \in M$  s.t.  $\text{dist}(m, z) > r - \varepsilon$ . Then the assertion follows directly from the inequality in (4.4.8.13).

1: The uniqueness follows immediately from the claim above, and for the existence, choose a sequence  $\{y_i\}_{i \in \mathbb{Z}_+} \in Y$  s.t.  $\text{diam}(y_i, x) \rightarrow \text{dist}(x, Y)$ , then it follows from the claim above that  $y_i$  is a Cauchy-sequence and converges to some  $y$  which is in  $Y$  because  $Y$  is closed. And then  $\text{diam}(x, y) = \text{dist}(x, Y)$  by continuity,

2: If  $(x_n)_{n \in \mathbb{Z}_+}$  is a sequence of points in  $X$  converges to  $x$ , then

$$\text{dist}(x, \text{pr}_Y(x)) \leq \text{dist}(x, \text{pr}_Y(x_n)) \leq \text{dist}(x, x_n) + \text{dist}(x_n, \text{pr}_Y(x_n)) \leq \text{dist}(x, x_n) + \text{dist}(x_n, \text{pr}_Y(x)),$$

Taking  $n \rightarrow \infty$  we get  $\text{dist}(x, \text{pr}_Y(x_n)) \rightarrow \text{dist}(x, \text{pr}_Y(x))$ , thus by the claim we proved above, we see  $\text{pr}_Y(x_n) \rightarrow \text{pr}_Y(x)$ . Thus  $\text{pr}_Y$  is continuous by (4.4.8.19).

3: For any isometry  $f : X \rightarrow X$  preserving  $Y$  and  $x \in X$ ,

$$\text{dist}(f(x), \text{pr}_Y(f(x))) < \text{dist}(f(x), f(\text{pr}_Y(x))) = \text{dist}(x, \text{pr}_Y(x)),$$

and the same inequality for  $f^{-1}$  shows that this is an equality. Thus  $\text{pr}_Y(f(x)) = f(\text{pr}_Y(x))$  by uniqueness in item1. □

### Compact Metric Spaces

**Lemma (4.4.8.35) [Lebesgue-Number Lemma].** For any open covering  $U_i$  of a compact metric space  $(X, \text{dist}) \in \text{MetSpa}$ , there exists a  $\delta > 0$  that any subset  $X$  of diameter smaller than  $\delta$  is contained in some  $U_i$ .  $\lrcorner$

*Proof:* If  $X$  is in the covering  $U_i$ , then there is nothing to prove, otherwise, it suffices to assume the covering is a finite covering, let  $C_i = X - U_i$ , and let  $f(x) = \frac{1}{n} \sum d(x, C_i)$ . Notice  $f(x) > 0$ , because  $x \notin C_i$  for some  $C_i$ . Now it is also continuous, so it has a minimal value  $\delta > 0$ .

Now if  $B$  has diameter smaller than  $\delta$ , then if  $x_0 \in B$ ,  $\delta < f(x_0) < d(x, C_{i_0})$ , where  $d(x, C_{i_0})$  is the maximal among  $d(x, C_k)$ , and  $B \in B(x, \delta)$ , thus  $B \in U_{i_0}$ .  $\square$

**Prop. (4.4.8.36) [Uniform Continuity Theorem].** If  $f : X \rightarrow Y$  is a continuous map between two metric spaces that  $X$  is compact, then  $f$  is uniformly continuous.  $\lrcorner$

*Proof:* Take an open covering of  $Y$  with balls  $B(y_i, \varepsilon/2)$  of diameter  $\varepsilon/2$ , and consider their inverse image, then choose the lebesgue number  $\delta$  for this covering (4.4.8.35), we see that for any  $d(x, y) < \delta$ ,  $d(f(x), f(y)) < \varepsilon$ .  $\square$

**Prop. (4.4.8.37).** If  $f$  is an isometry of a compact metric space  $X$ , then it is a bijection thus an homeomorphism.  $\lrcorner$

*Proof:* It is clearly injective. If it is not surjective, then choose a  $x \notin \text{Im}(f)$ , because  $\text{Im}(f)$  is compact hence closed in  $X$ ,  $d(x, \text{Im}(f)) = \varepsilon > 0$ . Now consider the minimal  $N$  that we can cover  $X$  with open subsets of diameter smaller than  $\varepsilon$ , this  $N$  exists because  $X$  is compact. Now if  $U_i$  covers  $X$ , then the one that contains  $x$  cannot intersect with  $\text{Im}(f)$ . But  $f^{-1}(U_i)$  is an open cover of  $X$  with smaller numbers of open subsets, contradiction.  $\square$

## 9 Baire-Spaces

**Def. (4.4.9.1) [Baire-Spaces].** For  $X \in \mathcal{T}\text{op}$ , a subset  $E \subset X$  is called of **Baire-first category** if  $E \subset \cup_{i \in \mathbb{Z}_+} Z_i$ , where each  $Z_i \subset X$  is a closed subset s.t.  $\text{Int}(Z_i) = \emptyset$ .  $E$  is called of **Baire-second category** if it is not of first category.

A **Baire-space** is a topological space s.t. any nonempty open subsets of it is of second category.  $\lrcorner$

**Prop. (4.4.9.2) [Baire-Category Theorem].** Any complete metric space (4.4.8.23) or locally-compact Hausdorff space is a Baire space.  $\lrcorner$

*Proof:* Choose consecutively (precompact) open subsets that don't intersect  $\overline{E_n}$  to look for a limit point.  $\square$

## 10 Uniform Spaces

**Def. (4.4.10.1) [Uniform Spaces].**  $\lrcorner$

**Def. (4.4.10.2) [Cauchy Filter in the Topological Group Case].** A **Cauchy filter** is a topological Abelian group is a filter  $\mathcal{F}$  that for any nbhd  $U$  of 0, there exists  $E \in \mathcal{F}$  that  $x - y \in U$  if  $x, y \in E$ .  $\lrcorner$

**Def. (4.4.10.3) [Complete Uniform Spaces].** A topological Abelian group is called **complete uniform space** iff it is Hausdorff, and any Cauchy filter has a limit.  $\lrcorner$

## 11 Manifolds

**Def. (4.4.11.1) [Locally-Euclidean Spaces].** ┘

**Prop. (4.4.11.2) [Paracompactness for Manifolds].** For a connected Hausdorff locally-Euclidian space (4.4.11.1), the condition of paracompact, second countable and existence of a compact exhaustion is equivalent. ┘

*Proof:* Cf. [Paracompactness and second countable]. □

**Def. (4.4.11.3) [Manifolds].** For  $n \in \mathbb{N}$ , a (topological) **manifold** of dimension  $n$  is a topological space that is Hausdorff, second-countable and locally-Euclidean (4.4.11.1). By (4.4.11.2), the last condition is equivalent to saying it is paracompact.

The category of topological manifolds of dimension  $n$  is denoted by  $\text{Mani}^n$ . Denote  $\text{Mani} \triangleq \bigcup_{n \in \mathbb{N}} \text{Mani}^n$ . ┘

**Thm. (4.4.11.4) [Annulus Theorem, Rado-Moise-Quinn-Kirby].** For any map  $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$  that maps  $\mathbb{D}^n$  into its interior, there is a homeomorphism  $\mathbb{D}^n \setminus h(\mathbb{D}^n) \cong \mathbb{S}^{n-1} \times \mathbb{I}$  that identifies  $\partial \mathbb{D}^n$  and  $h(\partial \mathbb{D}^n)$  with  $\mathbb{S}^{n-1} \times \{0\}, \mathbb{S}^{n-1} \times \{1\}$  resp. ┘

*Proof:* □

**Def. (4.4.11.5) [Jordan-Curves].** A **Jordan-curve** in a topological space  $X$  is an injective closed curve  $\gamma : [a, b] \rightarrow X$ . ┘

**Thm. (4.4.11.6) [Jordan-Curve Theorem, Jordan1887].** Let  $\gamma$  be a Jordan curve in  $\mathbb{R}^2$ , then  $\mathbb{R}^2 \setminus \gamma$  has exactly two components, one is bounded, denoted by  $\text{int} \gamma$ , while the other is unbounded. ┘

*Proof:* □

### Triangularizations

**Def. (4.4.11.7) [Triangularizations].** A **triangularizable space** is a topological space that is homeomorphic to the geometrization of a simplicial set  $X$  (4.6.3.8) s.t.

- $X$  is generated by its  $n$ -simplices for some  $n \in \mathbb{Z}$ .
- For any  $A, B \in X_n$ ,  $A \cap B = \emptyset$  or  $A \cap B \in X_{n-1}$ .
- For any  $Z \in X_{n-1}$ , there are at most two  $A \in X_n$  s.t.  $Z \subset A$ .

Any such homeomorphism is called a **triangularization** of this space. ┘

**Thm. (4.4.11.8) [Triangularization of Surfaces, Rado1920].** Any compact connected topological surface (4.4.11.12) is triangularizable. ┘

*Proof:* ? □

**Def. (4.4.11.9) [Pachner Moves].** Let  $M$  be a triangulated  $n$ -dimensional manifold, and let  $C \subset M$  be a sub-triangulated manifold of dimension  $n$  with an isomorphism  $\varphi : C \rightarrow C' \subset \partial \Delta_{n+1}$ . Then the **Pachner move** on  $N$  associated to  $C$  is the triangulated manifold

$$N' = (N \setminus C) \cup_{\varphi} (\partial \Delta_{n+1} \setminus C').$$

┘

**Example (4.4.11.10).** For surfaces, there are two Pachner moves (4.4.11.9) (and their inverses):

- $(ABC) \cup (ACD) \mapsto (ABD) \cup (BCD)$ .
- $(ABC) \mapsto (ABD) \cup (BCD) \cup (CAD)$ .

┘

**Thm. (4.4.11.11) [Pachner1991].** Given any closed surface  $\Sigma$ , any two triangulations of  $\Sigma$  (4.4.11.7) is related by a finite sequence of isotopies and Pachner moves (4.4.11.9). ┘

*Proof:* ┘

### Surfaces

**Def. (4.4.11.12) [Topological Surfaces].** A **topological surface** (with boundaries) is a topological manifold of dimension 2 (with boundaries). ┘

**Def. (4.4.11.13) [Planer Glueing Diagram].** A **planer glueing diagram** is a polygon with some glueing of its edges with orientations clockwise such that no three edges are glued together. We can label the edges by symbols and their inverses. different symbols are considered not glued together. For example,  $aba^{-1}b^{-1}, abab$ .

A **compact planer glueing diagram** is a planer glueing diagram that every symbol appear exactly twice in the the edges of the polygon. ┘

**Prop. (4.4.11.14).** The space correspond to a planer glueing diagram is a compact topological surface with boundaries.

The space correspond to a compact planer glueing diagram is a compact topological surface. ┘

*Proof:* It is compact as a quotient of a compact space. It can verified that it is locally Euclidean at each space. And it is clearly Hausdorff. ┘

**Lemma (4.4.11.15).**  $T^2 \# RP^2 \cong RP^2 \# RP^2 \# RP^2$ . ┘

*Proof:* By (4.4.11.13), it suffices to show that  $aabcb^{-1}c^{-1}$  is homeomorphic to  $eeffgg$ . For this, use cut-and-paste. ? ┘

**Thm. (4.4.11.16) [Classification of Surfaces].** Any compact topological surface is homeomorphic to exactly one of the following:  $S^2, \#^n T^2$  or  $\#^n RP^2$ ,  $n \in \mathbb{Z}_+$ . The corresponding planer diagram is  $aa^{-1}, a_1b_1a_1^{-1}b_1^{-1} \dots a_nb_na_n^{-1}b_n^{-1}, e_1e_1e_2e_2 \dots e_ne_n$ . ┘

*Proof:* By triangularization theorem (4.4.11.8), it is easy to see that any compact surface is the space corresponding to a compact planer diagram. Then it suffices to show any compact planer diagram is isomorphic to either  $aa^{-1}, a_1b_1a_1^{-1}b_1^{-1} \dots a_nb_na_n^{-1}b_n^{-1}, e_1e_1e_2e_2 \dots e_ne_n$ .

Firstly we can reduce to the case where all vertices are glued together: Take a vertex  $P$ , then we can spot all the vertices that are glued to  $P$ . If these are not all the vertices, we can find an edge  $e = PQ$ , and let  $RPQ$  be a triangle, then we can cut this triangle out to paste along the edge pairing  $e$ . Then we can reduce the number of vertices glued to  $P$ . Eventually, we can eliminate  $P$ .

Secondly we can collet all edges with the same orientation (i.e.  $\dots a \dots a \dots$ ) to be adjacent: (i.e.  $\dots aa \dots$ ). To do this, suppose  $a = P_1Q_1 = P_2Q_2$ , we simply cut along the line  $P_1P_2$  and glue  $a$  together.

Now if all edges glued together are of the same orientations, then it is clearly isomorphic to  $\#^n RP^2$ . Otherwise there are edged  $a, a^{-1}$  glued together with the opposite orientation. Then

there must be another pair  $b, b^{-1}$  s.t. these four pairs are ordered by  $a, b, a^{-1}, b^{-1}$ . This is because otherwise there will be two vertices. Then by some cut-and-paste, we can move these edges together as  $\dots aba^{-1}b^{-1} \dots$ . By induction, we can make all edges with the opposite orientations together and in pairs.

Thus our diagram is a connected sum of  $\mathbb{T}^2$  and  $\mathbb{RP}^2$ . Then we can reduce the diagram to one of the three kinds by (4.4.11.15).

Finally, it suffices to show these three kinds are different: Their Euler characteristics are

$$\chi(\mathbb{S}^2) = 2, \quad \chi(\#^n \mathbb{T}^2) = 2 - 2n, \quad \chi(\#^n \mathbb{RP}^2) = 2 - n.$$

So the only case their Euler characters equal are  $\chi(\#^n \mathbb{T}^2) = \chi(\#^{2n} \mathbb{RP}^2)$ . But in this case, they are still not homeomorphic, as  $H_2(\#^n \mathbb{T}^2) = \mathbb{Z}$  and  $H_2(\#^{2n} \mathbb{RP}^2) = 0$ .  $\square$

## 12 Commonly Used Spaces

**Def. (4.4.12.1) [Torus].** The ( $n$ -dimensional) **torus**  $\mathbb{T}^n$  is defined to be  $\mathbb{T}^n = (\mathbb{S}^1)^n$ .  $\lrcorner$

**Def. (4.4.12.2) [ $\mathbb{RP}^n$ ].** The **real projective space**  $\mathbb{RP}^n$  is the  $\mathbb{R}$ -points of  $\mathbb{P}_{\mathbb{R}}^n$  with the canonical topology.  $\lrcorner$

**Def. (4.4.12.3) [Möbius-Band].** The **Möbius-band** is the space defined to be the planer glueing diagram  $abac$ .  $\lrcorner$

**Def. (4.4.12.4) [Klein-Bottle].** The **Klein-bottle**  $\mathbb{K}^2$  is the space defined by the planer glueing diagram  $aba^{-1}b$ .  $\lrcorner$

## 4.5 Model Categories

Main references are [Model Category and Simplicial Methods, Goerss], [Model Categories, Kan-  
tor], [Homotopy Theories and Model Categories, Dwyer/Spalinski], [Lur09], [Hovey, Model Cate-  
gories].

**Def. (4.5.0.1).** If  $\mathcal{C} \in \mathbf{Cat}$  and  $S$  is a class of morphisms in  $\mathcal{C}$ , we denote  $l(S)$  the set of morphisms that has left lifting property w.r.t. all morphisms in  $\mathcal{C}$ , and  $r(S)$  the set of morphisms that have right lifting property w.r.t. all morphisms in  $\mathcal{C}$ . Then  $l(S)$  is stable under pushout and  $r(S)$  is stable under pull backs.  $\lrcorner$

**Def. (4.5.0.2) [Model Structure].** A **model structure** on a category  $\mathcal{C}$  is three classes of morphisms: **fibrations**, **cofibrations** and **weak equivalences** that satisfy the following axioms: (Denote a **A trivial (co)fibration** is a (co)fibration that is also a weak equivalence.)

M1  $\mathcal{C}$  has finite limits and colimits.

M2 (two out of three) If two of  $f, g, fg$  is weak equivalence, then so is the third.

M3 (retracts) Fibrations, cofibrations and weak equivalences are closed under retract.

M4 (lifting property) We have a lifting property with a cofibration  $i$  and fibration  $p$  when either of them is a weak equivalence.

M5 (factorization) Any map  $f$  can be factored as  $pi$  where  $i$  is trivial cofibration and  $p$  is a fibration, and also as  $pi$  where  $i$  is a cofibration and  $p$  is a trivial fibration.

**Remark (4.5.0.3).** Notice the axioms are symmetric in fibrations and cofibrations, thus the opposite category  $\mathcal{C}^{\text{op}}$  has a natural model structure. So whenever we write a theorem, we should always remember its dual counterpart.  $\lrcorner$

**Lemma (4.5.0.4) [Closedness].** A model category satisfies the retraction axiom iff:

- fibration =  $r(\text{trivial cofibrations})$ ,
- cofibration =  $l(\text{trivial fibrations})$ ,
- weak equivalence =  $uv$ , where  $v \in l(\text{fibrations})$  and  $u \in r(\text{cofibrations})$ .

*Proof:* If these are satisfied, retraction axiom is easy: A retract satisfies the same lifting properties. Hence retraction of a (co)fibration is a (co)fibration. For retracts weak equivalences, Cf.[Quillen, Homotopical Algebra, Chap5.2].

Conversely, using (4.2.2.12), we first factorize a  $p = f \circ i$ , where  $i$  is a trivial cofibration, then because  $p \in r(i)$ ,  $p$  is a retraction of  $f$  hence a fibration. And similarly for cofibrations and weak equivalences.  $\square$

**Cor. (4.5.0.5).** In a model category,  
trivial fibrations =  $r(\text{cofibrations})$ ,  
trivial cofibrations =  $l(\text{fibrations})$ .  $\lrcorner$

*Proof:* The proof is the same as that of (4.5.0.4).  $\square$



**Cor. (4.5.0.6).** In a model category, the class of (trivial)fibrations is stable under base change and the class of (trivial)cofibrations is stable under cobase change.  $\lrcorner$

**Prop. (4.5.0.7).** Let  $p$  be a fibration in  $C_{cf}$ , then  $p \in r(\text{Cof})$  iff  $\gamma(p)$  is an isomorphism, Cf.[Quillen 5.2]. So if conditions of(4.5.0.4) are satisfied (i.e.  $C$  is a closed model category),  $\gamma(f)$  is an isomorphism iff  $f$  is a weak equivalence by the characterization of weak-equivalence of(4.5.0.4).  $\lrcorner$

*Proof:*  $\square$

**Def. (4.5.0.8) [Left Proper Model Categories].** A model category is called **left proper** if weak equivalences are stable under cobase change by cofibrations. Dually it is called **right proper** if weak equivalences are stable under base change by fibrations.  $\lrcorner$

**Prop. (4.5.0.9) [Cofibration is Left Proper].** For a pushout diagram

$$\begin{array}{ccc} A & \xrightarrow{i} & B \\ \downarrow j & & \downarrow j' \\ A' & \xrightarrow{i'} & B' \end{array}$$

in a model category  $\mathcal{C}$ , if  $i$  is cofibration and  $A, A'$  are cofibrant, and  $j$  is weak equivalence, then  $j'$  is also weak equivalence.  $\lrcorner$

*Proof:* It suffices to show  $j'$  is an isomorphism in  $\text{Ho}(\mathcal{C})$ (4.5.1.15). For this, by Yoneda lemma, it suffices to show  $\text{Hom}_{\text{Ho}(\mathcal{C})}(B', Z) \cong \text{Hom}_{\text{Ho}(\mathcal{C})}(B, Z)$  for any fibrant object  $Z$ .

For surjectivity, by(4.5.1.18), it suffices to show  $\pi(B', Z) \cong \pi(B, Z)$ . Given a map  $f : B \rightarrow Z$ , because  $j$  is weak equivalence, there is a map  $g : A' \rightarrow Z$  that  $g \circ j \sim f \circ i$ . Then by(4.5.1.10), there is a  $f' \sim f$  that  $f' \circ i = g \circ j$ , which determines a morphism  $B' \rightarrow Z$ .

For injectivity, If  $P$  is a path object that  $H : B \rightarrow P$  induces a homotopy between maps  $s \circ j', s' \circ j'$ , then we need to extend this homotopy to  $\tilde{H} : B' \rightarrow P$ , and the method is the argument is the same as above.  $\square$

**Cor. (4.5.0.10).** If  $\mathcal{M}$  is a model category s.t. each object is cofibrant, then  $\mathcal{M}$  is left proper.  $\lrcorner$

## 1 Homotopies

**Def. (4.5.1.1) [Cylinder Objects].** A **cylinder object** for an object  $X$  is an object  $X \wedge I$  which gives a factorization of the natural map  $X \amalg X \rightarrow X$  as  $X \amalg X \xrightarrow{i} X \wedge I \xrightarrow{j} X$ , where  $j \in W$ . It is called a **good cylinder object** if  $i$  is a cofibration, and **very good cylinder object** if  $j$  is trivial fibration. By factorization axiom, every object has a very good cylinder object.

There are two natural morphisms  $X \mapsto X \wedge I$ , denoted by  $\partial_0$  and  $\partial_1$ .

Dually we can define **path object**  $Y^I$  for  $Y$ , and every object has a very good path object.  $\lrcorner$

**Prop. (4.5.1.2).** If  $A$  is cofibrant and  $A \wedge I$  is a cylinder object for  $A$ , then  $\partial_i : A \rightarrow A \times I$  are trivial cofibrations.  $\lrcorner$

*Proof:* Because it's pushout of  $\emptyset \rightarrow A$  and  $\sigma \circ \partial_i = \text{id}_A$ .  $\square$

**Cor. (4.5.1.3).** if  $f \sim^l g$ , then  $f$  is a weak equivalence iff  $g$  is a weak equivalence.  $\lrcorner$

*Proof:* This is because  $f = H \circ \partial_0, g = H \circ \partial_1$ , and we can use(4.5.1.2).  $\square$



**Def. (4.5.1.4) [Homotopies].** Two morphisms  $f, g : X \rightarrow Y$  are called **(good/very good) left homotopic**, denoted by  $f \sim^l g$  iff there is a (good/very good) cylinder object  $X \amalg X \rightarrow X \wedge I$  with  $X \wedge I \rightarrow Y$  that induce  $(f, g) : X \amalg X \rightarrow Y$ . Dually for right homotopies. And we denote by  $\pi^l(A, B)/\pi^r(A, B)$  the equivalence classes of  $\text{Hom}(A, B)$  under the equivalence relation generated by left/right homotopies.

If  $f, g \in \text{Hom}(X, Y)$  and  $\varphi \in \text{Hom}(Y, Z)$  that  $\varphi \circ f = \varphi \circ g$ , then  $f, g$  is called **left homotopic over  $Z$**  if there is a homotopy  $H$  of  $f \sim^l g$  that  $\varphi H$  is the trivial homotopy. Dually for **right homotopic under  $X$** .  $\lrcorner$

**Lemma (4.5.1.5) [Very Good Homotopies].** For  $f, g \in \text{Hom}(X, Y)$ , if  $f \sim^l g$ , then  $f, g$  are good left homotopic. And if  $Y$  is fibrant, then  $f, g$  are moreover very good left homotopic.  $\lrcorner$

*Proof:* The first assertion is each, just choose a factorization of the cylinder object  $X \amalg X \xrightarrow{i} X \wedge I' \xrightarrow{j} X \wedge I$  that  $i$  is cofibrant. If  $Y$  is fibrant, then we further factorize  $X \wedge I' \xrightarrow{i} X \wedge I'' \xrightarrow{j} X$ , where  $i$  is cofibrant and  $j$  is trivial fibration, then by two out of three,  $i$  is also trivial cofibration, and it suffices to extend the homotopy  $X \wedge I' \rightarrow Y$  to  $X \wedge I'' \rightarrow Y$ , and this is because  $Y$  is fibrant.  $\square$

**Prop. (4.5.1.6) [Homotopy is Equivalence Relation].** If  $A$  is cofibrant, then the left homotopy is an equivalence relation on  $\text{Hom}(A, B)$ .  $\lrcorner$

*Proof:* Reflexivity and symmetry is trivial, the only problem is transitivity, so we construct a glueing  $A \wedge I''$  as the pushout of  $\partial_1 : A \rightarrow A \wedge I$  and  $\partial'_0 : A \rightarrow A \wedge I'$ .  $A \wedge I'' \rightarrow A$  is a weak equivalence by the universal property and (4.5.0.4), so this is a cylinder object. The rest is easy.  $\square$

**Prop. (4.5.1.7) [Properties of Left Homotopies].** If  $A$  is cofibrant and  $f, g \in \text{Hom}(A, B)$ , then

1. If  $f, g$  are right homotopic, then  $s \rightarrow B^I$  can be chosen to be trivial Cof.
  2. If  $f, g$  are right homotopic, then so does  $uf \sim ug$  or  $fv \sim gv$ . Thus if  $A$  is cofibrant, there is a composition map:  $\pi^r(A, B) \times \pi^r(B, C) \rightarrow \pi^r(A, C)$ .
  3. For any trivial fibration  $X \rightarrow Y$ ,  $\pi^l(A, X) \rightarrow \pi^l(A, Y)$  is a bijection.
- And dual arguments hold for fibrant objects.  $\lrcorner$

*Proof:*

1. factorize  $B \rightarrow B^I$  to  $B \rightarrow B^{I'} \rightarrow B^I$  where  $B \rightarrow B^{I'} \in \text{TCof}$  and  $B^{I'} \rightarrow B^I \in W$ , so  $B^{I'}$  is also a cylinder object and the homotopy  $A \rightarrow B^I$  can be lifted to  $A \rightarrow B^{I'}$ .

2. there is a diagram

$$\begin{array}{ccc} B & \xrightarrow{su} & C^I \\ \downarrow s & & \downarrow (d_0, d_1) \\ B^I & \xrightarrow{(d_0 u, d_1 u)} & C \times C \end{array}$$

which has a lifting  $\varphi$ , then composed with  $A \rightarrow B^I$

will give the desired homotopy.

3. the map is well-defined, it is surjective because of lifting property, and it is injective because  $A \amalg A \rightarrow A \times I \in \text{Cof}$  so the homotopy can be lifted to  $X$ .  
Cf. [Homotopy Theories and Model Categories, P20, 21].  $\square$

**Lemma (4.5.1.8) [Left and Right Homotopies].** If  $X$  is cofibrant, then for  $f, g \in \text{Hom}(X, Y)$ , if  $f \sim_l g$ , then  $f \sim_r g$ . And the dual conclusion holds for  $Y$  fibrant.

In particular, for morphisms between bifibrant objects, left and right homotopic are equivalent.

⌋

*Proof:* Consider a cylinder object  $j : X \wedge I \rightarrow X$  for  $X$  and a path object for  $Y$ . Suppose  $f, g$  are left homotopic via a map  $H : X \wedge I \rightarrow Y$ , then we consider the diagram

$$\begin{array}{ccccc} X & \xrightarrow{f} & Y & \xrightarrow{\quad} & Y^I \\ \downarrow \partial_1 & & \searrow & \nearrow & \downarrow \\ X \wedge I & \xrightarrow{(f \circ j) \times H} & Y \times Y & & Y^I \end{array}.$$

Then it can be solved by some  $\tilde{H}$  because  $\partial_1$  is trivial cofibration (4.5.1.2), and then it can be checked  $H \circ \partial_1$  gives the desired right homotopy.  $\square$

**Prop. (4.5.1.9) [Whitehead's Theorem].** Let  $X, Y \in \mathcal{C}$  be bifibrant, then a map  $f : X \rightarrow Y$  is an equivalence iff there is a  $g : X \rightarrow Y$  that  $fg$  and  $gf$  are homotopic to id.  $\square$

*Proof:* Cf. [Homotopy Theories and Model Categories, P23].  $\square$

**Prop. (4.5.1.10) [Lifting Criterion].** Let  $\mathcal{C}$  be a model category and  $i : A \rightarrow B$  be a cofibration between cofibrant objects, and  $X$  is fibrant,  $g : B \rightarrow X, f : A \rightarrow X$  satisfies  $g \circ i \sim f$ , then there is a  $g' \sim g$  that  $g' \circ i = f$ .  $\square$

*Proof:* Choose a good cylinder object  $C(A)$  for  $A$  and factorize

$$C(A) \coprod_{A \coprod A} (B \coprod B) \rightarrow C(B) \rightarrow B$$

where the first map is cofibration and the second is trivial fibration, then  $C(B)$  is a good cylinder object for  $B$ .

The homotopy is given by a map  $C(A) \coprod_A B \rightarrow X$ , and we check  $(A) \coprod_A B \rightarrow C(B)$  is a trivial cofibration (check  $C(A) \coprod_A B \rightarrow B$  is weak equivalence using (4.5.0.6) and check  $C(A) \coprod_A B \rightarrow C(A) \coprod_A \coprod_A (B \coprod B)$  is a cofibration as a cobase change of  $A \rightarrow B$  and  $C(A) \coprod_A \coprod_A (B \coprod B) \rightarrow C(B)$  is a cofibration by definition), so the homotopy extends to a homotopy  $C(B) \rightarrow X$ , which is a homotopy between  $g$  and some  $g'$  and  $g' \circ i = f$ .  $\square$

**Def. (4.5.1.11).** Let  $C_c, C_f, C_{cf}$  denote the full subcategory of cofibrant, fibrant and cofibrant-fibrant objects. And we define  $\pi C_c$  as the category module right homotopy equivalence between morphisms, dually for  $\pi C_f$ .

Notice (4.5.1.7) assures  $\pi C_c, \pi C_f$  are truly categories.

Notice for  $C_{cf}$ , left homotopy is equivalent to right homotopy by (4.5.1.8), so  $\pi C_{cf}$  is full subcategory for both  $\pi C_c$  and  $\pi C_f$ .  $\square$

**Lemma (4.5.1.12) [Fibrant and Cofibrant Replacement].** For an object  $X$  in a model category  $\mathcal{C}$ , the axioms show there is a cofibrant object  $QX$  and a trivial fibration  $QX \rightarrow X$ . Also there is a fibrant object  $RX$  and a trivial cofibration  $X \rightarrow RX$ . We fix choices of  $Q, R$  that is identity on bifibrant objects, and consider it a mapping from  $\mathcal{C}$  to  $\mathcal{C}$ .

Then given any morphism  $f : X \rightarrow Y$ , there is a morphism  $\tilde{f} : QX \rightarrow QY$  lifting  $f$ , and  $\tilde{f}$  depends up to left and right homotopy only on  $f$ . And if  $Y$  is fibrant, then it depends up to left and right homotopy only on the left homotopy classes of  $f$ .

Dually assertions also holds, so we have functors:  $Q : \mathcal{C} \rightarrow \pi C_c$  and  $R : \mathcal{C} \rightarrow \pi C_f$ .  $\square$

*Proof:* The existence of the lifting follows from the fact  $QX$  is fibrant and  $QY \rightarrow Y$  is trivial fibration. The uniqueness of left homotopy follows from (4.5.1.7), and also right homotopy, because  $QX$  is cofibrant and use (4.5.1.8). For the last assertion, notice when  $Y$  is fibrant, (4.5.1.7) shows the left homotopy class of  $QX \rightarrow Y$  is determined, and use (4.5.1.7) again, the class of  $f$  is also determined.  $\square$

**Cor. (4.5.1.13).** The restrictions define functors  $Q' : \pi\mathcal{C}_c \rightarrow \pi\mathcal{C}_{cf}$  and  $R' : \pi\mathcal{C}_f \rightarrow \pi\mathcal{C}_{cf}$ .  $\lrcorner$

**Def. (4.5.1.14) [Homotopy Category].** For any model category  $\mathcal{C}$ , we construct a **homotopy category**  $\text{Ho}(\mathcal{C})$  whose objects are the same as  $\mathcal{C}$ , but  $\text{Hom}_{\text{Ho}(\mathcal{C})}(X, Y) = \text{Hom}_{\pi\mathcal{C}_{cf}}(RQX, RQY) = \pi(RQX, RQY)$ .

There is a functor  $\gamma : \mathcal{C} \rightarrow \text{Ho}(\mathcal{C})$  by sending  $X$  to  $RQX$ , by (4.5.1.12).  $\lrcorner$

**Prop. (4.5.1.15) [Weak Equivalence and Isomorphisms].** A morphism in  $\mathcal{C}$  maps to an isomorphism in  $\text{Ho}(\mathcal{C})$  iff it is a weak equivalence. The morphisms in  $\text{Ho}(\mathcal{C})$  are generated by the image of morphisms in  $\mathcal{C}$  and the inverse of images of weak equivalences in  $\mathcal{C}$ .  $\lrcorner$

*Proof:* If  $f \in \mathcal{C}$  is a weak equivalence, then  $f' = RQ(f)$  is also a weak equivalence, by two out of three lemma. Then Whitehead theorem (4.5.1.9) shows  $f'$  is an isomorphism in  $\pi\mathcal{C}_{cf}$  hence in  $\text{Ho}(\mathcal{C})$ . Conversely, if  $f'$  has an inverse in  $\text{Ho}(\mathcal{C})$ , then  $f'$  is a weak equivalence by Whitehead (4.5.1.9) again, and so is  $f$ .

For the last assertion, just notice  $\text{Hom}(RQX, RQY) \rightarrow \text{Hom}_{\text{Ho}(\mathcal{C})}(X, Y)$  is a surjection, and  $X \rightarrow RQX, Y \rightarrow RQY$  are weak equivalence hence are isomorphisms in  $\text{Ho}(\mathcal{C})$ .  $\square$

**Cor. (4.5.1.16).** If  $F, G : \text{Ho}(\mathcal{C}) \rightarrow \mathcal{D}$  are two functors and  $t : F \circ \gamma \rightarrow G \circ \gamma$  is a natural transformation, then  $t$  also gives a natural transformation  $F \rightarrow G$ .  $\lrcorner$

*Proof:* This is because the objects of  $\text{Ho}(\mathcal{C})$  are the same as that of  $\mathcal{C}$ , and the morphisms are generated by  $\gamma(f)$  and  $\gamma(g)^{-1}$  where  $g$  is a weak equivalence. Then the desired transformation commutative diagrams commute.  $\square$

**Lemma (4.5.1.17).** Let  $\mathcal{C}$  be a model category and  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor taking weak equivalences to isomorphisms, then if  $f \sim^l g$  or  $f \sim^r g$ ,  $F(f) = F(g)$ .  $\lrcorner$

*Proof:* We only prove for left homotopy and the right homotopy is dual: given the cylinder object  $A \wedge I$ , just need to prove that  $F(\partial_0) = F(\partial_1)$ .  $\square$

**Prop. (4.5.1.18).** Suppose  $A$  is cofibrant and  $X$  is fibrant, then the map  $\gamma : \text{Hom}(A, X) \rightarrow \text{Hom}_{\text{Ho}(\mathcal{C})}(A, X)$  is surjective, and induces a bijection  $\pi(A, X) \cong \text{Hom}_{\text{Ho}(\mathcal{C})}(A, X)$ .  $\lrcorner$

*Proof:* (4.5.1.17) shows  $\gamma$  identifies homotopic maps. Consider the following commutative diagram:

$$\begin{array}{ccc} \pi(RA, QX) & \longrightarrow & \pi(A, X) \\ \downarrow \gamma & & \downarrow \gamma \\ \text{Hom}_{\text{Ho}(\mathcal{C})}(RA, QX) & \longrightarrow & \text{Hom}_{\text{Ho}(\mathcal{C})}(A, X) \end{array} \quad .$$

The second vertical arrow is isomorphism by (4.5.1.15), the first arrow is isomorphism by (4.5.1.7).

The left vertical arrow is identity by construction, so the right vertical arrow is also isomorphism.  $\square$

**Cor. (4.5.1.19).** There is a natural isomorphism  $\pi\mathcal{C}_{cf} \cong \text{Ho}(\mathcal{C})$ .  $\lrcorner$

**Prop. (4.5.1.20) [Homotopy Category as Localizing Category].** Let  $\mathcal{C}$  be a model category and  $W$  the class of weak equivalences, then the functor  $\gamma : \mathcal{C} \rightarrow \text{Ho}(\mathcal{C})$  is the localizing category of  $\mathcal{C}$  w.r.t  $W$ .  $\lrcorner$

*Proof:* Cf. [Homotopy Theories and Model Categories, P29]. ?  $\square$

## 2 Quillen Adjunctions and Derived Functors

**Def. (4.5.2.1) [Quillen Adjunctions].** An adjunction  $(F, G) : \mathcal{C} \rightleftarrows \mathcal{D}$  between model categories is called a **Quillen adjunction** if  $F$  preserves cofibrations and  $G$  preserves fibrations. By adjointness and (4.5.0.4), in fact  $F$  preserves also trivial cofibrations and  $G$  preserves trivial fibrations.

Moreover, it is called a **Quillen equivalence** if for any cofibrant object  $C \in \mathcal{C}$  and fibrant object  $D \in \mathcal{D}$ , a map  $C \rightarrow G(D)$  is a weak equivalence iff the adjoint map  $F(C) \rightarrow D$  is a weak equivalence.  $\lrcorner$

**Def. (4.5.2.2) [Derived Functors].** Let  $\mathcal{C}$  be a model category and  $F : \mathcal{C} \rightarrow \mathcal{D}$  a functor, then a **left derived functor** of  $F$  is a left Kan extension of  $F$  along  $\gamma : \mathcal{C} \rightarrow \text{Ho}(\mathcal{C})$ .

Dually we can define **right derived functors**.  $\lrcorner$

**Prop. (4.5.2.3) [Existence of Derived Functors].** In the situation of (4.5.2.2), if  $F$  maps weak equivalences between cofibrant objects to isomorphisms in  $\mathcal{D}$ , then the left derived functor  $(LF, t)$  exists, and for each cofibrant object  $X$ , the morphism  $t_X : LF(X) \rightarrow F(X)$  is an isomorphism.

Dually for the right derived functor case.  $\lrcorner$

*Proof:* Cf. [Homotopy Theories and Model Categories, P42].  $\square$

**Lemma (4.5.2.4).** Let  $\mathcal{C}$  be a model category and  $F : \mathcal{C}_c \rightarrow \mathcal{D}$  be a functor that maps trivial cofibrations in  $\mathcal{C}_c$  to isomorphisms, then  $F$  maps right-homotopic morphisms to the same morphism.  $\lrcorner$

*Proof:* Let  $H : A \rightarrow B^I$  be a right homotopy between  $f$  and  $g$ , where  $B^I$  is a very good path object (4.5.1.5), then  $B \rightarrow B^I$  is a trivial cofibration, and thus mapped by  $F$  to an isomorphism. Then we can show  $F(\partial_0) = F(\partial_1)$ , and then  $F(f) = F(g)$ .  $\square$

**Def. (4.5.2.5) [Total Left Derived Functors].** Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a morphism of model categories, then a **total derived functor**

$$LF : \text{Ho}(\mathcal{C}) \rightarrow \text{Ho}(\mathcal{D})$$

is defined to be a left derived functor of the morphism  $\mathcal{C} \rightarrow \mathcal{D} \rightarrow \text{Ho}(\mathcal{D})$ .  $\lrcorner$

**Lemma (4.5.2.6) [Brown].** Let  $F$  be a morphism of model categories that maps trivial cofibration between cofibrant objects to weak equivalences, then it preserves weak equivalences between cofibrant objects.  $\lrcorner$

*Proof:* If  $f : A \rightarrow B$  is a weak equivalence between cofibrant objects, then we can factor the morphism  $(f, \text{id}) : A \amalg B \rightarrow B$  as  $A \amalg B \xrightarrow{q} C \xrightarrow{p} B$  that  $q$  is cofibration and  $p$  is trivial fibration. It can be shown that  $q \circ \partial_i : B \rightarrow C$  are trivial fibrations and  $C$  is cofibrant, thus  $F(q \circ \partial_i)$  are weak equivalences, and hence also  $F(p)$  is weak equivalence and so does  $F(f)$ .  $\square$

**Prop. (4.5.2.7) [Total Derived Functors and Quillen Equivalence].** If  $(F, G)$  is a pair of Quillen functors between two model categories  $\mathcal{C}, \mathcal{D}$ , then total derived functors

$$LF : \text{Ho}(\mathcal{C}) \rightleftarrows \text{Ho}(\mathcal{D}) : RG$$

exists and form an adjunction pair. And if  $(F, G)$  is a Quillen equivalence, then  $(LF, RG)$  defines an equivalence of homotopy categories.  $\lrcorner$

*Proof:* By (4.5.2.3) and its dual and (4.5.2.6), the total derived functors  $LF, RG$  exist.

Next for  $A$  cofibrant in  $\mathcal{C}$  and  $X$  fibrant in  $\mathcal{D}$ , we can show the adjunction map  $\text{Hom}(A, G(X)) \cong \text{Hom}(F(A), X)$  preserves homotopy equivalence relations (4.5.1.8) and induces an isomorphism  $\pi(A, G(X)) \cong \pi(F(A), X)$ : If  $H : A \wedge I \rightarrow X$  is a good homotopy between  $f, g$ , then  $A \wedge I$  is cofibrant, and because  $F$  preserves colimits and because of (4.5.2.6),  $F(A \wedge I)$  is cylinder object for  $F(A)$ , thus  $f^\flat \sim g^\flat$ . A dual argument shows the converse.

Now for any  $A \in \mathcal{C}$  and  $X \in \mathcal{D}$ , there is a bijection

$$\text{Hom}_{\text{Ho}(\mathcal{C})}(A, RG(X)) \cong \pi(QA, G(SX)) \cong \pi(F(QA), SX) \cong \text{Hom}_{\text{Ho}(\mathcal{D})}(LFA, X)$$

where the first isomorphism is due to the fact  $QA \rightarrow RQA$  is trivial cofibration and  $G(SX)$  is fibrant, thus we can use (4.5.1.7), dually for the last isomorphism.

Finally if  $(F, G)$  is a Quillen equivalence, then consider the unit map:

$$A \rightarrow RG(LF(A)).$$

If  $A$  is cofibrant, then this is  $A \rightarrow G(SF(A))$  which is a weak equivalence because  $F(A) \rightarrow SF(A)$  does, so it is an isomorphism in  $\text{Ho}(\mathcal{C})$ . Now any object in  $\text{Ho}(\mathcal{C})$  is isomorphic to a cofibrant object, we know the unit map is an isomorphism. Dually the counit map is an isomorphism, thus  $LF, GF$  are a pair of equivalences.  $\square$

### 3 Combinatorial Model Structure

Cf. [HTT, A.2.6]

**Def. (4.5.3.1) [Cofibrantly-Generated Model Categories].** A cofibrantly-generated model category is a model category  $\mathcal{C}$  that

- there is a small set  $I$  of generating cofibrations that generates the class of cofibrations as the minimal weakly saturated class containing  $I$ .
- there is a small set  $J$  of generating trivial cofibrations that generates the class of trivial cofibrations as the minimal weakly saturated class containing  $J$ .

┘

**Def. (4.5.3.2) [Combinatorial Model Categories, Smith].** A combinatorial model category is a cofibrantly generated (4.5.3.1) and locally presentable (4.1.2.5) model category.  $\square$

**Prop. (4.5.3.3) [Smith].** Let  $\mathcal{M}$  be a combinatorial model category and  $\mathcal{M}^{[1]}$  the category of arrows, then the full subcategory generated by fibrations, weak equivalences, or trivial fibrations are all full accessible (4.1.2.4) subcategories of  $\mathcal{M}^{[1]}$ .  $\square$

*Proof:* Cf. [Lur09]P818.  $\square$

**Prop. (4.5.3.4) [Constructing Combinatorial Model Categories].** Let  $\mathcal{M}$  be a locally presentable category and  $W, C$  be classes of morphisms in  $\mathcal{M}$  s.t.

- $C$  is a weakly saturated class generated by a small subset  $C_0$ .
- $C \cap W$  is a weakly saturated class.
- $W \subset A^{[1]}$  is a full accessible subcategory.
- $W$  satisfies the 2-out-of-3 property.

- $r(C) \subset W$ .

Then  $\mathcal{M}$  admits a combinatorial model category with

- Cofibrations:  $C$ .
- Weak equivalences:  $W$ .
- Fibrations:  $r(C \cap W)$ .

┘

*Proof:* Cf. [Lur09]P821.

□

**Lemma (4.5.3.5).** Situation as in (4.5.3.4), the class  $C \cap W$  is a weakly saturated class generated by a small subset  $S \subset C \cap W$ .

┘

*Proof:* Cf. [Lur09]P819.

□

**Prop. (4.5.3.6) [Constructing Left Proper Model Categories].** Let  $\mathcal{M}$  be a locally presentable category with a class  $W$  of morphisms and a small set of morphisms  $C_0$  s.t.

- $W$  is a perfect class ?,
- $C_0$  is stable under cobase change.
- $W$  is stable under cobase change by  $C_0$ .
- $r(C_0) \subset W$ .

Then there is a left proper combinatorial model category on  $\mathcal{M}$  with

- Cofibrations: The weakly saturated closure  $C$  of  $C_0$ .
- Weak Equivalences:  $W$ .
- Fibrations:  $r(C \cap W)$ .

Moreover, any left proper combinatorial model category arises in this way.

┘

*Proof:* Cf. [Lur09]P823.

□

## 4 Generating new Model Categories

**Prop. (4.5.4.1) [Overcategories and Undercategories].** If  $\mathcal{C}$  is a model category and  $A \in \mathcal{C}$ , then the undercategory  $\mathcal{C}_{A/}$  and the overcategory  $\mathcal{C}_{/A}$  have natural model structures.

┘

*Proof:*

□

**Prop. (4.5.4.2) [Transfer Model Structures via Left Adjoint].** Let

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G$$

be an adjoint pair of categories and

- $\mathcal{C}, \mathcal{D}$  are complete and cocomplete,
- $\mathcal{C}$  is a cofibrantly generated model category,
- $\mathcal{C}, \mathcal{D}$  are presentable categories and  $G$  is an accessible functor.
- If we define a morphism  $f$  in  $\mathcal{B}$  a fibration/weak equivalent iff  $G(f)$  is a fibration/weak equivalence, and a cofibration iff it has left lifting property w.r.t. trivial fibrations, then  $\mathcal{B}$  has a path object factorization and a fibrant replacement operator.

Then this defines a cofibrantly generated model category on  $\mathcal{B}$ , and makes  $(F, G)$  a Quillen adjunction.  $\lrcorner$

*Proof:* It suffices to show that the factorization property holds: In fact, it suffices to show if  $I, J$  are generating classes of cofibrations and trivial cofibrations, then  $F(I), F(J)$  are generating classes of cofibrations and trivial cofibrations. But this suffices to show  $F(J)$  is a weak equivalence. For this, show that any morphism in  $\mathcal{B}$  that has left lifting properties w.r.t. all fibrations is a weak equivalence using hypothesis item4. ?  $\square$

### Bousfield Localizations

Cf.[P.S. Hirschhorn. Model categories and their localizations].

**Def. (4.5.4.3) [Bousfield Localizations].** Let  $\mathcal{M}, \mathcal{M}'$  be model categories with the same underlying category, then  $\mathcal{M}'$  is called a **Bousfield localization** of  $\mathcal{M}$  if

- $\text{Cof}(\mathcal{M}) = \text{Cof}(\mathcal{M}')$ .
- $\text{Weak}(\mathcal{M}) \subset \text{Weak}(\mathcal{M}')$ .

$\lrcorner$

## 5 Enriched and Monoidal Model Categories

**Def. (4.5.5.1) [Left Quillen Bifunctor].** Let  $\mathcal{A}, \mathcal{B}, \mathcal{C}$  be model categories, then a functor  $\mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$  is called a **left Quillen bifunctor** if

- For any cofibrations  $i : A \rightarrow A' \in \mathcal{A}, B \rightarrow B' \in \mathcal{B}$ , the induced map

$$i \wedge j : F(A', B) \coprod_{F(A, B)} F(A, B') \rightarrow F(A', B')$$

is a cofibration in  $\mathcal{C}$ .

- $F$  preserves small colimits separably in each variables.

$\lrcorner$

**Def. (4.5.5.2) [Monoidal Model Category].** A **monoidal model category** is a monoidal category  $\mathcal{S}$  equipped with a model structure that:

- The tensor product  $\otimes : \mathcal{S} \times \mathcal{S} \rightarrow \mathcal{S}$  is a left Quillen bifunctor.
- The unit objects  $1 \in \mathcal{S}$  is cofibrant.
- The monoidal structure is closed.

$\lrcorner$

**Def. (4.5.5.3) [Enriched Model Category].** Given a monoidal model category  $\mathcal{S}$ , an  **$\mathcal{S}$ -enriched model category** is an  $\mathcal{S}$ -enriched category  $\mathcal{A}$  with a model structure satisfying:

- $\mathcal{A}$  is tensored and cotensored over  $\mathcal{S}$ . ?
- the tensor product  $\mathcal{A} \times \mathcal{S} \rightarrow \mathcal{S}$  is a left Quillen bifunctor (4.5.5.1).

$\lrcorner$

**Prop. (4.5.5.4).** The second condition in (4.5.5.3) is equivalent to the following: For a cofibration  $i : U \rightarrow V$  and a fibration  $p : X \rightarrow Y$ , the induced map

$$\text{Map}(V, X) \xrightarrow{(i^*, p_*)} \text{Map}(U, X) \times_{\text{Map}(U, Y)} \text{Map}(V, Y)$$

is a fibration in  $\mathcal{S}$ , and trivial fibration if any of  $i, p$  is weak equivalence.  $\lrcorner$

*Proof:* Use the adjunction relations to write it out?  $\square$

**Def. (4.5.5.5) [Fibrant Enriched Categories].** Let  $\mathcal{A}$  be a  $\mathcal{S}$ -enriched category, then we denote  $\mathcal{A}^\circ$  the subcategory of bifibrant objects of  $\mathcal{A}$ , which is also a  $\mathcal{S}$ -enriched category.  $\lrcorner$

## 6 Diagram Categories

[Lur09]A.2.8. A.3.3., A.3.5.

**Prop. (4.5.6.1) [(Injective)Projective Model Categories].** Let  $\mathcal{S}$  be an excellent model category and  $\mathcal{A}$  a combinatorial  $\mathcal{S}$ -enriched model category,  $\mathcal{C}$  a small  $\mathcal{S}$ -enriched category, then there are two model structures on  $\mathcal{A}^{\mathcal{C}}$ :

- The **projective model category**  $\text{Func}(\mathcal{C}, \mathcal{A})_{\text{Proj}}$  with
  - Fibrations: **projective fibrations**  $F$  s.t.  $F(C) \rightarrow G(C)$  is a fibration in  $\mathcal{A}$  for each  $C \in \mathcal{C}$ .
  - Weak equivalences:  $F$  s.t.  $F(C) \rightarrow G(C)$  is a weak equivalence in  $\mathcal{A}$  for each  $C \in \mathcal{C}$ .
  - Cofibrations: **projective cofibrations** determined by the above two.
- The **injective model category**  $\text{Func}(\mathcal{C}, \mathcal{A})_{\text{inj}}$  with
  - Cofibrations: **injective cofibrations**  $F$  s.t.  $F(C) \rightarrow G(C)$  is a cofibration in  $\mathcal{A}$  for each  $C \in \mathcal{C}$ .
  - Weak equivalences:  $F$  s.t.  $F(C) \rightarrow G(C)$  is a weak equivalence in  $\mathcal{A}$  for each  $C \in \mathcal{C}$ .
  - Fibrations: **injective fibrations** determined by the above two.

$\lrcorner$

*Proof:* Cf. [Lur09]P868, P828.  $\square$

**Cor. (4.5.6.2).** Both model categories are left/right proper iff  $\mathcal{A}$  is.  $\lrcorner$

**Cor. (4.5.6.3).** Projective cofibrations are injective cofibrations, and injective fibrations are projective fibrations.  $\lrcorner$

**Prop. (4.5.6.4).** Let  $\mathcal{C}$  be a model category and let

$$\begin{array}{ccc} A & \xrightarrow{j} & A_1 \\ \downarrow i & & \downarrow \\ A_0 & \longrightarrow & A_0 \amalg_A A_1 \end{array}$$

be a pushout diagram, then it is a homotopy pushout diagram if either of the following is satisfied:

- $j$  is a cofibration and  $A, A_0$  are cofibrant.
- $j$  is a cofibration and  $\mathcal{C}$  is left proper.

$\lrcorner$

*Proof:*

$\square$



## Reedy Model Categories

### Homotopy Colimits and Limits

Should be redone in the general language of diagram categories. ?

**Def. (4.5.6.5) [Homotopy Colimits and Limits].** In a model category  $\mathcal{M}$ , for any diagram  $A_0 \leftarrow A \rightarrow A_1$ , there exists a commutative diagram

$$\begin{array}{ccccc} A'_0 & \xleftarrow{i} & A' & \xrightarrow{j} & A'_1 \\ \downarrow & & \downarrow & & \downarrow \\ A_0 & \xleftarrow{\quad} & A & \xrightarrow{\quad} & A_1 \end{array} .$$

such that  $A'$  is cofibrant, and  $i, j$  are both cofibrant, and the vertical arrows are weak equivalences.

Then it can be shown that  $A'_0 \coprod_{A'} A'_1$  only depends on  $A_0 \leftarrow A \rightarrow A_1$  up to weak equivalence. Then a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & A_1 \\ \downarrow & & \downarrow \\ A_0 & \longrightarrow & C \end{array}$$

is called a **homotopy colimit** if for any such cofibrant replacement,

$$A'_0 \coprod_{A'} A'_1 \rightarrow A_0 \coprod_A A_1 \rightarrow C$$

is a weak equivalence.

Dually we can define homotopy limits. ┘

*Proof:* ? □

**Cor. (4.5.6.6).** Weakly equivalent diagrams induce weakly equivalent homotopy colimits. ┘

**Prop. (4.5.6.7).** In a model category  $\mathcal{M}$ , a diagram

$$\begin{array}{ccc} A & \xrightarrow{j} & A_1 \\ \downarrow & & \downarrow \\ A_0 & \longrightarrow & A_0 \coprod_A A_1 \end{array}$$

is a homotopy limit if  $j$  is cofibrant, and either

- $A, A_0$  are cofibrant, or
  - $\mathcal{M}$  is left proper.
- ┘

*Proof:* ? □

## 7 Model Structures on $\mathcal{Cat}_S$

**Def. (4.5.7.1) [Homotopy Category].** Let  $S$  be a monoidal model category, then there is a natural monoidal structure on the homotopy category  $hS$  (4.5.1.14), and the functor  $S \mapsto hS$  is monoidal, thus we can transfer from a category  $\mathcal{C}$  enriched over  $S$  to an  $hS$ -enriched category, called the **homotopy category** of  $\mathcal{C}$ .  $\lrcorner$

**Def. (4.5.7.2) [Weak Equivalences of Enriched Categories].** Let  $S$  be a monoidal model category and  $\mathcal{Cat}_S$  be the category of categories enriched over  $S$ , then a morphism  $F : \mathcal{C} \rightarrow \mathcal{D} \in \mathcal{Cat}_S$  is called a **weak equivalence** if it induced an isomorphism of their homotopy categories, or equivalently,

- $F$  is essentially surjective,
  - For any  $X, Y \in \mathcal{C}$ ,  $\text{Map}_{\mathcal{C}}(X, Y) \rightarrow \text{Map}_{\mathcal{D}}(FX, FY)$  is a weak equivalence in  $S$ .
- $\lrcorner$

**Def. (4.5.7.3) [Generating Cofibrations in  $\mathcal{Cat}_S$ ].** Let  $S$  be a monoidal model category,  $A$  is an object of  $S$ , then we can denote  $[1]_A$  the  $S$ -enriched category consists of objects  $\{X, Y\}$  and that  $\text{Hom}(X, X) = \text{Hom}(Y, Y) = 1_S$ ,  $\text{Hom}(X, Y) = A$ ,  $\text{Hom}(Y, X) = \emptyset$ . And if  $1_S$  is the initial object of  $S$ , then we denote  $[1]_{1_S}$  by  $[1]_S$ . Also we denote  $[0]_S$  the  $S$ -enriched category consisting of one element and the morphism space is  $1_S$ .

Let  $[1]_S^\sim$  be the category consisting of two objects  $\{X, Y\}$  that  $\text{Hom}(Z_1, Z_2) = 1_S$  for any  $Z_1, Z_2 \in S$ .

Let  $C_0$  be the class of morphisms in  $\mathcal{Cat}_S$  consisting of

- $\emptyset \rightarrow [0]_S$ .
  - The induced map  $[1]_S \rightarrow [1]_{S'}$  where  $S \rightarrow S'$  ranges over a generating class of cofibrations of  $S$ .
- $\lrcorner$

**Prop. (4.5.7.4) [Model Category on  $\mathcal{Cat}_S$ ].** Let  $S$  be a combinatorial monoidal model category that every object of  $S$  is cofibrant and the collection of weak equivalences of  $S$  is stable under filtered colimits, then there exists a left proper combinatorial model structure on  $\mathcal{Cat}_S$  that

- The class of cofibrations in  $\mathcal{Cat}_S$  is the smallest weakly saturated class generated by  $C_0$  defined in (4.5.7.3),
  - The weak equivalences are as defined in (4.5.7.2).
- $\lrcorner$

*Proof:* Cf. [HTT, P856].  $\square$

**Cor. (4.5.7.5).** Let

$$f : S \rightleftarrows S' : g$$

be a Quillen adjunction between monoidal model categories satisfying conditions in (4.5.7.4), then they induces a Quillen adjunction

$$F : \mathcal{Cat}_S \rightleftarrows \mathcal{Cat}_{S'} : G$$

and this is a Quillen equivalence if  $(f, g)$  is.  $\lrcorner$

*Proof:*

$\square$

**Prop. (4.5.7.6).** Let  $\mathcal{C}, \mathcal{D}$  be  $\mathcal{S}$ -enriched model categories and

$$F : \mathcal{C} \xrightleftharpoons{\text{Quillen}} \mathcal{D} : G$$

is a Quillen adjunction of underlying model categories. Assume every objects of  $\mathcal{C}$  is cofibrant and the maps  $\beta_{X,S} : S \otimes F(X) \rightarrow F(S \otimes X)$  is a weak equivalence for  $X \in \mathcal{C}, S \in \mathcal{S}$  cofibrant, then the following are equivalent:

- $(F, G)$  is a Quillen equivalence.
- $G$  determines a weak equivalence (4.5.7.2) of the underlying  $\mathcal{S}$ -enriched categories  $\mathcal{D}_{cf} \rightarrow \mathcal{C}_{cf}$ .

┘

*Proof:* Cf.[HTT, P853].

□

**Def. (4.5.7.7) [Local Fibrations].** Let  $\mathcal{C}$  be an  $\mathcal{S}$ -enriched category where  $\mathcal{S}$  is a monoidal model category, then a morphism  $f \in \mathcal{C}$  is called an **equivalence** if it maps to an isomorphism in  $h\mathcal{C}$ .

$\mathcal{C}$  is called **locally fibrant** if for any  $X, Y \in \mathcal{C}$ , the mapping space  $\text{Map}(X, Y)$  is fibrant in  $\mathcal{S}$ .

An  $\mathcal{S}$ -enriched functor  $F : \mathcal{C} \rightarrow \mathcal{C}'$  is called a **local fibration** if the following conditions are satisfied:

- for any  $X, Y \in \mathcal{C}$ , the induced map  $\text{Map}(X, Y) \rightarrow \text{Map}(FX, FY)$  is a fibration in  $\mathcal{S}$ .
- the induced map  $h\mathcal{C} \rightarrow h\mathcal{C}'$  is a quasi-fibration of categories.

┘

**Def. (4.5.7.8) [Invertibility Hypothesis].** We say a monoidal model category  $\mathcal{S}$  satisfies the **invertibility hypothesis** if: For any cofibrant morphism  $[1]_{\mathcal{S}} \rightarrow \mathcal{C}$  (4.5.7.3) of  $\mathcal{S}$ -enriched categories, and maps to a morphism  $f$  which is invertible in the homotopy category  $h\mathcal{C}$ , take the pushout:

$$\begin{array}{ccc} [1]_{\mathcal{S}} & \xrightarrow{i} & \mathcal{C} \\ \downarrow & & \downarrow j \\ [1]_{\mathcal{S}}^{\sim} & \longrightarrow & \mathcal{C}\langle f^{-1} \rangle \end{array} .$$

then  $j$  is a weak equivalence of  $\mathcal{S}$ -enriched categories (4.5.7.2).

┘

**Def. (4.5.7.9) [Excellent Model Category].** An **excellent model category** is a monoidal model category  $\mathcal{S}$  that The monoidal structure is symmetric.

- $\mathcal{S}$  is combinatorial,
- Every monomorphism in  $\mathcal{S}$  is a cofibration, and the collection of cofibrations is stable under products,
- The class of weak equivalences in  $\mathcal{S}$  is stable under filtered colimits,
- $\mathcal{S}$  satisfies the invertibility condition (4.5.7.8)

┘

**Lemma (4.5.7.10).** Let  $T : \mathcal{S} \rightarrow \mathcal{S}'$  be a monoidal functor between monoidal model categories satisfies axioms besides invertibility hypothesis, that is also a left Quillen functor, then if  $\mathcal{S}'$  satisfies invertibility hypothesis, so does  $\mathcal{S}$ .

┘

*Proof:* Cf.[HTT, P862]. ?

□

**Prop. (4.5.7.11) [Fibration and Local Fibration].** If  $\mathcal{S}$  is an excellent model category, then

- An  $\mathcal{S}$ -enriched category  $\mathcal{C}$  is a fibrant object in  $\text{Cat}_{\mathcal{S}}$  iff it is locally fibrant (4.5.7.7).
- Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be an  $\mathcal{S}$ -enriched functor and  $\mathcal{D}$  is fibrant in  $\text{Cat}_{\mathcal{S}}$ , then  $F$  is fibrant in  $\text{Cat}_{\mathcal{S}}$  iff it is a local fibration.

┘

*Proof:* Cf. [HTT, P863].

□

## Path Spaces

### Homotopy Colimits of Enriched Categories

## 8 Examples

**Prop. (4.5.8.1) [Kan-Quillen Model Structure].** By (4.6.3.48),  $s\text{Set}$  is a combinatorial left and right proper Kan-Quillen model category with

- Weak equivalences: weak equivalences,
- Cofibrations: inclusions,
- Fibrations: Kan fibrations.

┘

**Prop. (4.5.8.2) [Dwyer, Kan].**  $s\text{Set}$  has an excellent model category with the Kan model structure and the Cartesian monoidal structures.

┘

*Proof:*

□

**Prop. (4.5.8.3) [ $q$ -Model Structure].** For a unital ring  $R$ , then the category  $\text{Ch}_{\mathbb{N}} R$  has the structure of a model category with a morphism  $f : M_{\bullet} \rightarrow N_{\bullet}$  being

- a weak equivalence if  $H_n(f)$  is isomorphism for any  $n$ .
- a fibration if  $M_n \rightarrow N_n$  is surjective for any  $n \geq 1$ .
- a cofibration if  $M_n \rightarrow N_n$  is injective with projective cokernel for any  $n \geq 0$ .

┘

*Proof:* [Model category and simplicial methods, P5] or [Homotopy Theories and Model Categories].

□

**Prop. (4.5.8.4) [Serre-Quillen].** By (4.13.7.28), the category  $\mathcal{CG}$  can be given a **Serre-Quillen model structure** with

- Weak equivalences: weak homotopy equivalence,
- Fibrations: Serre fibrations,
- Cofibrations: Retracts of morphisms  $X \rightarrow Y$  where  $Y$  is obtained from  $X$  by attaching cells.

And this restricts to a model category on the category  $\mathcal{CG}\mathcal{H}$  of compactly generated weak Hausdorff spaces.

┘

**Prop. (4.5.8.5) [Hurewicz-Strøm].** The category  $\mathcal{Top}$  can be given a **Hurewicz-Strøm model structure** with

- Weak equivalences: homotopy equivalences.
- Cofibrations: closed Hurewicz cofibrations.

- Fibrations: Hurewicz fibrations.

┘

*Proof:* See(4.13.7.31).

□

**Prop.(4.5.8.6) [Derived Categories Model Structure].** If  $\mathcal{A}$  is an Abelian category with enough injectives, then  $K^+(\mathcal{A})$  is a model category with

- Weak equivalence: quasi-isomorphisms,
- Fibration: epimorphisms with  $\ker$  in  $K^+(\mathcal{I})$ ,
- Cofibration: monomorphisms.

┘

*Proof:*

□

**Prop.(4.5.8.7) [Joyal Model Structures].** By(4.6.4.19), there is a Joyal Model category structure on  $s\mathbf{Set}$  with

- Cofibrations: monomorphisms.
- Weak equivalences: categorical equivalences defined in(4.6.4.12).
- Fibrations: **categorical fibrations** or **Joyal fibrations** which has the right lifting property w.r.t. trivial cofibrations.

┘

**Prop.(4.5.8.8) [Reedy Model Structures].** Cf.[HTT, A.2.9].

┘

**Prop.(4.5.8.9) [Bergner-Model Structure].** There is a Bergner model structure on  $s\mathbf{Cat}$ ?(4.6.4.4)

┘

## 4.6 Simplicial Homotopy Theory

Main references are [Jardine Simplicial Homotopy Theory], [Lur09].

**Notation (4.6.0.1).**

- Use notations from [Model Categories](#).

┘

### 1 Simplicial Objects

**Def. (4.6.1.1) [Simplex Category].** The **simplex category**  $\Delta$  consists of simplicial objects  $[n]$  for each  $n \geq 0$  and there maps are order-preserving maps.

$\Delta$  has a subcategory  $\Delta_+$  consisting of the same objects but the morphisms are all surjective order-preserving maps.

For  $\mathcal{C} \in \mathcal{T}\text{op}$ , a **simplicial object** in  $\mathcal{C}$  is a functor  $\Delta^{\text{op}} \rightarrow \mathcal{C}$ . A **cosimplicial object** in  $\mathcal{C}$  is a simplicial object in  $\mathcal{C}^{\text{op}}$ . The category of simplicial objects in  $\mathcal{C}$  is denoted by  $s\mathcal{C}$ .

Given a simplicial or cosimplicial object in  $\mathcal{A}$ , its **underlying degeneracy map** is defined to be?

┘

**Prop. (4.6.1.2).** If  $\mathcal{C}$  is complete or cocomplete, then so is  $s\mathcal{C}$ .

┘

**Def. (4.6.1.3).**  $\Delta^n$  is the simplicial set  $\Delta^n([m]) = \text{Hom}([m], [n])$ .

┘

**Def. (4.6.1.4) [Augmentation].** If  $X$  is a simplicial object in a category, then an **augmentation** of  $X$  is a morphism  $d : A \rightarrow X$  that  $dd_0 = dd_1$ . In case  $\mathcal{C}$  is  $\text{Mod}_R$ , this is equivalent to a morphism  $\pi_0(X) \rightarrow A$  (5.8.2.2).

┘

**Def. (4.6.1.5) [s-Free Simplicial Objects].**  $X \in s\mathcal{C}$  is called **s-free** if the underlying category  $\Delta_+^{\text{op}} \rightarrow \mathcal{C}$  is  $\Delta^{\text{op}}$ -free (4.1.3.16).

Equivalently, an  $s$ -free object is a simplicial object  $X$  that there are objects  $Z_n \in \mathcal{C}$  that  $X_n = \coprod_{\varphi: [n] \rightarrow [k]} \varphi^* Z_k$ . Moreover, a simplicial morphism of simplicial objects are called  $s$ -free if the underlying diagram  $X_+ \rightarrow Y_+$  is of the form  $X_+ \rightarrow X_+ \coprod Y_0$  where  $Y_0$  is  $s$ -free.

┘

**Prop. (4.6.1.6) [ $s\mathcal{C}$  is a Simplicial Category].** Let  $s\mathcal{C}$  be the category of simplicial  $\mathcal{C}$ -objects, then it can be made into a simplicial category which is also tensored and cotensored (4.2.2.7) over  $s\text{Set}$ .

┘

*Proof:* We define first a action of  $s\text{Set}$  on  $s\mathcal{C}$ :

$$\otimes : s\text{Set} \times s\mathcal{C} \rightarrow s\mathcal{C} : (K, X) \mapsto (K \otimes X)_n = \coprod_{K_n} X_n,$$

with the simplicial maps determined by that of  $K$  and  $X$ .

Also there is a action of  $s\text{Set}^{\text{op}}$  on  $s\mathcal{C}$ :

$$(-)^- : s\text{Set}^{\text{op}} \times s\mathcal{C} \rightarrow s\mathcal{C} : (K \otimes X)_n = \coprod_{K_n} X_n$$

with the simplicial maps determined by that of  $K$  and  $X$ .

Then there is an adjointness

$$\text{Hom}_{s\mathcal{C}}(K \otimes X, Y) \cong \text{Hom}_{s\mathcal{C}}(X, Y^K)$$

for any simplicial set  $K$ .

Next we define  $\text{Map}_{s\mathcal{C}}(X, Y) \subset \text{Set}_\Delta$  as  $(\text{Map}_{s\mathcal{C}}(X, Y))_n = \text{Hom}_{s\mathcal{C}}(X \otimes \Delta^n, Y)$ , then there are functorial isomorphisms

$$\text{Map}_{s\mathcal{C}}(K \otimes X, Y) \cong \text{Map}_{s\mathcal{C}}(X, Y^K) \cong \text{Map}_{\text{Set}_\Delta}(K, \text{Map}_{s\mathcal{C}}(X, Y))$$

(easy to check). So we are done.  $\square$

**Remark (4.6.1.7) [Realization Functor].** For  $\mathcal{C} \in \text{Cat}$  that has countable colimits, there is a **simplicial realization functor**

$$|-| : s\mathcal{C} \rightarrow \mathcal{C} : X \mapsto \varinjlim_{\Delta_+^{\text{op}}} X_n.$$

$\lrcorner$

## 2 Topological Categories

**Def. (4.6.2.1) [Topological Categories].** A **topological category** is a category that is enriched over the category  $\mathcal{C}\mathcal{G}$  of compactly generated and Hausdorff spaces. The category of topological categories is denoted by  $\text{Cat}_{\mathcal{C}\mathcal{G}}$ .

Two topological categories is called **strongly equivalent** if they are equivalent as enriched categories.  $\lrcorner$

**Def. (4.6.2.2) [Homotopy Category of a Topological Category].** Given a topological category  $\mathcal{C}$ , the **homotopy category**  $h\mathcal{C}$  of  $\mathcal{C}$  is defined to be the category transferred from  $\mathcal{C}$  (4.2.2.4) by the right-lax monoidal functor  $\pi_0$  (4.1.5.15).  $\lrcorner$

**Def. (4.6.2.3) [Homotopy Category of Spaces].** Let  $\mathcal{C}$  be the category of CW complexes that the morphisms are given the compact-open topology, then its homotopy category  $\mathcal{H}$  is called the **homotopy category of spaces**.  $\lrcorner$

## 3 Simplicial Sets

### Simplicial Sets

**Remark (4.6.3.1).** The fact that any simplicial set  $X$  is a colimit of  $\Delta^n$  (4.1.3.13) is important in proving properties of constructions of simplicial set.  $\lrcorner$

**Def. (4.6.3.2) [Objects and Morphisms].** Given a simplicial set  $S$ , its **objects** are just objects in  $S([0])$ , and its **morphisms** are objects in  $S([1])$ .  $\lrcorner$

**Def. (4.6.3.3) [Simplicial Sets and Categories].** There is an embedding  $\Delta \rightarrow \text{Cat} : [n] \rightarrow [n]$  regarding  $[n]$  as a category, which by Yoneda extension (4.1.3.14) and (4.1.1.47) corresponds to an adjunction

$$\tau_1 : s\text{Set} \rightleftarrows \text{Cat} : N$$

where  $\tau_1$  is called the **fundamental category functor** and  $N : \text{Cat} \rightarrow s\text{Set}$  is the **nerve functor**.  $\lrcorner$

**Prop. (4.6.3.4) [Nerve Functor is Fully Faithful].** The nerve functor  $N : \text{Cat} \rightarrow s\text{Set}$  is fully faithful. Equivalently, by (4.1.1.32), for any  $\mathcal{C} \in \text{Cat}$ , there is a natural isomorphism  $\tau_1(N(\mathcal{C})) \cong \mathcal{C}$ .  $\lrcorner$

**Prop. (4.6.3.5) [Natural Transformation and Homotopy].** A natural transformation will induce homotopic nerve map. thus a pair of adjoint functors will induce a simplicial homotopy between their nerve.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.6.3.6) [Characterizing Nerves].** For  $K \in s\text{Set}$ ,

- there exists  $\mathcal{C} \in \text{Cat}$  that  $K \cong N(\mathcal{C})$  iff for each  $0 < i < n$  and each diagram

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & K \\ \downarrow & \nearrow \text{dotted} & \\ \Delta^n & & \end{array}$$

there exists uniquely a dotted arrow.

- there exists  $\mathcal{C} \in \text{Grpd}$  that  $K \cong N(\mathcal{C})$  iff for each  $0 \leq i \leq n$  and each diagram

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & K \\ \downarrow & \nearrow \text{dotted} & \\ \Delta^n & & \end{array}$$

there exists uniquely a dotted arrow.  $\lrcorner$

*Proof:* [HTT, P9].  $\square$

**Def. (4.6.3.7) [Fundamental Groupoid Functor].** Let  $\pi_1 = \text{Grp} \circ \tau_1 : s\text{Set} \rightarrow \text{Grpd}$ , called the **fundamental groupoid functor**, then by (4.6.3.3) and (4.1.1.36), there is an adjunction

$$\pi_1 : s\text{Set} \rightleftarrows \text{Grpd} : N$$

**Def. (4.6.3.8) [Topological Realization Functor].** There is a functor  $\Delta \rightarrow \mathcal{CG} : [n] \mapsto \Delta^n$ , which by Yoneda extension (4.1.3.14) corresponds to an adjunction

$$|\cdot| : s\text{Set} \rightleftarrows \mathcal{CG} : \text{Sing}$$

called the **topological geometrization** functor and the singular complex functor.  $\lrcorner$

**Def. (4.6.3.9) [Classifying Spaces of a Category].** For  $\mathcal{C} \in \text{Cat}$ , the **classifying space** of  $\mathcal{C}$  is defined to be  $\mathbb{B}\mathcal{C} = |N(\mathcal{C})| \in \mathcal{CG}$ .  $\lrcorner$

**Prop. (4.6.3.10).** The geometrization as a functor from  $s\text{Set} \rightarrow \mathcal{CG}$  preserves finite limits.

The three kinds of geometrization of a bisimplicial set is the same: geometrization the diagonal simplicial set, the twice geometrization of left(resp. right) simplicial set.  $\lrcorner$

*Proof:* Cf.[Jardine P9]. ?  $\square$

**Def. (4.6.3.11) [Weak (Homotopy) Equivalences].** A morphism of simplicial sets  $S \rightarrow T$  is called a **weak equivalence** if the induced map  $|S| \rightarrow |T|$  (4.6.3.8) is a weak homotopy equivalence.  $\lrcorner$



### Constructing Simplicial Sets

**Def. (4.6.3.12) [Opposite Simplicial Sets].** There is an involution in the simplex category  $\iota : \Delta \rightarrow \Delta$  that maps any ordered set to its reverse order. Then for any simplicial set  $S : \Delta^{\text{op}} \rightarrow \text{Set}$ , there is another simplicial set  $S^{\text{op}} = S \circ \iota$ , called the **opposite simplicial set**.  $\lrcorner$

**Prop. (4.6.3.13).** For  $\mathcal{C} \in \text{Cat}$ , there is a natural isomorphism of simplicial sets  $N(\mathcal{C})^{\text{op}} \cong N(\mathcal{C}^{\text{op}})$ .  $\lrcorner$

**Def. (4.6.3.14) [Mapping Spaces].** For  $X, Y \in s\text{Set}$ , the **mapping space**  $\text{Map}(X, Y)$  or  $Y^X$  is a simplicial set s.t.

$$\text{Map}(X, Y)_n = \text{Hom}_{s\text{Set}}(\Delta^n \times X, Y).$$

$\lrcorner$

**Prop. (4.6.3.15) [Closed Cartesian Monoidal Structure].**  $s\text{Set}$  is a closed Cartesian monoidal category, and for any  $X, Y, Z \in s\text{Set}$ , there is an isomorphism of simplicial sets

$$\text{Map}(X, \text{Map}(Y, Z)) \cong \text{Map}(X \times Y, Z).$$

$\lrcorner$

*Proof:*

$\square$

**Prop. (4.6.3.16).** For  $\mathcal{A}, \mathcal{B} \in \text{Cat}$ , there is a natural isomorphism of simplicial sets  $N(\text{Func}(\mathcal{A}, \mathcal{B})) \cong \text{Map}(N(\mathcal{A}), N(\mathcal{B}))$ .  $\lrcorner$

*Proof:* There are natural bijections

$$\begin{aligned} N(\text{Func}(\mathcal{A}, \mathcal{B}))_n &= \text{Func}([n] \times \mathcal{A}, \mathcal{B}) \\ (4.6.3.4) \quad &\cong \text{Map}(N([n] \times \mathcal{A}), N(\mathcal{B})) \\ &\cong \text{Map}(\Delta^n \times N(\mathcal{A}), N(\mathcal{B})) \\ &= \text{Map}(N(\mathcal{A}), N(\mathcal{B}))_n, \end{aligned}$$

and they form an isomorphism of simplicial sets.  $\square$

**Cor. (4.6.3.17).** For any  $x \in S_m, x' \in S'_n$ , there is a unique  $y \in (S * S')_{m+n+1}$  s.t.  $y|_{[0, \dots, m]} \cong x, y|_{[m+1, \dots, m+n+1]} \cong x'$ .  $\lrcorner$

**Def. (4.6.3.18) [Joins].** Let  $S, S' \in s\text{Set}$ , then the **join** of  $S, S'$  is defined to be the simplicial set that for all any finite ordered set  $J$ ,

$$(S \star S')(J) = \cup_{J=I \amalg I'} S(I) \times S'(I').$$

where  $I, I'$  satisfies  $i < i'$  for any  $i \in I, i' \in I'$ . And the glueing is natural.  $\lrcorner$

**Prop. (4.6.3.19).**

- $\Delta^i \star \Delta^j \cong \Delta^{i+j+1}$ .
- The join operation is a functor  $s\text{Set} \rightarrow s\text{Set}$  that preserves colimits in each coordinate,
- $s\text{Set}$  is a symmetric monoidal category under the join operation.
- If  $\mathcal{C}, \mathcal{C}' \in \text{Cat}$ , there is a natural isomorphism  $N(\mathcal{C}) \star N(\mathcal{C}') \cong N(\mathcal{C} \star \mathcal{C}')$ .

┘

*Proof:* All these are clear. □

**Def. (4.6.3.20) [Cones].** For a simplicial set  $K$ , the **left/right cone** of  $K$  are defined to be the join  $K^{\triangleleft} = \Delta^0 \star K$  and  $K^{\triangleleft} = K \star \Delta^0$ .

For a map  $f : X \rightarrow S$ , the **left/right cone** of  $f$  are defined to be  $S \coprod_X X^{\triangleleft}$  and  $S \coprod_X X^{\triangleleft}$ . ┘

**Example (4.6.3.21).**  $(\Lambda_2^2)^{\triangleleft} \cong (\Lambda_0^2)^{\triangleleft} \cong \Delta^1 \times \Delta^1$ . ┘

**Def. (4.6.3.22) [Overcategories and undercategories].** For  $p : K \rightarrow S \in s\mathbf{Set}$ , there is a simplicial set  $S_{/p}$  that

$$\mathrm{Hom}(Y, S_{/p}) \cong \mathrm{Hom}_p(Y \star K, S) \triangleq \mathrm{Hom}_{s\mathbf{Set}_{K/}}(Y \star K, S)$$

And dually we can define the **undercategories**.

There are canonical maps  $S_{/p} \rightarrow S$  and  $S_{p/} \rightarrow S$ . ┘

*Proof:* We just define  $(S_{/p})_n = \mathrm{Hom}_p(K \star \Delta^n, S)$ , then the condition holds for  $\Delta^n$ , and use the fact every simplicial set is a colimit of  $\Delta^n$ s (4.6.3.1), and both sides commutes with colimits in  $Y$  by (4.6.3.19). □

**Def. (4.6.3.23).** For  $\mathcal{C}_0 \subset \mathcal{C} \in s\mathbf{Set}$  and any diagram  $p : K \rightarrow \mathcal{C}$ , denote  $\mathcal{C}_{/p}^0 = \mathcal{C}^0 \times_{\mathcal{C}} \mathcal{C}_{/p}$ . ┘

**Prop. (4.6.3.24).** For  $p : A \rightarrow B \in \mathbf{Cat}$ , there is a natural isomorphism of simplicial sets

$$N(B_{/p}) \cong N(B)_{/N(p)}.$$

┘

*Proof:* There is a natural map from LHS to RHS, and we also have natural isomorphisms

$$\begin{aligned} N(B_{/p})_n &= \mathrm{Hom}([n], B_{/p}) \\ &= \mathrm{Hom}_{\mathbf{Cat}_{A/}}(A \rightarrow [n] \star A, A \xrightarrow{p} B) \\ (4.6.3.4) &= \mathrm{Hom}_{s\mathbf{Set}_{N(A)/}}(N(A) \rightarrow N([n] \star A), N(A) \xrightarrow{N(p)} N(B)) \\ (4.6.3.19) &= \mathrm{Hom}_{s\mathbf{Set}_{N(A)/}}(N(A) \rightarrow \Delta^n \star N(A), N(A) \xrightarrow{N(p)} N(B)) \\ &= \mathrm{Hom}_{s\mathbf{Set}}(\Delta^n, N(B)_{/N(p)}) = (N(B)_{/N(p)})_n \end{aligned}$$

□

**Prop. (4.6.3.25) [Generating Simplicial Sets].** Let  $\mathcal{U}$  be a collection of simplicial sets that:

- $\mathcal{U}$  is stable under isomorphisms,
- $\mathcal{U}$  is stable under disjoint union,
- $\Delta^n \subset \mathcal{U}$  for any  $n$ ,

- If there is a pushout diagram 
$$\begin{array}{ccc} X & \longrightarrow & X' \\ \downarrow f & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$$
 and  $X, X', Y \in \mathcal{U}$  and  $f$  is a monomorphism, then

$$Y' \in \mathcal{U},$$

- suppose we are given a sequence of monomorphisms between objects in  $\mathcal{U}$  indexed over  $\mathbb{N}$ , then the colimit belongs to  $\mathcal{U}$ .

Then  $\mathcal{U} = s\text{Set}$ . ┘

*Proof:* Use induction on the dimension of  $S$ , and notice  $S$  is glued together using their simplexes. ┘

### Fibrations and Anodynes

**Def. (4.6.3.26) [Fibrations].** A morphism of simplicial sets is called a

- **Kan fibration** iff it has right lifting property w.r.t all  $\Lambda_i^n \rightarrow \Delta^n, 0 \leq i \leq n$ .
- **left fibration** iff it has right lifting property w.r.t. all inclusions  $\Lambda_i^n \subset \Delta^n, 0 \leq i < n$ .
- **right fibration** iff it has right lifting property w.r.t all inclusions  $\Lambda_i^n \subset \Delta^n, 0 < i \leq n$ .
- **inner fibration** iff it has right lifting property w.r.t all inclusions  $\Lambda_i^n \subset \Delta^n, 0 < i < n$ .

So a morphism between topological spaces  $X \rightarrow Y$  is a Serre fibration iff  $S(X) \rightarrow S(Y)$  is a Kan fibration (4.6.3.8). ┘

**Def. (4.6.3.27) [Anodynes].** A morphism of simplicial set is called a

- **anodyne** iff it has left lifting property w.r.t. all Kan fibrations.
  - **left anodyne** iff it has left lifting property w.r.t. all left fibrations.
  - **right anodyne** iff it has left lifting property w.r.t all right fibrations.
  - **inner anodyne** iff it has right lifting property w.r.t all inner fibrations.
- ┘

**Def. (4.6.3.28) [Trivial (Kan) Fibrations].** A morphism  $X \rightarrow S$  of simplicial sets that has right lifting property w.r.t. all inclusions  $\partial\Delta^n \rightarrow \Delta^n$  is called a **trivial fibration**.

A **cofibration** is a morphism that has left lifting property w.r.t all trivial fibrations. By (4.2.2.15) a cofibration of simplicial sets is just an inclusion. ┘

**Lemma (4.6.3.29) [Join and Anodynes].** If  $f : A_0 \subset A$  and  $g : B_0 \subset B$  are inclusions of simplicial sets, then

$$h : (A_0 \star B) \coprod_{A_0 \star B_0} (A \star B_0) \subset A \star B$$

is a(n)

- inner anodyne if either  $f$  is right anodyne or  $g$  is left anodyne, then
  - left anodyne if  $f$  is left anodyne.
- ┘

*Proof:* 1: By symmetry we assume  $f$  is right anodyne. Notice the class of all morphisms  $f$  that the conclusion holds is weakly saturated because  $\star$  commutes with colimits, so it suffice to check for  $f : \Lambda_j^n \subset \Delta^n$ . Then similarly it suffices to check for  $g : \partial\Delta^m \subset \Delta^m$ , but then the inclusion is just  $\Lambda_j^{m+n+1} \subset \Delta^{m+n+1}$ , which is left anodyne.

2 is similar. ┘

**Prop. (4.6.3.30) [Anodynes].**

- The saturated class generated by either of the following three classes of monomorphisms are both left anodynes:

1.  $\Lambda_k^n \subset \Delta^n$ ,  $0 \leq k < n$ .
2.  $(\Delta^m \times \{0\}) \coprod (\partial \Delta^m \times \Delta^1) \subset \Delta^m \times \Delta^1$ .
3.  $(S' \times \{0\}) \coprod (S \times \Delta^1) \subset S' \times \Delta^1$ , where  $S \subset S'$ .

Similar conclusion holds for right anodynes and together implies the similar conclusion for anodynes.

•

┘

*Proof:*

- 2 and 3 are equivalence because any inclusion comes from attaching cells (4.6.3.3).  
Now inclusions in 2 are compositions of pushouts of inclusions  $\Lambda_k^{n+1} \subset \Delta^{n+1}$ , where  $0 \leq k \leq n$ , thus it is generated by 1. Conversely,  $\Lambda_i^n \subset \Delta^n$  is a retract of  $(\Delta^n \times \{0\}) \coprod (\Lambda_i^n \times \Delta^1) \subset \Delta^n \times \Delta^1$ : Cf.[HTT, P64].

•

□

**Cor.(4.6.3.31) [Products and Anodynes].** Let  $A \subset A'$  be a(n) left(inner) anodyne and  $B \subset B'$ , then so does the induced map

$$(A \times B') \coprod_{A \times B} (A' \times B) \rightarrow A' \times B'.$$

┘

*Proof:* For left anodyne, the proof is similar to that of (4.6.3.29), just check for classes 3 of (4.6.3.30), and use the fact

$$(S' \times \Delta^1) \times B \coprod (S' \times \{0\}) \coprod (S \times \Delta^1) \times B' \rightarrow (S' \times \Delta^1) \times B'$$

is just

$$(S' \times B \coprod S \times B') \times \Delta^1 \coprod (S' \times B') \times \{0\} \rightarrow (S' \times B') \times \Delta^1.$$

which is left(inner) anodyne. And similarly for inner anodynes.

□

### Left Fibration

**Remark(4.6.3.32) [Left and Right Fibrations Dual].** The theory of left fibrations is dual to the theory of right fibrations, thus we don't study right fibrations.

┘

**Prop.(4.6.3.33) [Left Fibration and CoFibered in Groupoids].** Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor, then  $\mathcal{C}$  is a category cofibered in groupoids over  $\mathcal{D}$  iff the induced functor  $N(F) : N(\mathcal{C}) \rightarrow N(\mathcal{D})$  is a left fibration of simplicial sets.

┘

*Proof:* By (4.6.3.6),  $N(F)$  is an inner fibration, thus it suffices to check for  $\Lambda_0^n \rightarrow \Delta^n$ . For  $n = 1$ , this is the definition of cofibered category (4.2.3.8), and  $n = 2$  is just the surjectivity of the map defining CoCartesian arrows (4.2.3.5), and  $n = 3$  is equivalent to the injectivity of the map defining Cocartesian arrows. And for  $n > 3$ , then extension is automatic for nerves.

□

**Remark(4.6.3.34) [Right Fibrations and Fibered Categories].** The (left)right fibration is the  $\infty$ -categorical analogue of (co)fibered categories in usual category theory.

┘

**Remark (4.6.3.35).** Given a left fibration  $X \rightarrow S$  is more or less similar to given a functor from the homotopy category  $hS$  to the  $\infty$ -category  $\mathcal{H}$  of spaces.  $\lrcorner$

*Proof:* Cf.[HTT, P58]. ?  $\square$

**Prop. (4.6.3.36) [Over(Under)categories and Fibrations].** Given a diagram of simplicial sets:

$$A \subset B \xrightarrow{p} X \xrightarrow{q} S.$$

Let  $r = q \circ p$ ,  $p_0 = p|_A$ ,  $r_0 = r|_A$ , and  $q$  is an inner fibration, then

- the induced map  $X_{p/} \rightarrow X_{p_0/} \times_{S_{r_0/}} S_{r/}$  is a left fibration. And dual argument holds for overcategories.
- If  $q$  is a left fibration or  $A \subset B$  is right anodyne, then  $X_{p/} \rightarrow X_{p_0/} \times_{S_{r_0/}} S_{r/}$  is a trivial fibration.
- If  $q$  is moreover a left fibration, then the induced map  $X_{/p} \rightarrow X_{/p_0} \times_{S_{/r_0}} S_{/r}$  is a left fibration.  $\lrcorner$

*Proof:* These just follow from (4.6.3.29).  $\square$

**Prop. (4.6.3.37) [Homotopy Extension Lifting Property].** For  $p : X \rightarrow S, i : A \subset B \in s\text{Set}$ , consider the map

$$q : X^B \rightarrow X^A \times_{S^A} S^B.$$

- If  $p$  is a left fibration, then  $q$  is a left fibration.
- If  $p$  is a left fibration  $i$  is a left anodyne, then  $q$  is a trivial fibration.
- If  $i : \{0\} \subset \Delta^1$ , then  $p$  is a left fibration iff  $q$  is a trivial fibration.
- If  $p$  is an inner fibration, then  $q$  is an inner fibration.
- If  $p$  is an inner fibration and  $i$  is an inner anodyne, then  $q$  is a trivial fibration.  $\lrcorner$

*Proof:* First notice that a right lifting of  $q$  w.r.t a map  $Z \rightarrow Z'$  is equivalent to a right lifting of  $p$  w.r.t the map  $Z \times B \coprod Z' \times A \rightarrow Z' \times B$ . Then the conclusions follow from (4.6.3.31) and (4.6.3.30).  $\square$

**Prop. (4.6.3.38) [Homotopy Section and Left Fibration].** Let  $p : X \rightarrow S \in s\text{Set}$  and  $s : S \rightarrow X$  be section of  $p$ , and let  $h \in \text{Hom}_S(X \times \Delta^1, X)$  that  $h|_{X \times \{0\}} = s \circ p$  and  $h|_{X \times \{1\}} = \text{id}$ , then  $s$  is a left anodyne.  $\lrcorner$

*Proof:* Cf.[HTT, P65].  $\square$

**Prop. (4.6.3.39) [Left Fibration and Functor to Spaces].** Let  $X \rightarrow S$  be a left fibration, then the fibers are all Kan complexes by (4.7.1.27), and for any edge  $f : s \rightarrow s' \in S$ , we can solve the lifting diagram

$$\begin{array}{ccc} \{0\} \times X_s & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \Delta^1 \times X_s & \xrightarrow{\quad} & \Delta^1 \xrightarrow{f} S \end{array}$$

because the left hand side is left anodyne by (4.6.3.31), thus getting a morphism  $f_! : X_s \rightarrow X_{s'}$ .

Then this determines a functor from  $hS$  to  $\mathcal{H}$  the homotopy category of spaces.  $\lrcorner$

*Proof:* First notice  $f_!$  is uniquely defined up to homotopy: given to dotted arrow solving the diagram, we can use the lifting property w.r.t. the map

$$\left(\Delta^1 \times \partial\Delta^1 \times X_s\right) \coprod_{\{0\} \times \partial\Delta^1 \times X_s} \left(\{0\} \times \Delta^1 \times X_s \rightarrow \Delta^1 \times \Delta^1 \times X_s\right),$$

which is a left anodyne by (4.6.3.31), to get a homotopy between them.

Next if  $\eta \in \text{Hom}_{\mathcal{H}}(K, X_s), \eta' \in \text{Hom}_{\mathcal{H}}(K, X_{s'})$ , then  $\eta' = f_! \circ \eta$  iff there is a map  $q : K \times \Delta^1 \rightarrow X$  that  $q \circ p$  is given by the mapping  $X_s \times \Delta^1 \rightarrow \Delta^1 \xrightarrow{f} S$ , and  $p|_{K \times \{0\}}, p|_{K \times \{1\}}$  have homotopy types  $\eta, \eta'$  resp..

Then for any  $g \circ f \cong h \in S$ , which is depicted by a 2-complex, we can use the left anodyne  $X_u \times \{0\} \subset X_u \times \Delta^2$  to get a morphism  $p : X_u \times \Delta^2 \rightarrow X$ , then  $p|_{X_u \times \{1\}} \cong f_!, p|_{X_u \times \{2\}} \cong h_!$ , and the map  $p|_{X_u \times \Delta\{1,2\}}$  witnesses the fact  $g_! \circ f_! \cong h_!$ .  $\square$

### Kan Fibrations

**Remark (4.6.3.40) [Kan Complexes and Groupoids].** Kan complexes are  $\infty$ -categorical analogy of groupoids, by (4.7.1.27) and (4.6.3.6).  $\lrcorner$

**Lemma (4.6.3.41).** If a left fibration  $p : S \rightarrow T$  is a weak homotopy equivalence of Kan complexes, then it is a surjection on vertices.  $\lrcorner$

*Proof:* Because it is homotopy equivalence, for any  $t \in T$ , there is a morphism  $p(s) \rightarrow t$  for some  $s \in S$  (Kan complex used), thus it lifts to a morphism in  $S$ , so surjective on vertices.  $\square$

**Lemma (4.6.3.42) [Fibrations of Kan Complexes].** If  $S \rightarrow T$  is a left fibration and  $T$  is a Kan complex, then  $p$  is a Kan fibration.  $\lrcorner$

*Proof:* Firstly  $S$  is a Kan complex. Let  $A \subset B$  be anodyne morphisms, we need to show  $p : S^B \rightarrow S^A \times_{T^A} T^B$  is surjective on vertices. Since  $S, T$  are complexes,  $S^B \rightarrow S^A$  and  $T^B \rightarrow T^A$  are trivial fibrations by (4.6.3.37). Cf. [HTT, P66]. ?

Notice this is an immediate consequence of (4.6.3.47), because the homotopy category of a Kan complex is a groupoid, by (4.7.1.27), so  $f_!$  must be isomorphisms.  $\square$

**Prop. (4.6.3.43) [Examples of Kan Complexes].**

- The singular complex of topological space is a Kan complex.
- The nerve of a category is a Kan complex iff the category is a groupoid, by (4.6.3.33).
- Simplicial  $R$ -module are Kan complexes, by (5.8.1.1).

$\lrcorner$

*Proof:*

$\square$

**Prop. (4.6.3.44).** The bar resolution  $BG$  is a Kan fibration for every group  $G$ .  $\lrcorner$

*Proof:*

$\square$

**Prop. (4.6.3.45).** A principal  $G$  fibration, i.e.  $X \rightarrow X/G$  where  $X$  is a simplicial object of  $G$ -sets that  $G$  acts freely on  $X_n$ , is a Kan fibration.  $\lrcorner$

**Lemma (4.6.3.46) [Left Fibrations and Trivial Kan Fibrations].** Let  $p : S \rightarrow T \in s\mathbf{Set}$  be a left fibration that all the fibers are contractible, then  $p$  is a trivial Kan fibration.  $\lrcorner$

*Proof:* By duality, it suffices to prove for right fibrations. Because fiber is nonempty, it has right lifting property w.r.t  $\emptyset \subset \Delta^0$ , and for  $n > 0$ , let  $\partial\Delta^n \rightarrow S$  be any map, then to show the lifting property, we may take fiber product and assume  $T = \Delta^n$ , thus  $S$  is an  $\infty$ -category. Cf.[HTT, P66].  $\square$

**Prop. (4.6.3.47) [Characterizing Kan-Fibrations].** Let  $p : S \rightarrow T \in s\mathbf{Set}$  be a left fibration, then  $p$  is a Kan-extension iff the morphism  $f_!$  defined in (4.6.3.39) is an isomorphism in  $\mathcal{H}$  for any morphism  $f \in T$ .  $\lrcorner$

*Proof:* Cf.[HTT, P66].  $\square$

**Def. (4.6.3.48) [Kan-Quillen Model Structure].** There is a combinatorial left and right proper model structure on  $s\mathbf{Set}$  called the **Kan-Quillen model structure** with

- Weak equivalences: weak equivalences (4.6.3.11).
- Cofibrations: inclusions,
- Fibrations: Kan fibrations.

$\lrcorner$

*Proof:* Cf.[Jardine P62]. ?  $\square$

**Prop. (4.6.3.49) [Quillen].** The geometrization functor and the singular complex functor (4.6.3.8) defines a Quillen equivalence:

$$|\cdot| : s\mathbf{Set} \rightleftarrows \mathcal{CG} : \mathbf{Sing},$$

where the RHS is Serre-Quillen model category (4.13.7.28) and the LHS is the Kan-Quillen model category (4.6.3.48).

The functor  $|\cdot| \circ \mathbf{Sing}$  is also denoted by  $\Gamma : \mathcal{CG} \rightarrow \mathcal{CG}$ .  $\lrcorner$

*Proof:* Cf.[May, P125]. ?  $\square$

**Cor. (4.6.3.50).** The localized category of  $\mathcal{CG}$  and  $s\mathbf{Set}$  at weak homotopy equivalence classes are the same, and it is just the homotopy category of spaces  $\mathcal{H}$ , by (4.5.2.7).  $\lrcorner$

**Cor. (4.6.3.51).** By (4.6.3.43), for any  $S \in s\mathbf{Set}$ ,  $S \rightarrow \mathbf{Sing} |S|$  is a fibrant replacement w.r.t. the Kan-Quillen model category.  $\lrcorner$

### Marked Simplicial Sets

**Def. (4.6.3.52) [Marked Simplicial Sets].** A **marked simplicial set** is a pair  $(X, \mathcal{E}_X)$  where  $X \in s\mathbf{Set}$  and  $\mathcal{E}_X \subset X_1$  is a set of edges containing all the degenerate ones. The category of marked simplicial sets is denoted by  $s\mathbf{Set}^+$ .  $\lrcorner$

## 4 Simplicial Categories

**Def. (4.6.4.1) [Simplicial Categories].** The category  $s\mathbf{Cat}$  of **simplicial categories** consists of categories enriched over the Cartesian monoidal category  $s\mathbf{Set}$ .  $s\mathbf{Cat}$  is complete and cocomplete, by (4.2.2.3) and (4.6.3.15)(4.6.1.2).  $\lrcorner$

**Def. (4.6.4.2) [Homotopy Category].** There are singular complex functor and geometrization functor that induce isomorphism of  $h(s\mathbf{Set}) \cong h(\mathcal{CG})$  by (4.6.3.50), thus the theory of simplicial categories and topological categories are the same.  $\lrcorner$

### Bergner Model Structure

**Def. (4.6.4.3) [Dwyer-Kan Equivalences].** A **Dwyer-Kan equivalence of simplicial categories** is a weak equivalence of simplicial categories as enriched categories(4.6.4.16).  $\lrcorner$

**Def. (4.6.4.4) [Bergner Model Structure].** There is a left proper combinatorial model structure **Bergner model structure** on  $s\mathcal{Cat}$  called the **Bergner model structure** with

- Weak equivalences: Dwyer-Kan equivalences(4.6.4.3).
- Fibrations:  $F : \mathcal{C} \rightarrow \mathcal{D}$  that is an quasi-fibration(4.2.3.7) and for any  $x, y \in \mathcal{C}$ ,  $\text{Map}_{\mathcal{C}}(x, y) \rightarrow \text{Map}_{\mathcal{D}}(Fx, Fy)$  is a Kan-fibration.
- Cofibrations:

$\lrcorner$

*Proof:* ? Cf.[Bergner. A model category structure on the category of simplicial categories] or [Lurie].  $\square$

### Simplicial Nerves

**Def. (4.6.4.5) [Thickened Finite Ordered Sets].** Let  $J$  be a finite ordered set, define a simplicial category  $\mathfrak{C}[\Delta^J]$  as follows:

- The objects are elements of  $J$ .
- For  $i, j \in J$ ,  $\text{Mor}(i, j) = N(P_{i,j})$ , where  $P_{i,j}$  is the poset  $\{I \subset [i, j] | i, j \in I\}$ .
- The composition of simplicial sets is induced by the union of partially ordered sets  $P_{i,j} \times P_{j,k} \subset P_{i,k}$ .

$\lrcorner$

**Def. (4.6.4.6) [Coherent Nerves].** by Yoneda-extension and(4.6.4.1), the functor  $\Delta \rightarrow s\mathcal{Cat} : [n] \mapsto \mathfrak{C}[\Delta^n]$  corresponds to an adjunction

$$\mathfrak{C} : s\mathcal{Set} \rightleftarrows s\mathcal{Cat} : N_{\Delta}$$

where  $N_{\Delta}$  is called the **coherent nerve** functor, and

$$(N_{\Delta}(\mathcal{C}))_n = \text{Hom}_{\mathcal{Cat}_{\Delta}}(\mathfrak{C}[\Delta^n], \mathcal{C}).$$

If  $\mathcal{C}$  is a topological category, then define the **topological nerve** to be the simplicial nerve of  $\text{Sing}(\mathcal{C})$ .

By the definition and the adjointness of(4.6.3.8), the nerve functor  $N_{\Delta}$  is right adjoint to the functor  $|\cdot| \circ \mathfrak{C}$  or  $\mathfrak{C}$ .  $\lrcorner$

**Remark (4.6.4.7).** It should be checked that to give a 2-complex in  $N_{\Delta}(\mathcal{C})$  is equivalent to giving morphisms  $f, g, h \in \mathcal{C}$  and a path from  $g \circ f$  to  $h$ .  $\lrcorner$

**Thm. (4.6.4.8)**  $[\mathfrak{C} : s\mathcal{Set} \xrightleftharpoons{\text{Quillen}} s\mathcal{Cat} : N_{\Delta}]$ . The adjunction(4.6.4.6) is a Quillen adjunction w.r.t. the Joyal model category on  $s\mathcal{Set}$ (4.6.4.19) and the Bergner model category on  $s\mathcal{Cat}$ (4.6.4.4).

In particular, a theory of  $(\infty, 1)$ -categories given by the model of simplicial sets and the model of simplicial categories are the same?.  $\lrcorner$



*Proof:* Cf.[HTT, P89].?

Firstly we show  $\mathfrak{C}$  preserves cofibrations. It suffices to show that  $\mathfrak{C}[\partial\Delta^n] \subset \mathfrak{C}[\Delta^n]$  is a cofibration(4.5.7.4). But notice this two simplicial category only differ at  $\text{Hom}_{\mathfrak{C}[\partial\Delta^n]}(0, n)$  is the boundary of the simplicial cube  $(\Delta^1)^{n-1} \cong \text{Hom}_{\mathfrak{C}[\Delta^n]}(0, n)$ , thus the inclusion is a pushout of the inclusion  $[1]_{\partial(\Delta^1)^{n-1}} \subset [1]_{(\Delta^1)^{n-1}}$ , which is a cofibration by(4.5.7.3).

Left properness is clear(4.5.0.9).  $\mathfrak{C}$  preserves weak equivalences by(4.6.4.12) and(4.5.7.4), so  $(\mathfrak{C}, N)$  is a Quillen adjunction. To show it is a Quillen equivalence: It suffices to check for each simplicial set  $S$  and fibrant simplicial category  $\mathcal{C}$ , a map  $u : S \rightarrow N(\mathcal{C})$  is a categorical equivalence iff the adjoint map  $v : \mathfrak{C}[S] \rightarrow \mathcal{C}$  is an equivalence of simplicial categories. But  $v$  factors as

$$\mathfrak{C}[S] \xrightarrow{\mathfrak{C}[u]} \mathfrak{C}[N(\mathcal{C})] \xrightarrow{w} \mathcal{C}$$

and the counit map  $w$  is an equivalence by(4.6.4.14).  $\square$

**Cor.(4.6.4.9).** The coherent nerve of a Bergner-fibrant(4.6.4.4) simplicial category is an  $\infty$ -category.  $\lrcorner$

**Prop.(4.6.4.10)[Kan Fibrations Nerve Inner Fibrations].** Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a map of simplicial categories that for any  $C, C' \in \mathcal{C}$ ,  $\text{Map}(C, C') \rightarrow \text{Map}(F(C), F(C'))$  is a Kan fibration, then the induced map of simplicial sets  $N(\mathcal{C}) \rightarrow N(\mathcal{D})$  is an inner fibration.  $\lrcorner$

*Proof:* This is because a lifting of  $N(\mathcal{C}) \rightarrow N(\mathcal{D})$  w.r.t.  $\lambda_j^n \subset \Delta^n$  is equivalent to a lifting of  $\mathcal{C} \rightarrow \mathcal{D}$  w.r.t.  $\mathfrak{C}[\Lambda_j^n] \subset \mathfrak{C}[\Delta^n]$ . But this lifting is equivalent to a lifting of  $\text{Map}(F(0), F(n)) \rightarrow \text{Map}(F'(0), F'(n))$  w.r.t. the anodyne map  $\text{Map}_{\mathfrak{C}[\Lambda_j^n]} \subset \text{Map}_{\mathfrak{C}[\Delta^n]}$ , which is a cube removing the interior and a face.  $\square$

**Cor.(4.6.4.11).** The topological nerve of a topological category  $\mathcal{C}$  is an  $\infty$ -category, as the singular complex of a topological space is always a Kan complex, by(4.6.3.43).  $\lrcorner$

**Def.(4.6.4.12)[Homotopy Category].** For  $S \in s\text{Set}$ , the **homotopy category**  $hS$  is defined to be the homotopy category(4.6.3.50) of the simplicial category  $\mathfrak{C}[S]$ (4.6.4.6), which is an  $\mathcal{H}$ -enriched category. A map of simplicial sets is called a **categorical equivalence** if their homotopy categories are equivalent as  $\mathcal{H}$ -enriched categories.  $\lrcorner$

**Remark(4.6.4.13).**  $f : S \cong T \in s\text{Set}$  is a categorical equivalence iff  $\mathfrak{C}(f) : \mathfrak{C}[S] \rightarrow \mathfrak{C}[T]$  is a Dwyer-Kan equivalence(4.6.4.3).  $\lrcorner$

**Lemma(4.6.4.14).** Let  $\mathcal{C}$  be fibrant simplicial category, then the counit map  $u : \text{Map}_{\mathfrak{C}[N_{\Delta}(\mathcal{C})]}(x, y) \rightarrow \text{Map}_{\mathcal{C}}(x, y)$  is a weak homotopy equivalence of simplicial sets.  $\lrcorner$

*Proof:* Cf.[HTT, P72].  $\square$

**Lemma(4.6.4.15).** Let  $\mathcal{C}$  be a topological category, then the counit map  $|\mathfrak{C}(N(\mathcal{C}))| \cong \mathcal{C}$ (4.6.4.6) is an equivalence of topological categories.  $\lrcorner$

*Proof:* By the Quillen equivalence between  $s\text{Set}$  and  $\mathcal{C}\mathcal{G}$ (4.6.3.49), this follows from(4.6.4.14), as by(4.5.7.11) and(4.6.3.43),  $\text{Sing}(\mathcal{C})$  is a fibrant simplicial category.  $\square$

**Prop.(4.6.4.16) [Topological Category and  $\infty$ -Category Equivalent].** The adjunction pair  $(|\mathfrak{C}[\cdot]|, N)$  defines a bijection between equivalent classes of topological(or simplicial) categories and  $\infty$ -categories.  $\lrcorner$

*Proof:* It suffices to show the units and counits are equivalences:

$$|\mathfrak{C}[N(\mathcal{C})]| \cong \mathcal{C}, \quad S \mapsto N(|\mathfrak{C}[S]|).$$

The first is (4.6.4.15), and the second follows from the first by remark (4.6.4.13).  $\square$

**Prop. (4.6.4.17).** For  $S, S' \in s\text{Set}$ , the natural map  $\mathfrak{C}[S \times S'] \rightarrow \mathfrak{C}[S] \times \mathfrak{C}[S']$  is an equivalence of simplicial categories.  $\lrcorner$

*Proof:* If  $S, S'$  are nerves of fibrant simplicial categories  $\mathcal{C}, \mathcal{C}'$ , then we have a diagram  $\mathfrak{C}[S \times S'] \rightarrow \mathfrak{C}[S] \times \mathfrak{C}[S'] \rightarrow \mathcal{C} \times \mathcal{C}'$ . Then by the two out of three axiom, the assertion follows from the fact that for any fibrant simplicial category  $\mathcal{D}$ ,  $\mathfrak{C}[N(\mathcal{D})] \rightarrow \mathcal{D}$  is an equivalence (4.6.4.15).

Now for general  $S, S'$ , we can find a categorical equivalences  $S \rightarrow N(|\mathfrak{C}[S]|) = T$ , and then  $S \times S' \rightarrow T \times T'$  is also categorical equivalence by (4.6.4.17), and we are done, by (4.6.4.19).  $\square$

### Joyal Model Structure

**Lemma (4.6.4.18)[Inner Anodyne is Categorical Equivalence].** Every inner anodyne map  $A \rightarrow B$  of simplicial sets is a categorical equivalence.  $\lrcorner$

*Proof:* The class of morphisms  $f$  that  $\mathfrak{C}(f)$  is a trivial cofibration is weakly saturated (because  $\mathfrak{C}$  is a left adjoint (4.6.4.6) and (4.5.0.5)), then it suffices to check for  $\Lambda_j^n \subset \Delta^n$ . Then  $\mathfrak{C}[\Lambda_j^n] \subset \mathfrak{C}[\Delta^n]$  is a pushout of  $[i]_K \subset [1]_{(\Delta^1)^{n-1}}$ , where  $K$  is obtained from  $(\Delta^1)^{n-1}$  by moving a face and the interior. Thus it is a trivial cofibration (4.5.7.4).  $\square$

**Prop. (4.6.4.19)[Joyal Model Structure].** There is a left proper combinatorial model structure called **Joyal model structure** on  $s\text{Set}$ , with:

- Cofibrations: monomorphisms.
- Weak equivalences: categorical equivalences defined in (4.6.4.12).
- Fibrations: **categorical fibrations** or **Joyal-fibrations** which has the right lifting property w.r.t. trivial cofibrations.  $\lrcorner$

*Proof:* Cf. [HTT, P89]. ?  $\square$

**Cor. (4.6.4.20).** if  $K, A, B \in s\text{Set}$  and  $f : A \rightarrow B$  is a categorical equivalence, then  $A \times K \rightarrow B \times K$  is also a categorical equivalence.  $\lrcorner$

*Proof:* Choose a factorization  $B \rightarrow Q$ , which is an inner anodyne and  $Q \in \infty\text{-Cat}$ , by small object argument (4.2.2.13), then  $B \times K \rightarrow Q \times K$  is also an inner anodyne map (4.6.3.31), hence a categorical equivalence (4.6.4.18), so we can assume  $B$  is an  $\infty$ -category. Similarly we can reduce to the case  $A, K$  are also  $\infty$ -categories.

For the rest, Cf. [HTT, P92].  $\square$

**Prop. (4.6.4.21)[ $\infty$ -Category Fibrant in Joyal Model].**  $C \in s\text{Set}$  is Joyal-fibrant iff it is an  $\infty$ -category.  $\lrcorner$

*Proof:* Fibrant objects are  $\infty$ -categories, by (4.6.4.18). For the converse, fix an  $\infty$ -category and an inclusion  $A \subset B$ , given a map  $A \rightarrow \mathcal{C}$ , the inclusion  $\mathcal{C} \subset C \coprod_A B$  is also a categorical equivalence because Joyal model structure is left proper (4.6.4.19), thus  $\mathcal{C}$  is a retract of  $\mathcal{C} \coprod_A B$  by (4.6.4.22), which gives an extension  $B \rightarrow \mathcal{C}$ .  $\square$

**Lemma (4.6.4.22).** Let  $\mathcal{C} \subset \mathcal{D} \in s\mathbf{Set}$  be a categorical equivalence and  $\mathcal{C} \in \mathbf{Cat}_\infty$ , then  $\mathcal{C}$  is a retract of  $\mathcal{D}$ .  $\lrcorner$

*Proof:* Include  $\mathcal{D}$  into an  $\infty$ -category by small object argument and (4.6.4.18), we may assume  $\mathcal{D}$  is also an  $\infty$ -category. So we finish by applying (4.7.1.25) for  $A = \mathcal{C}$  and  $B = \mathcal{D}$ .  $\square$

**Cor. (4.6.4.23).** Any  $S \in s\mathbf{Set}$  is weakly equivalent to an  $\infty$ -category.  $\lrcorner$

**Def. (4.6.4.24) [Joyal Joins].** For  $X, Y \in s\mathbf{Set}$ , define the **Joyal Join**

$$X \diamond Y = X \coprod_{X \times Y \times \{0\}} (X \times Y \times \Delta^1) \coprod_{X \times Y \times \{1\}} Y = (X \coprod Y) \coprod_{X \times Y \times \partial \Delta^1} (X \times Y \times \Delta^1).$$

By (4.5.6.7), this is a homotopy colimit w.r.t. the Joyal model category, so if  $X \rightarrow X', Y \rightarrow Y'$  are categorical equivalences,  $X \diamond Y \rightarrow X' \diamond Y'$  is also a categorical equivalence (4.5.6.6).  $\lrcorner$

**Prop. (4.6.4.25).** There is a projection map  $X \diamond Y \rightarrow \Delta^1$ , and a map  $X \times Y \times \Delta^1 \rightarrow X \star Y$  that is compatible with projection onto  $\Delta^1$ , thus inducing a map

$$X \diamond Y \rightarrow X \star Y$$

that is compatible with projection onto  $\Delta^1$ . Then this map is a categorical equivalence.  $\lrcorner$

*Proof:* Because both  $\diamond$  and  $\star$  commutes with filtered colimits, by (4.5.3.3), it suffices to show for  $X$  with only f.m. non-degenerate simplexes. Then we use induction. The case  $X = \emptyset$  is trivial, and if  $X = X' \coprod_{\partial \Delta^n} \Delta^n$ , because Joyal model category is left proper (4.6.4.19), by (4.5.6.7),  $X \diamond Y \rightarrow X \star Y$  is a homotopy pushout of  $X' \diamond Y \rightarrow X' \star Y$ . Then it suffices to show for  $X = \Delta^n$ .

By similar reason, because  $\Delta^{\{0,1\}} \coprod_{\{1\}} \Delta^{\{1,2\}} \coprod_{\{2\}} \coprod \dots \coprod_{\{n-1\}} \Delta^{\{n-1,n\}} \rightarrow \Delta^n$  is an anodyne ?, and anodyne are categorical equivalences (4.6.4.18), and homotopy limits of categorical equivalences are categorical equivalences by (4.5.6.6), it suffices to prove for  $X = \Delta^0$  or  $X = \Delta^1$ . The same argument then shows it suffices to prove for  $Y = \Delta^0$  or  $Y = \Delta^1$ . And in each case the desired map is an isomorphism.  $\square$

**Cor. (4.6.4.26).** If  $S \rightarrow S', T \rightarrow T'$  are categorical equivalences, then  $S \star T \rightarrow S' \star T'$  is also a categorical equivalence, by (4.6.4.24).  $\lrcorner$

**Cor. (4.6.4.27).** For  $X, Y \in s\mathbf{Set}$ , there is a natural equivalence of simplicial categories

$$\mathfrak{C}[X \star Y] \cong \mathfrak{C}[X] \star \mathfrak{C}[Y].$$

$\lrcorner$

*Proof:* Cf. [Lur09]P240.  $\square$

## 5 Simplicial Model Categories

**Def. (4.6.5.1) [Simplicial Model Categories].** A **simplicial model category** is a  $s\mathbf{Set}$ -enriched model category (4.5.5.3).  $\lrcorner$

**Prop. (4.6.5.2) [Simplicial Model Category Criterion].** Let  $\mathcal{C}$  be a simplicial category that is equipped with a model structure that every object of  $\mathcal{C}$  is cofibrant and the collection of weak equivalences is stable under filtered colimits, then  $\mathcal{C}$  is a simplicial model category iff the following conditions holds:

- $\mathcal{C}$  is both tensored and cotensored over  $\mathbf{Set}_\Delta$ .
- Given a cofibration of simplicial sets  $i : K \rightarrow L$  and a cofibration  $C \rightarrow D \in \mathcal{C}$ , the induced map  $(C \otimes L) \coprod_{C \otimes K} D \otimes K \rightarrow D \otimes L$  is a cofibration in  $\mathcal{C}$ .
- The natural map  $C \otimes \Delta^n \rightarrow C \otimes \Delta^0 \cong C$  is a weak equivalence in  $\mathcal{C}$ .

┘

*Proof:* Cf.[HTT, P850].

□

**Prop.(4.6.5.3).** Let  $\mathcal{C}$  be a simplicial model category and  $X$  cofibrant and  $Y$  fibrant, then  $K = \text{Map}(X, Y)$  is a Kan complex, and there is a canonical bijection  $\pi_0 K \cong \text{Hom}_{\text{Ho}(\mathcal{C})}(X, Y)$ .

┘

*Proof:* Use(4.5.5.4).

□

### Model-Categorical Yoneda Extensions

**Def.(4.6.5.4)**  $[U(\mathcal{C})]$ . For  $\mathcal{C} \in \mathbf{Cat}$ , denoted  $U(\mathcal{C}) = \mathcal{P}\text{Sh}(\mathcal{C}, s\mathbf{Set})_{\text{Proj}}$ , which is a left and right proper combinatorial model category, by(4.6.3.48) and(4.5.6.1)(4.5.6.2). And there is a natural functor

$$s \downarrow : \mathcal{C} \mapsto \mathcal{P}\text{Sh}(\mathcal{C}, \mathbf{Set}) \rightarrow \mathcal{P}\text{Sh}(\mathcal{C}, s\mathbf{Set}).$$

┘

**Prop.(4.6.5.5)** **[Model-Categorical Yoneda Extensions]**. Let  $\mathcal{C} \in \mathbf{Cat}$ ,  $\mathcal{M}$  be a model category, then for any functor  $Q : \mathcal{C} \rightarrow \mathcal{M}$ , there is a Quillen adjunction  $(L, R) : U(\mathcal{C}) \xrightleftharpoons{\text{Quillen}} \mathcal{M}$  together with a natural weak equivalence  $L \circ s \downarrow \cong Q \in \text{Func}(\mathcal{C}, \mathcal{M})$ . And the category of such extensions is contractible.

┘

*Proof:* Cf.[D. Dugger. Universal homotopy theories].

□

### Exponentiation in Model Categories

#### Localizations and Presentations

**Def.(4.6.5.6)** **[S-Local and S-Equivalences]**.

┘

**Prop.(4.6.5.7)** **[Localizations of Simplicial Model Categories]**. Let  $\mathcal{M}$  be a left proper combinatorial simplicial model category and  $S \subset \text{Cof}(\mathcal{M})$  be a small subset, then there is a left proper combinatorial simplicial model category  $S^{-1}\mathcal{M}$  with the same underlying category as  $\mathcal{M}$  and

- Cofibrations:  $\text{Cof}(\mathcal{M})$ .
- Weak Equivalences:  $S$ -equivalences in  $\mathcal{M}$ .
- Fibrations: defined by the above two.

And  $X \in \mathcal{M}$  is fibrant in  $S^{-1}\mathcal{M}$  iff  $X$  is  $S$ -local and fibrant in  $\mathcal{M}$ .

┘

*Proof:* Cf.[Lur09]P906.

□

**Prop.(4.6.5.8).** Let  $\mathcal{M}$  be a left proper combinatorial simplicial model category, then

- Any combinatorial Bousfield localization(4.5.4.3) of  $\mathcal{M}$  is of the form  $S^{-1}\mathcal{M}$ , where  $S \subset \text{Cof}(\mathcal{M})$  is a small subset.
- For any two small subset  $S, T \subset \text{Cof}(\mathcal{M})$ ,  $S^{-1}\mathcal{M}$  and  $T^{-1}\mathcal{M}$  coincide iff the class of  $S$ -local objects and  $T$ -local objects coincide.

┘

*Proof:* Cf.[Lur09]P908. □

**Prop.(4.6.5.9) [Combinatorial Model Categories have Presentations].** Every combinatorial model category  $\mathcal{M}$  has a presentation, i.e. there exists a  $\mathcal{C} \in \mathbf{Cat}$  a set  $S$  of morphisms in  $U(A)$  and a Quillen equivalence

$$U(\mathcal{C})[S^{-1}] \xrightleftharpoons{\text{Quillen}} \mathcal{M}(4.6.5.7).$$

┘

*Proof:* [D. Dugger. Combinatorial model categories have presentations]. □

## 6 Covariant Model Structure

**Prop.(4.6.6.1) [Covariant Model Structure].** For  $S \in s\mathbf{Set}$ , a map  $X \rightarrow Y \in (\mathbf{Set}_\Delta)_S$  is called a

- **covariant cofibration** if it is a monomorphism.
- **covariant equivalence** if the induced map  $X^\triangleleft \coprod_X S \rightarrow Y^\triangleleft \coprod_Y S$  is a categorical equivalence(4.6.4.12).

Then these define a left proper combinatorial model structure on  $(\mathbf{Set}_\Delta)_S$ . ┘

*Proof:* Cf.[HTT, P69]. □

**Lemma(4.6.6.2).** Every left anodyne map in  $(\mathbf{Set}_\Delta)_S$  is a covariant equivalence. ┘

*Proof:* Cf.[HTT, P69]. □

**Prop.(4.6.6.3) [Covariant Model Structure].**  $(\mathbf{Set}_\Delta)_S$  is a simplicial model category with the contravariant model structure and the simplicial structure where

$$\text{Map}(X, Y) = Y^X \times_{S^X} \{\varphi\} \in \mathbf{Set}_\Delta$$

where  $\varphi : X \rightarrow S$  is the structure map. ┘

*Proof:* We use(4.6.5.2), it suffices to check that  $X \times \Delta^n \rightarrow X \times \Delta^0$  is a covariant equivalence. But it has a section, which is a left anodyne by(4.6.3.31), thus it is a covariant equivalence by(4.6.6.2). □

**Cor.(4.6.6.4) [Contravariant Model Structure].** Let  $S$  be a simplicial set, then the covariant model is usually not self-dual, and we can define a **contravariant model structure** as follows:

- A **contravariant cofibration** is a monomorphism of simplicial sets.
- $f$  is a **contravariant equivalence** in  $(\mathbf{Set})_S$  iff  $f^{op}$  is a covariant equivalence in  $(\mathbf{Set})_{S^{op}}$ .
- $f$  is a **contravariant fibration** in  $(\mathbf{Set})_S$  iff  $f^{op}$  is a covariant fibration in  $(\mathbf{Set})_{S^{op}}$ .

┘

**Prop.(4.6.6.5) [Base Change].** For  $S \rightarrow S' \in s\mathbf{Set}$ , the forgetful functor and base change functor  $j_!, j^*$  defines a Quillen adjunction of covariant models:

$$j_! : (\mathbf{Set}_\Delta)_S \xrightleftharpoons{\text{Quillen}} (\mathbf{Set}_\Delta)_{S'} : j^*$$

┘

*Proof:* it is clearly a pair of adjoints, and  $j_!$  preserves cofibrations.  $j_!$  also preserves covariant equivalences: Cf.[HTT, P71] ?. Thus it is a Quillen adjunction.  $\square$

**Lemma (4.6.6.6).** Let  $S' \subset S$  be simplicial sets, let  $p : X \rightarrow S$  be a map and  $q : Y \rightarrow S$  be a right fibration. Let  $X' = X \times_S S', Y' = Y \times_S S'$ , then the restriction map

$$\varphi : \text{Map}_{(\text{Set}_\Delta)_{/S}}(X, Y) \rightarrow \text{Map}_{(\text{Set}_\Delta)_{/S'}}(X', Y')$$

is a Kan fibration.  $\lrcorner$

*Proof:* Firstly it is a right fibration because it has right lifting property w.r.t. right anodyne inclusion  $A \rightarrow B$ : this is because  $(A \times X') \coprod (A \times X) \subset B \times X$  is also a right anodyne (4.6.3.31). Next we apply this to the inclusion  $\emptyset \subset S'$  to see that  $\text{Map}_{(\text{Set}_\Delta)_{/S'}}(X', Y')$  is a Kan complex (4.7.1.27), and then  $\varphi$  is a Kan fibration by (4.6.3.42).  $\square$

**Lemma (4.6.6.7).** Let  $p : X \rightarrow S$  be an object of  $(\text{Set}_\Delta)_{/S}$ , then  $p$  is a right fibration iff it is a covariant fibrant object in  $(\text{Set}_\Delta)_{/S}$ .  $\lrcorner$

*Proof:* Cf.[HTT, P85].  $\square$

**Def. (4.6.6.8) [Pointwise Equivalence].** Let  $X \rightarrow Y$  be a map in  $\text{RFib}(S)$ , then  $f$  is called a **pointwise equivalence** iff the induced map  $X_s \rightarrow Y_s$  is a homotopy equivalence of Kan complexes (4.7.1.27) for any  $s \in S$ .  $\lrcorner$

**Lemma (4.6.6.9).** Let  $f : X \rightarrow Y$  be a morphism in  $\text{RFib}(S)$ , then the following are equivalent:

- $f$  is a pointwise equivalence.
- $f$  is an equivalence in the simplicial category  $(\text{Set})_{/S}$ .
- For any  $A \in (\text{Set})_{/S}$ ,  $f$  induces a homotopy equivalence of Kan complexes:  $\text{Map}_{(\text{Set}_\Delta)_{/S}}(A, X) \rightarrow \text{Map}_{(\text{Set}_\Delta)_{/S}}(A, Y)$ .  $\lrcorner$

*Proof:* Cf.[HTT, P82].  $\square$

**Prop. (4.6.6.10) [Equivalences].** In the situation of (4.6.6.8),  $f$  is a pointwise equivalence iff it is a contravariant equivalence iff it is a categorical equivalence.  $\lrcorner$

*Proof:* Cf.[HTT, 2.2.3.13, 3.3.1.5] ?.  $\square$

**Prop. (4.6.6.11) [Contravariant Fibration as Right Fibration].** Let  $f : X \rightarrow Y$  be a map in  $\text{RFib}(S)$ , then  $f$  is a contravariant fibration in  $(s\text{Set})_{/S}$  iff it is a right fibration.  $\lrcorner$

*Proof:* Cf.[HTT, P86].  $\square$

### Straightening and Unstraightening

**Def. (4.6.6.12) [Straightening and Unstraightening].** Fix a simplicial set  $S$ , a simplicial category  $\mathcal{C}$  and a functor  $\mathcal{C}[S] \rightarrow \mathcal{C}^{\text{op}}$ . Given an object  $X \in (\text{Set}_\Delta)_{/S}$ , let  $v$  be the cone point of  $X^\triangleright$ . Then the simplicial category  $\mathcal{M} = \mathcal{C}[X^\triangleright] \coprod_{\mathcal{C}[X]} \mathcal{C}^{\text{op}}$  can be viewed as a correspondence ? between  $\mathcal{C}^{\text{op}}$  and  $\Delta^0$ , thus giving a simplicial functor

$$\text{St}_\varphi X : \mathcal{C} \rightarrow \text{Set}_\Delta : C \mapsto \text{Map}_{\mathcal{M}}(C, v).$$

Then  $St_\varphi$  is called the **straightening functor** associated to  $\varphi$ . And we denote by  $St_S$  the functor  $St_\varphi$  where  $\varphi : \text{id}_{\mathfrak{C}[S]}$ .

By the adjoint functor theorem (4.1.1.35),  $St_\varphi$  has a left adjoint called **unstraightening functor**  $UnSt_\varphi$ .  $\lrcorner$

**Prop. (4.6.6.13).** Let  $S$  be a simplicial set,  $\mathfrak{C}$  a simplicial category and  $\varphi : \mathfrak{C}[S] \rightarrow \mathfrak{C}^{\text{op}}$  a simplicial functor, then the straightening and unstraightening functor determines a Quillen adjunction

$$St_\varphi : (\text{Set}_\Delta)_S \xrightleftharpoons{\text{Quillen}} \text{Set}_\Delta^\mathfrak{C} : UnSt_\varphi$$

determines a Quillen adjunction, where the LHS has the contravariant model structure and the RHS has the projective model structure. And if  $\varphi$  is an equivalence of simplicial categories, then  $(St_\varphi, Un_\varphi)$  is a Quillen equivalence.  $\lrcorner$

*Proof:*  $\square$

### Unstraightening of Right Fibrations

**Prop. (4.6.6.14).** For every simplicial set  $S$ , the unstraightening  $Un_S$  induces an equivalence of simplicial categories

$$(\text{Set}_\Delta^{\mathfrak{C}[S]^{\text{op}}})_{cf} \rightarrow \text{RFib}(S).$$

(4.5.5.5), where the RHS is the category of fibrations  $X \rightarrow S$ .  $\lrcorner$

*Proof:* Cf. [HTT, P83].  $\square$

## 7 Cartesian Fibrations

**Remark (4.6.7.1).** The theory of Cartesian fibrations is an analogue of the theory of fibered categories.  $\lrcorner$

**Def. (4.6.7.2) [ $p$ -Cartesian].** Let  $p : X \rightarrow S$  be an inner fibration and  $f : x \rightarrow y$  is an edge of  $X$ , then  $f$  is called a  **$p$ -Cartesian** if the induced map

$$X_{/f} \rightarrow X_{/y} \times_{S_{/p(y)}} S_{/p(f)}$$

is a trivial Kan fibration.  $\lrcorner$

**Remark (4.6.7.3).** For  $\mathfrak{C} \in \text{Cat}$  and  $p : N(\mathfrak{C}) \rightarrow \Delta^1 \in s\text{Set}$ , a morphism  $f \in \mathfrak{C}$  is  $p$ -Cartesian iff it is Cartesian in the usual sense.  $\lrcorner$

**Prop. (4.6.7.4) [Characterization of Cartesian Fibrations].** Let  $p : \mathfrak{C} \rightarrow \mathcal{D}$  be an inner fibration of  $\infty$ -categories, then an edge  $f : Y \rightarrow Z \in \mathfrak{C}$  is  $p$ -Cartesian iff for every object  $X \in \mathfrak{C}$ , there is a Cartesian diagram

$$\begin{array}{ccc} \text{Map}(X, Y) & \longrightarrow & \text{Map}(X, Z) \\ \downarrow & & \downarrow \\ \text{Map}(p(X), p(Y)) & \longrightarrow & \text{Map}(p(X), p(Z)) \end{array} .$$

*Proof:* Cf. [HTT, P131].  $\square$

### Cartesian Fibrations

**Def. (4.6.7.5) [Cartesian Fibration].** A **Cartesian fibration** is an inner anodyne map  $p : X \rightarrow S$  that for any edge  $f : x \rightarrow y \in S$  and a vertex  $\tilde{y}$  mapping to  $y$ , there is a  $p$ -Cartesian edge  $\tilde{f}$  with  $p(\tilde{f}) = f$ . The dual of a Cartesian fibration is called a **coCartesian fibration**.  $\lrcorner$

**Prop. (4.6.7.6).** The class of Cartesian fibrations is stable under compositions and base change.  $\lrcorner$

**Prop. (4.6.7.7) [Cartesian Fibration and Right Fibration].** Let  $p : X \rightarrow S$  be an inner fibration, then the following are equivalent:

- $p$  is a Cartesian fibration and the every edge of  $X$  is  $p$ -Cartesian.
- $p$  is a right fibration.
- $p$  is a Cartesian fibration and every fiber  $X_s$  is a Kan complex.

$\lrcorner$

*Proof:* Cf.[HTT, P122]. ?

$\square$

**Def. (4.6.7.8) [Locally Cartesian Fibration].** A map  $X \rightarrow S$  of simplicial sets is called a **locally Cartesian fibration** if it is an inner fibration and for every edge  $\Delta^1 \rightarrow S$ , the pullback  $X \times_S \Delta^1 \rightarrow \Delta^1$  is a Cartesian fibration.  $\lrcorner$

**Prop. (4.6.7.9) [Cartesian and Locally Cartesian].** Let  $p : X \rightarrow S$  be a locally Cartesian fibration, then the following are equivalent:

- $p$  is a Cartesian fibration.
- Given a composition  $fg \cong h$  in the homotopy category, if  $f, g$  are both locally  $p$ -Cartesian, then  $h$  is also locally  $p$ -Cartesian.
- Every locally  $p$ -Cartesian edge in  $X$  is  $p$ -Cartesian.

$\lrcorner$

*Proof:* Cf.[HTT, P124].

$\square$

**Prop. (4.6.7.10).** Given maps of  $\infty$ -categories:  $\mathcal{C} \xrightarrow{p} \mathcal{D} \xrightarrow{q} \mathcal{E}$ , if  $q, q \circ p$  are both locally Cartesian fibrations and  $p$  maps locally  $(q \circ p)$ -Cartesian maps to locally  $q$ -Cartesian maps and for any  $Z \in \mathcal{E}$ ,  $p$  induces a categorial equivalence  $\mathcal{C}_Z \rightarrow \mathcal{D}_Z$ , then  $p$  is a categorial equivalence.  $\lrcorner$

*Proof:* Cf.[HTT, P132]. ?

$\square$

**Prop. (4.6.7.11).** Categorical equivalences between  $\infty$ -categories are stable under base change of Cartesian fibrations of  $\infty$ -categories.  $\lrcorner$

*Proof:* Cf.[HTT, P132].

$\square$

## 8 Simplicial Homology

**Prop. (4.6.8.1).** For a Kan fibration  $X$ , there can be defined a homotopy groups  $\pi_n$  that they agree with  $\pi_i(|X|)$  thus also  $\pi_i(S|X|)$ , Cf.[Weibel P263]. Thus we see that  $|BG|$  is truly the Eilenberg-MacLane spaces  $BG$ .  $\lrcorner$

## 9 Cyclic Homology Theory

References are [欧阳恩林, 循环同调].



### Combinatorial Category

**Def. (4.6.9.1).** The **Segal category**  $\text{Fin}_*$  is the category of pointed finite sets. A morphism is called **inert** iff  $|f^{-1}(\{i\})| = 1$  for all  $i \neq *$ . It is called **active** iff  $f^{-1}(\{*\}) = \{*\}$ .

A morphism can be uniquely factorized as a composition  $gh$ , where  $h$  is inert and  $g$  is active.  $\lrcorner$

**Prop. (4.6.9.2).** There is a morphism  $\text{Cut} : \Delta^{\text{op}} \rightarrow \text{Fin}_*$  where we interpret  $[n] \in \text{Fin}_*$  as the set of cut in  $[n]$ , and

$$\text{Cut}(\alpha)(i) = \begin{cases} j & , \alpha(j-1) < i \leq \alpha(j) \exists j \in [n]_+ \\ 0 & , \text{otherwise} \end{cases}$$

$\lrcorner$

**Prop. (4.6.9.3).** The category of functors from the  $E_\infty = \text{Fin}_*$  to  $\text{Cat}$  that

$$X([n]) \xrightarrow{\prod_{i=2}^n X(\rho^i)} \prod_{i=1}^n X([1]) \quad n \geq 0$$

and  $X([0])$  is the final object, is equivalent to the category of symmetric unital monoidal categories with base category  $(X([1]))$ . (Because the commutativity of morphisms encodes the fact that the tensor action is symmetric).

Similarly, the category of functors from the  $\Delta^{\text{op}}$  to  $\text{Cat}$  that

$$X([n]) \xrightarrow{\prod_{i=2}^n X(\rho^i)} \prod_{i=1}^n X([1]) \quad n \geq 0$$

is equivalent to the category of symmetric unital monoidal categories  $(X([1]))$ . And it is symmetric iff it factors through  $\text{Cut} : \Delta^{\text{op}} \rightarrow \text{Fin}_*$ .  $\lrcorner$

**Def. (4.6.9.4).** The **Conne cyclic category**  $\Delta_C$  is a category containing  $\Delta$  that  $\text{Aut}_{\Delta_C}([n])$  is  $C_{n+1}$ . And every morphism  $[n] \rightarrow [m]$  in  $\Delta_C$  can be uniquely written as the form  $\varphi g$ , where  $\varphi \in \text{Hom}_\Delta([n], [m])$  and  $g \in \text{Aut}_{\Delta_C}([n])$ .

$\Delta_C^{\text{op}}$  is isomorphic to  $\Delta_C$  Cf.[杨恩林循环同调 P31], thus  $\Delta$  and  $\Delta^{\text{op}}$  are all subcategories of  $\Delta_C$ .

$\lrcorner$

**Def. (4.6.9.5).** The category  $\Delta_S$  is the category that  $\text{Aut}_{\Delta_S}([n]) \cong S^n$  and every morphism  $[n] \rightarrow [m]$  in  $\Delta_S$  can be uniquely written as the form  $\varphi g$ , where  $\varphi \in \text{Hom}_\Delta([n], [m])$  and  $g \in \text{Aut}_{\Delta_S}([n])$ .  $\lrcorner$

**Def. (4.6.9.6).** For a category  $C$ , a **cyclic object** in  $C$  is a functor  $\Delta_C^{\text{op}} \rightarrow C$ .

For example, the functor that maps  $[n]$  to  $C_{n+1}$  and the functor maps to the pull back of the order of the cyclic, is a cyclic object.  $\lrcorner$

### Hochschild Homology(Jeremy Hahn)

**Def. (4.6.9.7)[Hochschild Homology Group].** Let  $R$  be a commutative ring and  $A$  a flat  $R$ -algebra,  $A^{\text{env}} = A \otimes_R A^{\text{op}}$ . Then an  $A^{\text{env}}$ -module is equivalent to an  $(A, A)$ -bimodule.

If  $M$  is an  $(A, A)$ -bimodule, then we define **Hochschild homology group**  $HH_n(A/R, M) = \text{Tor}_n^{\text{env}}(M, A)$ . And also we denote  $HH_n(A/R) = HH_n(A/R, M)$ .

$H_n(A, M)$  is a  $Z(A)$  module by the action of  $Z(A)$  on  $M$  and  $HH_*$  defines a functor  $\mathcal{C}\text{Ring}_R \rightarrow \text{Mod}_R$ .  $\lrcorner$

**Def. (4.6.9.8)[Hochschild Complex].** Let  $R$  be a commutative ring and  $A$  a flat  $R$ -algebra, we define  $A^{env} = A \otimes_R^L A^{op}$ , and  $HH(A/R) = A \otimes_{A^{env}}^L A \in D(A)$ .  $\lrcorner$

**Def. (4.6.9.9)[Flat Case].** For a flat  $R$ -algebra  $A$  and a  $(A, A)$ -bimodule  $M$ , there is a simplicial module  $C(A, M)$  called the **Hochschild complex** of  $A$  with coefficient in  $M$ , with  $M_n = M \otimes A^n$  that

$$d_i(m, a_1, \dots, a_n) = \begin{cases} (m_0 a_1, a_2, \dots, a_n) & i = 0 \\ (a_n m_0, a_1, \dots, a_{n-1}) & i = n \\ (m_0, a_1, \dots, a_i a_{i+1}, \dots, a_n) & \text{otherwise} \end{cases}$$

$$s_j(m, a_1, \dots, a_n) = (m, a_1, \dots, a_{j-1}, 1, a_{j+1}, \dots, a_n)$$

The homology group of the Moore complex associated to the Hochschild complex is just  $HH_n(A, M)$ . The Moore complex is of the form

$$\dots \rightarrow M \otimes A \otimes A \otimes A \xrightarrow{\partial_3} M \otimes A \otimes A \xrightarrow{\partial_2} M \otimes A \xrightarrow{\partial_1} M \xrightarrow{\partial_0} 0 \rightarrow 0 \rightarrow \dots$$

where

$$\partial_1(m \otimes a) = ma - am, \quad \partial_2(m \otimes a_1 \otimes a_2) = ma_1 \otimes a_2 - m \otimes a_1 a - 2 + a_2 m \otimes a_1$$

$$\partial_3(m \otimes a_1 \otimes a_2 \otimes a_3) = ma - 1 \otimes a_2 \otimes a_3 - m \otimes a_1 a_2 \otimes a_3 + m \otimes a_1 \otimes a_2 a_3 - a_3 m \otimes a_1 \otimes a_2.$$

$\lrcorner$

*Proof:*

$\square$

**Example (4.6.9.10).**

- $HH_n(R, R) = R$  if  $n = 0$  and 0 otherwise.
- $HH_0(A/R, A/R) = A_{ab}$ .
- If  $A \in \mathcal{Ab}$ ,  $HH_1(A, A) \cong \Omega_{A/R}^1$  is given by  $a \otimes x \mapsto adx$  by (5.4.3.4).
- For a symmetric  $(A, A)$ -module  $M$ , thus we have  $H_1(A, M) = M \otimes_A A^{ab}$  and  $H_1(A, M) = M \otimes_A \Omega_{A/R}^1$ . And if  $M$  is flat,  $H_n(A, M) = M \otimes_A H_n(A, A)$ . ?

$$\bullet \quad HH_n(R[X]/R) = \begin{cases} R[X] & n = 0 \\ \Omega_{R[X]/R}^1 & n = 1 \\ 0 & \text{otherwise} \end{cases}.$$

$\lrcorner$

**Example (4.6.9.11)[ $HH(\mathbb{F}_p/\mathbb{Z})$ ].** Because  $\mathbb{F}_p \otimes_{\mathbb{Z}}^L \mathbb{F}_p = (\mathbb{F}_p \xrightarrow{0} \mathbb{F}_p)$ . ?

$$HH_*(\mathbb{F}_p, \mathbb{Z}) \cong \mathbb{F}_p[X_1, X_2, \dots, ] / (X_i X_j = \binom{i+j}{i} X_{i+j})$$

where  $\deg(X_i) = 2i, \partial X_i = 0$ .

$\lrcorner$

**Prop. (4.6.9.12).** Suppose  $A, B$  are  $R$ -algebras, then

$$HH(A \otimes_R^L B/R) = HH(A/R) \otimes_R^L HH(B/R).$$

$\lrcorner$

**Cor. (4.6.9.13).**  $HH(R[X_1, \dots, X_n]/R) = \Omega_{R[X_1, \dots, X_n]/R}^*$ .  $\lrcorner$

**Prop. (4.6.9.14).** If  $A$  is a commutative  $R$ -algebra, then  $HH(A/R)$  is naturally a commutative dga. In particular,  $HH_*(A/R)$  is a graded ring.  $\lrcorner$

**Prop. (4.6.9.15) [Spectral Sequence].** For a commutative ring  $A$  and a symmetric  $A$ -bimodule  $M$ , there is a spectral sequence

$$E_{pq}^2 = \text{Tor}_p^R(H_p(A, A), M) \Rightarrow H_{p+q}(A, M).$$

$\lrcorner$

### Hochschild Homology

**Prop. (4.6.9.16) [Hochschild-Kostant-Rosenberg].** The isomorphism  $\Omega_{A/R}^1 \cong HH_1(A)$  extends to a graded ring map

$$\Psi : \Omega_{A/k}^* \rightarrow H_*(A, A)$$

. If  $A/R$  be smooth algebra and  $R$  Noetherian, then  $\Psi$  is an isomorphism of graded algebra. Cf. [Weibel P322], [阳恩林循环同调 P133].  $\lrcorner$

**Def. (4.6.9.17) [Tsygan's Double Complex].** For a cyclic object  $M$  in an Abelian category, let  $t_*$  be the cyclic morphism and  $\partial_n = \sum_{i=0}^n (-1)^i d_i$ ,  $\partial'_n = \sum_{i=0}^{n-1} (-1)^i d_i$ ,  $N_n = \sum_{k=0}^n ((-1)^n t_n)^k$ , then there is a double complex  $CC(M)$ :

$$\begin{array}{ccccc} \downarrow \partial & & \downarrow -\partial' & & \downarrow \partial \\ M_1 & \xleftarrow{1-(-1)^1 t} & M_1 & \xleftarrow{N} & M_1 & \xleftarrow{1-(-1)^1 t} \\ \downarrow \partial & & \downarrow -\partial' & & \downarrow \partial \\ M_0 & \xleftarrow{1-(-1)^0 t} & M_0 & \xleftarrow{N} & M_0 & \xleftarrow{1-(-1)^0 t} \end{array}$$

That the column are 2-cyclic. Cf. [Weibel P337]. The first column is called the **Hochschild complex of  $M$** :  $C^h(M)$ , the second column is called **acyclic complex of  $M$**  (4.6.9.18)  $C^a(M)$ . And we can even augment a cokernel column on the left, which is the complex of  $M$  modulo the cyclic action, called the **Conne complex**  $C^\lambda(M)$ .

We define the **Cyclic Homotopy Group**  $HC_n(M) = H_n(\text{Tot} CC(M))$  and when  $M$  is the cyclic module  $C(A)$  (4.6.9.9), denote  $CC(C(A)) = CC(A)$ ,  $HC_n(A) = HC_n(C(A))$ .  $\lrcorner$

**Lemma (4.6.9.18).** The second column is exact and  $h = t_{n+1}s_n$  is a null-homotopy. Cf. [阳恩林循环同调 P122].  $\lrcorner$

**Lemma (4.6.9.19).** Notice the rows are in fact a group homology  $\text{Hom}(\mathbb{Z}/(n+1)\mathbb{Z}, M_n)$ , thus when  $\mathbb{Q} \in R$ , we have the rows are acyclic because the group homology is killed by  $|G|$  (9.8.1.8), thus  $HC_*(M) \cong H_*^\lambda(M)$  are isomorphisms by spectral sequence.  $\lrcorner$

**Prop. (4.6.9.20) [Conne SBI Sequence].** For a cyclic module  $M$ , there is a long exact sequence

$$\cdots \rightarrow HH_n(M) \xrightarrow{I} HC_n(M) \xrightarrow{S} HC_{n-2}(M) \xrightarrow{B} HH_{n-1}(M) \rightarrow \cdots$$

$\lrcorner$

*Proof:* shift the diagram 2 column right, then there is an exact sequence of double complexes and notice the second column is exact(4.6.9.18), thus we have the kernel is quasi-isomorphic to  $C^h(M)$ . So the sequence follows.  $\square$

**Cor. (4.6.9.21).**  $HC_0(A) = HH_0(A) = A_{ab}$ .

When  $A \in \mathcal{A}b$ ,  $HC_1(A) = \text{Coker}(HC_0(A) \xrightarrow{B} HH_1(A)) = \Omega_{A/R}^1/dA$  as a  $R$  module, because we can verify that  $B(a) = a \otimes 1 - 1 \otimes a$ .  $\lrcorner$

**Cor. (4.6.9.22).** For a morphism of two cyclic objects,  $HH_*(M) \cong HH_*(M')$  iff  $HC_n(M) \cong HC_n(M')$ . (Use five lemma).  $\lrcorner$

**Def. (4.6.9.23).** A **mixed complex**  $(M, b, B)$  is a complex with  $b : M_n \rightarrow M_{n-1}$  and  $B : M_n \rightarrow M_{n+1}$  that makes  $M$  into a double chain complex. And there is a **Conne double complex** associated with this mixed complex. And similarly there is a same **SBI** sequence associated to the following diagram:

$$\begin{array}{ccccc}
 & \downarrow & & \downarrow & & \downarrow \\
 M_2 & \xleftarrow{B} & M_1 & \xleftarrow{B} & M_0 \\
 \downarrow b & & \downarrow b & & \\
 C_1 & \xleftarrow{B} & C_0 & & \\
 \downarrow b & & & & \\
 C_0 & & & & 
 \end{array}$$

From a cyclic object  $M$ , we notice that the  $2k$ -th column is acyclic(4.6.9.18), thus there is a snake-like connection homomorphism  $B$  that makes  $M$  into a mixed complex  $BM$ . Cf.[Weibel P344]. And the Conne double complex will compute the same cyclic homology with previous defined cyclic homology, Cf.[Weibel P345].

Notice for this  $B$ ,  $B_*$  on homology is exactly the composition  $BI$ .  $\lrcorner$

**Prop. (4.6.9.24).** Let  $R \rightarrow A \in \mathcal{C}Ring$  and  $M \in \text{Mod}_A$  then there is a natural morphism

$$M \otimes_A \Omega_{A/R}^n \xrightarrow{\varepsilon_n} H_n(A, M) \xrightarrow{\pi_n} M \otimes_A \Omega_{A/R}^n.$$

such that  $\pi_n \circ \varepsilon_n = n!$ .

We first define a map  $\varepsilon_n : M \otimes \wedge^n A \rightarrow H_n(A, M)$  that

$$\varepsilon_n(m, a_1, \dots, a_n) = \sum \text{sgn}(\sigma)(m, a_{\sigma^{-1}(1)}, \dots, a_{\sigma^{-1}(n)})$$

then define  $\varepsilon_n(m \otimes x da_1 \wedge \dots \wedge da_n) = \varepsilon_n(mx, a_1, \dots, a_n)$ . And we verify that this map is well-defined and maps into  $Z_n(C(A, M))$ , Cf.[阳恩林循环同调 P99].

Then we define  $\pi_n(m, a_1, \dots, a_n) = m \otimes da_1 \wedge \dots \wedge da_n$  and verify easily that this vanish on  $B_n(C(A, M))$ . And it is easy to verify  $\pi_n \circ \varepsilon_n = n!$ .  $\lrcorner$

**Prop. (4.6.9.25).** When  $A$  is a unital  $R$ -algebra, there is a commutative diagram

$$\begin{array}{ccc}
 \Omega_{A/R}^n & \xrightarrow[(d)]{(n+1)d} & \Omega_{A/R}^{n+1} \\
 \uparrow \scriptstyle \pi_n \downarrow \scriptstyle \varepsilon_n & & \uparrow \scriptstyle \pi_{n+1} \downarrow \scriptstyle \varepsilon_{n+1} \\
 HH_n(A) & \xrightarrow{B_*} & HH_{n+1}(A)
 \end{array}$$

$\lrcorner$

*Proof:* We notice  $B = (1 - (-1)^n t) sN$  :

$$(m, a_1, \dots, a_n) \mapsto \sum_{i=0}^n (-1)^{in} (1, a_i, \dots, a_n, m, a_1, \dots, a_{i-1}) - \sum_{i=0}^n (-1)^{in} (a_i, 1, a_{i+1}, \dots, a_n, m, a_1, \dots, a_{i-1}).$$

Cf.[阳恩林循环同调 P128]. □

**Cor.(4.6.9.26).** For a commutative unital  $R$ -algebra  $A$ , there is a functorial  $\varepsilon_n : \Omega_{A/R}^n / d\Omega_{A/R}^{n-1} \rightarrow HC_n(A)$  making the following diagram commutative:

$$\begin{array}{ccccccc} \xrightarrow{0} \Omega^{n-1}/d\Omega^{n-2} & \xrightarrow{d} & \Omega^n & \longrightarrow & \Omega^n/d\Omega^{n-1} & \xrightarrow{0} & \Omega^{n-2}/d\Omega^{n-3} \longrightarrow \dots \\ & & \downarrow \varepsilon_{n-1} & & \downarrow \varepsilon_n & & \downarrow \varepsilon_{n-2} \\ \longrightarrow & HC_{n-1} & \xrightarrow{B} & HH_n & \xrightarrow{I} & HC_n & \xrightarrow{S} HC_{n-2} \xrightarrow{B} \dots \end{array}$$

which is induced by the cokernel. Cf.[阳恩林循环同调 P130]. When  $\mathbb{Q} \in R$ ,  $\varepsilon_n$  is a split injection. ┘

**Prop.(4.6.9.27).** When  $\mathbb{Q} \in R$ ,  $\frac{1}{n!}\pi_n$  induces a morphism of mixed complexes  $(BA, \partial, B) \rightarrow (\Omega_{A/R}^*, 0, d)$  by(4.6.9.24), thus there is a natural map

$$HC_n(A) \rightarrow \Omega_{A/R}^n / d\Omega_{A/R}^{n-1} \bigoplus_{i>0} H_{dR}^{n-2i}(A).$$

┘

**Prop.(4.6.9.28) [Morita Invariance].**  $\text{Tr} : HH_*(M_r(A), M_r(M)) \cong HH_*(A, M)$  by the trace and inclusion functors. Cf.[阳恩林循环同调 Morita Invariance]. In particular, there is an isomorphism  $HH_*(M_r(A)) \cong HH_*(A)$ , thus also  $HC_*(M_r(A)) \cong HC_*(A)$  by(4.6.9.20). ┘

**Prop.(4.6.9.29) [Karoubi].**  $BG$  is a cyclic group, and then the cyclic homology group  $HC_n(G, A) \cong \bigoplus_{k \geq 0} H_{n-2k}(G, A)$ . Cf.[Weibel P339]. ┘

## 4.7 $\infty$ -Categories

References are [Joy02], [Lur09], [Lur11], [Infinity Categories from Scratch], [A whirlwind tour of the world of  $(\infty, 1)$ -categories], [Gro15], [Introduction to  $\infty$ -Categories].

**Notation (4.7.0.1).**

- Use notations defined in [Simplicial Homotopy Theory](#).

┘

### 1 $\infty$ -Categories

**Def. (4.7.1.1) [ $\infty$ -Category].** An  $\infty$ -category is a simplicial set that has lifting property w.r.t any  $\Lambda_i^n \rightarrow \Delta^n$ , where  $0 < i < n$ . ┘

**Cor. (4.7.1.2).** By (4.6.3.6), the nerve of a category is an  $\infty$ -category. ┘

**Def. (4.7.1.3) [Sub- $\infty$ -Categories].** Let  $\mathcal{C} \in \text{Cat}_\infty$  and  $\mathcal{D}_0 \subset \mathcal{C}_0$  be a subset of vertices, then there is a sub-simplicial set  $\mathcal{D} \subset \mathcal{C}$  consisting of simplexes with all vertices in  $\mathcal{C}$ . Then it is also an  $\infty$ -category, called the  $\infty$ -category spanned by  $\mathcal{D}_0$ . ┘

**Prop. (4.7.1.4) [Characterizing  $\infty$ -Categories].**  $\mathcal{C} \in s\text{Set}$  is an  $\infty$ -category iff the restriction map

$$\text{Map}(\Delta^2, \mathcal{C}) \rightarrow \text{Map}(\Lambda_1^2, \mathcal{C})$$

is a trivial Kan fibration. ┘

*Proof:* This follows immediately from (4.6.3.30). □

**Def. (4.7.1.5) [Homotopy Categories of  $\infty$ -Categories].** For  $\mathcal{C} \in \text{Cat}_\infty$ , the homotopy category of  $\mathcal{C}$  (4.6.4.12) has a simpler description: Let  $f, g : X \rightarrow Y \in \mathcal{C}$  be called **homotopic maps** if there is

a 2-complex of the form 
$$\begin{array}{ccc} & X & \\ \text{id} \nearrow & & \searrow f \\ X & \xrightarrow{g} & Y \end{array}$$
.  $f : X \rightarrow Y$  is called an **equivalence** if there exists a map  $g : Y \rightarrow X$  s.t.  $f \circ g \cong \text{id}_Y$  and  $g \circ f \cong \text{id}_X$ .

Then the homotopy relation is an equivalence relation, and the composition of homotopic maps are homotopic, and we get a category  $\text{Ho}(\mathcal{C})$ . Then this category is naturally isomorphic to  $\tau_1(\mathcal{C})$  (4.6.3.3). ┘

*Proof:* [Lur09]P32. □

**Prop. (4.7.1.6) [Equivalence of  $\infty$ -Categories].** Categorical equivalence (4.6.4.12) between  $\infty$ -categories is an equivalence relation (4.6.4.12), by Joyal model category (4.6.4.19). ┘

**Def. (4.7.1.7) [Underlying  $\infty$ -Categories of Simplicial Model Categories].** Let  $\mathcal{M}$  be a simplicial model category (4.6.5.1), then  $N_\Delta(\mathcal{M}_{cf})$  is an  $\infty$ -category by (4.6.4.4) ?, called the **underlying  $\infty$ -category** of  $\mathcal{M}$ . ┘

**Lemma (4.7.1.8).** Let  $\mathcal{S} = s\text{Set}$ , then the map  $\beta_{X, \mathcal{S}}$  is a weak equivalence for every cofibrant object  $X \in \mathcal{C}$ . ┘

*Proof:* Cf. [HTT, P853]. □

**Prop. (4.7.1.9)[Quillen Equivalent Simplicial Model Categories Give Equivalent Underlying  $\infty$ -Categories].** Let  $\mathcal{C}, \mathcal{D}$  be simplicial model categories and

$$F : \mathcal{C} \xrightarrow[\text{Quillen}]{} \mathcal{D} : G$$

is a Quillen equivalence, every element of  $\mathcal{C}$  is cofibrant, and  $G$  is a simplicial functor, then  $G$  induces an equivalence of their underlying  $\infty$ -categories  $N_{\Delta}(\mathcal{D}_{cf}) \cong N_{\Delta}(\mathcal{C}_{cf})$ .  $\lrcorner$

*Proof:* This is because  $G : \mathcal{D}_{cf} \rightarrow \mathcal{C}_{cf}$  is an equivalence of simplicial categories, by (4.5.7.6) and (4.7.1.8), and then (4.6.4.16) shows  $N_{\Delta}(G) : N_{\Delta}(\mathcal{D}_{cf}) \rightarrow N_{\Delta}(\mathcal{C}_{cf})$  is an equivalence of  $\infty$ -categories.  $\square$

**Def. (4.7.1.10)[Homotopy Coherent Diagrams].** Let  $\mathcal{C} \in \text{Cat}_{\infty}$  and  $\mathcal{J} \in \text{Cat}$ , then a **homotopy coherent  $\mathcal{I}$ -diagram** is a map  $N(\mathcal{J}) \rightarrow \mathcal{C} \in s\text{Set}$ .  $\lrcorner$

### Categorical Constructions

**Def. (4.7.1.11)[Categorical Constructions].** A **categorical construction** is a functorial construction  $T : (s\text{Set})^m \times (s\text{Set}^{\text{op}})^n \rightarrow s\text{Set}$  s.t.

- For  $\mathcal{C}_i \in \text{Cat} \subset s\text{Set}$ ,  $T(\mathcal{C}_1, \dots, \mathcal{C}_n) \subset \text{Cat} \subset \text{Cat}_{\infty}$ .
- For  $\mathcal{C}_i \in \text{Cat}_{\infty} \subset s\text{Set}$ ,  $T(\mathcal{C}_1, \dots, \mathcal{C}_n) \subset \text{Cat}_{\infty}$ .
- If for each  $i$ ,  $\mathcal{C}_i, \mathcal{D}_i \in \text{Cat}_{\infty} \subset s\text{Set}$  are categorically equivalent, then  $T(\mathcal{C}_1, \dots, \mathcal{C}_n)$  and  $T(\mathcal{D}_1, \dots, \mathcal{D}_n)$  are equivalent  $\infty$ -categories.  $\lrcorner$

**Prop. (4.7.1.12)[Opposite  $\infty$ -Categories].** For  $\mathcal{C}, \mathcal{D} \in \text{Cat}_{\infty}$ ,

- The opposite (4.6.3.12)  $\mathcal{C}^{\text{op}}$  is also an  $\infty$ -category.
- If  $\mathcal{C} \rightarrow \mathcal{D}$  is a categorical equivalence, then  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$  is also a categorical equivalence.

Thus the opposite construction is a categorical construction, by (4.6.3.13).  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.7.1.13)[Mapping Spaces of  $\infty$ -Categories].** Let  $\mathcal{C}, \mathcal{D} \in \text{Cat}_{\infty}$  and  $K, K' \in s\text{Set}$ , then

- $\text{Map}(K, \mathcal{C})$  is also an  $\infty$ -category.
- If  $f : \mathcal{C} \rightarrow \mathcal{D}$  is a categorical equivalence, then the induced map  $\text{Map}(K, \mathcal{C}) \rightarrow \text{Map}(K, \mathcal{D})$  is also a categorical equivalence.
- If  $g : K \rightarrow K'$  is a categorical equivalence, then the induced map  $\text{Map}(K', \mathcal{C}) \rightarrow \text{Map}(K, \mathcal{C})$  is also a categorical equivalence.

In particular, mapping space is a categorical construction, by (4.6.3.16).  $\lrcorner$

*Proof:* 1 follows from (4.6.3.31). 2, 3 follow from [HTT, P94]. ?  $\square$

**Prop. (4.7.1.14)[Joins of  $\infty$ -Categories].** Let  $\mathcal{C}, \mathcal{D} \in \text{Cat}_{\infty}$ , then

- The join (4.6.3.18)  $\mathcal{C} * \mathcal{D}$  is also an  $\infty$ -category.
- If  $\mathcal{C} \rightarrow \mathcal{C}', \mathcal{D} \rightarrow \mathcal{D}' \in \text{Cat}_{\infty}$  are categorical equivalences, then  $\mathcal{C} * \mathcal{D} \rightarrow \mathcal{C}' * \mathcal{D}'$  is also a categorical equivalence.

In particular, the join is a categorical construction by (4.6.3.19).  $\lrcorner$

*Proof:* 1: Given a morphism  $p : \Lambda_i^n \rightarrow S \star S'$ , if the image is in  $S$  or  $S'$ , then it can be extended to  $\Delta^n$  by hypothesis. Thus we may assume that it maps  $\{0, \dots, j\}$  into  $S$  and  $\{j+1, \dots, n\}$  into  $S'$ , then we restrict  $p$  to get a morphism  $\Delta^{\{0, \dots, j\}} \rightarrow S, \Delta^{\{j+1, \dots, n\}} \rightarrow S'$ , which determines a map  $\Delta^n \cong \Delta^j * \Delta^{n-j-1}$  (4.6.3.19)  $\rightarrow S \star S'$ , and it extends  $p$  by (4.6.3.17).

2 follows from (4.6.4.26).  $\square$

**Prop. (4.7.1.15) [Homotopic Maps].** If  $X$  is an  $\infty$ -category, then so does  $X^B$  for any simplicial set  $B$ , by (4.6.3.37).

And we can call two maps in  $\text{Map}(B, X)$  **homotopic** if they are equivalent as vertices in  $X^B$  (4.7.1.5).  $\lrcorner$

**Lemma (4.7.1.16).** Let  $p : \mathcal{C} \rightarrow \mathcal{D}$  be a left fibration of  $\infty$ -categories and  $f : X \rightarrow Y$  be a morphism that  $p(f)$  is an equivalence in  $\mathcal{D}$ , then  $f$  is an equivalence in  $\mathcal{C}$ . Compare with (4.2.3.6).  $\lrcorner$

*Proof:* Let  $\bar{g}$  be a homotopy inverse to  $f$ , then there is a 2-complex

$$\begin{array}{ccc} & p(Y) & \\ p(f) \nearrow & & \searrow \bar{g} \\ p(X) & \xrightarrow{\quad} & p(X) \end{array}$$

and by left fibration property lifts to a 2-complex

$$\begin{array}{ccc} & Y & \\ f \nearrow & & \searrow g \\ X & \xrightarrow{\quad} & X \end{array}.$$

So  $f$  admits a left homotopy inverse, and by the same reason,  $g$  admits a left homotopy inverse, thus  $g$  has a left homotopy inverse, and it can be chosen to be  $f$ .  $\square$

**Lemma (4.7.1.17) [Equivalence Lifts via Left Fibrations].** Let  $p : \mathcal{C} \rightarrow \mathcal{D}$  be a left fibration of  $\infty$ -categories, if  $\bar{X} \in \mathcal{D}$  and  $Y \in \mathcal{C}$  and  $\bar{f} : \bar{X} \rightarrow p(Y)$  is an equivalence, then  $\bar{f}$  can be lifted to a morphism in  $\mathcal{C}$ . (Which is also an equivalence by (4.7.1.16)).  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.7.1.18) [Equivalence and Left Extension].** Let  $\mathcal{C}$  be an  $\infty$ -category and  $\varphi$  an edge, then  $\varphi$  is an equivalence iff for any  $n \geq 2$  and every map  $\Lambda_0^n \rightarrow \mathcal{C}$  that  $f_0|_{\Delta^{\{0,1\}}} = \varphi$ , there exists an extension of  $f_0$  to  $\Delta^n$ .  $\lrcorner$

*Proof:* If  $\varphi$  is an equivalence, then consider the diagram

$$\begin{array}{ccc} \{0\} & \longrightarrow & \mathcal{C}/\Delta^{n-2} \\ \downarrow & \nearrow \varphi' & \downarrow q \\ \Delta^1 & \xrightarrow{\quad} & \mathcal{C}/\partial\Delta^{n-2} \end{array}$$

Because  $\mathcal{C}/\partial\Delta^{n-2} \rightarrow \mathcal{C}$  is right fibration (4.6.3.36), and by the dual of (4.7.1.16),  $\varphi'$  is an equivalence, thus by the dual of (4.7.1.17) the dotted arrow exists.

Conversely, if the condition holds, then we can use a diagram  $\Lambda_0^2 \rightarrow \mathcal{C}$  to find a morphism  $\psi$  that  $\psi \circ \varphi \cong \text{id}$ , and we can also use a diagram  $\Lambda_0^3 \rightarrow \mathcal{C}$  to witness the fact  $\varphi \circ \psi \cong \text{id}$ , so  $\varphi$  is an equivalence.  $\square$



**Cor. (4.7.1.19).** An equivalence in  $\text{Map}(K, \mathcal{C})$  is equivalent to a map  $K \times \Delta^1 \rightarrow \mathcal{C}$  that  $\{x\} \times \Delta$  are all mapped to equivalences in  $\mathcal{C}$ .  $\square$

*Proof:* ? Cf.[HTT, P106].  $\square$

**Def. (4.7.1.20) [Space of Morphisms].** For vertices  $x, y$  in a simplicial set  $S$ , we want to define a representative for  $\text{Map}_{hS}(x, y)$  other than  $\text{Map}_{\mathcal{C}[S]}(x, y)$ . We define the **space of right morphisms**

$$\text{Hom}_S^R(x, y) = S_{/y} \times_S \{x\}.$$

The definition is not symmetric, instead, we define the **space of left morphisms**  $\text{Hom}_S^L(x, y) = (\text{Hom}_{S^{op}}^R(x, y))^{op}$ .

Also we can define  $\text{Hom}_S(x, y) = \{x\} \times_S S^{\Delta^1} \times_S \{y\}$ , then there are natural inclusions:

$$\text{Hom}_S^R(x, y) \hookrightarrow \text{Hom}_S(x, y) \hookleftarrow \text{Hom}_S^L(x, y).$$

**Prop. (4.7.1.21).** If  $\mathcal{C} \in \text{Cat}_\infty$  and  $x, y \in \mathcal{C}$ , then  $\text{Hom}_{\mathcal{C}}^R(x, y)$  is a Kan complex, and the inclusions defined in (4.7.1.20) are weak equivalences.  $\square$

*Proof:* This is obvious, because the right lifting diagram w.r.t.  $\Lambda_j^n \subset \Delta^n, 0 < j \leq n$  is equivalent to an extension  $\Lambda_j^n \star \Delta^0 \subset \Delta^{n+1}$  that satisfies  $\tilde{u}|_{\Delta_{\{0, \dots, n\}}} = x$ . It can be solved by a two-step extension where the first is by identity extension and then extend using inner fibration property.

For the last assertion, Cf.[HTT, 4.2.1.8]. ?  $\square$

**Prop. (4.7.1.22).** Let  $p : \mathcal{C} \rightarrow \mathcal{D}$  be an inner fibration of  $\infty$ -categories, then the induced maps on the spaces of right morphisms are Kan fibrations.  $\square$

*Proof:* Since  $p$  is an inner fibration, the induced map  $\tilde{\varphi} : \mathcal{C}_{/Y} \rightarrow \mathcal{D}_{/p(Y)} \times_{\mathcal{D}} \mathcal{C}$  is a right fibration by (4.6.3.36), and the morphism on  $\text{Hom}_{\mathcal{C}}^R(X, Y)$  is obtained from  $\tilde{\varphi}$  by restricting to fiber over  $X$ , thus also a right fibration. And by (4.7.1.21) and (4.6.3.42).  $\square$

**Lemma (4.7.1.23).** Let  $\mathcal{C} \rightarrow \mathcal{D}$  be a fully faithful map of  $\infty$ -categories and  $p : K \rightarrow \mathcal{C} \in s\text{Set}$ , then the map of  $\infty$ -categories (4.7.1.24)

$$\mathcal{C}_{j/} \times_{\mathcal{C}} \{x\} \rightarrow \mathcal{D}_{pj/} \times_{\mathcal{D}} \{p(x)\}$$

is a homotopy equivalence.  $\square$

*Proof:* Cf.[HTT, P134].  $\square$

**Prop. (4.7.1.24) [Overcategories and Undercategories].**

- If  $\mathcal{C}$  be an  $\infty$ -category and  $p : K \rightarrow \mathcal{C}$  be a morphism, then the projection  $\mathcal{C}_{p/} \rightarrow \mathcal{C}$  is a left fibration. In particular,  $\mathcal{C}_{p/}$  is itself an  $\infty$ -category. Dually,  $\mathcal{C}_{/p}$  is also an  $\infty$ -category.
- Let  $p : \mathcal{C} \rightarrow \mathcal{D} \in \text{Cat}_\infty$  be an equivalence of  $\infty$ -categories and let  $j : K \rightarrow \mathcal{C} \in s\text{Set}$ , then the induced map  $\mathcal{C}_{j/} \rightarrow \mathcal{D}_{pj/}$  is an equivalence of  $\infty$ -categories. The dual holds for undercategories. In particular, the overcategories and undercategories are categorical constructions, by (4.6.3.24).  $\square$

*Proof:* 1: We use the proposition (4.6.3.36) in case  $S = \Delta^0, A = \emptyset, X = \mathcal{C}$ .

2: There is a factorization  $\mathcal{C}_{j/} \xrightarrow{f} \mathcal{D}_{pj/} \times_{\mathcal{D}} \mathcal{C} \xrightarrow{g} \mathcal{D}_{pj/}$ . (4.7.1.23)(4.7.1.24) shows  $\mathcal{C}_{j/}$  and  $\mathcal{D}_{pj/} \times_{\mathcal{D}} \mathcal{C}$  are fiberwise equivalent left fibrations over  $\mathcal{C}$ , thus by (4.6.7.7) and (4.6.7.10),  $f$  is a categorical equivalence. Also,  $g$  is a categorical equivalence by (4.6.7.11). So we are done. ? Cf.[HTT, P135].  $\square$

**Prop. (4.7.1.25)[Lifting of Homotopies].** Let  $p : \mathcal{C} \rightarrow \mathcal{D}$  be a categorical equivalence of  $\infty$ -categories and  $A \subset B$  be an inclusion of simplicial sets. Let  $f_0 : A \rightarrow \mathcal{C}, g : B \rightarrow \mathcal{D}$  be any maps that  $h_0 : A \times \Delta^1 \rightarrow \mathcal{D}$  be an equivalence between  $g|_A$  and  $p \circ f_0$ , then there exists a map  $B \rightarrow \mathcal{C}$  and an equivalence  $h : B \times \Delta^1 \rightarrow \mathcal{D}$  between  $g$  and  $p \circ f$  that  $h_0 = h|_{A \times \Delta^1}$ .  $\lrcorner$

*Proof:* Working with simplexes, it suffices to prove for  $A = \partial\Delta^n \subset B = \Delta^n$ . The case  $n = 0$  is true because categorical equivalence is essentially surjective. For  $n > 0$ , we need to construct  $h$  from  $h|_{\Delta^n \times \{0\}} \coprod \partial\Delta^n \times \Delta^1$ , and this is a composition of pushout of  $\Lambda_k^{n+1} \subset \Delta^{n+1}$ . For  $k \neq 0$ , the extension is clear because  $\mathcal{D}$  is  $\infty$ -category, and for  $k = 0$ , we need to use [HTT, P136].  $\square$

### $\infty$ -Groupoids(or Kan Complexes/Spaces)

**Def. (4.7.1.26) [ $\infty$ -Groupoid].** An  $\infty$ -groupoid or an **anima** is an  $\infty$ -category  $\mathcal{C}$  that  $\mathrm{Ho}(\mathcal{C})$  is a groupoid(4.7.1.5), or equivalently, all morphisms are equivalences. The full subcategory of  $\mathrm{Cat}_\infty$  consisting of  $\infty$ -groupoids are denoted by  $\mathrm{Grpd}_\infty$ .  $\lrcorner$

**Prop. (4.7.1.27) [ $\infty$ -Groupoids  $\iff$  Kan Complex].** For  $\mathcal{C} \in s\mathrm{Set}$ , the following are equivalent:

- $\mathcal{C}$  is an  $\infty$ -groupoid.
- $\mathcal{C} \rightarrow \Delta^0$  is a left fibration.
- $\mathcal{C} \rightarrow \Delta^0$  is a right fibration.
- $\mathcal{C}$  is a Kan complex.

$\lrcorner$

*Proof:* 1, 2 are equivalent by(4.7.1.18), and dually 1, 3 are equivalent, and 4 = 2 + 3.  $\square$

### $n$ -Categories

**Def. (4.7.1.28) [ $n$ -Categories].** For  $\mathcal{C} \in s\mathrm{Set}$  and  $n \geq -1$ , then  $\mathcal{C}$  is called an  $n$ -category if it is an  $\infty$ -category and:

- Given any maps  $f, f' : \Delta^n \rightarrow \mathcal{C}$  that are homotopic(4.7.1.15) relative to  $\partial\Delta^n$ , then  $f = f'$ .
- For any  $m > n$  and maps  $f, f' : \Delta^m \rightarrow \mathcal{C}$  that coincide on  $\partial\Delta^m$ , then  $f = f'$ .

Also  $\mathcal{C}$  is called an  $(-2)$ -category iff it is isomorphic to  $\Delta^0$ .

The definition of an  $n$ -category is equivalent to the following: if  $f, f' : K \rightarrow \mathcal{C}$  satisfies  $f|_{sk^n K}$  is homotopic to  $f'|_{sk^n K}$  relative to  $sk^{n-1}K$ , then  $f = f'$ .  $\lrcorner$

**Cor. (4.7.1.29).** If  $\mathcal{C}$  is an  $n$ -category and  $m > n$ , then the restriction map  $\mathrm{Hom}(\Delta^m, \mathcal{C}) \rightarrow \mathrm{Hom}(\partial\Delta^m, \mathcal{C})$  is bijective.(use  $\infty$ -category property to extend).  $\lrcorner$

**Prop. (4.7.1.30).** A  $(-1)$ -category is seen to be isomorphic to  $\emptyset$  or  $\Delta^0$ . A 0-category is equivalent to the nerve of a partially ordered set.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.7.1.31)[ $\mathrm{Cat}$  and  $\mathrm{Cat}_1$ ].** For a simplicial set  $S$ , the following are equivalent:

- $u : S \rightarrow N(hS)$ (4.6.3.3) is an isomorphism of simplicial sets.
- There is a small category  $\mathcal{C}$  that  $S \cong N(\mathcal{C})$ .
- $S$  is a 1-category.

$\lrcorner$

*Proof:* It suffices to show  $3 \rightarrow 1$ : we induct on the dimension:  $n = 0$  is trivial and  $n = 1$  follows from the definition of 1-category (4.7.1.28). For  $n > 1$ , the injectivity of  $u$  follows from induction hypothesis and (4.7.1.28), and for surjectivity, for a map  $\Delta^n \rightarrow N(hS)$ , choose  $0 < i < n$  and let lift  $\Lambda_i^n$  to  $S$ , then use the fact  $S$  is an  $\infty$ -category to lift to  $\Delta^n$ , and now it coincide on  $N(hS)$  because it is a nerve of a category.  $\square$

**Prop. (4.7.1.32).** If  $\mathcal{C}$  is an  $n$ -category, then for any simplicial set  $X$ ,  $\mathcal{C}^X$  is also an  $n$ -category.  $\lrcorner$

*Proof:* This is because  $sk^p(K \times X) \subset sk^p(K) \times X$  for any simplicial set  $K$  and integer  $p$ , and use (4.7.1.28).  $\square$

**Prop. (4.7.1.33).** Let  $n \geq 1$  and  $\mathcal{C}$  an  $\infty$ -category, then  $\mathcal{C}$  is an  $n$ -category iff it satisfies the unique lifting property w.r.t. the inclusion  $\Lambda_i^m \subset \Delta^m$ , where  $0 < i < m$ .  $\lrcorner$

*Proof:* Cf.[HTT, P109].  $\square$

**Def. (4.7.1.34) [ $n$ -Truncated Kan Complexes].** Let  $X$  be a Kan complex and  $k \geq -1$ , then a Kan complex is called  **$k$ -truncated** if for every  $i > k$  and every point  $x \in X$ , we have  $\pi_i(X, x) \cong *$ . And it is called  **$(-2)$ -truncated** if it is contractible.  $\lrcorner$

**Prop. (4.7.1.35).** A  $(-1)$ -truncated Kan complex is either empty or contractible. A 0-contractible Kan complex is a Kan complex that  $X \rightarrow \pi_0(X)$  is a homotopy equivalence.  $\lrcorner$

*Proof:* ?.  $\square$

**Prop. (4.7.1.36) [Equivalent to an  $n$ -Category].** Let  $\mathcal{C}$  be an  $\infty$ -category and  $n \geq -1$ , then the following conditions are equivalent:

- There is a minimal model  $\mathcal{C}' \subset \mathcal{C}$  that is  $n$ -truncated.
- $\mathcal{C}$  is categorically equivalent to an  $n$ -truncated category.
- For any  $X, Y \in \mathcal{C}$ , the mapping space  $\text{Map}(X, Y)$  is  $(n - 1)$ -truncated.

$\lrcorner$

*Proof:* Cf.[HTT, P112].  $\square$

**Cor. (4.7.1.37).** A Kan complex is categorically equivalent to an  $n$ -category iff it is  $n$ -truncated.  $\lrcorner$

*Proof:* Cf.[HTT, P113] ?.  $\square$

**Cor. (4.7.1.38).** Let  $\mathcal{C}$  be an  $\infty$ -category and  $K$  a simplicial set, if  $\text{Map}(C, D)$  is  $n$ -truncated for any objects  $C, D \in \mathcal{C}$ , then the  $\infty$ -category  $\mathcal{C}^K$  has the same property.  $\lrcorner$

*Proof:* Cf.[HTT, P114].  $\square$

## 2 $\infty$ -Category of $\infty$ -Categories

**Def. (4.7.2.1) [Models on  $s\text{Set}^+$ ].** There is a simplicial model structure on  $s\text{Set}^+$  (4.6.3.52) s.t. the fibrant and cofibrant objects are exactly objects of the form  $\mathcal{C}^\natural$  where  $\mathcal{C}$  is an  $\infty$ -category, and  $\mathcal{C}^\natural = (\mathcal{C}, \mathcal{E}_{\mathcal{C}})$  where  $\mathcal{E}_{\mathcal{C}}$  is the set of equivalences in  $\mathcal{C}$  (4.7.1.5).  $\lrcorner$

*Proof:* ?  $\square$

**Def. (4.7.2.2) [ $\text{Cat}_\infty$ ].** The underlying  $\infty$ -category  $\text{Cat}_\infty = N_\Delta(s\text{Set}_{cf}^+)$  of  $s\text{Set}^+$  is called the **category of  $\infty$ -category of  $\infty$ -categories**. It can be verified that the fundamental category of  $\text{Cat}_\infty$  consists of equivalent classes of  $\infty$ -categories.  $\lrcorner$

**Def. (4.7.2.3) [Relative Mapping Spaces].** Let  $p : X \rightarrow S, q : Y \rightarrow S \in s\text{Set}$ , then define  $\text{Map}_S(X, Y) = \text{Map}_{\text{Cat}_\infty/T}(X, Y) \in s\text{Set}$ .  $\lrcorner$

### $(\infty, n)$ -Categories

**Def. (4.7.2.4) [  $(\infty, n)$ -Categories ].** For  $n \in \mathbb{N}$ , an  $(\infty, n)$ -category is an  $\infty$ -category s.t. for  $k > n$ , any  $k$ -maps is invertible<sup>?</sup>. In particular, an  $(\infty, 0)$ -category is just an  $\infty$ -groupoid.  $\lrcorner$

**Def. (4.7.2.5) [  $\mathcal{G}rpd_\infty$  ].** The underlying  $\infty$ -category  $N_\Delta(\mathbf{Kan})$  of  $s\mathbf{Set}$  with the Joyal model structure (4.6.4.19) is denoted by  $\mathcal{G}rpd_\infty$ . Then it is a sub- $\infty$ -category of  $\mathbf{Cat}_\infty$ , and it is an  $(\infty, 1)$ -category.  $\lrcorner$

*Proof:* <sup>?</sup>  $\square$

### 3 Limits and Colimits

**Remark (4.7.3.1) [Universal Properties].** The objects in an  $\infty$ -category characterized by a universal property will be a contractible Kan complex.  $\lrcorner$

**Def. (4.7.3.2) [Final Objects].** For  $\mathcal{C} \in \mathbf{Cat}_\infty$ ,  $x \in \mathcal{C}$  is called a **final object** if the canonical map  $\mathcal{C}_{/x} \rightarrow \mathcal{C} \in s\mathbf{Set}$  is a trivial Kan fibration. Equivalently, for any  $y \in \mathcal{C}$ ,  $\mathrm{Map}_{\mathcal{C}}^R(y, x)$  is a contractible Kan complex. It is clear that the sub- $\infty$ -category of final objects in  $\mathcal{C}$  is either empty or a contractible Kan complex.

Dually we can define initial objects.  $\lrcorner$

*Proof:* If  $x$  is a final object, then  $\mathrm{Map}_{\mathcal{C}}^R(y, x)$  is the fiber of  $\mathcal{C}_{/x} \rightarrow \mathcal{C}$  over  $x$  (4.7.1.20), so it is a contractible Kan-complex by (4.7.1.27). The converse follows from (4.6.3.46).  $\square$

**Cor. (4.7.3.3).** For  $\mathcal{C} \in \mathbf{Cat}_\infty$  with a final object  $*$ , we fix a final object, then there is a section  $\mathcal{C} \rightarrow \mathcal{C}_{/*}$ , which maps each  $C \in \mathcal{C}$  to a morphism  $C \rightarrow * \in \mathcal{C}$ . We fix such a morphism. The dual is true for initial objects.  $\lrcorner$

**Prop. (4.7.3.4).** If  $\mathcal{C} \in \mathbf{Cat}_\infty$  and  $*$   $\in \mathcal{C}$  is final, then for any diagram  $p : K \rightarrow \mathcal{C}$ ,  $*$  is also final in  $\mathcal{C}_{p/}$ .<sup>?</sup>  $\lrcorner$

**Def. (4.7.3.5) [Limits and Colimits].** For  $\mathcal{C} \in \mathbf{Cat}_\infty$  and  $p : K \rightarrow \mathcal{C} \in \mathbf{Cat}$ , a **colimit** of  $p$  is defined to be an initial object of  $\mathcal{C}_{p/}$ , and a **limit** of  $p$  is defined to be a final object of  $\mathcal{C}_{/p}$ . By (4.7.3.2), (co)limits are defined up to a contractible choice, and we may use  $(\varinjlim_p) \varprojlim_p$  to denote any one of them.  $\lrcorner$

**Def. (4.7.3.6).** For  $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}_\infty$ , let  $\mathrm{Func}^L(\mathcal{C}, \mathcal{D})$  denote the sub- $\infty$ -category of  $\mathrm{Func}(\mathcal{C}, \mathcal{D})$  consisting of functors preserving colimits, and let  $\mathrm{Func}^R(\mathcal{D}, \mathcal{C})$  denote the sub- $\infty$ -category of  $\mathrm{Func}(\mathcal{D}, \mathcal{C})$  consisting of functors preserving limits.  $\lrcorner$

**Prop. (4.7.3.7).** For  $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}_\infty$ , there is a natural equivalence of  $\infty$ -categories  $\mathrm{Func}^L(\mathcal{C}, \mathcal{D}) \cong \mathrm{Func}^R(\mathcal{D}, \mathcal{C})^{\mathrm{op}}$ .  $\lrcorner$

*Proof:* Cf. [Lur09]P356.  $\square$

### Cofinal Diagrams

**Def. (4.7.3.8) [Cofinal Maps].** A **cofinal map**  $p : S \rightarrow T \in s\mathbf{Set}$  is a map s.t. for any right fibration  $X \rightarrow T \in s\mathbf{Set}$ ,

$$\mathrm{Map}_T(T, X) \rightarrow \mathrm{Map}_T(S, X) \in s\mathbf{Set} \text{ (4.7.2.3)}$$

is a homotopy equivalence.  $\lrcorner$

**Prop. (4.7.3.9) [Cofinal Maps give Same Colimits].** For  $v : K' \rightarrow K \in s\mathbf{Set}$ , the following are equivalent:

- $v$  is cofinal.
- for any  $\mathcal{C} \in \mathbf{Cat}_\infty$  and  $p : K \rightarrow \mathcal{C} \in s\mathbf{Set}$ , the induced map  $\mathcal{C}_{p/} \rightarrow \mathcal{C}_{p \circ v/}$  is an equivalence of  $\infty$ -categories.

┘

*Proof:* Cf. [Lur09]P226. □

### Computing Diagrams

**Prop. (4.7.3.10) [Reducing to Coherent Colimits].** For any  $K \in s\mathbf{Set}$ , there exists a category  $\mathcal{J} \subset \mathbf{Cat}$  and a cofinal map  $N(\mathcal{J}) \rightarrow K \in s\mathbf{Set}$ . ┘

*Proof:* Cf. [Lur09]P255. □

**Prop. (4.7.3.11) [Colimits in Overcategories].** Let  $\mathcal{C} \in \mathbf{Cat}_\infty$  and  $q : T \rightarrow \mathcal{C} \in s\mathbf{Set}$ ,  $p : K \rightarrow \mathcal{C}_{/q}$ , and  $p_0 : K \xrightarrow{p} \mathcal{C}_{/q} \rightarrow \mathcal{C}$ . If  $p_0$  has a colimit, then

- $p$  also has a colimit, and the colimit is preserved by the  $\mathrm{pr} : \mathcal{C}_{/q} \rightarrow \mathcal{C}$ .
- $x \in \mathcal{C}_{/q}$  is a colimit of  $p$  iff  $\mathrm{pr}(x) \in \mathcal{C}$  is a colimit of  $p_0$ .

┘

*Proof:* Cf. [Lur09]P48. □

**Cor. (4.7.3.12) [Colimits in Diagram Categories].** For  $K, S \in s\mathbf{Set}$ , let  $\mathcal{C}$  be an  $\infty$ -category that admits  $K$ -indexed colimits, then

- the  $\infty$ -category  $\mathrm{Func}(S, \mathcal{C})$  (4.7.1.13) also admits  $K$ -indexed colimits.
- $p : K^\triangleright \rightarrow \mathrm{Func}(S, \mathcal{C})$  is a colimit diagram iff for each  $C \in \mathcal{C}$ , the evaluation  $p_C : K^\triangleright \rightarrow \mathcal{C}$  is a colimit diagram.

┘

*Proof:* Cf. [Lur09]P315. □

### Kan Extensions

**Def. (4.7.3.13) [ $p$ -Left Kan Extensions].** For a commutative diagram of  $\infty$ -categories

$$\begin{array}{ccc} \mathcal{C}^0 & \xrightarrow{F_0} & \mathcal{D} \\ \downarrow & \nearrow F & \downarrow p \\ \mathcal{C} & \longrightarrow & \mathcal{D}' \end{array},$$

$F$  is called a  **$p$ -left Kan extension** of  $F_0$  if for any  $C \in \mathcal{C}$ , the induced diagram

$$\begin{array}{ccc} \mathcal{C}_{/C}^0 & \xrightarrow{F_C} & \mathcal{D} \\ \downarrow & \nearrow F & \downarrow p \\ (\mathcal{C}_{/C}^0)^\triangleright & \longrightarrow & \mathcal{D}' \end{array},$$

exhibits  $C$  as a  $p$ -colimit ? of  $F_C$ . ┘

**Lemma (4.7.3.14) [Existence of  $p$ -Left Kan Extensions].** Given a commutative diagram of  $\infty$ -categories

$$\begin{array}{ccc} \mathcal{C}^0 & \xrightarrow{F_0} & \mathcal{D} \\ \downarrow & \nearrow F & \downarrow p \\ \mathcal{C} & \longrightarrow & \mathcal{D}' \end{array},$$

then there exists a  $p$ -left Kan extension  $F$  iff for any  $C \in \mathcal{C}$  the composition

$$\mathcal{C}_{/p}^0 \rightarrow \mathcal{C}^0 \xrightarrow{F_0} \mathcal{D}$$

admits a  $p$ -colimit. ┘

*Proof:* Cf. [Lur09]P282. □

**Lemma (4.7.3.15).** Let  $\mathcal{C} \rightarrow \mathcal{D}' \xleftarrow{p} \mathcal{D} \in \mathbf{Cat}_\infty$  where  $p$  is a Joyal-fibration, and  $\mathcal{C}^0 \subset \mathcal{C}$  a full subcategory, the the restriction functor

$$i^* : \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}, \mathcal{D}) \rightarrow \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}^0, \mathcal{D})$$

has right lifting property w.r.t. all  $\partial\Delta^n \rightarrow \Delta^n$  s.t. the map  $\partial\Delta^n$  maps  $\{0\}$  to a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  which is a  $p$ -left Kan extension (4.7.3.13) of  $F|_{\mathcal{C}^0}$ , ┘

*Proof:* Cf. [Lur09]P279. □

**Prop. (4.7.3.16).** Let  $\mathcal{C} \rightarrow \mathcal{D}' \xleftarrow{p} \mathcal{D} \in \mathbf{Cat}_\infty$  where  $p$  is a Joyal-fibration, and  $\mathcal{C}^0 \subset \mathcal{C}$  a full subcategory. Let

- $\mathcal{K} \subset \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}, \mathcal{D})$  be the full subcategory of functors  $F : \mathcal{C} \rightarrow \mathcal{D}$  which are  $p$ -left Kan extensions (4.7.3.13) of  $F|_{\mathcal{C}^0}$ ,
- $\mathcal{K}' \subset \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}^0, \mathcal{D})$  be the category of functors  $F_0 : \mathcal{C}^0 \rightarrow \mathcal{D}$  s.t. for any  $C \in \mathcal{C}$ , the induced functor  $\mathcal{C}_{/C}^0 \rightarrow \mathcal{D}$  has a  $p$ -colimit.

Then the restriction functor  $i^* : \mathcal{K} \rightarrow \mathcal{K}'$  is a trivial Kan fibration. ┘

*Proof:* This follows immediately from (4.7.3.14) and (4.7.3.15). □

**Cor. (4.7.3.17) [Functorial Left Kan Extensions].** Let  $\mathcal{C} \rightarrow \mathcal{D}' \xleftarrow{p} \mathcal{D} \in \mathbf{Cat}_\infty$  where  $p$  is a Joyal-fibration, and  $\mathcal{C}^0 \subset \mathcal{C}$  a full subcategory. Suppose that every functor  $F_0 : \mathcal{C}^0 \rightarrow \mathcal{D}$  over  $\mathcal{D}'$  admits a  $p$ -left Kan extension (4.7.3.13), then the restriction

$$i^* : \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}, \mathcal{D}) \rightarrow \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}^0, \mathcal{D})$$

admits a section  $i_!$  whose essential image are exactly those  $F$  which are  $p$ -left Kan extension of  $F|_{\mathcal{C}^0}$ . Any such  $i_!$  is called the **left Kan extension functor**. Moreover, this  $i_!$  is left adjoint to  $i^*$ :

$$i^* : \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}, \mathcal{D}) \rightleftarrows \mathrm{Func}_{\mathcal{D}'}(\mathcal{C}^0, \mathcal{D}) : i_!$$

┘

*Proof:* Cf. [Lur09]P284. □

## 4 Adjunctions

### Correspondences

**Def. (4.7.4.1) [Correspondence].** Let  $\mathcal{C}, \mathcal{D}$  be  $\infty$ -categories, an **correspondence** between  $\mathcal{C}, \mathcal{D}$  is defined to be an  $\infty$ -category  $\mathcal{M}$  and a map  $\mathcal{M} \rightarrow \Delta^1$  s.t.  $\mathcal{C} \cong \mathcal{M}_0$  and  $\mathcal{D} \cong \mathcal{M}_1$ . ┘

### Adjunctions

**Def. (4.7.4.2) [Adjunctions].** An **adjunction** between  $\infty$ -categories  $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}_\infty$  is a map  $q : \mathcal{M} \rightarrow \Delta^1 \in s\mathbf{Set}$  that is both Cartesian fibration and coCartesian fibration, together with equivalences  $\mathcal{M}_{\{0\}} \cong \mathcal{C}, \mathcal{M}_{\{1\}} \cong \mathcal{D}$ .

If  $\mathcal{M} \rightarrow \Delta^1$  is an adjunction and  $f : \mathcal{C} \rightarrow \mathcal{D}, g : \mathcal{D} \rightarrow \mathcal{C}$  are the associated functors, then  $f$  is said to be **left adjoint** to  $g$  and  $g$  is said to be **right adjoint** to  $f$ .  $\lrcorner$

**Def. (4.7.4.3) [Unit Transformations].** For a pair of functors  $f : \mathcal{C} \rightarrow \mathcal{D}, g : \mathcal{D} \rightarrow \mathcal{C}$  between  $\infty$ -categories, a **unit transformation** for  $(f, g)$  is a morphism  $u : \mathrm{id} \rightarrow g \circ f \in \mathrm{Func}(\mathcal{C}, \mathcal{C})$  s.t. for any  $C \in \mathcal{C}, D \in \mathcal{D}$ , the composition

$$\mathrm{Map}_{\mathcal{D}}(f(C), D) \xrightarrow{g} \mathrm{Map}_{\mathcal{C}}(gf(C), g(D)) \xrightarrow{u(C)} \mathrm{Map}_{\mathcal{C}}(C, g(D))$$

is a homotopy equivalence.  $\lrcorner$

**Prop. (4.7.4.4) [Unit Transformations and Adjunctions].** For a pair of functors  $f : \mathcal{C} \rightarrow \mathcal{D}, g : \mathcal{D} \rightarrow \mathcal{C}$ ,  $f$  is left adjoint to  $g$  iff there is a unit transformation  $u : \mathrm{id} \rightarrow g \circ f$ .  $\lrcorner$

*Proof:* Cf. [Lur09]P339.  $\square$

**Prop. (4.7.4.5) [Adjunction and Limits].** Left adjoints between  $\infty$ -categories preserve small colimits and right adjoints between  $\infty$ -categories preserves small limits.  $\lrcorner$

*Proof:* Cf. [Lur09]P345.  $\square$

### Localizations

**Def. (4.7.4.6) [Localization of  $\infty$ -Categories].** A functor  $\mathcal{C} \rightarrow \mathcal{D} \in \mathbf{Cat}_\infty$  is called a **localization of  $\infty$ -category** if it admits a fully faithful right adjoint.  $\lrcorner$

## 5 Presentable $\infty$ -Categories

### $\infty$ -Categories of Presheaves

**Def. (4.7.5.1) [ $\infty$ -Category of Presheaves].** For  $S \in s\mathbf{Set}$ , there exists an  $\infty$ -category  $\mathcal{P}\mathrm{Sh}_\infty^{\mathrm{Set}}(S) = \mathrm{Func}(S^{\mathrm{op}}, \mathcal{S}) \subset \mathbf{Cat}_\infty$  by (4.7.1.13), called the  **$\infty$ -category of presheaves on  $S$** . More generally, for any  $\mathcal{C} \in \mathbf{Cat}_\infty$ ,  $\mathcal{P}\mathrm{Sh}_\infty^{\mathrm{Set}}(S; \mathcal{C}) = \mathrm{Func}(S^{\mathrm{op}}, \mathcal{C})$  is called the  **$\infty$ -category of  $\mathcal{C}$ -valued presheaves on  $S$** .  $\lrcorner$

**Prop. (4.7.5.2) [Cocompleteness].** For  $S \in s\mathbf{Set}$  and  $\mathcal{C} \in \mathbf{Cat}_\infty$  that is cocomplete, then  $\mathcal{P}\mathrm{Sh}(S; \mathcal{C})$  is also cocomplete by (4.7.3.12). In particular,  $\mathcal{P}\mathrm{Sh}_\infty^{\mathrm{Set}}(S)$  is cocomplete.  $\lrcorner$

**Def. (4.7.5.3) [Yoneda Embedding].** For  $K \in s\mathbf{Set}$ , let  $\mathcal{C} = \mathfrak{C}[K]$ , then the mapping space construction defines a map

$$\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathbf{Kan} : (X, Y) \mapsto \mathrm{Sing} | \mathrm{Map}_{\mathcal{C}}(X, Y) |$$

by (4.6.3.43). And there is a natural map  $\mathfrak{C}[K^{\mathrm{op}} \times K] \rightarrow \mathcal{C}^{\mathrm{op}} \times \mathcal{C}$ , so we have a map

$$\mathfrak{C}[K^{\mathrm{op}} \times K] \rightarrow \mathbf{Kan}$$

which corresponds to a map

$$K^{\mathrm{op}} \times K \rightarrow N_\Delta(\mathbf{Kan}) = \mathbf{Grpd}_\infty.$$

and induces a map

$$\jmath_{\Delta} : K \mapsto \text{Func}(K^{\text{op}}, \text{Grpd}_{\infty}) = \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(K),$$

When  $\mathcal{C} \in \text{Cat}$  and  $K = N(\mathcal{C})$  this is compatible with the usual Yoneda embedding composed with simplicial enhancement, so it is called the **Yoneda embedding**.  $\lrcorner$

**Prop. (4.7.5.4) [ $\infty$ -Categorical Yoneda Lemma].** For  $K \in s\text{Set}$ , the Yoneda embedding (4.7.5.3)  $\jmath : K \rightarrow \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(K)$  is fully faithful.  $\lrcorner$

*Proof:* Cf. [Lur09]P316.  $\square$

**Prop. (4.7.5.5).** let  $\mathcal{C} \in \text{Cat}_{\infty}$ , then  $\jmath_{\Delta} : \mathcal{C} \rightarrow \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(\mathcal{C})$  preserves all small limits.  $\lrcorner$

*Proof:* Cf. [Lur09]P316.  $\square$

**Lemma (4.7.5.6).** Let  $S \in s\text{Set}$ , and  $\mathcal{C} \in \text{Cat}_{\infty}$ , then

- Any functor  $F : \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S) \rightarrow \mathcal{C}$  is a left Kan extension of  $f|_S$  iff  $f$  preserves all small colimits.
- If  $\mathcal{C}$  is cocomplete, then any functor  $S \rightarrow \mathcal{C}$  has a left Kan extension  $\mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S) \rightarrow \mathcal{C}$ .

$\lrcorner$

*Proof:* Cf. [Lur09]P322.  $\square$

**Thm. (4.7.5.7) [ $\infty$ -Categorical Yoneda Extensions].** Let  $S \in s\text{Set}$  and  $\mathcal{C}$  be a cocomplete  $\infty$ -category, the Yoneda embedding  $\jmath_{\Delta} : S \rightarrow \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S)$  induces an equivalence of  $\infty$ -categories

$$\text{Func}^L(\mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S), \mathcal{C}) \cong \text{Func}(S, \mathcal{C}).$$

$\lrcorner$

*Proof:* This follows from (4.7.3.17) and (4.7.5.6).  $\square$

**Cor. (4.7.5.8).** For  $S \in s\text{Set}$ ,  $\mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S)$  is generated by  $S$  under small colimits. In particular,  $\text{Grpd}_{\infty}$  is generated by  $\Delta^0$  under small colimits.  $\lrcorner$

*Proof:* If  $\mathcal{C} \subset \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S)$  is a strictly full subcategory stable under small colimits containing  $\jmath_{\Delta}(S)$ , then  $\mathcal{C}$  is cocomplete by (4.7.5.2), and then  $\jmath_{\Delta} : S \rightarrow \mathcal{C}$  is of the form  $F \circ \jmath_{\Delta}$  by (4.7.5.7) where  $F : \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S) \rightarrow \mathcal{C}$  preserves small colimits. Then we can regard  $F$  as a self-map of  $\mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S)$  that is identity on  $\text{id} : S \rightarrow S$ . Thus by (4.7.5.7) again,  $F$  is equivalent to  $\text{id}_S$ . Thus every object of  $\mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S)$  is equivalent some elements in  $\mathcal{C}$ , so  $\mathcal{C} = \mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(S)$ .  $\square$

### Accessible $\infty$ -Categories

### Presentable $\infty$ -Categories

**Def. (4.7.5.9) [Presentable  $\infty$ -Category].** An  $\infty$ -category is called a **presentable  $\infty$ -category** if it is accessible and cocomplete.  $\lrcorner$

**Prop. (4.7.5.10) [Simpson].** For  $\mathcal{C} \in \text{Cat}_{\infty}$  is presentable iff it is an accessible reflective localization **?** of an  $\mathcal{P}\text{Sh}_{\infty}^{\text{Set}}(\mathcal{D})$  for some  $\mathcal{D} \in \text{Cat}_{\infty}$ .  $\lrcorner$

*Proof:* Cf. [HTT, P5.5.1.1].  $\square$

**Prop. (4.7.5.11).** A presentable  $\infty$ -category is complete.  $\lrcorner$



*Proof:* Cf.[HTT, P5.5.2.4]. □

**Prop. (4.7.5.12) [Presentable  $\infty$ -Categories as Simplicial Model Categories].**  $\mathcal{C} \in \mathcal{C}at_\infty$  is presentable iff it is equivalent to the underlying category  $N_\Delta(\mathcal{M}_{cf})$  for some combinatorial simplicial model category  $\mathcal{M}$ . ┘

*Proof:* □

**Thm. (4.7.5.13) [Adjoint Functor Theorem].** Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor between presentable  $\infty$ -categories, then

- $F$  admits a right adjoint iff it preserves small colimits.
  - $F$  admits a left adjoint iff it preserves small limits and  $\kappa$ -filtered colimits for some regular cardinal  $\kappa$ .
- ┘

*Proof:* [HTT. 5.5.2.9]. □

### Compactly Generated $\infty$ -Categories

**Def. (4.7.5.14).** ┘

## 6 $\infty$ -Topoi

## 7 Topological Cyclic Homology(Scholze)

## 4.8 Abelian Categories and Triangulated Categories

References are [Sta], [Ver96] and [G-M03].

### 1 Additive Categories

**Def. (4.8.1.1) [Preadditive Categories].**  $\mathcal{A} \in \mathbf{Cat}$  is called a **preadditive category** if:

- **A1:**  $\mathcal{A}$  is enriched over  $(\mathbf{Ab}, \oplus)$ .

┘

**Def. (4.8.1.2) [Zero Element].** Let  $\mathcal{A}$  be a preadditive category and  $x \in \mathcal{A}$ , then the following are equivalent:

- $x$  is initial.
- $x$  is final.
- $\mathrm{id}_x = 0 \in \mathrm{Mor}(x, x)$ .

Such an element is called a **zero element** in  $\mathcal{A}$ , denoted by  $0$ . If  $0$  exists, then a morphism  $\alpha : x \rightarrow y$  factors through  $0$  iff  $\alpha = 0$ .

┘

*Proof:* Cf. [Sta]00ZZ.

□

**Cor. (4.8.1.3).** An additive functor transforms a zero object to a zero object.

┘

**Prop. (4.8.1.4) [Finite Direct Sums].** If  $\mathcal{A}$  is a preadditive category and  $x, y \in \mathcal{A}$ . If one of  $x \times y$  and  $x \coprod y$  exists, then so does the other, and they are isomorphic, called the **direct sum** of  $x, y$ , denoted by  $x \oplus y$ .

┘

*Proof:* Cf. [Sta]0101.

□

**Def. (4.8.1.5) [Additive Functors].** A functor between preadditive categories  $F : \mathcal{C} \rightarrow \mathcal{D}$  is called an **additive functor** if it is a morphism of  $\mathbf{Ab}$ -enriched categories.

┘

**Def. (4.8.1.6) [Left and Right Orthogonal Subcategories].** Let  $\mathcal{B} \subset \mathcal{A}$  be a preadditive subcategory, then the right **orthogonal subcategory** of  $\mathcal{B}$  in  $\mathcal{A}$  is defined to be the full subcategory  $\mathcal{B}^\perp \subset \mathcal{A}$  consisting of objects  $C \in \mathcal{A}$  s.t.  $\mathrm{Hom}_{\mathcal{A}}(B, C) = 0$  for any  $B \in \mathcal{B}$ . Similarly we can define **left orthogonal subcategories**  ${}^\perp\mathcal{B}$ .

┘

### Additive Categories

**Def. (4.8.1.7) [Additive Categories].** A preadditive category  $\mathcal{A}$  (4.8.1.1) is called an **additive category** iff moreover it satisfies

- **A2:** There exists a zero element (4.8.1.2).
- **A3:**  $\mathcal{A}$  admits finite sums and finite products (so they are equal by (4.8.1.4)). Also, the sum induce the Abelian structure of  $\mathrm{Hom}(X, Y)$ .

We use the symbol  $\mathcal{A} \in \mathbf{AddCat}$  to mean that  $\mathcal{A}$  is an additive category.

┘

**Prop. (4.8.1.8) [Characterizing Direct Sum Decompositions].** If  $\mathcal{A}$  is a preadditive category with zero object,  $x, y, z \in \mathcal{A}$ , then  $z$  is the product and sum of  $x, y$  in  $\mathcal{A}$  iff there are four morphism that satisfies some identities.

┘

*Proof:* Cf. [Sta]0102.

□

**Cor. (4.8.1.9).** An additive functor between additive categories transforms finite direct sums to direct sums.  $\lrcorner$

**Def. (4.8.1.10) [Kernels, Cokernels, Images and Coimages].** Let  $\mathcal{A}$  be a preadditive category and  $f : X \rightarrow Y$  is a map, then

- the **kernel** of  $f$  is the fiber product
 
$$\begin{array}{ccc} \ker(f) & \longrightarrow & X \\ \downarrow & & \downarrow f \\ 0 & \longrightarrow & Y \end{array}$$
- the **cokernel** of  $f$  is the fiber pushforward
 
$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Coker}(f) \end{array}.$$
- the **image** of  $f$  is the cokernel of the kernel.
- the **coimage** of  $f$  is the kernel of the cokernel.

$\lrcorner$

**Cor. (4.8.1.11).** If they exist, then kernel and coimage are monomorphisms, and cokernel and image are epimorphisms, by general non-sense.  $\lrcorner$

**Def. (4.8.1.12) [Strict Morphisms].** If the image and coimage of  $f$  exist, then there is a natural decomposition  $f : X \rightarrow \text{Im}(f) \rightarrow \text{Coim}(f) \rightarrow Y$ , by the universal property. Then  $f$  is called a **strict morphism** if  $\text{Im}(f) \rightarrow \text{Coim}(f)$  is an isomorphism.  $\lrcorner$

**Def. (4.8.1.13) [Exact Sequence].** A diagram  $0 \rightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \rightarrow 0$  is called an **exact sequence** if  $f$  is the kernel of  $g$  and  $g$  is the cokernel of  $f$ .

Clearly for such an exact sequence,  $f$  is a strict monomorphism and  $g$  is a strict epimorphism. Conversely, any strict monomorphism of strict epimorphism can be extended to an exact sequence.  $\lrcorner$

**Def. (4.8.1.14) [Kernel of an Additive Functor].** The **kernel of an additive functor** (4.8.1.5) between additive categories is the full subcategory of objects that are mapped to 0.  $\lrcorner$

**Def. (4.8.1.15) [Compact Objects].** Let  $\mathcal{D}$  be an additive category with arbitrary direct sums, then a **compact object**  $K \in \mathcal{D}$  is an object that

$$\bigoplus_i \text{Hom}(K, E_i) \rightarrow \text{Hom}(K, \bigoplus_i E_i)$$

is bijective for any set  $I$  and objects  $E_i \in \mathcal{D}$ .  $\lrcorner$

### Karoubian Categories

**Def. (4.8.1.16) [Karoubian Categories].** A **Karoubian category** is an additive category (4.8.1.7)  $\mathcal{A}$  that satisfies the following equivalent conditions:

- Every idempotented endomorphism of an object of  $\mathcal{A}$  has a kernel.
- Every idempotented endomorphism of an object of  $\mathcal{A}$  has a cokernel.
- Every idempotented endomorphism  $p : z \rightarrow z$  induces a direct sum decomposition  $z = x \oplus y$  (exists by A3) that  $p$  corresponds to the projection  $z \rightarrow x$ .

┘

*Proof:*  $1 \rightarrow 3$ : Let  $p : z \rightarrow z$  be an idempotent, let  $x = \ker(p)$ ,  $y = \ker(1 - p)$ , then there are maps  $x \rightarrow z$ ,  $y \rightarrow z$ . Then  $p : z \rightarrow z$  factors through  $z \rightarrow y \rightarrow z$ . Similarly  $(1 - p) : z \rightarrow z$  factors through  $z \rightarrow x \rightarrow z$ . Then it can be verified that  $z = x \oplus y$  is a direct sum decomposition (4.8.1.8).

$2 \rightarrow 3$  is dual.

$3 \rightarrow 1$ ,  $3 \rightarrow 2$  are easy by (4.8.1.8). □

**Prop. (4.8.1.17).** Let  $\mathcal{D}$  be a preadditive category,

- if  $\mathcal{A}$  has countable products and kernels of morphisms that have a right inverse, then  $\mathcal{A}$  is Karoubian.
- Dually, if  $\mathcal{A}$  has countable coproducts and cokernels of morphisms that have a left inverse, then  $\mathcal{A}$  is Karoubian.

┘

*Proof:* Given any idempotent morphism  $e : X \rightarrow X$ ,  $e$  has a kernel iff  $W \mapsto \ker(\text{Mor}(M, X) \xrightarrow{e} \text{Mor}(M, X))$  is representable. Notice that for any Abelian group  $A$ ,

$$\ker(e : A \rightarrow A) = \ker(\Phi : \prod_{\mathbb{Z}} A \rightarrow \prod_{\mathbb{Z}} A)$$

where

$$\Phi(a_1, a_2, \dots) = (ea_1 + (1 - e)a_2, ea_2 + (1 - e)a_3, \dots)$$

and it has a right inverse

$$\Psi(a_1, a_2, \dots) = (a_1, (1 - e)a_1 + ea_2, (1 - e)a_2 + ea_3, \dots)$$

thus the kernel exists. □

### Coherent Functors

**Def. (4.8.1.18) [Coherent Functors].** For  $\mathcal{A} \in \text{AddCat}$ , a **coherent functor** on  $\mathcal{A}$  is a functor  $F \in \text{PSh}(\mathcal{A})$  s.t. there exists a morphism  $s \rightarrow t \in \mathcal{A}$  and  $F$  is the cokernel of the map  $\mathcal{A}(s) \rightarrow \mathcal{A}(t)$ .

Then the class of coherent functors on  $\mathcal{A}$  form an additive category with cokernels, denoted by  $\text{Coh}(\mathcal{A})$ . ┘

*Proof:* Cf. [Neeman, Triangulated Categories]. □

**Prop. (4.8.1.19).** Any additive functor between additive categories  $f : \mathcal{A} \rightarrow \mathcal{B}$  extends via Yoneda map  $\mathcal{A} \rightarrow \text{Coh}(\mathcal{A})$  to a right exact functor  $f^* : \text{Coh}(\mathcal{A}) \rightarrow \text{Coh}(\mathcal{B})$  (4.8.1.18). ┘

## 2 Exact Categories

Main references are [Exact categories, Theo].

**Def. (4.8.2.1) [Exact Categories].** Let  $\mathcal{C}$  be an small additive category and  $\mathcal{E}$  be a set of short sequences

$$0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$$

in  $\mathcal{C}$ . If  $0 \rightarrow X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z \rightarrow 0$  is in  $\mathcal{E}$ , we call  $\varphi$  an **admissible monomorphism** and  $\psi$  an **admissible epimorphism**. Then  $(\mathcal{C}, \mathcal{E})$  is called an **exact category** if it satisfies:

- **Ex1:** For  $X \in \mathcal{C}$ ,  $\text{id}_X$  is both an admissible monomorphism and an admissible epimorphism.
- **Ex2:** For any triangle  $0 \rightarrow X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z \rightarrow 0$  in  $\mathcal{E}$ ,  $\varphi$  is the kernel of  $\psi$  and  $\psi$  is the cokernel of  $\varphi$ .
- **Ex3:** If  $f, g$  are admissible monomorphisms, then so is  $gf$ . If  $f, g$  are admissible epimorphisms, then so is  $gf$ .
- **Ex4:** Admissible monomorphisms are stable under pushouts and admissible epimorphisms are stable under pullbacks.

┘

**Cor. (4.8.2.2).** If  $(\mathcal{C}, \mathcal{E})$  is an exact category and  $X, Y \in \mathcal{C}$ , then  $0 \rightarrow X \rightarrow X \oplus Y \rightarrow Y \rightarrow 0$  is in  $\mathcal{E}$ . ┘

*Proof:* The first is a pushout of  $0 \rightarrow A$ , and the second is a pullback of  $B \rightarrow 0$ . ┐

**Cor. (4.8.2.3).** If  $\mathcal{C} \in \mathcal{Ab}$  and  $\mathcal{E}$  the set of exact sequences in  $\mathcal{C}$ , then  $(\mathcal{C}, \mathcal{E})$  is an exact category. And if  $\mathcal{E}'$  the set of split exact sequences in  $\mathcal{C}$ , then  $(\mathcal{C}, \mathcal{E}')$  is also an exact category. ┘

**Thm. (4.8.2.4) [Obscure Axiom].** If  $(\mathcal{C}, \mathcal{E})$  is an exact category, and if  $f : X \rightarrow Y \in \mathcal{C}$  is a morphism having a kernel and there is a morphism  $g : Z \rightarrow X$  that  $fg$  is an admissible epimorphism, then so is  $f$ . Dual assertion holds for admissible monomorphisms. ┘

*Proof:* ? Cf. [Keller, Chain complexes and stable categories, P28], [Exact Categories, Theo]P10. ┐

**Cor. (4.8.2.5).** If  $(i, p), (i', p')$  are triangles, then  $(i \oplus i', p \oplus p')$  is exact iff  $(i, p), (i', p')$  are both exact. ┘

*Proof:* If  $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0, 0 \rightarrow B' \rightarrow B \rightarrow B'' \rightarrow 0$  are both exact, then notice for any  $C \in \mathcal{C}$ ,  $0 \rightarrow A' \oplus C \rightarrow A \oplus C \rightarrow A'' \rightarrow 0$  is exact ( $A \oplus C \rightarrow A''$  is a composition of  $A \oplus C \rightarrow A'' \oplus C$  and  $A'' \oplus C \rightarrow A''$ ). Hence by composition,  $A' \oplus B' \rightarrow A \oplus B$  is an admissible monomorphism. So  $0 \rightarrow A' \oplus B' \rightarrow A \oplus B \rightarrow A'' \oplus B'' \rightarrow 0$  is exact because it is clearly a kernel-cokernel triangle.

For the converse, it is clear that  $(i, p), (i', p')$  are kernel-cokernel pairs. And notice  $\begin{bmatrix} 1 \\ 0 \end{bmatrix} i = \text{diag}(i, i') \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  is an admissible monomorphism, so  $i$  is an admissible monomorphism by the obscure axiom. ┐

**Prop. (4.8.2.6).** The pullback of an admissible monomorphism along an admissible epimorphism is an admissible monomorphism. Dual assertion holds for admissible epimorphisms. ┘

*Proof:* Let  $A \rightarrow B$  be an admissible monomorphism with kernel  $C$ , and  $B' \rightarrow B$  be an admissible epimorphism, then the pullback  $A'$  exists by the axiom. Then we can prove that  $A' \rightarrow B'$  is the kernel of the admissible epimorphism  $B' \rightarrow B \rightarrow C$ , thus an admissible epimorphism. ┐

### Quasi-Abelian Categories

**Def. (4.8.2.7) [Quasi-Abelian Categories].** A **quasi-Abelian category** is an additive category (4.8.1.7) s.t. strict epimorphisms are stable under pullbacks and strict monomorphisms are stable under pushforwards. ┘

**Prop. (4.8.2.8).** A quasi-Abelian category is naturally an exact category. ┘

*Proof:* It suffices to show that strict monomorphisms and strict epimorphisms are stable under compositions. Suppose  $u : E \rightarrow F, v : F \rightarrow G$  are strict epimorphisms, and  $w = v \circ u$ , then there is a commutative diagram

$$\begin{array}{ccccc}
 & & \ker(w) & \longrightarrow & \ker(v) \\
 & \nearrow & \downarrow i_w & & \downarrow i_v \\
 \ker(u) & \xrightarrow{i_u} & E & \xrightarrow{u} & F \\
 & & \searrow w & & \downarrow v \\
 & & & & G
 \end{array} ,$$

and the upper-right square is a pullback square. Then it suffices to show that  $w$  is the cokernel of  $i_w$ , which is easy.

The argument for strict monomorphisms are dual.  $\square$

**Prop.(4.8.2.9) [Characterizing Quasi-Abelian Categories].** An additive category  $\mathcal{A}$  is quasi-Abelian iff there are Abelian categories  $\mathcal{A}^\sharp$  and  $\mathcal{A}^\flat$  and fully faithful additive embeddings  $\mathcal{A} \subset \mathcal{A}^\sharp, \mathcal{A} \subset \mathcal{A}^\flat$  s.t.

- If  $A \rightarrow E$  is a monomorphism in  $\mathcal{A}^\sharp$  and  $E \in \mathcal{A}$ , then  $A \in \mathcal{A}$  too.
- If  $E \rightarrow B$  is an epimorphism in  $\mathcal{A}^\flat$  and  $E \in \mathcal{A}$ , then  $B \in \mathcal{A}$  too.

If this is the case, then the short exact sequences in  $\mathcal{A}$  are exactly those which are exact in  $\mathcal{A}^\sharp$  and/or  $\mathcal{A}^\flat$ .  $\lrcorner$

*Proof:* Cf.[Schneider]Lemma1.2.34.  $\square$

**Example(4.8.2.10).** The category of torsion-free sheaves on a smooth projective variety is a quasi-Abelian category.  $\lrcorner$

*Proof:*  $\square$

### Geometric Exact Categories

**Def.(4.8.2.11)[Geometric Exact Categories].** A **geometric exact category** consists of an exact category  $(\mathcal{C}, \mathcal{E})$  and a mapping  $A$  from  $\text{Ob}(\mathcal{C})$  to  $\text{Set}$  together with morphisms:

- a morphism  $f^* : A(X) \rightarrow A(Y)$  for any admissible monomorphism  $f : X \rightarrow Y$ .
- a morphism  $g_* : A(X) \rightarrow A(Z)$  for any admissible epimorphism  $g : X \rightarrow Z$ .

that satisfies the following axioms:

- **A1:**  $A(0) = \text{pt}$ .
- **A2:** If  $i, j$  are admissible monomorphisms, then  $(ji)^* = i^*j^*$ .
- **A3:** If  $p, q$  are admissible epimorphisms, then  $(qp)_* = q_*p_*$ .
- **A4:**  $\text{id}_X^* = (\text{id}_X)_* = \text{id}_{A(X)}$ .
- **A5:** If  $f : X \rightarrow Y$  is an isomorphism, then  $f^*f_* = f_*f^* = \text{id}$ .

- **A6:** For any Cartesian and Cocartesian diagram
 
$$\begin{array}{ccc}
 X & \xrightarrow{u} & Y \\
 \downarrow p & & \downarrow q \\
 Z & \xrightarrow{v} & W
 \end{array}$$
 , if  $u$  is admissible monomorphism

or  $q$  is admissible epimorphism, then  $v^*q_* = p_*u^*$ .

- **A7:** If  $X \xrightarrow{u} Y \xrightarrow{v} Z$  is a diagram in  $\mathcal{C}$  that  $u$  is an admissible epimorphism and  $v$  is an admissible monomorphism, and if  $h_X \in A(X), h_Z \in A(Z)$  satisfy  $u_*(h_X) = v^*(h_Z)$ , then there exists  $h \in A(X \oplus Z)$  that  $(\text{id}, vu)^*(h) = h_X$  and  $\pi_{2*}(h) = h_Z$ . (Notice  $(\text{id}, vu)$  is an admissible monomorphism because it is composition of  $X \rightarrow X \oplus Z$  with the isomorphism  $(\pi_1, vu\pi_1 + \pi_2) : X \oplus Z \rightarrow X \oplus Z$ ).

For any  $X \in \mathcal{C}$ , an element of  $A(X)$  is called a **geometric structure** on  $X$ .

Exact categories can be viewed as geometric exact categories by asserting  $A(X) = \text{pt}$  for all  $X \in \mathcal{C}$ .  $\lrcorner$

**Def.(4.8.2.12) [Morphisms compatible with the Geometric Structure].** Let  $(\mathcal{C}, \mathcal{E}, A)$  be a geometric exact category, if  $(X', h'), (X'', h'')$  are two geometric objects, then a morphism  $f : X' \rightarrow X''$  is said to be a **morphism compatible with the geometric structure** if there exists a geometric object  $(X, h)$  and an admissible monomorphisms  $u : X' \rightarrow X$  and an admissible epimorphism  $v : X \rightarrow X''$  that  $h' = u^*(h)$  and  $h'' = v_*(h)$  and  $f = vu$ .

The composition of two morphisms compatible with the geometric structure is also a morphism compatible with the geometric structure.

We denote  $\mathcal{C}_A$  the category of geometric objects of  $\mathcal{C}$ , and  $\mathcal{E}_A$  the set of diagrams of geometric objects  $0 \rightarrow X' \xrightarrow{u} X \xrightarrow{v} X'' \rightarrow 0$  that the underlying diagram is in  $\mathcal{E}$  and  $u, v$  are compatible with the geometric structures.  $\lrcorner$

**Prop.(4.8.2.13) [Hermitian Spaces].** The f.d. Hermitian spaces over  $\mathbb{C}$  or f.d. normed vector spaces over  $\mathbb{R}$  form a geometric exact category.  $\lrcorner$

*Proof:* Cf.[Harder-Narasimhan Categories, P4].  $\square$

**Prop.(4.8.2.14) [F.D. Ultranormed Banach Spaces].** Let  $K$  be a complete valued field, then the category of f.d. ultranormed Banach spaces over  $K$  (19.1.6.5) is a geometric exact category.  $\lrcorner$

*Proof:* It suffices to check axiom A7, but  $vu$  has  $\text{norm} \leq 1$ , and for any  $\varphi : E \rightarrow F$  of  $\text{norm} \leq 1$ , we can endow  $E \oplus F$  with the maximum norm, then in the decomposition  $E \xrightarrow{(\text{id}, \varphi)} E \oplus F \xrightarrow{\pi_2} F$ , we have  $(\text{id}, \varphi)^*(h) = h_E$  and  $(\pi_2)_*(h) = h_F$ .  $\square$

**Prop.(4.8.2.15) [Filtrations in an Abelian Category].** The filtrations (4.3.3.1) in an Abelian category form a geometric exact category.  $\lrcorner$

*Proof:* Cf.[Harder-Narasimhan Categories, P5].  $\square$

### 3 Abelian Categories

**Def.(4.8.3.1) [Abelian Categories].** An **Abelian category** is a preadditive category  $\mathcal{A} \in \text{Cat}(4.8.2.7)$  s.t. every morphism is strict (4.8.1.12). We use the symbol  $\mathcal{A} \in \text{AbCat}$  to mean that  $\mathcal{A}$  is an Abelian category.

We use the notation  $\mathcal{A} \in \text{AbCat}$  to denote that  $\mathcal{A}$  is an Abelian category.  $\lrcorner$

**Remark(4.8.3.2).** WARNING: An additive category that epimorphism+monomorphism is isomorphism need not be an Abelian category. Cf.[<https://mathoverflow.net/questions/41722/is-every-balanced-pre-abelian-category-abelian>] for a counter-example.  $\lrcorner$

**Prop.(4.8.3.3).** For  $\mathcal{A} \in \text{AbCat}$ , the functors

$$\mathcal{C} \rightarrow \mathcal{A}b : Y \mapsto \text{Hom}(X, Y),$$

and

$$\mathcal{C}^{\text{opp}} \rightarrow \mathcal{A}b : Y \mapsto \text{Hom}(Y, X)$$

are both left exact. (Note that left and right is seen on the image).  $\lrcorner$

**Def. (4.8.3.4) [Injectives and Surjectives].** For  $\mathcal{A} \in \mathcal{A}b\mathcal{C}at$ , a morphism  $f : X \rightarrow Y \in \mathcal{A}$  is called an **injection** if  $\ker f = 0$ , in which case  $X$  is called a **subobject** of  $Y$ . And  $f$  is called a **surjection** if  $\text{Coker } f = 0$ , in which case  $Y$  is called a **quotient** of  $X$ .

$f$  is an injection iff it is a monomorphism, it is a surjection iff it is an epimorphism.  $\lrcorner$

**Def. (4.8.3.5) [Simple Objects].** For  $\mathcal{A} \in \mathcal{A}b\mathcal{C}at$ , an object  $x \in \text{Ob}(\mathcal{A})$  is called a **simple object** if  $x \neq 0$  and the only subobjects of  $x$  are  $0$  and  $x$  (4.8.3.4).  $\lrcorner$

**Prop. (4.8.3.6).** Axiom A3 asserts the good existence of product and sum of objects as we wanted, and it can be used to prove that monomorphism and epimorphism are stable under pushout and pullback.

But this uses A4 strongly, Cf. [MacLane Categories for working mathematicians P203]. (For epimorphism, first prove  $0 \rightarrow X \times_U Y \rightarrow X \times Y \rightarrow U \rightarrow 0$  is exact when  $X \rightarrow U$  is epi).  $\lrcorner$

**Prop. (4.8.3.7).** equalizer and finite product derives finite limit, thus finite limits and finite colimits exists in Abelian categories.  $\lrcorner$

**Prop. (4.8.3.8) [Mitchell's embedding theorem].** If  $\mathcal{A} \in \mathcal{A}b\mathcal{C}at$  is small, then there exists  $R \in \text{Ring}$ , not necessary commutative and a fully faithful and exact functor  $\mathcal{A} \rightarrow \text{Mod}_R$  that preserves kernel and cokernel. WARNING: it may not preserve sum and product, let alone limits and colimits.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.8.3.9).** If  $\mathcal{A} \in \mathcal{A}b\mathcal{C}at$  and  $\mathcal{C} \in \mathcal{C}at$ , then  $\text{Hom}(\mathcal{C}, \mathcal{A}) \in \mathcal{A}b\mathcal{C}at$ . In particular,  $\text{Ch}^{\mathbb{Z}}(\mathcal{A}) \in \mathcal{A}b\mathcal{C}at$  (4.8.7.1).  $\lrcorner$

### Localization

**Prop. (4.8.3.10) [Localization Category].** If  $\mathcal{C}$  is a preadditive category and  $S$  is a left or right localizing system of  $\mathcal{C}$ , then there exists a natural additive structure on  $S^{-1}\mathcal{C}$  and a localizing functor  $\mathcal{C} \rightarrow S^{-1}\mathcal{C}$  that is additive.  $\lrcorner$

*Proof:* Cf. [Sta]05QD.  $\square$

**Lemma (4.8.3.11).** If  $\mathcal{C}$  is additive and  $S$  is localizing, let  $X$  be an element of  $\mathcal{C}$ , then:  $Q(X) = 0$  iff there is a morphism  $0 : X \rightarrow Y$  that is an element of  $S$  iff there is a morphism  $0 : Z \rightarrow X$  that is an element of  $S$  iff there is a morphism  $0 : Z \rightarrow X$  that is an element of  $S$   $\lrcorner$

*Proof:* If such  $0 : X \rightarrow Y \in S$ , then it maps to isomorphisms in  $S^{-1}(\mathcal{C})$  by (4.1.1.59), so  $Q(X) = 0$ . If  $Q(X) = 0$ , then the morphism  $0 \rightarrow X$  is mapped to an isomorphism, so by (4.1.1.61), there are  $g, h$  that  $fg = hf = 0$ , so  $Z \rightarrow 0 \rightarrow X \in S$ . Dually for the other direction.  $\square$

**Prop. (4.8.3.12) [Localized Abelian Categories].** If  $\mathcal{A}$  is Abelian and  $S$  is localizing, then  $S^{-1}\mathcal{A}$  is an Abelian category and  $\mathcal{A} \rightarrow S^{-1}\mathcal{A}$  is exact.  $\lrcorner$

*Proof:* By (4.1.1.59) and its dual,  $\mathcal{A} \rightarrow S^{-1}\mathcal{A}$  preserves finite limits and colimits.  $\square$



**Serre-Subcategories**

**Def. (4.8.3.13)[Serre-Subcategories].** A **Serre-subcategory** of an Abelian category is a non-empty full subcategory  $\mathcal{C}$  that if

$$A \rightarrow B \rightarrow C$$

is exact and  $A, C \in \text{Ob}(\mathcal{C})$ , then  $B \in \text{Ob}(\mathcal{C})$ .

A **weak Serre-subcategory** of an Abelian category is a non-empty full subcategory  $\mathcal{C}$  s.t. if

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$$

is exact and  $A, B, D, E \in \mathcal{C}$ , then  $C \in \mathcal{C}$ . ┘

**Prop. (4.8.3.14).**

- A Serre category is equivalent to a full subcategory  $\mathcal{A}$  that contains 0, all the subobjects and quotient objects of  $\mathcal{A}$ , and extensions of objects of  $\mathcal{A}$  are in  $\mathcal{A}$ .
- A weak Serre category is equivalent to a full subcategory  $\mathcal{A}$  that contains 0, and all the kernels, cokernels between objects in  $\mathcal{A}$ , and all the extensions of objects in  $\mathcal{A}$ .

In these cases,  $\mathcal{A}$  is an Abelian category and the functor  $i : \mathcal{A} \rightarrow \mathcal{C}$  is exact. ┘

*Proof:* One direction of these two are trivial, it suffices to prove the converse. For the first,  $0 \rightarrow \text{Im } A \rightarrow B \rightarrow \text{Im } B \rightarrow 0$ , so  $B \in \mathcal{C}$ . For the second,  $0 \rightarrow \text{Coker}(A \rightarrow B) \rightarrow C \rightarrow \ker(D \rightarrow E) \rightarrow 0$ , so  $C \in \mathcal{C}$ . □

**Prop. (4.8.3.15)[Quotients by Serre-Subcategories].** For an exact functor  $F$  between Abelian categories, the kernel of  $F$  is a Serre subcategory. And any Serre subcategory is the kernel of an essentially surjective exact functor  $Q : \mathcal{A} \rightarrow \mathcal{A}/\mathcal{C}$ , and this functor satisfies the universal property that any exact functor between Abelian categories  $F : \mathcal{A} \rightarrow \mathcal{C}$  that  $\mathcal{C} \subset \ker(F)$  factors uniquely through  $\mathcal{A}/\mathcal{C}$ . ┘

*Proof:* The full subcategory of  $\ker(F)$  is clearly a Serre subcategory by checking the definition. Conversely, consider  $S = \{ \text{all the morphisms that has kernel and cokernel in } \mathcal{C} \}$ , first we prove it is a localizing system (4.1.1.55).

The long exact sequence (4.8.8.4) shows that if  $f, g \in S$ , then  $gf \in S$ . For other verifications, Cf. [Sta]02MS?.

Next we construct  $\mathcal{C}/\mathcal{A}$  as  $S^{-1}\mathcal{C}$ . Consider which objects are mapped to 0 in  $\mathcal{C}/\mathcal{A}$ , use (4.8.3.11) and consider the kernel and cokernel, it is easy to see that  $\ker(Q) = \mathcal{C}$ . If another category  $\mathcal{D}$  and  $F : \mathcal{C} \rightarrow \mathcal{D}$  satisfies  $\mathcal{C}$  is mapped to 0, then it is clear that elements in  $S$  is mapped to isomorphism, so it factors through  $\mathcal{C}/\mathcal{A}$  by universal property (4.1.1.59). □

**Prop. (4.8.3.16)[ $K_0$  Group of Serre-Subcategories].** Let  $\mathcal{A}$  be an Abelian category and  $\mathcal{C}$  a Serre subcategory, with  $\mathcal{A}/\mathcal{C} = \mathcal{B}$  (4.8.3.15). Then

- The exact functors  $\mathcal{C} \rightarrow \mathcal{A} \rightarrow \mathcal{B}$  induces an exact sequence

$$K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B}) \rightarrow 0,$$

- The kernel of  $K_0(\mathcal{C}) \rightarrow K_0(\mathcal{A})$  is generated by elements of the form

$$[\ker(\psi)/\text{Im}(\varphi)] - [\ker(\varphi)/\text{Im}(\psi)]$$

where  $\varphi, \psi : M \rightarrow M$  are pairs of maps that  $\varphi \circ \psi = \psi \circ \varphi = 0$ . ┘

*Proof:* Cf. [Sta]02MX. □

### Artinian Abelian Categories

**Def. (4.8.3.17) [Artinian Abelian Categories].** For  $\mathcal{C} \in \mathbf{AbCat}$ ,

- $\mathcal{C}$  is called an **Artinian Abelian category** if every object of  $\mathcal{C}$  satisfies descending chain condition for subobjects (4.8.3.4).
- $\mathcal{C}$  is called a **Noetherian Abelian category** if every object of  $\mathcal{C}$  satisfies the ascending chain condition for subobjects.

┘

**Prop. (4.8.3.18) [Artinian Noetherian Abelian Categories].** For  $\mathcal{C} \in \mathbf{AbCat}$ ,  $\mathcal{C}$  is Artinian and Noetherian iff for any object  $A \in \mathbf{Ob}(\mathcal{C})$  there is a filtration of subobjects

$$0 = A_0 \subset A_1 \subset \dots \subset A_n = A$$

s.t.  $A_i/A_{i-1} \in \mathcal{C}$  are simple objects (4.8.3.5) for each  $i \in [n]_+$ .

┘

*Proof:* Cf. [Sta]0FCJ.

□

**Prop. (4.8.3.19) [Jordan-Holder].** Let  $\mathcal{C} \in \mathbf{AbCat}$  be Artinian (4.8.3.17), then any maximal length filtration of an element  $X$  has the same length, and the set of quotients  $X_{k+1}/X_k$  is the same, up to order.

┘

*Proof:* Cf. [Sta]0FCK.

□

### Grothendieck Abelian Category

References are [Rings of Quotients]

**Def. (4.8.3.20) [Grothendieck-Abelian Categories].** A **Grothendieck-Abelian category** is an Abelian category  $\mathcal{A}$  satisfying the following axioms:

**AB3:** It is a locally small cocomplete Abelian category.

**AB5:** Small filtered colimits are exact. This is equivalent to { for any family of subobjects  $\{A_i\}$  of  $A$  to  $B$  indexed by inclusion can induce a morphism  $\sum A_i \rightarrow B$  (internal sum) } ?

**GEN:** It has a **generator**, which is an object  $U \in \mathcal{A}$  s.t. for any proper subobject  $N \subsetneq M$ , there is a map  $U \rightarrow M$  that doesn't factor through  $N$ .

┘

**Def. (4.8.3.21) [Further Grothendieck-Axioms].** For a Grothendieck Abelian category  $\mathcal{A}$ , we can also formulate the following axioms:

**AB4:** Arbitrary direct sums are exact.

**AB6:** For any index set  $J$  and filtered categories  $I_j, j \in J$  and diagrams indexed over  $I_j$ , the natural map

$$\varinjlim_{i_j \in I_j} \prod_{j \in J} M_{i_j} \rightarrow \prod_{j \in J} \varinjlim_{i_j \in I_j} M_{i_j}$$

is an isomorphism.

**Dual Axioms:** Axioms with an \* meaning that the dual category satisfies something.

┘

**Prop. (4.8.3.22).** If  $\mathcal{A}$  is a Grothendieck Abelian category, then so is  $\mathcal{PSh}^{\text{Set}}(\mathcal{A})$ .

┘

*Proof:* For the presheaf, the only problem is the existence of generator, for that, just construct a family of presheaves and sum them. Take  $Z_X = i_{f_X}(U)$ , where  $U$  is the generator of  $\mathcal{A}$  and  $f = \text{pt} \rightarrow \mathcal{A}^C : \text{pt} \rightarrow U$ . Then  $F(X) = \text{Hom}(Z_X, F)$  by adjointness(7.1.2.9). So they are a family of generators.  $\square$

**Prop. (4.8.3.23).** Any Grothendieck Abelian category has a functorial injective embedding.  $\lrcorner$

*Proof:* Cf.[Sta]079H. ?  $\square$

**Prop. (4.8.3.24) [Representability on Grothendieck Category].** A contravariant functor from a Grothendieck category to  $\text{Set}$  is representable iff it takes colimits to limits.  $\lrcorner$

*Proof:*  $M \oplus M \rightarrow M$  with induce a map  $F(M) \times F(M) \rightarrow F(M)$  thus  $F(M)$  is a semigroup, and the inverse of  $\text{id}_M$  in  $\text{Hom}(M, M)$  maps to a  $F(M) \rightarrow F(M)$  which is the inverse, Thus in fact  $F$  is a left adjoint functor to  $\mathcal{A}b$ .

Let  $U$  be a generator,  $A = \sum_{s \in F(U)} U$ , let  $s_{univ} = (s) \in F(A) = \prod_{s \in F(U)} F(U)$ . let  $A'$  be the largest objects that  $s_{univ}$  restricts to 0 in  $A'$ , let  $\bar{s}_{univ}$  be in  $F(A/A')$  that maps to  $s_{univ}$  in  $F(A)$  (because  $F$  is left exact). Then we claim  $(A/A', \bar{s}_{univ})$  represents  $F$ . Cf.[Sta]07D7. ?  $\square$

**Cor. (4.8.3.25) [Grothendieck Categories are Cocomplete].** Any Grothendieck category satisfies  $\text{AB3}^*$ .  $\lrcorner$

*Proof:* It suffices to show that small direct products exist. And this is because  $F = \prod_i \text{Hom}(-, M_i)$  commutes with colimits.  $\square$

**Prop. (4.8.3.26) [Grothendieck Categories are Locally Presentable].** Any Grothendieck category is locally presentable. In fact, for an Abelian category with exact filtered colimits, it is a Grothendieck Abelian category iff it is locally presentable.  $\lrcorner$

*Proof:* Cf.[Colimits and homological algebra, Krause]Cor5.2. ?  $\square$

**Cor. (4.8.3.27).** If  $F : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$  commutes with small colimits, then  $F$  is a left adjoint.  $\lrcorner$

*Proof:* ?  $\square$

### Examples of Grothendieck Category

**Prop. (4.8.3.28) [Modules].** For  $R \in \text{Alg}$ ,  $\text{Mod}_R$  is a Grothendieck Abelian category. More generally, for  $R, S \in \text{Alg}$ ,  $\text{Mod}_{R-S} = \text{Mod}_{R \otimes S^{\text{op}}}$  is a Grothendieck Abelian category.  $\lrcorner$

*Proof:* This is clear, by(5.1.1.27).  $\square$

**Prop. (4.8.3.29) [Sheaf of Modules].**  $\text{Sh}(\mathcal{O})$  for a ringed site  $(\mathcal{C}, \mathcal{O})$  is a Grothendieck Abelian category.  $\lrcorner$

*Proof:* It is obviously an Abelian category and have filtered colimits as presheaves, which are exact because colimits in the category of Abelian groups are exact, and for a family of generators, take  $j_! \mathcal{O}_U$  as the representative for  $\Gamma(U, -)$ , which is the sheaf associated to the sheaf  $Z_U$  in the proof of(4.8.3.22).  $\square$

**Cor. (4.8.3.30).** Let  $\mathcal{C}$  be a site, the categories  $\mathcal{P}\text{Sh}(\mathcal{C})$  and  $\text{Sh}(\mathcal{C})$  are Grothendieck Abelian categories.  $\lrcorner$

*Proof:* For the presheaf, Cf.(4.8.3.22). For the sheaf, it follows from (4.8.3.29).  $\square$

**Remark(4.8.3.31).** The category of Abelian sheaves doesn't satisfy AB4\*, i.e. not every limit of epimorphisms is epimorphism.  $\lrcorner$

*Proof:* Consider the constant sheaf  $\prod_{q \in [0,1]} B(\frac{p}{q}, \frac{1}{q})$  on  $[0, 1]$ .  $\square$

**Prop.(4.8.3.32) [Quasi-Coherent Sheaves].** For  $X \in \mathcal{S}ch$ ,  $\mathcal{QCoh}(X)$  is Grothendieck category, and there is a **coherentor** left adjoint to the forgetful functor.  $\lrcorner$

*Proof:* Firstly  $\mathcal{QCoh}(X) \in \mathcal{AbCat}$  by(7.5.1.3), and on affine open set, the colimit is an  $\mathcal{Qco}$  sheaf, thus the colimit exists in  $\mathcal{Qco}$  and equals the colimit in the category of sheaves, thus filtered colimits is exact because  $\mathcal{Mod}(\mathcal{O}_X)$  is Grothendieck(4.8.3.29). The generator exists, Cf.[Sta]077P.

The coherentor exists by the fact that  $h_{\mathcal{F}}$  commutes with colimits and by the property of Grothendieck category(4.8.3.24).  $\square$

### Others

**Def.(4.8.3.33)[Essential Morphism].** In an Abelian category, an injection  $A \rightarrow B$  is called **essential** iff every non-zero subobject of  $B$  intersects  $A$ . A surjection is called **essential** iff every proper subobject of  $A$  is not mapped to  $B$ .  $\lrcorner$

**Def.(4.8.3.34) [Noetherian Abelian Category].** In a Grothendieck Abelian category  $\mathcal{A}$ , an object  $M$  is called **finitely generated** if for every ascending chain

$$M_1 \subset M_2 \subset \dots \subset M$$

with  $\cup_i M_i = M$ , we have  $M_i = M$  for some  $i$ .

$\mathcal{A}$  is called **Noetherian** iff a subobject of a f.g. object is f.g..  $\mathcal{A}$  is called **Artinian** iff every f.g. object has finite length<sup>?</sup>.  $\lrcorner$

## 4 Triangulated Categories

References are [Triangulated Categories, Neeman].

**Notation(4.8.4.1).**

- Let  $\mathcal{D} \in \mathcal{TriCat}$ (4.8.4.2).

**Def.(4.8.4.2) [Triangulated Categories].** A **triangulated category** is an additive category  $\mathcal{T}$ (4.8.1.7) with an additive automorphism  $\Sigma : \mathcal{T} \cong \mathcal{T}$  denoted by  $X \mapsto X[1]$  and a set of **distinguished triangles**  $\text{Tri}(\mathcal{T})$  that is stable under isomorphism, and satisfying the following axioms:

**(TR1):**  $X \xrightarrow{\text{id}} X \rightarrow 0 \rightarrow X[1] \in \text{Tri}(\mathcal{D})$ . Any morphism  $X \xrightarrow{u} Y$  can be completed to some  $X \xrightarrow{u} Y \rightarrow C(u) \rightarrow X[1] \in \text{Tri}(\mathcal{T})$ . Such an object  $C(u)$  is called the **mapping cone** of  $u$ .

**(TR2):**  $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1] \in \text{Tri}(\mathcal{D})$  iff  $Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-\Sigma(f)} Y[1] \in \text{Tri}(\mathcal{D})$ .

**(TR3):** Any two consecutive morphisms of two distinguished triangles can be extended to a morphism of distinguished triangles.

**(TR4):** If  $X \xrightarrow{f} Y \xrightarrow{g} Z$  are morphisms, then there are maps  $C(f) \rightarrow C(gf)$ ,  $C(gf) \rightarrow C(g)$  by TR3, Then  $C(f) \rightarrow C(gf) \rightarrow C(g) \rightarrow C(f)[1] \in \text{Tri}(\mathcal{D})$ .

A **triangulated subcategory** is an additive subcategory  $\mathcal{D}'$  stable under  $[1]$  and  $[-1]$  together with a subclass of triangles in  $\mathcal{D}'$  that forms a triangulated category.

We use the symbol  $\mathcal{A} \in \text{TriCat}$  to denote that  $\mathcal{A}$  is a triangulated category.  $\lrcorner$

**Def. (4.8.4.3) [Notations].** We use the tuple  $(X, Y, Z, f, g, h)$  to represent a distinguished triangle  $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ .  $\lrcorner$

**Prop. (4.8.4.4).** Given  $(X, Y, Z, f, g, h), (X', Y', Z', f', g', h')$ , they are both in  $\text{Tri}(\mathcal{D})$  iff  $(X \oplus X', Y \oplus Y', Z \oplus Z', f \oplus f', g \oplus g', h \oplus h')$  is in  $\text{Tri}(\mathcal{D})$ .  $\lrcorner$

*Proof:* [Sta]05QS.  $\square$

**Cor. (4.8.4.5).**  $(X, X \oplus Y, Y, (1, 0), (0, 1), 0)$  is a distinguished triangle.  $\lrcorner$

**Prop. (4.8.4.6).** There is a natural notation of a product of triangulated categories.  $\lrcorner$

**Def. (4.8.4.7) [Cohomological Functors and Exact Functors].** A functor from a triangulated category to an Abelian category is called a **cohomological functor** if it maps a distinguished triangle to a long exact sequence. Similarly, we can define homological functors.

Conversely, a  $\delta$ -**functor** is a functor from an Abelian category to a triangulated category together with a map from the category of exact sequences to the category of distinguished triangles.

A functor  $F$  between two triangulated category is called an **exact functor** if it maps distinguished triangles to distinguished triangles, and there is an isomorphism of functors  $\xi : F \circ [1] \rightarrow [1] \circ F$ .  $\lrcorner$

**Prop. (4.8.4.8).** An exact functor between triangulated categories is additive.  $\lrcorner$

*Proof:* Cf. [Sta]05QY.  $\square$

**Def. (4.8.4.9) [Bi-Exact Functors].** Let  $\mathcal{D}, \mathcal{D}', \mathcal{E} \in \text{TriCat}$ , a **bi-exact bifunctor**  $F : \mathcal{D} \times \mathcal{D}' \rightarrow \mathcal{E}$  is a functor that for any  $X \in \mathcal{D}$ ,  $F(X, -) : \mathcal{D}' \rightarrow \mathcal{E}$  is an exact functor and for any  $Y \in \mathcal{D}'$ ,  $F(-, Y) : \mathcal{D} \rightarrow \mathcal{E}$  is an exact functor. By (4.8.4.8), any bi-exact functor is bi-additive.  $\lrcorner$

**Prop. (4.8.4.10) [Adjunction of Exact Functors].** If  $(F, G) : \mathcal{D} \rightleftharpoons \mathcal{D}'$  is an adjunction pair between triangulated categories and  $F$  is exact, then  $G$  is also an exact functor between triangulated categories.  $\lrcorner$

*Proof:* Use adjunction, we can show that  $G$  commutes with  $[1]$ , and if  $A \rightarrow B \rightarrow C \rightarrow A[1] \in \text{Tri}(\mathcal{D}')$ , choose a distinguished triangle  $G(A) \rightarrow G(B) \rightarrow X \rightarrow G(A)[1] \in \text{Tri}(\mathcal{D})$ , then by (TR3) we get a map of distinguished triangles  $(F(G(A)), F(G(B)), F(X)) \rightarrow (A, B, C)$ , which by adjunction defines a map of distinguished triangles  $(G(A), G(B), X) \rightarrow (G(A), G(B), G(C))$ , which shows  $X \cong G(C)$ , and then  $(G(A), G(B), G(C))$  is in  $\text{Tri}(\mathcal{D})$ .  $\square$

**Prop. (4.8.4.11).** If  $(F, G) : \mathcal{D} \rightleftharpoons \mathcal{D}'$  is an adjunction pair between triangulated categories and  $F, G$  are exact,  $F$  is fully faithful and  $\ker(G) = 0$ , then this is an equivalence of categories.  $\lrcorner$

*Proof:* By (4.1.1.32),  $u : \text{id} \rightarrow gf$  is an isomorphism. Now for any  $X \in \mathcal{D}'$ , choose a distinguished triangle  $(F(G(X)), X \rightarrow Y) \in \text{Tri}(\mathcal{D}')$ , which corresponds to  $(G(F(G(X))), G(X), G(Y)) \in \text{Tri}(\mathcal{D})$ , and by (4.1.1.30),  $G(X) \xrightarrow{u_X} GFG(X) \rightarrow G(X)$  is  $\text{id}_X$ , so we get an isomorphism of triangles  $(G(F(G(X))), G(X), G(Y)) \cong (G(X), G(X), 0)$ , so  $G(Y) \cong 0$ , and  $Y = 0$  by hypothesis. So  $v : FG \rightarrow \text{id}$  is also an isomorphism, so  $(F, G)$  is an equivalence.  $\square$

**Lemma (4.8.4.12).** If  $(X, Y, Z, f, g, h) \in \text{Tri}(\mathcal{D})$ , then  $g \circ f = 0$ .  $\lrcorner$

*Proof:* By TR1  $(X, X, 0, \text{id}, 0, 0)$  is a distinguished triangle, and by TR3 there is a map of distinguished triangles

$$\begin{array}{ccccccc} X & = & X & \longrightarrow & 0 & \longrightarrow & X[1] \\ \parallel & & \downarrow f & & \downarrow & & \parallel \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & X[1] \end{array},$$

so  $g \circ f = 0$ . □

**Prop. (4.8.4.13) [Long Exact Sequences].** Let  $C \in \mathcal{D}$ , then  $\text{Hom}_{\mathcal{D}}(-, C)$  (resp.  $\text{Hom}_{\mathcal{D}}(C, -)$ ) is a homological (resp. cohomological) functor (4.8.4.7). ┘

*Proof:* By (4.8.4.4), for any distinguished triangle  $(X, Y, Z, f, g, h)$ ,  $\text{Hom}(C, X) \rightarrow \text{Hom}(C, Y) \rightarrow \text{Hom}(C, Z)$  is 0, and if  $a \in \text{Hom}(C, Y)$  is mapped to  $0 \in \text{Hom}(C, Z)$ , then the morphism  $(a, 0) : (C, 0) \mapsto (Y, Z)$  extends to a morphism of distinguished triangles  $(b, a, 0) : (C, C, 0, \text{id}, 0, 0) \rightarrow (X, Y, Z, f, g, h)$ , thus  $f \circ b = a$ . □

The converse case is dual. □

**Cor. (4.8.4.14).** If  $(a, b, c) : (X, Y, Z, f, g, h) \rightarrow (X', Y', Z', f', g', h') \in \text{Tri}(\mathcal{D})$  that two of  $a, b, c$  are isomorphisms, then this is an isomorphism of triangles.

In particular, the completion in TR1 is unique (may up to non-unique isomorphism) by TR3. ┘

*Proof:* By 5-lemma,  $\text{Hom}(C, X) \rightarrow \text{Hom}(C, X')$  is an isomorphism, then  $X \rightarrow X'$  is an isomorphism by Yoneda lemma. □

**Prop. (4.8.4.15).**  $f : X \rightarrow Y \in \mathcal{D}$  is an isomorphism iff  $C(f) = 0$ . ┘

*Proof:* There is a morphism  $(\text{id}_X, f, 0) : (X, X, 0, \text{id}, 0, 0) \rightarrow (X, Y, C(f), f, g, h)$  by (TR1) and (TR3). By (4.8.4.14),  $f$  is an isomorphism iff  $0 \rightarrow C(f)$  is an isomorphism, which is equivalent to  $Z = 0$ . □

**Prop. (4.8.4.16).** In  $\mathcal{D}$ , any commutative diagram

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & Y' \end{array}$$

can be extended to a diagram

$$\begin{array}{ccccccc} X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & X[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X' & \longrightarrow & Y' & \longrightarrow & Z' & \longrightarrow & X'[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X'' & \longrightarrow & Y'' & \longrightarrow & Z'' & \longrightarrow & X''[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X[1] & \longrightarrow & Y[1] & \longrightarrow & Z[1] & \longrightarrow & X[2] \end{array}$$

where the lower right is anti-commutative. ┘

*Proof:* Let  $(X, Y, Z), (X', Y', Z'), (X, X', X''), (Y, Y', Y''), (X, Y, A)$  be distinguished triangles, then we can find maps  $a : Z \rightarrow A, b : A \rightarrow Y', a' : X'' \rightarrow A, b' : A \rightarrow Z$  by TR3(4.8.4.2). Then TR4 says  $(Z \rightarrow A, Y''), (X'' \rightarrow Z \rightarrow Z')$  is distinguished.

Now let  $(X'', Y'', Z'')$  be distinguished, then we use TR4 again to  $(X'', A, Y'')$ , then  $(Z', Z'', Z[1])$  is distinguished, thus so does  $(Z \rightarrow Z' \rightarrow Z'')$ .

Now it is left to verify the anti-commutativity of the righthdown square, for this, Cf.[Sta]05R0.

□

**Def. (4.8.4.17) [Exceptional Objects].** If  $k \in \text{Field}$  and  $\mathcal{D} \in \text{TriCat}_k$ , then an **exceptional object** in  $\mathcal{D}$  is an object  $E \in \mathcal{D}$  s.t.

$$\text{Hom}^i(E, E) = \begin{cases} 0 & , i \in \mathbb{Z}^\times \\ k & , i = 0 \end{cases}$$

┘

**Def. (4.8.4.18) [Generating Objects].** A family of objects  $\{X_\alpha\}_\kappa$  is said to **generate**  $\mathcal{D}$  if any triangulated subcategory of  $\mathcal{C}$  containing  $\{X_\alpha\}_\kappa$  is equivalent to  $\mathcal{D}$  through the inclusion. ┘

**Prop. (4.8.4.19) [Equivalence and Generators].** Let  $F : \mathcal{C} \rightarrow \mathcal{D} \in \text{TriCat}$  be an exact functor, Suppose that

- $\{X_\alpha\}_\kappa$  is a family of objects generating  $\mathcal{C}$ .
- $\{F(X_\alpha)\}_\kappa$  generate  $\mathcal{D}$ .
- $F$  induces isomorphisms  $\text{Hom}(X_i, X_j[l]) \cong \text{Hom}(F(X_i), F(X_j)[l])$  for any  $i, j \in \kappa$  and  $l \in \mathbb{Z}$ .

Then  $F$  is an exact equivalence. ┘

*Proof:* Let  $\mathcal{C}' \subset \mathcal{C}$  be the full subcategory consisting o objects  $Y$  s.t.

$$\text{Hom}(X_i[l], Y) \xrightarrow{F} \text{Hom}(F(X_i)[l], FY)$$

is an isomorphism for any  $i \in \kappa$  and  $l \in \mathbb{Z}$ . Then  $\mathcal{C}'$  is a strictly full triangulated subcategory of  $\mathcal{C}$ . Since  $\mathcal{C}'$  contains all  $X_i$ , it is just equal to  $\mathcal{C}$ . The same arguing then shows that the full subcategory consisting o objects  $Y$  s.t.

$$\text{Hom}(Y, Z) \xrightarrow{F} \text{Hom}(FY, FZ)$$

for any  $Z \in \mathcal{C}$  is also a full triangulated subcategory thus also  $\mathcal{C}$ . Finally,  $F(\mathcal{C})$  is a subcategory of  $\mathcal{D}$  equivalent to  $\mathcal{D}$ , thus  $F$  is an equivalence of categories, which is also an exact equivalence by(4.8.4.10). □

### Triangulated Subcategories

**Def. (4.8.4.20) [Saturated Triangulated Subcategories].** A full triangulated subcategory  $\mathcal{D}' \subset \mathcal{D}$  is called a **saturated triangulated subcategory** if whenever  $X \oplus Y$  is isomorphic to an object of  $\mathcal{D}'$ ,  $X, Y$  are all isomorphic to an object of  $\mathcal{D}'$ . ┘

**Def. (4.8.4.21) [Thick Triangulated Subcategories].** A full triangulated subcategory  $\mathcal{C} \subset \mathcal{D}$  is called a **thick triangulated subcategory** if objects  $X, Y \in \mathcal{D}$  must lie in  $\mathcal{C}$ , provided there exists a morphism  $f : X \rightarrow Y$  s.t.

- $f$  factors through an object in  $\mathcal{C}$ , and
- $f$  has a mapping cone  $C(f) \in \mathcal{C}$ .

⌋

**Prop. (4.8.4.22) [Orthogonal Triangulated Subcategories].** Let  $\mathcal{B} \subset \mathcal{D}$  a full triangulated subcategory, then  ${}^\perp\mathcal{B}$  and  $\mathcal{B}^\perp$  are also full triangulated subcategories of  $\mathcal{D}$ . ⌋

*Proof:*

□

**Def. (4.8.4.23) [Semi-Orthogonal Decompositions].** A **semi-orthogonal decomposition** of  $\mathcal{D}$  is a sequence  $(\mathcal{B}_0, \dots, \mathcal{B}_n)$  of full triangulated subcategories s.t.  $\mathcal{B}_j \in \mathcal{B}_i^\perp$  (4.8.4.22) whenever  $0 \leq j < i \leq n$ , and  $\mathcal{D}$  is generated by  $\{\mathcal{B}_i\}$ .

And if each  $\mathcal{B}_i$  consists of a single exceptional object of  $\mathcal{D}$  (4.8.4.17), then this sequence is called a **full exceptional sequence** of  $\mathcal{A}$ . ⌋

**Prop. (4.8.4.24) [Kernels are Saturated Triangulated Subcategories].** For  $\mathcal{D}, \mathcal{D}' \in \text{TriCat}$  and  $\mathcal{A} \in \text{AbCat}$ ,

- Let  $F : \mathcal{D} \rightarrow \mathcal{D}'$  be an exact functor, then the full subcategory of  $\mathcal{D}$  consisting of objects  $X$  that  $F(X) = 0$  is a strictly full saturated triangulated subcategory of  $\mathcal{D}$ , called the **kernel** of  $F$ .
- Let  $H : \mathcal{D} \rightarrow \mathcal{A}$  be a (co)homological functor. then the full subcategory of  $\mathcal{D}$  consisting of objects  $X$  that  $F(X[n]) = 0$  for all  $n$  is a strictly full saturated triangulated subcategory of  $\mathcal{D}$ , called the **kernel** of  $H$ .

⌋

*Proof:* Cf. [Sta]05RC, 05RD.

□

### Localizations of Triangulated Category

**Def. (4.8.4.25) [Compatible Localizing Systems].** A localizing system (4.1.1.55)  $S \subset \mathcal{D}$  is said to be compatible with the triangulated structure if it satisfies the following axioms:

**(LS4):** For  $s \in S$ ,  $s[n] \in S$  for any  $n \in \mathbb{Z}$ .

**(LS5):** Any two consecutive morphisms of two distinguished triangle can be extended to a morphism of distinguished triangles by morphisms in  $S$ .

⌋

**Cor. (4.8.4.26).** Let  $\mathcal{D}$  be a triangulated category. If a class of morphisms  $S$  satisfy (LS1), (LS5) and (LS6), then (LS2) holds as well. ⌋

*Proof:* Let  $f : X \rightarrow Y \in \mathcal{D}$  and  $s : X \rightarrow X' \in S$ , we can use (TR1) and (TR2) to extend these to a morphism of distinguished triangles

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \longrightarrow & Z & \longrightarrow & X[1] \\ \downarrow s & & \downarrow s' & & \parallel & & \downarrow s[1] \\ X' & \longrightarrow & Y' & \longrightarrow & Z & \longrightarrow & X'[1] \end{array}$$

The right extension is dual.

□

**Prop. (4.8.4.27) [Exact Functors and Saturated Localizing Systems].** For  $\mathcal{D}, \mathcal{D}' \in \text{TriCat}$  and  $\mathcal{A} \in \text{AbCat}$ ,



- If  $F : \mathcal{D} \rightarrow \mathcal{D}'$  is an exact functor and

$$S = \{f \in \mathcal{D} \mid F(f) \text{ is an isomorphism}\},$$

then  $S$  is a saturated localizing system compatible with the triangulated structure (4.8.4.25).

- If  $H : \mathcal{D} \rightarrow \mathcal{A}$  is a (co)homological functor and

$$S = \{f \in \mathcal{D} \mid H^i(f) \text{ is an isomorphism}\},$$

then  $S$  is a saturated localizing system compatible with the triangulated structure (4.8.4.25).  $\lrcorner$

*Proof:* Cf. [Sta]05R4, 05R6.  $\square$

**Prop. (4.8.4.28) [Localized Triangulated Categories].** Let  $S \subset \mathcal{D}$  be a localizing system compatible with the triangulated structure, then there is a unique triangulated category structure on  $S^{-1}\mathcal{D}$  that the localization functor  $Q : \mathcal{D} \rightarrow S^{-1}\mathcal{D}$  (4.1.1.54) is exact.

Moreover,  $S^{-1}\mathcal{D}$  satisfies the following universal properties:

- If  $\mathcal{A}$  is an Abelian category and  $H : \mathcal{D} \rightarrow \mathcal{A}$  is a cohomological functor that  $H(s)$  are isomorphisms for all  $s \in S$ , then the unique factorization  $H' : S^{-1}\mathcal{D} \rightarrow \mathcal{A}$  s.t.  $H' \circ Q = H$  (4.1.1.59) is also a cohomological functor.
- If  $\mathcal{D}'$  is a triangulated category and  $F : \mathcal{D} \rightarrow \mathcal{D}'$  is an exact functor that  $F(s)$  are isomorphisms for all  $s \in S$ , then the unique factorization  $F' : S^{-1}\mathcal{D} \rightarrow \mathcal{D}'$  s.t.  $F' \circ Q = F$  (4.1.1.59) is also an exact functor.  $\lrcorner$

*Proof:* Cf. [Sta]05R6.  $\square$

**Cor. (4.8.4.29) [Kernel of Localizing Functor].** An object  $Z \in \mathcal{D}$  is in the kernel of the localizing functor  $Q : \mathcal{D} \rightarrow S^{-1}\mathcal{D}$  iff it satisfies the following equivalent conditions:

- There exists  $Z'$  s.t.  $0 : Z \rightarrow Z' \in S$ .
- There exists  $Z'$  s.t.  $0 : Z' \rightarrow Z \in S$ .
- There exists  $Z'$  and a distinguished triangle  $(X, Y, Z \oplus Z', f, g, h)$  that  $f \in S$ .  $\lrcorner$

*Proof:* Cf. [Sta]05R8.  $\square$

**Prop. (4.8.4.30) [Triangulated Subcategories of Localized Triangulated Categories].** Let  $\mathcal{D}$  be a triangulated category and  $S$  is a localizing system compatible with the triangulated structure. Let  $\mathcal{D}'$  be a full triangulated subcategory of  $\mathcal{D}$  and  $S'$  is a localizing system of  $\mathcal{D}'$ . If either of the following holds:

- For any  $s : X \rightarrow Y \in S$  and  $X \in \mathcal{D}'$ , there is a morphism  $t : Y \rightarrow Z$  that  $Z \in \mathcal{D}'$  and  $t \circ s \in S'$ .
- The same as in item1 but with arrows reversed.

Then the natural functor  $(S')^{-1}\mathcal{D}' \rightarrow S^{-1}\mathcal{D}$  is fully faithful.  $\lrcorner$

*Proof:* This is immediate from (4.1.1.62).  $\square$

### Quotients of Triangulated Categories

**Prop. (4.8.4.31) [Full Triangulated Subcategories and Localizing Systems].**

- If  $\mathcal{D}' \subset \mathcal{D}$  is a full triangulated subcategory, then

$$S = \{f : X \rightarrow Y \in \mathcal{D}' \mid \text{there exists a distinguished triangle } (X, Y, Z, f, g, h), Z \in \mathcal{D}'\}$$

is a localizing system compatible with the triangulated structure of  $\mathcal{D}$ . And  $S$  is saturated iff  $\mathcal{D}'$  is saturated.

- If  $S$  is a localizing system of  $\mathcal{D}$ , then  $\mathcal{D}' = \ker(\mathcal{D} \rightarrow S^{-1}\mathcal{D})$  (4.8.4.28)(4.8.4.24) is a saturated full triangulated subcategory of  $\mathcal{D}$ .

┘

*Proof:* 1: Cf. [Sta]05RH. □

**Def. (4.8.4.32) [Quotient Triangulated Categories].** Let  $\mathcal{B} \subset \mathcal{D}$  be a full triangulated subcategory, define the **quotient triangulated category**  $\mathcal{D}/\mathcal{B}$  as the localized triangulated category  $S^{-1}\mathcal{D}$ , where  $S$  is the localizing system associated to  $\mathcal{B}$  (4.8.4.31).

Moreover,  $\mathcal{D}/\mathcal{B}$  satisfies the following universal properties:

- If  $\mathcal{A} \in \text{AbCat}$  and  $H : \mathcal{D} \rightarrow \mathcal{A}$  is a cohomological functor s.t.  $\mathcal{B} \subset \ker(H)$ , then  $H$  factors uniquely through  $\mathcal{D}/\mathcal{B}$  by a functor  $H'$ , and  $H'$  is also a cohomological functor.
- If  $\mathcal{D}' \in \text{TriCat}$  and  $F : \mathcal{D} \rightarrow \mathcal{D}'$  is an exact functor s.t.  $\mathcal{B} \subset \ker(F)$ , then  $F$  factors uniquely through  $\mathcal{D}/\mathcal{B}$  by a functor  $F'$ , and  $F'$  is also an exact functor.

┘

*Proof:* The universal properties follow from the universal properties of localized triangulated categories (4.8.4.28) and the definition of  $S$  (4.8.4.31), using the long exact sequence or (4.8.4.15). □

**Prop. (4.8.4.33) [Saturation].** Let  $\mathcal{D}' \subset \mathcal{D}$  be a full triangulated subcategory, then the kernel of the quotient map  $\mathcal{D} \rightarrow \mathcal{D}/\mathcal{D}'$  (4.8.4.32) is a strictly full saturated subcategory consisting of objects  $Z \in \mathcal{D}$  that  $Z \oplus Z'$  is an object of  $\mathcal{D}'$  for some object  $Z' \in \mathcal{D}$ . In particular, it is the smallest saturated triangulated subcategory containing  $\mathcal{D}'$ , called the **saturation of  $\mathcal{D}'$** . ┘

*Proof:* The kernel is a strictly full saturated subcategory by (4.8.4.24). The description of the objects of kernel follows from the definition (4.8.4.32)(4.8.4.31) and (4.8.4.29). □

### Representability

References are [Triangulated Categories, Neeman] and [Brown Representation, Krause].

**Def. (4.8.4.34) [Generators].** Let  $\mathcal{D}$  be a triangulated category, a **generator of  $\mathcal{D}$**  is an object of  $\mathcal{D}$  s.t. for any object  $K$  of  $\mathcal{D}$ , there exists some integer  $n$  and a non-zero map  $E[n] \rightarrow K$ . ┘

**Prop. (4.8.4.35).** For  $\mathcal{T} \in \text{TriCat}$ , the compact objects of  $\mathcal{D}$  form a Karoubian, saturated, strictly full triangulated category of  $\mathcal{T}$ . ┘

*Proof:* Cf. [Sta]09QH. □

**Def. (4.8.4.36) [Compactly Generated Triangulated Categories].** Let  $\mathcal{D} \in \text{TriCat}$  with arbitrary small direct sums, then  $\mathcal{D}$  is said to be a **compactly generated triangulated category** if there exists a set of compact objects  $\{E_i\}$  that  $\bigoplus E_i$  is a generator of  $\mathcal{D}$  (4.8.4.34). ┘

**Thm. (4.8.4.37) [Brown’s Representability Theorem, Neeman[?Nee96]/Krause].** Suppose  $\mathcal{T} \in \mathbf{TriCat}$  has arbitrary small direct sums and is generated by a set of perfect objects, then a functor  $F : \mathcal{T} \rightarrow \mathcal{Ab}^{\mathrm{op}}$  is cohomological and transforms direct sums into products, then  $F$  is representable.

In particular, the hypothesis is satisfied when  $\mathcal{T}$  is compactly generated (4.8.4.36) (e.g. when  $\mathcal{T} = D(\mathcal{A})$  for some Grothendieck Abelian category  $\mathcal{A} \in \mathcal{AbCat}$ ).  $\lrcorner$

*Proof:* Cf. [Sta]0A8F, 0F5X or [Krause, A Brown representability theorem].  $\square$

**Cor. (4.8.4.38) [Adjointness Lemma].** Let  $\mathcal{T} \in \mathbf{TriCat}$  with arbitrary small direct sums that is compactly generated (4.8.4.36) and  $F : \mathcal{T} \rightarrow \mathcal{T}'$  an exact functor of triangulated categories that preserves direct sums, then  $F$  has an exact right adjoint.

The hypothesis is satisfied when  $\mathcal{T} = D(\mathcal{A})$  for some Grothendieck Abelian category  $\mathcal{A} \in \mathcal{AbCat}$ .  $\lrcorner$

*Proof:* By Brown’s representability theorem (4.8.4.37), for any  $Y \in \mathcal{T}'$ , there is an object  $G(Y) \in \mathcal{T}$  that represents the contravariant cohomological functor  $\mathcal{T} \rightarrow \mathcal{Ab} : X \mapsto \mathrm{Hom}_{\mathcal{T}'}(F(X), Y)$  (4.8.4.13). Then  $G$  is a functor by Yoneda’s lemma. It is exact by (4.8.4.10).  $\square$

**Prop. (4.8.4.39) [Vector Space Representability].** Let  $k \in \mathbf{Field}$  and  $\mathcal{D} \in \mathbf{TriCat}_k$  has direct sums and is compactly generated. Denote  $\mathcal{D}_c$  the subcategory of compact objects in  $\mathcal{D}$  (4.8.1.15). If  $H : \mathcal{D}_c^{\mathrm{op}} \rightarrow \mathcal{Vect}_k$  is a  $k$ -linear cohomological functor s.t.  $\dim_k(H(X)) < \infty$  for any  $X \in \mathcal{D}_c$ , then  $H = \mathcal{Y}_Y$  for some  $Y \in \mathcal{D}$ .  $\lrcorner$

*Proof:* Cf. [Sta]0FYG.  $\square$

## Serre Functors

References are [Representable functors, Serre functors, and mutations, Bondal].

**Def. (4.8.4.40) [Serre Functors].** Let  $k \in \mathbf{Field}$ ,  $\mathcal{A} \in \mathbf{TriCat}_k$  s.t.  $\dim_k \mathrm{Hom}(A, B) < \infty$  for any  $A, B \in \mathcal{A}$ , then a **Serre functor** on  $\mathcal{A}$  is defined to be an additive equivalence  $F : \mathcal{A} \cong \mathcal{A}$  s.t. there are functorial isomorphisms

$$\varphi_{E,G} : \mathrm{Hom}_{\mathcal{A}}(E, G) \cong \mathrm{Hom}_{\mathcal{A}}(E, F(G))^*$$

and the composition

$$(\varphi_{F(E), F(G)})^* \circ \varphi_{E,G} : \mathrm{Hom}(E, G) \rightarrow \mathrm{Hom}(E, F(G))^* \rightarrow \mathrm{Hom}(F(E), F(G))$$

coincides with the isomorphism induced by  $F$ .  $\lrcorner$

**Prop. (4.8.4.41).** Let  $k \in \mathbf{Field}$ ,  $\mathcal{A} \in \mathbf{TriCat}_k$  s.t.  $\dim_k \mathrm{Hom}(A, B) < \infty$  for any  $A, B \in \mathcal{A}$ , then

- Any Serre functor on  $\mathcal{A}$  is exact.
- A Serre functor on  $\mathcal{A}$  (4.8.4.40) is unique up to a graded natural isomorphism if it exists.  $\lrcorner$

*Proof:* Cf. [B-K89] ?  $\square$

## 5 Sibling Functors

**Def. (4.8.5.1) [Sibling Functors].** For  $\mathcal{A} \in \mathcal{AbCat}$ , two exact functors  $F, F' : D^b(\mathcal{A}) \rightarrow \mathcal{D}$  are called **sibling functors** if  $F|_{\mathcal{A}} \cong F'|_{\mathcal{A}}$  and  $F(K) \cong F'(K)$  for any  $K \in D^b(\mathcal{A})$ .  $\lrcorner$

## 6 T-Structures and Slicings

**Def. (4.8.6.1)[T-Structures].** A **T-structure** on a triangulated category  $\mathcal{D}$  (4.8.4.2) is a pair of strictly full subcategories  $\mathcal{D}^{\leq 0}$  and  $\mathcal{D}^{\geq 0}$  s.t. if we denote  $\mathcal{D}^{\leq n} = \mathcal{D}^{\leq 0}[-n]$  and  $\mathcal{D}^{\geq n} = \mathcal{D}^{\geq 0}[-n]$ , then

1. If  $A \in \mathcal{D}^{\leq 0}$  and  $B \in \mathcal{D}^{\geq 1}$ , then  $\text{Hom}(A, B) = 0$ .

2.  $\mathcal{D}^{\leq 0} \subset \mathcal{D}^{\leq 1}$ ,  $\mathcal{D}^{\geq 1} \subset \mathcal{D}^{\geq 0}$ .

3. For any  $E \in \mathcal{D}$ , there is a distinguished triangle  $(A, E, B)$  with  $A \in \mathcal{D}^{\leq 0}$  and  $B \in \mathcal{D}^{\geq 1}$ .

And  $\mathcal{D}$  is said to be **bounded w.r.t. this T-structure** iff each object in  $\mathcal{D}$  is contained in  $\mathcal{D}^{\geq a} \cap \mathcal{D}^{\leq b}$  for some  $a, b \in \mathbb{Z}$ .

We use the symbol  $(\mathcal{D}, \mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0}) \in t\text{-TriCat}$  or  $\mathcal{D} \in t\text{-TriCat}$  to denote that  $(\mathcal{D}, \mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$  is a triangulated category with the T-structure  $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ .  $\lrcorner$

**Def. (4.8.6.2) [Truncations].** For  $\mathcal{D} \in t\text{-TriCat}$  and  $E \in \mathcal{D}$ , by item3 of (4.8.6.1) there exists a distinguished triangle  $(A, E, B)$  with  $A \in \mathcal{D}^{\leq 0}$  and  $B \in \mathcal{D}^{\geq 1}$ . Then it follows from the long exact sequence (4.8.4.13) that for any  $X \in \mathcal{D}^{\leq 0}$ ,

$$\text{Hom}(X, A) \cong \text{Hom}(X, E).$$

In other words,  $A$  is initial w.r.t. morphisms from  $\mathcal{D}^{\leq 0}$  to  $E$ . Thus by using the axiom of choice, we can define a **truncation functor**

$$\tau_{\leq 0} : \mathcal{D} \rightarrow \mathcal{D}^{\leq 0}$$

right adjoint to the inclusion functor  $\mathcal{D}^{\leq 0} \subset \mathcal{D}$ .

Similarly, we can define the truncation functor

$$\tau_{\geq 0} : \mathcal{D} \rightarrow \mathcal{D}^{\geq 0}$$

left adjoint to the inclusion functor  $\mathcal{D}^{\geq 0} \subset \mathcal{D}$ . And by shifting, we can define all  $\tau_{\leq n}$  and  $\tau_{\geq n}$  for  $n \in \mathbb{Z}$ .

In particular, there are canonical **truncation distinguished triangles**

$$(\tau_{\leq n} E, E, \tau_{\geq n+1} E).$$

$\lrcorner$

**Prop. (4.8.6.3) [Orthogonality].** For  $\mathcal{D} \in t\text{-TriCat}$  and  $E \in \mathcal{D}, n \in \mathbb{Z}_+$ , the following are equivalent:

- $E \in \mathcal{D}^{\geq n+1}$ .
- $\text{Hom}(A, E) = 0$  for any  $A \in \mathcal{D}^{\leq n}$ .

$\lrcorner$

*Proof:* We may assume  $n = 0$ . And  $1 \rightarrow 2$  follows from the definition. For  $2 \rightarrow 1$ , it suffices to show that  $\tau_{\leq n} E = 0$ . For this, use the adjunction

$$\text{Hom}(\tau_{\leq 0} E, \tau_{\leq 0} E) = \text{Hom}(\tau_{\leq 0} E, E) = 0.$$

$\square$

**Cor. (4.8.6.4) [Extension Property].** Suppose  $(X, Y, Z)$  is a distinguished triangle and  $n \in \mathbb{Z}$ .

- If  $X, Z \in \mathcal{D}^{\leq n}$ , then  $Y \in \mathcal{D}^{\leq n}$ .
- If  $X \in \mathcal{D}^{\leq n}$  and  $Y \in \mathcal{D}^{\leq n+1}$ , then  $Z \in \mathcal{D}^{\leq n}$ .

┘

*Proof:* These follow from the orthogonality property above.  $\square$

**Cor. (4.8.6.5).** Let  $\mathcal{D} \in \text{TriCat}$  and  $\alpha, \beta$  are two T-structures s.t.  ${}^\alpha\mathcal{D}^{\leq 0} \subset {}^\beta\mathcal{D}^{\leq 0}$  and  ${}^\alpha\mathcal{D}^{\geq 0} \subset {}^\beta\mathcal{D}^{\geq 0}$ , then the two T-structures are the same.  $\square$

**Prop. (4.8.6.6) [Compatibility].** Let  $\mathcal{D}$  be a triangulated category with a T-structure, then for  $m, n \in \mathbb{Z}$ ,

$$\tau_{\geq m}(\mathcal{D}^{\leq n}) \subset \mathcal{D}^{\leq n}, \quad \tau_{\leq m}(\mathcal{D}^{\geq n}) \subset \mathcal{D}^{\geq n}.$$

And if  $m > n$ ,

$$\tau_{\geq m}(\mathcal{D}^{\leq n}) = \tau_{\leq n}(\mathcal{D}^{\geq m}) = 0.$$

┘

*Proof:* If  $m > n$  and  $X \in \mathcal{D}^{\leq n}$ , then  $\text{Hom}(X, \tau_{\geq m}(X)) = 0$ , and then by adjunction  $\text{Hom}(\tau_{\geq m}(X), \tau_{\geq m}(X)) = 0$ , so  $\tau_{\geq m}(X) = 0$ .

If  $m \leq n$  and  $X \in \mathcal{D}^{\leq n}$ , then there is a truncation distinguished triangle  $(\tau_{\leq m-1}X, X, \tau_{\geq m}X)$ , so it follows from the extension property 2 that  $\tau_{\geq m}X \in \mathcal{D}^{\leq n}$ .  $\square$

**Def. (4.8.6.7) [Cores].** For  $\mathcal{D} \in t\text{-TriCat}$ , the **core** of  $\mathcal{D}$  is defined to be the full subcategory

$$\mathcal{D}^\heartsuit \triangleq \mathcal{D}^{\geq 0} \cap \mathcal{D}^{\leq 0} \subset \mathcal{D}.$$

┘

**Prop. (4.8.6.8) [Cores are Abelian Categories].** If  $\mathcal{D} \in t\text{-TriCat}$ , then  $\mathcal{D}^\heartsuit$  is an Abelian category, and a sequence

$$0 \rightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \rightarrow 0$$

in  $\mathcal{D}^\heartsuit$  is exact iff there exists some distinguished triangle  $(X[0], Y[0], Z[0], u, v, w)$  in  $\mathcal{D}$ .  $\square$

*Proof:* ?  $\square$

**Prop. (4.8.6.9).** For  $\mathcal{D} \in t\text{-TriCat}$  and  $X, Z \in \mathcal{D}^\heartsuit$ ,

$$\text{Ext}_{\mathcal{D}^\heartsuit}^1(Z, X) = \text{Hom}(Z, X[1]).$$

┘

*Proof:* ?  $\square$

**Prop. (4.8.6.10) [Core of Bounded T-structures].** Let  $\mathcal{D} \subset \text{TriCat}$  and  $\mathcal{A} \subset \mathcal{D}$  a full additive subcategory, then  $\mathcal{A}$  is the core of a bounded T-structure iff the following conditions are true:

- If  $k_1 > k_2 \in \mathbb{Z}$ , then  $\text{Hom}_{\mathcal{D}}(A[k_1], B[k_2]) = 0$  for any  $A, B \in \mathcal{A}$ .
- For any  $E \neq 0 \in \mathcal{D}$ , there is a finite sequence of integers  $k_1 > k_1 > \dots > k_n \in \mathbb{Z}$  and a collection of triangles

$$\begin{array}{ccccccc} 0 = E_0 & \xrightarrow{\quad} & E_1 & \longrightarrow & \dots & \longrightarrow & E_{n-1} & \xrightarrow{\quad} & E_n = E \\ & \swarrow \text{dotted} & \searrow & & & & \swarrow \text{dotted} & \searrow & \\ & & A_1 & & & & & & A_n \end{array}$$

with  $A_i \in \mathcal{A}[k_i]$  for any  $i \in [n]_+$ .

*Proof:* If  $\mathcal{A}$  is the core of a bounded T-structure, then clearly these two conditions are true. Conversely, these conditions suffice to verify the conditions in (4.8.6.1).  $\square$

**Def. (4.8.6.11)[Cohomology Functors].** For  $\mathcal{D} \in t\text{-TriCat}$  and  $n \in \mathbb{Z}$ , the  $n$ -th cohomology functor is defined to be

$$H^n : \mathcal{D} \rightarrow \mathcal{D}^\vee : X \mapsto \tau_{\leq 0} \tau_{\geq 0}(X[n]) = (\tau_{\leq n} \tau_{\geq n} X)[n].$$

### Slicings

**Def. (4.8.6.12)[Slicings].**

**Prop. (4.8.6.13).** Let  $\mathcal{P}$  be a slicing of a triangulated category  $\mathcal{D} \in \text{TriCat}$ , then for any interval  $I \subset \mathbb{R}$  of length  $< 1$ , the full subcategory  $\mathcal{P}(I) \subset \mathcal{D}$  is quasi-Abelian.

## 7 Chain Complexes

**Def. (4.8.7.1)[Chain Complexes].** A **chain complex** over an additive category  $\mathcal{A}$  is  $?$ . The category of complexes over  $\mathcal{A}$  is denoted by  $\text{Ch}^{\mathbb{Z}}(\mathcal{A})$ .

**Prop. (4.8.7.2).** For any Abelian category  $\mathcal{A}$ ,  $\text{Ch}^{\mathbb{Z}}(\mathcal{A})$  is an Abelian category.

**Prop. (4.8.7.3).** The natural inclusion  $\mathcal{A} \subset \text{Ch}^{\mathbb{Z}}(\mathcal{A})$  embeds  $\mathcal{A}$  as a full subcategory of  $\text{Ch}^{\mathbb{Z}}(\mathcal{A})$ , and  $H^0$  is just the left adjoint.

An object  $K \in \text{Ch}^{\mathbb{Z}}(\mathcal{A})$  is called **discrete** if it is in the essential image of this embedding.

**Def. (4.8.7.4)[Shifting of Complexes].** For  $\mathcal{A} \in \text{AddCat}$ , there is a **shifting of complex functor**  $[1] : \text{Ch}^{\mathbb{Z}}(\mathcal{A}) \rightarrow \text{Ch}^{\mathbb{Z}}(\mathcal{A})$  s.t. for  $K^\bullet \in \text{Ch}^{\mathbb{Z}}(\mathcal{A})$ ,

$$K^\bullet[1]_n = K^\bullet_{n+1}, \quad d_{K^\bullet[1]}^n = -d_{K^\bullet}^{n+1}.$$

**Def. (4.8.7.5)[Truncation of Complexes].** Let  $\mathcal{A}$  be an Abelian category and  $A^\bullet \in \text{Ch}^{\mathbb{Z}}(\mathcal{A})$ , there are several ways to truncate  $A^\bullet$ :

- The **stupid truncation**  $\sigma_{\leq n} A = \dots \rightarrow A_{n-1} \rightarrow A_n \rightarrow 0 \rightarrow \dots$ . There is a morphism  $A \rightarrow \sigma_{\leq n} A$ .
- The stupid truncation  $\sigma_{\geq n} A = \dots \rightarrow 0 \rightarrow A_n \rightarrow A_{n+1} \rightarrow \dots$ . There is a morphism  $\sigma_{\geq n} A \rightarrow A$ .
- The **canonical truncation**  $\tau_{\leq n} A = \dots \rightarrow A_{n-1} \rightarrow \ker(d_n) \rightarrow 0 \rightarrow \dots$ . There is a natural morphism  $\tau_{\leq n} A \rightarrow A$  that induces isomorphism on cohomology groups on degree  $\leq n$ .
- The canonical truncation  $\tau_{\geq n} A = \dots \rightarrow 0 \rightarrow \text{Coker}(d_{n-1}) \rightarrow A_{n+1} \rightarrow \dots$ . There is a natural morphism  $A \rightarrow \tau_{\geq n} A$  that induces isomorphism on cohomology groups on degree  $\geq n$ .

**Cor. (4.8.7.6).** There are exact sequences of complexes

$$0 \rightarrow \tau_{\leq n} A^\bullet \rightarrow A^\bullet \rightarrow \tau_{\geq n+1} A^\bullet \rightarrow 0$$

$$0 \rightarrow \sigma_{\geq n+1} A^\bullet \rightarrow A^\bullet \rightarrow \sigma_{\leq n} A^\bullet \rightarrow 0$$

**Def. (4.8.7.7)[Cone & Cylinder].** The **mapping cone** of  $f : K^\bullet \rightarrow L^\bullet$  is the complex  $C(f)^\bullet$  that:

$$C(f) = K[1]^\bullet \oplus L^\bullet, \quad d(k^{i+1}, l^i) = (-d_K k^{i+1}, f(k^{i+1}) + d_L l^i)$$

The **mapping cylinder** of  $f : K^\bullet \rightarrow L^\bullet$  is the complex  $\text{Cyl}(f)$  that:

$$\text{Cyl}(f) = K^\bullet \oplus K[1]^\bullet \oplus L^\bullet, \quad d(k^i, k^{i+1}, l^i) = (d_K k^i - k^{i+1}, -d_K k^{i+1}, f(k^{i+1}) + d_L l^i)$$

It is a shame I haven't see clearly the similarity of this with the topological cone and cylinder, should study it further.  $\perp$

**Def. (4.8.7.8)[Double Complexes].** A **double complex** over an Abelian category is a complex over  $\text{Comp}(\mathcal{A})$  (4.8.7.2).  $\perp$

**Def. (4.8.7.9)[Totalization].** Given a double complex  $K^{\bullet,\bullet}$  over an Abelian category  $\mathcal{A}$ , the associated **total complexes** is defined to be

$$\begin{aligned} (\text{Tot}^\Pi(K))^n &= \prod_{n=p+q} K^{p,q}, & d^n &= \prod_{n=p+q} (d_1^{p,q} + (-1)^p d_2^{p,q}) \\ (\text{Tot}^\oplus(K))^n &= \bigoplus_{n=p+q} K^{p,q}, & d^n &= \sum_{n=p+q} (d_1^{p,q} + (-1)^p d_2^{p,q}) \end{aligned}$$

$\perp$

**Def. (4.8.7.10)[Hom Complexes].** Let  $\mathcal{A}$  be an Abelian category and  $P^\bullet, Q^\bullet \in K(\mathcal{A})$ , we define **Hom complex**  $\text{Hom}^\bullet(P^\bullet, Q^\bullet)$  to be

$$\text{Hom}^n(P^\bullet, Q^\bullet) = \prod \text{Hom}_i(P^i, Q^{n+i}),$$

with the differential giving by  $d(\{f_k\})_i = \{df_i - (-1)^i f_{i+1}d\}$  and suitable signatures.

It is clear that  $H^n(\text{Hom}^\bullet(P^\bullet, Q^\bullet)) = \text{Hom}_{K(\mathcal{A})}(P^\bullet, Q^\bullet[n])$ .  $\perp$

### Homotopy Category $K(\mathcal{A})$

**Def. (4.8.7.11)[Homotopies of Complexes].**  $\perp$

**Def. (4.8.7.12)[ $K(\mathcal{A})$ ].** Let  $\mathcal{A}$  be an additive category, the **homotopy category of complexes**  $K(\mathcal{A})$  is the category whose objects are complexes in  $\mathcal{A}$  and whose morphisms are homotopy classes of morphism between complexes.  $\perp$

**Prop. (4.8.7.13)[Distinguished Triangle of  $K^*(\mathcal{A})$ ].** For any morphism  $K^\bullet \rightarrow L^\bullet$ , there exists a termwise-splitting exact sequence of Complexes commuting in  $K(\mathcal{A})$ .

$$\begin{array}{ccccccc} K^\bullet & \longrightarrow & L^\bullet & & & & \\ \parallel & & \downarrow \alpha & & & & \\ 0 & \longrightarrow & K^\bullet & \longrightarrow & \text{Cyl}(f) & \longrightarrow & C(f) \longrightarrow 0 \\ & & & & \downarrow \beta & & \parallel \\ & & 0 & \longrightarrow & L^\bullet & \longrightarrow & C(f) \longrightarrow K^\bullet[1] \longrightarrow 0 \end{array}$$

where  $\beta\alpha = \text{id}$  and  $\alpha\beta \sim \text{id}$ . And  $K^\bullet \rightarrow L^\bullet \rightarrow C(f) \rightarrow K^\bullet[1]$  is called a distinguished triangle. Any exact triple of complexes in  $\text{Kom}(\mathcal{A})$  is quasi-isomorphic to a distinguished triangle. In fact, we can define the distinguished triangle in  $K(\mathcal{A})$  as that induced by a split exact sequence, Cf.[Sta]014L.

Notice all this can imitate the similar parallel construction in the category of topological spaces.

$\perp$

*Proof:* Cf.[Gelfand P157] □

**Cor.(4.8.7.14)[Long Exact Sequences].** A distinguished triangle will induce a long exact sequence of cohomology groups, for this, just need to verify that the  $\delta$ -homomorphism and the morphism  $C(f) \rightarrow K^\bullet[1]$  induce the same map of cohomology groups. ┘

**Cor.(4.8.7.15).** A morphism of complexes  $f : K^\bullet \rightarrow L^\bullet$  is quasi-iso iff  $C(f)$  is acyclic. It is homotopic to 0 iff  $f$  can be extended to a morphism  $C(f) \rightarrow L$ . ┘

**Prop.(4.8.7.16)[ $K(\mathcal{A})$  is Triangulated].** If  $\mathcal{A}$  is an additive category, then  $K^*(\mathcal{A}) \in \text{TriCat}(4.8.4.2)$ , with shifting functors defined in(4.8.7.4) and distinguished triangles defined in(4.8.7.13).

And an additive functor will induce exact functor between  $K^*$  because distinguished is split. ┘

*Proof:* Cf.[Sta]014S. or [Gelfand P246]. □

**Prop.(4.8.7.17).** An additive functor  $F : \mathcal{A} \rightarrow \mathcal{B}$  between additive categories induces an exact functor  $K(\mathcal{A}) \rightarrow K(\mathcal{B})$ . Moreover, this functor maps quasi-isomorphisms to quasi-isomorphisms(4.8.8.1). ┘

*Proof:* Cf.[Sta]014X. □

**Def.(4.8.7.18)[Bounded Subcategories].** ┘

**Prop.(4.8.7.19).** Let  $\mathcal{A}$  be an Abelian category,

- If  $A^\bullet \subset K^+(\mathcal{A})$ , then  $\tau_{\leq n}(A^\bullet) \rightarrow A^\bullet$  is a quasi-isomorphism for  $n$  sufficiently large.
  - If  $A^\bullet \subset K^-(\mathcal{A})$ , then  $A^\bullet \rightarrow \tau_{\geq n}A^\bullet$  is a quasi-isomorphism for  $n$  sufficiently small.
- ┘

## Unbounded Complexes

**Lemma(4.8.7.20)[Left Resolutions of Unbounded Complexes].** Let  $\mathcal{A}$  be an Abelian category and  $\mathcal{P}$  be a subset of objects of  $\mathcal{A}$ . Assume that every object of  $\mathcal{A}$  is a quotient of an object of  $\mathcal{P}$ , then for any complex  $K^\bullet$ , there exists a commutative diagram

$$\begin{array}{ccccccc} P_1^\bullet & \longrightarrow & P_2^\bullet & \longrightarrow & \dots & & \\ \downarrow & & \downarrow & & & & \\ \tau_{\leq 1}K^\bullet & \longrightarrow & \tau_{\leq 2}K^\bullet & \longrightarrow & \dots & & \end{array}$$

where the vertical arrows are quasi-isomorphisms, and each  $P_n^\bullet$  is a bounded above complex with terms in  $\mathcal{P}$ , and each  $P_n^\bullet \rightarrow P_{n+1}^\bullet$  are termwise-split injections and the cokernel is also a complex with terms in  $\mathcal{P}$ . ┘

*Proof:* Cf.[Sta]06XX. □

## 8 Cohomology of Complexes

**Def.(4.8.8.1)[Quasi-isomorphism].** ┘

**Prop.(4.8.8.2)[Five lemma].** In an Abelian category, if there is a diagram

$$\begin{array}{ccccccccc} * & \longrightarrow & * & \longrightarrow & * & \longrightarrow & * & \longrightarrow & * \\ \downarrow s & & \downarrow g & & \downarrow f & & \downarrow h & & \downarrow i \\ * & \longrightarrow & * & \longrightarrow & * & \longrightarrow & * & \longrightarrow & * \end{array}$$

Where the rows are exact and  $g, h$  are isomorphisms. If  $i$  is injective, then  $f$  is surjective; if  $s$  is surjective, then  $f$  is injective. ┘



*Proof:* Rotate the diagram counterclockwise  $90^\circ$ . Then use the two different filtration both converge (4.10.7.8).  $\square$

**Prop. (4.8.8.3)[Snake lemma].** In an Abelian category, if there is a diagram

$$\begin{array}{ccccccc} & & * & \xrightarrow{i} & * & \longrightarrow & * \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & * & \longrightarrow & * & \xrightarrow{s} & * \end{array}$$

where the rows are exact, then there is a long exact sequence

$$\ker f \rightarrow \ker g \rightarrow \ker h \rightarrow \operatorname{Coker} f \rightarrow \operatorname{Coker} g \rightarrow \operatorname{Coker} h$$

And if  $i$  is injective, then the first one is injective; if  $s$  is surjective, then the last one is surjective.  $\lrcorner$

*Proof:* Rotate the diagram counterclockwise  $90^\circ$ . Then use the two different filtration both converge (4.10.7.8).  $\square$

**Cor. (4.8.8.4).** In an Abelian category, if  $f : A \rightarrow B, g : B \rightarrow C$ , then there is a long exact sequence:

$$0 \rightarrow \ker f \rightarrow \ker gf \rightarrow \ker g \rightarrow \operatorname{Coker} f \rightarrow \operatorname{Coker} gf \rightarrow \operatorname{Coker} g \rightarrow 0.$$

$\lrcorner$

*Proof:* Use snake lemma(as modules), there is a diagrams:

$$\begin{array}{ccccccc} & & A & \xrightarrow{f} & B & \longrightarrow & \operatorname{Coker} f \longrightarrow 0 \\ & & \downarrow gf & & \downarrow g & & \downarrow \\ 0 & \longrightarrow & C & = & C & \longrightarrow & 0 \end{array}$$

So by Snake lemma,

$$\ker gf \rightarrow \ker g \rightarrow \operatorname{Coker} f \rightarrow \operatorname{Coker} gf \rightarrow \operatorname{Coker} g \rightarrow 0.$$

As Abelian category is dual, we can do this dually to get:

$$0 \rightarrow \ker f \rightarrow \ker gf \rightarrow \ker g \rightarrow \operatorname{Coker} f \rightarrow \operatorname{Coker} gf.$$

They splint together to get the desired long exact sequence.  $\square$

**Prop. (4.8.8.5).** For a  $3 \times 3$  diagram of complexes, the connection homomorphism satisfies an anti-commutative diagram:

$$\begin{array}{ccc} H^{q-1}(Z'') & \xrightarrow{\delta} & H^q(X'') \\ \downarrow \delta & & \downarrow -\delta \\ H^q(Z) & \xrightarrow{\delta} & H^{q+1}(X) \end{array}$$

by (4.8.4.16) as the category  $K(\mathcal{A})$  is triangulated.  $\lrcorner$

**Prop. (4.8.8.6)[Universal Coefficient Theorem].** Should be somewhere in [Weibel].  $\lrcorner$

**Def. (4.8.8.7) [Herbrand Quotient].** For a complex of  $R$ -modules cyclic of order 2, we define the **additive Herbrand quotient** as  $\text{length}_R(H^0) - \text{length}_R(H^1)$ , when both are definable and the **multiplicative Herbrand quotient** as  $|H^0|/|H^1|$  when they are both finite.  $\lrcorner$

**Prop. (4.8.8.8).** For an exact sequence  $0 \rightarrow M \rightarrow N \rightarrow K \rightarrow 0$  of complexes of cyclic order 2, we have  $h(N) = h(M) + h(K)$  and  $h^*(N) = h^*(M)h^*(K)$  in the sense that if two of them are definable, then so is the third. This is an easy consequence of long exact sequence.  $\lrcorner$

**Prop. (4.8.8.9).** If each term of this complex has finite length, then  $h(M) = 0$ . If each term is finite, then  $h^*(M) = 0$ . This is a consequence of isomorphism theorem. So we have, if a morphism of complexes has kernel and cokernel finite, then it induces an isomorphism on  $h$  or  $h^*$ .  $\lrcorner$

*Proof:*

$\square$

## 9 Injectives & Projectives

**Remark (4.8.9.1).** The use of injection resolutions can be replaced by the use of  $\infty$ -categories. ??  $\lrcorner$

**Def. (4.8.9.2) [Injective Objects].** An **injective object** in a Abelian category is a  $I$  s.t.  $\text{Hom}(-, I)$  is an exact functor, equivalently, maps to  $I$  can be extended along injections.

A **projective object** in a Abelian category is a  $I$  s.t.  $\text{Hom}(I, -)$  is an exact functor, equivalently, maps to  $I$  can be pulled back along surjections.  $\lrcorner$

**Prop. (4.8.9.3).** Product of injective elements are injective, sums of projective elements are projective.  $\lrcorner$

**Prop. (4.8.9.4).** In an Abelian category, the direct summand of a projective object is projective. (The summand has definition in an Abelian category).  $\lrcorner$

**Prop. (4.8.9.5).** If a functor  $f$  between Abelian categories is left adjoint to an exact functor, then it preserves injectives. Dually for projectives.  $\lrcorner$

**Prop. (4.8.9.6).** If  $A$  is an Abelian category, the chain complex category  $Ch(A)$  is abelian by (4.8.3.9). A chain complex  $P$  is projective iff it is a split exact complex of projective objects. The same is true by dual argument for injectives.  $\lrcorner$

*Proof:* If  $K$  is projective, use the surjection  $C(\text{id}_K) \rightarrow K[1]$ , there is a homotopy between  $\text{id}_K$  and 0. Thus we have  $x = dhx + hdx$ . And if  $dhx = hdy$ , then  $dhd y = 0$ , thus  $dy = 0$ , so  $K = dhK \oplus hdK$  and thus  $K[n] = B_n \oplus B_{n+1}$ . Thus  $K$  is a direct product of  $0 \rightarrow B \rightarrow B \rightarrow 0$ . And this one is projective if  $B$  is projective.  $\square$

**Prop. (4.8.9.7) [Check Injectives].** In a Grothendieck Abelian category with generator  $U$ , an object is injective iff it is extendable over subobjects of  $U$ . (AB5 assures we can extend by Zorn's lemma. Then use GEN, Cf. [Sta]079G ?). If it is a family of objects, it suffices to extend over each one of them.  $\lrcorner$

*Proof:*

$\square$

### Injective Resolutions

**Prop. (4.8.9.8) [Horseshoe Lemma].** For an exact sequence  $0 \rightarrow X_1 \rightarrow X \rightarrow X_2 \rightarrow 0$  and an injective resolution of  $X_1$  and  $X_2$ , there is an injective resolution of  $X$  commuting with them. (Choose them one-by-one, in fact,  $I_n = I_n^1 \oplus I_n^2$  using the injectivity of  $I_n^1$ . Snake lemma told us that the cokernel is an exact sequence, use that to define the next one.  $\lrcorner$

**Prop. (4.8.9.9).** For two liftings of morphisms  $X_1 \rightarrow Y_1$  and  $X_2 \rightarrow Y_2$ , there is a lifting of the morphism  $X \rightarrow Y$  compatible with that. Cf. [Weibel P2.4.6].  $\lrcorner$

**Prop. (4.8.9.10) [Cartan-Eilenberg Resolution].** For a complex  $K \in K^+(\mathcal{A})$ , a **Cartan-Eilenberg resolution** of  $K$  consists of a 2-complex  $I^{\bullet, \bullet}$  and a map of complexes  $K \rightarrow I^{\bullet, 0}$  that the induced complexes:

$$\begin{aligned} 0 \rightarrow K^i \rightarrow I^{i,0} \rightarrow I^{i,2} \rightarrow \dots \\ 0 \rightarrow B^i(K) \rightarrow B_x^i(I^{\bullet,0}) \rightarrow B_x^i(I^{\bullet,1}) \rightarrow \dots \\ 0 \rightarrow Z^i(K) \rightarrow Z_x^i(I^{\bullet,0}) \rightarrow Z_x^i(I^{\bullet,1}) \rightarrow \dots \\ 0 \rightarrow H^i(K) \rightarrow B_x^i(I^{\bullet,0}) \rightarrow H_x^i(I^{\bullet,1}) \rightarrow \dots \end{aligned}$$

are all injective resolutions, and the exact sequences

$$\begin{aligned} 0 \rightarrow B_x^i(I^{\bullet,j}) \rightarrow Z_x^i(I^{\bullet,j}) \rightarrow H_x^i(I^{\bullet,j}) \rightarrow 0 \\ 0 \rightarrow Z_x^i(I^{\bullet,j}) \rightarrow I^{\bullet,j} \rightarrow Z_x^i(I^{\bullet,j}) \rightarrow 0 \end{aligned}$$

split.

Then if  $\mathcal{I}_{\mathcal{B}}$  is sufficiently large, for any  $K$  in  $K(\mathcal{B})$  there is a Cartan-Eilenberg resolution.  $\lrcorner$

*Proof:* Cf. [Gelfand P210], [Weibel P146]. ?  $\square$

**Cor. (4.8.9.11).** For a CE resolution of a complex  $K \in K^+(\mathcal{B})$ , the spectral sequence can be applied and shows  $K \rightarrow \text{Tot}(L)$  is a quasi-isomorphism, i.e.  $\text{Tot}(L)$  is an injective resolution of  $K$ .  $\lrcorner$

**Cor. (4.8.9.12) [Functoriality of Cartan-Eilenberg Resolution].** If  $f : A \rightarrow B$  is a chain map and  $A \rightarrow P, B \rightarrow Q$  are Cartan-Eilenberg resolutions, then there is a double complex map  $\tilde{f} : P \rightarrow Q$  extending  $f$ . And if  $f$  is homotopic to  $g$ , then  $\tilde{f}$  is homotopic to  $\tilde{g}$ . In other words, we have a functor  $K(\mathcal{A}) \rightarrow K(\mathcal{I}_{\mathcal{A}}^{\bullet, \bullet})$

In particular, for any two Cartan-Eilenberg resolutions  $P, Q$  of  $A$  and an additive functor  $F$ , the chain complex  $\text{Tot}^{\Pi}(F(P))$  and  $\text{Tot}^{\Pi}(F(Q))$  are chain homotopy equivalent.  $\lrcorner$

**Def. (4.8.9.13) [Injective Amplitude].** Let  $\mathcal{A}$  be an Abelian category with sufficiently injectives, then  $K \in D(\mathcal{A})$  is said to have **finite injective dimension** if  $K \subset D^b(\mathcal{I}_{\mathcal{A}})$ . It is said to have **injective amplitude** in  $[a, b]$  iff  $K \subset D^{[a,b]}(\mathcal{I}_{\mathcal{A}})$ .  $\lrcorner$

**Prop. (4.8.9.14).** Suppose  $\mathcal{A}^{\bullet} \rightarrow \mathcal{I}^{\bullet}$  is an injective resolution of sheaves on  $(\mathcal{C}, \mathcal{O})$ , then the induced map of presheaves with values in  $D_{\infty}(\mathbb{Z})$ :

$$|\mathcal{A}^{\bullet}| \rightarrow |\mathcal{I}^{\bullet}|$$

identifies  $|\mathcal{I}^{\bullet}|$  with the shiftification of  $|\mathcal{A}^{\bullet}|$ .  $\lrcorner$

*Proof:* ?  $\square$

## 10 Tensor Category Case

In this subsection we consider complexes over a tensor category (4.2.1.2)  $\mathcal{A}$ .

**Lemma (4.8.10.1) [Koszul Sign Rule].** There is an isomorphism of complexes

$$\sigma : C^\bullet \otimes D^\bullet \rightarrow D^\bullet \otimes C^\bullet : \sigma(x \otimes y) = (-1)^{mn} y \otimes x$$

where  $x \in C^m$  and  $y \in D^n$ . ┘

*Proof:*

$$\partial(\sigma(x \otimes y)) = \partial((-1)^{mn} y \otimes x) = (-1)^{mn} (\partial y \otimes x) + (-1)^{mn+n} (y \otimes \partial x) = (-1)^n \sigma(x \otimes \partial y) + \sigma(\partial x \otimes y) = \sigma(\partial(x \otimes y))$$

□

**Prop. (4.8.10.2) [Commutative Monoidal Structure on  $\text{Ch}^{\mathbb{Z}}(\mathcal{A})$ ].** The functor

$$\text{Ch}^{\mathbb{Z}}(\mathcal{A}) \times \text{Ch}^{\mathbb{Z}}(\mathcal{A}) \rightarrow \text{Ch}^{\mathbb{Z}}(\mathcal{A}) : (C^\bullet, D^\bullet) \mapsto \text{Tot}^\oplus(C^\bullet \otimes D^\bullet)$$

endows  $\text{Ch}^{\mathbb{Z}}(\mathcal{A})$  with a (non-strict) commutative monoidal structure. ┘

*Proof:* Commutativity follows from (4.8.10.1). Associativity follows from the definition of totalization (4.8.7.9). □

**Prop. (4.8.10.3) [Commutative Monoidal Structure on  $K^*(\mathcal{A})$ ].** This monoidal functor descends to a strict monoidal structure on  $K^*(\mathcal{A})$ . ┘

*Proof:* □

**Prop. (4.8.10.4) [Tensoring is Exact].** Let  $L^\bullet \in K(\mathcal{A})$ , then tensoring functor  $\text{Tot}^\oplus(- \oplus L^\bullet)$  is an exact functor between triangulated categories. ┘

*Proof:* □

**Prop. (4.8.10.5).** An  $R$ -module  $I$  is injective iff for any injective homomorphism from  $I$  to any  $R$ -module splits. ┘

*Proof:* The critical point is that we can always embed  $I$  to an injective hull  $J$  by (4.8.3.23), then  $J = I \oplus J'$ , so  $I$  is clearly injective. □

**Prop. (4.8.10.6) [K-Injective under Change of Rings].** If  $R \rightarrow S$  is a ring map, then

1. If  $R \rightarrow S$  is flat and  $I^\bullet$  is a  $K$ -injective complex of  $S$ -modules, then  $I^\bullet$  is  $K$ -injective as a complex of  $R$ -modules.
2. If  $R \rightarrow S$  is surjective and  $I^\bullet$  is a complex of  $S$ -modules that is  $K$ -injective as a complex of  $R$ -modules, then it is  $K$ -injective as a complex of  $S$ -modules.
3. If  $I^\bullet$  is a  $K$ -injective complex of  $R$ -modules, then  $\text{Hom}_R(S, I^\bullet)$  is  $K$ -injective as a complex of  $S$ -modules.

┘

*Proof:* 1: This is because  $\text{Hom}_{K(R)}(M^\bullet, I^\bullet) = \text{Hom}_{K(S)}(M^\bullet \otimes_R S, I^\bullet)$  and (4.10.2.1), as tensoring  $S$  is exact.

2: This is because  $\text{Hom}_{K(R)}(N^\bullet, I^\bullet) = \text{Hom}_{K(S)}(N^\bullet, I^\bullet)$  for a complex of  $S$ -modules  $N^\bullet$  and (4.10.2.1).

3: This is because  $\text{Hom}_{K(S)}(N^\bullet, \text{Hom}_R(S, I^\bullet)) = \text{Hom}_{K(R)}(N^\bullet, I^\bullet)$ , and (4.10.2.1). □

## 4.9 Stable $\infty$ -Categories

References are [nLab], [Lur11]

**Notation(4.9.0.1).**

- Use notations defined in  $\infty$ -Categories.

┘

### 1 Stable $\infty$ -Categories

**Def.(4.9.1.1) [Zero Objects].** A **zero object** in an  $\infty$ -category is an object that is both initial and final. A **pointed  $\infty$ -category** is an  $\infty$ -category with a zero object.

In a pointed  $\infty$ -category  $\mathcal{C}$ , the subcategory of zero objects is a trivial Kan complex. And for any  $x, y \in \mathcal{C}$ ,  $\text{Map}(x, y)$  (4.7.1.20) is a trivial Kan complex, by (4.5.8.3) and (4.7.1.21). So there is a **zero morphism**  $0 : x \rightarrow y$  that is unique up to contractible choice.

┘

**Def.(4.9.1.2) [ $\infty$ -Category of Pointed Objects].** For  $\mathcal{C} \in \text{Cat}_\infty$  with a final object  $*$ , denote  $\mathcal{C}^{\text{pt}} = \mathcal{C}_{*/}$ , called the  **$\infty$ -category of pointed objects** in  $\mathcal{C}$ . Then  $\mathcal{C}_*$  is pointed, and there is a map  $(-)_- : \mathcal{C}_* \rightarrow \mathcal{C}$ , and there is also a map  $(-)_+ : \mathcal{C} \rightarrow \mathcal{C}_*$  given by adding a final object. Then we have an adjunction between  $\infty$ -categories:

$$(-)_+ : \mathcal{C} \rightleftarrows \mathcal{C}^{\text{pt}} : (-)_-$$

┘

*Proof:*

□

**Prop.(4.9.1.3).** Let  $\mathcal{C}$  be a pointed presentable  $\infty$ -category, then evaluation at  $S^0 \in \text{Grpd}_\infty^{\text{pt}}$  induces an equivalence of  $\infty$ -categories

$$\text{Func}^L(\text{Grpd}_\infty^{\text{pt}}, \mathcal{C}) \cong \mathcal{C}.$$

┘

*Proof:*

□

**Def.(4.9.1.4) [Triangles in  $\infty$ -Categories].** A **triangle** is a pointed  $\infty$ -category  $\mathcal{C}$  is a diagram

$$\tau : \Delta^1 \times \Delta^1 \rightarrow \mathcal{C} : \begin{array}{ccc} x & \longrightarrow & y \\ \downarrow & \searrow & \downarrow \\ 0 & \longrightarrow & z \end{array}$$

where  $0$  is a zero object. It is called a **fiber sequence** if this diagram is a pullback diagram, and called a **cofiber sequence** if this diagram is a pushout diagram. Denote  $\text{Tri}(\mathcal{C}) \subset \text{Func}(\Delta^1 \times \Delta^1, \mathcal{C})$  the full sub- $\infty$ -category of triangles.

┘

**Def.(4.9.1.5) [Cofiber Maps and Fiber Maps].** If  $\mathcal{C}$  is a pointed  $\infty$ -category which admits cofibers, then by (4.7.3.16) and (4.7.3.17), there is a **fiber sequence map**

$$\text{Cof} : \text{Func}(\Delta^1, \mathcal{C}) \rightarrow \text{Func}(\Delta^1 \times \Delta^1, \mathcal{C}).$$

unique up to contractible choice. And its composition with  $ev_{(1,1)}$  is also denoted by  $\Sigma : \text{Func}(\Delta^1, \mathcal{C}) \rightarrow \mathcal{C}$ . The dual is true for fibrations.

┘

**Prop. (4.9.1.6) [Cofiber Map Preserves Colimits].** If  $\mathcal{C}$  is a pointed  $\infty$ -category which admits cofibers, choose a zero object  $0$ , then any  $\text{Cof} : \text{Func}(\Delta^1, \mathcal{C}) \rightarrow \mathcal{C}$  is left adjoint to a functor  $\mathcal{C} \rightarrow \mathcal{C}_{0/} \rightarrow \text{Func}(\Delta^1, \mathcal{C})$  which maps each  $C \in \mathcal{C}$  to a morphism  $0 \rightarrow C$ . In particular,  $\text{Cof}$  preserves small colimits by (4.7.4.5). Dually,  $\text{Fib}$  preserves small limits.  $\lrcorner$

*Proof:*  $\square$

**Def. (4.9.1.7) [Suspension and Loop Diagrams].** For  $\mathcal{C} \in \text{Cat}_\infty$ , denote  $\mathcal{C}^\Sigma \subset \text{Func}(\Delta^1 \times \Delta^1, \mathcal{C})$  the full  $\infty$ -subcategory of cofiber sequences of the form

$$\tau : \Delta^1 \times \Delta^1 \rightarrow \mathcal{C} : \begin{array}{ccc} x & \longrightarrow & 0' \\ \downarrow & \searrow & \downarrow \\ 0 & \longrightarrow & z \end{array}$$

where  $0, 0'$  are zero objects.

Dually, define  $\mathcal{C}^\Omega \subset \text{Func}(\Delta^1 \times \Delta^1, \mathcal{C})$  the full  $\infty$ -subcategory of exact triangles of the form

$$\tau : \Delta^1 \times \Delta^1 \rightarrow \mathcal{C} : \begin{array}{ccc} x & \longrightarrow & 0' \\ \downarrow & \searrow & \downarrow \\ 0 & \longrightarrow & z \end{array}$$

where  $0, 0'$  are zero objects.  $\lrcorner$

**Def. (4.9.1.8) [Suspension Functors and Loop Functors].** If  $\mathcal{C} \in \text{Cat}_\infty$  is a pointed  $\infty$ -category with admits cofibers, then same argument as in (4.9.1.5) shows that  $\mathcal{C}^\Sigma \rightarrow \mathcal{C}$  is a trivial Kan fibration, so there is a section  $\Sigma : \mathcal{C} \rightarrow \mathcal{C}^\Sigma$ , called a **suspension functor**. And its composition with  $ev_{(1,1)}$  is also denoted by  $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$ .

Dually, if  $\mathcal{C}$  is a pointed  $\infty$ -category which admits fibers, we can define **loop functors**. If  $\mathcal{C}$  admits both fibers and cofibers, for  $C \in \mathcal{C}$  and  $n \in \mathbb{N}$ , denote  $X[n] = \Sigma^n(X)$ ,  $X[-n] = \Omega^n(X)$ .  $\lrcorner$

**Def. (4.9.1.9) [Stable  $\infty$ -Category].** A **stable  $\infty$ -category** is a pointed  $\infty$ -category  $(\mathcal{C}, 0)$  s.t.

- $\mathcal{C}$  admits cofibers and fibers.
- A triangle in  $\mathcal{C}$  is a fiber sequence iff it is a cofiber sequence.

Notice that  $\mathcal{C}$  is stable iff  $\mathcal{C}^{\text{op}}$  is stable.  $\lrcorner$

**Prop. (4.9.1.10).** If  $\mathcal{C}$  is a pointed category which admits both cofibers and fibers, then there is an adjunction

$$\Sigma : \text{Ho}(\mathcal{C}) \rightleftarrows \text{Ho}(\mathcal{C}) : \Omega.$$

And if  $\mathcal{C}$  is stable, they are mutually inverse to each other.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.9.1.11) [HA.1.1.3.4].** A stable  $\infty$ -category is complete and cocomplete. And pullback squares and pushout squares coincide.  $\lrcorner$

**Prop. (4.9.1.12).** Let  $\mathcal{C}$  be a pointed  $\infty$ -category, then the following are equivalent:

- $\mathcal{C}$  is stable.
- $\mathcal{C}$  admits finite colimits, and the suspension functor  $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$  is an equivalence.
- $\mathcal{C}$  admits finite limits, and the loop functor  $\Omega : \mathcal{C} \rightarrow \mathcal{C}$  is an equivalence.

$\lrcorner$

*Proof:* Cf. [HA, 1.4.2.27].  $\square$

### Homological Algebra

**Prop. (4.9.1.13) [Homotopy Groups].** Let  $\mathcal{C}$  be a pointed  $\infty$ -category that admits cofibers, then for any  $X, Y \in \mathcal{C}$ , there is a bijection

$$\mathrm{Hom}_{\mathrm{Ho}(\mathcal{C})}(\Sigma^n(X), Y) \cong \pi_n \mathrm{Map}_{\mathcal{C}}(X, Y).$$

┘

*Proof:* ?

□

**Def. (4.9.1.14) [Ext Groups].** For a pointed  $\infty$ -category  $\mathcal{C}$  and  $X, Y \in \mathcal{C}$ , denote  $\mathrm{Ext}^n(X, Y) = \mathrm{Hom}_{\mathrm{Ho}(\mathcal{C})}(X[-n], Y)$  (4.9.1.8), called the  $n$ -th **Ext group**. ┘

**Prop. (4.9.1.15).** Let  $\mathcal{C}$  be a pointed  $\infty$ -category that admits cofibers, then any diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & 0 \\ \downarrow & & \downarrow f' \\ 0' & \longrightarrow & Y \end{array}$$

in  $\mathcal{C}$  corresponds to a homotopy class of morphisms  $\theta \in \mathrm{Hom}_{\mathrm{Ho}(\mathcal{C})}(X[1], Y)$ . Then in this way, the diagram

$$\begin{array}{ccc} X & \xrightarrow{f'} & 0' \\ \downarrow & & \downarrow f' \\ 0 & \longrightarrow & Y \end{array}$$

corresponds to  $-\theta \in \mathrm{Hom}_{\mathrm{Ho}(\mathcal{C})}(X[1], Y)$ , where the group structure is given by (4.9.1.13). ┘

*Proof:* Cf. [Lur11]P25. □

**Lemma (4.9.1.16).** Let  $\mathcal{C}$  be a pointed  $\infty$ -category that admits cofibers, and the suspension functor  $\Sigma$  is an equivalence, then  $\mathrm{Ho}(\mathcal{C})$  is an additive category (4.8.1.7). ┘

*Proof:* It follows from (4.9.1.13) and the compatibility of group structures on  $\pi_n$  that  $\mathcal{C}$  is preadditive. To show it is additive, Cf. [Lur11]P24. ? □

**Def. (4.9.1.17) [Distinguished Triangles].** Let  $\mathcal{C}$  be a pointed  $\infty$ -category which admits cofibers, then a diagram  $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$  in  $\mathrm{Ho}(\mathcal{C})$  is called a **distinguished triangle** if there exists a diagram  $\Delta^1 \times \Delta^2 \rightarrow \mathcal{C}$  as shown:

$$\begin{array}{ccccc} X & \xrightarrow{\tilde{f}} & Y & \longrightarrow & 0 \\ \downarrow & & \downarrow \tilde{g} & & \downarrow \\ 0' & \longrightarrow & Z & \xrightarrow{\tilde{h}} & W \end{array}$$

s.t.

- Both  $0, 0'$  are zeros.
- both square are pushout squares.
- $\tilde{f}$  lifts  $f$ ,  $\tilde{g}$  lifts  $g$ .
- $h : Z \rightarrow X[1]$  equals  $\tilde{h}$  composed with an equivalence  $W \cong X[1]$  determined by the outer rectangle.

┘

**Prop. (4.9.1.18) [Stable  $\infty$ -Categories and Triangulated Categories].** Let  $\mathcal{C}$  be a stable  $\infty$ -category, then the translation functor defined (4.9.1.8) and distinguished triangles defined in (4.9.1.17) endow  $\mathrm{Ho}(\mathcal{C})$  with the structure of a triangulated category (4.8.4.2). ┘

*Proof:* Cf. [Lur11]P27. ┐

**Def. (4.9.1.19) [Exact Functors].** An **exact functor** between stable  $\infty$ -categories is a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  that is left and right exact and preserves fiber sequences. ┘

**Prop. (4.9.1.20).** Let  $\mathrm{Cat}_{\infty}^{\mathrm{Ex}} \subset \mathrm{Cat}_{\infty}$  be the subcategory of all stable  $\infty$ -categories and exact functors, then it admits all small limits and small filtered colimits, and they are preserved by the inclusion. ┘

*Proof:* Cf. [HA1.1.4.4., 1.1.4.6.] ┐

**Def. (4.9.1.21) [T-Structures].** A  $T$ -structure on a stable  $\infty$ -category  $\mathcal{C}$  is a pair of full subcategories  $\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0}$  that ┘

**Prop. (4.9.1.22).** For any  $n \in \mathbb{Z}$ ,  $\mathcal{C}_{\leq n} \subset \mathcal{C}$  is a localization, thus admits a left adjoint  $\tau_{\leq n}$ , called the **truncation functor**. Dually for  $\tau_{\geq n}$ . ┘

*Proof:* Cf. [HA, P1.2.1.5]. ┐

**Def. (4.9.1.23) [Heart].** The **heart**  $\mathcal{C}^{\heartsuit} \subset \mathcal{C}$  is defined to be the full subcategory of  $\mathcal{C}_{\leq 0} \cap \mathcal{C}_{\geq 0} \subset \mathcal{C}$ . ┘

### Spectra Objects and Stabilization

**Def. (4.9.1.24) [Spectra].** A **spectrum** is an object in the universal  $(\infty, 1)$ -category  $\mathrm{Sp} = \mathrm{Sp}(\mathrm{Top}) \cong \mathrm{Sp}(\mathrm{Grpd}_{\infty})$  ┘

**Def. (4.9.1.25) [Homology Groups of Spectra].** ┘

**Def. (4.9.1.26) [Connective Spectra].** A **connective spectrum** is a spectrum (4.9.1.24) that the negative homotopy groups vanish. ┘

**Prop. (4.9.1.27).** There are equivalences between the following categories **?**:

- connected spectra.
- infinite loop spaces.
- group-like  $\mathbb{E}_{\infty}$ -spaces.

┘

*Proof:* ┐

┐

**Def. (4.9.1.28) [ $\mathbb{E}_{\infty}$ -Rings].** An  **$\mathbb{E}_{\infty}$ -ring** is an object in  $\mathcal{C}\mathrm{Mon}_{\infty}(\mathrm{Sp}(\mathrm{Grpd}_{\infty}))$  (4.9.1.24). ┘

**Cor. (4.9.1.29).** For  $(\mathcal{C}, 0) \in \mathrm{Cat}_{\infty}^{\mathrm{pt}}$ , the following are equivalent:

- $\mathcal{C}$  is stable.
- $\mathcal{C}$  admits finite finite colimits and the suspension functor  $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$  is an equivalence.
- $\mathcal{C}$  admits finite finite limits and the loop functor  $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$  is an equivalence.

┘

*Proof:* Cf. [Lur11]P149. ┐

┐



### Presentable Stable $\infty$ -Categories

#### 2 Monoidal $\infty$ -Categories

**Def. (4.9.2.1) [Monoidal  $\infty$ -Categories].** A **monoidal  $(\infty, 1)$ -category** is a tuple  $(\mathcal{C}, \otimes)$  where

- $\mathcal{C}$  is a simplicial set.
- $\otimes : \mathcal{C} \rightarrow N(\Delta)^{\text{op}}$  is a coCartesian fibration.
- For  $n \in \mathbb{N}$  and  $0 \leq i \leq n-1$ , the induced  $(\infty, 1)$ -functor  $\mathcal{C}_{[n]} \rightarrow \mathcal{C}_{\{i, i+1\}}$  determines an equivalence of  $(\infty, 1)$ -categories

$$\mathcal{C}_{[n]} \rightarrow \mathcal{C}_{\{0,1\}} \times \dots \times \mathcal{C}_{\{n-1,n\}} \cong (\mathcal{C}_{[1]})^n$$

⌋

**Def. (4.9.2.2) [ $\mathbb{A}_\infty$ -Rings].** Let  $\mathcal{C}$  be a stable monoidal  $(\infty, 1)$ -category (4.9.2.1) with the  $(\infty, 1)$ -functor

$$p_\infty : \mathcal{C} \rightarrow N(\Delta)^{\text{op}},$$

an  **$\mathbb{A}_\infty$ -Ring** is a lax monoidal section of  $p_\infty$ .

⌋

#### Monoidal $(\infty, 1)$ -Categories

**Def. (4.9.2.3) [Symmetric Monoidal  $(\infty, 1)$ -Categories].** A symmetric monoidal  $(\infty, 1)$ -category is an  $(\infty, 1)$ -category which is  $\infty$ -tuply monoidal. Equivalently, it is a commutative algebra in the  $(\infty, 1)$ -category of  $(\infty, 1)$ -categories.

Or equivalently, it is a coCartesian fibration of simplicial sets

$$\pi : \mathcal{C}^\otimes \rightarrow N(\text{finSet}_*)$$

s.t. for any  $n \in \mathbb{N}$ , the associated functor  $\mathcal{C}_{[n]}^\otimes \rightarrow \mathcal{C}_{[1]}^\otimes$  determines an equivalence of  $(\infty, 1)$ -categories.

⌋

*Proof:* ?

□

**Def. (4.9.2.4) [ $\mathbb{E}_\infty$ -Algebras (Commutative Monoids)].** Let  $\mathcal{C}$  be a symmetric monoidal  $(\infty, 1)$ -category, then an  **$\mathbb{E}_\infty$ -algebra** of a **commutative monoid** in  $\mathcal{C}$  is a lax monoidal  $(\infty, 1)$ -functor  $*$   $\rightarrow \mathcal{C}$ ?. The  $(\infty, 1)$ -category of commutative monoids in  $\mathcal{C}$  is denoted by  $\mathcal{C}\text{Mon}_\infty(\mathcal{C})$ .

⌋

**Def. (4.9.2.5) [ $\mathbb{E}_\infty$ -Space].** An  **$\mathbb{E}_\infty$ -space** is a commutative  $\infty$ -monoid in  $\text{Grpd}_\infty$  (4.7.2.5), i.e.

$$\mathbb{E}_\infty\text{-Spa} = \mathcal{C}\text{Mon}_\infty(\text{Grpd}_\infty).$$

It is also denoted by  $\mathcal{C}\text{Mon}_\infty$ .

⌋

**Def. (4.9.2.6) [Commutative Ring Spectra ( $\mathbb{E}_\infty$ -Rings)].** An  **$\mathbb{E}_\infty$ -ring** or a **commutative ring spectra** is a commutative monoid in the stable  $(\infty, 1)$ -category of spectra?.

⌋

**Prop. (4.9.2.7) [ $\infty$ -Abelianization].** There is an  $(\infty, 1)$ -functor

$$\mathcal{A}b_\infty : \text{Grpd}_\infty \rightarrow \mathcal{C}\text{Mon}_\infty \text{ (4.9.2.5)}$$

that is left adjoint to the

⌋

## 4.10 Derived Categories

Main references are [G-M03] and [Sta]. Need to be refreshed by the language of  $\infty$ -categories ?

### 1 Derived Categories

**Def. (4.10.1.1) [Derived Categories].** For  $\mathcal{A} \in \mathbf{AbCat}$ , the full subcategory  $Ac(\mathcal{A})$  of  $K(\mathcal{A})$  consisting of acyclic complexes is a strictly full saturated triangulated subcategory, and its associated saturated localizing system is the class  $\mathbf{QIso}(\mathcal{A})$  of quasi-isomorphisms.

Thus the kernel of the localizing functor  $Q_{\mathcal{A}} : K(\mathcal{A}) \rightarrow \mathbf{QIso}(\mathcal{A})^{-1}K(\mathcal{A})$  is  $Ac(\mathcal{A})$ , and  $H^0$  factors through  $Q_{\mathcal{A}}$ . Then the quotient triangulated category (4.8.4.32)

$$D(\mathcal{A}) = K(\mathcal{A})/Ac(\mathcal{A}) = \mathbf{QIso}(\mathcal{A})^{-1}K(\mathcal{A})$$

is called the **derived category** of  $\mathcal{A}$ . ┘

*Proof:* As  $H^0$  is a cohomological functor on  $K(\mathcal{A})$  (4.8.7.14), these follow from ?? and (4.8.4.33). □

**Cor. (4.10.1.2) [Universal Properties of Derived Categories].** For  $\mathcal{A} \in \mathbf{AbCat}$ ,

- By (4.8.4.28), the derived category  $D(\mathcal{A})$  has the universal property that any exact functor between Distinguished categories  $F : K(\mathcal{A}) \rightarrow \mathcal{D}$  s.t. quasi-isomorphisms is mapped to isomorphisms uniquely factors through  $D(\mathcal{A})$ .
- Let  $F : \mathcal{A} \rightarrow \mathcal{B} \in \mathbf{AbCat}$ , then  $F$  induces an exact functor  $F : K(\mathcal{A}) \rightarrow K(\mathcal{B}) \in \mathbf{TriCat}$  (4.8.7.17), and it maps quasi-isomorphisms to quasi-isomorphisms, so by item1 induces a morphism of categories  $\mathcal{D}(\mathcal{A}) \rightarrow \mathcal{D}(\mathcal{B})$ . ┘

**Prop. (4.10.1.3) [Universal  $\delta$ -Functors].** The functor  $\mathrm{Ch}^{\mathbb{Z}}(\mathcal{A}) \rightarrow D(\mathcal{A})$  is a  $\delta$ -functor. ┘

*Proof:* Cf. [Sta] 014Z. ? □

**Prop. (4.10.1.4) [T-structures].** Let  $\mathcal{A} \in \mathbf{AbCat}$  and  $K^{\bullet} \in K(\mathcal{A})$ , then for any  $a \in \mathbb{Z}$ , truncations induces a distinguished triangle

$$\tau_{\leq a}K^{\bullet} \rightarrow K^{\bullet} \rightarrow \tau_{\geq a+1}K^{\bullet} \rightarrow \tau_{\leq a}K^{\bullet}[1].$$

In particular,  $K(\mathcal{A})$  and  $D(\mathcal{A})$  are triangulated categories with T-structures (4.8.6.1). ┘

*Proof:* This distinguished triangle comes from the exact sequence  $0 \rightarrow \tau_{\leq a}K^{\bullet} \rightarrow K^{\bullet} \rightarrow K^{\bullet}/\tau_{\leq a}K^{\bullet}$  via the  $\delta$ -functor (4.10.1.3) and the fact  $K^{\bullet}/\tau_{\leq a}K^{\bullet} \rightarrow K^{\bullet} \rightarrow \tau_{\geq a+1}K^{\bullet}$  is a quasi-isomorphism thus an isomorphism in  $D(\mathcal{A})$ . □

### Triangulated Subcategories of $D(\mathcal{A})$

**Prop. (4.10.1.5).** Let  $\mathcal{A} \in \mathbf{AbCat}$  and  $\mathcal{L}$  a full triangulated subcategory of  $K(\mathcal{A})$ , let  $S = \mathcal{L} \cap \mathbf{QIso}(\mathcal{A})$ , then  $S$  is the saturated localizing system compatible with the triangulated structure associated to the cohomological functor  $H^0$  restricted to  $\mathcal{L}$  (4.8.4.27), then we can form the localizing triangulated category  $S^{-1}\mathcal{L}$  (4.8.4.28) and there is a natural exact functor of triangulated categories  $S^{-1}\mathcal{L} \rightarrow D(\mathcal{A})$  by the universal property (4.8.4.28). ┘

**Prop. (4.10.1.6) [Full Subcategories of  $D(\mathcal{A})$ ].** Let  $\mathcal{L} \subset \tilde{\mathcal{L}}$  be full triangulated subcategories of  $K(\mathcal{A})$ , and  $S = \mathcal{L} \cap \text{QIso}(\mathcal{A})$ ,  $\tilde{S} = \tilde{\mathcal{L}} \cap \text{QIso}(\mathcal{A})$ . If either of the following holds:

- For any  $s : L_1 \rightarrow \tilde{L}_1 \in \tilde{S}$  and  $L_1 \in \mathcal{L}$ , there is a morphism  $t : \tilde{L}_1 \rightarrow L_2$  that  $L_2 \in \mathcal{L}$  and  $t \circ s \in \tilde{S}$ .
- The same as in item1 but with arrows reversed.

then the natural functor  $S^{-1}\mathcal{L} \subset \tilde{S}^{-1}\tilde{\mathcal{L}}$  is fully faithful.  $\lrcorner$

*Proof:* This follows immediately from (4.10.1.5) and (4.8.4.30).  $\square$

**Cor. (4.10.1.7) [Bounded Derived Categories].** For  $*$  =  $-$ ,  $+$ ,  $b$ , the triangulated categories  $D^*(\mathcal{A})$  are the localized category of  $K^*(\mathcal{A})$  at the classes of isomorphism in  $K^*(\mathcal{A})$  (4.10.1.5). Then they are naturally full subcategories of  $D(\mathcal{A})$ , by (4.10.1.6), as the condition is satisfied by (4.8.7.19).  $\lrcorner$

**Def. (4.10.1.8) [Derived Category of Serre Subcategories].** If  $\mathcal{B}$  is a Serre subcategory of an Abelian category  $\mathcal{A}$ , let  $D_{\mathcal{B}}^*(\mathcal{A})$  be the full subcategory of  $D^*(\mathcal{A})$  consisting of objects  $X$  that  $H^n(X) \in \mathcal{B}$  for all  $n$ , then  $D_{\mathcal{B}}^*(\mathcal{A})$  is a strictly full saturated triangulated subcategory of  $D^*(\mathcal{A})$ , as it is just the kernel of the cohomological functor  $H^0 : D^*(\mathcal{A}) \rightarrow \mathcal{A} \rightarrow \mathcal{A}/\mathcal{B}$  (4.8.3.15) (4.8.4.24).

Also there are natural exact functors  $D^*(\mathcal{B}) \rightarrow D_{\mathcal{B}}^*(\mathcal{A})$ , and  $D^*(\mathcal{A})/D_{\mathcal{B}}^*(\mathcal{A}) \rightarrow D^*(\mathcal{A}/\mathcal{B})$  (4.8.4.32).  $\lrcorner$

**Prop. (4.10.1.9).** The map  $D^*(\mathcal{A}) \rightarrow D^*(\mathcal{A}/\mathcal{B})$  is essentially surjective.  $\lrcorner$

*Proof:* Cf. [Sta]06XL.  $\square$

**Prop. (4.10.1.10).** Let  $\mathcal{B} \subset \mathcal{A}$  be a Serre subcategory and suppose that for any surjection  $X \rightarrow Y \in \mathcal{A}$  with  $Y \in \mathcal{B}$ , there is a subobject  $X' \subset X$  that  $X' \rightarrow Y$  is surjective, then the exact functor  $D^*(\mathcal{B}) \rightarrow D_{\mathcal{B}}^*(\mathcal{A})$  is an equivalence for  $*$  =  $-$  or  $b$ .  $\lrcorner$

*Proof:* Cf. [Sta]0FCL.  $\square$

**Prop. (4.10.1.11).** If  $(\mathcal{C}, \mathcal{O})$  is a ringed site and  $\mathcal{A} \subset \text{Mod}_{\mathcal{O}}$  is the Serre subcategory of torsion modules, then the functor  $D(\mathcal{A}) \rightarrow D_{\mathcal{A}}(\mathcal{O})$  is an equivalence.  $\lrcorner$

*Proof:* Cf. [Sta]0DD7.  $\square$

**Prop. (4.10.1.12) [Embedding of  $\mathcal{A}$  in  $D(\mathcal{A})$ ].** For  $A \in \text{AbCat}$ , the natural inclusion  $i : \mathcal{A} \subset D(\mathcal{A})$  induces an equivalence of  $\mathcal{A}$  with the subcategory of  $D(\mathcal{A})$  consisting of complexes with cohomology concentrated at degree 0.

An object  $K \in D(\mathcal{A})$  is called **discrete** if it is in the essential image of this inclusion.  $\lrcorner$

*Proof:* The natural map  $H^0 : D(\mathcal{A}) \rightarrow \mathcal{A}$  is inverse to  $i$ , so  $i$  is faithful. To show it is full, let  $(L, f, s^{-1})$  be a morphism from  $i(M)$  to  $i(N)$ , then we get a morphism  $(H^0(L), H^0(f), H^0(s)^{-1})$ , and these two morphisms are both dominated by the morphism  $(\sigma^{\geq n} L, \sigma^{\leq n} \circ f, (\sigma^{\leq n} \circ s)^{-1})$ , so they are equal morphisms in  $D(\mathcal{A})$ . But  $(H^0(L), H^0(f), H^0(s)^{-1}) = i(s^{-1}f)$  is in the image of  $i$ , so  $i$  is full.

The assertion that any complex with cohomology groups concentrated at degree 0 is true by using truncation functors (4.8.7.19).  $\square$

**Prop. (4.10.1.13) [K-Groups].** For  $A \in \text{AbCat}$ , the embedding (4.10.1.12) induces an isomorphism of  $K$ -groups  $K_0(\mathcal{A}) \cong K_0(D^b(\mathcal{A}))$  (11.1.2.1).  $\lrcorner$

*Proof:* The map  $\mathcal{A} \rightarrow D^b(\mathcal{A})$  is a  $\delta$ -functor by (4.10.1.3), thus by (11.1.2.2) induces a map  $K_0(\mathcal{A}) \cong K_0(D^b(\mathcal{A}))$ . There is a reverse map  $K_0(D^b(\mathcal{A})) \rightarrow K_0(\mathcal{A}) : X \mapsto \sum_i (-1)^i H^i(X)$ , which is inverse the the map above: one direction is clear, for the other, use induction and the truncation triangles (4.10.1.4).  $\square$

**Def. (4.10.1.14) [Perfect Complexes].** For  $\mathcal{A} \in \mathbf{AbCat}$ , a **perfect complex** in  $D(\mathcal{A})$  is an object that is isomorphic to a bounded complex.  $\lrcorner$

### Operations on the Derived Category

**Lemma (4.10.1.15) [Direct Sum].** If  $\mathcal{A} \in \mathbf{AbCat}$  has exact countable direct sums, then  $D(\mathcal{A})$  has countable direct sums given by term-wise direct sums.  $\lrcorner$

*Proof:* A system of morphisms  $K_i^\bullet \rightarrow L^\bullet$  is a system of quasiisomorphisms  $M_i^\bullet \rightarrow K_i^\bullet$  and  $M_i \rightarrow L^\bullet$ . Then by hypothesis  $\oplus M_i^\bullet \rightarrow \oplus K_i^\bullet$  is a quasi-iso, thus defines a morphism  $\oplus K_i^\bullet \rightarrow L^\bullet$ . It can be verified that this morphism is unique.  $\square$

**Lemma (4.10.1.16) [Termwise Colimit as Hocolim].** Let  $\mathcal{A} \in \mathbf{AbCat}$  and  $L_n^\bullet$  be a system of complexes of  $\mathcal{A}$ . Assume colimits over  $\mathbb{N}$  exists and are exact over  $\mathcal{A}$ , then the termwise colimit  $L^\bullet$  is a derived colimit in  $D(\mathcal{A})$ .  $\lrcorner$

*Proof:* We have an exact sequence

$$0 \rightarrow \oplus L_n^\bullet \rightarrow \oplus L_n^\bullet \rightarrow L^\bullet \rightarrow 0$$

and the termwise direct sum is the direct sum in  $D(\mathcal{A})$  by (4.10.1.15), and then  $L^\bullet$  is a derived colimit, by (4.8.7.13).  $\square$

### Bounded (Co)Homological Dimensions

**Def. (4.10.1.17) [Bounded Cohomological Dimensions].** An exact functor between two derived categories  $F : D(\mathcal{A}) \rightarrow D(\mathcal{B})$  is called has **cohomological dimension bounded by  $N$**  if  $F(D^{\leq m}(\mathcal{A})) \subset D^{\leq m+N}(\mathcal{B})$ . Dually, it has **homological dimension bounded by  $N$**  if  $F(D^{\geq m}(\mathcal{A})) \subset D^{\geq m-N}(\mathcal{B})$ .  $\lrcorner$

**Prop. (4.10.1.18).** If  $(F, G) : D(\mathcal{A}) \rightleftharpoons D(\mathcal{B})$  is an adjunction pair of exact functors, then  $F$  has cohomological dimension bounded by  $N$  iff  $G$  has homological dimension bounded by  $N$ .  $\lrcorner$

*Proof:* If  $F$  has cohomological dimension bounded by  $N$ , let  $K \in D^{\geq m}(\mathcal{B})$ , then

$$\mathrm{Hom}_{D(\mathcal{A})}(\tau_{\leq m-N-1} GK, GK) = \mathrm{Hom}_{D(\mathcal{B})}(F\tau_{\leq m-N-1} GK, K) = 0.$$

The dual case is similar.  $\square$

## 2 K-injectives and K-Adapted Classes

**Prop. (4.10.2.1) [K-injectives].** For  $\mathcal{A} \in \mathbf{AbCat}$ , a complex  $I^\bullet$  in  $K(\mathcal{A})$  is called a **K-injective** object iff it satisfies the following equivalent conditions:

- $\mathrm{Hom}_{K(\mathcal{A})}(S^\bullet, I^\bullet) = 0$  for any acyclic  $S^\bullet$  in  $K(\mathcal{A})$ .
- $\mathrm{Hom}_{K(\mathcal{A})}(N^\bullet, I^\bullet) \cong \mathrm{Hom}_{K(\mathcal{A})}(M^\bullet, I^\bullet)$  for quasi-iso  $M^\bullet \rightarrow N^\bullet$ .

- Dually, we can define **K-projective** objects.

└

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For the second assertion, Cf.[Sta]079P.?

7

### K-Adapted Classes

**Def. (4.10.2.11)[K-Adapted Classes].** Let  $\mathcal{A}$  be an Abelian category,  $\mathcal{D}$  a triangularized category and  $F : K(\mathcal{A}) \rightarrow \mathcal{D}$  a triangularized functor, a **K-adapted class** for  $F$  is a full triangulated subcategory  $\mathcal{R} \subset K(\mathcal{A})$  that

- If  $I \rightarrow I'$  is a quasi-isomorphism in  $\mathcal{R}$ , then  $F(I) \rightarrow F(I')$  is an isomorphism in  $\mathcal{D}$ .
- Every  $A^\bullet \in K(\mathcal{A})$  admits some quasi-isomorphism  $A^\bullet \rightarrow I^\bullet$  where  $I^\bullet \in \mathcal{R}$ .

┘

**Cor. (4.10.2.12)[K-Injective is K-Adapted].** If  $\mathcal{A}$  has sufficiently many K-injectives, then the class of K-injectives is K-adapted to any left exact functor  $F$ , by (4.10.2.2) and (4.10.2.1). ┘

### 3 Derived Functors

**Def. (4.10.3.1)[Right Derived Functors].** Let  $\mathcal{A}$  be an Abelian category,  $E$  a triangularized category and  $F : K(\mathcal{A}) \rightarrow E$  an exact functor between triangulated categories. Then a **right derived functor** of  $F$  is an exact functor  $RF : D(\mathcal{A}) \rightarrow E$  together with a natural transformation  $\eta : F \rightarrow RF \circ Q_{\mathcal{A}}$  that satisfies the following universal property: For any exact functor  $G : D(\mathcal{A}) \rightarrow E$  and a natural transformation  $\eta' : F \rightarrow G \circ Q_{\mathcal{A}}$ , there is a natural transformation  $\theta : RF \rightarrow G$  that  $\eta' = (\theta \star Q_{\mathcal{A}}) \circ \eta$ .

In particular, if  $RF$  exists, then it is unique up to unique isomorphism of exact functors. ┘

**Prop. (4.10.3.2)[Derived Functors via K-Adapted Classes].** Let  $\mathcal{A}$  be an Abelian category,  $\mathcal{D}$  a triangularized category and  $F : K(\mathcal{A}) \rightarrow E$  an exact functor, if there exists a K-adapted class  $\mathcal{R}$  for  $F$ , then  $RF$  exists. Moreover, for  $I \in \mathcal{R}$ , the morphism  $\eta_I : F(I) \rightarrow RF \circ Q_{\mathcal{A}}(I)$  is an isomorphism in  $\mathcal{D}$ . ┘

*Proof:* Let  $S = \mathcal{R} \cap \text{QIso}(\mathcal{A})$ , then  $\mathcal{R}, K(\mathcal{A})$  and the localizing systems  $S, \text{QIso}(\mathcal{A})$  satisfy the conditions in (4.10.1.6), so  $\iota : S^{-1}\mathcal{R} \subset D(\mathcal{A})$  is a full triangulated subcategory, and it is also essentially surjective, thus it is an equivalence of categories. Let  $I$  be a right adjoint of the inclusion, then  $I$  is also an exact functor (4.8.4.10), and there is a natural isomorphism  $\zeta : \text{id}_{D(\mathcal{A})} \cong \iota \circ I$ .

By the universal property, there is an exact functor  $F_{\mathcal{R}} : S^{-1}\mathcal{R} \rightarrow E$  extending  $F|_{\mathcal{R}}$ . Define

$$RF = F_{\mathcal{R}} \circ I : D(\mathcal{A}) \rightarrow E.$$

To construct  $\eta : F \rightarrow RF \circ Q_{\mathcal{A}}$ , for any  $X \in \text{Ob}(K(\mathcal{A})) = \text{Ob}(D(\mathcal{A}))$ , the isomorphism  $\zeta$  gives an isomorphism  $X \cong \iota(I(X))$ , which is represented by a roof  $(R, f, s^{-1}), f, s \in \text{QIso}(\mathcal{A})$ , and we can assume  $R \in \text{Ob}(\iota(S^{-1}\mathcal{R})) = \text{Ob}(\mathcal{R})$  by hypothesis, thus  $s \in S$ , and  $F(s)$  is an isomorphism in  $E$  by hypothesis. Thus we get a morphism  $\eta_X : F(s)^{-1} \circ F(f) : F(X) \rightarrow F(\iota(I(X))) = RF(Q_{\mathcal{A}}(X))$ . This  $\eta_X$  is independent of the representation given, because if there is another dominant roof  $(R', t \circ f, (t \circ s)^{-1})$ , where  $t : R \rightarrow R' \in S$ , then  $F(t \circ s)^{-1} \circ F(t \circ f) = F(s)^{-1} \circ F(f) \in E$ .

And for a morphism  $f : X \rightarrow Y \in K(\mathcal{A})$ ,  $\zeta$  gives a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\zeta_X} & \iota(I(X)) \\ \downarrow f & & \downarrow \iota(I(f)) \\ Y & \xrightarrow{\zeta_Y} & \iota(I(Y)) \end{array}$$

in  $D(\mathcal{A})$  where  $\zeta_X, \zeta_Y$  are represented by  $(R_X, f_X, s_X^{-1}), (R_Y, f_Y, s_Y^{-1})$ , and  $I(f)$  is represented  $(R_0, f_0, s_0^{-1})$ , then we can construct roof to realize this commutative diagram in  $K(\mathcal{A})$  and show

there is a commutative diagram

$$\begin{array}{ccc}
 F(X) & \xrightarrow{F(s_X)^{-1} \circ F(f_X)} & RF(Q_{\mathcal{A}}(X)) \\
 \downarrow F(f) & & \downarrow F(s_0)^{-1} \circ F(f_0) = RF \circ Q_{\mathcal{A}}(f) \\
 F(Y) & \xrightarrow{F(s_Y)^{-1} \circ F(f_Y)} & RF(Q_{\mathcal{A}}(Y))
 \end{array}$$

in  $E$ , which means  $\eta : F \rightarrow RF \circ Q_{\mathcal{A}}$  is a natural isomorphism.

Notice that  $\eta_I$  is an isomorphism for  $I \in \mathcal{R}$ , because in this case  $F(f_I), F(s_I)$  are both isomorphisms.

It remains to show the universal properties of  $\eta$ : Given any exact functor  $G : D(\mathcal{A}) \rightarrow E$  and a natural transformation  $\eta' : F \rightarrow G \circ Q_{\mathcal{A}}$ , for any  $X \in \text{Ob}(K(\mathcal{A})) = \text{Ob}(D(\mathcal{A}))$ , we get a morphism  $\eta'_X : F(X) \rightarrow G(X)$  in  $E$ . Notation as before, because  $\eta'$  is a natural transformation, we get a commutative diagram in  $K(\mathcal{A})$

$$\begin{array}{ccc}
 F(X) & \xrightarrow{\eta'_X} & G(X) \\
 \downarrow F(f) & & \downarrow G(Q_{\mathcal{A}}(f)) \\
 F(R_X) & \xrightarrow{\eta'_{R_X}} & G(R_X) \\
 \uparrow F(s) & & \uparrow G(Q_{\mathcal{A}}(s)) \\
 F(\iota(I(X))) & \xrightarrow{\eta'_{\iota(I(X))}} & G(\iota(I(X)))
 \end{array}$$

In  $E$ . As  $\zeta_X = (R_X, f, s^{-1})$  is an isomorphism between  $X$  and  $\iota(I(X))$ ,  $f, s$  are both quasi-isomorphisms, thus the vertical arrows are both isomorphisms, and we get a commutative diagram

$$\begin{array}{ccc}
 F(X) & \xrightarrow{\eta'_X} & G(X) \\
 \downarrow \eta_X & & \downarrow G(\zeta_X) \\
 RF(Q_{\mathcal{A}}(X)) & \xrightarrow{\eta'_{\iota(I(X))}} & G(\iota(I(X)))
 \end{array}$$

So we can define

$$\theta_X = (G(\zeta_X))^{-1} \circ \eta'_{\iota(I(X))} : RF(Q_{\mathcal{A}}(X)) \rightarrow G(X),$$

then it is functorial in  $X$  as both  $\eta'_{\iota(I(X))}$  and  $G(\zeta_X)$  are functorial in  $X$ , so defines a natural transformation  $RF \circ Q_{\mathcal{A}} \rightarrow G$  that  $(\theta \star Q_{\mathcal{A}}) \circ \eta = \eta'$ .

For the uniqueness of  $\eta$ : For any  $X \in \text{Ob}(S^{-1}\mathcal{R}) = \text{Ob}(\mathcal{R})$ ,  $\eta_X$  is an isomorphism, thus  $\eta_X$  is determined, but  $\iota : S^{-1}\mathcal{R} \rightarrow D(\mathcal{A})$  is essentially surjective, so  $\eta_X$  is determined for any  $X \in D(\mathcal{A})$ .

□

**Cor.(4.10.3.3) [Derived Functors via K-Injective Classes].** Let  $\mathcal{A} \in \text{AbCat}$ ,  $E \in \text{TriCat}$  and  $F : K^*(\mathcal{A}) \rightarrow E$  an exact functor, if  $K^*(\mathcal{A})$  has enough K-injectives, then the class of K-injectives is K-adapted to  $F$  (4.10.2.12), so  $RF : D^*(\mathcal{A}) \rightarrow E$  exists. Moreover, for any K-injective  $I^\bullet$ , the morphism  $\eta_{I^\bullet} : F(I^\bullet) \rightarrow RF \circ Q_{\mathcal{A}}(I^\bullet)$  is an isomorphism in  $E$  ┘



**Cor. (4.10.3.4) [Derived Functors via Injectives].** Let  $\mathcal{A}, \mathcal{B} \in \mathcal{AbCat}$  s.t  $\mathcal{A}$  has enough injective and exact countable products, and  $F : \mathcal{A} \rightarrow \mathcal{B}$  an additive functor, then  $K^*(F) : K^*(\mathcal{A}) \rightarrow D^*(\mathcal{A}) \rightarrow D^*(\mathcal{B})$  is an exact functor by (4.8.7.16), and  $K^*(\mathcal{A})$  has enough K-injectives by (4.10.2.9).

However, this is useless unless  $F$  is left exact in which case we see  $R^0 F(A) = F(A)$  for any  $A \in \mathcal{A}$  by (4.10.3.3).  $\lrcorner$

**Cor. (4.10.3.5) [Naturality].** If  $\eta : F \rightarrow F'$  is a natural transformation of left-exact functors, then by universal property (4.10.3.1), there is a natural transformation  $R\eta : RF \rightarrow RF'$  inducing a long exact sequence about cohomology groups extending  $\eta$ .  $\lrcorner$

**Prop. (4.10.3.6) [Composition of Derived Functors].** Let  $\mathcal{A}, \mathcal{B} \in \mathcal{AbCat}$  and  $\mathcal{D} \in \mathcal{TriCat}$ . If  $F : K^*(\mathcal{A}) \rightarrow K^*(\mathcal{B}), G : K^*(\mathcal{B}) \rightarrow \mathcal{D}$  are exact functors between triangulated categories, and  $\mathcal{R}_{\mathcal{A}}$  is K-adapted to  $F$  and  $G \circ F$ ,  $\mathcal{R}_{\mathcal{B}}$  is K-adapted to  $G$ , then the natural transformation  $R(G \circ F) \rightarrow RG \circ RF$  is an isomorphism.  $\lrcorner$

*Proof:*  $RF$  is isomorphic to  $F$  on  $\mathcal{R}_{\mathcal{A}}$ , so for any  $A^\bullet \in K^*(\mathcal{A})$ , choose some quasi-isomorphism  $A^\bullet \rightarrow R$ , where  $R \in \mathcal{R}_{\mathcal{A}}$ , then there is a commutative diagram

$$\begin{array}{ccccc} R(G \circ F)(A^\bullet) & \xrightarrow{\cong} & R(G \circ F)(R) & \xrightarrow{\cong} & G \circ F(R) \\ \downarrow & & \downarrow & & \downarrow \text{id} \\ RG \circ RF(A^\bullet) & \xrightarrow{\cong} & RG \circ RF(R) & \xrightarrow{\cong} & G \circ F(R) \end{array}$$

$$G \circ F(R) \cong G(F(R)) \rightarrow RG(F(I^\bullet)) \cong RG \circ RF(A^\bullet),$$

and we are done.  $\square$

**Def. (4.10.3.7) [Universal  $\delta$ -Functors].** A **universal  $\delta$ -functor** between Abelian categories is one that any natural transformation from  $T^0$  to another  $\delta$ -functor will generate a  $\delta$ -map. A **effaceable  $\delta$ -functor** is one that for any  $n > 0$  and any object  $A$ , there is an injection  $A \rightarrow B$  that  $T^n(A) \rightarrow T^n(B) = 0$ .  $\lrcorner$

**Prop. (4.10.3.8) [Grothendieck].** A  $\delta$ -functor is universal if it is effaceable.  $\lrcorner$

*Proof:* We construct by induction on  $n$ . choose a  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  such that  $T^{n+1}(A) \rightarrow T^{n+1}(B) = 0$  then there is an isomorphism  $T^{n+1}(A) \cong \text{Coker}(T^n(B) \rightarrow T^n(C))$ , and so we can construct the map on  $T^{n+1}$  induces by

$$\text{Coker}(T^n(B) \rightarrow T^n(C)) \rightarrow \text{Coker}(G^n(B) \rightarrow G^n(C)) \rightarrow G^{n+1}(A).$$

This can be verified to be a  $\delta$  map.  $\square$

**Prop. (4.10.3.9).** The derived functors form a universal  $\delta$ -functor (when it exists).  $\lrcorner$

*Proof:* It is a  $\delta$  functor by (4.10.1.1), it is universal by (4.10.3.8).  $\square$

**Prop. (4.10.3.10).** Derived functor commutes with filtered colimits on a Grothendieck Abelian category, this is by AB5.  $\lrcorner$

**Prop. (4.10.3.11) [Hypercohomology].** Given an Abelian category  $\mathcal{A}$  with enough injectives,  $\mathcal{B}$  a complete Abelian category,  $F : \mathcal{A} \rightarrow \mathcal{B}$  a left exact functor, and  $K \in \mathcal{K}(\mathcal{A})$ , we can define the **right**



**hyper-derived functor** of  $F$  at  $K$  as  $\mathbb{R}F(K) = \text{Tot}^\Pi F(P) \in K(\mathcal{B})$  where  $K \rightarrow P$  is a Cartan-Eilenberg resolution of  $K$ . and the **hypercohomologies** of  $F$  at  $K$  as  $R^n F(K) = H^n(\text{Tot}^\Pi F(P))$ . Dually we can define the left hyper-derived functor and hyperhomologies.

For complexes in  $K^+(\mathcal{A})$ , there is no restriction and the right derived-hyper functor descends to a functor from  $D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ .

When the Abelian category  $\mathcal{A}$  satisfies AB3\* and AB4\*, i.e. the direct product is exact, then  $\text{Tot}^\Pi$  of the Cartan-Eilenberg resolution of any complex is a quasi-isomorphism to it by the dual of (4.8.9.11). (Take horizontal filtration, AB4\* assures it collapse).  $\lrcorner$

### Derived Bifunctors

**Def. (4.10.3.12) [Right Derived Functors of Bi-Exact Bi-Functors].** Let  $\mathcal{A}, \mathcal{B}$  be Abelian categories and  $F : K(\mathcal{A}) \times K(\mathcal{B}) \rightarrow \mathcal{D}$  be bi-exact bifunctor between triangulated categories, a **right derived functor** of  $F$  is a bi-exact bifunctor  $RF : D(\mathcal{A}) \times D(\mathcal{B}) \rightarrow \mathcal{D}$  together with a natural transformation  $\eta : F \rightarrow RF \circ (Q_{\mathcal{A}} \times Q_{\mathcal{B}})$  that satisfies the following universal property: For any bi-exact bifunctor  $G : \mathcal{D}(\mathcal{A}) \times \mathcal{D}(\mathcal{B}) \rightarrow \mathcal{D}$  and a natural transformation  $\eta' : F \rightarrow G \circ (Q_{\mathcal{A}} \times Q_{\mathcal{B}})$ , there is a natural transformation  $\theta : RF \rightarrow G$  that  $\eta' = (\theta \star (Q_{\mathcal{A}} \times Q_{\mathcal{B}})) \circ \eta$ .

In particular, if  $RF$  exists, then it is unique up to unique isomorphisms of bi-exact bi-functors.  $\lrcorner$

**Prop. (4.10.3.13) [Deriving Bi-Exact Functors].** Let  $\mathcal{A}_1, \mathcal{A}_2$  be Abelian categories and  $F : K(\mathcal{A}_1) \times K(\mathcal{A}_2) \rightarrow \mathcal{D}$  be bi-exact bifunctor between triangulated categories, if  $\mathcal{R}_i$  are full subcategories of  $\mathcal{A}_i$  s.t.

- If  $R_i \rightarrow R'_i$  are quasi-isomorphisms in  $\mathcal{R}_i$ , the induced map  $F(R_1, R_2) \rightarrow F(R'_1, R'_2)$  is an isomorphism in  $\mathcal{D}$ .
- Any  $A_i^\bullet \in K(\mathcal{A}_i)$  admits some quasi-isomorphism  $A_i^\bullet \rightarrow I_i^\bullet$  where  $I_i^\bullet \in \mathcal{R}_i$ .

Then the right derived functor (4.10.3.12)  $RF$  exists. Moreover, for any  $R_i \in \mathcal{R}_i$ ,  $\eta_{R_1, R_2} : F(R_1, R_2) \rightarrow RF \circ (Q_{\mathcal{A}_1} \times Q_{\mathcal{A}_2})(R_1, R_2)$  is an isomorphism.  $\lrcorner$

*Proof:* Let  $S_i = \mathcal{R}_i \cap \text{QIso}(\mathcal{A}_i)$ , then  $\mathcal{R}_i, K(\mathcal{A}_i)$  and the localizing systems  $S_i, \text{QIso}(\mathcal{A}_i)$  satisfy the conditions in (4.10.1.6), so  $\iota : S_i^{-1}\mathcal{R}_i \subset D(\mathcal{A}_i)$  is a full triangulated subcategory, and it is also essentially surjective, thus it is an equivalence of categories. Let  $I_i$  be a right adjoint of the inclusion, then  $I_i$  is also an exact functor (4.8.4.10), and there is a natural isomorphism  $\zeta_i : \text{id}_{D(\mathcal{A}_i)} \cong \iota_i \circ I_i$ .

By hypothesis of universal properties, the functor  $F$  extends to an exact functor  $F_{\mathcal{R}} : \prod_i S_i^{-1}\mathcal{R}_i \rightarrow \mathcal{D}$ . Define

$$RF = F_{\mathcal{R}} \circ \prod_i \mathcal{I}_i : \prod_i D(\mathcal{A}_i) \rightarrow \mathcal{D}.$$

The rest of the proof is verbatim as that of (4.10.3.2).  $\square$

**Remark (4.10.3.14).** This proposition can be naturally extended to any multi-exact multi-functors. In fact, I suppose this can be deduced from the usual derived functors by considering the tensor product category  $\mathcal{A} \otimes \mathcal{B}$ , Cf. [Basic Concepts of Enriched Category Theory, Kelly].  $\lrcorner$

**Prop. (4.10.3.15) [Derived Functors of Adjunctions].** Let  $\mathcal{A}_1, \mathcal{A}_2$  be Abelian categories and  $(F, G)$  be an adjunction between  $K(\mathcal{A}_1)$  and  $K(\mathcal{A}_2)$ , and  $\mathcal{R}_i$  are full subcategories of  $\mathcal{A}_i$  s.t.

- If  $R_i \rightarrow R'_i$  are quasi-isomorphisms in  $\mathcal{R}_i$ , the induced map  $F(R'_1, R_2) \rightarrow \text{Hom}_{\mathcal{B}}(R_1, R'_2)$  is an isomorphism in  $D(\mathcal{A}_b)$ .
- Any  $A_1^\bullet \in K(\mathcal{A}_1)$  admits some quasi-isomorphism  $A_1^\bullet \rightarrow I_1^\bullet$  where  $I_1^\bullet \in \mathcal{R}_1$ .

• Any  $A_2^\bullet \in K(\mathcal{A}_2)$  admits some quasi-isomorphism  $I_1^\bullet \rightarrow A_2^\bullet$  where  $I_2^\bullet \in \mathcal{R}_2$ .  
then there is a functorial isomorphism

$$R\mathrm{Hom}(LF(A^\bullet), B^\bullet) \cong R\mathrm{Hom}(A^\bullet, RG(B^\bullet)) \in D(\mathcal{A}b)$$

for  $A^\bullet \in K(\mathcal{A}), B^\bullet \in K(\mathcal{B})$ . In particular,

$$\mathrm{Hom}(LF(A^\bullet), B^\bullet) \cong \mathrm{Hom}(A^\bullet, RG(B^\bullet)) \in \mathcal{A}b$$

┘

*Proof:* Regard both sides as right derived functors of the isomorphic bi-exact bi-functors

$$K(\mathcal{A})^{op} \times K(\mathcal{B}) \rightarrow D(\mathcal{A}b) : (A^\bullet, B^\bullet) \mapsto \mathrm{Hom}_B^\bullet(F(A^\bullet), B^\bullet) \cong \mathrm{Hom}_A^\bullet(A^\bullet, G(B^\bullet)) \in D(\mathcal{A}b).$$

Then the functorial isomorphism follows from universal properties. □

**Prop. (4.10.3.16).** Situation as in (4.10.3.15), for any  $K^\bullet \in K(\mathcal{A}_2)$ , there is a commutative diagram of adjunction maps

$$\begin{array}{ccc} LF \circ G(K^\bullet) & \longrightarrow & F \circ G(K^\bullet) \\ \downarrow & & \downarrow \\ LF \circ RG(K^\bullet) & \longrightarrow & K^\bullet \end{array}$$

┘

*Proof:* Cf. [Sta]0FPI. □

### Internal Hom

**Def. (4.10.3.17) [Internal Hom].** If  $\mathcal{A}$  is an Abelian category and  $K(\mathcal{A})$  has enough K-injectives or enough K-projectives, thus by (4.10.3.13), we can define the **internal Hom**

$$R\mathrm{Hom} : D(\mathcal{A})^{op} \times D(\mathcal{A}) \rightarrow D(\mathcal{A})$$

as the right derived functor of the bi-exact bifunctor  $\mathrm{Hom}^\bullet : K(\mathcal{A})^{op} \times K(\mathcal{A}) \rightarrow D(\mathcal{A}b)$ . ┘

**Prop. (4.10.3.18) [Ext Groups].** For  $X, Y \in D(\mathcal{A})$ , define the **ext groups**

$$\mathrm{Ext}_{\mathcal{A}}^i(X, Y) = H^i(R\mathrm{Hom}(X, Y)) = \mathrm{Hom}_{D(\mathcal{A})}(X[0], Y[i]).$$

┘

*Proof:* To show  $H^i(R\mathrm{Hom}(X, Y)) = \mathrm{Hom}_{D(\mathcal{A})}(X[0], Y[i])$ , choose a K-injective resolution  $Y \rightarrow I^\bullet$  of  $Y$ , then

$$H^i(R\mathrm{Hom}(X, Y)) = H^i(\mathrm{Hom}^\bullet(X, I^\bullet)) = \mathrm{Hom}_{K(\mathcal{A})}(X, I^\bullet[n]) = \mathrm{Hom}_{D(\mathcal{A})}(X, I^\bullet[n]) = \mathrm{Hom}_{D(\mathcal{A})}(X, Y[n])$$

by (4.8.7.10) and the definition (4.10.2.1). □

**Prop. (4.10.3.19).** If  $P^\bullet \rightarrow X^\bullet$  is a projective resolution, then  $\mathrm{Ext}^i(X^\bullet, Y^\bullet) = \mathrm{Hom}_{K(\mathcal{A})}(P^\bullet[-i], Y^\bullet)$ .

If  $Y^\bullet \rightarrow I^\bullet$  is an injective resolution, then  $\mathrm{Ext}^i(X^\bullet, Y^\bullet) = \mathrm{Hom}_{K(\mathcal{A})}(X^\bullet, I^\bullet)$ . ┘

**Prop. (4.10.3.20).** If  $X \in D^{\leq a}(\mathcal{A})$  and  $Y \in D^{\geq b}(\mathcal{A})$ , then for  $i < b - a$ ,  $\text{Ext}^i(X, Y) = 0$ , and  $\text{Ext}^{b-a}(X, Y) = \text{Hom}(H^a(X), H^b(Y))$ .

In particular, for  $A, B \in \mathcal{A}$ ,  $\text{Ext}^i(A, B) = 0$  for  $i < 0$  and  $\text{Ext}^0(A, B) = \text{Hom}(A, B)$ .  $\lrcorner$

*Proof:* This is because  $X$  can be represented by  $K^{\leq a}(\mathcal{P})$  or  $Y$  can be represented by  $K^{\geq b}(\mathcal{I})$ .  $\square$

**Prop. (4.10.3.21) [Long Exact Sequences].** Let  $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$  be an exact sequence in  $\mathcal{A}$ , then it is distinguished in  $D(\mathcal{A})$ , so by (4.8.4.13), for any  $B \in \mathcal{A}$ , there is a long exact sequence

$$0 \rightarrow \text{Hom}(B, A_1) \rightarrow \text{Hom}(B, A_2) \rightarrow \text{Hom}(B, A_3) \rightarrow \text{Ext}^1(B, A_1) \rightarrow \text{Ext}^1(B, A_2) \rightarrow \dots,$$

and similarly a long exact sequence

$$0 \rightarrow \text{Hom}(A_3, B) \rightarrow \text{Hom}(A_2, B) \rightarrow \text{Hom}(A_1, B) \rightarrow \text{Ext}^1(A_3, B) \rightarrow \text{Ext}^1(A_2, B) \rightarrow \dots$$

$\lrcorner$

**Def. (4.10.3.22) [Yoneda Extensions].** Let  $\mathcal{A}$  be an Abelian category and  $A, B \in \mathcal{A}$ , a **Yoneda extension** of  $B$  by  $A$  of degree  $d$  is an exact sequence

$$0 \rightarrow A \rightarrow Z_{d-1} \rightarrow \dots \rightarrow Z_0 \rightarrow B \rightarrow 0.$$

One Yoneda extension is said to **dominate** another if there is a map of extensions s.t. restricts to  $\text{id}_A$  and  $\text{id}_B$ . Two Yoneda extensions of degree  $d$  is called **equivalent** if they are dominated by a common Yoneda extension. (Notice this is an equivalence relation by (4.10.3.23)).  $\lrcorner$

**Prop. (4.10.3.23) [Ext and Yoneda Extensions].** There is a map from the equivalence classes of Yoneda extensions of degree  $d$  of  $B$  over  $A$  to  $\text{Ext}^d(B, A)$ , which maps the exact sequence  $0 \rightarrow A \rightarrow Z_{d-1} \rightarrow \dots \rightarrow Z_0 \rightarrow B \rightarrow 0$  to  $f s^{-1} \in \text{Hom}_{D(\mathcal{A})}(B[0], A[d])$ , where  $f : (A \rightarrow Z_{d-1} \rightarrow \dots \rightarrow Z_0) \rightarrow A[d]$ , and  $s : (A \rightarrow Z_{d-1} \rightarrow \dots \rightarrow Z_0) \rightarrow B[0]$ .

and corresponds to the set of  $i$ -term extensions of  $Y$  by  $X$ . There is a natural map

$$\text{Ext}_{\mathcal{A}}^i(X, Y) \times \text{Ext}_{\mathcal{A}}^i(Y, Z) \rightarrow \text{Ext}_{\mathcal{A}}^{i+j}(X, Z)$$

by composition or equivalently the conjunction of extensions.  $\lrcorner$

*Proof:* Cf. [Sta]06XU.  $\square$

**Prop. (4.10.3.24) [Explicit Addition as Extensions].** In an Abelian category with enough injectives, the extension  $\text{Ext}^1(N, M)$  is equivalent with the Abelian group of extensions with Baer sum as addition.  $\lrcorner$

*Proof:* We choose a projective resolution  $0 \rightarrow K \rightarrow P \rightarrow N \rightarrow 0$ , so  $\text{Hom}(K, M) \rightarrow \text{Ext}^1(N, M)$  is surjective, so choose a lifting and the pushout  $0 \rightarrow M \rightarrow L \rightarrow N \rightarrow 0$  with be the corresponding extension, Now the Baer sum is easy to define and verify.  $\square$

**Prop. (4.10.3.25).** If  $\mathcal{A}$  is an Abelian category and  $p \geq 0$  s.t.  $\text{Ext}^p(A, B) = 0$  for any  $A, B \in \mathcal{A}$ , then  $\text{Ext}^i(A, B) = 0$  for any  $i \geq p$  and  $A, B \in \mathcal{A}$ .  $\lrcorner$

*Proof:* Any Yoneda extension of degree  $i$  is a conjunction of extensions of degree  $p$  and degree  $i - p$ .  $\square$

**Cor. (4.10.3.26).** If  $\mathcal{A}$  is an Abelian category s.t.  $\text{Ext}^2(A, B) = 0$  for any  $A, B \in \mathcal{A}$ , then each object of  $D^b(\mathcal{A})$  is isomorphic to a direct sum of cohomologies.  $\lrcorner$

*Proof:* Let  $K$  be represented by  $K^\bullet \in D^{[a,b]}(\mathcal{A})$ . We use induction on  $b - a$ . If  $b - a > 0$ , then there is a distinguished triangle  $\tau_{\leq b-1}K \rightarrow K^\bullet \rightarrow H^b(K)[-b] \rightarrow \tau_{\leq b-1}K^\bullet[1]$  (4.10.1.4). If we can prove  $H^b(K)[-b] \rightarrow \tau_{\leq b-1}K^\bullet[1]$  is 0, then we finish by (4.8.4.5). But by induction and the hypothesis,

$$\text{Hom}_{D(\mathcal{A})}(H^b(K)[-b], \tau_{\leq b-1}(K^\bullet)[1]) = \oplus_{i < b} \text{Ext}_{\mathcal{A}}^{b-i+1}(H^b(K), H^i(K)),$$

which vanishes by hypothesis.  $\square$

### Acyclic Objects

**Prop. (4.10.3.27) [*F*-Acyclic Objects].** For a left exact functor  $F$ , an object  $X$  is (right) $F$ -acyclic if  $RF$  is defined for  $X$ , and the natural map  $F(X) \rightarrow RF(X)$  is an isomorphism, or equivalently  $R^iF(X) = 0$  for all  $i > 0$ .

Then there is an adapted class of  $F$  iff the class of  $F$ -acyclic objects  $\text{Acy}(F)$  is sufficiently large, and in this case adapted classes of  $F$  are exactly sufficiently large subclasses of  $\text{Acy}(F)$ , and  $\text{Acy}(F)$  contains all injectives (But the class of injectives may not be sufficiently large!).  $\lrcorner$

*Proof:* Cf. [Gelfand P195].  $\square$

**Prop. (4.10.3.28) [Leray's Acyclicity Lemma].** Let  $F : \mathcal{A} \rightarrow \mathcal{B}$  be an additive functor and  $RF$  is everywhere defined, then for a complex  $A^\bullet$  in  $K^+(\mathcal{A})$  consisting of  $F$ -acyclic objects, the natural map  $F(A^\bullet) \rightarrow RF(A^\bullet)$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]015E. This may be equivalent to (4.10.3.27).  $\square$

**Prop. (4.10.3.29) [Acyclic Criterion].** Let  $F$  be a left exact functor from an Abelian category  $\mathcal{C}$  of enough injectives to another Abelian category,  $T$  is a class of objects of  $\mathcal{C}$  that satisfies:

- $T$  is sufficiently large.
- Cokernel of maps between elements of  $T$  is in  $T$  and  $0 \rightarrow F(A) \rightarrow F(A') \rightarrow F(\text{Coker}) \rightarrow 0$  is exact. (To use induction).

Then every object of  $T$  is  $F$ -acyclic.  $\lrcorner$

*Proof:* Cf. [[Sta]05T8].  $\square$

**Prop. (4.10.3.30) [Injectives are adapted].** By (4.10.2.5), if  $\mathcal{A}$  contains sufficiently many injectives, then injective objects are adapted to any left exact functor  $F$ . (Because  $\text{id}$  on acyclic injective complexes is homotopic to 0 by the lemma).  $\lrcorner$

## 4 $D_\infty(R)$

### 5 (Co)Homological Dimension

**Prop. (4.10.5.1).** If  $\mathcal{A}$  has enough projectives, then the projective dimension of an object  $X$  is the length of projective resolutions. (Use resolution and long sequence).  $\lrcorner$

**Prop. (4.10.5.2) [Hilbert Theorem].** For an Abelian category  $\mathcal{A}$ , the category  $\mathcal{A}[T]$  is an Abelian category. If  $\mathcal{A}$  has enough projectives and have infinite direct sum, then  $\text{dhp}_{\mathcal{A}[T]}(X, t) \leq \text{dhp}_{\mathcal{A}}(X) + 1$  and equality with  $t = 0$ .  $\lrcorner$

**Cor. (4.10.5.3).** The Categories  $\mathcal{A}b$  and  $K[X]\text{-mod}$  have homological dimension 1.  $K[X_i, \dots, X_k]$  has homological dimension  $k$ .  $\lrcorner$

**Def. (4.10.5.4) [Injective Amplitude].** Let  $\mathcal{A}$  be an Abelian category with enough injectives,  $K \in D(\mathcal{A})$  is said to have **finite injective dimension** if  $K \in D^b(\mathcal{I})$ . It is said to have **injective amplitude in  $[a, b]$**  if  $K \in D^{[a, b]}(\mathcal{I})$ .  $\lrcorner$

**Prop. (4.10.5.5).** Let  $\mathcal{A}$  be an Abelian category and  $K \in D(\mathcal{A})$ ,

- If  $K \in D^b(\mathcal{A})$  and  $H^i(K)$  all have finite injective dimensions, then  $K$  also has finite injective dimension.
- If  $K$  is represented by  $K^\bullet \in K^b(\mathcal{A})$ , and  $K^i$  all have finite injective dimensions, then  $K$  also has finite injective dimension.

$\lrcorner$

*Proof:* 1 follows from the Grothendieck spectral sequence applied to the functor  $\text{Hom}(N, -)$  for any  $N \in \mathcal{A}$  and the CE resolution of  $K$  (4.8.9.10).

2 follows from 1 as we can use induction to show all  $H^i(K)$  has finite injective dimensions.  $\square$

## 6 Derived Limits and Colimits

**Def. (4.10.6.1) [Derived (Co)Limits].** Let  $\mathcal{D}$  be a triangulated category, and  $(K_n, f_n)$  is an inverse system of objects in  $\mathcal{D}$ , then an object  $K$  is called the **derived colimit** of it iff there  $\oplus K_n$  exists and there is a distinguished triangle

$$\oplus K_n \rightarrow \oplus K_n \rightarrow K \rightarrow \oplus K_n[1]$$

where the first map is given by  $(1 - f_n)$ . By TR1, the derived colimit exists as long as  $\oplus K_n$  exists, and by TR3 (4.8.4.13), the colimit is unique if it exists. And by TR3 again a morphism of systems induces a morphism of colimits.

The definition of **derived limit** is dual.  $\lrcorner$

**Prop. (4.10.6.2) [Cofinality of Hocolim].** Let  $\mathcal{D}$  be a triangulated category and  $(K_n, f_n)$  be a system, if  $0 \leq n_{i_0} < n_{i_1} < \dots$  be a sequence of integers, then there is an isomorphism  $ho \text{ colim } K_{n_i} \rightarrow ho \text{ colim } K_n$ .  $\lrcorner$

*Proof:* Cf. [[Sta]0CRJ].  $\square$

**Lemma (4.10.6.3).** Let  $\mathcal{A}$  be an Abelian category with countable products and enough injectives, then the derived limit  $R \lim$  for any inverse system in  $D^+(\mathcal{A})$  exists.  $\lrcorner$

*Proof:* It suffices to show  $\prod K_i^\bullet$  exists in  $D(\mathcal{A})$ . But every  $K_n^\bullet$  has a  $K$ -injective resolution  $I_n^\bullet$ , by (4.8.9.10)(4.10.2.5). And then  $\prod K_n^\bullet$  is represented by  $\prod I_n^\bullet$ , by (4.10.2.7).  $\square$

**Def. (4.10.6.4) [Rlim].** Rlim on an Abelian category  $\mathcal{A}$  with countable products and enough injectives is defined to be the right derived functor of  $\lim : \mathcal{A}^{\mathbb{N}} \rightarrow \mathcal{A}$ . Equivalently, it is just the derived limit (4.10.6.1)(4.10.6.3) in  $D(\mathcal{A})$  restricted to the case where each  $K_n$  is discrete.  $\lrcorner$

**Prop. (4.10.6.5).** Let  $R \lim \mathcal{A}$  exists on  $\mathcal{A}$ , if  $K_n, f_n$  is a system of objects in  $D^+(\mathcal{A})$ , then there are exact sequences

$$0 \rightarrow R^1 \lim(H^m(K_n), f_n) \rightarrow R^{m+1} \lim(K_n, f_n) \rightarrow \lim(H^m(K_n), f_n) \rightarrow 0.$$

Immediately from the definition (4.10.6.1).  $\lrcorner$

## 7 Spectral Sequence

Reference for this section is [Weibel Ch5]. All the definition below is dual for homology and cohomology, just rotate the diagram 180 degree.

We work in an Abelian category.

**Def. (4.10.7.1).** A convergent **Spectral Sequence** is a three-dimensional arrange of entries  $E_r^{p,q}$  that:

1.  $d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$  that  $d_r d_r = 0$ .
2.  $H^{p,q}(E_r^{p,q}) \cong E_{r+1}^{p,q}$ . And  $E_r^{p,q}$  has a direct limit  $E_\infty^{p,q}$ .
3. There is a complex  $E^\bullet$  and a decreasing bounded filtration  $F^p E^n$  on each  $E^n$  and  $E_\infty^{p,q} \cong F^p E^{p+q} / F^{p+1} E^{p+q}$ .

┘

**Def. (4.10.7.2) [Notations For Cohomological Spectral Sequence].**

- The cohomology filtration is called **bounded below**  $F^{n_s} E^n = 0$  for some  $n_s$ , it is called **bounded above**  $F^{n_s} E^n = E^n$  for some  $n_s$ .
- The cohomology filtration is called **exhaustive** iff  $\cup F^i E^n = E^n$ .
- The spectral sequence is called **regular** iff  $d_{pq}^r = 0$  for sufficiently large  $r$ .
- A spectral sequence is said to **weakly converges** to  $E^\bullet$  if there is a filtration

$$\dots \subset F^t H^n \subset F^{t-1} H^n \subset \dots \subset F^s H^n \subset \dots \subset H^n$$

that  $E_\infty^{pq} \cong F^p H^{p+q} / F^{p+1} H^{p+q}$ .

- A spectral sequence **approaches**  $E^\bullet$  if it weakly converges to  $E^\bullet$ .
- A spectral sequence **converges** to  $E^\bullet$  if it approaches  $E^\bullet$ , it is regular, and  $E^n = \varprojlim (E^n / F^p E^n)$ .
- If a first quadrant spectral sequence converges to  $E^\bullet$ , then the morphisms  $E_0^{n,0} \rightarrow E_\infty^{n,0} \subset E^n$  and  $E^n \rightarrow E_\infty^{0,n} \rightarrow E_0^{0,n}$  are called the **edge morphisms**.

┘

**Prop. (4.10.7.3) [Notation for Filtrations on Homological Complexes].** Let  $C_\bullet$  be a complex and  $\dots \subset F_{p-1} C \subset F_p C \subset \dots \subset C$  be filtrations of complexes. Then it is called **exhaustive** if  $C = \cup F_p C$ . It is called **Hausdorff** if  $\cap F_p C = 0$ . It is called **complete** if  $C = \varprojlim C / F_p C$ .

┘

**Def. (4.10.7.4) [Spectral Sequence of a Filtered Complex].** For a complex  $K^\bullet$  and a filtration  $F^p K^n$  on  $K^n$ , we have a natural spectral sequence

$$E_1^{pq} = H^{p+q}(F^p E^{p+q} / F^{p+1} E^{p+q}), \quad E^n = H^n(K^\bullet), \quad F^p E^n = H^n(F^p K^\bullet).$$

For a morphism of filtered complexes that are isomorphism for some  $r$ , induction on the exact sequence  $0 \rightarrow F^p E^n \rightarrow F^{p+1} E^n \rightarrow E_\infty^{p,n-p}$  and use five-lemma shows it induces isomorphism on  $H^* E$ .

┘

**Prop. (4.10.7.5) [Comparison Theorem].** For a morphism  $F$  between two convergent spectral sequences, if it is an isomorphism for some  $r$ , then it induce isomorphism on the infinite homologies.

┘

*Proof:* Clearly  $F$  induces isomorphisms on  $E_\infty^{p,q}$ . Because there are exact sequence

$$0 \rightarrow F^{p+1}H^n \rightarrow F^pH^n \rightarrow E_\infty^{p,n-p} \rightarrow 0$$

we can use five lemma and induction to show that  $F$  induces isomorphisms on  $F^pH^n/F^sH^n$ . Then because  $H^n = \cup F^pH^n$ , we can take colimit to show  $F$  induces isomorphisms on  $H^n/F^sH^n$ , then take inverse limits, we are done.  $\square$

**Prop. (4.10.7.6) [Classical Convergence].** If the filtration on a complex  $C_\bullet$  is bounded below and exhaustive for all  $C_n$ , then there is a spectral sequence that is also bounded below and converges to  $H_\bullet(C_\bullet)$ .  $\lrcorner$

*Proof:* Cf[Gelfand P203] for cohomological case and [Weibel P135] for homological case.  $\square$

**Prop. (4.10.7.7) [Complete convergence].** If the filtration is complete, exhaustive and the spectral sequence is regular, then the spectral sequence weakly converges to  $H_\bullet(C_\bullet)$ . And if it is also bounded above, then it converges to  $H_\bullet(C_\bullet)$ .  $\lrcorner$

*Proof:* Cf.[Weibel, P139].  $\square$

There are two examples, the stupid filtration and the canonical filtration, the canonical filtration is natural and factors through  $D(\mathcal{A})$ .

**Prop. (4.10.7.8) [Spectral Sequence of a Double Complex].** A double complex has two natural filtration of the total complex, they defines two spectral sequence, one has

$$E_{2,x}^{p,q} = H_x^p(H_y^{q,\bullet}(L^{\bullet,\bullet}))$$

and the other has

$$E_{2,y}^{p,q} = H_y^p(H_x^{q,\bullet}(L^{\bullet,\bullet})).$$

Cf.[Gelfand P209]. In fact under reflection, there is only one spectral sequence. For the horizontal filtration, the differential goes vertical first, for the vertical filtration, the differential goes horizontal first. The differential goes one way, the convergence goes reversely.

If both the filtration is finite and bounded, in particular if  $E$  is in the first quadrant, then they both converges to  $H^n(E)$ , this will generate important consequences.  $\lrcorner$

**Cor. (4.10.7.9).** If a double complex in the first quadrant has its all column acyclic (3rd-quadrant pointing), then the total complex is acyclic. Thus a morphism of double complex inducing quasi-isomorphism on each column induces a quasi-isomorphism on the total complex.

If a double complex has  $H_p(C_{*,q}) = 0, \forall p > 0, q$ , then

$$H_n(\text{Tot}C_{*,*}) = H_n(\text{Coker}(C_{1,*} \rightarrow C_{0,*}))$$

$\lrcorner$

**Prop. (4.10.7.10) [Horizontal Filtration].** For a second-quadrant-free homology double complex, the filtration is bounded below and exhaustive for  $\text{Tot}^\oplus$ , so the classical convergence(4.10.7.6) applies and there is a convergence

$$E_{p,q}^2 = H_p^h H_q^v(C) \Rightarrow H_{p+q} \text{Tot}^\oplus(C).$$

For a fourth-quadrant free homology double complex, the filtration is complete and exhaustive for  $\text{Tot}^\Pi$ , so the complete convergence(4.10.7.7) applies and there is a weak convergence

$$E_{p,q}^2 = H_p^h H_q^v(C) \Rightarrow H_{p+q} \text{Tot}^\Pi(C).$$

$\lrcorner$

**Cor. (4.10.7.11)[Grothendieck Spectral Sequence].** If  $\mathcal{A}, \mathcal{B}, \mathcal{C}$  be Abelian categories,  $\mathcal{B}$  have enough injectives and  $F : K^+(\mathcal{A}) \rightarrow K^+(\mathcal{B}), G : K^+(\mathcal{B}) \rightarrow K^+(\mathcal{C})$  are exact functors between triangulated categories. If  $\mathcal{R}_{\mathcal{A}}$  is adapted to  $F, G \circ F$ , then for any  $X^\bullet \in K^+(\mathcal{A})$ , there is a spectral sequence convergence(to upper left)

$$E_2^{p,q} = R^p G(R^q F(X)) \implies E^\infty = R^n(G \circ F)(X).$$

And if  $\mathcal{R}_{\mathcal{A}} = \mathcal{I}_{\mathcal{A}}$ , this spectral sequence is functorial in  $X^\bullet$ .

In particular, this applies to the case that  $F$  is right adjoint to an exact functor and both  $\mathcal{A}, \mathcal{B}$  have enough injectives.  $\lrcorner$

*Proof:* Choose a functorial injective resolution  $X \mapsto I_X^\bullet$  (4.10.2.9), let  $K^\bullet = F(I_X^\bullet) = RF(X)$ , and choose a functorial CE resolution of  $K^\bullet$  (4.8.9.10), because the resolutions for  $B^i \rightarrow Z^i \rightarrow H^i$  and  $Z^i \rightarrow K^i \rightarrow B^{i+1}$  split and  $G$  is additive, we have

$$H_x^{q,\bullet}(G(L^{\bullet,\bullet})) = G(H_x^{q,\bullet}(L^{\bullet,\bullet})) = RG(H^q(K))$$

So

$$E_{2,y}^{p,q} = H_y^p(H_x^{q,\bullet}(L^{\bullet,\bullet})) = R^p G(H^q(K)) = R^p G(R^q F(X))$$

and

$$E^\bullet = RG(\text{Tot}(L)) = G(\text{Tot}(L)) = RG(K) = RG \circ RF(X) \cong R(G \circ F)(X) \text{ (4.10.3.6)}.$$

$\square$

**Cor. (4.10.7.12).** The low degree parts read:

$$0 \rightarrow R^1 G(F(A)) \rightarrow R^1(G \circ F)(A) \rightarrow G(R^1 F(A)) \rightarrow R^2(G)(F(A)) \rightarrow R^2(G \circ F)(A).$$

(Check definition). More generally, if  $R^p G(R^q F(A)) = 0, 0 < q < n$ , then

$$R^m G(F(A)) \cong R^m(G \circ F)(A) \quad m < n$$

And

$$0 \rightarrow R^n G(F(A)) \rightarrow R^n(G \circ F)(A) \rightarrow G(R^n F(A)) \rightarrow R^{n+1} G(F(A)) \rightarrow R^{n+1}(G \circ F)(A).$$

$\lrcorner$

**Remark (4.10.7.13).** The Grothendieck spectral sequence is very important.  $\lrcorner$

**Cor. (4.10.7.14)[Spectral Sequence for Hypercohomologies].** For chain complex  $K$  in  $K^+(\mathcal{A})$  and a left exact functor  $F$ , the CE resolution will generate two spectral sequences by (4.10.7.10):

$$E_{2,x}^{p,q} = H_x^p(R^q F(A)) \Rightarrow \mathbb{R}^{p+q}(A), \text{ when } A \text{ is bounded below}$$

$$E_{2,y}^{p,q} = (R^p F)(H^q(A)) \Rightarrow \mathbb{R}^{p+q}(A). \text{ weakly convergent}$$

where the RHS is the hypercohomologies (4.10.3.11).  $\lrcorner$



## 4.11 Differential Graded Algebras

Main references are [Ker] and [Sta].

**Def. (4.11.0.1) [Differential Graded Algebras].** A **differential graded algebra** or *DGA* is a chain complex  $A^\bullet$  of  $R$ -modules with  $R$ -linear maps  $A^m \times A^n \rightarrow A^{m+n}$  that

$$d(ab) = d(a)b + (-1)^n ad(b).$$

that makes  $\oplus A^n$  into an associative and unital  $R$ -algebra.

Notice the first condition is equivalent to giving a map  $\text{Tor}(A^\bullet \times_R A^\bullet) \rightarrow A$ .

For a differential algebra  $A^\bullet$ , a right **differential module** is defined naturally. The tensor operation gives a closed symmetric monoidal structure  $\mathcal{M}_A$ .

Notice a usual  $R$ -algebra  $A$  can be seen as a differential graded algebra as  $A^0 = A$  and  $A^n = 0$  for  $n > 0$ .

And as in the case of chain complexes, the category of differential modules over  $A$  can be given a derived category.  $\lrcorner$

**Def. (4.11.0.2).** A differential graded algebra  $A^\bullet$  is called **commutative** if  $ab = (-1)^{\deg(a)\deg(b)}ba$ . It is called **strictly commutative** if moreover  $a^2 = 0$  for  $\deg(a)$  odd.  $\lrcorner$

**Def. (4.11.0.3).** For two differential graded algebras  $A, B$ , the tensor graded algebra  $A^\bullet \otimes B^\bullet$  is given by the  $\text{Tor}(A^\bullet \otimes_R B^\bullet)$ .  $\lrcorner$

### 1 dg-Categories

**Def. (4.11.1.1) [dg-Categories].** Given a DGA  $A$ , a **dg-category** over  $A$  is a category enriched over the monoidal category  $\mathcal{M}_A$  (4.11.0.1). Let  $dgCat_A$  denote the category of small dg-categories over  $A$  where morphisms are given by monoidal functors.  $\lrcorner$

**Def. (4.11.1.2) [Homotopy Categories].** Because  $H^0$  and  $Z^0$  are right-lax monoidal functors from  $\text{Ch}(R)$  to  $\text{Mod}(R)$ , given a dg-category  $\mathcal{C}$ , by transferring, we can get categories  $H^0(\mathcal{C})$  and  $Z^0(\mathcal{C})$  enriched over  $\text{Mod}_{A^0}$ .  $\lrcorner$

**Def. (4.11.1.3) [Equivalences].** A morphism between dg-categories are called an **equivalence** if induces quasi-isomorphisms on all hom-complexes.  $\lrcorner$

**Prop. (4.11.1.4) [Model Category of dg-Categories].** There is a cofibrantly generated model category on  $dgCat_A$ , where weak equivalences are quasi-equivalences and the fibrations are morphisms  $F : \mathcal{A} \rightarrow \mathcal{B}$  that:

- induces component-wise surjections on hom-complexes.
- given an isomorphism  $g : F(X) \rightarrow Y \in H^0(\mathcal{B})$ , there is an isomorphism in  $H^0(\mathcal{A})$  lifting  $g$ .

This monoidal structure is induced from that of the case  $A = R$  and the right-lax monoidal functor  $\text{Ch}(R) \rightarrow \mathcal{M}(A)$  given by  $M^\bullet \mapsto A \otimes M$ .  $\lrcorner$

*Proof:* Cf. [Tabuada, Goncalo. Une structure de catgorie de modles de Quillen sur la catgorie des dg- catgories. C. R. Acad. Sci. Paris Sr. I Math. 340 (1) (2005), 15-19. (2005), 3309?3339.]  $\square$

### 2 Sheaves of DGAs

## 4.12 Topology II

**Notation(4.12.0.1).**

- Use notations on Topology Theory defined in [Topology I](#).
- Use notations on Group Theory defined in [Group Theory](#).
- Use notations on Category Theory defined in ??.
- Let  $X \in \mathcal{T}\text{op}$ .
- Let  $G \in \mathcal{T}\text{opGrp}$ ([4.12.1.1](#)).

┘

### 1 Topological Groups

**Def.(4.12.1.1) [Topological Groups].** A **topological group** is a group object in the category of groups([4.1.1.67](#)). ┘

**Prop.(4.12.1.2).** If  $U$  is a nbhd of  $\mathbf{1}$  in a topological group, then there is a nbhd  $V$  of  $\mathbf{1}$  that  $VV \subset U$ . ┘

*Proof:* Consider the map  $G \times G \rightarrow G$  continuous, and it maps  $(\mathbf{1}, \mathbf{1})$  to  $\mathbf{1} \in U$ , so the preimage of  $U$  contains a nbhd of  $(\mathbf{1}, \mathbf{1})$ , thus some  $V_1 \times V_2$ , then choose  $V = V_1 \cap V_2$ . ┘

**Prop.(4.12.1.3).** Let  $G$  be a connected topological group, then for any nbhd  $U$  of  $e$ ,  $G = \bigcup_{n=1}^{\infty} (U \cup U^{-1})^n$ .  
In particular, any open subgroup of  $G$  equals  $G$ . ┘

*Proof:* This is because the RHS is an open subgroup of  $G$ , so all its cosets in  $G$  are also open, so it equals  $G$  as  $G$  is connected. ┘

**Prop.(4.12.1.4) [Separating Axioms].** The following are equivalent:

- $e$  is a closed pt.
- $G$  is  $T_1$ .
- $G$  is Hausdorff( $T_2$ ).
- $G$  is regular.
- $G$  is completely regular

┘

*Proof:* ┘

**Prop.(4.12.1.5).** A Hausdorff topological group is completely regular. ┘

*Proof:* Use a sequence of neighbourhood of identity to construct a uniform metric on  $G$ . Then set  $\phi(x) = \min\{1, 2\sigma(a, x)\}$ . Cf.[Abstract Harmonic Analysis Ross §8.4] ┘

**Prop.(4.12.1.6).** For a compact subset  $K$  and a nbhd  $U$  of  $e$  in a topological group, there exists a nbhd  $V$  of  $e$  that  $xVx^{-1} \subset U$  for any  $x \in K$ . ┘

*Proof:* For any  $x$ , there exists a nbhd  $W_x$  of  $x$  and a nbhd  $V_x$  of  $e$  that  $txt^{-1} \in U$  for any  $t \in W_x$  and  $y \in V_x$ . Let f.m.  $W_{x_i}$  cover  $K$ , then  $V = \bigcap V_{x_i}$  satisfies the condition. ┘

**Prop. (4.12.1.7).** A compact open nbhd of  $e$  in a Hausdorff topological group contains an open subgroup of  $G$ .  $\lrcorner$

*Proof:* Cf.[Etale Cohomology Fulei P147]  $\square$

**Prop. (4.12.1.8)[Homogenous Space].** Let  $G$  be a topological group and  $H$  a closed subgroup.  $G/H$  is the quotient space in the quotient topology(4.4.1.9), then it is Hausdorff.  $\lrcorner$

*Proof:* If  $\bar{x} \neq \bar{y}$ , then consider a preimage  $xy^{-1} \in G \setminus H$ , then we can find some open subset  $V$  that  $VV \subset G \setminus H$  by(4.12.1.2), thus  $\bar{x} + \bar{V} \cap \bar{y} + \bar{V} = \emptyset$ . Hence  $G/H$  is Hausdorff.  $\square$

### Group Actions

**Prop. (4.12.1.9)[Quotient by Group Action is Open].** Let  $X \in \text{Top}$  and  $G \times X \rightarrow X$  be a group action, then the quotient map  $\pi : X \rightarrow X/G$  is open.  $\lrcorner$

*Proof:* This is because the  $\pi^{-1}(\pi(U)) = \bigcup_{g \in G} gU$ .  $\square$

**Def. (4.12.1.10)[Regular Action].** An **regular action** is an action  $\gamma : G \times X \rightarrow X$  that satisfies the following equivalent conditions:

- the graph of  $\gamma$  in  $X \times X$  is closed.
- The diagonal  $\Delta_{X/G} \subset X/G \times X/G$  is closed.
- $X/G$  is Hausdorff.

$\lrcorner$

*Proof:*  $2 \iff 3$  is clear, for 1, 2, notice  $X/G \times X/G \cong (X \times X)/(G \times G)$ , and the inverse image of  $\Delta$  in  $X \times X$  is just the graph of  $\gamma$ .  $\square$

**Def. (4.12.1.11)[Proper Action].** A **proper action** is an action  $\gamma : G \times X \rightarrow X$  that the graph map  $\Gamma : G \times X \rightarrow X \times X$  is a proper map.  $\lrcorner$

**Prop. (4.12.1.12)[Proper Action is Regular].** A proper action(4.12.1.11) of  $G$  on  $X \in \text{LCH}$  is a regular action.  $\lrcorner$

*Proof:* This follows from(4.4.2.12).  $\square$

**Def. (4.12.1.13)[Proper Discontinuous Actions].** A group action is called **proper discontinuous** iff any elements  $x, y \in H$  there are nbhds  $U_x, U_y$  that  $\{g \in G | g(U_x) \cap U_y \neq \emptyset\}$  is finite.  $\lrcorner$

**Def. (4.12.1.14)[Covering Space Action].** A **covering space action** is action of a topological group  $G$  on a topological space  $Y$  is called if for any  $y \in Y$ , there is a nbhd  $U$  that  $g(U) \cap U = \emptyset$  if  $g \neq 1$ .  $\lrcorner$

**Prop. (4.12.1.15)[Characterization of Proper Actions].** Let  $\gamma : G \times X \rightarrow X$  be a group action that  $X$  is Hausdorff, then  $\gamma$  is a proper action(4.12.1.11) iff any  $K \subset X$  compact, the set  $G_K = \{g \in G | g(K) \cap K \neq \emptyset\}$  is compact.

In particular, if  $H \leq G$  is an open subgroup of finite index, then  $G$  acts properly on  $X$  iff  $H$  acts properly on  $X$ .  $\lrcorner$

*Proof:* Let  $\Gamma : G \times X \rightarrow X \times X$  be the graph.

1  $\rightarrow$  2:  $G_K = \pi_1(\Gamma^{-1}(K \times K))$ , thus it is compact.

2  $\rightarrow$  1: Let  $L \subset M \times M$  be compact, then  $L \subset \pi_1(L) \times \pi_2(L)$ , and  $L$  is closed. Let  $K = \pi_1(L) \cup \pi_2(L)$ , then  $\Gamma^{-1}(L) \subset \Gamma^{-1}(\pi_1(L) \times \pi_2(L)) \subset G_K \times K$ , which is a closed subset of a compact set, so  $\Gamma^{-1}(L)$  is compact, and  $\Gamma$  is a proper map.  $\square$

**Prop. (4.12.1.16).** If  $G$  is a compact topological group, then any group action  $\gamma : G \times X \rightarrow X$  on a Hausdorff space  $X$  is proper.  $\lrcorner$

**Prop. (4.12.1.17) [Orbit of Proper Maps].** Let  $\theta$  be a proper action of  $G$  on a Hausdorff space  $X$ , then each orbit map  $\theta^{(p)}$  is proper. In particular, if  $X$  is locally compact, then the orbits are all closed (4.4.2.12).  $\lrcorner$

*Proof:* For any compact subset  $K \subset X$ ,  $(\theta^{(p)})^{-1}(K)$  is closed by continuity and is contained in  $G_{K \cup \{p\}}$ , thus is compact.  $\square$

**Prop. (4.12.1.18) [Properly Discontinuous Map is Proper].** If  $G$  acts properly discontinuously on a topological space  $H$ , then for any compact subsets  $K_1, K_2 \in H$ ,  $\{g \in G \mid K_2 \cap g(K_1) \neq \emptyset\}$  is finite. In particular, if  $H$  is Hausdorff, then it is a proper action.  $\lrcorner$

*Proof:* Notice for any two points we can find nbhds that f.m.  $g$  intersects these two nbhds, so we can use the compactness to find f.m. pair of nbhds to cover  $K_2$ , and then use these nbhds to cover  $K_1$  and finish the proof.  $\square$

**Prop. (4.12.1.19) [Proper Free Action is a Covering Space Action].** Let  $G \times X \rightarrow X$  be a proper free action on a locally compact Hausdorff space, then it is a covering space action.  $\lrcorner$

*Proof:* For any  $p \in X$ , choose a precompact nbhd  $U$  of  $x$ , then  $G_{\overline{U}}$  is finite, then shrink  $U$ .  $\square$

**Prop. (4.12.1.20) [Continuity].** Let a topological group  $G$  acts freely and properly on a space  $X$ . If  $G$  and  $X/G$  are both connected, then  $X$  is connected.  $\lrcorner$

*Proof:* If  $U, V$  are open subsets of  $X$  that  $U \cap V = \emptyset, U \cup V = X$ , then  $\pi(U), \pi(V)$  are open by (4.12.1.9). Now  $\pi(U) \cap \pi(V) = \emptyset$ , because otherwise there is a  $G$ -orbit of  $X$  intersecting both  $U$  and  $V$ , contradiction the fact  $G$  is connected. Then  $\pi_1(U) = \emptyset$  or  $\pi(V) = \emptyset$ , thus  $U = \emptyset$  or  $V = \emptyset$ , and  $X$  is connected.  $\square$

**Def. (4.12.1.21) [Constructible Action].** An action of a topological group  $G$  on a topological space  $X$  is called **constructible** if its graph is constructible in  $X \times X$  (4.12.2.10).  $\lrcorner$

**Prop. (4.12.1.22).** Let  $\gamma : G \times X \rightarrow X$  be an action

- $\gamma$  is constructible iff the diagonal  $\Delta_{X/G} \subset X/G \times X/G$  is constructible.
  - If  $\gamma$  is constructible and  $X$  is non-empty, then there is a  $G$ -invariant open subset  $U \subset X$  that  $G$  acts regularly.
  - For a constructible action, each  $G$ -orbit is locally closed.
- $\lrcorner$

*Proof:* Cf. [Bernstein-Zelevinsky, P54].  $\square$

### Totally Disconnectedness

**Prop. (4.12.1.23).** A compact topological group is totally disconnected iff the intersection of all compact open nbhds of  $e$  equals  $\{e\}$ .  $\lrcorner$

*Proof:* If it is totally disconnected, then  $\{1\}$  is closed, so  $G$  is Hausdorff(4.12.1.4), so by(4.4.1.25), the assertion is true. Conversely, if the intersection of all compact open nbhds of  $e$  equals  $\{e\}$ , then  $\{1\}$  is closed because  $G$  is a group.  $\square$

**Prop. (4.12.1.24).** A precompact nbhd of a  $e$  in a totally disconnected topological group contains a compact open subgroup.  $\lrcorner$

*Proof:* Cf.[Etale Cohomology Fulei P147].  $\square$

## 2 Spaces from Algebraic Geometry

### Noetherian Space

**Def. (4.12.2.1) [Noetherian Spaces].** A **Noetherian space** is a space  $X \in \mathcal{T}_{\text{op}}$  that any descending chain of closed subsets stabilizes. A **locally Noetherian space** is a space  $X \in \mathcal{T}_{\text{op}}$  that every point has a nbhd  $U$  s.t.  $U$  is a Noetherian space.  $\lrcorner$

**Prop. (4.12.2.2).** A Noetherian space is quasi-compact and all subsets of it in the induced topology is Noetherian hence quasi-compact.  $\lrcorner$

*Proof:* Let  $T \subset X$ , for a chain of closed subsets  $Z_i \cap T$  of  $T$ ,  $Z_1, Z_1 \cap Z_2, \dots$  stabilize in  $X$ , hence the chain stabilize in  $T$ .  $\square$

**Prop. (4.12.2.3).** If  $X$  can be covered by f.m. Noetherian subspaces, then  $X$  is Noetherian.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.12.2.4) [Noetherian Space F.M. Irreducible Components].** A Noetherian space has only f.m. irreducible component, hence it has only f.m. connected components.  $\lrcorner$

*Proof:* Let  $\mathcal{C}$  be the family of closed subset that has infinitely many component, then there is a minimal object, but it is not irreducible, one of the component has infinitely many components and be smaller.  $\square$

### Quasi-Separated

**Def. (4.12.2.5) [Quasi-Separated].** A space  $X$  is called **quasi-separated** if the diagonal morphism is quasi-compact(4.4.2.2). If  $X$  has a basis consisting of quasi-compact open subsets, then this is equivalent to any intersection of two quasi-compact open subsets is quasi-separated open.  $\lrcorner$

### Specialization & Generalization

**Def. (4.12.2.6) [Specializations and Generalizations].** Let  $X \in \mathcal{T}_{\text{op}}$  then  $x$  is said to be a **specialization of  $y$**  if  $x \in \overline{\{y\}}$ . And in this case  $y$  is said to be a **generalization of  $x$** .

And they are called **immediate specializations/generalizations** if there are no other points  $z \in X$  s.t.  $y \rightarrow z \rightarrow x$ .  $\lrcorner$

**Def. (4.12.2.7)[Going Up and Down].** A map  $f$  of spaces is said to satisfy the **going-up property** iff specialization lifts along  $f$ . It is said to satisfy the **going-down property** iff generalization lifts along  $f$ .  $\lrcorner$

**Prop. (4.12.2.8).** A closed map satisfies going-up.  $\lrcorner$

*Proof:* If  $y \rightarrow y'$ ,  $f(x) = y$ , consider  $f(\overline{\{x\}})$ , it is closed and contains  $y$ , so it contains  $y'$ , thus the result.  $\square$

### Constructible Sets

**Def. (4.12.2.9)[Retrocompact Subsets].** A subset of  $X$  is called **retrocompact** if the inclusion map is quasi-compact (4.4.2.2).  $\lrcorner$

**Def. (4.12.2.10)[Constructible Subset].** A subset of  $X$  is called **constructible** if it is a finite union of sets of the form  $U \cap V^c$  where  $U, V$  are open and retrocompact in  $X$ . In the case when  $X$  is Noetherian, by (4.12.2.2), all subsets are retrocompact hence constructible sets are just union of locally closed subsets of  $X$ .

A set of  $X$  is called **locally constructible** if locally it is constructible. If  $X$  is quasi-compact, then a locally constructible set is just a constructible set.  $\lrcorner$

**Prop. (4.12.2.11).** The set of constructible subsets of  $X$  is a Boolean algebra.  $\lrcorner$

*Proof:* Cf. [Sta]005H.  $\square$

**Prop. (4.12.2.12)[Constructible and Subsets].**

- If  $U$  is open in  $X$ , then for any  $E$  constructible in  $X$ ,  $E \cap U$  is constructible in  $U$ .
- If  $U$  is retrocompact open and  $E$  is constructible in  $U$ , then  $E$  is constructible in  $X$ .

$\lrcorner$

*Proof:* Easy.  $\square$

**Prop. (4.12.2.13).** Any constructible subsets of  $X$  is retrocompact.  $\lrcorner$

*Proof:* It suffices to prove  $U_i \cap V_i^c \cap W$  is quasi-compact for  $W$  quasi-compact, but this is because it is a closed subspace of the quasi-compact subspace  $U_i \cap W$ .  $\square$

**Cor. (4.12.2.14).** An open subset of  $X$  is constructible iff it is retrocompact, a closed subset of  $X$  is constructible iff its complement is retrocompact by (4.12.2.11).  $\lrcorner$

**Prop. (4.12.2.15).** If  $X = V_1 \cup \dots \cup V_m$  is a finite union of retrocompact open subsets, then a subset  $E \subset X$  is constructible in  $X$  iff  $E \cap V_i$  is constructible in  $V_i$  for each  $i$ .  $\lrcorner$

*Proof:* Cf. [Sta]053W.  $\textcolor{red}{?}$   $\square$

**Def. (4.12.2.16)[Constructible topology].** The **constructible topology**  $X_{cons}$  on a quasi-compact space  $X$  is generated by the open subsets  $U, U^c$ , where  $U$  is a quasi-compact open.

Notice that the space is quasi-compact, so the constructible topology is the coarsest topology that every constructible subset of  $X$  is both open and closed.  $\lrcorner$

**Prop. (4.12.2.17).** Let  $X$  be quasi-compact and quasi-separated, then any constructible subset of  $X$  is quasi-compact. In particular, if  $Y$  is closed in  $X$ , then  $Y$  is constructible iff it is quasi-compact.  $\lrcorner$

*Proof:* For  $Y = \bigcup_{i=1}^n (U_i - V_i)$ , with  $U_i, V_j$  quasi-compact open in  $X$ , then  $U_i - V_i$  is closed in  $U_i$  thus quasi-compact, and then  $Y$  is quasi-compact.  $\square$

**Prop. (4.12.2.18).** Let  $E$  be a constructible subset of a space  $X$ , if  $E$  is dense in then  $E$  contains some open dense subset of its closure.  $\lrcorner$

*Proof:* Let  $Y = \bigcup_{i=1}^k Y_i$  where  $Y_i$  are locally closed. Denote  $Z_i = \overline{Y_i} \setminus Y_i$ ,  $Z = \bigcup Z_i$ ,  $W = \overline{Y} \setminus Z$ . Then  $W \subset Y$ , and we show that  $W$  is open dense in  $\overline{Y}$ : As  $Y_i$  are locally closed,  $Y_i$  is open in  $\overline{Y_i}$ , thus  $Z_i$  is closed, and  $Z$  is closed, so  $W$  is open in  $\overline{Y}$ . To show  $W$  is dense in  $\overline{Y}$ : If some open subset  $U$  of  $\overline{Y}$  satisfies  $U \cap Z_i$ , then  $U$  cannot be contained in any  $Z_i$ , so we can inductively show  $U \setminus Z_1, U \setminus (Z_1 \cup Z_2), \dots$  are non-empty, which is a contradiction.  $\square$

### Irreducible

**Def. (4.12.2.19) [Irreducible Space].** A space is called **irreducible** iff there are no two nonempty nonintersecting open subsets. Thus an open subset of an irreducible set is dense and irreducible.  $\lrcorner$

**Prop. (4.12.2.20).** If  $Y$  is irreducible in  $X$ , then  $\overline{Y}$  is also irreducible.  $\lrcorner$

*Proof:* Any two nonempty open sets of  $\overline{Y}$  must intersect  $Y$  thus must intersect.  $\square$

**Prop. (4.12.2.21).** If  $X \in \mathcal{T}_{\text{op}}$  and  $U \subset X$  is open, then  $Y \mapsto \overline{Y}$  is a order preserving bijection between irreducible closed subspaces of  $U$  and irreducible closed subspaces of  $X$  meeting  $U$ .  $\lrcorner$

### Jacobson Space

**Def. (4.12.2.22).** Let  $X$  be a space and  $X_0$  the set of closed pts of  $X$ , then  $X$  is called **Jacobson** iff  $\overline{Z \cap X_0} = Z$  for every closed subset  $Z$  of  $X$ . This is equivalent to every non-empty locally closed subset of  $X$  contains a closed pt.

Thus there is a correspondence between closed subsets of  $X_0$  and closed subsets of  $X$ , so they have the same Krull dimension.  $\lrcorner$

**Prop. (4.12.2.23).** Being Jacobson is local. And for an open covering  $U_i$  of  $X$ ,  $X_0 = \bigcup U_{i,0}$ .  $\lrcorner$

*Proof:* Firstly, if  $X = \bigcup U_i$  where  $U_i$  are Jacobson,  $X_0 \cap U_i = U_{i,0}$ . One direction is trivial, for the other, let  $x$  be closed in  $U_i$ , then consider  $\{x\} \cap U_j$ . If  $x \notin U_j$ , this is empty, if  $x \in U_j$ , consider  $T = \{x\} \cup (U_j - U_i \cap U_j)$ , then  $T$  is closed in  $U_j$ , so by hypothesis, closed pts of  $U_j$  are dense in  $T$ , so  $x$  must be closed in  $U_j$ , so  $x$  is closed in  $X$ . Now clearly  $X$  is Jacobson.

Conversely, if  $X$  is Jacobson, for a closed subset  $Z$  of  $U_i$ ,  $X_0 \cap \overline{Z}$  is dense in  $\overline{Z}$ , so  $X_0 \cap Z$  is dense in  $Z$ , then clearly  $U_i$  is Jacobson.  $\square$

**Cor. (4.12.2.24).** If  $X$  is Jacobson, then any locally constructible sets of  $X$  is Jacobson. And its closed pts are closed in  $X$ .  $\lrcorner$

*Proof:* By the proposition, we only have to prove for constructible sets. For  $T = \bigcup T_i$  where  $T_i$  is locally closed, then a locally closed set in  $T$  has a non-empty intersection  $T \cap T_i$  which is also locally closed for some  $i$ .

Hence it has a closed pt in  $X$  hence in  $T$ , so  $T$  is Jacobson. The second assertion is implicit in the proof.  $\square$

**Prop. (4.12.2.25).** If  $X$  is Jacobson, then an open set  $U$  of  $X$  is compact iff  $U \cap X_0$  is compact, hence an open set  $U$  is retrocompact iff  $U \cap X_0$  is retrocompact.

Hence the constructible sets of  $X$  correspond to the constructible sets of  $X_0$ .

And Irreducible closed subsets correspond to irreducible subsets of  $X_0$ .  $\lrcorner$

### Krull-Dimension

**Def. (4.12.2.26) [Krull-Dimensions].** The **Krull-dimension** of a topological space is the length of the longest chain of closed irreducible subsets.

The **local Krull-dimension**  $\dim_x(X) = \min\{\dim U \mid x \in U \subset X \text{ open in } X\}$ .  $\lrcorner$

**Prop. (4.12.2.27).** If  $Y \subset X$ , then  $\dim Y \leq \dim X$ , because the closure of any chain of  $Y$  is a chain of  $X$  by (4.12.2.20).

For an open covering  $\{U_i\}$  of  $X$ ,  $\dim X = \sup \dim U_i$ , because for any chain of closed irreducible subsets, if  $U_i$  intersects the minimal one, then  $\dim U_i = \text{length of this chain}$ .  $\lrcorner$

**Prop. (4.12.2.28).**  $\dim X = \sup \dim_x(X)$ .  $\lrcorner$

*Proof:* The right is smaller than the left by (4.12.2.27), and for any chain  $Z_0 \subset Z_1 \subset \dots \subset Z_n$  of irreducible closed subset of  $X$ , if I choose a point  $x \in Z_0$ , then  $\dim_x(X) \geq n$ .  $\square$

**Prop. (4.12.2.29).** In case  $X = \text{Spec } A$  for a Noetherian ring  $A$ ,  $\dim X = \sup \dim A_p$ , because  $A$  is of finite ?  $\lrcorner$

**Def. (4.12.2.30) [Codimensions].** Let  $X \in \mathcal{T}\text{op}$  and  $Y \subset X$  be an irreducible closed subspace, then the **codimension** of  $Y$  in  $X$  is defined to be the supremum of lengths  $e$  of chains  $Y = Y_0 \subset Y_1 \subset \dots \subset Y_e = X$ , denoted by  $\text{codim}(Y, X)$ .  $\lrcorner$

**Prop. (4.12.2.31) [Codimensions are Local].** Let  $X \in \mathcal{T}\text{op}$  and  $Y \subset X$  be an irreducible closed subspace. If  $U \subset X$  is an open subset s.t.  $Y \cap U \neq \emptyset$ , then  $\text{codim}(Y, X) = \text{codim}(Y \cap U, U)$ .  $\lrcorner$

*Proof:* Cf. [Sta]02I4. ?  $\square$

### Catenary spaces and Dimension Functions

**Def. (4.12.2.32) [Catenary Space].**  $X \in \mathcal{T}\text{op}$  is called **catenary space** iff for any inclusion of irreducible closed subsets of  $X$ , their codimension is finite and every maximal chain of irreducible closed subsets has the same dimension. This is equivalent to  $\text{codim}(X, Y) + \text{codim}(Y, Z) = \text{codim}(X, Z)$ .  $\lrcorner$

**Prop. (4.12.2.33).** Catenary is a local property, by (4.12.2.21).  $\lrcorner$

**Def. (4.12.2.34) [Dimension Function].** For  $X \in \mathcal{T}\text{op}$ , consider the specialization relation (4.12.2.6), a **dimension function** is a function  $\delta : |X| \rightarrow \mathbb{Z}$  that

- if  $y$  is a specialization of  $x$  in  $X$ , then  $\delta(y) < \delta(x)$ .
- if it is a direct specialization, then  $\delta(y) = \delta(x) - 1$ .

The dimension function is usually considered only when the space is sober.  $\lrcorner$

**Prop. (4.12.2.35).** Let  $X \in \mathcal{T}\text{op}$  be sober and catenary with a catenary function, then  $X$  is catenary, and if  $x$  is a specialization of  $x$  in  $X$ , then

$$\delta(x) - \delta(y) = \text{codim}(\overline{\{y\}}, \overline{\{x\}}).$$

$\lrcorner$

*Proof:* This is clear from the definitions (4.12.2.30).  $\square$

**Prop. (4.12.2.36).** If  $X \in \mathcal{T}\text{op}$  be locally Noetherian and sober, and  $\delta, \delta'$  are two dimension functions on  $X$ , then  $\delta - \delta'$  is locally constant on  $X$ .  $\lrcorner$

*Proof:* We may assume  $X$  is Noetherian, so it has only f.m. irreducible components by (4.12.2.4), then  $\delta - \delta'$  is locally constant on the  $X$  minus the irreducible components not passing through  $x$  by (4.12.2.35).  $\square$



### Sober Spaces

**Def. (4.12.2.37) [Sober Spaces].** A **sober space** is a topological space  $X \in \mathcal{Top}$  s.t. every irreducible closed subset has a unique generic point.  $\lrcorner$

**Prop. (4.12.2.38).** A sober space (4.12.2.37) is  $T_1$ . Conversely, a finite  $T_0$  space is sober.  $\lrcorner$

*Proof:* The first assertion is because if  $x \in \overline{\{y\}}$  and  $y \in \overline{\{x\}}$ , then  $\overline{\{x\}} = \overline{\{y\}}$ , and this irreducible closed subset has two generic point, contradiction.

If the space is finite, then for a closed irreducible subset  $T = \{x_1, \dots, x_n\}$ ,  $T = \cup \overline{\{x_i\}}$ , as it is irreducible,  $T = \overline{\{x_i\}}$  for some  $x_i$ , and  $i$  is unique as it is  $T_1$ , so  $X$  is sober.  $\square$

**Prop. (4.12.2.39) [Soberization].** There is a left adjoint  $t$  to the forgetful functor from the Sober spaces.  $t(X)$  consists of irreducible closed subsets of  $X$ , and use  $t(Y)$  for  $Y$  closed as closed subsets. for a map  $f : X \rightarrow Z$  to a sober space  $Z$ , the extension maps the generic point of an irreducible  $Y$  to the generic point of the closure of  $f(Y)$ .  $\lrcorner$

**Def. (4.12.2.40) [Zariski-Space].** A Noetherian sober space (4.12.2.37) is called a **Zariski-space**.  $\lrcorner$

**Prop. (4.12.2.41) [Catenary and Sober Spaces].** Let  $X \in \mathcal{Top}$  be locally Noetherian, sober, then  $X$  is catenary iff any point  $x \in X$  has a nbhd  $U$  which has a dimension function.  $\lrcorner$

*Proof:* Cf. [Sta]02IC.

The other direction follows from (4.12.2.35) and (4.12.2.33).  $\square$

### 3 Spectral Spaces

References are [Sta]5.23 and [Adic Spaces].

**Def. (4.12.3.1) [Spectral Space].** A **spectral space** is a qcqs (4.12.2.5) sober space s.t. the quasi-compact opens form a basis for the topology.

A space is called **locally spectral** iff it has an open covering by spectral spaces.

A morphism  $f : X \rightarrow Y$  between locally spectral spaces is called a **spectral map** if for any open spectral spaces  $U \subset f^{-1}(V)$ ,  $f : U \rightarrow V$  is quasi-compact.  $\lrcorner$

**Prop. (4.12.3.2) [Connected Components of Spectral Spaces].** If  $X$  is a spectral space (4.12.3.1), then any connected component of  $X$  is an intersection of clopen subsets.  $\lrcorner$

*Proof:* Let  $x \in X$  and  $S$  be the intersection of all clopen subsets of  $X$  containing  $x$ , then it suffices to show  $S$  is connected. Suppose  $S = B \amalg C$  with  $B, C$  closed, then  $B, C$  are compact, thus there exist quasi-compact opens  $U, V \subset X$  that  $B = U \cap S, C = V \cap S$ . Then  $U \cap V \cap S = \emptyset$ . Now  $U \cap V$  is quasi-compact also, so there exists some clopen  $Z_\alpha$  containing  $x$  that  $Z_\alpha \cap U \cap V = \emptyset$ . Similarly, there exists some clopen  $Z_\beta$  containing  $x$  that  $Z_\beta \subset U \cup V$ . Then  $Z_\gamma = Z_\alpha \cap Z_\beta$  is clopen and contained in  $U \Delta V$ , Then Both  $Z_\gamma \cap U$  and  $Z_\gamma \cap V$  is clopen, so  $U = \emptyset$  or  $V = \emptyset$ .  $\square$

**Cor. (4.12.3.3).** Let  $X$  be a spectral space, then for a subset  $T$  of  $X$ ,  $T$  is an intersection of clopen subsets of  $X$  iff  $T$  is closed in  $X$  and is a union of connected components of  $X$ .  $\lrcorner$

*Proof:* If  $T$  is an intersection of clopen subsets, then  $T$  is clearly a union of connected components of  $X$ . Conversely, if  $T$  is a union of connected components of  $X$ , if  $x \notin T$ , let  $C$  be a connected components containing  $x$ . Then  $C$  is an intersection of clopen subsets, by (4.12.3.2). These subsets are closed under finite intersections, so by the compactness of  $T$ , there is a clopen subset containing  $T$  but not  $x$ , so we are done.  $\square$

**Prop. (4.12.3.4).** The product of two spectral spaces(4.12.3.1) is spectral.  $\lrcorner$

*Proof:* Cf.[Sta]0907? $\square$

**Prop. (4.12.3.5).** If  $X$  is quasi-compact and every connected component of  $X$  is an intersection of clopen subsets, then  $\pi_0(X)$  is a profinite space. In particular, this applies when  $X$  spectral, by(4.12.3.2).  $\lrcorner$

*Proof:* Cf. [Sta]0900. $\square$

### Constructible Topology

**Lemma (4.12.3.6).** If  $X$  is a finite  $T_0$  space, then it is spectral(4.12.3.1) and every subset of  $X$  is constructible.  $\lrcorner$

*Proof:* Cf.[Adic Space Morel, P26].  $\square$

**Prop. (4.12.3.7).** If  $X$  is a spectral space, then the constructible topology(4.12.2.16) is Hausdorff, totally disconnected and quasi-compact.  $\lrcorner$

*Proof:* The space is sober hence  $T_0$ (4.12.2.38), and then the constructible topology is Hausdorff and totally disconnected.

To show quasi-compactness, it suffices to show that the family  $\mathcal{C}$  of quasi-compact open and complement quasi-compact open subsets has the finite intersection property(4.4.2.3). Notice elements in  $\mathcal{C}$  are all quasi-compact. Now if there is a family that has the finite intersection property by has intersection 0, by Zorn's lemma, there is a maximal one of them,  $\mathcal{B}$ . Now let  $Z$  be the intersection of all the closed subsets in  $\mathcal{B}$ , then it is non-empty as  $X$  is quasi-compact. And we claim  $Z$  is irreducible: otherwise  $Z = Z_1 \cup Z_2$ , thus there are quasi-compact open sets  $U_1, U_2$  that  $U_1 \cap Z_1 \neq \emptyset, U_1 \cap Z_2 = \emptyset, U_2 \cap Z_2 \neq \emptyset, U_2 \cap Z_1 = \emptyset$ . then let  $B_i = X - U_i$ , then  $B_1, B_2$  cannot be added to  $\mathcal{B}$  by maximality, so there is a finite intersection  $T_1, T_2$  that  $B_i \cap T_i = \emptyset$ . But then  $Z \cap T_1 \cap T_2 = \emptyset$ , but  $Z$  is an intersection of closed subsets, thus some finite intersection of closed subsets in  $\mathcal{B}$  will  $\cap T_1 \cap T_2 = \emptyset$ , contradiction.

So now  $Z$  is irreducible, but then for every element  $B \in \mathcal{B}$ ,  $Z \cap B$  contains the generic point of  $Z$ , thus the intersection of  $B$  is not empty, contradiction.  $\square$

**Cor. (4.12.3.8).** Let  $X$  be a spectral space, then

- The constructible topology is finer than the original topology.
- A subset  $X$  is constructible iff it is clopen in the constructible topology of  $X$ .
- If  $U$  is open in  $X$ , then the constructible topology induces the constructible topology on  $U$ .

$\lrcorner$

*Proof:*

1: Every open subset of  $X$  is a union of its quasi-compact open subsets, so it is open in the constructible topology.

2: Clearly constructible subset is clopen in the constructible topology. Conversely, if  $Y$  is clopen, then  $Y$  is a union of constructible subsets, but also it is quasi-compact, so it is a finite union of constructible subsets, thus constructible.

3: Cf.[Mor19] P28.  $\square$

**Prop. (4.12.3.9).** If  $E \subset X$  is closed in the constructible topology, then it is a spectral space with the induced topology, and the inclusion map is spectral.  $\lrcorner$

*Proof:* Cf. [Mor19]P34. □

**Prop. (4.12.3.10).** For a set  $E$  closed in the constructible topology in a spectral space,

- Every point of  $\overline{E}$  is a specialization of elements in  $E$ . Thus if  $E$  is stable under specialization, then it is closed.
- If  $E$  is open in the constructible topology and stable under generalization, then it is open. ┘

*Proof:* Cf. [Sta]0903? □

**Prop. (4.12.3.11).** For a map between spectral spaces  $f : X \rightarrow Y$ , the following are equivalent:

- $f$  is spectral.
- $f$  is quasi-compact.
- $f : X_{\text{cons}} \rightarrow Y_{\text{cons}}$  is continuous.

And if this is true, then  $f : X_{\text{cons}} \rightarrow Y_{\text{cons}}$  is proper. ┘

*Proof:*  $1 \rightarrow 2 \rightarrow 3$  is trivial, an open subset of  $X$  is quasi-compact iff it is clopen in the constructible topology (4.12.3.8), so  $3 \rightarrow 2$ . For  $2 \rightarrow 1$ , notice that if  $U \subset f^{-1}(V)$  are open spectrals, and  $W \subset U$  is quasi-compact open, then  $f^{-1}W \cap U$  is quasi-compact open, because  $X$  is quasi-separated.

Finally,  $f$  is proper because  $X_{\text{cons}}, Y_{\text{cons}}$  is compact Hausdorff (4.12.3.7), then use (4.4.2.10). □

### Criterion of Spectralness

**Lemma (4.12.3.12).** Let  $X$  be a quasi-compact  $T_0$  space that is has a subbasis consisting of quasi-compact open subsets that is stable under finite intersections. Let  $X'$  be the topology generated by the quasi-compact open subsets and their complements, then the following are equivalent:

- $X$  is spectral.
- $X'$  is compact Hausdorff, and its topology has a basis consisting of open and closed subsets.
- $X'$  is quasi-compact. ┘

*Proof:* Cf. [Adic Space Morel P30]. □

**Prop. (4.12.3.13) [Hochster's Criterion of Spectrality].** Let  $X' = (X_0, \mathcal{T}')$  be a quasi-compact topological space, and let  $\mathcal{U}$  be the family of clopen subsets of  $\mathcal{T}'$ , let  $\mathcal{T}$  be the topology generated by  $\mathcal{U}$ , let  $X = (X_0, \mathcal{T})$ .

Then if  $X$  is  $T_0$ , then it is spectral, and every element of  $\mathcal{U}$  is quasi-compact open in  $X$ , and  $X' = X_{\text{cons}}$ . ┘

*Proof:* Cf. [Adic Space Morel, P29]. □

**Prop. (4.12.3.14) [Spectral and Inverse Limit].** A space is spectral iff it is an inverse limit of finite sober (finite  $T_1$  (4.12.2.38)) spaces. ┘

*Proof:* Cf. [Sta]09XX. □

**Prop. (4.12.3.15) [Inverse Limits of Spectral Spaces].** An inverse limit of spectral spaces by spectral maps is a spectral space. ┘

*Proof:* ? □

**Prop. (4.12.3.16) [Spec and Spectral Space].** A spectral space is exactly the underlying space of spectrum of some ring.  $\lrcorner$

*Proof:* The spectrum of a ring is qc: if  $\cup D(f_i) = \text{Spec } A$ , then  $(f_i) = (1)$ , so f.m. of them generate (1). And similarly it is quasi-separable and  $D(f) = \text{Spec } A_f$  is quasi-compact. For the other direction, Cf. [M. Hochster. Prime ideal structure in commutative rings, Thm6] ?.  $\square$

**Cor. (4.12.3.17).** Every quasi-compact irreducible scheme is homeomorphic to an affine scheme.  $\lrcorner$

**Cor. (4.12.3.18) [Characterization of Spectral Spaces].** The following are equivalent for a topological space  $T$ :

1.  $T \cong \text{Spec } R$  for some ring  $R$ .
2.  $T \cong \varprojlim T_i$  where  $\{T_i\}$  is an inverse system of finite  $T_0$  spaces.
3.  $T$  is spectral.

$\lrcorner$

*Proof:* This follows from (4.12.3.16) and (4.12.3.14).  $\square$

### w-localness

**Def. (4.12.3.19) [pro-Zariski Localization].** A map of spectral spaces  $f : W \rightarrow V$  is called a **Zariski localization** if  $W = \coprod_i U_i$  where  $U_i \rightarrow V$  is a quasi-compact open immersion. A **pro-Zariski localization** is a cofiltered limit of Zariski localizations of  $V$ .  $\lrcorner$

**Def. (4.12.3.20) [w-Local].** A spectral space is called **w-local** if the set of closed pts of  $X$  is closed and any point of  $X$  specializes to a unique closed pt. A morphism of  $w$ -local spaces are called **w-local** if it is spectral and maps closed pts to closed pts.  $\lrcorner$

**Prop. (4.12.3.21).** If  $X$  is  $w$ -local and  $Y \subset X$  is a closed subset, then  $Y$  is also  $w$ -local.  $\lrcorner$

*Proof:*  $Y$  is spectral by (4.12.3.9).  $Y_0$  is closed because  $Y_0 = Y \cap X_0$ . And the second assertion is also trivial.  $\square$

**Prop. (4.12.3.22).** Let  $X$  be a spectral space and  $T$  profinite, then  $Y = X \times_{\pi_0(X)} T$  is also spectral and  $T = \pi_0(Y)$ . If moreover  $X$  is  $w$ -local, then  $Y$  is also  $w$ -local and  $Y \rightarrow X$  is  $w$ -local.  $\lrcorner$

*Proof:* Cf. [Sta]096C.  $\square$

**Def. (4.12.3.23) [Localization Along a Closed Set].** Given a closed set  $Z$  of a spectral set  $X$ , the pro-open subset of  $X$  consisting of all points that specializes to a point of  $Z$  is called the **localization of  $X$  along  $Z$** . And  $X$  is called **local along  $Z$**  if  $X_0 \subset Z$ .  $\lrcorner$

**Prop. (4.12.3.24).** A spectral space that is local along a closed  $w$ -local subset  $Z \subset X$  with  $\pi_0(Z) \cong \pi_0(X)$ , is also  $w$ -local.  $\lrcorner$

*Proof:*  $X_0 = Z_0$  is clearly closed, and if a pt  $x$  of  $X$  specializes to two closed pts of  $Z$ , then the  $\pi_0$  map is not injective, contradiction.  $\square$

## 4.13 Algebraic Topology I: Homotopies

References are [May99, AGP02, Hat02], [同调论, 姜伯驹], [<https://ncatlab.org/nlab/show/Introduction+to+Homotopy+Theory#HomotopyGroupsOfTopologicalSpaces>].

**Notation (4.13.0.1).**

- Use notations on topology defined in [Topology I](#).
- All spaces are assumed to be compactly generated, and products and mapping spaces are assumed to be compact generated versions.

┘

### 1 Homotopy Types

**Def. (4.13.1.1) [Retraction].** A **retraction** of a space  $X$  to a subspace  $A$  is a map  $r : X \rightarrow A$  that  $r|_A = \text{id}_A$ .

┘

**Def. (4.13.1.2) [Homotopy].** A **homotopy**  $f_t : X \rightarrow Y$  is a family of maps  $f_t$  for every  $t \in I$  that  $f : X \times I \rightarrow Y$  is continuous. For two homotopies  $F : X \times I \rightarrow Y$  and  $F' : X \times I \rightarrow Y$  s.t.  $F_1 = F'_0$ , we can define the **composite homotopy**  $F' \cdot F : X \times I \rightarrow Y$ .

Two maps  $f_0, f_1 : X \rightarrow Y$  are called **homotopic** if there is a homotopy  $f_t : X \rightarrow Y$  connecting them. Homotopy relations are denoted by  $f_0 \cong f_1$ .

Let  $A$  be a subspace of  $X$ , then a **homotopy relative to  $A$**  is a homotopy  $f_t : X \rightarrow Y$  whose restriction to  $A$  is fixed.

Let  $E_1, E_2$  be spaces over  $B$ , then maps  $f_0, f_1 : E_1 \rightarrow E_2$  over  $B$  are called **fiber homotopic** if there is a homotopy  $f_t : E_1 \rightarrow E_2$  connecting them that each  $f_t$  are maps over  $B$ . Homotopy relations over  $B$  are denoted by  $f_0 \cong_B f_1$ .

┘

**Def. (4.13.1.3) [Homotopy Equivalences].** A map  $f : X \rightarrow Y$  is called a **homotopy equivalence** if there is a map  $g : Y \rightarrow X$  that  $f \circ g \cong \text{id}$  and  $g \circ f \cong \text{id}$ .

A space having the homotopy type of a point is called **contractible**.

A map  $f : X \rightarrow Y$  over  $B$  is called a **fiber homotopy equivalence** if there is a map  $g : Y \rightarrow X$  over  $B$  that  $f \circ g \cong_B \text{id}$  and  $g \circ f \cong_B \text{id}$ .

┘

**Def. (4.13.1.4) [Deformation Retraction].** A **deformation retraction** of a space  $X$  onto a subspace  $A$  is a homotopy  $f_t : X \rightarrow X, t \in I$  that  $f_0 = \text{id}, f_1(X) = A$  and  $f_t|_A = \text{id}_A$  for all  $t$ .

┘

**Prop. (4.13.1.5).** A map  $X \rightarrow Y$  is a homotopy equivalence iff the mapping cylinder deformation retracts onto  $X$ .

┘

*Proof:* ?

□

**Def. (4.13.1.6) [Excisive Triads].** A **triad of spaces** is a triple  $(X; A, B)$  s.t.  $A \subset X, B \subset X$ . An **excisive triad** is a triad  $(X; A, B)$  s.t.  $A^\circ \cup B^\circ = X$ .

┘

### Relative Spaces

**Def. (4.13.1.7) [Relative Spaces].** The category  $\mathcal{CG}^{\text{rel}}$  is the category of pairs  $(X, A)$  where  $A \subset X \in \mathcal{CG}$ , and morphisms in  $\mathcal{CG}^{\text{rel}}$  are equivariant maps.

┘

**Def. (4.13.1.8) [Smash Products].** **smash products** in  $\mathcal{CG}^{\text{rel}}$  are defined to be  $(X, A) \wedge (Y, B) = (X \times Y, (X, A) \vee (Y, B))$ , where  $(X, A) \vee (Y, B) = X \times B \cup A \times Y \subset X \times Y$ .  $\lrcorner$

**Def. (4.13.1.9) [Cones and Suspensions].** For  $(X, A) \in \mathcal{Top}^{\text{rel}}$ , the **cone** over  $(X, A)$  is defined to be  $C(X, A) = (X, A) \wedge (\mathbb{I}, \{1\})$ . The **suspension** of  $(X, A)$  is defined to be  $\Sigma(X, A) = (X, A) \wedge \mathbb{S}^1$ .  $\lrcorner$

**Prop. (4.13.1.10) [ $\mathcal{CG}^{\text{rel}}$  is Closed].** For  $(X, A), (Y, B), (Z, C) \in \mathcal{CG}^{\text{rel}}$ , by (4.4.3.15) there is a homeomorphism

$$\text{Map}((X, A) \wedge (Y, B), (Z, C)) \cong \text{Map}((X, A), \text{Map}((Y, B), (Z, C))).$$

In particular,  $\mathcal{CG}^{\text{rel}}$  is a closed symmetric monoidal category.  $\lrcorner$

**Def. (4.13.1.11) [Pointed Spaces].** The category  $\mathcal{CG}^{\text{pt}}$  is the subcategory of  $\mathcal{CG}^{\text{rel}}$  consisting of pairs  $(X, A)$  s.t.  $A \cong \text{pt}$ . It is stable under finite products.

We can define similarly homotopies of maps between pointed spaces, and denote  $\langle X, Y \rangle$  the homotopy classes of maps from  $X$  to  $Y$ .  $\lrcorner$

**Def. (4.13.1.12) [Smash Products].** For  $X, Y \in \mathcal{CG}^{\text{pt}}$ , the **smash product**  $X \wedge Y$  is defined to be  $X \wedge Y = X \times Y / X \vee Y$ , where  $X \vee Y = X \coprod_* Y$ , called the **wedge sum** of  $X$  and  $Y$ . Similar to (4.13.1.9), we can define cones and suspensions of pointed spaces.

Then the smash product and wedge sums are commutative and associative, and they satisfy

$$(X \vee Y) \wedge Z = X \wedge Z \vee Y \wedge Z.$$

Because of (4.13.1.14), the smash product and wedge sums are also denoted by  $X \otimes Y$  and  $X \oplus Y$ .  $\lrcorner$

*Proof:* For the identities, use the fact product commutes with colimits (4.4.3.16).  $\square$

**Prop. (4.13.1.13) [Smash Products, and Homotopy].** Suspensions preserves homotopy, as  $\Sigma(X \wedge I) = X \wedge I \wedge \mathbb{S}^1 \cong (X \wedge \mathbb{S}^1) \wedge I$ .  $\lrcorner$

**Prop. (4.13.1.14) [ $\mathcal{CG}^{\text{pt}}$  is Closed].** For  $X, Y, Z \in \mathcal{CG}^{\text{pt}}$ , by (4.13.1.10), there is a bijection

$$\text{Map}(X \wedge Y, Z) \cong \text{Map}(X, \text{Map}(Y, Z)).$$

In particular,  $\mathcal{CG}^{\text{pt}}$  is a closed symmetric monoidal category.  $\lrcorner$

**Def. (4.13.1.15) [Well-Pointed Spaces].** A pointed space  $(X, *) \in \mathcal{CG}^{\text{pt}}$  is called a **well-pointed space** if  $*$   $\rightarrow X$  is a cofibration. The category of well-pointed spaces is denoted by  $\mathcal{CG}^{\text{well-pt}}$ .  $\lrcorner$

**Def. (4.13.1.16) [Augmentation].** There is a functor  $\mathcal{CG} \rightarrow \mathcal{CG}^{\text{well-pt}} : X \mapsto X_+ = (X \coprod \text{pt}, \text{pt})$ .  $\lrcorner$

**Prop. (4.13.1.17).** The cone construction

$$\mathcal{Top}^{\text{rel}} \rightarrow \mathcal{Top} : (X, A) \mapsto C(A \rightarrow X)$$

preserves homotopy equivalence.  $\lrcorner$

**Prop. (4.13.1.18) [Path Spaces and Loop Spaces].** For  $X \in \mathcal{CG}^{\text{pt}}$ , define the **path space**  $P(X) = \text{Map}(\mathbb{I}, \{0\}, X)$ , and define the **loop space**  $\Omega(X) = \text{Map}(\mathbb{S}^1, X)$ .

Then by (4.13.1.14), there are homeomorphism of spaces

$$\text{Map}(\Sigma(X), Y) \cong \text{Map}(X, \Omega(Y)).$$

Thus also

$$\langle \Sigma(X), Y \rangle \cong \langle X, \Omega(Y) \rangle.$$

$\lrcorner$

**Def. (4.13.1.19) [Weak Products].** For  $\{X_\alpha\}_\Sigma \in \mathcal{CG}^{\text{pt}}$ , define the **weak product**  $\prod' X_\alpha$  as

$$\prod' X_\alpha = \varinjlim_{S \subset \Sigma, \#S < \infty} \bigvee_{i \in S} X_i$$

with base point  $\prod *$ . ┘

## 2 Isotopies

**Def. (4.13.2.1) [Isotopies].** For  $V, M \in \mathcal{Mani}$ , an **isotopy** from  $V$  to  $M$  is a map  $V \times I \rightarrow M$  s.t. for any  $t \in I$ , the map  $F_t : V \rightarrow M$  is an embedding. ┘

**Prop. (4.13.2.2) [Alexander's Trick].** If  $f, g : \mathbb{D}^n \rightarrow \mathbb{D}^n$  be two self-homeomorphisms that there is an isotopy  $F : \mathbb{D}^n \times I \rightarrow \mathbb{D}^n$  between  $f|_{\partial\mathbb{D}^n}$  and  $g|_{\partial\mathbb{D}^n}$ , then there exists an isotopy between  $f$  and  $g$  extending  $F$ . ┘

*Proof:* Firstly, if  $g = \text{id}_{\mathbb{D}^n}$  and  $f$  fixes every points on the boundary, then an isotopy connecting  $f$  to the identity is given by

$$J(x, t) = \begin{cases} tf(x/t) & , 0 \leq \|x\| < t \\ x & , t \leq \|x\| \leq 1 \end{cases}.$$

In general, isotopy  $g^{-1}F : \mathbb{D}^n \times I \rightarrow \mathbb{D}^n$  between  $g^{-1}f|_{\partial\mathbb{D}^n}$  and  $\text{id}_{\partial\mathbb{D}^n}$  can be extended to an isotopy from  $F' : \text{Cone}(g^{-1}f|_{\partial\mathbb{D}^n})$  to  $\text{id}_{\mathbb{D}^n}$ . Then  $\text{Cone}(g^{-1}f|_{\partial\mathbb{D}^n})$  and  $g^{-1}f$  are identical on  $\partial\mathbb{D}^n$ , so by the argument above, they are isotopic. Then  $g^{-1}f \cong \text{id}_{\mathbb{D}^n}$ , which gives an isotopy between  $f$  and  $g$ . □

**Cor. (4.13.2.3) [Homogeneity Lemma].** If  $M \in \mathcal{Mani}_{\text{cntd}, \partial}$  and  $x, y \in M$  are interior points, then there exists a self-homeomorphism of  $M$  taking  $x$  to  $y$  that is isotopic to  $\text{id}_M$ . ┘

**Prop. (4.13.2.4).** Any orientation-preserving self-homeomorphism  $f$  of  $\mathbb{S}^d$  is isotopic to  $\text{id}_{\mathbb{S}^d}$ . ┘

*Proof:* By the decomposition of  $\mathbb{S}^d$  into two hemispheres and using the annulus theorem (4.4.11.4), we see we can isotopy  $f$  to a map that fixes the equator. Then we can use Alexander's trick (4.13.2.2). □

## 3 CW Complexes

**Def. (4.13.3.1) [CW Complexes].** A **CW complex** is a space  $X$  that  $\emptyset \rightarrow X$  is cofibrantly generated by  $\mathbb{S}^n \rightarrow \mathbb{D}^n$ .

For a CW complex  $X$ , a **representation of  $X$**  is given as follows:

- $\text{sk}_0 X$  is a discrete set, whose elements are regarded as 0-cells.
- Inductively, form the  **$n$ -skeleton**  $\text{sk}_n X$  from  $\text{sk}_{n-1} X$  by adding  **$n$ -cells**  $e_\alpha^n$  via maps  $\varphi_\alpha : \mathbb{S}^{n-1} \rightarrow \text{sk}_{n-1} X$ .
- Let  $X = \bigcup \text{sk}_n X$  be given the quotient topology from  $\coprod_n \text{sk}_n X$ .

We will also call the image of  $e_\alpha^n$  as an  $n$ -cell. The category of CW complexes is denoted by  $\mathcal{CW}$ . Any CW complex is compactly generated, by definition. ┘

**Def. (4.13.3.2) [Sub-CW Complexes].** For a CW complex  $X$ , a **subcomplex**  $Y \subset X$  is a subspace  $Y \subset X$  s.t. the cells with images in  $Y$  makes  $Y$  a CW complex. ┘



**Def. (4.13.3.3)[Finite CW Complexes].**

- A **CW complex of finite dimension** is a CW complex  $X$  s.t.  $X = \text{sk}_n X$  for some  $n \in \mathbb{N}$ .
- A **CW complex of finite type** is a CW complex  $X$  that has only finitely many  $n$ -cells for each  $n$ .
- A **finite CW complex** is a CW complex that is both of finite dimensional and of finite type.
- A **locally finite CW complex** is a CW complex that each cell intersect only f.m. other cells.

⌋

**Prop. (4.13.3.4)[Dimension of CW Complexes].** Given a representation of a CW complex of finite dimension  $X$ ,  $\dim X$  is defined to be the minimal  $n$  s.t.  $X = \text{sk}_n X$ . Then this dimension is independent of the presentation given.

⌋

*Proof:*

□

**Prop. (4.13.3.5).** Any compact subspace  $K$  of a CW complex  $X$  is contained in a finite subcomplex. In particular, a CW complex is compact iff it is finite.

⌋

*Proof:* Firstly  $K$  intersects with only f.m. interiors of cells of  $X$  (where we assume the interior of a point in  $\text{sk}_0 X$  is the point itself): otherwise take  $S = \{x_1, x_2, \dots\}$  be an infinite sequence of points lying in different cells, then  $S \cap e_\alpha$  is closed for any cell  $e_\alpha$  of  $X$ , so  $S$  is closed. But the same argument show any single point of  $S$  is also closed, so  $S$  is discrete, so  $S$  is finite, contradiction.

Then it suffices to show that any cell meets only f.m. cells: for this we can use induction on  $\dim X$ , and notice that for any cell  $e_\alpha$ , the image of its attaching map is compact.

□

**Prop. (4.13.3.6)[Miyazaki].** Any CW complex is paracompact Hausdorff, thus also normal by (4.4.7.6).

⌋

*Proof:* To show it is Hausdorff, for any two points  $x, y$ , it is not hard to find disjoint precompact nbhds of  $x, y$  by induction on cells.

To show it is paracompact, Cf. [Rudolph Fritsch and Renzo Piccinini, Cellular Structures in Topology]Thm1.3.5.?

□

**Prop. (4.13.3.7).** A CW complex is locally compact iff it is locally finite. A connected CW complex is metrizable iff it is locally finite.

⌋

*Proof:* Cf. [Rudolph Fritsch and Renzo Piccinini, Cellular Structures in Topology, Cambridge University Press, 1990.]Prop1.5.7.

□

**Prop. (4.13.3.8)[Product of CW Complexes].** For  $X, Y \in \mathcal{CW}$ , then the compactly generated product space  $X \times_c Y$  admits a CW complex structure with

$$\text{sk}_n X = \cup_{i+j=n} \text{sk}_i X \times \text{sk}_j Y.$$

And if either  $X$  or  $Y$  is locally compact or both  $X, Y$  have countably many cells, then  $X \times_c Y = X \times Y$ .

⌋

*Proof:* Given presentations of CW complexes of  $X, Y$ , we can define the CW complex structure on  $X \times Y$  by choosing homeomorphisms of pairs

$$(\mathbb{D}^n, \mathbb{S}^n) \cong (\mathbb{D}^p \times \mathbb{D}^q, \mathbb{D}^p \times \mathbb{S}^{q-1} \cup \mathbb{S}^{p-1} \times \mathbb{D}^q),$$

where  $p + q = n$ . Then it is a CW structure on  $X \times_c Y$  by (4.4.3.16).

For the last assertion, the case  $X$  is locally compact follows from (4.4.2.33). The case that  $X, Y$  both have countably many cells follow from [Hatcher, P524].?

□



**Prop. (4.13.3.9) [Cellular Maps].** Given  $X, Y \in \mathcal{CW}$ , a **cellular map**  $f : X \rightarrow Y \in \mathcal{CW}$  is a map s.t.  $f(\text{sk}_n X) \subset \text{sk}_n Y$  for any  $n \in \mathbb{N}$ . And a **cellular homotopy** between two cellular maps  $f, g : X \rightarrow Y$  is a homotopy  $F : X \times I \rightarrow Y$  between  $f$  and  $g$  s.t.  $F$  is a cellular map.  $\lrcorner$

**Prop. (4.13.3.10) [CW Structure on Compact Manifolds].**  $\text{Mani}_{\text{cpct}} \in \mathcal{CW}^{\text{fin}}$ .  $\lrcorner$

*Proof:*

$\square$

**Def. (4.13.3.11) [Relative CW complexes].** A **relative CW complex** is a pair  $(A, X)$  s.t.  $A \rightarrow X$  is cofibrantly generated by  $\mathbb{S}^n \rightarrow \mathbb{D}^n$ . The category of relative CW complexes is denoted by  $\mathcal{CW}^{\text{rel}}$ . A **pointed CW complex** is a pointed space  $(X, *) \in \mathcal{Top}^{\text{pt}}$  s.t.  $X$  has a presentation as a CW complex with  $*$   $\in X^0$ . The category of pointed CW complexes is denoted by  $\mathcal{CW}^{\text{pt}}$ . Cellular maps in  $\mathcal{CW}^{\text{rel}}$  is defined similarly as that of (4.13.3.9).  $\lrcorner$

**Prop. (4.13.3.12) [Pathspace is CW Complex].** The homotopy fibers (4.4.3.18) of any map  $f : A \rightarrow B$  between CW complexes are homotopic to a CW complex.  $\lrcorner$

*Proof:* [Milnor, 1959]. or [Rudolph Fritsch and Renzo Piccinini, Cellular Structures in Topology].  $\square$

**Cor. (4.13.3.13) [Loop Space].** For  $X \in \mathcal{CW}^{\text{pt}}$ , the loop space (4.4.3.19)  $\Omega X$  is homotopic to a pointed CW complex. In particular, if  $X$  is of f.t., then so does  $\Omega X$ .  $\lrcorner$

*Proof:* This is a special case of (4.13.3.12) applied to  $A = \{*\}$ , by (4.4.3.20).  $\square$

### Common CW complexes

**Def. (4.13.3.14) [Infinite Vector Spaces].** Let  $\mathbb{K} = \mathbb{R}, \mathbb{C}$  or  $\mathbb{H}$ , then the **infinite vector space**  $\mathbb{K}^\infty$  is the CW complex  $\varinjlim_n \mathbb{K}^n$ .

There can be given a norm map on  $\mathbb{K}^\infty$ , as the canonical embedding is norm preserving.  $\lrcorner$

**Prop. (4.13.3.15) [Grassmannian].** Let  $\mathbb{K} = \mathbb{R}, \mathbb{C}$  or  $\mathbb{H}$ , then the **Grassmannian**  $\text{Gr}(k, \mathbb{K}^n)$  admits a CW complex structure.  $\lrcorner$

*Proof:* Cf. [Algebraic Topology Miller, P46] or [Characteristic Classes, Milnor]. ?  $\square$

**Def. (4.13.3.16) [Infinite Grassmannian].** Let  $\mathbb{K} = \mathbb{R}, \mathbb{C}$  or  $\mathbb{H}$ , then the **infinite Grassmannian**  $\text{Gr}(k, \mathbb{K}^\infty)$  is the CW complex  $\varinjlim_n \text{Gr}(k, \mathbb{K}^n)$  (16.5.2.12).

In particular,  $\text{Gr}(1, \mathbb{K}^\infty) \cong \varinjlim_n \mathbb{K}\mathbb{P}^n$  is defined to be the **infinite projective space**  $\mathbb{K}\mathbb{P}^\infty$ .  $\lrcorner$

**Def. (4.13.3.17) [Infinite Stiefel Manifold].** Let  $\mathbb{K} = \mathbb{R}, \mathbb{C}$  or  $\mathbb{H}$ , then the **infinite Stiefel Manifold**  $V_k(\mathbb{K}^\infty)$  is the CW complex  $\varinjlim_n V_k(\mathbb{K}^n)$  where  $V_k(\mathbb{K}^n)$  is the set of orthonormal  $k$ -frames (16.5.2.12).  $V_1(\mathbb{K}^\infty) = \varinjlim_n \mathbb{S}^n$  is called the **infinite sphere**. It is isomorphic to

$$\mathbb{S}^\infty = \varinjlim_k S^{2k+1} = \varinjlim_k \mathbb{C}^{2k+1} \setminus \{0\} / \mathbb{R}^+ = (\mathbb{C}^\infty \setminus \{0\}) / \mathbb{R}_+^\times.$$

$\lrcorner$

**Prop. (4.13.3.18).**  $V_k(\mathbb{K}^\infty)$  is contractible.  $\lrcorner$

*Proof:* We prove for  $\mathbb{S}^\infty$ , the proof for  $V_k(\mathbb{K}^\infty)$  is verbatim, using Smith orthogonalization. There is a map  $H : \mathbb{S}^\infty \times I \rightarrow \mathbb{S}^\infty$  defined by

$$H((x_1, x_2, \dots), t) = ((1-t)x_1, tx_2 + (1-t)x_1, tx_3 + (1-t)x_2, \dots)/N$$

where  $N$  is the norm of the non-zero element  $((1-t)x_1, tx_2 + (1-t)x_1, tx_3 + (1-t)x_2, \dots)$ . The image of  $H_1$  is in the subspace  $\mathbb{S}_1^\infty$  of elements  $(x_1, x_2, \dots)$  with  $x_1 = 0$ . Notice there is another map  $H' : \mathbb{S}_1^\infty \times I \rightarrow \mathbb{S}^\infty$  defined by

$$H'((0, x_2, \dots), t) = (t, (1-t)x_2, (1-t)x_3, \dots)/N'$$

where  $N'$  is the norm of the non-zero element  $(t, (1-t)x_2, (1-t)x_3, \dots)$ . Then the composition homotopy  $(H' \circ H_1) \cdot H$  gives the desired homotopy from  $\text{id}_{\mathbb{S}^\infty}$  to a constant map.  $\square$

**Prop. (4.13.3.19).**  $\mathbb{RP}^\infty = \mathbb{S}^\infty / \{\pm 1\} = (\mathbb{C}^\infty \setminus \{0\}) / \mathbb{R}^\times$ .  $\lrcorner$

**Def. (4.13.3.20) [James Reduced Product].** Let  $X$  be a space with a basepoint  $e$ , let the **James reduced product** space  $J(X)$  be the quotient of  $\coprod_n X^n$  under the identification  $(x_1, \dots, x_j, \dots, x_n) \sim (x_1, \dots, \hat{x}_j, \dots, x_n)$  if  $x_j = e$ .

$J(X)$  is a union of subspaces  $J_m(X)$ , where  $J_m(X)$  is the quotient space of  $X^m$  under the identification  $(x_1, \dots, x_j, e, \dots, x_n) \sim (x_1, \dots, e, x_j, \dots, x_n)$ . If  $(X, e)$  is a CW-pair, then  $J_m$  is obtained from  $X^m$  by glueing together its  $m$  subcomplexes where one of the coordinates is  $e$ . It is then clear  $J$  is a CW complex.  $\lrcorner$

**Def. (4.13.3.21) [Infinite Symmetric Product].** Let  $X$  be a space with a basepoint  $e$ , define the **infinite symmetric product** as the quotient of  $J(X)$  by permutations. If  $X$  is a simplicial complex (4.13.4.1), then  $SP(X)$  is a CW-complex.  $\lrcorner$

*Proof:* Cf. [Hat02]P482.  $\square$

## 4 Simplicial Complexes

**Def. (4.13.4.1) [Simplicial Complexes].** A **simplicial complex** is a pair  $X = (|X|, \mathcal{C})$  where  $|X| \in \text{Set}$  and  $\mathcal{C} \subset \mathcal{P}(X)$  s.t. if  $\sigma \in \mathcal{C}$  and  $\tau \subset \sigma$ , then  $\tau \in \mathcal{C}$ .

A **subcomplex** of  $(|X|, \mathcal{C})$  is a simplicial complex  $(X', \mathcal{C}')$  with an injection  $i : X' \hookrightarrow X$  s.t.  $i(\mathcal{C}') \subset \mathcal{C}$ .

A **simplicial map** of simplicial complexes  $f : (|X'|, \mathcal{C}') \rightarrow (|X|, \mathcal{C})$  is a map of sets  $|f| : |X'| \rightarrow |X| \in \text{Set}$  s.t.  $f(\mathcal{C}') \subset \mathcal{C}$  and  $\# \sigma = \# f(\sigma)$  for each complex  $\sigma$  in  $X$ .

A **simplicial retraction** is a simplicial map  $f : X \rightarrow Y$  where  $Y$  is a simplicial subcomplex of  $X$  and  $f|_Y = \text{id}_Y$ .  $\lrcorner$

**Def. (4.13.4.2) [Chamber Complexes].** For a simplicial complex  $X = (|X|, \mathcal{C})$  (4.13.4.1),

- A **simplex** in  $X$  is an element in  $\mathcal{C}$ .
- The **dimension of simplex** of a simplex  $\sigma \in \mathcal{C}$  is defined to be  $\dim(\sigma) = \# \sigma - 1$ .
- The **dimension of simplicial complex** of  $X$  is defined to be the supremum of dimension of simplexes in  $\mathcal{C}$ .
- Two simplexes of the same dimension  $n \in \mathbb{Z}_+$  are called **adjacent simplexes** if their intersection has cardinality  $n - 1$ .

- A **simplicial chamber** of  $X$  is a simplex that is not contained in other simplexes of  $X$ .
- A **simplicial gallery** of simplicial chambers is a sequence  $(C_0, \dots, C_n)$  of chambers s.t.  $C_i$  and  $C_{i+1}$  are adjacent for each  $i \in [n-1]$ . A **non-stutter gallery** is a simplicial gallery s.t.  $C_i \neq C_{i+1}$  for any  $i$ .
- $X$  is called a **chamber complex** if each two chambers in  $X$  is connected by a gallery (in particular, all chambers in  $X$  has the same dimension).
- $(|X'|, C') \subset X$  is called a **chamber subcomplex** if it is a chamber complex of the same dimension as  $X$ .
- $X$  is called a **thin chamber complex** if it is a chamber complex of dimension  $n \in \mathbb{Z}_+$  and each simplex of dimension  $n-1$  is contained in exactly 2 chambers.
- $X$  is called a **thick chamber complex** if it is a chamber complex of dimension  $n \in \mathbb{Z}_+$  and each simplex of dimension  $n-1$  is contained in at least 3 chambers.

┘

## 5 Homotopy Groups

**Def. (4.13.5.1) [Homotopy Groups].** For  $(X, *) \in \mathcal{Top}^{\text{pt}}$  and  $n \in \mathbb{N}$ , define the **homotopy groups**  $\pi_n(X) = \pi_n(X, *) = \langle \mathbb{S}^n, X \rangle$ , which is a pointed set.

And  $\pi_1(X)$  is a group as  $\mathbb{S}^1$  has a cogroup structure:  $\mathbb{S}^1 \rightarrow \mathbb{S}^1 \wedge \mathbb{S}^1$ . Similarly,  $\pi_n(X) = \pi_1(\Omega^{n-1}X)$  are also groups, and the group structure are given by the cogroup structure  $\mathbb{S}^n \rightarrow \mathbb{S}^n \wedge \mathbb{S}^n$ . ┘

**Prop. (4.13.5.2) [Loop Spaces and Homotopy Groups].**  $\pi_{i+1}(M) \cong \pi_i(\Omega(M))$ . ┘

*Proof:* This follows from the fact  $\mathbb{S}^n = \Sigma^n(\mathbb{S}^0)$  and (4.13.1.18). ┐

**Prop. (4.13.5.3) [Homotopy Groups are Abelian].** For  $X \in \mathcal{CG}^{\text{pt}}$ , the homotopy group  $\pi_n(X)$  is an Abelian group for  $n \geq 2$ . ┘

*Proof:* By (4.13.5.2), it suffices to show for  $n = 2$ , and  $\pi_2(X) = \pi_1(\Omega X)$  is Abelian by (4.14.2.30) and (4.14.2.24). ┐

**Cor. (4.13.5.4).** For any  $X \in \mathcal{CG}$ ,  $\langle \Sigma(X), Y \rangle$  is a group and  $\langle \Sigma^2(X), Y \rangle$  is an Abelian group. ┘

**Prop. (4.13.5.5) [Homotopy Groups and Change of Basepoints].** Let  $X \in \mathcal{CG}$  and  $\gamma : \mathbb{I} \rightarrow X$  be a path from  $a$  to  $b$ , then for any  $n \in \mathbb{Z}_+$ , we can define a map  $\gamma_{\#} : \pi_n(X, a) \rightarrow \pi_n(X, b)$  as follows: For any  $f : (\mathbb{I}^n, \partial I^n) \rightarrow (X, a)$ , identify  $\mathbb{I}^n$  with the subspace of  $\mathbb{R}^n$  consisting of vectors  $\mathbf{x}$  s.t.  $d(\mathbf{x}) \leq \frac{1}{2}$ , where  $d((x_1, \dots, x_n)) = \min(|x_i|)$ , and

$$\gamma_{\#}(f) : (\mathbb{I}^n, \partial I^n) \rightarrow X : \mathbf{x} \mapsto \begin{cases} f(2\mathbf{x}) & , \|\mathbf{x}\| \leq \frac{1}{4} \\ \gamma(4\|\mathbf{x}\| - 1) & , \frac{1}{4} \leq \|\mathbf{x}\| \leq 1 \end{cases}$$

Then for any  $x_0 \in X$ , this defines an actions of  $\pi_1(X, x_0)$  on  $\pi_n(X, x_0)$  by group homomorphisms for any  $n \in \mathbb{Z}_+$ . And for  $n = 1$ , this is just the conjugation action. ┘

*Proof:* It is a well-defined action because any homotopy between two maps  $f_1, f_2 : (\mathbb{I}^n, \partial I^n) \rightarrow (X, a)$  can generate a homotopy between  $\gamma_{\#}(f_1)$  and  $\gamma_{\#}(f_2)$ , and it is an action by group homomorphism because there is a homotopy between  $\gamma_{\#}(f) \cdot \gamma_{\#}(g)$  and  $\gamma_{\#}(f \cdot g)$ , which is easy to write out. ┐

**Def. (4.13.5.6) [Simple Spaces].** A space  $X \in \mathcal{CG}$  is called a **simple topological space** if for any  $x_0 \in X$ , the action of  $\pi_1(X, x_0)$  defined in (4.13.5.5) is trivial.  $\lrcorner$

**Def. (4.13.5.7) [Weak Homotopy Equivalences].**  $f : X \rightarrow Y \in \mathcal{CG}$  is called a **weak homotopy equivalence** iff it induces isomorphism  $\pi_n((X, x_0)) \cong \pi_n(Y, f(x_0))$  for any  $n \in \mathbb{N}$  and  $x_0 \in X$ .  $\lrcorner$

**Prop. (4.13.5.8).** Let  $f_0, f_1 : X \rightarrow Y \in \mathcal{CG}$  be maps and  $h : \mathbb{I} \times X \rightarrow Y$  be a homotopy  $x_0 \in X$ , then if  $\gamma : \mathbb{I} \rightarrow Y : t \mapsto h(t, x_0)$ , then

$$\gamma_{\#} \circ \gamma_{\#}(f_0)_* = (f_1)_*.$$

$\lrcorner$

*Proof:* For  $f : f : (\mathbb{I}^n, \partial \mathbb{I}^n) \rightarrow (X, x_0)$  is easy to write out a homotopy between  $\gamma_{\#}(f_0)_*(f)$  and  $(f_1)_*(f)$ .  $\square$

**Cor. (4.13.5.9).** Any homotopy equivalence is a weak homotopy equivalence.  $\lrcorner$

**Def. (4.13.5.10) [Relative Homotopy Groups].** For an inclusion  $i : (A, *) \subset (X, *) \in \mathcal{Top}^{\text{pt}}$ ,  $n \in \mathbb{Z}_+$ , define the **relative homotopy groups**  $\pi_n(X, A) = \pi_n(X, A, *) = \pi_{n-1}(P(X; *, A))$ , where  $P(X; *, A)$  is the homotopy fiber of  $i$  ?. Equivalently, if we denote  $\mathbb{J}^n = \partial \mathbb{I}^{n-1} \times \mathbb{I} \cup \mathbb{I}^{n-1} \times \{0\} \subset \mathbb{I}^n$  for  $n \geq 2$  and  $\mathbb{J}^1 = \{0\} \subset \mathbb{I}^1$ , then

$$\pi_n(X, A) = [(\mathbb{I}^n, \partial \mathbb{I}^n, \mathbb{J}^n), (X, A, *)].$$

This is a pointed set for  $n = 1$ , a group for  $n = 2$ , and an Abelian group for  $n \geq 3$ .  $\lrcorner$

**Prop. (4.13.5.11) [Long Exact Sequence of Relative Homotopy Groups].**  $\lrcorner$

*Proof:*  $\square$

**Cor. (4.13.5.12) [Covering Spaces and Homotopy Groups].** If  $E \rightarrow B$  is a covering space, then  $\pi_n(E) \rightarrow \pi_n(B)$  is an isomorphism for  $n \geq 2$ .  $\lrcorner$

**Prop. (4.13.5.13).** If  $X \in \mathcal{CG}^{\text{pt}}$  is contractible, then  $\pi_n(X) = 0$  for any  $n \geq 0$ .  $\lrcorner$

**Prop. (4.13.5.14) [Products and Homotopy Groups].** For  $X, Y \in \mathcal{CG}^{\text{pt}}$ ,  $\pi_n(X \times Y) = \pi_n(X) \times \pi_n(Y)$  for any  $n \in \mathbb{N}$ .  $\lrcorner$

**Prop. (4.13.5.15) [Colimits and Homotopy Groups].** If  $X = \varinjlim_i X_i \in \mathcal{CG}^{\text{pt}}$  is a filtered colimits, there are natural isomorphisms

$$\varinjlim_i \pi_n(X_i) \cong \pi_n(X)$$

for each  $n \in \mathbb{N}$ .  $\lrcorner$

*Proof:* This follows from the fact any compact subset of  $X$  is contained in some  $X_i$  ?.  $\square$

**Def. (4.13.5.16) [ $n$ -Connectedness].** For  $n \in \mathbb{N}$ ,  $(X, A) \in \mathcal{CG}^{\text{rel, pt}}$  is called  **$n$ -connected** if  $\pi_i(X, A) = *$  for any  $i \leq n$ .  $X \in \mathcal{Top}^{\text{pt}}$  is called  **$n$ -connected** if  $\pi_i(X) = *$  for any  $i \leq n$ .  $\lrcorner$

**Def. (4.13.5.17) [ $n$ -Equivalences].** For  $n \in \mathbb{Z}_+$ ,  $f : (X, A) \rightarrow (Y, B) \in \mathcal{Top}^{\text{rel}}$  is called an  **$n$ -equivalence** if  $f_* : \pi_p(X, A) \rightarrow \pi_p(Y, B)$  are isomorphisms for  $p < n$ , and surjective for  $p = n$ .  $\lrcorner$

**Thm. (4.13.5.18) [Excision for Homotopy Groups].** if  $m, n \geq 0$ ,  $(X; A, B)$  is an excisive triad s.t.  $(A, A \cap B)$  is  $m$ -connected and  $(B, A \cap B)$  is  $n$ -connected, then  $(A, A \cap B) \rightarrow (A \cup B, A)$  is an  $m + n$ -equivalence.  $\lrcorner$

*Proof:* Cf.[Hatcher P360].  $\square$

**Cor. (4.13.5.19).** For  $n > 1$ ,  $\pi_n(\bigvee_{\alpha} S^{\alpha})$  is free Abelian with  $\pi_n(S^n)$  as generators. This is because  $(\prod_{\alpha} S^n, \bigvee_{\alpha} S^n)$  is  $(2n - 1)$ -connected thus use excision, because  $\pi_n(\prod_{\alpha} S^n)$  is easy to calculate.  $\lrcorner$

**Prop. (4.13.5.20).** If  $f : X \rightarrow Y$  is an  $n$ -equivalence between  $(n - 1)$ -connected spaces, then the quotient map  $\text{pr} : (M(f), X) \rightarrow (C(f), *)$  is an  $2n$ -equivalence. And if  $X, Y$  are both  $n$ -connected, then  $\text{pr}$  is an  $2n + 1$ -equivalence.  $\lrcorner$

*Proof:*  $\square$

**Cor. (4.13.5.21) [Quotients and Homotopy Groups].** For  $n \in \mathbb{Z}_+$ , if  $i : A \rightarrow X$  is a cofibration that is a  $n$ -equivalence of  $(n - 1)$ -connected space, then the map  $(X, A) \rightarrow (X/A, *)$  is a  $2n$ -equivalence. And if  $A, X$  are  $n$ -connected, then this is a  $2n + 1$ -equivalence.  $\lrcorner$

*Proof:* This follows from the commutative diagram

$$\begin{array}{ccc} (M(i), A) & \longrightarrow & (C(i), *) \\ \downarrow & & \downarrow \\ (X, A) & \longrightarrow & (X/A, *) \end{array}$$

where the vertical maps are homotopy equivalences, by (4.13.7.7).  $\square$

**Prop. (4.13.5.22) [Suspension and Homotopy Groups, Freudenthal].** For  $X \in \mathcal{CG}^{\text{well-pt}}$ , there are maps

$$\Sigma_* : \pi_p(X) \rightarrow \pi_{p+1}(\Sigma(X)) : f \in \langle \mathbb{S}^p, X \rangle \mapsto \Sigma(f) \in \langle \mathbb{S}^{p+1} \cong \Sigma(\mathbb{S}^p), \Sigma(X) \rangle \quad (4.13.1.13).$$

Then if  $n \in \mathbb{N}$ ,  $X \in \mathcal{CG}^{\text{well-pt}}$  and  $X$  is  $n$ -connected, then  $\Sigma_*$  is bijective for  $p \leq 2n$  and surjective for  $p = 2n + 1$ .

In particular,  $\Sigma^n(X)$  is  $(n - 1)$ -connected.  $\lrcorner$

*Proof:* Cf.[May, P85].  $\square$

**Def. (4.13.5.23) [Stable Homotopy].** For  $X \in \mathcal{CG}^{\text{well-pt}}$ , using the suspension (4.13.5.22), we can form the colimit

$$\pi_p^s(X) = \varinjlim_{n \in \mathbb{N}} \pi_{p+n}(\Sigma^n(X)).$$

Then by (4.13.5.22),

$$\pi_p^s(X) = \pi_{p+n}(\Sigma^n(X)), \quad n \geq p + 2.$$

$\lrcorner$

**Prop. (4.13.5.24).** The homotopic direct limit of a family of homotopy equivalence is a homotopy equivalence. Cf.[Morse Theory Milnor].  $\lrcorner$

### Fundamental Groups

**Def. (4.13.5.25) [Simply Connected].** A space is called **simply connected** if it is connected and  $\pi_1(X, x) = 0$  for some point  $x \in X$ .  $\lrcorner$

**Prop. (4.13.5.26).** For a connected subset  $K \subset \mathbb{C}$ , the following are equivalent:

- $K$  is simply-connected,
- For any Jordan curve  $\gamma \subset K$ ,  $\text{int}(\gamma) \subset K$ .
- $\mathbb{P}^1(\mathbb{C}) \setminus K$  is connected.

Is this really true?  $\lrcorner$

*Proof:*  $3 \rightarrow 2$ : If  $\gamma \subset K$  is a Jordan curve,  $\text{int}(\gamma)$  and  $\text{Ext}(\gamma)$  is an open partition of  $\mathbb{P}^1(\mathbb{C}) \setminus K$ . As  $\mathbb{P}^1(\mathbb{C}) \setminus K$  is connected,  $\text{int}(\gamma) \setminus K = \emptyset$ .  $\lrcorner$

?

**Def. (4.13.5.27) [Semilocally Simply Connected].** A space is called **semilocally simply connected** if for any point  $x \in X$ , there is a nbhd  $U$  of  $x$  that the image of the inclusion-induced map  $\pi_1(U, x) \rightarrow \pi_1(X, x)$  is trivial.  $\lrcorner$

**Prop. (4.13.5.28) [Van Kampen].** If  $X$  is a union of path-connected subsets  $A_\alpha$  all containing  $x_0$  that  $A_\alpha \cap A_\beta$  and  $A_\alpha \cap A_\beta \cap A_\gamma$  are all path-connected, then  $*\pi_1(A_\alpha) / \sim$  where  $\sim$  is generated by  $i_*(\pi_1(A_\alpha \cap A_\beta)) \in \pi_1(A_\alpha) \sim i_*(\pi_1(A_\alpha \cap A_\beta)) \in A_\beta$  for every  $\alpha, \beta$ .  $\lrcorner$

*Proof:* Cf.[Hatcher P52].  $\lrcorner$

**Prop. (4.13.5.29) [Wedge Sums].** If  $X_i \in \mathcal{CG}^{\text{well-pt}}$  and  $X = \bigvee X_i$ , then  $\pi_1(X) = *_i \pi_1(X_i)$ .  $\lrcorner$

## 6 CW Approximations

**Prop. (4.13.6.1) [CW Complex has Homotopy Extension Property].** For  $(X, A) \in \mathcal{CW}^{\text{rel}}$ ,  $X \times \{0\} \cup A \times I$  is a deformation retract of  $X \times I$ , thus  $(X, A)$  has the homotopy extension property (4.13.7.1).  $\lrcorner$

*Proof:* Cf.[Hat02]P15.  $\lrcorner$

**Lemma (4.13.6.2) [Compression and Homotopies].** For  $n \in \mathbb{Z}_+$ ,  $f : B \rightarrow Y \in \mathcal{CG}$ , the following are equivalent:

- $f_*$  induces an injection on  $\pi_n$  and a surjection on  $\pi_{n+1}$ .
- Any map  $(\mathbb{D}^n, \mathbb{S}^n) \rightarrow (Y, B) \in \mathcal{Top}^{\text{rel}}$  is homotopic rel  $\mathbb{S}^n$  to a map  $\mathbb{D}^n \rightarrow B$ .

*Proof:* Cf.[May, P70].?  $\lrcorner$

**Thm. (4.13.6.3) [Compression Theorem].** If  $n \in \mathbb{Z}_+$ ,  $(X, A) \in \mathcal{CW}^{\text{rel}}$  has relative dimension  $\leq n$  and  $f : B \rightarrow Y$  an  $n$ -equivalence, then any map  $(X, A) \rightarrow (Y, B) \in \mathcal{CG}^{\text{rel}}$  is homotopic rel  $A$  to a map  $X \rightarrow B$ .

In particular, (relative) homotopy doesn't depend on (2-degree)higher dimensional cells, (but might on lower one).  $\lrcorner$

*Proof:* Use (4.13.6.2) on each cells.  $\lrcorner$

**Thm. (4.13.6.4) [Whitehead].** If  $f : Y \rightarrow Z \in \mathcal{CG}$  is an  $n$ -equivalence, then  $[X, Y] \rightarrow [X, Z]$  is bijective for  $X \in \mathcal{CW}^{\text{pt}}$  of dimension  $\leq n - 1$ , and surjective for  $\dim X = n$ .  $\lrcorner$

*Proof:* The surjectivity follows from the compression theorem for the pair  $(\emptyset, X) \in \mathcal{CW}^{\text{rel}}$ , and the injectivity follows from the compression theorem applied to  $(X \times \mathbb{I}, X \times \partial\mathbb{I}) \in \mathcal{CW}^{\text{rel}}$ .  $\square$

**Cor. (4.13.6.5).** If  $Y \rightarrow Z \in \mathcal{Top}^{\text{pt}}$  is a weak-equivalence, then  $\langle X, Y \rangle \rightarrow \langle X, Z \rangle$  is an equivalence for any  $X \in \mathcal{CW}^{\text{pt}}$ .  $\lrcorner$

**Cor. (4.13.6.6) [Whitehead].** For  $n \in \mathbb{Z}_+$ , an  $n$ -equivalence of CW complexes of dimension  $\leq n - 1$  is a homotopy equivalence.

A weak equivalence of CW complexes is a homotopy equivalence.  $\lrcorner$

*Proof:* Let  $e : Y \rightarrow Z \in \mathcal{CW}$  satisfies the hypothesis, then because  $e_* : [Z, Y] \rightarrow [Z, Z]$  is a bijection by compression theorem (4.13.6.3), there exists  $f : Z \rightarrow Y$  s.t.  $e \circ f \cong \text{id}$ . Thus  $e \circ f \circ e \cong e$ . Then since  $e_* : [Y, Y] \rightarrow [Y, Z]$  is also a bijection by compression theorem,  $f \circ e \cong \text{id}$ .  $\square$

**Cor. (4.13.6.7).** If  $X \in \mathcal{CW}^{\text{pt}}$  and  $\pi_n(X) = 0$  for all  $n$ , then  $X$  is contractible.  $\lrcorner$

**Cor. (4.13.6.8) [Whitehead Combinatorial Homotopy Theorem I].** ? If  $M$  and  $K$  is dominated by CW complexes, then any weak homotopy equivalence is an homotopy equivalence. If the map is an inclusion, then it is a deformation retract. In particular, if  $M$  is manifold, then it is dominated by its tubular nbhd, so this theorem is applied.  $\lrcorner$

*Proof:* For inclusion, use compression, and in general use mapping cylinder and cellular approximation.  $\square$

### Cellular and CW Approximations

**Lemma (4.13.6.9).** For  $n \in \mathbb{Z}_+$ ,  $f : (X, A) \in \mathcal{CW}^{\text{rel}}$  has no  $m$ -cells for  $m \leq n$ , then  $(X, A)$  is  $n$ -connected. In particular,  $(X, X^n)$  is  $n$ -connected.  $\lrcorner$

*Proof:* It suffices to show that any map  $f : (\mathbb{I}^n, \partial\mathbb{I}^n, \mathbb{J}^n) \rightarrow (X, A, *)$  is homotopic to a map  $(\mathbb{I}^n, \mathbb{J}^n) \rightarrow (A, *) \text{ rel } \mathbb{J}^n$  (4.13.5.1). For this, notice that the image is compact, thus we can assume that  $(X, A)$  is finite ?. Thus by doing one cell by one cell, it suffices to show for  $X = A \coprod_{f, \mathbb{S}^n} \mathbb{D}^n$ . In this case, notice that by simplicial approximation, we can find a homotopic map  $f \cong f'$  that  $f'$  avoids some point in  $(\mathbb{D}^n)^o$  ?. Then we can use projection to construct a homotopy.  $\square$

**Thm. (4.13.6.10) [Cellular Approximationss].** Any map  $f : (X, A) \rightarrow (Y, B) \in \mathcal{CW}^{\text{rel}}$  is homotopic rel  $A$  to a cellular map.

In particular, any  $f : X \rightarrow Y \in \mathcal{CW}$  is homotopic to a cellular map.  $\lrcorner$

*Proof:* This follows from (4.13.6.9).  $\square$

**Remark (4.13.6.11).** The cellular approximation makes the computation of homotopy theoretically easier, but the difficulty comes from the complexity of the homotopy group of the sphere.  $\lrcorner$

**Remark (4.13.6.12).** In fact, all the rest of this subsubsection can be rewritten by the geometric realization functor  $\Gamma(X) \rightarrow X$  (4.6.3.8). It is functorial in the level of spaces. ?  $\lrcorner$



**Prop. (4.13.6.13) [ $n$ -Connected CW Models].** For a pair  $(A, X)$ , if  $A$  is CW complex, then there is a  $n$ -connected (4.13.5.16) CW pairs  $(Z, A) \rightarrow (X, A)$  that is identity on  $A$ , and  $\pi_i(Z) \rightarrow \pi_i(X)$  is isomorphism for  $i > n$  and injection for  $i = n$ .

Such a pair  $(Z, A)$  is called a  $n$ -connected CW model of  $(X, A)$ , and moreover it can be constructed from  $A$  by attaching cells of dimension greater than  $n$ .  $\lrcorner$

*Proof:* Cf.[Hatcher P353].  $\square$

**Cor. (4.13.6.14).** For an  $n$ -connected CW pair  $(X, A)$ , there exist a homotopic  $(Z, A) \cong (X, A) \text{ rel } A$  that  $Z \setminus A$  has only cells of dimension greater than  $n$ .  $\lrcorner$

*Proof:* Choose the  $n$ -connected approximation as above. The map induce and isomorphism on  $\pi_{>n}$  by definition and on  $\pi_{<n}$  because  $\pi_i(A) \rightarrow \pi_i(Z)$  and  $\pi_i(A) \rightarrow \pi_i(X)$  are isomorphisms. And on  $\pi_n$ , it is injective by definition and surjective because  $\pi_n(A) \rightarrow \pi_n(Z) \rightarrow \pi_n(X)$  is isomorphism. Hence we know that the collapsed mapping cylinder at is weak-homotopy equivalent to  $Z$ , thus it deforms into  $Z$  by (4.13.6.8), thus  $Z \rightarrow X \text{ rel } A$  by (4.13.1.5).  $\square$

**Cor. (4.13.6.15) [Functorial CW Approximations].** A **CW approximation** of a space  $X$  is a CW complex  $Z$  and a weak homotopy equivalence  $Z \rightarrow X$ . A **CW approximation** of a pair  $(X, A)$  is pair of CW complexes  $(Z, Z_0)$  and a morphism  $(Z, Z_0) \rightarrow (X, A)$  that induces isomorphisms on both relative and absolute homotopy groups.

Thus there exists a CW approximation for any space  $A$ , and also there exists a CW approximation for any pair  $(X, X_0)$ .  $\lrcorner$

*Proof:* Just choose  $A$  to be a set containing a point for each connected component of  $X$ , then  $\pi_0(Z) \rightarrow \pi_0(X)$  is surjective hence injective.

For pairs, first approximate  $X_0$  and use the mapping cylinder to get a embedding. ?  $\square$

**Prop. (4.13.6.16) [Functoriality of CW Models].** Given  $n$ -connected CW model  $f : (Z, A) \rightarrow (X, A)$  and  $f' : (Z', A') \rightarrow (X', A')$ , then any map of pairs  $g : (X, A) \rightarrow (X', A')$  can be extended to a map of pairs  $h : (Z, A) \rightarrow (Z', A')$  that  $gf \cong f'h \text{ rel } A$ . And such a map  $h$  is unique up to homotopy  $\text{rel } A$ .  $\lrcorner$

*Proof:* Cf.[Hatcher, P355].  $\square$

**Cor. (4.13.6.17) [Uniqueness of CW approximation].** The CW approximation is unique up to homotopy.  $\lrcorner$

**Cor. (4.13.6.18) [Localizing Category].** Together with Whitehead combinatorial homotopy theorem (4.13.6.8) the homotopy category of spaces defined in (4.6.2.3) is the category of spaces  $\mathcal{CG}$  localized by the class of weak homotopy equivalence classes.  $\lrcorner$

**Prop. (4.13.6.19) [CW Approximation of Excisive Triads].** Cf.[May, 79].  $\lrcorner$

## 7 Fibrations and Cofibrations

### Cofibrations

**Def. (4.13.7.1) [Homotopy Extension Property].**  $(X, A) \in \mathcal{Top}^{\text{rel}}$  is said to satisfy the **homotopy extension property** if every map  $X \times \{0\} \coprod_{A \times \{0\}} A \times I \rightarrow Y$  can be extended to a map  $X \times I \rightarrow Y$ .  $\lrcorner$



**Def. (4.13.7.2) [Cofibrations].** A **cofibration** is a map  $j : A \rightarrow X$  that for every map  $X \times \{0\} \coprod_{A \times \{0\}} A \times I \rightarrow Y$  can be extended to a map  $X \times I \rightarrow Y$ .

This implies  $X \times \{0\} \cup A \times I \rightarrow X \times I$  has a left inverse. And the converse is clearly true.

Then a cofibration is just a topological embedding with closed image that  $(X, A)$  has the homotopy extension property (4.13.7.1).

Cofibrations is stable under cobase change and coproducts.  $\perp$

*Proof:* ?  $\square$

**Def. (4.13.7.3) [NDR-Pairs].** An neighborhood deformation retraction pair, or an **NDR-pair** is a pair  $(X, A) \in \mathcal{T}op^{rel}$  s.t. there is a map  $u : X \rightarrow I$  s.t.  $u^{-1}(0) = A$ , and a homotopy  $f_t : X \rightarrow X \text{ rel } A$  s.t.  $f_0 = id_X$ , and  $f_1(u^{-1}([0, 1])) \subset A$ .  $\perp$

**Prop. (4.13.7.4) [NDR Pairs and Cofibrations].** If  $A \subset X$  is closed, then  $(X, A)$  is a NDR-pair (4.13.7.3) iff  $A \rightarrow X$  is a cofibration.  $\perp$

*Proof:* Cf. [May, P45]. ?  $\square$

**Prop. (4.13.7.5) [CW-Pairs are Cofibrations].** By compression theorem (4.13.6.3), for any CW-pair  $(A, X)$ ,  $A \rightarrow X$  is a cofibration.  $\perp$

**Def. (4.13.7.6) [Hurewicz Cofibration].** A **closed Hurewicz cofibration**  $i : A \subset B$  is a closed inclusion of spaces that  $B \times \{0\} \coprod A \times I \rightarrow B \times I$  has left extension property w.r.t any map  $Y \rightarrow pt$ .  $\perp$

**Prop. (4.13.7.7) [Quotient a Contractible Cofibration].** If  $A \rightarrow X$  is a cofibration and  $A$  is contractible, then the quotient map  $X \rightarrow X/A$  is a homotopy equivalence. In particular, this applies to  $(X, A) \in \mathcal{CW}^{rel}$ , by (4.13.6.1).  $\perp$

*Proof:* Let  $f_t : X \rightarrow X$  be a homotopy extending a contraction of  $A$  to a point, with  $f_0 = id$ . Since  $f_t(A) \subset A$ , they descends to a homotopy  $\bar{f}_t : X/A \rightarrow X/A$ . Because  $f_1(A)$  is a point, there is a map  $g$  in the following diagram

$$\begin{array}{ccc} X & \xrightarrow{f_1} & X \\ \downarrow \pi & \nearrow g & \downarrow \pi \\ X/A & \xrightarrow{\bar{f}_1} & X/A \end{array} .$$

So  $g$  and  $\pi$  are inverse homotopy equivalences, because  $f_1 \cong f_0 = id$  and  $\bar{f}_1 \cong \bar{f}_0 = id$ .  $\square$

**Prop. (4.13.7.8).** Let  $X$  be a normal space, then an inclusion  $j : A \hookrightarrow X$  is a cofibration iff  $A \hookrightarrow V$  is a cofibration for some open nbhd  $V$  of  $j(A) \subset X$ .  $\perp$

*Proof:* Cf. [AGP02]P92.  $\square$

**Prop. (4.13.7.9) [Homotopic Glueing Functions].** Let  $A \rightarrow X_1$  be a cofibration with  $X$  Hausdorff, and we have attaching maps  $f, g : A \rightarrow X_0$  that is homotopic, then  $X_0 \coprod_f X_1 \cong X_0 \coprod_g X_1 \text{ rel } X_0$ .  $\perp$

*Proof:* Now choose a homotopy  $H : A \times I \rightarrow X_0$  connecting  $f$  and  $g$ , then  $H$  induces a quotient space  $Z = X_0 \coprod_H (X_1 \times I)$ . Let  $X = X_0 \coprod_f X_1, Y = X_0 \coprod_g X_1$ , then there are natural inclusion maps  $i : X_1 \rightarrow Z, j : Y \rightarrow Z$ , and there are also deformation retractions  $Z \rightarrow X, Z \rightarrow Y$  constructed as follows:

Choose a deformation retraction  $r$  of  $X_1 \times I$  onto  $X_1 \times \{0\} \amalg A \times I$  (4.13.7.1), and  $H$  induces a map  $\overline{H} : D^n \times \{0\} \amalg S^n \times I \rightarrow S^n \amalg_f D^n$ , making the following diagram commutative

$$\begin{array}{ccc}
 A \times I & \xrightarrow{\quad} & X_1 \times I \\
 \downarrow H & & \downarrow r \\
 & A \times I \amalg X_1 \times \{0\} & \\
 & \downarrow \overline{H} & \\
 X_0 & \xrightarrow{\text{id}} & X_0 \amalg_f X_1
 \end{array}$$

which by definition defines a deformation retraction  $r_1 : X_0 \amalg_H (X_1 \times I) \rightarrow X_0 \amalg_f X_1$ . Similarly we have deformation retraction  $r_2 : Z \rightarrow Y$ . So  $r_2 \circ i$  and  $r_1 \circ j$  induce an homotopy equivalence between  $X$  and  $Y$ .  $\square$

**Prop. (4.13.7.10).** If  $A \rightarrow X, A \rightarrow Y$  are cofibrations, and  $f : X \rightarrow Y$  is a homotopy equivalence that  $f|_A = \text{id}_A$ , then  $f$  is a homotopy equivalence rel  $A$ .  $\lrcorner$

*Proof:* Cf.[Hatcher] P16.  $\square$

**Cor. (4.13.7.11).** If  $j : A \rightarrow X$  is a cofibration which is also a homotopy equivalence, and  $X$  is Hausdorff, then  $A$  is a deformation retraction of  $X$ .  $\lrcorner$

### Serre Fibrations

**Def. (4.13.7.12) [Serre Fibration].** A **Serre fibration** is the right lifting class of  $D^n \times \{0\} \rightarrow D^n \times I$  for every  $n$ . This is equivalent to: for any homotopy of  $\partial D^n$  and a image  $D^n$ , there is a homotopy of  $D^n$ .

In particular, Serre fibrations are stable under base change, by (4.5.0.1).  $\lrcorner$

**Prop. (4.13.7.13).** Being a Serre fibration is local on the target.  $\lrcorner$

*Proof:* Cf.[Homotopical Point of View]P127.  $\square$

**Prop. (4.13.7.14) [Long Exact Sequence of Serre Fibration].** Let  $\pi : E \rightarrow X$  be a Serre fibration, let  $b_0 \in B$  and  $x_0 \in F = \pi^{-1}(b_0)$ , then the map  $\pi_* : \pi_n(E, F, x_0) \rightarrow \pi_n(B, b_0)$  is an isomorphism for all  $n \geq 1$ . In particular, there is a long exact sequence

$$\dots \rightarrow \pi_n(F, x_0) \rightarrow \pi_n(E, x_0) \rightarrow \pi_n(B, b_0) \rightarrow \pi_{n-1}(F, x_0) \rightarrow \dots \rightarrow \pi_0(B, b_0) \rightarrow 0.$$

by (4.13.5.11).  $\lrcorner$

*Proof:* Cf.[Hat02]P376. [nLab].  $\square$

### Hurewicz Fibrations

**Def. (4.13.7.15) [Hurewicz Fibrations].** A **Hurewicz fibration** is a map  $p : X \rightarrow Y \in \mathcal{CG}$  that has right lifting property w.r.t maps  $A \times \{0\} \rightarrow A \times [0, 1]$  for any  $A \in \mathcal{CG}$ . in particular, a Hurewicz fibration is a Serre fibration (4.13.7.12).

Hurewicz fibrations are stable under base change.  $\lrcorner$

**Prop. (4.13.7.16) [Comparing Fibers of a Hurewicz Fibration].** If  $\pi : E \rightarrow B$  is a Hurewicz fibration, then for any arc  $\gamma$  in  $B$  with  $\gamma(0) = a, \gamma(1) = b$ , there is a homotopy equivalence  $f_\gamma : \pi^{-1}(a) \cong \pi^{-1}b$ , and the homotopy class of  $f_\gamma$  only depends on the homotopy class of  $\gamma$ .

In other words, a Hurewicz fibration over  $B$  determines a contravariant functor  $\Pi_1(B) \rightarrow \text{Ho}(\mathcal{CG})$ .

┘

*Proof:* This follows from the homotopy lifting property.  $\square$

**Cor. (4.13.7.17).** For a Hurewicz fibration  $E \rightarrow B$ , if  $B$  is contractible, then it is fiber homotopy equivalent to a trivial fiber bundle.  $\perp$

**Prop. (4.13.7.18).** Let  $\pi : E \rightarrow B, \pi' : E' \rightarrow B$  be Hurewicz fibrations and  $f : E \rightarrow E'$  be a map over  $B$  that is also a homotopy equivalence, then  $f$  is a fiber homotopy equivalence.  $\perp$

*Proof:* Cf. [Miller, P141]  $\color{red}?$   $\square$

**Prop. (4.13.7.19) [Homotopy Invariance of Pullbacks].** Let  $\pi : E \rightarrow X$  be a Hurewicz fibration and  $f_0, f_1 : Y \rightarrow X \in \mathcal{CG}$  are two maps that are homotopic, then the two Hurewicz fibrations  $f_0^*E/Y, f_1^*E/Y$  are fiber homotopy equivalent.  $\perp$

*Proof:* Cf. [Miller, P140]  $\color{red}?$   $\square$

**Prop. (4.13.7.20) [Homotopy Fiber of Contractible Fibration].** Let  $f : E \rightarrow B$  be a Hurewicz fibration with  $E$  contractible, then the homotopy fibers (4.4.3.18)  $F$  over  $b_0$  is weak homotopy equivalent to  $\Omega(B, b_0)$ .  $\perp$

*Proof:* Let  $x_0 \in \pi^{-1}(b_0)$ . If we compose the contraction of  $E$  to  $x_0$  with  $\pi$ , then we get for each  $x \in E$  a path  $\gamma_x$  from  $\pi(x)$  to  $b_0$ . Then these give a map  $E \rightarrow PB : x \mapsto \pi\gamma_x^{-1}$ , which is a lift of

$\pi$ . Then this map gives a commutative diagram

$$\begin{array}{ccccc} F & \longrightarrow & E & \longrightarrow & B \\ \downarrow & & \downarrow & & \parallel \\ \Omega(B, b_0) & \longrightarrow & PB & \longrightarrow & B \end{array}$$

that gives a map of their

corresponding long exact sequences (4.13.7.14). Thus  $F \rightarrow \Omega(B, b_0)$  is a weak homotopy equivalence because  $E, PB$  are both contractible (4.4.3.20).  $\square$

**Prop. (4.13.7.21) [Pathspaces are Hurewicz Fibrations].** For any map  $f : A \rightarrow B$ , the map  $\pi : E_f \rightarrow B : (a, \gamma) \mapsto \gamma(1)$  from the pathspace (4.4.3.18) is a Hurewicz fibration.

In particular, take  $A = \{x_0\}$ , then the path space  $PB \rightarrow B$  is a Hurewicz fibration.  $\perp$

*Proof:* Firstly this map is continuous by (4.4.3.5). To verify the homotopy lifting property, let  $g_t : X \rightarrow B$  be a homotopy and  $\tilde{g}_0 : X \rightarrow E_f$  be a lifting, let  $\tilde{g}_0(x) = (h(x), \gamma_x)$ . Define a lift  $\tilde{g}_t : X \rightarrow E_f$  by  $\tilde{g}_t(x) = (h(x), \gamma_x \cdot g_{[0,t]}(x))$ . The second term is concatenation, which can be defined because  $g_0(x) = \pi\tilde{g}_0(x) = \gamma_x(1)$ .

To check this is a continuous homotopy, by (4.4.3.7), it suffices to show  $A \times I \times I \rightarrow B : (x, s, t) \mapsto$

$$\gamma_x \cdot g_{[0,s]}(x)(t) = \begin{cases} \gamma(x, (1+t)s) & s \leq \frac{1}{1+t} \\ g_{(1+t)s-1}(x) & s \geq \frac{1}{1+t} \end{cases} \text{ is continuous.} \quad \square$$

**Cor. (4.13.7.22) [Homotopy Fibers].** We can embed  $A$  into  $E_f$  by mapping  $x$  to  $(x, \gamma_x)$ , where  $\gamma_x$  is the trivial loop at  $x$ . Then  $A$  is a deformation contraction of  $E_f$ , by restricting to shorter and shorter initial segments:  $H_t : E_f \rightarrow E_f : (a, \gamma) \mapsto (a, \gamma_t)$ , where  $\gamma_t(x) = \gamma(tx)$ .

Then we factored  $f$  as a homotopy equivalence followed by a fibration:  $A \rightarrow E_f \rightarrow B$ .

If  $x_0 \in X$  and  $F_f$  is the fiber of  $E_f$  over  $x_0$ , then a map  $(I^{i+1}, \partial I^{i+1}, J^i) \rightarrow (A, B, x_0)$  is the same as a map  $(I^i, \partial I^i) \rightarrow (F_f, \gamma_0)$ , where  $\gamma_0$  is the trivial loop at  $x_0$ . Thus  $\pi_{i+1}(A, B, x_0) = \pi_i(F_f, \gamma_0)$ .  $\perp$

**Prop. (4.13.7.23) [Pathspace of a Hurewicz Fibration].** If  $\pi : E \rightarrow B$  is a fibration, then the inclusion  $E \rightarrow E_\pi$  (4.13.7.22) is a fiber homotopy equivalence. In particular, the homotopy fibers of  $\pi$  are homotopy equivalent to the actual fibers.  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.13.7.24) [Fibration Sequence].** Given a fibration  $\pi : E \rightarrow B$  with  $F = \pi^{-1}(b_0)$ ,  $x_0 \in F$ , there is a sequence

$$\dots \rightarrow \Omega^2(B, b_0) \rightarrow \Omega(F, x_0) \rightarrow \Omega(E, x_0) \rightarrow \Omega(B, b_0) \rightarrow F \rightarrow E \rightarrow B \rightarrow 0$$

where any two consecutive maps form a fibration, up to homotopy.  $\lrcorner$

*Proof:* By (4.13.7.23), the inclusion  $i$  of  $F$  to the homotopy fiber  $F_\pi$  over  $\pi$  over  $p$  is a homotopy equivalence, and it extends to a map  $i : F_p \rightarrow E : (x, \gamma) \mapsto x$ , which is also a Hurewicz fibration, because it is the pullback of the fibration  $PB \rightarrow B$  (4.13.7.21).

Thus we can take the homotopy fiber  $F_i$  of  $i : F_p \rightarrow E$  over  $x_0$ , and similarly there is a fibration  $j : F_i \rightarrow F$ , and  $F$  is naturally homotopic to the actual fiber of  $i$ , which is just  $\Omega(B, b_0)$ .  $\square$

### Model Category Structures

**Lemma (4.13.7.25) [Cofibrant Replacement].** Any map  $f : X \rightarrow Y$  is a composition  $X \rightarrow M_f \rightarrow Y$ . Notice  $M_f \rightarrow Y$  is a homotopy equivalence and  $X \rightarrow M_f$  is cofibrant by (4.13.7.4).  $\lrcorner$

**Cor. (4.13.7.26) [Homotopy equivalence and mapping cylinder].** A map  $f : X \rightarrow Y$  is a homotopy equivalence iff  $X$  is a deformation retraction of the mapping cylinder  $M_f$ . In particular, two spaces are homotopically equivalent iff there is a third space containing both of them as deformation retractions.  $\lrcorner$

*Proof:* Cf. [Hatcher] P16.  $\square$

**Lemma (4.13.7.27) [Fibrant Replacement].** Any map  $f : X \rightarrow Y$  is a composition  $X \rightarrow N(f) \rightarrow Y$ , where  $N(f) = X \times_Y Y^I$ , where  $Y^I \rightarrow Y$  is evaluation at  $\{0\}$ , and  $X \rightarrow N(f)$  is given by  $x \mapsto (x, c_{f(x)})$ , where  $c_{f(x)} \in Y^I$  is the constant map  $f(x)$ .

It is true that  $\lrcorner$

*Proof:*  $X \rightarrow N(f)$  is homotopy inverse to the projection map: We can use the homotopy  $N(f) \times I \rightarrow N(f) : (\chi, a) \mapsto \chi_a : t \mapsto \chi(at)$ .

To show that  $N(f) \rightarrow Y$  is a fibration, Cf. [May, 50] ?.  $\square$

**Prop. (4.13.7.28) [Serre-Quillen].** The category  $\mathcal{CG}$  can be given a **Serre-Quillen model structure** with

- Weak equivalences: weak homotopy equivalence,
- Fibrations: Serre fibrations,
- Cofibrations: Retracts of morphisms  $X \rightarrow Y$  where  $Y$  is obtained from  $X$  by attaching cells.  $\lrcorner$

*Proof:* Cf. [Homotopy Theories and Model Categories, Chap8].  $\square$

**Prop. (4.13.7.29).** The homotopy category  $\text{Ho}(\text{Top})$  of the Serre-Quillen model structure is equivalent to the homotopy category of spaces  $\mathcal{H}$ .  $\lrcorner$

*Proof:*

□

**Lemma (4.13.7.30).** Every map can be decomposed as a homotopy equivalence followed by a fibration, by the construction of homotopy fibers. ┘

*Proof:*

□

**Prop. (4.13.7.31) [Hurewicz-Strøm].** The category  $\mathcal{CG}$  can be given a **Hurewicz-Strøm model structure** with

- Weak equivalences: homotopy equivalences.
- Cofibrations: closed Hurewicz cofibrations.
- Fibrations: Hurewicz fibrations.

┘

*Proof:* Cf.[strum paper].

□

**Prop. (4.13.7.32).** The homotopy category of the Hurewicz-Strøm model structure is equivalent to the usual category of homotopy types. ┘

## 8 Calculations of Homotopy Groups

**Prop. (4.13.8.1).** For  $0 < k < n \in \mathbb{Z}$ ,  $\pi_k(\mathbb{S}^n) = 0$ . And for  $0 < n$ ,  $\pi_0(\mathbb{S}^n) = *$ . ┘

*Proof:* This follows from cellular approximation(4.13.6.10) and the canonical CW structure on  $\mathbb{S}^n$ . □

**Prop. (4.13.8.2).** For  $i \geq 2$ ,  $\pi_1(\mathbb{RP}^i) \cong \mathbb{Z}/(2)$ , and  $\pi_n(\mathbb{RP}^i) \cong \pi_n(\mathbb{S}^i)$  for  $n \neq 1$ . ┘

*Proof:* This follows from the fiber sequence  $\mathbb{Z}/(2) \rightarrow \mathbb{S}^n \rightarrow \mathbb{RP}^n$  and(4.13.5.12). □

**Lemma (4.13.8.3).** There is a covering space  $\mathbb{R} \rightarrow \mathbb{S}^1$  with fiber  $\mathbb{Z}$ , thus  $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$ . ┘

**Prop. (4.13.8.4) [Hopf Fibration].** By(4.15.1.12) and(4.13.7.14), there is a long exact sequence

$$\dots \rightarrow \pi_i(\mathbb{S}^1) \rightarrow \pi_i(\mathbb{S}^3) \rightarrow \pi_i(\mathbb{S}^2) \rightarrow \pi_{i-1}(\mathbb{S}^1) \rightarrow \dots \rightarrow \pi_0(\mathbb{S}^2) \rightarrow 1.$$

Thus  $\pi_i(\mathbb{S}^3) \cong \pi_i(\mathbb{S}^2)$  for  $i \geq 3$ , and  $\pi_2(\mathbb{S}^2) \cong \pi_1(\mathbb{S}^1) \cong \mathbb{Z}$ (4.13.8.3). ┘

**Prop. (4.13.8.5) [Finiteness of Sphere Homotopy].** For  $p > n > 1$ ,  $\pi_p(\mathbb{S}^n)$  are all finite except for  $\pi_{4n-1}(\mathbb{S}^{2n})$ . ┘

*Proof:*

□

**Prop. (4.13.8.6) [Basic Sphere Homotopy].** For  $n \in \mathbb{Z}_+$ , there are maps  $\pi_i(\mathbb{S}^n) \rightarrow \pi_{i+1}(\mathbb{S}^{n+1})$  that is an isomorphism for  $i < 2n - 1$  and surjective for  $i = 2n - 1$ .

In particular,  $\pi_n(\mathbb{S}^n) \cong \pi_2(\mathbb{S}^2) \cong \mathbb{Z}$  for any  $n \geq 2$ . ┘

*Proof:* This follows from Freudenthal suspension theorem(4.13.5.22). The last assertion follows from(4.13.8.3). □

**Prop. (4.13.8.7).**  $\pi_4(\mathbb{S}^3) \cong \pi_4(\mathbb{S}^2) \cong \mathbb{Z}/(2)$ . ┘

*Proof:*

□

**Prop. (4.13.8.8).** for  $i \leq 2m$ ,  $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i-1} U(m)$ , and

$$\pi_{i-1} U(m) \cong \pi_{i-1} U(m+1) \cong \dots$$

and for  $j \neq 1$ ,  $\pi_j U(m) \cong \pi_j SU(m)$ .

Similarly,  $\pi_i \Omega_1(2m) \cong \pi_{i+1} O(2m)$  for  $i \leq n-4$ . (16.4.2.12), Cf[Morse Theory Milnor Prop23.4]. ⊥

**Cor. (4.13.8.9)[Bott Periodicity theorem for Unitary Groups].** The stable homotopy group  $\pi_i U$  has period 2.  $\pi_{2k+1} U \cong 0$  and  $\pi_{2k} U \cong \mathbb{Z}$ . ⊥

*Proof:* Use the last proposition and long exact sequence to show that for  $1 \leq i \leq 2m$ ,

$$\pi_{i-1} U = \pi_{i-1} U(m) \cong \pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m) \cong \pi_{i+1} U.$$

Notice that  $U(m) \rightarrow U(2m)/U(m) \rightarrow G_m(\mathbb{C}^{2m})$

□

**Prop. (4.13.8.10) [Bott Periodicity for O].** For the infinite dimensional orthogonal space  $O$ ,  $\Omega_8(16r) \cong O(r)$ ,  $\Omega_4(8r) \cong Sp(2r)$ . So  $\Omega_8 \cong O$  and  $\Omega_4 O \cong Sp$ . Thus by (4.13.5.2),

$$\pi_i(O) = \mathbb{Z}/(2), \mathbb{Z}/(2), 0, \mathbb{Z}, 0, 0, 0, \mathbb{Z}, \dots, \quad \pi_i(Sp) = 0, 0, 0, \mathbb{Z}, \mathbb{Z}/(2), \mathbb{Z}/(2), 0, \mathbb{Z}, \dots$$

respectively. (Use (16.4.2.13)) Cf.[Morse Theory Prop24.7].

⊥

**Prop. (4.13.8.11).** If a simply-connected finite complex  $X$  is not contractible, then infinitely many of its homotopy groups are non-zero. ⊥

*Proof:* Cf.[Jean-Pierre Serre, Cohomologie modulo 2 des complexes d'Eilenberg-MacLane]

□

## 4.14 Algebraic Topology II: Homologies and Cohomologies

**Notation (4.14.0.1).**

- Use notations on homotopy theory defined in [Algebraic Topology I: Homotopies](#).

⌋

### 1 Homologies

#### Axiomatic Homologies

**Def. (4.14.1.1) [Eilenberg-Steenrod Homology Theories, Eilenberg-Steenrod 1945].** A **Eilenberg-Steenrod homology theory** is given by the following data:

- A functor  $E_p(-) : \text{Ho}(\mathcal{T}\text{op}^{\text{rel}}) \rightarrow \mathcal{A}\text{b}$  for any  $p \in \mathbb{Z}$ . For  $X \in \mathcal{T}\text{op}$ , denote  $H_p(X, \emptyset)$  by  $E_p(X)$ .
- For any  $p \in \mathbb{Z}$ , an equivariant **boundary map**  $\partial : E_p(X, A) \rightarrow E_{p-1}(A)$  for any  $(X, A) \in \text{Ho}(\mathcal{T}\text{op}^{\text{rel}})$ .

that satisfies

**Exact Sequence:** For  $(X, A) \in \mathcal{T}\text{op}^{\text{rel}}$ , there is a natural long exact sequence

$$\cdots \rightarrow E_p(A) \rightarrow E_p(X) \rightarrow E_p(X, A) \xrightarrow{\partial} E_{p-1}(A) \rightarrow \cdots$$

**Excision:** For any excisive triad  $(X; A, B)$  (4.13.1.6), the inclusion  $(A, A \cap B) \rightarrow (X, B)$  induces an isomorphism

$$E_p(A, A \cap B) \cong E_p(X, B)$$

**Additivity:** If  $(X, A) = \coprod_{\alpha} (X_{\alpha}, A_{\alpha})$ , then there are natural isomorphisms

$$\bigoplus_{\alpha} E_p(X_{\alpha}, A_{\alpha}) \cong E_p(X, A).$$

By functorial CW approximations on relative spaces (4.13.6.15), CW approximation of excisive triads (4.13.6.19) and Whitehead theorem (4.13.6.6), such a theory is equivalent to a theory restricted to all CW complexes with cellular maps as morphisms. ⌋

**Prop. (4.14.1.2) [Cofibrations].** Given an Eilenberg-Steenrod homology theory  $E_*$ , for any cofibration  $i : A \rightarrow X \in \mathcal{T}\text{op}$ , there is a natural isomorphism

$$E_*(X, A) \cong E_*(X/A, *).$$

⌋

*Proof:* As  $i$  is a cofibration,  $C(A) \rightarrow C(i) = X \coprod_{A \times \{0\}} C(A)$  is also a cofibration. Consider the commutative diagram

$$\begin{array}{ccc} E_*(X \coprod_{A \times \{0\}} (A \times [0, 2/3]), A \times [1/3, 2/3]) & \longrightarrow & E_*(C(i), (A \times [1/3, 1]) / (A \times \{1\})) \\ \downarrow & & \downarrow \\ E_*(X, A) & \longrightarrow & E_*(X/A, *) \end{array}$$

Then the upper horizontal map is an isomorphism by excision, and the vertical maps are isomorphisms by homotopy invariance: The right vertical map is

$$(C(i), (A \times [1/3, 1]) / (A \times \{1\})) \cong (C(i), C(A)) \cong (X/A, *)$$

by (4.13.7.7). Then the lower horizontal map is also an isomorphism. □

**Def. (4.14.1.3) [Reduced Homology Theories].** A **reduced Eilenberg-Steenrod homology theory** is given by the following data: A functor  $\tilde{E}_*(-) : \text{Ho}(\mathcal{CG}^{\text{well-pt}}) \rightarrow \mathcal{Ab}$  for any  $p \in \mathbb{Z}$  that satisfies

**Exactness:** For any cofibration  $i : A \rightarrow X \in \mathcal{CG}^{\text{well-pt}}$ , the sequences

$$\tilde{E}_p(A) \rightarrow \tilde{E}_p(X) \rightarrow \tilde{E}_p(X/A)$$

are exact.

**Suspension:** For any  $X \in \mathcal{CG}^{\text{well-pt}}$ , there are natural isomorphisms

$$\Sigma_* : \tilde{E}_p(X) \cong \tilde{E}_{p+1}(\Sigma X).$$

**Additivity:** If  $X = \bigvee_{\alpha} X_{\alpha}$ , then there are natural isomorphisms

$$\bigoplus_{\alpha} \tilde{E}_p(X_{\alpha}) \cong \tilde{E}_p(X).$$

By functorial CW approximations on pointed spaces(4.13.6.15), such a theory is equivalent to a theory restricted to all CW complexes with cellular maps as morphisms.  $\lrcorner$

**Prop. (4.14.1.4) [Reduction to Reduced Homology Theories].** Giving an Eilenberg-Steenrod homology theory is equivalent to giving a reduced Eilenberg-Steenrod homology theory. Thus from now on we will not distinguish from a reduced ES-homology theory and an ES-homology theory.  $\lrcorner$

*Proof:* Given an ES homology theory  $E_*$ , for  $X \in \mathcal{T}\text{op}^{\text{pt}}$ , define  $\tilde{E}_*(X) = E_*(X, *)$ , then the maps  $* \rightarrow X \rightarrow *$  induce a natural splitting  $E_*(X) \cong \tilde{E}_*(X) \oplus E_*(*)$ . Then it is easy to show  $\tilde{E}_*$  is a reduced ES homology theory by(4.14.1.2): For suspension, because  $C(X)$  is contractible and  $X \rightarrow C(X)$  is a cofibration, the long exact sequence for  $C(X)/X \cong \Sigma(X)$  and(4.14.1.2) implies the isomorphism

$$\Sigma^{-1} : \tilde{E}_{p+1}(\Sigma(X)) \cong \tilde{E}_{p+1}(C(X)/X) \cong E_{p+1}(C(X), X) \xrightarrow{\partial} \tilde{E}_p(X).$$

Conversely, if  $\tilde{E}_*$  is a reduced ES homology theory, define  $E_*(X, A) = \tilde{E}_*(C(A \rightarrow X))$  where  $C(A \rightarrow X)$  is pointed over the cone point. In particular,  $E_*(X) = \tilde{E}_*(X_+)$ (4.13.1.16). We also define the boundary map as

$$\partial : E_*(X, A) \cong \tilde{E}_*(C(i_+)) \xrightarrow{\partial} \tilde{E}_*(\Sigma(A_+)) \xrightarrow{\Sigma^{-1}} E_{*-1}(A_+),$$

where  $\partial$  comes from  $\Sigma(A_+) \cong C(i_+)/X_+$ .

To show  $E_*$  is a ES homology theory: additivity is clear. The homotopy invariance follows from(4.13.1.17). To show exactness?. To show excision?. Cf.[May, P110].

To show that these two constructions are inverse to each other, it suffices to show that the  $\partial$  and  $\Sigma$  defined are inverse to each other?.  $\square$

**Lemma (4.14.1.5).** Let  $E$  be an ES-homology theory and  $B \subset A \subset X$ , then there is a functorial long exact sequence

$$\cdots \rightarrow E_p(A, B) \rightarrow E_p(X, B) \rightarrow E_p(X, A) \xrightarrow{\partial} E_{p-1}(A, B) \rightarrow \cdots,$$

where  $\partial : E_p(X, A) \rightarrow E_{p-1}(A) \rightarrow E_{p-1}(A, B)$ .  $\lrcorner$

*Proof:* This follows from diagram chase or using the functorial CW approximations(4.13.6.15).  $\square$



**Lemma (4.14.1.6).** If  $E$  be an ES-homology theory and  $(X; A, B)$  is an excisive triad, then the natural map

$$E_*(A, A \cap B) \oplus E_*(B, A \cap B) \rightarrow E_*(X, A \cap B)$$

is an isomorphism.  $\lrcorner$

*Proof:* This is done by using the CW approximations for excisive triads (4.13.6.19).  $\square$

**Prop. (4.14.1.7) [Mayer-Vietoris Sequences].** Let  $E$  be an ES-homology theory and  $(X; A, B)$  be an excisive triad, then there is a long exact sequence

$$\cdots \rightarrow E_p(A \cap B) \xrightarrow{((i_1)_*, (i_2)_*)} E_p(A) \oplus E_p(B) \xrightarrow{(\tilde{j}_1)_* - (\tilde{j}_2)_*} E_p(X) \xrightarrow{\Delta} E_{p-1}(A \cap B) \rightarrow \cdots,$$

where  $\Delta$  is the composite

$$E_p(X) \rightarrow E_p(X, B) \cong E_p(A, A \cap B) \xrightarrow{\partial} E_{p-1}(A \cap B).$$

$\lrcorner$

*Proof:* Cf. [May, P112].  $\square$

**Prop. (4.14.1.8) [Relative Mayer-Vietoris Sequences].** The relative MV-sequence is related to the MV-sequence (4.14.1.7) by

$$\begin{array}{ccccccc} \cdots & \longrightarrow & E_p(Y, A \cap B) & \longrightarrow & E_p(Y, A) \oplus E_p(Y, B) & \longrightarrow & E_p(Y, X) \longrightarrow E_{p-1}(Y, A \cap B) \longrightarrow \cdots \\ & & \downarrow \partial & & \downarrow \partial \oplus \partial & & \downarrow \partial \\ \cdots & \longrightarrow & E_p(A \cap B) & \longrightarrow & E_p(A) \oplus E_p(B) & \longrightarrow & E_p(X) \longrightarrow E_{p-1}(A \cap B) \longrightarrow \cdots \end{array}$$

$\lrcorner$

**Prop. (4.14.1.9) [Colimits].** If  $X = \varinjlim_{i \in \mathbb{N}} X_i$ , then there is a natural isomorphism

$$\varinjlim E_*(X_i) \cong E_*(X).$$

$\lrcorner$

*Proof:* Cf. [May, P115].  $\square$

**Cor. (4.14.1.10).** For a reduced ES-homology theory  $\tilde{E}_*$  and  $(X_i)_{i \in \mathbb{N}} \in \mathcal{T}^{\text{pt}}$ , there are natural isomorphisms

$$\varinjlim \tilde{E}_*(X_i) \cong \tilde{E}_*\left(\coprod_i X_i\right) \quad (4.13.1.19).$$

$\lrcorner$

### The Ordinary Homology Theory

**Def. (4.14.1.11) [Ordinary Homology Theories].** For  $\Lambda \in \mathcal{A}b$ , an **ordinary homology theory** with coefficients in  $\Lambda$  is a generalized homology theory (4.14.2.1) that satisfies the additional axiom

$$\textbf{Dimension: } E_p(\text{pt}) = \begin{cases} \Lambda & , p = 0 \\ 0 & , p \neq 0 \end{cases}.$$

┘

**Def. (4.14.1.12) [Ordinary Reduced Homology Theories].** For  $\Lambda \in \mathcal{A}b$ , an **ordinary reduced homology theory** with coefficients in  $\Lambda$  is a reduced homology theory (4.14.1.3) that satisfies the additional axiom

$$\text{Dimension: } \tilde{E}_p(S^0) = \begin{cases} \Lambda & , p = 0 \\ 0 & , p \neq 0 \end{cases}.$$

┘

**Def. (4.14.1.13) [Hurewicz Homomorphism].** For an ordinary homology theory  $H_*$ , if we denote the generator of  $\tilde{H}_0(\mathbb{S}^0)$  by  $a_0$ , and  $a_n = \Sigma^n(a) \in \tilde{H}_n(\mathbb{S}^n)$ , then for any  $X \in \mathcal{CG}^{\text{pt}}$ , define the **Hurewicz homomorphism**

$$\text{Hur}_X : \pi_n(X) \rightarrow \tilde{H}_n(X) : f \mapsto f_*(a_n).$$

┘

**Prop. (4.14.1.14).** For  $n \in \mathbb{Z}_+$ ,  $X \in \mathcal{Top}^{\text{pt}}$ ,  $\text{Hur}_X$  is a homomorphism of groups. ┘

*Proof:* Cf.[May, P117]. ┘

**Prop. (4.14.1.15) [Generalized Hurewicz theorem].** If  $n \geq 2$  and  $(X, A)$  is a  $(n-1)$ -connected pair of spaces, then the Hurewicz map induces an isomorphism

$$\pi_n(X, A) / \sim \pi_1(A) \cong H_n(X, A),$$

and  $H_k(X, A) = 0, k < n$ . Moreover, the Hurewicz map  $\pi_{n+1}(X, A) \rightarrow H_{n+1}(X, A)$  is surjective. ┘

*Proof:* Cf.[Hatcher P371 and 390Ex23]. ? ┘

**Cor. (4.14.1.16) [Converse Whitehead].** If  $f : X \rightarrow Y$  is a map of simple pointed spaces that induces isomorphisms on all homology groups, then  $f$  is a weak homotopy equivalence. ┘

*Proof:* We may change  $Y$  to the mapping cylinder of  $f$ , then notice all the relative homology groups  $H_n(Y, X)$  vanishes by long exact sequence, and then by the generalized Hurewicz theorem,  $\pi_n(Y, X) = 0$ , and then  $f$  is a weak homotopy equivalence, by (4.13.5.11). ┘

**Thm. (4.14.1.17) [Hurewicz].** For  $n \in \mathbb{Z}_+$ , if  $X \in \mathcal{CG}^{\text{pt}}$  is  $(n-1)$ -connected, then  $\text{Hur}_X : \pi_n(X) \rightarrow \tilde{H}_n(X)$  induces an isomorphism  $\pi_n(X)^{\text{ab}} \cong \tilde{H}_n(X)$ . In particular, if  $n > 1$ ,  $\pi_n(X) \cong \tilde{H}_n(X)$  (4.13.5.1). ┘

*Proof:* Cf.[May, P118]. ┘

### The Cellular and Singular Realizations

**Def. (4.14.1.18) [Degree of Maps].** For any  $f : \mathbb{S}^n \rightarrow \mathbb{S}^n \in \mathcal{Top}$ , we can choose a base point and use cellular approximation to make it homotopic to a cellular map  $(\mathbb{S}^n, *) \rightarrow (\mathbb{S}^n, *)$ , and define the **degree of  $f$**  to be the multiplying factor of  $f_* : \pi_n(\mathbb{S}^n) \cong \mathbb{Z} \rightarrow \pi_n(\mathbb{S}^n) \cong \mathbb{Z}$  (4.13.8.6).

Then this degree is independent of the base point chosen simply by rotation. ┘

**Prop. (4.14.1.19).** The antipodal map  $\mathbb{S}^n \rightarrow \mathbb{S}^n$  has degree  $(-1)^{n-1}$ . ┘

*Proof:* This is because the antipodal map is a composition of  $n+1$  reflections, and each reflection is an  $n$ -suspension of a reflection on  $\mathbb{S}^1 \rightarrow \mathbb{S}^1$ , which clearly has degree  $-1$ . Thus we finish by (4.13.8.6). ┘

**Def. (4.14.1.20) [Cellular Complex on CW].** Given  $X \in \mathcal{CW}$  with presentation  $X = \cup_i X^i$ , we can define the **cellular homology groups** as follows: Let  $C_\bullet(X)$  be the complex with  $C_p(X)$  the free Abelian group generated by the  $p$ -cells of  $X$ , and the map  $C_p(X) \rightarrow C_{p-1}(X)$  is given by  $e_\alpha \mapsto a_{\alpha\beta} e_\beta$ , where for any  $n$ -cell  $\alpha$  and  $n-1$ -cell  $\beta$ ,  $a_{\alpha\beta}$  is the degree(4.14.1.18) of the map

$$S^{p-1} \xrightarrow{\alpha} X^{p-1} \rightarrow X^{p-1}/X^{p-2} \xrightarrow{\beta^{-1}} S^{p-1}.$$

Then  $C_\bullet(X)$  is truly a complex. More generally, for any  $\Lambda \in \mathcal{Ab}$ , we can define  $C_\bullet(X; \Lambda) = C_\bullet(X) \otimes \Lambda$ .  $\lrcorner$

*Proof:* Cf. [May, P99] ?.  $\square$

**Def. (4.14.1.21) [Cellular Homology Groups].** For  $X \in \mathcal{CW}$  and a subcomplex  $A \subset X$ , there is an inclusion  $C_*(A) \rightarrow C_*(X)$ . If we define  $C_*(X, A) = C_*(X)/C_*(A)$ . Define the homology groups  $H_p(X) = H_p(C_\bullet(X))$ ,  $H_p(X, A) = H_p(C_\bullet(X, A))$ .

then the long exact sequence associated to the exact sequence  $0 \rightarrow C_*(A) \rightarrow C_*(X) \rightarrow C_*(X, A) \rightarrow 0$  gives a boundary map  $\partial : H_p(X, A) \rightarrow H_{p-1}(A)$ .

More generally, for any  $\Lambda \in \mathcal{Ab}$ , we can define  $C_\bullet(X, A; \Lambda)$  and also  $H_p(X; \Lambda)$ ,  $H_p(X, A; \Lambda)$ .  $\lrcorner$

**Def. (4.14.1.22) [Cellular Complex of Products].** For  $X, Y \in \mathcal{CW}$ , with the CW structure on  $X \times Y$  given in(4.13.3.8), then

$$C_\bullet(X) \otimes C_\bullet(Y) \cong C_\bullet(X \times Y).$$

$\lrcorner$

*Proof:* The map  $C_\bullet(X) \otimes C_\bullet(Y) \cong C_\bullet(X \times Y)$  is given by  $e_\alpha \otimes e_\beta \mapsto (-1)^{|\alpha||\beta|} e_{\alpha \times \beta}$ .

It is clear that this is an isomorphism up to sign. Fo the sign problem, see [May, P101].  $\square$

**Prop. (4.14.1.23).** The cellular homology groups in(4.14.1.21) defines a ES-homology theory(4.14.2.1).  $\lrcorner$

*Proof:* For homotopy invariance, notice a homotopy  $X \times I \rightarrow Y$  between  $f, g : X \rightarrow Y$  will induce a map

$$C_\bullet(X) \otimes C_\bullet(I) \rightarrow C_\bullet(Y)$$

by(4.14.1.22), and with the canonical CW structure,  $C_\bullet(I)$  is just  $0 \rightarrow \mathbb{Z} \xrightarrow{(1, -1)} \mathbb{Z} \oplus \mathbb{Z} \rightarrow 0$ . By writting it out, this gives exactly a homotopy between the two maps  $f_*, g_*$ . Additivity, excision and exactness is clear.  $\square$

**Def. (4.14.1.24) [Singular Homologies].** For  $X \in \mathcal{Top}$ ,  $\Lambda \in \mathcal{Ab}$ , the **singular cohomology groups** with coefficients in  $R$  is defined to be  $H_{\text{sing},*}(X; \Lambda) = H_*(\Gamma(X); R)$ (4.6.3.8). This is the just the homology on  $\mathcal{Top}^{\text{rel}}$  corresponding to the cellular homology on CW-pairs.  $\lrcorner$

**Thm. (4.14.1.25) [Uniqueness of Ordinary Homology Theories].** There exists a unique ordinary homology theory, i.e. the one defined in(4.14.1.21).  $\lrcorner$

*Proof:* Cf. [May, P119].  $\square$

## 2 Cohomologies

**Remark (4.14.2.1) [Eilenberg-Steenrod Cohomology Theories].** The whole basic theory of cohomologies is dual to that of homologies, so it is omitted here.

We only focus on new features appearing.  $\lrcorner$

**Def. (4.14.2.2) [Cohomological Operators].** Given reduced ES-cohomology theories  $\tilde{E}^*, \tilde{F}^*, p, n \in \mathbb{Z}$ , a **cohomological operator** of type  $p$  and degree  $n$  is a natural homomorphism  $\tilde{E}^p \rightarrow \tilde{F}^{p+n}$ .

A **stable cohomological operator** of degree  $n$  is a sequence of cohomological operators  $\Phi_p^n$  of type  $p$  and degree  $n$  that commutes with the suspension isomorphism.  $\lrcorner$

### Ordinary Cohomology Theories

**Thm. (4.14.2.3) [Uniqueness of Ordinary Homology Theories].** There exists a unique ordinary cohomology theory, i.e. the one give by cellular cohomology groups.  $\lrcorner$

*Proof:* The proof is the same as that of (4.14.1.25).  $\square$

**Prop. (4.14.2.4) [Universal Coefficient Theorem].** See (4.8.8.6).  $\lrcorner$

**Cor. (4.14.2.5).** A map between topological spaces that induce isomorphism on arbitrary homology group induce isomorphisms on cohomology groups.  $\lrcorner$

**Def. (4.14.2.6) [Cup Product].** For  $X, Y \in \mathcal{CW}$ , as  $C_\bullet(X) \otimes C_\bullet(Y) \cong C_\bullet(X \times Y)$  (4.14.1.22), for any  $R \in \mathcal{CAlg}$ , there is a chain homomorphism

$$\mathrm{Hom}^\bullet(C_\bullet(X), R) \otimes \mathrm{Hom}^\bullet(C_\bullet(Y), R) \rightarrow \mathrm{Hom}^\bullet(C_\bullet(X \times Y), R),$$

which induces a natural **cross product map**

$$H^*(X; R) \otimes H^*(Y; R) \rightarrow H^*(X \times Y; R).$$

And composing with the diagonal map  $\Delta_X : X \rightarrow X \times X$  gives a **cup product map**

$$H^*(X; R) \times H^*(X; R) \rightarrow H^*(X \times X; R)$$

that makes  $H^*(X; R)$  into a unital commutative graded  $R$ -algebra.  $\lrcorner$

*Proof:* Cf. [May, P139].  $\square$

**Def. (4.14.2.7) [Bockstein Homomorphism].**  $\lrcorner$

**Prop. (4.14.2.8).** the square of the Bockstein homomorphism  $H^n(X, \mathbb{F}_p) \rightarrow H^{n+2}(X, \mathbb{F}_p)$  is trivial.  $\lrcorner$

*Proof:* There is a commutative diagram of commutative rings

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\times m} & \mathbb{Z} & \longrightarrow & \mathbb{Z}/(m) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z}/(m) & \xrightarrow{\times m} & \mathbb{Z}/(m^2) & \longrightarrow & \mathbb{Z}/(m) \longrightarrow 0 \end{array}$$

which induces commutative diagrams

$$\begin{array}{ccc} H^n(X, \mathbb{Z}/(m)) & \xrightarrow{\tilde{\beta}} & H^{n+1}(X, \mathbb{Z}) \\ \parallel & & \downarrow \rho \\ H^n(X, \mathbb{Z}/(m)) & \xrightarrow{\beta} & H^n(X, \mathbb{Z}/(m)) \end{array}$$

so  $\beta = \rho\tilde{\beta}$ , and  $\beta^2 = \rho\tilde{\beta}\rho\tilde{\beta} = 0$ , as  $\tilde{\beta}\rho = 0$  by long exact sequence.  $\square$

**Prop. (4.14.2.9) [Lefschetz Fixed Point Theorem].**  $\lrcorner$

**Prop. (4.14.2.10) [Wedge Sums].** For  $M, N \in \mathcal{T}_{\text{op}}$ ,  $\tilde{H}^i(M \wedge N) = \tilde{H}^i(M) \otimes \tilde{H}^i(N)$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.14.2.11) [Connected Sums].** Let  $M, N \in \mathcal{M}_{\text{ani}, \text{cntd}, \text{cpct}}^d$ , the cohomology of  $M \# N$  is ?  $\lrcorner$

And if  $M, N$  are orientable, then  $H^*(M \# N) = 1 \oplus [\tilde{H}^*(M) \oplus \tilde{H}^*(N)/([M] - [N])]$ .  $\lrcorner$

*Proof:* As  $M \# N / S^{d-1} = M \vee N$ , there is a long exact sequence

$$\dots \rightarrow \tilde{H}^i(M \vee N) \rightarrow \tilde{H}^i(M \# N) \rightarrow \tilde{H}^i(S^{d-1}) \rightarrow \tilde{H}^{i+1}(M \# N) \rightarrow \dots$$

Thus by (4.14.2.10),  $\tilde{H}^i(M \# N) = \tilde{H}^i(M) \oplus \tilde{H}^i(N)$  for  $i \neq d-1, d$ . And there is an exact sequence

$$0 \rightarrow \tilde{H}^{d-1}(M \vee N) \rightarrow \tilde{H}^{d-1}(M \# N) \rightarrow \mathbb{Z} \rightarrow \tilde{H}^d(M \vee N) \rightarrow \tilde{H}^d(M \# N) \rightarrow 0.$$

If  $M, N$  are both orientable, then so does  $M \# N$ , and the sequence looks like

$$0 \rightarrow \tilde{H}^{d-1}(M \vee N) \rightarrow \tilde{H}^{d-1}(M \# N) \xrightarrow{0} \mathbb{Z} \rightarrow \mathbb{Z} \otimes \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0.$$

and  $\tilde{H}^{d-1}(M \vee N) \cong \tilde{H}^{d-1}(M \# N)$ .

If only one of  $M, N$  is orientable, then similarly,  $\tilde{H}^{d-1}(M \vee N) \cong \tilde{H}^{d-1}(M \# N)$ .

If neither  $M, N$  are orientable, then nor does  $M \# N$ , and the sequence looks like

$$0 \rightarrow \tilde{H}^{d-1}(M \vee N) \rightarrow \tilde{H}^{d-1}(M \# N) \rightarrow \mathbb{Z} \rightarrow 0 \otimes 0 \rightarrow 0 \rightarrow 0.$$

which splits ?, so  $\tilde{H}^{d-1}(M \# N) \cong \tilde{H}^{d-1}(M \vee N) \oplus \mathbb{Z}$ .  $\square$

### Cup and Cap Products

Main references are [Hat02] Chap3.2.

**Prop. (4.14.2.12).** The cup product will restrict to a relative version:

$$H^*(X, A) \times H^*(X, B) \rightarrow H^*(X, A \cup B),$$

This implies that if  $X$  is a union of  $n$  contractible open set, then the cup product of  $n$ -elements vanish. In particular, the cup product in a suspension vanishes.  $\lrcorner$

**Prop. (4.14.2.13) [Künneth Formula].** The cross product  $H^*(X, \mathbb{R}) \otimes_R H^*(Y, \mathbb{R}) \rightarrow H^*(X \times Y, \mathbb{R})$  is an isomorphisms of rings if  $X, Y$  are CW complexes and  $H^*(Y, \mathbb{R})$  are a finite free  $R$ -modules for any  $k$ .  $\lrcorner$

*Proof:* Cf. [Hat02] P219.  $\square$

**Def. (4.14.2.14) [Cap Products].**  $\lrcorner$

### Cohomology of Fiber Bundles

**Prop. (4.14.2.15) [Leray-Hirsch].** For a fiber bundle  $F \rightarrow E \rightarrow B$  and a ring  $R$  s.t.  $H^n(F, R)$  is f.g free for all  $n$ , and there exist classes  $c_j$  of  $H^*(E)$  that constitute a basis for each fiber  $F$ , then

$$H^*(B, R) \otimes H^*(F, R) \rightarrow H^*(E, R)$$

is an isomorphism of  $H^*(B, R)$ -modules. ┘

*Proof:* □

**Cor. (4.14.2.16).**

- $H^*(U(n); \mathbb{Z}) = \Lambda_{\mathbb{Z}}[x_1, x_3, \dots, x_{2n-1}]$ .
  - $H^*(SU(n, \mathbb{R}); \mathbb{Z}) = \Lambda_{\mathbb{Z}}[x_3, \dots, x_{2n-1}]$ .
  - $H^*(Sp(n); \mathbb{Z}) = \Lambda_{\mathbb{Z}}[x_3, x_7, \dots, x_{4n-1}]$ .
- ┘

**Prop. (4.14.2.17).**  $H^*(\mathbb{G}(n, \mathbb{K}^\infty); \mathbb{Z})$  where  $\mathbb{K} = \mathbb{R}, \mathbb{C}, \mathbb{H}$  is generated by the symmetric polynomials, where for  $\mathbb{R}$  the coefficient is  $\mathbb{Z}_2$ . ┘

*Proof:* Use the flag variety and first calculate for  $\infty$ . Then use Poincare duality to show it is mapped onto the symmetric polynomials. Cf.[Hatcher P435]. □

**Prop. (4.14.2.18) [Leray-Serre].** For a Serre fibration, e.g. fiber bundle,  $F \rightarrow E \rightarrow B$ , that  $B$  is simply connected, then there is a spectral sequence

$$E_2^{pq} = H_p(B, H_q(F)) \Rightarrow H_{p+q}(E) \quad E_2^{pq} = H^p(B, H^q(F)) \Rightarrow H^{p+q}(E)$$
┘

**Cor. (4.14.2.19) [Wang Sequence].** When  $B = S^n$ , there is a long exact sequence:

$$\cdots \rightarrow H_q(F) \rightarrow H_q(E) \rightarrow H_{q-n}(F) \rightarrow H_{q-1}(F) \rightarrow H_{q-1}(E) \rightarrow \cdots$$
┘

**Cor. (4.14.2.20) [Gysin Sequence].** When  $F = S^n$ , there is a long exact sequence:

$$\cdots \rightarrow H_{p-n}(B) \rightarrow H_p(E) \rightarrow H_p(B) \rightarrow H_{p-n-1}(B) \rightarrow H_{p-1}(E) \rightarrow \cdots$$
┘

### H-Spaces

**Def. (4.14.2.21) [H-Spaces].** An **H-space** is a unital magma object in  $\text{Ho}(\mathcal{CG}^{\text{pt}})$ . Equivalently, an H-space is a pointed space  $X \in \mathcal{CG}^{\text{pt}}$  with a map  $\mu : X \times X \rightarrow X \in \text{Ho}(\mathcal{CG}^{\text{pt}})$  s.t.  $X \rightarrow \{*\} \times X \rightarrow X$  and  $X \rightarrow X \times \{*\} \rightarrow X$  is homotopic to  $\text{id}_X$ .

An **associative H-space** is an H-space s.t.  $\mu_X$  is associative in  $\mathcal{Top}^{\text{pt}} / \sim$ . Similarly, we can define group-H-spaces and commutative H-spaces.

An H-space is called strictly associative(or a monoid space) if it comes from a unital associative object in the category of spaces. ┘

**Remark (4.14.2.22).** The definition of a H-space structure can be modified when  $X$  is a CW complex. In fact, when  $(X, e)$  is a CW-pair, if there exists a map  $\mu : X \times X \rightarrow X$  and a point  $e \in X$  that  $X \rightarrow \{e\} \times X \rightarrow X$  and  $X \rightarrow X \times \{e\} \rightarrow X$  is homotopic to  $\text{id}_X$ , then  $\mu$  can be homotoped that  $e$  is a strict identity.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (4.14.2.23).** If  $H$  is an H-space, then for any  $X \in \mathcal{T}\text{op}^{\text{pt}}$ ,  $\langle X, H \rangle$  is a magma. Moreover, if  $H$  is a commutative associative group H-space, then  $\langle X, H \rangle \in \mathcal{A}\text{b}$ .  $\lrcorner$

**Prop. (4.14.2.24)[Loop Spaces are H-Spaces].** For any  $X \in \mathcal{T}\text{op}^{\text{pt}}$ ,  $\Omega(X)$  is an associative group-H-space, and  $\Omega^2(X)$  is a associative commutative group-H-space.  $\lrcorner$

*Proof:* The multiplication is given by concatenation of loops. It is continuous by adjunction arguments.  $\square$

**Prop. (4.14.2.25).**  $\mathbb{C}\mathbb{P}^\infty$  can be given a commutative strictly associative H-space structure. More generally,  $B(\mathbb{Z}/n) \cong (\mathbb{C}^\infty \setminus \{0\})/(\mathbb{R}^+ \times \mu_n)$  (4.15.3.16) can be given a commutative strictly associative H-space structure. In particular this applies to  $\mathbb{R}\mathbb{P}^\infty$ .  $\lrcorner$

*Proof:* We can regard  $\mathbb{C}^\infty$  as the space of polynomials with complex coefficients, then the polynomial multiplication gives a map  $(\mathbb{C}^\infty \setminus \{0\}) \times (\mathbb{C}^\infty \setminus \{0\}) \rightarrow \mathbb{C}^\infty \setminus \{0\}$  that descends to a map  $B(\mathbb{Z}/n) \times B(\mathbb{Z}/n) \rightarrow B(\mathbb{Z}/n)$ . And  $e = (1, 0, \dots, 0, \dots)$  is the unit.  $\square$

**Prop. (4.14.2.26).** The James reduced product  $J(X)$  (4.13.3.20) is a strictly associative H-space with multiplication given by  $(x_1, \dots, x_n)(y_1, \dots, y_m) = (x_1, \dots, x_n, y_1, \dots, y_m)$ , and the identity  $e$ . This H-space structure also descends to a H-space structure on  $SP(X)$  (4.13.3.21).  $\lrcorner$

**Prop. (4.14.2.27).** The universal cover of a H-space is a H-space.  $\lrcorner$

*Proof:* Take an arbitrary lift that maps  $(\tilde{e}, \tilde{e})$  to  $\tilde{e}$ . Notice the homotopy can also be lifted.  $\square$

**Prop. (4.14.2.28)[Cohomology Ring].** The cohomology ring of a H-space is a topologists' Hopf algebra, by Kunneth formula and naturality.  $\lrcorner$

**Cor. (4.14.2.29).**  $\mathbb{C}\mathbb{P}^n$  is not a H-space.  $\lrcorner$

*Proof:* This is because  $H^*(\mathbb{C}\mathbb{P}^n, \mathbb{Z}) = \mathbb{Z}[\alpha]/\alpha^n$ ,  $|\alpha| = 2$ , which is not a topologist's Hopf algebra, by (3.11.3.13).  $\square$

**Prop. (4.14.2.30).** The fundamental group of an H-space is Abelian.  $\lrcorner$

*Proof:* This is because  $\pi_1$  preserves products, so takes unital magma space to unital magma objects (4.1.1.72). And the unital magma objects in the category of groups is the Abelian groups (4.1.1.71).  $\square$

**Prop. (4.14.2.31)[Adam].**  $\mathbb{S}^0, \mathbb{S}^1, \mathbb{S}^3, \mathbb{S}^7$  are the only spheres that have H-structures.  $\lrcorner$

*Proof:* Firstly  $S^1 \subset \mathbb{C}, S^3 \subset \mathbb{H}, S^7 \subset \mathbb{O}$  are submonoids, so they have H-structures. ?  $\square$

**Cor. (4.14.2.32).**  $\mathbb{R}\mathbb{P}^n$  has a H-structure iff  $n = 1, 3, 7$ .  $\lrcorner$

*Proof:* This is because the universal cover of a H-space is a H-space (4.14.2.27). Also  $\mathbb{S}^1, \mathbb{S}^2, \mathbb{S}^3, \mathbb{S}^7$  are monoid spaces, and  $-1$  are in their center, so the quotients are also monoid spaces.  $\square$

### Examples of Calculations

**Cor. (4.14.2.33).**

$$H^*(\mathbb{RP}^n, \mathbb{F}_2) = \mathbb{F}_2[a]/a^{n+1}, |a| = 1, \quad H^*(\mathbb{CP}^n, \mathbb{Z}) = \mathbb{Z}[\alpha]/\alpha^{n+1}, |\alpha| = 2$$

┘

*Proof:* We prove for  $\mathbb{RP}^n$ , the  $\mathbb{CP}^n$  case is similar. Use induction. For  $n = 1$ , this is clear. For  $n > 1$ , notice  $\mathbb{RP}^n$  is the  $n-1$ -skeleton of  $\mathbb{RP}^n$ , thus the map of rings  $H^*(\mathbb{RP}^n, \mathbb{F}_2) \rightarrow H^*(\mathbb{RP}^{n-1}, \mathbb{F}_2)$  maps the generator of  $H^2(\mathbb{RP}^n, \mathbb{F}_2)$  to the generator  $a$  of  $H^2(\mathbb{RP}^{n-1}, \mathbb{F}_2)$ . Then by hypothesis  $a^{n-1} \neq 0$ . The  $H^{n-1}(\mathbb{RP}^n, \mathbb{F}_2) \cong H_1(\mathbb{RP}^n, \mathbb{F}_2) \cong \mathbb{Z}/(2)$  is generated by  $\alpha^{n-1}$ . Then the Poincaré duality shows that  $\alpha \cup \alpha^{n-1} \neq 0$ , so we are done.  $\square$

**Cor. (4.14.2.34).** For  $n > 1$ , the natural homomorphism  $H^n(\mathbb{RP}^n, \mathbb{Z}/2\mathbb{Z}) \rightarrow H^n(S^n, \mathbb{Z}/2\mathbb{Z})$  is 0.  $\quad \text{┘}$

*Proof:* Because the cohomology ring map maps  $a \in H^1(\mathbb{RP}^n, \mathbb{Z}/2\mathbb{Z})$  to  $0 \in H^1(S^n, \mathbb{Z}/2\mathbb{Z}) = 0$ .  $\square$

**Prop. (4.14.2.35)** [ $H^*(K, \mathbb{F}_2)$ ]. The cohomology ring of the Klein bottle  $K$  is  $H^*(K, \mathbb{Z}/(2)) = \mathbb{F}_2[x, y]/(xy, x^2 - y^2, x^3, y^3)$ .  $\quad \text{┘}$

*Proof:* Let  $\varphi \in C^0(K, \mathbb{Z}/(2))$  be the dual of  $v$ ,  $\alpha, \beta, \gamma \in C^1(K, \mathbb{Z}/(2))$  be the dual of  $a, b, c$ , and  $\mu, \lambda \in C^2(K, \mathbb{Z}/(2))$  be the dual of  $A, B$ , then

$$\partial(\varphi)(a) = \partial(\varphi)(b) = \partial(\varphi)(c) = 0$$

$$\delta(\alpha)(A) = \alpha(\partial(A)) = \alpha(a + b - c) = 1, \quad \delta(\alpha)(B) = \alpha(\partial(B)) = \alpha(b + c - a) = -1$$

$$\delta(\beta)(A) = \beta(\partial(A)) = \beta(a + b - c) = 1, \quad \delta(\beta)(B) = \beta(\partial(B)) = \beta(b + c - a) = 1$$

$$\delta(\gamma)(A) = \gamma(\partial(A)) = \gamma(a + b - c) = -1, \quad \delta(\gamma)(B) = \gamma(\partial(B)) = \gamma(b + c - a) = 1$$

So

$$H^0(K, \mathbb{Z}/(2)) = \mathbb{Z}/(2)\varphi, \quad H^1(K, \mathbb{F}_2) = \mathbb{F}_2(\alpha + \beta) \oplus \mathbb{F}_2(\alpha + \gamma), \quad H^2(K, \mathbb{F}_2) = (\mathbb{F}_2\mu \oplus \mathbb{F}_2\lambda)/(\mu + \lambda).$$

Now we calculate the cup product:

$$(\alpha + \beta) \cup (\alpha + \beta)(A) = (\alpha + \beta)(a) \cdot (\alpha + \beta)(b) = 1$$

$$(\alpha + \beta) \cup (\alpha + \gamma)(A) = (\alpha + \beta)(a) \cdot (\alpha + \gamma)(b) = 0$$

$$(\alpha + \gamma) \cup (\alpha + \gamma)(A) = (\alpha + \gamma)(a) \cdot (\alpha + \gamma)(b) = 0$$

$$(\alpha + \beta) \cup (\alpha + \beta)(B) = (\alpha + \beta)(b) \cdot (\alpha + \beta)(c) = 0$$

$$(\alpha + \beta) \cup (\alpha + \gamma)(B) = (\alpha + \beta)(b) \cdot (\alpha + \gamma)(c) = 1$$

$$(\alpha + \gamma) \cup (\alpha + \gamma)(B) = (\alpha + \gamma)(b) \cdot (\alpha + \gamma)(c) = 0$$

Then  $(\alpha + \beta) \cup (\alpha + \beta) = (\alpha + \beta) \cup (\alpha + \gamma) = \mu \in H^2(K, \mathbb{Z}/(2))$ . Now if we set  $\alpha + \beta = x, \beta + \gamma = y$ , then the cohomology ring

$$H^*(K, \mathbb{Z}/(2)) = \mathbb{Z}/(2)[x, y]/(xy, x^2 - y^2, x^3, y^3).$$

$\square$

**Prop. (4.14.2.36)** [ $H^*(L, \mathbb{Z}/(m))$ ]. Let  $L$  be the Lens space, then the cohomology ring is calculated at [Hat02]P304.  $\quad \text{┘}$

*Proof:*  $\square$



### 3 Manifolds

#### Orientations

**Prop. (4.14.3.1).** Let  $d \in \mathbb{Z}_+$ ,  $R \in \mathcal{CAlg}$ ,  $M \in \text{Mani}^d$ ,  $X \subset M$ , then for any  $x \in M$ , choose a nbhd  $U$  of  $x \in M$  s.t.  $U \cong \mathbb{R}^n$ , then by excision and exactness,

$$H_d(M, M \setminus \{x\}) \cong H_d(U, U \setminus \{x\}) \cong \tilde{H}_{d-1}(U \setminus \{x\}) \cong \tilde{H}_{d-1}(\mathbb{S}^{d-1}) \cong R.$$

┘

**Prop. (4.14.3.2) [Vanishing].** For  $M \in \text{Mani}_{\text{cpct}, \text{cntd}, \partial}^d$ ,  $\Lambda \in \mathcal{Ab}$ ,  $H_i(M; \Lambda) = 0$  for  $i > d$ , and  $\tilde{H}_d(M; \Lambda) = 0$  unless  $M$  is compact without boundary.

┘

*Proof:*

□

**Def. (4.14.3.3) [ $R$ -Fundamental Class].** Let  $R \in \mathcal{CRing}$ ,  $M \in \text{Mani}_{\text{cntd}}^d$ ,  $X \subset M$ , an  $R$ -fundamental class of  $M$  at  $X$  is an element  $z \in H_d(M, M \setminus X)$  s.t. for any  $x \in X$ , the map

$$H_d(M, M \setminus X) \rightarrow H_d(M, M \setminus \{x\})$$

maps  $z$  to a generator of  $H_d(M, M \setminus \{x\}) \cong R$  (4.14.3.1).

┘

**Def. (4.14.3.4) [ $R$ -Orientation].** Let  $R \in \mathcal{CAlg}$ ,  $M \in \text{Mani}_{\text{cntd}}^d$ ,  $X \subset M$ , an  $R$ -orientation of  $M$  is an open cover  $\{U_i\}$  of  $M$  and  $R$ -fundamental classes  $z_i$  of  $M$  at  $U_i$  s.t.  $z_i, z_j$  restricts to the same element in  $H_d(M, M \setminus (U_i \cup U_j))$ .

For  $M \in \text{Mani}_{\text{cntd}, \partial}^d$ , an  $R$ -orientation is an  $R$ -orientation of  $M^\circ$ .

┘

**Prop. (4.14.3.5) [ $R$ -Fundamental Classes and  $R$ -Orientations].** For  $R \in \mathcal{CAlg}$ ,  $M \in \text{Mani}^d$  and  $K \subset M$  compact, then

- $H_i(M, M \setminus K, R) = 0$  for  $i > d$ .
- Any  $R$ -orientation of  $M$  defines an  $R$ -fundamental class of  $M$  at  $K$  that is compatible for  $x \in K$ .

In particular, if  $M$  is compact, then an  $R$ -orientation is equivalent to an  $R$ -fundamental class.

┘

*Proof:* Firstly, if  $K \subset U$  and  $U \cong \mathbb{R}^n$  is a coordinate chart on which the  $R$ -orientation is defined, then by excision and exactness,

$$H_i(M, M \setminus K, R) \cong H_i(U, U \setminus K, R) \cong H_{i-1}(U \setminus K, R)$$

which vanishes for  $i > n$  by vanishing theorem (4.14.3.2). And the  $R$ -orientation restricts to  $K$ .

In general,  $K$  can be written as a sum of compact subsets each contained in a coordinate char on which the  $R$ -orientation is defined. Then using induction, it suffices to show that if the results hold for  $K, L, K \cap L$ , then it holds for  $K \cup L$ : item1 follows from the MV-sequence

$$H_{i+1}(M, M \setminus (K \cup L)) \rightarrow H_i(M, M \setminus (K \cup L)) \rightarrow H_i(M, M \setminus K) \oplus H_i(M, M \setminus L) \rightarrow H_i(M, M \setminus (K \cap L)).$$

For item2, the same MV-sequence with  $R$ -coefficients and  $i = d$  together with the definition of  $R$ -orientation classes (4.14.3.4) shows there exists a fundamental class at  $K \cap L$  that restricts to the fundamental classes at  $K$  and  $L$ .

□

**Prop. (4.14.3.6) [Orientation Dichotomy].** For  $d \in \mathbb{Z}_+$ ,  $M \in \text{Mani}_{\text{cntd}, \text{cpct}}^d$ , there are only two cases:

- $M$  is non-orientable, and  $H_d(M) = 0$ .
- $M$  is orientable, and the map  $H_d(M) \rightarrow H_d(M, M \setminus \{x\}) \cong \mathbb{Z}$  (4.14.3.1) is an isomorphism for any  $x \in M$ .

┘

*Proof:* For any  $R \in \mathcal{CAlg}$ , as  $M \setminus \{x\}$  is connected and non-compact,  $H_d(M \setminus \{x\}; R) = 0$  by (4.14.3.2), thus

$$H_d(M, R) \rightarrow H_d(M, M \setminus \{x\}; R) \cong R$$

is injective. Thus  $H_d(M) \cong 0$  or  $\mathbb{Z}$ . In particular, by universal coefficients theorem,

$$H_d(M) \otimes \mathbb{F}_p \rightarrow H_d(M, M \setminus \{x\}) \otimes \mathbb{F}_p$$

is injective for any  $p \in \text{Prime}$ . Then it is clear to see that  $H_d(M) \rightarrow H_d(M, M \setminus \{x\})$  is an isomorphism for any  $x \in M$ . □

**Prop. (4.14.3.7) [Fundamental Classes on Manifolds with Boundaries].** For  $M \in \text{Mani}_{\partial}^d$ ,  $\partial M \neq 0$ , an  $R$ -orientation on  $M$  determines an  $R$ -orientation of  $\partial M$ . Moreover, the fundamental class  $[\partial M]_R$  defined by (4.14.3.5) comes from the partial of a unique element  $[M]_R \in H_d(M, \partial M; R)$ , called the  **$R$ -fundamental class determined the orientation**. ┘

*Proof:* By collar nbhd theorem,  $M \cong M^o$ , and by vanishing theorem,  $H_d(M) \cong H_d(M^o) = 0$ , thus  $H_d(M, \partial M; R) \rightarrow H_{d-1}(\partial M; R)$  is injective. Let  $N$  be an open collar of  $\partial M$ , then  $H_d(M^o, M^o \cap N; R) \cong H_n(M, \partial M; R)$ . Also,  $M \setminus N$  is compact, so the orientation on  $M^o$  defines a fundamental class in  $H_d(M, M^o \cap N; R) \cong H_n(M, \partial M; R)$ . Then this class determined the orientation on every point of  $M^o$ , and determines the orientation on  $\partial M$ , Cf. [May, P170] ?. □

**Prop. (4.14.3.8) [Orientation Coverings].** ┘

*Proof:* □

**Cor. (4.14.3.9) [Two Orientations].** If  $M \in \text{Mani}_{\text{cntd}}^d$ , then if  $M$  is orientable, it has exactly two orientations. And if  $M$  is simply connected or  $\pi_1(M)$  has no subgroup of index 2, then  $M$  is orientable. ┘

**Def. (4.14.3.10) [Connected Sums].** For  $M, N \in \text{Mani}_{\text{cntd}, \text{cpct}}^d$ , define the **connected sum**  $M \# N$  as follows: Take a subset  $D \subset M$ ,  $D' \subset N$  that are contained in coordinate charts (orientable coordinates charts if  $M, N$  are oriented).  $\varphi : U \cong \mathbb{R}^d$ ,  $\varphi' : U' \cong \mathbb{R}^d$  s.t.  $\varphi(D) \cong \mathbb{D}^d \cong \varphi'(D')$ . Then define

$$M \# N = (M \setminus D) \coprod_{\partial D \cong \partial D'} (N \setminus D')$$

where  $\partial D \cong \partial D'$  is an orientation-reversing homeomorphism when  $M, N$  are both orientable. This is independent of  $D, D'$  and the homeomorphism  $\partial D \cong \partial D'$  chosen. ┘

*Proof:* Firstly we show that  $M \setminus D$  is independent of  $D$  chosen: As  $M$  is locally path connected and connected, it is path connected, Then we can consider a chain of charts isomorphic to  $\mathbb{R}^d$  that connects any two such subsets  $D, D'$ . Then reducing to affine charts, it suffices to show for the case  $M \cong \mathbb{R}^d$ . In this case, we can use isotopy of  $M$  to make  $D \subset D'$ , and then use the coordinate charts and

annulus theorem(4.4.11.4) to show that  $D' \setminus D$  is isomorphic to  $\mathbb{S}^{d-1} \times \mathbb{I}$ . Then the homeomorphism  $M \setminus D \cong M \setminus D'$  can be easily given.

Now we show the connected sum is independent of the homeomorphism chosen:  $\partial D$  and  $\partial D'$  are given orientations by the coordinate charts, and in the non-orientable case, use the argument above on the orientation covering of  $M$ (4.14.3.8), we know there exists a self-homeomorphism of  $M$  that is isotopic to  $\text{id}_M$  that reverses the orientation on  $\partial D$ . Thus in both cases, we can assume  $\partial D \cong \partial D'$  is orientation-preserving. Now(4.13.2.4) shows any such homeomorphism is isotopic, and using the coordinate charts, we can extend the isotopy to a nbhd of nbhd of  $\partial D$ . Thus the connected sum is well-define.  $\square$

**Remark(4.14.3.11).** WARNING: The connected sum may be different in the orientable case if we change one of the orientations:  $\mathbb{CP}^2 \# \mathbb{CP}^2$  is not homeomorphic to  $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$ .  $\lrcorner$

*Proof:*

$\square$

**Prop.(4.14.3.12).** For  $M \in \text{Mani}_{\text{cpct}}$ ,  $\text{id}_M$  is not homotopic to a constant map, because they induce different maps on  $[M]_2$ .  $\lrcorner$

### Poincaré Duality

**Lemma(4.14.3.13) [Poincaré Duality].** For  $R \in \mathcal{CAlg}$ , if  $M \in \text{Mani}_{\text{cntd, cpct}}^d$  is  $R$ -oriented, then for any  $p \in \mathbb{Z}$ , there is an isomorphism

$$D = (-) \cap [M]_R : H^p(M; R) \cong H_{n-p}(M; R).$$

In particular,  $H^n(X, \mathbb{F}_2) \cong \mathbb{Z}/(2)$ , and the non-trivial element is called the **fundamental homology class mod 2**, denoted by  $[M]_2$ .  $\lrcorner$

*Proof:* which follows immediately from(7.9.2.29) and(7.3.5.9). (Should also attain the compact cohomology case if we know the relation of compact sheaf cohomology better).  $\square$

**Prop.(4.14.3.14) [Relative Poincaré Duality].** Let  $R \in \mathcal{CAlg}$ ,  $M \in \text{Mani}_{\text{cpct}, \partial}^d$  be  $R$ -orientable with fundamental class  $[M]_R \in H_d(M, \partial M; R)$ , then for any  $\Lambda \in \text{Mod}_R$ , capping with  $[M]_R$  defines isomorphisms

$$H^p(M, \partial M; \Lambda) \cong H_{d-p}(M), \quad H^p(M; \Lambda) \cong H_{d-p}(M, \partial M; \Lambda).$$

$\lrcorner$

*Proof:* Cf.[May, P170] ?.

$\square$

**Cor.(4.14.3.15).** For  $M \in \text{Mani}_{\text{cntd, cpct, orntd}}^d$ , for any  $p \in \mathbb{Z}$ , there is a perfect pairing

$$H^p(M)_{\text{lf}} \times H^{n-p}(M)_{\text{lf}} \rightarrow \mathbb{Z} : (\alpha, \beta) \mapsto \langle \alpha \cup \beta, [M] \rangle.$$

$\lrcorner$

*Proof:* By universal coefficient theorem ?,  $H^p(M)_{\text{lf}} \cong \text{Hom}(H_p(M), \mathbb{Z})$ . So if  $\alpha \in H^p(M)$  is non-zero in  $H^p(M)_{\text{lf}}$ , there exists  $a \in H_p(M)$  s.t.  $\langle a, \alpha \rangle = 1$ . Then by Poincaré duality(4.14.3.13), there exists  $\beta \in H^{d-p}(M)$  s.t.  $\beta \cap [M] = a$ . Thus  $\langle \alpha \cup \beta, [M] \rangle = \langle \alpha, \beta \cap [M] \rangle = 1$ .  $\square$

**Def. (4.14.3.16) [Intersection Pairing].** Let  $k \in \mathbb{N}$ ,  $M \in \text{Mani}_{\text{cpct,orntd,cntd}}^{2k}$ . The Poincaré duality (4.14.3.15) shows the cup product pairing

$$H^k(M; \mathbb{Q}) \times H^k(N; \mathbb{Q}) \rightarrow \mathbb{Q} : (\alpha, \beta) \mapsto \langle \alpha \cup \beta, [M] \rangle$$

is non-degenerate, called the **intersection pairing** of  $M$ . Notice if  $k$  is odd, it is skew-symmetric, and if  $k$  is even, it is symmetric.  $\lrcorner$

**Def. (4.14.3.17) [Index].** For  $M \in \text{Mani}_{\text{cpct,orntd,cntd}}^d$ , define the **index of**  $M$  as the index of the symmetric intersection pairing of  $M$  if  $d = 4k$ , and 0 otherwise. The index of  $M$  is denoted by  $I(M)$ .  $\lrcorner$

**Prop. (4.14.3.18).** For  $M, N \in \text{Mani}_{\text{cpct,orntd,cntd}}$ ,  $I(M \times N) = I(M)I(N)$ .  $\lrcorner$

*Proof:* Cf. [May, P167].  $\square$

**Prop. (4.14.3.19).** For  $k \in \mathbb{N}$ ,  $k \in \text{Field}$ ,  $M \in \text{Mani}_{\text{cpct}, \partial}^{2k+1}$ , then

$$\dim_k H_k(\partial M; k) = 2 \dim \ker(H_k(\partial M; k) \xrightarrow{i_*} H_k(M; k)) = 2 \dim \text{Im}(H^k(\partial M; k) \xrightarrow{i^*} H^k(M; k)).$$

$\lrcorner$

*Proof:* There is a commutative diagram

$$\begin{array}{ccccc} H^k(\partial M; k) & \xrightarrow{i^*} & H^k(M; k) & \xrightarrow{\partial} & H^{k+1}(M, \partial M; k) \\ \downarrow D & & \downarrow D & & \downarrow D \\ H_{k+1}(M, \partial M; k) & \xrightarrow{\partial} & H_k(M; k) & \xrightarrow{i_*} & H_k(\partial M; k) \end{array}$$

where the vertical arrows are isomorphisms and  $i_*$  is the vector dual of  $i^*$  by universal coefficient theorem. Thus the theorem follows.  $\square$

**Prop. (4.14.3.20) [Gysin Sequence].** Cf. [姜伯驹同调论].  $\lrcorner$

### Euler Characteristic

**Def. (4.14.3.21) [Euler Characteristic].** For any  $X \in \mathcal{T}\text{op}$  with finite  $\mathbb{Z}$ -homologies, define the **Euler character** of  $X$  to be

$$\chi(X) = \sum_i (-1)^i H_i(X; \mathbb{Z}).$$

Then by universal coefficients theorem (4.14.2.4),

$$\chi(X) = \sum_i (-1)^i H_i(X; k)$$

for any  $k \in \text{Field}$ .

The condition holds for  $X \in \mathcal{CW}^{\text{fin}}$ , in particular any  $X \in \text{Mani}_{\text{cpct}}$  (4.13.3.10).  $\lrcorner$

**Prop. (4.14.3.22).** For any  $i \in \mathbb{N}$ ,  $X \in \text{Mani}_{\text{cpct,cntd}}^{2i+1}$ ,  $\chi(X) = 0$ .  $\lrcorner$

*Proof:* This follows from Poincaré duality for  $\mathbb{F}_2$ -coefficients.  $\square$

**Prop. (4.14.3.23)[Euler Characteristic and Boundary].** For  $i \in \mathbb{N}$ ,  $M \in \text{Mani}_{\text{cpct}, \partial}^{2i+1}$ , then  $\chi(\partial M) = 2\chi(M)$ . ┘

*Proof:* Consider the “double”  $\widetilde{M}$  of  $M$  along  $\partial M$ , then  $\widetilde{M} \in \text{Mani}_{\text{cpct}}^{2i+1}$ . It is clear by excision and collar neighborhood theorem that  $\chi(\widetilde{M}) + \chi(\partial M) = 2\chi(M)$ . Then we use (4.14.3.22). □

**Cor. (4.14.3.24).** If  $N \in \text{Mani}_{\text{cpct}}$  is a boundary of some other  $M \in \text{Mani}_{\text{cpct}, \partial}$ , then  $\chi(N) = 0$ . ┘

*Proof:* If  $\dim N$  is odd, this is (4.14.3.22). If  $\dim N$  is even, this is (4.14.3.23). □

**Prop. (4.14.3.25).** For any  $i \in \mathbb{N}$ ,  $M \in \text{Mani}_{\text{cpct}, \text{orntd}}^{4i+2}$ ,  $\chi(M)$  is even. ┘

*Proof:* The parity of  $\chi(M)$  is the same as  $H^{2i+1}(M, \mathbb{Q})$ , which is odd by (4.14.3.16) and (3.5.11.26). □

**Prop. (4.14.3.26)[Morse Inequality].** For any  $X \in \mathcal{CW}$ ,

$$\chi(X) \leq \sum (-1)^i c_i(X),$$

where  $c_i(X)$  is the number of  $i$ -dimensional cells. (Use the dimension counting of the long exact sequence). ┘

**Prop. (4.14.3.27).** For  $M \in \text{Mani}_{\text{cpct}, \text{orntd}, \text{cntd}}^d$ ,  $I(M) \equiv \chi(M) \pmod{2}$ . ┘

*Proof:* If  $d$  is odd,  $I(M) = 0 = \chi(M)$  is even by (4.14.3.22). If  $d = 4k + 2$ , then  $I(M) = 0$  and  $\chi(M)$  is even by (4.14.3.25). if  $d = 4k$ , then  $I(M) \equiv \chi(M) \equiv \dim H^{2k}(M, \mathbb{Q})$ . □

### deRham Cohomology

**Prop. (4.14.3.28)[de Rham Comparison].** For  $X \in \text{Mani}_{\text{sm}}$ ,  $G \in \mathcal{Ab}$ , by (7.3.5.11).

$$H_{\text{dR}}^*(X, G) \cong H^*(X, \underline{G}).$$

┘

**Prop. (4.14.3.29) [Homotopy Axiom for deRham Cohomology].** Two homotopic map between smooth manifolds induce the same map on deRham Cohomology. ┘

*Proof:* We only have to prove the case of  $M \times \mathbb{R} \rightarrow M$ , where any constant section map induces an isomorphism  $H_{\text{dR}}^*(M \times I) \cong H_{\text{dR}}^*(M)$ . Because any homotopy is a morphism  $M \times I \rightarrow N$  where  $f$  and  $g$  are the sections 0 and 1.

For the zero section, we define  $K : a + bdt \mapsto \int_0^t b$ . This is the desired homotopy, Cf.[Differential Forms in Algebraic Topology Bott Tu]. □

## 4 Applications

**Prop. (4.14.4.1)[No Contraction to the Boundary].** For  $M \in \text{Mani}_{\text{cpct}, \partial}^d$ , there are no retraction from  $M$  onto  $\partial M$ . ┘

*Proof:* We may assume  $\partial M$  is connected and non-empty, otherwise clearly there are no retraction. If it has a retraction,  $H_{d-1}(\partial M, \mathbb{F}_2) \rightarrow H_{d-1}(M, \mathbb{F}_2)$  has a left inverse. Thus It suffices to show that  $H_{d-1}(\partial M, \mathbb{Z}/(2)) \rightarrow H_{d-1}(M, \mathbb{Z}/(2))$  is 0. So it suffices to show that  $H_d(M, \partial M, \mathbb{Z}/(2)) \rightarrow H_{d-1}(\partial M, \mathbb{Z}/(2))$  is surjective. And this follows from (4.14.3.7) and (4.14.3.6) as the image of  $[M]_2$  is  $[\partial M]_2$ , which generates  $H_{d-1}(\partial M, \mathbb{F}_2)$ . □

**Prop. (4.14.4.2) [Brouwer Fixed-Point Theorem].** For any  $f : \mathbb{D}^n \rightarrow \mathbb{D}^n \in \mathcal{Top}$ ,  $\text{Fix}(f) \neq \emptyset$ .  $\lrcorner$

*Proof:* If  $\text{Fix}(f) = \emptyset$ , consider the intersection of the ray  $f(x)x$  with  $S^{n-1}$ , then it depends continuously on  $x$ , and this defines a function from  $\mathbb{D}^n$  to  $S_n$  that is identity on  $S^n$ , but then this is a retraction from  $\mathbb{D}^n \rightarrow S^n$ , which is impossible by (4.14.4.1).  $\square$

**Prop. (4.14.4.3).** If  $M \in \text{Mani}_{\text{cnd}, \partial}^d$  is contractible and  $\partial M \neq \emptyset$ , then  $\partial M$  is a homotopy sphere.  $\lrcorner$

*Proof:* As  $M^o \sim M$  is contractible, it is orientable by (4.14.3.9). Then by relative Poincaré duality (4.14.3.14) and the long exact sequence,

$$H_p(\partial M) \cong H_{p+1}(M, \partial M) \cong H^{d-p-1}(M) = 0.$$

$\square$

**Lemma (4.14.4.4).** If  $m > n \in \mathbb{Z}_+$ , any map  $f : \mathbb{RP}^m \rightarrow \mathbb{RP}^n$  induces  $f_* = 0 : \pi_1(\mathbb{RP}^m) \rightarrow \pi_1(\mathbb{RP}^n)$ .  $\lrcorner$

*Proof:* As  $\pi_1(\mathbb{RP}^m) = \begin{cases} \mathbb{Z}/(2) & , m > 1 \\ \mathbb{Z} & , m = 1 \end{cases}$ . The assertion is clearly true for  $n = 1$ . And if  $n \in \mathbb{Z}_+$ ,  $f_* \neq 0$ , then by the naturality of Hurewicz homomorphism (4.14.1.17),  $f_* \neq 0 : H_1(\mathbb{RP}^m) \rightarrow H_1(\mathbb{RP}^n)$ . Then by universal coefficient theorem, so is  $f^* \neq 0 : H^1(\mathbb{RP}^n, \mathbb{F}_2) \rightarrow H^1(\mathbb{RP}^m, \mathbb{F}_2)$ . But  $f^*$  is a ring homomorphism and  $H^*(\mathbb{RP}^m)$  is generated by  $a = f^*(a') \in H^1(\mathbb{RP}^m)$  (4.14.2.33), so  $a^{n+1} = f^*((a')^{n+1}) = 0$ , contradiction.  $\square$

**Lemma (4.14.4.5).** For  $m > n \in \mathbb{Z}_+$ , there are no antipodal maps  $f : \mathbb{S}^m \rightarrow \mathbb{S}^n$ , i.e.  $f(x) = -f(-x)$ .  $\lrcorner$

*Proof:* Any such map induces a map  $\mathbb{RP}^m \rightarrow \mathbb{RP}^n$ , which induces 0 on  $\pi_1$ , so by covering space theory, there exists a lifting  $\mathbb{RP}^m \rightarrow \mathbb{S}^n$ . The composition  $\mathbb{S}^m \rightarrow \mathbb{RP}^m \rightarrow \mathbb{S}^n$  is the original map, by the theory of covering spaces, as  $\mathbb{S}^n$  is connected and they agree at least on one point. But then  $f$  takes antipodal maps to the same point, contradiction.  $\square$

**Thm. (4.14.4.6) [Borsuk-Ulam].** For ant map  $f : \mathbb{S}^n \rightarrow \mathbb{R}^n$ , there exists  $x \in \mathbb{S}^n$  s.t.  $f(x) = f(-x)$ .  $\lrcorner$

*Proof:* If  $f(x) \neq f(-x)$  for any  $x \in \mathbb{S}^n$ , then we can define a map  $\mathbb{S}^n \rightarrow \mathbb{S}^{n-1}$  that maps  $x$  to the intersection of the ray  $0, f(x) - f(-x)$  with  $\mathbb{S}^n$ . Then this map is antipodal, which contradicts (4.14.4.5).  $\square$

**Thm. (4.14.4.7) [Brouwer Degree Theorem].** Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n \in \mathcal{Top}$  and  $0 \in \Omega \subset \mathbb{R}^n$  be an open compact subset with continuous boundary s.t.  $\langle x, f(x) \rangle > 0$  for any  $x \in \partial\Omega$ , then there exists  $x \in \Omega$  s.t.  $f(x) = 0$ .  $\lrcorner$

*Proof:*

$\square$

## 5 Obstruction Theory & General Cohomology Theories

### Towers

**Prop. (4.14.5.1) [Towers].** There are Whitehead Towers and Postnikov Towers for a CW complex  $X$ .

$$\cdots \rightarrow Z_2 \rightarrow Z_1 \rightarrow Z_0 \rightarrow X \rightarrow \cdots \rightarrow X_2 \rightarrow X_1 \rightarrow X_0$$

$Z_n$  annihilate  $\pi_{\leq n}(X)$ ,  $X_n$  remains only  $\pi_{\leq n}(X)$ . The towers can be chosen to be fibrations, with fibers  $K(\pi_n X, n)$  by (4.13.7.30).  $\lrcorner$

**Prop. (4.14.5.2).** There is a Postnikov towers of :

$$\dots \rightarrow \mathbf{B}\text{FiveBrane} \rightarrow \mathbf{B}\text{String}(n) \rightarrow \mathbf{B}\text{Spin}(n) \rightarrow \mathbf{B}\text{SO}(n) \rightarrow \mathbf{B}\text{O}(n),$$

with corresponding obstructions  $w_1(X), w_2(X)$  and  $p_1(X)/2$ .  $\lrcorner$

**Prop. (4.14.5.3)[Obstructions].** If a connected abelian CW complex  $X$  ( $\pi_1(X)$  abelian and action on higher homotopy trivial) and  $(W, A)$  satisfies  $H^{n+1}(W, A; \pi_n X) = 0$  for all  $n$ , then  $A \rightarrow X$  can extend to a map  $M \rightarrow X$ .  $\lrcorner$

*Proof:* Cf.[Hatcher P417].  $\square$

**Cor. (4.14.5.4).** A map between Abelian CW complexes that induce isomorphisms on homology is a homotopy equivalence.  $\lrcorner$

*Proof:* Notice that  $\pi_1(X)$  acts trivially on  $\pi_1(Y, X)$  and use Hurewicz.  $\square$

## 6 Brown Representability

**Prop. (4.14.6.1).** For any  $Z \in \mathcal{T}\text{op}^{\text{pt}}$ , the functor

$$\langle -, Z \rangle : \mathcal{CW}^{\text{pt}} \rightarrow \text{Set}^{\text{pt}}$$

satisfies:

**Exactness:** If  $A \subset X$  is a subcomplex, then there is an exact sequence

$$\langle X/A, Z \rangle \rightarrow \langle X, Z \rangle \rightarrow \langle A, Z \rangle.$$

**Additivity:** If  $X = \bigvee_i X_i \in \mathcal{CW}^{\text{pt}}$ , then

$$\langle X, Z \rangle \cong \prod_i \langle X_i, Z \rangle.$$

$\lrcorner$

*Proof:* 1 follows from the fact that  $A \rightarrow X$  is a cofibration(4.13.7.5). 2 is trivial.  $\square$

## Spectrum

**Def. (4.14.6.2) [Prespectrums].** A **prespectrum** is a sequence of pointed spaces  $(T_n)_{n \in \mathbb{Z}}$  together with maps  $\sigma_n \in \langle \Sigma T_n \rightarrow T_{n+1} \rangle \cong \langle T_n, \Omega T_{n+1} \rangle$ .

An  $\Omega$ -**Spectrum** is an prespectrum  $(T_n)_{n \in \mathbb{Z}}$  s.t.  $\sigma_n$  are all weak-homotopy equivalences.  $\lrcorner$

**Def. (4.14.6.3) [Suspension Prespectrums].** For  $X \in \mathcal{T}\text{op}^{\text{pt}}$ , the sequence of pointed spaces  $(\Sigma^n(X))_{n \in \mathbb{Z}}$  (where we define  $\Sigma^{-1} = \Omega$ ) is a prespectrum, called the **suspension prespectrum** of  $X$ . If  $X = S^0$ , this is called the **sphere spectrum**.  $\lrcorner$

**Prop. (4.14.6.4)[Prespectrum and Homotopy Theories].** Let  $(T_n)_{n \in \mathbb{N}}$  be a prespectrum consisting of pointed CW complexes s.t.  $T_n$  is  $(n-1)$ -connected for any  $n \in \mathbb{N}$ , then

$$\tilde{E}_p(X) = \varinjlim_n \pi_{p+n}(X \wedge T_n)$$

is a reduced homology theory on  $\mathcal{CW}^{\text{pt}}$ .  $\lrcorner$

*Proof:* Cf.[May, P176]. □

**Prop. (4.14.6.5) [ $\Omega$ -Prespectrum and ES-Cohomology theories].** If  $K_n$  is an  $\Omega$ -spectrum, i.e.  $K_n \cong \Omega K_{n+1}$  weak equivalence, then the functors  $X \mapsto \langle X, K_n \rangle$  define a reduced ES-cohomology theory on  $\mathcal{CW}^{\text{pt}}$  ┘

*Proof:* By(4.14.6.1), these functors satisfy the additivity and exactness. The natural suspension isomorphism is given by

$$\Sigma : \langle X, T_n \rangle \rightarrow \langle \Sigma(X), \Sigma T_n \rangle \cong \langle \Sigma(X), \Sigma T_n \rangle,$$

which is an isomorphism by(4.13.6.5). □

**Thm. (4.14.6.6) [Brown Representability].** Any reduced ES-cohomology theory on  $\mathcal{CW}^{\text{pt}}$  is represented by a  $\Omega$ -prespectrum. ┘

*Proof:* ? □

### Eilenberg-MacLane Spaces

**Def. (4.14.6.7) [Eilenberg-MacLane Spaces].** For  $\Lambda \in \mathcal{Ab}$ , an **Eilenberg-MacLane space** is defined to be a pointed CW complex  $K(\Lambda, n) \in \mathcal{CW}^{\text{pt}}$  s.t.  $\pi_k(K(\Lambda, n)) = \begin{cases} \Lambda & , k = n \\ 0 & , \text{otherwise} \end{cases}$ . ┘

**Prop. (4.14.6.8) [Eilenberg-MacLane Spaces and Homologies].** For  $X \in \mathcal{CW}^{\text{pt}}, \Lambda \in \mathcal{Ab}$ , there are isomorphisms

$$\tilde{H}_p(X; \Lambda) \cong \varinjlim_{p+n} \pi_{p+n}(X \wedge K(\Lambda, n)).$$

┘

*Proof:* Cf.[May, P176]. □

**Prop. (4.14.6.9) [Eilenberg-MacLane Spaces and Cohomologies].** For  $\Lambda \in \mathcal{Ab}, X \in \mathcal{CW}^{\text{pt}}, n \in \mathbb{N}$ ,  $K(\Lambda, n)$  are unique up to weak homotopy, and there are natural isomorphisms

$$\tilde{H}^n(X; \Lambda) \cong \langle X, K(\Lambda, n) \rangle.$$

┘

*Proof:* Given a  $K(\Lambda, n)$ , if we define  $K(\Lambda, m) = \Sigma^{m-n}(K(\Lambda, n))$ , then these are all CW complexes by(4.13.3.13), and it defines an ordinary cohomology theory: The dimension axiom follows from the definition of  $K(\Lambda, n)$ , and it is a reduced ES-cohomology theory by(4.13.6.5) and(4.14.6.10). Thus the asserted isomorphism follows from(4.14.2.3).

This shows in particular that  $K(\Lambda, n)$  are unique up to homotopy by Yoneda lemma. □

**Cor. (4.14.6.10) [Uniqueness of Eilenberg-MacLane Spaces].**  $K(\Lambda, n)$  is unique up to homotopy, and  $\Omega(K(\Lambda, n)) \sim K(\Lambda, n-1) \in \mathcal{CW}^{\text{pt}}$ . ┘

**Prop. (4.14.6.11) [Eilenberg-MacLane Spaces Exist].** Take  $K(\Lambda, n) = \Gamma(B(\Lambda, n))$ (4.15.3.10), where  $\Lambda$  is regarded as a discrete subgroup.

Note  $K(G, 1)$  is constructed the same as by(4.6.8.1). ┘



*Proof:* □

**Prop. (4.14.6.12) [Cohomology Operators].** Cohomological operators  $\tilde{H}^p(-; \Lambda) \rightarrow \tilde{H}^{p+n}(-; \Lambda')$  are in bijection with elements in  $\tilde{H}^{p+n}(K(\Lambda, p); \Lambda')$ , by (4.14.6.9) and Yoneda lemma. ┘

**Prop. (4.14.6.13) [Examples].**

- $K(\mathbb{Z}, 1) = S^1 = U(1)$ .
  - $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$ .
  - $K(\mathbb{Z}/(2), 1) = \mathbb{RP}^\infty$ .
- ┘

*Proof:* 1 is clear.

2: This is because by (4.15.1.17),  $S^\infty \rightarrow \mathbb{CP}^\infty$  is a locally trivial bundle with fiber  $S^1$ , and  $S^\infty$  is contractible by (4.13.3.18) thus by (4.13.7.14),  $\pi_2(\mathbb{CP}^\infty) = \mathbb{Z}$  and  $\pi_n(\mathbb{CP}^\infty) = 0$  for  $n \neq 2$ .

3: This is because by (4.15.1.17),  $S^\infty \rightarrow \mathbb{CP}^\infty$  is a locally trivial bundle with fiber  $\{\pm 1\}$ , and  $S^\infty$  is contractible by (4.13.3.18) thus by (4.13.7.14),  $\pi_2(\mathbb{CP}^\infty) = \mathbb{Z}/2\mathbb{Z}$  and  $\pi_n(\mathbb{CP}^\infty) = 0$  for  $n \neq 2$ . □

### Steenrod Powers

**Prop. (4.14.6.14) [Steenrod Powers].** There are stable cohomology operators

$$\text{Sq} : H^*(X, \mathbb{F}_2) \rightarrow H^*(X, \mathbb{F}_2) : x \in H^k(X, \mathbb{F}_2) \mapsto \sum_{i \in \mathbb{N}} \text{Sq}^i(x), \quad \text{Sq}^i(x) \in H^{k+i}(X, \mathbb{F}_2)$$

called **Steenrod Powers** that satisfies

- $\text{Sq}(\alpha \cup \beta) = \text{Sq}(\alpha) \cup \text{Sq}(\beta)$ .
- $\text{Sq}^i(\alpha) = \alpha^2$  if  $i = |\alpha|$ , and 0 if  $i > |\alpha|$ .

For  $p \in \mathbf{P}$ , the total Steenrod powers  $P$  is a similar map from  $H^n(X, \mathbb{F}_p) \rightarrow H^{n+*}(X, \mathbb{F}_p)$  that  $P^i(\alpha) = \alpha^p$  if  $2i = |\alpha|$  and 0 if  $2i > |\alpha|$ . ┘

*Proof:* Cf. [Hatcher P497]. ? □

**Prop. (4.14.6.15) [Adam Relations].** For  $0 < i < 2j \in \mathbb{Z}$ ,

$$\text{Sq}^i \text{Sq}^j = \sum_{0 \leq k \leq \lfloor \frac{i}{2} \rfloor} \binom{j-k-1}{i-2k} \text{Sq}^{i+j-k} \text{Sq}^k \in \text{End}(H^*(-, \mathbb{F}_2))$$

There are Adam relation calculators in terms of Serre-Cartan basis at <https://math.berkeley.edu/~kruckman/adem/>. ┘

*Proof:* □

**Cor. (4.14.6.16).** The subalgebra of  $\text{End}(H^*(-, \mathbb{F}_2))$  generated by  $\{\text{Sq}^{\mathbb{Z}}\}$  is generated respectively by elements  $\text{Sq}^{2^k}$ .

and for  $p \in \mathbf{P}$  The subalgebra of  $\text{End}(H^*(-, \mathbb{F}_p))$  generated by  $\{P^{\mathbb{Z}}\}$  is generated by elements  $P^{p^k}$  and  $\beta$ . ┘

*Proof:* ? □

**Prop. (4.14.6.17) [H\*(K(F<sub>2</sub>, p), F<sub>2</sub>)].**  $H^*(K(\mathbb{F}_2, p), \mathbb{F}_2)$  can be calculated, Cf. [May, P185]. ┘

*Proof:* □

## 7 Stable Homotopy Theory

## 4.15 Fiber Bundles & Topological K-Theory

References are [Ati64, M-S74, AGP02].

**Remark (4.15.0.1).** Any base space  $B$  in this section is assumed to be paracompact. Instead, we may require in the definition of locally trivial bundles s.t. there exists a trivialization that is dominated by a partition of unity on  $B$ .  $\lrcorner$

### 1 Fiber Bundles

**Def. (4.15.1.1) [Fiber Bundles].**  $\pi : E \rightarrow X \in \mathcal{CG}$  is called a **fiber bundle** with fiber  $F$  if every fiber  $\pi^{-1}(x)$  is homeomorphic to  $F$ . If any  $x \in X$  has a nbhd  $U$  together with a homeomorphism  $\pi^{-1}(U) \cong U \times F$  over  $U$ , then it is called a **locally trivial bundle**. And a **numerical locally trivial bundle** is a locally trivial bundle  $\pi : E \rightarrow X$  s.t. there is a numerical covering  $\{U_i \rightarrow X\}$  s.t. there are homeomorphisms  $\pi^{-1}(U_i) \cong U_i \times F$  over  $U_i$  for each  $i$ .  $\lrcorner$

**Prop. (4.15.1.2) [Dold].** If  $\pi : E \rightarrow X \in \mathcal{CG}$  and there is a numerical covering  $\{U_i \rightarrow X\}$  s.t.  $\pi^{-1}(U_i) \rightarrow U_i$  are Hurewicz fibrations, then  $\pi$  is a Hurewicz fibration.  $\lrcorner$

*Proof:* Cf. [Tammo tom Dieck, Algebraic Topology] Chap13?  $\square$

**Cor. (4.15.1.3) [Fiber Bundles are Hurewicz Fibrations].** Every numerical locally trivial bundle is a Hurewicz fibration.  $\lrcorner$

**Prop. (4.15.1.4) [Pullback Bundle].** Let  $\pi : E \rightarrow X$  be a fiber bundle with fiber  $F$  and  $f : Y \rightarrow X$  a map, then the pullback space  $f^*E \rightarrow Y$  (4.4.1.8) is also a fiber bundle over  $Y$  with fiber  $F$ , called the **pullback bundle**.  $\lrcorner$

**Prop. (4.15.1.5).** If  $E \rightarrow B, E' \rightarrow B'$  are fiber bundles both with compact Hausdorff fibers/or both with discrete fibers/, and  $f : E/B \rightarrow E'/B'$  is a bundle map that induces isomorphisms on the fibers, then  $E \cong f^*E'$  over  $B$ .  $\lrcorner$

*Proof:*  $\square$

**Lemma (4.15.1.6).** Suppose  $\pi : E \rightarrow B \times I$  is a fiber bundle whose restriction to  $B \times [0, a]$  and  $B \times [a, 1]$  are all trivial for some  $a \in I$ , then  $E \rightarrow B \times I$  is a trivial bundle.  $\lrcorner$

*Proof:* Choose trivializations  $\varphi_1 : B \times [0, a] \times B \times F \rightarrow \pi^{-1}(B \times [0, a])$ , and  $\varphi_2 : B \times [a, 1] \times B \times F \rightarrow \pi^{-1}(B \times [a, 1])$ , then these induces a map

$$B \times \{a\} \times F \xrightarrow{\varphi_1|} \pi^{-1}(B \times \{a\}) \xrightarrow{\varphi_2^{-1}|} B \times \{a\} \times F,$$

of the form  $(b, a, v) \mapsto (b, a, g(b)v)$ , where  $g : B \rightarrow \text{Homeo}(F)$  is continuous. Then we get a trivialization

$$B \times I \times F \rightarrow E : \varphi(b, t, v) = \begin{cases} \varphi_1(b, t, v) & t \leq a \\ \varphi_2(b, t, g(b)v) & t \geq a \end{cases}.$$

$\square$

**Lemma (4.15.1.7).** Let  $E \rightarrow B \times I$  be a fiber bundle, then there exists a covering  $\{U_i\}$  of  $B$  that  $E$  is trivial on each  $U_i \times I$ .  $\lrcorner$

*Proof:* For each  $b \in B$ , we can find a nbhd  $U_b$  and a division  $0 = s_0 < s_1 < \dots < s_n = 1$  that  $E$  is trivial on each of  $U_b \times [s_i, s_{i+1}]$ . Then by (4.15.1.6),  $E$  is trivial on  $U_b \times I$ . Then these  $\{U_b\}$  is a covering of  $X$  that  $E$  is trivial on each  $U_b \times I$ .  $\square$

**Lemma (4.15.1.8).** Let  $\pi : E \rightarrow B \times I$  be a fiber bundle, where  $B$  is a paracompact space, and  $r : B \times I \rightarrow B \times I$  defined by  $r(b, x) = (b, 1)$ , then there exists a bundle morphism  $f$  over  $r$  that induces an isomorphism  $r^*E \cong E$  over  $B \times I$ .  $\lrcorner$

*Proof:* By (4.15.1.7), we can choose a covering  $\{U_\alpha\}$  of  $B$  that  $E$  is trivial over each  $U_\alpha \times I$ , and let  $\psi_\alpha$  be a partition of unity  $1 = \sum \psi_i$  that  $\text{Supp}(\psi_i) \subset U_i$  and  $\{\text{Supp}(\psi_i)\}$  is locally finite. We also define  $\mu_\alpha(x) = \frac{\psi_\alpha(x)}{\max\{\psi_\beta(x)\}}$ , then  $\mu_\alpha$  are all continuous and subordinate to  $\{U_\alpha\}$ , and for each  $x \in B$ ,  $\max\{\mu_\alpha(x)\} = 1$ .

Let  $\varphi_\alpha : U_i \times I \times F \rightarrow \pi^{-1}(U_i \times I)$  be the local trivializations. We define a bundle map  $f_\alpha : E/B \times I \rightarrow E/B \times I$  by identity outside  $\pi^{-1}(U_i \times I)$  and  $f_\alpha(\varphi_\alpha(b, t, v)) = \varphi_\alpha(b, \max(\mu_\alpha(x), t), v)$ , then  $f_\alpha$  is continuous and induces an isomorphism  $f_\alpha^*E \cong E$  over  $B \times I$ . Now choose a well-ordering on  $\alpha$ , by local finiteness, for each  $v \in E$ , there is a nbhd  $W_v \times I$  of  $\pi(v) \in B \times I$  that  $W_v \cap U_\alpha \neq \emptyset$  for only  $\alpha$  in a finite set  $\{\alpha_1, \dots, \alpha_m\}$  with  $\alpha_1 < \alpha_2 < \dots < \alpha_m$ . Then we define a bundle map  $f : E \rightarrow E$  that  $f|_{\pi^{-1}(W_v \times I)} = f_{\alpha_m} \circ \dots \circ f_{\alpha_1}$ . Then this is well-defined, and it is a bundle map over  $r$  that induces an isomorphism  $f^*E \cong E$ .  $\square$

**Prop. (4.15.1.9) [Homotopy Invariance of Fiber Bundles].** Let  $E' \rightarrow B'$  be fiber bundles. If  $f, g : B \rightarrow B'$  are two homotopic maps with  $B$  paracompact, then there is a bundle isomorphism  $f^*E' \cong g^*E'$ .  $\lrcorner$

*Proof:* Let  $F : B \times I \rightarrow B'$  be a homotopy from  $f$  to  $g$ , and let  $i_v : B \rightarrow B \times I : i_v(b, x) = (b, v)$  for  $v = 0, 1$ . Let  $r : B \times I \rightarrow B \times I$  be the retraction defined by  $r(b, x) = (b, 1)$ , then by (4.15.1.8), there is an isomorphism of fiber bundles

$$f^*E' \cong (F \circ i_0)^*E' \cong i_0^*F^*E' \cong i_0^*r^*F^*E' \cong i_1^*F^*E' \cong g^*E'.$$

$\square$

**Prop. (4.15.1.10) [Cone Bundles].** Let  $E/X$  be a fiber bundle with fiber  $F$ , then we can construct a new fiber bundle  $\text{Cone}(E)/X$  with bundle  $C(F)$ .  $\lrcorner$

**Cor. (4.15.1.11).** This is because we can use the transformation characterization to extend maps  $U_i \cap U_j \rightarrow \text{Aut}(F)$  to maps  $U_i \cap U_j \rightarrow \text{Aut}(C(F))$ .  $\lrcorner$

**Prop. (4.15.1.12) [Hopf Fibration].** There is a locally trivial fiber bundle  $S^3 \rightarrow S^2$  with fiber  $S^1$ , called the **Hopf fibration**.  $\lrcorner$

*Proof:* Cf. [AGP02]P129.  $\square$

**Prop. (4.15.1.13) [Ehresmann].** Let  $f : E \rightarrow B \in \text{Mani}_{\text{sm}}$  be a proper submersion, then it is a locally trivial bundle.  $\lrcorner$

*Proof:* Cf. [Björn Dundas, A Short Course in Differential Topology]. ?  $\square$

**Cor. (4.15.1.14).** There is a locally trivial bundle  $S^{2n-1} \rightarrow \mathbb{CP}^n$  with fiber  $S^1$ .  $\lrcorner$

*Proof:*

$\square$

**Cor. (4.15.1.15).** There is a locally trivial bundle  $\mathbb{RP}^{2n-1} \rightarrow \mathbb{CP}^n$  with fiber  $S^1$ . ┘

*Proof:*

□

**Prop. (4.15.1.16) [Classifying Space Bundles].** There is a locally trivial bundle  $U(n, \mathbb{K}) \rightarrow V_n(\mathbb{K}^\infty) \rightarrow \text{Gr}(n, \mathbb{K}^\infty)$ . for  $\mathbb{K} = \mathbb{R}, \mathbb{C}$  or  $\mathbb{H}$ . ┘

*Proof:*

□

**Cor. (4.15.1.17).** There is a locally trivial bundle  $S^\infty \rightarrow \mathbb{CP}^\infty$  with fiber  $S^1$ . ┘

**Prop. (4.15.1.18).** There is a locally trivial bundle  $S^n \rightarrow \mathbb{RP}^n$  with fiber  $\{\pm 1\}$ . ┘

*Proof:*

□

**Prop. (4.15.1.19).** There is a locally trivial fiber bundle  $S^\infty \rightarrow \mathbb{RP}^\infty$  with fiber  $\{\pm 1\}$ . ┘

*Proof:* Cf. [AGP02]P335.

□

### Covering Space

**Def. (4.15.1.20) [Covering Space].** A **covering space** is a fiber bundle  $E \rightarrow X$  with discrete fibers. ┘

**Prop. (4.15.1.21).** if  $X$  and  $Y$  are Hausdorff spaces,  $f : X \rightarrow Y$  is a local homeomorphism,  $X$  is compact, and  $Y$  is connected, then  $f$  a covering map. ┘

*Proof:* First,  $f$  is surjective (using the connectedness), and that for each  $y \in Y$ ,  $f^{-1}(y)$  is finite. Because  $X$  is compact, there exists a finite open cover of  $X$  by  $\{U_i\}$  such that  $f(U_i)$  is open and  $f|_{U_i} : U_i \rightarrow f(U_i)$  is a homeomorphism. For  $y \in Y$ , let  $\{x_1, \dots, x_n\} = f^{-1}(y)$  (the  $x_i$  all being different points). Choose pairwise disjoint neighborhoods  $U_1, \dots, U_n$  of  $x_1, \dots, x_n$ , respectively (using the Hausdorff property).

By shrinking the  $U_i$  further, we may assume that each one is mapped homeomorphically onto some neighborhood  $V_i$  of  $y$ .

Now let  $C = X \setminus (U_1 \cup \dots \cup U_n)$  and set

$$V = (V_1 \cap \dots \cap V_n) \setminus f(C)$$

$V$  should be an evenly covered nbhd of  $y$ . □

**Prop. (4.15.1.22).** If  $\pi : \tilde{B} \rightarrow B$  is a local onto homeomorphism with the property of lifting arcs. Let  $\tilde{B}$  be arcwise connected and  $B$  simply connected, then  $\pi$  is a homomorphism. ┘

*Proof:* only need to prove injective. If  $p_1$  and  $p_2$  map to the same point, then they can be connected, and the image is a loop thus contractable, contradiction. □

**Cor. (4.15.1.23).** If  $\tilde{B}$  is locally arcwise connected and  $B$  is locally simply connected, then  $\pi$  is a covering map. ┘

*Proof:* Choose the connected components of a simply connected nbhd of a point  $p$  and use (4.15.1.22). □

**Prop. (4.15.1.24)[Homotopy Lifting Property].** Given a covering space  $\pi : \tilde{X} \rightarrow X$ , and a homotopy  $f_t : Y \rightarrow X$ , and a map  $f_0 : Y \rightarrow \tilde{X}$  lifting  $f_0$ , then there is a unique homotopy  $\tilde{f}_t : Y \rightarrow \tilde{X}$  of  $\tilde{f}_0$  lifting  $f_t$ .  $\lrcorner$

*Proof:* Let  $U_\alpha$  be a covering of  $X$  that the We first construct a lift  $\tilde{F} : N \times I \rightarrow \tilde{X}$  for  $N$  a nbhd near some point  $y_0 \in Y$ . Because  $f$  is continuous, there is a nbhd  $N$  of  $y_0$  and a partition  $0 = t_0 < t_1 < \dots < t_m = 1$  of  $I$  that each  $N \times [t_i, t_{i+1}]$  is mapped into some  $U_\alpha$ . Then we can construct a lifting  $\tilde{F} : N \times I \rightarrow \tilde{X}$  by induction using the local homeomorphism property of covering space.

Next we show the uniqueness in the special case that  $Y$  is a point. This can also be done using a partition of  $I$  and induction.

Finally we can construct lifting near every point  $y \in Y$ , and also they coincide on the overlap because of the uniqueness we just proved. So these liftings glue together to give a lifting  $\tilde{f}_t : Y \rightarrow \tilde{X}$ .  $\square$

**Cor. (4.15.1.25).** The map  $\pi_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$  induced by the covering map is injective. And the image of this map consists of homotopy types of loops that based at  $x_0$  whose lift starting at  $\tilde{x}_0$  are also loops.  $\lrcorner$

*Proof:* This is because a homotopy of a the image of a loop to trivial loop in  $\tilde{X}$  can be lifted to a homotopy of the loop itself to trivial loop. And this homotopy also fixes the endpoint, because the lifting of a trivial loop must be a trivial loop.

For the second assertion, one direction is easy, for the other, if a loop is homotopic to the image of a loop of  $\tilde{X}$ , then it is itself the image of a loop of  $\tilde{X}$ .  $\square$

**Prop. (4.15.1.26)[Degree of a Covering].** Let  $\pi : \tilde{X} \rightarrow X$  be a covering map, then the cardinality of  $\pi^{-1}(x)$  is a locally constant function of  $x$ . Thus if  $X$  is constant, this cardinality is fixed for any  $x \in X$ , and it is called the degree of the covering.

The number of sheets of a covering with  $\tilde{X}$  path-connected equals the index of  $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$  in  $\pi_1(X, x_0)$ .  $\lrcorner$

*Proof:* For a loop  $g$  in  $X$  based at  $x_0$ , let  $\tilde{g}$  be its lift to  $\tilde{X}$  starting at  $\tilde{x}_0$ . Now if  $h \in \pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ , then the loop  $h \cdot g$  has lift that has the same ending as  $\tilde{g}$ . So we get a map from the quotient set  $\pi_1(X, x_0)/\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$  to  $p^{-1}(x_0)$  mapping  $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))[g]$  to  $\tilde{g}(1)$ . This map is injective, and it is surjective because  $\tilde{X}$  is path-connected. Then we are done.  $\square$

**Prop. (4.15.1.27) [Unique Lifting Property].** Let  $\pi : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$  be a covering space and  $f : (Y, y_0) \rightarrow (X, x_0)$  be a map with  $Y$  path-connected and locally path-connected, then a lift  $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$  exists iff  $f_*(\pi_1(Y, y_0)) \subset \pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ . And when  $Y$  is connected, this lifting is unique.

In particular, a covering space has unique path lifting property.  $\lrcorner$

*Proof:* One direction is clear, for the other, to construct a lifting, choose a path  $\gamma$  from  $y_0$  to  $y$ , the path  $f\gamma$  has a unique lifting  $\tilde{f}\gamma$  starting from  $\tilde{x}_0$ . Define  $\tilde{f}(y) = \tilde{f}\gamma$ . This is a well-defined map: if  $\gamma'$  is another path from  $y_0$  to  $y$ , then  $f\gamma^{-1}f\gamma'$  is a loop that is homotopic to the image of a loop at  $\tilde{x}_0$ . Now we can lift this homotopy, and then  $f\gamma^{-1}f\gamma'$  is also the image of a loop at  $\tilde{x}_0$ , which must be  $\tilde{f}\gamma^{-1}\tilde{f}\gamma'$  by uniqueness. So  $\tilde{f}$  is well-defined.

It can be verified that  $\tilde{f}$  is continuous.

The uniqueness is clear, because if there are two lifts, the points that they are equal and the points that they are not are both open in  $Y$ .  $\square$

**Prop. (4.15.1.28) [Galois Theory of Covers].** Let  $(X, x_0)$  be a path-connected and locally path-connected and semilocally simply-connected space (4.13.5.27), then

- there is a connected and simply-connected covering space  $\tilde{X}$  of  $X$ , called a **universal cover** of  $X$ .
- $\pi_1(X, x_0)$  acts continuously and properly on  $\tilde{X}/X$ .
- For any subgroup  $H$  of  $\pi_1(X, x_0)$ , there is a connected covering space  $\pi : X_H \rightarrow X$  that  $\pi_*(\pi_1(X_H, \tilde{x}_0)) = H$  for a suitably chosen base point  $\tilde{x}_0$ . And this covering space is unique up to isomorphism over  $(X, x_0)$ . Thus by (4.15.1.27), there is an inclusion-preserving bijection between isomorphism classes of covering spaces over  $X$  and the the set of conjugacy classes of subgroups of  $\pi_1(X, x_0)$ .

┘

*Proof:* Cf. [Hat02]P64, P67. ?

□

**Def. (4.15.1.29) [Normal Covering Spaces].** A **Galois covering space** is a covering space  $\pi : \tilde{X} \rightarrow X$  that for any  $x \in X$  and two elements  $\tilde{x}, \tilde{x}' \in \pi^{-1}(x)$ , there is a covering isomorphism of  $\tilde{X}/X$  taking  $\tilde{x}$  to  $\tilde{x}'$ .

┘

**Prop. (4.15.1.30).** Let  $\pi : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$  be a path-connected covering space of a path-connected, locally path-connected space  $X$ , and let  $H$  be the subgroup  $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subset \pi_1(X, x_0)$ , then

- The covering space is Galois iff  $H$  is normal in  $\pi_1(X, x_0)$ .
- The group  $G(\tilde{X})$  of covering transformations of  $\tilde{X}$  is isomorphic to  $N(H)/H$ .

┘

*Proof:* 1: Let  $\tilde{x}_1, \tilde{x}_2 \in \pi^{-1}(x_0)$  and  $\gamma$  a path from  $\tilde{x}_1$  to  $\tilde{x}_2$  corresponding to an element of  $\pi_1(X, x_0)$ , then  $H$  is normal is equivalent to  $\pi_*(\pi_1(\tilde{X}, \tilde{x}_1)) = \pi_*(\pi_1(\tilde{X}, \tilde{x}_2))$ . Then the lifting criterion shows there is a covering transformation taking  $\tilde{x}_1$  to  $\tilde{x}_2$ . The converse is also true.

2: From the above argument, we can define a map  $N(H) \rightarrow G(\tilde{x})$  by mapping a  $\gamma \in N(H)$  to a covering transformation mapping  $\tilde{x}_0$  to  $\tilde{x}_1$ . And the kernel of this map is exactly those  $\gamma$  lifting to a loop at  $\tilde{x}_0$ , which are exactly the elements of  $H$ .

□

**Prop. (4.15.1.31) [Covering Space Action].** If  $G \in \text{Grp}$  and  $G \times Y \rightarrow Y$  is a covering space action (4.12.1.14), then the quotient map  $Y \mapsto Y/G$  is a Galois covering space. And if  $Y$  is path-connected,  $G$  is the group of covering transformations.

┘

*Proof:* The condition on the action shows it is locally a homeomorphism, thus it is a covering space. And it is a normal covering space because  $g_1 g_2^{-1}$  takes any  $g_1(x)$  to  $g_2(x)$ . The group of covering transformations is just  $G$ , because the covering transformation on a path-connected space is determined by its action on a single point.

□

## 2 Vector Bundles

### Basics

**Def. (4.15.2.1) [Vector Bundle].** Let  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ , a  **$\mathbb{K}$ -vector bundle** of dimension  $n$  over a topological space  $X$  is a fiber bundle over  $X$  with fiber  $\mathbb{K}^n$  that each trivialization  $\varphi_\alpha$  restricts to  $\mathbb{K}$ -linear isomorphisms on the fibers. The category of vector bundles over  $X$  is denoted by  $\text{Vect}_{\mathbb{K}}(X)$ . A **vector bundle homomorphism**  $E \rightarrow F$  is a map of spaces over  $X$  that the maps on the fibered are all  $\mathbb{K}$ -linear.

Then a  $\mathbb{K}$ -vector bundle of dimension  $n$  over  $X$  is just an associated  $GL(n, \mathbb{K})$ -bundle with fiber  $\mathbb{K}^n$  over  $X$ . In particular, there is a bijection  $\text{Vect}_{\mathbb{K}}^n(X) \cong \mathcal{P}_{GL(n, \mathbb{K})}(X)$  by (4.15.3.7).  $\lrcorner$

**Def. (4.15.2.2) [Trivial Vector Bundles].** For any  $n \in \mathbb{N}$ , the **trivial vector bundle** of rank  $n$  is denoted by  $e^n$ .  $\lrcorner$

**Prop. (4.15.2.3) [Constructions of Vector Bundles].** Let  $T : (\text{Vect}^f / \mathbb{K})^{\otimes n} \rightarrow \text{Vect}^f / \mathbb{K}$  be a functor that is either covariant or contravariant for each of its factor that  $T : \prod_i \text{Hom}(V_i, W_i) \rightarrow \text{Hom}(T(V_i), T(W_i))$  is continuous, then we have a functor  $T : \text{Vect}(X)^n \rightarrow \text{Vect}(X)$  that is either covariant or contravariant for each of its factor.  $\lrcorner$

*Proof:* Cf. [Ati64]P6.  $\square$

**Cor. (4.15.2.4).** In this way, given a vector bundles  $E, F$  on  $X$ , we can construct

$$E \oplus F, \quad E \otimes F, \quad \text{Hom}(E, F) \cong E^* \otimes F, \quad E^*, \quad T^n E, \quad \wedge^i E.$$

$\lrcorner$

**Prop. (4.15.2.5) [Existence of Hermitian Metric].** There exists a Hermitian (Riemannian) metric on any bundle  $E$  over a paracompact space  $X$ . In this way,  $E^* \cong E(\overline{E})$ .  $\lrcorner$

*Proof:* Choose a metric on each trivialization open subset and use partition of unity to glue.  $\square$

**Cor. (4.15.2.6).** A vector bundle over a paracompact space can have its transform maps  $\in O(n)$  (or  $U(n)$ ).  $\lrcorner$

*Proof:* We can choose the metric on it compatible with the given metric. In this way, the transform map is  $\in O(n)$  (or  $U(n)$ ).  $\square$

**Cor. (4.15.2.7) [Semisimplicity of  $\text{Vect}(B)$ ].** Any exact sequence of vector bundles over a paracompact space  $B$  splits.  $\lrcorner$

*Proof:* Because we can take the orthogonal complement.  $\square$

**Cor. (4.15.2.8).** If  $f : X \rightarrow Y$  is a homotopy equivalence, then  $f^* : \text{Vect}(Y) \rightarrow \text{Vect}(X)$  is an isomorphism.  $\lrcorner$

In particular, if  $X$  is contractible, then every bundle over  $X$  is trivial.  $\lrcorner$

**Def. (4.15.2.9) [Orientations of Vector Bundles].** Let  $\pi : E \rightarrow X$  be a vector bundle of rank  $n$  over a space  $X$  and  $\mathbb{R} \cong \mathbb{Z}$  or  $\mathbb{Z}/2\mathbb{Z}$ , then a **orientation of  $E$**  with coefficient  $R$  is a functions that assigns a generator  $u_x$  of  $H^n(E_x, 0, R) \cong R$  for any  $x \in X$  s.t. for any  $x \in X$ , there is a nbhd  $U$  of  $x \in X$  and an element  $u \in H^n(\pi^{-1}U, \pi^{-1}(0), R)$  s.t.  $u$  pulls back to  $u_x$  via  $(E_x, 0) \subset (\pi^{-1}U, \pi^{-1}(0))$  for any  $x \in U$ . In particular, any vector bundle is  $\mathbb{F}_2$ -orientable.  $\lrcorner$

**Prop. (4.15.2.10) [Orientation of Complex Vector Bundles].** For  $E/X \in \text{Vect}_{\mathbb{C}}(X)$ , the underlying real bundle has a preferred orientation, which is compatible with direct sums.  $\lrcorner$

In particular, the tangent bundle of a complex manifold gives rise to a unique orientation of the underlying real manifold.  $\lrcorner$

*Proof:* Let  $a_1, \dots, a_n$  be a complex basis for  $E$ , then take the real basis to be  $a_1, ia_1, \dots, a_n, ia_n$ . This orientation is stable under  $GL(n, \mathbb{C})$  transformation, as  $GL(n, \mathbb{C})$  is connected.  $\square$



**Prop. (4.15.2.11).** If  $E$  is a complex vector bundle over  $X$ , then  $E \otimes_{\mathbb{R}} \mathbb{C} \cong E \oplus \overline{E}$  as complex vector bundles.  $\lrcorner$

**Lemma (4.15.2.12).** Suppose  $\pi : E \rightarrow B \times I$  is a fiber bundle whose restriction to  $B \times [0, a]$  and  $B \times [a, 1]$  are all trivial for some  $a \in I$ , then  $E \rightarrow B \times I$  is a trivial bundle.  $\lrcorner$

*Proof:* The proof is similar to that of (4.15.1.6)  $\square$

**Lemma (4.15.2.13).** Let  $E \rightarrow B \times I$  be a fiber bundle, then there exists a covering  $\{U_i\}$  of  $B$  that  $E$  is trivial on each  $U_i \times I$ .  $\lrcorner$

*Proof:* The proof is similar to that of (4.15.1.7)  $\square$

**Lemma (4.15.2.14).** Let  $\pi : E \rightarrow B \times I$  be a fiber bundle, where  $B$  is a paracompact space, and  $r : B \times \mathbb{I} \rightarrow B \times I$  defined by  $r(b, x) = (b, 1)$ , then there exists a bundle morphism  $f$  over  $r$  that induces an isomorphism  $r^*E \cong E$  over  $B \times \mathbb{I}$ .  $\lrcorner$

*Proof:* The proof is similar to that of (4.15.1.8)  $\square$

**Prop. (4.15.2.15) [Homotopy Invariance of Vector Bundles].** Let  $E' \rightarrow B'$  be a vector bundle. If  $f, g : B \rightarrow B'$  are two homotopic maps with  $B$  paracompact, then there is a bundle isomorphism  $f^*E' \cong g^*E'$ .  $\lrcorner$

*Proof:* The proof is similar to that of (4.15.1.9)  $\square$

### Bundles of Finite Type

**Def. (4.15.2.16) [Bundles of Finite Type].** Let  $X$  be a paracompact Hausdorff space, then a **vector bundle of finite type** over  $X$  is a vector bundle over  $X$  that has a covering by f.m. trivialization maps. The category of  $k$ -dimensional vector bundles over  $X$  of f.t. is denoted by  $\text{Vect}_k^{\text{ft}}(X)$ . Trivially, any vector bundle over a compact space is of f.t..  $\lrcorner$

**Prop. (4.15.2.17) [Vector Bundles on the Quotient].** Let  $Y$  be a closed subspace of  $X$ ,  $E$  a vector bundle over  $X$ , then a trivialization  $\alpha : E|_Y \cong Y \times V$  defines a bundle  $E/\alpha$  over  $X/Y$ . The isomorphism class of  $E/\alpha$  only depends on the homotopy type of  $\alpha$ .  $\lrcorner$

*Proof:* To show it is a vector bundle, notice the trivialization  $\alpha$  extends to a  $\square$

**Prop. (4.15.2.18) [Splitting Principle].** For a vector bundle  $E \rightarrow X$ , there is a space  $Y \rightarrow X$  that  $p^*$  is injective on  $H^*(-, \mathbb{Z})$  and  $p^*E$  splits as a sum of line bundles. This proposition is useful when proving theorems about characteristic classes.  $\lrcorner$

*Proof:* It suffices to find a  $Y$  that  $p^*E$  has a subbundle, then choose its orthogonal part, and use induction. For this, choose  $Y = P(E)$ , then  $Y$  has a tautological bundle, which is a subbundle of  $p^*E$ , and  $Y$  is fibered over  $X$  with fiber  $\mathbb{P}^n$ , and we want to use Leray-Hirsch, so check the fact  $H^*(\mathbb{P}^n)$  is free and generated by the first Chern class, by (4.15.4.17) and (7.7.2.1). And Chern class is functorial, so the powers of Chern class of  $f^*E$  will generate the cohomology ring of any stalks.  $\square$

**Prop. (4.15.2.19).** For any bundle  $E$  over a compact Hausdorff space  $X$ , there is a surjective bundle map  $X \otimes \mathbb{R}^n \rightarrow E$  for some  $n \geq 0$ .  $\lrcorner$

*Proof:* Choose a finite cover of trivialization of  $E$ , then we can glue these maps together via a partition function.  $\square$

**Cor. (4.15.2.20) [Negation of Bundles].** For any vector bundle  $E$  on a compact space  $X$ , there is a vector bundle  $F$  that  $E \oplus F$  is a trivial bundle.  $\lrcorner$

*Proof:* Choose a bundle map  $\mathbb{R}^n \times X \rightarrow E$  that is surjective, then the kernel of this map is a bundle  $F$ , such that  $E \oplus F \cong \mathbb{R}^n$  (By taking a Hermitian metric (4.15.2.5) and taking the orthogonal bundle).  $\square$

**Cor. (4.15.2.21) [Global Transversal Sections].** For a vector bundle over a compact manifold, there exists a global section transversal to the zero section, in particular, if  $\dim E > M$ , then it has no zero.  $\lrcorner$

*Proof:* Choose a bundle map  $\mathbb{R}^n \times X \rightarrow E$  that is surjective, and then use parametric transversality theorem (16.1.1.53) to prove there is a section that is transversal.  $\square$

**Prop. (4.15.2.22) [Constructing Vector Bundles].**  $\lrcorner$

**Cor. (4.15.2.23).** There is a natural isomorphism  $\text{Vect}_n(S(X)) \cong [X, GL(n, \mathbb{C})]$ .  $\lrcorner$

*Proof:* Write  $\Sigma(X) = C^+(X) \amalg C^-(X)$ , and  $C^\pm(X)$  are both contractible, thus  $E$  are trivial restricted to them (4.15.2.8). Let  $\alpha^\pm$  be the trivialization isomorphism, then  $\alpha^+ \circ \alpha^-$  is a bundle map of  $X \times \mathbb{R}^n$ , which is equivalent to a map  $\alpha : X \rightarrow GL(n, \mathbb{C})$ . The homotopy type of  $\alpha$  is determined because  $C^\pm(X)$  are both contractible, and vice versa.  $\square$

**Prop. (4.15.2.24) [Vector Bundles as Modules].**  $\Gamma$  induces an equivalence between the category of vector bundles over  $X$  and the category of finitely projective modules over  $C(X)$ .  $\lrcorner$

*Proof:* Clearly a bundle induces a module over  $C(X)$ . And it is a fully faithful functor. Now the image is the subcategory of finite projective modules, because every bundle is a direct summand of a trivial bundle, and a trivial bundle corresponds to a finite free  $C(X)$ -modules.  $\square$

### Thom isomorphism

**Prop. (4.15.2.25) [Thom Class].** Let  $R = \mathbb{Z}$  or  $\mathbb{F}_2$  and an  $R$ -orientable vector bundle  $E$  over base  $B$  of rank  $n$ .

Then there exists uniquely **Thom class**  $u_E \in H^n(E, E \setminus B, R)$  that induce the preferred generator  $H^n(E_x, E_x \setminus \{0\}, R)$  (4.15.2.9) on every fiber. Then the relative Leray-Hirsch will give an isomorphism

$$\varphi_R : H^i(B, R) \cong H^{i+n}(E, E \setminus B, R) : x \mapsto \pi^*(x) \cup u_E.$$

For  $\mathbb{F}_2$  coefficient there exists a Thom class, and for orientable bundle there exists a  $\mathbb{Z}$ -Thom class. Notice that fiber bundle over a simply connected base is orientable.  $\lrcorner$

*Proof:*  $\square$

**Cor. (4.15.2.26) [Naturality].**

- Let  $B' \rightarrow B$  be a map and  $E/B$  be an  $R$ -orientable vector bundle, which induces a map  $f^* : H^n(E, E_0, R) \rightarrow H^n(f^*(E), E'_0, R)$ . Then  $f^*u_E \cong u_{f^*(E)}$ .
- The Thom class  $u_E$  maps to  $u_{E,2}$  under the change of coefficients  $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ .  $\lrcorner$

**Prop. (4.15.2.27).** Similarly, for a orientable fiber bundle  $S^{n-1} \rightarrow E \rightarrow B$ , make it a  $D^n \rightarrow E' \rightarrow B$  bundle, then  $E'$  is homotopy equivalent to  $B$  so there is a Gysin sequence

$$\rightarrow H^{i-n}(B) \xrightarrow{\smile e} H^i(B) \rightarrow H^i(E) \rightarrow H^{i-n+1}(B) \rightarrow$$

Where the Euler class  $e$  is chosen to commute with the Thom isomorphism.  $\lrcorner$

### Examples of Vector Bundles

**Def. (4.15.2.28) [ $n$ -Universal  $k$ -Vector Bundle].** Let  $K = \mathbb{R}$  or  $\mathbb{C}$ , endow  $\mathbb{K}^n$  with the canonical Hermitian metric, define  $E_k(\mathbb{K}^n)$  be the subspace of  $\text{Gr}(k, \mathbb{K}^n) \times \mathbb{K}^n$  consisting of pairs  $(W, v)$  that  $v \in W$ . Then this is a vector bundle over the Grassmannian  $\text{Gr}(k, \mathbb{K}^n)$  (16.5.2.12), called the  **$n$ -universal  $k$ -vector bundles**, or the **tautological bundles**.  $\lrcorner$

*Proof:* We construct localization maps: endow  $\mathbb{K}^n$  with the natural metric. For  $W_0 \in G_k(\mathbb{K}^n)$ , then the subspace  $U$  of  $G_k(\mathbb{K}^n)$  consisting of  $W$  that  $W \cap W_0^\perp = \emptyset$  is a nbhd of  $W_0$ , and it is naturally homeomorphic to  $\text{Hom}(W_0, W_0^\perp)$ . There is an isomorphism

$$\pi^{-1}(U) \cong U \times W_0 : (f, (f + \text{id})w_0) \mapsto (f, w_0).$$

□

**Prop. (4.15.2.29) [Universal Vector Bundle].** Because by (4.15.3.23)  $E_k(\mathbb{K}^\infty)/\text{Gr}(k, \mathbb{K}^\infty)$  is the universal bundle, there is a map  $i_k : \text{Gr}(k, \mathbb{K}^\infty) \rightarrow \text{Gr}(k+1, \mathbb{K}^\infty)$  s.t.  $i_k^*(E_{k+1}(\mathbb{K}^\infty))$  is the bundle  $E_k(\mathbb{K}^\infty) \oplus \mathbb{K}$ . Then it can be shown

$$BU(\mathbb{K}) = \text{colim}_k (\text{Gr}(k, \mathbb{K}^\infty) \xrightarrow{i_k} \text{Gr}(k+1, \mathbb{K}^\infty))$$

is a CW complex.

Moreover, there are maps  $w_{k,l} : \text{Gr}(k, \mathbb{K}^\infty) \times \text{Gr}(l, \mathbb{K}^\infty) \rightarrow \text{Gr}(k+l, \mathbb{K}^\infty)$  that corresponds to the bundle  $E_k(\mathbb{K}^\infty) \times E_l(\mathbb{K}^\infty)$ , and these maps induces an H-space structure on  $BU$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (4.15.2.30) [Tautological Line Bundles].** For  $\text{Gr}(1, \mathbb{R}^{n+1}) \cong \mathbb{RP}^n$ , the line bundle  $E_1(\mathbb{R}^{n+1})$  is denoted by  $\gamma_n^1$ .

Then the tangent bundle  $\tau_n$  of  $\mathbb{RP}^n$  is isomorphic to  $\text{Hom}(\gamma_n^1, \gamma_n^\perp)$ , where  $\gamma_n^\perp$  is the orthogonal complement of  $\gamma_n^1$  in  $e_{\mathbb{RP}^n}^{n+1}$  (with the canonical norm).  $\lrcorner$

*Proof:* This is because a tangent vector at  $x \in \mathbb{RP}^n$  is equivalent to a homomorphism  $[x]$  to  $[x]^\perp$ .

? How to prove this rigorously.  $\square$

**Cor. (4.15.2.31).**  $\tau_n \oplus e^1 \cong (\gamma_n^1)^{n+1}$ .  $\lrcorner$

*Proof:* By (4.15.2.30),  $\tau_n \oplus e^1 \cong \text{Hom}(\gamma_n^1, \gamma_n^1) \oplus \text{Hom}(\gamma_n^1, \gamma_n^\perp) = \text{Hom}(\gamma_n^1, \mathbb{R}^n) = (\gamma_n^1)^{*n}$ . Then we are done because  $\gamma_n^1$  is self-dual because it has a Euclidean metric (4.15.2.5).  $\square$

**Prop. (4.15.2.32) [Pullback the Universal Bundle].** For  $V \in \text{Vect}/\mathbb{K}$ , for any continuous map  $\varphi : X \rightarrow \text{Gr}(k, V)$ , we get a subspace  $E_\varphi = \{(x, v) \in X \times V \mid v \in \varphi(x)\} \subset X \times V$ . This is a vector bundle over  $X$ , and it is a subbundle of the trivial bundle  $X \times V \rightarrow X$ . In fact,  $E_\varphi \cong f^*E_k(V)$  (4.15.2.28).  $\lrcorner$

**Cor. (4.15.2.33).** Let  $f : X' \rightarrow X$  and  $\varphi : X \rightarrow \text{Pr}(V)$ , then  $f^*(E_\varphi) = E_{\varphi \circ f}$ .  $\lrcorner$

**Prop. (4.15.2.34) [Infinite Universal  $k$ -Vector Bundle].** The  $n$ -universal  $k$ -vector bundles  $E_k(\mathbb{K}^n)$  (4.15.2.28) for various  $n$  gets together to a bundle  $E_k(\mathbb{K}^\infty)$  on  $\text{Gr}(k, \mathbb{K}^\infty)$  that the pullback of  $E_k(\mathbb{K}^\infty)$  via  $i : \text{Gr}(k, \mathbb{K}^n) \rightarrow \text{Gr}(k, \mathbb{K}^\infty)$  is  $E_k(\mathbb{K}^n)$ .  $\lrcorner$

**Def. (4.15.2.35) [Hopf Bundle].** Define a map  $\varphi : \mathbb{C}P^n \rightarrow G_1(\mathbb{C}^{n+1})$  that  $\varphi([z]) = [z]$ , then this defines a vector bundle on  $\mathbb{C}P^n$  by (4.15.2.32), called the **dual of Hopf bundle**. The **Hopf bundle** is defined to be the dual of the dual of Hopf bundle.  $\lrcorner$

### 3 Principal Bundles

Reference are [Principal Bundles and Classifying Spaces] and [L-W89].

**Def. (4.15.3.1) [Principal Bundles].** For  $G \in \mathcal{TopGrp}$ , a **principal  $G$ -bundle** is a fiber bundle  $pr : P \rightarrow X$  with a right  $G$ -action over  $X$  that is simply transitive on each fiber.

For  $B \in \mathcal{Top}$ , denote  $\mathcal{P}_G(B)$  the isomorphism classes of principal bundles on  $B$ .  $\lrcorner$

**Example (4.15.3.2).** Let  $X \in \mathcal{TopGrp}$  be paracompact Hausdorff.

- If  $E \in \mathcal{Vect}^n(X)$ , then the point-wise bases of  $E$  form a principal  $GL(n)$ -bundle over  $X$ , denoted by  $P_{GL}(E)$ .
- If  $E \in \mathcal{Vect}_{\text{orntd}}^n(X)$ , then the point-wise orientation-compatible bases of  $E$  form a principal  $GL(n)$ -bundle over  $X$ , denoted by  $P_{GL^+}(E)$ .
- If  $E \in \mathcal{Vect}_{\text{Riem}}^n(X)$ , then the point-wise orthonormal bases of  $E$  form a principal  $O(n)$ -bundle over  $X$ , denoted by  $P_O(E)$ .
- If  $E \in \mathcal{Vect}_{\text{Riem,orntd}}^n(X)$ , then the point-wise orientation-compatible orthonormal bases of  $E$  form a principal  $SO(n)$ -bundle over  $X$ , denoted by  $P_O(E)$ .
- The universal covering space  $\tilde{X} \rightarrow X$  is a principal  $\pi_1(X)$ -bundle.

$\lrcorner$

**Prop. (4.15.3.3) [Classifying Principal Bundles].** It is clear(or follows from(7.3.2.18)) that there is a bijection

$$\mathcal{P}_G(B) \cong \check{H}^1(X, G)(9.8.4.1).$$

$\lrcorner$

**Def. (4.15.3.4) [Associated Bundles].** Let  $G \in \mathcal{TopGrp}$ . For a principal bundle(4.15.3.1)  $\pi : P \rightarrow X$  and any  $F \in \mathcal{Top}$  with an action  $\rho : G \rightarrow \text{Homeo}(F)$ , the **associated bundle** of  $(pr, \rho)$  is the total space of  $P \times F$  module the equivalence  $[gg_0, v] = [g, g_0v]$ . This is a  $F$ -fiber bundle over  $X$ .  $\lrcorner$

**Prop. (4.15.3.5) [Homogenous Space].** For  $G \in \mathcal{LieGrp}$  and  $H \leq G$  a closed subgroup, the quotient  $H \backslash G$  can be given a structure of a  $G$ -homogenous space and  $G \rightarrow H \backslash G$  is a principal  $H$ -bundle.  $\lrcorner$

*Proof:*

$\square$

**Prop. (4.15.3.6).** The projection  $\mathbb{S}^{2n+1} \rightarrow \mathbb{CP}^n$  is a principal  $\mathbb{S}^1$ -bundle.  $\lrcorner$

*Proof:*

$\square$

**Prop. (4.15.3.7) [Associated Bundles].** If  $G \in \mathcal{TopGrp}$  and  $G$  acts freely on  $F \in \mathcal{Top}$ , then an **associated  $G$ -bundle with fiber  $F$**  is a locally trivial  $F$ -bundles with a  $G$ -action $?$ .

Then the isomorphism classes of associated  $G$ -bundles with fiber  $F$  over  $B$  is in bijection with  $\mathcal{P}_G(B)$ .  $\lrcorner$

*Proof:*  $?$

$\square$

### Classifying Space

**Def. (4.15.3.8)[Classifying Spaces].** For  $G \in \mathcal{TopGrp}$ , a **classifying space** of  $G$  is a space  $B(G) \in \mathcal{Top}$  together with a principal  $G$ -bundle  $E(G) \rightarrow B(G)$  s.t.  $E(G)$  is contractible.

Notice  $\pi_{n+1}(BG) = \pi_n(G)$  by (4.13.7.14). ┘

**Thm. (4.15.3.9)[Classifying Spaces and Principal Bundles].** If  $G \in \mathcal{TopGrp}$  and  $E(G) \rightarrow B(G)$  is a classifying space for  $G$ , then for any  $B \in \mathcal{Top}$  that is paracompact, the pullback induces a bijection

$$[B, B(G)] \cong \mathcal{P}_G(B).$$

In particular, the classifying spaces is uniquely defined up to homotopy, if it exists. ┘

*Proof:* ?? □

**Prop. (4.15.3.10)[Classifying Spaces Exist].** For  $G \in \mathcal{TopGrp}$ , define

$$\begin{aligned} E_*(G) &\in s\mathcal{Top} : E_n(G) = G^{n+1} \\ d_i((g_1, \dots, g_{n+1})) &= \begin{cases} (g_2, \dots, g_{n+1}) & , i = 0 \\ (g_1, g_{i-1}, g_i g_{i+1}, g_{i+2}, \dots, g_{n+1}) & , 1 \leq i \leq n \end{cases} \\ s_i((g_1, \dots, g_n)) &= (g_1, \dots, g_i, e, g_{i+1}, \dots, g_n), \quad 0 \leq i \leq n \end{aligned}$$

And

$$\begin{aligned} B_*(G) &\in s\mathcal{Top} : B_n(G) = G^n \\ d_i((g_1, \dots, g_n)) &= \begin{cases} (g_2, \dots, g_n) & , i = 0 \\ (g_1, g_{i-1}, g_i g_{i+1}, g_{i+2}, \dots, g_n) & , 1 \leq i \leq n-1 \\ (g_1, \dots, g_n) & i = n \end{cases} \\ s_i((g_1, \dots, g_{n-1})) &= (g_1, \dots, g_i, e, g_{i+1}, \dots, g_{n-1}), \quad 0 \leq i \leq n-1. \end{aligned}$$

Then there is a map

$$\text{pr} : E_*(G) \rightarrow B_*(G) \in s\mathcal{Top} : (g_1, \dots, g_{n+1}) \in E_n(G) \mapsto (g_1, \dots, g_n) \in B_n(G).$$

Denote  $E(G) = |E_*(G)|$ ,  $B(G) = |B_*(G)|$ , then  $\text{pr}$  induces a map  $\text{pr} : E(G) \rightarrow B(G)$ .  $B(G)$  is called the **classifying space** of  $G$ . ┘

**Cor. (4.15.3.11).** Any group  $G$  is a fundamental group of a topological space. ┘

**Prop. (4.15.3.12).** Situation as in (4.15.3.10), there is a right action of  $G$  on  $E_*(G)$  on the last coordinates, which induces an action of  $G$  on  $E(G)$ , and  $B(G) \cong E(G)/G$ . Then if  $(G, e)$  is well-pointed,  $E(G) \rightarrow B(G)$  is a locally trivial bundle with fiber  $G$ , and  $E(G)$  is contractible. In particular,  $G \cong \Omega(BG)$ , by (4.13.7.20). ┘

*Proof:* ?. Cf.[Milnor, Construction of Universal Fiber Bundles, I, II]. □

**Cor. (4.15.3.13).** For  $G, G' \in \mathcal{TopGrp}$ ,

$$E(G \times G') \cong E(G) \times E(G'), \quad B(G \times G') \cong B(G) \times B(G').$$

┘

**Cor. (4.15.3.14).** If  $G$  is commutative, then  $G \times G \rightarrow G$  is a homomorphism that makes  $B(G)$  a commutative topological group. Then we can define for any  $n \in \mathbb{N}$ ,

$$B(G, n) = B^{\circ n}(G).$$

┘

**Prop. (4.15.3.15).**  $[X, BG] \cong G$ -bundles on  $X$ . And  $BG$  is Abelian if  $G$  is Abelian. Thus the classifying space  $BG$  is unique up to homotopy equivalence because they all represent the functor from the CW homotopy category to the set of  $G$ -bundles on it.

┘

*Proof:* Cf.[Principal Bundles and Classifying Space P13].

□

**Prop. (4.15.3.16)[Examples of Classifying Spaces].**

- $B(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{S}^\infty/(\mathbb{Z}/n) = (\mathbb{C}^\infty \setminus \{0\})/(\mathbb{R}^+ \times \mu_n)$  (4.13.3.17). In particular,  $B(\mathbb{Z}/2) \cong \mathbb{RP}^\infty$  (4.13.3.19).
- $BSU(2, \mathbb{R}) = \mathbb{HP}^\infty$ .
- $B(\mathbb{Z}^g) = \mathbb{T}^g$  because  $\mathbb{R} \rightarrow \mathbb{T}^1$  is a universal cover, this can be seen observing only has to satisfy the sum of inner angle is  $\pi$ .
- $BO(n), BU(n), BSp(n)$  are respectively the infinite Grassmannians  $\text{Gr}(n, \mathbb{R}^\infty), \text{Gr}(n, \mathbb{C}^\infty), \text{Gr}(n, \mathbb{H}^\infty)$ , because there is a locally trivial fiber bundle (4.15.1.16)  $U(n, \mathbb{K}) \rightarrow V_n(\mathbb{K}^\infty) \rightarrow \text{Gr}(n, \mathbb{K}^\infty)$ , and  $V_n(\mathbb{K}^\infty)$  is contractible (4.13.3.18). In particular,  $B(\mathbb{Z}/2) = \mathbb{RP}^\infty$  and  $BS^1 = \mathbb{CP}^\infty$ .

┘

*Proof:*

□

**Def. (4.15.3.17)[Admissible Subgroups].** An **admissible subgroup** of a topological group  $G$  is a subgroup  $H \leq G$  s.t.  $G \rightarrow G/H$  is a principal  $H$ -bundle. In particular, this is the case for  $G \in \mathcal{L}\text{ieGrp}$  and  $H \leq G$  closed, by (4.15.3.5).

┘

**Prop. (4.15.3.18).** If  $H$  is an admissible subgroup of  $G$ , then there is a homotopy fiber sequence  $G/H \rightarrow BH \rightarrow BG$ .

┘

*Proof:* Cf.[Principal Bundles and Classifying Space P22].

□

**Cor. (4.15.3.19).** There are homotopy equivalences  $\Omega BK \cong K$  and  $B\Omega K \cong K$ .

┘

**Prop. (4.15.3.20).** If  $H$  is an admissible normal subgroup of  $G$ , then there is a homotopy fiber sequence  $BH \rightarrow BG \rightarrow B(G/H)$ .

┘

*Proof:*

□

**Cor. (4.15.3.21).**

- there are fiber bundles  $\mathbb{S}^0 \rightarrow BSO(n) \rightarrow BO(n)$  and similarly for  $BSU(n)$  and  $BSp(n)$ .
- there are fiber bundles  $\mathbb{S}^n \rightarrow BO(n) \rightarrow BO(n+1)$ .
- there are fiber bundles  $U(n)/T^n \rightarrow (\mathbb{CP}^\infty)^n \rightarrow BU(n)$ , where  $U(n)/T^n$  is the variety of complete flags in  $\mathbb{C}^n$ .
- for a discrete group  $H \subset G$ ,  $BH \rightarrow BG$  is a covering map.

- there are fiber bundles  $BSO(n) \rightarrow BO(n) \rightarrow \mathbb{RP}^\infty$  and similarly for  $\mathbb{C}$  and  $\mathbb{H}$ .
- there are fiber bundles  $\mathbb{RP}^\infty \rightarrow B\text{Spin}(n) \rightarrow BSO(n)$ .

┘

*Proof:* These are all classifying spaces of Lie groups. □

**Prop. (4.15.3.22) [Classifying Line Bundles].** Note that  $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty = B(K(\mathbb{Z}, 1)) = BU(1)$  (4.15.3.16), thus we have a bijection  $H^2(X, \mathbb{Z}) \cong [X, K(\mathbb{Z}, 2)] \cong \text{Vect}_{\mathbb{C}}^1(X)$ . Similarly, we have a bijection  $H^1(X, \mathbb{Z}/2\mathbb{Z}) \cong \text{Vect}_{\mathbb{R}}^1(X)$ . ┘

**Prop. (4.15.3.23) [Universal Real Bundle].** Let  $X$  be paracompact, then there is a natural bijection

$$[X, \text{Gr}(k, \mathbb{K}^\infty)] \rightarrow \text{Vect}_{\mathbb{K}}^k(X) : f \mapsto f^* E_k(\mathbb{K}^\infty). \quad (4.15.3.16)(4.15.2.34)$$

┘

*Proof:* Because  $E$  is of f.t., we can find a f.d. vector space  $W$  with a metric and a vector bundle epimorphism  $\varphi : X \times W \rightarrow E$  via partition of unity. Then we can take the map  $\varphi : X \rightarrow G_k(W) : x \mapsto \ker(\varphi_x)^\perp$ . Then  $f^* E_k(\mathbb{K}^\infty) \cong E$  via restriction of  $\varphi$ . The last assertion follows from the definition of  $E_k(\mathbb{K}^\infty)$  (4.15.2.34) and (4.15.2.15).

Cf. [AGP02]P284. ? □

**Cor. (4.15.3.24).** By (4.14.6.13), there are isomorphisms

$$\text{Vect}_1^{\mathbb{R}}(X) \cong [X, \text{Gr}(1, \mathbb{R}^\infty)] = [X, K(\mathbb{Z}/(2), 1)] = H^1(X, \mathbb{Z}/(2)).$$

$$\text{Vect}_1^{\mathbb{C}}(X) \cong [X, \text{Gr}(1, \mathbb{C}^\infty)] = [X, K(\mathbb{Z}, 2)] = H^2(X, \mathbb{Z}).$$

┘

### Cohomology of Classifying Spaces

**Prop. (4.15.3.25).**  $H_*(\mathbf{BG}, \mathbb{Z}) \cong H_*(\Lambda, \mathbb{Z})$  and  $H^*(\mathbf{BG}, \mathbb{Z}) \cong H^*(G, \mathbb{Z})$ . ┘

*Proof:* Because  $EG$  is weakly contractible,  $S_*(EG)$  is a free  $\mathbb{Z}[G]$ -module resolution of  $\mathbb{Z}$  and  $S_*(EG)_G$  is identified with  $S_*(BG)$ . The rest is easy. □

**Def. (4.15.3.26) [Whitney Sums].** ┘

**Prop. (4.15.3.27).** The diagonal map  $O(1)^n \rightarrow O(n)$  induces a map

$$\omega : (\mathbb{RP}^\infty)^n \cong (\mathbf{BO}(1))^n \rightarrow \mathbf{BO}(n).$$

The conjugation by permutation matrices preserves  $O(1)^n$ , thus induces a covariant action on this map. The action on  $\mathbf{BO}(n)$  is identity ?, and the action on  $(\mathbf{BO}(1))^n$  is by permutation.

Then it induces a map

$$H^*(\mathbf{BO}(n)) \rightarrow H^*((\mathbf{BO}(1))^n)^{\mathfrak{S}_n} \cong \mathbb{F}_2[\sigma_1, \sigma_2, \dots, \sigma_n]$$

where  $\sigma_i$  are the elementary symmetric functions. This is injective and thus surjective when tensoring any field, thus it is a bijection ?.

Denote the inverse image of  $\sigma_i$  by  $w_i$ , then

$$H^*(\mathbf{BO}(n)) \cong \mathbb{F}_2[w_1, \dots, w_n], \quad |w_i| = i$$

and these elements satisfy

- $c_0 = 1$ .
- $i_1^*(c_1)$  is the canonical generator of  $H^2(\mathbf{BU}(1))$ .
- $i_n^*(w_i) = c_i$ .
- $p^*(w_i) = \sum_{j=0}^i w_j \otimes w_{i-j}$ .

┘

*Proof:* Cf.[Cohomology of Classifying Space Toda P82].

□

**Prop. (4.15.3.28)**  $[H^*(\mathbf{BSO}(n), \mathbb{F}_2)]$ .  $\text{pr} : \mathbf{BSO}(n) \rightarrow \mathbf{BO}(n)$  is the universal covering with fiber  $\mathbb{Z}/(2)$ , and

$$H^*(\mathbf{BSO}(n), \mathbb{F}_2) \cong \mathbb{F}_2[\text{pr}^* \omega_2, \dots, \text{pr}^* \omega_n].$$

┘

*Proof:*

□

**Prop. (4.15.3.29).** The diagonal map  $U(1)^n \rightarrow U(n)$  induces a map

$$\omega : (\mathbb{CP}^\infty)^n \cong (\mathbf{BU}(1))^n \rightarrow \mathbf{BU}(n).$$

The conjugation by permutation matrices preserves  $U(1)^n$ , thus induces a covariant action on this map. The action on  $\mathbf{BU}(n)$  is identity ?, and the action on  $(\mathbf{BU}(1))^n$  is by permutation.

Then it induces a map

$$H^*(\mathbf{BU}(n)) \rightarrow H^*((\mathbf{BU}(1))^n)^{\mathfrak{S}_n} \cong \mathbb{Z}[\sigma_1, \sigma_2, \dots, \sigma_n]$$

where  $\sigma_i$  are the elementary symmetric functions. This is injective and thus surjective when tensoring any field, thus it is a bijection ?.

Denote the inverse image of  $\sigma_i$  by  $c_i$ , then

$$H^*(\mathbf{BU}(n)) \cong \mathbb{Z}[c_1, \dots, c_n], \quad |c_i| = 2i$$

and these elements satisfy

- $c_0 = 1$ .
- $i_1^*(c_1)$  is the canonical generator of  $H^2(\mathbf{BU}(1))$ .
- $i_n^*(c_i) = c_i$ .
- $p^*(c_i) = \sum_{j=0}^i c_j \otimes c_{i-j}$ .

┘

*Proof:* Cf.[Cohomology of Classifying Space Toda P81].

□

**Prop. (4.15.3.30).**

$$H^*(\mathbf{BO}(2n), \mathbb{Z}[\frac{1}{2}]) \cong H^*(\mathbf{BO}(2n+1), \mathbb{Z}[\frac{1}{2}]) \cong H^*(\mathbf{BSO}(2n+1), \mathbb{Z}[\frac{1}{2}]) = \mathbb{Z}[\frac{1}{2}][p_1, p_2, \dots, p_n], \quad |p_i| = 4i$$

$$H^*(\mathbf{BSO}(2n), \mathbb{Z}[\frac{1}{2}]) = \mathbb{Z}[\frac{1}{2}][p_1, p_2, \dots, p_n, e], \quad e^2 = p_n, \quad |p_i| = 4i$$

Cf.[Cohomology of Classifying Space Toda P81].

┘



**Prop. (4.15.3.31)[Complexifications].** As  $\mathbf{BSO}(n)$  is the set of oriented real  $n$ -planes in  $\mathbb{R}^\infty$ ,  $\mathbf{BU}(n)$  is the set of complex  $n$ -planes in  $\mathbb{C}^\infty$ , regarding a complex  $n$ -plane as an oriented real  $2n$ -plane induces a map

$$r : \mathbf{BU}(n) \rightarrow \mathbf{BSO}(2n).$$

And complexification of a real plane induces a map

$$c : \mathbf{BO}(n) \rightarrow \mathbf{BU}(n).$$

Then

$$p_k = (-1)^k c^*(c_{2k}) \in H^{4k}(\mathbf{BO}(n)), \quad c^*(c_k) = w_k^2 \in H^{2k}(\mathbf{BO}(n), \mathbb{F}_2), \quad p_k = w_{2k}^2 \in H^{4k}(\mathbf{BO}(n), \mathbb{F}_2)$$

$$r^*(w_{2k}) = c_k \in H^{2k}(\mathbf{BU}(n), \mathbb{F}_2), \quad r^*(e) = \overline{c_n} \in H^{2n}(\mathbf{BU}(n)).$$

$$\mathbf{Bk}^*(p_k) = \sum_{i+j=k} (-1)^i c_i c_j, \quad .$$

Cf.[Cohomology of Classifying Space Toda P81]. ┘

## 4 Characteristic Classes

References are [Cohomology of Classifying Space, Toda] and [M-S74, L-W89, May99].

**Def. (4.15.4.1) [Characteristic Classes].** For an ES-cohomology theory  $E^*$ ,  $p \in \mathbb{Z}, n \in \mathbb{N}$ , a **characteristic class** functor of degree  $p$  for  $n$ -dimensional  $\mathbb{K}$ -bundles is a natural assignment  $c : \mathcal{V}\text{ect}_{\mathbb{K}}^n(X) \rightarrow E^p(X)$  for any  $X \in \mathcal{T}\text{op}$  paracompact Hausdorff that is functorial w.r.t. pullbacks.

By (4.15.3.23), there is a bijection between characteristic class functors of degree  $p$  with elements in  $E^p(\mathbf{BO}(n))$ . ┘

### Stiefel-Whitney Classes

**Def. (4.15.4.2) [Stiefel-Whitney Classes].** A **Stiefel-Whitney class** is a total characteristic class functor  $w : \mathcal{V}\text{ect}_{\mathbb{R}}(X) \rightarrow H^*(X, \mathbb{F}_2)$  for each  $X \in \mathcal{T}\text{op}$  paracompact Hausdorff s.t. the following hold:

- $w(E) = 1 + w_1(E) + \dots + w_n(E) \in H^*(X, \mathbb{F}_2)$ ,  $|w_i| = i, n = \text{rank}(E)$ .
- $f^*(w(E)) = w(f^*(E))$ .
- $w(E \oplus F) = w(E)w(F)$ .
- On the tautological bundle  $\gamma_1^1$  over  $\mathbb{RP}^1$ ,  $w(\gamma_1^1) = 1 + w_1(\gamma_1^1)$  and  $w_1(\gamma_1^1)$  is the unique non-trivial element in  $H^1(\mathbb{RP}^1, \mathbb{F}_2)$ . ┘

**Prop. (4.15.4.3) [Existence and Uniqueness of Stiefel-Whitney Classes].** There exists uniquely a Stiefel-Whitney class functor, and they can be defined using the Thom isomorphism

$$\varphi_E : H^i(B; \mathbb{F}_2) \cong H^{i+n}(E, E \setminus B; \mathbb{F}_2) \quad (4.15.2.25)$$

as

$$w(E) = \varphi_E^{-1} \text{Sq}(\varphi_E(1)) \in H^*(B; \mathbb{F}_2) = \varphi_E^{-1} \text{Sq}(u_E).$$
┘

*Proof:* Cf.[May, P191] and [Milnor, P92].?

□

**Prop. (4.15.4.4).** For  $B \in \mathcal{T}\text{op}$  paracompact Hausdorff,  $E, E' \in \mathcal{V}\text{ect}_{\mathbb{R}}(B)$ , then

- If  $E$  is trivial,  $w(E) = 1$ .
- If  $E \oplus E'$  is trivial, then

$$w(E') = w(E)^{-1} = 1 + w_1 + (w_1^2 + w_2) + (w_1^3 + w_3) + (w_1^4 + w_1^2 w_2 + w_2^2 + w_4) + \dots$$

- Let  $\gamma_n^1$  be the tautological line bundle over  $\mathbb{R}P^n$  (4.15.2.30), then  $w(\gamma_n^1) = 1 + a$ .
- Let  $\xi, \eta$  be vector bundles on  $M, N$ , then

$$w(\xi \times \eta) = \pi_1^* w(\xi) \cup \pi_2^* w(\eta) = w(\xi) \times w(\eta)$$

in  $H^*(M \times N, \mathbb{F}_2)$ .

┘

*Proof:* 1: A trivial bundle is the pullback of a bundle over pt, thus  $w(f^*(E)) = w^*(e(E)) = 1$ .

2 is trivial.

3: This is because  $E_1(\mathbb{R})/\mathbb{R}P^1 \rightarrow E_1(\mathbb{R}^n)/\mathbb{R}P^n$  is a bundle map, thus  $w_1(E_1(\mathbb{R}^n))$  pulls back to  $w_1(E_1(\mathbb{R})) \neq 0$ , thus  $w_1(E_1(\mathbb{R}^n)) = a$ .

4: This follows from the definition that  $\xi \times \eta = \pi_1^* \xi \oplus \pi_2^* \eta$  and Künneth formula (4.14.2.13). □

**Cor. (4.15.4.5) [Tangent and Normal Bundles].** Let  $M$  be a submanifold of a smooth manifold  $N$ , let  $\mathcal{T}_M, \mathcal{T}_N$  be the tangent bundles and  $\nu$  the normal bundle, then

$$w(\mathcal{N}_{M/N}) = w(\mathcal{T}_N)w(\mathcal{T}_M)^{-1}$$

┘

*Proof:* This follows from the smooth nbhd theorem.?

□

**Prop. (4.15.4.6) [Čech Cohomology Interpretation].** By (4.15.3.3), each  $V \in \mathcal{V}\text{ect}_{\mathbb{R}}^n(X)$  corresponds to an element  $\xi_V \in H^1(X, \mathcal{O}(n))$ , and the map  $w_1 : H^1(X, \mathcal{O}(n)) \rightarrow H^1(X; \mathbb{Z}/(2))$  induced from (7.3.2.19) the exact sequence

$$0 \rightarrow \mathcal{SO}(n) \rightarrow \mathcal{O}(n) \rightarrow \mathbb{Z}/(2) \rightarrow 0$$

just maps  $\xi_V$  to the first Siefel-Whitney class of  $V$  (4.15.4.2).

And if  $V$  is orientable, then  $V$  corresponds to an element  $\xi_V \in H^1(X, \mathcal{O}(n))$ , and the map  $w_2 : H^1(X, \mathcal{SO}(n)) \rightarrow H^2(X; \mathbb{Z}/(2))$  induced from (7.3.2.19) the exact sequence

$$0 \rightarrow \mathbb{Z}/(2) \rightarrow \text{Spin}(n) \rightarrow \mathcal{SO}(n) \rightarrow 0$$

just maps  $\xi_V$  to the second Siefel-Whitney class of  $V$  (4.15.4.2). And then we have a Čech representation of  $w_2(V)$ .┘

*Proof:* To prove this, we should show the definitions are natural and check on  $\mathbf{BSO}(n)$  or  $\mathbf{BO}(n)$ , basically because  $V_n(\mathbb{R}^\infty)$  is contractible. □

**Prop. (4.15.4.7) [Wu Formula].** For  $E \in \mathcal{V}\text{ect}_{\mathbb{R}}(B)$ ,

$$\text{Sq}^i(w_j) = \sum_{t=0}^i \binom{j+t-i-1}{t} w_{i-t} w_{j+t}$$

┘

*Proof:* Use splitting principal. □

**Def. (4.15.4.8) [Characteristic Classes of Smooth Manifolds].** For  $M \in \mathcal{M}\text{ani}_{\text{sm}}$ , denote  $w(M) = w(TM)$ . ┘

**Prop. (4.15.4.9) [Wu Formula].** For  $M \in \mathcal{M}\text{ani}_{\text{sm}, \text{cpct}}^d$ , define the **Wu class**  $v = \sum v_i \in H^*(M, \mathbb{F}_2)$  s.t.

$$\langle \nu(M) \cup x, [M]_2 \rangle = \langle \text{Sq}(x), [M]_2 \rangle$$

for any  $x \in H^*(M, \mathbb{F}_2)$ . In particular,  $\nu_k(M)_k \cup x = \text{Sq}^k(x)$  for any  $x \in H^{d-k}(M, \mathbb{F}_2)$ . Notice such a class exists by Poincaré duality.

Then the Stiefel-Whitney class of  $M$  is given by the Wu class:

$$w(M) = \text{Sq}(\nu(M)) \text{ (4.14.6.14)}.$$

┘

*Proof:* Cf. [Milnor, P132]. □

**Cor. (4.15.4.10) [Homotopy Invariance of Stiefel-Whitney Classes for Manifolds].** The Stiefel-Whitney classes only depends on the homotopy type of  $M$ , and pullbacks of Stiefel-Whitney classes along smooth maps between smooth manifolds only depends on the homotopy type of the map. ┘

**Def. (4.15.4.11) [Stiefel-Whitney Numbers].** Let  $M$  be a closed smooth manifold of dimension  $n$ , then for each tuple  $(i_1, \dots, i_k)$  with  $i_1 + \dots + i_k = n$ , define the **Stiefel-Whitney number**

$$w_{i_1, \dots, i_k}(M) = (w_{i_1}(TM) \cup \dots \cup w_{i_k}(TM), [M]) \in \mathbb{Z}/2\mathbb{Z}.$$

┘

### Euler Classes

**Def. (4.15.4.12) [Euler Classes].** Axioms for **Euler classes** for orientable real bundles  $E/B$ :

- $e(e^1) = 0$ .
- For any map  $f : B' \rightarrow B$ ,  $f^*(e(E)) = e(f^*(E))$ .
- $e(E \oplus F) = e(E)e(F)$ .
- for the opposite orientation  $-E$ ,  $e(-E) = -e(E)$ .

┘

**Prop. (4.15.4.13) [Existence of Euler Classes].** For any oriented  $E/B \in \mathcal{V}\text{ect}_{\mathbb{R}}(B)$ , let  $e(E)$  be defined as the image of the Thom class (4.15.2.25) under the maps

$$\varphi_E^{-1} : H^n(E, E_0, \mathbb{Z}) \rightarrow H^n(E, \mathbb{Z}) \cong H^n(B, \mathbb{Z}).$$

or equivalently,  $e(E) = \varphi^{-1}(u_E \cup u_E)$  where  $u_E$  is the Thom class and  $\varphi_{\mathbb{Z}}$  is the Thom isomorphism (4.15.2.25). then it is the desired Euler class. ┘

*Proof:* ?

1: If  $E$  has a non-zero section  $s : B \rightarrow E_0$ , then  $B \xrightarrow{s} E_0 \subset E \rightarrow B$  is identity, thus

$$H^n(B) \xrightarrow{\pi^*} H^n(E) \rightarrow H^n(E_0) \xrightarrow{s^*} H^n(B)$$

is the identity. But  $H^n(B) \xrightarrow{\pi^*} H^n(E) \rightarrow H^n(E_0)$  equals the restriction of  $u_E$  to  $E_0$ , which is 0.

2: This  $e(E)$  is compatible with base change by (4.15.2.26) and the definition.

3: For products, Cf.[Milnor, P100].

4: If the orientation is reversed, then  $u_{-E} = -u_E$ . □

**Cor. (4.15.4.14).**

- For a trivial line bundle,  $e(E) = 0$ .
- For an odd dimensional vector space  $E/B$ ,  $2e(E) = 0$ .

┘

*Proof:* 1: A trivial bundle is the pullback of a bundle over pt, thus  $e(f^*(E)) = f^*(e(E))$  is trivial.

2: This is because there is an orientation preserving isomorphism  $(-1) : E/B \cong (-E)/B$  that is isomorphism on the base, so  $e(-E) = e(E)$ . □

**Cor. (4.15.4.15) [Euler classes and Whitney Classes].** For a vector bundle  $E/B$  of rank  $n$ , the natural map  $H^n(B, \mathbb{Z}) \rightarrow H^n(B, \mathbb{Z}/2\mathbb{Z})$  maps the Euler class  $e(E)$  to the top Whitney class  $w_n(E)$ .

┘

*Proof:*  $e(E) = \varphi^{-1}(u_E \times u_E)$ , which maps to  $\varphi_2^{-1}(u_{E,2} \cup u_{E,2}) = \varphi_2^{-1}(Sq^n(u_{E,2})) = w_n(E)$  by (4.15.2.26). □

**Prop. (4.15.4.16) [Euler Class and Euler Characteristic].** For  $M \in \text{Mani}_{\text{cpct,sm,orntd}}$ , then

$$\langle e(\mathcal{T}_M), [M] \rangle = \chi(M).$$

┘

*Proof:* ?

□

### Chern Classes

**Def. (4.15.4.17) [Chern Classes].** Axioms for **Chern classes** for complex bundles:

- $c(E) = 1 + c_1(E) + \dots + c_n(E) \in H^*(X, \mathbb{Z})$ ,  $|c_i| = 2i, n = \deg(E)$ .
- $f^*(c(E)) = c(f^*(E))$ .
- $c(E \oplus F) = c(E)c(F)$ .
- For the tautological bundle  $\eta$  over  $\mathbb{CP}^\infty$ ,  $c_1(\eta)$  corresponds to the element  $\text{id} \in [\mathbb{CP}^\infty, \mathbb{CP}^\infty] \cong H^2(\mathbb{CP}^\infty)$ .

┘

**Prop. (4.15.4.18).** There exists uniquely a natural transformation  $c : \text{Vect}_{\mathbb{C}}(X) \rightarrow H^*(X, \mathbb{Z})$  satisfying these axioms. (For this, it suffice to calculate the cohomology ring of  $\mathbf{BGL}(n)(\mathbb{C})$ , Cf.[Cohomology of Classifying Space Toda]). ┘

**Cor. (4.15.4.19).** For a trivial bundle  $E = \underline{\mathbb{C}}$ ,  $c_i(E) = 0$  for  $i > 0$ , because  $E$  is a pullback from a bundle on pt.

In particular, for any complex vector bundle  $E$ ,  $c(E \oplus \underline{\mathbb{C}}) = c(E)$ .  $\lrcorner$

**Prop. (4.15.4.20) [First Chern Class Map].** A complex line bundle can be seen as an element of  $H^1(X, \underline{\mathbb{C}}^*)$ , by (7.3.2.14), by the exact sequence

$$0 \rightarrow \underline{\mathbb{Z}} \rightarrow \underline{\mathbb{C}} \xrightarrow{e^{2\pi i}} \underline{\mathbb{C}}^* \rightarrow 0$$

( $\underline{\mathbb{C}}$  is sheaf of smooth functions from  $X$  to  $\mathbb{C}$ ) which gives a map  $H^1(X, \underline{\mathbb{C}}^*) \rightarrow H^2(X, \underline{\mathbb{Z}})$ , called the **first Chern class map**. It is called so because it gives the first Chern class of this complex line bundle. It is also an isomorphism because  $\underline{\mathbb{C}}$  is fine sheaf so acyclic.  $\lrcorner$

*Proof:* Only have to prove they are equal in  $H^2(X, \mathbb{C})$ . We choose a totally convex covering  $U_i$  of  $X$  by (16.3.4.21), then it is a fine cover, so by (7.3.2.15) the Čech cohomology and sheaf cohomology equal.

Use the Chern-Weil map definition of the Chern class, a connection on a line bundle satisfies  $\nabla e_\alpha = \omega_\alpha e_\alpha$ , and if  $e_\beta = e_\alpha g_{\alpha\beta}$ , then  $\omega_\beta = g_{\alpha\beta}^{-1} \omega_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} dg_{\alpha\beta} = \omega_\alpha + d(\log g_{\alpha\beta})$ . So  $\Omega_\alpha = d\omega_\alpha$  locally, and the first Chern class is giving by  $\Omega_\alpha$  in  $H^2(X, \mathbb{C})$ .

Then we need to understand the deRham isomorphism. For the exact sequence  $0 \rightarrow \underline{\mathbb{C}} \rightarrow \mathcal{A}^0 \rightarrow \mathcal{A}^1 \rightarrow \dots$ , it has a splitting:  $0 \rightarrow \underline{\mathbb{C}} \rightarrow \mathcal{K}^1 \rightarrow 0$  and  $0 \rightarrow \mathcal{K}^1 \rightarrow 0 \rightarrow \mathcal{A}^1 \rightarrow \mathcal{K}^2 \rightarrow 0$ , this gives

$$0 \rightarrow H^1(X, \mathcal{K}^1) \xrightarrow{\delta} H^2(X, \underline{\mathbb{C}}) \rightarrow 0, \quad \mathcal{A}^1(X) \rightarrow \mathcal{K}^2(X) \xrightarrow{\delta} H^1(X, \mathcal{K}^1) \rightarrow 0.$$

because  $\mathcal{A}^k$  are fine sheaves. The composite of them is just the de Rham isomorphism (Here we are identifying  $H^2(X, \underline{\mathbb{C}})$  to  $H^2(X, \mathbb{C})$  by (7.3.5.9)). Tracking the lifting, we notice  $\Omega$  is mapped to the cocycle  $\{\log g_{\alpha\beta} + \log g_{\beta\gamma} - \log g_{\alpha-\beta}\}$ , which is exactly the image of the first Chern class map.  $\square$

**Cor. (4.15.4.21).** Complex line bundles are characterized by the first Chern class up to smooth isomorphism, because  $H^1(X, \underline{\mathbb{C}}^*) \rightarrow H^2(X, \underline{\mathbb{Z}})$  is an isomorphism.  $\lrcorner$

**Prop. (4.15.4.22) [Chern Classes and Stiefel-Whitney Classes].** For any  $V \in \text{Vect}_{\mathbb{C}}^n(X)$ , denote by  $V_{\mathbb{R}}$  the underlying  $\mathbb{R}$ -vector space of  $V$ , then

$$w_{2i}(V_{\mathbb{R}}) \equiv c_i(V) \in H^{2i}(X, \mathbb{Z}/(2)).$$

$\lrcorner$

*Proof:* This follows from (4.15.3.31).  $\square$

### Pontryagin Classes

**Def. (4.15.4.23) [Pontryagin Classes].** The Pontryagin classes are defined as  $p_k(E) = (-1)^k c_{2k}(E_{\mathbb{C}}) \in H^{4k}(X, \mathbb{Z})$ .  $\lrcorner$

**Def. (4.15.4.24) [Pontryagin Number].** Let  $M$  be a closed manifold of dimension  $4n$ , then for each tuple  $I = (i_1, \dots, i_k)$  with  $i_1 + \dots + i_k = n$ , define the **Pontryagin numbers**

$$p_I(M) = (p_{i_1}(TM) \cup \dots \cup p_{i_k}(TM), [M]) \in \mathbb{Z}.$$

$\lrcorner$

**Prop. (4.15.4.25) [Pontryagin Number of Product Manifolds].** Let  $M, N$  be closed submanifolds of dimension  $4m, 4n$  resp., then for any tuple  $I = (i_1, \dots, i_k)$  with  $i_1 + \dots + i_k = m + n$ ,

$$p_I(M \times N) = \sum_{I_1, I_2 | I_1 \amalg I_2 = I} p_{I_1}(M) p_{I_2}(N)$$

where the summation is over all partitions  $I_1, I_2$  of  $m, n$  resp.. ┘

*Proof:* Cf. [Milnor, P193]. □

### Combinatorial Pontryagin Classes

See [Milnor]Chap20.

### Calculations and Applications

**Prop. (4.15.4.26).**  $w(\mathbb{RP}^n) = (1 + a)^{n+1}$ . In particular,  $w(\mathbb{RP}^n) = 1$  iff  $n = 2^k - 1$  for some  $k$ . ┘

*Proof:* This follows from (4.15.2.31). □

**Cor. (4.15.4.27).** Let  $n = 2^k - 1 - r$ , where  $r \leq 2^{k-1} - 1$ , then  $\mathbb{RP}^n$  can be immersed into  $\mathbb{R}^{n+d}$  iff  $d \geq r$ . ┘

*Proof:* By (4.15.4.5), the normal bundle  $\nu$  satisfies  $w(\nu) = (1+a)^r$ , and  $r < n$ , thus  $d = \text{rank}(\nu) \geq r$ . □

**Cor. (4.15.4.28).** If  $n+1 = 2^r m$ , then there doesn't exist  $2^r$ -linearly independent vector fields on  $\mathbb{RP}^n$ . ┘

*Proof:* This is because if there is, then  $w(\mathbb{RP}^n)$  has degree  $\leq n - 2^r = 2^r(m-1) - 1$ , but in fact it equals  $(1 + a^{2^r})^m$  has leading term  $a^{2^r(m-1)}$ . □

**Remark (4.15.4.29).** By considering irreducibility, we can prove stronger results, like  $\tau_{\mathbb{RP}^4}$  doesn't contain a subbundle of rank 2. ┘

## 5 K-Theory

**Def. (4.15.5.1) [Topological K-Groups].** For  $X \in \mathcal{T}_{\text{op}_{\text{cpt}}}$ , the **K-group**  $K(X)$  is defined to be  $K_0(\text{Vect}(X))$  (11.1.2.6), which is a ring under sum and tensor. Two vector bundles  $E, F$  are called **stably equivalent** if  $[E] = [F]$ .

There is a degree map  $\deg K(X) \rightarrow \mathbb{Z}$ , and the kernel is denoted by  $\widetilde{K}(X)$ . There is a canonical splitting  $K(X) \cong \widetilde{K}(X) \oplus \mathbb{Z}$ . ┘

**Prop. (4.15.5.2).** Any  $f : X \rightarrow Y \in \mathcal{T}_{\text{op}}$  induces a group morphism  $f^* : K(Y) \rightarrow K(X)$ . By (4.15.1.9), this map only depends on  $f \in [X, Y]$ . ┘

**Prop. (4.15.5.3).** By (4.15.2.20), over a compact Hausdorff space,  $E, F$  is stably equivalent iff  $E \oplus \mathbb{R}^n \cong F \oplus \mathbb{R}^n$  for some  $n$ . ┘

**Thm. (4.15.5.4) [Periodicity Theorem].** Let  $L$  be a line bundle over  $X$ , then as a  $K(X)$ -algebra,  $K(P(L \oplus 1))$  is generated by  $[H]$ , and is subject to the single relation  $([H] - [1])([L][H] - [1]) = 0$ . ┘

*Proof:* Cf.[K theory, Atiyah, P46].  $\square$

**Cor. (4.15.5.5).**  $K(S^2)$  is generated by  $[H]$  as a  $K(\text{pt})$ -module, and is subject to the single relation  $([H] - [1])^2 = 0$ .  $\lrcorner$

**Cor. (4.15.5.6).** There is an isomorphism  $\mu : K(X) \otimes K(S^2) \rightarrow K(X \times S^2)$ , where  $\mu(a \otimes b) = (\pi_1^* a)(\pi_2^* b)$ .  $\lrcorner$

**Def. (4.15.5.7) [Setup for Proof of Periodicity Theorem].** Given a line bundle  $E$  over  $X$ , we can associate a projective bundle  $P(E)$  that  $P(E)_x = P(E_x)$ . Now denote  $P^0$  the subspace of  $P(E)$  consisting of all vectors of length  $\leq 1$  and  $P_\infty$  the subspace consisting of all vectors of length  $\geq 1$  together with the infinity section. There are projections  $\pi_0 : P_0 \rightarrow X$  and  $\pi_\infty : P_\infty \rightarrow X$ , which are homotopy equivalences.

Now by (4.15.2.8),  $\lrcorner$

## 6 Adam Operators

## 7 Cobordism

**Def. (4.15.7.1) [Bordism Groups].** For  $X \in \text{Top}$ , define the **bordism group**  $\Omega_n(X)$  to be the set of pairs  $(M, f)$  where  $M$  is a closed smooth manifold of dimension  $n$  and  $f : M \rightarrow X$  is a continuous map, under the equivalence relation that  $(M_0, f_0) \sim (M_1, f_1)$  iff there is an  $n+1$  manifold  $N$  and a map  $F : N \rightarrow X$  that  $\partial F = f_1 \amalg f_2$ . This is a vector space over  $\mathbb{F}_2$ .

Let  $\Omega_n = \Omega_n(\text{pt})$ , then for any  $m, n$ , there is a map  $\Omega_m \times \Omega_n \rightarrow \Omega_{m+n} : (M, N) \mapsto M \times N$ , so  $\Omega_*$  is a graded commutative ring.  $\lrcorner$

**Prop. (4.15.7.2) [Thom].** If  $N \in \text{Mani}_{\text{sm, cpct}, \partial}^{n+1}$  and  $\partial(N) = M$ , then all the Stiefel-Whitney numbers of  $M$  vanish.

Conversely, if  $M \in \text{Mani}_{\text{sm, cpct}}^n$  with all Stiefel-Whitney numbers 0, then it is the boundary of a compact smooth manifold with boundary.  $\lrcorner$

*Proof:* Let  $[N] \in H_{n+1}(N, M)$  be the fundamental homology class of the pair, then  $\partial([N]) = [M]$ . Let  $\tau_N, M$  be the tangent bundle of  $N, M$ , then  $\tau_N|_M = \tau_M \oplus \mathbb{R}^1$ , because there is an outward pointing normal bundle. Thus the Stiefel-Whitney classes of  $M$  comes from restriction of that of  $N$ . Now the composite  $H^n(N) \rightarrow H^n(M) \xrightarrow{\delta} H^n(N, M)$  is 0, so  $\delta(w_{i_1, \dots, i_k}(M)) = 0$ . thus

$$\langle w_{i_1, \dots, i_k}(M), [M] \rangle = \langle \delta(w_{i_1, \dots, i_k}(M)), [N] \rangle = 0.$$

Conversely, Cf.[Stong, Notes on Cobordism Theory]. ?  $\square$

**Prop. (4.15.7.3).** For  $m \in \mathbb{Z}_+$ ,  $\mathbb{RP}^{2m-1}$  is a boundary and  $\mathbb{RP}^{2m}$  is not a boundary.  $\lrcorner$

*Proof:* Consider the fiber bundle  $S^1 \rightarrow \mathbb{RP}^{2n-1} \rightarrow \mathbb{CP}^n$ , its cone bundle is  $D^1 \rightarrow M^n \rightarrow \mathbb{CP}^{2n-1}$ , which has boundary  $\mathbb{RP}^{2n-1}$ .

$\mathbb{RP}^{2m}$  is not a boundary by (4.15.7.2) and the fact its top Stiefel-Whitney class is non-zero.  $\square$

**Thm. (4.15.7.4) [Thom].**

$$\Omega_* = \mathbb{F}_2[t_2, t_4, t_5, \dots]$$

is a graded polynomial algebra where there's one generator  $t_n \in \Omega_n$  for each  $n \neq 2^k - 1$ .

Also  $\Omega_*(X) = H_*(X, \Omega_*) \cong H_*(X, \mathbb{F}_2) \otimes_{\mathbb{F}_2} \Omega_*$ .  $\lrcorner$

*Proof:*

□

**Cor. (4.15.7.5).** A map  $f : X \rightarrow Y$  induces a map  $f_* : \Omega_n(X) \rightarrow \Omega_n(Y)$ . This map only depends on the homotopy type of  $f$ . ┘

### Oriented Cobordism

**Def. (4.15.7.6)[Oriented Bordism Groups].** When  $X$  is an orientable manifold, define the **oriented bordism group**  $\Omega_n^{so}(X)$  to be the set of pairs  $(M, f)$  where  $M$  is an oriented smooth manifold of dimension  $n$  and  $f : M \rightarrow X$  is a continuous map preserving orientation, under the equivalence relation that  $(M_0, f_0) \sim (M_1, f_1)$  iff there is an  $n+1$  manifold  $N$  and a map  $F : N \rightarrow X$  that  $\partial F = (-f_1) \amalg f_0$ .

Let  $\Omega_n^{so} = \Omega_n^{so}(\text{pt})$ , then for any  $m, n$ , there is a map  $\Omega_m^{so} \times \Omega_n^{so} \rightarrow \Omega_{m+n}^{so} : (M, N) \mapsto M \times N$ , and  $M \times N \cong (-1)^{mn} N \times M$ , so  $\Omega_*^{so}$  is a graded anti-commutative ring. ┘

**Prop. (4.15.7.7).** The relation of (oriented) bordism is truly an equivalence class. It suffices to show transitivity, and this is because we can piece together two manifold with boundaries by collar neighborhood theorem(16.1.1.7). ┘

**Prop. (4.15.7.8).** If  $N \in \text{Mani}_{\text{sm, cpct, orntd}, \partial}^{n+1}$  and  $\partial(N) = M$ , then all the Pontryagin numbers of  $M$  vanish. ┘

*Proof:* Let  $[N] \in H_{n+1}(N, M)$  be the fundamental homology class of the pair, then  $\partial([N]) = [M]$ . Let  $\tau_N, M$  be the tangent bundle of  $N, M$ , then  $\tau_N|_M = \tau_M \oplus \mathbb{R}^1$ , because there is an outward pointing normal bundle. Thus the Stiefel-Whitney classes of  $M$  comes from restriction of that of  $N$ . Now the composite  $H^n(N) \rightarrow H^n(M) \xrightarrow{\delta} H^n(N, M)$  is 0, so  $\delta(w_{i_1, \dots, i_k}(M)) = 0$ . thus

$$\langle w_{i_1, \dots, i_k}(M), [M] \rangle = \langle \delta(w_{i_1, \dots, i_k}(M)), [N] \rangle = 0.$$

□

**Prop. (4.15.7.9)[Mayer-Vitories].** Let  $X = U \cup V$  where  $U, V$  are open subsets of  $X$ , ? ┘

**Thm. (4.15.7.10)[Thom].**

$$\Omega_*^{so} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}[u_1, \dots, u_n, \dots],$$

where  $u_i \in \Omega_{4i}^{so}$  and is represented by  $[\mathbb{CP}^{2i}]$ . Moreover,

- $\Omega_0^{so} = \mathbb{Z}$ .
- $\Omega_1^{so} = 0$ .
- $\Omega_2^{so} = 0$ .
- $\Omega_3^{so} = 0$ .
- $\Omega_4^{so} = \mathbb{Z}$ , generated by  $\mathbb{CP}^2$ .
- $\Omega_5^{so} = \mathbb{Z}/2\mathbb{Z}$ , generated by  $Y^5$ .
- $\Omega_6^{so} = 0$ .
- $\Omega_7^{so} = 0$ .
- $\Omega_8^{so} = \mathbb{Z} \oplus \mathbb{Z}$ , generated by  $\mathbb{CP}^4$  and  $\mathbb{CP}^2 \times \mathbb{CP}^2$ .



- $\Omega_9^{so} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ , generated by  $Y^9$  and  $Y^5 \times \mathbb{CP}^2$ .
- $\Omega_{10}^{so} = \mathbb{Z}/2\mathbb{Z}$ , generated by  $Y^5 \times Y^5$ .
- $\Omega_{11}^{so} = \mathbb{Z}/2\mathbb{Z}$ , generated by  $Y^{11}$ .
- $\Omega_n^{so} \neq 0$  for  $n \geq 8$ .

┘

*Proof:* Cf.[Milnor, P203].

By (4.15.7.8), for any partition  $I$  of  $k$ , there is a map  $\Omega_{4k} \rightarrow \mathbb{Z} : M \mapsto p_I(M)$ . These maps can be used to show that the products  $\{\mathbb{CP}^{2i_1} \times \mathbb{CP}^{2i_r} \mid \sum i_r = k\}$  are linearly independent in  $\Omega_{4k}$  by (4.15.4.25).  $\square$

**Prop.(4.15.7.11) [Index and Oriented Cobordism].** If  $k \in \mathbb{N}$ ,  $W \in \text{Mani}_{\text{cntd,orntd,cpct}}^{4k+1}$ , then the index  $I(M) = 0$ .  $\square$

*Proof:* This follows from the fact the image of  $H^{2k}(W) \xrightarrow{i^*} H^{2k}(M)$  is a half-dimensional subspace (4.14.3.19) that the intersection product is trivial:

$$\langle i^* \alpha \cup i^* \beta, \partial[W] \rangle = \langle \alpha \cup \beta, i_* \partial[W] \rangle = 0.$$

Then it is by linear algebra? that the signature of this pairing is 0.  $\square$

**Def.(4.15.7.12) [Stably Framed Manifold].** A **stably framed manifold** is a smooth manifold  $M$  of dimension  $n$  together with an isomorphism  $TM \oplus \mathbb{R}^N \cong \mathbb{R}^{N+n}$  for some  $N$ .  $\square$

**Def.(4.15.7.13) [Framed Bordism Groups].** Because  $T(\partial N) \oplus \mathbb{R} \cong TN|_{\partial N}$ , we have the notion of a stably framed bordism  $\Omega_n^{fr}(X)$ . Denote  $\Omega_n^{fr} = \Omega_n^{fr}(\text{pt})$ .  $\square$

**Prop.(4.15.7.14).**  $S^3 \cong SU(2)$  is naturally framed, and this generates  $\Omega_3^{fr} \cong \mathbb{Z}/24$ .  $\square$

**Thm.(4.15.7.15) [Pontryagin].**  $\Omega_n^{fr} \cong \pi_{n+N}(S^N)$ .  $\square$

## 8 Applications

**Prop.(4.15.8.1).** A simply connected manifold is orientable. (Use the orientable double cover).  $\square$

## 4.16 Stable Homotopy Theory

Main references are [Higher Algebra, Lurie].

## 5 | Commutative Algebras

### 5.1 Commutative Algebra I

References are [A-M69, Mat80], [Sta]Chap10, [Commutative Algebra with a View Towards Algebraic Geometry] and [Wei94]Chap4.

Commutative rings are studied in this subsection.

**Notation (5.1.0.1).**

- All rings and algebras in this section is assumed to be commutative.
- Let  $0 \neq A, 0 \neq R \in \mathcal{CRing}$ .

┘

#### 1 Basics

**Def. (5.1.1.1) [Integral Monoids].** An **integral monoid** is a monoid  $M \in \mathcal{CMon}^{0,1}$  s.t. for any  $a, b \in M$ ,  $ab = 0$  implies  $a = 0$  or  $b = 0$ . ┘

**Def. (5.1.1.2) [Ideals].** For an arbitrary  $R \in \mathcal{CMon}^{0,1}$ , an **ideal** of  $R$  is a subgroup  $I \leq R$  that is a left ideal. The category of ideals of  $R$  is denoted by  $\text{Ideal}(R)$ . ┘

**Prop. (5.1.1.3) [Quotients].** For  $I \in \text{Ideal}(R)$ , there is a **quotient ring**  $R/I \in \mathcal{CRing}_R$  with the universal property: any  $f : R \rightarrow R' \in \mathcal{CRing}$  that vanishes on  $I$  factors through  $R/I$ . ┘

*Proof:* ┘

**Prop. (5.1.1.4).** The quotient map induces an order-preserving bijection of ideals of  $A/I$  and ideals of  $A$  containing  $I$ . ┘

**Prop. (5.1.1.5) [Prime Avoidance].** Let  $n \in \mathbb{Z}_+$ ,  $I \in \text{Ideal}(R)$  and  $\{P_i\}_{i \in [n]_+} \in \text{Spec}(R)$  s.t.  $I \not\subseteq P_i$  for any  $i$ , then  $I \not\subseteq \cup P_i$ . ┘

*Proof:* Use induction on  $n$ . For  $n = 1$  this is trivial. For  $n \geq 2$ , let  $z_i \in I \setminus \cup_{j \neq i} P_j$ . Now consider  $z = z_1 \cdot z_{n-1} + z_n$ . If  $z \in P_i$  for some  $i < n$ , then  $z_n \in P_i$ , contradiction. If  $z \in P_n$ , then some  $z_i, i < n$  is in  $P_n$  because  $P_n$  is a prime ideal, contradiction. ┘

**Prop. (5.1.1.6) [Existence of a Maximal Ideal].**  $R$  has a maximal ideal. ┘

*Proof:* Use Zorn's lemma, the union of a chain of ideals is an ideal. ┘

**Cor. (5.1.1.7).** Any non-trivial ideal is contained in a maximal ideal. ┘

*Proof:* If  $I \subset A$  is a non-trivial ideal, then  $A/I$  is a non-zero ring, thus  $A/I$  has a maximal ideal, which corresponds to a maximal ideal of  $A$  containing  $I$  (5.1.1.4).  $\square$

**Thm. (5.1.1.8) [Chinese Remainder Theorem, S. Tzu300-400].** Let  $n \in \mathbb{Z}_+$  and  $\mathfrak{a}_1, \dots, \mathfrak{a}_n \in \text{Ideal}(A)$  be pairwise coprime ideals, then for any  $x_1, \dots, x_n \in A$ , there exists  $x \in A$  s.t.  $x \equiv x_i \pmod{\mathfrak{a}_i}$ .  $\square$

*Proof:* The case  $n = 1$  is trivial, and if  $n = 2$ , then by hypothesis  $1 = x_1 + x_2$  where  $x_i \in \mathfrak{a}_i$ . Then for any  $x_1, x_2 \in A$ , we can consider  $x = x_1 a_1 + x_2 a_2$ .

Suppose  $n \geq 3$ . Then for any  $i \in \mathbb{Z}_{\geq 2}$ , there exists  $a_i \in \mathfrak{a}_1, b_i \in \mathfrak{a}_i$  s.t.  $1 = a_i + b_i$ . Then

$$1 = \prod_{i=2}^n (a_i + b_i) \in \mathfrak{a}_1 + \prod_{i=2}^n \mathfrak{a}_i.$$

So by the  $n = 2$  case, there exists  $y_1 \in A$  s.t.

$$y_1 \equiv 1 \pmod{\mathfrak{a}_1}, \quad y_1 \equiv 0 \pmod{\mathfrak{a}_i}, \forall i \geq 2.$$

By the same argument, for any  $j \in [n]_+$ , there exists  $y_j \in A$  s.t.

$$y_j \equiv 1 \pmod{\mathfrak{a}_j}, \quad y_j \equiv 0 \pmod{\mathfrak{a}_i}, \forall i \neq j.$$

So for any  $x_1 \in A$ , we can just take  $x = \sum_{i=1}^n x_i y_i$ .  $\square$

**Prop. (5.1.1.9).** If  $r(I)$  and  $r(J)$  are coprime, then  $I, J$  are coprime.  $\square$

*Proof:* As  $a + b = 1$ , and  $a^m \in I, b^n \in J, 1 = (a + b)^{m+n} \in I + J$ .  $\square$

**Def. (5.1.1.10) [Local Ring].**  $R$  is called a **local ring** if it has only one maximal ideal. Equivalently, there is a prime ideal  $\mathfrak{m}$  that any element in  $R \setminus \mathfrak{m}$  is invertible. A local ring is denoted by  $(R, \mathfrak{m}) \in \text{locCRing}$ .

$R$  is called a **semilocal ring** if it has only f.m. maximal ideals.  $\square$

*Proof:* If  $\mathfrak{m}$  is the maximal ideal, then for any  $x \in R$ , if  $(x)$  is not all of  $R$ , then  $x$  is contained in a maximal ideal by (5.1.1.6), which can only be  $\mathfrak{m}$ . Conversely, if there is a prime ideal  $\mathfrak{m}$  that any element in  $R \setminus \mathfrak{m}$  is invertible, then clearly every non-trivial ideal is included in  $\mathfrak{m}$ .  $\square$

**Prop. (5.1.1.11).** Any quotient of a local ring is also a local ring.  $\square$

**Def. (5.1.1.12) [Local Ring Map].** A map between two local rings are called **local ring map** iff it maps non-invertible elements to non-invertible elements, equivalently,  $f^{-1}(\mathfrak{m}_S) = \mathfrak{m}_R$ .  $\square$

**Def. (5.1.1.13) [Ideal of Definition].** In a Noetherian local ring  $(R, \mathfrak{m})$ , an ideal  $I \subset R$  is called an **ideal of definition** if  $\sqrt{I} = \mathfrak{m}$ .  $\square$

**Prop. (5.1.1.14) [Ideals of Products and Filters].** If  $F_i, i \in I$  is a collection of fields, then the prime ideals in the ring  $\prod F_i$  is in bijection with the ultrafilters on  $I$ , where the ultrafilter  $\mathcal{F}$  corresponds to the ideal  $p_{\mathcal{F}} = \{(a_i) | \text{the set of coordinates that } a_i = 0 \text{ is in } \mathcal{F}\}$ . And in the same way, ideals of  $\prod F_i$  corresponds to the filters on  $I$ .  $\square$

*Proof:* Clearly  $p_{\mathcal{F}}$  is an ideal, and if  $\mathcal{F}$  is an ultrafilter, let  $Z(a)$  be the coordinates that  $a$  is zero on, and notice  $Z(ab) = Z(a) \cup Z(b)$ , then  $ab \in p$  iff  $a \in p$  or  $b \in p$ , by (2.2.10.7), so it is a prime ideal.

Conversely, notice that any two  $a, b$  with  $Z(a) = Z(b)$  differs by a unit, so  $\mathcal{F}_p = \{Z(a) | z \in p\}$  is easily checked to be a filter. And if  $p$  is a prime, then for any  $A \subset I$ , let  $a, b \in \prod F_i$  be that  $Z(a) = A, Z(b) = I - A$ , then  $ab = 0 \in p$ , so  $a \in p$  or  $b \in p$ .  $\square$

**Def. (5.1.1.15)[Torsion-Free Modules].** Let  $S \subset A$  be a set in a commutative ring, then an  $A$ -module  $M$  is called  **$S$ -torsion-free** if for any  $\{x \in M \mid Sx = 0\} = 0$ .  $\lrcorner$

**Prop. (5.1.1.16)[Maximal Torsion-Free Quotient].** Let  $S \subset A$  be a set, the functor from the category of torsion-free  $A$ -modules to the category of  $A$ -modules has a left adjoint, called the **maximal  $S$ -torsion-free quotient**.  $\lrcorner$

*Proof:* A quotient of  $M$  is determined by the kernel. It suffices to prove if  $M/N_1, M/N_2$  are both  $S$ -torsion-free, then  $M/N_1 \cap N_2$  is also  $S$ -torsion-free: This is easy.  $\square$

**Prop. (5.1.1.17).** Let  $A$  be a ring and  $B$  a finite  $A$ -algebra. if  $A \rightarrow B$  is epimorphism in the category of rings, then  $A \rightarrow B$  is surjective.  $\lrcorner$

*Proof:* Notice that  $h^B \rightarrow h^A$  is injective iff  $h^B \times_{h^A} h^B \cong h^B$ , or equivalently,  $B \times_A B \rightarrow B$  is an isomorphism. Now we can localize  $A$  at maximal ideals, thus we can assume  $A$  is local, with maximal ideal  $\mathfrak{m}$  and residue field  $k$ . And then use Nakayama lemma, it suffices to show that  $k \rightarrow B/\mathfrak{m} = C$  is surjective. But  $C \times_k C \cong C$ , so  $\dim_k C = 1$  or  $0$ , which means  $k \rightarrow C$  is surjective.  $\square$

**Def. (5.1.1.18) [Locally Nilpotent].** A **locally nilpotent ideal** is an ideal consisting of nilpotent elements.  $\lrcorner$

**Prop. (5.1.1.19).** If  $I \in \text{Ideal}(A)$  is a locally nilpotent(5.1.1.18), then  $1 + I \rightarrow 1 + I : x \mapsto x^n$  is an isomorphism.  $\lrcorner$

*Proof:* The converse is given by  $x \mapsto (1 + x)^{1/n} = 1 + \binom{1/n}{1}x + \binom{1/n}{2}x^2 + \dots$   $\square$

**Cor. (5.1.1.20).** If  $I$  is a locally nilpotent ideal of  $A$ , then a unit in  $A$  is an  $n$ -th power iff it is an  $n$ -th power in  $A/I$ .  $\lrcorner$

*Proof:* If  $a \equiv b^n \pmod{I}$ , then  $b$  is also a unit, and  $ab^{-n} = c^n$  for some  $n$  by(5.1.1.19), thus  $a$  is a  $n$ -th power.  $\square$

**Def. (5.1.1.21)[Ring Maps Essentially of Finite Type(Finite Presentations)].**  $f : R \rightarrow S$  is said to be **essentially of finite type** if  $S$  is a localization of a ring  $S' \in \mathcal{CRing}/R$  locally of f.t..

$f : R \rightarrow S$  is said to be **essentially of finite presentation** if  $S$  is a localization of a ring  $S' \in \mathcal{CRing}/R$  locally of f.p..  $\lrcorner$

### Tensor Product, Limits and Colimits

**Remark (5.1.1.22)[Tensor Product].** Tensor product is defined in(3.2.4.13). Notice that in case the rings are all commutative, there are no need to distinguish between left and right modules.  $\lrcorner$

**Def. (5.1.1.23)[Tensor Algebras].** For a module  $M$  over a commutative ring  $R$ , we define the

- **tensor algebra** operator from  $\text{Mod}_R$  to graded algebras over  $R$  that is left adjoint to the forgetful functor. It can be defined as follows:

$$T(M) = \bigoplus_{n \geq 0} \otimes^n M$$

as the module, and the algebra structure determined by the canonical map  $\otimes^m M \times \otimes^n M \rightarrow \otimes^{m+n} M$ .

- **exterior product**  $\wedge^k M$  as the module with the universal property that  $\text{Hom}_B(\wedge^k M, N)$  is the set of all morphisms  $M^n \rightarrow N$  that vanishes on all elements that have two equal coordinates.
- **exterior algebra** operator  $\wedge$  from  $\text{Mod}_R$  to the category of strict graded commutative algebras over  $R$  that is left adjoint to the forgetful functor. It can be defined as follows:  $\wedge(M) = T(M)/(x \otimes x)$ , where  $x \in M$ , or equivalently  $\wedge(M) = \bigoplus_{k \geq 0} \wedge^k(M)$ .
- **symmetric algebra** operator  $S$  from  $\text{Mod}_R$  to  $\mathcal{C}\text{Ring}_R$  that is left adjoint to the forgetful functor. It can be defined as follows:  $\wedge(M) = T(M)/(x \otimes y - y \otimes x)$  where  $x, y \in M$ .

┘

**Cor. (5.1.1.24).** The construction of  $T(M)$ ,  $\wedge M$  and  $\text{Sym}(M)$  commutes with all colimits, because they are all left adjoints.

┘

**Prop. (5.1.1.25) [Tensor Product and Quotient].** Let  $R$  be a commutative ring and  $I, J$  be ideals of  $R$ , then  $R/I \otimes_R R/J \cong R/(I + J)$ .

┘

*Proof:* This follows from the universal property of quotient and tensoring.

□

**Prop. (5.1.1.26).** There is a pullback square

$$\begin{array}{ccc} R/I \cap J & \longrightarrow & R/I \\ \downarrow & & \downarrow \\ R/J & \longrightarrow & R/(I + J) \end{array}$$

┘

*Proof:* The pullback is just the elements in  $R/I \times R/J$  that map to the same element in  $R/(I + J)$ . If  $(x + I, y + J)$  maps to the same element  $z + I + J$ , then  $x = y + i + j$ , so  $x - i = y + j$ , and the pullback lies in the image of  $\Delta : R \mapsto R/I \times R/J$ . Now the kernel is just  $I \cap J$ .

□

**Prop. (5.1.1.27) [Filtered Colimits of Modules are Exact].** If  $R \in \text{Ring}$  and  $\mathcal{I}$  be an index category that each connected components of  $\mathcal{I}$  is filtered, then taking colimits over  $\mathcal{I}$  is exact in the category  $\text{Mod}_R$ .

┘

*Proof:* It is clearly right exact. To check left exactness, Cf. [Sta]04B0.?

□

**Cor. (5.1.1.28) [Filtered Colimits of Abelian Groups are Exact].** Filtered colimits are exact in  $\text{Ab}$ .

┘

**Prop. (5.1.1.29).** Let  $k$  be a field and  $A, B$  be  $k$ -algebras and let  $\mathfrak{b} \subset A \otimes_k B$  be an ideal. Then among the ideals  $\mathfrak{a} \subset A$  that  $\mathfrak{b} \subset \mathfrak{a} \otimes_k B$ , there exists a smallest one.

┘

*Proof:* Choose a  $k$ -basis of  $B$ , then the smallest ideal  $\mathfrak{a}$  is just the ideal generated by all the  $A$ -coefficients of elements of  $\mathfrak{b}$ .

□

## Localization

**Def. (5.1.1.30) [Localization as Filtered Colimit].** Let  $A$  be a commutative ring and  $S$  be a multiplicatively closed subset of  $A$  containing 1 and not containing 0, the localization  $S^{-1}A$  is defined to be a ring over  $A$  that any ring map  $A \rightarrow B$  that maps elements of  $S$  to units factors through  $S^{-1}A$ .

$S^{-1}A$  can be constructed as

$$\varinjlim_{s \in S} A$$

where the ordering is defined to be  $s < t$  if  $t = sr$  for some  $r \in S$ , and if  $t = sr$ , there is a map from  $A_s$  to  $A_t$  defined by multiplying by  $r$ . This is easily seen to be a filtered colimit. There is easily seen to be the localization.  $\lrcorner$

**Prop. (5.1.1.31) [Localization is exact].**  $S^{-1}$  is an exact functor from  $\text{Mod}_R$  to  $\text{Mod}_R$ . Because it is a filtered colimit (4.8.3.28)(5.1.1.27).  $\lrcorner$

**Cor. (5.1.1.32).**  $(R/I)_{\bar{P}} \cong R_P/IR_P$ , in particular,  $k(R/P) \cong R_P/PR_P$ .  $\lrcorner$

**Def. (5.1.1.33) [Total Ring of Fractions].** For any ring  $R$ , the set of non-zerodivisors in  $R$  is a multiplicatively closed set  $S$ , and the localization  $\text{Frac}(R) = S^{-1}R$  is called the **total ring of fractions** of  $R$ .  $\lrcorner$

**Prop. (5.1.1.34) [Localization along an Ideal].** Let  $I$  be an ideal of  $A$ , then the localization of  $A$  along  $I$  is the ring  $\tilde{A}$  that  $\text{Spec } \tilde{A}$  is the localization of  $\text{Spec } A$  along  $V(I)$  as defined in (4.12.3.23) (because  $\text{Spec } A$  is spectral, by (4.12.3.16)).

Equivalently, it can be defined as  $\tilde{A} = S^{-1}A$ , where  $S = A \setminus (\cup_{\mathfrak{p} \in V(I)} \mathfrak{p})$ . (It can be checked that  $S$  is multiplicatively closed).

Notice then  $I \subset \text{rad } \tilde{A}$ .

For  $f \in A$ , we call the localization of  $A$  along  $f$  the  **$f$ -localization** of  $A$ . In fact, it is the universal  $A$ -algebra that  $f \in \text{rad } \tilde{A}$ .  $\lrcorner$

**Prop. (5.1.1.35).** Let  $R$  be a commutative ring,  $f_1, \dots, f_n \in R$ , and  $M$  an  $R$ -module, then  $M \rightarrow \oplus_i M_{f_i}$  is injective iff  $M \rightarrow \oplus_i M : m \mapsto (mf_1, \dots, mf_n)$  is injective.  $\lrcorner$

*Proof:* Cf. [Sta]0565.  $\square$

**Prop. (5.1.1.36).** For any ring  $A$ ,  $A \rightarrow \prod A_{\mathfrak{m}}$  is injective, where  $\mathfrak{m}$  are maximal ideals of  $A$ .

For a domain  $A$ ,  $A = \bigcap A_{\mathfrak{m}}$  inside the fraction field of  $A$ , where  $\mathfrak{m}$  are maximal ideals of  $A$ .  $\lrcorner$

*Proof:* if  $g \in \text{Frac}(A)$  is in the RHS, then  $I = \{x \in A \mid xg \in A\}$  is an ideal of  $A$  not contained in any maximal ideal, thus  $I = 1$  (5.1.1.7), and thus  $g \in A$ .  $\square$

**Lemma (5.1.1.37).** Let  $R$  be a ring and  $\mathfrak{p}$  be a prime, then there exists an  $f \in R$ ,  $f \notin \mathfrak{p}$  that  $R_f \subset R_{\mathfrak{p}}$ , if any of the following holds:

- $R$  is a domain.
- $R$  is Noetherian.
- $R$  is reduced and has f.m. irreducible components.

$\lrcorner$

*Proof:* Cf. [Sta]0BX1.  $\square$

**Def. (5.1.1.38) [Identifying Local Rings].** A ring map  $A \rightarrow B$  is said to **identify local rings** if for every prime  $\mathfrak{q} \subset B$ , the map  $A_{\varphi^{-1}(\mathfrak{q})} \rightarrow B_{\mathfrak{q}}$  is an isomorphism.  $\lrcorner$

**Prop. (5.1.1.39).** The property of identifying local rings is stable under base change and composition. (This is immediate from (5.1.1.40)).  $\lrcorner$

**Prop. (5.1.1.40) [Tensor Product and Localization].** For a ring map  $R \rightarrow S$ , let  $q \subset \text{Spec } S$ ,  $p = q \cap R$ , then  $(M \otimes_R S)_q = M_p \otimes_{R_p} S_q$  for any  $R$ -module  $M$ .  $\lrcorner$

*Proof:*  $(M \otimes_R S)_q = M \otimes_R S_q = M \otimes_R R_p \otimes_{R_p} S_q = M_p \otimes_{R_p} S_q$ .  $\square$

### Noetherian

**Def. (5.1.1.41) [Noetherian Module].** Let  $R \in \mathcal{CRing}$  and  $M \in \text{Mod}_R$ , then  $M$  is called a **Noetherian module** if any ascending chain of submodules stabilizes.

$R$  is called a **Noetherian ring** iff  $R$  is Noetherian over itself.  $\lrcorner$

**Prop. (5.1.1.42).** Let  $R \in \mathcal{CRing}$  and  $0 \rightarrow M_1 \rightarrow M \rightarrow M_2$  be an exact sequence in  $\text{Mod}_R$ , then  $M$  is Noetherian iff  $M_1, M_2$  are both Noetherian.  $\lrcorner$

**Prop. (5.1.1.43) [Hilbert Basis Theorem, Hilbert1888].** Let  $A$  be a Noetherian ring, then quotient ring, f.g. module, f.g. algebra, localization and power series of  $A$  are Noetherian. Hence graded algebra of a Noetherian ring  $A$  by an ideal  $I$  is Noetherian. Products of Noetherian rings are Noetherian.  $\lrcorner$

*Proof:* Only need to prove  $A[X]$  and  $A[[X]]$ , localization and others are quotients of these. For an ascending chain of ideal  $I_j$  of  $A[X]$ , we consider the coefficients ideal  $I_{i,j}$  of  $X^i$  of  $I_j$ , then there are only f.m. different  $I_{i,j}$ s, so we have  $I_j$  stabilize as well.

Similarly for  $A[[X]]$ , we prove any ideal  $I$  is f.g. Consider the lowest terms coefficient ideal at degree  $i$ , then it is ascending and stabilize, then a set of generators as a whole generate  $I$ .  $\square$

**Remark (5.1.1.44).** The subring of a Noetherian ring is NOT necessarily Noetherian, by the example of  $k[X_1, \dots, X_n, \dots] \subset k(X_1, \dots, X_n, \dots)$ .  $\lrcorner$

**Prop. (5.1.1.45).** When  $A$  is Noetherian and is quipped with  $I$ -adic topology, then  $I$  is f.g., and there is a surjective ring map  $A[[X]] \rightarrow A^\wedge$ , mapping to the generators of  $I$ , hence the completion is Noetherian. (It is surjective can be seen by the Cauchy sequence construction of completion).  $\lrcorner$

**Prop. (5.1.1.46).** If  $R \rightarrow R'$  is ring map of f.t., then if  $S \in \mathcal{CRing}/R$  and  $S$  is Noetherian, then  $S \otimes_R R'$  is Noetherian, because  $S \otimes_R R'$  is of f.t. over  $S$ , and use (5.1.1.43).  $\lrcorner$

**Cor. (5.1.1.47).** For  $k \in \text{Field}$ ,  $S \in \mathcal{CRing}/k$ , then for any f.g. field extension  $K/k$ ,  $S \otimes_k K$  is Noetherian. (Because there is a f.g. algebra  $B$  over  $k$  that  $K$  is the localization of  $B$ , and use (5.1.1.43)).  $\lrcorner$

**Prop. (5.1.1.48).** If  $R$  is Noetherian and  $M$  is a f.g.  $R$ -module, then there is a filtration  $\{M_i\}$  of  $M$  that the quotients are all isomorphic to  $R_{\mathfrak{p}_i}$  where  $\mathfrak{p}_i$  are primes.  $\lrcorner$

*Proof:*  $M$  is generated by  $x_i$ , so  $(x_1) \cong R/I_1$ , and so we modulo  $x_i$ , then the result follows by induction. So we may assume  $M = R/I$ . We use Noetherian condition to choose a maximal element  $J$  that is a counterexample, then  $J$  is not a prime, so there are  $a, b \notin J$  that  $ab \in J$ . Then we have a filtration  $0 \subset aR/(J \cap aR) \subset R/J$ . Notice  $R/(J + bR) \rightarrow aR/(J \cap aR) \rightarrow 0$ , and the second quotient is  $R/(J + aR)$ , so they all can be factorized.  $\square$

**Prop. (5.1.1.49).** A Noetherian ring has only f.m. minimal prime ideals.  $\lrcorner$

*Proof:* This is a consequence of (7.4.1.20) (7.4.1.21) and (4.12.2.4).  $\square$

**Prop. (5.1.1.50) [Cohen].** If  $R \in \mathcal{CRing}$  and every prime ideal is f.g., then  $R$  is Noetherian.  $\lrcorner$

*Proof:* Suppose  $P$  is not Noetherian. Firstly the set of non-finitely generated ideals has the chain property: if  $I_i$  is a chain of non-f.g. ideals of  $R$ , then  $I = \cup_{i \in \Phi} I_i$  is non-f.g., otherwise there are



$(f_i) = I$ , but  $\{f_i\} \in I_h$  for some  $h$ , thus  $I_h$  is f.g.. Then we use the Zorn's lemma to find a maximal non-f.g. ideal  $I$ . We show that  $I$  is a prime ideal:

$I \neq R$  because  $R = (1)$ . So if  $a, b \in R \setminus I$  that  $ab \in I$ , then  $I + (a)$  and  $I + (b)$  is f.g. by  $p_i + r_i a$ , by maximality, and let  $K = (P : a)$ , then  $I \subset I + (b) \subset K$ , thus  $K$  is f.g., so does  $aK$ .

Now I claim  $I = (p_1) + \dots + (p_n) + aK$ : one direction is clear, and if  $r \in I \subset I + (a)$ , then  $r = \sum c_i(p_i + r_i a)$ , thus  $(\sum c_i r_i) a = r - \sum c_i p_i \in I$ , thus  $\sum c_i r_i \in K$ , thus  $r = \sum c_i p_i + (\sum c_i r_i) a \in (p_1) + \dots + (p_n) + aK$ .

So now  $I$  is f.g., contradiction, which shows  $I$  is a prime, but this contradicts the hypothesis.  $\square$

**Prop. (5.1.1.51) [Modules over Noetherian Ring are Noetherian].** Let  $R$  be a Noetherian ring, then any submodule of a finite module  $M$  over  $R$  is finite. Thus any module over  $R$  is a Noetherian module (5.1.1.41). In particular, any module over  $R$  is of f.p.  $\lrcorner$

*Proof:* it suffices to prove the first assertion: we use induction on the minimal number of generators of  $M$ : if it is generated by 1 element, then  $M \cong R/I$  for some ideal  $I$ , thus  $N \subset M$  is isomorphic to some  $J/I$ , so it is finite because  $J$  is finite. If the minimal number of generators of  $M$  is greater than 1, then there exists an exact sequence

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

where  $M', M''$  has fewer number of generators. Now there is also an exact sequence

$$0 \rightarrow N \cap M' \rightarrow N \rightarrow \overline{N} \rightarrow 0,$$

and the minimal number of generators of  $N$  is smaller than the sum of that of  $M'$  and  $M''$ , thus it is also finite.  $\square$

**Prop. (5.1.1.52) [Artin-Tate [A-T51]].** Let  $R$  be a Noetherian ring and  $S$  a f.g.  $R$ -algebra. If  $T \subset S$  is an  $R$ -subalgebra that  $S$  is a finite module over  $T$ , then  $T$  is f.g. over  $R$ .  $\lrcorner$

*Proof:* Let  $x_1, \dots, x_n \in S$  generate  $S$  as an  $R$ -algebra, and let  $y_1, \dots, y_m$  generate  $S$  as an  $T$ -module. Then there exists  $a_{ij} \in T$  s.t.  $x_i = \sum_j a_{ij} y_j$ . There also exists  $b_{ijk}$  s.t.  $y_i y_j = \sum_k b_{ijk} y_k$ . Let  $T' \subset T$  be the subalgebra generated by the elements  $\{a_{ij}\}$  and  $\{b_{ijk}\}$ , then it is f.g. over  $R$  thus a Noetherian ring (5.1.1.43).

$S$  is a surjective image of  $S' = T'[Y_1, \dots, Y_m]/(Y_i Y_j - \sum_k b_{ijk} Y_k)$ , which is a finite module over  $T'$ , so  $S$  is also finite module over  $T'$ . Then  $T \subset S$  is also finite over  $T'$ , thus f.g. over  $R$ .  $\square$

**Prop. (5.1.1.53) [Krull-Akizuki].** If  $R$  is a Noetherian domain of dimension 1 with fraction field  $K$ ,  $L/K$  a finite extension of fields, then for any ring  $A$  s.t.  $R \subset A \subset L$ ,  $A$  is Noetherian.  $\lrcorner$

*Proof:* Cf. [Sta]00PG.  $\square$

## 2 Lengths

**Def. (5.1.2.1) [Lengths].** The **length** of a  $R$ -module  $M$  is the supremum of lengths of chains of submodules of  $M$ , denoted by  $\text{length}_R(M)$ .  $\lrcorner$

**Prop. (5.1.2.2).** Length is an additive function on  $\text{Mod}_R$ .  $\lrcorner$

*Proof:* Cf. [Sta]00IV.  $\square$

**Prop. (5.1.2.3).** If  $\text{length}_R(M) < \infty$ , then any maximal chain of submodules has the same length.  $\lrcorner$

*Proof:* Let  $l(M)$  be the minimal length of a maximal chain, then if  $M \subsetneq N$ , then firstly  $l(M) < l(N)$ , because a maximal chain of  $M$  restricts to a maximal chain of  $N$ , and if the length is the same, then each term is in  $M$ , so  $N \subset M$ , contradiction. Now any chain has length  $l(M)$ , because if there is a chain  $M_i$ , then  $l(M_0) < l(M_1) < \dots < l(M)$ .  $\square$

**Prop. (5.1.2.4).** Let  $R \in \mathcal{CRing}$  and  $S$  a multiplicative set of  $R$ , then  $\text{length}_R(M) \geq \text{length}_{S^{-1}R}(S^{-1}M)$ .  
 $\lrcorner$

*Proof:* This is because any  $S^{-1}R$ -submodule of  $S^{-1}M$  is of the form  $S^{-1}N$  where  $N$  is a  $R$ -submodule of  $M$ .  $\square$

**Prop. (5.1.2.5).** If  $R$  is a ring with a maximal ideal  $\mathfrak{m}$ ,  $M$  is an  $R$ -module that  $\mathfrak{m}M = 0$ , then  $\text{length}_R(M) = \dim_{k(\mathfrak{m})}(M)$ .  
 $\lrcorner$

**Prop. (5.1.2.6) [Length over a Local Ring].** Let  $(R, \mathfrak{m})$  be a local ring, and  $M \in \text{Mod}_R$  is of finite length, then  $\mathfrak{m}^n M = 0$  for some  $n \in \mathbb{N}$ .

Conversely, if  $\mathfrak{m}$  is f.g. and  $M$  is a finite  $R$ -module, and  $\mathfrak{m}^n M = 0$  for some  $n \in \mathbb{N}$ , then  $\text{length}_R(M) < \infty$ .  
 $\lrcorner$

*Proof:*  $M$  is clearly a finite module over  $R$ . Take  $n = \text{length}_R(M)$ . If  $f_1, \dots, f_n \in \mathfrak{m}$  and  $x \in M$  s.t.  $f_1 \dots f_n x \neq 0$ , then by Nakayama the submodules

$$0 \subset (f_1 \dots f_n)x \subset (f_2 \dots f_n)x \subset (f_n)x \subset M$$

are all different, so  $\text{length}_R(M) \geq n$ , contradiction.

For the converse, use additivity on the filtration  $0 = \mathfrak{m}^n M \subset \mathfrak{m}^{n-1} M \subset \dots \subset M$ , where each quotient is finite over  $R$  and annihilated by  $\mathfrak{m}$ , so we conclude by (5.1.2.5).  $\square$

**Prop. (5.1.2.7).** If  $(A, \mathfrak{m})$  is a local ring and  $B$  is a ring over  $A$  with f.m. maximal ideals  $\mathfrak{m}_i$  s.t. each  $\mathfrak{m}_i$  is over  $\mathfrak{m}$  and  $k(\mathfrak{m}_i)/k(\mathfrak{m})$  is finite, then there for any  $M \in \text{Mod}_B$  with  $\text{length}_B M < \infty$ ,

$$\text{length}_A M = \sum_i [k(\mathfrak{m}_i) : k(\mathfrak{m})] \text{length}_{B_{\mathfrak{m}_i}} M_{\mathfrak{m}_i}.$$

$\lrcorner$

**Prop. (5.1.2.8).** Let  $A \rightarrow B$  be a flat local maps of local rings, then for any  $M \in \text{Mod}_A$ ,

$$\text{length}_A(M) \text{length}_B(B/\mathfrak{m}_A B) = \text{length}_B(M \otimes_A B).$$

$\lrcorner$

*Proof:* Cf. [Sta]02M1. ?  $\square$

**Cor. (5.1.2.9).** Let  $A \rightarrow B \rightarrow C$  be local maps of local rings,  $C/B$  is flat, and  $\mathfrak{m}_A$  the maximal ring of  $A$ , then

$$\text{length}_B(B/\mathfrak{m}_A B) \text{length}_C(\mathfrak{m}_B C) = \text{length}_C(C/\mathfrak{m}_B C).$$

$\lrcorner$

**Prop. (5.1.2.10) [Order of Vanishing].** If  $R$  is a semi-local Noetherian domain of dimension 1 and  $a, b$  are not zero-divisors, then  $f(a) = \text{length}(R/(a))$  satisfies  $f(a) + f(b) = f(ab) < \infty$ .

So this  $R \subset K$  and has fraction field  $K$ , then  $f$  extends to an additive function on  $K^*$ , denoted by  $\text{ord}_R(f)$ .  $\lrcorner$

*Proof:* It is finite by [Sta]00PF. It is additive because length is additive (5.1.2.1) and  $0 \rightarrow R/(a) \rightarrow R(ab) \rightarrow R(b) \rightarrow 0$ .  $\square$

### 3 Artinian Ring

**Def. (5.1.3.1) [Artinian Rings].** A ring  $A$  is called **Artinian** if any descending chain of ideals of  $A$  stabilizes.

For example, a f.g.  $k$ -algebra that is a finite  $k$ -module is an Artinian ring.  $\lrcorner$

**Lemma (5.1.3.2).** Let  $A$  be an Artinian ring, then  $A$  has f.m. maximal ideals.  $\lrcorner$

*Proof:* Suppose for the contrary that there are infinitely many maximal ideals  $\mathfrak{m}_1, \dots, \mathfrak{m}_n, \dots$ , then we can consider the chain of ideals  $\mathfrak{m}_1 \supset \mathfrak{m}_1 \cap \mathfrak{m}_2 \supset \dots$ . Then it is an infinite descending chain, by Chinese remainder theorem, contradicting the fact that  $A$  is Artinian.  $\square$

**Lemma (5.1.3.3).** If  $A$  is an Artinian ring, then the Jacobson radical is nilpotent.  $\lrcorner$

*Proof:* Consider the Jacobson radical  $I$ ,  $I^n = I^{n+1}$  for some  $n$ , let  $J = \text{Ann}(I^n)$ , it suffices to show  $J = A$ . If not, choose a minimal  $J'$  that contains  $J$  but not  $J$  (exists by Artinian property), then  $J' = J + Ax$ , and  $IJ' \subset J$  by Nakayama, so  $xI^{n+1} \subset JI^n = 0$ , so  $x \in J$ , contradiction.  $\square$

**Prop. (5.1.3.4) [Characterization of Artinian Rings].** The following are equivalent:

1.  $A$  is Artinian.
2.  $A$  is Noetherian of dimension 0.
3.  $\text{length}_A A < \infty$ .
4.  $A$  is a finite product of local Artinian rings.
5.  $A$  is Noetherian, Jacobson (5.2.6.4), and has f.m. maximal ideals.

$\lrcorner$

*Proof:* 1  $\iff$  3: if  $\text{length}_A A < \infty$ , then  $A$  is clearly Artinian. Conversely, if  $A$  is Artinian, then by (5.1.3.2)(5.1.3.3)(5.2.6.7)  $A$  a finite product of its localization of maximal ideals, so we may assume  $A$  is local with maximal ideal  $\mathfrak{m}$ . Then  $\mathfrak{m}^n = 0$  for some  $n$  by (5.1.3.3), and  $\mathfrak{m}^i/\mathfrak{m}^{i+1}$  has length the same as their dimension as a  $A/\mathfrak{m}$  vector space, by (5.1.2.5), which is finite because  $A$  is Artinian, so  $\text{length}_A A < \infty$ .

1 + 3  $\rightarrow$  5:  $A$  has f.m. maximal ideals by (5.1.3.2). It is Jacobson by (5.1.3.3).  $\text{length}_A A < \infty$  clearly implies  $A$  is Noetherian.

5  $\rightarrow$  2: By (5.2.6.7).

2  $\rightarrow$  5: all prime ideals are maximal, so  $\text{Spec } A$  is discrete, so  $A$  has f.m. maximal ideals, and it is clearly Jacobson.

5  $\rightarrow$  3: By (5.2.6.7),  $R$  is a product of its local rings, and the local rings are all Noetherian and Jacobson (5.2.6.6), so by Nakayama, they have finite lengths. So also  $R$  has finite length.

5  $\rightarrow$  4: By lemma (5.2.6.7) below,  $A$  is a product of its localization, and its localizations also satisfies 5, and by 5  $\rightarrow$  3  $\rightarrow$  1, they both have descending conditions.

4  $\rightarrow$  5: An Artinian ring is Noetherian and Jacobson by 1 + 3  $\rightarrow$  5, then so does their product.

$\square$

**Cor. (5.1.3.5) [Reduced Artinian Ring].** A reduced local Artinian ring is a field. In particular, A reduced Artinian ring is a product of fields.  $\lrcorner$

*Proof:* An Artinian local ring  $A$  is Jacobson (5.1.3.4) so the maximal ideal  $\mathfrak{m} = 0$  as  $A$  is reduced.

$\square$

**Prop. (5.1.3.6).** For an Artinian local ring  $A$ , the following are equivalent:

1.  $A$  is a PID.
2. the maximal ideal  $\mathfrak{m}$  is principal.
3.  $\dim_k(\mathfrak{m}/\mathfrak{m}^2) \leq 1$ .

┘

*Proof:* It suffices to prove  $3 \rightarrow 1$  : If  $\mathfrak{m} = \mathfrak{m}^2$ , then  $\mathfrak{m} = 0$  by Nakayama, so  $A$  is a field. If  $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$ , then  $\mathfrak{m}$  is principle by Nakayama. And  $\mathfrak{m}$  is nilpotent by (5.1.3.4), so for any ideal  $a$  there is a minimal  $n$  that  $a \subset \mathfrak{m}^n$ . Now choose  $y \in a - \mathfrak{m}^{n+1}$ , then  $y = ux^r$ , and  $u \notin (x)$ , so  $u$  is a unit, thus  $x^r \in a$ , meaning  $a = \mathfrak{m}^n$  hence principal.  $\square$

## 4 Local Properties

**Def. (5.1.4.1)[Local properties].** A property  $P$  of rings or modules over a ring is called **local property** iff  $X$  has  $P$  iff  $X_{f_i}$  all has  $P$  for a covering  $(f_1, \dots, f_n) = 1$ .

A property of morphisms of rings is called **local on the target** iff  $R \rightarrow S$  has  $P$  iff  $R_{f_i} \rightarrow S_{f_i}$  has  $P$  for a covering  $(f_1, \dots, f_n) = 1$  in  $R$ .  $\square$

**Prop. (5.1.4.2)[Stalkwise Properties].** For a commutative ring  $R$ , a property  $P$  is called stalkwise if  $A$  satisfies  $P$  iff all  $A_{\mathfrak{m}}$  satisfies  $P$  where  $\mathfrak{m}$  are maximal ideals, and iff all  $A_{\mathfrak{p}}$  satisfies  $P$  where  $\mathfrak{p}$  are prime ideals of  $A$ .

1. Trivial is stalkwise for modules over  $R$ . Hence so does injectivity and surjectivity because localization is exact.
2. Torsion-free is stalkwise for modules over  $R$  if  $R$  is integral.
3. Flatness for modules over  $R$ .
4. Flatness for rings over  $R$  on the source.
5. Formal unramifiedness for rings over  $R$ , both on the target and source.
6. (universally)catenary is stalkwise.
7. reducedness is stalkwise.
8. Integral+integrally closed is stalkwise.
9. normal is stalkwise.
10. regular is stalkwise.

┘

*Proof:*

1. It suffice to prove an element is trivial on every localization then it is 0. For this, consider the annihilator  $\text{Ann}(x)$ , it is not contained in any maximal ideal so it contains 1.
2. if  $xf = 0$  but  $f \neq 0$ , then  $x \in \text{Ann}(f) \neq (1)$ , so  $\text{Ann}(f) \subset \mathfrak{m}$  maximal, so  $f$  is torsion in  $M_{\mathfrak{m}}$  over  $R_{\mathfrak{m}}$ . Conversely, if  $f$  is torsion in  $R_{\mathfrak{m}}$ , then it is clearly torsion over  $R$ .
3. We use the definition (5.4.1.2). Notice  $(IM)_{\mathfrak{p}} = I_{\mathfrak{p}}M_{\mathfrak{q}}$  and every ideal of  $R_{\mathfrak{p}}$  is of the form  $I_{\mathfrak{p}}$ . Then use the fact injective is stalkwise (5.1.4.2).
4. We use the definition (5.4.1.2). Notice  $(I \otimes_R S)_{\mathfrak{q}} = I_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} S_{\mathfrak{q}}$  for all primes  $\mathfrak{q}$  of  $S$  and  $\mathfrak{p} = \mathfrak{q} \cap R$ . And every ideal of  $R_{\mathfrak{p}}$  is of the form  $I_{\mathfrak{p}}$ . Then use the fact injective is stalkwise (5.1.4.2)

5. Because formally unramified is equivalent to  $\Omega_{R/S} = 0$  (5.4.6.1), so we get the result by functorial properties of  $\Omega_{S/R}$  (5.4.3.6) and triviality is stalkwise (5.1.4.2).
6. For any two prime ideals  $\mathfrak{p} \subset \mathfrak{q}$ , we can choose a maximal ideal containing them.
7. use (5.1.1.36).
8. If  $A$  is integrally closed, then clearly any localization of  $A$  is integrally closed. Conversely, if  $r \in K(A)$  is integral over  $A$  but  $r \notin A$ , let  $I = \{s \in A \mid rs \in A\}$ , then  $I \neq A$ , so  $I \subset \mathfrak{m}$  for some maximal ideal  $\mathfrak{m}$ , then  $r \notin A_{\mathfrak{m}}$ , so  $A_{\mathfrak{m}}$  is not integrally closed, contradiction.
9. By definition.
10. By definition.

□

**Remark (5.1.4.3).** Our main technique of proving local properties are using affine communication theorem (7.4.1.2). ┘

**Prop. (5.1.4.4) [Local Properties].** For a fixed ring  $R$ ,

1. Every property that is stalkwise is a local property. (5.1.4.2) The properties listed below should be not stalkwise.
2. Every property that satisfies faithfully flat descent is a local property. The properties listed below should not satisfy faithfully flat descent.
3. Noetherian.
4. F.t. ring maps on the source.
5. F.p. ring maps on the source.
6. N-1 and N-2 and universally Japanese for rings.
7. Nagata for rings.

┘

*Proof:*

- 1.
- 2.
3. If  $A$  is Noetherian, then  $A_{f_i}$  are Noetherian by (5.1.1.43). Conversely, if  $A_{f_i}$  are all Noetherian and  $I_1 \subset I_2 \subset \dots$  is an ascending chain of ideals of  $A$ , consider  $A \rightarrow \prod A_{f_i}$  faithfully flat, thus  $I_1 \otimes_A (\prod A_{f_i}) \subset I_2 \otimes_A (\prod A_{f_i}) \subset \dots$  is an ascending chain of  $\prod A_{f_i}$ . Now  $\prod A_{f_i}$  are Noetherian by (5.1.1.43), so this chain stabilizes. But this ring map is faithfully flat, so the original chain must also stabilize.
4. Let  $(g_1, \dots, g_n) = 1$ , choose  $\sum h_i g_i = 1$ , and let  $x_{ij} = y_{ij}/g_i^{n_{ij}}$  generates  $S_{g_i}$ . Now let  $S'$  be the sub- $R$ -algebra of  $S$  generated by  $y_{ij}, g_i, h_j$ . Then  $(S')_{f_i} \rightarrow S_{f_i}$  is surjective for any  $i$ , so  $S' \rightarrow S$  is also surjective, by (5.1.4.2). Then  $S' = S$ , and  $S$  is f.g. over  $R$ .
5. Cf. [Sta]00EP.
6. If  $A$  is N-1, then  $A_{f_i}$  are N-1 because taking integral closure commutes with localization. The same for N-2. Conversely, if all  $A_{f_i}$  are N-1 or N-2, so is  $A$  because finiteness is local (5.1.4.4). The universal Japanese case follows from the N-2 case.
7. This follows from the localness of Noetherian and N-2 (5.1.4.4).

□

## 5 Fitting Ideals

**Def. (5.1.5.1) [Fitting Ideals].** Let  $R \in \mathcal{CRing}$  and  $M$  be a finite  $R$ -module, then for any presentation

$$\bigoplus_{j \in J} R \rightarrow R^{\oplus n} \rightarrow M \rightarrow 0$$

given by a  $n \times J$  matrix  $A$ , the ideal generated by the  $(n - k) \times (n - k)$  matrices of  $A$  is independent of the presentation chosen, called the  $k$ -th **Fitting ideal** of  $A$ , denoted by  $\text{Fitt}_k(M)$ .  $\lrcorner$

## 6 Rings and Categories

### Quotient by Equivalence Relations

**Prop. (5.1.6.1) [Quotients by Equivalence Relations].** Let  $u_0, u_1 : A_0 \rightarrow A_1$  be an equivalence in the dual category of  $\mathcal{CRing}_{R_0}$  (4.1.1.24). If  $u_0$  is locally free of constant rank  $r$ , then a quotient  $u : A \rightarrow A_0$  exists, and  $u$  is locally free of constant rank  $r$ .  $\lrcorner$

*Proof:* Cf. [Mil17b]P592.  $\square$

## 7 Spectra

**Lemma (5.1.7.1).** In  $\text{Spec } A$ , a subset  $U$  is retrocompact iff it is quasi-compact iff it is a finite union of standard opens  $D(f_i)$ . In particular, any constructible subset of  $\text{Spec } A$  is a finite union of morphisms of the form  $D(f_i) \cap V(g_1, \dots, g_m)$ .  $\lrcorner$

*Proof:* Retrocompact is quasi-compact because  $\text{Spec } A$  is quasi-compact. A quasi-compact subset is equivalent to a finite union of  $D(f_i)$ . Now for any quasi-compact subset  $V = \bigcup_j D(g_j)$ ,  $U \cap V = \bigcup_{ij} D(f_i g_j)$  is also quasi-compact, so  $\bigcup_i D(f_i)$  is retrocompact.  $\square$

**Lemma (5.1.7.2).** Let  $R$  be a ring, then

- the image of any standard open subsets of  $\text{Spec } R[X]$  is qc open in  $\text{Spec } R$ .
- for  $g, f \in R[X]$  with  $g$  monic, the image of  $D(f) \cap V(g)$  is qc open in  $\text{Spec } R$ .

 $\lrcorner$ 

*Proof:* 1: For any prime  $\mathfrak{p}$  of  $R$ , the primes mapping to  $\mathfrak{p}$  has a minimal one,  $\mathfrak{p}[X]$ , so  $\mathfrak{p}$  is in the image of  $\text{Spec } R[X] \rightarrow \text{Spec } R$  iff  $f \in \mathfrak{p}[X]$ , which means the image is  $D(a_0, a_1, \dots, a_n)$ , where  $f = a_0 + a_1 X + \dots + a_n X^n$ .

2:  $R[X]/g$  is finite free over  $R$ , let  $P(T) = T^d + r_{d-1}T^{d-1} + \dots + r_0$  be the characteristic polynomial of  $f$  acting on  $R[X]/g$  by left multiplication, then  $\mathfrak{p} \in V(r_0, \dots, r_{d-1})$  iff  $f$  acts nilpotently on  $R[X]/g \otimes_R k(\mathfrak{p})$ , which is equivalent to  $\mathfrak{p}$  being in the image of  $D(f) \cap V(g)$  (by base change to  $k(\mathfrak{p})$  argument).  $\square$

**Lemma (5.1.7.3) [Affine Chevalley].** The  $\text{Spec}$  map of a f.p. ring map maps constructible sets to constructible sets.  $\lrcorner$

*Proof:* Suppose  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_m)$ , then it suffices to show for the case  $S = R/(f_1, \dots, f_m)$  and  $S = R[X]$ .

If  $S = R/(f_1, \dots, f_m) = R/I$ , it suffices to show the image of  $D(\bar{g}) \cap V(\bar{h}_1, \dots, \bar{h}_m)$  is retrocompact in  $\text{Spec } R$ : in fact its image is just  $D(g) \cap V(h_1, \dots, h_m) \cap V(I)$ , where  $g, h_i$  are any inverse images in  $R$ .

For the second case, we first prove some localizing result: If  $S = R_f$ , then the compact open subsets of  $\text{Spec } S$  are compact open subsets of  $\text{Spec } R$ , so  $\text{Spec } S \rightarrow \text{Spec } R$  maps constructible sets to constructible sets. Now for any  $c \in R$ ,  $\text{Spec } R = \text{Spec } R/c \amalg \text{Spec } R_c$ , so to prove the proposition for  $R$ , it suffices to prove for  $R_c$  and  $R/c$ .

Let  $D(f) \cap V(g_1, \dots, g_n) \subset \text{Spec } R[X]$ , where  $f, g_i$  are polynomials. We can use induction on the degree series  $\deg(g_1) \leq \deg(g_2) \leq \dots \leq \deg(g_n)$ . If the leading coefficient of  $g_1$  is invertible, then we can use reduction to  $R/c$  and  $R_c$  to either reduce the degree of  $g_1$  or reduce to the case it is invertible. If it is invertible, then we can use Euclidean division. Eventually, it can be reduced to the case  $D(f) \cap V(g)$  where  $g$  is monic, or  $D(f)$ . This is proved in the above lemma(5.1.7.2).  $\square$

**Lemma (5.1.7.4).** For a Noetherian local ring  $(A, \mathfrak{m})$ ,  $\text{Spec } A - \mathfrak{m}$  is affine iff  $\dim A \leq 1$ .  $\lrcorner$

*Proof:* if  $\dim A = 0$ , this is true, if  $\dim A = 1$ , let  $f \in \mathfrak{m}$  not in any other minimal primes of  $A$ , then  $\text{Spec } A - \mathfrak{m} = \text{Spec } A_f$ .

Conversely, Cf.[Sta]0BCR.  $\square$

### Idempotents

**Prop. (5.1.7.5)[Clopen Subsets].** The clopen subsets of  $\text{Spec } A$  corresponds to idempotents in  $A$ .  $\lrcorner$

*Proof:* Notice that  $A$  is the global section of  $\text{Spec } A$ , so for any clopen subset  $U \subset \text{Spec } A$ , we can take the section  $e$  which is 1 on  $U$  and 0 on  $\text{Spec } A \setminus U$ . Then  $e^2 = e$ .  $\square$

**Cor. (5.1.7.6).** A local ring has no non-trivial idempotents, and then an idempotent is defined by the maximal ideals that it vanishes.  $\lrcorner$

**Cor. (5.1.7.7).** If  $I$  is an ideal of  $R$  that  $I = I^2$ , and  $I$  is f.g., then  $V(I)$  is open and closed in  $\text{Spec } R$ , and  $V(I) = R_e$  for some idempotent  $e$ .  $\lrcorner$

*Proof:* By Nakayama, there is a  $f = 1 - e$  with  $e \in I$  that  $fI = 0$ . So  $e - e^2 = 0$  and  $f^2 = f$ .  $V(I) = D(f) = D(e)$ .  $\square$

**Lemma (5.1.7.8).** If  $I$  is a locally nilpotent ideal, then  $R \rightarrow R/I$  induces a bijection on idempotents.  $\lrcorner$

*Proof:* Because  $R \rightarrow R/I$  induces a homeomorphism on the spectra, and clopen subsets of the spectrum corresponds to the idempotents(5.1.7.5).  $\square$

**Lemma (5.1.7.9).** Let  $R$  be a ring and  $T \subset \text{Spec } R$  is a set. Then the following are equivalent:

- $T$  is closed and is a union of connected components of  $\text{Spec } R$ .
- $T$  is an intersection of clopen subsets.
- $T = V(I)$  where  $I$  is generated by idempotents.

$\lrcorner$

*Proof:* 1 and 2 are equivalent by(4.12.3.3), and  $2 \rightarrow 3 \rightarrow 1$  are easy.  $\square$

**Prop. (5.1.7.10).** Let  $R$  be a ring, then any connected component of  $\text{Spec } R$  is of the form  $V(I)$ , where  $I$  is an ideal generated by idempotents that any idempotent of  $R$  maps to either 0 or 1 in  $R/I$ .  $\lrcorner$

*Proof:* By(4.12.3.2) and(4.12.3.16), a connected component of  $\text{Spec } R$  is an intersection of clopen subsets, so it is of the form  $V(I)$  where  $I$  is generated by idempotents. The last assertion is equivalent to  $V(I)$  being connected.  $\square$

### Going-up and down

**Def. (5.1.7.11) [Going-up and Going Down].** Going-up and down for topological spaces is defined in (4.12.2.7). A ring map  $R \rightarrow S$  is said to satisfy the going-up property iff its Spec map does, equivalently, for any prime ideal  $\mathfrak{q} \subset S$  and prime ideal  $\mathfrak{p} \subset \mathfrak{q} \cap R$ , there exists a prime ideal  $\mathfrak{q}' \subset \mathfrak{q}$  that  $\mathfrak{q}' \cap R = \mathfrak{p}$ .

It is said to satisfy the going-down property iff its Spec map does, equivalently, for any prime ideal  $\mathfrak{q} \subset S$  and prime ideal  $\mathfrak{q} \cap R \subset \mathfrak{p}$ , there exists a prime ideal  $\mathfrak{q}' \subset \mathfrak{q}$  that  $\mathfrak{q}' \cap R = \mathfrak{p}$ .  $\lrcorner$

**Prop. (5.1.7.12).** Going-up and Going-down are stable under composition, trivially.  $\lrcorner$

**Prop. (5.1.7.13) [Integral Map satisfies Going-Up].** Integral ring map satisfies going-up (5.2.1.5). Flat ring map satisfies going-down (5.4.1.18).  $\lrcorner$

**Lemma (5.1.7.14).** If the image of the Spec map of a ring map is closed under specialization, then this image is closed.  $\lrcorner$

*Proof:* Let it be  $R \rightarrow S$ , let  $I$  be the kernel, then the image is contained in  $V(I)$ , so we may replace  $R$  be  $R/I$ , then  $R \subset S$ . Now we show the image contains all the minimal primes of  $R$ : for a minimal prime  $\mathfrak{p}$ ,  $A_{\mathfrak{p}} \subset B_{\mathfrak{p}}$ , thus  $B_{\mathfrak{p}}$  is not-empty, and thus has a maximal ideal, whose intersection with  $A_{\mathfrak{p}}$  can only be  $\mathfrak{p}$  by hypothesis, thus  $\mathfrak{p}$  is in the image of Spec. Then the image is all of Spec  $R$  by hypothesis, thus closed.  $\square$

**Cor. (5.1.7.15) [Going-up and Spec Closed].** Going-up is equivalent to Spec map closed.  $\lrcorner$

*Proof:* If going-up holds, then Spec map is closed by (5.1.7.14). Conversely, a closed map satisfies going-up, by (4.12.2.8).  $\square$

**Prop. (5.1.7.16).** If  $R \rightarrow S$  is a ring map that satisfies going-up, and  $P \subset S$  is a maximal ideal, then  $P \cap R$  is also a maximal ideal.  $\lrcorner$

**Prop. (5.1.7.17) [Krull].** If  $A \subset B$  is an integral extension of integral domains, and  $A$  is normal, then going-down holds.  $\lrcorner$

*Proof:* Let  $L_1, K$  be the fraction fields of  $B, A$  resp., and let  $L$  be the normal extension of  $K$  contained in  $L_1$ ,  $C$  the integral closure of  $A$  in  $L$ . Let  $P \in \text{Spec } B$  and  $\mathfrak{p} = P \cap A$ ,  $\mathfrak{p}' \subset \mathfrak{p}$ . Take a prime ideal  $Q' \in \text{Spec } C$  lying over  $\mathfrak{p}'$ , and by going-up applied to  $A \subset C$  (5.1.7.13), there is a prime ideal  $Q_1$  lying over  $\mathfrak{p}$  that  $Q' \subset Q_1$ . Take  $Q \in \text{Spec } C$  lying over  $P$ , then by (5.3.5.13) there is a  $\sigma \in G_{L/K}$  that  $\sigma(Q_1) = Q$ . Set  $P' = \sigma(Q') \cap B$ , then  $P' \subset P$  is a prime of  $B$  lying over  $\mathfrak{p}'$ , so going-down holds for  $A \subset B$ .  $\square$

**Prop. (5.1.7.18) [Going-down and Spec Open].** If Spec map is open, then going-down holds.  $\lrcorner$

*Proof:*  $\square$

### Minimal Primes and Irreducible Components

**Prop. (5.1.7.19) [Minimal Primes Exists].** Every nonzero ring contains a minimal prime ideal.  $\lrcorner$

*Proof:* Firstly prime ideal exists, by (5.1.1.6), and we use Zorn's lemma to find a minimal prime ideal: it suffices to show the intersection of a chain of prime ideals is a prime ideal, this is not hard.  $\square$



**Lemma (5.1.7.20).** If  $\mathfrak{p}$  is a minimal prime of  $R$ , then  $\mathfrak{p}R_{\mathfrak{p}}$  is locally nilpotent by (5.2.6.1). In particular, if  $R$  is reduced, then  $R_{\mathfrak{p}}$  is a field.  $\square$

**Prop. (5.1.7.21) [Zerodivisors in a Reduced Ring].** Let  $R$  be a reduced ring, then

- $R \rightarrow \prod_{\mathfrak{p} \text{ minimal}} R_{\mathfrak{p}}$  is an embedding into a product of fields.
- $\cup_{\mathfrak{p} \text{ minimal}} \mathfrak{p}$  is the set of zerodivisors of  $R$ .

$\square$

*Proof:* 1: By (5.1.7.20),  $R_{\mathfrak{p}}$  are fields. In particular, the kernel of  $R \rightarrow R_{\mathfrak{p}}$  is  $\mathfrak{p}$ . Then the kernel of the map  $R \rightarrow \prod_{\mathfrak{p} \text{ minimal}} R_{\mathfrak{p}}$  is  $\cap_{\mathfrak{p} \text{ minimal}} \mathfrak{p} = 0$  by (5.2.6.1).

2: If  $xy = 0$  and  $x \neq 0$ , then  $x \notin \mathfrak{p}$  for some minimal prime  $\mathfrak{p}$  by (5.2.6.1), thus  $y \in \mathfrak{p}$ . Conversely, if  $y \in \mathfrak{p}$  for some minimal prime  $\mathfrak{p}$ , then  $y$  is mapped to  $0 \in R_{\mathfrak{p}}$ , which means there are some  $x \notin \mathfrak{p}$  that  $xy = 0$ .  $\square$

**Prop. (5.1.7.22).** If  $R$  is a ring with f.m. minimal primes  $\mathfrak{q}_i$  and  $\cup_i \mathfrak{q}_i$  is the set of zerodivisors of  $R$ , then the ring of fractions of  $R$  (5.1.1.33) is equal to  $\prod_i R_{\mathfrak{q}_i}$ .  $\square$

*Proof:* Cf. [Sta]02LX.  $\square$

**Prop. (5.1.7.23) [Irreducible Components of Spectrum].** The irreducible closed subsets of  $\text{Spec } R$  are exactly the sets of the form  $V(\mathfrak{p})$  for some prime  $\mathfrak{p} \subset R$ . The irreducible components of  $\text{Spec } R$  are exactly the sets of the form  $V(\mathfrak{p}_i)$  for some minimal prime  $\mathfrak{p}_i$ .  $\square$

**Prop. (5.1.7.24).** If  $R$  be a ring and  $\mathfrak{p}$  a minimal prime of  $R$ . If  $W \subset \text{Spec } R$  is a quasi-compact open subset of  $\text{Spec } R$  not containing  $\mathfrak{p}$ , then there exists some  $f \in R$  that  $\mathfrak{p} \subset D(f)$  and  $D(f) \cap W = \emptyset$ .  $\square$

*Proof:*  $W$  is of the form  $\cup_{i=1}^r D(f_i)$ . As  $\mathfrak{p} \notin D(f_i)$ ,  $f_i \in \mathfrak{p}$  for each  $i$ . Then (5.1.7.20) says  $f_i$  are nilpotent in  $R_{\mathfrak{p}}$ , so there is some  $g \in R$  that  $gf_i$  are nilpotent in  $R$  for any  $i$ , which means  $g$  satisfies the requirement.  $\square$

**Prop. (5.1.7.25).** For  $R \subset S$ , all the minimal primes of  $R$  are in the image of the Spec map of a minimal prime of  $S$ .  $\square$

*Proof:* Localize w.r.t. to the minimal prime  $\mathfrak{p}$ , then it is a local ring with only one prime. And  $S_{\mathfrak{p}}$  is nonzero because localization is exact, so it has a maximal ideal  $\mathfrak{q}$ . Now we choose a minimal prime of  $S$  contained in  $\mathfrak{q}$ , then it is also mapped to  $\mathfrak{p}$ .  $\square$

### Universal Homeomorphism

Cf. [Sta]10.45 and [Sta]28.44.

**Prop. (5.1.7.26).** If  $\varphi : R \rightarrow S \in \mathcal{C}\text{Ring}$  and  $p$  is a prime number that satisfies:

- $S$  is generated over  $R$  by elements  $x$  that there is  $n$  that  $x^{p^n} \in \varphi(R)$  and  $p^n x \in \varphi(x)$ .
- $\ker(\varphi)$  is locally nilpotent.

then  $\text{Spec } S \rightarrow \text{Spec } R$  is a homeomorphism, and any base change of  $\varphi$  satisfies the above conditions, so it is a universal homeomorphism.

In particular, this applies to any base change of a field extension  $k'/k$  that is purely inseparable, because it is f.f. hence injective.  $\square$

*Proof:* Cf. [Sta]0BRA.  $\square$

## 5.2 Commutative Algebra II

References are [Sta].

**Notation (5.2.0.1).**

- Use notations as in 3.2.
- Let  $A, R \in \mathcal{CRing}$ .

┘

### 1 Integral Extensions

**Def. (5.2.1.1) [Totally Integrally Closed].** For  $A \rightarrow B \in \mathcal{CRing}$ ,  $f \in B$  is called **almost integral** (or totally integral when almost mathematics is performed:) over  $A$  if  $f^{\mathbb{N}}$  lies in a f.g.  $A$ -module of  $B$ . It is clear that the elements of totally integral elements of  $B$  is a subring. And  $A$  is called **totally integrally closed** in  $B$  iff any  $f \in B$  totally integral over  $A$  is in  $A$ . ┘

**Prop. (5.2.1.2).** For a ring map  $\varphi : A \rightarrow B$ , an element  $x$  is integral over  $A$  iff  $x$  is contained in a finite  $A$ -module in  $B$ . In particular, the elements of  $B$  that are integral over  $A$  is a ring containing  $\varphi(A)$ . ┘

*Proof:* If  $x$  is integral, then  $\varphi(A)[x]$  is finite. If  $\varphi(A)[x]$  is finite, then there is a set of generators of polynomials in  $x$ . Then for  $m$  large,  $x^m = \sum a_i f_i(x)$ , so  $x$  is integral over  $A$ . ┐

**Prop. (5.2.1.3) [Integral Extension of Field].** For  $A \subset B$ , if  $B$  is integral over  $A$ , then  $A$  is a field iff  $B$  is a field. ┘

*Proof:* If  $A$  is a field,  $y^{-1} = -a_n^{-1}(y^{n-1} + \dots + a_{n-1}) \in B$ . If  $B$  is a field,  $x^{-1} = -(b_1 + b_2x + \dots + b_mx^{m-1}) \in A$ . ┐

**Cor. (5.2.1.4).** If  $B$  is integral over  $A$ , then a prime  $\mathfrak{p} \subset B$  is maximal iff  $\mathfrak{p} \cap A$  is maximal. ┘

*Proof:* Look at the integral extension  $A/(\mathfrak{p} \cap A) \rightarrow B/\mathfrak{p}$ . ┐

**Prop. (5.2.1.5) [Going-Up].** Let  $A \rightarrow B$  integral. Then:

1. There is no inclusion relation between prime ideals of  $B$  lying over a fixed prime ideal of  $A$ .
2. if  $A \subset B$ , then the Spec map is surjective. In particular, for any  $\mathfrak{p} \subset A$ ,  $\mathfrak{p}B \cap A = \mathfrak{p}$ .
3. The going-up holds. In particular, the Spec map of an integral ring map is closed, by (5.1.7.15). ┘

*Proof:*

1. If  $\mathfrak{p} \cap A = \mathfrak{p}' \cap A = \mathfrak{q}$ , Localize at  $\mathfrak{q}$ , then  $\mathfrak{p}, \mathfrak{p}'$  are both maximal ideals of  $B_{\mathfrak{q}}$  by (5.2.1.4), they cannot contain each other.
2. For any prime  $\mathfrak{p}$  of  $A$ , since  $A_{\mathfrak{p}} \subset B_{\mathfrak{p}}$ ,  $B_{\mathfrak{p}} \neq 0$ , so it has a maximal ideal (5.1.1.6), and use (5.2.1.4).
3. for any prime ideal  $\mathfrak{q}$  of  $B$  and  $\mathfrak{p} = \mathfrak{q} \cap A$ , replace  $A \rightarrow B$  by  $A/\mathfrak{p} \subset B/\mathfrak{q}$ , then we can use 2. ┐

┘

## 2 Graded Rings

Cf.[Matsumura Ch11].

**Def. (5.2.2.1) [Graded Rings].** A **graded ring** is a ring  $A = \bigoplus_{n \in \mathbb{Z}} A_n$  that  $A_m A_n \subset A_{m+n}$ . A **graded module** over a graded ring  $A$  is a module  $M = \bigoplus_{n \in \mathbb{Z}} M_n$  that  $A_m M_n \subset M_{m+n}$ .

Notice that often we mean  $\mathbb{Z}^{\geq 0}$ -graded rings when we say graded rings. For a  $\mathbb{Z}^{\geq 0}$ -graded ring  $A$ , the subset  $A_+ = \bigoplus_{n=1}^{\infty} A_n$  is an ideal of  $A$ , called the **irrelevant ideal**.  $\lrcorner$

**Def. (5.2.2.2) [Twisted Modules].** Let  $A$  be a graded ring and  $M$  a graded  $A$ -module, denote  $M(n)$  the graded  $A$ -module s.t.  $M(n)_m = M_{m+n}$ .  $\lrcorner$

**Lemma (5.2.2.3).** Let  $A = \bigoplus_{n=0}^{\infty} A_n$  be a graded ring, then a set of homogenous elements  $f_i \in A_+$  generate  $A$  as an algebra over  $A_0$  iff they generate  $A_+$  as an ideal of  $A$ .  $\lrcorner$

*Proof:* If  $f_i$  generate  $A$  as algebra over  $A_0$ , then every element of  $A_+$  is a polynomial in  $f_i$  with constant coefficients in  $A_0$ , thus  $f_i$  generates  $A_+$  as an ideal. Conversely, if  $f_i$  generate  $A_+$  as an ideal, then for any homogenous element  $f$  we can use induction on the degree of  $f$  to show that  $f$  is a polynomial in  $f_i$ .  $\square$

**Prop. (5.2.2.4) [Noetherian Graded Rings].** A graded ring  $A = \bigoplus_{n=0}^{\infty} A_n$  is Noetherian iff  $A_0$  is Noetherian and  $A_+$  is f.g. as an ideal of  $A$ .  $\lrcorner$

*Proof:* If  $A$  is Noetherian, then clearly  $A_+$  is f.g. and  $A_0 = A/A_+$  is Noetherian. Now if  $A_+$  is f.g. as an ideal of  $A$ , then it is generated by f.m. homogenous elements  $f_i$ , so we see  $f_i$  generates  $A$  as an algebra over  $A_0$ , which means  $A$  is a quotient of a polynomial ring over  $A_0$ , thus Noetherian by (5.1.1.43).  $\square$

**Def. (5.2.2.5) [Homogenous Ideals].** Let  $A_{\bullet}$  be a graded ring, then a **homogenous ideal**  $I_{\bullet}$  of  $A_{\bullet}$  is a ring that is generated by homogenous elements.  $\lrcorner$

**Prop. (5.2.2.6) [Equivalent definition of Homogenous Ideals].** Let  $S_{\bullet}$  be a graded ring, then

- an ideal  $I$  of  $S_{\bullet}$  is homogenous iff it contains the degree  $n$  part of each of its element for any  $n$ .
- The set of homogenous ideals of  $S_{\bullet}$  is stable under sum, product, intersection and radical.
- A non-trivial homogenous ideal  $I$  of  $S_{\bullet}$  is a prime ideal iff for any homogenous elements  $a, b$ , if  $ab \in I$ , then  $a \in I$  or  $b \in I$ .

$\lrcorner$

*Proof:* 1: If  $I$  contains the degree  $n$  part of each of its element for any  $n$ , then clearly it is generated by homogenous elements. Conversely, if it is generated by homogenous elements, then any element  $f = \sum a_i f_i$ , where  $f_i$  is homogenous. Then we can see  $[f]_n = \sum [a_i]_{n-\deg f_i} f_i$  is also in  $I$ .

2: Use 1 and the definition. For radicals, we show  $\sqrt{I}$  contains all its homogenous parts: if  $f \in \sqrt{I}$ , then  $f^n \in I$  for some  $n$ , then we see that the minimal degree part  $[f]_m$  of  $f$  also satisfies  $[f]_m^n \in I$ , because  $I$  contains the homogenous parts of each of its elements. Then we can use induction to show that all the homogenous parts of  $f$  is in  $\sqrt{I}$ .

3: One direction is trivial, for the other, if  $a = \sum a_i, b = \sum b_i$  satisfies  $ab \in I$ , and  $a \notin I, b \notin I$ , and  $a_i, b_j$  homogenous. Let  $i_0, j_0$  be the minimal numbers that  $a_{i_0} \notin I, b_{j_0} \notin I$ , then  $a_{i_0} b_{j_0}$  is not in  $I$ , contradicting the fact  $I$  is homogenous.  $\square$

**Prop. (5.2.2.7).** Let  $R \rightarrow S$  be a homomorphism of graded rings, then the integral closure of  $R$  in  $S$  is a homogenous ideal of  $S$ .  $\lrcorner$

*Proof:* consider the base change  $\varphi : R \otimes_{R_0} R_0[t, t^{-1}] \rightarrow S \otimes_{S_0} S_0[t, t^{-1}]$ , where  $\deg(t) = 0$ , and the integral closure is denoted by  $A$ . Then there is an automorphism of  $\varphi: s \mapsto t^{\deg s} s$ . This automorphism thus preserves the integral closure. if  $s = s_n + s_{n+1} + \dots + s_m \in S$  is integral over  $R$ , to show each  $s_i$  are integral over  $S$ . We may assume  $n > 0$  because  $s_0$  is clearly integral over  $R_0$ . Now we use induction on  $m$ . If  $m > n$ , consider  $t^n s_n + \dots + t^m s_m$  is also in  $A$ , we see  $(t^m - t^n)s_i \in A$  by induction hypothesis.

Notice  $S \subset S[t, t^{-1}]/(t^m - t^n - 1) = S[t]/(t^m - t^n - 1)$  is injective, and the image of  $(t^m - t^n)s_i$  is  $s_i$ , which is integral over  $R[t]/(t^m - t^n - 1)$ , and this ring is finite over  $R$ , so  $s_i$  is also integral over  $R$ .  $\square$

**Prop. (5.2.2.8).** Let  $A$  be a graded ring that is f.g. over  $A_0$  and  $M$  be a f.g. graded  $A$ -module, then each  $M_n$  is a finite  $S_0$ -module.  $\lrcorner$

**Def. (5.2.2.9) [Reductions of Graded Rings].** Let  $S$  be a graded ring and  $d \geq 1$ , define  $S^{(d)}$  the graded ring  $\oplus_{n \geq 0} S_{nd}$ . And for a graded  $S$ -module  $M$ , define  $M^{(d)} = \oplus_{n \in \mathbb{Z}} M_{nd}$ .  $\lrcorner$

**Prop. (5.2.2.10).** If  $S$  is a graded ring that is f.g. over  $S_0$ , then for  $d$  sufficiently divisible,  $S^{(d)}$  is generated by the degree 1 part over  $S_0$ .  $\lrcorner$

*Proof:* Let  $S$  be generated by homogenous elements  $f_1, \dots, f_r$ . Let  $M = \text{lcm}(\deg(f_1), \dots, \deg(f_r))$ , then for any  $f \in S_{Nm}$  and  $N \geq r$ , by pigeonhole principle, there are some  $i$  s.t.  $f_i^{M/\deg(f_i)} | f$ . Thus any monomials of degree  $nrM$  is a product of polynomials of degree  $rM$ . So  $d = rM$  satisfies the hypothesis.  $\square$

### Topology Defined by Ideals

**Def. (5.2.2.11) [Filtrations on Graded Modules].** Let  $A$  be a ring,  $M$  a graded  $A$ -module, and  $\mathfrak{a}$  be an ideal of  $A$ , an  $\mathfrak{a}$ -filtration of  $M$  is a descending sequence of submodules  $M = M_0 \supset M_1 \supset \dots$  that  $\mathfrak{a}M_n \subset M_{n+1}$ . It is called a **stable filtration** iff there is an  $N$  that  $\mathfrak{a}M_n = M_{n+1}$  for  $n \geq N$ .

For an ideal  $\mathfrak{a} \subset A$ , there can be associated a graded ring  $A^* = \oplus \mathfrak{a}^n$ , and an  $\mathfrak{a}$ -filtration  $M$  can be associated a graded module over  $A^* : M^* = \oplus M_n$ . When  $A$  is Noetherian, then so is  $A^*$ , because it is a quotient of a polynomial ring over  $A$  (5.1.1.43).  $\lrcorner$

**Lemma (5.2.2.12).** If  $A$  is a Noetherian ring and  $M$  is a f.g.  $A$ -module that has a  $\mathfrak{a}$ -filtration  $M_n$ , then  $M^*$  is f.g. over  $A^*$  iff  $M_n$  is a stable filtration.  $\lrcorner$

*Proof:* As every  $M_n$  is finite over  $A$ , if it is stable, then  $M^*$  is generated over  $A^*$  by all the generator of  $M_n, n \leq N$ , so it is f.g.. Conversely, if it is f.g., then it is clear that  $M_n$  is a stable filtration.  $\square$

**Prop. (5.2.2.13) [Artin-Rees].** For  $A$  Noetherian and  $I$  an ideal, let  $N \subset M$  be finite  $A$ -modules, then if  $M_n$  is a stable filtration of  $M$ , then  $M_n \cap N$  is a stable filtration of  $N$ .

In particular, let  $M_n = I^n M$ , then  $I^n M \cap N = I^{n-r}(I^r M \cap N)$ , hence the  $I$ -adic topology on  $M$  induce the  $I$ -adic topology on  $N$ .  $\lrcorner$

*Proof:* This is immediate from the lemma above, as  $N^*$  is an  $A^*$ -submodule of  $M^*$ , and  $A^*$  is Noetherian (5.2.2.11).  $\square$

**Cor. (5.2.2.14) [Krull's Intersection Theorem].** Notation as in (5.2.2.13), let  $N = \bigcap_{n=0}^{\infty} I^n M$ , then the  $I$ -adic topology on  $N$  is trivial, by Artin-Rees, thus  $IN = N$ . So Nakayama tells us there is an element  $a \in 1 + I$  that  $aN = 0$ . Thus if  $I \subset \text{rad}(A)$  or  $A$  is an integral domain,  $N = 0$ . This can be used to use induction to prove some theorem.

In particular, for any prime ideal  $\mathfrak{p}$  containing  $I$ , use the above on  $R_{\mathfrak{p}}$  shows  $N_{\mathfrak{p}} = 0$ . But also  $N$  is f.g., so there exists an element  $g \notin \mathfrak{p}$  that  $N_g = 0$ .  $\lrcorner$

**Cor. (5.2.2.15) [Krull].** For  $A$  Noetherian, if  $I \subset \text{rad}(A)$  or  $A$  is a domain, then  $\bigcap_{n=0}^{\infty} I^n = 0$ .  $\lrcorner$

**Prop. (5.2.2.16).** Notice for any ring  $A$  and a non-zero-divisor  $f$ , if  $I = \bigcap_{n=0}^{\infty} f^n A$ , then  $fI = I$ , needless of the Noetherian property.  $\lrcorner$

*Proof:* If  $x \in I$ ,  $x = fy$ , because  $x \in f^n A$ ,  $fy = f^n t$  for some  $t$ , so  $y = f^{n-1}t$ , so  $f \in I$ . Thus  $I = fI$ .  $\square$

**Def. (5.2.2.17) [Hilbert-Serre].** Let  $A$  be a Noetherian graded ring with  $A_0$  Artinian that  $A_+$  is generated by  $A_1$ . For a f.g. graded  $A$ -module  $M = \bigoplus M_n$ , we have  $l(M_n)$  is a numerical polynomial of  $n$  (2.6.1.14) for  $n$  sufficiently large, called the **Hilbert Polynomial**. Its degree is the dimension of  $\text{Supp } M \subset \text{Proj}(A)$ .  $\lrcorner$

*Proof:* We prove by induction on the minimal number of generators of  $A_1$  (it is finite by (5.2.2.4)). If it is 0, then  $M_n = 0$  for  $n$  large and the result holds. Now choose  $x \in S_1$  as one of the minimal set of generators, then the induction hypothesis applies to  $S/(x)$ .

Firstly, if  $x$  acts nilpotently on  $M$ , then we do induction on the minimal number  $r$  that  $x^r M = 0$ . If  $r = 1$ , then  $M$  is a module over  $S/(x)$  and the assertion holds. If  $r > 1$ , then we can find an exact sequence  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  that  $M', M''$  has smaller  $r$ , then we have the desired result, because  $l$  is additive.

Next, if  $x$  doesn't act nilpotently on  $M$ , let  $M' \subset M$  is the largest submodule that  $x$  acts nilpotently, then there is an exact sequence  $0 \rightarrow M' \rightarrow M \rightarrow M/M' \rightarrow 0$ . So we can assume multiplication by  $x$  is injective on  $M$ .

Let  $\overline{M} = M/xM$ , then for any  $d$ , there are exact sequences

$$0 \rightarrow M_d \xrightarrow{x} M_{d+1} \rightarrow \overline{M}_{d+1}.$$

so  $l(M_{d+1}) - l(M_d) = l(\overline{M}_{d+1})$ . Then we finish by (2.6.1.15).  $\square$

**Cor. (5.2.2.18).** Let  $k$  be a field,  $I \subset k[X_1, \dots, X_n]$  be a non-zero graded ideal, and  $M = k[X_1, \dots, X_n]/I$ , then the numerical polynomial  $n \mapsto \dim_k(M_n)$  has degree  $< d - 1$ .  $\lrcorner$

*Proof:* The numerical polynomial associated to  $k[X_1, \dots, X_n]$  is  $n \mapsto \binom{n-1+d}{d-1}$ , and for any non-zero homogenous element  $f \in I$  of degree  $e$ ,  $f \cdot k[X_1, \dots, X_n]_{d-e} \subset I_d$ , thus  $\dim_k(M_n) < \binom{n-1+d}{d-1} - \binom{n-e-1+d}{d-1}$ , which means the numerical polynomial has degree  $< d - 1$ .  $\square$

**Prop. (5.2.2.19) [Hilbert Polynomial and Dimension].** For a Noetherian local ring  $A$ , the Hilbert polynomial of a f.g. module  $M$  w.r.t  $\mathfrak{m}$  has degree  $\dim M$ . And  $\dim M$  is the smallest integer  $r$  s.t. there exists  $x_1, \dots, x_r$  that  $l(M/x_1 M + \dots, x_r M) < \infty$ .  $\lrcorner$

*Proof:* Cf. [Mat P76].  $\square$

### 3 Completions

This subsection should be combined with the derived completion.

**Prop. (5.2.3.1).** Let the topology on a  $A$ -module be defined by countable filtration of submodules, then iff  $M$  is complete, then  $M/N$  is complete in the quotient topology.  $\lrcorner$

*Proof:* Write  $x_{i+1} - x_i = y_i + z_i$  with  $y_n \in M_n$  and  $z_n \in N$ , then the image of the limit of  $\sum y_i$  is the limit of  $\overline{x_i}$ .  $\square$

**Def. (5.2.3.2) [Completeness].** Let  $I$  be an ideal of  $R$ , the  $I$ -adic completion of a  $R$ -module is a functor  $\varphi : M \mapsto \widehat{M} = \lim M/I^n M$ . An  $R$ -module is called  $I$ -adically complete if the natural map  $M \rightarrow \widehat{M}$  is an isomorphism.

This is compatible with the general notion of completion of a topological Abelian groups (12.2.1.7).  $\lrcorner$

**Prop. (5.2.3.3).** Let  $R$  be a ring and  $I \subset R$  be an ideal,  $\varphi : M \rightarrow N$  be a map of  $R$ -modules. Then

- If  $M/IM \rightarrow N/IN$  is surjective, then  $\widehat{M} \rightarrow \widehat{N}$  is surjective. In particular, this holds for  $M \rightarrow N$  surjective.
- If  $0 \rightarrow K \rightarrow M \rightarrow N \rightarrow 0$  is exact and  $N$  is flat, then  $0 \rightarrow \widehat{K} \rightarrow \widehat{M} \rightarrow \widehat{N} \rightarrow 0$  is exact.
- $M \otimes_R \widehat{R} \rightarrow \widehat{M}$  is surjective for any finite  $R$ -module  $M$ .

$\lrcorner$

*Proof:* Cf. [Sta]0315.  $\square$

**Prop. (5.2.3.4).** Let  $I \in \text{Ideal}(A)$  be a f.g. ideal and  $M \in \text{Mod}_A$ , then  $\widehat{M}$  is  $I$ -adically complete and  $I^n \widehat{M} = \ker(\widehat{M} \rightarrow M/I^n M) = (I^n M)^\wedge$ . In particular,  $R^\wedge/I^n R^\wedge \cong R/I^n$ .  $\lrcorner$

*Proof:* Because  $I$  is f.g., so does  $I^n$ . If  $I^n = (f_1, \dots, f_r)$ . Applying (5.2.3.3) to  $(f_1, \dots, f_r) : M^r \rightarrow I^n M$  shows

$$\widehat{M}^r \rightarrow (I^n M)^\wedge = \varprojlim_{m \geq n} I^n M/I^m M = \ker(\widehat{M} \rightarrow M/I^n M),$$

but the image is clearly  $I^n \widehat{M}$ , so  $\widehat{M}/I^n \widehat{M} \cong M/I^n M$ . Taking inverse limit yields  $(M^\wedge)^\wedge = M^\wedge$ .  $\square$

**Cor. (5.2.3.5) [F.g. Completions are Complete].** Let  $I \in \text{Ideal}(A)$  is f.g. and  $(M_n)$  an inverse system of  $A$ -modules that  $I^n M_n = 0$ , then  $M = \lim M_n$  is  $I$ -adically complete.  $\lrcorner$

*Proof:* We have maps  $M \rightarrow M/I^n \rightarrow M_n$ , taking limit, we get  $M \rightarrow \widehat{M} \rightarrow M$ , so  $M$  is a direct summand of  $\widehat{M}$ . Since  $\widehat{M}$  is  $I$ -adically complete by (5.2.3.4), so does  $M$ .  $\square$

**Prop. (5.2.3.6).** If  $I$  is a f.g. ideal of  $A$  and  $(M_n)_{n \in \mathbb{Z}_+}$  is an inverse system of  $A$ -modules s.t.  $M_n = M_{n+1}/I^n M_{n+1}$ , then  $M = \lim M_n$  is  $I$ -adically complete and  $M/I^n M = M_n$ .  $\lrcorner$

*Proof:*  $\widehat{M}$  is  $I$ -adically complete by (5.2.3.5), and  $M \rightarrow M_n$  are all surjective because the transition maps are surjective. Consider the inverse system  $N_n = \ker(M \rightarrow M_n)$ . Since  $M_n = M_{n+1}/I^n M_{n+1}$ , the map  $N_{n+1} + I^n M \rightarrow N_n$  is surjective, and thus  $N_{n+1}/(N_{n+1} \cap I^{n+1} M) \rightarrow N_n/(N_n \cap I^n M)$  is surjective.

Taking the inverse limit of the exact sequences

$$0 \rightarrow N_n/(N_n \cap I^n M) \rightarrow M/I^n M \rightarrow M_n \rightarrow 0,$$

we get an exact sequence

$$0 \rightarrow \varprojlim N_n / (N_n \cap I^n M) \rightarrow \widehat{M} \rightarrow M.$$

As  $M$  is  $I$ -adically complete,  $\widehat{M} = M$ , thus  $\varprojlim N_n / (N_n \cap I^n M) = 0$ , thus  $N_n / (N_n \cap I^n M) = 0$  for any  $n$  as  $n$  the transition maps are surjective. Then  $M/I^n M = M_n$ , as desired.  $\square$

**Cor. (5.2.3.7) [Spectrum Map of Completions].**  $\text{Spec } R^\wedge \rightarrow \text{Spec } R$  has image  $\text{Spec } R/I \subset \text{Spec } R$ . This follows from (5.2.3.16) and  $R/I \cong R^\wedge/I$ .  $\lrcorner$

**Prop. (5.2.3.8).** If  $I$  is an ideal of  $R$  and  $0 \rightarrow M \rightarrow N \rightarrow Q \rightarrow 0$  is an exact sequence that  $Q$  is annihilated by a power of  $I$ , then completion produces an exact sequence

$$0 \rightarrow \widehat{M} \rightarrow \widehat{N} \rightarrow Q \rightarrow 0$$

$\lrcorner$

*Proof:* If  $I^c Q = 0$ , then  $Q/I^n Q = Q$  for  $n \geq c$ , and  $I^n M \subset M \cap I^n N \subset I^{n-c} M$  because of this. Then  $\widehat{M} = \varprojlim M / (M \cap I^n N)$  by (4.1.1.42), and we apply (5.9.3.2) to the inverse system of exact sequences

$$0 \rightarrow M / (M \cap I^n N) \rightarrow N / I^n N \rightarrow Q \rightarrow 0$$

to conclude.  $\square$

**Cor. (5.2.3.9).** If  $t \in A$  is a non-zero-divisor and  $0 \rightarrow M \rightarrow N \rightarrow Q \rightarrow 0 \in \text{Mod}_A$  is an exact sequence s.t.  $tQ = 0$ , then  $M$  is  $t$ -adically complete iff  $N$  is  $t$ -adically complete.  $\lrcorner$

*Proof:* Use snake lemma.  $\square$

**Cor. (5.2.3.10) [Non-zero-divisors].** Take  $I = (f)$  and  $M = N = R$ , then we see that if  $f$  is a nonzero-divisor in  $R$  then  $f$  is a nonzero-divisor in  $\widehat{R}$ .  $\lrcorner$

**Prop. (5.2.3.11).** The completion of a submodule  $N \subset M$  is the closure of  $\varphi(N)$  (By direct construction). The completion of  $M/N$  is  $M^*/N^*$  because it is right exact.  $\lrcorner$

**Cor. (5.2.3.12).** If  $N$  is open in  $M$  then  $M/N \cong M^\wedge/N^\wedge$  because  $M/N$  is discrete hence complete.  $\lrcorner$

**Lemma (5.2.3.13).** Let  $A$  be a ring and  $I = (f_1, \dots, f_r)$  be a f.g. ideal. If  $M \rightarrow \varprojlim M/f_i^n M$  is surjective for each  $i$ , then  $M \rightarrow \varprojlim M/I^n M$  is also surjective.  $\lrcorner$

*Proof:* Note that  $\varprojlim M/I^n M = \varprojlim M/(f_1^n, \dots, f_r^n)M$ , as  $I^n \subset (f_1^n, \dots, f_r^n) \subset I^n$ , and elements in  $\varprojlim M/(f_1^n, \dots, f_r^n)M$  can be written as an infinite sum  $\xi = \sum_n \sum_i f_i^n x_{n,i}$ . There is an element  $x_i$  mapping to  $\sum_n f_i^n x_{n,i}$  for any  $i$ , thus  $\sum_i x_i$  maps to  $\xi$ .  $\square$

**Lemma (5.2.3.14).** For  $I \subset J \in \text{Ideal}(A)$ , if  $M$  is  $J$ -adically complete and  $I$  is f.g., then  $M$  is  $I$ -adically complete.  $\lrcorner$

*Proof:* It is clearly  $I$ -adically Hausdorff, and for completeness, by (5.2.3.13) it suffices to show for  $I = (f)$ : Let  $x_n \in M$  with  $x_n - x_{n+1} \in f^n M$ , then  $\{x_n\}$  is  $J$ -adically Cauchy, thus there is an element  $x$  that  $x - x_n \in J^n$ , and we can replace  $x_n$  by  $x_n - x$  to assume  $x_n \in J^n$ . Now we prove  $x_n \in (f^n)$ : assume  $x_n - x_{n+1} = f^n z_n$ , then

$$x_n = f^n(z_n + f z_{n+1} + \dots).$$

This equation is true because it is  $J$ -adically Cauchy.  $\square$



**Prop. (5.2.3.15).** If  $(A, \mathfrak{m}_A) \rightarrow (B, \mathfrak{m}_B)$  is a local homomorphism of local rings s.t.  $\mathfrak{m}_A, \mathfrak{m}_B$  are f.g. and  $\dim S \otimes \kappa(\mathfrak{m}) < \infty$ , then

- $\widehat{B}$  is equal to the  $\mathfrak{m}$ -adic completion of  $B$ .
- $\widehat{B}$  is a finite module over  $\widehat{A}$ .

┘

*Proof:* 1: Because  $B/\mathfrak{m}_A B$  is Artinian local (5.1.3.1), we have  $\mathfrak{m}_B \subset \sqrt{\mathfrak{m}_A B}$  because Artinian rings are Jacobson (5.1.3.4). Because  $\mathfrak{m}_B$  is f.g., we have  $\mathfrak{m}_B^n \subset \mathfrak{m}_A B$  for some  $n \in \mathbb{Z}_+$ . So the assertion follows.

2: This follows from (5.2.3.5) and (5.2.3.17). □

### Properties of Complete Rings

**Prop. (5.2.3.16).** If  $A$  is  $I$ -adically complete, then  $I \subset \text{rad } A$ . ┘

**Prop. (5.2.3.17).** Let  $I \in \text{Ideal}(A)$  s.t.  $A$  is  $I$ -adically complete and  $M \in \text{Mod}_A$ . Suppose  $M$  is  $I$ -adically separated and  $M/IM$  is a finite  $A/I$ -module, then  $M$  is a finite  $A$ -module. ┘

*Proof:* Cf. [Sta]031D. □

**Prop. (5.2.3.18).** Let  $A$  be a ring with a non-zero-divisor  $t$ , then any limit of  $t$ -adically complete algebras is  $t$ -adically complete. ┘

*Proof:* Check the definition directly. □

**Prop. (5.2.3.19) [Zariski Rings].** A Noetherian  $I$ -adic ring is called **Zariski ring** if it satisfies the following equivalent conditions:

- Every finite module is Hausdorff in the  $I$ -adic topology.
- Every submodule in a finite module is closed in the  $I$ -adic topology.
- Every ideal is closed.
- $I \subset \text{rad } A$ .
- $A^\wedge/A$  is f.f.

Hence every complete Hausdorff ring is Zariski. ┘

*Proof:* 1  $\rightarrow$  2: Apply it to the submodule  $M/N$ .

3  $\rightarrow$  4: If  $I \not\subset m$ , then  $I^n + m = A$ , thus  $\overline{M} = A$ , contradiction.

4  $\rightarrow$  1: by intersection theorem (5.2.2.14).

4  $\rightarrow$  5: for any maximal ideal  $m$ ,  $I \subset m$  so it is open, thus  $A^*/mA^* = A/m \neq 0$  by (5.2.3.12) thus f.f. by (5.4.1.5).

5  $\rightarrow$  1: by (5.4.1.20), for any  $m$  maximal, there is a maximal ideal  $m'$  lying over  $m$ , so  $IA^* \subset m'^*$  by (5.2.3.24), thus  $I \subset m$ , hence  $I \subset \text{rad } A$ . □

**Cor. (5.2.3.20).** In a Zariski ring  $A$ , maximal ideals are open, thus  $A/m \cong A^*/mA^*$  by (5.2.3.12), thus  $\text{Spec } A^* \rightarrow \text{Spec } A$  is bijection on closed pt. ┘

**Prop. (5.2.3.21) [Cohen Structure Theorem].** If  $A$  is a complete local ring containing a field  $k$  that the residue field is separably generated over  $k$ , then there is a field  $K$  containing  $k$  that is a Cohen ring, i.e. complete local ring with a prime number as a uniformizer, that has the same residue field as  $A$ . ┘



*Proof:*

□

**Lemma (5.2.3.22) [Complete Interchanging Lemma].** If  $R$  is a commutative ring,  $x, y \in R$ , if  $x$  is not a zero-divisor in  $R$  and  $R$  is  $x$ -adically complete, and  $y$  is not a zero-divisor in  $R/x$  and  $R/x$  is  $y$ -adically complete, then the same is true with  $x, y$  interchanged.  $\lrcorner$

*Proof:* ?

□

### Completion of Noetherian Rings

**Prop. (5.2.3.23).** When  $A$  is Noetherian and  $M \in \text{Mod}_A$  is finite, the natural map  $M \otimes_A A^\wedge \rightarrow M^\wedge$  is an isomorphism (use  $M$  is finite presentation and tensor & completion is right exact), and five lemma.  $\lrcorner$

**Cor. (5.2.3.24) [Flatness of Noetherian Completions].** When  $A$  is Noetherian,  $A \rightarrow A^\wedge$  is flat (because flatness is check for finite module).

And when  $A$  is complete Hausdorff, any finite module  $M \in \text{Mod}_A$  is complete Hausdorff and hence any its submodule is complete thus closed in it. Hence the the completion of a submodule  $N \subset M$  is  $\varphi(N)A^\wedge \subset M^\wedge = MA^\wedge$ . In fact this implies complete Hausdorff adic-ring is Zariski.  $\lrcorner$

**Remark (5.2.3.25).** WARNING: If  $A$  is not Noetherian, in general  $A \rightarrow A^\wedge$  is not flat, Cf. [Sta]0AL8.  $\lrcorner$

**Prop. (5.2.3.26).** Let  $A$  be Noetherian and  $I, J \in \text{Ideal}(A)$ . If  $A$  is  $I$ -adically complete and  $A/I$  is  $J$ -adically complete, then  $A$  is  $J$ -adically complete.  $\lrcorner$

*Proof:* Cf. [Sta]0DYC.

□

## 4 Dimension

**Def. (5.2.4.1) [Dimensions and Heights].** For a  $A$ -module  $M$ ,  $\dim(M)$  is defined as  $\dim(A/\text{Ann}(M))$ .

The **height** of an ideal  $I$  in  $A$  is defined as the infimum of heights of the prime ideals over  $I$ .

The **dimension (Commutative Rings)** of  $R$  is defined to be the supremum of heights of prime ideals of  $R$ .  $\lrcorner$

**Prop. (5.2.4.2).**  $\dim A = \sup \dim A_{\mathfrak{p}}$ .  $\lrcorner$

**Def. (5.2.4.3) [Catenary Rings].**  $A$  is called **catenary ring** if for any pair of primes  $\mathfrak{p} \subset \mathfrak{q} \subset A$ , any maximal chain of primes  $\mathfrak{p} = \mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \dots \subset \mathfrak{p}_e = \mathfrak{q}$  has the same length. A Noetherian ring is called **universally catenary ring** if all f.g. algebras over it are catenary.  $\lrcorner$

**Prop. (5.2.4.4).**  $A$  is catenary iff  $\text{Spec } A$  is a catenary space (4.12.2.32).  $\lrcorner$

**Prop. (5.2.4.5).** Any quotient ring and localization of a (universally)catenary ring is catenary. Any quotient of a Noetherian universally catenary ring is universally catenary. Catenary and universally catenary are stalkwise properties (5.1.4.2).  $\lrcorner$

**Prop. (5.2.4.6).** If  $(A, \mathfrak{m})$  is a Noetherian local ring, then  $A$  is catenary iff  $\mathfrak{p} \rightarrow \dim(A/\mathfrak{p})$  is a dimension function on  $\text{Spec } A$ .  $\lrcorner$

*Proof:* This follows from (4.12.3.16) (4.12.2.41) and (4.12.2.35).  $\lrcorner$

□

**Example (5.2.4.7) [Universally Catenary Rings].** The following are same examples of universally catenary Rings:

- A f.g. algebra over a universally catenary ring.
- A Noetherian C.M. rings.
- 1-dimensional Noetherian domains,
- Fields.

┘

*Proof:* Cf. [Sta]00NM. ?

□

**Def. (5.2.4.8) [Hilbert Polynomials].** Let  $(A, \mathfrak{m})$  be a Noetherian local ring,  $I$  an ideal of definition, then

┘

### Dimension of Noetherian Local Rings

**Prop. (5.2.4.9).** For a Noetherian local ring  $R$ , the following three numbers are equal:

- $\dim R$ .
- $d(R)$ .
- the minimal number of elements needed to generate an ideal of definition of  $R$  (5.1.1.13).

A **system of parameters** is  $d$  elements  $g_1, \dots, g_d$  that generate an ideal of definition of  $R$ , where  $d = \dim R$ .

┘

*Proof:* Cf. [Sta]00KQ. ?

□

**Cor. (5.2.4.10).** The dimension of a Noetherian local ring is finite, by item3. Thus the codimension of a subscheme in a Noetherian scheme is finite.

┘

**Cor. (5.2.4.11).** If  $A$  is a Noetherian local ring with maximal ideal  $\mathfrak{m}$ , then  $\dim A \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$ .

┘

*Proof:* By Nakayama, if  $x_1, \dots, x_d \in \mathfrak{m}$  generate  $\mathfrak{m}/\mathfrak{m}^2$ , then  $(x_1, \dots, x_d) = \mathfrak{m}$ .

□

### Dimension and Ring Extensions

**Prop. (5.2.4.12) [Dimension and Going-Up(Down)].** If  $A \rightarrow B$  is a ring map that Spec map is surjective and  $A \rightarrow B$  satisfies either going-up or going-down, then  $\dim B \geq \dim A$ .

┘

*Proof:* The hypothesis implies any chain of primes in  $A$  can lifted to a chain of primes in  $B$ . □

**Prop. (5.2.4.13) [Dimensions and Noetherian Ring Extensions].** Let  $A \rightarrow B$  be a map between Noetherian rings,  $P$  a prime ideal of  $B$ ,  $\mathfrak{p} = P \cap A$ , then:

- $\text{ht}(P) \leq \text{ht}(\mathfrak{p}) + \text{ht}(P/\mathfrak{p}B)$ , in other words  $\dim(B_P) \leq \dim(A_{\mathfrak{p}}) + \dim(B_P/\mathfrak{p}B_P)$ .
- equality holds if going-down holds. For example, if it is flat (5.1.7.13).

┘

*Proof:* 1: Localize at  $\mathfrak{p}$  and  $P$ , we may assume  $A, B$  are local rings with maximal ideals  $\mathfrak{p}$  and  $P$ , then use the characterization (5.2.4.9) of dimension, if  $x_1, \dots, x_d$  generate an ideal of definition of  $A_{\mathfrak{p}}$  and  $y_1, \dots, y_e$  generate an ideal of definition of  $B_P/\mathfrak{p}B_P$ , then  $x_1, \dots, x_d, y_1, \dots, y_e$  generate an ideal of definition of  $B_P$ .

2: If going down holds, for any chain of primes in  $B_P$  containing  $\mathfrak{p}B_P$ , we can lift the chain of primes in  $A_{\mathfrak{p}}$  to a chain of primes in  $B_P$  to get a longer chain, thus we get the other direction of inequality.  $\square$

**Prop. (5.2.4.14) [Dimension of Integral Extensions].** Let  $A \rightarrow B$  be an integral ring map, then:

1. Spec maps closed points to closed points, and  $\dim(A) \geq \dim(B)$ , which equality if  $A \subset B$ .
2. If  $A, B$  is Noetherian,  $\text{ht}(P) \leq \text{ht}(P \cap A)$
3. If  $A, B$  is Noetherian and going down holds, then  $\text{ht}(J) = \text{ht}(J \cap A)$  for any ideal  $J \subset B$ .  $\lrcorner$

*Proof:* 1: By (5.2.1.5), there is no inclusion relation between prime over a fixed prime, so  $\dim(B) \leq \dim(A)$ . On the other hand, if  $A \subset B$ , then Spec map is surjective going-up holds (5.2.1.5), so  $\dim(B) \geq \dim(A)$  (5.2.4.12).

2: Follows from (5.2.4.13)(1) since  $\text{ht}(P/(P \cap A)B) = 0$  by (5.2.1.5).

3: In this case,  $\text{ht}(P) = \text{ht}(P \cap A)$  holds by (5.2.4.13)(2), then use the surjectiveness of Spec for the integral extension  $A/J \cap A \subset B/J$  (5.2.1.5).  $\square$

**Cor. (5.2.4.15).** if  $A \rightarrow B$  is integral and faithfully flat, then  $\dim A = \dim B$ .  $\lrcorner$

*Proof:* This follows from (5.2.4.14) and (5.4.1.27).  $\square$

**Prop. (5.2.4.16) [Dimension and Completion].** For a local ring  $A$ ,  $\dim A = \dim \hat{A}$ .  $\lrcorner$

*Proof:*  $\square$

### Noetherian Normalization

**Prop. (5.2.4.17).** For a Noetherian ring  $A$ ,  $\dim A[X] = \dim A + 1$ .  $\lrcorner$

*Proof:* Let  $\mathfrak{p}$  be a prime ideal of  $A$  and let  $\mathfrak{q}$  be a prime ideal of  $A[X]$  maximal among primes lying over  $\mathfrak{p}$ , then  $\text{ht}(\mathfrak{q}/\mathfrak{p}A[X]) = 1$ . In fact, by localizing, we can assume  $\mathfrak{p}$  is a maximal ideal, then  $A[x]/\mathfrak{p}A[x]$  is a polynomial ring over a field thus a PID and  $\text{ht}(\mathfrak{q}/\mathfrak{p}A[X]) = 1$ . Thus  $\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{p}) + 1$  by (5.2.4.13). Now we are done, because  $\text{Spec } A[X] \rightarrow \text{Spec } A$  is surjective.  $\square$

**Prop. (5.2.4.18) [Krull's Height Theorem].** In a Noetherian domain  $R$ , the height of an ideal generated by  $n$  elements is at most  $n$ .  $\lrcorner$

*Proof:* Let  $\mathfrak{p}$  be a minimal ideal containing  $(f_1, \dots, f_n)$ , then it suffices to show  $\dim(R_{\mathfrak{p}}) \leq n$ . In this case,  $(f_1, \dots, f_n)$  is an ideal of definition of  $R_{\mathfrak{p}}$ , thus we can use (5.2.4.9).  $\square$

**Prop. (5.2.4.19) [Number of Generators].** To show an ideal  $I \subset A$  cannot be generated by smaller than  $n$  element, choose a maximal ideal  $\mathfrak{m}$ , then show that  $\dim_{A/\mathfrak{m}} I/\mathfrak{m}I \geq n$ .  $\lrcorner$

**Cor. (5.2.4.20).**

- $(x, z)$  is not principal in  $k[x, y, z]$ .
- $(wz - xy)$  is not principal in  $k[x, y, z]$ .
- $(xy, yz, xz)$  is not generated by two elements in  $k[x, y, z]$ .  $\lrcorner$

**Lemma (5.2.4.21).** Let  $A = k[X_1, \dots, X_n]$  be a polynomial ring over a field  $k$ , and  $I$  is an ideal of  $A$  of height  $r$ , then we can choose  $Y_1, \dots, Y_n \in A$  that  $A$  is integral over  $k[Y_1, \dots, Y_n]$  and  $I \cap k[Y_1, \dots, Y_n] = (Y_1, \dots, Y_r)$ .  $\lrcorner$

*Proof:* We use induction on  $r$ .  $r = 0$  is easy. For  $r = 1$ , let  $f(X)$  be any non-zero polynomial in  $I$ , then we can assign suitable integral weights  $d_1 = 1, d_2, \dots, d_n$  to  $X_i$  that monomials of  $f$  have different weights. Put  $Y_i = X_i - X_1^{d_i}$  for  $i \geq 2$ , and

$$Y_1 = f(X) = f(X_1, Y_2 + X_1^{d_2}, \dots, Y_n + X_1^{d_n}) = a_1 X_1^N + g(X_1, Y_2, \dots, Y_n)$$

where  $g$  has degree in  $X_1$  lower than  $N$ . Then  $X_1$  is integral over  $k[Y_1, \dots, Y_n]$ , and hence  $X_i = Y_i + X_1^{d_i}$  is also integral over  $k[Y]$ .

Now  $(Y_1)$  is a prime ideal in  $k[Y]$  of height 1 and  $(Y_1) \in I \cap k[Y]$ . Also notice  $I \cap k[Y]$  has height 1 by (5.2.4.14) (Because going-down holds by (5.1.7.17)), so  $(Y_1) = I \cap k[Y]$ .

For  $r \geq 2$ , let  $J \subset I$  be an ideal with height  $r - 1$ , let  $J$  be an ideal of  $k[X]$  contained in  $I$  that  $\text{ht}(J) = r - 1$ . (This is possible by choosing  $f_i$  out of all minimal primes containing  $(f_1, \dots, f_{r-1})$  and use Krull's Height Theorem). By induction hypothesis, there exists  $Z_1, \dots, Z_n$  that  $k[X]$  is integral over  $k[Z]$ , and  $J \cap k[Z] = (Z_1, \dots, Z_{r-1})$ . Now  $\text{ht}(I \cap k[Z]) = r$  by the same argument above, thus there exist  $f \in I \cap k[Z] \setminus (Z_1, \dots, Z_{r-1})$ , and do the same for  $r = 1$  again, we can find the desired  $Y_i$  that  $Y_k = Z_k$  for  $k \leq r - 1$ .  $\square$

**Prop. (5.2.4.22) [Noetherian Normalization Theorem].** If  $k \in \mathbf{Field}$  and  $A \in \mathcal{CRing}/k$  is f.g. over  $k$ . then there are  $r$  alg. independent elements  $y_i$  that  $A$  is integral over  $k[y_i]$ .  $\lrcorner$

*Proof:* Let  $A = k[X_1, \dots, X_n]/I$  and  $\text{height}(I) = n - r$ , then by the lemma we can choose  $Y_1, \dots, Y_n$  that  $k[X_1, \dots, X_n]$  is integral over  $k[Y_1, \dots, Y_n]$  (so  $Y_1, \dots, Y_n$  are algebraically independent) and  $I \cap k[Y_1, \dots, Y_n] = (Y_{r+1}, \dots, Y_n)$ . Now we can just choose  $y_i = Y_i$  for  $i \leq r$ .  $\square$

**Cor. (5.2.4.23) [Dimension and Transcendental Degree].** If  $A$  is a f.g. integral ring over a field  $k$ , then  $\dim A = \text{tr. deg}_k A$ .  $\lrcorner$

*Proof:* This is because integral extensions of integral Noetherian rings have the same dimensions (5.2.4.14) and their fraction fields have the same transcendental degrees.  $\square$

**Cor. (5.2.4.24).** Let  $A, B$  be f.g. algebras over a field  $k$ , then  $\dim(A \otimes_k B) = \dim A + \dim B$ .  $\lrcorner$

**Cor. (5.2.4.25) [Dimension and Field Base Change].** Let  $K/k$  be a field extension and  $S$  a f.g. algebra over  $k$ , then  $\dim S = \dim S \otimes_k K$ .  $\lrcorner$

*Proof:* By Noetherian normalization, there exists a finite injective map  $k[d_1, \dots, d_n] \rightarrow S$  where  $n = \dim S$ . Then there exists a finite injective map  $K[d_1, \dots, d_n] \rightarrow S_K$ , so  $\dim S \otimes_k K = n$ , by (5.2.4.14) and (5.2.4.17).  $\square$

**Prop. (5.2.4.26) [Codimensions and Field Base Change].** Let  $K/k$  be a field extension and  $S$  a f.g.  $k$ -algebra. Let  $\mathfrak{q}$  be a prime of  $S$  and  $\mathfrak{q}_K$  be a prime of  $S_K$  lying over  $\mathfrak{q}$ , then

$$\dim(S_K \otimes_S k(\mathfrak{q}))_{\mathfrak{q}_K} = \dim(S_K)_{\mathfrak{q}_K} - \dim S_{\mathfrak{q}} = \text{tr. deg}_k k(\mathfrak{q}) - \text{tr. deg}_K k(\mathfrak{q}_K).$$

Moreover, for any  $\mathfrak{q}$ , we can choose  $\mathfrak{q}_K$  so that this number is 0.  $\lrcorner$

*Proof:* Cf. [Sta]0CWE.  $\square$

### Local Dimension over Fields

**Prop. (5.2.4.27) [Local Dimension].** Let  $S$  be an algebra f.g. over a field  $k$ ,  $X = \operatorname{Spec} S$  and  $x \in X$ , then the following three numbers are equal:

- the local dimension (4.12.2.26)  $\dim_x(X)$ .
- $\max \dim(Z)$  where  $Z$  runs through irreducible components of  $X$  passing through  $x$ .
- $\min \dim(S_{\mathfrak{m}})$ , where  $\mathfrak{m}$  are maximal ideals containing  $\mathfrak{p}_x$ .

┘

*Proof:* Cf. [Sta]00OT. □

**Lemma (5.2.4.28).** Let  $k$  be a field and  $S$  a f.g.  $k$ -algebra,  $X = \operatorname{Spec} S$ ,  $x \in X$ , then

$$\dim_x(X) = \dim S_{\mathfrak{p}} + \operatorname{tr. deg}_k k(\mathfrak{p}).$$

┘

*Proof:* Cf. [Sta]00P1. □

**Cor. (5.2.4.29).** Let  $S' \rightarrow S$  be a surjection of f.g. algebras over a field  $k$ ,  $\mathfrak{p}$  a prime ideal of  $S$  and  $\mathfrak{p}'$  its inverse image in  $S'$ , corresponding to  $x, x'$  in  $X = \operatorname{Spec} S$ ,  $X' = \operatorname{Spec} S'$  resp., then

$$\dim_{x'}(X') - \dim_x(X) = \operatorname{ht}(\mathfrak{p}') - \operatorname{ht}(\mathfrak{p}).$$

┘

**Def. (5.2.4.30) [Relative Dimension].** Let  $R \rightarrow S$  be a ring map of f.t., and  $\mathfrak{q} \subset S$  be a prime over  $\mathfrak{p} \subset R$ , then we define the **relative dimension** of  $S/R$  at  $\mathfrak{q}$  to be  $\dim_{\mathfrak{q}}(\operatorname{Spec} S)_{\mathfrak{p}}$ . The supremum of all these numbers over  $\mathfrak{q} \subset \operatorname{Spec} S$  is called the relative dimension of  $S/R$ , denoted by  $\dim(S/R)$ . ┘

**Lemma (5.2.4.31) [Local Dimension and Field Extension].** Let  $K/k$  be a field extension,  $S$  be a f.g.  $k$ -algebra, and  $X = \operatorname{Spec} S$ . Now if  $\mathfrak{p}_K$  is an element of  $S_K$  lying over  $\mathfrak{p} \subset S$ , then  $\dim_{\mathfrak{p}}(S) = \dim_{\mathfrak{p}_K}(S_K)$ . ┘

*Proof:* The proof is by reduction to polynomial ring. Let  $S = k[X_1, \dots, X_n]/I$ , let  $\mathfrak{p}'_K, \mathfrak{p}'$  be primes of  $k[X_1, \dots, X_n]$  and  $K[X_1, \dots, X_n]$  that is the image of  $x$  and  $x_K$ , then there is a commutative diagram

$$\begin{array}{ccc} K[X_1, \dots, X_n]_{\mathfrak{q}'_K} & \longrightarrow & (S_K)_{\mathfrak{q}_K} \\ \downarrow & & \downarrow \\ k[X_1, \dots, X_n]_{\mathfrak{q}'} & \longrightarrow & S_{\mathfrak{q}} \end{array}.$$

The vertical arrows are flat because they are local morphisms of flat maps, and their fibers are the same, so by (7.6.3.17),  $\operatorname{ht}(\mathfrak{q}'_K) - \operatorname{ht}(\mathfrak{q}') = \operatorname{ht}(\mathfrak{q}_K) - \operatorname{ht}(\mathfrak{q})$ . Also use (5.2.4.29) on the horizontal maps, then we get the desired assertion. □

**Prop. (5.2.4.32) [Semicontinuity of Dimensions].** Let  $f : R \rightarrow S$  be a ring map of f.t., then the map  $\mathfrak{q} \mapsto \dim_{\mathfrak{q}}(S/R)$  is a upper-semicontinuous function on  $\operatorname{Spec}(S)$ .

Moreover, if  $f$  is of f.p., then the set  $\{\mathfrak{q} \mid \dim_{\mathfrak{q}}(S/R) \leq n\}$  is quasi-compact open in  $\operatorname{Spec}(S)$ . ┘

*Proof:* Cf. [Sta]00QH, 00QJ. □

## 5 Support and Associated Primes

**Def. (5.2.5.1) [Support of a Module].** The **support**  $\text{Supp}(M)$  of a module  $M \in \text{Mod}_R$  is the set of all  $\mathfrak{p} \in \text{Spec}(R)$  that  $M_{\mathfrak{p}} \neq 0$ . When  $M$  is f.g.,  $\text{Supp}(M) = V(\text{Ann}(M))$ .  $\lrcorner$

**Prop. (5.2.5.2) [Support is Non-Empty].** The support of a nonzero module is not empty, because triviality is stalkwise by (5.1.4.2).  $\lrcorner$

**Prop. (5.2.5.3).** If  $0 \rightarrow N \rightarrow M \rightarrow Q \rightarrow 0$ , then we have  $\text{Supp}(M) = \text{Supp}(N) \cup \text{Supp}(Q)$ , this is because localization is exact.  $\lrcorner$

**Cor. (5.2.5.4).**  $\lrcorner$

**Prop. (5.2.5.5).** Let  $(R, \mathfrak{m})$  be a Noetherian local ring and  $M$  a finite  $R$ -module,  $f \in \mathfrak{m}$ , then

$$\dim \text{Supp}(M) - 1 \leq \dim(\text{Supp}(M/fM)) \leq \dim(\text{Supp}(M))$$

$\lrcorner$

*Proof:* Cf. [Sta]0B52.  $\square$

**Prop. (5.2.5.6).** Let  $0 = M_0 \subset M_1 \subset \dots \subset M_n = M$  be a filtration of  $M$  that  $M_i/M_{i-1} \cong R/\mathfrak{p}_i$  where  $\mathfrak{p}_i$  are primes, by (5.1.1.48), then  $\text{Supp}(M) = \cup_i V(\mathfrak{p}_i)$ .

In particular, the minimal primes in  $\{\mathfrak{p}_i\}$  are the same as the minimal primes of  $\text{Supp}(M)$ . Moreover, the multiplicity of a prime  $\mathfrak{p}_i$  equals  $\text{length}_{R_{\mathfrak{p}}} M_{\mathfrak{p}}$ .  $\lrcorner$

*Proof:* Cf. [Sta]00L7.  $\square$

**Prop. (5.2.5.7).** Let  $(R, \mathfrak{m})$  be a Noetherian local ring and  $M$  a non-zero finite  $R$ -module, then  $\text{Supp}(M) = V(\mathfrak{m})$  iff  $\text{length}_R(M) < \infty$ .  $\lrcorner$

*Proof:* Cf. [Sta]00L5.  $\textcolor{red}{?}$   $\square$

**Prop. (5.2.5.8).** Let  $A$  be Noetherian and  $I$  be an ideal, then  $I^n M = 0$  for some  $n$  iff  $\text{Supp}(M) \subset V(I)$ .  $\lrcorner$

*Proof:* If  $I^n M = 0$ , then if  $I \not\subseteq P$ , then  $M_P = 0$ . Conversely, we have a filtration of  $M$ , and by (5.2.5.3) we have all the  $P_i$  include  $I$ , so  $I^n$  annihilate  $M$ .  $\square$

**Prop. (5.2.5.9).** If  $R$  is a ring and  $M$  is a f.p.  $R$ -module, then  $\text{Supp}(M)$  is a closed subset of  $\text{Spec } R$  whose complement is quasi-compact.  $\lrcorner$

*Proof:* Let  $R^m \rightarrow R^n \rightarrow M \rightarrow 0$ , then the support of  $M$  is just the locus that some minor of the linear map from  $R^m \rightarrow R^n$  doesn't vanish. Then its complement is quasi-compact.  $\square$

**Prop. (5.2.5.10).** Let  $(R, \mathfrak{m})$  be a Noetherian local ring and  $M$  is a finite  $R$ -module, then  $d(M) = \dim(\text{Supp}(M))$ .  $\lrcorner$

*Proof:* Cf. [Sta]00L8.  $\square$

**Prop. (5.2.5.11).** Let  $R$  be a Noetherian ring and  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  be an exact sequence of finite  $R$ -modules, then  $\dim \text{Supp}(M) = \max(\dim \text{Supp}(M'), \dim \text{Supp}(M''))$ .  $\lrcorner$

*Proof:* Cf. [Sta]0B51.  $\square$

### Associated Primes of a Module

**Def. (5.2.5.12) [Associated Primes of a Module].** The **weakly associated primes**  $\text{Ass}(M)$  of an  $A$ -module  $M$  is the set of minimal primes of  $A/\text{Ann}(m)$  for some  $m \in M$ .

The **associated primes**  $\text{Ass}(M)$  is the set of primes  $\{p = \text{Ann}(m)\}$  where  $m \in M$ .  $\lrcorner$

**Prop. (5.2.5.13).**  $\text{Ass}_R(M) \subset \text{WeakAss}_R(M)$ , and if  $R$  is Noetherian, the converse is also true.  $\lrcorner$

*Proof:* Cf. [Sta]058A.  $\square$

**Prop. (5.2.5.14) [Associated Primes and Exact Sequence].** For an exact sequence  $0 \rightarrow M_1 \rightarrow M \rightarrow M_2 \rightarrow 0$ ,  $\text{WeakAsso}(M) \subset \text{WeakAsso}(M_1) \cup \text{WeakAsso}(M_2)$  and  $\text{WeakAsso}(M_1) \subset \text{WeakAsso}(M)$ .

$\lrcorner$

*Proof:* Cf. [Sta]0548.  $\square$

**Cor. (5.2.5.15) [F.M. Associated Primes].** For a finite module  $M$  over a Noetherian ring  $A$ ,  $\text{Ass}_A(M)$  is finite by (5.1.1.48).  $\lrcorner$

**Prop. (5.2.5.16) [Associated Primes and Support].**  $\text{WeakAsso}(M) \subset \text{Supp } M$ , and their minimal elements are the same.

If  $M \neq 0$ ,  $\text{WeakAsso}(M) \neq \emptyset$  by (5.2.5.2), and  $\text{WeakAsso}(A/I)$  contains all the minimal primes over  $I$ .  $\lrcorner$

*Proof:* If  $\mathfrak{p} = \text{Ann}(m)$ , then  $m$  is nonzero in  $M_{\mathfrak{p}}$ , so  $M_{\mathfrak{p}}$  is nonzero, i.e.  $\mathfrak{p} \in \text{Supp}(M)$ .

For the second assertion, we first prove for  $M$  finite, and then write any module as sum of finite submodules, and use the fact  $\text{Supp}$  and  $\text{ass}$  are all unions of those of the submodules. Cf. [Sta]05C4 0588.  $\square$

**Prop. (5.2.5.17) [Weakly Associated Primes and Zero-divisors].** Let  $M$  a  $R$ -module, then the union of the weakly associated primes of  $M$  is the set of zero-divisors in  $M$ .  $\lrcorner$

*Proof:* Elements in associated points are zero-divisors obviously, and conversely, if  $xm = 0$ , then  $x \in \text{Ann}(m)$  and  $\text{Ann}(m)$  has an associated point  $\mathfrak{q}$  by (5.2.5.16). Now  $x$  must be in  $\mathfrak{q}$  and  $\mathfrak{q}$  is also an associated point of  $M$  by (5.2.5.16). Cf. [Sta]05C3.  $\square$

**Cor. (5.2.5.18).** Use the prime avoidance (5.1.1.5), we can prove if  $R$  is Noetherian and  $M$  is a finite  $R$ -module, then  $I \subset \mathfrak{p}$  for some  $\mathfrak{p} \in \text{WeakAss}(M)$  iff  $I$  consists of zero-divisors.  $\lrcorner$

**Prop. (5.2.5.19) [Associated Primes and Maps].** For a ring map  $\varphi : R \rightarrow S$  and a  $S$ -module  $M$ , then

$$\text{Spec}(\varphi)(\text{Ass}_S(M)) \subset \text{Ass}_R(M) \subset \text{WeakAss}_R(M) \subset \text{Spec}(\varphi)(\text{WeakAss}_S(M)).$$

Equalities hold if  $S$  is Noetherian. Also  $\text{WeakAss}_R(M) = \text{Spec}(\varphi)(\text{WeakAss}_S(M))$  if  $\varphi$  is a finite ring map.  $\lrcorner$

*Proof:* Cf. [Sta]05C7, 05E1.

We prove it is equal. If  $\mathfrak{p} = \text{Ann}_R(m)$ , then we let  $I = \text{Ann}_S(m)$ , then  $R/\mathfrak{p} \subset S/I \subset M$ , so by (5.1.7.25), there is a minimal prime of  $S$  over  $I$  that are mapped to  $\mathfrak{p}$ , now this prime is in  $\text{Ass}(S/I)$  by (5.2.5.16) and also in  $\text{Ass}_S(M)$  by (5.2.5.14).  $\square$

**Prop. (5.2.5.20) [Associated Primes and Localization].** Let  $\varphi : \operatorname{Spec} S^{-1}A \rightarrow \operatorname{Spec} A$  and  $M$  an  $A$ -module, then

$$\operatorname{Ass}_A(S^{-1}M) = \operatorname{Spec}(\varphi)(\operatorname{WeakAsso}_{S^{-1}A}(S^{-1}M)) = \operatorname{WeakAsso}_A(M) \cap \varphi(\operatorname{Spec}(S^{-1}R)).$$

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*Proof:* Cf. [Sta]05C9?.

The first equality is by (5.2.5.19). For the second, if  $\operatorname{Ann}_A(x) = \mathfrak{p}$  and  $\mathfrak{p} \cap S = \emptyset$ , then  $\operatorname{Ann}_{S^{-1}A}(x/1) = S^{-1}(\mathfrak{p})$ . Conversely, if  $\operatorname{Ann}_{S^{-1}A}(x/s) = S^{-1}\mathfrak{p}$ , then  $\mathfrak{p} \cap S = \emptyset$ , and  $\operatorname{Ann}_A(x) \subset S^{-1}\mathfrak{p} \cap A = \mathfrak{p}$ .  $\square$

**Cor. (5.2.5.21) [Associated Primes are Stalkwise].** Let  $R$  be a ring and  $M$  an  $R$ -module,  $\mathfrak{p} \subset R$ , then the following are equivalent:

- $\mathfrak{p} \subset \operatorname{WeakAss}(M)$ .
- $\mathfrak{p}R_{\mathfrak{p}} \subset \operatorname{WeakAss}(M_{\mathfrak{p}})$ .
- $M_{\mathfrak{p}}$  contains some element  $m$  that  $\sqrt{\operatorname{Ann}(m)} = \mathfrak{p}R_{\mathfrak{p}}$ .

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*Proof:*  $1 \rightarrow 2$ :  $\mathfrak{p}$  is a minimal prime of  $I = \operatorname{Ann}(m)$  for some  $m \in M$ , so  $I_{\mathfrak{p}}$  is the minimal prime of  $\operatorname{Ann}(m) \subset R_{\mathfrak{p}}$ .

$2 \rightarrow 3$ : As  $\mathfrak{p}R_{\mathfrak{p}}$  is the maximal prime, it is the only prime over  $\operatorname{Ann}(m)$ , so  $\mathfrak{p}R_{\mathfrak{p}} = \sqrt{\operatorname{Ann}(m)}$ .

$3 \rightarrow 1$ : This means there are some  $m \in M$  that  $\sqrt{I_{\mathfrak{p}}} = \mathfrak{p}R_{\mathfrak{p}}$ , which means that  $\mathfrak{p}$  is a minimal prime over  $\operatorname{Ann}(m)$ .  $\square$

**Prop. (5.2.5.22).** If  $M$  is an  $R$ -module, then  $M \rightarrow \prod_{\mathfrak{p} \in \operatorname{WeakAss}(M)} M_{\mathfrak{p}}$  is injective.  $\square$

*Proof:* Cf. [Sta]05CB.

If  $m \neq 0 \in M$ , there is an associated prime  $\mathfrak{p}$  of  $Rm$  (5.2.5.16), then it is an associated prime of  $M$ , and then  $(x)_{\mathfrak{p}} \subset M_{\mathfrak{p}}$  is not zero.  $\square$

**Def. (5.2.5.23) [Embedded Primes].** A non-minimal prime in  $\operatorname{Ass}_R(M)$  is called a **embedded prime**. Equivalently, it is an associated point that is not a generic point of  $\operatorname{Supp}(M)$ .  $\square$

**Prop. (5.2.5.24).** For a reduced ring  $R$ ,  $\operatorname{WeakAss}_R(R)$  is just the set of minimal primes of  $R$ .  $\square$

*Proof:* Cf. [Sta]0EMA.  $\square$

**Cor. (5.2.5.25) [Reduced Ring No Embedded Primes].** A reduced ring has no embedded primes, because it has no nilpotent elements. Hence all its associated primes are just the minimal primes.  $\square$

**Def. (5.2.5.26) [Unmixed Ideals].**  $I$  is called **unmixed** if primes in  $\operatorname{Ass}(A/I)$  all have the same height. In particular, they don't contain each other.  $\square$

### Primary Decomposition

**Def. (5.2.5.27).** For  $R$  Noetherian, a  $R$ -module  $M$  is called **coprimary** iff it has only one associated primes. A submodule  $N$  of  $M$  is called  **$p$ -primary** iff  $\operatorname{Ass}(M/N) = \{p\}$ . A ring is called  **$p$ -primary** iff  $(0)$  is  $p$ -primary.

Notice coprimary is equivalent to the following: if  $a \in A$  is a zero divisor for  $M$ , then for each  $x \in M$ , there is a  $n$  that  $a^n x = 0$ , i.e. **locally nilpotent**. And for ideals in a Noetherian ring, this is equivalent to  $r(I)$  is a prime.  $\square$



*Proof:* If  $M$  is  $p$ -primary, if  $x \in M$  is nonzero, then  $\text{Ass}(Rx) = \{p\}$ , so  $p$  is the unique minimal element of  $\text{Supp}(Rx) = V(\text{Ann}(x))$  by (5.2.5.16). So  $p$  is the radical of  $\text{Ann}(x)$ , i.e.  $a^n x = 0$  for some  $n$  (5.2.6.2).

Conversely, we know the ideal  $p$  of locally nilpotent elements equals the union of the associated primes (5.2.5.17), so if  $q \in \text{Ass}M = \text{Ann}(x)$ , then by definition,  $p \subset q$ . So  $p = q$ , and thus  $\text{Ass}M = \{p\}$ .  $\square$

**Lemma (5.2.5.28).** A primary ring has no nontrivial idempotent element, because  $e$  and  $1 - e$  will all belong to the same minimal ideal  $p$ .  $\perp$

**Lemma (5.2.5.29).** The intersection of  $p$ -primary submodules are  $p$ -primary. (Because there is a injection  $M/Q_1 \cap Q_2 \rightarrow M/Q_1 \oplus M/Q_2$ ).  $\perp$

**Lemma (5.2.5.30)[Associated Prime and Primary Decomposition].** If  $N = \cap Q_i$  is an irredundant primary decomposition and if  $Q_i$  belongs to  $p_i$ , then we have  $\text{Ass}(M/N) = \{p_1, \dots, p_r\}$ .  $\perp$

*Proof:* There is a injection  $M/N \rightarrow M/Q_1 \oplus \dots M/Q_r$  which shows  $\text{Ass}(M/N) \subset \{p_1, \dots, p_r\}$ . And for the inverse, notice  $Q_2 \cap \dots \cap Q_r/N$  is a submodule of  $M/Q_1$ , which shows  $\text{Ass}(Q_2 \cap \dots \cap Q_r/N) = \{p_1\}$  by (5.2.5.16).  $\square$

**Prop. (5.2.5.31).** If  $N$  is a  $p$ -primary submodule of a  $R$ -module  $M$ , and  $p'$  is a prime ideal, then

- $N_{p'} = M_{p'}$  if  $p \not\subset p'$ .
- $N = M \cap N_{p'}$  if  $p \subset p'$ .

$\perp$

*Proof:*  $M_{p'}/N_{p'} = (M/N)_{p'}$ , and  $\text{Ass}((M/N)_{p'}) = \text{Ass}(M/N) \cap \{\text{primes contained in } p'\} = \emptyset$  by (5.2.5.20). So  $M_{p'} = N_{p'}$  by (5.2.5.16).

For the second, notice it suffices to show  $M/N \rightarrow M_{p'}/N_{p'}$  is injective. But this is because  $A - p'$  contains no nonzero-divisor, by (5.2.5.17).  $\square$

**Cor. (5.2.5.32) [Second Uniqueness of Primary Decomposition].** For an irredundant primary decomposition  $N = \cap Q_i$ , if  $Q_1$  corresponds to  $p_1$  and  $p_1$  is minimal in  $\text{Ass}(M/N)$ , then  $Q_1 = M \cap N_{p_1}$ . In particular, the minimal prime part of a irredundant primary decomposition is uniquely determined.  $\perp$

*Proof:* By the above proposition, there are elements  $u_i$  of  $Q_i$ ,  $i \neq 1$  that are mapped to units in  $M_{p_1}$ , so  $Q_1 \cdot u_2 u_3 \dots u_r$  is mapped onto the image of  $Q_1 \rightarrow M_{p_1}$ . Then  $Q_1 = M \cap (Q_1)_{p_1} = M \cap N_{p_1}$ .  $\square$

**Prop. (5.2.5.33).** If  $R$  is Noetherian and  $M$  is a  $R$ -module, there are  $p$ -primary submodules  $Q(p)$  for each  $p \in \text{Ass}M$  that  $(0) = \bigcap_{p \in \text{Ass}M} Q(p)$ .  $\perp$

*Proof:* For a  $p \in \text{Ass}M$ , we seek  $Q(p)$  to be the maximal submodule  $N$  that  $p \notin \text{Ass}N$ . This has a maximal ideal because of Zorn and the fact  $\text{Ass}(\cup N_\lambda) = \cup \text{Ass}(N_\lambda)$ . Then We have  $\text{Ass}(M/Q(p)) = \{p\}$ , otherwise there is another  $p'$ , then there is a  $Q'/Q(p) \cong A/p'$ . Now  $Q'$  is bigger than  $Q(p)$ . Finally,  $(0) = \bigcap_{p \in \text{Ass}M} Q(p)$  because it has no associated primes.  $\square$

**Cor. (5.2.5.34)[Primary Decomposition].** If  $M$  is f.g. over a Noetherian ring  $R$ , then any submodule has a primary decomposition. (Notice  $M$  has only f.m. associated primes).  $\perp$

**Def. (5.2.5.35) [Symbolic Power].** For a prime ideal  $\mathfrak{p}$  in a Noetherian ring, The  $n$ -th **symbolic power**  $\mathfrak{p}^{(n)}$  is defined to be the  $\mathfrak{p}$ -primary component of  $\mathfrak{p}^n$ , who has only one minimal prime(hence one associated prime). The symbolic power is giving by  $\mathfrak{p}^n A_{\mathfrak{p}} \cap A$  by (5.2.5.32).  $\perp$

## 6 Jacobson Radical and Nilradical

### Nilradical

**Def. (5.2.6.1) [Nilradical].** For  $R \in \mathcal{CRing}$ , the **nilradical** of  $R$  is defined to be the ideal consisting of nilpotent elements.  $\lrcorner$

**Prop. (5.2.6.2).** The nilradical  $\mathfrak{n}$  of a ring  $A$  (5.2.6.1) is the intersection of all prime ideals.  $\lrcorner$

*Proof:* Every nilpotent element is contained in every prime, and if  $a$  is not nilpotent, then the localization  $A_a$  is nonzero, hence there is a maximal ideal, i.e. there is a prime of  $A$  not containing  $a$ .  $\square$

**Cor. (5.2.6.3).** In particular,  $\text{Spec } A/\mathfrak{n} \rightarrow \text{Spec } A$  is a homeomorphism, and  $A \rightarrow A/\mathfrak{n}$  induces a bijection on idempotents and units.  $\lrcorner$

### Jacobson Ring

**Def. (5.2.6.4) [Jacobson Ring].** A commutative ring is called **Jacobson** if every prime ideal is an intersection of maximal ideals. In particular, the Jacobson radical equals the nilradical. This is equivalent to every radical ideal is an intersection of maximal primes.  $\lrcorner$

**Prop. (5.2.6.5).**  $R$  is Jacobson iff  $\text{Spec } R$  is Jacobson space (4.12.2.22). In particular, the closed pts are dense in any closed subsets (Hilbert's Nullstellensatz satisfied).  $\lrcorner$

*Proof:* We need to show that a locally closed subset contains a closed pt, we assume this set is of the form  $V(I) \cap D(f)$ ,  $I$  is radical, then  $f \notin I$ , then by the condition, there is a  $I \subset \mathfrak{m}$  that  $f \notin \mathfrak{m}$ , thus the result.

Conversely, for a radical ideal, let  $J = \cap_{I \subset \mathfrak{m}} \mathfrak{m}$ , then  $J$  is radical and  $V(J)$  is the closure of  $V(I) \cap X_0$ ,  $V(I) = V(J)$ , and because they are both radical,  $I = J$ .  $\square$

**Cor. (5.2.6.6).** Being Jacobson is a local property, and quotient of Jacobson ring is Jacobson, and maximal ideals of  $R_f$  are maximal in  $R$ . (Immediate from (5.2.6.5) (4.12.2.23) and (4.12.2.24)).  $\lrcorner$

**Prop. (5.2.6.7).** If a Jacobson ring  $A$  has f.m. maximal ideals, then it is the product of its localizations at maximal primes and  $\dim A = 0$ .  $\lrcorner$

*Proof:* Any prime ideal  $\mathfrak{p}$  is a finite intersection of maximal ideals, so it equals one of them, so  $\dim A = 0$ . Now  $A/I = \oplus A/\mathfrak{m}_i$  by Chinese remainder theorem, so  $\text{Spec } A/I$  is discrete with  $n$  pts, so by (5.1.7.8) there are  $n$  idempotents  $e_i$  that  $e_i \equiv \delta_{ij} \pmod{\mathfrak{m}_j}$ ,  $\sum e_i = 1$ . Thus  $R = \prod Re_i$ . And  $Re_i$  is just the localization at a maximal prime.  $\square$

**Lemma (5.2.6.8).** If  $R$  is a Jacobson domain and  $R \subset K$  where  $K$  is a field, and  $K$  is f.g. over  $R$ , then  $R$  is a field and  $K/R$  is a finite field extension.  $\lrcorner$

*Proof:* By induction, it suffices to consider the monogenic case  $A = R[a]$ . So  $a$  is algebraic over quotient field of  $R$  because  $A$  is a field. Let  $\sum r_i t^i$  be a polynomial satisfied by  $a$ , and let  $\mathfrak{m}$  be a maximal ideal of  $R$  that  $r_n \notin \mathfrak{m}$  (exists because  $\text{rad } R = 0$ ). Then Nakayama says  $\mathfrak{m}A \not\subseteq A$ . Then  $\mathfrak{m} = 0$  because  $A$  is a field, hence  $R$  is a field.  $\square$

**Lemma (5.2.6.9).** Let  $R \subset A$  be commutative domains s.t.  $A$  is f.g. over  $R$ , then  $\text{rad } A = 0$  if  $\text{rad } R = 0$ .  $\lrcorner$

*Proof:* By induction, it suffices to consider the case  $A = R[a]$ . If  $a$  is transcendental over quotient field of  $R$ , then we finish by (3.6.3.12). Now assume  $a$  is algebraic over quotient field of  $R$ , let  $\sum r_i t^i, \sum s_i t^i$  be the polynomials satisfied by  $a, b$  of minimal degrees, then  $s_0 = -\sum_{i=1}^m s_i b^i \neq 0 \in \text{rad } A$ , and  $r_n s_0 \neq 0$ .

From the fact  $\text{rad } R = 0$ , we can find a maximal ideal  $\mathfrak{m}$  that  $r_n s_0 \notin \mathfrak{m}$ . Then Nakayama says

$$\mathfrak{m} \cdot S^{-1}A \not\subseteq S^{-1}A.$$

In particular,  $\mathfrak{A} \not\subseteq A$ . Choose a maximal ideal of  $A$  containing  $\mathfrak{m}A$ , then it cannot contain  $s_0$ , contradicting  $s_0 \in \text{rad } A$ .  $\square$

**Prop. (5.2.6.10) [Generalized Nullstellensatz].** If  $R$  is Jacobson and  $S$  is a finitely generated  $R$ -algebra, then:

- $S$  is Jacobson.
- The maximal ideal of  $S$  intersect with  $R$  a maximal ideal, and the quotient ring extension is finite, (in particular algebraic).

In particular, a f.g. algebra over a ring of dimension 0, (e.g. Artinian ring or field) is Jacobson.

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*Proof:* To show  $S$  is Jacobson, consider for any prime  $\mathfrak{p} \subset A$ ,  $A/\mathfrak{p}$  is a f.g. domain over  $R/\mathfrak{p} \cap R$ . Because  $R$  is Jacobson,  $\text{rad}(R/\mathfrak{p} \cap R) = 0$ , so  $\text{rad}(A/\mathfrak{p}) = 0$ , by (5.2.6.9). And this shows  $A$  is Jacobson.

If  $\mathfrak{m}$  is maximal in  $S$ , then  $R/\mathfrak{m} \cap R \rightarrow S/\mathfrak{m}$  satisfies the condition of (5.2.6.8), by (5.2.6.6), so the first two assertions are proved.  $\square$

**Cor. (5.2.6.11).** If  $R$  is Jacobson and  $S \in \mathcal{CAlg}^{\text{fg}}(R)$  is reduced, then  $\bigcap_{\mathfrak{m} \subset R \text{ maximal}} \mathfrak{m}S = 0$ .  $\square$

*Proof:* This is because

$$\bigcap_{\mathfrak{m} \subset R \text{ maximal}} \mathfrak{m}S \subset \bigcap_{\mathfrak{M} \subset S \text{ maximal}} (\mathfrak{M} \cap R)S \subset \bigcap_{\mathfrak{M} \subset S \text{ maximal}} \mathfrak{M} = 0.$$

$\square$

### Zariski-Pairs

**Def. (5.2.6.12) [Zariski-Pairs].** A pair  $(A, I)$  is called a **Zariski-pair** iff  $I$  is contained in the Jacobson radical of  $A$ .  $\square$

**Prop. (5.2.6.13).** If  $(A, I)$  is a Zariski-pair (5.2.6.12), then the map  $A \rightarrow A/I$  induces a bijection between the idempotents.  $\square$

*Proof:* idempotents are determined by the maximal ideals that it vanishes (5.1.7.6), and  $A \rightarrow A/I$  induces a bijection on the maximal ideals.  $\square$

## 7 Dedekind-Domains

**Def. (5.2.7.1) [Dedekind-Domains].** A **Dedekind-domain** is an integrally-closed Noetherian domain of dimension 1. A UFD is a Dedekind domain by (5.3.5.2).  $\square$

**Prop. (5.2.7.2) [Characterizing].** For a domain  $R$ , the following are equivalent:

1.  $R$  is a Dedekind domain.
2.  $R$  is Noetherian and each  $R_{\mathfrak{m}}$  is DVR for any maximal ideal  $\mathfrak{m}$ .
3. each ideal of  $R$  can be written as a product of prime ideals uniquely.

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*Proof:* 1  $\iff$  2 as normal is a stalkwise and (5.3.5.21).

3  $\rightarrow$  2: If 3 is true, then  $\mathfrak{p} \neq \mathfrak{p}^2$  for each prime  $\mathfrak{p}$ , so choose  $x \in \mathfrak{p} - \mathfrak{p}^2$ , then for each  $y \in \mathfrak{p}$ ,  $(x, y) = \prod p_i$ , then exactly one  $p_i$  (may assume  $p_1$ ) is contained in  $\mathfrak{p}$ , so  $(x, y)R_{\mathfrak{p}} = p_1R_{\mathfrak{p}}$ . Now in fact  $(x)R_{\mathfrak{p}} = p_1R_{\mathfrak{p}}$ , because  $(x, y^2)R_{\mathfrak{p}}$  is also a prime, so  $y = ax + by^2$  in  $R_{\mathfrak{p}}$ ,  $(1 - by)y = ax \in (x)R_{\mathfrak{p}}$  is a prime, so  $y \in (x)R_{\mathfrak{p}}$ . This is for all  $y \in \mathfrak{p}$ , so  $(x)R_{\mathfrak{p}} = p_1R_{\mathfrak{p}}$ .

Now if  $(x) = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ , then  $p_1R_{\mathfrak{p}} = pR_{\mathfrak{p}}$ , so  $p_1 = p$ , and  $p$  is f.g. by the lemma (5.2.7.3) below.  $\mathfrak{p}$  is arbitrary, so  $R$  is Noetherian and  $R_{\mathfrak{m}}$  is DVR for  $\mathfrak{m}$  maximal, by (5.3.5.21).

1  $\rightarrow$  3 : if 1 is true, then any ideal is a unique intersection of primary ideals, and primary ideals are their radical are different, so they are coprime (5.1.1.9), so this is in fact a unique decomposition into products of primary ideals. And any primary ideal is a power of its radical, because this is the case after localization.  $\square$

**Lemma (5.2.7.3).**  $I, J$  be ideals in a ring  $A$  and  $IJ = (f)$  where  $f$  is a non-zero-divisor, then  $I, J$  are f.g. and finitely locally free of rank 1 as  $A$ -modules.  $\square$

*Proof:* The second assertion implies the first, by (5.3.1.7).  $f = \sum x_i y_i$ , and  $x_i y_i = a_i f$ , so  $\sum a_i = 1$  as  $f$  is non-zero-divisor. Now we show  $I_{a_i}$  as  $J_{a_i}$  is free of rank 1. Now after localization,  $f = xy$ , so  $x, y$  are non zero divisors. Now if  $x' \in I$ , then  $x'y = af = axy$  for some  $a$ , so  $x' = ax$ .  $\square$

### Fractional Ideals

**Def. (5.2.7.4) [Fractional Ideals].** For an integral domain  $(\mathcal{O}, K)$ , then an  $\mathcal{O}$ -submodule  $M$  of  $K$  is called a **fractional ideal** if  $xM \subset A$  for some  $x \in K^\times$ .

Every f.g. submodule in  $K$  is a fractional ideal, and if  $\mathcal{O}$  is Noetherian, then the converse is true, because it is of the form  $x^{-1}\mathfrak{a}$ .  $\square$

**Prop. (5.2.7.5).** Situation as in (5.2.7.4), an  $A$ -submodule  $M \subset K$  is called an **invertible ideal** if there is an  $A$ -submodule  $N \subset K$  that  $MN = A$ . If this is the case, then  $M, N$  are f.g., because there are  $\sum x_i y_i = 1$ , so  $M$  is generated by  $x_i$  and  $N$  is generated by  $y_i$ .  $\square$

**Prop. (5.2.7.6).** Situation as in (5.2.7.4), Invertibility is a stalkwise property.  $\square$

*Proof:* Notice  $(A : M)_{\mathfrak{p}} = (A_{\mathfrak{p}} : M_{\mathfrak{p}})$ , and  $M$  is invertible iff  $M(A : M) = A$ . Then use the fact isomorphism is stalkwise (5.1.4.2).  $\square$

**Lemma (5.2.7.7).** A local domain  $A$  is a DVR iff every non-zero fractional ideal of  $A$  is invertible.  $\square$

*Proof:* If is a DVR, let  $\mathfrak{m} = (x)$ , for any fractional ideal  $M$  let  $yM \subset A = (x^r)$ , then  $M = (x^{r-s})$ , where  $v(y) = s$ . Conversely, if every non-zero fractional ideal of  $A$  is invertible, then they are all f.g. (5.2.7.5), so  $A$  is Noetherian. Now it suffices to prove that every ideal of  $A$  is a power of  $\mathfrak{m}$ , by (5.3.5.21). If this is not true, choose a maximal element  $\mathfrak{a}$  in the set of ideals that is not a power of  $\mathfrak{m}$  (by Noetherian), then  $\mathfrak{m}^{-1}\mathfrak{a} \subset \mathfrak{m}^{-1}\mathfrak{m} = A$ , and  $\mathfrak{m}^{-1}\mathfrak{a} \supset \mathfrak{a}$ , but it is not  $\mathfrak{a}$ , so  $\mathfrak{m}^{-1}\mathfrak{a} = \mathfrak{m}^k$  for some  $k$ , so  $\mathfrak{a} = \mathfrak{m}^{k+1}$ , contradicton.  $\square$

**Prop. (5.2.7.8) [Dedekind Domain Fractional Ideals are Invertible].** An integral domain is a Dedekind domain iff every non-zero fractional ideal is invertible.  $\lrcorner$

*Proof:* Immediate from (5.2.7.7) and (5.2.7.2)(5.2.7.6).  $\square$

**Def. (5.2.7.9) [Class Group].** For an integral domain  $\mathcal{O}$ , denote  $\text{Cl}(\mathcal{O})$  the Abelian group of invertible fractional ideals of  $\mathcal{O}$  modulo principal fractional ideals, called the **class group** of  $\mathcal{O}$ .  $\lrcorner$

**Prop. (5.2.7.10) [Localizations of Dedekind Domains].** Let  $(\mathcal{O}, F)$  be a Dedekind domain and  $S \subset \text{Spec } \mathcal{O}$  be a finite subset of maximal ideals of  $\mathcal{O}$ , let  $\mathcal{O}_S$  denote the localization of  $\mathcal{O}$  at all such  $\mathfrak{p}_i$ . Then there is an exact sequence of Abelian groups:

$$1 \rightarrow \mathcal{O}^* \rightarrow \mathcal{O}_S^* \rightarrow \bigoplus_{\mathfrak{p} \in S} F^\times / \mathcal{O}_{\mathfrak{p}}^* \rightarrow \text{Cl}(\mathcal{O}) \rightarrow \text{Cl}(\mathcal{O}_S) \rightarrow 1.$$

$\lrcorner$

*Proof:* The only non-trivial map is the map  $\bigoplus_{\mathfrak{p} \in S} F^\times / \mathcal{O}_{\mathfrak{p}}^* \rightarrow \text{Cl}(\mathcal{O})$  given by  $(\overline{a_{\mathfrak{p}}})$  mapsto  $\prod_{x \in S} \mathfrak{p}^{v_{\mathfrak{p}}(a_x)}$ .  $\square$

### Extensions of Dedekind Domains

References are [Neu99]Chap1.8..

**Prop. (5.2.7.11) [Krull-Akizuki].** If  $A$  is a Noetherian domain of dimension 1 with fraction field  $K$  and  $K \subset_{\text{fin}} L \in \mathbf{Field}$ , then the integral closure  $B$  of  $A$  in  $L$  is a Dedekind domain with fraction field  $L$ , and  $\text{Spec } B \rightarrow \text{Spec } A$  is surjective, and have finite fibers and induces finite residue field extension.  $\lrcorner$

*Proof:* Cf. [Sta]09IG or [Neu99]P77. ?  $\square$

**Cor. (5.2.7.12).** The integral closure of a Dedekind domain in a finite extension fields of its quotient fields is again a Dedekind domain.  $\lrcorner$

**Def. (5.2.7.13) [Situation].** Let  $(\mathcal{O}_K, K)$  be a Dedekind domain,  $K \subset_{\text{fin}} L \in \mathbf{Field}$ , and  $\mathcal{O}_L$  the integral closure of  $\mathcal{O}_K$  in  $L$ , which is also a Dedekind domain by (5.2.7.11).  $\lrcorner$

**Prop. (5.2.7.14).** Situation as in (5.2.7.13), if  $\alpha_1, \dots, \alpha_n$  is a basis of  $L/K$  that is contained in  $\mathcal{O}_L$ , and  $d = d(\alpha_1, \dots, \alpha_n)$  (3.2.6.33), then

$$d\mathcal{O}_L \subset \mathcal{O}_K \{\alpha_1, \dots, \alpha_n\}.$$

$\lrcorner$

*Proof:* For any  $\alpha = a_1\alpha_1 + \dots + a_n\alpha_n \in \mathcal{O}_L$ ,  $a_i \in K$ ,  $\alpha$  satisfies the following equations

$$\text{tr}_{L/K}(\alpha_i \alpha) = \sum_j \text{tr}_{L/K}(\alpha_i \alpha_j) a_j$$

for any  $i$ . Notice  $\text{tr}(\alpha_i \alpha) \in \mathcal{O}_K$ , so  $da_j \in \mathcal{O}_K$  by linear algebra.  $\square$

**Prop. (5.2.7.15).** Situation as in (5.2.7.13), if  $L/K$  is separable and  $\mathcal{O}_K$  is a PID, then every f.g.  $\mathcal{O}_L$ -module in  $L$  is a free  $\mathcal{O}_K$ -module of rank  $[L : K]$ . In particular,  $\text{rank}_{\mathcal{O}_K} \mathcal{O}_L = [L : K]$ .  $\lrcorner$

*Proof:* Let  $\alpha_1, \dots, \alpha_n$  be a basis of  $L/K$  that is contained in  $\mathcal{O}_L$ , and let  $d = \text{disc}(\alpha_1, \dots, \alpha_n) \neq 0$  by (3.2.6.34), then by (5.2.7.14),  $d\mathcal{O}_L \subset \mathcal{O}_K\{\alpha_1, \dots, \alpha_n\}$ . Notice any generator of the torsion-free module  $\mathcal{O}_L$  over  $\mathcal{O}_K$  is a generator of the field  $L/K$ , so  $\text{rank}_{\mathcal{O}_K} \mathcal{O}_L \geq [L : K]$ . Now if  $M \subset L$  is a free  $\mathcal{O}_L$ -module with a set of generators  $\mu_1, \dots, \mu_r$ , then we can take  $a \in \mathcal{O}_K$  s.t.  $a\mu_i \in \mathcal{O}_L$ . Then

$$adM \subset d\mathcal{O}_L \subset \mathcal{O}_K\{\alpha_1, \dots, \alpha_n\}.$$

And then

$$[L : K] \leq \text{rank}_{\mathcal{O}_K} \mathcal{O}_L \leq \text{rank}_{\mathcal{O}_K} M = \text{rank}_{\mathcal{O}_K}(adM) \leq [L : K].$$

and the assertion follows.  $\square$

**Prop. (5.2.7.16)[Integral Basis of Joins].** Situation as in (5.2.7.13), if  $L/K, L'/K$  are two finite Galois extensions in  $\bar{K}$  s.t.  $L \cap L' = K$ , and let  $\omega_1, \dots, \omega_n$  (resp.  $\omega'_1, \dots, \omega'_{n'}$ ) be an integral basis of  $L/K$  (resp.  $L'/K$ ) with discriminants  $d$  (resp.  $d'$ ) (3.2.6.33). Suppose that  $(d, d') = \mathcal{O}_K$ , then  $\omega_i \omega'_j$  is an integral basis of  $LL'/K$  with discriminant  $d^{n'}(d')^n$ .  $\square$

*Proof:* The hypothesis implies that  $[LL' : K] = nn'$ , so  $\{\omega_i \omega'_j\}$  is a basis of  $LL'/K$  that is integral. To show it is an integral basis, if  $\alpha = \sum_{i,j} a_{ij} \omega_i \omega'_j \in \mathcal{O}_{LL'}$ , we need to show that  $a_{ij} \in \mathcal{O}_K$ . Let  $\beta_i = \sum a_{ij} \omega_j$ , and let  $\text{Gal}(LL'/L) = \{\sigma_1, \dots, \sigma_n\}$ ,  $\text{Gal}(LL'/L) = \{\sigma'_1, \dots, \sigma'_{n'}\}$ ,

$$T = (\sigma'_i \omega'_j)_{i,j}, \quad \underline{a} = (\sigma'_1 \alpha, \dots, \sigma'_{n'} \alpha)^t, \quad \underline{b} = (\beta_1, \dots, \beta_{n'})^t,$$

then  $\det(T)^2 = d'$ , and  $\underline{a} = T\underline{b}$ . So  $\det(T)\underline{a} = T^* \underline{b}$  has integral entries, and  $d'b$  is integral. Thus  $d'a_{ij} \in \mathcal{O}_K$  for each  $i, j$ . Dually,  $da_{ij} \in \mathcal{O}_K$ . Thus by hypothesis,  $a_{ij} \in \mathcal{O}_K$ .

To calculate the discriminant,

$$d = d(\{\omega_i \omega'_j\}) = \det(\{\sigma_k \sigma'_l \omega_i \omega'_j\}_{k,\ell,i,j})^2 = \det((\sigma_k \omega_i)_{k,i} \otimes (\sigma'_l \omega'_j)_{l,j})^2 = \det((\sigma_k \omega_i)_{k,i})^{2n'} \det((\sigma'_l \omega'_j)_{l,j})^{2n} = d^{n'}(d')^n$$

$\square$

**Def. (5.2.7.17)[Conductors].** Let  $R \subset S \in \mathcal{C}\text{Ring}$  the **conductor(ring)** of  $R$  in  $S$  is defined to

$$\mathfrak{c} = \{\alpha \in R \mid \alpha S \subset R\}.$$

And if  $(R, K)$  is an integral domain, the **conductor of  $R$**  is defined to be the conductor of  $R$  in the integral closure of  $R$  w.r.t.  $K$ .  $\square$

**Prop. (5.2.7.18).** Let  $R$  be a Noetherian integral domain with integral closure  $S$ , then the conductor of  $R$  in  $S$  is nonzero iff  $S$  is f.g. as a  $R$ -module.  $\square$

*Proof:* This follows from (5.2.7.4).  $\square$

**Def. (5.2.7.19)[Conductors of an Element].** In situation (5.2.7.13) and  $\alpha \in \mathcal{O}_L$ , then the **conductor(integral element)** of  $\alpha$  w.r.t  $K$  is the conductor of  $\mathcal{O}_K[\alpha]$  in  $\mathcal{O}_L$  (5.2.7.17), denoted by  $\mathfrak{d}(\alpha)$ . It is non-zero by (5.2.7.4) as  $\mathcal{O}_L$  is f.g. over  $\mathcal{O}_K[\alpha]$ .  $\square$

**Prop. (5.2.7.20)[Inertia and Ramification].** Situation as in (5.2.7.13), let  $\mathfrak{p}$  be a maximal ideal of  $\mathcal{O}_K$ , then  $\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}_L \neq 0$  by (5.2.1.5). Let  $\mathfrak{p}\mathcal{O}_L = \prod_i \mathfrak{P}_i^{e_i}$  be a decomposition. Denote  $e_i$  the **ramification degree** of  $\mathfrak{P}_i$  over  $\mathfrak{p}$ , and  $f_i = [\mathcal{O}_L/\mathfrak{P}_i : \mathcal{O}_K/\mathfrak{p}]$  the **inertia degree** of  $\mathfrak{P}_i$  over  $\mathfrak{p}$ .  $\square$

**Prop. (5.2.7.21) [Fundamental Identity].** Situation as in (5.2.7.13), if  $\mathcal{O}_L$  is a finite  $\mathcal{O}_K$ -module, then for any  $\mathfrak{p} = \prod_i \mathfrak{P}_i^{e_i}$ ,

$$\sum e_i f_i = [L : K].$$

And this is the case when  $L/K$  is separable.  $\lrcorner$

*Proof:* By hypothesis,  $\mathcal{O}_{L,\mathfrak{p}}$  is a finite  $\mathcal{O}_{K,\mathfrak{p}}$ -module of rank  $[L : K]$ , thus  $\mathcal{O}_L/\mathfrak{p}$  is an  $\mathcal{O}_K/\mathfrak{p}\mathcal{O}_L$ -module of rank  $n$ . Notice there is an isomorphism  $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \prod_i \mathcal{O}_L/\mathfrak{P}_i^{e_i}$ , thus

$$[L : K] = \sum_i \dim_{\mathcal{O}_L/\mathfrak{p}} \mathcal{O}_L/\mathfrak{P}_i^{e_i} = \sum_i e_i f_i.$$

If  $L/K$  is separable, let  $\{\alpha_1, \dots, \alpha_n\}$  be a basis of  $L/K$  with  $\alpha_i \in \mathcal{O}_K$ , then as  $d(\alpha_1, \dots, \alpha_n) \neq 0$  by (3.2.6.34),  $\mathcal{O}_L \subset d^{-1}(\alpha_1 \mathcal{O}_K + \dots + \alpha_n \mathcal{O}_K)$  by (5.2.7.14). So it is a finite  $\mathcal{O}_K$ -module, as  $\mathcal{O}_K$  is Noetherian.  $\square$

**Prop. (5.2.7.22) [Decompositions in a Galois Extension].** Situation as in (5.2.7.13), if  $L/K$  is Galois, then for any  $\mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$ ,  $\text{Gal}(L/K)$  acts transitively on the set of primes over  $\mathfrak{p}$ .  $\lrcorner$

*Proof:* If  $\mathfrak{P}, \mathfrak{P}'$  are two primes over  $\mathfrak{p}$  but  $\mathfrak{P}' \neq \sigma \mathfrak{P}$  for any  $\sigma \in \text{Gal}(L/K)$ , then by Chinese remainder theorem, there exists  $x \in \mathcal{O}_L$  s.t.

$$x \equiv 0 \pmod{\mathfrak{P}'}, \quad x \equiv 1 \pmod{\sigma \mathfrak{P}}, \forall \sigma \in \text{Gal}(L/K).$$

But then  $\text{Nm}_{L/K}(x) = \prod_{\sigma \in \text{Gal}(L/K)} \sigma(x) \in \mathfrak{P}' \cap \mathfrak{O}_K = \mathfrak{p}$ , but none of  $\sigma(x)$  is in  $\mathfrak{P}$ , so  $\prod_{\sigma \in \text{Gal}(L/K)} \sigma(x) \notin \mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$ , contradiction.  $\square$

**Thm. (5.2.7.23) [Dedekind-Kummer].** Situation as in (5.2.7.13), suppose  $L = K(\theta)$ , where  $\theta \in \mathcal{O}_L$  has minimal polynomials  $p(X) \in \mathcal{O}_K[X]$  (such a  $\theta$  exists if  $L/K$  is separable, by (3.2.6.18)). Let  $\mathfrak{d}$  be the conductor of  $\theta$ , then for any prime ideal  $\mathfrak{p}$  of  $\mathcal{O}_K$  prime to  $\mathfrak{d}$ , let

$$\bar{p}(X) = \prod_i \bar{p}_i(X)^{e_i}$$

be the decomposition of  $\bar{p}(X)$  into irreducible factors in  $\mathcal{O}_K/\mathfrak{p}[X]$ , where  $\bar{p}_i(X) \in \mathcal{O}_K[X]$  are monic polynomials, then

- $\mathfrak{P}_i = \mathfrak{p}\mathcal{O} + p_i(\theta)\mathcal{O}_L$  are different primes ideals in  $\mathcal{O}_L$ .
- $\mathfrak{p}\mathcal{O}_L = \prod_i \mathfrak{P}_i^{e_i}$ .
- The inertia degree  $f_i$  equals  $\deg(\bar{p}_i(X))$ .

$\lrcorner$

*Proof:* Cf. [Neu99] P48.  $\square$

**Def. (5.2.7.24) [Ramification Notations].** Situation as in (5.2.7.20), let  $\mathfrak{p}\mathcal{O}_L = \prod_{i=1}^r \mathfrak{P}_i^{e_i}$ , then

- $\mathfrak{p}$  **splits completely** in  $L$  if  $r = [L : K]$ .
- $\mathfrak{p}$  is **non-split** in  $L$  if  $r = 1$ .
- $\mathfrak{P}_i$  is **unramified over**  $\mathfrak{p}$  if  $e_i = 1$  and  $\kappa(\mathfrak{P}_i)/\kappa(\mathfrak{p})$  is separable.
- $\mathfrak{P}_i$  is **ramified in** over  $\mathfrak{p}$  if it is not unramified over  $\mathfrak{p}$ .
- $\mathfrak{P}_i$  is **totally ramified over**  $\mathfrak{p}$  if it is ramified over  $\mathfrak{p}$  and  $f_i = 1$ .



- $\mathfrak{p}$  is called **unramified in  $L$**  if all  $\mathfrak{P}_i$  are unramified over  $\mathfrak{p}$ .
- $\mathfrak{p}$  is called **ramified in  $L$**  if some  $\mathfrak{P}_i$  is ramified over  $\mathfrak{p}$ .
- $L/K$  is called a **unramified extension** if all maximal ideals of  $\mathcal{O}_K$  are unramified.

┘

**Prop. (5.2.7.25) [Almost Every Prime is Unramified].** Situation as in (5.2.7.13), then a.e. maximal prime  $\mathfrak{p}$  of  $\mathcal{O}_K$  is unramified in  $L$ , by (5.2.7.39). ┘

**Prop. (5.2.7.26).** Situation as in (5.2.7.13),  $\mathcal{O}_L \otimes_{\mathcal{O}_K} \mathcal{O}_{K,\mathfrak{p}} = \prod_i \mathcal{O}_{L,\mathfrak{P}_i}$ . ┘

*Proof:* Cf. [MIT notes, 11.7]. ┘

**Prop. (5.2.7.27).** If a prime  $\mathfrak{p}$  splits completely in two separable extension  $LM$  of  $K$ , then it also splits completely in the composite  $LM$ . ┘

*Proof:* We use the language of valuation. The extension of a valuation  $v$  of  $K$  corresponds to the set of equivalent classes of algebra map from  $L$  to  $\overline{K_v}$  module conjugacy over  $K_v$ . So We only need to show that two different maps of  $LM$  are not conjugate over  $K_v$ . But the restrict of them to  $L$  or  $M$  is different, thus not conjugate over  $K_v$  by the assumption. ┘

**Cor. (5.2.7.28).** A prime splits completely in a separable extension  $L/K$  if it splits completely in  $\tilde{L}/K$ . ┘

*Proof:* This is because the Galois closure is the composite of the conjugates of  $L$ . ┘

### Differents and Discriminants

**Def. (5.2.7.29) [Differents].** Situation as in (5.2.7.13), if  $L/K$  is separable, then there is a **trace form** on  $L$ :  $(x, y) \rightarrow \text{tr}(xy)$ , which is non-degenerate by (3.2.6.32).

We define the **dual module** for a fractional ideal  $I$  of  $\mathcal{O}_L$  as  $I^\vee = \{x \in L \mid \text{tr}(xI) \in \mathcal{O}_K\}$ . This is truly a fractional ideal because if  $\alpha_i \in \mathcal{O}_L$  is a basis of  $L/K$  and  $d = \det(\text{tr}(\alpha_i \alpha_j)) \in \mathcal{O}_K^\times$  (3.2.6.34), then for any  $a \in I \cap \mathcal{O}_L$ ,  $ad \cdot I^\vee \in \mathcal{O}_L$ , because if  $x = \sum x_i \alpha_i \in I^\vee$ , then  $\sum ax_i \text{tr}(\alpha_i \alpha_j) = \text{tr}(xa \alpha_j) \in \mathcal{O}_K$ , so solve the equation shows  $da x_i \in \mathcal{O}_K$ .

The **different of  $K/L$**  is defined to be  $\mathfrak{d}_{L/K} = ((\mathcal{O}_L)^\vee)^{-1} \in \text{Ideal}(\mathcal{O}_L)$ . ┘

**Prop. (5.2.7.30) [Properties of Differents].** Different is compatible with composition, localization w.r.t. a prime ideal and completion. ┘

*Proof:* Cf. [Neu99]P195. ┘

**Cor. (5.2.7.31).**  $\mathfrak{d}_{L/K} = \prod_{\mathfrak{P}} \mathfrak{d}_{L_{\mathfrak{P}}/K_{\mathfrak{P}}}$ . ┘

**Prop. (5.2.7.32).** If  $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ , then  $\mathfrak{d}_{L/K} = (f'(\alpha))$ , where  $f(X) = \text{Irr}(\alpha, K; X)$ . ┘

*Proof:* By (3.2.6.30), if  $f(X)/(X - \alpha) = \beta_0 + \beta_1 X + \dots + \beta_{n-1} X^{n-1}$ , then  $\mathcal{O}_L^\vee = f'(\alpha)^{-1}(\beta_0, \dots, \beta_{n-1})$ . Now the result follows if  $(\beta_0, \dots, \beta_{n-1}) = \mathcal{O}_L$ , which is easy to see if we write  $\beta_i$  as polynomials of  $\alpha$ . ┘

**Cor. (5.2.7.33).** If  $L/K$  is finite separable extension of local fields, then

$$v_L(\mathfrak{d}_{L/K}) = \sum_{\sigma \in G, \sigma \neq 1} i_{L/K}(\sigma) = \int_{-1}^{\infty} (|\text{Gal}(L/K)_t| - 1) dt.$$

Notation as in (19.1.4.23). ┘



**Prop. (5.2.7.34) [Different as Annihilator of Kähler Differential].** Situation as in (5.2.7.13), if  $L/K$  is separable, the different  $\mathfrak{d}_{L/K}$  is the annihilator of  $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$ .  $\square$

*Proof:* It suffices to show the exact sequence

$$0 \rightarrow \mathfrak{d}_{L/K} \rightarrow \mathcal{O}_L \rightarrow \Omega_{\mathcal{O}_L/\mathcal{O}_K},$$

but because exactness is stalkwise (5.1.4.2), we can localized at a maximal ideal, then by (19.1.4.2),  $\mathcal{O}_L = \mathcal{O}_K[x]$  is monogenous, thus  $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$  is cyclic, and the annihilator of  $dx$  is  $f'(x)$ . So by (5.2.7.32) we are done.  $\square$

**Prop. (5.2.7.35) [Ramification and Different].** Situation as in (5.2.7.13), if  $L/K$  is separable, a maximal ideal  $\mathfrak{P} \in \text{Spec}(\mathcal{O}_L)$  is ramified over  $\mathcal{O}_K$  (5.2.7.24) iff  $\mathfrak{P} \mid \mathfrak{d}_{L/K}$ .

Let  $e$  be the ramification of  $\mathfrak{P}$ , then the power  $s$  of  $\mathfrak{P}$  in  $\mathfrak{d}_{L/K}$  is

$$\begin{cases} s = e - 1 & \mathfrak{P} \text{ tamely ramified} \\ e \leq s \leq e - 1 + v_{\mathfrak{P}}(e) & \mathfrak{P} \text{ wildly ramified} \end{cases}.$$

$\square$

*Proof:* Cf. [Neu99]P199. ?  $\square$

**Def. (5.2.7.36) [Discriminants].** Situation as in (5.2.7.13), if  $L/K$  is separable, the **discriminant of field extension**  $L/K$  is defined to be the set  $\text{disc}(L/K)$  consisting of discriminants  $\text{disc}(\alpha_1, \dots, \alpha_n)$  (3.2.6.33) where  $\{\alpha_i\}$  is a basis of  $L/K$  s.t.  $\alpha_i \in \mathcal{O}_L$ .

Because  $\text{disc}(\alpha_1, \dots, \alpha_n) \in \mathcal{O}_K^\times$  (3.2.6.34) and  $\text{tr}_{L/K}$  is  $\mathcal{O}_K$ -linear,  $\text{disc}(L/K) \in \text{Ideal}(\mathcal{O}_K)^\times$ .  $\square$

**Prop. (5.2.7.37) [Differents and Discriminants].** Situation as in (5.2.7.13), we have

$$\text{disc}(L/K) = \text{Nm}_{L/K} \mathfrak{d}_{L/K}.$$

$\square$

*Proof:* Cf. [Neu99]P201.  $\square$

**Cor. (5.2.7.38) [Compositions and Discriminants].** Situation as in (5.2.7.13), if  $K \subset_{\text{fin}} L \subset_{\text{fin}} M \in \text{Field}$  are separable extensions, we have

$$\text{disc}(M/K) = \text{disc}(L/K)^{[M:L]} \text{Nm}_{L/K}(\text{disc}(M/L)).$$

$\square$

*Proof:* Apply  $\text{Nm}_{M/K} = \text{Nm}_{L/K} \circ \text{Nm}_{M/L}$  to the equation  $\mathfrak{d}_{M/K} = \mathfrak{d}_{M/L} \mathfrak{d}_{L/K}$  (5.2.7.30), we get

$$\text{disc}(M/K) = \text{Nm}_{L/K}(\text{disc}(M/L)) \text{Nm}_{L/K}(\mathfrak{d}_{L/K}^{[M:L]}) = \text{Nm}_{L/K}(\text{disc}(M/L)) \text{disc}(L/K)^{[M:L]}.$$

$\square$

**Cor. (5.2.7.39) [Ramification and Discriminant].** Situation as in (5.2.7.13), if  $L/K$  is separable, then a maximal ideal  $\mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$  ramifies in  $L$  (5.2.7.24) iff  $\mathfrak{p} \mid \text{disc}(L/K)$ .

In particular, only f.m. maximal ideals of  $\mathcal{O}_K$  are ramified in  $L/K$ , and the extension is unramified iff  $\text{disc}(L/K) = 1$ .  $\square$

*Proof:* This follows from (5.2.7.35) and (5.2.7.37).  $\square$

**Cor. (5.2.7.40).**  $\text{disc}(L/K) = \prod_{\mathfrak{P}} \text{disc}(L_{\mathfrak{P}}/K_{\mathfrak{P}})$ .  $\square$

*Proof:* This follows from (5.2.7.31) and (5.2.7.37).  $\square$

### 5.3 Commutative Algebra III

**Notation(5.3.0.1).**

- Let  $R \in \mathcal{CRing}$ .

┘

#### 1 Projective

References are [Projective Modules], [Sta] and [Vak17].

**Def. (5.3.1.1)[Projective Modules].** A module  $P$  over  $R$  is called **projective module** iff  $\text{Hom}(P, -)$  is exact, or equivalent, for any surjective map of modules  $F \rightarrow Q \rightarrow 0$ ,  $\text{Hom}(P, F) \rightarrow \text{Hom}(P, Q)$  is surjective.

┘

**Prop. (5.3.1.2).** Localization and tensor product preserves projective because they are left adjoints. And when tensoring f.f. map, then the converse is also true(5.4.2.1).

┘

**Prop. (5.3.1.3).** A module over a ring is projective iff it is a direct summand of a free module, in particular, it is flat. Moreover, there is a free module  $Q$  that  $P \oplus Q = F$  free.

┘

*Proof:* For the second assertion, we can choose an arbitrary  $Q$  that  $P \oplus Q$  free, and see  $\bigoplus_{\mathbb{N}}(P \oplus Q)$  is free. □

**Lemma (5.3.1.4).** A projective module is a direct sum of countably generated projective modules. ┘

*Proof:* This follows from(3.2.4.28). □

**Prop. (5.3.1.5)[Projective over Local Ring].** A projective module  $P$  over a local ring  $R$  or a PID is free. ┘

*Proof:* Local ring case: By(5.3.1.4) and(3.2.4.29), it suffices to show that any element  $x$  of  $P$  is contained in a free direct summand of  $P$ . Because  $P$  is projective, it is a direct summand of a free module  $F$ ,  $F = P \oplus Q$ . Let  $B$  be a basis of  $F$  that the number of basis element in the expression of  $x$  is minimal. Let  $x = \sum a_i e_i$ . Then no  $a_i$  is contained in the ideal generated by other  $a_j$ , otherwise we can choose another basis to show this is minimal. Let  $e_i = y_i + c_i$  be decompositions into  $P$  and  $Q$  components, and write  $y_i = \sum a_{ij} e_j + t_i$ , where  $t_i$  is combination of elements in  $B$  other than  $e_i$ . Now it suffices to show  $\det(a_{ij})$  is invertible, because in this way  $\{y_i\} \cup (B \setminus \{x_1\})$  is a basis of  $F$  and  $x = \sum a_i y_i$  because  $x \in P$ . And  $N = \text{span}\{y_i\}$  is a summand of  $P$  because  $N \subset P$  and both  $N, P$  are summands of  $F$ .

To show  $\det(a_{ij})$  is invertible, notice that by plugging  $y_i = \sum b_{ij} e_j + t_i$  into  $\sum a_i e_i = \sum a_i y_i$  shows  $a_j = \sum a_i a_{ij}$ , thus by the argument before,  $a_{ij}$  are non-invertible for  $i \neq j$  and  $1 - a_{ii}$  is non-invertible, so  $a_{ii}$  is invertible. Because  $R$  is a local ring, we can easily see  $\det(a_{ij})$  is invertible.

PID case: directly from(3.2.4.21). □

**Prop. (5.3.1.6).** If  $I \in \text{Ideal}(R)$  is nilpotent and  $\overline{P}$  is a projective  $R/I$ -module, then there exists a projective  $R$ -module  $P$  that  $P/IP \cong \overline{P}$ . ┘

*Proof:* Cf.[Sta]P07LV. □

### Finite Projective Modules

**Prop. (5.3.1.7) [Finite Projective Modules].** For  $M \in \text{Mod}_R$ , the following are equivalent:

1.  $M$  is finite projective.
2.  $M$  is f.p. and flat.
3.  $M$  is f.p. and all its localizations at (maximal) primes are free.
4.  $M$  is finite locally free.
5.  $M$  is finite and locally free.
6.  $M$  is finite and all its localizations at primes are free and the function  $x \rightarrow \text{rank}_{\kappa(x)}(M \otimes_R \kappa(x))$  is a locally constant function on  $\text{Spec } R$ .

The category of finite projective modules over  $R$  is denoted by  $\text{Mod}^{\text{f.proj}}/R$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 2$ :  $M \otimes K = R^m$  for some  $K$  and  $m$ , so  $K$  is finite and  $M = R^m/K$  is f.p. And  $M$  is flat because it is a summand of  $R^n$  (5.4.1.4).

$2 \rightarrow 4$ : For any prime  $p$ , choose a basis for the  $k(p)$ -vector space  $M \otimes k(p)$ , then by Nakayama, their inverse image generate  $M_g$  for some  $g \notin p$  (3.2.4.8), and the kernel  $K$  of this generation is finite because  $M_g$  is f.p. And  $K \otimes k(p) = 0$  by the flatness of  $M_g$ . Then by Nakayama again there is a  $g' \notin p$  that  $M_{gg'} = 0$  (3.2.4.8).

$4 \rightarrow 3$ : Because f.p. is local (5.1.4.4).

$3 \rightarrow 2$ : Because flatness is trivial.

$4 \rightarrow 5$ : Because finite is local (5.1.4.4).

$5 \rightarrow 4, 4 \rightarrow 6$ : Trivial.

$6 \rightarrow 4$ : Cf. [Sta]00NX.?

$2 + 3 + 4 + 5 + 6 \rightarrow 1$ : Cf. [Sta]00NX.?

Consider the stalk, it is all free by (5.3.1.2) and (5.3.1.5), thus by (7.5.1.38), it is locally free.  $\square$

**Cor. (5.3.1.8) [Partially Stalkwise].** If  $P$  is f.p., then finite projectiveness is a stalkwise property for  $P$ .  $\lrcorner$

**Cor. (5.3.1.9) [Projective and Flat].** A finite module over a Noetherian ring is projective iff it is flat.  $\lrcorner$

**Cor. (5.3.1.10).** If  $M$  is finite projective, then the canonical map  $\text{Hom}(M, N) \otimes L \rightarrow \text{Hom}(M, N \otimes L)$  is an isomorphism.  $\lrcorner$

*Proof:* By proposition above  $M$  is f.p. and finite locally free, so by (5.3.7.7) and tensor commutes with localization, we can check locally, where  $M$  is finite free so the isomorphism is obvious.  $\square$

**Def. (5.3.1.11) [Characteristic Polynomials for Finite Projective Modules].** Let  $M$  be a finite projective module over a ring  $A$ . then we can define a characteristic polynomial in  $A[X]$  for any map of  $A$ -modules  $M \rightarrow M$ : if  $M$  is free, then this map is defined as usual. In general, we can find an open covering  $\text{Spec } A_{f_i}$  of  $\text{Spec } A$  that  $M_{f_i}$  is free over  $A_{f_i}$ . Thus we can define the characteristic polynomial locally and glue them together to get a characteristic polynomial in  $A[X]$ .

In particular, we can also define trace and norm of a  $A$ -module map  $M \rightarrow M$ . And when  $B$  is a locally free  $A$ -algebra, then there are trace and norm maps  $\text{tr} : B \rightarrow A$  and  $Nm : B \rightarrow A$ .  $\lrcorner$

**Prop. (5.3.1.12).** Let  $R$  be a ring and  $I$  be a locally nilpotent ideal. If  $\overline{P}$  is a finite projective module over  $R/I$ , then there exists a finite projective  $R$ -module  $P$  that  $P/IP \cong \overline{P}$ .  $\lrcorner$

*Proof:* Cf.[Sta]0D47. □

**Prop. (5.3.1.13).** Let  $M$  be a  $R$  module and  $I$  a nilpotent ideal of  $R$ . If  $M/IM$  is a projective  $R/I$ -module and  $M$  is flat over  $R$ , then  $M$  is a projective  $R$ -module. ┘

*Proof:* Cf.[Sta]05CG. □

**Thm. (5.3.1.14) [Serre-Suslin].** Every finite projective module over the polynomial ring  $k[x_1, \dots, x_n]$  is free. ┘

*Proof:* Cf.[?Lan05]P850. ? □

### Duality of Projective Modules

**Prop. (5.3.1.15) [Basis Criterion of Projectiveness].** An  $A$ -module  $P$  is projective iff there are elements  $x_i$  in  $P$  and  $f_i$  in  $P^*$  that for any  $x$ ,  $f_i(x) = 0$  a.e.  $i$ , and  $\sum f_i(x)x_i = x$ . Moreover,  $P$  is finite projective iff there are f.m. of them. ┘

*Proof:* If  $P$  is projective, as a summand of a free module, then we can choose the coordinates of the inclusion map as  $f_i$ , and choose the image of the quotient map of the coordinate as  $x_i$ . The converse is verbatim. □

**Cor. (5.3.1.16) [Finite Projective Duality].** If  $P$  is projective, then  $P \rightarrow P^{**}$  is injective, and if  $P$  is finite projective, then it is an isomorphism. ┘

*Proof:* If  $f(x) = 0$  for all  $f \in P^*$ , then the proposition says  $x = 0$ . And if  $P$  is finite projective, it can be seen  $x_i, f_i$  forms a "basis" of  $P^*$  (finiteness used), so  $f_i$  generate  $P^*$ , and similarly  $x_i$  generate  $P$ , so  $P \rightarrow P^{**}$  is surjective. □

**Cor. (5.3.1.17).** If  $P$  is projective over  $R$ , then  $P^* \neq 0$ . ┘

**Cor. (5.3.1.18).** In the meanwhile of the proof, we already get: if  $P$  is finite projective, then  $P^*$  is finite projective, by (5.3.1.15). ┘

**Cor. (5.3.1.19).** If  $P$  is finite projective, the the map  $P \otimes M \rightarrow \text{Hom}(P^*, M)$  is an isomorphism. ┘

*Proof:* In (5.3.1.10), let  $N = R$  and let  $M = P^*$ , then use the fact  $P \cong P^{**}$ . □

**Thm. (5.3.1.20) [Quillen-Suslin].** For  $k \in \text{Field}$ , any finite projective module over  $k[X_1, \dots, X_k]$  is free. (Highly nontrivial). ┘

*Proof:* Cf.[?Lan05]P850. □

**Prop. (5.3.1.21).**  $\prod^{\mathbb{N}} \mathbb{Z}$  is not free over  $\mathbb{Z}$ , by (3.1.3.4). ┘

## 2 Injective

**Prop. (5.3.2.1) [Baer's Criterion].** A right  $R$ -module  $I$  is injective iff for every right ideal  $J$  of  $R$ , every map  $J \rightarrow I$  can be extended to a map  $R \rightarrow I$ . (Directly from (4.8.9.7)). ┘

**Cor. (5.3.2.2).** A module over a PID is injective iff it is divisible. ┘

**Cor. (5.3.2.3).**  $A$  is injective iff  $\text{Ext}^1(R/I, A) = 0$  for every ideal  $I$  of  $R$ . ┘

**Cor. (5.3.2.4)[Rank].** Let  $M$  a finite projective  $R$ -module, then  $M$  is said to have rank  $n$  if  $M/\mathfrak{m}M$  is of rank  $n$  over the field  $R/\mathfrak{m}$  for any arbitrary maximal ideal  $\mathfrak{m}$  of  $R$ .  $\lrcorner$

**Prop. (5.3.2.5).** The category of  $R$ -mod has enough injectives by (4.8.3.28), and it has enough projectives trivially.  $\lrcorner$

**Prop. (5.3.2.6).** If  $I$  is an injective  $A$ -module, then for any ideal  $\alpha$  of  $A$ ,  $\Gamma_\alpha(I) = \{m | \alpha^n m = 0\}$  for some  $n$  is injective.  $\lrcorner$

*Proof:* Use Baer criterion, for any ideal  $b$  of  $A$ , it is f.g. so there is a  $n$  that  $\phi(\alpha^n b) = 0$ , and Artin-Rees tells us that  $\phi(\alpha^N \cap b) = 0$  for some  $N$ . So we have an extension of  $\phi$  over  $b/b \cap \alpha^N$  to  $A/\alpha^N \rightarrow I$ , and this obviously factor through  $\Gamma_\alpha(I)$ , so it is done.  $\square$

**Prop. (5.3.2.7).** For an injective module  $A$ -module  $I$ ,  $I \rightarrow I_f$  is surjective.  $\lrcorner$

*Proof:* we have the sheaf of modules  $\tilde{I}$  is flabby (7.7.1.5), thus the map to the stalk is surjective.  $\square$

### Pontryagin-Duality

References are [Weibel Homological Algebra].

**Def. (5.3.2.8)[Pontryagin-Duals].** The **Pontryagin-dual**  $M^\vee$  of a left  $R$ -module  $M \in \text{Mod}_R$  is the right  $R$ -module  $\text{Hom}_{\text{Ab}}(M, \mathbb{Q}/\mathbb{Z})$ , where  $(fr)(b) = f(rb)$ .

It is easily verified that if  $A \neq 0$ , then  $A^\vee \neq 0$ , and  $\mathbb{Q}/\mathbb{Z}$  is an injective  $\mathbb{Z}$ -module, thus the Pontryagin dual is faithfully exact.  $\lrcorner$

**Prop. (5.3.2.9).**  $M$  is flat  $R$ -module iff  $M^\vee$  is an injective right  $R$ -module (Because  $\text{Hom}(-, \mathbb{Q}/\mathbb{Z})$  is exact).  $\lrcorner$

## 3 Homological Dimension

**Def. (5.3.3.1).** For a  $R$ -mod  $A$ , the **projective dimension**  $\text{pd}(A)$  is the minimal length of a projective resolution of  $A$ . The **injective dimension**  $\text{id}(A)$  is the minimal length of an injective resolution of  $A$ . The **flat dimension**  $\text{fd}(A)$  is the minimal length of a flat resolution of  $A$ .  $\lrcorner$

**Prop. (5.3.3.2).** If  $R$  is Noetherian, then  $\text{fd}(A) = \text{pd}(A)$  for every f.g. module  $A$ .  $\lrcorner$

*Proof:* Use (5.3.3.3), we see that if we choose a syzygy and look at the  $n$ -th term, then it is f.p and flat, so we have it is projective by (5.4.1.14).  $\square$

**Lemma (5.3.3.3)[pd].** If  $\text{Ext}^{d+1}(A, B) = 0$  for every  $B$ , then for every resolution

$$0 \rightarrow M \rightarrow P_{d-1} \rightarrow \dots, P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$$

where  $P_k$  is projective, then  $M$  is projective. Hence we have  $\text{pd}(A) \leq d$ . (Use dimension shifting, the following two are the same).  $\lrcorner$

**Lemma (5.3.3.4)[id].** If  $\text{Ext}^{d+1}(A, B) = 0$  for every  $A$ , then for every resolution

$$0 \rightarrow B \rightarrow P_0 \rightarrow \dots, P_{n-1} \rightarrow M \rightarrow 0$$

where  $P_k$  is injectives, then  $M$  is injective. Hence we have  $\text{id}(B) \leq d$   $\lrcorner$

**Lemma (5.3.3.5)[fd].** If  $\text{Tor}_{d+1}(A, B) = 0$  for every  $B$ , then for every resolution

$$0 \rightarrow M \rightarrow F_{d-1} \rightarrow \dots, F_1 \rightarrow F_0 \rightarrow A \rightarrow 0$$

where  $F_k$  is flat, then  $M$  is flat. Hence we have  $fd(A) \leq d$  ┘

**Prop. (5.3.3.6)[Global Dimension Theorem].** The following are the same for any ring  $R$  and called the **left global dimension** of  $R$ :

1.  $\sup\{id(B)\}$
  2.  $\sup\{pd(A)\}$
  3.  $\sup\{pd(R/I)\}$
  4.  $\sup\{d : \text{Ext}_R^d(A, B) \neq 0 \text{ for some module } A, B\}$ .
- ┘

*Proof:* This follows from (5.3.3.3), (5.3.3.4) and (5.3.2.3). □

**Prop. (5.3.3.7).** A  $\mathbb{Z}$  has global dimension 1 because injective is equivalent to divisible, and this shows that a quotient of an injective is injective. ┘

**Prop. (5.3.3.8)[Tor Dimension Theorem].** The following are the same for any ring  $R$  and called the **Tor dimension** of  $R$ :

1.  $\sup\{fd(A)\}$  for  $A$  a left module.
  2.  $\sup\{fd(B)\}$  for  $B$  a right module.
  3.  $\sup\{pd(R/I)\}$  for  $I$  a left ideal.
  4.  $\sup\{pd(R/J)\}$  for  $J$  a right ideal.
  5.  $\sup\{d : \text{Tor}_d^R(A, B) \neq 0 \text{ for some module } A, B\}$ .
- ┘

*Proof:* This follows from (5.3.3.5) applied to  $R$  and  $R^{op}$  and also (5.4.1.2). □

**Prop. (5.3.3.9)[Change of Rings].** Let  $S \rightarrow R$  be a ring map, let  $A \in \text{Mod}_R$ , then we have  $pd_S(A) \leq pd_R(A) + pd_S(R)$ . ┘

*Proof:* Use the Cartan-Eilenberg resolution and the total complex has length  $pd_R(A) + pd_S(R)$ . □

## 4 Depth

### Regular sequences

**Def. (5.3.4.1) [Regular Sequences].** If  $R$  is a commutative ring and  $M$  is an  $R$ -module, then a sequence  $(f_1, \dots, f_n)$  of elements of  $R$  is called a **M-regular sequence** if  $f_k$  is a nonzero-divisor of  $M/(f_1, \dots, f_{k-1})$  and  $M/(f_1, \dots, f_n) \neq 0$ . If  $M = R$ , then it is simply called a **regular sequence**. ┘

**Prop. (5.3.4.2).** If  $R$  is a **local ring**,  $M$  a finite  $R$ -module, and  $(f_1, \dots, f_n)$  is a  $M$ -regular sequence, then a permutation of this sequence is also an  $M$ -regular sequence. ┘

*Proof:* By transposition of adjacent indices, we can assume  $n = 2$ . Then  $x$  is a non-zero-divisor, and  $x \in \mathfrak{m}$ , so Now  $(x, y)$  is an  $M$ -regular sequence iff  $M \otimes_R^L R/x$  is discrete and  $M \otimes_R^L R/x \otimes_R^L R/y$  is discrete. Then it suffices to prove  $y$  is injective on  $M$ : If  $ym = 0$ , then  $m = xm'$  for some  $m'$ , because  $y$  is injective on  $M/x$ , and then  $ym' = 0$  also because  $x$  is injective on  $M$ . Then  $x$  is surjective on  $\ker(y)$ , thus  $\ker(y) = 0$  by Nakayama.  $\square$

**Prop. (5.3.4.3).** Let  $R$  be a ring and  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$  be an exact sequence of  $R$ -modules. Then if a sequence  $(f_1, \dots, f_r) \in R^r$  is both  $M_1$ -regular and  $M_3$ -regular, then it is also  $M_2$ -regular.  $\lrcorner$

*Proof:* Use the snake lemma,  $\ker(f_1|_M) = 0$ , and  $0 \rightarrow M_1/fM_1 \rightarrow M_2/fM_2 \rightarrow M_3/fM_3 \rightarrow 0$ , then use induction.  $\square$

**Prop. (5.3.4.4).** For any integers  $e_i \geq 1$ ,  $(f_1, \dots, f_n)$  is a  $M$ -regular iff  $(f_1^{e_1}, \dots, f_n^{e_n})$  is  $M$ -regular.  $\lrcorner$

*Proof:* Use induction on  $n$ .  $n = 1$  is trivial, and it suffices to show that  $(f_1^e, f_2, \dots, f_n)$  is  $M$ -regular. Then we use induction on  $e$ : There is an exact sequence  $0 \rightarrow M/f_1M \rightarrow M/f_1^eM \rightarrow M/f_1^{e-1}M \rightarrow 0$ , so by (5.3.4.3), if  $(f_1, \dots, f_n)$  is a  $M$ -regular, then  $(f_1^e, f_2, \dots, f_n)$  is  $M$ -regular. Conversely, if  $(f_1^e, f_2, \dots, f_n)$  is  $M$ -regular, then  $f_2$  is injective on  $M/f_1^eM$ , so also injective on  $M/f_1M$ , thus injective on  $M/f_1^{e-1}M$  by induction hypothesis, and also there is a further exact sequence  $0 \rightarrow M/(f_1, f_2)M \rightarrow M/(f_1^e, f_2)M \rightarrow M/(f_1^{e-1}, f_2)M \rightarrow 0$ , so we can consider  $f_3$ , and then so on, thus show  $(f_1^e, f_2, \dots, f_n)$  is  $M$ -regular.  $\square$

**Prop. (5.3.4.5).** Let  $R$  be a ring, then the following are equivalent:

- any permutation of  $(f_1, \dots, f_r)$  is a regular sequence.
- Any subsequence of  $(f_1, \dots, f_r)$  is a regular sequence.
- $(f_1X_1, \dots, f_rX_r)$  is a regular sequence of  $R[X]$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2$ : Trivial.  $2 \rightarrow 1$ : Use induction on  $r$ : If  $r = 2$  and  $(x, y)$  is regular, we have  $M \otimes_R^L R/x, M \otimes_R^L R/y$  are both discrete, and  $M \otimes_R^L R/x \otimes_R^L R/y$  is also discrete, thus  $M \otimes_R^L R/y \otimes_R^L R/x$  is also discrete and  $(y, x)$  is regular. For  $r > 2$ , it suffices to show  $f_{\sigma(r)}$  is regular in  $M/(f_{\sigma(1)}, \dots, f_{\sigma(r-1)})$ . If  $\sigma(r) = r$ , then we are done, otherwise,  $f_r$  and  $f_{\sigma(r)}$  are both injective on  $M/(f_1, \dots, \widehat{f_{\sigma(r)}}, \dots, f_{r-1})$  by induction hypothesis, and  $(f_{\sigma(r)}, f_r)$  is a regular sequence for this ring, then so is  $(f_r, f_{\sigma(r)})$  by the  $r = 2$  case, thus we are done.

$3 \iff 2$ : Notice as a  $R$ -module,  $R[X_1, \dots, X_r]/(f_1X_1, \dots, f_iX_i)$  is a direct sum of the modules  $R/I_E X_1^{e_1} \dots X_r^{e_r}$ , where  $I_E$  is generated by those  $f_j$  that  $j \leq i$  and  $e_j > 0$ . Then  $f_{i+1}X_{i+1}$  is injective on this iff  $f_{i+1}$  is injective on  $R/I_E$  for any  $E$ . Then it is clear that this is equivalent to 2.  $\square$

**Def. (5.3.4.6) [Quasi-Regular Sequence].** Let  $f_1, \dots, f_c \in R$  and  $J = (f_1, \dots, f_c)$ , let  $M$  be a  $R$ -module, then there is a canonical map  $M/JM \otimes_{R/J} R/J[X_1, \dots, X_c] \rightarrow \otimes_{n \geq 0} J^n M/J^{n+1}M$ . Then  $(f_1, \dots, f_c)$  is called  $M$ -quasi-regular sequence iff this is an isomorphism.  $\lrcorner$

### Depth

**Prop. (5.3.4.7) [Rees].** For a f.g. module  $M$  and  $IM \neq M$ ,

$$\text{depth}_I(M) = \min\{i \mid \text{Ext}_A^i(A/I, M) \neq 0\} = \min\{i \mid \text{Ext}_A^i(N, M) \neq 0\}$$

where  $\text{depth}_I(M)$  is the length of the maximal  $M$ -regular sequence in  $I$ ,  $N$  is a finite  $A$ -module with  $\text{Supp}(N) \subset V(I)$ .  $\lrcorner$



*Proof:* If No elements of  $I$  are  $M$ -regular, then  $i \subset \cup \text{Ass}(M)$  thus in one of them, so  $\text{Hom}_{A_p}(k, M_p) \neq 0$ , and we have  $N_p/PN_p = N \otimes_A k_p$  nonzero by Nakayama, thus  $\text{Hom}_k(N \otimes_A k_p, k_p) \neq 0$ , thus  $\text{Hom}_{A_p}(N_p, M_p) = (\text{Hom}_A(N, M))_p \neq 0$ , so  $\text{Ext}_A^0(N, M) \neq 0$ . Other dimensions follows by induction, consider the cokernel of  $M \xrightarrow{a_1} M$ .

Conversely, use induction, then we have an injection  $\text{Ext}_A^i(N, M) \xrightarrow{a_1} \text{Ext}_A^i(N, M)$  for  $i < n$ . And the condition shows that  $I \subset \sqrt{\text{Ann}(M)}$ , so  $a_1^n N = 0$ , thus the result.  $\square$

**Cor. (5.3.4.8).** Two maximal regular sequence in a f.g. module have the same length.  $\lrcorner$

**Cor. (5.3.4.9).** For a module  $M$  over a Noetherian ring  $A$ , we know  $\Gamma_I(M) = \{m \mid I^n m = 0 \text{ for some } n\}$ , and  $H_I^n$  is its right derived functor, then we have  $\text{depth}_I(M) \geq n \iff H_I^i(M) = 0$  for  $i < n$ . (Because derived functor commutes with colimits, consider  $N = A/I^k$ ).  $\lrcorner$

**Lemma (5.3.4.10)[Ischebeck].** For a Noetherian local ring  $A$ , if  $M, N$  are finite modules, then we have  $\text{Ext}_A^i(N, M) = 0$  for  $i < \text{depth}(M) - \dim N$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (5.3.4.11).** Let  $A$  be a local ring and  $M$  is finite  $A$ -module, then  $\text{depth}(M) \leq \dim A/P \leq \dim M$  for every  $P \in \text{Ass}(M)$ . (Because  $\text{Hom}(A/P, M) \neq 0$ .)  $\lrcorner$

*Proof:*  $\square$

**Prop. (5.3.4.12)[Auslander-Buchsbaum Formula].** For a local ring  $R$ , if  $M$  is a finitely generated  $R$ -mod, if  $\text{pd}(M) < \infty$ , then we have  $\text{depth}(R) = \text{depth}(M) + \text{pd}(M)$ .  $\lrcorner$

*Proof:* Cf.[Weibel P109].  $\square$

### Cohen-Macaulay

**Def. (5.3.4.13)[Cohen-Macaulay Modules].** For  $A$  Noetherian local, a f.g.  $A$ -module  $M$  is called **Cohen-Macaulay** if  $\text{depth}(M) = \dim M$ . In view of (5.3.4.11), this is equivalence to  $\text{depth}(M) = \dim A/P$  for all  $P \in \text{Ass}(M)$ . A **Cohen-Macaulay ring** is a ring  $A$  that is Cohen-Macaulay over itself.

A localization of a C.M local ring is C.M, so we call a ring **Cohen-Macaulay** if all its localization at primes are C.M.  $\lrcorner$

**Prop. (5.3.4.14)[Gorenstein Ring].** A ring  $R$  is called **Gorenstein** iff  $\text{id}_R R < \infty$ . A Gorenstein local ring is C.M. In this case,  $\text{depth}(R) = \text{id}_R R = \dim R$ , and  $\text{Ext}_R^q(R/m, R) \neq 0 \iff q = \dim R$ .  $\lrcorner$

*Proof:* Cf.[Weibel P107].  $\square$

**Prop. (5.3.4.15).** A ring is C.M. iff for all ideals, the associated primes of  $A/I$  all have the same height as  $I$ , i.e. unmixed.  $\lrcorner$

*Proof:*  $\square$

**Prop. (5.3.4.16).** If a local ring is C.M. and  $I = (x_1, \dots, x_r)$  is a regular sequence, then  $(x_1, \dots, x_r)$  is  $\lrcorner$

*Proof:* ? Isn't this always true?  $\square$



**Prop. (5.3.4.17).** Let  $A$  is a Noetherian local ring and  $M$  a f.g. module, if a set of elements  $(x_1, \dots, x_r)$  forms a regular sequence for  $M$ , then  $\dim M/(x_1, \dots, x_r) = \dim M - r$ . The converse is also true when  $A$  is C.M. If this is the case, then  $A/(x_1, \dots, x_r)$  is also C.M.  $\lrcorner$

*Proof:* By (5.2.2.19), we have  $<$ , for the converse,  $\text{Supp}(M/fM) = \text{Supp}(M) \cap \text{Supp}(A/fA) = \text{Supp}(M) \cap V(f)$ , and when  $f$  is  $M$ -regular,  $V(f)$  doesn't contain any  $\text{Ass}(M)$  thus no minimal elements of  $\text{Supp}(M)$ , so  $\dim(M/fM) < \dim M$ , thus we have  $>$ .  $\square$

When  $A$  is C.M.:?  $\square$

**Prop. (5.3.4.18).** Let  $R \rightarrow S$  be a local homomorphism of Noetherian local rings, if  $R$  is C.M. and  $S$  is finite flat over  $R$  or  $S$  is flat over  $R$  and  $\dim S \leq \dim R$ , then  $S$  is C.M., and  $\dim R = \dim S$ .  $\lrcorner$

*Proof:* Cf. [Sta]00R5.  $\square$

## 5 Normal & Regular Rings

### Normal Ring

**Def. (5.3.5.1) [Normal Rings].** A **normal domain** is a domain and is integrally closed in its fraction field.

A domain is normal iff all its localizations are normal, so we can define a **normal ring** to be a ring that all its stalks are normal local rings. In particular, a normal ring is reduced.  $\lrcorner$

*Proof:* The localization of a normal domain is normal, and the converse follows from  $A = \bigcap A_{\mathfrak{m}}$  (5.1.1.36).  $\square$

**Prop. (5.3.5.2) [UFD is Normal].** A UFD is a normal domain.  $\lrcorner$

*Proof:* If  $A$  is a UFD, then for any  $x \in \text{Frac}(A)$  integral over  $A$ , checking the primes dividing the coefficients, we see  $x \in A$ .  $\square$

**Prop. (5.3.5.3).** A normal ring  $R$  is integrally closed in its ring of fractions.  $\lrcorner$

*Proof:* Let  $x \in Q(R)$  be integral over  $R$ , and  $I = \{f \in R \mid fx \in R\}$ , then for any prime  $\mathfrak{p}$  of  $R$ ,  $R \rightarrow R_{\mathfrak{p}}$  is injective, so  $R_{\mathfrak{p}} \subset \text{Frac}(R) \otimes_R R_{\mathfrak{p}}$ , and  $x \otimes 1$  is integral over  $R_{\mathfrak{p}}$ , thus  $x \otimes 1 \in R_{\mathfrak{p}}$ , which means  $x \otimes 1 = 1 \otimes a/f$  for some  $a, f \in R, f \notin \mathfrak{p}$ . This means  $f'(fx - a) = 0 \in Q(R)$  for some  $f' \notin \mathfrak{p}$ , so  $ff' \in I$ , and thus  $I$  is not contained in any prime ideal, so  $I = R$  and  $x \in R$ .  $\square$

**Prop. (5.3.5.4).** Let  $R$  be a reduced ring with f.m. minimal prime ideals, then the following are equivalent:

- $R$  is a normal ring.
- $R$  is integrally closed in its ring of fractions.
- $R$  is a finite product of normal domains.

In particular, a Noetherian normal domain is a finite product of normal domains (5.1.1.49).  $\lrcorner$

*Proof:*  $3 \rightarrow 1$  is trivial,  $1 \rightarrow 2$  is by (5.3.5.3), for  $2 \rightarrow 3$ : let  $\mathfrak{p}_i$  be the minimal prime ideals of  $R$ , then  $\text{Frac}(R) = \prod_{i=1}^r Q_{\mathfrak{p}_i}$  (5.1.7.22), with each  $Q_{\mathfrak{p}_i}$  field because  $R$  is reduced. Denote the idempotents of  $\text{Frac}(R)$  by  $e_i$ . Then  $e_i$  is integral thus in  $R$ . These idempotents make  $R$  into a product of domains, which are just  $R/\mathfrak{p}_i$ , because the kernel of the map  $R \rightarrow R_{\mathfrak{p}_i}$  is  $\mathfrak{p}_i$ . Now  $R$  is integrally closed in  $\text{Frac}(R)$  implies each  $R/\mathfrak{p}_i$  is integrally closed in  $R_{\mathfrak{p}_i}$ , thus  $R$  is a finite product of normal domains.  $\square$

**Def. (5.3.5.5) [Normalization].** The **normalization** of an integral domain is the alg.closure of it in its quotient field. It commutes with localization.  $\lrcorner$

**Example (5.3.5.6).** The normalization of  $\mathbb{R}[X, Y]/(X^2 + Y^2)$  is  $\mathbb{C}[X]$ , because the map

$$\mathbb{R}[X, Y]/(X^2 + Y^2) \rightarrow \mathbb{C}[X] : X \mapsto X, Y \mapsto iX$$

is injective, and  $i, X$  are both integral over  $\mathbb{R}[X, Y]/(X^2 + Y^2)$ .  $\lrcorner$

**Def. (5.3.5.7) [Completely Normal Domains].** A domain is called **completely normal** iff all almost normal elements are in  $A$ , i.e.  $\{u | \exists a, au^n \in A \ \forall n\} \in A$ . For Noetherian ring, completely normal is equivalent to normal.  $\lrcorner$

*Proof:* Cf. [Sta]00GX.  $\square$

**Prop. (5.3.5.8) [Power Series over Domain Domain].**  $A$  is a normal domain, then so does  $A[[X]]$ . If  $A$  is Noetherian normal domain, then so does  $A[[X]]$ .  $\lrcorner$

*Proof:* Cf. [Sta]030A, 0BI0.  $\square$

**Prop. (5.3.5.9).** Direct limits of normal rings are normal.  $\lrcorner$

*Proof:* Let  $\mathfrak{p}$  be an ideal of  $R = \varinjlim R_i$ ,  $\mathfrak{p}_i = \mathfrak{p} \cap R_i$ , then  $R_{\mathfrak{p}} = \varinjlim (R_i)_{\mathfrak{p}_i}$ , so it suffices to prove for normal domains, the rest is easy.  $\square$

**Prop. (5.3.5.10) [Closure in Separable Extension].** Let  $R$  be a Noetherian normal domain with field of fraction  $K$  and  $L/K$  is a finite separable field extension, then the integral closure  $S$  of  $R$  in  $L$  is finite over  $R$ .  $\lrcorner$

*Proof:* Let  $\text{tr} : L \times L \rightarrow K : (x, y) \mapsto \text{tr}(xy)$  be the trace pairing, then as  $L/K$  is finite separable, this is non-degenerate. Now if  $x \in L$  is integral over  $R$ , then  $\text{tr}(x) \in R$ . So if  $x_1, \dots, x_n$  are integral and form a basis of  $L$  over  $K$ , then  $M = \{y \in L | \langle x_i, y \rangle \in R\}$  is an  $R$ -module and  $M \cong R^n$ , and  $S \subset M$ , so  $S$  is finite over  $R$  as  $R$  is Noetherian.  $\square$

**Prop. (5.3.5.11) [Field Extension].** Let  $A$  be a  $K$ -algebra and  $L/K \in \mathbf{Field}$ , then if  $A_L$  is a normal integral domain, so is  $A$ . And the converse is also true if  $L/K$  is separable.  $\lrcorner$

**Prop. (5.3.5.12) [Algebraic Hartogs's Lemma].** A principal ideal  $A$  in a Noetherian normal domain is unmixed and  $A = \bigcap_{\text{ht } p=1} A_p$ .  $\lrcorner$

*Proof:* Cf. [Matsumura P124].  $\textcolor{red}{?}$  [Rising Sea, P320], [Sta]031T.  $\square$

**Prop. (5.3.5.13) [Krull].** If  $A$  is a normal domain with fraction field  $K$ ,  $L$  is a normal extension field of  $K$  and  $B$  is the integral closure of  $A$  in  $L$  that  $A \subset B$  is integral, then any two prime ideals of  $B$  lying over a prime in  $A$  are conjugate by an action of  $G_{L/K}$ .  $\lrcorner$

*Proof:* Firstly if  $G_{L/K}$  is finite, and  $P, P'$  be primes of  $B$  that  $P \cap A = P' \cap A$ . Let  $P_i = \sigma_i(P)$ . If  $P' \neq P_i$  for any  $i$ , then  $P' \not\subseteq P_i$  for any  $i$  by (5.2.1.5), so there is some  $x \in P'$  not in any of  $P_i$ . Now let  $y = (\prod_i \sigma_i(x))^q$  where  $q = 1$  if  $\text{char}(K) = 0$  and  $q = p^r$  for  $r$  large if  $\text{char}(K) = p$ , then  $y \in K$  thus in  $A$  because  $A$  is normal, and it is not in  $P$  by hypothesis. But  $y \in P' \cap A = P \cap A$ , contradiction.

For the infinite field extension case, let  $K'$  be the fixed field of  $G_{L/K}$  so that  $K'/K$  is purely inseparable, then there is clearly exactly one prime of  $K'$  over any prime of  $A$ . So we can assume  $L/K$  is Galois, then use profinite group technique to find a  $\sigma \in G_{L/K}$  that  $\sigma(P) = P'$ .  $\square$

**Prop. (5.3.5.14) [Hironaka].** Let  $A$  be a local Noetherian domain that is a localization of an algebra of f.t. over a field  $k$ . Let  $t \in A$  that

- $tA$  has only one minimal associated prime ideal  $p$ .
- $t$  generate the maximal ideal of  $A_p$ .
- $A/p$  is normal.

Then  $p = tA$  and  $A$  is normal. ┘

*Proof:* Cf.[Hartshorne P264]. □

**Prop. (5.3.5.15) [Quadratic Extension is Normal].** If  $A$  is a UFD that 2 is invertible in  $A$  and  $f \in A \setminus A^2$ , then  $A[T]/(T^2 - f)$  is integral and normal. ┘

*Proof:* It is integral by (3.2.3.14). To show it is normal, assume  $F(T) \in B[T]$  that  $f(\alpha) = 0$ , where  $\alpha \in K(B) \setminus B$ , then by replacing  $F$  by  $\overline{F}F$ , we can assume  $F(T) \in A[T]$ . We can assume  $\alpha \notin K(A)$ , thus  $\alpha = g + hZ$ , then the minimal polynomial of  $\alpha$  is  $Q(T) = T^2 - 2gT + (g^2 - h^2f) \in K(A)[T]$ . Then  $F(T) = P(T)Q(T)$  in  $K(A)[T]$ . By Gauss's lemma,  $2g, g^2 - h^2f \in A$ , so  $g, f \in A$  by hypothesis. □

**Cor. (5.3.5.16).** If  $k$  is a field of characteristic  $\neq 2$ , then

- $k[x_1, \dots, x_n]/(x_1^2 + \dots + x_m^2)$  is integral normal for  $n \geq m \geq 3$ .
  - $k[x, y, z, w]/(wz - xy)$  is normal. (Diagonalize).
- ┘

**Prop. (5.3.5.17).**  $\mathbb{Z}[\sqrt{n}]$  is integrally closed for  $n \equiv 3 \pmod{4}$ , and  $\mathbb{Z}[\frac{1+\sqrt{n}}{2}]$  is integrally closed for  $n \equiv 1 \pmod{4}$ , by (19.3.1.46). ┘

### Regular Ring

**Def. (5.3.5.18) [Regular Rings].** A Noetherian local ring  $(A, \mathfrak{m}, k)$  is called **regular local** iff it  $\text{rank}_k \mathfrak{m}/\mathfrak{m}^2 = \dim A$ . This is equivalent to  $\text{rank}_k \mathfrak{m}/\mathfrak{m}^2 \leq \dim A$ , or  $\text{gr } A \cong k[X_1, \dots, X_d]$  by (5.2.2.19).

Localization of a regular local ring at primes are regular local, Cf.[Sta]0AFS. Hence we define a **regular ring** to be a Noetherian ring that all its localization at primes are regular local. ┘

*Proof:* Cf.[Matsumura P139]. □

**Prop. (5.3.5.19).** If  $A$  is regular, then  $A[X_1, \dots, X_n]$  is regular, and  $A[[X_1, \dots, X_n]]$  is regular. ┘

*Proof:* Cf.[Matsumura P176]. □

**Prop. (5.3.5.20) [Auslander-Buchsbaum].** A regular local ring is UFD. In particular it is a normal domain. Thus a regular ring is normal and thus reduced (5.3.5.1). ┘

*Proof:* Cf.[Matsumura P142],[Weibel P106]. □

**Cor. (5.3.5.21).** A regular local ring of dimension 0 is a field, and a regular local ring of dimension 1 is a DVR. ┘

**Prop. (5.3.5.22).** A regular local ring is Gorenstein hence C.M.. ┘

*Proof:* □

**Prop. (5.3.5.23)[Regular Local Ring and Regular Sequences].** Let  $(R, \mathfrak{m})$  be a regular local ring of dimension  $d$ ,

- If  $x_1, \dots, x_c$  be a sequence that maps to linearly independent elements in  $\mathfrak{m}/\mathfrak{m}^2$ , then  $(x_1, \dots, x_c)$  is a regular sequence, and  $R/(x_1, \dots, x_c)$  is a regular local ring of dimension  $d - c$ .
- If  $I \subset \mathfrak{m}$  and  $R/I$  is a regular local ring, then  $I = (x_1, \dots, x_c)$  where  $(x_1, \dots, x_c)$  is a regular sequence.
- If  $(x_1, \dots, x_c)$  is a regular sequence in  $\mathfrak{m}$  and  $R/(x_1, \dots, x_c)$  is a regular local ring, then  $R$  is also a regular local ring.

┘

*Proof:* 1: We can complete it to a sequence  $(x_1, \dots, x_d)$  that  $(x_1, \dots, x_d)$  generate  $\mathfrak{m}/\mathfrak{m}^2$ , then by Nakayama  $(x_1, \dots, x_d) = \mathfrak{m}$ . Now by Krull's height theorem(5.2.4.18),  $R/(x_1)$  has dimension  $\geq d - 1$  and  $(x_2, \dots, x_d)$  generate the maximal ideal, so  $R/(x_1)$  is a regular local ring by definition. Now  $x_1 \neq 0$  because  $R$  is a domain(5.3.5.20), thus we can use induction to show  $(x_1, \dots, x_c)$  is a regular sequence.

2: Let  $\dim(R/I) = d - c$ , then the hypothesis shows  $(I + \mathfrak{m}^2)/\mathfrak{m}^2 \cong I/\mathfrak{m}I$  has rank  $c$ , thus we can choose  $(x_1, \dots, x_c) \in I$  that generate  $I/\mathfrak{m}I$ . Thus by Nakayama  $I = (x_1, \dots, x_c)$ , and  $(x_1, \dots, x_c)$  is a regular sequence by item1.

3: By induction, it suffices to prove the  $c = 1$  case, then any lift of  $x_1$  together with  $d - 1$  generator of the maximal ideal of  $R/\mathfrak{m}$  is a set of generator of  $\mathfrak{m}$ . □

**Cor. (5.3.5.24).** A regular local ring is C.M., by item1. ┘

**Prop. (5.3.5.25).** If a quotient of a Noetherian local ring by a non-zerodivisor is regular, then it is itself regular. ┘

**Prop. (5.3.5.26)[Serre].** A Noetherian local ring  $A$  is regular iff the global dimension of  $A$  is finite. ┘

*Proof:* Cf.[Mat P139]. □

**Prop. (5.3.5.27).** For  $A$  a regular local ring and  $M$  a f.g.  $A$ -module,

$$pd(M) + \text{depth } M = \dim A.$$

Cf.[Hartshorne P237]. ┘

**Cor. (5.3.5.28).** For a f.g. module  $M$  over a regular local ring  $A$ ,  $pd(M) \leq n$  iff  $\text{Ext}^i(M, A) = 0$  for all  $i > n$ . ┘

*Proof:* This is because we can use dimension shifting to show  $\text{Ext}^i(M, N) = 0$  for all  $N$  f.g., then(5.3.3.3) says that  $pd(M) \leq n$ . □

**Prop. (5.3.5.29)[Regular and Regular Sequence].** Let ┘

**Prop. (5.3.5.30).** Let  $R \rightarrow S$  be a flat local homomorphism of Noetherian local rings that  $R$  is regular,  $S/\mathfrak{m}_S$  is regular, then  $S$  is also regular. ┘

*Proof:* Cf.[Sta]031E. □

**Serre's Regularity Conditions  $R_k$  &  $S_k$** 

**Def. (5.3.5.31) [Serre's Conditions].**  $R \in \mathcal{CRing}$  is called  $R_k$  iff for all prime  $p$  of height  $\leq k$ ,  $A_p$  is regular. And a ring is called **regular in codimension 1** if it is  $R_1$ .

$R \in \mathcal{CRing}$  is called  $S_k$  iff  $\text{depth}(A_p) \geq \min(k, \text{ht}(p))$  for all prime  $p$ .

A module  $M$  is called  $S_k$  iff  $\text{depth}(M_p) \geq \min(k, \dim \text{Supp } M_p)$  for all prime  $p$ . ┘

**Prop. (5.3.5.32).**

- $M$  is  $S_1$  iff  $M$  has no associated embedded primes. Cf. [Sta]031Q.
- A Noetherian ring is reduced iff it is  $R_0$  and  $S_1$ . Cf. [Sta]031R.
- (Serre Criterion) A Noetherian ring is normal iff it is  $R_1$  and  $S_2$ . Cf. [Sta]031S.
- A ring is C.M. iff it is  $S_N$ . ┘

*Proof:* □

**Cor. (5.3.5.33) [Regular and Normal].** A regular ring is normal, and normal ring is regular in codimension 1. ┘

*Proof:* By (5.3.5.32), it suffices to prove that a regular ring satisfies  $R_1$  and  $S_2$ . A regular ring is C.M. (5.3.5.22) so it is  $S_2$  by (5.3.5.32), it is  $R_1$  by (5.3.5.18) □

**Cor. (5.3.5.34) [Normal and Regular of Dimension 1].** A Noetherian local ring of dim 1 is normal iff it is regular. i.e. integral domain and integrally closed iff maximal ideal is principal. **This is practical when verifying a curve is regular.** ┘

**6 Geometric Properties**

**Def. (5.3.6.1).**

- A  $k$ -algebra  $S$  is called **geometrically reduced/integral/connected**... over a field  $k$  iff for any field extension  $k'/k$ ,  $\text{Spec } S_{k'}$  is reduced/integral/connected....
- A Noetherian  $k$ -algebra  $S$  is called **geometrically regular** iff for any f.g. field extension  $K/k$ ,  $S_K$  is regular (Notice  $A \otimes_k k'$  is Noetherian (5.1.1.47), so this makes sense). ┘

**Prop. (5.3.6.2) [Geometrically reduced].** If  $S$  is a  $k$ -algebra, the following are equivalent.

1.  $S$  is geometrically reduced.
2.  $S \otimes_k \bar{k}$  is reduced.
3.  $S \otimes_k k^{\text{perf}}$  is reduced.
4.  $S \otimes_k k'$  is reduced for any finite purely inseparable field extension  $k'/k$ .
5.  $S \otimes_k k^{1/p}$  is reduced.
6. residue fields of  $S$  at maximal points are reduced.
7.  $S \otimes_k R$  is reduced for every reduced  $k$ -algebra  $R$ . ┘

*Proof:*  $1 \rightarrow 7$ : We can assume  $R$  is f.g., thus  $R$  is contained in a finite product of fields Cf.[Sta]030V?, and then we can assume  $R$  is a product of fields, and we are done.

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4$  is clear.  $3 \rightarrow 5$  is clear.

$4 \rightarrow 1$ : For any field extension  $K/k$ , we can assume WLOG  $K/k$  is f.g., thus?

$5 \rightarrow 1$ : ? 6: ? Cf.[Sta]030V and [Gortz 135]. □

**Prop. (5.3.6.3)[Geometrically Irreducible].** Let  $S$  be a  $k$ -algebra, the following are equivalent:

1.  $S$  is geometrically irreducible.
  2. For any finite separable field extension  $k'/k$ , the spectrum of  $S_{k'}$  is irreducible.
  3. The spectrum of  $S_{k^{\text{sep}}}$  is irreducible.
  4. The spectrum of  $S_{\bar{k}}$  is irreducible.
- ┘

*Proof:* Cf.[Sta]037K. ? □

**Prop. (5.3.6.4).** If  $R, S \in \mathcal{C}\text{Ring}/k$  and  $S/k$  is geometrically irreducible, then the map

$$\text{Spec}(R \otimes_k S) \rightarrow \text{Spec } R$$

induces a bijection on irreducible components. ┘

*Proof:* Cf.[Sta]037O. ? □

**Cor. (5.3.6.5)[Geometrically Irreducible Normal Integral Domains].** Let  $(R, K) \in \mathcal{C}\text{Ring}/k$  be an integral domain. If  $K$  contains no non-trivial finite separable field extension  $k'$  of  $k$ , then  $R$  is geometrically integral. The converse holds if  $R$  is normal. ┘

*Proof:* If  $R$  is not geometrically irreducible, then for some separable field extension  $k'/k \in \mathbf{Field}$ ,  $R \otimes_k k' \subset K \otimes_k k'$  is not integral. Thus  $K$  must contain a non-trivial separable field extension of  $k$ . Conversely, if  $R$  is normal, for any non-trivial finite separable field extension  $k'/k \in \mathbf{Field}$ ,  $k' \subset R$  because  $R$  is normal. Thus  $R \otimes_k k'$  is not integral. □

**Prop. (5.3.6.6)[Geometrically Integral].** For  $S \in \mathcal{C}\text{Ring}/k$ , the following are equivalent:

1.  $S$  is geometrically integral.
  2. For any finite separable field extension  $k'/k$ ,  $S_{k'}$  is an integral domain.
  3.  $S_{\bar{k}}$  is an integral domain.
  4.  $S \otimes_k R$  is an integral domain for any integral domain  $R$  over  $k$ .
- ┘

*Proof:* This follows from (5.3.6.3)(5.3.6.2) and (5.3.6.4). □

**Prop. (5.3.6.7).** It suffices to check geometrically regular for  $k'/k$  finite purely inseparable. ┘

*Proof:* Cf.[Sta]0381. ? □

## 7 Finitely Presentedness

### Finite Presented Modules

**Def. (5.3.7.1)[Finitely Presented Modules].** A finitely presented module is a module of the form  $R^m/R^n$ .

Finite presentation is stable under base change because tensoring is right exact.  $\lrcorner$

**Prop. (5.3.7.2).** Given an exact sequence of  $R$ -modules  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ ,

- If  $M_1, M_3$  are f.p., then so does  $M_2$ .
- If  $M_3$  is f.p. and  $M_2$  is f.g., then  $M_1$  is f.g.
- If  $M_2$  is f.p. and  $M_1$  is f.g., then  $M_3$  is f.p.

$\lrcorner$

*Proof:* 1: e can find an commutative diagram 
$$\begin{array}{ccccccc} 0 & \longrightarrow & R^m & \longrightarrow & R^{m+n} & \longrightarrow & R^m \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 \longrightarrow 0 \end{array}$$
 and use the

snake lemma to see the kernel is f.g..

2: Use the diagram 
$$\begin{array}{ccccccc} R^m & \longrightarrow & R^n & \longrightarrow & M & \longrightarrow & 0 \\ \downarrow \alpha & & \downarrow & & \downarrow \text{id} & & \\ 0 & \longrightarrow & M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 \longrightarrow 0 \end{array}$$
 and snake lemma, then image and

cokernel of  $\alpha$  are all finite, so  $M_1$  is finite.

3: Choose a presentation  $R^m \rightarrow R^n \rightarrow M_2$  and a surjection  $f : R^k \rightarrow M_1$ , then we can lift  $f$  to  $R^k \rightarrow R^n$ , and then  $M_3$  can be written as a quotient  $R^{m+k} \rightarrow R^n \rightarrow M_3$ .  $\square$

**Cor. (5.3.7.3).** A direct summand of a f.p. module is f.p..  $\lrcorner$

**Prop. (5.3.7.4).** If  $R \rightarrow S$  is a f.g. ring map and a  $S$ -module  $M$  is f.p. over  $R$ , then it is f.p. over  $S$ .  $\lrcorner$

*Proof:* Let  $S = R[x_1, \dots, x_n]$ , and  $M = R[y_1, \dots, y_m]/(\sum a_{ij}y_j), 1 \leq i \leq t$ , then as  $M$  is a  $S$ -module, we let  $x_i y_j = \sum a_{ijk} y_k$ , and forms a quotient  $S^{mn+t} \rightarrow S^m \rightarrow N \rightarrow 0$ , where  $S^{mn+t}$  corresponds to the relations  $\sum a_{ij}y_j$  and  $x_i y_j - \sum a_{ijk} y_k$ . Then there is a surjective  $A$ -module map  $N \rightarrow M$ , and we check it is injective: if  $z = \sum b_j y_j$  are mapped to 0, where  $b_j \in S$ , then we can transform  $z$  into the shape  $\sum c_j y_j$ , where  $c_j \in R$  by relations  $x_i y_j - \sum a_{ijk} y_k$ . Thus it is zero by definition.  $\square$

**Prop. (5.3.7.5)[Direct Limits of F.P. Modules].** Any module is a direct limit of f.p. modules. This can be seen by considering all finite submodules and f.m relations between them.  $\lrcorner$

**Prop. (5.3.7.6)[Characterizing Finite and F.P. Modules].** Let  $N$  be an  $R$ -module, then

- $N$  is finite  $R$ -module iff for any filtered colimits  $M = \varinjlim M_i$  of  $R$ -modules, the map  $\varinjlim \text{Hom}(N, M_i) \rightarrow \text{Hom}(N, \varinjlim M_i)$  is injective.
- $N$  is a f.p.  $R$ -module iff for any filtered colimits  $M = \varinjlim M_i$  of  $R$ -modules, the map  $\varinjlim \text{Hom}(N, M_i) \rightarrow \text{Hom}(N, \varinjlim M_i)$  is a bijection.

$\lrcorner$

*Proof:* 1: If  $N$  is generated by  $x_i$  and a map  $f : N \rightarrow M_i$  maps to  $0 \in \text{Hom}(N, \varinjlim M_i)$ , then there is a  $j$  that  $f : M \rightarrow M_i \rightarrow M_j$  is 0. Thus  $f = 0$ . Conversely,  $N$  is the sum of its f.g. submodules  $N'$ ,

thus  $N \rightarrow \varinjlim N/N_i = 0$ , which implies the identity map  $N \rightarrow N$  vanishes for some  $N/N'$  where  $N'$  is a finite submodule of  $N$ , so  $N = N'$  and  $N$  is a finite.

2: If  $M$  is f.p., we can get the assertion by writing  $M$  as a quotient of free modules and use the fact filtered colimit is exact (5.1.1.27). Conversely, write  $N$  as a filtered colimits of f.p. modules (5.3.7.5), then  $\text{id} : M \rightarrow M$  factors through some f.p. module, so it is a direct summand of a f.p. module, thus f.p. by (5.3.7.3).  $\square$

**Cor. (5.3.7.7) [FP and Localization].** For  $M$  f.p.,  $S^{-1} \text{Hom}_R(M, N) = \text{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$  for any  $R$ -module  $N$ . (Use the presentation and Hom is left exact).  $\square$

*Proof:*  $\text{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N) \cong \text{Hom}_R(M, S^{-1}N)$  by duality, and then we can use (5.3.7.6), as localization is a filtered colimit.  $\square$

### Finitely Presented Ring Map

**Def. (5.3.7.8) [Finitely Presented Ring Map].** A ring map is called **of finite presentation** iff it is a quotient of a free algebra by a free algebra.  $\square$

**Prop. (5.3.7.9).** Finite presentation is stable under composition (choose a presentation form to see) and base change because tensoring is right exact.

It is local on the source and target by (5.1.4.4).  $\square$

**Prop. (5.3.7.10).** For  $S$  f.p. over  $R$ , then the kernel of any surjective ring map  $R[X_1, \dots, X_n] \xrightarrow{\alpha} S$  is f.g..  $\square$

*Proof:* Let  $S = R[Y_1, \dots, Y_m]/(f_1, \dots, f_k)$ , then if  $\alpha(X_i) \cong g_i(Y)$ , then  $\alpha : R[X_1, \dots, X_n] \rightarrow R[X_1, \dots, X_m, Y_1, \dots, Y_m]/(f_1, \dots, f_k, X_i - g_i)$ . And the  $Y_i$  are in the image, thus we let  $Y_i$  are mapped onto by  $h_j(X)$ , then  $\ker \alpha = (f_i(h_j(X)), X_i - g_i(X))$ .  $\square$

**Prop. (5.3.7.11).** If  $g \circ f : R \rightarrow S' \rightarrow S$  is of finite presentation and  $f$  is of finite type, then  $g$  is of finite presentation.  $\square$

*Proof:* Let  $S' = R[y_1, \dots, y_a]$  and  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_m)$ , then let  $h_i(X) \cong y_i$  in  $S$ , then  $S = S'[X_1, \dots, X_n]/(f_1, \dots, f_m, h_i - y_i)$ .  $\square$

**Prop. (5.3.7.12) [Normal Form of F.P.].** If  $S$  is f.p. over  $R$  that  $S$  has a presentation  $S = R[X_1, \dots, X_n]/I$  that  $I/I^2$  is free over  $S$ , then  $S$  has a presentation  $R[X_1, \dots, X_m]/(f_1, \dots, f_c)$  that  $(f_1, \dots, f_c)/(f_1, \dots, f_c)^2$  is freely generated by  $f_1, \dots, f_c$ .  $\square$

*Proof:* Cf. [Sta]07CF.  $\square$

**Prop. (5.3.7.13) [Finite Type Locally of Finite Presentation].** If  $R \rightarrow S$  is a injective map of f.t. of domains, then there are  $f \neq 0 \in R, g \neq 0 \in S$  that  $R_f \rightarrow S_{fg}$  is of f.p.  $\square$

*Proof:* Use induction on the number of generators of  $S/R$ . If  $S = R[x]$ , then  $S = R[X]/\mathfrak{q}$ . If  $q = 0$ , then  $S$  is of f.p.. If  $g = fx^d + a_{d-1}x^{d-1} + \dots + a_0$  be a polynomial of minimal degree in  $\mathfrak{q}$ , then  $R \rightarrow S_f$  is of f.p.

The more generator case can be reduced to the single generator case, because f.p. ring map is stable under composition (5.3.7.9).  $\square$

**Lemma (5.3.7.14) [Filtered Colimits and F.P.].** Let  $R \rightarrow A$  be a ring map, then the category of f.p.  $R$ -algebras  $A'$  with an  $R$ -algebra map  $A' \rightarrow A$  is filtered, and the colimit is just  $A$ .  $\square$

*Proof:* Cf. [0BUF].  $\square$



## 8 Nagata & Excellent Rings

**Def. (5.3.8.1)[Japanese & Nagata Rings].** Let  $R$  be a domain with quotient field  $K$ , then  $R$  is called **N-1** iff the integral closure of  $R$  in  $K$  is a finite  $R$ -module.

$R$  is called **N-2** or **Japanese** iff for any finite field extension  $L/K$ , its integral closure in  $L$  is a finite  $R$ -module.

A ring  $R$  is called **universally Japanese** if for any domain  $S$  of f.t. over  $R$ ,  $S$  is Japanese.

A ring  $R$  is called **Nagata** if it is Noetherian and for any prime  $p$ ,  $R/p$  is Japanese.

All these properties are local properties(5.1.4.4). And they are in fact stable under any localizations. ┘

**Prop. (5.3.8.2)[Nagata].** Let  $R$  be a ring, the following are equivalent:

- $R$  is Nagata,
  - any f.g.  $R$ -algebra is Nagata.
  - $R$  is universally Japanese and Noetherian.
- ┘

*Proof:* Cf.[Sta]0334. □

**Prop. (5.3.8.3)[Nagata and Formal Fibers].** Let  $(A, \mathfrak{m})$  be a Noetherian local ring, then  $A$  is Nagata iff the formal fibers of  $A$  are geo.reduced. ┘

*Proof:* Cf.[Sta]0BJ0. □

### Excellent Rings

**Def. (5.3.8.4)[G-Rings].** A **G-ring** is a Noetherian ring s.t. for any prime  $\mathfrak{p} \subset R$ , the map  $R_{\mathfrak{p}} \rightarrow R_{\mathfrak{p}}^{\wedge}$  is regular. ┘

**Def. (5.3.8.5)[Excellent Rings].** A **quasi-excellent ring** is a ring that is Noetherian, G-ring and J-2. A **excellent ring** is a quasi-excellent ring that is universally catenary. ┘

**Prop. (5.3.8.6).** Quasi-excellent rings are Nagata. ┘

*Proof:* Cf.[Sta]07QV. □

**Prop. (5.3.8.7)[Examples].** The following rings and f.g. algebras over them are excellent rings:

- fields.
- Noetherian complete local rings
- Dedekind domain with fraction field of characteristic 0.

In particular, ring of integers in all local fields and number fields are excellent. ┘

*Proof:* Cf.[Sta]0335. □

**Cor. (5.3.8.8).** The above rings are all Nagata, by(5.3.8.6). ┘

## 9 Separability

Main references are [Matsumura Ch10], [Weibel Chap P309] and [Sta]10.41, 10.43.

**Def. (5.3.9.1) [Separable Algebra].** A f.d semisimple algebra  $R$  over a field  $k$  is called **separable** iff for every field extension  $K/k$ ,  $R \otimes_k K$  is semisimple.  $\lrcorner$

**Def. (5.3.9.2).** A field extension  $K/k$  is called **separably generated** iff it  $K$  is a separable algebraic extension of a purely transcendental field  $L/k$ .

A field extension  $K/k$  is called **separable** iff all f.g. subextensions are separably generated.

An algebra  $A/k$  is called **separable** iff  $A \otimes_k k'$  is reduced for any  $k'/k$  algebraic.  $\lrcorner$

**Prop. (5.3.9.3).** If  $k \subset K$  is a f.g. field extension, then there is a finite purely inseparable field extension  $k \subset k'$  that  $k' \subset K$  is separable.  $\lrcorner$

*Proof:*  $\square$

**Prop. (5.3.9.4) [Separable and Geo.Reduced].** Let  $K/k$  be a field extension, then  $K/k$  is separable iff  $K/k$  is geometrically reduced.  $\lrcorner$

*Proof:* Cf. [Sta]030W.  $\square$

**Cor. (5.3.9.5).** If  $K/k$  is a separable field extension and  $S$  is a reduced  $k$ -algebra, then  $S \otimes_k K$  is reduced.  $\lrcorner$

*Proof:* Cf. [Sta]030U.  $\square$

**Cor. (5.3.9.6).** A separably generated field extension is separable.  $\lrcorner$

## 10 Henselian Ring

References are [Sta]Chap10.148.

**Def. (5.3.10.1) [Henselian Local Rings].** A local ring  $(\mathcal{O}_K, \mathfrak{m}_K, \kappa)$  is called a **Henselian local ring** iff for every  $f \in R[X]$  and  $a_0 \in k$  that  $\bar{f}(a_0) = 0$  and  $\bar{f}'(a_0) \neq 0$ , then there is a root  $\alpha$  of  $f$  lifting  $a_0$ . It is called **strictly-Henselian** if moreover  $\kappa = \kappa^{\text{sep}}$ .  $\lrcorner$

### Henselian Pairs

**Def. (5.3.10.2) [Henselian Pairs].** A **Henselian pair** is a Zariski-pair  $(A, I)$  (5.2.6.12) s.t. : For any  $f, g \in A[T]$  monic s.t.  $\bar{f} = \bar{g}\bar{h} \in A/I[T]$  and  $\bar{g}, \bar{h}$  are coprime and monic, there is a factorization  $f = gh$  lifting the decomposition.  $\lrcorner$

In particular, if  $f$  has a simple root  $\bar{x}$  in  $A/I$ , then it has a root  $x \in A$  lifting  $\bar{x}$ .  $\lrcorner$

**Prop. (5.3.10.3).** Filtered limits of Henselian pairs is Henselian, this is clear from the definition (5.3.10.2).  $\lrcorner$

**Lemma (5.3.10.4).** If  $A$  is a ring with ideal  $I$ , if  $\bar{f} = \bar{g}\bar{h}$  be a factorization of a polynomial  $f \in A[T]$  in  $A/I[T]$ , then there is an étale ring map  $A \rightarrow A'$  that  $A/IA \cong A'/IA'$ , and a factorization  $f = g'h' \in A'[T]$  lifting the factorization.  $\lrcorner$

*Proof:* Cf. [Sta]0ALH.  $\square$

**Prop. (5.3.10.5) [Topological Invariance of Étale Sites].** If  $I$  is locally nilpotent, then  $(A, I)$  is Henselian, in particular  $A_{\text{ét}} \cong (A/I)_{\text{ét}}$  (5.3.10.9).  $\lrcorner$

*Proof:* First if  $A \rightarrow S$  is étale, then  $A/I \rightarrow S/IS$  is étale by base change (5.4.7.5) and the map is essentially surjective by (5.4.7.15). And any map  $B/IB \rightarrow B'/IB'$  can be lifted to  $B \rightarrow B'$  because étale is smooth and use (5.4.5.16). And the lifting is unique, otherwise if  $f, g$  are two lifting, because étale is unramified, so if we choose an idempotent  $e$  generating the kernel of  $B \otimes_A I \rightarrow B \rightarrow B$  (5.4.6.10), then  $f \otimes g(e) \in IB'$ , which is locally nilpotent, thus  $f \otimes g(e) = 0$ , thus  $f = g$ .

For then Henselian,  $I$  is clearly contained in the Jacobson radical, and for the decomposition, by (5.3.10.4) there is an étale map  $A \rightarrow A'$  that  $A/IA \cong A'/IA'$  that lifts the factorization, but  $A = A'$ , by what we have seen above.  $\square$

**Cor. (5.3.10.6) [Complete Pair is Henselian].** If  $(A, I)$  is a pair that  $A$  is  $I$ -adically complete, then  $(A, I)$  is Henselian.  $\lrcorner$

*Proof:*  $I$  is in the Jacobson radical because  $1 + I$  consists of units, and by (5.3.10.5) and (5.3.10.4) we can lift the decomposition to  $A/I^n$  inductively. As  $A = \lim A/I^n$ , we are done.  $\square$

**Prop. (5.3.10.7) [Equivalent Definitions of Henselian Pair].** The following are all equivalent to  $(A, I)$  being Henselian:

- Given any étale ring map  $A \rightarrow A'$ , then any  $A' \rightarrow A/I$  lifts to an  $A$ -algebra map  $A' \rightarrow A$ .
- For any finite/integral  $A$ -algebra  $B$ , the map  $B \rightarrow B/IB$  induces a bijection on idempotents.
- (Gabber)  $(A, I)$  is Zariski and every monic polynomial  $f(T) \in A[T]$  of the form  $T^n(T-1) + a_n T^n + \dots + a_1 T + a_0$  with  $a_i \in I$  has a root  $\alpha \in 1 + I$ .

Moreover, root in item3 is unique.  $\lrcorner$

*Proof:* Cf. [Sta]09XI.  $\square$

**Cor. (5.3.10.8).** if  $(A, I)$  is Henselian and  $A \rightarrow B$  is integral, then  $(B, IB)$  is also Henselian.  $\lrcorner$

**Prop. (5.3.10.9) [Henselian Liftings].** If  $(A, I)$  is a Henselian pair (5.3.10.2), then the base change functor  $A_{\text{ét}} \cong (A/I)_{\text{ét}}$  is an equivalence of categories.  $\lrcorner$

*Proof:* Cf. [Sta]09ZL. ?  $\square$

**Prop. (5.3.10.10).** A Zariski-pair  $(R, I)$  (5.2.6.12) is Henselian (5.3.10.2) iff the pair  $(\mathbb{Z} \oplus I, I)$  is Henselian. In particular, the property of being Henselian only depends on the non-unital ring  $I$ .  $\lrcorner$

*Proof:* Cf. [Almost Ring Theory, 5.1.9].  $\square$

## 5.4 Commutative Algebra IV

**Notation (5.4.0.1).**

- Let  $R \in \mathcal{CRing}$ .

┘

### 1 Flatness

**Def. (5.4.1.1) [Flatness].** A module  $M$  over a ring  $R$  is called **flat** if  $M \otimes_R - : \text{Mod}_R \rightarrow \text{Mod}_R$  is an exact functor. This is compatible with the definition (7.2.2.15).

And  $R \rightarrow S \in \mathcal{CRing}$  is called a **flat ring map** if  $S$  is flat as an  $R$ -module.

┘

**Prop. (5.4.1.2).** Flatness need only be checked for finite modules, and it is equivalent to  $\text{Tor}_1(M, A/I) = 0$  for any f.g. ideal  $I$  (i.e.  $I \otimes M \rightarrow M$  is injective).

┘

*Proof:* This is because of (7.3.3.12) and the fact tensor product commutes with colimit.

□

**Cor. (5.4.1.3).** If  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ , then  $M'$  and  $M''$  flat implies  $M$  is flat.

┘

**Prop. (5.4.1.4).** If  $M$  is flat then  $\text{Tor}_i^A(M, N) = 0$  for all  $i > 0$ , because we have: if  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ ,  $M_2, M_3$  flat, then  $M_1$  is flat (Use 9 entry sequence and the fact that Tor is symmetric (5.9.2.5)). So  $\text{Tor}_{n+1}(M_3, N) = \text{Tor}_n(M_1, N) = 0$  by induction.

And a direct summand of a flat module is flat. Thus we have the class of flat modules is adapted  $- \otimes N$  for all  $N$  (because free is flat).

┘

**Prop. (5.4.1.5) [Faithfully Flatness].** The following are equivalent:

- $M$  is flat and for any  $N \neq 0$ ,  $N \otimes_R M \neq 0$ .
- $M$  is flat and for any (maximal) prime ideal  $\mathfrak{m}$  of  $R$ ,  $k(\mathfrak{m}) \otimes_R M \neq 0$ . (When  $\mathfrak{m}$  is maximal, which means  $\mathfrak{m}M \neq M$ ).

And such a  $M$  is called **faithfully flat** over  $R$ .

And  $R \rightarrow S \in \mathcal{CRing}$  is called a **faithfully flat ring map** if  $S$  is faithfully flat as an  $R$ -module.

┘

*Proof:*  $1 \rightarrow 2$  is easy.  $2 \rightarrow 1$ : any nonzero module has a submodule  $R/I$ , choose a maximal ideal  $\mathfrak{m}$  containing  $I$ , then  $(A/I) \otimes_A M \subset N \otimes_A M$  surjects to  $k(\mathfrak{m}) \otimes_A M \neq 0$ .

□

**Prop. (5.4.1.6) [Flatness and Base Change].**

- (Faithfully) Flatness is stable under base change.
- Flatness satisfies f.f. descent (5.4.2.1).
- Flatness is stable under filtered colimit because filtered colimit commutes with tensoring (3.2.4.13) and is exact (5.1.1.27). In particular,  $S^{-1}A$  is flat (5.1.1.30).
- Let  $S \rightarrow S'$  be a map of  $R$ -algebras,  $M$  is an  $S$ -module,  $M' = M \otimes_S S'$ , then if  $M$  is flat over  $R$ , so does  $M'$ . The converse also holds if  $S \rightarrow S'$  is f.f..
- If  $R \rightarrow S$ , and a  $S$ -module  $M$  is  $R$ -flat and  $S$ -f.f., then  $R \rightarrow S$  is flat.

┘

*Proof:*

4: For any injection of  $R$ -modules  $N \rightarrow N'$ , use the fact  $\ker(N \otimes_R M \rightarrow N' \otimes_R M) \otimes_S S' = \ker(N \otimes_R M' \rightarrow N' \otimes_R M')$ .

5: For any injection of  $R$ -modules  $N \rightarrow N'$ , use the fact  $\ker(N \otimes_R S \rightarrow N' \otimes_R S) \otimes_S M = \ker(N \otimes_R M \rightarrow N' \otimes_R M)$ .  $\square$

**Prop. (5.4.1.7) [Equational Criterion of Flatness].** For a  $R$ -module  $M$ , a relation  $\sum f_i x_i = 0$  where  $f_i \in R, x_i \in M$  are called **trivial** iff  $\vec{x} = A \vec{y}$  for some  $A \in M_{n \times n}(R)$ ,  $\vec{y} \in M^n$ , and  $\vec{f}^t A = 0$ . Then  $M$  is flat iff all relations of elements of  $M$  is trivial.  $\lrcorner$

*Proof:* Assume  $M$  is flat over  $R$ , and  $\sum f_i x_i = 0$  is a relation in  $M$ . Let  $I = (f_1, \dots, f_n)$  and let  $K = \ker(R^n \rightarrow I : (x_i) \mapsto \sum x_i f_i)$ , then  $\sum f_i \otimes x_i = 0 \in I \otimes_R M$  by flatness, then  $\sum e_i \otimes x_i \in K \otimes_R M$ . Let  $\sum e_i \otimes x_i = \sum k_j \otimes y_j$ , and  $k_j = \sum_i a_{ij} e_i$ , then this is the desired relations.

Conversely, suppose every relation is trivial, and  $I$  is an ideal of  $R$ , let  $x = \sum f_i \otimes x_i \in I \otimes M$  be an element mapping to  $0 \in R \otimes M = M$ , then  $\sum f_i x_i$  is a relation, so it is trivial, and then  $x = \sum f_i \otimes x_i = \sum f_i \otimes (a_{ij} y_j) = \sum (f_i a_{ij}) y_j = 0$ .  $\square$

**Prop. (5.4.1.8) [Gorodov-Lazard].** Any flat  $A$ -module is isomorphic to a direct limit of finite free modules.  $\lrcorner$

*Proof:* Cf. [Sta]058G.  $\square$

**Prop. (5.4.1.9) [Flat Modules are Torsion-Free].** Flat modules are torsion-free.  $\lrcorner$

*Proof:* If  $x \in R$  is a nonzero-divisor,  $R \xrightarrow{x} R$  is injective, thus  $M \xrightarrow{x} M$  is also injective.  $\square$

**Prop. (5.4.1.10) [Flat over Local Rings].** A finite module  $M$  over a local ring  $A$  is flat iff it is free. In particular, finite modules over a field are all flat.  $\lrcorner$

*Proof:* Let  $A/\mathfrak{m} = k$ , choose a  $k$ -basis  $x_i$  of  $M/\mathfrak{m}M$ , then they generate  $M$  by Nakayama. It suffices to prove that  $x_i$  are independent over  $R$ . For this, use equational criterion of flatness (5.4.1.7), we prove that if  $x_i$  is independent over  $k$ , then they are independent over  $A$ . Use induction, if  $x \neq 0$  in  $M/\mathfrak{m}M$ , if  $fx = 0$  for some  $f \in A$ , then  $x = \sum a_j y_j$  that  $fa_j = 0$ , but then some  $a_j$  is a unit, so  $f = 0$ .

If  $\sum f_i x_i = 0$ , then by hypothesis,  $f_i \in \mathfrak{m}$ , and there are  $y_j$  that  $x_i = \sum a_{ij} y_j$ ,  $\sum f_i a_{ij} = 0$ . As  $x_n \notin \mathfrak{m}M$ , there is a  $a_{nj} \notin \mathfrak{m}$ , so  $f_n = \sum (-a_{ij}/a_{nj}) f_i$ . then  $\sum_{i \neq n} f_i (x_i - a_{ij}/a_{nj} x_n) = 0$ , but  $x_i - a_{ij}/a_{nj} x_n$  is also independent over  $k$ , so by induction,  $f_i = 0$ , also does  $f_n$ , so we are done.  $\square$

**Prop. (5.4.1.11) [Flatness over Bézout-Domains].** If  $M \in \text{Mod}_R$  and  $R$  is a Bézout-domain, then  $M$  is flat over  $R$  iff it is torsion-free.  $\lrcorner$

*Proof:* One direction is clear by (5.4.1.9). If it is torsion-free, we use the equational criterion of flatness (5.4.1.7):

Let  $\sum a_i x_i = 0$  where  $a_i \in R^*, x_i \in M$ , then consider  $(a_1, \dots, a_n) = (a)$ , thus  $a_i = ab_i$  for some  $b_i \in R$ , and  $\sum c_i b_i = 1$  for some  $c_i \in R$  as  $R$  is a domain. Because  $M$  is torsion-free,  $\sum_i b_i x_i = 0$ . Notice  $\vec{x} = (1 - \vec{c} \vec{b}^t) \vec{x}$ , so we can take  $\vec{y} = \vec{x}$ ,  $A = 1 - \vec{c} \vec{b}^t$ . Then  $\vec{b}^t A = \vec{b}^t - \vec{b}^t \vec{c} \vec{b}^t = 0$ .  $\square$

**Cor. (5.4.1.12) [Flatness over Valuation Rings].** A module over a valuation ring is flat iff it is torsion-free. In particular, if  $(A, \mathfrak{m}, \varpi) \in \text{DVR}$ , then  $M \in \text{Mod}_A$  is flat iff  $\varpi : M \rightarrow M$  is injective.  $\lrcorner$

*Proof:* Because valuation ring is Bézout (12.2.2.8), we can use (5.4.1.11).  $\square$

**Cor. (5.4.1.13) [Flat Modules over Dedekind-Domains].** If  $A$  is a Dedekind-domain (5.2.7.1) and  $M \in \text{Mod}_A$ , then  $M$  is flat over  $A$  flat iff it is torsion-free over  $A$ .  $\lrcorner$

*Proof:* Because flatness and torsion-freeness is stalkwise (5.1.4.2), so it suffices to prove for its localizations, which are DVRs (5.2.7.2), so the result follows from (5.4.1.12).  $\square$

**Prop. (5.4.1.14) [Finite Flat is Locally Free].** Finitely presented flat module is equivalent to finite projective and equivalent to finite locally free. (Immediate from (5.3.1.7)).  $\lrcorner$

**Prop. (5.4.1.15).** if  $M$  is a flat  $R$ -module, then  $IM \cap JM = (I \cap J)M$  for ideals of  $A$ .  $\lrcorner$

*Proof:* Tensoring the exact sequence  $0 \rightarrow I \cap J \rightarrow I \oplus J \rightarrow J \cup I \rightarrow 0$  with  $M$ .  $\square$

**Prop. (5.4.1.16) [Local & Infinitesimal Criterion of Flatness].** Suppose  $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n}) \in \text{NlocCRing}$  and  $M \in \text{Mod}_{B/\mathfrak{m}^n}$ , then the following are equivalent:

- $M$  is flat over  $A$ .
- $\text{Tor}_1^A(M, A/\mathfrak{m}) = 0$ .
- $M/\mathfrak{m}^n M$  is flat over  $A/\mathfrak{m}^n$  for sufficiently large  $n$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2, 1 \rightarrow 3$  is trivial. For the converse, we need to prove  $\text{Tor}_1^A(M, A/I) = 0$  for any ideal  $I \subset A$ , or  $I \otimes M \rightarrow M$  is injective.

If 2 holds, then  $\text{Tor}_1^A(M, N) = 0$  for any  $A$ -module of finite length, because by devissage we can reduce  $N$  to  $A/\mathfrak{m}$ .

Tensoring the exact sequence  $0 \rightarrow \mathfrak{m}^n \cap I \rightarrow I \rightarrow I/I \cap \mathfrak{m}^n \rightarrow 0$  with  $M$ , we get

$$\mathfrak{m}^n \cap I \otimes M \rightarrow I \otimes M \rightarrow (I/I \cap \mathfrak{m}^n) \otimes M \rightarrow 0$$

and also the exact sequence  $0 \rightarrow I/I \cap \mathfrak{m}^n \rightarrow R/\mathfrak{m}^n \rightarrow R/(I + \mathfrak{m}^n) \rightarrow 0$  gives

$$0 \rightarrow (I/I \cap \mathfrak{m}^n) \otimes M \rightarrow M/\mathfrak{m}^n M \rightarrow M/(I + \mathfrak{m}^n)M \rightarrow 0$$

by the fact  $R/(I + \mathfrak{m}^n)$  has finite length in case2 or the fact they are all  $R/\mathfrak{m}^n M$  modules in case3.

Thus the kernel of  $I \otimes M \rightarrow M$  is contained in  $(\mathfrak{m}^n \cap I) \otimes M$  for any  $n$ , which means it is contained in  $\mathfrak{m}^n(I \cap M)$  for any  $n$  by Artin-Rees (5.2.2.13). Thus the kernel is trivial by Krull's intersection theorem (5.2.2.14) as  $I \otimes M$  is finite over  $S$ .  $\square$

### Flat Ring Maps

**Prop. (5.4.1.17) [Flatness is Local].** Flatness for ring maps (5.4.1.1) is stalkwise both on the target and source, thus flatness is local both on the target and the source (5.1.4.2).  $\lrcorner$

**Cor. (5.4.1.18) [Going-down].** Going-down holds for flat ring map.  $\lrcorner$

*Proof:* The ring map  $R_{\mathfrak{p}'} \rightarrow S_{\mathfrak{q}'}$  is flat by (5.4.1.17), thus it is f.f. by (5.4.1.22). Then (5.4.1.20) says  $\mathfrak{p} \subset \mathfrak{p}'$  is in the image.  $\square$

**Prop. (5.4.1.19).** If  $f : A \rightarrow B, g : B \rightarrow C \in \text{CRing}$  are flat ring maps (5.4.1.1), then so is  $g \circ f : A \rightarrow C \in \text{CRing}$ .  $\lrcorner$

*Proof:* Cf. [GAGA Serre P26].  $\square$

**Prop. (5.4.1.20).** The following are equivalent:

- $A \rightarrow B$  is f.f.
- It is flat and  $\text{Spec } B \rightarrow \text{Spec } A$  is surjective.
- It is flat and  $\text{Spec}$  map contains all the closed pts.

┘

*Proof:* This follows from (5.4.1.5) as we see that  $\mathfrak{p}$  is in the image of  $\text{Spec}$  map iff  $k(\mathfrak{p}) \otimes_A B \neq 0$ .  $\square$

**Cor. (5.4.1.21).** Integral flat injective ring extension is f.f., by (5.2.1.5). ┘

**Cor. (5.4.1.22).** Flat local ring map of local rings is f.f.. ┘

**Cor. (5.4.1.23).** Filtered colimits of f.f. rings over  $R$  is f.f. ┘

*Proof:* It is flat by (5.4.1.6), and for a maximal ideal  $\mathfrak{m}$  of  $R$ ,  $S_i/\mathfrak{m}S_i$  is non-zero, hence the direct limit is non-zero because 1 is contained. So  $\mathfrak{m}$  is in the image, hence it is f.f. by (5.4.1.20).  $\square$

**Cor. (5.4.1.24) [Filtered Colimit of Flat Ring Maps].** If  $I$  is filtered and  $R_i \rightarrow S_i$  are (faithfully) flat ring maps, then  $\text{colim}_I R_i \rightarrow \text{colim}_I S_i$  is (faithfully) flat. ┘

*Proof:* For any  $\text{colim } R_i$ -module  $M$ ,  $(\text{colim } S_i) \otimes_{\text{colim } R_i} M = \text{colim}(S_i \otimes_{R_i} M)$ , so it is flat, because  $\text{colim}$  is exact. For the faithfully flatness, for any maximal ideal  $\mathfrak{m}$  of  $\text{colim } R_i$ , let  $\mathfrak{m}_i = \mathfrak{m} \cap R_i$ , then  $S_i/\mathfrak{m}_i S_i \neq 0$ , thus the direct limit is also  $\neq 0$ , so  $\mathfrak{m}$  is in the image, hence it is f.f. by (5.4.1.20).  $\square$

**Prop. (5.4.1.25).** If  $R \rightarrow S$  is (faithfully) flat ring map and  $M$  is a (faithfully) flat  $S$ -module, then  $M$  is a (faithfully) flat  $R$ -module. In particular, (faithfully) flatness is stable under composition. Also (faithfully) flatness is stable under base change. ┘

**Prop. (5.4.1.26).** If  $B$  is flat over  $A$ , then

$$\text{Tor}_i^A(M, N) \otimes B = \text{Tor}_i^B(M_{(B)}, N_{(B)}), \quad \text{Ext}_i^A(M, N) \otimes B = \text{Ext}_i^B(M_{(B)}, N_{(B)}).$$

┘

**Prop. (5.4.1.27) [Faithfully Flat Ring Maps are Injective].** A f.f. ring map  $R \rightarrow S$  is universally injective. In particular, tensoring with  $R/I$ , we get  $R \cap IS = I$  for an ideal  $I$  of  $R$ . ┘

*Proof:* Because  $R \rightarrow S$  is f.f., we only need to show that  $N \otimes_R S \rightarrow N \otimes_R S \otimes_R S$  is injective for any  $N$ , but this is true because it has a left inverse.  $\square$

**Prop. (5.4.1.28).** A f.f. map between valuation rings is equivalent to an injective local homomorphism. ┘

**Prop. (5.4.1.29).** A flat ring map maps a non-zero-divisor to a non-zero-divisor, because if we consider the principal ideal generated by it, then (5.4.1.2) shows the ideal in  $M$  is also injective, so it is not a zero-divisor. ┘

**Thm. (5.4.1.30) [Flat Maps are Open].** The  $\text{Spec}$  map of a ring map  $R \rightarrow S$  of f.p. that satisfies going-down (e.g. flat), is open. ┘

*Proof:*  $S \rightarrow S_f$  satisfies going-down and is of f.p, so we see that  $R \rightarrow S_f$  satisfies going down. It suffices to prove the image of this map is open. By Chevalley, the image is constructible, and it is stable under generalization. So it is open by (4.12.3.10).  $\square$

**Cor. (5.4.1.31).** The Spec map  $f$  of a f.f. ring map is a quotient map.  $\lrcorner$

**Prop. (5.4.1.32) [Noetherian Completions are Flat].** If  $A$  is Noetherian and  $I$  is an ideal, the the  $I$ -adic completion  $\hat{A}$  is flat over  $A$  by (5.2.3.24).  $\lrcorner$

**Cor. (5.4.1.33) [Flatness and Completions].** For  $f : A \rightarrow B \in \mathcal{NlocCRing}$ ,  $f$  is flat iff  $\hat{f} : \hat{A} \rightarrow \hat{B}$  is flat.  $\lrcorner$

*Proof:* This follows from the fact that  $A \rightarrow \hat{A}, B \rightarrow \hat{B}$  are f.f. (5.4.1.32)(5.4.1.30) and faithfully flat descent (5.4.2.1).  $\square$

**Prop. (5.4.1.34) [Generic Freeness+F.P.].** If  $R$  is reduced,  $S$  a f.g.  $R$ -algebra and  $M \in \mathcal{Mod}_S^{\text{fg}}$ , then there exists an open dense subset  $U \subset \text{Spec } R$  that there is a covering of  $U$  by standard opens  $D(f)$  s.t.

- $M_f$  and  $S_f$  are free over  $R_f$ .
- $S_f$  is a f.p.  $R_f$ -algebra.
- $M_f$  is a f.p.  $S_f$ -module.

In particular, it is generically flat.  $\lrcorner$

*Proof:* Cf. [Sta]051Z. ?  $\square$

**Prop. (5.4.1.35) [Miracle Flatness].** Let  $f : A \rightarrow B \in \mathcal{NlocCRing}$  s.t.

- $A$  is regular.
- $B$  is Cohen-Macaulay.
- $\dim B = \dim A + \dim(A/f(\mathfrak{m}_A)B)$ ,

then  $f$  is flat.  $\lrcorner$

*Proof:* Cf. [Sta]00R4.  $\square$

**Prop. (5.4.1.36) [Slicing Criterion for Flat Modules on the Target].** Let  $R \rightarrow S$  be a local homomorphism of local rings s.t.

- $S$  is essentially of f.p. over  $R$ ,
- $M$  is of f.p. over  $S$ ,
- $\text{Tor}_1^R(M, R/I) = 0$ .
- $M/IM$  is flat over  $R/I$ .

Then  $M$  is flat over  $R$ .  $\lrcorner$

*Proof:* Cf. [Sta]0471. ?  $\square$

**Prop. (5.4.1.37) [Slicing Criterion for Flatness on the Source].** Let  $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n}) \in \mathcal{locCRing}$  and  $t \in S$  s.t.

- $S$  is essentially of f.p. over  $R$ ,
- $S$  is flat over  $R$ ,
- $t \in (S/\mathfrak{m}S)^\times$ .

Then  $S/tS$  is flat over  $R$ , and  $t \in S^\times$ .  $\lrcorner$

*Proof:* Cf. [Sta]046Z.  $\square$



**Prop. (5.4.1.38) [Fibral Criterion of Flat Modules].** Let  $R \rightarrow S \rightarrow S'$  be local homomorphisms of local rings and  $\mathfrak{m}$  is the maximal ideal of  $R$ . If

- $R \rightarrow S'$  is essentially of f.p.
- $R \rightarrow S$  is essentially of f.t.
- $0 \neq M \in \text{Mod}_{S'}$  is of f.p. over  $S$ .
- $M/\mathfrak{m}M$  is flat over  $S/\mathfrak{m}S$ .
- $M$  is flat over  $R$ .

Then  $S$  is essentially of f.p. and flat over  $R$  and  $M$  is flat over  $S$ . ┘

*Proof:* [Sta]05UV. □

## 2 Faithfully Flat Descent

**Prop. (5.4.2.1) [Faithfully Flat Descent].** List of properties that descent through faithfully flat morphism. Let  $R \rightarrow R'$  be a f.f. ring map.

1. Finiteness for modules over a ring.
2. F.p. for modules over a ring.
3. Flatness for modules over a ring.
4. Finite locally freeness and invertibility for modules over a ring.
5. Mittag-Leffler for modules over a ring.
6. Projectiveness for modules over a ring.
7. F.g. for ring maps.
8. F.p. for ring maps, both on the target and source.
9. (Formal)Smoothness for ring maps.
10. Noetherian for rings.
11. Reducedness for rings.
12. Normal for rings.
13. Regular for rings. (But notice regular is only étale-local(5.4.7.29) but not fppf-local).
14. Being Noetherian and has property  $(R_k)$  for rings.
15. Local Complete Intersection ring maps.

┘

*Proof:*

1. Cf. [Sta]03C4.
2. Cf. [Sta]03C4.
3. Let  $M$  be a  $R$ -module, and  $M' = M \otimes_R R'$ , if  $M'$  is  $R$ -flat, then for any  $R$ -module  $N$ ,  $(N \otimes_R M) \otimes_R S = (N \otimes_R S) \otimes_S M'$ , so as  $\cdot \otimes_R S$  is exact and reflects exactness,  $\cdot \otimes_R M$  is exact, so  $M$  is  $R$ -flat.
4. This follows from f.f. descent for f.p. and flatness and(5.3.1.7).
5. Cf. [Sta]05A5.

6. Cf. [Sta]05A9.
7. Cf. [Sta]00QP.
8. Cf. [Sta]00QQ, 00EP.
9. Use criterion (5.4.5.3), we see by flatness that the sequence  $I/I^2 \rightarrow \Omega_{P/R} \otimes_P S \rightarrow \Omega_{S/R} \rightarrow 0$  commutes with flat base change, and when it is f.f., then use (5.4.3.6) and descent for projectiveness (5.4.2.1) that  $\Omega_{S/R}$  is projective, so it is a split exact sequence. The smooth case follows from definition (5.4.5.12) as f.p. can descend.
10. Because for  $S \rightarrow S'$  faithfully flat and a chain of ideals  $I_k$  in  $S$ ,  $I_k S' = I_k \otimes_S S'$ , and  $I_k S'$  is stable if  $S'$  is Noetherian, so also  $I_k$  is stable because it is faithfully flat.
11.  $S \rightarrow S'$  is f.f. hence injective (5.4.1.27).
12. Normality is stalkwise, so it suffices to assume  $R \rightarrow S$  is f.f. and  $R, S$  are both local. Then  $S$  is normal integral thus by (5.4.1.27),  $R$  is also normal integral. Now if  $a/b \in R$ , then  $a/b \in S$ ,  $a \in bS$ , thus  $a \in bS \cap R = bR$  by (5.4.1.27), so  $a/b \in R$ .
13. Cf. [Sta]07NG.?
14. Cf. [Sta]0353.
15. flatness and f.p. both satisfies f.f. descent, so it suffices to show that if  $\mathfrak{p}' \subset R'$  is a prime lying over  $\mathfrak{p} \subset R$ , then  $S \otimes_R k(\mathfrak{p})$  is a local complete intersection iff  $S' \otimes_{R'} k(\mathfrak{p}') = S \otimes_R k(\mathfrak{p}) \otimes_{k(\mathfrak{p})} k(\mathfrak{p}')$  is a local complete intersection, and this follows from [Sta]00SI.

□

**Prop. (5.4.2.2) [fpqc-Poincaré Lemma].** If a ring map  $A \rightarrow B$ , either has a section  $B \rightarrow A$ , or it is faithfully flat, then the Amitsur complex  $s(M)$  for the canonical descent datum (with augmentation):

$$0 \rightarrow M \rightarrow B \otimes_A M \rightarrow B \otimes_A B \otimes_A M \rightarrow \dots$$

with Čech-like maps, is exact.

┘

*Proof:* In the case  $A \rightarrow B$  has a section  $s$ , It suffices to construct a nullhomotopy of the case  $M = A$ . Then we can just let  $h(e_0 \otimes e_1 \otimes \dots \otimes e_r) = s(e_0)e_1 \otimes \dots \otimes e_r$ .

The f.f. case can be reduced to the first case by tensoring  $B$  to consider  $B \rightarrow B \otimes_A B$ , because it has a section.

□

**Cor. (5.4.2.3) [Glueing Functions].** If  $R \in \mathcal{CRing}$ ,  $M \in \text{Mod}_R$  and  $f_1, \dots, f_n \in R$  s.t.  $(f_1, \dots, f_n) = (1)$ , then there is an exact sequence

$$0 \rightarrow M \rightarrow \prod_i M_{f_i} \rightrightarrows \prod_{i,j} M_{f_i f_j}.$$

In particular this holds for  $M = R$ .

┘

*Proof:* This is just (5.4.2.2) applied to  $A \rightarrow \prod_i A_{f_i}$ , which is faithfully flat.

□

### Formal Glueing of Modules

References are [Sta]Chap15.88 and 15.89.

**Lemma (5.4.2.4).** Let  $f : R \rightarrow S \in \mathcal{C}\text{Ring}$  and  $I = (f_1, \dots, f_r) \subset R$  be an ideal, then for any  $M \in \text{Mod}_R$  we can define a complex

$$0 \rightarrow M \xrightarrow{\alpha} M \otimes_R S \times \prod M_{f_i} \xrightarrow{\beta} \prod (M \otimes_R S)_{f_i} \times \prod M_{f_i f_j},$$

where  $\alpha(m) = (m \otimes 1, m, \dots, m)$ ,  $\beta(m', m_1, \dots, m_t) = (m' - m_1 \otimes 1, m' - m_2 \otimes 1, \dots, m' - m_t \otimes 1, m_1 - m_2, \dots, m_{t-1} - m_t)$ .

Suppose that  $R \rightarrow S$  is flat and  $R/I \rightarrow S/IS$  is an isomorphism, then this complex is exact.  $\lrcorner$

*Proof:* Cf. [Sta]05EK.  $\square$

**Def. (5.4.2.5) [Category of Gluing Data].** Let  $S \in \mathcal{C}\text{Ring}/R$  and  $I = (f_1, \dots, f_r) \subset R$  be an ideal, then we define the category  $\text{Glu}(R \rightarrow S, f_1, \dots, f_r)$  of gluing data consisting of objects  $M = (M', M_i, \alpha_i, \alpha_{ij})$  where  $M'$  is a  $S$ -module,  $M_i$  are  $R_{f_i}$ -modules,  $\alpha_i : (M')_{f_i} \rightarrow M_i \otimes_R S$  and  $\alpha_{ij} : (M_i)_{f_j} \rightarrow (M_j)_{f_i}$  are isomorphisms s.t.

- $\alpha_{ij} \circ \alpha_i = (\alpha_j)_{f_i}$ .
- $\alpha_{jk} \circ \alpha_{ij} = \alpha_{ik}$ .

There is a canonical functor  $\text{Can} : \text{Mod}_R \rightarrow \text{Glu}(R \rightarrow S, f_1, \dots, f_r)$  and also a morphism  $H^0 : \text{Glu}(R \rightarrow S, f_1, \dots, f_r) \rightarrow \text{Mod}_R$  where

$$H^0(M) = \ker(M' \times \prod M_i \rightarrow \prod M'_{f_i} \times \prod (M_i)_{f_j}).$$

$H^0$  is a left inverse of  $\text{Can}$ , by (5.4.2.4).  $\lrcorner$

**Lemma (5.4.2.6).** If  $R \rightarrow S$  is flat, then  $\text{Glu}(R \rightarrow S, f_1, \dots, f_r)$  is an Abelian category, and  $\text{Can}$  is an exact functor that commutes with arbitrary colimits.

If moreover  $(f_1, \dots, f_r) = R$ , then  $\text{Can}$  and  $H^0$  induces an equivalence of categories.  $\lrcorner$

*Proof:* The kernels and cokernels can be constructed because  $- \otimes_R S$  is exact, and  $\text{Can}$  is exact because  $R_{f_i}$  and  $S$  are flat over  $R$ , and also tensoring commutes with taking colimits.

For the last assertion, by (5.4.2.5) it suffices to show that  $\text{Can}$  is essentially surjective. For this, just use (5.4.2.3) on both  $R$  and  $S$ .  $\square$

**Prop. (5.4.2.7).** In the setting of (5.4.2.5), if  $R/I \rightarrow S/IS$  is an isomorphism, then  $\text{Can}$  and  $H^0$  induces an equivalence of categories.  $\lrcorner$

*Proof:* Cf. [Sta]05ER.  $\square$

**Cor. (5.4.2.8).** If  $R \rightarrow S$  is a flat ring map and  $f \in R$  that  $R/fR \cong S/fS$  is an isomorphism, then there is a pullback diagram of categories:

$$\begin{array}{ccc} \text{Mod}_R & \longrightarrow & \text{Mod}_{R_f} \\ \downarrow & & \downarrow \\ \text{Mod}_S & \longrightarrow & \text{Mod}_{S_f} \end{array}$$

and we can also restrict to the category of f.g./f.p/flat/projective(any property satisfying f.f. descent) modules.  $\lrcorner$

*Proof:* For the last assertion, notice  $R \rightarrow R^\wedge \times \prod R_{f_i}$  is f.f. by (5.2.3.7), then use f.f. descent (5.4.2.1).  $\square$

**Cor. (5.4.2.9).** If  $R$  is a Noetherian ring,  $f \in R$  and  $R^\wedge$  the  $f$ -adic completion of  $R$ , then there is an pullback of categories:

$$\begin{array}{ccc} \text{Mod}_R & \longrightarrow & \text{Mod}_{R_f} \\ \downarrow & & \downarrow \\ \text{Mod}_{R^\wedge} & \longrightarrow & \text{Mod}_{R_f^\wedge} \end{array}$$

and we can also restrict to the category of f.g./f.p/flat/projective(any property satisfying f.f. descent) modules.  $\lrcorner$

*Proof:* This satisfies the hypothesis of (5.4.2.7) by (5.2.3.24).  $\square$

**Lemma (5.4.2.10).** If  $f \in R$  and  $R \rightarrow R' \in \mathcal{CRing}$  induces an isomorphism  $R/(f^n) \cong R'/(f^n)$  for any  $n \in \mathbb{Z}_+$ , then for any  $f$ -power torsion  $R$ -module  $M$ , the map  $M \rightarrow M \otimes_R R'$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]0BNK.  $\square$

**Lemma (5.4.2.11).** If  $f \in R$  and  $R \rightarrow R' \in \mathcal{CRing}$  induces an isomorphism  $R/(f^n) \cong R'/(f^n)$  for any  $n \in \mathbb{Z}_+$ , then for any  $M \in \text{Mod}_R$ ,  $M$  is finite iff  $M \otimes_R (\hat{R} \oplus R[f^{-1}])$  is finite over  $\hat{R} \oplus R[f^{-1}]$ .  $\lrcorner$

**Def. (5.4.2.12) [Gluing Pairs and Glueing Modules].** For  $R' \in \mathcal{CRing}/R$  and  $f \in R$  s.t.  $R/(f^n) \cong R'/(f^n)$  for any  $n \in \mathbb{Z}_+$ ,  $(R \rightarrow R', f)$  is called a **gluing pair** if the sequence

$$0 \rightarrow R \rightarrow R' \oplus R[f^{-1}] \rightarrow R'[f^{-1}] \rightarrow 0$$

is exact. The pair  $(R, f)$  is called a gluing pair if  $(R \rightarrow \hat{R}, f)$  is a gluing pair (This makes sense by (5.2.3.6)).

This is equivalent to  $R[f^\infty] \rightarrow R'[f^\infty]$  is bijective.

$M \in \text{Mod}_R$  is called a **glueable module** for  $(R \rightarrow R^\wedge, f)$  if the sequence

$$0 \rightarrow M \rightarrow M_{R'} \oplus M_{R[f^{-1}]} \rightarrow M_{R'[f^{-1}]} \rightarrow 0$$

is exact.

This is equivalent to  $M[f^\infty] \rightarrow M_{R'}[f^\infty]$  is a bijection. And when  $(R \rightarrow R', f)$  is a gluing pair, this is equivalent to  $M[f^\infty] \rightarrow M_{R'}[f^\infty]$  is injective.  $\lrcorner$

*Proof:* Cf. [Sta]0BNR, 0BNW, 08NX.  $\square$

**Prop. (5.4.2.13) [Flatness, Non-Zero-Divisors and Gluing].** For  $R' \in \mathcal{CRing}/R$  and  $f \in R$ ,  $(R \rightarrow R', f)$  is a gluing pair (5.4.2.12) when  $R \rightarrow R^\wedge$  is flat. In particular  $(R, f)$  is a gluing pair then  $R$  is Noetherian or  $f$  is not a zero-divisor (5.2.3.10).

If  $(R \rightarrow R', f)$  is a gluing pair, then  $M$  is glueable if  $\text{Tor}_1^R(M, R')$  is  $f$ -power torsion, or equivalently  $\text{Tor}_1^R(M, R'_f) = 0$ . In particular this is the case when  $M$  is flat over  $R$  or  $f$  is not a zero-divisor for  $M$ . And when  $R \rightarrow R'$  is flat, any  $R$ -module  $M$  is glueable, in particular this is the case for  $(R, f)$  when  $R$  is Noetherian.  $\lrcorner$

*Proof:* If  $R \rightarrow R^\wedge$  is flat, then tensoring  $R^\wedge$  gives an exact sequence

$$0 \rightarrow R[f^n] \otimes_R R^\wedge \rightarrow R^\wedge \xrightarrow{f^n} R^\wedge,$$

and  $R[f^n] \otimes_R R^\wedge \cong R[f^n]$  by (5.4.2.10), so  $R[f^\infty] \cong R^\wedge[f^\infty]$  and  $(R \rightarrow R^\wedge, f)$  is a glueing pair (5.4.2.12).

If  $R$  is Noetherian, then  $R \rightarrow R^\wedge$  is flat by (5.4.2.10).

If  $f$  is not a zero-divisor, then  $f$  is also not a zero-divisor in  $R^\wedge$  by (5.2.3.10), so  $(R \rightarrow R^\wedge, f)$  is a glueing pair (5.4.2.12).

For the second assertion, Cf. [Sta]0BNX.  $\square$

**Thm. (5.4.2.14) [Beauville-Laszlo].** Let  $(R \rightarrow R', f)$  be a glueing pair (5.4.2.12), then there are pull-back diagrams of categories:

$$\begin{array}{ccccc} \{\text{glueing } R\text{-modules}\} \subset \text{Mod}_R & \longrightarrow & \text{Mod}_{R'[\frac{1}{f}]} & & \text{Mod}_R^{\text{flat}} \longrightarrow \text{Mod}_{R'[\frac{1}{f}]}^{\text{flat}} & & \text{Mod}_R^{\text{fin,lf}} \longrightarrow \text{Mod}_{R'[\frac{1}{f}]}^{\text{fin,lf}} \\ \downarrow & & \downarrow & , & \downarrow & & \downarrow \\ \text{Mod}_{R'} & \longrightarrow & \text{Mod}_{R'[\frac{1}{f}]} & & \text{Mod}_{R'}^{\text{flat}} \longrightarrow \text{Mod}_{R'[\frac{1}{f}]}^{\text{flat}} & & \text{Mod}_{R'}^{\text{fin,lf}} \longrightarrow \text{Mod}_{R'[\frac{1}{f}]}^{\text{fin,lf}} \end{array}$$

Similar diagram holds for  $\text{Mod}^{f\text{-regular}}$ .  $\lrcorner$

*Proof:* ? For glueable  $R$ -module,  $M \mapsto M \otimes (R' \oplus R[f^{-1}])$  is faithfully flat by the exact sequences in (5.4.2.12). Thus it suffices to check this map is essentially surjective.

Firstly, for  $M' \in \text{Mod}_{R'}$  and  $M_1 \in \text{Mod}_{R[f^{-1}]}$  with an isomorphism  $\alpha : M_1 \otimes R \cong M'[f^{-1}]$ , the map of  $R$ -modules  $d : M' \oplus M[f^{-1}] \rightarrow M'[f^{-1}]$  is surjective: For  $z \in M'[f^{-1}]$ , write  $\alpha^{-1}(z) = \sum y_i \otimes g_i$  where  $g_i \in R'$  and  $y_i \in M'$ . Then there exists  $n \in \mathbb{Z}_+$  and  $y'_i \in M'$  s.t.  $\alpha(y_i \otimes 1) = y'_i / f^n$  for each  $i$ . Write  $g_i = a_i + f^n b_i$  with  $a_i \in R$  and  $b_i \in R'$  (which can be done because  $R/(f^n) \cong R'/(f^n)$ ), then for  $y = \sum a_i y_i \in M_1$  and  $x = \sum b_i y'_i$ , we have  $d(x, -y) = z$ .

Then we have an exact sequence

$$0 \rightarrow M \rightarrow M' \oplus M[f^{-1}] \xrightarrow{d} M'[f^{-1}] \rightarrow 0.$$

for some  $M \in \text{Mod}_R$ . It suffices to show that this induces an isomorphism  $M \otimes (R' \oplus R[f^{-1}]) \cong M' \oplus M[f^{-1}]$ . For this, Cf. [Sta]0BP7. ?

For the second diagram, we need to show that  $M \in \text{Mod}_R$  is flat if  $M \otimes R[f^{-1}]$  is flat over  $R[f^{-1}]$  and  $M \otimes R'$  is flat over  $R'$ . Cf. [Sta]0BP7. ?

For the second diagram, we need to show that  $M \in \text{Mod}_R$  is finite presented and flat if  $M \otimes R[f^{-1}]$  is finitely presented and flat over  $R[f^{-1}]$  and  $M \otimes R'$  is finitely presented and flat over  $R'$ .  $M$  is flat by the second diagram, and it is finite by (5.4.2.11). Let  $0 \rightarrow K \rightarrow R^{\oplus n} \rightarrow M \rightarrow 0$  be an exact sequence, then this exact sequence is also exact after tensoring  $R' \oplus R[f^{-1}]$  because  $M$  is flat, so  $K \otimes_R (R' \oplus R[f^{-1}])$  is finite over  $R' \oplus R_f$ , thus  $K$  is finite over  $R$  by (5.4.2.11). So we proved that  $M$  is finitely presented over  $R$ .

For the last assertion, use (5.4.2.13).  $\square$

**Cor. (5.4.2.15).** If  $R \in \mathcal{C}\text{Ring}$  and  $f \in R$  is a nonzero-divisor, denote by  $\hat{R}$  the  $f$ -adic completion of  $R$ , then there is a pullback diagram of categories:

$$\begin{array}{ccccc} \text{Mod}_R & \longrightarrow & \text{Mod}_{R[\frac{1}{f}]} & & \text{Mod}_R^{\text{flat}} \longrightarrow \text{Mod}_{R[\frac{1}{f}]}^{\text{flat}} & & \text{Mod}_R^{\text{fin,lf}} \longrightarrow \text{Mod}_{R[\frac{1}{f}]}^{\text{fin,lf}} \\ \downarrow & & \downarrow & , & \downarrow & & \downarrow \\ \text{Mod}_{\hat{R}} & \longrightarrow & \text{Mod}_{\hat{R}[\frac{1}{f}]} & & \text{Mod}_{\hat{R}}^{\text{flat}} \longrightarrow \text{Mod}_{\hat{R}[\frac{1}{f}]}^{\text{flat}} & & \text{Mod}_{\hat{R}}^{\text{fin,lf}} \longrightarrow \text{Mod}_{\hat{R}[\frac{1}{f}]}^{\text{fin,lf}} \end{array}$$

Similar diagram holds for  $\text{Mod}^{f\text{-regular}}$ .  $\lrcorner$

*Proof:* By (5.2.3.4),  $R/(f^n) \rightarrow \hat{R}/(f^n)$  is an isomorphism for any  $n \in \mathbb{Z}_+$ , so  $(R \rightarrow R^\wedge, f)$  is a glueing pair, and any  $M \in \text{Mod}_R$  a glueing module by (5.4.2.13), so we can use (5.4.2.14) above.  $\square$

### 3 Kähler Differentials

**Def. (5.4.3.1) [Derivations].** For  $R \in \mathcal{A}lg_S$  and  $M \in \mathcal{M}od_R$ , a **derivation** over  $S$  from  $R$  to  $M$  is a morphism  $\delta : R \rightarrow M \in \mathcal{M}od_S$  s.t.  $\delta(ab) = a\delta(b) + b\delta(a)$  for any  $a, b \in R$ . The set of all derivatives from  $R$  to  $M$  over  $S$  is denoted by  $\text{Der}_S(R, M)$ , which is an  $S$ -module.  $\lrcorner$

**Def. (5.4.3.2) [ $A \ltimes_R M$ ].** For any  $A \in \mathcal{A}lg_R$ , there is a functor

$$A \ltimes_R (-) : \mathcal{M}od_A \rightarrow (\mathcal{A}lg_R)_{/A}$$

s.t.  $A \ltimes_R M = A \otimes_R M$  as a module with algebra structure given by

$$(a, x)(b, y) = (ab, ay + bx), \text{ where } a, b \in A, x, y \in M.$$

Then for any  $X \in (\mathcal{A}lg_R)_{/A}$  and  $M \in \mathcal{M}od_A$ , there is a bijection of sets

$$\text{Hom}_{(\mathcal{A}lg_R)_{/A}}(X, A \ltimes_R M) \cong \text{Der}_R(X, M),$$

where  $M$  is an  $X$ -algebra via  $X \rightarrow A \rightarrow M$ .  $\lrcorner$

**Def. (5.4.3.3) [Kähler Differential].** For  $S \in \mathcal{A}lg_R$ , the **Kähler Differential**  $\Omega_{S/R}$  is defined as a left  $S$ -module s.t.  $\text{Der}_R(S, M) \cong \text{Hom}_S(\Omega_{S/R}, M)$  for any  $M \in \mathcal{M}od_S$ . In particular,  $\text{Der}_R(S, S)$  is the  $S$ -dual of  $\Omega_{S/R}$ .  $\lrcorner$

**Prop. (5.4.3.4).** One construction is by the free group generated by elements of  $R$  module some relations.

It can also be constructed as follows: there are two ring maps  $\lambda_i$  from  $S$  to  $S \otimes_R S$ , and one map  $\varepsilon$  from  $S \otimes_R S$  to  $S$ . Let  $I = \ker \varepsilon$  as a  $R$  module by  $\lambda_1$ , then  $I/I^2 \cong \Omega_{S/R}$  by (5.4.3.8) that

$$0 \rightarrow I/I^2 \rightarrow \Omega_{S \otimes_R S/S} \otimes_{S \otimes_R S} S \rightarrow \Omega_{S/S} \rightarrow 0.$$

So  $I/I^2 \cong \Omega_{S/R} \otimes_S (S \otimes_R S) \otimes_{S \otimes_R S} S \cong \Omega_{S/R}$ . And it can be verified that  $a \otimes 1 - 1 \otimes a$  corresponds to  $da$ .  $\lrcorner$

**Prop. (5.4.3.5) [Adjointness].** The functor  $X \rightarrow A \otimes_X \Omega_{X/R}$  from  $(\mathcal{A}lg_R)_{/A}$  to  $\mathcal{M}od_A$  is left adjoint to the functor  $A \ltimes_R (-)$  defined in (5.4.3.2).  $\lrcorner$

*Proof:* Because they are both equivalent to  $\text{Der}_R(X, M)$ .  $\square$

**Cor. (5.4.3.6) [Functoriality].** From the first construction, we can see directly that for a family of morphisms  $R_i \rightarrow S_i$ ,

$$\Omega_{\text{colim } S_i / \text{colim } R_i} = \text{colim } \Omega_{S_i / R_i}.$$

In particular, we have:

$$T^{-1}\Omega_{B/A} = \Omega_{T^{-1}B/A}, \quad \Omega_{S^{-1}B/S^{-1}A} = S^{-1}\Omega_{B/A}.$$

Moreover, we have

$$\Omega_{S/R} \otimes_R R' = \Omega_{S \otimes_R R' / R'}, \quad (S \otimes_R \Omega_{T/R}) \oplus (T \otimes_R \Omega_{S/R}) \cong \Omega_{S \otimes_R T / R}$$

by universal properties.  $\lrcorner$

*Proof:* We prove for the localization: it suffices to show the following two assertions:

1.  $S^{-1}\Omega_{A/B} \cong \Omega_{S^{-1}A/B}$ .
2. If  $T \subset B$  is a multiplicatively closed subset that  $i(t)$  are all invertible in  $A$ , then  $\Omega_{A/T^{-1}B} \cong \Omega_{A/B}$ .

We check the universal properties: For any  $S^{-1}A$ -module  $M$ ,

$$\mathrm{Hom}_{S^{-1}A}(S^{-1}\Omega_{A/B}, M) \cong \mathrm{Hom}_{S^{-1}A}(S^{-1}A \otimes_A \Omega_{A/B}, M) \cong \mathrm{Hom}_A(\Omega_{A/B}, M) \cong \mathrm{Der}_B(A, M),$$

$$\mathrm{Hom}_{S^{-1}A}(\Omega_{S^{-1}A/B}, M) \cong \mathrm{Der}_B(S^{-1}A, M)$$

There is a map  $\mathrm{Der}_B(S^{-1}A, M) \rightarrow \mathrm{Der}_B(A, M)$  by restriction, and the converse is given by

$$d \mapsto d\left(\frac{a}{s}\right) = \frac{sda - ads}{s^2}.$$

This is well-defined as

$$d\left(\frac{at}{st}\right) = \frac{std(at) - atd(st)}{s^2t^2} = \frac{sda - ads}{s^2},$$

and it satisfies

$$\begin{aligned} d\left(\frac{a_1}{t_1} + \frac{a_2}{t_2}\right) &= d\left(\frac{a_1t_2 + a_2t_1}{t_1t_2}\right) \\ &= \frac{(a_1d(t_2) + t_2d(a_1) + a_2d(t_1) + t_1d(a_2))t_1t_2 - (a_1t_2 + a_2t_1)(t_1d(t_2) - t_2d(t_1))}{t_1^2t_2^2} \\ &= \frac{t_1t_2^2d(a_1) + t_1^2t_2d(a_2) - a_2t_1^2d(t_2) + a_1t_2^2d(t_1)}{t_1^2t_2^2} \\ &= d\left(\frac{a_1}{t_1}\right) + d\left(\frac{a_2}{t_2}\right) \end{aligned}$$

$$\begin{aligned} d\left(\frac{a_1a_2}{t_1t_2}\right) &= \frac{(a_1d(a_2) + a_2d(a_1))t_1t_2 - a_1a_2(t_1d(t_2) + t_2d(t_1))}{t_1^2t_2^2} \\ &= \frac{a_1(t_2d(a_1) - a_2d(t_2))}{t_1t_2^2} + \frac{a_2(t_1d(a_2) - a_1d(t_1))}{t_2t_1^2} \\ &= \frac{a_1}{t_1}d\left(\frac{a_2}{t_2}\right) + \frac{a_2}{t_2}d\left(\frac{a_1}{t_1}\right) \end{aligned}$$

Thus this  $d$  is an extension of the derivative to  $S^{-1}A$ . Thus we get the desired isomorphism by Yoneda lemma.  $\square$

**Prop. (5.4.3.7) [Jacobi-Zariski Sequences].** For a sequence of commutative rings:  $A \rightarrow B \rightarrow C$ , there is an exact sequence

$$C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0$$

of  $C$ -modules. It has a left inverse and splits iff any derivation over  $A$  from  $B$  to a  $C$ -module can be extended to a derivation over  $A$  from  $C$  to  $M$ . This is trivially true when  $B \rightarrow C$  has a retraction, and true when  $C/B$  is formally smooth by (5.4.5.5).  $\lrcorner$

*Proof:* Taking  $\mathrm{Hom}$  with an arbitrary  $C$ -module  $M$ , by universal property, we need to check the exactness of  $0 \rightarrow \mathrm{Der}_B(C, M) \rightarrow \mathrm{Der}_A(C, M) \rightarrow \mathrm{Der}_A(B, M)$ , which is easy.  $\square$

**Prop. (5.4.3.8)[Second Exact Sequence].** (This is a special case of (8.1.1.5)). If  $S' = S/I$ , then there is an exact sequence of  $R'$ -modules:

$$I/I^2 \rightarrow \Omega_{S/R} \otimes_S S' \rightarrow \Omega_{S'/R} \rightarrow 0.$$

Where  $f \in I$  is mapped to  $df \otimes 1$  and it has a left inverse and splits iff  $S/I^2 \rightarrow S'$  has a right inverse. And in fact  $\Omega_{S/R} \otimes_S S' \cong \Omega_{(S/I^2)/R} \otimes_S S'$ .  $\lrcorner$

*Proof:* For a  $S/I$ -module  $M$ , we check:

$$0 \rightarrow \text{Der}_R(S/I, M) \rightarrow \text{Der}_R(S, M) \rightarrow \text{Hom}_{S/I}(I/I^2, M)$$

To prove  $\Omega_{S/R} \otimes_S S' \cong \Omega_{(S/I^2)/R} \otimes_S S'$ , we apply  $\text{Hom}$  for a  $S'$ -module  $M$ .

So to prove the left exactness, we may assume  $I^2 = 0$ . If we have an inverse  $\Omega_{S/R} \otimes_S S' \rightarrow I$ , then it gives a derivation  $D : A \rightarrow I$  that is identity on  $I$ , so  $a - D(a)$  gives a  $R$ -ring map  $S \rightarrow S'$  that is trivial on  $I$  (because  $I^2 = 0$ ). Hence it gives a  $S/I \rightarrow S'$  that is inverse to the projection.

For the converse, if  $d : S/I \rightarrow S'$  is a right inverse, then  $a - d(\bar{a})$  is a derivation  $S \rightarrow I$ , which is identity on  $I$ , so it gives an inverse map  $\Omega_{S/R} \otimes_S S' \rightarrow I$  by universal property.  $\square$

**Cor. (5.4.3.9).** If  $R \rightarrow S$  is of f.p., then  $\Omega_{S/R}$  is of f.p. over  $S$ . If  $R \rightarrow S$  is of f.t., then  $\Omega_{S/R}$  is of f.t. over  $S$ . (Follows from the second exact sequence (5.4.3.8) and (5.4.3.10)).  $\lrcorner$

**Cor. (5.4.3.10)[Examples].**

1.  $\Omega_{A[X_1, \dots, X_n]/A} = A[X_1, \dots, X_n]\{dX_1, \dots, dX_n\}$ .
2. If  $S = A[X_i]/\{f_j\}$ , then  $\Omega_{S/A} = S[dX_i]/\{df_j\}$ .
3.  $\Omega_{A[X_i]/k} = \Omega_{A/k} \otimes_A A[X_i] \oplus A[X_i]\{dX_1, \dots, dX_n\}$ .
4. (Standard Étale Algebra) For  $A = R[x]_g/(f)$ , where  $f'$  has image invertible in  $A$ ,  $\Omega_{A/R} = 0$ .
5. The differential for the inclusion  $k[y^2, y^3] \rightarrow k[y]$  is  $k[y]/(2y, 3y^2)\{dy\}$ .

$\lrcorner$

**Cor. (5.4.3.11).** 1: Use the differential operator and universal property.

2: Use item 1 and (5.4.3.8).

3: Use item 1 and the fact (5.4.3.8) splits because any derivative of  $A/k$  can be extended to derivative of  $B/k$  by acting on the coefficients.

4:

5: Use definition.  $\lrcorner$

**Cor. (5.4.3.12).** If  $S/I$  is a field  $k$  that embeds in  $S$ , then  $I/I^2 \cong \Omega_{S/k} \otimes_S k$ .  $\lrcorner$

**Prop. (5.4.3.13).** Let  $k \subset K \subset L$  be fields, and  $L/K$  f.g., then

$$\dim_L \Omega_{L/k} \geq \dim_K \Omega_{K/k} + \text{tr. deg}(L/K).$$

Equality holds if  $L/K$  is separably generated, i.e. separable over a transcendental basis. If  $K = k$ , then the equality holds iff  $L/k$  is separably generated. In particular,  $L/k$  is separable algebraic extension iff  $\Omega_{L/k} = 0$ .  $\lrcorner$



*Proof:* Take a subfield  $K \subset K(t_1, \dots, t_n) \subset L$  that  $L$  is separable algebra over  $K(t_1, \dots, t_n)$ . Then it suffices to add one element a time, so we may assume  $L = K(\alpha)$ .

1: If  $\alpha$  is transcendental over  $K$ , then  $\Omega_{K[\alpha]/t} \cong \Omega_{K/k} \otimes K[\alpha] \oplus K[\alpha]dt$  by (5.4.3.10), and by localization (5.4.3.6) we get  $\Omega_{L/K} \cong \Omega_{K/k} \otimes_K L \oplus Ldt$ , thus  $\text{rank } \Omega_{L/k} = \text{rank } \Omega_{K/k} + 1$ .

2: If  $\alpha$  is separable over  $K$ , then there is a monic polynomial  $f \in K[X]$  that  $K[\alpha] \cong K[X]/(f)$ . Then  $f'$  is invertible in  $K[\alpha]$ , and by (5.4.3.10)  $\Omega_{K[\alpha]/k} = \Omega_{K/k} \otimes_K L \oplus LdX/(d(f) + f'dX) \cong \Omega_{K/k} \otimes_K L$ . Thus  $\text{rank } \Omega_{L/k} = \text{rank } \Omega_{K/k}$ .

3: If  $K$  has characteristic  $p$  and  $L = K[X]/(X^p - a)$  and  $d_{K/k}(a) = 0$ , then  $\Omega_{K[\alpha]/k} = \Omega_{K/k} \otimes_K L \oplus LdX/(d(a)) \cong \Omega_{K/k} \otimes_K L \oplus LdX$ . Thus  $\text{rank } \Omega_{L/k} = \text{rank } \Omega_{K/k} + 1$ .

4: If  $K$  has characteristic  $p$  and  $L = K[X]/(X^p - a)$  and  $d_{K/k}(a) \neq 0$ , then  $\Omega_{K[\alpha]/k} = \Omega_{K/k} \otimes_K L \oplus LdX/(d(a))$  has rank  $\text{rank } \Omega_{K/k}$ .

Thus the assertion is clear.  $\square$

**Prop. (5.4.3.14) [Differential and Regularity].** Let  $B$  be a Noetherian local ring containing its residue field  $k$  and  $k$  is perfect, then  $\Omega_{B/k}$  is a free  $B$ -module of rank  $\dim B$  iff  $B$  is regular. [Hartshorne Ex.2.8.1] has a generalization of this fact.  $\lrcorner$

*Proof:* One way is by (5.4.3.12). Conversely, if  $B$  is regular, then it is integral (5.3.5.20), so  $\Omega_{B/k} \otimes_B K = \Omega_{K/k}$  (5.4.3.6) is of  $K$ -dimension  $\text{tr. deg } K/k = \dim B$ , where  $K$  is the quotient field of  $B$ , and  $\Omega_{B/k} \otimes k \cong m/m^2$  is of  $k$ -dimension  $\dim B$  once again. These two facts shows that  $\Omega_{B/k}$  is free  $B$ -module of rank  $\dim B$  by (5.4.8.1).  $\square$

**Prop. (5.4.3.15).** The Kähler differential  $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$  for an extension of number fields is cyclic.  $\lrcorner$

*Proof:* Because it is locally cyclic (5.2.7.34).  $\square$

## 4 Complete Intersections

### Koszul Complex

**Prop. (5.4.4.1) [Koszul Complex].** The complex of  $\mathbb{Z}[X_1, \dots, X_n]$ -modules  $K_\bullet$ , where

$$K_r = \wedge^{r+1} \mathbb{Z}[dX_1, \dots, dX_n] \otimes_{\mathbb{Z}} \mathbb{Z}[X_1, \dots, X_n]$$

and the morphism is given linearly by

$$\iota : dX_{i_0} \wedge dX_{i_1} \wedge \dots \wedge dX_{i_r} \mapsto \sum_k X_{i_k} dX_{i_0} \wedge \dots \wedge dX_{i_{k-1}} \wedge dX_{i_{k+1}} \wedge \dots \wedge dX_{i_r}$$

Then this is a free resolution of  $\mathbb{Z}$  over  $\mathbb{Z}[X_1, \dots, X_n]$ .  $\lrcorner$

*Proof:* We can use induction. If  $r = 1$ , then this is clear. For the induction process, if  $\iota(dX_1 \wedge \alpha + \beta) = 0$ , and  $\beta \in K_i$ , where  $\beta, \alpha$  has no  $dX_1$  involved. Notice if  $dX_1 \wedge \alpha \in K_0$ , then we may assume  $\alpha$  has no constant coefficient, because this is impossible.

then  $X_1\alpha - dX_1 \wedge \iota(\alpha) + \iota(\beta) = 0$ . Then we see that  $\iota(\alpha) = 0$ . We can write  $\alpha = \alpha_0 + X_1\alpha_1 + X_1^2\alpha_2 + \dots$ , where  $\alpha_i$  has no  $X_i$  involved, then we see that  $\iota(\alpha_i) = 0$ , then by induction hypothesis  $\alpha = \iota(\alpha')$  (notice  $\alpha_0$  has no constant coefficient), and  $\iota(\beta) + X_1\iota(\alpha') = 0$ . If  $\beta = X_1\beta_1 + \beta_2$ , where  $\beta_2$  has no  $X_1$  involved, then  $\iota(\beta_2) = 0$ , so  $\beta_2 = \iota(\beta'_2)$ , and  $\iota(\beta_1 + \alpha') = 0$ , so  $\beta_1 + \alpha' = \iota(\beta'_1)$ . So

$$dX_1 \wedge \alpha + \beta = dX_1 \wedge \iota(\alpha') + \iota(X_1\beta'_1 + \beta'_2) - X_1\alpha' = \iota(X_1\beta'_1 + \beta'_2 - dX_1 \wedge \alpha').$$

And in degree 0, this is clear.  $\square$

**Def. (5.4.4.2) [Koszul Complex].** Let  $A$  be a ring and  $I = (f_1, \dots, f_n) \in A$  is an ideal, then the **Koszul complex** for  $Kos(A, f_1, \dots, f_n)$  is an object in  $D(A)$  defined by

$$Kos(A, f_1, \dots, f_n) = A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z},$$

where  $A$  is a  $\mathbb{Z}[X_1, \dots, X_n]$ -algebra by mapping  $X_i \rightarrow f_i$ . If  $M$  is an  $A$ -module, then we define  $Kos(M, f_1, \dots, f_n) = M \otimes_A^L Kos(A, f_1, \dots, f_n)$ .  $\lrcorner$

**Prop. (5.4.4.3).** If  $f_1, \dots, f_n \in R$ , then  $I = (f_1, \dots, f_n)$  annihilates  $H^*(K(f_1, \dots, f_n))$ . In particular,  $Kos(A, f_1, \dots, f_n)$  is in the image of  $D(A/I) \subset D(A)$ .  $\lrcorner$

*Proof:* This is because every  $H^i(A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z})$  is an  $H^0(A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z}) = A/I$ -algebra, because it is a simplicial algebra.  $\square$

**Prop. (5.4.4.4).**  $K(A, f_1, \dots, f_n, g_1, \dots, g_m) = K(A, f_1, \dots, f_n) \otimes_A K(A, g_1, \dots, g_m)$ . (Easy).  $\lrcorner$

**Prop. (5.4.4.5).** The cone of the map

$$f_n : K(f_1, \dots, f_{n-1}) \rightarrow K(f_1, \dots, f_{n-1})$$

is isomorphic to  $K(f_1, \dots, f_n)$ .  $\lrcorner$

*Proof:* This is because  $\mathbb{Z}[X] \xrightarrow{X} \mathbb{Z}[X] \rightarrow \mathbb{Z}$  is an exact triangle, so  $A \xrightarrow{f_n} A \rightarrow A \otimes_{\mathbb{Z}[X]}^L \mathbb{Z}$  is an exact triangle, so  $Kos(A, f_1, \dots, f_{n-1}) \xrightarrow{f_n} K(A, f_1, \dots, f_{n-1}) \rightarrow K(A, f_1, \dots, f_n)$  is an exact triangle.  $\square$

**Prop. (5.4.4.6).** Let  $A$  be a ring and  $M$  be an  $A$ -module. Let  $f_1, \dots, f_{r-1}, f, g$  be elements of  $A$ , then there is a natural distinguished triangle

$$Kos(M, f_1, \dots, f_{r-1}, f) \rightarrow Kos(M, f_1, \dots, f_{r-1}, fg) \rightarrow Kos(M, f_1, \dots, g).$$

$\lrcorner$

*Proof:* We use (5.4.4.5) to consider  $Kos(M, f_1, \dots, f_{r-1}, f)$  as a cone of  $f_n : Kos(M, f_1, \dots, f_{r-1}) \rightarrow Kos(M, f_1, \dots, f_{r-1})$ . Then this distinguished triangle is given by (4.8.4.16) applied to the diagram

$$\begin{array}{ccc} Kos(M, f_1, \dots, f_{r-1}) & \xrightarrow{\quad} & Kos(M, f_1, \dots, f_{r-1}) \\ \downarrow f & & \downarrow fg \\ Kos(M, f_1, \dots, f_{r-1}) & \xrightarrow{g} & Kos(M, f_1, \dots, f_{r-1}) \end{array}$$

$\square$

**Prop. (5.4.4.7) [Koszul Complex and Čech Complex].** Let  $A$  be a commutative ring and  $I = (f_1, \dots, f_n) \subset A$ ,  $K_n^\bullet = Kos(A, f_1^n, \dots, f_r^n) = A \otimes_{\mathbb{Z}[X_1, \dots, X_r]}^L \mathbb{Z}$ . Then there are natural maps

$$\dots \rightarrow K_3^\bullet \rightarrow K_2^\bullet \rightarrow K_1^\bullet$$

compatible with the inverse system  $H^0(K_n^\bullet) = A/(f_1^n, \dots, f_r^n)$ . Then there is a description of  $R \operatorname{colim} K_n^\bullet$  (5.9.5.6) as the alternating Čech complex

$$R \rightarrow \bigoplus_{i_0} R_{f_{i_0}} \rightarrow \bigoplus_{i_0 < i_1} R_{f_{i_0} f_{i_1}} \rightarrow \dots \rightarrow R_{f_1 f_2 \dots f_r}$$

where  $R$  sits in degree 0.  $\lrcorner$

*Proof:* Cf.[Sta]0913, which is not hard.  $\square$

**Def. (5.4.4.8) [Koszul-Regular Sequence].** Let  $A$  be a ring and  $(f_1, \dots, f_n) \in A$  be a sequence, then  $f_1, \dots, f_n$  is called  **$M$ -Koszul-regular** iff  $\text{Kos}(M, f_1, \dots, f_n) = 0$ . It is called **Koszul-regular** iff  $\text{Kos}(A, f_1, \dots, f_n) = 0$ .  $\lrcorner$

**Prop. (5.4.4.9).** If  $(f_1, \dots, f_r)$  is  $(M\text{-Koszul})$ regular and  $n_i > 0$ , then  $(f_1^{n_1}, \dots, f_r^{n_r})$  is also  $(M\text{-Koszul})$ regular.  $\lrcorner$

*Proof:* This follows from (5.4.4.6).  $\square$

**Prop. (5.4.4.10) [Regular and Koszul-Regular].** A  $M$ -regular sequence (5.3.4.1) is  $M$ -Koszul-regular. A regular sequence is Koszul-regular.  $\lrcorner$

*Proof:* Let  $(f_1, \dots, f_r)$  be a regular sequence. The assertion is clear when  $r = 1$ . For the induction:

$$\begin{aligned} \text{Kos}(M, f_1, \dots, f_n) &= \text{Kos}(M, f_2, \dots, f_n) \otimes_{\mathbb{Z}[X]}^L \mathbb{Z} = \text{Kos}(A, f_2, \dots, f_n) \otimes_A^L M \otimes_{\mathbb{Z}[X]}^L \mathbb{Z} \\ &= \text{Kos}(A/f_1, f_2, \dots, f_n) \otimes_{A/f_1}^L M/f_1 M = \text{Kos}(M/f_1 M, f_2, \dots, f_n) \end{aligned}$$

so we can use induction.  $\square$

### Complete Intersections

**Def. (5.4.4.11) [Global Complete Intersections].**  $R \rightarrow S \in \mathcal{C}\text{Ring}$  is called a **global intersection** if  $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$  that every non-empty fiber of  $\text{Spec } S \rightarrow \text{Spec } R$  has dimension  $n - c$ .  $\lrcorner$

**Prop. (5.4.4.12).** Global intersection is stable under base change and composition. And  $S \rightarrow S_f$  a global intersection, so it is stable under localization.  $\lrcorner$

*Proof:* Base change: this is because flatness, f.p. is stable under base change, and the fibers are the same.

Localization:  $S_f = S[X]/(fX - 1)$  has trivial fibers.

Composition: It suffices to assume  $R$  is a field and calculate the dimension of the fibers, which is true by dimension formula (5.2.4.13).  $\square$

**Prop. (5.4.4.13) [Noetherian Approximation].** Let  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$  is a global intersection over  $R$ , then there is a f.g.  $\mathbb{Z}$ -algebra  $R_0 \subset R$  that  $f_i \in R_0[X_1, \dots, X_n]$ , and  $S_0 = R_0[X_1, \dots, X_n]/(f_1, \dots, f_c)$  is a global intersection over  $R$ .  $\lrcorner$

*Proof:* Cf.[Sta]00SU.  $\square$

**Prop. (5.4.4.14) [Complete Intersection is Regular].** Let  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$  be a global complete intersection, then  $\text{Spec } S \rightarrow \text{Spec } R[X_1, \dots, X_n]$  is a regular embedding (7.6.7.1), and  $S \in \mathcal{C}\text{Ring}^{\text{flat}}/R$ .  $\lrcorner$

*Proof:* Cf.[Sta]00SV.  $\square$

### Local Complete Intersections

**Def. (5.4.4.15) [Local Complete Intersections].**  $R \rightarrow S$  is called a **local complete intersection** locally is a global complete intersection (5.4.4.11). Local complete intersections are flat, by (5.4.4.14).  $\lrcorner$

**Cor. (5.4.4.16).** Local complete intersection is local on the source and target. In fact satisfies f.f. descent (5.4.2.1).  $\lrcorner$

**Prop. (5.4.4.17).** If  $S$  is a f.t.  $k$ -algebra and  $K/k$  is a field extension, then  $S$  is a local complete intersection iff  $S_K$  is a local complete intersection.  $\lrcorner$

**Prop. (5.4.4.18) [Global Intersections and Fibers].** Let  $R \rightarrow S$  be a ring map and  $\mathfrak{q} \subset S$  be a prime lying over  $\mathfrak{p} \subset R$ , then the following are equivalent:

- $R \rightarrow S$  is a global complete intersection around  $\mathfrak{q}$ .
- $R \rightarrow S$  is a f.p. around  $\mathfrak{q}$ ,  $S_{\mathfrak{q}}/R_{\mathfrak{p}}$  is flat, and (the fiber)  $S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}}$  is a local complete intersection ring over  $k(\mathfrak{p})$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 2$  follows from (5.4.4.14). For  $2 \rightarrow 1$ , Cf. [Sta]00SY.  $\square$

**Prop. (5.4.4.19) [Local Complete Intersection over Fields is Stalkwise].** For a f.g.  $k$ -algebra  $S$ ,  $S$  is a local complete intersection iff all localization at (maximal) primes are complete intersection local rings over  $k$ . In particular, being local complete intersection is a stalkwise property.  $\lrcorner$

*Proof:* This is because over field everything is f.p. and flat, and use (5.4.4.18).  $\square$

**Prop. (5.4.4.20).** If  $S$  is a f.g. local complete intersection over a field  $k$ , then it is a CM ring.  $\lrcorner$

*Proof:* Cf. [Sta]00SB.  $\square$

**Def. (5.4.4.21) [Complete Intersection Local Rings].** Let  $A$  be a local ring essentially of f.t. over a field  $k$ , then  $A$  is called a **complete intersection local ring** over  $k$  if there exists a surjection  $R \rightarrow A$  from a regular local ring essentially of f.t. over  $k$  that the kernel is generated by a regular sequence.  $\lrcorner$

**Prop. (5.4.4.22).** When  $A$  is f.g. local ring over a field  $k$ , then  $A$  is a complete intersection local ring iff  $A$  is local complete intersection over  $k$  (5.4.4.11).  $\lrcorner$

*Proof:* Cf. [Sta]00SC.  $\square$

See more in [Sta]Chap 23.8.

## 5 Smooth Maps

### Formally Smooth Ring Maps

**Def. (5.4.5.1) [Formally Smooth Ring Maps].**  $R \rightarrow S \in \mathcal{CRing}$  is called a **formally smooth ring map** if has right lifting property w.r.t all ring maps  $A \rightarrow A/I$  where  $I \in \text{Ideal}(A)$  s.t.  $I^2 = 0$ .

Formal smooth is stable under base change and composition, by universal property.

For  $R \rightarrow R[T_1, \dots, T_n]$  is formally smooth for any  $n \in \mathbb{Z}_+$ .  $\lrcorner$

**Prop. (5.4.5.2).** Let  $S \in \mathcal{C}\text{Ring}/R$  with a presentation  $S = P/J$  where  $P \in \mathcal{C}\text{Ring}/R$  is formally smooth (e.g. polynomial algebra) and  $J \in \text{Ideal}(P)$  is stable under  $R$ . Then  $S$  is formally smooth over  $R$  iff there is a map  $S \rightarrow P/J^2$  that is right converse to the obvious projection.  $\lrcorner$

*Proof:* One way is from the definition of formally smooth applied to  $P/J^2$  and  $J$ . Conversely, for any  $A$  and  $I$ , we notice the map  $P \rightarrow S \rightarrow A/I$  can be lifted to  $P \rightarrow A$ , and  $J$  is mapped to  $I$ , so  $J^2$  is mapped to 0, so we have a map  $P/J^2 \rightarrow A$ . Then  $S \rightarrow P/J^2 \rightarrow A$  is the lifting.  $\square$

**Cor. (5.4.5.3).** If  $P \rightarrow S$  is a presentation of  $S/R$  by polynomial algebra with kernel  $I$ , then  $S/R$  is formal smooth iff

$$0 \rightarrow I/I^2 \rightarrow \Omega_{P/R} \otimes_P S \rightarrow \Omega_{S/R} \rightarrow 0$$

is split exact as in (5.4.3.8).  $\lrcorner$

*Proof:* This sequence is split exact iff  $P/J^2 \rightarrow S$  has a right converse, by (5.4.3.8).  $\square$

Now we consider the relation of formal smoothness and Kähler differentials.

**Cor. (5.4.5.4) [Equivalence Definition].**  $S/R$  is formally smooth iff  $NL_{S/R}$  is quasi-isomorphic to a projective  $S$ -module at degree 0.  $\lrcorner$

*Proof:* If  $S/R$  is formally smooth, then choose a presentation will suffice by (5.4.5.3). The converse is also true by projectiveness and (5.4.5.3).  $\square$

**Cor. (5.4.5.5).** If  $C/B$  is formally smooth, then the Jacobi-Zariski sequence

$$0 \rightarrow C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0$$

as in (5.4.3.7) is split exact, by (8.1.1.5). In particular, any derivation of  $B$  to a  $C$ -module can be extended to a derivation  $C$  to a  $C$ -module.  $\lrcorner$

**Cor. (5.4.5.6).** If  $A \rightarrow B \rightarrow C$  with  $A \rightarrow C$  formally smooth and  $B \rightarrow C$  surjective with kernel  $I$ , then there is a split sequence

$$0 \rightarrow I/I^2 \rightarrow \Omega_{B/A} \otimes_B C \rightarrow \Omega_{C/A} \rightarrow 0$$

by (8.1.2.6).  $\lrcorner$

### Standard Smooth Algebra

**Def. (5.4.5.7) [Standard Smooth Ring Maps].** A **standard smooth ring** over  $R$  is a algebra  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$ , where  $c \leq n$  and  $J(\frac{f_1, \dots, f_c}{X_1, \dots, X_c})$  is invertible in  $S$ .  $\lrcorner$

**Prop. (5.4.5.8) [Standard Smooth Localization].** If  $R \rightarrow S$  is standard smooth, then  $R \rightarrow S_g$  is standard smooth, and  $R_f \rightarrow S_f$  is standard smooth (because stable under base change (5.4.5.9)).  $\lrcorner$

*Proof:* For localization at  $g \in S$ , let  $h$  be an inverse image of  $g$  in  $R[X_1, \dots, X_n]$ , then  $S_g = R[X_1, \dots, X_n, X_{n+1}]/(f_1, \dots, f_c, X_{n+1}h - 1)$ , and it is standard smooth.  $\square$

**Prop. (5.4.5.9).** Standard smoothness is stable under base change and composition.  $\lrcorner$

*Proof:* For base change, notice the Jacobi matrix is the base change of the Jacobi matrix, so it is also invertible. For composition, write out the presentation, the determinant is the product of the presentation.  $\square$

**Lemma (5.4.5.10) [Kähler Differential of Smooth Algebra is Free].** The Kähler differential of a standard smooth algebra  $S$  over a field  $k$  is free of rank  $\dim S$ , and  $(f_1, \dots, f_c)/(f_1, \dots, f_c)^2$  is free over  $S$  with basis  $f_1, \dots, f_c$ . Moreover,  $S$  is pure dimensional.  $\lrcorner$

*Proof:* The naive cotangent complex for  $S/R$  is

$$d : (f_1, \dots, f_c)/(f_1, \dots, f_c)^2 \rightarrow S[dX_1, \dots, dX_n].$$

By hypothesis and linear algebra it is a split injection, and  $\Omega_{S/R} = S[dX_{c+1}, \dots, dX_n]$ , so it is free of rank  $n - c = \dim S$ , because  $S$  is a global complete intersection (5.4.5.11).  $\square$

**Prop. (5.4.5.11) [Standard Smooth and Complete Intersection].** A standard smooth algebra  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$  is a global complete intersection (5.4.4.11).  $\lrcorner$

*Proof:* It suffices to show any fiber of  $S$  has dimension  $n - c$ . For this, notice  $S \otimes_R k(\mathfrak{p})$  is also standard smooth, then we reduce to the field case. Now  $I = (f_1, \dots, f_c)$  satisfies  $I/I^2 \rightarrow \oplus S dx_i$  is a split injection. For any maximal ideal  $\mathfrak{m}$  containing  $I$ , tensoring  $k(\mathfrak{m})$ , we get an injection  $I/\mathfrak{m}I \rightarrow \oplus k(\mathfrak{m}) dx_i$ . Notice there is a commutative diagram

$$\begin{array}{ccc} I & \longrightarrow & S/\mathfrak{m} \otimes I \cong I/\mathfrak{m}I \\ \downarrow d & & \downarrow \text{incl} \\ \oplus S dx_i & \xrightarrow{dx_i \mapsto 1 \otimes x_i} & S/\mathfrak{m} \otimes \mathfrak{m} \cong \mathfrak{m}/\mathfrak{m}^2 \end{array} .$$

And the lower horizontal map is an isomorphism by Hilbert's Nullstellensatz, so the image of  $f_i$  in  $\mathfrak{m}_i/\mathfrak{m}_i^2$  are linearly independent over  $k(\mathfrak{m})$ , thus we can use (5.3.5.23) to show that  $\dim S = n - c$ .  $\square$

### Smoothness

**Def. (5.4.5.12) [Smooth Ring Map].**  $R \rightarrow S \in \mathcal{C}\text{Ring}$  is called a **smooth ring map** if it satisfies the following equivalent conditions:

- It is of f.p. and the naive cotangent complex  $NL_{S/R}$  is quasi-isomorphic to a finite locally free  $S$ -module placed at degree 0. In other words,

$$0 \rightarrow I/I^2 \rightarrow \Omega_{P/R} \otimes_P S \rightarrow \Omega_{S/R} \rightarrow 0$$

is exact and  $\Omega_{S/R}$  is finite locally free. By (8.1.1.1), we only need to prove for a single presentation of  $S$ .

- It is locally standard smooth.
- It is formally smooth and of f.p..

We say  $S$  is smooth at  $x$  if it is smooth on a nbhd of  $x$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 3$ : by (5.4.5.4).  $3 \rightarrow 1$ : By (5.4.5.4),  $\Omega_{S/R}$  is f.p. and projective, so it is finite projective.

At this point we already know that the first definition is stable under base change and composition, because f.p. and formal smoothness both do (5.4.5.1)(5.3.7.9).

And also the first definition is local on source because f.p. does (5.1.4.4) and  $NL$  commutes with localization (8.1.1.6) so we can use the local properties of triviality (5.1.4.2) and finite projectiveness (5.3.1.7).

Now it is also local on the source because it is stable under base change and composition and  $R \rightarrow R_{f_i}$  does by locality on the source.

2  $\rightarrow$  1: Now the property are all local on source. It suffices to prove a standard smooth map is smooth. This follows from (5.4.5.10).

1  $\rightarrow$  2: We need to prove, assuming the first definition, it is locally standard smooth. For this, Cf. [Sta]00TA ? .  $\square$

**Cor. (5.4.5.13).** Smoothness is stable under composition and base change. Smoothness is local on the source and target (In particular,  $R \rightarrow R_f$  is smooth). (Already proved in the proof of (5.4.5.12)).  $\lrcorner$

**Prop. (5.4.5.14) [Smooth Maps are Complete-Intersections].** A smooth map is a local complete-intersection (5.4.4.15), hence flat.  $\lrcorner$

*Proof:* Clear from (5.4.5.12).  $\square$

**Prop. (5.4.5.15) [Noetherian Descent].** A smooth ring map  $R \rightarrow S$  is a base change of smooth ring map over a ring f.g. over  $\mathbb{Z}$ .  $\lrcorner$

*Proof:* Use the equivalence definition (5.4.5.2), we know that there is a map

$$S = R[X_1, \dots, X_n]/(f_1, \dots, f_c) \rightarrow R[X_1, \dots, X_n]/(f_1, \dots, f_c)^2,$$

which if we write  $\sigma(X_i) = h_i$ , then must satisfy

$$f_i(h_1, \dots, h_n) = \sum a_{ijk} f_j f_k.$$

Then we consider the subalgebra generated by  $f_i, h_i, a_{ijk}$ , then by the same reason, they form a smooth algebra over  $\mathbb{Z}$ , and its tensor with  $R$  gives out  $S$ .  $\square$

**Cor. (5.4.5.16) [Strong Lifting Property].** Smooth ring maps satisfy right lifting property w.r.t. any ring map  $A \rightarrow A/I \in \mathcal{C}\text{Ring}$ , where  $I \in \text{Ideal}(A)$  is locally nilpotent.  $\lrcorner$

*Proof:* By (5.4.5.15),  $R \rightarrow S$  is a base change of a smooth ring map  $R_0 \rightarrow S_0$  where  $R_0$  is f.g. over  $\mathbb{Z}$ . Now if  $S_0$  is generated by  $x_1, \dots, x_n$  and  $a_1, a_n \in A$  maps to the image of  $x_1, \dots, x_n$  in  $A/I$ , then consider the subring  $A_0$  generated by  $R_0$  and  $a_i$ , and let  $I_0 = A_0 \cap I$ , then it suffices to prove this case followed by base change. But now  $A_0$  is f.g. over  $\mathbb{Z}$ , so it is Noetherian, and then  $I$  is nilpotent, thus we have a desired lifting.  $\square$

**Prop. (5.4.5.17) [Stalkwise].** If  $R \rightarrow S$  is f.p., then it is smooth iff it is  $S_q/R_p$  is smooth for every (maximal) prime  $q$  of  $S$  and  $p$  under it.  $\lrcorner$

*Proof:* Because of f.p., we only need to check triviality of  $H_1(NL)$  and finite projectivity of  $\Omega_{S/R}$  (5.4.5.12). But both triviality and finite projectivity is stalkwise (5.1.4.2). (Notice  $R \rightarrow R_p$  is smooth).  $\square$

**Cor. (5.4.5.18) [Smooth Locus and Flat Base Change].** If  $R \rightarrow S$  is of f.p. and  $R \rightarrow R'$  is flat. Then the smooth locus of  $S' = S \otimes_R R'/R'$  is the inverse image of smooth locus of  $S/R$ .  $\lrcorner$

*Proof:* One direction is because smooth is stable under base change. Conversely, the local ring map is f.f., so  $H_1(NL_{S'/R'}, q) = H_1((NL_{S/R} \otimes_S S')_q) = H_1(NL_{S/R, p} \otimes_{S_p} S'_q)$ . Then the result follows as  $S'_q/S_p$  is f.f. and triviality and finite projective descents for f.f. map (5.4.2.1).  $\square$

**Lemma (5.4.5.19).** A global complete intersection  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$  is smooth at a point  $\mathfrak{p}$  iff the Jacobian has rank  $\geq n - \dim_{\mathfrak{q}}(S)$  at  $\mathfrak{q}$ , i.e.  $J(\frac{f_1, \dots, f_c}{X_1, \dots, X_c})$  is not in  $\mathfrak{q}$  for some permutation of  $X_1, \dots, X_n$ .  $\lrcorner$

*Proof:* This is a special case of (5.4.5.24). However, we need to prove it first here. If it is smooth at  $\mathfrak{q}$ , then  $\Omega_{S/R}$  is locally free of dimension  $\geq n - \dim_{\mathfrak{q}}(S)$  at  $\mathfrak{q}$  by (5.4.5.10), so the Jacobian has rank  $\geq n - \dim_{\mathfrak{q}}(S)$ . Conversely, if  $g = J(\frac{f_1, \dots, f_c}{X_1, \dots, X_c}) \notin \mathfrak{q}$ , then  $S_g = R[X_1, \dots, X_n, X_{n+1}]/(f_1, \dots, f_c, gX_{n+1} - 1)$  is a standard smooth map.  $\square$

**Prop. (5.4.5.20)[Fiberwise].** For a ring map  $R \rightarrow S$  and  $\mathfrak{q}$  is a prime of  $S$  over  $\mathfrak{p}$ . Then  $S/R$  is smooth at  $\mathfrak{q}$  iff  $S/R$  is of f.p. around  $\mathfrak{q}$  and  $S_{\mathfrak{q}}/R_{\mathfrak{p}}$  is flat and  $S \otimes_R k(\mathfrak{p})/k(\mathfrak{p})$  is smooth at  $\mathfrak{q}$ .  $\lrcorner$

*Proof:* One direction is because smooth is flat, f.p. and stable under base change. Conversely, by (5.4.5.14) and (5.4.4.18), change  $R$  to  $R_g$  for some  $g \notin \mathfrak{p}$ , we may assume  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$  is a global complete intersection. Then we may use (5.4.5.19) to see the map is smooth on the standard open subset defined by the product of Jacobians of  $f_i$ .  $\square$

**Prop. (5.4.5.21).** If  $A \rightarrow A[X_1, \dots, X_n] \rightarrow R$  is smooth, then  $A[X_1, \dots, X_n] \rightarrow R$  is smooth.  $\lrcorner$

*Proof:* The desired map is firstly of f.p. by (5.3.7.11), and it can be verified to be formally smooth, because  $A[X_1, \dots, X_n]$  is free.  $\square$

### Smooth over Fields

**Lemma (5.4.5.22).** Let  $S$  be f.g. over a alg.closed field  $k$  and  $\mathfrak{m}$  a maximal ideal, then the following are equivalent:

- $S_{\mathfrak{m}}$  is regular.
- $\dim_k \Omega_{S/k} \otimes_S k \leq \dim S_{\mathfrak{m}}$
- $\dim_k \Omega_{S/k} \otimes_S k = \dim S_{\mathfrak{m}}$
- $S/k$  is smooth at  $\mathfrak{m}$ .

$\lrcorner$

*Proof:* Cf. [Sta]00TS.  $\square$

**Prop. (5.4.5.23)[Differential Criterion of Smoothness].** For a ring  $S$  f.g. over a field  $k$ ,  $S$  is smooth in a nbhd of  $\mathfrak{q}$  iff  $\dim_{k(\mathfrak{q})} \Omega_{S/k} \otimes k(\mathfrak{q}) \leq \dim_{\mathfrak{q}}(S)$ .

And in this case, equality hold, and  $S_{\mathfrak{q}}$  is regular.  $\lrcorner$

*Proof:* Cf. [Sta]00TT.

If  $S$  is smooth at  $x$ , then  $\Omega_{S/R}$  is finite free on a nbhd of  $x$  of rank equals to the dimension (5.4.5.10), so the equation holds.

Conversely, if  $\dim_{k(x)} \Omega_{S/k} \otimes k(x) \leq \dim_x(X)$ , then  $\square$

**Cor. (5.4.5.24)[Jacobian Criterion of Smoothness].** For a f.p. ring  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$  and  $\mathfrak{q} \subset S$ ,  $S$  is smooth at  $\mathfrak{q}$  iff the Jacobian has rank  $\geq n - \dim_{\mathfrak{q}}(S)$  at  $\mathfrak{q}$  iff the Jacobian has rank  $= n - \dim_{\mathfrak{q}}(S)$  at  $\mathfrak{q}$ .  $\lrcorner$

**Lemma (5.4.5.25).** Let  $k$  be a field and  $(R, \mathfrak{m}, \kappa)$  be a Noetherian local ring containing  $k$ . If the residue field of  $R$  is a f.g. field extension of  $k$ , then the derivation map

$$\mathfrak{m}/\mathfrak{m}^2 \rightarrow \Omega_{R/k} \otimes_R \kappa$$

is injective.  $\lrcorner$



*Proof:* Cf.[Sta]00TU. □

**Prop. (5.4.5.26)[Smooth and Regular at Geometric Points].** Let  $S$  be f.g. over a field  $k$ , if  $k(\mathfrak{q})/k$  is separable (e.g. char 0) for  $\mathfrak{q}$  a prime of  $S$ , then  $S$  is smooth at  $\mathfrak{q}$  iff it is regular at  $\mathfrak{q}$ . ┘

*Proof:* Let  $R = S_{\mathfrak{q}}$  with maximal ideal  $\mathfrak{m}$ . By (5.4.5.25) and (5.4.3.8) there is an exact sequence

$$0 \rightarrow \mathfrak{m}/\mathfrak{m}^2 \rightarrow \Omega_{R/k} \otimes_R k(\mathfrak{q}) \rightarrow \Omega_{k(\mathfrak{q})/k} \rightarrow 0.$$

Since  $k(\mathfrak{q})/k$  is separable,  $\dim_{k(\mathfrak{q})} \Omega_{k(\mathfrak{q})/k} = \text{tr. deg}(k(\mathfrak{q})/k)$ . So

$$\dim_{k(\mathfrak{q})}(\Omega_{R/k} \otimes_R k(\mathfrak{q})) = \dim_{k(\mathfrak{q})} \mathfrak{m}/\mathfrak{m}^2 + \text{tr. deg}(k(\mathfrak{q})/k) \geq \dim R + \text{tr. deg}(k(\mathfrak{q})/k) = \dim_{\mathfrak{q}}(S)$$

with identity iff  $R$  is regular (The last identity comes from (7.6.3.7)). So we are done by differential criterion of smoothness (5.4.5.23). □

**Prop. (5.4.5.27) [Smooth over Fields and Geo.Regular].** Let  $S$  be f.g. over a field  $k$ , then  $S$  is smooth over  $k$  iff  $S$  is geo.regular (5.3.6.1). In particular, by (5.3.6.7), if  $k = k^{\text{perf}}$ , then smooth over  $k$  is equivalent to regular. ┘

*Proof:* If  $S$  is smooth at  $x$ , then all its base change is smooth at  $x$  (5.4.5.13), and the stalk is regular by (5.4.5.23), so it is geometrically regular at  $x$ .

Conversely, if  $X$  is geometrically regular, then for any point  $x \in X$ ,  $k(x)$  is f.g. over  $k$ , so by (5.3.9.3) there is a finite purely inseparable extension  $k'/k$  that the compositum  $k'k(x)$  is separable over  $k'$ . Then by (5.1.7.26),  $\text{Spec } A \otimes_k k'$  is homeomorphic to  $\text{Spec } A$ , so there is a unique prime  $p'$  of  $X_{k'}$  over  $X$ , and its residue field is  $k'k(x)$ . So by (5.4.5.26), as  $k'k(x)/k'$  is separable,  $X_{k'}$  is smooth over  $k'$  at  $p'$ . And f.f. descent for smoothness (5.4.2.1) says  $X$  is also smooth over  $k$  at  $p$ . □

**Cor. (5.4.5.28)[Differential and Smoothness].** Let  $k$  be a field of characteristic 0 and  $S$  a f.g. algebra over  $k$ , and  $\mathfrak{q}$  a prime ideal of  $S$ , if  $\Omega_{S/k, \mathfrak{q}}$  is free over  $S_{\mathfrak{q}}$ , then  $S$  is smooth in a nbhd of  $\mathfrak{q}$ . ┘

*Proof:* Cf.[Sta]00TX. □

**Prop. (5.4.5.29)[Generic Smoothness].** Let  $R \rightarrow S$  be an injective ring map of f.t. with  $R, S$  domains, then it is smooth at (0) iff the quotient field map is separable. ┘

*Proof:* If  $S$  is smooth at 0, then replacing  $S$  by  $S_g$  for some  $g$ , we can assume  $R \rightarrow S$  is smooth. Then  $K \rightarrow S \otimes_R K$  is also smooth (5.4.5.13), and also for any field extension  $K'$  of  $K$ . Then  $S \otimes_R K'$  is regular, by (5.4.5.23), a priori reduced (7.4.1.5). Thus  $S \otimes_R K$  is geometrically reduced. Hence also  $L$  is geometrically reduced over  $K$ , thus separable, by (5.3.9.4).

Conversely, by (5.4.1.34), we may assume  $R \rightarrow S$  is of f.p., thus to show it is smooth at (0), it suffices to show  $S \otimes_R K$  is smooth at (0), by (5.4.5.20). Then this follows from (5.4.5.26). □

### Smoothing Ring Maps

**Prop. (5.4.5.30).** A regular homomorphism of Noetherian rings is a filtered colimit of smooth ring maps. ┘

*Proof:* Cf.[Sta]07GC. □

## 6 Unramified Maps

### Formally Unramified Ring Maps

**Def. (5.4.6.1) [Formally Unramified Ring Maps].**  $R \rightarrow S \in \mathcal{CRing}$  is called **formally unramified** if for any  $A \in \mathcal{CRing}/R$  and  $I \in \text{Ideal}(A)$  s.t.  $I^2 = 0$ , a map  $S \rightarrow A/I$  has at most one extension to a map  $S \rightarrow A$ .

Formally unramifiedness is equivalent to  $\Omega_{S/R} = 0$ . So it is stable under composition by Jacobi-Zariski sequence (5.4.3.7).  $\lrcorner$

*Proof:* Let  $J = \ker(S \otimes_R S \rightarrow S)$ , let  $A_{univ} = S \otimes_R S/J^2$ , then  $J/J^2 \cong \Omega_{S/R}$  (5.4.3.4), so we have two natural map from  $S$  to  $A_{univ}$ , they differ by the universal differential  $S \rightarrow \Omega_{S/R}$ . If  $S/R$  is unramified, then  $ds = 0$  for all  $s \in S$ , so  $\Omega_{S/R} = 0$ .

Conversely, if there is a  $A$  and  $A/J$  that there are two liftings  $\tau_1, \tau_2$ , then we let  $A_{univ} \rightarrow A$  defined by  $s_1 \otimes s_2 \rightarrow \tau_1(s_1)\tau_2(s_2)$ , this is well-defined, and because  $A_{univ} \cong S$ , this map descends to  $S$ , so  $\tau_1(s_1s_2) = \tau_2(s_1s_2)$ .  $\square$

**Prop. (5.4.6.2) [Formally Unramified are Stalkwise].** Formally unramified is stalkwise both on the source and target (5.1.4.2).  $\lrcorner$

**Prop. (5.4.6.3).** Colimits of formally unramified rings over  $R$  is formally unramified.  $\lrcorner$

*Proof:* Trivial from the definition.  $\square$

### Unramified Ring Maps

**Def. (5.4.6.4) [Unramified Maps].** An **unramified ring map** is a ring map that is formally unramified and f.g..

A **G-unramified ring map** is a ring map that is formally unramified and of f.p..

These two notions are stable under composition and base change. These two notions are local on the source and target.  $R \rightarrow R_f$  is  $G$ -unramified. (5.4.6.1)(5.1.4.2)  $\lrcorner$

**Prop. (5.4.6.5).**  $R \rightarrow R/I$  is unramified, and if  $I$  is f.g., then it is  $G$ -unramified. (Trivial).  $\lrcorner$

**Prop. (5.4.6.6) [Stalkwise and Fiberwise].** If  $R \rightarrow S$  is of f.t(f.p.), then it is unramified ( $G$ -unramified) at a prime  $q$  of  $S$  iff  $(\Omega_{S/R})_q = 0$  iff  $\Omega_{S/R} \otimes_S k(q) = 0$  iff  $(\Omega_{S \otimes k(p)/k(p)})_q = 0$  iff  $\Omega_{S \otimes k(p)/k(p)} \otimes k(q) = 0$ .  $\lrcorner$

*Proof:* By Nakayama, two pair of them are equivalent, and if  $\Omega_{S/R,q} = 0$ , then  $\Omega_{S/R,g} = 0$  for some  $g \notin q$  (because support of finite module is open), so  $R \rightarrow S_g$  is ( $G$ -)unramified. And notice in fact  $\Omega_{S/R} \otimes_S k(q) = \Omega_{S \otimes k(p)/k(p)} \otimes_{k(p)} k(q)$ .  $\square$

**Prop. (5.4.6.7) [Unramified F.g. Ring Maps].** For  $R \rightarrow S \in \mathcal{CRing}$ , if  $S$  is unramified at  $\mathfrak{q} \in \text{Spec}(S)$  over  $\mathfrak{p} \in \text{Spec}(R)$ , then  $\mathfrak{p}S_{\mathfrak{q}} = \mathfrak{q}S_{\mathfrak{q}}$  and  $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$  is finite separable. And the converse is true if  $S$  is of f.t. over  $R$ .  $\lrcorner$

*Proof:* Suppose  $R \rightarrow S_g$  is unramified with  $g \in S \setminus \mathfrak{q}$ , then  $S \otimes \kappa(\mathfrak{p})$  is unramified over  $\kappa(\mathfrak{p})$ , hence by (5.4.5.23), it is also smooth, so it is étale, and (5.4.7.9) gives the result.

For the converse, Cf[Sta]02FM.  $\color{red}?$   $\square$

**Prop. (5.4.6.8) [Unramified Maps are Quotient of Standard Étale Maps].** A ring map is unramified iff it is locally a quotient of a standard Étale map.  $\lrcorner$

*Proof:* Cf. [Sta]0395.  $\square$

**Prop. (5.4.6.9).** Any  $G$ -unramified map is a base change of a  $G$ -unramified map over a ring  $R_0$  f.g. over  $\mathbb{Z}$ . And similarly any unramified map is a quotient of a base change of a  $G$ -unramified map over a ring  $R_0$  f.g. over  $\mathbb{Z}$ .  $\lrcorner$

*Proof:* Let  $S = R[X_1, \dots, X_n]/(g_1, \dots, g_c)$ , then we have  $dX_i = \sum a_{ij}dg_j + a_{ijk}g_jdX_k$ , so we let  $R_0$  be generated by  $g_i, a_{ij}, a_{ijk}$ , so  $S_0 = R_0[X_1, \dots, X_n]/(g_1, \dots, g_c)$  is  $G$ -unramified.  $\square$

**Prop. (5.4.6.10) [Unramifiedness and Idempotent].** If  $R \rightarrow S$  is of f.t., then it is unramified iff  $S \times_R S \rightarrow S$  is isomorphic to  $S \otimes_R S \rightarrow (S \otimes_R S)_e$  for some diagonal idempotent  $e \in S \otimes_R S$  that  $e \ker(\mu) = 0$ , i.e.  $S \otimes_R S \cong S \times S'$ .  $\lrcorner$

*Proof:* If it is  $G$ -unramified, the kernel  $I$  satisfies  $I/I^2 = 0$ , and  $I$  is f.g. (by  $x_i \otimes 1 - 1 \otimes x_i$ ) so we can use (5.1.7.7).  $\square$

Conversely, the existence of the diagonal idempotent  $e$  implies that  $I = I^2$ .  $\square$

### Loca-Unramified Ring Maps

**Def. (5.4.6.11) [Local-Unramified Ring Maps].** Let  $f : (A, \mathfrak{m}_A, \kappa_A) \rightarrow (B, \mathfrak{m}_B, \kappa_B) \in \mathcal{CRing}$  be a local ring map between Noetherian local rings, then  $f$  is called a **local-unramified ring map** if

- $\mathfrak{m}_A B = \mathfrak{m}_B$ ,
- $\kappa_B/\kappa_A$  is finite separable.
- $B$  is essentially of f.t. over  $A$  (5.1.1.21).  $\lrcorner$

**Prop. (5.4.6.12) [Local-Unramified and Unramified].** Let  $f : (A, \mathfrak{m}_A, \kappa_A) \rightarrow (B, \mathfrak{m}_B, \kappa_B) \in \mathcal{CRing}$  be a local ring map between Noetherian local rings. If  $f$  is unramified, then  $f$  is local-unramified (5.4.6.11), and the converse is true if  $f$  is of f.t..  $\lrcorner$

*Proof:* This follows from (5.4.6.7).  $\square$

**Prop. (5.4.6.13) [Properties of Local-Unramified Ring Maps].** Let  $f : (A, \kappa_A) \rightarrow (B, \kappa_B) \in \mathcal{CRing}$  be a local ring map between Noetherian local rings.

- If  $f$  is local-unramified (5.4.6.11), then  $\hat{B}$  is a finite  $\hat{A}$ -module.
- If  $f$  is local-unramified and  $\kappa_A = \kappa_B$ , then  $\hat{B}$  is a quotient of  $\hat{A}$ .
- If  $f$  is local-unramified and  $\kappa_A = \kappa_A^{\text{sep}}$ , then  $\hat{B}$  is a quotient of  $\hat{A}$ .  $\lrcorner$

*Proof:* 1: This follows from (5.2.3.15).

2: This follows from item1 and Nakayama's lemma.

3: This follows from item2.  $\square$

**Prop. (5.4.6.14) [Local-Unramified Map between Noetherian Complete Local Rings].** If  $f : (A, \kappa_A) \rightarrow (B, \kappa_B) \in \mathcal{CRing}$  is a local ring map between Noetherian local rings essentially of f.t. (5.1.1.21), then  $f$  is local-unramified iff  $\hat{f} : \hat{A} \rightarrow \hat{B}$  is local-unramified iff  $\hat{f}$  is unramified.  $\lrcorner$

*Proof:*  $f$  is local-unramified iff  $\hat{f}$  is local-unramified, because  $\mathfrak{m}_{\hat{A}} = \mathfrak{m}_A \hat{A}$  and  $B \rightarrow \hat{B}$  is faithfully flat (5.4.1.32).

If  $\hat{f}$  is local-unramified, then  $\hat{A} \rightarrow \hat{B}$  is finite by (5.4.6.13), thus  $\hat{f}$  is unramified by (5.4.6.12).

If  $\hat{f}$  is unramified, then  $\hat{f}$  is local-unramified, by (5.4.6.12).  $\square$

## 7 Étale Maps

### Formally Étale

**Def. (5.4.7.1) [Étale Ring Maps].** A ring map  $R \rightarrow S$  is called **formally étale ring map** iff it is formally smooth(5.4.5.1) and formally unramified(5.4.6.1).  $\lrcorner$

**Prop. (5.4.7.2).** Colimits of formally étale rings over  $R$  is formally étale. (The liftings are compatible because of uniqueness).  $\lrcorner$

**Prop. (5.4.7.3).** For any multiplicative set  $S \subset R$ ,  $R \rightarrow S^{-1}R$  is formally étale.  $\lrcorner$

*Proof:* It suffice to prove that if  $\varphi(s)$  is invertible modulo  $I$ , then  $\varphi(s)$  is invertible, but this is true because  $I$  is nilpotent.  $\square$

### Étale Map

**Def. (5.4.7.4) [Étale Ring Maps].**  $R \rightarrow S \in \mathcal{CRing}$  is called **étale ring map** if it is of f.p. and the naive cotangent complex is exact, i.e.  $I/I^2 \cong \Omega_{P/R} \otimes_P S$ .

In particular, étale is equivalent to smooth & formally unramified( $\Omega_{R/S} = 0$ ).  $\lrcorner$

**Cor. (5.4.7.5) [Properties of Étale Ring Maps].**

1. Étalemap is stable under base change and composition.
2. Étalemap is local on the source and target. In particular,  $R \rightarrow R_f$  is étale.
3. If  $R \rightarrow S$  is of f.p. and  $R \rightarrow R'$  is flat. Then the set of primes in  $S' = S \otimes_R R'$  that has a nbhd that is étale over  $R'$  is the inverse image of set of primes in  $S$  that has a nbhd that is étale over  $R$ . (The same as(5.4.5.18)).
4. Étale map are smooth, hence flat(5.4.5.14).
5. Any étale map is a base change of an étale map over a ring  $R_0$  f.g. over  $\mathbb{Z}$ . (Cf.[Sta]00U2).  $\lrcorner$

**Cor. (5.4.7.6) [Étale Localness of Differential].** For ring morphisms  $A \rightarrow R \rightarrow S$ , if  $R \rightarrow S$  is étale, then  $\Omega_{S/A} = \Omega_{R/A} \otimes_R S$ .  $\lrcorner$

*Proof:* This follows from(5.4.7.4) and(5.4.5.5).  $\square$

**Prop. (5.4.7.7) [Jacobson Criterion for Étaleness].** Étale ring maps are standard smooth(5.4.5.7).

Conversely,  $S = R[X_1, \dots, X_n]/(f_1, \dots, f_n) \in \mathcal{CRing}/R$  is étale iff  $\text{Jac} \left( \frac{f_1, \dots, f_n}{X_1, \dots, X_n} \right)$  is invertible in  $S$ .  $\lrcorner$

*Proof:*  $I/I^2 \cong \Omega_{P/R} \otimes_P S$ , so  $I/I^2$  is free, so by(5.3.7.12), there is a presentation of  $S$  that  $f_1, \dots, f_c$  freely generate  $I/I^2$ , then obviously  $c = n$  and  $J(\frac{f_1, \dots, f_c}{X_1, \dots, X_n})$  is invertible in  $S$ , i.e.  $S$  is standard smooth.  $\square$

**Cor. (5.4.7.8) [Example of Étale Maps].** For any  $n \in \mathbb{Z}_+$ ,

$$S = R[X_1, \dots, X_n, X_{n+1}]/\left(f_1, \dots, f_n, X_{n+1} \det\left(\frac{f_1, \dots, f_n}{X_1, \dots, X_n}\right) - 1\right) \in \mathcal{CRing}/R$$

is étale over  $R$ .  $\lrcorner$

**Prop. (5.4.7.9).** If  $R \rightarrow S$  is étale at a nbhd of a prime  $q$  of  $S$  over  $p$ , then  $pS_q = qS_q$ , and  $k(q)/k(p)$  is finite separable.  $\lrcorner$

*Proof:* We can replace  $S$  by  $S_q$  so  $S_q/R$  is étale. Then  $S \otimes k(p)/k(p)$  is étale, that is  $S_p/pS_p$  is a finite product of finite separable fields, so  $S_q/pS_q = (S_p/pS_p)_q =$  some separable closed field.  $\square$

**Lemma (5.4.7.10).** If  $R \rightarrow S \in \mathcal{CRing}$  is étale and  $q$  is a prime of  $S$  over  $p$ , then  $S/R$  is étale in a nbhd of  $q$  if

- $R \rightarrow S$  is of f.p.
- $R_p \rightarrow S_q$  is flat.
- $pS_q = qS_q$ .
- $k(q)/k(p)$  is a finite separable field extension.

 $\lrcorner$ 

*Proof:* Cf. [Sta]00U6.  $\square$

**Prop. (5.4.7.11) [Étale=Flat & G-Unramified].**  $R \rightarrow S \in \mathcal{CRing}$  is étale (5.4.7.4) iff it is flat and G-unramified (5.4.6.4) (i.e. of f.p. and  $\Omega_{S/R} = 0$ ).  $\lrcorner$

*Proof:* One direction is by definition, and the converse is by (5.4.7.10) and (5.4.6.7).  $\square$

**Prop. (5.4.7.12).** A ring map of f.p. is formally étale iff it is étale.  $\lrcorner$

*Proof:* In the f.p. case, formally smooth is equivalent to smooth (5.4.5.12).  $\square$

**Prop. (5.4.7.13).** If  $S, S' \in \mathcal{CRing}/R$  are étale, then any  $S \rightarrow S' \in \mathcal{CRing}/R$  is étale.  $\lrcorner$

*Proof:*  $S \rightarrow S'$  is of f.p. by (5.3.7.11), the rest Cf. [Sta]00U7.  $\square$

**Prop. (5.4.7.14) [Étale Algebra seen explicitly as Finite Projective Modules].** Étale algebras are finite projective, by (5.3.1.7). And we can see this clearly as follows: There is an diagonal idempotent as it is unramified (5.4.6.10), If  $e = \sum a_i \otimes b_i$ , then we can realize  $S$  as a direct command of  $R^n$  through maps

$$S \xrightarrow{\alpha} R^n \xrightarrow{\beta} S$$

where  $\alpha(f) = (\text{tr}_{S/R}(fa_i))$ , and  $\beta((g_i)) = \sum g_i b_i$ .  $\lrcorner$

*Proof:* We check that  $\beta \circ \alpha = \text{id}$ : Notice first that  $\text{tr}_{i_2}(e) = \text{tr}_{S/S}(1) = 1$ , following from the decomposition above, so  $\sum \text{tr}_{S/R}(a_i) b_i = 1$ , thus shows that  $\beta \alpha(1) = 1$ .

Now for general  $f$ , using the formula  $(f \otimes 1)e = (1 \otimes f)e$ , we get  $\sum \text{tr}(fa_i) b_i = \sum \text{tr}(a_i) b_i f = f$ .  $\square$

**Prop. (5.4.7.15).** For  $I \in \text{Ideal}(R)$ , any étale ring map  $R/I \rightarrow \bar{S} \in \mathcal{CRing}$  comes from an étale ring map  $R \rightarrow S$ .  $\lrcorner$

*Proof:* Use (5.4.7.7), an étale map is of the form  $\bar{S} = R/I[X_1, \dots, X_n]/(\bar{f}_1, \dots, \bar{f}_n)$  that  $\delta = J(\frac{f_1, \dots, f_n}{X_1, \dots, X_n})$  is invertible in  $S$ , then we take  $S = R[X_1, \dots, X_n, X_{n+1}]/(f_1, \dots, f_n, X_{n+1}\delta - 1)$ , then it is étale by (5.4.7.8) and maps to  $\bar{S}$ .  $\square$

**Prop. (5.4.7.16) [Étale Ring Maps of Dedekind-Domains].** If  $A \rightarrow B \in \mathcal{CRing}$  is étale and  $A$  is a Dedekind-domain (5.2.7.1), then  $B$  is a finite product of Dedekind-domains.  $\lrcorner$

*Proof:* Cf. [Sta]0AP2.  $\square$

### Standard Étale Ring Maps

**Def. (5.4.7.17) [Standard Étale Ring Maps].** a **standard étale ring map** is a ring map  $R \rightarrow R' = R[X]_{gf}/(f) \in \mathcal{CRing}$  where  $f$  is monic and  $g \in R[X]$ . A standard étale ring map is étale by Jacobian criterion(5.4.7.7).

Standard étale is stable under base change and principal localization, but not stable under composition.  $\lrcorner$

**Prop. (5.4.7.18) [Étale and Standard Étale].** A ring map is étale iff it is locally standard étale on the source.  $\lrcorner$

*Proof:* Cf.[Sta]00UE. ?  $\square$

**Prop. (5.4.7.19).** Giving any ring  $R$  and a prime  $p$ , if there is a finite separable extension  $L/k(p)$ , then there is a standard étale map  $R \rightarrow R'$  that for some  $q'$ ,  $k(q') \cong L$  over  $k$ .  $\lrcorner$

*Proof:*  $L = k(p)[\alpha]$  by primitive element theorem, so the minimal polynomial of  $\alpha$  is separable, and if we change  $\alpha$  to  $c\alpha$  for some  $c \in k(p)$ , we can assume  $f$  can be lifted to a  $f \in R[X]$ . Now  $f'(\alpha)$  is invertible in  $L$ , so there is a map from  $R[X]_{f'}/(f)$  to  $L$ , whose kernel gives the desired prime  $q$ .  $\square$

### Local-Étale Ring Maps

**Def. (5.4.7.20) [Local-Étale Ring Maps].** A **local-étale ring map** is a local homomorphism  $f : A \rightarrow B$  between Noetherian local rings that is flat and local-unramified(5.4.6.11).  $\lrcorner$

**Prop. (5.4.7.21) [Local-Étale and Étale].** If  $f : A \rightarrow B$  is a local homomorphism between Noetherian local rings of f.t., then  $f$  is étale iff  $f$  is local-étale(5.4.7.20).  $\lrcorner$

*Proof:* This follows from(5.4.6.12).  $\square$

**Prop. (5.4.7.22) [Local-Étale Map between Noetherian Complete Local Rings].** If  $f : (A, \kappa_A) \rightarrow (B, \kappa_B) \in \mathcal{CRing}$  is a local ring map between Noetherian local rings essentially of f.t.(5.1.1.21), then  $f$  is local-étale(5.4.7.20) iff  $\hat{f} : \hat{A} \rightarrow \hat{B}$  is local-étale iff  $\hat{f}$  is étale. And in this case,  $\hat{B} = \hat{A}^{[\kappa_B : \kappa_A]}$  as  $\hat{A}$ -modules.  $\lrcorner$

*Proof:* By(5.4.6.14) and(5.4.7.11), it suffices to show that  $f$  is flat iff  $\hat{f}$  is flat. But this follows from(5.4.1.33).

For the last assertion, notice that  $\hat{B}$  is finite flat over  $\hat{A}$ (5.4.6.13), thus finite free.  $\square$

### Étale over Fields

**Prop. (5.4.7.23) [Étale Algebra over Fields].**  $R \in \mathcal{CRing}/k$  is étale iff it is a finite product of finite separable extensions of  $k$ .  $\lrcorner$

*Proof:* If  $k'/k$  is finite separable, then  $k' = k(\alpha)$  for some  $\alpha$  by primitive element theorem, thus  $k' = k[X]/(f)$  that  $f'$  is invertible in  $k'$ , thus it is étale by(5.4.7.7).

Conversely, Cf.[Sta]00U3.  $\square$

**Cor. (5.4.7.24) [Étale over Perfect Fields].** Let  $k$  be a perfect field. If  $R$  is a  $k$ -algebra that is a finite as a  $k$ -module, then it is étale over  $k$  iff it is reduced.  $\lrcorner$

*Proof:* If it is étale, then it is reduced, by (5.4.7.23). Conversely, if it is finite and reduced, then it is a reduced Artinian ring (5.1.3.4), so a product of fields over  $k$ , so étale as  $k$  is perfect.  $\square$

**Prop. (5.4.7.25) [Étale and Trace Form].** Let  $A$  be a f.d.  $k$ -algebra, then  $A$  is étale over  $k$  iff the trace form:  $A \times A \rightarrow k : (a, b) \mapsto \text{tr}_{A/k}(ab)$  is non-degenerate.  $\perp$

*Proof:* If it is étale, then the trace form is non-degenerate by (5.4.7.23) and (3.2.6.32). Conversely, if the trace form is non-degenerate, then  $A$  is reduced, because if  $a \in A$  is nilpotent, then  $ab$  are nilpotent for any  $b \in A$ , and  $\text{tr}_{A/k}(ab) = 0$ . Then as a reduced Artinian ring,  $A$  is isomorphic to a product of fields, then by (3.2.6.32), all the fields are separable over  $k$ .  $\square$

**Prop. (5.4.7.26) [Étale and Unramified over Fields].** For  $k \in \text{Field}$ , a f.g.  $k$ -algebra is étale iff it is (G-)unramified over  $k$ , by (5.4.5.23).  $\perp$

**Prop. (5.4.7.27) [Largest Étale Subalgebras].** For  $k \in \text{Field}$  and  $A \in \mathcal{C}\text{Ring}^{\text{ft}}/k$  there exists a maximal étale  $k$ -subalgebra of  $A$ . Also this subalgebra commutes with arbitrary field base change.  $\perp$

*Proof:* If  $R$  is an étale subalgebra of  $A$ , then  $R_{\bar{k}}$  is étale over  $\bar{k}$ , thus isomorphic to  $(\bar{k})^n$  for some  $n$ . Now  $n$  is smaller than the number of connected components of  $\text{Spec } A_{\bar{k}}$ , which is finite. So it suffices to show the composite of two étale subalgebras of  $A$  is étale. For this, notice  $RR'$  is a quotient of  $R \otimes_k R'$ , which is a finite product of finite separable fields over  $k$ , thus its quotient is also a finite product of finite separable fields over  $k$ , which is étale.  $\square$

### Étale Descents

**Prop. (5.4.7.28) [Dimension is Étale-Local].** If  $A \rightarrow B \in \mathcal{C}\text{Ring}$  is étale and  $\mathfrak{q} \in \text{Spec}(B)$  with  $\mathfrak{q} \cap A = \mathfrak{p}$ , then  $\dim(A_{\mathfrak{p}}) = \dim(B_{\mathfrak{q}})$ .  $\perp$

*Proof:* Cf. [Sta]07QP. ?  $\square$

**Prop. (5.4.7.29) [Regular is Étale-Local].** If  $A \rightarrow B \in \mathcal{C}\text{Ring}$  is étale and  $\mathfrak{q} \in \text{Spec}(B)$  with  $\mathfrak{q} \cap A = \mathfrak{p}$ , then  $A_{\mathfrak{p}}$  is regular iff  $B_{\mathfrak{q}}$  is regular.  $\perp$

*Proof:* Cf. [Sta]0AH0.  $\square$

## 8 Local Algebras

References are [Local Algebra, Serre].

**Prop. (5.4.8.1).** If  $A$  is a Noetherian local integral domain with residue field  $k$  and quotient field  $K$ , if  $M$  is a f.g.  $A$ -module that  $\dim_k M \otimes_A k = \dim_K M \otimes_A K = r$ , then  $M$  is free of rank  $r$ .

In other words, if the rank of  $M$  at the generic point and closed pt of  $B$  are the same, then  $M$  is free.  $\perp$

*Proof:* First  $M$  is generated by  $r$  elements by Nakayama and the kernel  $R$  of  $A^r \rightarrow M$  vanishes when tensoring  $K$ , thus vanish because it is torsion-free.  $\square$

**Prop. (5.4.8.2).** Let  $A \rightarrow B$  be a local ring map of local rings that

- $B$  is finite as an  $A$ -module.
- $\mathfrak{m}_B$  is a f.g. ideal.

- $A/\mathfrak{m}_A \cong B/\mathfrak{m}_B$ .
- $\mathfrak{m}_A/\mathfrak{m}_A^2 \cong \mathfrak{m}_B/\mathfrak{m}_B^2$ .

Then  $A \rightarrow B$  is surjective. ┘

*Proof:* By Nakayama, to show it is surjective, it suffices to show  $A/\mathfrak{m}_A \rightarrow B/\mathfrak{m}_A B$  is surjective, then it suffices to show  $\mathfrak{m}_A \otimes_A B \rightarrow \mathfrak{m}_B$  is surjective. For this, use Nakayama again on  $B$  to reduce to the fact  $\mathfrak{m}_A \otimes_A B/\mathfrak{m}_B \rightarrow \mathfrak{m}_B \otimes B/\mathfrak{m}_B$  is surjective, which is satisfied because this is just  $\mathfrak{m}_A/\mathfrak{m}_A^2 \rightarrow \mathfrak{m}_B/\mathfrak{m}_B^2$ . □



## 5.5 $p$ -adic Commutative Algebras

**Notation (5.5.0.1).**

- Let  $p \in \text{Prime}$ .

┘

### 1 $\mathbb{F}_p$ -Algebras

**Def. (5.5.1.1) [Relative Frobenius Maps].** Let  $S \rightarrow R$  be a ring map, then the **relative Frobenius**  $\varphi_{R/S}$  is the map  $R \otimes_{S, \text{Frob}} S \rightarrow R$  induced by universal property. ┘

**Def. (5.5.1.2) [Perfect Rings].**  $R \in \mathcal{CRing}/\mathbb{F}_p$  is called a **perfect ring** iff the Frobenius map  $\varphi : P \rightarrow P$  is an isomorphism. It is called **semi-perfect** if  $\varphi$  is surjective. This is compatible with the definition (3.2.6.19) when  $R$  is a field. ┘

**Prop. (5.5.1.3) [Perfection and Coperfection].** If  $R$  is of char  $p$ , we define  $R_{\text{perf}} = \varinjlim_{\varphi} R$  and  $R^{\text{perf}} = \varprojlim_{\varphi} R$ .

The  $(\cdot)_{\text{perf}}$  and  $(\cdot)^{\text{perf}}$  are respectively the left and right adjoint to the forgetful functor from the category of perfect rings to the category of rings of characteristic  $p$ .

In particular, the category of perfect rings admits limits and colimits, and it equals the limits and colimits in the category of rings. ┘

*Proof:* Firstly, both  $R_{\text{perf}}$  and  $R^{\text{perf}}$  are perfect: for  $R_{\text{perf}}$ , every element in  $R_{\text{perf}}$  is represented by an element  $a_n \in R_n$ , and this element is equivalent to  $a_n^p \in R_{n+1}$ , so its  $p$ -th root is  $a_n \in R_{n+1}$ . For  $R^{\text{perf}}$ , an element  $(\dots, x_n, \dots, x_0)$  has  $p$ -th root  $(\dots, x_{n+1}, \dots, x_1)$ .

Second it is easily checked to be a functor because  $\varphi$  is natural. The universal property is easy.

□

**Prop. (5.5.1.4) [Perfection Kills Nilextensions].** If  $f : R \rightarrow S \in \mathcal{CRing}/\mathbb{F}_p$  is surjective with nilpotent kernel, then  $R_{\text{perf}} \rightarrow S_{\text{perf}}$  and  $R^{\text{perf}} \rightarrow S^{\text{perf}}$  are both isomorphisms. ┘

*Proof:*  $-\text{perf}$  map is clearly surjective, and it is injective because if  $a$  maps to 0, then  $\text{Frob}^k(a) \in \ker f$  for some  $k$ , so it is nilpotent, so  $\text{Frob}^{k+n}(a) = 0$ .

$-\text{perf}$  is clearly injective, and it is surjective because: suppose  $\ker f^n = 0$ , then for a  $(s_n) \in S$ , let  $t_m$  be the inverse image of  $s_{mn}$ , for each  $m$ , and let  $x = (x_n), x_{mn-k} = \text{Frob}^k t_m$ , then  $(x) \in R^{\text{perf}}$  and  $x$  maps to  $s$ . ┘

**Def. (5.5.1.5) [Perfectly Finitely Presented].** A map of perfect  $\mathbb{F}_p$ -algebras  $B \rightarrow A$  are called **perfectly finitely presented** if  $A = (A_0)_{\text{perf}}$  for some f.p.  $B$ -algebra  $A_0$ . ┘

**Prop. (5.5.1.6) [Aberbach-Hochster].** Let  $R$  be a perfect  $\mathbb{F}_p$ -algebra,  $f_1, \dots, f_r \in R$ , and consider the ideal  $I = \sqrt{(f_1, \dots, f_r)} \in R$ . Then  $R/I$  has flat dimension  $\leq r$  as an  $R$ -module. ┘

*Proof:* We only prove for  $r = 1$ .

We show  $I = \varinjlim_{f^{1/p^n-1/p^{n+1}}} R$ . The map is given by  $(a_n) \mapsto f^{1/p^n} a_n$ . This is injective because if  $f^{1/p^n} a = 0$ , then  $f^{1/p^{n+1}} a = 0$  by perfectness, thus  $a$  is killed by the transition map  $f^{1/p^n-1/p^{n+1}}$ . □

**Prop. (5.5.1.7) [Perfect Algebras are Tor Independent].** For any two perfect  $A$ -algebras  $B, C$  where  $A$  is a perfect  $\mathbb{F}_p$ -algebra,  $\text{Tor}_i^A(B, C) = 0$  for  $i > 0$ . ┘

*Proof:*  $A \rightarrow B$  can be written as a composition of a perfection of a free  $A$ -algebra and a quotient. The perfection of free algebra is flat, thus we can assume  $B = A/I$ . By a filtered colimit argument again, we can assume  $I = (f_1^{\frac{1}{p^\infty}}, \dots, f_r^{\frac{1}{p^\infty}})$  is perfectly f.p. By induction, we can assume  $r = 1$ . Now the lemma (5.5.1.6) applied to  $R = C$  shows that  $IC = \varinjlim_{f^{1/p^n} - 1/p^{n+1}} C = I \otimes_A^L C$ , so  $B \otimes_A^L C = C/IC$  is discrete.  $\square$

**Prop. (5.5.1.8).** If  $R$  is a perfect  $\mathbb{F}_p$ -algebra,  $I$  is a radical ideal, and  $J = R[I] \subset R$ , then  $J$  and  $I + J$  are both radical, and the square

$$\begin{array}{ccc} R & \longrightarrow & R/I \\ \downarrow & & \downarrow \\ R/J & \longrightarrow & R/I + J \end{array}$$

is both a fiber pullback and pushout square of commutative rings.  $\square$

*Proof:*  $J$  is clearly a radical, and notice that  $I + J$  is the kernel of the map  $R \rightarrow R/I \otimes_R R/J = R/I + J$  (5.1.1.25), and the target is a colimit of perfect rings thus perfect, by (5.5.1.3). Thus  $I + J$  is also perfect, thus a radical ideal.

By (5.1.1.26), to show the map is a pullback square, it suffices to show that  $I \cap J = 0$ . If  $x \in I \cap J$ , then  $x^2 = 0$ , thus  $x^p = 0$ , so  $x = 0$ .  $\square$

### Tilting

**Def. (5.5.1.9) [Tilting].** For any  $R \in \mathcal{CRing}$ , the  $(p$ -adic)**tilting** of  $R$  is defined to be  $R^b = (R/(p))^{\text{perf}}$ , endowed with the profinite topology.  $\square$

**Prop. (5.5.1.10).** If  $R$  is a f.g. algebra over an alg.closed field  $k$  of char  $p$ , then  $R^b \cong k^{\pi_0(\text{Spec } R)}$ .  $\square$

*Proof:* It suffice to prove for  $\text{Spec } R$  connected and reduced, because by (5.5.1.4). We first prove the case  $\text{Spec } R$  is irreducible, i.e.  $R$  is integral:

In this case, choose a closed point  $x$ , then there is a map  $R \rightarrow \widehat{R}_x$ , where  $\widehat{R}_x$  is the  $\mathfrak{m}_x$ -adic completion. By Krull (5.2.2.15) and the fact  $R$  is integral, this map is injective, so it suffices to show that  $0 = (\widehat{R}_x)^{\text{perf}} = \varinjlim (R_x/\mathfrak{m}_x^n)^{\text{perf}}$ . But  $(-)^{\text{perf}}$  is a right adjoint so commutes with colimits and  $\widehat{R}_x = \varinjlim R_x/\mathfrak{m}_x^n$ . But  $(R_x/\mathfrak{m}_x^n)^{\text{perf}} = (R_x/\mathfrak{m}_x)^{\text{perf}} = k^{\text{perf}} = k$ , by (5.5.1.4) again.

If  $R$  is not irreducible, ?  $\square$

**Prop. (5.5.1.11) [Examples of Tilting].**

- $\mathbb{F}_p[t]_{\text{perf}} = \mathbb{F}_p[t^{\frac{1}{p^\infty}}]$ ,  $\mathbb{F}_p[t]^b = \mathbb{F}_p[t]^{\text{perf}} = \mathbb{F}_p$ .
- $(\mathbb{Z}_p)^b = \mathbb{F}_p$ .
- If  $R$  is a perfect ring of char  $p$  and  $f \in R$  is a non-zerodivisor, then  $(R/f)^{\text{perf}}$  is the  $f$ -adic completion of  $R$ . In particular,  $(\mathbb{F}_p[t^{\frac{1}{p^\infty}}]/(t))^{\text{perf}} \cong \widehat{\mathbb{F}_p[t^{\frac{1}{p^\infty}}]}$ .
- $(\widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]})^{\text{perf}} \cong \widehat{\mathbb{F}_p[t^{\frac{1}{p^\infty}}]} \cong \widehat{\mathbb{F}_p[t]_{\text{perf}}} \cong (\mathbb{F}_p[t]_{\text{perf}}/(t))^{\text{perf}}$ .

$\square$

*Proof:* The first two are trivial, for the third, notice  $\widehat{R}_f = \varprojlim_n R/f^n = \varprojlim_n R/f^{p^n}$ , and there are

$$\begin{array}{ccc} R/f & \xrightarrow{\varphi} & R/f \\ \downarrow \varphi^{k+1} & & \downarrow \varphi^k \\ R/f^{p^{k+1}} & \xrightarrow{i} & R/f^{p^k} \end{array}, \text{ so } \varprojlim_n R/f^{p^n} \cong (R/f)^{\text{perf}}.$$

For the fourth, only the first equivalence needs proving, the others are consequences of the first three items. Then notice

$$(\widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]})^b \cong \varprojlim (\widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]}/p^k)^{\text{perf}} \cong (\widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]}/p)^{\text{perf}} \cong (\mathbb{F}_p[t^{\frac{1}{p^\infty}}]/t)^{\text{perf}} \cong \widehat{\mathbb{F}_p[t^{\frac{1}{p^\infty}}]}$$

The last isomorphism by item3.  $\square$

**Prop. (5.5.1.12).** If  $R \in \mathcal{C}\text{Ring}$  is a  $p$ -adically complete,  $\pi \in R^\times, p \in (\pi)$ , then the map  $R \rightarrow R/p$  induces an homeomorphism of monoids:

$$\varprojlim_{x \rightarrow x^p} R \cong \varprojlim_{\varphi} R/\pi \stackrel{(5.5.1.4)}{=} \varprojlim_{\varphi} R/p = R^b$$

┘

*Proof:* Injectivity: if  $(a_n), (b_n) \in \varprojlim_{x \rightarrow x^p} R$  satisfies  $a_n \equiv b_n \pmod{\pi}$  for all  $n$ , then applying power lifting(2.6.1.10),  $a_n \equiv b_n \pmod{\pi^{n+k}}$  for all  $k$ , so  $a_n = b_n$ .

Surjectivity: for  $(\bar{a}_n) \in R^b$ , choose arbitrary lifting  $a_n$ , then  $a_{n+k+1}^p \equiv a_{n+k} \pmod{\pi}$  for all  $n+k$ , so  $k \mapsto a_{n+k}^{p^k}$  is a Cauchy sequence by power lifting(2.6.1.10) again, thus converging to some point  $b_n$ . then it's easily checked that  $b_{n+1}^p = (\lim a_{n+1+k}^{p^k})^p = \lim a_{n+1+k}^{p^{k+1}} = b_n$ . so  $(b_n)$  maps to  $(\bar{a}_n)$ .

For the topology: it is clearly continuous, and for the reverse, if  $(a_i), (b_i)$  satisfies that  $a_i \equiv b_i \pmod{\pi}$  for  $i < k$ , then the image in  $\varprojlim_{x \rightarrow x^p} R$  satisfies  $x_i \equiv y_i \pmod{p^{k-i}}$  for  $i < k$ , thus it is open.  $\square$

**Cor. (5.5.1.13) [Sharp Map].** From this proposition, we get a multiplicative **sharp map**:

$$\sharp : R^b \rightarrow R : (\bar{a}_n) \mapsto \lim_{k \rightarrow \infty} a_k^{p^k},$$

and its image is just the elements that has a compatible system of  $p^k$ -th roots  $x^{\frac{1}{p^k}}$ . These elements are also called **perfect**.  $\square$

**Cor. (5.5.1.14) [Addition in  $R^b$ ].** From the isomorphism(5.5.1.12) above, we can read what the addition looks like in the presentation  $\varprojlim_{\varphi} R$ : if  $(f_n), (g_n)$  are two elements, then their addition is given by  $(h_n)$ , where  $h_n = \lim_k (f_{n+k} + g_{n+k})^{p^k}$ .  $\square$

**Cor. (5.5.1.15) [Fontaine's Map].** By(8.1.4.4), the natural map  $R^b \rightarrow R/p$  induces a map  $\theta_R : W(R^b) \rightarrow R$  of rings, called the **Fontaine's map**, which writes as  $\sum [a_i]p^i \mapsto \sum a_i^\sharp p^i$ . And we denote  $A_{\text{inf}}(R) = W(R^b)$  the **infinitesimal Fontaine's ring** of  $R$ .  $\square$

**Prop. (5.5.1.16).** If  $R$  is  $p$ -complete, the Fontaine's functor  $\theta_R$  is surjective iff  $R/p$  is semiperfect.  $\square$

*Proof:* As  $R$  is  $p$ -complete,  $\theta$  is surjective iff it is surjective modulo  $p$ . Because its reduction modulo  $p$  is  $R^b \rightarrow R/p$  is surjective as  $\varphi : R/p \rightarrow R/p$  does.  $\square$

**Prop. (5.5.1.17) [Tilting as a Valuation Ring].** If  $R$  is a domain or a valuation ring, then the same is true for  $R^b$ . In the valuation case, the valuation of  $R^b$  can in fact be chosen to be  $|\cdot| \circ \sharp$ , so in particular, the rank of  $R^b$  is no more than the rank of  $R$ .  $\lrcorner$

*Proof:* Use the isomorphism  $\varprojlim_{x \rightarrow x^p} R \cong \varprojlim_{\varphi} R/p = R^b$  (5.5.1.12).

For the domain case, if  $(a_n)(b_n) = 0$ , then  $a_nb_n = 0$ , so  $a_0 = 0$  or  $b_0 = 0$ , so  $(a_n) = 0$  or  $(b_n) = 0$ . Similarly, if  $R$  is a valuation ring, then  $R^b$  is firstly a domain, and it suffices to prove that for any  $(a_n), (b_n) \in R^b$ , the quotient of one by another is in  $R^b$ , by (12.2.2.3). For this, because  $R$  is valuation ring, we may assume  $a_0/b_0 \in R$ , so  $a_n/b_n$  is also in  $R$ , because their power do, and  $R$  is normal (12.2.2.6), thus  $(a_n)/(b_n) \in R^b$ .

For the valuation given, notice in the above proof,  $|(a_n)| \leq |(b_n)|$  iff  $|a_0| \leq |b_0|$ , so the valuation are equivalent to  $|\cdot| \circ \sharp$  by (12.2.3.14), so it can be chosen to be so.  $\square$

**Prop. (5.5.1.18) [Tilting and Completion].** If  $R = A/I$  with  $A/p$  perfect, then  $R^b$  identifies with the  $I$ -adic completion of  $A/p$ .  $\lrcorner$

*Proof:*  $R^b = \varprojlim_{\varphi} R/p = \varprojlim_{\varphi} A/(p, I)$ . But there are commutative diagrams

$$\begin{array}{ccc} A/(p, I) & \xrightarrow{\varphi} & A/(p, I) \\ \downarrow \varphi^{n+1} & & \downarrow \varphi^n \\ A/(p, I^{n+1}) & \longrightarrow & A/(p, I^n) \end{array}$$

where the vertical arrows are isomorphisms because  $A/p$  is perfect. So the conclusion follows.  $\square$

## 2 Witt-Theory

References are [Wit36, Rab14] and [Michiel Hazewinkel, Formal Groups and Applications], [ $p$ -divisible groups, lecture, Zink].

### Witt-Vectors

**Prop. (5.5.2.1) [ $\Lambda(A)$ ].** For  $A \in \mathcal{CRing}$ , define  $\Lambda(A) \triangleq 1 + TA[[T]]$ . Then there exists  $X_n, Y_n \in \mathbb{Z}[T_1, \dots, T_n]$  independent of  $A$  for each  $n \in \mathbb{Z}_+$ , s.t. every element  $f = 1 + \sum_{n \in \mathbb{Z}_+} x_n T^n \in \Lambda(A)$  can be written in the form

$$f = \prod_{n \in \mathbb{Z}_+} (1 - y_n T^n)$$

s.t.  $y_n = Y_n(x_1, \dots, x_n)$  and  $x_n = X_n(y_1, \dots, y_n)$  for each  $n \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* By functoriality and universality, it suffices to prove for  $A = \mathbb{Z}[\{X_n\}_{n \in \mathbb{Z}_+}]$ . In this case, we can find  $y_n$  one by one, which are clearly polynomials only depending on the first  $n$  coordinates. The existence of  $X_n$  is clear from the equation  $1 + \sum_{n \in \mathbb{Z}_+} x_n T^n = \prod_{n \in \mathbb{Z}_+} (1 - y_n T^n)$ .  $\square$

**Def. (5.5.2.2) [Divisor Stable Subsets].** A non-empty subset  $P \subset \mathbb{Z}_+$  is called **divisor-stable** if it is stable under taking divisors.

If  $P = \mathbb{Z}_+$ , denote  $\Delta_P = \widehat{\Delta}$ ,  
 if  $P = \{1, p, p^2, \dots\}$ , denote  $\Delta_P = \Delta_p$ ,  
 if  $P = \{1, p, \dots, p^k\}$ , denote  $\Delta_P = \Delta_{p,k}$ .  $\lrcorner$

**Def. (5.5.2.3) [Witt-Polynomials, Witt[[Wit36](#)]].** For  $n \in \mathbb{Z}_+$ , the  $n$ -th **Witt-polynomial** is defined to be

$$\mathbb{W}_n = \sum_{d|n} dX_d^{n/d} \in \mathbb{Z}[\{X_d : d|n\}].$$

For example,

$$\mathbb{W}_{p^k} = X_1^{p^k} + pX_p^{p^{k-1}} + \dots + p^n X_{p^n}.$$

┘

**Def. (5.5.2.4) [Witt-Vectors, Witt[[Wit36](#)]].** For a divisor-stable set  $P \subset \mathbb{Z}_+$  ([5.5.2.2](#)), define  $\Delta_P = \mathbb{Z}[\{X_n | n \in P\}]$  and consider the functor

$$W_P : \mathcal{CRing} \rightarrow \mathbf{Set} : A \mapsto W_P(A) = \mathrm{Hom}(\Delta_P, A) \cong \prod_P A \text{ ([5.5.2.6](#))},$$

then  $W_P$  is an exact functor.

If  $P = \mathbb{Z}_+$ ,  $W_P(A)$  is denoted by  $\widehat{W}(A)$ .

If  $P = \{1, p, p^2, \dots\}$ ,  $W_P(A)$  is denoted by  $W_p(A)$ , called the  **$p$ -typical Witt-vector** on  $A$ .

If  $P = \{1, p, \dots, p^k\}$ ,  $W_P(A)$  is denoted by  $W_{p,k}(A)$ , called the **truncated  $p$ -typical Witt-vector** on  $A$ . ┘

**Prop. (5.5.2.5) [ $\widehat{W}(A)$  and  $\Lambda(A)$ ].** For each  $A \in \mathcal{CRing}$ , there is a natural bijection of sets:

$$\widehat{W}(A) \cong \Lambda(A) : (x_1, \dots, x_n, \dots) \mapsto f_x = \prod_{n \in \mathbb{Z}_+} (1 - x_n T^n) \text{ ([5.5.2.1](#))},$$

And if  $\mathbb{Q} \subset A$ , there is a natural isomorphism of Abelian groups

$$D = T \frac{d \log}{dT} : \Lambda(A) \cong TA[[T]].$$

Moreover,

$$D(f_x(T)) = \sum_{n \in \mathbb{Z}_+} \mathbb{W}_n(x) t^n, \text{ where } t \in \widehat{W}(A) \text{ ([5.5.2.3](#))}. \quad \text{┘}$$

*Proof:* For each  $x \in \widehat{W}(A)$ ,

$$D(f_x(T)) = \sum_{n \in \mathbb{Z}_+} \frac{nx_n T^n}{1 - x_n T^n} = \sum_{n \in \mathbb{Z}_+} (nx_n T^n + nx_n^2 T^{2n} + nx_n^3 T^{3n} + \dots),$$

so  $D(f_x(T)) = \sum_{n \in \mathbb{Z}_+} \mathbb{W}_n(x) t^n$ . ┘

**Prop. (5.5.2.6) [Ring Structure on  $W_P(A)$ , Witt[[Wit36](#)]].** Situation as in ([5.5.2.4](#)), there is a unique commutative ring scheme structure on  $\mathbf{W}_P \stackrel{\Delta}{=} \mathrm{Spec} \Delta_P$  s.t. the map

$$W_* : \mathbb{Z}[\{X_n\}_{n \in P}] \rightarrow \Delta_P : X_n \mapsto \mathbb{W}_n \in \Delta_P \text{ ([5.5.2.3](#))}$$

induces a homomorphism of ring schemes  $\mathbf{W}_P \rightarrow \mathrm{Spec} \mathbb{Z}[X_P]$ , where the ring structure on  $\mathrm{Spec} \mathbb{Z}[X_P]$  is given by

$$\times : \mathbb{Z}[X_P] \rightarrow \mathbb{Z}[X_P] \otimes \mathbb{Z}[Y_P] : X_n \mapsto X_n \otimes Y_n, \quad + : \mathbb{Z}[X_P] \rightarrow \mathbb{Z}[X_P] \otimes \mathbb{Z}[Y_P] : X_n \mapsto X_n \otimes 1 + 1 \otimes Y_n.$$

Moreover, for any divisor-stable set  $P' \subset P$ , the natural map  $\Delta_{P'} \rightarrow \Delta_P$  induces a homomorphism of ring schemes.

Then  $W_P : \mathcal{C}\text{Ring} \rightarrow \text{Set}$  prolongs to a functor  $W_P : \mathcal{C}\text{Ring} \rightarrow \mathcal{C}\text{Ring}$ , and there is a natural ring homomorphism

$$W_* : W_P(A) \rightarrow \prod_P A : f \mapsto (f(W_n))_{n \in P} \quad (5.5.2.3).$$

For  $f \in W_P(A)$ ,  $W_*(f) = (f(W_n))_{n \in P}$  is called the **ghost-coordinate** of  $x$ .  $\lrcorner$

*Proof:* Firstly we show that there exists polynomials  $S_n, P_n \in \mathbb{Z}[\{X_n, Y_n\}_{n \in P \cap [n]_+}]$  s.t.

$$W_n(\{S_m\}_{m \in P}) = W_n(\{X_m\}_{m \in P}) + W_n(\{Y_m\}_{m \in P}) \in \mathbb{Z}[\{X_m, Y_m\}_{m \in P}],$$

$$W_n(\{P_m(X_i, Y_i)\}_{m \in P}) = W_n(\{X_m\}_{m \in P}) \cdot W_n(\{Y_m\}_{m \in P}) \in \mathbb{Z}[\{X_m, Y_m\}_{m \in P}].$$

For this, Cf. [\[Rab14\]Section3](#)?

After this, we show this is a ring scheme. This follows from the definition as all coefficients are in  $\mathbb{Z}$ .  $\square$

**Cor. (5.5.2.7).** Situation as in the proof of (5.5.2.6),

- For each  $n \in P$ , there exists  $f_n \in \mathbb{Z}[\{X_{P \cap [n-1]_+}, Y_{P \cap [n-1]_+}\}]$  s.t.

$$S_n(\{X_n, Y_n\}_{n \in P}) = X_n + Y_n + f_n(\{X_{P \cap [n-1]_+}, Y_{P \cap [n-1]_+}\}).$$

- $S_1 = X_1 + Y_1$ .

$\lrcorner$

*Proof:*  $\square$

**Prop. (5.5.2.8) [Witt-Coordinates].** Situation as in (5.5.2.4), there is another map

$$\varphi_* : W_P(A) \rightarrow \prod_P A : f \mapsto (f(X_n))_{n \in P}$$

which is an isomorphism of sets but not an isomorphism of groups.

For  $x \in W_P(A)$ ,  $\varphi_*(x)$  is called the **Witt-coordinate** of  $x$ . All coordinates of elements in  $x$  will be assumed to be in the Witt-coordinate by default.

Then the map  $W_* \circ \varphi_*^{-1}$  is given by

$$W_* \circ \varphi_*^{-1} : \prod_P A \rightarrow \prod_P A : \mathbf{x} = \varphi(x_n)_{n \in P} \mapsto {}^W(W_n(\mathbf{x}))_{n \in P}.$$

And calculating in the natural coordinates are operated by the polynomials in the proof of (5.5.2.6):

$$\varphi(x_n)_{n \in P} + \varphi(y_n)_{n \in P} = \varphi(S_n(\mathbf{x}, \mathbf{y}))_{n \in P},$$

$$\varphi(x_n)_{n \in P} \cdot \varphi(y_n)_{n \in P} = \varphi(P_n(\mathbf{x}, \mathbf{y}))_{n \in P}.$$

In particular,  $\varphi(0)$  is the zero element and  $\varphi(1, 0, \dots)$  is the unit element.  $\lrcorner$

**Cor. (5.5.2.9) [Ghost-Coordinates are Injective].** Situation as in (5.5.2.4),  $W_* : W_P(A) \rightarrow \prod_P A$  is injective if  $A$  is  $n$ -torsion-free for any  $n \in P$ , and an isomorphism iff  $n$  is invertible in  $A$  for any  $n \in P$ .

This is very useful because it can prove equations by checking on each  $W_n$ . Also when checking universal equations, it is even not necessary that  $A$  is  $n$ -torsion-free, because any ring  $A$  is a quotient of a free  $\mathbb{Z}$ -algebra.  $\lrcorner$

*Proof:* It suffices to prove for  $W_* \circ \varphi_*^{-1}$ . Using the Witt polynomials and prove inductive on  $n \in P$  s.t. if  $x = \varphi(x_n)$  is mapped to 0, then  $x_n = 0$  for any  $n \in P$ .

In case  $n$  is invertible in  $A$  for any  $n \in P$ , solve  $\varphi(x_n)$  out of  $W(x_n)$  inductively.  $\square$

**Def. (5.5.2.10) [Topology on  $W_P(A)$ ].** For a divisor-stable set  $P \subset \mathbb{Z}_+$  (5.5.2.2) and any  $A \in \mathcal{CRing}$ , the natural map

$$W_P(A) \rightarrow \varprojlim_{n \in \mathbb{Z}_+} W_{P \cap [n]_+}(A)$$

is an isomorphism of rings, so we can define the **natural topology** on  $W_P(A)$  is as the profinite topology.

Then this topology makes  $W_P(A)$  a topological ring, and it is discrete iff  $\#P < \infty$ , and  $W_P$  is a functor from  $\mathcal{CAlg}$  to the category of profinite rings.  $\lrcorner$

**Prop. (5.5.2.11).** Let  $P$  be a divisor-stable set and  $A \in \mathcal{CAlg}$ , then for  $x = (x_n), y = (y_n) \in W_P(A)$ , if  $x_n y_n = 0$  for any  $n \in P$ , then

$$x + y = \varphi(x_n + y_n) \in W_P(A).$$

$\lrcorner$

*Proof:* It suffices to prove on  $\Delta_P$ . By (5.5.2.9), it suffices to prove  $W_n((x_n + y_n)) = W_n(x) + W_n(y)$  for any  $n \in P$ . This is true because

$$W_n((x_n + y_n)) = \sum_{d|n} d(x_d + y_d)^{n/d} = \sum_{d|n} dx_d^{n/d} + \sum_{d|n} dy_d^{n/d} = W_n(x) + W_n(y).$$

$\square$

**Prop. (5.5.2.12) [Teichmüller -Lifts].** For a divisor-stable set  $P \subset \mathbb{Z}_+$  (5.5.2.2) and  $A \in \mathcal{CRing}$ , denote  $[a] = \varphi(a, 0, \dots) \in W_P(A)$  for each  $a \in A$ . Then for any  $x = \varphi(x_n) \in W_P(A)$ ,

$$[a]x = \varphi(a^n x_n)_{n \in P}.$$

In particular,  $[\cdot] : A \rightarrow W_P(A)$  is multiplicative, called the **Teichmüller-lift** function of  $A$ .  $\lrcorner$

*Proof:* It suffices to prove for  $\Delta_P$  (5.5.2.6). By (5.5.2.9) and taking a torsion-free cover of  $A$ , it suffices to prove  $W_n((a^n x_n)) = W_n([a]x)$  for any  $n \in P$ . This is because

$$W_n((a^n x_n)) = \sum_{d|n} d(a^d x_d)^{n/d} = a^n \sum_{d|n} dx_d^{n/d} = W_n([a])W_n(x) = W_n([a]x).$$

$\square$

### Frobenius and Verschiebung-Maps

**Def. (5.5.2.13) [Verschiebung-Maps].** Let  $P \subset \mathbb{Z}_+$  be a divisor-stable subset and  $n \in \mathbb{Z}_+$ . For each  $A \in \mathcal{CRing}$ , define

$$V_n : W_P(A) \rightarrow W_P(A) : \varphi(x_m) \mapsto \varphi(y_m), \text{ where } y_m = \begin{cases} 0 & , n \nmid m \\ x_{m/n} & , n|m \end{cases}.$$

Then  $V_n$  is a functorial continuous additive map, called the **Verschiebung-map**. Moreover, for each  $m \in P$ , we have

$$W_m(V_n(x)) = \begin{cases} 0 & , n \nmid m \\ nW_{m/n}(x) & , n|m \end{cases}.$$

And if  $P = \mathbb{Z}_+$ , under the bijection  $\widehat{W}(A) \cong \Lambda(A)$  (5.5.2.5)  $V_n : \Lambda(A) \rightarrow \Lambda(A)$  is given by  $V_n(f(t)) = f(t^n)$ .  $\lrcorner$

*Proof:* By compatibility with change of  $P$  (5.5.2.6), it suffices to prove for  $P = \mathbb{Z}_+$ , and then the assertion is clear and the continuity follows by compatibility with change of  $P$  and taking inverse limits.

For  $n, m \in \mathbb{Z}_+$  denote  $y = V_n(x)$ , then when  $n \nmid m$ , then

$$W_m(V_n(x)) = \sum_{d|m} dy_d^{m/d} = 0$$

because whenever  $d|m$ , we have  $n \nmid d$ . And if  $m = nm'$ , then

$$W_m(V_n(x)) = \sum_{d|m} dy_d^{m/d} = \sum_{d|m'} (nd)y_{nd}^{m'/d} = n \sum_{d|m'} dx_d^{m'/d} = nW_{m'}(x).$$

$\square$

**Def. (5.5.2.14) [Frobenius-Maps, Cartier[Car67]].** Let  $P \subset \mathbb{Z}_+$  be a divisor-stable subset and  $n \in \mathbb{Z}_+$  s.t.  $nP \subset P$ , then there is a natural transformation  $F : W_P \mapsto W_P$  of topological-ring-valued functors called the **Frobenius-map for Witt-rings** s.t. for each  $x \in W_P(A)$  and  $m \in P$  we have

$$W_m(F_n(x)) = W_{mn}(x).$$

And when  $P = \mathbb{Z}_+$ , under the bijection  $\widehat{W}(A) \cong \Lambda(A)$  (5.5.2.5)  $F_n$  is given by

$$F_n(f) = \text{Nm}_{A[[T]]/A[[T^n]]}(f).$$

$\lrcorner$

*Proof:* Cf. [Rab14]P17?  $\square$

**Prop. (5.5.2.15) [Frobenius and Verschiebung-Maps].** Let  $m, n \in \mathbb{Z}_+$  and  $P \subset \mathbb{Z}_+$  is a divisor-stable subset s.t.  $mP \subset P, nP \subset P$ . Let  $A \in \mathcal{CAlg}$ , let  $x, y \in W_P(A)$ , then

- $V_n \circ V_m = V_m \circ V_n = V_{mn}$ .
- $F_n \circ V_n = n \cdot \text{id}$ .
- $V_n(F_n(x)y) = xV_n(y)$ .
- $F_n([a]) = [a^n]$  for each  $a \in A$ .
- If  $(m, n) = 1$ , then  $V_m \circ F_n = F_n \circ V_m$ .
- $(V_n(x))^m = n^{m-1}V_n(X^m)$ .

$\lrcorner$

*Proof:* These are all formal identities of polynomials, so by functoriality and universality, it suffices to prove for  $A \in \mathcal{CRing}/\mathbb{Q}$ . in this case, it suffices to check on the ghost-coordinate (5.5.2.9), which follows easily from (5.5.2.13) and (5.5.2.14).  $\square$



**Prop. (5.5.2.16)** [ $F_p$  and  $V_p$ ]. If  $P \subset \mathbb{Z}_+$  is a divisor-stable subset and  $p \in \mathbf{Prime}$  s.t.  $pP \subset P$ , then for each  $A \in \mathcal{CRing}$ ,  $x \in W_P(A)$  and  $y = F_p(x)$ , we have

- $y_m \equiv x_m^p \pmod{pA}$  for each  $m \in P$ .
  - $F_p(x) \equiv x^p \pmod{pW_P(A)}$ .
- In particular, if  $A \in \mathcal{CRing}^p$ , then
- $F_p \circ V_p = V_p \circ F_p = p$ .
  - $F_p((x_n)_{n \in P}) = (x_n^p)_{n \in P}$ , which is an automorphism if  $A$  is perfect.
  - $p = F_p \circ V_p(1) = (0, 1, 0, \dots)$ .

⌋

*Proof:* Cf. [Rab14]P19?.

□

**Prop. (5.5.2.17)** [Cartier-Homomorphisms]. Let  $P \subset \mathbb{Z}_+$  be a divisor-stable subset and  $A \in \mathcal{CRing}$  s.t. no element of  $P$  is a zero-divisor of  $A$ . Suppose  $A$  is equipped with ring homomorphism  $\varphi_p : A \rightarrow A$  for each  $p \in P \cap \mathbf{Prime}$ , s.t.

- $\varphi_p(x) \equiv x^p \pmod{pA}$  for each  $x \in A$ .
- $\varphi_p \circ \varphi_q = \varphi_q \circ \varphi_p$  for any  $p, q \in P \cap \mathbf{Prime}$ .

For each  $n = \prod_i p_i^{e_i} \in P$ , we can define

$$\varphi_n = \varphi_{p_1}^{(e_1)} \circ \dots \circ \varphi_{p_k}^{(e_k)},$$

then there exists a unique ring homomorphism  $\delta : A \rightarrow W_P(A)$  s.t.  $W_n \circ \delta = \varphi_n$  for each  $n \in P$ , called the **Cartier-homomorphism** of  $A$ . ⌋

*Proof:* Let  $T \subset \mathbb{Z}_+$  be the multiplicative subset generated by  $P$ , and define  $A' = T^{-1}A$ , then  $W_* : W_P(A') \cong \prod_P A'$  (5.5.2.9). then each  $\varphi_n$  extends to  $A'$ , and we can define

$$\delta = W_*^{-1} \left( \prod_P \varphi_n \right) : A' \rightarrow \prod_P A' \rightarrow W_P(A'),$$

which satisfies the requirement. Then it suffices to show that  $\delta(A) \subset W_P(A)$ .

For  $x \in A$  and  $y = \delta(x)$ , we will show by induction on  $n \in P$  s.t.  $y_n \in A$ . For  $n = 1$ ,  $y_1 = w_1(y) = \sigma_1(x) = x \in A$ . Now let  $1 \leq n = \prod_i p_i^{e_i} \in P$ , take a prime factor  $p_i | n$ , and denote  $m = n/p_i$ , then by induction we have

$$\varphi_m(y) = \sum_{d|m} dy_d^{m/d} \in A$$

with  $y_d \in A$ . Then

$$\varphi_n(y) = \varphi_{p_i} \left( \sum_{d|m} dy_d^{m/d} \right) = \sum_{d|m} d\varphi_{p_i}(y_d)^{m/d}.$$

For each  $d|m$ , let  $v_p(d) = s$ , then

$$\varphi_{p_i}(y_d)^{m/d} \equiv y_d^{m/d} \pmod{p^{e_i-s}A}$$

by (2.6.1.10). So  $d\varphi_{p_i}(y_d)^{m/d} \equiv dy_d^{m/d} \pmod{p^{e_i}A}$ , and

$$\varphi_n(y) \equiv \sum_{d|m} dy_d^{m/d} \equiv \sum_{d|n, d \neq n} dy_d^{n/d} \pmod{p^{e_i}A}.$$

Then Chinese remainder theorem implies that

$$\varphi_n(y) \equiv \sum_{d|n, d \neq n} dy_d^{n/d} \pmod{nA},$$

and

$$y_n = \frac{1}{n} \left( \varphi_n(y) = \sum_{d|n, d \neq n} dy_d^{n/d} \right) \in A.$$

□

### $p$ -Typical Witt Vectors

**Notation (5.5.2.18).**

- We abbreviate  $W_p(A)$  to  $W(A)$  and  $W_{p,k}(A)$  to  $W_k(A)$ . And we write

$$\varphi(x_0, x_1, \dots) \triangleq \varphi(x_1, x_p, x_{p^2}, \dots).$$

And we write

$$\mathbf{w}_0 \triangleq \mathbf{W}_1 = X_0, \quad \mathbf{w}_1 \triangleq \mathbf{W}_p = X_0^p + pX_1, \quad \mathbf{w}_2 \triangleq \mathbf{W}_{p^2} = X_0^{p^2} + pX_1 + p^2X_0, \dots$$

┘

**Def. (5.5.2.19) [Frobenius and Verschiebung-Maps].** For each  $A \in \mathcal{CRing}$ , we denote the Frobenius-map  $F_p : W(A) \rightarrow W(A)$  (5.5.2.14) by  $\varphi_A$ , and the Verschiebung-map  $V_p : W(A) \rightarrow W(A)$  by  $\text{Ver}_A$ . Then by (5.5.2.15) and (5.5.2.16), for each  $x, y \in W(A)$ , we have

- $\varphi \circ \text{Ver} = p$ .
- $\text{Ver}(\varphi(x)y) = x \text{Ver}(y)$ .
- $\varphi([a]) = [a^p]$ .
- $\text{Ver}(x)^p = p^{p-1} \text{Ver}(x^p)$ .
- $\varphi((x_0, x_1, \dots, x_n, \dots)) = (y_0, y_1, \dots, y_n, \dots)$  where  $y_i \equiv x_i^p \pmod{pA}$  for each  $i \in \mathbb{N}$ .
- $\varphi(x) \equiv x^p \pmod{pW(A)}$ .

And if  $A \in \mathcal{CRing}^p$ , then

- $\varphi \circ \text{Ver} = \text{Ver} \circ \varphi = p$ .
- $\varphi((x_0, x_1, \dots, x_n, \dots)) = (x_0^p, x_1^p, \dots, x_n^p, \dots)$ , which is an automorphism if  $A$  is perfect.
- $p = \varphi(\text{Ver}(1)) = (0, 1, 0, \dots)$ .

┘

**Prop. (5.5.2.20) [Cartier-Homomorphism].** If  $A \in \mathcal{CRing}^p$  is equipped with a lift of Frobenius  $\varphi$ , then there exists a unique ring homomorphism  $\delta : A \rightarrow W(A)$  s.t.  $\mathbf{w}_n(\delta(a)) = \varphi^{(n)}(a)$  for each  $a \in A$  and  $n \in \mathbb{N}$ .

┘

*Proof:* This follows from (5.5.2.17).

□

**Prop. (5.5.2.21)** [Structure of  $\Delta_p$ ]. The ring structure on  $\text{Spec } \Delta_p$  (5.5.2.6) is given by

$$+ : \Delta_p \rightarrow \Delta_p \otimes \Delta_p : X_{p^k} \mapsto S_k(X_P, Y_P), \quad \times : \Delta_p \rightarrow \Delta_p \otimes \Delta_p : X_{p^k} \mapsto P_k(X_P, Y_P)$$

where

$$S_1 = X_1 + Y_1, \quad S_2 = X_p + Y_p + \sum_{i=1}^{p-1} \frac{1}{p} \binom{p}{i} X_1^i Y_1^{p-i},$$

$$P_1 = X_1 Y_1, \quad P_2 = X_1^p Y_p + X_p Y_1^p + p X_p Y_p.$$

┘

*Proof:*

□

**Cor. (5.5.2.22)** [ $W_2$ ]. Let  $W_2 = W_P$  where  $P = \{1, p\}$ , then  $W_2(A)$  is the set  $A \times A$  (as the natural coordinates) with the addition and multiplication given by

$$(x_0, x_1) + (y_0, y_1) = (x_0 + y_0, x_1 + y_1 + \frac{x_0^p + y_0^p - (x_0 + y_0)^p}{p}), \quad (x_0, x_1)(y_0, y_1) = (x_0 y_0, x_0^p y_1 + y_0^p x_1 + p x_1 y_1)$$

There are two natural morphism of rings  $\mathbf{w}_1, \mathbf{w}_2 : W_2(A) \rightarrow A$ :

$$\mathbf{w}_1((x_0, x_1)) = x_0, \quad \mathbf{w}_2((x_0, x_1)) = x_0^p + p x_1.$$

┘

**Prop. (5.5.2.23)**. For  $k \in \mathbb{Z}_+$  and  $A \in \mathcal{CAlg}$ , the kernel of  $\mathbf{w}_1 : W_{p,k}(A) \rightarrow A$  is a nilpotent ideal.

┘

*Proof:* Cf. [Basic Algebra 2, Jacobson] P508. ?

□

**Cor. (5.5.2.24)**. An element  $x \in W_{p,k}(A)$  is a unit iff  $W_1(x) \in A$  is a unit.

┘

**Example (5.5.2.25)** [ $p$ -Typical Witt Vectors].

- $W_p(\mathbb{F}_q)$  is the unramified extension of  $\mathbb{Z}_p$  of degree  $\log_p q$ .
- $W_p(\overline{\mathbb{F}_p})$  is the completion of the maximal unramified extension of  $W_p(\mathbb{F}_p)$ .

┘

*Proof:*

□

**Def. (5.5.2.26)** [Witt-Vectors over Valued Rings]. If  $\kappa \in \text{Perf}^p$  has a complete valuation  $v$ , then we can endow  $W(\kappa)$  with a finer topology: we let  $w_k(x) = \inf_{i \leq k} v(x_i)$ , where  $x = \sum p^i f(x_i)$ . Now  $w_k(x + y) \geq \inf(w_k(x), w_k(y))$  by (5.5.3.10). The **weak topology** of  $W(\kappa)$  is defined by the semi-valuations  $w_k$ .

┘

**Prop. (5.5.2.27)**. If  $a, b \in \mathcal{O}_R + p^{n+1}W(R)$ , then

$$p^n v(a_n - b_n) \geq w_n(a - b) \geq \inf_{k \leq n} p^{-k} v(a_{n-k} - b_{n-k}).$$

So we see that a sequence is Cauchy in  $W(R)$  if each coordinate is Cauchy in  $R$ , so  $W(R)$  is complete in the weak topology.

┘

*Proof:* Firstly the last proposition follows from the first because we can always multiply by a  $f(\alpha)$  to make the first  $n$  coordinate in  $\mathcal{O}_R$ .

The first is nearly an immediate consequence of (5.5.3.10).  $\square$

**Prop. (5.5.2.28).**  $\mathcal{O}_{\mathcal{E}} = W(K^{\frac{1}{p^\infty}})$  is a complete ring with maximal ideal  $p\mathcal{O}_{\mathcal{E}}$ . And  $\mathcal{O}_{\mathcal{E}}[\frac{1}{p}] = \mathcal{E}$  is complete ring of character  $p$ . And the same construction of  $\overline{K^{\frac{1}{p^\infty}}}$  yields the completion of maximal unramified extension of  $\mathcal{O}_{\mathcal{E}}$ , and the Galois group is the same as  $G_K$ .  $\lrcorner$

**Prop. (5.5.2.29) [van Der Kallen].** If  $A \rightarrow B \in \mathcal{C}\text{Ring}$  is étale, then  $W_r(A) \rightarrow W_r(B)$  is also étale. Moreover, if  $A \rightarrow A' \in \mathcal{C}\text{Ring}$  with  $B' = B \otimes_A A'$ , then the natural map

$$W_r(A') \otimes_{W_r(A)} W_r(B) \rightarrow W_r(B')$$

is also an isomorphism.  $\lrcorner$

*Proof:* Cf. [Integral  $p$ -adic Hodge, BMS].  $\square$

### Cohen-Rings

**Def. (5.5.2.30) [Cohen-Rings].** For any  $k \in \text{Field}^p$ , there exists a unique absolutely-unramified (i.e. maximal ideal generated by  $p$ ) DVR of type  $(0, p)$  and residue field  $k$ , called the **Cohen-ring** for  $k$ , denoted by  $\text{Coh}(k)$ .  $\lrcorner$

*Proof:* Cf. [Fontaine-OuYang]P185. ?  $\square$

**Cor. (5.5.2.31).**  $W(k) \subset \text{Coh}(k) \subset W(k^{\text{perf}})$ .  $\lrcorner$

### 3 $p$ -Local Rings

**Def. (5.5.3.1) [ $p$ -Local Rings].** A  **$p$ -local commutative ring** is a commutative ring  $A \in \mathcal{C}\text{Ring}$  s.t.  $p \in \text{rad } A$ .  $\lrcorner$

**Prop. (5.5.3.2).** Let  $A$  be a  $p$ -adically complete ring, then  $A$  is  $p$ -local, by (5.2.3.16).  $\lrcorner$

**Prop. (5.5.3.3).** If  $A$  is  $p$ -adically complete and has bounded  $p$ -torsions, the  $p$ -completion of a smooth  $A$ -algebra is  $p$ -completely smooth.  $\lrcorner$

*Proof:* This follows from (5.9.7.4) and (5.9.6.16).  $\square$

**Def. (5.5.3.4) [ $p$ -Normal Rings].** A  $p$ -torsion-free ring  $R$  is called  **$p$ -normal** if  $R$  is  $p$ -root closed in  $R[\frac{1}{p}]$ .  $\lrcorner$

### Complete Discrete Valuation Rings

Structure of complete  $p$ -local DVRs will be studied in this subsubsection.

References are [Ser79], [Integral  $p$ -adic Hodge, BMS].

**Def. (5.5.3.5) [Strict  $p$ -Ring].** A  **$p$ -ring** is a topological ring  $A \in \text{Top}\mathcal{C}\text{Ring}$  which is complete Hausdorff in the topology defined by the decreasing chain of ideals  $\mathfrak{a}_1 \supset \mathfrak{a}_2 \supset \cdots$  s.t.  $\mathfrak{a}_m \mathfrak{a}_n \subset \mathfrak{a}_{m+n}$  and  $\kappa \triangleq A/\mathfrak{a}_1 \in \text{Perf}^p$ .

It is called a **strict  $p$ -ring** if moreover  $\mathfrak{a}_n = (p^n)$  and  $p \in A^\times$ .  $\lrcorner$

**Prop. (5.5.3.6)[Teichmüller-Lifts].** For a  $p$ -ring  $A \in \mathcal{CRing}$ (5.5.3.5), there exists a unique section map  $[-] : k \rightarrow A$  that is multiplicative, called the **Teichmüller-lift** function.

If  $\text{char } A = p$ , then the Teichmüller-lift is also additive (but not in general). And an element is in the image of the Teichmüller-lift function iff it is a  $p^n$ -th power for any  $n \in \mathbb{Z}_+$ .

If  $A$  is a strict  $p$ -ring, then each element of  $A$  is of the form

$$x = \sum_{n \in \mathbb{N}} [x_n] p^n.$$

┘

*Proof:* For any  $\lambda \in k$ , the  $\lambda^{p^{-n}}$  is unique in  $k$ , and if we consider  $U_n$  the set of all  $x^{p^n}$  where  $x$  is a lift of  $\lambda^{p^{-n}}$ , then  $U_n$  is a descending set. Moreover, the diameter converges to 0, because  $a \equiv b \pmod{\mathfrak{a}_1}$  implies  $a^{p^n} \equiv b^{p^n} \pmod{\mathfrak{a}_{n+1}}$  as  $p \in \mathfrak{a}_1$ . So it converses to a unique point  $f(\lambda)$  in  $A$ . And we see that any other  $f'$  maps  $\lambda$  to a  $p^n$ -th root hence in  $U_n$  for any  $n$ , hence it map be equal to  $f(\lambda)$ . The rest is easy.  $\square$

**Cor. (5.5.3.7) [Equal Characteristic case].** For  $(A, \kappa) \in \text{CDVR}$ , if  $\text{char } \kappa = \text{char } A$  and  $\kappa = \kappa^{\text{perf}}$ , then  $A \cong \kappa[[T]]$ .  $\square$

**Def. (5.5.3.8) [(0, p)-type case].** For  $(A, \kappa, K) \in \text{CDVR}$  of  $(0, p)$ -type, we have  $e = v(p) \geq 1$ , called the **absolute ramification index** of  $A$ . It is called **absolutely unramified** iff  $e = 1$ .  $\square$

**Prop. (5.5.3.9) [Universal Strict  $p$ -Ring].** The **universal strict  $p$ -ring** is the ring  $S_p = \mathbb{Z}_p \left[ \left\{ X_i^{p^{-n}} \right\}_{i \in \mathbb{N}, n \in \mathbb{Z}_+} \right] \in \mathcal{CRing}$ . Its residue ring is  $\mathbb{F}_p \left[ \left\{ X_i^{p^{-n}} \right\}_{i \in \mathbb{N}, n \in \mathbb{Z}_+} \right]$  which is perfect.  $X_i$  are all Teichmüller-lifts, as they have all  $p^n$ -roots(5.5.3.6).

Now we consider the  $*$   $\in \{+, -, \times\}$  in  $\hat{S}$ . then there are polynomials  $Q_i^* \in \mathbb{F}_p \left[ \left\{ X_i^{p^{-n}} \right\}_{i \in \mathbb{N}, n \in \mathbb{Z}_+} \right]$  s.t.

$$x * y = \sum_{n \in \mathbb{N}} [Q_n^*(x, y)] p^n.$$

In fact, we have

$$Q_n^+ = S_{p^n}(X_0^{p^{-n}}, Y_0^{p^{-n}}, X_1^{p^{-n+1}}, Y_1^{p^{-n+1}}, \dots, X_{n-1}^{p^{-1}}, Y_{n-1}^{p^{-1}}, X_n, Y_n) \quad (5.5.2.6),$$

$$Q_n^\times = P_{p^n}(X_0^{p^{-n}}, Y_0^{p^{-n}}, X_1^{p^{-n+1}}, Y_1^{p^{-n+1}}, \dots, X_{n-1}^{p^{-1}}, Y_{n-1}^{p^{-1}}, X_n, Y_n)$$

┘

*Proof:* ? Cf. [Rab14]Thm1.5.  $\square$

**Prop. (5.5.3.10)[Universal Law of  $p$ -Rings].** For any strict  $p$ -ring  $(A, \kappa)$ (5.5.3.5) and  $*$   $\in \{+, -, \times\}$ , the calculation of  $*$  in  $A$  is dominated by  $Q_i^*$  defined in(5.5.3.9), i.e.

$$\left( \sum_{i \in \mathbb{N}} [\alpha_i] p^i \right) * \left( \sum_{i \in \mathbb{N}} [\beta_i] p^i \right) = \sum_{i \in \mathbb{N}} [Q_i^*(\alpha_0, \alpha_1, \dots, \beta_0, \beta_1, \dots)] p^i.$$

In particular, if  $A$  is a strict  $p$ -ring, then the structure of  $A$  is determined by these polynomials  $\{Q_i^*\}$ .  $\square$

*Proof:* There is a surjective local ring map  $\theta : \mathbb{Z}_p \left[ \left\{ X_i^{p^{-n}}, Y_i^{p^{-n}} \right\}_{i \in \mathbb{N}, n \in \mathbb{Z}_+} \right] \rightarrow A$  induced by  $X_i \mapsto [\alpha_i], Y_i \mapsto [\beta_i]$  as Teichmüller-lifts all have  $p^{-n}$ -th roots (5.5.3.6). So the assertion follows from (5.5.3.9) and the fact  $\theta([a]) = [\bar{\theta}(a)]$  for each  $a \in \kappa$ , which follows from the construction of Teichmüller-lifts (5.5.3.6).  $\square$

**Prop. (5.5.3.11) [Universal Properties of Strict  $p$ -Rings].** For two  $p$ -ring  $(A, \kappa), (A', \kappa')$  that  $A$  is strict (5.5.3.5), any  $\varphi : \kappa \rightarrow \kappa' \in \mathcal{C}\text{Ring}$  lifts to a unique ring homomorphism  $g : A \rightarrow A'$ . In particular, two strict  $p$ -ring with the same residue ring is canonically isomorphic.  $\square$

*Proof:* We have already seen that ring homomorphism commutes with Teichmüller lift. Now we define

$$g(a) = \sum g([\alpha_i])p^i = \sum [\varphi(\alpha_i)]p^i,$$

and this is the unique choice. It is a ring homomorphism by universal law of (5.5.3.10).  $\square$

**Lemma (5.5.3.12) [Strict  $p$ -Rings and Witt-Vectors].** If  $(A, \kappa)$  is strict  $p$ -ring (5.5.3.5) s.t.  $\kappa = \kappa^{\text{perf}}$ , then the map

$$f : W_p(k) \rightarrow A : {}^\varphi(x_1, x_p, x_{p^2}, \dots) \mapsto \sum_{n \in \mathbb{N}} [x_{p^n}^{1/p^n}] p^n \text{ (5.5.2.4) (5.5.3.6)}$$

is an isomorphism of topological rings.  $\square$

*Proof:* This follows by combining (5.5.3.6) (5.5.2.8) and (5.5.3.9).  $\square$

**Prop. (5.5.3.13) [ $W_p$  For Perfect Rings].** For  $\kappa \in \text{Perf}^p$ , there exists uniquely a strict  $p$ -ring  $W(\kappa)$  (5.5.3.5) with residue ring  $\kappa$ , which is just the ring of Witt vectors  $W_p(\kappa)$ .

Then  $W_p$  is a faithful functor from  $\text{Perf}^p$  to the category of  $p$ -rings with perfect residue fields that is left adjoint to the functor mapping a  $p$ -ring with perfect residue fields to its residue field, by (5.5.3.11).  $\square$

*Proof:* For a canonical ring  $\mathbb{F}_p[X_\alpha^{p^{-n}}], \mathbb{Z}_p[X_\alpha^{p^{-n}}]$  is a strict  $p$ -ring. Now arbitrary perfect  $p$ -ring is a quotient of  $\mathbb{F}_p[X_\alpha^{p^{-n}}]$ , so we can construct its strict  $p$ -ring  $W(k)$  as the quotient of  $\widehat{\mathbb{Z}}[X_\alpha^{p^{-n}}]$ . Uniqueness follows from (5.5.3.11).

Notice it is nothing mysterious, it is just the set of all formal sum  $\sum f(x_i)p^i$  under the operation defined in (5.5.3.9). See also (5.5.3.12).  $\square$

**Cor. (5.5.3.14).** For each  $k \in \mathbb{Z}_+, W_{p,k}(\mathbb{F}_p) \cong \mathbb{Z}/(p^k)$ . And  $W_p(\mathbb{F}_p) \cong \mathbb{Z}_p$ .  $\square$

**Prop. (5.5.3.15) [Witt-Rings and  $p$ -Rings].** Suppose  $K \in \text{Perf}^p$  and  $(A, \kappa)$  is a  $p$ -ring (5.5.3.5), then any  $f : K \rightarrow \kappa$  lifts uniquely to a commutative diagram

$$\begin{array}{ccc} W_p(K) & \xrightarrow{F} & R \\ \downarrow \mathbf{w}_{p^0} & & \downarrow \\ K & \xrightarrow{f} & \kappa \end{array} .$$

$\square$

*Proof:* Define

$$F : W_p(K) \rightarrow R : x = \sum_{n \in \mathbb{N}} [x_{p^n}^{1/p^n}] p^n \mapsto \sum_{n \in \mathbb{N}} [f(x_{p^n}^{1/p^n})] p^n,$$

then it is clearly a continuous bijection. So it suffices to show that  $F$  is a ring homomorphism. This follows by combining (5.5.3.6) (5.5.2.8) and (5.5.3.9).

The uniqueness follows from that fact that each  $[a] \in W_p(K)$  is a Teichmüller-lift, and any such  $F$  must commute with taking Teichmüller-lifts, by the proof of (5.5.3.6).  $\square$

#### 4 $\delta$ -Rings

**Def. (5.5.4.1) [ $\delta$ -Ring].** For  $R \in \mathcal{CRing}$ , a  $(p)$ - $\delta$ -ring structure on  $R$  characterizes the deficit in lifting the Frobenius action on  $R/p$ . i.e.  $\varphi(x) = x^p + p\delta(x)$ . A  $\delta$ -ring is a pair  $(R, \delta)$  where  $R \in \mathcal{CRing}$  and  $\delta : R \rightarrow R \in \mathbf{Set}$  is a map s.t.  $\delta(0) = \delta(1) = 0$ , and satisfies:

$$\delta(xy) = x^p\delta(y) + y^p\delta(x) + p\delta(x)\delta(y), \quad \delta(x+y) = \delta(x) + \delta(y) - \frac{(x+y)^p - x^p - y^p}{p}.$$

$\delta$ -rings naturally form a category, denoted by  $\mathcal{CRing}^\delta$ . And in case  $A$  is  $p$ -torsionfree, a  $\delta$ -structure on  $A$  is the same as a lifting of the Frobenius on  $A/p$ .  $\lrcorner$

**Def. (5.5.4.2) [ $\delta$ -Pairs].** The category of  $\delta$ -pairs consists of pairs  $(A, I)$  where  $A$  is a  $\delta$ -ring and  $I$  is an ideal of  $A$  that morphisms  $\varphi : (A, I) \rightarrow (B, J)$  are  $\delta$ -ring maps  $A \rightarrow B$  that  $\varphi(I) \subset J$ .  $\lrcorner$

**Prop. (5.5.4.3) [ $\delta$ -Rings and  $W_2$ ].** A  $\delta$ -ring structure on  $A$  is the same as a section of the map  $W_1 : W_2(A) \rightarrow A$ , and a morphism of  $\delta$ -rings is a commutative diagram of sections.  $\lrcorner$

*Proof:* By the description of  $W_2$  in (5.5.2.22), this is clear, the morphism is given by  $A \rightarrow W_2(A) : x \mapsto (x, \delta(x))$ .  $\square$

**Lemma (5.5.4.4) [Initial  $\delta$ -Algebra].** We usually work with  $\delta$ -algebras over  $\mathbb{Z}_{(p)}$ . Then there is an initial object in the category of  $\delta$ -rings, given by  $(\mathbb{Z}_{(p)}, \delta)$  with  $\delta(x) = \frac{x-x^p}{p}$ .  $\lrcorner$

**Prop. (5.5.4.5).** For  $(A, \delta) \in \mathcal{CRing}^\delta$ ,  $\varphi : x \mapsto x^p + p\delta(x)$  commutes with  $\delta$ .  $\lrcorner$

*Proof:* We need to check  $\delta(x^p + p\delta(x)) = \delta(x)^p + p\delta(\delta(x))$ . This is hard to check, but we can check  $\varphi(\frac{\varphi(x)-x^p}{p}) = \frac{\varphi(\varphi(x))- \varphi(x)^p}{p}$ , so the conclusion is true when  $A$  is  $p$ -torsion-free. But by (5.5.4.10), every  $\delta$ -ring is a quotient of a free thus  $p$ -torsion-free ring, thus the equation is also true for arbitrary  $A$ .  $\square$

**Def. (5.5.4.6) [Commutative Coalgebra  $\mathbb{Z}[\delta]$ ].** Denote  $\mathbb{Z}[\delta] = \mathbb{Z}[e, \delta_1, \dots, \delta_n, \dots]$ . Using the formulas in (5.5.4.1), we can write  $\delta^{\circ n}(xy)$  and  $\delta^{\circ n}(x+y)$  as functions of  $\delta^{\circ i}(x), \delta^{\circ i}(y)$  for  $0 \leq i \leq n$ , i.e.

$$\delta^n(xy) = M_n(x, \delta(x), \delta^{\circ 2}(x), \dots, \delta^{\circ n}(x), y, \delta(y), \dots, \delta^{\circ n}(y))$$

$$\delta^n(x+y) = S_n(x, \delta(x), \delta^{\circ 2}(x), \dots, \delta^{\circ n}(x), y, \delta(y), \dots, \delta^{\circ n}(y))$$

Then we can change  $\delta^{\circ n}$  to  $\delta_n$  to define a commutative ring scheme structure on  $\mathrm{Spec} \mathbb{Z}[\delta]$ :

$$\times : \mathbb{Z}[\delta] \rightarrow \mathbb{Z}[\delta] \otimes \mathbb{Z}[\delta] : \delta_n \mapsto M_n, \quad + : \mathbb{Z}[\delta] \rightarrow \mathbb{Z}[\delta] \otimes \mathbb{Z}[\delta] : \delta_n \mapsto S_n.$$

it is easily prove by induction that if we denote  $\mathbb{Z}[\varphi] = \mathbb{Z}[e, \varphi, \varphi_2, \dots, \varphi_n, \dots]$ , and  $\text{Spec } \mathbb{Z}[\varphi]$  the commutative ring scheme with the ring structure given by

$$\times : \mathbb{Z}[\varphi] \rightarrow \mathbb{Z}[\varphi] \otimes \mathbb{Z}[\varphi] : \varphi_n \mapsto \varphi_n \otimes \varphi_n, \quad + : \mathbb{Z}[\varphi] \rightarrow \mathbb{Z}[\varphi] \otimes \mathbb{Z}[\varphi] : \varphi_n \mapsto \varphi_n \otimes 1 + 1 \otimes \varphi_n,$$

then the map

$$\mathbb{Z}[\varphi] \rightarrow \mathbb{Z}[\delta] : \varphi_n \mapsto \text{expansion of } \varphi^{\circ n} \in \mathbb{Z}[e, \delta_1, \dots, \delta_n]$$

induces a homomorphism of ring schemes  $\text{Spec } \mathbb{Z}[\delta] \rightarrow \text{Spec } \mathbb{Z}[\varphi]$ .

There is a natural functor:

$$[\delta] : \mathbb{Z}[\delta] \rightarrow \mathbb{Z}[\delta] : \delta_i \mapsto \delta_{i+1}$$

┘

**Lemma (5.5.4.7) [ $\delta$ -Component].** Let  $\varphi = e^p + p\delta$  be a polynomial in  $e, \delta$ , then there are polynomials  $\Theta_n \in \mathbb{Z}[e, \delta_1, \delta_2, \dots, \delta_n]$  s.t.

$$\varphi^{\circ n} = \Theta_0^{p^n} + p\Theta_1^{p^{n-1}} + \dots + p^n \Theta_n = W_{p^n}(\Theta_0, \dots, \Theta_n), \forall n$$

In particular,  $\Theta_0 = e, \Theta_1 = \delta$ .

Moreover,  $\mathbb{Z}[\Theta_0, \Theta_1, \dots, \Theta_n] = \mathbb{Z}[e, \delta_1, \delta_2, \dots, \delta_n]$  for any  $n$ .

┘

*Proof:* Use equation  $\varphi \circ \varphi^n = \varphi^n \circ \varphi$  and module  $p^n \mathbb{Z}[\Theta_0, \Theta_1, \dots, \Theta_n]$ . ?

□

**Prop. (5.5.4.8) [Witt Vectors as  $\delta$ -Rings].** By (5.5.4.7), there is an isomorphism of algebras

$$\Delta_p \cong \mathbb{Z}[\delta] : X_{p^n} \mapsto \Theta_n,$$

but it is also an isomorphism of ring schemes because of the uniqueness property of (5.5.2.6).

Then for any  $A \in \mathcal{CAlg}$ , every element of  $W_p(A)$  has a  $\delta$ -coordinates:

$$f \in W_p(A) = \text{Hom}(\Delta_p, A) \mapsto (f(e), f(\delta_1), f(\delta_2), \dots)_\delta.$$

Let  $W_p(A)$  be defined as in (5.5.2.4), then  $[\delta] : \mathbb{Z}[\delta] \rightarrow \mathbb{Z}[\delta]$  induces a homomorphism  $\delta : W_p(A) \rightarrow W_p(A)$  that can be checked to be a  $\delta$ -functor from our definition of coalgebra structure on  $\mathbb{Z}[\delta]$ . Thus  $W_p$  is a functor  $\mathcal{CAlg} \rightarrow \mathcal{CAlg}^\delta$ .

┘

**Prop. (5.5.4.9) [Witt Vectors as an Right Adjoints].**  $W_p$  is right adjoint to the forgetful functor  $\mathcal{CAlg}^\delta \rightarrow \mathcal{CAlg}$ .

┘

*Proof:* Given a ring homomorphism  $f : B \rightarrow A$ , let

$$f^\delta : B \rightarrow W_p(A) : x \mapsto (f(x), f(\delta(x)), f(\delta^{\circ 2}(x)), \dots)_\delta,$$

Then  $f^\delta$  is a  $\delta$ -ring homomorphism: It is a homomorphism by our definition of the ring structure on  $\text{Spec } \mathbb{Z}[\delta]$  (5.5.4.6), and it is a  $\delta$ -ring homomorphism by our definition  $\delta$ -ring structure (5.5.4.8).

And it is easy to see  $f^\delta$  is the unique  $\delta$ -ring homomorphism  $B \rightarrow W(A)$  that restricts to  $f : B \rightarrow A$ , thus

$$\text{Hom}^\delta(B, W(A)) \cong \text{Hom}(B, A).$$

□



**Prop. (5.5.4.10) [Free  $\delta$ -Rings].** The ring  $\mathbb{Z}\{x_i\}$  is a ring on the free generators  $\{x, \delta(x_i), \delta^2(x_i), \dots\}$  and the Frobenius morphism defined by asserting  $\varphi(\delta^i(x)) = \delta^i(x)^p + p\delta^{i+1}(x)$ .

Generally we can define the **free  $\delta$ -ring** generated by  $\{x_i\}$  over a  $\delta$ -ring  $A$  as the tensor  $A \otimes \mathbb{Z}\{x_i\}$ , and it satisfies the universal property.

Then the Frobenius action is f.f. ⌋

*Proof:* It is easily verified that for any  $\delta$ -ring  $R$  over  $A$  and a map of sets  $\{x_i\} \rightarrow R$ , there is a unique morphism  $\mathbb{Z}_{(p)}\{x_i\} \rightarrow R$  sending  $x$  to  $f$ , thus verifying the universal property.

For the f.f., notice it is a colimit of  $\varphi_n : \mathbb{Z}[x, \delta(x), \dots, \delta^n(x)] \rightarrow \mathbb{Z}[x, \delta(x), \dots, \delta^{n+1}(x)]$ , so by (5.4.1.24), it suffices to show  $\varphi_n$  are all f.f.. We decompose this map as  $n$  maps:

$$\mathbb{Z}[x, \delta(x), \dots, \delta^n(x)] \cong \mathbb{Z}[x, \delta(x), \dots, (\delta^i(x))^p, \dots, \delta^n(x)] \subset \mathbb{Z}[x, \delta(x), \dots, \delta^n(x)]$$

which are all f.f., so it is f.f. □

**Cor. (5.5.4.11) [Frobenius is Fpqc locally Surjective].** For a  $\delta$ -ring  $A$  and an element  $x \in A$ , there is a faithfully flat morphism of  $\delta$ -rings  $A \rightarrow B$  that the image of  $x$  in  $B$  is of the form  $\varphi(y)$  for some  $y \in B$ . ⌋

**Cor. (5.5.4.12).** Set  $B$  as the pushout of the diagram  $\mathbb{Z}_{(p)}\{s\} \leftarrow \mathbb{Z}_{(p)}\{t\} \rightarrow A$ , where the arrows sends  $t$  to  $\varphi(s)$  and  $x$ .  $B$  exists by (5.5.4.13) and the underlying ring is the same as the ring pushout, thus  $A \rightarrow B$  is faithfully flat by (5.5.4.10). ⌋

**Prop. (5.5.4.13) [Limits and Colimits of  $\delta$ -Rings].**  $\mathcal{CAlg}^\delta$  admits limits and colimits, and their underlying rings are just the ring-theoretical limit and colimit, as the forgetful functor  $\mathcal{CAlg}^\delta \rightarrow \mathcal{CAlg}$  has both left and right adjoints (5.5.4.9)(5.5.4.10). ⌋

*Proof:* Use (5.5.4.3), the construction of limits is straightforward, as  $W_2$  commutes with limits. To construct colimits, notice the morphisms  $A_i \rightarrow W_2(A_i)$  induces a morphism

$$\varinjlim A_i \rightarrow \varinjlim W_2(A_i) \rightarrow W_2(\varinjlim A_i)$$

and clearly this map is a section of  $W_2(\varinjlim A_i) \rightarrow \varinjlim A_i$ , thus given a  $\delta$ -ring structure on  $\varinjlim A_i$ .

It is a colimit because any commutative diagrams  $\varphi_i : (A_i \rightarrow W_2(A_i)) \rightarrow (B \rightarrow W_2(B))$  induces a unique commutative diagram

$$\begin{array}{ccccc} \varinjlim_i A_i & \longrightarrow & \varinjlim W_2(A_i) & \xrightarrow{\varinjlim W_2(\varphi_{ij})} & W_2(\varinjlim A_i) \\ \downarrow \varinjlim \varphi_i & & \downarrow & \swarrow W_2(\varinjlim \varphi_i) & \\ B & \longrightarrow & W_2(B) & & \end{array} .$$

□

### Extension of $\delta$ -Structures

**Lemma (5.5.4.14) [Quotients].** Let  $A$  be a  $\delta$ -ring and  $I$  be an ideal, then if  $I$  is stable under  $\delta$ , then there is a natural  $\delta$ -structure on  $A/I$  compatible with  $A$ . In general, if  $J$  is an ideal of  $A$ , then there is a universal  $\delta$ - $A$ -algebra  $B = A/J$ , where  $J = \cup_{n \geq 0} \delta^n(I)$ . It is the universal  $\delta$ - $A$ -algebra that the image of  $I$  is 0. ⌋

**Lemma (5.5.4.15) [Localization].** Let  $A$  be a  $\delta$ -ring and  $S$  be a multiplicative set of  $A$  that  $\varphi(S) \in S$ , then there is a unique  $\delta$ -structure on  $S^{-1}A$ , and it satisfies the universal property.  $\lrcorner$

*Proof:* Firstly if  $A$  is  $p$ -torsionfree, in this case a  $\delta$ -structure is the same as a lifting of the Frobenius on  $A/p$ , thus the proposition is clear because  $\varphi_{S^{-1}A} : S^{-1}A \rightarrow S^{-1}A$  is uniquely determined.

Generally, we choose a free  $\delta$ -ring  $F$  and a surjection  $\alpha : F \rightarrow A$ , then  $T = \alpha^{-1}S$  is multiplicative, and  $T^{-1}F$  admits a unique  $\delta$ -structure. But now  $S^{-1}A = T^{-1}F \otimes_F A$ , so there is a  $\delta$ -structure on  $S^{-1}A$  as the colimit, so compatible with that of  $S^{-1}A$ . Then it's also the unique one (because if there is another one, the colimit properties gives a morphism of  $\delta$ -rings above  $\text{id}_{S^{-1}A}$ , which must be identity).  $\square$

**Lemma (5.5.4.16) [ $p$ -adic Localization].** If  $A$  is a  $\delta$ -ring with  $p \in \text{rad}(A)$ , then the formula  $\varphi(f) = f^p + p\delta(f)$  shows if  $f$  is a unit, then  $\varphi(f)$  is also a unit (3.6.3.2), so  $S^{-1}A = T^{-1}A$ , where  $T = \{S, \varphi(S), \varphi^2(S), \dots\}$ .

Thus for any  $\delta$ -ring  $A$  and a multiplicative set  $S$ , the  $p$ -localization (5.1.1.34)  $(S^{-1}A)_{(p)}$  is the same as the  $p$ -localization of  $T^{-1}A$ . Then (5.5.4.15) shows that  $(S^{-1}A)_{(p)}$  carries a unique  $\delta$ -structure compatible with that of  $A$ .  $\lrcorner$

**Lemma (5.5.4.17) [Completions].** For a  $\delta$ -ring  $A$  and a f.g. ideal  $I$ , the  $I$ -adic completion of  $A$  has a unique  $\delta$ -structure compatible with that of  $A$ .  $\lrcorner$

*Proof:* Let  $A \rightarrow W_2(A)$  corresponds to the  $\delta$ -structure by (5.5.4.3), then there is a natural map  $A \rightarrow W_2(A) \rightarrow W_2(\tilde{A})$ , then by the universal property of complete, it extends to a map  $\tilde{A} \rightarrow W_2(\tilde{A})$ , which is a ring map and is a section, all by universal properties.  $\square$

**Lemma (5.5.4.18) [Derived Completion].** If  $A$  is a  $\delta$ -ring and  $I \subset A$  is an ideal containing  $p$ , then the derived  $I$ -completion ring  $\hat{A}$  of  $A$  admits a unique  $\delta$ -structure extending that of  $A$ .  $\lrcorner$

*Proof:* The proof is similar as (5.5.4.17).  $\square$

**Prop. (5.5.4.19) [Étale Extension].** Let  $A$  be a  $\delta$ -ring with a f.g. ideal  $I$  containing  $p$ . Assume  $B$  is a derived  $I$ -complete and  $I$ -completely étale  $A$ -algebra, then  $B$  admits a unique  $\delta$ -structure compatible with that of  $A$ .

In particular, any  $\delta$ -structure on an algebra  $A$  passes uniquely to its derived  $I$ -completion for any ideal  $I \subset A$  containing  $p$ .  $\lrcorner$

*Proof:* By Elkik's algebraization (5.9.7.9), we can write  $B$  as derived  $I$ -completion of some étale  $A$ -algebra  $B'$ . Then  $W_2(A) \rightarrow W_2(B')$  is étale by van der Kallen's theorem (5.5.2.29). Then  $W_2(B)$  is the derived  $I$ -completion of  $W_2(B')$ , with the  $A$ -algebra structure given by  $A \rightarrow W_2(A) \rightarrow W_2(B')$ . For the rest, Cf. [Scholze, Prism, 2.18].  $\square$

### Distinguished Elements

**Def. (5.5.4.20) [Distinguished Elements].** In a  $\delta$ -ring  $A$ , an element  $d$  is called a **distinguished element** if  $\delta(d)$  is a unit in  $A$ . A distinguished element is preserved by a  $\delta$ -ring map.  $\lrcorner$

**Lemma (5.5.4.21) [Distinguished up to units].** If  $A$  is a  $\delta$ -ring,  $d$  is distinguished and  $u$  is a unit, then  $ud$  is also distinguished, if  $d, p \in \text{rad}(A)$ .  $\lrcorner$

*Proof:*  $\delta(ud) = u^p\delta(d) + d^p\delta(u) + p\delta(u)\delta(d)$  is a unit.  $\square$

**Lemma (5.5.4.22) [Irreducibility of Distinguished Elements].** Let  $A$  be a  $\delta$ -ring and  $d$  be distinguished element in  $A$ . If  $d = fh$  for some  $f, h \in A$  that  $f, p \in \text{rad}(A)$ , then  $f$  is also distinguished and  $h$  is a unit.  $\lrcorner$

*Proof:* Notice that  $\delta(d) = f^p\delta(h) + h^p\delta(f) + p\delta(f)\delta(h)$ ,  $\delta(d)$  is a unit,  $f^p\delta(h) + p\delta(f)\delta(h) \in \text{rad}(A)$ , thus  $h^p\delta(f)$  is a unit, so we are done.  $\square$

**Prop. (5.5.4.23) [Characterization of Distinguished Elements].** Fix a  $\delta$ -ring  $A$  and an element  $d$  that  $d, p \in \text{rad}(A)$ , then  $d$  is distinguished iff  $p \in (d, \varphi(d))$ . In particular, distinguished elements is stable under units.  $\lrcorner$

*Proof:* If  $d$  is distinguished, then  $\delta(d)$  is a unit, thus  $\varphi(d) = d^p + d^p + p\delta(d)$  shows immediately  $p \in (d, \varphi(d))$ . Conversely, if  $p = ad + b\varphi(d)$ , we show  $\delta(d)$  is invertible. It suffices to show it is invertible modulo  $(d, p)$  as  $d, p \in \text{rad}(A)$ , or equivalently  $(p, d, \varphi(d)) = A$ . If it is not the case, then we may take a  $(p, d, \varphi(d))$ -adic completion to assume  $p, d, \varphi \in \text{rad}(A)$ , thus the equation simplifies to  $p(1 - b\delta(d)) = cd$ . The left side is distinguished, by (5.5.4.21), and then  $d$  is also distinguished, by (5.5.4.22), so truly  $(p, d, \varphi(d)) = A$ .  $\square$

**Prop. (5.5.4.24) [Examples of Distinguished Elements].** The element  $d$  is distinguished in the following cases:

- (Crystalline cohomology) Take  $A = \mathbb{Z}_{(p)}$  and  $d = p$ , then  $\delta(p) = 1 - p^{p-1}$  is a unit.
- ( $q$ -de Rham cohomology) Take  $A = \mathbb{Z}_p[[q-1]]$  and  $d = [p]_q = \sum_{i=0}^{p-1} q^i \in A$ , with the  $\delta$ -structure determined by  $\varphi(q) = q^p$ .
- (Breuil-Kisin cohomology) Fix a discretely valued field  $K/Q_p$  with uniformizer  $\pi$ ,  $W$  the maximal unramified subring of  $\mathcal{O}_K$ . Take  $A = W[[u]]$  with  $\delta(u) = u^p$ , then any generator of the kernel of the map of  $A \rightarrow \mathcal{O}_K : u \mapsto \pi$  is distinguished?  $\lrcorner$
- ( $A_{\text{inf}}$ -cohomology) Let  $A$  be the  $(p, q-1)$ -completion of  $\mathbb{Z}_p[q^{\frac{1}{p^\infty}}]$ . Then  $A$  is  $p$ -torsion free and  $\varphi(q) = q^p$  gives a  $\delta$ -structure. Then  $d = [p]_q$  defined in item2 is also distinguished. And  $\varphi^n(d)$  is distinguished for any  $n \in \mathbb{Z}$  by (5.5.4.5).  $\lrcorner$

*Proof:* 2: It is clear  $\varphi$  is continuous and  $\delta$  stabilizes  $(q-1)$ , and  $d$  is distinguished because the image of  $\delta(d)$  in  $A/(q-1) \cong \mathbb{Z}_q$  is  $\delta(p) = 1 - p^{p-1}$  is a unit, thus it is also a unit in  $A$ .  $\square$

### Perfect $\delta$ -Ring

**Def. (5.5.4.25) [Perfect  $\delta$ -Ring].** A **perfect  $\delta$ -ring** is a  $\delta$ -ring that  $\varphi$  is an isomorphism.  $\lrcorner$

**Prop. (5.5.4.26) [Perfections].** The inclusion of the category of perfect  $\delta$ -rings to the category of  $\delta$ -rings admits left and right adjoints,  $A_{\text{perf}}$  and  $A^{\text{perf}}$  with definition similar to (5.5.1.9).  $\lrcorner$

*Proof:* We use (5.5.4.3), the map  $A \rightarrow W_2(A) \rightarrow W_2(A_{\text{perf}})$  extends uniquely to a map  $A_{\text{perf}} \rightarrow W_2(A_{\text{perf}})$  lifting the  $\delta$ -action of  $A$ . Similarly, because  $(-)^{\text{perf}}$  is a limit and  $W_2(-)$  is a right adjoint, then  $W_2(-)$  commutes with  $(-)^{\text{perf}}$ . In particular, there is a natural map  $A^{\text{perf}} \rightarrow W_2(A^{\text{perf}})$ .  $\square$

**Lemma (5.5.4.27) [Frobenius Kills  $p$ -Torsion].** If  $A$  is a  $\delta$ -ring and  $x \in A$  satisfies  $px = 0$ , then  $\varphi(x) = 0$ . In particular, if  $A$  is perfect, then  $A$  is  $p$ -torsionfree.  $\lrcorner$

*Proof:* Applying  $\delta$  to  $px = 0$ , we have  $0 = p^p\delta(x) + x^p\delta(p) + p\delta(x)\delta(p) = p^p\delta(x) + \varphi(x)\delta(p)$ . As  $\delta(p)$  is a unit, and  $p^p\delta(x) = p^{p-1}(\varphi(x) - x^p) = \varphi(p^{p-1}x) - p^{p-1}x^p = 0$ , thus  $\varphi(x) = 0$ .  $\square$

**Prop. (5.5.4.28) [Perfect  $p$ -Complete  $\delta$ -Rings].** The following categories are equivalent:

- The category of perfect  $p$ -adically complete  $\delta$ -rings.
- The category  $p$ -adically complete and  $p$ -torsionfree rings  $A$  with  $A/p$  perfect.
- The perfect  $\mathbb{F}_p$ -algebras.

In particular, every perfect  $p$ -complete  $\delta$ -ring is of the form  $W(k)$  thus has Teichmüller expansions.  $\lrcorner$

*Proof:* 2 and 3 are equivalent by Witt vector construction, by (8.1.4.3), noticing that there is a natural lifting on  $W(k)$  lifting the Frobenius of  $k/\mathbb{F}_p$  that induces a  $\delta$ -functor<sup>?</sup>. There is a forgetful functor from 1 to 2, by (5.5.4.27) (notice  $A/p$  is also perfect because if  $\varphi(x) \in (p)$ , then  $x \in \varphi^{-1}(p) = (p)$  as  $\varphi(p) = p\varphi(1) = p$ ) and it is faithful. Now there is an equivalence from  $3 \rightarrow 1 \rightarrow 2$ , thus  $1 \rightarrow 2$  is essentially surjective thus is an equivalence.  $\square$

**Prop. (5.5.4.29) [Perfect Element has Rank 1].** Fix a  $\delta$ -ring  $A$  and an element  $x \in A$ , then  $\delta(x^{p^n}) \in p^n A$  for any  $n$ . In particular, if  $A$  is  $p$ -adically separated and  $y$  is perfect in  $A$ , then  $\delta(y) = 0$ .  $\lrcorner$

*Proof:* By formal calculation, it suffices to show that  $p\delta(x^{p^n}) \in p^{n+1}A$ , which is equivalent to  $\varphi(x^{p^n}) \equiv x^{p^{n+1}} \pmod{p^{n+1}A}$ , which is true by (2.6.1.10).  $\square$

**Prop. (5.5.4.30) [Distinguished Elements in Perfect  $\delta$ -Rings].** Let  $A$  be a perfect  $p$ -complete  $\delta$ -ring (or perfect  $\mathbb{F}_p$ -algebra by (5.5.4.28)), and  $d \in A$ , then  $d$  is distinguished iff its coefficient of  $p$  in the Teichmüller expansion (5.5.4.28) is a unit.

If  $d$  is distinguished, then it is a nonzero-divisor, and  $A/d[p^\infty] = A/d[p]$ .  $\lrcorner$

*Proof:* Let  $d = \sum_{i \geq 0} [a_i]p^i$ , then

$$\delta(d) = \frac{1}{p} \left( \sum_{i \geq 0} [a_i^p]p^i - \left( \sum_{i \geq 0} [a_i]p^i \right)^p \right) \equiv [a_1^p] \pmod{pA}$$

thus it is a unit iff  $a_1$  is a unit, because  $A$  is  $p$ -complete.

Now if  $d$  is distinguished, and  $fd = 0$ . If  $f \neq 0$ , we may assume  $p \nmid f$ , because  $A$  is  $p$ -torsionfree and  $p$ -adically complete (5.5.4.28). Now

$$\varphi(f)\delta(fd) = \varphi(f)(f^p\delta(d) + \delta(f)\varphi(d)) = \varphi(f)f^p\delta(d) = 0,$$

so  $f^p\varphi(f) = 0$ , and  $f^{2p} \equiv 0 \pmod{p}$ . Hence  $f \equiv 0 \pmod{p}$ , but then  $p|f$ , contradiction.

For the last assertion, it suffices to show that  $A/d[p^2] = A/d[p]$ . If  $p^2f = dg$ , then  $\varphi(g)\delta(gd) = \varphi(g)(\delta(d)g^p + \varphi(d)\delta(g)) = \varphi(g)\delta(d)g^p + \varphi(dg)\delta(g) \in pA$ , thus  $\varphi(g)g^p \in pA$ , hence  $g^{2p} \in pA$ , and hence  $g \in pA$ , showing  $pf \in dA$ .  $\square$

## 5.6 Divided Power Algebras

**Notation (5.6.0.1).**

- Let  $p \in \text{Prime}$ .

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### 1 Basics

**Def. (5.6.1.1) [PD-Structures, Cartan/Roby].** For  $A \in \mathcal{CRing}$  and  $I \in \text{Ideal}(A)$ , a **divided power structure** or **pd-structure** on  $I$  is a collection of maps  $I_n : I \rightarrow A, n \in \mathbb{N}$  that

- $\gamma_0(x) = 1, \gamma_1(x) = x, \gamma_n(x) \in I$  for any  $n \in \mathbb{N}$ .
- $\gamma_n(x + y) = \sum \gamma_{n-i}(x)\gamma_i(y)$ .
- $\gamma_n(\lambda x) = \lambda^n \gamma_n(x), \lambda \in A, x \in I$ .
- $\gamma_m(x)\gamma_n(x) = \binom{m+n}{n} \gamma_{m+n}(x)$ .
- $\gamma_n(\gamma_m(x)) = \frac{(mn)!}{(m!)^n n!} \gamma_{mn}(x)$ .

A **divided power ring** or a **pd-structure** is a triple  $(A, I, \gamma)$  where  $A \in \mathcal{CRing}, I \in \text{Ideal}(A)$  and  $\gamma$  is a pd-structure on  $I$ . A morphism of divided power rings is a morphism of pairs  $(A, I)$  that preserves pd-structures.

For a pd-structure  $(A, I)$ , denote  $I^{[n]}$  the ideal generated by  $\prod_i \gamma_{n_i}(x_i)$  where  $x_i \in I$  and  $\sum n_i \geq n$ . And  $(A, I)$  is called a **nilpotent pd-structure** if  $I^{[n]} = 0$  for some  $n \in \mathbb{Z}_+$ . If  $(A, I)$  is a nilpotent pd-structure, then  $I$  is nilpotent, because  $n! \gamma^n(x) = x^n$ . ┘

**Remark (5.6.1.2).** A pd-structure is a simulation of the divided powers  $\gamma_n(x) = \frac{x^n}{n!}$  in case  $1/n!$  is not definable. ┘

**Example (5.6.1.3) [PD-Structures].** For  $A \in \mathcal{CRing}$ ,

- There is a unique pd-structure on  $(0)$  given by  $\gamma_n = 0$  for any  $n \in \mathbb{N}$ .
- If  $A \in \mathcal{CRing}/\mathbb{Q}$ , then for any  $I \in \text{Ideal}(A)$  there is a unique pd-structure given by  $\gamma_n(x) = x^n/n!$ , by (5.6.1.11).
- If  $m \in \mathbb{Z}_+$  s.t.  $m!$  is invertible in  $A$  and  $I \in \text{Ideal}(A)$  s.t.  $I^{m+1} = 0$ , then there is a pd-structure on  $I$  given by  $\gamma_n = \begin{cases} x^n/n! & n \in [m] \\ 0 & n \in \mathbb{Z}_{>m} \end{cases}$ .

┘

**Def. (5.6.1.4) [Exponentials and Logarithms].** If  $(A, I, \gamma)$  is a nilpotent divided power ring (5.6.1.1), then we can define the **divided power exponential** and the **divided power logarithm** maps

$$\exp : I \mapsto 1 + I : x \mapsto \sum_{n \in \mathbb{N}} \gamma_n(x).$$

$$\log : 1 + I \mapsto I : 1 + x \mapsto \sum_{n \in \mathbb{Z}_+} (-1)^{n-1} (n-1)! \gamma_n(x).$$

Then they are mutually inverse isomorphisms between the additive group  $I$  and the multiplicative group  $(1 + I)^\times$ . ┘

**Prop. (5.6.1.5)[Limits and Colimits].** The category of divided power rings has all limits and colimits, the limits commute with forget functors but the colimits don't. However, the colimit always commutes with the functor taking  $(A, I)$  to  $A/I$ . This can be seen from the universal property of colimit applied to the pd-structures that  $I = 0$ .  $\lrcorner$

*Proof:* The construction of the limit is clear. For the colimits, we use representability criterion(4.1.1.28), Cf.[Sta]07GX?.  $\square$

**Prop. (5.6.1.6).** For a divided power ring  $(A, I, \gamma)$ (5.6.1.1), we have  $n!\gamma(x) = x^n$ .  $\lrcorner$

*Proof:*  $n\gamma_n(x) = \gamma_1(x)\gamma_{n-1}(x)$ , so we can use induction.  $\square$

**Prop. (5.6.1.7).** If  $I, J \in \text{Ideal}(A)$  and  $\gamma$  a pd-structure on  $I$  and  $\delta$  a pd-structure on  $J$ , then

- $\gamma, \delta$  agree on  $IJ$ .
- If  $\gamma, \delta$  agree on  $I \cap J$ , then they extends to a pd-structure on  $I + J$ .

$\lrcorner$

*Proof:* 1: for  $x \in I, y \in J$ ,  $\gamma_n(xy) = y^n \gamma_n(x) = n! \delta_n(y) \gamma_n(x) = \delta(xy)$ .  
2: direct calculation.  $\square$

**Prop. (5.6.1.8)[ $p$ -Nilpotent and Thickening].** Let  $(A, I, \gamma)$  a pd-structure s.t.  $p$  is nilpotent in  $A/I$ , then  $I$  is locally nilpotent iff  $p$  is nilpotent in  $A$ , or equivalently  $(A, I, \gamma)$  is a pd-thickening(27.5.1.2).  $\lrcorner$

*Proof:* If  $p^N = 0 \in A$ , then for any  $x \in I$ ,  $x^{p^N} = (pN)!\gamma_{pN}(x) = 0$ . Then converse is trivial.  $\square$

### Constructing PD-Structures

**Prop. (5.6.1.9)[ $\mathbb{Z}_{(p)}$ -Algebras].** Cf.[Sta]07GN.  $\lrcorner$

**Prop. (5.6.1.10)[PD-Structure on  $\mathbb{Z}_{(p)}$ -Algebras].** For  $A \in \mathcal{CRing}/\mathbb{Z}_{(p)}$  and  $I \in \text{Ideal}(A)$ , two pd-structures  $\gamma, \gamma'$  on  $I$  are equal iff  $\gamma_p = \gamma'_p$ . Moreover, given any map  $\delta : I \rightarrow I$  s.t.

- $p!\delta(x) = x^p$ ,
- $\delta(ax) = a^p \delta(x)$  for any  $a \in A, x \in I$ ,
- $\delta(x + y) = \delta(x) + \delta(y) + \sum_{i+j=p, i>0, j>0} \frac{1}{i!j!} x^i y^j$ .

Then there exists a unique pd-structure on  $I$  that  $\gamma_p = \delta$ .  $\lrcorner$

*Proof:* Just notice that  $\gamma_n(x) = cx\gamma_{n-1}(x)$  for some  $c$  invertible in  $\mathbb{Z}_{(p)}$ , and also  $\gamma_{pm}(x) = c\gamma_m(\gamma_p(x))$  for some  $c$  invertible in  $\mathbb{Z}_{(p)}$ , thus  $\gamma$  is uniquely determined, and we can also define  $\gamma_n$  inductively in this way, and the verification of axioms is in [Sta]07GS?.  $\square$

**Prop. (5.6.1.11).** Let  $A \in \mathcal{CRing}$  be torsion-free and  $I \in \text{Ideal}(A)$ , then:

- $I$  has at most one pd-structure(5.6.1.1).
- if  $\{\gamma_n : I \rightarrow I\}_{n \in \mathbb{N}}$  are maps, then  $\gamma$  is a pd-structure iff  $n!\gamma_n(x) = x^n$ .
- $I$  has a pd-structure iff there is a set of generators  $\{x_i\}$  of  $I$  that  $x_i^n \in n!I$ .

$\lrcorner$

*Proof:* 1 is clear from(5.6.1.6).

2: because  $A \subset A \otimes_{\mathbb{Z}} \mathbb{Q}$ , we can verify in  $A \otimes_{\mathbb{Z}} \mathbb{Q}$ , then the verifications are trivial.

3: Use the axioms to extend linearly and additively.  $\square$

**Prop. (5.6.1.12) [DVR].** If  $(\mathcal{O}_K, \mathfrak{m}_K)$  is a DVR of characteristic  $(0, p)$ , ramification index  $e$ , then  $\mathfrak{m}_K$  has a pd-structure iff  $e \leq p - 1$ .  $\lrcorner$

*Proof:*  $\mathfrak{m}_K$  has at most one pd-structure by (5.6.1.11). It suffices to show that  $x^n/n!$  is in  $\mathfrak{m}$  for any  $x \in \mathfrak{m}$ . Thus the assertion follows from (2.6.1.33).  $\square$

### Extending PD-Structure

**Def. (5.6.1.13) [Extending PD-Structure].** Let  $(A, I, \gamma)$  be a pd-structure and  $B$  is an  $A$ -algebra, we say that  $\gamma$  extends to  $B$  if  $A \rightarrow B$  extends to a morphism of pd-structures  $(A, I, \gamma) \rightarrow (B, IB, \gamma')$ .

Let  $(A, I), (B, J)$  be two pd-structures and  $B$  is an  $A$ -algebra, then these two pd-structures are said to be compatible iff the pd-structure on  $A$  extends to  $B$  and the pd-structure on  $J$  and  $IB$  coincides, or equivalently, there is a pd-structure on  $IB + J$  compatible with  $IB$  and  $J$ , by (5.6.1.7).  $\lrcorner$

**Prop. (5.6.1.14) [Extendability, Berthlot].** Let  $(A, I)$  is a pd-structure and  $B$  an  $A$ -algebra, if any of the following holds:

- $IB = 0$ ,
- $I$  is principal,
- $B[I] = 0$ , (e.g.  $A \rightarrow B$  is flat).

then  $\gamma$  extends to  $B$ .  $\lrcorner$

*Proof:* 1 is trivial.

2: if  $I = (x)$ , we define  $\gamma_n(bx) = b^n \gamma_n(x)$ . This is well defined: if  $(b - b')x = 0$ , then  $(b^n - (b')^n) \gamma_n(x) = 0$  because  $\gamma_n(x) \in (x)$ . Verifications of axioms is routine.

3: The condition shows  $I \otimes_A B \cong IB$ , thus it suffices to define  $\gamma$  on  $I \otimes_A B$ . For this we define on  $I \times B$  and descend: let  $\gamma_n((x, b)) = b^n \gamma_n(x)$  and extend by freeness and axioms in (5.6.1.1), then it is easy to show it is bi-additive and  $A$ -linear, so descend to  $I \otimes B$  by (3.2.4.13).  $\square$

**Prop. (5.6.1.15) [PD-Structure and Completions].** Let  $(A, I, \gamma)$  be a pd-structure that  $p$  is nilpotent in  $A/I$ , then each  $\gamma_n$  is continuous in the  $p$ -adic topology and extends to a pd-structure  $\hat{\gamma}$  on  $\hat{I}$ .

If moreover  $A$  is a  $\mathbb{Z}_{(p)}$ -algebra, then for  $e$  large,  $p^e A \in I$  is preserved by  $\gamma$  and

$$(\hat{A}, \hat{I}, \hat{\gamma}) = \text{colim}_e (A/p^e A, I/p^e A, \gamma).$$

$\lrcorner$

*Proof:* Let  $p^t \in I$ , then 1 follows from (5.2.3.8).  $\gamma_n$  is clearly continuous, and  $\gamma_n$  preserves  $p^e A$  because

$$\gamma_n(p^e a) = p^n \gamma_n(p^{e-1} a) = \frac{p^n}{n!} p^{n(e-1)} a^n \in p^e A.$$

The limit equation follows from (5.6.1.5).  $\square$

**Prop. (5.6.1.16) [Quotient].** Let  $(A, I, \gamma)$  be a pd-structure and  $\mathfrak{a} \subset A$  is an ideal, and  $I' = I \cap \mathfrak{a}$ , then the following are equivalent:

- $\delta$  extends to  $A/\mathfrak{a}$ .
- $I'$  is preserved by  $\gamma$ .
- There is a set of generator  $x_i$  of  $I'$  that  $\gamma_n(x_i) \in I'$  for all  $n, i$ .



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*Proof:*  $1 \rightarrow 2 \rightarrow 3$  is clear.  $2 \rightarrow 1$ : we can just define  $\gamma(x + I) = \gamma(x) + I$ , which is well defined by axiom2 of(5.6.1.1).  $3 \rightarrow 2$  is clear.  $\square$

**Def. (5.6.1.17) [Free PD-Algebra].** For a pd-structure  $(A, I, \delta)$ , the **free pd-algebra**  $A\langle t_1, \dots, t_n \rangle$  is defined to be the  $A$ -algebra generated by symbols  $t_i^{[n_i]}$ , where  $n_i > 0$ , modulo the algebraic relations  $t_i^{[m]} t_i^{[n]} = \binom{m+n}{n} t_i^{[m+n]}$ . Denote by  $A\langle t_1, \dots, t_n \rangle_+$  the ideal generated by  $t_i^{[n_i]}$  where  $n_i > 0$ .

Then the ideal  $J$  generated by  $I$  and  $A\langle t_1, \dots, t_n \rangle_+$ , where  $n_i > 0$  has a unique pd-structure that  $\gamma_n(t_i) = t_i^{[n]}$  and  $(A, I, \delta) \rightarrow (A\langle t_1, \dots, t_n \rangle, J, \gamma)$  is a morphism of pd-structures.

It has a universal property that  $\text{Hom}((A\langle t_1, \dots, t_n \rangle, J, \gamma), (C, K, \varepsilon))$  is the same as  $\text{Hom}((A, I, \delta), (C, K, \varepsilon))$  with specified  $n$  elements in  $K$ .  $\square$

*Proof:* Because  $IA\langle t_i \rangle \cap A\langle t_1, \dots, t_n \rangle_+ = IA\langle t_i \rangle + A\langle t_1, \dots, t_n \rangle_+$ , by(5.6.1.7), it suffices to construct pd-structures on  $IA\langle t_i \rangle$  and  $A\langle t_1, \dots, t_n \rangle_+$ . The former is by(5.6.1.14) and for the latter: if  $A$  is torsion-free, then we can use(5.6.1.11) because  $\gamma_m(x)^n = n! \gamma_n(\gamma_m(x)) = \frac{(mn)!}{(m!)^n} \gamma_{mn}(x) \in n! A\langle t_1, \dots, t_n \rangle_+$ . In general, we write  $A = R/\mathfrak{a}$  where  $R$  is a torsion-free pd-structure(can choose  $\mathbb{Z}\langle A \rangle$ ), so there is a pd-structure on  $R\langle t_i \rangle$ , and  $R\langle t_i \rangle / \mathfrak{a}\langle t_i \rangle = A\langle t_i \rangle$ , then we can use(5.6.1.16) to construct a pd-structure on  $A\langle t_i \rangle$  compatible with that of  $A$ .

The verification of universal property is omitted.  $\square$

### PD-Envelopes

**Prop. (5.6.1.18) [PD-Envelopes].** Let  $(A, I, \gamma)$  be a pd-structure, then there is a **pd-envelope** functor

$$(B, J) \rightarrow (D_B(J), \overline{J}, \overline{\delta})$$

from the the category of pairs over  $(A, I)$  to the category of pd-structures over  $(A, I, \gamma)$  that is left adjoint to the forgetful functor.

In particular, by the universal property of pd-envelope, there is a morphism  $(B, J) \rightarrow (D_B(J), \overline{J}) \rightarrow (B/J, 0)$  of pairs, so in particular  $D/\overline{J} \rightarrow B/J$  is surjective.  $\square$

*Proof:* We use adjoint functor theorem(4.1.1.35), the forgetful functor preserves limit by(5.6.1.5), and it satisfies the set-theoretical condition: for any pair  $(B, J)$  over  $(A, I)$  and a morphism  $\psi : (B, J) \rightarrow (C, K)$  over  $(A, I)$  where  $\varphi : (A, I, \gamma) \rightarrow (C, K, \delta)$  is a pd-morphism, then we can consider the subring  $C' \subset C$  generated by all  $\varphi(A)$ ,  $\psi(B)$  and  $\delta_m(J)$ , and  $K' \subset K \cap C'$  the ideal of  $C'$  generated by  $\varphi(I)$ ,  $\delta_n(\psi(J))$ , then  $|C'| < |A| \otimes |B|^{\aleph_0}$  and its type is bounded by a cardinal, so does  $(C, I)$ .  $\square$

**Prop. (5.6.1.19) [PD-Envelope of Quotients].** Let  $(A, I, \gamma)$  be a pd-structure and  $\varphi : B' \rightarrow B$  be a surjection of  $A$ -algebras with kernel  $K$ . Let  $IB \subset J \subset B$  an ideal and  $J' = \varphi^{-1}(J)$  and  $D_{B', \gamma}(J') = (D', \overline{J}', \overline{\gamma})$ , then  $D_{B, \gamma}(J) = (D'/K', \overline{J}'/K', \overline{\gamma})$  where  $K'$  is the ideal generated by all  $\overline{\gamma}_n(k)$  for  $n \geq 0$  and  $k \in K$ .  $\square$

*Proof:* There is a pd-structure on  $(D'/K', \overline{J}'/K', \overline{\gamma})$  by(5.6.1.16). A map of pairs  $(B, J) \rightarrow (T, I', \gamma)$  is equivalent to a map of pairs  $(B', J') \rightarrow (T, I', \gamma)$  that vanishes on  $K$ , or a map of pd-structures  $(D', \overline{J}', \overline{\gamma}) \rightarrow (T, I', \gamma)$  that vanishes on  $\gamma_n(K)$ , thus this is clearly represented by  $(D'/K', \overline{J}'/K', \overline{\gamma})$ .  $\square$



**Prop. (5.6.1.20).** If  $(A, I, \gamma)$  is a pd-structure and  $(B, J)$  is a pair over  $(A, I)$ , then

$$D_{B[X_i], \gamma}(JB[X_i] + (X_i)) \cong D_{B, \gamma}(J) \langle X_i \rangle.$$

⌋

*Proof:* This follows from the universal property of free pd-structure (5.6.1.17). □

**Prop. (5.6.1.21) [Extension Properties of PD-Envelopes].** Let  $(A, I, \gamma)$  be a pd-structure and  $(B, J)$  is a pair over  $(A, I)$ , then

- For each surjection of pd-structures  $(A, I, \gamma) \rightarrow (A', I', \gamma')$ , let  $B' = B \otimes_A A'$ , then we have

$$D_{B', \gamma'}(JB) = D_{B, \gamma}(J) \otimes_A A'.$$

- If  $B'$  is flat over  $B$ , then we have

$$D_{B', \gamma'}(JB') = D_{B, \gamma}(J) \otimes_B B'.$$

⌋

*Proof:* ? □

### PD-Structures and $\delta$ -Rings

**Lemma (5.6.1.22).** If  $A$  is a  $p$ -torsionfree  $\mathbb{Z}_{(p)}$ - $\delta$ -ring, denote  $\gamma_n(z) = \frac{z^n}{n!}$ . If  $z \in A$  satisfies  $\gamma_p(z) \in A$ , then  $\gamma_n(z) \in A$  for any  $n$ . ⌋

*Proof:* WARNING: this is not an easy consequence of power counting. We first prove for  $n = p^2$ : as  $A$  is a  $\delta$ -ring,  $\delta(\frac{z^p}{p}) \in A$

$$\delta\left(\frac{z^p}{p}\right) = \frac{1}{p} \left( \frac{\varphi(z)^p}{p} - \frac{z^{p^2}}{p^p} \right) = \frac{(z^p + p\delta(z))^p}{p^2} - \frac{z^{p^2}}{p^{p+1}} \in A.$$

The first term is in  $A$  by assumption, thus the second term is also in  $A$ , proving the case for  $n = p^2$ .

Now for general  $n$ , it suffices to prove for  $n = kp$ . But it can be checked that  $\gamma_{nk}(z) = u\gamma_k(\gamma_p(z))$  where  $u$  is a unit. Now by what we just proved, we can use induction hypothesis for  $z = \gamma_p(z)$ , and conclude that  $\gamma_{nk}(z) \in A$ . □

**Prop. (5.6.1.23).** The ring  $C = \mathbb{Z}_{(p)}\{x, \frac{\varphi(x)}{p}\}$  (5.5.4.10) identifies with the pd-envelope  $D = D_{\mathbb{Z}_{(p)}\{x\}}(x) = \mathbb{Z}_{(p)}\{x\}[\{\gamma_n(x)\}]$  where  $\gamma_n(x) = \frac{x^n}{n!}$ . Moreover, it also equals to  $\mathbb{Z}_{(p)}[X_1, X_2, \dots]/(pX_1 - x^p, pX_2 - X_1^p, \dots)$ . ⌋

*Proof:* It suffices to show that the smallest  $\delta$ -ring of  $\mathbb{Z}_{(p)}\{x\}[\frac{1}{p}]$  containing  $\mathbb{Z}_{(p)}\{x\}$  and  $\frac{\varphi(x)}{p}$  is the same as the smallest ring of  $\mathbb{Z}_{(p)}\{x\}[\frac{1}{p}]$  containing  $\mathbb{Z}_{(p)}\{x\}$  and  $\frac{\varphi(x)}{n!}$ .

$D \subset C$  is immediate from (5.6.1.22). To show  $C \subset D$ , notice  $\frac{x^p}{p} \in D$ , it suffices to show  $\varphi$  preserves  $D$ , or equivalently,  $\varphi(y) - y^n \in pD$  for any  $y \in D$ . Now

$$\varphi\left(\frac{x^n}{n!}\right) = \frac{(x^p + p\delta(x))^n}{n!} = \frac{\sum_{i=0}^n \binom{n}{i} (pi)! p^{n-i} \frac{x^{pi}}{(pi)!} \delta(x)^{n-i}}{n!}$$

The coefficients

$$\frac{\binom{n}{i}(pi)!p^{n-i}}{n!}$$

are all in  $p\mathbb{Z}_{(p)}$ , thus  $\varphi(\frac{x^n}{n!}) \in pD$ . On the other hand,

$$(\frac{x^n}{n!})^p = \gamma_p(\gamma_n(x)) = u\gamma_{pn}(x) \cdot p! \in pD$$

where  $u$  is a unit in  $\mathbb{Z}_{(p)}$  by (5.6.1.1), thus we are done.  $\square$

**Lemma (5.6.1.24).** If  $A$  is a  $p$ -torsionfree (equivalently, flat)  $\mathbb{Z}_{(p)}$ -algebra and  $(a, p)$  is a regular sequence in  $A$ , then  $D_A((a)) \cong A \otimes_{\mathbb{Z}_{(p)}\{x\}} D_{\mathbb{Z}_{(p)}}((x)) = A[X_1, X_2, \dots]/(pX_1 - a^p, pX_2 - X_1^p, \dots)$ .  $\square$

*Proof:* By (5.6.1.23),

$$A \otimes_{\mathbb{Z}_{(p)}\{x\}} D_{\mathbb{Z}_{(p)}}((x)) = A \otimes_{\mathbb{Z}_{(p)}\{x\}} [\{\gamma_n(x)\}] = A[X_1, X_2, \dots]/(a^p - pX_1, X_1^p - pX_2, X_2^p - pX_3, \dots)$$

thus there is a natural map from  $A \otimes_{\mathbb{Z}_{(p)}\{x\}} D_{\mathbb{Z}_{(p)}}((x))$  to  $D_A((a))$  given by

$$X_k \mapsto \frac{a^{p^k}}{p^{\text{ord}_p(n!)}} = \frac{a^{p^k}}{p^{1+p+\dots+p^{k-1}}} \in D_A((a)).$$

It is surjective, and it is an isomorphism when inverting  $p$ . Thus the kernel are all  $p^\infty$ -torsions. Then to show it is an isomorphism, it suffices to show  $A[X_1, X_2, \dots]/(a^p - pX_1, X_1^p - pX_2, X_2^p - pX_3, \dots)$  is  $p$ -torsion-free. It is a filtered colimit, so it suffices to show  $A[X_1, \dots, X_n]/(a^p - pX_1, X_1^p - pX_2, \dots, X_{n-1}^p - pX_n)$  is  $p$ -torsionfree.

For this, it suffices to prove  $A' = A[X_1, \dots, X_k]/(px_1 - a^k, px_2 - x_1^k, \dots, px_k - x_{k-1}^p)$  is  $p$ -torsion-free. If we denote  $K(R) = K(R[X_1, \dots, X_n], px_1 - a^k, px_2 - x_1^k, \dots, px_k - x_{k-1}^p)$  for any ring  $R$ , then we have a distinguished triangle

$$K(A) \xrightarrow{p} K(A) \rightarrow K(Kos(A[X_1, \dots, X_k], p)) = K(A/p)$$

as  $A$  is  $p$ -torsionfree. Now we can consider the spectral sequence associated to this distinguished triangle. Notice first that  $(px_1 - a^k, px_2 - x_1^k, \dots, px_k - x_{k-1}^p)$  is a regular sequence in  $A'/p$ , which is clear. So the  $E_1$  page looks like

$$\begin{array}{ccccc} A' & \xrightarrow{p} & A' & \longrightarrow & A'/p \\ \uparrow & & \uparrow & & \uparrow \\ * & \longrightarrow & * & \longrightarrow & 0 \\ \uparrow & & \uparrow & & \uparrow \\ * & \longrightarrow & * & \longrightarrow & 0 \\ \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & \dots & \longrightarrow & \dots \end{array}$$

and the  $E_2$  page is of the form

$$\begin{array}{ccccc} A'[p] & \xrightarrow{p} & 0 & \longrightarrow & 0 \\ \uparrow & & \uparrow & & \uparrow \\ * & \longrightarrow & * & \longrightarrow & 0 \\ \uparrow & & \uparrow & & \uparrow \\ * & \longrightarrow & * & \longrightarrow & 0 \\ \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & \dots & \longrightarrow & \dots \end{array}$$

This spectral sequence converges to 0, so  $A'[p] = 0$ , we win.  $\square$

**Prop. (5.6.1.25) [PD-Envelope for Regular Sequence].** Let  $A$  be a  $p$ -torsionfree  $\delta$ -ring and  $p, f_1, \dots, f_r$  define a regular sequence in  $A$ , then  $A\{\frac{\varphi(f_1)}{p}, \dots, \frac{\varphi(f_r)}{p}\}$  identifies with the pd-envelope  $D_A(I)$  of  $I = (f_1, \dots, f_r)$  as a subring of  $A[\frac{1}{p}]$ .  $\lrcorner$

*Proof:* In case  $r = 1$ ,  $A\{\frac{\varphi(f_1)}{p}\} = A \otimes_{\mathbb{Z}_{(p)}\{x\}} \mathbb{Z}_{(p)}\{x, \frac{\varphi(x)}{p}\} = A \otimes_{\mathbb{Z}_{(p)}\{x\}} D_{\mathbb{Z}_{(p)}}((x)) \cong D_A((f_1))$  by (5.6.1.24).

The general case follows from this, by considering the tower

$$\begin{aligned} (A, (f_1)) &\rightarrow (D_A(f_1), (f_2)) = (A\{\frac{\varphi(f_1)}{p}\}, (f_2)) \\ &\rightarrow (D_{A\{\frac{\varphi(f_1)}{p}\}}(f_2), (f_3)) = (A\{\frac{\varphi(f_1)}{p}, \frac{\varphi(f_2)}{p}\}, (f_3)) \\ &\rightarrow \dots = A\left\{\frac{\varphi(f_1)}{p}, \dots, \frac{\varphi(f_r)}{p}\right\} \end{aligned}$$

The equations are true because  $(p, f_k)$  are regular in  $A\{\frac{\varphi(f_1)}{p}, \dots, \frac{\varphi(f_{k-1})}{p}\}$ :

$$\begin{aligned} A\left\{\frac{\varphi(f_1)}{p}, \dots, \frac{\varphi(f_{k-1})}{p}\right\}/p &= A/p[X_{11}, X_{12}, \dots, X_{21}, X_{22}, \dots, \dots, X_{k-1,1}, X_{k-1,2}, \dots]/ \\ &\quad \left(f_1^p, f_2^p, \dots, f_{k-1}^p, X_{11}^p, X_{12}^p, \dots, X_{21}^p, X_{22}^p, \dots, \dots, X_{k-1,1}^p, X_{k-1,2}^p, \dots\right) \end{aligned}$$

and  $f_k$  is a nonzero-divisor in it because it is a non-zero divisor in  $A/(p, f_1^p, \dots, f_{k-1}^p)$ , as  $(p, f_1^p, \dots, f_k^p)$  is also a regular sequence by (5.3.4.4). then a map of pairs of rings  $(A, I) \rightarrow (C, J)$  will lift through this tower uniquely by the universal property of pd-envelop.

then we see  $A\left\{\frac{\varphi(f_1)}{p}, \dots, \frac{\varphi(f_r)}{p}\right\}$  is just the pd-envelop of  $(A, I)$  (5.6.1.18), by the universal property.  $\square$

**Lemma (5.6.1.26).** Let  $A \rightarrow B \in \mathcal{C}\text{Ring}^p$ . If  $(x_1, \dots, x_n) \in B$  is regular w.r.t.  $A$  (5.8.3.7), then  $D_B((x_1, \dots, x_n))$  is  $A$ -flat.  $\lrcorner$

*Proof:* By (5.6.1.23) and base change, it is clear that  $D_B(I)$  is a free  $B/I^p$ -algebra. Thus we need to show that  $B/I^p$  is a flat  $A$ -module. And this is true as the sequence  $x_1^p, \dots, x_n^p$  is also regular w.r.t.  $A$  (5.8.3.8).  $\square$

**Prop. (5.6.1.27) [Flatness of PD-Envelope].** If  $A \rightarrow B$  is a map of simplicial rings,  $A$  is naturally a simplicial pd-structure with  $I_\bullet = pA_\bullet$ , and  $B$  is  $p$ -completely flat over  $A$  (5.8.3.5), and  $(x_1, \dots, x_n) \in \pi_0(B)$  is  $p$ -completely regular w.r.t.  $A$  (5.8.3.7), then the completed pd-envelope  $D = D_B((x_1, \dots, x_n))^\wedge$  is  $p$ -completely flat over  $A$ .  $\lrcorner$

*Proof:* By definition (5.8.3.5), to check it is  $p$ -completely flat, it suffices to check

$$\text{Kos}(A, p) \rightarrow \text{Kos}(A, p) \otimes_A^L D$$

is flat. and by (5.8.3.4) it suffices to check that

$$\pi_0(\text{Kos}(A, p)) \rightarrow \pi_0(\text{Kos}(A, p)) \otimes_A^L D$$

is flat. The formation of pd-envelope commutes with derived base change by universal property, and tensor with  $A/p$ -algebra undoes the completions, so we are reduced to the the case  $A' = \pi_0(Kos(A, p))$  and  $B' = A' \otimes_A^L B$  is flat over  $A'$  and to show that

$$A' \rightarrow D_{B'}((x_1, \dots, x_n))$$

is flat. Notice  $B'$  is a discrete flat  $A'$ -algebra by definition(5.8.3.5), and  $(x_1, \dots, x_n)$  is a sequence in  $B'$  that  $Kos(B, x_1, \dots, x_n)$  is a sequence regular w.r.t.  $A'$ :

$$\begin{aligned} Kos(B, x_1, \dots, x_n) &= Kos(B' = A' \otimes_A^L B, x_1, \dots, x_n) = A' \otimes_A^L Kos(B, x_1, \dots, x_n) \\ &= A' \otimes_{Kos(A, p)}^L Kos(A, p) \otimes_A Kos(B, x_1, \dots, x_n) \\ &= A' \otimes_{Kos(A, p)}^L Kos(B, p, x_1, \dots, x_n) \end{aligned}$$

is flat because  $Kos(A, p) \rightarrow Kos(B, p, x_1, \dots, x_n)$  does by definition(5.8.3.7) and(5.8.3.3). So we are done by(5.6.1.26).  $\square$

**Cor. (5.6.1.28).** If  $A$  is a  $p$ -complete simplicial  $\delta$ -ring and  $B$  is a  $p$ -completely flat simplicial  $\delta$ - $A$ -algebra, and if  $x_1, \dots, x_r \in \pi_0(B)$  is  $p$ -completely regular w.r.t.  $A$ , then

$$C_\bullet = B_\bullet \left\{ \frac{x_1}{p}, \dots, \frac{x_r}{p} \right\}$$

is  $p$ -completely flat over  $A$ .  $\lrcorner$

*Proof:* Let  $C'_\bullet = B_\bullet \left\{ \frac{\varphi(x_1)}{p}, \dots, \frac{\varphi(x_r)}{p} \right\}^\wedge$ , then  $C'_\bullet$  is  $p$ -completely flat over  $A$ , by(5.6.1.25)(? why  $B$  is  $p$ -torsionfree and  $(p, x_1, \dots, x_n)$  is regular sequence) and(5.6.1.27). Now there is a commutative diagram

$$\begin{array}{ccccccc} A & \longrightarrow & A\{x_1, \dots, x_r\} & \longrightarrow & B & \longrightarrow & B\left\{ \frac{x_1}{p}, \dots, \frac{x_r}{p} \right\} \longrightarrow C_\bullet \\ \parallel & & \downarrow \psi & & \downarrow \psi_B & & \downarrow \psi' & \downarrow \psi_C \\ A & \longrightarrow & A\{\varphi(x_1), \dots, \varphi(x_r)\} & \longrightarrow & B' & \longrightarrow & B'\left\{ \frac{\varphi(x_1)}{p}, \dots, \frac{\varphi(x_r)}{p} \right\} \longrightarrow C'_\bullet \end{array}$$

where  $\psi$  is the relative Frobenius and  $\psi_B, \psi_C$  is the derived base change and completed derived base change. Then  $\psi$  is f.f. as it is the base change of the Frobenius on the free  $\delta$ -ring  $\mathbb{Z}\{x_1, \dots, x_r\}$ (5.5.4.10), thus so does  $\psi'$ . Then  $\psi_C$  is  $p$ -completely flat, because it is the completion(use(5.9.7.2)(5.9.7.4)). Now the conclusion follows, by completely f.f. descent(5.9.7.2).  $\square$

## 2 PD-deRham-Complexes

**Def. (5.6.2.1)[Modules of PD-Differentials].** Let  $B$  be an  $A$ -algebra and  $(B, J, \delta)$  be a pd-structure and  $M$  a  $B$ -module, then a **pd- $A$ -derivation** from  $B$  to  $M$  is a an element  $\theta$  of  $\text{Der}_A(B, M)$  that  $\theta(\delta_n(x)) = \delta_{n-1}(x)\theta(x)$  for  $n \geq 1$  and  $x \in J$ .

As in(5.4.3.4), there is a **module of pd-differential**  $\Omega_{B/A, \delta} \in \text{Mod}_A$  with a **universal pd-differential**  $d_{B/A, \delta} : B \rightarrow \Omega_{B/A, \delta}$  s.t.

$$\text{Hom}_B(\Omega_{B/A, \delta}, M)$$

is isomorphic to the set of pd- $A$ -derivations of  $B$  to  $M$ , functorially in  $M$ .  $\lrcorner$

**Prop. (5.6.2.2) [PD-Differential of PD-Envelope].** Let  $(A, I, \gamma)$  be a pd-structure and  $(B, J)$  be a pair over  $(A, I)$ , and let  $D_{B/A, \gamma}(J) = (D, \bar{J}, \bar{\gamma})$  be the pd-envelope, then  $\Omega_{D/A, \bar{\gamma}} = \Omega_{B/A} \otimes_B D$ .  $\lrcorner$

*Proof:* It suffices to show that for any  $M \in \text{Mod}_A$ , the set of  $A$ -derivations  $B \rightarrow M$  is isomorphic to the set of pd- $A$ -derivations  $D \rightarrow M$ .

Let  $D \otimes M$  be the ring that  $M^2 = 0$  and a pd-structure on  $\bar{J} \oplus M$  is given by  $\delta_n(x+m) = \delta_n(x) + \delta_{n-1}(x)m$ . Then a pd- $A$ -derivations  $D \rightarrow M$  is equivalent to a pd-ring map  $(D, \bar{J}) \rightarrow (D \oplus M, \bar{J} \oplus M)$ , and an  $A$ -derivations  $B \rightarrow M$  is also equivalent to a map of pairs  $(B, J) \rightarrow (D \oplus M, \bar{J} \oplus M)$ , thus we are done by the universal property of  $D$ .  $\square$

**Prop. (5.6.2.3).** Let  $B \in \mathcal{C}\text{Ring}/A$  and  $(B, J, \delta)$  be a pd-structure, then

- if  $(B[X], JB[X], \delta')$  is the  $\delta$ -structure extended from that of  $(B, J, \delta)$  as in (5.6.1.14), then

$$\Omega_{B[X]/A, \delta'} = \Omega_{B/A, \delta} \otimes_B B[X] \oplus B[X]dx.$$

- Let  $B\langle x \rangle$  be the free pd-algebra over  $B$  (5.6.1.17), then

$$\Omega_{B\langle x \rangle/A, \delta'} = \Omega_{B/A, \delta} \otimes_B B\langle x \rangle \oplus B\langle x \rangle dx.$$

- Let  $K \subset J$  be an ideal preserved by  $\delta$  and then consider the quotient  $(B' = B/K, \bar{J} = J/K, \bar{\delta})$ , then  $\Omega_{B'/A, \bar{\delta}}$  is quotient of the module  $\Omega_{B/A, \bar{\delta}} \otimes_B B'$  by the  $B'$ -submodule generated by  $dk$  where  $k \in K$ .  $\lrcorner$

*Proof:* These are all somewhat trivial.  $\square$

**Prop. (5.6.2.4) [PD-Differential and Completion].** Let  $A$  be a  $\mathbb{Z}_{(p)}$ -algebra,  $B$  be an  $A$ -algebra and  $(B, J, \delta)$  be a pd-structure and  $p$  is nilpotent in  $B/J$ , then

$$\lim_e \Omega_{B_e/A, \bar{\delta}} = (\Omega_{B/A, \delta})^\wedge = (\Omega_{\widehat{B}/A, \widehat{\delta}})^\wedge.$$

where  $B_e = B/p^e$ .  $\lrcorner$

*Proof:* By (5.6.1.15), the terms make sense. Now by (5.6.2.3) and the observation  $d(p^e) = 0$ , we have  $\Omega_{B_e/A, \bar{\delta}} = \Omega_{B/A, \delta}/p^e = \Omega_{\widetilde{B}/A, \widetilde{\delta}}/p^e$ , thus we are done.  $\square$

**Def. (5.6.2.5) [PD-de Rham Complex].** Let  $\Omega_{B/A, \delta}^i = \wedge^i \Omega_{B/A, \delta}$ , then the surjection  $\Omega_{B/A} \rightarrow \Omega_{B/A, \delta}$  satisfies the condition of (9.2.1.2), thus there is a **pd-de Rham complex**  $\Omega_{B/A, \delta}^\bullet$ .  $\lrcorner$

### PD-Poincaré Lemma

**Lemma (5.6.2.6) [PD-Poincaré Lemma].** Let  $A \in \mathcal{C}\text{Ring}$  and  $P = A\langle X_i \rangle$  is the free PD-algebra over  $A$ , then for any  $A$ -module, the complex

$$0 \rightarrow M \rightarrow M \otimes_A P \rightarrow M \otimes_A \Omega_{P/A, \delta}^\wedge \rightarrow \dots$$

is exact. And if  $D = \widehat{P}$  and let  $\Omega_D^n = \Omega_{P/A, \delta}^n$ , then for any  $p$ -complete  $A$ -module  $M$ , the complex

$$0 \rightarrow M \rightarrow M \widehat{\otimes}_A P \rightarrow M \widehat{\otimes}_A \Omega_{P/A, \delta}^\wedge \rightarrow \dots$$

is exact.  $\lrcorner$

*Proof:* It suffices to show that  $0 \rightarrow M \rightarrow M \otimes_A P \rightarrow M \otimes_A \Omega_{P/A,\delta}^1 \rightarrow \dots$  is homotopic to 0. For this, notice every element of  $\Omega_{P/A,\delta}^n$  is of the form  $\sum P \prod_{j=0}^n dx_{i_0} dx_{i_1} \dots dx_{i_n}$ , so we can let

$$f(\omega) = f(\gamma_{n_{i_0}}(x_{i_0}) dx_{i_0} \wedge \omega) = \gamma_{n_{i_0}+1}(x_{i_0}) \omega$$

where  $\omega$  doesn't divide  $dx_k$  for  $k < i_0$ . Then it can be checked that  $df + fd = \text{id}$ , so we are done.  $\square$

**Prop. (5.6.2.7).** If  $A$  is a ring and  $(B, J, \delta)$  is a pd-structure, and let  $P = B\langle X_i \rangle$  be the free pd-structure (5.6.1.17). Let  $M$  be a  $B$ -module endowed with an integrable connection  $\nabla : M \rightarrow M \otimes_B \Omega_{B/A,\delta}^1$ , then the map of de Rham complexes

$$M \otimes_B \Omega_{B/A,\delta}^* \rightarrow M \otimes_B \Omega_{P/A,\delta}^*$$

is a quasi-isomorphism. And if we denote  $D, D'$  the  $p$ -adic completions of  $B, P$ , and  $\Omega_D, \Omega_{D'}$  the  $p$ -adic completion of  $\Omega_{B/A,\delta}, \Omega_{P/A,\delta}$ , and  $M$  is a  $p$ -complete  $B$ -module endowed with an integrable connection  $\nabla : M \rightarrow M \otimes_D \Omega_D$ , then the map of de Rham complexes

$$M \otimes_D \Omega_D^* \rightarrow M \otimes_{D'} \Omega_{D'}^*$$

is a quasi-isomorphism.  $\square$

*Proof:* Consider the filtration  $F^*$  on  $\Omega_{B/A,\delta}^\bullet$  given by the stupid truncation  $\sigma_{\geq i} \Omega_{B/A,\delta}^\bullet$ , and consider the filtration on  $\Omega_{P/A,\delta}^\bullet$  given by

$$F^*(\Omega_{P/A,\delta}^\bullet) = F^*(\Omega_{B/A,\delta}^\bullet) \wedge \Omega_{P/A,\delta}^\bullet.$$

Notice that we have a split exact sequence

$$0 \rightarrow \Omega_{B/A,\delta}^1 \otimes_B P \rightarrow \Omega_{P/A,\delta}^1 \rightarrow \Omega_{P/B,\delta}^1 \rightarrow 0$$

and  $\Omega_{P/B,\delta}^1$  is free on  $X_i$  over  $B$  (pondering the universal property, this is for the same reason as (5.4.3.7)).

Then we see that  $F^i(\Omega_{P/A,\delta}^\bullet) \rightarrow \Omega_{P/A,\delta}^\bullet$  is termwise split injection for any  $i$ , and the graded is  $\Omega_{B/A,\delta}^i \otimes_B \Omega_{P/B,\delta}^\bullet$ . Thus if we let  $F^i(M \otimes_B \Omega_{P/A,\delta}^\bullet) = M \otimes F^i(\Omega_{P/A,\delta}^\bullet)$ , then the graded is  $M \otimes_B \Omega_{B/A,\delta}^i \otimes_B \Omega_{P/B,\delta}^\bullet$ , which is quasi-isomorphic to  $M \otimes_B \Omega_{B/A,\delta}^i$  by (5.6.2.6). Then the original map is a filtered complexes that induces quasi-isomorphism on gradeds, so it induces a quasi-isomorphism, because it induces an morphism between two convergent spectral sequences, by (4.10.7.6) and (4.10.7.5).  $\square$

## 5.7 Almost Ring Theory

References are [Almost Ring Theory Gabber/Ramero] and [Almost Ring Theory Foundations Gabber/Ramero].

**Notation (5.7.0.1).**

- The setup of almost mathematics is an flat ideal  $I \in \text{Ideal}(R)$  s.t.  $I^2 = I$ . In particular,  $I \otimes I \cong I^2 = I$ .
- Denote  $i : R \rightarrow R/I$ . Then there is a map  $i_* : M \mapsto M_R : \text{Mod}_{R/I} \rightarrow \text{Mod}_R$ , which has a left adjoint  $i^* : N \mapsto N/IN$ , and a right adjoint  $i^! : N \mapsto \text{Hom}(R/I, N)$ .

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### 1 Homological Theory

#### Almost Modules

**Prop. (5.7.1.1) [Examples].**

- If  $K \in \text{PerfdField}$ ,  $R = K^0$ ,  $I = K^{00}$ , then  $I$  is flat over  $R$ , because if  $\varpi$  is a pseudo-uniformizer of  $K$  (12.2.8.9), then  $I = (\pi^{\frac{1}{p^\infty}})$ , which is a colimit of free modules thus flat, and  $I^2 = I$  clearly.
- Let  $R \in \mathbb{C}\text{Ring}$  and  $f$  is an arbitrary element with compatible  $p^n$ -th roots, let  $I = (f^{\frac{1}{p^\infty}})$ , then  $I^2 = I$ . To show  $I$  is flat, consider:

$$M_0 \xrightarrow{f^{1-\frac{1}{p}}} M_1 \xrightarrow{f^{\frac{1}{p}-\frac{1}{p^2}}} M_2 \rightarrow \dots \rightarrow M_n \xrightarrow{\frac{1}{p^n}-\frac{1}{p^{n+1}}} M_{n+1} \rightarrow \dots$$

where  $M_i \cong R$ , and  $M = \text{colim } M_i$ , then  $M$  is flat, and there is a map  $M \rightarrow I : 1 \in M_n \rightarrow f^{\frac{1}{p^n}}$ , then this map is surjective, and it is injective: if  $\alpha$  maps to 0, then  $\alpha f^{\frac{1}{p^n}} = 0$ , so  $\alpha p^m f = 0$  for all  $m \geq n$ , and by perfectness of  $R$ ,  $\alpha f^{\frac{1}{p^m}} = 0$ , so in particular,  $\alpha = 0 \in M_{n+1}$ .

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**Prop. (5.7.1.2) [The Category of Almost  $R$ -Modules in disguise].** Let  $\mathcal{A} \subset \text{Mod}_R$  be the category of all  $R$ -modules  $M$  that the action  $I \otimes M \rightarrow M$  is an isomorphism (By  $I \otimes I = I$  this is equivalent to  $M = I \otimes N$  for some  $N$ ) then:

- The inclusion  $j_! : \mathcal{A} \rightarrow \text{Mod}_R$  is exact, i.e. the cokernel, kernel of objects in  $\mathcal{A}$  are also in  $\mathcal{A}$ .
- $j_!$  has a right adjoint  $j^* : M \mapsto I \otimes M$ , and the unit map  $N \rightarrow j^* j_! N$  is an isomorphism on  $\mathcal{A}$ .
- $j^*$  has its right adjoint  $j_*(M) = \text{Hom}(I, M)$ , and the counit  $j^* j_* M \rightarrow M$  is an isomorphism on  $\mathcal{A}$ .

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*Proof:* 1: an easy consequence of five-lemma.

2: We need to show for  $N \in \mathcal{A}$ ,  $\text{Hom}(N, I \otimes M) \cong \text{Hom}(N, M)$ . Notice there is a distinguished triangle

$$I \otimes M \rightarrow M \rightarrow M \otimes R/I,$$

as  $- \otimes_R^L M$  is a derived functor and  $I$  is flat. So it suffices to show

$$R\text{Hom}_R(N, M \otimes_R^L R/I) = 0 = R\text{Hom}_{R/I}(N \otimes_R^L R/I, M \otimes_R^L R/I).$$

And in fact  $N \otimes_R^L R/I = 0$ , because  $N \otimes_R^L R/I = N \otimes_R^L I \otimes_R^L R/I$ , and  $I \otimes_R^L R/I = I \otimes_R R/I = I/I^2 = 0$  by flatness and hypothesis.

$N \cong j^* j_! N$  is an easy consequence of  $I \otimes I = I$ .

3: The adjointness is just Tor-Hom-adjunction, and for the isomorphism  $I \otimes \text{Hom}(I, M) \cong M$ , as  $I$  is flat, it suffices to prove the stronger result that  $I \otimes_R^L R \text{Hom}(I, M) = M[0]$ . As there is an exact triangle

$$R \text{Hom}(R/I, M) \rightarrow M \rightarrow R \text{Hom}(I, M),$$

so it suffices to show  $I \otimes_R^L R \text{Hom}(R/I, M) = 0$ , because  $I \otimes_R^L M = M$ . But this is because  $I \otimes_R^L R \text{Hom}(R/I, M) = I \otimes_R^L R/I \otimes_R^L R \text{Hom}(R/I, M)$ , and  $I \otimes_R^L R/I = 0$  as before.  $\square$

**Prop. (5.7.1.3)[Category of Almost  $R$ -modules].**

- The image of the functor  $i_* : \text{Mod}_{R/I} \rightarrow \text{Mod}_R$  is a Serre subcategory of  $\text{Mod}_R$ , so the quotient  $\text{Mod}_R^a = \text{Mod}_R / \text{Mod}_{R/I}$  exists by (4.8.3.15),
- The quotient  $q : \text{Mod}_R \rightarrow \text{Mod}_R^a$  admits fully faithful left and right adjoints. In particular,  $q$  preserves all limits and colimits.
- The image of  $i$  is a 'tensor ideal' of  $\text{Mod}_R$ , so the quotient  $\text{Mod}_R^a$  inherits a natural symmetric monoidal  $\otimes$ -product structure.
- There is a functor  $\text{alHom} : (\text{Mod}_R^a)^{\text{op}} \times \text{Mod}_R^a \rightarrow \text{Mod}_R^a : (X, Y) \rightarrow \text{alHom}(X, Y)$  that  $\text{alHom}(X, -)$  is right adjoint to  $- \otimes X$ :

$$\text{Hom}(Z \otimes X, Y) \cong \text{Hom}(Z, \text{alHom}(X, Y)).$$

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*Proof:* 1: the image of  $i_*$  is just the category of modules killed by  $I$ , if  $M$  is killed by  $I$ , then subobjects and quotients of  $M$  is killed by  $I$ , and if  $M$  is an extension of two elements killed by  $I$ , then  $IM = I^2M = 0$ .

2: In fact we show that the category  $\mathcal{A}$  in (5.7.1.2) and the functor  $j^*$  is just equivalent to  $\text{Mod}_R^a$ : First:  $j^*(\text{Mod}_{R/I}) = 0$ , because  $I \otimes M = I \otimes_R R/I \otimes_{R/I} M = I/I^2 \otimes_{R/I} M = 0$  as  $I = I^2$ , and  $j^*$  is exact because  $I$  is flat.

And for any  $R$ -module  $M$ , consider  $I \otimes M \rightarrow M$ , it has kernels and cokernels, then tensoring  $I$ , it becomes  $I \otimes M \rightarrow I \otimes M$  (5.7.0.1). as  $I$  is flat, the kernel and cokernels are killed by  $I$ , so for any functor  $q$  to another category that kills  $\text{Mod}_{R/I}$ ,  $q(M) = q(I \otimes M) = qj_!j^*(M)$ , so  $q$  factors through  $M$ , uniquely, as  $j^*$  is surjective.

Now the left/right adjoints exist by (5.7.1.2).

3: if  $IM = 0$ , then  $IM \otimes N = 0$ , so the tensor products pass to the quotient, and  $j^*$  is a map between symmetric monoidal categories.

4:  $\text{alHom}$  is defined by  $\text{alHom}(j^*M, j^*N) = j^*(\text{Hom}(M, N)) = \text{Hom}(M, N)^a$ . This is well defined, because if  $IM = 0$  or  $IN = 1$ , then  $I \text{Hom}(M, N) = 0$ .  $\square$

**Cor. (5.7.1.4).**

- $i^* j_! = 0$ .
- $i^! j_* = 0$ .
- $j^* i_* = 0$ , and the kernel of  $j^*$  is just  $i_*(\text{Mod}_{R/I})$ .

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*Proof:* 1:  $R/I \otimes I \otimes M = 0$ , because  $I \otimes R/I = 0$ .  
 2:  $\text{Hom}(R/I, \text{Hom}(I, M)) = 0$ , because  $I \otimes R/I = 0$ .  
 3: This is by 2 of the proposition(5.7.1.3).  $\square$

**Remark(5.7.1.5).** The construction above can be summarized as the following diagram:

$$\text{Mod}_{R/I} \begin{array}{c} \xleftarrow{i^*} \\ \xrightarrow{i_*} \\ \xleftarrow{i^!} \end{array} \text{Mod}_R \begin{array}{c} \xleftarrow{j^!} \\ \xrightarrow{j^*} \\ \xleftarrow{j_*} \end{array} \text{Mod}_R^a$$

with four adjoint pair and three vanishing. This should be seen as an analogy of the case of topology:  $X$  is a space and  $i : U \rightarrow X$  is open in  $X$ , and  $j : Z \rightarrow X$  is closed,  $Z = X - U$ , then the defined sheaf operations are the same as written above.

However, one should not consider  $\text{Mod}_R^a$  as the sheaf of modules on the open subscheme  $\text{Spec } R_f$  for some pseudo-uniformizer, because the map  $M \rightarrow M \otimes R_f$  factors through  $\text{Mod}_R^a$  as it vanishes on  $\text{Mod}_{R/I}$ , but it is not  $\text{Mod}_{R/I}^a$ . For example, if  $k$  is a perfect field, and consider  $R = k[t^{\frac{1}{p^\infty}}]$ , then the module  $M = R/(t)$  is also killed by  $\otimes R_f$ , but it is not killed by  $I$ .

Then one may consider it is the category of Qco sheaves on  $D(I)$ , but first this is not an affine scheme, and second this is false, anyway. And we should imagine an non-existent open subscheme  $\overline{U}$  bigger than  $U$ , as it contains any affine opens of  $U$ .  $\lrcorner$

**Remark(5.7.1.6).** Notice  $j^*$  is both left exact and right exact, so it preserves both arbitrary limits and colimits, so almostification nearly loses anything. In particular, the category  $\text{Mod}_R^a$  has all colimits and limits.  $\lrcorner$

**Def.(5.7.1.7)[Almost Commutative Algebras].** As  $\text{Mod}_R^a$  has a symmetric monoidal structure, it is possible to define the category of **almost commutative algebras** as the category of commutative unitary monids in  $\text{Mod}_R^a$ , denoted by  $\mathcal{C}\text{Ring}_R^a$ . Notice that its unit object is  $I = R^a$ .

There is an obvious map

$$(-)^a : \mathcal{C}\text{Ring}_R \rightarrow \mathcal{C}\text{Ring}_R^a,$$

and yet another functor

$$(-)_* : \mathcal{C}\text{Ring}_R^a \rightarrow \mathcal{C}\text{Ring}_R,$$

because  $M \rightarrow M_*$  is lax symmetric monoidal, i.e. there are natural map  $M_* \otimes N_* \rightarrow (M \otimes N)_*$ . This is a right adjoint of  $(-)^a$ , as  $j^*$  and  $j_*$  is adjoint.

Finally there is a functor

$$(-)_{!!} : \mathcal{C}\text{Ring}_R^a \rightarrow \mathcal{C}\text{Ring}_R,$$

whose construction is a little complicated, first notice the functor  $(-)_!$  preserves multiplication but it has no units, so in order to give it a unit, consider the module pushout:  $(A_! \oplus V)/I$ , which has a natural multiplicative structure that can be made into a  $R$ -module, and  $(-)_!$  is left adjoint to  $(-)^a$ , Cf.[Almost Ring Theory P22].  $\lrcorner$

**Prop.(5.7.1.8).**  $(-)_!$  preserves faithfully flatness.  $\lrcorner$

*Proof:* Cf.[Almost Ring theory P52] ?  $\square$

**Def.(5.7.1.9).** For an almost commutative algebra  $A$ , a **left module** is an almost module  $M \subset \text{Mod}_R^a$  that has a left action  $A \otimes M \rightarrow M$  that has natural commutative diagrams as one expects. And for any  $R$ -algebra  $A$ , there are natural maps  $\text{Mod}_A \rightarrow \text{Mod}_A^a$ .  $\lrcorner$

## Almost Homological Algebra

### 2 Almost Commutative Algebra

**Def. (5.7.2.1)[Almost Notations].** Given a  $R$ -module  $M$ , an element  $f \in M$  is called **almost zero** if  $I \cdot f = 0$ , and  $M$  is called **almost zero** if all  $f \in M$  is almost zero.

Denote

$$M^a = j^* M \in \text{Mod}_R^a, \quad M_* = j_* M^a = \text{Hom}(I, M), \quad M_{\dagger} = j_{\dagger} M^a = I \otimes M.$$

Then there are morphisms  $M_{\dagger} \rightarrow M \rightarrow M_*$ , which becomes isomorphisms after almostification.  $\lrcorner$

**Prop. (5.7.2.2).** If  $I = (f^{\frac{1}{p^\infty}})$ , then  $M_* = \{x \in M[f^{-1}] \mid f^{\frac{1}{p^n}} x \in A\}$  for all  $n$ .  $\lrcorner$

**Prop. (5.7.2.3).** If  $M \rightarrow N$  is almost surjective maps of  $K^0$ -algebras that  $M/I \rightarrow N/I$  is surjective, then  $M \rightarrow N$  is surjective.  $\lrcorner$

*Proof:* As  $I$  is flat over  $K^0$ , if  $M \rightarrow N \rightarrow Q \rightarrow 0$  is the cokernel, tensoring  $A/I$ , as  $M/I \rightarrow N/I$  is surjective,  $Q/IQ = 0$ , but  $Q$  is almost zero thus  $IQ = 0$ , so  $Q = 0$ .  $\square$

**Def. (5.7.2.4)[Almost Properties].** Something is called **almost XXX** if it is XXX when passed to the category of almost  $R$ -modules. For example,

- elements of  $M_*$  are called **almost elements** of  $M$ .
- $M$  is called **almost flat** iff  $M^a \otimes -$  is exact on  $\text{Mod}_R^a$ , equivalently,  $\text{Tor}_{>0}^R(M, N)$  is almost zero for all  $N$ .
- $M$  is called **almost projective** iff  $\text{alHom}(M, -)$  is exact on  $\text{Mod}_R^a$ , which is equivalent to  $\text{Ext}_R^{>0}(M, N)$  is almost zero for all  $N$ .  
Notice this is not equivalent to projective in  $\text{Mod}_R^a$ , because  $R$  is almost projective, but  $\text{Hom}_{R^a}(R^a, M^a) = \text{Hom}(I, M)$  is not exact as  $I$  is not projective:  $\text{Ext}_{R^a}^1(R^a, R^a) = \text{Ext}_R^1(I, R) = \text{Ext}^2(k, R)$ , which is not 0 if  $R$  is the valuation ring of a non-spherically complete perfectoid field  $K$ , like  $\widehat{\mathbb{Q}_p}$ ?.
- $M$  is called **almost finitely generated/almost finitely presented** if for any  $\varepsilon \in I$ , there is a f.g./f.p.  $M_\varepsilon \rightarrow M$  with  $N_\varepsilon$  generators that the kernel and cokernel are killed by  $\varepsilon$ . It is called **uniformly almost finitely generated** iff  $N_\varepsilon$  is independent of  $\varepsilon$ .  
Notice this definition doesn't depends on  $M$  chosen?.
- If  $\text{char } S > 0$ ,  $S$  is called **almost perfect** iff  $S_*$  is perfect.  $\lrcorner$

**Prop. (5.7.2.5)[Enough Almost Injectives].** The category  $\text{mod}_R^a$  has enough injectives. In fact  $j^*, j_*$  both preserves injectives, because they has exact left adjoints, so  $I$  is injective  $R$ -module iff  $I^a$  is injective  $R^a$ -module, and  $J$  is injective  $R^a$ -module iff  $J_*$  is injective  $R$ -module. So to construct an injective resolution in  $R^a$ , pass to  $R$ -modules using either  $(-)_*$  or  $(-)_{\dagger}$  and find an injective resolution, then almostificate it.  $\lrcorner$

**Prop. (5.7.2.6)[Derived Functors of  $(-)_*$ ].** Notice that  $\text{Hom}_{R^a}(M^a, N^a) = \text{Hom}_R(I \otimes M, N)$  by adjointness, so using (5.7.2.5),

$$\text{Ext}_{R^a}^k(M^a, N^a) = \text{Ext}_R^k(M_*, N) = \text{Ext}_R^k(M, R \text{Hom}(I, N)),$$

then as  $M_* = \text{Hom}(I, M)$ , the derived functor of  $(-)_*$  is just  $\text{Ext}_R^k(I, M) = \text{Ext}_{R^a}^k(R^a, M^a)$ .

Notice that  $\text{Ext}_{R^a}^k(M, N)$  are all almost zero, as  $j^a j_* = \text{id}$ , and use trivial Grothendieck spectral sequence.  $\perp$

**Prop. (5.7.2.7) [(Example) A Quadratic Extension of a Perfectoid Field].** If  $K = \widehat{Q_p[p^{\frac{1}{p^\infty}}]}$  and  $L = K(\sqrt{p})$  with  $p \neq 2$ , then  $L^0$  is a uniformly almost f.p. projective  $K^0$ -module.  $\perp$

*Proof:* It suffices to find for each  $n$  a  $K^0$ -module  $R_n$  of rank 2 that  $R_n \rightarrow L^0$  is injective with cokernel annihilated by  $p^{\frac{1}{p^n}}$ . For this, consider  $R_n = K^0 \oplus K^0 p^{\frac{1}{2p^n}}$ , then  $L^0 = \widehat{\text{colim}_n R_n}$ .

Notice that the cokernel of  $R_n \rightarrow R_{n+1}$  is killed by  $p^{\frac{1}{p^n}}$ , because

$$p^{\frac{1}{p^n}} \cdot p^{\frac{1}{2p^{n+1}}} = p^{\frac{(p+1)/2}{p^{n+1}}} \cdot p^{\frac{1}{2p^n}} \subset R_n.$$

So by killing one by one, the cokernel of  $R_n \rightarrow \text{colim}_n R_n$  is killed by any  $p$ -power with power larger than  $\sum \frac{1}{p^n}$ , in particular by  $p^{\frac{1}{p^{n-1}}}$ . So  $\text{colim}_n R_n$  is an extension of  $R_0$  by a cokernel killed by  $p$ , so it is also  $p$ -adically complete, and  $L^0 = \text{colim}_n R_n$ . Now Consider  $0 \rightarrow R_n \rightarrow \text{colim}_n R_n \rightarrow \text{Coker}$ , then  $\text{Ext}^n(\text{colim}_n R_n, N) = \text{Ext}^n(\text{Coker}, N)$  is killed by  $p^{\frac{1}{p^{n-1}}}$  for all  $n$ , so it is killed by  $I$ , thus  $\text{colim}_n R_n$  is almost projective.  $\square$

### Completions and Closures

**Prop. (5.7.2.8) [prc and Completion].** If  $A$  is a ring with a nonzero-divisor  $f$  that  $A \subset A[f^{-1}]$  is  $p$ -root closed(prc), then:

- $\widehat{A} \subset \widehat{A}[f^{-1}]$  is  $p$ -root closed.
- If  $f$  admits a compatible  $p$ -power roots, then  $A_* \subset A_*[f^{-1}]$  is  $p$ -root closed (where almost mathematics is performed w.r.t  $(f^{\frac{1}{p^\infty}})$ ).  $\perp$

*Proof:* We first replace  $A$  with its maximal separated quotient  $A/(\cap_n f^n A = I)$ :  $f$  is still non-zero-divisor, because if  $fg \in I$ , then  $fg \in f^n A$  for all  $n$ , so  $g \in f^{n-1} A$  as  $f$  is non-zero-divisor. And it is  $p$ -root closed, because if  $a^p \in A/I[f^{-1}]$ , then  $a^p = b + f^{-c}d$  for  $c$  integer and  $d \in I$ . Notice  $I = fI = f^c I$  by (5.2.2.16), so  $f^{-c}d \in I$  as well, so  $a \in A$ .

Now  $A$  is  $f$ -separated, in particular,  $A \hookrightarrow \widehat{A}$ .

1: If  $g \in \widehat{A}[f^{-1}]$  and  $g^p \in \widehat{A}$ , then  $f^N g \in \widehat{A}$  for some  $N$  and choose a  $m \geq N(p-1)$ , then by the density,  $g = g_0 + f^m g_1$  for some  $g_0 \in A[f^{-1}]$ ,  $g_1 \in \widehat{A}$ . Notice  $f^N g_0 \in \widehat{A}$ , now

$$g^p = g_0^p + p g_0^{p-1} f^m g_1 + \dots + (f^m g_1)^p,$$

By definition of  $m$ , all terms except  $g_0^p$  are in  $\widehat{A}$ , so  $g_0^p \in A$ , so  $g_0 \in A$ , and  $g \in \widehat{A}$ .

2: Use the convention (5.7.2.2), if  $g \in A_*[f^{-1}]$  that  $g^p \in A_*$ , then  $f^{\frac{1}{p^n}} g^p \in A$  for all  $n$ , so  $(f^{\frac{1}{p^{n+1}}} g)^p \in A$ , thus  $f^{\frac{1}{p^{n+1}}} g \in A$ , hence  $g \in A_*$ .  $\square$

**Prop. (5.7.2.9) [ic and Completion].** Let  $A$  be a ring with a non-zero-divisor  $f$ , if  $A \subset A[f^{-1}]$  is integrally closed, then:

- $\widehat{A} \subset \widehat{A}[f^{-1}]$  is integrally closed.
- If  $f$  admits a compatible  $p$ -power roots, then  $A_* \subset A_*[f^{-1}]$  is integrally closed (where almost mathematics is performed w.r.t  $(f^{\frac{1}{p^\infty}})$ ).

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*Proof:* We first replace  $A$  with its maximal separated quotient  $A/(\cap_n f^n A = I)$ :  $f$  is still non-zero-divisor and  $I$  is  $f$ -divisible as in the proof of (5.7.2.8). And it is integrally closed, because if  $g$  satisfies a monic polynomial  $h(X) \in A/I[f^{-1}][X]$ , then choose a lifting,  $h(g) \in I[f^{-1}] = I \subset A$ , so  $g$  is integral over  $A$  thus  $g \in A$ , and  $g \in A/I$ . Now  $A$  is  $f$ -separated and  $A \hookrightarrow \hat{A}$ .

1: If  $g \in \hat{A}[f^{-1}]$  satisfies a polynomial  $H \in \hat{A}[X]$ , then  $g = f^{-c}h$  for  $h \in \hat{A}$ , and then  $h$  satisfies a polynomial  $H(f^c x)$ , and choose an approximation of coefficients of  $H(x)$  and  $h_0$  of  $h \bmod f^{cn}$ , then it is clear that  $H(f^c h_0) \in f^{cn} \hat{A} \cap A = f^{cn} A$ , so when dividing back,  $g_0 = f^{-c} h_0$  is integral over  $A$  thus  $g_0 \in A$ , thus  $h_0 \in f^c A$ , and  $h \equiv h_0 \bmod f^{cn}$ , thus  $h \in f^c \hat{A}$ , and  $g \in A$ .

2: Use the convention (5.7.2.2), if  $g \in A_*[f^{-1}]$  is integral over  $A_*$ , then there are polynomial  $H$  that  $H(g) = 0$ , now if  $\varepsilon = f^{\frac{1}{p^k}}$  consider another polynomial  $H(x/\varepsilon)$ , then its coefficients are all in  $A$ , thus  $\varepsilon g$  is integral over  $A$  thus  $\varepsilon g \in A$ , and then  $g \in A_*$ .  $\square$

**Prop. (5.7.2.10) [tic and Completion].** Let  $A$  be a ring with a non-zero-divisor  $f$  that admits a compatible system of  $p$ -power roots  $f^{\frac{1}{p^n}}$  for all  $n > 0$ , and  $A$  is totally integrally closed(tic) in  $A[f^{-1}]$ , then  $\hat{A}$  is totally integrally closed in  $\hat{A}[f^{-1}]$  and  $A = A_*$ .  $\square$

*Proof:* 1: Notice totally integrally closed is  $p$ -root closed, so  $\hat{A} \subset \hat{A}[f^{-1}]$  is  $p$ -root closed. Now if  $f^k g^{\mathbb{N}} \subset \hat{A}$  for some  $k$ , then by prc,  $f^{\frac{k}{p^n}} g \in \hat{A}$  for all  $n$ , thus  $g$  in an almost zero element in  $\hat{A}[f^{-1}]/\hat{A} \cong A[f^{-1}]/A$ , and then  $g$  is totally integrally closed over  $A$ , because for any  $n$ , let  $n < p^k$ , then  $f^{\frac{1}{p^k}} g \in A$ , thus  $f^{\frac{n}{p^k}} g \in A$ , and  $fg^n \in A$ .

2: Because  $f^{\frac{1}{p^k}} A_* \subset A$  by convention (5.7.2.2), clearly  $A_*$  is totally integrally closed in  $A$ , thus  $A_* \subset A$ .  $\square$

### Almost Étale Map

**Def. (5.7.2.11).** A map  $A \rightarrow B \in \mathcal{C}\text{Ring}_R$  is called **almost étale** iff:

- $B$  is almost f.p. projective over  $A$ .
- (Unramifiedness (5.4.6.10)) There exists a diagonal idempotent  $e \in (B \otimes_A B)_*$ . i.e.  $e^2 = e$  and  $\mu_*(e) = 1$ , and  $\ker(\mu)_* \cdot e = 0$ , where  $\mu : B \otimes_A B \rightarrow B$  is the multiplication map.

┘

**Prop. (5.7.2.12) [Example of Almost Étale Maps].** Let  $K = \widehat{Q_p[p^{\frac{1}{p^\infty}}]}$  and  $L = K(\sqrt[p]{p})$  with  $p \neq 2$ , then  $L^0/K^0$  is uniformly almost f.p projective  $K^0$ -module, by (5.7.2.7). We show it is finite étale: flatness is clear, as  $L^0/K^0$  is torsion-free and  $K^0$  is a valuation ring and use (5.7.3.3).

For unramifiedness, notice that

$$L \otimes_K L \cong L \times L : (a, b) \mapsto (ab, a\sigma(b)).$$

by (3.2.8.8), the diagonal idempotent  $e$  is given by

$$e = \frac{1}{2p^{\frac{1}{2p^n}} \otimes 1} (1 \otimes p^{\frac{1}{2p^n}} + p^{\frac{1}{2p^n}} \otimes 1)$$

for any  $n \geq 0$ , then we see  $p^{\frac{1}{p^n}} e \in L^0 \otimes_{K^0} KL^0$  for all  $n$ , thus  $e \in (L^0 \otimes_{K^0} L^0)_*$ .  $\square$

**Lemma (5.7.2.13)** [Lemma for Almost Purity in Characteristic  $p$ ]. If  $\eta : R \rightarrow S$  is an integral map of perfect rings. If  $\eta[t^{-1}]$  is finite étale for some  $t \in R$ , then  $\eta$  is almost finite étale w.r.t the ideal  $I = (t^{\frac{1}{p^\infty}})$ .  $\lrcorner$

*Proof:* Firstly, we may assume  $R, S$  are both  $t$ -torsion-free, because the  $t$ -torsion part  $R[t^\infty]$  and  $S[t^\infty]$  is almost zero: if  $t^c \alpha = 0$ , then  $t^c \alpha^{p^n} = 0$ , so  $t^{\frac{c}{p^n}} \alpha = 0$ . So we reduce to  $R/R[t^\infty] \rightarrow S/S[t^\infty]$ , which doesn't change anything.

Now we reduce to the case that  $R, S$  are integrally closed in  $R[t^{-1}]$  and  $S[t^{-1}]$ : it suffices to show that  $R_{int} \subset R_*$ , thus they are almost isomorphic. For this, an element  $f \in R_{int}$  satisfies  $f^{\mathbb{N}} t^k \in R$  for some  $k$ , so by perfectness,  $f t^{\frac{k}{p^n}} \in R$  for all  $n$ , so  $f \in R_*$ .

Now check unramifiedness: let  $e \in (S \otimes_R S)[t^{-1}]$  be a diagonal idempotent, then  $t^c e \in S \otimes_R S$  for some  $c$ , now  $e^2 = e$ , so easily  $e \in (S \otimes_R S)_*$ .

Now check almost finite projective: for  $m > 0$ , represent  $t^{\frac{1}{p^m}} e = \sum a_i \otimes b_i \in S \otimes_R S$ , then use the map  $S \xrightarrow{\alpha} R^n \xrightarrow{\beta} S$  as in (5.4.7.14), then  $\beta \alpha = t^{\frac{1}{p^m}}$  on  $S$ , as  $S$  is  $t$ -torsion free,  $R^n \rightarrow S$  is injective with  $t^{\frac{1}{p^m}}$ -torsion cokernel, for any  $m$ . So  $S$  is almost finite projective.  $\square$

**Prop. (5.7.2.14)** [Almost Purity in Characteristic  $p$ ]. If  $R \in \text{Perfd}_p$ , then using the almost mathematics w.r.t.  $I = (t^{\frac{1}{p^\infty}})$ ,  $S \rightarrow S_*[t^{-1}]$  gives an isomorphism of categories:  $R_{af\acute{e}t} \cong R[t^{-1}]_{f\acute{e}t}$ .  $\lrcorner$

*Proof:* As in the proof of (5.7.2.13), we may assume  $R$  is  $t$ -torsion-free. Notice that any integral extension of  $R[t^{-1}]$  comes from an integral extension of  $R$  (choose the integral closure), so the lemma above (5.7.2.13) tells us the functor is essentially surjective.

Now we construct an inverse functor,  $S_*[t^{-1}]$  maps to  $T^a$ , where  $T$  is the integral closure of  $R$  in  $S_*[t^{-1}]$ . By lemma (5.7.2.15) below,  $S$  is almost perfect. So  $S_*$  is  $t$ -torsion-free, as if  $t^c f = 0$ , then  $t^{\frac{c}{p^n}} f = 0$  for all  $n$ , so  $f = 0 \in (S_*)_* = S_*$ . So now  $S_* \subset S_*[t^{-1}]$ . Clearly  $T$  is also perfect and  $t$ -torsion-free. So  $R \rightarrow T$  is an integral extension that is identified with  $R[t^{-1}] \rightarrow S_*[t^{-1}]$  after inversion of  $t$ .

To show that  $T^a = S$ , it suffices to show  $T_* = S_*$ . for  $f \in T$ ,  $f^{\mathbb{N}}$  spans a finite module of  $T[t^{-1}] = S_*[t^{-1}]$ , so  $t^c f^{\mathbb{N}} \subset S_*$ , then by perfectness,  $f \in (S_*)_* = S_*$ , so  $T_* \in S_*$ . Conversely, if  $g \in S_*$ , then  $tg^{\mathbb{N}}$  lies in a f.g.  $R$ -module of  $S_*[t^{-1}]$ , by almost f.g.. So  $t^c g^{\mathbb{N}} \subset T$ , and then by perfectness  $g \in T_*$ .  $\square$

**Lemma (5.7.2.15)**. Almost finite étale map of rings of char  $p$  is almost relatively perfect.  $\lrcorner$

*Proof:* Cf. [Bhatt notes on Perfectoid Spaces P28].  $\square$

### 3 Almost Mathematics on Perfectoid Fields

**Prop. (5.7.3.1)** [Almost Elements]. If  $K \in \text{PerfdField}$ ,  $R = K^0$ ,  $I = K^{00}$  and  $M \in \text{Mod}_R$ , then

- If  $M$  is torsion-free, then  $M_* = \{m \in M \otimes_{K^0} K \mid Im \in M\} = \{m \in M \otimes_{K^0} K \mid t^{\frac{1}{p^n}} m \in M\}$ , by (12.2.8.12) and (5.7.2.2).
- $I_* = R_* = R$ . More generally, for an ideal  $J \subset R$ , let  $c = \sup\{|x| \mid x \in J\}$ , then  $J_* = \{a \in K, |a| \leq c\}$ .

$\lrcorner$

**Prop. (5.7.3.2).** Let  $K$  be a perfectoid field with a pseudo-uniformizer  $\varpi$ . If  $\alpha : M \rightarrow N$  is an almost surjective map of  $K^0$ -algebras that  $M$  is  $\pi$ -adically separated and  $N$  is  $\pi$ -torsion-free that  $\alpha \bmod \pi$  is an almost isomorphism, then  $\alpha$  is an almost isomorphism.  $\square$

*Proof:* We may replace  $N$  with the image of  $\alpha$  as to assume  $\alpha$  is surjective. Now if  $L = \ker \alpha$ , then the  $\pi$ -torsion-freeness of  $N$  shows  $L/\pi$  is the kernel of  $(\alpha \bmod \pi)$ , and  $L/\pi$  is almost zero, thus  $L$  is almost  $\pi$ -divisible, but it is also  $\pi$ -separated, thus it is almost zero (using  $t \sum_i \frac{1}{p^{a_i}} m \in \cap_n t^n L$ ).  $\square$

**Prop. (5.7.3.3) [Almostification and Completeness].** Let  $K$  be a perfectoid field with a pseudo uniformizer  $t$  and  $R = K^0, I = K^{00}$ , let  $M \in \text{Mod}_R^a$ , then:

- $M$  is almost flat iff  $M_*$  is  $R$ -flat iff  $M_!$  is  $R$ -flat.
- Assume  $M$  is almost flat, then  $M$  is  $t$ -adically complete iff  $M_*$  does.
- Assume  $M$  is almost flat, then for each  $f \in K^0$ ,  $fM_* \cong (fM)_*$ , and  $M_*/fM_* \subset (M/fM)_*$ . And for any  $\varepsilon \in I$ , the image of  $(M/f\varepsilon M)_*$  and  $M_*/fM_*$  in  $(M/fM)_*$  are identical.

$\square$

*Proof:* 1:  $R$  is a valuation ring, so  $M_*$  is  $R$ -flat iff  $M_*[t]$  is flat by (5.4.1.12), as  $t$  is a pseudo uniformizer. As  $(-)_*$  is left exact,  $M_*[t] = (M[t])_*$ , so if  $M$  is almost flat, then  $M[t] = 0$  as  $t$  is nonzero-divisor, so  $M_*$  is  $R$ -flat. The converse is true as  $M = (M_*)^a$ , and the tensor is compatible.

For  $(-)_!$ , this follows from the observation that  $(-)_!$  and  $(-)^a$  are both exact and commute with tensor products, and notice  $M_! \otimes N = (M \otimes N^a)_!$ .

2: As  $(-)^a$  commutes with all limits and colimits, if  $M_*$  is  $t$ -adically complete then so does  $M = (M_*)^a$ . Conversely, if  $M$  is  $R$ -flat and  $t$ -adically complete, then  $M_!, M_*$  are also  $R$ -flat, and consider the commutative diagram:

$$\begin{array}{ccc} M_! & \xrightarrow{a} & \lim(M/t^n M)_! = \lim M_!/t^n M_! = \widehat{M_!} \\ \downarrow d & & \downarrow b \\ M_* & \xrightarrow{c} & \lim(M/t^n M)_* \end{array}$$

then  $d$  is almost isomorphism by (5.7.1.2) and so does  $b$  because  $(-)^a$  commutes with all limits, and  $c$  is an isomorphism as  $(-)_*$  commutes with limits and  $M$  is  $t$ -adically complete. So  $a$  is also almost isomorphism.

Now notice  $M_!$  is flat hence  $t$ -torsion-free, so the kernel of  $a, d$  must be 0, with almost zero cokernels. Now (5.2.3.9) shows first  $M_!$  is complete and next  $M_*$  is complete.

3: Notice  $(fM_*)^a = fM$  as  $(-)^a$  is exact, so

$$(fM)_* = \text{Hom}(I, fM_*) = \{y \in M_*[t^{-1}] \mid Iy \subset fM_*\} = f\{y \in M_*[t^{-1}] \mid Iy \subset M_*\} = fM_*$$

and  $M_*/fM_* \subset (M/fM)_*$  follows from the left exactness of  $(-)_*$ .

For the last assertion, consider the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \xrightarrow{f} & M & \longrightarrow & M/fM \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & M/\varepsilon M & \xrightarrow{f} & M/f\varepsilon M & \longrightarrow & M/fM \longrightarrow 0 \end{array}$$

and apply  $(-)_* = \text{Hom}_{R^a}(R^a, -)$  and use (5.7.2.6), then

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_*/fM_* & \xrightarrow{a} & (M/fM)_* & \longrightarrow & \text{Ext}_{R^a}^1(R^a, M)[f] \longrightarrow 0 \\ & & \downarrow & & \parallel & & \downarrow c \\ 0 & \longrightarrow & (M/f\varepsilon M)_* & \xrightarrow{b} & (M/fM)_* & \longrightarrow & \text{Ext}_{R^a}^1(R^a, M/\varepsilon M) \end{array}$$

To show  $a, b$  has the same image, it suffices to show that  $c$  is injective. For this, it suffices to show  $\text{Ext}_{R^a}^1(R^a, M) \rightarrow \text{Ext}_{R^a}^1(R^a, M/\varepsilon M)$  is injective. Consider the exact sequence  $0 \rightarrow M \xrightarrow{\varepsilon} M \rightarrow M/\varepsilon M \rightarrow 0$ , it suffices to show that  $\varepsilon \text{Ext}_{R^a}^1(R^a, M) = 0$ , and this is obvious as  $\varepsilon \in I$  (5.7.2.6).  $\square$

**Prop. (5.7.3.4)[General Completeness and Almostification].** More generally, if  $J = (f_1, \dots, f_r) \subset R$  is a f.g. ideal, then an  $R^a$ -module  $M$  is  $J$ -adically complete iff  $M_*$  does.  $\lrcorner$

*Proof:* Cf.[Perfectoid Spaces Bhatt P32].  $\square$

### Banach Space

**Prop. (5.7.3.5)[Uniform Banach  $K$ -Algebra].** If  $K \in \text{PerfdField}$  with a pseudo uniformizer  $t$ , then the following categories are equivalent:

- The category of uniform Banach  $K$ -algebras.
- The category  $\mathcal{D}_{tic}$  of  $t$ -adically complete and  $t$ -torsionfree  $K^0$ -algebras  $A$  with  $A$  totally integrally closed in  $A[t^{-1}]$  (5.2.1.1).
- The category  $\mathcal{D}_{ic}$  of  $t$ -adically complete and  $t$ -torsionfree  $K^0$ -algebras that  $A$  is integrally closed in  $A[t^{-1}]$  and  $A = A_*$ .
- The category  $\mathcal{D}_{prc}$  of  $t$ -adically complete and  $t$ -torsionfree  $K^0$ -algebras that  $A$  is  $p$ -root closed in  $A[t^{-1}]$  and  $A = A_*$ .

$\lrcorner$

*Proof:* The last three are equivalent, because if  $A \in \mathcal{D}_{tic}$ , then  $A = A_*$  by (5.7.2.10), as  $K$  is perfect by (12.2.8.4). So  $\mathcal{D}_{tic} \subset \mathcal{D}_{ic} \subset \mathcal{D}_{prc}$ , so it suffices to show that  $\mathcal{D}_{prc} \subset \mathcal{D}_{tic}$ . Now for any  $f$  that  $f^{\mathbb{N}} \subset t^{-k}A$ , then  $t^k f^{p^n} \subset A$ , and  $A$  is  $p$ -root closed, so  $t^{\frac{k}{p^n}} f \subset A$  for all  $n$ , so  $f \in A_*$  (5.7.2.3), but  $A_* = A$ .

The equivalence of 1, 2 is general, by (19.1.6.8).  $\square$

**Prop. (5.7.3.6).** If  $K \in \text{PerfdField}$ , then the category of uniform Banach spaces has all colimits and limits.  $\lrcorner$

*Proof:* Cf.[Bhatt P38].  $\square$

## 5.8 Simplicial Commutative Algebras

Main references are [Model Categories and Simplicial Methods, Paul Goerss and Kristen Schemmerhorn], [Simplicial Commutative Rings, Mathew].

### 1 Simplicial Groups

**Prop. (5.8.1.1).** A morphism of simplicial groups, when regarded as simplicial set, is a Kan fibration iff

$$X \rightarrow \pi_0 X \otimes_{\pi_0(Y)} Y$$

is surjective. In particular, any simplicial group is a Kan complex.  $\lrcorner$

*Proof:* Cf.[Simplicial Homology Theory Jardine P12]  $\square$

**Def. (5.8.1.2) [Simplicial Modules].** A **simplicial module** over a simplicial ring  $R_\bullet$  is a map of simplicial map  $R_\bullet \times M_\bullet \rightarrow M_\bullet$  that is a ring action.  $\lrcorner$

### 2 Simplicial $R$ -Modules and Resolutions

**Def. (5.8.2.1) [Moore Complex].** Giving a simplicial object in an Abelian category, we can have a **Moore chain complex** with Čech-like differentials.  $\partial_n = \sum_1^n (-1)^i d_i$ . And we have  $\partial^2 = 0$ .  $\lrcorner$

*Proof:* Should use  $d_i d_j = d_{j-1} d_i$  for  $i < j$ .  $\square$

**Def. (5.8.2.2) [Normalized Moore complex of a simplicial  $R$ -Module].** The **normlized Moore complex** of a simplicial  $R$ -module  $M$  is the chain complex

$$NM : \cdots \rightarrow NM_n \xrightarrow{(-1)^n d_n} NM_{n-1} \rightarrow \cdots$$

where  $NM_n = \bigcap_{i=0}^{n-1} \ker(d_i) \in M_n$ . This is a chain complex because  $d_{n-1} d_n = d_{n-1} d_{n-1}$  is 0 on  $NM_n$ . In fact  $NM$  is preserved by all injections.

The **homotopy groups**  $\pi_*(M)$  of  $M$  is defined to be the homology of the normalization of  $M$ . And it can be shown that as a set  $\pi_n(M)$  is just the  $n$ -th homotopy group of the geometrization of the

The **degenerate complex** of a Moore complex  $DM$  is the chain complex that  $D_n = \sum_{i=0}^{n-1} s_i M_{n-1}$  is a sub chain complex of  $M$  by the relation of  $d_i, s_j$ .  $\lrcorner$

**Def. (5.8.2.3).** A morphism of simplicial Abelian groups is called a **weak equivalence** is it induces an isomorphism on the homotopy groups.  $\lrcorner$

**Prop. (5.8.2.4) [Differential Graded Structures].** For any simplicial commutative  $R$ -algebra  $A$ , the homotopy groups  $\pi_*(A)$  form a graded commutative  $R$ -ring.  $\lrcorner$

*Proof:* The group structure on  $\pi_*(A)$  is given by smash products. Cf.[Simplicial Commutative Algebras, Mathew, P2].  $\square$

**Cor. (5.8.2.5) [ $\pi_0$  is an Algebra].** If  $Y$  is a simplicial commutative  $R$ -algebra, then  $\pi_0(Y) = Y_0 / (\text{Im}(d_0 - d_1))$ , which is an algebra.  $\lrcorner$



*Proof:* It suffices to show  $\text{Im}(d_0 - d_1)$  is an ideal of  $Y_0$ : If  $a \in Y_0$ , then

$$(d_0 - d_1)(s_0(a)y) = d_0 s_0(a) d_0(y) - d_1 s_0(a) d_1(y) = a(d_0 - d_1)(y).$$

□

**Prop. (5.8.2.6).** if  $R_\bullet$  is a simplicial ring and  $M_\bullet$  is a simplicial  $R_\bullet$ -ring, then  $\pi_*(M)$  is a graded  $\pi_*(R)$ -module. ┘

**Prop. (5.8.2.7).** The simplicial homology of the Moore complex of the bar resolution  $BG$  of group homology with coefficient in  $R$  is just the group homology  $H_n(G, R)$  for the trivial module  $R$ . And it has the same homology with the geometrization  $|BG|$ . ┘

**Lemma (5.8.2.8).**  $A_* \cong NA_* \oplus DA_*$  as a complex,  $NA_*$ ,  $A_*$ ,  $(A/DA_*)$  are all homotopically equivalent. ┘

*Proof:* We define similarly  $N_k A_*$  and  $D_k A_*$  and induct on  $k$ , our conclusion is the case  $k = n - 1$ . When  $k = 0$ ,  $\text{Im } d_0 \oplus \ker s_0 A_n = A_n$  because  $d_0 s_0 = id_{n-1}$  thus  $A_{n-1} \xrightarrow{s_0} A_n$  is a split injection.

There are two split exact rows by simplicial relations:

$$\begin{array}{ccccccc} 0 & \longrightarrow & N_{k-1} A_{n-1} & \xrightarrow{s_k} & N_{k-1} A_n & \xrightarrow{1-s_k d_k} & N_k A_n \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A_{n-1}/D_{k-1} A_{n-1} & \xrightarrow{s_k} & A_n/D_{k-1} A_n & \longrightarrow & A_n/D_k A_n \longrightarrow 0 \end{array}$$

The first one split because it has a right section, the second one split because it has a left section. So by induction,  $N_k A_n \rightarrow A_n/D_k A_n$  is an isomorphism, thus  $N_k A_n \oplus D_k A_n = A_n$  because it splits.

For the homotopy equivalence, Cf.[Jardine P150]. □

**Prop. (5.8.2.9)[Dold-Kan Correspondence].** For  $R \in \text{Ring}$ , the normalized Moore complex(5.8.2.2) functor  $N$  gives an equivalence of categories:

$$N_* : s\text{Mod}_R \cong Ch_{\geq 0}(\mathcal{A}).$$

and the inverse is given by

$$(\sigma C_\bullet)_n = \bigoplus_{[n] \rightarrow [k]} C_k$$

and a morphism  $\sigma_n \rightarrow \sigma_m$  for a morphism  $[m] \rightarrow [n]$  is defined as follows: For  $[n] \rightarrow [k]$ , write  $[m] \rightarrow [n] \rightarrow [k]$  as  $[m] \rightarrow [r] \xrightarrow{\psi} [k]$  where  $\varphi$  is injective, thus maps  $a \in C_k$  in  $\sigma C_n$  to  $\psi^*(a) \in C_r$  in  $\sigma C_m$ , where  $\psi^*$  is zero unless  $\psi = d^n : \Delta[n-1] \rightarrow \Delta[n]$ . And homotopy groups and homology groups correspond via this equivalence, so does weak equivalences. ┘

*Proof:*  $\sigma(\mathcal{C}_\bullet)$  defines a simplicial Abelian group because of the uniqueness of the the canonical decomposition. There is a natural map from  $\sigma(NA)$  to  $\mathcal{A}$ .

Now the task is to show that  $\sigma(NA) \cong A$  and  $N(\sigma C) \cong C$ . We has  $N(\sigma C)_n = C_n$  because  $d^i C_n$  is 0 for  $i \neq n$  and the other components are all degeneracies thus are not in  $N(\sigma C)_n = C_n$  by(5.8.2.8).

Then we prove  $\sigma(NA) \cong A$ . It is a surjection by(5.8.2.8) and induction. For the injectivity, if  $(a_\varphi) \neq 0$  is mapped to 0,  $a_{\text{id}_n}$  is 0 by(5.8.2.8). And we choose an ordering on the  $\varphi : [n] \rightarrow [k]$  by dominating, and suppose  $\psi$  is a minimal one. Now choose a section  $\xi$  of  $\psi$  that  $\xi$  is the maximal section, thus  $\varphi \xi$  cannot be  $\text{id}_k$  for any other  $\varphi$ . Now by induction we have  $a_\psi = 0$ , contradiction. □

**Cor. (5.8.2.10) [Trivial Simplicial Algebra].** There is a functor from an  $R$ -algebra  $S$  to a trivial simplicial  $R$ -algebra  $s(S)$ , it is a fully faithful embedding and  $\pi_0$  is left adjoint to it.  $\lrcorner$

*Proof:* This is the adjointness of (4.10.1.12) under the equivalence  $\sigma$  (5.8.2.9).  $\square$

**Cor. (5.8.2.11) [Model Structure on  $s\text{Mod}_R$ ].** By (4.5.4.2) applied to the equivalence with  $Ch_{\geq 0}R$ ,  $s\text{Mod}_R$  has the structure of a model category where a morphism  $X \rightarrow Y$  is

- an equivalence if it is a weak equivalence,
- a fibration if  $NX_n \twoheadrightarrow NY_n$  for any  $n \geq 1$ .
- a cofibration if the maps of the degenerate diagrams is of the form

$$X_n \rightarrow Y_n = X_n \oplus \bigoplus_{\varphi: [n] \rightarrow [k]} \varphi^* P_k$$

compatible with the differential, and  $P_k$  are all projectives.  $\lrcorner$

**Prop. (5.8.2.12) [Fibrations].** A fibration of simplicial  $R$ -modules is a fibration iff it is a fibration of simplicial sets. Moreover, by (5.8.1.1), this is equivalent to  $X \rightarrow \pi_0 X \otimes_{\pi_0(Y)} Y$  is surjective.  $\lrcorner$

*Proof:*  $\square$

**Prop. (5.8.2.13) [Simplicial Model Structure].**  $s\text{Mod}_R$  admits a simplicial model category structure.  $\lrcorner$

*Proof:* Firstly  $s\text{Mod}_R$  is a simplicial category tensored and cotensored over  $\text{Set}_\Delta$  (4.2.2.7) by (4.6.1.6), then it suffices to show (4.5.5.3)  $?$ .  $\square$

**Prop. (5.8.2.14) [Model Structure on  $s\mathcal{C}\text{Ring}_R$ ].** By (4.5.4.2) applied to the forgetful functor to  $\text{Set}_\Delta$ , Let  $R$  be a commutative algebra, then the category of simplicial commutative algebras  $s\mathcal{C}\text{Alg}_R$  has a simplicial model category structure where a morphism is

- a weak equivalence if it is a weak equivalence of simplicial sets.
- a fibration if it is fibration of simplicial sets.

*Proof:*  $\square$

**Cor. (5.8.2.15).** For a ring map  $R \rightarrow S$ , the tensor product  $S \otimes_R -$  and forgetful functor form a Quillen adjunction between  $s\mathcal{C}\text{Ring}_R$  and  $s\mathcal{C}\text{Ring}_S$ .  $\lrcorner$

**Def. (5.8.2.16) [Free Morphisms].** A morphism of simplicial  $R$ -algebras is called **free** if it is  $s$ -free (4.6.1.5) on the a set of objects  $P_k$  where  $P_k$  are projective  $R$ -modules.  $\lrcorner$

**Prop. (5.8.2.17) [Cofibrations in  $s\text{Alg}_R$ ].** A morphism in  $s\text{Alg}_R$  is a cofibration iff it is a retraction of a free morphism (5.8.2.16). In particular, a cofibrant simplicial  $R$ -algebra is the symmetrization of a chain of projective  $R$ -modules.  $\lrcorner$

*Proof:* This follows from (4.5.4.2) (4.2.2.10) and notice the free morphisms just corresponds free commutative algebra applied to the attaching cell morphism in the category of sets.  $\square$

### Simplicial Resolutions

**Def. (5.8.2.18) [Simplicial Resolutions].**

- Let  $M \in \text{Mod}_R$ , a **simplicial resolution** of  $M$  is a cofibrant replacement of  $M$ , or equivalently, it is an augmented simplicial  $R$ -module  $X \rightarrow M$  that  $NX \rightarrow M$  is a projective resolution.
- Let  $M \in \text{CRing}_R$ , a **free resolution** of  $M$  is free cofibrant replacement of  $M$ , or equivalently, it is an augmented simplicial commutative  $R$ -algebras  $X \rightarrow M$  that  $NX \rightarrow M$  is a resolution of  $R$ -modules, and  $P_n$  are all projective (4.1.1.22) in  $\text{Alg}_R$ . Notice we can always choose a free resolution only only cofibrant resolution by (4.5.8.3) and Dold-Kan complex.

┘

**Def. (5.8.2.19) [Bar Resolution].** Let  $\mathcal{C}$  be a category and  $T : \mathcal{C} \rightarrow \mathcal{C}$  a monad, and  $X$  is an algebra over  $T$  (4.3.1.2), then we can form a simplicial  $T$ -algebras where  $B(T, X)_n = T^{n+1}X$  where the simplicial operators come from the action of  $T$  on itself and the action of  $T$  on  $X$ .

Then there is a simplicial morphism  $B(T, X) \rightarrow X$ , which is a simplicial homotopy.

┘

**Remark (5.8.2.20).** If  $\mathcal{C}$  is the category of sets,  $R$  is an algebra and  $T$  is a functor that sends a set  $S$  to  $R[S]$ , then a  $T$ -algebra is just an  $R$ -algebra, and the bar resolution is just the canonical resolution, and it is a cofibrant replacement, by (5.8.2.17).

┘

**Prop. (5.8.2.21).** If  $f, g : A^\bullet \rightarrow B^\bullet$  are two homotopic maps of cosimplicial Abelian groups, then  $f, g$  induces an isomorphism between their totalizations.

┘

*Proof:* Cf. [[Sta]019S].

□

### 3 Properties

**Def. (5.8.3.1).** If  $A$  is a ring and  $f_1, \dots, f_n \in A$ , the Koszul complex is defined to be

$$\text{Kos}(A, f_1, \dots, f_n) = A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z}.$$

We want to extend this definition to the case of simplicial commutative rings.

Now if  $A$  is a simplicial ring and  $f_1, \dots, f_n \in \pi_0(A)$ , let  $g_1, \dots, g_n$  in  $A_0$  lifting  $f_i$ , then we define

$$\text{Kos}(A, f_1, \dots, f_n) = A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z}.$$

Then we need to check that this is independent of the lifting: if there is another set of lifting  $h_i$ , because we have identities

$$A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z} \cong (A \otimes_{\mathbb{Z}[X_1, \dots, X_n]}^L \mathbb{Z}[X_1]) \otimes_{\mathbb{Z}[X]}^L \mathbb{Z}$$

it suffices to prove for  $n = 1$ . Then there is a  $\gamma \in A_1$  that  $d_0(\gamma) = g, d_1(\gamma) = h$ . Then we consider the evaluating maps

$$e_0, e_1 : \underline{\text{Hom}}(\Delta_1, A) \rightarrow A$$

are weak equivalences ?, and then the maps

$$e_0 : \underline{\text{Hom}}(\Delta_1, A) \otimes_{\gamma, \mathbb{Z}[X]}^L \mathbb{Z} \rightarrow A \otimes_{g, \mathbb{Z}[X]}^L \mathbb{Z}$$

$$e_1 : \underline{\text{Hom}}(\Delta_1, A) \otimes_{\gamma, \mathbb{Z}[X]}^L \mathbb{Z} \rightarrow A \otimes_{h, \mathbb{Z}[X]}^L \mathbb{Z}$$

are also weak equivalences, so we are done.

And if  $M$  is a simplicial  $A$ -module, then we define

$$Kos(M, f_1, \dots, f_n) = M \otimes_A^L Kos(A, f_1, \dots, f_n)$$

┘

**Def. (5.8.3.2) [Flatness].** A map  $A \rightarrow B$  of simplicial rings is called **(faithfully)flat** if  $\pi_0(B)$  is (faithfully)flat over  $\pi_0(A)$  and  $\pi_i(A) \otimes_{\pi_0(A)} \pi_0(B) \rightarrow \pi_i(B)$  is an isomorphism for any  $i$ . ┘

**Prop. (5.8.3.3).** Flatness is stable under base change. ┘

*Proof:* Cf.[Emerton note, completely flatness, P12]. □

**Prop. (5.8.3.4).** If  $\pi_0(A) \otimes_A^L M$  is (faithfully)flat over  $\pi_0(A)$ , then  $M$  is (faithfully)flat over  $A$ . ┘

*Proof:* Cf.[Emerton note, completely flatness, P12]. □

**Def. (5.8.3.5).** If  $A$  is a simplicial ring and  $M$  is a simplicial  $A$ -module, if  $I = (f_1, \dots, f_n)$  is an ideal of  $\pi_0(A)$ , then  $M$  is called  **$I$ -completely flat** over  $A$  if  $Kos(M, f_1, \dots, f_n)$  is flat over  $Kos(A, f_1, \dots, f_n)$ .

Clearly if  $M$  is  $A$ -flat, then it is  $I$ -completely flat. ┘

**Prop. (5.8.3.6) [Flatness and Derived Completion].** If  $A$  is a simplicial ring,  $M$  is a flat simplicial  $A$ -module, and  $I = (f_1, \dots, f_n)$  is an ideal of  $\pi_0(A)$ , then its derived  $I$ -completion  $\widehat{M} = \text{holim}_n Kos(M, f_1^N, \dots, f_n^N)$  is  $I$ -completely flat. The proof is similar to that of (5.9.7.4). ┘

**Prop. (5.8.3.7) [Relative Regular Sequence].** If  $A \rightarrow B$  is a map of simplicial rings, then  $x_1, x_2, \dots, x_n \in \pi_0(B)$  is called **regular with respect to  $A_\bullet$**  if

$$A \rightarrow Kos(B, x_1, \dots, x_n)$$

is flat. And if  $I = (f_1, \dots, f_m) \in \pi_0(A)$ , then it is called  **$I$ -completely regular** if

$$Kos(A, f_1, \dots, f_m) \rightarrow Kos(A, f_1, \dots, f_m) \otimes_A^L Kos(B, x_1, \dots, x_n) = Kos(B, f_1, \dots, f_m, x_1, \dots, x_n)$$

is flat. In particular, regular relative to  $A_\bullet$  implies  $I$ -completely regular relative to  $A_\bullet$ . ┘

**Prop. (5.8.3.8).** If  $(f_1, \dots, f_{r-1}, f)$  is regular w.r.t.  $A$  and  $(f_1, \dots, f_{r-1}, g)$  is regular w.r.t.  $A$ , then  $(f_1, \dots, f_{r-1}, fg)$  is also regular w.r.t.  $A$ . In particular, for any  $n_i > 0$ ,  $f_1^{n_1}, \dots, f_r^{n_r}$  is also regular w.r.t.  $A$ . ┘

*Proof:* Similarly as in (5.4.4.6), we have a distinguished triangle

$$Kos(M, f_1, \dots, f_{r-1}, f) \rightarrow Kos(M, f_1, \dots, f_{r-1}, fg) \rightarrow Kos(M, f_1, \dots, g).$$

Then we can use induction. □

**Prop. (5.8.3.9) [Regular and Derived Completion].** If  $x_1, \dots, x_n \in \pi_0(B)$  is regular w.r.t.  $A$ , then they are  $I$ -completely regular w.r.t.  $A$  in the derived  $I$ -completion  $\widehat{B}$ . The proof is similar to that of (5.8.3.6). ┘

#### 4 Non-Abelian Derived Functors

**Def. (5.8.4.1) [ $\infty$ -Category of Chain Complexes].** Let  $\mathcal{A}$  be an additive category, then  $\mathrm{Ch}(\mathcal{A})$  is enriched over  $\mathrm{Ch}(\mathcal{A}b)$ , and the Dold-Kan correspondence can be made into a right-lax monoidal functor by [?](#) [S. Schwede and B. Shipley. Equivalences of monoidal model categories.], so  $N_*(\mathrm{Ch}(\mathcal{A}))$  is enriched over  $s\mathcal{A}b$ , which consists of Kan complexes [\(5.8.1.1\)](#). Thus  $N_*(\mathrm{Ch}(\mathcal{A}))$  is Bergner-fibrant, and we can define the  $\infty$ -category of chain complexes

$$\mathrm{Ch}_\infty(\mathcal{A}) = N_\Delta(N_*(\mathrm{Ch}(\mathcal{A}))),$$

which is an  $\infty$ -category by [\(4.6.4.9\)](#). ⌋

**Prop. (5.8.4.2) [Left Derived Functor].** Let  $F : \mathrm{Poly}_A \rightarrow \mathcal{C}$  be a functor where  $\mathcal{C}$  is any  $\infty$ -category admitting all colimits (e.g.  $D_\infty(\mathcal{A}b)$ ), then there exists a left Kan extension  $LF$  of  $F$  along  $\mathrm{Poly}_A \subset \mathcal{C}\mathrm{Ring}_A$  that

- $LF$  commutes with filtered colimit.
- $LF$  commutes with geometric realization of simplicial resolutions: given  $B \in \mathcal{C}\mathrm{Ring}_A$  and a simplicial resolution  $P_\bullet \rightarrow B$  by  $A$ -algebras, the geometric realization  $|LF(P_\bullet)|$  is equivalent to  $LF(B)$ .

which is called the **left derived functor** of  $F$ . ⌋

*Proof:* Cf. [Bhatt, Prism, 7.1.2]. □

## 5.9 Derived Commutative Algebras

References are [Sta].

[Sta]Chap15 contains many beautiful results working on the derived category of rings, and is used heavily in Scholze's Thesis.

The basic construction of  $R\text{tensor}$  and  $R\text{Hom}$  should be redone at the level of ringed sites, Cf. [Sta]Chap21.

This section is obsolete and should be redone in the language of  $\infty$ -categories.

### 1 Basics

**Prop. (5.9.1.1) [Product in  $D(R)$ ].** Let  $R$  be a ring and  $K_n \in D(R)$ , then the product in  $D(R)$  of  $K_n$  is given by  $\prod I_n$ , where  $I_n$  are  $K$ -injective resolutions of  $K_n$ .  $\lrcorner$

*Proof:* This is immediate from (4.10.2.7).  $\square$

**Def. (5.9.1.2) [Homotopy Fiber Square].** A square of Abelian groups is called a **homotopy fiber square** if it is a homotopy fiber square in the derived category, or equivalently, the kernel of the two rows (or the two columns) are isomorphic.

This notion is identical to the notion of pullback square when the rows or the columns are surjective.  $\textcolor{red}{?}$   $\lrcorner$

**Prop. (5.9.1.3) [Bockstein Differential].** Let  $I$  be an invertible ideal of  $A$ , for any  $M^\bullet \in D(A)$ , we use the Breuil-Kisin Twist notation (27.6.5.1) and consider the exact triangle

$$M^\bullet \otimes_A^L A/I\{i+1\} \rightarrow M^\bullet \otimes_A^L I^i/I^{i+2} \rightarrow M^\bullet \otimes_A^L A/I\{i\}$$

obtained from the exact triangle

$$I^{n+1}/I^{n+2} \rightarrow I^n/I^{n+2} \rightarrow I^n/I^{n+1}$$

tensoring  $\mathcal{O}_\Delta$ . Then we get a Bockstein differential

$$\beta^n : H^n(M^\bullet \otimes_A^L A/I\{n\}) \rightarrow H^{n+1}(M^\bullet \otimes_A^L A/I\{n+1\})$$

Then these maps satisfy  $\beta^{n+1} \circ \beta^n = 0$ .  $\lrcorner$

*Proof:* Consider the morphism of distinguished triangles:

$$\begin{array}{ccccc} I^{n+1}/I^{n+3} & \longrightarrow & I^n/I^{n+3} & \longrightarrow & I^{n+1}/I^{n+1} \\ \downarrow & & \downarrow & & \downarrow \\ I^{n+1}/I^{n+2} & \longrightarrow & I^n/I^{n+2} & \longrightarrow & I^{n+1}/I^{n+2} \end{array}$$

then we see for any  $M^\bullet \in D(A)$ ,  $\beta^n$  factors as

$$H^n(M^\bullet \otimes_A^L I^n/I^{n+1}) \rightarrow H^{n+1}(M^\bullet \otimes_A^L I^{n+1}/I^{n+3}) \rightarrow H^{n+1}(M^\bullet \otimes_A^L I^{n+1}/I^{n+2})$$

and also we consider the distinguished triangle

$$I^{n+2}/I^{n+3} \rightarrow I^{n+1}/I^{n+3} \rightarrow I^{n+1}/I^{n+2}$$

to see that the composition

$$H^{n+1}(M^\bullet \otimes_A^L I^{n+1}/I^{n+3}) \rightarrow H^{n+1}(M^\bullet \otimes_A^L I^{n+1}/I^{n+2}) \xrightarrow{\beta^{n+1}} H^{n+2}(M^\bullet \otimes_A^L I^{n+2}/I^{n+3})$$

is 0, and this two observation gives the result.  $\square$

### Injective Amplitude

**Prop. (5.9.1.4) [Injective Amplitude].** For  $K \in D(A)$ , the following are equivalent:

- $K$  has finite amplitude in  $[a, b]$ .
- $\text{Ext}^i(N, K) = 0$  for any  $N \in D^0(A)$  and  $i \notin [a, b]$ .
- $\text{Ext}^i(A/I, K) = 0$  for any ideal  $I$  of  $A$ .

┘

*Proof:*  $1 \rightarrow 2 \rightarrow 3$  is clear. For  $3 \rightarrow 1$ : Notice  $\text{Ext}^n(A, K) = H^n(K)$ ,  $H^i(K) = 0$  for any  $i \notin [a, b]$ . Then  $K$  is represented by a complex

$$0 \rightarrow I^a \rightarrow I^a \rightarrow I^{a+1} \rightarrow \dots \rightarrow I^b \rightarrow \dots$$

Let  $J = \ker(I^b \rightarrow I^{b+1})$ , then  $K$  is also represented by

$$0 \rightarrow I^a \rightarrow I^a \rightarrow I^{a+1} \rightarrow \dots \rightarrow I^b \rightarrow J \rightarrow 0.$$

Let  $K' = (I^a \rightarrow \dots \rightarrow I^b) \in D(A)$ , then there is an distinguished triangle

$$J[-b] \rightarrow K \rightarrow K' \rightarrow J[1-b],$$

which induces an exact sequence  $\text{Ext}^b(R/I, K') \rightarrow \text{Ext}^1(R/I, J) \rightarrow \text{Ext}^{b+1}(R/I, K)$  for any ideal  $I \subset A$ . Then by  $1 \rightarrow 2$ ,  $\text{Ext}^1(R/I, J) = 0$ , implying  $J$  is injective.  $\square$

**Prop. (5.9.1.5) [Dedekind Domain].** Let  $R$  be a Dedekind domain, then every ideal  $I$  is finite torsion-free thus projective over  $R$ , so every  $R$ -module has injective dimension  $\leq 1$ . In particular,  $\text{Ext}^i(M, N) = 0$  for any  $i \geq 2$  and  $M, N \in \text{Mod}_R$ . In particular, by (4.10.3.26), any  $K \in D^+(R)$  is isomorphic to a direct sum of their cohomology groups.  $\square$

**Prop. (5.9.1.6).** Let  $(R, \mathfrak{m}, k)$  be a Noetherian local ring and  $K \in D^+(R)$  have finite cohomology modules, then  $K$  has finite injective dimension iff  $\text{Ext}_R^i(k, K) = 0$  for  $i$  large.  $\square$

*Proof:* Cf. [Sta]0AVJ.  $\square$

## 2 Derived Tensor and Tor

**Prop. (5.9.2.1) [Differential Graded Structure].** If  $K$  is a commutative  $A$ -algebra object in  $D(A)$  in the monoidal structure defined in (7.3.3.8), then  $\oplus_{n \geq 0} H^n(K^\bullet)$  carries a natural graded commutative  $A$ -algebra structure.  $\square$

*Proof:* Compare with (5.8.2.4).

We may replace  $K$  by a  $K$ -flat resolutions  $L^1$  (7.3.3.3) that the algebra structure map for  $K$  is represented by a morphism  $L_1^\bullet \otimes_A L_1^\bullet \rightarrow L^2$  of complexes where  $L^2$  is a complex quasi-isomorphic to  $K$ , hence the graded  $A$ -algebra structure is clear by (4.8.10.2), and it is commutative by (4.8.10.1). (How to check the structure is uniquely determined?).  $\square$

**Tor**

**Def. (5.9.2.2) [Tor].** Let  $M, N$  be  $A$ -module, then the torsion group  $\text{Tor}_n^A(M, N)$  is defined to be  $H^n(M \otimes_{\mathcal{O}}^L N)$ , compatible with the definition in (7.3.3.11).  $\lrcorner$

**Def. (5.9.2.3) [Torsion Group].** Let  $A$  be a commutative ring,  $B$  an  $A$ -algebra and  $I$  be an ideal, then the  $I$ -torsion of  $B$  is defined to be  $\text{Tor}_1^A(A/I, B)$ , denoted by  $B[I]$ . In case  $I = (f)$ , it can be checked that  $B[f]$  is the set of elements of  $B$  that killed by  $f$ .

Also we denote  $B[I^\infty] = \text{colim}_{n \rightarrow \infty} B[I^n]$ . And  $B$  is said to have **bounded  $f$ -torsion** iff  $B[f^\infty] = B[f^n]$  for some  $n$ .  $\lrcorner$

**Prop. (5.9.2.4).** If  $A$  is a commutative ring with bounded  $I$ -torsions and  $B$  is a flat  $A$ -module, then  $B$  also has bounded  $I$ -torsions. (Because tensoring  $B$  is exact).  $\lrcorner$

**Prop. (5.9.2.5) [Balancing Tor].** In  $\mathcal{C}\text{Ring}$ ,  $\text{Tor}_n(A, B) = \text{Tor}_n(B, A)$ . This can be seen using spectral sequence of the double complex of flat resolutions of  $A$  and  $B$ . Similarly, we have two definitions of  $\text{Ext}^i(M, N)$  are compatible.  $\lrcorner$

*Proof:*  $\square$

**Prop. (5.9.2.6) [Base Change].** For a ring extension  $R \rightarrow S$ , using projective resolution and spectral sequence, there is a first quadrant homology spectral sequence:

$$E_{pq}^2 = \text{Tor}_p^S(\text{Tor}_q^R(A, S), B) \Rightarrow \text{Tor}_{p+q}^R(A, B).$$

Similarly, for  $\text{Ext}$ ,

$$E_2^{pq} = \text{Ext}_S^p(A, \text{Ext}_R^q(S, B)) \Rightarrow \text{Ext}_R^{p+q}(A, B).$$

$\lrcorner$

**Prop. (5.9.2.7) [Universal Coefficient Theorem].** Let  $P$  be a free  $R$ -module so  $d(P_n)$  are all flat, then  $Z(P_n)$  are also flat and

$$0 \rightarrow d(P_{n+1}) \rightarrow Z_n \rightarrow H_n \rightarrow 0$$

is a free resolution. we have an exact sequence:

$$0 \rightarrow H_n(P) \otimes_R M \rightarrow H_n(P \otimes_R M) \rightarrow \text{Tor}_1^R(H_{n-1}(P), M).$$

$$0 \rightarrow \text{Ext}_R^1(H_{n-1}(P), M) \rightarrow H^n(\text{Hom}_R(P, M)) \rightarrow \text{Hom}(H_n(P), M) \rightarrow 0$$

and these exact sequences non-canonically split because  $Z_n$  is a direct summand of  $P_n$ , thus  $Z_n \otimes M$  is a direct summand of  $P_n \otimes M$  and a fortiori  $Z_n(P_n \otimes M)$ . so  $H_n(P) \otimes M$  is a direct summand of  $H_n(P \otimes_R M)$ .  $\lrcorner$

**Internal Hom and Derived Tensor**

**Prop. (5.9.2.8).** If  $R$  is a ring and  $K, L, M \in D(R)$ , then

$$R\text{Hom}_R(K, R\text{Hom}(L, M)) = R\text{Hom}_R(K \otimes_R^L L, M).$$

$\lrcorner$

*Proof:* This is a special case of (7.3.3.29).  $\square$



**Cor. (5.9.2.9).**

$$\mathrm{Hom}_{D(R)}(K, R\mathrm{Hom}(L, M)) = \mathrm{Hom}_{D(R)}(K \otimes_R^L L, M),$$

i.e., derived tensor is left adjoint to internal Hom.  $\lrcorner$

*Proof:* This follows from taking  $H^0$ , by (4.10.3.18).  $\square$

**Prop. (5.9.2.10) [Derived Base Change Adjunction].** For a ring map  $R \rightarrow S$  and any  $L \subset D(R)$ ,  $M \subset D(S)$ , there is an isomorphism

$$\mathrm{Hom}_R(L, M) \cong \mathrm{Hom}_S(L \otimes_R^L S, M)$$

$\lrcorner$

*Proof:* This follows from (7.3.3.16).  $\square$

### 3 Rlim

**Def. (5.9.3.1) [Rlim].** Rlim is the derived limit (4.10.6.1) in  $D(\mathcal{A}b)$  restricted to the inverse systems consisting of discrete complexes.  $\lrcorner$

**Lemma (5.9.3.2).** The set of Mittag-Leffler Complexes in  $\mathcal{A}b(\mathbb{N})$  is adapted for  $R\mathrm{lim}$ .  $\lrcorner$

*Proof:* Firstly, for any complex  $(A_n)$ , we can associate to it the complex  $(B_n)$  where  $B_n = A_n \oplus A_{n-1} \oplus \dots \oplus A_1$ , then  $(B_n)$  is a Mittag-Leffler complex and  $(A_n) \hookrightarrow (B_n)$ . So ML complexes are sufficiently large.

Now for any exact sequence of complexes  $0 \rightarrow (A_n) \rightarrow (B_n) \rightarrow (C_n) \rightarrow 0$ , if  $A_n$  is ML, then  $\lim B_i \rightarrow \lim C_i$  is surjective: for an element  $(c_i) \in \lim C_i$ , let  $E_i = \pi_i^{-1}(c_i) \in B_i$ , then  $(E_i)$  is an inverse system of nonempty sets, and it suffices to show  $(E_i)$  is ML, because then (4.1.1.46) will show there is a element  $(e_i) \in \lim E_i \subset \lim B_i$  that maps to  $(c_i)$ .

For this, Cf. [Sta]0598.  $\square$

**Prop. (5.9.3.3) [Rlim].**

- If  $(A_n)$  is Mittag-Leffler, then  $R^1 \mathrm{lim}((A_n)) = 0$ .
- $R\mathrm{lim}((A_n))$  is represented by the complex in degree 0, 1:

$$\prod_n A_n \rightarrow \prod_n A_n : (x_n) \mapsto (x_n - f_{n+1}(x_{n+1}))$$

.

- for any  $(A_n) \in \mathcal{A}b(\mathbb{N})$  we have  $R^p \mathrm{lim}((A_n)) = 0$  for  $p > 1$ .  $\lrcorner$

*Proof:* 1 follows from (5.9.3.2) and (4.10.3.2).

2, 3: We use (4.10.3.2) again. Notice the complex  $(B_n)$  where  $B_n = A_n \oplus A_{n-1} \oplus \dots \oplus A_1$  and the complex  $(C_n)$  where  $C_n = A_{n-1} \oplus A_{n-2} \oplus \dots \oplus A_1$  form an exact sequence of complexes

$$0 \rightarrow (A_n) \rightarrow (B_n) \rightarrow (C_n) \rightarrow 0$$

where  $B_n \rightarrow C_n : (x_i) \mapsto (x_i - f_{i+1}(x_{i+1}))$ , and  $(B_n), (C_n)$  are both ML, so we are done.  $\square$

## 4 Lifting Complexes

**Prop. (5.9.4.1) [Lifting Projective Complex Along Thickening].** Let  $R$  be a ring and  $I$  be a nilpotent ideal, and  $K \in D(R)$ . Now if  $K \otimes_R^L R/I$  is represented by a bounded above complex of projective  $R/I$ -modules, then there is a complex  $P$  of bounded above complex of projective  $R$ -modules that  $P \cong K \in D(R)$ , and  $P \otimes_R R/I \cong E$ .  $\lrcorner$

*Proof:* Cf. [Sta]09AR.  $\square$

## 5 Pseudo-Coherent and Perfect Modules

**Def. (5.9.5.1) [Pseudo-Coherent Modules].** Let  $R$  be a ring,  $m \in \mathbb{Z}$ , then  $K \in D(R)$  is called an  **$m$ -pseudo-coherent module** iff there exists a complex  $E^\bullet \in K^b(R)$  and a morphism  $\alpha : E^\bullet \rightarrow K^\bullet$  where  $K^\bullet$  represents  $K$ , s.t.  $H^i(\alpha)$  is an isomorphism for  $i > m$ , and surjective for  $i = m$ .

$K \in D(R)$  is called a **pseudo-coherent module** if it is represented by a bounded above complex of finite free  $R$ -modules.  $\lrcorner$

**Prop. (5.9.5.2).** If  $A$  is Noetherian and  $C^\bullet$  is a complex of  $A$ -modules bounded above that every cohomology group  $H^i$  is a finite  $A$ -module, then there is a complex  $L^\bullet$  of finite free  $A$ -modules, that  $g : L^\bullet \rightarrow C^\bullet$  is a quasi-isomorphism.

Moreover, if  $C^i$  are all flat  $A$ -modules, then  $L^\bullet \otimes_A M \rightarrow C^\bullet \otimes_A M$  is quasi-isomorphism for every  $M$ .  $\lrcorner$

*Proof:*  $C^\bullet$  is bounded above so we choose  $L^n = 0$ , and use induction to construct  $L^n$  that  $H^i(L) \rightarrow H^i(C)$  is isomorphism for  $i > n+1$  and surjection for  $i = n+1$ . For this, choose a generator  $x_1, \dots, x_r$  of  $H^n(C)$  in  $Z^n(C)$ , and let  $y_{r+1}, \dots, y_s$  be a generator of  $g^{-1}(B^{n+1}(C))$  (Noetherian used), and let  $g(y_i) = dx_i$  for  $x_i \in C^n$ .

Now let  $L^n$  be freely generated by  $e_1, \dots, e_s$  and  $de_i = 0$  for  $i \leq r$  and  $de_i = y_i$  for  $i > r$ , and let  $g : L^n \rightarrow C^n$  be  $ge_i = x_i$ . Then it can be verified to be a quasi-isomorphism.

If  $C^i$  are all flat, we check isomorphism for all f.g. modules  $M$ , because  $\otimes$  and cohomology all commutes with direct limits. Use induction, for  $n$  large, both are 0, and if we write  $0 \rightarrow R \rightarrow F \rightarrow M \rightarrow 0$ , for  $F$  finite free, then there is a commutative diagram of long exact sequences, and for  $F$ ,  $H^i$  are obviously isomorphism, so I can use five lemma.  $\square$

**Def. (5.9.5.3) [Perfect Complexes of Modules].** Let  $R$  be a ring, then  $K \in D(R)$  is called a **perfect module** if  $K$  is quasi-isomorphic to a bounded complex of finite projective  $R$ -modules. An  $R$ -module  $M$  is called perfect iff  $M[0]$  is perfect.  $\lrcorner$

**Prop. (5.9.5.4) [Perfectness and Pseudo-Coherence].** An object  $K \in D(R)$  is perfect iff it is pseudo-coherent and has finite Tor amplitude.  $\lrcorner$

*Proof:* Cf. [Sta]0658.  $\square$

**Prop. (5.9.5.5).** If  $R$  is a regular ring of finite dimension, then an object  $K \in D(R)$  is perfect iff  $K \in D^b(R)$  and each  $H^i(K)$  is a finite  $R$ -module.  $\lrcorner$

*Proof:* Cf. [Sta]066Z.  $\square$

**Prop. (5.9.5.6) [Duality of Perfect Complexes].** Let  $K$  be a perfect complex of  $D(A)$ , then the **dual complex**  $K^\vee = R\mathrm{Hom}(K, A)$  is also a perfect complex and  $(K^\vee)^\vee \cong K$ . Also, there is a functorial isomorphism

$$L \otimes_A^L K^\vee = R\mathrm{Hom}_A(K, L)$$

$\lrcorner$

*Proof:* Cf. [Sta]07VI. □

**Prop. (5.9.5.7).** If  $A$  is a ring and  $K_n$  is a system of perfect objects in  $D(A)$ , then for any  $E \in D(A)$ , there is an isomorphism

$$R\mathrm{Hom}_A(\mathrm{hocolim} K_n, E) \cong R\mathrm{lim} E \otimes_A^L K_n^\vee$$

┘

*Proof:* By (5.9.5.6),  $R\mathrm{lim} E \otimes_A^L K_n^\vee = R\mathrm{lim} R\mathrm{Hom}(K_n, E)$  which fits into a distinguished triangle

$$R\mathrm{lim} R\mathrm{Hom}(K_n, E) \rightarrow \prod \mathrm{Hom}(K_n, E) \rightarrow \prod \mathrm{Hom}(K_n, E)$$

. So it suffices to show that

$$\prod \mathrm{Hom}(K_n, E) \cong R\mathrm{Hom}_A(\oplus K_n, E).$$

This follows from Yoneda lemma and (7.3.3.28)(7.3.3.29). □

**Prop. (5.9.5.8) [Perfectness and Thickening].** If  $R$  is a ring,  $I \subset R$  is a nilpotent ideal, and  $K \in D(R)$ . If  $K \otimes_R^L R/I$  is perfect in  $D(R/I)$ , then  $K$  is perfect in  $R$ . Moreover, if  $K \otimes_R^L R/I = 0$ , then  $K = 0$ . ┘

*Proof:* Let  $\overline{P}^\bullet \cong K \otimes_R^L R/I$  where  $P$  is a complex of finite projective  $R/I$ -modules, then by (5.9.4.1) there is a complex of projective  $R$ -modules  $P$  that  $P/IP \cong \overline{P}$ . Then it follows from Nakayama that  $P$  is bounded. □

### Pseudo-Coherent Modules

**Def. (5.9.5.9) [Pseudo-Coherent Complexes of Modules].** Let  $R \in \mathcal{CAlg}$ ,  $K \in D(R)$  is called an  **$m$ -pseudo-coherent module** if there exists a perfect object  $E \in D(R)$  and a morphism  $E \rightarrow K$  that induces isomorphisms on  $H^i$  for  $i > m$  and surjection on  $H^m$ .

$K \in D(R)$  is called a **pseudo-coherent module** if it is represented by a bounded above complex of finite free  $R$ -modules. ┘

## 6 Derived Completeness

Cf. [Sta]Chap15.90.

**Def. (5.9.6.1).** For a ring  $A$ ,  $f \in A$  and a complex  $K \in D(A)$ , we denote by  $T(K, f)$  a derived limit of the system

$$\dots \rightarrow K \xrightarrow{f} K \xrightarrow{f} K$$

┘

**Prop. (5.9.6.2) [Properties of  $T(K, f)$ ].** For a ring  $A$ ,  $f \in A$  and  $K \in D(A)$ , the following are equivalent:

1.  $T(K, f) = 0$ .
2.  $R\mathrm{Hom}_A(A_f, K) = 0$ .
3.  $\mathrm{Ext}_A^n(A_f, K) = 0$  for all  $n$ .

4.  $\text{Hom}_{D(A)}(E, K) = 0$  for all  $E \in D(A_f)$ .
5. For any  $p \in \mathbb{Z}$ ,  $\text{Hom}_A(A_f, H^p(K)) = 0$  and  $\text{Ext}_A^1(A_f, H^p(K)) = 0$ .
6. For any  $p \in \mathbb{Z}$ ,  $T(H^p(K), f) = 0$ .

┘

*Proof:* 2, 3 is clearly equivalent.

$4 \rightarrow 3$  is clear, and for  $3 \rightarrow 4$ : Let  $I^\bullet$  be a complex representing  $K$ , then 3 says  $\text{Hom}_A(A_f, I^\bullet)$  is acyclic, and  $\text{Hom}_{D(A)}(E, K) = \text{Hom}_{K(A)}(E, I^\bullet) = \text{Hom}_{K(A_f)}(E, \text{Hom}_A(A_f, I^\bullet))$ . As  $\text{Hom}_A(A_f, I^\bullet)$  is both acyclic and  $K$ -injective(4.8.10.6), we get it is homotopic to 0 by(4.10.2.1), thus we get 4.

$1 \iff 3$ : There is a free resolution of  $A_f$  given by

$$0 \rightarrow \bigoplus_n A \rightarrow \bigoplus_n A \rightarrow A_f \rightarrow 0$$

where the first map is  $(a_n) \mapsto (a_n - fa_{n-1})$ , and the second map is  $(a_n) \mapsto \sum a_i/f^i$ . Applying  $\text{Hom}_A(-, I^\bullet)$ , we get a distinguished triangle

$$R\text{Hom}_A(A_f, K) \rightarrow \prod K \rightarrow \prod K.$$

So this shows  $R\text{Hom}_A(A_f, K)$  is just  $T(K, f)$ , so we get  $1 \iff 3$ .

$1 \iff 5 \iff 6$ : There is a spectral sequence convergence(choose a finite free resolution of  $A_f$  then rotate and use(4.10.7.10)):

$$E_2^{p,q} = \text{Ext}_A^q(A_f, H^p(K)) \Rightarrow \text{Ext}^{p+q}(A_f, K)$$

This spectral sequence degenerates at  $E_2$  because  $A_f$  has a length 1 resolution by free  $A$ -modules hence the  $E_2$  page has only 2 rows. So there is an exact sequence

$$0 \rightarrow \text{Ext}_A^1(A_f, H^{p-1}(K)) \rightarrow \text{Ext}_A^1(A_f, K) \rightarrow \text{Hom}_A(A_f, H^p(K)) \rightarrow 0.$$

Then we are done. □

**Lemma (5.9.6.3).** Let  $A$  be a ring,  $K \in D(A)$ , then the set  $I$  of  $f$  that  $T(K, f) = 0$  is a radical ideal of  $A$ . ┘

*Proof:* If  $T(K, f) = 0$  and  $g \in A$ , then  $A_{gf}$  is a  $A_f$ -module, then

$$\text{Ext}_A^n(A_{gf}, K) = \text{Hom}_{D(A)}(A_{gf}[-n], K) \text{ (7.3.3.32)} = 0$$

by(5.9.6.2) item4. Then  $T(K, gf) = 0$  by(5.9.6.2) again. And if  $f, g \in I$ , there is an exact sequence

$$0 \rightarrow A_{f+g} \rightarrow A_{f(f+g)} \oplus A_{g(f+g)} \rightarrow A_{fg(f+g)} \rightarrow 0$$

by(5.4.2.3) and a easy check that the last term is surjective. Then from the long exact sequence of  $\text{Ext}$ , we get  $\text{Ext}^n(A_{f+g}, K) = 0$  for any  $n$ . Finally if  $f^n \in I$ , then  $f \in I$ , because  $A_f = A_{f^n}$ . □

**Def. (5.9.6.4) [Derived Completeness].** Let  $A$  be a ring,  $K \in D(A)$ ,  $I$  is an ideal of  $A$ , then  $K$  is said to be **derived complete** w.r.t  $I$  if  $T(K, f) = 0$  for any  $f \in A$ . Let  $D_{\text{comp}}(A, I)$  denote the subcategory consisting of derived  $I$ -complete objects in  $D(A)$ .

Let  $M$  be an  $A$ -module, then  $M$  is called **derived complete** w.r.t  $I$  if  $M[0] \in D(A)$  is derived complete w.r.t  $I$ . ┘

**Prop. (5.9.6.5).** A  $\aleph_0$ -filtered colimit of derived  $I$ -complete rings is also derived  $I$ -complete.  $\lrcorner$

*Proof:* Cf.[Bhatt, Prism, 5.4.3].  $\square$

**Prop. (5.9.6.6) [Complete and Derived Complete].** Let  $A$  be a ring and  $I$  be an ideal,  $M$  an  $A$ -module, then

- If  $M$  is  $I$ -adically complete, then  $T(M, f) = 0$  for any  $f \in I$ .
- If  $T(M, f) = 0$  for all  $f \in I$  and  $I$  is f.g., then  $M \rightarrow \lim M/I^n M$  is surjective.

In particular, if  $I$  is f.g.,  $M$  is  $I$ -adically complete iff  $M$  is derived  $I$ -adically complete and  $\cap I^n M = 0$ .

In particular, when  $M$  is f.g. over  $A$  Noetherian and  $I \subset \text{rad}(A)$ , derived  $I$ -complete is equivalent to  $I$ -complete(5.2.2.14).  $\lrcorner$

*Proof:* If  $M$  is  $I$ -adically complete, by(5.9.6.2), it suffices to show that  $\text{Hom}(A_f, M) = 0$  and  $\text{Ext}^1(A_f, M) = 0$ . But  $M = \varprojlim_n M/I^n M$ , and  $\text{Hom}(A_f, M/I^n M) = 0$ , because  $f \in I$ . For  $\text{Ext}$ , use(4.10.3.24), for any extension

$$0 \rightarrow M \rightarrow E \rightarrow A_f \rightarrow 0,$$

chose arbitrary  $e_n$  that maps to  $1/f^n$ . Then  $\delta_n = fe_{n+1} - e_n \in M$ . We consider

$$e'_n = e_n + \delta_n + f\delta_{n+1} + f^2\delta_{n+2} + \dots,$$

which exist because  $M$  is  $I$ -adically complete. Then  $fe'_{n+1} = e'_n$ , so this gives a splitting of the extension.

Conversely, if  $I = (f_1, \dots, f_r)$  and  $T(M, f_i) = 0$  for any  $i$ , then by(5.2.3.13), we can assume  $I = (f)$ . Then consider the extension

$$0 \rightarrow M \rightarrow E \rightarrow A_f \rightarrow 0$$

where  $E = (M \oplus \bigoplus Ae_n)/(x_n - fe_{n+1} + e_n) \rightarrow A_f$  that maps  $M$  to 0 and  $e_n$  to  $1/f^n$ . This extension splits by(5.9.6.2) and(4.10.3.24), thus there is an element  $x + e_0$  that generate a copy of  $A_f$  in  $E$ .

But then  $x + e_0 = x - x_0 + fe_1 = x - x_0 - fx_1 + f^2e_2 \dots$ , which implies  $x - x_0 - fx_1 - f^2x_2 - \dots - f^{n-1}x_{n-1} \in f^n E + A_f$  for any  $n$ . Then  $x - x_0 - fx_1 - f^2x_2 - \dots - f^{n-1}x_{n-1} \in f^n M$ , because  $E = M \oplus A_f$ . Then we are done.  $\square$

**Prop. (5.9.6.7).** If  $M \in D(A/I) \subset D(A)$ , then  $M$  is derived- $I$ -complete. (This follows from the definition of  $T(M, f)$ ).  $\lrcorner$

**Prop. (5.9.6.8) [Category of Derived Complete Modules].** Let  $I$  be an ideal of  $A$ , then the derived  $I$ -complete  $A$ -modules form a weak Serre subcategory of  $\text{Mod}_A$ . In particular,  $D_{\text{comp}}(A, I)$  is also a weak Serre subcategory.  $\lrcorner$

*Proof:* If  $f : M \rightarrow N$  is a map of derived  $I$ -complete  $A$ -modules, then we consider the complex  $K = (M \rightarrow N)$ , then there is an exact sequence  $0 \rightarrow M[1] \rightarrow K \rightarrow N \rightarrow 0$ , so we have  $\text{Ext}^n(A_f, K) = 0$  for any  $f \in I, n \in \mathbb{Z}$  because  $M, N$  does(5.9.6.2), so  $K$  is derived  $I$ -complete by(5.9.6.2) again. Then we have  $\ker(f), \text{Coker}(f)$  are derived  $I$ -complete, by(5.9.6.2) again. Extension is also clear.  $\square$

**Lemma (5.9.6.9).** If  $R$  is ring,  $I$  is an ideal, and  $K \in D(R)$  that  $K \otimes_R^L R/I = 0$ , then  $K \otimes_R^L M$  for any  $M \in D^b(R)$  with all the cohomology groups  $I$ -power torsions.  $\lrcorner$

*Proof:* We use the truncation(4.8.7.6), then it suffices to prove for  $M$  discrete. Now  $M = \cup M[I^n]$ , and we have  $K \otimes_R^L M = \text{hocolim} K \otimes_R^L M[I^n]$ , so we may assume  $I^n M = 0$  for some  $n$ . Consider the  $R$ -algebra  $R' = R/I^n \oplus M$ , where  $M^2 = 0$ , then it suffices to show  $K' = K \otimes_R^L R' = 0$ . Now  $0 = K \otimes_R^L R/I = K' \otimes_{R'}^L R/I$ , so by(5.9.5.8)  $K' = 0$ .  $\square$

**Prop. (5.9.6.10) [Derived Nakayama].** the derived tensor product  $-\otimes_A^L A/I$  reflects isomorphism on  $D_{\text{comp}}(A, I)$ , i.e. if  $M \otimes_A^L A/I = 0$ , then  $M = 0$ .  $\lrcorner$

*Proof:* Let  $I = (f_1, \dots, f_r)$ , by (5.9.6.9),  $M \otimes_A^L K_n = 0$  for any  $K_n = \text{Kos}(A, f_1^n, \dots, f_r^n)$ , so  $K = R \lim K \otimes_A^L K_n = 0$ .  $\square$

**Cor. (5.9.6.11).** If  $I$  is f.g. and  $M$  is a derived  $I$ -complete  $A$ -module that  $M/IM = 0$ , then  $M = 0$ .  $\lrcorner$

*Proof:* ? This should be an immediate corollary.

Let  $I = (f_1, \dots, f_r)$ , if  $M \neq 0$ , let  $i$  be the largest integer that  $M/(f_1, \dots, f_i)M \neq 0$ , then  $N$  is also derived  $I$ -complete by (5.9.6.8). But  $f_{i+1} : N \rightarrow N$  is surjective, so  $T(N, f_{i+1}) \neq 0$ , contradiction.  $\square$

**Prop. (5.9.6.12).** If  $A$  is derived  $I$ -complete, then  $(A, I)$  is a Henselian pair.  $\lrcorner$

*Proof:* Cf. [[Sta]0G3H] ?.  $\square$

**Prop. (5.9.6.13) [Derived  $I$ -Completion].** If  $I = (f_1, \dots, f_n)$  is f.g. in  $A$ , the inclusion of categories  $D_{\text{comp}}(A, I) \subset D(A)$  has a left adjoint, which maps  $K$  to

$$\widehat{K} = R \text{Hom}((A \rightarrow \prod_{i_0} A_{f_{i_0}} \rightarrow \prod_{i_0 < i_1} A_{f_{i_0} f_{i_1}} \rightarrow \dots \rightarrow A_{f_1, \dots, f_r}), K).$$

called the **derived  $I$ -completion** of  $K$ .

Moreover, this construction is identical to  $K \mapsto K^\bullet = R \lim(K \otimes_A^L K_n^\bullet)$ , by (5.9.5.7) and (5.4.4.7).  $\lrcorner$

*Proof:* There is a map  $(A \rightarrow \prod_{i_0} A_{f_{i_0}} \rightarrow \prod_{i_0 < i_1} A_{f_{i_0} f_{i_1}} \rightarrow \dots \rightarrow A_{f_1, \dots, f_r}) \rightarrow A$ , which induces a morphism  $K \rightarrow \widehat{K}$ . Now by (7.3.3.28),  $R \text{Hom}(A_f, \widehat{K})$  is isomorphic to

$$R \text{Hom}((A_f \rightarrow \prod_{i_0} A_{f f_{i_0}} \rightarrow \prod_{i_0 < i_1} A_{f f_{i_0} f_{i_1}} \rightarrow \dots \rightarrow A_{f f_1, \dots, f_r}), K)$$

as  $A_f$  is  $A$ -flat. Now this one is 0 for any  $f \in I$ , by (5.4.4.7), so  $\widehat{K}$  is derived  $I$ -complete.

Conversely, if  $\widehat{K}$  is derived  $I$ -complete, then  $R \text{Hom}(A_f, K) = 0$  for any  $f \in I$ , thus  $K \rightarrow \widehat{K}$  is an isomorphism as we inductively use the stupid truncation (4.8.7.6).  $\square$

**Cor. (5.9.6.14).** If  $M$  is an  $A$ -module, then  $H^0(\widehat{M})$  is the derived- $I$ -completion in the category of modules, by (4.10.1.12).  $\lrcorner$

**Cor. (5.9.6.15).** (5.9.6.2) show the notion of derived  $I$ -complete and derived  $I$ -completion only depends on  $\text{rad } I$ .  $\lrcorner$

### Principal Ideal Case

**Prop. (5.9.6.16) [Bounded Torsion and Derived Completion].** Let  $A \in \mathcal{C}\text{Ring}$  and  $f \in A$ . If  $M \in \text{Mod}_A$  has bounded  $f^\infty$ -torsion, then the derived  $f$ -completion of  $M$  as a complex is a module and coincides with the classical  $f$ -adic completion.  $\lrcorner$

*Proof:* The derived  $f$ -completion is defined to be

$$\widehat{M} = R \lim_n (M \otimes_{\mathbb{Z}[X]}^L \mathbb{Z}[X]/(x^n)) = R \lim_n (M \xrightarrow{f^n} M).$$

So by (4.10.6.5), there are exact sequences

$$\begin{aligned} R^1 \lim_n M/f^n M &\cong H^1(\widehat{M}) \\ 0 \rightarrow R^1 \lim_n M[f^n] &\rightarrow H^0(\widehat{M}) \rightarrow \lim_n M/f^n \rightarrow 0 \\ H^{-1}(\widehat{M}) &\cong \lim_n M[f^n] \end{aligned}$$

Now the hypothesis implies that  $(M[f^n])$  is Mittag-Leffler and  $\lim_n M[f^n] = 0$ , so we have the desired result.  $\square$

**Cor. (5.9.6.17).** Let  $R$  be a perfect  $\mathbb{F}_p$ -algebra, then the derived  $p$ -completion and  $p$ -adic completion of  $R$  coincide.  $\lrcorner$

## 7 Derived Completely Properties

**Def. (5.9.7.1) [Derived Completely Properties].** Let  $A$  be a commutative ring and  $I$  is a f.g. ideal, then  $M \subset D(A)$  is called  **$I$ -completely (faithfully) flat/smooth/étale/...** iff  $M \otimes_A^L A/I$  is discrete and is a (faithfully) flat  $A/I$ -module.

$M$  is said to have **finite  $I$ -completely Tor amplitude** if  $M \otimes_A^L A/I$  is a bounded complex in  $D(A/I)$ .

Clearly, any flat/smooth/étale  $A$ -module  $M$  is  $I$ -completely flat/smooth/étale/... for any  $I$ . And  $M$  has finite  $I$ -completely amplitude if  $M$  has a finite resolution of flat  $A$ -modules.  $\lrcorner$

**Prop. (5.9.7.2) [ $I$ -Completely F.F. Descent].** If  $A \rightarrow B \rightarrow B'$  are ring maps and  $I$  is an ideal of  $A$ ,

1. If  $C$  is a  $B$ -algebra and  $B$  is  $I$ -completely f.f. over  $A$ , then  $C$  is  $I$ -completely (f.)f. over  $A$  iff  $C$  is  $I$ -completely (f.)f. over  $B$ .
2. If  $B \rightarrow B'$  is  $I$ -completely flat,  $M$  is a  $B$ -module, then  $M$  is  $I$ -completely flat over  $A$  iff  $M \otimes_B^L B'$  is  $I$ -completely flat over  $A$ .

$\lrcorner$

*Proof:* 1: One direction is easy, for the other, if  $C$  is  $I$ -completely (f.)f. over  $A$ , then

$$C \otimes_A^L A/I = C \otimes_B^L B \otimes_A^L A/I = (C \otimes_B^L B/I) \otimes_{B/I} (B \otimes_A^L A/I)$$

is a discrete and (f.)f.  $C/I$ -module iff  $(C \otimes_B^L B/I)$  does, as  $(B \otimes_A^L A/I)$  is f.f. over  $B/I$ .

2: This is because  $(B' \otimes_B^L M) \otimes_A^L A/I \cong B' \otimes_B^L (M \otimes_A^L A/I)$ .  $\square$

**Prop. (5.9.7.3).** If  $I$  is generated by a Koszul-regular sequence, then any  $A$ -module  $M$  has finite  $I$ -completely Tor amplitude.  $\lrcorner$

*Proof:* This is because  $A/i \cong A \otimes_{\mathbb{Z}[X_1, \dots, X_r]}^L \mathbb{Z}$  in this case, so  $M \otimes_A^L A/I \cong M \otimes_{\mathbb{Z}[X_1, \dots, X_r]}^L \mathbb{Z}$ , which has f.m. homology groups because  $\mathbb{Z}$  has a finite free  $\mathbb{Z}[X_1, \dots, X_n]$ -resolution (5.4.4.1).  $\square$

**Prop. (5.9.7.4) [ $I$ -Completely Flatness and Derived Completion].** Let  $A$  be a commutative ring and  $I$  be f.g., then the derived  $I$ -completion of a  $I$ -completely (faithfully) flat/étale  $M$  is  $I$ -completely (faithfully) flat/étale, both the complex and the module. In fact,  $H^0(\widehat{M}) \otimes_A^L A/I \cong \widehat{M} \otimes_A^L A/I \cong M \otimes_A^L A/I$ .  $\lrcorner$



*Proof:* Because objects in the image of  $D(A/I) \rightarrow D(A)$  are all derived  $I$ -complete, by (5.9.6.7), so there is an isomorphism  $M \otimes_A^L A/I \cong \widehat{M} \otimes_A^L A/I$  for any  $M$ , because they are both left adjoint to  $\text{Mod}_{A/I} \subset D(A)$ , by the definition of derived  $I$ -completion and (5.9.2.10). So  $\widehat{M} \otimes_A^L A/I \cong M \otimes_A^L A/I = M \otimes_A A/I$ , which is a flat  $A/I$ -module.  $\square$

**Prop. (5.9.7.5) [Flatness and Completed Derived Tensor].** The completed derived tensor of a  $I$ -completely flat/étale module is discrete and  $I$ -completely flat/étale.  $\square$

*Proof:* If  $M$  is an  $I$ -completely flat  $A$ -module, then  $M \widehat{\otimes}_A^L B$  is  $I$ -complete and  $I$ -completely flat by (5.9.7.4), and  $(M \widehat{\otimes}_A^L B) \otimes_B^L B/I = H^0(M \widehat{\otimes}_A^L B) \otimes_B^L B/I = M \otimes_A^L B/I$ , because they are both left adjoint of the forgetful functor  $\text{Mod}_{B/I} \subset D(A)$ . Then we have  $M \widehat{\otimes}_A^L B \cong H^0(M \widehat{\otimes}_A^L B)$  is discrete, as they are both  $I$ -complete, by (5.9.6.2) and derived Nakayama.  $\square$

**Prop. (5.9.7.6).**  $A$  is derived  $I$ -completely flat iff  $A$  is  $I^n$ -completely flat for any  $n > 0$ . Moreover, in this case,  $A$  is derived  $J$ -completely flat for any ideal  $J$  that  $I \subset \text{rad } J$ .  $\square$

*Proof:* The exact sequence

$$0 \rightarrow I/I^2 \rightarrow A/I^2 \rightarrow A/I \rightarrow 0$$

induces distinguished triangles

$$(M \otimes_A^L A/I) \otimes_{A/I}^L I/I^2 \rightarrow M \otimes_A^L I/I^2 \rightarrow M \otimes_A^L A/I^2 \rightarrow M \otimes_A^L A/I$$

which shows  $M \otimes_A^L I/I^2$  is discrete,  $= N$ . Then  $N$  is an  $A/I^2$ -module that  $N \otimes_{A/I^2}^L A/I = M/IM$  is  $A/I$ -flat. Then  $N$  is  $A/I^2$  flat: for any  $A/I^2$ -module  $L$ , let  $L' = IL$  and  $L'' = L/IL$  which are both  $A/I$ -modules, then there is a distinguished triangle

$$N \otimes_{A/I^2}^L L' \rightarrow N \otimes_{A/I^2}^L L \rightarrow N \otimes_{A/I^2}^L L''$$

and

$$\begin{aligned} N \otimes_{A/I^2}^L L' &= (N \otimes_{A/I^2}^L A/I) \otimes_{A/I}^L L' = (M/IM) \otimes_{A/I} L' \\ N \otimes_{A/I^2}^L L'' &= (N \otimes_{A/I^2}^L A/I) \otimes_{A/I}^L L'' = (M/IM) \otimes_{A/I} L'' \end{aligned}$$

are both discrete, hence so does  $N \otimes_{A/I^2}^L L$ , implying it is  $A/I^2$ -flat.

In a similar fashion, we can show  $M$  is  $I/I^n$ -flat for any  $n$ . And if  $I \subset \text{rad } J$ , then  $I^n \subset J$ , so  $M \otimes_A^L A/J = (M \otimes_A^L A/I^n) \otimes_{A/I^n}^L A/J$  is discrete and  $A/J$ -flat.  $\square$

**Prop. (5.9.7.7).** Let  $A$  be a ring and  $I$  be an invertible ideal, then any derived  $(p, I)$ -complete and  $(p, I)$ -completely flat  $A$ -complex  $M \in D(A)$  is discrete and  $(p, I)$ -complete. Moreover, for any  $n \geq 0$ , we have  $M[I^n] = 0$  and  $M/I^n M$  has bounded  $p^\infty$ -torsion.  $\square$

*Proof:*  $M$  is  $(I^n, p)$ -completely flat by (5.9.7.6), so we find  $M \otimes_A^L A/I^n$  is  $p$ -completely flat in  $D(A/I^n)$ . Notice that  $A/I^n$  has bounded  $p^\infty$ -torsion by induction ( $I$  invertible used), (5.9.7.11) says  $M \otimes_A^L A/I^n$  is discrete in  $D(A/I^n)$  and has bounded  $p^\infty$ -torsion, and is  $p$ -adically complete. In particular,

$$M \otimes_A^L A/I^n = (M \otimes_A^L A/I^{n+1}) \otimes_{A/I^{n+1}}^L A/I^n$$

So if we denote  $M \otimes_A^L A/I^n = M_n$ , then  $M_n = M_{n+1}/I^n M_{n+1}$ , as  $M$  is derived  $I$ -complete, we have  $M = R\varprojlim (M \otimes_A^L A/I^n)$ , so clearly  $M$  is discrete. And then we have  $M \otimes_A^L AA/I^n = M/I^n M$ , which means  $M[I^n] = 0$ , and  $M/I^n M$  has bounded  $p^\infty$ -torsion. Now because  $M$  is derived  $(p, I)$ -complete,

$$M = R\varprojlim_{m,n} ((M \otimes_A I^n \rightarrow M) \otimes_{A/I^n} (A/I^n \xrightarrow{p^m} A/I^n)) = R\varprojlim_{m,n} (M/I^n \xrightarrow{p^m} M/I^n) = R\varprojlim_{m,n} (M/(I^n, p^n))$$



is  $(p, I)$ -complete.  $\square$

**Lemma (5.9.7.8).** Let  $C$  be a commutative ring with a f.g. ideal  $J$ , and  $D$  a  $C$ -algebra that has finite  $J$ -complete Tor amplitude, then the  $J$ -completed base change operator  $-\hat{\otimes}_C^L D$  commutes with totalization in  $D^{\geq 0}(C)$  and  $D^{\geq 0}(D)$ , i.e. if  $M^\bullet$  is a cosimplicial object with in  $D^{\geq 0}(C)$  with totalization  $M$ , then

$$\mathrm{Tot}(M^\bullet) \hat{\otimes}_C^L D \cong \mathrm{Tot}(M^\bullet \hat{\otimes}_C^L D)$$

via the natural map.  $\lrcorner$

*Proof:* Cf.[Scholze, Prism, 4.22].  $\color{red}?$   $\square$

**Prop. (5.9.7.9)[Elkik's Algebrization Theorem].** Let  $A$  be a commutative ring and  $I$  is a f.g. ideal, then an  $A$ -algebra is derived  $I$ -completely étale/smooth iff it is the derived  $I$ -completion of some étale/smooth  $A$ -algebra.  $\lrcorner$

*Proof:* if it is the derived  $I$ -completion of some étale/smooth algebra, then it is derived  $I$ -completely étale/smooth by (5.9.7.4). Converse, Cf.[A. Arabia, "Relèvements des algèbres lisses et de leurs morphismes", Commentarii Mathematici Helvetici 76 (2001), 607–639.].  $\square$

**Lemma (5.9.7.10).** Let  $A$  be a ring, if  $M$  is an  $A$ -module with bounded  $f^\infty$ -torsion, i.e.  $M[f^\infty] = M[f^c]$  for some  $c > 0$ , then there are maps

$$(M \xrightarrow{f^n} M) \rightarrow M/f^n M, \quad M/f^{n+c} \rightarrow (M \xrightarrow{f^n} M)$$

in  $D(A)$  inducing an equivalence between two pro-objects  $\{M \xrightarrow{f^n} M\}$  and  $\{M/f^n\}$ .  $\lrcorner$

*Proof:* The first map is obvious, for the second map, use the following commutative diagram:

$$\begin{array}{ccc} M/M[f^c] & \xrightarrow{f^{n+c}} & M \\ \downarrow f^c & & \downarrow \\ M & \xrightarrow{f^n} & M \end{array}$$

the upper row is injective thus isomorphic to  $M/f^{n+c}M$ , then this gives the map. It can be checked that this is an equivalence of pro-objects  $\color{red}?$ .  $\square$

**Prop. (5.9.7.11).** Let  $A$  be a commutative ring that has bounded  $f^\infty$ -torsion, then for a  $M \in D(A)$ , the following is equivalent:

- $M$  is derived  $f$ -complete and  $f$ -completely flat.
- $M$  is discrete and is represented by a  $f$ -adically complete module that  $M/f^n M$  is  $A/f^n$ -flat for any  $n > 1$  and  $M$  has bounded  $f^\infty$ -torsion.

Furthermore, in this case,  $M \otimes_A A[f^\infty] = M[f^\infty]$ .  $\lrcorner$

*Proof:* By (5.9.7.10),  $\{A/f^n A\}$  and  $\{Kos(A, f^n)\}$  are two equivalence pro-objects in  $D(A)$ . So if 1 or 2 holds, then  $M$  is derived  $f$ -complete, so

$$M = R\lim(M \otimes_A^L Kos(A, f^n)) = R\lim(M \otimes_A^L A/f^n A).$$

Now if 1 holds, then  $M_n = M \otimes_A^L A/f^n A$  is discrete by (5.9.7.6), and  $M_n = M_{n+1}/f^n M_{n+1}$  is surjective:

$$M \otimes_A^L A/f^n A = (M \otimes_A^L A/f^{n+1} A) \otimes_{A/f^{n+1} A}^L A/f^n.$$

So  $M = R\lim(M \otimes_A^L A/f^n A)$  is discrete. Then  $M \otimes_A A/f^n = M \otimes_A^L A/f^n$  is flat over  $A/f^n$ .

Next we prove  $M \otimes_A A[f^\infty] = M[f^\infty]$ : There is an exact sequence

$$0 \rightarrow (A[f^n])[1] \rightarrow \text{Kos}(A, f^n) \rightarrow A/f^n \rightarrow 0$$

Then tensoring  $M \otimes_A^L$  gives a distinguished triangle

$$(M \otimes_A^L A[f^n])[1] \rightarrow \text{Kos}(M, f^n) \rightarrow M \otimes_A^L A/f^n.$$

Notice that

$$M \otimes_A^L A[f^n] = (M \otimes_A^L A/f^n) \otimes_{A/f^n}^L A[f^n] = (M \otimes_A A/f^n) \otimes_{A/f^n}^L A[f^n] = M \otimes_A A[f^n]$$

by flatness, so the distinguished triangle shows  $M \otimes_A A[f^n] \cong H^{-1}(\text{Kos}(M, f^n)) = M[f^n]$ .

Conversely, if 2 holds, then there are equivalences of pro-objects

$$\{M \otimes_A^L A/f^n\} \cong \{M \otimes_A^L \text{Kos}(A, f^n)\} = \{\text{Kos}(M, f^n)\} \cong \{M/f^n M\}$$

by (5.9.7.10) as  $M$  has bounded  $f^\infty$ -torsion. So

$$\widehat{M} = R\lim\{M \otimes_A^L \text{Kos}(A, f^n)\} = R\lim\{M/f^n M\} = M[0]$$

so  $M$  is derived  $p$ -complete. And the constant system

$$\{M \otimes_A^L A/f\} = \{M \otimes_{A/f^n}^L A/f^n \otimes_{A/f^n}^L A/f\} \cong \{M/f^n M \otimes_{A/f^n}^L A/f\} = \{M/fM\}$$

where we used  $M/f^n M$  is  $A/f^n$ -flat. So  $M \otimes_A^L A/f \cong M/fM$  is flat over  $A/f$ .  $\square$

## 8 Duality

**Def. (5.9.8.1) [Dualizing Complexes].** Let  $A$  be a Noetherian ring, then a **dualizing complex** for  $A$  is a complex  $\omega_A \in D(A)$  s.t.

- $\omega_A$  has finite injective dimension.
- $H^i(\omega_A)$  are all finite  $A$ -modules.
- $A[0] \rightarrow R\text{Hom}(\omega_A, \omega_A)$  is a quasi-isomorphism.

┘

**Prop. (5.9.8.2) [Dualizing Complex is Local].** Let  $A$  be a Noetherian ring,

- If  $B = S^{-1}A$  and  $\omega_A \in D(A)$  is a dualizing complex for  $A$ , then  $\omega_A \otimes_A^L B$  is a dualizing complex for  $B$ .
- If  $(f_1, \dots, f_n) = (1) \in A$ , and  $\omega_A \in D(A)$  satisfies  $(\omega_A)_{f_i}$  are dualizing complexes for  $A_{f_i}$  for any  $i$ , then  $\omega_A$  is a dualizing complex for  $A$ .

┘

*Proof:* Cf. [Sta]0A7G, 0A7H.  $\square$

**Def. (5.9.8.3) [Dualizing Modules].** Let  $(A, \mathfrak{m}, k)$  be a Noetherian local ring and  $\omega_A$  a normalized dualizing complex for  $A$ , then  $[\omega_A] = H^{-\dim A}(\omega_A)$  is called a **dualizing module** for  $A$ .  $\square$

**Def. (5.9.8.4) [Relative Dualizing Complex].** Cf. [Sta]0E9M.  $\square$

### Quasi-Finite Case

Cf. [Sta]Chap49.4.

**Prop. (5.9.8.5).**  $\square$

## 6 | Homotopy Theory

### 6.1 Operads

References are [Operads in algebra, topology and physics, Stasheff].

#### 1 Topological Operads

**Def. (6.1.1.1) [Endomorphism Operads].** For  $X \in \mathbf{Set}$ , then **endomorphism operad**  $\mathrm{End}_X$  is defined to be the collection of sets  $\{\mathrm{Func}(X^n, X)\}_{n \in \mathbb{Z}_+}$  together with maps

$$\circ_i : \mathrm{Func}(X^n, X) \times \mathrm{Func}(X^m, X) \rightarrow \mathrm{Func}(X^{m+n-1}, X)$$

for  $m, n \in \mathbb{Z}_+, i \in [n]_+$  given by

$$(f \circ_i g)(x_1, \dots, x_{m+n-1}) = f(x_1, \dots, x_{i-1}, g(x_i, \dots, x_{i+m-1}), x_{i+m}).$$

┘

**Def. (6.1.1.2) [Operads].** For  $\mathcal{C} \in \mathbf{Cat}$  be a monoidal category, an **operad**  $(\mathcal{O}, \circ_i)$  in  $\mathcal{C}$  is defined to be a collection of objects  $\{\mathcal{O}(n)\}_{n \in \mathbb{Z}} \in \mathcal{C}$  and maps  $\circ_i : \mathcal{O}(n) \times \mathcal{O}(m) \rightarrow \mathcal{O}(m+n-1)$ , s.t. they satisfy relations manifested in the endomorphism operads (6.1.1.1). ? Cf. [Operads in algebra, topology and physics, Stasheff]P

┘

**Example (6.1.1.3) [Operads].**

- The **tree operad** is defined to be the operad  $(\mathcal{T}(n), \circ_i) \in \mathbf{Set}$  where  $\mathcal{T}(n)$  is the set of trees with one root and  $n$  leaves labeled from 1 to  $n$ . And  $f \circ_i g$  is just the tree obtained by grafting the root of  $g$  to the  $i$ -th leaf of  $f$ .
- The  $r$ -dimensional **little cube operad** is defined to be the operad  $(\mathcal{C}_r(n), \circ_i) \in \mathbf{Set}$  where  $\mathcal{C}_r(n)$  is the set of an ordered collection of  $j$   $r$ -cubes linearly embedded in the standard  $r$ -dimensional unit cube  $\mathbb{I}^r$  with disjoint interiors and axes parallel to those of  $\mathbb{I}^r$ . And  $f \circ_i g$  is just the collection obtained by scaling  $g$  into the  $i$ -th cube of  $f$ .

┘

#### 2 Algebras over Operads

**Def. (6.1.2.1) [Algebras over Operads].** Let  $(\mathcal{O}, \circ_i)$  be an operad in  $\mathcal{C}$  (6.1.1.2), then an **algebra over operad**  $\mathcal{O}$  is an object  $A \in \mathcal{C}$  with a map of operads  $(\mathcal{O}, \circ_i) \rightarrow \mathrm{End}_A$ .

┘

### 3 Operads in Topology

**Notation (6.1.3.1).**

- All topological spaces are assumed to be Hausdorff and compactly generated.

┘

**Thm. (6.1.3.2) [Loop Spaces and Operads, Boardman-Vogt/May].** If  $Y \in \mathcal{T}\text{op}$  is connected and has the homotopy type of a CW-complex, then  $Y$  has the homotopy type of a  $n$ -fold loop space  $\Omega^n X$  iff  $Y$  is an algebra over the operad  $\mathcal{C}^r$  (6.1.1.3). ┘

*Proof:*

□

### 4 Operads in Algebra

### 5 Operads in Geometry

Part II

# Algebraic Geometry



## 7 | Algebraic Geometry I: Scheme Theory

### 7.1 Sites, Sheaves, Topoi and Stacks

*“I can illustrate the second approach with the same image of a nut to be opened. The first analogy that came to my mind is of immersing the nut in some softening liquid, and why not simply water? From time to time you rub so the liquid penetrates better, and otherwise you let time pass. The shell becomes more flexible through weeks and months—when the time is ripe, hand pressure is enough, the shell opens like a perfectly ripened avocado! A different image came to me a few weeks ago. The unknown thing to be known appeared to me as some stretch of earth or hard marl, resisting penetration... the sea advances insensibly in silence, nothing seems to happen, nothing moves, the water is so far off you hardly hear it... yet it finally surrounds the resistant substance.”*  
 –A. Grothendieck

References are [D-M69, Vis08, Ols16, Sta] and [Fibered Category to Algebraic Stacks, Lamb].

#### Notation(7.1.0.1).

- Use notations on categories defined in [Categories II](#).

┘

#### 1 Sites

**Def. (7.1.1.1)[Sites].** A **site** is given by a category  $\mathcal{C} \in \mathbf{Cat}$  and a set  $\mathbf{Cov}(\mathcal{C})$  of families of morphisms with fixed target, called the **coverings** of  $\mathcal{C}$  that:

- An isomorphism is a covering.
- Coverings of coverings are coverings.
- Base change of a covering is a covering.

Sometimes a site is wrongly called a **topology**, the difference is that the morphism of site is looks like a reverse of a morphism of topology(7.1.1.7). We never talk about the category of sites, but we use  $\mathcal{C} \in \mathbf{Site}$  to mean that  $\mathcal{C}$  is a site.

┘

**Def. (7.1.1.2)[Discrete Topology].** A **discrete topology** is a site that the only coverings are identities. In this way, we can regard any category as a site.

┘

**Def. (7.1.1.3)[Topological Basis].** For  $\mathcal{C} \in \mathbf{Site}$ , a **topological basis** for  $\mathcal{C}$  is a subset  $\mathfrak{G} \subset \mathbf{Ob}(\mathcal{C})$  s.t. for any  $U \in \mathcal{C}$ , there is a covering  $\{U_i \rightarrow U\} \in \mathbf{Cov}(\mathcal{C})$  where each  $U_i \in \mathfrak{G}$ .

┘

**Def. (7.1.1.4)[Quasi-Compact Objects].** In a site  $\mathcal{C} \in \mathbf{Site}$ , an object  $U \in \mathcal{C}$  is called **quasi-compact** if each covering of  $U$  has a finite subcovering. The site  $\mathcal{C}$  is called **Noetherian** if each object of  $\mathcal{C}$  is quasi-compact.

Given a site  $\mathcal{C}$ , we can define a new site  $\mathcal{C}^f$  whose coverings are coverings of  $\mathcal{C}$  that are finite. Then this is truly a site and it is Noetherian.  $\lrcorner$

**Def. (7.1.1.5) [Quasi-Separated Objects].** In a site  $\mathcal{C} \in \text{Site}$ , an object  $X \in \mathcal{C}$  is called **quasi-separated** if for any pair of morphisms  $U \rightarrow X \leftarrow V$  where  $U, V$  are quasi-compact object (7.1.1.4), the fiber product  $U \times_X V$  is also quasi-compact.  $\lrcorner$

**Def. (7.1.1.6) [Comma Topology].** For a site  $\mathcal{C}$  and an object  $S$ , we have the comma category  $\mathcal{C}/S$  (4.1.1.17), and we can define a topology on it where the coverings are coverings of  $\mathcal{C}$  that is compatible over  $S$ .  $\lrcorner$

**Def. (7.1.1.7) [Continuous Functor].** A **continuous functor** between sites  $\mathcal{C} \rightarrow \mathcal{D}$  is a functor that preserves coverings and any base change by morphisms in a covering?  $\lrcorner$

A **morphism of sites**  $\mathcal{C} \rightarrow \mathcal{D}$  is given by a continuous functor  $u : \mathcal{D} \rightarrow \mathcal{C}$  that  $u_s$  (7.1.2.11) is exact.  $\lrcorner$

This exact condition is easy to be satisfied, by (7.1.2.14).  $\lrcorner$

**Def. (7.1.1.8) [Cocontinuous functors].** A **cocontinuous functor** between sites  $u : \mathcal{C} \rightarrow \mathcal{D}$  is a functor that for any  $U \in \mathcal{C}$  and any covering  $\{V_i \rightarrow u(U)\}$  in  $\mathcal{D}$ , there is a covering  $\{U_i \rightarrow U\} \in \mathcal{C}$  that refines  $\{V_i \rightarrow u(U)\}$  after the functor  $u$ .  $\lrcorner$

### Topologies and Sieves

**Def. (7.1.1.9) [Sieves].** For a covering  $\mathcal{U} = \{U_i \rightarrow U\}$  in a category  $\mathcal{C}$ , define a subfunctor  $h_{\mathcal{U}} \subset h_U$ , where for each  $X$   $h_{\mathcal{U}}(X)$  consists of elements in  $\text{Hom}(X, U)$  that factor through some  $U_i \rightarrow U$ .  $\lrcorner$

A **sieve**  $S$  on  $U$  is a subfunctor of  $h_U$ . Notice that any sieve must be of the form  $h_{\mathcal{U}}$ , by choosing  $\mathcal{U}$  to consist of all arrows in  $\{S(T)\}_{T \in \mathcal{C}}$ .  $\lrcorner$

**Def. (7.1.1.10).** If  $\mathcal{T}$  is a Grothendieck topology on a category  $\mathcal{C}$ , then a sieve  $S \subset h_U$  over  $U$  is said to **belong to**  $\mathcal{T}$  or just a sieve of the site  $\mathcal{C}$  if there exists a covering  $\mathcal{U}$  of  $U$  that  $h_{\mathcal{U}} \subset S$ .  $\lrcorner$

### G-Spaces

**Def. (7.1.1.11) [G-Spaces].** A **G-space** is a set  $X$  with a family of subsets of  $X$  that they form a site w.r.t inclusions and that covering are all set-theoretic coverings (but not necessarily conversely). These subsets are called **admissible opens** of  $X$  and covers are called **admissible covers**. (In other words, a G-topological space is a "topological space without unions"). Morphisms of G-spaces are maps that are simultaneously a continuous map and a morphism of sites.  $\lrcorner$

**Def. (7.1.1.12) [Completeness].** The completeness of a G-topological space  $X$ :

- G0:  $\emptyset$  and  $X$  are admissible open.
- G1: Let  $\{U_i \rightarrow U\}$  be an admissible cover, then a subset  $V \subset U$  is admissible if  $V \cap U_i$  are all admissible.
- G2: Let  $\{U_i \rightarrow U\}$  be a cover of admissible opens for  $U$  admissible, then the cover is admissible if it has an admissible cover as a refinement.  $\lrcorner$

**Lemma (7.1.1.13) [Admissible is Local].** If G<sub>2</sub> is satisfied for a G-topological space  $X$ , then for an admissible covering  $\{X_i \rightarrow X\}$  and another covering  $\{U_i \rightarrow X\}$  between admissible opens, it is admissible iff  $U_i \cap X_j$  is an admissible covering for  $X_j$  for each  $j$ . (By composition,  $\{U_i \cap X_i \rightarrow X\}$  is admissible, and it refines  $\{U_i \rightarrow X\}$ ).  $\lrcorner$



**Prop.(7.1.1.14) [Glue of Complete G-topological spaces].** For sets  $\cup X_i = X$ , if there are Grothendieck category  $\mathcal{I}_i$  on  $X_i$  making  $X_i$  into a G-topological space, and they all satisfies the completeness conditions  $G_0, G_1, G_2$  of (7.1.1.12). Assume that  $X_i \cap X_j$  is  $\mathcal{I}_i$ -open in  $X_i$  and  $\mathcal{I}_i, \mathcal{I}_j$  restrict to the same topology on  $X_i \cap X_j$ , then there is a unique Grothendieck category  $\mathcal{I}$  on  $X$  making  $X$  a G-topological space that:

- $X_i$  is  $\mathcal{I}$ -open and  $\mathcal{I}$  restricts to  $\mathcal{I}_i$  on  $X_i$ ,
- $\mathcal{I}$  satisfies the completeness conditions  $G_0, G_1, G_2$ .
- $X_i$  is a  $\mathcal{I}$ -covering of  $X$ .

┘

*Proof:* By (7.1.1.12) and (7.1.1.13), the uniqueness is straightforward, for the existence,

- check Grothendieck first: Composition, base change.
- check condition 1: by hypothesis, and (7.1.1.12) applied to  $X_i \cap X_j \rightarrow X_i$  (this is admissible because  $\text{id}_{X_i}$  refined it).
- check condition 2:  $G_0$  obvious,  $G_1$  by if  $V \cap U_i \cap X_i$  admissible, then  $V \cap X_i$  admissible by admissibility of  $U_i \rightarrow U$ , then  $V$  is admissible,  $G_2$ : obvious
- check condition 3: because  $X_i \cap X_j \rightarrow X_i$  is admissible.

□

**Def.(7.1.1.15).** A G-topological space is called **connected** iff there isn't two nonempty admissible open subset  $X_1, X_2$  that  $X_1 \cap X_2 = \emptyset$  and  $\{X_1, X_2 \rightarrow X\}$  is an admissible cover. ┘

### G-torsor

**Def.(7.1.1.16) [Torsors].** Let  $\mathcal{C}$  be a site and  $\mathcal{G} \in \text{Sh}^{\text{grp}}(\mathcal{C})$ , then a **pseudo G-torsor** is a sheaf  $\mathcal{F} \in \text{Sh}^{\text{set}}(\mathcal{C})$  endowed with an action  $\mathcal{G} \times \mathcal{F} \rightarrow \mathcal{F}$  s.t.  $\mathcal{G} \times \mathcal{F} \rightarrow \mathcal{F} \times \mathcal{F} : (g, f) \mapsto (gf, f)$  is an isomorphism.

A pseudo G-torsor is called a **G-torsor** if for any  $U \in \mathcal{C}$ , there is a covering  $\{U_i \rightarrow U\}$  that  $\mathcal{F}(U_i) \neq \emptyset$  for each  $i$ . ┘

**Prop.(7.1.1.17).** Situation as in (7.1.1.16), a G-torsor on  $\mathcal{C}$  is trivial iff  $\Gamma(\mathcal{C}, \mathcal{F}) \neq \emptyset$  (7.3.1.1). ┘

*Proof:* This is because the transitive action of  $\mathcal{G}$  on the global section induces an isomorphism  $\mathcal{G} \rightarrow \mathcal{F}$ . □

**Prop.(7.1.1.18).** If  $\mathcal{C}$  is a subcanonical site, and  $\mathcal{G} \in \text{Sh}^{\text{grp}}(\mathcal{C})$ , then  $\mathcal{F} \in \text{Sh}^{\text{set}}(\mathcal{C})$  together with an action  $\alpha : \mathcal{G} \times \mathcal{F} \rightarrow \mathcal{F}$  is a G-torsor iff for any  $U \in \mathcal{C}$ , there is a covering  $\{U_i \rightarrow U\}$  that the restrictions to  $U_i$  are trivial torsors, i.e.  $\alpha|_{\mathcal{C}/U_i} = \pi_2 : \mathcal{G}|_{U_i} \times \mathcal{F}|_{U_i} \rightarrow \mathcal{F}|_{U_i}$ . ┘

*Proof:* If  $\mathcal{F}$  is a G-torsor, then the restrictions of the torsor on  $U_i$  are trivial because they have global sections in  $\Gamma(\mathcal{C}/U_i, \mathcal{F}) = \mathcal{F}(U_i)$ , by (7.1.1.17). Conversely, if there is a covering  $\{U_i \rightarrow U\}$  that the restrictions to  $U_i$  are trivial torsors, then the map  $\mathcal{G} \times \mathcal{F} \rightarrow \mathcal{F} \times \mathcal{F} : (g, f) \mapsto (gf, f)$  are isomorphisms when restricted to  $U_i$ , which means  $\mathcal{G}(U) \times \mathcal{F}(U) \rightarrow \mathcal{F}(U) \times \mathcal{F}(U)$  is an isomorphism for any  $U$ , because they are sheaves, so it is an isomorphism, and  $\mathcal{F}$  is a G-torsor. □

**Cor.(7.1.1.19) [Representable G-Torsor].** If  $\mathcal{C} \in \text{Site}$  and  $G \in \text{Grp}(\mathcal{C})$ , then

- $X \rightarrow Y$  is a G-torsor (in the category  $\mathcal{C}/Y$ ) (7.1.1.16) iff  $X \rightarrow Y$  is a G-equivariant map (where the action of  $G$  on  $Y$  is trivial),  $G \times X \rightarrow X \times_Y X : (g, x) \mapsto (gx, x)$  is an isomorphism, and  $\{X \rightarrow Y\}$  is refined by a covering of  $Y$ .

- If  $\mathcal{C}$  is a subcanonical site, then  $X \rightarrow Y$  is a  $G$ -torsor in the category  $\mathcal{C}/Y$  iff  $X \rightarrow Y$  is a  $G$ -equivariant map (where the action of  $G$  on  $Y$  is trivial), and there exists a covering  $\{Y_i \rightarrow Y\}$  that each  $Y_i \times_Y X \rightarrow Y_i$  is a trivial torsor, i.e.  $G$ -equivariantly isomorphic to  $G \times Y_i \rightarrow Y_i$ .  $\lrcorner$

**Cor. (7.1.1.20).** If  $\mathcal{C} \in \text{Site}$ ,  $G \in \text{Grp}(\mathcal{C})$  and  $X \rightarrow Y$  is a  $G$ -torsor (7.1.1.16), then the map  $G \times G \times X \rightarrow X \times_Y X \times_Y X : (g, h, x) \mapsto (ghx, hx, x)$  is an isomorphism.  $\lrcorner$

### Presheaves

**Def. (7.1.1.21) [Presheaves].** Let  $\mathcal{C}, \mathcal{D} \in \text{Cat}$ , then a **presheaf** of objects in  $\mathcal{D}$  on  $\mathcal{C}$  is a functor  $\mathcal{C}^{op} \rightarrow \mathcal{D}$ . The category of presheaves of objects in  $\mathcal{D}$  on  $\mathcal{C}$  is denoted by  $\mathcal{PSh}(\mathcal{C}; \mathcal{D})$ . For any  $U \in \mathcal{C}$ ,  $\Gamma(U, -)$  is the functor  $\mathcal{PSh}(\mathcal{C}; \mathcal{D}) \rightarrow \mathcal{D} : F \mapsto F(U)$ .  $\lrcorner$

**Prop. (7.1.1.22).** If  $\mathcal{D}$  is complete or cocomplete, and the same is true for  $\mathcal{PSh}(\mathcal{C}; \mathcal{D})$ .  $\lrcorner$

**Def. (7.1.1.23) [Coherent Sheaves].** For  $\mathcal{C} \in \text{Cat}$ , denote  $\mathcal{PSh}_\infty(\mathcal{C}) = \mathcal{PSh}(\mathcal{C}; \text{Grpd}_\infty)$  (4.7.2.5), called the category of **coherent sheaves** on  $\mathcal{C}$ . It is complete and cocomplete.  $\lrcorner$

**Def. (7.1.1.24) [Points].** A **point of a site** is a Cf. [Sta]00Y3.  $\lrcorner$

## 2 Sheaves and Topoi

**Def. (7.1.2.1) [Sheaves].** Let  $\mathcal{C}$  be a site and  $\mathcal{D} \in \text{Cat}_\infty$ , then  $\mathcal{F} \in \mathcal{PSh}(\mathcal{C}; \mathcal{D})$  is called a **sheaf** of objects in  $\mathcal{D}$  on  $\mathcal{C}$  iff for any sieve  $\mathcal{S}$  on  $U$  belong to  $\mathcal{C}$ , the natural map

$$\text{Map}(\downarrow_U, \mathcal{F}) \rightarrow \text{Map}(\downarrow_{\mathcal{U}}, \mathcal{F})$$

is an equivalence ?.  $\lrcorner$

**Def. (7.1.2.2) [Separate Presheaves].** Let  $\mathcal{C}$  be a site, then  $\mathcal{F} \in \mathcal{PSh}^{\text{Set}}(\mathcal{C})$  is called a **separated presheaf** if  $F(U) \hookrightarrow \prod_i F(U_i)$  is injective for any covering  $\mathcal{U} = \{U_i \rightarrow U\} \in \text{Cov}(\mathcal{C})$ .  $\lrcorner$

**Def. (7.1.2.3) [Effective Epimorphisms].** An epimorphism  $\{U_i \rightarrow V\}$  in a category is called a **family of effective epimorphisms** if

$$\text{Hom}(V, Z) \rightarrow \prod \text{Hom}(U_i, Z) \rightrightarrows \prod \text{Hom}(U_i \times_V U_j, Z)$$

is exact for each  $Z$ . Similarly for a **family of universal effective epimorphisms**.  $\lrcorner$

**Prop. (7.1.2.4) [Subcanonical Site].** The class of all families of universal effective epimorphisms in a category forms a Grothendieck topology, called the **canonical topology**. It is the finest topology that all representable presheaves are sheaves.

Topologies that are coarser than the canonical topology are called **subcanonical topology**. Equivalently, a subcanonical topology is a topology that every representable presheaf is a sheaf.  $\lrcorner$

*Proof:* We only need to verify that family of universal effective epimorphisms is closed under composition. For this, first prove epimorphism, then use epimorphism to prove effectiveness. Universal follows routinely. Cf. [Tamme].  $\square$

**Prop. (7.1.2.5).** For a subcanonical topology on a site  $\mathcal{C}$ , its restriction on a localizing category  $\mathcal{C}/S$  is subcanonical.  $\lrcorner$

*Proof:* The only nontrivial part is that the glued morphism is a morphism over  $S$ . For this, consider its composition that maps to  $S$ , then the uniqueness of the exact sequence (7.1.2.3) will show that it is truly a  $S$ -morphism.  $\square$

**Prop. (7.1.2.6).** Let  $\mathcal{C}$  be a subcanonical site, and  $f : X \rightarrow Y$  is an arrow in  $\mathcal{C}/S$ , suppose there is a covering  $\{S_i \rightarrow S\}$  that the pullback of  $f$  to  $\mathcal{C}/S_i$  are all isomorphisms, then  $f$  is an isomorphism.  $\lrcorner$

*Proof:* This follows from (7.1.3.7) and (7.1.3.9).  $\square$

**Prop. (7.1.2.7) [Sheafification].** Let  $\mathcal{C}$  be a site, consider the functor

$$(\cdot)^+ : \mathcal{PSh}^*(\mathcal{C}) \rightarrow \mathcal{Sh}^*(\mathcal{C}) : F^+(U) = \varinjlim \ker\left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \times_U U_j)\right) = \check{H}^0(U, F).$$

Then

- For  $F \in \mathcal{PSh}^*(\mathcal{C})$ ,  $F^+$  is separated.
- If  $F \in \mathcal{PSh}^*(\mathcal{C})$  is separated,  $F \rightarrow F^+$  is injective and  $F^+ \in \mathcal{Sh}^*(\mathcal{C})$ . (The problem of separated is that the cover may not be identical in  $U_i \times_U U_j$  but only on a cover of it).
- $(\cdot)^+$  is left exact.
- **sheafification functor**  $(\cdot)^\# : \mathcal{PSh}^* \rightarrow \mathcal{Sh}^* : F \mapsto F^\# = F^{++}$  is exact and it is left adjoint to the forgetful functor.

So the forgetful functor is left exact and it preserves injectives. Thus the sheaf cokernel is the sheafification of the presheaf kernel, the sheaf kernel is the presheaf kernel.  $\lrcorner$

*Proof:* The separatedness is simple. For sheaf condition, an element of  $F^+(U_i)$  is represented by a covering  $\{V_{ij} \rightarrow U_i\}$ , and there restriction to  $U_i \times_U U_j$  coincide by separatedness hence the covering  $\{V_{ij} \rightarrow U\}$  is an element of  $F^+(U)$ .

$\mathcal{Sh}$  is left exact because  $(-)^+$  is left exact from  $\mathcal{PSh}^*$  to  $\mathcal{PSh}^*$  by (7.3.2.4) checked on every element  $U$ . It is right exact trivially, hence it is exact.  $\square$

**Def. (7.1.2.8) [Constant Sheaves].** The **constant sheaf**  $\underline{S}$  for a set  $S$  is the sheafification of the constant presheaf  $U \mapsto S$ .  $\lrcorner$

### Transfer of Sheaves under Morphisms of Sites

**Def. (7.1.2.9) [Functoriality of Presheaves].** Given a functor of sites  $u : \mathcal{C} \rightarrow \mathcal{C}'$ , which should be regarded as an inverse map, there are maps

$$u^p F'(U) = F'(u(U)) : \mathcal{PSh}(\mathcal{C}') \rightarrow \mathcal{PSh}(\mathcal{C}), \quad u_p(F)(U') = \varinjlim_{U_i | U' \rightarrow u(U_i)} F(U_i) : \mathcal{PSh}(\mathcal{C}) \rightarrow \mathcal{PSh}(\mathcal{C}')$$

Then  $u_p$  is left adjoint to  $u^p$ .

We can also define a functor

$${}_p u : {}_p u(F)(U') = \varprojlim_{\{U_i | u(U_i) \rightarrow U'\}^{op}} F(U_i) : \mathcal{PSh}(\mathcal{C}) \rightarrow \mathcal{PSh}(\mathcal{C}').$$

Then this functor is right adjoint to  $u^p$ , by duality.  $\lrcorner$

*Proof:* A map  $f \in \text{Mor}(u_p(F), G)$  is represented by compatible maps

$$\varinjlim_{U_i: U' \rightarrow u(U_i)} F(U_i) \rightarrow G(U'),$$

and this is represented by compatible maps  $F(U_i) \rightarrow G(U')$  which is indexed over  $\prod_{U' \in \mathcal{C}'} \mathcal{I}_{U'}$ , where  $\mathcal{I}_{U'} = \{U_i : U' \rightarrow u(U_i)\}$ . Now this is equivalent to compatible maps  $F(U_i) \rightarrow G(u(U_i))$ , which is a map  $g \in \text{Mor}(F, u^p(G))$ .  $\square$

**Cor. (7.1.2.10).**  $u^p$  is exact.  $\lrcorner$

**Def. (7.1.2.11) [Functoriality of Sheaves].**

- Given a continuous functor  $u : \mathcal{C} \rightarrow \mathcal{C}'$  between sites, there are maps

$$u_s = \sharp \circ u_p \circ \iota : \mathcal{S} \rightarrow \mathcal{S}', \quad u^s = u^p \circ \iota : \mathcal{S}' \rightarrow \mathcal{S}.$$

$u_s$  is left adjoint to  $u^s$ , by adjointness of  $u_p, u^p$  and  $\sharp, \iota$ .

- Given a cocontinuous functor  $u : \mathcal{C} \rightarrow \mathcal{C}'$  between sites, there are maps

$$u^s = \sharp \circ u^p \circ \iota : \mathcal{S}' \rightarrow \mathcal{S}, \quad {}_s u = {}_p u \circ \iota : \mathcal{S} \rightarrow \mathcal{S}'.$$

$u^s$  is left adjoint to  ${}_s u$ , by adjointness of  $u^p, {}_p u$  and  $\sharp, \iota$ . Moreover,  $u^s$  is exact.  $\lrcorner$

*Proof:* 1: Notice if  $\mathcal{F}$  is a sheaf, then  $u^s \mathcal{F}$  is also a sheaf, by continuity (7.1.1.7).

2:  ${}_p u \mathcal{F}$  is a sheaf by [Sta]00XK?.  $u^p$  is clearly right exact, and it is left exact because  $\iota, \sharp$  do, and  $u^p$  is exact by (7.1.2.10).  $\square$

**Cor. (7.1.2.12).** When  $u$  is continuous,  $(u_p(G))^\sharp \cong (u_p(G^\sharp))^\sharp$  for any presheaf  $G$  on  $T$ .

When  $u$  is cocontinuous,  $(u^p G)^\sharp \cong (u^p(G^\sharp))^\sharp$  for any presheaf  $G$  on  $T$ .  $\lrcorner$

*Proof:* Use Yoneda lemma.  $\square$

**Prop. (7.1.2.13).** For  $Z \in T$ ,  $u_p h_Z = h_{u(Z)}$ .  $\lrcorner$

*Proof:* Use the adjointness of  $u_p, u^p$  (7.1.2.9), then for any presheaf  $F$ ,

$$\text{Hom}(u_p h_Z, F) = \text{Hom}(h_Z, u^p F) = f^p(F)(Z) = F(f(Z)),$$

thus we are done by Yoneda lemma.  $\square$

**Prop. (7.1.2.14) [When is  $u_s$  Exact].** If  $u : \mathcal{C} \rightarrow \mathcal{C}'$  is a continuous functor between sites that  $\mathcal{I}_{U'}$  is cofiltered for any  $U' \in \mathcal{C}'$ , where  $\mathcal{I}_{U'}$  is the category of all pairs  $(U, \varphi)$  where  $U \in \mathcal{C}$  and  $\varphi : U' \rightarrow u(U)$ , then  $u_s$  is exact.

In particular, this is the case when  $\mathcal{C}, \mathcal{C}'$  both have weakly final objects and finite fiber products and  $u$  preserves them. Notice the condition of weakly final objects can be released if we can show  $\mathcal{I}_{U'}$  is nonempty for any  $U'$ .  $\lrcorner$

*Proof:* It suffices to show the left exactness of  $u_p$ . By definition,  $u_p(U') = \varinjlim_{\mathcal{I}_{U'}^{op}} F_{U'}$  where  $F_{U'}$  is the covariant functor  $(U, \varphi) \rightarrow F(U)$ . Because  $\mathcal{I}_{U'}^{op}$  is filtered, this colimit process is exact form  $\text{Hom}(\mathcal{I}_{U'}^{op}, \mathcal{A}b)$  to  $\mathcal{A}b$  and  $u_p$  is exact. Now shification is also exact (7.1.2.7), so we conclude.

The last assertion is clear. (Notice the weakly-final object are used to assure  $\mathcal{I}_{U'}$  is nonempty.)  $\square$

**Prop. (7.1.2.15) [Localization Site].** For a site  $T$  and  $Z \in T$ , there is a site  $T/Z$  as objects over  $T$ , and  $i : T/Z \rightarrow T$  is continuous. Then  $i^s$  is exact.  $\lrcorner$

*Proof:*  $R^q i^s(F) = (i^p(\mathcal{H}^q(F)))^\sharp$  (7.3.1.7), and  $(\mathcal{H}^q(F))^\sharp = 0$  (7.3.2.14), so it suffices to show  $i^p$  and  $\sharp$  commutes. But  $i^s$  and  $+$  commutes obviously.  $\square$

**Prop. (7.1.2.16) [Sheaf Condition is Local].** To check sheaf condition for presheaf w.r.t. a topology, it suffice to show that for any covering, there is a refinement covering of it that sheaf condition hold, because by the definition of sheafification functor,  $F^+ = F$ , so  $F$  is a sheaf.  $\lrcorner$

**Cor. (7.1.2.17).** For two topology on a same category that  $\mathcal{I}'$  is finer than  $\mathcal{I}$ , then any  $\mathcal{I}'$ -sheaf is a  $\mathcal{I}$ -sheaf. And if any covering in  $\mathcal{I}'$  can be refined by a covering in  $\mathcal{I}$ , then  $\mathcal{S} \rightarrow \mathcal{S}'$  is an equivalence of categories. In particular, if  $T$  is Noetherian,  $\mathcal{S}(T)$  and  $\mathcal{S}(T^f)$  (7.1.1.4) are equivalent.  $\lrcorner$

### Topoi

**Def. (7.1.2.18) [Topoi].** A **topos** is an Abelian category that is equivalent to the category of sheaves  $\text{Sh}^{\text{Set}}(\mathcal{C})$  on a site  $\mathcal{C}$ . A **morphism of topoi**  $f : \text{Sh}^{\text{Set}}(\mathcal{C}) \rightarrow \text{Sh}^{\text{Set}}(\mathcal{D})$  is an adjunction

$$f^{-1} : \text{Sh}^{\text{Set}}(\mathcal{D}) \rightleftarrows \text{Sh}^{\text{Set}}(\mathcal{C}) : f_*$$

s.t.  $f^{-1}$  is exact. Compositions of morphisms of topoi are defined routinely. A **2-morphism of topoi** between two morphisms of topoi  $f, g : \text{Sh}^{\text{Set}}(\mathcal{C}) \rightarrow \text{Sh}^{\text{Set}}(\mathcal{D})$  is a natural transformation  $t : f_* \rightarrow g_*$ .  $\lrcorner$

**Prop. (7.1.2.19) [Presheaves and Topoi].**  $\mathcal{C} \in \text{Cat}$  is a Grothendieck category iff it is a left exact, reflective accessible localization (4.1.2.10) of  $\mathcal{P}\text{Sh}^{\text{Set}}(\mathcal{A})$  for some  $\mathcal{A} \in \text{Cat}$ .  $\lrcorner$

### Sites and Topoi

**Prop. (7.1.2.20) [Continuous Maps and Topoi].** A morphism of sites  $f : \mathcal{D} \rightarrow \mathcal{C}$  consists of a continuous functor  $u : \mathcal{C} \rightarrow \mathcal{D}$  that  $u_s$  is exact, so it induces a functor of topoi  $f : \text{Sh}^{\text{Set}}(\mathcal{C}) \rightarrow \text{Sh}(\mathcal{D})$ , if we define  $f^{-1} = u_s, f_* = u^s$ , by (7.1.2.11).  $\lrcorner$

**Prop. (7.1.2.21) [Cocontinuous Maps and Topoi].** Let  $u : \mathcal{C} \rightarrow \mathcal{D}$  be a cocontinuous functor between sites, then this defines a morphism of topoi  $g : \text{Sh}^{\text{Set}}(\mathcal{C}) \rightarrow \text{Sh}^{\text{Set}}(\mathcal{D})$ , if we define  $g^{-1} = u^s$  and  $g_* = u_s$ , by (7.1.2.11).  $\lrcorner$

**Prop. (7.1.2.22) [Cocontinuous Map with a Right Adjoint].** Let  $u : \mathcal{C} \rightarrow \mathcal{D}$  be a cocontinuous functor between sites with a right adjoint  $v : \mathcal{D} \rightarrow \mathcal{C}$ , then the morphism of topoi  $g : \text{Sh}^{\text{Set}}(\mathcal{C}) \rightarrow \text{Sh}^{\text{Set}}(\mathcal{D})$  is pretty simple,

$$g^{-1}\mathcal{G}(U) = \mathcal{G}(u(U)), \quad g_*\mathcal{F}(V) = \mathcal{F}(v(V)).$$

This holds in particular for localization of sites.  $\lrcorner$

*Proof:* It follows from adjunction that  $u^p h_V = h_{v(V)}$ , so  $g^{-1}(h_V^\sharp) = (u^p h_V^\sharp)^\sharp = (u^p h_V)^\sharp = h_{v(V)}^\sharp$ , and

$$(g_*\mathcal{F})(V) = \text{Hom}(h_V^\sharp, g_*\mathcal{F}) = \text{Hom}(g^{-1}h_V^\sharp, \mathcal{F}) = \text{Hom}(h_{v(V)}^\sharp, \mathcal{F}) = \mathcal{F}(v(V)).$$

The other identity is clear.  $\square$

**Prop. (7.1.2.23) [Continuous and Cocontinuous Maps and Topoi].** Let  $u : \mathcal{C} \rightarrow \mathcal{D}$  be a functor between sites that satisfies

- $u$  is continuous and cocontinuous,
- $u$  is fully faithful,
- $\mathcal{C}$  has final objects and fiber products and  $u$  preserves them.

then there are two maps of topoi:  $f = (u_s, u^s) : \mathrm{Sh}^{\mathrm{Set}}(\mathcal{D}) \rightarrow \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C})$ ,  $g = (u^s, {}_s u) : \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C}) \rightarrow \mathrm{Sh}^{\mathrm{Set}}(\mathcal{D})$  induced by  $u$  (7.1.2.20)(7.1.2.14)(7.1.2.21). They satisfy  $f \circ g = \mathrm{id}_{\mathrm{Sh}^{\mathrm{Set}}(\mathcal{C})}$ , and  $f^{-1} = u_s$  is fully faithful.

In particular, if we define  $g_! = u_s$ , then there are two adjunctions

$$(g_!, g^{-1}) : \mathrm{Sh}(\mathcal{C}) \rightleftarrows \mathrm{Sh}(\mathcal{D}), \quad (g^{-1}, g_*) : \mathrm{Sh}(\mathcal{D}) \rightleftarrows \mathrm{Sh}(\mathcal{C}).$$

□

*Proof:* We need to show that for any  $\mathcal{F} \in \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C})$ ,  $\mathcal{F} \cong g^{-1}g_!\mathcal{F}$  and  $g^{-1}g_*\mathcal{F} \cong \mathcal{F}$ . For this, Cf. [Sta]00XT.?

Then  $f^!$  is fully faithful follows from the equality  $\mathcal{F} \cong g^{-1}g_!\mathcal{F}$ . □

**Prop. (7.1.2.24) [Special Cocontinuous Maps and Topoi].** A functor  $u : \mathcal{C} \rightarrow \mathcal{D}$  between sites is called a **special cocontinuous functor** if:

- $u$  is continuous and cocontinuous,
- Given any  $a, b : U' \rightarrow U \in \mathcal{C}$  s.t.  $u(a) = u(b) : u(U') \rightarrow u(U)$ , there is a covering  $\{f_i : U'_i \rightarrow U'\}$  s.t.  $a \circ f_i = b \circ f_i$ .
- Given any  $U', U \in \mathcal{C}$  and a morphism  $c : u(U') \rightarrow u(U) \in \mathcal{D}$ , there exists a covering  $\{f_i : U'_i \rightarrow U'\} \in \mathcal{C}$  and morphisms  $c_i : U'_i \rightarrow U$  that  $u(c_i) = c \circ u(f_i)$ .
- Given any  $V \in \mathcal{D}$ , there is a covering of the form  $\{u(U_i) \rightarrow V\}$  in  $\mathcal{D}$ .

Then the induced morphism of topoi  $g : \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C}) \cong \mathrm{Sh}^{\mathrm{Set}}(\mathcal{D})$  (7.1.2.21) is an equivalence of topoi. □

*Proof:* Cf. [Sta]03A0.?

□

**Cor. (7.1.2.25) [Comparing Topologies].** Let  $\mathcal{C}'$  be a fully subcategory of  $\mathcal{C}$ , and

- $i : \mathcal{C}' \rightarrow \mathcal{C}$  is continuous, i.e.  $\mathrm{Cov}(\mathcal{C}') \rightarrow \mathrm{Cov}(\mathcal{C})$ .
- $i : \mathcal{C}' \rightarrow \mathcal{C}$  is cocontinuous, i.e. any covering  $\{U_i \rightarrow U'\} \in \mathrm{Cov}(\mathcal{C})$  s.t.  $U' \in \mathcal{C}'$  has a refinement  $\{U'_j \rightarrow U'\} \in \mathrm{Cov}(\mathcal{C}')$ .
- Each  $U \in \mathcal{C}$  has a covering  $\{U_i \rightarrow U\} \in \mathrm{Cov}(\mathcal{C})$  with  $U_i \in \mathcal{C}'$ .

then  $i : \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C}') \rightarrow \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C})$  is an equivalence of topoi.  $i^{-1}$  is just the restriction functor, and  $i_*$  is called the **extension functor of sheaves**.

In particular, this applies to the case  $i : \mathcal{C}' = \mathcal{C}/Z \rightarrow \mathcal{C}$ , the localization category, in which case for  $Z' \in \mathcal{C}/Z$ ,  $i^{-1}F(Z') = F(Z')$  is called the **restriction sheaf**. □

**Prop. (7.1.2.26) [Localizing at Sheaves].** Let  $\mathcal{C}$  be a site and  $\mathcal{F}_i$  be a set of topoi on  $\mathcal{C}$ , then there is an equivalence  $\mathrm{Sh}^{\mathrm{Set}}(\mathcal{C}) \cong \mathrm{Sh}^{\mathrm{Set}}(\mathcal{C}')$  induced by a special cocontinuous functor  $\mathcal{C} \rightarrow \mathcal{C}'$  (7.1.2.24) s.t.

- $\mathcal{C}'$  has the subcanonical topology,
- A family of morphisms  $\{V_i \rightarrow V\}$  are a covering of  $\mathcal{C}'$  iff  $\coprod h_{V_i} \rightarrow h_V$  is surjective.
- $\mathcal{C}'$  has fiber products and a final object.

- Every subsheaf of a representable sheaf is representable,
- Each  $g_*\mathcal{F}_i$  is a representable sheaf.

┘

*Proof:* Cf. [Sta]03CI. □

### 3 Stacks

**Notation (7.1.3.1).**

- Let  $\mathcal{C} \in \text{Site}$ .

┘

**Def. (7.1.3.2) [Stacks].**  $\mathcal{F} \in \text{FibCat}(\mathcal{C})$  (4.2.3.8) is called a **prestack** (resp. **stack**) over  $\mathcal{C}$  if for each covering  $\{U_i \rightarrow U\}$  in  $\mathcal{C}$ , the functor  $\text{Hom}(\textstyle\bigvee_U, \mathcal{F}) \rightarrow \text{Hom}(\textstyle\bigvee_U, \mathcal{F})$  (7.1.1.9) is fully faithful (resp. an equivalence of categories).

Denote the category of prestacks (resp. stacks) over  $\mathcal{C}$  by  $\mathcal{P}\text{Sta}(\mathcal{C})$  (resp.  $\text{Sta}(\mathcal{C})$ ). ┘

**Def. (7.1.3.3) [Category of Descent Datum].** Let  $\mathcal{F} \in \text{FibCat}(\mathcal{C})$ ,  $U \in \mathcal{C}$  and  $\mathcal{U} \in \text{Cov}_{\mathcal{C}}(U)$ . Given a choice of fibered products  $U_{ij}, U_{ijk}$  in  $\mathcal{C}$ , we can define the **category of descent datum**  $\mathcal{F}_{\text{desc}}(\mathcal{U})$  to be the category of tuples  $(\xi_i, \xi_{ij}, \xi_{ijk})$  where  $\xi_\alpha \in \mathcal{F}(U_\alpha)$  with Cartesian morphisms between them that are commutative. A morphism in  $\mathcal{F}_{\text{desc}}(\mathcal{U})$  is a family of morphisms  $(\varphi_i, \varphi_{ij}, \varphi_{ijk})$  commuting with the Cartesian morphisms.

Then there is an equivalence of categories  $\text{Hom}(h_{\mathcal{U}}, \mathcal{F}) \cong \mathcal{F}_{\text{desc}}(\mathcal{U})$ . ┘

*Proof:* For any  $F : h_{\mathcal{U}} \rightarrow \mathcal{F}$ ,  $U_i \rightarrow U \in h_{\mathcal{U}}(U_i)$ , applying  $F$  to the arrows  $\text{id}_{U_i}, U_{ij} \rightarrow U_i, U_{ijk} \rightarrow U_{ij} \in h_{\mathcal{U}}$ , we get an element in  $\mathcal{F}_{\text{desc}}(\mathcal{U})$ . Also a natural transformation  $F \rightarrow G$  maps to a morphism in  $\mathcal{F}_{\text{desc}}(\mathcal{U})$ , so we get a functor  $T : \text{Hom}(h_{\mathcal{U}}, \mathcal{F}) \rightarrow \mathcal{F}_{\text{desc}}(\mathcal{U})$ .

Conversely, choose an arbitrary choice of pullbacks (that coincide with  $\xi_{ij} \rightarrow \xi_i$ ), for any arrow  $f : T \rightarrow U \in h_{\mathcal{U}}(T)$ , we choose a  $U_i$  that  $T \rightarrow U$  factors as  $T \rightarrow U_i \rightarrow U$  (also for  $f = \text{id}_{U_i}$ , choose  $U_i$ ), then define  $F(f)$  as the pullback of  $\xi_i$  along  $T \rightarrow U_i$ . For any morphisms  $T' \rightarrow T \rightarrow U \in h_{\mathcal{U}}$  and their choice of  $U_i, U_j$  that  $T' \rightarrow U$  factors through  $U_j \rightarrow U$  and  $T \rightarrow U$  factors as  $U_i \rightarrow U$ , then  $T' \rightarrow U_j$  factors through  $U_{ij}$ , i.e. we have a commutative diagram

$$\begin{array}{ccccc} T' & \longrightarrow & U_{ij} & \longrightarrow & U_j \\ \downarrow & & \downarrow & & \downarrow \\ T & \longrightarrow & U_i & \longrightarrow & U \end{array}.$$

Then by Cartesian properties, we get a unique map  $F(T') \rightarrow F(T)$ , which is Cartesian by (4.2.3.6). It can be shown these maps make  $F$  a functor, using Cartesian property and the existence of  $\xi_{ijk}$ . And also for any map of descent datum, we get a natural transformation in  $\text{Hom}(h_{\mathcal{U}}, \mathcal{F})$ . Thus we get a functor  $S : \mathcal{F}_{\text{desc}}(\mathcal{U}) \rightarrow \text{Hom}(\textstyle\bigvee_U, \mathcal{F})$ .

The construction of  $S$  shows  $T$  is essentially surjective and full, and also faithful, so  $(S, T)$  is an equivalence of categories. □

**Def. (7.1.3.4) [Everyday Descent Datums].** Let  $\mathcal{F} \in \text{FibCat}(\mathcal{C})$ ,  $U \in \mathcal{C}$  and  $\mathcal{U} \in \text{Cov}_{\mathcal{C}}(U)$ . Given a choice of fibered products  $U_{ij}, U_{ijk}$  in  $\mathcal{C}$  and a cleavage of  $\mathcal{F}/\mathcal{C}$ , let  $\mathcal{F}(\mathcal{U})$  be the category of tuples  $(\xi_i, \varphi_{ij})$  where  $\xi_i \in \mathcal{F}(U_i)$  and  $\varphi_{ij}$  are isomorphisms  $\varphi_{ij} : \text{pr}_1^* \xi_i \cong \text{pr}_2^* \xi_j \in \mathcal{F}(U_{ij})$  s.t.

$$\text{pr}_{13}^* \varphi_{ik} = \text{pr}_{23}^* \varphi_{jk} \circ \text{pr}_{12}^* \varphi_{ij} : \text{pr}_1^* \xi_i \cong \text{pr}_3^* \xi_k.$$



**WARNING:** Notice this doesn't make sense unless we insert three isomorphisms such as  $\mathrm{pr}_{12}^* \mathrm{pr}_1^* \xi_i \cong \mathrm{pr}_{13}^* \mathrm{pr}_1^* \xi_i$ .

Then there is a (non-canonical!) equivalence of categories  $\mathcal{F}_{desc}(\mathcal{U}) \rightarrow \mathcal{F}(\mathcal{U})$ .  $\lrcorner$

*Proof:* For  $(\xi_i, \xi_{ij}, \xi_{ijk}) \in \mathcal{F}_{desc}(\mathcal{U})$ , there are isomorphisms  $\xi_{ij} \cong \mathrm{pr}_1^* \xi_i$  and  $\xi_{ij} \cong \mathrm{pr}_2^* \xi_j$ , which gives an isomorphism  $\varphi_{ij} : \mathrm{pr}_1^* \xi_i \rightarrow \mathrm{pr}_2^* \xi_j$ . This gives an element of  $\mathcal{F}(\mathcal{U})$ , by comparison with  $\xi_{ijk}$ .

Conversely, given  $(\xi_i, \varphi_{ij}) \in \mathcal{F}(\mathcal{U})$ , choose an ordering on  $I$ , for  $i < j$ , let  $\xi_{ij} = \mathrm{pr}_1^* \varphi_i$ , and  $\mathrm{pr}_2^* \varphi_{ij} : \xi_{ij} \rightarrow \xi_j$  is Cartesian. And for  $i < j < k$ , let  $\xi_{ijk} = \mathrm{pr}_1^* \xi_i$ , then there are Cartesian morphisms  $\xi_{ijk} \rightarrow \xi_{ij}, \xi_{ijk} \rightarrow \xi_{ik}, \xi_{ijk} \rightarrow \xi_{jk}$  by Cartesian properties, and it can be verified that the cocycle condition guarantees the diagram can be completed.  $\square$

**Cor. (7.1.3.5) [Cocycle Conditions].** If  $\mathcal{F} \in \mathrm{FibCat}(\mathcal{C})$  is a rigid fibered category, then there is no need to check the cocycle condition, because every automorphism over  $\mathrm{id}_{\mathcal{U}}$  is trivial.  $\lrcorner$

**Cor. (7.1.3.6) [Stacks and Sheaves].** A rigid (pre)stack over  $\mathcal{C}$  is equivalent to a (pre)sheaf on  $\mathcal{C}$ , by (4.2.3.26).  $\lrcorner$

**Cor. (7.1.3.7).** A site is subcanonical iff any representable fibered category  $\mathcal{Y}_{\mathcal{U}} \rightarrow \mathcal{C}$  is a stack.  $\lrcorner$

**Prop. (7.1.3.8) [Equivalence Categories and Stacks].** If  $\mathcal{F} \rightarrow \mathcal{G} \in \mathrm{FibCat}(\mathcal{C})$  is an equivalence of fibered categories over  $\mathcal{C}$ , then  $\mathcal{F}$  is a prestack/stack(in groupoid) iff  $\mathcal{G}$  is.  $\lrcorner$

*Proof:* There is a strict commutative diagram of categories

$$\begin{array}{ccc} \mathrm{Hom}(h_U, \mathcal{F}) & \longrightarrow & \mathrm{Hom}(h_U, \mathcal{F}) \\ \downarrow & & \downarrow \\ \mathrm{Hom}(h_U, \mathcal{G}) & \longrightarrow & \mathrm{Hom}(h_U, \mathcal{G}) \end{array}$$

that the vertical arrows are equivalences of categories, then we are done.  $\square$

**Prop. (7.1.3.9) [Prestack and Hom Functor].** Let  $\mathcal{F}$  be a fibered category over a site  $\mathcal{C}$ , then  $\mathcal{F}$  is a prestack iff for any object  $S$  of  $\mathcal{C}$  and two objects  $\xi, \eta \in \mathcal{F}(S)$ , the presheaf  $\underline{\mathrm{Hom}}_S(\xi, \eta) : (\mathcal{C}/S)^{op} \rightarrow \mathrm{Set}$  (4.2.3.17) is a sheaf in the comma site  $\mathcal{C}/S$  (7.1.1.6).  $\lrcorner$

*Proof:* By (7.1.3.8), it suffices to show  $\mathrm{Hom}_S(\xi, \eta)$  is a stack in the comma topology  $\mathcal{C}/S$ . Then it can be shown that  $\mathrm{Hom}(h_U, \underline{\mathrm{Hom}}_S(\xi, \eta)) \rightarrow \mathrm{Hom}(h_U, \underline{\mathrm{Hom}}_S(\xi, \eta))$  is an equivalence of categories iff  $\mathrm{Hom}(h_U, \mathcal{F}) \rightarrow \mathrm{Hom}(h_U, \mathcal{F})$  is fully faithful.  $\square$

**Lemma (7.1.3.10).** Let  $\mathcal{F} \in \mathcal{P}\mathrm{Sta}(\mathcal{C})$  and  $S, S'$  be sieves belonging to the topology of  $\mathcal{C}$  that  $S' \subset S$ , then the restriction functor

$$\mathrm{Hom}_{\mathcal{C}}(S, \mathcal{F}) \rightarrow \mathrm{Hom}_{\mathcal{C}}(S', \mathcal{F})$$

is faithful.  $\lrcorner$

*Proof:* it suffices to show for  $S' = h_U$  for some covering  $\mathcal{U}$ . Let  $F, G \in \mathrm{Hom}_{\mathcal{C}}(S', \mathcal{F})$  and  $\varphi, \psi$  be two natural transformations from  $F$  to  $G$  that induce the same natural transformation from the restriction of  $F$  to  $h_U$  to the restriction of  $G$  to  $h_U$ , then  $\varphi = \psi$ . For this, just notice there are commutative diagrams

$$\begin{array}{ccc} F(T \times_U U_i/U) & \xrightarrow{\varphi_{T \times_U U_i/U}} & G(T \times_U U_i/U) \\ \downarrow & & \downarrow \\ F(T/U) & \xrightarrow{\varphi_{T/U}} & G(T/U) \end{array},$$



where the vertical arrows are Cartesian, and the hypothesis implies  $\varphi_{T \times_U U_i/U} = \psi_{T \times_U U_i/U}$ . Then we deduce  $\varphi_{T/U} = \psi_{T/U}$  as  $\underline{\text{Hom}}_T(F(T/U), G(T/U))$  is a sheaf, as  $\mathcal{F}$  is a presheaf (7.1.3.9).  $\square$

**Prop. (7.1.3.11)[Stack and Sieves].** A prestack  $\mathcal{F} \in \mathcal{P}\text{Sta}(\mathcal{C})$  is stack iff for any  $U \in \mathcal{C}$  and any sieve  $\mathcal{S}$  on  $U$  belonging to  $\mathcal{T}$ ,

$$\text{Hom}(\mathcal{Y}_U, \mathcal{F}) \rightarrow \text{Hom}(\mathcal{S}, \mathcal{F})$$

is an equivalence of categories.  $\lrcorner$

*Proof:* Let  $\mathcal{S}$  be a sieve on  $U$  belong to  $\mathcal{C}$ , choose a covering  $\mathcal{U}$  of  $U$  that  $h_{\mathcal{U}} \subset \mathcal{S} \subset h_U$ , then there is a factorization

$$\text{Hom}(h_{\mathcal{U}}, \mathcal{F}) \rightarrow \text{Hom}(\mathcal{S}, \mathcal{F}) \rightarrow \text{Hom}(h_U, \mathcal{F}).$$

Then we are done by (7.1.3.10).  $\square$

**Cor. (7.1.3.12).** let  $\mathcal{T}, \mathcal{T}'$  be two topologies on a category  $\mathcal{C}$  that  $T'$  is subordinate to  $\mathcal{T}$  and  $\mathcal{F} \rightarrow \mathcal{C}$  is a fibered category, then if  $\mathcal{F}$  is a prestack/stack relative to  $\mathcal{T}$ , it is also true for  $\mathcal{T}'$ .  $\lrcorner$

**Prop. (7.1.3.13)[2-Fiber Products of Stacks].** There is a natural 2-category of stacks over  $\mathcal{C}$  defined as a sub-2-category of the 2-category of fibered-categories over  $\mathcal{C}$ , and the  $(2, 1)$ -category of stacks over  $\mathcal{C}$  has 2-fibered products, which coincides with that of (4.2.3.13).  $\lrcorner$

*Proof:* For  $\mathcal{X} \rightarrow \mathcal{S}, \mathcal{Y} \rightarrow \mathcal{S} \in \text{Sta}(\mathcal{C})$ , the category  $\mathcal{X} \times_{\mathcal{S}} \mathcal{Y}$  is a fibered category over  $\mathcal{C}$ , by (4.2.3.13). It remains to show that the morphism presheaves sheaves and descent datum are effective.

For this, Cf. [Sta]026G. ?  $\square$

**Prop. (7.1.3.14)[Associated Stack Fibered in Groupoids].** Let  $\mathcal{C} \in \text{Site}$ ,  $\mathcal{F} \in \mathcal{P}\text{Sta}(\mathcal{C})$  and  $\mathcal{F}_{\text{Cart}} \in \text{Sta}^{\text{grpd}}(\mathcal{C})$  is the associated category fibered in groupoids (4.2.3.22), then  $\mathcal{F}_{\text{Cart}}$  is also a prestack. And in this case,  $\mathcal{F}$  is a stack iff  $\mathcal{F}_{\text{Cart}}$  is a stack.  $\lrcorner$

*Proof:* The categories  $\mathcal{F}(\mathcal{U})$  and  $\mathcal{F}_{\text{Cart}}(\mathcal{U})$  have the same isomorphism classes of objects, as isomorphisms are Cartesian, so it suffices to show  $\mathcal{F}$  is a prestack iff  $\mathcal{F}_{\text{Cart}}$  is a prestack. For this, use (7.1.3.9) and consider  $\xi, \eta \in \mathcal{F}(U)$  and a covering  $\{U_i \rightarrow U\}$ , if there are arrows  $\alpha_i : \xi_i \rightarrow \eta_i$  that are compatible, then there is a unique arrow  $\alpha : \xi \rightarrow \eta$  restricting to  $\alpha_i$ , thus it suffices to show  $\alpha$  is Cartesian. But since  $\mathcal{F}_{\text{Cart}}$  is a category fibered in groupoids, each  $\alpha_i$  is invertible, and their inverses glue together to a morphism  $\beta : \eta \rightarrow \xi$  which is the inverse of  $\alpha$ , so  $\alpha$  is also an isomorphism thus Cartesian.  $\square$

**Prop. (7.1.3.15)[2-Fibered Products of Stacks Fibered in Groupoids].** The 2-category of stacks fibered in groupoids over  $\mathcal{C}$  (automatically a  $(2, 1)$ -category because any 2-morphism is an isomorphism) has 2-fiber products, and coincide with that of (7.1.3.13).  $\lrcorner$

*Proof:* This is clear from (7.1.3.13) and (4.2.3.24).  $\square$

**Prop. (7.1.3.16)[Equivalence of Stacks].** Let  $F : \mathcal{S}_1 \rightarrow \mathcal{S}_2 \in \text{Sta}(\mathcal{C})$  be fully faithful, then  $F$  is an equivalence of stacks iff for any  $x \in \mathcal{S}_{2,U}$ , there exists a covering  $\{f_i : U_i \rightarrow U\}$  s.t.  $f_i^* x$  is in the essential image of the functor  $F : \mathcal{S}_{1,U} \rightarrow \mathcal{S}_{2,U}$ .  $\lrcorner$

*Proof:* Easy.  $\square$

**Prop. (7.1.3.17)[Subcanonical Site Prestack].** If  $\mathcal{C}$  is a subcanonical site (7.1.2.4) and  $\mathcal{P}$  is a class of arrows in  $\mathcal{C}$  stable under base change, then the corresponding fibered category  $\mathcal{P} \rightarrow \mathcal{C}$  is a prestack.  $\lrcorner$

*Proof:* By (7.1.3.9), we need to prove for any covering  $\{U_i \rightarrow U\}$  and arrows  $X \rightarrow U, Y \rightarrow U$ ,  $X_i = U_i \times_U X$ , and  $X_{ij} = U_{ij} \times_U X$  and analogous for  $Y$ , if there are arrows  $f_i : X_i \rightarrow Y_i$  over  $U_i$  that the arrows  $X_{ij} \rightarrow Y_{ij}$  induced by  $f_i$  and  $f_j$  coincide, then there is a unique arrow  $f : X \rightarrow Y$  over  $U$  that induces  $f_i$ .

Notice that the composite  $X_i \xrightarrow{f_i} Y_i \rightarrow Y$  give sections  $g_i \in h_Y(X_i)$ , and the pullback of  $g_i, g_j$  to  $X_{ij}$  coincide by hypothesis. Now  $h_Y$  is a sheaf by hypothesis, so there is an arrow  $f : X \rightarrow Y$  that pulls back to  $f_i$  on  $X_i$ . Finally  $f$  is compatible over  $U$  because  $(Y \rightarrow U) \circ f$  and  $(X \rightarrow U)$  coincide when composed with  $X_i \rightarrow X$ , and  $h_U$  is a sheaf.  $\square$

**Prop. (7.1.3.18) [Category of Sheaves is a Stack].** For  $\mathcal{C} \in \text{Site}$ , we denote  $(\text{Sh}/\mathcal{C})(X) = \text{Sh}(\mathcal{C}/X)$ , then  $\text{Sh}/\mathcal{C} \in \text{Sta}/\mathcal{C}$ .  $\lrcorner$

*Proof:* To show  $\text{Sh}/\mathcal{C}$  is a prestack, by (7.1.3.9), it suffices to show for any  $F, G \in \text{Sh}(\mathcal{C}/X)$ ,  $\underline{\text{Hom}}_X(F, G)$  is a sheaf. For this, let  $\{U_i \rightarrow U\}$  be a covering, and  $\varphi_i : F_{U_i} \rightarrow G_{U_i}$  be morphisms of sheaves that their restrictions to  $F_{U_{ij}} \rightarrow G_{U_{ij}}$  are compatible, then for any  $T \rightarrow U$ , there are commutative diagrams

$$\begin{array}{ccccc} F(T) & \longrightarrow & \prod_i F_i(T_i) & \longrightarrow & \prod_{i,j} F_{ij}(T_{ij}) \\ \downarrow & & \downarrow & & \downarrow \\ G(T) & \longrightarrow & \prod_i G_i(T_i) & \longrightarrow & \prod_{i,j} G_{ij}(T_{ij}) \end{array}$$

where  $\varphi_T : F(T) \rightarrow G(T)$  is the unique function of sets that makes the diagram commutative. And it can be shown that these  $\varphi_T$  defines a natural transformation  $F_U \rightarrow G_U$ .

Now for any covering  $\{U_i \rightarrow U\}$  and a descent datum  $(F_i, F_{ij})$ , we need to show it is effective. We define a function  $F$  on  $\mathcal{C}/U$ :  $F(T) = \text{equal}(\prod_i F(T_i) \rightrightarrows \prod_{i,j} F(T_{ij}))$ , then it can be shown that this is a sheaf by spectral sequence.

Then it suffices to check  $F_{U_k} = F_k$ . For any  $T \rightarrow U_k$ ,  $T_i$  maps to  $U_{ik}$ , so  $F_i(T_i) = F_k(T_i)$ , thus for any  $s \in F_k(T)$ , we can produce an element  $(s_{T_i}) \in \prod_i F_i(T_i)$  that satisfies compatibility conditions, which gives us an element of  $F(T)$ . It can be shown this is a natural transformation  $F_k \rightarrow F_{U_k}$ , and it is an isomorphism of sheaves.  $\square$

**Prop. (7.1.3.19) [Descent Along Torsors].** Let  $\mathcal{C} \in \text{Site}$ ,  $G \in \text{Grp}(\mathcal{C})$  and  $X \rightarrow Y \in \text{Tors}_G(Y)$ ,  $\mathcal{F} \in \text{Sta}/\mathcal{C}$ , then there exists a canonical equivalence of categories between  $\mathcal{F}(Y)$  and the category of  $G$ -equivariant objects  $\mathcal{F}^G(X)$  (4.2.3.15).  $\lrcorner$

*Proof:* By (7.1.1.19)  $X \rightarrow Y$  is refined by a covering, thus by (7.1.3.11),  $\mathcal{F}(Y)$  is equivalent to  $\mathcal{F}(X \rightarrow Y)$ . And we check  $\mathcal{F}(X \rightarrow Y) \cong \mathcal{F}^G(X)$ . Then by (7.1.1.20),  $\mathcal{F}(X \rightarrow Y)$  consists of elements  $\xi \in \mathcal{F}(X), \eta \in \mathcal{F}(G \times X)$  and Cartesian arrows  $\varphi_1, \varphi_2$  over  $\alpha, \pi_2$ . Cf. [Vis08]P106. ?  $\square$

### Stackifications

**Prop. (7.1.3.20) [Stackifications].** For  $p : \mathcal{F} \rightarrow \mathcal{C} \in \text{FibCat}()$ , there exists a  $\mathcal{F}' \in \text{Sta}(\mathcal{C})$  and  $G : \mathcal{F} \rightarrow \mathcal{F}' \in \text{FibCat}(\mathcal{C})$  s.t. for any stack  $\mathcal{X} \in \text{Sta}(\mathcal{C})$ , any morphism  $F : \mathcal{F} \rightarrow \mathcal{X} \in \text{FibCat}(\mathcal{C})$  factors through  $G$  2-commutatively and uniquely up to 2-isomorphisms. In other words, there is a canonical equivalence of categories:

$$\text{Mor}_{\text{FibCat}(\mathcal{C})}(\mathcal{F}, \mathcal{X}) \cong \text{Mor}_{\text{Sta}(\mathcal{C})}(\mathcal{F}', \mathcal{X}).$$

In particular, such a stack  $\mathcal{F}'$  is determined up to unique 2-isomorphisms, and is called the **stackification** of  $\mathcal{F}$ .  $\lrcorner$

*Proof:* By (4.2.3.18), we may assume  $\mathcal{F}$  is split, thus  $\mathcal{F}$  corresponds to the functor  $\mathcal{C}^{op} \rightarrow \mathcal{Cat} : U \mapsto \text{Hom}(\mathcal{Y}_U, \mathcal{F})$ . Then we define a functor

$$\mathcal{F}' : \mathcal{C}^{op} \rightarrow \mathcal{Cat} : U \mapsto \varinjlim_{\mathcal{U} \in \text{Cov}_{\mathcal{C}}(U)} \text{Hom}(\mathcal{Y}_{\mathcal{U}}, \mathcal{F}).$$

then there is a natural map of fibered categories  $\mathcal{F} \rightarrow \mathcal{F}'$ . For any  $\mathcal{X} \in \text{Sta}(\mathcal{C})$ , because  $\text{Hom}(\mathcal{Y}_U, \mathcal{X}) \rightarrow \text{Hom}(\mathcal{Y}_{\mathcal{U}}, \mathcal{X})$  is an isomorphism for any  $\mathcal{U} \in \text{Cov}_{\mathcal{C}}(U)$ , the desired equivalence of categories follows.  $\square$

**Prop. (7.1.3.21) [Characterizing Stackifications].** Let  $\mathcal{S} \in \text{FibCat}(\mathcal{C})$  and  $G : \mathcal{S} \rightarrow \mathcal{S}'$  be the stackification (7.1.3.20), then

- For any  $U \in \mathcal{C}$  and  $x, y \in \mathcal{S}_U$ , the map

$$\mathcal{M}or(x, y) \rightarrow \mathcal{M}or(G(x), G(y))$$

identifies the RHS as the shiffication of the LHS.

- For any  $U \in \mathcal{C}$  and  $x \in \mathcal{S}'_U$ , there exists a covering  $\{U_i \rightarrow U\}$  that for any  $i$ ,  $x|_{U_i}$  is in the essential image of  $G_U : \mathcal{S}_U \rightarrow \mathcal{S}'_U$ .

⌋

*Proof:* Cf. [Sta]0435.  $\square$

**Cor. (7.1.3.22).** Let  $\mathcal{S} \in \text{FibCat}(\mathcal{C})$  and  $G : \mathcal{S} \rightarrow \mathcal{S}'$  be the stackification (7.1.3.20), then  $\mathcal{S}'$  is a stackification in groupoids if  $\mathcal{S}$  is a fibered in groupoids.  $\square$

*Proof:* Check that  $\mathcal{S}'_U$  is a groupoid for any  $U \in \mathcal{U}$ , using (7.1.3.21).  $\square$

**Prop. (7.1.3.23) [Stackifications Commute with 2-Fibered Products].** Stackifications commute with 2-fibered products.  $\square$

*Proof:* Cf. [Sta]04Y1.  $\square$

**Prop. (7.1.3.24).** Suppose  $X \in \mathcal{T}op$  and  $\eta : \mathcal{F} \rightarrow \mathcal{G} \in \mathcal{P}Sta(X)$  s.t.

- $\mathcal{F} \in \text{Sta}/X$ .
- $\eta$  induces isomorphisms on stalks.
- $\eta(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  is fully faithful.

Then  $\eta$  is an equivalence of prestacks. In particular,  $\mathcal{G}$  is also a stack.  $\square$

*Proof:* Let  $\mathcal{H}$  be the stackification of  $\mathcal{G}$ , then  $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H}$  is a morphism of stacks that is isomorphism on the stalk, so it is an isomorphism ?. But  $\mathcal{G}$  is separated, so for any open  $U$ ,  $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ ,  $\mathcal{G}(U) \rightarrow \mathcal{H}(U)$  is fully faithful, and their composition is an equivalence, thus both of them are equivalences.  $\square$

## Gerbes

**Def. (7.1.3.25) [Gerbes over Sites].** For  $\mathcal{C} \in \text{Site}$ , a **gerbe** over  $\mathcal{C}$  is a  $\mathcal{S} \in \text{Sta}^{\text{Grpd}}(\mathcal{C})$  s.t.

- For any  $U \in \mathcal{C}$ , there is a covering  $\{U_i \rightarrow U\}$  that  $\mathcal{S}_{U_i}$  is nonempty for any  $i$ .
- For any  $U \in \mathcal{C}$  and  $x, y \in \mathcal{S}_U$ , there exists  $\{U_i \rightarrow U\} \in \text{Cov}(\mathcal{C})$  s.t.  $x|_{U_i} = y|_{U_i}$  for any  $i$ .

We use  $\mathcal{S} \in \text{Gerb}(\mathcal{C})$  to denote that  $\mathcal{S}$  is a gerbe over  $\mathcal{C}$ .  $\square$

**Prop. (7.1.3.26).** Let  $\mathcal{C} \in \text{Site}$  and  $\mathcal{S}_1, \mathcal{S}_2 \in \text{Cat}/\mathcal{C}$ . Suppose  $\mathcal{S}_1 \cong \mathcal{S}_2 \in \text{Cat}/\mathcal{C}$ , then  $\mathcal{S}_1$  is a gerbe over  $\mathcal{C}$  iff  $\mathcal{S}_2$  is a gerbe over  $\mathcal{C}$ .  $\lrcorner$

*Proof:* If  $\mathcal{S}_1$  is a gerbe, then  $\mathcal{S}_2 \in \text{Sta}(\mathcal{C})$  by (7.1.3.8). The other properties of gerbes are easy to verify for  $\mathcal{S}_2$ , using (4.2.3.12).  $\square$

**Prop. (7.1.3.27).** If  $\mathcal{C} \in \text{Site}$  and  $\mathcal{S} \in \text{Gerby}(\mathcal{C})$  s.t. for all  $U \in \mathcal{C}$  and  $x \in \mathcal{S}_U$ , the sheaf of groups  $\underline{\text{Aut}}(x)$  on  $\mathcal{C}/U$  is Abelian, then there exists a sheaf of Abelian groups over  $\mathcal{C}$  and for any  $x \in \mathcal{S}_U$  an isomorphism  $\mathcal{G}|_U \rightarrow \underline{\text{Aut}}(x)$  that for any morphism  $\varphi : x \rightarrow y \in \mathcal{S}_U$ , the diagram

$$\begin{array}{ccc} \mathcal{G}|_U & \xlongequal{\quad} & \mathcal{G}|_U \\ \downarrow & & \downarrow \\ \underline{\text{Aut}}(x) & \xrightarrow{\alpha \mapsto \varphi \circ \alpha \circ \varphi^{-1}} & \underline{\text{Aut}}(y) \end{array}$$

is commutative.  $\lrcorner$

*Proof:* It can be checked by using the fact  $\underline{\text{Aut}}(x)$  is Abelian that there are canonical morphisms  $\underline{\text{Aut}}(x) \rightarrow \underline{\text{Aut}}(y)$  induced by any morphism  $\varphi : x \rightarrow y$ .

If there is no morphism from  $x$  to  $y$ , then we can use the condition of gerbe to obtain morphisms  $\underline{\text{Aut}}(x)|_{U_i} \rightarrow \underline{\text{Aut}}(y)|_{U_i}$  locally, and then glue together. Similarly, if  $\mathcal{S}_U$  is empty, then we can restrict to a covering and then glue.

Finally, notice this gives an Abelian sheaf  $\mathcal{G} = \underline{\text{Aut}}$  on  $\mathcal{C}$ .  $\square$

**Def. (7.1.3.28) [Gerbes over Stacks in Groupoids].** For  $F : \mathcal{X} \rightarrow \mathcal{Y} \in \text{Sta}^{\text{grp}}(\mathcal{C})$ , the following are equivalent:

- $F$  factors as  $F = F' \circ a$  where  $a : \mathcal{X} \rightarrow \mathcal{X}' \in \text{Cat}(\mathcal{C})$  is an equivalence, and  $F' : \mathcal{X}' \rightarrow \mathcal{Y} \in \text{Gerby}(\mathcal{Y})$ , where  $\mathcal{Y}$  is equipped with the topology inherited from  $\mathcal{C}$ ?
- If  $F$  factors as  $F = F' \circ a$  where  $a : \mathcal{X} \rightarrow \mathcal{X}' \in \text{Cat}(\mathcal{C})$  is an equivalence, and  $F' : \mathcal{X}' \rightarrow \mathcal{Y} \in \text{FibCat}^{\text{grp}}(\mathcal{Y})$ , then  $\mathcal{X}' \in \text{Gerby}(\mathcal{Y})$ , where  $\mathcal{Y}$  is equipped with the topology inherited from  $\mathcal{C}$ ?
- The following two conditions are satisfied:
  - For any  $U \in \mathcal{C}$  and  $y \in \mathcal{Y}_U$ , there exists  $\{U_i \rightarrow U\} \in \text{Cov}_{\mathcal{C}}(U)$  s.t. each  $y|_{U_i}$  is in the essential image of  $F_{U_i} : \mathcal{X}_{U_i} \rightarrow \mathcal{Y}_{U_i}$ .
  - For any  $U \in \mathcal{C}$  and  $x, y \in \mathcal{X}_U$ ,

$$\underline{\text{Isom}}(x, y) \rightarrow \underline{\text{Isom}}(F(x), F(y))$$

is a surjection of sheaves.

If the above conditions are satisfied, we say that  $\mathcal{X}$  is a **gerbe(over stack)** over  $\mathcal{Y}$ .  $\lrcorner$

*Proof:* Cf. [Sta]06P1.  $\square$

**Prop. (7.1.3.29).** For  $\mathcal{X} \rightarrow \mathcal{Y} \rightarrow \mathcal{Z} \in \text{Sta}^{\text{grp}}(\mathcal{C})$ , if  $\mathcal{X}$  is a gerbe over  $\mathcal{Y}$  and  $\mathcal{Y}$  is a gerbe over  $\mathcal{Z}$ , then  $\mathcal{X}$  is a gerbe over  $\mathcal{Z}$ .  $\lrcorner$

*Proof:* This follows easily from (7.1.3.28).  $\square$

**Prop. (7.1.3.30).** If  $\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{Y}' \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{Y} \end{array} \in \text{Sta}^{\text{grp}}(\mathcal{C})$  is a 2-fibered product, and  $\mathcal{X}$  is a gerbe over  $\mathcal{Y}$  (7.1.3.28), then  $\mathcal{X}'$  is a gerbe over  $\mathcal{Y}'$ . ┘

*Proof:* Cf. [Sta]06P3. □

**Prop. (7.1.3.31).** Let  $\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{Y}' \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{Y} \end{array} \in \text{Sta}^{\text{grp}}(\mathcal{C})$  is a 2-fibered product. If

- for each  $U \in \mathcal{C}$  and  $x \in \mathcal{Y}_U$ , there exists  $\{U_i \rightarrow U\} \in \text{Cov}_{\mathcal{C}}(U)$  s.t. each  $x|_{U_i}$  is in the essential image of  $\mathcal{Y}'_{U_i} \rightarrow \mathcal{Y}_{U_i}$ ,
- $\mathcal{X}'$  is a gerbe over  $\mathcal{Y}'$ .

Then  $\mathcal{X}$  is a gerbe over  $\mathcal{Y}$ . ┘

*Proof:* Cf. [Sta]06P4. □

**Prop. (7.1.3.32).** Let  $\mathcal{S} \in \text{Ger}(\mathcal{C})$  s.t. for each  $U \in \mathcal{C}$  and each  $x \in \mathcal{S}_U$  we have  $\underline{\text{Isom}}(x, x) \in \text{Sh}(\mathcal{C}/U)$  (an Abelian sheaf), then

- there exists  $\mathcal{G} \in \text{Sh}(\mathcal{C})$ ,
- For each  $U \in \mathcal{C}$  and  $x \in \mathcal{S}_U$ , there exists an isomorphism  $\mathcal{G}|_U \cong \underline{\text{Isom}}(x, x) \in \text{Sh}(\mathcal{C}/U)$ ,

s.t. for each  $U \in \mathcal{C}$  and each  $\varphi : x \cong y \in \mathcal{S}_U$ , the following commutative diagram holds:

$$\begin{array}{ccc} \mathcal{G}|_U & \xlongequal{\quad} & \mathcal{G}|_U \\ \downarrow & & \downarrow \\ \underline{\text{Isom}}(y, y) & \xrightarrow{\varphi^*} & \underline{\text{Isom}}(x, x). \end{array}$$

┘

*Proof:* Cf. [Sta]0CJY. □

## Bands

### 4 Sites over Schemes

**Notation (7.1.4.1).**

- Let  $S \in \text{Sch}$ .

┘

**Prop. (7.1.4.2).** Fiber products exist in the category of schemes, by (7.2.7.15). ┘

## Zariski Topology

**Def. (7.1.4.3) [Zariski Topology].** The **Zariski topology** has the covering of a scheme  $T$  as classes of open immersions  $\{T_i \rightarrow T\}$  that their images cover  $T$ .

The **Zariski site**  $\text{Sch}_{\text{Zar}}/S$  has the objects as all schemes over  $S$ .

The **small Zariski site**  $S_{\text{Zar}}$  has the objects as all open subschemes over  $S$ .

The **restricted Zariski site**  $S_{\text{Zar pf}}$  has the objects as all schemes that are qcqs open subschemes of  $S$ .

The **big affine Zariski site**  $\mathcal{A}ff_{\text{Zar}}/S$  has the objects as all schemes affine over  $S$ .

These are all topologies because open immersions satisfies base change trick(7.4.5.11).

In particular when the cover has only one element and is affine, the descent datum is equivalent to compatible isomorphisms

$$\varphi_{13} : N \otimes_A B \otimes_A B \xrightarrow{\varphi_{12}} B \otimes_A M \otimes_B \xrightarrow{\varphi_{23}} B \otimes_A B \otimes_A M.$$

┘

**Def. (7.1.4.4) [Zariski Stacks].** The category of sheaves on  $\mathcal{S}ch_{\text{Zar}}/S$  is denoted by  $\mathcal{S}h_{\text{Zar}}/S$ . The category of stacks on  $\mathcal{S}ch_{\text{Zar}}/S$  is denoted by  $\mathcal{S}ta_{\text{Zar}}/S$ . ┘

**Prop. (7.1.4.5) [Affine and Full Sites].** The inclusion functor  $\mathcal{A}ff_{\text{Zar}}/S \rightarrow \mathcal{S}ch_{\text{Zar}}/S$  is a special cocontinuous functor, so by(7.1.2.24), it induces an equivalence of topoi  $\mathcal{S}h(\mathcal{A}ff_{\text{Zar}}/S) \cong \mathcal{S}h_{\text{Zar}}/S$ . ┘

**Def. (7.1.4.6) [Restriction to Small Sites].** The inclusion functor  $S_{\text{Zar}} \rightarrow \mathcal{S}ch_{\text{Zar}}/S$  satisfies the hypothesis of(7.1.2.23), thus induces morphisms of topoi

$$\pi_S : \mathcal{S}h_{\text{Zar}}/S \rightarrow \mathcal{S}h(S_{\text{Zar}}), \quad i_S : \mathcal{S}h(S_{\text{Zar}}) \rightarrow \mathcal{S}h_{\text{Zar}}/S$$

that satisfies  $\pi_S \circ i_S = \text{id}_{S_{\text{Zar}}}$ . In particular,  $i_S^{-1}\mathcal{F}$  is called the **restriction to small sites** of  $\mathcal{F}$ . ┘

**Prop. (7.1.4.7).** A sheaf w.r.t the small Zariski topology is equivalent to a sheaf on  $S$ , trivially, so the sheaf cohomology on  $\mathcal{A}ff_{\text{Zar}}/S$  is equivalent to usual sheaf cohomology on  $S$ . ┘

**Prop. (7.1.4.8) [Transfer of Big Sites].** Let  $f : T \rightarrow S \in \mathcal{S}ch_{\text{Zar}}/S$ , then  $\mathcal{S}ch_{\text{Zar}}/T \rightarrow \mathcal{S}ch_{\text{Zar}}/S$  is a localization, so by(7.1.2.22) it induces a morphism of topoi

$$f : \mathcal{S}h_{\text{Zar}}/T \rightarrow \mathcal{S}h_{\text{Zar}}/S : f^{-1}\mathcal{G}(U/T) = \mathcal{G}(U/S), \quad f_*\mathcal{F}(U/S) = \mathcal{F}(U \times_S T/T).$$

┘

**Prop. (7.1.4.9) [Transfer of Small Sites].** Let  $f : T \rightarrow S \in \mathcal{S}ch_{\text{Zar}}/S$ , then the base change functor  $S_{\text{Zar}} \rightarrow T_{\text{Zar}}$  is continuous, and by(7.1.2.14) induces a morphism of topoi

$$f : \mathcal{S}h_{\text{Zar}}/T \rightarrow \mathcal{S}h_{\text{Zar}}/S.$$

┘

**Prop. (7.1.4.10).** By(7.3.1.15), if  $X$  is qs, then  $\mathcal{S}h(X_{\text{Zar}}) \rightarrow \mathcal{S}h(X_{\text{Zar}fp})$  is an equivalence by  $i_s$  and  $i^s$ . ┘

### Étale Topology

**Def. (7.1.4.11) [Étale Topology].** The **étale topology** has the covering of a scheme  $T$  as classes of étalemorphisms that their images cover  $T$ .

The **étale site**  $\mathcal{S}ch_{\text{ét}}/S$  has the objects as all schemes over  $S$ .

The **small étale site**  $S_{\text{ét}}$  has the objects as all schemes that are étale over  $S$ .

The **restricted étale site**  $S_{\text{ét}fp}$  has the objects as all schemes that are étale and qcqs over  $S$ .

The **big affine étale site**  $\mathcal{A}ff_{\text{ét}}/S$  has the objects as all schemes affine over  $S$ .

These are truly topologies because étale is stable under base change and composition. ┘

**Prop. (7.1.4.12).** Zariski covering is étale, because open immersions are étale.  $\lrcorner$

**Prop. (7.1.4.13).** For a family of maps in  $S_{\text{ét}}$  to be a covering, it suffices to check their image is adjointly surjective, by (7.6.6.4).  $\lrcorner$

**Prop. (7.1.4.14) [Affine and Full Sites].** The inclusion functor  $\mathcal{A}\text{ff}_{\text{ét}}/S \rightarrow \text{Sch}_{\text{ét}}/S$  is a special cocontinuous functor, so by (7.1.2.24), it induces an equivalence of topoi  $\text{Sh}(\mathcal{A}\text{ff}_{\text{ét}}/S) \cong \text{Sh}_{\text{ét}}/S$ .  $\lrcorner$

**Def. (7.1.4.15) [Restriction to Small Sites].** The inclusion functor  $S_{\text{ét}} \rightarrow \text{Sch}_{\text{ét}}/S$  satisfies the hypothesis of (7.1.2.23), thus induces morphisms of topoi

$$\pi_S : \text{Sh}_{\text{ét}}/S \rightarrow \text{Sh}(S_{\text{ét}}), i_S : \text{Sh}(S_{\text{ét}}) \rightarrow \text{Sh}_{\text{ét}}/S$$

that satisfies  $\pi_S \circ i_S = \text{id}_{S_{\text{ét}}}$ . In particular,  $i_S^{-1}\mathcal{F}$  is called the **restriction to small sites** of  $\mathcal{F}$ .  $\lrcorner$

**Prop. (7.1.4.16).** Any étale covering of a qc scheme can be refined a finite covering by affine étale schemes, this is because étale map are open (7.6.6.3).  $\lrcorner$

**Def. (7.1.4.17) [Étale Stacks].** The category of sheaves on  $\text{Sch}_{\text{ét}}/S$  is denoted by  $\text{Sh}_{\text{ét}}/S$ . The category of stacks on  $\text{Sch}_{\text{ét}}/S$  is denoted by  $\text{Sta}_{\text{ét}}/S$ .  $\lrcorner$

**Prop. (7.1.4.18).** For a qc scheme  $X$ ,  $X_{\text{ét}fp}$  is a Noetherian topology, because étale map is open, and any object in  $X_{\text{ét}fp}$  is qc.  $\lrcorner$

**Prop. (7.1.4.19) [Transfer of Big Sites].** Let  $f : T \rightarrow S \in \text{Sch}_{\text{ét}}/S$ , then the functor  $u : \text{Sch}_{\text{ét}}/T \rightarrow \text{Sch}_{\text{ét}}/S$  is cocontinuous, and has base change as a right adjoint, so by (7.1.2.22) it induces a morphism of topoi

$$f : \text{Sh}_{\text{ét}}/T \rightarrow \text{Sh}_{\text{ét}}/S : f^{-1}\mathcal{G}(U/T) = \mathcal{G}(U/S), \quad f_*\mathcal{F}(U/S) = \mathcal{F}(U \times_S T/T).$$

**Prop. (7.1.4.20) [Transfer of Small Sites].** Let  $f : T \rightarrow S \in \text{Sch}_{\text{ét}}/S$ , then the base change functor  $S_{\text{ét}} \rightarrow T_{\text{ét}}$  is continuous, and by (7.1.2.14) induces a morphism of topoi

$$f : \text{Sh}(T_{\text{ét}}) \rightarrow \text{Sh}(S_{\text{ét}})$$

**Prop. (7.1.4.21).** If  $X$  is qs, then  $\text{Sh}(X_{\text{ét}}) \rightarrow \text{Sh}(X_{\text{ét}fp})$  is an equivalence by  $i_s$  and  $i^s$ .  $\lrcorner$

*Proof:* Want to use (7.1.2.25), one condition is satisfied by (7.1.4.16), so it suffice to check any  $X' \in X_{\text{ét}}$  has an étale covering by schemes étale and qcqs over  $X$ . For any point  $p \in X'$ , there is an affine nbhd  $U'$  that maps to an affine nbhd  $U'$  of  $X$ , so  $U' \rightarrow U$  is étale and qcqs, and  $U \rightarrow X$  is open immersion and qs, it is qc because  $X$  is qs and  $U$  is qc (7.4.4.27). So these affine nbhds  $U'$  cover  $X'$ .  $\square$

**Prop. (7.1.4.22) [Cohomology Big and Small Sites].** The inclusion of small sites to the big sites has no infection on the sheaf cohomology, by (7.3.1.4). This is applicable to all topologies  $\tau$  considered here.  $\lrcorner$

**Prop. (7.1.4.23) [Topological Invariance of Étale Sites].** If  $f : S' \rightarrow S$  is universally homeomorphism, then  $f : S'_{\text{ét}} \rightarrow S_{\text{ét}}$  is an equivalence of sites. And it induces an isomorphism of categories

$$f^* : S'_{\text{fét}} \rightarrow S_{\text{fét}}.$$

In particular this applies to  $S' = S_{\text{red}}$ .  $\lrcorner$

*Proof:* Cf. [Sta]04DZ. 0BQZ. ?  $\square$



### Smooth Topology

This topology will be shown to be identical to the étale topology, so it is not so important.

**Def. (7.1.4.24) [Smooth Topology].** The **smooth topology** has the covering of a scheme  $T$  as classes of smooth morphisms that their images cover  $T$ .

The **big smooth site**  $\mathcal{S}ch_{sm}/S$  has the objects as all schemes over  $S$ .

The **small smooth site**  $S_{sm}$  has the objects as all schemes that are smooth over  $S$ .

The **restricted smooth site**  $S_{\acute{e}t,fp}$  has the objects as all schemes that are smooth and qcqs over  $S$ .

The **big affine smooth site**  $Aff_{\acute{e}t}/S$  has the objects as all schemes affine over  $S$ .

These are truly topologies because smoothness is stable under base change and composition.  $\lrcorner$

### Syntomic Topology

**Def. (7.1.4.25) [Syntomic Topology].** The **syntomic topology** has the covering of a scheme  $T$  as classes of syntomic morphisms that their images cover  $T$ .  $\lrcorner$

### fppf Topology

**Def. (7.1.4.26) [Fppf Topology].** The **fppf topology** has the covering of a scheme  $T$  as classes of flat locally of finite presentation morphisms that their images cover  $T$ . (f.f.+locally of f.p.).

The **big Zariski site**  $\mathcal{S}ch_{fppf}/S$  has the objects as all schemes over  $S$ .

The **big affine Zariski site**  $Aff_{fppf}/S$  has the objects as all schemes affine over  $S$ .

They are all topologies because flatness and finite presentedness satisfy base change trick by (7.6.2.1) and (7.6.1.1).  $\lrcorner$

**Prop. (7.1.4.27) [Syntomic Covering is Fppf].** A syntomic covering is fppf by definition (5.4.4.18).  $\lrcorner$

**Prop. (7.1.4.28).** A fppf covering of an affine scheme can be refined a finite affine fppf covering, because fppf map are open (7.6.2.10).  $\lrcorner$

**Def. (7.1.4.29) [Fppf Stacks].** The category of sheaves on  $\mathcal{S}ch_{fppf}/S$  is denoted by  $\mathcal{S}h_{fppf}/S$ . The category of stacks on  $\mathcal{S}ch_{fppf}/S$  is denoted by  $\mathcal{S}ta_{fppf}/S$ .  $\lrcorner$

### fpqc Topology

**Def. (7.1.4.30) [Fpqc Topology].** The **fpqc topology** has the covering of a scheme  $T$  as classes of flat morphisms s.t. their images cover  $T$  and for any affine open  $U \subset T$ , the restriction on  $T$  can be refined by a finite affine cover of open affine subschemes of the covering (f.f.+qc). It is a topology by (7.6.2.1) and (7.4.4.25).

When the covering consists of affine schemes, it is called a **standard fpqc covering**.  $\lrcorner$

**Def. (7.1.4.31) [Fpqc Stacks].** Defining fpqc sites has inescapable set-theoretic difficulties, thus we don't consider fpqc sites and fpqc cohomologies. Cf. [Sta]0BBK.

Nevertheless, we will denote the category of presheaves on  $\mathcal{S}ch/S$  satisfying the sheaf condition w.r.t. the fpqc topology by  $\mathcal{S}h_{fpqc}/S$ , and denote the category of fibered categories over  $\mathcal{S}ch/S$  satisfying the sheaf condition w.r.t. the fpqc topology by  $\mathcal{S}ta_{fpqc}/S$ .  $\lrcorner$

**Prop. (7.1.4.32) [fppf is fpqc].** Fppf coverings are fpqc.  $\lrcorner$



*Proof:* Use (7.6.2.10), we see that fppf covering consists of open morphisms, thus it is qc because affine scheme is quasi-compact.  $\square$

**Prop. (7.1.4.33).** A covering consisting of flat morphisms refined by a fpqc covering is a fpqc covering. Hence being fpqc is local on the target, because a Zariski cover is a fpqc covering.

If  $U$  is a covering consisting of flat morphisms that there is a fpqc covering  $V$  that  $U \times V \rightarrow V$  is a fpqc covering, then  $U$  is fpqc, because  $U \times V$  does and it refines  $U$ .  $\lrcorner$

**Lemma (7.1.4.34) [Checking Sheaf Condition].** Let  $S \in \text{Sch}$ ,  $F \in \mathcal{P}\text{Sh}^{\text{Set}}((\text{Sch}_S)_{\text{fpqc}})$  is a sheaf iff it is a sheaf w.r.t the Zariski topology and satisfies sheaf property w.r.t the single covering  $V \rightarrow U$  f.f. between affine schemes.  $\lrcorner$

*Proof:* This follows from (7.1.3.6)(7.1.3.7) and (7.1.5.7).  $\square$

**Prop. (7.1.4.35) [fpqc Site is Subcanonical].** The coverings in  $(\text{Sch}_S)_{\text{fpqc}}$  are families of universal effective epimorphisms. In other words,  $(\text{Sch}_S)_{\text{fpqc}}$  is subcanonical.

In particular, for any covering  $\mathcal{U} = \{U_i \rightarrow X\} \in \text{Cov}((\text{Sch}_S)_{\text{fpqc}})$  and  $\{V_{ijk} \rightarrow U_i \times_X U_j\} \in \text{Cov}((\text{Sch}_S)_{\text{fpqc}})$ ,

$$X = \text{Coeq}\left(\coprod_{i,j,k} V_{ijk} \rightrightarrows \coprod_i U_i\right) \in \text{Sch}_S.$$

$\lrcorner$

*Proof:* By (7.1.4.34), it suffices to show that any representable presheaf is a sheaf w.r.t Zariski topology and f.f. affine morphisms. The Zariski case follows from (7.1.5.3), for the second,  $\text{Spec } B \rightarrow \text{Spec } Z$ , for any scheme  $X$ , the morphism corresponds to  $0 \rightarrow \text{Hom}(R, A) \rightarrow \text{Hom}(R, B) \rightarrow \text{Hom}(R, B \otimes_A B)$ , but this follows immediately from (5.4.2.2), with  $M = A$ .  $\square$

**Cor. (7.1.4.36).** For  $f : Y \rightarrow X$  a morphism of schemes, if  $Z \in X_\tau$  for the above topologies  $\tau$ , then  $f^*(\text{Hom}_X(-, Z)) \cong \text{Hom}(-, Z \otimes_X Y)$ , in other words, the inverse sheaf of a representable sheaf is representable.  $\lrcorner$

*Proof:* By definition,  $f^*(\text{Hom}_X(-, Z))$  is the sheaf associated to the presheaf  $f_p(\text{Hom}_X(-, Z))$ , which by (7.1.2.13) is just the presheaf represented by  $Z \otimes_X Y$ , but by the proposition, it is already a sheaf.  $\square$

**Prop. (7.1.4.37) [Coherent Sheaves on  $\text{Sch}_{\text{fpqc}}/X$ ].** Let  $\mathcal{F} \in \mathcal{Q}\text{Coh}(X)$ , then the functor  $X' \rightarrow \Gamma(X', \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'})$  satisfies the axiom for Abelian sheaves on  $\text{Sch}_{\text{fpqc}}/X$ , by (7.1.5.13).  $\lrcorner$

**Prop. (7.1.4.38) [Qco Sheaves on Sites].** For  $X \in \text{Sch}$ ,  $\tau \in \{\text{fppf}, \text{étale}, \text{smooth}, \text{syntomic}, \text{Zariski}\}$ , restriction defines an equivalence of categories

$$\mathcal{Q}\text{Coh}(X) \cong \mathcal{Q}\text{Coh}(\text{Sch}_\tau/X).$$

And if  $\tau \in \{\text{Zariski}, \text{étale}\}$ , restriction defines an equivalence of categories

$$\mathcal{Q}\text{Coh}(X) \cong \mathcal{Q}\text{Coh}(X_\tau).$$

$\lrcorner$

*Proof:* Cf. [Sta]03DX.  $\square$

PH-Covering

**Def. (7.1.4.39)** [Standard PH-Covering]. ┘

**Def. (7.1.4.40)** [PH-Topology]. ┘

**Prop. (7.1.4.41)** [Zariski Covering is PH-Covering]. A Zariski covering is a PH-covering. ┘

*Proof:* Cf. [[Sta]0DBH]. □

**Prop. (7.1.4.42)**. A proper surjective morphism is a ph-covering. ┘

*Proof:* Cf. [[Sta]0DES]. □

V-Topology

**Def. (7.1.4.43)** [Standard V-Covering]. A finite family of morphisms  $T_i \rightarrow X$  of affine schemes is a covering in the **standard v-topology** if for any morphism  $\text{Spec } V \rightarrow X$  where  $V$  is a valuation ring, there is an extension of valuation rings (12.2.2.11)  $V \rightarrow W$  and a morphism  $\text{Spec } W \rightarrow \text{Spec } V \times_X T_i$  for some  $i$ . ┘

**Def. (7.1.4.44)** [V-Topology]. A family of morphisms  $\{T_i \rightarrow T\}$  is called a **v-covering** in the **v-topology** if for any open affine subscheme  $U$  of  $T$ , the base change is refined by a standard v-covering of  $U$ .

The v-coverings form a topology, by [[Sta]0ETJ]. ┘

**Lemma (7.1.4.45)**. A standard fpqc covering is a standard v-covering. ┘

*Proof:* Cf. [[Sta]022E]. □

**Prop. (7.1.4.46)** [fpqc Covering is v-Covering]. A fpqc covering is a v-covering. ┘

*Proof:* This follows immediately from (7.1.4.45). □

**Prop. (7.1.4.47)**. A standard ph-covering is a standard v-covering. ┘

*Proof:* Cf. [[Sta]0ETD]. □

**Prop. (7.1.4.48)** [ph-Covering is V-Covering]. A ph-covering is a v-covering. ┘

*Proof:* This follows immediately from (7.1.4.47). □

**Cor. (7.1.4.49)**. A proper and surjective map is a v-covering, by (7.1.4.48) and (7.1.4.42). ┘

Arc-Topology

**Def. (7.1.4.50)** [Arc-Topology]. A finite family of morphisms  $\{T_i \rightarrow X\}$  of schemes is a covering in the **arc-topology** if for any morphism  $\text{Spec } V \rightarrow X$  where  $V$  is a rank1-valuation ring, there is a rank1-valuation ring  $W$  and a morphism  $\text{Spec } W \rightarrow \text{Spec } V \times_X T_i$  for some  $i$  that  $V \rightarrow W$  is f.f.. ┘

**Prop. (7.1.4.51)**. V-coverings are arc coverings. ┘

*Proof:* This is by the definition and (12.2.2.11). □

## 5 Descent for Schemes

References are [Sta]Chap34 and 10.158.

### General Principal

**Prop. (7.1.5.1).** A property of schemes is called **local** in a topology if for any covering  $\{U_i \rightarrow S\}$ ,  $S$  has  $P$  iff  $U_i$  has  $P$ . A property of morphisms is called **local** in a topology if for any covering  $\{U_i \rightarrow S\}$ ,  $X \rightarrow S$  has  $P$  iff  $X \times_S U_i \rightarrow U_i$  has  $P$ .  $\lrcorner$

**Prop. (7.1.5.2)[Twists and Čech Cohomology].** Let  $\xi$  be an object of a stack  $\mathcal{F}$  over  $\mathcal{C} \in \text{Site}$  lying over an object  $U \in \mathcal{C}$ , we call an object  $\xi' \in \mathcal{F}(U)$  a **twist** of  $\xi$  if there is some covering  $\{U_i \rightarrow U\}$  that the pullback of  $\xi$  and  $\xi'$  to  $U_i \rightarrow U$  are isomorphic.

Then there is a natural bijection between  $\mathcal{F}(U)$ -isomorphism classes of twists of  $\xi$  with  $\check{H}^1(U, \underline{\text{Aut}}(\xi))$ .  $\lrcorner$

*Proof:* Cf.[Appendix of Lamb].  $\square$

### Zariski Descent

**Lemma (7.1.5.3)[Zariski Descent of Qco Sheaves].** The fibered category  $X \mapsto (\mathcal{Q}\text{Coh}/X)$  is a stack over  $\text{Sch}_{\text{Zar}}/S$ .  $\lrcorner$

*Proof:* This is a consequence of (7.1.3.18) and the fact quasi-coherentness is a Zariski local property.  $\square$

**Prop. (7.1.5.4)[Zariski Descent of Schemes].** The fibered category  $X \mapsto \text{Sch}/X$  is in  $\text{Staz}_{\text{Zar}}/S$ .  $\lrcorner$

*Proof:* Firstly it is a prestack by (7.1.3.17) and (7.1.4.35).

To show any descent datum of effective, let  $\{U_i \rightarrow U\}$  be a Zariski covering, and  $X_i \rightarrow U_i$  be schemes with descent datum  $\varphi_{ij} : X_i \times_{U_i} U_{ij} \cong X_j \times_{U_j} U_{ij}$ , then we define  $X = \coprod X_i / \sim$ , where  $x_i \sim x_j$  if  $x_i \in U_{ij}, x_j \in U_{ji}$  and  $\varphi_{ij}(x_i) = \varphi_{ji}(x_j)$ . It can be shown this is an equivalence relation. Denote  $\varphi_i : X_i \rightarrow X$  the natural map, and  $U_i = \varphi_i(X_i)$ . Define the topology on  $X$  as the quotient topology. In particular,  $\varphi_i$  is a homeomorphism onto the image. Then we can use (7.1.3.18) to glue the sheaves of rings  $(\varphi_i)_*(\mathcal{O}_{X_i})$  to a sheaf of rings  $\mathcal{O}_X$  on  $X$  that  $\varphi_i^* \mathcal{O}_X = \mathcal{O}_{X_i}$ . Also we have a map  $f : X \rightarrow U$  by set-theoretical and topological consideration. For the structure map  $f^{-1}(\mathcal{O}_U) \rightarrow \mathcal{O}_X$ , use (7.1.3.18).  $\square$

**Cor. (7.1.5.5).** If  $\mathcal{P}$  is a subclass of arrows of schemes that is stable under base change and local on the target, then  $\mathcal{P}/S$  is a stack over  $(\text{Sch}/S)_{\text{Zar}}$ .  $\lrcorner$

**Cor. (7.1.5.6)[Zariski Descent of Schemes with a Qco Sheaf].** The fibered category over  $\text{Sch}/S$

$$X \mapsto \{\text{Schemes over } X \text{ with a Qco sheaf } \mathcal{F}\}$$

is a stack in  $\text{Staz}_{\text{Zar}}/S$ .  $\lrcorner$

*Proof:* This is a combination of (7.1.5.4) and (7.1.5.3).  $\square$

### Fpqc Descent

**Prop. (7.1.5.7) [Reduction to Affine Case].** Let  $S \in \text{Sch}$ ,  $\mathcal{F} \in \text{FibCat}(\text{Sch}/S)$ . Suppose that

- $\mathcal{F} \in \text{Stazar}/S$ .
- When  $V \rightarrow U$  is a f.f. morphism of affine  $S$ -schemes,  $\mathcal{F}(U) \rightarrow \mathcal{F}(V \rightarrow U)$  is an equivalence of categories.

then  $\mathcal{F} \in \text{Sta}_{\text{fpqc}}/S$ .  $\lrcorner$

*Proof:* Firstly, to show  $\mathcal{F}$  is a prestack, using (7.1.3.9), it suffices to show for an  $S$ -scheme  $T \rightarrow S$  and objects  $\xi, \eta \in \mathcal{F}(T)$ , the functor

$$\underline{\text{Hom}}_T(\xi, \eta) : (\text{Sch}/T)^{op} \rightarrow \text{Set}$$

is a sheaf. But then we can use (7.1.4.34) to achieve this.

Next, according to (4.2.3.18) and (7.1.3.8), we may assume  $\mathcal{F}$  is splitting.

Notice that  $\mathcal{F}(\emptyset)$  is equivalent to the category pt. This is because  $\mathcal{F}(\emptyset)$  is equivalent to  $\mathcal{F}(\mathcal{U})$ , where  $\mathcal{U}$  is the null Zariski covering of  $\emptyset$  (with no mapping or objects at all!). Then for any disjoint union of open subschemes  $U = \coprod U_i$ , there is a natural isomorphism of categories  $\mathcal{F}(U) \cong \prod_i \mathcal{F}(U_i)$ .

Thus for any covering  $\mathcal{U} = \{U_i \rightarrow X\}$ , to show  $\mathcal{F}(X) \rightarrow \mathcal{F}(\mathcal{U})$  is an equivalence of categories, it suffices to show that  $\mathcal{F}(X) \rightarrow \mathcal{F}(\coprod U_i \rightarrow X)$  is an equivalence of categories: this is because  $\coprod U_i \times_X \coprod U_i \cong \coprod U_i \times_X U_j$ , so  $\mathcal{F}(\coprod U_i \rightarrow X) \rightarrow \mathcal{F}(\mathcal{U})$  is an equivalence of categories.

If  $\mathcal{U} = \{V \rightarrow U\}$  is a covering of  $F$  with a single morphism that is qc and  $U$  is affine, then by the proof of (7.1.3.11), we can choose a finite affine cover of  $V$  reduce to the case of covering of f.m. affine maps, then we finish by the case above.

If  $\mathcal{U} = \{V \rightarrow U\}$  is a covering of  $F$  with a single morphism and  $U$  is affine, then we can find a Zariski covering  $\{V_i \rightarrow V\}$  that each  $V_i$  is qc. and surjects onto  $U$ . Then there are maps of categories  $\mathcal{F}(U) \rightarrow \mathcal{F}(\{V \rightarrow U\}) \rightarrow \mathcal{F}(\{V_i \rightarrow V\})$ , where  $\mathcal{F}(U) \rightarrow \mathcal{F}(\{V_i \rightarrow V\})$  is an equivalence and  $\mathcal{F}(U) \rightarrow \mathcal{F}(\{V \rightarrow U\})$  is fully faithful. Thus to show  $\mathcal{F}(U) \rightarrow \mathcal{F}(\{V \rightarrow U\})$  is an equivalence, it suffices to show  $\mathcal{F}(\{V \rightarrow U\}) \rightarrow \mathcal{F}(\{V_i \rightarrow V\})$  is faithful, which is true by (7.1.3.10).

For general case, Cf. [Vis08]P88.  $\square$

**Prop. (7.1.5.8).**  $\text{Mor}/S : X \mapsto \text{Sch}/X$  is a prestack over  $\text{Sch}_{\text{fpqc}}/S$ , by (7.1.4.35) and (7.1.3.17).  $\lrcorner$

**Def. (7.1.5.9) [Affine Fpqc Descent Datum].** For  $A \rightarrow B \in \mathcal{C}\text{Alg}$ , define a category  $\text{Mod}_{A \rightarrow B}$  as follows: its objects are pairs  $(N, \psi)$ , where  $N \in \text{Mod}_B$  and  $\psi : N \otimes_A B \rightarrow B \otimes_A N$  is an isomorphism of  $B \otimes_A B$ -modules s.t. that

$$\psi_{12} \circ \psi_{01} = \psi_{02} : N \otimes_A B \otimes_A B \rightarrow B \otimes_A B \otimes_A N$$

where  $\psi_{ij}$  is permuting the  $i, j$ -parts using  $\psi$ . The morphisms in  $\text{Mod}_{A \rightarrow B}$  are maps in  $\text{Mod}_B$  that is compatible with  $\psi$ .

There is a natural functor  $\text{Mod}_{A \rightarrow B} \rightarrow \text{Mod}_B$ .  $\lrcorner$

**Lemma (7.1.5.10) [Affine Fpqc Descent].** For  $A \rightarrow B \in \mathcal{C}\text{Alg}$ , there is a functor  $F : \text{Mod}_A \rightarrow \text{Mod}_{A \rightarrow B}$ , where  $M$  is mapped to  $(B \otimes_A M, \psi_M)$  with

$$\psi_M : (B \otimes_A M) \otimes_A B \rightarrow B \otimes_A (B \otimes_A M) : b \otimes m \otimes b' \mapsto b \otimes b' \otimes m.$$

Then when  $A \rightarrow B$  is f.f., this is an equivalence of categories.  $\lrcorner$

*Proof:* We construct an inverse  $T$  that maps  $(N, \psi)$  to  $\{n | \psi(n \otimes 1) = 1 \otimes n\}$ . Then  $TF \cong \text{id}$  because of the first exactness of (5.4.2.2).

And for  $FT \cong \text{id}$ , let  $T((N, \psi)) = M$ . Notice if  $\psi(n \otimes 1) = \sum_i b_i \otimes n_i$ , then by cocycle condition,

$$\sum_i b_i \otimes 1 \otimes n_i = \sum_i b_i \otimes \psi(n_i \otimes 1).$$

so  $\psi(N \otimes 1) \subset \ker(\text{id}_B \otimes (n \mapsto \psi(n \otimes 1) - 1 \otimes n)) = B \otimes M$  because  $B/A$  is flat.

So we defined a map  $N \xrightarrow{\Psi} B \otimes_A M \in \text{Mod}_B$ , and the composition  $B \otimes_A M \xrightarrow{\text{mul.}} N \xrightarrow{\Psi} B \otimes_A M$  is identity, because

$$\psi(bm \otimes 1) = (b \otimes 1)\psi(m \otimes 1) = (b \otimes 1)(1 \otimes m) = b \otimes m.$$

This shows  $\Psi$  is surjective. And  $\Psi$  is injective because  $n \mapsto n \otimes 1$  is injective as  $B/A$  is f.f., and  $\psi$  is an isomorphism. So  $\Psi$  is an isomorphism of  $N \cong FT(N)$ .

Finally, to show  $\psi = \psi_M$ , as now  $B \otimes_A M \xrightarrow{\text{mul.}} N$  is an isomorphism, we check

$$\psi((bm) \otimes b') = (b \otimes b')\psi(m \otimes 1) = (b \otimes b')(1 \otimes m) = b \otimes (b'm).$$

□

**Remark (7.1.5.11).** In fact, a descent datum is always effective iff  $A \rightarrow B$  is universally injective.

Cf. [Sta]. And f.f. extension is u.i. (5.4.1.27). ┘

**Prop. (7.1.5.12) [fpqc Descent for Qco Sheaves].** For  $S \in \text{Sch}$ , we have  $\text{QCoh}/S \in \text{Sta}_{\text{fpqc}}/S$ . ┘

*Proof:* We use (7.1.5.7), the first condition is satisfied by (7.1.5.3), and for the second condition, let  $\text{Spec } B \rightarrow \text{Spec } A$  be a f.f. morphism of affine schemes, then  $\text{QCoh}(\text{Spec } B \rightarrow \text{Spec } A)$  is equivalent to  $\text{Mod}_{A \rightarrow B}$ , and  $\text{QCoh}(\text{Spec } A)$  is equivalent to  $\text{Mod}_A$ , so the conclusion follows from (7.1.5.10). □

**Cor. (7.1.5.13).** For any  $\mathcal{F} \in \text{QCoh}(S)$ , the functor  $(\text{Sch}/S)^{\text{op}} \rightarrow \text{Ab} : T \rightarrow \Gamma(T, f^* \mathcal{F}) \in \text{Sta}_{\text{fpqc}}/S$ , hence are also sheaves w.r.t. the fppf, étale, Zariski topologies. ┘

*Proof:* Because this functor is just  $\underline{\text{Hom}}_S(\mathcal{O}_S, \mathcal{F})$ , and it is a sheaf (7.1.3.9). □

**Cor. (7.1.5.14).** If  $\mathcal{P}$  is a property of Qco sheaves that is stable under base change and fpqc local, then  $\text{QCoh}^{\mathcal{P}}/S \in \text{Sta}_{\text{fpqc}}/S$ . ┘

**Prop. (7.1.5.15) [Descending Affine Morphisms].** For  $S \in \text{Sch}$ , let  $\mathcal{P}$  be the class of affine arrows in  $\text{Sch}/S$  that denote by  $\text{Mor}^{\text{aff}}/S$  the resulting fibered category, then  $\text{Mor}^{\text{aff}}/S \in \text{Sta}_{\text{fpqc}}/S$ . ┘

*Proof:* Firstly  $\text{Mor}^{\text{aff}}/S \in \text{Sta}_{\text{Zar}}/S$ , and it satisfies the affine fpqc descent condition of (7.1.5.7) (Notice the  $\{n | \psi(n \otimes 1) = 1 \otimes n\}$  is now a ring, because  $\psi$  is a ring homomorphism), so we are done by (7.1.5.7). □

**Cor. (7.1.5.16).** If  $\mathcal{P}$  is a subclass of affine arrows stable under base change and fpqc local on the target, then  $\text{Mor}^{\mathcal{P}}/S \in \text{Sta}_{\text{fpqc}}/S$ . ┘

**Prop. (7.1.5.17) [Descent via Ample Invertible Sheaves].** Let  $S$  be a scheme and  $\mathcal{P}$  be a class of flat proper morphisms of f.p. in  $\text{Sch}/S$  that is local in the fpqc topology. Assume that for each object  $\xi : X \rightarrow U \in \mathcal{P}$ , we have an invertible sheaf  $\mathcal{L}_\xi$  on  $X$  that is ample relative to  $X \rightarrow U$ , and for an arrow  $f : (X \xrightarrow{\xi} U) \rightarrow (Y \rightarrow \eta V)$ , we have an isomorphism  $\rho_f : f^* \mathcal{L}_\eta \cong \mathcal{L}_\xi$  that satisfies  $\rho_{gf} = \rho_f \circ f^* \rho_g$ , then  $\text{Mor}^{\mathcal{P}}/S \in \text{Sta}_{\text{fpqc}}/S$ . ┘

*Proof:* Cf. [Vistoli, P96]. ?

□

### Étale Descent

**Prop. (7.1.5.18).** ┘

**Prop. (7.1.5.19) [Galois-Descent].** Let  $L/K \in \mathbf{GalField}^{\text{fin}}$ , then  $\text{Spec } L \rightarrow \text{Spec } K$  is a  $\text{Gal}(L/K)$ -torsor in the étale topology. so Galois descent is a special case of étale descent along torsors (7.1.3.19).

Notice this is also true for arbitrary Galois field extensions with continuity condition added, because we can take direct limits of categories over its finite normal subextensions. ┘

*Proof:* Consider the locally constant group scheme  $\underline{G} = \text{Spec}(\prod_{g \in G} K)$ , let  $X = \text{Spec } L, Y = \text{Spec } K$ , then  $\{X \rightarrow Y\}$  is an étale cover, and the action  $\underline{G} \times X \rightarrow X$  is given by

$$L \rightarrow \prod_{g \in G} L : x \mapsto \prod_{g \in G} (g(x)).$$

Thus  $X \rightarrow Y$  is a  $G$ -equivariant map, and there is an isomorphism  $G \times X \cong X \times_Y X : (g, x) \mapsto (gx, x)$  that corresponds to the isomorphism

$$L \otimes_K L \cong \prod_{g \in G} L : (a, b) \mapsto \prod_{g \in G} (g(a)b)$$

□

**Cor. (7.1.5.20) [Galois Descent of Closed Immersions].** Let  $K/k \in \mathbf{GalField}$ ,  $X \in \mathbf{Sch}/k$ . Then the category of closed subschemes of  $X$  is equivalent to the category of closed subschemes of  $X_K$  that is base change from some  $X_{k'}$  where  $k'/k$  is finite, and is stable under the action of  $\text{Gal}(K/k)$ . This weird finiteness condition can be removed when  $X$  is locally algebraic over  $k$ . ┘

*Proof:* This is because the class of closed immersions is a stack (7.1.5.16). □

**Remark (7.1.5.21).** When  $Y \subset X$  is a subvariety and  $K = k^{\text{sep}}$ , to check  $Y$  is stable under action of  $\text{Gal}_k$ , it suffices to check that the geometric points is closed under action of  $\Gamma$ . This is because the geometric points are dense in  $Y'$  (7.12.1.10). ┘

**Cor. (7.1.5.22) [Galois Descent for Ideals of Algebras].** Let  $A \in \mathbf{CRing}_k$ , and  $K/k$  be a Galois field extension with Galois group  $\Gamma$ , then the category of ideals of  $A$  is equivalent to the category of ideals of  $A_K$  that is base change from some  $A_{k'}$  where  $k'/k$  is finite, and is stable under the action of  $\Gamma$ . This weird finiteness condition can be removed when  $A$  is f.g. over  $k$ . ┘

**Cor. (7.1.5.23) [Galois Descent of Morphisms].** Let  $X, Y \in \mathbf{Sch}^{\text{loc.ft}}/k$  and  $K/k \in \mathbf{GalField}$  with Galois group  $\Gamma$ ,  $X' = X \otimes_k K, Y' = Y \otimes_k K$ . If  $Y$  is separated, then a morphism  $\varphi' : X' \rightarrow Y'$  arises from a morphism  $X \rightarrow Y$  iff its graph  $\Gamma_{\varphi'} \subset X' \times_{k'} Y'$  is stable under action of  $\Gamma$ . In this case  $\varphi$  is unique.

And when  $X, Y \in \mathbf{Var}/k$  and  $K = k^{\text{sep}}$ , then it suffices to check the map

$$\varphi'(k^{\text{sep}}) : X'(k^{\text{sep}}) \rightarrow Y'(k^{\text{sep}})$$

commutes with action of  $\text{Gal}_k$ . ┘

**Cor. (7.1.5.24) [Galois Descent for Qco Sheaves].** For  $K/k \in \mathbf{GalField}$  and  $X \in \mathbf{Sch}/k$ , the base change map

$$\mathcal{QCoh}/X \rightarrow \mathcal{QCoh}^{\text{Gal}(K/k)}/X_K$$

is an equivalence of categories, with inverse  $\mathcal{F} \mapsto \mathcal{F}^{\text{Gal}(K/k)}$ .

In particular, the functor  $V \mapsto K \otimes_k V$  induces an equivalence  $\mathbf{Vect}_k \cong \mathbf{Rep}_K(\text{Gal}(K/k))$ . ┘

*Proof:* Because  $\mathcal{Q}\text{Coh}/\mathcal{S}\text{ch}$  is a fpqc stack (7.1.5.12), this follows from (7.1.3.19).  $\square$

**Prop. (7.1.5.25)** [Failure of Étale Descent for proper smooth morphisms]. Cf. [Vis08]P107.  $\lrcorner$

*Proof:*  $\square$

### Descending Properties

**Prop. (7.1.5.26)** [Properties of Morphisms Local in Fpqc Topology]. The following properties of morphisms are local on the target w.r.t. the fpqc topology.

1. quasi-compact.
2. (quasi-)separated.
3. Universally closed.
4. universally open.
5. universally submersive.
6. surjective.
7. universally injective.
8. universally homeomorphism.
9. (locally)of f.t.
10. (locally)of f.p.
11. properness.
12. flatness.
13. (closed/open)immersion.
14. isomorphism/monomorphism.
15. (quasi-)affineness.
16. Being quasi-compact immersions.
17. Being integral morphisms.
18. (locally)(quasi-)finiteness.
19. Being syntomic morphisms.
20. Being smooth/unramified/étale.
21. Being finite locally free.

$\lrcorner$

*Proof:* Cf. [Sta]Chap34.20.

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- 17.
- 18.
19. Cf. [\[Sta\]](#)00SM.
- 20.
- 21.

□

**Prop. (7.1.5.27) [Properties of Schemes Local in the Fppf Topology].** •

┘

*Proof:* Cf. [\[Sta\]](#)Chap34.13.

□

**Prop. (7.1.5.28) [Descending Properties that is not Fpqc Local].**

- If  $X \rightarrow Y$  is a faithfully flat morphism of schemes and  $X$  is (geo.)reduced, then  $Y$  is also (geo.)reduced.
- If  $X \rightarrow Y$  is a faithfully flat morphism of f.p. between schemes and  $X$  is (geo.)regular, then  $Y$  is also (geo.)regular.

┘

*Proof:* Looking stalkwise, these follows from [\(5.4.2.1\)](#).

□

**Prop. (7.1.5.29) [Descending Properties along Torsors].** Let  $G$  be a group object in  $(\text{Sch}_S)_\tau$  and  $X \rightarrow S$  a  $G$ -torsor. Then if  $\tau$  is a subcanonical topology and  $\mathcal{P}$  is a property that is local on  $(\text{Sch}_S)_\tau$  and  $G \rightarrow S$  has  $\mathcal{P}$ , then  $X \rightarrow S$  has  $\mathcal{P}$ .

┘

*Proof:* By [\(7.1.1.19\)](#), there is a covering  $\{Y_i \rightarrow S\}$  s.t. each  $X \times_S Y_i \rightarrow Y_i \cong G \times_S Y_i$  is trivial, thus each  $X \times_S Y_i \rightarrow Y_i$  has  $\mathcal{P}$ , and then  $X \rightarrow S$  has  $\mathcal{P}$ . □



**Weil-Restrictions**

References are [BLR90]Section7.6. and [Conrad-Gabber-Prasad, Pseudo-Reductive Groups]Section A.5.

**Def. (7.1.5.30) [Weil-Restrictions].** For  $S'/S \in \text{Sch}$  and  $X \in \text{Sch}/S$ , then the **Weil restriction** is the scheme  $\text{Res}_{S'/S}(X) \in \text{Sch}/S$  representing the functor

$$\underline{\text{Hom}}_S(S', X) : \text{Sch}/S \rightarrow \text{Set} : T \mapsto X(T \times_S S'),$$

if it exists. ┘

**Thm. (7.1.5.31).** Let  $S' \rightarrow S \in \text{Sch}$  be finite locally-free, and  $X \in \text{Sch}/S'$  satisfying that: for any  $s \in S$  and every finite subset of  $X_{\kappa(s)}$  is contained in some affine open subset of  $X_{\kappa(s)}$ , then  $\text{Res}_{S'/S}(X)$  exists. In particular, this applies to the case that  $X$  is quasi-projective. ┘

*Proof:* Cf.[BLR90, Chap7.6. Thm4]. □

**Thm. (7.1.5.32)[Quasi-Projectiveness].** Let  $S' \rightarrow S \in \text{NSch}$  be finite locally-free, and  $X \in \text{Sch}/S'$  is quasi-projective, then  $\text{Res}_{S'/S}(X)$  is represented by a quasi-projective scheme over  $S$ . ┘

*Proof:* Cf.[Weil Restriction for Schemes]P5. □

**Example (7.1.5.33) [Deligne-Torus].** The **Deligne-torus** is defined to be

$$\mathbb{S} = \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m.$$

Then

1. There are isomorphisms

$$\mathbb{S}(\mathbb{R}) \cong \mathbb{C}^\times, \quad \mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_m \times \mathbb{G}_m,$$

where for any  $T \in \mathcal{C}\text{Ring}/\mathbb{C}$ , the second isomorphism is given by

$$(T \otimes_{\mathbb{R}} \mathbb{C})^\times \mapsto T^\times \times T^\times : t \otimes a \mapsto (t \otimes a, t \otimes \bar{a}).$$

2. Under the isomorphism  $\mathbb{S}_{\mathbb{C}}(\mathbb{C}) \cong \mathbb{C}^\times \times \mathbb{C}^\times$ , the complex conjugation acts by  $\overline{(z_1, z_2)} = (\bar{z}_2, \bar{z}_1)$ .
3. There is a **weight homomorphism**  $w : \mathbb{G}_m \rightarrow \mathbb{S}$  s.t. for any  $T \in \mathcal{C}\text{Ring}/\mathbb{R}$ , the map is given by

$$T^\times \rightarrow (T \otimes \mathbb{C})^\times : t \mapsto t^{-1}.$$

4. There is an exact sequence

$$1 \rightarrow \mathbb{G}_m \xrightarrow{w} \mathbb{S} \xrightarrow{z \mapsto z/\bar{z}} \text{U}(1) \rightarrow 1.$$

(This can be checked on  $\mathbb{C}$ -points by(10.1.5.6), as  $\text{U}(1)$  is a torus). ┘

**6 Étale Torsors**

**Prop. (7.1.6.1).** For  $X \in \text{Sch}$ ,  $\check{H}^1(X, \text{GL}(n)) \cong \text{Coh}_{\text{lf}}^n(X)$  as pointed sets, by(7.1.5.2). ┘

## 7.2 Ringed Topoi, Ringed Sites, Ringed $G$ -Spaces and Schemes

*“L’algèbre n’est qu’une géométrie écrite; la géométrie n’est qu’une algèbre figurée.”*

–S. Germain

### 1 Ringed Topoi, Ringed Sites

**Def. (7.2.1.1) [Ringed Topoi].** A **ringed topos** is a pair  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O})$  where  $\mathcal{C}$  is a site and  $\mathcal{O}$  is a unital object in  $\mathrm{Sh}(\mathcal{C})$ , called the **structure sheaf**. A morphism of ringed topoi  $(f, f^\sharp) : (\mathcal{C}, \mathcal{O}) \rightarrow (\mathcal{C}', \mathcal{O}')$  consists of a morphism of topoi (7.1.2.18)  $f : \mathrm{Sh}(\mathcal{C}) \rightarrow \mathrm{Sh}(\mathcal{C}')$  and a map of sheaves of rings  $f^\sharp : f^{-1}\mathcal{O}' \rightarrow \mathcal{O}$ . A composition of morphism of topoi is defined to be  $(g, g^\sharp) \circ (f, f^\sharp) = (g \circ f, f^\sharp \circ f^{-1}(g^\sharp))$ .  $\lrcorner$

**Def. (7.2.1.2) [Ringed Sites].** A **ringed site** is pair  $(\mathcal{C}, \mathcal{O})$  where  $\mathcal{C}$  is a site and  $\mathcal{O}$  is a sheaf of rings on  $\mathcal{C}$ , called the **structure sheaf**. A ringed site induces a ringed topos. A morphism of ringed sites is a morphism of sites that the induced morphism of topoi (7.1.2.20) is a morphism of ringed topoi (7.2.1.1).  $\lrcorner$

A site is naturally a ringed site where  $\mathcal{O} = \underline{\mathbb{Z}}$  the constant sheaf (7.1.2.8). So we only consider ringed sites afterwards, then a morphism of ringed sites is naturally a morphism of ringed sites. So whenever we say  $\mathcal{C}$  is a site, it is understood as a ringed site  $(\mathcal{C}, \underline{\mathbb{Z}})$ .  $\lrcorner$

**Prop. (7.2.1.3) [Ringed Topoi and Ringed Sites].** Let  $(f, f^\sharp) : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \rightarrow \mathrm{Sh}(\mathcal{D}, \mathcal{O}_{\mathcal{D}})$  be a morphism of ringed topoi, then we can find ringed sites  $(\mathcal{C}', \mathcal{O}_{\mathcal{C}'})$  and  $(\mathcal{D}', \mathcal{O}_{\mathcal{D}'})$  and a diagram

$$\begin{array}{ccc} (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) & \xrightarrow{(f, f^\sharp)} & (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}}) \\ \downarrow (g, g^\sharp) & & \downarrow (e, e^\sharp) \\ (\mathrm{Sh}(\mathcal{C}'), \mathcal{O}_{\mathcal{C}'}) & \xrightarrow{(h, h^\sharp)} & (\mathrm{Sh}(\mathcal{D}'), \mathcal{O}_{\mathcal{D}'}) \end{array}$$

where

- $(g, g^\sharp)$  and  $(e, e^\sharp)$  are equivalence of ringed topoi.
- $\mathcal{C}', \mathcal{D}'$  have final objects and finite products.
- $(h, h^\sharp)$  is induced by a continuous functor  $\mathcal{D}' \rightarrow \mathcal{C}'$  that the preserves the final object and finite products, (thus induces a morphism of sites by (7.1.2.14)).

Moreover, given a set of sheaves  $\mathcal{F}_i$  on  $\mathcal{C}$  and a set of sheaves  $\mathcal{G}_i$  on  $\mathcal{D}$ , we may choose  $\mathcal{C}'$  and  $\mathcal{D}'$  that these sheaves are representable by objects in  $\mathcal{C}'$  or  $\mathcal{D}'$ .  $\lrcorner$

*Proof:* [Sta]03CR.  $\square$

**Prop. (7.2.1.4) [Multiplicative Structure Sheaf].** Given a ringed topoi  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O})$ , the presheaf  $U \mapsto \mathcal{O}^*(U)$  is a sheaf of groups, called the **multiplicative structure sheaf**  $\mathcal{O}^*$ .  $\lrcorner$

*Proof:* This comes from the sheaf property of  $\mathcal{O}$  and the fact the inverse of an element is unique.  $\square$

**Def. (7.2.1.5) [Local Ringed Site].** A ringed site  $(\mathcal{C}, \mathcal{O})$  is called a **local ringed site** if

$$\emptyset^\sharp \rightarrow \mathrm{Equalizer}(0, 1 : \mathrm{pt} \rightarrow \mathcal{O})$$

is an isomorphism of sheaves, and for any  $U \in \mathcal{C}$  and  $f \in \mathcal{O}(U)$ , there exists a covering  $\{U_i \rightarrow U\}$  s.t. for any  $j$ , either  $f|_{U_i}$  is invertible or  $(1 - f)|_{U_i}$  is invertible.  $\lrcorner$

**Prop. (7.2.1.6) [Characterizing Local Ringed Sites].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, the following are equivalent:

- $(\mathcal{C}, \mathcal{O})$  is a local ringed site.
- (Partition of Unity) For any  $U \in \mathcal{C}$ ,  $f_1, \dots, f_n \in \mathcal{O}(U)$  that  $(f_1, \dots, f_n) = (1)$ , there is a covering  $\{U_i \rightarrow U\}$  that for each  $j$ , there exists an  $i$  that  $f_i$  is invertible on  $U_j$ .
- The map of sheaves of sets:

$$(\mathcal{O} \otimes \mathcal{O}) \coprod (\mathcal{O} \times \mathcal{O}) \rightarrow \mathcal{O} \times \mathcal{O}$$

which maps  $(f, a)$  in the first component to  $(f, af)$  and  $(f, b)$  in the second component to  $(f, b(1 - f))$  is surjective.

┘

*Proof:* Cf. [Sta]04ES. □

**Def. (7.2.1.7) [Local Ringed Topoi].** If  $f : \mathrm{Sh}(\mathcal{C}') \rightarrow \mathrm{Sh}(\mathcal{C})$  is a morphism of topoi and  $(\mathcal{C}, \mathcal{O})$  is a local ringed site, then  $(\mathrm{Sh}(\mathcal{C}'), f^{-1}\mathcal{O})$  is also a local ringed site. In particular, being a local ringed site is an intrinsic property, so we can define **local ringed topoi** to be ringed topoi that the underlying ringed sites are local ringed. ┘

*Proof:* Because  $f^{-1}$  is exact (7.1.2.18) and commutes with products and equalizers, it maps the isomorphism

$$\emptyset^\# \rightarrow \mathrm{Equalizer}(0, 1 : \mathrm{pt} \rightarrow \mathcal{O})$$

to the corresponding isomorphism of  $\mathcal{C}'$ , and also the sejection of

$$(\mathcal{O} \otimes \mathcal{O}) \coprod (\mathcal{O} \times \mathcal{O}) \rightarrow \mathcal{O} \times \mathcal{O}$$

in (7.2.1.6) to that of  $\mathcal{C}'$ , thus  $(\mathrm{Sh}(\mathcal{C}'), f^{-1}\mathcal{O})$  is also a local ringed site. □

**Def. (7.2.1.8) [Morphisms of Local Ringed Topoi].** A **morphism of local ringed topoi** is a morphism of ringed topoi  $(f, f^\#) : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_\mathcal{C}) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_\mathcal{D})$  that the diagram of sheaves

$$\begin{array}{ccc} f^{-1}(\mathcal{O}_\mathcal{D}^*) & \xrightarrow{f^\#} & \mathcal{O}_\mathcal{C}^* \\ \downarrow & & \downarrow \\ f^{-1}(\mathcal{O}_\mathcal{D}) & \xrightarrow{f^\#} & \mathcal{O}_\mathcal{C} \end{array}$$

is Cartesian, where  $\mathcal{O}_\mathcal{C}^*$  is the multiplicative structure sheaf (7.2.1.4). Morphisms of local ringed topoi are stable under compositions. ┘

## Ringed Spaces

**Def. (7.2.1.9) [Ringed Spaces].** A **ringed space** is a pair  $(X, \mathcal{O}_X)$  consisting of a topological space  $X$  and a sheaf of rings  $\mathcal{O}_X$  on  $X$ . A morphism of ringed spaces  $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  consists of a pair  $(f, f^\#)$  where  $f$  is a continuous map  $X \rightarrow Y$  and  $f^\#$ , which induces a map of sites  $(f^{-1}, f_*) : X_{\mathrm{Zar}} \rightarrow Y_{\mathrm{Zar}}$  (7.2.6.7), and  $f^\#$  is a map  $f^\# : f^{-1}(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$  (7.2.6.2), such that  $(f, f^\#)$  is a map of ringed topoi (7.2.1.1). ┘

**Def. (7.2.1.10) [Local Ringed Space].** A **local ringed space** is a topological space  $X$  with a sheaf of rings  $\mathcal{O}_X$  that  $(X, \mathcal{O}_X)$  forms a local ringed site (7.2.1.5). A morphism of local ringed space is a morphism of ringed spaces that the corresponding morphism of ringed topoi  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$  is a morphism of local ringed topoi (7.2.1.8).  $\lrcorner$

**Prop. (7.2.1.11).** A ringed space  $(X, \mathcal{O}_X)$  is a locally ringed space iff any stalks  $\mathcal{O}_{X,x}$  is either 0 or a local rings.  $\lrcorner$

*Proof:* Cf. [Sta]04ET.  $\square$

## 2 Modules on Ringed Topoi

Main References are [Sta]Chap 18.

**Def. (7.2.2.1) [Modules on Ringed Topoi].** Let  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O})$  be a ringed topos, then a sheaf of  $\mathcal{O}$ -modules is a presheaf of  $\mathcal{O}$ -modules that the underlying presheaf of Abelian groups is a sheaf.  $\lrcorner$

**Def. (7.2.2.2) [Support].** Let  $(X, \mathcal{O}_X)$  be a ringed space,  $\mathcal{F}$  a  $\mathcal{O}_X$ -module, then the **support** of  $\mathcal{F}$  is the set of points  $x$  that  $\mathcal{F}_x \neq 0$ . It is denoted by  $\mathrm{Supp}(\mathcal{F})$ . For a section  $s \in \Gamma(X, \mathcal{F})$ ,  $\mathrm{Supp}(s)$  is defined to be the set of points  $x$  that  $s_x \neq 0 \in \mathcal{F}_x$ .  $\lrcorner$

**Prop. (7.2.2.3).** Glueing sheaves is available for ringed spaces, similar to (7.1.5.3).  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.2.2.4).** Glueing ringed spaces is available.  $\lrcorner$

*Proof:*  $\square$

**Def. (7.2.2.5) [Local Properties of Modules].** On a ringed site  $(\mathcal{C}, \mathcal{O})$ , an  $\mathcal{O}$ -module  $\mathcal{F}$  is called locally has property  $P$  if for any object  $U \in \mathcal{C}$ , there is a covering  $\{U_i \rightarrow U\}$  that  $\mathcal{F}|_{U_i}$  has property  $P$ .  $\lrcorner$

**Def. (7.2.2.6) [Intrinsic Properties of Modules].** An **intrinsic property** of sheaves of modules in a ringed topos  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O})$  that is invariant under equivalence of topoi.  $\lrcorner$

**Def. (7.2.2.7) [Tensor Products Sheaf].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{F}, \mathcal{G}$  be  $\mathcal{O}$ -modules, then the **tensor product**  $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$  is defined to be the shification of the presheaf  $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}(U)} \mathcal{G}(U)$ .

The tensor product is easily seen to be an intrinsic notion (7.2.2.6), so it can be defined on any ringed topoi.  $\lrcorner$

**Prop. (7.2.2.8) [Base Change].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{O}_2$  be a sheaf  $\mathcal{O}_1$ -algebras,  $\mathcal{G}$  be a sheaf of  $\mathcal{O}_1$ -module and  $\mathcal{F}$  a sheaf of  $\mathcal{O}_2$ -module, then

$$\mathrm{Hom}_{\mathcal{O}_1}(\mathcal{G}, \mathcal{F}_{\mathcal{O}_1}) = \mathrm{Hom}_{\mathcal{O}_2}(\mathcal{G} \otimes_{\mathcal{O}_1} \mathcal{O}_2, \mathcal{F}).$$

$\lrcorner$

*Proof:* This can be seen from the definition of tensor product and the fact shification doesn't bother because  $\mathcal{F}$  is a sheaf.  $\square$

**Def. (7.2.2.9) [Transfer of Modules on Ringed Sites].** Let  $(f, f^\#)$  be a morphism of ringed topoi  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O}) \rightarrow (\mathrm{Sh}(\mathcal{C}'), \mathcal{O}')$ , then there are functor:

- the **pushforward**:  $f_* : \text{Mod}(\mathcal{O}) \rightarrow \text{Mod}(\mathcal{O}') : f_*\mathcal{F} = f_*\mathcal{F}$  as a  $\mathcal{O}'$ -module via  $\mathcal{O}' \rightarrow f_*\mathcal{O} \rightarrow f_*\mathcal{F}$ .
- the **pullback**  $f^* : \text{Mod}(\mathcal{O}') \rightarrow \text{Mod}(\mathcal{O}) : f^*\mathcal{G} = \mathcal{O} \otimes_{f^{-1}\mathcal{O}'} f^{-1}\mathcal{G}$  via  $f^\# : f^{-1}\mathcal{O}' \rightarrow \mathcal{O}$ .  $f^*$  is left adjoint to  $f_*$ , by the adjointness of  $f^{-1}$  and  $f_*$  (7.1.2.18).
- (Extension by zero): For the localization map of sites  $j_U : (\mathcal{C}/U, \mathcal{O}_U) \rightarrow (\mathcal{C}, \mathcal{O})$ ,  $j_U^* = j_U^{-1}$  has a left adjoint  $j_{U!}$ , called the **extension by zero** functor, defined by shiffication of the presheaf

$$\mathcal{G} \mapsto j_{U!}(\mathcal{G}) : j_{U!}(\mathcal{G})(V) = \bigoplus_{\varphi: V \rightarrow U} \mathcal{G}(V \xrightarrow{\varphi} U \in \mathcal{C}/U)$$

with obvious restriction mappings and an obvious  $\mathcal{O}$ -module structure. ┘

*Proof:* To show  $j_{U!}$  is left adjoint to  $j_U^*$ , Cf.[Sta]03DI. □

**Remark (7.2.2.10).** The  $f^*$  may not be exact.  $f^{-1}$  is exact, but we tensored with  $\mathcal{O}_X$ , it is exact when  $f$  is flat (7.2.2.15). ┘

**Prop. (7.2.2.11) [Extension by Zero is Exact].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $U \in \mathcal{C}$ , then the extension by zero functor  $j_{U!} : \text{Mod}(\mathcal{O}_U) \rightarrow \text{Mod}(\mathcal{O})$  is exact and reflects exactness. ┘

*Proof:* It is right exact because is a left adjoint, and it is left exact by direct inspection and the fact that shiffication is exact.

For refecton of exactness, Cf.[Sta]0E8G. ? □

**Prop. (7.2.2.12) [Tensor Products and Pullbacks].** Tensor products commute with pullbacks, in particular with taking stalks. So tensoring with a locally free sheaf is exact. ┘

*Proof:* By (7.2.3.3) and (7.2.3.5), for any  $\mathcal{O}$ -module  $\mathcal{H}$ ,

$$\text{Hom}(f^*\mathcal{F} \otimes f^*\mathcal{G}, \mathcal{H}) = \text{Hom}(\mathcal{F}, f_*\text{Hom}(f^*\mathcal{G}, \mathcal{H})) = \text{Hom}(\mathcal{F}, \text{Hom}(\mathcal{G}, f_*\mathcal{H})) = \text{Hom}(f^*(\mathcal{F} \otimes \mathcal{G}), \mathcal{H}).$$

□

**Prop. (7.2.2.13) [ $j_{U!}$  Commutes with Restriction].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $U \in \mathcal{C}, \mathcal{G} \in \text{Mod}(\mathcal{O}_U), \mathcal{F} \in \text{Mod}(\mathcal{O})$ , then there is a natural isomorphism  $j_{U!}\mathcal{G} \otimes_{\mathcal{O}} \mathcal{F} \cong j_{U!}(\mathcal{G} \otimes_{\mathcal{O}_U} \mathcal{F}|_U)$  ┘

*Proof:* By (7.2.3.3) and (7.2.3.5), for any  $\mathcal{H} \in \text{Mod}(\mathcal{O})$ ,

$$\begin{aligned} \text{Hom}_{\mathcal{O}}(j_{U!}(\mathcal{G} \otimes_{\mathcal{O}_U} \mathcal{F}|_U), \mathcal{H}) &= \text{Hom}_{\mathcal{O}_U}(\mathcal{G} \otimes_{\mathcal{O}_U} \mathcal{F}|_U, \mathcal{H}|_U) = \text{Hom}_{\mathcal{O}_U}(\mathcal{G}, \text{Hom}_{\mathcal{O}}(\mathcal{F}|_U, \mathcal{H}|_U)) \\ &= \text{Hom}_{\mathcal{O}_U}(\mathcal{G}, \text{Hom}_{\mathcal{O}_U}(\mathcal{F}, \mathcal{H})|_U) = \text{Hom}_{\mathcal{O}}(j_{U!}\mathcal{G}, \text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{H})) \\ &= \text{Hom}_{\mathcal{O}}(j_{U!}\mathcal{G} \otimes_{\mathcal{O}} \mathcal{F}, \mathcal{H}) \end{aligned}$$

then use Yoneda lemma. □

**Prop. (7.2.2.14) [Properties of Tensor Products].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{F}, \mathcal{G}$  be sheaves of  $\mathcal{O}$ -modules, then

1. If  $\mathcal{F}, \mathcal{G}$  are locally free, then so does  $\mathcal{F} \otimes \mathcal{G}$ .
2. If  $\mathcal{F}, \mathcal{G}$  are locally finite free, then so does  $\mathcal{F} \otimes \mathcal{G}$ .
3. If  $\mathcal{F}, \mathcal{G}$  are locally generated by sections, then so does  $\mathcal{F} \otimes \mathcal{G}$ .

4. If  $\mathcal{F}, \mathcal{G}$  are of f.t., then so does  $\mathcal{F} \otimes \mathcal{G}$ .
5. If  $\mathcal{F}, \mathcal{G}$  are quasi-coherent., then so does  $\mathcal{F} \otimes \mathcal{G}$ .
6. If  $\mathcal{F}, \mathcal{G}$  are of f.p., then so does  $\mathcal{F} \otimes \mathcal{G}$ .
7. If  $\mathcal{F}$  is of f.p. and  $\mathcal{G}$  is coherent, then  $\mathcal{F} \otimes \mathcal{G}$  is coherent. In particular, if  $\mathcal{F}, \mathcal{G}$  are coherent, then so does  $\mathcal{F} \otimes \mathcal{G}$  (7.2.2.27).

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*Proof:* Cf. [Sta]03L6.

□

### Flat Modules

**Def. (7.2.2.15) [Flat Modules and Flat Morphisms].** For  $\mathcal{C} \in \text{Site}$  and  $\mathcal{O}$  a presheaf of rings,  $\mathcal{F} \in \mathcal{P}\text{Sh}(\mathcal{O})$  is called **flat** if the functor  $P\text{Mod}(\mathcal{O}) \rightarrow P\text{Mod}(\mathcal{O}) : \mathcal{G} \mapsto \mathcal{G} \otimes_{\mathcal{O}} \mathcal{F}$  is exact.

Let  $\mathcal{C}$  be a ringed site, and  $\mathcal{F}$  is a sheaf of  $\mathcal{O}$ -modules, then it is called **flat** if the functor  $\text{Mod}(\mathcal{O}) \rightarrow \text{Mod}(\mathcal{O}) : \mathcal{G} \mapsto \mathcal{G} \otimes_{\mathcal{O}} \mathcal{F}$  is exact.

A morphism  $(f, f^{\sharp})$  is called a **flat morphism** if the ring map  $f^{\sharp} : f^{-1}\mathcal{O}' \rightarrow \mathcal{O}$  is flat, or equivalently, the pullback functor (7.2.2.9)  $f^*$  is exact.

If  $(f, f^{\sharp})$  is a morphism of ringed topoi  $(\text{Sh}(\mathcal{C}), \mathcal{O}) \rightarrow (\text{Sh}(\mathcal{C}'), \mathcal{O}')$ , and  $\mathcal{F}$  is a sheaf of  $\mathcal{O}$ -modules, then  $\mathcal{F}$  is **flat** over  $(\text{Sh}(\mathcal{C}'), \mathcal{O}')$  if  $\mathcal{F}$  is flat over  $f^{-1}\mathcal{O}'$ .

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**Prop. (7.2.2.16).** Let  $\mathcal{C}$  be a site and  $\mathcal{O}$  a presheaf of rings with shification  $\mathcal{O}^{\sharp}$ .

- If  $\mathcal{F}$  is a presheaf of  $\mathcal{O}$ -modules that each  $\mathcal{F}(U)$  is flat  $\mathcal{O}(U)$ -modules, then  $\mathcal{F}$  is flat.
- If  $\mathcal{F}$  is a flat presheaf of  $\mathcal{O}$ -modules, then  $\mathcal{F}^{\sharp}$  is a flat  $\mathcal{O}^{\sharp}$ -modules.
- A filtered colimits of flat presheaves of modules is flat. A direct sum of flat presheaves of modules is flat.
- A filtered colimits of flat sheaves of modules is flat. A direct sum of flat sheaves of modules is flat.

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**Prop. (7.2.2.17) [Flatness is Stalkwise].** Let  $(X, \mathcal{O}_X)$  be a ringed space, then an  $\mathcal{O}_X$ -module  $\mathcal{F}$  is flat iff the stalks  $\mathcal{F}_x$  are all flat  $\mathcal{O}_{X,x}$ -modules.

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*Proof:* Cf. [Sta]05NE.

□

**Prop. (7.2.2.18) [Flat Morphism and Support].** Let  $f : (X, \mathcal{O}_X) \rightarrow (X', \mathcal{O}_{X'})$  be a flat morphism of local ringed spaces,  $\mathcal{F}$  an  $\mathcal{O}_{X'}$ -module, then  $\text{Supp}(f^*(\mathcal{F})) = f^{-1}(\text{Supp}(\mathcal{F}))$ .

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*Proof:* Use the fact flat ring map of local rings is faithfully flat.

□

### Modules of Finite Type & Finite Presentation

**Def. (7.2.2.19) [Finite Type].** An  $\mathcal{O}$ -module is called if **finite type** iff locally a quotient of a finite free sheaf.

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**Prop. (7.2.2.20) [Extension of F.T. Sheaves].** if  $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$  is an exact sequence of  $\mathcal{O}$ -modules and  $\mathcal{F}_1, \mathcal{F}_3$  are of f.t., then  $\mathcal{F}_2$  is of f.t..

┘

*Proof:* For any  $U \in \mathcal{C}$ , choose a covering  $\{U_i \rightarrow U\}$  that  $\mathcal{F}_3(U_i)$  is generated by f.m. sections, then by passing to a covering, we may assume these sections come from  $\mathcal{F}_2$ . Pass to another covering that  $\mathcal{F}_1$  is generated by f.m. sections, then on this covering,  $\mathcal{F}_2$  is generated by f.m. sections.  $\square$

**Def. (7.2.2.21)[Finite Presentation].** A sheaf of modules  $\mathcal{F}$  is called **of finite presentation** iff locally it is a cokernel of two finite free modules. The pullback of a f.p. sheaf is f.p, by the left adjointness of  $f^*$ .  $\lrcorner$

**Prop. (7.2.2.22)[FP-FT-FT].** If  $f : \mathcal{G} \rightarrow \mathcal{F}$  is a surjection of  $\mathcal{O}$ -modules,  $\mathcal{F}$  is of f.p. and  $\mathcal{G}$  is of f.t., then the kernel is of finite type.  $\lrcorner$

*Proof:* We first show for  $\mathcal{G} = \mathcal{O}^n$ : By pass to covering, we can construct a diagram

$$\begin{array}{ccccccc} \mathcal{O}_{U_{ij}}^m & \longrightarrow & \mathcal{O}_{U_{ij}}^n & \longrightarrow & \mathcal{F}|_{U_{ij}} & \longrightarrow & 0 \\ \downarrow \alpha & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \ker(f)|_{U_{ij}} & \longrightarrow & \mathcal{G}|_{U_{ij}} & \longrightarrow & \mathcal{F}|_{U_{ij}} \longrightarrow 0 \end{array}$$

and then use snake lemma. The image and cokernel of  $\alpha$  are all of f.t., then  $\ker(f)|_{U_{ij}}$  is of f.t. by (7.2.2.20).

For general  $\mathcal{G}$ , locally choose a surjection  $\varphi : \mathcal{O}_{U_i}^n \rightarrow \mathcal{G}|_{U_i}$ , then  $\ker(f|_{U_i}) = \varphi(\ker(\varphi \circ f|_{U_i}))$ , which is of f.t..  $\square$

**Prop. (7.2.2.23).** Pullbacks of a module of finite type is of finite type. Pullback of a module of finite presentation is of finite presentation.

Finite type and finite presentation are local on the target.  $\lrcorner$

*Proof:* This is because pullback is a left adjoint thus right exact. They are local on the target because they are defined locally.  $\square$

**Prop. (7.2.2.24)[Support is Zariski Closed].** If  $(X, \mathcal{O}_X)$  is a ringed space and  $f : \mathcal{G} \rightarrow \mathcal{F}$  is surjective at a point  $x$  and  $\mathcal{F}$  is of f.t., then it is surjective on a Zariski nbhd of  $x$ . Thus the support of a f.t. sheaf is closed (look at  $0 \rightarrow \mathcal{F}$ ).  $\lrcorner$

*Proof:* Choose a nbhd of  $x$  that  $\mathcal{F}(U)$  is generated by  $s_1, \dots, s_n \in \mathcal{F}(U)$ , because  $f$  is surjective at the stalk of  $x$ , after shrinking  $U$ , we may assume  $s_i = f(t_i)$  for  $t_i \in \mathcal{G}(U)$ , so  $f$  is surjective on  $U$ .  $\square$

### (Quasi-)Coherent Sheaves

**Def. (7.2.2.25)[Quasi-Coherent Sheaf].** An  $\mathcal{O}$ -module  $\mathcal{F}$  on a ringed site is called **quasi-coherent** iff locally it is a cokernel of two free modules. A locally f.p. sheaf of modules is Qco.  $\lrcorner$

**Prop. (7.2.2.26)[Associated Qco Sheaves].** And for a ringed space  $(X, \mathcal{O}_X)$  and a  $R = \Gamma(X, \mathcal{O}_X)$ -module  $M$ , we have a coherent sheaf  $\mathcal{F}_M$  on  $X$ , defined as  $\pi^*(M)$ , where  $M$  is seen as a qco sheaf on  $(\text{pt}, R)$ . It is the sheaf associated to the presheaf  $U \mapsto \mathcal{O}_X(U) \otimes M$ .

This construction is a functor from the category of  $R$ -module to the category of Qco  $\mathcal{O}_X$ -modules, and it commutes with colimits because  $\pi^*$  does. And it is left adjoint to  $\Gamma$  by (7.2.2.9):

$$\text{Hom}_A(M, \Gamma(X, \mathcal{G})) \cong \text{Hom}_{\mathcal{O}_X}(\mathcal{F}_M, \mathcal{G})$$

$\lrcorner$

**Def. (7.2.2.27) [Coherent Sheaves].** A **coherent sheaf** is a  $\mathcal{O}$ -module  $\mathcal{F}$  that is of f.t. and for any object  $U$  and for any set of elements of  $\Gamma(U, \mathcal{F})$ , the kernel of  $\oplus \mathcal{O}_U \rightarrow \mathcal{F}|_U$  is of f.t..

A coherent sheaf is of f.p., by base change to a smaller covering, and it is Qco.  $\lrcorner$

**Prop. (7.2.2.28) [Properties of Coherent Sheaves].** Any f.t. subsheaf of a coherent sheaf is coherent, by definition. Any kernel of a morphism from a f.t. sheaf to a coherent sheaf is of f.t..

$\text{Coh}(X)$  is a weak Serre subcategory of  $\text{Mod}_{\mathcal{O}_X}$ . In particular, if  $\mathcal{O}_X$  is coherent, then a sheaf is coherent iff it is f.p.  $\lrcorner$

*Proof:* Let  $\mathcal{G}$  be a sheaf of f.t. and  $\mathcal{F}$  be a coherent sheaf. For any map  $f : \mathcal{G} \rightarrow \mathcal{F}$  and  $U \in \mathcal{C}$ , let  $\{U_i \rightarrow U\}$  be a covering that there are surjections  $\varphi_i : \mathcal{O}_{U_i}^{n_i} \rightarrow \mathcal{G}|_{U_i}$ . Then  $\ker(f \circ \varphi_i)$  is a  $\mathcal{O}_{U_i}$ -module of f.t.. Now  $\varphi_i : \ker(f \circ \varphi_i) \rightarrow \ker(f)|_{U_i}$  is a surjection, so  $\ker(f)|_{U_i}$  is of f.t., and  $\ker(f)$  is of f.t..

The kernel of a map between coherent sheaves is of f.t. by the result above, thus it is coherent.

Let  $\varphi : \mathcal{F}_1 \rightarrow \mathcal{F}$  be a map between coherent sheaves, then  $\text{Im}(\varphi)$  and  $\text{Coker}(\varphi)$  are of f.t.. For any  $U \in \mathcal{C}$  and sections  $\bar{s}_i$  of  $\text{Coker}(\varphi)(U)$  inducing a map  $\bar{\Phi} : \mathcal{O}_U^n \rightarrow \text{Coker}(\varphi)|_U$ , we can choose a covering  $\{U_i \rightarrow U\}$  that  $\bar{s}_i$  comes from  $s_i \in \mathcal{F}(U_i)$  for any  $i$ . Now we can choose coverings  $\{U_{ij} \rightarrow U_i\}$  that there are surjections  $\mathcal{O}_{U_{ij}}^{n_{ij}} \rightarrow \text{Im}(\varphi)(U_{ij})$ . Consider the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_{U_{ij}}^{n_{ij}} & \longrightarrow & \mathcal{O}_{U_{ij}}^{n_{ij}} \oplus \mathcal{O}_{U_{ij}}^n & \longrightarrow & \mathcal{O}_{U_{ij}}^n \longrightarrow 0 \\ & & \downarrow & & \downarrow \Phi & & \downarrow \\ 0 & \longrightarrow & \text{Im}(\varphi)|_{U_{ij}} & \longrightarrow & \mathcal{F}|_{U_{ij}} & \longrightarrow & \text{Coker}(\varphi)|_{U_{ij}} \longrightarrow 0 \end{array},$$

then snake lemma gives a surjection  $\ker \Phi \rightarrow \ker(\bar{\Phi}) \rightarrow 0$ . Because  $\ker \Phi$  is of f.t., so does  $\ker(\bar{\Phi})$ , and  $\text{Coker}(\varphi)$  is coherent.

Let  $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$  be an exact sequence that  $\mathcal{F}_1, \mathcal{F}_3$  is coherent, then by (7.2.2.20),  $\mathcal{F}_2$  is of f.t.. For any object  $U \in \mathcal{C}$ , consider the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & 0 & \longrightarrow & \mathcal{O}_U^n & \longrightarrow & \mathcal{O}_U^n \longrightarrow 0 \\ & & \downarrow & & \downarrow \varphi_1 & & \downarrow \varphi_2 \\ 0 & \longrightarrow & \mathcal{F}_1|_U & \longrightarrow & \mathcal{F}_2|_U & \longrightarrow & \mathcal{F}_3|_U \longrightarrow 0 \end{array},$$

the snake lemma gives an exact sequence  $0 \rightarrow \ker(\varphi_1) \rightarrow \ker(\varphi_2) \rightarrow \mathcal{F}_1|_U$ . So  $\ker(\varphi_1)$  is of f.t., so  $\mathcal{F}_2$  is coherent.  $\square$

**Prop. (7.2.2.29).** The pullback of a (quasi-)coherent module is a (quasi-)coherent, because  $f^*$  is a left adjoint.  $\lrcorner$

**Prop. (7.2.2.30).** (Quasi-)Coherence is local on the target.  $\lrcorner$

*Proof:* This is because they are defined locally.  $\square$

**Prop. (7.2.2.31).** Let  $(X, \mathcal{O}_X)$  be a ringed space and  $x \in X$ .

- Let  $f : \mathcal{G} \rightarrow \mathcal{F}$  be a map of  $\mathcal{O}_X$ -modules. If  $\mathcal{G}$  is of f.t. and  $\mathcal{F}$  is coherent, and  $f$  is injective at the stalk of  $x$ , then there exists a nbhd  $U$  of  $x$  that  $f|_U$  is injective.
- Let  $f : \mathcal{G} \rightarrow \mathcal{F}$  be a map of coherent  $\mathcal{O}_X$ -modules that is surjective at the stalk of  $x$ , then there exists a nbhd  $U$  of  $x$  that  $f|_U$  is surjective.



- Let  $f : \mathcal{G} \rightarrow \mathcal{F}$  be a map of coherent  $\mathcal{O}_X$ -modules that is isomorphism at the stalk of  $x$ , then there exists a nbhd  $U$  of  $x$  that  $f|_U$  is an isomorphism.  $\lrcorner$

*Proof:* 1: Consider the kernel of  $f$ , then it is of f.t. by definition (7.2.2.27). Then  $\ker(f)_x = 0$ , so there is a nbhd  $U$  of  $x$  that  $\ker(f)|_U = 0$ , by (7.2.2.24), which means  $f|_U$  is injective.

2: this is immediate from (7.2.2.24).

3 follows from 1 and 2.  $\square$

### 3 Construction of Sheaves

#### Internal Hom

**Def. (7.2.3.1) [Internal Hom].** Let  $\mathcal{C}$  is a category and  $\mathcal{O}$  is a presheaf of rings,  $\mathcal{F}, \mathcal{G}$  be presheaves of  $\mathcal{O}$ -modules, then  $U \mapsto \text{Hom}_{\mathcal{O}|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$  defines a presheaf  $\mathcal{H}om(\mathcal{F}, \mathcal{G})$  of  $\mathcal{O}$ -modules, and there is a natural evaluation map

$$\mathcal{F} \otimes_{\mathcal{O}} \mathcal{H}om(\mathcal{F}, \mathcal{G}) \rightarrow \mathcal{G}.$$

Now if  $\mathcal{C}$  is a site and  $\mathcal{O}$  is a sheaf of rings,  $\mathcal{F}, \mathcal{G}$  be sheaves of  $\mathcal{O}$ -modules, then  $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$  is a sheaf of  $\mathcal{O}$ -modules by (7.1.3.18), called the **internal Hom sheaf**. Denote  $\mathcal{H}om(\mathcal{F}, \mathcal{O})$  by  $\mathcal{F}^\vee$ .  $\lrcorner$

**Prop. (7.2.3.2).** Internal Hom sheaf commutes with localization:  $\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U) = \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})|_U$ . This follows from the definition.  $\lrcorner$

**Prop. (7.2.3.3) [Tensoring and Inner Hom].** If  $\mathcal{C}$  is a site,  $\mathcal{O}$  is a sheaf of rings and  $\mathcal{F}, \mathcal{G}, \mathcal{H}$  are sheaves of  $\mathcal{O}$ -modules, then there is a canonical isomorphism of sheaves:

$$\mathcal{H}om_{\mathcal{O}}(\mathcal{H} \otimes_{\mathcal{O}} \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}}(\mathcal{H}, \mathcal{G})).$$

In particular, taking limit over  $\mathcal{C}$ , we see  $- \otimes_{\mathcal{O}} \mathcal{H}$  is left adjoint to  $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, -)$ :

$$\text{Hom}_{\mathcal{O}}(\mathcal{H} \otimes_{\mathcal{O}} \mathcal{F}, \mathcal{G}) \cong \text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}}(\mathcal{H}, \mathcal{G})).$$

In particular, the monoidal category of  $\mathcal{O}_X$ -modules is closed (7.2.2.15).  $\lrcorner$

*Proof:* Omitted (Recall the definition of tensor product sheaf (7.2.2.7)).  $\square$

**Prop. (7.2.3.4).**

$$\mathcal{H}om(\varinjlim A_i, B) \cong \varprojlim \mathcal{H}om(A_i, B) \quad \mathcal{H}om(A, \varprojlim B_i) \cong \varprojlim \mathcal{H}om(A, B_i)$$

$\lrcorner$

*Proof:* This is immediate from (4.1.5.11).  $\square$

**Prop. (7.2.3.5).**  $f^*$  is left adjoint to  $f_*$  by (7.2.2.9):  $\text{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \cong \text{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$ . In fact

$$f_*(\mathcal{H}om_{\mathcal{O}}(f^*\mathcal{G}, \mathcal{F})) \cong \mathcal{H}om_{\mathcal{O}'}(\mathcal{G}, f_*\mathcal{F}).$$

by checking on every open subset  $U \subset Y$ .  $\lrcorner$

**Prop. (7.2.3.6).** Let  $\mathcal{C}$  be a site and  $\mathcal{O} \rightarrow \mathcal{O}'$  is a map of sheaves of rings, then for any  $\mathcal{G} \in \text{Mod}(\mathcal{O}')$  and  $\mathcal{F} \in \text{Mod}(\mathcal{O})$ , there is a natural isomorphism

$$\text{Hom}_{\mathcal{O}}(\mathcal{G}, \mathcal{F}) = \text{Hom}_{\mathcal{O}'}(\mathcal{G}, \text{Hom}_{\mathcal{O}}(\mathcal{O}', \mathcal{F})).$$

by checking on every open subset  $U \subset Y$ . ┘

**Prop. (7.2.3.7).** Let  $f_* : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  be a morphism of ringed spaces, and  $\mathcal{F}, \mathcal{G}$  are  $\mathcal{O}_Y$ -modules. If  $\mathcal{F}$  is f.p. and  $f$  is flat, then the canonical map

$$f^* \text{Hom}_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{G}) \rightarrow \text{Hom}_{\mathcal{O}_X}(f^* \mathcal{F}, f^* \mathcal{G})$$

is an isomorphism. ┘

*Proof:* ? □

**Prop. (7.2.3.8).** Let  $(X, \mathcal{O}_X)$  be a ringed space and  $\mathcal{F}, \mathcal{G}$  be  $\mathcal{O}_X$ -modules. If  $\mathcal{F}$  is f.p. or locally free, then the canonical map

$$\text{Hom}(\mathcal{F}, \mathcal{G})_x \rightarrow \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$$

is an isomorphism. ┘

*Proof:* Choose a presentation of  $\mathcal{F}$ . This follows from the exactness of taking stalks and (7.2.3.4). □

**Prop. (7.2.3.9).** Let  $(X, \mathcal{O}_X)$  be a ringed site and  $\mathcal{F}, \mathcal{G}$  be  $\mathcal{O}_X$ -modules. If  $\mathcal{F}$  is f.p. and  $\mathcal{G}$  is coherent, then  $\text{Hom}(\mathcal{F}, \mathcal{G})$  is also coherent. In particular, this applies to  $\mathcal{F}, \mathcal{G}$  both coherent. ┘

*Proof:* This follows from (7.2.3.4) and (7.2.2.28). □

### Tensor Sheaves

**Def. (7.2.3.10) [Tensor Sheaves].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{F}$  an  $\mathcal{O}$ -module, then we define

- $T(\mathcal{F})$  to be the sheafification of the presheaf  $U \mapsto T_{\mathcal{O}_X(U)}(\mathcal{F}(U))$  (5.1.1.23).
  - $\wedge \mathcal{F}$  to be the sheafification of the presheaf  $U \mapsto \wedge_{\mathcal{O}_X(U)}(\mathcal{F}(U))$  (5.1.1.23).
  - $\text{Sym}(\mathcal{F})$  to be the sheafification of the presheaf  $U \mapsto \text{Sym}_{\mathcal{O}_X(U)}(\mathcal{F}(U))$  (5.1.1.23).
- ┘

**Cor. (7.2.3.11).** Over a ringed space  $(X, \mathcal{O}_X)$ , the construction of  $T(\mathcal{F})$ ,  $\wedge(\mathcal{F})$  and  $\text{Sym}(\mathcal{F})$  commutes with taking stalks, because the construction of tensor algebras and shiffication are both left adjoints (5.1.1.24). Also they commutes with pullbacks, because they satisfy the same universal properties. ┘

**Prop. (7.2.3.12).** let  $\mathcal{F}$  be an  $(\mathcal{C}, \mathcal{O})$ -module, then the following properties are preserved under the construction of  $T(\mathcal{F})$ ,  $\wedge(\mathcal{F})$  and  $\text{Sym}(\mathcal{F})$ :

- Locally generated by sections.
  - Finite Type.
  - Finite Presented.
  - Coherent.
  - Quasi-coherent.
  - Locally free.
- ┘

*Proof:* Cf. [Sta]01CL. □

#### 4 Sheaf of Differentials

**Prop. (7.2.4.1).** If  $\mathcal{C}$  is a site and  $\mathcal{O}_1 \rightarrow \mathcal{O}_2$  is a homomorphism of sheaves of rings and  $\mathcal{F}$  is a sheaf of  $\mathcal{O}_2$ -modules, then an  $\mathcal{O}_1$ -**derivation** from  $\mathcal{O}_2$  to  $\mathcal{F}$  is a map that for any  $U \in \mathcal{C}$ , the map  $\mathcal{O}_2(U) \rightarrow \mathcal{F}(U)$  is a  $\mathcal{O}_1(U)$ -derivation (5.4.3.1).  $\square$

**Prop. (7.2.4.2) [Sheaf of Differentials].** Let  $\mathcal{C}$  be a site and  $\mathcal{O}_1 \rightarrow \mathcal{O}_2$  is a homomorphism of sheaves of rings and  $\mathcal{F}$  is a sheaf of  $\mathcal{O}_2$ -modules, then the functor  $\text{Mod}(\mathcal{O}_2) \rightarrow \text{Ab} : \mathcal{F} \rightarrow \text{Der}_{\mathcal{O}_1}(\mathcal{O}_2, \mathcal{F})$  is representable by a sheaf of modules  $\Omega_{\mathcal{O}_2/\mathcal{O}_1}$ , called the **sheaf of differentials**. and the map  $d : \mathcal{O}_2 \rightarrow \Omega_{\mathcal{O}_2/\mathcal{O}_1}$  is called the universal derivation.  $\square$

*Proof:* The construction is similar to that of (5.4.3.4): if for any sheaf  $\mathcal{F}$  we denote  $\mathcal{O}_2[\mathcal{F}]$  generated by shification of the presheaf  $U \mapsto \mathcal{O}_2(U)[\mathcal{F}(U)]$ , then  $\Omega_{\mathcal{O}_2/\mathcal{O}_1}$  is the cokernel of

$$\mathcal{O}_2[\mathcal{O}_2 \times \mathcal{O}_2] \oplus \mathcal{O}_2[\mathcal{O}_2 \times \mathcal{O}_2] \oplus \mathcal{O}_2[\mathcal{O}_1] \rightarrow \mathcal{O}_2[\mathcal{O}_2]$$

 $\square$ 

**Prop. (7.2.4.3) [Localness of Sheaf of Differentials].** If  $\mathcal{O}_1 \rightarrow \mathcal{O}_2$  is a homomorphism of presheaves of rings, then  $\Omega_{\mathcal{O}_2^\#/\mathcal{O}_1^\#}$  is the sheafification of the presheaf  $U \mapsto \Omega_{\mathcal{O}_2(U)/\mathcal{O}_1(U)}$ .  $\square$

*Proof:* This is because the construction of  $\Omega_{\mathcal{O}_2^\#/\mathcal{O}_1^\#}$  (5.4.3.4) for all  $U$  gives an exact sequence of presheaves, and the shification of which is just the construction in (7.2.4.2), so we are done because shification is exact (7.1.2.7).  $\square$

**Prop. (7.2.4.4) [Change of Sites].** Let  $f : \text{Sh}(\mathcal{D}) \rightarrow \text{Sh}(\mathcal{C})$  be a morphism of topoi and  $\varphi : \mathcal{O}_1 \rightarrow \mathcal{O}_2$  a homomorphism of rings on  $\mathcal{C}$ , then there is a canonical isomorphism  $f^{-1}\Omega_{\mathcal{O}_2/\mathcal{O}_1} \cong \Omega_{f^{-1}\mathcal{O}_2/f^{-1}\mathcal{O}_1}$  compatible with the universal derivations.  $\square$

*Proof:* This follows from the construction (7.2.4.2) and the fact  $f^{-1}$  is exact (7.2.4.2).  $\square$

**Prop. (7.2.4.5) [Functoriality of  $\Omega$ ].** Let  $\varphi : (\mathcal{O}_1 \rightarrow \mathcal{O}_2) \rightarrow (\mathcal{O}'_1 \rightarrow \mathcal{O}'_2)$  be a commutative diagram of sheaves of rings over a site  $\mathcal{C}$ , then the map  $\mathcal{O}'_1 \rightarrow \mathcal{O}'_2$  composed with the derivative  $\mathcal{O}'_2 \rightarrow \Omega_{\mathcal{O}'_2/\mathcal{O}'_1}$  is a  $\mathcal{O}'_1$ -derivative, thus we obtain a map of  $\mathcal{O}_2$ -modules  $\Omega_{\mathcal{O}_2/\mathcal{O}_1} \rightarrow \Omega_{\mathcal{O}'_2/\mathcal{O}'_1}$ , or equivalently a map of  $\mathcal{O}'_2$ -modules  $\Omega_{\mathcal{O}_2/\mathcal{O}_1} \otimes_{\mathcal{O}_2} \mathcal{O}'_2 \rightarrow \Omega_{\mathcal{O}'_2/\mathcal{O}'_1}$ . Thus  $\Omega_{-/-}$  is a functor of arrows.  $\square$

Moreover, if  $\mathcal{O}'_2 = \mathcal{O}_2 \otimes_{\mathcal{O}_1} \mathcal{O}'_1$ , then this map is an isomorphism, by (7.2.4.3) and (5.4.3.6).  $\square$

**Prop. (7.2.4.6).** Let  $\mathcal{O}_1 \rightarrow \mathcal{O}_2 \rightarrow \mathcal{O}'_2$  be a map of sheaves of rings that  $\mathcal{O}_2 \rightarrow \mathcal{O}'_2$  is surjective with kernel  $\mathcal{I} \subset \mathcal{O}_2$ , then there is a canonical exact sequence of  $\mathcal{O}'_2$ -modules

$$\mathcal{I}/\mathcal{I}^2 \rightarrow \Omega_{\mathcal{O}_2/\mathcal{O}_1} \otimes_{\mathcal{O}_2} \mathcal{O}'_2 \rightarrow \Omega_{\mathcal{O}'_2/\mathcal{O}_1} \rightarrow 0,$$

where the first map is characterized by mapping local sections  $f$  of  $\mathcal{I}$  to  $df \otimes 1$ .  $\square$

*Proof:* The first map is well-defined if  $d(\mathcal{I}^2) = 0$ . To show the exactness, let  $\mathcal{O}''_2 \subset \mathcal{O}'_2$  to be the presheaf of  $\mathcal{O}_1$ -algebras that  $\mathcal{O}''_2(U)$  the image of  $\mathcal{O}_2(U) \rightarrow \mathcal{O}'_2(U)$ . Then there is an exact sequence

$$\mathcal{I}(U)/\mathcal{I}(U)^2 \rightarrow \Omega_{\mathcal{O}_2(U)/\mathcal{O}_1(U)} \otimes_{\mathcal{O}_2(U)} \mathcal{O}''_2(U) \rightarrow \Omega_{\mathcal{O}''_2(U)/\mathcal{O}_1(U)} \rightarrow 0$$

by (5.4.3.8). Now shification of these presheaves gives use the desired result by (7.2.4.3).  $\square$

**Def. (7.2.4.7) [Sheaf of Differentials].** Let  $(X, \mathcal{O}_X) \rightarrow (S, \mathcal{O}_S)$  be a morphism of ringed sites,

- Let  $\mathcal{F}$  be an  $\mathcal{O}_X$ -module, an  $S$ -derivation from  $\mathcal{O}_X$  to  $\mathcal{F}$  is a derivation over  $f^{-1}\mathcal{O}_S$ . The set of  $S$ -derivations is denoted by  $\text{Der}_S(\mathcal{O}_X, \mathcal{F})$ .
- the **sheaf of differentials**  $\Omega_{X/S}$  is defined to be a sheaf of modules  $\Omega_{\mathcal{O}_X/f^{-1}\mathcal{O}_S}$  (7.2.4.2), with a universal derivation  $d_{X/S} : \mathcal{O}_X \rightarrow \Omega_{X/S}$ .

 $\square$

## 5 Locally Free sheaves

**Prop. (7.2.5.1).** Pullbacks of (finite)locally free sheaves are (finite)locally free. Sub- $\mathcal{O}_X$ -modules of a (finite)locally free sheaf is (finite)locally free, by (3.2.4.21).  $\lrcorner$

**Prop. (7.2.5.2) [Finite Locally Free Sheaves and  $\mathcal{H}om$ ].** For any finite locally free sheaf  $\mathcal{E}$  on a ringed site  $(\mathcal{C}, \mathcal{O})$ :

- $\mathcal{E}^{\vee\vee} \cong \mathcal{E}$ .
- $\mathcal{H}om_{\mathcal{O}}(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^{\vee} \otimes_{\mathcal{O}} \mathcal{F}$ .
- $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \otimes \mathcal{H} \cong \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G} \otimes_{\mathcal{O}} \mathcal{H})$  if  $\mathcal{F}$  or  $\mathcal{H}$  is finite locally free.
- $\mathcal{H}om_{\mathcal{O}}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{E}^{\vee} \otimes \mathcal{G})$ , by the first and (7.2.3.3).  $\lrcorner$

*Proof:* We define the map, and verify locally, which is by 1.  $\square$

**Prop. (7.2.5.3).**  $\mathcal{H}om(\mathcal{H}, -)$  is exact for any locally free sheaf  $\mathcal{H}$ .  $\lrcorner$

### Invertible Sheaves

**Def. (7.2.5.4) [Invertible Sheaf].** An **invertible sheaf**  $\mathcal{L}$  on a ringed topoi  $(\mathcal{S}h(\mathcal{C}), \mathcal{O})$  is an invertible object in the symmetric monoidal category  $\mathcal{M}od_{\mathcal{O}}$  (4.1.5.22).  $\lrcorner$

**Prop. (7.2.5.5).** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{L}$  an  $\mathcal{O}$ -module, the following are equivalent:

- $\mathcal{L}$  is an invertible sheaf.
- there exists some  $\mathcal{O}$ -module  $\mathcal{N}$  that  $\mathcal{L} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{O}$ .

And in this case,  $\mathcal{L}$  is flat and of finite presentation, and  $\mathcal{N} \cong \mathcal{H}om_{\mathcal{O}}(\mathcal{L}, \mathcal{O})$ .  $\lrcorner$

*Proof:*  $\mathcal{L}$  is flat because tensoring  $\mathcal{L}$  is an equivalence thus exact. Let  $\psi : \mathcal{L} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{O}$  the isomorphism,  $U$  an element of  $\mathcal{C}$ , then by construction of  $\otimes$ , after localization, we may assume there exists sections  $x_i \in \mathcal{L}(U), y_i \in \mathcal{N}(U)$  that  $\psi(x_i \otimes y_i) = 1$ . Then there is an automorphism of  $\mathcal{L}|_U : x \mapsto \sum \psi(x \otimes y_i) x_i$ . This automorphism factors through

$$\mathcal{L}|_U \rightarrow \mathcal{O}_U^{\oplus n} \rightarrow \mathcal{L}|_U,$$

thus  $\mathcal{L}|_U$  is a direct summand of a finite free  $\mathcal{O}_U$ -module, thus  $\mathcal{L}$  is of finite presentation.

Assume  $\mathcal{L}$  is invertible, consider the evaluation map

$$\mathcal{L} \otimes_{\mathcal{O}} \mathcal{H}om(\mathcal{L}, \mathcal{O}) \rightarrow \mathcal{O},$$

and by (7.2.3.3),

$$\mathcal{H}om(\mathcal{O}, \mathcal{O}) = \mathcal{H}om_{\mathcal{O}}(\mathcal{L} \otimes \mathcal{N}, \mathcal{O}) \rightarrow \mathcal{H}om_{\mathcal{O}}(\mathcal{N}, \mathcal{H}om_{\mathcal{O}}(\mathcal{L}, \mathcal{O})).$$

The image of 1 gives a morphism  $\mathcal{N} \rightarrow \mathcal{H}om_{\mathcal{O}}(\mathcal{L}, \mathcal{O})$ . Tensoring  $\mathcal{L}$  gives the inverse of the evaluation map.  $\square$

**Cor. (7.2.5.6).** The pullback of an invertible sheaf is an invertible sheaf, because tensoring commutes with pullbacks (7.2.2.12).  $\lrcorner$

**Def. (7.2.5.7) [Picard Groups].** For any ringed site  $(\mathcal{C}, \mathcal{O})$ , there is a set of invertible modules over  $\mathcal{C}$  that any invertible module is isomorphic to exactly one of them. Then this set forms an Abelian group, called the **Picard group**  $\text{Pic}(\mathcal{O})$ .  $\lrcorner$

**Prop. (7.2.5.8) [Invertible Sheaves and Locally Free Sheaves of Rank 1].** If  $(X, \mathcal{O}_X)$  is a ringed space, then any locally free  $\mathcal{O}_X$ -module of rank 1 is invertible. And when  $(X, \mathcal{O}_X)$  is a local ringed space, the converse holds as well.  $\lrcorner$

*Proof:* Assume  $\mathcal{L}$  is locally free of rank 1 and consider the evaluation map (7.2.3.1)

$$\mathcal{L} \otimes_{\mathcal{O}} \text{Hom}(\mathcal{L}, \mathcal{O}) \rightarrow \mathcal{O}.$$

This map is an isomorphism when restricting to any trivializing covering of  $\mathcal{L}$ , so it is an isomorphism. Thus  $\mathcal{L}$  is invertible by (7.2.5.5).

Assume  $(\text{Sh}(\mathcal{C}), \mathcal{O})$  is a local ringed topoi and  $\mathcal{L}$  is invertible, the proof of (7.2.5.5) shows there exists a covering  $\{U_i \rightarrow U\}$  that  $\mathcal{L}|_{U_i}$  is a direct summand of a finite free  $\mathcal{O}_{U_i}$ -module. Replacing  $U_i$  by  $U$ , let  $\pi$  be the projection of  $\mathcal{O}_U^r$  onto  $\mathcal{L}|_U$  which corresponds to a matrix with entries in  $\mathcal{O}(U)$ . The image of  $\pi$  acting on  $\mathcal{O}(U)^r$  is a finite free  $\mathcal{O}(U)$ -module  $M$ , thus there are  $f_1, \dots, f_t$  generating unit ideal of  $\mathcal{O}(U)$  such that  $M_{f_i}$  is finite free. Now by definition of local ringed topoi (7.2.1.6), after replacing  $U$  by a covering, we may assume  $M$  is finite free, which means  $\mathcal{L}|_U$  is free summand of  $\mathcal{O}_U^r$ . But  $\mathcal{L}$  is invertible, thus rank of  $\mathcal{L}$  is 1.  $\square$

**Prop. (7.2.5.9).** Let  $X$  be a locally ringed space and  $\mathcal{L} \in \text{Pic}(X)$  that is generated by global sections  $s_0, \dots, s_n$ . If  $\mathcal{F}$  is the kernel of the map  $\mathcal{O}_X^{\oplus n+1} \rightarrow \mathcal{L}$ , then  $\mathcal{F} \otimes \mathcal{L}$  is globally generated.  $\lrcorner$

*Proof:*  $\mathcal{F}$  is finite locally free of rank  $n$  by (7.2.5.1). The elements

$$s_{ij} = (0, \dots, 0, s_j, 0, \dots, 0, s_i, \dots, 0) \in \Gamma(X, \mathcal{L}^{\oplus n+1})$$

is in  $\Gamma(X, \mathcal{F} \otimes \mathcal{L})$ , and it can be verified locally s.t. they generate  $\mathcal{F} \otimes \mathcal{L}$ .  $\square$

## 6 Sheaves on Spaces

### Sheaves on Topological Spaces

**Remark (7.2.6.1).** A topological space can be regarded as a ringed space by assigning the locally constant sheaf  $\mathbb{Z}$  as the structure sheaf.  $\lrcorner$

**Def. (7.2.6.2) [Grothendieck's Six Operators].** Let  $f : X \rightarrow Y$  be a continuous map of topological spaces, then the inverse image defines a continuous map between sites  $Y_{Zar} \rightarrow X_{Zar}$ , so by (7.1.2.9) and (7.1.2.11) we can define

- the **pushforward**  $f^p F$ ,  $f^p F(U) = F(f^{-1}(U))$  sends presheaf to presheaf.
- the **direct image**  $f_* \mathcal{F}$ ,  $f_* \mathcal{F}(U) = \mathcal{F}(f^{-1}(U))$  sends sheaf to sheaf.
- the **inverse image**  $f_p \mathcal{G}$ ,  $f_p \mathcal{G}(U) = \varinjlim_{f(U) \subset V} \mathcal{G}(V)$  that sends presheaf to presheaf.
- the **inverse image**  $f^{-1} \mathcal{G} = f_s(\mathcal{G})$  that sends sheaf to sheaf.
- For a morphism of locally compact spaces, we can define a **proper direct image**:

$$f_!(\mathcal{F})(U) = \{s \in \Gamma(f^{-1}(U), \mathcal{F}) \mid \text{Supp}(s) \rightarrow U \text{ proper}\}$$

This is a subsheaf of  $f_* \mathcal{F}$  and it is left exact. we denote  $\Gamma_c(X, \mathcal{F})$  as the group  $f_!(\mathcal{F})$  where  $f : X \rightarrow \text{pt}$ . And the stalk  $f_!(\mathcal{F})_y = \Gamma_c(f^{-1}(y), \mathcal{F}|_{f^{-1}(y)})$  Cf. [Gelfand P224 P225].

- the **proper inverse image** (special case)  $i^!$  for a closed immersion  $Z \subset X$  defined by

$$i^!(\mathcal{F})(U = V \cap Z) = \{s \in \Gamma(V, \mathcal{F}) \mid \text{Supp}(s) \in Z\}.$$

sends Abelian sheaves to Abelian sheaves.

- the internal tensor product.
- the internal Hom.

⌋

*Proof:* Check that  $f_!$  is a sheaf: it is separated clearly, it suffices to show that for a covering  $\cup U_i = W$  and  $\xi_i \in F(f^{-1}(U_i))$ , the section  $\xi \in F(f^{-1}(W))$  they generated by sheaf property of  $F$  is in  $f_!F(W)$ . For a compact subset  $K$ , there is a finite cover  $\cup_i U_i$  of it, thus  $K - \cup_{i \neq j} U_j$  is compact in  $U_j$ , thus its inverse image is compact in  $\text{Supp}(\xi)$ . there are f.m.  $U_j$ , thus the inverse image of  $K$  is compact in  $\text{Supp}(\xi)$ .  $\square$

**Def. (7.2.6.3)[Stalks].** The convenience of rings spaces compared to the case of sites is the it has stalk functors: For any presheaf  $\mathcal{F}$  on  $X$  and a point  $i : x \rightarrow X$ , define the **stalk**  $\mathcal{F}_x = i_p(\mathcal{F})$  (7.2.6.2). ⌋

**Prop. (7.2.6.4)[Stalks Commutes with Shiffication].** Taking stalks commutes with shiffication. ⌋

*Proof:* Cf. [Sta]007Z.  $\square$

**Prop. (7.2.6.5).** If a sheaf on a ringed space has only one non-vanishing stalk, then it is a skyscraper sheaf. (Because the restriction map to that point for every open set is an isomorphism). ⌋

**Prop. (7.2.6.6)[Stalks].** Taking stalks is a left adjoint to the skyscraper sheaf from Presheaves to Sets, thus it preserves cokernel. Moreover, for a map of sheaves  $\mathcal{F} \rightarrow \mathcal{G}$  on  $X$ ,

- $\varphi$  is a monomorphism iff  $\varphi_x$  is injective for all  $x \in X$ .
- $\varphi$  is an epimorphism iff  $\varphi_x$  is surjective for all  $x \in X$ .
- $\varphi$  is an isomorphism iff  $\varphi_x$  is surjective for all  $x \in X$ .

⌋

*Proof:* 1: If  $\varphi$  is a monomorphism, then  $\varphi_x$  is clearly. Conversely, if  $\varphi_x$  are all injective, if  $s \in \mathcal{F}(U)$  mapsto  $0 \in (U)$ , then  $s_x$  mapsto  $0 \in \mathcal{G}_x$  for all  $x$ , thus  $s_x = 0$  for all  $x$ , thus  $s = 0$ .

2: If  $\varphi$  is an epimorphism, then  $\varphi_x$  is surjective by definition. The converse is also true.

3: If  $\varphi_x$  is isomorphism for all  $x$ , then  $\varphi$  is monomorphism by 1, and for any  $t \in \mathcal{G}(U)$ ,  $t$  is locally coming from some section of  $s$ , and these sections are compatible on their intersections because of monomorphism, so they glue together to a section  $s \in \mathcal{F}(U)$  that  $\varphi(U)(s) = t$ .  $\square$

**Prop. (7.2.6.7)[Topological Spaces and Sites].** Let  $f : X \rightarrow Y$  be a continuous map of topological spaces, then  $f$  induces a map of sites  $X \rightarrow Y$  because  $f^{-1}$  is exact (7.1.2.14), thus induces a map of topoi  $f : \text{Sh}(X) \rightarrow \text{Sh}(Y)$  (7.1.2.20). ⌋

**Cor. (7.2.6.8).** Let  $f : X \rightarrow Y$  be a continuous map of topological spaces,

- Let  $\mathcal{G}$  be a presheaf on  $Y$ , then there is a canonical bijection of stalks  $(f_p(\mathcal{G}))_x = \mathcal{G}_{f(x)}$ . If  $\mathcal{G}$  is a sheaf on  $Y$ , then there is a canonical bijection of stalks  $(f^{-1}(\mathcal{G}))_x = \mathcal{G}_{f(x)}$ .
- $f^{-1}$  is left adjoint to  $f_*$ .
- $f_!$  is left exact when  $X, Y$  are locally compact. And  $j_!$  is left adjoint to the functor  $j^{-1}$  for an inclusion of open subset  $j : U \subset X$ .

- $i^!$  is right adjoint to  $i_*$  for a closed immersion  $i : Z \rightarrow X$ , in particular  $i_*$  is exact when  $i$  is a closed immersion.  $\lrcorner$

*Proof:* 1: This is because  $(-)_p$  commutes with composition (7.2.6.3), and also shiffication commutes with  $(-)_p$  (7.2.6.4).

2: This is immediate.

3:

4: The adjointness follows from the fact that any section under a homomorphism  $i_*\mathcal{G} \rightarrow \mathcal{F}$  has support contained in  $Z$ .  $\square$

**Prop. (7.2.6.9).** Let  $i : Z \rightarrow X$  be a closed immersion, then the functor  $i_* : \mathcal{A}b(Z) \rightarrow \mathcal{A}b(X)$  is exact, fully faithful, with the essential image those sheaves with support in  $Z$ .  $\lrcorner$

**Prop. (7.2.6.10) [Canonical Exact Sequences].** We have a canonical exact sequences of sheaves of modules:

$$0 \rightarrow j_!(\mathcal{F}|_U) \rightarrow \mathcal{F} \rightarrow i_*(\mathcal{F}|_Z) \rightarrow 0$$

$$0 \rightarrow i_*i_Y^!\mathcal{F} \rightarrow \mathcal{F} \rightarrow j_*(\mathcal{F}|_U) \rightarrow 0$$

(check on stalks), which is important to use reduction to calculate sheaf cohomology. The latter induces long exact sequences:

$$0 \rightarrow H_Y^0(X, \mathcal{F}) \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(U, \mathcal{F}|_U) \rightarrow H_Y^1(X, \mathcal{F}) \rightarrow \dots$$

$\lrcorner$

*Proof:* Cf. [Sta]02UT.  $\square$

**Prop. (7.2.6.11).** On a topological space  $X$ , for a qc open subset  $U$ ,  $(\oplus \mathcal{F}_i)(U) = \oplus \mathcal{F}_i(U)$ . This uses the compactness of  $U$ .  $\lrcorner$

### Morphisms of Local Ringed Spaces

**Def. (7.2.6.12) [Open Immersion of Ringed Spaces].** A morphism  $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  of ringed spaces is called an **open immersion** if  $f$  is a homeomorphism of  $X$  onto an open subset of  $Y$ , and the map  $f^\# : f^{-1}(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$  is an isomorphism.  $\lrcorner$

**Prop. (7.2.6.13).** Let  $(X, \mathcal{O}_X)$  be a ringed space,  $U \subset X$  an open subset, and  $\mathcal{O}_U = \mathcal{O}_X|_U$  is a sheaf of rings on  $U$ , then  $(U, \mathcal{O}_U) \rightarrow (X, \mathcal{O}_X)$  is an open immersion, and  $(U, \mathcal{O}_U)$  is called the **open subspace** associated to  $U$ .  $\lrcorner$

**Prop. (7.2.6.14) [Universal Property of Open Immersions].** Let  $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  be an open immersion of ringed spaces, then it has the universal property that any morphism of ringed spaces  $(T, \mathcal{O}_T) \rightarrow (Y, \mathcal{O}_Y)$  that factors set-theoretically through  $f(X)$  factors uniquely through  $(X, \mathcal{O}_X)$ .  $\lrcorner$

**Def. (7.2.6.15) [Closed Immersion].** Let  $i : Z \rightarrow X$  be a morphism of local ringed spaces, then  $i$  is called a **closed immersion of locally ringed spaces** if:

- $i$  is a homeomorphism of  $Z$  onto a closed subspace of  $X$ .
- the map  $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z$  corresponding to  $f^\#$  is surjective with kernel  $\mathcal{I}$ .
- the  $\mathcal{O}_X$ -module  $\mathcal{I}$  is locally generated by sections.



And for a closed immersion,  $\mathcal{I}$  is called the **ideal sheaf** of  $i$ .  $\square$

**Def. (7.2.6.16) [Closed Immersion Defined by Ideals].** Let  $(X, \mathcal{O}_X)$  be a local ringed space, and  $\mathcal{I} \subset \mathcal{O}_X$  a sheaf of ideals on  $X$  locally generated by sections. Let  $Z$  be the support of the sheaf of rings  $\mathcal{O}_X/\mathcal{I}$ .  $Z$  is closed in  $X$  because it is the support of 1. by (7.2.6.9), there is a unique sheaf of rings  $\mathcal{O}_Z$  on  $Z$  that  $i_*\mathcal{O}_Z = \mathcal{O}_X/\mathcal{I}$ . For any  $z \in Z$ , the stalk  $\mathcal{O}_z = \mathcal{O}_{X,z}/\mathcal{I}_z$  is a quotient of a local ring and is non-zero, thus a local ring. Then  $(Z, \mathcal{O}_Z)$  is a local ringed space and  $i : (Z, \mathcal{O}_Z) \rightarrow (X, \mathcal{O}_X)$  is a closed immersion, called the **closed immersion defined  $\mathcal{I}$** .  $\square$

**Prop. (7.2.6.17) [Closed Immersions are Equivalent to Ideals].** Let  $f : X \rightarrow Y$  be a closed immersion of local ringed spaces with ideal sheaf  $\mathcal{I}$ . Let  $i : Z \rightarrow X$  be the closed immersion defined by  $\mathcal{I}$  (7.2.6.16), then  $f$  is isomorphic to  $i$ .  $\square$

*Proof:* Because  $f_*\mathcal{O}_Z \cong \mathcal{O}_X/\mathcal{I}$  on  $X$ .  $\square$

**Prop. (7.2.6.18).** If  $f : X \rightarrow Y$  is a closed immersion of local ringed spaces with ideal sheaf  $\mathcal{I}$ , then  $f_*$  is fully faithful on  $\text{Mod}_{\mathcal{O}_X}$ , with image those modules annihilated by  $\mathcal{I}$ .  $\square$

*Proof:* Cf. [Sta]08KS.  $\square$

## 7 Spec and Schemes

**Def. (7.2.7.1) [Spectrum].** Given a commutative ring  $A$ , the **spectrum of ring** of  $A$   $\text{Spec } A$  is a locally ringed space whose underlying space is the set of primes of  $A$ , and the topology is generated by the standard open subsets  $D(f) = \{\mathfrak{p} \mid f \notin \mathfrak{p}\}$ .

To define the structure sheaf  $\mathcal{O}_X$ , we first define a sheaf on the site of standard open subsets, which takes value  $R_f$  on  $D(f)$ . This is truly a sheaf by (5.4.2.3), and then we can use (7.1.2.25) to extend this sheaf to a sheaf on  $\text{Spec } A$ , called the structure sheaf  $\mathcal{O}_X$ .

A locally ringed space of the form  $\text{Spec } A$  is called an **affine scheme**. The category of affine schemes is denoted by  $\text{Aff}$ .  $\square$

**Def. (7.2.7.2) [Schemes].** The category  $\text{Sch}$  of **schemes** is the full subcategory of the category of local ringed spaces (7.2.1.10) consisting of local ringed spaces that are locally isomorphic to  $\text{Spec } A$ .  $\square$

**Lemma (7.2.7.3).** If  $X$  is a local ringed space,  $x \in X$ , and  $Y = \text{Spec } A$  an affine scheme,  $f : X \rightarrow Y$  is a morphism, consider the ring map  $\Gamma(X, \mathcal{O}_X) \xrightarrow{f^\#} \Gamma(Y, \mathcal{O}_Y) \rightarrow \mathcal{O}_{Y,x}$ , and consider the inverse image  $\mathfrak{p}$  of  $\mathfrak{m}_x$ , which corresponds to  $y \in Y$ , then  $f(x) = y$ .  $\square$

*Proof:* There are commutative diagrams

$$\begin{array}{ccc} \Gamma(Y, \mathcal{O}_Y) & \longrightarrow & \mathcal{O}_{Y,f(x)} \\ \downarrow & & \downarrow \\ \Gamma(X, \mathcal{O}_X) & \longrightarrow & \mathcal{O}_{X,x} \end{array}$$

and the map of local rings is a local ring map. So the inverse image of  $\mathfrak{m}_x$  is just  $\mathfrak{m}_{f(x)}$ , so  $\mathfrak{m}_{f(x)} = \mathfrak{m}_y$ .  $\square$

**Prop. (7.2.7.4).** Let  $X$  be a local ringed spaces and  $Y = \text{Spec } A$  an affine scheme, then the map  $\text{Hom}(X, \text{Spec } A) \rightarrow \text{Hom}(A, \Gamma(X, \mathcal{O}_X))$  is an isomorphism.  $\square$



*Proof:* The inverse map is constructed as follows: for any  $\varphi : A \rightarrow \Gamma(X, \mathcal{O}_X)$  and  $x \in X$ , define  $\Phi(x)$  to be the point corresponding to the inverse image of  $\mathfrak{m}_x$  in  $A \rightarrow \Gamma(X, \mathcal{O}_X) \rightarrow \mathcal{O}_{X,x}$ . In this way,  $\Phi^{-1}(D(f))$  is just  $D(\varphi(f)) \subset X$  which is open, thus  $\Phi$  is continuous. Now we want to construct a sheaf homomorphism  $f^{-1}(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$ , and it suffices to construct compatible maps on the affine open basis  $D(f)$ , by (7.1.3.18). Now  $\Gamma(D(f), \mathcal{O}_Y) = A_f$ , and because  $f$  is invertible on  $D(\varphi(f))$ , there is by universal property a unique map  $A_f \rightarrow D(\varphi(f))$  extending  $\varphi$ . Then by universal property these maps are compatible. Notice the construction here also shows the homomorphism is determined by the map set-theoretically.

Finally, we need to show this induces a local ring map on the stalks, and this is quite obvious from the definition.

Then we show these two maps are inverse to each other: It suffices to show any ring map  $A \rightarrow \Gamma(X, \mathcal{O}_X)$  comes uniquely from a map  $X \rightarrow Y$ : the uniqueness is proven by (7.2.7.3), and the sheaf homomorphism is determined by the set-theoretical map by the above argument.  $\square$

**Cor. (7.2.7.5) [Adjointness of Spec and  $\Gamma$ ].** The Spec operator  $\text{Spec} : \mathcal{CAlg}^{op} \rightarrow \text{Sch}$  is right adjoint to  $X \rightarrow \Gamma(X, \mathcal{O}_X)$ :

$$\text{Hom}_{\text{Sch}}(X, \text{Spec}(A)) \cong \text{Hom}_{\mathcal{CAlg}}(A, \Gamma(X, \mathcal{O}_X)).$$

Notice the category of schemes is a full subcategory of the category of locally ringed spaces.  $\lrcorner$

**Cor. (7.2.7.6).** Aff is equivalent to  $\mathcal{CAlg}^{op}$ .  $\lrcorner$

**Prop. (7.2.7.7) [Points of Schemes].** Let  $X \in \text{Sch}$  and  $R \in \mathcal{CRing}$  is local, then there is a natural bijection between morphisms  $\text{Spec } R \rightarrow X$  and pairs  $(p, \varphi)$  where  $p \in X$  is a point and  $\varphi : \mathcal{O}_{X,p} \rightarrow R$  is a local ring map.  $\lrcorner$

*Proof:* Consider where the closed point of  $\text{Spec } R$  is mapped to and choose an affine open nbhd of that point, then we reduce to the affine case, which is by (7.2.7.5).  $\square$

**Cor. (7.2.7.8).** if  $f : Y \rightarrow X$  is a morphism of schemes that  $f(y) = x$ , then we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_{Y,y} & \longrightarrow & Y \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,x} & \longrightarrow & X \end{array}$$

$\lrcorner$

**Cor. (7.2.7.9).** The points of  $X$  are in bijection with equivalent classes of morphisms from the spectra of fields to  $X$ , and each equivalent class contains a minimal element  $\text{Spec } k(x) \rightarrow X$ .  $\lrcorner$

*Proof:* This is because a local ring map from  $\mathcal{O}_{X,x}$  to a field factors through  $k(x)$ .  $\square$

**Prop. (7.2.7.10).** The closure of a subset  $T$  of  $\text{Spec}(A)$  is  $V(\cap_{\mathfrak{p} \in T} \mathfrak{p})$ .  $\lrcorner$

**Prop. (7.2.7.11) [Scheme is Sober].** The underlying space of a scheme is sober.  $\lrcorner$

*Proof:* Firstly this is true for affine schemes, by (4.12.3.16). Then notice for any affine open subscheme  $U$ , the generic point for  $Z \cap U$  is the generic point for  $Z$ .  $\square$

### Construction of Schemes

**Prop. (7.2.7.12) [Global Spec].** There is an  $S$ -scheme  $f : \mathbf{Spec}_S \mathcal{A} \rightarrow S$  for every Qco sheaf of  $\mathcal{O}_S$ -algebras  $\mathcal{A}$  on  $S$  that for any affine open subscheme  $U \subset X$ ,  $f^{-1}(U) \cong \mathrm{Spec} \mathcal{A}(U)$  over  $U$ . This construction is right adjoint to the direct image map:

$$\mathrm{Hom}_{\mathcal{C}\mathrm{Alg}_{\mathcal{O}_S}}(\mathcal{A}, \pi_* \mathcal{O}_X) \cong \mathrm{Hom}_{\mathrm{Sch}/S}(X, \mathbf{Spec}_S \mathcal{A}).$$

and defines an equivalence of affine morphisms over  $S$  and Qco  $\mathcal{O}_S$ -algebras. Moreover, this defines an equivalence of the category of  $\mathcal{A}$ -modules and the category of  $\mathcal{O}_{\mathbf{Spec}_S \mathcal{A}}$ -modules.  $\lrcorner$

*Proof:* Choose an affine open covering  $\{U_i \rightarrow X\}$  of  $X$ , and consider the schemes  $\mathrm{Spec} \mathcal{A}(U_i) \rightarrow U_i$  over  $U_i$ , then their restrictions to  $U_{ij}$  are compatible, because this is true after further restriction to an affine open covering of  $U_{ij}$ , we can use (7.1.5.4). Then we can use (7.1.5.4) to get an  $S$ -scheme  $f : \mathbf{Spec}_S \mathcal{A} \rightarrow S$  that  $f^{-1}(U_i) \cong \mathrm{Spec} \mathcal{A}(U_i)$ .

Now we check that for any affine open subset  $U \subset X$ ,  $f^{-1}(U) \cong \mathrm{Spec} \mathcal{A}(U)$  over  $U$ . But this is true after base change to  $U_i \cap U$  for any  $i$ , so it is true, by (7.1.5.4).

To show the adjointness condition, by (7.1.5.4), it suffices to show canonical (thus compatible) isomorphism for  $S$  affine. In this case, this reduces to (7.2.7.5).  $\square$

**Cor. (7.2.7.13).** Let  $S$  be a scheme and  $\mathcal{A}$  is a Qco sheaf of  $\mathcal{O}_S$ -algebras, then

- For any morphism  $g : S' \rightarrow S$ ,  $S' \times_S \mathbf{Spec}_S(\mathcal{A}) \cong \mathbf{Spec}_{S'}(g^* \mathcal{A})$ .
- The natural map  $\mathcal{A} \rightarrow \pi_* \mathcal{O}_{\mathbf{Spec}_S(\mathcal{A})}$  is an isomorphism of  $\mathcal{O}_S$ -algebras.

$\lrcorner$

*Proof:* 1: It can be checked that  $S' \times_S \mathbf{Spec}_S(\mathcal{A})$  and  $\mathbf{Spec}_{S'}(g^* \mathcal{A})$  satisfy the same universal property.

2: It suffices to check on affine opens, then it is trivial.  $\square$

**Lemma (7.2.7.14) [Affine Case].** Fiber product of affine schemes is also affine that corresponds to the tensor product of their corresponding rings, by (7.2.7.5).  $\lrcorner$

**Prop. (7.2.7.15) [Finite Limits of Schemes].** Sch admits fiber products, and there is a final object  $\mathrm{Spec} \mathbb{Z}$ , so arbitrary limits exist in the category of schemes (4.1.1.48).

In fact, finite limits exist in the category of ringed spaces, and finite limits of schemes coincide with finite limits as ringed spaces.  $\lrcorner$

*Proof:* Let  $f : X \rightarrow S, g : Y \rightarrow S$ , and  $U_i$  is an affine open covering of  $S$ ,  $V_{ij}$  is an affine open covering of  $f^{-1}(U_i)$  and  $W_{ik}$  is an affine open covering of  $g^{-1}(U_i)$ , then we can check  $h_{V_{ij}} \times_{h_{U_i}} h_{W_{ik}}$  is a covering of  $h_X \times_{h_S} h_Y$  by representable open subfunctors (10.9.1.4), by (7.2.7.14), thus it is representable.  $\square$

**Cor. (7.2.7.16) [Open Subschemes].** Let  $X \rightarrow S, Y \rightarrow S$ , and  $V \subset X, W \subset Y$  be open subschemes mapping into open subscheme  $U \subset S$ , then there is a natural open immersion  $V \times_U W \rightarrow X \times_S Y$  with image  $\pi_1^{-1}V \cap \pi_2^{-1}(W)$ .  $\lrcorner$

*Proof:* There is a natural map  $V \times_U W \rightarrow X \times_S Y$  by Yoneda lemma, and this map has the universal property that any map  $f : (T, \mathcal{O}_T) \rightarrow X \times_S Y$  that  $\pi_1 \circ f$  has image in  $V$  and  $\pi_2 \circ f$  has image in  $W$  factors uniquely through  $V \times_U W$ . But so does the open immersion  $\pi_1^{-1}V \cap \pi_2^{-1}(W) \rightarrow X \times_S Y$  (7.2.6.14), so they are equal.  $\square$

**Cor. (7.2.7.17).** Let  $f : X \rightarrow S, g : Y \rightarrow S$ , and  $U_i$  is an affine open covering of  $S$ ,  $V_{ij}$  is an affine open covering of  $f^{-1}(U_i)$  and  $W_{ik}$  is an affine open covering of  $g^{-1}(U_i)$ , then

$$X \times_S Y = \cup_i \cup_{j,k} V_{ij} \times_{U_i} W_{ik}$$

is an affine open covering of  $X \times_S Y$ .

Also, the structure sheaf of  $X \times_S Y$  is given by  $\mathcal{O}_{X \times_S Y} = \pi_1^{-1} \mathcal{O}_X \otimes_{\pi^{-1} \mathcal{O}_S} \pi_2^{-1} \mathcal{O}_Y$ .  $\lrcorner$

**Cor. (7.2.7.18).** The equalizer of two morphisms from  $X$  to  $Y$  exists, it is a locally closed subscheme of  $X$ , and it is a closed subscheme of  $X$  if  $Y$  is separated.  $\lrcorner$

*Proof:* because it is the base change of  $\Delta : Y \rightarrow Y \times Y$  (4.1.1.48), then use (7.4.5.27).  $\square$

**Remark (7.2.7.19) [Infinite Product of Schemes Doesn't Exist].** WARNING: infinite products of schemes may not exist, Cf. [Sta]0CNH. Intuitively, if you want to glue affine products together, you will notice you can identify only those products that are equal a.e..  $\lrcorner$

**Remark (7.2.7.20).** If  $S \in \text{Sch}$  and  $X, S' \in \text{Sch}/S$ , we sometimes denote  $X \times_S S'$  by  $X_{S'}$ , if no confusion is caused.  $\lrcorner$

**Def. (7.2.7.21) [Generic Fibers and Special Fibers].** If  $X$  is a scheme over an integral ring  $A$ , then the **generic fiber** of  $X/A$  is the stalk  $X_\eta$ , where  $\eta = (0) \in \text{Spec } A$ .

If  $S \in \text{Sch}$  and  $X \in \text{Sch}/S$ , a **special fiber** of  $X/S$  is the stalk of  $X$  over a closed point  $s \in |S|_0$ .  $\lrcorner$

## 8 Rational Maps

**Def. (7.2.8.1) [Rational Maps].** Let  $X, Y \in \text{Sch}/S$  s.t.  $Y/S$  is separated, a **rational map** over  $S$   $f : X \rightarrow Y$  is an equivalence class of maps  $U \rightarrow Y$  over  $S$  where  $U$  is an open dense subset of  $X$ . A **rational function** on  $X$  is a rational map  $X \rightarrow \mathbb{A}^1$ . It has a ring structure. The ring of rational functions is denoted by  $R(X)$ .

Because two rational maps are equivalent iff they are compatible on the intersection of their domain as  $Y/S$  is separated, a rational map  $\varphi : X \rightarrow Y$  has a maximal domain of definition, denoted by  $\text{dom}(\varphi)$ .  $\lrcorner$

**Prop. (7.2.8.2).** If  $X$  is a scheme with f.m. generic points  $\eta_i$ , then

$$R(X) = \prod \mathcal{O}_{X, \eta_i}.$$

$\lrcorner$

*Proof:* Cf. [Sta]01RV.  $\square$

**Def. (7.2.8.3) [Birational Maps].**  $X, Y \in \text{Sch}/S$  are called **birational** over  $S$  if they are isomorphic in the category of  $S$ -schemes with dominant rational maps.  $\lrcorner$

**Prop. (7.2.8.4).** Two schemes  $X, Y$  over  $S$  are birational over  $S$  iff there are nonempty open subschemes  $U \subset X, V \subset Y$  that are isomorphic over  $S$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (7.2.8.5) [ $S$ -Dense Subsets].** Let  $X \in \text{Sch}/S$ , a subscheme  $U \subset X$  is called  **$S$ -dense** if  $U_s$  is dense in  $X_s$  for any  $s \in S$ .  $\lrcorner$

**Prop. (7.2.8.6).** Let  $X \in \text{Sch}^{\text{pf}}/S$  and  $V \subset X$  is a qc open subscheme, then the set of points  $s \in S$  s.t.  $V_s \subset X_s$  is not dense is locally constructible in  $S$ . And if  $V$  is  $S$ -dense in  $X$ , then it is also schematically dense in  $X$ .  $\lrcorner$

*Proof:* Cf.[?BLR]P56.  $\square$

**Prop. (7.2.8.7).** If  $X \in \text{Sch}_{\text{qc}}/S$  and  $U \subset X$  is an  $S$ -dense open subscheme, then  $U$  contains an  $S$ -dense open subscheme of  $S$  that is qc.  $\lrcorner$

*Proof:* Cf.[?BLR]P56.  $\square$

**Def. (7.2.8.8) [ $S$ -Rational Maps].** Let  $X, Y \in \text{Sch}^{\text{sep,loc.pf}}/S$  an  $S$ -**rational map**  $f : X \rightarrow Y$  is an equivalence class of maps  $U \rightarrow Y$  over  $S$  where  $U$  is an open subset of  $X$  that is  $S$ -dense.

Because two rational maps are equivalent iff they are compatible on the intersection of their domain as  $Y/S$  is separated, a rational map  $X \rightarrow Y$  has a maximal domain of definition.  $\lrcorner$

**Def. (7.2.8.9) [ $S$ -Birational Maps].** Let  $X, Y \in \text{Sch}^{\text{sep,loc.pf}}/S$  are called  $S$ -**birational** if they are isomorphic in the category of  $S$ -schemes with dominant  $S$ -rational maps.  $\lrcorner$

**Prop. (7.2.8.10).** The notion of  $S$ -rational and  $S$ -birational maps are stable under base change, because a base change of fields is flat locally of f.p. thus open.  $\lrcorner$

**Prop. (7.2.8.11)[Faithfully Flat Descent].** Let  $X', X, Y \in \text{Sch}^{\text{sep,loc.pf}}/S$  and  $\varphi : Y \rightarrow S$  an  $S$ -rational map,

- If  $f : X' \rightarrow X$  flat, then  $\varphi \circ f$  is an  $S$ -rational map  $X' \rightarrow Y$ , and  $\text{dom}(\varphi \circ f) = f^{-1}(\text{dom}(\varphi))$ . In particular, if  $f$  is f.f. and  $\varphi \circ f$  is a morphism, then  $\varphi$  is also a morphism.
- If  $T \rightarrow S$  is flat, then the base change  $\varphi_T : X_T \rightarrow Y_T$  satisfies  $\text{dom}(\varphi_T) = \text{dom}(\varphi) \times_S T$ .  $\lrcorner$

*Proof:* Cf.[BLR90]P58.  $\square$

## 9 Associated Points

References are [Sta]Chap30.

**Def. (7.2.9.1) [Associated Points].** For  $X \in \text{Sch}$  and  $\mathcal{F} \in \mathcal{QCoh}(X)$ , a point  $x \in X$  is said to be **associated to  $\mathcal{F}$**  iff  $\mathfrak{m}_x$  is associated to  $\mathcal{F}_x$ , which is equivalent to  $\mathfrak{m}_x$  are all zero-divisors in  $M$  by (5.2.5.18). When  $\mathcal{F} = \mathcal{O}_X$ ,  $x$  is called an **associated point of  $X$** .  $\lrcorner$

**Prop. (7.2.9.2).** If  $X \in \text{NSch}$  and  $\mathcal{F} \in \mathcal{QCoh}(X)$ , then for any  $x \in X$ ,  $x$  is associated to  $\mathcal{F}$  iff  $\mathfrak{m}_x$  is an associated prime of  $\Gamma(X, \mathcal{O}_X)$  of  $\Gamma(U, \mathcal{F})$  for an affine nbhd  $U$  of  $x \in X$ .  $\lrcorner$

*Proof:* Cf.[Sta]02OK.  $\square$

**Prop. (7.2.9.3).** Same results of associated points are parallel to the discussion of associated primes:

- relations of  $\text{Asso}(\mathcal{F})$  w.r.t exact sequences (5.2.5.14).
- $\text{Asso}(\mathcal{F}) \subset \text{Supp}(\mathcal{F})$  (5.2.5.16).
- When  $X \in \text{NSch}$  and  $\mathcal{F}$  is coherent, for a quasi-compact open set  $U$  of  $X$ , the number of associated points in  $U$  is finite (5.2.5.16).
- When  $X \in \text{NSch}$ ,  $\mathcal{F} = 0$  iff  $\text{Asso}(\mathcal{F})$  is empty (5.2.5.16).

- When  $X \in \mathbf{NSch}$ , If  $\text{Asso}(\mathcal{F}) \subset$  an open subset  $U$ , then  $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F})$  is injective (5.2.5.22).
- If  $X \in \mathbf{NSch}$ , then the minimal elements (under specialization) of  $\text{Supp}(\mathcal{F})$  are associated points of  $\mathcal{F}$ . in particular, any generic point of an irreducible component of  $X$  is an associated points of  $X$ .
- If  $X \in \mathbf{NSch}$ , then if a map  $\varphi : \mathcal{F} \rightarrow \mathcal{G} \in \mathcal{QCoh}(X)$  that is injective at all the stalks of  $\text{Asso}(\mathcal{F})$ , then  $\varphi$  is injective.

┘

## 10 Frobenius-Morphisms

**Def. (7.2.10.1) [Frobenius-Morphisms].** Let  $k_0 \in \mathbf{Field}^{\text{fin}}$ ,  $k = \overline{k_0}$ ,  $X_0 \in \mathbf{Sch}/k_0$ ,  $X = X_0 \otimes_{k_0} k$ ,

- The **absolute Frobenius-morphism** for  $X$  (resp.  $X_0$ ) is the automorphism  $\varphi_X : X \rightarrow X$  (resp.  $\varphi_{X_0} : X_0 \rightarrow X_0$ ) that is the  $\#k$ -th power on  $\mathcal{O}_X$  (resp.  $\mathcal{O}_{X_0}$ ).
- $F_X = \text{id}_{X_0} \times_{k_0} \varphi_{k_0}^{-1} : X \rightarrow X \in \mathbf{Sch}$ , called the **arithmetic Frobenius-morphism**, which is not  $k$ -linear!
- $\text{Fr}_X = \varphi_{X_0} \times_{k_0} \text{id}_k : X \rightarrow X \in \mathbf{Sch}/k$ , which is  $k$ -linear, called the **geometric Frobenius-morphism**. When  $X_0$  is affine,  $\text{Fr}_X(k) : X(k) \rightarrow X(k)$  is given by  $(a_1, \dots, a_n) \mapsto (a_1^{\#k_0}, \dots, a_n^{\#k_0})$ . Geometric Frobenius-morphisms are usually preferred because it is  $k$ -linear.
- If  $U_0 \in \mathbf{Sch}/X_0$ , then the **relative Frobenius-morphism**  $F_{U/X_0} : U \rightarrow \varphi_{X_0}^*(U) \in \mathbf{Sch}/X_0$  is defined by the universal property of the base change of  $U$  by  $\varphi_{X_0}^r$ .

┘

**Prop. (7.2.10.2).** Situation as in (7.2.10.1),  $\text{Fr}_X = \varphi_{r,X} \circ F_X : X \rightarrow X$ .

┘

*Proof:* Easy.

□

**Prop. (7.2.10.3) [Relative Frobenius are Universal Homeomorphisms].**  $F_{U/X_0}$  is a universal homeomorphism. In particular, if  $U \rightarrow X_0$  is étale, then it is an isomorphism.

This also holds for  $U \rightarrow X_0 \in \mathbf{AlgSp}/k_0$ , Cf. [Zhu17]P465.

┘

*Proof:* Cf. [Sta]0EBS?.

□

**Prop. (7.2.10.4).** If  $k \in p\text{-Field}$  and  $X \in \mathbf{Sch}/k$ , then  $X^{(p)}$  is reduced iff  $X$  is geo.reduced. This follows from (7.4.3.2).

┘

## 7.3 Cohomology on Ringed Sites

*“The question you raise, ‘How can such a formulation lead to computations?’ doesn’t bother me in the least! Throughout my whole life as a mathematician, the possibility of making explicit, elegant computations has always come out by itself, as a byproduct of a thorough conceptual understanding of what was going on. Thus I never bothered about whether what would come out would be suitable for this or that, but just tried to understand - and it always turned out that understanding was all that mattered.”*

–A. Grothendieck

References are [Sta], [Har77] and [Sheaf Cohomology, Anonymous]. Should be refreshed with the language of  $\infty$ -categories?

**Notation (7.3.0.1).**

- We use notations defined in Sites, Sheaves, Topoi and Stacks.

┘

### 1 Derived Cohomology

$D(\text{Mod}(\mathcal{O}))$

**Def. (7.3.1.1) [Setups].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, write  $K(\mathcal{O}) = K(\text{Mod}(\mathcal{O}))$ ,  $D(\mathcal{O}) = D(\text{Mod}(\mathcal{O}))$ . The Abelian category  $\text{Mod}(\mathcal{O})$  contains enough injectives by (4.8.3.29) and (4.8.3.23), so we can consider right derived functor for any left exact functor.

1. (Hom) Let  $K$  be a presheaf of sets on  $\mathcal{C}$ , then  $\mathcal{F} \mapsto \text{Hom}_{\mathcal{P}\text{Sh}(\mathcal{C})}(K, \mathcal{F})$  is a left exact functor  $\text{Mod}(\mathcal{O}) \rightarrow \text{Ab}$ , thus we denote its derived functors as  $H^i(K, \mathcal{F})$ .
2. (Section) The **section functor**  $\Gamma(U, \mathcal{F})$  is the left exact functor  $\text{Mod}(\mathcal{O}) \rightarrow \text{Mod}(\mathcal{O}(U))$  and call the derived functors  $H^i(U, \mathcal{F}) = H^i(R\Gamma(U, \mathcal{F}))$  as the  **$i$ -th cohomology** of  $\mathcal{F}$  at  $U$ . In fact, this functor is just  $\text{Mor}_{\mathcal{P}\text{Sh}(\mathcal{C})}(h_U, \mathcal{F})$  defined in (7.3.1.1).
3. (Global Section) Let  $e$  be the final object in  $\mathcal{P}\text{Sh}(\mathcal{C})$ , then we define the **global section functor**  $\Gamma(\mathcal{C}, -)$  to be the left exact functor  $\text{Mod}(\mathcal{O}) \rightarrow \text{Ab} : \mathcal{F} \mapsto \text{Mor}_{\mathcal{P}\text{Sh}(\mathcal{C})}(e, \mathcal{F}) = \varprojlim_{X \in \mathcal{C}^{\text{op}}} \Gamma(X, \mathcal{F})$ , then we define its derived functor  $R(\mathcal{C}, \mathcal{F})$ , and call the derived functors  $H^i(\mathcal{C}, \mathcal{F}) = H^i(R\Gamma(\mathcal{C}, \mathcal{F}))$  the  **$i$ -th cohomology group** of  $\mathcal{F}$  on  $\mathcal{C}$ .
4. (Pushforward) Let  $(\text{Sh}(\mathcal{C}), \mathcal{O}) \rightarrow (\text{Sh}(\mathcal{D}), \mathcal{O}')$  be a morphism of topoi, then  $f_*$  is a left exact functor from  $\text{Mod}(\mathcal{O})$  to  $\text{Mod}(\mathcal{O}')$  (7.1.2.18), and we call its derived functors  $R^i f_*$  the  **$i$ -th higher direct images**.
5. (Shification) Let shification functor  $\iota : \mathcal{P}\text{Sh}(\mathcal{C}) \rightarrow \text{Sh}(\mathcal{C})$  is left exact, and we call the derived functors  $\mathcal{H}^p(F)$  the **sheaf-cohomology presheaves** of  $F$ .

┘

**Def. (7.3.1.2)  $[\mathbb{G}_a, \mathbb{G}_m]$ .** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, denote  $\mathbb{G}_{a,\mathcal{C}} = \mathcal{O}$ ,  $\mathbb{G}_{m,\mathcal{C}} = \mathcal{O}^*$ .

┘

**Prop. (7.3.1.3) [Global Section as Pushforward].** For a ringed site  $(\mathcal{C}, \mathcal{O})$ , if we endow  $\mathcal{C}$  with the discrete topology, then there is a functor  $\mathcal{C} \rightarrow \text{pt}$  which is cocontinuous, thus induce a morphism of ringed topoi  $\pi : (\text{Sh}(\mathcal{C}), \mathcal{O}) \rightarrow (\text{Sh}(\text{pt}) \cong \text{Ab}, \Gamma(\mathcal{C}, \mathcal{O}))$  (7.1.2.21). Then  $\pi_*$  is exactly  $\mathcal{F} \mapsto \Gamma(\mathcal{C}, \mathcal{F})$ , so  $R\pi_* \mathcal{F} = H^i(\mathcal{C}, \mathcal{F})$ .

┘

**Prop. (7.3.1.4) [Change of Topologies].** Let  $\mathcal{C}, \mathcal{C}'$  be sites and  $i : \mathcal{C}' \rightarrow \mathcal{C}$  be a fully subcategory of  $\mathcal{C}$ ,  $i : \mathcal{C}' \rightarrow \mathcal{C}$  is continuous and cocontinuous satisfying the hypothesis of (7.1.2.23), then for  $\mathcal{F} \in \text{Sh}(\mathcal{C}'), \mathcal{G} \in \text{Sh}(\mathcal{C})$ , there are functorial isomorphisms

$$H^p(\mathcal{C}', U, i^{-1}\mathcal{G}) \cong H^p(\mathcal{C}, U, \mathcal{G}).$$

And if  $i$  satisfies the hypothesis of (7.1.2.25), then moreover there are functorial isomorphisms

$$H^p(\mathcal{C}', U, F') \cong H^p(\mathcal{C}; U, i_*F') = H^p(\mathcal{C}; i(U), i_!F')$$

┘

*Proof:* For the first assertion, notice  $g^{-1}$  preserves injectives because it is right adjoint to  $g_! = f^{-1}$  is exact (7.1.2.23).

The second assertion follows immediate from (7.1.2.25). □

### Calculations

**Prop. (7.3.1.5) [Locality of Cohomologies].** Let  $(\mathcal{C}, \mathcal{O})$  be a ring site and  $U \in \mathcal{C}$ , then for  $\mathcal{F} \in \text{Mod}(\mathcal{O}_X)$ ,  $H^n(U, \mathcal{F}) = H^n(\mathcal{C}/U, \mathcal{F}|_U)$ , by (7.3.4.2) and the composition of derived functors applied to the pullback  $j^* : \mathcal{C}/U \rightarrow \mathcal{C}$ . ┘

**Prop. (7.3.1.6) [Sheaf-Cohomological Presheaves].** The forgetful functor is right adjoint to the exact shification functor, the composition of derived functors applied to the functor  $\Gamma(U, -) \circ \iota$  from  $\mathcal{P}\text{Sh}(\mathcal{C})$  to  $\mathcal{A}\text{b}$  shows its right derived functor is

$$\mathcal{H}^p(F) = R^p\iota(F) : U \rightarrow H^p(U, F).$$

┘

**Prop. (7.3.1.7) [Higher Direct Images].** For  $f : (\mathcal{C}', \mathcal{O}') \rightarrow (\mathcal{C}, \mathcal{O})$  a morphism of ringed topoi and  $\mathcal{F} \in \text{Sh}(\mathcal{C})$ , the composition of derived functors applied to the functor  $(\sharp \circ (f_*)^{\mathcal{P}\text{Sh}}) \circ \iota$  (because  $\sharp, (f_*)^{\mathcal{P}\text{Sh}}$  are exact (7.1.2.10)) shows that  $R^p f_* \mathcal{F} = (f_* \mathcal{H}^p(\mathcal{F}))^\sharp$ . So flasque sheaves thus flabby sheaves are right acyclic for  $f_*$ .

In particular,  $R^p f_* \mathcal{F}$  can be calculated locally on the base. ┘

**Prop. (7.3.1.8) [Relative Leray's Spectral Sequence].** Let

$$(f, f^\sharp) : (\text{Sh}(\mathcal{C}), \mathcal{O}) \rightarrow (\text{Sh}(\mathcal{C}'), \mathcal{O}'), \quad (g, g^\sharp) : (\text{Sh}(\mathcal{C}'), \mathcal{O}') \rightarrow (\text{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$$

be morphisms of ringed topoi, then the natural transformation  $R(g \circ f) \rightarrow Rg_* \circ Rf_*$  is an isomorphism, and for any  $\mathcal{F}^\bullet \in K^+(\mathcal{O})$ , there is a spectral sequence convergence

$$E_2^{p,q} = R^p g_* R^q f_* (\mathcal{F}^\bullet) \implies E^n = R^n (g \circ f)_* \mathcal{F}^\bullet.$$

┘

*Proof:* This is just the Grothendieck spectral sequence (4.10.7.11), where the condition is satisfied by (7.3.1.7) and (7.3.4.12). □



**Cor. (7.3.1.9) [Leray's Spectral Sequence].** Let  $(f, f^\sharp) : (\mathcal{S}h(\mathcal{C}), \mathcal{O}) \rightarrow (\mathcal{S}h(\mathcal{C}'), \mathcal{O}')$  be a morphism of ringed topoi, then for any  $\mathcal{F}^\bullet \in K^+(\mathcal{O})$ , then  $R\Gamma(\mathcal{C}, -) \rightarrow R\Gamma(\mathcal{C}', -) \circ Rf_*$  is an isomorphism, and there is a spectral sequence convergence

$$E_2^{p,q} = H^p(\mathcal{C}', R^q f_*(\mathcal{F}^\bullet)) \implies E^n = H^n(\mathcal{C}, \mathcal{F}^\bullet).$$

┘

**Prop. (7.3.1.10) [Relative Mayer-Vietoris Sequence].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{U} : \{U \rightarrow X, V \rightarrow X\}$  is a covering s.t.  $U \rightarrow X$  is a monomorphism, then for any  $\mathcal{F} \in \mathcal{S}h(\mathcal{O})$  and any morphism of ringed sites  $f : (\mathcal{C}, \mathcal{O}) \rightarrow (\mathcal{C}', \mathcal{O}')$ , there is a long exact sequence of sheaves in  $\mathcal{S}h(\mathcal{O}')$ :

$$0 \rightarrow (f|_X)_*(\mathcal{F}|_X) \rightarrow (f|_U)_*(\mathcal{F}|_U) \oplus (f|_V)_*(\mathcal{F}|_V) \rightarrow (f|_{U \cap V})_*(\mathcal{F}|_{U \cap V}) \rightarrow R^1(f|_X)_*(\mathcal{F}|_X) \rightarrow \dots$$

that is functorial in  $\mathcal{F}$ .

┘

*Proof:* Choose a functorial injection resolution  $\mathcal{F} \rightarrow \mathcal{I}^\bullet$  (4.10.2.9), then there is an exact sequence of complexes

$$0 \rightarrow (f|_X)_*(\mathcal{I}^\bullet|_X) \rightarrow (f|_U)_*(\mathcal{I}^\bullet|_U) \oplus (f|_V)_*(\mathcal{I}^\bullet|_V) \rightarrow (f|_{U \cap V})_*(\mathcal{I}^\bullet|_{U \cap V}) \rightarrow 0$$

because injective sheaves are flabby (7.3.4.9) and  $(U \times_X V) \times_X f^{-1}W \rightarrow V \times_X f^{-1}W$  is a monomorphism for any  $W \in \mathcal{C}'$ . Then we take the long exact sequence of cohomology.  $\square$

**Cor. (7.3.1.11) [Mayer-Vietoris Sequence].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{U} : \{U \rightarrow X, V \rightarrow X\}$  is a covering s.t.  $U \rightarrow X$  is a monomorphism, then for any  $\mathcal{F} \in \mathcal{S}h(\mathcal{O})$ , there is a long exact sequence

$$0 \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(U, \mathcal{F}) \oplus H^0(V, \mathcal{F}) \rightarrow H^0(U \times_X V, \mathcal{F}) \rightarrow H^1(X, \mathcal{F}) \rightarrow \dots$$

that is functorial in  $\mathcal{F}$ .

┘

**Prop. (7.3.1.12) [Compatibility with Algebraic Structures].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{F} \in \mathcal{S}h(\mathcal{C})$ ,  $K \in \mathcal{P}Sh(\mathcal{C})$ , then  $H^*(K, \mathcal{F})$  is the same calculated as  $\mathcal{O}$ -modules or Abelian sheaves.  $\square$

*Proof:* Denote  $(C, \mathbb{Z})$  the trivial ringed site and  $ab : (C, \mathcal{O}) \rightarrow (C, \mathbb{Z})$  the forgetful functor, then  $ab$  is exact and  $\text{Hom}(K, \mathcal{F}_{ab}) = \text{Hom}(K, \mathcal{F})$ , thus we can use the Leray spectral sequence (7.3.1.9).  $\square$

**Prop. (7.3.1.13) [Direct Products].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{F}_i$  be a family of sites indexed over a set  $I$ , then for any presheaf of sets  $K$  on  $\mathcal{C}$ , there is a Leray spectral sequence convergence (7.3.1.8)

$$H^p(K, R^q(\prod_i \mathcal{F}_i)) \rightarrow R^n(\prod_i \text{Hom}(K, \cdot))(\mathcal{F}_i) \cong \prod_i H^n(K, \mathcal{F}_i).$$

In particular,  $H^1(K, \prod_i \mathcal{F}_i) \rightarrow \prod_i H^1(K, \mathcal{F}_i)$  is injective.

┘

**Prop. (7.3.1.14) [Filtered Colimits].**  $H^n(U, -)$  commutes with filtered colimits if  $T$  is a Noetherian topology.  $\square$

*Proof:*  $n = 0$  case follows from the fact the colimit presheaf is already a sheaf, because for any finite cover, the Čech complex of the limit sheaf is the filtered colimit of Čech complexes, and filtered colimit is exact.

And a filtered colimits of injective sheaves is flask, because flask need only be checked for finite coverings at this case (because of the fact  $T$  and  $T^f$  have equivalent category of sheaves (7.1.1.4) and definition of flask (7.3.4.8)), and a filtered colimit of exact Čech complexes is exact. So we can choose a functorial injective resolution (4.10.2.9) and use the colimit resolution to calculate the cohomology.  $\square$



**Lemma (7.3.1.15).** If  $X$  is a qs ringed space, then  $\mathrm{Sh}(X) \rightarrow \mathrm{Sh}(X_{fp})$  is an equivalence by  $i_s$  and  $i^s$ , where  $fp$  is the subtopology of  $X$  generated by qcqs open subsets.  $\lrcorner$

*Proof:* The proof is the same proof as that of (7.1.4.21).  $\square$

**Prop. (7.3.1.16) [Filtered Colimits].**  $H^i(U, -)$  commutes with direct limits if  $X$  is a qs ringed space and  $U \subset X$  is qc.  $\lrcorner$

*Proof:* This follows from (7.3.1.15) the same way (9.4.1.8) follows from (7.1.4.21).  $\square$

### Low Dimensions

Cf. [Sta]Chap21.5-7.

**Prop. (7.3.1.17) [ $H^1$  and Picard Group].** Let  $(\mathcal{C}, \mathcal{O})$  be a local ringed site, then there is a canonical isomorphism of Abelian groups

$$H^1(\mathcal{C}, \mathcal{O}^*) = \mathrm{Pic}(\mathcal{O}).$$

$\lrcorner$

*Proof:* Let  $\mathcal{L}$  be an invertible sheaf, then there exists a subsheaf

$$\mathcal{L}^* : U \mapsto \{s \in \mathcal{L}(U) \mid s \cdot : \mathcal{O}(U) \rightarrow \mathcal{L}(U) \text{ is an isomorphism}\}.$$

Notice if  $f \in \mathcal{O}^*(U)$  and  $s \in \mathcal{L}^*(U)$ , then  $fs \in \mathcal{L}^*(U)$ , and any two  $s, s' \in \mathcal{L}^*(U)$  differ by an element of  $\mathcal{O}^*(U)$ , so  $\mathcal{L}^*$  is a pseudo- $\mathcal{O}^*$ -torsor. Moreover, as  $\mathcal{L}$  is locally free of rank 1 by (7.2.5.8), so  $\mathcal{L}^*(U)$  has sections locally, so it is an  $\mathcal{O}^*$ -torsor.

In this way, we get a map

$$\mathrm{Pic}(\mathcal{O}) \rightarrow \mathrm{Tor}(\mathcal{O}_X^*) \cong H^1(X, \mathcal{O}_X^*) \text{ (7.3.2.18)}.$$

This map is injective: if  $\mathcal{L}$  corresponds to a trivial torsor, then  $\mathcal{L}^*$  has a global section, and then  $\mathcal{L}$  is also trivial. This map is also surjective, because if  $\mathcal{F}$  is an  $\mathcal{O}_X^*$ -torsor, then we can define

$$\mathcal{L}_1 : U \mapsto [\mathcal{F}(U) \otimes \mathcal{O}(U)] / \mathcal{O}(U)^*,$$

where the action is given by  $f \cdot (s, g) = (fs, f^{-1}g)$ , and  $\mathcal{L}_1$  is an  $\mathcal{O}$ -module given the addition  $(s, g) + (s', g') = (s, g + \frac{s'}{s}g)$ , where  $\frac{s'}{s} \in \mathcal{O}(U)$  satisfies  $\frac{s'}{s} \cdot s = s'$ . Then the sheafification  $\mathcal{L}$  of  $\mathcal{L}_1$  is a locally trivial bundle that maps  $\mathcal{F}$ .  $\square$

**Prop. (7.3.1.18) [ $H^2$  and Objects of Gerbes].** Let  $\mathcal{C}$  be a site and  $\mathcal{S} \rightarrow \mathcal{C}$  be a gerbe whose automorphism sheaves are Abelian. Let  $\mathcal{G}$  be the sheaf defined in (7.1.3.27). If  $U$  is an object of  $\mathcal{C}$  that

- there exists a cofinal system of coverings  $\{U_i \rightarrow U\}$  that for any such covering,  $H^1(U_i, \mathcal{G}) = 0$ ,  $H^1(U_i \times_U U_j, \mathcal{G}) = 0$ ,
- $H^2(U, \mathcal{G}) = 0$ .

Then  $\mathcal{S}_U$  is non-empty.  $\lrcorner$

*Proof:* By hypothesis, there is a covering  $\{U_i \rightarrow U\}$  and  $x_i$  in  $\mathcal{S}$  lying over  $U_i$ . By item1, after refining the covering, we may assume  $H^1(U_i, \mathcal{G}) = 0$  and  $H^1(U_{ij}, \mathcal{G}) = 0$ . Consider the sheaf

$$\mathcal{F}_{ij} = \mathrm{Isom}(x_i|_{U_{ij}}, x_j|_{U_{ij}})$$

on  $\mathcal{C}/U_{ij}$ , then there is an action  $\mathcal{G}_{U_{ij}} \times \mathcal{F}_{ij} \rightarrow \mathcal{F}_{ij}$ . Then  $\mathcal{F}_{ij}$  is a pseudo  $\mathcal{G}|_{U_{ij}}$ -torsor and clearly a torsor because any two objects of a gerbe is locally isomorphic.

By (7.3.2.18), these torsors are trivial, thus having a global section. In other words, there are isomorphisms  $\varphi_{ij} : x_i|_{U_{ij}} \rightarrow x_j|_{U_{ij}}$ . To get an object  $x$  over  $U$ , it suffices to manage the choices of  $\varphi_{ij}$  to get a descent datum. For this, use the fact  $H^2(U, \mathcal{G}) = 0$  and  $\check{H}^2(\mathcal{U}, \mathcal{G}) \rightarrow H^2(U, \mathcal{G})$  is injective by Čech to Derived spectral sequence (7.3.2.13).  $\square$

### Others

**Prop. (7.3.1.19).** Let  $K' \rightarrow K$  be a map of presheaves of sets on  $\mathcal{C}$  whose shification is surjective. Set  $K'_p = K' \times_K \times \dots \times_K K'$ , then for any  $\mathcal{F} \in \text{Sh}(\mathcal{C})$ , there is a spectral sequence convergence

$$E_1^{p,q} = H^q(K'_p, \mathcal{F}) \Rightarrow H^{p+q}(K, \mathcal{F}).$$

┘

*Proof:* Since shification is exact,  $(K'_p)^\sharp = (K')^\sharp_p$ . Then we use (7.1.2.26) to change to a larger site  $\mathcal{C}'$  where the topoi are equivalent and  $K', K$  are objects in  $\mathcal{C}'$  and  $K' \rightarrow K$  is a covering, then we use the  $E_1$  page of the Čech to derived spectral sequence (7.3.2.13). Notice this need modification, the modification goes back to the proof of the Grothendieck spectral sequence, where we choose the natural Čech complex resolution in place of the CE resolution, because we have (7.3.2.3).  $\square$

**Cor. (7.3.1.20)[Čech-Alexander Resolution].** If  $\mathcal{C}$  is a site with the indiscrete topology,  $X$  a weakly final object (4.1.1.8) of  $\mathcal{C}$ , then for any Abelian sheaf  $\mathcal{F}$  on  $\mathcal{C}$ , the total cohomology  $R\Gamma(\mathcal{C}, \mathcal{F})$  is represented by the Čech complex

$$\mathcal{F}(X) \rightarrow \mathcal{F}(X \times X) \rightarrow \mathcal{F}(X \times X \times X) \rightarrow \dots$$

┘

*Proof:* By (7.3.4.10),  $H^q(X^p, \mathcal{F}) = 0$  for  $q > 0$ . The assumption says  $h_X \rightarrow *$  is surjective, thus the conclusion is a special case of (7.3.1.19).  $\square$

## 2 Čech Cohomology

**Def. (7.3.2.1)[Čech Complex and Čech Cohomology].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{U} : \{U_i \rightarrow U\}$  be a covering, we have a canonical complex of presheaves  $\mathbb{Z}_{\mathcal{U}, \bullet}$  defined to be

$$\dots \rightarrow \bigoplus_{i_0, i_1, i_2} j_! \mathbb{Z}_{U_{i_0 i_1 i_2}} \rightarrow \bigoplus_{i_0, i_1} j_! \mathbb{Z}_{U_{i_0 i_1}} \rightarrow \bigoplus_{i_0} j_! \mathbb{Z}_{U_{i_0}} \rightarrow 0.$$

And for any presheaf of  $\mathcal{O}_X$ -module  $\mathcal{F}$ , the complex

$$\text{Hom}_{Psh(\mathcal{O})}^\bullet(\mathbb{Z}_{\mathcal{U}, \bullet} \otimes \mathcal{O}, \mathcal{F}) \cong \text{Hom}_{PAb}^\bullet(\mathbb{Z}_{\mathcal{U}, \bullet}, \mathcal{F})$$

is called the **Čech complex**  $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$  of  $\mathcal{F}$ . The cohomology  $\check{H}^*(\mathcal{U}, \mathcal{F})$  of  $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$  is called the **Čech cohomology** of  $\mathcal{F}$  w.r.t  $\mathcal{U}$ .

$\mathbb{Z}_{\mathcal{U}, \bullet}$  is exact except in degree 0, where the homology is  $j_! \mathbb{Z}_U$ . This is because we have a homotopy: choose a fixed  $i_0$ , for a  $s \in \Gamma(U_{i_1 \dots i_n}, \mathcal{F})$ , we map it to  $(hs)_{ii_1 \dots i_n} = \delta_{i, i_0} s$  ?. In particular, an injective sheaf is Čech acyclic.  $\square$

**Lemma (7.3.2.2).** The Čech complexes  $\check{C}^\bullet(\mathcal{U}, -)$  induces a functor from  $\mathcal{PSh}(\mathcal{C})$  to  $K^+(\mathcal{A}b)$ , which is an exact functor.  $\lrcorner$

*Proof:* Because in each degree this functor is a sum of functors of the form  $\mathcal{F} \mapsto \mathcal{F}(U)$ , which are exact functors on  $\mathcal{PSh}(\mathcal{C})$ .  $\square$

**Prop. (7.3.2.3) [Čech Complex as Derived Functors].** Let  $\mathcal{C}$  be a site and  $\mathcal{U} : \{U_i \rightarrow U\}$  be a covering, then  $\check{H}^0(\mathcal{U}, -)$  is left exact, and for  $\mathcal{F} \in \mathcal{Sh}(\mathcal{C})$ , there are functorial quasi-isomorphisms

$$\check{C}^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow R\check{H}^0(\mathcal{U}, \mathcal{F})$$

which is functorial in  $\mathcal{F}$ .  $\lrcorner$

*Proof:* Choose a functorial injective resolution of presheaves  $\mathcal{I}^\bullet$  of  $\mathcal{F}$ , and consider the double complex  $\check{C}(\mathcal{U}, \mathcal{I}^\bullet)$ . There are maps of complexes

$$\check{C}(\mathcal{U}, \mathcal{F}) \rightarrow \text{Tot}(\check{C}(\mathcal{U}, \mathcal{I}^\bullet)), \quad \check{H}^0(\mathcal{U}, \mathcal{I}^\bullet) \rightarrow \text{Tot}(\check{C}(\mathcal{U}, \mathcal{I}^\bullet))$$

which are both quasi-isomorphism by an application of spectral sequence and the fact the columns and rows are exact in positive degrees: The columns are exact because of (7.3.2.2) and the rows are exact because  $\mathbb{Z}_{\mathcal{U}, \bullet}$  is exact in positive degrees (7.3.2.1) and  $\mathcal{I}^p$  are injective. Then we have the desired quasi-isomorphism, and it is functorial in  $\mathcal{F}$ .  $\square$

**Cor. (7.3.2.4) [Čech Cohomologies].** If we take filtered colimit for coverings,  $\mathcal{F} \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}) = H^0(\mathcal{U}, \mathcal{F})$  is a left exact functor from presheaves to sets, the derived complex is just  $\varinjlim_{\mathcal{U}} \check{C}^\bullet(\mathcal{U}, \mathcal{F})$ , and the derived functors are just  $\varinjlim_{\mathcal{U}} \check{H}^q(\mathcal{U}, \mathcal{F})$ .  $\lrcorner$

*Proof:* This is because we can take colimit of the conclusion of (7.3.2.3), because the colimit is filtered by (7.3.2.5) so exact, so the Čech complex also represents the derived complex.  $\square$

**Lemma (7.3.2.5).** The refinement morphism of Čech cohomologies of two coverings doesn't depends on the refinement map chosen.  $\lrcorner$

*Proof:* For two refinement map, there is a commutative diagram

$$\begin{array}{ccc} \prod F(U_i) & \xrightarrow{d^0} & \prod F(U_i \times_U U_j) \\ \downarrow f-g & \swarrow \Delta^1 & \\ \prod F(U'_j) & & \end{array}$$

so it induce the same map on the kernel.  $\square$

**Def. (7.3.2.6) [Alternating Čech Complexes].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{U} : \{U_i \rightarrow U\} \in \text{Cov}(\mathcal{C})$ ,  $\mathcal{F} \in \mathcal{Sh}(\mathcal{O})$ , then the **alternating Čech complex** of  $\mathcal{F}$  is defined to be the subcomplex

$$\check{C}_{\text{alt}}^\bullet(\mathcal{U}, \mathcal{F}) = \{s \in \check{C}^\bullet(\mathcal{U}, \mathcal{F}) \mid s_{i_0, \dots, i_p} = 0 \text{ if } i_m = i_n, m \neq n, s_{i_{\sigma(0)}, \dots, i_{\sigma(p)}} = \text{sgn}(\sigma) s_{i_0, \dots, i_p}\} \subset \check{C}^\bullet(\mathcal{U}, \mathcal{F}).$$

It is truly a subcomplex.  $\lrcorner$

**Prop. (7.3.2.7) [Alternating and Usual Complexes].** Let  $(\mathcal{X}, \mathcal{O}_X)$  be a ringed space,  $\mathcal{U} : \{U_i \rightarrow U\} \in \text{Cov}(\mathcal{C})$ ,  $\mathcal{F} \in \mathcal{Sh}(\mathcal{O}_X)$ , then the inclusion

$$\check{C}_{\text{alt}}^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F})$$

is a homotopy equivalence.  $\lrcorner$

*Proof:* Cf. [Sta]01FM. The proof is rather complicated. ?  $\square$

**Remark (7.3.2.8).** WARNING: This is not right for ringed sites, e.g. the étale sites, for example, cf. (9.4.1.25).  $\lrcorner$

### Comparison Theorems

**Prop. (7.3.2.9).** If two coverings are refinements of each other, then their Čech cohomology is isomorphic.  $\perp$

*Proof:* Because the refinement morphism doesn't depend on the refinement map (7.3.2.5).  $\square$

**Prop. (7.3.2.10) [Comparison Theorem for Čech Acyclicity].** If there are two coverings  $\mathfrak{U}, \mathfrak{V}$  and a presheaf  $\mathcal{F}$ , then we can construct a double Čech complex with the  $(p, q)$ -term being  $\mathcal{F}(U_{i_1, \dots, i_p} \cap V_{j_1, \dots, j_q})$ . Then the vertical and horizontal arrays calculate the Čech cohomology  $\prod_j H^*(\mathfrak{U}|_{V_{j_1, \dots, j_q}}, \mathcal{F})$ ,  $\prod_i H^*(\mathfrak{V}|_{U_{i_1, \dots, i_q}}, \mathcal{F})$  respectively.

So by Spectral sequence (4.10.7.8), if both higher Čech cohomology group  $H^k(\mathfrak{U}|_{V_{j_1, \dots, j_q}}, \mathcal{F})$ ,  $H^k(\mathfrak{V}|_{U_{i_1, \dots, i_q}}, \mathcal{F})$  vanish, i.e., they are both  $\mathcal{F}$ -acyclic, then  $H^*(\mathfrak{U}, \mathcal{F}) \cong H^*(\mathfrak{V}, \mathcal{F})$ .  $\perp$

**Cor. (7.3.2.11).** If  $\mathfrak{V}$  is a refinement of  $\mathfrak{U}$ , and  $\mathfrak{V}|_{U_{i_1, \dots, i_p}}$  are all  $\mathcal{F}$ -acyclic, then  $H^*(\mathfrak{U}, \mathcal{F}) \cong H^*(\mathfrak{V}, \mathcal{F})$ .  $\perp$

*Proof:* It suffices to prove  $\mathfrak{U}|_{V_{i_1, \dots, i_q}}$  is  $\mathcal{F}$ -acyclic. But  $\mathfrak{U}|_{V_{i_1, \dots, i_q}}$  and  $\text{id}_{V_{i_1, \dots, i_q}}$  are refinements of each other, so (7.3.2.9) settles the proof.  $\square$

**Cor. (7.3.2.12).** If  $\mathfrak{V}|_{U_{i_1, \dots, i_p}}$  is  $\mathcal{F}$ -acyclic, then the covering  $H^*(\mathfrak{U} \times \mathfrak{V}, \mathcal{F}) = H^*(\{U_i \cap V_j\}, \mathcal{F}) \cong H^*(\mathfrak{U}, \mathcal{F})$ .  $\perp$

*Proof:* Because  $\mathfrak{V}|_{U_{i_1, \dots, i_p}}$  and  $\mathfrak{U} \times \mathfrak{V}|_{U_{i_1, \dots, i_p}}$  are refinement of each other, so  $\mathfrak{U} \times \mathfrak{V}|_{U_{i_1, \dots, i_p}}$  are  $\mathcal{F}$ -acyclic by (7.3.2.9), and  $\mathfrak{U} \times \mathfrak{V}$  refines  $\mathfrak{U}$ , so (7.3.2.11) can be applied.  $\square$

**Prop. (7.3.2.13) [Čech to Derived].** For any  $\mathcal{F} \in \text{Sh}(\mathcal{C})$ , the Grothendieck spectral sequence applied to  $\Gamma(U, -) = H^0(\{U_i \rightarrow U\}, -) \circ \iota = \check{H}^0(U, -) \circ \iota$  gives us:

$$H^p(\{U_i \rightarrow U\}, \mathcal{H}^q(\mathcal{F})) \implies H^{p+q}(U, \mathcal{F}).$$

$$\check{H}^p(U, \mathcal{H}^q(\mathcal{F})) \implies H^{p+q}(U, \mathcal{F}).$$

$\perp$

**Cor. (7.3.2.14) [Higher Cohomologies Vanish Locally].** For any  $\mathcal{F} \in \text{Sh}(\mathcal{C})$ ,  $\mathcal{H}^p(\mathcal{F})^{++} = \mathcal{H}^p(\mathcal{F})^\# = 0$  for  $p > 0$ . In particular

$$\mathcal{H}^p(\mathcal{F})^+(U) = \check{H}^0(U, \mathcal{H}^p(\mathcal{F})) = 0 \quad p > 0.$$

because  $\mathcal{H}^p(\mathcal{F})^+$  is separated by (7.1.2.7). In particular, for any  $s \in H^p(U, \mathcal{F})$ ,  $p > 0$ , there exists a covering  $\mathcal{U}$  of  $U$  s.t.  $s|_{U_i} = 0$  for any  $i$ .

Thus the low degree of Čech-to-sheaf cohomology says (4.10.7.12):

$$0 \rightarrow \check{H}^1(U, \mathcal{F}) \rightarrow H^1(U, \mathcal{F}) \rightarrow 0 \rightarrow \check{H}^2(U, \mathcal{F}) \rightarrow H^2(U, \mathcal{F}) \rightarrow \check{H}^1(U, \mathcal{H}^1(\mathcal{F})) \rightarrow \check{H}^3(U, \mathcal{F}) \rightarrow H^3(U, \mathcal{F}).$$

$\perp$

*Proof:* By Grothendieck spectral sequence applied to forgetful functor and exact  $^\#$  functor, where the condition are satisfied by (7.3.4.3).  $\square$

**Cor. (7.3.2.15) [Acyclic Covering Calculates Derived Cohomologies].** If  $H^q(U_{i_0 i_1 \dots i_r}, \mathcal{F}) = 0$  for  $q \in \mathbb{Z}_+$ , then  $H^p(\{U_i \rightarrow U\}, \mathcal{F}) = H^p(U, \mathcal{F})$  as  $\mathcal{O}_X(U)$ -modules.  $\perp$

*Proof:* Because  $H^p(\{U_i \rightarrow U\}, \mathcal{H}^q(F))$  vanish for  $q > 0$ .  $\square$

**Prop. (7.3.2.16) [Čech Acyclic Čech Comparison].** If  $\mathcal{C}$  be a ringed site,  $\mathfrak{G} \subset \mathcal{C}$ ,  $\text{Cov} \subset \text{Cov}(\mathcal{C})$  and  $\mathcal{F} \in \text{Sh}(\mathcal{C})$  that

- For any  $\{U_i \rightarrow U\} \in \text{Cov}$ ,  $U_i, U \in \mathfrak{G}$ , and  $U_{i_0, \dots, i_p} \in \mathfrak{G}$ .
- $\text{Cov}|_U$  is cofinal in  $\text{Cov}(\mathcal{C}/U)$  for any  $U \in \mathfrak{G}$ .
- $\check{H}^q(U, \mathcal{F}) = 0$  for any  $U \in \mathfrak{G}$  and  $q > 0$ .

Then  $H^p(X, \mathcal{F}) = \check{H}^p(X, \mathcal{F}) = H^p(\{U_i \rightarrow X\}, \mathcal{F})$  for any  $\{U_i \rightarrow X\} \in \text{Cov}$ .  $\lrcorner$

*Proof:* By (7.3.2.15), we only have to show that  $H^q(U, \mathcal{F}) = 0$  for  $U \in \mathfrak{G}$  and  $q > 0$ . Use induction on  $q$ , use Čech to sheaf2:  $\check{H}^p(U, \mathcal{H}^q(\mathcal{F})) \Rightarrow H^{p+q}(U, \mathcal{F})$ . The case  $q \neq 0$  is by condition 2, 3, and induction hypothesis. For  $p = 0$ , use (7.3.2.14).  $\square$

### Non-Abelian Čech Cohomologies

**Def. (7.3.2.17) [Non-Abelian Cohomology].** Let  $\mathcal{C}$  be a site and  $\mathcal{G} \in \text{Sh}^{\text{grp}}(\mathcal{C})$ , for any covering  $\mathcal{U} = \{U_i \rightarrow U\} \in \text{Cov}(\mathcal{C})$ , we can define a **non-Abelian Čech cohomology**  $\check{H}^1(\mathcal{U}, \mathcal{G})$  as follows: Define  $Z^1(\mathcal{U}, \mathcal{G})$  = sets of families

$$\{(c_i) | c_i \in \Gamma(U_{ij}, \mathcal{G}) : c_{jk} c_{ik}^{-1} c_{ij} = 1\}.$$

If  $c \in Z^1(G, M)$ , then for any  $g \in \Gamma(U, \mathcal{G})$ ,

$$c_g = (c'_{ij} = g|_{U_i}^{-1} c_{ij} g|_{U_j})$$

is also in  $Z^1(\mathcal{U}, \mathcal{G})$ . This defines an equivalence relation on  $Z^1(\mathcal{U}, \mathcal{G})$ , the equivalence classes are called  $\check{H}^1(G, M)$ . This is compatible with the commutative case.

Taking filtered colimit over all coverings of  $U$ , we can also define  $\check{H}^1(U, \mathcal{G})$ .  $\lrcorner$

**Prop. (7.3.2.18) [ $H^1$  and Torsors].** Let  $\mathcal{C}$  be a site and  $\mathcal{H} \in \text{Sh}(\mathcal{C})$ , then there is a canonical isomorphism of  $\mathcal{H}$ -torsors (7.1.1.16) and  $H^1(\mathcal{C}, \mathcal{H})$ .  $\lrcorner$

*Proof:* Cf. [Sta]03AJ. Should have a non-commutative version.  $\square$

**Prop. (7.3.2.19) [Long Exact Sequence of Non-Abelian Čech Cohomologies].** Let  $\mathcal{C}$  be a site and  $1 \rightarrow \mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{C} \rightarrow 1$  is an exact sequence in  $\text{Sh}^{\text{grp}}(\mathcal{C})$ , then for any  $U \in \mathcal{C}$ , there is a long exact sequence of pointed sets

$$1 \rightarrow \Gamma(U, \mathcal{A}) \rightarrow \Gamma(U, \mathcal{B}) \rightarrow \Gamma(U, \mathcal{C}) \xrightarrow{\delta} \check{H}^1(U, \mathcal{A}) \rightarrow \check{H}^1(U, \mathcal{B}) \rightarrow \check{H}^1(U, \mathcal{C}) \xrightarrow{\Delta} \check{H}^2(U, \mathcal{A})$$

the last term is defined only when  $\mathcal{A}$  is in the center of  $\mathcal{B}$  and  $U$  satisfies: For any covering  $\mathcal{U}$  and a refinement  $\mathcal{W} \rightarrow \mathcal{U} \times \mathcal{U}$ , there exists a refinement  $\mathcal{U}' \rightarrow \mathcal{U}$  s.t.  $\mathcal{U}' \times \mathcal{U}'$  is a refinement of  $\mathcal{W}$ .

$\delta$  is defined as follows: for  $c \in \Gamma(\mathcal{C}, \mathcal{C})$ , by taking a covering  $\{U_i \rightarrow U\}$ , we may assume that  $c|_{U_i} = b_i \in \Gamma(U_i, \mathcal{B})$ , then  $a_{ij} = b_i^{-1} b_j$  is a cocycle, and a different choice differ by a cocycle, so this is well-defined.

$\Delta$  is defined as: for any covering  $\mathcal{U}$  of  $U$  and  $c = (c_{ij})$  a cocycle in  $\check{H}^1(\mathcal{U}, \mathcal{C})$ , by passing to a refinement, we may assume that  $c_{ij} \in \Gamma(U_{ij}, \mathcal{B})$ , then  $\delta(c)$  is represented by the 2-cocycle  $a_{ijk} = c_{jk} c_{ik}^{-1} c_{ij} \in \Gamma(U, \mathcal{A})$ .  $\lrcorner$

*Proof:* **??** The verification of well-definedness of  $\Delta$  is checked at [Serre Local Fields P124]. **?**

For the exactness at  $C^G$ , the definition of  $\delta$  shows that  $\delta(c) = 1$  iff there is an inverse image  $b$  that  $b^{-1}\sigma(b) = 1$  for all  $\sigma$ .

For the exactness at  $H^1(G, A)$ ,  $a_\sigma = b^{-1}\sigma(b)$  if  $a_\sigma$  is in the image of  $\delta$ . Conversely, the image of  $b$  in  $C$  is in  $C^G$ , so it is in the image of  $\delta$ .

For the exactness at  $H^1(G, B)$ , one way is clear, and for the other, if  $\pi(b_\sigma) = c^{-1}\sigma(c)$ , then if  $t$  is an inverse image of  $c$ , then  $tb_\sigma\sigma(t)^{-1}$  is a cocycle in  $A$  cohomologous to  $b_\sigma$ .

For the exactness at  $H^1(G, C)$ , one way is clear, and if  $b_s$  is an inverse image of  $b_s$  and  $a_{\sigma,\tau} = b_\sigma\sigma(b_\tau)b_{\sigma\tau}^{-1}$  is a coboundary, then it is  $a_\sigma\sigma(a_\tau)a_{\sigma\tau}^{-1}$ , so we change  $b$  to  $a_\sigma^{-1}b_\sigma$ , as  $A$  is in the center of  $B$ , this lifts  $c$  to a cocycle in  $B$ .  $\square$

**Prop. (7.3.2.20) [Hilbert's Theorem 90].** For  $L/K \in \text{GalField}$ ,  $H^1(\text{Gal}(L/K), \text{GL}(n; L)) = 1$ , where  $L$  is equipped with the discrete topology.  $\lrcorner$

*Proof:* We prove any cocycle is a coboundary, for this, notice any cocycle factor through a finite quotient, and the images of it is contained in a finite extension of  $K$ , hence it reduce to the case of  $L/K$  finite.

By definition, this is equivalent to any  $B$ -semi-linear representation of  $G$  free of finite rank is trivial, which is by (24.1.1.15).  $\square$

**Cor. (7.3.2.21) [semi-linear representations].** The proposition implies that any semi-linear  $L$ -representation of  $\text{Gal}(L/K)$  is trivial.  $\lrcorner$

**Cor. (7.3.2.22).**  $H^1(\text{Gal}(L/K), \text{SL}_n(L)) = 1$ . This is seen from the exact sequence  $1 \rightarrow \text{SL}(n, L) \rightarrow \text{GL}(n, L) \rightarrow L^\times \rightarrow 1$ .  $\lrcorner$

### 3 Derived Homology

#### K-Flat Complexes

**Def. (7.3.3.1) [K-flat Complexes].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, a complex  $\mathcal{K}^\bullet$  of  $\mathcal{O}$ -modules is called  $K$ -flat if for any acyclic complex  $\mathcal{F}^\bullet$  of  $\mathcal{O}$ -modules, the total complex  $\text{Tot}^\oplus(\mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{K}^\bullet)$  is acyclic, or equivalently, tensoring with  $\mathcal{K}^\bullet$  maps quasi-iso to quasi-iso, by the long exact sequence and the fact tensoring is an exact functor of triangulated categories (4.8.10.4).  $\lrcorner$

**Prop. (7.3.3.2).** If  $K, K'$  are  $K$ -flat complexes of  $\mathcal{O}$ -modules,

- $\text{Tot}^\oplus(K \otimes_{\mathcal{O}} K')$  is  $K$ -flat.
- If  $(K_1, K_2, K_3)$  is a distinguished triangle in  $K(\mathcal{O})$ , if two of them is  $K$ -flat, then the third is also  $K$ -flat.
- Any bounded above complex of flat  $\mathcal{O}$ -modules is  $K$ -flat.
- Any filtered colimits of  $K$ -flat complexes are  $K$ -flat.

$\lrcorner$

*Proof:* 1: This follows from (4.8.10.2).

3: use (4.8.10.4), and the long exact sequence.

4: Cf. [Sta]06YQ.

5: because we are taking termwise-colimit, and Tot and tensor all commute with filtered colimits.

$\square$

**Prop. (7.3.3.3) [K-Flat Resolutions].** Any complex  $P^\bullet$  of  $\mathcal{O}$ -modules has a  $K$ -flat resolution  $K^\bullet \rightarrow P^\bullet$ , moreover, each term of  $K^\bullet$  is a flat  $\mathcal{O}$ -module and  $K^\bullet \rightarrow P^\bullet$  is termwise surjective.  $\lrcorner$

*Proof:* Cf. [Sta]06Y4.  $\square$

**Lemma (7.3.3.4).** Let  $P \rightarrow Q$  be a quasi-iso of  $K$ -flat complexes of  $\mathcal{O}$ -modules, then for any complex  $L$  of  $\mathcal{O}$ -modules,  $\text{Tot}(L \otimes P) \rightarrow \text{Tot}(L \otimes Q)$  is a quasi-isomorphism.  $\lrcorner$

*Proof:* Choose a  $K$ -flat resolution (7.3.3.3)  $K$  of  $L$ , then notice

$$\text{Tot}(L \otimes P) \cong \text{Tot}(K \otimes P) \cong \text{Tot}(K \otimes Q) \cong \text{Tot}(L \otimes Q)$$

by definition of  $K$ -flatness (7.3.3.1).  $\square$

**Prop. (7.3.3.5) [Pullback of K-Flat is K-Flat].** Let  $f : (\mathcal{S}h(\mathcal{C}'), \mathcal{O}_{\mathcal{C}'}) \rightarrow (\mathcal{S}h(\mathcal{C}), \mathcal{O}_{\mathcal{C}})$  be a morphism of ringed topoi, then  $f^*$  maps  $K$ -flat complexes to  $K$ -flat complexes.  $\lrcorner$

*Proof:* Cf. [Sta]0G7E.  $\square$

### Derived Tensor Product and Tor

**Def. (7.3.3.6) [Derived Tensor Product].** Let  $\mathcal{A} \rightarrow \mathcal{B}$  be a homomorphism of sheaves of rings over a site  $\mathcal{C}$ , then the functor  $\text{Tot}^\oplus(- \otimes_{\mathcal{O}} -)$  is a bi-exact bifunctor of triangulated categories  $K(\mathcal{A}) \times K(\mathcal{B}) \rightarrow D(\mathcal{B})$  by (4.8.10.4)(4.8.4.9), and the class of  $K$ -flat complexes in  $K(\mathcal{A})$  and  $K$ -flat complexes in  $K(\mathcal{B})$  satisfies the condition of (the dual of) ?? by (7.3.3.1) and (7.3.3.3), so we get a left derived functor

$$- \otimes_{\mathcal{A}}^L - : D(\mathcal{A}) \times D(\mathcal{B}) \rightarrow D(\mathcal{B}),$$

called the **derived tensor product**. And if any of  $M^\bullet, N^\bullet \in K(\mathcal{A})$  is  $K$ -flat, the natural map  $M^\bullet \otimes_{\mathcal{A}}^L N^\bullet \rightarrow \text{Tot}^\oplus(M^\bullet \otimes_{\mathcal{A}} N^\bullet)$  is an isomorphism in  $D(\mathcal{B})$ .  $\lrcorner$

**Prop. (7.3.3.7).** Let  $\mathcal{A} \rightarrow \mathcal{B}$  be a homomorphism of sheaves of rings over a site  $\mathcal{C}$ . Let  $N^\bullet$  be a  $\mathcal{B}$ -module, then for  $M^\bullet \in D(\mathcal{A})$ , there are functorial isomorphisms

$$M^\bullet \otimes_{\mathcal{A}}^L N^\bullet \cong (M^\bullet \otimes_{\mathcal{A}}^L \mathcal{B}) \otimes_{\mathcal{B}}^L N^\bullet$$

$\lrcorner$

*Proof:* Consider both sides as right derived functors of the multi-exact multifunctor. As this is true for  $\text{Tot}^\oplus$ , these follow from the universal property and (4.10.3.14), the conditions are satisfied by (7.3.3.3)(7.3.3.4)(7.3.3.5).  $\square$

**Cor. (7.3.3.8) [Commutative Monoidal Structure].** If  $\mathcal{A} = \mathcal{B} = \mathcal{C} = \mathcal{O}$  and  $K, L, M \in K(\mathcal{O})$ , there are natural isomorphisms

$$K \otimes^L L \cong L \otimes^L K, \quad (K \otimes^L L) \otimes^L M \cong K \otimes^L (L \otimes^L M).$$

Thus  $D^*(\mathcal{O})$  has a commutative monoidal structure.  $\lrcorner$

**Remark (7.3.3.9) [WARNING].** If  $M, N$  are two  $\mathcal{A}$ -modules, then we can define  $M_R \otimes_R^L N$  and  $M \otimes_R^L N_R$ , but there are no reason for them to be isomorphic.  $\lrcorner$

$$(M \otimes_A^L N) \otimes_B^L K = M \otimes_A^L (N \otimes_B^L K) = (M \otimes_A^L C) \otimes_C^L (N \otimes_B^L K)$$

and

$$(M \otimes_A^L N) \otimes_B^L K \cong (M \otimes_A^L K) \otimes_C^L (N \otimes_B^L C)$$

└

*Proof:* Consider both sides as right derived functors of the multi-exact multifunctor. As these are true for  $\text{Tot}^\oplus$ , these follow from the universal property and (4.10.3.14), the conditions are satisfied by (7.3.3.3)(7.3.3.4)(7.3.3.5).  $\square$

**Def. (7.3.3.11)[Tor Modules].** Let  $\mathcal{F}, \mathcal{G}$  be  $\mathcal{O}$ -modules, then the **Tor modules**  $\mathrm{Tor}_p^{\mathcal{O}}(\mathcal{F}, \mathcal{G})$  is defined to be  $H^{-p}(\mathcal{F} \otimes_{\mathcal{O}}^L \mathcal{G})$ .  $\square$

**Prop. (7.3.3.12)[Flatness and Tor].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, and  $\mathcal{F}$  is an  $\mathcal{O}$ -module. Then  $\mathcal{F}$  is a flat  $\mathcal{O}$ -modules iff  $\mathrm{Tor}_1^{\mathcal{O}}(\mathcal{F}, \mathcal{G}) = 0$  for any  $\mathcal{O}$ -module  $\mathcal{G}$ .  $\square$

*Proof:* If  $\mathcal{F}$  is flat, then clearly  $\mathrm{Tor}_1^{\mathcal{O}}(\mathcal{F}, \mathcal{G}) = 0$  for any  $\mathcal{O}$ -module  $\mathcal{G}$ . Conversely, use the long exact sequence associated to  $\mathcal{F} \otimes_{\mathcal{O}}^{\mathbf{L}} \cdot$  (4.8.10.4).  $\square$

**Def. (7.3.3.13)[Relative Cup Product].** Let  $f : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$  be a morphism of ringed topoi, then for any  $K, L \in D(\mathcal{O}_{\mathcal{C}})$ , there is a canonical functorial **relative cup product**

$$Rf_*K \otimes_{\mathcal{O}_{\mathcal{D}}}^L Rf_*L \rightarrow Rf_*(K \otimes_{\mathcal{O}_{\mathcal{C}}}^L L)$$

in  $D(\mathcal{O}_{\mathcal{D}})$  that is the adjunction map of the map

$$Lf^*(Rf_*K \otimes_{\mathcal{O}_{\mathcal{D}}}^L Rf_*L) \rightarrow Lf^*Rf_*K \otimes_{\mathcal{O}_{\mathcal{C}}}^L Lf^*Rf_*L \rightarrow K \otimes_{\mathcal{O}_{\mathcal{C}}}^L L$$

by (7.3.3.15). This map is symmetric and associative.

In particular, if  $(\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}}) = (\mathrm{Sh}(\mathrm{pt}), \Gamma(\mathcal{C}, \mathcal{O}_{\mathcal{C}}) = A)$  (7.3.1.3), we get a map

$$R\Gamma(\mathcal{C}, K) \otimes_A^L R\Gamma(\mathcal{C}, L) \rightarrow \Gamma(\mathcal{C}, K \otimes_{\mathcal{O}_{\mathcal{C}}}^L L).$$

L

## Derived Pullback

**Def. (7.3.3.14) [Derived Pullback].** Let  $f : (\mathcal{S}h(\mathcal{C}'), \mathcal{O}') \rightarrow (\mathcal{S}h(\mathcal{C}), \mathcal{O})$  be a morphism of ringed site, define the **derived pullback**

$$Lf^* : D(\mathcal{O}) \rightarrow D(\mathcal{O}') : \mathcal{F}^\bullet \mapsto f^{-1}\mathcal{F}^\bullet \otimes_{f^{-1}\mathcal{O}}^L \mathcal{O}' \quad (7.3.3.6).$$

In particular, if  $\mathcal{F}^\bullet$  is K-flat, then there are natural isomorphisms  $Lf^*\mathcal{F}^\bullet \cong f^*\mathcal{F}^\bullet$ .

Moreover,  $Lf^*$  is also naturally isomorphic to the left derived functor of the pullback morphism  $f^*$ , by composition of derived functors applied to the  $f^* = \text{Tot}(\cdot \otimes_{f^{-1}\mathcal{O}} \mathcal{O}') \circ f^{-1}$  where  $f^{-1}$  is exact. In particular,  $Lf^*$  only depends on the underlying map  $f^*$  of ringed topoi.  $\square$

**Prop. (7.3.3.15).** Let  $f : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$ ,  $g : (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}}) \rightarrow (\mathrm{Sh}(\mathcal{E}), \mathcal{O}_{\mathcal{E}})$  be a morphism of ringed topoi, then



- There is a natural isomorphism  $L(g \circ f)^* \cong Lf^* \circ Lg^*$ . In particular, as localizing map  $j^*$  is exact,  $Lf^*$  can be calculated locally on the base.
- For  $\mathcal{F}^\bullet, \mathcal{G}^\bullet \in \text{Mod}_{\mathcal{O}_{\mathcal{D}}}$ , there are natural functorial isomorphisms

$$Lf^*(\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{D}}}^L \mathcal{G}^\bullet) \cong Lf^* \mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{E}}}^L Lf^* \mathcal{G}^\bullet.$$

In particular, as localizing map  $j^*$  is exact, the derived tensor product can be calculated locally on the base.

- For  $\mathcal{F}^\bullet \in \text{Mod}_{\mathcal{O}_{\mathcal{E}}}, \mathcal{G}^\bullet \in \text{Mod}_{\mathcal{O}_{\mathcal{D}}}$ , there are natural functorial isomorphisms

$$\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{E}}}^L Lf^* \mathcal{G}^\bullet \cong \mathcal{F}^\bullet \otimes_{f^{-1}\mathcal{O}_{\mathcal{D}}}^L f^{-1} \mathcal{G}^\bullet$$

- For  $\mathcal{F}^\bullet, \mathcal{G}^\bullet \in \text{Mod}_{\mathcal{O}_{\mathcal{D}}}$ , the natural isomorphisms satisfy a commutative diagram

$$\begin{array}{ccc} Lf^*(\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{D}}}^L \mathcal{G}^\bullet) & \longrightarrow & Lf^* \text{Tot}^\oplus(\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{D}}} \mathcal{G}^\bullet) \\ \downarrow & & \downarrow \\ Lf^* \mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{E}}}^L Lf^* \mathcal{G}^\bullet & & f^* \text{Tot}^\oplus(\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{D}}} \mathcal{G}^\bullet) \\ \downarrow & & \downarrow \\ f^* \mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{E}}}^L f^* \mathcal{G}^\bullet & \longrightarrow & \text{Tot}^\oplus(f^* \mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{E}}} f^* \mathcal{G}^\bullet) \end{array}$$

⌋

*Proof:* 1, 2, 3: These follow from (7.3.3.10).

4: By a universal argument,  $(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \mapsto Lf^*(\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{D}}}^L \mathcal{G}^\bullet)$  is the derived functor of the bi-exact bifunctor

$$F : K(\mathcal{O}_{\mathcal{D}}) \times K(\mathcal{O}_{\mathcal{D}}) \rightarrow D(\mathcal{O}_{\mathcal{E}}) : (\mathcal{F}^\bullet, \mathcal{G}^\bullet) \mapsto f^* \text{Tot}^\oplus(\mathcal{F}^\bullet \otimes_{\mathcal{O}_{\mathcal{D}}} \mathcal{G}^\bullet).$$

Then by the universal property, both natural transformations corresponds to a natural transformation  $\theta_i : LF \rightarrow LF$ . So it suffices to show that they are isomorphic. But K-flat morphisms are essentially surjective in  $D(\mathcal{O}_{\mathcal{D}})$ , thus it suffices to prove the diagram is commutative for  $\mathcal{F}, \mathcal{G}$  K-flat. But in this case, there are commutative diagrams

$$\begin{array}{ccc} LF(\mathcal{F}^\bullet, \mathcal{G}^\bullet) & \xrightarrow{\eta_{(\mathcal{F}, \mathcal{G})}} & F(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \\ \downarrow \theta_i \star (Q_{\mathcal{O}_{\mathcal{D}}} \times Q_{\mathcal{O}_{\mathcal{D}}}) & & \downarrow \text{id} \\ LF(\mathcal{F}^\bullet, \mathcal{G}^\bullet) & \xrightarrow{\eta_{(\mathcal{F}, \mathcal{G})}} & F(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \end{array}$$

and  $\eta_{(\mathcal{F}, \mathcal{G})}$  is an isomorphism, so  $\theta_1 = \theta_2$ . □

**Prop. (7.3.3.16) [Adjunction].**  $Lf^*$  is left adjoint to  $Rf_*$ , by (4.10.3.15) and (7.2.2.9). ⌋

**Prop. (7.3.3.17) [Base Change Map].** Let

$$\begin{array}{ccc} (\text{Sh}(\mathcal{C}'), \mathcal{O}_{\mathcal{C}'}) & \xrightarrow{g'} & (\text{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \\ \downarrow f' & & \downarrow f \\ (\text{Sh}(\mathcal{D}'), \mathcal{O}_{\mathcal{D}'}) & \xrightarrow{g} & (\text{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}}) \end{array}$$

be a commutative diagram of ringed topoi, then for any  $K \in D(\mathcal{O}_{\mathcal{C}})$ , there is a canonical base change map

$$Lg^*Rf_*K \rightarrow R(f')_*L(g')^*K$$

functorial in  $K$ , and this map is compatible with composition of diagrams.  $\lrcorner$

*Proof:* By adjunction(7.3.3.16), this follows from the canonical map

$$Rf_*K \rightarrow Rg_*R(f')_*L(g')^*K = Rf_*R(g')_*L(g')^*K,$$

which comes from the adjunction map  $K \rightarrow R(g')_*L(g')^*K$ .  $\square$

**Cor. (7.3.3.18) [Flat Base Change Morphism].** Let

$$\begin{array}{ccc} (\mathrm{Sh}(\mathcal{C}'), \mathcal{O}_{\mathcal{C}'}) & \xrightarrow{g'} & (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \\ \downarrow f' & & \downarrow f \\ (\mathrm{Sh}(\mathcal{D}'), \mathcal{O}_{\mathcal{D}'}) & \xrightarrow{g} & (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}}) \end{array}$$

be a commutative diagram of ringed sites. Assume both  $g, g'$  are flat, then for any  $\mathcal{F}^\bullet \in K^+(\mathcal{O}_{\mathcal{C}})$ , there is a canonical base change map

$$g^*Rf_*\mathcal{F}^\bullet \rightarrow Rf'_*(g')^*\mathcal{F}^\bullet.$$

and when  $(\mathrm{Sh}(\mathcal{D}'), \mathcal{O}_{\mathcal{D}'}) = (\mathrm{Sh}(\mathcal{C}/U), \mathcal{O}_U)$  is the localizing site, then this map is an isomorphism

$$(Rf_*\mathcal{F}^\bullet)_U \cong Rf'_*\mathcal{F}_U^\bullet.$$

$\lrcorner$

*Proof:* The last assertion follows from the fact the restriction of a K-injective is also a K-injective(7.3.4.2).  $\square$

**Def. (7.3.3.19) [Projection Map].** Let  $f : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$  be a morphism of ringed topoi, then for any  $E \in D(\mathcal{O}_{\mathcal{C}})$  and  $K \in D(\mathcal{O}_{\mathcal{D}})$ , there is a canonical functorial **projection map**

$$Rf_*E \otimes_{\mathcal{O}_{\mathcal{D}}}^L K \rightarrow Rf_*(E \otimes_{\mathcal{O}_{\mathcal{C}}}^L Lf^*K)$$

which is the adjunction of the map

$$Lf^*(Rf_*E \otimes_{\mathcal{O}_{\mathcal{D}}}^L K) \cong Lf^*Rf_*E \otimes_{\mathcal{O}_{\mathcal{C}}}^L Lf^*K \rightarrow E \otimes_{\mathcal{O}_{\mathcal{C}}}^L Lf^*K$$

by(7.3.3.15).  $\lrcorner$

**Prop. (7.3.3.20) [Projection Formula].** In situation(7.3.3.19), if  $K$  is perfect(7.3.4.17), then the projection map is an isomorphism.  $\lrcorner$

*Proof:* To check it is an isomorphism, it suffice to find a covering  $\{U_i \rightarrow V\}$  for each  $V \in \mathcal{C}$  that this map is an isomorphism on  $U_i$ . (To see this, look at  $H^i$ , and notice  $U \mapsto \mathrm{Mod}(\mathcal{O}_U)$  is a stack). Then we may assume  $K$  is a finite complex of  $\mathcal{O}_U$ -modules consisting of finite free  $\mathcal{O}_U$ -modules. And then we use truncation to reduce to the case  $K$  is discrete, in which case it is trivial.  $\square$

**Inner Hom**

**Def. (7.3.3.21) [Sheaf Hom Complexes].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $P^\bullet, Q^\bullet \in K(\mathcal{O})$ , we define the **Sheaf Hom complex**  $\mathcal{H}om^\bullet(P^\bullet, Q^\bullet)$  to be

$$\mathcal{H}om^n(P^\bullet, Q^\bullet) = \prod_i \mathcal{H}om_{\mathcal{O}}(P^i, Q^{n+i}) \quad (7.2.3.1),$$

with the differential giving by  $d(\{f_k\})_i = \{df_i - (-1)^i f_{i+1}d\}$  and suitable signatures.

It is clear that

$$\Gamma(U, \mathcal{H}om_{\mathcal{O}}^\bullet(P^\bullet, Q^\bullet)) = \text{Hom}_{\mathcal{O}_U}^\bullet(P^\bullet|_U, Q^\bullet|_U), \quad \Gamma(\mathcal{C}, \mathcal{H}om_{\mathcal{O}}^\bullet(P^\bullet, Q^\bullet)) = \text{Hom}_{\mathcal{O}}^\bullet(P^\bullet|_U, Q^\bullet|_U)$$

and

$$H^n(\Gamma(U, \mathcal{H}om^\bullet(P^\bullet, Q^\bullet))) = \text{Hom}_{K(\mathcal{O}_U)}(P^\bullet|_U, Q^\bullet|_U[n]), \quad H^n(\Gamma(\mathcal{C}, \mathcal{H}om^\bullet(P^\bullet, Q^\bullet))) = \text{Hom}_{K(\mathcal{O})}(P^\bullet, Q^\bullet[n]).$$

┘

**Prop. (7.3.3.22) [Adjunction].** For  $K^\bullet, M^\bullet, L^\bullet \in K(\mathcal{O})$ , there is a canonical functorial isomorphism:

$$\mathcal{H}om^\bullet(K, \mathcal{H}om^\bullet(L, M)) = \mathcal{H}om^\bullet(\text{Tot}(K \otimes_R L), M).$$

┘

*Proof:* Cf. [Sta]0A5Y. □

**Prop. (7.3.3.23).** For  $K^\bullet, M^\bullet, L^\bullet \in K(\mathcal{O})$ , there are canonical functorial morphisms:

- $\text{Tot}(\mathcal{H}om(K^\bullet, L^\bullet) \otimes_{\mathcal{O}} K^\bullet) \rightarrow L^\bullet$ ,
- $\text{Tot}(\mathcal{H}om^\bullet(L, M) \otimes_{\mathcal{O}} \mathcal{H}om^\bullet(K, L)) \rightarrow \mathcal{H}om^\bullet(K, M)$ ,
- $\text{Tot}(\mathcal{H}om^\bullet(L, M) \otimes_{\mathcal{O}} K) \rightarrow \mathcal{H}om^\bullet(\mathcal{H}om^\bullet(K, L), M)$ ,
- $\text{Tot}(K \otimes_{\mathcal{O}} \mathcal{H}om^\bullet(M, L)) \rightarrow \mathcal{H}om^\bullet(M, \text{Tot}(K \otimes_{\mathcal{O}} L))$ ,
- $K \rightarrow \mathcal{H}om^\bullet(L, \text{Tot}(K \otimes_{\mathcal{O}} L))$ .

┘

*Proof:* All these are consequences of (7.3.3.22). □

**Lemma (7.3.3.24).** Let  $(\mathcal{C}, \mathcal{O})$  be a site and  $(\mathcal{I}^\bullet)' \rightarrow \mathcal{I}^\bullet$  be a quasi-isomorphism of K-injective  $\mathcal{O}$ -modules and  $(\mathcal{L}^\bullet)' \rightarrow \mathcal{L}^\bullet$  be a quasi-isomorphism of  $\mathcal{O}$ -modules, then the natural map

$$\mathcal{H}om^\bullet(\mathcal{L}^\bullet, (\mathcal{I}^\bullet)') \rightarrow \mathcal{H}om^\bullet((\mathcal{L}^\bullet)', \mathcal{I}^\bullet)$$

is a quasi-isomorphism. ┘

*Proof:*  $H^n(\mathcal{H}om^\bullet(\mathcal{L}^\bullet, (\mathcal{I}^\bullet)'))$  is the sheaf  $U \mapsto \text{Hom}_{K(\mathcal{O}_U)}(\mathcal{L}_U^\bullet, (\mathcal{I}^\bullet)'_U[n])$ , and  $(\mathcal{I}^\bullet)'_U[n]$  is K-injective by (7.3.4.2). □

**Def. (7.3.3.25) [Internal Hom].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site,  $K(\mathcal{O})$  has enough K-injectives, thus by (4.10.3.13), we can define the **internal Hom**

$$R\mathcal{H}om : D(\mathcal{O})^{op} \times D(\mathcal{O}) \rightarrow D(\mathcal{O})$$

as the right derived functor of the bi-exact bifunctor  $\mathcal{H}om^\bullet : K(\mathcal{O})^{op} \times K(\mathcal{O}) \rightarrow D(\mathcal{O})$ , where the conditions are satisfied by (7.3.3.24). ┘

**Prop. (7.3.3.26)[Internal Hom and Localization].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, for any  $\mathcal{G}, \mathcal{F} \in K(\mathcal{O})$  and  $U \in \mathcal{C}$ , the natural transformation

$$R\mathcal{H}om(\mathcal{F}, \mathcal{G})|_U \rightarrow R\mathcal{H}om(\mathcal{F}|_U, \mathcal{G}|_U)$$

is an isomorphism. In particular, we can calculate  $\mathcal{E}xt$  locally.  $\lrcorner$

*Proof:* This follows from (7.3.3.21)(7.3.4.2) and trivial Grothendieck duality applied to restrictions.  $\square$

**Prop. (7.3.3.27) [Derived Hom and Internal Hom].** For a ringed site  $(\mathcal{C}, \mathcal{O})$  and  $\mathcal{F}, \mathcal{G} \in D(\mathcal{O})$ ,  $U \in \mathcal{C}$ ,

$$R\Gamma(\mathcal{C}, R\mathcal{H}om(\mathcal{F}, \mathcal{G})) = R\mathcal{H}om(\mathcal{F}, \mathcal{G}), \quad R\Gamma(U, R\mathcal{H}om(\mathcal{F}, \mathcal{G})) = R\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$

In particular,

$$\Gamma(U, R\mathcal{H}om(\mathcal{F}, \mathcal{G})) = \mathcal{H}om_{D(\mathcal{O}_U)}(\mathcal{F}|_U, \mathcal{G}|_U),$$

and

$$H^0(\mathcal{C}, R\mathcal{H}om(\mathcal{F}, \mathcal{G})) = \mathcal{H}om_{D(\mathcal{O})}(\mathcal{F}, \mathcal{G}), \quad H^p(\mathcal{C}, R\mathcal{H}om(\mathcal{F}, \mathcal{G})) = \mathcal{E}xt_{\mathcal{O}}^p(\mathcal{F}, \mathcal{G})$$

by (7.3.3.21)(7.3.4.2)(7.3.4.13) and the composition of derived functors.  $\lrcorner$

**Prop. (7.3.3.28).** If  $\mathcal{K}, \mathcal{L}, \mathcal{M} \in D(\mathcal{O})$ , then there is a natural isomorphism

$$R\mathcal{H}om_{\mathcal{O}}(\mathcal{K}, R\mathcal{H}om(\mathcal{L}, \mathcal{M})) \cong R\mathcal{H}om_{\mathcal{O}}(\mathcal{K} \otimes_{\mathcal{O}}^L \mathcal{L}, \mathcal{M}).$$

*Proof:* Consider both sides as the derived functor of the triple-exact triple-functor

$$K(\mathcal{O})^{op} \times K(\mathcal{O})^{op} \times K(\mathcal{O}) \rightarrow D(\mathcal{O}) : (\mathcal{K}^\bullet, \mathcal{L}^\bullet, \mathcal{M}^\bullet) \mapsto \text{Tot}^\oplus(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \text{Tot}^\oplus(\mathcal{L}^\bullet \otimes_{\mathcal{O}} \mathcal{M}^\bullet)),$$

the conditions are satisfied by (7.3.4.13)(7.3.3.3). As this is true for  $\text{Tot}$  by (7.3.3.22), this follows from the universal properties and (4.10.3.14).  $\square$

**Cor. (7.3.3.29)[Adjunction].** By (7.3.3.28), taking  $H^0(R\Gamma(\mathcal{C}, -))$ , we get:

$$\mathcal{H}om_{D(\mathcal{O})}(\mathcal{K}, R\mathcal{H}om(\mathcal{L}, \mathcal{M})) = \mathcal{H}om_{D(\mathcal{O})}(\mathcal{K} \otimes^L \mathcal{L}, \mathcal{M}).$$

that is, derived tensor is left adjoint to internal hom.  $\lrcorner$

**Prop. (7.3.3.30).** Given complexes  $\mathcal{K}, \mathcal{L}, \mathcal{M} \in D(\mathcal{O})$ , there are canonical functorial morphisms:

- $\mathcal{H}om(\mathcal{K}, \mathcal{L}) \otimes_{\mathcal{O}}^L \mathcal{K} \rightarrow \mathcal{L}$ ,
- $R\mathcal{H}om(\mathcal{L}, \mathcal{M}) \otimes_{\mathcal{O}}^L R\mathcal{H}om(\mathcal{K}, \mathcal{L}) \rightarrow R\mathcal{H}om(\mathcal{K}, \mathcal{M})$ ,
- $R\mathcal{H}om(\mathcal{L}, \mathcal{M}) \otimes_{\mathcal{O}}^L \mathcal{K} \rightarrow R\mathcal{H}om(R\mathcal{H}om(\mathcal{K}, \mathcal{L}), \mathcal{M})$ ,
- $\mathcal{K} \otimes_{\mathcal{O}}^L R\mathcal{H}om(\mathcal{M}, \mathcal{L}) \rightarrow R\mathcal{H}om(\mathcal{M}, \mathcal{K} \otimes_{\mathcal{O}}^L \mathcal{L})$ ,
- $\mathcal{K} \rightarrow R\mathcal{H}om(\mathcal{L}, \mathcal{K} \otimes_{\mathcal{O}}^L \mathcal{L})$ .

*Proof:* These are direct consequences of (7.3.3.29).  $\square$

**Prop. (7.3.3.31).** Let  $(\mathrm{Sh}(\mathcal{C}), \mathcal{O}) \rightarrow (\mathrm{Sh}(\mathcal{C}'), \mathcal{O}')$  be a morphism of ringed site and  $\mathcal{K}, \mathcal{L} \in D(\mathcal{O})$ , there is a natural morphism

$$Rf_* R\mathcal{H}om_{\mathcal{O}}(L, K) \rightarrow R\mathcal{H}om_{\mathcal{O}'}(Rf_* L, Rf_* K).$$

┘

*Proof:* This follows from the adjunction (7.3.3.28)(7.3.3.30) and the relative cup product (7.3.3.13).

□

**Def. (7.3.3.32) [Internal Ext Groups].** For any  $\mathcal{G}, \mathcal{F} \in K(\mathcal{O})$ , we define the **internal Ext groups**

$$\mathcal{E}xt_{\mathcal{O}}^n(\mathcal{G}, \mathcal{F}) = H^i(R\mathcal{H}om(\mathcal{G}, \mathcal{F})) \in \mathrm{Mod}(\mathcal{O}).$$

┘

**Prop. (7.3.3.33).** For  $\mathcal{E}, \mathcal{F}, \mathcal{L} \in \mathrm{Mod}(\mathcal{O}_X)$  and  $\mathcal{L}$  finite locally free,

$$\mathcal{E}xt^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) \cong \mathcal{E}xt^i(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G}) \cong \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \otimes \mathcal{L}^\vee$$

because there are maps between them (7.2.5.2), and  $\mathcal{E}xt$  is local, so check locally. In particular,

$$\mathrm{Ext}^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) = \mathrm{Ext}^i(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G}).$$

┘

**Prop. (7.3.3.34) [Spectral Sequences for Internal Ext].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{K}^\bullet \in K^-(\mathcal{O}), \mathcal{F}^\bullet \in K^+(\mathcal{O})$ , then there are spectral sequence convergence

$$E_2^{i,j} = \mathcal{E}xt_{\mathcal{O}}^i(H^{-j}(\mathcal{K}^\bullet), \mathcal{F}^\bullet) \implies \mathcal{E}xt_{\mathcal{O}}^{i+j}(\mathcal{K}^\bullet, \mathcal{F}^\bullet).$$

$$E_1^{i,j} = \mathcal{E}xt_{\mathcal{O}}^j(\mathcal{K}^{-i}, \mathcal{F}^\bullet) \implies \mathcal{E}xt_{\mathcal{O}}^{i+j}(\mathcal{K}^\bullet, \mathcal{F}^\bullet)$$

$$E_2^{i,j} = H^i(\mathcal{C}, \mathcal{E}xt_{\mathcal{O}}^j(\mathcal{K}^\bullet, \mathcal{F}^\bullet)) \implies \mathrm{Ext}_{\mathcal{O}}^{i+j}(\mathcal{K}^\bullet, \mathcal{F}^\bullet)$$

┘

*Proof:* 1, 2: Choose a (bounded below) injective resolution  $\mathcal{I}^\bullet$  of  $\mathcal{F}^\bullet$ , and these are the two spectral sequences associated to the double complex  $\mathcal{H}om_{\mathcal{O}}(\mathcal{K}^\bullet, \mathcal{I}^\bullet)$ .

3: Use Grothendieck spectral sequence on (7.3.3.27). □

## 4 Acyclic Sheaves

**Prop. (7.3.4.1) [Pushforward of Injectives].** Let  $f : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_C) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_D)$  be a flat map of ringed topoi, then  $f$  preserves injectives, as  $f^*$  is exact. ┘

**Prop. (7.3.4.2) [Restriction of (K-)Injectives].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site,  $U \in \mathcal{C}$ , then

- If  $\mathcal{K}^\bullet$  be a K-injective complex  $\mathcal{O}$ -modules, then  $(\mathcal{K}^\bullet)|_U$  is a K-injective complex of  $\mathcal{O}_U$ -modules.
- If  $\mathcal{I}$  is an injective  $\mathcal{O}$ -module, then  $\mathcal{I}|_U$  is an injective  $\mathcal{O}$ -module.

┘

*Proof:* Use (4.10.2.6) and the fact  $j_U^*$  is right adjoint to the exact functor  $j_{U!}$  (7.2.2.9)(7.2.2.9). □

**Prop. (7.3.4.3).** If  $\mathcal{I} \in \mathrm{Sh}(\mathcal{C})$  is injective, then  $\mathcal{I}$  is also injective in  $\mathcal{P}\mathrm{Sh}(\mathcal{C})$ . ┘

*Proof:* Because restriction is right adjoint to the exact shification functor. □

**Prop. (7.3.4.4).** For an injective sheaf  $\mathcal{F}$  on a site  $(\mathcal{C}, \mathcal{O})$ ,  $F(U)$  is injective Abelian group for every  $U \in \mathcal{C}$ . ┘

*Proof:* This is because for the morphism  $i : \mathrm{pt} \rightarrow \mathcal{C} : \mathrm{pt} \mapsto U$ ,  $i_p$  is exact ( $i_p A(V) = \oplus_{\mathrm{Hom}(V, U)} A$ ), hence  $i^p$  preserves injectives. □

### Flask Sheaves

**Def. (7.3.4.5) [Totally Acyclic Sheaves].** A **totally acyclic sheaf** on a ringed site  $(\mathcal{C}, \mathcal{O})$  is an object in  $\mathrm{Sh}(\mathcal{C})$  that is acyclic for any functor  $\mathrm{Hom}_{\mathcal{P}\mathrm{Sh}(\mathcal{C})}(K, -)$ .  $\lrcorner$

**Prop. (7.3.4.6) [Characterization of Totally Acyclic Sheaves].** Let  $\mathcal{C}$  be a site and  $\mathcal{F} \in \mathrm{Sh}(\mathcal{C})$ , then  $\mathcal{F}$  is totally acyclic iff

- $\mathcal{F}$  is acyclic for  $\Gamma(U, -)$  for any  $U \in \mathcal{C}$ .
- For every surjection  $K' \rightarrow K$  of sheaves of sets the extended Čech complex

$$0 \rightarrow H^0(K, \mathcal{F}) \rightarrow H^0(K', \mathcal{F}) \rightarrow H^0(K' \times_K K', \mathcal{F}) \rightarrow \dots$$

is exact.  $\lrcorner$

*Proof:* Cf. [Sta]07A1.  $\square$

**Prop. (7.3.4.7).** A totally acyclic sheaf is acyclic for any map of ringed topoi  $f : (\mathrm{Sh}(\mathcal{C}), \mathcal{O}_C) \rightarrow (\mathrm{Sh}(\mathcal{D}), \mathcal{O}_D)$ .  $\lrcorner$

*Proof:* This follows from the description of  $R^i f_*$  in (7.3.1.7).  $\square$

**Def. (7.3.4.8) [Flask Sheaves].** A **flask sheaf** on a ringed site  $(\mathcal{C}, \mathcal{O})$  is an object of  $\mathrm{Sh}(\mathcal{C})$  that satisfies the following equivalent conditions:

- it is acyclic for  $\iota$ .
- It is acyclic for any  $\check{H}^0(\{U_i \rightarrow U\}, -)$
- It is acyclic for all  $\Gamma(U, -)$ .

In particular a totally acyclic sheaf is flask.

A **flabby sheaf** is an object of  $\mathrm{Sh}(\mathcal{C})$  that for any monomorphism  $U \rightarrow V$ ,  $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$  is surjective;  $\lrcorner$

*Proof:* 1  $\iff$  3 is by (7.3.1.6),

3  $\rightarrow$  2: use Čech to sheaf1 (7.3.2.13), notice the  $q > 0$  terms vanish, thus  $\check{H}^p(\{U_i \rightarrow U\}, \mathcal{F}) = H^p(U, \mathcal{F}) = 0$ .

2  $\rightarrow$  3 follows from (7.3.2.16).  $\square$

**Prop. (7.3.4.9).** Injective sheaves are flabby, Flabby sheaves on a ringed space are flask.  $\lrcorner$

*Proof:* For the first assertion, use (7.2.2.9) and the fact  $j_! \mathcal{O}_U \rightarrow j_! \mathcal{O}_V$  is a monomorphism for  $V \rightarrow U$  a monomorphism by definition (7.2.2.9).

For the last assertion, use (4.10.3.29): Injectives are flabby, so it is sufficiently large. For an exact sequence  $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$  of sheaves, if  $\mathcal{F}$  is flabby, then  $\mathcal{H}$  is just the presheaf cokernel. (It reduces to  $\check{H}^1(\{U_i \rightarrow U\}, \mathcal{F}) = 0$ , and this is done by Zorn's lemma). Thus if  $\mathcal{F}$  is flabby,  $\mathcal{G}$  is flabby iff  $\mathcal{H}$  is flabby (by five lemma).  $\square$

**Prop. (7.3.4.10).** On a discrete site, all sheaves are flask, because  $\iota$  is the identity functor.  $\lrcorner$

**Prop. (7.3.4.11).** Filtered colimits of flabby sheaves are flabby. (This is because filtered colimits are exact).  $\lrcorner$

Filtered colimits of injective sheaves over a Noetherian topological space are injective. (Use Baer criterion, then notice every sub-object of  $\mathbb{Z}_U$  is finitely generated because it has only f.m. connected component (4.12.2.4) so it maps to some  $F_\alpha$ ).  $\lrcorner$

**Prop. (7.3.4.12) [Pushforward of Totally Acyclic/Flask Sheaves].** Let  $f : (\mathcal{Sh}(\mathcal{C}), \mathcal{O}_C) \rightarrow (\mathcal{Sh}(\mathcal{D}), \mathcal{O}_D)$  be a map of ringed topoi (resp. ringed site), then  $f_*$  maps totally acyclic (resp. flask) sheaves to totally acyclic (resp. flask) sheaves (7.3.4.5)(7.3.4.8). In particular, it maps injectives to totally acyclic sheaves.  $\lrcorner$

*Proof:* By (7.2.1.3) we may assume  $f$  is a map of ringed sites, and  $K = h_U$  for some  $U \in \mathcal{C}$ . Then we notice  $H^*(\{U_i \rightarrow U\}, f_*\mathcal{F}) = H^*(\{f^{-1}(U_i) \rightarrow f^{-1}(U)\}, \mathcal{F})$ , then we can use (7.3.2.16) to show  $f_*\mathcal{F}$  is totally acyclic.  $\square$

**Prop. (7.3.4.13) [Technical Lemma].** If  $\mathcal{K}^\bullet$  is  $K$ -flat  $\mathcal{O}$ -modules and  $\mathcal{I}^\bullet$  is  $K$ -injective  $\mathcal{O}$ -modules, then  $\mathcal{H}om^\bullet(\mathcal{K}^\bullet, \mathcal{I}^\bullet)$  is  $K$ -injective.  $\lrcorner$

*Proof:* Use definitions (7.3.3.1)(4.10.2.1) and (7.3.3.22).  $\square$

**Prop. (7.3.4.14).** If  $\mathcal{I}$  is an injective  $\mathcal{O}_X$ -module, then for a coherent locally free sheaf  $\mathcal{L}$ ,  $\mathcal{L} \otimes \mathcal{I}$  is also injective, because tensoring with  $\mathcal{L}$  is adjoint to tensoring with  $\mathcal{L}^\vee$  (7.2.5.2), which is exact.  $\lrcorner$

### Pseudo-Coherent Sheaves

**Def. (7.3.4.15) [Strictly Perfect Complexes of Modules].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, a **strictly perfect complex of  $\mathcal{O}$ -modules** of  $\mathcal{O}$ -modules is a finite complex of  $\mathcal{O}$ -modules that each term is a direct summand of a finite free  $\mathcal{O}$ -module.  $\lrcorner$

**Def. (7.3.4.16) [Pseudo-Coherent Complexes of Modules].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, then  $\mathcal{K} \in D(\mathcal{O}_X)$  is called an  **$m$ -pseudo-coherent  $\mathcal{O}_X$ -module** if for any  $U \in \mathcal{C}$ , there exists a covering  $\{U_i \rightarrow U\}$  s.t. for any  $i$ , there are strictly perfect object  $E_i \in D(\mathcal{O}_{U_i})$  and a morphism  $E_i \rightarrow \mathcal{K}|_{U_i}$  that induces isomorphisms on  $H^i$  for  $i > m$  and surjection on  $H^m$ .

$\mathcal{K} \in D(R)$  is called a **pseudo-coherent  $\mathcal{O}_X$ -module** if it is  $m$ -pseudo-coherent for any  $m$ .  $\lrcorner$

### Perfect Sheaves

**Def. (7.3.4.17) [Perfect Complexes of Modules].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, a **perfect complex of  $\mathcal{O}$ -modules** of  $\mathcal{O}$ -modules is an object  $E$  in  $\mathcal{D}(\mathcal{O})$  that for any  $U \in \mathcal{C}$  there is a covering  $\{U_i \rightarrow U\}$  that  $E|_{U_i}$  can be represented by a strictly perfect complexes (7.3.4.15).  $\lrcorner$

**Prop. (7.3.4.18).** Every mapping from a strictly perfect complex to an acyclic complex has a cover of open sets that on each open set the map is nullhomotopic.  $\lrcorner$

*Proof:* This is true for a single direct summand of a finite free sheaf, and we can use induction to prove, Cf. [Sta]08C7.  $\square$

**Cor. (7.3.4.19).** The strictly perfect complexes are fake " $K$ -projective" objects in  $K(\mathcal{O}_X)$ . Note it is not technically  $K$ -projective, but it has all the properties of  $K$ -projective when proven, noticing the fact it is irrelevant when taken shification.  $\lrcorner$

**Def. (7.3.4.20) [Perfect Sheaves].** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site, an object  $K^\bullet$  in  $K(\mathcal{O})$  **perfect** if there is a covering  $\mathcal{U}$  that on each  $U_i$  there is a quasi-iso  $K_i^\bullet \rightarrow K^\bullet|_{U_i}$  with  $K_i^\bullet$  strictly perfect.

This is equivalent to  $K^\bullet$  is locally represented by perfect objects in  $D(\mathcal{O})$  by the fact that perfect object is fake  $K$ -projective.  $\lrcorner$

**Prop. (7.3.4.21).** When  $X$  is local ringed space, perfectness is equivalent to the fact that it is locally a finite free  $\mathcal{O}_{U_i}$ -module.  $\lrcorner$

*Proof:* This is because direct summand of a finite free module is free, Cf.[Sta]0BCI.  $\square$

**Prop. (7.3.4.22) [Strictly Perfect Modules are “Acyclic” for  $\text{Hom}^\bullet$ ].** If  $(\mathcal{C}, \mathcal{O})$  is a ringed site and  $\mathcal{E}^\bullet, \mathcal{F}^\bullet \in K(\mathcal{O})$ , then  $R\mathcal{H}om(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$  can be calculated directly by  $\mathcal{H}om^\bullet(\mathcal{E}, \mathcal{F})$  in the following cases:

- $\mathcal{E}^\bullet$  is strictly perfect.
- $\mathcal{E}^\bullet \in D^-(\mathcal{O}_X), \mathcal{F}^\bullet \in D^+(\mathcal{O}_X)$ , and each term  $\mathcal{E}^n$  is a direct summand of a finite free  $\mathcal{O}_X$ -module.

┘

*Proof:* Cf.[Sta]08I5, 08DM.  $\square$

**Prop. (7.3.4.23) [Duals].** Let  $K$  be a perfect object in  $D(\mathcal{O})$ , then

- $K^\vee = R\mathcal{H}om(K, \mathcal{O})$  is also a perfect object, and  $(K^\vee)^\vee \cong K$ .
- For any  $M \in D(\mathcal{O})$ , there are functorial isomorphisms

$$M \otimes^{\mathbb{L}} K^\vee \cong R\mathcal{H}om(K, M), \quad H^0(\mathcal{C}, M \otimes^{\mathbb{L}} K^\vee) \cong \text{Hom}_{D(\mathcal{O})}(K, M).$$

┘

*Proof:* Cf.[Sta]08JJ.  $\square$

**Def. (7.3.4.24) [Relative Perfect Modules].** Let  $(X, \mathcal{O}_X) \rightarrow (S, \mathcal{O}_S)$  be a morphism of ringed sites that is flat and locally of f.p., then  $E \in D(\mathcal{O}_X)$  is called a **perfect object relative to  $S$**  if  $E$  is pseudo-coherent and  $E$  locally has finite tor dimension as an object in  $D(f^{-1}\mathcal{O}_S)$ . Cf.[Sta]08CG.  $\perp$

## 5 Topological Sheaves

References are [Sheaf Cohomology, Anonymous].

A topological space can be seen as a ringed site, so the theory of ringed sites applies to topological spaces.

### Acyclic sheaves

**Def. (7.3.5.1).** An Abelian sheaf on a paracompact Hausdorff topological space  $X$  is called

**soft** iff is and  $\forall$  closed  $V, \mathcal{F}(X) \rightarrow \mathcal{F}(V)$  is surjective. A flabby sheaf is soft.

**fine** iff the sheaf of rings  $\text{Hom}(\mathcal{F}, \mathcal{F})$  is soft.

Fine and soft are local properties (Use Zorn’s lemma to construct one-by-one).

┘

**Prop. (7.3.5.2).** For a sheaf of **unital rings** over a paracompact Hausdorff space  $X$ , the following are equivalent,

1. it is a soft sheaf.
2. for any disjoint closed sets  $V, W$ , there is a section of  $X$  that is 0 on  $V$ , and 1 on  $W$ .
3. it possesses a partition of unity.
4. it is a fine sheaf.

Note any soft sheaf possesses a partition of unity.  $\perp$



*Proof:*  $1 \iff 2$  is easy and  $1 \rightarrow 3$  is the to choose a closed locally finite subcover and use Zorn's lemma to construct one-by-one. For  $3 \rightarrow 1$ , notice a closed section can extend to a slightly larger nbhd.

Because for a sheaf of rings  $\mathcal{F}$ , a partition of unity is equivalent to a partition of unity  $\text{Hom}(\mathcal{F}, \mathcal{F})$ , so 34 are equivalent because 13 are equivalent.  $\square$

**Cor. (7.3.5.3).**

- Note that a fine sheaf possesses a decomposition of section because the previous proposition applies to  $\text{Hom}(\mathcal{F}, \mathcal{F})$ , and a partition of unity in  $\text{Hom}(\mathcal{F}, \mathcal{F})$  yields a decomposition of section in  $\mathcal{F}$ . Thus a fine sheaf is soft. (extend to a small nbhd and use partition of unity).
- The sheaf of modules over a soft sheaf of rings is soft, by partition of unity.
- The continuous function sheaf on a paracompact Hausdorff space or the smooth function sheaf on a smooth manifold is fine, thus any smooth module is fine (Use bump function).  $\lrcorner$

**Prop. (7.3.5.4).** Soft sheaf, e.g. fine sheaf is adapted to  $\Gamma(X, -)$ . (Similar as in (7.3.4.9), notice flabby is soft and the others are the same as before).  $\lrcorner$

**Prop. (7.3.5.5).** Let  $X$  be a locally compact space of finite compact dimension, when  $S$  is a soft sheaf, and one of  $S$  and  $\mathcal{F}$  is flat, then  $S \otimes_k \mathcal{F}$  is soft. Cf.[Cohomology of Sheaves Iversen P319].  $\lrcorner$

**Prop. (7.3.5.6).** Over a locally compact space of finite dimension, any flat sheaf  $\mathcal{F}$  on  $X$  has a resolution of soft flat sheaves.  $\lrcorner$

*Proof:* Cf.[Gelfand P232].  $\square$

**Lemma (7.3.5.7).** A constant sheaf on an irreducible topological space is flabby, thus flask.  $\lrcorner$

### Basics

**Prop. (7.3.5.8).** If  $i : Z \rightarrow X \in \mathcal{T}\text{op}$  is a closed immersion, then for any  $\mathcal{F} \in \text{Sh}(Z)$ ,  $H^*(Z, \mathcal{F}) = H^*(Z, i_* \mathcal{F})$ .  $\lrcorner$

*Proof:* This is because  $i_*$  is exact (7.2.6.7), so we can apply the Leray spectral sequence (7.3.1.9).  $\square$

### Comparison Theorems

**Prop. (7.3.5.9) [Singular].** For any locally contractible topological space  $X$  and  $G \in \mathcal{A}\text{b}$ , there are canonical isomorphisms

$$H_{\text{sing}}^*(X, G) \cong H^*(X, \underline{G}).$$

$\lrcorner$

*Proof:* Shiffication of the singular cochain complex is a flabby presheaf resolution of  $\underline{G}$  because it is locally contractible, check on stalks. Then we only have to prove  $C^\bullet(X) \rightarrow (C/V)^\bullet(X)$  is quasi-isomorphism, where  $V$  is the presheaf of locally vanishing cochain. It suffice to prove  $V^\bullet(X)$  is exact.

For any  $i$ -cocycle  $\varphi$ , for any  $i-1$ -complex  $\sigma$ , use barycentric subdivision, we can construct a  $c_\sigma$  whose boundary is  $\sigma$  and other simplexes on which  $\phi$  vanishes, so we have the coboundary of  $\eta : \sigma \rightarrow \varphi(c_\sigma)$  is  $\varphi$ .  $\square$

**Lemma (7.3.5.10) [Poincaré].** For a smooth manifold  $X$  of dimension  $n$ , there is an exact sequence

$$0 \rightarrow \underline{\mathbb{R}}_X \xrightarrow{d} \Omega - X^0 \xrightarrow{d} \Omega - X^1 \xrightarrow{d} \dots \xrightarrow{d} \Omega_X^n \rightarrow 0$$

┘

*Proof:*

□

**Prop. (7.3.5.11) [DeRham].** For any smooth manifold  $X$ ,

$$H_{\text{dR}}^*(X, \underline{\mathbb{R}}_X) \cong H^*(X, \underline{\mathbb{R}})$$

Where the right is constant sheaf cohomology.

┘

*Proof:* Use the fact that smooth sheaf is fine so adapted to sheaf cohomology (7.3.5.4), and Poincaré lemma (7.3.5.10). □

**Prop. (7.3.5.12) [Period Maps].** For a smooth manifold  $X$ , by a similar method as (7.3.5.9), we can define a **differentiable singular cohomology**  $H_{\text{sing},\infty}^*(X)$ , and prove a canonical isomorphism

$$H_{\text{sing},\infty}^*(X, \mathbb{R}) \cong H^*(X, \overline{\mathbb{R}}).$$

Then combining with (7.3.5.11), we get a canonical isomorphism

$$H_{\text{dR}}^*(X, \underline{\mathbb{R}}_X) \cong H_{\text{sing},\infty}^*(X, \mathbb{R})$$

which can be described as follows: there is a map of sheaves

$$\Omega^k \rightarrow C_{\text{sing},\infty}^\bullet(X)$$

that is locally defined to be  $\omega \mapsto \sigma \mapsto \int_\sigma \omega$ , then this map gives a map of complexes

$$\Omega^\bullet \rightarrow C_{\text{sing},\infty}^\bullet(X)$$

that induces the isomorphism.

┘

*Proof:* ? Cf.[Warner, P206].

□

**Prop. (7.3.5.13) [Čech and Sheaf Cohomology].** For a paracompact Hausdorff space  $X$ , there are isomorphisms

$$\check{H}^i(X, \mathbb{Z}) \cong H^i(X, \mathbb{Z}).$$

┘

*Proof:* Cf.[Godement, Prop5.10.1]. ?

□

### Cohomology with Proper Support

References are [Cohomology of Sheaves Iversen].

**Prop. (7.3.5.14).** Soft sheaf is adapted to  $f_!$  when  $X, Y$  are locally compact. Cf.[Gelfand P226]. So we can use soft resolution to define  $R^i f_!$ , in particular, when  $Y = \text{pt}$ , we denote it by  $H_c^i(X, \mathcal{F})$ . Using (7.2.6.2), we get the stalk of  $R^i f_!(\mathcal{F})$  at  $y$  is just  $H_c^i(f^{-1}(y), \mathcal{F}|_{f^{-1}(y)})$ .  $\lrcorner$

**Def. (7.3.5.15).** The **compact dimension** of a locally compact topological space is the smallest  $n$  that  $H_c^i(X, \mathcal{F}) = 0$  for  $i > n$ . It is also the maximal length of minimal soft resolution.

$\dim_c \mathbb{R}^n = n$ , and when  $Y$  is an open or closed subset of  $X$ ,  $\dim_c Y \leq \dim_c X$ .  $\dim_c$  is local in the sense if every point has a nbhd of dimension  $\leq n$ , then  $\dim_c X \leq n$ . Cf.[Iversen].  $\lrcorner$

**Prop. (7.3.5.16)[Proper Pushforward Commutes with Pullback].** For a pullback diagram

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{\tau'} & X \\ \downarrow \pi' & & \downarrow \pi \\ Z & \xrightarrow{\tau} & Y \end{array}$$

we have  $\tau^{-1}(\pi_! \mathcal{F}) = \pi'_!(\tau')^{-1} \mathcal{F}$ .  $\lrcorner$

*Proof:*  $\square$

### Cohomology on Noetherian Spaces

There are three basic objects, the derived functor for  $f_*$  as an Abelian sheaf,  $f_*$  as a  $\mathcal{O}_X$ -module,  $\Gamma(U, -)$  as an Abelian sheaf. Notice that an Abelian group is just a  $\mathbb{Z}$ -module.

**Prop. (7.3.5.17)[Grothendieck Vanishing].** The sheaf cohomology of an Abelian sheaf over a Noetherian topological space of dimension  $n$  vanish for  $k > n$ .  $\lrcorner$

*Proof:* Use (7.2.6.10) and (7.3.5.8) and long exact sequence, we can reduce to the case of  $X$  irreducible. Then we induct on dimension. Notice first any sheaf is a filtered colimits of sheaf generated by f.m sections, thus we can use (7.3.1.14) to reduce to f.m sections case. And notice  $\mathcal{F}_{\alpha'} \rightarrow \mathcal{F}_{\alpha} \rightarrow \mathcal{G}$ , then  $G$  is generated by at most  $|\alpha| - |\alpha'|$  elements, so reduce to the one section case.

Now it is a quotient sheaf of  $\mathbb{Z}$ , look at the kernel  $R$ . If the kernel is  $d\mathbb{Z}$  at the generic pt, then  $R|_V \cong \mathbb{Z}$  on some nbhd, and  $R|_V/\mathbb{Z}$  supports on a lower dimension set, then we only need to consider the pushout of constant sheaf  $\mathbb{Z}_U$ .

Now there is an exact sequence  $0 \rightarrow \mathbb{Z}_U \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}_Y \rightarrow 0$  (7.2.6.10),  $\mathbb{Z}$  is flabby (7.3.5.7) so flask, and the conclusion follows by induction.

Cf.[Sta]02UU. ?  $\square$

**Prop. (7.3.5.18).** For  $f : X \rightarrow Y$ , if  $\mathcal{I}$  is an injective module on  $X$ , then  $\check{H}^p(\{U_i \rightarrow U\}, f_* \mathcal{I}) = 0$  for every open cover for an open subset  $U$  (7.3.4.8). This is because Čech cohomology is a derived functor. (Notice  $f_* \mathcal{I}$  may not be injective when  $f$  is not flat).  $\lrcorner$

**Cor. (7.3.5.19)[Mayer-Vietoris].** For  $X = U \cup V$ , there is a long exact sequence

$$0 \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(U, \mathcal{F}) \oplus H^0(V, \mathcal{F}) \rightarrow H^0(U \cap V, \mathcal{F}) \rightarrow H^1(X, \mathcal{F}) \rightarrow \dots$$

derived from the Čech to sheaf because it has only two column, just wrap out the definition.  $\lrcorner$

## 7.4 Properties of Schemes

*“Algebra is the offer made by the devil to the mathematician...All you need to do, is give me your soul: give up geometry.”*

–M. Atiyah

References are [Har77, Tin20, Sta].

**Notation (7.4.0.1).**

- Let  $X, S \in \text{Sch}$ .

┘

### 1 Basic Scheme Properties

#### Affine Local Properties of Schemes

**Lemma (7.4.1.1) [Nike’s Trick].** For  $X \in \text{Sch}$  and two affine open subschemes  $U, V \subset X$ . If  $x \in U \cap V$ , then  $x$  has an open nbhd in  $U \cap V$  that are distinguished in both  $U$  and  $V$ . ┘

*Proof:* Choose a nbhd of  $x$  that is distinguished in  $\text{Spec } A$  that is contained in  $\text{Spec } A \cap \text{Spec } B$ , then because distinguished of distinguished is distinguished, we may assume  $i : \text{Spec } A \subset \text{Spec } B$ . Now let  $f \in B$  be an element that  $D(f) \subset \text{Spec } A$ , then I claim  $D(i^\#(f)) = D(f)$ , this will finish the proof, but this is equivalent to  $i^{-1}(\text{Spec } B_f) = \text{Spec } A_{i^\#(f)}$ , which is true for ideal-theoretical reason. ┐

**Prop. (7.4.1.2) [Affine Communication Theorem].** A property  $P$  of affine open subsets is called **affine local** if:  $\text{Spec}(A)$  has  $P \Rightarrow$  all  $\text{Spec}(A_f)$  has  $P$ , and any cover of  $\text{Spec}(A_{f_i})$  has  $P \Rightarrow \text{Spec}(A)$  has  $P$ . Notice a stalk-wise property is obviously affine-local.

Now if we call  $X$  has  $\tilde{P}$  if  $X = \bigcup_i \text{Spec } A_i$  that  $A_i$  has  $P$ . Then the following is equivalent:

- any open affine subscheme of  $X$  has  $P$ .
- any open subscheme of  $X$  has  $\tilde{P}$ .
- $X$  has a cover of open subschemes that has  $\tilde{P}$ .
- $X$  has  $\tilde{P}$ .

┘

*Proof:*  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$  is obvious. It suffices to prove  $4 \rightarrow 1$ : if  $X = \bigcup \text{Spec } A_i$ , for any open affine subscheme of  $X$ , by (7.4.1.1), it can be covered by distinguished opens that are also distinguished in some  $\text{Spec } A_i$ , so by hypothesis it has  $P$ . ┐

**Remark (7.4.1.3).** When proving locality of morphism properties using affine communication theorem, one usually resort to [Local Properties](#). ┘

**Prop. (7.4.1.4) [List of Stalkwise Properties].** All properties defined by a stalkwise ring-theoretic property that is stalkwise. (5.1.4.2) ┘

**Prop. (7.4.1.5) [List of properties affine local on the target].** (All the property besides the  $H$ -projectiveness is local on the target).

1. Because affineness is local on the target (7.1.5.26), all properties defined by a ring-theoretic property local on the target is local on the target (5.1.4.4).

2. All properties that is stalkwise.
3. All properties that satisfies faithfully flat descent.(7.1.5.26)
4. Locally projective morphism.

┘

**Prop. (7.4.1.6)** [List of properties affine local on the source]. (not complete)

1. All properties defined to be local ring map property local on the source.(5.1.4.4)
2. Openness.

┘

*Proof:*

1. Trivial.
2. Trivial.

□

### Irreducible

**Def. (7.4.1.7)** [Irreducible Schemes]. A scheme is called **irreducible** iff its underlying topological space is irreducible.

┘

**Prop. (7.4.1.8)** [Nearly Affine Local]. For a scheme, the following are equivalent:

1. It is irreducible.
2. There is an affine cover  $U_i$  of  $X$  that  $U_i$  are all irreducible and  $U_i \cap U_j \neq \emptyset$ .
3. Every affine open subset of  $X$  is irreducible.

┘

*Proof:* As scheme is sober(7.2.7.11), if  $X$  is irreducible, then  $X$  has a unique generic pt  $\eta$  that  $\overline{\{\eta\}} = X$ , then 2, 3 all holds. If 2 holds, then for a decomposition  $X = Z_1 \cup Z_2$ , any  $U_i$  belongs to  $Z_1$  or  $Z_2$ , so it is easy to see  $Z_1 = X$  or  $Z_2 = X$ . If 3 holds, then choose an affine cover  $U_i$  of  $X$ , then  $U_i \cap U_j \neq \emptyset$ , otherwise  $U_i \amalg U_j$  is affine and not irreducible, contradiction, so 2 holds. □

**Cor. (7.4.1.9).** The fiber product of irreducible schemes is irreducible, because .

┘

### Reducedness

**Def. (7.4.1.10)** [Reduced Schemes].  $X$  is called **reduced scheme** if  $\mathcal{O}_X(U)$  is reduced for every open set  $U$ . Reduced is a stalk-wise property(7.4.1.5), it suffices to check reducedness at closed pts.

The full subcategory of reduced schemes is denoted by  $\text{Sch}_{\text{red}}$ .

┘

**Prop. (7.4.1.11).** For  $X \in \text{Sch}_{\text{red}}$ , the map  $\Gamma(X, \mathcal{O}_X) \rightarrow \prod_{x \in X} k(x)$  is injective.

┘

**Prop. (7.4.1.12).** For  $X \in \text{NSch}$ (7.4.1.20), the set of points with reduced stalk is open in  $X$ .

┘

*Proof:* This set is just the set of points  $x$  that  $\mathcal{N}_x = 0$ , where  $\mathcal{N}$  is the sheaf of nilradicals, which is coherent, so it has closed supports(7.5.1.38). □

**Prop. (7.4.1.13)[Reduction].** For any  $X \in \text{Sch}$ , define  $X_{\text{red}} = \text{Spec}(\mathcal{O}_X/\mathcal{N}) \in \text{Sch}/X$  where  $\mathcal{N}$  is the sheaf of nilpotent elements. This construction is right adjoint to the forgetful functor by the adjoint property of **Spec**(7.2.7.12).  $X_{\text{red}} \rightarrow X$  is a closed immersion.

It's useful to change to  $X_{\text{red}}$  when the proposition only involve topology because  $X_{\text{red}}$  has the same topology as  $X$ . A map can induce a map on their reduced structure.  $\lrcorner$

**Prop. (7.4.1.14)[Induced Reduced Scheme Structure].** If  $X \in \text{Sch}$  and  $Z \subset |X|$  is a locally closed subset, then there is a unique reduced subscheme  $Z_{\text{red}}$  of  $X$  with underlying topological space  $Z$ , called the **induced reduced scheme structure** of  $Z$ . It has the universal property that any morphism from a reduced scheme  $Y$  to  $X$  that has image in  $Z$  factors through this subscheme (By virtue of reducedness).

In particular, there is a closed subscheme structure  $X_{\text{red}}$  of  $X$ , called the **underlying reduced subscheme** of  $X$ .  $\lrcorner$

*Proof:* The uniqueness is clear by the universal property. The existence is clear when  $X$  is affine and  $Z$  is closed in  $X$ . Then we can use the uniqueness to glue them to a global subscheme structure.  $\square$

### Integral

**Def. (7.4.1.15)[Integral Schemes].** A scheme  $X$  is called **integral** if  $\mathcal{O}_X(U)$  are all integral. This is equivalent to reduced and irreducible. So a scheme is integral iff there is an integral open affine cover that are pairwise-intersect(7.4.1.7).

The category of integral schemes is denoted by  $\text{Sch}^{\text{int}}$ .  $\lrcorner$

*Proof:* If  $X$  is irreducible and reduced, then so does any affine subscheme  $\text{Spec } R$ , so  $R$  is integral as  $(0)$  is the generic prime, because it has only one minimal prime consisting of nilpotent elements. Conversely, if  $X$  is reduced, then any affine subscheme  $\text{Spec } R$  is integral so reduced, and is irreducible by the presence of prime  $(0)$ .  $\square$

**Cor. (7.4.1.16).** The projective space over an integral scheme is integral. (Check the affine covers are dense). The projective space  $P_{\mathbb{Z}}^n$  is integral.  $\lrcorner$

**Prop. (7.4.1.17)[Integral is Almost Stalkwise].** Let  $X$  be a non-empty and connected scheme, then  $X$  is integral iff all the  $\lrcorner$

**Def. (7.4.1.18)[Function Field].** Let  $X$  be an integral scheme with generic point  $\eta$ , then  $R(X) \cong \mathcal{O}_{X,\eta}$  is a field(7.2.8.2), called the **function field of  $X$** , denoted by  $K(X)$ . Then any rational function on  $X$  is defined on an open dense subset of  $X$ .  $\lrcorner$

**Prop. (7.4.1.19).** If  $X$  is an integral scheme and  $Z_1, Z_2$  are closed subschemes of  $X$  with generic points  $\eta_1, \eta_2$ , then  $\mathcal{O}_{X,\eta_1} \not\subseteq \mathcal{O}_{X,\eta_2}$ . In particular, if  $Z = \{x\}$  consists of a closed point, then there is a rational function defined near  $x$  that is not in  $\mathcal{O}_{X,\eta_2}$ .  $\lrcorner$

*Proof:* [Sta]02NF.  $\square$

### Noetherian

**Def. (7.4.1.20)[Noetherian Scheme].** A scheme is called **locally Noetherian** if it can be covered by open affine spectra of noetherian rings. It is called **Noetherian** if moreover it is quasi-compact.

(Locally)Noetherian is affine local(7.4.1.5).

Denote the full subcategory of locally Noetherian schemes by  $\text{NSch}$ .  $\lrcorner$

**Prop. (7.4.1.21) [Noetherian Scheme is Noetherian].** The underlying space of a Noetherian scheme is a Noetherian space.  $\lrcorner$

*Proof:* By (4.12.2.3), we are reduced to the affine case. Now it is clear from the definition.  $\square$

**Prop. (7.4.1.22).** Any locally closed subscheme of a (locally) Noetherian scheme is (locally) Noetherian. In particular, an subset of a Noetherian scheme is quasi-compact.  $\lrcorner$

*Proof:* This is because any localization and quotients of a Noetherian ring is Noetherian (5.1.1.43), and any subset of a Noetherian space is quasi-compact (4.12.2.2)(7.4.1.21).  $\square$

**Prop. (7.4.1.23) [Finitely Many Irreducible Components].** For a closed subscheme in a locally Noetherian space, the collection of its irreducible components is locally finite in  $X$ , because a Noetherian space has f.m. irreducible components (4.12.2.4).

In particular, a locally Noetherian space is locally connected.  $\lrcorner$

**Prop. (7.4.1.24).** Let  $k'/k$  be a f.g. field extension, then a scheme  $X$  over  $k$  is locally Noetherian iff  $X_{k'}$  is locally Noetherian.  $\lrcorner$

*Proof:* Locally Noetherian is affine local, so the problem is totally ring-theoretic. If  $X_{k'}$  is Noetherian, then so does  $X$  by ff descent (5.4.2.1). If  $X$  is Noetherian, then so does  $X_{k'}$  by (5.1.1.47).  $\square$

### Jacobson

**Def. (7.4.1.25).** An scheme is called **Jacobson** iff its underlying topological space is Jacobson (4.12.2.22). In particular, an affine scheme  $\text{Spec } R$  is Jacobson iff  $R$  is Jacobson (5.2.6.5).

So by (4.12.2.23), being Jacobson is a local property.  $\lrcorner$

**Prop. (7.4.1.26) [Locally Algebraic Scheme is Jacobson].** For a scheme locally algebraic over a field  $k$ , the set of closed points  $X_0$  is dense in every closed subset of  $X$ , Because it is a Jacobson space by (4.12.2.23) and (5.2.6.10). Equivalently, every locally closed subset of  $X$  contains a closed point.

Moreover, the residue field of a closed point is finite over  $k$  by (5.2.6.10), and the converse is also true. In particular, by (7.2.7.9), the closed points of  $X$  are just the geometric points.  $\lrcorner$

*Proof:* For the converse, because  $k \subset A/\mathfrak{p}_x \subset k(x)$  are finite hence integral extensions, by (5.2.1.3)  $A/\mathfrak{p}_x$  is a field, thus  $x$  is a closed point.  $\square$

**Cor. (7.4.1.27) [Algebraic Scheme Preserves Closed Points].** A morphism between algebraic schemes over a field  $k$  maps closed points to closed points.  $\lrcorner$

**Remark (7.4.1.28).** When  $X$  is geo.reduced, a stronger statement shows the set of separable closed points of  $X$  is dense in  $X$ , Cf. (7.4.3.3).  $\lrcorner$

**Cor. (7.4.1.29) [Check Surjectiveness on Closed Points].** If a morphism of locally algebraic schemes over a field  $k$  is surjective on closed points, then it is surjective.  $\lrcorner$

*Proof:* By Chevalley theorem (7.6.1.5), the image is a locally constructible set, thus the supplement set is also locally constructible. Now if it is not surjective, then there is an open closed subset  $U \cap Z$  not in the image. But this set contains a closed point by (7.4.1.26), which is a contradiction.  $\square$

### Cohen-Macaulay

**Def. (7.4.1.30) [Cohen-Macaulay Schemes].** A **Cohen-Macaulay scheme** of a C.M. scheme is a scheme  $X \in \text{Sch}$  s.t. all its stalks is C.M. local. Thus being C.M. is a stalkwise property.  $\lrcorner$

### Dimensions and Catenary Schemes

**Def. (7.4.1.31) [Dimensions of Schemes].** For  $x \in X$ , define

$$\dim(X) \triangleq \dim(|X|), \quad \dim_x(X) \triangleq \dim_x(|X|) \quad (4.12.2.26)$$

$\lrcorner$

**Prop. (7.4.1.32).** We have  $\dim(X) = \sup_{x \in X} \dim(\mathcal{O}_{X,x}) = \sup_{x \in X} \dim_x(X)$  (7.4.1.31).  $\lrcorner$

*Proof:* Cf. [Sta]04MU.  $\square$

**Prop. (7.4.1.33).** If  $Y \subset X$  is an irreducible closed subset with generic point  $\xi$ , then  $\text{codim}(Y, X) = \dim(\mathcal{O}_{X,\xi})$  (4.12.2.30).  $\lrcorner$

*Proof:* By (4.12.2.31), we may assume that  $X$  is affine, then the assertion follows from [Sta]00ET?  $\square$

**Prop. (7.4.1.34) [Separated Schemes of Dimension 0, Gabber2016].** If  $\dim(X) = 0$ , then the following are equivalent:

- $X$  is quasi-separated.
- $X$  is separated.
- $X$  is Hausdorff.
- Every affine open subsets of  $X$  is closed.

And in this case, each connected components of  $X$  is a point.  $\lrcorner$

*Proof:* Cf. [Sta]0CKV.  $\square$

**Prop. (7.4.1.35) [Noetherian Schemes of Dimension 0].** If  $X \in \text{NSch}$  and  $\dim(X) = 0$ , then  $X$  is a disjoint union of spectra of Artinian local rings.  $\lrcorner$

*Proof:* Cf. [Sta]0AAX. ?  $\square$

**Def. (7.4.1.36) [Catenary Schemes].** A **catenary scheme** is a scheme that its underlying space is catenary (4.12.2.32). A **universally catenary scheme** is a scheme  $S \in \text{NSch}$  s.t. any  $X \in \text{Sch}^{\text{loc.ft}}/S$  is catenary.

(Universally)catenary is a stalkwise property, by (5.1.4.2).  $\lrcorner$

**Prop. (7.4.1.37) [Catenary and Dimension Functions].** Let  $X \in \text{Sch}$  be locally Noetherian, then  $X$  is catenary iff locally around every point there is a dimension function, by (4.12.2.41).  $\lrcorner$

**Example (7.4.1.38) [Universally Catenary Schemes].** The following are some examples of universally catenary schemes, by (5.2.4.7).

- A scheme locally of f.t. over a universally catenary scheme.
- A C.M. scheme.
- Spectrum of a 1-dimensional Noetherian domains,
- Spectrum of a fields.

$\lrcorner$



### Japanese & Nagata Schemes

**Def. (7.4.1.39) [Japanese & Nagata Schemes].** A (universally) **Japanese scheme** is a scheme that can be covered by open affine spectrum of (universally) Japanese rings (5.3.8.1). A **Nagata scheme** is a scheme that can be covered by open affine spectrum of Nagata rings (5.3.8.1).  $\lrcorner$

**Prop. (7.4.1.40).** Being (universally) Japanese or Nagata is a local property, and are stable under taking open subschemes, by (5.3.8.1).  $\lrcorner$

**Prop. (7.4.1.41).** Let  $X$  be a locally Noetherian scheme, then  $X$  is Nagata iff every integral closed subscheme  $Z$  of  $X$  is Japanese.  $\lrcorner$

*Proof:* One direction is clear. If every integral closed subscheme  $Z$  of  $X$  is Japanese, let  $U = \text{Spec } A \subset X$  be an affine open subscheme and  $Z = V(\mathfrak{p}) \subset U$ , we need to show  $A/\mathfrak{p}$  is Japanese. Consider  $\bar{Z}$  with the reduced induced structure is an integral closed subscheme of  $X$  and  $Z$  is an open subscheme of  $\bar{Z}$ , thus  $\bar{Z}$  is Japanese and so is  $Z$  by (7.4.1.39).  $\square$

## 2 Normal & Regular

**Def. (7.4.2.1) [Normal & Regular Schemes].** A scheme is called **normal** if all its stalks are normal domains (5.3.5.1), or equivalently all its affine sections are normal rings by (5.3.5.1). In particular, a normal scheme is reduced.

A locally Noetherian scheme is called **regular** iff all its stalks are regular local rings (5.3.5.18), i.e. all affine opens are regular rings. Regular only have to be checked at close pt by (5.3.5.18).  $\lrcorner$

**Def. (7.4.2.2) [Non-normal Loci].** For  $X \in \text{Sch}$ , the set of points  $x \in X$  s.t.  $\mathcal{O}_{X,x}$  is a closed subset by (5.3.5.1). Its induced reduced structure is called the **non-normal locus** of  $X$ .  $\lrcorner$

**Def. (7.4.2.3) [Singular Loci].** For  $X \in \text{NSch}$ , the set of points  $x \in X$  s.t.  $\mathcal{O}_{X,x}$  is a regular local ring is called the **regular locus** of  $X$ , and the complement of the regular locus is called the **singular locus** of  $X$ .  $\lrcorner$

**Prop. (7.4.2.4) [Noetherian and Integral].** For  $X \in \text{NSch}$ ,  $X$  is normal iff it is a disjoint union of integral normal schemes.  $\lrcorner$

*Proof:* It suffices to show that a connected locally Noetherian scheme is integral. Thus it suffices to show it is irreducible. Suppose there are several irreducible components, and let  $p$  be a intersection point, then we may assume  $X$  is affine, then this follows from (5.3.5.4).  $\square$

**Cor. (7.4.2.5).** A normal scheme is integral iff it is connected.  $\lrcorner$

**Prop. (7.4.2.6).** If  $X$  is an integral normal scheme, then  $\Gamma(X, \mathcal{O}_X)$  is a normal ring.  $\lrcorner$

*Proof:* If  $X$  is integral, then  $R = \Gamma(X, \mathcal{O}_X)$  is integral. If  $f = a/b \in K(R)$  that is integral over  $R$ , then for any affine open  $U \subset X$ ,  $b|_U$  is non-zero as  $X$  is integral, thus  $f|_U$  is integral over  $\mathcal{O}_X(U)$  thus  $f|_U \in \mathcal{O}_X(U)$ , thus  $f \in R$ .  $\square$

**Prop. (7.4.2.7) [Normalization].** For an integral scheme  $X$ , there is a  $X_{\text{nom}} \rightarrow X$  which is  $\text{Spec}(\mathcal{O}_{X, \text{nom}})$ , any dominant morphism  $f$  from a normal integral scheme to  $X$  will factor through  $X_{\text{nom}}$ . (Use the adjointness for  $\text{Spec}$  and notice  $f$  maps generic to generic).  $\lrcorner$

*Proof:*

$\square$

**Prop. (7.4.2.8) [Normalizations are Birational].** The normalization of an integral scheme  $X$  is a finite birational map.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (7.4.2.9) [Normalization is not flat].** Non-trivial normalization is never flat.  $\lrcorner$

*Proof:* Use (7.6.2.6). ?  $\square$

**Prop. (7.4.2.10) [Regular and Normal].** Regular scheme is C.M and locally factorial, hence normal, by (5.3.5.22) and (5.3.5.20). A Normal scheme is regular in codimension 1, by (5.3.5.33).  $\lrcorner$

**Cor. (7.4.2.11).** A regular connected scheme  $X$  is irreducible, by (7.4.2.5).  $\lrcorner$

**Prop. (7.4.2.12).** For a locally Noetherian scheme of dimension  $\leq 1$ , normal is equivalent to regular, by (12.2.3.5).  $\lrcorner$

### Factorial Schemes

**Def. (7.4.2.13) [Factorial Schemes].** A **factorial scheme** is a scheme that all the local rings are UFD.  $\lrcorner$

**Prop. (7.4.2.14).** If  $A$  is a UFD, then  $\text{Spec } A$  is factorial.  $\lrcorner$

### Dedekind Scheme

**Def. (7.4.2.15) [Dedekind Scheme].** A **Dedekind scheme** is an integral Noetherian normal scheme of dimension 1.  $\lrcorner$

## 3 Geometrical properties

**Def. (7.4.3.1) [Geometric Properties over Fields].**

- A scheme  $X$  is called **geometrically integral/reduced/separated/irreducible**... over a field  $k$  iff for any field extension  $k'/k$ ,  $X_{k'}$  is integral/reduced/separated/...
- A locally Noetherian scheme is called **geometrically regular** iff for any f.g. field extension  $K/k$ ,  $X_K$  is regular. It is stalkwise by (7.4.3.18).  $\lrcorner$

### Geo.reducedness

**Prop. (7.4.3.2) [Geo.Reduced].** For a scheme  $X$  over a field  $k$ , the following are equivalent:

1.  $X$  is geometrically reduced.
2. For every reduced  $k$ -scheme  $Y$ , the product  $X \otimes_k Y$  is reduced.
3. All stalks are geometrically reduced ring.
4.  $X$  is reduced and for every maximal point  $\eta$  of  $X$ , the residue field  $k(\eta)$  is separable over  $k$ .
5.  $X_{k^{\text{perf}}}$  is reduced.
6.  $X_K$  is reduced for every finite purely inseparable field extension  $K/k$ .
7.  $X_{k^{1/p}}$  is reduced.

In particular, if  $k$  has characteristic 0, then  $\text{geo.reduced} \iff \text{reduced}$ .  $\lrcorner$

*Proof:* As reduced is local, these all follows from (5.3.6.2).  $\square$

**Prop. (7.4.3.3) [Geo.Reduced and Arithmetic Points].** Let  $X \in \text{Sch}^{\text{loc.ft}}/k$  be geo.reduced, then the set of closed points with finite separable field extensions  $\kappa(x)/k$  is dense in  $X$ .  $\lrcorner$

*Proof:* Combine (7.6.4.18) and (7.6.4.20).  $\square$

**Def. (7.4.3.4) [Density of Points].** Let  $X$  be an algebraic scheme over a field  $k$  and  $k'/k$  be a field extension, then a subset  $S \subset X(k')$  is said to be **schematically dense** in  $X$  if the only closed subscheme  $Z \subset X$  over  $k$  that  $S \subset Z(k')$  is  $X$  itself.

$\color{red}{?}$  Should merge with the definition of scheme-theoretical image.  $\lrcorner$

**Prop. (7.4.3.5) [Schematically Dense Subset].** Let  $X$  be an algebraic scheme over a field  $k$ ,  $S \subset X(k)$  be a subset. Then the following are equivalent:

1.  $S$  is schematically dense in  $X$ .
2.  $X$  is reduced and  $S$  is dense in  $|X|$ .
3. The family of homomorphisms  $\mathcal{O}_X \rightarrow k : f \mapsto f(s)$  is jointly injective.

$\lrcorner$

*Proof:*  $1 \rightarrow 2$ : Let  $\bar{S}$  be the induced reduced structure of the closure  $S$  in  $X$  (7.4.1.14), then  $\bar{S} = X$ , so  $X$  is reduced with  $X = \bar{S}$ .

$2 \rightarrow 3$ :  $\color{red}{?}$  Cf. [Milne Algebraic Groups, P10].  $\square$

**Cor. (7.4.3.6).** A schematically dense subset remains schematically dense after field base changes.  $\lrcorner$

**Cor. (7.4.3.7).** The schematic closure of a subset commutes with base change.  $\lrcorner$

*Proof:* Use the third definition in (7.4.3.5), notice that the valuation maps are also jointly injective because  $k'/k$  is flat.  $\square$

**Cor. (7.4.3.8).** If  $X$  admits a schematically dense subset  $S$ , then  $X$  is geo.reduced.  $\lrcorner$

**Prop. (7.4.3.9).** If  $X(k')$  is dense in  $X$ , then  $X$  is reduced. Conversely, if  $X(k')$  is dense in  $|X_{k'}|$  and  $X$  is geo.reduced, then  $X(k')$  is dense in  $X$ .  $\lrcorner$

*Proof:*  $X$  is reduced because  $X_{\text{red}}(k') = X(k')$ . Conversely if  $Z \subset X$  is a closed subscheme that  $Z(k') = X(k')$ , then  $|Z_{k'}| = |X_{k'}|$  by condition, and then  $Z_{k'} = X_{k'}$  as  $X_{k'}$  is reduced. Thus  $Z = X$  by flatness.  $\square$

**Cor. (7.4.3.10) [Geometric Points Schematically Dense].** If  $X$  is locally algebraic and geo.reduced, then  $X(k')$  is schematically dense in  $X$  for any separably closed field  $k'$  containing  $k$ .  $\lrcorner$

*Proof:*  $X(k')$  is dense in  $|X_{k'}|$  by (7.4.3.3), thus it is schematically dense in  $X$  by (7.4.3.9).  $\square$

**Cor. (7.4.3.11).** If  $Z, Z'$  are closed subvarieties of  $X \in \text{Sch}^{\text{loc.ft}}/k$  s.t.  $Z(k') = Z'(k') \subset X(k')$  for some separably closed field  $k'$  containing  $k$ , then  $Z = Z'$ . In other words, a closed subvariety of  $X$  is determined by the subset  $Z(k^s) \subset X(k^s)$ .  $\lrcorner$

*Proof:* The closed subscheme  $Z \cap Z'$  satisfies  $Z \cap Z'(k') = Z(k')$ , so  $Z \cap Z' = Z$  by (7.4.3.10). Similarly  $Z \cap Z' = Z'$ .  $\square$

Geo.Connected and Geo.Irreducible

**Prop. (7.4.3.12) [Geo.Connectedness].** For a scheme  $X$  over a field  $k$ , the following are equivalent:

- For every connected  $k$ -scheme  $Y$ , the product  $X \otimes_k Y$  is connected.
- $X$  is geometrically connected (7.4.3.1).
- $X_{K^s}$  is connected.
- $X_K$  is connected for any finite separable extension  $K/k$ .

┘

*Proof:* Cf. [Sta]0385, 0389. ?

□

**Prop. (7.4.3.13) [Invariance of Base Change].** Let  $X$  be a scheme over a field  $k$  and  $k'/k$  a field extension, then  $X$  is geo.connected iff  $X_{k'}$  is geo.connected.

┘

*Proof:* Cf. [Sta]054N.

□

**Prop. (7.4.3.14).** Let  $T \rightarrow X$  be a map of schemes over a field  $k$ , if  $T$  is geo.connected and  $X$  is connected, then  $X$  is geo.connected.

┘

*Proof:* Cf. [Sta]056R.

□

**Cor. (7.4.3.15) [Connected with a Rational Point].** Let  $X$  be a scheme over a field  $k$ . Assume  $X$  is connected and has a point  $x$  that  $k$  is alg.closed in  $k(x)$ , then  $X$  is geo.connected. In particular, if  $X$  is connected and has a rational point, then  $X$  is geo.connected.

┘

*Proof:* Cf. [Sta]04KV.

□

**Prop. (7.4.3.16) [Geometrically Irreducible].** For a scheme  $X$  over a field  $k$ , the following are equivalent:

- For every irreducible  $k$ -scheme  $Y$ , the product  $X \otimes_k Y$  is irreducible.
- $X$  is geometrically irreducible.
- $X_{k^s}$  is irreducible.
- $X$  is irreducible and if  $\eta$  is the generic pt of  $X$ , then  $k$  is separably closed in  $k(\eta)$ .
- $X_K$  is irreducible for any finite separable extension  $K/k$ .

┘

*Proof:* Cf. [Gortz P136]. ?

□

Geo.Integral

**Cor. (7.4.3.17) [geometrically Integral].** For a scheme  $X$  over a field  $k$ , the following are equivalent:

- For every integral  $k$ -scheme  $Y$ , the product  $X \otimes_k Y$  is integral.
- $X$  is geometrically integral.
- $X_K$  is irreducible for any finite extension  $K/k$ .
- $X_{\bar{k}}$  is integral.
- $X$  is integral and if  $\eta$  is the generic pt of  $X$ , then  $k$  is alg.closed in  $k(\eta)$  and  $k(\eta)/k$  is separable.

┘

*Proof:*  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$  is easy, Cf. [Gortz P136]. ?

□

### Geometrically Regular

**Prop. (7.4.3.18).** Let  $X$  be a locally Noetherian scheme over a field  $k$ , then  $X$  is geometrically regular iff the local ring  $\mathcal{O}_{X,x}$  is geometrically regular over  $k$ . Thus it suffice to check for finite purely inseparable field extensions  $k'/k$ , by (5.3.6.7).  $\lrcorner$

*Proof:* For a finite purely inseparable field extension,  $\mathcal{O}_{X,x} \otimes_k k'$  is also a local ring because their spectra are the same (5.1.7.26), so  $\mathcal{O}_{X,x}$  is geometrically regular by (5.3.6.7).

Conversely, if  $\mathcal{O}_{X,x}$  is geo.regular, then for any field extension  $k'/k$ , stalks of  $X_{k'}$  are localization of  $\mathcal{O}_{X,x} \otimes_k k'$ , so it is regular by (5.3.5.18).  $\square$

**Cor. (7.4.3.19).** A geometrically regular ring is geometrically reduced, by (5.3.5.20) and (7.4.3.2).  $\lrcorner$

**Prop. (7.4.3.20) [Partially Invariance Under Base Change].** If  $k'/k$  is a f.g. field extension, then  $X_{k'}$  is geo.regular over  $k'$  iff  $X_k$  is geo.regular over  $k$ .

In fact, this is true for any field extension  $k'/k$  if  $X$  is locally algebraic over  $k$ , because in this case geo.regular is equivalent to smoothness.  $\lrcorner$

*Proof:* One direction is trivial, for the other, Cf. [Sta]038W<sup>?</sup>.  $\square$

## 4 Basic Morphism Properties

Main references are [Sta]02WE.

### Base Change Trick

**Prop. (7.4.4.1) [Base Change Trick].** If a property  $P$  of morphisms satisfy:

- Closed immersion has  $P$ .
- Stable under base change and composition.

Then

- it is stable under product.
- $g \circ f : X \rightarrow Y \rightarrow Z$  has  $P$  and  $g$  separated implies  $f$  has  $P$ .
- it is stable under  $f_{\text{red}}$ . (Notice  $X_{\text{red}} \rightarrow X$  is closed immersion).

$\lrcorner$

*Proof:* For the product, we may assume one of them is identity and use composition, but then the product is just base change, so it has  $P$ .

For the second, factorize  $f : X \rightarrow X \times_Z Y \rightarrow Y$ , the first is base change of  $\Delta : Y \rightarrow Y \times_Z Y$ , so it satisfies  $P$  because  $g$  is separable, and the second map is a base change of  $X \rightarrow Z$ , so it satisfies  $P$ , so  $f$  satisfies  $P$ .

$X_{\text{red}} \rightarrow X \rightarrow Y = X_{\text{red}} \rightarrow Y_{\text{red}} \rightarrow Y$  has  $P$  because  $X_{\text{red}} \rightarrow X$  is closed immersion, and  $Y_{\text{red}} \rightarrow Y$  is separable because closed immersion is separable (checked directly), so by what has been proved,  $X_{\text{red}} \rightarrow Y_{\text{red}}$  has  $P$ .  $\square$

**Prop. (7.4.4.2).** Lists of properties satisfying the base change trick (7.4.4.1) (not complete):

1. Universal closed/universal injective morphisms.
2. Affineness.
3. Morphisms (locally) of finite type.

4. Finite Morphisms.
5. Integral Morphisms.
6. Morphisms (locally) of finite presentation.
7. Quasi-affine morphisms.
8. Closed Immersions.
9. Quasi-compactness.
10. (Quasi-)Separatedness.
11. Proper.
12. Unramified.
13. Monomorphisms.
14. (Locally) Quasi-finiteness.
15. H-projectiveness.

┘

*Proof:*

1. Trivial.
2. Because affineness is local on the target (7.4.1.5), this follows from (7.2.7.17) and (7.2.7.14).
3. Trivial.
4. Trivial.
- 5.
6. By (5.3.7.9).
7. Affine morphism is quasi-compact: because quasi-compactness is local on the target, we can reduce to the affine case, thus it is quasi-compact, by (7.4.4.26). To show a base change of quasi-compact morphism is quasi-compact, because quasi-compactness is local on the target (7.4.1.5), then we can choose a cover by affine opens that the image is contained in an affine open, thus it reduces to show a map between affine schemes is quasi-compact, which is (7.4.4.26).
8. For closed immersions, use (7.2.7.16) and check locally, for open immersions, use (7.2.7.16).
9. It suffices to show an affine map is quasi-compact.
10. Closed immersion is separated is checked directly. Composition: For  $X \rightarrow Y \rightarrow Z$ , the diagonal map decomposes as  $X \rightarrow X \times_Y X \rightarrow X \times_Z X$ , the second one is closed immersion (or quasi-compact) by (7.4.5.29), so this follows from that of closed immersion and qc. Base change: The diagonal commutes with base change (4.1.1.49), so this follows from that of closed immersion and qc.
11. Because universally closed, f.t. and separatedness both do (7.4.4.2).
- 12.
- 13.
- 14.

□

### Injectivity and Monomorphisms

**Def. (7.4.4.3) [Injectivity and Monomorphisms].** A morphism of schemes is called **injective morphism** if it is injective topologically. A morphism of schemes is called a **monomorphism morphism** if it is a monomorphism in the category of schemes.  $\lrcorner$

**Prop. (7.4.4.4) [Universally Injective].** For a morphism of schemes  $X \rightarrow S$ , the following are equivalent:

- It is universally injective.
- It is injective and the residue field extension are all purely inseparable.
- The diagonal map is surjective.
- For any field  $K$ ,  $\text{Hom}(\text{Spec } K, X) \rightarrow \text{Hom}(\text{Spec } K, S)$  is injective.

In particular, any monomorphism is universally injective.  $\lrcorner$

*Proof:* 1  $\rightarrow$  4: For  $s \in \text{Hom}(\text{Spec } K, S)$ , a  $x \in \text{Hom}(\text{Spec } K, X)$  mapping to  $s$  is a section of the injective map  $X \times_S \text{Spec } K \rightarrow \text{Spec } K$ , which is unique if it exists.

4  $\rightarrow$  1: If  $S' \rightarrow S$  is a morphism,  $X' = X \times_S S'$ , and  $x, x' \in X'$  map to the same point  $s' \in S'$ , then we can choose a common field extension  $K/k(s')$  of  $k(x)/k(s)$  and  $k(x')/k(s)$ . Then we get two elements in  $\text{Hom}(\text{Spec } K, X')$ , and the hypothesis says they map to the same element in  $\text{Hom}(\text{Spec } K, X)$ . Thus they are the same, as  $X'$  is the fiber product.

1  $\rightarrow$  2: If the residue field extension is not purely inseparable, then tensoring with this field extension, we get two points mapping to the same, contradiction.

2  $\rightarrow$  4: If the residue field extension is purely inseparable, then  $k(x) \rightarrow K$  is determined by the composition,  $k(s) \rightarrow k(x) \rightarrow K$ , which means 4 is true.

1  $\rightarrow$  3: If  $X \rightarrow S$  is universally injective, then  $X \times_S X \rightarrow X$  is injective, and  $\Delta_{X/S}$  is a section of this map, thus it must be surjective.

3  $\rightarrow$  1: If  $\Delta_{X/S}$  is surjective, then for any base change  $S' \rightarrow S$ ,  $\Delta_{X'/S'}$  is surjective, thus  $X' \rightarrow S'$  is injective, thus 1 holds.  $\square$

**Prop. (7.4.4.5) [Monomorphism and Injectivity].** A morphism of schemes  $j : X \rightarrow Y$  is a monomorphism if  $j$  is an injective map and for any  $x \in X$ ,  $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$  is surjective.  $\lrcorner$

*Proof:* First check topologically, then check the local ring map.  $\square$

**Cor. (7.4.4.6).** For any scheme  $X$  and a point  $x$ ,  $\text{Spec } \mathcal{O}_{X,x} \rightarrow X$  is a monomorphism.  $\lrcorner$

### Closed Map

**Prop. (7.4.4.7) [Universal Closed].** Universal closedness is local on the basis and satisfies the base change trick(7.4.4.2).  $\lrcorner$

**Prop. (7.4.4.8).** If  $g$  is surjective, then  $f \circ g$  is universally closed iff  $f$  is universally closed (because surjective is S.u.B).  $\lrcorner$

**Prop. (7.4.4.9) [Closed Map and Specialization].** The image of a quasi-compact morphism is closed iff it is stable under specialization. And it is a closed map iff specializations lift along  $f$ .  $\lrcorner$

*Proof:* For the first, the question is local, so reduce to  $Y$  affine, and then  $X$  is qc =  $\cup U_i$ , then we can replace  $X$  by an affine  $\coprod U_i$ , then reduce to the affine case(5.1.7.14).

For the second, for any closed subset of  $X$  with its induced reduced structure, the restriction to it is still qc and specialization lifts, so we prove the image is closed. Now the image is stable under specialization, so the result follows from the first assertion.  $\square$

### Affine Map

**Prop. (7.4.4.10).**  $X$  is affine if there is a finite set of elements  $f_i \in \Gamma(X, \mathcal{O}_X)$  that generate the unit ideal and  $X_{f_i}$  is affine.  $\lrcorner$

*Proof:* First prove that  $X_{f_i} \cap X_{f_j} = X_{f_i f_j}$  is affine because affine intersect  $X_{f_i}$  is affine. Second, prove  $\Gamma(X_f, \mathcal{O}_X) = \Gamma(X, \mathcal{O}_X)_f$ , finally glue them to get a map  $X \rightarrow \text{Spec}(A)$  and use the fact isomorphism is local on the target (7.4.1.5).  $X$  is affine scheme if  $X \rightarrow \text{Spec}(\Gamma(X))$  is affine.  $\square$

**Cor. (7.4.4.11).** Affineness is affine local on the target, and it satisfies the base change trick (7.4.4.2).  $\lrcorner$

**Lemma (7.4.4.12) [Serre Criterion of Affineness].**

- A qc scheme  $X$  is isomorphic to an affine scheme iff  $H^1(X, \mathcal{I}) = 0$  for every Qco sheaf of ideals  $\mathcal{I}$ .
- If  $X$  is qcqs, then it suffices to show  $H^1(X, \mathcal{I}) = 0$  for every Qco sheaf of ideals  $\mathcal{I}$  of f.t..
- If  $f : X \rightarrow Y$  is a quasi-compact morphism between quasi-separated schemes, then  $f$  is affine iff for any Qco sheaf of ideals  $\mathcal{I}$  on  $X$ ,  $R^1 f_* \mathcal{I} = 0$ .

$\lrcorner$

*Proof:* ? Cf. [Sta]01XF.

The case of affine scheme is proven by (7.7.1.1) and (7.7.1.2). The converse: For every point  $p$ , choose an open affine nbhd  $U$ , let  $Y = X - U$ , by the exact sequence

$$0 \rightarrow \mathcal{I}_{Y \cup \{p\}} \rightarrow \mathcal{I}_Y \rightarrow k(P) \rightarrow 0,$$

we have a surjective map  $\Gamma(X, \mathcal{I}_Y) \rightarrow \Gamma(X, k(P))$  thus there is a  $f \in A = \Gamma(X, \mathcal{O}_X)$  that  $P \in X_f \subset U$  is affine. So using (7.4.4.10), we only have to show that for f.m  $f_i$ , they generate  $\Gamma(X, \mathcal{O}_X)$ . This is by considering the kernel  $F$  of  $\mathcal{O}_X^r \rightarrow \mathcal{O}_X : (a_1, \dots, a_r) \rightarrow \sum f_i a_i$ , and there is a filtration on  $F$ , the quotient of which are all coherent sheaves because kernel and cokernel are Qco, and there by induction and hypothesis,  $H^1(X, F) = 0$ , thus the result.

2: In the qcqs case, by (7.7.1.6), we can use filtered colimit to show that  $H^1(X, \mathcal{I}) = 0$  for any Qco sheaf of ideals.

3: For any affine open  $U$  of  $Y$ , the inclusion  $U \rightarrow Y$  is qc, thus  $f^{-1}(U) \rightarrow X$  is also qc. Now any Qco sheaf of ideals  $\mathcal{I}$  on  $U$  extends to a Qco sheaf of ideals on  $X$  by (7.5.1.8), then we can use Leray spectral sequence to conclude that  $H^1(f^{-1}(U), \mathcal{I}) = 0$ , so  $f^{-1}(U)$  is affine.  $\square$

**Cor. (7.4.4.13).** For a Noetherian scheme  $X$ ,  $X$  is affine iff  $X_{\text{red}}$  is affine.  $\lrcorner$

*Proof:* The canonical exact sequence (7.2.6.10) reads:  $0 \rightarrow \mathcal{N}\mathcal{F} \rightarrow \mathcal{F} \rightarrow i_* i^* \mathcal{F} \rightarrow 0$ , so iff  $X_{\text{red}}$  is affine, then we have  $H^i(\mathcal{F}) \cong H^i(\mathcal{N}\mathcal{F})$ , and notice  $\mathcal{N}^k = 0$  for some  $k$ .  $\square$

**Cor. (7.4.4.14).** For a Noetherian reduced scheme  $X$ ,  $X$  is affine iff each irreducible component is affine. (The same as the above, notice that  $\prod p_i = 0$ , for the minimal primes of  $A$ ). (The reducedness can be dropped by the last proposition).  $\lrcorner$

**Lemma (7.4.4.15).** If a morphism  $X \rightarrow Y$  is a homeomorphism onto a closed subset of  $Y$ , then  $f$  is affine.  $\lrcorner$

*Proof:* Cf. [[Sta]04DE].  $\square$



Quasi-affine

**Def. (7.4.4.16) [Quasi-Affine Morphism].** A scheme is called **quasi-affine** iff it is isomorphic to a qc open subscheme of an affine scheme. A morphism is called **quasi-affine** iff the inverse of any affine scheme is quasi-affine.  $\lrcorner$

**Prop. (7.4.4.17).** Quasi-affine is local on the target and satisfies the base change trick.  $\lrcorner$

*Proof:* Cf. [Sta]01SN.  $\square$

**Prop. (7.4.4.18).** Any qc immersion  $i$  of a quasi-affine scheme is quasi-affine.  $\lrcorner$

*Proof:* As  $i$  is qc,  $i$  factors through a map  $Z \rightarrow U \rightarrow X$ , where  $U$  is a quasi-compact open subset of  $X$ . Thus it suffices to consider the closed immersion case, which is clear.  $\square$

**Prop. (7.4.4.19).** A morphism  $f : X \rightarrow S$  is quasi-affine iff  $\mathcal{O}_X$  is  $f$ -ample. In particular, A scheme is quasi-affine iff  $\mathcal{O}_X$  is ample.  $\lrcorner$

*Proof:* Cf. [Sta]01QE.  $\square$

**Cor. (7.4.4.20).** Let  $X$  be a quasi-affine scheme and  $H^p(X, \mathcal{O}_X) = 0$  for  $p > 0$ , then  $X$  is affine.  $\lrcorner$

*Proof:* By (7.4.4.19)  $\mathcal{O}_X$  is ample, then by (7.5.4.20),  $X$  is affine.  $\square$

**Prop. (7.4.4.21) [Descent].** Let  $X$  be a scheme over a field  $k$  and  $K/k$  be a field extension, then  $X$  is quasi-affine iff  $X_K$  is quasi-affine.  $\lrcorner$

*Proof:* Cf. [Sta]0BDD.  $\square$

Dominant

**Prop. (7.4.4.22).** Let  $f : X \rightarrow S$  be a map of schemes.

- If every generic point of irreducible components of  $S$  is in the image of  $f$ , then  $f$  is dominant.
- If  $f$  is quasi-compact, then the converse is also true. More precisely, if a generic point  $\eta$  is not in the image, then it is not in the closure of the image.
- If  $X$  has only f.m. irreducible component, then the converse is also true.

$\lrcorner$

*Proof:* Cf. [Sta]Chap28.8.  $\square$

**Prop. (7.4.4.23) [Dominant Map between Integral Schemes].** If  $f : X \rightarrow S$  is a map between integral schemes, then the following are equivalent:

- $f$  is dominant.
- $f(\eta_X) = \eta_Y$ .
- for some(all) affine open subset  $U \subset X, V \subset Y$  with  $f(U) \subset V$ , the ring map  $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(U)$  is injective.
- for some(all)  $x \in X$ , the local ring map  $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$  is injective.

$\lrcorner$

*Proof:* Cf. [Sta]0CC1.  $\square$

**Def. (7.4.4.24) [Dominant Rational Maps].** A rational map between irreducible schemes is called a **dominant map** if it maps the generic point to the generic point.  $\lrcorner$

### Quasi-Compact

**Prop. (7.4.4.25) [Quasi-Compact Morphism].** A morphism  $f : X \rightarrow S$  of schemes is quasi-compact iff the inverse image of any quasi-compact open subsets is quasi-compact, because affine opens form a basis of  $X$ .

Quasi-compactness is local on the target and satisfies the base change trick(7.4.4.2).  $\lrcorner$

**Prop. (7.4.4.26).** A map between affine schemes is quasi-compact.  $\lrcorner$

*Proof:* Because quasi-compactness is local on the target(7.4.1.5), it suffices to show the inverse image of a distinguished open subset is quasi-compact, and this is true.  $\square$

**Prop. (7.4.4.27).** Let  $f : X \rightarrow Y$ ,  $g : Y \rightarrow Z$ . If  $g \circ f$  is quasi-compact and  $g$  is qs, then  $f$  is qc.  $\lrcorner$

*Proof:* Factor it through  $X \rightarrow X \times_Z Y \rightarrow Y$ . The second map is a base change of  $X \rightarrow Z$  hence qc, the first map is a section of  $X \times_Z Y \rightarrow X$ , which is a base change of  $Y \rightarrow Z$ , hence qs, so by(7.4.5.31), the first map is also qc.  $\square$

**Prop. (7.4.4.28).** Any map between Noetherian schemes is quasi-compact, by(7.4.1.22).  $\lrcorner$

### Finite Type Morphisms

**Def. (7.4.4.29) [Morphisms of Finite Type].** A morphism  $f : X \rightarrow S$  is called of **locally of finite type** if for there exists an affine open cover  $\{\text{Spec}(B_i)\}$  of  $S$  that  $f^{-1}(U_i)$  has an affine open cover of spec of finite generated  $B_i$ -algebras. It is called **finite type** if moreover it is quasi-compact.

Let  $k \in \mathbf{Field}$ , then  $S \in \mathbf{Sch}/k$  is called **(locally)algebraic** iff  $S$  is (locally)of finite type over  $\text{Spec } k$ .

For  $S \in \mathbf{Sch}$ , denote  $\mathbf{Sch}^{\text{loc.ft}}/S$ (resp.  $\mathbf{Sch}^{\text{ft}}/S$ ) the full subcategory of  $\mathbf{Sch}/S$  consisting of (resp. locally) of finite type schemes over  $S$ .

(Locally)Finite type is affine local on the target and on the source, and satisfies the base change trick(7.4.4.2).  $\lrcorner$

**Prop. (7.4.4.30) [Closed Subschemes Descending Chain Stablizes].** The closed subschemes of an algebraic scheme  $X$  over a field  $k$  satisfy the descending chain condition.  $\lrcorner$

*Proof:* As  $X$  is Noetherian, the topological chain  $|Z_1| \supset |Z_2| \supset \dots$  stablizes, so we may assume  $|X| = |Z_1| = |Z_2| = \dots$ . Now choose a finite affine cover  $\text{Spec } A_i$  of  $X$ , then  $Z_i$  corresponds to ideals  $I_i$  of  $A_i$ , so they stablizes, as  $A_i$  are Noetherian.  $\square$

**Cor. (7.4.4.31).** Arbitrary intersection of closed subschemes of an algebraic scheme  $X$  is well-defined.  $\lrcorner$

**Prop. (7.4.4.32) [Algebraic Schemes of Dimension 0 over Fields].** If  $k \in \mathbf{Field}$  and  $X \in \mathbf{Sch}^{\text{loc.ft}}/k$  s.t.  $\dim(X) = 0$ , then  $X$  is a disjoint union of f.d. local Artinian  $k$ -algebras.  $\lrcorner$

*Proof:* This follows from(7.4.1.35).  $\square$

**Prop. (7.4.4.33) [Dimensions for Locally Algebraic  $k$ -Schemes].** For  $k \in \mathbf{Field}$  and  $X \in \mathbf{Sch}^{\text{loc.ft}}/k$ , if  $X$  is irreducible, then  $\dim X = \dim U$  for any nonempty open  $U \subset X$ .  $\lrcorner$

*Proof:* It suffices to show any two affine open subsets  $U, U'$  of  $X$  have the same dimension, then use(4.12.2.27). Now  $U \cap U' \neq \emptyset$  as  $X$  is irreducible, and then it contains a maximal pt  $x$  by Hilbert's Nullstellensatz, and then  $\dim U = \dim(\mathcal{O}_{X,x}) = \dim U'$  because f.g. algebras over a field is catenary.  $\square$

### Integral & Finite Map

**Def. (7.4.4.34) [Finite and Integral Map].** A morphism  $f : X \rightarrow S$  is called **integral** if it is affine and there is a affine open covering  $U_i = \text{Spec } A_i$  of  $S$  that  $f^{-1}(U_i) = \text{Spec } B_i$  and  $A_i \rightarrow B_i$  is integral.

Integral is affine local on the target and satisfies the base change trick(7.4.4.2).

A morphism  $f : X \rightarrow S$  is called **finite** if it is affine and there is a affine open covering  $U_i = \text{Spec } A_i$  of  $S$  that  $f^{-1}(U_i) = \text{Spec } B_i$  and  $B_i$  are finite modules over  $A_i$ .

Finiteness is affine local on the target and satisfies the base change trick(7.4.4.2).

A morphism  $f : X \rightarrow S$  is called **(locally) quasi-finite** if it is (locally) of f.t. and the inverse of a point is a discrete set.  $\lrcorner$

**Prop. (7.4.4.35) [Integral Morphism is Closed].** Specialization lifts along an integral morphism. In particular, an integral morphism is closed, by(7.4.4.9).  $\lrcorner$

*Proof:* If  $f(x) = y, y \rightarrow y'$ , then we can choose an open affine  $U$  containing  $y'$ , which also contains  $y$ . So we can choose an open affine containing  $x$  mapping into  $U$ , then it reduces to the affine case(5.1.7.13).  $\square$

**Prop. (7.4.4.36) [Chevalley].** If  $f : Y \rightarrow X$  is integral surjective,  $Y$  is affine, then  $X$  is affine.  $\lrcorner$

*Proof:* Cf.[Sta]05YU.  $\square$

**Lemma (7.4.4.37).** If  $f : Y \rightarrow X$  is finite surjective,  $Y$  is affine, then  $X$  is affine.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.4.4.38) [Finite and Integral].** A morphism is finite iff it is integral of f.t..  $\lrcorner$

**Prop. (7.4.4.39) [Integral and Affine u.c.].** Integral map is equivalent to u.c. and affine.  $\lrcorner$

*Proof:* Integral is stable under base change. And if it is integral, then it is closed by(5.2.1.5).

Conversely, we need to prove if  $\text{Spec } B \rightarrow \text{Spec } A$  is u.c., then  $A \rightarrow B$  is integral. For any  $a \in B$ , let  $J$  be the kernel of  $A[X] \rightarrow B[X]/(aX - 1)$ , then if  $f \in J$ , then  $f(X) = (aX - 1)q(X)$ , and  $X^{\deg(f)} f(\frac{1}{X}) = (a - X)X^{\deg(q)} q(\frac{1}{X})$  vanishes at  $a$ , so it suffices to find a  $f \in J$  with constant term 1. It suffices to show that  $\text{Spec}(A[X]/J + (x))$  is empty:

As  $f$  is universally closed,  $\text{Spec } B[X] \rightarrow \text{Spec } A[X]$  is closed, thus the image of  $\text{Spec}(B[X]/(aX - 1))$  is closed in  $\text{Spec } A$ , which is the underlying set of  $\text{Spec } A/J$ , by(7.4.5.15). Now notice  $T = \text{Spec}(A[X]/J + (x)) \times_{\text{Spec}(A[X]/J)} \text{Spec}(B[X]/(aX - 1))$  is empty, because  $(A[X]/J + (x)) \otimes_{A[X]/J} (B[X]/(aX - 1)) = \text{Spec } A \otimes_{\text{Spec } A[X]} \text{Spec}(B[X]/(aX - 1)) = 0$ , as  $X$  is invertible in  $B[X]/(aX - 1)$  but vanish in  $A$ . But  $T \rightarrow \text{Spec}(A[X]/J + (x))$  is surjective, as a base change of  $\text{Spec}(B[X]/(aX - 1)) \rightarrow \text{Spec } A/J$ , thus we are done.  $\square$

**Def. (7.4.4.40) [Degree of Finite Morphisms at a Point].** Suppose  $\pi : X \rightarrow Y$  is a finite morphism, then  $\pi_* \mathcal{O}_X$  is a  $\mathcal{O}_Y$ -module of f.t.. Let  $p \in Y$ , then the **degree of  $\pi$  at  $p$**  is the rank of this sheaf on  $Y$ (7.5.1.11), which is an upper-semicontinuous function(7.5.1.41).  $\lrcorner$

**Prop. (7.4.4.41).** The degree of a finite morphism  $\pi : X \rightarrow Y$  at  $p$  is the dimension over  $k(p)$  of the fiber of  $\pi$  at  $p$ . In particular, the degree is zero iff  $\pi^{-1}(p) = \emptyset$ .  $\lrcorner$

*Proof:* Look affine locally.  $\square$

**Lemma (7.4.4.42).** For  $f : Y \rightarrow X$  finite surjective and  $X$  locally Noetherian, for every integral subscheme  $Z$  of  $X$  with generic point  $\xi$ , there is a coherent sheaf  $\mathcal{F}$  on  $Y$  that the support of  $f_*\mathcal{F}$  is  $Z$  and  $(f_*\mathcal{F})_\xi$  is annihilated by  $\mathfrak{m}_\xi$ .  $\lrcorner$

*Proof:* We consider an inverse image of  $\xi = \xi'$ , and let  $Z' = \overline{\{\xi'\}}$  with the induced reduced structure, then let  $\mathcal{F} = i_*\mathcal{O}_{Z'}$  on  $Y$ ,  $\mathcal{F}$  is coherent, then we need to show that  $(f_*\mathcal{F})_\xi$  is annihilated by  $\mathfrak{m}_\xi$ . This is because it factors through  $Z$ .  $\color{red}{?}$  Cf. [Sta]01YO.  $\square$

### Quasi-Finite Morphisms

**Def. (7.4.4.43) [Quasi-Finite Morphisms].** A morphism  $f : X \rightarrow S \in \text{Sch}$  is called a **locally quasi-finite morphism** if it is locally spec map of quasi-finite ring maps of f.t.  $\color{red}{?}$ .

And  $f$  is called a **quasi-finite morphism** if  $f$  is of f.t. and any  $x \in X$  is isolated in  $f^{-1}(f(x))$ . (Locally) Quasi-finite morphisms are local on the target and satisfies base change trick.  $\lrcorner$

**Prop. (7.4.4.44) [Characterization of Quasi-Finiteness].** For  $f : X \rightarrow S \in \text{Sch}$ , the following are equivalent:

- $f$  is quasi-finite.
- $f$  is locally quasi-finite and quasi-compact.
- $f$  is locally of f.t., qc, and has finite fibers.

 $\lrcorner$ 

*Proof:* Cf. [Sta]01TJ, 02NH.  $\square$

**Prop. (7.4.4.45).** Immersions are locally quasi-finite.  $\lrcorner$

**Prop. (7.4.4.46).** Let  $f : X \rightarrow S$  be a morphism of schemes and  $s \in S$ . Assume that  $f$  is locally of f.t. and  $f^{-1}(s)$  is a finite set, then  $X_s$  is a finite discrete topological space, and  $f$  is quasi-finite at every point of  $f^{-1}(s)$ .  $\lrcorner$

*Proof:* Cf. [Sta]02NG.  $\square$

**Prop. (7.4.4.47) [Zariski's Main Theorem].** Let  $f : Y \rightarrow X$  be an affine morphism of f.t.. Let  $Y \xrightarrow{f'} X' \xrightarrow{\nu} X$  where  $X'$  is the normalization of  $X$  in  $Y$ , then there exists an open subscheme  $U' \subset X'$  that  $(f')^{-1}(U) \rightarrow U$  is an isomorphism and  $(f')^{-1}(U)$  is the set of points at which  $f$  is quasi-finite.  $\lrcorner$

*Proof:* Cf. [Sta]03GT.  $\square$

**Prop. (7.4.4.48) [Quasi-Finite Locus is Open].** For  $f : X \rightarrow S \in \text{Sch}$ , the set of points of  $X$  that  $f$  is quasi-finite is an open subset  $U \subset X$ , and  $U \rightarrow S$  is locally quasi-finite.  $\lrcorner$

*Proof:* Cf. [Sta]01T1.  $\square$

**Prop. (7.4.4.49).** If  $f : Y \rightarrow X$  is a quasi-finite morphism of schemes, and  $T \subset Y$  is nowhere dense, then  $f(T) \subset X$  is also nowhere dense.  $\lrcorner$

*Proof:* Cf. [Sta]03J2.  $\square$

## 5 Zariski's Main Theorem

References are [Sta]Chap36.38.

**Thm. (7.4.5.1) [Zariski's Main Theorem].** For a morphism  $X \rightarrow S$  that is quasi-finite and separated, if  $S$  is qcqs, Then there is a factorization  $X \rightarrow T \rightarrow S$  that  $X \rightarrow T$  is a qc open immersion and  $T \rightarrow S$  is finite.  $\lrcorner$

*Proof:* Cf. [Sta]05K0.  $\square$

### Generically Finiteness

**Def. (7.4.5.2) [Generically Finite Morphisms].** A **generically finite morphism** is a morphism  $f : X \rightarrow Y$  locally of f.t. and there exists a dense open subset  $U \subset X$  s.t.  $U \rightarrow Y$  is locally quasi-finite.  $\lrcorner$

**Lemma (7.4.5.3).** Let  $R \rightarrow S$  be a ring map of f.t., if  $\mathfrak{p} \subset R$  is a minimal prime that there are f.m. primes in  $S$  lying over  $\mathfrak{p}$ , then there is a  $g \in R$ ,  $g \notin \mathfrak{p}$  that  $R_g \rightarrow S_g$  is finite.  $\lrcorner$

*Proof:* The condition means  $S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}}$  has only f.m. primes and  $\mathfrak{p}S_{\mathfrak{p}}$  is locally nilpotent. Then  $\kappa(\mathfrak{p}) \rightarrow S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}}$  is finite. Let  $x_i$  generate  $S$  over  $R$ , then there are polynomials  $P_i \in R[X]$  that  $P_i(x_i) = 0 \in S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}}$ , and thus  $P_i^{e_i}(x_i) = 0 \in S_{\mathfrak{p}}$ . Now choose  $g \notin \mathfrak{p}$  that  $g$  divides the common divisors of the coefficients of  $P_i$  and  $P_i(x_i) = 0 \in S_g$ , then  $R_g \rightarrow S_g$  is finite.  $\square$

**Prop. (7.4.5.4) [Generically Finiteness].** Let  $f : X \rightarrow Y \in \text{Sch}$  be of finite type. If  $\eta_Y \in Y$  is a generic point, then:

- $f^{-1}(\eta_Y)$  is a finite set iff there are affine opens  $U_i \subset X, i = 1, \dots, n$  and  $V \subset Y$  that  $f(U_i) \subset V, \eta \in V$  and  $f^{-1}(\eta) \subset \cup U_i$  and  $U_i \rightarrow V$  are finite.
- If  $f$  is qs, then we can further restrict to  $n = 1$ .
- If  $f$  is qcqs, then we can further restrict to  $U_1 = f^{-1}(V)$ .  $\lrcorner$

*Proof:* These are local on  $Y$ , so we can change  $Y$  to an affine open subscheme containing  $\eta$ .

1: If these affine opens exist, then  $\#f^{-1}(\eta) < \infty$  because finite maps are quasi-finite (7.4.6.5).

If  $f^{-1}(\eta) = \{\xi_1, \dots, \xi_n\}$ , choose affine opens  $U_i$  around each  $\xi_i$ , then we reduce to the affine case (7.4.5.3).

2: Cf. [Sta]02NW.

3:  $\square$

**Prop. (7.4.5.5) [Extension of Fraction Fields].** Let  $f : X \rightarrow Y$  be a dominant morphism locally of f.t. between integral schemes, then the following are equivalent:

1.  $\text{tr.deg } K(X)/K(Y) = 0$ .
2.  $[K(X) : K(Y)] < \infty$ .
3. there exists non-empty affine opens  $U \subset X, V \subset Y$  s.t.  $f(U) \subset V$  and  $f : U \rightarrow V$  is finite.
4.  $\eta_X$  is the only point mapping to  $\eta_Y$ .
5. If  $f$  is qc or qs, in item3 we can assume  $U = f^{-1}(V)$ .  $\lrcorner$

*Proof:* Cf. [Sta]02NX.  $\square$

**Def. (7.4.5.6) [Degree of Generically Finite Dominant Morphisms].** Let  $f : X \rightarrow Y$  be a dominant morphism locally of f.t. between integral schemes, then by (7.4.4.23),  $f$  induces an injective map of function fields  $R(Y) \rightarrow R(X)$ , the degree of field extension  $[R(X) : R(Y)]$  is called the **degree of  $f$** , denoted by  $\deg(f)$ . It is a positive integer or  $\infty$ .

When  $\deg(f)$  is finite (i.e. the conditions of (7.4.5.5) hold), it is called a **separable/purely inseparable morphism** if  $R(X)/R(Y) \in \mathbf{Field}$  is separable/purely inseparable. And also we can define the separable, inseparable degree of  $f$ .  $\lrcorner$

### Immersions

**Def. (7.4.5.7) [Immersion]s.** A **closed immersion** of schemes is a closed immersion of local ringed spaces (7.2.6.15). A **closed subscheme** of a scheme  $X$  is a closed sub-ringed space (7.2.6.15) that is also a scheme.

An **open immersion** of schemes is an open immersion of local ringed spaces (7.2.6.15). An **open subscheme** of a scheme  $X$  is an open subspace (7.2.6.15) that is also a scheme.

Open and closed immersions are affine local on the target (7.4.1.5).

An **immersion** is a morphism that is a closed immersion of an open immersion.  $\lrcorner$

**Lemma (7.4.5.8) [Closed Immersion for Schemes].** Let  $f : Y \rightarrow X$  be a morphism of schemes that induces a homeomorphism of  $Y$  onto a closed subset of  $X$ , and  $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$  is surjective, then it is a closed immersion (7.2.6.15).  $\lrcorner$

*Proof:* It suffices to show that the kernel of  $f^\#$  is Qco. For this, notice that  $f$  is quasi-compact, and it is a monomorphism by (7.4.4.5), in particular separated by (7.4.5.37). Then (7.5.1.6) shows  $f_*\mathcal{O}_Y$  is Qco. Then the kernel is Qco by (7.5.1.3).  $\square$

**Prop. (7.4.5.9) [Closed Subschemes of Affine Schemes].** The closed sub-ringed spaces of  $X = \text{Spec } A$  are all closed subschemes, and they corresponds to ideals  $I \subset A$ :

- If  $I \subset A$  is an ideal, then the morphism  $Z = \text{Spec } A/I \rightarrow \text{Spec } A$  is a homeomorphism of  $Z$  onto a closed subspace  $V(I)$  of  $X$ , and also the stalk map at a point  $\mathfrak{p} \subset Z$  is  $R_{\mathfrak{p}} \rightarrow (R/I)_{\mathfrak{p}/I} = R_{\mathfrak{p}}/IR_{\mathfrak{p}}$ , which is surjective. So this is a closed immersion by (7.4.5.8).
- By (7.2.6.17), for any closed subscheme  $Z$  of  $X$  with sheaf of ideals  $\mathcal{I}$ ,  $Z$  is isomorphic to the closed subscheme of  $X$  defined by  $\mathcal{I}$ . Now  $\mathcal{I}$  is locally generated by sections, so the quotient sheaf  $\mathcal{O}_X/\mathcal{I}$  is a Qco sheaf, so it is of the form  $\tilde{S}$  for some  $A$ -module  $S$ , by (7.5.1.2). Then  $\mathcal{I}$ , as the kernel of  $\mathcal{O}_X \rightarrow \tilde{S}$ , is also Qco (7.5.1.3), so it equals  $\tilde{I}$  for some ideal  $I \subset A$ . Thus  $S = R/I$ , and we are done.  $\lrcorner$

**Prop. (7.4.5.10) [Closed Subscheme of Schemes].** The closed sub-ringed spaces of a scheme  $X$  are all closed subschemes, and they corresponds to Qco  $\mathcal{O}_X$ -sheaves of ideals via the ideal sheaf (7.2.6.15):  $\lrcorner$

*Proof:* Let  $i : Y \rightarrow X$  be a closed immersion, for any  $x \in X$ , choose an open affine nbhd  $U$  of  $x \in X$ , then  $i : i^{-1}(U) \rightarrow U$  is also a closed immersion, so it corresponds to  $\text{Spec } A/I \rightarrow \text{Spec } A$  for some ideal  $I$  by (7.4.5.9). So  $Z$  is a scheme, and the ideal sheaf  $\mathcal{I}$  is Qco.  $\square$

**Prop. (7.4.5.11).** Closed immersion satisfies the base change trick (7.4.4.2). Open immersions are stable under base change and composition. Immersions are stable under base change and composition.  $\lrcorner$

*Proof:* For immersion, shrink the open subset.  $\square$

**Remark (7.4.5.12).** A open immersion followed by a closed immersion can be written as a closed immersion followed by an open immersion, but not reversely. The reverse happens if the immersion is quasi-compact or the source is reduced (use the reduced induced structure).  $\lrcorner$

**Def. (7.4.5.13) [Schematic Images].** For a morphism  $f : X \rightarrow Y$ , there is a closed scheme called **scheme-theoretic image** or the **schematic image** that is the smallest closed subscheme of  $Y$  that  $f$  factors through  $Z$ .

For an immersion of schemes, the scheme-theoretic image of the immersion is called the **scheme-theoretic closure** or the **schematic closure**.  $\lrcorner$

*Proof:* Consider the kernel of the structural map, and the kernel contains a maximal Qco sheaf of ideals  $\mathcal{I}$  by (7.5.1.12).  $\square$

**Prop. (7.4.5.14).** If  $\begin{array}{ccc} X_1 & \xrightarrow{f_1} & Y_1 \\ \downarrow & & \downarrow \\ X_2 & \xrightarrow{f_2} & Y_2 \end{array}$  be a commutative diagram of schemes and  $Z_i \subset Y_i$  be the scheme-

theoretic image of  $f_i$ , then it induces a commutative diagram  $\begin{array}{ccccc} X_1 & \longrightarrow & Z_1 & \longrightarrow & Y_1 \\ \downarrow & & \downarrow & & \downarrow \\ X_2 & \longrightarrow & Z_2 & \longrightarrow & Y_2 \end{array}$ .  $\lrcorner$

*Proof:* It suffices to show  $Z_1$  factors through the closed subscheme  $Z_2 \times_{Y_2} Y_1$  of  $Y_1$ , which follows from the universal property of  $Z_1$ .  $\square$

**Prop. (7.4.5.15).** Let  $f : X \rightarrow Y$  be a qc morphism of schemes and  $Z$  be the schematic image, then

- the kernel  $\mathcal{I} = \ker(\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X)$  is Qco, thus  $Z$  is the closed subscheme determined by  $\mathcal{I}$ .
- For any open subscheme  $U \subset Y$ , the scheme-theoretical image of  $f|_{f^{-1}(U)}$  is equal to  $Z \cap U$ .
- $f(X)$  is dense in  $Z$ .

$\lrcorner$

*Proof:* 1: As being Qco is local, it suffices to show for  $Y$  affine. Then  $X = \cup U_i$  is a finite union of affine schemes. Now take  $X' = \coprod U_i$ , then there are maps  $X' \xrightarrow{f'} X \rightarrow Y$ . Then  $\mathcal{O}_X$  is a subsheaf of  $f'_*\mathcal{O}_{X'}$ . So  $\mathcal{I} = \ker(\mathcal{O}_Y \rightarrow f_*f'_*\mathcal{O}_{X'})$ . Now  $f \circ f'$  is qcqs, thus by (7.5.1.6),  $f_*f'_*\mathcal{O}_{X'}$  is Qco, thus also  $\mathcal{I}$  is Qco.

2 follows from 1 as the formation of  $\mathcal{I}$  commutes with restriction to open subschemes.

3 follows from 2 as the scheme-theoretical image of empty set is empty.  $\square$

**Prop. (7.4.5.16).** If  $f : X \rightarrow Y$  is a morphism and  $X$  is reduced, then the scheme-theoretical image of  $f$  is the induced-reduced structure (7.4.1.14) of  $\overline{f(X)} \subset Y$ .  $\lrcorner$

*Proof:* This is clear.  $\square$

**Def. (7.4.5.17) [Scheme-Theoretically Dense].** An open subscheme  $U \subset X$  is called **scheme-theoretically dense** if for any open subscheme  $V$  of  $X$ , the scheme-theoretical closure of  $U \cap V$  in  $V$  is equal to  $V$ .  $\lrcorner$

**Prop. (7.4.5.18).** If the inclusion  $U \rightarrow X$  is qc, then  $U$  is scheme-theoretically dense in  $X$  iff the scheme-theoretical closure of  $U$  is  $X$ , by (7.4.5.15).  $\lrcorner$



**Prop. (7.4.5.19).** Let  $j : U \rightarrow X$  be an open immersion of schemes, then  $U$  is scheme-theoretically dense in  $X$  iff  $\mathcal{O}_X \rightarrow j_*\mathcal{O}_U$  is injective.  $\lrcorner$

*Proof:* If it is not injective, then we can find an open subscheme  $V$  of  $X$  that the kernel is non-zero, thus it contains a non-zero  $\mathcal{Q}\text{co}$  ideal sheaf, which means the scheme-theoretical closure of  $U \cap V$  is not  $V$ .  $\square$

**Cor. (7.4.5.20).** If  $U, V$  are open subschemes of  $X$  scheme-theoretically dense in  $X$ , then  $U \cap V$  is also scheme-theoretically dense in  $X$ .  $\lrcorner$

*Proof:*  $\mathcal{O}_X \rightarrow \mathcal{O}_X(U) \rightarrow \mathcal{O}_X(U \cap V)$  is injective.  $\square$

**Prop. (7.4.5.21)[Scheme-Theoretical Image of Immersions].** Let  $f : Z \rightarrow X$  be an immersion and either  $f$  is qc or  $Z$  is reduced. Let  $\overline{Z}$  be the scheme-theoretical image of  $f$ , then the morphism  $Z \rightarrow \overline{Z}$  is an open immersion that identifies  $Z$  with a scheme-theoretical dense open subscheme of  $\overline{Z}$ .  $\lrcorner$

*Proof:* Cf. [Sta]01RG.  $\square$

**Prop. (7.4.5.22)[Immersion to be Closed].** An immersion  $f$  is a closed immersion iff the image is closed.  $\lrcorner$

*Proof:* Let  $i : Y \rightarrow U$  be a closed immersion where  $j : U \subset X$  is an open subscheme, then  $i$  is associated to an ideal sheaf  $\mathcal{I}$  of  $\mathcal{O}_U$ . Now because  $\mathcal{I}|_{U \setminus f(Y)} = \mathcal{O}_{U \setminus f(Y)}$ , we can glue  $\mathcal{I}$  and  $\mathcal{O}|_{X \setminus f(Y)}$  to a sheaf of ideals  $\mathcal{J} \subset \mathcal{O}_X$ . Now  $j^*\mathcal{O}_X/\mathcal{J} = \mathcal{O}_U/\mathcal{I} \cong i_*\mathcal{O}_Y$ , thus  $\mathcal{O}_X \rightarrow f_*\mathcal{O}_Y = j_*(\mathcal{O}_U/\mathcal{I}) = j_*j^*(\mathcal{O}_X/\mathcal{J})$  is surjective, because  $\mathcal{O}_U/\mathcal{I}$  is supported on a closed subset of  $U$ . Thus  $f$  is a closed immersion by (7.4.5.8).  $\square$

**Prop. (7.4.5.23)[Immersion are monomorphisms].** Immersions are monomorphisms.  $\lrcorner$

*Proof:* It is easy to check (7.4.4.5) for both open and closed immersions.  $\square$

**Prop. (7.4.5.24)[Equivalent Definitions of Closed Immersion].** The following are equivalent for a morphism  $f$ :

- $f$  is a closed immersion.
- $f$  is a proper monomorphism.
- $f$  is proper, unramified and u.i..
- $f$  is a u.c., unramified monomorphism.
- $f$  is u.c., unramified and u.i..
- $f$  is u.c., locally of f.t. and a monomorphism.
- $f$  is u.c., u.i., locally of f.t. and formally unramified.

$\lrcorner$

*Proof:* 4 – 7 are equivalent by (7.6.5.13). For the rest, Cf. [Sta], 04XV.  $\square$



### Universal Homeomorphism

**Prop. (7.4.5.25) [Criterion for Universal Homeomorphisms].** A morphism is a universal homeomorphism iff it is integral, surjective and universally injective.  $\square$

*Proof:* A universally homeomorphism is affine by (7.4.4.15). It is clearly u.c, so it is integral by (7.4.4.39). Conversely, it is integral hence u.c, and universally bijective, so it is universally homeomorphism.  $\square$

**Cor. (7.4.5.26).** The reduction  $X_{\text{red}} \rightarrow X$  is a universal homeomorphism, as closed immersion is u.c..  $\square$

### Separatedness

**Def. (7.4.5.27) [Separated Morphisms].**  $f : X \rightarrow Y \in \text{Sch}$  is called **separated morphism** if the diagonal  $\Delta : X \rightarrow X \times_Y X$  is a closed immersion. It is called **quasi-separated** if the diagonal is quasi-compact.

In fact  $\Delta$  is always an immersion because maps between affine scheme is separated so  $\Delta(X)$  is closed in  $\cup U_{ij} \otimes_{V_i} U_{ij}$  where  $U, V$  are affine open, hence it suffice to check the image is closed.  $\square$

**Prop. (7.4.5.28).** (Quasi-)Separatedness is local on the target because closed immersion and quasi-compact is local on the target (7.4.1.5).

(Quasi-)Separatedness satisfies base change trick by (7.4.4.2).  $\square$

**Prop. (7.4.5.29) [Graphs].** By (4.1.1.51), for  $X \rightarrow S$  and  $Y \rightarrow S$ , the map  $X = X \times_Y Y \rightarrow X \times_S Y$  is an immersion. It is closed immersion if  $Y \rightarrow S$  is separated, and it is qc if  $Y \rightarrow S$  is quasi-separated.  $\square$

**Cor. (7.4.5.30).** For  $X \rightarrow Y \in \text{Sch}/S$ , the map  $X = X \times_Y Y \rightarrow X \times_S Y$  is an immersion. It is closed immersion if  $Y \rightarrow S$  is separated, and it is qc if  $Y \rightarrow S$  is quasi-separated.  $\square$

**Cor. (7.4.5.31).** If  $s : S \rightarrow X$  is a section of  $f : X \rightarrow S$ , the above proposition applies to this case, because  $S = S \times_X X \rightarrow S \times_S X = X$ .  $\square$

**Prop. (7.4.5.32) [Characterizing (Quasi-)Separated Morphisms].** A morphism is quasi-separated iff for any two affine open that mapped into an affine open, their intersection is quasi-compact. This is because quasi-compact is local on the target.

A morphism is separated iff for any two affine open that mapped into an affine open, their intersection is affine and  $\mathcal{O}(U) \otimes_{\mathcal{O}(W)} \mathcal{O}(V) \rightarrow \mathcal{O}(U \cap V)$  is surjective. This is because closed immersion is local on the target (7.4.1.5).  $\square$

**Cor. (7.4.5.33).** Any  $X \in \text{NSch}$  is quasi-separated.  $\square$

**Cor. (7.4.5.34).** If  $g \circ f$  is (quasi-)separated, then so is  $f$ .  $\square$

**Cor. (7.4.5.35).** If  $X$  is (quasi-)separated, then  $X \rightarrow Y$  is (quasi-)separated.  $\square$

**Prop. (7.4.5.36) [Injective Maps are Separated].** Injective maps of schemes are separated.  $\square$

*Proof:* Let  $f : X \rightarrow Y$  be an injective map. Firstly  $X \times_Y X$  is a union of affine subschemes of the form  $U \times_V U$  where  $U, V$  are affine and  $f(U) \subset V$ : let  $z \in X \times_Y X$ , then  $\pi_1(z) = \pi_2(z)$ , because they map to the same point in  $Y$ , thus we can choose affine nbhds  $U, V$  of  $\pi_1(z)$  and  $f(\pi_1(z))$ . Now for each of these  $U \times_V U$ ,  $\Delta : U \rightarrow U \times_V U$  is closed immersion, thus  $\Delta_X$  is also closed immersion (7.4.1.5).  $\square$

**Cor. (7.4.5.37).** monomorphisms are separated because they are universal injective (7.4.4.4), and im-  
mersions are separated.  $\lrcorner$

**Prop. (7.4.5.38) [Affine Maps are Separated].** (Quasi-)Affine morphism is separated.  $\lrcorner$

*Proof:* Check the diagonal is a closed immersion directly.  $\square$

**Prop. (7.4.5.39).** if  $f : X \rightarrow S$  is affine,  $h : Y \rightarrow S$  is separated, and  $g : X \rightarrow Y$  is a morphism over  $S$ , then  $g$  is affine.  $\lrcorner$

*Proof:* Decompose  $g$  as  $X \rightarrow X \times_S Y \rightarrow Y$ , where the first map is base change of  $\Delta_Y$  and the second map is base change of  $f$ , so they are both affine.  $\square$

**Prop. (7.4.5.40) [Scheme-Theoretic Equalizer].** If  $X, Y$  are schemes over  $S$  and  $a, b : X \rightarrow S$  are morphisms, then there is a largest locally closed subscheme  $Z$  of  $X$  that  $a|_Z = b|_Z$ . And if  $Y/S$  is separated,  $Z$  is a closed subscheme of  $X$ .  $\lrcorner$

*Proof:* By definition,  $Z$  should be the fibered product:

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow (a,b) \\ Y & \xrightarrow{\Delta_Y} & Y \times_S Y \end{array}$$

then the theorem follows from the definition and base change trick of locally closed morphisms.  $\square$

**Prop. (7.4.5.41) [Qsqc Lemma].** If  $X \in \text{Sch}_{\text{qcqs}}$  and  $s \in \Gamma(X, \mathcal{O}_X)$ , then there is a natural isomorphism  $\Gamma(X, \mathcal{O}_X)_s \cong \Gamma(X_s, \mathcal{O}_X)$ .  $\lrcorner$

*Proof:* Firstly this is true for  $X$  affine by definition (7.2.7.1). In general, let  $U_i$  be an affine open cover of  $X$  that  $U_i \cap U_j = \cup_{k=0}^{c_{ij}} U_{ijk}$  be a finite cover by affine opens, then there is an exact sequence

$$\Gamma(X, \mathcal{O}_X) \rightarrow \prod_i \Gamma(U_i, \mathcal{O}_X) \rightarrow \prod_{ijk} \Gamma(U_{ijk}, \mathcal{O}_X).$$

Then we can take localization by  $s$  to get an exact sequence

$$\Gamma(X, \mathcal{O}_X)_s \rightarrow \prod_i \Gamma(D(s, U_i), \mathcal{O}_X)_s \rightarrow \prod_{ijk} \Gamma(D(s, U_{ijk}), \mathcal{O}_X)_s.$$

but  $D(s, U_i)$  is an open cover of  $X_s$  with  $D(s, U_i) \cap D(s, U_j) = \cup_k D(s, U_{ijk})$ , thus we get a natural isomorphism  $\Gamma(X, \mathcal{O}_X)_s \cong \Gamma(X_s, \mathcal{O}_X)$ .  $\square$

**Prop. (7.4.5.42).** If  $S \in \text{Sch}$ ,  $X \in \text{Sch}^{\text{sep}}/S$  and  $\varphi : T \rightarrow T' \in \text{Sch}/S$  is scheme-theoretically dominant, then the map  $\text{Hom}_S(T', X) \rightarrow \text{Hom}_S(T, X)$  is injective.  $\lrcorner$

*Proof:* For  $f, g : T \rightarrow X$ , consider  $(f, g) : T \rightarrow X \times X$ . Let  $Z \subset T$  be the inverse image of the diagonal  $\Delta_X \subset X \times X$ , and similarly consider the composite map  $(f', g') : T' \rightarrow T \rightarrow X \times X$  and let  $Z' \subset T'$  be the inverse image of the diagonal  $\Delta_X \subset X \times X$ . Then  $Z, Z'$  are closed subschemes of  $T, T'$  resp., and  $Z' = \varphi^{-1}(Z)$ . Now if  $f' = g'$ , then  $Z' = T'$ , and then  $Z = T$  because  $T \rightarrow T'$  is dominant.  $\square$

**Cor. (7.4.5.43).** If  $R \hookrightarrow R' \in \mathcal{C}\text{Ring}$  is an inclusion and  $X \in \text{Sch}^{\text{sep}}/R$ , then  $X(R) \rightarrow X(R')$  is injective.  $\lrcorner$

### Constructible Subsets of Schemes

**Prop. (7.4.5.44).** For  $X \in \text{Sch}$ , a subset  $E \subset |X|$  is locally constructible(4.12.2.10) iff  $E \cap |U|$  is constructible in  $|U|$  for any affine open subset  $U \subset X$ .  $\lrcorner$

*Proof:* Cf.[Sta]054C. ?  $\square$

**Cor. (7.4.5.45).** For  $X \in \text{Sch}_{\text{qcqs}}$ , any locally constructible subset  $E \subset |X|$ (4.12.2.10) is constructible.  $\lrcorner$

*Proof:* This follows from(4.12.2.12) and(4.12.2.15).  $\square$

**Prop. (7.4.5.46) [Qcqs Schemes are Spectral].** For any  $X \in \text{Sch}_{\text{qcqs}}$ ,  $|X|$  is a spectral space, by(7.2.7.11) and(27.2.3.20).  $\lrcorner$

**Prop. (7.4.5.47).** Suppose  $X \in \text{Sch}$  and  $E \subset |X|$  is a constructible subset. If  $\xi$  is the generic fiber of an irreducible component of  $X$ , then

- If  $\xi \in E$ , then some open nbhd of  $\xi$  is contained in  $E$ .
- If  $\xi \notin E$ , then some open nbhd of  $\xi$  is disjoint from  $E$ .

$\lrcorner$

*Proof:* Because the complement of a constructible subset is constructible(4.12.2.11), it suffices to show item2. We may assume  $X$  is affine, so  $E$  is constructible by(7.4.5.44). In this case  $X$  is qcqs thus a spectral space by(7.4.5.46), and  $\xi \notin \text{cl}(E)$  by(4.12.3.10).  $\square$

## 6 Proper & Projective

**Prop. (7.4.6.1)[Proper].** A **proper morphism(Schemes)** is a morphism of schemes that is separated, of finite type and universally closed.

Properness is local on the target, because all these three properties do(7.4.1.5). Properness satisfies the base change trick(7.4.4.2).  $\lrcorner$

**Prop. (7.4.6.2).** The class of proper morphisms satisfies the base change trick(7.4.4.1), by valuation criterion(7.4.6.13) and fibered products tricks.  $\lrcorner$

*Proof:* Closed immersion is proper because it is f.t. and is affine so separated(7.4.5.27), and it is universally closed because immersions are stable under base change(7.4.5.11).  $\square$

**Prop. (7.4.6.3) [Image of Proper Map].** If  $f : X \rightarrow Y$  is morphism between separated schemes f.t over  $S$ , then if  $X$  is proper, then  $f$  is proper(by base change trick) thus the image is closed, and is proper over  $S$  in its scheme-theoretic structure.  $\lrcorner$

*Proof:* Notice proper is qc, so by(7.4.5.13), the scheme-theoretic closure has the same underlying space as the image. Then use(7.4.4.8) to show it is u.c..  $\square$

**Cor. (7.4.6.4) [Connected Proper to Affine Constant].** A morphism from a connected proper scheme to an Noetherian affine scheme  $\text{Spec } A$  is constant.  $\lrcorner$

*Proof:* Because the image is proper, by(7.7.4.12), the global section of its image is a finite module over  $k$  thus Artinian(5.1.3.4) so has finitely many points and is discrete. But  $X$  is connected, thus it is constant.  $\square$

**Prop. (7.4.6.5) [Chevalley].** Let  $f : X \rightarrow Y$  be a morphism of schemes, the following are equivalent:

- $f$  is finite.
- $f$  is proper and affine.
- $f$  is proper with finite fibers.
- $f$  is proper and locally quasi-finite.
- $f$  is separated, u.c., locally of f.t. and has finite fibres.

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*Proof:* The fiber of  $f : X \rightarrow S$  is  $\text{Spec}(k(y) \otimes_A B)$ , which is Artinian(5.1.3.4), so it has finitely many primes and they are all closed. Finite morphism is proper because it is integral(7.4.4.39).

The converse follows from Zariski's main theorem(7.4.5.1).

Cf.[Sta]02LS.

□

**Cor. (7.4.6.6) [Quasi-Affine+Proper  $\iff$  Affine].** Let  $f : X \rightarrow \text{Spec } A$  be a quasi-affine and proper map, then  $f$  is affine.

┘

*Proof:* This follows from(7.7.2.8)(7.4.4.20) and the fact  $H^p(X, \mathcal{O}_X) = 0$  for  $p > 0$ .

□

**Prop. (7.4.6.7) [Quasi-Finite+Proper  $\iff$  Finite].** Let  $f : X \rightarrow S \in \text{Sch}$  be proper, and  $s \in S$  that  $\#f^{-1}(s) < \infty$ , then there exists a nbhd  $V$  of  $s \in S$  that  $f^{-1}(V) \rightarrow V$  is finite.

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*Proof:* By(7.4.4.46),  $f$  is quasi-finite at the points of  $f^{-1}(s)$ . By(7.4.4.48) the set  $U$  of points that  $f$  is quasi-finite is open. Denote  $Z = X \setminus U$ , then  $f(Z)$  is closed in  $S$  and doesn't contain  $s$ . Then  $V = S \setminus f(Z)$  satisfies the requirement by(7.4.6.5).

□

**Prop. (7.4.6.8) [Generically Finite+Proper  $\implies$  Finite].** Let  $f : X \rightarrow S$  be a proper morphism of locally Noetherian schemes and  $y \in Y$  satisfies  $\dim \mathcal{O}_{Y,y} \leq 1$ , and if one of the following holds:

- For every generic point  $\eta$  of  $X$  s.t.  $f(\eta)$  generalizes  $y$ ,  $k(\eta)/k(f(\eta))$  is algebraic.
- $f$  is quasi-finite at every generic point of  $x$ .
- $f$  is quasi-finite on a dense subset of  $X$ .

then there exists an open nbhd  $V$  of  $y \in Y$  s.t.  $f^{-1}(V) \rightarrow V$  is finite.

┘

*Proof:* Cf.[Sta]0AB7.

□

**Example (7.4.6.9) [Non-Proper Schemes].** If  $k \in \text{Field}$  and  $X \in \text{Sch}^{\text{sep}}/k$  with  $\dim X \geq 1$ , and  $x \in |X|_0$ , then  $X \setminus \{x\}$  is not proper over  $k$ .

┘

*Proof:* Cf.[Sta]0A24.

□

**Def. (7.4.6.10) [Modification and Alteration].** A **modification(of schemes)** of  $X \in \text{Sch}$  is a scheme  $X'$  together with a proper birational map  $X' \rightarrow X$ .

A **alteration(of schemes)** of  $X$  is a scheme  $X'$  together with a proper dominant morphism  $f : X' \rightarrow X$  that for some non-empty open  $U$  of  $X$ ,  $f^{-1}(U) \rightarrow U$  is finite.

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**Prop. (7.4.6.11) [Modifications and Alterations].** Any modification is an alteration(7.4.6.10). And if  $Y \rightarrow X \in \text{Sch}$  be an alteration, then it can be decomposed into  $Y \rightarrow W \rightarrow X$  where  $Y \rightarrow X$  is a modification and  $W \rightarrow X$  is finite surjective.

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*Proof:* ? Use Stein factorization.

□

### Valuation Criteria

**Lemma (7.4.6.12) [Valuation Criteria Lemma].** If  $X$  is a scheme and  $x \rightarrow y$  is a specialization of pts, then for any field extension  $K/k(x)$ , there is a valuation ring  $A \subset K$  and a morphism  $\text{Spec } A \rightarrow X$  that maps the generic pt  $\eta$  to  $x$  and the unique closed pt to  $y$ .

Moreover, if  $X$  is locally Noetherian, then  $A$  can be chosen to be a DVR.  $\lrcorner$

*Proof:* There is a morphism  $\mathcal{O}_{X,y} \rightarrow k(x) \rightarrow K$ , so there is a valuation ring  $A$  with field of fractions  $K$  that dominate the image of  $\mathcal{O}_{X,y}$  by (12.2.2.1), which is also a local ring (5.1.1.11). Then this is what we desire.

For the locally Noetherian case, add (12.2.3.8).  $\square$

**Prop. (7.4.6.13) [Valuation Criteria].** The valuation criterion for  $\text{Spec } K \rightarrow \text{Spec } R$  where  $R$  is a valuation ring with field of fractions  $K$ : Given a morphism  $f : X \rightarrow S$ ,

1. If it is qc, then it is universally closed iff there is at least one lifting.
2. it is separated iff it is quasi-separated and there is at most one lifting.
3. it is proper iff it is finite type, quasi-separated and lifting exists uniquely.

$\lrcorner$

Moreover, if  $S$  is locally Noetherian and  $f$  is locally of f.t., then it suffices to check for discrete valuation rings. *Proof:*

1. Firstly, in this case, by (7.4.4.9), it suffices to prove that: specializations lift along any base change of  $f$  iff it has at least one lifting. If specializations lift along any base change of  $f$ , change  $S$  to  $\text{Spec } A$  and  $X$  to  $X \times_S \text{Spec } A$ . Let  $x'$  be the image of  $\text{Spec } K \rightarrow X$ , then by hypothesis there is a specialization  $x' \rightarrow x$  where  $x$  maps to the closed pt of  $\text{Spec } A$ . Then we get a map  $A \rightarrow \mathcal{O}_{X,x} \rightarrow k(x') \rightarrow K$ , which is exactly the quotient map  $A \rightarrow K$ . So the image of  $\mathcal{O}_{X,x}$  in  $K$  dominates  $A$ , which means it is just  $A$ . Thus we get a map  $\mathcal{O}_{X,x} \rightarrow A$ , which gives a map  $\text{Spec } A \rightarrow X$  that commutes  $\text{Spec } K$ .

Conversely, if  $f$  has at least one lifting, then any base change of  $f$  also has at least one lifting by categorical reason. Thus it suffices to show specializations lift along  $f$ . Let  $s' \rightarrow s$  be a specialization in  $S$  and  $x' \in X$  maps to  $s'$ , we can apply (7.4.6.12) to  $k(s') \subset k(x') = K$ , then we get a lifting diagram, and the image of the closed pt of  $\text{Spec } A$  under the lifting is a point mapping to  $s$ .

2. If it is separated, then if there are two liftings, then consider their equalizer, it is a closed subscheme of  $\text{Spec } A$  by (7.2.7.18), and it contains the generic pt, so it equals  $\text{Spec } A$ , as desired. Conversely, if there are at most one lifting, then we want to prove the diagonal is closed. But by (7.4.5.27) and (7.4.5.22) and the valuation criterion for u.c., it suffices to prove the existence of a lifting for the diagonal map (7.4.6.13). But in fact, a valuation diagram for the diagonal correspond to two liftings of a valuation criterion for  $X \rightarrow S$ , then they are the same, and  $\text{Spec } A \rightarrow X \times_S X$  lifts along the diagonal.

3. follows from the above two.

$\square$

**Prop. (7.4.6.14) [Extension of Rational Maps].** Let  $X, Y$  be schemes over  $S$ ,  $X$  is locally Noetherian and  $Y/S$  is proper. If there is a morphism from an open subset  $U$  of  $X$  to  $Y$ , and there is a point  $x$  in the closure of  $U$  with the stalk being a valuation ring, then the morphism can be extended to an open set containing  $x$ .  $\lrcorner$

*Proof:* We can replace  $X$  by an affine open nbhd of  $X$ . By (7.4.7.1), we assume  $X$  is affine and  $\Gamma(X, \mathcal{O}_X) \subset \mathcal{O}_{X,x}$ . In particular  $X$  is integral with generic pt  $\xi$  with residue field  $K$ . Then  $U$  contains  $\xi$ . By the valuation criterion (7.4.6.13), the morphism  $\text{Spec } K \xrightarrow{\xi} U \rightarrow Y$  can be lifted to a morphism  $\text{Spec } \mathcal{O}_{X,x} \rightarrow Y$ , thus lemma (7.4.7.2) shows there is a morphism on a nbhd  $V$  of  $X$  spreading this morphisms.

Now because  $Y/S$  is separated, the equalizer of this morphism and  $f$  on their intersection is a closed subscheme by (7.4.5.40), but it contains  $\xi$ , so they coincide on the intersection, so we are done.  $\square$

**Prop. (7.4.6.15) [Singularity in Codimension1].** Let  $\varphi : X \rightarrow X'$  be a rational map from a locally Noetherian scheme  $X/K$  regular in codimension1 to a proper scheme  $X'/K$  with maximal domain  $U$ , then

$$\text{codim}(X \setminus U, X) \geq 2.$$

In particular, if  $X$  is non-singular curve, then  $\varphi$  is a morphism.  $\lrcorner$

*Proof:* Use (7.4.6.14), noticing that the stalk at a point of codimension1 is a DVR (5.3.5.21).  $\square$

### Projective Morphism

**Def. (7.4.6.16) [Projective Morphisms].** A **projective morphism**  $X \rightarrow Y$  is a closed immersion  $X \rightarrow \text{Proj}(\mathcal{E})$  for some  $\mathcal{E} \in \mathcal{QCoh}^{\text{ft}}/Y$ .

A **strongly projective morphism**  $X \rightarrow Y$  is a closed immersion  $X \rightarrow \text{Proj}(\mathcal{E})$  for some  $\mathcal{E} \in \mathcal{QCoh}^{\text{lf,ft}}/Y$ .

An **H-projective morphism**  $X \rightarrow Y$  is a closed immersion  $X \rightarrow \mathbb{P}_Y^n$  for some  $n \in \mathbb{Z}_+$ .

An **H-quasi-projective morphism** is an H-projective morphism composed with a quasi-compact open immersion.

A **locally projective morphism** is a morphism  $f : X \rightarrow Y \in \text{Sch}$  s.t. there exists a covering  $U_i$  of  $Y$  that  $f^{-1}(U_i) \rightarrow U_i$  is projective.  $\lrcorner$

**Prop. (7.4.6.17).** For a morphism  $f : X \rightarrow S$ , the following are equivalent:

- $f$  is locally projective.
- There is a covering  $U_i$  of  $S$  that  $f^{-1}(U_i) \rightarrow U_i$  is H-projective.

$\lrcorner$

*Proof:* Clearly 2 implies 1, and for the converse, it suffices to show that projective morphism is locally H-projective. Locally on each affine open nbhd  $U = \text{Spec } R$ ,  $X_U$  is isomorphic to a closed subscheme of  $P(\mathcal{E})$  for some f.t. Qco sheaf of  $\mathcal{O}_U$ -modules  $\mathcal{E}$ . Write  $\mathcal{E} = \widetilde{M}$  for some f.t.  $R$ -module  $M$ , and choose a set of generators  $x_1, \dots, x_n$  for  $M$ , which induces a surjection of graded  $R$ -algebra  $R[X_0, \dots, X_n] \rightarrow \text{Sym}_R(M)$ , then the corresponding morphism  $P(\mathcal{E}) \rightarrow \mathbb{P}^n$  is a closed immersion, so  $f^{-1}(U)$  is a H-projective scheme over  $U$ .  $\square$

**Prop. (7.4.6.18).** H-(quasi-)projectiveness is stable under base change and composition. (because Segre embedding is closed). Disjoint union of f.m. projective morphisms is projective (embed into the Segre embedding).  $\lrcorner$

*Proof:* [Sta]01WE.  $\square$

**Cor. (7.4.6.19) [Projective Maps are Proper].** Projective morphisms are locally projective and locally projective morphisms are proper. Thus a quasi-projective morphism is locally of f.t. and separated (7.4.5.37).  $\lrcorner$

*Proof:* Because locally projective and proper are both local on the base (7.4.4.2), it suffices to show that H-projective morphism is proper by (7.4.6.17).

Because properness satisfies base change trick (7.4.4.2), it suffices to show  $\mathbb{P}_S^n \rightarrow S$  is proper. Also this is base change of  $\mathbb{P}_{\mathbb{Z}}^n \rightarrow \mathbb{Z}$ , it suffices to check this one.  $\mathbb{P}_{\mathbb{Z}}^n$  is clearly separated by (27.2.3.20) (looking at the natural affine covering), and qc. Finally we show it is u.c. using valuation criterion (7.4.6.13):

Let 
$$\begin{array}{ccc} \text{Spec } K & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \text{Spec } R & \longrightarrow & \text{Spec } \mathbb{Z} \end{array}$$
 be a diagram, by induction on  $n$ , we may assume the image  $\xi_1$  of  $\text{Spec } K$

is not contained in any of the hypersurface  $V(x_i)$ , then  $x_i$  are all invertible in  $\mathcal{O}_{\xi_1}$ , and there is a morphism  $\varphi : k(\xi_1) \rightarrow K$ . Let  $f_{ij}$  be the image of  $x_i/x_j$  under  $\varphi$ , and choose  $k$  which  $f_{k0}$  has the minimal valuation, then  $f_{ik} \in R$  for any  $i$ , which means there is a map

$$\mathbb{Z}[x_0/x_k, \dots, x_n/x_k] \rightarrow R$$

compatible with  $\varphi$ , or equivalently a map  $\text{Spec } R \rightarrow D(x_k) \subset X$  commuting the diagram.  $\square$

**Prop. (7.4.6.20) [Descent for Projectiveness].** Let  $X$  be a scheme over a field  $k$ , and  $K/k$  is a field extension, then  $X_k$  is (quasi-)projective iff  $X_K$  is (quasi-)projective.  $\lrcorner$

*Proof:* This follows from (7.5.4.21) and f.f. descent.  $\square$

**Prop. (7.4.6.21).** H-Projective scheme over  $\text{Spec } A$  is of the form  $\text{Proj } S$  where  $S_0 = A$  and  $S$  is f.g over  $S_0$  by  $S_1$  (7.5.2.11).  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.4.6.22) [Projective Morphisms and Closed Embeddings].** Let  $k$  be an alg.closed field and  $\pi : X \rightarrow Y$  is a projective morphism of algebraic schemes over  $k$  that is injective on closed points and injective on tangent vectors at closed points, then  $\pi$  is a closed embedding.  $\lrcorner$

*Proof:* Cf. [Vak17]P506.  $\square$

**Prop. (7.4.6.23) [Chow's Lemma].** Let  $X \rightarrow S$  be separated of f.t over a qcqs  $S$ , then there is a birational, H-projective map  $\pi : X' \rightarrow X$  over  $S$  that  $X'$  is quasi-projective over  $S$ .

If  $X$  is proper, then  $X'$  is projective over  $S$ . And if  $X$  is integral/irreducible/reduced over  $S$ ,  $X'$  can also be chosen to be so.  $\lrcorner$

*Proof:* We only prove for  $S$  Noetherian, for the general case, Cf. [Sta]0203.

In this case all schemes here are Noetherian. Suppose  $X_i, i \leq r$  are irreducible components of  $X$  with generic points  $\eta_i$ , by (7.4.7.5) and the fact  $X$  is qc we can find a finite affine cover  $X = \cup U_i$  that each  $U_i$  contains all the generic points of  $X$ , thus  $U = U_1 \cap \dots \cap U_n$  is dense in  $X$ . Let  $X^*$  be the schematic-closure of  $U$  in  $X$ , and  $U_i^* = U_i \cap X^*$ , then we may replace  $X$  by  $X^*$  and assume that  $U$  is schematically dense in  $X$  by (7.4.5.18) as  $X$  is qs. By (7.5.4.24) and (7.4.4.19), there are immersions



$U_i \rightarrow \mathbb{P}_S^{n_i}$  together with a closed subscheme  $Z_i \subset \mathbb{P}_S^{n_i}$  that  $U_i \rightarrow Z$  is a scheme-theoretically dense open immersion (7.4.5.21).

Consider the map  $(j_1, \dots, j_m) : U \rightarrow \mathbb{P}_S^{n_1} \times \dots \times \mathbb{P}_S^{n_m}$  with scheme-theoretic image  $Z$ , and the commutative diagrams

$$\begin{array}{ccc} U & \xrightarrow{(j_1, \dots, j_m)} & \mathbb{P}_S^{n_1} \times \dots \times \mathbb{P}_S^{n_m} \\ \downarrow & & \downarrow \pi_i \\ U_i & \xrightarrow{j_i} & \mathbb{P}_S^{n_i} \end{array},$$

which induces a map  $p_i : Z \rightarrow Z_i$  (7.4.5.14) which is proper.

Consider  $p_i^{-1}(j_i(U_i)) = V_i$  and  $X' = \cup_i V_i$  which is an open subscheme of  $Z \subset \mathbb{P}_S^{n_1} \times \dots \times \mathbb{P}_S^{n_m}$  thus quasi-projective over  $S$ .

Finally, we prove that the morphism  $p_i : V_i \rightarrow U_i$  glue together to a proper birational morphism  $\pi : X' \rightarrow X$ : this is because they are compatible on  $U$ , which is scheme-theoretically dense in  $V_i$  for each  $i$  thus also scheme-theoretically dense in  $V_i \cap V_j$ , thus  $p_i$  and  $p_j$  are compatible on  $V_i \cap V_j$  as the target  $X$  is separated over  $S$ .

To show  $\pi$  is proper, firstly notice  $\pi^{-1}(U_i) = V_i$ : there are decompositions  $V_i \rightarrow \pi^{-1}(U_i) \rightarrow U_i$ , where  $V_i \rightarrow U_i$  is proper, thus  $V_i \rightarrow \pi^{-1}(U_i)$  is also proper and  $V_i$  is schematically dense in  $\pi^{-1}(U_i)$  because it contains  $U$ , so it is an isomorphism.

For the same reason  $U \rightarrow \pi^{-1}(U)$  is an isomorphism. Finally  $\pi$  is projective because it factors through some map  $X' \rightarrow X \times_S \mathbb{P}_S^n = \mathbb{P}_X^n$  and it is proper.

If  $X$  is reduced, then  $X'$  is reduced by (7.4.5.16), and if  $X$  is irreducible, then  $X' = Z$  is the closure of  $j(X)$  by (7.4.5.21), which is irreducible.  $\square$

## 7 Technical Lemmas

**Lemma (7.4.7.1).** Let  $X$  be a scheme and  $x$  a point, then there exists an open affine nbhd  $U$  of  $x$  that  $\mathcal{O}_X(U) \rightarrow \mathcal{O}_{X,x}$  is injective, if any of the follows holds:

- $X$  is integral.
- $X$  is locally Noetherian.
- $X$  is reduced with f.m. irreducible components.

$\perp$

*Proof:* This problem is clearly local hence follows from the algebra case (5.1.1.37).  $\square$

**Lemma (7.4.7.2) [Spread Out Stalk Morphism].** Let  $X, Y$  be schemes over  $S$ ,  $s \in S$  and  $x, y$  be pts over  $S$ , then:

- Let  $f, g : X \rightarrow Y$  be morphisms over  $S$  that  $f(x) = g(x) = y$  and  $f_x^\# = g_x^\#$ , then  $f = g$  on a nbhd  $U$  of  $x$  if any of the following holds:
  - (a)  $Y/S$  is locally of f.t..
  - (b)  $X$  is integral.
  - (c)  $X$  is locally Noetherian.
  - (d)  $X$  is reduced with f.m. irreducible components.
- Let  $\varphi : \mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$  be a local ring map over  $\mathcal{O}_{S,s}$ , then there is a morphism  $f$  from a nbhd  $U$  of  $x$  mapping to  $Y$  that  $f(x) = y$ , and  $f_x^\# = \varphi$ , if any of the following holds:



- (a)  $Y/S$  is locally of f.p..
- (b)  $Y/S$  is locally of f.t. and  $X$  is integral.
- (c)  $Y/S$  is locally of f.t. and  $X$  is locally Noetherian.
- (d)  $Y/S$  is local of f.t. and  $X$  is reduced with f.m. irreducible components.

┘

*Proof:* Cf. [\[Sta\]0BX6](#).

□

**Prop. (7.4.7.3) [Base Change of Fields is Quotient Map].** For any scheme  $X$  over a field  $k$  and algebraic extensions  $K/k$ ,  $X_K \rightarrow X$  is a quotient map, as it is surjective [\(7.6.2.1\)](#), continuous and closed [\(7.4.4.39\)](#).

┘

**Lemma (7.4.7.4).** Let  $X$  be a qs scheme and  $Z_i$  be a finite set of irreducible components of  $X$ . Let  $\eta_i$  be the generic point of  $Z_i$ , then there are open affine subsets  $U_i \in \mathcal{U}_i$  that  $U_1, \dots, U_n$  are pairwise disjoint.

┘

*Proof:*

□

**Lemma (7.4.7.5).** Let  $X$  be a qs scheme and  $Z_i$  be a finite set of irreducible components of  $X$ . Let  $\eta_i$  be the generic point of  $Z_i$  and  $x \in X$ , then there is an affine open subset of  $X$  containing  $x$  and  $\eta_i$ .

┘

*Proof:* Let  $x \in Z_1, \dots, Z_r$  but  $x \notin Z_{r+1}, \dots, Z_n$ , then we can find an arbitrary affine open nbhd  $W$  of  $x$  that contains  $\eta_1, \dots, \eta_r$  but not  $\eta_{i+1}, \dots, \eta_n$ . By [\(7.4.7.4\)](#), we may choose pairwise disjoint affine open nbhds  $U_{r+1} \in \mathcal{U}_{r+1}, \dots, U_n \in \mathcal{U}_n$ . Now  $U_i \cap W$  is quasi-compact and doesn't contain  $\eta_i$ , so we can shrink  $U_i$  s.t.  $W \cap U_i = \emptyset$  by [\(5.1.7.24\)](#). Then  $U = W \coprod (\coprod U_i)$  satisfies the desired condition. □

References are [Sta] and [Vak17].

## 7.5 Quasi-Coherent Sheaves on Schemes

### 1 (Quasi-)Coherent Sheaves

**Lemma (7.5.1.1) [Associated Qco Sheaves on Affine Scheme].** On an affine scheme  $\text{Spec } A$ , there is a sheaf  $\widetilde{M}$ , that is  $M_f$  on  $\text{Spec } A_f$ . To check it is a sheaf, we only need to check to affine coverings, and this follows from (5.4.2.2).  $\lrcorner$

**Prop. (7.5.1.2) [Qco Sheaves on Affine Schemes].**  $M \mapsto \widetilde{M}$  is an equivalence to the category of quasi-coherent sheaves over  $\text{Spec } A$ . In particular,

$$\text{Hom}_A(M, \Gamma(X, \mathcal{G})) \cong \text{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathcal{G}),$$

so  $M \mapsto \widetilde{M}$  commutes with colimits and is exact, and commutes with pullbacks.

When  $A$  is Noetherian, this also induces an equivalence between finite  $A$ -modules and coherent sheaves over  $\text{Spec } A$ , because finiteness for modules is local (7.4.1.5).  $\lrcorner$

*Proof:* For any  $A$ -module  $M$ , there is a sheaf of modules  $\mathcal{F}_M$  on  $X = \text{Spec } A$  by (7.2.2.26). This is left adjoint to  $\Gamma$  and defines a functor from the category of  $A$ -modules to the category of  $\mathcal{O}_{\text{Spec } A}$ -modules.

By universal property of  $\mathcal{F}_M$  (7.2.2.26), there is a natural map  $\mathcal{F}_M \rightarrow \widetilde{M}$  corresponding to the ring map  $M \mapsto \Gamma(\text{Spec } A, \widetilde{M}) = M$ . The induced maps on the stalk at a point  $x$  is  $M \otimes_A \mathcal{O}_{X,x} \rightarrow M_{\mathfrak{p}}$ , which is isomorphism, so  $\mathcal{F}_M \cong \widetilde{M}$ .

From the universal property of  $\widetilde{M} = \mathcal{F}_M$ ,  $\text{Hom}(\widetilde{M}, \widetilde{N}) = \text{Hom}(M, N)$ , thus  $\sim$  is fully faithful, to show it is an equivalence, it suffices to show for any Qco sheaf  $\mathcal{F}$  on  $\text{Spec } A$ , the natural map  $\Gamma(\widetilde{X}, \mathcal{F}) \rightarrow \mathcal{F}$  is an isomorphism: ? Cf. [Sta]01IA.

If  $0 \rightarrow M_1 \rightarrow M \rightarrow M_2 \rightarrow 0$  is exact, then  $0 \rightarrow \widetilde{M}_1 \rightarrow \widetilde{M} \rightarrow \widetilde{M}_2 \rightarrow 0$  is exact, because localization is exact.  $\square$

**Prop. (7.5.1.3) [properties of (Quasi-)Coherent Sheaves on Schemes].**

- $\mathcal{QCoh}(X)$  and  $\text{Coh}(X)$  are weak Serre subcategories of  $\text{Mod}_{\mathcal{O}_X}$ , and  $\text{Coh}(X)$  is a Serre subcategory of  $\mathcal{QCoh}(X)$ .
- Colimits in  $\text{Mod}(\mathcal{O}_X)$  preserves  $\mathcal{QCoh}(X)$ , because localization is exact.
- Tensor product of two (Q)co sheaf is (Q)co, and locally free if they are locally free. More explicitly,  $\widetilde{M \otimes_A N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$  as tensor product commutes with  $\pi^*$ .
- If  $\mathcal{F}$  is Qco, then so does  $T(\mathcal{F})$ ,  $\text{Sym}(\mathcal{F})$  and  $\wedge(\mathcal{F})$ , by (7.2.3.12).
- Given  $\mathcal{F}, \mathcal{G} \in \mathcal{QCoh}(\mathcal{F})$  that  $\mathcal{F}$  is f.p.,  $\mathcal{H}om(\mathcal{F}, \mathcal{G}) \in \mathcal{QCoh}(X)$ , by (7.2.3.2) and (7.5.1.2). More explicitly, affine locally  $\mathcal{H}om_X(\widetilde{M}, \widetilde{N}) = \text{Hom}_A(M, N)$ .
- pullback preserves  $\mathcal{QCoh}(X)$  and  $\text{Coh}(X)$ , by (7.2.2.29). More explicitly, if  $\text{Spec } B \subset Y$  is mapped into  $\text{Spec } A \subset X$ , then  $f^*(\mathcal{F})(\text{Spec } B) = \mathcal{F}(\text{Spec } A) \otimes_A B$ , using the fact  $\widetilde{M} = \mathcal{F}_M = \pi^* M$ .

$\lrcorner$

*Proof:* 1:  $\text{Coh}(X)$  follows from (7.2.2.28). For  $\text{QCoh}(X)$ , it follows from (7.5.1.2) that the kernel and cokernel of  $\varphi : \widetilde{M} \rightarrow \widetilde{N}$  is just  $\widetilde{\ker \varphi}$  and  $\widetilde{\text{Coker } \varphi}$  which is Qco, for the extension of Qco, use (7.7.1.3) that the global section is exact, so there is a morphism of exact sequences  $\Gamma(\widetilde{X}, \widetilde{\mathcal{F}}_i)(U) \rightarrow \mathcal{F}_i(U)$ , and five lemma gives the result.  $\square$

**Prop. (7.5.1.4) [Qco Sheaves of F.T./F.P].** Let  $X = \text{Spec } A$  and  $\mathcal{F} = \widetilde{M}$  a Qco sheaf on  $X$ , then  $\mathcal{F}$  is of f.t./f.p. if  $M$  is of f.t./f.p. over  $A$ .  $\square$

*Proof:* These follows from the fact finiteness/f.p. are local properties for modules over a ring (5.1.4.4).  $\square$

**Def. (7.5.1.5) [Locally Projective Qco Sheaves].** Let  $X$  be a scheme, then a **locally projective Qco sheaf** is a Qco sheaf  $\mathcal{F}$  on  $X$  that is affine locally a locally projective module sheaf.

Being locally projective for Qco sheaves satisfies fpqc descent, by (5.4.2.1).  $\square$

**Prop. (7.5.1.6) [Qcqs Pushforward].** If  $f$  is qcqs, then the pushforward of a Qco sheaf is Qco. (Used in??).  $\square$

*Proof:* The question is local so we let  $Y$  be affine, and then  $X$  is qcqs, so we cover it with affine opens  $U_i$  and their intersections are  $U_{ijk}$ . Then we see by sheaf property

$$0 \rightarrow f_* \mathcal{F} \rightarrow \oplus_i f_*(\mathcal{F}|_{U_i}) \rightarrow \oplus_{ijk} f_*(\mathcal{F}|_{U_{ijk}})$$

The last two are Qco because two are maps between affine schemes, so the first is Qco.  $\square$

**Prop. (7.5.1.7) [Qco Sheaves on Qcqs Schemes].** For a qcqs scheme  $X$  and  $s \in \Gamma(X, \mathcal{O}_X)$ , and a Qco module  $\mathcal{F}$ ,  $(\Gamma(X, \mathcal{F}))_s \cong \Gamma(X_s, \mathcal{F})$ .  $\square$

*Proof:* The canonical map  $f : X \rightarrow \text{Spec } \Gamma(X, \mathcal{O}_X)$  (7.2.7.5) is qcqs, so  $f_* \mathcal{F}$  is Qco on  $\text{Spec } \Gamma(X, \mathcal{O}_X)$  by (7.5.1.6). Then the result follows from the fact  $f^{-1}(\text{Spec } (\Gamma(X, \mathcal{O}_X)_s)) = X_s$  and the definition of  $f_*$ .  $\square$

**Lemma (7.5.1.8) [Extending Qco Sheaves].** Let  $i : U \rightarrow X$  be a quasi-compact open immersion of schemes,  $\mathcal{F}$  a Qco  $\mathcal{O}_X$ -module and  $\mathcal{G} \subset \mathcal{F}|_U$  a Qco  $\mathcal{O}_U$ -submodule, then there exists a Qco  $\mathcal{O}_X$ -submodule  $\mathcal{G}' \subset \mathcal{F}$  that  $\mathcal{G}'|_U = \mathcal{G}$ .  $\square$

*Proof:* immersion is separated (7.4.5.37), so  $i_* \mathcal{G}$  is a Qco  $\mathcal{O}_X$ -sheaf by (7.5.1.6), and it is a submodule of  $i_* i^* \mathcal{F}$ , so the kernel

$$\mathcal{H} = \ker(\mathcal{F} \oplus i_* \mathcal{G} \rightarrow i_* i^* \mathcal{F})$$

is also Qco by (7.5.1.3), and  $\mathcal{H} \subset \mathcal{F}$ ,  $\mathcal{H}|_U = \mathcal{G}$ .  $\square$

**Prop. (7.5.1.9) [Extending Qco Sheaves of F.T.].** Let  $X$  be a qcqs scheme, and  $U \subset X$  a qc open subset,  $\mathcal{F}$  a Qco  $\mathcal{O}_X$ -module and  $\mathcal{G} \subset \mathcal{F}|_U$  a Qco  $\mathcal{O}_U$ -submodule of f.t., then there exists a Qco  $\mathcal{O}_X$ -submodule  $\mathcal{G}' \subset \mathcal{F}$  of finite type that  $\mathcal{G}'|_U = \mathcal{G}$ .  $\square$

*Proof:* Let  $n$  be the minimal number of affine open subsets  $U_i$  that  $X = U \cup \bigcup U_i$ , by induction on  $n$ , it suffices to prove for  $n = 1$ . Thus we may assume  $X = U \cup V$  where  $U, V$  are affine opens. Now  $U \cap V$  is qc because  $X$  is qs. Then we can change  $(X, U)$  to  $(V, U \cap V)$ , because we can glue the resulting sheaf. Then we reduce to the case  $X$  is affine.

Let  $X = \text{Spec } R$  and  $\mathcal{F} = \widetilde{M}$ , then by (7.5.1.8), there exists a Qco sheaf  $\widetilde{N}$  that  $\widetilde{N}|_U = \mathcal{G}$ . By hypothesis we can cover  $U$  by f.m. open affine  $D(f_i)$  that  $N_{f_i}$  is f.g., by element  $x_{ij}/f_j^{n_i}$ . Let  $N'$  be the submodule of  $N$  generated by these elements  $x_{ij}$ , then  $\widetilde{N}'$  meets our requirement.  $\square$

**Cor. (7.5.1.10)[Qco Sheaf is a Direct Union of Qco Sheaves of F.T.].** Let  $X$  be a qcqs scheme, then any Qco sheaf  $\mathcal{F}$  on  $X$  is a direct colimit of its Qco subsheaves of f.t..  $\lrcorner$

*Proof:* It is a direct colimit because the sum of two Qco sheaves of f.t. is also Qco of f.t.. Now for any affine open  $U \subset X$  and  $s \in \mathcal{O}_X(U)$ ,  $s$  generates a Qco  $\mathcal{O}_U$ -submodule of f.t. of  $\mathcal{F}|_U$ , and by (7.5.1.9) this extends to a Qco  $\mathcal{O}_X$ -submodule of  $\mathcal{F}$ . Then we see that the direct colimit of Qco subsheaves of f.t. of  $\mathcal{F}$  contains elements of  $\mathcal{F}(U)$  for any affine open subset  $U$ , thus it is just  $\mathcal{F}$ .  $\square$

**Def. (7.5.1.11)[Rank of Sheaves].** Let  $X$  be a scheme and  $\mathcal{F}$  a Qco sheaf on  $X$ ,  $p \in X$ , then the **rank of  $\mathcal{F}$  at  $p$**  is  $\text{rank}_p(\mathcal{F}) = \dim_{k(p)} \mathcal{F}_p / \mathfrak{m}_p \mathcal{F}_p$ .  $\lrcorner$

**Prop. (7.5.1.12)[Maximal Qco Submodule].** For  $X$  a scheme and any  $\mathcal{O}_X$ -module  $\mathcal{F}$ , there is a Qco submodule of  $\mathcal{F}$  maximal among all Qco submodules of  $\mathcal{F}$ . This is because a direct colimit of Qco sheaves is Qco (7.5.1.3).  $\lrcorner$

**Prop. (7.5.1.13)[Integral and Finite Modules].** Let  $X$  be a qcqs scheme and  $\mathcal{A}$  an integral Qco  $\mathcal{O}_X$ -submodule, then

- $\mathcal{A}$  is the directed colimits of its finite Qco  $\mathcal{O}_X$ -modules.
- $\mathcal{A}$  is a direct colimit of finite and finitely presented Qco  $\mathcal{O}_X$ -modules.

$\lrcorner$

*Proof:* Cf. [Sta]0817.  $\square$

**Def. (7.5.1.14). ?**

- for a closed immersion  $Y \rightarrow X$ , there is  $i^! : \mathcal{Q}\text{Coh}(X) \rightarrow \mathcal{Q}\text{co}(Y)$  that is right adjoint to  $i_*$ :  $i^! \mathcal{G} = i^*(\mathcal{H}_Z(\mathcal{G})')$ , where  $\mathcal{H}_Z(\mathcal{G})$  is the sheaf of sections annihilated by  $\mathcal{I}$  and  $\mathcal{F}'$  is the maximal Qco sheaf of  $\mathcal{F}$ .
- For  $f$  proper between locally Noetherian scheme, there is a inverse sheaf  $f^! \mathcal{G} = \mathcal{H}\text{om}_Y(f_* \mathcal{O}_X, \mathcal{G})$ , which maps  $\mathcal{Q}\text{co}(Y)$  to  $\mathcal{Q}\text{Coh}(X)$  by (7.5.1.33) and??. When  $f$  is affine, in particular when it is finite, then  $f^!$  is right adjoint to  $f_*$  on  $\mathcal{Q}\text{co}(7.9.2.12)$ .

$\lrcorner$

### Associated Points

**Def. (7.5.1.15)[Weakly Associated Points].** Let  $X \in \text{Sch}$  and  $\mathcal{F} \in \mathcal{Q}\text{Coh}(X)$ , then a **weakly associated point of  $\mathcal{F}$**  is the set of points  $x \in X$  that  $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$  is weakly associated to the module  $\mathcal{F}_x$ . The set of w.ass points are denoted by  $\text{WeakAsso}(\mathcal{F})$ . The **weakly associated points of  $X$**  is the weakly associated points of  $\mathcal{O}_X$ .

Similarly we can stalkwise define the set of **associated points** of  $X$ .  $\lrcorner$

**Def. (7.5.1.16)[Embedded Points].** Let  $X$  be a scheme and  $\mathcal{F}$  a Qco sheaf on  $X$ , then an **embedded point** of  $\mathcal{F}$  is an associated point that is a specialization of another associated point.  $\lrcorner$

**Prop. (7.5.1.17)[Properties of Associated Points].** Let  $X$  be a scheme and  $\mathcal{F}_i$  be Qco sheaves on  $X$ , then

1.  $\text{Ass}(\mathcal{F}) \subset \text{WeakAsso}(\mathcal{F}) \subset \text{Supp}(\mathcal{F})$ .
2. If  $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$  is an exact sequence, then  $\text{WeakAss}(\mathcal{F}_1) \subset \text{WeakAss}(\mathcal{F}_2)$ , and  $\text{WeakAss}(\mathcal{F}_2) \subset \text{WeakAsso}(\mathcal{F}_1) \cup \text{WeakAsso}(\mathcal{F}_3)$ .
3.  $\text{WeakAss}(\mathcal{F}) = \emptyset$  iff  $\mathcal{F} = 0$ .

4. The generic points of  $\text{Supp}(\mathcal{F})$  are in  $\text{WeakAsso}(\mathcal{F})$ . In particular, generic points of  $X$  are in  $\text{WeakAss}(X)$ .
5. If  $X$  is locally Noetherian, then  $\text{Ass}(\mathcal{F}) = \text{WeakAsso}(\mathcal{F})$ .
6. If  $X$  is reduced, then  $\text{WeakAsso}(X)$  is just the generic points of  $X$ . In particular,  $X$  has no embedded points.
7. If  $X$  is locally Noetherian, then an associated point is an embedded point iff it is not a generic point of  $\text{Supp}(\mathcal{F})$ .
8. If  $X$  is locally Noetherian and  $\mathcal{F}$  is coherent, then  $\mathcal{F}$  has no embedded points iff it satisfies  $(S_1)$ .
9. If  $X$  is locally Noetherian of dimension  $\leq 1$ , then  $X$  is C.M. iff it has no embedded points.

┘

*Proof:* By (5.2.5.21), these reduces to the affine case, so they follows from

- (5.2.5.13)(5.2.5.16).
- (5.2.5.14).
- (5.2.5.16).
- (5.2.5.16).
- (5.2.5.21).
- (5.2.5.24).
- (5.2.5.23).
- [Sta]031Q.
- [Sta]0BXG.

□

**Prop. (7.5.1.18).** Let  $X$  be a scheme and  $\mathcal{F} \in \mathcal{QCoh}(X)$ . If  $U$  is an open subset of  $X$  that contains  $\text{WeakAsso}(\mathcal{F})$ , then  $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F})$  is injective.

┘

*Proof:* If  $s \in \Gamma(X, \mathcal{F})$  be a section that restricts to 0 on  $U$ , let  $\mathcal{F}'$  be the subsheaf generated by  $s$ , then  $\text{WeakAss}(\mathcal{F}') \subset \text{WeakAsso}(\mathcal{F})$ , but  $\text{Supp}(\mathcal{F}') \subset X \setminus U$ , thus  $\text{WeakAss}(\mathcal{F}') = \emptyset$ , so  $\mathcal{F}' = 0$  by (7.5.1.17). □

**Prop. (7.5.1.19) [Schematically Dense and Associated Points].** Let  $X$  be a locally Noetherian scheme and  $U \subset X$  an open subset, then the following are equivalent:

- $U$  is schematically dense in  $X$ .
- $U$  is dense in  $X$  and contains all the embedded points of  $X$ .
- $U$  contains  $\text{Ass}(X)$ .

┘

*Proof:* The problem is local, so we assume  $X = \text{Spec } A$ , then 2, 3 are clearly equivalent.

Let  $U = \cup_i D(f_i)$ , then by (7.4.5.15),  $U$  is schematically dense in  $A$  iff  $A \rightarrow \prod A_{f_i}$  is injective. If  $\mathfrak{p} = \text{Ann}(x)$  for some  $x \in A$ , then  $x$  maps to some non-zero element of  $A_{f_i}$ , then  $f_i \notin \mathfrak{p}$ , so  $\mathfrak{p} \subset D(f_i) \subset U$ . Conversely, if  $\text{Ass}(X) \subset U$ , then every map  $A \rightarrow A_{\mathfrak{p}}$  factors through  $A \rightarrow A_{f_i}$  for some  $i$ , so injectivity of  $A \rightarrow \prod_{\mathfrak{p} \in \text{Ass}(X)} A_{\mathfrak{p}}$  implies injectivity of  $A \rightarrow \prod_i A_{f_i}$ . □

**Prop. (7.5.1.20).** Let  $X$  be a scheme and  $\varphi : \mathcal{F} \rightarrow \mathcal{G} \in \mathcal{QCoh}(X)$  that  $\mathcal{F}_x \rightarrow \mathcal{G}_x$  is injective for any  $x \in \text{WeakAss}(\mathcal{F})$ , then  $\varphi$  is injective.  $\lrcorner$

*Proof:* The hypothesis says  $\text{WeakAss}(\ker(\varphi)) = 0$ , so  $\ker(\varphi) = 0$  by (7.5.1.17).  $\square$

### Fitting Ideals

**Def. (7.5.1.21) [Fitting Ideals].** Let  $X \in \text{Sch}$  and  $\mathcal{F} \in \mathcal{QCoh}^{\text{ft}}(X)$ , the **fitting ideal** of  $\mathcal{F}$  is defined to be  $\mathfrak{f}(\mathcal{F})$ . Cf. [Sta]0C3C.  $\lrcorner$

### Locally Free Sheaves

**Prop. (7.5.1.22) [Locally Free is Stalkwise].**  $\mathcal{F} \in \mathcal{QCoh}^{\text{pf}}(X)$  is locally free iff its stalks are all free, by (5.3.1.7).  $\lrcorner$

**Def. (7.5.1.23) [Locally Free Sheaves].** Let  $X \in \text{Sch}$  and  $\delta : X \rightarrow \mathbb{N}$  a locally constant function, then the category of locally free sheaves of rank  $\delta$  is denoted by  $\text{Coh}_{\text{lf}}^{\delta}(X)$ . The category of locally free coherent sheaves is denoted by  $\text{Coh}_{\text{lf}}(X)$ .  $\lrcorner$

**Prop. (7.5.1.24) [Locally Free Sheaves].** Suppose  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$  is an exact sequence of sheaves on a scheme  $X$ , then

- If  $\mathcal{F}', \mathcal{F}''$  are both locally free, then so is  $\mathcal{F}$ .
- If  $\mathcal{F}, \mathcal{F}'$  are both locally free of finite rank, then so is  $\mathcal{F}''$ .

$\lrcorner$

*Proof:* Firstly of all in each case all sheaves are Qco by (7.5.1.3).

1: on an affine open  $U = \text{Spec } A \subset X$ , the exact sequence is induced by  $0 \rightarrow A^I \rightarrow \Gamma(U, \mathcal{F}) \rightarrow A^J \rightarrow 0$ , so it splits and  $\Gamma(U, \mathcal{F}) \cong A^{I+J}$  is free.

2: on an affine open  $U = \text{Spec } A \subset X$ , the exact sequence is induced by  $0 \rightarrow \Gamma(U, \mathcal{F}') \rightarrow A^m \rightarrow A^n \rightarrow 0$ , where the map  $A^m \rightarrow A^n$  is represented by a  $n \times m$  matrix  $M$ . Then  $U$  is covered by open subsets  $U_i$  that some  $n \times n$  minor of  $M$  is invertible. Then after a change of coordinates on each subset,  $\mathcal{F}(U_i) \cong A^{m-n}$ .  $\square$

**Prop. (7.5.1.25).** For a exact sequence  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$  in  $\mathcal{QCoh}^{\text{free}}(X)$ , there is a filtration of  $\text{Sym}^r \mathcal{F}$ :

$$0 = \mathcal{G}^{r+1} \subset \mathcal{G}^r \subset \dots \subset \mathcal{G}^0 = \text{Sym}^r \mathcal{F}$$

that

$$\mathcal{G}^p / \mathcal{G}^{p+1} \cong \text{Sym}^p \mathcal{F}' \otimes \text{Sym}^{r-p} \mathcal{F}''.$$

$\lrcorner$

*Proof:* On any affine open subset, choose a splitting of the exact sequence, then use coordinates.  $\square$

**Prop. (7.5.1.26).** For a exact sequence  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$  in  $\text{Coh}_{\text{lf}}(X)$ , there is a filtration of  $\wedge^r \mathcal{F}$ :

$$0 = \mathcal{G}^{r+1} \subset \mathcal{G}^r \subset \dots \subset \mathcal{G}^0 = \wedge^r \mathcal{F}$$

that

$$\mathcal{G}^p / \mathcal{G}^{p+1} \cong \wedge^p \mathcal{F}' \otimes \wedge^{r-p} \mathcal{F}''.$$

In particular,

$$\wedge \mathcal{F}' \otimes \wedge \mathcal{F}'' \cong \wedge \mathcal{F}.$$

and when  $\mathcal{F}'' \cong \mathcal{L}$  is a line bundle, there is an exact sequence

$$0 \rightarrow \wedge^r(\mathcal{F}') \rightarrow \wedge^r(\mathcal{F}) \rightarrow \wedge^{r-1}(\mathcal{F}') \otimes \mathcal{L} \rightarrow 0$$

┘

*Proof:* On any affine open subset, choose a splitting of the exact sequence, then use coordinates.

□

**Prop. (7.5.1.27) [Perfect Pairing Wedge Product Sheaf].** Let  $\mathcal{F}$  be a locally free sheaf of rank  $n$ , then there is a perfect pairing  $\wedge^r \mathcal{F} \otimes \wedge^{n-r} \mathcal{F} \rightarrow \wedge \mathcal{F}$  which is a perfect pairing, i.e. it induces an isomorphism  $\wedge^r \mathcal{F} \cong (\wedge^{n-r} \mathcal{F})^\vee \otimes \wedge \mathcal{F}$ .

┘

*Proof:* The map is natural, and the isomorphism can be seen at the level of stalks, by (7.2.3.11). □

**Prop. (7.5.1.28).** Let  $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$  is an exact sequence in  $\mathcal{Q}\text{Coh}(X)$  where  $\mathcal{H}$  is a locally free sheaf, then for any Qco sheaf  $\mathcal{E}$  of  $X$ , then for any Qco sheaf  $\mathcal{E}$  on  $X$ , there is an exact sequence of sheaves

$$0 \rightarrow \mathcal{H}om(\mathcal{H}, \mathcal{E}) \rightarrow \mathcal{H}om(\mathcal{G}, \mathcal{E}) \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{E}) \rightarrow 0.$$

In particular,  $0 \rightarrow \mathcal{H}^\vee \rightarrow \mathcal{G}^\vee \rightarrow \mathcal{F}^\vee \rightarrow 0$  is exact.

┘

*Proof:* As  $\mathcal{H}om(-, \mathcal{E})$  is left exact, it suffices to show the last one is surjective. This is local, so we may assume  $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$  is  $0 \rightarrow M \rightarrow N \rightarrow A^n \rightarrow 0$ , so this sequence splits, and then  $\mathcal{H}om(\mathcal{G}, \mathcal{E}) \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{E})$  is surjective, by (7.5.1.3). □

**Prop. (7.5.1.29) [Grothendieck].** Every object in  $\text{Coh}^{\text{free}, r}(\mathbb{P}_k^1)$  is isomorphic to  $\bigoplus_{i=1}^r \mathcal{O}(a_i)$  for a unique non-decreasing sequence of integers  $a_1, \dots, a_r$ .

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*Proof:* Use induction on  $r$ . The  $r = 1$  case follows from (7.5.3.16). For a general  $r$ , let  $\mathcal{E} \in \text{Coh}^{\text{free}, r}(\mathbb{P}_k^1)$ , then  $\mathcal{E}(m)$  is generated by global sections for  $m$  large as  $\mathcal{O}_{\mathbb{P}^1}$  is ample, and  $H^0(\mathcal{E}(-m)) = H^1(\mathcal{E}^\vee(m)) = 0$  by Serre duality and (7.7.2.5), so there is a maximal  $m$  s.t. there is a non-zero map  $\mathcal{O}(m) \rightarrow \mathcal{E}$ . The image of this map is also locally free and has degree  $\geq m$  by (7.13.2.8), so it must have degree  $m$  and  $\mathcal{O}_X(m) \rightarrow \mathcal{E}$  is injective by (7.13.2.8). Now the cokernel  $\mathcal{F}$  is also locally free, because otherwise let  $\mathcal{F}_{\text{tor}}$  be the torsion part of  $\mathcal{F}$  (7.13.2.16), and let  $\mathcal{N}$  be the inverse image of  $\mathcal{F}_{\text{tor}}$  in  $\mathcal{E}$ , then it is locally free thus invertible, and  $\mathcal{O}(m) \hookrightarrow \mathcal{N}$ , so  $N = \mathcal{O}(m)$  by (7.13.2.8), and  $\mathcal{F}_{\text{tor}} = 0$ .

There is an exact sequence

$$0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{E}(-m-1) \rightarrow \mathcal{F}(-m-1) \rightarrow 0$$

which induces  $H^0(\mathcal{E}(-m-1)) = H^0(\mathcal{F}(-m-1)) = 0$  as  $H^1(\mathbb{P}_k^1, \mathcal{O}(-1)) = 0$ , so by induction hypothesis  $\mathcal{F} = \bigoplus_{i=1}^{r-1} \mathcal{O}(a_i)$  and  $a_i \leq m$ . To show this sequence splits, notice  $\text{Ext}^1(\mathcal{F}(-m-1), \mathcal{O}(-1)) = \bigoplus_{i=1}^{r-1} H^1(\mathcal{O}(m-a_i)) = 0$ . □

**Prop. (7.5.1.30) [Splitting Principal].** Let  $X$  be an integral scheme and  $\mathcal{E} \in \text{Coh}^{\text{free}}(X)$ , then there exists a modification  $X' \rightarrow X$  s.t.  $f^* \mathcal{E}$  has a filtration by invertible sheaves.

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*Proof:* Use induction on  $\text{rank}(\mathcal{E})$ . If  $\text{rank}(\mathcal{E}) \leq 1$ , this is trivial, otherwise let  $\text{rank}(\mathcal{E}) = r$ ,  $P = \mathbf{P}(\mathcal{E})$ , then  $\pi : P \rightarrow X$  is proper and there is a canonical surjection  $\pi^* \mathcal{E} \rightarrow \mathcal{O}_P(1)$ , with kernel in  $\text{Coh}^{\text{free}, r-1}(P)$ . Let  $U$  be an open subset of  $X$  s.t.  $\mathcal{E}$  is trivial, then  $\pi^{-1}(U) \cong \mathbb{P}_U^{r-1}$ . Let  $s : U \rightarrow \mathbb{P}^{r-1}(U)$  be a section, and let  $X'$  be the scheme-theoretic closure of  $U$ , which is integral. Then  $X' \rightarrow X$  is proper and thus a modification. Now  $f^* \mathcal{E}$  has an invertible quotient, and we are done by induction hypothesis. □

### Coherent Sheaves

**Prop. (7.5.1.31) [Coh(X)].** The category of coherent sheaves on a scheme  $X$  is denoted by  $\text{Coh}(X)$ . When  $\mathcal{O}_X \in \text{Coh}(X)$ ,  $\text{Coh}(X) = \mathcal{Q}\text{Coh}^{\text{pf}}(X)$  (7.2.2.28).

In particular when  $X$  is locally Noetherian,  $\text{Coh}(X) = \mathcal{Q}\text{Coh}^{\text{ft}}(X)$ . And the notion of coherence are used usually only when  $X$  is locally Noetherian.

(Quasi-)coherence is an affine local property by (7.2.2.30).  $\mathcal{Q}\text{Coh}(X)$  is an Abelian category, by (7.2.2.28).  $\lrcorner$

**Lemma (7.5.1.32).** If  $f$  is finite, then  $f_*$  maps coherent sheaves to coherent sheaves.  $\lrcorner$

*Proof:* This is trivial.  $\square$

**Prop. (7.5.1.33) [Proper Pushforward].** If  $f$  is a proper morphism between locally Noetherian schemes, then  $f_*$  maps coherent sheaves to coherent sheaves.  $\lrcorner$

*Proof:* Immediate from Grothendieck's coherence theorem (7.7.4.11).  $\square$

**Prop. (7.5.1.34) [Artin-Rees].** Let  $X$  be a Noetherian scheme,  $\mathcal{F}$  a coherent  $\mathcal{O}_X$ -module,  $\mathcal{G}$  a Qco subsheaf of  $\mathcal{F}$ ,  $\mathcal{I} \subset \mathcal{O}_X$  a Qco sheaf of ideals, then there exists some  $c > 0$  that for all  $n \geq c$

$$\mathcal{I}^{n-c}(\mathcal{I}^c \mathcal{F} \cap \mathcal{G}) = \mathcal{I}^n \mathcal{F} \cap \mathcal{G}.$$

$\lrcorner$

*Proof:* Cover  $X$  by f.m. affine open subsets, then this follows from the affine case (5.2.2.13).  $\square$

**Cor. (7.5.1.35) [Vanish Analytically].** Let  $X$  be a Noetherian scheme,  $\mathcal{F}$  a coherent  $\mathcal{O}_X$ -module, for any element  $f \in \bigcap_n \mathfrak{m}_x^n \mathcal{F}$ ,  $f$  vanishes at a nbhd of  $x$ .  $\lrcorner$

*Proof:* This follows from the intersection theorem (5.2.2.14).  $\square$

**Prop. (7.5.1.36) [Deligne].** On a Noetherian scheme  $X$ , let  $\mathcal{F}$  be a Qco sheaf,  $\mathcal{G}$  be a coherent sheaf and  $\mathcal{I}$  be a Qco sheaf of ideals corresponding to  $Z$ ,  $U = X - Z$ , then we have

$$\varinjlim \text{Hom}_{\mathcal{O}_X}(\mathcal{I}^n \mathcal{G}, \mathcal{F}) \cong \text{Hom}_{\mathcal{O}_U}(\mathcal{G}|_U, \mathcal{F}|_U).$$

In particular,

$$\varinjlim \text{Hom}_{\mathcal{O}_X}(\mathcal{I}^n, \mathcal{F}) \cong \Gamma(U, \mathcal{F}).$$

$\lrcorner$

*Proof:* Cf. [Sta]01YB.  $\square$

**Prop. (7.5.1.37) [Kleinmann].** If  $X$  is a Noetherian integral separated locally factorial scheme, then every coherent sheaf on  $X$  is a quotient of a finite locally free sheaf.  $\lrcorner$

*Proof:* Cf. [Hartshorne P238].  $\textcolor{red}{?}$   $\square$

**Prop. (7.5.1.38) [Support of Modules].** For  $X \in \text{Sch}$  and  $\mathcal{F} \in \mathcal{Q}\text{Coh}^{\text{ft}}(X)$ , the support (7.2.2.2)  $\text{Supp}(\mathcal{F})$  is closed by (7.2.2.24).

For a flat morphism  $f$ ,  $\text{Supp}(f^*(\mathcal{F})) = f^{-1}(\text{Supp } \mathcal{F})$ , by (7.2.2.18).  $\lrcorner$



*Proof:* because affine locally, for a set of generators  $x_i$  of  $M$ ,  $\text{Ann}(\mathcal{F}) = \cup \text{Ann}(x_i)$ , and  $\text{Ann}(x_i)$  is closed.  $\square$

**Cor. (7.5.1.39).** Any coherent sheaf on an integral scheme is locally free over a dense open subset.  $\lrcorner$

**Cor. (7.5.1.40) [Geometric Nakayama].** If  $\mathcal{F} \in \mathcal{QCoh}^{\text{ft}}(X)$  and for some  $p \in U \subset X$  and  $a_1, \dots, a_n \in \mathcal{F}(U)$  generate  $\mathcal{F}_p \otimes k(p)$ , then there is an affine nbhd  $V \subset U$  that  $a_1, \dots, a_n$  generate  $\mathcal{F}(V)$ .  $\lrcorner$

**Cor. (7.5.1.41) [Upper-Semicontinuity of Ranks].** For a Qco sheaf  $\mathcal{F}$  of f.t. over  $X$ , the rank function  $p \mapsto \text{rank}_p(\mathcal{F})$  is an upper semicontinuous function on  $X$ .  $\lrcorner$

*Proof:* By Nakayama,  $\varphi(y)$  is equal to the minimal number of generators of the  $\mathcal{O}_y$ -module  $\mathcal{F}_y$ . But these generators extends to a nbhd of  $y$ , so  $\varphi \leq n$  on this nbhd.  $\square$

**Def. (7.5.1.42) [Schematic Support].** Let  $X$  be a scheme and  $\mathcal{F}$  a Qco sheaf of f.t. over  $X$ , then the **annihilating ideal** is the ideal  $\mathcal{I} \subset \mathcal{O}_X$  that is affine-locally defined by  $\text{Ann}(\mathcal{F}(U)) \subset \mathcal{O}_X(U)$ , and the **schematic support**  $\text{Supp}(\mathcal{F})$  of  $\mathcal{F}$  is the closed subscheme of  $X$  corresponding to  $\text{Ann}(\mathcal{F})$ .  $\lrcorner$

**Prop. (7.5.1.43).** Let  $\mathcal{F}$  be a Qco sheaf of f.t. over a reduced scheme that the rank function  $\text{rank}_p(\mathcal{F})$  is locally constant, then it is locally free.  $\lrcorner$

*Proof:* Let  $\text{rank}_p(\mathcal{F}) = n$ , by (7.5.1.38), for any  $x \in X$ , there exists an affine nbhd  $U = \text{Spec } A$  of  $x$  and a surjection  $f : \mathcal{O}_U^n \rightarrow \mathcal{F}_U$ . Consider the kernel of  $f$ . If  $(r_1, \dots, r_n)$  is in the kernel and  $r_1 \neq 0$ , then  $r_1 \notin \mathfrak{p}$  for some prime  $\mathfrak{p}$  of  $A$ , as  $A$  is reduced. Then  $\text{rank}_{x_{\mathfrak{p}}}(\mathcal{F}) < n$ , contradiction.  $\square$

### Torsion-Free Sheaves

**Def. (7.5.1.44) [Torsion Sheaves].** Let  $X$  be an integral scheme, then a **torsion sheaf** is a Qco sheaf that its stalk at the generic point of  $X$  vanishes. Equivalently, for any affine open  $U \subset X$ ,  $\mathcal{F}(U)$  is a torsion  $\mathcal{O}_X(U)$ -module.  $\lrcorner$

**Prop. (7.5.1.45).** A torsion  $\mathcal{O}_X$ -module of f.t. on an integral scheme vanish on a dense open subset.  $\lrcorner$

**Prop. (7.5.1.46).** For  $X \in \text{Sch}_{\text{int}}$ , any  $\mathcal{F} \in \mathcal{QCoh}(X)$  factors as  $0 \rightarrow \mathcal{F}_{\text{tor}} \rightarrow \mathcal{F} \rightarrow \mathcal{F}_{\text{tf}} \rightarrow 0$ , where  $\mathcal{F}$  is a torsion sheaf (7.5.1.44) and  $\mathcal{F}_{\text{tf}}$  is Qco and torsion-free.  $\lrcorner$

**Cor. (7.5.1.47).** The subpresheaf  $U \mapsto \{s \in \mathcal{F}(U) | s_{\eta} = 0\}$  is a sheaf, and it is also Qco as on an affine open  $U \subset X$ , it is just  $\mathcal{F}(U)_{\text{tor}}$ . So the quotient is clearly Qco and torsion-free.  $\lrcorner$

**Prop. (7.5.1.48) [Formal Glueing of Vector Bundles].** If  $X$  be a Dedekind scheme and  $x \in |X|_0$ , let  $\hat{X}_x = \text{Spec}(\hat{\mathcal{O}}_{X,x}) \rightarrow X$  be the completion of  $X$  at  $x$ , then there is a pullback of categories:

$$\begin{array}{ccc} \text{Coh}_{\text{lf}}(X) & \longrightarrow & \text{Coh}_{\text{lf}}(X \setminus \{x\}) \\ \downarrow & & \downarrow \\ \text{Coh}_{\text{lf}}(\hat{X}_x) & \longrightarrow & \text{Coh}_{\text{lf}}(\hat{X}_x \setminus \{x\}) \end{array}$$

$\lrcorner$

*Proof:* By glueing, we may study locally near  $x$ , then we can assume that  $X = \text{Spec } A$  is affine. Now shrink  $X$  even more, we can assume that  $x$  is defined by a single  $f \in A$  (localized at the maximal ideal defined by  $x$ ), then we finish by Beauville-Laszlo theorem (5.4.2.15).  $\square$

### Devissage of Coherent Sheaves

**Lemma (7.5.1.49).** Let  $X$  be a Noetherian scheme and  $\mathcal{F} \in \text{Coh}(X)$ , let  $\mathcal{I}$  be a sheaf of ideals that correspond to  $Z$ , then  $\text{Supp}(\mathcal{F}) \subset Z$  iff  $\mathcal{I}^n \mathcal{F} = 0$  for some  $n$ . (This follows easily from Noetherian and (5.2.5.8)).  $\lrcorner$

**Lemma (7.5.1.50).** Let  $X$  be a Noetherian scheme and  $\mathcal{F} \in \text{Coh}(X)$  s.t.  $\text{Supp}(\mathcal{F}) = Z_1 \cup Z_2$  where  $Z_1, Z_2$  are closed, then there is an exact sequence of coherent sheaves  $0 \rightarrow \mathcal{G}_1 \rightarrow \mathcal{F} \rightarrow \mathcal{G}_2 \rightarrow 0$  that  $\text{Supp}(\mathcal{G}_i) \subset Z_i$ .  $\lrcorner$

*Proof:* Let  $\mathcal{I}$  be the ideal defining the induced reduced structure of  $Z_2$ , we use the exact sequence  $0 \rightarrow \mathcal{I}^n \mathcal{F} \rightarrow \mathcal{F} \rightarrow \text{Coker} \rightarrow 0$  and use (7.5.1.49) on  $X \setminus Z_2$  to find  $n$  large that  $\text{Supp}(\mathcal{I}^n \mathcal{F}) \subset Z_1$ , and notice  $\text{Coker}$  has support in  $Z_2$ .  $\square$

**Lemma (7.5.1.51).** Let  $X$  be an integral scheme and  $Z \subset X$  an integral closed subscheme with generic point  $Z$ . If  $\mathcal{F} \in \text{Coh}(X)$  satisfies  $\mathcal{F}_\xi$  is annihilated by  $\mathfrak{m}_\xi$ , then there exists some  $r \geq 0$  and sheaf of ideals  $\mathcal{I}$  on  $Z$  and an injection  $i_*(\mathcal{I}^{\oplus r}) \hookrightarrow \mathcal{F}$  that is an isomorphism at  $\xi$ .  $\lrcorner$

*Proof:* Cf. [Sta]01YE.  $\square$

**Prop. (7.5.1.52).** Let  $X$  be a Noetherian scheme and  $\mathcal{F} \in \text{Coh}(X)$ , then there is a filtration of coherent sheaves that the quotients are pushforward of ideal sheaves on integral subschemes of  $X$ . This is analogous to the filtration in the module case.  $\lrcorner$

*Proof:* We consider the set of these counterexamples and their  $\text{Supp}$ , then use Noetherian induction. The minimal one  $Z$  is irreducible, otherwise from (7.5.1.50) we find a filtration for it. Let the ideal of sheaf of the induced reduced structure of  $Z$  be  $\mathcal{I}$ , then  $\mathcal{I}^n \mathcal{F} = 0$  for some  $n$  by (7.5.1.49), then we may assume  $\mathcal{I} \mathcal{F} = 0$ . Then we use (7.5.1.51) to finish the proof.  $\square$

**Cor. (7.5.1.53) [Basic Dévissage].** Let  $X \in \text{NSch}_{\text{cpct}}$  and  $\mathcal{P}$  be a property of coherent sheaves on  $X$  s.t.

- (1) for an exact sequence of sheaves:  $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_2 \rightarrow 0$ , if  $\mathcal{F}_i$  has  $\mathcal{P}$ , then  $\mathcal{F}$  has  $\mathcal{P}$ .
- (2) for any integral closed subscheme  $i : Z \subset X$  and any Qco sheaf of ideals  $\mathcal{I}$  on  $Z$ ,  $\mathcal{P}$  holds for  $i_* \mathcal{I}$ .

then  $\mathcal{P}$  holds for any coherent sheaf of  $X$ .  $\lrcorner$

**Lemma (7.5.1.54).** Cf. [Sta]01YH.  $\lrcorner$

**Prop. (7.5.1.55) [Dévissage of Coherent Sheaves I].** Let  $X$  be a Noetherian scheme and  $P$  be a property of coherent sheaves on  $X$  s.t.

- (1) for an exact sequence  $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_2 \rightarrow 0 \in \text{Coh}(X)$ , if two of them have  $P$ , then the third also has  $P$ .
- (2) For every integral closed subscheme  $Z$  of  $X$  with generic point  $\xi$ , there is a  $\mathcal{G} \in \text{Coh}(X)$  that
  - (a)  $\text{Supp } \mathcal{G} = Z$ .
  - (b)  $\mathcal{G}_\xi \cong \mathcal{O}_X / \mathfrak{m}_x$ .
  - (c)  $P$  holds for  $\mathcal{G}$ .

Then  $P$  holds for every coherent sheaf on  $X$ .  $\lrcorner$

*Proof:* Suppose otherwise, let  $Z$  be the minimal counterexample of item2 of (7.5.1.55). Then it is easily seen that  $P$  holds for any  $\mathcal{F}$  with support strictly contained in  $\mathbb{Z}$ . Then take  $\mathcal{G}$  as in item2, and let  $0 \rightarrow i_*(\mathcal{I}^{\oplus r}) \hookrightarrow \mathcal{G} \rightarrow \text{Coker} \rightarrow 0$ , then  $\text{Supp}(\text{Coker})$  is strictly contained in  $Z$ , and  $r = 1$  by hypothesis, so  $i_*(\mathcal{I})$ . Then for any other sheaf of ideals  $\mathcal{I}'$  on  $Z$ , if  $\mathcal{I}'$  is supported on a smaller subscheme, then it has  $P$ , otherwise there are two exact sequences

$$0 \rightarrow i_*\mathcal{I} \rightarrow i_*(\mathcal{I} + \mathcal{I}') \rightarrow \mathcal{Q} \rightarrow 0,$$

$$0 \rightarrow i_*\mathcal{I}' \rightarrow i_*(\mathcal{I} + \mathcal{I}') \rightarrow \mathcal{Q} \rightarrow 0,$$

where  $\mathcal{I}, \mathcal{Q}$  are supported on smaller subschemes, so  $P$  also holds for  $\mathcal{I}'$ .  $\square$

**Prop. (7.5.1.56) [Dévissage of Coherent Sheaves II].** Let  $X$  be a Noetherian scheme and  $P$  be a property of coherent sheaves on  $X$  s.t.

- (1) for an exact sequence of sheaves:  $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_2 \rightarrow 0$ , if  $\mathcal{F}_i$  has  $P$ , then  $\mathcal{F}$  has  $P$ .
- (2) If  $\mathcal{F}^{\oplus r}$  has  $P$ , then  $\mathcal{F}$  has  $P$ .
- (3) For every integral closed subscheme  $Z$  of  $X$  with generic point  $\xi$ , there is a  $\mathcal{G} \in \text{Coh}(X)$  that
  - (a)  $\text{Supp } \mathcal{G} = Z$ .
  - (b)  $\mathcal{G}_\xi$  is annihilated by  $\mathfrak{m}_\xi$ .
  - (c) For every sheaf of ideals  $\mathcal{I}$  on  $X$  that  $\mathcal{I}_\xi = \mathcal{O}_{X,\xi}$ , there is a Qco subsheaf  $\mathcal{G}' \subset \mathcal{I}\mathcal{G}$  that  $\mathcal{G}'_\xi = \mathcal{G}_\xi$  and  $P$  holds for  $\mathcal{G}'$ .

Then  $P$  holds for every coherent sheaf on  $X$ .  $\lrcorner$

*Proof:* Cf. [Sta]01YM.  $\square$

## 2 Projective Spaces

**Prop. (7.5.2.1).** For any graded ideal  $I \subset S$ ,  $V_+(I) = \emptyset$  iff  $S_+ \subset \sqrt{I}$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (7.5.2.2) [Projective Schemes].** For a graded ring  $S$ , we have a scheme  $\text{Proj}(S)$  that consists of homogenous primes of  $S$  minus  $S_+$  and the affine cover is  $D(f) = \{p | f \notin p\}$ , and  $\mathcal{O}(D(f)) = \text{Spec } S_{(f)}$ , where  $S_{(f)}$  is the degree zero part of  $T^{-1}S$ . It has  $\mathcal{O}_p = S_{(p)}$ .  $\lrcorner$

*Proof:* Cf. [Sta]01M5?  $\square$

**Cor. (7.5.2.3).** For any graded ring  $S$ ,  $\text{Proj}(S)$  is a separated scheme.  $\lrcorner$

*Proof:* Check that for standard affine opens  $D_+(f)$  and  $D_+(g)$ ,  $D_+(f) \cap D_+(g) = D_+(fg)$  is affine open, and  $S_{(f)} \otimes_{\mathbb{Z}} S_{(g)} \rightarrow S_{(fg)}$  is surjective, which are both clear.  $\square$

**Prop. (7.5.2.4) [Representing Functor of Projective Schemes].** Let  $S$  be a graded ring generated by  $S_1$  over  $S_0$ , then  $\text{Proj}(S)$  represents the functor that maps a scheme  $Y$  to the set of pairs  $(\mathcal{L}, \psi)$ , where  $\mathcal{L}$  is an invertible sheaf on  $Y$ , and  $\psi : S \rightarrow \Gamma_*(Y, \mathcal{L})$  is a graded ring homomorphism that  $\mathcal{L}$  is generated by the global sections  $\psi(S_1)$ , up to strict equivalences.  $\lrcorner$

*Proof:* Cf. [Sta]01NA.  $\square$

**Prop. (7.5.2.5) [Associated Qco Sheaves].** Let  $M$  be a graded  $S$ -module, then

- there is a unique Qco sheaf  $\widetilde{M} \in \mathcal{QCoh}(\text{Proj}(S))$  s.t.  $\Gamma(D_+(f), \widetilde{M}) = M_f$  and the restrictions are compatible with base changes..
- For a point  $x \in \text{Proj}(S)$  corresponding to a homogenous prime not containing  $S_+$ ,  $\widetilde{M}_x = M_{(p)}$ .
- $M \mapsto \widetilde{M}$  is an exact functor from the category of graded  $S$ -modules to  $\mathcal{QCoh}(\text{Proj}(S))$ .
- There is a canonical ring map  $S_0 \rightarrow \Gamma(\text{Proj}(S), \mathcal{O}_{\text{Proj}(S)})$  and canonical  $S_0$ -module map  $M_0 \rightarrow \Gamma(\text{Proj}(S), \widetilde{M})$ .

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*Proof:* ?

4: This follows from the fact short exact sequences can be checked on stalks, and item2.  $\square$

**Cor. (7.5.2.6).** There is a canonical morphism of schemes  $\text{Proj}(S) \rightarrow \text{Spec } S_0$ .  $\square$

**Prop. (7.5.2.7).** If  $X \subset \mathbb{P}_k^n$  is a closed subvariety disjoint from a  $d$ -dimensional subspace  $L \subset \mathbb{P}_k^n$ , then the projection  $\pi : X \rightarrow \mathbb{P}^{n-d-1}$  with center  $L$  induces a finite map  $X \rightarrow \pi(X)$ .  $\square$

*Proof:* Cf.[Shafarovich 1, P63].  $\square$

**Cor. (7.5.2.8).** If  $F_0, \dots, F_s$  are forms of degree  $m > 0$  on  $\mathbb{P}_k^n$  having no common zero on a closed subvariety  $X \subset \mathbb{P}^n$ , then  $\varphi(x) = [F_0(x), \dots, F_s(x)]$  defines a finite map  $\varphi : X \rightarrow \varphi(X)$ .  $\square$

*Proof:* Let  $v_m : \mathbb{P}^n \rightarrow \mathbb{P}^N$  be the Veronese embedding of degree  $m$ , then  $X \rightarrow v_m(X)$  is an isomorphism, and  $\varphi$  is a composition of  $v_m$  and a projection  $\mathbb{P}_k^N \rightarrow \mathbb{P}_s^n$ , thus it is a finite map, by(7.5.2.7).  $\square$

**Prop. (7.5.2.9)[Tensor and Proj].** For two graded ring with the same  $S_0 = A$ ,  $\text{Proj}(S \times_A T) \cong X \times_A Y$ , where  $(S \times_A T)_n = S_n \times_A T_n$   $\square$

*Proof:*  $\square$

**Def. (7.5.2.10)[Relative Proj].** The **relative Proj**  $\mathcal{S}$  over locally Noetherian  $Y$  of a Qco graded  $\mathcal{O}_Y$ -algebra  $\mathcal{S}$  f.g. over  $\mathcal{S}_0$  by coherent  $\mathcal{S}_1$  is the glueing of locally  $\text{Proj } \mathcal{S}$ .  $\text{Proj } \mathcal{S} \rightarrow Y$  is locally projective thus proper. It is equipped with invertible sheaf  $\mathcal{O}(1)$  by glueing.  $\square$

**Prop. (7.5.2.11)[Closed Subscheme of Projective Scheme].** The closed scheme of  $X = \mathbb{P}_A^n$  corresponds to the saturated homogenous ideal  $\mathcal{I}_Y$ , (i.e. for any  $s$ , if there is an  $n$  that for any  $i$ ,  $x_i^n s \in \mathcal{I}_Y$ , then  $s \in \mathcal{I}_Y$ ).

So projective scheme over  $\text{Spec } S_0$  corresponds to  $\text{Proj } S$ , where  $S$  are f.g. over  $S_0$  by  $S_1$  saturated in the sense above.  $\square$

*Proof:* A closed immersion is proper, thus the kernel  $\mathcal{I}_Y$  of the structural map is a Qco(7.5.1.3), so it must be an ideal on every affine open, because Qco is affine local. Then we should use(7.5.3.6),  $\Gamma_*(\mathcal{I}_Y)$  will suffice. Cf.[Hartshorne Ex2.5.10].  $\square$

**Prop. (7.5.2.12).** The global section of a projective space  $\text{Proj } S \rightarrow \text{Spec } S_0$  is just  $S_0$ , this is by(7.5.3.6).  $\square$

**Prop. (7.5.2.13).** A quasi-projective scheme  $X$  over a field  $k$  of dimension  $r$  can be covered by  $r + 1$  open affine subsets. This is because there are  $r$  hyperplane that intersect  $X$  non-empty. This can happen by choosing a hyperplane non-intersecting the generic point of  $X$ , otherwise we choose many hyperplane, then their intersection is empty.  $\square$

### Sheaves on Proj

**Def. (7.5.2.14) [Serre Twisting Sheaves].** Let  $S$  be a graded ring and  $X = \text{Proj}(S)$ , then we can define **Serre twisting sheaves**  $\mathcal{O}_X(n) = \widetilde{S(n)} \in \mathcal{Q}\text{Coh}(X)$ . For  $\mathcal{F} \in \mathcal{Q}\text{Coh}(\text{Proj}(S))$ , the **Serre twisting sheaf** of  $\mathcal{F}$  is the sheaf  $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$ . ┘

**Prop. (7.5.2.15).** Let  $S$  be a graded ring and  $X = \text{Proj}(S)$ ,

1. there are canonical maps

$$\mathcal{O}_X(m) \otimes_{\mathcal{O}_X} \mathcal{O}_X(n) \rightarrow \mathcal{O}_X(m+n),$$

inducing a map of graded rings  $S \rightarrow \Gamma_*(X, \mathcal{O}_X) = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n))$ .

2. For any  $\mathcal{F} \in \text{Sh}(\mathcal{O}_X)$ , there are canonical maps

$$\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{F} \rightarrow \mathcal{F}(n).$$

inducing a map of graded module structure of  $\Gamma_*(X, \mathcal{O}_X)$  on  $\Gamma_*(X, \mathcal{F}) = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{F}(n))$ .

3. There is a canonical map  $\Gamma_*(X, \mathcal{F}) \rightarrow \mathcal{F}$  that is identity on global sections.
4. For any graded  $S$ -module  $M$ , let  $M(n) = M \otimes_S S(n)$ , there are maps  $\widetilde{M}(n) \rightarrow \widetilde{M(n)}$ . (Use property of Qco sheaves).
5. For graded rings  $M, N$  over  $S$ , there are canonical maps  $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} \rightarrow \widetilde{M \otimes_S N}$ . ┘

*Proof:* □

**Prop. (7.5.2.16).** Let  $S$  be a graded ring generated by  $S_1$  over  $S_0$ , then

1.  $(M \otimes_S N)_{(f)} \cong M_{(f)} \otimes_{S_{(f)}} N_{(f)}$  for  $f \in S_1$ .
2. There canonical maps in item 1, 4, 5 of (7.5.2.15) are isomorphisms.
3. For a graded ring map  $S \rightarrow T$ , we have the corresponding Proj map  $f : U \rightarrow T$  that  $f^*(\widetilde{M}) \cong \widetilde{(M \otimes_S T)|_U}$  and  $f_*(\widetilde{N}|_U) \cong \widetilde{N_S}$ . That's to say,  $f^*(\widetilde{M}(n)) = f^*(\widetilde{M})(n)$  and  $f_*(\widetilde{M}(n)) = f_*(\widetilde{M})(n)$ . ┘

*Proof:* 1:

- 2: By 1 and check locally that tensoring  $f^n$  is an isomorphism on  $D_+(f)$ . □

**Cor. (7.5.2.17).**  $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$  for any scheme  $X$  projective over  $Y$ . ┘

**Prop. (7.5.2.18) [Twisting of Proj].** With notation as in (7.5.2.10), Let  $S' = S * \mathcal{L} : S'_d = S_d \otimes \mathcal{L}^d$ , then  $\varphi : \text{Proj } S' \rightarrow \text{Proj } S$  is an isomorphism that induces

$$\mathcal{O}'(1) \cong \varphi^* \mathcal{O}(1) \otimes \pi'^* \mathcal{L}.$$

**Prop. (7.5.2.19).** If  $Y$  is Noetherian and admits an ample invertible sheaf, then by definition, we have  $S_1 \otimes \mathcal{L}^n$  is base point free for some  $n$ , thus we have a morphism  $\text{Proj}(S * \mathcal{L}^n) \rightarrow \mathbb{P}_Y^N$ , so  $P = \text{Proj } S$  is  $H$ -quasi-projective with  $\mathcal{O}(1) = \mathcal{O}_P(1) \otimes \pi^* \mathcal{L}^n$ . ┘

### Relative Projective Spaces

**Def. (7.5.2.20)** [Relative Projective Spaces]. ┘

**Def. (7.5.2.21)** [Quasi-Coherent Vector Bundles]. For  $X \in \text{Sch}$ , a map  $\pi : V \rightarrow X \in \text{Sch}$  is called an **Quasi-coherent vector bundle** if it is affine, and  $\pi_* \mathcal{O}_V$  endowed with the structure of a graded  $\mathcal{O}_S$ -algebra structure  $\pi_* \mathcal{O}_V = \bigoplus \mathcal{E}_n$ , where  $\mathcal{E}_0 = \mathcal{O}_S$ , and  $\text{Sym}^n \mathcal{E}_1 \rightarrow \mathcal{E}_n$  is an isomorphism for any  $n$ . The category of Qco vector bundles on  $S$  is denoted by  $\text{Vect}^{\text{Qcoh}}(S)$ .

A morphism of affine bundles is a map  $E' \rightarrow E$  over  $X$  s.t. the associated map of  $\mathcal{O}_X$ -modules  $f_* : \pi_* \mathcal{O}_V \rightarrow \pi'_* \mathcal{O}_{V'} \in \text{Alg}(\mathcal{O}_X)$  is compatible with grading.

For  $\mathcal{E} \in \text{QCoh}(X)$ , we can define the **associated vector bundle**  $V(\mathcal{E})$  as  $\text{Spec}_X \text{Sym}(\mathcal{E})$  (7.2.3.10)(7.2.7.12). In this way, the category of Qco vector bundles over  $S$  is anti-equivalent to  $\text{QCoh}(X)$ .

The category of locally free vector bundles over  $X$  is denoted by  $\text{Vect}(X)$ . And the category of finite locally free vector bundles over  $X$  is denoted by  $\text{Vect}^{\text{fin}}(X)$ . ┘

**Def. (7.5.2.22)** [Projective Bundles]. For  $X \in \text{Sch}$  and  $\mathcal{E} \in \text{QCoh}(X)$ , we can define the **associated projective bundle**  $\mathbf{P}(\mathcal{E})$  as  $\text{Proj}_S \text{Sym}(\mathcal{E})$  (7.2.3.10)(7.2.7.12). It is equipped with a Serre twisting sheaf  $\mathcal{O}(1)$ , which is the glue of locally the Serre sheaf in projective space. There is a surjective morphism  $\pi^*(\mathcal{E}) \rightarrow \mathcal{O}(1)$  (local check). ┘

**Prop. (7.5.2.23)**. For  $g : Y \rightarrow X \in \text{Sch}$  and  $E \in \text{QCoh}(X)$ , a morphism  $Y \rightarrow \mathbf{P}(\mathcal{E}) \in \text{Sch}/X$  is equivalent to an invertible sheaf  $\mathcal{L} \in \text{Pic}(Y)$  together with a surjective map  $g^* \mathcal{E} \rightarrow \mathcal{L}$ .

In particular, giving a morphism  $X \rightarrow \mathbf{P}_A^n$  is equivalent to a base point free invertible sheaf with  $n$  generators on  $X$ . ┘

*Proof:* If there is a morphism, it will pullback  $\pi^*(\mathcal{E}) \rightarrow \mathcal{O}(1)$  into  $g^* \mathcal{E} \rightarrow \mathcal{L}$ . For the converse, construct locally and glue, we have the natural morphisms  $A[x_1/x_i, \dots, x_n/x_i] \rightarrow \mathcal{O}_{X_{s_i}} : x_j/x_i \rightarrow s_j/s_i$  in a homogenous sense. It is natural hence glue together. For the module, maps  $x_i \rightarrow s_i$ . □

**Cor. (7.5.2.24)**. All automorphisms of  $\mathbf{P}_k^n$  is linear. ┘

*Proof:* The Picard group of  $\mathbf{P}_k^n$  is  $\mathbb{Z}$  and is generated by  $\mathcal{O}(1)$  (7.5.3.16), so the automorphism will map  $\mathcal{O}(1)$  to  $\mathcal{O}(\pm 1)$  and  $\mathcal{O}(-1)$  has no global section (7.5.3.5). And the global section is  $n$ -dimensional and determines the morphism by the prop. □

**Def. (7.5.2.25)** [Projective Space]. Let  $A$  be a ring, the **projective space**  $\mathbf{P}_A^n$  is defined to be

$$\mathbf{P}_A^n = \text{Proj}(A[T_0, \dots, T_n])$$

with  $\deg(T_i) = 1$ . It represents the functor that maps a scheme  $T$  to the equivalence classes of pairs  $(\mathcal{L}, (s_0, \dots, s_n))$ , where  $\mathcal{L}$  is an invertible sheaf on  $T$ , and  $s_0, \dots, s_n \in \Gamma(T, \mathcal{L})$  that generate  $\mathcal{L}$  by (7.5.2.23). For any scheme  $S$ ,  $\mathbf{P}^n \times_{\mathbb{Z}} S$  is called the projective space over  $S$ . ┘

**Prop. (7.5.2.26)**. Let  $R$  be a ring and  $X = \mathbf{P}_R^n$ ,  $\mathcal{F} \in \text{QCoh}(X)$ , then the canonical map  $\widetilde{\Gamma_*(X, \mathcal{F})} \rightarrow \mathcal{F}$  (7.5.2.15) is an isomorphism. ┘

*Proof:* Cf. [Sta]03GM. ? This is a corollary. □

**Prop. (7.5.2.27)** [Closed Subschemes of  $\mathbf{P}_R^n$ ]. Let  $Z$  be a closed subscheme of  $\mathbf{P}_R^n$ , then it is of the form

$$Z = \text{Proj}(R[X_0, \dots, X_n]/I) \subset \text{Proj}(R[X_0, \dots, X_n])$$

where  $I = \bigoplus I_n$  and  $I_n = \ker(R[X_0, \dots, X_n]_d \rightarrow \Gamma(Z, \mathcal{O}_Z(d)))$ . ┘

*Proof:* Cf.[Sta]03GI.?

□

**Prop. (7.5.2.28) [Segre Embedding].** Let  $S$  be a scheme, there is a natural closed immersion

$$\mathbf{P}_S^m \times_S \mathbf{P}_S^n \rightarrow \mathbf{P}_S^{mn+m+n}$$

called the **Segre embedding**.

┘

*Proof:* It suffices to prove for  $S = \mathbb{Z}$ , and in this case, it suffices to write down an invertible sheaf on  $\mathbf{P}_S^m \times_S \mathbf{P}_S^n$  with  $(n+1)(m+1)$  global sections that generate it. Then we take the invertible sheaf  $\mathcal{L} = \pi_1^* \mathcal{O}_{\mathbf{P}^m}(1) \otimes \pi_2^* \mathcal{O}_{\mathbf{P}^n}(1)$ . and the sections  $X_i Y_j$ , where  $(X_0, \dots, X_m)$  generate  $\mathcal{O}_{\mathbf{P}^m}(1)$  and  $(Y_0, \dots, Y_n)$  generate  $\mathcal{O}_{\mathbf{P}^n}(1)$ .

It is a closed immersion by [Sta]01WD.?

□

**Prop. (7.5.2.29) [Venerose Embedding].**

┘

### 3 Invertible Sheaves

General invertible sheaves on a ringed site is treated in 5.

**Prop. (7.5.3.1) [Faithfully Flat Descent].** To show a quasi-coherent sheaf is a line bundle, it suffices to show fpqc-locally, by (5.4.2.1).

┘

**Def. (7.5.3.2) [Basepoint-Free Line Bundles].** Let  $k \in \mathbf{Field}$ ,  $X \in \mathbf{Sch}/k$  and  $\mathcal{L} \in \mathbf{Pic}(X)$ , then  $\mathcal{L}$  is called **basepoint-free** if the intersection of elements in  $\{\mathrm{div}(s) \mid s \in H^0(V)\}$  is empty.

┘

**Prop. (7.5.3.3).** If  $X$  is qcqs over a field  $k$  and  $K/k$  is a field extension, then  $\mathcal{L}$  is basepoint free iff  $\mathcal{L}_K$  is basepoint free over  $X_K$ .

┘

*Proof:* By flat base change (7.7.5.1),  $H^0(X_K, \mathcal{L}_K) = H^0(X, \mathcal{L}) \otimes_k K$ .

□

**Prop. (7.5.3.4) [Relative Triviality].** Let  $f : X \rightarrow Y$  be a finite morphism of schemes and  $\mathcal{L} \in \mathbf{Pic}(X)$ , then for any  $y \in Y$ , there exists a nbhd  $U$  of  $y \in Y$  s.t.  $\mathcal{L}|_{f^{-1}(U)}$  is trivial.

┘

*Proof:* Cf.[Sta]0BUT.

□

**Prop. (7.5.3.5) [Global Sections].** Let  $\mathcal{L}$  be an invertible sheaf over qcqs scheme  $X$ , for  $\mathcal{F} \in \mathbf{QCoh}(X)$ , let the **global section functor**  $\Gamma_*(\mathcal{F}) = \bigoplus \Gamma(X, \mathcal{F} \otimes \mathcal{L}^n)$ , then

$$\Gamma_*(\mathcal{F})_{(f)} \cong \mathcal{F}(X_f).$$

where  $s \in \Gamma(X, \mathcal{L})$ . In particular that if there is a section  $f$  of  $\mathcal{F}$  on  $X_s$ , then for some  $n$ ,  $f \otimes s^n$  is a global section of  $\mathcal{F} \otimes \mathcal{L}^n$ .

┘

*Proof:* This is nearly the same as the proof that  $(\mathrm{Spec} A)_f = \mathrm{Spec} A_f$ , Cf.[Sta]01PW.?

□

**Cor. (7.5.3.6).** When  $X = \mathrm{Proj} S$  projective over  $\mathrm{Spec} S_0$  and  $\mathcal{F} \in \mathbf{QCoh}(X)$ ,  $\widehat{\Gamma_*(\mathcal{F})} \cong \mathcal{F}$ , where  $\Gamma_*(\mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F}(n))$ , which is a graded  $S$ -module. In particular,  $\Gamma_*$  for projective space  $\mathbb{P}_A^n$  equals  $A[x_1, \dots, x_n]$ .

┘

**Def. (7.5.3.7) [Regular Sections].**

┘



**Prop. (7.5.3.8) [Complete Series].** Let  $H^0(X, \mathcal{L})$  be the sections that corresponds to injective maps  $\mathcal{L}^{-1} \rightarrow \mathcal{O}_X$ , then there is a canonical isomorphism

$$H^0(X, \mathcal{L})_{reg}/H^*(X, \mathcal{O}_X^*) \cong |\mathcal{L}|.$$

Notice when  $X$  is not integral,  $H^0(X, \mathcal{L})_{reg}$  may not equal  $H^0(X, \mathcal{L})$ . ┘

*Proof:* Cf. [Kle05]P22. □

**Prop. (7.5.3.9) [Meromorphic Sections].** Let  $X$  be a locally Noetherian scheme having no embedded points, then every invertible sheaf  $\mathcal{L} \in \text{Pic}(X)$  has a meromorphic section. In particular, this applies for  $X$  integral, by (7.5.1.17). ┘

*Proof:* Cf. [Sta]0EMI. □

### Picard Groups

**Remark (7.5.3.10) [Picard Groups].** The Picard group  $\text{Pic}(X)$  of a local ringed space  $(X, \mathcal{O}_X)$  is defined in (7.2.5.7), and it is isomorphic to  $H^1(X, \mathcal{O}_X^*)$  by (7.3.1.17). ┘

**Prop. (7.5.3.11) [Class Group].** If  $X = \text{Spec } \mathcal{O}$  where  $\mathcal{O}$  is a Dedekind domain, then by (5.2.7.8), the isomorphism class of invertible sheaves on  $X$  is equivalent to the isomorphism class of fractional ideals modulo principle ideals. Thus  $\text{Pic}(\mathcal{O})$  equals the class group of  $\mathcal{O}$  (5.2.7.9). In particular, if  $R$  is a UFD, then  $\text{Pic}(\text{Spec } R) = 0$ . ┘

**Def. (7.5.3.12) [Invertible Sheaf Associated to Cartier Divisors].** For a Cartier divisor on  $X \in \text{Sch}$  (7.8.1.1), define  $\mathcal{O}_X(D)$  (or  $\mathcal{L}(D)$ ) the **Line bundle associated to  $D$**  as the sub  $\mathcal{O}_X$ -module of  $\mathcal{K}$  locally generated by  $(f_i^{-1})$ , where  $D = (f_i)$  locally. Equivalently, it is the line bundle  $\mathcal{I}_D^{-1}$ . ┘

**Def. (7.5.3.13) [Linear Series].** Let  $\mathcal{L} \in \text{Pic}(X)$ , then denote  $|\mathcal{L}|$  the **complete linear series** of effective Cartier divisors  $D$  s.t.  $\mathcal{L}(D) \cong \mathcal{L}$ . ┘

**Def. (7.5.3.14) [Weil divisor of an invertible Sheaf].** For  $X$  a locally Noetherian integral scheme and  $\mathcal{L}$  an invertible sheaf in  $\mathcal{K}$ , if  $s \in \Gamma(X, \mathcal{K} \otimes \mathcal{L})$  is a meromorphic section of  $\mathcal{L}$  (which exists by (7.5.3.9)), for any prime Weil divisor  $Z$  with generic pt  $\eta$ , define  $\text{ord}_Z(s) = \text{ord}_{\mathcal{O}_{X,\eta}}(s/s_\eta)$  (5.1.2.10), for any  $s_\eta$  a generator of  $\mathcal{L}_\eta$  over  $\mathcal{O}_{X,\eta}$ . This is independent of  $s_\eta$  chosen.

The prime Weil divisors that  $\text{ord}_Z(s) \neq 0$  is locally finite, the same as in (9.1.1.18). And any two different sections  $s_i$  defines Weil divisors up to a difference of  $\text{div}(f)$  (9.1.1.18). So we can define the **Weil divisor class associated to  $\mathcal{L}$**  as  $\sum \text{ord}_Z(s)[Z]$  for any meromorphic section  $s$  of  $\mathcal{L}$ .

It is easy to verify that this induces a homomorphism from  $\text{Pic}(X)$  to  $\text{Cl}(X)$ . ┘

**Prop. (7.5.3.15) [Cl-Pic].** For a normal integral locally Noetherian scheme, the above map  $\text{Pic}(X) \rightarrow \text{Cl}(X)$  (7.5.3.14) is an injection. It is an isomorphism iff all local rings of  $X$  are UFD.

Explicitly, the inverse image of a prime Weil divisor  $D$  is the sheaf  $\mathcal{O}_X(D) = \text{Hom}(\mathcal{I}_D, \mathcal{O}_X) = \mathcal{I}_D^{-1}$ , and there is an exact sequence of sheaves on  $X$ :

$$0 \rightarrow \mathcal{O}_X(-D) = \mathcal{I}_D \rightarrow \mathcal{O}_X \rightarrow i_*\mathcal{O}_D \rightarrow 0$$

In particular, this applies to non-singular prevarieties over a field  $k$ , by (7.4.2.10). ┘



*Proof:* If it is not injective, then some meromorphic section on  $\mathcal{L}$  has no associated Weil divisors, then it suffices to show  $\mathcal{L}$  is trivialized by  $s$ . Consider on an affine subscheme  $\text{Spec } A$ , then  $\text{ord}_{A_{\mathfrak{p}}}(s) = 0$  for each minimal prime  $\mathfrak{p}$  of  $A$ , but  $A_{\mathfrak{p}}$  is DVR by (5.3.5.21), so  $s \in A_{\mathfrak{p}}^*$  for each minimal prime  $\mathfrak{p}$ , so  $s \in A^*$  by (5.3.5.12). This shows  $s$  trivialize  $\mathcal{L}$ .

To show it is surjective, it suffices to show any Weil divisor  $D$  is in the image: notice  $D$  is an effective Cartier divisor by (7.8.1.5), and by definition (7.8.1.6) the vanishing of the canonical section  $1_D \in \mathcal{O}_X(D)$  is exactly  $D$ .  $\square$

**Prop. (7.5.3.16) [Examples of Picard Groups].**

- If  $X$  is a locally Noetherian normal integral separated scheme, then so are  $X \times \text{Spec } \mathbb{Z}[T]$  and  $\mathbb{P}_X^n$ , and  $\text{Cl}(X \times \text{Spec } \mathbb{Z}[T]) = \text{Cl}(X)$  and  $\text{Cl}(\mathbb{P}_X^n) = \mathbb{Z} \oplus \text{Cl}(X)$ .
- For a UFD  $R$ ,  $\text{Pic}(\mathbb{A}_R^n) \cong \text{Cl}(\mathbb{A}_R^n) = 0$ .
- For a UFD  $R$ ,  $\text{Pic}(\mathbb{P}_R^n) \cong \text{Cl}(\mathbb{P}_R^n) \stackrel{\text{deg}}{\cong} \mathbb{Z}$ , and it is generated by  $\mathcal{O}_{\mathbb{P}_R^n}(1)$ .
- For any UFD  $R$ ,  $\text{Pic}(\mathbb{P}_R^1 \times \mathbb{P}_R^1) \cong \text{Cl}(\mathbb{P}_R^1 \times \mathbb{P}_R^1) \cong \mathbb{Z} \oplus \mathbb{Z}$ .
- If  $k$  is a field and  $Y \subset \mathbb{P}_k^n$  is a hypersurface of degree  $d$ , then  $\text{Pic}(\mathbb{P}_k^n \setminus Y) \cong \text{Cl}(\mathbb{P}_k^n \setminus Y) \cong \mathbb{Z}/d\mathbb{Z}$ .  $\lrcorner$

*Proof:* By (7.5.3.15) and (9.1.4.7)(9.1.6.10)(9.1.4.8)(9.1.4.9).  $\square$

**Prop. (7.5.3.17).** Let  $X$  be a complete prevariety over a field  $k$  of characteristic  $p > 0$ , then for any  $n$  prime to  $p$  and any purely inseparable field extension  $k'/k$ , the natural map

$$\text{Pic}(X)[n] \rightarrow \text{Pic}(X_{\bar{k}})[n]$$

is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]0CD5.  $\square$

## 4 Ample Invertible Sheaves

**Prop. (7.5.4.1).** Let  $X \in \text{Sch}$ ,  $\mathcal{L} \in \text{Pic}(X)$ ,  $s \in \Gamma(X, \mathcal{L})$ , then for any affine open  $U \subset X$ ,  $X_s \cap U$  is affine.  $\lrcorner$

*Proof:* It suffices to show: If  $R$  is a ring,  $N$  is an invertible  $R$ -module and  $s \in N$ , then  $U = \{\mathfrak{p} | s \notin \mathfrak{p}N\}$  is an affine open subset of  $\text{Spec } R$ . For this, let  $R' = \varinjlim_{-\otimes s} N^{\otimes n}$ , then in fact  $U = \text{Spec } R'$ . Because locally,  $R'$  is just  $R_s$  via an isomorphism  $N_f \cong R_f$ .  $\square$

**Def. (7.5.4.2) [Ample Invertible Sheaves].** On a quasi-compact scheme  $X$ , an **ample invertible sheaf** is a line bundle  $\mathcal{L} \in \text{Pic}(X)$  s.t. there is some  $n \in \mathbb{Z}_+$  and sections  $s_i \in \Gamma(X, \mathcal{L}^n)$  that  $X_{s_i}$  is an affine cover of  $X$ . In particular, an ample invertible sheaf is globally generated.

For a qc morphism  $f : X \rightarrow Y$ , an invertible sheaf on  $X$  is called (relatively)  **$f$ -ample** iff it is ample restricted to every open subscheme  $f^{-1}(V)$ , where  $V$  is an affine open in  $Y$ . In particular, an invertible sheaf  $\mathcal{L}$  on a quasi-compact scheme  $X$  is ample iff it is  $f$ -ample where  $f : X \rightarrow \text{Spec } \mathbb{Z}$ .  $\lrcorner$

**Prop. (7.5.4.3).** For  $m \in \mathbb{Z}_+$ , an invertible sheaf  $\mathcal{L}$  is ( $f$ -)ample iff  $\mathcal{L}^m$  is ( $f$ -)ample.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.5.4.4) [Ample Implies Separatedness].** When there is a  $f$ -ample invertible sheaf for  $f : X \rightarrow Y$  qc, then  $f$  is separated. In particular, if  $X \in \text{Sch}_{qc}$  has an ample invertible sheaf, then  $X$  is separated.  $\lrcorner$

*Proof:* [Sta]09MP. ?  $\square$

**Prop. (7.5.4.5) [Characterizing Ampleness].** Let  $X \in \text{Sch}_{qc}$ ,  $\mathcal{L} \in \text{Pic}(X)$ ,  $S = \Gamma_*(X, \mathcal{L})$ , then the following are equivalent:

1.  $\mathcal{L}$  is ample.
2. The open subsets  $X_s$ , where  $s \in \Gamma_*(X, \mathcal{L})$  homogeneous, cover  $X$ , and the associated morphism  $X \rightarrow \text{Proj } S$  is an open immersion.
3. The open subsets  $X_s$  where  $s \in \Gamma_*(X, \mathcal{L})$  homogeneous, form a topological basis for  $X$ .
4. The affine open subsets of the form  $X_s$  where  $s \in \Gamma_*(X, \mathcal{L})$  homogeneous, form a topological basis for  $X$ .
5. For any  $\mathcal{F} \in \mathcal{QCoh}(X)$ , the sum of images of the canonical maps  $\Gamma(X, \mathcal{F} \otimes \mathcal{L}^n) \otimes \mathcal{L}^{\otimes -n} \rightarrow \mathcal{F}$  is surjective.
6.  $X$  is quasi-separated, and for any  $\mathcal{F} \in \mathcal{QCoh}^{\text{ft}}(X)$ ,  $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$  is generated by global sections for  $n$  sufficiently large.
7.  $X$  is quasi-separated, and for any  $\mathcal{F} \in \mathcal{QCoh}^{\text{ft}}(X)$ , there exists  $n \in \mathbb{Z}$  s.t.  $\mathcal{F}$  is a quotient of a direct sum of f.m. copies of  $\mathcal{L}^{\otimes -n}$ .  $\lrcorner$

*Proof:* Cf. [Sta]01Q3. ?  $\square$

**Cor. (7.5.4.6).** The pullback of an ample invertible sheaf along a qc immersion is still ample.  $\lrcorner$

**Cor. (7.5.4.7).** Let  $S$  be a quasi-separated scheme and  $X, Y$  be schemes over  $S$ . If  $\mathcal{L}$  is an ample invertible sheaf over  $X$  and  $\mathcal{N}$  an ample invertible sheaf over  $Y$ , then  $\mathcal{M} = \text{pr}_1^* \mathcal{L} \otimes_{\mathcal{O}_{X \times_S Y}} \text{pr}_2^* \mathcal{N}$  is ample over  $X \times_S Y$ .  $\lrcorner$

*Proof:* Because  $X \times_S Y \rightarrow X \times Y$  is a qc immersion, by (7.5.4.6), it suffices to show for  $S = \text{Spec } Z$ . Then if  $X_s$  is an affine nbhd of  $x$  and  $Y_t$  is an affine nbhd of  $Y$ , then  $(X \times Y)_{\pi_1^* s \otimes \pi_2^* t}$  is an affine nbhd of  $x \times y$ .  $\square$

**Cor. (7.5.4.8) [Tensor Product of Ample Invertible Sheaves is Ample].** Let  $X \in \text{Sch}$ ,  $\mathcal{L}, \mathcal{M} \in \text{Pic}(X)$ . If  $\mathcal{M}$  is globally-generated and  $\mathcal{L}$  is ample, then  $\mathcal{L} \otimes \mathcal{M}$  is ample. In particular if  $\mathcal{L}, \mathcal{M}$  are ample invertible sheaves, then  $\mathcal{L} \otimes \mathcal{M}$  is ample.  $\lrcorner$

*Proof:* For any  $x \in X$  and  $U$  a nbhd of  $x$ , choose  $s \in \Gamma(X, \mathcal{L}^n)$  that  $x \in X_s \subset U$ , and choose  $t \in \Gamma(X, \mathcal{M})$  that  $t_x \neq 0$ , then  $x \in X_{s \otimes t^n} \subset U$ , thus  $X_r$  form a basis for  $X$  where  $r \in \Gamma(X, (\mathcal{L} \otimes \mathcal{M})^n)$ , so  $\mathcal{L} \otimes \mathcal{M}$  is ample.  $\square$

**Cor. (7.5.4.9).** Let  $X \in \text{Sch}$ ,  $\mathcal{L}, \mathcal{M} \in \text{Pic}(X)$ . If  $\mathcal{L}$  is ample, then  $\mathcal{M} \otimes \mathcal{L}^{\otimes n}$  is ample for  $n$  sufficiently large.  $\lrcorner$

*Proof:* This is because  $\mathcal{L} \otimes \mathcal{M}^n$  is generated by global sections for  $n$  sufficiently large, thus  $\mathcal{L} \otimes \mathcal{M}^{n+1}$  is ample, by (7.5.4.8).  $\square$

**Prop. (7.5.4.10) [Gluing Ample Invertible Sheaves].** Let  $X \in \text{Sch}_{qc}$  s.t. there is an affine open covering of the form  $X_{s_i}$  where  $s_i$  are sections of  $\mathcal{L}_i$  and  $\mathcal{L}_i$  are globally generated line bundles in  $\text{Pic}(X)$ , then  $X$  has an ample line bundle.  $\lrcorner$

*Proof:* Let  $X_{s_i}$  be an affine open covering of  $X$  where  $s_i$  are sections of globally generated invertible sheaves  $\mathcal{L}_i$ ,  $i = 1, \dots, r$ . As  $\mathcal{L}_i$  are globally generated, let  $X = \cup_j X_{t_{ij}}$  for  $j = 0, 1, \dots, m_i$  where  $t_{i0} = s_i$ . Then for the line bundle  $\otimes_i \mathcal{L}_i$ , the sections  $t_{1,j_1} \otimes \dots \otimes t_{r,j_r}$  where at least one of  $j_i$  equals 0, covers  $X$ , and they are all affine by (7.5.4.1). So  $\otimes_i \mathcal{L}_i$  is ample.  $\square$

**Prop. (7.5.4.11).**  $f : X \rightarrow Y$ , let  $\mathcal{L}$  be  $f$ -ample on  $X$  and  $\mathcal{M}$  ample on  $Y$ , then  $\mathcal{L} \otimes f^* \mathcal{M}^n$  is ample for  $n$  large.  $\lrcorner$

*Proof:* Cf. [Sta]0892.  $\square$

**Cor. (7.5.4.12).** If  $f : X \rightarrow Y$  is quasi-affine, then the pullback of an ample invertible sheaf is ample, by (7.4.4.19) and (7.5.4.3).  $\lrcorner$

**Prop. (7.5.4.13) [Pullback of Ampleness].** If  $f : Y \rightarrow X$  is finite and surjective morphism between schemes proper over a Noetherian affine scheme, then for any invertible sheaf  $\mathcal{L}$  on  $X$ ,  $\mathcal{L}$  is ample iff  $f^* \mathcal{L}$  is ample.  $\lrcorner$

*Proof:* Cf. [Sta]0B5V.  $\square$

One direction follows from (7.5.4.12). For the other we use Serre criterion (7.7.2.8) and devissage (7.5.1.56). We only verify 3: By (7.4.4.42), there exists such coherent sheaf  $f_* \mathcal{F}$  for any integral subscheme, and for a any Qco sheaf of ideals  $\mathcal{I}$ ,  $\mathcal{I} f_* \mathcal{F} = f_*(f^{-1} \mathcal{I} \mathcal{F})$  because  $f$  is affine, thus

$$H^p(X, \mathcal{I} f_* \mathcal{F}) = H^p(X, f_*(f^{-1} \mathcal{I} \mathcal{F} \otimes \mathcal{L}^n)) = H^p(Y, f^{-1} \mathcal{I} \mathcal{F} \otimes \mathcal{L}^n)$$

by projection formula, and  $f$  is affine. This vanish for  $n$  large.  $\square$

**Cor. (7.5.4.14) [Ampleness and Irreducible Components].** Let  $X$  be a scheme proper over a Noetherian affine scheme, then an invertible sheaf  $\mathcal{L}$  on  $X$  is ample iff it is ample on the induced reduced structure of irreducible components of  $X$ .  $\lrcorner$

*Proof:* This is (7.5.4.13) applied to the case  $\coprod_i X_i \rightarrow X$ .  $\square$

**Prop. (7.5.4.15) [Ampleness Restricted to Reduced Case].** If  $i : Z \rightarrow X$  is a closed immersion of schemes that induce homeomorphism on the underlying topology, then  $\mathcal{L}$  is ample iff  $i^* \mathcal{L}$  is ample.

In particular, this applies to  $X_{\text{red}} \rightarrow X$ .  $\lrcorner$

*Proof:* Cf. [Sta]09MW.  $\square$

**Prop. (7.5.4.16).** Let  $f : X \rightarrow Y$  be a proper morphism of schemes and  $\mathcal{L}$  an invertible sheaf on  $X$ . If  $y \in Y$  satisfies  $\mathcal{L}_y$  is ample on  $X_y$ , then there is a nbhd  $U$  of  $y \in Y$  that  $\mathcal{L}|_{f^{-1}(U)}$  is  $f$ -ample.  $\lrcorner$

*Proof:* Cf. [Sta]0D2S.  $\square$

**Prop. (7.5.4.17) [Ampleness is Fiberwise].** For  $f : X \rightarrow T \in \text{Sch}$  proper and  $\mathcal{L} \in \text{Pic}(X)$ ,  $\mathcal{L}$  is  $f$ -ample iff for any  $t \in T$ ,  $\mathcal{L}_t$  is ample over  $X_t$ .  $\lrcorner$

*Proof:* [Positivity in Algebraic Geometry]P97.  $\square$

**Prop. (7.5.4.18) [Finite Map and Ample].** Let  $\mathcal{L}$  be a basepoint-free line bundle on a proper scheme  $X$ , then the associated map  $X \rightarrow P\Gamma(X, \mathcal{L})$  is finite iff  $\mathcal{L}$  is ample.  $\lrcorner$

*Proof:* Cf. [Positivity in Algebraic Geometry, P28].  $\square$

**Prop. (7.5.4.19) [Ampleness is Étale-Local].** Ampleness satisfies étale descent. Cf. [Sta]0D35.  $\lrcorner$

**Prop. (7.5.4.20).** Let  $X$  be a scheme. Let  $\mathcal{L}$  be an ample invertible  $\mathcal{O}_X$ -module. Let  $n_0$  be an integer. If  $H^p(X, \mathcal{L}^{-n}) = 0$  for  $n \geq n_0$  and  $p > 0$ , then  $X$  is affine.  $\lrcorner$

*Proof:* Cf. [Sta]0EBD.  $\square$

**Prop. (7.5.4.21) [Gluing Ample Line Bundles].** Let  $K/k \in \text{Field}$ ,  $X \in \text{Sch}/k$  s.t.  $X_K$  has an ample line bundle, then  $X$  also has an ample line bundle.  $\lrcorner$

*Proof:* Cf. [Sta]0BDC.  $\square$

### Very Ample Invertible Sheaves

**Def. (7.5.4.22) [Very Ampleness].** Let  $f : X \rightarrow S$  be a morphism, a ***f*-very ample** invertible sheaf on  $X$  is the pullback of  $\mathcal{O}(1)$  along some immersion  $X \rightarrow \text{Proj}(\mathcal{E})$  for some  $\mathcal{Q}\text{co}$  module  $\mathcal{E}$  over  $Y$ , Cf. (7.5.2.14). It is called **H-very ample** iff  $\mathcal{E}$  is trivial. Notice when  $X$  is proper, this immersion must be closed by (7.4.6.3).

When  $S$  is affine and  $f : X \rightarrow S$  is of f.t., *f*-very ample is equivalent to H-very ample.  $\lrcorner$

*Proof:* Cf. [Sta]02NP.  $\square$

**Prop. (7.5.4.23) [Tensor Product of Very Ample Line Bundles].** Let  $f : X \rightarrow \text{Spec } A$  be a morphism. If  $\mathcal{L}$  is H-very ample and  $\mathcal{M}$  is generated by global sections, then  $\mathcal{L} \otimes \mathcal{M}$  is H-very ample. In particular, the tensor product of two H-very ample invertible sheaves is H-very ample.  $\lrcorner$

*Proof:* The hypothesis means  $\mathcal{L} = \varphi^* \mathcal{O}(1)$ , where  $\varphi : X \rightarrow \mathbb{P}_A^n$  is an immersion, and  $\mathcal{M} = \psi^* \mathcal{O}(1)$ , where  $\psi : X \rightarrow \mathbb{P}_k^m$  is a morphism. Then the product  $T : X \rightarrow \mathbb{P}_k^n \times \mathbb{P}_k^m$  is also an immersion, by base change trick (7.4.4.2), as  $\mathbb{P}_k^n \times \mathbb{P}_k^m \rightarrow \mathbb{P}_k^n$  is separated. Then  $S \circ T : X \rightarrow \mathbb{P}^{mn+m+n}$ , where  $S : \mathbb{P}_k^n \times \mathbb{P}_k^m \rightarrow \mathbb{P}^{mn+m+n}$  is the Segre embedding, is also an immersion, and  $(ST)^* \mathcal{O}(1) \cong \mathcal{L} \otimes \mathcal{M}$ , thus it is H-very ample.  $\square$

**Prop. (7.5.4.24) [Ample and H-Very Ample].** If  $f : X \rightarrow S$  is locally of f.t. and  $\mathcal{L}$  is an ample invertible sheaf on  $X$ , then  $\mathcal{L}^{\otimes m}$  is H-very ample for  $m$  sufficiently large.  $\lrcorner$

*Proof:* Choose an affine open cover  $\{V_i\}$  of  $S$ . By (7.5.4.5), there are f.m. affine opens  $X_{s_i}$  that cover  $X$  refining a inverse image of  $\{V_i\}$ . Now  $\mathcal{O}_X(X_{s_i})$  is f.g. over  $\mathcal{O}_S(V_i)$ , so we can find f.m.  $f_{ij} \in \mathcal{O}_X(X_{s_i})$  that generates it over  $\mathcal{O}_S(V_i)$ . By (7.5.3.5), we can write each  $f_{ij} = s_{ij}/s_i^{e_{ij}}$  for some  $a_{ij}$  homogenous. We can multiply by a factor to make all the  $s_i^{e_{ij}}$  the same degree  $N$ , and  $f_{ij} = s_{ij} s_i^{N/\deg(s_i) - e_{ij}}$ , then all the elements  $s_i, s_{ij}$  generates the invertible sheaf  $\mathcal{L}$ , thus inducing a map  $j : X \rightarrow \mathbb{P}_k^m$ . This map is an immersion, because  $j^{-1}(D(T_i)) = X_{s_i}$  and the function  $T_{ij}/T_i$  on  $D(T_i)$  pulls back to  $s_{ij}/s_j$ . Thus  $j$  is locally a closed immersion, thus an immersion.

Now  $\mathcal{L}^{\otimes d_1}$  is H-very ample for some  $d_1$ , in particular it is separated, and by (7.5.4.5) there is some  $d_2$  that  $\mathcal{L}^{\otimes d}$  is generated by global sections for all  $d \geq d_2$ , then by (7.5.4.23),  $\mathcal{L}^{\otimes d}$  is H-very ample for  $d \geq d_1 + d_2$ .  $\square$

**Prop. (7.5.4.25) [ $f$ -Very Ample Implies  $f$ -Ample].** If  $f : X \rightarrow S$  is qc, then  $f$ -very ample implies  $f$ -ample.  $\lrcorner$

*Proof:* Cf. [Sta]01VN. ?  $\square$

**Cor. (7.5.4.26) [Serre].** If  $S \in \mathcal{A}ff$ ,  $f : X \rightarrow S$  is of f.t. and  $\mathcal{L} \in \text{Pic}(X)$ , then the following are equivalent:

- $\mathcal{L}$  is ample .
- $\mathcal{L}$  is  $f$ -ample.
- $\mathcal{L}^{\otimes n}$  is (H-)  $f$ -very ample for some (all large)  $n$ .

$\lrcorner$

*Proof:* This follows from (7.5.4.22) (7.5.4.24) and (7.5.4.25).  $\square$

**Cor. (7.5.4.27).** If  $f : X \rightarrow S$  is of f.t. and  $S$  is quasi-compact,  $\mathcal{L}$  is an invertible sheaf on  $X$ , then the following are equivalent:

- $\mathcal{L}$  is  $f$ -ample.
- $\mathcal{L}^{\otimes n}$  is (H-)  $f$ -very ample for some (all large)  $n$ .

$\lrcorner$

*Proof:* Cf. [Sta]01VU.  $\square$

## 5 Sheaf of Differentials

**Prop. (7.5.5.1) [Differentials on Schemes].** Consider a morphism of schemes  $X \rightarrow Y$ , we define the sheaf of differentials  $\Omega_{X/Y}$  together with an  $S$ -derivative  $\mathcal{O}_X \rightarrow \Omega_{X/S}$  as for ringed sites (7.2.4.7). Then  $\Omega_{X/S} \in \mathcal{QCoh}(X)$  by (7.2.4.5). In fact, If  $U = \text{Spec } A$  is mapped into  $\text{Spec } B \subset S$ , then  $\Omega_{X/S}(U) \cong \widetilde{\Omega_{A/B}}$ .

In particular, the stalk of  $\Omega_{X/S}$  at a point  $x \in X$  is  $\Omega_{\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}}$ .

Thus when  $X \rightarrow Y$  is locally of f.t. (or locally of f.p.), then  $\Omega_{X/S}$  is an  $\mathcal{O}_X$ -module of f.t. (or of f.p.).  $\lrcorner$

**Prop. (7.5.5.2) [Base change and Differentials].** Let  $\begin{array}{ccc} X' & \xrightarrow{f} & X \\ \downarrow g' & & \downarrow g \\ S' & \longrightarrow & S \end{array}$  be a commutative diagram of

schemes, then there is a canonical map

$$f^* \Omega_{X/S} \rightarrow \Omega_{X'/S'}$$

which is an isomorphism if the diagram is a fiber product square.  $\lrcorner$

*Proof:* Such a diagram gives a diagram  $(f^{-1}g^{-1}\mathcal{O}_S \rightarrow f^{-1}\mathcal{O}_X) \rightarrow ((g')^{-1}(\mathcal{O}_{S'}) \rightarrow \mathcal{O}_{X'})$  of sheaves of rings on  $X'_{Zar}$ , thus the conclusion follows from (7.2.4.5) and (7.2.7.17).  $\square$

**Prop. (7.5.5.3).** Let  $X, Y$  be schemes over another scheme  $S$ , then

$$\pi_1^* \Omega_{X/S} \oplus \pi_2^* \Omega_{Y/S} \cong \Omega_{X \times_S Y/S},$$

where the maps are given by (7.5.5.2).  $\lrcorner$

*Proof:* It suffices to check on affine subschemes, so we may assume  $X, Y, S$  are affine, thus the map is given by

$$\Omega_{A/S} \oplus_S B \oplus_{A \otimes_S B} \Omega_{B/S} \otimes_S A \rightarrow \Omega_{A \otimes_S B/S}$$

which is an isomorphism by (5.4.3.6).  $\square$

**Prop. (7.5.5.4).** Let  $X, Y$  be schemes over another scheme  $S$ , then  $\Omega_{X \times_S Y/S} \cong \pi_1^* \Omega_{X/S} \otimes \pi_2^* \Omega_{Y/S}$ .  $\lrcorner$

**Prop. (7.5.5.5) [Jacobi-Zariski Sequence].** Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ , then there is an exact sequence of sheaves on  $X$ :

$$f^* \Omega_{Y/Z} \rightarrow \Omega_{X/Z} \rightarrow \Omega_{X/Y} \rightarrow 0.$$

Where the maps come from (7.5.5.2).  $\lrcorner$

*Proof:* Immediate from (5.4.3.8).  $\square$

**Prop. (7.5.5.6).** The stalk of the differential sheaf  $\Omega_{X/k}$  at a rational point  $x$  of a scheme over a field  $k$  is just the Zariski cotangent space  $\mathfrak{m}_x/\mathfrak{m}_x^2$  (7.6.4.21).  $\lrcorner$

*Proof:* Using the Jacobi exact sequence (5.4.3.8) on an affine nbhd  $\text{Spec } A$  of  $x$  for  $A$  and  $\mathfrak{m}_x$ . Then we verified that there is a right inverse of  $A/\mathfrak{m}_x^2 \rightarrow k(x) = x$ , then it follows that  $\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \Omega_{A/k} \otimes_A k(x) = \Omega_{A/\mathfrak{m}_x/k}$  which is the stalk of  $\Omega_{X/k}$  by (7.5.5.1).  $\square$

**Prop. (7.5.5.7) [Euler Exact Sequence].** If  $X = \mathbb{P}_A^n = \text{Proj}(A[x_0, \dots, x_n])$  over  $Y = \text{Spec } A$ , then there is an exact sequence

$$0 \rightarrow \Omega_{X/A} \rightarrow (\mathcal{O}_X(-1))^{n+1} \rightarrow \mathcal{O}_X \rightarrow 0.$$

$\Omega_{X/A}$  is locally free by (7.12.1.16), so by taking dual and exterior powers,

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X(1)^{n+1} \rightarrow \mathcal{T}_X \rightarrow 0, \quad \mathcal{K}_X \cong \mathcal{O}_X(-n-1).$$

$\lrcorner$

*Proof:* Let  $S = A[x_0, \dots, x_n]$ ,  $E = S(-1)^{\oplus n+1}$  with a basis  $e_0, \dots, e_n$ , then there is a map of  $S$ -graded modules  $E \rightarrow S$  with kernel  $M$ .  $E \rightarrow S$  is surjective in all dimension  $\geq 1$ , so we have an exact sequence

$$0 \rightarrow \widetilde{M} \rightarrow \widetilde{E} \rightarrow \mathcal{O}_X \rightarrow 0.$$

Notice  $E_{(x_i)} \rightarrow S_{(x_i)}$  is given by  $e_j \mapsto x_j/x_i$ , so  $\widetilde{M}|_{D(x_i)}$  is a free sheaf generated by sections  $(1/x_i)e_j - (x_j/x_i^2)e_i, j \neq i$ . So we can define  $\varphi_i : \Omega_{X/A}|_{D(x_i)} \rightarrow \widetilde{M}|_{D(x_i)} : d(x_j/x_i) \mapsto (1/x_i)e_j - (x_j/x_i^2)e_i, j \neq i$  mapping isomorphically onto the kernel. It suffices to show this map glue to a map  $\Omega_{X/A} \rightarrow \widetilde{E}$ : On  $D(x_i x_j)$ ,  $x_k/x_i = (x_k/x_j) \cdot (x_j/x_i)$ , so

$$d\left(\frac{x_k}{x_i}\right) - \frac{x_k}{x_j} d\left(\frac{x_j}{x_i}\right) = \frac{x_j}{x_i} d\left(\frac{x_k}{x_j}\right).$$

Applying  $\varphi_i$  to the LHS and  $\varphi_j$  to the RHS gives the same element  $(1/x_i)e_k - (x_k/x_i x_j)e_j$ , which shows compatibility.  $\square$

### Conormal Sheaves

**Def. (7.5.5.8) [Conormal Sheaf of an Immersion].** Let  $i : Z \rightarrow X$  be a closed immersion with corresponding sheaf of ideals  $\mathcal{I}$ . Consider the Qco sheaf  $\mathcal{I}/\mathcal{I}^2$ , which is annihilated by  $\mathcal{I}$ , thus corresponds to a sheaf on  $Z$  by (7.2.6.18), called the **conormal sheaf**  $\mathcal{C}_{Z/X}$  of  $Z$ .

More generally, if  $i$  is any immersion, we can define the conormal sheaf as the conormal sheaf of the closed immersion  $i : Z \rightarrow X \setminus \partial Z$ . And also the **normal sheaf**  $\mathcal{N}_{Z/X}$  is defined to be  $\mathcal{N}_{Z/X} = \text{Hom}_{\mathcal{O}_Z}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Z)$ .  $\lrcorner$

**Prop. (7.5.5.9) [Pullback of Conormal Sheaf].** Let 
$$\begin{array}{ccc} Z' & \xrightarrow{i} & X \\ \downarrow f & & \downarrow g \\ Z' & \xrightarrow{i'} & X' \end{array}$$
 be a fiber product square where  $i, i'$

are immersions, then the canonical map

$$f^* \mathcal{C}_{Z'/X'} \rightarrow \mathcal{C}_{Z/X}$$

is surjective, and if  $g$  is flat, it is an isomorphism.  $\lrcorner$

*Proof:* Change  $X'$  to  $X' \setminus \partial Z'$  and  $X$  to  $X \setminus (g^{-1} \partial Z' \cup \partial Z)$ , we may assume  $i$  is a closed immersion. Then we may localize to the case  $X'$  and  $X$  is affine. Then we notice if  $R' \rightarrow R$  is a ring map and  $I' \subset R'$  is an ideal, with  $I = I'R$ , then  $(I'/(I')^2) \otimes_{R'} R \rightarrow I/I^2$  is surjective, and if  $R/R'$  is flat, then  $I \cong I' \otimes_{R'} R$ , and the map is an isomorphism.  $\square$

**Prop. (7.5.5.10).** Let  $Z \xrightarrow{i} Y \rightarrow X$  be immersions of schemes, then there is a canonical exact sequence

$$i^* \mathcal{C}_{Y/X} \rightarrow \mathcal{C}_{Z/X} \rightarrow \mathcal{C}_{Z/Y} \rightarrow 0$$

*Proof:* By changing  $Y$  to  $Y \setminus \partial Z$  and  $X$  to  $X \setminus (\partial(Y \setminus \partial Z))$ , we can assume the immersions are closed immersion. Now by restricting to affine subsets, it suffices to show that for surjective ring maps  $C \rightarrow B \rightarrow A$ , if  $I = \ker(B \rightarrow A)$ ,  $J = \ker(C \rightarrow A)$ ,  $K = \ker(C \rightarrow B)$ , then there is an exact sequence

$$K/K^2 \otimes_B A \rightarrow J/J^2 \rightarrow I/I^2 \rightarrow 0.$$

But this follows from the observation  $K = \ker(J \rightarrow I)$ .  $\square$

**Prop. (7.5.5.11) [Conormal Sheaf of the Diagonal].** Let  $f : X \rightarrow S$  be a morphism, then there is a canonical isomorphism between  $\Omega_{X/S}$  and the conormal sheaf of the diagonal  $\Delta : X \rightarrow X \otimes_S X$ .  $\lrcorner$

*Proof:* Cf. [Sta]08S2. ?  $\square$

**Cor. (7.5.5.12).** If  $f : X \rightarrow S$  is a monomorphism, e.g. an immersion, then  $\Omega_{X/S} = 0$ .  $\lrcorner$

**Prop. (7.5.5.13).** If  $f : Z \rightarrow X$  is an immersion of schemes over  $S$ , then there is an exact sequence of sheaves on  $Z$ :

$$\mathcal{C}_{Z/X} \rightarrow i^* \Omega_{X/S} \rightarrow \Omega_{Z/S} \rightarrow 0.$$

*Proof:* Replace  $X$  by  $X \setminus \partial Z$ , we can assume  $f$  is a closed immersion. This follows immediate from (7.2.4.6).  $\square$

**Prop. (7.5.5.14).** If  $i : Z \rightarrow X$  is an immersion over  $S$  that locally has a left inverse, then the canonical sequence

$$0 \rightarrow \mathcal{C}_{Z/X} \rightarrow i^* \Omega_{X/S} \rightarrow \Omega_{Z/S} \rightarrow 0.$$

is locally split exact. In particular, if  $s : S \rightarrow X$  is a section of the structure morphism  $X \rightarrow S$ , then the map  $\mathcal{C}_{S/X} \rightarrow s^* \Omega_{X/S}$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]0474. ?

□

**Prop. (7.5.5.15).** Let  $\begin{array}{ccc} Z & \xrightarrow{i} & X \\ & \searrow j & \downarrow \\ & & Y \end{array}$  be a commutative diagram where  $i, j$  are immersions, then there is a

canonical exact sequence

$$\mathcal{C}_{Z/Y} \rightarrow \mathcal{C}_{Z/X} \rightarrow i^* \Omega_{X/Y} \rightarrow 0,$$

where the first arrows comes from (7.5.5.9) and the second map comes from (7.5.5.13).  $\lrcorner$

*Proof:* By replacing  $Y$  by  $Y \setminus \partial Z$  and  $X$  by  $X \setminus (\partial i(Z) \cup \partial j(Z))$ , then we may assume  $i, j$  are closed immersion. Then we check locally, the exactness follows from (8.1.1.5).  $\square$



## 7.6 More Properties of Morphisms between Schemes

*“One should never try to prove anything that is not almost obvious.”*

–A. Grothendieck

References are [Sta].

**Notation(7.6.0.1).**

- Use notations on properties of morphisms between schemes defined in [Properties of Schemes](#).
- Let  $S \in \text{Sch}$

┘

### 1 Finitely Presentedness

**Def.(7.6.1.1) [Locally of Finite Presentation].** A morphism between schemes  $f : Y \rightarrow X$  is called **of locally finite presentation** iff for any point  $x \in X$ , there is an open affine mapped into an open affine that the ring map is of finite presentation. It is called **of finite presentation** iff moreover it is qcqs.

locally of finite presentation is local on the source and target and it is stable under composition and base change but it doesn't satisfy the base change trick by [\(7.4.1.6\)](#) [\(7.4.1.5\)](#) and [\(5.3.7.9\)](#). ┘

**Prop.(7.6.1.2).** Open immersion is locally of finite presentation. ┘

**Prop.(7.6.1.3).** When the target is locally Noetherian, (locally)finite type and (locally)finite presentation is equivalent. ┘

**Prop.(7.6.1.4).** For  $f : X \rightarrow Y$  over  $S$ , if  $X/S$  is locally of f.p. and  $Y/S$  is locally of f.t., then  $f$  is locally of f.p.. If moreover  $X$  is of f.t. and  $Y$  is qc, then  $f$  is of f.t.. ┘

*Proof:* The first follows from [\(5.3.7.11\)](#), the second needs to check qcqs. Qc follows from [\(7.4.4.27\)](#).

□

**Prop.(7.6.1.5) [Chevalley].** A qc morphism locally of f.p. maps locally constructible subset to locally constructible subset. ┘

*Proof:* We prove  $f(E) \cap U_i$  is constructible for every  $U_i$  affine open in  $X$ . The inverse image of  $U_i$  is qc, hence a locally constructible set is constructible [\(4.12.2.10\)](#). So we reduce to the affine case [\(5.1.7.3\)](#). □

**Cor.(7.6.1.6).** As in the proposition, if  $Y$  is qc, and the image is dense in  $Y$ , then it contains an open dense subset of  $Y$ , by [\(4.12.2.18\)](#). ┘

### 2 Flatness

**Def.(7.6.2.1) [Flatness for Schemes].** Flat modules and flat morphisms over schemes are defined in the same way as that of ringed spaces [\(7.2.2.15\)](#). ┘

**Prop.(7.6.2.2).** Flatness is stalkwise by [\(7.2.2.17\)](#), it is stable under base change and compositions.  $\mathcal{F} \in \text{Coh}(X)$  is flat over  $X$  iff it is locally free, by [\(5.4.1.10\)](#)

Thus for  $\mathcal{F} \in \text{QCoh}(X)$ , flatness is equivalent to: For any affine open subsets  $\text{Spec } A \subset X$ ,  $\Gamma(\text{Spec } A, \mathcal{F})$  is flat over  $A$ , because flatness is also stalkwise for modules [\(5.1.4.2\)](#). Similarly, a morphism of schemes  $f : X \rightarrow Y$  is flat iff for any affine opens  $\text{Spec } B \subset X, \text{Spec } A \subset Y$  that  $f(\text{Spec } B) \subset \text{Spec } A$ ,  $B$  is flat over  $A$ . ┘

**Prop. (7.6.2.3).** For a flat morphism of ringed space,  $f^*$  is exact. ┘

*Proof:* Because it is  $f^{-1}$  followed by tensoring with  $\mathcal{O}_X$ , check on stalks. □

**Prop. (7.6.2.4) [Faithfully Flat Descent for Flat Modules].** Let  $f : X \rightarrow Y$  be a morphism of schemes over  $S$  and  $\mathcal{G} \in \mathcal{QCoh}(Y)$ , then  $\mathcal{G}$  is flat over  $S$  iff  $f^*\mathcal{G}$  is flat over  $S$ . In particular,  $Y$  is flat over  $S$  iff  $X$  is flat over  $S$ . ┘

*Proof:* This follows from (5.4.1.6). □

**Prop. (7.6.2.5) [Finite Locally Free Modules].** For  $X \in \mathcal{Sch}$  and  $\mathcal{F} \in \mathcal{QCoh}(X)$ , the following are equivalent, by (5.3.1.7).

1.  $\mathcal{F}$  is finite projective.
  2.  $\mathcal{F}$  is f.p. and flat.
  3.  $\mathcal{F}$  is f.p. and all its localizations at (maximal)primes are free.
  4.  $\mathcal{F}$  is finite locally free.
  5.  $\mathcal{F}$  is finite and locally free.
  6.  $\mathcal{F}$  is finite and all its localizations at primes are free and the function  $p \rightarrow \dim_{k(p)} \mathcal{F} \otimes_R k(p)$  is a locally constant function on  $\text{Spec } R$ .
- ┘

**Cor. (7.6.2.6).** Let  $f : X \rightarrow Y \in \mathcal{NSch}$  be a finite morphism where  $Y$  is reduced, then the following are equivalent:

- $f$  is flat.
  - $f_*\mathcal{O}_X$  is locally free.
  - $\dim_{k(x)}(\pi_*\mathcal{O}_X)_x \otimes k(x)$  is a locally constant function for  $x \in Y$ .
- ┘

*Proof:* 3 follows from (7.5.1.43). □

**Prop. (7.6.2.7) [Going-Down].** Generalizations lift along a flat morphism. ┘

*Proof:* We can find an affine nbhd, then choose a nbhd of the inverse image, then a generalization in an affine open is a true generalization, so it reduce to the affine case. The rest follows from going-down (5.4.1.18). □

**Cor. (7.6.2.8) [Flat Map and Irreducible Components].** A flat morphism maps generic points to generic points. ┘

**Prop. (7.6.2.9) [Flat Map and Associated Points].** Let  $f : X \rightarrow S$  be a morphism of schemes that  $S$  is locally Noetherian, and  $\mathcal{F}$  a Qco sheaf on  $X$ . If  $\mathcal{F}$  is flat over  $S$ , then  $f$  maps  $\text{WeakAss}(\mathcal{F})$  to  $\text{Ass}(S)$ . In particular, if  $X$  is flat over  $S$ , then  $f$  maps  $\text{WeakAss}(X)$  to  $\text{Ass}(S)$ . ┘

*Proof:* Let  $x \in X, f(x) = s$  that  $s \in \text{Ass}(S)$ , then we get a map  $(\mathcal{O}_{S,s}, \mathfrak{m}_s) \rightarrow (\mathcal{O}_{X,x}, \mathfrak{m}_x)$ . If  $\mathfrak{m}_s$  is not associated point, then by prime avoidance (5.1.1.5), there is some  $m \in \mathfrak{m}_s$  that is not a non-zero divisor, by (5.2.5.17), so  $f^\sharp(m)$  is also a non-zero divisor on  $\mathcal{F}_x$ , so  $\mathfrak{m}_x$  is not a weakly associated point of  $\mathcal{F}$ . □

**Prop. (7.6.2.10) [Flat Map is Open].** A flat morphism locally of f.p. is (universally) open, hence it is qc.

And a qc f.f. morphism of schemes is a quotient map.  $\lrcorner$

*Proof:* It suffices to consider the affine case. Then the assertion follows from (5.4.1.30).

For the second, by (7.6.2.7), a subset whose inverse image is closed is stable under specialization (surjectiveness used), then it is closed by (7.4.4.9)  $\square$

**Prop. (7.6.2.11) [Flat Pullback of Closed Subschemes].** Let  $Z \subset X$  be a closed subscheme corresponding to a Qco sheaf of ideals  $\mathcal{I}$ , and  $f : X' \rightarrow X$  be a flat morphism, then the pullback  $Z' \subset X'$  is a closed subscheme that corresponds to the ideal sheaf  $f^*\mathcal{I}$ .  $\lrcorner$

*Proof:* This is because  $f^*$  is exact.  $\square$

**Prop. (7.6.2.12) [Pullback of Flat Closed Subschemes].** Let  $X$  be a scheme over  $S$ ,  $S' \rightarrow S$  be a morphism of schemes, and  $Z \subset X$  be a closed subscheme corresponding to a Qco sheaf of ideals  $\mathcal{I}$  flat over  $S$ , then the pullback  $Z' \subset X'$  is a closed subscheme that corresponds to the ideal sheaf  $f^*\mathcal{I}$ .  $\lrcorner$

**Prop. (7.6.2.13) [Flat Loci is Open].** For a morphism  $f : X \rightarrow S$  locally of f.p., and  $\mathcal{F} \in \mathcal{QCoh}^{\text{pf}}(X)$ , then the set of points of  $X$  that  $\mathcal{F}$  is flat over  $S$  is open. In particular, if  $f$  is a closed map, then the set of points of  $Y$  that  $\mathcal{F}$  is flat is open.  $\lrcorner$

*Proof:* Cf. [Sta]00RC. ?

The last assertion follows from the first one.  $\square$

**Prop. (7.6.2.14) [Generic Flatness].** For a morphism  $f : X \rightarrow S$  of f.t., if  $S$  is reduced and  $\mathcal{F}$  is a Qco f.t.  $\mathcal{O}_X$ -module, then there exists an open dense subset  $U$  of  $S$  that  $X_U \rightarrow U$  is flat, of f.p., and  $\mathcal{F}_{X_U}$  is flat over  $U$  and of f.p. over  $\mathcal{O}_U$ .  $\lrcorner$

*Proof:* As flat and f.p. is local on the base, it suffices to show for  $S$  affine. Then this almost immediately reduces to the affine case by choosing an affine cover of  $X$ , except that we need  $X_U$  to be qs over  $U$ . To achieve this, it suffices to do it second time and let  $X = \cup_{i \leq n} \text{Spec } B_i = \cup X_i$ , and  $X_i \cap X_j = D(I_{ij})$ ,  $M_{ij} = B_i/I_{ij}$ , then choose  $f$  large enough that over  $S_f$ , all  $M_{ij}$  are f.p. over  $B_i$ , then by (5.2.5.9),  $X_{i,f} \cap X_{j,f}$  is qc, thus  $X_f$  is f.p. over  $S_f$ .  $\square$

**Thm. (7.6.2.15) [Fibral Criterion of flatness].** Let  $S \in \text{Sch}$  and  $f : X \rightarrow Y$  in  $\text{Sch}/S$ ,  $\mathcal{F} \in \mathcal{QCoh}^{\text{pf}}(X)$ . Let  $x \in X, y = f(x)$  and  $x$  maps to  $s \in S$ . Suppose  $X, Y$  are both locally Noetherian or  $X$  is locally of f.p. over  $S$  and  $Y$  is locally of f.t. over  $S$ , then the following are equivalent:

- $\mathcal{F}$  is flat over  $S$  at  $x$  and  $\mathcal{F}_s$  is flat over  $Y_s$  at  $x$ .
- $Y$  is flat over  $S$  at  $y$  and  $\mathcal{F}$  is flat over  $Y$  at  $x$ .

$\lrcorner$

*Proof:* Cf. [Sta]039B, 039C.  $\square$

**Prop. (7.6.2.16) [Hilbert Polynomial Constant in Flat Family].** For  $X/T$  projective, where  $T$  is an integral Noetherian scheme and  $X \subset \mathbb{P}_T^n$ . Then for each point  $T$ ,  $X_t$  is a closed subscheme of  $\mathbb{P}_{k(t)}^n$ , so we can consider its Hilbert Polynomial  $P_t$ . Then  $X/T$  is flat iff  $P_t$  is independent of  $T$ .

? Needs huge improvement.  $\lrcorner$

*Proof:*  $P_t(m) = \dim_{k(t)} H^0(X_t, \mathcal{O}_{X_t}(m))$  for  $m$  large by (7.7.3.10). And we may let  $X = \mathbb{P}_T^n$  and prove for any coherent sheaf  $\mathcal{F}$ . Moreover, we may let  $T$  be a affine local Noetherian, because flatness is local and we only need to compare Hilbert polynomial with the generic point. Now we prove a stronger assertion: The following are equivalent:

- $\mathcal{F}$  is flat over  $T$ .
- $H^0(X, \mathcal{F}(m))$  is a free  $A$ -module of finite rank, for  $m$  large.
- The Hilbert polynomial  $P_t$  of  $\mathcal{F}_t$  on  $X_t = \mathbb{P}_{k(t)}^n$  is independent of  $t$ .

1  $\rightarrow$  2: Use the canonical cover and Čech cohomology, then we notice when  $m$  is large,  $H^0(X, \mathcal{F}(m))$  is a kernel of the Čech resolution, so it is flat. And it is also finite by (7.7.4.12). Then it is free because it is flat by (5.4.1.10).

2  $\rightarrow$  1: Let  $M = \bigoplus_{m \geq m_0} H^0(X, \mathcal{F}(m))$ , then  $\widetilde{M} = \mathcal{F}$  (7.5.3.6), notice that the truncation doesn't affect.

2  $\rightarrow$  3: It suffice to prove that for any  $t \in T$ , when  $m$  is large,

$$H^0(X_t, \mathcal{F}_t(m)) \cong H^0(X, \mathcal{F}(m)) \otimes_A k(t).$$

For this, we may use (7.7.5.1) to pass to the localization and assume  $t$  is the closed pt of  $T$ . Then  $A \rightarrow k(t)$  is surjective and we may let  $A^q \rightarrow A \rightarrow k \rightarrow 0$ , then by (7.7.4.8), we have  $H^0(X_t, \mathcal{F}_t(m))$  is the cokernel of  $H^0(X, \mathcal{F}(m))^q \rightarrow H^0(X, \mathcal{F}(m))$ , but this cokernel is  $H^0(X, \mathcal{F}(m)) \otimes_k$  because tensoring is right-adjoint, so we are done.

3  $\rightarrow$  2: We have the rank of  $H^0(X, \mathcal{F}(m))$  at the generic and closed point of  $T$  are the same (still use  $H^0(X_t, \mathcal{F}_t(m)) \cong H^0(X, \mathcal{F}(m)) \otimes_A k(t)$ .) Now (5.4.8.1) gives  $H^0(X, \mathcal{F}(m))$  is free. It is f.g. automatically.  $\square$

**Cor. (7.6.2.17).** For a flat morphism to a connected scheme  $T$ , the dimension, degree, and arithmetic genus of the fibers are independent of  $t$ .  $\lrcorner$

*Proof:* By (7.7.3.6) and (7.7.3.12).  $\square$

**Def. (7.6.2.18).** For a surjective map of varieties  $f : X \rightarrow T$  over an alg.closed field  $k$ , its fibers over closed points with induced reduced structure  $X_{(t)}$  is called a **algebraic family of varieties parametrized by  $T$**  if

1.  $f^{-1}(t)$  is irreducible of dimension  $\dim X - \dim T$  for every closed point  $t$ .
2. If  $\zeta$  is the generic point of  $f^{-1}(t)$ , then  $F^\sharp \mathfrak{m}_t$  generates the maximal ideal  $\mathfrak{m}_\zeta \subset \mathcal{O}_{\zeta, X}$ .

$\lrcorner$

**Prop. (7.6.2.19).** if  $X_{(t)}$  is an algebraic family of normal varieties over an alg.closed field  $k$  parametrized by a nonsingular curve  $T$ , then it is a flat family of schemes.  $\lrcorner$

*Proof:* By (7.13.1.21),  $X \rightarrow T$  is flat. So what we need to do is to prove  $X_t$  is reduced so  $X_t = X_{(t)}$ . Let  $A = \mathcal{O}_{x, X}$ , let  $u_t$  be a uniformizer of  $\mathcal{O}_{t, T}$ , then  $A/tA$  is the local ring of  $x$  on  $X_t$ . By hypothesis  $X_t$  is irreducible so  $tA$  has a unique minimal prime  $p$  in  $A$ , and  $t$  generate the maximal ideal of  $A_p$  by hypothesis. The local ring of  $X_{(t)}$  is  $A/p$ , so  $A/p$  is normal by hypothesis. Then the result follows from (5.3.5.14).  $\square$

**Cor. (7.6.2.20) [Igusa].** Let  $X_{(t)}$  be an algebraic family of normal varieties in  $\mathbb{P}_k^n$  for  $k$  alg.closed parametrized a variety  $T$ , then the Hilbert polynomials of  $X_{(t)}$  are independent of  $t$ .  $\lrcorner$

*Proof:* ? Why is  $X/T$  projective? Cf. [Hartshorne P265].  $\square$

### Degenerating Techniques

#### Finite Locally Free Morphism

**Def. (7.6.2.21) [Finite Locally Free].** A morphism  $f : X \rightarrow Y$  is called **finite locally free of rank  $d$**  iff it is affine, and  $f_*\mathcal{O}_X$  is a finite locally free  $\mathcal{O}_Y$ -module of rank  $d$ . ┘

**Cor. (7.6.2.22).** If  $f$  is finite locally free of rank  $n$ , then for any locally free sheaf  $E$  of rank  $k$  on  $X$ ,  $f_*E$  is locally free of rank  $nk$ . ┘

**Prop. (7.6.2.23).**  $f : X \rightarrow Y \in \text{Sch}$  is finite locally free (7.6.2.21) iff it is finite, flat and of f.p.. In particular, when  $Y$  is locally Noetherian, this is equivalent to  $f$  is finite flat. ┘

*Proof:* Both notions are local on the target, so we reduce to the ring case, which is (5.3.1.7). □

**Cor. (7.6.2.24).** Finite locally freeness is stable under composition and base change, and it is local on the target. ┘

**Prop. (7.6.2.25) [Trace and Norm].** Let  $f : Y \rightarrow X$  be a finite locally free map of constant rank, then there are trace and norm maps  $\text{tr} : f_*\mathcal{O}_Y \rightarrow \mathcal{O}_X$ ,  $\text{Nm} : f_*\mathcal{O}_Y \rightarrow \mathcal{O}_X$  compatible with arbitrary base change. ┘

*Proof:* The proof is the same as that of (5.3.1.11). □

**Prop. (7.6.2.26).** Let  $f : Y \rightarrow X$  be a finite locally free map of constant rank, and  $b \in \Gamma(Y, \mathcal{O}_Y)$ , then  $f(Z(b)) = Z(\text{Nm}(b))$ . ┘

*Proof:* We can assume  $X$  is affine, then we need to show that for a prime  $\mathfrak{p}$  with inverse images  $\mathfrak{p}_i$ ,  $b \in \cup \mathfrak{p}_i$  iff  $\text{Nm}(b) \in \mathfrak{p}$ . We localize at  $\mathfrak{p}$ , then  $\text{Nm}(b) \in \mathfrak{p}$  iff  $\text{Nm}(b)$  is non-invertible iff multiplication by  $b$  is non-invertible iff  $b$  is non-invertible iff  $b \notin \cup \mathfrak{p}_i$ , because  $\mathfrak{p}_i$  are all the maximal ideals of  $B_{\mathfrak{p}}$ . □

### 3 Dimensions

References are [Mat80] and [Vak17]Chap11.

**Prop. (7.6.3.1) [Locally Algebraic Scheme is Catenary].** If  $X$  is a locally algebraic scheme over a field  $k$  purely of dimension  $n$ , and  $Y$  an irreducible subscheme of  $X$ , then  $\dim Y + \text{codim}(Y, X) = \dim X$ . ┘

*Proof:* Choose an affine open of the generic point of  $Y$ , then we are reduced to the affine case (5.2.4.3) (5.2.4.7). □

**Prop. (7.6.3.2).** For any scheme,  $\dim \mathcal{O}_x = \text{codim}(\overline{\{x\}}, X)$ . ┘

**Prop. (7.6.3.3).** For an integral scheme algebraic over a field  $k$ ,

$$\dim X = \dim \mathcal{O}_{p,X} = \dim U = \text{tr.deg } K(X)/k$$

for any closed point  $p$  and any open subscheme  $U$ . ┘

*Proof:* Use closed point are dense (7.4.1.26) and  $k$  is universal catenary to prove it is true for some  $U$  and all the closed point in it, so other  $U$ 's because  $X$  is irreducible. The last equation follows from (5.2.4.23). □

**Prop. (7.6.3.4) [Finite Surjection Preserves Dimension].** Let  $X \rightarrow Y$  be a surjective finite morphism of algebraic integral schemes over a field  $k$ , then  $\dim X = \dim Y$ .  $\lrcorner$

*Proof:* The hypothesis implies that for any affine open  $\text{Spec } A \subset Y$ , the inverse image is  $\text{Spec } B$  that  $A \rightarrow B$  is an injective integral ring extension, so we can use (7.6.3.3) and (5.2.4.14).  $\square$

**Prop. (7.6.3.5).** Let  $X$  be a locally Noetherian scheme, if  $U \subset X$  is an open subscheme that  $U \rightarrow X$  is affine, then every irreducible components of  $X - U$  has codimension  $\leq 1$ . And if  $U$  is dense, then equality must hold.  $\lrcorner$

*Proof:* Cf. [Sta]0BCU.  $\square$

**Prop. (7.6.3.6) [Local Dimensions].** Let  $X$  be a locally algebraic scheme over a field  $k$  and  $x \in X$ , then the local dimension  $\dim_x(X)$  equals the maximal dimension of irreducible components of  $X$  passing through  $x$ , by (5.2.4.27).  $\lrcorner$

**Prop. (7.6.3.7).** If  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$ ,  $x \in X$  and  $s = f(x)$ , then

$$\dim_x(X_s) = \dim \mathcal{O}_{X_s, x} + \text{tr.deg}_{k(s)} k(x).$$

$\lrcorner$

*Proof:* This reduces to the case  $Y = \text{Spec } k(s)$ , and it follows from (5.2.4.28).  $\square$

**Prop. (7.6.3.8) [Semicontinuity of Dimension].** Let  $f : X \rightarrow S$  be a morphism of schemes locally of f.t., then the function  $x \mapsto \dim_x(X_{f(x)})$  is upper-semicontinuous on  $X$ .

Moreover, if  $f$  is of f.p., then the open subsets  $\{x \mid \dim_x(X_{f(x)}) \leq n\}$  is retrocompact.  $\lrcorner$

*Proof:* This follows directly from (5.2.4.32).  $\square$

**Prop. (7.6.3.9) [Local Dimension and Base Change].** Let

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

be a fiber product diagram of schemes, and  $f$  is locally of f.t.. Suppose  $x' \in X'$ ,  $x = g'(x')$ ,  $s = g(s')$ . Then

- $\dim_x(X_s) = \dim_{x'}(X_{s'})$ .

- 

$$\dim \mathcal{O}_{F, x'} = \dim \mathcal{O}_{X'_{s'}, x'} - \dim \mathcal{O}_{X_s, x} = \text{tr.deg}_{k(s)}(k(x)) - \text{tr.deg}_{k(s')} (k(x'))$$

where  $F$  is the fiber of the morphism  $X'_{s'} \rightarrow X_s$  over  $x$ . In particular,  $\dim \mathcal{O}_{X'_{s'}, x'} \geq \dim \mathcal{O}_{X_s, x}$  and  $\text{tr.deg}_{k(s')} (k(x')) \leq \text{tr.deg}_{k(s)}(k(x))$ .

- Given  $s', s, x$  that  $f(x) = g(s')$ , there exists an  $x' \in X'$  that  $\dim \mathcal{O}_{X'_{s'}, x'} = \dim \mathcal{O}_{X_s, x}$  and  $\text{tr.deg}_{k(s')} (k(x')) = \text{tr.deg}_{k(s)}(k(x))$ .

$\lrcorner$

*Proof:* It can be reduced to the case that  $S = \text{Spec } k(s)$ ,  $S' = \text{Spec } k(s')$  and  $X, X'$  affine. Then 1 follows from (5.2.4.31), and 2, 3 follows from (5.2.4.26).  $\square$

**Cor. (7.6.3.10)[Dimension and Field Extension].** Let  $K/k$  be a field extension,  $X$  a locally algebraic scheme over  $k$  purely of dimension  $n$ , then  $X_K$  is a scheme purely of dimension  $n$ .  $\lrcorner$

**Remark (7.6.3.11).** This proposition shows in particular local dimension behaves better than the dimension of the stalk.  $\lrcorner$

**Def. (7.6.3.12)[Relative Dimensions].** A morphism of schemes which locally of f.t. is called of **relative dimension**  $n$  iff all fibers  $X_s$  are equidimensional of dimension  $n$ .  $\lrcorner$

**Prop. (7.6.3.13).** Being a morphism of relative dimension  $n$  is stable under field extension, by (7.6.3.10).  $\lrcorner$

### Dimension and Flatness

**Prop. (7.6.3.14) [Faithfully Flat Morphism].** If  $f : Y \rightarrow X$  is a faithfully flat morphism, then  $\dim Y \geq \dim X$ .  $\lrcorner$

*Proof:* This is easy from (7.6.2.7).  $\square$

**Prop. (7.6.3.15)[Integral Flat Morphisms].** If  $f : X' \rightarrow X$  is an integral flat morphism of schemes, and  $X$  is pure of dimension  $n$ , then so does  $X'$ . The converse holds if  $f$  is faithfully flat.  $\lrcorner$

*Proof:* By (7.6.2.8) and (7.4.4.35),  $f$  maps an irreducible component of  $X'$  onto an irreducible component of  $X$ , Then by (4.12.2.27), the proposition reduces to the affine case (5.2.4.15). If  $f$  is faithfully flat, then every irreducible component of  $X$  is in the image.  $\square$

**Cor. (7.6.3.16)[Dimension and Field Extension].** If  $K/k$  is an algebraic extension,  $X$  a scheme over  $k$  purely of dimension  $n$ , then  $X_K$  is a scheme purely of dimension  $n$ . Compare with (7.6.3.10).  $\lrcorner$

**Prop. (7.6.3.17)[Dimension Extension and Flatness].** Let  $f : X \rightarrow Y, g : Y \rightarrow S$  be locally of f.t.,  $x \in X, y = f(x), s = g(y)$ , then

$$\dim_x(X_s) \leq \dim_x(X_y) + \dim_y(Y_s).$$

Moreover, equality holds if  $\mathcal{O}_{X,x}/\mathcal{O}_{Y,y}$  is flat.

In particular, if  $S = \operatorname{Spec} k$  and  $X, Y$  are irreducible and  $X$  is flat over  $Y$ , then  $\dim X_y = \dim X - \dim Y$  for any  $y \in Y$ .  $\lrcorner$

*Proof:* By (7.6.3.7) and the fact transcendental degree is additive, this reduces to

$$\dim \mathcal{O}_{X_s,x} \leq \dim \mathcal{O}_{X_y,x} + \dim \mathcal{O}_{Y_s,y}.$$

We can assume  $X, Y$  is affine and  $S = \operatorname{Spec} k(s)$ , so the rest follows from (5.2.4.13).  $\square$

**Cor. (7.6.3.18).** If  $f : X \rightarrow Y, g : Y \rightarrow Z$  are of relative dimension  $m$  and  $n$  (7.6.3.12), and  $f$  is flat, then  $g \circ f$  is of relative dimension  $m + n$ .  $\lrcorner$

**Cor. (7.6.3.19)[Generic Dimension Equation].** If  $f : X \rightarrow Y$  is a dominant morphism of irreducible algebraic schemes over  $K$  that  $X$  is reduced, then there is a dense open subset  $U$  of  $Y$  that for any  $y \in U$ ,

$$\dim(X_y) = \dim X - \dim Y.$$

$\lrcorner$



*Proof:* This is a combination of the above proposition and generic flatness(7.6.2.14).  $\square$

**Prop. (7.6.3.20) [Relative Dimension and Base Change].** By(7.6.3.9), the base change of a morphism of locally algebraic schemes over a field  $k$  of relative dimension  $n$  is still of relative dimension  $n$ . In particular, for a variety over  $K$  a field, the dimension is invariant under base change of fields.  $\lrcorner$

**Cor. (7.6.3.21).** Let  $Y, Z$  be irreducible locally algebraic schemes over  $k$ , then  $Y \times_k Z$  is pure of dimensional  $\dim(Y) + \dim(Z)$ .  $\lrcorner$

*Proof:* Combine(7.6.3.20) and(7.6.3.17).  $\square$

**Prop. (7.6.3.22).** For a morphism  $f : X \rightarrow Y$  between locally Noetherian schemes which is flat and locally of f.t and of relative dimension  $n$ , then if  $y = f(x)$ , we have  $\dim_x(X_y) = \dim_x(X) - \dim_y(Y)$ .  $\lrcorner$

*Proof:* Shrinking the nbhd, we may assume  $\dim_x(X) = \dim X$  and  $\dim_y(Y) = \dim Y$  and  $X, Y$  are affine. Now  $f$  is locally of f.p. and flat, so it is open(7.6.2.10). So we may assume  $f$  is surjective. Then  $\dim \mathcal{O}_{X,a} = \dim \mathcal{O}_{Y,b} + \dim \mathcal{O}_{X_b,a} = \dim \mathcal{O}_{Y,b} + n$  by(5.2.4.13), then taking supremum??, the result follows.  $\square$

**Cor. (7.6.3.23).** For a morphism of schemes that is flat and of f.t., if  $Y$  is irreducible, then  $X$  is equidimensional of dimension  $\dim Y + n$  iff  $X_y$  is equidimensional of dimension  $n$  for every  $y \in Y$ .  $\lrcorner$

*Proof:* The proof highly relies on(7.6.3.3).

If  $X$  is equidimensional of dimension  $\dim Y + n$ , for  $Z \subset X_y$  an irreducible component, choose a closed pt  $x$  of  $Z$  not contained in any other irreducible component, then

$$\dim_x Z = \dim_x X - \dim_y Y = \dim X - \dim \overline{\{x\}} - \dim Y + \dim \overline{\{y\}}.$$

The two closures are of the same dimension because by(5.2.6.10), their quotient field extension is finite, and use(5.2.4.14).

Conversely, for an irreducible component of  $X$ , choose a closed pt  $x$  of  $Z$  not contained in any other irreducible component, then the image is also closed, by(7.4.1.27), so the result is immediate.  $\square$

**Prop. (7.6.3.24).** If  $f : X \rightarrow Y$  is a proper flat morphism of schemes of f.p., then the dimension of fibers of  $f$  is a locally constant function.  $\lrcorner$

*Proof:* Cf.[Sta]04DJ. ?  $\square$

## 4 Smoothness

**Def. (7.6.4.1) [Smooth Morphisms].** A **smooth morphism** is a morphism  $f : X \rightarrow Y \in \mathcal{S}ch$  s.t. there is an open affine cover  $\{U_i\}$  of  $S$  and an open affine cover  $\{V_{ij}\}$  of  $f^{-1}(\{U_i\})$  s.t. the ring maps  $\Gamma(U_i) \rightarrow \Gamma(V_{ij})$  are all smooth(5.4.5.12). In particular, it is locally of f.p..

A **standard smooth morphism** is the Spec map of a standard smooth ring map. Smooth morphisms are locally standard smooth by(5.4.5.12).

Smoothness is local on the source and target(5.4.5.13). Smoothness is stable under base change and composition(5.4.5.13).  $\lrcorner$



**Prop. (7.6.4.2).** For a smooth morphism  $X \rightarrow S$ , the morphism of differential  $\Omega_{X/S}$  is locally free and  $\dim_x \Omega_{X/S} = \dim_x(X_{f(x)})(\text{local dimension}(4.12.2.26))$ .  $\lrcorner$

*Proof:* We can assume that  $X \rightarrow S$  is standard smooth, so by the proof in (5.4.5.12),  $\Omega_{X/S}$  is free of dimension  $n - c$ , and also standard smooth is relative global complete intersection (5.4.5.11), so  $U_{f(x)}$  is equidimensional of dimension  $n - c$ , thus the result.  $\square$

**Cor. (7.6.4.3) [Differential Criterion of Smoothness].** If  $X \rightarrow S$  is a flat of relative dimension  $n$ , then  $X$  is smooth over  $S$  iff it is locally of f.p. and  $\Omega_{X/S}$  is locally free of dimension  $n$ , by (5.4.5.23).  $\lrcorner$

**Prop. (7.6.4.4) [Smooth Morphisms are Open].** Smooth morphisms are syntomic hence flat. Smooth morphism is locally of f.p. Hence smooth morphism is universally open (7.6.2.10).  $\lrcorner$

**Prop. (7.6.4.5) [Fiberwise and Stalkwise].** For  $f : X \rightarrow S \in \text{Sch}$  locally of f.p. and  $x \in X$ , the following are equivalent:

- $f$  is smooth at  $x$ .
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $X_{f(x)}/\kappa(f(x))$  is smooth at  $x$ , by (5.4.5.20).
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $\Omega_{X/S,x}$  can be generated by  $\dim_x(X_{f(x)})$  elements, by (7.6.4.2) and (5.4.5.23).
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $\Omega_{X_s/s,x} \otimes_{\mathcal{O}_{X_s,x}} k(x) = \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$  can be generated by  $\dim_x(X_{f(x)})$  elements, by Nakayama, because  $\Omega_{X/S,x}$  is of f.p. by (5.4.3.9).

In particular, a smooth morphism can be seen as a family of smooth schemes.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.6.4.6).** If  $f : X \rightarrow Y \in \text{Sch}/S$  is smooth, then by (7.5.5.5)(5.4.5.5), there is an exact sequence of sheaves on  $X$ :

$$0 \rightarrow f^* \Omega_{Y/S} \xrightarrow{df^*} \Omega_{X/S} \rightarrow \Omega_{X/Y} \rightarrow 0.$$

$\lrcorner$

**Prop. (7.6.4.7).** If  $Z \rightarrow X \rightarrow S$ ,  $Z/S$  is smooth and  $Z \rightarrow X$  is an immersion, then there is an exact sequence of sheaves (7.5.5.13)(5.4.5.6):

$$0 \rightarrow \mathcal{C}_{Z/X} \rightarrow i^* \Omega_{X/S} \rightarrow \Omega_{Z/S} \rightarrow 0$$

$\lrcorner$

**Prop. (7.6.4.8).** If  $X \rightarrow Y \rightarrow S$ , and  $X \rightarrow Y$  are faithfully flat and locally of f.p.,  $X/S$  is smooth, then  $Y/S$  is smooth.  $\lrcorner$

*Proof:* Cf. [Sta]05B5.  $\square$

**Prop. (7.6.4.9).** If  $f : X \rightarrow S$  is faithfully flat and locally of f.p., then the set of points of  $S$  s.t.  $f$  is smooth is table under base change.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.6.4.10).** By (7.6.4.5)(7.6.4.2), A morphism is smooth of relative dimension  $n$  is equivalent to fppf+fibers equidimensional of dimension  $n$  and  $\Omega_{X/S}$  is locally free of dimension  $n$ .  $\lrcorner$

**Prop. (7.6.4.11) [Jacobian Criterion for Projective Schemes].** Let  $X$  be a closed subscheme of  $\mathbb{P}_R^n$  defined by  $r$  polynomials  $F_1(X_0, \dots, X_n), \dots, F_r(X_1, \dots, X_n)$ , then for  $x \in X$ ,  $X$  is smooth at  $x$  iff the Jacobian has rank  $\geq n - \dim_x X$  at  $x$  iff the Jacobian has rank  $= n - \dim_x X$  at  $x$ .  $\lrcorner$

*Proof:* Assume that  $x$  is in the standard open  $X_0 \neq 0$ , then we can use Euler's identity  $\sum X_i \frac{\partial F}{\partial X_i} = \dim(F)F$  to eliminate the first row, then divide by  $X_0^{\dim F}$  to get the Jacobian on the affine open  $\text{Spec } R[\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}]$ , so we finish by (5.4.5.24).  $\square$

**Prop. (7.6.4.12) [Generic Smoothness on the Source].** Let  $\pi : X \rightarrow Y$  be a dominant map of integral schemes of f.t. that  $R(Y) \rightarrow R(X)$  is separable, then there is a non-empty open subscheme  $U \subset X$  that  $\pi|_U$  is smooth of relative dimension  $\dim X - \dim Y$ .  $\lrcorner$

*Proof:* This reduces to the affine case, and follows from (5.4.5.29) and (7.6.3.17).  $\square$

**Prop. (7.6.4.13) [Generic Smoothness on the Target].** Let  $\pi : X \rightarrow Y$  be a morphism of  $k$ -varieties where  $\text{char } k = 0$ , and  $X$  is smooth over  $k$ , then there is a dense open subset  $U \subset Y$  that  $\pi^{-1}(U) \rightarrow U$  is smooth.  $\lrcorner$

*Proof:* Cf. [Vak17]P681.  $\square$

### Smooth over Fields

**Prop. (7.6.4.14) [Differential Criterion of Smoothness].** Let  $X$  be a scheme algebraic over a field  $k$ .

- If  $X$  is equidimensional of dimension  $n$ , then  $X$  is smooth over  $k$  iff  $\Omega_{X/S}$  is locally free of dimension  $n$ .
- If  $\Omega_{X/k}$  is locally free, and ( $k$  is of char 0) or ( $k$  is perfect and  $X$  is reduced), then  $X$  is smooth over  $k$ .  $\lrcorner$

*Proof:* 1: This follows from (7.6.4.3).

2: If  $k$  is of characteristic 0, then this follows from (5.4.5.28).

perfect case: [Sta]04QP. ?  $\square$

**Prop. (7.6.4.15) [Smooth over Field and Geo.Regular].** For a scheme locally algebraic over a field  $k$ ,  $X$  is geometrically regular iff it is smooth over  $k$ . In particular, if  $k$  is perfect, then smoothness is equivalent to regularity, by (7.4.3.18).  $\lrcorner$

*Proof:* The question is local around  $x$ , so may assume  $X$  is affine. Then this follows from (5.4.5.27).  $\square$

**Cor. (7.6.4.16) [Hartshorne Definition].** By (7.6.4.5) and (7.6.4.15), a morphism between schemes algebraic over a field  $k$  is smooth of relative dimension  $n$  iff  $f$  is flat and every fiber of  $f$  is geometrically regular of dimension  $n$ .  $\lrcorner$

**Cor. (7.6.4.17).** A smooth scheme over a field  $k$  is regular hence normal.  $\lrcorner$

**Prop. (7.6.4.18) [Smoothness and Arithmetic Points].** For  $X \in \text{Sch}^{\text{sm}}/k$ , the set of closed points of  $X$  with finite separable residue field  $\kappa(x)/k$  is dense in  $X$ .  $\lrcorner$

*Proof:* It suffices to show there exists one such points. By (7.6.6.7), we can assume  $X \xrightarrow{\pi} \mathbb{A}_k^d \rightarrow k$ , where  $\pi$  is étale. Thus  $X$  is open in  $\mathbb{A}_k^d$  by (7.6.6.3). Then we can choose a closed point of  $\pi(X)$  s.t. the residue field is finite separable, as  $k^{\text{sep}}$  is infinite. Then choose an inverse image, then the residue is finite étale over  $k$ , by (7.6.6.8). This point is clearly closed.  $\square$

**Prop. (7.6.4.19) [Smoothness at Rational Points].** Let  $X$  be a locally algebraic scheme over a field  $k$ . Let  $x \in X$  that  $k(x)/k$  is finite separable, then  $X$  is smooth at  $x$  iff it is a regular point, by (5.4.5.26).  $\lrcorner$

**Prop. (7.6.4.20) [Geo.Reduced Scheme Generic Smooth].** Let  $X \in \text{Sch}^{\text{loc.ft}}/k$  be geometrically reduced, then it contains an open dense subset that is smooth over  $k$ .  $\lrcorner$

*Proof:* The problem is local, so we may assume  $X$  is affine, consider its irreducible components, all their intersections can be removed, because they are nowhere dense, so we may assume  $X$  is irreducible. So  $X$  is integral, let  $\eta$  be the generic pt, then  $k(\eta)/k$  is separable, by (7.4.3.2). Then choose an affine subscheme  $\text{Spec } A \subset X$ , then  $A$  is smooth at  $(0)$  over  $k$ , by (5.4.5.29), then by definition, it is smooth on some dense open subscheme of  $X$ .  $\square$

### Tangent Spaces

**Def. (7.6.4.21) [Relative Tangent Spaces].** For  $X \in \text{Sch}/S$  and  $x \in X$ , define  $T_{X/S,x}^* = \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} k(x)$ , and  $T_{X/S,x} = \text{Hom}_{k(x)}(T_{X/S,x}^*, k(x)) = \text{Hom}_{\mathcal{O}_{X,x}}(\Omega_{X/S,x}, k(x))$ , called the **relative tangent space** of  $X$  over  $S$  at  $x$ .

When  $x \in X$  mapping to  $s \in S$ ,  $T_{X/S,x} = \text{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x))$ , where  $\mathfrak{m}_x$  is the maximal ideal of the stalk of  $x \in X_s$ .

Then it can be verified that for a  $k$ -scheme and a rational point  $x \in X$ ,  $T_{X,x}$  is in bijection with  $\text{Mor}_k(\text{Spec } k[\varepsilon], X)$  that maps the closed point to  $x$ .  $\lrcorner$

**Prop. (7.6.4.22) [Tangent Map].** Let  $f : X \rightarrow Y$  be a morphism of schemes over  $S$ ,  $x \in X$ ,  $f(x) = y$ . Assume  $k(x) = k(y)$ , then  $f$  defines a natural linear map

$$df : T_{X/S,x} \rightarrow T_{Y/S,y} : \text{Hom}_{\mathcal{O}_{X,x}}((\Omega_{X/S})_x, k(x)) \rightarrow \text{Hom}_{\mathcal{O}_{X,x}}((f^*\Omega_{Y/S})_x, k(x)) \quad (7.5.5.5) = \\ \text{Hom}_{\mathcal{O}_{X,x}}(\Omega_{Y/S,y} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}, k(y)) = \text{Hom}_{\mathcal{O}_{Y,y}}(\Omega_{Y/S,y}, k(y))$$

$\lrcorner$

**Cor. (7.6.4.23).** Via similar argument, if  $k(x) = k(s) = k(y)$ , there is an isomorphism

$$T_{X/S,x} \oplus T_{Y/S,y} \cong T_{X \times_S Y, (x,y)}.$$

$\lrcorner$

**Prop. (7.6.4.24).** Let  $X$  be a scheme over a field  $k$  and  $x \in X$  with residue field  $k(x)$ , then the tangent space is canonical isomorphic to  $\text{Mor}(k(x)[\varepsilon], X)$  that maps the closed point to  $x$ . And the vector bundle structure is given by the cogroup structure on  $\text{Spec } k[\varepsilon] : \mu : k[\varepsilon] \otimes k[\varepsilon] \rightarrow k[\varepsilon]$ .  $\lrcorner$

**Prop. (7.6.4.25) [Krull's Principal Tangent Theorem].** Let  $Z \subset X$  be a closed subscheme with ideal of definition  $\mathcal{I}$ ,  $x \in Z$ , then  $T_{Z,x}$  is the subspace of  $T_{X,x}$  cut out by  $(\mathcal{I} \bmod \mathfrak{m}_x^2)$ .

In particular,  $\mathcal{T}_{Y \cap Z,p} = T_{Y,p} \cap T_{Z,p} \subset T_{X,p}$ .  $\lrcorner$

**Prop. (7.6.4.26) [Tangent Criterion of Smoothness].** By (7.6.4.15)(7.6.4.19) and the definition of regular local ring, if  $X$  is a locally algebraic scheme over a field  $k$ , then  $X$  is smooth at a point  $x$  with residue field  $k(x)/k$  separable iff  $\dim_{k(x)} T_{X,x}$  equals ( $\leq$ ) the dimension of  $\mathcal{O}_{X,x}$ .  $\lrcorner$

### Bertini's Theorem

**Prop. (7.6.4.27) [Bertini].** For any  $k \in \mathbf{Field}$  and a quasi-projective scheme  $X \subset \mathbb{P}_k^n$  smooth away from f.m. points, there is an open dense subset  $U$  of the dual projective space  $\mathbb{P}_k^{n\vee}$  s.t. for any closed point  $[H] \in U$ ,  $H$  doesn't contain any component of  $X$ , and the scheme  $H \cap X$  is smooth over  $k$ .

Moreover, if  $X$  is a variety and  $\dim X \geq 2$ , the smooth cut is even a smooth variety by (7.9.2.24) and (7.4.2.11).  $\square$

*Proof:* Let  $\mathbb{P}^n = \text{Proj } k[x_0, \dots, x_n]$ ,  $(\mathbb{P}^n)^\vee = \text{Proj } k[a_0, \dots, a_n]$ , Let  $\bar{X}$  be cut out by equations  $f_1, \dots, f_r$ , we define a projective scheme  $Z \subset \mathbb{P}^n \times (\mathbb{P}^n)^\vee$  be cut by the equations

- $f_r$ .
- $f_{r+1} = \sum a_i x_i$ .
- determinants of  $(r+1) \times (r+1)$ -minors of  $(\frac{\partial f_i}{\partial x_j})_{(r+1) \times n}$ .

Now for any  $(x, [H]) \in Z$ ,  $x \in X$ ,  $x \in H$  that  $[H]$  is closed in  $(\mathbb{P}^n)^\vee$ , the last equation means  $X \cap H$  is non-smooth at  $p$  or contains the irreducible component passing  $p$ , by Jacobian criterion (7.6.4.11). Now consider the projection  $Z \rightarrow X$ , for each closed point  $p \in X$  that  $\dim_p X = d$ , the codimension of the fiber of  $Z$  over  $x$  is of dimension  $r-1 = n - \dim_x X - 1$  (To see this, one way is use regularity and tangent space, another way is to use the fact  $(x_i)$  is orthogonal to  $(\frac{\partial f_i}{\partial x_j})_j$  for all  $i$  by Euler identity, so the  $r+1$  restrictions of  $(a_i)$  are linearly independent), then by (7.6.3.17),  $\dim Z = \dim \pi_1^{-1}(Z) \leq n-1$ , Thus the image of  $Z$  in  $\mathbb{P}_k^{n\vee}$  is also a closed subscheme of dimension  $\leq n-1$ .

If  $X$  has f.m. singular points, we also need to cut out the surfaces  $N_{k(x)/k}(\sum x_i^k a_i)$ , where  $(x_i^j)$  are the f.m. closed singular points of  $x$ . They are also of dimension  $n-1$ .  $\square$

**Cor. (7.6.4.28) [Bertini's Theorem for Surfaces].** Similarly, for any  $d \geq 0$ , a generic dimension  $d$  surface intersect  $X$  at a smooth subscheme, by using Veronese embedding.  $\square$

**Cor. (7.6.4.29).** When  $X \subset \mathbb{P}_k^n$  is a projective  $k$ -variety where  $k$  has characteristic 0, the scheme  $Z$  in  $\mathbb{P}_k^{n\vee}$  is called the **dual variety** of  $X$ . When  $X$  is irreducible and smooth, it is of dimension  $n-1$ .  $\square$

*Proof:*  $\square$

**Prop. (7.6.4.30).** The dual variety of the dual variety is  $X$  itself.  $\square$

*Proof:* Cf. [Joe Harris, Algebraic Geometry, 15.24].  $\square$

**Prop. (7.6.4.31) [Kleinman-Bertini].** Let  $X$  be a  $k$ -variety that is homogenous space for a  $k$ -variety  $G$  over a field  $k$ , suppose  $\alpha : Y \rightarrow X, \beta : Z \rightarrow X$  are morphisms of varieties, then

- There is a non-zero open subset  $V \subset G$  that for any closed point  $\sigma \in V$ ,  $\dim \sigma(Y) \times_X Z = \dim Y + \dim Z - \dim X$ .
- If  $Y, Z$  are smooth over  $k$  of characteristic 0, then there is a non-empty open subset  $V \subset G$  that  $(G \times_k Y) \times_X Z \rightarrow G$  is smooth. In particular, for any closed point  $\sigma \in V$ ,  $\sigma(Y) \times_X Z$  is smooth over  $k(\sigma)$  of dimension  $\dim Y + \dim Z - \dim X$ .  $\square$

*Proof:* Let  $\Gamma = (G \times_k Y) \times_X Z$ , then there is a map  $\Gamma \rightarrow Y \times_k Z$ .  $Y \times_k Z$  has dimension  $\dim Y + \dim Z$  by (7.6.3.21), and there is a base change diagram

$$\begin{array}{ccc} (G \times_k Y) \times_X Z & \longrightarrow & Y \times_k Z \\ \downarrow & & \downarrow \\ G \times_k X & \longrightarrow & X \times_k X \end{array},$$

and  $G \times X \rightarrow X \times X$  is flat as it is equivariant under  $G \times G$ -action, by (7.6.2.14), thus  $\Gamma \rightarrow Y \times_k Z$  is flat of relative dimension  $\dim G - \dim X$ , thus it has dimension  $\dim Y + \dim Z + \dim G - \dim X$ .

1: This follows from (7.6.3.19) applied to the map  $G \times_k Y \times_X Z \rightarrow G$ .

2: As  $G \times X \rightarrow X \times X$  is flat, and its fiber are isomorphic to  $G_x$ , which is the fiber of the map  $o_x : G \rightarrow X$ , and this map is smooth by (7.6.4.13), thus  $G \times X \rightarrow X \times X$  is smooth, and then the assertion follows from (7.6.4.13).  $\square$

**Cor. (7.6.4.32).** Let  $Z$  be a smooth  $k$ -variety where  $\text{char } k = 0$ . Let  $V$  be a f.d. basepoint-free line series on  $Z$ , then the section of a general element of  $V$  is smooth.  $\lrcorner$

*Proof:* Apply Kleinman-Bertini to  $\varphi_V : Z \rightarrow \mathbb{P}V^\vee$ ,  $Y = V(\sum a_i x_i) \in PV \times PV^\vee \rightarrow PV^\vee$ ,  $X = PV^\vee$ .  $\square$

## 5 Unramified

**Def. (7.6.5.1) [Unramified Morphisms].** A morphism is called **(G-)unramified** iff there is an open affine cover  $U_i$  and an open affine cover of  $f^{-1}(U_i)$  that the induced ring map is (G-)unramified (5.4.6.4). Equivalently,  $\Omega_{X/S} = 0$  and it is locally of f.t.(f.p.).

(G-)unramifiedness is local on the source and target (7.4.1.5)(7.4.1.6). (G-)unramifiedness is stable under base change and composition (5.4.6.4). Moreover, unramifiedness satisfies the base change trick.  $\lrcorner$

**Prop. (7.6.5.2).** An unramified morphism (7.6.5.1) is locally quasi-finite.  $\lrcorner$

*Proof:* Cf. [Sta]02V5.  $\square$

**Prop. (7.6.5.3) [Fiberwise].** A morphism is (G-)unramified iff it is locally of f.t.(f.p.) and all the fibers  $X_s$  are disjoint unions of spectra of finite separable extensions of  $k(p)$ .  $\lrcorner$

*Proof:* By (5.4.6.7), Notice  $pS_q = qS_q$  is equivalent to every  $q$  is minimal in  $X_p$ , which is equivalent to  $X_p$  is discrete.  $\square$

**Cor. (7.6.5.4) [Unramified over Fields].** A scheme over a field  $k$  is unramified iff it is a disjoint union of spectra of finite separable extensions of  $k$ , because locally of f.p. is trivially satisfied.  $\lrcorner$

**Prop. (7.6.5.5) [Diagonal is Open].** A morphism  $X \rightarrow S$  is (G-)unramified iff it is of f.t.(f.p.) and the diagonal is a clopen immersion of  $X$ , thus all of  $X$ .  $\lrcorner$

*Proof:* If it is unramified, then the diagonal is an open immersion by (5.4.6.10). Conversely,  $\Omega_{X/S}$  is just the conormal sheaf of the diagonal map, so it is zero.  $\square$

**Cor. (7.6.5.6) [Sections of Unramified Morphism].** Any section of an unramified morphism is an open immersion. In particular, a section of a separable unramified morphism is a clopen immersion.  $\lrcorner$

*Proof:* This follows from the proposition and the fact  $S \rightarrow X$  is a base change of  $\Delta_{X/S}$ .  $\square$

**Cor. (7.6.5.7).** For  $f, g : X \rightarrow Y \in \text{Sch}/S$ , if  $Y/S$  is unramified and  $f, g$  are equal on a pt  $x$  of  $X$  (both on image and residue field), then there is a nbhd of  $x$  that  $f, g$  are equal.  $\lrcorner$

*Proof:* This follows as  $\Delta_{Y/S}$  is open immersion, so the set that  $f, g$  are equal is open in  $X$ .  $\square$

**Prop. (7.6.5.8) [Fiberwise].** For a morphism  $f : X \rightarrow S$  locally of f.t.(f.p.),  $x \in X$  and  $s = f(x)$ , the following are equivalent:

- It is  $(G-)$ unramified at  $x$ ,
- The fiber  $X_s$  is unramified over  $k(s)$  at  $x$ , by (7.6.5.3).
- $\Omega_{X_s, x} = 0$ .
- $\Omega_{X_s/s, x} \otimes_{\mathcal{O}_{X_s, x}} k(x) = \Omega_{X/S, x} \otimes_{\mathcal{O}_{X, x}} k(x) = 0$  by (5.4.6.6).
- $\mathfrak{m}_s \mathcal{O}_{X, x} = \mathfrak{m}_x$  and  $k(x)/k(f(x))$  is separable.

┘

*Proof:*  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$  is clear.  $4 \rightarrow 1$ : Nakayama implies  $\Omega_{X/S, x} = 0$ , thus  $\Omega_{X/S}$  vanishes on a nbhd of  $x$  as it is Qco of f.t..

$4 \iff 5$  follows from (5.4.6.7). □

**Cor. (7.6.5.9) [Generic Unramifiedness].** Let  $f : X \rightarrow Y$  be a finite separable dominant morphism of integral schemes, then there exists an open dense nbhd  $U$  of  $Y$  s.t.  $f : f^{-1}(U) \rightarrow U$  is unramified.

┘

**Prop. (7.6.5.10).** If  $X \rightarrow Y \rightarrow S$  is unramified, then  $X/Y$  is also unramified. And if  $X/S$  is  $G$ -unramified and  $Y/S$  is of f.t., then  $X/Y$  is  $G$ -unramified. ┘

*Proof:* By (7.6.1.4) and (7.5.5.5). □

**Prop. (7.6.5.11) [Base Change of Unramified Points].** If  $f : X \rightarrow S \in \text{Sch}$  is of f.t.(f.p.), then the set of points of  $S$  that  $f$  is unramified is stable under base change. ┘

*Proof:* □

**Prop. (7.6.5.12) [Tangent Criterion of unramifiedness].** Let  $f : X \rightarrow Y$  be a morphism of schemes locally of f.t. over  $S$ ,  $x \in X, y = f(x)$ , then the following are equivalent:

- $df : T_{X/S, x} \rightarrow T_{Y/S, y}$  is an injection (7.6.4.22).
- $f$  is unramified at  $x$ .

┘

*Proof:* It follows from (7.6.4.22) that  $df$  is an isomorphism iff  $(f^* \Omega_{Y/S})_x \otimes_{\mathcal{O}_{X, x}} k(x) \rightarrow \Omega_{X/S, x} \otimes_{\mathcal{O}_{X, x}} k(x)$  is surjective, which by Jacobi-Zariski (7.5.5.5) is equivalent to  $\Omega_{X/Y, x} \otimes_{\mathcal{O}_{X, x}} k(x) = 0$ , and this is equivalent to  $f$  is unramified at  $x$  by (7.6.5.8). □

**Prop. (7.6.5.13) [Unramified Universally Injective Morphisms].** For a morphism  $f$  of schemes, the following are equivalent:

- $f$  is unramified and a monomorphism.
- $f$  is unramified and universal injective.
- $f$  is locally of f.t., formally unramified (7.8.6.12) and universal injective.
- $f$  is locally of f.t. and a monomorphism.
- $f$  is locally of f.t. and  $X_y$  is either empty or  $X_y \rightarrow y$  is an isomorphism for all  $y \in Y$ .

┘

*Proof:* Cf. [Sta], 05VH. □

**Prop. (7.6.5.14) [Unramified and Smoothness].** Let  $f : X \rightarrow Y \in \text{Sch}/S$  s.t.  $X/S$  is smooth (of relative dimension  $n$ ) and  $Y/S$  is unramified, then  $f$  is also smooth (of relative dimension  $n$ ).  $\lrcorner$

*Proof:* This follows from differential criterion (7.6.4.3) and (7.6.1.4). Notice to show it is flat, use the fact  $\Delta_{Y/S}$  and  $X \times Y \rightarrow Y$  are both flat (7.6.5.5).  $\square$

### Noetherian Case

**Prop. (7.6.5.15).** Let  $S$  be a Noetherian scheme,  $X \rightarrow S$  a qc unramified morphism and  $Y \rightarrow S$  a morphism with  $Y$  Noetherian, then  $\text{Mor}_S(Y, X)$  is a finite set.  $\lrcorner$

*Proof:* Cf. [Sta] 0AKI.  $\square$

**Prop. (7.6.5.16) [Unramified Morphisms and DVR].** Let  $R_v$  be a DVR with fraction field  $K$  and  $\varphi : X \rightarrow X'$  be a morphism of schemes of f.t. over  $R_v$ . Let  $Q \in X'(K)$  and  $P \in X(\bar{K})$  with  $\varphi(P) = Q$ . Let  $w|v$  be a valuation of  $K(P)$  extending  $v$ . If  $P$  extends to an  $R_w$ -valued point  $\bar{P}$  of  $X$ , then using the fact  $R_w \cap K = R_v$ , we see  $Q$  also extends to a  $R_v$ -valued point of  $X'$ .

Denote the image of the maximal ideal of  $R_w$  under  $\bar{P}$  by  $P(w)$ , then if  $\varphi$  is unramified at  $P(w)$ , then  $K(P)/K$  is unramified in  $w$ .  $\lrcorner$

*Proof:* Since the unramified point is open,  $\varphi$  is also unramified at  $P$ , thus  $K(P)/K$  is separable (7.6.5.8). For the rest, Cf. [Diophantine Geometry, P598] ?  $\square$

## 6 Étale

**Def. (7.6.6.1) [Étale Morphisms].**  $f : X \rightarrow Y \in \text{Sch}$  is called an **étale morphism** if there is an open affine cover  $\{U_i\}$  of  $S$  and an open affine cover  $V_{ij}$  of  $f^{-1}(U_i)$  for each  $i$  s.t.  $\Gamma(U_i) \rightarrow \Gamma(V_{ij})$  are étale ring maps (5.4.7.4).

A **standard étale morphism** is the Spec map of a standard étale ring map (5.4.7.17). Étale morphisms are locally standard étale on the source, by (5.4.7.18).

Étale is local on the source and target (5.4.7.5).

Étale is stable under base change and composition (5.4.7.5).  $\lrcorner$

**Prop. (7.6.6.2) [Properties of Étale Morphisms].**

- Étale at a point  $x$  is equivalent to smooth and unramified at a  $x$  (5.4.7.4).
- Étale at a point  $x$  is equivalent to flat and  $G$ -unramified at that point, by (5.4.7.11). So étale over field is equivalent to  $G$ -unramified, because over a field it is obviously flat.
- Étale at a point  $x$  is equivalent to locally standard étale at that point (5.4.7.18).
- A morphism is étale iff it is smooth of relative dimension 0, by definition (7.6.4.10).
- Étale is equivalent to flat, locally of f.p. and formally unramified, by (5.4.7.11).  $\lrcorner$

**Cor. (7.6.6.3).** Étale morphisms are smooth, hence syntomic, flat.

Étale morphisms are universally open because it is flat and locally of f.p. (7.6.2.10).  $\lrcorner$

**Prop. (7.6.6.4) [Cancellation Law for Étale Maps].** For  $X, Y \in \text{Sch}_{\text{ét}}/S$ , any map  $X \rightarrow Y \in \text{Sch}/S$  is étale, by (5.4.7.13) and (7.6.6.2).  $\lrcorner$



**Prop. (7.6.6.5) [Fiberwise].** A morphism of schemes is étale iff it is flat, locally of f.p., and every fiber  $X_s$  is a disjoint union of spectra of finite separable field extensions of  $k(s)$ .  $\lrcorner$

*Proof:* Follows from (7.6.6.2)(7.6.5.3) and (5.4.7.10).  $\square$

**Cor. (7.6.6.6) [Étale Schemes over Fields].** For  $k \in \mathbf{Field}$ ,

- $X \in \mathbf{Sch}/k$  is étale iff it is a disjoint union of spectra of finite separable field extensions of  $k$ .
- étale over fields is equivalent to unramified over fields.
- The functor

$$(\mathrm{Spec} k)_{\mathrm{\acute{e}t}} \rightarrow \mathrm{Set}_{\mathrm{Gal}_k} : X \mapsto X(k^{\mathrm{sep}})$$

is an equivalence of categories.  $\lrcorner$

*Proof:* This follows from ?? and (3.2.8.12).  $\square$

**Prop. (7.6.6.7) [Zariski-Local Structure of Smooth Morphisms].** If  $f : X \rightarrow Y \in \mathbf{Sch}$  is smooth at  $x \in X$ , then there exist an affine open nbhd  $U$  of  $x \in X$  and affine open nbhd  $V$  of  $f(x) \in Y$  s.t.  $f|_U$  factors through  $U \xrightarrow{f'} V \times \mathbb{A}^n \xrightarrow{\mathrm{pr}_1} V$ , where  $f'$  is standard étale (7.6.6.1).  $\lrcorner$

*Proof:* Any standard smooth morphism (7.6.4.1) can be factorized as an étale map over a polynomial algebra, as easily seen. And then we can change it to a standard étale map by (5.4.7.18).  $\square$

**Prop. (7.6.6.8) [Stalkwise and Fiberwise].** For  $f : X \rightarrow S \in \mathbf{Sch}^{\mathrm{loc.pf}}/S$ , by (7.6.5.8)(7.6.4.5) and (7.6.6.6), the following are equivalent:

- It is étale at a point  $x \in X$ .
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $X_{f(x)}/\kappa(f(x))$  is étale at  $x$ .
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $X_{f(x)}/\kappa(f(x))$  is unramified at  $x$ .
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $\Omega_{X/S,x} = 0$ .
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat and  $\Omega_{X_{f(x)}/f(x),x} \otimes_{\mathcal{O}_{X_{f(x)},x}} \kappa(x) = \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x) = 0$ .
- $\mathcal{O}_{X,x}/\mathcal{O}_{S,f(x)}$  is flat,  $\mathfrak{m}_{f(x)}\mathcal{O}_{X,x} = \mathfrak{m}_x$  and  $\kappa(x)/\kappa(f(x))$  is separable.

In particular, an étale morphism can be seen as a covering space.  $\lrcorner$

**Prop. (7.6.6.9).** For  $X \rightarrow Y \rightarrow S \in \mathbf{Sch}$ , if  $X \rightarrow Y$  are faithfully flat and locally of f.p. and  $X \rightarrow S$  is étale, then  $Y \rightarrow S$  is also étale.  $\lrcorner$

*Proof:* Cf. [Sta]05B5.  $\square$

**Cor. (7.6.6.10).** If  $f : X \rightarrow S$  is faithfully flat and locally of f.p., then the set of points of  $S$  s.t.  $f$  is étale is table under base change.  $\lrcorner$

*Proof:* This follows from (7.6.5.11) and (7.6.4.9).  $\square$

**Def. (7.6.6.11) [Étale Neighborhoods].** For  $x \in X$ , an **étale neighborhood** of  $x \in X$  is defined to be an étale morphism  $f : U \rightarrow X \in \mathbf{Sch}$  together with  $u \in U$  s.t.  $f(u) = x$ .  $\lrcorner$

**Prop. (7.6.6.12).** If  $k \in \mathbf{Field}$ ,  $X, Y \in \mathbf{Sch}/k$  are étale and  $f : Y \rightarrow X \in \mathbf{Sch}$ , then  $f$  is surjective iff  $Y(k^{\mathrm{sep}}) \rightarrow X(k^{\mathrm{sep}})$  is surjective.  $\lrcorner$



*Proof:* If  $Y \rightarrow X$  is surjective, then ? □

**Prop. (7.6.6.13) [Fundamental Property of Étale Morphisms, Grothendieck[Gro71]].** For  $f : X \rightarrow Y \in \text{Sch}$ , the following are equivalent:

- $f$  is an open immersion.
- $f$  is universally injective and étale.
- $f$  is a flat monomorphism(7.4.4.3) locally of f.p..

┘

*Proof:* Cf.[Sta]025G. □

### Finite Étale Morphisms

**Prop. (7.6.6.14) [Étale-Local Structure of Finite étale Morphisms].** If  $f : X \rightarrow S \in \text{Sch}$  is finite étale and  $s \in S$ , then there exists an étale nbhd  $(U, u)$  of  $(S, s)$ (7.6.6.11) s.t.

$$X_U \cong \coprod_{j \in [n]_+} U$$

for some  $n \in \mathbb{Z}_+$ . ┘

*Proof:* Cf.[Sta]4HN. ? □

**Prop. (7.6.6.15) [Base Change of Finite Étale Morphisms].** For  $X \in \text{Sch}$ , the category  $X_{\text{fét}}$  has finite limits and colimits, and for any  $X' \rightarrow X \in \text{Sch}/X$ , the base change functor  $X_{\text{fét}} \rightarrow X'_{\text{fét}}$  is exact. ┘

*Proof:* Cf.[Sta]0BN9. □

## 7 Regular Immersions

**Def. (7.6.7.1) [Regular Immersions].** A **regular immersion** of codimension  $r$  is a locally closed embedding  $X \rightarrow Y \in \text{Sch}$  s.t. for any  $p \in X$ , the ideal of  $X$  in the local ring  $\mathcal{O}_{Y,p}$  is generated by a regular sequence of length  $r$ . ┘

## 8 Complete Intersection

Should be refreshed with intrinsic definition of locally complete intersection, Cf.[Sta].

**Def. (7.6.8.1) [Complete Intersection].** A **complete intersection** of codimension  $r$  in  $Y$  is a closed subscheme  $X$  that is the intersection of  $r$  effective Cartier divisors  $D_i$  that at each point  $p \in X$ , the equations defining  $D_i$  form a regular sequence. ┘

**Def. (7.6.8.2) [Locally Complete Intersection].** A closed subscheme  $Y$  of a nonsingular variety  $X$  over a field  $k$  is called **locally complete intersection** iff  $Y$  is locally generated by  $r = \text{codim}(Y, X)$  elements. By(5.3.4.17)  $Y$  is C.M.. In particular, by(7.12.1.16), a regular variety is always a locally complete intersection. ┘

**Def. (7.6.8.3) [Syntomic Morphisms].** A **standard syntomic morphism** is the Spec map of a global complete intersection ring map(5.4.4.11). A **syntomic morphism** is a morphism that is locally a standard syntomic morphism. ┘

**Prop. (7.6.8.4).** Syntomic is local on the source and target, stable under base change and composition, by (5.4.4.12).  $\lrcorner$

**Prop. (7.6.8.5).** Syntomic is equivalent to flat, locally of f.p.+fibers being local complete intersections.  $\lrcorner$

*Proof:* This follows from (5.4.4.18).  $\square$

**Cor. (7.6.8.6).** Syntomic morphisms are universally open.  $\lrcorner$

**Prop. (7.6.8.7).** An open immersion is syntomic, because localizations are global complete intersections (5.4.4.12).  $\lrcorner$

**Prop. (7.6.8.8).** If  $f : X \rightarrow S$  is a syntomic map, then the function  $x \mapsto \dim_x(X_{f(x)})$  is locally constant on  $X$ . If it is standard syntomic, then it is constant.  $\lrcorner$

*Proof:* It suffices to prove for syntomic maps, Cf. [Sta]02K0.  $\square$

**Prop. (7.6.8.9).** A local complete intersection has its ideal sheaf  $\mathcal{I}$ , then  $\mathcal{I}/\mathcal{I}^2$  locally free by (5.3.4.16).  $\lrcorner$

**Prop. (7.6.8.10).** If  $Y$  is a complete intersection in  $\mathbb{P}_k^n$  of hypersurfaces of degree  $d_1, \dots, d_r$ , then  $\omega_Y = \mathcal{O}_Y(\sum d_i - n - 1)$ .  $\lrcorner$

*Proof:* Use the exact sequence  $0 \rightarrow \mathcal{O}_Z(n-d) \rightarrow \mathcal{O}_Z \rightarrow i_*\mathcal{O}_Y \rightarrow 0$  and (7.12.1.17).  $\square$

**Prop. (7.6.8.11).** For a complete intersection of dimension  $q$ ,  $H^i(Y, \mathcal{O}_Y(n)) = 0$  for  $0 < i < q$ . And the natural map  $\Gamma(P, \mathcal{O}_P(n)) \rightarrow H^0(Y, \mathcal{O}_Y(n))$  is a surjection for every  $n$ . In particular,  $Y$  is connected, and the arithmetic genus  $p_a(Y) = \dim H^q(Y, \mathcal{O}_Y)$ .  $\lrcorner$

*Proof:* We use induction, the case  $Y = P$  follows from (7.7.2.1), let  $Y = Z \cap H$ , where  $H$  has degree  $d$ , then

$$0 \rightarrow \mathcal{O}_Z(n-d) \rightarrow \mathcal{O}_Z \rightarrow i_*\mathcal{O}_Y \rightarrow 0$$

thus use long exact sequence. The rest is easy.  $\square$

## 7.7 Cohomology of Schemes

Main references are [Sta]Chap29.

**Notation (7.7.0.1).**

- Use notations defined in [Cohomology on Ringed Sites](#).

┘

### 1 Zariski Cohomology

#### Qco Sheaves

**Lemma (7.7.1.1) [Zariski-Poincare].** A Qco sheaf on an affine scheme  $X$  is Čech-acyclic. ┘

*Proof:* Because the principal affine covers are cofinal in the ordering of covers, we only need to consider principal affine covers. Let  $R \rightarrow A = \prod R_{f_i}$ , then it is f.f., so we can use (5.4.2.2), just notice the higher term is  $\prod_{i_0, \dots, i_n} R_{f_{i_0} \dots f_{i_n}}$ .  $\square$

**Prop. (7.7.1.2) [Čech Cohomology on Separated Schemes].** If  $X$  is separated and  $\mathcal{F} \in \mathcal{QCoh}(X)$ ,  $H^p(X, \mathcal{F}) = \check{H}^p(X, \mathcal{F}) = H_{\text{alt}}^p(\{U_i \rightarrow X\}, \mathcal{F})$  for any open affine covering  $\{U_i \rightarrow X\}$ . ┘

*Proof:* Use (7.3.2.16) and (7.3.2.6), the family of affine open subsets of  $X$  satisfies the requirement because  $X$  is separated and (7.7.1.1), thus the result It can be calculated by alternating complexes by (7.3.2.7).  $\square$

**Cor. (7.7.1.3) [Affine Cohomological Vanishing].** If  $X$  is affine and  $\mathcal{F} \in \mathcal{QCoh}(X)$ ,  $H^i(X, \mathcal{F}) = 0$  for  $i > 0$ .

For a qcqs scheme  $X$ , choose a finite affine cover  $U_1, \dots, U_t$  of  $X$ , then for any  $\mathcal{F} \in \mathcal{QCoh}(X)$ ,  $H^n(X, \mathcal{F}) = 0$  for  $n \geq d$ , where

$$d = \max_{I \subset \{1, \dots, t\}} (\#I + t(U_I))$$

and  $t(Y)$  is the minimal cardinality of an affine open cover of  $Y$ . ┘

*Proof:* The last assertion follows from Čech to sheaf (7.3.2.13) and (7.7.1.2), as  $U_I$  is separated for any  $I \neq \emptyset$ .  $\square$

**Remark (7.7.1.4).** Compare with (8.2.0.4) ?. ┘

**Prop. (7.7.1.5) [ $\mathcal{QCoh}(X)$  vs.  $\text{Mod}(\mathcal{O}_X)$ ].** If  $X$  is a Noetherian scheme, then injective objects in  $\mathcal{QCoh}(X)$  are all flabby (7.3.4.8), thus nearly calculating all cohomologies are legitimate in the category  $\mathcal{QCoh}(X)$ .

However, in general, this is not true, as there are examples of injective  $A$ -module  $\mathcal{I}$  s.t.  $\tilde{\mathcal{I}}$  is not flabby on  $\text{Spec } A$ , by [Sta]0274. ┘

*Proof:* We use the Deligne formula (7.5.1.36) and the definition of injective, by considering the sheaf of ideals of the corresponding induced reduced structure.  $\square$

**Prop. (7.7.1.6) [Filtered Colimits].** By (7.3.1.16), if  $X$  is qcqs, then sheaf cohomology on  $X$  commutes with filtered colimits. ┘

**Prop. (7.7.1.7) [Ext For Coherent Sheaves].** On a locally Noetherian scheme, for  $\mathcal{F} \in \mathcal{QCoh}(X)$  and  $\mathcal{G} \in \mathcal{Coh}(X)$ ,  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \in \mathcal{QCoh}(X)$ , and affine locally are given by  $\text{Ext}^i(\mathcal{F}(U), \mathcal{G}(U))$ .  
Moreover  $\mathcal{G} \in \mathcal{Coh}(X)$ , then  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \in \mathcal{Coh}(X)$ . ┘

*Proof:* This follows from (7.9.1.13) and (7.9.1.10). □

**Prop. (7.7.1.8).** When  $X$  is locally Noetherian and  $\mathcal{F} \in \mathcal{Coh}(X)$ ,

$$\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})_x \cong \text{Ext}_{\mathcal{O}_x}^i(\mathcal{F}_x, \mathcal{G}_x).$$

┘

*Proof:* Taking stalk is exact. ? □

**Cor. (7.7.1.9).** If  $X$  is locally Noetherian, suppose that every coherent sheaf is a quotient of a locally free sheaf, we can define the **homological dimension**  $hd(\mathcal{F})$  of a coherent sheaf  $\mathcal{F}$  as the minimal length of a flat resolution of  $\mathcal{F}$ . Then  $hd(\mathcal{F}) \leq n \iff \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$  for every  $\mathcal{G}$  and every  $i > n$ . And  $hd(\mathcal{F}) = \sup pd_{\mathcal{O}_{X,x}} \mathcal{F}_x$ , by (7.7.1.8). ┘

**Prop. (7.7.1.10) [Künneth Formula].** If  $X, Y$  are qcqs over a field  $k$  and  $\mathcal{F}, \mathcal{G}$  be Qco  $\mathcal{O}_X, \mathcal{O}_Y$ -modules, then there is a canonical isomorphism:

$$H^n(X \times_{\text{Spec } k} Y, \text{pr}_1^* \mathcal{F} \otimes_{\mathcal{O}_{X \times_{\text{Spec } k} Y}} \text{pr}_2^* \mathcal{G}) \cong \bigoplus_{p+q=n} H^p(X, \mathcal{F}) \otimes_k H^q(Y, \mathcal{G}).$$

┘

*Proof:* Cf. [Sta]0BEF. ? □

**Prop. (7.7.1.11).** On a locally Noetherian scheme  $X$ , any Qco sheaf  $\mathcal{F}$  admits a resolution of Qco sheaves that are flabby. ┘

*Proof:* Because  $\mathcal{QCoh}(X)$  is Serre subcategory, ? □

**Lemma (7.7.1.12) [Gabber].** Let  $X$  be a scheme, then there exists a cardinal  $\kappa$  that every Qco sheaf is a colimit of its  $\kappa$ -generated Qco subsheaves. ┘

*Proof:* Cf. [Sta]077N. □

**Prop. (7.7.1.13) [Qco Cohomology Comparison].** For  $X \in \text{Sch}$ ,  $\mathcal{F} \in \mathcal{QCoh}(X)$ ,  $\tau \in \{\text{fppf}, \text{étale}, \text{smooth}, \text{syntomic}, \text{Zariski}\}$ , there are canonical isomorphisms

$$H^p(X, \mathcal{F}) \cong H^p(\text{Sch}_\tau/X, \mathcal{F}) \cong H^p(X_{\text{Zar}}, \mathcal{F}) \cong H^p(X_{\text{ét}}, \mathcal{F}).$$

┘

*Proof:* Let  $\mathcal{C} = \text{Sch}_\tau/X$  or  $X_{\text{ét}}, X_{\text{Zar}}$ . We use (7.3.2.16) with  $\mathfrak{G}$  the set of affine schemes and  $\text{Cov}$  the subset of  $\text{Cov}(\mathcal{C})$  consisting of morphisms between affine schemes. Then Čech vanishing is clear in the affine case by (5.4.2.2). Thus (7.3.2.16) says  $H^p(\mathcal{C}, \mathcal{F}_{\mathcal{C}}) = 0$  for  $p > 0$ .

Next, if  $U \subset X$  is an affine open s.t.  $U \rightarrow X$  is separated. Let  $\mathcal{U}$  be an affine open covering of  $U$ , then all  $U_{i_0, \dots, i_p}$  are affine, thus

$$H^p(\mathcal{C}, \mathcal{F}_{\mathcal{C}}) = \check{H}(\mathcal{U}, \mathcal{F}_{\mathcal{C}}) = H^p(U, \mathcal{F}|_U)$$

by (7.3.2.15).

Finally, for  $X$ , take an injective resolution  $\mathcal{F} \rightarrow \mathcal{I}^\bullet$ , and an injective resolution  $\mathcal{F}_\mathcal{C} \rightarrow \mathcal{J}^\bullet$ . Then the latter restricts to a chain complex  $\mathcal{F} \rightarrow \mathcal{J}^\bullet|_X$ , which is exact because it is exact on any affine open  $U \subset X$  as  $H^p(\mathcal{C}, \mathcal{F}_\mathcal{C}) = 0$  for  $p > 0$ . Thus by (4.10.2.5), there is a map  $\mathcal{J}^\bullet|_X \rightarrow \mathcal{I}^\bullet$ , inducing a map  $H^p(\mathcal{C}, \mathcal{F}_\mathcal{C}) = H^p(X, \mathcal{J}^\bullet) \rightarrow H^p(X, \mathcal{I}^\bullet) \cong H^p(X, \mathcal{F})$ .

To show these maps are isomorphisms, take an affine open covering  $\mathcal{U}$  of  $X$ , then there are Čech-to-derived spectral sequences

$${}^\tau E_2^{p,q} = \check{H}^p(\mathcal{U}, \mathcal{H}^q(\mathcal{F}^a)) \Rightarrow H^{p+q}(\mathcal{C}, \mathcal{F}_\mathcal{C}), \quad E_2^{p,q} = \check{H}^p(\mathcal{U}, \mathcal{H}^q(\mathcal{F}^a)) \Rightarrow H^{p+q}(\mathcal{C}, \mathcal{F}).$$

The map  $\mathcal{J}^\bullet|_X \rightarrow \mathcal{I}^\bullet$  induces a map  $\check{C}^\bullet(\mathcal{U}, \mathcal{J}) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{I})$  which induces a map of the spectral sequences. But as each affine open of  $X$  is separated over  $X$ , the  $E^2$ -page is an isomorphism by what we have proved. Thus  $\mathcal{H}^q(\mathcal{F}^a) \rightarrow H^{p+q}(\mathcal{C}, \mathcal{F})$  is an isomorphism.  $\square$

**Prop. (7.7.1.14).** For  $X \in \text{Sch}$ ,  $\tau \in \{\text{fppf, étale, smooth, syntomic, Zariski}\}$ ,

$$H_\tau^1(X, \mathbb{G}_m) \cong \text{Pic}(X) \quad (7.5.3.10).$$

⌋

*Proof:* As these  $X_\tau$  are all locally ringed sites, by (7.3.1.17),  $H_\tau^1(X, \mathbb{G}_m) \cong \text{Pic}(X_\tau)$ . Thus it suffices to prove that

$$\text{Pic}(X_{\text{Zar}}) \rightarrow \text{Pic}(X_{\text{ét}}) \rightarrow \dots \rightarrow \text{Pic}(X_{\text{fppf}})$$

are isomorphisms. Then this is because any invertible sheaf is quasi-coherent by (7.2.5.8), so  $\text{Pic}(X_\tau)$  are in fact the group of invertible objects in  $\mathcal{QCoh}(X_\tau)$ . But  $\mathcal{QCoh}(X_\tau)$  are all equivalent categories, by fpqc descent for Qco sheaves (7.1.5.12).  $\square$

## 2 Proper Schemes

**Prop. (7.7.2.1)[Cohomology of Projective Space].** Let  $X = \mathbb{P}_A^r$  we have:

(1)

$$H^q(X, \mathcal{O}_X(d)) = \begin{cases} (A[T_0, \dots, T_r])_d & q = 0 \\ 0 & q \neq 0, r \\ (\frac{1}{T_0 \dots T_r} A[\frac{1}{T_0}, \dots, \frac{1}{T_r}])_d & q = r \end{cases}$$

(2) The cup product defines a perfect pairing

$$H^0(X, \mathcal{O}_X(n)) \times H^r(X, \mathcal{O}_X(-n-r-1)) \rightarrow H^r(X, \mathcal{O}_X(-r-1)) \cong A.$$

⌋

*Proof:*  $X$  is separated, we use Čech cohomology. Let  $\mathcal{F} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_X(d)$ , then  $\mathcal{F}$  is a Qco graded ring. Let  $U_{i_0, \dots, i_k}$  be the affine open subset  $U(x_{i_0}) \cap \dots \cap U(x_{i_k})$  of  $X$ , then  $\mathcal{F}(U_{i_0, \dots, i_k}) = A[T_0, \dots, T_r, T_{i_0}^{-1}, \dots, T_{i_k}^{-1}]$  as graded rings, and the cohomology  $H^\bullet(\mathcal{F})$  is calculated by the Čech complex (7.3.2.1)

$$\check{C}^\bullet = \prod_i A[T_0, \dots, T_r, T_i^{-1}] \rightarrow \prod_{i,j} A[T_0, \dots, T_r, T_i^{-1}, T_j^{-1}] \rightarrow \dots \rightarrow A[T_0, \dots, T_r, T_1^{-1}, \dots, T_r^{-1}].$$

This complex has a natural  $\mathbb{Z}^r$ -grading, and the differential is natural inclusion thus preserves the grading, i.e.

$$\check{C}^\bullet = \bigoplus_{v \in \mathbb{Z}^r} \check{C}^\bullet(v)$$

where if we set  $NEG(v) = \{i \in \{0, \dots, r\} | v_i < 0\}$ , then

$$\check{C}^\bullet(v) = \prod_{NEG(v) \subset \{i\}} AT_0^{v_0} \dots T_r^{v_r} \rightarrow \prod_{NEG(v) \subset \{i, j\}} AT_0^{v_0} \dots T_r^{v_r} \rightarrow \dots \rightarrow AT_0^{v_0} \dots T_r^{v_r}$$

is the subcomplex of  $\check{C}^\bullet$ . So it suffices to calculate the cohomology of  $\check{C}^\bullet(v)$  for each  $v$ .

If  $NEG(v) = \{0, 1, \dots, n\}$ , then there is only one term, so

$$H^\bullet(\check{C}^\bullet(v)) = \begin{cases} AT_0^{v_0} \dots T_r^{v_r} & q = r \\ 0 & \text{otherwise} \end{cases}.$$

The sum of all such  $v$  clearly contribute to  $\frac{1}{T_0 \dots T_r} A[\frac{1}{T_0}, \dots, \frac{1}{T_r}]$  in degree  $r$ .

If  $NEG(v) = \emptyset$ , then the complex is isomorphic to the Čech complex calculating cohomology of  $\text{Spec } A$  using the trivial cover  $\{V_i \rightarrow \text{Spec } A\}$ ,  $V_i \cong \text{Spec } A$  times  $T_0^{v_0} \dots T_r^{v_r}$ . As  $\text{Spec}$  is separated, it is just to the sheaf cohomology of  $\text{Spec } A$ , which is  $A$  in degree 0 and 0 otherwise by (7.7.1.1). The sum of all such  $v$  clearly contribute to  $A[T_0, \dots, T_r]$  in degree 0.

Finally, for other  $v$ , choose a  $j \in \{0, 1, \dots, n\} \setminus NEG(v)$ , then the maps

$$h : \check{C}^{p+1}(v) \rightarrow \check{C}^p(v) : h(s)_{i_0 \dots i_p} = s_{ji_0 i_p}$$

(where we are using the alternating Čech complex) induces a homotopy between 0 and id, so  $\check{C}^{p+1}(v)$  have trivial cohomologies.

The pairing given by cup product makes  $T_0^{v_0} \dots T_r^{v_r}$  dual to  $T_0^{-1-v_0} \dots T_r^{-1-v_r}$ , thus it is a perfect pairing.  $\square$

**Cor. (7.7.2.2).** when  $n > 0$ ,  $H^r(X, \mathcal{O}_X(n-r-1)) = 0$ . Notice this an instance of the Kodaira vanishing theorem when  $k$  has characteristic 0 (16.10.7.3).  $\lrcorner$

**Prop. (7.7.2.3).** Let  $X = \mathbb{P}_k^n$  and  $0 \leq p, q \leq n$ , then  $H^q(X, \Omega_X^p) = 0$  for  $p \neq q$  and when  $p = q$ ,  $H^q(X, \Omega_X^p) = k$ .  $\lrcorner$

*Proof:* By (7.5.1.26) and (7.5.5.7), there is an exact sequence  $0 \rightarrow \Omega^q \rightarrow \wedge^q \mathcal{O}(-1) \rightarrow \Omega^{q-1} \rightarrow 0$ , and the middle has vanishing  $q$ -th cohomology by (7.7.2.1), thus we can induct and (7.7.2.1) gives the result.  $\square$

**Prop. (7.7.2.4) [Cohomology of Complete Intersections].** Let  $X$  be a closed subscheme of  $\mathbb{P}_k^n$  defined by a single homogenous equation  $f(x_0, x_1, x_2)$  of degree  $d$ , then  $\dim H^0(X, \mathcal{O}_X) = 1$ , and if  $n \geq 2$ , we have  $\dim H^1(X, \mathcal{O}_X) = (d-1)(d-2)/2$ .  $\lrcorner$

*Proof:* By (7.5.3.15), there is an exact sequence of sheaves on  $\mathbb{P}_k^n$ :

$$0 \rightarrow \mathcal{O}_{\mathbb{P}_k^n}(-d) \cong \mathcal{O}_{\mathbb{P}_k^n}(-X) \rightarrow \mathcal{O}_{\mathbb{P}_k^n} \rightarrow i_* \mathcal{O}_X \rightarrow 0$$

which induces a long exact sequence (7.7.4.2):

$$H^0(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n}(-d)) \rightarrow H^0(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n}) \rightarrow H^0(X, \mathcal{O}_X) \rightarrow H^1(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n}(-d))$$

$$\rightarrow H^1(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n}) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow H^2(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n}(-d)) \rightarrow H^2(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n})$$

By (7.7.2.1), this reads:

$$0 \rightarrow k \rightarrow H^0(X, \mathcal{O}_X) \rightarrow 0 \rightarrow 0 \rightarrow H^1(X, \mathcal{O}_X) \rightarrow H^2(\mathbb{P}_k^n, \mathcal{O}_{\mathbb{P}_k^n}(-d)) \rightarrow 0,$$

and if  $n = 2$ ,  $\dim H^2(\mathbb{P}_k^2, \mathcal{O}_{\mathbb{P}_k^2}(-d)) = (d-1)(d-2)/2$ .

? It is faster using Hilbert Polynomial. □

**Lemma (7.7.2.5) [Cohomology of Projective Space].** Let  $R$  be a Noetherian ring and  $n \geq 0$ , then for any coherent sheaf  $\mathcal{F}$  on  $\mathbb{P}_R^n$ ,

- $H^i(\mathbb{P}_R^n, \mathcal{F}(n)) = 0$  for  $i > 0$  and  $n$  large enough.
- For any  $i$ ,  $H^i(\mathbb{P}_R^n, \mathcal{F})$  is a finite  $R$ -module.
- For any  $k \in \mathbb{Z}$ ,  $\bigoplus_{d \geq k} H^0(\mathbb{P}_R^n, \mathcal{F}(d))$  is a finite  $R[T_0, \dots, T_n]$ -module.

┘

*Proof:* 1: The assertions are true for  $\mathcal{O}_X(n)$  by (7.7.2.1), and for general  $\mathcal{F}$ , we use descending induction on  $i$ . This is true for  $i > n$  by Čech cohomology (7.7.1.2). For general  $i > 0$ , choose a surjection  $\bigoplus \mathcal{O}_X(n_i) \rightarrow \mathcal{F}$  with coherent kernel  $\mathcal{R}$  (7.5.4.5), then there is an exact sequence

$$H^i(X, \bigoplus \mathcal{O}_X(n_i + n)) \rightarrow H^i(X, \mathcal{F}(n)) \rightarrow H^{i+1}(X, \mathcal{R}(n)),$$

and the left term vanish for  $n$  large (7.7.2.1), and the right term vanish by induction hypothesis.

2: Notation as above, we use descending induction on  $i$ . This is true for  $i > n$  by Čech cohomology (7.7.1.2). For general  $i > 0$ ,  $H^i(X, \bigoplus \mathcal{O}_X(n_i))$  is finite by (7.7.2.1), and  $H^{i+1}(X, \mathcal{R})$  is finite by induction hypothesis, thus  $H^i(X, \mathcal{F}(n))$  is also finite.

3: Notation as above, for  $n$  large,  $H^i(X, \bigoplus \mathcal{O}_X(n_i + n)) \rightarrow H^i(X, \mathcal{F}(n))$  is surjective. Thus  $M_k = \bigoplus_{d \geq k} H^0(\mathbb{P}_R^n, \mathcal{F}(d))$  is a quotient of  $N_k = \bigoplus_{d \geq k} H^0(\mathbb{P}_R^n, \mathcal{O}_X(d))$  for  $k$  large. Notice for  $k$  small enough,  $N_k \cong \bigoplus_i R[T_0, \dots, T_n][i]$  is a finite graded  $R[T_0, \dots, T_n]$ -module, thus  $N_k$  is finite for any  $k$  as  $R[T_0, \dots, T_n]$  is Noetherian. Then  $M_k$  is finite for  $k$  large, and each  $H^0(\mathbb{P}_R^n, \mathcal{F}(d))$  is itself finite by item 2, thus  $M_k$  is a finite  $R[T_0, \dots, T_n]$ -module for any  $k$ . □

**Prop. (7.7.2.6) [Cohomology of Proper Schemes].** Let  $X$  be a proper scheme over a Noetherian ring  $A$ ,  $\mathcal{L}$  an ample invertible sheaf on  $X$  and  $\mathcal{F} \in \text{Coh}(X)$ , then

- $H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes d}) = 0$  for  $i > 0$  and  $d$  large enough.
- The graded ring  $A = \bigoplus_{n \geq 0} H^0(X, \mathcal{L}^{\otimes n})$  is a f.g.  $R$ -algebra.
- For any  $k \in \mathbb{Z}$ ,  $\bigoplus_{d \geq k} H^0(X, \mathcal{F} \otimes \mathcal{L}^{\otimes d})$  is a finite  $A$ -module.

┘

*Proof:* By (7.5.4.26), there exists a  $d > 0$  and some immersion  $i : X \rightarrow \mathbb{P}_A^n$  that  $i^* \mathcal{O}_{\mathbb{P}_A^n}(1) \cong \mathcal{L}^{\otimes d}$ , and  $i$  is a closed immersion because  $X$  is proper. Let  $S = R[T_0, \dots, T_n]$ .

1: By projection formula (7.7.4.6),

$$i_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes nd+q}) = i_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes q})(n).$$

Then by (7.7.4.2) and (7.7.2.5), for  $n$  large enough,  $H^p(X, \mathcal{F} \otimes \mathcal{L}^{\otimes q}) = 0$  for any  $0 \leq q \leq d-1$ ,  $p > 0$ .

2: By proof of item 1 and (7.7.2.5), we see  $\bigoplus_{n \geq 0} A_{nd+q}$  is a finite graded  $S$ -module for any  $q$ , thus  $A = \bigoplus_{q=0}^{d-1} \bigoplus_{n \geq 0} A_{nd+q}$  is a f.g.  $R$ -module.

3: Similarly, we see  $\bigoplus_{d \geq k} H^0(X, \mathcal{F} \otimes \mathcal{L}^{\otimes d})$  is a finite graded  $S$ -module, and the  $S$ -module structure factors through  $S \rightarrow A$ , thus it is a finite  $A$ -module. □

**Lemma (7.7.2.7).** For an invertible sheaf  $\mathcal{L}$  on a qc scheme  $X$ , if for each Qco sheaf of ideals  $\mathcal{I} \in \mathcal{O}_X$ , there exists an  $n$  that  $H^1(X, \mathcal{I} \otimes \mathcal{L}^n) = 0$ , then  $\mathcal{L}$  is ample.  $\lrcorner$

*Proof:* For any closed pt  $P$ , choose an open affine nbhd  $U$  that  $\mathcal{L}$  is trivial, let  $Y = X - U$ , by the exact sequence  $0 \rightarrow \mathcal{I}_{Y \cup \{P\}} \rightarrow \mathcal{I}_Y \rightarrow k(P) \rightarrow 0$ , for each  $n$ ,

$$0 \rightarrow \mathcal{I}_{Y \cup \{P\}} \otimes \mathcal{L}^n \rightarrow \mathcal{I}_Y \otimes \mathcal{L}^n \rightarrow k(P) \otimes \mathcal{L}^n \rightarrow 0.$$

Thus by assumption for some  $n$  the map  $\Gamma(X, \mathcal{I}_Y \otimes \mathcal{L}^n) \rightarrow \Gamma(X, k(P) \otimes \mathcal{L}^n)$  is surjective. Now  $k(P) \otimes \mathcal{L}^n \cong A/\mathfrak{m}_P$ , so we let  $s \in \Gamma(X, \mathcal{I}_Y \otimes \mathcal{L}^n)$  maps to a section in  $\Gamma(X, k(P) \otimes \mathcal{L}^n)$  that corresponds to  $1 \in A/\mathfrak{m}_P$ , then  $P \in \text{Supp}(s) \subset U$ , and  $\text{Supp}(s)$  is affine. So we find an affine nbhd  $X_s$  for every closed pt of  $X$ .

Finally these  $X_s$  cover  $X$ , because the complement of  $\cup X_s$  is closed in  $X$  thus qc, then it contains a closed point by (4.4.6.1).  $\square$

**Prop. (7.7.2.8)[Serre's Cohomological Criterion of Ampleness].** If  $X$  is proper over a Noetherian affine scheme,  $\mathcal{L}$  is an invertible sheaf, then the following is equivalent.

- $\mathcal{L}$  is ample
  - For any  $\mathcal{F} \in \text{Coh}(X)$ , for  $n$  large enough,  $H^p(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$  for  $p > 0$ .
  - For any Qco sheaf of ideals  $\mathcal{I} \in \mathcal{O}_X$ , there exists an  $n \geq 1$  that  $H^1(X, \mathcal{I} \otimes \mathcal{L}^{\otimes n}) = 0$
- (Notice in this case  $H$ -ample  $\iff$  ample).  $\lrcorner$

*Proof:*  $1 \rightarrow 2$ : This follows from (7.7.2.6).  $2 \rightarrow 3$  is trivial.  $3 \rightarrow 1$ : (7.7.2.7).  $\square$

### Lichtenbaum's theorem

**Lemma (7.7.2.9).** Let  $X$  is a variety and  $\mathcal{F} \in \text{Coh}(\mathcal{O}_X)$ . If  $H^d(X, \mathcal{F}) \neq 0$ , then  $\dim X \geq d$ , and if equality holds,  $X$  is proper.  $\lrcorner$

*Proof:* Cf. [Sta]0G5E.  $\square$

**Prop. (7.7.2.10) [Lichtenbaum].** Let  $X$  be a non-empty separated scheme of f.t. over a field  $k$  of dimension  $d$ , then the following are equivalent:

- $H^d(X, \mathcal{F}) = 0$  for all  $\mathcal{F} \in \text{Coh}(X)$ .
- $H^d(X, \mathcal{F}) = 0$  for all  $\mathcal{F} \in \text{QCoh}(X)$ .
- No irreducible component of  $X$  of dimension  $d$  is proper over  $k$ .

$\lrcorner$

*Proof:* Cf. [Sta]0G5F.  $\square$

## 3 Euler Characteristics

**Def. (7.7.3.1)[Euler Characteristic].** Let  $X$  be proper over a field  $k$  and  $\mathcal{F} \in \text{Coh}(X)$ , then the **Euler characteristic** of  $\mathcal{F}$  is defined to be:

$$\chi(\mathcal{F}) = \sum (-1)^i \dim_k H^i(X, \mathcal{F}).$$

It is definable by Grothendieck vanishing (7.3.5.17) and Grothendieck coherence theorem (7.7.4.12), and It is clearly an additive functor on  $\text{Coh}(X)$ .  $\lrcorner$



**Prop. (7.7.3.2).** For a proper scheme  $X$  over a field  $k$  and  $\mathcal{L}_i$  be invertible sheaves on  $X$ . Then for any  $\mathcal{F} \in \text{Coh}(X)$ ,

$$\chi(X, \mathcal{F} \otimes \mathcal{L}_1^{n_1} \otimes \cdots \otimes \mathcal{L}_r^{n_r})$$

is a polynomial in  $(n_1, \dots, n_r)$  of total degree at most  $\dim \text{Supp } \mathcal{F}$ .  $\lrcorner$

*Proof:* Cf. [Sta]0BEM.  $\square$

**Prop. (7.7.3.3).** Let  $f : Y \rightarrow X$  be morphism between schemes proper over field  $k$  and  $\mathcal{F} \in \text{Coh}(X)$ ,

$$\chi(Y, \mathcal{F}) = \sum (-1)^i \chi(X, R^i f_* \mathcal{F}).$$

In particular, if  $f$  is affine(finite), then  $\chi(Y, \mathcal{F}) = \chi(X, f_* \mathcal{F})$ .  $\lrcorner$

**Prop. (7.7.3.4).** This formula makes sense by (7.7.4.4) and (7.7.4.12), and it is true by Leray spectral sequence (7.3.1.9).  $\lrcorner$

**Prop. (7.7.3.5).** If  $X$  is a proper scheme and  $\mathcal{F} \in \text{Coh}(X)$  with  $\dim \text{Supp}(\mathcal{F}) = 0$ , then  $\mathcal{F}$  is generated by global sections,  $H^i(X, \mathcal{F}) = 0$  for  $i > 0$ , and

$$\chi(X, \mathcal{F} \otimes \mathcal{E}) = n \chi(X, \mathcal{F})$$

for any  $\mathcal{E} \in \text{Vect}^n(X)$ .  $\lrcorner$

*Proof:* The first two are clear as  $\mathcal{F} = i_* \mathcal{G}$  where  $i : \text{Supp}(\mathcal{F}) \rightarrow X$ . The last assertion follows from the projection formula (7.7.4.6)  $i_*(\mathcal{G} \otimes i^* \mathcal{E}) \cong \mathcal{F} \otimes \mathcal{E}$ .  $\square$

**Def. (7.7.3.6) [Arithmetic Genus].** The **arithmetic genus** of a proper scheme of dimension  $r$  over a field is defined to be  $p_a(X) = (-1)^r (\chi(\mathcal{O}_X) - 1)$  (7.7.3.1). In particular, when  $X$  is a complete curve over a field  $k$ , then  $p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$  (7.12.1.12).

The arithmetic genus is stable under base change of fields, by flat base change theorem.  $\lrcorner$

**Prop. (7.7.3.7) [Arithmetic Genus of Product].** By Künneth formula, if  $X, Y$  are proper schemes of dimension  $r, s$  over a field  $k$ , then

$$H^n(X \otimes_k Y, \mathcal{O}_{X \otimes_k Y}) \cong \bigoplus_{0 \leq k \leq n} H^k(X, \mathcal{O}_X) \otimes H^{n-k}(Y, \mathcal{O}_Y).$$

Thus  $\chi(X \times_k Y) = \chi(X) \chi(Y)$ . In particular, we have

$$p_a(X \times Y) = p_a(X) p_a(Y) + (-1)^s p_a(X) + (-1)^r p_a(Y).$$

$\lrcorner$

**Prop. (7.7.3.8) [Arithmetic Genus of Complete Intersections].** Let  $C$  be a smooth complete intersection of a degree  $m$  surface  $S_1$  and a degree  $n$  surface  $S_2$  in  $\mathbb{P}_k^3$ , then  $p_a(C) = \frac{1}{2}(mn(m+n-4)+2)$ .  $\lrcorner$

*Proof:* There are exact sequences  $0 \rightarrow \mathcal{O}_{\mathbb{P}_k^3}(-S_i) \rightarrow \mathcal{O}_{\mathbb{P}_k^3} \rightarrow \mathcal{O}_{S_i} \rightarrow 0$ , which gives

$$0 \rightarrow \mathcal{O}_{\mathbb{P}_k^3}(-S_2)|_{S_1} \rightarrow \mathcal{O}_{S_1} \rightarrow \mathcal{O}_{S_1 \cap S_2} \rightarrow 0.$$

So by adjunction formula,

$$\mathcal{K}_C = \mathcal{K}_{S_1}(S_1 \cap S_2)|_C = \mathcal{K}_{\mathbb{P}_k^3}(S_1)(S_2)|_C = \mathcal{O}_C(m+n-4).$$

So  $2g - 2 = mn(m+n-4)$ .  $\square$

**Prop. (7.7.3.9) [Asymptotic Riemann-Roch].** If  $X$  is a proper scheme over a field  $k$  of dimension  $d$  and  $\mathcal{L}$  is an ample invertible sheaf, then  $\dim \Gamma(X, \mathcal{L}^n) \sim cn^d$ .  $\lrcorner$

*Proof:* Cf. [Sta]0BJ8.  $\square$

### Hilbert Polynomials

References are [Hartshorne I.7] and [Vak17].

**Prop. (7.7.3.10) [Hilbert Polynomial].** For a projective scheme over a field  $k$  and a coherent sheaf  $\mathcal{F}$ , there is a polynomial **Hilbert polynomial**  $P \in \mathbb{Q}[\lambda]$  that  $\chi(\mathcal{F}(n)) = P(n)$ , and  $\deg P \leq \dim \text{Supp}(\mathcal{F})$ .

This Hilbert polynomial is compatible with the definition in (7.7.3.12), because by (7.7.4.7), the higher cohomology group vanishes for  $n$  large, so  $\chi(\mathcal{F}(n)) = \Gamma(\mathcal{F}(n)) = \Gamma_*(\mathcal{F})_n$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.7.3.11).** Let  $X \hookrightarrow Y \hookrightarrow \mathbb{P}_k^n$  be a sequence of closed embeddings, then  $P_X(m) \leq P_Y(m)$  for  $m$  large, and if equality holds for  $m$  large, then  $X = Y$ .  $\lrcorner$

*Proof:* Cf. [Vak17] P490.  $\square$

**Def. (7.7.3.12) [Hilbert Polynomial].** For a scheme projective over a field  $k$  of dimension  $r$ , we define the **Hilbert polynomial**  $P_Y$  as the Hilbert polynomial of its homogenous coordinate ring  $\Gamma_*(Y)$ . It has dimension  $r$  by (5.2.2.17).

The **degree** of  $Y$  is defined as the  $r!$  times the leading coefficients of  $P_Y$ .  $\lrcorner$

**Prop. (7.7.3.13).**

- The degree is a positive integer.
  - If  $Y = Y_1 \cup Y_2$  and  $\dim Y_1 \cap Y_2 < r$ , then  $\deg Y = \deg Y_1 + \deg Y_2$ .
  - If  $H$  is a hypersurface whose ideal is generated by a homogeneous polynomial of degree  $d$ , then  $\deg H = d$ .
- $\lrcorner$

*Proof:* Cf. [Hartshorne P52].  $\square$

**Prop. (7.7.3.14).** For a variety of degree  $k$  and a general linear space, the intersection has  $k$  points.  $\lrcorner$

*Proof:*  $\square$

## 4 Relative Cohomology

**Prop. (7.7.4.1) [Filtered Colimits].** If  $f : X \rightarrow Y$  is qcqs, then  $R^i f_*$  commutes with filtered colimits, by (7.3.1.6) and (7.7.1.6).  $\lrcorner$

**Prop. (7.7.4.2) [Sheaf Cohomology Commutes with Affine Map].** For  $f : X \rightarrow Y$  affine and  $\mathcal{F} \in \mathcal{Q}\text{Coh}(X)$ ,  $H^n(Y, \mathcal{F}) = H^n(X, f_* \mathcal{F})$ .  $\lrcorner$

*Proof:* Because  $R^i f_* \mathcal{F}(U) = 0$  by (7.7.1.3) and (7.3.1.7), we can then use (7.3.1.8) to conclude.  $\square$

**Prop. (7.7.4.3) [Higher Direct Image Preserves Qco Sheaves].** If  $f : X \rightarrow S$  is qcqs then  $R^n f_*$  maps  $\mathcal{Q}\text{Coh}(X)$  to  $\mathcal{Q}\text{Coh}(S)$ , and for  $U \subset S$  affine open,  $\underline{(R^p f_* \mathcal{F})|_U} = \underline{(H^p(f^{-1}(U), \mathcal{F}))^\sim}$ .  $\lrcorner$

*Proof:* Firstly by (7.3.3.18),  $(R^p f_* \mathcal{F})|_U = R^p f_{U*}(\mathcal{F}|_U)$ , so it suffices to prove for  $S$  affine. Then  $R^p f_* \mathcal{F}$  is the sheafification of the presheaf  $G : U \mapsto H^p(f^{-1}(U), \mathcal{F})$  by (7.3.1.7).

If  $f$  is separated, then we can use Čech cohomology (7.7.1.2) to see that  $H^p(f^{-1}(D(f)), \mathcal{F}) = H^p(X, \mathcal{F})_f$  thus  $G$  is itself a sheaf and the conclusion follows.

In general we can choose a finite affine cover  $U_i$  of  $X$ , then every intersection of  $U_i$  is quasi-compact and separated, and there is a spectral sequence convergence  $H^p(\{U_i \rightarrow X\}, \mathcal{H}^q(F)) \implies H^{p+q}(X, F)$ , where the left can be calculated using Čech cohomology. Taking localization w.r.t.  $f$ , we get the desired isomorphism  $H^p(f^{-1}(D(f)), \mathcal{F}) = H^p(X, \mathcal{F})_f$  also by comparison(4.10.7.5).  $\square$

**Prop. (7.7.4.4)[Cohomological Boundedness of  $Rf_*$ ].** For a qcqs morphism  $f : X \rightarrow S$ , if  $S$  is qc, there is an  $N$  that for every base change  $f'$  of  $f$ , we have  $R^n f'_* \mathcal{F} = 0$  for every  $\mathcal{F} \in \mathcal{Q}\text{Coh}(X)$  and  $n \geq N$ .

In particular, if  $f$  is affine, then  $R^n f'_* \mathcal{F} = 0$  for  $n > 0$ . And if  $f$  is projective, then  $R^n f'_* \mathcal{F} = 0$  for  $n$  bigger than the maximal dimension of the fiber of  $f$ .  $\lrcorner$

*Proof:* Check affine locally on  $S$  and use(7.7.4.3), choose a finite affine cover  $\mathcal{U}$  of  $X$ . Then when  $n$  is large,  $(R^n f'_* \mathcal{F})|_{U_i} = 0$  by(7.7.4.3) and(7.7.1.3) for any  $i$ , thus  $R^n f'_* \mathcal{F} = 0$ . For base changes, notice the cardinality of the affine cover are the same.  $\square$

**Cor. (7.7.4.5).** For a qc separated scheme  $X$ , the cohomology vanish for  $n$  large. And when  $X$  is separated and can be covered by  $r$  affine opens, then  $N$  can be chosen to be  $r$ .  $\lrcorner$

**Cor. (7.7.4.6)[Projection Formula].**

- Let  $f : X \rightarrow Y$  be a morphism of schemes, and  $\mathcal{E}$  a locally free  $\mathcal{O}_Y$ -module, then for any  $\mathcal{F} \in \text{Mod}(\mathcal{O}_X)$  and any  $i$ , there are natural isomorphisms.

$$R^i f_*(\mathcal{F} \otimes f^* \mathcal{E}) \cong R^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

- If  $f : X \rightarrow Y$  is a qcqs morphism of schemes, then for any  $\mathcal{F} \in \text{Coh}(\mathcal{O}_X)$ ,  $\mathcal{E} \in \text{Coh}(\mathcal{O}_Y)$ , there is a natural isomorphism

$$Rf_*(\mathcal{F} \otimes^{\mathbb{L}} Lf^* \mathcal{E}) \cong Rf_*(\mathcal{F}) \otimes^{\mathbb{L}} \mathcal{E}.$$

$\lrcorner$

*Proof:* 1 follows from(7.3.3.20) and 2 follows from(7.9.1.8).  $\square$

### Proper Morphism and $\text{Coh}(X)$

**Prop. (7.7.4.7)[Relative Serre Vanishing].** If  $f : X \rightarrow Y$  is a proper morphism of locally Noetherian schemes,  $\mathcal{I}$  an invertible sheaf on  $X$ , then the following are equivalent:

- $\mathcal{L}$  is  $f$ -ample.
- For any  $\mathcal{F} \in \text{Coh}(X)$ , for  $n$  sufficiently large,  $R^i f_*(\mathcal{F} \otimes \mathcal{I}^{\otimes n}) = 0$  and  $i > 0$ .
- For any Qco sheaf of ideals  $\mathcal{I} \subset \mathcal{O}_X$ , there exists an  $n \geq 1$  that  $R^1 f_*(\mathcal{I} \otimes \mathcal{L}^{\otimes n}) = 0$ .

$\lrcorner$

*Proof:* 1  $\rightarrow$  2: By(7.7.4.3), this follows from(7.7.2.8). 2  $\rightarrow$  3 is trivial as  $X$  is Noetherian. 3  $\rightarrow$  1: Notice for any affine open subset  $U$  of  $Y$ ,  $U \rightarrow Y$  is qc, thus  $f^{-1}(U) \rightarrow X$  is quasi-compact, thus by(7.5.1.8), a Qco sheaf of ideals  $\mathcal{I}$  on  $f^{-1}(U)$  can be extended to a Qco sheaf of ideals on  $X$ . Then we can use(7.7.4.3) and Leray spectral sequence to reduce to(7.7.2.8).  $\square$

**Cor. (7.7.4.8).** Let  $X$  be proper over a Noetherian affine scheme with an ample invertible sheaf  $\mathcal{I}$ , then for any finite exact sequence in  $\text{Coh}(X)$ , if tensoring it with  $\mathcal{I}^{\otimes n}$  for large  $n$ , the resulting global section is exact.  $\lrcorner$

**Cor. (7.7.4.9).** Let  $X$  be proper over a Noetherian affine scheme with an ample invertible sheaf  $\mathcal{I}$ , and  $\mathcal{F}, \mathcal{G} \in \text{Coh}(X)$ ,  $i \geq 0$ , then for  $n$  large (depending on  $\mathcal{F}, \mathcal{G}, i$ ),

$$\text{Ext}^i(\mathcal{F}, \mathcal{G}(n)) \cong \Gamma(X, \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}(n))).$$

┘

*Proof:* By (7.3.3.34)(7.7.1.7), there is a spectral sequence s.t. for  $n$  large, all the small terms vanish.  $\square$

**Lemma (7.7.4.10).** If  $f : X \rightarrow Y$  is projective and  $Y$  locally Noetherian, then for any  $n \in \mathbb{N}$ ,  $R^n f_*$  maps coherent sheaves to coherent sheaves.  $\square$

*Proof:* By (7.7.4.3), this problem is local on the target, so we may assume that  $Y = \text{Spec } R$  and  $X = \mathbb{P}_R^n$ , in which case this follows from (7.7.2.5).  $\square$

**Prop. (7.7.4.11) [Grothendieck's Coherence Theorem].** If  $f : X \rightarrow Y$  is proper and  $Y$  locally Noetherian, then for any  $n \in \mathbb{N}$ ,  $R^n f_*$  maps coherent sheaves to coherent sheaves.  $\square$

*Proof:* By (7.7.4.3), this problem is local on the target, so we may assume  $Y$  is Noetherian, and then  $X$  is also Noetherian. We prove by devissage (7.5.1.55): 1 is trivial, for 2, for any closed subscheme  $i : Z \subset X$ , denote  $g = f|_Z$ , then it suffices to find a coherent sheaf  $\mathcal{G}$  on  $Z$  s.t.

1.  $\mathcal{G}_\xi \cong k(\xi)$ .

2.  $R^p g_* \mathcal{G}$  are coherent for any  $p \geq 0$ .

Because then  $i_* \mathcal{G}$  is a coherent sheaf on  $X$  (7.5.1.32) that  $R^p f_*(i_* \mathcal{G}) = R^p g_* \mathcal{G}$  for any  $p$  by relative Leray spectral sequence (7.3.1.8) and (7.7.4.4), and also  $(i_* \mathcal{G})_\xi = \mathcal{G}_\xi$ .

As  $g : Z \rightarrow Y$  is proper, by Chow's lemma (7.4.6.23), there is a birational, H-projective map  $\pi : Z' \rightarrow Z$  over  $Y$  that  $Z'$  is projective over  $Y$ . Then there is a closed immersion  $j : Z' \rightarrow \mathbb{P}_Y^n$  and an induced closed immersion  $j' : Z' \rightarrow \mathbb{P}_Z^n$ . Then  $\mathcal{L} = j^* \mathcal{O}_{\mathbb{P}_Y^n}(1) = (j')^* \mathcal{O}_{\mathbb{P}_Z^n}(1)$  is both  $g \circ \pi$ -relatively ample and  $\pi$ -relatively ample.

Hence by relative Serre vanishing (7.7.4.7) there exists an  $n$  that  $R^p \pi_* \mathcal{L}^{\otimes n} = 0$  for any  $p > 0$ . Let  $\mathcal{G} = \pi_* \mathcal{L}^{\otimes n}$ , then this  $\mathcal{G}$  satisfies the conditions:  $\mathcal{G}_\xi = \kappa(\xi)$  as  $\pi^{-1}(U) \rightarrow U$  is an isomorphism, and from relative Leray spectral sequence

$$R^p g_* R^q \pi_* \mathcal{L}^{\otimes n} \implies R^n (g \circ \pi)_* \mathcal{L}^{\otimes n},$$

we see  $R^p g_* \mathcal{G} \cong R^p g'_* \mathcal{L}^{\otimes n}$ , which are coherent by (7.7.4.10).  $\square$

**Cor. (7.7.4.12) [Coherent Cohomology Finite].** If  $\pi : X \rightarrow \text{Spec } A$  is proper over a Noetherian affine scheme  $A$  and  $\mathcal{F}$  is a coherent sheaf on  $X$ ,  $H^i(X, \mathcal{F})$  are finite  $A$ -modules.  $\square$

*Proof:* By (7.7.4.3),  $R^p \pi_* \mathcal{F}$  is the Qco sheaf  $\widetilde{H^p(X, \mathcal{F})}$ , which is coherent iff  $H^p(X, \mathcal{F})$  is a finite  $A$ -module.  $\square$

**Prop. (7.7.4.13).** Given a proper morphism between locally Noetherian schemes  $f : X \rightarrow Y$ , a coherent sheaf  $\mathcal{F}$  on  $X$  and a coherent sheaf of ideals  $\mathcal{I}$  of  $\mathcal{O}_Y$ . Then  $\mathcal{M} = \bigoplus_{n \geq 0} R^p f_*(\mathcal{I}^n \mathcal{F})$  is a Qco graded  $\mathcal{A} = \bigoplus_{n \geq 0} \mathcal{I}^n$ -module of f.t..  $\square$

*Proof:* By (7.7.4.3), this is local on  $Y$ , so we may assume  $Y$  is affine, Cf. [Sta]02O8.  $\square$

**Prop. (7.7.4.14).** Let  $f : X \rightarrow Y$  be a proper morphism between locally Noetherian schemes and  $\mathcal{F}$  a coherent sheaf on  $X$ ,  $y \in Y$ . Consider the infinitesimal nbhds  $X_n = \operatorname{Spec}(\mathcal{O}_{Y,y}/\mathfrak{m}_y^n) \times_Y X \xrightarrow{i_n} X$  of the fiber  $X_1 = X_y$ , and set  $\mathcal{F}_n = i_{n*}\mathcal{F}$ , then

$$(R^n f_* \mathcal{F})_y^\wedge \cong \varprojlim_n H^n(X_n, \mathcal{F}_n)$$

as  $\mathcal{O}_{Y,y}^\wedge$ -modules. ┘

*Proof:* Cf. [Sta]02OD. □

**Cor. (7.7.4.15) [Support of Higher Direct Images].** Let  $f : X \rightarrow Y$  be a proper morphism between locally Noetherian schemes and  $y \in Y$  s.t.  $\dim(X_y) = d$ , then for any coherent sheaf  $\mathcal{F}$  on  $X$ ,  $(R^p f_* \mathcal{F})_y = 0$  for  $p > d$ . ┘

*Proof:* Cf. [Sta]02V7. □

## 5 Base Change

**Prop. (7.7.5.1) [Flat Base Change].** For a Cartesian diagram of schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

If  $g$  is flat and  $f$  is qcqs, then for every Qco sheaf  $\mathcal{F}$  on  $X$  with base change  $\mathcal{F}'$  on  $X'$ , there is a canonical isomorphism

$$g^* Rf_* \mathcal{F} \cong Rf'_* \mathcal{F}'.$$

By (7.3.3.18), when  $S, S'$  is affine, this reads:

$$H^i(X, \mathcal{F}) \otimes_A B \cong H^i(X \otimes_A B, (g')^* \mathcal{F}).$$

┘

*Proof:* Firstly by (7.3.3.18)(7.7.4.3), it suffices to show for  $S, S'$  affine. If  $X$  is separated, then then we can use Čech cohomology (7.7.1.2), and the Čech complex of  $K'$  is just the cohomology of the Čech complex tensored with  $B$ , so it commutes with taking cohomology because  $B$  is  $A$ -flat.

Now if  $X$  is only qs, then we choose a finite affine open cover  $\{U_i\}$ , then every intersection of  $U_i$  is quasi-compact and separated. and there is a spectral sequence convergence  $H^p(\{U_i \rightarrow X\}, \mathcal{H}^q(F)) \implies H^{p+q}(X, F)$ . Tensoring with  $B$ , we also get the desired isomorphism  $H^i(X, \mathcal{F}) \otimes_A B \cong H^i(X \otimes_A B, (g')^* \mathcal{F})$  by comparison (4.10.7.5). □

**Prop. (7.7.5.2) [Finite Locally Free Base Change].** For a Cartesian diagram of schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

If  $S = \operatorname{Spec} A$ ,  $S' = \operatorname{Spec} B$  and  $B$  is finite locally free over  $A$ , for every Qco sheaf  $\mathcal{F}$  on  $X$  with base change  $\mathcal{F}'$  on  $X'$ , there is a canonical isomorphism

$$H^i(X, \mathcal{F}) \otimes_A B \cong H^i(X \otimes_A B, (g')^* \mathcal{F}).$$

┘

*Proof:* If  $X$  is separated, then the then we can use Čech cohomology(7.7.1.2), and the Čech complex of  $K'$  is just the cohomology of the Čech complex tensored with  $B$ , so it commutes with taking cohomology because  $B$  is  $A$ -flat and tensoring a f.p. ring map commutes with colimits of rings.

In general we choose an affine open cover  $\{U_i\}$ , then every intersection of  $U_i$  is separated. and there is a spectral sequence convergence  $H^p(\{U_i \rightarrow X\}, \mathcal{H}^q(F)) \implies H^{p+q}(X, F)$ . Tensoring with  $B$ , we also get the desired isomorphism  $H^i(X, \mathcal{F}) \otimes_A B \cong H^i(X \otimes_A B, (g')^* \mathcal{F})$  by comparison(4.10.7.5).  $\square$

**Prop. (7.7.5.3) [Representing Higher Direct Image].** Let  $f : X \rightarrow S$  be a qcqs morphism of schemes. If  $S$  is qc and separated and  $\mathcal{F}$  is a Qco sheaf on  $X$ , there exists a  $\mathcal{K}^\bullet \in K^+(\mathcal{QCoh}(\mathcal{O}_S))$  s.t. for any morphism  $g : S' \rightarrow S$ , the complex  $g^* \mathcal{K}^\bullet$  is a representative for  $Rf'_* \mathcal{F}'$ , where the notation is as in(7.7.5.1).  $\lrcorner$

*Proof:* We only prove the case  $X$  is separated, the general case in [Sta]01XN.

Choose a finite affine open covering  $\mathcal{U}$  of  $X$ , consider the complex of sheaves  $\mathcal{C}^\bullet(\mathcal{U}, \mathcal{F}) = \text{Hom}^\bullet((\mathbb{Z}_{\mathcal{U}, \bullet}^\sharp, \mathcal{F}))$ (7.3.2.1), then  $\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \bigoplus_{i_0 \dots i_p} j_{i_0 \dots i_p *} \mathcal{F}_{i_0 \dots i_p}$ . Then let  $\check{\mathcal{C}}^\bullet(\mathcal{U}, f, \mathcal{F}) = f_* \mathcal{C}^\bullet(\mathcal{U}, \mathcal{F})$ . There is a natural map  $\mathcal{F} \rightarrow \mathcal{C}^\bullet(\mathcal{U}, \mathcal{F})$ . Now  $j_{i_0 \dots i_p}$  and  $f_{i_0 \dots i_p}$  are both affine by(7.4.5.39), so are their base changes, thus the higher direct image vanish. So by relative Leray spectral sequence, we see each term  $j_{i_0 \dots i_p *} \mathcal{F}'_{i_0 \dots i_p}$  is  $f'_*$ -acyclic. Then by Leray's acyclicity theorem,

$$Rf'_*(\mathcal{F}') = Rf'_*(\mathcal{C}^\bullet(\mathcal{U}, \mathcal{F})) \cong \mathcal{C}^\bullet(\mathcal{U}', f', \mathcal{F}') = g^* \mathcal{C}^\bullet(\mathcal{U}, f, \mathcal{F}).$$

The last equation follows from the fact  $f_{i_0 \dots i_p}$  are all affine thus the base change are isomorphisms.  $\square$

**Prop. (7.7.5.4) [Proper Flat Base Change].** If  $f : X \rightarrow S$  is a proper morphism of f.p., then

- If  $E$  a perfect object in  $D(\mathcal{O}_X)$ ,  $\mathcal{G} \in D(\mathcal{O}_X)$  is representable by a bounded complex of f.p.  $\mathcal{O}_X$ -modules flat over  $S$ , then

$$Rf_*(E \otimes_{\mathcal{O}_X}^L \mathcal{G}^\bullet), \quad Rf_* R\mathcal{H}om(E, \mathcal{G}^\bullet)$$

are perfect complexes in  $D(\mathcal{O}_S)$ , and its formation commutes with base change.

- If  $E$  a pseudo-coherent object in  $D(\mathcal{O}_X)$ ,  $\mathcal{G} \in D(\mathcal{O}_X)$  is representable by a bounded above complex of f.p.  $\mathcal{O}_X$ -modules flat over  $S$ , then  $Rf_*(E \otimes_{\mathcal{O}_X}^L \mathcal{G}^\bullet)$  is a pseudo-coherent complex in  $D(\mathcal{O}_S)$ , and its formation commutes with base change.  $\lrcorner$

*Proof:* Cf.[Sta]0A1H, 0A1J, 0CSC. ?  $\square$

**Cor. (7.7.5.5).** Let  $f : X \rightarrow S$  be a proper morphism of f.p., then

- If  $E \in D(\mathcal{O}_X)$  is perfect(pseudo-coherent) and  $f$  is flat, then  $Rf_* E$  is a perfect(pseudo-coherent) object in  $D(\mathcal{O}_S)$ , and its formation commutes with base change.
- If  $\mathcal{G} \in \mathcal{QCoh}^{\text{pf}}(\mathcal{O}_X)$  and is flat over  $S$ , then  $Rf_* \mathcal{G}$  is a perfect object of  $D(\mathcal{O}_S)$  and its formation commutes with base change.  $\lrcorner$

**Cor. (7.7.5.6).** If  $A \in \mathcal{CAlg}$ ,  $X$  be a proper scheme of f.p. over  $A$ , then if  $\mathcal{G} \in \mathcal{QCoh}^{\text{pf}}(\mathcal{O}_X)$  and is flat over  $A$ , there exists a finite complex of finite projective  $A$ -modules  $L^\bullet$  that for any  $A' \in \mathcal{CAlg}_A$ ,  $M \in \text{Mod}_A$ , by(7.7.5.5) and projection formula(7.7.4.6),

$$H^i(X_{A'}, \mathcal{F}_{A'}) = H^i(L^\bullet \otimes_A A'), \quad H^i(X, \mathcal{F} \otimes_A M) = H^i(L^\bullet \otimes_A M)$$

$\lrcorner$

## 6 Semicontinuity

**Prop. (7.7.6.1).**  $T^i$  is left exact iff  $\text{Coker } d^{i-1}$  is a projective  $A$ -module, iff it is representable by a finite  $A$ -module.  $\square$

*Proof:* Denote  $W^i = \text{Coker } d^{i-1}$ , then  $\text{Coker } d^{i-1} \otimes_A M = W^i \otimes M$ , because tensoring is right exact. Thus  $T^i(M) = \ker(W^i \otimes M \rightarrow L^{i+1} \otimes M)$ . Then for  $M' \subset M$ , there is a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & T^i(M) & \longrightarrow & W^i \otimes M' & \longrightarrow & L^{i+1} \otimes M' \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & T^i(M) & \longrightarrow & W^i \otimes M & \longrightarrow & L^{i+1} \otimes M \end{array}$$

$\gamma$  is injective, so using spectral sequence, its clear  $\alpha$  is injective iff  $\beta$  is injective, i.e.  $W^i$  is flat, which is equivalent to finite projective (5.3.1.7).

To prove  $T^i$  is representable, let  $Q = \text{Coker}(L^{i+1,*} \rightarrow W^{i,*})$ , then  $Q$  is finite because  $W^i$  is finite (5.3.1.18), and  $0 \rightarrow \text{Hom}(Q, M) \rightarrow \text{Hom}(W^{i,*}, M) \rightarrow \text{Hom}(L^{i+1,*}, M)$ , but by (5.3.1.19), the last two are just  $W^i \otimes M$  and  $L^{i+1} \otimes M$ ,  $\text{Hom}(Q, M) = T^i(M)$  by what has already be proved.  $\square$

**Prop. (7.7.6.2).**  $T^i$  is right exact iff the cup product  $H^i(X, \mathcal{F}) \otimes_A M \rightarrow H^i(X, \mathcal{F} \otimes_A M)$  is an isomorphism for any  $A$ -module  $M$ .  $\square$

*Proof:* Because  $T^i$  and  $\otimes$  commutes with direct limit, it suffices to prove for  $M$  finite. In this case, choose a finite presentation  $A^r \rightarrow A^s \rightarrow M \rightarrow 0$ , then there is a diagram

$$\begin{array}{ccccccc} T^i(A) \otimes A^r & \longrightarrow & T^i(A) \otimes A^s & \longrightarrow & T^i(A) \otimes M & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ T^i(A^r) & \longrightarrow & T^i(A^s) & \longrightarrow & T^i(M) & & \end{array}$$

The first two vertical arrows are isomorphisms, so if  $T^i$  is right exact, so does the third vertical arrow. Conversely, if  $T^i(A) \otimes_A M \rightarrow T^i(M)$  are all isomorphisms, then by a similar diagram, we can show  $T^i(M) \rightarrow T^i(M')$  are surjective for  $M \rightarrow M'$  surjective.  $\square$

**Cor. (7.7.6.3).**  $T^i$  is exact iff it is right exact and  $T^i(A) = H^i(X, \mathcal{F})$  is a finite projective  $A$ -modules.  $\square$

*Proof:* When  $T^i$  is right exact,  $H^i(X, \mathcal{F} \otimes_A M) \cong H^i(X, \mathcal{F}) \otimes_A M$  by (7.7.6.2), so it is exact iff  $H^i(X, \mathcal{F})$  is flat. Because it is in priori finite, this is equivalent to finite projective (5.3.1.7).  $\square$

**Def. (7.7.6.4).** For a point  $y \in \text{Spec } A$ , define  $T_y^i(N) = H^i(L_y^\bullet \otimes N)$ , then  $T^i$  is (left/right)exact at  $y$  iff  $T_y^i$  are all (left/right)exact(exact sequence is stalkwise (5.1.4.2)).  $\square$

**Prop. (7.7.6.5).** If  $T^i$  is (left/right)exact at a point  $y$ , then the same is true on a nbhd of  $y$ .  $\square$

*Proof:* From (7.7.6.1),  $(\text{Coker } d^{i-1})_y$  is a finite projective  $A_p$  module, so it is free, and it is a coherent sheaf, so it is free at a nbhd of  $y$  by (7.5.1.38), so the same is true on a nbhd of  $y$ . Now right exactness of  $T^i$  is equivalent to left exactness of  $T^{i+1}$ , and exact is left exact+right exact, so we are done.  $\square$

**Prop. (7.7.6.6).**  $T^i$  is right exact at  $y$  iff  $H^i(X, \mathcal{F}) \otimes k(y) \rightarrow H^i(X, \mathcal{F} \otimes_A k(y))$  is surjective.  $\square$

*Proof:* Cf. [Hartshorne P289]. ?  $\square$

**Prop. (7.7.6.7)[Cohomology and Stalks].** Let  $f : X \rightarrow Y$  be a proper morphism of locally Noetherian schemes and  $\mathcal{F}$  is a coherent sheaf on  $X$  flat over  $Y$ ,  $y \in Y$ , then

- If the natural map

$$\varphi^i(y) : R^i f_*(\mathcal{F}) \otimes k(y) \rightarrow H^i(X_y, \mathcal{F}_y)$$

is surjective, then it is an isomorphism, and the same is true for  $y'$  in an open nbhd of  $y$ .

- Assume  $\varphi^i(y)$  is surjective, then  $\varphi^{i-1}(y)$  is also surjective iff  $R^i f_*(\mathcal{F})$  is finite projective in a nbhd of  $y$ .

┘

*Proof:* 1: This follows from (7.7.6.2)(7.7.6.6) and (7.7.6.5).

2:  $\varphi^{i-1}(y), \varphi^i(y)$  are both surjective iff  $T^i$  and  $T^{i-1}$  are both right exact at  $y$  (7.7.6.6), which is equivalent to  $T^i$  exact at  $y$ . Then we finish by (7.7.6.3)  $\square$

**Remark (7.7.6.8).** How's this related to (7.7.5.5)?

┘

**Prop. (7.7.6.9)[Semicontinuity of Cohomology].** Let  $X \rightarrow Y$  be a projective morphism of locally Noetherian schemes and  $\mathcal{F}$  is a coherent sheaf on  $X$ , flat over  $Y$ , then for each  $i$ ,  $h^i(y, \mathcal{F}) = \dim_{k(y)} H^i(X_y, \mathcal{F}_y)$  is an upper semicontinuous function on  $Y$ .  $\square$

*Proof:* The question is local on  $Y$ , so we may assume  $Y$  is affine Noetherian. By (7.7.5.1),  $H^i(y, \mathcal{F}) = \dim_{k(y)} T^i(k(y))$ . And as in the proof of (7.7.6.1),  $T^i(M) = \ker(W^i \otimes M \rightarrow L^{i+1} \otimes M)$ , and  $W^i \rightarrow L^{i+1} \rightarrow W^{i+1} \rightarrow 0$  is exact, so  $0 \rightarrow T^i(k(y)) \rightarrow W^i \otimes k(y) \rightarrow L^{i+1} \otimes k(y) \rightarrow W^{i+1} \otimes k(y) \rightarrow 0$ , and counting dimension,  $h^i(y, \mathcal{F}) = \dim W^i \otimes k(y) + \dim W^{i+1} \otimes k(y) - \dim L^{i+1} \otimes k(y)$ . Notice the last term is constant as  $L^{i+1}$  is free  $A$ -module and the first two terms are upper semi-continuous by (7.5.1.41), thus  $h^i(y, \mathcal{F})$  is upper-semicontinuous.  $\square$

**Cor. (7.7.6.10)[Grauert].** If  $Y$  is integral and  $h^i(y, \mathcal{F})$  is constant on  $Y$ , then  $R^i f_*(\mathcal{F})$  is locally free on  $Y$  and  $R^i f_*(\mathcal{F}) \otimes k(y) \cong H^i(X_y, \mathcal{F}_y)$ .  $\square$

*Proof:* Cf.[Hartshorne P288].  $\square$



## 7.8 Topics in Schemes

References are [Sta].

### 1 Cartier Divisors

**Def. (7.8.1.1) [Cartier Divisor].** A **Cartier divisor** on  $X \in \text{Sch}$  is an element in  $\Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*)$ .

An **effective Cartier divisor** is a Cartier divisor that is locally defined as  $\{(U_i, f_i)\}$  where  $f_i \in \Gamma(U_i, \mathcal{O}_{U_i})$  are non-zero-divisors, equivalently, it is a closed subscheme  $D \subset X$  s.t.  $\mathcal{I}(D) \in \text{Pic}(X)$ . (Notice by definition,  $\mathcal{K}$  is the localization w.r.t. non-zero-divisors, and  $f_i$  is invertible in  $\mathcal{K}^*$  so  $f_i$  must be nonzero-divisors.)

The monoid of effective Cartier divisors is denoted by  $\text{Cart}^{\text{eff}}(X)$ .

The **Cartier divisor group**  $\text{CaCl}(X)$  is the quotient of  $\Gamma(X, \mathcal{K}_X^*) \rightarrow \Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*)$ .  $\perp$

**Prop. (7.8.1.2) [Cartier Divisor and Flat Base Change].**

- The flat base change of Cartier divisor is also a Cartier divisor.
- The flat base change of a regular immersion (7.6.7.1) is also a regular immersion.

$\perp$

*Proof:* ?

$\square$

**Prop. (7.8.1.3) [Cartier Divisor is Nowhere Dense].** Let  $D \subset X$  be an effective Cartier divisor, then it is nowhere dense in  $X$ , i.e.  $X \setminus D \rightarrow X$  is scheme-theoretically dense.  $\perp$

*Proof:* It suffices to check affine-locally, it is qc so the scheme-theoretic closure of  $\text{Spec } A_f \rightarrow \text{Spec } A$  is  $V(\ker(A \rightarrow A_f)) = V(0)$  as  $f$  is a nonzero-divisor.  $\square$

**Prop. (7.8.1.4) [Closed Subschemes and Effective Cartier Divisors].** Let  $X \in \text{NSch}$  and  $D \subset X$  is a closed subscheme corresponding to a Qco sheaf of ideals  $\mathcal{I} \subset \mathcal{O}_X$ . If for any  $x \in D$ , the ideal  $\mathcal{I}_x \subset \mathcal{O}_{X,x}$  is generated by a single nonzero-divisor, then  $D$  is an effective Cartier divisor.  $\perp$

*Proof:* Cf. [Sta] 0AG8.  $\square$

**Cor. (7.8.1.5) [Prime Divisor and Effective Cartier Divisor].** Let  $X \in \text{NSch}$  and  $D$  is a prime Weil divisor on  $X$  and  $\mathcal{O}_{X,x}$  are UFDs for any  $x \in D$  (e.g.  $X$  is locally factorial), then  $D$  is an effective Cartier divisor.  $\perp$

*Proof:* For any  $x \in D$ , let  $A = \mathcal{O}_{X,x}$ , and  $\mathfrak{p}$  be the prime corresponding to the generic point of  $D$ , then  $\dim A_{\mathfrak{p}} = 1$ , so  $\mathfrak{p}$  is principle as  $A$  is a UFD, so  $D$  is an effective Cartier divisor by (7.8.1.4).  $\square$

**Prop. (7.8.1.6) [Cartier-Pic].** For an integral scheme  $X$ , the homomorphism  $\text{CaCl}(X) \rightarrow \text{Pic}(X)$  (7.8.1.1)(7.5.3.10) induced from long exact sequence of  $0 \rightarrow \mathcal{O}_X^* \rightarrow \mathcal{K}_X^* \rightarrow \mathcal{K}_X^*/\mathcal{O}_X^* \rightarrow 0$  is an isomorphism.

Explicitly, for a Cartier divisor  $D = \{(U_i, f_i)\}$ , the image is the invertible sheaf  $\mathcal{O}_X(D)$  (7.5.3.12).  $\perp$

*Proof:* It is clearly injective by definition, so it suffices to show any invertible sheaf can embed into the constant sheaf: for any invertible sheaf  $\mathcal{L}$ , tensor with  $\mathcal{K}$  and restrict to the stalk of the generic point, i.e. there is a compatible choice of homomorphisms into  $K(X)$ .  $\square$

**Prop. (7.8.1.7).** If  $X \in \text{NSch}$  and the diagonal map is affine, for a dense affine open  $U$ , if all the stalk of  $X \setminus U$  are UFD, then  $U$  is the complement of an effective Cartier divisor.  $\lrcorner$

*Proof:* The irreducible complements of  $X \setminus U$  is finite and has codimension 1 by (7.6.3.5) because  $U \rightarrow X$  is affine, and it is an effective Cartier divisor by (9.1.4.1), so their sum will suffice.  $\square$

**Prop. (7.8.1.8) [Pullback of Effective Cartier Divisors].** Let  $f : X \rightarrow Y \in \text{Sch}$  and  $D \in \text{Cart}^{\text{eff}}(Y)$ , then the pullback of  $D$  via  $f$  is an effective Cartier divisor on  $X$  if one of the following holds:

- $f(x) \notin D$  for any  $x \in \text{WeakAsso}(X)$ .
- $X, Y \in \text{Sch}^{\text{int}}$  and  $f$  is dominant.
- $f$  is flat.

$\lrcorner$

*Proof:* 2 is a special case of 1 as a reduced scheme has no embedded points.

1 follows from (5.2.5.17).

3 is easy.  $\square$

**Prop. (7.8.1.9) [Effective Cartier Divisors].** If  $X \in \text{NSch}_{\text{cpct}}$  with an ample invertible sheaf  $\mathcal{L}$ , then any line bundle in  $\text{Pic}(X)$  is isomorphism to  $\mathcal{O}_X(D - D')$  for some effective Cartier divisors  $D, D'$  on  $X$ .  $\lrcorner$

*Proof:* Cf. [Sta]0AYM.  $\square$

### Relative Effective Cartier Divisors

**Def. (7.8.1.10) [Relative (Effective) Cartier Divisors].** For  $X/S \in \text{Sch}$ , a **relative effective (Cartier) divisor** on  $X/S$  is an effective Cartier divisor  $D$  on  $X$  (7.8.1.1) that is flat over  $S$ .  $\lrcorner$

**Prop. (7.8.1.11).** If  $S \in \text{NSch}$  and  $X \in \text{Var}/S$ , and  $D$  is a relative Cartier divisor on  $X$ , then  $f_*(\mathcal{O}_D)$  is flat and locally free on  $S$ .  $\lrcorner$

*Proof:* Firstly  $f_*(\mathcal{O}_D)$  is f.p. by (7.12.5.2) and the fact  $R^1 f_* \mathcal{O}_X$  is locally free on  $S$ ?, and then it is locally free by (7.6.2.5).  $\square$

**Prop. (7.8.1.12) [Fibral Criterion of Relative Effective Divisor].** Let  $f : X \rightarrow S \in \text{NSch}$  be locally of f.t.,  $D \subset X$  be a closed subscheme and  $x \in D$ ,  $s = f(x)$ , then the following are equivalent:

- $D$  is a relative effective divisor at a nbhd of  $x$ .
- $X$  and  $D$  are flat over  $S$  at  $x$ , and the fiber  $D_s$  is a Cartier divisor on  $X_s$  at  $x$ .
- $X$  is flat over  $S$  at  $x$ , and  $D$  is cut out by an element that is regular on the fiber  $X_s$ .

In particular, a relative Cartier divisor pair can be regarded as a family of Cartier divisors pairs.  $\lrcorner$

*Proof:* 1  $\rightarrow$  2: Let  $\mathcal{O}_{D,x} = b\mathcal{O}_{X,x}$ , then there is an exact sequence  $0 \rightarrow \mathcal{O}_{X,x} \xrightarrow{b} \mathcal{O}_{X,x} \rightarrow \mathcal{O}_{D,x} \rightarrow 0$ , which induces a long exact sequence

$$\text{Tor}_1^{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x}, k(s)) \rightarrow \text{Tor}_1^{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x}, k(s)) \rightarrow \text{Tor}_1^{\mathcal{O}_{D,x}}(\mathcal{O}_{D,x}, k(s)) \rightarrow \mathcal{O}_{X_s,x} \otimes k(s) \xrightarrow{b} \mathcal{O}_{X_s,x} \otimes k(s).$$

Because  $D$  is flat over  $S$ ,  $\text{Tor}_1^{\mathcal{O}_{D,x}}(\mathcal{O}_{D,x}, k(s)) = 0$ , thus  $b$  is regular in  $\mathcal{O}_{X_s,x}$  and  $D_s$  is an effective divisor on  $X_s$  at  $x$ . Also the map  $\text{Tor}_1^{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x}, k(s)) \rightarrow \text{Tor}_1^{\mathcal{O}_{X_s,x}}(\mathcal{O}_{X_s,x}, k(s))$  is surjective, which is

multiplication by  $b$ , and  $b$  is in the maximal ideal, thus by Nakayama lemma,  $\mathrm{Tor}_1^{\mathcal{O}_{S,s}}(\mathcal{O}_{X,x}, k(s)) = 0$ . Then  $\mathcal{O}_{X,x}$  is  $\mathcal{O}_{S,s}$ -flat by local criterion(5.4.1.16).

2  $\rightarrow$  3: Let  $D$  corresponds to an ideal  $I$  in  $\mathcal{O}_{X,x}$ , and because  $D_s$  is effective divisor on  $X_s$ , there exists  $b \in I$  s.t. the image of  $b$  in  $\mathcal{O}_{X,x} \otimes k(s)$  generate the ideal of  $D_s$ . Then it suffices to show that  $b$  generates  $I$ . Notice there is an exact sequence  $0 \rightarrow I \rightarrow \mathcal{O}_{X,x} \rightarrow \mathcal{O}_{D,x} \rightarrow 0$ , and  $\mathcal{O}_{D,x}$  is flat over  $\mathcal{O}_{S,s}$ , thus  $I \otimes k(s) \rightarrow \mathcal{O}_{X,x} \otimes k(s)$  is injective, and the image is just the ideal of  $D_s$ , thus  $b$  generates  $I \otimes k(s)$ . Thus by Nakayama, it also generates  $I$ .

3  $\rightarrow$  1: The exact sequence  $0 \rightarrow I \rightarrow \mathcal{O}_{X,x} \rightarrow \mathcal{O}_{D,x} \rightarrow 0$  induces the long exact sequence

$$\mathrm{Tor}_1^{\mathcal{O}_{S,s}}(\mathcal{O}_{X,x}, k(s)) \rightarrow \mathrm{Tor}_1^{\mathcal{O}_{S,s}}(\mathcal{O}_{D,x}, k(s)) \rightarrow I \otimes k(s) \rightarrow \mathcal{O}_{X_s,x} \otimes k(s).$$

$I \otimes k(s) \rightarrow \mathcal{O}_{X_s,x} \otimes k(s)$  is injective because its composition with  $\mathcal{O}_{X_s,x} \otimes k(s) \rightarrow I \otimes k(s) \rightarrow \mathcal{O}_{X_s,x} \otimes k(s)$  is multiplication by  $b$ , thus injective as  $b$  is regular on the fiber  $X_s$ . Thus  $\mathrm{Tor}_1^{\mathcal{O}_{S,s}}(\mathcal{O}_{D,x}, k(s)) = 0$ , thus  $D$  is flat at  $x$  by local criterion(5.4.1.16). And also  $I$  is flat over  $\mathcal{O}_{S,s}$ . Notice  $b : \mathcal{O}_{X,x} \rightarrow I$  is isomorphism after tensoring  $k(x)$  and  $I$  is flat, so  $\ker(b) = 0$  by Nakayama, and  $b$  is regular.  $\square$

**Cor. (7.8.1.13).** If  $D, E$  are relative effective Cartier divisors on  $X/S$ , then  $D + E$  is also a relative Cartier divisor on  $X/S$ .  $\lrcorner$

*Proof:* This follows from item3 of(7.8.1.12).  $\square$

**Prop. (7.8.1.14) [Pullback of Divisors].** For  $S \in \mathrm{Sch}$ ,  $X \in \mathrm{Sch}/S$  and  $Z$  an relative Cartier divisor on  $X/S$ , then for any  $T \in \mathrm{Sch}/S$ ,  $Z_T$  is a relative Cartier divisor on  $X_T/T$ .

And if  $f : X' \rightarrow X$  is a flat morphism over  $S$ , then  $f^*Z$  is also a relative Cartier divisor on  $X'/S$ .  $\lrcorner$

*Proof:* If  $0 \rightarrow \mathcal{O}_X \xrightarrow{\mathcal{I}} \mathcal{O}_X \rightarrow i_*\mathcal{O}_Z \rightarrow 0$ , then  $0 \rightarrow \mathcal{O}_{X_T} \xrightarrow{\mathcal{I}_T} \mathcal{O}_{X_T} \rightarrow i_*\mathcal{O}_{Z_T} \rightarrow 0$  because  $\mathcal{O}_Z$  is flat over  $\mathcal{O}_S$ , so  $\mathcal{I}_T$  is also an effective divisor on  $X_T/T$ .

The second case is similar.  $\square$

**Def. (7.8.1.15) [Degree of Relative Cartier Divisors].** For  $X/S \in \mathrm{Sch}$  where  $S$  is connected, if  $D$  is a relative Cartier divisor on  $X$ , the **degree of relative Cartier divisor** of  $D$  is defined to be  $\deg(D) = \deg(\mathcal{L}(D)) = \mathrm{rank}_S f_*(\mathcal{O}_D)$ (7.8.1.11).  $\lrcorner$

### Normal-Crossing Divisors

**Def. (7.8.1.16) [Strictly Normal-Crossing Divisors].** For  $X \in \mathrm{NSch}$ , a **strictly normal-crossing divisor** on  $X$  is a effective Cartier divisor  $D \in \mathrm{Cart}^{\mathrm{eff}}(X)$ (7.8.1.1) s.t. for any  $p \in D$ ,  $\mathcal{O}_{X,p}$  is regular, and there exists a regular system of parameters  $x_1, \dots, x_d \in \mathfrak{m}_p$  s.t.  $D$  is cut out by  $x_1, \dots, x_r$  in  $\mathcal{O}_{X,p}$ ?  $\lrcorner$

**Def. (7.8.1.17) [Normal-Crossing Divisors].** For  $X \in \mathrm{NSch}$ , a **normal-crossing divisor** on  $X$  is an effective Cartier divisor  $D \in \mathrm{Cart}^{\mathrm{eff}}(X)$ (7.8.1.1) s.t. for each  $p \in D$ , there exists an étale nbhd  $f : (U, u) \rightarrow (X, p)$ (7.6.6.11) s.t.  $f^*D \in \mathrm{Cart}^{\mathrm{eff}}(U)$ (7.8.1.8) is a strictly normal-crossing divisor on  $U$ (7.8.1.16).  $\lrcorner$

**Prop. (7.8.1.18) [Characterizing Strictly Normal-Crossing Divisors].** For  $X \in \mathrm{NSch}$  and  $D \in \mathrm{Cart}^{\mathrm{eff}}(X)$ , let  $\{D_i\}_{i \in I}$  be the set of irreducible components of  $D$  viewed as reduced closed subschemes of  $X$ , then the following are equivalent:

- $D$  is a strictly normal-crossing divisor.

- $D$  is reduced and  $\cap_{j \in J} D_j$  is a regular scheme whose irreducible components has codimension  $\#J$  in  $X$ , for each  $J \subset I$ .

⌋

*Proof:* Cf. [Sta]0BIA.?

□

**Prop. (7.8.1.19) [Smooth Pullback of Normal-Crossing Divisors].** If  $f : X \rightarrow Y \in \mathbf{NSch}$  is smooth and  $D \in \mathbf{Cart}^{\text{eff}}(Y)$  is a (strictly)normal-crossing divisor (7.8.1.16)(7.8.1.17), then  $f^*D \in \mathbf{Cart}^{\text{eff}}(X)$  (7.8.1.8) is also a (strictly)normal-crossing divisor on  $X$ .

⌋

*Proof:* It suffices to prove for strictly normal-crossing divisors. For this, Cf. [Sta]0CBP? □

**Prop. (7.8.1.20) [Characterizing Normal-Crossing Divisors].** For  $X \in \mathbf{NSch}$  and a closed subscheme  $D \subset X$ , the following are equivalent:

- $D$  is a normal-crossing divisor on  $X$ .
- $D$  is reduced, the normalization  $D^\nu \rightarrow D$  is unramified, and for each  $n \in \mathbb{Z}_+$ , the scheme

$$Z_n = D^\nu \times_D \times \dots \times D^\nu \setminus \{(p_1, \dots, p_n) | \exists (i \neq j) p_i = p_j\}$$

is regular, and the morphism  $Z_n \rightarrow X$  is a local complete intersection morphism whose conormal sheaf is locally free of rank  $n$ ?

⌋

*Proof:* Cf. [Sta]0CBR.

□

**Prop. (7.8.1.21).** If  $X \in \mathbf{NSch}$  is J-2 or Nagata, then for a closed subscheme  $D \subset X$ ,  $D$  is a normal-crossing divisor on  $X$  (7.8.1.17) iff the pullback of  $D$  to  $\text{Spec } \mathcal{O}_{X,p}^{\text{sh}} \rightarrow X$  is a strictly normal-crossing divisor on  $\text{Spec } \mathcal{O}_{X,p}^{\text{sh}}$ , for each  $p \in D$ .

⌋

*Proof:* Cf. [Sta]0CBS.

□

## 2 Blowing-Ups

**Def. (7.8.2.1) [Blowing-Ups].** For  $X \in \mathbf{Sch}$  and a closed subscheme  $Z \subset X$ , the **blowingup** of  $X$  along  $Z$  is defined as the map  $\beta : \mathbf{Proj}_X(\oplus_{d \geq 0} \mathcal{I}(Z)^d) \rightarrow X$ , denoted by  $\mathbf{Bl}_Z(X)$ .

$\mathbf{E}_Z X \triangleq \beta^{-1}(Z)$  is an effective Cartier divisor of  $\mathbf{Bl}_Z(X)$  (7.8.1.1), called the **exceptional divisor**, the line bundle  $\mathcal{O}_{\tilde{X}}(E)$  associated with it is called the **exceptional invertible sheaf**.

Then  $\mathbf{Bl}_Z(X)$  has the universal property that any  $Y \rightarrow X \in \mathbf{Sch}$  that pulls back  $Z$  to an effective Cartier divisor in  $Y$  uniquely factors through  $\mathbf{Bl}_Z X$ .

Blowing-ups commute with flat base change, by universal property and (7.8.1.2)(7.8.1.2). ⌋

*Proof:* Notice first an effective divisor is equivalent to an invertible sheaf of ideal. And any morphism  $Z \rightarrow X$  pulls back  $I$  to the image of  $f^{-1}I \otimes_{\mathcal{O}_X} \mathcal{O}_Z \rightarrow f^{-1}I \cdot \mathcal{O}_Z$ . This is just  $\mathcal{O}(1)$  on  $\mathbf{Bl}_Z X$  so invertible.

For the construction, the local uniqueness will implies the existence. Notice locally  $\tilde{X}_I$  is projective over  $X$ . Now because the  $Z \rightarrow X$  pulls back  $I$  to an invertible sheaf and it is generated by  $f^{-1}(a_i)$ , we use?? to get another  $Z \rightarrow \text{Proj}_X^n$  and it factors through the closed subscheme  $\tilde{X}_I$ . If there is another morphism  $g$ , then  $f^{-1}I \cdot \mathcal{O}_Z = g^{-1}(\pi^{-1}I \cdot \mathcal{O}_{\tilde{X}_I}) \cdot \mathcal{O}_Z = g^{-1}(\mathcal{O}_{\tilde{X}_I}) \cdot \mathcal{O}_Z$  surjective, and a surjective morphism between two invertible sheaf is an isomorphism, and they are both ideal sheaves, thus is the same, so this morphism is unique as it is determined by the map on  $\mathcal{O}_X$ ?? □

**Cor. (7.8.2.2) [Properties of Blowing-Ups].** Situation as in (7.8.2.1),

- If  $Z$  is itself an effective Cartier divisor in  $X$  (7.8.1.1), then the  $\mathbf{Bl}_Z(X) = X$ .
- If  $U \subset X$  is an open subscheme, then  $\mathbf{Bl}_Z(U) = \beta^{-1}(U) \subset \mathbf{Bl}_Z(X)$ . In particular, blowing-up is a local construction.
- If  $Z = X$ , then  $\mathbf{Bl}_X(X) = \emptyset$ .
- $\beta : \beta^{-1}(X \setminus Z) \rightarrow X \setminus Z$  is an isomorphism.
- If  $X$  is reduced, then  $\mathbf{Bl}_Z(X)$  is also reduced.
- If  $X$  is irreducible and  $Z$  doesn't contain the generic point of  $X$ , then  $\mathbf{Bl}_Z(X)$  is irreducible.

┘

*Proof:* 1: By universal property.

2: open immersion are flat.

3: By universal property.

4: By item2 and 3.

5: ?

6: ?

□

**Cor. (7.8.2.3).** Situation as in (7.8.2.1),  $\beta : \mathbf{Bl}_Y(X) \rightarrow X$  is birational, proper thus surjective. If  $X$  is a (complete) variety, then so does  $\mathbf{Bl}_Y(X)$ . ┘

**Def. (7.8.2.4) [Strict Transforms].** Let  $Z \subset X$  is a closed subscheme and  $f : Y \rightarrow X$  be a morphism, then  $Y \times_X \mathbf{Bl}_Z X$  is called the **total transform** of  $Y$ , and the **strict transform** of  $Y$  is the schematic closure of  $Y \times_X \mathbf{Bl}_Z X \setminus Y \times_X E_Z X$  in  $Y \times_X \mathbf{Bl}_Z X$ , or equivalently the closed subscheme of  $Y \times_X \mathbf{Bl}_Z X$  cut out by the  $\mathcal{Q}_{\text{co}}$  ideal of sections supported on  $Y \times_X E_Z Y$ . ┘

**Prop. (7.8.2.5) [Strict Transformation].** Let  $Z \subset X$  is a closed subscheme and  $f : Y \rightarrow X$  be a morphism, then the strict transform of  $Y$  is the blowing-up of  $Y$  at the closed subscheme  $f^{-1}Z$ . ┘

*Proof:* Cf. [Sta]080E. ? (Recall the definition of fiber product, we only need to check for  $Z, X$  affine and glue. For this, check  $B \otimes_A (\oplus I^d) \rightarrow \oplus (IB)^d$  defines the fiber map). □

**Prop. (7.8.2.6).** If  $X$  is  $H$ -(quasi-)projective, then so does  $\tilde{X}_I$  and  $\pi$  is  $H$ -projective (7.5.2.19). And any birational projective morphism from another variety  $Z$  to  $X$  comes from a blowing-up. ┘

*Proof:* Cf. [Hartshorne P166]. □

**Prop. (7.8.2.7) [Exceptional Divisor].** There are canonical coordinates near  $E$ : let  $\tilde{U}_i$  be  $\tilde{U} - \{(l_i = 0)\}$ , then endow  $\tilde{U}_i$  with the coordinate  $z(i) = (l_j/l_i, \dots, z_i, \dots, l_n/l_i)$ , it is holomorphic to  $\mathbb{C}^n$ .  $\pi$  in this coordinate is written as  $(z(1), \dots, z(n)) \mapsto (z(i)z(1), \dots, z(i), \dots, z(n)z(i))$ .

The transition function can be written, it is

$$\varphi_j \circ \varphi_i^{-1}((z(i)_1, \dots, z(i)_n)) = \left( \frac{z(i)_1}{z(i)_j}, \dots, \frac{1}{z(i)_j}, \dots, z(i)_i z(i)_j, \dots, \frac{z(i)_n}{z(i)_j} \right).$$

Notice it is somewhat tricky because it has two different coordinates.

The defining function of  $E$  in this coordinate is  $(z(i)) = (z_i)$ . So the line bundle  $\mathcal{O}_{\tilde{X}}(E)$  has transition function  $g_{ij} = z(i)/z(j)$ , and it can be thought of as the line bundle that has line  $[l]$  at the point  $(z, [l]) \in \tilde{U}$ . So it is kind of tautological, in fact its restriction on  $E \cong \mathbb{CP}^{n-1}$  is just the tautological line bundle. ┘

**Prop. (7.8.2.8).** Situation as in (7.8.2.1), the canonical line bundle  $\mathcal{K}_{\mathbf{Bl}_Y(X)} = \pi^* \mathcal{K}_X + (\dim X - 1) \mathbf{E}_Y(X)$ .  
 $\perp$

*Proof:* Away from  $E$ , the  $\pi$  is a holomorphism, so It suffices to compare the two transition function of the two canonical maps near  $E$  using the coordinates in (7.8.2.7), with the local section given by  $dz_1 \wedge \dots \wedge dz_n$  and  $dz(i)_1 \wedge \dots \wedge dz(i)_n$  respectively. On  $\tilde{U}_i$ , locally  $dz_1 \wedge \dots \wedge dz_n$  is pulled by  $\pi^*$  to the trivial bundle on  $U'$ , and by calculation,  $dz(j)_1 \wedge \dots \wedge dz(j)_n = z(i)_j^{n-1} dz(i)_1 \wedge \dots \wedge dz(i)_n$ , so  $\mathcal{K}_{\tilde{X}} - (n-2)E$  has a global section  $z(i)_i^{n-1} dz(i)_1 \wedge \dots \wedge dz(i)_n$ , so it is also trivial on  $\tilde{U}$ , so  $\mathcal{K}_{\tilde{X}} = \pi^* \mathcal{K}_X + (n-1)E$  is true.  $\square$

### Blowing up along a regular variety

**Prop. (7.8.2.9) [Blowing-ups along Regular Immersions].** If  $X$  is a regular variety over  $k$  and  $Y$  is a regular closed subvariety, then  $\mathbf{Bl}_Y(X)$  is also regular (7.8.2.1), and  $\mathbf{E}_Y(X) \subset \mathbf{Bl}_Y(X)$  is locally principal. In fact,  $\mathbf{E}_Y(X) \rightarrow Y$  is isomorphic to  $\mathbf{Proj}(\mathcal{I}(Y)/\mathcal{I}(Y)^2)$ , and the normal sheaf  $\mathcal{N}_{\mathbf{E}_Y(X)} \mathbf{Bl}_Y(X) \cong \mathcal{O}_{\mathbf{Proj}(\mathcal{I}_Y/\mathcal{I}_Y^2)}(-1)$ .  
 $\perp$

*Proof:* (Imagine the blowing up of  $\mathbb{A}^2$  along  $\{0\}$ ).  $X' \cong \mathbf{Proj} \oplus \mathcal{I}^d$  and  $Y' \cong \mathbf{Proj} \oplus \mathcal{I}^d / \mathcal{I}^{d+1}$ . Then since  $Y$  is regular, (5.3.4.17) tells us  $\mathcal{I}$  is locally generated by a regular sequence and (5.3.4.16) tells  $Y' = \mathbb{P}(\mathcal{I}/\mathcal{I}^2)$ .  $Y'$  is regular by (5.3.5.19), and then (5.3.5.25) shows that  $X'$  is regular also. For the normal sheaf, the defining sheaf  $\mathcal{I}' = \mathcal{O}_{X'}$ , and then  $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{O}_Y(1)$ , thus  $\mathcal{N}_{Y'/X'} \cong \mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)}(-1)$ .  $\square$

**Prop. (7.8.2.10) [Picard Group of Blowing-Ups].** In a blowing up along a regular variety of codimension  $r \geq 2$ , There is an isomorphism  $\mathrm{Pic} X' \cong \mathrm{Pic} X \oplus \mathbb{Z}$  induced by the Weil divisor exact sequence of  $Y' \subset X'$ . This is because  $r \geq 2$  and (7.8.2.2).  
 $\perp$

We also have  $\omega_{X'} = f^* \omega_X \otimes \mathcal{L}((r-1)Y')$  because  $\mathcal{L}(Y') = \mathcal{O}(-1)$  and  $\omega_{Y'} \cong \omega_X \otimes \mathcal{L}(D) \otimes \mathcal{O}_Y$  by (7.12.1.17), so it suffice to prove  $\omega_{Y'} \cong f^* \omega_X \otimes \mathcal{O}_{Y'}(-r)$ . For this, notice for a closed pt of  $Y$ , the fiber is a  $\mathbb{P}^{r-1}$  because  $\mathcal{I}/\mathcal{I}^2$  is locally free of rank  $r$  by (7.12.1.16) and the functoriality of  $\mathcal{O}(1)$ .  $\perp$

## 3 Compatifications

**Def. (7.8.3.1) [Compatifications].** Let  $S \in \mathrm{Sch}$ ,  $X \in \mathrm{Sch}/S$ , then the category of **compactifications of schemes** of  $X/S$  is the category of pairs  $(i, \bar{X})$ , where  $i : X \rightarrow \bar{X}$  is a dominant open immersion and  $\bar{X} \rightarrow S$  is proper. If a compactification of  $X/S$  exists, then  $X/S$  is called a **compactifiable morphism**.  
 $\perp$

**Thm. (7.8.3.2) [Nagata-Compactification].** If  $S \in \mathrm{Sch}_{\mathrm{qcqs}}$ ,  $X \in \mathrm{Sch}^{\mathrm{sep}, \mathrm{ft}}/S$ , then  $X/S$  is compactifiable.  
 $\perp$

*Proof:* Cf. [Sta]0F41.  $\square$

## 4 Limits of Schemes

**Prop. (7.8.4.1) [Noetherian Reduction].** For any  $S \in \mathrm{Sch}_{\mathrm{qcqs}}$ , there exists a direct system  $(S_i, f_{ii'})$  indexed over a set  $I$  that

- $f_{ii'}$  are affine.
- $S_i$  are of f.t. over  $\mathbb{Z}$ .
- $S = \varprojlim_i S_i$ .

┘

*Proof:* Cf. [Sta]01ZA. □

**Prop. (7.8.4.2) [Characterizing Morphisms of F.P.].** Let  $f : X \rightarrow S$  be a morphism of schemes, then the following are equivalent:

- $f$  is locally of finite presentation.
- For any directed inverse system  $(T_i, f_{ii'})$  in  $\mathcal{A}ff_S$ , we have

$$\mathrm{Mor}_S(\varprojlim_i T_i, X) = \varinjlim_i \mathrm{Mor}_S(T_i, X)$$

- For any directed inverse system  $(T_i, f_{ii'})$  in  $\mathrm{Sch}_S$  with  $T_i$  qcqs and  $f_{ii'}$  affine, we have

$$\mathrm{Mor}_S(\varprojlim_i T_i, X) = \varinjlim_i \mathrm{Mor}_S(T_i, X)$$

┘

*Proof:* Cf. [Sta]01ZC. □

**Prop. (7.8.4.3) [Reduction to Finite Presented Morphisms].** Let  $f : X \rightarrow S$  be a morphism of schemes, if  $X$  is qcqs and  $S$  is qs, then  $X = \varprojlim_i X_i$  is a limit of a directed system of schemes  $X_i$  of f.p. over  $S$  with affine morphisms over  $S$ . ┘

*Proof:* Cf. [Sta]09MV. □

**Prop. (7.8.4.4) [Integral and Finite].** Let  $f : X \rightarrow S$  be an integral morphism of schemes with  $S$  qcqs, then  $X = \varprojlim_i X_i$  is a limit of a directed system of schemes  $X_i$  finite of f.p. over  $S$  with affine morphisms over  $S$ . ┘

*Proof:* Consider  $\mathcal{A} = f_* \mathcal{O}_X \in \mathcal{Q}\mathrm{Coh}(\mathcal{O}_S)$ , then  $\mathcal{A} = \varinjlim_i \mathcal{A}_i$  is a filtered colimit of finite and f.p.  $\mathcal{O}_S$ -modules by (7.5.1.13). Then  $X_i = \mathbf{Spec}_S(\mathcal{A}_i)$  satisfies the requirement by (7.2.7.12). □

### Descending Properties

**Prop. (7.8.4.5) [Descending l.p. Schemes].** ┘

### Descending Properties of Morphisms

**Prop. (7.8.4.6) [Descending Properness].** Cf. [Sta]0CNV. ┘

**Prop. (7.8.4.7) [Descending Morphisms between l.p. schemes].** ┘

### Sheaves

## 5 Discriminant and Different

### Trace Elements

**Def. (7.8.5.1) [Traces].** For  $f : X \rightarrow Y \in \mathrm{Sch}$  finite locally free, the **trace of  $f$**  is defined to be

$$\mathrm{tr}_f : f_* \mathcal{O}_X \rightarrow \mathcal{O}_Y.$$



Then  $\mathrm{tr}_f \circ f^\# = [\deg(f)] : \mathcal{O}_Y \rightarrow \mathcal{O}_Y$ . Let the **trace pairing** be

$$Q_f : f_*\mathcal{O}_X \otimes f_*\mathcal{O}_X \rightarrow \mathcal{O}_Y : (s, t) \mapsto \mathrm{tr}_f(st),$$

which is equivalent to a map  $Q_f : f_*\mathcal{O}_X \rightarrow f_*\mathcal{O}_X^\vee$ , so the determinant of which is

$$\det(Q_f) : \wedge(f_*\mathcal{O}_X) \rightarrow \wedge(f_*\mathcal{O}_X)^{-1},$$

a section of the line bundle  $\wedge(f_*\mathcal{O}_X)^{-2}$ . Then the **discriminant of  $f$**  is defined to be the closed subscheme  $D_f$  of  $Y$  cut out by this section  $\det(Q_f)$ .  $\lrcorner$

**Def. (7.8.5.2) [Trace Element of Rings].** Let  $A \rightarrow B$  be a flat quasi-finite map of Noetherian rings,  $[\omega_{B/A}]$  the dualizing module (5.9.8.4), then there exists a unique element  $\tau_{A/B} \in \omega_{B/A}$  s.t. for any Noetherian  $A$ -algebra  $A_1$  s.t.  $B \otimes_A A_1 = C \times D$  with  $C$  finite over  $A_1$ , the image of  $\tau_{B/A}$  in  $[\omega_{C/A_1}]$  is  $\mathrm{tr}_{C/A_1}$ , called the **trace element**.  $\lrcorner$

**Def. (7.8.5.3) [Trace Element of Morphisms].** For  $f : X \rightarrow Y \in \mathrm{NSch}$  be flat quasi-finite, denote  $\tau_{X/Y} \in \Gamma(X, [\omega_{X/Y}])$  the **trace element of  $f$**  affine locally given by the trace element  $\tau_{B/A}$  (7.8.5.2).  $\lrcorner$

**Prop. (7.8.5.4) [Discriminant and Étale Loci].** If  $f : X \rightarrow Y \in \mathrm{Sch}$  is finite locally free, then  $f$  is étale iff  $D_f = \emptyset$ .  $\lrcorner$

*Proof:*  $f$  is flat, so it suffices to check fiberwise. Then this follows from (5.4.7.25).  $\square$

### Differents

**Def. (7.8.5.5) [Kähler Different].** Let  $f : X \rightarrow Y$  be a morphism of schemes locally of f.t., the **Kähler different** of  $f$  is the 0-th fitting ideal  $\mathrm{Fit}_0(\Omega_{X/Y}) \in \mathcal{Q}\mathrm{Coh}(X)$  (7.5.1.21).  $\lrcorner$

**Prop. (7.8.5.6).** Let  $f : X \rightarrow Y$  be a morphism of schemes locally of f.t., then the closed subscheme cut out by the Kähler different is stable under base change.  $\lrcorner$

*Proof:* Cf. [Sta]0BVX.  $\square$

**Prop. (7.8.5.7).** If  $X = \mathrm{Spec} A[X_1, \dots, X_n]/(f_1, \dots, f_m)$ , then the Kähler different of  $X \rightarrow \mathrm{Spec} A$  is just the ideal generated by the Jacobians  $\det\left(\frac{\partial f_{i(j)}}{\partial X_j}\right)_{0 \leq j \leq n}$ .  $\lrcorner$

*Proof:* Because  $\Omega_{X/A}$  has a presentation

$$0 \rightarrow \bigoplus_{0 \leq i \leq m} f_i \xrightarrow{d} \bigoplus_{0 \leq j \leq n} dX_j \rightarrow \Omega_{X/A} \rightarrow 0.$$

$\square$

**Prop. (7.8.5.8) [Kähler Different and Unramified Locus].** Let  $f : X \rightarrow Y$  be a morphism of schemes locally of f.t., then the closed subscheme cut out by the Kähler different contains exactly the points that  $f$  is not unramified.  $\lrcorner$

*Proof:* Cf. [Sta]0C3J.  $\square$

**Def. (7.8.5.9) [Differents].** For  $f : X \rightarrow Y \in \mathrm{NSch}$  flat and quasi-finite, define the **different(schemes)** to be the annihilator of  $[\omega_{X/Y}]/\tau_{X/Y}$ , which is a coherent ideal  $\mathfrak{D}_f \subset \mathcal{O}_X$ .  $\lrcorner$



**Prop. (7.8.5.10) [Differents and Étale(Unramified) Locus].** Let  $f : X \rightarrow Y$  be a flat quasi-finite morphism between locally Noetherian schemes, then the closed subscheme of  $X$  cut out by the different  $\mathfrak{D}_f$  contains exactly the set of points that  $f$  is not étale(unramified).  $\lrcorner$

*Proof:* Cf. [Sta]0BW9.  $\square$

**Prop. (7.8.5.11).** If  $f : X \rightarrow Y$  is a quasi-finite syntomic morphism between locally Noetherian schemes, then the different  $\mathfrak{D}_f$  equals the Kähler different of  $f$  (7.8.5.5).  $\lrcorner$

*Proof:* Cf. [Sta]0BWG.  $\square$

**Prop. (7.8.5.12) [Differents for Smooth Schemes].** Let  $S$  be a locally Noetherian scheme and  $X, Y$  be smooth scheme of relative dimension  $n$  over  $S$ , and  $f : X \rightarrow Y$  is a quasi-finite morphism, then  $f$  is syntomic, and the closed subscheme  $R$  cut out by the different  $\mathfrak{D}_f$  of  $f$  is the locally principal vanishing locus of

$$\wedge^n(df^*) \in \text{Hom}(f^*\Omega_{Y/S}^n, \Omega_{X/S}^n) = \Gamma(Y, (f^*\Omega_{Y/S}^n)^{-1} \otimes \Omega_{X/S}^n).$$

And if  $f$  is étale at the associated points of  $X$ , then  $R$  is an effective Cartier divisor, and

$$f^*\Omega_{Y/S}^n \otimes \mathcal{O}(R) \cong \Omega_{X/S}^n.$$

$\lrcorner$

*Proof:* Cf. [Sta]0BWJ.  $\square$

## 6 Deformation Theory

References are [Sta]Chap36.

**Def. (7.8.6.1) [Thickenings].** For  $X, X' \in \text{Sch}$ ,  $X'$  is called a **thickening** of  $X$  iff  $X$  is a closed subscheme of  $X'$  that their underlying topological space are the same. Morphisms of thickenings are defined routinely.

A thickening is said to be **thickening of order  $n$**  iff the ideal sheaf  $\mathcal{I}$  satisfies  $\mathcal{I}^{n+1} = 0$ .

Base change and composition of a (order  $n$ )thickening is also a (order  $n$ )thickening, because closed immersion and surjective do.  $\lrcorner$

**Prop. (7.8.6.2).** Any thickening of an affine scheme is also affine.  $\lrcorner$

*Proof:* This is a special case of (7.4.4.36).  $\square$

**Prop. (7.8.6.3) [Picard Group of Thickenings].** Let  $X \subset X'$  be a first order thickening with ideal sheaf  $\mathcal{I}$ , then there is a canonical long exact sequence of Abelian groups:

$$0 \rightarrow H^0(X, \mathcal{I}) \rightarrow H^0(X', \mathcal{O}_{X'}^*) \rightarrow H^0(X, \mathcal{O}_X^*) \rightarrow H^1(X, \mathcal{I}) \rightarrow \text{Pic}(X') \rightarrow \text{Pic}(X) \rightarrow H^2(X, \mathcal{I}) \rightarrow \dots$$

$\lrcorner$

*Proof:* This follows from taking cohomology of the exact sequence  $0 \rightarrow \mathcal{I} \rightarrow \mathcal{O}_{X'} \rightarrow i_*\mathcal{O}_X \rightarrow 0$ , where  $\mathcal{I} \rightarrow \mathcal{O}_{X'}$  is given by  $a \mapsto 1 + a$ .  $\square$

**Prop. (7.8.6.4).** Let  $X \subset X'$  be a thickening with ideal sheaf  $\mathcal{I}$  and  $n$  is invertible in  $\mathcal{O}_X$ , then  $\text{Pic}(X)[n] \rightarrow \text{Pic}(X')[n]$  is an isomorphism.  $\lrcorner$

*Proof:* By taking cohomology of the exact sequence  $0 \rightarrow (1 + \mathcal{I})^* \rightarrow \mathcal{O}_{X'} \rightarrow i_*\mathcal{O}_X \rightarrow 0$ , it suffices to show that  $n : (1 + \mathcal{I})^* \rightarrow (1 + \mathcal{I})^*$  is an isomorphism, which is true by (5.1.1.19).  $\square$

**Def. (7.8.6.5) [Infinitesimal Neighbourhood].** Let  $Z \subset U \subset X$  be a closed immersion of the open subscheme  $U$  that  $Z$  corresponds to the ideal  $\mathcal{I}$  on  $U$ , then a  **$n$ -th infinitesimal neighbourhood** of  $Z$  in  $X$  is the closed subscheme of  $U$  corresponding to  $\mathcal{I}^n$ .

The infinitesimal neighbourhood of  $Z$  in  $X$  has the universal property s.t. for any infinitesimal thickening  $T \subset T'$  of order  $n$  over  $X$  and a map  $T \rightarrow Z \in \text{Sch}_X$  extension to a morphism of infinitesimal thickenings  $(T \subset T') \rightarrow (Z \rightarrow Z')$  over  $X$ .  $\lrcorner$

**Def. (7.8.6.6) [Infinitesimal Extension].** For  $k \in \text{Field}$ ,  $X \in \text{Sch}^{\text{ft}}/k$  and  $\mathcal{F} \in \text{Coh}(X)$ , a **infinitesimal extension** of  $X$  by the sheaf  $\mathcal{F}$  is a scheme  $X'$  over  $k$  that has a sheaf of ideals  $\mathcal{I}$  that  $\mathcal{I}^2 = 0$  and  $(X', \mathcal{O}_{X'}/\mathcal{I}) \cong (X, \mathcal{O}_X)$ , and moreover,  $\mathcal{I}$  with the  $\mathcal{O}_X$ -structure is isomorphic to  $\mathcal{F}$ .

There is a trivial extension, that is  $(X', \mathcal{O}_{X'}) \cong (X, \mathcal{O}_X \oplus \mathcal{F})$ , where the multiplication is  $(a, f)(a', f') = (aa', af' + a'f)$ .  $\lrcorner$

**Def. (7.8.6.7) [Deformation].** Let  $X$  be a scheme algebraic over a field  $k$ , an **infinitesimal deformation** of  $X$  is a scheme  $X'$  flat over  $D = k[t]/(t^2)$  that  $X' \otimes_D k = X$ . A infinitesimal deformation is a first order thickening, by (7.8.6.1).

If  $Y$  is a closed subscheme of  $X$ , then we define the **infinitesimal deformation of  $Y$  in  $X$**  to be a closed subscheme  $Y' \subset X \otimes_k D$  which is flat over  $D$  and  $Y' \otimes_D k = Y$ .

A scheme algebraic over a field  $k$  is called **rigid** if it has no infinitesimal deformations.  $\lrcorner$

**Prop. (7.8.6.8) [Affine Case].** Any thickening of an affine scheme is affine. (Immediate from (7.4.4.36)).  $\lrcorner$

**Prop. (7.8.6.9).** Let  $X$  be a nonsingular variety over an alg.closed field  $k$ , infinitesimal deformation of  $X$  is the same as an infinitesimal extension of  $X$  by the sheaf  $\mathcal{O}_X$ . Thus we get the set of infinitesimal deformations of  $X$  is parametrized by  $H^1(X, \mathcal{T}_X)$ , by (7.8.6.11) below.  $\lrcorner$

*Proof:* For an infinitesimal deformation, tensoring  $\mathcal{O}_{X'}$  with the exact sequence  $0 \rightarrow k \xrightarrow{t} D \rightarrow k \rightarrow 0$ , we get (by flatness)

$$0 \rightarrow \mathcal{O}_X \xrightarrow{t} \mathcal{O}_{X'} \rightarrow \mathcal{O}_X \rightarrow 0,$$

, and conversely, an extension is locally free (because it is f.g. so flat over  $D$  is equivalent to free).  $\square$

**Prop. (7.8.6.10).** If  $X$  is an affine regular scheme algebraic over an alg.closed field  $k$ , then any extension by coherent sheaf is trivial.  $\lrcorner$

*Proof:* For any infinitesimal extension, the morphism  $X \rightarrow X'$  is a closed immersion and surjection, so  $X'$  is also affine by (7.8.6.8),  $= \text{Spec } A'$ . Now the rest follows from ??  $\square$

**Cor. (7.8.6.11) [Infinitesimal Extension and Cohomology].** Let  $X$  be a nonsingular variety over an alg.closed field  $k$ , then the set of infinitesimal extensions by a coherent sheaf  $\mathcal{F}$  is parametrized by  $H^1(X, \mathcal{F} \otimes \mathcal{T}_X)$ .

If  $Y$  is a closed subscheme of  $X$ , then the set of infinitesimal deformation of  $Y$  in  $X$  is parametrized by  $H^0(Y, \mathcal{N}_{Y/X})$ .  $\lrcorner$

*Proof:* By the proposition, we know that an infinitesimal extension is locally isomorphic to  $(U, \mathcal{O}_X(U) \otimes \mathcal{F}(U))$ , by a section  $\mathcal{F}(U) \rightarrow \mathcal{O}_{X'}(U)$ .

But there is a twist, because there can be different sections. But the different sections are different at a  $\text{Hom}_{\mathcal{O}_X(U)}(\Omega_{\mathcal{O}_X(U)/k}, \mathcal{F}(U)) = (\mathcal{T} \otimes \mathcal{F})(U)$ . These forms a Čech cocycle for  $\mathcal{F} \otimes \mathcal{T}_X$ , and the converse is also true. Finally, use the fact that  $X$  is separated so Čech and sheaf cohomology coincide.

For the subscheme, ?

□

### Formal Properties

**Def. (7.8.6.12) [Formal Properties of Morphisms].** For  $f; X \rightarrow S \in \text{Sch}$ ,  $f$  is called a **formally unramified/smooth/étale** if for any first order thickening of affine schemes  $T \rightarrow T'$  and a morphism  $(T \rightarrow T') \rightarrow (X \rightarrow S)$ , there exists at most one/exists one/exists exactly one lifting  $T' \rightarrow X$ .

Formally unramified/smooth/étale morphisms are stable under base change and compositions.  $\lrcorner$

**Prop. (7.8.6.13).** For  $f : X \rightarrow S \in \mathcal{A}\text{ff}$ ,  $f$  is formally unramified/smooth/étale iff  $\Gamma(S) \rightarrow \Gamma(X)$  is formally unramified/smooth/étale (5.4.6.1)(5.4.5.1)(5.4.7.1).  $\lrcorner$

**Cor. (7.8.6.14).**

- $f : X \rightarrow S \in \text{Sch}$  is formally unramified iff  $\Omega_{X/S} = 0$ , by (5.4.6.1).  $\lrcorner$

**Prop. (7.8.6.15).**

- A morphism is (G-)unramified iff it is formally unramified and locally of f.t.(f.p.).
- A morphism is étale iff it is formally étale and locally of f.p.
- A morphism is smooth iff it is formally smooth and locally of f.p.  $\lrcorner$

*Proof:* 1: This follows from (7.6.5.1).

2:

3:  $\square$

## 7.9 Derived Categories of Coherent Sheaves and Birational Geometry

References are [Har77, B-O02], [Sta]Chap56 and references therein, [Huybrechts, Fourier-Mukai transforms in algebraic geometry].

**Notation (7.9.0.1).**

- Let  $S \in \text{Sch}$ .

┘

### 1 $\mathcal{D}_{\text{Qcoh}}(X)$

**Def. (7.9.1.1) [Notations].** Denote  $\mathcal{D}_{\text{Qcoh}}^*(X) = \mathcal{D}_{\text{Qcoh}(X)}^*(\text{Mod}(\mathcal{O}_X))$  (4.10.1.8). There is a natural functor  $\mathcal{D}^*(\text{Qcoh}(X)) \rightarrow \mathcal{D}_{\text{Qcoh}}^*(X)$ .

Denote  $\mathcal{D}_{\text{coh}}^*(X) = \mathcal{D}_{\text{coh}(X)}^*(\text{Mod}(\mathcal{O}_X))$  (4.10.1.8). There is a natural functor  $D^*(\text{Coh}(X)) \rightarrow D_{\text{coh}}^*(X)$ .

By (7.3.1.5), these notions are affine local.

┘

**Prop. (7.9.1.2) [Qco Direct Sum].** Direct sum exists in  $D_{\text{Qcoh}}(X)$ , and equals that in  $D(X)$ .

┘

*Proof:* By (4.10.1.15), direct sum exists in  $D(X)$  and are given by term-wise direct sums. Notice the direct sum of elements in  $D_{\text{Qcoh}}(X)$  is also in  $D_{\text{Qcoh}}(X)$  as direct sums are exact functor and  $\text{Qcoh}(X)$  is stable under direct sums.  $\square$

**Prop. (7.9.1.3) [Affine Case].** Let  $X = \text{Spec } A$ , there is a natural equivalence  $\sim: D(A) \rightarrow D(\text{Qcoh}(X))$ . Then the functors  $R\Gamma(X, -) : D_{\text{Qcoh}}(X) \rightarrow D(A)$  is quasi-inverse to the inclusion functor  $D^*(\text{Qcoh}(X)) \rightarrow D_{\text{Qcoh}}^*(X)$  (7.9.1.1), and they are both isomorphism of triangulated categories.

┘

*Proof:* Cf. [Sta]06Z0.

$\square$

**Prop. (7.9.1.4) [Qco Derived Pullbacks].** Let  $f : X \rightarrow S \in \text{Sch}$  be qcqs, then  $Lf^*$  maps  $D_{\text{Qcoh}}(X)$  into  $D_{\text{Qcoh}}(S)$ , and affine locally it is just the derived tensor functor  $- \otimes_A^L B : D(A) \rightarrow D(B)$  via identifications in (7.9.1.3), by (7.3.3.14)(7.3.3.15).

┘

**Prop. (7.9.1.5) [Qco Derived Tensor Products].** For  $X \in \text{Sch}$ ,  $D_{\text{Qcoh}}(X)$  is stable under derived tensor products, and affine locally it is just the derived tensor product of rings via identifications in (7.9.1.3), by (7.3.3.6)(7.3.3.15).

┘

**Prop. (7.9.1.6) [Qco Derived Pushforward].** Let  $f : X \rightarrow S$  be a qcqs morphism of schemes, then  $Rf_*$  maps  $D_{\text{Qcoh}}(X)$  into  $D_{\text{Qcoh}}(S)$ . And if  $f$  is affine, affine locally it is just the restriction  $D(B) \rightarrow D(A)$  via identifications in (7.9.1.3), by (7.3.1.9).

Moreover, if  $S$  is qc, then there exists a  $N \in \mathbb{Z}$  s.t.  $Rf_*(D_{\text{Qcoh}}^{\leq m}(X)) \subset D_{\text{Qcoh}}^{\leq m+N}(S)$ , and this  $N$  is invariant under base change.

┘

*Proof:* Cf. [Sta]08D5. ?

$\square$

**Prop. (7.9.1.7).** For  $f : X \rightarrow Y$  qcqs,  $Rf_* : D_{\text{Qcoh}}(X) \rightarrow D_{\text{Qcoh}}(Y)$  preserves direct sums.

┘

*Proof:* Cf. [Sta]08DZ.

$\square$

**Prop. (7.9.1.8) [Projection Formula].** Let  $f : X \rightarrow Y$  be a qcqs morphism of schemes and  $E \in D_{\mathcal{Q}\text{Coh}}(X)$ ,  $K \in D_{\mathcal{Q}\text{Coh}}(Y)$ , then the projection map (7.3.3.19)

$$Rf_*(E) \otimes_{\mathcal{O}_Y}^L K \rightarrow Rf_*(E \otimes_{\mathcal{O}_Y}^L Lf^*K)$$

is an isomorphism. ┘

*Proof:* Cf. [Sta]08EU. □

### Pseudo-Coherent and Perfect Complexes

**Prop. (7.9.1.9) [Affine Case].** If  $X = \text{Spec } A$ ,  $M \in D(A)$  and  $E$  is the corresponding element in  $D(X)$ , then

- $E$  is an  $(m-)$ pseudo-coherent object in  $D(X)$  iff  $M$  is an  $(m-)$ pseudo-coherent object in  $D(A)$  (5.9.5.9).
- $E$  has Tor-amplitude in  $[a, b]$  iff  $M$  has Tor amplitude in  $[a, b]$ .
- $E$  is a perfect object in  $D(X)$  iff  $M$  is a perfect object in  $D(A)$  (5.9.5.3). ┘

*Proof:* Cf. [Sta]08E7, 08EB, 08E9,. □

**Prop. (7.9.1.10) [Noetherian Case].** Let  $X \in \text{NSch}_{\text{qc}}$  and  $\mathcal{E} \in \mathcal{D}_{\mathcal{Q}\text{Coh}}(X)$ , then the following are equivalent:

- $\mathcal{E}$  is  $m$ -pseudo-coherent.
- $\mathcal{E} \in \mathcal{D}_{\mathcal{Q}\text{Coh}}^-(X)$  and  $H^i(\mathcal{E}) \in \mathcal{Coh}(X)$  for  $i \geq m$ .

In particular,  $E$  is pseudo-coherent iff  $E \in \mathcal{D}_{\mathcal{Coh}}^-(X)$ . ┘

*Proof:* Cf. [Sta]08E8. □

**Prop. (7.9.1.11) [Ext].** Let  $X \in \text{Sch}$ .

1. If  $X = \text{Spec } A$  and  $K, L \in D(A)$ , then  $\mathcal{E}xt^n(\widetilde{K}, \widetilde{L})$  is the sheaf extending the sheaf on the basis  $D(f) \mapsto \text{Ext}_{A_f}^n(K_f, L_f)$ .
2. If  $X = \text{Spec } A$  and  $K, L \in D(A)$ , then  $R\mathcal{H}om(\widetilde{K}, \widetilde{L}) = (R\text{Hom}_A(K, L))^\sim$  iff  $K$  is pseudo-coherent and  $L \in D^+(A)$ , or  $K$  is perfect.
3. If  $\mathcal{L} \in D_{\mathcal{Q}\text{Coh}}(X)$  and  $\mathcal{K} \in D(X)$  perfect, then  $R\mathcal{H}om(\mathcal{K}, \mathcal{L}) \in D_{\mathcal{Q}\text{Coh}}(X)$ .
4. If  $\mathcal{L} \in D_{\mathcal{Q}\text{Coh}}^+(X)$  and  $\mathcal{K} \in D(X)$  pseudo-coherent,  $R\mathcal{H}om(\mathcal{K}, \mathcal{L}) \in D_{\mathcal{Q}\text{Coh}}(X)$  and are locally bounded below. ┘

*Proof:* 1: This follows from (7.3.3.27) as

$$H^n(R\mathcal{H}om(X_f, \widetilde{K}, \widetilde{L})) = \text{Hom}_{D(\mathcal{O}_{X_f})}(\widetilde{K}_f, \widetilde{L}_f[n]) = \text{Hom}_{D(A_f)}(K_f, L_f[n]) = \text{Ext}^n(K_f, L_f).$$

2: This follows from (7.3.4.22).

3, 4 follows from item2, (7.3.4.22) and locality of cohomology. □

**Prop. (7.9.1.12) [Criterion for Relative Perfectness over Affine Base].** Let  $A \in \mathcal{CAlg}$  and  $X$  be a separated, flat of f.p. scheme over  $A$ ,  $K \in D_{\mathcal{Q}\text{Coh}}(X)$ . If  $R\Gamma(X, E \otimes^L K)$  is perfect in  $D(A)$  for any perfect object  $E \in D(X)$ , then  $K$  is perfect over  $A$ . ┘

*Proof:* Cf. [Sta]0GET. □

$D_{\text{coh}}(X)$ 

**Prop. (7.9.1.13) [ $R\mathcal{H}om$  Preserves  $D_{\text{coh}}(X)$ ].** If  $X \in \text{NSch}$ ,  $L \in D_{\text{coh}}^+(X)$  and  $K \in D_{\text{coh}}^-(X)$ , then  $R\mathcal{H}om(K, L) \in D_{\text{coh}}^+(X)$ .  $\lrcorner$

*Proof:* Cf. [Sta]0D0C.  $\square$

**Prop. (7.9.1.14).** If  $X \in \text{NSch}_{\text{qc}}$ , then there are natural equivalences

$$D^-(\text{Coh}(X)) \cong D_{\text{coh}(X)}^-(\mathcal{Q}\text{Coh}(X)) \cong D_{\text{coh}}^-(X), \quad D^b(\text{Coh}(X)) \cong D_{\text{coh}}^b(X).$$

 $\lrcorner$ 

*Proof:* Cf. [Sta]0FDA, 0FDB.  $\square$

**Lemma (7.9.1.15).** Let  $k \in \text{Field}$  and  $X \in \text{Sch}^{\text{proper}}/k$ , then  $\sum_{i \in \mathbb{Z}} \dim_k \text{Ext}^i(E, K) < \infty$  for any  $E \in D_{\text{perf}}(X)$  and  $K \in D_{\text{coh}}^b(X)$ .  $\lrcorner$

*Proof:* Cf. [Sta]0FYC.  $\square$

**Prop. (7.9.1.16) [Characterizing Coherent Sheaves].** Let  $k \in \text{Field}$  and  $X \in \text{Sch}^{\text{Proj}}/k$ , then for  $K \in D_{\mathcal{Q}\text{coh}}(X)$ , the following are equivalent:

- $K \in D_{\text{coh}}^b(X)$ .
- $\sum_{i \in \mathbb{Z}} \dim_k \text{Ext}^i(E, K) < \infty$  for any  $E \in D_{\text{perf}}(X)$ .

 $\lrcorner$ 

*Proof:*  $1 \rightarrow 2$  follows from (7.9.1.15). For  $2 \rightarrow 1$ , Cf. [Sta]0FYD.  $\square$

**Prop. (7.9.1.17) [Coherent Derived Proper Pushforwards].** Let  $S \in \text{NSch}_{\text{qc}}$  and  $f : X \rightarrow S$  be a proper morphism, then  $Rf_*$  maps  $D_{\text{coh}}^b(X)$  to  $D_{\text{coh}}^b(S)$ .  $\lrcorner$

*Proof:* Cf. [Sta]08E2.  $\square$

**Prop. (7.9.1.18) [Perfect Complexes for Regular Schemes].** If  $X \in \text{NSch}_{\text{qc}}$  is regular of finite dimension, then  $D_{\text{coh}}^b(X) = D_{\text{perf}}(X)$ , by (5.9.5.5) and (7.9.1.9).  $\lrcorner$

 $K_0$  Groups

**Def. (7.9.1.19) [ $K$ -Groups of Schemes].** The **Grothendieck  $K$ -group (Scheme)** of  $X$  is defined to be  $K_0(X) = K_0(D_{\text{perf}}(X))$ . If  $X \in \text{NSch}$ , we also define the **Grothendieck group of coherent sheaves** on  $X$  to be  $K'_0(X) = K_0(\text{Coh}(X))$ .  $\lrcorner$

**Prop. (7.9.1.20).** If  $X \in \text{NSch}_{\text{qc}}$  then

$$K'_0(X) = K_0(\text{Coh}(X)) = K_0(D^b(\text{Coh}(X))) = K_0(D_{\text{coh}}^b(X)).$$

In particular, there is a map  $K_0(X) \rightarrow K'_0(X)$ .  $\lrcorner$

*Proof:* This follows from (4.10.1.13) and (7.9.1.14).  $\square$

**Prop. (7.9.1.21) [ $K_0 \cong K'_0$ ].** If  $X$  is a Noetherian regular of finite dimension, then the map  $K_0(X) \rightarrow K'_0(X)$  (7.9.1.20) is an isomorphism, by (7.9.1.18).  $\lrcorner$

**Prop. (7.9.1.22).** For  $X \in \text{Sch}$ ,  $-\otimes^{\mathbf{L}}-$  defines a ring structure on  $K_0(X)$ , by (11.1.2.2) and (4.10.3.12).  
 $\perp$

**Prop. (7.9.1.23).** If  $X = \text{Spec } R$ , then  $K_0(X) = K_0(R)$ . And if  $R$  is Noetherian, then  $K'_0(X) = K'_0(R)$ .  
 $\perp$

*Proof:* Cf. [Sta]0FDH. □

**Prop. (7.9.1.24) [Push and Pull].** Let  $f : X \rightarrow Y \in \text{NSch}$  be a proper morphism, then there is a map

$$f_* : K'_0(X) \rightarrow K'_0(Y) : [\mathcal{F}] \mapsto \oplus_{i \geq 0} [R^{2i} f_* \mathcal{F}] - \oplus_{i \geq 0} [R^{2i+1} f_* \mathcal{F}],$$

which is definable by (7.9.1.17). And there is an obvious map  $f^* : K'_0(Y) \rightarrow K'_0(X)$ . They satisfy

$$f_*(\alpha \cdot f^* \beta) = f_* \alpha \cdot \beta, \alpha \in K'_0(X), \beta \in K'_0(Y).$$

$\perp$

*Proof:* The first assertion follows from long exact sequence for  $Rf_*$ . The last assertion follows from projection formula (7.9.1.8). □

**Prop. (7.9.1.25) [Lambda Operators].** Let  $X \in \text{Sch}$ , there are functors  $\lambda^r : K_0(\text{Coh}^{\text{free}}(X)) \rightarrow K_0(\text{Coh}^{\text{free}}(X))$  that sends  $[\mathcal{E}]$  to  $[\wedge^r \mathcal{E}]$ , called the **Lambda operators**.  $\perp$

*Proof:* Consider a map  $c : \text{Coh}^{\text{free}}(X) \rightarrow K_0(\text{QCoh}^{\text{free}}(X))[t] : \mathcal{E} \mapsto \sum_{i=0}^{\infty} [\wedge^i \mathcal{E}] t^i$ . It suffices to prove that for any exact sequence  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$ ,  $c(\mathcal{E}) = c(\mathcal{E}')c(\mathcal{E}'')$ , and then take  $\lambda^r$  as

$$c(M) = \sum_{r \geq 0} \lambda^r(M) t^r.$$

To show this, notice that there is a filtration of  $\wedge^r \mathcal{E}$  with quotients

$$\wedge^r \mathcal{E}', \wedge^{r-1}(\mathcal{E}') \otimes \mathcal{E}'', \dots, \wedge^r(\mathcal{E}''),$$

by (7.5.1.26). □

**Prop. (7.9.1.26) [Adam Operators].** Let  $X \in \text{Sch}$ , then there are **Adam operators**

$$\psi^{-1}, \psi^1, \psi^2, : K_0(X) \rightarrow K_0(X)$$

s.t. for  $\mathcal{L} \in \text{Pic}(X)$ ,

$$\psi^{-1}[\mathcal{L}] = [\mathcal{L}^{-1}], \quad \psi^1[\mathcal{L}] = [\mathcal{L}], \quad \psi^2[\mathcal{L}] = [\mathcal{L}^{\otimes 2}].$$

$\perp$

*Proof:* For any  $L \in K_0(X)$ , there is an action of  $\{\pm 1\}$  on  $L \otimes^{\mathbf{L}} L$  by switching factors. Denote  $(L \otimes^{\mathbf{L}} L)^+, (L \otimes^{\mathbf{L}} L)^-$  the fixed and anti-fixed parts ? of  $L \otimes^{\mathbf{L}} L$ , and define  $\psi^2[L] = [(L \otimes^{\mathbf{L}} L)^+] - [(L \otimes^{\mathbf{L}} L)^-]$ . ? □

### Coherator

**Prop. (7.9.1.27)**  $[R\mathcal{Q}_X]$ . ┘

**Prop. (7.9.1.28)**  $[D\mathcal{Q}_X]$ . Let  $X \in \text{Sch}^{\text{qcqs}}$ , then the inclusion functor  $D_{\mathcal{Q}\text{Coh}}(X) \rightarrow D(X)$  has a right adjoint, called the **coherator**, denoted by  $D\mathcal{Q}_X$ . ┘

*Proof:* Use (4.8.4.38). The conditions are satisfied as  $D_{\mathcal{Q}\text{Coh}}(X)$  is compactly generated and has direct sums that is preserved by inclusion by (7.9.1.32)(7.9.1.2). □

**Prop. (7.9.1.29)**. Let  $f : X \rightarrow Y \in \text{Sch}^{\text{qcqs}}$ , then

$$Rf_* \circ D\mathcal{Q}_X = D\mathcal{Q}_Y \circ Rf_* : D(X) \rightarrow D_{\mathcal{Q}\text{Coh}}(Y).$$

┘

*Proof:* They are both right adjoint to  $Lf^* : D_{\mathcal{Q}\text{Coh}}(Y) \rightarrow D(X)$ , as  $Lf^*$  maps  $D_{\mathcal{Q}\text{Coh}}(Y)$  into  $D_{\mathcal{Q}\text{Coh}}(X)$  (7.9.1.4). □

**Prop. (7.9.1.30) [Cohomological Boundedness of  $D\mathcal{Q}_X$ ]**. For  $X \in \text{Sch}^{\text{qcqs}}$ , there exists  $N \in \mathbb{Z}$  s.t. if  $K \in D(X)$  satisfies  $K|_U \in D^{[a,b]}(\mathcal{O}_U)$  for any affine open  $U \subset X$ , then  $D\mathcal{Q}_X(K) \in D_{\mathcal{Q}\text{Coh}}^{[a,b+N]}(X)$ . ┘

*Proof:* Cf. [Sta]0CSA. □

### Compact Generators

**Prop. (7.9.1.31) [Perfect and Compact Objects]**. For  $X \in \text{Sch}^{\text{qcqs}}$ ,  $K \in D(X)$  is perfect iff it is a compact object in  $D_{\mathcal{Q}\text{Coh}}(X)$ . ┘

*Proof:* Cf. [Sta]09M1. □

**Prop. (7.9.1.32) [ $D_{\mathcal{Q}\text{Coh}}(X)$  is Compactly Generated]**. There is a perfect object  $P \subset D_{\mathcal{Q}\text{Coh}}(X)$  that is a generator of  $D_{\mathcal{Q}\text{Coh}}(X)$  (4.8.4.34). ┘

*Proof:* Cf. [Sta]09IS. □

### Resolution Property

**Def. (7.9.1.33) [Resolution Properties]**.  $X \in \text{Sch}$  is said to have **resolution property** iff every  $\mathcal{F} \in \mathcal{Q}\text{Coh}^{\text{ft}}(X)$  is a quotient of a locally free sheaf. ┘

**Prop. (7.9.1.34)**. If  $X$  has an ample invertible sheaf, then  $X$  has the resolution property by (7.5.4.5). In fact, every coherent sheaf is a quotient of a finite direct sum of  $\mathcal{O}_X(-n)$ . ┘

**Prop. (7.9.1.35) [Regular Scheme has Resolution property]**. If  $X$  is qc regular scheme with an affine diagonal, then  $X$  has the resolution property, Cf. [Sta]0F8A<sup>?</sup>. Conversely, if  $X$  is qcqs with the resolution property, then  $X$  has affine diagonal. Cf. [Sta]0F8C. ┘

**Prop. (7.9.1.36) [Kleiman]**. If  $X$  is a qc irreducible and locally factorial scheme with affine diagonal map, then  $X$  has the resolution property. ┘

*Proof:* By (7.8.1.7), we have an basis of the form  $X_s$  for  $s \in \Gamma(X, \mathcal{L})$  for various invertible sheaves, then for any coherent sheaf, it is generated by f.m. sections in  $\Gamma(U_i, \mathcal{F})$  and  $U_i = X_s$  for  $s \in \Gamma(X, \mathcal{L})$ , and for each of them, we can use (7.5.3.5), we can extend these to global sections on  $\Gamma(\mathcal{F} \otimes \mathcal{L}_i^{n_i})$  for  $n_i$  large. Then tensoring  $\mathcal{L}_i^{-n_i}$ , we find a  $\oplus L_i^{-n_i} \rightarrow \mathcal{F}$  surjective. □



**Prop. (7.9.1.37).** When  $X$  has the resolution property,  $\mathcal{E}xt^\bullet(-, \mathcal{G})$  is an universal  $\delta$ -functor for every  $\mathcal{G} \in \mathcal{Q}\mathcal{C}oh(X)$ , because locally free sheaf is adapted to  $\mathcal{E}xt^\bullet(-, \mathcal{G})$  by (7.3.3.25), so we can calculate  $\mathcal{E}xt(\mathcal{F}, \mathcal{G})$  using a finite locally free resolution of  $\mathcal{F}$ .  $\lrcorner$

**Prop. (7.9.1.38).** If  $X$  is a qcqs scheme with the resolution property, then the map  $K_0(\mathcal{V}ect(X)) \rightarrow K_0(X)$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]0FDJ.  $\square$

## 2 Duality for Schemes

References are [Hartshorne Residues and Duality, Hartshorne], [Sta]Chap48 and [Grothendieck Duality and Base Change, Conrad]

### Right Adjoints of Pushforwards

**Prop. (7.9.2.1) [Right Adjoints of Pushforward].** Let  $f : X \rightarrow Y$  be a morphism between qcqs schemes, then the functor  $Rf_* : D_{\mathcal{Q}\mathcal{C}oh}(X) \rightarrow D_{\mathcal{Q}\mathcal{C}oh}(Y)$  has a right adjoint, denoted by  $f^\times$ .  $\lrcorner$

*Proof:* Use (4.8.4.38). The conditions are satisfied as  $D_{\mathcal{Q}\mathcal{C}oh}(X)$  is compactly generated and has direct sums by (7.9.1.32)(7.9.1.2), and  $Rf_*$  preserves direct limits (7.9.1.7).  $\square$

**Prop. (7.9.2.2).** Let  $f : X \rightarrow Y \in \mathcal{S}ch_{qcqs}$ , then  $f^\times$  maps  $D_{\mathcal{Q}\mathcal{C}oh}^+(Y)$  into  $D_{\mathcal{Q}\mathcal{C}oh}^+(X)$ .  $\lrcorner$

*Proof:* This follows from the fact  $Rf_*$  has finite cohomological dimension  $N$  (7.9.1.6) and (4.10.1.18).  $\square$

**Prop. (7.9.2.3).** Let  $f : X \rightarrow Y$  be a morphism of qcqs schemes, then for  $K \in D_{\mathcal{Q}\mathcal{C}oh}(X), L \in D_{\mathcal{Q}\mathcal{C}oh}(Y)$ , there is a canonical map

$$Rf_* R\mathcal{H}om_{\mathcal{O}_X}(L, f^\times K) \rightarrow R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* L, Rf_* f^\times K) \rightarrow R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* L, K),$$

by (7.3.3.31) and adjunction.

Then this map becomes isomorphism after applying the coherator  $D\mathcal{Q}_X$  (7.9.1.28).  $\lrcorner$

*Proof:* By Yoneda lemma, it suffices to show that for any  $M \in D_{\mathcal{Q}\mathcal{C}oh}(Y)$ ,

$$\mathrm{Hom}_Y(M, Rf_* R\mathcal{H}om_{\mathcal{O}_X}(L, f^\times K)) \cong \mathrm{Hom}_Y(M, R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* L, K)) :$$

$$\begin{aligned} \mathrm{Hom}_Y(M, Rf_* R\mathcal{H}om_{\mathcal{O}_X}(L, f^\times K)) &= \mathrm{Hom}_X(Lf^* M, R\mathcal{H}om_{\mathcal{O}_X}(L, f^\times K)) \\ &= \mathrm{Hom}_X(Lf^* M \otimes^L L, f^\times K) \\ (\text{ as } Lf^* M \otimes^L L &\in D_{\mathcal{Q}\mathcal{C}oh}(X) \text{ by (7.9.1.4)(7.9.1.5)}) = \mathrm{Hom}_Y(Rf_*(Lf^* M \otimes^L L), K) \\ &= \mathrm{Hom}_Y(M \otimes Rf_* L, K) \\ &= \mathrm{Hom}_Y(M, R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* L, K)) \end{aligned}$$

$\square$

**Cor. (7.9.2.4).** By (7.9.1.6), this adjointness is true without coheration if both  $R\mathcal{H}om_{\mathcal{O}_X}(L, f^\times K)$  and  $R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* L, K)$  are in  $D_{\mathcal{Q}\mathcal{C}oh}(Y)$ . This is the case if  $L, Rf_* L$  are perfect or  $K \in D_{\mathcal{Q}\mathcal{C}oh}^+(Y)$  and  $L, Rf_* L$  are pseudo-coherent, by (7.9.1.11).

In particular, this holds if  $f : X \rightarrow Y$  is a proper morphism of Noetherian schemes and  $L \in D_{\mathcal{C}oh}^-(X)$  and  $S \in D_{\mathcal{Q}\mathcal{C}oh}^+(S)$ , by (7.9.1.17) and (7.9.1.10).  $\lrcorner$

**Cor. (7.9.2.5) [Global Sections].** Let  $f : X \rightarrow Y \in \text{Sch}_{\text{qcqs}}$ , then for  $K \in D_{\text{Qcoh}}(X), L \in D_{\text{Qcoh}}(Y)$ , there is a canonical isomorphism

$$R\text{Hom}_{\mathcal{O}_X}(L, f^\times K) \rightarrow R\text{Hom}_{\mathcal{O}_Y}(Rf_*L, Rf_*f^\times K) \rightarrow R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*L, K)$$

┘

*Proof:* This is because for any  $E \in D(X)$ ,  $H^i(X, E) = \text{Ext}_X^i(\mathcal{O}_X, E) = \text{Hom}(\mathcal{O}_X[-i], E) = \text{Hom}(\mathcal{O}_X[-i], DQ(E))$  only depends on  $DQ(E)$ .  $\square$

**Prop. (7.9.2.6) [Proper Flat  $f^\times$ ].** Let  $f : X \rightarrow Y$  be a proper flat morphism of f.p. between qcqs schemes, then  $f^\times$  is compatible with base change between qcqs schemes.  $\square$

*Proof:* Cf. [Sta]0AAB.  $\square$

### Upper Shriek

**Prop. (7.9.2.7) [Lower Shriek].** Let  $f : X \rightarrow Y \in \text{Sch}_S^{\text{ft,sep}}$ , take a compactification  $\bar{f} : \bar{X} \rightarrow Y$  of  $X/Y$ , then we can define the functor

$$f^! : D_{\text{Qcoh}}^+(Y) \rightarrow D_{\text{Qcoh}}^+(X) : f^!K = \bar{f}^\times(K)|_X \text{ (7.9.2.1)}.$$

and this functor is independent of the compactification chosen.  $\square$

*Proof:* Cf. [Sta]0AA0.  $\square$

**Prop. (7.9.2.8).** If  $f : X \rightarrow Y, g : Y \rightarrow Z \in \text{Sch}_S^{\text{ft,sep}}$ , then there is a canonical isomorphism of functors  $(g \circ f)^! \cong f^! \circ g^!$ .  $\square$

*Proof:* Cf. [Sta]0ATX.  $\square$

**Prop. (7.9.2.9) [Properties of  $f^!$ ].** Let  $S \in \text{Sch}$  be Noetherian, then for  $X, Y \in \text{Sch}^{\text{ft,sep}}/S$ ,

- If  $j : X \rightarrow Y$  is an open immersion, then  $j^! = j^*$ .
- If  $i : X \rightarrow Y$  is a closed immersion, then  $i^! = R\mathcal{H}om(\mathcal{O}_X, -)$ .
- $f^!$  maps  $D_{\text{Qcoh}}^+(Y)$  into  $D_{\text{Qcoh}}^+(X)$ .
- If  $f : X \rightarrow Y$  is a local complete intersection morphism, then  $f^!\mathcal{O}_Y$  is an invertible object of  $D(X)$ , and  $f^!$  preserves perfect complexes.
- If  $f : X \rightarrow Y$  is finite, then  $f_*f^!(-) = R\mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, -)$ .
- If  $f : X \rightarrow Y$  is an effective Cartier divisor, then  $f^!(-) = Lf^*(-) \otimes_{\mathcal{O}_X}^L \mathcal{O}_Y(-X)[-1]$ .
- If  $f : X \rightarrow Y$  is a Koszul regular immersion of codimension  $c$ , then  $f^!(-) = Lf^*(-) \otimes^L \wedge^c \mathcal{N}[-c]$ .
- If  $f : X \rightarrow Y$  is smooth proper of relative dimension  $d$ , then  $f^!(-) = Lf^*(-) \otimes^L \Omega_{X/Y}^d[d]$ .

┘

*Proof:* Cf. [Sta]0AU0, 0AA2, 0AU1, 0B6V, 0AU3. ?  $\square$

### Dualizing Complexes

**Def. (7.9.2.10) [Dualizing Complex].** For  $X \in \mathbf{NSch}$ , a complex  $K \in D_{\mathcal{Q}\mathrm{coh}}(X)$  is called a **dualizing complex** if it satisfies the following equivalent conditions:

- For any affine open  $U = \mathrm{Spec} A \subset X$ ,  $K|_U$  is a dualizing complex in  $D(A)$ .
- There exists an affine open covering  $U_i = \mathrm{Spec} A_i$  of  $X$  s.t.  $K|_{U_i}$  is a dualizing complex in  $D(A_i)$  for any  $i$ .

┘

*Proof:* This follows from (5.9.8.2). □

**Prop. (7.9.2.11) [Dualizing].** Let  $X \in \mathbf{NSch}$  and  $\omega_X$  be a dualizing complex, then  $D = R\mathcal{H}om_{\mathcal{O}_X}(-, \omega_X)$  is an anti-equivalence of  $D_{\mathcal{Q}\mathrm{coh}}(X)$  with itself, and there is a canonical isomorphism  $\mathrm{id} \cong D^{\circ 2}$ .

Moreover, if  $X$  is qc, then  $D$  exchanges  $D_{\mathcal{Q}\mathrm{coh}}^+(X)$  and  $D_{\mathcal{Q}\mathrm{coh}}^-(X)$ , and induces an equivalence  $D : D_{\mathcal{Q}\mathrm{coh}}^b(X) \rightarrow D_{\mathcal{Q}\mathrm{coh}}^b(X)$ . ┘

*Proof:* Cf. [Sta]0A89. □

**Thm. (7.9.2.12) [Grothendieck Duality].** Let  $S \in \mathbf{NSch}_{\mathrm{qc}}$ , then

- $f^!$  makes  $D_{\mathcal{Q}\mathrm{coh}}^+$  a pseudo-functor on  $\mathcal{S}\mathrm{ch}^{\mathrm{ft}, \mathrm{sep}}/S$ .
- If  $f : X \rightarrow Y \in \mathcal{S}\mathrm{ch}_S^{\mathrm{ft}, \mathrm{sep}}$  is proper, then  $f^!$  is the right adjoint of  $Rf_*$ , and there is a canonical isomorphism

$$Rf_* R\mathcal{H}om_{\mathcal{O}_X}(K, f^! M) \cong R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* K, M)$$

for any  $K \in D_{\mathrm{coh}}^-(X)$  and  $M \in D_{\mathcal{Q}\mathrm{coh}}^+(Y)$ , by??

- If  $X \in \mathcal{S}\mathrm{ch}_S^{\mathrm{ft}, \mathrm{sep}}$  has a dualizing complex  $\omega_X$ , then  $D_X = R\mathcal{H}om(-, \omega_X)$  defines an involution of  $D_{\mathrm{coh}}(X)$  switching  $D_{\mathrm{coh}}^+(X)$  and  $D_{\mathrm{coh}}^-(X)$  and fixing  $D_{\mathrm{coh}}^b(X)$ .
- If  $Y$  has a dualizing complex  $\omega_Y$ , then
  - $\omega_X = f^! \omega_Y$  is a dualizing complex for  $X$ ,
  - for  $M \in D_{\mathrm{coh}}^+(Y)$ , there is a canonical isomorphism  $D_X(f^! M) \cong Lf^* D_Y(M)$ .
  - If moreover  $f$  is proper, then  $Rf_* R\mathcal{H}om_{\mathcal{O}_X}(K, \omega_X) = R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* K, \omega_Y)$  for  $K \in D_{\mathrm{coh}}^-(X)$ .

┘

*Proof:* Cf. [Sta]0AU3. □

**Thm. (7.9.2.13) [Over a Dualizing Basis].** ┘

*Proof:* Cf. [Sta]0AUE. □

### Dualizing Modules

**Prop. (7.9.2.14) [Relative Dualizing Modules].** Let  $f : X \rightarrow Y \in \mathbf{NSch}$  be a locally quasi-finite morphism, then there exists a unique coherent  $\mathcal{O}_Y$ -module  $\omega_{X/Y} \in \mathcal{C}\mathrm{oh}(Y)$  that affine locally is given by the dualizing sheaf  $\omega_{B/A}$  (5.9.8.4). ┘

*Proof:* Cf. [Sta]0BVG. □

### Relative Dualizing Complex

**Def. (7.9.2.15) [Relative Dualizing Complexes].** For a separable f.t. morphism of Noetherian schemes  $f : X \rightarrow S$ , the **relative dualizing complex** is defined to be  $\omega_{X/S} = f^! \mathcal{O}_S$ .  $\lrcorner$

**Prop. (7.9.2.16).** Let  $Y$  be a qcqs scheme and  $f : X \rightarrow Y$  be a proper flat morphism of f.p., then

- $\omega_{X/Y}$  is perfect over  $Y$ .
- $R^i f_* \omega_{X/Y} = 0$  for  $i > 0$ .
- $\mathcal{O}_X \rightarrow \mathcal{H}om_{\mathcal{O}_X}(\omega_{X/S}, \omega_{X/S})$  is an isomorphism.

$\lrcorner$

*Proof:* As  $f^\times$  commutes with base change (7.9.2.6), it suffices to assume that  $Y = \text{Spec } A$ .

1: By (7.9.2.4)(7.3.4.23) and (7.7.5.5), for any perfect object  $E \in D(X)$ , there are canonical isomorphisms:

$$Rf_*(E \otimes^L \omega_{X/Y}) = Rf_* R\mathcal{H}om_{\mathcal{O}_X}(E^\vee, \omega_{X/Y}) = R\mathcal{H}om_{\mathcal{O}_Y}(Rf_* E^\vee, \mathcal{O}_Y) = (Rf_* E^\vee)^\vee,$$

which is perfect. So  $\omega_{X/Y}$  is perfect over  $S$  by (7.9.1.12).

2: By (7.9.1.8),

$$\mathcal{H}om_Y(\mathcal{O}_Y[-i], Rf_* \omega_{X/Y}) = \text{Hom}_Y(Rf_* Lf^* \mathcal{O}_Y[-i], \mathcal{O}_Y) = \text{Hom}_Y((Rf_* \mathcal{O}_X)[-i], \mathcal{O}_Y).$$

By proper flat base change (7.7.5.5),  $Rf_* \mathcal{O}_X$  is perfect in  $Y$ , so it can be represented by a finite complex of finite projective  $A$ -modules, so by (7.3.4.22),  $R^i f_* \omega_{X/Y} = 0$  for  $i > 0$ .

3: For any perfect object  $E \in D(X)$ , by (7.9.2.4)(7.3.4.23)(7.7.5.5)(7.3.4.23) and (7.3.1.9), there are canonical isomorphisms

$$\begin{aligned} \text{Hom}_X(E, R\mathcal{H}om_{\mathcal{O}_X}(\omega_{X/Y}, \omega_{X/Y})) &= \text{Hom}_Y(Rf_*(E \otimes^L \omega_{X/Y}), \mathcal{O}_Y) \\ &= \text{Hom}_Y(Rf_* \mathcal{H}om(E^\vee, \omega_{X/Y}), \mathcal{O}_Y) \\ &= \text{Hom}_Y(\mathcal{H}om(Rf_* E^\vee, \mathcal{O}_Y), \mathcal{O}_Y) \\ &= R\Gamma(Y, Rf_* E^\vee) \\ &= \text{Hom}_X(E, \mathcal{O}_X) \end{aligned}$$

So by (7.9.1.32), perfect objects generate  $D_{\text{Qcoh}}(X)$ , so  $R\mathcal{H}om_{\mathcal{O}_X}(\omega_{X/Y}, \omega_{X/Y}) \cong \mathcal{O}_X$ .  $\square$

### Proper over Field case

**Prop. (7.9.2.17) [Serre Duality(Proper over Fields case)].** Let  $k \in \text{Field}$  and  $X \in \text{Sch}^{\text{proper}, \dim=d}/k$ , then there exists a unique dualizing complex  $\omega_X \in D_{\text{Qcoh}}(X)$  with the following properties:

- $H^i(\omega_X) \neq 0$  only for  $i \in [-\dim(X), 0]$ .
- $[\omega_X] = H^{-d}(\omega_X)$  is a coherent  $(S_2)$ -module whose support is the irreducible components of  $X$  of dimension  $d$ .
- $\dim \text{Supp}(H^i(\omega_X)) \leq -i$ .
- For  $x \in |X|_0$ ,  $H^i(\omega_{X,x}) \oplus \dots \oplus H^0(\omega_{X,x}) \neq 0$  iff  $\text{depth}(\mathcal{O}_{X,x}) \leq -i$ .

- For  $K \in D_{\text{Qcoh}}(X)$ , there is a functorial isomorphisms

$$\text{Ext}_X^{-i}(K, \omega_X) \cong \text{Hom}_k(H^i(X, K), k)$$

compatible with shifts and distinguished triangles, which characterizes  $\omega_X$  uniquely.

- There are functorial isomorphisms  $\text{Hom}(\mathcal{F}, [\omega_X]) = \text{Hom}_k(H^d(X, \mathcal{F}), k)$  for  $\mathcal{F} \in \text{Coh}(X)$ .
- If  $X$  is C.M., equidimensional, then  $\omega_X = [\omega_X][d]$ .
- If  $X$  is smooth over  $k$ , then  $\omega_X = \mathcal{K}_{X/k}[d]$ .

┘

*Proof:* Cf. [Sta]0FVV, 0AWT. ?

□

**Cor. (7.9.2.18) [Serre Duality (Smooth over Fields case)].** If  $k \in \text{Field}$  and  $X \in \text{Sch}^{\text{SmPrpr}, \dim=d}/k$ , then for any  $\mathcal{F} \in \text{Vect}(\mathcal{O}_X)$ , there is a functorial isomorphism:

$$H^i(X, \mathcal{F}) \cong (H^{n-i}(X, \mathcal{F}^\vee \otimes \mathcal{K}_{X/k}))^\vee.$$

┘

**Cor. (7.9.2.19).** For a smooth proper variety  $X$  over a field  $k$  of dimension  $n$ ,  $H^n(X, \mathcal{K}_X) = k$ , by (7.12.1.12).

┘

**Cor. (7.9.2.20).** For  $X \in \text{SmPrprVar}^n/k$ ,  $\Omega_{X/k}$  is locally free by (7.6.4.14), thus by (7.5.1.25),  $\Omega_{X/k}^{n-p} \cong (\Omega_{X/k}^p)^\vee \otimes \mathcal{K}_{X/k}$ . So by (7.9.2.18):

$$H^q(X, \Omega_{X/k}^p) \cong (H^{n-q}(X, \Omega_{X/k}^{n-p}))^\vee.$$

┘

**Cor. (7.9.2.21).** If  $X$  is a closed subscheme of  $\mathbb{P}_k^n$  of codimension  $r$ , then  $X$  has a dualizing sheaf  $[\omega_X] = \mathcal{E}xt_P^r(i_* \mathcal{O}_X, \mathcal{K}_{\mathbb{P}_k^n/k})$ , and  $\mathcal{E}xt^i(i_* \mathcal{O}_X, \mathcal{K}_{\mathbb{P}_k^n/k}) = 0$  for  $i < r$

┘

*Proof:* This follows from (7.9.2.17) and (7.9.2.12).

□

**Prop. (7.9.2.22) [Characterizing Cohen-Macaulay Schemes].** Let  $k \in \text{Field}$  and  $X \in \text{Sch}^{\text{Proj}, \dim n}/k$  and  $[\omega_X]$  be the dualizing sheaf, then for  $\mathcal{F} \in \text{Coh}(X)$ , there is a natural map

$$\text{Ext}^i(\mathcal{F}, [\omega_X]) \rightarrow (H^{n-i}(X, \mathcal{F}))^\vee$$

And the following are equivalent:

- For any  $\mathcal{F}$  locally free on  $X$ ,  $H^i(X, \mathcal{F}(-q)) = 0$  for  $i < n$  and  $q$  large.
- $H^i(X, \mathcal{O}_X(-q)) = 0$  for  $i < n$  and  $q$  large.
- This is an isomorphism of  $\delta$ -functors.
- $X$  is C.M. and equidimensional.

┘

*Proof:* Notice the left side is an universal  $\delta$ -functor in  $\mathcal{F}$  by (7.9.1.37), so the map exist, and  
 2  $\rightarrow$  3: This implies that the right is also universal by (7.9.1.34).  
 3  $\rightarrow$  1: For  $\mathcal{F}$  locally free, by (7.3.3.33),

$$H^i(X, \mathcal{F}(-q)) = (\text{Ext}^{n-i}(\mathcal{F}(-q), \omega_X))^\vee = (H^{n-i}(X, \mathcal{F}^\vee \otimes \omega_X(q)))^\vee$$

which is 0 for  $q$  large.

4  $\rightarrow$  1: Embed  $X$  in  $P = \mathbb{P}_k^N$ , for  $\mathcal{F}$  locally free, since  $X$  is catenary, equidimensional is equivalent to  $\dim \mathcal{F}_x = n$  for all closed pt  $x$ , and C.M. says  $\text{depth } \mathcal{F}_x = n$ . Thus by (5.3.5.27),  $pd_{\mathcal{O}_{P,x}} \mathcal{F}_x = N - n$ . Thus  $\mathcal{E}xt_P^k(\mathcal{F}, -)$  vanish for  $k > N - n$  checked on stalks.

Now  $H^i(X, \mathcal{F}(-q))$  is dual to  $\text{Ext}_P^{N-i}(\mathcal{F}, \omega_P(q))$  by the proof of (7.9.2.21), which is isomorphic to  $\Gamma(P, \mathcal{E}xt_P^{N-i}(\mathcal{F}, \omega_P(q)))$  for  $q$  large by (7.7.4.9), so it vanish when  $i < n$  by what we proved.

1  $\rightarrow$  4: The same as the proof of 4  $\rightarrow$  1, then for  $i < n$ ,

$$\Gamma(P, \mathcal{E}xt_P^{N-i}(\mathcal{F}, \omega_P(q))) = 0 = \Gamma(P, \mathcal{E}xt_P^{N-i}(\mathcal{F}, \omega_P)(q))$$

for  $q$  large, so  $\mathcal{E}xt_P^{N-i}(\mathcal{F}, \omega_P) = 0$  as it is coherent. Then the stalk is  $\text{Ext}_{\mathcal{O}_{P,x}}^{N-i}(\mathcal{O}_{X,x}, \mathcal{O}_{P,x})$ , so  $pd_{\mathcal{O}_{P,x}} \mathcal{F}_x \leq N - n$  by (5.3.5.28), so  $\text{depth } \mathcal{O}_{X,x} \geq n$ , we must have equality, thus  $X$  is C.M. and equidimensional, as it suffice to check at closed pts.  $\square$

**Cor. (7.9.2.23)[Enriques-Severi-Zariski].** Let  $X$  be a normal projective scheme that every irreducible component has dimension  $\geq 2$ , then for any  $\mathcal{F} \in \text{Vect}(X)$ ,  $H^1(X, \mathcal{F}(-q)) = 0$  for  $q$  large.  $\lrcorner$

*Proof:* Just notice that  $\dim \mathcal{F}_x \geq 2$ , and Serre criterion shows  $\text{depth } \mathcal{F}_x \geq 2$ , the rest is the same as 4  $\rightarrow$  1 in the proof of (7.9.2.22).  $\square$

**Prop. (7.9.2.24)[Ample Effective Divisor Connected].** Let  $X$  be a proper connected CM. equidimensional scheme over  $k$  of dimension at least 2 and  $D$  is an ample effective Cartier divisor, then  $D$  is connected.

In particular, if  $X$  is a smooth complete variety of dimension  $\geq 2$ ,  $D$  is also a complete variety.  $\lrcorner$

*Proof:* Let  $D = V(s)$ , where  $s$  is a section of an ample invertible sheaf, then there is an exact sequence

$$0 \rightarrow \mathcal{O}_X(-nD) \rightarrow \mathcal{O}_X \rightarrow i_* \mathcal{O}_{V(s^n)} \rightarrow 0,$$

which gives a long exact sequence

$$0 \rightarrow H^0(X, \mathcal{O}_X(-nD)) \rightarrow H^0(X, \mathcal{O}_X) \rightarrow H^0(V(s^n), \mathcal{O}_{V(s^n)}) \rightarrow H^1(X, \mathcal{O}_X(-nD)).$$

But by Serre duality (7.9.2.18) and (7.7.2.6),  $H^0(X, \mathcal{O}_X(-nD)) = H^{\dim X}(X, [\omega_X] \otimes \mathcal{O}_X(nD)) = 0$  for  $n$  sufficiently large, and  $H^1(X, \mathcal{O}_X(-nD)) = H^{\dim X-1}(X, [\omega_X] \otimes \mathcal{O}_X(nD)) = 0$  for  $n$  sufficiently large, so  $H^0(V(s^n), \mathcal{O}_{V(s^n)}) \cong H^0(X, \mathcal{O}_X)$  has no idempotents for  $n$  sufficiently large, so  $V(s^n)$  is connected, and  $D$  is also connected.  $\square$

**Cor. (7.9.2.25).** Any global complete intersection in  $\mathbb{P}^n$  is connected.  $\lrcorner$

### Topological Sheaves

**Prop. (7.9.2.26) [Global Verdier Duality].** If  $f : X \rightarrow Y$  is a map between locally compact spaces with finite dimension, then there exists a functor  $f^! : D^+(SAb_Y) \rightarrow D^+(SAb_X)$  that

$$R\mathrm{Hom}(Rf_! \mathcal{F}^\bullet, \mathcal{G}^\bullet) \cong R\mathrm{Hom}(\mathcal{F}, f^! \mathcal{G}^\bullet).$$

In particular,  $f^!$  is right adjoint to  $Rf_!$ . Cf.[Gelfand P228].

There is also a local form of Verdier duality, which implies the global version by taking global section, Cf.[Cohomology of Sheaves Iversen P330].  $\lrcorner$

**Prop. (7.9.2.27).** When  $X \rightarrow Y$  is an inclusion of open subset,  $f_!$  is just  $j_!$  defined in (7.2.6.2) and  $f^!$  is the restriction. When it is an inclusion of closed subset of locally compact spaces, it is the direct image  $f_*$  and  $f^!$  is the  $j^!$  previously defined in (7.2.6.2). They are not barely defined on  $D^+(SAb)$  but on  $SAb$ .  $\lrcorner$

**Prop. (7.9.2.28).** We consider the case where  $f : X \rightarrow \mathrm{pt}$ , and let  $G = \mathbb{Z}$ , denote  $f^!(\mathbb{Z})$  by  $\mathcal{D}_X^\bullet$ , called the **dualizing complex**, then there is a duality:

$$R\mathrm{Hom}(R\Gamma_c(X, \mathcal{F}^\bullet), \mathbb{Z}) \cong R\mathrm{Hom}(\mathcal{F}^\bullet, \mathcal{D}_X^\bullet).$$

for  $\mathcal{F}^\bullet \in D^+(\mathrm{Sh}(X))$ .  $\lrcorner$

**Prop. (7.9.2.29).** When  $X$  is a  $n$  dimensional topological manifold with boundary, then  $\mathcal{D}_X^\bullet = \omega_X[n]$ , where the sheaf  $\omega_X$  is defined by

$$\Gamma(U, \omega_X) = \mathrm{Hom}_{\mathcal{A}b}(H_c^n(U, \mathbb{Z}), \mathbb{Z}).$$

Cf.[Gelfand P234]. If we replace  $\mathbb{Z}$  by a field  $k$ , then  $\omega_X$  is the sheaf of  $k$ -orientations of  $\mathrm{int}(X)$ , thus the constant sheaf when  $X$  is oriented or  $\mathrm{char} k = 2$  ?.

In particular, place  $k$  in dimension  $i$  then we get an isomorphism

$$\mathrm{Hom}_k(H_c^i(X, \mathcal{F}), k) = \mathrm{Ext}^{n-i}(\mathcal{F}, \omega_X)$$

(because  $k$  is a field thus injective). Gelfand even gives an interpretation of this pair in [Gelfand P236].

And if  $\mathcal{F} = \omega_X$  and  $X$  oriented or  $\mathrm{char} k = 2$ , we have  $\mathrm{Ext}^{n-i}(k_X, k_X[n]) = H^{n-i}(X, k_X)$  using the adjointness of constant sheaf, so we get the Poincare duality:

$$H_c^i(X, k_X)^\vee \cong H^{n-i}(X, k_X).$$

$\lrcorner$

**Prop. (7.9.2.30).** Compact cohomology commute with colimits, Cf.[Cohomology of Sheaves Iversen P173].  $\lrcorner$

### 3 Fourier-Mukai Transforms

**Notation (7.9.3.1).**

- Let  $S \in \mathrm{Sch}$ .

$\lrcorner$

Cf. [Sta]Chap56.

**Def. (7.9.3.2) [Fourier-Mukai Transform].** Let  $X, Y \in \text{Sch}/S$ , then for any  $K \in D(X \times_S Y)$ , the **Fourier-Mukai transform** associated to  $K$  is defined to be the exact functor

$$\Phi_K : D(\mathcal{O}_X) \rightarrow D(\mathcal{O}_Y) : M \rightarrow R\text{pr}_{2*}(L\text{pr}_1^* M \otimes_{X \times_S Y}^L K).$$

And  $K$  is called the **kernel** of  $\Phi_K$ . ┘

**Prop. (7.9.3.3) [Qco Fourier-Mukai Transform].** Situation as in (7.9.3.2), if  $X \in \text{Sch}^{\text{qcs}}/S$ , then  $\Phi_K(D_{\text{Qcoh}}(X)) \subset D_{\text{Qcoh}}(Y)$ . ┘

*Proof:* This follows from (7.9.1.6)(7.9.1.5) and (7.9.1.6). □

**Prop. (7.9.3.4) [Compositions].** Let  $X, Y, Z \in \text{Sch}^{\text{qcs}}/S$  s.t.  $X, Z$  is tor-independent over  $S$  and  $Y/S$  is flat, then for any  $K \in D(X \times_S Y), K' \in D(Y \times_S Z)$ , we have

$$\Phi_{K'} \circ \Phi_K = \Phi_{K''}$$

where  $K'' = R\text{pr}_{13*}(L\text{pr}_{12}^* K \otimes_{X \times_S Y \times_S Z}^L L\text{pr}_{23}^* K') \in D(X \times_S Y \times_S Z)$ . ┘

*Proof:* Cf. [Sta]0FYS. □

**Prop. (7.9.3.5) [Perfect Fourier-Mukai Transform].** Situation as in (7.9.3.2), there are situations that  $\Phi_K(D_{\text{perf}}(X)) \subset D_{\text{perf}}(Y)$ . Cf. [Sta]0FYT. ? ┘

*Proof:* □

**Prop. (7.9.3.6) [Coherent Fourier-Mukai Transform].** Let  $X, Y \in \text{Sch}/S$  s.t.  $X/S$  is proper, then for any  $K \in D_{\text{coh}}(\mathcal{O}_{X \times_S Y})$ ,  $\Phi_K(D_{\text{coh}}(X)) \subset D_{\text{coh}}(Y)$ . when one of the following holds:

- $K$  has finite tor-dimension as an object in  $D(\text{pr}_1^* \mathcal{O}_X)$ .
  - $K$  can be presented by a bounded complex  $\mathcal{K}^\bullet$  of coherent  $\mathcal{O}_{X \times_S Y}$ -modules that are flat over  $X$ .
  - $X$  is regular.
  - $K$  is perfect and  $Y/S$  is flat.
- ┘

*Proof:* Cf. [Sta]0FYU. ? □

## Equivalence criteria for Fourier-Mukai transforms

### Abelian Varieties

### K3 Surfaces

## 4 Derived Categories of Varieties over Fields

References are [Orl97, B-O01, B-O02].

**Notation (7.9.4.1).**

- Let  $k \in \text{Field}$ .
- ┘



**Prop. (7.9.4.2).** If  $X \in \mathcal{S}mPrpr\mathcal{V}ar^n/k$ , then tensoring with  $\omega_X = \mathcal{K}_{X/k}[n]$  is a Serre functor on  $D_{\text{coh}}^b(X)$  (4.8.4.40).  $\lrcorner$

*Proof:* It follows from Serre duality (7.9.2.17) that

$$\text{Ext}^i(F, G) = (\text{Ext}^{-i}(G, F \otimes \omega_X))^*$$

?

□

### Representability

**Thm. (7.9.4.3) [Representability Lemma, Bondal-Bergh [B-B03]].** Let  $X \in \mathcal{S}ch^{\text{Proj}}/k$  and  $F : D_{\text{perf}}(X)^{\text{op}} \rightarrow \mathcal{V}ect_k$  a  $k$ -linear cohomology functor s.t.

$$\sum_{n \in \mathbb{Z}} \dim_k F(E[n]) < \infty$$

for any  $E \in D_{\text{perf}}(X)$ . Then  $F = \mathcal{Y}_K$  for some  $K \in D_{\text{coh}}^b(X)$ .  $\lrcorner$

*Proof:*  $D_{\mathcal{Q}coh}(X)$  has direct sums by (7.9.1.2) and is compactly generated by (7.9.1.32), and  $D_{\text{perf}}(X) = D_{\text{cpct}}(X)$  (7.9.1.31), so we can use (4.8.4.39) to show that  $F = \mathcal{Y}_K$  for some  $K \in D_{\mathcal{Q}coh}(X)$ . But then  $K \in D_{\text{coh}}^b(X)$  by (7.9.1.16). □

**Prop. (7.9.4.4) [Perfect Representability for Regular Schemes].** Let  $X \in \mathcal{S}ch/k$  be regular, then any  $k$ -linear cohomological functor  $F : D_{\text{perf}}(X)^{\text{op}} \rightarrow \mathcal{V}ect_k$  satisfying

$$\sum_{n \in \mathbb{Z}} \dim_k F(E[n]) < \infty$$

for any  $E \in D_{\text{perf}}(X)$  must be of the form  $F = \mathcal{Y}_K$  for some  $K \in D_{\text{perf}}(X)$ .

Dual results holds for homological functors.  $\lrcorner$

*Proof:* Cf. [Sta] 0FYL. □

**Prop. (7.9.4.5) [Existence of Adjoints].** For  $X, Y \in \mathcal{S}ch^{\text{proper}}/k$ , If  $X$  is regular, then any  $k$ -linear exact functor  $F : D_{\text{perf}}(X) \rightarrow D_{\text{perf}}(Y)$  has an exact left adjoint and an exact right adjoint.  $\lrcorner$

*Proof:* This follows formally from (7.9.4.4) and (7.9.1.15) □

### Fully Faithful Functors

**Thm. (7.9.4.6) [Bondal-Orlov [B-O01]].** Let  $X, X' \in \mathcal{V}ar^{\text{sm}}/k$  s.t.  $X$  is projective with  $\mathcal{K}_X$  or  $-\mathcal{K}_X$  ample, and  $D^b(X) \cong D^b(X')$  as graded categories, then  $X \cong X'$ .  $\lrcorner$

*Proof:* Cf. [B-O01]. ?

The idea of the proof is that for varieties with ample canonical or anticanonical sheaf we can recognize the skyscraper sheaves of closed points in  $\text{Db}(X)$  by means of the Serre functor (4.8.4.40). □

**Thm. (7.9.4.7) [Bondal-Orlov [B-O01]].** Let  $X, X' \in \mathcal{V}ar^{\text{sm}}/k$  s.t.  $X$  is projective with  $\omega_X$  or  $-\omega_X$  ample, then

$$\text{Aut}_{\mathcal{T}riCat}(D^b(X)) = \text{Aut}(X) \ltimes (\text{Pic}(X) \oplus \mathbb{Z}).$$

□

*Proof:* Cf.[B-O01].? □

**Prop. (7.9.4.8).** For  $X, Y \in \text{Sch}^{\text{SmPrpr}}/k$ , if  $F, G : D_{\text{coh}}^b(X) \rightarrow D_{\text{coh}}^b(Y)$  are two  $k$ -linear exact functors s.t.

- $F(\mathcal{F}) \cong G(\mathcal{F})$  for any  $\mathcal{F} \in \text{Coh}(\mathcal{O}_X)$  with  $\dim \text{Supp}(f) = 0$ .
- $F$  is fully faithful.
- $G$  is a Fourier-Mukai transform whose kernel is in  $D_{\text{coh}}^b(X \times Y)$ .

then there exists a Fourier-Mukai transform  $F'$  with kernel in  $D_{\text{coh}}^b(X \times Y)$  s.t.  $F, F'$  are siblings. ⌋

**Thm. (7.9.4.9)** [Orlov[Orl97]]. Let  $X, M \in \text{Sch}^{\text{SmPrpr}}/k$  and  $F : D_{\text{coh}}^b(M) \rightarrow D_{\text{coh}}^b(X)$  a fully faithful functor, then there exists a unique element  $E \in D_{\text{coh}}^b(M \otimes X)$  s.t.  $F = \Phi_E$  (7.9.3.2). ⌋

*Proof:* Cf.[Equivalences of derived categories and K3 surfaces, Orlov]. □

**Thm. (7.9.4.10)** [Bondal-Orlov[B-O01]]. Let  $X, M \in \text{SmPrprVar}/k$  and  $E \in D_{\text{coh}}^b(M \otimes X)$ , then

$$\Phi_E \triangleq R\text{pr}_{2*}(E \otimes^L \text{pr}_1^*(-)) : D_{\text{coh}}^b(M) \rightarrow D_{\text{coh}}^b(X)$$

is fully faithful iff the following are satisfied:

- $\text{Hom}_X^i(\Phi_E(\mathcal{O}_M(t)), \Phi_E(\mathcal{O}_M(t'))) = 0$  for any  $i \in \mathbb{Z}$  and any  $t \neq t' \in |M|_0$ .
  - $\text{Hom}_X^0(\Phi_E(\mathcal{O}_M(t)), \Phi_E(\mathcal{O}_M(t))) = k$  for any  $t \in |M|_0$ .
  - $\text{Hom}_X^i(\Phi_E(\mathcal{O}_M(t)), \Phi_E(\mathcal{O}_M(t))) = 0$  for any  $t \in |M|_0$  and  $i \notin [0, \dim M]$ .
- ⌋

### Derived Categories of Complete Intersections of Quadrics

**Thm. (7.9.4.11)** [Derived Categories of Complete Intersections of Quadrics, Kapranov[Kap88]/Bondal-Orlov]. For  $m \in \mathbb{N}, n \in \mathbb{Z}_+$  s.t.  $2m \leq n$ , consider a system of  $m$  quadrics in  $\mathbb{P}^{n-1}$  of the form  $\varphi : U \rightarrow \text{Sym}^2 V^*$  where  $U \cong k^m \in \text{Vect}_k, V \cong k^n \in \text{Vect}_k$ . Suppose the intersection  $X \subset \mathbb{P}^{n-1}$  is smooth of dimension  $n - m - 1$ , then

- There is a sheaf of finite algebras  $\mathcal{B}$  over  $\mathcal{O}_{\mathbb{P}(U)}$  together with a fully faithful embedding  $D^b(\text{Coh}(\mathcal{B})) \rightarrow D_{\text{coh}}^b(X)$ .
- There is an semi-orthogonal decomposition (4.8.4.23)

$$D_{\text{coh}}^b(X) = \langle \mathcal{O}_X(-n + 2m + 1), \mathcal{O}_X(-n + 2m + 2), \dots, \mathcal{O}_X, D^b(\text{Coh}(\mathcal{B})) \rangle$$

In particular, if  $n = 2m$ , then  $D^b(\text{Coh}(\mathcal{B})) \cong D_{\text{coh}}^b(X)$ . ⌋

*Proof:* Cf.[Semiorthogonal decomposition for algebraic varieties, Bondal-Orlov] and [B-O02].? □

**Cor. (7.9.4.12)** [Coherent Sheaves on  $\mathbb{P}^n$ , Beilinson[Bei78]]. There is an semi-orthogonal decomposition

$$D_{\text{coh}}^b(\mathbb{P}^n) = \langle \mathcal{O}_{\mathbb{P}^n}(-n), \mathcal{O}_{\mathbb{P}^n}(-n + 1), \dots, \mathcal{O}_{\mathbb{P}^n} \rangle$$
⌋

*Proof:* Direct proof is given in [Huybrechts, Fourier-Mukai transforms in algebraic geometry]Cor8.29. where Beilinson constructed a resolution of  $\Delta_{\mathbb{P}^n}$  using Koszul complexes. □

**Cor. (7.9.4.13).** If  $n \in 2\mathbb{Z}_+ + 1$  and  $X \in \text{SmPrprVar}/k$  is an intersection of two projective quadrics in  $\mathbb{P}^{n-1}$  and let  $C$  be the hyperelliptic curve which is the double cover of  $\mathbb{P}^1$  that parameterizes the pencil of quadrics, with ramification in the points corresponding to degenerate quadrics, then there is an semi-orthogonal decomposition

$$D_{\text{coh}}^b(X) = \langle \mathcal{O}_X(-n+4), \mathcal{O}_X(-n+4), \dots, \mathcal{O}_X, D_{\text{coh}}^b(C) \rangle$$

┘

*Proof:* Cf. [B-O02]. ?

In this case  $\text{Coh}(\mathcal{B}) \cong \text{Coh}(\mathcal{B}')$  where  $\mathcal{B}'$  is an Azumaya algebra over  $C$ , which is trivial because  $k = \bar{k}$ . so  $D^b(\text{Coh}(\mathcal{B})) \cong D_{\text{coh}}^b(C)$ .  $\square$

### Derived Categories and Blow-Ups

**Prop. (7.9.4.14).** Let  $X \in \text{Var}^{\text{sm}}/k$  and  $i : Y \subset X$  be a smooth variety of codimension  $r$ , denote  $\tilde{X} = \text{Bl}_Y(X) \xrightarrow{\text{pr}} X$  and  $\tilde{Y} = \text{pr}^{-1}(Y) \xrightarrow{j} X$ . As  $\tilde{Y} = \mathbb{P}(N_{Y/X})$ , we can define the relative Serre sheaf  $\mathcal{O}_{\tilde{Y}}(1)$  on  $\tilde{Y}$ .

Then for any  $\mathcal{L} \in \text{Pic}(\tilde{Y})$ , the functors

$$L\pi^* : D_{\text{coh}}^b(X) \rightarrow D_{\text{coh}}^b(\tilde{X}), \quad Rj_*(L \otimes \text{pr}^*(-)) : D_{\text{coh}}^b(Y) \rightarrow D_{\text{coh}}^b(\tilde{X})$$

are fully faithful.  $\square$

*Proof:* Cf. [Projective bundles, monoidal transformations, and derived categories of coherent sheaves, Orlov].  $\square$

**Thm. (7.9.4.15).** Situation as in (7.9.4.14), for  $r \in \mathbb{Z}$ , denote

$$D_X^b(\tilde{X}) = L\pi^*(D_{\text{coh}}^b(X)) \subset D_{\text{coh}}^b(\tilde{X}), \quad D_Y^b(\tilde{X})(r) = Rj_*(\mathcal{O}_{\tilde{Y}}(r) \otimes \text{pr}_Y^*(D_{\text{coh}}^b(Y))),$$

then there is a semi-orthogonal decomposition of triangulated categories (4.8.4.23):

$$D_{\text{coh}}^b(\tilde{X}) = \langle D_Y^b(\tilde{X})(-r+1), \dots, D_Y^b(\tilde{X})(-1), D_X^b(\tilde{X}) \rangle$$

┘

*Proof:* [Projective bundles, monoidal transformations and derived categories of coherent sheaves, Orlov] and [Semiorthogonal decomposition for algebraic varieties, Bondal-Orlov].  $\square$

**Conj. (7.9.4.16) [Flips and Derived Categories, Bondal-Orlov].** For any generalized flip  $X \rightarrow X^+ \in \text{SmProjVar}/k$  (7.9.5.1), there is an exact fully faithful functor  $F : D_{\text{coh}}^b(X^+) \rightarrow D_{\text{coh}}^b(X)$ , and for generalized flop, this is an equivalence.  $\square$

*Proof:*  $\square$

**Thm& Conj. Cor. (7.9.4.17) [Bridgeland [Bri02]].** This conjecture holds true when  $\tilde{X} \rightarrow X$  is a flop of 3-dimensional varieties.  $\square$

*Proof:*  $\square$

**Thm& Conj. Cor. (7.9.4.18) [Simple Flips, Bondal-Orlov].** Situation as in (7.9.5.2), then

$$R\pi_* L\pi^{+*} : D_{\text{coh}}^b(X^+) \rightarrow D_{\text{coh}}^b(X)$$

is fully faithful for  $l \leq m$ , and is an equivalence if  $l = m$ .  $\square$

*Proof:* Cf. [Semiorthogonal decomposition for algebraic varieties, Bondal-Orlov].  $\square$

### Categorical Desingularizations

**Def. (7.9.4.19) [Categorical Desingularizations].** For  $\mathcal{D} \in \text{TriCat}$ , a **categorical desingularization** or a **categorical resolution of singularity** is a pair  $(\mathcal{C}, \mathcal{K})$  where  $\mathcal{C} \in \text{AbVar}$  and  $\mathcal{K} \subset D^b(\mathcal{C})$  a thick triangulated subcategory  $\mathcal{K} \subset D^b(\mathcal{C})$  (4.8.4.21) s.t.  $\mathcal{D} = D^b(\mathcal{C})/\mathcal{K}$  (4.8.4.32).  $\lrcorner$

**Prop. (7.9.4.20).** If  $\text{pr} : \tilde{X} \rightarrow X$  is a desingularization (7.11.1.1) and  $X$  has rational singularities (7.11.1.2), then

$$R\text{pr}_* : D_{\text{coh}}^b(\tilde{X}) \rightarrow D_{\text{coh}}^b(X)$$

makes  $(\text{Coh}(\tilde{X}), \ker(R\text{pr}_*))$  a categorical desingularization of  $D_{\text{coh}}^b(X)$  (7.9.4.19).  $\lrcorner$

*Proof:* ?  $\square$

## 5 Birational Geometry

### Flips and Flops

**Def. (7.9.5.1) [generalized Flips and Flops].**  $\lrcorner$

**Prop. (7.9.5.2) [Simple Flips].** If  $Y \subset X \in \text{Var}^{\text{sm}} \cong \mathbb{P}_k^m$  and  $N_{X/Y} \cong \mathcal{O}_Y(-1)^{\oplus(l+1)}$ , then the blow-up  $\tilde{X} = \text{Bl}_{Y^+}(X^+) \xrightarrow{\text{pr}^+} X$  can be contracted to a smooth variety  $\pi : \tilde{X} \rightarrow X$  s.t. the inverse image  $\tilde{Y} = \text{pr}^{-1}(Y) \cong \mathbb{P}^m \times \mathbb{P}^l$  projects to the second component  $\mathbb{P}^l$ , giving a simple flip  $X \rightarrow X^+$ .  $\lrcorner$

## 7.10 Logarithmic Geometry

References are [DLLZ19], [Log p-divisible groups], [Logarithmic structures of Fontaine-Illusie, Kato], [Toric singularities, Kato].

## 7.11 Resolution of Singularities

References are [Sta], [Resolution of Singularities, Hauser-Lipman-Oort-Quiros] and [K-M85].

### 1 Desingularizations and Alterations

**Def. (7.11.1.1)[Desingularizations].** Let  $X \in \text{Sch}$  be reduced and locally Noetherian, then a **desingularization** of  $X$  is a modification (7.4.6.10)  $Z \rightarrow X$  s.t.  $Z$  is regular. A **strong desingularization** is a desingularization  $\text{pr} : Z \rightarrow X$  that  $\text{pr}$  is an isomorphism over any regular point  $x \in X$ .  $\lrcorner$

**Def. (7.11.1.2)[Rational Singularities].** For  $k \in \text{Field}^0$  any  $X \in \text{Sch}/k$ , then  $X$  is said to have **rational singularities** if  $X$  normal, of f.t. over  $k$ , and there exists a desingularization  $Y \rightarrow X$  (7.11.1.1) s.t.

$$R^i f_* \mathcal{O}_Y = 0, \quad \forall i \in \mathbb{Z}_+$$

$\lrcorner$

**Thm. (7.11.1.3) [Hironaka].** Let  $X$  be a reduced scheme locally of f.t. over an excellent, reduced, locally Noetherian scheme  $S$  of characteristic 0 (i.e.  $\text{char } \kappa(x) = 0$  for any  $x \in X$ ), then  $X$  admits a strong desingularization over  $S$ .  $\lrcorner$

*Proof:* [Hironaka, Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II, Ann. Math. 79 (1964), 109–203; 205–326.] or [Resolution of Singularities, Hauser, Lipman, Oort, Quiros].  $\square$

**Thm. (7.11.1.4) [de Jong].** Let  $R$  be a CDVR or a field,  $X \in \text{Sch}_{\text{int}}^{\text{ft,sep}}/R$ , then  $X$  admits an alteration (7.4.6.10)  $Y \rightarrow X$  over  $R$  s.t.  $Y$  is regular.  $\lrcorner$

*Proof:* [de Jong, Smoothness, Semi-stability and Alterations, Publ. Math. IHES 83 (1996), 51–93.] ?

Proof ideas: Replace  $X$  by a fibration  $X \rightarrow B$  where all fibers are curves. By induction on dimensions, we can find an alteration  $B' \rightarrow B$  s.t.  $B'$  is regular. Then  $X' = X \times_B B' \rightarrow X$  is an alteration. Then we use resolution of curves, which can be made effective in a non-singular family.

$\square$

## 7.12 Varieties

*“If there is one thing in mathematics that fascinates me more than anything else (and doubtless always has), it is neither ‘number’ nor ‘size,’ but always form.”*

–A. Grothendieck

References are [Sta] and [Har77].

The materials distinguishes here in the fact that most schemes considered have geo.reduced, geo.connected or geo.integral properties in nature.

### Notation (7.12.0.1).

- Let  $k \in \mathbf{Field}$ .
- Use notations on properties of morphisms between schemes defined in [Properties of Schemes](#) and [More Properties of Morphisms between Schemes](#).
- Use notations on quasi-coherent sheaves on schemes defined in ??.

┘

## 1 Varieties

### Classical varieties

**Prop. (7.12.1.1) [Soberization Functor].** Suppose  $k = k^s$ , the soberization functor  $t$  induce a fully faithful functor from classical varieties over  $k$  to quasi-projective integral schemes over  $k$ . It maps projective varieties to projective integral schemes and preserves fiber products ?.

┘

*Proof:* We assign the irreducible closed subsets space  $t(X)$  and show that this embeds  $X$  in  $t(X)$ , and for an affine variety  $(V, \mathcal{O}_V)$ , the regular function sheaf is isomorphic to the pullback sheaf on  $t(V) = \mathrm{Spec}(A)$ .

By definition  $t(X)$  is quasi-projective, which is separated, the set of geometric pts of any closed variety is dense so  $t(V)$  is homeomorphic to  $X$ . And because they are both reduced, they are isomorphic. So it is essentially surjective.

It is fully faithful by (7.12.1.11).

□

**Prop. (7.12.1.2).** The soberization of a classical variety  $X$  is regular at a closed point iff the local defining functions has rank  $n - \dim X$ .

┘

*Proof:* Consider the space of closed point of  $X$ , they correspond to classical points because  $k$  is alg.closed. Let  $a_p = (x_1 - a_1, \dots, x_n - a_n)$  and  $b$  be the locally defining ideal. Then the differential defines an isomorphism of vector space  $a_p/a_p^2 \cong k^n$ , and the local ring at  $p$  is  $m/m^2 \cong (a_p/b)/(a_p/b)^2 \cong a_p/(b + a_p^2)$ . The rank of the defining functions is  $b + a_p^2/a_p^2$ . Counting dimension gives us the result. (Use (7.6.3.3) also). □

### Varieties

**Def. (7.12.1.3) [Varieties].**

- A **variety** over  $k$  is a geo.integral separated scheme algebraic over  $k$ .
- An **prevariety** over  $k$  is an integral separated scheme algebraic over  $k$ .

- A **complete variety** over  $k$  is a variety over  $k$  that is also proper (i.e. universally closed) over  $k$ .
- A classical variety over  $k$  is an abstract variety over  $k$  because quasi-projective is f.t. and separated (7.4.6.19).
- A **non-singular variety** over  $k$  is a regular variety over  $k$ .
- A (pre)**curve** over  $k$  is a (pre)variety of dimension 1 over  $k$ .

The category of varieties over  $k$  is denoted by  $\text{Var}_k$ .

If  $X/k$  is a (complete)variety and  $K/k$  is a field extension, then  $X_K$  is a (complete)variety over  $K$ , by (7.4.3.17).  $\lrcorner$

**Remark (7.12.1.4).** Notice the prevariety is the same as the variety defined in [Sta].  $\lrcorner$

**Cor. (7.12.1.5).** Any variety is birational to an integral H-quasi-projective scheme. A complete variety is birational to an integral projective scheme by Chow's lemma (7.4.6.23)(7.4.6.3).  $\lrcorner$

**Prop. (7.12.1.6).** By valuation criterion, for a complete variety, every valuation of the function fields of  $K/k$  dominate a unique point of  $X$ . So the points of  $X$  correspond to valuations of  $K$  containing  $k$ .  $\lrcorner$

**Prop. (7.12.1.7) [Generically Smoothness].** A variety is generically smooth, by (7.6.4.20).  $\lrcorner$

**Prop. (7.12.1.8) [Nagata].** By Nagata compactification (7.8.3.2), any variety can be embedded as an open subset of a complete variety.  $\lrcorner$

*Proof:*

$\square$

**Prop. (7.12.1.9) [Product of Varieties].** The product of two (complete)varieties over  $k$  is also a (complete)variety.  $\lrcorner$

*Proof:* It is geometrically integral by (7.4.3.17), it is separated because separatedness is stable under composition and base change (7.4.4.2). So does properness.  $\square$

**Def. (7.12.1.10) [Arithmetic Points].** An **arithmetic point** of a scheme  $X$  is an element of  $X(k^{\text{sep}})$ . For  $X \in \text{Var}/k$ ,  $X(k^{\text{sep}})$  is dense in  $X$ , by (7.4.3.3).

An **geometric point** of a scheme  $X$  is an element of  $X(\bar{k})$ .  $\lrcorner$

**Prop. (7.12.1.11).** To verify two morphisms  $f, g$  between two varieties  $X$  and  $Y$  are equal, it suffices to prove that they are equal on the set of arithmetic points of an open subscheme  $U$  (7.12.1.10).  $\lrcorner$

*Proof:* Because the equalizer is a closed subscheme of  $X$  (7.2.7.18), and it contains all geometric pts of an open subset of  $X$ , so it must be  $X$ , as the geometric-points are dense in  $U$  (7.12.1.10), and  $X$  is reduced and irreducible.  $\square$

**Prop. (7.12.1.12) [Global Sections].** Let  $X \in \text{Sch}^{\text{proper}}/k$ , then

- $A = \Gamma(X, \mathcal{O}_X)$  is a finite  $k$ -algebra hence is a finite product of Artinian local  $k$ -algebras, one for each connected component of  $X$ .
- If  $X$  is reduced, then  $A = \prod k_i$  is a finite product of finite field extensions of  $k$ .
- If  $X$  is geo.reduced, then  $k_i$  are all separable over  $k$ .



- If  $X$  is geo.connected, then  $A$  is geo.irreducible over  $k$ (5.3.6.1).
- If  $X$  is geo.integral(i.e. a complete variety), then  $\Gamma(X, \mathcal{O}_X) = k$ .

┘

*Proof:* 1:  $H^0(X, \mathcal{O}_X)$  is finite  $k$ -algebra by(7.7.4.11), thus it is Artinian(5.1.3.4). The connected components of  $X$  clearly corresponds to idempotents of  $A$ , which corresponds to Artinian local  $k$ -algebras, as a local ring is connected.

2: This follows from(5.1.3.5).

3:  $A \otimes_k \bar{k} = H^0(X_{\bar{k}}, \mathcal{O}_{X_{\bar{k}}})$  is also reduced by flat base change(7.7.5.1), so each  $k'/k$  is geo.reduced(5.3.6.2), thus separable by(5.3.9.4).

4: By hypothesis  $A \otimes_k \bar{k} = H^0(X_{\bar{k}}, \mathcal{O}_{X_{\bar{k}}})$  is an Artinian local ring, thus it has only one point, so irreducible, so it is geo.irreducible(5.3.6.3).

5: By item3, 4,  $A = k'$  is a finite separable field extension of  $k$  that  $\text{Spec}(k' \otimes_k \bar{k})$  is irreducible, but  $k' \otimes_k \bar{k}$  is a finite product of  $\bar{k}$ , thus  $k' = k$ .  $\square$

**Prop. (7.12.1.13)[Check Properties on Geometric/Closed Points].** A nice property of varieties is that identity of two morphisms of products of varieties can be checked at the geometric pts(7.12.1.10), by(7.12.1.11) and(7.12.1.9).

Surjectiveness of a map between varieties can be checked on closed points, by(7.4.1.29).

Also surjective and injective of Qco sheaves need only be checked at closed pts by(7.5.1.38)(7.4.1.26).  $\square$

### Canonical Sheaves

**Prop. (7.12.1.14)[Canonical Sheaves].** For  $X \in \text{SmVar}/k$  and  $Y$  a local complete intersection of  $X$  of codimension  $r$ , by(7.6.4.2) and(7.6.4.15) and(7.6.8.9),  $\mathcal{K}_{X/k}$  and  $\mathcal{I}/\mathcal{I}^2$  is locally free, so we can define the following locally free sheaves:

- The **canonical sheaf**  $\mathcal{K}_{X/k} = \wedge^n \Omega_{X/k}$  on  $X$ .
- The **tangent sheaf**  $\mathcal{T}_X = (\mathcal{K}_{X/k})^{-1}$  on  $X$ .
- The conormal sheaf  $\mathcal{C}_{Y/X} = \mathcal{I}/\mathcal{I}^2$  on  $Y$ .
- The normal sheaf  $\mathcal{N}_{Y/X} = \mathcal{C}_{Y/X}^{-1}$  on  $Y$ .

┘

**Prop. (7.12.1.15)[Kodaira-Spencer map].** There is another characterization of tangent vector fields. (Note: this should be a special case of Prop8.5.9 in [FGA]).

Let  $X$  be a variety over  $k$  and  $S = k[\varepsilon]$  the dual numbers. Then  $H^0(X, \mathcal{T}_X) \cong \text{Aut}^{(1)}(X_S/S)$ , where  $\text{Aut}^{(1)}(X_S/S)$  means that the automorphisms of  $X_S$  over  $S$  that is identity on  $X$ (inclusion to  $X_S$  induced by  $\text{Spec } k \subset \text{Spec } S$ ).  $\square$

*Proof:* First the case  $X = \text{Spec } A$  is affine, then because  $H^0(X, \mathcal{T}_X) = \text{Hom}_k(\mathcal{K}_{A/k}, A) = \text{Der}(A, A)$ , so this is equivalent to  $\text{Der}(A, A) \cong$  automorphisms of  $A[\varepsilon]$  that is identity under pass to quotients to  $A$ . For this, a  $d \in \text{Der}(A, A)$  is mapped to  $a + b\varepsilon \mapsto a + b\varepsilon + d(a)\varepsilon$ . This is checked to be a ring morphism, and any desired morphism are like these.

The above construction is natural and functorial in  $A$ , so it glue together to give the global case.

 $\square$

**Prop. (7.12.1.16)[Smoothness and Conormal Sheaves].** ? Let  $X$  be a smooth variety over a field  $k$ , then an irreducible closed subscheme  $Y$  of codimension  $r$  in  $X$  is smooth iff  $\mathcal{K}_{Y/k}$  is locally free and (7.5.5.13) is exact on the left.

In this case,  $\mathcal{I}$  is locally generated by  $r$  elements and  $\mathcal{C}_{Y/X}$  is a locally free sheaf of rank  $r$  on  $Y$  by (7.6.8.9). ┘

*Proof:* Cf.[Hartshorne P178]. Should has something to do with (7.6.4.2), (7.6.4.15) and (7.6.4.14). □

**Prop. (7.12.1.17)[Adjunction Formulas].** For  $X \in \text{SmVar}/k$  and  $Y$  a smooth subvariety of codimension  $r$ . There is an exact sequence

$$0 \rightarrow \mathcal{I}/\mathcal{I}^2 \rightarrow \mathcal{K}_{X/k} \otimes \mathcal{O}_Y \rightarrow \mathcal{K}_{Y/k} \rightarrow 0$$

by (7.5.5.14). Taking the highest exterior power (7.5.1.25), we get:

$$\mathcal{K}_Y = \mathcal{K}_X \otimes \wedge^r \mathcal{N}_{Y/X} = \mathcal{K}_X \otimes (\wedge \mathcal{I}/\mathcal{I}^2)^{-1}$$

In particular, if  $r = 1$  then  $Y$  is a divisor  $D$  in  $X$ , the canonical sheaf

$$\mathcal{K}_Y \cong (\mathcal{K}_X \otimes \mathcal{L}(D))_Y, \quad \mathcal{K}_{\mathbb{P}_k^n/k} = \mathcal{O}(-n-1) \text{ (7.5.5.7).}$$

because  $\mathcal{I}_Y \cong \mathcal{L}(-Y)$  in this case so  $\mathcal{I}_Y/\mathcal{I}_Y^2 = L(-Y) \otimes \mathcal{O}_Y$ .

Taking dual, we get:

$$0 \rightarrow \mathcal{T}_Y \rightarrow \mathcal{T}_X \otimes \mathcal{O}_Y \rightarrow \mathcal{N}_{Y/X} \rightarrow 0$$

┘

**Prop. (7.12.1.18)[Geometric Genus].** For  $X \in \text{SmVar}/k$ , the **geometric genus**  $p_g$  is defined as the rank of the global section of the invertible canonical sheaf  $\mathcal{K}_X = \wedge^n \mathcal{K}_{X/k}$ . It is a birational invariance. With the same methods, we can prove the rank of global sections of any other functorially defined bundles of  $\mathcal{K}_X$  is birational invariance, e.g. Hodge numbers. ┘

*Proof:* For any rational map  $U \rightarrow Y$ , there is a subset  $V \in U$  and a local isomorphism  $V$  and  $f(V)$ , that will define an isomorphism of global sections. Because a nonzero section of an invertible sheaf cannot vanish on a dense open set  $f(V)$ , the morphism of global sections is injective into  $\Gamma(U, \mathcal{O}_U)$ . Now we find a  $U$  that  $\text{codim}(X - U) > 1$ , then we can use (5.3.5.12) to get  $\Gamma(U) = \Gamma(X)$ , then  $p_g(X) \geq p_g(X')$ , and the converse is also true. For this, we use valuation criterion of properness, then for any codimension 1 point, the stalk is a DVR, thus we find a  $\text{Spec } \mathcal{O}_p \rightarrow X'$ , this extends to a nbhd of  $p$  because  $X'$  is of f.t.. □

**Cor. (7.12.1.19).** By (7.5.5.7),  $\mathcal{K}_{\mathbb{P}_k^n} \cong \mathcal{O}(-n-1)$ , so it has no global section by (7.7.2.1),  $p_g(\mathbb{P}_k^n) = 0$ . Hence every rational variety over a field  $k$ , i.e. one that is birational to  $\mathbb{P}_k^n$ , has geometric genus 0. ┘

### Complete Varieties

**Lemma (7.12.1.20)[Rigidity Lemma].** Let  $X, Y \in \text{Var}/k, Z \in \text{Sch}^{\text{sep}}/k$  and  $f : X \times Y \rightarrow Z \in \text{Sch}/k$ . If  $X$  is complete and  $p \in X(k), y \in Y(k)$  s.t.  $f(\cdot, y) : X \rightarrow Z$  is constant, then  $f$  factors through the projection  $\text{pr}_Y : X \times Y \rightarrow Y$ . ┘

*Proof:* The equalizer is a closed subscheme of  $X \times Y$  as  $Z$  is separated (7.4.5.40), and  $X \times Y$  is a variety (7.12.1.9), thus we can check on closed points. Let  $g(y) = f(p, y) : Y \rightarrow Z$ , then we want to show  $f = g \circ \text{pr}_Y$ .

Let  $U$  be affine open in  $Z$ , then because  $X$  is universally closed,  $\text{pr}_Y$  is closed, so  $V = \text{pr}_Y(f^{-1}(Z \setminus U))$  is closed in  $Y$ . But if  $y \notin V(\bar{k})$ , then  $f(X, y) \subset U$ , and  $X$  is complete and connected, so  $f(\cdot, y)$  is constant (7.4.6.4), and  $f(x, y) = f(p, y)$ . Thus  $f = g \circ \text{pr}_Y$  on a non-empty open subset of  $X \times Y$ , which is a variety, so this is true on all of  $X \times Y$  by (7.12.1.11).  $\square$

**Prop. (7.12.1.21)[See-Saw Principle].** Let  $X$  be a complete variety over a field  $k$  and  $Y$  a  $k$ -scheme, then for any line bundle  $\mathcal{L}$  on  $X \times Y$ , there is a closed subscheme  $Y_1 \subset Y$  s.t. for any morphism  $f : S \rightarrow Y$ ,  $(1 \times f)^*\mathcal{L}$  is trivial on  $X \times S$  iff  $f$  factors through  $Y_1$ .  $\lrcorner$

*Proof:* This  $\mathcal{L}$  corresponds to a morphism  $Y \rightarrow \underline{\text{Pic}}_{X/k}$ , and clearly  $Y_1$  is the fiber of  $Y$  over  $e \in \underline{\text{Pic}}_{X/k}$ .  $\square$

**Cor. (7.12.1.22).** Let  $X$  be a complete variety over a field  $k$  and  $Y$  a reduced locally algebraic  $k$ -scheme, if  $\mathcal{L}, \mathcal{M}$  are two line bundles on  $X \times Y$  s.t.  $\mathcal{L}_y \cong \mathcal{M}_y$  for all closed points  $y \in Y$ , and for some  $x \in X(k)$ ,  $\mathcal{L}_x \cong \mathcal{M}_x$ , then  $\mathcal{L} \cong \mathcal{M}$ .  $\lrcorner$

**Prop. (7.12.1.23)[Theorem of the Cube].** If  $X, Y$  are complete varieties over a field  $k$ , and  $Z$  is a connected, locally Noetherian  $k$ -scheme, if  $x, y$  are rational points of  $X, Y$  resp. and  $z \in Z$ . Supposed  $\mathcal{L} \in \text{Pic}(X \times Y \times Z)$  that is trivial on  $x \times Y \times Z, X \times y \times Z, X \times Y \times z$ , then  $\mathcal{L}$  is trivial.  $\lrcorner$

*Proof:* Let  $Z'$  be the maximal closed subscheme of  $Z$  given by (7.12.1.21). We show that  $Z'$  is open, thus it is all of  $Z$ : If  $\zeta \in Z'$ , let  $I \subset \mathcal{O}_{Z, \zeta}$  be the ideal defining  $Z'$ , we show  $I = (0)$ , which is equivalent to  $Z'$  containing a nbhd of  $\zeta$  (locally Noetherian used). If not, then because  $\cap \mathfrak{m}^n = 0$  by Krull's theorem (locally Noetherian used), there is an  $n \geq 1$  that  $I \subset \mathfrak{m}^n, I \not\subset \mathfrak{m}^{n+1}$ . Let  $\mathfrak{a}_1 = (I, \mathfrak{m}^{n+1})$ , and  $\mathfrak{m}^{n+1} \subset \mathfrak{a}_2 \subset \mathfrak{a}_1$  that  $\dim_{k(\zeta)}(\mathfrak{a}_1/\mathfrak{a}_2) = 1$  (so  $\mathfrak{a}_1 = \mathfrak{a}_2 + k(\xi)a$  for some  $a \in \mathfrak{a}_2$ ), and let  $Z_i \subset \text{Spec } \mathcal{O}_{Z, \xi}$  be the closed subscheme defined by  $\mathfrak{a}_i$ . Let  $\mathcal{L}_i$  be the restriction of  $\mathcal{L}$  on  $X \times Y \times Z_i$ . If we show that  $\mathcal{L}_2$  is trivial, then  $Z_2$  is contained in  $Z'$ , which is contradiction because  $I \not\subset \mathfrak{a}_2$ .

For this, notice that  $\mathcal{L}_1$  is trivial, and to show that  $\mathcal{L}_2$  is trivial, it suffices to lift a non-vanishing global section  $s$  of  $\mathcal{L}_1$  to  $\mathcal{L}_2$ , because  $Z_1, Z_2$  has the same underlying set.

For this, notice there is an exact sequence  $0 \rightarrow k(\xi) \xrightarrow{a} \mathcal{O}_{Z_2} \rightarrow \mathcal{O}_{Z_1} \rightarrow 0$ , where  $k(\xi)$  is the skyscraper sheaf at  $\xi$ . So the obstruction of the lifting is an element  $\xi \in H^1((X \times Y)_{k(\xi)}, \mathcal{O}_{(X \times Y)_{k(\xi)}})$ . But now the conditions show that  $\xi$  is zero under the pullback along  $x \times Y \hookrightarrow X \times Y$  and  $X \times y \hookrightarrow X \times Y$ . So by Kunneth formula (7.7.1.10) and (7.12.1.12),  $\xi$  vanishes.  $\square$

## 2 Projective Varieties

**Example (7.12.2.1)[Non-Projective Smooth Proper Varieties].** There are proper smooth complex 3-folds that are not projective. Examples are given in [Har77]P443<sup>?</sup>.  $\lrcorner$

**Prop. (7.12.2.2)[Affine Dimension Theorem].** Suppose  $X, Y$  are subschemes of  $\mathbb{A}_k^n$  of codimensions  $d$  and  $e$  resp., and  $d + e \leq n$ , then every non-empty irreducible component of  $X \cap Y$  has dimension  $\geq n - d - e$ .  $\lrcorner$

*Proof:* If  $Y$  is an intersection of  $e$  hypersurfaces, then this follows from Krull's height theorem (5.2.4.18). In general, notice the diagonal map  $\Delta : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n \times \mathbb{A}_k^n$  is an isomorphism onto the diagonal  $T$  defined by  $\{X_i = Y_i\}_{i=1, \dots, n}$ , thus it is a complete intersection in  $\mathbb{A}_k^n \rightarrow \mathbb{A}_k^n \times \mathbb{A}_k^n$ , and  $\Delta$  induces an isomorphism  $X \cap Y \cong X \times Y \cap T$ , so we are done.  $\square$

**Cor. (7.12.2.3) [Projective Dimension Theorem].** Suppose  $X, Y$  are subvarieties of  $\mathbb{P}_k^n$  of codimensions  $d$  and  $e$  resp., and  $d + e \leq n$ , then  $X$  and  $Y$  intersect.  $\lrcorner$

*Proof:* Take the affine cone, then item2 shows any irreducible component of  $\overline{X} \cap \overline{Y}$  containing the origin has dimension  $\geq 1$ , which means  $X \cap Y \neq \emptyset$ .  $\square$

**Prop. (7.12.2.4) [Transversal Intersection].** If  $X \subset \mathbb{P}_k^n$  is a closed subscheme of dimension  $r$ , then there is an intersection of  $r + 1$  non-empty hypersurfaces missing  $X$ . And if  $k$  is infinite, these hypersurfaces can be chosen to be hyperplanes.

If  $\#k = \infty$ , there is an intersection of  $r$  non-empty hypersurfaces intersecting  $X$  at f.m. points.  $\lrcorner$

*Proof:* Let  $\eta_1, \dots, \eta_n$  be the generic points of  $X$ , we want to find a hypersurface  $F$  that doesn't contain any of these generic points. We use induction on  $n$ . If  $n = 1$ , then there is clearly a hyperplane missing  $\eta_1$ . If we find a hypersurface missing  $\eta_1, \dots, \eta_{n-1}$ , if it also misses  $\eta_n$ , we are done, if it contains  $\eta_n$ , we can change  $\eta_n$  to  $\eta_i$  and consider again. If we find polynomials  $F_1, \dots, F_n$  that  $F_i(\eta_i) = 0, F_i(\eta_j) \neq 0$ , we may assume  $\deg(F_i)$  are the same, so  $\sum_i F_1 \dots \widehat{F_i} \dots F_n$  is a polynomial non-zero on each of  $\eta_i$ . If moreover  $k$  is infinite, then we want to find a hyperplane missing every generic point of  $X$ . But each generic point corresponds to a graded ideal  $I_i$  of  $k[T_0, \dots, T_n]$ , and  $(I_i)_1 \neq k\{T_0, \dots, T_n\}$  are proper linear subspaces. As  $k$  is infinite, we can choose a  $a_0T_0 + \dots + a_nT_n$  not in these subspaces, thus non-zero on any generic point of  $X$ .

Now  $X \cap F$  is a closed subscheme of  $\mathbb{P}_k^n$  of dimension  $r - 1$ , thus we can use induction on  $r$  to finish.  $\square$

### 3 Birational Geometry

**Notation (7.12.3.1).**

- Let  $k \in \text{Field}$ .

$\lrcorner$

**Prop. (7.12.3.2).** Any  $X \in \text{Var}/k$  is birational to a hypersurface in  $\mathbb{A}_k^n$  for some  $n$ .  $\lrcorner$

*Proof:* Cf.[Diophantine Geometry, P575].  $\square$

**Prop. (7.12.3.3) [Prevarieties and Function Fields].** The following categories are equivalent.

- The category of prevarieties over  $k$  with dominant rational morphisms.
- The dual category of f.g. field extensions over  $k$ .

$\lrcorner$

*Proof:* Cf.[Sta]0BXN. ?  $\square$

**Prop. (7.12.3.4).** Let  $\varphi : X \rightarrow X'$  be a rational map of  $K$ -varieties with  $X$  smooth. If the base change  $\varphi_{\overline{K}}$  extends to a morphism  $X_{\overline{K}} \rightarrow X'_{\overline{K}}$ , then  $\varphi$  extends to a morphism  $X \rightarrow X'$ .  $\lrcorner$

*Proof:* Let  $U$  be an open dense subset that  $\varphi$  is defined. For a point  $x$ , let  $x' = \varphi_{\overline{K}}(x)$ , then  $x'$  is in the closure of  $\varphi(U)$ . By (7.4.7.2), it suffices to construct a morphism  $\mathcal{O}_{X', x'} \rightarrow \mathcal{O}_{X, x}$ , i.e., to prove for any rational function  $f$  regular at  $x'$ ,  $\varphi \circ f$  is regular at  $x$ . The argument is the same as that of (7.4.6.14).

For any such  $f$ ,  $f_{\overline{K}} \circ \varphi_{\overline{K}}$  is regular at  $x$ , thus no pole of  $\text{div}(f_{\overline{K}} \circ \varphi_{\overline{K}})$  passes through  $x$ . For the rest, Cf.[Diophantine Geometry, P576]. ?  $\square$

**Prop. (7.12.3.5) [Generic Separable Degree].** Let  $\varphi : X \rightarrow X'$  be a dominant morphism of varieties over a field  $k$  of the same dimension, then there exists an open dense subscheme  $U'$  of  $X'$  s.t.  $\varphi^{-1}(U') \rightarrow U'$  is finite and the fibers all have cardinality  $\deg_s(\varphi)$ .  $\lrcorner$

*Proof:* By (7.12.3.3), we can decompose  $\varphi$  and assume it is either separable or purely inseparable, and also primitive. If it is separable, let it be generated by  $t$ . After shrinking (and cutting closure of images), we may assume the coefficients of  $t$  are regular functions, and  $\Gamma(X) = \Gamma(X')[t]/(q(t))$ . As  $q$  is separable, there exists  $a, b \in K(X')$  s.t.  $aq + bq' = 1$ . We may shrink  $X'$  and assume  $a, b \in \Gamma(X')$ , so  $q$  is also separable over any fiber of  $X$ , which means the fibers all have cardinality  $\deg_s(\varphi)$ .

If it is purely inseparable, let  $q(t) = t^{p^k} - h$  be the minimal polynomial of  $t$  over  $K(X')$ ,  $h \in K(X')$ . Then after shrinking, we may assume  $h \in \Gamma(X')$ , thus  $q(t)$  is purely inseparable over any fiber of  $X$ , which means  $\square$

**Def. (7.12.3.6) [Kodaira Dimensions].** For  $X \in \text{SmPrprVar}^n/k$ , the **Kodaira dimension**  $\text{Kod}(X)$  of  $X$  is defined to be  $-\infty$  if  $h^0(X, \mathcal{K}_X^d) = 0$  for  $d$  large, or

$$\text{Kod}(X) = \sup\{k \in \mathbb{R}_+ | h^0(X, \mathcal{K}_X^d)/d^k \text{ is bounded}\}.$$

Then  $\text{Kod}(X) \in \{-\infty\} \cup (\mathbb{Z} \cap [0, n])$ .  $\lrcorner$

## 4 Others

**Prop. (7.12.4.1).** Varieties are triangulable.  $\lrcorner$

*Proof:* Cf. [Triangulation of Algebraic Sets, Hironaka].  $\square$

## Fano Varieties

**Def. (7.12.4.2) [Fano Varieties].** A **Fano variety** is a complete smooth variety  $X$  over a field  $K$  s.t.  $\mathcal{K}_X^*$  is ample.  $\lrcorner$

**Prop. (7.12.4.3).** a smooth complete intersection of hypersurfaces in  $\mathbb{P}_K^n$  is Fano if and only if the sum of their degrees is at most  $n$ .  $\lrcorner$

**Prop. (7.12.4.4) [Kollár-Miyaoka-Mori].** Fano varieties over an alg.closed field are rationally chain connected.  $\lrcorner$

*Proof:* Cf. [Rational connectedness and boundedness of Fano manifolds, Kollar].  $\square$

## Rationally Connected Varieties

**Def. (7.12.4.5) [Rationally Connected Varieties].**  $X \in \text{Var}/K$  is called **rationally connected** if any two points of  $X(\overline{K})$  can be connected by a rational curve (7.13.1.1) over  $\overline{K}$ .  $\lrcorner$

## 5 Relative Varieties

**Def. (7.12.5.1) [Varieties over Schemes].** Let  $S \in \text{Sch}$ , a (proper/smooth) **variety over  $S$**  is a (proper/smooth) flat morphism  $f : X \rightarrow S$  s.t. all the geometric fibers of  $f$  are geo.integral over the resp. residue field  $k(s)$ . It can be regarded as a family of (proper/smooth) varieties parametrized by  $S$ . the category of varieties over  $S$  is denoted by  $\text{Var}/S$ .

Being a (proper/smooth) variety is stable under base change.  $\lrcorner$

**Prop. (7.12.5.2) [Global Sections].** For  $S \in \mathbf{NSch}$  and  $X \in \mathbf{Var}^{\text{proper}}/S$ ,  $\mathcal{O}_S \rightarrow f_*\mathcal{O}_X$  is an isomorphism.  $\lrcorner$

*Proof:* For any closed point  $s \in S$ ,  $k(s) \rightarrow H^0(X_s, \mathcal{O}_{X_s})$  is an isomorphism by (7.12.1.12), and this isomorphism factors through  $k(s) \rightarrow f_*(\mathcal{O}_X) \otimes k(s) \rightarrow H^0(X_s, \mathcal{O}_{X_s})$ . So  $f_*(\mathcal{O}_X) \otimes k(s) \rightarrow H^0(X_s, \mathcal{O}_{X_s})$  is surjective, thus it is an isomorphism by item 4  $\rightarrow$  3 of (10.9.3.17). Thus the first map is also an isomorphism.

Now  $\mathcal{O}_S \rightarrow f_*(\mathcal{O}_X)$  is a surjection at  $s$  by Nakayama. Let  $\mathcal{Q}$  be the coherent sheaf on  $S$  associated to  $\mathcal{F}$ , then by (10.9.3.17), it is free at  $s$ , and  $\mathcal{Q}_s = H^0(X_s, \mathcal{O}_{X_s})$  is of rank 1, but also  $\mathcal{Q}^\vee \cong f_*(\mathcal{O}_X)$ , so  $\mathcal{O}_S \rightarrow f_*(\mathcal{O}_X)$  is in fact a surjection at  $s$ . Now  $s$  is arbitrary, so  $\mathcal{O}_S \rightarrow f_*(\mathcal{O}_X)$  is an isomorphism.  $\square$

**Cor. (7.12.5.3).** If  $S \in \mathbf{NSch}$  and  $f : X \rightarrow S \in \mathbf{SmPrprVar}^d/S$ , then  $f_*\Omega_{X/S} \cong R^1f_*\mathcal{O}_X$  is finite locally free of rank  $d$ , by Grothendieck duality and (7.6.4.2).  $\lrcorner$

**Prop. (7.12.5.4).** Let  $(\mathcal{O}_K, K, k)$  be a DVR and  $X \in \mathbf{Sch}^{\text{sm}}/R$  s.t.  $X_K$  is geo.integral and  $X_k$  is proper, then  $X \in \mathbf{SmPrprVar}/\mathcal{O}_K$ .  $\lrcorner$

*Proof:* Cf. [Good Reduction of Abelian Varieties, P495] ?  $\square$

**Prop. (7.12.5.5) [Generic Geometrically Irreducibility].** For  $f : X \rightarrow Y \in \mathbf{Sch}$  of finite type, if  $Y$  is irreducible with generic point  $\eta$ , and  $X_\eta$  is geometrically irreducible, then there exists a non-empty open subscheme  $V \subset Y$  s.t.  $f : X_V \rightarrow V$  has geometrically irreducible fibers.  $\lrcorner$

*Proof:* Cf. [Sta]0559.  $\square$

## 7.13 Curves

*“Everyone knows what a curve is, until he has studied enough mathematics to become confused through the countless number of possible exceptions.”*

–F. Klein

References are [黎曼曲面, 伍鸿熙], [Vak17]Chap19, [?G–H78], [Har77]Chap4 and [Sta]Chap53. In this section, properties to morphisms of relative dimension  $\leq 1$  are studied.

**Notation (7.13.0.1).**

- Let  $k \in \text{Field}$ .
- Use notations defined in [Cohomology of Schemes](#).
- Use notations defined in [Varieties](#).

┘

### 1 Basics

**Def. (7.13.1.1) [Rational Curve].** A **rational curve** over a field  $k$  is a curve that is birational to  $\mathbb{P}_k^1$ .  
┘

**Lemma (7.13.1.2).** If  $X$  is an integral separated scheme and  $U \subset X$  is an non-empty affine open that  $X \setminus U$  is a finite set of points with  $\mathcal{O}_{X, x_i}$  Noetherian of dimension 1, then there exists a base-point free invertible sheaf  $\mathcal{L} \in \text{Pic}(X)$  and a section  $s$  s.t.  $U = X_s$ .  
┘

*Proof:* Cf. [Sta]09NB. □

**Prop. (7.13.1.3).** A Noetherian separated scheme of dimension  $\leq 1$  has an ample invertible sheaf. ┘

*Proof:* First reduce to the case when  $X_{\text{red}}$ , because (7.13.2.19) shows any invertible sheaf on  $X_{\text{red}}$  is a pullback of a sheaf of  $X$  and (7.5.4.15) shows this sheaf is ample.

Second we reduce to the case  $X$  is integral. Let  $X_i$  are the integral irreducible components of  $Z$ , Cf. [Sta]09NX? □

Finally, for  $X$  integral, the assertion follows from (7.13.1.2) and (7.5.4.10). □

**Cor. (7.13.1.4) [Complete Precurves are Projective].** A separated algebraic scheme  $X$  of dimension 1 over  $k$  is  $H$ -(quasi)projective, by (7.13.1.3) and (7.5.4.24). If  $X$  is proper, then it is projective. ┘

**Prop. (7.13.1.5) [Completion of Curves].** For a separated algebraic  $k$ -scheme  $X$  of dimension  $\leq 1$ , there is an open immersion  $j : X \rightarrow \overline{X}$  that

- $\overline{X}$  is projective over  $k$ .
- $j(X) \subset \overline{X}$  is dense and schematically dense open subscheme.
- $\overline{X} \setminus X$  consists of f.m. closed points  $\{x_i\}$  of  $\overline{X}$ .

This  $\overline{X}$  is called a **completion curve** of  $X$ . And when  $X$  is reduced, the stalk at  $x_i$  are DVRs. In particular, it is non-singular if  $X$  is non-singular. ┘

*Proof:* By (7.13.1.3), we can assume  $X$  is a locally closed subscheme of  $\mathbb{P}_k^n$ . Let  $\overline{X}$  be the scheme theoretic image (7.4.5.13) of the inclusion, then 1, 2 holds by (7.4.5.21). 3 holds because  $\overline{X} \setminus X$  is Noetherian of dimension 0.

For the last assertion, Cf. [Sta]0BXW. □



**Cor. (7.13.1.6).** A morphism of prevarieties  $X \rightarrow Y$  with  $X$  a precurve (thus reduced) and  $Y$  proper over a field  $k$  factors through the completion  $\overline{X}$  of  $X$  by (7.4.6.14). In particular, the completion curves of  $X$  are unique.  $\lrcorner$

**Prop. (7.13.1.7) [Affine or Projective].** A precurve over a field  $k$  is either affine (not proper) or H-projective (proper).  $\lrcorner$

*Proof:* Cf. [Sta]0A27?, (Hard).  $\square$

**Cor. (7.13.1.8).** Let  $X$  be a separated scheme algebraic over a field  $k$ . If  $\dim X \leq 1$  and no irreducible component of  $X$  is proper of dimension 1, then  $X$  is affine.  $\lrcorner$

*Proof:* Let  $X_i$  be f.m. irreducible components of  $X$ , then they are precurves in the induced reduced structure, so they are affine by (7.13.1.7). Now  $\coprod X_i \rightarrow X$  is a finite surjective morphism, so  $X$  is affine by (7.4.4.36).  $\square$

**Prop. (7.13.1.9).** A map from a proper connected scheme to a precurve is either constant or surjective.  $\lrcorner$

*Proof:* Because the closed subset of a precurve is either itself or f.m. closed points.  $\square$

**Prop. (7.13.1.10) [Constant or Finite].** Let  $f : X \rightarrow Y$  be a morphism of schemes over a field  $k$  that  $Y$  is separated and  $X$  is proper of dimension  $\leq 1$ . If the image of every irreducible component of  $X$  is not a pt, then  $f$  is finite. If  $Y$  is a precurve, then it is moreover surjective.  $\lrcorner$

*Proof:* Cf. [Sta]0CCL. ?  $\square$

**Def. (7.13.1.11) [Separable Morphisms].** Let  $f : X \rightarrow Y$  be a non-constant morphism of precurves, then  $f$  is finite surjective by (7.13.1.10). so  $\deg(f)$  is finite (7.4.5.6), and we can define separable/purely separable morphisms as in (7.4.5.6).  $\lrcorner$

### Non-Singular Curves

**Example (7.13.1.12) [Non-Singular and Smooth Curves].** A non-singular curve  $C/k$  may not be smooth, for example, the curve  $tx^p + z^{p-1}y + y^p = 0$  is regular and geo.reduced, but not smooth.  $\lrcorner$

*Proof:* To show it is regular, check directly that at the closed point  $[1, t^{1/p}, 0]$ , the maximal ideal is just  $(y^p - t, z)$  and it just generated by  $z$ .  $\square$

**Def. (7.13.1.13) [Uniformizer].** Let  $C/k$  be a precurve, then the local rings of  $C$  at a non-singular closed point  $p$  is a DVR by (12.2.3.5). Then an element  $t \in K(C)$  with valuation 1 is called a **uniformizer at  $p$** .  $\lrcorner$

**Prop. (7.13.1.14).** Let  $C$  be a precurve over  $k$ , and  $t \in K(C)$  is a uniformizer at some non-singular point, then  $K(C)$  is a finite separable extension of the subfield generated by  $t$ .  $\lrcorner$

*Proof:*  $K(C)$  is a finite extension of  $K(t)$ , because  $\text{tr. deg}(K(C)/k) = 1$  and  $t$  is not algebraic over  $K$  by valuation reasons. Then it suffices to show for any  $x \in K(C)$ ,  $x$  is separable over  $K(t)$ . Let  $\Phi(X, T) = \sum a_{ij} T^i X^j$  be a minimal polynomial of  $x$  over  $K(T)$ . It is separable iff for some  $(j, p) = 1$  and some  $i$ ,  $a_{ij} \neq 0$ . If it is not separable then we can write  $\Phi(X, T) = \sum_{k=0}^{p-1} \varphi_k(X, T)^p T^k$ . But each  $\varphi_k(x, t)^p t^k$  has distinct valuations, unless they are all zero, contradicting the fact  $\Phi$  is a minimal polynomial.  $\square$



**Prop. (7.13.1.15) [Non-singular Complete Precurves and DVRs].** Let  $C$  be a non-singular complete precurve over  $k$ , then all valuation rings of  $K(C)$  containing  $k$  are DVRs, and the set of closed points of  $C$  correspond to the set of DVRs in  $K(C)$  containing  $k$ , which is denoted by  $\Sigma_{K(C)}^0$ .  $\lrcorner$

*Proof:* This is a consequence of the valuation criterion (7.4.6.13). Notice that the local rings of  $C$  at closed points are all DVRs (12.2.3.5), thus they must equal to the valuation ring given.  $\square$

**Prop. (7.13.1.16) [Extension of Rational Maps].** Rational map from a non-singular precurve to a complete prevariety is the same as a morphism, by (7.4.6.15).  $\lrcorner$

**Cor. (7.13.1.17).** Two birationally equivalent normal proper precurves over a field is isomorphic.

Thus if a normal precurve is birationally equivalent to another normal complete curve, then it is an open immersion, by (7.13.1.5).  $\lrcorner$

**Prop. (7.13.1.18) [Category of Non-singular Projective Precurves].** Let  $k$  be a field, the following categories are equivalent:

1. The opposite category of f.g. field extensions of  $k$  of trans.deg 1 with injective  $k$ -homomorphisms.
2. The category of precurves and dominant rational maps.
3. The category of normal complete precurves over  $k$  with non-constant morphisms.
4. The category of non-singular projective precurves over  $k$  with non-constant morphisms.

$\lrcorner$

*Proof:* 1 and 2 are equivalent by (7.12.3.3).

3 and 4 are equivalent by (7.13.1.7) and the fact normal and regular are the same (7.4.2.12).

For the rest, Cf. [Sta]0BY1. ?

$\square$

**Cor. (7.13.1.19) [Non-singular Projective Model].** Comparing this and (7.12.3.3), we see that every precurve over  $k$  is birational to a unique non-singular proper precurve over  $k$  with the same function field, which is called the **non-singular projective model**.  $\lrcorner$

**Prop. (7.13.1.20) [Flatness and Associated Points].**  $f : X \rightarrow Y$  with  $Y$  integral and regular of dimension 1. Then  $f$  is flat iff every associated prime of  $X$  is mapped to the generic point of  $Y$ .

In particular when  $X$  is reduced, this is equivalent to every irreducible component of  $X$  dominants  $Y$ , by (5.2.5.25).  $\lrcorner$

*Proof:* If  $x$  is mapped to a closed pt of  $Y$ , then  $\mathcal{O}_{y,Y}$  is a DVR, let  $t$  be a uniformizer, then  $t$  is not a zero-divisor, and  $f^\sharp(t) \in \mathfrak{m}_x$  is also not a zero-divisor. So  $x$  is not an associated point.

Conversely, to show  $f$  is flat, if  $y$  is the generic point, then  $\mathcal{O}_{y,Y}$  is a field, so it is flat. When  $y$  is a closed pt,  $\mathcal{O}_{y,Y}$  is a DVR, so by (5.4.1.11), we need to show that it is torsion free. If it is not, then  $f^\sharp(t)$  must be a zero-divisor for a uniformizer  $t$  of  $\mathcal{O}_{y,Y}$ . But then it is contained in some associated prime  $p$  of  $\mathcal{O}_{x,X}$  (5.2.5.17). Now  $p$  is mapped to  $y$ , which is a contradiction.  $\square$

**Cor. (7.13.1.21) [Morphism to a Non-singular Curve is Flat].** If  $f : X \rightarrow Y$  is a dominant morphism from a prevariety to a non-singular curve over  $k$ , then  $f$  is flat.  $\lrcorner$

**Cor. (7.13.1.22) [Flat Specializations along Curves].** Let  $Y$  be integral and regular of dimension 1 and  $P$  a closed pt.  $X$  is a closed subscheme in  $\mathbb{P}_{Y-P}^n$  that is flat over  $Y - P$ , then there is a unique closed subscheme  $\overline{X}$  closed in  $\mathbb{P}_Y^n$  that is flat over  $Y$  and restrict to  $X$  on  $\mathbb{P}_{Y-P}^n$ .  $\lrcorner$

*Proof:* Choose the scheme-theoretic closure of  $X$  in  $\mathbb{P}_Y^n$ . Cf.[Hartshorne P258].  $\square$

**Cor.(7.13.1.23) [Finite Flatness].** Any non-constant morphism from a precurve to a nonsingular precurve is finite locally free, by (7.13.1.20) and (7.13.1.10).  $\lrcorner$

**Prop.(7.13.1.24).** A projective non-degenerate non-singular curve of degree  $d$  in  $\mathbb{P}_k^n$  is isomorphic to the  $n$ -tuple embedding, and  $d = n$ .

This has easy generalizations to surfaces and higher dimensions.  $\lrcorner$

*Proof:* (7.5.3.16) shows  $\mathcal{O}_X(1) \cong \mathcal{O}(d)$  over  $\mathbb{P}_k^1$ , and the restriction of global sections is injective. So the global section is an isomorphism, and it defines the embedding up to a linear automorphism.  $\square$

**Prop.(7.13.1.25) [Genera Equal].** For a complete smooth curve  $X$  over a field  $k$ ,

$$p_a(X) = p_g(X) = \dim_k H^1(X, \mathcal{O}_X)$$

by Serre duality (7.9.2.18) and (7.12.1.18) (7.7.3.6).

So from now on we use genus to denote the arithmetic genus.  $\lrcorner$

**Cor.(7.13.1.26) [Topological Genus].** By GAGA, for a complex complete smooth curve, the genus also equals the topological genus.  $\lrcorner$

### Morphisms Between Non-singular Curves

**Prop.(7.13.1.27) [Degree and Analytic Degree].** Let  $f : C \rightarrow C'$  be a non-constant morphism between non-singular complete curves over  $\mathbb{C}$ , then the degree defined in (7.4.5.6) is the same as the degree as map of Riemann surfaces.  $\lrcorner$

*Proof:* This is because the map is finite locally free (7.13.1.23), thus for an affine open subset  $U = \text{Spec } A$  of  $C'$ , the  $f^{-1}(U) = \text{Spec } B$  where  $B \cong A^{\otimes n}$  as  $A$ -modules, where  $n$  is the degree. Then clearly for most  $x \in U$ ,  $\#f^{-1}(x) = n$  (look at the minimal polynomial of a family of generators for  $B$  over  $A$ ).  $\square$

**Def.(7.13.1.28) [Ramification Degrees].** Let  $f : X \rightarrow Y$  be a morphism of non-singular precurves over a  $k$ , let  $P \in X$  be a closed point and  $f(P) = Q$ . Because the local rings are DVRs, we define the **ramification degree**  $e_P(f)$  where  $f^\#(\mathfrak{m}_Q)\mathcal{O}_{X,P} = \mathfrak{m}_P^{e_P(f)}$ .

$f$  is called **weakly unramified at  $P$**  if  $e_P(f) = 1$ , and it is called **unramified at  $P$**  if moreover  $\kappa(P)/\kappa(f(P))$  is separable.  $f$  is called **tamely ramified at  $P$**  if  $\text{?}$

Notice that unramifiedness just means the morphism is unramified, by (7.6.5.8) and (7.13.1.23).  $\lrcorner$

**Prop.(7.13.1.29).** Let  $f : X \rightarrow Y$  be a non-constant morphism of non-singular precurves over  $k$ , then

- If  $g : Y \rightarrow X$  be another morphism of non-singular precurves, then for any  $P_1 \in X$ ,  $e_P(g \circ f) = e_P(f)e_{f(P)}(g)$ .
- for any closed point  $Q \in C_2$ ,  $\sum_{P \in f^{-1}(Q)} e_P(f)[k(P) : k(Q)] = \deg(f)$ .
- If  $k$  is alg.closed, then for a.e. closed point  $Q \in C_2$ ,  $\#f^{-1}(Q) = \deg_s(f)$ .

$\lrcorner$

*Proof:* 1: Trivial.

2: This follows from (7.13.1.46).

3: This follows from (7.12.3.5).  $\square$

**Prop. (7.13.1.30).** If  $f : X \rightarrow Y$  is a non-constant morphism of percurves over  $k$  s.t.  $X$  is smooth and  $Y$  is non-singular, then  $Y$  is also smooth, by (7.1.5.28).  $\lrcorner$

**Prop. (7.13.1.31).** Let  $X$  be a smooth curve over  $k$ ,  $x \in X$  and  $\bar{x} \in X_{\bar{k}}$  is a point mapping to  $x$ , then the ramification degree of  $\mathcal{O}_{X,x} \subset \mathcal{O}_{X_{\bar{k}},\bar{x}}$  equals the inseparable degree of  $k(x)/k$ .  $\lrcorner$

*Proof:* By (7.6.6.7), we can find a étale map  $X \rightarrow \mathbb{A}_k^1$ , which has ramification degree 1 and separable, so we may assume  $X = \mathbb{A}_k^1$ . Then the assertion is clear.  $\square$

**Prop. (7.13.1.32) [Separable Morphisms].** Let  $f : X \rightarrow Y$  be a morphism of smooth curves over  $k$ , then the following are equivalent:

1.  $f$  is finite separable.
2.  $df^* : f^*\Omega_{Y/k} \rightarrow \Omega_{X/k}$  is non-zero.
3.  $\Omega_{X/Y}$  is supported on a proper closed subscheme of  $X$ .
4. There exists a non-empty open subset  $U \subset X$  s.t.  $f$  is unramified.
5. There exists a non-empty open subset  $U \subset X$  s.t.  $f$  is étale.

$\lrcorner$

*Proof:* As  $X, Y$  are smooth,  $\Omega_{X/k}, \Omega_{Y/k}$  are invertible sheaves, and the exact sequence

$$f^*\Omega_{Y/k} \rightarrow \Omega_{X/k} \rightarrow \Omega_{X/Y} \rightarrow 0$$

shows 2, 3 are both equivalent to  $\Omega_{X/Y, \xi} = 0$ .

3, 4, 5 are equivalent as  $f$  is automatically flat (7.13.1.21).

1, 5 are equivalent by (7.6.5.9).  $\square$

**Prop. (7.13.1.33) [Riemann-Hurewitz for Separable Maps].** If  $f : X \rightarrow Y$  is a non-constant separable map between two complete non-singular curves over a field  $k$ , then

$$2g_X - 2 = \deg(f)(2g_Y - 2) + \sum_{x \in X} d_x [\kappa(x) : k],$$

where  $d_x = \text{length}_{\mathcal{O}_{X,x}}(\Omega_{X/Y,x}) \geq 0$  satisfies  $d_x \geq e_x(f) - 1$ , and equality holds iff  $f$  is tamely unramified at  $P$ .  $\lrcorner$

*Proof:* By (7.8.5.12), in this case the vanishing locus  $R$  of  $df^*$  is an effective Cartier divisor, and  $f^*\mathcal{K}_{Y/k} \otimes \mathcal{O}(R) \cong \mathcal{K}_{X/k}$ , so by (7.13.2.10)(7.13.2.3)(7.13.2.7),

$$2g_X - 2 = \deg(\mathcal{K}_{X/k}) = \deg(f^*\mathcal{K}_{Y/k} \otimes \mathcal{O}(R)) = \deg(f) \deg(\mathcal{K}_{Y/k}) + \deg(R) = \deg(f)(2g_Y - 2) + \deg(R).$$

For analysis of  $d_x$ , Cf. [Sta]0C1F. ?  $\square$

**Cor. (7.13.1.34).** The only geo.integral unramified finite covering of  $\mathbb{P}_k^1$  is itself.  $\lrcorner$

**Thm. (7.13.1.35) [De Franchis].** Let  $k \in \mathbf{Field}$  and  $C, C'$  are two complete smooth curves over  $k$ . Then if  $g(C) \geq 2$ , there are only f.m. non-constant maps  $C' \rightarrow C$ .

In particular,  $\# \text{Aut}(C) < \infty$ .  $\lrcorner$

*Proof:* We prove for  $k = \mathbb{C}$ ? Cf.[Mil08]P146.

Any automorphism of  $C$  fixes its set of Weierstrass sets, which is finite, so we only need to consider the case that it fixes all Weierstrass points.

If  $C$  is hyperelliptic, consider a hyperelliptic map, this covering has an involution of  $C$ , and this is just  $C \mapsto C/\tau$ . So modulo  $\tau$ , it suffices to show  $\mathbb{P}^1$  has f.m. automorphisms fixing the branch points. But this is true, as  $2g + 2 > 3$ .

If  $C$  is non-hyperelliptic, then there are more than  $2g + 2$  Weierstrass points. But we can find a function  $f$  on  $C$  with  $g + 1$  zeros and poles, by Riemann-Roch, then for an automorphism  $\varphi$  of  $C$ ,  $f - \varphi^*f$  has no more than  $2g + 2$  poles, so also has no more than  $2g + 2$  zeros. But  $\varphi$  fixes all Weierstrass points, so it has more than  $2g + 2$  zeros, contradiction.  $\square$

**Prop. (7.13.1.36) [Hurewitz's Automorphism Theorem].** If  $C$  is a complete smooth curve of genus  $g \geq 2$  over a field of characteristic 0, then  $\# \text{Aut}(C) \leq 84(g - 1)$ .  $\lrcorner$

*Proof:* We may pass to the alg.closure. By(7.13.1.35),  $\# \text{Aut}(C) < \infty$ . In fact, let  $G$  be a finite group acting on  $C$ , let  $C'$  be the non-singular precurve corresponding to  $C'$ , then by(3.2.8.7),  $C \rightarrow C'$  is a Galois cover with Galois group  $G$ . Now by Dedekind extension theory,  $G$  acts transitively on the preimage of a given point. Suppose there are  $n$  branched points with ramification degrees  $r_i$ , then by Riemann-Hurewitz(7.13.1.33)(notice in this case it is tamely unramified):

$$(2g - 2) = |G|(2g(C') - 2 + \sum_{i=1}^n \frac{r_i - 1}{r_i}).$$

Then a combinatorial argument shows the maximal possible  $|G|$  is obtained when  $g(C') = 0, n = 3$  and  $(r_1, r_2, r_3) = (2, 3, 7)$ .  $\square$

**Remark (7.13.1.37).** In the case  $\text{char } k \neq 0$ , there may be more automorphisms. For example, if  $p \in \mathbb{P} \setminus \{2\}$ , the completion of the affine curve  $y^{p^n} = x + x^{p^n+1}$  has genus  $g = p^n(p^n - 1)/12$  and  $\# \text{Aut}(C) = p^{3n}(p^{3n} + 1)(p^{2n} - 1)$ , Cf.[H.Stichtenoth,Über die Automorphismengruppe eines algebraischen Funktionenkörpers von Primzahlcharakteristik. I. Eine Abschätzung der Ordnung der Automorphismengruppe, Arch. Math. (Basel) 24 (1973), 527–544.].  $\lrcorner$

**Prop. (7.13.1.38) [Frobenius Map].** Let  $k \in \text{Field}^p, C \in \text{Cur}^{\text{SmPrpr}}/k$  and  $C \xrightarrow{F_{X/k,r}} C^{(r)}$  be the Frobenius, where  $C^{(r)} = C \otimes_{k, \text{Frob}^r} k$ , then  $F_{X/k,r}$  is purely inseparable, and  $\deg(F_{X/k,r}) = p^r$ . And it is a topological homeomorphism by(5.1.7.26).  $\lrcorner$

*Proof:* To show inseparability, we can base change to  $\bar{k}$ , then the field map is  $f/g \mapsto f(\underline{X}^{p^r})/g(\underline{X}^{p^r})$ , which is just  $K(C_{\bar{k}})^{p^r}$ , so purely inseparable of degree  $p^r$ .  $\square$

**Prop. (7.13.1.39) [Inseparable Decomposition].** Let  $k \in \text{Field}^p$ . If  $C_1$  is a smooth complete precurve, then any map of non-singular complete precurves  $f : C_1 \rightarrow C_2$  over  $k$  factors as

$$C_1 \xrightarrow{F_{X/k,r}} C_1^{(r)} \xrightarrow{\lambda} C_2$$

where  $C_1^{(r)} = C_1 \otimes_{k, \text{Frob}^r} k$ , and  $\lambda$  is separable.  $\lrcorner$

*Proof:* By(7.13.1.18), it suffices to show any inseparable morphism is a Frobenius. It suffices to show that any subfield of  $K(X)$  of index  $p$  equals  $K(X^{(1)}) = kK(X)^p$ . For this, Cf.[Sta]0CCY?.  $\square$

**Cor. (7.13.1.40).** If  $C \rightarrow C'$  is a non-constant separable map between complete smooth curves, then  $g(C) \geq g(C')$ .  $\lrcorner$

*Proof:* This follows from (7.13.1.39)(7.13.1.33) and the fact  $C_1^{(r)}$  is smooth and have the same genus as  $C_1$  by flat base change.  $\square$

**Prop. (7.13.1.41) [Finiteness of Ramified Covers].** Let  $C \in \mathcal{C}ur^{\text{sm}, \text{Prpr}}/\mathbb{C}$  and  $B \subset C(\mathbb{C})$  a finite set, then for any  $n \in \mathbb{Z}_+$ , there exists only f.m. isomorphism classes of  $n$ -fold ramified coverings of  $C$  that are unramified outside  $B$ .  $\lrcorner$

*Proof:* The ramified coverings correspond to  $n$ -fold covering spaces of  $C(\mathbb{C}) \setminus B$ , which corresponds to index- $n$  subgroups of  $\pi_1(C(\mathbb{C}) \setminus B)$  by (4.15.1.28).  $\pi_1(C(\mathbb{C}) \setminus B)$  is f.g., and a subgroup corresponds to an action of  $\pi_1(C(\mathbb{C}) \setminus B)$  on  $[n]_+$ . So there are only f.m..  $\square$

### Divisors on Curves

**Def. (7.13.1.42) [Weil Divisors on Curves].** If  $k = \bar{k}$  and  $X \in \mathcal{S}ch_{\text{int}}^{\text{loc.ft, dim}=1}/k$ , then a **Weil divisor** on  $X$  is defined to be a locally finite formal sum of closed pts.

If  $X$  is integral algebraic of dimension 1 over  $k$ , then the sum is in fact finite, we we can define the **degree** of a Weil divisor  $D = \sum n_P P$  as  $\deg(D) = \sum n_P [k(P) : k]$ .

Similarly, for a Cartier divisor  $D$  on  $X$ , the **degree of  $D$**  is defined to be  $\dim_k \Gamma(D, \mathcal{O}_D)$ . By (9.1.4.1), a Cartier divisor on  $X$  is equivalent to an effective Weil divisor, and the definition of degrees are compatible.  $\lrcorner$

**Cor. (7.13.1.43) [Non-singular Case].** As a non-singular precurve  $C$  is locally factorial, (7.5.3.15) shows in this case a line bundles on  $C$  is equivalent to a Weil divisor on  $C$ .  $\lrcorner$

**Prop. (7.13.1.44) [Pullback of Divisor].** If  $k = \bar{k}$  and  $f : X \rightarrow Y \in \mathcal{C}ur^{\text{sm}}/k$  is a non-constant map, then for any Weil divisor  $D$  on  $X$  (7.13.1.42), we have  $\deg f^* D = \deg f \cdot \deg D$ .  $\lrcorner$

*Proof:* This is because  $f$  is finite locally free (7.13.1.23), thus this follows from [Sta]02RH<sup>?</sup>.  $\square$

**Prop. (7.13.1.45).** An element  $\notin k$  in the function fields of a projective non-singular curve over an alg.closed  $k$  defines a inclusion  $k(f) \subset K(X)$  thus a morphism from  $X$  to  $\mathbb{P}_k^1$  (7.12.3.3), and  $(f) = \varphi^*(\{0\} - \{\infty\})$ .  $\lrcorner$

**Prop. (7.13.1.46).** Let  $\pi : C \rightarrow C'$  be a non-constant separable morphism of precurves over a field  $k$  that  $C'$  is nonsingular,  $p \in C'$  is a closed point, then  $\pi$  is finite locally free by (7.13.1.23), and

- $\pi^{-1}(p) \subset C$  is a dimension 0 scheme.
- $\dim_k(\Gamma(\pi^{-1}(p))) = (\deg \pi)(\deg p)$ .
- Let  $\varpi$  be a uniformizer of the DVR  $\mathcal{O}_{C', p}$ , then

$$\deg \pi = \sum_{x \in \pi^{-1}(p)} [k(x) : k(p)] \text{ord}_{\mathcal{O}_{C, x}}(f^* \varpi).$$

$\lrcorner$

*Proof:* Look affine locally, then these follow from the fundamental identity (5.2.7.21).  $\square$

**Prop. (7.13.1.47).** For a 1-dimensional integral scheme  $c : X \rightarrow k$  proper over  $k$  and a function  $f \in R(X)^*$ ,

$$\sum_{x \text{ closed}} [k(x) : k] \text{ord}_{\mathcal{O}_{X,x}}(f) = 0.$$

In other words, the number of zeros of  $f$  equals the number of poles of  $f$ .  $\lrcorner$

*Proof:* It suffices to show that the pushforward  $c_* \text{div}(f) = 0 \in \text{CH}_0(\text{Spec } k)$ . consider  $Y$  the closure of the graph of  $f$  in  $X \times \mathbb{P}_k^1$ , then there is a commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{\text{pr}_1} & X \\ \downarrow \text{pr}_2 & & \downarrow c \\ \mathbb{P}_k^1 & \xrightarrow{c'} & \text{Spec } k \end{array},$$

and  $\text{div}_X(f) = \text{pr}_{1*} \text{div}_Y(f)$  by (9.1.1.22). We may assume  $f$  is not constant, then  $\text{pr}_2$  is finite locally free of degree  $d$ , and  $\text{div}_Y(f) = \text{pr}_2^*([(\infty)] - [(\infty)])$ , so  $\text{pr}_{2*} \text{div}_Y(f) = d([(\infty)] - [(\infty)])$  is mapped to  $0 \in \text{CH}_0(\text{Spec } k)$ .  $\square$

## 2 Vector Bundles

### Degrees and Riemann-Roch

**Def. (7.13.2.1) [Degrees].** The **degree of a locally free sheaf**  $\mathcal{E}$  of rank  $n$  on a proper scheme  $X$  of dimension  $\leq 1$  over  $k$  is defined to be  $\deg(\mathcal{E}) = \chi(X, \mathcal{E}) - n\chi(\mathcal{O}_X)$  (7.7.3.1).

If  $X$  is integral (e.g. a complete precurve), this definition can extend to any coherent sheaves  $\mathcal{F}$ , if we define  $\text{rank}(\mathcal{F}) = \dim_{k(\eta)} \mathcal{F}_\eta$ .  $\lrcorner$

**Prop. (7.13.2.2).** The degree function is additive, stable under base change of fields, and stable under birational equivalence of proper scheme  $X$  of dimension 1 over  $k$ .  $\lrcorner$

*Proof:* The base change follows from flat base change (7.7.5.1), the additivity follows from the additivity of rank and Euler characteristic (7.7.3.1).

For the birational equivalence, If  $f : X \rightarrow Y$  is a birational map between proper schemes of dimension  $\leq 1$  over  $k$ , then  $f$  is proper with finite fibres, so  $f$  is finite (7.4.6.5), thus for any  $\mathcal{E} \in \text{QCoh}^{\text{free}, n}(Y)$ ,  $f_* f^* \mathcal{E} = \mathcal{E} \otimes_{\mathcal{O}_Y} f_* \mathcal{O}_X$ , and there is an exact sequence

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X \rightarrow \mathcal{Q} \rightarrow 0$$

where  $\mathcal{K}, \mathcal{Q}$  are coherent sheaves on  $Y$  with supported dimension 0. Then by (7.7.3.5) and (7.7.3.3),

$$\begin{aligned} \chi(Y, \mathcal{E}) - \chi(X, f^* \mathcal{E}) &= \chi(Y, \mathcal{E}) - \chi(Y, f_* f^* \mathcal{E}) \\ &= \chi(Y, \mathcal{K} \otimes \mathcal{E}) - \chi(Y, \mathcal{Q} \otimes \mathcal{E}) \\ &= n\chi(Y, \mathcal{K}) - n\chi(Y, \mathcal{Q}) \\ &= n\chi(Y, \mathcal{O}_Y) - n\chi(X, \mathcal{O}_X) \end{aligned}$$

$\square$

**Prop. (7.13.2.3) [Non-reduced Riemann-Roch].** If  $X$  is a proper scheme over  $k$  of dimension  $\leq 1$  with integral components  $C_i$  of  $X$  of dimension 1 with generic points  $\eta_1, \dots, \eta_r$  and multiplicities  $m_i$ , and  $\mathcal{E} \in \text{Vect}^n(C)$ ,  $\mathcal{F} \in \text{Coh}(C)$ , then

$$\chi(\mathcal{E} \otimes \mathcal{F}) = \sum_i (\text{length}_{\mathcal{O}_{X, \eta_i}} \mathcal{F}_{\eta_i}) \deg(\mathcal{E}|_{C_i}) + n\chi(\mathcal{F}).$$

┘

*Proof:* We use dévissage (7.5.1.55), the condition 1 are true by additivity of  $\chi$ , and for condition 2, take  $\mathcal{G} = i_* \mathcal{O}_Z$ , the equation holds by (7.7.3.5) and projection formula.  $\square$

**Cor. (7.13.2.4).** Let  $X$  be a proper scheme of dimension  $\leq 1$  over  $k$ , and  $C_i$  are the irreducible components of  $X$  of dimension 1 with the induced reduced structure and multiplicity  $m_i$ , then for  $\mathcal{E} \in \text{Coh}^{\text{free}}(X)$ ,

$$\deg(\mathcal{E}) = \sum m_i \deg(\mathcal{E}|_{C_i}).$$

┘

**Cor. (7.13.2.5).** Let  $X$  be a proper scheme of dimension  $\leq 1$  over  $k$ , then

- If  $\mathcal{E} \in \text{Coh}^{\text{free}, m}(X)$ ,  $\mathcal{F} \in \text{Coh}^{\text{free}, n}(X)$ , then  $\deg(\mathcal{E} \otimes \mathcal{F}) = m \deg(\mathcal{F}) + n \deg(\mathcal{E})$ .
- If  $\mathcal{L}, \mathcal{M} \in \text{Pic}(X)$ , then  $\deg(\mathcal{L} \otimes \mathcal{M}) = \deg(\mathcal{L}) + \deg(\mathcal{M})$ .
- If  $\mathcal{L} \in \text{Pic}(X)$ ,  $\deg(\mathcal{L}) = -\deg(\mathcal{L}^{-1})$ .
- If  $\mathcal{E} \in \text{Coh}^{\text{free}}(X)$ ,  $\deg(\mathcal{E}) = \deg(\wedge \mathcal{E})$ .

┘

*Proof:* By (7.13.2.4), we can assume  $X$  is integral. Then 1, 2 follow from (7.13.2.3), and 3 follows from the fact there is a modification  $X' \rightarrow X$  s.t.  $f^* \mathcal{E}$  has a filtration with invertible sheaves  $\mathcal{L}_i$  as quotients (7.5.1.30). Then by (7.13.2.2), we can work on  $X'$ . Then  $\deg(\mathcal{E}) = \sum_i \deg(\mathcal{L}_i)$ , and the assertion follows from additivity and (7.13.2.3).  $\square$

**Prop. (7.13.2.6).** If  $D$  is an effective Cartier divisor on a proper scheme  $X$  of dimension  $\leq 1$  over  $k$ , then for  $\mathcal{E} \in \text{Coh}^{\text{free}, n}(X)$ ,

$$\deg(\mathcal{E}(D)) = n \deg(D) + \deg(\mathcal{E})$$

where  $\deg(D) = \dim_k \Gamma(X, \mathcal{O}_D)$ . In particular,  $\deg(\mathcal{L}(D)) = \deg(D)$ .  $\square$

*Proof:* By (7.8.1.3),  $D$  is nowhere dense in  $X$ , thus  $D$  is finite over  $k$ , and there is an exact sequence

$$0 \rightarrow \mathcal{E} \rightarrow \mathcal{E}(D) \rightarrow \mathcal{E}(D)|_D \rightarrow 0.$$

So the assertion follows from the additivity of Euler characteristics and (7.7.3.5).  $\square$

**Prop. (7.13.2.7).** Let  $f : X \rightarrow Y$  be a non-constant map of complete precurves over  $k$  and  $\mathcal{E} \in \text{Coh}^{\text{free}}(Y)$ , then  $\deg(f^* \mathcal{E}) = \deg(f) \deg(\mathcal{E})$ .  $\square$

*Proof:* By (7.13.1.10) and (7.13.2.3),

$$\chi(X, f^* \mathcal{E}) = \chi(Y, \mathcal{E} \otimes f_* \mathcal{O}_X) = \deg(f) \deg(\mathcal{E}) + \text{rank}(\mathcal{E}) \chi(X, \mathcal{O}_X).$$

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**Prop. (7.13.2.8) [Degree and Sections].** Let  $C$  be a complete precurve over  $k$  and  $\mathcal{L} \in \text{Pic}(C)$ , then

- If  $\mathcal{L}$  has a non-zero section, then  $\deg(\mathcal{L}) \geq 0$ .
- If  $\mathcal{L}$  has a non-zero section that vanishes at some point, then  $\deg(\mathcal{L}) \geq 1$ .
- If both  $\mathcal{L}, \mathcal{L}^{-1}$  have non-zero sections, then  $\mathcal{L} \cong \mathcal{O}_X$ .
- If  $\deg(\mathcal{L}) \leq 0$  and  $\mathcal{L}$  has a non-zero section, then  $\mathcal{L} \cong \mathcal{O}_X$ .
- If  $\mathcal{N} \rightarrow \mathcal{L}$  is a non-zero map of invertible sheaves, then  $\deg(\mathcal{L}) \geq \deg(\mathcal{N})$ , with equality iff this is an isomorphism.

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*Proof:* If  $s$  is a section of  $\mathcal{L}$  with vanishing locus  $D$ , then  $D$  is an effective Cartier divisor and  $\mathcal{L} \cong \mathcal{L}(D)$ , so  $\deg(\mathcal{L}) = \deg(D)$  by (7.13.2.6), so these are all simple now.  $\square$

**Prop. (7.13.2.9) [Riemann-Roch].** Let  $D$  be a Weil divisor on a complete non-singular precurve  $X$  of genus  $g$  over  $k$ , then

- If  $l(D) = \dim_k H^0(X, \mathcal{L}(D))$ ,  $l(D)$  is finite by (7.7.4.11).
- $[\omega_X]$  is an invertible sheaf by [Sta]0BFQ?.
- $l(D) - l([\omega_X] - D) = \deg D + 1 - g$ .
- $\deg(D) = \deg(\mathcal{L}(D))$

Notice when  $X$  is smooth,  $[\omega_X]$  is just  $\mathcal{K}_X$  (7.9.2.17).  $\square$

*Proof:* 4 is equivalent to 3 by Serre duality (7.9.2.17), and 3 follows from (7.13.2.6).  $\square$

**Cor. (7.13.2.10).**  $\deg([\omega_C]) = 2g - 2$ .  $\square$

*Proof:* By (7.13.2.9) and Serre-duality (7.9.2.17),

$$\deg([\omega_C]) = h^0(X, [\omega_X]) - h^0(X, \mathcal{O}_X) + g - 1 = h^1(X, \mathcal{O}_X) - h^0(X, \mathcal{O}_X) + g - 1 = 2g - 2.$$

$\square$

**Prop. (7.13.2.11) [Twisting Sheaves].** Let  $\mathcal{L}$  be a line bundle on a complete nonsingular precurve  $C$ , and  $p \in C$  is a closed point of degree  $d$ , then  $0 \leq h^0(\mathcal{L}) - h^0(\mathcal{L}(-p)) \leq d$  for any  $\mathcal{L} \in \text{Pic}(C)$ .

In particular, by (7.13.2.8), if  $C$  is a complete non-singular curve, then  $h^0(\mathcal{L}) \leq \deg L + 1$  for any  $\mathcal{L} \in \text{Pic}(C)$ .  $\square$

*Proof:* There is an exact sequence  $0 \rightarrow \mathcal{O}_C(-p) \rightarrow \mathcal{O}_C \rightarrow \mathcal{O}_C|_p \rightarrow 0$ , tensoring with  $\mathcal{L}$  and take the cohomology, we get an exact sequence  $0 \rightarrow h^0(\mathcal{L}, \mathcal{L}(-p)) \rightarrow h^0(\mathcal{L}, \mathcal{L}) \rightarrow H^0(C, \mathcal{L}|_p)$ , and notice that  $h^0(C, \mathcal{L}|_p) = d$ .  $\square$

**Prop. (7.13.2.12) [Riemann-Roch for High Degree  $D$ ].** If  $\deg(D) \geq 2g - 1$ , then  $\deg([\omega_X] - D) < 0$ , so by (7.13.2.8),  $l([\omega_X] - D) = 0$ , thus  $l(D) = d - g + 1$ .  $\square$

**Cor. (7.13.2.13) [Characterizing  $[\omega_C]$ ].** Any degree  $2g - 2$  divisor  $D$  satisfies  $l(D) = g - 1$  or  $g$ , and the latter case happens iff  $D = [\omega_C]$ .  $\square$

*Proof:*  $l([\omega_C] - D) = 1$  iff  $[\omega_C] = D$  by (7.13.2.8).  $\square$

**Def. (7.13.2.14) [Special Divisors].** A **special divisor** on a complete non-singular curve is a divisor  $D$  that  $h^0([\omega_C] - D) > 0$ .  $\square$



**Prop. (7.13.2.15).** For any complete smooth curve  $C$  of genus  $g > 1$  over a field  $k$ , there is a closed point on  $C$  of degree  $\leq 2g - 2$ . And if  $g \geq 2$ , we can assume this point is a geometric point.  $\lrcorner$

*Proof:* The canonical sheaf  $\mathcal{K}_C$  is a line bundle of degree  $2g - 2$  and  $h^0(\mathcal{K}_C) = g \geq 1$ , so we can assume  $\mathcal{K}_C$  is an effective divisor, then one of its support point has degree  $\leq 2g - 2$ .

For the last assertion, Cf. [Sta]0CD4.  $\square$

### Vector Bundles

**Prop. (7.13.2.16) [Torsion-Free Sheaves].** Let  $C$  be a non-singular precurves over  $k$ , then

- Any torsion-free coherent sheaf  $\mathcal{F}$  on  $C$  is locally free.
- Any  $\mathcal{F} \in \text{Coh}(X)$  factors as  $0 \rightarrow \mathcal{F}_{\text{tor}} \rightarrow \mathcal{F} \rightarrow \mathcal{F}_{\text{lf}} \rightarrow 0$ , where  $\mathcal{F}$  is a torsion sheaf (7.5.1.44) and  $\mathcal{F}_{\text{lf}}$  is Qco and locally free, by (7.5.1.46)  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.13.2.17) [Pic<sup>0</sup>(C)].** For a smooth complete curve  $C$  over  $k$  with a rational point,  $\text{Pic}^0(C)$  are exactly the set of line bundles of degree 0, by (20.9.10.2). In particular, there is an exact sequence

$$0 \rightarrow \text{Pic}^0(C) \rightarrow \text{Pic}(C) \rightarrow \mathbb{Z} \rightarrow 0.$$

$\lrcorner$

**Prop. (7.13.2.18) [Torsion Elements].** If  $k = \bar{k}$  and  $C$  is a smooth complete curve of genus  $g$  over  $k$ , then for  $m \in \mathbb{Z} \cap k^\times$ ,  $\text{Pic}(C)[m] = \text{Pic}^0(C) \cong (\mathbb{Z}/(m))^{2g}$ .  $\lrcorner$

*Proof:* This follows from (20.9.10.2)(7.13.2.17) and (20.9.6.18).  $\square$

**Prop. (7.13.2.19).** If  $Z \rightarrow X$  is a closed immersion and  $\dim X \leq 1$ , then  $\text{Pic } X \rightarrow \text{Pic } Z$  is a surjection.  $\lrcorner$

*Proof:* Use the exact sequence  $0 \rightarrow (1 + \mathcal{I}) \cap \mathcal{O}_X^* \rightarrow \mathcal{O}_X^* \rightarrow i_* \mathcal{O}_Z^* \rightarrow 0$ ,  $\dim X \leq 1$  and the Grothendieck vanishing theorem gives the desired result, also notice  $i$  is affine.  $\square$

### Ample Line Bundles

**Prop. (7.13.2.20).** A line bundle  $\mathcal{L}$  over a complete precurve  $C$  over  $k$  is ample iff  $\deg(\mathcal{L}) > 0$ .  $\lrcorner$

*Proof:* If  $C$  is non-singular, the proof is easy:  $\mathcal{L}$  is ample iff  $\mathcal{L}^{\otimes n}$  is very ample for  $n$  large (7.5.4.24). If  $\deg(\mathcal{L}) < 0$ , this cannot happen, by (7.13.2.8). For  $\deg L \geq 0$ ,  $\mathcal{L}^{\otimes n}$  is very ample for  $n$  large by (7.13.2.26).

In general, Cf. [Sta]0B5X?  $\square$

**Cor. (7.13.2.21) [Ample and Nef Line Bundles].** Let  $\mathcal{L}$  be an invertible  $\mathcal{O}_X$ -module over a proper scheme of dimension  $\leq 1$  over  $k$ , let  $C_i$  be the integral components of  $X$  of dimension 1, then  $\mathcal{L}$  is a ample iff  $\deg(\mathcal{L}|_{C_i}) > 0$  for all  $i$ .  $\lrcorner$

*Proof:* This follows from (7.5.4.14) applied to the reduced structure of the irreducible components of  $X$ , together with the fact a line bundle on an irreducible component  $\text{Spec } k(x)$  of dimension 0 is obviously ample.  $\square$

**Prop. (7.13.2.22).** Let  $\mathcal{L}$  be a line bundle on a complete non-singular precurve  $C$  of degree  $d$  that is basepoint-free, then it determines a morphism  $\pi : C \rightarrow \mathbb{P}_k^1$  of degree  $d$ .  $\lrcorner$

*Proof:* This follows from (7.13.1.46).  $\square$

**Prop. (7.13.2.23).** Let  $C$  be a complete precurve over  $k$  with genus  $g > 0$ , and  $p, q$  are rational points on  $C$ , then  $\mathcal{O}_C(p) \cong \mathcal{O}_C(q)$  iff  $p = q$ , and  $h^0(C, \mathcal{O}_C(p)) = 1$ .  $\lrcorner$

*Proof:* The hypothesis shows  $\mathcal{L} = \mathcal{O}_C(p)$  has degree 1 and is basepoint-free, thus defines a degree 1 map  $C \rightarrow \mathbb{P}_k^1$ , which is an isomorphism, by (7.13.1.16) and (7.13.1.18), contradiction.

If  $h^0(C, \mathcal{O}_C(p)) \geq 2$ , then for any section  $s$ ,  $\text{div}(s) = q$  for some rational point  $q$ , so by the above argument,  $q = p$ , so any two such  $f$  is proportional.  $\square$

**Prop. (7.13.2.24).** Let  $C$  be a complete curve over  $k$  with genus  $g > 0$ , and  $\mathcal{L}$  is a line bundle of degree 2, then  $h^0(C, \mathcal{L}) \leq 2$ , and if the equality holds, then it is basepoint-free.  $\lrcorner$

*Proof:* We can base change to  $\bar{k}$ .  $h^0(C, \mathcal{L}) \leq 3$  by (7.13.2.11). If equality holds, then then  $h^0(C, \mathcal{L}(-p)) = 2$  for some  $p$ , contradiction by (7.13.2.23). If  $h^0(C, \mathcal{L}) = 2$ , then  $h^0(C, \mathcal{L}(-p)) < 2$  for any  $p$ , so it is basepoint-free. Conversely, if it is basepoint free, then  $\mathcal{L} \cong \mathcal{O}(p)$  for some rational point  $p$ , and  $h^0(C, \mathcal{L}) = h^0(C, \mathcal{O}_C) + 1 = 2$ .  $\square$

**Prop. (7.13.2.25)[Criterion of Very Ampleness].** Let  $k = \bar{k}$  and  $\mathcal{L}$  be a line bundle on a curve over  $k$ , then

- $\mathcal{L}$  is basepoint-free iff  $h^0(\mathcal{L}) - h^0(\mathcal{L} - p) = 1$  for any closed point  $p \in \mathcal{L}$ .
- $\mathcal{L}$  is very ample iff  $h^0(\mathcal{L}) - h^0(\mathcal{L} - p - q) = 2$  for any closed points  $p, q \in \mathcal{L}$ .

$\lrcorner$

*Proof:* Cf. [Vakil, P509]. ?  $\square$

**Cor. (7.13.2.26).** Any line bundle  $\mathcal{L}$  on a complete non-singular curve of genus  $g$  that  $\deg(\mathcal{L}) \geq 2g$  is basepoint-free, and if  $\deg(\mathcal{L}) \geq 2g + 1$ , then it is very ample.  $\lrcorner$

*Proof:* When  $k = \bar{k}$ , this follows from (7.13.2.25) and Riemann-Roch (7.13.2.9). For general  $k$ , use the fact being basepoint-free, closed embedding and degree are all stable and reflective under base change of fields (7.13.2.2)(7.5.3.3)(7.1.5.26).  $\square$

**Prop. (7.13.2.27)[Very Ample Line Bundles].** Let  $X$  be a complete curve over a field  $k$ , then

- If  $\mathcal{L}$  is effective and  $H^1(X, \mathcal{L}) = 0$ , then  $\mathcal{L}^{\oplus 6}$  is very ample.
- If  $\mathcal{L}$  is globally generated and  $H^1(X, \mathcal{L}) = 0$ , then  $\mathcal{L}^{\oplus 2}$  is very ample.

$\lrcorner$

*Proof:* Cf. [Sta]0E8V, 0E8W. ?  $\square$

### Curves in Low Dimension

**Prop. (7.13.2.28)[Projection Along a Point].** Let  $C \subset \mathbb{P}_k^n$  be a non-singular precurve, then for any rational point  $p \in C$ , there is a **projection along**  $p$  map  $v_p : C \rightarrow \mathbb{P}_k^{n-1}$  that is the projection along  $p$  for  $q \in C \setminus \{p\}$ , and extend to whole  $C$  by (7.13.1.16), which corresponds to the line bundle  $\mathcal{O}_C(1)(-p)$ .  $\lrcorner$

**Prop. (7.13.2.29).** Let  $C$  be a smooth plane precurve of degree  $e > 2$  and  $D_1, D_2$  are two polynomials of degree  $d$  not vanishing on  $C$ . Suppose there is a divisor  $E$  on  $C$  of degree  $de - 1$  and rational points  $p_i$  on  $C$  that  $D_i|_C = E + p_i$ , then  $\mathcal{O}_C(-E)$  is not base-free.  $\lrcorner$

*Proof:* Notice by genus formula  $g > 0$ , and  $\mathcal{O}_C(-E) \cong \mathcal{O}_C(p_i)$  has degree 1, then  $p_1 = p_2$  by (7.13.2.23).  $\square$

**Prop. (7.13.2.30).** Let  $C$  be a smooth conic plane curve over a field of characteristic  $\neq 2$ , show the dual variety of  $C$  is also a smooth conic. In particular, for a general point in the plane, there are two tangents to  $C$ . (This can be also proved using Riemann-Hurewitz by projection through this point).  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.13.2.31).** The number of plane conics containing  $i$  generally chosen points and  $5 - i$  generally chosen lines is 1, 2, 4, 4, 2, 1 resp. for  $i = 0, 1, \dots, 4, 5$ . (The duality comes from the duality between the conic and the dual conic (7.13.2.30)).  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.13.2.32) [Curves in  $\mathbb{P}^1 \times \mathbb{P}^1$ ].** Let  $C$  be a curve in  $\mathbb{P}^1 \times \mathbb{P}^1$  given by a bi-homogenous polynomial of type  $(a, b)$ . Then  $g(C) = (a - 1)(b - 1)$ .  $\lrcorner$

*Proof:* We have  $\mathcal{K}_C = (\mathcal{K}_{\mathbb{P}^1 \times \mathbb{P}^1} + C)|_C$  (7.12.1.17). We have  $\text{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}H_1 \oplus \mathbb{Z}H_2$ , thus

$$\begin{aligned} \deg(\mathcal{K}_C) &= \deg((\mathcal{K}_{\mathbb{P}^1 \times \mathbb{P}^1} + C)_C) \\ &= \deg((-2H_1 - 2H_2 + aH_1 + bH_2)|_C) \\ &= \deg(((a - 2)H_1 + (b - 2)H_2)|_C) \\ &= (a - 2)b + (b - 2)a = 2ab - 2a - 2b \\ &= 2g - 2. \end{aligned}$$

Thus  $g = (a - 1)(b - 1)$ .

? For non-smooth case, see AG psets.  $\square$

**Prop. (7.13.2.33) [Genus Formula].** Let  $C$  be a closed subscheme of  $\mathbb{P}_k^2$  defined by a homogenous polynomial  $f(x_0, x_1, x_2)$  of degree  $d$ , then it has arithmetic genus  $p_a(C) = (d - 1)(d - 2)/2$ .  $\lrcorner$

*Proof:* This follows from (7.7.2.4).  $\square$

**Cor. (7.13.2.34).** Let  $C$  be a complete smooth plane curve of degree  $d$ , then it has genus  $g = (d - 1)(d - 2)/2$ .  $\lrcorner$

*Proof:* Alternative proof: By adjunction formula,  $\mathcal{K}_C = \mathcal{K}_{\mathbb{P}^2}(C)|_C = \mathcal{O}_C(d - 3)$  which has degree  $d(d - 3)$ . But this also equals  $2g - 2$  (7.13.2.10), thus  $g = (d - 1)(d - 2)/2$ .  $\square$

**Prop. (7.13.2.35) [Cubic Curves in  $\mathbb{P}^3$ ].** A twisted cubic (rational)  $C$  in  $\mathbb{P}^3$  is contained in a quadric.  $\lrcorner$

*Proof:* Calculate that  $h^0(\mathcal{O}_{\mathbb{P}^3}(2)) = C_5^2 = 10$  and  $\deg(\mathcal{O}_C(2)) = 6$ , thus  $h^0(\mathcal{O}_C(2)) = 6 + 1 = 7$  by Riemann-Roch. Then  $C$  is contained in 3 quadrics.  $\square$

### 3 Singularities and $\delta$ -Invariants

**Def. (7.13.3.1) [Different Kinds of Singularities].** Let  $X$  be a curve over an alg.closed field  $k$ , a point  $p \in X$  is called a

- **node** if the completion of  $\mathcal{O}_{X,p}$  at  $\mathfrak{m}_{X,p}$  is isomorphic to  $k[[x, y]]/(xy)$  as topological local rings.
- **cusp** if the completion of  $\mathcal{O}_{X,p}$  at  $\mathfrak{m}_{X,p}$  is isomorphic to the completion of  $k[x, y]/(x^2 - y^3)$  at  $(x, y)$ .
- **tacnode** if the completion of  $\mathcal{O}_{X,p}$  at  $\mathfrak{m}_{X,p}$  is isomorphic to the completion of  $k[x, y]/(x^2 - y^4)$  at  $(x, y)$ .
- **triple point** if the completion of  $\mathcal{O}_{X,p}$  at  $\mathfrak{m}_{X,p}$  is isomorphic to the completion of  $k[x, y]/(x^3 - y^3)$  at  $(x, y)$ .

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**Def. (7.13.3.2) [ $\delta$ -Invariants].**

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### 4 Linear Series

**Def. (7.13.4.1) [Moduli Spaces].** Define the **Hilbert scheme**  $\mathcal{H}_{d,g,r} = \{\text{Curves } C \text{ of degree } d \text{ and genus } g\}$ .

Define the **moduli space of curves**  $\mathcal{M}_g = \{\text{isomorphism classes of smooth projective curves of genus } g\}$ .

Define  $W_d^r(C) = \{L \in \text{Pic}^d(C), h^0(L) \geq r + 1\}$ .

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**Def. (7.13.4.2) [Linear Series].** A **linear series** is a line bundle  $L$  together with a vector space  $V \subset H^0(L)$ .

A  $\mathfrak{g}_d^r$  is a line bundle  $L$  together with a vector space  $V \subset H^0(L)$  of dimension  $r + 1$ .

For a line bundle  $\mathcal{L}$ , denote by  $|L|$  the linear series  $(L, H^0(L))$ .

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**Cor. (7.13.4.3) [ $r \leq d$ ].** If  $\mathfrak{g}_d^r$  exists on a complete curve  $C$ , then  $r \leq d$ . And if  $r = d$ , then  $g(C) = 0$ .

┘

*Proof:* Let  $s$  be a section of  $\mathcal{L}$ , then the vanishing locus of  $s$  is an effective Cartier divisor  $D$ ,  $\mathcal{L} \cong \mathcal{L}(D)$ , and there is an exact sequence

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{L} \rightarrow \mathcal{L}|_D \rightarrow 0.$$

As  $D$  is an Artinian scheme,  $\mathcal{L}|_D$  is trivial as  $D$  is discrete, so by (7.13.2.6),  $h^0(\mathcal{L}|_D) = \deg(D)$ , and  $h^0(X, \mathcal{O}_X) = 1$ , so  $r = h^0(X, \mathcal{L}) \leq d$ .

If the equality holds, then  $H^0(X, \mathcal{L}) \rightarrow H^0(X, \mathcal{L}|_D)$  is surjective, so to show  $g(C) = 0$ , it suffices to show that  $h^0(X, \mathcal{L}) = 0$ . As  $\mathcal{L}|_D$  is trivial, there is a section  $t$  of  $\mathcal{L}$  that generate  $\mathcal{L}|_D$ . Consider the exact sequence

$$0 \rightarrow \mathcal{L} \rightarrow \mathcal{L}^2 \rightarrow \mathcal{L}^2|_D \rightarrow 0,$$

$H^0(X, \mathcal{L}^2) \rightarrow H^0(X, \mathcal{L}|_D)$  is surjective because  $\sigma \otimes t$  is mapped to  $\sigma$ . Then by (7.13.2.5),

$$h^0(X, \mathcal{L}^2) = 2r + 1 = \deg(\mathcal{L}^2) + 1.$$

So we may replace  $\mathcal{L}$  by arbitrarily large powers of  $\mathcal{L}$ . Now  $\mathcal{L}$  is ample by (7.13.2.20), so for  $r$  large,  $H^1(X, \mathcal{L}^r) = 0$ . □

**Prop. (7.13.4.4) [Linear Series and Maps].** For any  $\mathfrak{g}_d^r(L, V)$  on a non-singular precurve  $C$ , there is a map  $\varphi : C \rightarrow \mathbb{P}^r$  and a map  $\varphi^* \mathcal{O}_{\mathbb{P}^r}(1) \rightarrow \mathcal{L}$  s.t. the coordinates  $T_0, \dots, T_r$  are mapped to a basis of  $V$ .

This map is injective iff for any  $p \neq q$ ,  $V_{p+q}$  has codimension 2. This map is an immersion iff for all  $p$ ,  $V_{2p}$  has codimension 2. So in particular, this map is an embedding iff for any effective divisor  $D$  of degree 2,  $V_D$  has codimension 2.  $\lrcorner$

*Proof:* Take a basis  $s_0, s_1, \dots, s_r \in V$ , as  $C$  is non-singular, the image of the map  $(s_0, \dots, s_r) : \mathcal{O}_X^{r+1} \rightarrow \mathcal{L}$  is an invertible sheaf as it is torsion-free. Then take the corresponding embedding induced by (7.5.2.4).  $\square$

**Cor. (7.13.4.5).** If  $\mathcal{L}$  is a very ample line bundle of degree  $d$  on a curve  $C$ , then  $\varphi_{|\mathcal{L}|}$  embeds  $C$  as a closed subscheme of degree  $d$ .  $\lrcorner$

*Proof:* Use (7.13.4.4), The pullback of  $\mathcal{O}_{\mathbb{P}^r}(1)$  under  $\varphi$  is just  $\mathcal{L}$ , so the assertion follows from (7.13.2.7).  $\square$

**Lemma (7.13.4.6).** Let  $\mathcal{L} \in \text{Pic}^d(C)$  be general, then  $h^0(\mathcal{L}) = \max\{1, d - g + 1\}$ .  $\lrcorner$

*Proof:* If  $D = \sum p_i$  is a general effective divisor, notice  $h^0(K) = g$ ,  $h^0(K - p_1) = g - 1$ ,  $h^0(K - p_1 - p_2) = g - 2$ , and repeating this, we get  $h^0(K - D) = \max\{0, g - d\}$ , when we choose  $p_i$  that are as independent as possible (any section has f.m. zeros). Then by Riemann-Roch,  $l(D) = d - g + 1 + \max\{0, g - d\} = \max\{1, d - g + 1\}$ . As every divisor of degree  $d \geq g$  is effective, this settles the  $d \geq g$  case.

If  $D$  is non-effective and  $d \leq g - 1$ , we need to show  $W_d^0$  is not dominant in  $\text{Pic}^d$ . But  $J$  has dimension  $g$  by ?? and  $W_d^0$  has dimension at most  $d$ , so this is true.  $\square$

**Prop. (7.13.4.7).** Suppose  $D$  is a divisor of degree  $g + 3$ , then for  $D$  general,  $\varphi_D = \varphi_{|\mathcal{L}(D)|}$  is an embedding.  $\lrcorner$

*Proof:* By (7.13.4.6), for a general  $D$ ,  $l(D) = 4$ . Thus  $\varphi_D$  being not an embedding is equivalent to the existence of a divisor  $D_0 = p + q$  that  $l(D - D_0) \geq 3$ . This means  $D - D_0 \in W_{g+1}^2 = K - W_{g-3}^0$ . So the divisor  $D = D_0 + (D - D_0) \in W_2^0 + (K - W_{g-3}^0)$  which has dimension at most  $2 + (g - 3) = g - 1$ . But a general divisor  $D$  doesn't lie on this  $g - 1$ -dimensional subvariety by (20.9.10.11), so a general  $D$  defines an embedding  $\varphi_D$ .  $\square$

**Def. (7.13.4.8) [Canonical Map].** For a smooth curve  $C$  of degree  $g$  over a field  $k$  of genus  $g \geq 1$ , the canonical divisor  $\mathcal{K}_C$  is basepoint-free, and defines a **canonical map** to  $\mathbb{P}_k^{g-1}$ .  $\lrcorner$

*Proof:* By (7.13.2.2)(7.5.3.3)(7.1.5.26), it suffices to prove for  $\bar{k}$ , thus for any closed point  $p$ ,  $h^0(\mathcal{K}_C - p) = 2g - 3 - g + 1 + h^0(\mathcal{O}_C(p)) \leq g - 1 < h^0(\mathcal{K}_C) = g$  by (7.13.2.23), so it is basepoint-free.  $\square$

**Prop. (7.13.4.9) [Base-Free Pencil Trick].** Let  $\mathcal{L}, \mathcal{M}$  be line bundles on  $C$ . Let  $s_1, s_2$  be sections of  $H^0(\mathcal{L})$  without common zero, then the kernel of the map

$$s_1 H^0(\mathcal{M}) \oplus s_2 H^0(\mathcal{M}) \rightarrow H^0(\mathcal{L} \otimes \mathcal{M})$$

is  $H^0(\mathcal{M} \otimes \mathcal{L}^{-1})$ .  $\lrcorner$

*Proof:* Indeed, there is an exact sequence of sheaves:

$$\mathcal{M} \otimes \mathcal{L}^{-1} \rightarrow \mathcal{M} \oplus \mathcal{M} \xrightarrow{(s_1, s_2)} \mathcal{M} \otimes \mathcal{L} \rightarrow 0.$$

Where the first map maps a section  $t$  to the pair  $(ts_2, -ts_1)$ . This is a Koszul regular sequence (on a common trivialization  $U$  of  $\mathcal{M}, \mathcal{L}$ ,  $s_1, s_2 \in C(U)$ , thus this is just  $0 \rightarrow A \rightarrow A \oplus A \xrightarrow{s_1, s_2} A \rightarrow 0$ ). Taking the global section functor gives the desired result.  $\square$

**Prop. (7.13.4.10) [Geometric Riemann-Roch].** Let  $C$  be a non-hyperelliptic curve of genus  $g \geq 2$ . Then the canonical map  $\varphi_K$  is an embedding, so we can assume  $C \subset \mathbb{P}^{g-1}$ . Then for a divisor  $D = \sum p_i$ ,

$$r(D) = d - 1 - \dim \overline{D},$$

where  $\overline{D}$  is the linear subspace generated by  $p_i$ . Thus  $r(D)$  can be interpreted as the number of linear relations between  $p_i$ .  $\lrcorner$

*Proof:* By the definition of the canonical embedding,  $l(K - D)$  is just the dimension of hypersurfaces containing  $p_i$ , thus it is equal to  $g - 1 - \dim \overline{D}$ . Now by Riemann-Roch,  $r(D) = d - g + l(K - D) = d - g + g - 1 - \dim \overline{D} = d - 1 - \dim \overline{D}$ .  $\square$

**Prop. (7.13.4.11).**  $\lrcorner$

**Cor. (7.13.4.12) [Special Divisors].** Let  $U^{(g)}$  be the open subscheme of  $C^{(g)}$  corresponding to the set of non-special divisors of degree  $g$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.13.4.13) [Clifford Theorem].** For a divisor  $D$  of degree  $0 \leq d \leq 2g - 2$  on a curve of genus  $g$ ,

$$r(D) \leq \frac{d}{2}.$$

with equality iff one of the following holds:

- $r = d = 0, D = 0$ .
- $d = 2g - 2, r = g - 1, D = \mathcal{K}_C$ .
- $C$  is hyperelliptic and  $D = mg_2^1$ .

$\lrcorner$

*Proof:* Cf. [Algebraic Curves, Harris, P32].  $\lrcorner$

## 5 Hyperelliptic Curves

**Def. (7.13.5.1) [Hyperelliptic Curves].** A **hyperelliptic curve** over  $k$  is a complete non-singular curve  $C$  together with a finite degree 2 map  $C \rightarrow \mathbb{P}_k^1$ . Such a map is called a **hyperelliptic map**. For  $g > 0$ , this is equivalent to the existence of a  $g_2^1$  on  $C$  (such a linear series must be basepoint-free, by (7.13.2.24)).  $\lrcorner$

**Cor. (7.13.5.2) [Genus 2 Curves are Hyperelliptic].** Any complete nonsingular curve of genus 2 is hyperelliptic, because  $\mathcal{K}_C$  has degree 2 and  $h^0(\mathcal{K}_C) = 2$  (7.13.2.10).  $\lrcorner$

**Prop. (7.13.5.3) [Equation of Hyperelliptic Curves].** Any image of a hyperelliptic map from a hyperelliptic curve of genus  $g$  over an alg.closed field  $k$  of characteristic  $k \neq 2$  is isomorphic to the projective curve that is the completion(7.13.1.19) of the affine curve in  $\mathbb{A}^2$  defined in an affine chart by  $y^2 = \prod_{i=0}^{2g+2} (x - \alpha_i)$  or  $y^2 = \prod_{i=0}^{2g+1} (x - \alpha_i)$ , depending on whether the  $\infty$  is branched or not. And given  $2g + 2$  points in  $\mathbb{P}_k^1$ , there is exactly one degree 2 covering of  $\mathbb{P}^1$  branched over these points.  $\perp$

*Proof:* Given any  $r$  points of  $\mathbb{P}_k^1$ , we can use a transformation of  $\mathbb{P}^1$  to assume that all branch points are finite, denoted by  $\alpha_1, \dots, \alpha_r$ . Consider the curve  $C'$  defined by  $y^2 = \prod_{i=0}^{2g+2} (x - \alpha_i)$ , then it is a smooth curve by Jacobian criterion. Now let  $C$  be the smooth model of  $C'$ , and consider  $\pi : C \rightarrow \mathbb{P}^1$  by projecting to the  $x$ -coordinates and extend to the whole  $C$  by(7.13.1.16). This is finite separable of degree 2, and  $\pi$  is simply branched at the points  $\alpha_i$ , and simply branched over  $\infty$  if  $r$  is odd, thus it has genus  $\lfloor \frac{r-1}{2} \rfloor$  by Riemann-Hurwitz(7.13.1.33).

Conversely, by Riemann-Hurwitz(7.13.1.33), any degree 2 covering of  $\mathbb{P}^1$  has  $2g + 2$  branch points, because the branch points must be simply branched. The map  $k(x) \rightarrow k(C)$  is Galois of degree 2, take a  $y \in k(C)$  that  $\sigma(y) = -y$ , then  $y^2 = g \in k(x)$ . We can modify  $y$  s.t.  $g$  is a monic polynomial with no repeated factors. Then it is of the form  $g(x) = \prod_{i=1}^r (x - x_i)$ . Then it is isomorphic to the curve constructed above, and branched over  $x_1, \dots, x_r$  and possibly  $\infty$ . In particular, it is determined by the set of branched points, so we get the desired assertion.  $\square$

**Remark (7.13.5.4).** The completion of  $C$  can be explicitly constructed, by gluing another affine chart defined by  $(\frac{y}{x^{g+1}})^2 = \prod_{i=0}^{2g+2} (1 - \alpha_i \frac{1}{x})$  or  $(\frac{y}{x^{g+1}})^2 = \frac{1}{x} \prod_{i=0}^{2g+1} (1 - \alpha_i \frac{1}{x})$ .  $\perp$

**Prop. (7.13.5.5).** If  $C$  is a nonsingular complete curve of genus  $g \geq 2$  over  $k$  and  $\mathcal{L}$  is a line bundle that corresponds to a hyperelliptic map  $C \rightarrow \mathbb{P}_k^1$ , then  $\mathcal{L}^{\otimes(g-1)} \cong \mathcal{K}_C$ .

In particular, the image of  $C$  under the canonical map is the rational curve in  $\mathbb{P}_k^{g-1}$ .  $\perp$

*Proof:* Consider the composition of the hyperelliptic map and the Veronese map

$$C \xrightarrow{|\mathcal{L}|} \mathbb{P}_k^1 \xrightarrow{v_{g-1}} \mathbb{P}_k^{g-1},$$

which corresponds to the line bundle  $|\mathcal{L}^{\otimes(g-1)}|$ . This line bundle has degree  $2g - 2$ , and the map  $H^0(\mathbb{P}^{g-1}, \mathcal{O}(1)) \rightarrow H^0(C, \mathcal{L}^{\otimes(g-1)})$  is injective, because the Veronese map is non-degenerate. Thus  $h^0(\mathcal{L}^{\otimes(g-1)}) \geq g$ , thus isomorphic to  $\mathcal{K}_C$ , by(7.13.2.13). Hence this map is just the canonical map.  $\square$

**Cor. (7.13.5.6).** For a hyperelliptic curve, the smallest degree of an embedding  $C \rightarrow \mathbb{P}^r$  is  $g + r$ . Note that hyperelliptic curves cannot be embedded into  $\mathbb{P}^2$ , i.e. smooth plane curves are non-hyperelliptic.  $\perp$

*Proof:* Firstly we show a special divisor cannot induce an embedding: A special divisor is a divisor that is contained in a hypersurface intersection of  $C$  under the canonical map, thus by geometric Riemann-Roch(7.13.4.10),  $r(D)$  is just the number of points of  $D$  mapped to the same point under the canonical map, because intersection with the rational normal curve are all linearly independent. Now this means  $D$  contains  $g_2^1$ , so  $\varphi_D$  factors through  $\pi$ , thus not an embedding. ? Cf.[Algebraic Curves, Harris, P21].  $\square$

**Prop. (7.13.5.7) [Hyperelliptic and Canonical Map].** A complete smooth curve of genus  $g \geq 1$  is hyperelliptic iff the canonical map(7.13.4.8) is not a closed embedding.  $\perp$

*Proof:* The canonical map is not an embedding iff there exists an effective divisor  $D$  of degree 2 that  $l(K - D) > l(K) - 2$ . Now  $l(K - D) = 2g - 4 - g + 1 + l(D)$ , so this is equivalent to  $l(D) > 1$ , which is equivalent to a  $g_2^1$ , which is equivalent to hyperelliptic(7.13.5.1).  $\square$



**Prop. (7.13.5.8)[Uniqueness of Hyperelliptic Maps].** For a hyperelliptic curve of genus  $g \geq 2$ , up to automorphism of  $\mathbb{P}^1$ , there are at most one hyperelliptic map, or equivalently there is exactly one  $g_2^1$  on  $C$ .

But for (hyperelliptic)curve of genus 1, thus is not unique: any divisor of degree 2 is effective, but they are not unique, because  $\text{Pic}^2(C) \cong J(C)$  (20.9.10.11).  $\lrcorner$

*Proof:* If  $\mathcal{L} \neq \mathcal{M}$  are two line bundles on  $C$  with  $h^0 = 2$ , then  $h^0(\mathcal{L} \otimes \mathcal{M}) \geq 4$  by base-free pencil trick (7.13.4.9), and in fact  $h^0(\mathcal{L} \otimes \mathcal{M}) = 4$  by (7.13.4.3). Now we can run the same argument again to  $\mathcal{L}$  and  $\mathcal{L} \otimes \mathcal{M}$ , to show that  $h^0(\mathcal{L}^2 \otimes \mathcal{M}) = 6$ . And inductively  $h^0(\mathcal{L}^n \otimes \mathcal{M}) = 2n + 2$ . But then for  $n$  large, Riemann-Roch shows  $2n + 2 = 2n + 2 - g + 1$ , thus  $g = 1$ , contradiction.  $\square$

### Gonal Curves

**Def. (7.13.5.9)[Gonal Curves].** A **trigonal curve** is a complete non-singular curve  $C$  with a degree 3 map  $C \rightarrow \mathbb{P}^1$ . Generally, a  **$k$ -gonal curve** is a complete non-singular curve  $C$  with a degree  $k$  map  $C \rightarrow \mathbb{P}^1_k$ . Being  $k$ -gonal is equivalent to having a basepoint-free  $g_k^1$ .

Notice also for non-hyperelliptic curves, any  $g_3^1$  must be basepoint-free.  $\lrcorner$

**Def. (7.13.5.10)[Hurewitz Spaces].** Let the **Hurewitz space** be

$$\mathcal{H}_{d,g} = \{(C, f) | C \in \mathcal{M}_g, f : C \rightarrow \mathbb{P}^1 \text{ of degree } d \text{ with simple branching}\}.$$

In particular,  $\mathcal{H}_{2,g}$  is just the space of hyperelliptic curves together with a hyperelliptic map.  $\lrcorner$

**Prop. (7.13.5.11)[Dimension of Hurewitz Spaces].**  $\dim \mathcal{H}_{d,g} = 2d + 2g - 2$ .  $\lrcorner$

*Proof:* Let  $b = 2d + 2g - 2$ , then the branch divisor will consist of an unordered  $b$ -tuple of distinct points. Then we obtain a map  $\mathcal{H}_{d,g} \rightarrow \mathbb{P}^b \setminus \Delta$ , where we regard  $\mathbb{P}^b$  as the set of polynomials of degree  $b$  and  $\Delta$  the determinant, and the fiber is finite by cut-paste technique.  $\square$

**Cor. (7.13.5.12)[Dimension of Moduli Space of Curves].**  $\dim \mathcal{M}_g = 3g - 3$ .  $\lrcorner$

*Proof:* There is a map  $\mathcal{H}_{d,g} \rightarrow \mathcal{M}_g$ . When  $d$  is large, we can analyze the fiber of this map, that is, given a curve of genus  $g$ , how many simply branched maps  $C \rightarrow \mathbb{P}^1$  of degree  $d$  are there? Such a map is equivalent to a line bundle of degree  $d$  and a pair of base free sections  $\sigma_0, \sigma_1 \in H^0(L)$ . The base-free condition is an open condition, thus the dimension of the fiber is  $g + 2(d - g - 1) - 1 = 2d + g - 1$ . Thus the dimension of  $\mathcal{M}_g = 2d + 2g - 2 - (2d + g - 1) = 3g - 3$ .  $\square$

**Cor. (7.13.5.13).** The space of hyperelliptic curves has dimension  $2g - 1$ .  $\lrcorner$

*Proof:*  $\mathcal{H}_{2,g}$  has dimension  $2g + 2$ , and for any hyperelliptic curve, there is exactly one hyperelliptic map up to automorphism of  $\mathbb{P}^1$ , by (7.13.5.8), thus the space of hyperelliptic curves has dimension  $2g + 2 - 3 = 2g - 1$ .  $\square$

**Cor. (7.13.5.14).** If  $g \geq 3$ , then not all curves of genus  $g$  is hyperelliptic.  $\lrcorner$

*Proof:* Because when  $g \geq 3$ ,  $2g - 1 < 3g - 3$ .  $\square$

**Lemma (7.13.5.15).** For a line bundle of degree 3 on a curve of genus  $g \geq 3$ ,  $h^0(L) \leq 2$ , by Clifford's theorem (7.13.4.13). Thus a trigonal map must be associated to a complete linear series  $g_3^1$ .  $\lrcorner$

**Prop. (7.13.5.16)[Hyperelliptic are not Trigonal].** A curve of genus  $g \geq 3$  cannot be both hyperelliptic and trigonal.  $\lrcorner$



*Proof:* Suppose there are two basepoint-free line bundles  $\mathcal{L}, \mathcal{M}$  of degree 2 and 3 on  $C$  that  $h^0(\mathcal{L}) = h^0(\mathcal{M}) = 2$  (7.13.5.15), then the base-free pencil trick (7.13.4.9) implies that  $h^0(\mathcal{M} \otimes \mathcal{L}) \geq 4$ . If  $g = 3$ , then this contradicts Riemann-Roch. If  $g \geq 4$ , then this contradicts Clifford's theorem (7.13.4.13).  $\square$

**Prop. (7.13.5.17) [Genus 3 Curves are Hyperelliptic or Trigonal].** Any genus 3 non-hyperelliptic complete smooth curve  $C$  with a rational point is trigonal.  $\lrcorner$

*Proof:* If  $C$  is non-hyperelliptic, then the canonical map realizes  $C$  as a plane curve of degree 4 (7.13.8.6), thus the projection of  $C$  along a point on  $C$  induces a map  $C \rightarrow \mathbb{P}^1$ . This map is finite of degree 3 because it is non-constant as  $C$  is non-degenerate, and has degree 3, by (7.13.2.28).  $\square$

**Prop. (7.13.5.18) [Uniqueness of Trigonal Divisors].** There exists at most one  $g_3^1$  on a curve of genus  $g \geq 5$ .  $\lrcorner$

*Proof:* If  $\mathcal{L} \neq \mathcal{M}$  are two base free line bundles of degree 3 that  $h^0 = 2$ , then we can use base-free trick to show that  $h^0(\mathcal{L} \otimes \mathcal{M}) = 4$ . But this contradicts Clifford's theorem (7.13.4.13). Notice the equality cannot hold, because the degree is too low to be the canonical bundle, and also  $C$  cannot be both hyperelliptic and trigonal (7.13.5.16).  $\square$

## 6 Castelnuovo's Theory

**Lemma (7.13.6.1) [Castelnuovo's Lemma].** Let  $\Gamma \in \mathbb{P}^n$  be a configuration of  $d \geq 2n + 3$  points in linear general position, and if  $h_\Gamma(2) = 2n + 1$ , then  $\Gamma$  lies in a rational normal curve.  $\lrcorner$

## 7 Plucker Formulas

**Def. (7.13.7.1) [Weierstrass Group].** Let  $C$  be a smooth of genus  $g \geq 2$ ,  $p \in C$ , then  $S_p = \{-\text{ord}_p(f) \mid f \in H^0(\mathcal{O}_C(-p))\}$  is a semigroup, called the **Weierstrass semigroup** of  $p \in C$ . And the **gap sequence** is  $\mathbb{N} \setminus S_p$ , which is the set of orders of pole of  $p$  that doesn't occur.  $\lrcorner$

**Lemma (7.13.7.2).** We have  $|G_p| = g$ .  $\lrcorner$

*Proof:* Notice that

$$G_p = \{m : h^0(mp) = h^0((m-1)p)\}, \quad S_p = \{m : h^0(mp) = h^0((m-1)p) + 1\}$$

and we know by (7.13.2.12) that  $h^p(mp) = m - g + 1$  for  $m$  large, thus there are exactly  $g$  jumps, which shows  $|G_p| = g$ .  $\square$

**Def. (7.13.7.3) [Weierstrass Points].** A point  $p \in C$  is called **Weierstrass point** if the gap sequence is not  $\{1, 2, \dots, g\}$ . It is called a **hyperelliptic Weierstrass point** if  $G_p = \{1, 3, \dots, 2g-1\}$ , and is called a **normal Weierstrass point** if  $G_p = \{1, 2, 3, \dots, g-1, g+1\}$ .

Define the **weight** of the  $p \in C$  to be the sum  $w(p) = \sum_{i \leq g} (a_i - i)$ , where the gap sequence is numbered  $\{a_1, \dots, a_g\}$ .  $\lrcorner$

**Prop. (7.13.7.4).** For any compact Riemann surface  $C$ ,  $\sum_{p \in C} w(p) = g(g-1)(g+1)$ .  $\lrcorner$

*Proof:* Cf. [G-H78], P274.  $\square$

**Cor. (7.13.7.5).** For general  $p \in C$ ,  $G_p = \{1, 2, \dots, g\}$ .  $\lrcorner$

**Prop. (7.13.7.6)[Hyperelliptic Weierstrass Points].** If  $C$  is hyperelliptic defined by  $y^2 = \prod_{i=0}^{2g+1} (x - \alpha_i)$ , then there are  $2g + 2$  branching point of  $x$ , where  $S_p = \{0, 2, 4, \dots, 2g, 2g + 1, \dots\}$ . In this way,  $w(p) = g(g - 1)/2$ . Thus there are  $2g + 2$  Weierstrass points, and these are all of them.

If  $C$  is non-hyperelliptic, then by Clifford's theorem(7.13.4.13),

$$h^0(kp) < \frac{k}{2} + 1, k = 1, \dots, g$$

Thus  $h^0((a_i - 1)p) < \frac{a_i - 1}{2} + 1$ , and  $h^0((a_i - 1)p) = 1 + (a_i - 1) - (i - 1)$ . Thus we get

$$a_i \leq 2i - 2, i = 2, \dots, g.$$

Then  $w(p) \leq \sum_{i=2}^g (i - 2) = \frac{(g-1)(g-2)}{2}$ . Thus there are at least

$$\frac{2g(g-1)(g+1)}{(g-1)(g-2)} \geq 2g + 6$$

Weierstrass points.

To sum up, there are no less than  $2g+2$  Weierstrass points, and there are exactly  $2g+2$  Weierstrass points iff  $C$  is hyperelliptic.  $\lrcorner$

**Prop. (7.13.7.7)[No Automorphisms].** A generic Riemann surface of genus  $g \geq 3$  has no non-trivial automorphisms.  $\lrcorner$

*Proof:*  $\square$

## 8 Curves of Low Genus

In this subsection complete smooth curves of low genus are considered.

**Prop. (7.13.8.1)[Genus 0 Curve].** All smooth complete curve  $C$  of genus 0 is isomorphic to a plane conic.  $\lrcorner$

*Proof:* The curve has a degree 2 line bundle  $\mathcal{K}_C^\vee$ , thus by Riemann-Roch  $h^0(\mathcal{O}(p)) = 3$  and by(7.13.2.26) it induces a closed embedding of  $C$  into  $\mathbb{P}^2$  of degree 2, thus it is a plane conic.  $\square$

**Prop. (7.13.8.2).** A nonsingular curve  $C$  of genus 0 with a  $k$ -rational point  $p$  is isomorphic to  $\mathbb{P}_k^1$ :  $\lrcorner$

*Proof:* By Riemann-Roch,  $h^0(\mathcal{O}(p)) = 2$  and  $\deg \mathcal{O}(p) = 1$ , thus by(7.13.2.26),  $\mathcal{O}(p)$  defines a closed embedding of  $C$  into  $\mathbb{P}^1$ , which must be an isomorphism.  $\square$

**Prop. (7.13.8.3)[Genus 1 Curve].** By(7.13.2.26), an effective divisor of degree 3 induces an embedding of  $C$  into  $\mathbb{P}^2$ , So it is a smooth plane cubic by genus formula(7.13.2.33). Conversely, any smooth plane cubic has genus 1.

Similarly, if we take an effective divisor of degree 4, then it gives an embedding  $C \rightarrow \mathbb{P}^3$ . We know that  $h^0(\mathcal{O}_{\mathbb{P}^3}(2)) = 10$  and  $h^0(\mathcal{O}_C(2)) = 8$ , thus  $C$  is contained in 2 quadrics. Then  $C$  is the intersection of these two quadrics, by Bezout's theorem.  $\lrcorner$

**Prop. (7.13.8.4)[Genus 2 Curves and Degree 4 Divisors].** Let  $C$  be a curve of genus 2, and  $D$  a divisor of degree 4. Then  $l(D - K_C) = 1 + l(2K_C - D) \geq 1$ , thus  $D - K_C$  is effective. Let  $D - K_C = p + q$ . Then  $\varphi_D$  maps  $p, q$  to the same point.

There are two situations, firstly if  $D \neq 2K_C$ , then  $D - K_C$  can be written uniquely as  $p + q$ . Then the image of  $\varphi_D$  is a degree 4 curve with a node (if  $p = q$ ) or a cusp (if  $p \neq q$ ). Counting genus, this has exactly arithmetic genus 2.

If  $D = 2K_C$ . Then notice  $\varphi_{2K_C}$  is equal to  $\varphi_K$  followed by the normal curve map  $\mathbb{P}^1 \rightarrow \mathbb{P}^2$ . This is because if  $\omega_1, \omega_2$  are a basis of  $H^0(K_C)$ , then  $\omega_1^2, \omega_1\omega_2, \omega_2^2$  is a basis of  $H^0(K_C^2)$ . So  $\varphi_D$  is a 2 to 1 map to a rational normal curve in  $\mathbb{P}^2$ .  $\lrcorner$

**Prop. (7.13.8.5) [Genus 2 Curve and degree 5 Divisors].** Let  $C$  be a curve of genus 2 and  $D$  a divisor of degree 5, then  $\varphi_D : C \rightarrow \mathbb{P}^3$  embeds  $C$  as a degree 5 curve, by (7.13.2.26).

Notice  $h^0(\mathcal{O}_{\mathbb{P}^3}(2)) = 10$  and  $h^0(\mathcal{O}_C(2)) = 10 - 2 + 1 = 9$ , thus  $C$  lies on at least one quadric  $Q$ . And it can in fact lie on only one quadric, because if it lies on two quadrics, then  $C$  is the intersection, and can have degree at most 4.

Next, notice  $h^0(\mathcal{O}_{\mathbb{P}^3}(3)) = 20$  and  $h^0(\mathcal{O}_C(3)) = 15 - 2 + 1 = 14$ , so  $C$  lies on at least 6 cubics. Without the 4 cubics containing the quadric, there are still at least 2 new cubics. Let  $S$  be such a cubic, then  $S \cap Q$  is a curve of degree 6, so  $S \cap Q = C \cup L$ , where  $L \cong \mathbb{P}^1$ .  $\lrcorner$

**Prop. (7.13.8.6) [Genus 3 Non-Hyperelliptic Curve].** For any non-hyperelliptic smooth curve  $C$  of genus 3 over a field  $k$ , the canonical map (7.13.4.8) embeds  $C$  as a smooth quartic curve in  $\mathbb{P}^2$ . This induces an isomorphism of smooth quartic plane curves up to automorphism of  $\mathbb{P}^2$ .  $\lrcorner$

*Proof:* The canonical embedding is a closed embedding by (7.13.5.7). For the last assertion, any plane curve of degree 4 has genus 3 by (7.13.2.33), and the sheaf  $\mathcal{O}_C(1)$  has degree 4 and 3 sections, thus by (7.13.2.13), so this comes from a canonical embedding. In particular, any such curve can be embedded in  $\mathbb{P}^2$  in a unique way.  $\square$

**Prop. (7.13.8.7).** Any hyperelliptic smooth curve of genus 3 is a flat limit of non-hyperelliptic smooth curves of genus 3.  $\lrcorner$

*Proof:* Cf. [Vak17]P520.  $\square$

**Prop. (7.13.8.8) [Genus 4 Non-Hyperelliptic Curves].** Any non-hyperelliptic smooth complete canonical curve  $C$  of genus 4 is a complete intersection of a quadric surface and a cubic surface. Conversely, any regular complete intersection of a quadric surface and a cubic surface is a canonically embedded non-hyperelliptic curve of genus 4.  $\lrcorner$

*Proof:* Looking at the map  $H^0(\mathcal{O}_{\mathbb{P}^3}(2)) \rightarrow H^0(\mathcal{O}_C(2))$ , we see  $C$  lies on a unique quadric  $Q$  (uniqueness follows from Bezout's theorem and (7.7.3.11), by the same reason as (7.13.8.5)).

Next looking at the map  $H^0(\mathcal{O}_{\mathbb{P}^3}(3)) \rightarrow H^0(\mathcal{O}_C(3))$ , then the kernel has dimension at least 5, thus there is a cubic containing  $C$  but not  $Q$ . Then  $S \cap Q$  is a complete intersection of degree 6 containing  $C$ , also it has the same arithmetic genus as  $C$  (7.7.3.8), thus the same Hilbert polynomial (linear function), so  $C = S \cap Q$  by (7.7.3.11)s. Conversely, any smooth curve of the form  $S \cap Q$  is a canonical curve of genus 4, by adjunction formula ( $\mathcal{K}_C = \mathcal{O}_C(1)$  has degree 6).

Conversely, for any smooth complete intersection of a quadric surface and a cubic surface has arithmetic genus 4 by (7.7.3.8),  $\mathcal{O}_C(1)$  has degree 6 and  $h^0(\mathcal{O}_C(1)) \geq 4$  because  $C$  is non-degenerate: there are no smooth plane curve of genus 4.

Now if  $Q$  is non-singular, then  $C$  is of type (3, 3) in  $Q$ , thus the two projections are two trigonal map from  $C$  to  $\mathbb{P}^1$ , and if  $Q$  is a cone, the projection from the vertex is a trigonal map from  $C$  to  $\mathbb{P}^1$ .  $\square$

**Prop. (7.13.8.9) [Non-Hyperelliptic Curve of Genus  $\geq 5$ ].** For a non-hyperelliptic curve  $C$  of genus 5, we have a canonical map  $\varphi : C \rightarrow \mathbb{P}^4$ . Firstly we consider what quadrics  $C$  lies on:  $h^0(\mathcal{O}_{\mathbb{P}^4}(2)) = 15$  and  $h^0(\mathcal{O}_C(2)) = 12$ , so  $C$  lies on at least 3 quadrics. There are two cases:

1.  $C = \cap Q_i$  where  $Q_i$  are the quadrics containing  $C$ .
2.  $C$  is a strict subset of  $\cap Q_i$ .

which corresponds to non-trigonal and trigonal curves.  $\lrcorner$

*Proof:* The case 1 does occur, because by Bertini's theorem (7.6.4.28), for general three quadrics  $Q_i$ ,  $\cap Q_i$  is a smooth curve, and thus  $\mathcal{K}_C = (\mathcal{K}_{\mathbb{P}^4}(-5 + 2 + 2 + 2))|_C = \mathcal{O}_C(1)$ . So  $d = 8, g(C) = 5$ .

In this case,  $C$  is not trigonal, because if  $C$  is trigonal, then  $C$  has a  $g_3^1$ , which means  $C$  has three colinear point. Then all the quadrics  $Q_i$  must contain this line, contradiction.

The case 2: Cf. [Algebraic Curves, Harris, P24]. ? This case corresponds to trigonal curves.  $\square$

**Prop. (7.13.8.10) [Non-Hyperelliptic Curve of Genus 5 is Tetragonal].** Let  $C$  be a canonical embedded curve which is not trigonal admits a map of degree 4 to  $\mathbb{P}^1$ .  $\lrcorner$

*Proof:* Let  $\mathbb{P}^2 = \{Q | C \subset Q\}$ . We can ask there are singular quadrics in this set. Inside  $\mathbb{P}^{14}$  which is the space of all quadrics in  $\mathbb{P}^4$ , there is a quintic hypersurface of singular quadrics. ? For the rest, Cf. [Algebraic Curves, Harris, P25].  $\square$

**Prop. (7.13.8.11).** If  $C \subset \mathbb{P}_k^{g-1}$  is a canonical smooth curve of genus  $g \geq 6$ , then  $C$  is not a complete intersection.  $\lrcorner$

*Proof:*

$\square$

## Correspondences

### 9 Brill-Noether Theory

### 10 Relative Curves

**Def. (7.13.10.1) [Relative Curves].** Let  $S \in \text{Sch}$ , a **smooth curve** over  $S$  is a smooth morphism  $C \rightarrow S$  of relative dimension 1 that is separated and of f.p..  $\lrcorner$

**Prop. (7.13.10.2).** Let  $S \in \text{Sch}$  and  $C$  a smooth curve over  $S$ , then  $\lrcorner$

## 7.14 K3 Surfaces

References are [Lectures on K3 Surfaces Huybrechts] and [General Introduction to K3 Surfaces].

### 1 Surfaces

**Prop. (7.14.1.1).** Any complete non-singular surface over an algebraically closed field is projective.  $\lrcorner$

*Proof:* Cf.[Badescu, L. Algebraic surfaces]Thm2.8.  $\square$

### 2 K3 Surfaces

**Def. (7.14.2.1) [K3 Surfaces].** For  $k \in \mathbf{Field}$ , a(n) (algebraic)**K3 surface** over  $k$  is a complete non-singular surface  $X$  over  $k$  s.t.  $\omega_X \cong \mathcal{O}_X$ , and  $H^1(X, \mathcal{O}_X) = 0$ .  $\lrcorner$

**Prop. (7.14.2.2).** For  $k \in \mathbf{Field}$ , the intersection of three quadrics in  $\mathbb{P}_k^5$  is a K3 surface of genus 5 and degree 8.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.14.2.3).** For  $k \in \mathbf{Field}$  and  $X/k$  a K3 surface, we have

- $\Omega_X \cong \mathcal{T}_X$ ,
- $H^2(X, \mathcal{O}_X) \cong H^0(X, \mathcal{O}_X)$ ,
- $\chi(\mathcal{O}_X) = 2 \dim H^0(X, \mathcal{O}_X) = 2$ .

$\lrcorner$

*Proof:* Easy by Poincaré duality.  $\square$

**Prop. (7.14.2.4) [Chow Ring of K3 Surfaces].** Let  $k \in \mathbf{Field}$  and  $X/k$  a K3 surface, then

- All points of  $X$  that lies in some rational(possibly singular)curve have the same class  $c_X \in \mathrm{CH}_0(X)$ .
- The image of the intersection  $\mathrm{Pic}(X) \times \mathrm{Pic}(X) \rightarrow \mathrm{CH}_0(X)$  is contained in  $\mathbb{Z}c_X$ .
- $c_2(X) = 24c_X$ .
- If  $c_X$  is represented by some  $c \in X$ , then for any  $x \in X$ ,

$$(x, x) - (x, c) - (c, x) + (c, c) = 0 \in \mathrm{CH}_0(X \times X).$$

$\lrcorner$

*Proof:* Cf.[Chow Ring of a K3 Surface, Beauville-Voisin].  $\square$

### K3 Surfaces over Finite Fields

**Prop. (7.14.2.5).** If  $k \in \mathbf{Field}^{\mathrm{fin}}$  and  $X \in \mathbf{K3}/k$ , then  $H_{\mathrm{ét}}^1(X_{\bar{k}}, \mathbb{Q}_\ell) = H_{\mathrm{ét}}^3(X_{\bar{k}}, \mathbb{Q}_\ell) = 0$ .  $\lrcorner$

*Proof:*  $\square$

### 3 Enriques Surfaces

**Def. (7.14.3.1) [Enriques Surfaces].** Enriques surfaces are algebraic surfaces  $X$  such that the irregularity  $q = 0$  and the canonical line bundle  $\omega_X$  is non-trivial but has trivial square.  $\lrcorner$

**Prop. (7.14.3.2) [Enriques Surfaces are Quotients of K3 Surfaces].** If  $\text{char } k \neq 2$ , then any Enriques surface over  $k$  is a quotient of a K3 surface by an involution.  $\lrcorner$

*Proof:* ?  $\square$

### 4 Complex K3 Surfaces

**Def. (7.14.4.1) [Complex K3 Surfaces].** a **complex K3 surface** is a compact connected Kähler complex surface  $X/\mathbb{C}$  such that  $\omega_X \cong \mathcal{O}_X$  and  $H^1(X, \mathcal{O}_X) = 0$ .

Notice that by [Lectures on K3 surfaces]Section7.3.2, the condition that  $X$  being Kähler is redundant ? .  $\lrcorner$

**Prop. (7.14.4.2) [Complex and Algebraic K3 Surfaces].** By GAGA and (7.14.1.1), an algebraic K3 surface over  $\mathbb{C}$  is equivalent to a projective complex K3 surface.  $\lrcorner$

**Prop. (7.14.4.3) [K3 Surfaces are Spin].** ?  $\lrcorner$

**Prop. (7.14.4.4) [K3 Lattice].** For  $X \in \text{K3}/\mathbb{C}$ , the cohomology group  $H^2(X, \mathbb{Z})$  endowed with the intersection form is isomorphic to  $E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$ . In particular,  $H^2(X, \mathbb{Z})$  is even of signature  $(3, 19)$ .

Such a lattice is called a **K3 lattice**, denoted by  $\Lambda^{\text{K3}}$ .  $\lrcorner$

*Proof:* Cf.[Lectures on K3 surfaces]Prop1.3.5. ?  $\square$

**Prop. (7.14.4.5).** If  $X$  is a complex K3 surface, then  $\text{Pic}(X)$  is torsion-free.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.14.4.6) [Complex K3 Surfaces are Deformation Equivalent].** All complex K3 surfaces are deformation equivalent.  $\lrcorner$

*Proof:* ?

Proof sketch:  $\square$

### 5 Period Maps

References are [Huybrechts]Section6.3

**Def. (7.14.5.1) [Polarized K3 Surfaces].** A **polarized K3 surface** is a tuple  $(X, \mathcal{L})$  where  $X/k$  is a K3 surface and  $\mathcal{L}$  is an ample line bundle on  $X$ . A **primitive polarized K3 surface** is a polarized K3 surface  $(X, \mathcal{L})$  s.t.  $\mathcal{L}$  is not a multiple of other ample line bundles.  $\lrcorner$

**Prop. (7.14.5.2) [Period Map].** For a smooth proper family of K3 surfaces  $X/S$  where  $S$  is connected, simply-connected and has distinguished point  $s_0$  with an isomorphism  $H_{\text{Betti}}^2(X_{s_0}, \mathbb{Z}) \cong \Lambda^{\text{K3}}$ , then we can assume that all  $H_{\text{Betti}}^2(X_s, \mathbb{Z})$  for  $s \in S$  are canonically isomorphic to  $\Lambda^{\text{K3}}$ . Then the **period map** is defined to be

$$\mathcal{P} : S \mapsto \mathbb{P}(\Lambda_{\mathbb{C}}^{\text{K3}}) : t \mapsto [H_{\text{Betti}}^{2,0}(X, \mathbb{C})] \in \mathbb{P}^1(\Lambda_{\mathbb{C}}^{\text{K3}}).$$

$$\mathbb{D} = \{ \langle z \rangle \in \mathbb{P}(\Lambda_{\mathbb{C}}^{\text{K3}}) \mid \langle z, z \rangle = 0, \langle z, \bar{z} \rangle < 0, z \perp v \in \text{Hom}(C^*, \text{SO}(v^\perp)) \}.$$

$\lrcorner$

## 6 Moduli Space of K3 Surfaces

**Thm. (7.14.6.1) [Moduli Space of K3 Surfaces].** Let  $\Lambda^{\text{K3}} \cong E_8^{\oplus 2} \oplus U^{\oplus 3}$ , and fix a vector  $v \in \Lambda$  of length  $d > 0$ . Then for any  $N \in \mathbb{Z}_+$ , there is a variety  $M_{N,d}^{\text{K3}}/\mathbb{Q}$  representing the functor that maps  $S \in \text{Sch}/\mathbb{Q}$  to the set of datum  $(X/S, E, \xi)$ , where

- $\pi : X \rightarrow S$  is a family of K3 surfaces.
- $E \in \text{NS}(X/S)$  is a polarization of degree  $d$ .
- $\xi : R^2\pi_{*,\text{ét}}(\mathbb{Z}/(N)) \cong (\Lambda^{\text{K3}}/(N))_S$  is an isomorphism.

┘

*Proof:*

□

**Thm. (7.14.6.2) [Global Tonelli Theorem].** If  $(S, \mathcal{L}), (S', \mathcal{L}') \in \text{K3}^{\text{polar}}/\mathbb{C}$  and there exists an isometry

$$\varphi : H_{\text{Betti}}^2(S', \mathbb{Z}) \cong H_{\text{Betti}}^2(S, \mathbb{Z})$$

s.t.  $\varphi(\mathcal{L}') = \varphi(\mathcal{L})$  and  $\varphi_{\mathbb{C}}(H_{\text{Betti}}^{2,0}(S')) = H_{\text{Betti}}^{2,0}(S)$ , then there exists an isomorphism  $\sigma : S \cong S'$  s.t.  $\sigma^* = \varphi$ .

┘

*Proof:* ?

□

**Prop. (7.14.6.3).** For  $g \in \mathbb{Z}_{\geq 2}$ , the period map  $\wp_g : M_g^{\text{K3}} \rightarrow \mathcal{P}_g(7.14.5.2)$  is an open immersion.

┘

*Proof:*

□

**Thm. (7.14.6.4) [K3 Surfaces and Shimura Varieties].** For suitable compact open subset  $\mathcal{K} \subset \text{SO}(2, 19)(\mathbf{A}^f)$ , let  $M_{g,\mathcal{K}}$  be the moduli space of K3 surfaces of genus  $g$  with level structure  $\mathcal{K}$ , then the period map is a morphism

$$\wp_{g,\mathcal{K}} : M_{g,\mathcal{K}}^{\text{K3}} \otimes \mathbb{C} \rightarrow \text{Sh}_{\mathcal{K}}(\text{SO}(2, 19), \mathbb{D})_{\mathbb{C}}.$$

┘

*Proof:*

□

**Thm. (7.14.6.5) [Rizov].** The period map in (7.14.6.4) is defined over  $\mathbb{Q}$ .

┘

*Proof:* Cf. [Rizov, Complex multiplication for K3 surfaces] ?

□

## 7.15 Tropical Geometry

References are [Tropical Geometry, Introduction] and [Mik06].

From a geometric point of view tropical geometry describes worst possible degenerations of the complex structure on an  $n$ -fold  $X$ .

### 1 Tropical Arithmetic

**Def. (7.15.1.1) [Tropical Semifield].** The **tropical semifield** or the **tropical line**  $\mathbb{T}$  is defined to be the topological space  $\{-\infty\} \cup \mathbb{R}$  with the operation of addition and multiplication given by

$$a \oplus b = \max a, b, \quad a \otimes b = a + b.$$

┘

**Def. (7.15.1.2) [Tropical Polynomials].** A **tropical polynomial** is a finite tropical sum of tropical monomials.

A  **$\mathbb{Z}$ -affine function** is a tropical quotient of two tropical monomials. Equivalently, it is a linear function with integer slopes.

A **tropical rational function** is a tropical quotient of two tropical polynomials. ┘

**Prop. (7.15.1.3) [Tropical Fundamental Theorem of Algebra].** Every tropical polynomial function can be written uniquely as a tropical product of tropical linear functions. ┘

*Proof:*

□

**Def. (7.15.1.4) [Regular Tropical Functions].** A **regular tropical function** on a Euclidean open subset  $U \subset \mathbb{T}^n$  is a function of the form  $p(y) \oplus \varphi(y)$  where  $p$  is a tropical polynomial and  $\varphi(y)$  is a  $\mathbb{Z}$ -affine function. ┘

**Prop. (7.15.1.5) [Tropical Sheaf].** For any Euclidean open subset  $U \subset \mathbb{T}^n$ , the set of regular tropical functions on  $U$  forms a semi-algebra, i.e. a tropical algebra. And in this way we get a structure sheaf  $\mathcal{O}$  on  $\mathbb{T}^n$ . ┘

*Proof:* ?

□

### 2 Tropical Varieties

**Def. (7.15.2.1) [Tropical Hypersurfaces].** For a tropical polynomial  $f$ , the tropical hypersurface  $V(f) \subset \mathbb{T}^n$  is just the set of points that  $f$  is not linear, i.e.  $\frac{1}{f}$  is not regular (7.15.1.4). ┘

**Def. (7.15.2.2) [Tropical Graphs].** For a tropical polynomial  $f : \mathbb{T}^n \rightarrow \mathbb{T}$ , we can define its **tropical graph** as

$$\Gamma_f \in \mathbb{T}^n \times \mathbb{T} = V(y + f(x)).$$

┘

**Def. (7.15.2.3) [Contractions].** For any tropical polynomial  $f$ , the projection  $\Gamma_f \rightarrow \mathbb{T}^n$  is called a **tropical principal contraction**. A **tropical contraction** is a composition of tropical principal contractions (a restriction of a tropical principal contraction to a tropical tropical subset in  $\mathbb{T}^n$ ). ┘



### Tropical Varieties

**Def. (7.15.2.4) [Tropical Varieties].** For  $n \in \mathbb{Z}_+$ , a **tropical variety** of dimension  $n$  is a topological space  $X \in \mathcal{Top}$  with atlas  $\varphi_\alpha : U_\alpha \rightarrow \mathbb{T}^n$  s.t.  $\cup_\alpha U_\alpha = X$ , and

- Each  $\varphi_\alpha$  is a contraction to an open subset of  $\mathbb{T}^n$ .
- The translation maps  $\Phi_{\alpha,\beta} = \varphi_\beta \circ \varphi_\alpha^{-1}$  are induced by the integer affine-linear maps.
- There exists a finite covering  $\{W_j\}$  of  $X$  s.t. for any  $j$ , there exists  $\alpha$  s.t.  $W_j \subset U_\alpha$  and the closure of  $\varphi_\alpha(W_j)$  is contained in  $\varphi_\alpha(U_\alpha)$ .

┘

**Def. (7.15.2.5) [Tropical Varieties are Balanced].** If  $\delta : V \subset \mathbb{T}^N \rightarrow \mathbb{T}^n$  is contraction map, then  $V$  is balanced (7.15.3.2), where the weight on each facet is given by the gcd of the two monomials equating at this facet.

Then by the definition (7.15.2.4), the image of  $X$  under any tropical chart is balanced.

┘

*Proof:* ?

□

**Def. (7.15.2.6) [Finite Points].** For a tropical variety  $X$ , a point  $x \in X$  is called a **finite tropical point** if  $x$  is mapped to  $\mathbb{R}^n \subset \mathbb{T}^n$  under some (thus all) tropical atlas on  $X$ .

At a finite point, we have the notion of **integer tangent vector** to  $X$ .

┘

**Def. (7.15.2.7) [Tropical Morphisms].** A map  $f : X \rightarrow Y$  of tropical varieties is called a **linear tropical map** is that is linear on the tropical coordinates. It is called a **regular tropical map** if it is given by regular functions on tropical coordinates.

┘

**Prop. (7.15.2.8).** The image of a map  $f : X \rightarrow Y$  of tropical varieties is a  $(\dim X)$ -dimensional tropical variety.

┘

**Def. (7.15.2.9) [Equivalent Tropical Varieties].** A **tropical contraction of varieties** is a tropical map  $f : X \rightarrow Y$  of tropical varieties of the same dimension that is locally a contraction map. Two tropical varieties are called **equivalent tropical varieties** if they are in the same equivalence class under the equivalence relation generated by contraction maps.

┘

**Prop. (7.15.2.10) [Regular Tropical Maps are Contractions of Linear Tropical Maps].** If  $f : X \rightarrow Y$  is a regular tropical morphism (7.15.2.7), then there exists a contraction  $\delta : \tilde{X} \rightarrow X$  s.t.  $f \circ \delta : \tilde{X} \rightarrow Y$  is a linear tropical morphism.

┘

*Proof:* ?

□

### Projective Tropical Varieties

**Def. (7.15.2.11) [Tropical Projective Spaces].** For  $n \in \mathbb{Z}_+$ , the **tropical projective space**  $\mathbb{TP}^n$  is defined by the quotient of  $\mathbb{T}^{n+1} \setminus (-\infty, \dots, -\infty)$  under the equivalence relation  $(x_0, \dots, x_n) \sim (\lambda \otimes x_0, \dots, \lambda \otimes x_n)$  for any  $\lambda \in \mathbb{T}^\times$ .

It has  $n + 1$  tropical charts (7.15.2.4).

┘

**Def. (7.15.2.12) [Tropical Varieties].** Let  $K = \overline{\mathbb{C}((t))} = \mathbb{C}\{\{t\}\}$ , then any  $p \in K^\times$  can be written as a Puiseux series

$$p = c_1 t^{q_1} + c_2 t^{q_2} + \dots,$$

where  $c_i \in \mathbb{C}^\times$  and  $q_1 < q_2 < \dots \in \mathbb{Q}$  with bounded denominators. Then we call  $q_1$  the order of  $p$ .

Thus there is a map

$$\text{ord} : (K^\times)^n \rightarrow \mathbb{Q}^n \subset \mathbb{R}^n,$$

Then a **tropical variety** in  $\mathbb{R}^n$  is defined to be a subset of the form

$$T(I) = \overline{\text{ord}(V(I))}$$

where  $I$  is an ideal of  $K[X_1^\pm, \dots, X_n^\pm]$  and  $V(I) \subset (K^\times)^n$  its vanishing points.  $\lrcorner$

**Def. (7.15.2.13) [Projective Tropical Varieties].** And a **projective tropical variety** in  $\mathbb{TP}^{n-1}$  is defined to be a subset of the form

$$T(I) = \overline{\text{ord}(V(I))} / \mathbb{R}(1, \dots, 1).$$

$\lrcorner$

**Def. (7.15.2.14).** And a **tropical linear subspace** in  $\mathbb{TP}^{n-1}$  is defined to be a projective tropical variety  $T(I)$  s.t.  $I$  is generated by linear forms.  $\lrcorner$

**Prop. (7.15.2.15) [Tropical Variety and Tropical Numbers].** Tropical varieties can be regarded as sets where at least two of the terms in the tropical polynomial attains minimum. ?  $\lrcorner$

**Def. (7.15.2.16) [Tropical Basis].** For any  $w \in \mathbb{Q}^n$  called a **weight**, we can define the weight of a tropical monomial, and thus define the initial terms of a tropical polynomial  $f$ , denoted by  $In_w(f)$ . For any ideal  $I \subset \mathbb{C}\{\{t\}\}[X_1^\pm, \dots, X_n^\pm]$ , let  $In_w(I)$  be the ideal generated by initial forms of its elements. Then for any ideal  $I$ , there exists a finite set  $G \subset I$  s.t.

- If  $w \in T(I)$ , then  $\{In_w(g)\}_{g \in G}$  generate  $In_w(I)$ .
- If  $w \notin T(I)$ , then  $\{In_w(g)\}_{g \in G}$  contains a monomial.

Such a set  $G$  is called a **tropical basis** of  $I$ .  $\lrcorner$

*Proof:* Cf. [Solving Systems of Polynomial Equations] Section 9 or [tropical Grassmannian].  $\square$

**Cor. (7.15.2.17).** It can be verified that if  $G$  is a tropical basis for an ideal  $I$ , then  $T(I) = \bigcap_{g \in G} T(g)$ .  $\lrcorner$

**Def. (7.15.2.18) [Tropical Prevarieties].** A **tropical prevariety** in  $\mathbb{R}^n$  or  $\mathbb{TR}^n$  is an intersection of f.m. tropical hypersurfaces.

Any tropical variety is a tropical prevariety, but not conversely.  $\lrcorner$

*Proof:* Every tropical variety is a tropical prevariety by (7.15.2.17). For the converse ?.

$$I = 0x + 0y + 0z, \quad I' = 0x + 0y + 1z.$$

Then  $\square$

## Differential Forms

**Def. (7.15.2.19) [Tropical Differential Forms].** For  $k \in \mathbb{N}$ , a **tropical  $k$ -form** on a tropical variety  $X$  is an exterior real-valued  $k$ -form of the integer tangent vectors at every finite point  $x \in X$  (7.15.2.6) s.t. on each chart  $U_\alpha$ , the form is induced from a constant linear  $k$ -form on  $\varphi_\alpha(U_\alpha)$ .  $\lrcorner$

**Def. (7.15.2.20)[Globally-Defined Differential Forms].** A globally-defined differential form on a tropical variety  $X$  is a differential form (7.15.2.19) s.t. for any non-finite point  $x \in X$ , if  $x \in U_\alpha$  and  $\varphi_\alpha(x) \in \mathbb{T}^{N_\alpha}$  has the  $j$ -th coordinate  $-\infty$ , then the constant differential form  $\omega$  on  $\mathbb{R}^{N_\alpha}$  doesn't depend on the  $j$ -th coordinate.  $\lrcorner$

**Example (7.15.2.21).** There are globally-defined differential forms on any tropical torus  $\mathbb{R}^n/\Lambda$ , but there are no globally-defined differential forms on  $\mathbb{TP}^n$ .  $\lrcorner$

**Prop. (7.15.2.22).** For any tropical contraction  $\tilde{X} \rightarrow X$ , any globally-defined differential forms vanishes on the direction of the local contraction maps, so there is a bijection between globally-defined differential forms on  $\tilde{X}$  and that on  $X$ .  $\lrcorner$

### 3 Polyhedra Structures

**Def. (7.15.3.1)[Integer Convex Polyhedron].** An integer convex polyhedron in  $\mathbb{T}^n$  is the closure of a convex polyhedron in  $\mathbb{R}^n$  (may be unbounded) s.t. the slopes of all its faces are integers.  $\lrcorner$

**Def. (7.15.3.2)[Balanced Complexes].** A balanced simplicial complex of integer convex polyhedra  $X$  in  $\mathbb{T}^n$  of dimension  $k$  is a simplicial complex of integer convex polyhedra with each facets assigned a weight that satisfy the following: For any  $(k-1)$ -dimensional face  $Q$  of  $X$  and facets  $P_1, \dots, P_l$  adjacent to  $Q$ , if we consider the map  $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}^{n-k+1}$  mapping the affine  $(k-1)$ -subspace containing  $Q$  to a point and each  $P_i$  to an interval, and let  $v_j$  be the primitive integer vector parallel to  $\lambda(P_j)$  multiplied by the weight of  $P_j$ , then

$$\sum_{i=1}^l v_j = 0.$$

$\lrcorner$

**Thm. (7.15.3.3)[Characterizing Tropical Curves].** Tropical curves in  $\mathbb{TP}^2$  are exactly the balanced rational graphs.  $\lrcorner$

*Proof:* Cf.[Tropical geometry, first steps]P300. ?  $\square$

**Prop. (7.15.3.4)[Planer Quadratic Tropical Curves].** There are classifications of planer quadratic tropical curves in [Tropical geometry, first steps]P297.  $\lrcorner$

*Proof:*  $\square$

**Thm. (7.15.3.5)[Speyer].** Any tropical variety is an integral, weighted, balanced polyhedral complex.  $\lrcorner$

*Proof:*  $\square$

**Prop. (7.15.3.6)[Tropical Lines].** Here is a description of tropical lines in  $\mathbb{TP}^{n-1}$ : A tropical line in  $\mathbb{TP}^{n-1}$  is an embedded tree whose edges are either bounded line segments or unbounded half rays, subject to the following three rules:

- The directions of all edges in the tree are spanned by integer vectors.
- There are precisely  $n$  unbounded half rays. Their directions are the  $n$  standard coordinate directions  $e_1, \dots, e_n$  in  $\mathbb{TP}^{n-1}$ .
- If  $u_1, u_2, \dots, u_r$  are the primitive integer vectors in the directions of all outgoing edges at any fixed vertex of the tree then  $u_1 + u_2 + \dots + u_r = 0$ .

$\lrcorner$

*Proof:* Cf.[Tropical Grassmannian]  $\square$

### Tropical Incidences

**Thm. (7.15.3.7) [Tropical Pappus Hexagon Theorem, Tabera].** Let 1, 2, 3, 4, 5 be five points in  $\mathbb{TP}^2$ , and define the cross products:

$$a = 1 \otimes 4, b = 2 \otimes 4, c = 3 \otimes 4, a' = 1 \otimes 5, b' = 2 \otimes 5, c' = 3 \otimes 5,$$

$$6 = b \otimes c', 7 = a' \otimes c, 8 = a \otimes b', a'' = 1 \otimes 6, b'' = 2 \otimes 7, c'' = 3 \otimes 8.$$

Then the three tropical lines  $a'', b'', c''$  are concurrent.  $\lrcorner$

*Proof:* Cf. [Tropical Constructive Pappus' Theorem].  $\square$

## 4 Intersection Theory

**Def. (7.15.4.1) [Tropical Cycles].** For a tropical variety  $X$  and  $k \in \mathbb{N}$ , a  **$k$ -tropical cycle** is a  $k$ -dimensional integer piece-wise linear polyhedral complex weighted by integers that satisfy the balancing property (7.15.3.2).  $\lrcorner$

**Prop. (7.15.4.2) [Linear Pushforward of Cycles].** For a linear map of tropical varieties  $f : X \rightarrow Y$  and a tropical  $k$ -cycle  $B \subset X$ ,  $f_*(B) \subset Y$  is also a tropical  $k$ -cycle.  $\lrcorner$

**Def. (7.15.4.3) [Cartier Divisors].** For a tropical variety  $X$ , a **tropical Cartier divisor** on  $X$  is a formal linear sum of  $(\dim X - 1)$ -cycles on  $X$  s.t. there exists an open covering  $\{U_\alpha\}$  of  $X$  and regular functions  $f_\alpha : U_\alpha \rightarrow \mathbb{T}$  that defines  $D \cap U_\alpha \subset U_\alpha$  as a hypersurface.  $\lrcorner$

**Prop. (7.15.4.4) [Tropical Intersections]. ?**

The intersection is well-defined and gives a commutative associative ring structure on  $\text{Ch}^*(X)$ .  $\lrcorner$

**Def. (7.15.4.5) [Linear Pullback of Cycles].** For a linear map  $f : X \rightarrow Y$  of tropical varieties and  $B \subset Y$  a  $k$ -cycle, the pullback cycle is the  $(k + \dim X - \dim Y)$ -dimensional cycle

$$f^*(B) = \pi_*^X((X \times B) \cdot \Gamma_f) \subset X \text{ (7.15.4.4).}$$

$\lrcorner$

**Def. (7.15.4.6) [Linear Equivalence of Cycles].** Two cycles in a tropical variety  $X$  are called **tropical linearly equivalent** if they appear in a family of cycles in  $X$  parametrized by  $\mathbb{TP}^1$ .  $\lrcorner$

**Prop. (7.15.4.7).** The degree of a 0-cycle in a compact tropical variety is a deformation invariant.  $\lrcorner$

*Proof:* ?  $\square$

## 5 Tropical Curves

References are [M-Z08].

### Metric Graphs

**Def. (7.15.5.1)[Metric Graphs].** In a graph, a **leaf** is an edge adjacent to a 1-valent vertex. A **finite** is an edge that is not a leaf.

A **metric graph** is a finite graph  $\Gamma$  s.t. each of its finite edge is assigned a positive real length.

┘

**Prop. (7.15.5.2)[Tropical Curves and Metric Graphs].** There is a bijection between compact tropical curves and metric graphs, and a map  $f : \Gamma \rightarrow \Gamma'$  between tropical graphs is a regular morphism iff there exists a presentation of  $\Gamma$  as a metric graph s.t. for each  $e \in E(\Gamma)$ , there exists  $n(e) \in \mathbb{N}$  s.t.

$$d_{\Gamma'}(f(x), f(y)) = n(e)d_{\Gamma}(x, y).$$

┘

*Proof:* ?

□

**Prop. (7.15.5.3).** The 1-forms on a compact tropical curve  $\Gamma$  form a real vector space  $\Omega(\gamma)$  of dimension  $b_1(\Gamma)$ .

┘

## 6 Enumerative Tropical Geometry

References are [Mikhalkin, Enumerative tropical algebraic geometry in  $\mathbb{R}^2$ ].

**Prop. (7.15.6.1).** For  $g \in \mathbb{N}$  and  $d \in \mathbb{Z}_+$ , let  $\mathcal{Z}$  be a set of  $3d - 1 + g$  points in  $\mathbb{CP}^2$  in general position, and denote  $N_{g,d}^{\mathbb{C}}$  the number of curves of degree  $d$  and genus  $g$  through them. On the other hand, let  $\mathcal{X}$  be a set of  $3d - 1 + g$  points in  $\mathbb{TP}^2$  in general position, and denote  $N_{g,d}^{\mathbb{T}}$  the number of curves of degree  $d$  and genus  $g$  through them. Then  $N_{g,d}^{\mathbb{T}} = N_{g,d}^{\mathbb{C}}$ .

┘

*Proof:* ?

□



## 8 | Algebraic Geometry II: Spectral Algebraic Geometry

### 8.1 Andre-Quillen-Cohomology

Basic references are [Andre-Quillen Cohomology of Commutative Algebras Iyenger]. See also [Quillen Cohomology of Commutative Rings] and [Quillen On the (Co-)homology of Commutative Rings]. [Smoothness, Regularity and Complete Intersections] is a must read.

#### 1 Naive Cotangent Complex

This subsection is obsolete.

**Prop. (8.1.1.1) [Polynomial Replacement].** For a morphism of ring morphisms  $(R \rightarrow S) \rightarrow (R' \rightarrow S')$ , let  $\alpha, \alpha'$  be two presentations, then there exists morphism of presentations, and different morphisms induce homotopic maps  $NL_{S/R} \rightarrow NL_{S'/R'}$ .  $\lrcorner$

*Proof:* Cf. [[Sta]00S1]. In fact, any surjective formally smooth representation will give the naive cotangent complex, up to quasi-isomorphism (8.1.1.4).  $\square$

**Cor. (8.1.1.2).** If  $A = R[X_i]$  be a polynomial algebras, then  $NL_{A/R}$  is homotopic to  $(0 \rightarrow \Omega_{B/A})$  because  $A \rightarrow A$  is a presentation with zero kernel.

If  $R \rightarrow A$  is surjective with kernel  $I$ , then  $NL_{A/R}$  is homotopic to  $(I/I^2 \rightarrow 0)$ .  $\lrcorner$

**Lemma (8.1.1.3) [Formally Smooth Replacement 1].** If  $A \rightarrow B$  is a ring map that has two surjective presentations  $C \rightarrow B, D \rightarrow B$  with kernels  $I, J$ . If there is a map  $C \rightarrow D$  commuting these two presentation,  $D$  formally smooth, and  $C \rightarrow D$  is surjective or  $C$  is formally smooth, then their corresponding naive cotangent complexes are quasi isomorphic.  $\lrcorner$

*Proof:* Cf. [Foundations of Perfectoid Geometry P123].  $\square$

**Prop. (8.1.1.4) [Formally Smooth Replacement 2].** If  $B$  is an  $A$ -algebra that has two formally smooth presentation  $C \rightarrow B, D \rightarrow B$  with kernels  $I, J$ . then their corresponding naive cotangent complexes are quasi isomorphic.  $\lrcorner$

*Proof:* It suffices to prove they are both quasi isomorphic to the canonical cotangent complex. For

this, we first consider the diagram 
$$\begin{array}{ccc} D & \longrightarrow & C \\ \downarrow & & \downarrow \\ A[B] & \longrightarrow & B \end{array}$$
, where  $D = A[S]$  and  $S = C \amalg A[B]$  as sets. The

two map  $D \rightarrow A[B]$  and  $D \rightarrow A[B]$  can be chosen because  $C \rightarrow B$  is surjective. So the results follows from (8.1.1.3).  $\square$

**Prop. (8.1.1.5) [Jacobi-Zariski Sequence].** Let  $A \rightarrow B \rightarrow C$  be a ring map. Choose a presentation  $\alpha : P \rightarrow B$  for  $B/A$  with kernel  $I$ , a presentation  $\beta : Q \rightarrow C$  for  $C/B$  with kernel  $J$ , a presentation  $\gamma : R \rightarrow C$  for the induced representation  $C/A$  with kernel  $K$ , then there is an exact sequence of complexes:

$$\begin{array}{ccccccc} I/I^2 \otimes_B C & \longrightarrow & K/K^2 & \longrightarrow & J/J^2 & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 \longrightarrow \Omega_{P/A} \otimes_B C & \longrightarrow & \Omega_{R/A} \otimes C & \longrightarrow & \Omega_{Q/B} \otimes C & \longrightarrow & 0 \end{array}$$

Applying snake lemma, we get

$$H_1(NL_{B/A} \otimes_B C) \rightarrow H_1(L_{C/A}) \rightarrow H_1(L_{C/B}) \rightarrow C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0.$$

┘

*Proof:* Cf. [[Sta]00S2].

□

**Prop. (8.1.1.6) [Localization].** Let  $A \rightarrow B$  be a ring map, for a multiplicative set  $S$  of  $B$ , we have  $NL_{B/A} \otimes_B S^{-1}B$  is quasi-isomorphic to  $NL_{S^{-1}B/A}$ .

┘

*Proof:* Because it commutes with colimit, it suffice to prove for  $S = f$ , and this is the content of lemma (8.1.1.7) below.

□

**Lemma (8.1.1.7).** If  $A \rightarrow B$  is a ring map and  $\alpha : P \rightarrow B$  is a presentation of  $B$  with kernel  $I$ , then  $\beta : P[X] \rightarrow B_g : X \rightarrow 1/g$  is a presentation of  $B_g$  with kernel  $J = I + (gX - 1)$ . Then we have

- $J/J^2 = (I/I^2)_g \oplus B_g(fX - 1)$ .
- $\Omega_{P[X]/A} \otimes_{P[X]} B_g = \Omega_{P/A} \otimes_P B_g \oplus B_g dX$ .
- $NL(\beta) \cong NL(\alpha) \otimes_B B_g \oplus (B_g \xrightarrow{g} B_g)$ .

Hence  $NL_{B/A} \otimes_B B_g \rightarrow NL_{B_g/A}$  is a homotopy equivalence.

┘

*Proof:* Cf. [[Sta]08JZ].

□

## 2 Cotangent Complex

**Def. (8.1.2.1) [Cotangent Complex].** The adjunction of  $A \ltimes -$  and  $A \otimes_- \Omega_{-/R}$  (5.4.3.5) extends to an adjunction between  $(sCAlg_R)_{/A}$  and  $sMod_A$ . These categories are model categories by (5.8.2.11)(5.8.2.14) and (4.5.4.1), and  $A \ltimes -$  preserves all weak equivalences and fibrations, so it is a Quillen adjunction (4.5.2.1). Then the **cotangent complex**  $L_{A/R}$  as a simplicial  $A$ -module is defined to be the total left derived functor applied to the trivial simplicial algebra  $A$ . Equivalently, it is

$$L_{A/R} = A \otimes_X \Omega_{X/R}$$

where  $X$  is a cofibrant replacement (4.5.1.12) of  $A$ .

Because of the Dold-Kan equivalence, we sometimes also call  $NL_{A/R}$  the cotangent complex. ┘

**Def. (8.1.2.2) [André-Quillen Homology and Cohomology].** The **André-Quillen homology** is defined to be

$$D_q(A/R) = \pi_q(L_{A/R}) = H_q N(L_{A/R}) \quad (5.8.2.2).$$

More generally, if  $M$  is an  $A$ -module, then let

$$D_q(A/R, M) = \pi_q(L_{A/R} \otimes_A M).$$



The **André-Quillen cohomology** is defined to be

$$D^q(A/R, M) = \text{Ext}^n(NL_{A/R}, M).$$

┘

**Prop. (8.1.2.3) [Functoriality].** The cotangent complex is functorial in arrows  $R \rightarrow A$ : If there is a commutative diagram

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

then there is a natural morphism  $L_{A/S} \otimes_A B \rightarrow L_{B/R}$ . This is because if  $X$  is a cofibrant replacement for  $A$ , then  $X \otimes_R S$  is also cofibrant object, because  $B \otimes_A -$  is a Quillen adjunction, by (5.8.2.15), then it factors through  $X \otimes_R S \rightarrow Y \rightarrow B$  where  $Y$  is a cofibrant replacement of  $B$ , thus  $Y$  is also cofibrant. Then the functor  $L_{A/S} \otimes_A B \rightarrow L_{B/R}$  is induced by

$$B \otimes_A (A \otimes_X \Omega_{X/R}) \rightarrow B \otimes_Y \Omega_{Y/S}.$$

The formation of Kähler differential commutes with arbitrary colimit as it is a left adjoint, so the formation of cotangent complex commutes with filtered colimits, both in  $A$  and  $B$ . Especially, it commutes with taking stalks, hence the sheaf of cotangent complexes of a map between schemes can be constructed as in the case of Kähler differentials, and it is a Qco sheaf. ┘

**Def. (8.1.2.4) [Canonical Resolution].** By the Dold-Kan correspondence, we will say that two simplicial  $A$ -modules are quasi-isomorphic iff their normalized nerves are quasi-isomorphic. Then  $P_A(B) \rightarrow B$  is a quasi-isomorphic resolution of  $B$ , where  $B$  is the trivial complex. ┘

*Proof:* There is a homotopy  $d$  between  $\text{id}$  and  $0$  for  $n \leq 0$ , where

$$d_n : F(GF)^n G(B) \rightarrow F(FG)^n \circ GFG(B)$$

using counit map, and on degree  $0, -1$ , it is  $A[A[B]] \xrightarrow{\partial_1 - \partial_2} A[B] \rightarrow B \rightarrow 0$ , which is clearly  $0$ , so this is a zero map.

Thus  $\text{Tot}(P_A(B)) \cong B$ , and  $N_*(A) \cong \text{Tot}(A)$  by Dold-Kan correspondence, so we are done. ┘

**Cor. (8.1.2.5).**

$$D_0(L_{B/A}, M) = \Omega_{B/A} \otimes M, \quad D^0(A/R, M) = \text{Der}_R(A, M)$$

by the generator-relation definition of Kähler differential. ┘

**Prop. (8.1.2.6) [The fundamental Distinguished Triangle].** If  $\mathcal{T}$  is a site and  $A \rightarrow B \rightarrow C$  are morphisms of sheaves of rings over  $\mathcal{T}$ , then there is a morphism of simplicial  $A$ -modules that corresponds to distinguished triangles in  $D^{\leq 0}(C)$  via Dold-Kan correspondence:

$$L_{B/A} \otimes_B^L C \rightarrow L_{C/A} \rightarrow L_{C/B}.$$

In particular, if  $M$  is a  $B$ -module, then there are long exact sequences

$$\begin{aligned} \dots &\rightarrow D_1(A/R, M) \rightarrow D_1(B/R, M) \rightarrow D_1(B/A, M) \rightarrow M \otimes_A \Omega_{A/R} \rightarrow M \otimes_A \Omega_{B/A} \rightarrow M \otimes_B \Omega_{B/A} \rightarrow 0 \\ \dots &\rightarrow D^1(A/R, M) \rightarrow D^1(B/R, M) \rightarrow D^1(B/A, M) \rightarrow \text{Der}_R(A, M) \rightarrow \text{Der}_A(B, M) \rightarrow \text{Der}_A(B, M) \rightarrow 0 \end{aligned}$$

┘

*Proof:* Choose a simplicial resolution  $X \rightarrow A$  where  $X$  is free, then we factor the morphism  $X \rightarrow A \rightarrow B$  to get a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & Y \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

where  $i$  is free, then we have an exact sequence (in  $Ch_{\geq 0}(R)$  via Dold-Kan) of simplicial modules

$$0 \rightarrow B \otimes_X \Omega_{X/R} \rightarrow B \otimes_Y \Omega_{Y/R} \rightarrow B \otimes_Y \Omega_{Y/X} \rightarrow 0$$

because each  $X_n \rightarrow Y_n = X_n \otimes \bigoplus_{\varphi: [n] \rightarrow [k]} \varphi^* P_k$  has a retraction, and use (5.4.3.7).

Notice we have a simplicial map  $A \otimes_X Y \rightarrow B$  which is a weak equivalence and  $A \otimes_X Y$  is a free simplicial  $A$ -algebra. Then it suffices to note that  $X_n \rightarrow Y_n = X_n \otimes \bigoplus_{\varphi: [n] \rightarrow [k]} \varphi^* P_k$  is projective implies

$$B \otimes_Y \Omega_{Y/X} \cong B \otimes_{A \otimes_X Y} \Omega_{A \otimes_X Y/A}$$

, and  $\Omega_{X/R}$  is termwise projective. □

**Prop. (8.1.2.7) [Properties of Cotangent Complexes].**

- If  $B = s(P)$  where  $P$  is a projective  $A$ -module, then  $L_{B/A}$  is weakly equivalent to  $\Omega_{P/A}^1[0]$ .
- (Kunneth Formula) If  $B, C$  are Tor independent over  $A$ , then

$$L_{B \otimes C/A} \cong (L_{B/A} \otimes_A C) \oplus (L_{C/A} \otimes_A B).$$

- (Flat Base Change) If  $B, C$  are Tor independent over  $A$ ,  $L_{B/A} \otimes_A C \cong L_{B \otimes_A C/C}$ . ┘

*Proof:*

- $S_R(P)$  is already cofibrant in  $(sCAlg_R)_A$ .
- Let  $X \rightarrow B, Y \rightarrow C$  be cofibrant replacement of  $A, B$  respectively, then  $X \otimes_R Y \rightarrow B \otimes_A C$  is a cofibrant replacement, as  $X \otimes_A Y$  is cofibrant, and Tor independence shows

$$\pi_*(X \otimes_A Y) = \pi_*(X) \otimes_A \pi_*(Y) = B \otimes_A C.$$

thus the result follows from (5.4.3.6).

- The same as Künneth formula, noticing that  $X \otimes_A C \rightarrow B \otimes_A C$  is a weak equivalence by Tor independence, and  $X \otimes_A C$  is cofibrant. □

### 3 Relations with Algebraic Properties

Cf. [Andre-Quillen Homology].

**Prop. (8.1.3.1) [Acyclicity for Smooth Algebras].** If  $A \rightarrow B$  is smooth, then  $L_{B/A} \cong \Omega_{B/A}^1[0]$ . In particular, if  $A \rightarrow B$  is étale, then  $L_{B/A} = 0$ , and  $L_{C/A} \cong L_{B/A} \otimes_B C$ , by distinguished triangle (8.1.2.6). ┘

*Proof:* The cotangent complex is local, so we may assume it is standard smooth, so it factors as  $A \rightarrow A[X_1, \dots, X_k] \xrightarrow{g} B$ , where  $g$  is étale, so using the distinguished triangle and polynomial case, the result follows.  $\square$

**Prop. (8.1.3.2) [Compatibility with  $p$ -adic Completion].** If  $A$  is a  $p$ -adically complete commutative ring with bounded  $p$ -torsion and  $B$  is a flat  $A$ -module, then  $B$  also has bounded  $p^\infty$ -torsion by (5.9.2.4), let  $\hat{B}$  be the  $p$ -adic completion of  $B$ , then the cotangent complex  $L_{\hat{B}/B}$  vanishes after derived  $p$ -completion.

In particular, by the distinguished triangle (8.1.2.6), if  $B$  is a smooth algebra, then  $L_{\hat{B}/A} \cong \Omega_{\hat{B}/A}^1$  is a finite projective  $\hat{B}$ -module.  $\lrcorner$

*Proof:* This is true after base change  $-\otimes_A^L A/p$  by flat base change (8.1.2.7), (5.9.7.4) and derived Nakayama (5.9.6.10).  $\square$

## 4 Deformations

**Prop. (8.1.4.1) [Topological Invariance of Étale Site].** Let  $A$  be a ring, consider the following category:  $\mathcal{C}_A$  of flat  $A$ -algebras  $B$  that  $L_{B/A} = 0$ , then if  $\tilde{A} \rightarrow A$  is surjective with locally nilpotent kernel, then the base change defines an isomorphism of categories  $\mathcal{C}_{\tilde{A}} \cong \mathcal{C}_A$ .

By (8.1.2.7),  $L_{B/A}$  vanish is equivalent to being étale, thus the properties characterize the invariance of étale site under infinitesimal thickening.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (8.1.4.2) [Relative Perfect Case].** If  $A$  is a ring of char  $p$  and  $B$  is an  $A$ -algebra which is relatively perfect, i.e.  $B^{(1)} = B \otimes_{A, \text{Frob}} A \rightarrow B$  is an isomorphism, then  $L_{B/A} = 0$ .  $\lrcorner$

*Proof:* Notice for any  $A$ -algebra  $C$ , the relative Frobenius induces zero map  $L_{C^{(1)}/A} \rightarrow L_{C/A}$ , because by using the canonical polynomial resolution,  $d(x^p) = px^{p-1}xs = 0$ . Now the relative Frobenius is an isomorphism  $B^{(1)} \rightarrow B$ , thus induces an isomorphism  $L_{B^{(1)}/A} \rightarrow L_{B/A}$  by Functoriality, thus  $L_{B/A} = 0$ .  $\square$

**Cor. (8.1.4.3) [Witt Vector Construction].** There is an equivalence of categories of  $\mathcal{C}_n = \text{flat } \mathbb{Z}/p^n$ -algebras that  $A/p$  is perfect and  $\mathcal{C}_1 = \text{perfect rings over } \mathbb{Z}/p$ .

moreover, taking limit, this is even equivalent to the category of flat  $p$ -adically complete  $\mathbb{Z}_p$ -algebras that  $A/p$  is perfect. Which is just the construction of Witt vectors.  $\lrcorner$

*Proof:* It suffices to show that  $\mathcal{C}_n \subset \mathcal{C}_{\mathbb{Z}/p^n}$ : By (8.1.4.2) and flat base change (8.1.2.7),  $L_{A/(\mathbb{Z}/p^n)} \otimes_{\mathbb{Z}/p^n} \mathbb{Z}/p \cong L_{(A/p)/(\mathbb{Z}/p)} = 0$ , so  $L_{A/(\mathbb{Z}/p^n)} \otimes_{\mathbb{Z}/p^n} \mathbb{Z}/p^k \cong 0$  by induction, and so  $L_{A/(\mathbb{Z}/p^n)} \cong 0$ .

For the last assertion, it is flat because it is torsion-free, which is because if  $p(x_n) = 0$ , then by  $0 \rightarrow p^n \mathbb{Z}/p^{n+1} \rightarrow \mathbb{Z}/p^{n+1} \xrightarrow{p} \mathbb{Z}/p^n \rightarrow 0$  and the flatness of  $A_{n+1}$ ,  $x_{n+1} \in p^n A_{n+1}$ , thus  $x_n = 0$ , and  $x = 0$ .  $\square$

**Prop. (8.1.4.4) [Adjointness of Witt Vectors].** Using a more careful analysis of cotangent complex (embedded deformation), we can show that if  $A \rightarrow B \in \mathcal{C}_A$  and there is a infinitesimal deformation  $C \rightarrow C'$  of  $A$ -algebra, then a map  $B \rightarrow C'$  can be lifted to an  $A$ -algebra map  $A \rightarrow C$ .

In particular, taking inverse image, we get that

$$\text{Hom}_{\mathbb{F}_p}(A, B/p) \cong \text{Hom}_{\mathbb{C}\text{Ring}_{\mathbb{Z}_p}}(W(A), B).$$

which is the usual adjointness of the Witt vector construction.  $\lrcorner$

## 5 Algebra Extension

Cf.[Perfectoid Geometry Appendix B].

**Def. (8.1.5.1) [Algebra Extensions].** Let  $A \rightarrow B$  be a ring map and  $M$  be a  $B$ -module, then an  **$A$ -algebra extension** of  $B$  by  $M$  is a short exact sequence of  $A$ -modules  $0 \rightarrow M \rightarrow B' \rightarrow B \rightarrow 0$  that  $B'$  is an  $A$ -algebra with  $M$  being an ideal of it.

The set of such extensions are denoted by  $\text{Exal}_A(B, M)$ . ┘

**Prop. (8.1.5.2).**  $\text{Exal}_A(B, M)$  is a group under Baer sum, where the sum of two extension is the extension given by pushout, i.e.  $(B_1 \oplus B_2)/\{(m, -m) | m \in M\}$ . Moreover, it is a  $B$ -module, where the multiplication is the pushout along multiplication of  $b$  on  $M$ . ┘

**Prop. (8.1.5.3).** There is a trivial extension given by  $D_B(M) = B \oplus M$  (5.4.3.5), and the automorphism of  $D_B(M)$  is isomorphic to  $\text{Der}_A(B, M)$  via  $d \mapsto \text{id} \oplus d$ . ┘

*Proof:* Cf.[Foundations of Perfectoid Spaces Masullo P118]. □

**Prop. (8.1.5.4).** Let  $A \rightarrow B \rightarrow C$  be ring maps, then for any  $C$ -module  $M$ , there is an exact sequence

$$0 \rightarrow \text{Der}_B(C, M) \rightarrow \text{Der}_A(C, M) \rightarrow \text{Der}_A(B, M) \xrightarrow{\partial} \text{Exal}_B(C, M) \rightarrow \text{Exal}_A(C, M) \rightarrow \text{Exal}_A(B, M)$$

functorial in  $M$ . Where  $\partial$  is given by (8.1.5.3). ┘

*Proof:* Cf.[Foundations of Perfectoid Spaces Masullo P119]. □

**Prop. (8.1.5.5).** Let  $A \rightarrow B$  be a ring map or a map of sheaves of rings, and let  $M$  be a  $B$ -module, then there is an isomorphism of  $B$ -modules that is natural in  $M$ :

$$\text{Exal}_A(B, M) = \text{Ext}_B^1(NL_{B/A}, M).$$

┘

*Proof:* Cf.[Foundations of Perfectoid Spaces, P127]. □

### Infinitesimal Deformation

**Def. (8.1.5.6).** An **infinitesimal deformation** of a f.g.  $k$ -algebra is defined as a algebra  $A'$  flat over  $D = k[t]/(t^2)$  that  $A' \otimes_D k = A$ .

A f.g.  $k$ -algebra is called **rigid** if it has no infinitesimal deformations. ┘

**Prop. (8.1.5.7).** Let  $A$  be a f.g.  $k$ -algebra, write  $A$  as a quotient of a polynomial ring over  $k$  with kernel  $J$ , then there is an exact sequence  $J/J^2 \rightarrow \Omega_{P/k} \otimes_P A \rightarrow \Omega_{A/k} \rightarrow 0$  by (5.4.3.7), then we apply  $\text{Hom}_A(-, A)$  and let  $T^1(A) = \text{Coker}(\text{Hom}_A(\Omega_{P/k} \otimes_P A, A) \rightarrow \text{Hom}_A(J/J^2, A))$ . Then  $T^1(A)$  parametrize infinitesimal deformations of  $A$ . ┘

## 8.2 Spectral Algebraic Geometry(Lurie)

Main references are Lurie's work, [Derived Algebraic Geometry, Thesis, Lurie].

**Notation(8.2.0.1).**

- Use notations from [Derived Commutative Algebras](#).

┘

**Prop.(8.2.0.2)** [ $\mathcal{D}_{\mathcal{QCoh}}(\mathrm{Spec} R)$ ]. For  $R \in \mathcal{CRing}$ , there exists a unique sheaf of  $\infty$ -categories  $\mathcal{D}_{\mathcal{QCoh}}$  on  $\mathrm{Spec} R$  s.t. for any affine open subset  $U$ ,

$$\mathcal{D}_{\mathcal{QCoh}}(U) = \mathcal{D}(\mathcal{O}(U)).$$

┘

*Proof:* Cf.[Clausen, deRham Cohomology]L8P3. □

**Cor.(8.2.0.3).** For  $R \in \mathcal{CRing}$  and any  $M \in \mathrm{Mod}(R)$ , there exists an associated sheaf of derived modules  $M^\sharp$  on  $\mathrm{Spec} R$  s.t. for any affine open  $U \subset \mathrm{Spec} R$ ,  $M^\sharp(U) = M \otimes_R \mathcal{O}_{\mathrm{Spec} R}(U)$ . (I think this sheaf is just the derived shification of the associated sheaf  $\widetilde{M}$  on  $\mathrm{Spec} R$  defined in(7.5.1.1).) ┘

*Proof:* Cf.[Clausen, deRham Cohomology]L8P3. □

**Cor.(8.2.0.4).** Let  $X \in \mathcal{Aff}$  and  $\mathcal{F} \in \mathcal{QCoh}(X)$ , then  $H^q(X; \mathcal{F}) = 0$  for  $q > 0$ . ┘

*Proof:* This is because the sheaf cohomology is the global section of the derived shification, and the derived shification of  $\mathcal{F}(X)^\sharp$  equals  $\widetilde{\mathcal{F}(X)}$  on any affine opens by(8.2.0.3). Thus  $H^*(X; \mathcal{F}) = \mathcal{F}(X) \in D(\mathcal{O}_X(X))$ . □

**Cor.(8.2.0.5).** For  $R \in \mathcal{CRing}$ , there is a sheaf of Abelian categories on  $X = \mathrm{Spec} R$  with assigns  $\mathrm{Mod}_{\mathcal{O}_X(U)}$  for any affine open subset  $U \subset X$ . ┘

*Proof:* It suffices to show the discrete elements in  $\mathcal{D}_{\mathcal{QCoh}}(\mathrm{Spec} R)$  form a sheaf, i.e. if  $M \in \mathcal{D}_{\mathcal{O}_X(U)}$  is locally discrete, then it is discrete. And this is because the ring extensions are flat. □

**Def.(8.2.0.6)** [ $\mathcal{D}(X)$ ]. Let  $X \in \mathcal{Sch}$ , then  $\mathcal{D}(X) = \mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Sh}(X; \mathcal{D}(Z)))$  is an  $\infty$ -category, and the assignment  $\mathcal{D}_X : U \mapsto \mathcal{D}(U)$  is a sheaf of  $\infty$ -categories on  $X$ . ┘

*Proof:* ? □

**Def.(8.2.0.7)** [ $\mathcal{D}_{\mathcal{QCoh}}(\mathrm{Spec} R)$ ]. Let  $X \in \mathcal{Sch}$ , then there is a sub- $\infty$ -category

$$\mathcal{D}_{\mathcal{QCoh}}(X) \subset \mathcal{D}(X)$$

consisting of elements  $\mathcal{M}$  s.t. for any inclusion of affine opens  $U \subset V$ , there is a natural isomorphism

$$\mathcal{M}(V) \otimes_{\mathcal{O}_X(V)} \mathcal{O}_X(U) \cong \mathcal{M}(U).$$

And the assignment  $\mathcal{D}_{\mathcal{QCoh}, X} : U \mapsto \mathcal{D}_{\mathcal{QCoh}}(X)$  is a sheaf of  $\infty$ -categories on  $X$ .

Moreover, when  $X$  is affine, the global section functor defines an equivalence  $\mathcal{D}(X) \cong D(\mathcal{O}_X(X))$ .

┘

*Proof:* ?

This follows from (8.2.0.6) and the fact that the  $\mathcal{Q}_{\text{co}}$  requirement is local.

The last assertion is trivial.  $\square$

**Prop. (8.2.0.8) [Compact Objects in  $\mathcal{D}(R)$ ].** For  $R \in \mathcal{C}\text{Ring}$  and  $M \in \mathcal{D}(R)$ , the following are equivalent:

- $M$  is a compact object.
- $M$  lies in the thick (stable- $\infty$ )subcategory generated by  $R$ .
- $M$  is dualizable w.r.t. the tensor product.
- $M$  can be represented by a bounded chain of finite projective  $R$ -modules.

In particular, there is a full sub- $\infty$ -category of  $\mathcal{D}_{\mathcal{Q}\text{coh}}(R)$  consisting of perfect objects, denoted by  $\text{Perf}(R)$ .  $\lrcorner$

*Proof:* Cf. [Clausen, deRham Cohomology]L8P8.  $\square$

**Cor. (8.2.0.9).** For  $R \in \mathcal{C}\text{Ring}$ ,  $X = \text{Spec } R$ , the assignment  $\text{Perf} : U \mapsto \text{Perf}(\mathcal{O}_X(U))$  on affine opens is a subsheaf of  $\mathcal{D}_{\mathcal{Q}\text{coh}, X}$  on affine opens.  $\lrcorner$

*Proof:* Cf. [Clausen, deRham Cohomology]L8P9.  $\square$

**Def. (8.2.0.10) [Perfect Objects].** For  $X \in \text{Sch}$ , a **perfect quasi-coherent sheaf** on  $X$  is an object  $\mathcal{F} \in \mathcal{D}_{\mathcal{Q}\text{coh}}(X)$  s.t. for any affine open  $U \subset X$ ,  $\mathcal{F}|_U \in \mathcal{D}(U)$  is a compact object (i.e.  $\text{Hom}(\mathcal{M}, -) : \mathcal{D}_{\mathcal{Q}\text{coh}}(X) \rightarrow \text{sSet}$  commutes with filtered colimits).  $\lrcorner$

**Prop. (8.2.0.11) [Pull and Push].** For  $f : X \rightarrow Y \in \text{Sch}$ , there is a pushforward functor  $f_* : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$  s.t.

- $f_* \mathcal{D}_{\mathcal{Q}\text{coh}}(X) \subset \mathcal{D}_{\mathcal{Q}\text{coh}}(Y)$ .
- $f_* : \mathcal{D}_{\mathcal{Q}\text{coh}}(X) \rightarrow \mathcal{D}_{\mathcal{Q}\text{coh}}(Y)$  preserves colimits and limits.

And it has a left adjoint  $f^* : \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$  s.t.

- $f^* \mathcal{D}_{\mathcal{Q}\text{coh}}(Y) \subset \mathcal{D}_{\mathcal{Q}\text{coh}}(X)$ .
- If  $X, Y \in \text{Aff}$ , then  $f^*$  is just the derived base change functor.
- $f_*$  commutes with base change ?.
- $f_*$  satisfies projection formula ?.

$\lrcorner$

*Proof:* Cf. [Clausen, deRham Cohomology]L8P7.  $\square$

**Prop. (8.2.0.12).** If  $f$  is smooth and proper,  $f_* : \mathcal{D}_{\mathcal{Q}\text{coh}}(X) \rightarrow \mathcal{D}_{\mathcal{Q}\text{coh}}(Y)$  preserves perfect objects.  $\lrcorner$

*Proof:* Cf. [Clausen, deRham Cohomology]L8P8.  $\square$

**Prop. (8.2.0.13) [Grothendieck Duality].** If  $f : X \rightarrow Y \in \text{Sch}$  is proper and smooth, then there is a right adjoint  $f^!$  to  $f_* : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$ , and it satisfies

- There is a natural isomorphism of functors:  $f^* \otimes f^!(\mathcal{O}_Y) \cong f^!$ .
- If  $f$  has relative dimension  $d$ , then there is a natural isomorphism  $f^!(\mathcal{O}_Y) \cong \Omega_{X/Y}^d[d]$ .

$\lrcorner$

*Proof:* ?  $\square$

### 8.3 Elliptic Cohomology Theory(Lurie)

Main references are [Elliptic Cohomology,1, 2, Lurie].





## 9 | Weil Cohomologies

### 9.1 Intersection Theory

References are [Sta]Chap43, 44 and [Ful98].

**Notation(9.1.0.1).**

- Let  $k \in \mathbf{Field}$ .
- Let  $S$  be a universally catenary(hence locally Noetherian) scheme(7.4.1.36) endowed with a dimension function  $\delta$ (4.12.2.34). There are some examples of  $(S, \delta)$  in(9.1.0.1):
  - $S = \mathrm{Spec} k$  and  $\delta(|S|) = 0$ .
  - $A$  is a Noetherian domain of dimension 1,  $\delta(\mathfrak{p}) = 0$  if  $\mathfrak{p}$  is a maximal ideal and  $\delta(\eta) = 1$  for the generic point  $\eta$ .
  - $S$  is a C.M. scheme and let  $\delta(s) = -\dim(\mathcal{O}_{S,s})$ .

These follow from(7.4.1.38).

┘

**Prop.(9.1.0.2).**  $S$  is Jacobian and  $\delta(s) = 0$  for any closed point  $s \in S$ . If  $Z \subset S$  is an integral closed subscheme with generic point  $\xi$ , then

$$\delta(\xi) = \dim(Z) = \dim(\mathcal{O}_{Z,z})$$

where  $z \in Z$  is a closed point.

┘

*Proof:* Cf.[Sta]02QO.

□

**Prop.(9.1.0.3)[ $\delta$ -Dimension].** For  $f : X \rightarrow S$  locally of f.t., the function

$$\delta(x) = \delta(f(x)) + \mathrm{tr.deg}_{k(f(x))} k(x)$$

is a dimension function on  $X$ . In particular, this equation is satisfied for any morphisms between schemes of f.t. over  $S$ .

For a closed subscheme  $Z$  of  $X$ , define  $\dim_\delta(Z) = \sup \dim_\delta(\eta)$  where  $\eta$  are generic pts of irreducible components of  $Z$ .

┘

*Proof:* Cf.[Sta]02JW.?

□

## 1 Chow Homologies

### Cycles

**Def. (9.1.1.1) [Cycles].** An **algebraic cycle** on a scheme  $X \in \text{Sch}^{\text{loc.ft}}/S$  is a formal sum of integral closed subschemes of  $X$  with integer coefficients that is locally finite. For  $k \in \mathbb{N}$ , a  **$k$ -cycle** is a cycle that is a sum of integral closed subschemes of dimension  $k$  that is locally finite. The group of  $k$ -cycles over  $X$  is denoted by  $Z_k(X)$ .

If  $\dim_\delta(X) = d$ , then we also denote  $Z^k(X) = Z_{d-k}(X)$ .  $\lrcorner$

**Prop. (9.1.1.2).** Let  $X$  be a scheme locally of f.t. over  $S$ , and  $X = X_1 \cup X_2$  is a decomposition as closed subschemes, then there are exact sequences

$$Z_k(X_1 \cap X_2) \rightarrow Z_k(X_1) \oplus Z_k(X_2) \rightarrow Z_k(X) \rightarrow 0$$

$\lrcorner$

**Prop. (9.1.1.3) [Cycle associated to a Closed Subscheme].** For a closed subscheme  $Z$  of a scheme  $X \in \text{Sch}^{\text{loc.ft}}/S$ , if  $\dim_\delta(Z) \leq k$  and  $\eta \in Z$  has dimension  $k$ , then  $\eta$  is a generic pt of an irreducible component  $Z'$  of  $Z$ , and  $m_{Z,Z'} = \text{length}_{\mathcal{O}_{X,\eta}} \mathcal{O}_{Z,\eta}$  is finite.

So we may define the  **$k$ -cycle associated to  $Z$**  as:  $[Z]_k = \sum_{Z' \subset Z} m_{Z,Z'} [Z']$ , where the sum is over all integral components of  $Z$  of  $\delta$ -dimension  $k$ .  $\lrcorner$

*Proof:*  $m_{Z,Z'}$  is finite because  $\text{length}_{\mathcal{O}_{Z,\eta}} \mathcal{O}_{Z,\eta} = \text{length}_{\mathcal{O}_{X,\eta}} \mathcal{O}_{Z,\eta} < \infty$  because it is Noetherian and have 0 dimension (5.1.3.4). The sum is locally finite by (7.4.1.23).  $\square$

**Prop. (9.1.1.4) [Cycle associated to a Coherent Sheaf].** For  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{F} \in \text{Coh}(X)$ , if  $\dim_\delta \text{Supp}(\mathcal{F}) \leq k$  and  $\eta \in \text{Supp}(\mathcal{F})$  has dimension  $k$ , then  $\eta$  is a generic pt of an irreducible component  $Z'$  of  $Z$ , and  $m_{Z,\mathcal{F}} = \text{length}_{\mathcal{O}_{X,\eta}} \mathcal{F}_\eta < \infty$ .

So we may define the  **$k$ -cycle associated to  $\mathcal{F}$**  as:  $[\mathcal{F}]_k = \sum_{Z \subset X} m_{Z,\mathcal{F}} [Z]$ , where the sum is over all integral components of  $\text{Supp } \mathcal{F}$  of  $\delta$ -dimension  $k$ .  $\lrcorner$

*Proof:*  $\text{length}_{\mathcal{O}_{X,\eta}} \mathcal{F}_\eta < \infty$  by (5.2.5.7).  $\square$

**Prop. (9.1.1.5).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $Z \subset X$  a closed subscheme with  $\dim_\delta(Z) \leq k$ , then  $[Z]_k = [\mathcal{O}_Z]_k$ .  $\lrcorner$

**Prop. (9.1.1.6).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ , the cycle map from  $\text{Coh}^{\leq k}(X)$  to  $Z_k(X)$  is additive.  $\lrcorner$

### Pushforward and Pullback

**Lemma (9.1.1.7) [Degree of Maps].** Let  $f : X \rightarrow Y$  be a map between schemes integral and locally of f.t. over  $S$ , if  $\dim_\delta X = \dim_\delta Y$ , then either  $f(X)$  not dominant or the function field extension is finite. If  $f$  is dominant, the the degree of  $f$  (7.4.5.6) is a finite number.  $\lrcorner$

*Proof:* Because  $X$  is irreducible, so does  $f(X)$  and  $\overline{f(X)}$ . If  $f(X)$  is dominant, then  $f$  maps the generic point of  $X$  to that of  $Y$ . Now  $\deg_{K(Y)}(K(X)) = 0$  and  $K(X)/K(Y)$  is f.g., thus it is a finite extension.  $\square$

**Lemma (9.1.1.8).** Let  $f : X \rightarrow Y$  be a qc map between schemes integral and locally of f.t. over  $S$ , and  $\{Z_i\}$  is a locally finite collection of closed subschemes of  $X$ , then  $\{\overline{f(Z_i)}\}$  is also a locally finite collection of closed subschemes of  $Y$ .  $\lrcorner$

*Proof:* This is a simple topological proof and omitted.  $\square$

**Def. (9.1.1.9) [Proper Pushforward].** Let  $f : X \rightarrow Y$  be a proper morphism in  $\text{Sch}^{\text{loc.ft}}/S$ , we define a map

$$p_* : Z_k(X) \rightarrow \text{CH}_k(X)$$

as follows: If  $Z \subset X$  is an integral closed subscheme of  $X$  that  $\dim_\delta(Z) = k$ , then we define

$$f_*[Z] = \begin{cases} 0 & \dim_\delta(f(Z)) \leq k \\ \deg(Z/f(Z))[f(Z)] & \dim_\delta(f(Z)) = k \end{cases} \quad (9.1.1.7)$$

where we regard  $f(Z)$  as an integral closed subscheme of  $Y$  using its scheme-theoretical image. In general

$$f_*(\sum n_Z[Z]) = \sum n_Z f_*([Z]).$$

The sum is locally finite by (9.1.1.8).

It can be easily verified that  $f_* \circ g_* = (f \circ g)_*$ .  $\square$

**Prop. (9.1.1.10) [Pushforward of Coherent Sheaves].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$ , then

- If  $Z \subset X$  is an integral closed subscheme of  $X$  that  $\dim_\delta(Z) \leq k$ , then

$$f_*[Z]_k = [\mathcal{O}_Z]_k.$$

- If  $\mathcal{F} \in \text{Coh}^{\leq k}(X)$ , then

$$f_*[\mathcal{F}]_k = [f_*\mathcal{F}]_k.$$

$\square$

*Proof:* 1 follows from 2 and (9.1.1.5). To show 2, by restricting to  $\text{Supp}(\mathcal{F})$  and taking the scheme-theoretic image, it suffices to show for both closed immersions and proper dominant maps. The closed immersions case are easy. For the proper dominant case, it suffices to show  $f_*[\mathcal{F}]_k$  and  $[f_*\mathcal{F}]_k$  have the same coefficients in each integral subscheme  $Z \subset X$  of dimension  $k$ . By looking at the inverse image of the generic point of  $Z$ , we may assume that  $f$  is finite by (7.4.5.4). Thus we can assume  $f$  is finite. Then it reduces to the affine case, which follows from (5.1.2.7).  $\square$

**Lemma (9.1.1.11).** Let  $f : X \rightarrow Y$  be a flat morphism of relative dimension  $r$  in  $\text{Sch}^{\text{loc.ft}}/S$ , then for any closed subscheme  $Z \subset Y$ ,  $\dim_\delta(f^{-1}(Z)) = \dim_\delta(Z) + r$  if  $f^{-1}(Z) \neq \emptyset$ . If  $Z$  is irreducible and  $Z' \subset f^{-1}(Z)$  is an irreducible component, then  $Z'$  dominates  $Z$  and  $\dim_\delta(Z') = \dim_\delta(Z) + r$ .  $\square$

*Proof:* By passing to the integral components, we may assume  $Z = Y$  is integral and  $X \rightarrow Y$  is surjective, then notice  $f$  is open, and use (9.1.0.3) and (7.6.3.7).  $\square$

**Def. (9.1.1.12) [Flat Pullbacks].** Let  $f : X \rightarrow Y$  be a flat morphism of relative dimension  $r$  in  $\text{Sch}^{\text{loc.ft}}/S$ , for  $Z \subset X$  an integral closed subscheme that  $\dim_\delta(Z) \leq k$ , define

$$f^* : Z_k(Y) \rightarrow Z_{k+r}(X) : f^*([Z]) = [f^{-1}(Z)]_{k+r}$$

which is definable by (9.1.1.11). In general, define  $f^*(\sum n_Z[Z]) = \sum n_Z f^*([Z])$ , which is locally finite.  $\square$

**Prop. (9.1.1.13).** If  $f : X \rightarrow Y, g : Y \rightarrow Z \in \text{Sch}^{\text{loc.ft}}/S$  is flat of relative dimensions  $r$  and  $s$ , then  $f^* \circ g^* = (g \circ f)^*$ .  $\square$

*Proof:* Firstly  $g \circ f$  is flat of relative dimension  $r + s$  by (7.6.3.18). And the assertion follows from (5.1.2.9).  $\square$

**Prop. (9.1.1.14).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $i : Y \rightarrow X$  a reduced closed subscheme of  $X$ ,  $j : U = X \setminus Z \rightarrow X$ , then for any  $k \in \mathbb{Z}$ , there is an exact sequence

$$Z_k(Y) \xrightarrow{i_*} Z_k(X) \xrightarrow{j^*} Z_k(U) \rightarrow 0.$$

$\lrcorner$

**Prop. (9.1.1.15) [Pullback of Coherent Sheaves].** If  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  is flat of relative dimensions  $r$ ,  $\mathcal{F} \in \text{Coh}^{\leq k}(\mathcal{O}_Y)$ , then  $f^*\mathcal{F} \in \text{Coh}^{\leq k+r}(\mathcal{O}_X)$ , and

$$f^*[\mathcal{F}] = [f^*\mathcal{F}]_{k+r}.$$

In particular, for a closed subscheme  $Z \subset X$  with  $\dim_\delta(Z) \leq k$ ,  $f^*[Z]_k = [f^{-1}Z]_{k+r}$ , by (5.1.2.8).

$\lrcorner$

*Proof:* This follows from (5.1.2.9).  $\square$

**Prop. (9.1.1.16) [Push and Pull].**

- Let

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a diagram in  $\text{Sch}^{\text{loc.ft}}/S$  where  $f$  is proper and  $g$  is flat of relative dimension  $r$ , then

$$g^*f_* = (g')^*f'_* : Z_k(X) \rightarrow Z_{k+r}(Y').$$

- Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  be a finite locally free morphism of degree  $d$ , then it is both proper and flat of relative dimension 0, and  $f_*f^* = [d] : Z_k(Y) \rightarrow Z_k(Y)$ .

$\lrcorner$

*Proof:* 1: It suffices to prove for a closed subscheme  $W \subset X$  of  $\delta$ -dimension  $k$ . Then  $[W] = [\mathcal{O}_W]$  (9.1.1.5). Then by (9.1.1.10) and (9.1.1.15), the assertion follows from flat base change.

2: Similarly this follows from the fact  $f_*f^*\mathcal{O}_Z$  is a finite locally free  $\mathcal{O}_Z$ -sheaf of rank  $d$ .  $\square$

### Rational Equivalences and Chow-Groups

**Def. (9.1.1.17) [Prime Weil Divisors].** Let  $X \in \text{Sch}^{\text{loc.ft,int}}/S$ , then an integral closed scheme  $W \subset X$  of  $\delta$ -codimension 1 is called a **prime Weil divisor**.  $\lrcorner$

**Prop. (9.1.1.18) [Principal Weil Divisor].** Let  $X \in \text{Sch}^{\text{loc.ft,int}}/S$ ,  $f \in \mathcal{K}_X$ , for any prime Weil divisor  $Z$  with generic pt  $\eta$ , we can define  $\text{ord}_Z(f) = \text{ord}_{\mathcal{O}_{X,\eta}}(f)$  (5.1.2.10). It is multiplicative, and the closed integral subschemes  $Z$  that  $\text{ord}_Z(f) \neq 0$  is locally finite.

So, we can define the **principle Weil divisor**  $\text{div}(f) = \sum_Z \text{ord}_Z(f)[Z]$ .  $\lrcorner$

*Proof:* There is an open subset  $U$  that  $f \in \Gamma(U, \mathcal{O}_X^*)$ , so all  $Z$  are irreducible components of  $X - U$ , which is locally finite because  $X$  is locally Noetherian and (7.4.1.23).  $\square$

**Def. (9.1.1.19) [Principal Divisors].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $W \subset X$  a  $(k+1)$ -dimensional integral closed subscheme and  $f \in K(W)^\times$ , then the **principle divisor associated to  $f$**  is defined to be

$$\text{div}(f) = \sum \text{ord}_Z(f)[Z] \in Z^1(X),$$

where  $Z$  runs through codimension-1 integral closed subschemes of  $W$ , as defined in (9.1.1.18). This is truly a  $k$ -cycle.  $\lrcorner$

**Def. (9.1.1.20) [Rational Equivalence].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  has  $\delta$ -dimension  $k+1$ . Given any locally finite collection of integrally closed subschemes  $j_i : W_i \subset X$  of  $\delta$ -dimension  $k+1$  and  $f_i \in K(W_i)^\times$ , we can consider the  $k$ -cycle  $\sum (j_i)_*(\text{div}(f_i)) \in Z_k(X)$  (9.1.1.19). This is a  $k$ -cycle because  $\coprod W_i \rightarrow X$  is proper. Two  $k$ -cycles are called **rational equivalent** if they differ by a  $k$ -cycle of the this form.

Define the **Chow-group of  $k$ -cycles**  $\text{CH}_k(X)$  to be  $Z_k(X)$  modulo the rational equivalence relation.  $\text{CH}_1(X)$  is also denoted by  $\text{Cl}(X)$ .  $\lrcorner$

**Lemma (9.1.1.21) [Push and Pull of Principle Divisors].**

- If  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft,int}}/S$  is flat of relative dimension  $r$ , and  $g \in K(Y)^*$ , then  $f^* \text{div}(f) = \text{div}(f)$ .
- If  $\varphi : X \rightarrow Y \in \text{Sch}^{\text{loc.ft,int},\delta=d}/S$  is a dominant proper morphism,  $f \in K(X)^*$ , and  $g = \text{Nm}_{K(X)/K(Y)}(f) \in K(Y)^*$ , then  $f_* \text{div}(f) = \text{div}(g)$ .

$\lrcorner$

*Proof:* 1: Cf. [Sta]0EPH?.

2: Cf. [Sta]02RT?.

$\square$

**Lemma (9.1.1.22).** If  $X \in \text{Sch}^{\text{loc.ft,int},\delta=n}/S$   $U \subset X$  be an open subscheme and  $f \in \Gamma(U, \mathcal{O}_X)^* \subset K(X)^*$ , let  $Y$  be the graph of  $f$  in  $X \times_S \mathbb{P}_S^1$ , then

- the projection  $\text{pr}_1 : Y \rightarrow X$  is an isomorphism  $\text{pr}_1^{-1}(U) \rightarrow U$ , thus  $\dim_\delta(Y) = n$ .
- the closed subschemes  $Y_0 = \text{pr}_2^{-1}(\{0\})$  and  $Y_\infty = \text{pr}_2^{-1}(\{\infty\})$  of  $Y$  are effective Cartier divisors. In particular, they have  $\delta$ -dimension  $n-1$  by (9.1.4.2).
- $\text{div}_Y(f) = [Y_0] - [Y_\infty]$ .
- $\text{div}_X(f) = \text{pr}_{1*} \text{div}_Y(f) = [Y_0] - [Y_\infty]$ .

$\lrcorner$

*Proof:* 1 is clear.

2 follows from (7.8.1.8) as  $\text{pr}_2 : Y \rightarrow \mathbb{P}^1$  is dominant.

3 is clear.

4 follows from item1 and (9.1.1.21).  $\square$

**Prop. (9.1.1.23) [Rational Equivalence via Rational Functions].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ , then  $\alpha \in Z_k(X)$  is rationally trivial iff

$$\alpha = \sum ([ (W_i)_0 ]_k - [ (W_i)_1 ]_k) = j_0^*(\sum [W_i]) - j_\infty^*(\sum [W_i])$$

where  $\{W_i\}$  is a locally finite family of integral closed subschemes of  $X \times_S \mathbb{P}_S^1$  of  $\delta$ -dimension  $k$ . ( $j_0^*, j_\infty^*$  are the Gysin maps, which will be defined in (9.1.3.2))  $\lrcorner$

*Proof:* Firstly such a  $\sum([(W_i)_0]_k - [(W_i)_\infty]_k)$  is locally finite, and each  $[(W_i)_0]_k - [(W_i)_\infty]_k$  is rationally trivial: Similar as in (9.1.1.22),  $[(W_i)_0] - [(W_i)_\infty]$  is rationally trivial on  $W_i$ , and then it pushforward via  $\text{pr}_1$  is also rationally trivial, by (9.1.1.21).

Conversely, if  $\alpha = \sum (V_i \rightarrow X)_* \text{div}(f_i)$ , where  $\{V_i\}$  is a locally finite family of integral closed subschemes of  $X$  of  $\delta$ -dimension  $k+1$  and  $f_i \in K(V_i)^*$ . Let  $W_i \subset V_i \times \mathbb{P}_k^1 \subset X_i \times \mathbb{P}_k^1$  be the the graph of  $f$ , then  $(V_i \rightarrow X)_* \text{div}(f_i)$  equals  $[(W_i)_0]_k - [(W_i)_1]_k$  by (9.1.1.22) again. (We are secretly using the fact Gysin map commutes with pushforwards (9.1.5.9), but in this case it is trivial trough).  $\square$

**Prop. (9.1.1.24) [Push and Pull for Chow-Groups].**

- If  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  is flat of relative dimension  $r$ , then  $f^*$  induces a map  $\text{CH}_k(Y) \rightarrow \text{CH}_{k+r}(X)$  for each  $k \in \mathbb{N}$ .
- If  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  is proper, then  $f_*$  induces a map  $\text{CH}_k(X) \rightarrow \text{CH}_k(Y)$  for each  $k \in \mathbb{N}$ .

┘

*Proof:* Cf. [Sta]02S2, 02S1.  $\square$

**Prop. (9.1.1.25) [Restriction of Divisors].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $i : Y \rightarrow X$  a reduced closed subscheme of  $X$ ,  $j : U = X \setminus Z \rightarrow X$ , then for any  $k \in \mathbb{Z}$ , there is an exact sequence

$$\text{CH}_k(Y) \xrightarrow{i_*} \text{CH}_k(X) \xrightarrow{j^*} \text{CH}_k(U) \rightarrow 0.$$

┘

*Proof:* Cf. [Sta]02RX?  $\square$

**Prop. (9.1.1.26) [Excision].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $X_1, X_2$  be closed subschemes of  $X$  s.t.  $X_1 \cup X_2 = X$  as sets, then for any  $k \in \mathbb{Z}$ , there is an exact sequence

$$\text{CH}_k(X_1 \cap X_2) \rightarrow \text{CH}_k(X_1) \oplus \text{CH}_k(X_2) \rightarrow \text{CH}_k(X) \rightarrow 0.$$

┘

*Proof:* Cf. [Sta]0F94.  $\square$

**Def. (9.1.1.27) [Degree of 0-Cycles].** Let  $X$  be a proper scheme over a field  $k$ , the **degree of a zero cycle** is given by the proper pushforward

$$p_* : \text{CH}_0(X) \rightarrow \text{CH}_0(\text{Spec } k) \cong \mathbb{Z}.$$

Equivalently, if  $\alpha = \sum n_i [Z_i] \in \text{CH}_0(X)$ ,  $\deg(\alpha) = \sum n_i \deg(Z_i)$ .  $\deg$  is also denoted by  $\int_X$ .  $\square$

**Def. (9.1.1.28) [Stratification by Dimension].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ , then the following are equivalent:

- There exists a decomposition  $X = \coprod_n X_n$  where  $X_n$  is pure of  $\delta$ -dimension  $n$ .
- For any  $x \in X$ , there exist s nbhd  $x \in U$  s.t.  $U$  is pure of  $\delta$ -dimension  $n$ .
- For an  $x \in X$ , the irreducible components of  $X$  containing  $x$  are all of the same  $\delta$ -dimension  $n_x$ .

These conditions are satisfied if  $X$  is normal or Cohen-Macaulay.  $\square$

*Proof:*  $1 \rightarrow 2 \rightarrow 3$  is trivial.  $3 \rightarrow 1$  follows from the fact  $x \mapsto n_x$  is continuous.

If  $X$  is normal, it is a disjoint union of integral schemes by (7.4.2.4). For  $X$  Cohen-Macaulay, Cf. [Sta]0FE3. ? □

**Def. (9.1.1.29) [Cohomological Chow Groups].** If  $X$  satisfies (9.1.1.28), we define

$$Z^p(X) = \prod_n Z_{n-p}(X_n), \quad Z^*(X) = \bigoplus_p Z^p(X)$$

and the **Chow group of codimension  $p$  cycle**

$$\mathrm{CH}^p(X) = \prod_n \mathrm{CH}_{n-p}(X_n), \quad \mathrm{CH}^*(X) = \bigoplus_p \mathrm{CH}^p(X).$$

┘

**Def. (9.1.1.30) [Fundamental Class].** If  $X$  satisfies (9.1.1.28), define

$$[X] = \prod [X_n]_n \in \mathrm{CH}^0(X)$$

to be the **fundamental class of  $X$** . ┘

**Prop. (9.1.1.31) [Exterior Products of Cycle Classes].** For  $X, Y \in \mathrm{Sch}^{\mathrm{ft}}/k$  and  $k, l \in \mathbb{N}$ , we can define the exterior product

$$Z_k(X) \times Z_l(Y) \rightarrow Z_{k+l}(X \times Y) : [Z] \times [W] \rightarrow [Z \times W].$$

Then we have

- If  $\alpha \sim_{\mathrm{rat}} 0$  or  $\beta \sim_{\mathrm{rat}} 0$ , then  $\alpha \times \beta \sim_{\mathrm{rat}} 0$ . In particular, we get a map  $\mathrm{CH}_k(X) \times \mathrm{CH}_l(Y) \rightarrow \mathrm{CH}_{k+l}(X \times Y)$ , which is called the **exterior product of cycle classes**.
- If  $f : X' \rightarrow X \in \mathrm{Sch}^{\mathrm{ft}}/k, g : Y' \rightarrow Y \in \mathrm{Sch}^{\mathrm{ft}}/k$  are proper, then

$$(f \times g)_*(\alpha \times \beta) = f_*(\alpha) \times g_*(\beta).$$

- If  $f : X' \rightarrow X \in \mathrm{Sch}^{\mathrm{ft}}/k, g : Y' \rightarrow Y \in \mathrm{Sch}^{\mathrm{ft}}/k$  are flat, then

$$(f \times g)^*(\alpha \times \beta) = f^*(\alpha) \times g^*(\beta).$$

┘

*Proof:* Cf. [Ful98]P25. □

## 2 Chow-Groups and $K$ -Groups

Cf. [Sta]42.22, 42.56 and 42.68.

### 3 Gysin-Maps

**Def. (9.1.3.1) [Gysin-Maps for Virtual Divisors].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $\mathcal{L} \in \text{Pic}(X)$ ,  $s \in H^0(X, \mathcal{L})$ , denote  $D = Z(s)$ , and  $i : D \rightarrow X$  the closed immersion. Then for any  $k \in \mathbb{Z}$ , there is a **Gysin homomorphism**

$$i^* : Z_{k+1}(X) \rightarrow \text{CH}_k(D)$$

that for an integral closed subscheme  $Z \subset X$  with  $\dim_\delta(Z) = k + 1$ ,

$$i^*([Z]) = \begin{cases} [D \cap Z]_k & , Z \not\subset D \\ i'_*(c_1(\mathcal{L}|_Z) \cap [Z]) & , Z \subset D \text{ via } i' : Z \rightarrow D \end{cases}$$

and extends linearly to  $Z_{k+1}(X)$ , which is also locally finite.

We will see this is map  $i^*$  in fact factor through rational equivalence relations in (9.1.5.9).  $\lrcorner$

*Proof:* To show this is well defined, Cf. [Sta]02TO. ?  $\square$

**Remark (9.1.3.2).** If  $\mathcal{L}|_D \cong \mathcal{O}_D$ , the  $c_1(\mathcal{L}|_Z)$  term is trivial, so the Gysin homomorphism factors through  $Z_k(D) \rightarrow \text{CH}_k(D)$ .  $\lrcorner$

**Cor. (9.1.3.3).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $(\mathcal{L}, s, i : D \rightarrow X)$  as in (9.1.3.1), then for any  $\alpha \in \text{CH}_*(X)$ ,

$$i_* i^* \alpha = c_1(\mathcal{L}) \cap \alpha$$

$\lrcorner$

**Def. (9.1.3.4) [Intersection with Cartier Divisors].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $D \in \text{Cart}^{\text{eff}}(X)$ , denote  $D \cap \alpha = c_1(\mathcal{O}_X(D)) \cap \alpha$  for  $\alpha \in Z_*(X)$ . ?  $\lrcorner$

**Def. (9.1.3.5) [Gysin Map for Local Complete Intersection Maps].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  be a local complete intersection map s.t.  $f = g \circ i$  where  $g$  is smooth and  $i$  is a regular immersion, then we define the **Gysin map** for  $f$  to be  $f^! = i^! \circ g^* \in A^*(X \rightarrow Y)$ . This is independent of the decomposition  $f = g \circ i$  chosen. In this case, we say the Gysin map for  $f$  exists.  $\lrcorner$

*Proof:* Cf. [Sta]0FF2. ?  $\square$

**Prop. (9.1.3.6) [ $f^!$  and  $f^*$ ].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  be a local complete intersection map, then if the Gysin map  $f^!$  exists and  $f$  is flat, then  $f^*$  can be defined and  $f^! = f^* \in A^*(X \rightarrow Y)$ .  $\lrcorner$

*Proof:* Cf. [Sta]0FF4. ?  $\square$

### 4 Weil Divisors

**Prop. (9.1.4.1) [Weil-Cartier].** For  $X \in \text{Sch}^{\text{loc.ft}}/S$ , there is a map from Cartier divisors on  $X$  to Weil divisors mapping  $(D, s_D)$  to  $[Z(s_D)]_k$ . Notice this is defined because a Cartier divisor is locally defined by a regular element and Krull's principal ideal theorem.

Moreover, if  $X$  is locally factorial, this map is an isomorphism  $\text{CaCl}(X) \rightarrow \text{Cl}(X)$  by (7.8.1.5), and (effective)Cartier divisors correspond (effective)Weil divisors.

This in particular applies to non-singular varieties over a field, by (5.3.5.20).  $\lrcorner$



**Prop. (9.1.4.2).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $Z \subset X$  be an integral closed subscheme, then  $Z$  is a Weil divisor iff  $\dim_\delta(Z) = \dim_\delta(X) - 1$ . In particular, this holds for any irreducible component of a Cartier divisor by (9.1.4.1).  $\lrcorner$

**Def. (9.1.4.3) [Q-Cartier Divisors].** A **Q-Cartier divisor** on a locally Noetherian integral scheme is a Weil divisor that some multiply of it is the image of a Cartier divisor.  $\lrcorner$

**Example (9.1.4.4) [Non-Q-Cartier Divisors].** It is easy to show some Weil divisor is not an Q-Cartier divisor (on a singular variety), by showing its complement is not affine (7.8.1.1). For example, on  $X = \text{Spec } k[x, y, z, w]/(xy - zw)$ , the Weil divisor cut out by  $(z, w)$  is not Q-Cartier, as the closed subscheme  $y = w = 0$  of its complement is isomorphic to  $\mathbb{A}_k^2 \setminus \{(0, 0)\}$ , which is not affine by calculating Čech cohomology.

In particular,  $k[x, y, z, w]/(xy - zw)$  is normal but not a UFD.  $\lrcorner$

**Prop. (9.1.4.5) [Rational Functions and Poles].** If  $X$  is an integral locally Noetherian normal scheme and  $f \in K(X)$  has no poles, then  $f \in \Gamma(X)$ , by (5.3.5.12).  $\lrcorner$

**Prop. (9.1.4.6) [UFD and Class Groups].** For  $A$  a Noetherian normal domain, it is a UFD iff  $\text{Cl}(\text{Spec } A) = 0$ .  $\lrcorner$

*Proof:* It suffices to show minimal primes of  $A$  is principal iff minimal primes of  $A$  are principal divisors. This is done by (5.3.5.12) and (3.2.3.9).  $\square$

**Prop. (9.1.4.7) [Picard Group of Projective Spaces].** Let  $R$  be a UFD, then  $\text{Cl}(\mathbb{P}_R^n) = \mathbb{Z}$ , and it is generated by  $[H]$  where  $H$  is any hyperplane of  $\mathbb{P}_R^n$ . And a hypersurface  $Y$  of degree  $d$  is mapped to  $d$ .  $\lrcorner$

*Proof:* These follow from (9.1.6.10) and (9.1.4.6). The last assertion follows from the fact  $[Y] \sim d[H]$  by direct verification.  $\square$

**Cor. (9.1.4.8).** If  $Y \subset \mathbb{P}_k^n$  is a hypersurface of degree  $d$ , then  $\text{Cl}(\mathbb{P}^2 \setminus Y) = \mathbb{Z}/d\mathbb{Z}$ .  $\lrcorner$

*Proof:* This follows from (9.1.1.25) and (9.1.4.7).  $\square$

**Cor. (9.1.4.9).** If  $k \in \text{Field}$ ,  $\text{Cl}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) = \mathbb{Z} \oplus \mathbb{Z}$ , by (9.1.6.10).  $\lrcorner$

**Prop. (9.1.4.10).** If  $X$  is a non-singular cubic surface in  $\mathbb{P}_k^3$ , then  $\text{Cl}(X) \cong \mathbb{Z}^7$ .  $\lrcorner$

*Proof:* Cf. [Har77] Chap 5.4.8.  $\square$

## 5 Bivariant Classes

**Def. (9.1.5.1) [Bivariant Classes].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  and  $p \in \mathbb{Z}$ , a **bivariant class**  $c$  of degree  $p$  for  $f$  is a class of maps

$$c \cap - : \text{CH}_k(Y') \rightarrow \text{CH}_{k-p}(Y' \times_Y X)$$

for any  $Y' \in \text{Sch}^{\text{loc.ft}}/Y$  and  $k \in \mathbb{Z}_+$  that satisfies

- If  $Y'' \rightarrow Y' \in \text{Sch}^{\text{loc.ft}}/Y$  is proper,  $c \cap -$  commutes with proper pushforwards.
- If  $Y'' \rightarrow Y' \in \text{Sch}^{\text{loc.ft}}/Y$  is flat of relative dimension  $r \in \mathbb{Z}$ ,  $c \cap -$  commutes with flat pullbacks.
- Let  $(\mathcal{L}, s, i : D \rightarrow Y')$  be a triple on  $Y'$  as in (9.1.3.1), then  $c \cap -$  commutes with Gysin homomorphisms  $i^*$ .

The set of bivariant classes of degree  $p$  for  $f$  is denoted by  $A^p(X \rightarrow Y)$ . And denote  $A^*(X \rightarrow Y) = \bigoplus_{p \in \mathbb{Z}} A^p(X \rightarrow Y)$ .  $\perp$

**Prop. (9.1.5.2).** For any  $f : X \rightarrow Y, g : Y \rightarrow X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $p, q \in \mathbb{Z}$ ,  $A^p(X \rightarrow Y)$  is an Abelian group, and the composition defines a bilinear map

$$A^q(Y \rightarrow Z) \times A^p(X \rightarrow Y) \rightarrow A^{p+q}(X \rightarrow Z)$$

that is associative w.r.t. compositions in  $\text{Sch}^{\text{loc.ft}}/S$ .  $\perp$

**Def. (9.1.5.3) [Chow Cohomologies].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ , denote  $A^p(X) = A^p(\text{id}_X)$ , and  $A^*(X) = \bigoplus_{p \in \mathbb{Z}} A^p(X)$  the graded ring, called the **Chow cohomology group** of  $X$ .  $\perp$

**Prop. (9.1.5.4) [Restriction of Classes].** Let  $X \rightarrow Y, f : Y' \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  and  $X' = X \times_Y Y'$ , then obviously there is a map  $f^* : A^p(X \rightarrow Y) \rightarrow A^p(X' \rightarrow Y')$ . In particular, if  $Z \subset X$  is a closed subscheme, then  $f^*(c) \cap \alpha = c \cap f_*(\alpha)$  for  $c \in A^p(Z \rightarrow X), \alpha \in A^p(Z)$  by proper pushforward (9.1.5.1).  $\perp$

**Cor. (9.1.5.5) [Product of Classes].** Let  $X \rightarrow Y, X' \rightarrow Y' \in \text{Sch}^{\text{loc.ft}}/S$ ,  $c \in A^p(X \rightarrow Y), c' \in A^q(X' \rightarrow Y')$ , then the product  $c \circ c'$  is defined to be the element in  $A^{p+q}(X \times_S X' \rightarrow Y \times_S Y')$  defined by the map

$$\text{CH}_*(Y \times_S Y') \xrightarrow{c' \cap -} \text{CH}_{*+q}(Y \times_S X') \xrightarrow{c \cap -} \text{CH}_{*+p+q}(X \times_S X')$$

and all base change variants of this.

Notice  $c' \circ c$  is also in  $A^{p+q}(X \times_S X' \rightarrow Y \times_S Y')$ , so it makes sense to talk about when two classes commute. Maybe  $c' \circ c$  can be defined on smaller base changes, but to say  $c, c'$  commute always means  $c \circ c' = c' \circ c \in A^{p+q}(X \times_S X' \rightarrow Y \times_S Y')$ .  $\perp$

**Lemma (9.1.5.6) [Gysin Factors Through Rational Equivalences].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ , Gysin homomorphism for a triple  $(\mathcal{L}, s, j : D \rightarrow X)$  on  $X$  that are base change of  $(\mathcal{O}(1), x, j : S \times (0) \rightarrow \mathbb{P}_S^1)$  factors through rational equivalences.  $\perp$

*Proof:* This follows from the characterization of rational equivalences in (9.1.1.23) and the fact in this case Gysin maps are easy to describe: They are just  $k$ -parts of the inverse images.  $\square$

**Prop. (9.1.5.7) [Weakening Bivariant Conditions].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  and  $c$  is a class of maps

$$c \cap - : Z_k(Y') \rightarrow \text{CH}_{k-p}(Y' \times_Y X)$$

for any  $Y' \in \text{Sch}^{\text{loc.ft}}/Y$  that is compatible with Gysin homomorphism for triples  $(\mathcal{L}, s, j : D \rightarrow Y')$  that are base change of  $(\mathcal{O}(1), x, j : S \times (0) \rightarrow \mathbb{P}_S^1)$ , then  $c$  factors through rational equivalences

And if moreover  $c$  commutes with proper pushforwards and flat pullbacks (up to rational homotopy), then it induces a bivariant class  $c \in A^p(X \rightarrow Y)$ .  $\perp$

*Proof:* For the first assertion: As  $\mathcal{L}|_D = \mathcal{O}_D$ , the Gysin homomorphism is defined on the level of cycles by (9.1.3.2) and pass to  $\text{CH}_*$  (9.1.5.6), so compositions with Gysin homomorphisms are well-defined. Then  $c$  factors through rational equivalence by the characterization of rational equivalences in (9.1.1.23).

For the last assertion, Cf. [Ful98]P321 or [Sta]0F9A. ?  $\square$

**Prop. (9.1.5.8).** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  and  $X = \coprod_I X_i, Y = \coprod_J Y_j$  be clopen subschemes and  $\alpha : I \rightarrow J$  is a map of sets s.t.  $f(X_i) \subset Y_{\alpha(i)}$ , then for any  $p \in \mathbb{Z}_+$ ,

$$A^p(X \rightarrow Y) = \prod_I A^p(X_i \rightarrow Y_{\alpha(i)}).$$

┘

*Proof:* Cf. [Sta]0FDZ?.

□

**Prop. (9.1.5.9) [Gysin Homomorphisms are Bivariant].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $(\mathcal{L}, s, i : D \rightarrow X)$  be a triple on  $X$  as in (9.1.3.1), then the Gysin homomorphism  $(i')^*$  associated to the base changes of  $i$  form a bivariant class in  $A^1(D \rightarrow X)$ .

┘

*Proof:* Cf. [Sta]02TA, 0B71, 0B73. ? and (9.1.5.7).

Let  $f : X' \rightarrow X \in \text{Sch}^{\text{loc.ft}}/S$  and  $(\mathcal{L}, s, i : D \rightarrow X)$  be a triple on  $X$  as in (9.1.3.1), we can define the pullback triple  $f^*(\mathcal{L}, s, i) = (\mathcal{L}', s', i' : D' \rightarrow X')$ , and then

- If  $f$  is proper, then Gysin maps commute with proper pushforwards.
- If  $f$  is flat of relative dimension  $r$ , then the Gysin maps commute with flat pullbacks.
- If  $(\mathcal{M}, t, f : X' \rightarrow X)$  is a triple on  $X$  as in (9.1.3.1), then the different Gysin maps are compatible.

□

**Prop. (9.1.5.10) [Flat Pullbacks are Bivariant].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{loc.ft}}/S$  be flat of relative dimension  $r$ , then the flat pullbacks along base changes of  $f$  form a bivariant class of degree  $-r$ .

┘

*Proof:* This follows from (9.1.1.13)(9.1.1.16)(9.1.5.9) and (9.1.5.7).

□

**Prop. (9.1.5.11) [Proper Pushforwards are Bivariant].** Let  $f : X \rightarrow Y, g : Y \rightarrow Z \in \text{Sch}^{\text{loc.ft}}/S$ ,  $f$  is proper, and  $c \in A^p(X \rightarrow Z)$ , then the base change of  $f_* \circ c$  form a bivariant class in  $A^p(Y \rightarrow Z)$ .

┘

*Proof:* This reduces to the fact proper pushforwards commutes with flat pullbacks, proper pushforwards and Gysin maps, by (9.1.5.10)(9.1.5.9)(9.1.1.9) and (9.1.5.7).

□

## 6 Chern Classes

### Invertible Sheaves

**Prop. (9.1.6.1) [First Chern Classes].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{L} \in \text{Pic}(X)$ , then for any  $k \in \mathbb{Z}$ , there is a map

$$c_1(\mathcal{L}) \cap - : Z_{k+1}(X) \rightarrow \text{CH}_k(X)$$

called intersection with the **first Chern class** of  $\mathcal{L}$  (the name will be made clear in (9.1.6.5)), defined as follows: If  $i : Z \rightarrow X$  is an integral subscheme of  $X$  with  $\dim_\delta(Z) = k + 1$ , then  $c_1(\mathcal{L}) \cap [Z]$  is the the pushforward of the image of  $i^*\mathcal{L}$  under the map  $\text{Pic}(Z) \rightarrow \text{Cl}(Z) \cong \text{CH}_k(Z)$  (7.5.3.14), i.e.

$$c_1(\mathcal{L}) \cap [Z] = i_*(c_1(i^*\mathcal{L}) \cap [W]).$$

In general, define  $c_1(\mathcal{L}) \cap (\sum n_Z [Z]) = \sum n_Z c_1(\mathcal{L}) \cap [Z]$ . We will see the first Chern class factors through rational equivalence relations in (9.1.6.5).

┘

**Lemma (9.1.6.2)**[The first Chern Class is Multiplicative]. Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{L}, \mathcal{N} \in \text{Pic}(X)$ , then

$$c_1(\mathcal{L}) + c_1(\mathcal{N}) = c_1(\mathcal{L} \otimes \mathcal{N}) : Z_{*+1}(X) \rightarrow \text{CH}_*(X).$$

In particular,  $c_1(\mathcal{L}) = -c_1(\mathcal{L}^{-1})$ , and  $c_1(\mathcal{O}_X) = 0$ .  $\lrcorner$

**Prop. (9.1.6.3)**[Chern Classes and Zero Cycles]. Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{L} \in \text{Pic}(X)$ ,  $Y \subset X$  a closed subscheme. If  $s \in \Gamma(Y, \mathcal{L}|_Y)$  satisfies

- $\dim_\delta(Y) \leq k + 1$ .
- $\dim_\delta(Z(s)) \leq k$ .
- For any generic point  $\xi$  of irreducible components of  $Z(s)$  of  $\delta$ -dimension  $k$ , multiplying by  $s$  induces an injection  $\mathcal{O}_{Y,\xi} \rightarrow (\mathcal{L}|_Y)_\xi$ .

Then  $c_1(\mathcal{L}) \cap [Y]_{k+1} = [Z(s)]_k$ .  $\lrcorner$

*Proof:* Cf. [Sta]02SQ. ?  $\square$

**Prop. (9.1.6.4)**[First Chern Classes Factor Through  $\text{CH}_*(X)$ ]. Situation in (9.1.6.1), then the first Chern class map  $c_1(\mathcal{L}) \cap - : Z_{k+1}(X) \rightarrow \text{CH}_k(X)$  factors through  $Z_{k+1}(X) \rightarrow \text{CH}_{k+1}(X)$ .  $\lrcorner$

*Proof:* Cf. [Sta]02TI.  $\square$

**Prop. (9.1.6.5)**[First Chern Classes Form Bivariant Classes]. Let  $X \in \text{Sch}^{\text{loc.ft}}/X$  and  $\mathcal{L} \in \text{Pic}(X)$ , then the first Chern class maps  $c_1(f^*\mathcal{L}) \cap - : Z_*(X') \rightarrow \text{CH}_{*-1}(X')$  factor through rational equivalences for each  $f : X' \rightarrow X \in \text{Sch}^{\text{loc.ft}}/X$ , and form a bivariant class in  $A^1(X)$ , called the **first Chern class**  $c_1(\mathcal{L})$ .  $\lrcorner$

*Proof:* Cf. [Sta]02SU, 02SS, 0B72. and (9.1.5.7).  $\square$

To show it commutes with pullbacks,  $\square$

**Prop. (9.1.6.6)**[First Chern Classes Commute with Bivariant Classes]. Let  $f : X' \rightarrow X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{L} \in \text{Pic}(X)$ , then  $c_1(\mathcal{L})$  commutes with every element of  $A^*(X' \rightarrow X)$ .  $\lrcorner$

In particular,  $c_1(\mathcal{L})$  is in the center of  $A^*(X)$ .  $\lrcorner$

*Proof:* Cf. [Sta]0B7B. ?  $\square$

**Prop. (9.1.6.7)**. Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{L} \in \text{Pic}(X)$  be an ample invertible sheaf. Assume  $d = \dim(X) < \infty$ , then for any  $\mathcal{L}_1, \dots, \mathcal{L}_{d+1} \in \text{Pic}(X)$ ,  $c_1(\mathcal{L}_1) \circ \dots \circ c_1(\mathcal{L}_{d+1}) = 0 \in A^{d+1}(X)$ .  $\lrcorner$

*Proof:* Use induction on  $d$ :  $d = 0$  case is trivial as  $X$  is discrete thus any invertible sheaf is trivial. In general, by (7.8.1.9) and (9.1.6.2), it suffices to prove for  $\mathcal{L}_i = \mathcal{O}_X(D_i)$  being Cartier divisors. If  $i : D_{d+1} \rightarrow X$ , then  $c_1(\mathcal{L}_{d+1}) = i_* \circ i^*$  by (9.1.3.3), and  $c_1(\mathcal{L}_i)$  commutes with both  $i_*$  and  $i^*$ , so we can replace  $X$  by  $D$  and then use induction hypothesis.  $\square$

**Lemma (9.1.6.8)**. If  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $0 \rightarrow \mathcal{L}_1 \rightarrow \mathcal{E} \rightarrow \mathcal{L}_2 \rightarrow 0$  is an exact sequence with  $\mathcal{L}_1, \mathcal{L}_2 \in \text{Pic}(X)$ , and there is a non-vanishing section  $s$  of  $\mathcal{E}$ , then  $c_1(\mathcal{L}_1) \cap c_1(\mathcal{L}_2) = 0 \in A^2(X)$ .  $\lrcorner$

*Proof:* Consider the image  $\bar{s}$  of  $s$  in  $\Gamma(X, \mathcal{L}_2)$ , then we can consider the Gysin map associated to  $(\mathcal{L}_2, \bar{s}, j)$ . Then

$$c_1(\mathcal{L}_1) \cap c_1(\mathcal{L}_2) \cap \alpha = j_*(c_1(j^*\mathcal{L}_1) \cap j^*\alpha).$$

$j^*\mathcal{L}_1$  is trivialized by  $s$  now, so this one is vanishes.  $\square$

**Lemma (9.1.6.9).** If  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r \in \mathbb{Z}_+$ ,  $\mathcal{L} \in \text{Vect}^r(X)$ , let  $\pi : \mathbf{P}(\mathcal{E}) \rightarrow X$  be the projective space of  $\mathcal{E}$ , then for any  $k \in \mathbb{Z}$  and  $\alpha \in \text{CH}_k(X)$ ,

$$\pi_*(c_1(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1))^s \cap \pi^* \alpha) = \begin{cases} 0 & , s < r-1 \\ \alpha & , s = r-1 \end{cases} \in \text{CH}_{k+r-1-s}(X).$$

┘

*Proof:* Cf. [Sta]02TW. □

**Prop. (9.1.6.10) [Projective Bundles Formula].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r \in \mathbb{Z}_+$ ,  $\mathcal{L} \in \text{Vect}^r(X)$ , let  $\pi : \mathbf{P}(\mathcal{E}) \rightarrow X$  be the projective space of  $\mathcal{E}$ , then for any  $k \in \mathbb{Z}$ , the map

$$\text{CH}_{k+r-1}(\mathbf{P}(\mathcal{E})) = \bigoplus_{i=0}^r c_i(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1))^i \cap \pi^* \text{CH}_{k+i}(X).$$

is an isomorphism.

In particular, if  $X$  is integral, then

$$\text{Cl}(\mathbf{P}(\mathcal{E})) = \pi^*(\text{Cl}(X)) \oplus [c_1(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1))].$$

┘

*Proof:* Cf. [Sta]02TX. ? □

**Prop. (9.1.6.11) [Vector Bundles].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r \in \mathbb{Z}_+$ ,  $\mathcal{L} \in \text{Vect}^r(X)$ , let  $\pi : \mathbf{E} = \text{Spec}(\text{Sym}(\mathcal{E})) \rightarrow X$  be the vector bundle of  $\mathcal{E}$ , then for any  $k \in \mathbb{Z}$ ,  $p^* : \text{CH}_k(X) \rightarrow \text{CH}_{k+r}(X)$  is an isomorphism. ┘

*Proof:* Cf. [Sta]02TY. □

**Prop. (9.1.6.12) [Diagonal Intersections].** For  $X \in \text{SmProj}^n/k$ ,

$$\Delta \cdot \Delta = \deg(c_n(\mathcal{T}_X)).$$

┘

*Proof:* Cf. [Mustata, P14]. □

### Vector Bundles

**Prop. (9.1.6.13) [Splitting Principle].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $\mathcal{E}_i \subset \text{Vect}(X)$  be a finite set of finite locally free sheaves, there exists a projective flat morphism  $\pi : P \rightarrow X$  of relative dimension  $d$  s.t.

- For any  $Y \in \text{Sch}^{\text{loc.ft}}/X$ , the map  $\pi_Y^* : \text{CH}^*(Y) \rightarrow \text{CH}^*(Y \times_X P)$  is injective.
- Each  $\pi^* \mathcal{E}_i$  has a filtration with invertible quotient sheaves.

This is useful in the way that when proving functorial properties of Chern classes of vector bundles, we can reduce to invertible sheaves. ┘

*Proof:* Cf. [Sta]02UL. □

**Def. (9.1.6.14)[Chern Classes of Vector Bundles].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r \in \mathbb{Z}_+$ ,  $\mathcal{E} \subset \text{Vect}^r(X)$  with projective bundle  $\pi : \mathbf{P}(\mathcal{E}) \rightarrow X$ , then for any  $k, p \in \mathbb{N}$ , there is a map

$$c_p(\mathcal{E}) \cap - : \text{CH}_k(X) \rightarrow \text{CH}_{k-p}(X)$$

called **intersection with Chern classes** as follows (the name will be made clear in (9.1.6.15)): Let  $\alpha \in \text{CH}_k(X)$ , by (9.1.6.10) there are unique elements  $c_p \in \text{CH}_{k-p}(X)$  s.t.  $c_0 = \alpha$  and

$$\sum_{p=0}^r (-1)^p c_1(\mathcal{O}_{\mathbf{P}(\mathcal{E})})^p \cap \pi^* c_{r-p} = 0 \in \text{CH}_{k-1}(\mathbf{P}(i^*\mathcal{E})).$$

Then define  $c_p(\mathcal{E}) \cap \alpha = c_p$ .

In particular, if  $\mathcal{E}$  is an invertible sheaf,  $c_1(\mathcal{E}) \cap$  is just the intersection with the first Chern class  $c_1(\mathcal{E}) \cap \alpha$  (9.1.6.1)(9.1.6.6).  $\lrcorner$

**Prop. (9.1.6.15) [Chern Classes Form Bivariant Classes].** Let  $X \in \text{Sch}^{\text{loc.ft}}/X$ ,  $r \in \mathbb{Z}_+$ ,  $\mathcal{E} \in \text{Vect}^r(X)$ , then the maps  $c_p(f^*\mathcal{E}) \cap - : \text{CH}_k(X') \rightarrow \text{CH}_{k-p}(X')$  for each  $f : X' \rightarrow X \in \text{Sch}^{\text{loc.ft}}/X$ ,  $k \in \mathbb{Z}_+$  form a bivariant class in  $A^p(X)$ , called the **Chern classes**  $c_p(\mathcal{E})$ . By convention, set  $c_p(\mathcal{E}) = 0$  for  $p > r$ .

And  $c(\mathcal{E}) = \sum_{i=0}^r c_i(\mathcal{E}) \in A^*(X)$  are called the **total Chern class** of  $\mathcal{E}$ .  $\lrcorner$

*Proof:* For commuting with proper pushforward, use the definition (9.1.6.14) and (9.1.1.24)(9.1.6.5), and push and pull (9.1.1.16).

For commuting with flat pullback, use the definition (9.1.6.14) and (9.1.1.24)(9.1.6.5) and properties of flat pullbacks (9.1.1.13).

For commuting with Gysin map, use the definition (9.1.6.14) and (9.1.1.24)(9.1.6.5) and properties of Gysin maps (9.1.5.9).  $\square$

**Prop. (9.1.6.16)[Chern Classes Commute with Bivariant Classes].** Let  $f : X' \rightarrow X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{E} \in \text{Vect}^r(X)$ , then for any  $p \in \mathbb{Z}_+$ ,  $c_p(\mathcal{E})$  commutes with every element of  $A^*(X' \rightarrow X)$ .

In particular,  $c_p(\mathcal{E})$  are in the center of  $A^*(X)$ , and for  $\mathcal{E} \in \text{Vect}^r(X)$ ,  $\mathcal{F} \in \text{Vect}^s(X)$ ,  $p, q \in \mathbb{Z}_+$ ,  $c_p(\mathcal{E}), c_q(\mathcal{F})$  commute.  $\lrcorner$

*Proof:* Reduce to  $c_1$  and (9.1.6.6) as we did in (9.1.6.15).  $\square$

**Prop. (9.1.6.17) [Chern Classes for Arbitrary Vector Bundles].** Let  $X \in \text{Sch}^{\text{loc.ft}}/X$ ,  $\mathcal{E} \in \text{Vect}^\delta(X)$ , then constancy of  $r$  induces a clopen partition of  $X = \coprod_{i=0}^\infty X_i$ , where  $X_i$  are the clopen subscheme s.t.  $\delta = i$ . Then by (9.1.5.8),  $A^p(X) = \prod_{i=0}^\infty A^p(X_i)$ , so we can define the Chern class  $c_p(\mathcal{E})$  to be the product of  $c_p(\mathcal{E}|_{X_i})$ . Also we can define a **rank class**  $r(\mathcal{E}) \in A^0(X) = \prod_{i=0}^\infty A^0(X_i)$  that is product of  $[i] \in A^0(X_i)$ .  $\lrcorner$

**Prop. (9.1.6.18)[Additivity].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $0 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{E} \rightarrow \mathcal{E}_2 \rightarrow 0 \in \text{Vect}(X)$ , then

$$c(\mathcal{E}) = c(\mathcal{E}_1)c(\mathcal{E}_2).$$

$\lrcorner$

*Proof:* It suffices to check this by their action on  $[X]$  where  $X \in \text{Sch}^{\text{loc.ft,int}}/S$ . Using splitting principle (9.1.6.13), we can easily reduce to the case  $\mathcal{E}_1, \mathcal{E}_2 \in \text{Pic}(X)$  by induction on  $\text{rank}(\mathcal{E}_1)$  and

$\text{rank}(\mathcal{E}_2)$ . In this case, let  $\mathbf{P}$  be the projective bundle of  $\mathcal{E}$  with  $\mathcal{O}(1) = \mathcal{O}_{\mathbf{P}}(1)$ . By tag02U6, it suffices to show that

$$c_1(\mathcal{O}(1))^2 \cap \pi^* \alpha - c_1(\mathcal{O}(1)) \cap \pi^*(c_1(\mathcal{E}_1) \cap \alpha + c_1(\mathcal{E}_2) \cap \alpha) + \pi^*(c_1(\mathcal{E}_1) \cap c_1(\mathcal{E}_2) \cap \alpha).$$

for which it suffices to show that

$$(c_1(\mathcal{O}(1)) - c_1(\pi^* \mathcal{E}_1)) \cap (c_1(\mathcal{O}(1)) - c_1(\pi^* \mathcal{E}_2)) = c_1(\pi^* \mathcal{E}_1^\vee(1)) \cap c_1(\pi^* \mathcal{E}_2^\vee(1)) = 0$$

There is a surjection  $\mathcal{E} \rightarrow \mathcal{O}(1)$ , which corresponds to a non-zero section of  $\mathcal{E}^\vee(1)$ . Notice there is an exact sequence

$$0 \rightarrow \mathcal{E}_2^\vee \rightarrow \mathcal{E}^\vee(1) \rightarrow \mathcal{E}_1^\vee \rightarrow 0,$$

so the assertion follows from (9.1.6.8).  $\square$

**Prop. (9.1.6.19).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{E} \in \text{Vect}^r(X)$ , then  $c_1(\wedge \mathcal{E}) = c_1(\mathcal{E})$ , and

$$\prod_{p=0}^r c(\wedge^p \mathcal{E})^{(-1)^n} = 1 - (r-1)!c_r(\mathcal{E}) + \dots$$

┘

*Proof:* Use splitting principle and Cf. [Sta]0FEE.  $\square$

**Prop. (9.1.6.20)[Degrees and First Chern Classes].** Let  $X$  be a proper scheme over a field  $k$ , then

- Let  $\mathcal{L}_1, \dots, \mathcal{L}_d$  be invertible sheaves on  $X$  and  $Z \subset X$  a closed subscheme of dimension  $d$ , then

$$(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_d; Z) = \deg(c_1(\mathcal{L}_1) \cap \dots \cap c_1(\mathcal{L}_d) \cap [Z]_d).$$

In particular, if  $\mathcal{L}$  is an ample invertible sheaf,

$$\deg_{\mathcal{L}}(Z) = \deg(c_1(\mathcal{L})^d \cdot [Z]_d).$$

- If  $\dim X \leq 1$ , for  $\mathcal{E} \in \text{Coh}^{\text{free}}(X)$ ,

$$\deg(\mathcal{E}) = \deg(c_1(\mathcal{E}) \cdot [X]_1).$$

┘

*Proof:* Cf. [Sta]0AZ3, 0BFI. ?  $\square$

**Prop. (9.1.6.21)[Chern Characters].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r$  a locally constant  $\mathbb{Z}$ -valued functions on  $X$  and  $\mathcal{E} \in \text{Vect}^r(X)$ , define the **Chern character** of  $\mathcal{E}$  to be

$$\begin{aligned} \text{ch}(\mathcal{E}) &= \text{rank}(\mathcal{E}) + \sum_{p \geq 0} \frac{P_p(c_1(\mathcal{E}), \dots, c_n(\mathcal{E}))}{p!} \in \prod_{i \geq 0} A^i(X)_{\mathbb{Q}} \\ &= \text{rank}(\mathcal{E}) + c_1(\mathcal{E}) + \frac{1}{2}(c_1(\mathcal{E})^2 - 2c_2(\mathcal{E})) + \frac{1}{6}(c_1(\mathcal{E})^3 - 3c_1(\mathcal{E})c_2(\mathcal{E}) + 3c_3(\mathcal{E})) \\ &\quad + \frac{1}{24}(c_1(\mathcal{E})^4 - 4c_1(\mathcal{E})^2c_2(\mathcal{E}) + 4c_1(\mathcal{E})c_3(\mathcal{E}) + 2c_2(\mathcal{E})^2 - 4c_4(\mathcal{E})) + \dots \end{aligned}$$

where  $P_p$  are Chern polynomials (3.3.4.3).  $\square$

**Prop. (9.1.6.22) [Tensor Products].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r, s, t$  are locally constant  $\mathbb{Z}$ -valued functions on  $X$  and  $\mathcal{E} \in \text{Vect}(X)$ ,  $\mathcal{F} \in \text{Vect}(X)$ ,  $\mathcal{G} \in \text{Vect}(X)$ , then

- If there is an exact sequence  $0 \rightarrow \mathcal{E} \rightarrow \mathcal{G} \rightarrow \mathcal{F} \rightarrow 0$ , then  $\text{ch}(\mathcal{E}) + \text{ch}(\mathcal{F}) = \text{ch}(\mathcal{G})$ .
- $\text{ch}(\mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{F}) = \text{ch}(\mathcal{E}) \text{ch}(\mathcal{F})$ .
- $\text{ch}_i(\mathcal{E}^\vee) = (-1)^i \text{ch}_i(\mathcal{E})$ .

In particular,  $\text{ch}$  defines a ring homomorphism  $\text{ch} : K_0(\text{Vect}(X)) \rightarrow \prod_{i \geq 0} A^i(X)_{\mathbb{Q}}$ .  $\lrcorner$

*Proof:* It suffices to prove for  $X$  connected, and then by (9.1.6.13), we can assume  $\mathcal{E}, \mathcal{F}$  have filtrations with invertible quotient sheaves  $\mathcal{E}_1, \dots, \mathcal{E}_r$  and  $\mathcal{F}_1, \dots, \mathcal{F}_s$  with first Chern characters  $a_1, \dots, a_r$  and  $b_1, \dots, b_s$ , then by additivity (9.1.6.18) and the definition of Chern polynomials (3.3.4.3), we see

$$\text{ch}(\mathcal{E}) = \sum_{i=1}^r \exp(a_i), \quad \text{ch}(\mathcal{F}) = \sum_{j=1}^s \exp(b_j),$$

and  $\mathcal{E} \otimes \mathcal{F}$  has a filtration with quotient sheaves  $\mathcal{E}_i \otimes \mathcal{F}_j$  with first Chern characters  $a_i + b_j$  by (9.1.6.2), so the assertions are now clear.  $\square$

**Cor. (9.1.6.23) [Chern Classes of Tensor Products].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $r, s$  are locally constant  $\mathbb{Z}$ -valued functions on  $X$  and  $\mathcal{E} \in \text{Vect}^r(X)$ ,  $\mathcal{F} \in \text{Vect}^s(X)$ , then

$$\begin{aligned} c_i(\mathcal{E}^\vee) &= (-1)^i c_i(\mathcal{E}) \in A^i(X), \\ c_1(\mathcal{E} \otimes \mathcal{F}) &= r c_1(\mathcal{E}) + s c_1(\mathcal{F}) \in A^1(X), \\ c_2(\mathcal{E} \otimes \mathcal{F}) &= r c_2(\mathcal{F}) + s c_2(\mathcal{E}) + \binom{r}{2} c_1(\mathcal{F})^2 + (rs - 1) c_1(\mathcal{E}) c_1(\mathcal{F}) + \binom{s}{2} c_1(\mathcal{E})^2 \in A^2(X), \\ c_2(\text{Hom}(\mathcal{E}, \mathcal{E})) &= 2r c_2(\mathcal{E}) - (r - 1) c_1(\mathcal{E})^2 \in A^2(X). \end{aligned}$$

$\lrcorner$

**Prop. (9.1.6.24) [Todd Classes].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{E} \in \text{Vect}(X)$ , define the **Todd class**  $\text{Todd}(\mathcal{E})$  of  $\mathcal{E}$  to be

$$\begin{aligned} \text{Todd}(\mathcal{E}) &= \text{Todd}(c_1(\mathcal{E}), c_2(\mathcal{E}), \dots) \text{ (3.3.4.4)} = 1 + \frac{1}{2} c_1(\mathcal{E}) + \frac{1}{12} (c_1(\mathcal{E})^2 + c_2(\mathcal{E})) \\ &\quad + \frac{1}{24} c_1(\mathcal{E}) c_2(\mathcal{E}) + \frac{1}{720} (-c_1^4(\mathcal{E}) + 4c_1^2(\mathcal{E}) c_2(\mathcal{E}) + 3c_2^2(\mathcal{E}) + c_1(\mathcal{E}) c_3(\mathcal{E}) - c_4(\mathcal{E})) + \dots \end{aligned}$$

Then if  $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0 \in \text{Vect}(X)$  be an exact sequence, then  $\text{Todd}(\mathcal{E}) \text{Todd}(\mathcal{E}'') = \text{Todd}(\mathcal{E}')$ .  $\lrcorner$

*Proof:* The proof is the same as that of (9.1.6.22).  $\square$

**Prop. (9.1.6.25) [Borel-Serre].** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{E} \in \text{Coh}^{\text{free}, r}(X)$ , then

$$\sum_{p=0}^r (-1)^p \text{ch}(\wedge^p \mathcal{E}^\vee) = c_r(\mathcal{E}) \text{Todd}(\mathcal{E})^{-1}.$$

$\lrcorner$

*Proof:* ?  $\square$



**Prop. (9.1.6.26) [Chern Classes of K-Groups].**  $X \in \text{Sch}^{\text{loc.ft}}/S$ , by additivity and multiplicativity (9.1.6.18)(9.1.6.22), we can extend Chern classes and Chern characters to the K-group  $K_0(\text{Vect}(X))$ :

$$c : K_0(\text{Vect}(X)) \rightarrow \prod_{p \geq 0} A^p(X), \quad \text{ch} : K_0(\text{Vect}(X)) \rightarrow \prod_{p \geq 0} A^p(X)_{\mathbb{Q}}$$

where  $c$  is a group homomorphism and  $\text{ch}$  is a ring homomorphism.  $\lrcorner$

**Prop. (9.1.6.27) [Adam Operators and Chern Characters].**  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\alpha \in K_0(\text{Vect}(X))$ , then  $c_i(\psi^k(\alpha)) = 2^{ki} c_i(\alpha)$  and  $\text{ch}_i(\psi^k(\alpha)) = 2^{ki} \text{ch}_i(\alpha)$ .  $\lrcorner$

*Proof:* It suffices to prove for  $\alpha = [\mathcal{E}]$  where  $\mathcal{E} \in \text{Vect}(X)$ , then by splitting principle it suffices to prove for line bundles, and this is clear from (9.1.6.23) and the definition (9.1.6.21)t.  $\square$

### Perfect Complexes

**Lemma (9.1.6.28).** Let  $X \in \text{Sch}^{\text{loc.ft}}/S$ , then for each perfect complex  $\mathcal{E}$  in  $D(\mathcal{O}_X)$ , we can define the Chern classes, Chern characters and ranks of  $\mathcal{E}$ . But we only define for a subset of  $E$  here? Cf. [Sta]0F9E: If  $\mathcal{E}$  is represented by a finite complex  $\mathcal{E}^\bullet$  of vector bundles on  $X$ , define

$$c(\mathcal{E}) = \prod_i c(\mathcal{E}^i)^{(-1)^i} \in \prod_{i \geq 0} A^i(X), \quad \text{ch}(\mathcal{E}) = \sum_i (-1)^i \text{ch}(\mathcal{E}^i) \in \prod_{i \geq 0} A^i(X)_{\mathbb{Q}}, \quad r(\mathcal{E}) = \sum_i (-1)^i r(\mathcal{E}^i).$$

$\lrcorner$

*Proof:* To show this is well defined,  $\square$

**Prop. (9.1.6.29) [Chern Classes via Envelopes].** Let  $Y \rightarrow X \in \text{Sch}^{\text{loc.ft}}/S$  be an envelope?,  $\mathcal{E}$  perfect in  $D(\mathcal{O}_X)$  s.t.  $Lf^*\mathcal{E}$  is representable by a finite complex of vector bundles on  $Y$ , then there are unique bivariant classes  $c(\mathcal{E}) \in \prod_{i \geq 0} A^i(X)$ ,  $\text{ch}(\mathcal{E}) \in \prod_{i \geq 0} A^i(X)_{\mathbb{Q}}$ ,  $r(\mathcal{E})$  s.t. their restrictions to  $Y$  are of the form defined in (9.1.6.28).

Moreover, these bivariant classes are invariant of the envelope chosen. In this case, we say the Chern classes of  $\mathcal{E}$  are defined.  $\lrcorner$

*Proof:* Cf. [Sta]0GUD.  $\square$

**Prop. (9.1.6.30).** If  $X \in \text{Sch}^{\text{loc.ft}}/S$  and each irreducible components of  $X$  are qc, then  $X$  has an envelope. In particular, this applies to  $X$  qc.  $\lrcorner$

*Proof:* Cf. [Sta]0GUE?.  $\square$

**Prop. (9.1.6.31) [Additivity].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{E}_1 \rightarrow \mathcal{E}_2 \rightarrow \mathcal{E}_3 \rightarrow \mathcal{E}_1[1]$  is a distinguished triangle of perfect complexes in  $D(\mathcal{O}_X)$ , and one of the following holds:

- There exists an envelope  $f : Y \rightarrow X$  s.t.  $Lf^*\mathcal{E}_1 \rightarrow Lf^*\mathcal{E}_2$  is representable by a map of finite complexes of vector bundles on  $Y$ .
- Each irreducible components of  $X$  is irreducible.

Then Chern classes of  $\mathcal{E}_i$  are defined, and

$$c(\mathcal{E}_2) = c(\mathcal{E}_1)c(\mathcal{E}_3), \quad \text{ch}(\mathcal{E}_2) = \text{ch}(\mathcal{E}_1) + \text{ch}(\mathcal{E}_3).$$

$\lrcorner$

*Proof:* Cf. [Sta]0F9F? .  
Se

□

**Prop. (9.1.6.32).** If  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $\mathcal{E}$  be a perfect object in  $D(\mathcal{O}_X)$  whose Chern classes are defined, then

- Let  $f : X' \rightarrow X \in \text{Sch}^{\text{loc.ft}}/S$ ,  $p \in \mathbb{Z}_+$ , then  $c_p(\mathcal{E})$  commutes with every element of  $A^*(X' \rightarrow X)$ , i.e.  $c \cdot c_p(\mathcal{E}) = c_p(Lf^*\mathcal{E}) \cdot c$ .  
In particular,  $c_p(\mathcal{E})$  are in the center of  $A^*(X)$ , and for  $\mathcal{E}, \mathcal{F}$  perfect in  $D(\mathcal{O}_X)$  whose Chern classes are defined,  $p, q \in \mathbb{Z}_+$ ,  $c_p(\mathcal{E}), c_q(\mathcal{F})$  commute.
- $\log(c(\mathcal{E})) = \sum_{p \geq 0} \frac{P_p(c_1(\mathcal{E}), \dots, c_n(\mathcal{E}))}{p}$  also holds, where  $P_p$  are Chern polynomials (3.3.4.3). Similarly for Chern character  $\text{ch}(\mathcal{E})$ .
- $c_i(\mathcal{E}^\vee) = (-1)^i c_i(\mathcal{E})$ ,  $\text{ch}_i(\mathcal{E}^\vee) = (-1)^i \text{ch}_i(\mathcal{E})$ .

┘

*Proof:* These follow from the splitting principle and the definition (9.1.6.29).

□

**Prop. (9.1.6.33) [Tensor Products].** If  $X \in \text{Sch}^{\text{loc.ft}}/S$  and  $E, F$  perfect in  $D(\mathcal{O}_X)$  whose Chern classes are defined, then

$$\text{ch}(\mathcal{E} \otimes_{\mathcal{O}_X}^L \mathcal{F}) = \text{ch}(\mathcal{E}) \text{ch}(\mathcal{F}),$$

in particular, formulas in (9.1.6.23) hold.

┘

*Proof:* This follows from the splitting principle and the definition (9.1.6.29).

□

## 7 Non-Singular Intersection Theory

**Lemma (9.1.7.1).** If  $X \in \text{Sch}_{\text{reg}, qc}^{\text{ft, sep}}/S$  with bounded  $\delta$ -dimension, then the composition

$$K_0(\text{Vect}(X)) \otimes \mathbb{Q} \xrightarrow{\text{ch}} \prod_{p \geq 0} A^p(X) \xrightarrow{-\cap[X]} \text{CH}^*(X) \otimes \mathbb{Q}$$

is an isomorphism.

┘

*Proof:* Firstly  $K_0(X) = K'_0(X) = K_0(\text{Vect}(X))$  by (7.9.1.35) and (7.9.1.38).

The rest follows from [Sta]0FEY? .

□

**Prop. (9.1.7.2) [Q-Intersection Products on Regular Schemes].** If  $X \in \text{Sch}_{\text{reg}, qc}^{\text{ft, sep}}/S$  with bounded  $\delta$ -dimension, then the isomorphism  $K_0(\text{Vect}(X)) \otimes \mathbb{Q} \cong \text{CH}^*(X) \otimes \mathbb{Q}$  endows  $\text{CH}^*(X)_{\mathbb{Q}}$  with a commutative associative ring structure: If  $\alpha = \text{ch}(\alpha') \cap [X], \beta = \text{ch}(\beta') \cap [X]$ , then

$$\alpha \cdot \beta = \text{ch}(\alpha) \cap \text{ch}(\beta) \cap [X] = \text{ch}(\alpha') \cap \beta = \text{ch}(\beta') \cap \alpha.$$

And this ring structure preserves the gradation on  $\text{CH}^*(X)_{\mathbb{Q}}$ . Also it is preserved under morphism in  $\text{Sch}_{\text{reg}, qc}^{\text{ft, sep}}$  flat of relative dimension  $r$ .

┘

*Proof:* To prove it preserves gradation, suppose  $\alpha \in \text{CH}^i(X), \beta \in \text{CH}^j(X)$ , then  $\alpha'$  and  $2^{-i}\psi^2(\alpha')$  (9.1.6.27) are both inverse images of  $\alpha$ , so they are equal. Then  $\text{ch}(\alpha') = \text{ch}(2^{-i}\psi^2(\alpha'))$ , which means  $\text{ch}(\alpha') \in A^i(X)_{\mathbb{Q}}$ , and then  $\text{ch}(\alpha') \cap \beta \in \text{CH}^{i+j}(X)_{\mathbb{Q}}$ .

□

Smooth over Dedekind Schemes CaseSmooth over Fields Case

**Prop. (9.1.7.3) [Exterior Products].** If  $k \in \mathbf{Field}$ ,  $S = \operatorname{Spec} k$ ,  $X, Y \in \operatorname{Sch}^{\operatorname{loc.ft}}/k$ , there is a **exterior product map**

$$\operatorname{CH}_n(X) \otimes \operatorname{CH}_m(Y) \rightarrow \operatorname{CH}_{n+m}(X \times_S Y)$$

defined by sending  $[X'] \otimes [Y']$  to  $[X' \times_S Y']_{n+m}$ , where  $X' \subset X, Y' \subset Y$  are integral closed subschemes of dimension  $n$  and  $m$  resp.  $\lrcorner$

*Proof:* To show this is well defined, consider  $i : X' \rightarrow X, c : X \rightarrow \operatorname{Spec} k$ , then by (9.1.5.11),  $c_* \circ (c \circ i)^* \in A^{-n}(X \rightarrow \operatorname{Spec} k)$ , which sends  $[Y']$  to  $[X' \times_k Y']_{n+m}$ , so this map factors through rational equivalences on  $Y$ , and similarly it factors through rational equivalences on  $X$ .  $\square$

**Prop. (9.1.7.4).** If  $k \in \mathbf{Field}$ ,  $S = \operatorname{Spec} k$  and  $X \in \operatorname{Sch}^{\operatorname{loc.ft}}/k$ , there is a natural isomorphism  $A^p(X \rightarrow \operatorname{Spec} k) \cong \operatorname{CH}_{-p}(X)$  for any  $p \in \mathbb{Z}$ .  $\lrcorner$

*Proof:* The map  $A^p(X \rightarrow \operatorname{Spec} k) \rightarrow \operatorname{CH}_{-p}(X)$  is given by  $c \mapsto c \cap [\operatorname{Spec} k]$ . Conversely, for any  $\alpha \in \operatorname{CH}_{-p}(X)$ , we can define for any  $X' \in \operatorname{Sch}^{\operatorname{loc.ft}}/k$  a map

$$\operatorname{CH}_n(X') \rightarrow \operatorname{CH}_{n-p}(X \times_k X') : \beta \mapsto \alpha \times \beta \text{ (9.1.7.3)}.$$

Then this is a bivariant class in  $A^p(X \rightarrow \operatorname{Spec} k)$ : Let  $\alpha = \sum n_i [X_i]$ , and let  $g : \coprod_i X_i \rightarrow X$ ,  $f : \coprod_i X_i \rightarrow \operatorname{Spec} k$ . Denote  $\nu^* \in A^0(\coprod_i X_i)$  the bivariant class that multiplies by  $n_i$  at each component  $X_i$ , then  $g_* \circ \nu \circ f^* \in A^p(X \rightarrow \operatorname{Spec} k)$  by (9.1.5.10)(9.1.5.11), and this is just the map we defined above.

To show these two maps are inverse to each other, one direction is clear. To show  $c(\beta) = (c \cap [\operatorname{Spec} k]) \times \beta$ , it suffices to show for  $\beta = [X']$  is integral, then  $\beta = (X' \rightarrow \operatorname{Spec} k)^* [\operatorname{Spec} k]$ , then it is easy to see they are equal because  $c$  commutes with flat pullbacks  $(X' \rightarrow \operatorname{Spec} k)^*$ .  $\square$

**Cor. (9.1.7.5) [Commutativity and Associativity].** If  $k \in \mathbf{Field}$ ,  $S = \operatorname{Spec} k$  and  $X \in \operatorname{Sch}^{\operatorname{loc.ft}}/k$ , then  $c \in A^p(X \rightarrow \operatorname{Spec}(k))$  commutes with any  $c' \in A^q(Y \rightarrow Z)$  for any  $f : Y \rightarrow Z \in \operatorname{Sch}^{\operatorname{loc.ft}}/k$ . In other words, for any  $\alpha \in \operatorname{CH}_*(X), \beta \in \operatorname{CH}_*(Z)$  and  $c \in A^*(Y \rightarrow Z)$ ,

$$\alpha \times (c \cap \beta) = c \cap (\alpha \times \beta) \in \operatorname{CH}_*(X \times_k Y).$$

In particular, if  $\gamma \in \operatorname{CH}_*(Y)$ , then

$$(\alpha \times \beta) \times \gamma = \alpha \times (\beta \times \gamma) \in \operatorname{CH}_*(X \times_k Y \times_k Z).$$

$\lrcorner$

*Proof:* This is because  $c = g_* \circ \nu \circ f^*$  as in the proof of (9.1.7.4).  $\square$

**Prop. (9.1.7.6) [Perturbation and Chow Rings].** Any two algebraic cycles  $\gamma_1, \gamma_2 \in Z_*(X)$  are rationally equivalent to cycles  $\gamma'_1, \gamma'_2$  s.t.  $\gamma'_1, \gamma'_2$  intersect properly. And the rational class of resulting intersection cycle is well-defined. So there is an intersection on  $\operatorname{CH}^*(X)$  making it a commutative ring, called the **Chow ring of  $X$** .  $\lrcorner$

*Proof:*

$\square$

**Prop. (9.1.7.7) [Bezout].** The Chow ring of  $\mathbb{P}_k^n$  is isomorphic to  $\mathbb{Z}[x]/(x^{n+1})$ . The degree of an irreducible closed variety corresponds to the coefficient of it.  $\lrcorner$

*Proof:*

$\square$

**Def. (9.1.7.8) [Euler Characteristic].** If  $X$  is a smooth scheme over a field  $k$  of dimension  $d$  with tangent bundle  $\mathcal{T}_{X/k} \in \text{Vect}(X)$ , the **Euler characteristic of  $X$**  is defined to  $\deg(c_d(\mathcal{T}_{X/k}) \cap [X])$ , and the **Todd characteristic of  $X$**  is defined to  $\deg(\text{Todd}_d(\mathcal{T}_{X/k}) \cap [X])$  (9.1.6.24).  $\lrcorner$

**Prop. (9.1.7.9).** If  $X$  is a smooth scheme over a field  $k$  and  $i : Y \rightarrow X$  is a regular closed immersion and  $Y$  is equidimensional of dimension  $e$ , then

$$[Y]_e \cdot \alpha = i_*(i^!(\alpha)).$$

$\lrcorner$

*Proof:* Cf. [Sta]0FFE?.

$\square$

### Serre's Approach

**Prop. (9.1.7.10).** If  $X$  is a integral scheme smooth over a field  $k$ , and  $W, V$  be two integral closed subschemes of  $X$ , then  $\text{codim}(W \cap V) \leq \text{codim}(W) + \text{codim}(V)$ .  $\lrcorner$

*Proof:* Cf. [Sta]0AZP.

$\square$

**Def. (9.1.7.11) [Proper Intersections].** Let  $X$  be a integral scheme smooth over a field  $k$ , then two cycles  $\alpha = \sum n_i [W_i]$  and  $\beta = \sum m_j [V_j]$  are said to intersect properly iff

$$\text{codim}(W_i \cap V_j) \geq \text{codim}(W_i) + \text{codim}(V_j).$$

And in fact equality holds, by (9.1.7.10).  $\lrcorner$

**Lemma (9.1.7.12) [Tor Sheaves].** Let  $X$  be a integral scheme smooth over a field  $k$  and  $\mathcal{F}, \mathcal{G} \in \text{Coh}(X)$ , then  $\text{Tor}_p^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \in \text{Coh}(X)$ , with stalk at  $x \in X$  being  $\text{Tor}_p^{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$ , and is supported on  $\text{Supp}(\mathcal{F}) \cap \text{Supp}(\mathcal{G})$ , and nonzero only for  $0 \leq p \leq \dim X$ .  $\lrcorner$

*Proof:* Cf. [Sta]0AZT.

$\square$

**Def. (9.1.7.13) [Intersection via Tor].** Let  $X$  be a regular scheme and  $W, V \subset X$  be integral closed subschemes intersecting properly, by (9.1.7.12) we can define the **intersection product**

$$W \cdot V = \sum_i [\text{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_W, \mathcal{O}_V)]_{r+s-\dim X},$$

and for any irreducible component  $Z$  of  $W \cap V$  the **intersection multiplicities**

$$e(X, W \cdot V, Z) = \sum_i (-1)^i \text{length}_{\mathcal{O}_{X,Z}} \text{Tor}_i^{\mathcal{O}_{X,Z}}(\mathcal{O}_{W,Z}, \mathcal{O}_{V,Z}).$$

$\lrcorner$

*Proof:* Why is this compatible with the definition given before? Why the multiplicity is given by this form?  $\square$

**Remark (9.1.7.14) [Serre Conjecture].** Serre conjectured that in (9.1.7.13), even if  $W, V$  doesn't intersect properly, for an irreducible component  $Z \subset W \cap V$  of dimension  $\geq \dim W + \dim V - \dim X$ , we have  $e(X, W \cdot V, P) = 0$ .  $\lrcorner$

**Prop. (9.1.7.15) [Discrete Case].** Let  $X$  be a integral scheme smooth over  $k$  and  $W, V \subset X$  be closed subvarieties that intersect properly. Let  $Z$  be an irreducible component of  $V \cap W$  with generic point  $\xi$  and  $\mathcal{O}_{W,\xi}, \mathcal{O}_{V,\xi}$  are both C.M., then

$$e(X, W \cdot V, Z) = \text{length}_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{V \cap W, \xi}).$$

 $\lrcorner$ 

*Proof:* Cf. [Sta]0B02?.

□

**Prop. (9.1.7.16) [Exterior Product of Subvarieties].** Let  $X, Y$  be integral schemes smooth over  $k$  and  $W \subset X, V \subset Y$  be subvarieties, then  $[W] \times [V] = ([W] \times [Y]) \cdot ([X] \times [V]) \in Z_*(X \times Y)$ .  $\lrcorner$

*Proof:* As  $W \times V$  is a variety with generic point  $\xi$ , and  $X$  is smooth over  $k$ ,  $X \times V$  is smooth over  $V$ , thus  $\mathcal{O}_{X \times V, \xi}$  is regular, hence C.M., and similarly is  $\mathcal{O}_{W \times Y, \xi}$ , thus (9.1.7.15) applies to show that  $([W] \times [Y]) \cdot ([X] \times [V]) = [W] \times [V]$ .  $\square$

**Prop. (9.1.7.17) [Serre].** If  $X$  is a integral scheme smooth over a field  $k$ ,  $\mathcal{F} \in \text{Coh}_{\leq r}(X), \mathcal{G} \in \text{Coh}_{\leq s}(X)$ , and  $\dim(\text{Supp}(\mathcal{F}) \cap \text{Supp}(\mathcal{G})) \leq r + s - \dim X$ . Then  $[\mathcal{F}]_r$  and  $[\mathcal{G}]_s$  intersect properly, and

$$[\mathcal{F}]_r \cdot [\mathcal{G}]_s = \sum_p (-1)^p [\text{Tor}_p^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})]_{r+s-\dim X}$$

 $\lrcorner$ 

*Proof:* Cf. [Sta]0B0W.

□

**Cor. (9.1.7.18).** If  $X$  is a smooth integral scheme over a field  $k$ , the intersection product on  $\text{CH}^*(X)$  makes it a commutative graded (associative, unital) ring.  $\lrcorner$

*Proof:* To show the intersection is associative, let  $U, V, W$  be closed subvarieties of  $X$  s.t.  $\text{codim}(U \cap V \cap W, X) = \text{codim}(U, X) + \text{codim}(V, X) + \text{codim}(W, X) = p$ , then it suffices to show that

$$\sum_i (-1)^{i+j} [\text{Tor}_j(\mathcal{O}_U, \text{Tor}_i(\mathcal{O}_V, \mathcal{O}_W))]_{\dim X - p} = \sum_i (-1)^{i+j} [\text{Tor}_j(\text{Tor}_i(\mathcal{O}_U, \mathcal{O}_V), \mathcal{O}_W)]_{\dim X - p}.$$

This is true because they are both equal to

$$\sum_k (-1)^k [H^k(\mathcal{O}_U \otimes^{\mathbb{L}} \mathcal{O}_V \otimes^{\mathbb{L}} \mathcal{O}_W)]_{\dim X - p}$$

which is because there is an spectral sequence convergence

$$E_2^{p,q} = \text{Tor}_{-p}(\mathcal{O}_U, \text{Tor}_{-q}(\mathcal{O}_V, \mathcal{O}_W)) \Rightarrow H^{p+q}(\mathcal{O}_U \otimes^{\mathbb{L}} \mathcal{O}_V \otimes^{\mathbb{L}} \mathcal{O}_W)$$

and use the fact length functions are additive.  $\square$

## 8 Numerical Geometry

Let  $k \in \mathbf{Field}$  and  $k = \bar{k}$ .

References are <http://www.math.columbia.edu/~chaoli/docs/IntersectionTheory.html#sec13>.

**Prop. (9.1.8.1)**[Schubert Cycles of the Grassmannian]. ┘

**Prop. (9.1.8.2).** Given 4 curves  $C_1, \dots, C_4$  in  $\mathbb{P}_k^3$  of degree  $d_1, \dots, d_4$ , there are  $2d_1d_2d_3d_4$  lines intersecting all of them. ┘

*Proof:* □

**Prop. (9.1.8.3).** Use the projective bundle formula to show that given 9 lines  $L_1, \dots, L_8 \in \mathbb{P}_k^3$  in general position, there are 92 conics meeting all  $L_i$ . ┘

*Proof:* □

**Prop. (9.1.8.4).** Show that given two general twisted cubic curves  $C, C' \in \mathbb{P}_k^3$ , they have 10 common chords. ┘

*Proof:* □

**Prop. (9.1.8.5).** Show that given a general quintic surface  $S \in \mathbb{P}_k^3$ , there are 575 lines meeting  $S$  at only one point. ┘

*Proof:* □

**Prop. (9.1.8.6).** Show that given 5 general conics  $S_1, \dots, S_r$  in  $\mathbb{P}_k^3$ , there are 3264 conics tangent to all  $S_i$ . ┘

*Proof:* □

## 9 Riemann-Roch

Main references are [Ful98] Chap15, 18.

**Prop. (9.1.9.1)**[Grothendieck-Riemann-Roch]. Let  $f : X \rightarrow Y \in \mathbf{Sch}^{\text{loc.ft}}/S$  be proper smooth, and  $\mathcal{E} \in \mathbf{Coh}^{\text{free}}(X)$ ,  $\mathcal{T}_{X/Y} \in \mathbf{Coh}^{\text{free}}(X)$  is the relative tangent bundle, then

$$f_*(\text{Todd}(\mathcal{T}_{X/Y}) \text{ch}(\mathcal{E})) = \sum_i (-1)^i \text{ch}(R^i f_* \mathcal{E}).$$

┘

**Remark (9.1.9.2).** This theorem also has an Arithmetic version. ┘

*Proof:* □

**Cor. (9.1.9.3)**[Hirzebruch-Riemann-Roch]. If  $k \in \mathbf{Field}$  and  $X \in \mathbf{SmPrpr}/k$ ,  $\mathcal{E} \in \mathbf{Coh}^{\text{free}}(X)$ , then

$$\chi(X, \mathcal{E}) = \int_X \text{Todd}(\mathcal{T}_{X/k}) \text{ch}(\mathcal{E}).$$

In particular,

$$\chi(X, \mathcal{O}_X) = \int_X \text{Todd}(\mathcal{T}_{X/k}).$$

┘

## Singular Case

### 10 Numerical Equivalences

**Def. (9.1.10.1)[Numerical Equivalences].** Two cycles  $\gamma, \gamma' \in \text{CH}^k(X)$  are called **numerically equivalent** if for any  $\delta \in Z_k(X)$ ,  $\gamma \cdot \delta = \gamma' \cdot \delta$ . And we can define  $\text{CH}_{\text{num}}^k(X)$  the **Grothendieck group of codimension- $k$ -cycles** to be  $Z^k(X)$  modulo the numerical equivalence relation.  $\lrcorner$

**Prop. (9.1.10.2)[Grothendieck Rings].** There is a group homomorphism  $\text{CH}^*(X) \rightarrow \text{CH}_{\text{num}}^*(X)$  and the ring structure on  $\text{CH}^*(X)$  descends to  $\text{CH}_{\text{num}}^*(X)$ , making it a ring, called the **Grothendieck ring of  $X$** .  $\lrcorner$

**Prop. (9.1.10.3).** Each  $\text{CH}_{\text{num}}^k(X)$  is a finite free Abelian group.  $\lrcorner$

*Proof:* Use the Poincaré pairing on the étale cohomology, and the fact  $\gamma_X(\text{CH}^{2r-2k}(X)) \subset H_{\text{ét}}^{2r-2k}(X, \overline{\mathbb{Q}}_\ell)$  is a finite  $\overline{\mathbb{Q}}_\ell$ -vector space.  $\square$

### 11 Intersection for Line Bundles

References are [FGA, Appendix B].

## Algebraic Equivalence

**Def. (9.1.11.1)[Algebraically Equivalent Line Bundles].** Let  $X$  be a scheme over a field  $k$ , then  $\mathcal{L}_1, \mathcal{L}_2 \in \text{Pic}(X)$  are called **algebraically equivalent** if they are equivalent in the equivalence relation  $\mathcal{M} \sim \mathcal{N}$  iff there is a connected scheme  $T$  and a line bundle  $\mathcal{L} \in \underline{\text{Pic}}_{X/k}(T)$  s.t.  $\mathcal{N}_{k(t_1)} = \mathcal{L}_{t_1}$  and  $\mathcal{M}_{k(t_2)} = \mathcal{L}_{t_2}$  where  $t_1, t_2 \in T$ .  $\lrcorner$

**Cor. (9.1.11.2)[ $\text{Pic}^0(X)$ ].** The elements in  $\text{Pic}(X)$  algebraically equivalent to 0 is a subgroup of  $\text{Pic}(X)$ , denoted by  $\underline{\text{Pic}}^0(X)$ . In (9.1.11.4), we will see when  $X$  has a rational point and  $\underline{\text{Pic}}_{X/k}$  is representable, this is just  $\underline{\text{Pic}}_{X/k}^0(k)$ , by (10.9.3.8).  $\lrcorner$

**Prop. (9.1.11.3)[Algebraically Equivalent Divisors].** If  $X$  be a regular  $K$ -prevariety, then  $\text{Pic}(X) \cong \text{CH}^1(X)$  (7.5.3.15). We call two divisors  $D_1, D_2$  on  $X$  **algebraically equivalent** if  $\mathcal{O}(D_1)$  and  $\mathcal{O}(D_2)$  are algebraically equivalent.  $\lrcorner$

*Proof:* Cf.[Diophantine Geometry, P563].  $\square$

**Prop. (9.1.11.4)[Moduli Characterization].** If  $X$  is a scheme over a field  $k$  s.t.  $\underline{\text{Pic}}_{X/k}$  is representable, then two line bundles are algebraically equivalent iff they corresponds to points of  $\underline{\text{Pic}}_{X/k}$  in the same connected component.  $\lrcorner$

*Proof:* By (10.9.3.21),  $\underline{\text{Pic}}_{X/k}$  is a locally algebraic group scheme over  $k$ , so by (10.1.4.15), every connected components of it is irreducible. Clearly two algebraically equivalent line bundles are in the same connected component of  $\underline{\text{Pic}}_{X/k}$ . Conversely, the inclusion of their common connected component  $P$  corresponds to a line bundle on some fppf covering  $T \rightarrow P$ . Then use the fact fppf covering is open map, and the fact  $P$  is irreducible to get an algebraic equivalence chain.  $\square$

**Prop. (9.1.11.5)[Algebraic Equivalence for Curves].** If  $C$  is a smooth complete curve over a field  $k$ , then two line bundles  $\mathcal{L}_1, \mathcal{L}_2 \in \text{Pic}(C)$  are algebraically equivalent iff they have the same degree.  $\lrcorner$

*Proof:* It suffices to show after base change to  $\bar{k}$ . Then it suffices to show for any two closed points  $x_1, x_2 \in C$ ,  $\mathcal{L}(x_1) \sim \mathcal{L}(x_2)$ . Consider the diagonal of  $C \times C$  is a Cartier divisor, and its restriction to  $C \times \{x_i\}$  is  $\mathcal{L}(x_i)$ .  $\square$

### Numerical Intersections

**Def. (9.1.11.6).** Let  $X$  be a proper scheme and  $\mathcal{F}$  be a coherent sheaf with  $\dim \text{Supp } \mathcal{F} \leq n$ , and  $\mathcal{L}_1, \dots, \mathcal{L}_n$  are invertible sheaves on  $X$ , then we define  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; \mathcal{F})$  to be

$$\sum_{\{i_1, \dots, i_m\} \subset \{1, \dots, n\}} (-1)^m \chi(X, \mathcal{L}_{i_1}^\vee \otimes \dots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}).$$

When  $\mathcal{F}$  is the structure sheaf of a closed subscheme  $Y$ , then denote  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; \mathcal{F})$  by  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; \mathcal{Y})$ . This intersection is stable under base change of fields.  $\lrcorner$

**Prop. (9.1.11.7).** If  $X$  is a complete curve and  $\mathcal{L}$  an invertible sheaf on  $X$ , then  $(\mathcal{L}; X) = \deg(\mathcal{L})$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (9.1.11.8).** If  $k$  is an infinite field and  $X = \mathbb{P}_k^n$ , and  $Y$  is a dimension  $n$  subvariety of  $X$ . If  $H_1, \dots, H_n$  are generic chosen hypersurfaces of degree  $d_1, \dots, d_n$  resp., then

$$(\mathcal{O}_X(H_1) \cdot \dots \cdot \mathcal{O}_X(H_n); Y) = d_1 \dots d_n \deg(Y).$$

$\lrcorner$

*Proof:*  $\square$

**Prop. (9.1.11.9).** If  $D$  is an effective Cartier divisor on  $X$  that doesn't contain any associated point of  $\mathcal{F}$ , then

$$(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n \cdot \mathcal{O}(D); \mathcal{F}) = (\mathcal{L}_1|_Y \cdot \dots \cdot \mathcal{L}_n|_Y; \mathcal{F}|_D)$$

$\lrcorner$

*Proof:*  $\square$

**Prop. (9.1.11.10).** For a fixed  $\mathcal{F}$ ,  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; \mathcal{F})$  is a symmetric multilinear function of  $\mathcal{L}_1, \dots, \mathcal{L}_n$ , where the addition is tensor production. Moreover, if  $\mathcal{F}$  is a coherent sheaf with support dimension  $n$ , then  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_{n+1}; \mathcal{F}) = 0$ .  $\lrcorner$

*Proof:* Cf. [Rising Sea, P544].  $\square$

**Prop. (9.1.11.11).** The numerical intersection  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; \mathcal{F})$  only depends on the numerical classes of  $\mathcal{L}_i$ .  $\lrcorner$

**Prop. (9.1.11.12) [Projection Formula].** Let  $\pi : X_1 \rightarrow X_2$  be a proper morphism of proper schemes, then

$$(\pi^* \mathcal{L}_1 \cdot \dots \cdot \pi^* \mathcal{L}_n; \mathcal{F}) = (\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; \pi_* \mathcal{F}).$$

In particular, when  $X_1, X_2$  are integral with the same dimensions and  $\pi$  is a finite map,

$$(\pi^* \mathcal{L}_1 \cdot \dots \cdot \pi^* \mathcal{L}_n) = \deg(\pi) (\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n).$$

$\lrcorner$

*Proof:*

For the last assertion, notice  $\pi_* \mathcal{O}_{X_1} = \deg(\pi) \mathcal{O}_{X_2} +$  coherent sheaves with smaller support dimensions.  $\square$



**Prop. (9.1.11.13).** Let  $k$  be a field and  $X$  a proper scheme over  $k$ , and  $Z \subset X$  a closed subscheme of dimension  $d$ . If  $\mathcal{L}_1, \dots, \mathcal{L}_d$  are ample invertible sheaves on  $X$ , then  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_d; Z)$  is positive.  $\lrcorner$

*Proof:*  $\square$

**Prop. (9.1.11.14).** Let  $X$  be a complex projective scheme, then  $(\mathcal{L}_1 \cdot \dots \cdot \mathcal{L}_n; Z)$  equals

$$(c_1((\mathcal{L}_1)_{an}) \cup \dots \cup c_1((\mathcal{L}_n)_{an}), Z)$$

where  $c_1$  is the complex Chern class.  $\lrcorner$

*Proof:* Cf. [Rising Sea, P547].  $\square$

**Def. (9.1.11.15) [First Chern Class].** Let  $X$  be a proper scheme and  $K$  be the Grothendieck group of  $\text{Coh}(X)$ . For  $\mathcal{L} \in \text{Pic}(X)$ , define  $c_1(\mathcal{L})$  to be the endomorphism of  $K$  defined by

$$c_1(\mathcal{L})\mathcal{F} = \mathcal{F} - \mathcal{L}^{-1} \otimes \mathcal{F}.$$

**Prop. (9.1.11.16).** If  $\mathcal{L}, \mathcal{M} \in \text{Pic}(X)$ , then

- $c_1(\mathcal{L})c_1(\mathcal{M}) = c_1(\mathcal{L}) + c_1(\mathcal{M}) - c_1(\mathcal{L} \otimes \mathcal{M})$ .
- $c_1(\mathcal{O}_X) = 0$ .
- $c_1(\mathcal{L})c_1(\mathcal{L}^{-1}) = c_1(\mathcal{L}) + c_1(\mathcal{L}^{-1})$ .

In particular,  $c_1(\mathcal{L})$  and  $c_2(\mathcal{M})$  commutes.  $\lrcorner$

**Prop. (9.1.11.17).** If  $X$  is a proper scheme over a field  $k$  and let  $K^d$  be the subgroup generated by coherent sheaves over  $X$  with support dimension  $\leq d$ , then if  $\mathcal{F} \in K^d$ , then  $c_1(\mathcal{L})\mathcal{F} \in K^{d-1}$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (9.1.11.18).** If  $Y \subset X$  is a closed subscheme, and  $\mathcal{L} \in \text{Pic}(X)$  satisfy  $\mathcal{L}_Y$  is an effective Cartier divisor  $D$ , then  $c_1(\mathcal{L})[Y] = [D]$ .  $\lrcorner$

*Proof:* There is an exact sequence  $0 \rightarrow \mathcal{O}_Y(-D) \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_D$ , and  $\mathcal{O}_Y(-D) \cong \mathcal{L}^{-1} \otimes \mathcal{O}_Y$ .  $\square$

### $\tau$ -Equivalences

**Def. (9.1.11.19) [ $\tau$ -Equivalence].** Let  $X \in \text{Sch}/k$ , then two line bundles  $\mathcal{L}_1, \mathcal{L}_2$  are called  **$\tau$ -equivalent** if there are some  $m \in \mathbb{Z}_+$  s.t.  $\mathcal{L}_1^{\otimes m} \sim \mathcal{L}_2^{\otimes m}$ .  $\lrcorner$

**Prop. (9.1.11.20) [Moduli Characterization].** Let  $X \in \text{Sch}/k$  and  $\underline{\text{Pic}}_{X/k}$  is representable, then  $\mathcal{L} \in \text{Pic}(X)$  is  $\tau$ -equivalent to  $\mathcal{O}_X$  iff its corresponding point  $\lambda \in \underline{\text{Pic}}_{X/k}$  is in  $\underline{\text{Pic}}_{X/k}^\tau$ .  $\lrcorner$

*Proof:* Clear from the definition of  $\underline{\text{Pic}}_{X/k}^\tau$ .  $\square$

**Def. (9.1.11.21) [Numerically Equivalence].** Two line bundles  $L_1, L_2$  on a complete prevariety over a field  $K$  is called **numerically equivalent** if  $c_1(L_1) \cdot \alpha = c_1(L_2) \cdot \alpha$  for any complete precurve  $C \subset X$ . Two divisors on  $X$  is called **numerically equivalent** if their corresponding line bundle do.

Numerically trivial line bundles form a group, and are stable under proper pullbacks, by (7.13.2.7)(7.13.2.5).

**?** How is this definition equivalent to that of (9.1.10.1)?  $\lrcorner$

**Prop. (9.1.11.22)[Algebraic and Numerical Equivalences].** Let  $\mathcal{L}_1, \mathcal{L}_2$  be algebraically equivalent line bundles on a proper scheme  $X/k$ , then  $\deg_{\mathcal{L}_1}(X) = \deg_{\mathcal{L}_2}(X)$ . In particular, algebraically equivalent divisors are numerically equivalent.  $\lrcorner$

*Proof:* This is because in this case  $\mathcal{L}_1, \mathcal{L}_2$  are in the same connected component of  $\text{Pic}_{X/k}$ .  $\square$

**Cor. (9.1.11.23).** Let  $X$  be a complete prevariety over  $K$  and  $L_i$  be line bundles and  $Z \in Z_r(X)$ , then  $\deg(c_1(L_1).c_1(L_2).\dots.c_r(L_r).Z)$  only depends on algebraic equivalence classes of  $L_i$ .  $\lrcorner$

*Proof:* Cf.[Diophantine Geometry, P562].  $\square$

**Def. (9.1.11.24)[Bounded Set of Line Bundles].** Let  $X$  be a scheme over  $S$ , then a set  $\Lambda \subset \text{Pic}(X_k)$  where  $\text{Spec } k \rightarrow S$  are points are called **bounded** if there is a  $T \in \text{Sch}^{\text{ft}}/S$  and a line bundle  $\mathcal{M}$  on  $X_T$  s.t. for any  $\mathcal{L} \in \Lambda$ , there is some schematic point  $\text{Spec } k \rightarrow T$  s.t.  $\mathcal{L} \cong \mathcal{M}_k$ .  $\lrcorner$

**Prop. (9.1.11.25) [Numerical and  $\tau$ -Equivalence].** If  $X$  is proper over an alg.closed field  $k$  and  $\mathcal{L} \in \text{Pic}(X)$ , then the following are equivalent:

- $\mathcal{L}$  is  $\tau$ -equivalent to  $\mathcal{O}_X$ .
- $\mathcal{L}$  is numerically equivalent to  $\mathcal{O}_X$ .
- The family  $\{\mathcal{L}^{\oplus p} | p \in \mathbb{Z}\}$  is bounded.
- For any  $\mathcal{F} \in \text{Coh}(X)$ ,  $\chi(\mathcal{F} \otimes \mathcal{L}) = \chi(\mathcal{F})$ .
- For any  $p \in \mathbb{Z}$ ,  $\mathcal{L}^{\oplus p}(1)$  is ample.
- 

$\lrcorner$

*Proof:* Cf.[Kle05]P52, 57. is this true for  $k$  separably closed?  $\square$

**Prop. (9.1.11.26).** If  $X$  is projective over an alg.closed field  $k$ , then the set of line bundles numerically equivalent to  $\mathcal{O}_X$  is bounded.  $\lrcorner$

*Proof:* See the proof of [Kle05]P52.  $\square$

**Prop. (9.1.11.27).** If  $X$  is projective over an alg.closed field  $k$ , then there exists an  $m \in \mathbb{Z}$  s.t. for any  $\mathcal{L} \in \text{Pic}(X)$  numerically equivalent to  $\mathcal{O}_X$ ,  $\mathcal{L}$  is  $m$ -regular and  $\chi(\mathcal{L}(n)) = \chi(\mathcal{O}_X(n))$  for any  $n \in \mathbb{Z}$ .  $\lrcorner$

*Proof:* Cf.[Kle05]P55.  $\square$

### Nef Line Bundles

**Def. (9.1.11.28)[Nef Line Bundles].** A **nef line bundle** or **numerically effective line bundle** on a complete prevariety over a field  $k$  is a line bundle  $\mathcal{L}$  s.t.  $c_1(\mathcal{L}).C \geq 0$  for any complete precurve  $C \subset X$ .

Nef line bundles form a semigroup, and are stable under proper pullbacks, by (7.13.2.7)(7.13.2.5).  $\lrcorner$

**Prop. (9.1.11.29).** Ample line bundles are nef, by (7.13.2.20).  $\lrcorner$

Surface Case

**Def. (9.1.11.30) [(-1)-Curves].** Let  $X$  be a complete curve over a field  $k$ , then a **(-1)-curve** on  $X$  is a curve  $C$  consisting of smooth points,  $C \cong \mathbb{P}_k^1$  and  $C \cdot C = -1$ . ┘

**Prop. (9.1.11.31) [Hodge Index Theorem].** Let  $X$  is a smooth surface over a field  $k$  and  $\mathcal{L}, \mathcal{H} \in \text{Pic}(X)$  with  $\mathcal{H} \cdot \mathcal{H} > 0$  and  $\mathcal{L} \cdot \mathcal{H} = 0$ , then  $\mathcal{L} \cdot \mathcal{L} = 0$ , and the equality holds iff  $\mathcal{L}$  is numerically trivial. ┘

*Proof:* Cf. [Rising Sea, P552]. □

**Prop. (9.1.11.32) [ $\mathbb{P}^1 \times \mathbb{P}^1$ ].** Let  $X = \mathbb{P}^1 \times \mathbb{P}^1$ ,  $l = \mathbb{P}^1 \times \{0\}$  and  $m = \{0\} \times \mathbb{P}^1$ , then  $l \cdot l = m \cdot m = 0$ , and  $l \cdot m = 1$ . ┘

## 9.2 Algebraic deRham Cohomology

References are [algebraic de Rham Cohomology, Clausen] and [Gro66a, Gro15, Sta].

**Notation (9.2.0.1).**

- Use notations from [Spectral Algebraic Geometry \(Lurie\)](#).

┘

### 1 deRham Complexes

**Def. (9.2.1.1) [Absolute deRham Complex].** For  $B \in \mathcal{CRing}$ , let  $\Omega_B = \Omega_{B/\mathbb{Z}}$ , and  $\Omega_B^i = \wedge^i \Omega_B$ , then there is a **total de Rham complex** of  $B$ :

$$B \rightarrow \Omega_B^1 \rightarrow \Omega_B^2 \rightarrow \dots$$

as  $B$ -modules, which is a complex, where  $d(b_0 db_1 \wedge \dots \wedge db_p) = db_0 \wedge db_1 \wedge \dots \wedge db_p$ .

┘

*Proof:*  $d$  is well-defined on  $\Omega_B^1$  because it vanishes on the element  $d(a+b) - da - db$  and  $d(ab) - ad(b) - bd(a)$  by Leibniz rule, and then we get a map

$$\bigotimes_1^p \Omega_B \rightarrow \Omega_B^{p+1} : \omega_1 \otimes \dots \otimes \omega_p \mapsto \sum (-1)^{i+1} \omega_1 \wedge \dots \wedge d(\omega_i) \wedge \dots \wedge \omega_p.$$

We want to descend this to a map on  $\Omega_B^p$  using (5.1.1.23): it is clearly alternating, and it suffices to show it is  $f$ -linear, and this is clear by direct calculation.

Finally  $d^2 = 0$ .

□

**Prop. (9.2.1.2) [Quotient of de Rham Complexes].** Let  $B$  be a ring and  $\pi : \Omega_B \rightarrow \Omega$  be a surjective map of  $B$ -modules. Denote  $d : B \rightarrow \Omega_B \rightarrow \Omega$ , and  $\Omega^i = \wedge^i \Omega$ . Assume that the kernel of  $\pi$  is generated as a  $B$ -module by elements  $\omega$  that  $\wedge^2(\pi)(d_B(\omega)) = 0$  in  $\Omega^2$ , then there is a de Rham complex

$$B \rightarrow \Omega^1 \rightarrow \Omega^2 \rightarrow \dots$$

whose differential is defined by the rules similar to that of (9.2.1.1).

┘

*Proof:* Because  $\pi$  is surjective, so do  $\wedge^i \pi$ , and it suffices to show  $\wedge \pi$  gives a connecting morphism between  $\Omega_B^\bullet$  and  $\Omega^\bullet$ , then the well-definedness of  $d$  is automatic. Cf. [[Sta]07HY].

□

**Cor. (9.2.1.3) [Relative de Rham Complex].** If  $B$  is an  $A$ -algebra, the surjection  $\Omega_B \rightarrow \Omega_{B/A}$  satisfies the condition of (9.2.1.2) thus we can define the **relative deRham complex**  $\Omega_{B/A}^\bullet$ .

┘

*Proof:* The verification of the condition is routine.

□

**Prop. (9.2.1.4) [Universality of deRham Complexes].** Let  $C$  be a  $B$ -algebra and  $(E^\bullet, d)$  a non-negatively graded commutative  $B$ -dga and we are given a  $B$ -algebra map  $\eta : C \rightarrow E^0$  that for every  $x \in C$ , the element  $d(\eta(x)) \in E^1$  satisfies  $d^2 = 0$ , then the map  $C \rightarrow E^0$  extends uniquely to a map  $\Omega_{C/B}^\bullet \rightarrow E^\bullet$  of  $B$ -dga.

┘

*Proof:* One direction is trivial as  $\Omega_{C/B}^\bullet$  is strict by construction. Conversely, the composite map  $C \rightarrow E^0 \rightarrow E^1$  is a  $B$ -derivation thus extends to a map  $\eta^1 : \Omega_{C/B}^1 \rightarrow E^1$ , then the universal property of the exterior product (5.1.1.23) gives maps  $\eta^i : \Omega_{C/B}^i \rightarrow E^i$ , and this gives the desired extension. □

**Def. (9.2.1.5) [Connection].** Let  $B$  be a ring and  $\Omega_B \rightarrow \Omega$  be a quotient satisfying the condition of (9.2.1.2), then a **connection** on  $M$  is an additive map

$$\nabla : M \mapsto M \otimes_B \Omega : \quad \nabla(b \otimes m) = b\nabla(m) + m \otimes db$$

Given a connection on  $M$ , we can define maps

$$\nabla : M \otimes_B \Omega^i \rightarrow M \otimes_B \Omega^{i+1}, \quad \nabla(b \otimes \omega) = \nabla(b) \wedge \omega + m \wedge d\omega$$

This is well defined because it commutes it commutes with  $B$ -action. The connection is called **integrable** if  $\nabla^2 = 0$ .  $\lrcorner$

**Def. (9.2.1.6) [deRham Complexes for Schemes].**  $\lrcorner$

### Cohomologies

**Def. (9.2.1.7) [Algebraic de Rham Cohomology].** Let  $X \rightarrow S$  be a morphism of rings, then we define the **algebraic de Rham cohomology**  $H_{\text{dR}}^\bullet(X/S)$  of  $X$  over  $S$  as the image of the de Rham complex  $\Omega_{X/S}^\bullet$  in  $D(\text{Mod}(\mathcal{O}_X))$ .

Notice that if  $X$  is affine, then by (7.5.5.1) and (7.7.1.3),  $H_{\text{dR}}^n(\Omega_X^\bullet) = H^n(\Gamma(X, \Omega_{X/S}^\bullet))$ .  $\lrcorner$

**Prop. (9.2.1.8).** There is a similar construction of connections on a f.g. projective  $R$ -module  $M$  and Weil-Chern theory parallel to that of 5 and 4.

But in this case, the trace map is defined only when  $M$  is f.g. projective, which is called the **Hattoris-Stallings trace**: If  $A$  is f.g. projective, the natural map  $\text{Hom}_R(A, R) \otimes_R A \rightarrow \text{End}_A(P)$  is an isomorphism (Because locally it is an isomorphism??), and the inverse composed with  $\text{Hom}_R(A, R) \otimes_R A \rightarrow A$ , we get the desired map.

Also, when  $M$  is f.g. projective, there is a **Levi-Cevita connection** induced by the  $A \rightarrow \Omega_{A/R}^1$  because  $M$  is a direct summand of some  $A^n$ . This is verified to be independent of  $n$ , or one can more algeoly use the fact that projective module is locally free.

The Chern character is important, it defines a ring map from  $K_0(R)$  to  $H_{\text{dR}}^{ev}(A)$ . In fact, this can be lifted to a morphism  $K_0(A) \rightarrow HC_0^{\text{perf}}(A) \rightarrow H_{\text{dR}}^{ev}(A)$ , Cf.[阳恩林 循环同调 Dennis trace].  $\lrcorner$

**Def. (9.2.1.9) [Cup Products].**  $\lrcorner$

**Prop. (9.2.1.10) [Grothendieck].** For  $X \in \text{Sch}^{\text{sm}}/\mathbb{C}$ , there is a functorial equivalence

$$R\Gamma(X; \Omega_{X/\mathbb{C}}^\bullet) \cong R\Gamma(X^{\text{an}}; \Omega_{X^{\text{an}}/\mathbb{C}}^{\text{an}, \bullet}).$$

*Proof:*  $\square$

**Remark (9.2.1.11).** Notice if  $X$  is smooth and proper, then this follows from GAGA.  $\lrcorner$

**Def. (9.2.1.12) [Hodge Cohomologies].** For a morphism  $X \rightarrow S \in \text{Sch}$ , define the **Hodge cohomology** to be the graded  $\Gamma(X, \mathcal{O}_X)$ -algebra

$$H_{\text{Hdg}}^*(X/S) = \bigoplus_{n \geq 0} H_{\text{Hdg}}^n(X/S) = \bigoplus_{n \geq 0} \bigoplus_{p+q=n} H^p(X, \Omega_{X/S}^q) \quad (9.2.1.6)$$

with the  $\Gamma(X, \mathcal{O}_X)$ -linear cup product

$$\cup : H^p(X, \Omega_{X/S}^q) \otimes H^{p'}(X, \Omega_{X/S}^{q'}) \rightarrow H^{p+p'}(X, \Omega_{X/S}^{q+q'})$$

given by [Sta]068G<sup>?</sup>. It is associative and graded commutative.  $\lrcorner$

**Cor. (9.2.1.13).** For each map of morphisms  $f : (X' \rightarrow S') \rightarrow (X \rightarrow S)$ , there are pullback maps

$$f^* : H_{\text{Hdg}}^*(X/S) \rightarrow H_{\text{Hdg}}^*(X'/S')$$

compatible with gradings and cup products and the homomorphism  $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X', \mathcal{O}_{X'})$ , which is induced by  $f^* : f^{-1}\Omega_{X/S} \rightarrow \Omega_{X'/S'}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (9.2.1.14) [Hodge-to-deRham Spectral Sequence and Conjugate Spectral Sequences].** For  $X \rightarrow S \in \text{Sch}$ , using the Cartan-Eilenberg resolution of  $\Omega_{X/S}^\bullet$ , we get a **Hodge-to-deRham spectral sequence** (Cf. [Sta]015J ?)

$$E_1^{p,q} = H^q(X; \Omega_{X/S}^q) \implies H_{\text{dR}}^n(X/S).$$

(with arrows direction from  $p$  to  $q$ ), and a **conjugate spectral sequence**

$$E_2^{p,q} = H^p(X, \mathcal{H}^q(\Omega_{X/S}^\bullet)) \implies H_{\text{dR}}^n(X/S),$$

(with arrows direction from  $p$  to  $q$ ).  $\lrcorner$

**Cor. (9.2.1.15).** Situation as in (9.2.1.14), the Hodge-to deRham spectral sequence gives:

$$H_{\text{dR}}^0(X/S) = \ker(d : H^0(X, \mathcal{O}_X) \rightarrow H^0(X, \Omega_{X/S}^1)),$$

and the conjugate spectral sequence gives  $H_{\text{dR}}^0(X/S) = H^0(X, \mathcal{H}^0(\Omega_{X/S}^\bullet))$  and a long exact sequence

$$0 \rightarrow H^1(X, \mathcal{H}^0(\Omega_{X/S}^\bullet)) \rightarrow H_{\text{dR}}^1(X/S) \rightarrow H^0(X, \mathcal{H}^1(\Omega_{X/S}^\bullet)) \rightarrow H^2(X, \mathcal{H}^0(\Omega_{X/S}^\bullet)) \rightarrow H_{\text{dR}}^2(X/S).$$

$\lrcorner$

**Def. (9.2.1.16) [Hodge-Filtration].** For  $X \rightarrow S \in \text{Sch}$ , the Hodge-to-deRham spectral sequence (9.2.1.14) induces a **Hodge-filtration** on each  $H_{\text{dR}}^*(X/S)$ , i.e.

$$\text{Fil}^p H_{\text{dR}}^n(X/S) = \text{Im} \left( H^n(X, \sigma_{\geq p} \Omega_{X/S}^\bullet) \rightarrow H_{\text{dR}}^n(X/S) \right).$$

which satisfies

$$\text{Fil}^i H_{\text{dR}}^n(X/S) \cup \text{Fil}^j H_{\text{dR}}^m(X/S) \subset \text{Fil}^{i+j} H_{\text{dR}}^{n+m}(X/S)??$$

$\lrcorner$

### Curve case

**Prop. (9.2.1.17).** Let  $X$  be a smooth curve over  $\mathbb{C}$  with a smooth completion  $\overline{X}$ , and denote  $S = \overline{X} \setminus X$ . Then there is an exact sequence of sheaves

$$0 \rightarrow \Omega_{\overline{X}}^\bullet \rightarrow \Omega_{\overline{X}}^\bullet(\log S) \xrightarrow{\text{res}} i_* \Omega_S^{\bullet-1} \rightarrow 0, ?$$

inducing a  $\lrcorner$

**Thm. (9.2.1.18) [Deligne].** Situation as in (9.2.1.17), there is a canonical isomorphism

$$\mathbb{H}^n(\Omega_X^\bullet(\log S)) \cong \mathbb{H}^n(\Omega_Y^\bullet).$$

$\lrcorner$

## 2 Infinitesimal Sites

## 3 Poincaré-Duality

**Def. (9.2.3.1) [Situation].** Let  $S$  be a qcqs scheme and  $f : X \rightarrow S$  is a proper smooth morphism of schemes whose fiber are all equidimensional of dimension  $d$ .  $\lrcorner$

**Prop. (9.2.3.2) [Relative Poincaré duality].** In situation (9.2.3.1) there is a canonical  $\mathcal{O}_S$ -module map

$$t : Rf_*\Omega_{X/S}^d[d] \rightarrow \mathcal{O}_S$$

s.t. for any  $p$ , the pairing

$$Rf_*\Omega_{X/S}^p \otimes_{\mathcal{O}_S}^L Rf_*\Omega_{X/S}^{n-p} \rightarrow \mathcal{O}_S$$

induced from relative cup product and  $t$  is a perfect pairing of perfect complexes in  $D(\mathcal{O}_S)$  that is compatible perfect under base change. And it also induces a perfect  $\mathcal{O}_S$ -bilinear pairing

$$f_*\mathcal{O}_X \otimes R^df_*\Omega_{X/S}^d \rightarrow \mathcal{O}_S$$

compatible with base change.  $\lrcorner$

*Proof:* Cf. [Sta]0G8I.  $\square$

**Cor. (9.2.3.3).** If  $S$  is Noetherian and  $X$  is a proper smooth variety over  $S$ , then there is an isomorphism  $R^df_*\Omega_{X/S}^d \cong \mathcal{O}_S$  that is compatible with base change.  $\lrcorner$

## 9.3 Étale Fundamental Groups

References are [Gro71, K-M85].

### 1 Étale Connected Components

**Def. (9.3.1.1) [Étale Connected Components].** Let  $k \in \mathbf{Field}$  and  $X \in \mathbf{Sch}/k$ , define  $\pi_0(X)$  to be the spectrum of the largest étale subalgebra  $\pi(X)$  of  $\Gamma(X, \mathcal{O}_X)$  (5.4.7.27), called the **étale connected component** of  $X$ .

Then  $\pi_0 : \mathbf{Sch}/k \rightarrow (\mathbf{Spec} k)_{\text{ét}}$  is a functor.  $\lrcorner$

**Prop. (9.3.1.2).** Let  $X$  be a locally algebraic scheme over a field  $k$ , then

- for any field extension  $k'/k$ ,  $\pi_0(X_{k'}) = \pi_0(X)_{k'}$ .
- Let  $Y$  be a schemes over a field  $k$ , then  $\pi_0(X \times Y) = \pi_0(X) \times \pi_0(Y)$  and  $\pi_0(X \amalg Y) = \pi_0(X) \amalg \pi_0(Y)$ .  $\lrcorner$

*Proof:* 1: Cf. [Mil17b] P15.

2: There is a map  $\pi(X) \times_k \pi(Y) \rightarrow \pi(X \times Y)$ . Because  $\pi$  commutes with base change, we can base change to separable closure. In this case, it suffices to show if  $X, Y$  is connected then  $X \times Y$  is connected, but this follows from (7.4.3.12).  $\square$

**Prop. (9.3.1.3).** For  $X \in \mathbf{Sch}^{\text{loc.ft}}/k$ ,

- The mapping  $\text{pr} : X \rightarrow \pi_0(X)$  induces a bijection between  $|\pi_0(X)|_0$  and connected components of  $X$ .
- For all  $x \in |\pi_0(X)|$ , the fiber  $\text{pr}^{-1}(x)$  is geo.connected over  $\kappa(x)$ .
- $X \rightarrow \pi_0(X)$  is faithfully flat.  $\lrcorner$

*Proof:*  $|\pi_0(X)|$  is discrete, so the inverse image of each point  $x \in |\pi(X)|_0$  is a sum of connected components of  $X$ . But  $\pi_0(X_x) = \kappa(x)$ , so  $X_x$  must be connected by (9.3.1.2). Also, this implies that  $\pi_0(X_{\bar{\kappa}(x)}) = \pi_0(X_x) \times_{\kappa(x)} \bar{\kappa}(x) = \bar{\kappa}(x)$ , thus  $X_x$  is geo.connected.  $\square$

### 2 Étale Fundamental Groups

References are [Sta]Chap53.

**Prop. (9.3.2.1) [Connected Finite Étale Schemes].** For  $X \in \mathbf{Sch}_{\text{cntd}}$ ,  $Y \in X_{\text{fét}}$  is a connected object (4.1.1.21) iff  $Y$  is a connected scheme.  $\lrcorner$

*Proof:* Cf. [Sta]0BNB. ?  $\square$

**Prop. (9.3.2.2) [Étale Fundamental Groups].** For  $X \in \mathbf{Sch}_{\text{cntd}}$  and  $\bar{x}$  a geometric point of  $X$ , then the fibre functor

$$F_{\bar{x}} : X_{\text{fét}} \rightarrow \mathbf{finSet} : Y \mapsto |Y_{\bar{x}}|$$

makes  $X_{\text{fét}}$  a Galois-category (4.3.4.4). And we denote

$$\pi_1(X, \bar{x}) \triangleq \text{Aut}(F_{\bar{x}}),$$

called the **étale fundamental group** of  $X$ .  $\lrcorner$



*Proof:* Identifying  $(\bar{x})_{\text{fét}}$  with  $\text{finSet}$  (7.6.6.6), we see that  $F_{\bar{x}}$  is just the base change functor for the morphism  $\bar{x} \rightarrow X$ , thus  $X_{\text{fét}}$  has finite limits and colimits and  $F_{\bar{x}}$  is exact by (7.6.6.15).

$Y \in X_{\text{fét}}$  is a connected object (4.1.1.21) iff  $Y$  is a connected scheme by (9.3.2.1), then we show that each  $Y \in X_{\text{fét}}$  is a finite co-product of connected schemes?.

To show  $F_{\bar{x}}$  reflects isomorphisms, suppose  $Y \rightarrow Z \in X_{\text{fét}}$  s.t.  $F_{\bar{x}}(Y) \rightarrow F_{\bar{x}}(Z)$  is an isomorphism, to show  $Y \rightarrow Z$  is an isomorphism, by the above we may assume that  $Z$  is connected. Because  $\mathcal{O}_Y$  is finite locally free over  $\mathcal{O}_Z$ , it suffices to show that it is locally free of rank 1. But the rank is constant, and when base changed to  $\bar{x}$ , it is of rank 1.  $\square$

**Prop. (9.3.2.3) [Properties of Étale Fundamental Groups].** For  $X \in \text{Sch}_{\text{cntd}}$  with a geometric point  $\bar{x}$ ,

- The fiber functor  $F_{\bar{x}}$  (9.3.2.2) defines an equivalence of categories

$$X_{\text{fét}} \cong \pi_1(X, \bar{x})\text{-finSet} \text{ (9.3.2.2)}.$$

- If  $\bar{x}'$  is another geometric point of  $X$ , then there exists an isomorphism  $t : F_{\bar{x}} \cong F_{\bar{x}'}$  which induces an isomorphism  $\pi_1(X, \bar{x}) \cong \pi_1(X, \bar{x}')$  compatible with the equivalences in item 1. This isomorphism is independent of  $t$  up to inner automorphisms.
- For  $f : X \rightarrow Y \in \text{Sch}_{\text{cntd}}$ , denote  $\bar{y} = f \circ \bar{x}$ , then there is a canonical continuous homomorphism  $f_* : \pi_1(X, \bar{x}) \rightarrow \pi_1(Y, \bar{y})$  s.t. the following commutative diagram holds:

$$\begin{array}{ccc} Y_{\text{fét}} & \xrightarrow{f^*} & X_{\text{fét}} \\ \downarrow F_{\bar{y}} & & \downarrow F_{\bar{x}} \\ \pi_1(X, \bar{x})\text{-finSet} & \xrightarrow{f_*} & \pi_1(Y, \bar{y})\text{-finSet} \end{array}$$

And if  $f$  is a universal homeomorphism, then  $f_* : \pi_1(X, \bar{x}) \rightarrow \pi_1(Y, f(\bar{x}))$  is an isomorphism by (7.1.4.23).  $\square$

*Proof:* 1: This follows from (9.3.2.2) and (4.3.4.9).

2: This follows from (4.3.4.10) and (7.6.6.15).

3: This follows from (4.3.4.10) and (7.6.6.15) applied to the commutative diagram of categories:

$$\begin{array}{ccc} Y_{\text{fét}} & \xrightarrow{f^*} & X_{\text{fét}} \\ \downarrow F_{\bar{y}} & & \downarrow F_{\bar{x}} \\ \text{finSet} & \xlongequal{\quad} & \text{finSet} \end{array}.$$

$\square$

**Cor. (9.3.2.4) [Étale Fundamental Groups of Fields].** For  $k \in \text{Field}$ ,  $X = \text{Spec } k$ , let  $\bar{x} = \text{Spec}(\bar{k}) \rightarrow X$  be a geometric point, then the functor in (7.6.6.6) induces an equivalence of categories

$$X_{\text{fét}} \cong \text{Gal}_k\text{-finSet}.$$

And there is a canonical isomorphism  $\text{Gal}_k \rightarrow \pi_1(X, \bar{x})$  constructed as follows: For each  $Y \in X_{\text{fét}}$ , there is a functorial continuous action

$$\text{Aut}(\bar{k}/k) \times F_{\bar{x}}(Y) \rightarrow F_{\bar{x}}(Y),$$

which induces the desired isomorphism  $\text{Gal}_k \rightarrow \pi_1(X, \bar{x})$ .  $\square$

*Proof:* This follows from (7.6.6.6).  $\square$

**Cor. (9.3.2.5) [Locally Constant Sheaves and Fundamental Group].** By (9.4.2.19), if  $X \in \text{Sch}_{\text{cntd}}$  and  $\bar{x}$  be a geometric point of  $X$ , then there is an equivalence of categories between finite locally constant Abelian sheaves on  $X$  and finite  $\pi_1(X, \bar{x})$ -modules.  $\lrcorner$

**Prop. (9.3.2.6) [Categories and Fundamental Groups].** If  $f : X \rightarrow Y \in \text{Sch}_{\text{qcqs}}$  s.t.  $f^* : Y_{\text{fét}} \rightarrow X_{\text{fét}}$  is an equivalence of categories, then  $f_* : \pi_0(X) \rightarrow \pi_0(Y)$  is a homeomorphism.

And if  $Y$  is connected, then for any geometric point  $\bar{x}$  of  $X$  with  $\bar{y} = f \circ \bar{x}$ ,  $f_* : \pi_1(X, \bar{x}) \rightarrow \pi_1(Y, \bar{y})$  is an isomorphism.  $\lrcorner$

*Proof:* Firstly  $f(X)$  meets every clopen subset  $U \subset Y$ , otherwise  $U \times_Y X \rightarrow X$  is an isomorphism but  $U \rightarrow Y$  is not.

Conversely, for any clopen subset  $X_1 \subset X$ ,  $X_1 = X \times_Y V$  for some  $V \in Y_{\text{fét}}$ . Then  $V$  must be a clopen subset of  $Y$ .

Thus  $f_* : \pi_0(X) \rightarrow \pi_0(Y)$  is a bijection. And because  $\pi_0(X)$  and  $\pi_0(Y)$  are profinite spaces by (7.4.5.46) and (4.12.3.5), we see that  $f_*$  is a homeomorphism.  $\square$

**Lemma (9.3.2.7) [Rigidity Lemma].** If  $f, g : S' \rightarrow S''$  are two  $S$ -morphisms where  $S''$  is a separated étale  $S$ -scheme and  $(S', \bar{s}')$  is a pointed scheme that  $f(\bar{s}') = g(\bar{s}')$ , then  $f = g$ .  $\lrcorner$

*Proof:* The diagonal  $S'' \rightarrow S'' \otimes_S S''$  is a closed immersion and also étale hence open (7.6.6.3), so the diagonal is an clopen subset. And now  $f \times g : S' \rightarrow S'' \otimes_S S''$  intersects the diagonal, and  $S'$  is connected, so  $f, g$  are identical on the diagonal.  $\square$

**Def. (9.3.2.8) [Galois-Covers].**  $Y \rightarrow X \in X_{\text{fét}}$  is called a **Galois-covering map (of schemes)** if  $Y$  defines a Galois-object in  $X_{\text{fét}}$  (4.3.4.6).

For any  $Y \in X_{\text{fét}}$  connected, there exists a finite étale Galois cover  $Y' \rightarrow X$  that factors through  $Y$ , by (9.3.2.1) and (4.3.4.7).  $\lrcorner$

**Prop. (9.3.2.9).** For  $k \in \text{Field}$  s.t.  $k = \bar{k}$ , we have  $\pi_1(\mathbb{P}_k^1) = 0$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (9.3.2.10) [Arithmetic and Geometric Fundamental Groups].** Suppose  $k \in \text{Field}$  and  $X \in \text{Sch}_{\text{qcqs}}/k$  is geometrically connected (7.4.3.1), then for any  $\bar{x} \in X(\bar{k})$ , there is an exact sequence

$$1 \rightarrow \pi_1(X_{\bar{k}}, \bar{x}) \rightarrow \pi_1(X, \bar{x}) \rightarrow \text{Gal}_k \rightarrow 1.$$

$\lrcorner$

*Proof:* Cf. [Sta]0BTX?  $\square$

**Prop. (9.3.2.11) [Proper Scheme over Henselian Local Rings].** If  $(A, \mathfrak{m}, \kappa)$  is a Henselian local ring and  $X \in \text{Sch}^{\text{proper}}/A$ , then the base change functor

$$X_{\text{fét}} \rightarrow (X_{\kappa})_{\text{fét}}$$

is an equivalence of categories.  $\lrcorner$

*Proof:* Cf. [Sta]0A48.  $\square$

**Prop. (9.3.2.12) [Change of Alg. Closed Fields].** If  $k \subset k'$  and  $k = \bar{k}, k' = \bar{k}'$ , and  $X \in \text{Sch}^{\text{proper}}/k$ , then the base change functor

$$X_{\text{fét}} \rightarrow (X_{k'})_{\text{fét}}$$

is an equivalence of categories.  $\lrcorner$

*Proof:* Cf. [Sta]0A49.?  $\square$

## 9.4 Étale Cohomology Theory

*“It is less than four years since cohomological methods (i.e. methods of Homological Algebra) were introduced into Algebraic Geometry in Serre’s fundamental paper, and it seems certain that they are to overflow the part of mathematics in the coming years, from the foundations up to the most advanced parts.”*

–A. Grothendieck

References are [Fu11, Mil13b, Con15], [Sta], [Étale Cohomology, Tamme], and [Notes on étale cohomology of number fields, Mazur].

### Notation(9.4.0.1).

- Let  $X, S \in \text{Sch}_{\text{qcqs}}$ .
- Use notations on Properties of Schemes defined in [More Properties of Morphisms between Schemes](#).
- Use notations on Cohomology of Schemes defined in [Cohomology of Schemes](#).

┘

### 1 Basics

**Prop.(9.4.1.1).** The étale site  $X_{\text{ét}}$  is a ringed site,  $\text{Sh}(X_{\text{ét}})$  is a Grothendieck Abelian category, and we can define right derived functors of any left exact functor, by(7.3.1.1). ┘

### Étale Topoi

**Prop.(9.4.1.2) [Zariski-Étale Comparison].** For  $X \in \text{Sch}$ , the inclusion  $X_{\text{Zar}} \rightarrow X_{\text{ét}}$  of topologies which is a morphism of sites  $\varepsilon : X_{\text{ét}} \rightarrow X_{\text{Zar}}$ , for any  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , there is a Leray spectral sequence(7.3.1.8)

$$E_2^{pq} = H_{\text{Zar}}^p(X, R^q \varepsilon_*(\mathcal{F})) \Rightarrow H_{\text{ét}}^{p+q}(X, \mathcal{F}).$$

┘

**Def.(9.4.1.3) [Pushforward & Pullback].** For  $f : X \rightarrow Y \in \text{Sch}$ , there is a continuous functor between sites  $f_{\text{ét}} : Y_{\text{ét}} \rightarrow X_{\text{ét}}$ , which preserves final objects and finite fiber products, so by(7.1.2.14), it induces a morphism of sites(7.1.1.7)  $f_{\text{ét}} : X_{\text{ét}} \rightarrow Y_{\text{ét}}$ , which induces a morphism of topoi

$$f_{\text{ét}} : \text{Sh}(X_{\text{ét}}) \rightarrow \text{Sh}(Y_{\text{ét}}).$$

$f_{\text{ét}}^*$  is called the inverse image, it is exact. By definition(7.1.2.11), for  $\mathcal{F} \in \text{Sh}(Y_{\text{ét}})$  and  $X \in X_{\text{ét}}$ ,  $f^* \mathcal{F}(X')$  equals the colimit over all pairs  $(Y', \varphi)$  where  $Y' \in Y_{\text{ét}}$  and  $\varphi : X' \rightarrow Y' \times_Y X$ , equivalently, all  $X' \rightarrow Y'$  over  $Y$ . ┘

**Cor.(9.4.1.4) [Localizations].** By(7.1.2.25), an object  $i : X' \rightarrow X \in X_{\text{ét}}$  induce a morphism  $({}_s i, i^s)$  of topoi  $X'_{\text{ét}} \rightarrow X_{\text{ét}}$ .

Denote  $i^s \mathcal{F} = F/X'$ , then for  $Z' \in X'_{\text{ét}}$ , by(7.3.1.4) and(7.1.2.25)

$$F/X'(Z') = F(Z'), \quad H^q(X_{\text{ét}}; Z', F) \cong H^q(X'_{\text{ét}}; Z', F/X').$$

In particular,

$$H^q(X_{\text{ét}}; X', \mathcal{F}) = H^q(X'_{\text{ét}}; X', \mathcal{F}|_{X'}),$$

thus we will omit the ambient sites and the restriction of sheaves, which should cause no confusion.  
 $\perp$

**Prop. (9.4.1.5).** If  $Z$  is étale over  $X$ , then the canonical morphism

$$f^* \operatorname{Hom}_X(-, Z) \rightarrow \operatorname{Hom}_Y(-, Z \times_X Y)$$

is an isomorphism.  $\perp$

*Proof:* By definition,  $f^* \operatorname{Hom}_X(-, Z)$  is the sheaf associated to the presheaf  $f_p \operatorname{Hom}_X(-, Z)$  (7.1.2.11), which is identical to the presheaf  $\operatorname{Hom}_Y(-, Z)$  on  $Y_{\text{ét}}$ , but it is already a sheaf (7.1.4.35).  $\square$

**Prop. (9.4.1.6) [Relative Leray Spectral Sequence].** If  $f : X \rightarrow Y, g : Y \rightarrow Z \in \operatorname{Sch}$ , then for any  $\mathcal{F} \in \operatorname{Sh}(X_{\text{ét}})$ , there is a relative Leray spectral sequence (7.3.1.8)

$$E_2^{pq} = R^p g_*(R^q f_*(\mathcal{F})) \Rightarrow R^{p+q}(gf)_*(\mathcal{F}).$$

$\perp$

**Cor. (9.4.1.7) [Leray Spectral Sequence].** For a morphism of schemes  $f : X \rightarrow Y$  and  $\mathcal{F} \in \operatorname{Sh}(X_{\text{ét}})$ ,  $Y' \in Y_{\text{ét}}$ , there is a Leray spectral sequence??:

$$E_2^p = H^p(Y', R^q f_*(\mathcal{F})) \Rightarrow H^{p+q}(Y' \times_Y X, \mathcal{F})$$

$\perp$

**Prop. (9.4.1.8) [Commutates with Colimits].** For  $X \in \operatorname{Sch}^{\text{qcqs}}$ , by (7.1.4.18)(7.1.4.21) and (7.3.1.14),  $H_{\text{ét}}^q(X, -)$  commutes with filtered colimits.  $\perp$

**Prop. (9.4.1.9).** Let  $X, Y \in \operatorname{Aff}$ , then for any morphism of ringed sites  $(g, g^\sharp) : (X_{\text{ét}}, \mathcal{O}_X) \rightarrow (Y_{\text{ét}}, \mathcal{O}_Y)$ , there exists a unique morphism of schemes  $f : X \rightarrow Y \in \operatorname{Sch}$  s.t.  $(g, g^\sharp)$  is 2-isomorphic to  $(f, f^\sharp)$ .  $\perp$

*Proof:* Cf. [Sta]04I6.  $\square$

### Field case

**Thm. (9.4.1.10) [Étale Sites over Fields].** The functor  $f : X' \rightarrow X'(k_s)$  is an equivalence of topologies from the small étale site  $(\operatorname{Spec}(k))_{\text{ét}}$  to the canonical topology  $T_G$  on the category of  $G$ -sets, where  $G = G(k_s/k)$ .

In particular, any Abelian sheaf on  $\operatorname{Spec}(k)_{\text{ét}}$  is representable by  $?$ .  $\perp$

*Proof:* First  $f$  maps a family of morphisms of schemes to a covering iff this family is a covering itself. This is because both are defined by set-theoretical surjectivity, and this is by (7.6.6.12).

Next we need to show this is an equivalence of categories.  $f$  has a left adjoint  $g$  because  $X' \rightarrow \operatorname{Hom}_G(U, X'(k_s))$  is representable for any  $G$ -set  $U$ , because any  $G$ -set is equivalent to disjoint sums of  $G/H$ , and both category has arbitrary sums, so it suffice to prove for  $G/H$ , but this is represented by  $\operatorname{Spec} k'$ , where  $k'$  is the fixed field of  $H$ .

To prove  $fg \cong \operatorname{id}$  and  $gf \cong \operatorname{id}$ , they commutes with direct sums, so the first one is true because  $G/H \rightarrow fg(G/H) = \operatorname{Spec}(k_s)(k)$  is an isomorphism, and the second follows from (7.6.6.6) as all étale schemes over field  $k$  is a disjoint union of spectra of finite separable field extensions of  $k$ .  $\square$

**Cor. (9.4.1.11)** [Étale and Galois Cohomologies]. By (9.8.2.2),

$$\mathrm{Sh}((\mathrm{Spec} k)_{\mathrm{\acute{e}t}}) \rightarrow \mathrm{Mod}_G : \mathcal{F} \rightarrow \varinjlim_{k \subset k' \subset k^s} \mathcal{F}(\mathrm{Spec} k')$$

is an equivalence of categories, so

$$H_{\mathrm{\acute{e}t}}^q(\mathrm{Spec} k, \mathcal{F}) \cong H^q(G, \varinjlim_{k \subset k' \subset k^s} \mathcal{F}(\mathrm{Spec} k_s)).$$

In particular, if  $k = k^s$ , then  $(\mathrm{Spec} k)_{\mathrm{\acute{e}t}}$  is equivalent to  $\mathcal{A}b$ , and  $H_{\mathrm{\acute{e}t}}^p(\mathrm{Spec}(k), \mathcal{F}) = 0$  for  $p > 0$ .  $\lrcorner$

### Stalks

**Def. (9.4.1.12)** [Stalk]. By (9.4.1.10), for any scheme  $X$  and an arithmetic point  $x : \mathrm{Spec} k \rightarrow X$ , the section functor  $F \rightarrow F(x)$  is an equivalence of categories from  $(\mathrm{Spec} k)_{\mathrm{\acute{e}t}}$  to  $\mathcal{A}b$ . Thus for any  $\mathcal{F} \in \mathrm{Sh}(X_{\mathrm{\acute{e}t}})$ , we can define the **stalk map**

$$\mathrm{Sh}(X_{\mathrm{\acute{e}t}}) \rightarrow \mathcal{A}b : \mathcal{F} \mapsto (x^* \mathcal{F})(x).$$

$\lrcorner$

**Prop. (9.4.1.13)**. For any geometric point  $P$  of  $X$ ,

- the stalk map is exact and commutes with colimits.
- For any morphism  $u : P' \rightarrow P$  of geometric points over  $X$ ,  $\mathcal{F}_P \cong \mathcal{F}_{P'}$ .
- If  $X \rightarrow Y$  is a morphism, then  $(f^* F)_P \cong F_P$ .

$\lrcorner$

*Proof:* 1: taking stalk is a composition of  $f^*$  and taking section over  $P$  (which is an equivalence), so it is exact and commutes with colimits (9.4.1.3).

2, 3: Trivial.  $\square$

**Prop. (9.4.1.14)** [Stalk is Defined Naturally]. By the definition of  $f^*$  (7.1.2.11), if  $X'$  be an étale nbhd of  $P$  in  $X$ , i.e.  $P \rightarrow X' \rightarrow X$ , then

$$(f_{\mathrm{\acute{e}t}})_P(\mathcal{F}(P)) = \varinjlim_{X'} \mathcal{F}(X')$$

and  $F_P = f^* F(P)$ , thus there is a natural map  $\varinjlim_{X'} F(X') \rightarrow F_P$ .

Then we have:

$$\varinjlim_{X'} G(X') \rightarrow (G^\sharp)_P$$

for any presheaf  $G$  on  $X_{\mathrm{\acute{e}t}}$ .  $\lrcorner$

*Proof:* Firstly  $(f^\sharp)^\sharp(G) \cong (f^* G)^\sharp$  by (7.1.2.12). Then it suffices to prove that  $G(P) \rightarrow G^\sharp(P)$  is an isomorphism for any presheaf  $G$  on  $P_{\mathrm{\acute{e}t}}$ . But this is because  $P_{\mathrm{\acute{e}t}}$  is just  $\mathcal{A}b$  (9.4.1.10), and  $P \xrightarrow{\mathrm{id}} P$  is cofinal in the category of coverings of  $P$ .  $\square$

**Cor. (9.4.1.15)**. For  $X \rightarrow Y \in \mathrm{Sch}$  and  $P$  is a geometric point of  $Y$ , we have

$$R^p f_*(F)_P \cong \varinjlim_{P \in Y'} H^p(X \times_Y Y', F).$$

$\lrcorner$

**Cor. (9.4.1.16).** For  $X = \operatorname{Spec} k$ , the equivalence (9.4.1.10) of  $X_{\text{ét}}$  with continuous  $G$ -modules are just induced by taking the stalk at  $\operatorname{Spec} k_s$ .  $\lrcorner$

**Prop. (9.4.1.17) [Exactness and Stalks].** The exactness, injectivity and surjectivity of maps of sheaves  $F' \rightarrow F$  on  $X_{\text{ét}}$  can be checked on stalks (9.4.1.12).  $\lrcorner$

*Proof:* It suffices to prove the isomorphism case, because taking stalks are exact (9.4.1.13) and other maps can be characterized by isomorphisms.

Monomorphism: suppose not, if  $s \in F'(X')$  is mapped to 0, by taking base change, we can assume  $X' = X$ , and then  $0 = v(s)_{\bar{x}} = v_{\bar{x}}(s_{\bar{x}})$ , thus  $s_{\bar{x}} = 0$  by assumption. Now by (9.4.1.14), for any  $x$  there is an étale nbhd of  $x$  that  $s$  vanishes on it. So we find an étale covering of  $X$  that  $s$  vanishes, thus  $s = 0$  because  $F'$  is a sheaf.

Epimorphism: Similarly, for any  $v \in F(X')$ , we can pass to the base change and assume  $X' = X$ , then find for each  $x$  a nbhd that comes from some  $v(s_x)$ , and they glue together to be a global section of  $F'(X)$ .  $\square$

**Prop. (9.4.1.18) [Finite Morphism is Exact].** For a finite morphism  $f$ ,  $f_*$  are exact on étale topos.  $\lrcorner$

*Proof:* Check on stalks,  $\square$

### Strict Henselization

**Def. (9.4.1.19) [Strict Henselization].**  $\lrcorner$

**Prop. (9.4.1.20) [Hensel's Lemma].** Let  $(\mathcal{O}, \kappa)$  be a Henselian local ring and  $Y \in \operatorname{Sch}^{\text{sm}}/\mathcal{O}$ , then the map  $Y(\mathcal{O}) \rightarrow Y(\kappa)$  is surjective.  $\lrcorner$

*Proof:* Cf. [BLR90] Section 2.3. Prop 5.  $\square$

### Properties of Étale Cohomologies

**Prop. (9.4.1.21) [Restriction to Small Sites].** If  $\mathcal{F} \in \operatorname{Sh}_{\text{ét}}/S$ , then the cohomology groups of  $\mathcal{F}$  on  $S$  agrees with the cohomology of its restriction on  $\operatorname{Sh}(S_{\text{ét}})$ , by (7.3.1.4).  $\lrcorner$

**Prop. (9.4.1.22) [Čech Comparison, Arin].** If  $X \in \operatorname{Sch}$  is compact and any finite subset of  $X$  is contained in an open affine (e.g.  $X$  is quasi-projective), then for any covering  $\mathcal{U} \in \operatorname{Cov}(\operatorname{Sch}_{\text{ét}}/X)$  of  $X$ ,  $\check{H}(\mathcal{U}, -)$  is exact.

In particular, by (7.3.2.16), in this case, we can use Čech cohomology to calculate the étale cohomologies.  $\lrcorner$

*Proof:* Cf. [Milne, Étale Cohomologies, Prop 3.2.17].  $\square$

**Prop. (9.4.1.23) [Inverse Limits].** Let  $(X_i)_I$  be a projective system in  $\operatorname{Sch}_{qc}$  with affine morphisms,  $X_{\infty} = \varprojlim_{i \in I} X_i$ . For any projective system of sheaves  $(\mathcal{F}_i)_I$ ,  $\mathcal{F}_i \in \operatorname{Sh}((X_i)_{\text{ét}})$ , let  $\mathcal{F}_{\infty} = \varprojlim_{i \in I} \mathcal{F}_i$ , then there are isomorphisms

$$\varprojlim_{i \in I} H_{\text{ét}}^p(X_i, \mathcal{F}_i) \cong H_{\text{ét}}^p(X_{\infty}, \mathcal{F}_{\infty}).$$

$\lrcorner$

*Proof:* Cf. SGA4, Prop 7.5.8, or Artin 1962, Chap 3.3. ?  $\square$

**Prop. (9.4.1.24) [Étale-Zariski Comparison for Qco Sheaves].** Recall by (7.1.4.37) if  $M \in \mathcal{QCoh}(X)$  then  $\widetilde{M}$  is a fpqc sheaf on  $X$ , in particular an étale sheaf on  $X$ . Now the edge map of the Zariski-étale comparison for  $\widetilde{M}$  is an isomorphism:

$$H_{\text{Zar}}^p(X, M) \cong H_{\text{ét}}^p(X, \widetilde{M})$$

In particular,  $H_{\text{ét}}^p(X, \mathbb{G}_a) \cong H^p(X, \mathcal{O}_X)$ , and the étale cohomology for Qco sheaves vanishes on affine schemes.  $\lrcorner$

*Proof:* Cf. [Sta].

We show that  $R^p \varepsilon^s(\widetilde{M}) = 0$  for  $p > 0$ , Cf. [Tamme P109]. Not hard.  $\square$

**Prop. (9.4.1.25) [Galois Covering and Group Cohomology].** If  $\mathcal{U} = \{Y \rightarrow X\}$  is a single Galois covering with Galois group  $G$ ,  $\mathcal{P}$  is a presheaf on  $\text{Sch}/X$  s.t.  $\text{?}$  then  $\check{H}^r(\mathcal{U}, \mathcal{P}) = H^r(G, \mathcal{P}(Y))$ .

In particular, we may not use alternating complex to calculate the étale cohomologies.  $\lrcorner$

*Proof:*  $\square$

**Prop. (9.4.1.26) [Hochschild-Serre Spectral Sequences].** Let  $X \in \text{Sch}$  and  $Y \in X_{\text{ét}}$  be a Galois covering with Galois group  $\Gamma$ , then for any  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , there is a spectral sequence  $\lrcorner$

**Prop. (9.4.1.27) [Mayer-Vietoris Sequences].** Let  $X \in \text{Sch}$  and  $f : Y \rightarrow X \in X_{\text{ét}}$ ,  $U \subset X$  s.t.  $f(Y) \cup U = X$ , then for any  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , there is a long exact sequence

$$0 \rightarrow H_{\text{ét}}^0(X, \mathcal{F}) \rightarrow H_{\text{ét}}^0(U, \mathcal{F}) \oplus H_{\text{ét}}^0(Y, \mathcal{F}) \rightarrow H_{\text{ét}}^0(f^{-1}(U), \mathcal{F}) \rightarrow H_{\text{ét}}^1(X, \mathcal{F}) \rightarrow \dots$$

that is functorial in  $\mathcal{F}$ , by (7.3.1.11).  $\lrcorner$

### Restricted Cohomologies

**Def. (9.4.1.28) [Restricted Cohomologies].** Let  $X \in \text{Sch}$  and  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , if  $\varphi : U' \rightarrow U \in X_{\text{ét}}$  and  $Z' \subset U'$  is a closed subschemes s.t.  $Z' \rightarrow U' \rightarrow U$  is a closed embedding with image  $Z \subset U$ , then there is a canonical bijection

$$\{s \in \mathcal{F}(U) \mid \text{Supp}(s) \in Z\} = \{s' \in \mathcal{F}(U') \mid \text{Supp}(s') \in Z'\}.$$

which is given by restriction when  $\varphi^{-1}(Z) = Z'$ .

In particular, if  $Z \subset X$  be a locally closed subscheme, then there is a functor

$$\Gamma_Z(X, -) : \text{Sh}(X_{\text{ét}}) \rightarrow \mathcal{A}b : \mathcal{F} \mapsto \{s \in \mathcal{F}(U) \mid \text{Supp}(s) \in Z\} = \ker(\mathcal{F}(U) \rightarrow \mathcal{F}(U \setminus Z))$$

where  $U \subset X$  is any open subscheme containing  $Z$  as a closed subscheme.

$\Gamma_Z(X, -)$  is left exact as both  $\Gamma(U, -)$  and  $\Gamma(U \setminus Z, -)$  are.

The right derived cohomology of  $H_Z^r(X, -)$ , called the **restricted étale cohomology** of  $\mathcal{F}$  with supports in  $Z$ .  $\lrcorner$

*Proof:* Denote  $Z'' = \varphi^{-1}(Z) \subset U'$ , then  $Z' \subset Z''$  is closed and also étale because  $Z', Z''$  are both étale over  $Z$  (7.6.6.4). Thus  $Z'' = Z' \amalg T$  for some closed subset  $T$  by (7.6.2.10).

The open covering  $U' = (U' \setminus T) \cup (U' \setminus Z')$  shows that

$$\{s' \in \mathcal{F}(U') \mid \text{Supp}(s') \in Z'\} = \{s' \in \mathcal{F}(U' \setminus T) \mid \text{Supp}(s') \in Z'\},$$

and the étale covering  $\{U' \setminus T \rightarrow U, U \setminus Z \rightarrow U\}$  shows that

$$\{s \in \mathcal{F}(U) \mid \text{Supp}(s) \in Z\} = \{s' \in \mathcal{F}(U' \setminus T) \mid \text{Supp}(s') \in Z'\}.$$

$\square$

**Prop. (9.4.1.29) [Excision Sequences].** Situation as in (9.4.1.28), if  $Z \subset X$  is a closed subschemes, then for any  $\mathcal{F} \in \mathcal{S}h(X_{\text{ét}})$ , there is a functorial long exact sequence

$$\cdots \rightarrow H_Z^r(X, \mathcal{F}) \rightarrow H_{\text{ét}}^r(X, \mathcal{F}) \rightarrow H_{\text{ét}}^r(U, \mathcal{F}) \rightarrow H_Z^{r+1}(X, \mathcal{F}) \rightarrow \cdots$$

┘

*Proof:* Choose a functorial injective resolution (4.10.2.9), and this follows from the fact an injective sheaf is flabby (7.3.4.9).  $\square$

**Prop. (9.4.1.30) [Restricted Cohomology at a point].** For  $x \in |X|_0$  and any  $\mathcal{F} \in \mathcal{S}h(X_{\text{ét}})$ , there is an isomorphism

$$H_x^p(X, \mathcal{F}) \cong H_{\text{ét}}^r(\text{Spec } \mathcal{O}_{X,x}^{\text{sh}}, i_x^* \mathcal{F}) \text{ (9.4.1.28)(9.4.1.19).}$$

┘

*Proof:* Take limits over the affine étale nbhds of  $x \in X$  and use (9.4.1.23).  $\square$

### Kummer Theory

**Prop. (9.4.1.31) [Kummer Sequence].** For  $X \in \text{Sch}/\mathbb{Z}[\frac{1}{n}]$ , there is an exact sequence in  $\mathcal{S}h(X_{\text{ét}})$ :

$$0 \rightarrow \mu_{n,X} \rightarrow \mathbb{G}_{m,X} \xrightarrow{n} \mathbb{G}_{m,X} \rightarrow 0$$

┘

*Proof:* The proof is similar to that of Artin-Schreier sequence (9.4.1.37), noticing that  $A \rightarrow A[T]/(T^n - s)$  is an étale map.  $\square$

**Cor. (9.4.1.32).** For  $X \in \text{Sch}/\mathbb{Z}[\frac{1}{n}]$ , by (7.7.1.14), there is an exact sequence

$$0 \rightarrow H^0(X, \mathcal{O}_X^*)/nH^0(X, \mathcal{O}_X^*) \rightarrow H^1(X, \mu_n) \rightarrow \text{Pic}(X)[n] \rightarrow 0$$

┘

**Cor. (9.4.1.33).** If  $X = \text{Spec } A$  where  $A$  is a local ring and  $n$  is invertible in  $A$ , then  $H^1(X, \mu_n) \cong A^*/(A^*)^n$ . ?

┘

*Proof:* Cf. [Tamme, P110].  $\square$

**Cor. (9.4.1.34).** If  $k \in \text{Field}$ ,  $k = k^{\text{sep}}$ ,  $X \in \text{Sch}_{\text{red}}^{\text{proper}}/k$ , and  $n \in \mathbb{Z} \cap k^\times$ , then  $H_{\text{ét}}^1(X, \mu_n) \cong \text{Pic}(X)[n]$ , by (7.12.1.12).  $\square$

### Artin-Schreier Theory

**Prop. (9.4.1.35) [Characters of Finite Fields].** Any character  $\psi : \mathbb{F}_q \rightarrow \overline{\mathbb{Q}}_\ell^\times$  can be extended to  $\mathbb{F}_{q^n}$  by

$$\mathbb{F}_{q^n} \xrightarrow{\text{tr}} \mathbb{F}_q \xrightarrow{\psi} \overline{\mathbb{Q}}_\ell^\times,$$

also denoted by  $\psi$ .  $\square$



**Def. (9.4.1.36) [Artin-Schreier Covering].** The **Artin-Schreier cover** of  $\mathbb{A}^1$  is defined to be the finite Galois cover

$$\varphi : \mathbb{A}_{\mathbb{F}_q}^1 \rightarrow \mathbb{A}_{\mathbb{F}_q}^1 : x \mapsto x^q - x,$$

with the Galois group isomorphic to  $\mathbb{F}_q$  via  $a \mapsto (x \mapsto x + a)$ .  $\lrcorner$

**Prop. (9.4.1.37) [Artin-Schreier Sequence].** For  $X \in \text{Sch}/\mathbb{F}_p$ , let  $P = \text{id} - \text{Frob}$ , then there is an **Artin-Schreier exact sequence** of étale sheaves

$$0 \rightarrow (\underline{\mathbb{Z}/(p)})_X \rightarrow (\mathbb{G}_a)_X \xrightarrow{P} (\mathbb{G}_a)_X \rightarrow 0$$

$\lrcorner$

*Proof:* If  $s \in \mathcal{O}_{X'}$  is in the kernel, then  $s = s^p$ , so it is locally constant and comes from the map  $\mathbb{Z}/p\mathbb{Z} \rightarrow \mathcal{O}_{X'}$ . Conversely, for any  $s \in \mathcal{O}_{X'}$ , it suffices to find an étale cover that  $s$  is a  $p$ -th power in  $\mathcal{O}_{X'_i}$ . For this, it suffices to notice that for any  $p$ -ring  $A$ ,  $A[t]/(t^p - t - s)$  is free of rank  $p$  and étale over  $A$ .  $\square$

**Cor. (9.4.1.38).** If  $X$  has char  $p$ , then by the long exact sequence and (9.4.1.24), there is an exact sequence

$$0 \rightarrow H^0(X, \mathcal{O}_X)/P(H^0(X, \mathcal{O}_X)) \rightarrow H^1(X, (\mathbb{Z}/p\mathbb{Z})) \rightarrow H^1(X, \mathcal{O}_X)^F \rightarrow 0$$

where the last one is the fixed elements.  $\lrcorner$

**Cor. (9.4.1.39).** If  $X = \text{Spec } A$  and  $pA = 0$ , then  $H_{\text{ét}}^q(X, \mathbb{Z}/p\mathbb{Z}) = \begin{cases} A/\text{Im}((\text{Frob} - 1)|_A) & , q = 0 \\ 0 & , q \geq 1 \end{cases}$ .  $\lrcorner$

**Cor. (9.4.1.40).** if  $k \in p\text{-Field}$ ,  $k = k^{\text{sep}}$  and  $X$  is a reduced proper  $k$ -scheme, then  $H^1(X, \mathbb{Z}/p\mathbb{Z}) = (H^1(X, \mathcal{O}_X))^F$ .  $\lrcorner$

## 2 Constructible Sheaves

### Torsion Sheaves

**Def. (9.4.2.1) [Torsion Sheaves].** For  $\mathcal{C} \in \text{Site}$ ,  $\mathcal{F} \in \text{Sh}(\mathcal{C}; \text{Set})$  is called a **torsion sheaf** iff it is associated to a presheaf of torsion Abelian groups. Equivalently, the canonical morphism  $\varinjlim_n \mathcal{F}[n] \rightarrow \mathcal{F}$  is an isomorphism. The category of torsion sheaves on  $\mathcal{C}$  is denoted by  $\text{Sh}_{\text{tor}}(\mathcal{C})$ .  $\lrcorner$

*Proof:* It  $\mathcal{F} = P^\sharp$ , then consider  $0 \rightarrow P[n] \rightarrow P \xrightarrow{n} P \rightarrow 0$ . Because  $\sharp$  is exact,  $\mathcal{F}[n] = (P[n])^\sharp$ . Then because  $\sharp$  commutes with inductive limits and  $P = \varinjlim_n P[n]$ , it follows  $\mathcal{F} = \varinjlim_n \mathcal{F}[n]$ .

Conversely, if  $\mathcal{F} = \varinjlim_n \mathcal{F}[n]$ , then  $\mathcal{F} = \varinjlim_n (\mathcal{F}^{\text{PSh}}[n])^\sharp = (\varinjlim_n \mathcal{F}[n])^\sharp$ , and  $\varinjlim_n \mathcal{F}^{\text{PSh}}[n]$  is presheaf of torsion Abelian groups.  $\square$

**Remark (9.4.2.2).** WARNING: For a torsion sheaf,  $\mathcal{F}(U)$  need not be torsion Abelian, but this is the case if  $U$  is quasi-compact, Cf.[Tamme P146].  $\lrcorner$

**Prop. (9.4.2.3) [Being Torsion is Local].** For  $X \in \text{Sch}$ ,  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$  is a torsion sheaf (9.4.2.1) iff all stalks  $\mathcal{F}_x$  are torsion.  $\lrcorner$

*Proof:* By definition  $\mathcal{F}$  is torsion iff  $\varinjlim_n \mathcal{F}[n] \rightarrow \mathcal{F}$  is an isomorphism. Then use the fact isomorphisms are checked on stalks (9.4.1.17) and stalk maps are exact (9.4.1.13).  $\square$

**Prop. (9.4.2.4).** For  $f : X \rightarrow Y \in \text{Sch}$ .

- If  $\mathcal{F} \in \text{Sh}(Y_{\text{ét}})$  is a torsion sheaf on  $Y$  (9.4.2.1), then  $f^*\mathcal{F}$  is torsion sheaf on  $X$ .
- If  $f$  is qcqs and  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$  is a torsion sheaf on  $X$ , then  $R^n f_* \mathcal{F}$  are torsion sheaves on  $Y$  for any  $n \in \mathbb{N}$ . In particular, if  $X$  is qcqs and  $\mathcal{F}$  is a torsion sheaf on  $X$ , then  $H_{\text{ét}}^n(X, \mathcal{F})$  are torsion for all  $n \in \mathbb{N}$ .

┘

*Proof:* 1: This follows immediately from (9.4.2.3) and (9.4.1.13).

2: For any  $y \in Y$ ,  $(R^q f_* \mathcal{F})_{\bar{y}} \cong H^q(\bar{X}, \bar{F})$ , where  $\bar{X} = X \otimes_Y \bar{Y}$  and  $\bar{Y}$  is the strict localization of  $Y$  in  $\bar{y}$  by ? . Now  $\bar{F}$  is torsion sheaf by item1, and  $\bar{X} \rightarrow \bar{Y}$  is also qcqs with  $\bar{Y}$  being affine, so  $H^q(\bar{X}, \bar{F})$  is torsion by item3, so  $R^q f_* \mathcal{F}$  is torsion by (9.4.2.3).  $\square$

**Prop. (9.4.2.5).** Let  $X \in \text{NSch}_{\text{qc}}$  and  $x \in X$ , then

- for any torsion sheaf  $\mathcal{F} \in \text{Sh}(\text{Spec}(\kappa(x))_{\text{ét}})$ , the sheaves  $R^p i_* \mathcal{F}$  are torsion sheaf for  $p > 0$ .
- $H_{\text{ét}}^p(X, i_* \mathcal{F})$  are torsion for all  $p > 0$ .

┘

*Proof:* 1: Cf. [Tamme P148]. Uses strict Henselization.

2: Consider the Leray spectral sequence  $H_{\text{ét}}^p(X, R^q i_* \mathcal{F}) \implies H_{\text{ét}}^{p+q}(\text{Spec}(\kappa(x)), \mathcal{F})$ , the left term vanishes for  $p \geq 0, q > 0$  by item1 and (9.4.2.4), and the right hand side vanish for  $p + q > 0$  by (9.4.2.4), then it can be checked that  $H_{\text{ét}}^p(X, i_* \mathcal{F})$  are torsion for  $p > 0$ .  $\square$

**Prop. (9.4.2.6).** For a closed subscheme  $i : Y \subset X$ ,  $R^p i^!$  preserves torsion sheaves.

┘

*Proof:* Cf. [Tamme P148].

$\square$

**Prop. (9.4.2.7).** For a regular Noetherian scheme  $X$ ,  $H_{\text{ét}}^n(X, \mathbb{G}_m)$  are torsion for any  $n \in \mathbb{Z}_{\geq 2}$ .

┘

*Proof:* Cf. [Tamme P149].

$\square$

**Prop. (9.4.2.8).** For a torsion sheaf  $F$ , define  $F(\ell) = \varinjlim_n F[\ell^n]$ , so it is an  $\ell$ -torsion sheaf, and in fact

$$\bigoplus_{\ell \in \text{Prime}} F(\ell) = F,$$

this is because this is true at the stalks, because stalks is exact and commutes with colimits (9.4.1.13).

So if  $X$  is qcqs, then  $H^p(X, F) \cong H^p(X, F(\ell))$ , which is the primary decomposition of  $H^p(X, F)$ .

┘

*Proof:*

$\square$

### Cohomological Dimensions

**Def. (9.4.2.9) [Cohomological Dimension].** For  $X \in \text{Sch}_{\text{qcqs}}/k$  and  $\ell \in \text{Prime}$ , define the  $\ell$ -**adic cohomological dimension** of  $X$  as the smallest number  $\text{cd}_{\ell}(X) = n$  that  $H^p(X, \mathcal{F})_{[\ell]} = 0$  for all  $p > n$  and  $\mathcal{F} \in \text{Sh}^{\text{tor}}(X_{\text{ét}})$ , and define the **cohomological dimension**  $\text{cd}(X)$  of  $X$  as the smallest number  $n$  that  $H^p(X, F) = 0$  for all  $p > n$  and  $\mathcal{F} \in \text{Sh}_{\text{tor}}(X_{\text{ét}})$ . Equivalently,  $\text{cd}(X) = \sup_{\ell \in \mathbf{P}} \{\text{cd}_{\ell}(X)\}$ .

┘

**Prop. (9.4.2.10).** If  $k \in \mathbf{Field}$  and  $X \in \mathbf{Sch}^{\text{ft}}/k$ , then

$$\text{cd}_{\ell}(X) \leq \begin{cases} 2 \dim X + \text{cd}_{\ell}(k) & \ell \neq \text{char } k \\ \dim X + 1 & \ell = \text{char } k \end{cases}.$$

┘

*Proof:*

□

**Cor. (9.4.2.11).** If  $k \in \mathbf{Field}$ ,  $k = k^s$ , then  $\text{cd}(X) \leq 2 \dim X$ .

┘

*Proof:*

□

**Thm. (9.4.2.12) [Artin Vanishing Theorem].** If  $k \in \mathbf{Field}$ ,  $k = k^{\text{sep}}$  and  $X \in \mathbf{Aff}^{\text{ft}}/k$ , then  $\text{cd}(X) \leq \dim X$ .

┘

*Proof:* Cf. [Mil13b] P105. ?

□

**Prop. (9.4.2.13) [Arc Descent for Étale Cohomology].** Let  $R$  be a ring and  $\mathcal{G} \in \mathbf{Sh}_{\text{ét}}^{\text{tor}}(\text{Spec } R)$ , and  $\mathcal{F} : (\mathbf{Sch}_{\text{qcqs}}/R)^{\text{op}} \rightarrow D^{\geq 0}(\Lambda)$  be the functor  $(f : X \rightarrow \text{Spec } R) \mapsto R\Gamma(X_{\text{ét}}, f^*\mathcal{G})$ , then  $\mathcal{F}$  satisfies arc-descent.

┘

*Proof:* Cf. [Arc Topology, Bhatt, 5.4.]

□

### Constructible Sheaves

References are [Conrad notes, L3].

**Prop. (9.4.2.14).** If  $G \in \mathbf{GrpSch}/X$  is commutative, finite and étale, the sheaf  $G_X$  represented by  $G$  is locally finite on  $X_{\text{ét}}$ .

Conversely, any locally constant sheaf  $\mathcal{F} \in \mathbf{Sh}(X_{\text{ét}})$  is represented by a unique commutative étale group scheme over  $X$ , and it is finite if  $\mathcal{F}$  has finite stalks.

┘

*Proof:* Cf. [Tamme P152].

□

**Def. (9.4.2.15) [Finite étale Sheaves].** For  $X \in \mathbf{Sch}$ , a **finite étale sheaf** is a sheaf  $\mathcal{F} \in \mathbf{Sh}(X_{\text{ét}}; \mathbf{Set})$  s.t. all its stalks are finite.

┘

**Def. (9.4.2.16) [Lcc and Constructible étale Sheaves].** For  $X \in \mathbf{NSch}$ , a **lcc étale sheaf** on  $X$  is an étale sheaf  $\mathcal{F} \in \mathbf{Sh}(X_{\text{ét}}; \mathbf{Set})$  that is locally constant and finite (9.4.2.15). The category of lcc étale sheaves on  $X$  is denoted by  $\mathbf{Sh}_{\text{lcc}}(X_{\text{ét}}; \mathbf{Set})$ .

And a **constructible étale sheaf** on  $X$  is an étale sheaf  $\mathcal{F} \in \mathbf{Sh}(X_{\text{ét}}; \mathbf{Set})$  s.t. there is a stratification  $\{X_i\}$  of  $X$  s.t.  $\mathcal{F}_{X_i} \in \mathbf{Sh}_{\text{lcc}}((X_i)_{\text{ét}}; \mathbf{Set})$ . The category of constructible sheaves on  $X$  is denoted by  $\mathbf{Sh}_{\text{const}}(X_{\text{ét}}; \mathbf{Set})$ .

┘

**Prop. (9.4.2.17).**  $\mu_{n,X}$  is étale over  $X$  iff  $n$  is prime to the characteristic of all local residue fields of  $X$ . (Only unramifiedness is concerned, and it is fiberwise (5.4.6.6)). And we can compute the Kahler differential of  $k[T]/(T^n - 1)$  vanish iff  $n \neq 0$  in  $k$ .

In this case,  $\mu_n$  is locally isomorphic to  $(\mathbb{Z}/(n))_X$ , because for any affine open  $U = \text{Spec } A$ ,  $U' = \text{Spec } A[t]/(t^n - 1) \rightarrow U$  is étale and surjective (5.4.1.21) and  $U'$  has all  $n$ -th roots of unity, so  $(\mu_n)_{\text{Spec } U'} \cong (\mathbb{Z}/n\mathbb{Z})_{\text{Spec } U'}$ .

┘

**Example (9.4.2.18).** For  $n \in \mathbb{Z}_+$ ,  $\mu_n \in \mathrm{Sh}_{\mathrm{const}}(\mathbb{Z}[\frac{1}{n}])$  but  $\mu_n \notin \mathrm{Sh}_{\mathrm{lcc}}(\mathbb{Z}[\frac{1}{n}])$ .  $\lrcorner$

*Proof:* It is locally constant after the étale base change  $\mathbb{Z}[\frac{1}{n}] \rightarrow \mathbb{Z}[\frac{1}{n}][\zeta_n]$ .  $\square$

**Prop. (9.4.2.19) [Properties of Constructible Sheaves].**

- If  $F \in \mathrm{Sh}(X_{\mathrm{\acute{e}t}})$  and  $X$  has a finite decomposition into constructible reduced subschemes  $X_i$  that  $F/X_i$  are locally constant, then  $F$  is constructible. The converse is also true if  $X$  is qcqs.
- Constructibility is a local property.
- Constructibility is stable under pullback, pushout and finite direct limits.
- Constructibility is stable  $j_!$  for  $j$  qc étale.
- Subsheaves of a constructible sheaf are constructible.

$\lrcorner$

*Proof:* Cf. [Tamme P155]. [Conrad L3 P2], [Étale Cohomology and Weil Conjecture P42] ?  $\square$

**Prop. (9.4.2.20) [Lcc Sheaves and Finite Étale Schemes].** The functor  $X \mapsto \mathrm{Hom}_S(-, X)$  defines an equivalence of categories

$$\mathrm{Sch}^{\mathrm{f\acute{e}t}}/S \cong \mathrm{Sh}_{\mathrm{lcc}}(S_{\mathrm{\acute{e}t}}; \mathrm{Set}).$$

$\lrcorner$

*Proof:* The Yoneda functor is fully faithful, thus we need to show the essentially surjectivity. Notice first  $\mathrm{Hom}_S(-, X)$  is locally constant finite: we can restrict to an open subset of  $S$  that the fiber are of fixed order  $n$ , and  $X \rightarrow X \times_S X$  is étale and a closed immersion, thus  $X \times_S X = X \amalg Y$ , and  $Y$  is finite étale over  $X$  through  $\pi_1$ . Now by induction on the order of the fiber,  $Y = X \otimes \Sigma'$  locally. So  $X = S \times \Sigma$  locally, which means  $X$  represents the constant sheaf  $\underline{\Sigma}$  locally.

To show that every locally constant finite étale sheaf is represented by a finite étale scheme, Cf. [Conrad Etale coh, P19] ?.  $\square$

**Prop. (9.4.2.21).** If  $G$  is a commutative étale group scheme over  $X$ , then the sheaf  $G_X$  represented by  $G$  is constructible iff  $G$  is f.p. over  $X$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (9.4.2.22) [Locally Constancy and Specializations].** For  $S \in \mathrm{NSch}$  and  $\mathcal{F} \in \mathrm{Sh}_{\mathrm{const}}(S_{\mathrm{\acute{e}t}})$ ,  $\mathcal{F}$  is locally constant iff all the specialization maps for geometric points  $\mathcal{F}_{\bar{s}} \rightarrow \mathcal{F}_{\bar{\eta}}$  are bijective.  $\lrcorner$

*Proof:* If  $\mathcal{F}$  is locally constant, because the conclusion is local, we may assume  $\mathcal{F}$  is constant, then  $\mathcal{F}_{\bar{s}} \rightarrow \mathcal{F}_{\bar{\eta}}$  are all identities.

Conversely, for any geometric point  $s$ ,  $\Sigma = \mathcal{F}_s$  is finite by definition, thus there is an étale nbhd  $U$  of  $s$  that the map  $\underline{\Sigma} \rightarrow \mathcal{F}$  induces an isomorphism on  $s$ -stalks, so this is an isomorphism for any geometric point linked to  $s$  by specialization, in particular the generic point of the irreducible component containing  $s$  and all the points in this irreducible component, thus  $\mathcal{F}$  is constant on an open nbhd of  $s$  (because  $X$  is Noetherian thus has f.m. irreducible components), so  $\mathcal{F}$  is locally constant because  $X$  is Noetherian.  $\square$

**Prop. (9.4.2.23) [Constructible Sheaves are Noetherian].**  $\mathrm{Sh}_{\mathrm{const}}(X_{\mathrm{\acute{e}t}})$  are exactly the Noetherian objects in  $\mathrm{Sh}_{\mathrm{tor}}(X_{\mathrm{\acute{e}t}})$ .  $\lrcorner$

*Proof:*  $\square$

### Derived Categories of Constructible Sheaves

References are [Lei Fu, Étale Cohomology Theory].

**Def. (9.4.2.24) [Derived Category of Constructible Sheaves].** For  $\Lambda \in \mathcal{A}b^{\text{fin}}$ , define  $D_{\text{const}}^b(X_{\text{ét}}; \Lambda)$  to be the category of **derived category of bounded complexes with constructible cohomology sheaves**.

And define  $D_{\text{csttf}}^b(X_{\text{ét}}; \Lambda)$  to be the category of **derived category of bounded complexes with constructible cohomology sheaves of finite Tor dimensions**.  $\lrcorner$

*Proof:* ?  $\square$

### 3 Base Changes

**Prop. (9.4.3.1) [Proper Base Change].** If there is a Cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & Y' \\ \downarrow f' & & \downarrow f \\ X & \xrightarrow{g} & Y \end{array}$$

that  $f$  is proper, then for any torsion sheaf  $\mathcal{F}$  on  $Y'$ , the base change maps (7.3.3.18)

$$g^* R^q f_* \mathcal{F} \rightarrow R^q f'_*(g'^* \mathcal{F})$$

are isomorphisms.  $\lrcorner$

*Proof:* Cf. [Conrad L6].  $\square$

**Prop. (9.4.3.2) [Smooth Base Change].** If there is a Cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & Y' \\ \downarrow f' & & \downarrow f \\ X & \xrightarrow{g} & Y \end{array}$$

that  $f$  is smooth, then for any  $\mathcal{K}^\bullet \in D^+(X, \mathbb{Z}/(n))$ , the base change map (7.3.3.18)

$$f^* Rg_* \mathcal{K}^\bullet \rightarrow Rg'_* f'^* \mathcal{K}^\bullet$$

is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Lei Fu, P391].  $\square$

### 4 Lower Shriek Functor

References are [Sta]Chap61.

**Def. (9.4.4.1) [Lower Shriek Functors].** For  $f : X' \rightarrow X \in \text{Sch étale}$ , the **lower shriek functor**  $f_! : \text{Sh}(X'_{\text{ét}}) \rightarrow \text{Sh}(X_{\text{ét}})$  is defined to be ? Cf. [Lei Fu, P205].  $\lrcorner$

**Prop. (9.4.4.2) [Derived Upper Shriek Functors].** For  $\Lambda \in \mathcal{C}\text{Ring}_{\text{tor}}$  and  $f : X \rightarrow Y \in \text{Sch}_{\text{qcqs}}$  that is separated of f.t., the functor  $Rf_! : D(X_{\text{ét}}, \Lambda) \rightarrow D(Y_{\text{ét}}, \Lambda)$  commutes with direct sums. In particular, it has an exact right adjoint  $Rf^! : D(Y_{\text{ét}}, \Lambda) \rightarrow D(X_{\text{ét}}, \Lambda)$  by adjointness theorem (4.8.4.38), called the **derived upper shriek functor**.  $\lrcorner$

*Proof:* Cf. [Sta]0G29.  $\square$

### Cohomology with Compact Support

Cf.[Weil1, P18].

**Lemma (9.4.4.3).** Extension by 0 commutes with pullback Cf.[KF Lemma4.9].  $\square$

**Def. (9.4.4.4) [Higher Direct Images with Compact Support].** Let  $Y \in \mathcal{S}ch_{qcqs}$ ,  $f : X \rightarrow Y \in \mathcal{S}ch^{sep,ft}/Y$ ,  $\mathcal{F} \in \mathcal{S}h(X_{\acute{e}t})$ , then  $f$  factors as  $f : X \xrightarrow{j} \overline{X} \xrightarrow{\overline{f}} Y$  where  $j : X \rightarrow \overline{X}$  is an open dense subscheme and  $\overline{f}$  is proper, by Nagata-compactification (7.8.3.2). So we can define the **higher direct image with compact support** as

$$Rf_! \triangleq R\overline{f}_* \circ j_! : D^+(X, \text{tor}) \rightarrow D^+(Y, \text{tor}).$$

Then this notion is well-defined.

And if  $Y = \text{Spec } k$ , define  $H_{\acute{e}t,c}^i(X, \mathcal{F}) = H_{\acute{e}t}^i(\text{Spec } k, Rf_! \mathcal{F})$ , called the **étale cohomology with compact support**.  $\square$

*Proof:* To show it is well-defined, notice for any two compactification, we can find a common compactification that dominates them both  $?$ , so using lemma (9.4.4.5), we easily show they are isomorphic.  $\square$

**Lemma (9.4.4.5) [ $j_!$  and Higher Pushforward].** If there is a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & \overline{X} \\ \downarrow f & & \downarrow \overline{f} \\ Y & \xrightarrow{j} & \overline{Y} \end{array}$$

where  $i, j$  are open immersion and  $f, \overline{f}$  are proper, then there is a natural transformation  $j_! f_* \rightarrow \overline{f}_* i_!$  that induces a natural transformation

$$j_! Rf_* \rightarrow R\overline{f}_* i_!,$$

which is an isomorphism iff  $\overline{f}$  is proper.  $\square$

*Proof:* If this is a Cartesian diagram, then the natural transformation is give by

$$j_! f_* \rightarrow \overline{f}_* \overline{f}^* j_! f_* \cong \overline{f}_* i_! f^* f_* \rightarrow \overline{f}_* i_!.$$

(the second isomorphism is by (9.4.4.3)). The rest is by proper base change Cf.[KF P88].

The general case is also easily reduced to the Cartesian case.  $?$   $\square$

**Prop. (9.4.4.6) [Proper Maps Induce Maps on Proper Pushforward].** If  $f : X \rightarrow S \in \mathcal{N}Sch$  is separated of f.t. and  $g : Y \rightarrow X \in \mathcal{S}ch/S$  is proper, then for any étale Abelian sheaf  $\mathcal{F}$  on  $X$ , there is a canonical map

$$g : Rf_!(\mathcal{F}) \rightarrow R(f \circ g)_!(g^* \mathcal{F}).$$

$\square$

*Proof:* Choose a compactification  $X_2 \xrightarrow{j} \overline{X}_2$ , then choose a compactification  $X_1 \xrightarrow{i} \overline{X}_1$  of  $i \circ g$ , now

Cf. <https://math.stackexchange.com/questions/3120833/proper-morphism-induces-a-map-between-cohomology-groups>  $?$ .  $\square$

**Prop. (9.4.4.7)[Properties of Compact Pushforwards].** Suppose  $S \in \text{Sch}_{\text{qcqs}}$  and  $f : X \rightarrow S \in \text{Sch}$  is separated of f.t., then:

- (Base Change) If there is a Cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

then there is a natural isomorphism

$$g^* Rf_! \cong Rf'_! h^*.$$

- (Composition) For two morphisms  $f_1, f_2$  separated of f.t,  $R(f_1 \circ f_2)_! = R(f_1)_! \circ R(f_2)_!$ , which induces a Leray spectral sequence.
- (Excision) If  $\mathcal{F} \in D_{\text{tor}}^+(X)$  and  $Z \subset X$  be a closed subscheme and  $U = X \setminus Z$ , then there is a long exact sequence

$$\cdots \rightarrow R^p(f_U)_!(\mathcal{F}|_U) \rightarrow R^p f_! \mathcal{F} \rightarrow R^p(f_Z)_!(\mathcal{F}|_Z) \rightarrow R^{p+1}(f_U)_!(\mathcal{F}|_U) \rightarrow \cdots$$

┘

*Proof:* 1: Choose a compactification of  $f$ , then it suffices to show  $Rf_*^c$  and  $j_!$  both commutes with base change, which is by proper base change(9.4.3.1) and(9.4.4.3).

2: Two compactification can be splinted, and use(9.4.4.5).

3: Use the long exact sequence applied to the exact sequence(checked on stalks(9.4.1.17))

$$0 \rightarrow j_! j^* \mathcal{F} \rightarrow \mathcal{F} \rightarrow i_* i^* \mathcal{F} \rightarrow 0.$$

□

**Prop. (9.4.4.8)[Proper Pushforward to Direct Image].** There is a natural map from  $R^i f_! \rightarrow R^i f_*$ , which is induced by

$$R^i f_! = R^i \bar{f}_* j_! \rightarrow R^i \bar{f}_* j_* \rightarrow R^i f_*$$

where the second one is edge map of Leray spectral sequence.

In particular, there is a canonical map  $H_{\text{ét},c}^i(X, \mathcal{F}) \rightarrow H_{\text{ét}}^i(X, \mathcal{F})$ .

┘

**Prop. (9.4.4.9)[Vanishing Result].** If  $f : X \rightarrow S \in \text{Sch}^{\text{sep,ft}}/S$  and let  $d = \sup_{s \in S} \dim X_s$ , then if  $\mathcal{F} \in D(X, \text{tor})$  satisfies  $H^p(\mathcal{F})[p] = 0$  for  $p \geq r$ , then

$$R^p f_! \mathcal{F} = 0, \quad p \geq r + 2d.$$

┘

*Proof:* Cf.[Conrad L10 P4].

□

**Thm. (9.4.4.10)[Projection Formula].** If  $f : X \rightarrow S$  is a quasi-projective morphism,  $\mathcal{F} \in D^-(S)$  and  $\mathcal{G} \in D(X)$ , then we have a natural isomorphism

$$\mathcal{F} \otimes_S^L Rf_! \mathcal{G} \cong Rf_! (f^* \mathcal{F} \otimes_X^L \mathcal{G})$$

┘

*Proof:* We may pass to the compactification, as  $j_!$  commutes with  $f^*$ ?

□

### Frobenius Actions

References are [Étale Cohomology Theory, Lei Fu]Chap10.3.

#### Notation(9.4.4.11).

- Use notations as in [Frobenius-Morphisms](#).

┘

**Prop.(9.4.4.12) [Frobenius action on Étale Sheaves].** Let  $k \in \mathbf{Field}^p, k \cong \mathbb{F}_{p^r}$  and  $X \in \mathbf{Sch}^{\text{sep,ft}}/k, \mathcal{F} \in \mathbf{Sch}(X_{\text{ét}})$ , we have an isomorphism  $\mathcal{F} \cong (\varphi_{r,X})_* \mathcal{F}$  which is the inverse of the isomorphisms

$$(\varphi_{r,X})_*(\mathcal{F})(U) = \mathcal{F}(\varphi_{r,X}^{-1}(U)) \xrightarrow{F_{U/X}^*} \mathcal{F}(U),$$

and its adjoint  $\varphi_{r,X}^* \mathcal{F} \rightarrow \mathcal{F}$  is denoted by  $\varphi_{\mathcal{F}}$ .

Then  $\varphi_{\mathcal{F}}$  commutes with tensor product and it is an isomorphism.

┘

*Proof:* The adjoint is an isomorphism because  $\varphi^*$  induces equivalence of categories of étale site of  $X_{\text{ét}}$  with itself, by [\(7.1.4.23\)](#). □

**Prop.(9.4.4.13) [Compatibility of  $\varphi_{\mathcal{F}}$  wit Pullbacks].** Let  $k \in \mathbf{Field}^p, k \cong \mathbb{F}_{p^r}$  and  $f : Y \rightarrow X \in \mathbf{Sch}^{\text{sep,ft}}/k$  be separated,  $\mathcal{F} \in \mathbf{Sch}(X_{\text{ét}})$ , then the morphism  $\varphi_Y^* f^* \mathcal{F} \cong f^* \varphi_{r,X}^* \mathcal{F} \xrightarrow{f^* \varphi_{\mathcal{F}}} f^* \mathcal{F}$  is just  $\varphi_{f^* \mathcal{F}}$ .

┘

*Proof:* Cf. [\[Fu11\]P586](#). □

**Prop.(9.4.4.14) [Compatibility of  $\varphi_{\mathcal{F}}$  with Higher Direct Image].** Let  $k \in \mathbf{Field}^{\text{fin}}$  and  $f : X \rightarrow Y \in \mathbf{Sch}^{\text{sep,ft}}/k$  be separated,  $\mathcal{F} \in \mathbf{Sch}(X_{\text{ét}})$ , so we have a Cartesian diagram about  $\varphi_{r,X}$  and  $\varphi_Y$ . Then the composition

$$\varphi_S^* R^i f_* \mathcal{F} \rightarrow R^i f_* \varphi_{r,X}^* \mathcal{F} \xrightarrow{R^i f_*(\varphi_{\mathcal{F}})} R^i f_* \mathcal{F}$$

is just  $\varphi_{R^i f_* \mathcal{F}}$ .

┘

*Proof:* Cf. [\[Conrad L18 P4\]](#). ? □

**Cor.(9.4.4.15) [Compatibility of  $\varphi_{\mathcal{F}}$  with Proper Pushforward].** If  $X \rightarrow S$  is a separated morphism of f.t. between  $k$ -schemes and  $\mathcal{F}$  is a torsion Abelian sheaf on  $X_{\text{ét}}$ , then the morphism

$$\varphi_{r,S}^* R^i f_! \mathcal{F} \rightarrow R^i f_! \varphi_{r,X}^* \mathcal{F} \xrightarrow{R^i f_!(\varphi_{\mathcal{F}})} R^i f_! \mathcal{F}$$

is just  $\varphi_{R^i f_! \mathcal{F}}$ .

┘

*Proof:* Choose a compactification, the  $j_!$  doesn't matter, so we finish by [\(9.4.4.14\)](#). □

**Def.(9.4.4.16) [Frobenius Action on Compact Cohomology].** Let  $k \cong \mathbb{F}_{p^r} \in p\text{-Field}, X_0 \in \mathbf{Sch}^{\text{sep,ft}}/k, \mathcal{F}_0 \in \mathbf{Sh}((X_0)_{\text{ét}}), X = X_0 \times_k \bar{k}, \mathcal{F} = \mathcal{F}_0 \times_k \bar{k}$ ,

- As  $\text{Fr}_X$  is finite, by [\(9.4.4.6\)](#) it induces a pullback map  $\text{H}_{\text{ét},c}^i(X, \mathcal{F}) \rightarrow \text{H}_{\text{ét},c}^i(X, \text{Fr}_X^* \mathcal{F})$ , which by composing with the natural isomorphism

$$\text{Fr}_{\mathcal{F}} : \text{Fr}_X^* \mathcal{F} = \text{Fr}_X^*(X \rightarrow X_0)^* \mathcal{F}_0 = (X \rightarrow X_0)^*(\varphi_{r,X})^* \mathcal{F}_0 \xrightarrow{\varphi_{\mathcal{F}_0} \text{ (9.4.4.12)}} (X \rightarrow X_0)^* \mathcal{F}_0 = \mathcal{F}$$

gives an endomorphism  $\text{Fr}_{\mathcal{F}}^* : \text{H}_{\text{ét},c}^i(X, \mathcal{F}) \rightarrow \text{H}_{\text{ét},c}^i(X, \mathcal{F})$ , called the **geometric Frobenius action on the étale cohomology**.



- For  $F_X$ , there is a natural map  $\mathcal{F} \rightarrow F_X^* \mathcal{F}$ , and a natural isomorphism

$$F_{\mathcal{F}} : F_X^* \mathcal{F} = F_X^*(X \rightarrow X_0)^* \mathcal{F}_0 \cong ((X \rightarrow X_0) \circ F_X)^* \mathcal{F}_0 = \mathcal{F},$$

so we can define the action of  $F_X$  on  $H_{\text{ét},c}^i(X, \mathcal{F})$  by

$$F_X^* : H_{\text{ét},c}^i(X, \mathcal{F}) \rightarrow H_{\text{ét},c}^i(X, F_X^* \mathcal{F}) \xrightarrow{F_{\mathcal{F}}} H_{\text{ét},c}^i(X, F_X^* \mathcal{F}),$$

called the **arithmetic Frobenius action on the étale cohomology**.

- Similarly, the natural isomorphisms  $\varphi_{r,\mathcal{F}} : \varphi_{r,X}^* \mathcal{F} \cong \mathcal{F}$  (9.4.4.12) defines an action of  $\varphi_{r,X}$  on  $H_{\text{ét},c}^i(X, \mathcal{F})$  by

$$\varphi_{r,X}^* : H_{\text{ét},c}^i(X, \mathcal{F}) \rightarrow H_{\text{ét},c}^i(X, \varphi_{r,X}^* \mathcal{F}) \xrightarrow{\varphi_{r,\mathcal{F}}} H_{\text{ét},c}^i(X, F_X^* \mathcal{F}),$$

called the **absolute Frobenius action on the étale cohomology**. ┘

**Lemma (9.4.4.17).** Situation as in (9.4.4.16),  $\text{Fr}_X^* \mathcal{F} = \varphi_{r,X}^* F_X^* \mathcal{F} \xrightarrow{\varphi_{r,X}^* F_{\mathcal{F}}} \varphi_{r,X}^* \mathcal{F} \xrightarrow{\varphi_{r,\mathcal{F}}} \mathcal{F}$  is just  $\text{Fr}_{\mathcal{F}}$ , by (7.2.10.2). ┘

**Prop. (9.4.4.18) [Frobenius Actions are Compatible].** Situation as in (9.4.4.16),  $\varphi_{r,X}^* = \text{id}$  on  $H_{\text{ét},c}^i(X, \mathcal{F})$ .

In particular, so  $F_X^*$  agrees with  $\text{Fr}_X^*$  on  $H_{\text{ét},c}^i(X, \mathcal{F})$ , because (9.4.4.17) shows  $\text{Fr}_X^* = \varphi_{r,X}^* \circ F_X^*$  on  $H_{\text{ét},c}^i(X, \mathcal{G})$ . We can calculate with either one of them, and denote it by  $F_X^*$ , called the **Frobenius action on étale cohomologies**. ┘

*Proof:* Choose an injective resolution  $\mathcal{F} \rightarrow \mathcal{I}^\bullet$ , it suffices to show that

$$\Gamma(X, \mathcal{I}^\bullet) \rightarrow \Gamma(X, \varphi_* \varphi^* \mathcal{I}^\bullet) \xrightarrow{\varphi_{\mathcal{F}}} \Gamma(X, \varphi_* \mathcal{I}^\bullet)$$

is the identity. But notice by definition  $\mathcal{I}^\bullet \rightarrow \varphi_* \varphi^* \mathcal{I}^\bullet \xrightarrow{\varphi_{\mathcal{F}}} \varphi_* \mathcal{I}^\bullet$  is the inverse of the isomorphism in (9.4.4.12). □

## 5 Torsion Cohomologies

### Finiteness Theorems

**Prop. (9.4.5.1).** If  $S \in \text{NSch}_{\text{cpct}}$ ,  $f : X \rightarrow S \in \text{Sch}^{\text{sep,ft}}/S$ , and  $\mathcal{F} \in \text{Sh}_{\text{const}}(X_{\text{ét}})$  whose torsion order is invertible in  $S$ , then  $R^p f_* \mathcal{F}$  are all constructible on  $Y$ . ┘

*Proof:* Cf. [Conrad L10 P5]. □

**Cor. (9.4.5.2).** If  $k \in \text{Field}$ ,  $k^{\text{sep}} = k$ ,  $X \in \text{Sch}^{\text{proper}}/k$  and  $\mathcal{F} \in \text{Sh}_{\text{const}}(X_{\text{ét}})$ , then  $\#H_{\text{ét}}^n(X, \mathcal{F}) < \infty$  for all  $n \in \mathbb{N}$ . ┘

*Proof:* □

**Lemma (9.4.5.3).** If  $f : X \rightarrow S$  is smooth and proper, and  $\mathcal{F}$  is a locally constant finite Abelian sheaf with torsion order invertible on  $S$ , suppose  $S$  is Noetherian, then all specialization maps for  $R^p f_* \mathcal{F}$  are isomorphisms. ┘

*Proof:* Cf.[Conrad L10 P5]. □

**Prop. (9.4.5.4) [Proper Smooth Higher Direct Image of Lcc Sheaves].** If  $f : X \rightarrow S$  is smooth and proper, and  $F$  is a lcc Abelian sheaf (9.4.2.15) with torsion order invertible on  $S$ , then  $R^n f_* \mathcal{F}$  are locally constant finite sheaves for any  $n \in \mathbb{N}$ . ┘

*Proof:* By (9.4.2.20), we may assume  $\mathcal{F} = \underline{X}'$  for some finite étale scheme  $X' \rightarrow X$ . By Noetherian descent? together with proper base change, we may reduce to the case  $S$  is Noetherian. Thus by (9.4.5.1),  $R^p f_* \mathcal{F} = R^p f_! \mathcal{F}$  are constructible, and (9.4.5.3) shows that the stalk maps are isomorphisms. So (9.4.2.22) shows that  $R^p f_* \mathcal{F}$  are locally constant finite. □

## 6 $\ell$ -adic Étale Cohomologies

References are [Ekedahl, on the Adic Formalism] and [Lei Fu, Étale Cohomology Theory].

**Notation (9.4.6.1).**

- Let  $\ell \in \text{Prime}$ .
  - Let  $(\Lambda_E, \mathfrak{m}, E, \kappa_E) \in \text{CDVR}$  be of mixed characteristic  $(0, \ell)$  and suppose  $\#\kappa < \infty$ .
  - Let  $S, X \in \text{Sch}/\mathbb{Z}[\frac{1}{\ell}]$ .
  - Let  $\Lambda_n = \Lambda_E / \mathfrak{m}^n$ .
- ┘

### Artin-Rees Formalism

**Def. (9.4.6.2) [pre Artin-Rees Categories].** The **pre Artin-Rees category** of  $A$ -modules is a category s.t.

- Its objects consists of projective systems of  $A$ -modules  $M_\bullet = (M_n)_{n \in \mathbb{Z}}$  s.t.  $M_n = 0$  for  $n \ll 0$ ,
- Its morphisms are the elements of the set

$$\text{Hom}_{A\text{-R}}(M_\bullet, N_\bullet) = \varinjlim \text{Hom}(M_\bullet[d], N_\bullet).$$

An object  $M_\bullet$  in the Artin-Rees category is called a **null system** if for some  $\geq 0$  the map  $M_{n+} \rightarrow M_n$  vanishes for all  $n$ . ┘

**Prop. (9.4.6.3) [Artin-Rees Category].** The pre Artin-Rees category (9.4.6.2) is an Abelian category, and the null systems form a weak Serre-subcategory. Then we define the **Arin-Rees category** as the quotient category, denoted by  $\text{Sh}(S_{\text{ét}}; \Lambda_E)$ . ┘

*Proof:* □

**Prop. (9.4.6.4).** If the kernel and cokernel of two systems are all null systems, then they induce isomorphism on inverse limit. ┘

*Proof:* Cf.[Conrad L15, P5]. □

**Def. (9.4.6.5) [Artin-Rees  $\ell$ -adic Modules].** An object  $M_\bullet$  in the A-R category (9.4.6.3) is called an **Artin-Rees  $I$ -adic module** if it is represented by a system  $M_n$  s.t.  $M_n = 0$  for  $n < 0$  and  $M_n$  is finite over  $A_n$ ,  $M_{n+1} \otimes_{A_{n+1}} A_n \rightarrow M_n$  is an isomorphism for  $n \in \mathbb{N}$ . ┘

**Prop. (9.4.6.6).** The full subcategory of Artin-Rees  $I$ -adic modules is an Abelian category, and it is equivalent to the category of finite  $A$ -modules by the stalk functor.  $\lrcorner$

**Prop. (9.4.6.7).** The category  $\mathrm{Sh}_{\Lambda\text{-lcc}}(S_{\text{ét}})$  is equivalent to the full subcategory of  $\mathrm{Sh}_{\mathbf{m}}(S_{\text{ét}})$  consisting of objects with Artin-Rees property and that the terminal stable image is constructible.  $\lrcorner$

*Proof:*  $\square$

**Def. (9.4.6.8) [Strict  $\mathbf{m}$ -adic Sheaves].** The category  $\mathrm{Sh}_{\mathbf{m}\text{-strict}}(S_{\text{ét}})$  of **strict  $\mathbf{m}$ -adic sheaves** are defined as before.  $\lrcorner$

**Prop. (9.4.6.9).** We can strictify any  $\mathbf{m}$ -adic sheaf, Cf.[Conrad, Etale Coh].  $\lrcorner$

### Constructible and Lisse $\mathbf{m}$ -adic Sheaves

**Def. (9.4.6.10) [Constructible and Lisse Adic Sheaves].** For  $S \in \mathrm{Sch}$ ,  $\mathcal{F} \in \mathrm{Sh}(S_{\text{ét}}; \Lambda)$  is called **constructible  $\mathbf{m}$ -adic sheaf** if it is isomorphic to a strict system  $\{\mathcal{F}_n\}$  that  $\mathcal{F}_n \in \mathrm{Sh}_{\text{const}}(S_{\text{ét}}; \Lambda_n)$  (9.4.2.16). The subcategory of  $\mathrm{Sh}_{\mathbf{m}}(S)$  consisting of constructible  $\mathbf{m}$ -adic sheaves is denoted by  $\mathrm{Sh}_{\mathbf{m}\text{-const}}(S)$ .

$\mathcal{F} \in \mathrm{Sh}(S; \Lambda)$  (9.4.6.3) is called a **lisse  $\mathbf{m}$ -adic sheaf** on  $X$  if it is isomorphic to a strict system  $\{\mathcal{F}_n\}$  (9.4.6.8) s.t.  $\mathcal{F}_n \in \mathrm{Sh}_{\text{lcc}}(S_{\text{ét}}; \Lambda_n)$  (9.4.2.16). The category of lisse  $\mathbf{m}$ -adic Sheaves is denoted by  $\mathrm{Sh}_{\text{lisse}}(S_{\text{ét}}; \Lambda)$ .  $\lrcorner$

**Def. (9.4.6.11) [Tate-Twist Sheaves].** The **Tate-twist sheaf**  $\Lambda(1)$  is defined to be  $?$ . It is invertible, thus we denote its dual by  $\Lambda(-1)$ . Then  $\Lambda(r) \in \mathrm{Sh}_{\Lambda\text{-lcc}}(X_{\text{ét}})$  if  $p$  is invertible on  $X$ .

For any  $\mathbf{m}$ -adic sheaf  $\mathcal{F}$ , denote  $\mathcal{F}(1)$  to be the sheaf  $\mathcal{F} \otimes \Lambda(1)$ . Also denote  $\Lambda(r) = \Lambda(1)^{\otimes r}$ .  $\lrcorner$

**Def. (9.4.6.12) [Stalk Maps].** For any  $k \in \mathbf{Field}$ ,  $k = k^{\text{sep}}$  and  $s \in S(k^{\text{sep}})$ , there is a **stalk map**

$$\mathrm{Sh}_{\mathbf{m}\text{-const}}(S) \rightarrow \mathrm{Mod}_{\Lambda}^{\text{fin}} : \mathcal{F}_{\bullet} \mapsto \varprojlim_n (\mathcal{F}_n)_s.$$

It is well-defined by [Conrad, Etale Coh]P72.  $\lrcorner$

**Prop. (9.4.6.13).** Constructibility/lisse (9.4.6.10) are étale-local and stratification-local properties for  $\mathcal{F} \in \mathrm{Sh}_{\mathbf{m}}(S_{\text{ét}})$ .  $\lrcorner$

*Proof:* Cf.[Conrad, Etale Coh]P71.  $\square$

**Prop. (9.4.6.14).** The full subcategory  $\mathrm{Sh}_{\text{lisse}}(S_{\text{ét}}; \Lambda_E) \subset \mathrm{Sh}(S_{\text{ét}}; \Lambda_E)$  (9.4.6.10) is stable under taking kernels and cokernels.  $\lrcorner$

**Prop. (9.4.6.15) [Constructible and Lisse  $\mathbf{m}$ -adic Sheaves].** Let  $S \in \mathbf{NSch}$  and  $\mathcal{F} \in \mathrm{Sh}_{\text{const}}(S_{\text{ét}}; \Lambda_E)$ , then there is a stratification of  $X$  that  $\mathcal{F}$  is locally constant and finite on each stratum.  $\lrcorner$

*Proof:* Cf.[Conrad, Etale Coh]P73  $?$ .

By the stalk criterion of locally constant finite (9.4.2.22), a constructible extension of locally constant finite sheaves is also locally constant finite. So by the exact sequence  $1 \rightarrow \ell^{n-1}\mathcal{F}_n \rightarrow \mathcal{F}_n \rightarrow \mathcal{F}_{n-1} \rightarrow 1$ , iff we show there is a stratification that all  $\ell^{n-1}\mathcal{F}_n$  are locally constant finite, then by induction all  $\mathcal{F}_n$  are locally constant finite. But then  $\ell^{n-1}\mathcal{F}_n$  is a descending chain of quotients of  $\mathcal{F}_1$ , thus the kernel is ascending thus stabilizes because  $\mathcal{F}_1$  is constructible (9.4.2.23), so there are only f.m. such  $\ell^n\mathcal{F}_n$ , so there is a common stratification.  $\square$

**Cor. (9.4.6.16).** Exactness/Isomorphism between constructible sheaves can be checked at stalks.  $\lrcorner$

**Def. (9.4.6.17) [Lisse and Constructible  $E$ -Sheaves].** Define the category of **lisse  $E$ -sheaves** on  $X$  is defined to be the subcategory

$$\mathrm{Sh}_{\mathrm{lisse}}(X_{\mathrm{\acute{e}t}}; E) \triangleq \mathrm{Sh}_{\mathrm{lisse}}(X_{\mathrm{\acute{e}t}}; \Lambda_E) \otimes_{\Lambda_E} E \quad (9.4.6.10).$$

And define the category of **constructible  $E$ -sheaves** on  $X$  as the category

$$\mathrm{Sh}_{\mathrm{const}}(X_{\mathrm{\acute{e}t}}; E) \triangleq \mathrm{Sh}_{\mathrm{const}}(X_{\mathrm{\acute{e}t}}; \Lambda_E) \otimes_{\Lambda_E} E \quad (9.4.6.10).$$

**Def. (9.4.6.18) [Étale Tate-Twist Sheaves].** Denote  $E(r) = \Lambda_E(r) \otimes_{\Lambda_E} E \in \mathrm{Sh}_{\mathrm{lisse}}(X_{\mathrm{\acute{e}t}}; E)$  (9.4.6.17), called the **Étale Tate-twist sheaf**.  $\lrcorner$

**Prop. (9.4.6.19) [Lisse  $K$ -Sheaves and  $\pi_1(S, \bar{s})$ -Representations].** Assume  $S$  is connected, then for any  $\bar{s} \in S$  with  $\kappa(\bar{s})$  separable, the stalk map  $\mathcal{F} \rightarrow \mathcal{F}_{\bar{s}}$  induces equivalences:

$$\mathrm{Sh}_{\mathrm{const}}(S_{\mathrm{\acute{e}t}}; \Lambda_E) \cong \mathrm{Rep}_{\Lambda_E}(\pi_1(S, \bar{s})), \quad \mathrm{Sh}_{\mathrm{const}}(S_{\mathrm{\acute{e}t}}; E) \cong \mathrm{Rep}_K^{\mathrm{fd}}(\pi_1(S, \bar{s})).$$

*Proof:* The point is that by (24.2.3.2), any representation of  $\pi_1(S, \bar{s})$  stabilizes some  $\Lambda$ -lattice. So by the equivalence (9.3.2.2)(9.4.2.20) and taking limit using (9.4.6.6), we get the result about  $\mathcal{O}_E$ -sheaves and representations of  $\pi_1(S, \bar{s})$  over  $\mathcal{O}_E$ .  $\square$

**Prop. (9.4.6.20) [Lisse and Specializations].** If  $S \in \mathrm{NSch}$  and  $\mathcal{F} \in \mathrm{Sh}_{\mathrm{const}}(S_{\mathrm{\acute{e}t}}; \Lambda_E)$ , then  $\mathcal{F}$  is lisse (9.4.6.10) iff all the specialization maps for geometric points  $\mathcal{F}_{\bar{s}} \rightarrow \mathcal{F}_{\bar{\eta}}$  are bijective.

And if  $S$  is moreover normal, then the same holds for  $\mathcal{F} \in \mathrm{Sh}_{K\text{-const}}(S_{\mathrm{\acute{e}t}})$ .  $\lrcorner$

*Proof:* We can assume  $\mathcal{F}$  is strictly  $\mathfrak{m}$ -adic (9.4.6.10) and then use (9.4.2.22) on each  $\mathcal{F}_n$ .

For the second assertion, Cf. [Conrad, Etale Coh]P77.  $\square$

**Cor. (9.4.6.21) [Étale Descent for Lisse  $K$ -Sheaves].** If  $S$  is Noetherian and normal, then étale-descent holds for lisse  $K$ -sheaves.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (9.4.6.22) [Constructible Sheaves are Noetherian].** Constructible  $\mathfrak{m}$ -adic sheaves are Noetherian, i.e. scending chain of subsheaves stabilizes.  $\lrcorner$

*Proof:* Cf. [Conrad L16 P3].  $\square$

**Prop. (9.4.6.23).** For any  $\mathcal{F} \in \mathrm{Sh}_{\mathfrak{m}\text{-const}}(S_{\mathrm{\acute{e}t}})$ , there exists some  $\mathcal{G} \in \mathrm{Sh}_{\Lambda\text{-const}}(S_{\mathrm{\acute{e}t}})$  s.t.  $\mathcal{G} \subset \mathcal{F}$  and  $\mathcal{F}/\mathcal{G}$  has  $\Lambda$ -flat stalks.

In particular, any  $\mathcal{F} \in \mathrm{Sh}_{K\text{-const}}(S_{\mathrm{\acute{e}t}})$  is  $K$ -isomorphic to some  $K \otimes \mathcal{F}'$  where  $\mathcal{F}'$  has  $\Lambda$ -flat stalks.  $\lrcorner$

*Proof:* Cf. [Conrad Etale Coh]P75. ?  $\square$

**Def. (9.4.6.24) [Extensions].** Let  $\Lambda'/\Lambda$  be an extension, then we can define extension functors

$$\Lambda' \otimes_{\Lambda} - : \mathrm{Sh}_{\Lambda\text{-const}}(X_{\mathrm{\acute{e}t}}) \rightarrow \mathrm{Sh}_{\Lambda'\text{-const}}(X_{\mathrm{\acute{e}t}}), \quad K' \otimes_K - : \mathrm{Sh}_{K\text{-const}}(X_{\mathrm{\acute{e}t}}) \rightarrow \mathrm{Sh}_{K'\text{-const}}(X_{\mathrm{\acute{e}t}}).$$

?  $\lrcorner$

**Constructible and Lisse  $\overline{\mathbb{Q}_\ell}$ -Sheaves**

**Def. (9.4.6.25) [ $\overline{\mathbb{Q}_\ell}$ -Sheaves].** The category  $\mathrm{Sh}_{\mathrm{const}}(X_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell})$  of **constructible  $\overline{\mathbb{Q}_\ell}$ -sheaves** on  $X$  is defined to be the direct limit of categories

$$\mathrm{Sh}_{\mathrm{const}}(X_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell}) \triangleq \varinjlim_{E \in \ell\text{-}\mathbf{NField}} \mathrm{Sh}_{\mathrm{const}}(X_{\acute{\mathrm{e}}\mathrm{t}}; E).$$

And the category of **lisse  $\overline{\mathbb{Q}_\ell}$ -sheaves** on  $X$  is defined to be

$$\mathrm{Sh}_{\mathrm{lisse}}(X_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell}) \triangleq \varinjlim_{E \in \ell\text{-}\mathbf{NField}} \mathrm{Sh}_{\mathrm{lisse}}(X_{\acute{\mathrm{e}}\mathrm{t}}; E) \quad (9.4.6.17).$$

┘

**Prop. (9.4.6.26) [Lisse  $\overline{\mathbb{Q}_\ell}$ -Sheaves and  $\pi_1(X, \bar{x})$ -Representations].** Assume  $S$  is connected, then for any  $\bar{s} \in S$  with  $\kappa(\bar{s})$  separable, the stalk map  $\mathcal{F} \rightarrow \mathcal{F}_{\bar{s}}$  induces equivalences:

$$\mathrm{Sh}_{\mathrm{const}}(X_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell}) \cong \mathrm{Rep}_{\overline{\mathbb{Q}_\ell}}^{\mathrm{fd}}(\pi_1(X, \bar{x})), \quad \mathrm{Sh}_{\mathrm{const}}(X_{\acute{\mathrm{e}}\mathrm{t}}; \Lambda) \cong \mathrm{Rep}_\Lambda(\pi_1(X, \bar{s})).$$

┘

*Proof:* This follows from (9.4.6.19) and (24.2.3.2) by taking direct limit.  $\square$

**Cor. (9.4.6.27) [Irreducible and Semisimple Lisse Sheaves].**  $\mathcal{F} \in \mathrm{Sh}_{\mathrm{lisse}}(S_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell})$  is called an **irreducible lisse sheaf** (resp. **semi-simple lisse sheaf**) if its corresponding representation (9.4.6.26) is.

And  $\mathcal{F}$  is called a **geometrically irreducible lisse sheaf** (resp. **geometrically semi-simple lisse sheaf**) if  $\mathcal{F}_{k^{\mathrm{sep}}} \in \mathrm{Sh}_{\mathrm{lisse}}((S_{k^{\mathrm{sep}}})_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell})$  is irreducible (resp. semi-simple), or equivalently, its corresponding representation is irreducible/semisimple as a  $\pi_1(S_{k^{\mathrm{sep}}}, \bar{s})$ -representation.  $\square$

 **$\ell$ -adic Cohomologies**

**Def. (9.4.6.28) [Derived Category of Constructible Sheaves].** Define the **derived category of bounded complexes with constructible cohomology sheaves**  $D_{\mathrm{const}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; \Lambda_E)$  and the **derived category of bounded complexes with constructible cohomology groups with finite Tor dimensions**  $D_{\mathrm{csttf}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; \Lambda_E)$ .

And denote

$$\begin{aligned} D_{\mathrm{const}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; E) &\triangleq D_{\mathrm{const}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; \Lambda_E) \otimes_{\Lambda_E} E, & D_{\mathrm{csttf}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; E) &\triangleq D_{\mathrm{csttf}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; \Lambda_E) \otimes_{\Lambda_E} E. \\ D_{\mathrm{const}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell}) &\triangleq \varinjlim_{E \in \ell\text{-}\mathbf{NField}} D_{\mathrm{const}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; E), & D_{\mathrm{csttf}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; \overline{\mathbb{Q}_\ell}) &\triangleq \varinjlim_{E \in \ell\text{-}\mathbf{NField}} D_{\mathrm{csttf}}^b(X_{\acute{\mathrm{e}}\mathrm{t}}; E). \end{aligned}$$

┘

*Proof:* ? Cf. [Ekedahl, on the Adic Formalism].  $\square$

**Prop. (9.4.6.29) [Direct Pushforward of  $m$ -adic Sheaves].** For a compactifiable morphism  $f : X \rightarrow S \in \mathrm{Sch}$  and  $\mathcal{F} \in \mathrm{Sh}_{\mathrm{const}}(X_{\acute{\mathrm{e}}\mathrm{t}}; \Lambda_E)$ , we can define  $R^i f_*$  and  $R^i f_!$  termwisely, and we have  $R^i f_* \mathcal{F}$  is a constructible sheaf, hence so is  $R^i f_!$ .  $\square$

*Proof:* Cf. [Weil Conjecture and Étale sheaves, P128].  $\square$

**Def. (9.4.6.30) [ $p$ -adic étale Cohomologies].** For  $X \in \mathrm{Sch}$ ,  $p \in \mathrm{Prime}$  and any  $i \in \mathbb{N}$ , define

$$H_{\acute{\mathrm{e}}\mathrm{t}}^i(X; \mathbb{Q}_p) = \left( \varprojlim_n H_{\acute{\mathrm{e}}\mathrm{t}}^i(X; \mathbb{Z}/(p^n)) \right) \left[ \frac{1}{p} \right],$$

called the  $i$ -th  $p$ -adic étale cohomology of  $X$ .  $\square$

## 6-operator Formalism

### Artin-Schreier Sheaves

**Def. (9.4.6.31) [Artin-Schreier Sheaves].** The Artin-Schreier cover (9.4.1.36) gives a map  $\pi_1(\mathbb{A}_{\mathbb{F}_q}^1, \bar{s}) \rightarrow \mathbb{F}_q$ , when composed with any character  $\psi : \mathbb{F}_q \rightarrow \overline{\mathbb{Q}_\ell}^\times$ , we get a rank 1  $\ell$ -adic étale sheaf  $\mathcal{L}(\psi)$  via (9.4.6.19), called an **Artin-Schreier sheaf** on  $\mathbb{A}_{\mathbb{F}_q}^1$ .

Moreover, if we extend  $\psi$  to  $\mathbb{F}_{q^n}$  by  $\psi_n = \psi \circ \text{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}$ , then the definition of  $\mathcal{L}(\psi)$  or even  $\mathcal{L}(\psi_x)$  is compatible with base change. (where  $\psi_x(y) = \psi(xy)$ .)  $\lrcorner$

*Proof:* This is because over a base change field  $\mathbb{F}_{q^n}$ , there is a map of Galois coverings:

$$\wp^{(n)} = \wp \circ \text{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q} : \mathbb{A}_{\mathbb{F}_{q^n}}^1 \rightarrow \mathbb{A}_{\mathbb{F}_q}^1,$$

so by a minute's thought, their corresponding maps  $\pi_1(\mathbb{A}_{\mathbb{F}_{q^n}}^1, \bar{s}) \rightarrow \overline{\mathbb{Q}_\ell}^\times$  are the same. Thus  $\mathcal{L}(\psi_n) \cong \mathcal{L}(\psi)$  over  $\mathbb{F}_{q^n}$ .

For  $\psi_x$ , just notice if  $x \in \mathbb{F}_q$ ,  $\psi_x \circ \text{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q} = (\psi_n)_x$ .  $\square$

**Prop. (9.4.6.32).** Situation as in (9.4.6.31), for  $1 \neq \psi \in \chi(\mathbb{F}_q)$  and  $x \in \mathbb{F}_q$ , denote  $\psi_x \in \chi(\mathbb{F}_q) : y \mapsto \psi(xy)$ , then

$$\wp_*(\overline{\mathbb{Q}_\ell}) = \bigoplus_{x \in \mathbb{F}_q} \mathcal{L}(\psi_x).$$

$\lrcorner$

*Proof:* Cf. [K-W01]P39.  $\square$

**Prop. (9.4.6.33) [Frobenius Action on Artin-Schreier Sheaves].** Situation as in (9.4.6.31), for any  $x \in \mathbb{A}^1(\mathbb{F}_{q^n})$ ,  $\text{Frob}_x^{-1}$  acts on the stalk  $\mathcal{L}(\psi^{-1})_{\bar{x}}$  via multiplication by  $\psi(\text{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(x))$ .  $\lrcorner$

*Proof:* By (9.4.6.31), we may base change to  $\mathbb{F}_{q^n}$ . Let  $\alpha \in \mathbb{A}^1(\overline{\mathbb{F}_q})$  s.t.  $\alpha^{q^n} - \alpha = x$ , then the arithmetic Frobenius at  $x$  maps  $\alpha$  to  $\alpha^{q^n} = \alpha + x$ . So by the definition of fundamental group (9.3.2.2), (9.6.2.4) and the definition of the Artin-Schreier sheaf (9.4.6.31),  $\text{Frob}_x^{-1}$  acts via multiplication by  $\psi(\text{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(x))$ .  $\square$

## Analytification

### Comparison Theorems

**Thm. (9.4.6.34) [Topological Comparison, Artin].** For  $f : X \rightarrow S \in \text{Sch}^{\text{sep, ft}}/\mathbb{C}$ ,

- If  $\mathcal{F} \in \text{Sch}_{\text{tor}}(S_{\text{ét}})$ , then the comparison morphism for  $R^\bullet f_!$  is an isomorphism.
- If  $\mathcal{F} \in \text{Sch}_{\text{const}}(S_{\text{ét}})$ , then the comparison morphism for  $R^\bullet f_*$  is an isomorphism.

$\lrcorner$

*Proof:* [Conrad, Etale Coh] ?  $\square$

**Cor. (9.4.6.35) [Comparison with Betti Cohomology].** For  $X \in \text{Sch}^{\text{ft, sep}}/\mathbb{C}$ ,  $A \in \text{Ab}^{\text{fin}}$ , there is a canonical isomorphism

$$H_{\text{ét}}^q(X, \underline{A}_X) \cong H_{\text{Betti}}^q(X; A).$$

$\lrcorner$

**Cor. (9.4.6.36) [Good Reduction cases].** For  $(K, \mathcal{O}_K, \kappa) \in p\text{-NField}$  and  $\mathcal{X} \in \text{SmPrpr}/\mathcal{O}_K$ , let  $X = \mathcal{X}_\kappa$ , then for  $\ell \neq p$ , we have

$$H_{\text{ét}}^i(X_{\bar{\kappa}}; \mathbb{Q}_\ell) \cong H^i(\mathcal{X}_{\bar{\eta}}; \mathbb{Q}_\ell) \cong H_{\text{Betti}}^i(\mathcal{X}_{\mathbb{C}})_{\mathbb{Q}_\ell}.$$

In particular,  $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$  is an unramified representation of  $\text{Gal}_K$ .  $\lrcorner$

*Proof:* Let  $\text{Spec } \mathcal{O}_{F_v} = \{s, \eta\}$ , where  $s$  is the special point and  $\eta$  the generic point. Let  $\bar{\eta}, \bar{s}$  be arithmetic points mapping to  $\eta, s$  resp. By proper base change (9.4.3.1), The stalk of higher direct image of  $X \rightarrow \mathcal{O}_{F_v}$  along  $\bar{s}, \bar{\eta}$  are  $H_{\text{ét}}^i(\bar{X}, \mathbb{Q}_\ell)$  and  $H^i(\mathcal{X}_{\bar{\eta}}, \mathbb{Q}_\ell)$  resp., and they are the same by (9.4.5.4). And  $H_{\text{ét}}^i(\mathcal{X}_{\bar{\eta}}, \mathbb{Q}_\ell) \cong H_{\text{Betti}}^i(X, \mathbb{Q}_\ell)$  by (9.4.6.35) and the fact  $\bar{\mathbb{Q}}_\ell$  is isomorphic to  $\mathbb{C}$ .  $\square$

**Prop. (9.4.6.37) [Étale and Fppf Comparison].** Let  $X \in \text{Sch}$  and  $G$  a smooth commutative group scheme over  $X$ , then there are natural isomorphisms

$$H_{\text{ét}}^*(X; G) \cong H_{\text{fppf}}^*(X; G).$$

$\lrcorner$

*Proof:* Cf. [Mil80Etale Cohomologies, Prop. III.3].  $\square$

### **A<sub>f</sub>-Cohomologies**

**Def. (9.4.6.38) [A<sub>f</sub>-Cohomologies].** Define

$$H_{\text{ét}}^n(X; \mathbf{A}_f) \triangleq \left( \varprojlim_m H_{\text{ét}}^n(X, \mathbb{Z}/(m)) \right) \otimes \mathbb{Q}.$$

Then if  $X$  is a smooth complete variety over  $\mathbb{C}$ , then by (9.4.6.35), there is a canonical isomorphism

$$H_{\text{Betti}}^n(X) \otimes \mathbf{A}_f \cong H_{\mathbf{A}_f}^n(X).$$

Thus  $H_{\text{Betti}}^n(X) \otimes \mathbf{A}_f$  is intrinsic in  $X$  and  $H_{\mathbf{A}_f}^n(X)$  free over  $\mathbf{A}_f$ .  $\lrcorner$

### **Conjectures**

**Conj. (9.4.6.39) [Semisimplicity Conjecture, Grothendieck-Serre].** For  $F \in \text{NField}$  and  $X \in \text{SmPrpr}/F$ , the action of  $\text{Gal}_F$  on  $H_{\text{ét}}^i(X_{\bar{F}}, \mathbb{Q}_p)$  is semisimple.  $\lrcorner$

*Proof:*

$\square$

**Thm& Conj. Cor. (9.4.6.40).** This is known for  $X \in \text{AbVar}/F$  and some Shimura varieties  $X/F$ .  $\lrcorner$

*Proof:* By work of Faltings?  $\square$

**Conj. (9.4.6.41) [F-Semisimplicity Conjecture].** For  $F \in \text{NField}$  and  $X \in \text{SmPrpr}/F$ , if  $v \nmid p$ , then the action of  $\sigma_v$  on  $H_{\text{ét}}^i(X_{\bar{F}}, \mathbb{Q}_p)$  is semisimple.

There is also a version with  $v|p$ , which says that the action of  $\varphi_{\text{crys}, v}^{\deg v}$  on  $D_{\text{crys}}(H_{\text{ét}}^i(X_{\bar{F}}, \mathbb{Q}_p)|_{\text{Gal}_{F_v}})$  is semisimple.  $\lrcorner$

*Proof:*

$\square$

**Thm& Conj. Cor. (9.4.6.42).**  $F$ -semi-simplicity conjecture (9.4.6.41) is true for  $X \in \text{AbVar}/F$ .  $\lrcorner$

*Proof:* by work of Tate?  $\square$

**Weil-Axioms**

**Prop. (9.4.6.43) [Compact  $\ell$ -adic Étale Cohomologies].** Let  $k \in \mathbf{Field}$ ,  $k = k^{\text{sep}}$ ,  $X \in \mathbf{Sch}^{\text{sep,ft}}/k$ , and  $\Lambda \in \mathcal{A}\mathbf{b}^{\text{tor}}$  with  $(\text{char } k, \#\Lambda) = 1$ , then

1.  $H_{\text{ét},c}^i(X, \underline{\Lambda})$  are finite  $\Lambda$ -modules.
2. For  $n \in \mathbb{Z}$ ,  $H_{\text{ét},c}^n(X, \underline{\Lambda}) = 0$  unless  $n \in [2 \dim X]$ . And if  $X$  is affine, then  $H_{\text{ét},c}^n(X, \underline{\Lambda}) = 0$  unless  $n \in [\dim X]$ .
3. There is a natural isomorphism  $H_{\text{ét},c}^{2d}(X, \underline{\Lambda}) \cong \Lambda[S]$ , where  $S$  is the set of irreducible components of  $X$  of dimension  $d$ .
4. there is a long exact sequence

$$\cdots \rightarrow H_{\text{ét},c}^p(U, \underline{\Lambda}) \rightarrow H_{\text{ét},c}^p(X, \underline{\Lambda}) \rightarrow H_{\text{ét},c}^p(Z, \underline{\Lambda}) \rightarrow H_{\text{ét},c}^{p+1}(U, \underline{\Lambda}) \rightarrow \cdots$$

$$5. \text{ For } d \in \mathbb{Z}_+, H_{\text{ét},c}^i(\mathbb{A}_k^d, \underline{\Lambda}) = \begin{cases} \Lambda & , i = 2d \\ 0 & , \text{otherwise} \end{cases}.$$

6. If  $G \in \mathcal{A}\mathbf{lgGrp}_{\text{cnd}}/k$  acting regularly on  $X$ , then  $G(k)$  acts trivially on  $H_{\text{ét},c}^i(X, \underline{\Lambda})$ .

┘

*Proof:* 1 follows from (9.4.5.1). 4 follows from excision (9.4.4.7).

2 follows from (9.4.2.11) and (9.4.2.12).

3: By item2 and item4, it suffices to show for  $X$  irreducible. ?

5: This follows from Künneth formula (9.4.6.46) and (9.4.7.5).

6: Cf. [Deligne-Lustig] Prop 6.4 ?.

□

**Def. (9.4.6.44) [Cup Product].**

┘

**Cor. (9.4.6.45) [Homotopy Axiom].** Two maps  $\varphi, \varphi' : X \rightarrow Y \in \mathbf{Sch}$  induce the same map on the  $\ell$ -adic cohomology if their graphs are rationally equivalent.

┘

*Proof:* This is because the maps on cohomology depends only on the rational equivalence classes of the graph. ?

□

**Prop. (9.4.6.46) [Künneth Formula].** Let  $X, Y$  be complete varieties over  $k$ , then there is a natural isomorphism

$$H_{\text{ét}}^*(X, \mathbb{Q}_\ell) \otimes H_{\text{ét}}^*(Y, \mathbb{Q}_\ell) \cong H_{\text{ét}}^*(X \times Y, \mathbb{Q}_\ell).$$

┘

*Proof:* ?

□

**Lemma (9.4.6.47) [Torsion Poincaré-Duality].**

┘

**Prop. (9.4.6.48) [Poincaré-Duality].** If  $X \in \mathbf{Sch}^{\text{sm}\mathcal{P}\text{rpr}}/k$ ,  $d = \dim X$  and  $\mathcal{F} \in \mathbf{Sh}_{\text{lisse}}(X_{\text{ét}})$ , then there is a natural isomorphism  $H_{\text{ét},c}^{2d}(X_{k^{\text{sep}}}, \mathbb{Q}_\ell(d)) \xrightarrow{\text{tr}_X} \mathbb{Q}_\ell$  and a perfect pairing

$$H_{\text{ét},c}^n(X_{k^{\text{sep}}}, \mathcal{F}) \times H_{\text{ét},c}^{2d-n}(X_{k^{\text{sep}}}, \mathcal{F}^\vee(d)) \xrightarrow{\cup} H_{\text{ét},c}^{2d}(X_{k^{\text{sep}}}, \mathbb{Q}_\ell(d)) \xrightarrow{\text{tr}_X} \mathbb{Q}_\ell,$$

That is  $\text{Gal}_k$ -equivariant.

┘



*Proof:* Cf.[Conrad L12-13], [Weil 2Bhatt P5].?

□

**Def. (9.4.6.49) [Cycle Maps].** For  $X \in \text{Sch}^{\text{SmPrpr}}/k$ , there exists an **étale cycle map**

$$\text{cyl}_\ell : \text{CH}^j(X) \rightarrow H_{\text{ét}}^{2j}(X_{k^s}, \mathbb{Q}_\ell)(j)$$

s.t.

$$x \cdot y = \text{tr}_X(\text{cyl}_\ell(x) \cup \text{cyl}_\ell(y)).$$

┘

*Proof:*

□

## 7 Curves and Surfaces

**Prop. (9.4.7.1).** Let  $k \in \text{Field}$  s.t.  $k = k^{\text{sep}}$  and  $X \in \text{Sch}^{\text{sep,ft,dim}=1}/k$ . If  $\mathcal{F} \in \text{Sh}_{\text{tor}}(X_{\text{ét}})$ , then  $H_{\text{ét}}^i(X, \mathcal{F}) = 0$  for  $i \in \mathbb{Z}_{\geq 3}$ . If  $\mathcal{F} \in \text{Sh}_{\text{const}}(X_{\text{ét}})$ , then  $H_{\text{ét}}^i(X, \mathcal{F})$  are finite.

Moreover if  $X$  is affine and  $\mathcal{F}$  is locally killed by  $n$  not divisible by  $\text{char } k$  or  $X$  is proper and sections of  $\mathcal{F}$  are locally  $p$ -torsions with  $p = \text{char } k > 0$ , then  $H_{\text{ét}}^2(X, \mathcal{F}) = 0$ . ┘

*Proof:* Cf.[Conrad L4 P4] and [Tamme].

□

**Lemma (9.4.7.2) [Smooth Complete Curves].** If  $k \in \text{Field}$ ,  $k = k^{\text{sep}}$ ,  $g \in \mathbb{N}$  and  $X \in \text{SmPrprCur}^g/k$  and  $n \in \mathbb{Z} \cap k^\times$ , then there are canonical identifications

$$H_{\text{ét}}^q(X, \mu_n) = \begin{cases} \mu_n(k) & , q = 0 \\ \text{Pic}^0(X)[n] & , q = 1 \\ \mathbb{Z}/n\mathbb{Z} & , q = 2 \\ 0 & , q \geq 3 \end{cases}, \quad \#H_{\text{ét}}^q(X, \mu_n) = \begin{cases} n & , q = 0 \\ n^{2g} & , q = 1 \\ n & , q = 2 \\ 0 & , q \geq 3 \end{cases}.$$

In particular, because  $\mathbb{Z}/(n) \cong \mu_n$ , we can get the corresponding cohomologies. ┘

*Proof:* Cf.[Sta]03RQ.

□

**Prop. (9.4.7.3) [Torsion-Freeness].** Let  $k \in \text{Field}$ ,  $L \in \ell\text{-NField}$ ,  $X \in \text{SmCur}/k$ , then  $H_{\text{ét},c}^i(X, \mathcal{O}_L)$  are  $\mathcal{O}_L$ -torsion-free. ┘

*Proof:* Cf. [SGA4 $\frac{1}{2}$ , Chap3.3].?

□

**Prop. (9.4.7.4) [Curve and Jacobians].** For  $k \in \text{Field}$  and  $C \in \text{SmPrprCur}/k$ , we have

$$H_{\text{ét}}^1(C, \mathbb{Q}_\ell) \cong H_{\text{ét}}^1(\text{Jac}(C), \mathbb{Q}_\ell).$$

┘

*Proof:*

□

**Prop. (9.4.7.5) [Non-Complete Curves].** If  $k \in \text{Field}$ ,  $k = k^{\text{sep}}$ ,  $\ell \in \text{Prime} \setminus \{\text{char } k\}$ , then

$$H_{\text{ét},c}^i(\mathbb{A}_k^1, \mathbb{Q}_\ell) = \begin{cases} \mathbb{Q}_\ell & , i = 2 \\ 0 & , \text{otherwise} \end{cases}, \quad H_{\text{ét},c}^i(\mathbb{A}_k^1 \setminus \{0\}, \mathbb{Q}_\ell) = \begin{cases} \mathbb{Q}_\ell & , i = 1, 2 \\ 0 & , \text{otherwise} \end{cases}$$

┘

*Proof:* These follow from (9.4.7.2) by excision and (9.4.2.12).

□

## 9.5 Pro-Étale Cohomology

References are [?B-S14] and [Sta]Chap56.

Deligne introduced a derived category of  $\ell$ -adic sheaves as a certain 2-limit of categories of complexes of sheaves of  $\mathbb{Z}/\ell^n\mathbb{Z}$ -modules on the étale site of a scheme  $X$ . This approach is used in the paper by Beilinson, Bernstein, and Deligne ([BBD82]) as the basis for the theory of perverse sheaves. In a paper entitled “Continuous Étale Cohomology” ([Jan88]) Jannsen discusses an important variant of the cohomology of a  $\ell$ -adic sheaf on a variety over a field. His paper is followed up by a paper of Torsten Ekedahl ([Eke90]) who discusses the adic formalism needed to work comfortably with derived categories defined as limits.

### 1 Ring-Theoretical Stuff

**Def. (9.5.1.1) [étale Local Isomorphism].** A ring map  $A \rightarrow B$  is called a **local isomorphism** if for every prime  $\mathfrak{q} \in \text{Spec } B$ , there is a nbhd  $\text{Spec } B_{\mathfrak{q}}$  that  $\text{Spec } B_{\mathfrak{q}} \rightarrow \text{Spec } A$  is an open immersion.  $\lrcorner$

**Prop. (9.5.1.2).** The class of local isomorphisms is stable under base change and compositions. (This follows from (7.4.5.11)).

Moreover, if  $A \rightarrow B \rightarrow C$  are ring maps that  $A \rightarrow B, A \rightarrow C$  are both local isomorphisms, then  $B \rightarrow C$  is also a local isomorphism.  $\lrcorner$

**Def. (9.5.1.3) [w-Local Rings].** A ring  $A$  is called **w-local** if  $\text{Spec } A$  is w-local (4.12.3.20). it is called **strictly w-local** if it is w-local and every f.f. étale map  $A \rightarrow B$  has a section. A map of rings is called **w-local** if it induces a w-local map (4.12.3.20) on the  $\text{Spec}$ .  $\lrcorner$

**Prop. (9.5.1.4).** A w-local ring  $A$  is strictly w-local iff all local rings of  $A$  at closed pts are strictly-Henselian (5.3.10.1).  $\lrcorner$

*Proof:* Cf. [Pro-Etale Cohomology, Scholze, P10].  $\square$

### Ind-Zariski Algebra

**Def. (9.5.1.5) [Ind-Zariski Algebra].** A ring map  $A \rightarrow B$  is called **ind-Zariski/smooth/étale** if  $B$  is a filtered colimit of local isomorphisms/smooth ring maps/étale ring maps  $A \rightarrow B_i$ .  $\lrcorner$

**Prop. (9.5.1.6) [Properties of Ind-Zariski Algebras].** Ind-Zariski ring maps are stable under base change and composition, by (9.5.1.2). ?

If  $A \rightarrow B \rightarrow C$  are ring maps that  $A \rightarrow B, A \rightarrow C$  are both ind-Zariski, then  $B \rightarrow C$  is also ind-Zariski.

$A \rightarrow B$  be ind-Zariski, then it identifies local rings.  $\lrcorner$

*Proof:*  $\square$

**Def. (9.5.1.7) [Ind-(Zariski Localization)].** A ring map  $A \rightarrow B$  is called a **Zariski localization** if  $B = \prod_i^n A_{f_i}$ . An **ind-(Zariski localization)** of  $A$  is a colimit of Zariski localizations of  $A$ .  $\lrcorner$

### Ind-Smooth Algebra

### Ind-Étale Algebra

## 9.6 Cohomologies over Finite Fields

Basic References are [Conrad Seminar note in Stanford], [Seminar on Gross-Zagier over Function Fields, Lei Fu], [Seminar notes on Weil 2 Bhatt], [K-W01], [Étale Cohomology and Weil Conjecture], [A course on Weil Conjectures, Szamuely], [Deligne's proof of the Weil Conjecture, Jannsen].

**Notation(9.6.0.1).**

- Let  $k_0 \in \mathbf{Field}^{\text{fin}}$  and  $k = \overline{k_0}$ .
- Denote  $k_n$  the unique extension of  $k_0$  of degree  $n$  in  $k$ .
- Let  $\ell \in \mathbf{Prime} \setminus \{\text{char } k\}$ .
- Let  $X_0 \in \mathbf{Sch}^{\text{ft}}/k$  and denote  $X = X_0 \otimes_{k_0} k$ .
- Use notations on Étale Cohomology Theory defined in [Étale Cohomology Theory](#).
- Use notations on Frobenius Morphisms defined in ??.
- If  $\mathcal{F}_0 \in \mathbf{Sh}(X_{\text{ét}})$ , then denote  $\mathcal{F} = \mathcal{F}_0 \otimes_k \overline{k}$ .
- If  $x_0 \in |X_0|_0$ , denote  $x = x_0 \otimes_k \overline{k}$ , and choose a arbitrary  $\overline{x} \in x$ .
- Fix a CDVR  $(\Lambda, K, \mathfrak{m}, \kappa_\Lambda)$  of mixed characteristic  $(0, \ell)$ .
- For a Noetherian ring  $A \in \mathbf{CRing}$  with a distinguished ideal  $I \in \mathbf{Ideal}(A)$ , denote  $A_n = A/I^n$ .

┘

### 1 Trace Formulae

References are [Pin92], [Lefschetz-Verdier Rigid geometry, Lefschetz-Verdier trace formula and Deligne's conjecture, Fujiwara]. [Lefschetz-Verdier Trace Formula and a Generalization of a Theorem of Fujiwara].

**Notation(9.6.1.1).**

- Let  $k \in \mathbf{Field}$  s.t.  $k = k^{\text{sep}}$ .
- Let  $X, X_1, X_2 \in \mathbf{Sch}^{\text{sep,ft}}/k$ .
- Let  $E_\lambda \in \ell\text{-NField}$ .
- Let  $\Lambda \in \{\mathcal{O}_\lambda\} \cup \{\mathcal{O}_\lambda/\mathfrak{m}_\lambda^n\}_{n \in \mathbb{Z}_+}$ .

┘

### Cohomological Correspondences

**Def.(9.6.1.2)[Correspondences].** A **correspondence of schemes** between  $X_1, X_2 \in \mathbf{Sch}^{\text{sep,ft}}/k$  is a map  $c : C \rightarrow X_1 \times X_2$ . For such a correspondence, we denote  $c_1 \triangleq \text{pr}_1 \circ c, c_2 \triangleq \text{pr}_2 \circ c$ .

┘

**Def.(9.6.1.3)[Morphisms and Correspondences].** Any  $f : X \rightarrow X \in \mathbf{Sch}/S$  induces a correspondence

$$\Gamma_f : X \xrightarrow{f \times \text{id}_X} X \times_S X \in \mathbf{Sch}/S.$$

┘

**Def. (9.6.1.4) [Fixed Points of Correspondences].** For a correspondence  $a : Y \rightarrow X \times X \in \text{Sch}/k$  (9.6.1.2), the **fixed point scheme of the correspondence** is defined to be  $\text{Fix}(a) \stackrel{\Delta}{=} Y \times_{X \times X} \Delta_X$ .

And we denote the natural maps  $\Delta'_c : \text{Fix}(c) \rightarrow C, c' : \text{Fix}(c) \rightarrow X$ .  $\lrcorner$

**Def. (9.6.1.5) [Proper Correspondences].** A morphism of correspondences  $[f] = (f^\natural, f_1, f_2)$  between two correspondences  $c : C \rightarrow X_1 \times X_2, b : B \rightarrow Y_1 \times Y_2 \in \text{Sch}/k$  is called a **proper morphism** if  $f^\natural, f_1, f_2$  are all proper. And  $c : C \rightarrow X_1 \times X_2$  is called a **proper correspondence** if  $C/k, X_1/k, X_2/k$  are all proper.  $\lrcorner$

**Def. (9.6.1.6) [Compactifications of Correspondences].** For a correspondence  $c : C \rightarrow X_1 \times X_2 \in \text{Sch}^{\text{sep,ft}}/k$ , a **compactification of correspondence** of  $c$  is a pair  $([j], \bar{c})$  where  $\bar{c} : \bar{C} \rightarrow \bar{X}_1 \times \bar{X}_2 \in \text{Sch}^{\text{proper}}/k$  is a proper correspondence and  $[j] = (j_c, j_1, j_2)$  where  $j_c : C \rightarrow \bar{C}, j_1 : X_1 \rightarrow \bar{X}_1, j_2 : X_2 \rightarrow \bar{X}_2$  are all compactifications (7.8.3.1).

If  $c : C \rightarrow X_1 \times X_2 \in \text{Sch}^{\text{sep,ft}}/k$ , then any compactifications of schemes  $j_1 : X_1 \hookrightarrow \bar{X}_1, j_2 : X_2 \hookrightarrow \bar{X}_2$  extends to a compactification of correspondences  $([j], \bar{c})$ .  $\lrcorner$

*Proof:* To show that the compactification of correspondences exists, take an arbitrary compactification  $j_c : C \rightarrow C'$ , and take  $\bar{C}$  to be the schematic image (7.4.5.13) of the immersion

$$C \rightarrow C' \times \bar{X}_1 \times \bar{X}_2.$$

Then  $\bar{c} \stackrel{\Delta}{=} \text{pr}_{2,3} : \bar{C} \rightarrow \bar{X}_1 \times \bar{X}_2$  is a proper correspondence extending  $c$ , by (7.4.5.21).  $\square$

**Def. (9.6.1.7) [Cohomological Correspondences].** For a correspondence  $c : C \rightarrow X_1 \times_S X_2 \in \text{Sch}/S$  (9.6.1.5) a  **$c$ -morphism** is a morphism  $u : R(c_2)_! c_1^* K_1 \rightarrow K_2 \in D_{\text{csttf}}^b((X_2)_{\text{ét}}; \overline{\mathbb{Q}}_\ell)$  for some  $K_1 \in D_{\text{csttf}}^b((X_1)_{\text{ét}}; \overline{\mathbb{Q}}_\ell)$  and  $K_2 \in D_{\text{csttf}}^b((X_2)_{\text{ét}}; \overline{\mathbb{Q}}_\ell)$ .  $\lrcorner$

**Prop. (9.6.1.8) [Base Change Cohomological Correspondences].** If

$$\begin{array}{ccccc} X_1 & \xleftarrow{c_1} & C & \xrightarrow{c_2} & X_2 \\ \downarrow f_1 & & \downarrow f^\natural & & \downarrow f_2 \\ Y_1 & \xleftarrow{b_1} & B & \xrightarrow{b_2} & Y_2 \end{array} \in \text{Sch}^{\text{sep,ft}}/k$$

is a map of correspondences (9.6.1.2), and either  $(C_2)_{\text{red}} \cong (B \times_{Y_2} X_2)_{\text{red}}$  or  $f^\natural, f_2$  are étale, then we have a base change map  $\text{BC} : R(c_2)_!(f^\natural)^* \rightarrow (f_2)^* R(b_2)_!$  ?. Thus any  $b$ -morphism  $u : R(b_2)_!(b_1)^* K_1 \rightarrow K_2$  (9.6.1.7) pulls back to a  $c$ -morphism

$$[f]^*(u) : R(c_2)_!(c_1)^*(f_1^* K_1) = R(c_2)_!(f^\natural)^*(b_1)^* \mathcal{F}_1 \xrightarrow{\text{BC}} (f_2)^* R(b_2)_! b_1^* K_1 \xrightarrow{f_2^* u} f_2^* K_2.$$

In particular, we can restricts cohomological correspondences to open subschemes.  $\lrcorner$

**Prop. (9.6.1.9) [Extension of Cohomological Correspondences].** For a proper correspondence  $a : Y \rightarrow X \times_k X \in \text{Sch}/\kappa$  (9.6.1.5), suppose

- $Y \hookrightarrow \bar{Y} \in \text{Sch}/\kappa, X \hookrightarrow \bar{X} \in \text{Sch}/\kappa$  are compactifications (7.8.3.1).
- $\bar{a} : \bar{Y} \rightarrow \bar{X} \times_k \bar{X} \in \text{Sch}/\kappa$  is a proper correspondence extending  $a$ ,
- $K \in D_{\text{csttf}}^b(X_{\text{ét}}; \Lambda)$  lifting to a cohomological correspondence  $c_K \in \text{Corr}^{\text{coh}}(a; K)$  (9.6.1.7).

Then  $c_K$  extends canonically to a cohomological correspondence for  $j_!K$  (resp.  $j_*K$ ) if  $a_1$  (resp.  $a_2$ ) is proper. And if  $Y \subset \bar{Y}$  is proper, then this extension is unique.  $\lrcorner$

*Proof:* Cf. [Fuj97]P496.  $\square$

**Prop. (9.6.1.10) [Pushforward Cohomological Correspondences].** If

$$\begin{array}{ccccc} X_1 & \xleftarrow{c_1} & C & \xrightarrow{c_2} & X_2 \\ \downarrow f_1 & & \downarrow f^\natural & & \downarrow f_2 \\ Y_1 & \xleftarrow{b_1} & B & \xrightarrow{b_2} & Y_2 \end{array} \in \text{Sch}^{\text{sep,ft}}/k$$

is a map of correspondences (9.6.1.2), and one of the following holds:

- $\begin{array}{ccc} X_1 & \xleftarrow{c_1} & C \\ \downarrow f_1 & & \downarrow f^\natural \\ Y_1 & \xleftarrow{b_1} & B \end{array}$  is Cartesian.
- $f_1, f^\natural$  are proper.
- $c_1, b_1$  are proper.

Then  $\text{?}$   $\lrcorner$

**Def. (9.6.1.11) [Action of Cohomological Correspondence on Cohomology].** If  $c : C \rightarrow X_1 \times X_2 \in \text{Sch}^{\text{sep,ft}}/k$  is a correspondence s.t.  $c_1$  is proper, and  $u : R(c_2)_!(c_1)^*K_1 \rightarrow K_2 \in D_{\text{csttf}}^b((X_2)_{\text{ét}}, \Lambda)$  is a  $c$ -morphism, then the structural map  $[\pi_c] : c \rightarrow 1_k$  satisfies the conditions in (9.6.1.10), and we get a map

$$R\Gamma_c(u) \triangleq [\pi_c]_!(u) : R\Gamma_c(X_1, K_1) \rightarrow R\Gamma_c(X_2, K_2).$$

More explicitly,  $R\Gamma_c(u)$  is the composition

$$R\Gamma_c(X_1, K_1) \xrightarrow{c_1^*} R\Gamma_c(C, c_1^*K_1) = R\Gamma_c(C, R(c_2)_!(c_1)^*K_1) \xrightarrow{R\Gamma_c(u)} R\Gamma_c(X_2, K_2) \text{ (9.4.4.6)}.$$

More generally, if  $c : C \rightarrow X_1 \times X_2 \in \text{Sch}^{\text{sep,ft}}/k$  is a correspondence and  $U \subset X_1$  is an open subscheme s.t.  $c_1|_{c_1^{-1}(U)}$  is proper, and  $u : R(c_2)_!(c_1)^*K_1 \rightarrow K_2 \in D_{\text{csttf}}^b((X_2)_{\text{ét}}, \Lambda)$  is a  $c$ -morphism s.t.  $K_1|_{X \setminus U} = 0$ . Denote  $c^0 : c_1^{-1}(U_1) \rightarrow U_1 \times X_2$  the restriction of  $c$  and  $u^0$  the restriction of  $u$  to  $c^0$ , then

$$R(j_1)_! : R\Gamma_c(U_1, K_1|_{U_1}) \rightarrow R\Gamma_c(X_1, K_1|_{U_1})$$

is an isomorphism by excision (9.4.4.7), and we define

$$R\Gamma_c(u)R\Gamma_c(u^0) \circ (R(j_1))^{-1} : R\Gamma_c(X_1, K_1) \rightarrow R\Gamma_c(X_2, K_2).$$

$\lrcorner$

**Def. (9.6.1.12) [Local Traces].**  $\text{?}$  For any open subset  $\beta \subset \text{Fix}(a)$ , define  $\text{Tr}_\beta$  to be the composition

$$\text{Tr}_\beta : \text{Hom}(R(c_2)_!c_1^*K, K) \rightarrow H^0(\text{Fix}(c), K_{\text{Fix}(c)}) \rightarrow H^0(\beta, K_\beta).$$

And when  $\beta/k$  is proper, define  $\text{locTr}_\beta(u) : \text{Hom}(R(c_2)_!c_1^*K, K) \rightarrow \Lambda$  to be the trace of  $\text{Tr}_\beta$ .

And when  $\beta$  is a connected component of  $\text{Fix}(a)$  that is proper over  $k$  and  $u \in \text{Hom}(R(c_2)_!c_1^*K, K)$ ,  $\text{locTr}_\beta(u) = (\text{pr}_\beta)_! \circ \text{Tr}_\beta(u) \in \Lambda$   $\text{?}$  is called the **local trace** of  $u$  on  $\beta$ .  $\lrcorner$

*Proof:* Cf.[Fuj97]P495.?

□

**Thm. (9.6.1.13) [Trace Maps and Proper Push-forwards].** Let  $[f] = (f^\natural, f_1, f_2)$  be a proper morphism between correspondences  $c : C \rightarrow X \times X, b : B \rightarrow Y \times Y \in \mathcal{S}ch/k$  (9.6.1.5), then the induced morphism  $f' : \text{Fix}(c) \rightarrow \text{Fix}(b) \in \mathcal{S}ch/k$  (9.6.1.4) is also proper, and for any  $K \in D_{\text{csttf}}^b(X_{\text{ét}}, \Lambda)$ , there is a commutative diagram

$$\begin{array}{ccc} \text{Hom}((c_2)_!(c_1)^* K, K) & \xrightarrow{\text{Tr}_c} & H^0(\text{Fix}(c), K_{\text{Fix}(c)}) \\ \downarrow [f]_! & & \downarrow f'_! \\ \text{Hom}((b_2)_!(b_1)^*(f_! K), f_! K) & \xrightarrow{\text{Tr}_b} & H^0(\text{Fix}(b), K_{\text{Fix}(b)}) \end{array} \quad (9.6.1.4)$$

┘

*Proof:* Cf.[Var07]Chap4.?

□

**Cor. (9.6.1.14) [Lefschetz-Verdier Trace Formulae for Proper Correspondences, Illusie[III94]/Varshavsky[Var07]].** If  $c : C \rightarrow X \times X \in \mathcal{S}ch^{\text{proper}}/k$  is a proper correspondence (9.6.1.5) and  $u : \text{R}(c_2)_! c_1^* K \rightarrow K \in D_{\text{csttf}}^b(X_{\text{ét}}, \Lambda)$  is a  $c$ -morphism corresponding  $c_K \in \text{Corr}^{\text{coh}}(a; K)$  (9.6.1.7), we have

$$\text{Lef}(u) \stackrel{\Delta}{=} \text{Tr}(\text{R}\Gamma_c(u)) = \sum_{\beta \in \pi_0(\text{Fix}(c))} \text{locTr}_\beta(u) \quad (9.6.1.11)(9.6.1.4)(9.6.1.12),$$

where the sum on the RHS is finite. The LHS is called the **Lefschetz-trace** of  $u$ .

┘

### Locally Invariant Subsets

**Def. (9.6.1.15) [Locally Invariant Subsets].** Let  $c : C \rightarrow X \times X \in \mathcal{S}ch^{\text{sep,ft}}/k$  be a correspondence and  $Z \subset |X|$  is a closed subset, then

- $Z$  is called a  **$c$ -invariant subset** if  $|c_1|(|c_2|^{-1}(Z)) \subset Z$ .
- $Z$  is called  **$c$ -invariant in a nbhd of fixed points** if there exists  $W \subset C$  of  $\text{Fix}(c)$  s.t.  $Z$  is  $c|_W$ -invariant.
- $Z$  is called **locally  $c$ -invariant** if for any  $x \in Z$ , there exists an open nbhd  $U$  of  $x \in X$  s.t.  $Z \cap |U|$  is  $c|_U$ -invariant.

┘

**Cor. (9.6.1.16).** Situation as in (9.6.1.15), if  $x \in |X|_0$  and  $\#c_2^{-1}(x) < \infty$ , then  $\{x\}$  is locally  $c$ -invariant. In fact, we can take the nbhd  $U = \left(X \setminus c_1\left(c_2^{-1}(x)\right)\right) \cup \{x\}$  of  $x \in X$ .

┘

**Prop. (9.6.1.17) [Compactification of Locally Invariant Subsets].** Let  $c : C \rightarrow X \times X \in \mathcal{S}ch^{\text{sep,ft}}/k$  be a correspondence and  $U \subset X$  is an open subscheme s.t.  $c_1^{-1}(U)$  is dense in  $C$ ,  $c_1|_{c_1^{-1}(U)}$  is proper, and  $X \setminus U$  is locally  $c$ -invariant (9.6.1.15), then there exists a compactification  $\bar{c} : \bar{C} \rightarrow \bar{X} \times \bar{X}$  of  $c$  (9.6.1.6) s.t.  $\bar{X} \setminus U$  is locally  $\bar{c}$ -invariant.

┘

*Proof:* ? Cf.[Var07]P285.

□

**Prop. (9.6.1.18) [Restriction of Correspondences to Closed Subsets].** Let  $c : C \rightarrow X \times X \in \mathcal{S}ch^{\text{sep,ft}}/k$  be a correspondence,  $u : (c_2)_! c_1^* K \rightarrow K \in D_{\text{csttf}}^b(X_{\text{ét}}, \Lambda)$  a  $c$ -morphism, and  $Z \subset X$  a reduced closed subscheme.

- If  $Z$  is  $c$ -invariant (9.6.1.15), denote by  $c|_Z : c_2^{-1}(Z)_{\text{red}} \rightarrow Z \times Z$  the restriction of  $c$  to  $Z$ , and the closed immersion of correspondences  $[i_Z] : c|_Z \rightarrow c$ . Then by (9.6.1.8),  $u$  restricts to a  $c|_Z$ -morphism  $u|_Z \stackrel{\Delta}{=} [i_Z]^*(u)$ .
- For general  $Z$ , we can also restrict, Cf. [Var07]P286 ?.

┘

**Cor. (9.6.1.19) [Naive Local Traces].** Situation as in (9.6.1.18), suppose  $c_2$  is quasi-finite. Then for any  $x \in |X|_0$ ,  $c|_x$  is just  $c_1^{-1}(x) \cap c_2^{-1}(x) \rightarrow \{x\} \times \{x\}$ , and  $c_1^{-1}(x) \cap c_2^{-1}(x) = \text{Fix}(c|_x)$  is Artinian.

For any  $y \in \text{Fix}(c|_x)$ , the restriction of  $u|_x$  to  $\{y\}$  gives an endomorphism  $u_y : K_x \rightarrow K_x$ , which is the restriction of

$$u_x : (R(c_2)!c_1^*K)_x = \bigoplus_{y' \in c_2^{-1}(x)} (c_1^*K)_{y'} \rightarrow K_x$$

to  $(c_1^*K)_y \cong K_x$ . In particular,  $\text{locTr}_y(u|_x) = \text{tr}(u_y)$  (9.6.1.12) ?, called the **naive local trace** at  $y$ .  
┘

### Contracting Correspondences

**Def. (9.6.1.20) [Contracting Correspondences].** Let  $c : C \rightarrow X \times X \in \text{Sch}^{\text{sep,ft}}/k$  be a correspondence and  $Z \subset X$  a closed subscheme, then

- $Z$  is said to be **stabilized by**  $c$  if  $c_1|_{c_1^{-1}(Z)}$  factors through  $Z$  (i.e.  $c_1^*(\mathcal{I}_Z) \subset c_2^*(\mathcal{I}_Z)\mathcal{O}_C$ ).
- $c$  is said to be **contracting near**  $Z$  if  $c$  stabilizes  $Z$  and there exists  $n \in \mathbb{Z}_+$  s.t.  $c_1^*(\mathcal{I}_Z)^n \subset c_2^*(\mathcal{I}_Z)^{n+1}\mathcal{O}_C$ .
- $c$  is said to be **contracting near**  $Z$  **in a nbhd of fixed points** if there exists an open subscheme  $W \subset C$  containing  $\text{Fix}(c)$  s.t.  $c|_W$  is contracting near  $Z$ .

┘

**Thm. (9.6.1.21) [Local Traces for Contracting Correspondences].** Let  $c : C \rightarrow X \times X \in \text{Sch}^{\text{sep,ft}}/k$  be a correspondence contracting near a closed subscheme  $Z \subset X$  (9.6.1.20) and let  $\beta$  be an open connected subset of  $\text{Fix}(c)$  s.t.  $c'(\beta) \cap Z \neq \emptyset$  (9.6.1.4), then

- $|c'|(\beta) \subset Z$ . Hence  $\beta$  is an open connected subset of  $\text{Fix}(c|_Z)$ .
- For any  $c$ -morphism  $(c_2)!c_1^*K \rightarrow K$ , we have  $\text{Tr}_\beta(u) = \text{Tr}_\beta(u|_Z)$  (9.6.1.18). In particular, if  $\beta/k$  is proper, then  $\text{locTr}_\beta(u) = \text{locTr}_\beta(u|_Z)$  (9.6.1.12).

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*Proof:* Cf. [Fuj97]P515. □

### Lefschetz-Verdier Trace Formulae over Finite Fields

**Def. (9.6.1.22) [Ramifications].** Suppose  $k_0 \in \text{Field}^{\text{fin}}$  s.t.  $k = k_0^{\text{sep}}$ , and  $f : Y \rightarrow X \in \text{NSch}_{\text{cpct}}/k$ .

- If  $Z \subset X$  is a closed subscheme, denote  $\text{Ram}(f, Z)$  the smallest  $m \in \mathbb{Z}_+$  s.t.  $(\mathcal{I}_{f^{-1}(Z)})^m \subset \mathcal{I}_{f^{-1}(Z)}$ .
- If  $f$  is quasi-finite, then  $\text{Ram}(f, x) \leq \dim_k \Gamma(f^{-1}(x))$  for any  $x \in |X|_0$ , which is bounded ?. So we can define  $\text{Ram}(f) = \max_{x \in |X|_0} \text{Ram}(f, x) \in \mathbb{Z}_+$ .

┘

**Prop. (9.6.1.23) [High-Power Twisting Correspondences are Contracting].** Suppose

- $k_0 \in \mathbf{Field}^{\text{fin}}$  s.t.  $k = k_0^{\text{sep}}$ ,
- $c_0 : C_0 \rightarrow X_0 \times X_0 \in \text{Sch}^{\text{sep,ft}}/k_0$  is a correspondence (9.6.1.2),
- $Z \subset X$  is a reduced closed subscheme and  $n \in \mathbb{N}$  s.t.  $\#k_0^n > \text{Ram}(c_2, Z)$  (9.6.1.22) and  $|Z|$  is  $\text{Fr}_X^n \circ c$ -invariant (9.6.1.15),

then  $\text{Fr}_X^n \circ c$  is contracting near  $Z$  (9.6.1.20).  $\lrcorner$

*Proof:* Cf. [Var07]P289.  $\square$

**Cor. (9.6.1.24).** Suppose  $k_0 \in \mathbf{Field}^{\text{fin}}, k = k_0^{\text{sep}}$  and  $c_0 : C_0 \rightarrow X_0 \times X_0 \in \text{Sch}^{\text{sep,ft}}/k_0$  is a correspondence (9.6.1.2).

- If  $Z_0 \subset X_0$  is a reduced closed subscheme s.t.  $|Z|$  is  $c$ -invariant (9.6.1.15), then  $\text{Fr}_X^n \circ c$  is contracting near  $Z$  in a nbhd of fixed points (9.6.1.20) for any  $n \in \mathbb{Z}$  s.t.  $\#k_0^n > \text{Ram}(c_2, Z)$  (9.6.1.22).
- If  $c_2$  is quasi-finite, then for any  $x \in |X|_0$ ,  $\text{Fr}_X^n \circ c$  is contracting near  $x$  in a nbhd of fixed points (9.6.1.20) for any  $n \in \mathbb{Z}$  s.t.  $\#k_0^n > \text{Ram}(c_2)$  (9.6.1.22).  $\lrcorner$

*Proof:* Cf. [Var07]P289.  $\square$

**Thm. (9.6.1.25) [Simple Lefschetz-Verdier Trace Formulae, Delignes' Conjecture, Illusie [Ill94]/Zink [Zin90]/ Pink/Shipz/Fujiwara [Fuji97]/Varshavsky [Var07]].** Suppose  $k_0 \in \mathbf{Field}^{\text{fin}}$  s.t.  $k = k_0^{\text{sep}}$  and  $c_0 : C_0 \rightarrow X_0 \times X_0 \in \text{Sch}^{\text{sep,ft}}/k_0$  is a correspondence, then

- If  $c_2$  is quasi-finite, then  $\text{Fix}(\text{Fr}^n \circ c)$  is Artinian for any  $n \in \mathbb{N}$  s.t.  $q^n > \text{Ram}(c_2)$  (9.6.1.22),
- If  $U_0 \subset X_0$  is an open subscheme s.t.  $c_1|_{c_1^{-1}(U)}$  is proper and  $c_2|_{c^{-1}(U \times U)}$  is quasi-finite, and  $X \setminus U$  is locally  $c$ -invariant, then there exists  $N \in \mathbb{Z}_+$  satisfying the following: For any  $K \in D_{\text{csttf}}^b(X, \Lambda)$  s.t.  $K|_{X \setminus U} = 0$ ,  $n \in \mathbb{Z}_{\geq \log_{\#k_0} N}$  and a  $\text{Fr}_X^n \circ c$ -morphism  $u : (c_2)_!(\text{Fr}_X^n \circ c_1)^* K \rightarrow K$ , we have

$$\text{Lef}(u) = \sum_{y \in |\text{Fix}(\text{Fr}_X^n \circ c) \cap c^{-1}(U \times U)|} \text{tr}(u_y) \quad (9.6.1.14)(9.6.1.19),$$

where the RHS is a finite sum.

- In item2, if  $X_0/k_0, C_0/k_0$  are proper, then we can take

$$N = \max \left( \text{Ram}(c_2|_{c^{-1}(U \times U)}), \text{Ram}(c_2, X \setminus U) \right) \quad (9.6.1.22).$$

$\lrcorner$

*Proof:* 1: Choose a closed point  $x \in (\text{Fr}^n \circ c)'(\text{Fix}(\text{Fr}^n \circ c)) \subset X$  (9.6.1.4), then  $\text{Fr}^n \circ c$  is contracting near  $x$  in a nbhd of fixed points by (9.6.1.24). Then  $|\text{Fix}(\text{Fr}^n \circ c)| \subset \text{Fix}(c|_x)$  by (9.6.1.21), which is Artinian by (9.6.1.19).

3: By Lefschetz-Verdier trace formula (9.6.1.14), we have

$$\text{Lef}(u) = \sum_{\beta \in \pi_0(\text{Fix}(\text{Fr}^n \circ c))} \text{locTr}_{\beta}(u).$$

For any  $\beta \in \pi_0(\text{Fix}(\text{Fr}^n \circ c))$ , if  $c_2(\beta) \in X \setminus U$ , then  $\text{Fr}^n \circ c$  is contracting near  $X \setminus U$  in a nbhd of fixed points by (9.6.1.24). Thus  $\beta$  is a connected component of  $\text{Fix}(\text{Fr}^n \circ c|_{X \setminus U})$  and  $\text{locTr}_{\beta}(u) = \text{locTr}_{\beta}(u|_{X \setminus U})$  by (9.6.1.21), But  $K|_{X \setminus U} = 0$ , thus  $\text{locTr}_{\beta}(u) = 0$ .



And if  $c_2(\beta) \cap U \neq \emptyset$ , then it follows from item1 that  $\beta$  is a single point  $y \in \text{Fix}(\text{Fr}_X^n \circ c) \cap c^{-1}(U \times U)$ , and  $\text{Fr}_X^n \circ c$  is contracting near  $c_2(y)$  in a nbhd of fixed points by (9.6.1.24), so  $\text{locTr}_\beta(y) = \text{locTr}_y(u|_{c_2(y)}) = \text{tr}(u_y)$  by (9.6.1.21) and (9.6.1.19).

2: Cf. [Var07]P292. □

**Cor. (9.6.1.26) [Grothendieck-Lefschetz Trace Formula for Frobenius Morphisms].** Let  $X_0 \in \text{Sch}^{\text{sep}, \text{ft}}/k_0$ ,  $K_0 \in D_{\text{const}}^b(X_0, \overline{\mathbb{Q}}_\ell)$ , then

$$\sum_{x_0 \in |X_0|_0} \text{tr}(F_{x_0}^* | K_{\overline{x}}) = \text{tr}(F_X^* | [H_{\text{ét}, c}^\bullet(X, K)]) \quad (9.4.4.16).$$

┘

*Proof:* Because  $F_X^* = \text{Fr}_X^*$  on  $H_{\text{ét}, c}^*(X, \mathcal{F})$  by (9.4.4.18), this follow from (9.6.1.25).

[Fu11]P596 has a direct proof of this using induction on  $\dim X$ . □

### Lefschetz-Numbers

**Def. (9.6.1.27) [Lefschetz-Numbers of Sheaves].** Let  $X \in \text{Sch}^{\text{sep}, \text{ft}}/k$ ,  $\mathcal{F} \in \text{Sh}_{\text{const}}(X_{\text{ét}}, \overline{\mathbb{Q}}_\ell)$ , then for any  $g \in \text{Aut}(X/k)$ , define the **alternating cohomology group**

$$[H_{\text{ét}, c}^\bullet(X; \mathcal{F})] = \sum_{i \geq 0} (-1)^i H_{\text{ét}, c}^i(X, \mathcal{F}).$$

and the **Lefschetz-number** for  $\mathcal{F}$  to be

$$\text{Lef}(g, X; \mathcal{F}) \triangleq \text{tr} \left( g | [H_{\text{ét}, c}^\bullet(X, \mathcal{F})] \right) = \sum_{i \geq 0} (-1)^i \text{tr}(g | H_{\text{ét}, c}^i(X, \mathcal{F})).$$

┘

**Prop. (9.6.1.28).** Let  $k \in \text{Field}$ ,  $k = k^{\text{sep}}$ ,  $X, C' \in \text{Sch}^{\text{sep}, \text{ft}}/k$ ,  $g, h \in \text{Aut}(X)$ ,  $g' \in \text{Aut}(X')$ , then

1.  $\text{tr}(g, X, -)$  is additive.
2. If  $Z \subset X$  is a closed subscheme and  $U = X \setminus Z$ , then  $\text{tr}(g, X, \mathcal{F}) = \text{tr}(g, U, \mathcal{F}) + \text{tr}(g, Z, \mathcal{F})$ .
3.  $\text{tr}(g \times g, X \times X'; \mathcal{F} \times \mathcal{F}') = \text{tr}(g, X, \mathcal{F}) \text{tr}(g', X'; \mathcal{F}')$ .
4. If  $g, h$  commute, then  $\text{tr}(gh, X, \mathcal{F}) = \text{tr}(g, X^h, \mathcal{F})$ .
5. If  $T$  is a tori acting on  $X$ , then  $\text{tr}(g, X, \mathcal{F}) = \text{tr}(g, X^S, \mathcal{F})$ .

┘

*Proof:* 1, 2, 3 follow from (9.4.6.43).

4, 5: [Deligne-Lustig, Thm3.2] and [Digne and Michel, Prop10.15]. ? □

**Prop. (9.6.1.29).** Cf. [Representations of Finite Groups of Lie type, Digne and Michel]. ┘

**Cor. (9.6.1.30) [Lefschetz-Numbers of Schemes].** For  $X \in \text{Sch}^{\text{sep}, \text{ft}}/k$ , the Lefschetz-numbers  $\text{Lef}(\text{id}_X, X, \mathbb{Q}_\ell)$  for  $\ell \in \text{Prime}$  (9.6.1.27) are independent of  $\ell$  chosen, and it is called the **Lefschetz-number** for  $X$ . ┘

**Prop. (9.6.1.31) [Grothendieck-Lefschetz Trace Formulae].** Suppose  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}}$  and  $k = \overline{\mathbb{F}}_p$ . If  $X \in \text{Sch}^{\text{sep}, \text{ft}}/\mathbb{F}_q$ , then

$$\text{Lef}(\sigma_{q, X}, X_k; \mathbb{Q}_\ell) = \#X(\mathbb{F}_p).$$

┘

## 2 Weil-Sheaves

### Weil-Sheaves

**Def. (9.6.2.1) [Weil-Groups].** Cf. [Conrad, L19]. ? ┘

**Def. (9.6.2.2) [Weil Sheaves].** By Galois-descent, the pullback  $(X \rightarrow X_0)^*$  induces an equivalence of categories between  $\mathrm{Sh}_{\mathrm{const}}(X_{\mathrm{\acute{e}t}}, \overline{\mathbb{Q}}_\ell)$  to the category of constructible  $\overline{\mathbb{Q}}_\ell$ -sheaves on  $X$  with an specified  $\mathrm{Gal}(X/X_0) = \mathrm{Gal}_{k_0} \cong \widehat{\mathbb{Z}}$ -actions. In practice, sometimes it is hard to verify the action of  $\mathbb{Z} \subset \widehat{\mathbb{Z}}$  is continuous, which leads to the following definition:

The category  $\mathrm{WSh}_\ell(X_0)$  of  $(\overline{\mathbb{Q}}_\ell)$ -**Weil-sheaves** on  $X_0$  consists of pairs  $\mathcal{G}_0 = (\mathcal{G}, F_{\mathcal{G}})$  where  $\mathcal{G} \in \mathrm{Sh}_{\mathrm{const}}(X_{\mathrm{\acute{e}t}}, \overline{\mathbb{Q}}_\ell)$  and an isomorphism  $F_{\mathcal{G}} : F_X^* \mathcal{G} \rightarrow \mathcal{G} \in \mathrm{Sh}_{\mathrm{const}}(X_{\mathrm{\acute{e}t}}, \overline{\mathbb{Q}}_\ell)$  (7.2.10.1). Notice  $F_X^*$  is not  $\bar{k}$ -linear! There are natural definition of morphisms of Weil-sheaves.

A **lisse Weil-sheaf** is a Weil-sheaf  $\mathcal{G}_0 = (\mathcal{G}, F_{\mathcal{G}}) \in \mathrm{WSh}(X_0, \overline{\mathbb{Q}}_\ell)$  s.t.  $\mathcal{G}$  is lisse (9.4.6.25). The category of lisse Weil-sheaves are denoted by  $\mathrm{WSh}_{\ell, \mathrm{lisse}}(X_0)$ . ┘

**Prop. (9.6.2.3) [Constructible  $\overline{\mathbb{Q}}_\ell$ -Sheaves as Weil Sheaves].** For any  $\mathcal{F}_0 \in \mathrm{Sh}_{\mathrm{const}}((X_0)_{\mathrm{\acute{e}t}}; \overline{\mathbb{Q}}_\ell)$  (9.4.6.10), the canonical map

$$F_X^* \mathrm{pr}^* \mathcal{F}_0 \cong (\mathrm{pr} \circ F_X)^* \mathcal{F}_0 = \mathrm{pr}^* \mathcal{F}_0$$

makes  $\mathcal{F} = (X \rightarrow X_0)^* \mathcal{F}_0$  a Weil-sheaf. ┘

**Prop. (9.6.2.4).**

- $\mathrm{WSh}_\ell(X_0) \in \mathrm{AbCat}$ , the subcategory of lisse  $\overline{\mathbb{Q}}_\ell$ -sheaves is an Abelian subcategory.
- The constructions like  $R^i f_*$ ,  $R^i f_!$ ,  $f^*$  is functorial thus is definable on  $\mathrm{WSh}_\ell(X_0)$  by (9.4.2.19).
- (Frobenius-Actions on Cohomologies) The specified isomorphism  $F_{\mathcal{G}} : F_X^* \mathcal{G} \rightarrow \mathcal{G}$  gives a map  $\mathcal{G} \rightarrow F_X^* \mathcal{G} \xrightarrow{F_{\mathcal{G}}} \mathcal{G}$ , thus gives an action  $F_X^*$  of  $F_X$  on  $H_{\mathrm{\acute{e}t}, c}^*(X, \mathcal{G})$  and  $H_{\mathrm{\acute{e}t}}^*(X, \mathcal{G})$ .
- (Frobenius-Actions on Stalks) Let  $x_0 \in X_0(\bar{k})$ ,  $x = x_0 \otimes_{\kappa(x)} \bar{k}$ ,  $\bar{x} \in x$ , there is an Frobenius action  $F_{x_0}^*$  of  $x_0$  on  $\mathcal{G}_{\bar{x}}$  given as follows: Pulling back  $\mathcal{G}_0$  to a Weil-sheaf on  $x_0$ , then  $F_x^* \mathcal{G}_x \cong \mathcal{G}_x$  gives an isomorphism  $\mathcal{G}_{F^{i+1}(\bar{x})} \cong \mathcal{G}_{F^i(\bar{x})}$ , whose  $\deg(x_0)$ -th iterate gives the action  $\mathcal{G}_{\bar{x}} \cong \mathcal{G}_{F^{\deg(x_0)-1}(\bar{x})} \cong \dots \cong \mathcal{G}_{F(\bar{x})} \cong \mathcal{G}_{\bar{x}}$ . For different choice of  $\bar{x}$ , the actions are conjugate.

And if  $\mathcal{G}_0 \in \mathrm{Sh}_{\mathrm{const}}((X_0)_{\mathrm{\acute{e}t}}, \overline{\mathbb{Q}}_\ell)$ , then  $F_{x_0}^*$  acts via  $\varphi_{x_0}^{-1}$  ?. ┘

**Prop. (9.6.2.5) [Weil Sheaves and Representation].** If  $X_0$  is geo.connected, then there is an equivalence of categories

$$\mathrm{WSh}_\ell(X_0) \cong \mathrm{Rep}_\ell(W(X_0, \bar{x})) : \mathcal{G}_0 \mapsto (\mathcal{G}_0)_{\bar{x}},$$

and the correspondence defined in (9.4.6.26) is a sub-correspondence of this.

Thus the notion of **geometric irreducible Weil-sheaves** and **semi-simple Weil-sheaves** are definables. ┘

*Proof:* Because by the correspondence (9.4.6.26),  $\mathcal{G}$  corresponds to a representation of  $\pi_1(X, \bar{x})$ , and  $\pi_1(X, \bar{x})$  acts trivially on the Galois cover  $X/X_0$ . Now a representation of  $W(X_0, \bar{x})$  is equivalent to an automorphism  $\rho(\sigma)$  (where  $\sigma \in W(X_0, \bar{x})$  satisfies  $\deg(\sigma)$  corresponds to the geometric Frobenius) that  $\rho(\sigma)\rho(\pi_1(X, \bar{x}))\rho(\sigma^{-1}) = \rho(\sigma\pi_1(X, \bar{x})\sigma^{-1})$ , which is equivalent to an isomorphism  $F_X^* \mathcal{G} \rightarrow \mathcal{G}$ .  $\square$

**Cor.(9.6.2.6) [Constant Weil-Sheaves].** For any  $b \in \overline{\mathbb{Q}_\ell}^\times$ , there is a **constant Weil-sheaf**  $\mathcal{L}_b \in \text{WSh}_\ell(\text{Spec } k_0)$  corresponding to the character  $W(X_0, \bar{x}) \rightarrow \overline{\mathbb{Q}_\ell}^\times : x \mapsto b^{\deg(x)}$ .  $\lrcorner$

**Prop.(9.6.2.7) [Lisse Weil Sheaves and  $\overline{\mathbb{Q}_\ell}$ -Sheaves].** If  $X_0$  is geo.connected,  $\mathcal{G}_0 \in \text{WSh}_{\ell, \text{lisse}}(X_0)$  comes from a  $\overline{\mathbb{Q}_\ell}$ -sheaf iff some  $\deg 1$  element  $\sigma$  in  $W(X, \bar{x})$  acts on  $\mathcal{G}_{0\bar{x}}$  with eigenvalues which are  $\ell$ -adic units.  $\lrcorner$

*Proof:* This is purely a Galois representation problem, concerning whether the representation of  $W(X_0, \bar{x})$  can be extended to a representation of  $\pi_1(X_0, \bar{x})$ , and it is a continuity problem.

Firstly the representation of  $\pi_1(X, \bar{x})$  stabilizes a lattice  $\mathcal{O}_E^n$  for some  $E/\mathbb{Q}_\ell$  finite, and then extends  $E$  to contain coefficients of  $\rho(\sigma)$  and even its rational form. Then notice  $\pi_1(X_0, \bar{x})$  is the profinite completion of  $W(X_0, \bar{x})$ , thus it suffices to see if the image  $\rho(W(X_0, \bar{x}))$  is compact, and this is equivalent to eigenvalues of  $\rho(\sigma)$  are units.  $\square$

**Prop.(9.6.2.8) [Determinential Criterion for Genuine Sheaves].** If  $X_0$  is normal and geo.connected, then a irreducible lisse Weil-sheaf on  $X_0$  comes from a constructible  $\overline{\mathbb{Q}_\ell}$ -sheaf(9.6.2.3) iff its determinant bundle does.  $\lrcorner$

*Proof:* Use geometric monodromy group. Cf.[Conrad L19 P7].

First assume that  $\mathcal{G}_0$  is geometrically irreducible, then(9.6.4.4) shows that there is a nonzero power  $\sigma^m = gz$  where  $g \in G_{\text{geo}}(\overline{\mathbb{Q}_\ell})$  and  $z \in Z(G(\overline{\mathbb{Q}_\ell}))$ . Now  $G_{\text{geo}}$  is a semisimple algebraic group(9.6.4.4), so the determinential character maps  $G_{\text{geo}}(G(\overline{\mathbb{Q}_\ell}))$  to a finite group, because connected semisimple algebraic group has no nontrivial character as  $[G, G] = G$ ?. So the determinant of  $g$  is an  $\ell$ -unit, and  $\det(\sigma^m) = \det(z)$  is a unit. But  $z$  is a scalar by Schur's lemma, thus  $z$  is an  $\ell$ -adic unit. Now it suffices to show the eigenvalue of  $g$  are all  $\ell$ -units.

Now consider  $\rho(\pi_1(X, \bar{x}))$  is a compact group in  $\text{End}(V)$ , thus it generates a finite  $\mathcal{O}_E$ -submodule  $A$ , which is full-rank lattice in  $\text{End}(V)$  by Jacobson density theorem? and the fact  $\rho$  is absolutely irreducible.  $g$  normalized  $A$ , because  $\sigma$  and  $z$  both normalizes  $\rho(\pi_1(X, \bar{x}))$ , so the eigenvalue of the conjugate action of  $g$  are all  $\ell$ -units, but its eigenvalue are of the form  $\lambda_i \lambda_j^{-1}$  where  $\lambda_i$  are eigenvalues of  $g$ , so this together with the fact  $\det(g)$  is  $\ell$ -units shows that all  $\lambda_i$  are  $\ell$ -units.

For the general case, Cf.[Conrad L19 P7].?  $\square$

**Cor.(9.6.2.9) [Filtration of Lisse Weil-Sheaves].** If  $X_0$  is a normal and geo.connected and  $\mathcal{G}_0 \in \text{WSh}_{\ell, \text{lisse}}(X_0)$ , then there is a filtration

$$0 = \mathcal{G}_0^{(0)} \subset \mathcal{G}_0^{(1)} \subset \dots \subset \mathcal{G}_0^{(r)} = \mathcal{G}_0$$

s.t. each sub-quotients are irreducible and of the form  $\mathcal{F}_0^{(i)} \otimes \mathcal{L}_{b_i}$  for some  $b_i \in \overline{\mathbb{Q}_\ell}^\times$  (9.6.2.6) and  $\mathcal{F}_0^{(i)} \in \text{Sh}_{\text{lisse}}^{\text{irr}}((X_0)_{\text{ét}}; \overline{\mathbb{Q}_\ell})$ .  $\lrcorner$

*Proof:* It suffices to prove for  $\mathcal{G}_0$  irreducible. Then we choose  $b = \chi_{\det}(\sigma)^{1/n}$ , where  $\deg(\sigma) = 1$ , then

$$\wedge(\mathcal{G}_0 \otimes \mathcal{L}_{b^{-1}}) \cong \wedge(\mathcal{G}_0) \otimes \mathcal{L}_{\chi_{\det}(\sigma)}^{-1}$$

which has unit eigenvalues thus  $\mathcal{F}_0 = \mathcal{G}_0 \otimes \mathcal{L}_{b^{-1}}$  is a lisse  $\overline{\mathbb{Q}_\ell}$ -sheaf by(9.6.2.8).  $\square$

### Zeta-Functions and Trace Formulae for Weil-Sheaves

**Def. (9.6.2.10) [Zeta-Functions].** For  $\mathcal{F}_0 \in \text{WSh}_\ell(X_0)$  (9.6.2.2), the **zeta-function** associated to  $\mathcal{F}_0$  is defined to be

$$Z(X_0, \mathcal{F}_0; T) = \prod_{x_0 \in |X_0|_0} \det\left(1 - F_{x_0}^* T^{\deg(x_0)} | \mathcal{F}_{\bar{x}}\right)^{-1} \in 1 + T\Lambda[[T]],$$

where  $F_{x_0}^*$  is defined in (9.6.2.4). Notice if  $\mathcal{F}_0 \in \text{Sh}((X_0)_{\text{ét}})$ , then  $F_{x_0}^* = \varphi_{\kappa(x_0)}^{-1}$  by (9.6.2.4).  $\lrcorner$

**Prop. (9.6.2.11).** Situation as in (9.6.2.10), we have

$$Z(X_0, \mathcal{F}_0; T) = \exp\left(\sum_{n \geq 1} \sum_{x_0 \in X_0(k_n)} \text{tr}(F_{x_0}^* | \mathcal{F}_{\bar{x}}) \frac{T^n}{n}\right).$$

$\lrcorner$

*Proof:* By (3.5.9.26),

$$\begin{aligned} Z(X_0, \mathcal{F}_0; T) &= \prod_{x_0 \in |X_0|_0} \exp\left(\sum_{n \geq 1} \text{tr}((F_{x_0}^*)^n | \mathcal{F}_{\bar{x}}) \frac{T^n \deg(x_0)}{n}\right) \\ &= \exp\left(\sum_{n \geq 1} \sum_{x_0 \in |X_0|_0} \deg(x_0) \text{tr}((F_{x_0}^*)^n | \mathcal{F}_{\bar{x}}) \frac{T^n}{n}\right) \\ &= \exp\left(\sum_{n \geq 1} \sum_{x_0 \in X_0(k_n)} \text{tr}(F_{x_0}^* | \mathcal{F}_{\bar{x}}) \frac{T^n}{n}\right) \end{aligned}$$

where in the last equality, notice there are exactly  $\deg(x_0)$ -many points in  $X_0(k)$  over  $x_0 \in |X_0|_0$ .  $\square$

**Cor. (9.6.2.12).** If  $\mathcal{F} = \mathbb{Q}_\ell$ , then

$$Z(X_0, \mathbb{Q}_\ell; T) = \prod_{x_0 \in |X_0|_0} \frac{1}{1 - T^{\deg(x_0)}}$$

is just the zeta-function defined in (31.1.3.1).  $\lrcorner$

**Thm. (9.6.2.13) [Grothendieck-Lefschetz Trace Formula for Weil-Sheaves].** Let  $S_0 \in \text{Sch}^{\text{ft}}/k_0$ ,  $f_0 : X_0 \rightarrow S_0 \in \text{Sch}$  be separated of f.t. and  $\mathcal{F}_0 = (\mathcal{F}, F_{\mathcal{F}}) \in \text{WSh}_\ell(X_0)$  (9.6.2.2). We have:

$$Z(X_0, \mathcal{F}_0; T) = \prod_{n=0}^{2 \dim X_0} Z(S_0, R^n f_{0!} \mathcal{F}_0; T)^{(-1)^n} \in \overline{\mathbb{Q}_\ell}[[T]] \text{ (9.6.2.10)}.$$

In particular, for  $S_0 = \text{Spec } k_0$ ,  $X_0 \in \text{Sch}^{\text{sep, ft}}/k_0$ , we have

$$Z(X_0, \mathcal{F}_0; T) = \prod_{n=0}^{2 \dim X_0} \det\left(1 - F_X^* T | H_{\text{ét}, c}^n(X, \mathcal{F})\right)^{(-1)^{n+1}} \in \overline{\mathbb{Q}_\ell}[[T]].$$

Notice by (9.4.4.9), the higher proper pushforward of  $\mathcal{F}_0$  vanish for  $n > 2 \dim X_0$ .  $\lrcorner$

*Proof:* Because  $Z(X_0, \mathcal{F}_0; T)$  is defined by Euler-product, base change isomorphism for  $Rf_!$  holds (9.4.4.7), and description of the fiber (9.6.2.4), it suffices to prove for  $S_0 = \text{Spec } k_0$ .

Use the filtration in (9.6.2.9), notice that the trace is additive, so we can reduce to the case  $\mathcal{G}_0 = \mathcal{F}_0 \otimes \mathcal{L}_b$  and  $\mathcal{G}_{\bar{x}} = \mathcal{F}_{\bar{x}} \otimes \mathcal{L}_{b, \bar{x}}$ , then the Euler-factors are

$$\det \left( 1 - b^{\deg(x_0)} \varphi_{\kappa(x_0)}^{-1} T^{\deg(x_0)} | \mathcal{F}_{\bar{x}} \right)$$

and the cohomology factor is

$$\det \left( 1 - F_X^* T | H_{\text{ét}, c}^i(X, \mathcal{F} \otimes \mathcal{L}_b) \right) = \det \left( 1 - b F_X^* T | H_{\text{ét}, c}^i(X, \mathcal{F}) \right),$$

where the projection formula (9.4.4.10) is used, noticing the  $\mathcal{L}_b$  is pulled back from  $\text{Spec } k_0$ . Then the assertion is clear from (9.6.1.26) and (9.6.2.11)(9.6.2.4).  $\square$

**Cor. (9.6.2.14).** Situation as in (9.6.2.13), we have  $Z(X_0, \mathcal{F}_0; T) \in \overline{\mathbb{Q}}_\ell(T)$ .  $\square$

### 3 Weights and Purity

#### Determinantal Weights

**Prop. (9.6.3.1) [Structure of Weil Group of Curves].** If  $X_0 \in \text{SmCur}/k$ , then the image of  $\pi_1(X, \bar{x})$  in  $W(X_0, \bar{x})_{\text{ab}}$  is a product of a finite group and a pro- $p$  group.  $\square$

*Proof:* Denote  $K = R(X_0)$ , and let  $\overline{X_0}$  be the regular completion of  $X_0$ , with  $S_0 = \overline{X_0} \setminus X_0$ . Then we have an isomorphism  $\pi_1(\overline{X_0}, \bar{x}) \cong \text{Gal}_K$ , Cf. [Étale Cohomology Lei Fu P136] ?. Recall the global class field theory:

$$\begin{array}{ccccccc} \pi_1(\overline{X}, \bar{x})^{ab} & \longrightarrow & \pi_1(\overline{X_0}, \bar{x})^{ab} & \longrightarrow & \text{Gal}_K \cong \widehat{\mathbb{Z}} & \longrightarrow & 0 \\ & \searrow & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & I_K & \longrightarrow & W(\overline{X_0}, \bar{x})^{ab} \cong W(K, k) & \longrightarrow & W(k) \cong \mathbb{Z} \longrightarrow 0 \quad \text{?} \\ & \downarrow \cong & & & \downarrow \cong & & \downarrow \cong \\ 0 & \longrightarrow & K^* \backslash (A_K^*)^1 / \prod_v \mathcal{O}_v^* & \longrightarrow & K^* \backslash A_K^* / \prod_v \mathcal{O}_v^* & \longrightarrow & q^{\mathbb{Z}} \longrightarrow 0 \end{array}$$

So the image of  $\pi_1(\overline{X}, \bar{x})$  factors through  $\pi_1(\overline{X}, \bar{x}) \rightarrow \pi_1(\overline{X}, \bar{x})_{\text{ab}} \rightarrow K^* \backslash A_K^* / \prod_v \mathcal{O}_v^*$  which is the class number of  $K$ , is finite.

In this diagram,  $W(X_0, \bar{x})$  corresponds to  $K^* \backslash A_K^* / \prod_{v \notin S_0} \mathcal{O}_v^*$ , so

$$0 \rightarrow \ker(W(X_0, \bar{x}) \rightarrow W(\overline{X_0}, \bar{x})) \rightarrow \text{Im}(\pi_1(X_0, \bar{x})) \rightarrow \text{Im}(\pi_1(\overline{X_0}, \bar{x})) \rightarrow 0$$

But the kernel is a quotient of  $\prod_{v \in S_0} \mathcal{O}_v^*$ , which is a pro- $p$  group times a finite group, so finally  $\text{Im}(\pi_1(X_0, \bar{x}))$  is a product of a pro- $p$ -group times a finite group.  $\square$

**Lemma (9.6.3.2) [Curve Rank 1 case].** If  $X_0 \in \text{SmCur}/k_0$  and  $\chi : W(X_0, \bar{x}) \rightarrow \overline{\mathbb{Q}}_\ell^\times$  be a continuous character, then there exists a  $c \in \overline{\mathbb{Q}}_\ell^\times$  that  $\chi$  is a product of a character of finite order and the character  $\sigma \mapsto c^{\deg(\sigma)}$ .

In particular, the Weil-sheaf corresponding to  $\chi$  is punctually  $\iota$ -pure of weight  $2 \log_q |\iota(c)|$ .  $\square$

*Proof:* By (24.2.3.2), the image of  $\chi$  is in  $\mathcal{O}_E^*$  for some  $E \in \ell\text{-LField}$ , so by (9.6.3.1), it has an open subgroup which is pro- $p$  and pro- $\ell$  so trivial, thus  $\pi_1(X_0, \bar{x})$  is mapped to a finite group.

In particular there is an  $n$  that  $\chi^n = \text{id}$  on  $\pi_1(X_0, \bar{x})$ , so there is some  $b$  that  $\chi^n = b^{\deg(\sigma)}$ , hence if  $c$  is an  $n$ -th roots of  $b$  and we let  $\chi' = \chi/c^{\deg(\sigma)}$ , then  $(\chi')^n = 1$ .  $\square$

**Cor. (9.6.3.3) [Rank 1 Lisse Sheaf is Pure].** If  $X_0 \in \text{SmCur}/k_0$ , then any  $\mathcal{F} \in \text{WSh}_{\ell, \text{lisse}}^1(X_0)$  is pure.  $\lrcorner$

**Def. (9.6.3.4) [Determinantal Weight].** Suppose  $X_0$  is geo.connected and smooth, and  $\mathcal{F}_0 \in \text{WSh}_{\text{lisse}}(X_0; \overline{\mathbb{Q}}_\ell)$  with a filtration of lisse sheaves

$$0 = \mathcal{F}_n \subset \mathcal{F}_{n-1} \subset \dots \subset \mathcal{F}_0 \in \text{WSh}_{\text{lisse}}(X_0, \overline{\mathbb{Q}}_\ell)$$

with irreducible quotients, then for any  $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ , we define the **determinantal  $\iota$ -weights** of  $\mathcal{F}_0$  to be that of the  $\iota$ -weights of the top wedge products of the successive quotients divided by their ranks, which exists by (9.6.3.3).

Notice that the determinantal  $\iota$ -weights are unchanged when  $\mathcal{F}_0$  is replaced by its semi-simplification  $\mathcal{F}_0^{\text{ss}} = \bigoplus_{i \in \mathbb{N}} (\mathcal{F}_i / \mathcal{F}_{i-1})$ .  $\lrcorner$

### Pure Weil-Sheaves

**Def. (9.6.3.5) [Pure Weil-Sheaves].** For an embedding  $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ ,  $\mathcal{F}_0 \in \text{WSh}_\ell(X_0)$  is called  **$\iota$ -pure** of weight  $w \in \mathbb{Z}$  if for any closed point  $x \in X$ , the  $\overline{\mathbb{Q}}_\ell$ -eigenvalues of  $F_{x_0}^*$  on the stalks  $\mathcal{F}_x$  (9.6.2.4) are algebraic and satisfy  $|\iota(\alpha_i)| = (q^{\deg(x)})^{w/2}$ .

It is called **pure of weight  $w$**  iff for any closed point  $x \in X$ , the  $\overline{\mathbb{Q}}_\ell$ -eigenvalues are  $q^{\deg(x)}$ -Weil numbers of weight  $w$  (19.3.1.16), i.e.  $\iota$ -pure for any embedding  $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ .

It is said to be **( $\iota$ )-mixed** with weights  $w_1, \dots, w_n$  if it has a filtration of constructible  $\overline{\mathbb{Q}}_\ell$ -sheaves that each quotient is pure of weight  $w_i$  respectively.  $\lrcorner$

**Example (9.6.3.6).**  $\mathbb{Q}_\ell(r)$  is pure of weight  $-2r$ .  $\lrcorner$

*Proof:* This is because the geometric Frobenius  $\varphi_{\kappa(x)}^{-1}$  acts by  $1/q^{d_x}$ -th power, which is additively multiplying by  $(q^{d_x})^{-2/2}$ .  $\square$

**Prop. (9.6.3.7) [Permanence Properties of Purity].** For an embedding  $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ ,

- If  $f_0 : X_0 \rightarrow Y_0 \in \text{Sch}^{\text{ft}}/k_0$  and  $\mathcal{G}_0 \in \text{WSh}_\ell(Y_0)$  is  $\iota$ -pure, then  $f_0^* \mathcal{G}_0$  is also  $\iota$ -pure, and the converse is also true if  $f$  is surjective.
- If  $f_0 : X_0 \rightarrow Y_0 \in \text{Sch}^{\text{ft}}/k_0$  is finite and  $\mathcal{G}_0 \in \text{WSh}_\ell(X_0)$  s.t.  $j_0^*(\mathcal{G}_0)$  is  $\iota$ -pure, then  $\mathcal{G}_0$  is  $\iota$ -pure.
- Let  $k'_0/k_0 \in \text{Field}$  be a finite extension, then  $\mathcal{G}_0 \in \text{WSh}_\ell(X_0)$  is pure of weight  $\beta$  iff  $(\mathcal{G}_0)_{k'_0} \in \text{WSh}_\ell((X_0)_{k'_0})$  is.

$\lrcorner$

*Proof:* 1 is because the stalk corresponds.

2: This is because the stalks can be calculated, by ?.

3: ?

$\square$

### Semicontinuity of Weights

**Def. (9.6.3.8) [Maximal Weight].** For  $\mathcal{G}_0 \in \text{WSh}_\ell(X_0)$ , we can also define the **maximal  $\iota$ -weight** of  $\mathcal{G}_0$  as

$$\text{wt}(\mathcal{G}_0) = \sup_{x \in |X_0|} \sup_{\alpha_i} 2 \log_{N(x)}(|\iota(\alpha_i)|).$$

┘

**Lemma (9.6.3.9).** There exists  $c \in \mathbb{R}_+$  s.t. for any  $n \in \mathbb{Z}_+$  we have  $\#X_0(k_n) \leq cq^{n \dim X_0}$

┘

*Proof:* We can pass to the reduced structure of  $X_0$ , then we can use excision to pass to the integral case. Then choose an open affine dense subset  $U_0$  of  $X_0$ , then by Noetherian normalization, it factors through a finite map  $f : U_0 \rightarrow \mathbb{A}_{k_n}^{\dim X_0}$ , so

$$|U(k_n)| \leq (\deg f) q^{n \dim X_0}$$

Then we can use induction on dimension, because  $\dim(X_0 \setminus U_0) < \dim X_0$ .

□

**Lemma (9.6.3.10).** Let  $\mathcal{G}_0 \in \text{WSh}(X_0, \overline{\mathbb{Q}_\ell})$  and  $\beta \in \mathbb{R}_{>\text{wt}(\mathcal{G}_0)}$ , then

$$\iota(Z(X_0, \mathcal{G}_0, T)) = \prod_{x \in |X_0|_0} \iota(\det(1 - T^{d_x} F_x, \mathcal{G}_{0,\bar{x}})^{-1}) \quad (9.6.2.10)$$

converges for  $|T| < q^{-\beta/2 - \dim X_0}$  and has no zero or pole there.

┘

*Proof:* We can show that it has no zero or pole using the fact that the logarithmic derivative has no poles (when it is convergent). We suppress the isomorphism  $\iota : \overline{\mathbb{Q}_\ell} \rightarrow \mathbb{C}$  and calculate:

$$\frac{d \log}{dt} L(X_0, \mathcal{G}_0, t) = \sum_{x \in |X_0|_0} \sum_{n \geq 1} \deg(x) (\text{tr}(F_x^n)) t^{d_x n - 1} = \sum_{n \geq 1} \sum_{x \in |X_0|_0, \deg(x)|n} \deg(x) (\text{tr}(F_x^{n/d_x})) t^{n-1}$$

Notice by assumption on  $\beta$ ,  $|\text{tr}(F_x^{n/\deg(x)})| \leq r q^{n\beta/2}$ , where  $r = \sup_{x \in |X_0|_0} \dim_{\overline{\mathbb{Q}_\ell}} \mathcal{G}_{0,x}$  is finite because it has a stratification by (9.4.6.15), so

$$\left| \frac{d \log}{dt} L(X_0, \mathcal{G}_0, t) \right| \leq \sum_{n \geq 1} \sum_{\substack{x \in |X_0|_0 \\ \deg(x)|n}} d_x r q^{n\beta/2} t^{n-1} = \sum_{n \geq 1} |X_0(k_n)| r q^{n\beta/2} t^{n-1}$$

converges for  $|t| < q^{-\beta/2 - \dim X_0}$  by (9.6.3.9).

□

**Lemma (9.6.3.11) [Semi-continuity of Weights for Curves].** If  $X_0 \in \text{SmCur}/k_0$  and  $j_0 : U_0 \hookrightarrow X_0$  be a non-empty open immersion with  $S_0 = X_0 \setminus U_0$ . Let  $\mathcal{G}_0$  be a Weil-sheaf on  $X_0$  s.t. the restriction  $j_0^* \mathcal{G}_0$  is lisse and  $H_S^0(X, \mathcal{G}) = 0$ , then  $\text{wt}(j_0^*(\mathcal{G}_0)) \leq \beta$  implies  $\text{wt}(\mathcal{G}_0) \leq \beta$ .

┘

*Proof:* For any point  $x$ , consider an affine open subset of  $X_0$ , then reduce to the affine case, and because  $H_S^0(X, \mathcal{G}) = 0$  and the excision sequence (9.4.4.7), we have  $\mathcal{G} \hookrightarrow j_* j^* \mathcal{G}$ , so the weights of  $\mathcal{G}_0$  are no more than that of  $j_* j^* \mathcal{G}$ , and replacing  $\mathcal{G}_0$  with  $j_{0*} j_0^* \mathcal{G}_0$ , we can assume  $\mathcal{G}_0 = j_{0*} j_0^* \mathcal{G}_0$ . Then

$$H_c^0(X, \mathcal{G}) = H_c^0(X, j_* j^* \mathcal{G}) = H_c^0(U, j^* \mathcal{G}) = 0$$

by Poincare duality and the fact  $j_*$  is exact because it is finite.

Now by Grothendieck-Lefschetz trace formula(9.6.2.13),

$$Z(X_0, \mathcal{G}_0, T) = L(U_0, j_0^*(\mathcal{G}_0), T) \cdot \prod_{s \in |X_0|_0} \det \left( 1 - T^{d_s} F_s, \mathcal{G}_{\bar{s}} \right)^{-1} = \frac{\det \left( 1 - F_X T \middle| H_{\text{ét},c}^1(X, \mathcal{G}) \right)}{\det \left( 1 - F_X T \middle| H_{\text{ét},c}^2(X, \mathcal{G}) \right)}$$

Denote  $\mathcal{F}_0 = j_0^* \mathcal{G}_0$ , then

$$H_{\text{ét},c}^2(X, \mathcal{G}) = H_{\text{ét},c}^2(U, \mathcal{F}) = (\mathcal{F}_{\bar{x}})_{\pi_1(U, \bar{x})}(-1)$$

So the weights of eigenvalues of  $F_X$  on  $H_{\text{ét},c}^2(X, \mathcal{G}) \leq$  weights of  $\mathcal{F} + 2$ , hence the  $L$ -function converges for  $|t| < q^{-\beta/2-1}$ . Now the LHS has  $L(U_0, j_0^*(\mathcal{G}_0), t)$  converges for  $|t| < q^{-\beta/2-1}$  because  $w(\mathcal{F}_0) \leq \beta$ , and so for the points in  $S_0$ , we also have  $\det \left( 1 - t^{d_s} F_s, \mathcal{G}_{\bar{s}} \right)$  has no zero there, which means they have weights  $\leq \beta + 1$ . Now consider replacing  $\mathcal{G}_0$  with  $\mathcal{G}_0^{\otimes k}$  and let  $k \rightarrow \infty$ , then their weights  $\leq \beta$ .  $\square$

**Prop. (9.6.3.12) [Semi-continuity of Weights].** Let  $X_0$  be geo.normal,  $\mathcal{G}_0 \in \text{WSh}_{\ell, \text{lis}}(X_0)$  and  $j_0 : U_0 \hookrightarrow X_0$  be an open dense subscheme, then

- $w(\mathcal{G}_0) = w(j_0^* \mathcal{G}_0)$ .
- If  $j_0^*(\mathcal{G}_0)$  is  $\iota$ -pure of weights  $\beta$ , then  $\mathcal{G}_0$  is also  $\iota$ -pure of weights  $\beta$ .
- Let  $X_0$  be irreducible and normal, and  $\mathcal{G}_0$  is irreducible, then if  $j_0^* \mathcal{G}_0$  is  $\iota$ -mixed, then  $\mathcal{G}_0$  is  $\iota$ -pure.

┘

*Proof:* 1: The weights is local so we may assume  $X_0$  is irreducible, and then for any closed point  $x$ , we can connect it with  $U_0$  with a curve (choose an affine open and use Noetherian Normalization to choose an irreducible component of an arbitrary curve in  $\mathbb{A}^n$ ). Notice  $H_S^0(X, \mathcal{G}) = 0$  because it is lisse thus  $H^0(X, \mathcal{G})$  is determined by stalk thus  $H^0(X, \mathcal{G}) \rightarrow H^0(U, \mathcal{G}|_U)$  is injective. So we finish by the curve case(9.6.3.11).

2: Apply item1 to  $\mathcal{G}_0$  and  $\mathcal{G}_0^\vee$ .

3: It is  $\iota$ -mixed so it has  $\iota$ -pure Weil-sheaf constituents. Now by(9.4.6.15) we can find an open dense  $U_0$  that restriction to  $U_0$  has constituents  $\iota$ -pure lisse sheaves. But it is also irreducible because  $\pi_1(U_0, \bar{a}) \rightarrow \pi_1(X_0, \bar{a})$  is surjective<sup>?</sup>, so it is  $\iota$ -pure and item2 shows  $\mathcal{G}_0$  is  $\iota$ -pure.  $\square$

### $L^2$ -Norms and Maximal Weights

**Def. (9.6.3.13) [ $L^2$ -Norm of Sheaves].** As in(9.6.8.1), for any  $\mathcal{G}_0 \in D_{\text{const}}^b(X_0, \bar{\mathbb{Q}}_\ell)$ , we have a function

$$f^{\mathcal{G}_0} : X_0(\mathbb{F}_{q^n}) \rightarrow \bar{\mathbb{Q}}_\ell : x \mapsto \sum_i (-1)^i \text{tr}(F_x^{n/d_x} | (H^i(\mathcal{G}_0))_{\bar{x}}),$$

Fix an arbitrary isomorphism  $\bar{\mathbb{Q}}_\ell \cong \mathbb{C}$ , we can consider the usual  $L^2$ -norm for functions on  $X_0(k_n)$ , denoted by  $(f, g)_n$ .  $\square$

**Def. (9.6.3.14) [Maximal Weights].** Notice the equation form(9.6.3.10) can be rewritten as

$$\frac{d \log}{dT} L(X_0, \mathcal{G}_0, T) = \sum_{n \geq 1} \sum_{\substack{x_0 \in |X_0|_0 \\ \deg(x_0) | n}} d_x(\text{tr}(F_x^{n/d_x})) T^{n-1} = \sum_n (f^{\mathcal{G}_0}, 1)_n T^{n-1}.$$



Now we define another closed related function

$$\varphi^{\mathcal{G}_0}(t) = \sum_n \left\| f^{\mathcal{G}_0} \right\|_n^2 t^{n-1} \quad (9.6.3.13),$$

which works better with Fourier-transform we are about to define later.  $\lrcorner$

**Lemma (9.6.3.15).** There is a constant  $C$  that  $\left\| f^{\mathcal{G}_0}(x) \right\|^2 \leq C q^{n(w(\mathcal{G}_0) + \dim X_0)}$ , so  $\varphi^{\mathcal{G}_0}(T)$  converges for  $|T| \leq q^{-w(\mathcal{G}_0) - \dim X_0}$ .  $\lrcorner$

*Proof:* The proof is similar to that of (9.6.3.10) thus omitted.  $\square$

**Def. (9.6.3.16) [Norm of Weil Sheaves].** Define the **norm of a Weil-sheaf** to be

$$\|\mathcal{G}_0\| = \sup \left\{ \rho \left| \lim_n \frac{\left\| f^{\mathcal{G}_0} \right\|_n^2}{q^{n(\rho + \dim X_0)}} > 0 \right. \right\} \quad (9.6.3.13)$$

Then  $q^{-\|\mathcal{G}_0\| - \dim X_0}$  is just the radius of convergence of the function  $\varphi^{\mathcal{G}_0}(t)$  (9.6.3.14), and  $\|\mathcal{G}_0\| \leq w(\mathcal{G}_0)$  by lemma (9.6.3.15) above.  $\lrcorner$

**Prop. (9.6.3.17) [Radius of Convergence].** Let  $\mathcal{G}_0$  be a  $\iota$ -mixed sheaf on a smooth curve  $X_0$  over  $k$ , and  $H_{\text{ét},c}^0(X, \mathcal{G}) = 0$ , then  $\|\mathcal{G}_0\| = w(\mathcal{G}_0) = \beta$ .  $\lrcorner$

*Proof:* It suffices to show  $w(\mathcal{G}_0) \leq \|\mathcal{G}_0\|$ . First notice we can assume  $X_0$  is reduced because the nilpotents corresponds to zero Frobenius eigenvalues, and also it is connected, because the function  $f^{\mathcal{G}_0}$  is additive in  $X$ . Now we study by cases:

1: If  $\mathcal{G}_0$  is a lisse  $\iota$ -pure sheaf on a smooth affine curve  $X_0$ , we may assume  $\mathcal{G}_0 \neq 0$ , then  $\mathcal{G}_0 \otimes \overline{\mathcal{G}_0}$  (9.6.5.2) is  $\iota$ -real of weight  $2\beta$  and  $(f^{\mathcal{G}_0 \otimes \overline{\mathcal{G}_0}}, 1)_n = \left\| f^{\mathcal{G}_0} \right\|_n^2$ , so  $\varphi^{\mathcal{G}_0}(t)$  is just the logarithmic derivative of the  $L$ -function  $L(X_0, \mathcal{G}_0 \otimes \overline{\mathcal{G}_0}, t)$ , thus (9.6.3.10) shows its convergence radius  $\geq q^{-\beta-1}$ . And notice the  $H_c^0$  terms vanish so the poles can only appear as the zeros of  $H_c^2$  term, so (9.6.5.3) shows the poles of the  $L(X_0, \mathcal{G}_0 \otimes \overline{\mathcal{G}_0}, t)$  has weight  $2\beta + 2$ , thus the poles can only appear on  $|t| = q^{-\beta-1}$ .

Now consider each local Euler factor  $\det \left( 1 - F_x t^{d_x} |(\mathcal{G}_0 \otimes \overline{\mathcal{G}_0})_x \right)^{-1}$  has non-negative coefficients, they have poles because  $\mathcal{G}_0 \neq 0$ , and their poles have weight  $\beta$  because of purity, thus their product also has (real) poles, by previous argument, the pole has weight  $\beta + 1$ , thus it has convergence radius at most  $q^{-\beta-1}$ , so we are done.

2: If  $\mathcal{G}_0$  is a  $\iota$ -mixed, consider its semisimplification  $\mathcal{G}_0^{ss} = \mathcal{F}_0 \oplus \mathcal{H}_0$ , where  $\mathcal{F}_0$  is  $\iota$ -pure of weight  $w(\mathcal{G}_0)$ , and  $w(\mathcal{H}_0) \leq w(\mathcal{F}_0)$ .

Then  $f^{\mathcal{G}_0} = f^{\mathcal{F}_0} + f^{\mathcal{H}_0}$ , and

$$\varphi^{\mathcal{G}_0}(T) = \varphi^{\mathcal{F}_0}(T) + \sum_{n \geq 1} 2 \operatorname{Re}(f^{\mathcal{F}_0}, f^{\mathcal{H}_0})_n T^{n-1} + \varphi^{\mathcal{H}_0}(T).$$

Then by item 1,  $\varphi^{\mathcal{F}_0}(t)$  has convergence radius  $q^{-w(\mathcal{G}_0)-1}$ , and by (9.6.3.15)  $\varphi^{\mathcal{H}_0}(t)$  has radius at least  $q^{-w(\mathcal{H}_0)-1} > q^{-w(\mathcal{G}_0)-1}$ , and by Cauchy's inequality the middle term satisfies

$$\left| 2 \operatorname{Re}(f^{\mathcal{F}_0}, f^{\mathcal{H}_0})_n \right| \leq 2 \left\| f^{\mathcal{F}_0} \right\|_n \left\| f^{\mathcal{H}_0} \right\|_n \leq C q^{n(w(\mathcal{F}_0) + w(\mathcal{H}_0)/2 + 1)}$$

So the middle term has convergence radius  $> q^{-w(\mathcal{G}_0)-1}$ , so their sum has convergence radius  $q^{-w(\mathcal{G}_0)-1}$ .  $\square$

#### 4 Geometric Monodromy

**Notation (9.6.4.1).**

- Let  $X_0/k$  be geometrically connected and normal.

┘

**Def. (9.6.4.2)[Geometric Monodromy Group].** Let  $\mathcal{G}_0$  be a Weil-sheaf associated to a representation  $(V, \rho) = \text{GL}(\mathcal{G}_{\bar{x}})$  of  $W(X_0, \bar{x})$ , the **geometric monodromy group**  $G_{geo}$  associated to  $\mathcal{G}_0$  is the Zariski Closure of  $\rho(\pi_1(X, \bar{x})) \subset \text{GL}(V)$ .

Every element in  $\rho(W(X_0, \bar{x}))$  normalizes  $G_{geo}$  by continuity, so choosing an arbitrary generator  $\sigma \in W(\bar{k}/k)$ , we have an action of  $W(\bar{k}/k)$  on  $G_{geo}$ . Define  $G = W(\bar{k}/k) \ltimes G_{geo}$  the **arithmetic monodromy group** of  $\mathcal{G}_0$ .

┘

**Lemma (9.6.4.3).** If  $G_{geo}$  is connected, then there is a positive integer  $N$  that the semidirect sequence

$$1 \rightarrow G_{geo} \rightarrow \text{deg}^{-1}(N\mathbb{Z}) \xrightarrow{\text{deg}} N\mathbb{Z} \rightarrow 1$$

is direct, i.e.  $\text{deg}^{-1}(N\mathbb{Z}) \cong G_{geo} \times \mathbb{Z}$ .

┘

*Proof:* Choose a  $\text{deg}(g) = 1$ . The representation  $G_{geo}$  splits as a characters of  $Z(G)$ , and then some  $g^n$  stablizes these characters, hence stablizes  $Z(G)$ , which then it descends to an action on  $G_{adj}$ , whose automorphism is the automorphism of the Dynkin diagram ?, so finite, so some  $g^m$  fixes  $G_{adj}$  after changing a semidirect product, thus induces a map  $\text{Hom}(G_{adj}, Z(G))$ , but  $G_{adj}$  is semisimple(10.4.2.23), so the connected component is mapped to 1 in  $Z(G)$ (10.4.2.18), so there are only f.m. such homomorphism, showing  $g^k$  is 1, so the product is exact for  $N = k$ .  $\square$

**Prop. (9.6.4.4)[Geometric Monodromy Group is Semisimple].** Let  $\mathcal{G}_0 \in \text{WSh}_{\text{lisse}}^{geo, ss}(X_0)$ (9.6.2.5), then

- $G_{geo}$  and  $G_{geo}^0$  are semisimple algebraic group.
- Let  $Z = Z(G(\bar{\mathbb{Q}}_\ell))$ , then the map  $\psi : Z \rightarrow W(\bar{k}/k)$  has finite kernel and cokernel. In particular,  $Z$  contains an element of finite degree, and it is surjective after a finite base change of fields.

And notice in fact if  $\mathcal{G}_0$  is semisimple, then it is automatically geometrically semisimple by(24.1.2.15).  $\square$

*Proof:* 1:  $L G_{geo}$  is semisimple iff  $G_{geo}^0$  is semisimple. Pass to a finite étale covering, we may assume  $G_{geo} = G_{geo}^0$ . Let  $R(G_{geo}^0)$  be the radical and  $R_u(G_{geo}^0)$  be the unipotent radical, then  $R$  is normal in  $G^0$  and  $G^0$  is normal in  $G$ , so by(24.1.2.15)  $V = \text{GL}(\mathcal{G}_{\bar{x}})$  is irreducible  $R(G_{geo}^0)$  representation, but it is solvable, so  $V$  is a direct sum of 1-dimensional representations, and  $R_u(G_{geo}^0)$  is trivial, in particular  $G_{geo}^0$  is reductive. So it is semisimple if the maximal Abelian quotient  $G_{geo}^{ab}$  is finite(10.4.2.18).

let  $T_1$  be the maximal central torus of  $G_{geo}^0$ , then lemma(9.6.4.3) shows after a finite base change of fields, we may assume  $G = G_{geo} \times \mathbb{Z}$ , consider the composite  $W(X_0, \bar{x}) \rightarrow G_{geo} \times \mathbb{Z} \rightarrow G_{geo} \rightarrow G_{geo}^{ab}$ , then  $\pi_1(X, \bar{x})$  is Zariski dense in  $G_{geo}^{ab}$ , and (9.6.3.3) shows clearly  $G_{geo}^{ab}$  has no maximal torus thus finite.

2:  $\ker \psi \subset Z(G_{geo}(\bar{\mathbb{Q}}_\ell))$  is finite since  $G_{geo}$  is semisimple. To find an element in  $Z(G)$  of positive degree, we may use the same method as before to find an element  $\zeta$  that commutes with  $G_{geo}^0$ , and pass to a power, we may assume it acts trivially on  $G_{geo}/G_{geo}^0$ .

For any  $g \in G_{geo}$ , consider  $vp_g(n) = g\zeta^n g^{-1}\zeta^{-n} \in G_{geo}^0$ , so  $\varphi_g(m+n) = \varphi_g(n)\zeta^n \varphi_g(m)\zeta^{-n} = \varphi_g(n)\varphi_g(m)$ , thus it is a homomorphism, and if  $g' \in G_{geo}^0$ , then

$$\varphi_g = \varphi_{g(g^{-1}g'g)} = \varphi_{g'g} = g'\varphi_g(g')^{-1}$$

so  $\varphi_g$  has image in  $Z(G_{geo}^0)$ , which is finite, so  $\varphi_g(n) = 1$  for some  $n$ , then  $\zeta^n$  commutes with  $G_{geo}$  so  $\zeta^n \in Z(G)$ .  $\square$

**Cor.(9.6.4.5) [Weights and Center Element Actions].** Let  $\mathcal{G}_0 \in \text{WSh}_{\text{isse}}^{\text{ss}}(X_0)$  (9.6.2.5), if  $z \in Z(G(\overline{\mathbb{Q}}_\ell))$  satisfies  $\deg(z) = n \neq 0$ , which exists by (9.6.4.4), then if  $z$  acts on  $V$  with eigenvalues  $\alpha_i$ , then  $\frac{2}{n} \log_q(|\iota(\alpha_i)|)$  is just the determinential  $\iota$ -weights of  $\mathcal{G}_0$ .  $\lrcorner$

*Proof:*  $z$  is in the center, thus by Shur's lemma, it acts on each irreducible part of  $\mathcal{G}_0$  by a constant. Thus the determinential weights are clear, by definition.  $\square$

**Cor.(9.6.4.6) [Properties of Determinential Weights].** Suppose  $X_0 \in \text{SmCur}/k$  and  $\mathcal{F}_0, \mathcal{G}_0 \in \text{WSh}_{\text{isse}}(X_0)$ , then

- If  $\alpha_i$  are the determinential  $\iota$ -weights of  $\mathcal{F}_0$  and  $\beta_j$  be that of  $\mathcal{G}_0$ , then  $\alpha_i + \beta_j$  are those of  $\mathcal{F}_0 \otimes \mathcal{G}_0$  with multiplicity.
- For  $\gamma \in \mathbb{R}$ , let  $r(\gamma)$  be the sum of ranks of all irreducible constituents of  $\mathcal{F}_0$  which have determinential weight  $\gamma$  w.r.t  $\iota$ , then the determinential weights of  $\wedge^r \mathcal{F}_0$  are the numbers  $\sum_\gamma n(\gamma)\gamma$  with  $\sum n(\gamma) = r$  and  $0 \leq n(\gamma) \leq r(\gamma)$ ,  $n(\gamma) \in \mathbb{Z}$  with multiplicity.

$\lrcorner$

*Proof:* Firstly notice the determinential weight is unchanged when we change  $\mathcal{F}_0, \mathcal{G}_0$  to their semisimplification  $\mathcal{F}_0^{ss}, \mathcal{G}_0^{ss}$  (9.6.3.4). And notice  $(\mathcal{F}_0 \otimes \mathcal{G}_0)^{ss} = ((\mathcal{F}_0)^{ss} \otimes (\mathcal{G}_0)^{ss})^{ss}$ , thus the determinential weights of  $\mathcal{F}_0 \otimes \mathcal{G}_0$  are also unchanged. Similarly for the wedge product.

2: We may assume  $\mathcal{F}_0, \mathcal{G}_0$  are irreducible, and let  $\mathcal{H}_0 = (\mathcal{F}_0 \otimes \mathcal{G}_0)^{ss}$ ,  $G_{geo}^\oplus, G_{geo}^{ss}$  be the geometric monodromy group of  $\pi_1(X, \bar{x})$  in  $GL(\mathcal{F}_{\bar{x}} \oplus \mathcal{G}_{\bar{x}})$  and  $GL(\mathcal{H}_{\bar{x}})$  correspondingly, then  $G_{geo}^\oplus \rightarrow G_{geo}^{ss}$  is surjective because they are both the geometric monodromy group of  $\mathcal{H}_{\bar{x}}$ . So also  $G^\oplus \rightarrow G^{ss}$  is surjective. So if  $g$  be an element in the center of  $G^\oplus$  that has nonzero degree, then it maps to the center of  $G^{ss}$  of nonzero degree. And the action of  $g$  on each factor  $\mathcal{F}_x, \mathcal{G}_x$  is a constant, so action on  $H_x$  is also a constant, so we are done.

3: Easy from 2.  $\square$

## 5 Real Weil-Sheaves

**Def.(9.6.5.1) [ $\iota$ -Real Weil-Sheaf].**  $\mathcal{F}_0 \in \text{WSh}(X_0; \overline{\mathbb{Q}}_\ell)$  is called a  $\iota$ -real Weil-Sheaf if for any  $x \in |X_0|$ , the characteristic polynomial  $\iota(\det(1 - F_x t; \mathcal{F}_{\bar{x}}))$  of  $F_x$  real coefficients.  $\lrcorner$

**Prop.(9.6.5.2).** Any  $\iota$ -pure Weil-sheaf of weight  $w$  is a direct sum of a  $\iota$ -real  $\iota$ -pure Weil-sheaf. In fact,  $\mathcal{F}_0 \oplus \mathcal{F}_0^\vee(-w) = \mathcal{F}_0 \oplus \overline{\mathcal{F}_0}$  is  $\iota$ -real.  $\lrcorner$

**Lemma(9.6.5.3) [Eigenvalue of Cohomology and Stalk in Curve case].** Let  $X_0/k$  be a smooth curve,  $\mathcal{F}_0 \in \text{WSh}(X_0)$  is lisse, then the eigenvalues of  $F_X$  on  $H^0(X, \mathcal{F})$  or  $H_{\text{ét},c}^2(X, \mathcal{F})$  is related to the determinential weights of  $\mathcal{F}_0$  and the eigenvalue of  $F_x$  on  $\mathcal{F}_{\bar{x}}$ .  $\lrcorner$

*Proof:* Let  $V = \mathcal{F}_{\bar{x}}$ , then

$$H^0(X, \mathcal{F}) = V^{\pi_1(X, \bar{x})}, \quad H_c^2(X, \mathcal{F}) = V_{\pi_1(X, \bar{x})}(-1).$$

Then the base change sheaf of the sheaf  $V^{\pi_1(X, \bar{x})}$  or  $V_{\pi_1(X, \bar{x})}(-1)$  on  $\text{Spec } k$  is the maximal subsheaf/quotient lisse sheaf of  $\mathcal{F}_0$  that is constant on  $X$ . Then it has determinential weights just the action of  $F_X$  on the stalk by (9.6.4.5), which are also determinential weights of  $\mathcal{F}_0$  by (9.6.4.6).  $\square$

**Lemma (9.6.5.4) [Rankin-Selberg Method].** If  $X_0 \in \text{SmCur}/k$  and  $\mathcal{F}_0 \in \text{WSh}_{\text{lisse}}(X_0)$ , and  $w$  be the largest determinential weight of  $\mathcal{F}_0$ , then for any  $x \in |X_0|$ ,  $w_{N(x)}(\alpha) \leq w$ .  $\lrcorner$

*Proof:* By the arbitrariness of  $x$ , we can replace  $X_0$  by an affine open nbhd of  $x$ . Then  $H_c^0(X, \mathcal{G}) = 0$  by Artin vanishing (9.4.2.12). By Grothendieck trace formula,

$$\prod_{x \in |X_0|} \iota \det \left( 1 - t^{d_x} F_x | \otimes^{2k} \mathcal{F}_{\bar{x}} \right)^{-1} = \frac{\iota \det \left( 1 - t F_X^* | H_c^1(X, \otimes^{2k} \mathcal{F}) \right)}{\iota \det \left( 1 - t F_X^* | H_c^2(X, \otimes^{2k} \mathcal{F}) \right)}$$

Now the weight of root  $t_0$  of  $\det \left( 1 - t F_X^* | H_c^2(X, \otimes^{2k} \mathcal{F}) \right)$  has weight  $\leq$  (determinential weight of  $\mathcal{G}_0^{\otimes 2k}$ ) + 2(9.6.5.3)  $\leq 2kw + 1$  (9.6.4.6), so  $|t_0| \geq q^{-k\beta-1}$ .

Now by the formula?? and noticing  $\text{tr}(F_x^n, \otimes^{2k} \mathcal{F}_{\bar{x}}) = (\text{tr}(F_x, \mathcal{F}_{\bar{x}}))^{2k}$ , so  $(1 - t^{d_x} F_x | \otimes^{2k} \mathcal{F}_{\bar{x}})^{-1}$  has non-negative coefficients, which means their convergence radius are no less than  $q^{-k\beta-1}$ , equivalently,  $(1 - t^{d_x} F_x | \otimes^{2k} \mathcal{F}_{\bar{x}})$  has no zeros with eigenvalue  $< q^{-k\beta-1}$ .

So for any eigenvalue  $\alpha$  of  $F_x$  acting on  $\mathcal{F}_{\bar{x}}$ ,  $|\iota(\alpha^{-2k/d_x})| \leq q^{-k\beta-1}$ , or equivalently,

$$|\iota(\alpha)|^2 \leq N(x)^{\beta+1/k}.$$

Now let  $k \rightarrow \infty$ , we are done.  $\square$

**Lemma (9.6.5.5) [Real Sheaf Mixed Curve case].** Let  $X_0/k$  be smooth curve and  $\mathcal{F}_0 \in \text{WSh}(X_0)$  is  $\iota$ -lisse, then all irreducible constituents of  $\mathcal{F}_0$  is  $\iota$ -pure, and their  $\iota$ -weights coincides with their determinential weights.  $\lrcorner$

*Proof:* For  $\beta \in \mathbb{R}$ , let  $\mathcal{F}_0(\beta)$  be the sum of constituents of  $\mathcal{F}_0$  of determinential weight  $\beta$ , and let  $n(\beta) = \text{rank}(\mathcal{F}_0(\beta))$ , then we need to show that  $w_{N(x)}(\alpha_i(\beta)) = \beta$  for any eigenvalue of  $F_x$  on  $\mathcal{F}(\beta)_{\bar{x}}$ .

By definition of determinential weight, for each  $\gamma$ , we have  $\sum w_{N(x)}(\alpha_j(\gamma)) = n(\gamma)\gamma$ . Now let  $N = \sum_{\gamma > \beta} n(\gamma)$ , then any determinential weight of  $\wedge^{N+1} \mathcal{F}_0$  has weight  $\leq \beta + \sum_{\gamma > \beta} n(\gamma)\gamma$ : This is clear by (9.6.4.6) as the determinential weights of  $\wedge^{N+1} \mathcal{F}_0$  is of the form  $\sum_{\gamma} a(\gamma)\gamma$  that  $0 \leq a(\gamma) \leq \gamma$  and  $\sum_{\gamma} a(\gamma) = N + 1$ .

But now  $\alpha_i(\beta) \prod_{\gamma > \beta} \prod_{i=1}^{n(\gamma)} \alpha_j(\gamma)$  is an eigenvalue of  $(\wedge^{N+1} \mathcal{F}_0)_{\bar{x}}$ , but by lemma (9.6.5.4),  $w_{N(x)}(\alpha_i(t)) \leq t$ . Thus we must have equality  $w_{N(x)}(\alpha_i(\beta)) = \beta$ .  $\square$

**Prop. (9.6.5.6) [Real Sheaf is Mixed].** For  $X_0 \in \text{Sch}^{\text{ft}}/\mathbb{F}_q$ ,

- Any  $\iota$ -real Weil-sheaf on  $X_0$  is  $\iota$ -mixed.
- If  $X_0$  is irreducible and normal, any irreducible constituent of a lisse of an  $\iota$ -real sheaf is  $\iota$ -pure.  $\lrcorner$

*Proof:* Cf. [Bhatt P28], [KW, P36].

We have the following devissages:

- Choose an open subset  $j_0 : U_0 \hookrightarrow X_0$ ,  $S_0 = X_0 - U_0$  and consider the fundamental excision sequence (9.4.4.6), we can reduce to an open affine subscheme  $U_0 \subset X_0$ .
- We may base change to a finite field extension. ?
- So we may reduce to the case  $X_0$  is smooth, irreducible affine, and  $\mathcal{G}_0$  is lisse, with all the irreducible constituents geometrically irreducible (by base change, because they are geometrically semisimple (9.6.4.4)). And we may assume  $\dim X_0 > 1$  because the curve case is proven.

- Change  $k$  to the alg.closure of  $k$  in the function field of  $X_0$ , we can assume  $X_0$  is geometrically irreducible by (7.4.3.16).

Embed  $X_0$  in some projective space  $\mathbb{P}_0^N$ , then by a suitable Bertini theorem, the linear subspaces of codimension  $\dim X - 1$  that intersects  $X$  with a non-empty smooth irreducible curve  $C_L$  is dense in the Grassmannian. Now the closed points in any  $C_L$  is a pure-point for the any irreducible component  $\mathcal{F}_0$  of  $\mathcal{G}_0$  of the same weights. Now let  $L$  vary, then there is a dense subset of a finite extension of  $X_0$  that  $\mathcal{F}_0$  is pure. So we are done.  $\square$

## 6 Deligne's Purity Theorem

**Thm. (9.6.6.1) [Deligne's Purity Theorem].** If  $f : X_0 \rightarrow Y_0 \in \text{Sch}^{\text{ft}}/k_0$  is separated, and  $\mathcal{F}_0 \in \text{WSh}_{\text{lis}}(X_0, \overline{\mathbb{Q}_\ell})$  that is mixed weights  $\leq n$ , then for any integer  $i \geq 0$ , the sheaf  $R^i f_{0!} \mathcal{F}_0$  is also  $\iota$ -mixed of weights  $\leq n + i$ .

Moreover, each  $\iota$ -weight of  $R^i f_! \mathcal{F}$  is equivalent modulo  $\mathbb{Z}$  to an  $\iota$ -weight of  $\mathcal{F}$ . In particular, if all  $\iota$ -weights of  $\mathcal{F}$  is integral, then so is  $R^i f_! \mathcal{F}$ .  $\lrcorner$

*Proof:* This follows from (9.6.6.6).  $\square$

**Cor. (9.6.6.2).** If  $X_0 \in \text{Sch}^{\text{sm,sep,ft}}/k_0$  and  $\mathcal{F}_0$  is mixed of weight  $\geq n$ , then  $H_{\text{ét}}^i(X, \mathcal{F})$  is mixed of weights  $\geq n + i$ .  $\lrcorner$

*Proof:* By Poincaré duality (9.4.6.48),  $H_{\text{ét},c}^{2d-n}(X, \mathcal{F}^\vee(d)) = (H_{\text{ét}}^n(X, \mathcal{F}))^\vee$  as Galois representation, and  $\mathcal{F}^\vee(d)$  is still a lisse sheaf pure of weight  $-w - 2d$ , thus Deligne's purity theorem (9.6.6.1) shows that  $H_{\text{ét},c}^{2d-n}(X_{\bar{k}}, \mathcal{F}^\vee(d))$  has weight  $\leq (-w - 2d) + (2d - n) = -w - n$ , thus we are done.  $\square$

**Cor. (9.6.6.3) [Weil-Deligne Conjecture].** Suppose  $X_0 \in \text{Sch}^{\text{sm,sep,ft}}/k$ , and  $\mathcal{F}_0 \in \text{WSh}_{\text{lis}}(X_0, \overline{\mathbb{Q}_\ell})$  is pure of weight  $w$ , then  $\text{Im} \left( H_{\text{ét},c}^n(X, \mathcal{F}) \rightarrow H_{\text{ét}}^n(X, \mathcal{F}) \right)$  (9.4.4.8) is pure of weight  $w + n$ .  $\lrcorner$

*Proof:* The morphism  $H_{\text{ét},c}^n(X, \mathcal{F}) \rightarrow H_{\text{ét}}^n(X, \mathcal{F})$  is compatible with Frobenius  $\text{?}$ , so from (9.6.6.2) we know the image has weights  $\geq w + n$ , so combined with Deligne's purity theorem (9.6.6.1), we know it is pure of weight  $w + n$ .  $\square$

**Cor. (9.6.6.4).** If  $f_0 : X_0 \rightarrow Y_0$  is a smooth proper map of algebraic schemes and  $\mathcal{F}_0$  is  $\iota$ -pure of weight  $\beta$ , then  $R^i f_{0,*} \mathcal{F}_0$  is  $\iota$ -pure of weight  $\beta + i$ .  $\lrcorner$

*Proof:* Use proper base change of (9.4.4.7) to reduce to the case of (9.6.6.3). Notice in the proper case,  $Rf_* = Rf_!$ .  $\square$

**Cor. (9.6.6.5) [Riemann Hypothesis].** If  $X_0 \in \text{SmPrpr}/k$ , then  $H_{\text{ét}}^n(X; \mathbb{Q}_\ell)$  as a Weil-sheaf over  $\text{Spec } k$ , is pure of weight  $n$ .  $\lrcorner$

### Reduction to Curve case

**Prop. (9.6.6.6).** Deligne's purity theorem (9.6.6.1) can be reduced to case that  $X_0$  is a smooth geo.connected affine curve in  $\mathbb{A}_{k_0}^1$  and  $\mathcal{F}_0 \in \text{WSh}_{\text{lis}}(X_0, \overline{\mathbb{Q}_\ell})$ .  $\lrcorner$

*Proof:* We have the following dévissages about the Deligne's :

1. It is trivial in case  $f_0$  is quasi-finite. This is because of (9.4.4.9), as the fiber has dimension 0.
2. We can replace  $X_0$  by an affine open  $U_0 \subset X_0$  by Noetherian induction and excision sequence (9.4.4.7), which commutes with Frobenius action.

3. If the conclusion is true for  $g_0, h_0$ , then it is true for  $f_0 = g_0 \circ h_0$ , this follows from the Leray spectral sequence(9.4.4.7), which is Frobenius equivariant by(9.4.4.15).
4. We can replace  $Y_0$  with an affine open  $U_0 \subset Y_0$ : If the image  $f_0$  is not dense, then trivial, if it is dense, then choose any affine open  $U_0$ , then it suffices to prove for  $f_0 : f_0^{-1}(U_0) \rightarrow Y_0$  by item2, then then by item3 it suffice to prove for  $f_0 : f_0^{-1}(U_0) \rightarrow U_0$ , because  $U_0 \hookrightarrow Y_0$  is quasi-finite and use item1.

Now we claim we can reduce to the case of  $f_0 : X_0 \rightarrow Y_0$  surjective affine smooth with the fibers being geometrically irreducible curves: By devissage2 and 4, we may assume  $X_0, Y_0$  is affine, thus  $f_0$  is affine. Take a generic point  $\eta$  of  $Y_0$ (How to pass to the reduced case?), then  $(X_0)_\eta \rightarrow \text{Spec } k(\eta)$  is affine hence by Noetherian normalization(5.2.4.22) there is a finite map  $X_\eta \rightarrow \mathbb{A}_{k(\eta)}^n$ , and this spread out to a finite morphism  $f_0^{-1}(U_0) \rightarrow U_0$  for some affine open  $U_0 \subset Y_0$  because  $f_0$  is of f.t.. Then by Devissage1 and 3 we are reduced to the case  $A_{Y_0}^1 \rightarrow Y_0$ . Now by(9.4.6.15), there is an affine open  $U_0 \subset A_{Y_0}^1$  that  $\mathcal{F}_0|_{U_0}$  is lisse, so by Devissage2 we may change  $X_0$  to  $U_0$ .

That is we reduced to the case that  $\mathcal{F}_0$  is lisse and  $X_0$  is open in  $\mathbb{A}_{Y_0}^1$  so  $f_0$  is smooth affine, in particular open(7.6.4.4), so we can replace  $Y_0$  by  $f(X_0)$  and assume  $f_0$  is surjective. Then the fiber are all geometrically irreducible curves.

Then the assertion about weights are clear from proper base change(9.4.4.7) and the curve case.

For the  $\iota$ -mixedness, we may use(9.6.5.2) and(9.6.5.6) to reduce to showing that  $R^i f_!$  maps  $\iota$ -real sheaves to  $\iota$ -real sheaves.

for a geometric point  $\bar{x} \rightarrow x \rightarrow X_0$ , let  $C \rightarrow C_0$  be the fiber, which is affine irreducible, so  $H_c^0(X, \mathcal{F}) = 0$  by Poincaré duality(9.4.6.48) and Artin vanishing theorem(9.4.2.12), so

$$\iota L(C_0, \mathcal{G}_0, t) = \frac{\iota \det(1 - tF_X^* | H_c^1(C, \mathcal{G}|_C))}{\iota \det(1 - tF_X^* | H_c^2(C, \mathcal{G}|_C))}$$

by Grothendieck-Lefschetz formula(9.6.2.13). Now we can use Poincare duality and the definition that  $\mathcal{G}_0$  is pure of weight  $\beta$ , we know  $H_c^2(C, \mathcal{G}|_C)$  is pure of weight  $\beta + 2$ , by(9.6.5.3). And  $H_c^2(C, \mathcal{G}|_C)$  has weights smaller than  $\beta + 1$  by the curve case, so the two polynomial is coprime, and both has constant coefficient 1, which shows they are both real. And then by proper base change(9.4.4.7), this just says  $R^i f_0! \mathcal{G}_0$  is  $\iota$ -real.  $\square$

### Third Reduction

**Prop. (9.6.6.7).** If  $X_0 \subset \mathbb{A}_{k_0}^1$  is a smooth curve and  $\mathcal{F}_0 \in \text{WSh}_{\text{lisse}}(X_0; \overline{\mathbb{Q}_\ell})$ , the Deligne's purity theorem(9.6.6.1) is true.  $\lrcorner$

*Proof:* We have the following devissages:

- We only need to check for  $H_{\text{ét},c}^1(X, \mathcal{F})$ , because  $H_c^0$  vanish by Poincaré duality(9.4.6.48) and Artin vanishing theorem(9.4.2.12) and ,  $H_{\text{ét},c}^2(X, \mathcal{F})$  is dealt with in(9.6.5.3).
- We are free to pass to finite base change.
- We may assume  $\mathcal{F}_0$  is geometrically irreducible: By(9.6.4.4), all the irreducible constituents of  $\mathcal{F}_0$  are geometrically semisimple, so pass to a finite base change, we may assume that its irreducible filtration is just the geometric irreducible filtration, then because  $H_{\text{ét},c}^0 = 0$ ,  $H_{\text{ét},c}^1$  is left exact.
- We can assume that  $\mathcal{F}_0$  can be extended to a lisse sheaf on  $\infty$ . This is because we can choose a closed point and move it to  $\infty$  by using Möbius transform, after a finite base change.



- We can assume  $\mathcal{F}_0$  is not geometrically constant: if  $\mathcal{F}_0 \cong \overline{\mathbb{Q}_\ell}$ , then let  $i : U_0 \rightarrow P_{k_0}^1$  and  $Z_0 = P_{k_0}^1 - U_0$ , then there is a short exact sequence

$$0 \rightarrow j_{0!}(\overline{\mathbb{Q}_\ell}) \rightarrow j_*(\overline{\mathbb{Q}_\ell}) \rightarrow \mathcal{Q} \rightarrow 0$$

where  $\mathcal{Q}$  is supported on  $S$ , so its higher compact cohomology vanish, and weights of  $H^0(\mathcal{Q}) = \prod_{s \in S} (j_{0*}(\mathcal{F}_0))_{\overline{s}}$  is no more than the maximal weight of  $\overline{\mathbb{Q}_\ell}$  on  $X_0$ , which is 0, by semicontinuity of weights for curves (9.6.3.11). And  $j_*(\overline{\mathbb{Q}_\ell})$  is also geometrically constant, thus its cohomology is  $\text{Pic}(\mathbb{P}^1)[n] = 0$  by (9.4.7.2), so  $H^1(P^1, j_{0!}(\overline{\mathbb{Q}_\ell}))$  has weights zero.

The actual proof will use the following lemma (9.6.6.8). After that, notice by (9.6.8.6)

$$(T_\psi(G_0))|_{\{0\}} = R\Gamma_c(\mathbb{A}^1, \mathcal{G})[1] = R\Gamma_c(U, \mathcal{F})[1] = H_{\text{ét},c}^1(U, \mathcal{F})$$

Then to understand the Frobenius eigenvalues of  $H_c^1(U, \mathcal{F})$ , it suffices to understand the weights of  $T_\psi(\mathcal{G}_0)$ , i.e.

$$\text{wt}(T_\psi(\mathcal{G}_0)) \leq w + 1$$

Then we use (9.6.3.17), notice the condition is satisfied by lemma (9.6.6.8), so  $w(T_\psi(\mathcal{G}_0)) = \|T_\psi(\mathcal{G}_0)\|$ , and also  $w(G_0) = \|G_0\|$  for the same reason as  $H_c^0(\mathbb{A}^1, \mathcal{G}) = H_{\text{ét},c}^0(U, \mathcal{F}) = 0$  by Poincaré duality. Now (9.6.8.9) gives the result.  $\square$

**Lemma (9.6.6.8) [key assertions of Weil Proof].** If  $\mathcal{G}_0 = j_{0!}(\mathcal{F}_0)$  where  $j_0 : U_0 \hookrightarrow A_{k_0}^1$ ,  $\psi : k_0 \rightarrow \overline{\mathbb{Q}_\ell}$  is a fixed non-trivial additive character, then

- $T_\psi(\mathcal{G}_0)$  is a sheaf placed at degree 0.
- $H_{\text{ét},c}^0(\mathbb{A}^1, T_\psi(\mathcal{G}_0)) = 0$ .
- $T_\psi(\mathcal{G}_0)$  is  $\iota$ -mixed.

┘

*Proof:* 1: By (9.6.8.6), we need to show  $H_{\text{ét},c}^i(\mathbb{A}^1, \mathcal{G}_0 \otimes \mathcal{L}(\psi_a)) = 0$  for  $i \neq 1$ , and this is equivalent to

$$H_{\text{ét},c}^i(\mathbb{A}^1, j_! \mathcal{F} \otimes \mathcal{L}(\psi_a)) = H_c^i(U, F \otimes \mathcal{L}(\psi_a)) = 0.$$

Notice by vanishing result (9.4.4.8), only need to show  $i = 0, i = 2$ ,  $i = 0$  case is done by Poincaré duality (9.4.6.48) and Artin vanishing (9.4.2.12) because it is smooth and  $\mathcal{F}$  is lisse.

$H_c^2(\mathcal{G}_0 \otimes \mathcal{L}(\psi_a)) = V_{\rho \otimes \chi_a}|_{\pi_1(U, \overline{x})}(-1)$  by (9.6.5.3), and  $\rho \otimes \chi_a$  irreducible as  $\rho$  does, so if  $V_{\rho \otimes \chi_a}|_{\pi_1(U, \overline{x})} \neq 0$ , then  $\rho \otimes \psi_a$  is trivial representation. Then  $\mathcal{G} \cong \mathcal{L}_{\psi_a}$  on  $\mathbb{P}_{\overline{k}}^1$  as an étale sheaf on  $\mathbb{A}^1 \cup \{\infty\} = \mathbb{P}^1$  by our reduction, so we have the character  $\psi_{-a}$  factors through  $\pi_1(\mathbb{P}^1, \overline{x})$ , i.e.

$$\begin{array}{ccc} \pi_1(\mathbb{A}^1, \overline{x}) & \longrightarrow & \pi_1(\mathbb{P}^1, \overline{x}) = 0 \\ \downarrow & & \downarrow \\ \pi_1(\mathbb{A}_0^1, \overline{x}) & \longrightarrow & \pi_1(\mathbb{P}_0^1, \overline{x}) \xrightarrow{\psi_{-a}} \overline{\mathbb{Q}_\ell} \end{array}.$$

But this is in contradiction with the fact the Artin-Schreier cover is geometrically irreducible ?.

2: Denote  $T_\psi(\mathcal{G}_0) = \mathcal{K}_0$ , then by (9.6.8.5) and Fourier inversion (9.6.8.8):

$$H_c^0(\mathbb{A}^1, \mathcal{K}) = \mathcal{H}^{-1}((T_{\psi^{-1}}(\mathcal{K}_0))_0) = \mathcal{H}^{-1}(T_{\psi^{-1}} \circ T_\psi(j_{0!}(\mathcal{F}_0))_0) = \mathcal{H}^{-1}(j_{0!}(\mathcal{F}_0)(-1))_0 = 0$$

because  $\mathcal{F}_0$  is placed at degree 0.

3: To show  $\iota$ -mixed, the only thing we can do is show it is embedded in a  $\iota$ -real sheaf: Consider the  $\iota$ -real sheaf

$$\mathcal{H}_0 = \mathrm{pr}_2^*(j_{0!}\mathcal{F}_0) \otimes \mathrm{mul}^*(\mathcal{L}(\psi)) \oplus \mathrm{pr}_2^*(j_{0!}\mathcal{F}_0^\vee) \otimes \mathrm{mul}^*(\mathcal{L}(\psi^{-1}))(-w)$$

Then

$$(R^i\pi_1^1(\mathcal{H}_0))_{\bar{x}} = H^i(\{\bar{x}\} \times \mathbb{A}^1, \mathcal{H}_0) = H^i(j_{0!}\mathcal{F}_0 \otimes \mathcal{L}(\psi_x)) \oplus H^i(j_{0!}\mathcal{F}_0^\vee \otimes \mathcal{L}(\psi_{x^{-1}}))(-w)$$

which we proved to vanish for  $i \neq 1$ . So using Poincaré duality on  $\{\bar{x}\} \times \mathbb{A}^1$ ,

$$\det\left(1 - tF_x^{d_x}|(R^1\pi_1^1(\mathcal{H}_0))_{\bar{x}}\right) = \det\left(1 - tF_X|H_c^1(\{\bar{x}\} \times \mathbb{A}^1, H_0|_{\{\bar{x}\} \times \mathbb{A}^1})\right) = \prod_{y \in \mathbb{F}_{q^n}} \det\left(1 - tF_y^{d_y}|(\mathcal{H}_0)_{\overline{(x,y)}}\right)^{-1}$$

which is real, so by (9.6.5.5), the direct summand  $T_\psi(\mathcal{G}_0)$  is  $\iota$ -mixed.  $\square$

**Remark (9.6.6.9).** If we use the machinery of perverse sheaf and show that Fourier transform preserves perversity, then item 1, 2 will be a direct consequence, Cf. [Bhatt notes, P39]. In fact, this is just the bigger picture, given in [Weil conjectures Perverse Sheaves and  $l$ -adic Fourier Transform Kiehl/Weissauer].  $\lrcorner$

## 7 Semisimplicity and Hard Lefschetz Theorem

**Prop. (9.6.7.1) [Semisimplicity Theorem].** If  $X_0/k_0$  is smooth and  $\mathcal{F}_0 \in \mathrm{WSh}_{\mathrm{lis}}(X_0; \overline{\mathbb{Q}_\ell})$  is  $\iota$ -pure, then  $\mathcal{F}_0$  is semi-simple (9.6.2.5), thus geometrically semisimple by (9.6.4.4).  $\lrcorner$

*Proof:* Let  $\mathcal{F}'$  be the sum of irreducible lisse subsheaves of  $\mathcal{F}$ , then it is the largest semisimple subsheaf of  $\mathcal{F}$ . It is stable under  $G(\overline{\mathbb{F}_q}/\mathbb{F}_q)$ , thus can be descended to a lisse subsheaf  $\mathcal{F}'_0$  of  $\mathcal{F}_0$ , and let  $\mathcal{F}'' = \mathcal{F}_0/\mathcal{F}'_0$ , we want to show the exact sequence

$$0 \rightarrow \mathcal{F}'_0 \rightarrow \mathcal{F}_0 \rightarrow \mathcal{F}'' \rightarrow 0$$

splits geometrically. Notice this exact sequence defines an element in  $\mathrm{Ext}_X^1(\mathcal{F}'', \mathcal{F}') = H^1(X, (\mathcal{F}'')^\vee \otimes \mathcal{F}')$ .  $\mathcal{F}_0$  is pure, hence so does  $(\mathcal{F}''_0)^\vee \otimes \mathcal{F}'_0$ , thus  $H^1(X, (\mathcal{F}'')^\vee \otimes \mathcal{F}')$  is  $\iota$ -mixed of weights  $\geq 1$ . But the exact sequence is compatible with Frobenius action, it defines a Frobenius fixed element, which then must vanish.  $\square$

**Cor. (9.6.7.2).** If  $f : X \rightarrow Y$  is proper between smooth algebraic schemes, then the sheaves  $R^i f_* \mathbb{Q}_\ell$  are semisimple.  $\lrcorner$

### Hard Lefschetz

**Situation (9.6.7.3).** Suppose  $F \in \mathrm{Field}^0$  and  $\mathcal{H}$  is a cohomology theory  $\mathrm{SmProj}/k \rightarrow \mathcal{C}\mathrm{Ring}^{\mathrm{gr}}/F$  which satisfies

- (Poincaré duality)  $\mathcal{H}^i(X) \otimes_F \mathcal{H}^{2n-i}(X) \rightarrow \mathcal{H}^{2n}(X)$  is a perfect pairing, and the Frobenius  $F_X^* = q^n$  on  $\mathcal{H}^{2n}(X)$ .
- (Weak Lefschetz) If  $\mathcal{L}$  is an ample line bundle on  $X$  and  $H \subset X$  is a smooth divisor in  $|\mathcal{L}|$ , then

$$f^* : \mathcal{H}^i(X) \rightarrow \mathcal{H}^i(Y)$$

is an isomorphism for  $i = n - 2$  and injective for  $i = n - 1$ .



- (Zeta Function) Let  $\mathcal{P}^i(X; T) = \det(1 - F_X^* T | \mathcal{H}^i(X))$ , then the Hasse-Weil zeta function

$$Z(X; T) = \prod_{i=0}^{2n} \mathcal{P}^i(X; T)^{(-1)^{i+1}}.$$

┘

**Prop. (9.6.7.4).** Crystalline and  $\ell$ -adic étale cohomologies satisfy the hypothesis (9.6.7.3). ┘

*Proof:* □

**Prop. (9.6.7.5).** Situation as in (9.6.7.3), then  $\mathcal{P}^i(X; T) = P_{\text{ét}}^i(X; T)$ . ┘

*Proof:* ? □

**Cor. (9.6.7.6).**

- The characteristic polynomial  $P_{\text{ét}}(X; T)$  is independent of  $\ell \in \mathbf{Prime}$  chosen.
- $\dim_F \mathcal{H}^i(X) = \dim_{\mathbb{Q}_\ell} H_{\text{ét}}^i(X)$ .
- (Hard Lefschetz) If  $\mathcal{L}$  is an ample line bundle on  $X$ , then

$$\mathcal{H}^{n-i}(X) \xrightarrow{c_1(\mathcal{L})^i} \mathcal{H}^{n+i}(X)$$

is an isomorphism. ┘

**Cor. (9.6.7.7).**

$$b_d \geq b_{d-2} \geq \dots, \quad b_{d-1} \geq b_{d-3} \geq \dots$$

┘

**Prop. (9.6.7.8)** [Hard-Lefschetz]. Cf. [Bhatt P42]. ┘

## 8 Deligne-Fourier Transformation

### Sheaf-Function Correspondence

**Def. (9.6.8.1)** [Sheaf-Function Correspondence]. For a complex  $K_0 \in D_{\text{const}}^b(X_0, \overline{\mathbb{Q}}_\ell)$ , we can associate a function

$$f^{K_0} : X_0(\mathbb{F}_{q^n}) \rightarrow \overline{\mathbb{Q}}_\ell : x_0 \mapsto \text{tr}((F_{x_0}^*)^{n/d_x} | (K_0)_{\overline{x}}) = \text{tr}((\varphi_{x_0}^{-1})^{n/\deg(x_0)} | (K_0)_{\overline{x}}) \quad (9.6.2.4).$$

┘

**Prop. (9.6.8.2)** [Sheaf-Function Dictionary]. We can use Grothendieck formula for a constructible sheaf (9.6.2.13) to relate the function  $f^{K_0}$  to the compact cohomologies of  $\mathcal{H}^i K_0$ , and we can translate many known theorems:

- $f^{f^* K_0} = f^{K_0} \circ f$ .
- 

$$f^{K_0} \cdot f^{T_0} = f^{K_0 \otimes^L T_0} ?$$

- (Base Change)(9.4.4.7) asserts that given a Cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

then it says in case  $Y'$  is a closed point of  $Y$ ,

$$f^{Rf!K_0}(y) = \sum_{x \in X_y(\mathbb{F}_{q^n})} f^{K_0}(x)$$

where  $y \in Y(\mathbb{F}_{q^n})$ , and more generally

$$\sum_{x' \in X'_{y'}} f^{K_0}(g'(x')) = \sum_{x \in X_{g(y')}} f^{K_0}(x)$$

- The projection formula(9.4.4.10) turns out to say something trivial:

$$\sum_{x \in X_y} (f^{K_0}(f(x)) \cdot f^{T_0}(x)) = f^{K_0}(y) \cdot \left( \sum_{x \in X_y} f^{T_0}(x) \right)$$

┘

**Prop. (9.6.8.3).**  $f^{\mathcal{L}_0(\psi)}(x) = \psi(-x)$ , by(9.4.6.33).

┘

### Fourier-Deligne Transforms on $\mathbb{A}^1$

Use notations as in [Artin-Schreier Theory](#).

**Def. (9.6.8.4) [Fourier-Deligne Transform].** Consider the multiplication map  $\text{mul.} : \mathbb{A}_0 \times \mathbb{A}'_0 \rightarrow \mathbb{A}_0$ , let the sheaf  $\mathcal{L}(\psi)$  be placed at  $A_0$ , and  $K_0 \in D_{\text{const}}^b(\mathbb{A}'_0, \overline{\mathbb{Q}}_\ell)$  be placed at  $\mathbb{A}'_0$ , then define the **Deligne-Fourier transform**

$$T_\psi : D_{\text{const}}^b(\mathbb{A}'_0, \overline{\mathbb{Q}}_\ell) \rightarrow D_{\text{const}}^b(\mathbb{A}_0, \overline{\mathbb{Q}}_\ell) : K_0 \mapsto R\pi_1^1(\text{pr}_2^* K_0 \otimes^L \text{mul.}^* \mathcal{L}_0(\psi))[1]$$

┘

**Lemma (9.6.8.5).**  $f^{T_\psi K_0}(x) = -\sum_{y \in \mathbb{F}_{q^n}} f^{K_0}(y) \psi^{-1}(xy)$  for any  $x \in \mathbb{F}_{q^n}$ .

┘

*Proof:* By(9.6.8.2),

$$\begin{aligned} f^{T_\psi K_0}(x) &= \sum_{y \in \mathbb{F}_{q^n}} f^{(\text{pr}_2^* K_0 \otimes^L \text{mul.}^* \mathcal{L}_0(\psi))[1]}((x, y)) \\ &= - \sum_{y \in \mathbb{F}_{q^n}} f^{\text{pr}_2^* K_0}((x, y)) \cdot f^{\text{mul.}^* \mathcal{L}_0(\psi)}((x, y)) \\ (9.6.8.3) &= - \sum_{y \in \mathbb{F}_{q^n}} f^{K_0}(y) \psi(-xy). \end{aligned}$$

□

**Prop. (9.6.8.6).** For  $a \in \mathbb{A}^1(\mathbb{F}_{q^n})$ ,

$$(T_\psi(K_0))_a = R\Gamma_c(K \otimes^L \mathcal{L}(\psi_a))[1]$$

where  $\psi_a : \mathbb{F}_{q^n} \rightarrow \overline{\mathbb{Q}}_\ell^\times$  maps  $x \mapsto \psi(ax)$ . In particular,  $\mathcal{H}^i((T_\psi(K_0))_0) = H_{\text{ét},c}^i(\mathbb{A}^1, K)$ , so we placed the complex into a family of deformations.  $\lrcorner$

*Proof:* By base change(9.4.4.7),

$$(T_\psi(K_0))_a = R\Gamma_c((\text{pr}_2^* K_0 \otimes^L m^* \mathcal{L}_0(\psi))|_{\{a\} \times \mathbb{A}^1})[1] = R\Gamma_c(K \otimes^L \mathcal{L}(\psi_a))[1].$$

□

**Lemma (9.6.8.7).** If  $\delta_0 = i_{0*} \overline{\mathbb{Q}}_\ell$  is the skyscraper sheaf where  $i_0 : \{0\} \hookrightarrow \mathbb{A}_0^1$ , then

$$T_\psi(\overline{\mathbb{Q}}_\ell[1]) = \delta_0(-1).$$

□

*Proof:* For the Artin-Schreier cover  $\wp : x \mapsto x^q - x$ , by(9.4.6.32),

$$\wp_* \overline{\mathbb{Q}}_\ell \cong \bigoplus_{x \in \mathbb{F}_q} \mathcal{L}(\psi_x).$$

As  $\wp$  is finite,  $\wp_*$  is exact(9.4.1.18), so using the Leray spectral sequence(9.4.4.7), we can calculate

$$H_{\text{ét},c}^1(\mathbb{A}^1, \mathcal{L}(\psi_x)) = 0 = H_{\text{ét},c}^1(\mathbb{A}^1, \overline{\mathbb{Q}}_\ell), \quad H_{\text{ét},c}^2(\mathbb{A}^1, \mathcal{L}(\psi_x)) = 0 \text{ (9.4.4.9)}, \quad H_c^2(\mathbb{A}^1, \mathcal{L}(\psi_x)) = \delta_0(x) \overline{\mathbb{Q}}_\ell(-1) \text{ (9.6.5.3)}.$$

So

$$\left( R\pi_1^1(m^* \mathcal{L}_0(\psi)[1])[1] \right)_x = R\Gamma_c(\mathcal{L}(\psi_x))[2] = \delta_0(-1).$$

□

**Prop. (9.6.8.8) [Fourier Inversion].** For  $K_0 \in D_c^b(\mathbb{A}_0, \overline{\mathbb{Q}}_\ell)$ ,  $T_{\psi^{-1}} T_\psi(K_0) = K_0(-1)$ .  $\lrcorner$

*Proof:* Consider

$$\begin{array}{ccc} \mathbb{A}_0^1 \times \mathbb{A}_0^1 \times \mathbb{A}_0^1 & \xrightarrow{\text{pr}_{23}} & \mathbb{A}_0^1 \times \mathbb{A}_0^1 \xrightarrow{\text{pr}_2} \mathbb{A}_0^1 \\ \downarrow \text{pr}_{12} & & \downarrow \text{pr}_1 \\ \mathbb{A}_0^1 \times \mathbb{A}_0^1 & \xrightarrow{\text{pr}_2} & \mathbb{A}_0^1 \\ \downarrow \text{pr}_1 & & \\ \mathbb{A}_0^1 & & \end{array}$$

And we will use the following Cartesian diagrams:

$$\begin{array}{ccc} \mathbb{A}_0^3 & \xrightarrow{\alpha: (x,y,z) \mapsto (y,z-x)} & \mathbb{A}_0^2 \\ \downarrow \pi^{13} & & \downarrow \text{pr}_2 \\ \mathbb{A}_0^2 & \xrightarrow{\beta: (x,z) \mapsto z-x} & \mathbb{A}_0^1 \end{array} \quad , \quad \begin{array}{ccc} \mathbb{A}_0^1 & \longrightarrow & * \\ \downarrow \Delta & & \downarrow i_0 \\ \mathbb{A}_0^2 & \xrightarrow{\beta} & \mathbb{A}_0^1 \end{array}$$

Then

$$T_{\psi^{-1}} T_\psi K_0$$

$$(T_{\psi^{-1}} T_\psi f)(x)$$

$$= R\pi_!^1(\mathrm{pr}_2^* R\pi_!^1(\mathrm{pr}_2^* K_0 \otimes m^* \mathcal{L}_0(\psi)) \otimes m^* \mathcal{L}_0(\psi^{-1}))[2] = \sum_y \left( \sum_z f(z) \psi(-yz) \right) \psi(xy)$$

By base change(9.4.4.7) :

$$= R\pi_!^1 \left( R\pi_!^{12} \mathrm{pr}_{23}^* (\mathrm{pr}_2^* K_0 \otimes m^* \mathcal{L}_0(\psi)) \otimes m^* \mathcal{L}_0(\psi^{-1}) \right) [2]$$

By projection formula(9.4.4.7) :

$$= R\pi_!^1 R\pi_!^{12} (\mathrm{pr}_{23}^* (\mathrm{pr}_2^* K_0 \otimes m^* \mathcal{L}_0(\psi)) \otimes \mathrm{pr}_{12}^* m^* \mathcal{L}_0(\psi^{-1}))[2] = \sum_y \sum_z f(z) \psi(-yz) \psi(xy)$$

Combine the character:

$$= R\pi_!^1 R\pi_!^{12} (\mathrm{pr}_{23}^* \mathrm{pr}_2^* K_0 \otimes \alpha^* m^* \mathcal{L}_0(\psi))[2] = \sum_y \sum_z f(z) \psi(-y(z-x))$$

Change order of summation:

$$= R\pi_!^1 R\pi_!^{13} (\pi^{13*} \mathrm{pr}_2^* K_0 \otimes \alpha^* m^* \mathcal{L}_0(\psi))[2] = \sum_z \sum_y f(z) \psi(-y(z-x))$$

By projection formula:

$$= R\pi_!^1 (\mathrm{pr}_2^* K_0 \otimes R\pi_!^{13} \alpha^* m^* \mathcal{L}_0(\psi))[2] = \sum_z f(z) \sum_y \psi(-y(z-x))$$

By base change:

$$= R\pi_!^1 \left( \mathrm{pr}_2^* K_0 \otimes \beta^* R\pi_!^2 (\mathrm{mul}^* \mathcal{L}_0(\psi)) \right) [2] = R\pi_!^1 \left( \mathrm{pr}_2^* K_0 \otimes \beta^* T_\psi \overline{\mathbb{Q}}_\ell[-1] \right) [2]$$

By(9.6.8.7) :

$$= R\pi_!^1 (\mathrm{pr}_2^* K_0 \otimes \beta^* \delta_0[-2])[2] = R\pi_!^1 (\mathrm{pr}_2^* K_0 \otimes \beta^* \delta_0(-1)) = \sum_z f(z) \#k_0^n \delta_0(z-x)$$

Use base change and noticing  $i_0$  is finite thus proper and exact:

$$= R\pi_!^1 (\mathrm{pr}_2^* K_0 \otimes R\Delta_! \overline{\mathbb{Q}}_\ell(-1)) = \sum_z \sum_{x=z} q^n$$

By projection formula:

$$\begin{aligned} &= R\pi_!^1 R\Delta_! (\Delta^* \mathrm{pr}_2^* K_0 \otimes \overline{\mathbb{Q}}_\ell)(-1) \\ &= K_0(-1) \end{aligned} \quad \begin{aligned} &= \#k_0^n \sum_{\{z|z=x\}} f(z) \\ &= \#k_0^n f(x) \end{aligned}$$

□

**Prop. (9.6.8.9) [Plancherel Formula].**

$$\left\| f^{T_\psi(K_0)} \right\|_n = \#k_0^{n/2} \left\| f^{K_0} \right\|_n \quad (9.6.3.13).$$

In particular,

$$\|T_\psi(K_0)\| \leq \|K_0\| + 1 \quad (9.6.3.16)$$

┘

*Proof:* By definition and using(9.6.8.5),

$$\left\| f^{T_\psi(K_0)} \right\|_n^2 = \sum_{x \in k_n} f^{T_\psi(K_0)}(x) \overline{f^{T_\psi(K_0)}(x)}$$

$$\begin{aligned}
&= \sum_{x,y,z} f^{K_0}(y) \overline{f^{K_0}(z)} \psi(-xy) \psi(xz) \\
&= \#k_0^n \sum_{z=y} f^{K_0}(y) \overline{f^{K_0}(z)} \\
&= \#k_0^n (f, f)_n
\end{aligned}$$

□

## 9 Integrality Problems

**Def. (9.6.9.1) [Integral Sheaves].** Let  $X_0 \in \text{Sch}^{\text{sep,ft}}/k_0$  and  $\mathcal{F}_0 \in \text{WSh}_{\text{lis}}(X_0; \overline{\mathbb{Q}_\ell})$ , then  $\mathcal{F}_0$  is called an **integral lisse sheaf** if for all  $x_0 \in |X_0|_0$ , all the eigenvalues of  $F_x^*$  acting on  $\mathcal{F}_{\overline{x}}$  are algebraic integers. ┘

**Prop. (9.6.9.2) [Deligne].** If  $\mathcal{F}_0$  is an integral lisse sheaf (9.6.9.1) on  $X_0$ , then all coefficients of  $H_{\text{ét},c}^n(X, \mathcal{F})$  are integral for any  $n \in \mathbb{N}$ . ┘

*Proof:* Cf. [Weil 1 Proof, P21]. □

**Cor. (9.6.9.3).** All Frobenius-eigenvalues of  $H_{\text{ét},c}^i(X, \mathcal{F})$  are algebraic integers. ┘

## 9.7 Intersection Homology and Perverse Sheaves

References are [Perverse Sheaves, Beilinson-Bernsterin-Deligne], [D-modules, perverse sheaves, and representation theory], [K-W01].

**Notation (9.7.0.1).**

- Let  $k \in \mathbf{Field}$  s.t.  $\#k < \infty$  or  $k = \bar{k}$ .
- Let  $\ell \in \mathbf{P} \setminus \{\text{char } k\}$ .

⌋

## 9.8 Profinite Cohomology

Reference are [B-S64, Lan96, Ser97a, Neu15], [Brown, Cohomology of Groups]Chap6 and the giant book [Neukirch Cohomology of Number Fields].

### 1 Cohomology of Finite Groups

**Notation(9.8.1.1).**

- Let  $G \in \mathcal{G}rp^{\text{fin}}$ .

┘

**Def.(9.8.1.2)[Group Cohomologies].** The **group cohomology**  $H^n(G, A)$  is the derived functor of the left exact functor

$$\text{Mod}_G \rightarrow \text{Ab} : H^0(G, A) = A^G = \text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, A),$$

so  $H^n(G, A) = \text{Ext}_{\mathbb{Z}[G]}^n(\mathbb{Z}, A)$ .

The **group homology**  $H_n(G, A)$  is the left derived functor of the right exact functor

$$\text{Mod}_G \rightarrow \text{Ab} : H_0(G, A) = A_G = \mathbb{Z} \otimes_{\mathbb{Z}[G]} A,$$

so  $H_n(G, A) = \text{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, A)$ .

┘

**Prop.(9.8.1.3).** For a normal subgroup  $H$  of  $G$ ,  $A \mapsto A^H$  is left exact from  $G\text{-mod}$  to  $G/H\text{-mod}$  and preserves injectives ?? because it is right adjoint to the exact inclusion functor as  $\text{Hom}_G(B, A) = \text{Hom}_{G/H}(B, A^H)$ . Dually for  $A_H$ .

┘

**Prop.(9.8.1.4).** For  $G = \mathbb{Z}$ , we have a free resolution  $0 \rightarrow \mathbb{Z}[t, t^{-1}] \xrightarrow{1-t} \mathbb{Z}[t, t^{-1}] \rightarrow \mathbb{Z} \rightarrow 0$ . In particular, thus  $H_n(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}$  iff  $n = 0, 1$  and vanish otherwise.

┘

**Prop.(9.8.1.5)[Tate-Cohomologies].** There is a standard resolution of the  $\mathbb{Z}[G]$ -module  $\mathbb{Z}$ :

$$\cdots \longleftarrow X_{-2} \longleftarrow X_{-1} \xleftarrow{\mu \circ \varepsilon} X_0 \longleftarrow X_1 \longleftarrow \cdots$$

that  $X_q = X_{-q-1}$  are  $\mathbb{Z}[G]$ -module generated by  $q$ -cells  $(\sigma_1, \dots, \sigma_q)$ ,  $X_0 = X_{-1} = \mathbb{Z}[G]$ .

It then can be verified that for  $G \in \text{Ab}^{\text{fin}}$ ,  $\text{Hom}$  from this resolution gives out the Tate cohomology

$$H_{\text{Tate}}^n(G, A) = \begin{cases} H^n(G, A) & n \geq 1 \\ A^G / \text{tr}_G(A) & n = 0 \\ \ker(\text{tr}_G|A) / I_G A & n = -1 \\ H_{-1-n}(G, A) & n \leq -2 \end{cases}$$

and  $H_T^n$  is a long exact sequence.

In particular, the  $\text{Hom}$  complex looks like:

$$\cdots \rightarrow A_{-2} \xrightarrow{\partial_{-1}} A_{-1} \xrightarrow{\partial_0} A_0 \xrightarrow{\partial_1} A_1 \xrightarrow{\partial_2} A_2 \rightarrow \cdots$$

where  $A_{-1} = A_0 = A$  and  $\partial_0 x = \text{tr}_G x$ ,  $(\partial_1 x)(\sigma) = \sigma x - x$ ,

$$\partial_2(x)(\sigma_1, \sigma_2) = \sigma_1 x(\sigma_2) - x(\sigma_1 \sigma_2) + x(\sigma_1).$$

┘

*Proof:* Cf.[Neukirch CFT P13]?

□

**Remark (9.8.1.6).** From now on, consider only Tate cohomology. ┘

**Prop. (9.8.1.7).**

$$H^{-2}(G, \mathbb{Z}) = G_{\text{ab}}, \quad H^{-1}(G, \mathbb{Z}) = 0, \quad H^0(G, \mathbb{Z}) = \mathbb{Z}/|G|\mathbb{Z}, \quad H^1(G, \mathbb{Z}) = 0, \quad H^2(G, \mathbb{Z}) = \chi(G).$$

┘

*Proof:*  $H^0$  is trivial and  $H^1(G, \mathbb{Z}) = H^0(G, \mathbb{Q}/\mathbb{Z}) = 0$ ,  $H^2(G, \mathbb{Z}) = H^1(G, \mathbb{Q}/\mathbb{Z}) = \text{Hom}(G, \mathbb{Q}/\mathbb{Z})$ .  
 $H^{-1}(G, \mathbb{Z}) = {}_{N_G}\mathbb{Z}/I_G A = 0$ .

For  $H^{-2}(G, \mathbb{Z})$ , use the dimension shifting  $0 \rightarrow I_G \rightarrow \mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow 0$ ,  $= H^{-1}(G, I_G) = I_G/I_G^2$ .  
 And  $G^{ab} \cong I_G/I_G^2$  by  $\sigma \mapsto \sigma - 1$ . □

**Prop. (9.8.1.8) [Group Cohomologies are Torsion].** For  $G \in \text{Ab}^{\text{fin}}$ ,  $\#G \cdot H^n(G, A) = 0$  for any  $A \in \text{Mod}_G$ . (True for  $H^0$  and use dimension shifting). In particular, a divisible  $G$ -module  $A$  has trivial cohomology. ┘

### Operations

**Prop. (9.8.1.9) [Dimension Shifting].** There are fundamental split exact sequence  $0 \rightarrow I_G \rightarrow \mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow 0$  and  $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}[G] \rightarrow J_G \rightarrow 0$ , thus  $A_G = A/I_G A$ . This can be used to tensor with  $A$  and define natural dimension shifting of cohomology  $\delta$ . ┘

**Lemma (9.8.1.10).** If  $H \leq G$ , by Grothendieck spectral sequence applied to  $\text{Mod}_G \xrightarrow{\text{res}} \text{Mod}_H \xrightarrow{(\cdot)^H} \text{Ab}$  shows that for  $A \in \text{Mod}_G$ ,  $H^*(H, A)$  is the same as the right derived functor of the functor  $(\cdot)^H$  on  $\text{Mod}_G$ . ┘

**Def. (9.8.1.11) [Restrictions, Corestrictions and Inflation].** Let  $H$  be a subgroup of  $G$ ,

- The **inflation** is the  $\delta$ -morphism  $H^*(G/H, A) \rightarrow H^*(G, A)$  on  $\text{Mod}_{G/H}$  induced by the natural transformation ? How to define it ?
- The **restriction** are the  $\delta$ -morphisms  $H^*(G, A) \rightarrow H^*(H, A)$  on  $\text{Mod}_G$  induced by the natural transformation  $A^G \rightarrow A^H$ .
- The **corestricton** are the  $\delta$ -morphisms  $H^*(H, A) \rightarrow H^*(G, A)$  on  $\text{Mod}_G$  induced by the natural transformation  $A^H \rightarrow A^G : a \mapsto N_{G/H}a$ .

┘

*Proof:* These exist by (4.10.3.5) and (9.8.1.10). □

**Cor. (9.8.1.12).** Let  $H$  be a subgroup of  $G$ , then  $\text{cor} \circ \text{res} = [G : H]\text{id}$ . ┘

**Prop. (9.8.1.13) [Kernel of Restriction].** If  $G \in \text{Grp}^{\text{fin}}$ ,  $H_1, H_2 \leq G$  are conjugate subgroups, and  $M \in \text{Mod}_G$ , then the kernel of the restriction maps  $H^1(G, M) \rightarrow H^1(H_i, M)$  are identical. ┘

*Proof:* If a 1-cycle  $\sigma \mapsto f(\sigma)$  is a cocycle, then it is a boundary when restricted to  $H_1$  iff  $f(\sigma) = \sigma(a) - a$  for some  $a \in M$  for any  $\sigma \in H_1$ . Thus

$$f(x^{-1}\sigma x) = f(x^{-1}) + x^{-1}f(\sigma x) = -x^{-1}f(x) + x^{-1}f(\sigma) + x^{-1}\sigma f(x) = x^{-1}(\sigma(f(x) + a) - (f(x) + a))$$

is also boundary when restricted to  $x^{-1}H_1x$ . □



**Thm. (9.8.1.14) [Serre-Hochschild Spectral Sequence].** If  $G \in \text{Grp}^{\text{fin}}$  and  $H \trianglelefteq G$ , by Grothendieck spectral sequence, the relation  $A^G = (A^H)^{G/H}$  gives us a spectral sequence  $E$  that

$$E_2^{p,q} = H^p(G/H, H^q(H, A)) \implies E^n = H^n(G, A).$$

The edge morphisms are:

- inflation maps  $H^k(G/H, A^H) \xrightarrow{\text{inf}} H^k(G, A)$ .
- restriction maps  $H^k(G, A) \xrightarrow{\text{res}} H^k(H, A)^{G/H}$ .

And the lower parts give us:

$$0 \rightarrow H^1(G/H, A^H) \xrightarrow{\text{inf}} H^1(G, A) \xrightarrow{\text{res}} H^1(H, A)^{G/H} \xrightarrow{\text{trans}} H^2(G/H, A^H) \xrightarrow{\text{inf}} H^2(G, A).$$

And dually for homology group.

Moreover if  $H^k(H, A) = 0$  for  $k = 1, \dots, n-1$ , then the rows are blank, thus the above lower part can change to dimension  $n$ .  $\lrcorner$

*Proof:* Prove the compatibility of inflation, restriction with the definition given in (9.8.1.11). ??  $\square$

**Cor. (9.8.1.15) [Hopf].** If  $G = F/R$ ,  $F$  is free, then use the homology spectral sequence,  $H_2(G, \mathbb{Z}) \cong \frac{R \cap [F, F]}{[F, R]}$ . Cf. [Weibel P198].  $\lrcorner$

**Prop. (9.8.1.16).** For an isomorphism  $(\sigma^*, \sigma)$  of a group and its cochain map in the sense that  $\sigma^*(g)(\sigma(a)) = g(a)$ , we have an isomorphism of Conjugation acts trivially on the group cohomology, because it does on  $H^0$  because  $H^0 = A^G$  fixed by  $G$ , and it commutes with dimension shifting. (Warning, if you count directly  $a(\sigma\tau\sigma^{-1}) - \sigma a(\tau)$ , you won't get 0, but a 1-coboundary).  $\lrcorner$

**Prop. (9.8.1.17) [Cup Products].** The cup product is defined by  $C^p(X, A) \times C^q(X, B) \rightarrow C^{p+q}(X, A \otimes B)$ :

$$(a \smile b)(\sigma_1, \dots, \sigma_{p+q}) = a(\sigma_1, \dots, \sigma_p) \otimes \sigma_1 \dots \sigma_p b(\sigma_{p+1}, \dots, \sigma_{p+q}).$$

It satisfies  $\partial(a \smile b) = \partial(a) \smile b + (-1)^p a \smile \partial(b)$ , thus defines a:

$$\smile: H^p(G, A) \times H^q(G, B) \rightarrow H^{p+q}(G, A \otimes B)$$

for  $p, q \geq 0$ . And in negative dimension this is also definable but not computable, Cf. [Neukirch Cohomology of Number Fields P42] or [Neukirch Class Field Theory 2015 P45].

- $a \smile b = a \otimes b$  for  $a \in H^0(G, A), b \in H^0(G, B)$ .
- $\delta(a \smile b) = \delta a \smile b, \delta(a \smile b) = (-1)^p (a \smile \delta b)$  for  $a \in H^p(G, A)$ .
- $\smile$  is associative and skew-symmetric (follows from dimension shifting and the last one.)

$\lrcorner$

*Proof:*  $\square$

**Prop. (9.8.1.18) [Duality and Cup Product].** Let  $0 \rightarrow A' \xrightarrow{i} A \xrightarrow{j} A'' \rightarrow 0$  and  $0 \rightarrow B' \xrightarrow{u} B \xrightarrow{v} B'' \rightarrow 0$  be exact and there is a pairing  $\varphi: A \times B \rightarrow C$  that  $\varphi(A' \times A') = 0$  hence induce a compatible pairing on  $A' \times B''$  and  $A'' \times B'$ , then we have

$$\delta(\alpha) \smile \beta + (-1)^p \alpha \smile \delta(\beta) = 0$$

for  $\alpha \in H^p(G, A'')$  and  $\beta \in H^q(G, B'')$ .  $\lrcorner$

*Proof:* Use the definition of  $\delta$ , let  $a, b$  be the preimage of  $\alpha, \beta$  in  $A$  and  $B$ , and  $ia' = \partial a, ub' = \partial b$ , then  $\delta(\alpha) \smile \beta + (-1)^p \alpha \smile \delta(\beta) = a' \smile vb + (-1)^p ja \smile b' = \partial a \smile b + (-1)^p a \smile \partial b = \partial(a \smile b)$  is a boundary.  $\square$

**Prop. (9.8.1.19) [Naturality of Cup Products].**

$$\text{res}(a \smile b) = \text{res}(a) \smile \text{res}(b), \quad \inf(a \cup b) = \inf(a) \cup \inf(b), \quad \text{cor}(\text{res } a \smile b) = a \smile \text{cor } b.$$

┘

*Proof:* Cf.[Neukirch CFT P48] or [Central Simple Algebras]P100. ?

$\square$

**Prop. (9.8.1.20).** Let  $\sigma \in G^{ab} = H^{-2}(G, \mathbb{Z})$  and  $a_1 \in H^1(G, A), a_2 \in H^2(G, A)$ , then

$$a_1 \smile \sigma = a_1(\sigma), \quad a_2(\sigma) = \sum_{\tau} a_2(\tau, \sigma).$$

Cf.[Neukirch CFT P50,P51].

┘

**Prop. (9.8.1.21) [Cyclic Group Cohomologies].** For cyclic group, the Tate cohomology is 2-cyclic.

In particular, if  $\sigma$  be a generator for  $\mathbb{Z}/(n)$ , then

$$H_{\text{Tate}}^p(\mathbb{Z}/(n), A) = \begin{cases} A^G / \text{tr}(A) & , p \in 2\mathbb{Z} \\ \text{tr } A / (\sigma - 1)A & , p \in 2\mathbb{Z} + 1 \end{cases}.$$

┘

*Proof:* There is an exact sequence  $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow 0$ , and this defines an isomorphism  $\delta^2 : H^0(G, \mathbb{Z}) \cong H^2(G, \mathbb{Z})$ . And this is also true for any  $A$  when tensored with it. The isomorphism is  $a \mapsto \delta^2 a = \delta^2(1) \smile a$ .  $\square$

**Prop. (9.8.1.22) [Duality].** The cup product induces an isomorphism  $H^i(G, A^\vee) \cong (H^{-i-1}(G, A))^\vee$ , i.e.,  $H^n(G, A^\vee)$  and  $H_n(G, A)$  are dual to each other when  $n > 0$ , where  $A^\vee = \text{Hom}(A, \mathbb{Q}/\mathbb{Z})$ .  $\square$

*Proof:* We only need to verify  $A^{*G}/N_G A^* \cong (N_G A/I_G A)^*$  and use dimension shifting. Should use the injectivity of  $\mathbb{Q}/\mathbb{Z}$  and the compatibility of cup product with dual.  $\square$

**Cor. (9.8.1.23).** When  $A$  is  $\mathbb{Z}$ -free, the cup product also induce an isomorphism  $H^i(G, \text{Hom}(A, \mathbb{Z})) \cong H^{-i}(G, A)^\vee$ .  $\square$

**Prop. (9.8.1.24) [Theorem of Cohomological Triviality].** For a  $G$ -module  $A$ , if there is a  $q$  s.t.  $H^q(g, A) = H^{q+1}(g, A) = 0$  for all subgroups of  $G$ , then  $H^p(g, A) = 0$  for any  $p$  and subgroup  $g$ . Cf.[Neukirch CFT P57].  $\square$

**Prop. (9.8.1.25) [Tate's Theorem].** Assume  $A$  is a  $G$ -module that  $H^1(G, A) = 0$  and  $H^2(g, A)$  is cyclic of order  $|g|$  for any subgroup  $g$  of  $G$ , then for a generator  $a$  of  $H^2(G, A)$ , there is an isomorphism

$$a \smile : H^q(G, \mathbb{Z}) \cong H^{q+2}(G, A).$$

Cf.[Neukirch CFT P79].

┘

**Cor. (9.8.1.26).** In particular, by dimension shifting, if  $A$  is a  $G$ -module that  $H^1(G, A) = 0$  and  $H^2(g, A)$  is cyclic of order  $|g|$  for any subgroup  $g$  of  $G$  this gives an isomorphism:

$$a \smile : H^q(G, \mathbb{Z}) \cong H^{q+2}(G, A).$$

for a generator  $a$  of  $H^2(G, A)$ , because cup product commutes with dimension shifting.  $\square$

### Miscellaneous

**Prop. (9.8.1.27) [ $H^2$  and Extensions, Schreier].** For a  $G$ -module  $A$ , there is a correspondence of equivalence classes of extension of  $G$  over  $A$  that are compatible with the  $G$  action and  $H^2(G, A)$ .  $\lrcorner$

*Proof:* Cf.[Weibel P183]. In fact there are also interpretations of  $H^3(G, A)$  as  $0 \rightarrow A \rightarrow N \rightarrow E \rightarrow G$  under some equivalences.  $\square$

**Prop. (9.8.1.28) [Herbrand Quotients for Cyclic Groups].** When  $G$  is a cyclic group and  $A$  is a  $G$ -module, let  $f = \sigma - 1$ ,  $g = 1 + \sigma + \dots + \sigma^{n-1}$ , then we can form a cyclic complex of order 2 and compute the Herbrand quotient(4.8.8.7). In this case,  $g_{f,g}$  is just  $|H^0(G, A)|/|H^{-1}(G, A)|$ . And by(4.8.8.9), if a  $G$ -morphism  $A \rightarrow B$  has finite kernel and cokernel, then they have the same Herbrand quotient.  $\lrcorner$

## 2 Cohomology of Profinite Groups

**Notation (9.8.2.1).**

- Let  $G \in \text{ProfGrp}$ .
- Let  $p \in \text{Prime}$ .
- Elements in  $\text{Mod}_G$  are assumed to be discrete.

$\lrcorner$

**Prop. (9.8.2.2) [Abelian Sheaves on  $T_G$ ].** The category of Abelian sheaves on the canonical topology(7.1.2.4)  $T_G$  of  $G$ -sets is equivalent to the the category  $\text{Mod}_G^{\text{alg}}$ , by  $Z \mapsto \text{Hom}(-, Z)$ . The inverse map is given by  $F \mapsto \varinjlim F(G/H)$ .  $\lrcorner$

*Proof:* The task is to prove  $F \cong h_{\varinjlim} F(G/H)$ . Cf.[Tamme P29].

The inverse of the Yoneda functor is the functor  $F \mapsto F(G)$  as a left  $G$ -set where  $gs = F(\cdot g)s$ . The task is to show that  $F \cong h_{F(G)}$ . For this, for any  $U$  we consider the covering  $\{G \xrightarrow{\varphi_u} U \text{ where } \varphi_U(g) = gu. \text{ Sheaf condition says}$

$$F(U) \rightarrow \prod_{u \in U} F(G) \rightrightarrows F(G \times_U G)$$

is exact, in other words,  $F(U) \cong \text{Hom}_G(U, F(G))$ .  $\square$

**Prop. (9.8.2.3) [Profinite Cohomologies].** The **profinite cohomology** is the derived functor of  $A \rightarrow A^G$  on the Abelian category  $\text{Mod}_G^{\text{alg}}$ (9.8.2.2)(It has enough injectives by(24.1.2.2)). It satisfies

$$H^*(G, A) \cong H^*(C^\bullet(G, A)) \cong \varinjlim H^*(G/U, A^U)$$

where  $C^\bullet(G, A)$  is the set of continuous cochain complex of morphisms from  $G$  to  $A$  and the colimit is taken over the transition maps defined by inflations. Moreover, for the same reason, when  $G = \varprojlim G_i$ , and  $A = \varinjlim A_i$ , then

$$H^*(G, A) \cong \varinjlim H^*(G_i, A_i).$$

$\lrcorner$

*Proof:* The second is an isomorphism because  $C^n(G, A) = \varinjlim C^n(G/U, A^U)$  and direct limit is exact.

For the first, the  $H^0$  obviously coincide, so it suffice to prove  $H^*(C(G, A))$  form a universal  $\delta$ -functor. It is effaceable because  $(-)^U$  preserves injective modules(9.8.1.3).

For the last one, we need to check  $C^n(G, A) = \varinjlim C^n(G_i, A_i)$ . Notice  $G$  has the profinite topology, thus must factor through some  $G_i$ , and the right through some  $A_i$  because the image of a morphism from  $G^n$  to  $A$  has finite image. Thus the result follows.  $\square$

**Prop.(9.8.2.4)[Cohomology Groups are Torsion].** For  $M \in \text{Mod}_G^{\text{alg}}, i \in \mathbb{N}$ ,  $H^i(G, M)$  are torsion Abelian groups. And if  $G$  is a pro- $p$ -group, then  $H^i(G, M)$  are  $p$ -primary torsion subgroups.  $\lrcorner$

*Proof:* Notice if  $M \in \text{Mod}_G^{\text{alg}}$ , this follows from(9.8.2.3) and(9.8.1.8). The last assertion is similar.  $\square$

**Def.(9.8.2.5)[Restrictions, Co-restrictions and Inflation].** Let  $H \leq G$  be a closed subgroup, for any  $A \in \text{Mod}_G^{\text{alg}}$ , taking colimits over all open subgroup  $U_\alpha$  of  $G$  of the restriction maps

$$\text{res} : H^i(G/U_\alpha, A^{U_\alpha}) \rightarrow H^i(H/(H \cap U_\alpha), A^{U_\alpha}) \rightarrow H^i(H/H \cap U_\alpha, A^{H \cap U_\alpha}),$$

by(3.1.12.6), we get a restriction map

$$\text{res} : H^i(G, A) \rightarrow H^i(H, A).$$

Similarly, if  $H$  is open in  $G$ , we can define a corestriction map

$$H^i(H, A) \rightarrow H^i(G, A).$$

And if  $H \triangleleft G$  is normal, we can define an inflation map

$$H^i(G/H, A^H) \rightarrow H^i(G, A).$$

as the colimit of the inflation maps

$$H^i(G/HU_\alpha, A^{HU_\alpha}) \rightarrow H^i(G/H, A^{U_\alpha}).$$

$\lrcorner$

**Prop.(9.8.2.6).** Situation as in(9.8.2.5), if  $H \leq G$  is open, then  $\text{cor} \circ \text{res} = [G : H]$  for a subgroup  $H$  is also true for profinite cohomology(9.8.1.12)  $\lrcorner$

*Proof:* This is because of(9.8.2.3).  $\square$

**Prop.(9.8.2.7).** If  $H$  is a closed subgroup of a profinite group  $G$ ,  $p \in \text{Prime}$  s.t.  $p \nmid [G : H]$ , then for any  $A \in \text{Mod}_G^{\text{alg}}, i \in \mathbb{N}$ ,  $\text{res} : H^i(G, A) \rightarrow H^i(H, A)$  is injective on the  $p$ -primary part of  $H^i(G, A)$ .  $\lrcorner$

*Proof:* This follows from(9.8.2.6) and(9.8.2.5) by taking filtered colimits.  $\square$

**Lemma(9.8.2.8)[Shapiro].**

$$H_*(G, \text{Ind}_H^G(A)) \cong H_*(H, A), \quad H^*(G, \text{ind}_H^G(A)) \cong H^*(H, A)$$

because (co)induction is adjoint to exact functors, so it preserves injectives(projectives) and it is exact because  $\mathbb{Z}[G]$  is free  $\mathbb{Z}[H]$ -module.

And in the finite case, this is also true for Tate cohomology using dimension shifting.  $\lrcorner$

**Prop. (9.8.2.9) [Serre-Hochschild Spectral sequence].** The same spectral sequence as in the finite case (9.8.1.14) also applies to profinite cohomology with  $H$  closed normal in  $G$ .  $\lrcorner$

**Def. (9.8.2.10) [Cup Products].** For any  $U \leq G$  open and  $A, B \in \text{Mod}_G^{\text{alg}}$ , there are natural cup product maps

$$H^*(G/U, A^U) \times H^*(G/U, B^U) \rightarrow H^*(G/U, A^U \otimes B^U) \rightarrow H^*(G/U, (A \otimes B)^U).$$

Then by the naturality of inflation and the fact inflation commutes with cup product (9.8.1.19), we get a natural cup product map

$$H^*(G, A) \times H^*(G, B) \rightarrow H^*(G, A \otimes B).$$

$\lrcorner$

**Prop. (9.8.2.11).** The cup products for profinite groups (9.8.2.10) is associative, graded-commutative, and commutes with inflations, restrictions and corestrictions as in (9.8.1.19).  $\lrcorner$

*Proof:*

$\square$

### Pro- $p$ -Groups

**Thm. (9.8.2.12) [Golod-Shafarevich Inequality].** If  $G$  is a non-trivial pro- $p$ -group, then

$$\dim H^2(G, \mathbb{Z}/(p)) > \frac{(\dim H^1(G, \mathbb{Z}/(p)))^2}{4}.$$

$\lrcorner$

*Proof:* Cf. [Ser97a]P66?.

$\square$

### Serre-Tate Duality

### Cohomological Dimensions

**Def. (9.8.2.13) [Cohomological Dimensions].** For  $p \in \text{Prime}$ , the  $p$ -cohomological dimension  $\text{cd}_p(G)$  is defined as the infimum of integers  $n \in \mathbb{Z}_+$  s.t.  $H^n(G, A)[p^\infty] = 0$  for any  $A \in \text{Mod}_G^{\text{tor}}$ . The **strict  $p$ -cohomological dimension**  $\text{scd}_p(G)$  is defined as the infimum of integers  $n \in \mathbb{Z}_+$  s.t.  $H^n(G, A)[p^\infty] = 0$  for any  $A \in \text{Mod}_G$ .

The **cohomological dimension**  $\text{cd}(G)$  is defined to be  $\text{cd}(G) \triangleq \sup_p(\text{cd}_p(G))$ . The **strict cohomological dimension**  $\text{scd}(G)$  is defined to be  $\text{scd}(G) \triangleq \sup_p(\text{scd}_p(G))$ .  $\lrcorner$

**Prop. (9.8.2.14) [Characterizing Cohomological Dimensions].** For  $p \in \text{Prime}$  and  $n \in \mathbb{N}$ , the following are equivalent:

- $\text{cd}_p(G) \leq n$ .
- $H^i(G, A) = 0$  for any  $i > n$  and any  $p$ -torsion  $G$ -module  $A$ .
- $H^{n+1}(G, A) = 0$  for any simple  $p$ -torsion  $G$ -module  $A$ .

And if  $G$  is pro- $p$ , then it suffice to check  $H^{n+1}(G, \mathbb{Z}/(p)) = 0$

$\lrcorner$

*Proof:*  $1 \iff 2$ : For any torsion  $G$ -module  $A$ , we have  $A = \bigoplus_{p \in \text{Prime}} A[p^\infty]$ , so  $H^i(G, A[p^\infty])$  is the  $p$ -primary part of  $H^i(G, A)$ .

$2 \rightarrow 3$  is trivial.

$3 \rightarrow 1$ : Use the fact cohomology commutes with colimits (9.8.2.3), reduce to the case of  $A$  finite, and then use the quotient tower ?.

The last assertion is by (3.1.12.15).  $\square$

**Cor. (9.8.2.15).** For  $A \in \text{Mod}_G$  s.t.  $p : A \rightarrow A$  is surjective, then  $H^n(G, A)[p^\infty] = 0$  for all  $n > \text{cd}_p(G)$ .

In particular, if  $A \in \text{Mod}_G$  is divisible, then  $H^n(G, A) = 0$  for all  $n > \text{cd}(G)$ .  $\lrcorner$

*Proof:* This follows from the exact sequence  $0 \rightarrow A[p] \rightarrow A \rightarrow A \rightarrow 0$ .  $\square$

**Prop. (9.8.2.16).**  $\text{cd}_p(G) \leq \text{scd}_p(G) \leq \text{cd}_p(G) + 1$ .  $\lrcorner$

*Proof:* There are exact sequences  $0 \rightarrow A[p] \rightarrow A \xrightarrow{p} pA \rightarrow 0$  and  $0 \rightarrow pA \rightarrow A \rightarrow A/(p) \rightarrow 0$ .  $A[p]$  and  $A/(p)$  are  $p$ -torsion  $G$ -modules, so if  $i > \text{cd}_p(G) + 1$ , then  $H^i(G, A[p]) = H^{i-1}(G, A/(p)) = 0$ . so  $H^i(G, A) \xrightarrow{p} H^i(G, pA)$  and  $H^i(G, pA) \rightarrow H^i(G, A)$  are injections, so their composition  $H^i(G, A) \xrightarrow{p} H^i(G, A)$  is injective, showing  $(H^i(G, A))_p = 0$ , so  $\text{scd}_p(G) \leq \text{cd}_p(G) + 1$ .  $\square$

**Cor. (9.8.2.17).** For any  $p \in \text{Prime}$ , we have  $\text{scd}_p(G) \geq 2$ .  $\lrcorner$

*Proof:* If  $\text{scd}_p(G) = 1$ , then  $\text{cd}_p(G) = 1$ , and ?  $\square$

**Prop. (9.8.2.18).** If  $H \leq G$  is closed, then  $\text{cd}_p(H) \leq \text{cd}_p(G)$  and  $\text{scd}_p(H) \leq \text{scd}_p(G)$ . And the equality holds if  $[G : H]$  is relatively prime to  $p$ .  $\lrcorner$

*Proof:* The first is because of Shapiro's lemma (9.8.2.8). For the equality, use (9.8.2.7).  $\square$

**Cor. (9.8.2.19).** For  $p \in \text{Prime}$ ,  $\text{cd}_p(G) = \text{cd}_p(G_p) = \text{cd}(G_p)$ ,  $\text{scd}_p(G) = \text{scd}_p(G_p) = \text{scd}(G_p)$ , where  $G_p$  is the Sylow- $p$ -subgroup of  $G$ .  $\lrcorner$

**Prop. (9.8.2.20) [Maximal Pro- $p$ -Quotients].** If  $G(p) = G/N$  is the maximal pro- $p$ -quotient of  $G$  and  $\text{cd}_p(N) \leq 1$ , then the canonical maps

$$H^n(G(p), \mathbb{Z}/(p)) \rightarrow H^n(G, \mathbb{Z}/(p))$$

are isomorphisms. In particular,  $\text{cd}(G(p)) \leq \text{cd}_p(G)$ , by (9.8.2.19)(9.8.2.14).  $\lrcorner$

*Proof:* Cf. [Ser97a]P74.  $\square$

**Cor. (9.8.2.21).** If  $\text{cd}_p(G) \leq 1$ , then  $\text{cd}_p(G(p)) \leq 1$ , where  $G(p) = G/N$  is the maximal pro- $p$ -quotient of  $G$ .  $\lrcorner$

*Proof:* It follows from (9.8.2.18) that  $\text{cd}_p(N) \leq 1$ .  $\square$

**Prop. (9.8.2.22).** If  $p \in \text{Prime}$  and  $H \trianglelefteq G$  is closed, then  $\text{cd}_p(G) \leq \text{cd}_p(H) + \text{cd}_p(G/H)$ , by Hochschild-Serre spectral sequence.  $\lrcorner$

**Example (9.8.2.23) [Cohomological Dimensions].**

- For any  $p \in \text{Prime}$ ,  $\text{cd}_p(\widehat{\mathbb{Z}}) = 1$ . But  $H^2(\widehat{\mathbb{Z}}, \mathbb{Z}) \cong H^1(\widehat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}) = \mathbb{Q}/\mathbb{Z}$ , so  $\text{scd}_p(\widehat{\mathbb{Z}}) = 2$ . ?
- If  $k \in \text{Field}^{\text{fin}}$ , then  $\text{cd}(k) = 1$ ,  $\text{scd}(k) = 2$ . ?

$\lrcorner$

### 3 Galois-Cohomologies

References are [Ser97a] and [Neukirch, Cohomology of Number Fields]Chap6.

**Prop. (9.8.3.1) [Hilbert's Additive Satz 90].** For  $K \in \mathbf{Field}$ , if  $L/K$  is a Galois-extension, then  $H^n(\mathrm{Gal}(L/K), L) = 0$  for any  $n \in \mathbb{Z}_+$ , where  $L$  is equipped with the discrete topology.  $\lrcorner$

*Proof:* From the normal basis theorem??, for finite Galois extension  $L/K$ ,  $L$  is an induced module over  $K$ , thus  $H^*(G, L) = H_*(G, L) = 0$  for  $* \neq 0$  and  $H_T^*(G, L) = 0$  by (9.8.2.8).

Hence the same is true, for arbitrary Galois-extension, when  $L$  is equipped with the discrete topology, the same as in the proof of (10.2.5.12).  $\square$

**Prop. (9.8.3.2) [Hilbert's Multiplicative Satz 90].**  $H^1(\mathrm{Gal}(L/K), L^\times) = 0$  for any  $L/K \in \mathbf{GalField}$ , where  $L$  is equipped with the discrete topology.  $\lrcorner$

*Proof:* This follows from (10.2.5.12).  $\square$

**Prop. (9.8.3.3) [Generalized Hilbert's Additive Satz 90].** If  $L/K \in \mathbf{GalField}$  and  $k \in \mathbb{Z}_+$ , then  $H^1(\mathrm{Gal}(L/K), W_{p,k}^+(L)) = 0$ .  $\lrcorner$

*Proof:* By linear independence of characters, if  $L/K$  is finite, then there is some  $x_0 \in L$  s.t.  $\mathrm{tr}_{L/K}(x_0) \neq 0$ . Then by (5.5.2.24),  $x = (x_0, 0, \dots, 0)$  is a unit in  $W_{p,k}^+(L)$ . Given any cocycle  $\mu : \sigma \mapsto \mu_\sigma$ , let

$$\theta = \mathrm{tr}_{L/K}(x)^{-1} \sum_{\sigma \in \mathrm{Gal}(L/K)} \mu_\sigma \sigma(x),$$

then for  $\tau \in \mathrm{Gal}(L/K)$ ,

$$\theta - \tau(\theta) = \mathrm{tr}_{L/K}(x)^{-1} \sum_{\sigma \in \mathrm{Gal}(L/K)} [\mu_{\tau\sigma}(\tau\sigma)(x) - \tau(\mu_\sigma)(\tau\sigma)(x)] = \mathrm{tr}_{L/K}(x)^{-1} \mu_\tau \sum_{\sigma \in \mathrm{Gal}(L/K)} (\tau\sigma)(x) = \mu_\tau.$$

Thus  $\mu$  is a coboundary.

Hence the same is true, for arbitrary Galois extension, when  $L$  is equipped with the discrete topology, the same as in the proof of (10.2.5.12).  $\square$

**Def. (9.8.3.4) [Galois Cohomologies].** For  $k \in \mathbf{Field}$ , for any  $M \in \mathbf{Mod}^{\mathrm{alg}}(\mathrm{Gal}_k)$ , denote  $H^i(k, M) = H^i(\mathrm{Gal}_k, M)$ .  $\lrcorner$

**Prop. (9.8.3.5).**  $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/(2)$ .  $\lrcorner$

*Proof:* By (9.8.1.21) and (9.8.1.5),  $H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times) \cong \{u \in \mathbb{C}^\times | \bar{u} = u\} / \{|u|^2, u \in \mathbb{C}^\times\} \cong \mathbb{Z}/(2)$ .  $\square$

**Prop. (9.8.3.6) [Kummer-Cohomologies].** For  $L/K \in \mathbf{GalField}$  and  $m \in \mathbb{Z} \cap k^\times$ , there exists an isomorphism

$$K^\times / (K^\times)^m \xrightarrow{\partial, \cong} H^1(K, \mathbb{Z}/(p^r)).$$

$\lrcorner$

*Proof:* This follows from the Kummer exact sequence  $1 \rightarrow \mu_m \rightarrow \mathbb{G}_m \xrightarrow{\times m} \mathbb{G}_m \rightarrow 1$  and Hilbert's theorem90 (9.8.3.2).  $\square$

**Prop. (9.8.3.7) [Restrictions and Co-restrictions on  $H^0$ ].** If  $K/k \in \mathbf{Field}$  is separable, then

$$\mathrm{res}_k^K = i : k \rightarrow K, \quad \mathrm{cor}_k^K = \mathrm{Nm}_{K/k} : K \rightarrow k$$

┘

*Proof:* This follows from the definition (9.8.1.11). □

**Cor. (9.8.3.8) [Change of Fields and Kummer Cohomology].** For  $K/k \in \mathbf{GalField}$  and  $m \in \mathbb{Z} \cap k^\times$ , there are commutative diagrams

$$\begin{array}{ccc} k^\times & \xrightarrow{\partial} & H^1(k, \mu_m) \\ \downarrow & & \downarrow \mathrm{res} \\ K^\times & \xrightarrow{\partial} & H^1(K, \mu_m) \end{array} \quad \begin{array}{ccc} K^\times & \xrightarrow{\partial} & H^1(K, \mu_m) \\ \downarrow \mathrm{Nm}_{K/k} & & \downarrow \mathrm{cor} \\ k^\times & \xrightarrow{\partial} & H^1(k, \mu_m) \end{array}$$

┘

*Proof:* Cf. [Central Simple Algebras] P131. ? □

**Prop. (9.8.3.9) [Artin-Schreier Cohomology].** If  $k \in \mathbf{Field}$ ,  $p = \mathrm{char} K \in \mathbf{Prime}$  and  $r \in \mathbb{Z}_+$ , then there exists an isomorphism

$$W_{p,r}(k)^+ / (\mathrm{Frob} - \mathrm{id}) W_{p,r}(k)^+ \cong H^1(k, \mu_{p^r}).$$

┘

*Proof:* This follows from the Artin-Schreier exact sequence (3.2.8.16) and Hilbert's theorem90 (9.8.3.3). □

**Prop. (9.8.3.10) [Unramified Classes].** For  $F \in \mathbf{GField}$ , a  $M \in \mathbf{Mod}_{\mathrm{Gal}_F}$  is called a **unramified class** at a place  $v \in \Sigma_F$  if it is trivial when restricted to  $H^k(I_{F_v}, M)$ .

Notice  $I_{F_v}$  is defined only up to conjugacy, but this notion is well-defined by (9.8.1.13). ┘

**Prop. (9.8.3.11) [Unramified Classes are Rare].** Let  $C$  be a proper Dedekind domain or a complete non-singular curve over a field  $k$  with fraction field  $F$  and  $G = \mathrm{Gal}_F$ , then for any finite  $G_F$ -module  $M$ , if  $S$  is a finite set of places in  $F$  and  $H^1(\mathrm{Gal}_F, M, S)$  the set of elements of  $H^1(F, M, S)$  that is unramified outside  $S$ , then  $\#H^1(G_F, M, S) < \infty$ . ┘

*Proof:* By the inflation restriction exact sequence (9.8.1.14), we can reduce  $G$  to a subgroup of finite index. And because  $M$  is a finite  $G$ -module, there is a finite index subgroup of  $G$  that acts trivially on  $M$ . Thus we can assume  $M$  is  $G$ -trivial. But then if  $m > 0$  s.t.  $mM = 0$ , then the assertion follows from the fact the maximal Abelian extension of exponent  $m$  unramified outside  $S$  is finite over  $K$  (19.7.4.5). □

### Cohomological Dimensions

**Def. (9.8.3.12) [Cohomological Dimensions].** For  $k \in \mathbf{Field}$ , define the **cohomological dimension** of  $F$  to be  $\mathrm{cd}(F) \triangleq \mathrm{cd}(\mathrm{Gal}_F)$ ,  $\mathrm{cd}_p(k) \triangleq \mathrm{cd}_p(\mathrm{Gal}_k)$  (9.8.2.13), and similarly the strictly cohomological dimensions to be

$$\mathrm{scd}_p(k) = \mathrm{scd}_p(\mathrm{Gal}_k), \quad \mathrm{scd}(k) = \mathrm{scd}(\mathrm{Gal}_k).$$

┘



**Prop. (9.8.3.13).** If  $p \in \mathbf{Prime}$  and  $k \in \mathbf{Field}^p$ ,

- $\mathrm{cd}_p(k) \leq 1$  and also  $\mathrm{cd}_p(\mathrm{Gal}_{K,p}) \leq 1$ , where  $\mathrm{Gal}_{K,p}$  is the maximal pro- $p$ -quotient of  $\mathrm{Gal}_k$ .
- If  $H^2(L, (k^{\mathrm{sep}})^\times) = 0$  for all  $L/k \subset k^{\mathrm{sep}}$ , then  $\mathrm{cd}(k) \leq 1$ . In particular  $H^i(k, (k^{\mathrm{sep}})^\times) = 0$  for  $i \geq 2$ .

┘

*Proof:* 1: Let  $G_p$  be the Sylow- $p$ -subgroup of  $\mathrm{Gal}_K$  and  $K_p = (K^{\mathrm{sep}})^{G_p}$ . There is an exact sequence

$$0 \rightarrow \mu_p(K) \rightarrow K^{\mathrm{sep}} \xrightarrow{x^p - x} K^{\mathrm{sep}} \rightarrow 0,$$

and combined with the fact that  $H^i(G_p, K_s) = H^i(\mathrm{Gal}(K^{\mathrm{sep}}/K_p), K^{\mathrm{sep}}) = 0$  for  $i \geq 1$  (9.8.3.1), so  $H^i(G_p, \mathbb{Z}/(p)) = 0$  for  $i \geq 2$ . Thus by (9.8.2.14) and (9.8.2.19),  $\mathrm{cd}_p(K) \leq 1$ .

2: similarly, for  $\ell \neq p$ ,  $H^2(G_\ell, \mu_\ell) = \varinjlim_L H^2(\mathrm{Gal}(K_s/L), \mu_\ell(k^{\mathrm{sep}})) = 0$ , so  $\mathrm{cd}_\ell(K) \leq 1$ . Then  $\mathrm{cd}(k) \leq 1$ , and  $\mathrm{scd}(k) \leq 2$ , so  $H^i(k, (k^{\mathrm{sep}})^\times) = 0$  for  $i \geq 1$ .  $\square$

**Cor. (9.8.3.14).** If  $K'/K \in \mathbf{Field}$  is purely-inseparable, then the canonical map  $\mathrm{Br}(K) \rightarrow \mathrm{Br}(K')$  is surjective.  $\square$

*Proof:* Cf. [Ser97a]P75?  $\square$

**Prop. (9.8.3.15) [Algebraic Extensions].** If  $K'/K$  is an algebraic extension and  $p \in \mathbf{Prime}$ , then  $\mathrm{cd}_p(K') \leq \mathrm{cd}_p(K)$ , which is an equality if one of the following holds:

- $p \nmid [K' : K]_{\mathrm{sep}}$ .
- $[K' : K]_{\mathrm{sep}} < \infty$ ,  $\mathrm{cd}_p(K) < \infty$ .

In fact, if  $[K' : K] < \infty$ , then  $\mathrm{cd}_p(K') = \mathrm{cd}(K)$  unless  $p = 2$ ,  $K$  is an ordered field, and  $\mathrm{cd}_2(K') < \infty$ .  $\square$

*Proof:* Cf. [Ser79a]P83?  $\square$

**Prop. (9.8.3.16) [Transcendental Extensions].** For  $K \subset_{\mathrm{fin}} L \in \mathbf{Field}$  and  $p \in \mathbf{Prime}$ , we have  $\mathrm{cd}_p(\mathrm{Gal}_L) \leq \mathrm{cd}_p(\mathrm{Gal}_K) + \mathrm{tr.deg}(L/K)$ .  $\square$

*Proof:* Cf. [Etale Cohomology Fulei P169].  $\square$

**Cor. (9.8.3.17).** If  $k = k^{\mathrm{sep}}$  and  $K$  be a function field over  $k$ , then  $\mathrm{cd}(K) \leq 1$ .

And if  $p = \mathrm{char} K \in \mathbf{Prime}$ , then  $H^2(K, (K^{\mathrm{sep}})^\times)$  is a  $p$ -torsion group.  $\square$

*Proof:* Th first one is clear, for the second, for any  $l \neq p$ , use the exact sequence  $\mu_l(K_s) \rightarrow K_s^* \xrightarrow{x \rightarrow x^l} K_s^* \rightarrow 0$ , then  $H^2(G(K_s/K), \mu_l(K_s)) = 0$ , and  $H^2(G(K_s/K), K_s^*) \xrightarrow{l} H^2(G(K_s/K), K_s^*)$  is injective.  $l$  is arbitrary, so  $H^2(G(K_s/K), K_s^*)$  must be a  $p$ -torsion group.  $\square$

### cd-1 Fields

**Def. (9.8.3.18) [cd-1 Fields].** For  $k \in \mathbf{Field}$ , the following are equivalent:

- $\mathrm{cd}(k) = 1$ , and when  $\mathrm{char} k > 0$ , we have moreover that  $\mathrm{Br}(K)[p^\infty] = 0$  for any algebraic extension  $K/k \in \mathbf{Field}$ .
- $\mathrm{Br}(K/k) = 0$  for all algebraic extension  $K/k \in \mathbf{Field}$ .
- If  $K/k \in \mathbf{Field}$  is an algebraic extension and  $L/K \in \mathbf{GalField}^{\mathrm{fin}}$ , then  $L^\times$  is  $\mathrm{Gal}(L/K)$ -cohomological trivial?  $\square$

- If  $K/k \in \mathbf{Field}$  is an algebraic extension and  $L/K \in \mathbf{GalField}^{\text{fin}}$ , then  $\text{Nm}_{L/K} : L^\times \rightarrow K^\times$  is surjective.

And moreover, It suffices to check any of these conditions for all finite separable extensions  $K/k \in \mathbf{Field}$ . If these hold, then  $k$  is said to be a **cd-1 field**.  $\lrcorner$

*Proof:* ? Cf. [Ser97a]P78.

By (9.8.3.14) it suffices to check item1 or item2 on separable extensions.  $\square$

**Prop. (9.8.3.19).** For  $K \in \mathbf{Field}$ ,

- If  $K$  is a cd-1 field (9.8.3.18), then every algebraic extension of  $K$  is a cd-1 field.
- If  $K = K^{\text{perf}}$ , then  $K$  is a cd-1 field iff  $\text{cd}(K) \leq 1$ .

$\lrcorner$

*Proof:* 1 is trivial.

2: If  $K = K^{\text{perf}}$ , then  $x \mapsto x^p$  is a bijection of  $\mathbb{G}_m$ , thus  $\text{Br}(K)[p^\infty] = 0$ . This can be applied to any algebraic extension of  $k$  by (3.2.6.20), so  $k$  is a cd-1 field.  $\square$

**Prop. (9.8.3.20) [ $C_1$  Fields are cd-1 Fields].** If  $K \in \mathbf{Field}$  is  $C_1$  (21.6.2.5),  $L/K \in \mathbf{GalField}^{\text{fin}}$ , then  $\text{Nm}_{L/K}(L^\times) = K^\times$ .

In particular,  $C_1$ -fields are cd-1 fields, by (21.6.2.6) and (9.8.3.18). And if  $p = \text{char } K \in \mathbf{Prime}$ , then  $[K : K^p] | p$ .  $\lrcorner$

*Proof:* For any  $a \in K^\times$ , consider the equation

$$\text{Nm}_{L/K}(x) = ax_0, \text{ where } x \in L, x_0 \in K.$$

Then it is homogenous of degree  $[L : K]$  with  $[L : K] + 1$  variables, so it has a non-trivial solution. Then  $x_0 \neq 0$  otherwise  $x = 0$ . So  $a = \text{Nm}_{L/K}(x/x_0)$  and the assertion follows.

For the last assertion, notice that if  $K'/K$  is purely-inseparable of degree  $p$ , then  $\text{Nm}_{K'/K}(K') = (K')^p = K$ , so  $[K : K^p] = [K' : K] = p$ .  $\square$

**Example (9.8.3.21) [cd-1 Fields].**

- Finite fields are cd-1 fields, by (21.6.2.7) and (9.8.3.20).
- Algebraic function fields over  $K = \overline{K} \in \mathbf{Field}$  are cd-1 fields, by (21.6.2.9) and (9.8.3.20).

$\lrcorner$

**Prop. (9.8.3.22) [Henselian DVR with Alg. Closed Residue Fields are cd-1 Fields].** If  $(\mathcal{O}_K, K) \in \mathbf{DVR}$  is Henselian and  $\kappa = \overline{\kappa}$ , then  $K$  is a cd-1 field (9.8.3.18).  $\lrcorner$

*Proof:* For any finite separable extension  $L/K \in \mathbf{Field}$  and  $E/L \in \mathbf{GalField}^{\text{fin}}$ , let  $T$  be the kernel of the map  $\text{Res}_{E/L} \mathbb{G}_m \rightarrow \mathbb{G}_m \in \mathbf{GrpSch}/L$ , then  $T$  is a torus, and  $H^1(E, L) = 0$  by (10.5.2.4). Then we see  $\text{Nm}_{E/L} E^\times = L^\times$ , and thus  $K$  is a cd-1 field.  $\square$

### Local Fields

**Prop. (9.8.3.23).** If  $K \in p\text{-LField}$ , then  $\text{cd}(K) = \text{scd}(K) = 2$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (9.8.3.24).**  $\text{cd}(\mathbb{R}) = \text{scd}(\mathbb{R}) = \infty$ .  $\lrcorner$

*Proof:* ? □

**Thm. (9.8.3.25).** If  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  is Henselian with  $\kappa = \bar{\kappa}$ , then  $K$  is a cd-1 field (9.8.3.18). In particular,  $\text{cd}(K) = 1$ . ┘

*Proof:* Cf. [K-P23]P79. ? □

**Thm. (9.8.3.26) [Tate].** For any  $p, \ell \in \text{Prime}$ ,  $\text{cd}_p(\text{Gal}_{\overline{\mathbb{Q}_\ell}}) = \text{scd}_p(\text{Gal}_{\overline{\mathbb{Q}_\ell}}) = 2$ . ┘

*Proof:* Cf. [Ser97a]Chap2.5.3.. □

### Global Fields

References are [Tat63].

**Notation (9.8.3.27).**

- Let  $F \in \text{GField}$ .

┘

**Def. (9.8.3.28) [Šafarevič-Tate Groups and Hasse-Principle].** For any  $A \in \text{Mod}_{\text{Gal}_F}$ , denote

$$\text{III}(A) = \ker \left( H^1(F, A) \xrightarrow{\text{res}} \prod_{v \in \Sigma_F} H^1(F_v, A) \right),$$

called the **Šafarevič-Tate group** of  $A$ . Notice the kernel is independent of the extension of  $v$  to  $F^{\text{sep}}$ , because any two such extensions induces conjugate groups  $\text{Gal}_{F_v}$  in  $\text{Gal}_F$  and use (9.8.1.13).

$A \mapsto \text{III}(A)$  induces a functor  $\text{III} : \text{Mod}_{\text{Gal}_F} \rightarrow \text{Ab}$ .

For any  $G \in \text{GrpSch}/F$ , denote  $\text{III}(G/F) \triangleq \text{III}(G(F^{\text{sep}}))$ , called the **Šafarevič-Tate group** of  $G$ . And if  $\text{III}(G/F) = 1$ , then we say **Hasse-principle** holds for  $G$ . ┘

## 4 Non-Abelian Cohomologies

**Def. (9.8.4.1) [Non-Abelian Cohomologies].** Let  $G, M \in \text{TopGrp}$  with a continuous action of  $G$  on  $M$  by group homomorphisms, then we define  $H^0(G, M) = M^G$ .

We define the set of **cocycles**  $Z^1(G, M)$  to be the set of continuous maps  $x : G \rightarrow M$  that

$$\sigma_1(x(\sigma_2))x(\sigma_1\sigma_2)^{-1}x(\sigma_1) = 1, \quad \text{i.e.} \quad x(gh) = x(g)g(x(h))$$

If  $x \in Z^1(G, M)$  and  $m \in M$ , then  $x_m : \sigma \rightarrow m^{-1}x(\sigma)\sigma(m) \in Z^1(G, M)$  too. This defines an equivalence relation on  $Z^1(G, M)$ , the equivalence classes are called  $H^1(G, M)$ . This is compatible with the commutative case. ┘

**Prop. (9.8.4.2).** Restriction map and inflation map is definable for  $H^0$  and  $H^1$ , and  $H^1(H, M)$  is a  $G/H$ -set where  $G$  acts on  $H^1(H, M)$  by  $g(c)(h) = g(c(g^{-1}hg))$ . ┘

**Prop. (9.8.4.3).** There is an exact sequence of pointed sets:

$$0 \rightarrow H^1(G/H, M^H) \xrightarrow{\text{inf}} H^1(G, M) \xrightarrow{\text{res}} H^1(H, M)^{G/H} \text{ (9.8.4.2)}.$$

┘

*Proof:* First  $\text{res}(H^1(G, M)) \subset H^1(H, M)^{G/H}$  because  $g(c)(h) = c(g)^{-1}c(h)h(c(g))$  is checked so  $g(c)$  is cohomologous to  $c$ .

$\text{res} \circ \text{inf} = 0$  is easy, if  $\text{res}(c) = 0$ , then  $c$  is trivial on  $H$ , hence  $c(gh) = c(g)$  and  $h(c(g)) = c(hg) = c(g \cdot g^{-1}hg) = c(g)$ , so  $c$  is inflated from  $H^1(G/H, M^H)$ .

For the injectivity of  $\text{inf}$ . If  $c(\bar{g}) = g^{-1}g(a)$ , then  $a \in M^H$ , so it is a coboundary in  $H^1(G/H, M^H)$ .

□

**Prop. (9.8.4.4).** Let  $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$  be an exact sequence of  $\mathcal{T}\text{opGrp}_G$ , then there is a long exact sequence of pointed sets

$$1 \rightarrow A^G \rightarrow B^G \rightarrow C^G \xrightarrow{\delta} H^1(G, A) \rightarrow H^1(G, B) \rightarrow H^1(G, C) \xrightarrow{\Delta} H^2(G, A)$$

the last term is defined only when  $A \leq Z(B)$ .

Where  $\delta$  is defined as follows: for  $c \in C^G$ , let  $b$  be an inverse image of  $c$  in  $B$ , then  $a_\sigma = b^{-1}\sigma(b) \in A$ , and it defines a cocycle in  $H^1(G, A)$ , different choice differ by a coboundary, so it is well-defined.

$\Delta$  is defined as follows: for  $c_\sigma$  a cocycle in  $H^1(G, C)$ , choose  $b_s$  inverse images of  $c_s$ , then  $a_{\sigma, \tau} = b_\sigma \sigma(b_\tau) b_{\sigma\tau}^{-1}$  is a cocycle in  $H^2(G, A)$ . ┘

*Proof:* Similar to (7.3.2.19), but need to show continuity? □

**Prop. (9.8.4.5) [Shapiro's Lemma].** If  $K/k \in \mathbf{Field}^{\text{fin}}$  and  $G \in \mathcal{A}\text{lgGrp}^{\text{sm}}/K$ , then

$$H^1(k, \text{Res}_{K/k} G) \cong H^1(K, G).$$

┘

*Proof:* Cf. [Kneser1969, Lectures on Galois cohomology of classical groups]1.3.? □

### Interpretation of $H^1$ and Torsors

**Def. (9.8.4.6) [A-Torsors].** Let  $A$  be a discrete  $G$ -group, an **A-torsor** is a discrete  $G$ -set with a right  $A$ -action that is simply transitive and semi-linear in  $G$ . ┘

**Prop. (9.8.4.7) [ $H^1$  and Torsors].** We have a canonical bijection of pointed sets:  $H^1(G, A) \cong \text{Tors}_G(A)$ . ┘

*Proof:* If  $X$  is an  $A$ -torsor, choose  $x \in X$ , then  $\sigma(x) = xa_\sigma$  for  $a_\sigma \in A$ . Now that  $\sigma \mapsto a(\sigma)$  is checked to be a cocycle, and change of  $x$  changes to  $\sigma \mapsto b^{-1}a_\sigma\sigma(b)$ .

Conversely, for an  $a \in H^1(G, A)$ , we let  $X = A$  be a right  $A$ -module, and let  $\sigma'(x) = a_\sigma\sigma(x)$ , i.e. regarding coming from  $x = 1$ , then this is an inverse map. □

**Prop. (9.8.4.8) [Extension of Rings].** ┘

**Prop. (9.8.4.9).** There is an isomorphism of pointed sets  $H^1(G, O(\varphi_L)) \cong E_\varphi(L/K)$ . ┘

*Proof:* Cf. [Neukirch Cohomology of Number Fields P346]. □

## 5 Continuous Cohomologies

In this subsubsection cohomology of  $G$ -modules with topology is studied. References are [Cohomology of Number Fields, Neukirch]Chap 2.7.

**Prop. (9.8.5.1).**  $H_{\text{cont}}^*(G, -)$  forms a long exact sequence for any  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  of continuous  $G$ -modules.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (9.8.5.2).** If  $A$  is a compact  $G$ -module which is an inverse limit of finite discrete  $G$ -modules  $A_n$  and  $H^i(G, A_n)$  is finite for all  $n$ , then

$$H_{\text{cont}}^{i+1}(G, A) = \varprojlim_n H^{i+1}(G, A_n).$$

$\lrcorner$

*Proof:* Cf.[Cohomology of Number Fields Neukirch P142].  $\square$

**Lemma (9.8.5.3).** Let  $\pi$  be a topologically nilpotent element of  $A$  which is complete in the  $\varpi$ -adic topology and  $\varpi$  is not a zero-divisor, let  $R = A/(\varpi)$  equipped with discrete topology. Let  $G$  be a group which acts continuously on  $A$  and fix  $\pi$ , then if  $H^1(G, R)$  is trivial, then  $H^1(G, A)$  is trivial, and if moreover  $H^1(G, GL(n; R))$  is trivial, then  $H^1(G, GL(n; A))$  is trivial.  $\lrcorner$

*Proof:* Cf.[Galois Representations Berger P15].  $\square$

**Prop. (9.8.5.4) [Cyclic Case].** if  $G$  is a topological cyclic group  $\overline{\langle g \rangle}$ , then the map  $H^1(G, M) \rightarrow M/(1 - g)$  is well-defined and injective. And when  $M$  is profinite,  $p$ -adically complete, then the map is also surjective.  $\lrcorner$

*Proof:* The surjection: there is only one choice:  $c(g^i) = (1 + g + \dots + g^{i-1})(m)$ . And we need to verify that it is continuous. The case of  $p$ -adic can be deduced from profinite case, because  $c(\gamma) \in p^{-k}M$  for some  $k$ , and  $p^{-k}M$  is then profinite. For any finite quotient  $N$  of  $M$ , there is a  $k$  that  $kM = 0$ , and a  $n$  that  $g^n = \text{id}$  on  $N$ , so  $c(g^{rkn}) = 0$  on  $N$ , which shows  $c$  is continuous.  $\square$

**Prop. (9.8.5.5) [Inf-Res Exact Sequence].**  $\lrcorner$

**Prop. (9.8.5.6) [Cohomology as Extensions].** Let  $G$  be a topological group acting on a field  $K$  and  $W \in \text{Rep}_K(G)$ , there exists a bijection continuous extensions

$$0 \rightarrow W \rightarrow W' \rightarrow K \rightarrow 0$$

and  $H_{\text{cont}}^1(G, W)$  that is  $K$ -linear.  $\lrcorner$

*Proof:* For any such extension, choose a  $K$ -linear splitting  $W' \cong W \oplus K$ , then

$$g(w, c) = (g(w) + g(c)\tau(g), g(c)),$$

where  $\tau : G \rightarrow W$  is a continuous 1-coboundary in  $H^1(G, W)$ , and clearly changing a splitting changes this coboundary by a 1-cocycle, and it is clearly a bijection.  $\square$



# 10 | Algebraic Geometry III: Group Theory

## 10.1 Group Schemes I: Structure Theory

*“Whenever you have to do with a structure-endowed entity  $S$  try to determine its group of automorphisms, the group of those element-wise transformations which leave all structural relations undisturbed. You can expect to gain a deep insight into the constitution of  $S$  in this way.”*

–H. Weyl

References are [Sta]Chap38 and [Mil17].

**Notation(10.1.0.1).**

- Let  $k \in \mathbf{Field}$ .
- Let  $R \in \mathbf{CRing}$ .
- Let  $S \in \mathbf{Sch}$ .

┘

### 1 Group Schemes

**Def.(10.1.1.1) [Group Scheme].** A **monoid scheme** over  $S$  is a monoid object in the Cartesian monoidal category  $\mathbf{Sch}/S$ . The category  $\mathbf{Grp}_S$  of **group schemes** over  $S$  consists of group objects in the Cartesian category  $\mathbf{Sch}/S$ (4.1.1.67). Let  $\mathbf{AlgGrp}/S$  denote the category of groups schemes over  $S$  that is also algebraic over  $S$ .

An **open/closed subgroup scheme** of a group scheme  $G/S$  is an open/closed subscheme of  $G/S$  that represents a subgroup functor of  $G/S$ .

A **smooth/flat/separated/... group scheme** is a group scheme  $G/S$  that  $G$  is smooth/flat/separated/... over  $S$ .

We have the left(right)translation for an elements in  $G(R)$ , equivalently, a natural transformation on  $G$ , and base change  $(G \otimes_S S')(T'/S') = G(T'/S)$ . ┘

**Remark(10.1.1.2)[Yoneda Interpretation].** We do not need to verify all the relations, whenever we have a functorial commutative group structure on all the set  $\mathbf{Hom}(T, G)$ , we immediately recover the map  $m : G \times G \rightarrow G$  as  $\mathrm{pr}_1 \cdot \mathrm{pr}_2 \subset G(G \times G)$ ,  $\mathrm{inv} : G \rightarrow G$  as  $\mathrm{inv}$  in  $G(G)$ ,  $u : S \rightarrow G$  as  $e$  in  $G(S)$ , by Yoneda lemma. ┘

**Cor.(10.1.1.3)[Group Schemes as Functors].** From the Yoneda-interpretation and strong Yoneda-lemma(10.8.1.1), a group scheme over  $S$  is equivalent to a contravariant functor

$$\mathbf{Aff}_S \rightarrow \mathbf{Grp} : R \mapsto \mathbf{Hom}(\mathbf{Spec} R, G)$$

(or equivalently a functor  $\text{Sch}_S \rightarrow \text{Grp}$ ) that is represented by a scheme over  $S$ .

In particular, the category of affine group scheme over  $\text{Spec } R$  is equivalent to the category of commutative Hopf algebras over  $R$ (3.11.4.1).  $\lrcorner$

**Def. (10.1.1.4) [Categorical Group Definitions].** Because we can regard group schemes as functors by (10.1.1.3), we can define categorical constructions of group schemes:

- (Trivial Group Scheme)  $S$  is a trivial group scheme over  $S$ . It is the zero object in  $\text{Grp}_S$ , denoted by  $e$ .
- (Commutative Group Schemes) A group scheme  $G$  is called a **commutative group scheme** if each  $G(R)$  is commutative.
- ( $[n]_G$ ) For any  $n \geq 1$ , the natural transformation of commutative groups functors  $x \mapsto x^n$  induces a morphism of commutative group schemes  $[n] : G \mapsto G$  for any group scheme  $G$ .
- (Semi-Direct Product) Let  $G$  be a group scheme acting on a group scheme  $H$ , then we can form a **semi-direct product group scheme**  $G \ltimes H$  representing the functor  $T \mapsto G(T) \ltimes H(T)$ . and  $G \ltimes H$  is isomorphic to  $G \times H$  as schemes.
- (Kernel) For a homomorphisms of group schemes  $\varphi : G \rightarrow H$  over  $R$ , we define the **kernel group scheme**  $\ker \varphi$  representing the functor  $R \mapsto \ker(\varphi(R))$ . it is represented by the fibered product

$$\begin{array}{ccc} \ker \varphi & \longrightarrow & \text{Spec } R \\ \downarrow & & \downarrow \varepsilon_H \\ G & \xrightarrow{\varphi} & H \end{array}$$

then it is a group scheme over  $R$ .

When  $G, H$  are affine group schemes, it corresponds to the cokernel Hopf algebra defined in (3.11.4.15).

- (Short Exact Sequences) A sequence of homomorphisms of algebraic groups  $e \rightarrow N \xrightarrow{i} G \xrightarrow{q} Q$  is called an **exact** if they are exact as sheaves in  $\text{Sh}(\text{Aff}_S)$  (or  $\text{Sh}(\text{Sch}_S)$ ). Exact sequences are stable under base change of fields
- (Quotient Scheme) Let  $H \subset G$  be subgroup schemes. If  $\tilde{G}/\tilde{H}$  is representable, then it is called the **quotient scheme**, denoted by  $G/H$ .
- (Monomorphism) A homomorphism  $H \rightarrow G$  of group schemes is called a **monomorphism** if  $\tilde{H} \rightarrow \tilde{G}$  is injective.
- (Quotient Map) A homomorphism  $H \rightarrow G$  of group schemes is called a **quotient map** if  $\tilde{G} \rightarrow \tilde{H}$  is surjective in  $\text{Sh}(\text{Aff}_{\text{fppf}}/k)$ .
- (Normal/Characteristic/Central Subgroups) A subgroup scheme  $H$  of  $G$  is called normal/characteristic/central if  $H(R)$  is normal/characteristic/central in  $G(R)$  for any  $k$ -algebra  $R$ .
- (Normalizing) Let  $H, N$  be subgroup schemes of  $G$ , then we say  $H$  normalizes  $N$  if  $H$  is contained in the normalizer of  $N$  in  $G$ .
- (Product Subgroup) Let  $H, N$  be algebraic subgroups of an algebraic group  $G$  such that  $H$  normalizes  $N$ , then we define  $NH \subset G$  be the algebraic subgroup that is the image of the homomorphism  $N \rtimes H \rightarrow G$ , if it is representable.
- (Generated Subgroup Scheme) Let  $\{\varphi_i : X_i \rightarrow G\}$  be a family of maps to a group scheme  $G$ , then the smallest subgroup sheaf functor  $H$  of  $G$  generated by images of  $\varphi_i$  is called the



**generated subgroup scheme** if it is representable, denoted by  $\langle X_i, \varphi_i \rangle$ . It is clear such generated subgroup scheme commutes with base change of fields.

- (Commutator Subgroup) If  $H_1, H_2$  are subgroups of  $G$ , let  $[H_1, H_2]$  be the subgroup generated by the commutator map of  $H_1, H_2$  if it is representable. Or equivalently, it is the subgroup generated by the map  $H_1 \times H_2 \rightarrow G : [h_1, h_2] \mapsto h_1 h_2 h_1^{-1} h_2^{-1}$ .
- (Derived Series) For a group scheme  $G$ , define  $G^{(1)} = [G, G]$  and define inductively  $G^{(n+1)} = [G^{(n)}, G^{(n)}]$  if they are representable.
- (Derived Central Series) For a group scheme  $G$ , define  $G^1 = [G, G]$  and define inductively  $G^{n+1} = [G^n, G^n]$  if they are representable.
- (Subnormal Series) For a group scheme  $G$ , a **subnormal series** is a finite sequence of algebraic subgroups  $G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_r = e$  s.t.  $G_{i+1}$  is normal in  $G_i$ .
- (Solvable Group Schemes) A **solvable group scheme** is a group scheme  $G$  that has a subnormal series  $G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_s = \{e\}$  that each quotient  $G_i/G_{i+1}$  is commutative.
- (Nilpotent Group Schemes) A **nilpotent group scheme** is a group scheme  $G$  that has a subnormal series  $G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_s = \{e\}$  that each quotient  $G_i/G_{i+1}$  is central in  $G/G_{i+1}$ . Such a sequence is called a **central normal series**.
- (Split Solvable(resp. Nilpotent) Group Schemes) A **split solvable**(resp. **split nilpotent group**) is a group scheme  $G$  that has a subnormal(resp. central) series  $G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_s = \{e\}$  that each quotient  $G_i/G_{i+1}$  is isomorphic to either  $\mathbb{G}_a$  or  $\mathbb{G}_m$ .
- (Perfect Group Schemes) A group scheme  $G$  is called **perfect group** if  $G = [G, G]$ .

┘

**Lemma (10.1.1.5) [Butterfly Lemma].** Let  $H_1, H_2$  be algebraic subgroups of an algebraic group  $G$ ,  $N_1, N_2$  are normal subgroups of  $H_1$  and  $H_2$ , then there is a canonical isomorphism of algebraic groups:

$$N_1(H_1 \cap H_2)/N_1(H_1 \cap N_2) \cong N_2(H_2 \cap H_1)/N_2(H_2 \cap N_1).$$

┘

*Proof:* Use the Butterfly lemma for groups (3.1.8.2) and shiffy. □

**Def. (10.1.1.6) [Translation Map].** For  $G \in \mathcal{Gp}/S$  and any  $a \in G(S)$ , there is a (left) translation map

$$l_a : G \cong S \times_S G \xrightarrow{(a, \text{id})} G \times_S G \xrightarrow{m} G.$$

and it satisfies  $l_a, l_b$  by associativity. ┘

**Prop. (10.1.1.7) [Common Group Schemes].**

- $\mathbf{D}(\Gamma) = \text{Spec } \mathbb{Z}[\Gamma]$  for a group  $\Gamma$ , representing  $\underline{\text{Hom}}(\Gamma, \mathbb{G}_m)$  (3.11.4.3).
- $\mu_n = \mathbb{Z}[T]/(T^n - 1)$  (3.11.4.5)
- $\mathbb{G}_a = \mathbb{Z}[T]$  (3.11.4.6)
- $\underline{\Gamma} = \text{Spec } \prod_{\gamma \in \Gamma} \mathbb{Z}$  (3.11.4.9).
- $\alpha_{p^r} = \text{Spec } \mathbb{F}_p[X]/X^{p^r}$  (3.11.4.13).
- $V_a = \text{Spec } \text{Sym}(V^\vee)$  (3.11.4.8).

⌋

**Prop. (10.1.1.8).**

- $\text{Hom}(\mathbb{G}_{m,R}, \mathbb{G}_{a,R}) = 0$ .
- $\text{Hom}(\mathbb{G}_{a,R}, \mathbb{G}_{m,R}) = 0$ .
- If  $R$  is reduced, then  $\text{Aut}(\mathbb{G}_{a,R}) \cong R$ .
- If  $k \in \mathbf{Field}^p$ , then  $\text{End}_k(\alpha_p) = \mathbb{G}_a$ .

⌋

*Proof:* 1: Such a homomorphism corresponds to an element  $f(T) = \sum a_i T^i \in R[T]$  s.t.  $\sum a_i T^i \otimes T^i = \sum a_i (T^i \otimes 1 + 1 \otimes T^i)$ , which implies  $a_i = 0$  for all  $i$ . A homomorphism  $\mathbb{G}_{m,R} \rightarrow \mathbb{G}_{a,R}$  corresponds to an element  $f(T) = \sum a_i T^i$  s.t.  $\sum a_i T^i \otimes T^i = \sum a_i (T^i \otimes 1 + 1 \otimes T^i)$ , which implies  $a_i = 0$  for all  $i$ .

2: The proof is the same as that of (10.1.1.8).

3: Any endomorphism of  $\mathbb{G}_{a,R}$  is of the form  $X \mapsto a_0 + a_1 X + \dots + a_n T^n$ . If it is an automorphism, then for any prime ideal  $\mathfrak{p} \in \text{Spec } R$ ,  $a_0 \notin \mathfrak{p}$  and  $a_i \in \mathfrak{p}, i > 0$ , thus  $a_0$  is a unit and  $a_i$  are nilpotent for  $i > 1$ , so  $a_i = 0$  for  $i > 1$ .

4: Any homomorphism  $(\alpha_p)_R \rightarrow (\alpha_p)_R$  corresponds to an element  $f(T) = \sum a_i T^i \in R[T]$  s.t.  $\sum a_i (T \otimes 1 + 1 \otimes T)^i = \sum a_i (T^i \otimes 1 + 1 \otimes T^i)$ . So  $a_i = 0$  unless  $i \in p^{\mathbb{N}}$ , and in which case  $f \equiv a_1 T \pmod{T^p}$ , so the assertion follows.  $\square$

**Prop. (10.1.1.9).** For  $G \in \mathcal{G}rp/S$ , there is a Cartesian diagram:

$$\begin{array}{ccc} G & \xrightarrow{\Delta_{X/S}} & G \times_S G \\ \downarrow & & \downarrow (g, g') \mapsto m(i(g), g') \\ S & \xrightarrow{e} & G \end{array}$$

This can be seen by a testing scheme  $T$ .  $\square$

**Cor. (10.1.1.10) [Separatedness].**  $G \in \mathcal{G}rp/S$  is (quasi-)separated iff  $e$  is qc(closed immersion). In particular, if  $S$  is a field, then  $G$  is separated.  $\square$

**Def. (10.1.1.11) [Character Groups].** A **character** of a group scheme  $G$  is a homomorphism  $G \rightarrow \mathbb{G}_m$ .

It is easy to see that a character of  $G$  is equivalent to a group-like element (3.11.4.17) in  $\Gamma(G)$ . The characters of  $G$  form a **character group**, denoted by  $X^*(G)$ . Also denote  $X^\bullet(G) = X^*(G_{k^{\text{sep}}})$ .

Similarly we define  $X_*(G)$  (resp.  $X_\bullet(G)$ ) as the set of cocharacters of  $G$  (resp.  $G_{k^{\text{sep}}}$ ).

In particular, if  $G \in \mathcal{A}lg\mathcal{G}rp/k$ , then the set of characters of  $G$  are linearly independent, by (3.11.4.18).  $\square$

**Prop. (10.1.1.12) [Sheaf of Differentials is Parallel].** For  $f : G \rightarrow S \in \mathcal{G}rp/S$ , there are canonical isomorphisms

$$\Omega_{G/S} \cong f^* \mathcal{C}_{G/S} \cong f^* e^* \Omega_{G/S}.$$

In particular, if  $S$  is the spectrum of a local ring, then  $\Omega_{G/S}$  is a free  $\mathcal{O}_G$ -module.  $\square$

*Proof:* By base change,  $\Omega_{G \otimes_S G/S} = \pi_0^* \Omega_{G/S}$ , and the transition map

$$\tau : G \otimes_S G \rightarrow G \otimes_S G : (g, h) \mapsto (m(g, h), h)$$

is an automorphism of  $G \otimes_S G$  over  $G$ , so there is an isomorphism

$$\tau^* \pi_0^* \Omega_{G/S} \cong \pi_0^* \Omega_{G/S}$$

but  $\pi_0 \circ \tau = m$ , showing this isomorphism is  $m^* \Omega_{G/S} \cong \pi_0^* \Omega_{G/S}$ . Now pulling this isomorphism along the isomorphism by  $(e \circ f, \text{id}) : G \rightarrow G \otimes_S G$  we obtain the isomorphism

$$\Omega_{G/S} \cong f^* e^* \Omega_{G/S}.$$

Finally  $e^* \Omega_{G/S} \cong \mathcal{C}_{S/G}$  by (7.5.5.14). □

**Prop. (10.1.1.13).** Any  $G \in \text{GrpSch}/k$  is geo.reduced. ┘

*Proof:* Cf. [Sta]047O. ? □

**Prop. (10.1.1.14) [Galois Descent].** Let  $k \in \text{Field}$ ,  $G \in \text{Grp}/k$  and  $K/k$  is a Galois extension with Galois group  $\Gamma$ . If  $H'$  is a subgroup of  $G \otimes_k K$ , then  $H'$  is stable under the action of  $\Gamma$  iff there exists a subgroup  $H$  of  $G$  that  $H \otimes_k K = H'$ . In this case,  $H$  is unique. ┘

*Proof:* Use (7.1.5.20) and (7.1.5.23). □

**Prop. (10.1.1.15).** Let  $X, Y$  be varieties over a field  $k$  that both have at least one  $K$ -point, and  $X$  is complete. Then any morphism  $X \times Y \rightarrow G$  to a group scheme  $G$  over  $k$  factorizes as  $f(x, y) = g(x)h(y)$ , where  $f : X \rightarrow G$  and  $h : Y \rightarrow G$ . ┘

*Proof:* Fix a  $y_0 \in Y(K)$  and define a morphism  $g : X \rightarrow G : x \mapsto f(x, y_0)$ , then the morphism  $F : X \times Y \rightarrow G : (x, y) \mapsto g(x)^{-1} f(x, y)$  is constant on  $X \times \{y_0\}$ . Then the rigidity lemma (7.12.1.20) and (10.1.1.10) shows  $F(x, y) = h(y)$  where  $h : Y \rightarrow G$  is a morphism. Then we are done. □

**Cor. (10.1.1.16).** Any morphism from a  $\mathbb{P}_K^1$  to a group scheme  $G$  is constant. ┘

*Proof:* Let  $(x_0, x_1)$  be a homogenous coordinate of  $\mathbb{P}^1$ , consider the morphism  $s : \mathbb{P}^1 \times \mathbb{A}^1 \rightarrow \mathbb{P}^1 : (x_0, x_1) \times y \mapsto (x_0, x_0 + x_1 y)$ . Let  $f : \mathbb{P}^1 \rightarrow G$  be a morphism, consider the composition

$$\mathbb{P}^1 \times \mathbb{A}^1 \xrightarrow{s} \mathbb{P}^1 \xrightarrow{f} G$$

then by (10.1.4.21),  $f \circ s$  factors as  $f(s(x, y)) = g(x)h(y)$ .

We take  $y = 0$ , then  $s(x, 0) = x$ , and  $g(x) = f(x)h(0)^{-1}$ . Thus  $f(s(x, y)) = f(x)h(0)^{-1}h(y)$ . Next we take  $x = (0, 1)$ , then  $s((0, 1), y) = (0, 1)$ , and  $f((0, 1)) = f((0, 1))h(0)^{-1}h(y)$ . This shows  $h(y) = h(0)$  is constant, thus  $f(s(x, y)) = f(x)$ . Finally, let  $x = (0, 1)$ , then  $s((0, 1), y) = y$ , and  $f(y) = h(0)$  is constant. □

**Cor. (10.1.1.17).** Let  $U$  be an open subset of  $\mathbb{P}^1$ , then any morphism from  $U$  to a group variety  $G$  is constant. In particular,  $G$  contains no rational curve, and any morphism from a rationally connected variety to  $G$  is constant, in particular  $\mathbb{P}_K^n$ . ┘

*Proof:* Any rational map from  $\mathbb{P}^1$  to  $G$  can be extended to a morphism, by (7.13.1.16), thus it is constant, by the proposition above. □

**Prop. (10.1.1.18) [Weil's Extension Theorem].** Let  $S \in \text{NSch}_{\text{nom, qc}}$ ,  $Z \in \text{Sch}^{\text{sm}}/S$ ,  $G \in \text{Grp}^{\text{sm, sep}}/S$ . If  $\varphi : Z \rightarrow G$  is an  $S$ -rational map, then every irreducible component of  $Z \setminus \text{dom}(\varphi)$  is of codimension 1. ┘

*Proof:* Cf. [BLR90]P109. ? □

## Classification of groups schemes of height 1 over a field

### Quotients of Group Schemes

**Thm. (10.1.1.19) [Grothendieck].** Suppose  $G \in \mathcal{G}rpSch^{\text{ft}}/S$  and  $H \leq G \in \mathcal{G}rpSch/S$  is a closed subgroup scheme. If  $H$  is proper flat over  $S$  and if  $G$  is quasi-projective over  $S$ , then the quotient sheaf  $\mathcal{G}/_{\text{fppf}}(G)/\mathcal{Y}o_{\text{fppf}}(H)$  is representable by a scheme over  $S$ , denoted by  $G/H$ .  $\lrcorner$

*Proof:* Cf. [Grothendieck, A. Technique de descente et theoremes d'existence en geometrie algebrique, III].  $\square$

**Cor. (10.1.1.20) [Existence of Quotient Schemes].** Let  $H \rightarrow G \in \mathcal{A}lg\mathcal{G}rp/k$  be a monomorphism (10.1.5.7), then the quotient sheaf  $\tilde{G}_{\text{fppf}}/\tilde{H}_{\text{fppf}}$  is representable by an algebraic scheme, and the natural map  $G \rightarrow G/H$  is faithfully flat.  $\lrcorner$

*Proof:* Cf. [Mil17b]P605.  $\square$

**Cor. (10.1.1.21) [Quotient by Normal Subgroups].** For  $G \in \mathcal{A}lg\mathcal{G}rp/k$ , every normal algebraic subgroup  $N \trianglelefteq G \in \mathcal{A}lg\mathcal{G}rp/k$  arises as the kernel of a quotient map  $G \rightarrow G/N$ . And if  $G$  is affine,  $G/N$  is also affine.  $\lrcorner$

*Proof:* This is a corollary of (10.1.1.20) and (10.1.1.24) and (10.8.1.12).

The second assertion follows from [Mil17]P103.  $\square$

**Cor. (10.1.1.22) [Quotient Map is a Cokernel].** Let  $q : G \rightarrow Q \in \mathcal{A}lg\mathcal{G}rp/k$  be a quotient map with kernel  $N$ , then every homomorphism  $G \rightarrow H$  whose kernel contains  $N$  factors uniquely through  $q$ .  $\lrcorner$

**Prop. (10.1.1.23) [Quotient as Torsors].** Let  $G \in \mathcal{G}rpSch/S$  and  $H \leq H' \leq G \in \mathcal{A}lg\mathcal{G}rp/S$  be subgroups schemes, then there is a Cartesian diagram

$$\begin{array}{ccc} G \times (H'/H) & \longrightarrow & G/H \\ \downarrow & & \downarrow \\ G & \longrightarrow & G/H' \end{array}.$$

In particular,  $G$  is a  $(\text{fppf})H$ -torsor over  $G/H$ .  $\lrcorner$

*Proof:* For any  $T \in \mathcal{S}ch/S$ ,  $G(T) \times H(T) \rightarrow G(T) \times_{G(T)/H(T)} G(T)$  is an isomorphism, and then use the fact shification commutes with limits.  $\square$

**Cor. (10.1.1.24).** Situation as in (10.1.1.23), for any  $T \in \mathcal{S}ch/S$ , the nonempty fibers of  $G(T) \rightarrow (G/H)(T)$  are cosets of  $H(T)$ . In particular, the fiber over  $\mathbf{1} \in G/H$  is just  $H$ .  $\lrcorner$

**Cor. (10.1.1.25) [Smoothness of the Quotients].** Situation as in (10.1.1.23), suppose  $S = k \in \mathbf{Field}$ . If  $H'/H$  is smooth (resp. finite/étale) over  $k$ , then the map  $G/H \rightarrow G/H'$  is smooth (resp. finite flat/étale).  $\lrcorner$

*Proof:* Because  $G \rightarrow G/H$  is faithfully flat (10.1.1.20), we can use faithfully flat descent (7.1.5.26) and (10.1.1.23).  $\square$

**Cor. (10.1.1.26) [Points of Smooth Quotients].** If  $H \leq G \in \mathcal{A}lg\mathcal{G}rp/k$  is a closed subgroup, then  $G(\bar{k}) \rightarrow (G/H)(\bar{k})$  is surjective. Moreover, if  $H$  is smooth, then  $G(k^{\text{sep}}) \rightarrow (G/H)(k^{\text{sep}})$  is surjective, and

$$(G/H)(k) = \{gH(k^{\text{sep}}) \in G(k^{\text{sep}})/H(k^{\text{sep}}), \sigma(g)g^{-1} \in H(k^{\text{sep}}) \forall \sigma \in \text{Gal}_k\}.$$

$\lrcorner$

*Proof:* For any  $x \in H(\bar{k})$  (resp.  $x \in H(k^{\text{sep}})$ ),  $G_x$  is non-empty and locally algebraic (resp. smooth) over  $\kappa(x)$  by (10.1.1.25), thus has a  $\bar{k}$  (resp.  $k^{\text{sep}}$ -point) by (7.6.4.18).  $\square$

**Prop. (10.1.1.27) [Additivity of Dimensions].** If  $G \in \text{AlgGrp}/k$  and  $H \leq G \in \text{AlgGrp}/k$ , then

$$\dim G = \dim H + \dim G/H.$$

┘

*Proof:* This follows from (7.6.3.17) and (10.1.1.24). (Notice  $G(\bar{k}) \rightarrow H(\bar{k})$  is surjective (10.1.5.6)).  $\square$

**Thm. (10.1.1.28) [Homomorphism Theorem].** Every homomorphism  $f : G \rightarrow H \in \text{AlgGrp}/k$  factors uniquely as

$$G \xrightarrow{q} I \xrightarrow{i} H,$$

where  $q$  is a quotient map and  $i$  is a monomorphism.  $I$  is called the **image of  $f$** .  $\lrcorner$

*Proof:* Let  $N = \ker(f)$ , and  $q : G \rightarrow I = G/N$  (10.1.1.21), then by (10.1.1.22),  $f$  factors through  $I$  via a monomorphism  $i$  by (10.1.1.19).  $\square$

**Cor. (10.1.1.29).** A homomorphism of algebraic groups is a quotient map iff it is an epimorphism in the category of algebraic schemes.  $\lrcorner$

*Proof:* For any homomorphism of group schemes  $\varphi : G \rightarrow H$ , factor it as in (10.1.1.28), then we can form the quotient space  $H/I$ , so if it is an epimorphism,  $I = H$ , and  $\varphi$  is a quotient map. Conversely, if it is a quotient map, then  $\mathcal{O}_H \rightarrow \varphi_* \mathcal{O}_G$  is injective by (10.1.5.4), thus it is an epimorphism.  $\square$

**Cor. (10.1.1.30).** If  $f : G \rightarrow H \in \text{AlgGrp}/k$  is surjective and  $H$  is reduced, then  $f$  is a quotient map.  $\lrcorner$

**Prop. (10.1.1.31).** Let  $H, N$  be subgroup schemes of  $G$  with  $N$  normal, then the canonical map

$$N \rtimes H \rightarrow G$$

is an isomorphism iff  $N \cap H = e$  and  $NH = G$ .  $\lrcorner$

*Proof:* This follows from the exact sequence  $e \rightarrow N \cap H \rightarrow N \rtimes H \rightarrow NH \rightarrow e$ .  $\square$

**Prop. (10.1.1.32) [Isomorphism Theorem].** Let  $H, N$  be algebraic subgroups of an algebraic group  $G$  such that  $H$  normalizes  $N$ , then  $H \cap N$  is a normal algebraic subgroup of  $H$ , and the natural map

$$H/H \cap N \rightarrow HN/N$$

is an isomorphism.  $\lrcorner$

*Proof:* The isomorphism is induced by shification of the natural isomorphism

$$T_R : H(R)/H(R) \cap N(R) \cong H(R)N(R)/N(R) \text{ (3.1.2.8)}.$$

$\square$

**Thm. (10.1.1.33) [Correspondence Theorem].** Let  $G \in \text{AlgGrp}/k$  and  $N \trianglelefteq G \in \text{AlgGrp}/k$  is a normal algebraic subgroup, then

- For any algebraic subgroup  $H \leq G \in \mathcal{AlgGrp}/k$ , the inverse image of the image of  $H$  in  $G/N$  is  $HN$ .
- The map  $H \mapsto H/N$  is a bijection between the set of algebraic subgroups of  $G$  containing  $N$  to the set of algebraic subgroups of  $G/N$ .
- An algebraic subgroup  $H$  of  $G$  containing  $N$  is normal in  $G$  iff  $H/N$  is normal in  $G/N$ , in which case the natural map  $G/H \rightarrow (G/N)/(H/N)$  is an isomorphism.

⌋

*Proof:* Cf. [Mil17]P113. ?

□

**Prop. (10.1.1.34) [Group Formation].** WARNING: the category  $\mathcal{AlgGrp}_k$  of algebraic groups is not Abelian, not even an exact category, but it is a group formation (4.3.2.1), and its subcategory  $\mathcal{CAlgGrp}_k$  of commutative algebraic groups is an Abelian category, by (10.1.1.21)(10.1.5.3)(10.1.1.4). Also the subcategory of affine algebraic groups is a Serre subcategory, by (10.1.1.21) and (10.1.5.33).

⌋

**Cor. (10.1.1.35).** By taking shiffication, it can be shown that the same categorical properties of  $\mathcal{Grp}$  holds for  $\mathcal{AlgGrp}_k$ , such as the nine lemma, five lemma, and the restrictive snake lemma.

⌋

**Prop. (10.1.1.36) [Extensions of Group Schemes].** If  $G', G'' \in \mathcal{GrpSch}^{\text{pf}}/S$  with  $G'$  affine flat over  $S$ , and

$$1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1 \in \text{Sh}(\mathcal{Sch}_{\text{fppf}}/S)$$

is an exact sequence, then  $G$  is representable by a f.p.  $S$ -scheme flat and affine over  $G''$ . Moreover,  $G', G''$  are finite group schemes over  $S$  (10.1.2.1) iff  $G$  is.

⌋

*Proof:* Cf. [CCO14]P35. ?

□

## 2 Finite Group Schemes

References are [Finite Flat Group Schemes and  $p$ -Divisible Groups, Jakob], [Finite Group Schemes, Pink], [Introduction to Finite Group Schemes], [Finite Flat Group Schemes, Tate].

**Def. (10.1.2.1) [Finite Locally Free Group Schemes].** For  $S \in \mathcal{Sch}$ , a **finite locally free group scheme** or a **finite group scheme** over  $S$  is a group scheme  $G \in \mathcal{GrpSch}/S$  that is finite locally free (7.6.2.21) over  $S$ . It is said to have **order/rank**  $d$  if  $G$  is finite locally free of rank  $d$  over  $S$ , where  $d$  is a locally constant  $\mathbb{N}$ -valued function on  $S$ .

The category of finite locally free group scheme over  $S$  of rank  $d$  is denoted by  $\text{finGrpSch}^d/S$ .

The category of commutative finite group schemes over  $S$  of rank  $d$  is denoted by  $\text{finAbSch}^d/S$ .

⌋

**Prop. (10.1.2.2) [Connected-étale Exact Sequences Split over Perfect Fields].** If  $k \in \text{Field}$  s.t.  $k = k^{\text{sep}}$  and  $G \in \text{finGrpSch}/k$ , then  $(G^0)_{\text{red}} = \mathbf{1}$ , and the connected-étale exact sequence (10.1.4.13)

$$1 \rightarrow G^0 \rightarrow G \rightarrow \pi_0(G) \rightarrow 1$$

splits.

⌋

*Proof:*  $(G^0)_{\text{red}} = \mathbf{1}$  because it is connected and étale (7.6.6.6). Then (10.1.5.6) implies that  $G(k^{\text{sep}}) = \pi_0(k^{\text{sep}})$ . Then we have  $G_{\text{red}} \cong \pi_0(G)$  by (10.1.4.4) because they are both étale over  $k$  (7.6.6.6). Thus  $G \cong G^0 \rtimes \pi_0(G)$ .

□

**Prop. (10.1.2.3) [Infinitesimal Finite Group Schemes].** For  $G \in \text{finGrpSch}/S$ ,  $G$  is called a **infinitesimal finite group** over  $S$  if it satisfies the following equivalent conditions:

- The augmentation ideal  $\mathcal{I}$  of  $G$  is locally nilpotent.
- For each  $s \in S$ ,  $G_s$  is connected.
- For each  $s \in S$ ,  $G_s$  doesn't contain isolated points.
- For each geometric point  $\bar{s}$  of  $S$ ,  $G_{\bar{s}}$  doesn't contain isolated points.

┘

*Proof:* ?

□

**Prop. (10.1.2.4) [Cartier].** For  $k \in \text{Field}^0$ ,

- Every affine finite commutative group scheme over  $k$  is finite étale.
- If  $k = \bar{k}$ , then there is an equivalence of categories between finite commutative group schemes over  $k$  and  $\mathcal{A}b$ , by  $G \mapsto G(k)$  and  $\Gamma \mapsto \underline{\Gamma}_k$ .

┘

*Proof:* 1: by (3.11.4.20) and (5.4.7.24).

2:  $\underline{\Gamma}_k$  is clearly affine commutative group schemes over  $k$ . If  $A$  is finite locally free Hopf algebra over  $k$ , then it is finite étale by item1 and isomorphic to a finite product of copies of  $k$ . Now the equivalence is clear.

□

**Prop. (10.1.2.5) [Finite Group Schemes of Order Invertible in  $S$  is Finite Étale].** A finite locally free group scheme  $G$  over  $S$  of order invertible in  $S$  is finite étale.

┘

*Proof:* Cf. [Jakob, P45]. ?

□

**Prop. (10.1.2.6) [Finite Étale Group Schemes].** Let  $X$  be a connected smooth scheme with a geometric point  $\bar{x}$ , then there is a equivalence of categories:

$$\text{Grp}(X_{\text{fét}}) \cong \text{Grp}^{\text{fin}} / \pi_1(X, \bar{x})$$

by (9.3.2.2). In particular, constant group schemes correspond to finite groups with trivial  $\pi_1(X, x)$  actions.

┘

**Cor. (10.1.2.7).** The category of finite étale commutative group schemes over  $X$  is Abelian.

The category of commutative group schemes over  $X$  of order invertible in  $X$  is Abelian, by (10.1.2.5).

┘

**Prop. (10.1.2.8) [Quotients Maps and Faithfully Flat Maps].** For  $f : X \rightarrow Y \in \text{finGrpSch}/S$ ,  $f$  is a quotient map iff it is faithfully flat.

┘

*Proof:* ? By fibral criterion of flatness (7.6.2.15), it suffices to check on geometric fibers, which follows from (10.1.5.4).

□

### Commutative Finite Group Schemes

**Thm. (10.1.2.9) [Cartier-Duality].** If  $G \in \text{finAbSch}/S$ , then  $\mathcal{O}_G$  is a finite locally free  $\mathcal{O}_S$ -Hopf algebra, and  $\text{Hom}_{\mathcal{O}_S}(\mathcal{O}_G, \mathcal{O}_S) = \mathcal{O}_G^\vee$  is again a finite locally free  $\mathcal{O}_S$ -Hopf algebra (7.2.5.2), and thus  $\text{Spec } \mathcal{O}_G$  is a finite locally free group scheme over  $S$ , called the **Cartier-dual**  $G^D$  of  $G$ .

Then  $G^D$  represents the Hom sheaf  $\underline{\mathrm{Hom}}_{\mathrm{Sch}/S}(G, \mathbb{G}_m)$ . In particular, there is a natural bi-commutative morphism  $G \times G^D \rightarrow \mathbb{G}_m \in \mathrm{GrpSch}/S$  inducing isomorphism  $G \cong \underline{\mathrm{Hom}}_{\mathrm{Sch}/S}(G^D, \mathbb{G}_m)$  and  $G^D \cong \underline{\mathrm{Hom}}_{\mathrm{Sch}/S}(G, \mathbb{G}_m)$ .

Cartier-duality commutes with base changes. And there is a natural isomorphism  $(G^D)^D = G$ , so the Cartier-duality is a contravariant auto-equivalence of  $\mathrm{finAbSch}/S$ .  $\square$

*Proof:* Cf.[Abelian Varieties, Geer]P42<sup>?</sup>.

It suffices to show that

$$G^D(T) \cong \mathrm{Hom}_T(G \otimes_S T, \mathbb{G}_{m,T})$$

for any  $T \in \mathrm{Sch}/S$ . Notice a  $g \in G^D(T)$  corresponds to an  $R$ -algebra morphism  $g \in \mathrm{Hom}_{\mathrm{Alg}/\mathcal{O}_S}(\mathcal{O}_G^\vee \rightarrow \mathcal{O}_T) = \mathcal{O}_G \otimes_{\mathcal{O}_S} \mathcal{O}_T$  (7.2.5.2) that satisfies

$$(\Delta \otimes \mathrm{id}_T)(g) = g \otimes_T g \in (\mathcal{O}_G \otimes_{\mathcal{O}_S} \mathcal{O}_G) \otimes_{\mathcal{O}_S} \mathcal{O}_T, \quad (\varepsilon \otimes \mathrm{id}_T)(g) = 1$$

Also,  $g$  is a unit, as

$$g \cdot (\iota \times \mathrm{id}_T)(g) = \mu \circ ((\mathrm{id}_{\mathcal{O}_G} \otimes \iota) \otimes \mathrm{id}_{\mathcal{O}_T})(g \otimes g) = \mu \circ ((\mathrm{id}_{\mathcal{O}_G} \otimes \iota) \otimes \mathrm{id}_T) \circ (\Delta \otimes \mathrm{id}_T)(g) = (\eta_{\mathcal{O}_G} \otimes \mathrm{id}_T)(\varepsilon \otimes \mathrm{id}_T)(g) = 1$$

so  $g$  corresponds to a  $\mathcal{O}_T$ -Hopf algebra map

$$\mathcal{O}_T[X, X^{-1}] \rightarrow \mathcal{O}_G \otimes_{\mathcal{O}_S} \mathcal{O}_T$$

which maps  $X$  to  $g$ , thus inducing a homomorphism in  $\mathrm{Hom}_T(G \otimes_S T, \mathbb{G}_{m,T})$ . This process is invertible.  $\square$

**Prop. (10.1.2.10)** [ $\underline{\Gamma}$  is Cartier Dual to  $D(\Gamma)$ ]. Let  $\Gamma \in \mathrm{Ab}^{\mathrm{fin}}$  and  $S$  be a scheme, then  $\underline{\Gamma}_S$  is Cartier dual to  $D(\Gamma)_S$ .  $\square$

*Proof:* By (10.1.2.9), it suffices to show for  $S = \mathrm{Spec} \mathbb{Z}$ . Now  $\underline{\Gamma} = \prod_{\gamma \in \Gamma} \mathbb{Z}$ , and  $\Delta(e_\gamma) = \sum_{gg'=\gamma} e_g \otimes e_{g'}$ . Let  $f_\gamma \in \underline{\Gamma}^\vee$  be dual to  $e_\gamma$ , then

$$f_\gamma \cdot f_{\gamma'} = \sum_{g \in \Gamma} \Delta^\vee(f_\gamma \otimes f_{\gamma'})(e_g) f_g = \sum_{g \in \Gamma} f_\gamma \otimes f_{\gamma'} \left( \sum_{st=g} e_s \otimes e_t \right) f_g = f_{\gamma\gamma'}$$

so  $\underline{\Gamma}^\vee \cong \mathbb{Z}[\Gamma]$ , and

$$\Delta(f_\gamma) = \sum_{g, g' \in \Gamma} \mu^\vee(f_\gamma)(e_g \otimes e_{g'}) f_g \otimes f_{g'} = f_\gamma \otimes f_\gamma.$$

$\square$

**Cor. (10.1.2.11).**  $\underline{\mathbb{Z}/n\mathbb{Z}}$  is Cartier-dual to  $\mu_n$ .  $\square$

**Prop. (10.1.2.12)** [ $\alpha_p$  is Self Cartier-Dual]. For any  $S \in \mathrm{Sch}^p$ ,  $\alpha_{p,S}$  is Cartier-dual to itself.  $\square$

*Proof:* By (10.1.2.9), it suffices to show for  $S = \mathrm{Spec} \mathbb{F}_p$ . Then  $\alpha_p = \mathbb{F}_p[X]/X^p$  with

$$\Delta(X) = X \otimes 1 + 1 \otimes X, \quad \varepsilon(X) = 0, \quad S(X) = -X.$$

Let  $Y_i \in \alpha_p^\vee$  dual to  $X^i$  then

$$Y_i \cdot Y_j = \sum_{k=0}^{p-1} \Delta^\vee(Y_i \otimes Y_j)(X^k) Y_k = \sum Y_i \otimes Y_j (\Delta(X^k)) Y_k = \sum Y_i \otimes Y_j \left( \sum_{a+b=k} \binom{k}{a} X^a \otimes X^b \right) Y_k = \binom{i+j}{i} Y_k$$



Now  $\binom{i+j}{i}$  is unit, so  $\alpha_p^\vee = \mathbb{F}_p[Y]/Y^p$  where  $Y = Y_1$ . and

$$\Delta(Y) = \sum_{a,b} \mu^\vee(Y)(X^a \otimes X^b)Y_a \otimes Y_b = \sum Y(X^{a+b}) = Y \otimes 1 + 1 \otimes Y,$$

so  $\alpha_p^\vee = \alpha_p$ . □

**Prop. (10.1.2.13).** For  $k \in \mathbf{Field}^p$  and  $G \in \mathbf{finGrpSch}/k$  is connected, then  $\mathrm{rank}(G) \in p^{\mathbb{N}}$ . ┘

*Proof:* Cf.[Van de Geer, P68]. □

**Cor. (10.1.2.14).** If  $k \in \mathbf{Field}^p$  and  $G \in \mathbf{finAbSch}/k$ , then  $p \nmid \mathrm{rank}(G)$  iff  $G$  and  $G^D$  are both étale over  $k$ . ┘

*Proof:* This follows from the connected-étale exact sequence (10.1.4.13) and (10.1.2.13). Conversely, if  $G$  and  $G^D$  are both étale, then we may base change to  $\bar{k}$ , and  $G$  is a constant group scheme. But if  $p \mid \mathrm{rank}(G)$ , then  $G$  has a factor  $\mathbb{Z}/(p^r)$ , thus  $G^D$  has a factor  $\mu_{p^r}$ , which is not étale. □

**Prop. (10.1.2.15).** For  $k \in \mathbf{Field}$  s.t.  $k = k^{\mathrm{perf}}$  and  $G \in \mathbf{finGrpSch}/k$ , there is a canonical decomposition

$$G = G_{cc} \times G_{ce} \times G_{ec} \times G_{ee},$$

where  $G_{cc}, G_{ce}, G_{cc}^D, G_{ec}^D$  are connected and  $G_{ec}, G_{ee}, G_{ce}^D, G_{ee}^D$  are étale over  $k$ .

And if  $\mathrm{rank}(G) \in p^{\mathbb{N}}$ , then  $G_{ee} = 1$ , by (10.1.2.14). ┘

*Proof:* This follows from Cartier-duality (10.1.2.9) and (10.1.2.2). □

**Prop. (10.1.2.16) [No Maps between étale and Connected Finite Schemes].** If  $G_1, G_2 \in \mathbf{finAbSch}/k$  s.t.  $G_1$  is étale and  $G_2$  is connected, then  $\mathrm{Hom}_k(G_1, G_2) = 1, \mathrm{Hom}_k(G_2, G_1) = 1$ . ┘

*Proof:*  $(G_2)_{\mathrm{red}} = 1$  because it is connected and étale. Any map from  $G_1$  to  $G_2$  factors through  $(G_2)_{\mathrm{red}}$ , which is trivial. Similarly, and map from  $G_2$  to  $G_1$  factors through  $G_1^0$ , which is trivial. □

**Prop. (10.1.2.17) [Multiplicativity of Ranks].** If  $e \rightarrow G_1 \rightarrow G \rightarrow G_2 \rightarrow e \in \mathbf{finGrpSch}/S$  is an exact sequence, then  $\mathrm{rank}(G) = \mathrm{rank}(G_1) \cdot \mathrm{rank}(G_2)$ . ┘

*Proof:* This follows from the fact that  $G$  is a  $G_1$ -torsor over  $G_2$  (10.1.1.23). □

**Prop. (10.1.2.18).** If  $S \in \mathbf{Sch}^p$ , the three group schemes  $\underline{\mathbb{Z}/p\mathbb{Z}}_S, \mu_{p,S}, \alpha_{p,S}$  are mutually non-isomorphic. ┘

*Proof:* We may take a fiber and assume  $S = \mathrm{Spec} K$ , then  $\underline{\mathbb{Z}/p\mathbb{Z}}_S$  is reduced,  $\mu_{p,S}$  is non-reduced and  $\alpha_{p,S}$  is non-reduced. Then we can look at the reducedness of the group scheme and its Cartier dual. □

**Prop. (10.1.2.19) [The Order Kills the Finite Group, Deligne].** For  $n \in \mathbb{Z}_+$  and  $G \in \mathbf{finAbSch}/S$  with constant rank  $n$ , we have  $[n]_G = 0 : G \rightarrow G$ . ┘

*Proof:* Cf.[Jakob P12] or [Tate-Oort, Group schemes of prime order]. ? □

**Prop. (10.1.2.20) [Properties of Commutative Finite Group Schemes].** For  $G \in \mathbf{finAbSch}/S$ ,

- The following are equivalent:

- $G$  is étale over  $S$ .
- For each  $s \in S$ ,  $G_s$  is étale over  $\kappa(s)$ .
- For each geometric point  $\bar{s}$  of  $S$ ,  $G_{\bar{s}}$  is reduced.
- $G$  is étale-locally constant.
- $G$  is unipotent iff  $G^D$  is infinitesimal(10.1.2.9)(10.1.2.3).
- The following are equivalent:
  - $G$  is of multiplicative-type over  $S$ .
  - $G^D$  is étale over  $S$ .
  - For each  $s \in S$ ,  $G_s$  is of multiplicative-type over  $\kappa(s)$ .
  - Étale-locally  $G$  embeds into  $G_m^r$  for some  $r \in \mathbb{Z}_+$ .
- $G$  is called a **bi-infinitesimal group scheme** if it is infinitesimal and unipotent.

┘

**Prop. (10.1.2.21).** If  $1 \rightarrow G_1 \rightarrow G \rightarrow G_2 \rightarrow 1 \in \text{finAbSch}/S$  is an exact sequence, then

- $G$  is étale over  $S$  iff  $G_1, G_2$  are both étale over  $S$ .
- $G$  is multiplicative-type over  $S$ (10.1.2.20) iff  $G_1, G_2$  are both multiplicative-type over  $S$ .

┘

*Proof:* 1: This follows from(7.6.6.2), the fibral criterion of étaleness(7.6.6.8) and the connected-étale exact sequence(10.1.4.13).

2: This follows from item1 and Cartier-duality. □

**Def. (10.1.2.22) [Verschiebung-Maps].** If  $\text{char } k = p \in \text{Prime}$ , then for any  $G \in \text{finAbSch}/k$ , we can define a **Verschiebung-map**

$$\text{Ver}_{G/k} : G^{(p)} \rightarrow G$$

which is the Cartier-dual of the relative Frobenius of the Cartier-dual of  $G$ (10.1.2.9).

Then we have

$$F_{G/k} \circ \text{Ver}_{G/k} = [p] = \text{Ver}_{G/k} \circ F_{G/k}.$$

┘

*Proof:* ? □

### Commutative $p$ -Group Schemes

**Notation (10.1.2.23).**

- Let  $p \in \text{Prime}$ .

┘

Cf.[Finite Flat Group Schemes, Tate]Section4.

**Def. (10.1.2.24) [ $p$ -Group Schemes].** The category of commutative group schemes over  $S$  annihilated by a power of  $p$  is denoted by  $p\text{-finAbSch}/S$ . ┘

**Prop. (10.1.2.25).** If  $\kappa \in \mathbf{Field}^p$  s.t.  $\kappa = \bar{\kappa}$ , then for each  $G \in p\text{-finAbSch}/\kappa$ , there is a composition series

$$G_n = 0 \leq G_{n-1} \leq \dots \leq G_1 \leq G_0 = G$$

s.t. each subquotient  $G_i/G_{i+1}$  is isomorphic to one of the groups  $\alpha_p, \mu_p, \mathbb{Z}/(p)$ . And

- $G$  is infinitesimal iff each subquotient is isomorphic to  $\alpha_p$  or  $\mu_p$ .
- $G$  is étale iff each subquotient is isomorphic to  $\mathbb{Z}/(p)$ .
- $G$  is unipotent iff each subquotient is isomorphic to  $\mathbb{Z}/(p)$  or  $\alpha_p$ .
- $G$  is of multiplicative-type iff each subquotient is isomorphic to  $\mu_p$ .
- $G$  is of bi-infinitesimal iff each subquotient is isomorphic to  $\alpha_p$ .

⌋

*Proof:* SGA4, XVII.1.7 and 4.2.1. ?

□

### Dieudonné-Theory

References are [Lazard, Commutative formal group], [Moduli of Abelian Varieties, Norman-Oort] and [CCO14].

**Notation (10.1.2.26).**

- Let  $p \in \mathbf{Prime}$ .
- Let  $\kappa \in \mathbf{Field}^p$  s.t.  $\kappa = \kappa^{\text{perf}}$ .

⌋

**Thm. (10.1.2.27) [Dieudonné-Theory for Finite Commutative  $p$ -Group Schemes].** There is an additive anti-equivalence between Abelian categories:

$$M^* : p\text{-finAbSch}/\kappa \rightarrow \text{Mod}_{D(\kappa)}^{W(\kappa)\text{-fin}} : G \mapsto M^*(G) \triangleq \text{Hom}_{\kappa}(G, \mathbf{CW}_k)$$

(?, [CCO14]P347) called the **contravariant Dieudonné functor**, with inverse functor given by

$$M \mapsto G(M) \triangleq \left( R \in \mathcal{CRing}/\kappa \mapsto \text{Hom}_{D(\kappa)}(M, \mathbf{CW}(R)) \right).$$

Moreover,

- $\text{length}_{W(k)} M^*(G) = \log_p \#|G|$ .
- If  $\kappa' \in \mathbf{Perf}^p/\kappa$ , then there are natural isomorphisms

$$M^*(G_{\kappa'}) = M^*(G) \otimes_{W(\kappa)} W(\kappa').$$

- The relative Frobenius-morphism  $F_{G/\kappa} : G \rightarrow G^{(p)}$  corresponds to

$$M^*(G^{(p)}) \cong \varphi_*(M^*(G)) \xrightarrow{F} M^*(G),$$

and the Verschiebung-morphism  $\text{Ver}_{G/\kappa} : G^{(p)} \rightarrow G$  corresponds to

$$M^*(G) \xrightarrow{V} \varphi_*(M^*(G)) = M^*(G^{(p)}).$$



**Prop. (10.1.2.31) [Comparison of Classical and Crystalline Dieudonné Theories].** There are functorial isomorphisms

$$E^*(G) \cong M^*(G)_{(p)}, \quad E_*(G) \cong M_*(G)^{(p)} \text{ (10.1.2.29)(10.1.2.27).}$$

for each finite commutative  $G \in \mathcal{G}\text{rpSch}/\kappa$  with  $p$ -power order. Taking projective limits, we get functorial isomorphism

$$E^*(X) \cong M^*(X)_{(p)}, \quad E_*(X) \cong M_*(X)^{(p)} \text{ (10.1.2.29)(10.1.2.27).}$$

for each  $X \in p\text{Div}/\kappa$ . ┘

*Proof:* Cf. [Berthelot-Breen-Messing, Theorie de Dieudonné cristalline 2]4.2.14. □

More differential invariants of  $G$  expressed via Dieudonné modules can be found in [CCO14]B.3.10?.

### Finite Group Schemes over Henselian Local Rings

**Notation (10.1.2.32).**

- Let  $(\mathcal{O}_K, \mathfrak{m}_K, \kappa)$  be a Henselian local ring. ┘

**Prop. (10.1.2.33).** ┘

*Proof:* □

**Cor. (10.1.2.34).** Let  $R$  be a equal characteristic Henselian local ring of characteristic  $p > 0$ , then every finite locally free group scheme over  $R$  of prime order is automatically commutative. ┘

*Proof:* Cf. [Shatz, P50]. □

**Cor. (10.1.2.35).** Let  $R$  be a strict Henselian local ring with residue field of characteristic  $p > 0$ , then any connected finite locally free group scheme over  $R$  has order  $p^t$  for some  $t > 0$ . ┘

*Proof:* Cf. [Shatz, P50]. □

**Prop. (10.1.2.36) [Raynaud].** Let  $(K, \mathcal{O}_K, \kappa) \in p\text{-NField}$  and  $G \in \mathcal{G}\text{rpSch}/\mathcal{O}_K$  be quasi-finite flat of finite type, then there is a canonical exact sequence

$$0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0 \in \mathcal{G}\text{rpSch}/\mathcal{O}_K$$

s.t.  $G'$  is finite flat over  $\mathcal{O}_K$  and  $G''$  is étale quasi-finite over  $\mathcal{O}_K$  with trivial special fiber. ┘

*Proof:* Cf. [Maz78]P134. □

**Prop. (10.1.2.37) [Specialization Lemma, Raynaud].** Let  $p \in \text{Prime}$ ,  $(K, \mathcal{O}_K, \kappa) \in p\text{-NField}$  s.t.  $e(K/\mathbb{Q}_p) < p - 1$ . If  $f : \mathcal{H} \rightarrow \mathcal{G} \in \mathcal{G}\text{rpSch}/\mathcal{O}_K$  is finite flat and  $f_K : \mathcal{H}_K \rightarrow \mathcal{G}_K$  is a monomorphism, then  $f$  is a closed embedding. And if  $f_K$  is an isomorphism, then  $f$  is also an isomorphism. ┘

*Proof:* ? Cf. [Ray74]. □

**Cor. (10.1.2.38).** Let  $p \in \text{Prime}$ ,  $(K, \mathcal{O}_K, \kappa) \in p\text{-NField}$  s.t.  $e(K/\mathbb{Q}_p) < p - 1$ . if  $\mathcal{G} \in \mathcal{G}\text{rpSch}/\mathcal{O}_K$  is finite flat and  $x \in \mathcal{G}(\mathcal{O}_K)$  is a section, then the order of  $x$  equals the order of the specialization of  $x$  in  $G(\kappa)$ . ┘

### 3 Groupoid Schemes

Cf. [Sta]Cha38, 39.

### 4 Algebraic Groups over Fields

**Notation (10.1.4.1).**

- Let  $G \in \text{locAlgGrp}/k$ .

┘

**Prop. (10.1.4.2) [Smoothness and Geo.Reducedness].** For  $x \in |G|_0$ ,  $G$  is smooth at  $x$  iff it is geo.reduced at  $x$ . ┘

*Proof:* If  $G$  is smooth, then it is geo.regular thus geo.reduced. The converse follows from (10.2.1.9).  
□

**Cor. (10.1.4.3) [Cartier].** If  $\text{char } k = 0$ , then  $G/k$  is smooth. ┘

*Proof:* This is a consequence of (10.1.1.13) and (10.1.4.2). Alternative proof:  $\Omega_{G/k}$  is free, by (10.1.1.12), so it is smooth by (7.6.4.14). □

**Prop. (10.1.4.4) [Reduced Structure].** If  $G_{\text{red}}$  is geo.reduced, then it is a subgroup of  $G$ . This is the case if  $k = k^{\text{perf}}$ . ┘

*Proof:* This is because  $G_{\text{red}} \times G_{\text{red}}$  is reduced so the multiplication factors through  $G_{\text{red}} \times G_{\text{red}}$ . □

**Prop. (10.1.4.5) [Smoothness in Characteristic  $p$ ].** Let  $\text{char } k = p \neq 0$ ,  $k = k^{\text{perf}}$  and let  $G$  be affine, and  $r \geq 0$ . Then the image of the relative Frobenius  $F_{G/k} : G \rightarrow G^{(p^r)}$  is geo.reduced group scheme when  $r$  is sufficiently large. ┘

*Proof:* To show it is a group scheme, notice  $F^r$  is a homomorphism and use homomorphism theorem (10.1.1.28). And It corresponds to

$$\Gamma(G) \otimes_{k, \varphi^r} k \rightarrow \Gamma(G) : a \otimes c \mapsto ca^{p^r}.$$

The image of which is just  $\Gamma(G)^{p^r}$  as  $k$  is perfect. To show it is geo.reduced, we can assume  $k$  is alg.closed, then the nilradical  $N$  of  $\Gamma(G)$  is nilpotent, so some  $N^m = 0$ , and then the image is reduced for any  $p^r > m$ . □

**Prop. (10.1.4.6) [Smoothness and Tangent Space].**  $G/k$  is smooth iff it is regular iff  $\dim_k T_e \leq \dim_e G$ , where  $e$  is the identity element. ┘

*Proof:* By homogeneity and the fact smooth locus is open,  $G$  is smooth iff it is smooth at  $e$ . Now  $e$  is a rational point, by (5.4.5.26),  $G$  is smooth at  $e$  iff it is regular at  $e$ . □

**Prop. (10.1.4.7) [Closed Subgroups and Points].** Let  $S \leq G(k)$  a closed subgroup, then there is a unique reduced closed subgroup  $H$  of  $X$  that  $H(k) = S$ . Moreover,  $H$  is geo.reduced. The algebraic subgroups of  $G$  arising in this way are exactly those that  $H(k)$  is schematically dense in  $H$ .

In particular, if  $k = k^{\text{sep}}$ , then  $H \mapsto H(k)$  is a bijection between closed subgroups of  $G$  and closed subgroups of  $G(k)$ , by (7.4.3.10). ┘

*Proof:* Let  $H$  be a reduced closed subscheme of  $G$  that  $H(k) = S$ , then  $S$  is dense in  $|H|$  and  $H$  is reduced, so by (7.4.3.5) and (7.4.3.8) shows  $S$  is schematically dense in  $H$  and  $H$  is geo.reduced. Therefore  $H \times H$  is reduced and thus multiplication map  $H \times H \rightarrow G$  factors through  $H$ , also does inversion and unit, so  $H$  is a subgroup of  $G$ .

The converse is also true.  $\square$

**Cor. (10.1.4.8) [Zariski Closure].** Let  $S \subset G(k)$  a subgroup, then there is a unique geo.reduced closed subscheme  $H$  of  $G$ , called the Zariski closure of  $S$  in  $G$ , such that  $H(k)$  is the Zariski closure of  $S \leq G(k)$ .  $\lrcorner$

**Prop. (10.1.4.9) [Algebraic Group Scheme is Quasi-Projective].** If  $G \in \mathcal{AlgGrp}/k$ , then  $G$  is quasi-projective.  $\lrcorner$

*Proof:* Cf. [Sta]0BF7?  $\square$

**Prop. (10.1.4.10) [Center Subgroup].** The center  $Z(G)$  of  $G$  is a closed subgroup scheme of  $G$ .  $\lrcorner$

*Proof:* Cf. [Sta]0BF8.  $\square$

**Prop. (10.1.4.11).** Every étale normal subgroup of a connected algebraic group is central.  $\lrcorner$

*Proof:* This is because the automorphism group scheme of an étale group scheme is étale?  $\square$

**Prop. (10.1.4.12) [Identity Components].** Consider the identity component  $G^0$  of  $G$ , then

- It commutes with the formation of identity component commutes with base change of fields. In particular,  $G^0$  is geo.connected.
- $G$  is locally connected (7.4.1.23), thus  $G^0$  is clopen in  $G$ .
- $G^0$  is a characteristic algebraic subgroup of  $G$

$\lrcorner$

*Proof:* 1, 2 follows from (7.4.3.15).

3: By (7.4.3.12),  $G^0 \times G^0$  is connected, thus  $G^0 \times G^0$  is mapped into  $G^0$ , and it is an open subscheme, thus it is an algebraic subgroup of  $G$ .  $\square$

**Cor. (10.1.4.13) [Connected-étale Exact Sequence].** Group structure induces a group structure on the maximal étale subalgebra of  $\Gamma(G)$ , thus  $G \rightarrow \pi_0(G)$  is a group homomorphism (9.3.1.1), which is faithfully flat by (9.3.1.3), and the stabilizer of the identity in  $\pi_0(G)$  is just  $G^0$  by (9.3.1.3) again. Thus  $\pi_0(G) = G/G^0$  by (10.2.1.10).

In fact,

$$1 \rightarrow G^0 \rightarrow G \rightarrow \pi_0(G) \rightarrow 1$$

is the unique exact sequence with  $G^0$  connected and  $\pi_0(G)$  étale.  $\lrcorner$

**Cor. (10.1.4.14).** If  $G \in \mathcal{AlgGrp}/k$  and  $H \leq G$  is a closed subgroup, then  $G/H \rightarrow G/H^0$  is finite smooth, by (10.1.4.13) and (10.1.1.25).  $\lrcorner$

**Prop. (10.1.4.15).** Every connected component of  $G$  is irreducible.  $\lrcorner$

*Proof:* By (10.1.4.12),  $G_k^0$  is connected. To show it is irreducible, it suffices to consider its reduced structure, and in this case it is smooth (Notice in this case  $G_{\text{red}}^0$  is an algebraic group because  $G_{\text{red}}^0 \times G_{\text{red}}^0$  is reduced). Thus no closed point is connected in two irreducible component, thus it is irreducible. Then  $G^0$ , as a quotient space of  $G_k^0$ , is also irreducible.

For another connected component  $H$ , choose a closed point  $h$ , let  $L$  be a finite normal extension of  $k$  containing  $k(h)$ , then each connected component of  $H_L$  maps surjectively onto  $H$ , thus contains one of the finitely many inverse images of  $h$  in  $H_L$ . And they are all rational points, thus isomorphic to  $G_L^0$ , which is irreducible, thus  $H$  is also irreducible.  $\square$

**Cor. (10.1.4.16) [Connected Algebraic Group is Geo.Irreducible].** If  $G$  is connected and qc, then it is geo.irreducible by (10.1.4.12) and (10.1.4.15).  $\lrcorner$

**Cor. (10.1.4.17).** If  $G \in \text{AlgGrp}/k$  is connected, then for any non-empty open subschemes  $U, V$  of  $G$ ,  $U \times V \rightarrow G$  is surjective.  $\lrcorner$

*Proof:* By (7.4.1.29), it suffices to check on closed points. Let  $p$  be a closed point, by base change, we may assume  $x \in G(k)$ , then  $xV^{-1} \cap U \neq \emptyset$  as  $G$  is geo.irreducible (10.1.4.16), thus  $x \in UV$ .  $\square$

**Cor. (10.1.4.18).** Every connected component of  $G$  is algebraic over  $k$ .  $\lrcorner$

*Proof:* For the identity component, take a non-empty affine open subset  $U$ , then  $U^2 = G^0$  by (10.1.4.17), thus  $G^0$  is qc. For the other components, the same proof as that of (10.1.4.15) shows they are also quasi-compact.  $\square$

**Def. (10.1.4.19) [Torsion Component].** Let  $G$  is commutative, then  $G^\tau = \cup_{n>0} [n]^{-1}G^0 \subset G$  is an open group subscheme, called the **torsion component** of  $G$ . Forming  $G^\tau$  commutes with change of fields, and if  $G^\tau$  is quasi-compact, it is a clopen subgroup.  $\lrcorner$

**Prop. (10.1.4.20).** Let  $G$  is commutative, then any algebraic subgroup of  $G$  is contained in  $G^\tau$ . In particular, if  $G^\tau$  is algebraic over  $k$ , then it is the maximal algebraic subgroup of  $G$ .  $\lrcorner$

*Proof:* As  $H$  is qc, it is covered by f.m. translates of  $G^0$ , thus  $G^0$  has finite index in  $G^0H$ , which means  $[n](H) \subset G^0$  for some  $n \in \mathbb{Z}_+$ .  $\square$

**Prop. (10.1.4.21).** Let  $X, Y$  be varieties over a field  $k$  that both have at least one  $K$ -point, and  $X$  is complete. Then any morphism  $X \times Y \rightarrow G$  to a group scheme  $G$  over  $k$  factorizes as  $f(x, y) = g(x)h(y)$ , where  $f : X \rightarrow G$  and  $h : Y \rightarrow G$ .  $\lrcorner$

*Proof:* Fix a  $y_0 \in Y(K)$  and define a morphism  $g : X \rightarrow G : x \mapsto f(x, y_0)$ , then the morphism  $F : X \times Y \rightarrow G : (x, y) \mapsto g(x)^{-1}f(x, y)$  is constant on  $X \times \{y_0\}$ . Then the rigidity lemma (7.12.1.20) and (10.1.1.10) shows  $F(x, y) = h(y)$  where  $h : Y \rightarrow G$  is a morphism. Then we are done.  $\square$

**Cor. (10.1.4.22).** If  $G$  is qc, then any morphism from a  $\mathbb{P}_K^1$  to  $G$  is constant.  $\lrcorner$

*Proof:* Let  $(x_0, x_1)$  be a homogenous coordinate of  $\mathbb{P}^1$ , consider the morphism  $s : \mathbb{P}^1 \times \mathbb{A}^1 \rightarrow \mathbb{P}^1 : (x_0, x_1) \times y \mapsto (x_0, x_0 + x_1y)$ . Let  $f : \mathbb{P}^1 \rightarrow G$  be a morphism, consider the composition

$$\mathbb{P}^1 \times \mathbb{A}^1 \xrightarrow{s} \mathbb{P}^1 \xrightarrow{f} G$$

then by (10.1.4.21),  $f \circ s$  factors as  $f(s(x, y)) = g(x)h(y)$ .

We take  $y = 0$ , then  $s(x, 0) = x$ , and  $g(x) = f(x)h(0)^{-1}$ . Thus  $f(s(x, y)) = f(x)h(0)^{-1}h(y)$ . Next we take  $x = (0, 1)$ , then  $s((0, 1), y) = (0, 1)$ , and  $f((0, 1)) = f((0, 1))h(0)^{-1}h(y)$ . This shows  $h(y) = h(0)$  is constant, thus  $f(s(x, y)) = f(x)$ . Finally, let  $x = (0, 1)$ , then  $s((0, 1), y) = y$ , and  $f(y) = h(0)$  is constant.  $\square$



**Cor. (10.1.4.23).** Let  $U$  be an open subset of  $\mathbb{P}^1$ , then any morphism from  $U$  to a group variety  $G$  is constant. In particular,  $G$  contains no rational curve, and any morphism from a rationally connected variety to  $G$  is constant, in particular  $\mathbb{P}_K^n$ .  $\lrcorner$

*Proof:* Any rational map from  $\mathbb{P}^1$  to  $G$  can be extended to a morphism, by (7.13.1.16), thus it is constant, by the proposition above.  $\square$

**Def. (10.1.4.24) [Anti-Affine Group Schemes].** An **anti-affine algebraic group** over  $k$  is an algebraic group  $G$  over  $k$  s.t.  $\Gamma(G) = k$ . For example, Abelian varieties are anti-affine by (7.12.1.12).  $\lrcorner$

**Prop. (10.1.4.25).** Let  $\varphi : G \rightarrow H$  be a homomorphism of algebraic groups over  $k$  that  $G$  is anti-affine and  $H$  is connected, then  $\varphi$  factors through the center of  $H$  (10.1.4.10).  $\lrcorner$

*Proof:* Cf. [Mil17]P151.  $\square$

**Cor. (10.1.4.26).** If  $G$  is a connected and qc, then every anti-affine algebraic subgroup is contained in the center of  $G$ .  $\lrcorner$

**Cor. (10.1.4.27).** If  $G$  is anti-affine and qc, then it is commutative and connected.  $\lrcorner$

*Proof:*  $G \rightarrow \pi_0(G)$  is surjective but  $\pi_0(G)$  is affine thus this map is trivial, so  $G$  is connected. Then it is also commutative by (10.1.4.26).  $\square$

### Constructing Algebraic Groups

**Prop. (10.1.4.28) [Generated Closed Subgroup].** Let  $\varphi_i : X_i \rightarrow G$  be a family of maps from algebraic schemes  $X$  to an algebraic group over a field  $k$ .

- If  $X_i$  and  $G$  are all affine, then the generated group scheme is representable by an algebraic subgroup  $H$ .
- If  $X_i$  are all geo.reduced. Then the generated group scheme is representable by a smooth subgroup  $H$ .

Moreover, in both cases, if there is only a single map  $\varphi : X \rightarrow G$  and  $X$  is geo.connected, and  $e \in \varphi(X)$ , then  $\langle X, \varphi \rangle$  is geo.connected.  $\lrcorner$

*Proof:* Cf. [Mil17]P54, 55, 56.  $\square$

**Def. (10.1.4.29) [Commutator Subgroups].** If  $G \in \text{AlgGrp}/k$  is affine or smooth, then  $[G, G]$  exists by (10.1.4.28), and:

- $[G, G]$  is the intersection of all normal subgroups  $H$  of  $G$  s.t.  $G/H$  is commutative.
- For a field extension  $k'/k$ ,  $[G_{k'}, G_{k'}] = [G, G]_{k'}$ .
- If  $G$  is connected/smooth, then  $[G, G]$  is also connected/smooth.
- If  $G$  is an affine smooth connected group scheme, then  $[G, G]$  is the unique smooth connected subgroup scheme that satisfies  $[G, G](\bar{k}) = [G(\bar{k}), G(\bar{k})]$ .  $\lrcorner$

*Proof:* 1: Cf. [Mil17]P129.

2: This is formal.

3: This follows from (10.1.4.28).

4: Cf. [Mil17]P130.  $\square$

**Cor. (10.1.4.30).** If  $G \in \mathcal{AlgGrp}/k$ , the derived series and central derived series of  $G$  (10.1.1.4) exist. And  $G$  is solvable (resp. nilpotent) iff  $G^{(n)} = e$  (resp.  $G^n = e$ ) for some  $n \in \mathbb{N}$ .  $\lrcorner$

**Def. (10.1.4.31) [Transporter].** Let  $G \times X \rightarrow X$  be an action of an algebraic group on a scheme, and  $Y, Z$  be subschemes of  $X$ ,  $Z$  is a closed subscheme and  $Y$  is an algebraic scheme, then the functor

$$R \mapsto \{g \in G(R) \mid gY_R \subset Z_R\}$$

is represented by a closed subscheme of  $G$ , called the **transporter**  $T_G(Y, Z)$ .  $\lrcorner$

*Proof:*  $T_G(Y, Z)$  is in fact the sheaf  $\mathcal{H}om(Y, Z) \times_{\mathcal{H}om(Y, X)} G$ , and notice  $\mathcal{H}om(Y, Z) \rightarrow \mathcal{H}om(Y, X)$  is a closed subfunctor (10.8.1.3), thus  $T_G(Y, Z)$  is a closed subfunctor of  $G$ , thus represented by a closed subscheme of  $G$ .  $\square$

**Cor. (10.1.4.32).** Let  $G \times X \rightarrow X$  be an action of an algebraic group on a scheme, and  $Y, Z$  be closed subschemes of  $X$ , then the functor

$$R \mapsto \{g \in G(R) \mid gY_R = Z_R\}$$

is represented by a closed subscheme of  $G$ .  $\lrcorner$

*Proof:* In this case, the functor is represented by the closed subscheme  $T_G(Y, Z) \cap \text{inv}(T_G(Z, Y))$ .  $\square$

**Cor. (10.1.4.33) [Stablizer].** Let  $G \times X \rightarrow X$  be an action of an algebraic group on a scheme, and  $Y$  be closed subschemes of  $X$ , then the functor

$$\text{Stab}_G(Y) : R \mapsto \{g \in G(R) \mid gY_R = Y_R\}$$

is represented by a closed subscheme of  $G$ , called the **stablizer subgroup of  $Y$** .  $\lrcorner$

**Def. (10.1.4.34) [Normalizer].**  $\lrcorner$

**Def. (10.1.4.35) [Centralizer].**  $\lrcorner$

### Group Varieties

**Def. (10.1.4.36) [Group Varieties].** A **group variety** over  $k$  is a  $k$ -variety (7.12.1.3) that is also a group scheme.

The category of group varieties over  $k$  is denoted by  $\text{GrpVar}/k$ .  $\lrcorner$

**Prop. (10.1.4.37) [Group Varieties are Smooth].** Any  $X \in \text{GrpVar}/k$  is smooth by (10.1.4.2). Conversely, any smooth connected algebraic group over  $k$  is a group variety, by (10.1.4.2)(10.1.4.16) and (10.1.1.10).  $\lrcorner$

**Prop. (10.1.4.38) [Tangent Bundle Trivial].** For  $G \in \text{GrpVar}/k$ ,  $T_{X,e}$  is the tangent space at  $e$ , then there is a natural isomorphism  $\Omega_{X/k} \cong T_{X,e}^* \otimes \mathcal{O}_X$ . Also true for  $\mathcal{T}_X$  (because  $\Omega_{X/k}$  is locally free as  $X$  is smooth (10.1.4.37)(7.12.1.14)).  $\lrcorner$

*Proof:* There should be another proof using relation in (7.5.5.14)?.

Use a dual number characterization of tangent spaces and tangent vector fields (7.6.4.21)(7.12.1.15), then notice a tangent vector  $\tau \in T_{X,e}$  is a  $S = k[\varepsilon]$ -point of  $X$ , then right translation gives a translation  $X_S \rightarrow X_S$  that is invariant on  $X$ , which gives a tangent vector on  $X$ .

So there is a map  $T_{X,e} \otimes \mathcal{O}_X \rightarrow \Omega_{X/k}$ . To check isomorphism, both are locally free of the same rank, so it suffices to show it is surjective. But on closed pts, pass to Nakayama, this is clearly true, so it is surjective by (7.12.1.13).  $\square$

**Prop. (10.1.4.39) [Dimension Theorem].** Let  $\varphi : G \rightarrow H$  be a surjective homomorphism of group varieties, then

$$\dim(G) = \dim(H) + \dim(\ker(\varphi)).$$

┘

*Proof:* This is a consequence of (7.6.3.19), as the fiber over any closed pt is isomorphic to a field base change of  $\ker(\varphi)$ .  $\square$

**Prop. (10.1.4.40).** Every binational homomorphism of a connected affine group varieties is an isomorphism.  $\square$

*Proof:* Such an isomorphism induces a homomorphism  $A \rightarrow B$  of integral Hopf algebras that is an isomorphism on the fraction field, then it is an isomorphism by (3.11.4.22).  $\square$

**Def. (10.1.4.41) [Quasi-Central Homomorphisms].** A **quasi-central homomorphism**  $\varphi : G' \rightarrow G$  of group varieties over  $k$  is a homomorphism the kernel of  $\varphi(\bar{k})$  is central in  $G'(\bar{k})$ .  $\square$

**Prop. (10.1.4.42).** If a homomorphism of group varieties  $\varphi : G' \rightarrow G$  is quasi-central and the kernel of  $\text{Lie}(\varphi)$  is central, then  $\ker(\varphi) \subset Z(G')$ .  $\square$

*Proof:* Cf. [Bruhat-Tits, 2.1].  $\square$

**Prop. (10.1.4.43) [1-Dimensional Group Varieties].** Suppose  $k = \bar{k}$  and  $G$  is a group variety of dimension 1, then either  $G \cong \mathbb{G}_a$ ,  $\mathbb{G}_m$  or an elliptic curve.  $\square$

*Proof:* Cf. [Sil99].  $\square$

## 5 Homomorphisms

**Notation (10.1.5.1).**

- Let  $G \in \text{AlgGrp}/k$ .

┘

**Prop. (10.1.5.2) [Connected-Étale Sequence].** Cf. [Mil17]P114, [Finite Flat Group Schemes, Tate]Section3.7 and [?Mil17b]P117.  $\square$

**Prop. (10.1.5.3) [Algebraic Subgroups are Closed].** Any algebraic subgroup  $H \leq G$  is a closed subgroup. In particular, an algebraic subgroup of an affine algebraic group is affine.

WARNING:  $H$  must first be an algebraic group, so we can use Chevalley theorem to show it is locally closed.  $\square$

*Proof:* As  $H_{k'} \rightarrow H$  is a quotient map (7.4.7.3), we can assume  $k$  is alg.closed, and also assume  $H, G$  are reduced. Now Chevalley shows that the image is a constructible set of  $G$ . Then we can consider all on the level of  $\bar{k}$  points, because  $G$  is Jacobson. Then  $H$  contains an open subset of  $\bar{H}$  by (4.12.2.18), which implies  $H(\bar{k})$  is open in  $\bar{H} \cap G(\bar{k})$ . Now  $\bar{H} \cap G(\bar{k})$  is the closure of  $H(\bar{k})$  in  $G(\bar{k})$ , thus it is also a subgroup of  $G(\bar{k})$ , and we can consider the coset of  $H(\bar{k})$  in  $\bar{H} \cap G(\bar{k})$ .  $H$  is open in  $\bar{H}$ , and  $\bar{H}$  is compact, and also  $\bar{H} \cap G(\bar{k})$  is compact because  $G$  is Jacobson, so there are only f.m. coset, thus  $H(\bar{k})$  is also closed in  $\bar{H} \cap G(\bar{k})$ , thus  $H$  is also closed in  $\bar{H}$ , so  $H = \bar{H}$ . ? Cf. [Mil17]P19.  $\square$

**Prop. (10.1.5.4) [Characterizing Quotient Maps].** The following conditions on a homomorphism  $\varphi : G \rightarrow Q \in \text{AlgGrp}/k$  are equivalent:

- $\varphi$  is faithful flat.
- $\varphi$  is a quotient map (10.1.1.4).
- The homomorphism  $\mathcal{O}_Q \rightarrow \varphi_* \mathcal{O}_G$  is injective.

┘

*Proof:*  $1 \rightarrow 2$  follows from the very definition of fat subfunctors, as f.f. map is a fppf cover. Cf. [Mil17]P109. ?  $\square$

**Cor. (10.1.5.5) [Surjective Map between Group Varieties].** Let  $\varphi : G \rightarrow H \in \text{AlgGrp}/k$  is surjective and  $H$  is reduced, then  $\varphi$  is a quotient map. In particular, a surjective map between group varieties are quotient maps.  $\square$

*Proof:* The hypothesis implies  $G(\bar{k})$  acts transitively on  $H(\bar{k})$ , thus  $\varphi$  is faithfully flat by (10.2.1.4).  $\square$

**Prop. (10.1.5.6) [Check Quotient Map on Closed Points].** Let  $\varphi : G \rightarrow H \in \text{AlgGrp}/k$ . If  $\varphi$  is a quotient map, then  $\varphi : G(\bar{k}) \rightarrow H(\bar{k})$  is surjective. The converse is also true if  $H$  is reduced.

In particular, if  $\mathbf{1} \rightarrow N \rightarrow G \rightarrow H \rightarrow \mathbf{1} \in \text{AlgGrp}/k$  is an exact sequence, then  $\mathbf{1} \rightarrow N(\bar{k}) \rightarrow G(\bar{k}) \rightarrow H(\bar{k}) \rightarrow \mathbf{1}$  is exact. Moreover, if  $N$  is smooth, then  $\varphi : G(k^{\text{sep}}) \rightarrow H(k^{\text{sep}})$  is also surjective, and

$$H(k) = \{gN(k^{\text{sep}}) \in G(k^{\text{sep}})/N(k^{\text{sep}}), \sigma(g)g^{-1} \in N(k^{\text{sep}}) \forall \sigma \in \text{Gal}_k\}.$$

┘

*Proof:* This follows from (10.1.1.26). For the converse, notice the image of  $\varphi$  has the same  $\bar{k}$ -points as  $H$ , thus it equals  $H$  as  $H$  is reduced and locally algebraic.  $\square$

**Prop. (10.1.5.7) [Characterizing Monomorphisms].** For a homomorphism  $\varphi : G \rightarrow H$  of algebraic groups over  $k$ , the following are equivalent:

- $\varphi$  is a monomorphism.
- $\ker(\varphi) = e$  (10.1.1.4).
- $\varphi$  is a monomorphism in the category of algebraic groups over  $k$ .
- $\varphi$  is a monomorphism in the category of algebraic schemes over  $k$ .

Moreover, a monomorphism is just a closed embedding.  $\square$

*Proof:*  $1 \rightarrow 4 \rightarrow 3$  is obvious.

$3 \rightarrow 2$ : the composition of  $\ker(\varphi) \rightarrow G$  with  $\varphi$  is trivial, thus  $\ker(\varphi)$  is trivial.

2  $\iff$  1: This follows from the definition of  $\ker(\varphi)$ (10.1.1.4).

For the last assertion, if  $\varphi$  is a closed embedding, then it is a monomorphism in the category of algebraic schemes over  $k$ (7.4.5.23). Conversely, if  $\varphi$  is a monomorphism, consider the quotient space  $G \rightarrow G/H$ (10.1.1.19), then by(10.1.1.24),  $H$  is the fiber over  $o \in G/H$ , thus a closed subscheme.  $\square$

**Def. (10.1.5.8) [Embeddings].** Following(10.1.5.7), define an **embedding of algebraic groups** to be a closed immersion of algebraic groups.  $\lrcorner$

**Def. (10.1.5.9) [Almost-Direct Products].**  $G$  is called an **almost-direct product** of f.m.  $G_1, \dots, G_n \in \text{AlgGrp}/k$  if there is a quotient map(10.1.5.4)

$$G_1 \times \dots \times G_n \rightarrow G \rightarrow 1$$

with finite kernel.  $\lrcorner$

### Isogenies

**Def. (10.1.5.10) [Finite-Index Subgroups].** An algebraic subgroup  $H \leq G$  is said to have **finite index** if the quotient  $G/H \in \text{AlgGrp}/k$  is a finite scheme.  $\lrcorner$

**Prop. (10.1.5.11).** The identity component  $G^0$  is of finite index in  $G$ (10.1.5.10).  $\lrcorner$

*Proof:* Notice an algebraic group  $H$  is finite iff  $\#H(\bar{k}) < \infty$ . And  $(G/G^0)(\bar{k}) = G(\bar{k})/G^0(\bar{k})$ (10.1.4.12) is finite as  $G$  is compact.  $\square$

**Def. (10.1.5.12) [Isogeny].** An **isogeny of algebraic groups** is a homomorphism  $f : G \rightarrow H \in \text{AlgGrp}/k$  s.t.  $\ker(f)$  is finite and  $\text{Im}(f) \leq H$  is of finite index(10.1.5.10).

When  $\text{char } k > 0$ , a **Frobenius-isogeny** is an isogeny s.t. the kernel equals the kernel of some power of the Frobenius map.

Any isogeny is a finite morphism by(10.1.1.25).  $\lrcorner$

**Def. (10.1.5.13) [Strongly Connected Groups].**  $G$  is called a **strongly connected group** if it has no non-trivial subgroup of finite index.  $\lrcorner$

**Def. (10.1.5.14) [Strongly Identity Components].** The **strongly identity component**  $G^{s0}$  of an algebraic group  $G$  is defined to be the intersection of the algebraic subgroups of finite index. Thus it is a characteristic subgroup of  $G$ .  $\lrcorner$

**Prop. (10.1.5.15).**  $G/G^{s0}$  is a finite scheme. In particular,  $G^{s0}$  is the smallest algebraic subgroup having the same dimension as  $G$ .  $\lrcorner$

*Proof:* As  $G$  is Noetherian,  $G^{s0} = H_1 \cap \dots \cap H_r$  for  $G/H_i$  finite schemes. Thus  $G/G^{s0} \hookrightarrow H_1 \cap \dots \cap H_r$  is a finite scheme.  $\square$

**Prop. (10.1.5.16) [Strongly Identity Components and Identity Components].**  $G^{s0}$  is connected, and the converse is true if  $G$  is smooth. In fact, if  $G$  is smooth,  $G^{s0} = G^0$ .

In particular, a group variety has no non-trivial algebraic subgroup of finite index. Thus an isogeny to a group varieties is surjective(thus a quotient map by(10.1.1.30)).  $\lrcorner$

*Proof:*  $G^{s0}$  is connected because the identity component is of finite index(10.1.5.11). Conversely,  $G^0/G^{s0}$  is smooth and connected and finite(10.1.5.33), thus it is a group variety of dimension 0, which is trivial.  $\square$

**Prop. (10.1.5.17).** If  $k = k^{\text{perf}}$  and  $G \in \mathcal{A}lgGrp/k$ , then  $G_{\text{red}}^0 = G^{s0}$ . ┘

*Proof:* Cf. [Mil17]P122. □

**Def. (10.1.5.18) [Isogenies].** A **central isogeny**(resp. **multiplicative Isogeny**) is an isogeny(10.1.5.12) that the kernel is central(resp. of multiplicative type). ┘

**Prop. (10.1.5.19).** Any isogeny with kernel order prime to the characteristic has étale kernel by(10.1.2.5), thus it is central, by(10.1.4.11). ┘

**Prop. (10.1.5.20) [Multiplicative Isogenies are Central].** Every multiplicative isogeny from a connected algebraic group is central, by(10.2.3.27). ┘

**Def. (10.1.5.21).** A composition of multiplicative isogenies is a multiplicative isogeny. ┘

*Proof:* This is because there is an exact sequence  $e \rightarrow \ker(\varphi_1) \rightarrow \ker(\varphi_2 \circ \varphi_1) \rightarrow \ker(\varphi_2) \rightarrow e$ , thus  $\ker(\varphi_2 \circ \varphi_1)$  is central by(10.2.3.30), hence of multiplicative type by(10.2.3.19). □

**Prop. (10.1.5.22) [Isogenies between Group Varieties].** Let  $\varphi : G \rightarrow H$  be a surjective homomorphism of group varieties, then  $\varphi$  is flat. And if  $\dim G = \dim H$ , then  $\varphi$  is finite, and the rank of  $\ker(\varphi)$  equals the separable degree of  $K(G)/K(H)$ . ┘

*Proof:*  $\varphi$  is flat by(10.2.1.4). By(7.12.3.5), there exists a dense open subscheme  $U'$  s.t.  $\varphi^{-1}(U') \rightarrow U'$  is finite, thus by homogeneity,  $\varphi$  is finite, and rank of  $\ker(\varphi)$  is  $[K(G) : K(H)]_s$ . □

### Subnormal Series

**Prop. (10.1.5.23) [Solvability].** If  $G$  is either affine or smooth, then  $G$  is solvable iff  $G^{(n)} = e$  for some  $n \in \mathbb{Z}_+$ (10.1.1.4). In particular, for  $k'/k \in \mathbf{Field}$ ,  $G$  is solvable iff  $G_{k'}$  is solvable. ┘

**Prop. (10.1.5.24) [Nilpotency].** If  $G$  is either affine or smooth, then  $G$  is nilpotent iff  $G^n = e$  for some  $n \in \mathbb{Z}_+$ (10.1.1.4). In particular, for  $k'/k \in \mathbf{Field}$ ,  $G$  is nilpotent iff  $G_{k'}$  is nilpotent.

In particular, if  $G$  is a nilpotent and geo.connected, then it contains a non-trivial geo.connected subgroup scheme in its center, by(10.1.4.29). ┘

**Prop. (10.1.5.25).** A split solvable algebraic group  $G$  is an affine group variety, by(10.1.5.33). ┘

**Lemma (10.1.5.26).** If  $G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_s = e$  is a subnormal series and  $\dim G = \dim G_i/G_{i+1}$  for some  $i$ , then  $G \sim G_i/G_{i+1}$ . ┘

*Proof:* The maps  $G_i \rightarrow G_i/G_{i+1}$  and  $G_i \rightarrow G_{i-1} \rightarrow \dots \rightarrow G_0 = G$  are isogenies(10.1.5.12). □

**Def. (10.1.5.27) [Composition Series].** A **composition series** of  $G$  is defined to be a maximal object among the subnormal series

$$G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_s = e$$

that satisfies  $\dim G_0 > \dim G_1 > \dots > \dim G_s$ . ┘

**Prop. (10.1.5.28).** Any two composition series of an algebraic group  $G$  has refinements that the quotients are isogenous. ┘

**Cor. (10.1.5.29) [Jordan-Holder].** ┘

**Example (10.1.5.30) [Unip( $n$ ) is Nilpotent].**  $\text{Unip}(n)$  has a central normal series of length  $\frac{n(n-1)}{2}$  s.t. each subquotient is isomorphic to  $\mathbb{G}_a$ , Cf. [Mil17]P136. ┘

**Maximal Subgroups with Properties****Notation(10.1.5.31).**

- Let  $G \in \text{AlgGrp}/k$ .

┘

**Def.(10.1.5.32)[Good Properties].** A property of algebraic groups over fields is called a **good property(algebraic groups)** s.t.

- $e_k$  has  $\mathcal{P}$ .
- Every extension of algebraic groups over  $k$  with property  $\mathcal{P}$  has  $\mathcal{P}$ .
- Every quotient of a algebraic group over  $k$  with property  $\mathcal{P}$  has  $\mathcal{P}$ .

┘

**Prop.(10.1.5.33)[List of Good Properties].** The following are group properties for algebraic groups over fields(10.1.5.32):

1. Strongly connectedness.
2. Solvability. Moreover, subgroups of solvable groups are solvable.
3. Smoothness.
4. Affineness. Moreover, subgroups of affine groups are affine.
5. Connectedness.
6. Finiteness. Moreover, subgroups of finite groups are finite.
7. Unipotency. Moreover, subgroups of unipotent groups are unipotent.
8. Being a group variety.

┘

*Proof:*

1. This follows from the isomorphism and correspondence theorems(10.1.1.32)(10.1.1.33).
2. This follows from standard argument and correspondence theorems(10.1.1.32)(10.1.1.33).
3. Let  $e \rightarrow N \rightarrow G \rightarrow H \rightarrow e$  be an exact sequence of algebraic groups, then if  $G$  is smooth, then so is  $H$  by(10.2.1.9) and the fact  $G/H$  is geo.reduced by(7.1.5.28) and(10.1.5.4). If  $N, H$  are smooth, then so is  $G$  by the fact smoothness is stalkwise.
4. Notice by(10.1.1.23) and(7.1.5.29).  $G \rightarrow Q$  is affine.
5. Let  $e \rightarrow N \rightarrow G \rightarrow H \rightarrow e$  be an exact sequence of algebraic groups. If  $N$  is connected, then  $G \rightarrow \pi_0(G)$  is trivial on  $N$  thus factors through  $H$ , and hence through  $\pi_0(H)$ , which is trivial if  $H$  is connected, so in this case  $G$  is connected. And if  $G$  is connected, the surjective homomorphism  $G \rightarrow H \rightarrow \pi_0(H)$  factors through  $\pi_0(G)$ , so  $\pi_0(H) = 1$ .
6. This follows from(10.1.1.27).
7. This follows from(10.3.4.5).
8. This follows from(10.1.4.37).

□

**Prop.(10.1.5.34) [Maximal Group Varieties Exist].** If  $\mathcal{P}$  is a good property of algebraic groups(10.1.5.32), then  $G$  contains a largest normal subgroup variety  $H \leq G$  with property  $\mathcal{P}$ . And also  $G/H$  has contains no non-trivial such subgroups.

┘

*Proof:*  $G$  contains at least one normal subgroup variety, namely  $e$ . There exists a maximal such one, by taking the one with maximal dimension (because smooth varieties are reduced), then it is the largest, because if  $H'$  is another, then  $HH'$  is a larger one, by (10.1.5.33) and the fact  $NH$  is a quotient of  $N \rtimes H$ .

And if  $G/H$  contains a normal subgroup variety  $H'$  that has  $P$ , then the inverse image of  $H'$  in  $G$  is also a normal subgroup variety that has  $P$ , contradiction.  $\square$



## 10.2 Group Schemes II: Solvable Groups

References are [Mil17].

**Notation(10.2.0.1).**

- Let  $k \in \mathbf{Field}$ .
- Let  $R \in \mathbf{CRing}$ .
- Use notations on group schemes defined in [Group Schemes I: Structure Theory](#).

┘

### 1 Group Actions

**Remark(10.2.1.1).** Group action of an group functor on an object is defined in [\(4.1.1.68\)](#).

┘

**Prop.(10.2.1.2)[Group Action].** An action of a group scheme  $G \in \mathbf{GrpSch}$  on a scheme  $X \in \mathbf{Sch}$  is equivalent to a right  $\Gamma(G)$ -comodule structure on  $\Gamma(X)$  as  $\Gamma(G)$ -modules. This action will induce a right comodule structure on  $\Gamma(X)$ .

The action of  $G$  on itself is called the **regular action** of  $G$ .

┘

**Prop.(10.2.1.3).** Let  $\mu : G \times X \rightarrow X$  be an action of an group scheme  $G$  on a scheme  $X$ , then it is faithfully flat, and it is called a smooth/finite/...action if  $\mu$  is smooth/finite/..

┘

*Proof:* We can see this from the commutative diagram [\(4.1.1.69\)](#).

□

**Prop.(10.2.1.4)[Images of Equivariant Maps].** Let  $G$  be a group functor and  $X, Y$  be non-empty algebraic schemes on which  $G$  acts, and  $f : X \rightarrow Y$  is an equivariant map.

- If  $Y$  is reduced and  $G(\bar{k})$  acts transitively on  $Y(\bar{k})$ , then  $f$  is faithfully flat.
- If  $G(\bar{k})$  acts transitively on  $X(\bar{k})$ , then the set  $f(|X|)$  is locally closed in  $|Y|$ , so we can let  $f(X)_{\text{red}}$  denote its reduced subscheme structure [\(7.4.1.14\)](#).
- If  $X$  is reduced and  $G(\bar{k})$  acts transitively on  $X(\bar{k})$ , then  $f$  factors into

$$X \xrightarrow{\text{faithfully flat}} f(X)_{\text{red}} \xrightarrow{\text{immersion}} Y.$$

Moreover,  $f(X)_{\text{red}}$  is stable under the action of  $G$ .

┘

*Proof:* Cf. [\[Mil17\]P26](#). ?

1:

2:

3:  $f$  factors through  $f(X)_{\text{red}}$  because  $X$  is reduced [\(7.4.1.14\)](#). Then the first assertion follows from 1 and 2. The last assertion follows from universal property again. □

**Def.(10.2.1.5)[Orbit Maps].** Let  $\mu : G \times X \rightarrow X$  be an action of  $G \in \mathbf{AlgGrp}/k$  on  $X \in \mathbf{Sch}^{\text{ft}}/k$ . For any  $x \in X(k)$ , the **orbit map**

$$\mu_x : G \rightarrow X : g \mapsto gx$$

is defined to be the restriction to  $\mu$  to  $G \times \{x\} \cong G$ . The image of the orbit map is locally closed in  $X$  by [\(10.2.1.4\)](#), and then its reduced structure subscheme is called the **orbit scheme**  $O(x)$  of  $x$ . ┘

**Prop. (10.2.1.6) [Fixed Subscheme].** Let  $\mu : G \times X \rightarrow X$  be an action of a group functor  $G$  on a separated algebraic scheme over  $k$ , then the functor

$$\tilde{X}^G : R \mapsto \{x \in X(R) \mid \mu(g, x_{R'}) = x_{R'}, \forall R - \text{algebra } R', g \in G(R')\}$$

is representable by a closed subscheme  $X^G$  of  $X$ , called the **fixed subscheme** of this action. Then it can be seen directly that the formation of fixed subscheme commutes with extension of base fields.  $\perp$

*Proof:* An element  $x \in X(R)$  defines two functors

$$G(R') \rightarrow X(R') : g \mapsto gx_{R'}$$

$$G(R') \rightarrow X(R') : g \mapsto x_{R'}$$

which are both natural in  $R'$ . Thus we get a map  $X(R) \rightarrow \text{Hom}(G_R, X_R \times X_R)$  which is also natural in  $R$ , thus induce a map  $X \mapsto \text{Hom}(G, X \times X)$ .

Then there is a Cartesian diagram

$$\begin{array}{ccc} \tilde{X}^G & \longrightarrow & \text{Hom}(G, \Delta_X) \\ \downarrow & & \downarrow \text{closed} \\ \tilde{X} & \longrightarrow & \text{Hom}(G, X \times X) \end{array} \quad .$$

The right vertical map is a closed subfunctor by (10.8.1.10), as  $\Delta_X$  is closed in  $X \times X$  because  $X$  is separated. Hence  $\tilde{X}^G$  is a closed subfunctor of  $\tilde{X}$ , thus represented by a closed subscheme of  $X$ , by (10.8.1.3).  $\square$

**Prop. (10.2.1.7) [Orbit Map is Faithfully Flat and Smooth].** Let  $\mu : G \times X \rightarrow X$  be an action of an algebraic group  $G$  on an algebraic scheme  $X$  and  $x \in X(k)$ .

- If  $X$  is reduced and  $G(\bar{k})$  acts transitively on  $X(\bar{k})$ , then the orbit map  $\mu_x : G \rightarrow X$  is faithfully flat.
- If  $G$  is reduced, then  $O_x$  is stable under  $G$ , and the map  $\mu_x : G \rightarrow O_x$  is faithfully flat. If  $G$  is smooth, then  $O_x$  is also smooth.

$\perp$

*Proof:* 1: this follows from (10.2.1.4).

2: The first statement follows from (10.2.1.4)3, and then  $\mathcal{O}_{O_x} \rightarrow \mu_{x*}(\mathcal{O}_G)$  is universally injective. Therefore if  $G$  is smooth, then  $\mathcal{O}_{O_x}$  is geometrically reduced. Then  $O_x$  is smooth by (10.2.1.9).  $\square$

**Def. (10.2.1.8) [Homogenous Space].** A non-empty algebraic scheme  $X$  with an action of a group scheme  $G$  is called a **homogeneous space** for  $G$  if  $G(\bar{k})$  acts transitively on  $X(\bar{k})$ , and for any (some)  $x \in X(\bar{k})$ ,  $\mu_x$  is faithfully flat (thus surjective).  $\perp$

**Prop. (10.2.1.9) [Smoothness for Homogenous Spaces].** If  $G$  is a group scheme and  $X$  is a generically geo.reduced  $G$ -homogeneous space, then  $X$  is smooth.  $\perp$

*Proof:* By (7.6.4.20), it has an open dense smooth locus. Now smoothness can be checked after base change to alg.closed field (5.4.2.1), but then because  $G(\bar{k})$  acts transitively on itself, thus all the geometric points are smooth. But geometric points are dense in  $G$  (7.4.1.26), thus  $G$  is smooth.  $\square$

### Orbits

**Prop. (10.2.1.10) [Homogenous Space as Quotient Spaces].** Let  $G \in \mathcal{AlgGrp}/k$  acts on  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$  with a rational point  $x \in X(k)$ , then  $X \cong G/G_x$  iff the orbit map  $\mu_x : G \rightarrow X$  is faithfully flat.  $\square$

*Proof:* If  $(X, x)$  is a quotient space, then by (10.1.1.20),  $\mu_x$  is faithfully flat. Conversely, if it is faithfully flat, then clearly  $\mu_x(\tilde{G})$  is a fat subfunctor of  $\tilde{X}$ , thus  $X$  represents the quotient functor  $\tilde{G}/\tilde{G}_x$ .  $\square$

**Cor. (10.2.1.11).** If  $G \in \mathcal{AlgGrp}_{\text{red}}/k$  and  $\mu : G \times X \rightarrow X$  is an action of  $G$  on a  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$  with a rational point  $x \in X(k)$ , then  $O(x) \cong G/G_x$ .  $\square$

*Proof:* Because  $G$  is reduced,  $\mu_x : G \rightarrow O(x)$  is faithfully flat by (10.2.1.7), and  $O(x)$  is stable under the action of  $G$  by (10.2.1.4), thus (10.2.1.10) applies to this case.  $\square$

**Prop. (10.2.1.12) [Orbit Map are Immersions].** If  $G \in \mathcal{AlgGrp}^{\text{sm}}/k$  acts on  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$ , then for any  $x \in X(k)$ ,  $G/G_x \rightarrow X$  is an immersion.  $\square$

*Proof:* Cf. [Mil17]P143.  $\square$

**Prop. (10.2.1.13).** Let  $G \in \mathcal{AlgGrp}^{\text{sm}}/k$  acts on  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$ , then:

- A reduced closed subscheme  $Y \subset X$  is stable under  $G$  iff  $\bar{Y}$  is stable under  $G(\bar{k})$ .
- Let  $Y$  be a locally closed subscheme of  $X$ , then if  $Y$  is stable under  $G$ , then  $(\bar{Y})_{\text{red}}$  and  $(\bar{Y}_{\bar{k}} \setminus Y)_{\text{red}}$  is also stable under  $G$ .

$\square$

*Proof:* 1: Because  $G$  is geo.reduced and  $Y$  is reduced,  $G \times Y$  is reduced (7.4.3.2), thus  $\mu : G \times Y \rightarrow X$  factors through  $Y$  iff  $\mu(\bar{k})$  factors through  $Y(\bar{k})$ .

2:  $|\bar{Y}|_{\text{red}}(\bar{k})$  is the closure of  $Y(\bar{k})$  in  $X(\bar{k})$ . So as  $G(\bar{k})$  acts continuously on  $X(\bar{k})$ , if it fixes  $Y(\bar{k})$ , then it also fixes  $(\bar{Y})(\bar{k})$  and  $(\bar{Y} \setminus Y)(\bar{k})$ , thus we finish by item1.  $\square$

**Cor. (10.2.1.14) [Orbits of Smallest Dimension are Closed].** Let  $G \in \mathcal{AlgGrp}^{\text{sm}}/k$  acts on  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$ , and let  $Y \subset X$  be a non-empty locally closed subscheme stable under the action of  $G$  of the smallest dimension, then it is closed.  $\square$

*Proof:* This is because  $\dim Y > \dim(\bar{Y} \setminus Y)_{\text{red}}$  (Because irreducible components of  $\bar{Y}$  is not contained in  $\bar{Y} \setminus Y$ ).  $\square$

**Cor. (10.2.1.15) [Orbit Lemma, Borel].** Let  $k = \bar{k}$  and  $G \in \mathcal{AlgGrp}^{\text{sm}}/k$  acts on  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$ , then every orbit of minimal dimension is closed.  $\square$

*Proof:* If  $Y$  is an orbit of minimal dimension, then  $(\bar{Y} \setminus Y)_{\text{red}}$  is stable under  $G$  and has smaller dimensions. If it is non-zero, then it contains an orbit.  $\square$

**Def. (10.2.1.16) [G-Orbits].** Let  $G \in \mathcal{AlgGrp}^{\text{sm}}/k$  acts on  $X \in \mathcal{Sch}^{\text{ft,sep}}/k$ , then a reduced subscheme  $O \leq X$  is called an  $G$ -orbit of  $X$  if  $(O_{\bar{k}})_{\text{red}} = O_{\gamma}$  for some  $\gamma \in X(\bar{k})$ .  $\square$

**Prop. (10.2.1.17) [Galois Theory of Orbits].** Situation as in (10.2.1.16), there is a bijection between the set of  $G$ -orbits in  $X$  and the set of  $\text{Aut}_k(\bar{k})$ -stable  $G(\bar{k})$ -orbits in  $X(\bar{k})$ .  $\square$

*Proof:* Cf. [Getz]P437.  $\square$

**Def. (10.2.1.18) [Relatively Semisimple and Relatively Regular Elements].** Situation as in (10.2.1.16),  $\gamma \in X(k)$  is called a **relatively semisimple element** if  $O(\gamma) \subset X$  is closed. And  $\gamma \in X(k)$  is called a **relatively regular element** if the dimension of  $O(\gamma) \subset X$  is maximal among all  $\gamma \in X(\bar{k})$ .  $\lrcorner$

**Prop. (10.2.1.19) [Semisimple Elements in Conjugation Actions].** If  $k = k^{\text{perf}}$  and  $G \in \text{RedGrp}/k$ , then under the conjugate action of  $G$  on itself,  $\gamma \in G(k)$  is relatively semisimple (resp. relatively regular) iff  $x$  is semisimple (resp. regular in the sense of [Steinberg. Regular elements of semisimple algebraic groups]).  $\lrcorner$

*Proof:* ? [Steinberg, Regular elements of semisimple algebraic groups] Prop 6.13.  $\square$

**Prop. (10.2.1.20) [Stabilizers are Reductive].** Situation as in (10.2.1.16), if  $G^0 \in \text{RedGrp}/k$  and  $\gamma \in X(k)$ , then  $O(\gamma)$  is affine iff  $(G_\gamma)^0 \in \text{RedGrp}/k$ . In particular, if  $X$  is affine, then for any relatively semisimple element  $\gamma \in X(k)$  (10.2.1.18),  $(G_\gamma)^0 \in \text{RedGrp}/k$ .  $\lrcorner$

*Proof:* Cf. [Getz] P438. ?  $\square$

### Linear Actions

**Prop. (10.2.1.21).** For  $G \in \text{linAlgGrp}/k$ , any  $(V, \rho) \in \text{Rep}(G)$  induces an action of  $G$  on  $\underline{V}$ , and also an action of  $G$  on the projective algebraic scheme  $\mathbb{P}(V)$ .  $\lrcorner$

**Prop. (10.2.1.22).** Let  $G \times X \rightarrow X$  be an action of  $G \in \text{linAlgGrp}/k$  on an  $X \in \text{Aff}^{\text{ft}}/k$  over  $k$ , then there exists  $V \in \text{Rep}^{\text{fd}}(G)$  and an equivariant closed embedding  $X \hookrightarrow \underline{V}$ .  $\lrcorner$

*Proof:* Cf. [Mil17b] P145.  $\square$

**Def. (10.2.1.23) [Linear Action].** An action of  $G \in \text{AlgGrp}/k$  on  $X \in \text{Sch}^{\text{ft}}/k$  is called a **linear action** if there exists  $(V, \rho) \in \text{Rep}^{\text{fd}}(G)$  and an equivariant non-degenerate immersion  $X \hookrightarrow \mathbb{P}(V)$ .  $\lrcorner$

**Prop. (10.2.1.24).** If  $G \times X \rightarrow X$  is a transitive action of  $G \in \text{linAlgGrp}/k$  on  $X \in \text{Sch}^{\text{ft, sep}}/k$  s.t.  $X(k)$  is non-empty, then this action is linear.  $\lrcorner$

*Proof:* Cf. [Mil17b] P145.  $\square$

**Def. (10.2.1.25) [Grassmannian Variety].** The **Grassmannian variety**  $\text{Gr}(n, k)$  is a scheme over  $\mathbb{Z}$  defined to be the quotient of  $\text{GL}(n)$  by the algebraic subgroup  $B$  fixing a subspace of dimension  $k$  (10.1.1.19).  $\lrcorner$

The Grassmannian varieties are projective by (10.9.2.16).  $\lrcorner$

**Remark (10.2.1.26).** For an explicit construction of  $\text{Gr}(n, k)$ , Cf. [Sta] 089T or [Nit05] P6.  $\lrcorner$

**Def. (10.2.1.27) [Flag Varieties].** The **flag variety** is a scheme over  $\mathbb{Z}$  defined to be the quotient of  $\text{GL}(n)$  by the algebraic subgroup  $B_F$  fixing a flag  $F$  (10.1.1.19).  $\lrcorner$

The flag varieties are projective by (10.9.2.19).  $\lrcorner$

## 2 Lie Algebras

**Notation(10.2.2.1).**

- Let  $G \in \text{locAlgGrp}^{\text{aff}}/k$ .

⌋

**Def.(10.2.2.2)[Lie Algebra of an Algebraic Group].** By(7.6.4.24), the tangent space of  $G$  at  $e$  is isomorphic to

$$\text{Lie}(G) = \ker(G(k[\varepsilon]/(\varepsilon^2)) \rightarrow G(k))$$

as a vector space. For any homomorphism  $f : G \rightarrow H$ , there is a Lie algebra map

$$\text{Lie}(f) : \text{Lie}(G) \rightarrow \text{Lie}(H)$$

induced by  $f$ .

In particular, if  $G$  is affine, it is the set of homomorphisms  $\Gamma(G) \rightarrow k[\varepsilon]$  that the composition with  $k[\varepsilon] \rightarrow k$  is the counit map  $\varepsilon : \Gamma(G) \rightarrow k$ (10.1.1.2).

If  $G$  is affine,  $\varphi$  maps the augmentation ideal  $I_G = \ker(\varepsilon)$  into  $(\varepsilon)$ , and thus is trivial on  $I_G^2$ . So  $\varphi$  factors through  $\Gamma(G)/I_G^2$ . Now  $\Gamma(G)/I_G^2 \cong k \oplus I_G/I_G^2$  by(3.11.4.11), so

$$\text{Lie}(G) \cong \text{Hom}_k(I_G/I_G^2, k).$$

And we define  $\text{Lie}(G)$  to be  $L(G)$ .

In general, for  $R \in \mathcal{CRing}/k$ , define  $\mathfrak{g}(R) = \ker(G(R[\varepsilon]) \rightarrow G(R))$ , then similarly

$$(G) \cong \text{Hom}_k(I_G/I_G^2, k).$$

$G(R[\varepsilon])$  acts on  $\mathfrak{g}(R)$  by inner automorphism, so also does  $G(R)$ , and we get a homomorphism of algebraic groups

$$\text{Ad} : G \rightarrow \mathbf{GL}(\mathfrak{g})$$

This homomorphism commutes with Lie algebra homomorphism: If  $f : G \rightarrow H$  is a homomorphism of algebraic groups, then there is a commutative diagram

$$\begin{array}{ccc} G \times \mathfrak{g} & \xrightarrow{(x,X) \mapsto \text{Ad}(x)X} & \mathfrak{g} \\ \downarrow f & & \downarrow \text{Lie}(f) \\ H \times \mathfrak{h} & \xrightarrow{(y,Y) \mapsto \text{Ad}(y)Y} & \mathfrak{h} \end{array}$$

Define a Lie bracket on  $\mathfrak{g}$  as follows:  $[X, Y] = \text{ad}(X)(Y) = \text{Lie}(\text{Ad})(X)(Y)$ . Then this is a Lie algebra structure on  $\mathfrak{g}$ , and it commutes with arbitrary base change.

For example,  $\mathfrak{gl}(n) = \text{Lie}(\text{GL}(n))$  is given the Lie algebra structure  $[X, Y] = XY - YX$ . ⌋

*Proof:* To verify this is truly a Lie algebra, we take a faithful embedding of  $G$  into some  $\mathbf{GL}(V)$ (10.3.1.22). Thus it suffices to prove the Lie algebra of  $\mathbf{GL}(V)$  is a Lie algebra. Now for  $A, B \in \text{Mat}(n; R)$ , pondering the definition shows

$$(1 + \delta A)(1 + \varepsilon B)(1 + \delta A)^{-1} = 1 + \varepsilon B + \varepsilon \delta [A, B] \in k[\varepsilon, \delta]/(\varepsilon^2, \delta^2).$$

So by calculation  $[A, B] = XY - YX$ , so it is truly a Lie algebra.

To show the group structure on  $\text{Lie}(G)$  equals the structure on the tangent space, use the Eckmann-Hilton argument(3.1.1.5), notice the hypothesis is satisfied because both composition is the morphism  $\text{Spec } k[\varepsilon] \rightarrow \text{Spec}(k[\varepsilon] \otimes k[\varepsilon]) \xrightarrow{\varphi} G \times G \xrightarrow{\mu} G$ , where  $\varphi = (a, b, c, d) : k[\varepsilon] \otimes k[\varepsilon] \xrightarrow{\varphi} G \times G$ . □

**Cor. (10.2.2.3) [Exponential Map].** As there are natural isomorphisms  $\mathfrak{g}(R) \cong \mathfrak{g} \otimes R$ , we can write  $e^{\varepsilon X}$  the element of  $G(R) \leq G(R[\varepsilon])$  corresponding to  $X \in \mathfrak{g} \otimes R$ . Then  $e^{\varepsilon X + \varepsilon Y} = e^{\varepsilon X} e^{\varepsilon Y}$ , and by functoriality, for any homomorphism  $f : G \rightarrow H$ ,

$$f(e^{\varepsilon X}) = e^{\varepsilon \operatorname{Lie}(f)(X)}.$$

Also

$$x \cdot e^{\varepsilon Y} x^{-1} = e^{\varepsilon \operatorname{Ad}(x)Y}$$

and also the commutative diagram in (10.2.2.2) means

$$f(e^{\varepsilon X}) = e^{\varepsilon \operatorname{Lie}(f)(X)}.$$

┘

**Cor. (10.2.2.4) [Lie algebra commutes with Limits].** It can be seen from the definition (10.2.2.2) that the Lie algebra construction commutes with limits of algebraic groups, because kernel commutes with limits. In particular it commutes with kernel map. ┘

**Prop. (10.2.2.5) [Lie Algebra of Smooth Subgroups].** Let  $H \leq G$  be an algebraic subgroup s.t.  $\operatorname{Lie}(H) = \operatorname{Lie}(G)$ . If  $H$  is smooth and  $G$  is connected, then  $H = G$ . ┘

*Proof:* Recall that  $\dim \mathfrak{g} \geq \dim G$ , with equality iff  $G$  is smooth (10.1.4.6), so the condition forces  $G$  to be smooth. Now  $G$  is smooth and connected thus irreducible (10.1.4.16) and  $\dim G = \dim H$ , so  $H = G$ .  $\square$

**Cor. (10.2.2.6).** Let  $H_1, H_2$  be connected algebraic subgroups of  $G$  and  $H_1 \cap H_2$  is smooth. If  $\operatorname{Lie}(H_1) = \operatorname{Lie}(H_2)$ , then  $H_1 = H_2$ . ┘

**Cor. (10.2.2.7).** If  $\operatorname{char} k = 0$  and  $G \in \operatorname{AlgGrp}/k$ , then there is a bijection between connected algebraic subgroups of  $G$  and Lie subalgebras of  $\operatorname{Lie}(G)$ , because every subgroup is smooth, by (10.1.4.3). ┘

**Cor. (10.2.2.8).** Let  $G \in \operatorname{AlgGrp}/k$  and  $\{H_i\}_{i \in I}$  are smooth algebraic subgroups of  $G$ . If  $\operatorname{Lie}(H_i)$  generate  $\operatorname{Lie}(G)$  as a Lie algebra, then  $H_i$  generates  $G$  (10.1.4.28). ┘

*Proof:* Let  $H$  be the subgroup they generate, then  $H$  is smooth (10.1.4.28) and  $\operatorname{Lie}(H) = \operatorname{Lie}(G)$ , thus  $H = G$  by (10.2.2.5).  $\square$

**Prop. (10.2.2.9) [Lie Algebra of Quotients].** For an exact sequence  $1 \rightarrow K \rightarrow G \rightarrow H \rightarrow 1 \in \operatorname{AlgGrp}/k$ ,  $\operatorname{Lie}(G) \rightarrow \operatorname{Lie}(H)$  is surjective iff  $K$  is smooth. ┘

*Proof:* ?  $\square$

### Stabilizers, Centers and Centralizers

**Prop. (10.2.2.10) [Lie Algebra of Stabilizer].**  $G \in \operatorname{AlgGrp}/k$  and  $V \in \operatorname{Rep}^{\operatorname{fd}}(G)$ , then it induces an action of  $\mathfrak{g}$  on  $V$  (10.2.2.2). Let  $W \subset V$  be a subspace, then the stabilizer  $\operatorname{Stab}_G(W)$  is a subgroup of  $G$  (10.3.1.3)

$$\operatorname{Lie}(\operatorname{Stab}_G(W)) = \operatorname{Stab}_{\mathfrak{g}}(W).$$

In particular,  $\dim(\operatorname{Stab}_G(W)) \leq \dim \operatorname{Stab}_{\mathfrak{g}}(W)$ , with equation iff  $\operatorname{Stab}_G(W)$  is smooth. ┘

*Proof:* By (10.2.2.3),

$$\begin{aligned}
 X \in \operatorname{Lie}(\operatorname{Stab}_G(W)) &\iff r(e^{\varepsilon X})W[\varepsilon] \subset W[\varepsilon] \\
 &\iff e^{\varepsilon \operatorname{Lie}(r)(X)}W_R[\varepsilon] \subset W_R[\varepsilon] \\
 &\iff (1 + \varepsilon \operatorname{Lie}(r)(X))(W_R + \varepsilon W_R) \subset (W_R + \varepsilon W_R) \\
 &\iff \operatorname{Lie}(r)(X)(W_R) \subset W_R \\
 &\iff X \in \operatorname{Stab}_{\mathfrak{g}}(W)
 \end{aligned}$$

□

**Prop. (10.2.2.11) [Lie Algebra of Center].** If  $G \in \operatorname{GrpVar}/k$ , then

$$\dim z(\mathfrak{g}) \geq \dim Z(G),$$

and if equality holds, then  $Z(G)$  is smooth and  $\operatorname{Lie}(Z(G)) = z(\mathfrak{g})$ . ┘

*Proof:* There are maps

$$\begin{aligned}
 \operatorname{Ad} : G &\mapsto \operatorname{GL}(\mathfrak{g}), Z(G) \subset \ker(\operatorname{Ad}), \\
 \operatorname{ad} : \mathfrak{g} &\rightarrow \mathfrak{gl}(\mathfrak{g}), \ker(\operatorname{ad}) = z(\mathfrak{g}).
 \end{aligned}$$

Because Lie algebra commutes with kernel (10.2.2.4),  $\operatorname{Lie}(Z(G)) \subset \operatorname{Lie}(\ker(\operatorname{Ad})) = \ker(\operatorname{ad})$ . So

$$\dim z(\mathfrak{g}) = \dim \ker(\operatorname{ad}) = \dim \operatorname{Lie}(\ker(\operatorname{Ad})) \geq \dim \ker(\operatorname{Ad}) \geq \dim Z(G)$$

with equality iff  $\ker(\operatorname{Ad})$  is smooth and  $\dim \ker(\operatorname{Ad}) = \dim Z(G)$  and thus  $\ker(\operatorname{Ad})^0 = Z(G)^0$ , so  $Z(G)$  is also smooth. Finally,  $\operatorname{Lie}(Z(G)) \subset z(\mathfrak{g})$ , so they are equal if they have the same dimensions. □

**Prop. (10.2.2.12) [Lie Algebra of Centralizer].** Let  $G \in \operatorname{AlgGrp}/k$  and  $H \leq G$  is an algebraic subgroup, then  $H$  acts on  $\mathfrak{g}$  by  $\operatorname{Ad}$ , and

$$\operatorname{Lie}(C_G(H)) = \mathfrak{g}^H, \quad \operatorname{Lie}(N_G(H))/\operatorname{Lie}(H) = (\operatorname{Lie}(G)/\operatorname{Lie}(H))^H.$$

┘

*Proof:*

$$\begin{aligned}
 X \in \operatorname{Lie}(C_G(H)) &\iff x(e^{\varepsilon X})_S x^{-1} = e^{\varepsilon X}_S, \quad \forall k[\varepsilon] \rightarrow S, x \in H(S), \\
 X \in \mathfrak{g}^H &\iff y e^{\varepsilon X_R} y^{-1} = e^{\varepsilon X_R}, \quad \forall k \rightarrow R, x \in H(R).
 \end{aligned}$$

And it can be shown these two are equal. Similarly, there is a natural map  $\operatorname{Lie}(N_G(H)) \rightarrow \operatorname{Lie}(G)/\operatorname{Lie}(H)$ , and the image lies in the fixed subgroup of  $H$ , because

$$X \in \operatorname{Lie}(N_G(H)) \iff (e^{\varepsilon X})_S x (e^{\varepsilon X})_S^{-1} \in H(S) \iff e^{\varepsilon \operatorname{ad}(x)X_S} \in H(S) e^{\varepsilon X_S}, \quad \forall k[\varepsilon] \rightarrow S, x \in H(S)$$

$$X \in (\operatorname{Lie}(G)/\operatorname{Lie}(H))^H \iff e^{\varepsilon \operatorname{ad}(x)X_R} \in e^{\varepsilon (X_R + \operatorname{Lie}(H)(R))}, \quad \forall k \rightarrow R, x \in H(R)$$

Then it can be shown that  $X$  satisfies condition in 1 iff  $\overline{X}$  satisfies condition in 2, thus we are done. □

**Cor. (10.2.2.13) [Kernel of Ad].** If  $\operatorname{char} k = 0$  and  $G \in \operatorname{AlgGrp}/k$  is connected, then  $\ker(\operatorname{Ad}) = Z(G)$ . ┘

*Proof:* Let  $H = \ker(\operatorname{Ad})$ , then  $\operatorname{Lie}(C_G(H)) = \mathfrak{g}^H = \mathfrak{g}$ , which implies  $H = G$  by (10.2.2.7). □

### 3 Groups of Multiplicative Type

**Notation(10.2.3.1).**

- Let  $G \in \text{linAlgGrp}/k$ .

┘

#### Diagonalizable Groups

**Def.(10.2.3.2)[Diagonalizable Groups].** A group scheme over  $R$  isomorphic to  $\mathbf{D}(M)$  for some f.g. commutative group  $M$  is called a **diagonalizable group**.

┘

**Prop.(10.2.3.3).**  $G \in \text{GrpSch}^{\text{fg,aff}}/R$  is diagonalizable(10.2.3.2) iff  $\Gamma(G)$  is a finite projective  $R$ -module generate by the group-like elements(3.11.2.4) in  $\Gamma(G)$ .

┘

*Proof:* For the group algebra  $k(M)$ , its group-like elements are just  $\{m|m \in M\}$  by(3.11.4.19), and they clearly span  $R(M)$  as a  $R$ -space. Conversely, if the set  $M$  of group-like elements in  $\Gamma(G)$  spans  $\Gamma(G)$ , then by(3.11.4.18) they form a basis of  $\Gamma(G)$ , so there is an isomorphism of vector spaces  $\mathbf{D}(M) \rightarrow \Gamma(G)$ . But this is also a homomorphism, because they are on a basis.  $\square$

**Cor.(10.2.3.4)[Classifying Diagonalizable Groups].** If  $\text{Spec}(R)$  is connected, then

- $M \mapsto \mathbf{D}(M)$  is a contravariant equivalence from  $\mathcal{CAlg}_R^{\text{fg}}$  to the category of diagonalizable algebraic groups, with inverse give by  $G \mapsto X^*(G)$ .
- This functor preserves exact sequences.
- Algebraic subgroups and quotient groups of diagonalizable groups are diagonalizable.

┘

*Proof:* 1: By(10.2.3.3), it suffices to show that  $\text{Hom}(M, M') \rightarrow \text{Hom}(\mathbf{D}(M'), \mathbf{D}(M))$  is an isomorphism. Because  $\mathbf{D}$  sends direct sums to direct products, it suffices to check the case that  $M, M'$  are cyclic. This is easy to check, just notice that a group homomorphism maps group-like elements to group-like elements.

2: If  $M' \rightarrow M$  is injective, then  $k[M'] \rightarrow k[M]$  is injective, thus faithfully flat by(3.11.4.21), and  $\mathbf{D}(M) \rightarrow \mathbf{D}(M')$  is a quotient map. Conversely,  $\mathbf{D}(M) \rightarrow \mathbf{D}(M')$  is a quotient map iff  $k[M'] \rightarrow k[M]$  is faithfully flat thus injective thus  $M' \rightarrow M$  is injective. Now the kernel of  $\mathbf{D}(M) \rightarrow \mathbf{D}(M')$  is represented by  $k[M]/I_{k[M']}$ , where  $I_{k[M']}$  is the augmentation ideal. Then it is isomorphic to  $k[M/M']$ .

3: Let  $H$  be an algebraic subgroup of  $G$ , then the map  $\Gamma(G) \rightarrow \Gamma(H)$  is surjective, and sends group-like elements to group-like elements, thus  $\Gamma(H)$  is also spanned by group-like elements, and  $H$  is diagonalizable.

if  $\mathbf{D}(M) \rightarrow Q$  is a quotient map, then its kernel is an algebraic subgroup thus equals  $\mathbf{D}(M'')$  for some quotient  $M''$  of  $M$ . Let  $M$  be the kernel of  $M \rightarrow M''$ , then  $\mathbf{D}(M) \rightarrow \mathbf{D}(M')$  and  $\mathbf{D}(M) \rightarrow Q$  are quotient maps with the same kernel, so they are isomorphic, by(10.1.1.22).  $\square$

**Prop.(10.2.3.5)[Representation of Diagonalizable Groups].** For  $R \in \mathcal{CRing}$  and  $G \in \text{GrpSch}^{\text{aff}}/R$ , the following conditions are equivalent:

1.  $G$  is diagonalizable.
2. Every representation of  $G$  is diagonalizable.
3. Every f.d. representation of  $G$  is diagonalizable.



┘

*Proof:* Cf.[Pseudo-Reductive Groups]P545.

1  $\rightarrow$  2: We need to show for any co-module  $\rho : V \rightarrow V \otimes \Gamma(G)$  (3.11.2.22),  $V$  is a sum of 1-dimensional representations, or equivalently, it is spanned by vectors  $u$  that  $\rho(u) \in Ru \otimes \Gamma(G)$ . Let  $v \in V$ , we can write  $\rho(v) = \sum u_i \otimes e_i$  where  $e_i$  are group-like elements in  $G$ .

Applying comodule relations (3.11.2.22), we get

$$\sum u_i \otimes e_i \otimes e_i = \sum \rho(u_i) \otimes e_i, \quad v = \sum u_i \text{ (3.11.4.17)}.$$

so  $\rho(u_i) = u_i \otimes e_i$  and they span  $V$ .

2  $\rightarrow$  1: The regular representation of  $G$  is diagonalizable, so  $\Gamma(G)$  is spanned by its eigenvectors, for any eigenvector  $f \in \Gamma(G)$ , so  $\mu(f) = f \otimes e$  where  $e$  is a group-like element. Applying  $\varepsilon \otimes \text{id}$  shows  $f = \varepsilon(f)e$ , so  $G$  is diagonalizable.

2  $\rightarrow$  3: trivial.

3  $\rightarrow$  2: Every representation of  $G$  is a sum of f.d. representation, so it is a sum of 1-dimensional representations, so it is diagonalizable by (10.3.1.17).  $\square$

### Tori

**Def. (10.2.3.6) [Linear Tori].** A **split torus** over  $k$  is a linear group scheme of the form  $T = \mathbb{G}_{m,k}^n$  for some  $n \in \mathbb{Z}_+$ , and a **linear torus** over  $k$  is defined to be  $T \in \text{linAlgGrp}/k$  s.t.  $T_{k^{\text{sep}}}$  is a split torus. If  $k = k^{\text{sep}}$ , then any torus is a split torus. In particular, any torus split over a finite separable extension.  $\square$

**Prop. (10.2.3.7).** By (10.2.3.4), a split torus is just a group scheme  $\mathbf{D}(\mathbb{Z}^n)$  for some  $n \in \mathbb{Z}_+$ , and quotient of a split torus is a split torus.

Thus a quotient of a torus is a torus, and an algebraic subgroup of a torus is a torus iff it is a group variety.  $\square$

**Prop. (10.2.3.8) [Classifying Tori].** For a torus  $T/k$  of dimension  $n$ ,  $X^\bullet(T) \cong \mathbb{Z}^n \cong X_\bullet(T)$  (10.1.1.11), and there is a natural perfect pairing  $X^\bullet(X) \times X_\bullet(X) \rightarrow \mathbb{Z}$ . In particular,

$$X^\bullet(-) : \{\text{tori over } k\} \rightarrow \{\text{finite free Abelian groups with a continuous action of } \text{Gal}_k\}$$

is an equivalence of categories, by Galois descent.  $\square$

*Proof:* This follows from (10.2.3.7) and (10.2.3.4).  $\square$

**Def. (10.2.3.9) [Monoidal Transformation].** For  $A \in \text{GL}(n; \mathbb{Z})$  with  $\det A = \pm 1$ , we define an isomorphism of  $\mathbb{G}_{n,k}$ :

$$\varphi_A(x) = (x_1^{a_{11}} x_2^{a_{12}} \dots x_n^{a_{1n}}, \dots, x_1^{a_{n1}} \dots x_n^{a_{nn}}).$$

these isomorphisms  $\varphi_A$  is called the **monoidal transformations**.  $\square$

**Prop. (10.2.3.10).** Let  $G \in \text{GrpVar}/k$  and  $T \leq G$  be a central torus, then

- $T \cap [G, G]$  is finite.
- If  $G/T$  is a perfect group, then the sequence

$$e \rightarrow T \cap [G, G] \rightarrow T \times [G, G] \rightarrow G \rightarrow e$$

is exact.

In particular,  $G/[G, G]$  is a torus.  $\lrcorner$

*Proof:* Cf. [Mil17]P246.  $\square$

**Def. (10.2.3.11) [Induced Tori].** An **induced torus** is a torus  $T$  over  $k$  s.t.  $X^\bullet(T)$  has a  $\mathbb{Z}$ -basis that is stable under the action of  $\text{Gal}_k$ . It is easy to see from (10.2.3.8) that this is equivalent to  $T$  being a finite product of torus of the form  $\text{Res}_{k'/k} \mathbb{G}_m$  where  $k'/k$  is a finite separable extension.  $\lrcorner$

**Cor. (10.2.3.12).** If  $T \in \text{Tori}/k$  is induced (10.2.3.11) and  $k \subset k' \subset k^{\text{sep}} \in \text{Field}$ , then  $T_{k'}$  is induced.  $\lrcorner$

**Prop. (10.2.3.13).** For any torus  $T/k$ , there exists an exact sequence

$$1 \rightarrow R_1 \rightarrow R_0 \rightarrow T \rightarrow 1$$

where  $R_1 \in \text{AlgGrp}/k$  is a tori and  $R_0 \in \text{AlgGrp}/k$  is an induced tori.  $\lrcorner$

*Proof:* Let  $k' \leq k^{\text{sep}}$  be a splitting field for  $T$ . Define  $R_0 = \text{Res}_{k'/k}(T_{k'})$  with a norm map  $\text{Nm}_{k'/k} : R_0 \rightarrow T$ , then  $\text{Nm}_{k'/k}$  is surjective with a connected kernel (by looking at based changed to  $k'$ ).  $\square$

### Groups of Multiplicative Type

**Def. (10.2.3.14) [Groups of Multiplicative Type].**  $G$  is called an **algebraic group of multiplicative type** over  $k$  if  $G_{k'}$  is diagonalizable (10.2.3.2) for some field extension  $k'/k$ . In this case, we say that  $G$  is split over  $k'$ .

Subgroups and quotient groups of groups of multiplicative type are also of multiplicative type, because this is true for diagonalizable groups (10.2.3.4).  $\lrcorner$

**Prop. (10.2.3.15) [Characterizing Groups of Multiplicative Type].** The following are equivalent:

- $G$  is of multiplicative type.
- $G$  is commutative and  $\text{Hom}(G, \mathbb{G}_a) = 0$ .
- $G$  is commutative and  $\Gamma(G)$  is coétale.
- $G_{k^{\text{sep}}}$  is diagonalizable.

$\lrcorner$

*Proof:* Cf. [Mil17]237.  $\square$

**Cor. (10.2.3.16).** An algebraic group over  $k$  becomes diagonalizable over some field extension of  $k$  iff it becomes diagonalizable over some finite separable extension of  $k$ .  $\lrcorner$

**Cor. (10.2.3.17).** If a group of multiplicative type splits over a purely inseparable extension of  $k$ , then it splits over  $k$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P238.  $\square$

**Cor. (10.2.3.18).** If  $G$  is smooth and commutative, then  $G$  is of multiplicative type iff  $G(\bar{k})$  consists of semisimple elements.  $\lrcorner$

*Proof:* We can assume that  $k = \bar{k}$ , and embed  $G$  into  $\text{GL}(n)$  for some  $n \in \mathbb{Z}_+$  (10.3.1.22). If  $G$  is of multiplicative type, then by (10.2.3.5), there is a basis that  $G \subset \mathbb{D}_n$ , so all the elements in  $G(k)$  is diagonalizable hence semisimple. Conversely, if  $G(k)$  are all semisimple, then they form a commutative family of semisimple elements, so  $G(k) \subset \mathbb{D}_n(k)$  in some basis. Because  $G$  is smooth thus reduced,  $G \subset \mathbb{D}_n$ .  $\square$

**Cor. (10.2.3.19).** An extension of algebraic groups of multiplicative type is of multiplicative type iff it is commutative.  $\lrcorner$

*Proof:* A exact sequence  $e \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow e$  of commutative group schemes gives rise to an exact sequence

$$0 \rightarrow \operatorname{Hom}(G'', \mathbb{G}_a) \rightarrow \operatorname{Hom}(G, \mathbb{G}_a) \rightarrow \operatorname{Hom}(G', \mathbb{G}_a)$$

of Abelian groups by (10.1.1.22), thus we can use characterization (10.2.3.15).  $\square$

**Thm. (10.2.3.20) [Classifying Groups of Multiplicative Type].** The functor  $X^*$  is an equivalence from the category of algebraic groups of multiplicative type over  $k$  to the category of f.g.  $\mathbb{Z}$ -modules equipped with a continuous action of  $\operatorname{Gal}_k$ , and the inverse is given by  $M \mapsto$  the Galois decent of  $\mathbf{D}(M)$ .  $\lrcorner$

*Proof:* This follows from (10.2.3.4) and Galois descent.  $\square$

**Cor. (10.2.3.21) [Largest Subtori].** If  $G$  is of multiplicative type, then there is a unique exact sequence

$$1 \rightarrow G_t \rightarrow G \rightarrow G'' \rightarrow 1$$

where  $G_t$  is a torus over  $k$  and  $G''$  is a finite group over  $k$ .  $G_t$  is the largest subtorus of  $G$ , and is also the unique subtorus of  $G$  with finite quotient, denoted by  $G_t$ . In particular, this exact sequence is compatible with base change of fields.

Then  $G_t = G_{\text{red}}^0$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P241.  $\square$

**Cor. (10.2.3.22).** A group variety of multiplicative type is a torus. And if  $H \leq G$  is a normal subgroup scheme, then  $H_t \leq G$  is a normal torus.  $\lrcorner$

**Prop. (10.2.3.23) [Representation of Groups of Multiplicative Type].** If  $G$  is of multiplicative type (e.g. a torus), then  $\operatorname{Rep}_k^{\text{fd}}(G)$  is a semisimple Abelian category, and the isomorphism classes of simple objects in  $\operatorname{Rep}_k^{\text{fd}}(G)$  are classified by the orbits of  $\operatorname{Gal}_k$  acting on  $X^\bullet(G)$ .

If  $(V, r) \in \operatorname{Rep}_k^{\text{fd}}(G)$  corresponds to an orbit  $\Sigma$ , and let  $\chi \in \Sigma$ , then  $\operatorname{End}(V) \cong k_\chi$ , where  $k_\chi$  is the subfield of  $k^s$  fixed by the stabilizer of  $\chi$  in  $\operatorname{Gal}_k$ . In particular,  $\dim_k k_\chi = \dim V_\chi$ .  $\lrcorner$

*Proof:* The group  $G$  is split by a finite Galois extension  $K/k$  by (10.2.3.16). Then  $\operatorname{Gal}(K/k)$  acts on  $\Gamma(G_K)$  through its action on  $K$ . Let  $(V, r)$  be a representation of  $G_K$  and let  $\rho$  be the corresponding co-action, then by (7.1.5.24), the functor  $V \mapsto V \otimes_k K$  induces an equivalence between  $\operatorname{Rep}(G)$  and  $\operatorname{Rep}_k^{\text{fd}}(G_K)$  with a semi-linear action of  $\operatorname{Gal}(K/k)$  fixing  $\rho$ .

Let  $V \in \operatorname{Rep}_k^{\text{fd}}(G)$ , then  $V_K$  decomposes as a representation of  $G_K$  into

$$K \otimes V = \bigoplus_{\chi \in X^*(G_K)} V_\chi.$$

and an element  $\gamma \in \operatorname{Gal}(K/k)$  maps  $V_\chi$  isomorphically onto  $V_{\sigma\chi}$ . Thus the set of  $\chi$  occurring in  $K \otimes V$  is stable under the action of  $\bar{\Gamma}$ .

Conversely, if  $\Sigma$  is an orbit of  $\operatorname{Gal}(K/k)$  in  $X^*(G_K)$  and  $V$  is a 1-dimensional  $K$  vector space, then  $\bigoplus_{\chi \in \Sigma} V_\chi$  has a natural semi-linear action of  $\operatorname{Gal}(K/k)$ , so it arises from a simple representation of  $G$  over  $k$ .  $\square$

**Prop. (10.2.3.24) [Density Theorem for Groups of Multiplicative Type].** Let  $G$  be a smooth of multiplicative type, then  $G$  is commutative.

- The only closed subscheme containing every  $G[n]$  is  $G$  itself.
- If  $G$  is smooth, then the only closed subscheme containing  $G[n]$  for  $n$  prime to characteristic of  $k$ , is  $G$  itself.

┘

*Proof:* 1: 2: Cf. [Mil17]P242. ?

□

**Cor. (10.2.3.25).** Suppose  $G$  is of multiplicative type, then any two homomorphisms from  $G$  to another algebraic group  $H$  that coincide on  $G[n]$  for all  $n \geq 1$  are equal.

┘

*Proof:* This is because the equalizer is a closed subscheme of  $G$ , as  $H$  is separated (10.1.1.10). □

**Prop. (10.2.3.26) [Rigidity Theorem for Groups of Multiplicative Type].** Let  $G, H$  be diagonalizable groups over  $k$ , and let  $X$  be a connected group scheme over  $k$ . Let  $\varphi : X \times G \rightarrow H$  be a morphism that for all  $k$ -algebra  $R$  and  $x \in X(R)$ , the map  $g \mapsto \varphi(x, g) : G(R) \rightarrow H(R)$  is a homomorphism. Then for any  $x_0 \in X(k)$ , we have  $\varphi(x, g) = \varphi(x_0, g)$  for any  $k$ -algebra  $R$  and  $(x, g) \in X(R) \times G(R)$ .

┘

*Proof:* Cf. [Mil17]P243. ?

□

**Cor. (10.2.3.27).** If  $G$  is connected, then any action of  $G$  on an algebraic group  $H$  of multiplicative type by group homomorphisms is trivial.

┘

**Cor. (10.2.3.28).** If  $G$  is connected, then every normal algebraic subgroup of multiplicative type of  $G$  is contained in the center of  $G$ .

┘

*Proof:* The action of  $G$  on  $N$  by inner automorphism is trivial.

□

**Cor. (10.2.3.29).** Let  $H$  be a subgroup of multiplicative type of an algebraic group  $G$ , then  $N_G(H)^0 = Z_G(H)^0$ , i.e.  $C_G(H)$  is an open subgroup of  $N_G(H)$ .

┘

*Proof:* The inner action of  $N_G(H)^0$  on  $H$  by inner automorphism is trivial.

□

**Cor. (10.2.3.30).** If  $N$  is a normal subgroup of an algebraic group  $H$  that  $N$  and  $H/N$  are of multiplicative type, then every action of a connected algebraic group  $G$  on  $H$  by group homomorphisms preserving  $N$  is trivial.

┘

*Proof:* The action of  $G$  on  $N$  is trivial, thus the action factors through  $G \times H/N \rightarrow H$ , thus also factors through  $G \times H/N \rightarrow N$ . Now the action is trivial, by (10.2.3.26). □

**Cor. (10.2.3.31) [Extension of Groups of Multiplicative Types].** An extension  $G$  of algebraic groups of multiplicative type is of multiplicative type if it is connected.

┘

*Proof:* The conjugation action of  $G$  on itself is trivial, by (10.2.3.30), thus  $G$  is commutative. Thus it is of multiplicative type by (10.2.3.19). □

**Cor. (10.2.3.32).** If  $G$  is connected and  $H' \leq H \leq G$  are normal subgroups of  $G$  s.t.  $H', H/H'$  are both of multiplicative type, then  $H$  is central and of multiplicative type.

┘

*Proof:* Consider the conjugate action of  $G$  on  $H$ , then  $H$  is central by (10.2.3.30). In particular  $H$  is commutative and thus of multiplicative type by (10.2.3.19). □

### Linearly Reductive Groups

**Def. (10.2.3.33) [Linearly Reductive Groups].**  $G$  is called **linearly reductive** if any  $\rho \in \mathcal{R}\text{ep}_k^{\text{fd}}(G)$  is totally decomposable.  $\lrcorner$

**Prop. (10.2.3.34).** Let  $k'/k \in \mathbf{Field}$ . If  $G_{k'}$  is linearly reductive, then so is  $G$ . Conversely, if  $G$  is linearly reductive and  $k'$  is separable over  $k$ , then  $G_{k'}$  is linearly reductive.  $\lrcorner$

*Proof:* Cf. [Mil17]P248.  $\square$

**Prop. (10.2.3.35).** If  $G$  is commutative, then it is linearly reductive iff it is of multiplicative type. In particular, a linear torus is linearly reductive, by (10.2.3.16).  $\lrcorner$

*Proof:* Cf. [Mil17]P248. ?  $\square$

**Prop. (10.2.3.36) [Hilbert].** Let  $G \subset \text{GL}(n)$  be linearly reductive and let  $A = k[T_1, \dots, T_n]$ , then  $A^G$  is f.g. as a  $k$ -algebra.  $\lrcorner$

*Proof:* Cf. [Mil17]P249. ?  $\square$

### 4 Actions of Tori

References are [CGP15].

**Notation (10.2.4.1).**

- Let  $G \in \text{linAlgGrp}/k$ .

$\lrcorner$

**Thm. (10.2.4.2) [Smoothness of Fixed Points].** If  $G \in \text{GrpVar}/k$  is linearly reductive (10.2.3.33) (e.g. a torus, by (10.2.3.35)) and acts on  $X \in \text{SmVar}/k$ , and  $S \subset G(k)$  is dense in  $|G|$ , then the fixed scheme  $X^S$  is smooth and equals  $X^G$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P254.  $\square$

**Cor. (10.2.4.3) [Fixed Points of Semisimple Elements].** If  $G \in \text{GrpVar}/k$  acts on  $X \in \text{SmVar}/k$  and  $s \in G(k)$  is semisimple, and let  $G' \leq G$  be the closed subgroup generated by  $s$ , then  $X^{G'} = X^s$  is smooth.  $\lrcorner$

*Proof:*  $G'$  is linearly reductive (10.2.3.33) because  $s$  is semisimple.  $\square$

### Dynamical Methods

**Notation (10.2.4.4).**

- Let  $G \in \text{GrpSch}^{\text{fg,aff}}/R$ .

$\lrcorner$

**Prop. (10.2.4.5).** For  $V \in \mathcal{R}\text{ep}^{\text{fd}}(\mathbb{G}_m)$  and  $v \in V$ , the orbit map

$$\mu_{[v]} : \mathbb{G}_m \rightarrow \mathbb{P}^1(V) : t \mapsto t[v]$$

extends uniquely to a map  $\mathbb{P}^1 \rightarrow \mathbb{P}^1(V)$ . And either  $[v]$  is a fixed point or the closure of its orbit has exactly two fixed points:  $\{\mu_{[v]}(0), \mu_{[v]}(\infty)\}$ .  $\lrcorner$

*Proof:* The case that  $[v]$  is fixed is easy, so we assume that  $[v]$  is not fixed. Write  $V = \oplus_{n \in \mathbb{Z}} V_n$  where  $\mathbb{G}_m$  acts on  $V_n$  via  $t \mapsto t^n$ , and write  $v = v_r + \dots + v_s$  where  $v_r \neq 0, v_s \neq 0$ . By looking at the affine charts, it is clear that  $\mu_{[v]}$  extends uniquely to a map  $\mathbb{P}^1 \rightarrow \mathbb{P}^1(V)$  s.t.  $\mu_{[v]}(0) = [v_r]$  and  $\mu_{[v]}(\infty) = [v_s]$ , and these are the only two fixed points.  $\square$

**Def. (10.2.4.6) [Concentrator Schemes].** For  $X \in \text{Aff}/R$  with an action of  $\mathbb{G}_m$  and  $Z \subset X$  a closed subscheme stable under the action, there is a  $\mathbb{Z}$ -gradation  $\Gamma(X) = \oplus_{n \in \mathbb{Z}} \Gamma(X)_n$  where  $\mathbb{G}_m$  acts on  $\Gamma(X)_n$  by  $t \mapsto t^n$  and similarly for  $\Gamma(Z)$ .  $\Gamma(X) \rightarrow \Gamma(Z)$  is a graded subjective homomorphism with kernel  $I(Z) = \oplus_{n \in \mathbb{Z}} I(Z)_n$ .

The **concentrator subscheme** of  $Z \subset X$  is the closed subscheme of  $X$  defined by the ideal generated by  $I(Z)_0 + \sum_{n \in \mathbb{Z}_-} \Gamma(X)_n$ , denoted by  $X_Z$ .  $\lrcorner$

**Prop. (10.2.4.7) [Dynamical Meaning of Concentrator Subschemes].** Situation as in (10.2.4.6), for  $R' \in \mathcal{C}\text{Ring}/R$ ,

$$X_Z(R') = \left\{ x \in X(R') : \lim_{t \rightarrow 0} tx \text{ exists and lies in } Z(R') \right\}.$$

And  $X_Z$  is unchanged if we replace  $\lambda$  by  $\lambda^n$  for some  $n \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* For  $x \in X(R')$ , the orbit map  $\mu_x : (\mathbb{G}_m)_{R'} \rightarrow X_{R'}$  corresponds to the map  $\sum_n f_n \rightarrow \sum_n x(f_n)T^n$ . Thus the limit exists iff  $x(f_n) = 0$  for any  $n \in \mathbb{Z}_{<0}$ , in which case the limit is the point in  $X(R)$  corresponding to the homomorphism  $\Gamma(X) \rightarrow R' : \sum f_n \mapsto x(f_0)$ . Thus the limits lies in  $Z(R')$  iff  $x \in X_Z(R')$ .

For the last assertion, Cf. [CGP] 2.1.7.  $\square$

**Prop. (10.2.4.8) [Smoothness of Concentrator Subschemes].** Situation as in (10.2.4.6), if  $X, Z$  are smooth, then  $X_Z$  is smooth.  $\lrcorner$

*Proof:* Cf. [Mil17] P261.  $\square$

**Prop. (10.2.4.9) [Concentrator Subgroups of Torus Actions].** If  $\lambda : \mathbb{G}_m \times G \rightarrow G \in \text{GrpSch}/R$  is an action of  $\mathbb{G}_m$  by group homomorphisms, then the concentrator subschemes (10.2.4.6)

$$P_G(\lambda) \triangleq G_G, \quad U_G(\lambda) \triangleq G_e$$

are subgroup schemes by (10.2.4.6). Then

1.  $P_G(\lambda) \leq G$  is a subgroup scheme,  $U_G(\lambda) \trianglelefteq P_G(\lambda)$  is a normal subgroup scheme.
- 2.

$$P(\lambda) \cap P(-\lambda) = Z(\lambda)$$

is the fixed point of  $\lambda$  on  $G$ , and  $P_G(\lambda) \cap U_G(-\lambda) = e$ .

3.  $P_G(\lambda), U_G(\lambda), Z_G(\lambda)$  are stable under base change, and if  $H \leq G$  is a subgroup scheme that is stable under action of  $T$ , then  $P_H(\lambda) = H \cap P_G(\lambda), U_H(\lambda) = H \cap U_G(\lambda), Z_H(\lambda) = H \cap Z_G(\lambda)$ .
4. If  $G/R$  is of f.t., then the fibers of  $U_G(\lambda)$  are unipotent.

$\lrcorner$

*Proof:* 1: This follows from (10.2.4.7).

2: If  $t.g = g$  for all  $t \in \mathbb{G}_m$ , then  $g \in P(\lambda) \cap P(-\lambda)$ . Conversely, if  $g \in P(\lambda) \cap P(-\lambda)$ , then  $\text{Orb}_g : \mathbb{G}_m \rightarrow G$  extends to a map  $\mathbb{P}^1 \rightarrow G$ , which can only be the constant map, thus  $t.g = g$  for all  $t \in \mathbb{G}_m$ . The last assertion is clear.

3: This follows from (10.2.4.7).

4: To show the fibers of  $U_G(\lambda)$  are unipotent, it suffices to show for  $R \in \mathbf{Field}$ . And then we can embed  $G$  into  $\text{GL}(n)$  for some  $n$  by (10.3.1.22). Then this follows from [CGP15]P2.1.1?  $\square$

**Cor. (10.2.4.10).** Situation as in (10.2.4.9), if  $G/R$  is of f.p., then

1.  $P_G(\lambda), U_G(\lambda), Z_G(\lambda)$  are also of f.p.
2. The fibers of  $U_G(\lambda)$  are connected, and if the fibers of  $G$  are connected, then the same hold for  $P_G(\lambda)$  and  $Z_G(\lambda)$ .
3.  $\text{Lie}(Z_G(\lambda)) = \text{Lie}(G)_0, \text{Lie}(U_G(\lambda)) = \text{Lie}(G)_+, \text{Lie}(P_G(\lambda)) = \text{Lie}(G)_{\geq 0}$ .
4. The multiplication map induces an group isomorphism  $Z_G(\lambda) \rtimes U_G(\lambda) \cong P_G(\lambda)$ .
5. If  $R \in \mathbf{Field}$ ,  $G$  is connected, and  $f : G \rightarrow H$  is an  $S$ -equivariant quotient map, then so are the maps  $P_G(\lambda) \rightarrow P_H(\lambda), U_G(\lambda) \rightarrow U_H(\lambda), Z_G(\lambda) \rightarrow Z_H(\lambda)$ .
6. If  $R = k \in \mathbf{Field}$ , the multiplication map  $U(-\lambda) \times P(\lambda) \rightarrow G$  is an open immersion. And if  $H \leq G$  is an algebraic subgroup stable under action of  $\mathbb{G}_m$ , then

$$H \cap (U_G(-\lambda') \times P_G(\lambda')) = U_H(-\lambda) \times P_H(\lambda).$$

And if moreover  $G/R$  is smooth, then

1.  $P_G(\lambda), U_G(\lambda), Z_G(\lambda)$  are smooth.
2. The multiplicative map  $U_G(\lambda) \times P_G(\lambda) \rightarrow G$  is an open immersion. This open immersion is an isomorphism if  $R \in \mathbf{Field}$  and  $G$  is connected and solvable.
3. If  $R \in \mathbf{Field}$ , then  $U_G(\lambda)$  is split unipotent.

┘

*Proof:* Cf. [CGP15]2.1.12, 2.2.9, 2.2.5.?

1: When  $G$  is of f.p.,  $G$  and the action  $\lambda$  descends to a Noetherian subring of  $R$ , then it is clear that  $P_G(\lambda), U_G(\lambda), Z_G(\lambda)$  are of f.p..

2: Cf. [CGP15]Prop2.1.8.?

3:  $\text{Lie}(P_G(\lambda)) = P_G(\lambda)(R[\varepsilon]) \cap \text{Lie}(G)$ , so? Cf. [CGP15]2.1.8.

And  $\text{Lie}(Z_G(\lambda)) = \text{Lie}(P_G(\lambda)) \cap \text{Lie}(P_G(-\lambda)) = \text{Lie}(G)_0$ .

4: Notice that the group homomorphism  $P_G(\lambda) \rightarrow Z_G(\lambda) : g \mapsto \lim_{t \rightarrow 0} t.g$  is a projection.

5: As  $f$  is flat, the induced map of open subschemes

$$U_G(-\lambda) \times Z_G(\lambda) \times U_G(\lambda) = U_G(-\lambda) \times P_G(\lambda) \rightarrow U_H(-\lambda) \times P_H(\lambda) = U_H(-\lambda) \times Z_H(\lambda) \times U_H(\lambda)$$

is flat. Thus the map on each factor is flat, thus open (5.4.1.30), but the image is also closed by (10.1.5.3). Now by item2, the maps are surjective, in particular they are quotient maps by (10.1.5.4).

6: Cf. [CGP15]Prop2.1.8.

For the smooth case:

1: Cf. [CGP15]Prop2.1.8.

2: Cf. [CGP15]Prop2.1.12.

3: Cf. [CGP15]Prop2.1.10.  $\square$



**Cor. (10.2.4.11).** If  $G \in \text{GrpVar}/k$  is non-solvable, then there exists a sub-torus  $T \leq G$  s.t.  $Z_G(T)$  is a subgroup variety of smaller dimension as  $G$ .  $\lrcorner$

*Proof:* If every maximal tori of  $G$  is central, take a maximal torus  $T \leq G$ , then  $G/T$  is unipotent by (10.3.4.18). Thus  $G$  is solvable, contradiction.  $\square$

**Prop. (10.2.4.12) [Weil Restrictions of Concentrator Subgroups].** Let  $R'/R \in \mathcal{CRing}$  is finite locally free,  $G' \in \text{GrpSch}^{\text{aff}, \text{pf}}/R$  with  $G = \text{Res}_{R'/R}(G')$ , and  $\lambda' \in X_*(G')$  with  $\lambda = \mathbb{G}_m \rightarrow \text{Res}_{R'/R} \mathbb{G}_m \rightarrow G \in X_*(G)$ , then we have

$$P_G(\lambda) = \text{Res}_{R'/R}(P_{G'}(\lambda')), \quad U_G(\lambda) = \text{Res}_{R'/R}(U_{G'}(\lambda')), \quad Z_G(\lambda) = \text{Res}_{R'/R}(Z_{G'}(\lambda')) \quad (10.2.4.9).$$

$\lrcorner$

*Proof:* Cf. [CGP15]P61.  $\square$

**Def. (10.2.4.13) [Roots].** Suppose  $R$  is local and  $S/R$  is a non-trivial split torus  $?$  and  $G/R$  is smooth with an action of  $T$ , then by (10.2.3.5),  $\mathfrak{g} = \text{Lie}(G)$  decomposes as

$$\mathfrak{g} = \mathfrak{g}_0 \otimes \bigotimes_{\alpha \in X^\bullet(T)} \mathfrak{g}_\alpha,$$

where  $\mathfrak{g}^0 = \mathfrak{g}^S = \text{Lie}(G^S) = \text{Lie}(Z_G(S)) = \text{Lie}(S) = \mathfrak{t}$  (10.2.2.12)(10.4.2.15) and  $S$  acts on  $\mathfrak{g}_\alpha$  by the character  $\alpha$  for  $\alpha \in X^\bullet(S)$ . The set of  $\alpha \in X^\bullet(S)$  s.t.  $\mathfrak{g}_\alpha \neq 0$  is denoted by  $\widehat{\Phi}(G, S)$ , and  $\Phi(G, S) = \widehat{\Phi}(G, S) \setminus \{0\}$  is called the set of **roots of**  $(G, S)$ , denoted by  $\Phi(G, S)$ .

Also denote  $\text{wt}(G, S)$  the sub-semigroup of  $X^\bullet(S)$  generated by  $\Phi(G, S)$ .  $\lrcorner$

**Prop. (10.2.4.14).** Let  $R = k$ , and  $G/k$  smooth with an action of a split torus  $S/k$ . If  $H \leq G$  is a subgroup variety that is stable under action of  $T$ , then  $\Phi(S, H) = \Phi(S, G) \cap \text{wt}(G, S)$  (10.2.4.13).  $\lrcorner$

*Proof:* Cf. [Pseudo-Reductive Groups]Chap3.3.  $\square$

**Prop. (10.2.4.15).** Let  $R = k$ , and  $G/k$  smooth with an action of a split torus  $S/k$ . If  $H_1, H_2 \leq G$  are subgroup varieties that are stable under action of  $T$ , and  $H = \langle H_1, H_2 \rangle$ ,  $\widehat{\Phi}(S, H) = \widehat{\Phi}(S, G) \cap \text{wt}(G, S)$  (10.2.4.13). then

- $\text{wt}(S, H)$  is the semigroup generated by  $\text{wt}(S, H_1)$  and  $\text{wt}(S, H_2)$ .
- $\text{wt}(S, [H_1, H_2]) \subset \text{wt}(S, H_1) + \text{wt}(S, H_2)$ .

$\lrcorner$

*Proof:* Cf. [Pseudo-Reductive Groups]Prop3.3.5.  $\square$

**Thm. (10.2.4.16) [Subgroup varieties and Weight Groups].** Let  $R = k$ , and  $G/k$  smooth with an action of a split torus  $S/k$ . For any sub-semigroup  $A \leq X^\bullet(S)$ , there exists a unique  $S$ -invariant subgroup variety  $H_A \leq G$  s.t.

$$\text{Lie}(H_A) = \bigoplus_{\alpha \in A \cap \widehat{\Phi}(G, S)} \text{Lie}(G)_\alpha.$$

And any  $S$ -invariant subgroup variety  $H' \leq G$  s.t.  $\widehat{\Phi}(H', S) \subset A$  is contained in  $H_A$ .

When  $0 \notin A$ ,  $H_A$  is unipotent, so also denoted by  $U_A$ .  $\lrcorner$

*Proof:* Cf. [Pseudo-Reductive Groups]Prop3.3.6.  $\square$



**Prop. (10.2.4.17) [Filtrations on Unipotent Subgroups].** Let  $R = k$ , and  $G/k$  smooth with an action of a split torus  $S/k$ . For any  $a \in X^\bullet(S) \setminus \{0\}$ , let  $A = \mathbb{Z}_+ \cdot a$ ,  $A^j = \mathbb{Z}_{\geq j} a$  where  $j \in \mathbb{Z}_+$ , and  $U_{ja} = U_{A^j}$  (10.2.4.16), then

- Each  $U_{(j+1)a}$  is normal in  $U_{ja}$ , and the quotients  $U_{ja}/U_{(j+1)a}$  are vector groups. If  $U_{ja}/U_{(j+1)a} \neq 1$ , then  $\widehat{\Phi}(U_{ja}/U_{(j+1)a}, S) = \{ja\}$ .

┘

*Proof:* Cf. [Pseudo-Reductive Groups] Prop 3.3.8. □

**Prop. (10.2.4.18).** Let  $R = k$ ,  $G \in \text{GrpVar}/k$  be solvable with an action of a split torus  $S/k$ . Suppose there is a decomposition  $\widehat{\Phi}(G, S) = \coprod_{i \in [n]_+} \Psi_i$  s.t. the sub-semigroups  $A_i$  generated by  $\Psi_i$  are pairwise-disjoint, then the natural multiplication map

$$H_{A_1} \times \dots \times H_{A_n} \rightarrow G$$

is a group isomorphism. ┘

**Prop. (10.2.4.19) [Integral Root Groups].** Let  $(\mathcal{O}_K, K, \kappa) \in p\text{-Field}$ ,  $R = \mathcal{O}_K$  and  $\mathcal{S} \in \text{GrpVar}/\mathcal{O}_K$  be a split torus, and  $\mathcal{G}/\mathcal{O}_K$  is smooth. For any  $\alpha \in X^\bullet(\mathcal{S})$ ,  $\mathcal{S}_\alpha = \ker(\alpha) \subset \mathcal{S}$  is isomorphic to  $\mathbf{D}(X^\bullet(\mathcal{S})/\mathbb{Z} \cdot \alpha)$ , thus flat over  $\mathcal{O}_K$ , and  $\mathcal{S}/\mathcal{S}_\alpha \cong \mathbb{G}_m$ . Denote  $S = \mathcal{S}_K$ ,  $G = \mathcal{G}_K$ , then  $X^\bullet(\mathcal{S}) \cong X^\bullet(S) \cong X^\bullet(\mathcal{S}_\kappa)$ .

Given any root  $\alpha \in \Phi(\mathcal{G}, \mathcal{S})$  (10.2.4.13), there is a unique closed  $\mathcal{O}_K$ -smooth subgroup scheme  $\mathcal{U}_\alpha \leq \mathcal{G}^0$  s.t.

- $\mathcal{U}_\alpha$  is stable under the action of  $\mathcal{S}$  and  $\mathcal{S}_\alpha$  acts trivially on it.
- $(\mathcal{U}_\alpha)_\kappa$  is connected and unipotent.
- 

$$\text{Lie}(\mathcal{U}_\alpha) = \bigoplus_{\beta \in \mathbb{Q}_+ \cdot \alpha \cap \Phi} \mathfrak{g}_\beta.$$

And this  $\mathcal{U}_\alpha$  also satisfies

- $\mathcal{U}_\alpha$  is the schematic closure of  $U_\alpha$  in  $\mathcal{G}$  (7.4.5.13).
- $\mathcal{U}_\alpha(\mathcal{O}_K) = U_\alpha(K) \cap \mathcal{G}(\mathcal{O}_K)$ .
- $(\mathcal{U}_\alpha)_\kappa$  is the  $\alpha$ -root group for the action of  $\mathcal{S}_\kappa$  on  $\mathcal{G}_\kappa$ .

This  $\mathcal{U}_\alpha$  is called the **integral  $\alpha$ -root group** of  $\mathcal{G}$  w.r.t.  $\mathcal{S}$ . ┘

*Proof:* Cf. [K-P23] P134. □

**Prop. (10.2.4.20) [Root Groups].** Suppose  $G \in \text{GrpVar}^{\text{aff}}/k$  and  $S \leq G$  is a split torus. For  $\alpha \in \Phi(G, T)$  (10.2.4.13), denote

- $S_\alpha = \ker(\alpha)_t$ .
- $G_\alpha = C_G(S_\alpha)$ , which is a subgroup variety of  $G$  with Lie algebra

$$\text{Lie}(G_\alpha) = \mathfrak{g}^{S_\alpha} = \mathfrak{g}^S \oplus \bigoplus_{\beta \in \mathbb{Q}_+ \cdot \alpha \cap \Phi(G, S)} \mathfrak{g}_\beta.$$

- $U_\alpha = U_{G_\alpha}(\lambda)$ , where  $\lambda \in X_\bullet(S)$  is any cocharacter s.t.  $\langle \alpha, \lambda \rangle > 0$ .
- $G^\alpha = [G_\alpha, G_\alpha]$ .

Then

- These are all group varieties, and  $U_\alpha$  is normalized under  $S$ .
- 

$$\mathrm{Lie}(U_\alpha) = \bigoplus_{\beta \in \mathbb{Z}_+, \alpha \cap \Phi(G, S)} \mathfrak{g}_\beta.$$

- For any  $T$ -normalized subgroup variety  $H \leq G$ ,  $H$  contains  $U_\alpha$  (resp. is contained in  $U_\alpha$ ) iff  $\mathrm{Lie}(U) \subset \mathrm{Lie}(U_\alpha)$  (resp.  $\mathrm{Lie}(U_\alpha) \subset \mathrm{Lie}(U)$ ). In particular  $U_\alpha$  is independent of  $\lambda$  chosen, and is called the **root group** of  $\alpha$ .

┘

*Proof:* Cf. [Mil17]P348? and [K-P23]P135.

□

**Cor. (10.2.4.21).**  $\dim G = \dim T + \#\Phi$ .

┘

*Proof:* Because both  $G, T$  are smooth, this follows from (10.1.4.6).

□

**Cor. (10.2.4.22).** Notation as in (10.2.4.20),  $G_\alpha$  is generated by  $U_\alpha, U_{-\alpha}$  and  $T$ , and  $G^\alpha$  is generated by  $U_\alpha$  and  $U_{-\alpha}$ .

┘

*Proof:* Because both  $G, T$  are smooth, we can check on the level of Lie algebras.? And use the fact  $\mathrm{SL}(2)$  is generated by  $U^+$  and  $U^-$ .

□

**Lemma (10.2.4.23).** If  $k = \bar{k}, G \in \mathrm{GrpVar}/k$  and  $T \leq G$  is a torus, then  $C_G(T) \leq \cup_{B \in \mathcal{B}^T} B$ , where  $\mathcal{B}^T$  is the set of Borel subgroups of  $G$  containing  $T$ .

┘

*Proof:* Cf. [Mil17]P363.

□

**Prop. (10.2.4.24).** If  $G \in \mathrm{GrpVar}$  is solvable and  $T \leq G$  is a torus, then  $Z_G(T)$  is connected and  $G = UZ_G(T)$  for some unipotent subgroup  $U \leq G$ .

┘

*Proof:* Cf. [Mil17]P399.?

□

### Białynicki-Birula Decomposition

**Prop. (10.2.4.25)** [Białynicki-Birula Decomposition]. Cf. [Mil17]P272.

┘

## 5 Cohomologies and Extensions

**Notation (10.2.5.1).**

- Let  $G \in \mathrm{AlgGrp}/k$ .
- Use notations on non-Abelian cohomologies defined in [Profinite Cohomology](#).

┘

**Def. (10.2.5.2)** [Galois Cohomology of Algebraic Groups]. For  $G \in \mathrm{AlgGrp}/k$  and  $L/k \in \mathrm{GalField}$ , denote

$$H^1(\mathrm{Gal}(K/k), G) \triangleq H^1(\mathrm{Gal}(K/k), G(K)), \quad H^1(k, G) \triangleq H^1(\mathrm{Gal}_k, G(k^{\mathrm{sep}})) \quad (9.8.4.1),$$

where  $G(K)$  or  $G(k^{\mathrm{sep}})$  is equipped with the discrete topology.

┘

**Prop. (10.2.5.3) [Long Exact Sequence].** Let  $e \rightarrow N \rightarrow G \rightarrow Q \rightarrow e$  be an exact sequence in  $\mathcal{AlgGrp}/k$  s.t.  $N/k$  is smooth (e.g.  $\text{char } k = 0$  (10.1.4.3)), then there is an exact sequence

$$1 \rightarrow N(k) \rightarrow G(k) \rightarrow Q(k) \rightarrow H^1(k, N) \rightarrow H^1(k, Q) \rightarrow H^1(k, Q).$$

┘

*Proof:* This follows from (10.1.5.6) and (9.8.4.4). □

**Def. (10.2.5.4) [Inner Automorphisms].** An **inner automorphism** of  $G$  is an automorphism that its restriction on  $G_{\bar{k}}$  is conjugation by some  $g \in G(\bar{k})$ . ┘

**Prop. (10.2.5.5).** The group of inner automorphisms of  $G$  are isomorphic to  $G_{\text{ad}}(k)$ . ┘

*Proof:* There is an embedding of groups  $G^{\text{ad}} \hookrightarrow \text{Aut}(G)$ , and  $x \in \text{Aut}(G)(k)$  is an inner automorphism iff it is contained in  $G_{\text{ad}}(\bar{k})$ , which means it is contained in  $G_{\text{ad}}(k)$ . □

### Forms of Algebraic Groups

**Thm. (10.2.5.6) [Forms of Vector Spaces and Galois Cohomologies].** Let  $V \in \mathcal{Vect}^{\text{fin}}/k$  and  $t_i \in V^{\otimes r_i} \otimes (V^{\vee})^{\otimes s_i}$ , and let  $G \leq \mathbf{GL}(V)$  is the algebraic subgroup fixing each  $t_i$ , then for any  $K/k \in \mathcal{GalField}$ ,  $H^1(\text{Gal}(K/k), G)$  classifies  $K/k$ -forms of  $(V, \{t_i\})$ . (This is a special case of (7.1.5.2)). ┘

*Proof:* Let  $(V', t')$  be a  $K/k$ -form of  $(V, t)$  with an isomorphism  $f : (V, \{t_i\})_K \cong (V', \{t'_i\})_K$ , then the map

$$\sigma \mapsto a_{\sigma}(f) = f^{-1} \circ \sigma(f), \text{ where } \sigma \in \text{Gal}(K/k)$$

defines a 1-cocycle in  $H^1(\text{Gal}(K/k), G)$ , and the cohomology class doesn't depend on  $f$  (9.8.4.1). And if another  $K/k$ -form  $(V_1, \{t_{1,i}\})$  with an isomorphism  $f_1 : (V, \{t_i\})_K \cong (V_1, \{t_{1,i}\})_K$  satisfies  $[a_{\sigma}(f)] = [a_{\sigma}(f_1)]$ , then there exists  $g \in G(K)$  s.t.

$$f_1^{-1} \circ \sigma(f_1) = g^{-1} f^{-1} \circ \sigma(f) \sigma(g), \text{ where } \sigma \in \text{Gal}(K/k).$$

Thus  $f \circ g \circ f_1^{-1} = \sigma(f \circ g \circ f_1^{-1})$  for any  $\sigma \in \text{Gal}(K/k)$ . Thus  $f \circ g \circ f_1^{-1}$  is an isomorphism from  $(V_1, \{t_{1,i}\})$  to  $(V, \{t_i\})$ .

Conversely, for any cohomology class  $(a_{\sigma})_{\sigma \in \text{Gal}(K/k)} \in H^1(\text{Gal}(K/k), G)$ , define a new semi-linear action of  $\text{Gal}(K/k)$  on  $V_K$  by

$$\sigma'(x) = a_{\sigma} \cdot \sigma(x),$$

and define

$$V' = \{x \in V_K \mid \sigma'(x) = x\}.$$

Then  $V'_K = V_K$  by (7.1.5.24), and  $\sigma'(t_i) = t_i$  for any  $\sigma \in \text{Gal}(K/k)$ , so  $t_i \in (V')^{\otimes r_i} \otimes (V'^{\vee})^{\otimes s_i}$ . Thus  $(V', \{t_i\})$  is a  $K/k$ -form of  $(V, t)$  with  $a_{\sigma}(\text{id}) = a_{\sigma}$ . □

**Def. (10.2.5.7) [Forms of Algebraic Groups].** For  $K/k \in \mathcal{Field}$ , two algebraic groups  $G, G' \in \mathcal{AlgGrp}/k$  are called  $K/k$ -forms of each other if  $G_K \cong G'_K$ . ┘

**Thm. (10.2.5.8) [Forms and Galois Cohomologies].** For  $G \in \mathcal{AlgGrp}/k$  and  $K/k \in \mathcal{GalField}$ , the cohomology group  $H^1(\text{Gal}(K/k), \underline{\text{Aut}}(G))$  classifies  $K/k$ -forms of  $G$  (10.2.5.7). The form  $G'$  corresponding to  $(a_{\sigma})_{\sigma \in \text{Gal}(K/k)}$  satisfies

$$G'(k) = \{x \in G(k^{\text{sep}}) \mid a_{\sigma}(\sigma(x)) = x, \forall \sigma \in \text{Gal}(K/k)\}.$$

┘

*Proof:* This is a special case of (7.1.5.2). Alternatively it can be proved the same way as the proof of (10.2.5.6). □

### Inner Forms

**Def.(10.2.5.9) [Inner Forms of Algebraic Groups].** For  $G \in \text{AlgGrp}/k$  and  $K/k \in \text{GalField}$ , a  $K/k$ -form  $G'$  of  $G$  (10.2.5.7) is called an  $K/k$ -**inner form** of  $G$  if its corresponding cohomology class in  $H^1(\text{Gal}(K/k), \underline{\text{Aut}}(G))$  (10.2.5.8) is contained in the image of the map

$$H^1(\text{Gal}(K/k), G/Z(G)) \rightarrow H^1(\text{Gal}(K/k), \underline{\text{Aut}}(G)).$$

And it is called a **pure inner form** if it is contained in the image of the map

$$H^1(\text{Gal}(K/k), G) \rightarrow H^1(\text{Gal}(K/k), \underline{\text{Aut}}(G)).$$

┘

**Cor.(10.2.5.10).** If  $G, G'$  are inner forms, then  $Z(G) \cong Z(G')$ . ┘

**Def.(10.2.5.11) [Inner Twistings, Vogan[Vog93]].** For  $G \in \text{AlgGrp}/k$  and  $K/k \in \text{GalField}$ , a  $K/k$ -**inner twisting** of  $G$  is a pair  $(G', \xi)$  where  $G' \in \text{AlgGrp}/k$  and  $\xi : G_K \rightarrow (G_1)_K \in \text{AlgGrp}/K$  an isomorphism s.t.  $\xi^{-1}\xi^\sigma \in G_{\text{ad}}(K)$  for all  $\sigma \in \text{Gal}(K/k)$ .

Two  $K/k$ -inner twistings  $(G_1, \xi_1), (G_2, \xi_2)$  are called **isomorphic inner twisting** if there exists an isomorphism  $f : G_1 \rightarrow G_2$  s.t.  $\xi_2^{-1} \circ f \circ \xi_1$  is an inner form of  $G$ . In this way, the set of isomorphism classes of  $K/k$ -inner twists of  $G$  are in bijection with  $H^1(\text{Gal}(K/k), G/Z(G))$ .

Similarly, a  $K/k$ -**pure inner twisting** of  $G$  is a pair  $(G', \xi, z)$  where  $G' \in \text{AlgGrp}/k$  and  $\xi : G_K \rightarrow (G_1)_K \in \text{AlgGrp}/K$  an isomorphism,  $z \in Z^1(\text{Gal}(K/k), G(K))$  s.t.  $\xi^{-1}\xi^\sigma = \text{Ad}(z_\sigma)$  for all  $\sigma \in \text{Gal}(K/k)$ .

Two pure  $K/k$ -inner twistings  $(G_1, \xi_1, z_1), (G_2, \xi_2, z_2)$  are called **isomorphic pure inner twisting** if there exists an isomorphism  $f : G_1 \rightarrow G_2$  and  $g \in G(K)$  s.t.  $\xi_2^{-1} \circ f \circ \xi_1 = \text{Ad}(g)$  and  $(z_2)_\sigma = g(z_1)_\sigma \sigma(g^{-1})$  for all  $\sigma \in \text{Gal}(K/k)$ . In this way, the set of isomorphism classes of pure  $K/k$ -inner twists of  $G$  are in bijection with  $H^1(\text{Gal}(K/k), G(K))$ . ┘

**Prop.(10.2.5.12) [Hilbert's Theorem 90].** For  $K/k \in \text{GalField}$ ,  $H^1(\text{Gal}(K/k), \text{GL}(n)) = 1$  (10.2.5.2). In particular,  $H^1(k, \text{GL}(n)) = 1$ .

More generally, if  $A \in \text{Ring}^{\text{fin}}/k$ , then  $H^1(\text{Gal}(K/k), A^\times) = 1$ . ┘

*Proof:* We prove any cocycle is a coboundary, for this, notice any cocycle factor through a finite quotient (because it maps  $e$  to  $1$ ), and the images of it is contained in a finite extension of  $K$ , hence it reduces to the case when  $[K : k] < \infty$ . Then the assertion follows from (10.2.5.6).

For the second assertion, notice that if  $W, W'$  are  $A$ -modules s.t.  $W_K \cong W'_K$  as  $A_K$ -modules, then  $W^{\oplus [K:k]} \cong (W')^{[K:k]}$ , thus  $W \cong W'$  as  $A$ -modules, by Krull-Schmidt theorem (Cf. [Krull-Schmidt categories and projective covers]?). ┘

**Cor.(10.2.5.13).** For  $K/k \in \text{GalField}$ , we have  $H^1(\text{Gal}(K/k), \text{SL}(n)) = 1$  (10.2.5.2). In particular,  $H^1(k, \text{SL}(n)) = 1$ . ┘

*Proof:* This is seen from the exact sequence  $1 \rightarrow \text{SL}(n; K) \rightarrow \text{GL}(n; K) \rightarrow K^\times \rightarrow 1$ . ┘

**Prop.(10.2.5.14) [No Pure Inner forms for  $\text{Sp}(n)$ ].** If  $\text{char } k \neq 2$  and  $K/k \in \text{GalField}$ , then  $H^1(\text{Gal}(K/k), \text{Sp}(n)) = 1$  (10.2.5.2). In particular,  $H^1(k, \text{Sp}(n)) = 1$ . ┘

*Proof:* Similar to the proof of (10.2.5.12), we may assume that  $[K : k] < \infty$ . Then this follows from (10.2.5.6) and the fact that any two non-degenerate alternating forms on  $k^n$  are congruent, by (3.5.11.26). ┘

**Prop. (10.2.5.15) [Non-degenerate Quadratic Spaces].** If  $\text{char } k \neq 2$  and  $V/k$  is a non-degenerate quadratic space, then  $H^1(k, \text{O}(V))$  classifies the isomorphism classes of non-degenerate quadratic spaces over  $k$  of dimension  $\dim V$ .  $\lrcorner$

*Proof:* This follows from (10.2.5.6) and the fact that any two non-degenerate quadratic forms on  $(k^{\text{sep}})^{\oplus n}$  are congruent.  $\square$

**Prop. (10.2.5.16) [No Inner Forms over Finite Fields].** If  $k \in \text{Field}^{\text{fin}}$  and  $G \in \text{GrpVar}/k$ , then  $H^1(k, G) = 1$ .  $\lrcorner$

*Proof:* For any 1-cocycle  $f : \text{Gal}_k \rightarrow G(\bar{k})$ , by Lang's theorem (10.6.0.2), there exists  $g \in G(\bar{k})$  s.t.  $f(\varphi_k) = g^{-1} \text{Frob}(g)$ . Thus  $f$  agrees with the principle cocycle defined by  $g$  on  $\varphi_k$ , thus they agree for any power of  $\varphi_k$ , and thus  $f$  is a coboundary by continuity.  $\square$

## 6 Triangularizable Algebraic Groups

**Notation (10.2.6.1).**

- Let  $k \in \text{Field}$ .
- Let  $G \in \text{linAlgGrp}/k$ .

$\lrcorner$

**Def. (10.2.6.2) [Triangularizable Algebraic Groups].** A **triangularizable algebraic group** is an algebraic group s.t. every simple representation has dimension 1. Equivalently, for any  $V \in \text{Rep}^{\text{fd}}(G)$  (or for some faithful representation  $V \in \text{Rep}^{\text{fd}}(G)$ ), there exists a basis of  $V$  s.t.  $G$  is mapped into  $\text{B}(n)$ .  $\lrcorner$

**Prop. (10.2.6.3).** Unipotent algebraic groups and diagonalizable algebraic groups are triangularizable by definitions (10.3.4.3) (10.2.3.5).  $\lrcorner$

**Prop. (10.2.6.4) [Characterizing Triangularizable Groups].** The following are equivalent:

- $G$  is triangularizable.
- Every  $V \in \text{Rep}^{\text{fd}}(G)$  is triangularizable.
- $G$  is isomorphic to an algebraic subgroup of  $\text{B}(n)$  for some  $n \in \mathbb{Z}_+$ .
- There exists a normal unipotent algebraic subgroup  $U \leq G$  s.t.  $G/U$  is diagonalizable.

$\lrcorner$

*Proof:* Cf. [Mil17] P325. ?  $\square$

**Cor. (10.2.6.5).** Subgroups, quotients and base change of triangularizable subgroups are triangularizable.  $\lrcorner$

**Prop. (10.2.6.6) [Largest Unipotent Subgroups].** The following are equivalent:

- $G_{k'}$  is triangularizable for some separable field extension  $k'/k$ .
- $G$  contains a normal unipotent algebraic subgroup  $G_u \trianglelefteq G$  s.t.  $G/G_u$  is of multiplicative type. And in these cases,  $G_u$  is unique and contains every unipotent algebraic subgroup of  $G$ , called the **largest unipotent subgroup** of  $G$ , and we have:

- $G_u$  is the unique normal unipotent subgroup of  $G$  with quotient of multiplicative type. In particular,  $G_u$  is compatible with base change of fields.

- $G_u$  is the smallest normal subgroup of  $G$  with quotient of multiplicative type. In particular, the formation of  $G_u$  commutes with base change of fields.
- $G_u(\bar{k}) = G(\bar{k})_u$ .

⌋

*Proof:* Cf. [Mil17]P325, 326. ?

□

**Prop. (10.2.6.7) [Structure of Triangularizable Groups].** If  $G$  is a triangularizable group, then there exists a normal series

$$G \triangleright G_0 \triangleright G_1 \triangleright \dots \triangleright G_r = e$$

s.t.

- $G_0 = G_u$  (10.2.6.6).
- For  $i \in \mathbb{N}$ , the action of  $G$  on  $G_i/G_{i+1}$  factors through  $G/G_u$ , and there exists an embedding  $G_i/G_{i+1} \hookrightarrow G_a$  that is equivariant w.r.t. some linear action of  $G/G_u$  on  $G_a$ .

⌋

*Proof:* Cf. [Mil17]P330.

□

**Cor. (10.2.6.8).** If  $k = k^{\text{perf}}$ ,  $G$  is smooth and  $G_u \in \text{GrpVar}/k$ , then the above normal series can be chosen s.t. each  $G_i$  is a group variety, and the embedding  $G_i/G_{i+1} \hookrightarrow G_a$  is an isomorphism.

⌋

*Proof:* Cf. [Mil17]P331.

□

**Cor. (10.2.6.9) [Unipotent Groups split over  $k^{\text{perf}}$ ].** If  $k = k^{\text{perf}}$  and  $U \in \text{GrpVar}/k$  is unipotent, then  $U$  has a central normal series with components isomorphic to  $G_a$ . Moreover, if  $T \cong G_m^r$  is a torus acting on  $U$ , then it is possible to choose the normal series s.t. the action of  $T$  on the components are linear.

⌋

*Proof:*  $G = U \rtimes T$  is a triangularizable group variety by (10.2.6.4), and the assertion follows from (10.2.6.8) for  $G$ .

□

**Prop. (10.2.6.10) [Splitting of Triangularizable Groups].** Let  $e \rightarrow G_u \rightarrow G \xrightarrow{q} D \rightarrow e$  be an exact sequence of algebraic groups over  $k$  s.t.  $G_u$  is unipotent and  $D$  is smooth of multiplicative type, then the sequence splits if either of the following holds:

- $G_u$  is commutative and  $q$  admits a section as a map of schemes.
- $k = \bar{k}$ .
- $k = k^{\text{perf}}$  and  $G_u$  is connected (e.g.  $G$  is connected, by (10.2.6.11)).
- $G_u$  is a split unipotent group.

And in each case, any two right inverse  $s, s' : D \rightarrow G \in \text{AlgGrp}/k$  of  $q$  are conjugate by an element of  $G_u(k)$ , and subgroups of multiplicative type in  $G$  that are maximal are of the form  $s(D)$  for some right inverse  $s : D \rightarrow G \in \text{AlgGrp}/k$  of  $q$ . In particular, they are conjugate under  $G_u(k)$ .

⌋

*Proof:* Cf. [Mil17]P334.

□

**Cor. (10.2.6.11) [Smoothness and Connectedness of  $G_u$ ].** Situation as in (10.2.6.10), if  $G$  is smooth (resp. connected), then so are  $G_u$  and  $G/G_u$ , by (10.1.4.17) and (10.1.4.2), because after base change to  $\bar{k}$ ,  $G \cong G_u \times (G/G_u)$  as schemes.

⌋

## 7 Solvable Groups

**Notation(10.2.7.1).**

- Let  $k \in \mathbf{Field}$ .
- Let  $G \in \mathbf{linAlgGrp}/k$ .

┘

**Thm.(10.2.7.2) [Solvable Group Varieties over Perfect Fields].** Suppose  $k = k^{\text{perf}}$  and  $G$  is a solvable group variety, then

- $G$  is triangularizable after a separable base change by(10.2.6.6), and the largest unipotent subgroup  $G_u \trianglelefteq G$  is a group variety, and the formation of  $G_u$  is compatible with base change of fields.
- The canonical exact sequence  $e \rightarrow G_u \rightarrow G \xrightarrow{q} D \rightarrow e$  splits, and  $D$  is a torus.
- Any two splittings  $s_1, s_2 : D \rightarrow G \in \mathbf{AlgGrp}/k$  are conjugate by action of  $G_u(k)$ .
- Diagonalizable subgroups of  $G$  that are maximal are maximal tori of  $G$  of the form  $s(D)$  for some splitting  $s : D \rightarrow G \in \mathbf{AlgGrp}/k$  of  $q$ . In particular, they are conjugate under  $G_u(k)$ .
- Let  $T \leq G$  be a maximal torus, then  $G = G_u \rtimes T$ .

┘

*Proof:* These follow from(10.2.6.6)(10.2.6.10) and(10.2.6.11). Notice  $D$  is a torus by(10.1.5.33) and(10.2.3.21). □

### Borel Fixed Point Theorem

**Prop.(10.2.7.3).** If  $G \in \mathbf{GrpVar}/k$  is solvable and  $H \leq G \in \mathbf{AlgGrp}$ , then  $G/H$  doesn't contain a proper subscheme of dimension  $> 0$ . In particular,  $G = H$  if  $G/H$  is proper. ┘

*Proof:* Cf.[Mil17]P353. □

**Cor.(10.2.7.4).** If  $G$  is a solvable group variety acting on a separated algebraic scheme  $X$  over  $k$ , then no orbits of  $X$  contains a proper subscheme of dimension  $> 0$ . ┘

*Proof:* By(10.2.1.11), the orbits are quotients of  $G$ . □

**Cor.(10.2.7.5) [Borel Fixed Point Theorem].** Let  $G$  is a split solvable group acting on a  $X \in \mathbf{Sch}^{\text{proper}}/k$  and  $X(k) \neq \emptyset$ , then  $X^G(k) \neq \emptyset$ . ┘

*Proof:* Cf.[Mil17]P343.

It suffices to change to  $\bar{k}$  and assume that  $X$  is reduced. Then by(10.2.1.15)  $X$  has a closed orbit, which is then proper, thus must be a point. □

**Cor.(10.2.7.6) [Borel Fixed Point Theorem].** If  $G$  is solvable and acts on a  $X \in \mathbf{Sch}^{\text{proper}}/k$  and  $X \neq \emptyset$ , then  $X^G \neq \emptyset$ . In particular,  $X^G(\bar{k}) \neq \emptyset$ . ┘

*Proof:* It suffices to base change to  $\bar{k}$ , and then the assertion follows from(10.2.7.5). □

**Thm.(10.2.7.7)[Lie-Kolchin].** Suppose  $k = \bar{k}$ , then  $G$  is solvable iff it is triangularizable. In particular, any solvable group variety over  $k$  is a triangularizable after a finite field extension. ┘

*Proof:* For any  $V \in \mathbf{Rep}^{\text{fd}}(G)$ ,  $G$  acts on the full flag variety of  $V$ . Thus by Borel's fixed point theorem(10.2.7.6), there is a flag in  $V$  that is fixed by  $G$ , which means  $G$  is mapped into  $\text{Unip}(n)$  in this basis.

The converse is true by(10.2.6.4), as  $B(n)$  is solvable. □



### Nilpotent Groups

**Thm. (10.2.7.8).** Let  $G$  be unipotent and connected. Then

- $Z(G)_s$  is the largest algebraic subgroup of  $G$  of multiplicative type, and  $G/Z(G)_s$  is unipotent.
- If  $G$  becomes trigonalizable over a separable extension of  $k$ , then it has a unique decomposition  $G = G_u \times Z(G)_s$  where  $G_u$  is unipotent and  $Z(G)_s$  is of multiplicative type.
- If  $G$  is smooth, then  $Z(G)_s \in \text{Tori}/k$ .

┘

### Split Solvable Groups

**Prop. (10.2.7.9) [Split Solvable Groups].** Split solvable groups (10.1.1.4) are group varieties by (10.1.5.33). Extensions and quotients of split solvable groups are split solvable. ┘

*Proof:* ?

□

**Prop. (10.2.7.10) [Split Solvable Groups are Triangularizable].** Suppose  $G \in \text{GrpVar}/k$ . If  $G$  is a split solvable group, then  $G$  is triangularizable (10.2.6.2), and the converse is true if  $k = k^{\text{perf}}$ . ┘

*Proof:* Take a faithful representation  $V \in \text{Rep}^{\text{fd}}(G)$ , then  $G$  acts on the full flag variety  $\text{Flag}(V)$ . Then  $G$  fixes a flag by Borel fixed point theorem (10.2.7.5), which implies that  $G$  is triangularizable (10.2.6.2). Conversely, if  $k = k^{\text{perf}}$  and  $G$  is triangularizable, then  $G_u$  (10.2.6.6) is a group variety by (10.2.6.11) and  $G/G_u$  is a split torus by (10.2.6.11).  $G_u$  is split unipotent by (10.2.6.9), and thus  $G$  is split solvable. □

**Cor. (10.2.7.11).** If  $k = k^{\text{perf}}$ , then unipotent group varieties over  $k$  are split solvable groups, by (10.2.6.3). ┘

**Prop. (10.2.7.12) [Characterizing Split Solvable Groups over  $\bar{k}$ ].** If  $k = \bar{k}$ , the following are equivalent:

- $G$  is a triangularizable group variety.
- $G$  is split solvable.
- $G$  is a group variety, and  $G(k)$  is a solvable group.
- $G$  is a solvable group variety.

┘

*Proof:* 1  $\iff$  2 follows from (10.2.7.10).

2  $\rightarrow$  3 follows from (10.2.7.9).

3  $\rightarrow$  4: By considering the derived series of  $G$  (10.1.4.30), this follows from (10.1.4.29).

4  $\rightarrow$  1 follows from Lie-Kolchin theorem (10.2.7.7). □

**Prop. (10.2.7.13) [Structure of Split Solvable Groups].** If  $G$  is split solvable, then

- $G$  is triangularizable by (10.2.7.10), and the largest unipotent subgroup  $G_u \trianglelefteq G$  (10.2.6.6) is split unipotent.
- The canonical exact sequence  $e \rightarrow G_u \rightarrow G \xrightarrow{q} D \rightarrow e$  splits, and  $D$  is a split torus.
- Any two splittings  $s_1, s_2 : D \rightarrow G \in \text{AlgGrp}/k$  are conjugate by action of  $G_u(k)$ .



- Subgroups of multiplicative type in  $G$  that are maximal are split maximal torus of the form  $s(D)$  for some right inverse  $s : D \rightarrow G \in \text{AlgGrp}/k$  of  $q$ . In particular, they are conjugate under  $G_u(k)$ .
- Let  $T \leq G$  be a split maximal torus, then  $G = G_u \rtimes T$ .

┘

*Proof:* We can prove  $G_u$  is split unipotent by induction on  $\dim G$ , and  $D$  is a torus by (10.1.5.33) and (10.2.3.21). Also we can prove  $D$  is a split torus by induction on  $\dim G$  using (10.2.3.20). So the assertions follow from (10.2.6.6) and (10.2.6.10).  $\square$

**Prop. (10.2.7.14) [Torsors for Split Solvable Groups].** Suppose  $G$  is split solvable.

- If  $X \in \text{Aff}^{\text{ft}}/k$  and  $\text{Pic}(X) = 0$ , then every  $G$ -torsor on  $X$  is trivial.
- If  $X \in \text{Aff}^{\text{ft}}/k$  and  $G$  is unipotent, then every  $G$ -torsor on  $X$  is trivial.
- Every  $G$ -torsor is Zariski-locally trivial.

┘

*Proof:* Cf. [Mil17]P344.  $\square$

**Cor. (10.2.7.15).** If  $G$  is a split solvable, then  $G \cong \mathbb{A}^r \times (\mathbb{A}^1 \setminus 0)^{\dim G - r}$ , where  $r = \dim G_u$  (10.2.7.13).

┘

*Proof:* Cf. [Mil17]P344.  $\square$

**Thm. (10.2.7.16) [Rosenlicht].** If  $G$  is a solvable group variety, then  $G$  is split solvable iff there exists a dominant map of  $k$ -schemes  $\varphi : (\mathbb{A}^1 \setminus 0)^n \rightarrow G$  for some  $n \in \mathbb{Z}_+$ , or equivalently, a homomorphism of  $k$ -algebras:  $\varphi_* : \Gamma(G) \rightarrow k[T_1, \dots, T_n, T_1^{-1}, \dots, T_n^{-1}]$  for some  $n \in \mathbb{Z}_+$ .  $\square$

*Proof:* Cf. [Mil17]P345.  $\square$

**Thm. (10.2.7.17) [Lazard].** The following are equivalent:

- There exists an isomorphism of  $k$ -schemes  $G \cong \mathbb{A}^{\dim G}$ .
- $G$  is split unipotent.
- $G$  is reduced and solvable, and there exists a dominant map of  $k$ -schemes  $\varphi : \mathbb{A}^n \rightarrow G$  for some  $n \in \mathbb{Z}_+$ .

┘

*Proof:* Cf. [Mil17]P345.

1  $\iff$  2 follows from (10.2.7.11) and (10.2.7.15).

1  $\rightarrow$  3 is trivial.

3  $\rightarrow$  1: ? We only prove for  $G$  geo.reduced:  $G$  is geo.reduced and connected, thus a group variety, and then  $G$  is split solvable by (10.2.7.16). Thus  $G \cong \mathbb{A}^r \times (\mathbb{A}^1 \setminus 0)^{\dim G - r}$ , where  $r = \dim G_u$  by (10.2.7.15). There are no dominant maps  $\mathbb{A}^n \rightarrow \mathbb{A}^1 \setminus 0$ , so  $G = G_u$ , which is split unipotent (10.2.7.13).  $\square$

### 10.3 Representations of Algebraic Groups I: Elementary Theory

References are [Mil17] and [Representations of Algebraic Groups, Jantzen].

**Notation (10.3.0.1).**

- Let  $k \in \mathbf{Field}$ .
- Let  $R \in \mathbf{CRing}$ .
- Let  $G \in \mathbf{linAlgGrp}^{\text{aff}}/k$ .

┘

#### 1 Basics

**Def. (10.3.1.1) [Linear Representations of Algebraic Groups].** Let  $V$  be a free  $R$ -module and let  $\text{GL}(V)$  be the group functor

$$\mathcal{C}\text{Alg}_R \rightarrow \mathbf{Grp} : R' \mapsto \text{Aut}(V \otimes_R R').$$

Then a **linear representation**  $(r, V)$  of an arbitrary group scheme  $G/R$  is a homomorphism of group functors  $r : G \rightarrow \text{GL}_R(V)$ . It is called **faithful representation** if  $r$  is a monomorphism as natural transformation of functors.

If  $V/R$  is finite free, this is equivalent to a homomorphism  $G \rightarrow \text{GL}(V)$ . And by (10.1.5.7), it is faithful iff  $G \rightarrow \text{GL}(V)$  is a closed embedding.

For a linear representation  $(r, V)$ , let  $\text{End}(V) = \text{Aut}(r)$ .

┘

**Prop. (10.3.1.2) [Representations and Co-modules].** A representation of an arbitrary group scheme  $G/R$  on a free  $R$ -module  $V/R$  is equivalent to a right  $\Gamma(G)$ -comodule structure on  $V$  (3.11.2.22). ┘

*Proof:* Let  $A = \Gamma(G)$ . For any representation  $r : G \rightarrow \text{GL}(V)$ , it induces a map  $G(A) \rightarrow \text{GL}_A(V \otimes A)$ , which maps  $\text{id}_A$  to a map  $\rho : V \rightarrow V \otimes A$ . Now if  $\rho(e_j) = \sum e_i \otimes a_{ij}$ , then by functoriality, the map  $G(R') \rightarrow \text{GL}_{R'}(V \otimes R')$  is given by  $g \mapsto (e_j \mapsto e_i \otimes a_{ij}(g))$ .

And it can be verified that this is a group homomorphism iff the comodule condition is satisfied.

□

**Def. (10.3.1.3) [Stabilizers].** Let  $(r, V)$  be a representation of  $G$ ,  $W$  a subspace of  $V$ , consider the functor

$$\text{Stab}_G(W) : R' \mapsto \{g \in G(R') \mid g(W_{R'}) = W_{R'}\},$$

then if  $V$  is of f.d.,  $\text{Stab}_G(W)$  is representable by a closed subgroup of  $G$ , called the **stabilizer group scheme** of  $W$  in  $V$ . ┘

*Proof:* Let  $\rho : \Gamma(G) \rightarrow V \otimes \Gamma(G)$  be the comodule action corresponding to  $r$ , let  $\{e_i\}_{i \in I}$  be a basis for  $W$ , and extends it to a basis  $\{e_i\}_J \coprod_I$  of  $V$ , and

$$\rho(e_j) = \sum e_i \otimes a_{ij}, \quad a_{ij} \in \Gamma(G).$$

Let  $g \in \text{Hom}(\Gamma(G), R')$ , then  $ge_j = \sum e_i \otimes g(a_{ij})$ , and then clearly  $G_W$  is represented by the quotient of  $\Gamma(G)$  by the ideal generated by  $\{a_{ij} \mid i \in I, j \in J\}$ . ┘

**Cor. (10.3.1.4).** If  $S \in \mathbf{Sch}$  and  $G$  is an affine group scheme over  $S$ ,  $\mathcal{V}$  a locally free sheaf on  $S$  and  $\mathcal{W}$  a locally free subsheaf of  $\mathcal{V}$ , then similarly  $\text{Stab}_G(\mathcal{W})$  is representable by a closed subgroup of  $G$  over  $S$ . ┘

*Proof:* By considering affine-locally, and modify the proof of (10.3.1.3) to replace  $k$  by an arbitrary  $R \in \mathcal{CRing}$ .  $\square$

**Cor. (10.3.1.5).** Let  $(\rho, V)$  be a f.d. representation of  $G$  and  $S \subset G(k)$  be a subset schematically dense in  $G$ , then a subspace  $W \subset V$  is stable under  $G$  iff it is stable under  $S$ .  $\lrcorner$

**Prop. (10.3.1.6) [Fixed Subspace].** Let  $(r, V) \in \text{Rep}_k^{\text{fd}}(G)$ , define the **fixed subspace** by  $G$ :

$$V^G = \{v \in V \mid gv_R = v_R, \forall R \in \mathcal{CRing}_k, g \in G(R)\}.$$

Then for any  $R \in \mathcal{CAlg}_k$ ,  $V^G \otimes R$  is the submodule of  $V \otimes R$  that is fixed by elements of  $G(R')$  for any  $R' \in \mathcal{CAlg}_R$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P96.  $\square$

**Cor. (10.3.1.7).** The fixed subspace is compatible with base change of fields.  $\lrcorner$

**Def. (10.3.1.8) [Subrepresentations].** For  $V \in \text{Rep}_k(G)$ , a subspace  $W \subset V$  is called a **subrepresentation** of  $V$  if  $G = \text{Stab}_G(W)$ .

A linear representation is called **simple** if has no non-trivial subrepresentations.  $\lrcorner$

**Prop. (10.3.1.9).** Let  $U$  be a normal subgroup of  $G$ , then for any representation  $V \in \text{Rep}_k^{\text{fd}}(G)$ ;  $V^U$  is a subrepresentation of  $V$ .  $\lrcorner$

**Prop. (10.3.1.10) [Union of F.D. Subrepresentations].** Any  $(\rho, V) \in \text{Rep}_k(G)$  is a filtered union of its f.d. subrepresentations.  $\lrcorner$

*Proof:* It suffices to consider comodules and prove any vector  $v \in V$  is contained in a f.d. submodule. Let  $\{e_i\}$  be a basis of  $\Gamma(G)$ , let

$$\Delta(e_i) = \sum_{j,k} r_{ijk} e_j \otimes e_k, \quad \rho(v) = \sum_i v_i \otimes e_i, \quad v_i \in V.$$

Because  $(\text{id}_V \otimes \Delta) \circ \rho = (\rho \otimes \text{id}_{\Gamma(G)}) \circ \rho$  (3.11.2.22), thus

$$\sum_{i,j,k} r_{ijk} v_i \otimes e_j \otimes e_k = \sum_k \rho(v_k) e_k$$

which means

$$\rho(v_k) = \sum_{j,k} r_{ijk} v_i \otimes e_j.$$

Thus  $\text{span}\{v_i\}$  is a f.d. comodule of  $V$  containing  $v$ .  $\square$

**Cor. (10.3.1.11).** Any simple representations of  $G$  is of f.d.  $\lrcorner$

**Cor. (10.3.1.12) [Schur's Lemma].**  $(V, r)$  is a simple representation of  $G$ , then  $\text{End}(V)$  is a f.d. division algebra  $D$  over  $k$ . In particular, if  $k = \bar{k}$ , then  $\text{End}(V) = k$ .  $\lrcorner$

**Prop. (10.3.1.13) [Simple Representations of Product Groups].** Let  $G_1, G_2 \in \text{AlgGrp}/k$  and  $V_1, V_2$  are simple representations of  $G_1, G_2$  resp., then  $V_1 \otimes V_2$  is a simple representation of  $G_1 \times G_2$ .

Conversely, if  $\text{End}(V) = k$  for any simple representation  $(V, r)$  of  $G_1$ , then any simple representation of  $G_1 \times G_2$  is of this form.  $\lrcorner$

*Proof:* Cf.[Mil17]P91. □

**Def. (10.3.1.14) [Semisimple Representations].** A linear representation  $(r, V)$  of an algebraic group is called **semisimple** if it is a direct sum of simple representations. Equivalently, it is a sum of simple subrepresentation. (The proof is the same).

Or equivalently, every submodule has a complement. ┘

**Prop. (10.3.1.15) [Base Change of Fields].** Let  $(V, r) \in \text{Rep}^{\text{fd}}(G)$ ,  $k'/k$  a field extension, then  $(V', r') = (V, r) \otimes_k k'$  is a representation of  $G_{k'}$ , and

- If  $(V', r')$  is simple(resp. semisimple), and  $(V, r)$  is simple(resp. semisimple).
  - If  $(V, r)$  is simple and  $\text{End}(V) = k$ , then  $(V', r')$  is simple.
  - If  $(V, r)$  is semisimple and  $(k'/k)$  is separable or  $\text{End}(V)$  is a separable algebra over  $k$ , then  $(V', r')$  is semisimple.
- ┘

*Proof:* Cf.[Mil17]P90.

1: This is trivial.

2:

3: □

**Prop. (10.3.1.16) [Constructing All F.D. Representations].** Let  $V \in \text{Rep}^{\text{fd}}(G)$  is faithful, then every  $W \in \text{Rep}^{\text{fd}}(G)$  is a subquotient of  $V^m \otimes (V^\vee)^n$ . ┘

*Proof:* Cf.[Mil17]P88. □

**Def. (10.3.1.17) [Diagonalizable Representation].** A representation of  $G$  is called a **diagonalizable representation** if it is a direct sum of 1-dimensional representations.

Let  $(\rho, V) \in \text{Rep}(G)$ . If  $V$  is a sum of 1-dimensional representations, then  $\rho$  is diagonalizable(10.3.1.17). ┘

*Proof:* Let  $V = \sum_{\chi \in X(G)} V_\chi$ . If the sum is not direct, then there is some relation  $v_1 + v_2 + \dots + v_m = 0$  for  $v_i \in V_{\chi_i}$ . Applying  $\rho$  shows

$$0 = v_1 \otimes a(\chi_1) + \dots + v_m \otimes a(\chi_m)$$

so any coordinate of  $v_i$  is 0 by(10.1.1.11). □

**Prop. (10.3.1.18) [Chevalley].** Any algebraic subgroup  $H \leq G$  arises as the stabilizer of a 1-dimensional subspace in a f.d. representation of  $G$ . ┘

*Proof:* Cf.[Mil17]P94. □

**Cor. (10.3.1.19).** If  $\text{char } k = 0$ , then an algebraic subgroup  $H$  of  $G$  is normal in  $G$  iff for any linear representation  $V$  of  $G$  and a character  $\chi \in X(H)$ , the eigenspace  $V_\chi$  is stable under  $G$ . ┘

*Proof:* □

**Prop. (10.3.1.20).** If  $\#\pi_0(G)$  is prime to  $\text{char } k$ , then any representation  $V$  of  $G$  is semisimple iff it's semisimple when restricted to  $G^0$ . ┘

### Linear Algebraic Group

**Def. (10.3.1.21) [Linear Algebraic Groups].** A **linear algebraic group** over  $R$  is a closed subgroup scheme of  $\mathrm{GL}(n)_R$  for some  $n \in \mathbb{Z}_+$ . The category of linear algebraic groups over  $R$  is denoted by  $\mathrm{linAlgGrp}/R$ .

If  $R = k \in \mathrm{Field}^0$ , then a linear algebraic group over  $k$  is automatically smooth, by Cartier theorem (10.1.4.3).  $\lrcorner$

**Prop. (10.3.1.22) [Affine Algebraic Group is Linear].** The regular representation of  $G$  (10.2.1.2) contains a faithful f.d. sub-representation. In particular, the regular representation is itself faithful, and  $G$  is a linear algebraic group (10.3.1.21).  $\lrcorner$

*Proof:* Let  $\{e_i\}$  be a set of generators of  $\Gamma(G)$  as an  $R$ -algebra. Let  $V$  be a f.d. subrepresentation of the regular representation containing  $\{e_i\}$  which exists by (10.3.1.10), let  $\{v_i\}$  be a basis for  $V$ , and suppose  $\Delta(e_j) = \sum e_i \otimes a_{ij}$ , then the image of  $\Gamma(\mathrm{GL}(V)) \rightarrow \Gamma(G)$  contains  $a_{ij}$ . Now because  $\varepsilon : A \rightarrow R$  is the counit,

$$e_j = (\varepsilon \otimes \mathrm{id})\Delta(e_j) = \sum \varepsilon(e_i)a_{ij},$$

so the image contains  $V$ , so it contains  $\Gamma(G)$  by (10.3.1.10). So  $\Gamma(\mathrm{GL}(V)) \rightarrow \Gamma(G)$  is a closed immersion, thus a faithful representation, by (10.3.1.1).  $\square$

## 2 Tannakian Duality

**Lemma (10.3.2.1).** Let  $G$  be an arbitrary affine group scheme over  $R$  corresponding to a Hopf algebra  $A$ . If  $u$  is an  $R$ -endomorphism of  $A$  s.t. the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \otimes A \\ \downarrow u & & \downarrow 1 \otimes u \\ A & \xrightarrow{\Delta} & A \otimes A \end{array}$$

then there exists a  $g \in G(R)$  that  $u = r_A(g)$ , where  $r_A$  is the regular action of  $G$  on  $A$ .  $\lrcorner$

*Proof:* Let  $\varphi : G \rightarrow G$  be the morphism corresponding to  $u$ , then the commutative diagram shows  $\varphi_S(xy) = x\varphi_S(y)$  for  $x, y \in G(S)$ . Then  $\varphi_S(x) = xg_S$  where  $g_S = \varphi_S(e)$ .

Then for  $f \in A, x \in G(R)$ ,  $x(uf) = \varphi_R(x)(f) = (xg)(f) = x(r_A(f))$ , thus  $u = r_A(g)$ .  $\square$

**Prop. (10.3.2.2).** Let  $R \in \mathcal{C}\mathrm{Ring}_k$ . Suppose that for any  $(V, r_V) \in \mathrm{Rep}_k^{\mathrm{fd}}(G)$ , we are given an  $R$ -linear map  $\lambda_R : V_R \rightarrow V_R$  that satisfies:

- $\lambda_{V \otimes W} = \lambda_V \otimes \lambda_W$ .
- $\lambda_1 = \mathrm{id}$ .
- For  $G$ -invariant maps  $u : V \rightarrow W$ ,  $\lambda_W \circ u_R = u_R \circ \lambda_V$ .

then there exists a unique  $g \in G(R)$  that  $\lambda_V = r_V(g)$  for any  $V$ .  $\lrcorner$

*Proof:* Cf. [Milne, P164]. ?  $\square$

**Cor. (10.3.2.3) [Reconstruction Theorem].** Let  $\omega : \mathrm{Rep}_k(G) \rightarrow \mathrm{Vect}_k$  be the forgetful functor, and for any  $k$ -algebra  $R$ , let  $\omega_R = R \otimes \omega$ , then this proposition says the canonical morphism  $G(R) \rightarrow \mathrm{End}^{\otimes}(\omega_R)$  is an isomorphism. Now if  $\underline{\mathrm{Aut}}^{\otimes}(\omega)$  is the functor  $R \mapsto \mathrm{End}^{\otimes}(\omega_R)$ , then  $G \cong \underline{\mathrm{Aut}}^{\otimes}(\omega)$ .  $\lrcorner$

**Cor. (10.3.2.4).** Let  $G, G'$  be affine algebraic groups over  $k$  and let  $F : \mathcal{R}\text{ep}_k(G') \rightarrow \mathcal{R}\text{ep}_k(G)$  be a tensor functor that  $\omega^G \circ F = \omega^{G'}$ , then there exists a unique homomorphism  $f : G \rightarrow G'$  that  $F = \omega^f$ .  $\lrcorner$

*Proof:* Such a tensor functor defines a homomorphism

$$F^* : \underline{\text{Aut}}^\otimes(\omega^G)(R) \rightarrow \underline{\text{Aut}}^\otimes(\omega^{G'})(R)$$

functorial in  $R$ , thus defines a homomorphism  $f : G \rightarrow G'$  by Yoneda lemma and (10.3.2.4).  $\square$

**Lemma (10.3.2.5).** Let  $\mu$  be a cocharacter  $\mathbb{G}_m \rightarrow G$  over  $\overline{\mathbb{Q}_p}$ , then the conjugacy class  $\{\mu\}$  of  $\mu$  is defined over a finite extension  $E/\mathbb{Q}_p$ .  $\lrcorner$

*Proof:* By Tannakian duality, two cocharacters are conjugate over a field  $K$  iff the filtrations they defined on  $V \otimes \overline{\mathbb{Q}_p}$  for some faithful  $V \in \mathcal{R}\text{ep}_{\overline{\mathbb{Q}_p}}(G)$  are isomorphic by an action of  $G(K)$ . Now this action of  $G(\overline{\mathbb{Q}_p})$  on  $V_{\overline{\mathbb{Q}_p}}$  is defined over a f.d. field extension  $E/\mathbb{Q}_p$ , so the filtrations are isomorphic by an action of  $G(E)$ .  $\square$

### Tannakian Reconstruction

**Prop. (10.3.2.6) [Tannakian Reconstruction].** Let  $(\mathcal{C}, \otimes)$  be a rigid Abelian tensor category that  $k = \text{End}(\mathbf{1})$  and  $\omega : \mathcal{C} \rightarrow \mathcal{V}\text{ect}_k^{\text{fin}}$  an exact faithful  $k$ -linear tensor functor, then the functor  $\underline{\text{Aut}}^\otimes(\omega)$  is representable by an affine groups scheme  $G$ , and  $\mathcal{C} \cong \mathcal{R}\text{ep}_k^{\text{fd}}(G)$ .  $\lrcorner$

*Proof:* Cf. [Milne, Tannakian category, P21]. ?  $\square$

**Cor. (10.3.2.7) [Tannakian Reconstruction].** Let  $\mathcal{C} \in \text{AbCat}_k$  and  $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  a  $k$ -bilinear functor. Suppose there are given a faithful exact  $k$ -linear functor  $F : \mathcal{C} \rightarrow \mathcal{V}\text{ect}_k$  and functorial isomorphisms

$$\varphi_{X,Y,Z} : X \otimes (Y \otimes Z) \rightarrow (X \otimes Y) \otimes Z, \quad \psi_{X,Y} : X \otimes Y \rightarrow Y \otimes X$$

s.t.

- $F$  commutes with  $\otimes$ , and maps  $\varphi$  and  $\psi$  to the natural associativity and commutativity isomorphism in  $\mathcal{V}\text{ect}_k$ .
- There exists an identity object  $\mathbf{1} \in \mathcal{C}$  that  $k \rightarrow \text{End}(\mathbf{1})$  is an isomorphism and  $\dim_k F(\mathbf{1}) = 1$ .
- Any object  $L \in \mathcal{C}$  s.t.  $\dim_k F(L) = 1$  is an invertible object.

Then  $\mathcal{C}$  is equivalent to  $\mathcal{R}\text{ep}_k^{\text{fd}}(G)$  for some affine group scheme  $G$  over  $k$ . In fact,  $G \cong \underline{\text{Aut}}^\otimes(\omega)$  as in (10.3.2.3)  $\lrcorner$

*Proof:* The proof of (10.3.2.6) shows  $F$  defines an equivalence of categories  $\mathcal{C} \rightarrow \mathcal{R}\text{ep}_k^{\text{fd}}(G)$  where  $G$  is an affine monoid scheme representing  $\text{End}_k^\otimes(\omega)$ . Thus we may assume  $\mathcal{C} = \mathcal{R}\text{ep}_k(G)$ . For the rest, Cf. [Tannakian Categories, Milne, P24].  $\square$

**Cor. (10.3.2.8) [Real Algebraic Envelopes].** For  $K \in \text{TopGrp}$ ,  $\mathcal{R}\text{ep}_{\mathbb{R}}^{\text{fd}}(K)$  together with the forgetful functor satisfies the hypothesis of (10.3.2.7), thus there is an affine group  $\widetilde{K} \in \text{GrpSch}^{\text{aff}}/\mathbb{R}$ , called the **real algebraic envelope** of  $K$ , and an equivalence of categories

$$\mathcal{R}\text{ep}_{\mathbb{R}}^{\text{fd}}(\widetilde{K}) \rightarrow \mathcal{R}\text{ep}_{\mathbb{R}}^{\text{fd}}(K)$$

induced by a homomorphism  $K \rightarrow \widetilde{K}(\mathbb{R})$ , which is an isomorphism when  $K$  is compact (? Cf. [Serre, Gerbes], 5.2)  $\lrcorner$

**Cor. (10.3.2.9) [Hochschild-Mostow Group].** Similar as (10.3.2.8), if  $G$  is a complex Lie group or a f.g. abstract group, then  $\text{Rep}^{\text{fd}}(G)$  satisfies the hypothesis of (10.3.2.7), thus there exists a unique  $A(G) \in \text{GrpSch}^{\text{aff}}/\mathbb{C}$  with a homomorphism  $P : G \rightarrow A(G)$  that induces an equivalence of categories

$$\text{Rep}^{\text{fd}}(A(G)) \cong \text{Rep}^{\text{fd}}(G).$$

This  $A(G)$  is called the **Hochschild-Mostow group** of  $G$ . ┘

**Cor. (10.3.2.10) [Lie Algebra Representations].** For  $\mathfrak{g} \in \text{LieAlg}_k$ ,  $\text{Rep}_k^{\text{fd}}(\mathfrak{g})$  together with the forgetful functor to  $\text{Vect}_k^{\text{fin}}$  satisfies the hypothesis of (10.3.2.7), thus there exists a unique  $G(\mathfrak{g}) \in \text{GrpSch}^{\text{aff}}/k$  with a homomorphism  $P : U(\mathfrak{g}) \rightarrow G(\mathfrak{g})$  that induces an equivalence of categories

$$\text{Rep}^{\text{fd}}(G(\mathfrak{g})) \cong \text{Rep}^{\text{fd}}(\mathfrak{g}).$$

┘

**Prop. (10.3.2.11).** Let  $\mathcal{C} \in \text{AbCat}_k$ , and let  $\omega : \mathcal{C} \rightarrow \text{Vect}_k^{\text{fin}}$  be an exact faithful  $k$ -linear functor, then there exists a coalgebra  $C$  s.t.  $\mathcal{C}$  is equivalent to the category of  $C$ -comodules of f.d. ┘

*Proof:* Cf. [Mil17] P175. □

### Properties of $G$ and $\text{Rep}_k^{\text{fd}}(G)$

**Prop. (10.3.2.12).** Let  $G/k$  be an arbitrary affine group scheme, then:

- $G$  is finite iff there exists an object  $X \in \text{Rep}_k^{\text{fd}}(G)$  that every object of  $\text{Rep}_k^{\text{fd}}(G)$  is isomorphic to a subquotient of  $X^n$  for some  $n > 0$ .
- $G$  is algebraic iff there exists an object  $X \in \text{Rep}_k^{\text{fd}}(G)$  that every object of  $\text{Rep}_k^{\text{fd}}(G)$  is isomorphic to a subquotient of  $X^n \otimes (X^\vee)^m$  for some  $m, n \geq 0$ .

┘

*Proof:* Cf. [Milne, Tannakian categories, P25]. □

**Prop. (10.3.2.13).** Let  $f : G \rightarrow G' \in \text{linAlgGrp}/k$ , and let  $\omega^f$  be the corresponding functor  $\text{Rep}_k(G') \rightarrow \text{Rep}_k(G)$ . Then

- $f$  is faithfully flat iff  $\omega^f$  is fully faithful and each  $\omega^f$  induces an equivalence of subobjects of  $X'$  and  $\omega^f(X')$ .
- $f$  is a closed immersion iff every object of  $\text{Rep}_k^{\text{fd}}(G)$  is isomorphic to a subquotient of an object  $\omega^f(X')$ .

┘

*Proof:* Cf. [Milne, Tannakian categories, P25]. □

**Cor. (10.3.2.14).** Suppose  $\text{char } k = 0$ , then  $G$  is connected iff for any  $1 \neq V \in \text{Rep}_k(G)$ ;  $\langle V \rangle$  is not stable under  $\otimes$ . ┘

*Proof:* Cf. [Milne, Tannakian categories, P25]. □

### 3 Semi-Simple Elements

References are [Conjugacy Classes in Semisimple Groups, Hymphreys], [Conjugacy Classes, Springer-Steinberg].

**Prop. (10.3.3.1) [Abstract Jordan-Decompositions].** Suppose  $k = k^{\text{perf}}$  and  $g \in G(k)$ , then there exists a unique pair of elements  $(g_s, g_u)$  in  $G(k)$  s.t. for any  $(\rho, V) \in \text{Rep}_k^{\text{fd}}(G)$ , we have  $\rho(g_s) = \rho(g)_s$  and  $\rho(g_u) = \rho(g)_u$  (3.5.10.7). Furthermore,

$$g = g_s g_u = g_u g_s.$$

The elements  $g_s, g_u$  are called the semisimple and unipotent parts of  $g$ , and this decomposition is called the abstract Jordan decomposition of  $g$ .  $g \in G(k)$  is called a **semisimple element** (resp. **unipotent element**) if  $g = g_s$  (resp.  $g = g_u$ ).  $\perp$

*Proof:* This follows from the functoriality of Jordan-decompositions (3.5.10.7) and the reconstruction theorem (10.3.2.2).  $\square$

**Cor. (10.3.3.2).** To check a decomposition is the Jordan-decomposition, it suffices to check for a single faithful representation of  $G$ .  $\perp$

**Remark (10.3.3.3).** If  $k = \bar{k}$  and  $G \in \text{GrpVar}/k$ . In general, the set  $G(k)_s$  of semisimple elements in  $G(k)$  is not closed for the Zariski topology, but the set  $G(k)_u$  of unipotent elements are closed for the Zariski topology. To see this, embed  $G$  into  $\text{GL}(n)$  for some  $n \in \mathbb{Z}_+$ , then the set of unipotent elements are the matrices with characteristic polynomial  $(T-1)^n$ , and this is a polynomial condition.  $\perp$

**Prop. (10.3.3.4) [Semi-Simple Elements and Tori].** Semi-simple elements of  $G$  are exactly elements contained in some maximal torus of  $G$ .  $\perp$

*Proof:* If  $x$  is contained in a torus  $T$ , then for any  $(\rho, V) \in \text{Rep}_k^{\text{fd}}(G)$ , we have  $\rho|_T$  is diagonalizable (10.2.3.23), thus  $\rho(x)$  is diagonalizable, thus semisimple. Conversely, if  $x$  is semisimple, for a faithful representation  $(\rho, V) \in \text{Rep}_k^{\text{fd}}(G)$  (10.3.1.22), then there is a basis of  $V$  s.t.  $\rho(x)$  is diagonal, thus it is contained in the maximal torus of  $G$ , by [Linear Algebraic Groups, Springer] P109?  $\square$

**Thm. (10.3.3.5) [Steinberg].** If  $G \in \text{RedGrp}/k$  and  $x \in G(k)$  is a semi-simple element (10.3.3.1), then  $Z_G(x)^0 \in \text{RedGrp}/k$ . Moreover, if  $[G, G]$  is simply-connected, then  $Z_G(x)$  is connected.  $\perp$

*Proof:* Now take a maximal torus  $T \leq G$  containing  $x$  (10.3.3.4), then  $Z_G(x) = Z_{[G, G]}(x).T$ . By (10.4.2.13), it suffices to prove for  $G$  semisimple. Cf. [Conjugacy Classes, Springer-Steinberg] 2.3.9.?. Use z-extension?  $\square$

### 4 Unipotent Groups

**Def. (10.3.4.1) [Unipotent Groups].**  $G$  is called a **unipotent algebraic group** if  $V^G \neq 0$  for any  $V \in \text{Rep}_k^{\text{fd}}(G)$ . Equivalently, every finite representation sends  $G$  into  $\text{Unip}(n)$  in some coordinates, by (10.3.1.10).  $\perp$

**Prop. (10.3.4.2).** For  $k'/k \in \text{Field}$ ,  $G$  is unipotent iff  $G_{k'}$  is unipotent.  $\perp$

*Proof:* This follows from the fact that for  $V \in \text{Rep}_k^{\text{fd}}(G)$ ;  $(V \otimes k')^{G_{k'}} = V^G \otimes k'$  (10.3.1.7).  $\square$



**Prop. (10.3.4.3) [Characterizing Unipotent Subgroups].** The following are equivalent:

- $G$  is unipotent.
- $G$  is isomorphic to a subgroup of  $\text{Unip}(n)$  for some  $n \geq 2$ .
- The Hopf algebra  $\Gamma(G)$  is coconnected ?.

┘

*Proof:* Cf. [Mil17]P281.

□

**Example (10.3.4.4) [ $\mathbb{G}_a$  is Unipotent].**  $\mathbb{G}_a$  and any subgroups of  $\mathbb{G}_a$  are unipotent, because  $\mathbb{G}_a \cong \text{Unip}(2)$ .

┘

**Prop. (10.3.4.5).** Subgroups, quotient groups, and extensions of unipotent groups are unipotent.

┘

*Proof:* quotient groups and extensions of unipotent groups are unipotent by easy manipulation of the definition (10.3.4.1). Subgroups of unipotent groups are unipotent by (10.3.4.3).

□

**Cor. (10.3.4.6).** If  $\text{char } k = 0$ , then any unipotent algebraic group  $U$  over  $k$  is a group variety.

┘

*Proof:*  $G$  is smooth by (10.1.4.3). Consider  $\pi_0(G) = G/G^0$ , then it is a finite unipotent group by (10.3.4.5), and contained in  $\text{Unip}(n)$  for some  $n \in \mathbb{Z}_+$ . But  $\text{Unip}(n; \bar{k})$  has no elements of finite order, thus  $\pi_0(G) = \mathbf{1}$ .

□

**Prop. (10.3.4.7).** Representations of  $\mathbb{G}_a$  over a field of characteristic 0 corresponds to locally nilpotent endomorphisms.

┘

*Proof:* <https://qchu.wordpress.com/2017/11/26/the-representation-theory-of-the-additive-group-s>

□

**Prop. (10.3.4.8).** If  $U$  is a normal unipotent subgroup of  $G$ , then for any  $V \in \text{Rep}^{\text{fd}}(G)$  semisimple,  $U$  acts trivially on  $V$ .

┘

*Proof:* Consider a simple subrepresentation  $W$  of  $V$ , then  $W^U \neq 0$  is a subrepresentation of  $G$  by (10.3.1.9), thus  $W^U = W$  and  $U$  acts trivially on  $W$ . As  $V$  is a sum of simple representations (10.3.1.14),  $U$  acts trivially on  $V$ .

□

**Cor. (10.3.4.9).** If  $G$  is both unipotent and of multiplicative type, then  $G = e$ .

┘

*Proof:* Let  $(\rho, V)$  be a faithful representation of  $G$  (10.3.1.22), then it is semisimple by (10.2.3.23), but then by definition of unipotency (10.3.4.1),  $G$  acts trivially on  $V$ , so  $G = e$ .

□

**Cor. (10.3.4.10).** The intersection of a unipotent subgroup and a subgroup of multiplicative type in  $G$  is trivial.

┘

*Proof:* This is because unipotency and multiplicative type are both properties that are stable under subgroups, by (10.2.3.14) and (10.3.4.5).

□

**Prop. (10.3.4.11).** If  $G \leq \text{GL}(n)$ , then  $G$  is unipotent iff there is a subgroup of  $G(k)$  that is dense in  $G$  and consists of unipotent elements (10.3.3.1).

┘

*Proof:* One direction is clear. For the other direction, if  $S \subset G(k)$  is such a dense subgroup, then by Kolchin's theorem (24.1.1.13), there is a basis s.t.  $S \subset \text{Unip}(n; k)$ . So  $\text{Unip}(n) \cap G = G$ , and  $G \subset \text{Unip}(n)$ , hence unipotent by (10.3.4.3).

□

**Cor. (10.3.4.12).** If  $G$  is smooth, then  $G$  is unipotent iff elements in  $G(\bar{k})$  are all unipotent (10.3.3.1).  
 $\lrcorner$

*Proof:* Cf. [Mil17]P283. ? □

**Prop. (10.3.4.13) [Unipotent Groups are Nilpotent].** Any unipotent algebraic group admits a central normal series whose subquotients are isomorphic to algebraic subgroups of  $\mathbb{G}_a$ . In particular, any unipotent group is nilpotent, and hence solvable. □

*Proof:* This is because  $\text{Unip}(n)$  has a central normal series of length  $\frac{n(n-1)}{2}$  and subquotients isomorphic to  $\mathbb{G}_a$  (10.1.5.30), and  $G$  is a subgroup of some  $\text{Unip}(n)$  by (10.3.4.3). □

**Def. (10.3.4.14) [Split Unipotent Groups].** A **split unipotent group** is a unipotent algebraic group with a central normal series whose subquotients are all isomorphic to  $\mathbb{G}_a$ . This agrees with the definition in (10.1.1.4). □

**Prop. (10.3.4.15).**  $G$  is unipotent iff any non-trivial subgroup  $U \leq G$  admits a non-trivial homomorphism to  $\mathbb{G}_a$ . □

*Proof:* Cf. [Mil17]P285. ? □

**Prop. (10.3.4.16).** If  $G$  is connected and  $N = \ker(\text{Ad} : G \rightarrow \mathbf{GL}(\mathfrak{g}))$ , then  $N/Z(G)$  is unipotent. □

*Proof:* Cf. [Mil17]P285. ? □

**Prop. (10.3.4.17) [Unipotent Group Varieties over Perfect Fields are Split].** If  $k = k^{\text{perf}}$  and  $G \in \text{GrpVar}/k$  is unipotent (10.3.4.1), then  $G$  is split. □

*Proof:* Cf. [Mil17]Prop14.53 ? or [Bor91]Cor15.5(2). □

**Prop. (10.3.4.18) [Tori and Unipotency].** If  $G \in \text{GrpVar}/k$ , then  $G$  is unipotent iff it contains no non-trivial tori. □

*Proof:* Cf. [Mil17]378. □

**Prop. (10.3.4.19) [Group Acting on Unipotent Groups].** If  $G \in \text{GrpVar}/k$  is unipotent and  $\Gamma \in \text{Grp}^{\text{fin}}$  acts by group homomorphisms on  $G$ . If  $\#\Gamma \neq 0 \in k$ , then  $U^\Gamma \in \text{GrpVar}/k$ . □

*Proof:* Cf. [K-P23]P285. □

**Prop. (10.3.4.20) [Galois Cohomology].** If  $k = k^{\text{sep}}$  and  $U \in \text{GrpVar}/k$  is unipotent, then  $E/F \in \text{GalField}$ , then  $H^1(\text{Gal}(E/F), U(E)) = 0$ . □

*Proof:* This follows from (10.3.4.17) and (9.8.3.1) by induction. □

## 5 Reductive Groups

**Prop. (10.3.5.1).** For  $G \in \text{GrpVar}/k$ , if  $G$  has a faithful semisimple representation, then  $R_u(G) = 1$ , by (10.3.4.8). In particular,  $G$  is reductive if it has a geometrically semisimple faithful representation. □

**Thm. (10.3.5.2) [Representations of Reductive Groups are Semisimple].** Suppose  $k \in \text{Field}^0$  and  $G$  is connected, then the following are equivalent:

- $G \in \text{RedGrp}/k$ .
- Every f.d. representation of  $G$  is semisimple(10.3.1.14).
- Some faithful representation of  $G$  is semisimple(10.3.1.14).

┘

*Proof:* Cf.[Mil17]P477.

□

**Thm. (10.3.5.3) [Semisimple Lie Algebras].** If  $\text{char } k = 0$  and  $\mathfrak{g} \in \text{LieAlg}_k$  is semisimple, there is a Tannakian reconstructed group  $G(\mathfrak{g})$ (10.3.2.10). Then:

- $\eta : \mathfrak{g} \rightarrow \text{Lie}(G(\mathfrak{g}))$  is an isomorphism.
- $G(\mathfrak{g}) \in \text{RedGrp}^{\text{ss}}/k$ .
- For any  $H \in \text{AlgGrp}/k$ , there is an isomorphism

$$\text{Hom}(G(\mathfrak{g}), H) \cong \text{Hom}(\mathfrak{g}, \text{Lie}(H)) : b \mapsto \text{Lie}(b) \circ \eta.$$

- If  $\mathfrak{g}$  is split with root system  $(V, \Phi)$ , then  $G(\mathfrak{g})$  is split semisimple and simply-connected.

┘

*Proof:* Cf.[Mil17]P507.?

□

**Thm. (10.3.5.4) [Representations of Semisimple Groups].** If  $\text{char } k = 0$  and  $G \in \text{RedGrp}^{\text{ss}}/k$ , then  $\mathfrak{g}$  is a semisimple Lie algebra, and the natural functor  $\text{Rep}^{\text{fd}}(G) \rightarrow \text{Rep}^{\text{fd}}(\mathfrak{g})$  is fully faithful. It is essentially surjective if  $G$  is simply-connected.

┘

*Proof:* To show  $\mathfrak{g}$  is semisimple, See [Mil17]P476?

And then  $G(\mathfrak{g})$ (10.3.5.3) is the universal covering of  $G$  by(10.4.5.11). Thus  $\text{Rep}^{\text{fd}}(G) \rightarrow \text{Rep}^{\text{fd}}(G(\mathfrak{g})) \cong \text{Rep}^{\text{fd}}(\mathfrak{g})$  is clearly fully faithful.

□

### Split Reductive Groups

**Notation (10.3.5.5).**

- Let  $(G, B, T) \in \text{RedGrp}/k$  be split with root datum  $(X^\bullet, \Phi, \Phi^\vee)$ .

┘

**Def. (10.3.5.6) [Dominant Weights].** A **dominant weight** is an element  $\lambda \in X^\bullet$  s.t.  $\langle \lambda, \alpha^\vee \rangle \geq 0$  for all  $\alpha \in \Phi^+$ . The set of dominant weights is denoted by  $X^+$ .

┘

**Prop. (10.3.5.7).** For  $\lambda \in X^+$ , the set of dominant weights  $\mu \in X^+$  s.t.  $\mu < \lambda$  is finite.

┘

*Proof:* Choose a scalar product on  $X_{\mathbb{R}}$  that is invariant under  $W$ (3.9.5.5), then for  $\mu \in X^+$  s.t.  $\mu < \lambda$ , we have

$$(\mu, \mu) < (\mu, \lambda) < (\lambda, \lambda)$$

by(3.9.2.4). As  $X$  is discrete in  $X_{\mathbb{R}}$ , the assertion follows.

□

**Thm. (10.3.5.8) [Classifying Simple Representations].** For each dominant weight  $\lambda \in X^\bullet$ (10.3.5.6), there exists a  $V_\lambda \in \text{Irr}(G)$  unique up to isomorphism, s.t.

$$V_\lambda = V_\lambda[\lambda] \oplus \bigoplus_{\mu < \lambda} V_\lambda[\mu],$$

where  $T$  acts on  $V_\lambda[\mu]$  by the character  $\mu$ , and  $\dim_k V_\lambda[\lambda] = 1$ . This is called the **simple representation with highest weight**  $\lambda$ .

Conversely, any irreducible representation of  $G$  is of this form for a unique dominant  $\lambda \in X^\bullet$ . ┘

*Proof:* ? □

**Cor. (10.3.5.9).**  $\text{End}_k(V_\lambda) = k$  because it is generated by  $V_\lambda[\lambda]$ . In particular,  $V_\lambda$  is absolutely simple(10.3.1.15). ┘

**Prop. (10.3.5.10) [Representations and Isogenies].** For  $\lambda \in X^+$ ,  $Z(\tilde{G}) \cong D(\mathbb{P}(\Phi)/Q(\Phi))$  acts on  $V_\lambda$  via the character  $\lambda + Q(\Phi) \in P(\Phi)/Q(\Phi)$ . And if  $(G, B, T) \rightarrow (G', B', T') \in \text{RedGrp}/k$  is a central isogeny of split reductive groups(10.4.5.8), then  $V_\lambda$  factors through  $G'$  iff  $\lambda \in X^\bullet(T')$ . ┘

**Prop. (10.3.5.11) [Dual Representations].** It is easy to see that for  $\lambda \in P^+(\Phi)$ ,  $V_\lambda^* \cong V_{-w_0(\lambda)}$ . ┘

**Def. (10.3.5.12) [Characters of Representations].** For  $V = \bigoplus_{\chi \in X^\bullet} V_\chi \in \text{Rep}^{\text{fd}}(G)$ , the **character** of  $V$  is define to be the element

$$\text{ch}(V) = \sum_{\chi \in X^\bullet} \dim_k(V[\chi])e^\chi \in \mathbb{Z}[X^\bullet].$$

This extends to a homomorphism  $\text{ch} : K_0(\text{Rep}^{\text{fd}}(G)) \rightarrow \mathbb{Z}[X^\bullet]$ . ┘

**Thm. (10.3.5.13).**  $\text{ch} : K_0(\text{Rep}^{\text{fd}}(G)) \rightarrow \mathbb{Z}[X^\bullet]$  is injective with image  $\mathbb{Z}[X^\bullet]^W$ . ┘

*Proof:* By(10.3.5.8), it suffices to show that  $\{\text{ch}(V_\lambda)\}_{\lambda \in X^+}$ (10.3.5.6) form a basis of  $\mathbb{Z}[X^\bullet]^W$ .

As  $W$  acts simply transitively on Weyl chambers, every orbit of  $W$  in  $X^\bullet$  contains exactly one dominant weight. For  $\lambda \in X^+$ ,  $\text{ch}(V_\lambda)$  is fixed by  $W$  because  $W$  acts by inner automorphisms. So  $\text{ch}(V_\lambda) = We^\lambda + \sum_i We^{\lambda_i}$  where  $\lambda_i \in X^+$  and  $\lambda_i < \lambda$ .

From this we see that the map is injective. And because the set of dominant weights smaller than  $\lambda$  are finite(10.3.5.7), it is easy to show by induction or Zorn's lemma that every element in  $\mathbb{Z}[X^\bullet]^W$  is in the image of  $\text{ch}$ . □

**Def. (10.3.5.14) [Infinitesimally Irreducible Representations].**  $(V, r) \in \text{Rep}^{\text{fd}}(G)$  is called **infinitesimally irreducible** if  $(V, \text{Lie}(r)) \in \text{Rep}^{\text{fd}}(\mathfrak{g})$  is irreducible.

By(10.3.5.3), if  $\text{char } k = 0$  and  $G \in \text{RedGrp}^{\text{ss}}/k$ , then  $V \in \text{Rep}^{\text{fd}}(G)$  is irreducible iff it is infinitesimally irreducible. ┘

### Partial-Orderings

**Prop. (10.3.5.15) [Partial-Orderings].** For  $G \in \text{RedGrp}$ ,  $T \leq G$  a maximal torus and  $\Phi^+ \subset \Phi(G, T)$  a system of positive roots, let the partial-ordering on  $X_\bullet(T)_\mathbb{R}$  be as defined in(3.9.5.34), then for  $x, x' \in X_\bullet(T)_\mathbb{R}$ ,  $x \prec x'$  iff for any  $\rho : G \rightarrow \text{GL}(n) \in \text{Rep}_k^{\text{fd}}(G)$ , if  $T' \leq \text{GL}(n)$  is a maximal torus containing  $\rho(T)$ , then

$$\rho(x) \prec \rho(x').$$

┘

*Proof:* Cf.[R-R96]P165. ? □

## 10.4 Group Theory III: Reductive Groups

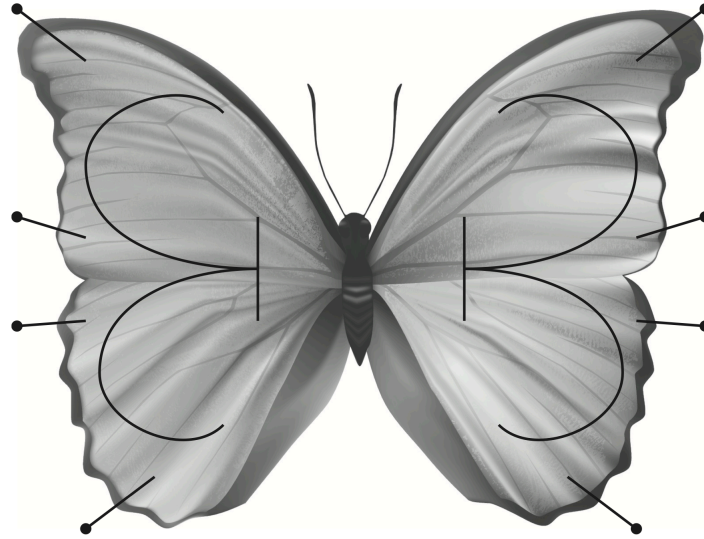


diagram borrowed from [Mil17]

*“Before Genesis, God gave the devil free reign to construct much of creation, but dealt with certain matters himself, including semisimple groups.”*

– C. Chevalley.

References are [B-T65, D-G70, D-G70a, D-G70b, Hum75, Bor91, P-R94, Spr98, Mil17, Con14].

### Notation(10.4.0.1).

- Let  $k \in \mathbf{Field}$ .
- Let  $R \in \mathbf{CRing}$ .
- Let  $\emptyset \neq S \in \mathbf{Sch}$ .
- Use notations on solvable groups defined in [Group Schemes II: Solvable Groups](#).
- Let  $G, G' \in \mathbf{GrpVar}^{\text{aff}}/k$ .

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### 1 Borel Subgroups and Parabolic Subgroups

**Def.(10.4.1.1)[Borel Subgroups].** A **Borel subgroup** of  $G$  is a solvable subgroup variety of  $G$  s.t.  $G/B$  is proper. The set of Borel subgroups of  $G$  is denoted by  $\mathcal{B}_G$ .

A **Borel pair** is a pair  $(B, T)$  where  $B$  is a Borel subgroup of  $G$  and  $T$  is a maximal torus of  $G$ (10.4.1.16) contained in  $B$ .

If  $k'/k \in \mathbf{Field}$ , then  $B \leq G$  is Borel iff  $B_{k'} \leq G_{k'}$  is Borel, by(10.1.5.23) and **?**.

$G$  is called a **quasi-split group** if it contains a Borel subgroup. And  $G$  is called a **split group** if it contains a Borel subgroup that is split(10.1.1.4)(If  $G$  is reductive, This is equivalent to  $G$  containing a split maximal torus by(10.4.4.2)).

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**Def. (10.4.1.2) [Parabolic Subgroups].** A **parabolic subgroup**  $P \leq G$  is a smooth closed subgroup s.t.  $G/P$  is proper. In particular, a Borel subgroup is parabolic.  $\lrcorner$

**Prop. (10.4.1.3) [Parabolic and Borel Subgroups].** If  $k = \bar{k}$ , then parabolic subgroups of  $G$  are exactly closed subgroups containing a solvable subgroup of maximal dimension.  $\lrcorner$

*Proof:* If  $P$  contains a solvable subgroup of maximal dimension, then  $G/P$  is proper by 1  $\rightarrow$  2 part of (10.4.1.4). Conversely, if  $G/P$  is proper, then by Borel fixed point theorem (10.2.7.6), if  $B$  is a solvable subgroup of  $G$  of maximal dimension, then  $B$  has a fixed point  $xP \in G/P$  for some  $x \in G(k)$ , which means  $x^{-1}Bx \leq P$ .  $\square$

**Cor. (10.4.1.4) [Characterizing Borel Subgroups over  $\bar{k}$ ].** If  $k = \bar{k}$  and  $H \leq G$  is a closed subgroup, then the following are equivalent:

- $H$  is a solvable subgroup variety of  $G$  of maximal dimension.
- $H$  is Borel (10.4.1.1).
- $H$  is minimally parabolic.

In particular,  $G$  has a Borel subgroup.  $\lrcorner$

*Proof:* 1  $\rightarrow$  2: If  $H$  is maximal solvable, there exists  $V \in \text{Rep}^{\text{fd}}(G)$  s.t.  $H$  is the stabilizer of a 1-dimensional subspace  $L \subset V$ . Now  $B$  acts on  $V/L$ , and by Lie-Kolchin theorem (10.2.7.7),  $B$  fixes a full flag of  $V$ , starting from  $L$ . Thus  $B$  is the stabilizer of this flag. Then because  $B$  is maximal,

2  $\rightarrow$  3: If  $H$  is solvable and  $G/H$  is proper, then it is parabolic by definition (10.4.1.2). And if  $P \leq H$  is a parabolic subgroup of  $G$ , then  $P$  contains a Borel subgroup  $B$  by (10.4.1.3). But because  $H$  is solvable, by definition of Borel subgroup, we have  $B = H$ , so  $H$  is a minimal parabolic subgroup.

3  $\rightarrow$  1: If  $H$  is minimally parabolic, then  $H$  contains a solvable subgroup  $B$  of maximal dimension which is parabolic by (10.4.1.3), and then  $H = B$  by minimality.  $\square$

**Cor. (10.4.1.5).**  $G$  is solvable iff  $G_{\bar{k}}$  contains no properly-contained parabolic subgroups, by (10.1.5.23) and (10.4.1.4).  $\lrcorner$

**Prop. (10.4.1.6) [Minimally Parabolic Subgroups].** Any two minimally parabolic subgroups of  $G$  are conjugate by an element of  $G(k)$ .

In particular, if  $G$  is quasi-split (10.4.1.1), then the minimally parabolic subgroups of  $G$  coincide with Borel subgroups of  $G$ , and any two Borel subgroups of  $G$  are conjugate by an element of  $G(k)$ .  $\lrcorner$

*Proof:* Cf. [Mil17] P546.

The last assertion follows from (10.4.1.4).  $\square$

**Prop. (10.4.1.7) [Split Borel Pairs are Conjugate].**

1. If  $H \leq G$  is a split solvable subgroup and  $B \leq G$  is a split Borel subgroup, then  $gHg^{-1} \leq B$  for some  $g \in G(k)$ .
2. Any two split Borel subgroups of  $G$  (10.4.1.1) are conjugate by an element of  $G(k)$ .
3. (Borel-Tits) Any two maximal split tori of  $G$  are conjugate by an element of  $G(k)$ .
4. If  $G$  is a split group (10.4.1.1), any two split Borel pairs (10.4.1.1) are conjugate by an element of  $G(k)$ .
5. Any split maximal torus is contained in a split Borel pair, and any split Borel subgroup is contained in a split Borel pair.

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*Proof:* 1: Let  $H$  acts on  $G/B$  by left multiplication, then by Borel fixed-point theorem(10.2.7.5), there exists a fixed point  $P \in G/B(k)$ . the inverse image of  $P$  in  $G$  is a  $B$ -torsor over  $k$ , which is trivial by(10.2.7.14). Thus  $HgB \subset gB$  for some  $g \in G(k)$ , and then  $H \leq gBg^{-1}$ .

2: This follows from item1.

3: ? Cf.[pesudo-reductive groups]Theorem C.2.3 or [Bor91]Theorem 15.14.

4: For any two split Borel pairs  $(B, T)$  and  $(B', T')$ ,  $B$  and  $B'$  are conjugate by item2, and then use item2 applied to  $B$ .

5: Let  $T$  be a split maximal torus, then  $G$  has a split Borel subgroup by(10.4.4.2), thus  $T$  is contained in a split Borel subgroup by item1. And any split Borel subgroup  $B \leq G$  is conjugate to this Borel subgroup by item2, so  $B$  contains a split maximal torus.  $\square$

**Cor.(10.4.1.8).** Suppose  $k = \bar{k}$  and  $T \leq G$  is a maximal torus. Then for any subgroup variety  $H \leq G$  containing  $T$ ,  $N_G(T)(k)$  acts transitively on the set of  $G(k)$ -conjugates of  $H$  containing  $T$ . In particular,  $N_G(T)(k)$  acts transitively on the set of Borel-subgroups containing  $T$  by(10.4.1.7).  $\square$

*Proof:* Cf.[Mil17]P355. ?  $\square$

**Prop.(10.4.1.9) [Homomorphisms and Borel Subgroups].** Let  $R \in \mathcal{C}\text{Ring}/k$  and  $\varphi_1, \varphi_2 : G_R \rightarrow G'_R$  be homomorphisms that agree on  $B_R$  for some Borel subgroup  $B \leq G$ , then  $\varphi_1 = \varphi_2$ .  $\square$

*Proof:* Cf.[Mil17]P358.  $\square$

**Cor.(10.4.1.10) [Centralizer of Borel Subgroups].** Let  $B \leq G$  be a Borel subgroup, then  $Z(B) = Z_G(B) = Z(G)$ .  $\square$

*Proof:*  $Z_G(B) = Z(G)$  follows from(10.4.1.9).

To show that  $Z(G) \leq B$ , by(10.1.4.10), we may pass to  $\bar{k}$ , and let  $T$  be a maximal torus contained in  $B$ (10.4.1.7). Then  $Z(G) \subset Z_G(T) \leq B$  by(10.4.1.20).  $\square$

**Thm.(10.4.1.11) [Normalizer Theorem, Chevalley].** If  $P \leq G$  is a parabolic subgroup(10.4.1.2), then  $P$  is connected and  $N_G(P) = P$ .  $\square$

*Proof:* Cf.[Mil17]P367. ?  $\square$

**Prop.(10.4.1.12) [Borel Subgroups and Quotient Maps].** Suppose  $k = \bar{k}$ ,  $q : G \rightarrow G'/k$  is a quotient map and  $H \leq G$  is a subgroup variety. If  $H$  is a parabolic subgroup/Borel subgroup/maximally unipotent subgroup/maximal torus, then so is  $q(H)$ , and any such subgroup of  $G'$  arises in this way.  $\square$

*Proof:* Cf.[Mil17]P357. ?  $\square$

**Cor.(10.4.1.13).** Quotients of split groups are split groups(10.4.1.1), by(10.4.1.12) and(10.2.7.9).  $\square$

### Density Theorem

**Thm.(10.4.1.14) [Density Theorem].** Suppose  $k = \bar{k}$ .

- If  $T \leq G$  is a maximal torus, and let  $C = C_G(T)^0$ , then  $\cup_{g \in G(k)} g|C|g^{-1}$  contains a dense open subset of  $|G|$ .
- If  $B \leq G$  is a Borel subgroup, then  $|G| = \cup_{g \in G(k)} g|B|g^{-1}$ .



- Every element of  $G(k)$  is contained in a Borel subgroup of  $G$ . Every normal Borel subgroup equals  $G$ .
- Every semisimple element  $s \in G(k)$  is contained in a maximal torus.

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*Proof:* 1, 2: Cf. [Mil17]P362. ?

3 follows from item 1, 2.

4: Let  $B$  be a Borel subgroup containing  $s$ , and let  $T \leq G$  be a maximal torus of  $G$  contained in  $B$  (10.4.1.7). Then the Zariski closure  $S = \overline{\{s\}}$  is diagonalizable by (10.2.3.5). Thus  $S$  is contained in a conjugate of  $T$  by (10.2.7.2).  $\square$

### Maximal Tori and Cartan Subgroups

**Def. (10.4.1.15) [Maximally-Split Tori].** A **maximally-split torus** of  $G$  is a torus  $T \leq G$  s.t.  $T$  contains a maximal split torus of  $G$ .  $\perp$

**Def. (10.4.1.16) [Maximal Tori].** For  $G \in \text{GrpSch}^{\text{aff}}/S$ , a **maximal torus** of  $G$  is an  $S$ -torus  $T \leq G$  s.t.  $T_{\bar{s}} \leq G_{\bar{s}}$  is torus not contained in any other torus of  $G_{\bar{s}}$ , for any geometric point  $\bar{s} \in S$ .  $\perp$

**Prop. (10.4.1.17) [Maximal Tori Exist, Grothendieck [D-G70b]].** Let  $k'/k \in \text{Field}$ , then  $T \leq G$  is a torus that is maximal iff  $T_{k'} \leq G_{k'}$  is a torus that is maximal. In particular, maximal tori exist.  $\perp$

*Proof:*  $T$  is maximal iff it is maximal in  $Z_G(T)$ , which is also a group variety by (10.2.4.10). If  $T' \leq C_G(T)/T$  is a torus, let  $G'$  be the inverse image of  $Z_{Z_G(T)/T}(T')$  in  $C_G(T)$  and let  $T''$  be the inverse image of  $T'$  in  $C_G(T)$ , which are both group varieties, and  $T''$  is normal in  $G'$ . Thus (10.2.3.32) implies that  $T''$  is also a torus.

Thus  $T$  is maximal iff  $Z_G(T)/T$  contains no tori, which is equivalent to  $Z_G(T)/T$  being unipotent by (10.3.4.18). This property and centralizer, quotients are both invariant under base change by (10.3.4.2)(10.2.1.6), thus the assertion follows.  $\square$

**Def. (10.4.1.18) [Ranks and Semisimple Ranks].** The (reductive)**rank of a group variety** of  $G$  is the dimension of a maximal torus in  $G$ . The **semisimple rank** of  $G$  is rank of  $G/R(G)$ .

These are well-defined because maximal tori exist and are conjugate over  $G(k^{\text{sep}})$  by (10.4.1.7).  $\perp$

**Prop. (10.4.1.19) [Nilpotent Groups and Maximal Tori].** Let  $k = \bar{k}$  and  $G \in \text{GrpVar}/k$ , then the following are equivalent:

- $G$  has only one maximal torus.
- Some Borel-subgroup of  $G$  is unipotent.
- $G$  is unipotent.
- Every maximal torus of  $G$  is central.

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*Proof:* Cf. [Mil17]P359.  $\square$

**Prop. (10.4.1.20) [Centralizer of Maximal Tori].** If  $T \leq G$  is a maximal torus, then

- $Z_G(T) = N_G(T)^0$ . In particular,  $Z_G(T) \in \text{GrpVar}/k$  by (10.2.4.2).
- $Z_G(T)$  is nilpotent.
- $Z_G(T)$  is contained in any Borel subgroup containing  $T$ .



- $Z_G(T) = Z_G(T)_u \times T$  if  $k = k^{\text{perf}}$ . ┘

*Proof:* 1:  $Z_G(T)$  is connected by (10.2.4.10), so this follows from the rigidity theorem (10.2.3.29). ┘

2: We may pass to  $\bar{k}$  and this follows from (10.4.1.19). ┘

3: We may pass to  $\bar{k}$ . As  $Z_G(T)$  is connected and nilpotent, it is contained in a Borel-subgroup by (10.4.1.4), thus it is contained in any Borel-subgroup by (10.4.1.8). ┘

4: This follows from (10.2.7.8). □

**Cor. (10.4.1.21).** ┘

**Prop. (10.4.1.22).** If  $B \leq G$  is a Borel subgroup and  $T \leq B$  is a torus, then  $Z_G(T) \cap B$  is a Borel subgroup of  $Z_G(T) \in \text{GrpVar}/k$  (10.4.1.20). And if  $k = \bar{k}$ , then any Borel subgroup of  $Z_G(T)$  is of this form. ┘

*Proof:* Cf. [Mil17]P365. ? □

**Def. (10.4.1.23) [Weyl-Groups].** For a maximal split torus  $S \leq G$ , the **Weyl-group**  $W(G, S)$  is defined to be the finite group ┘

$$W(G, S) \triangleq N_G(S)/Z_G(S) \cong \pi_0(N_G(S)) \text{ (10.1.4.13) (10.4.1.20).}$$

$W(G, S)$  acts faithfully on  $S$ . For more about Weyl-groups, Cf. [Bor91]Section21. ? ┘

**Def. (10.4.1.24) [Cartan-Subgroups].** A **Cartan-subgroup** of  $G$  is the centralizer of a maximal torus (10.4.1.16). ┘

Every Cartan-subgroup of  $G$  is a nilpotent group subvariety, by (10.2.4.10)(10.4.1.19)(10.2.4.2). And if  $k = \bar{k}$ , they are all conjugate by action of  $G(k)$  (10.4.1.7), and their union contains a dense open subset of  $|G|$  by (10.4.1.14). ┘

**Prop. (10.4.1.25).**  $G$  is generated by its Cartan subgroups (10.4.1.24). ┘

*Proof:* Cf. [Mil17]P381. □

### The Variety of Minimal Parabolic Subgroups

**Prop. (10.4.1.26).** If  $G$  is a quasi-split group, then any Borel subgroup  $B \leq G$  defines a  $k$ -structure of the variety of Borel subgroups:  $(G/B)_{\bar{k}} \cong \mathcal{B}_G$ , and this is independent of  $B$  chosen, by (10.4.1.6). ┘

### Radical Subgroups and Unipotent Radical Subgroups

**Def. (10.4.1.27) [Radical Subgroups].** The **radical subgroup scheme**  $R(G)$  of  $G$  is the largest solvable normal subgroup variety of  $G$ , which exists by (10.1.5.33) and (10.1.5.34). ┘

**Def. (10.4.1.28) [Unipotent Radical Subgroups].** The **unipotent radical subgroup** of  $G$  is the largest normal unipotent subgroup variety  $R_u(G)$ , which exists by (10.3.4.5) and (10.1.5.34). ┘

**Prop. (10.4.1.29).** For arbitrary  $G \in \text{GrpVar}/k$  and  $k'/k \in \text{Field}$  is separable, then  $R(G_{k'}) = R(G)_{k'}$ . ┘

*Proof:*  $R(G)_{k'} \subset R(G_{k'})$  by definition, and it suffices to prove they are the same when  $k'/k$  is Galois. But then this follows from Galois descent (7.1.5.20) and the fact connectedness, smoothness, normality and solvability is reflexive (10.1.5.23). □

**Thm. (10.4.1.30) [Kostant-Rosenlicht].** If  $U \in \text{GrpVar}/k$  is unipotent and  $U$  acts on  $X \in \text{Aff}^{\text{ft}}/k$ , then every  $U$ -orbit in  $X$  (10.2.1.16) is closed. In particular, any  $\gamma \in X(k)$  is relatively semisimple (10.2.1.18).  $\lrcorner$

*Proof:* Cf. [Mil17]P373.  $\square$

**Thm. (10.4.1.31) [Chevalley].** If  $k = \bar{k}$  and  $T$  is a maximal torus of  $G$ , then

$$R_u(G)T = \left( \bigcap_{B \in \mathcal{B}^T} B \right)_{\text{red}}^0, \quad R_u(G) = \left( \bigcap_{B \in \mathcal{B}^T} B_u \right)_{\text{red}}^0 \quad (10.2.7.13),$$

where  $\mathcal{B}^T$  is the finite set of Borel subgroups of  $G$  containing  $T$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P371.  $\square$

**Cor. (10.4.1.32).** If  $k = \bar{k}$  and  $S/k$  is a torus acting on  $G$ , then  $R_u(G^S) = R_u(G)^S$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P371. ?  $\square$

## 2 Reductive Groups

### Semisimple Groups

**Def. (10.4.2.1) [Semisimple Algebraic Groups].**  $G$  is called a **semisimple algebraic group** iff it is an affine group variety and  $R(G_{\bar{k}}) = e$  (10.4.1.27). The category of semisimple algebraic groups over  $k$  is denoted by  $\text{RedGrp}^{\text{ss}}/k$ .

$G$  is called an **adjoint reductive group** if  $G$  is semisimple and  $Z(G) = 1$ .  $\lrcorner$

**Prop. (10.4.2.2).** For arbitrary  $G \in \text{AlgGrp}/k$  and  $k'/k \in \text{Field}$ ,  $G$  is semisimple iff  $G_{k'}$  is semisimple.  $\lrcorner$

*Proof:* Affineness, smoothness and geo.connectedness satisfies field descent by (7.1.5.26), so it suffices to prove for  $k', k$  alg.closed. But if  $N$  is non-trivial normal solvable subgroup variety of  $G_{k'}$ , it is defined on a f.g. field extension  $K = k(x_1, \dots, x_n)$  of  $k$ , thus  $N_K$  is a normal solvable subgroup variety of  $G_K$ , and they extends to smooth group varieties  $\mathcal{G}$  and  $\mathcal{N}$  on some open subscheme  $\text{Spec } A \subset \text{Spec } k[x_1, \dots, x_n]$ . But on some maximal ideal  $\mathfrak{m}$ ,  $\mathcal{N}_{k(\mathfrak{m})}$  is non-zero (5.2.6.11) normal open subgroup of  $\mathcal{G}_{k(\mathfrak{m})}$ , and  $k(\mathfrak{m}) \cong k$  as  $k = \bar{k}$ , contradiction.  $\square$

**Prop. (10.4.2.3) [Semisimple Quotients].** If  $k = k^{\text{perf}}$ , then  $G$  is semisimple iff  $R(G) = e$  (10.4.1.27). In particular,  $G/R(G)$  is semisimple, by (10.1.5.34).  $\lrcorner$

**Cor. (10.4.2.4).** If  $G$  is semisimple, then it contains no non-trivial commutative normal subgroup varieties. The converse is also true if  $k = k^{\text{perf}}$ .  $\lrcorner$

**Prop. (10.4.2.5) [Radical Subgroup and Borel Subgroups].** If  $k = \bar{k}$ , then

$$R(G) = \left( \bigcap_{B \leq G \text{ Borel}} B \right)_{\text{red}}^0.$$

$\lrcorner$

*Proof:* The RHS is a solvable normal subgroup variety of  $G$ , thus contained in  $R(G)$ . Conversely,  $R(G)$  is contained in a Borel subgroup by (10.4.1.4), and because it is normal, it is contained in every Borel subgroup by (10.4.1.7).  $\square$

**Def. (10.4.2.6)[Simple Groups].** A **simple algebraic group** is a semisimple non-commutative group with no non-trivial normal algebraic subgroups.

An **almost-simple algebraic group** is a semisimple non-commutative group s.t. every properly-contained normal algebraic subgroup is finite.

A **geometrically (almost-)simple algebraic group** is an algebraic group  $G$  s.t.  $G_{\bar{k}}$  is (almost-)simple.

A(n) **(almost-)pseudo-simple algebraic group** is a non-commutative group variety  $G$  s.t. every properly-contained normal algebraic subgroup is trivial (resp. finite). (May not be semisimple).  $\perp$

### Reductive Groups

**Def. (10.4.2.7) [Unipotent Radical Subgroups].**  $G$  is called a (connected) **reductive group** iff  $R_u(G_{\bar{k}}) = e$  (10.4.1.28). The category of reductive group schemes over  $k$  is denoted by  $\text{RedGrp}/k$ .  $G$  is called a **pseudo-reductive group** if  $R_u(G) = e$ .  $\perp$

**Prop. (10.4.2.8).** Suppose  $G \in \text{AlgGrp}/k$  and  $k'/k \in \text{Field}$ , then  $G$  is reductive iff  $G_{k'}$  is reductive.  $\perp$

*Proof:* The proof is verbatim as that of (10.4.2.2).  $\square$

**Prop. (10.4.2.9)[Pseudo-Reductive Quotients].**  $R/R_u(G)$  is pseudo-reductive, by (10.1.5.34), called the **largest pseudo-reductive quotient** of  $G$ .  $\perp$

**Prop. (10.4.2.10).** If  $k'/k \in \text{Field}$  is separable, then  $R_u(G_{k'}) = R_u(G)_{k'}$ . In particular,  $R_u(G_k) = R_u(G_{\bar{k}})$  if  $k = k^{\text{perf}}$ .  $\perp$

*Proof:*  $R(G)_{k'} \subset R(G_{k'})$  by definition, and it suffices to prove they are the same when  $k'/k$  is Galois. But then this follows from Galois descent (7.1.5.20) and the fact connectedness, smoothness, normality and unipotency is reflexive (10.3.4.2).  $\square$

**Cor. (10.4.2.11)[Reductive Quotients].** If  $k = k^{\text{perf}}$ , then  $G$  is reductive iff  $R_u(G) = e$ . In particular,  $G/R_u(G)$  is reductive, by (10.1.5.34), called the **largest reductive quotient** of  $G$ .  $\perp$

**Prop. (10.4.2.12).** If  $k = k^{\text{perf}}$ , then  $G(k) \rightarrow (G/R_u(G))(k)$  is surjective.  $\perp$

*Proof:* This follows from (10.3.4.20) and (10.4.1.28).  $\square$

**Prop. (10.4.2.13)[Extensions of Reductive Groups].** If  $1 \rightarrow G_1 \rightarrow G \rightarrow G_2 \rightarrow 1 \in \text{AlgGrp}/k$  is an exact sequence and  $G_1, G_2 \in \text{RedGrp}/k$ , then  $G \in \text{RedGrp}/k$ .  $\perp$

*Proof:* It suffices to prove for  $k = \bar{k}$ . Then  $R_u(G) \cap G_1$  is finite and  $R_u(G)/(R_u(G) \cap G_1) \subset H$  is a unipotent group variety (10.3.4.5), thus trivial, and then  $R_u(G) = 1$ .  $\square$

**Prop. (10.4.2.14)[Centralizers of Tori are Reductive].** If  $G \in \text{RedGrp}/k$  and  $S$  is a torus acting on  $G$ , then  $G^S \in \text{RedGrp}/k$ . In particular, the centralizer of a torus in  $G$  is a reductive group.  $\perp$

*Proof:*  $G^S$  is a group variety by (10.4.1.20), and  $R_u(G_{\bar{k}}^S) = R_u(G_{\bar{k}})^S = e$  by (10.4.1.32).  $\square$

**Prop. (10.4.2.15) [Maximal Tori of Reductive Groups].** If  $G \in \mathcal{R}\text{edGrp}/k$ , then

- For any maximal torus  $T \leq G$  (10.4.1.16),  $Z_G(T) = T$ . In particular,  $W(G, T) = N_G(T)/T$ . Thus Cartan subgroups of  $G$  are just maximal tori of  $G$ .
- $Z(G) \leq T$  for any maximal torus  $T \leq G$ , and  $Z(G) = \cap_{T \text{ maximal tori}} T$ .
- If  $T \leq G$  is a maximal torus, then  $Z(G) = \ker(\text{Ad} : T \rightarrow \mathbf{GL}(\mathfrak{g}))$ .

┘

*Proof:* 1: We may assume  $k = \bar{k}$ . Because  $C_G(T)$  is a group variety (10.2.4.10),  $C_G(T) \leq (\cap_{B \in \mathcal{B}^T} B)_{\text{red}}^0 = R_u(G)T = T$  (10.2.4.23)(10.4.1.31).

2:  $Z(G) \leq T$  by item1. And any  $g \in \cap_{T \text{ maximal tori}} T(\bar{k})$  commutes with all Cartan subgroups, which generate  $G$  by (10.4.1.25), so  $g \in Z(G)(\bar{k})$ .

3:  $Z(G) \leq \ker(\text{Ad} : T \rightarrow \mathbf{GL}(\mathfrak{g}))$  by (10.4.2.15). The quotient  $\ker(\text{Ad})/Z(G)$  is unipotent by (10.3.4.16), thus the image of  $\ker(\text{Ad} : T \rightarrow \mathbf{GL}(\mathfrak{g}))$  in  $\ker(\text{Ad})/Z(G)$  is trivial by (10.2.3.14)(10.3.4.9). □

**Prop. (10.4.2.16) [Radicals of Reductive Groups].** If  $G \in \mathcal{R}\text{edGrp}/k$ , then

1.  $Z(G)$  is of multiplicative type.
2.  $R(G) = Z(G)_t = Z(G)_{\text{red}}^0$  (10.2.3.21).
3.  $R(G_{k'}) = R(G)_{k'}$  for any  $k'/k \in \mathbf{Field}$ .
4.  $G/R(G) \in \mathcal{R}\text{edGrp}^{\text{ss}}/k$ .
5.  $G_{\text{ad}} \triangleq G/Z(G)$  is an adjoint group (10.4.1.27).
6.  $Z(G) \cap [G, G]$  is finite.

┘

*Proof:* 1: Let  $T$  be a maximal torus in  $G$  (10.4.1.17), then  $Z(G) \leq C_G(T) = T$  (10.4.2.15), thus  $Z(G)$  is of multiplicative type.

2: Cf. [Mil17]P373?

3: This follows from 3 because taking center and taking largest torus commute with base change of fields (10.2.3.21).

4: By item3 we can base change to  $\bar{k}$ , and then the assertion follows from (10.4.2.3).

5:  $G_{\text{ad}}$  is semisimple by item4, and if  $Z'$  be the inverse image of  $Z(G/Z(G))$  in  $G$ , then  $Z'$  is an extension of  $Z(G/Z(G))$  by  $Z(G)$ , which are both of multiplicative type by item1, thus the conjugate action of  $G$  on  $Z'$  is trivial by (10.2.3.30), thus  $Z' = Z(G)$  and  $Z(G/Z(G)) = 1$ .

6:  $R(G) \cap [G, G]$  is finite by (10.2.3.10), and  $R(G)$  has finite index in  $Z(G)$  by item2, so  $Z(G) \cap [G, G]$  is finite. □

**Cor. (10.4.2.17) [Central and Multiplicative Isogenies].** Central and multiplicative isogenies from a reductive group are the same thing, by (10.1.5.20) and (10.4.2.16). ┘

**Prop. (10.4.2.18) [Reductive Groups and Semisimple Groups].** A semisimple group is reductive, because  $R_u(G) \subset R(G)$ , because a unipotent group is solvable. Conversely, if  $G \in \mathcal{R}\text{edGrp}/k$ , then the following are equivalent:

- $G$  is semisimple.
- $R(G) = e$ .
- $Z(G)$  is finite.

- $G/[G, G]$  is finite.

┘

*Proof:* 1  $\iff$  2 follows from (10.4.2.16).

2  $\iff$  3: As  $R(G)$  is the maximal subtorus of the group  $Z(G)$  of multiplicative type, the assertion follows from (10.2.3.21).

4: ?

□

**Prop. (10.4.2.19).** If  $G \in \text{RedGrp}/k$ , then every commutative normal subgroup variety of  $G$  is a torus, and the converse is also true if  $k = k^{\text{perf}}$ .

┘

*Proof:* If  $N$  is a commutative normal subgroup variety of  $G$ , then  $N \subset R(G)$ , which is a torus by (10.4.2.16) and (10.2.3.7). The converse follows from the fact  $R_u(G) = e$  because  $\text{Unip}(n)$  has no non-zero subtorus by (10.5.3.8).

□

**Prop. (10.4.2.20).** Any commutative affine group variety  $G$  is of the form  $U \times T$  where  $U$  is unipotent and  $T$  is a torus.

┘

*Proof:* ?

□

**Prop. (10.4.2.21) [Semisimple and Reductive under Isogenies].** Let  $\varphi : G' \rightarrow G \in \text{GrpVar}/k$  be an isogeny. If  $G$  is reductive or semisimple, then so is  $G'$ .

┘

*Proof:* By (10.4.2.8), it suffices to prove for  $k = \bar{k}$ . Let  $U$  be a normal unipotent/solvable subgroup variety of  $G$ , then  $\varphi(U)$  is also a unipotent/solvable group variety by (10.1.5.33), and it is normal because  $\varphi$  is a quotient map (10.1.5.33). Thus  $\varphi(U) = e$ , which implies  $U$  is a finite group variety, thus trivial.

□

**Prop. (10.4.2.22) [Matsushima's Criterion].** If  $G \in \text{RedGrp}/k$  and  $H \leq G$  is a smooth algebraic subgroup, then  $G/H$  is affine iff  $H^0$  is reductive.

┘

*Proof:* Cf. [Borel, On affine algebraic homogeneous spaces].

□

**Prop. (10.4.2.23) [Derived Groups are Semisimple].** For  $G \in \text{RedGrp}/k$ ,

- The semisimple rank of  $G$  equals  $\text{rank}(G/R(G)) = \text{rank } G - \dim Z(G) = \text{rank}(G^{\text{ad}})$ .
- $[G, G] \rightarrow G/R(G) \rightarrow G^{\text{ad}}$  are central isogenies.
- $[G, G]$  is semisimple with rank equal to the semisimple rank of  $G$ .

┘

*Proof:* 1: The kernel is finite by (10.4.2.16), and the image is of finite index by (10.4.2.16)(10.4.5.17) and (10.2.3.10).

2: The semisimple rank of  $G$  equals  $\text{rank}(G_{\bar{k}}/R(G_{\bar{k}})) = \text{rank}(G_{\bar{k}}/R(G)_{\bar{k}}) = \text{rank}(G/R(G))$  (10.4.2.16). Let  $T$  be a maximal torus of  $G$ , then  $Z(G) \leq T$  by (10.4.2.15), and  $T/R(G)$  is a maximal torus of  $G/R(G)$ , so  $\text{rank}(G/R(G)) = \dim(T/R(G)) = \dim T - \dim Z(G) = \text{rank } G - \dim Z(G)$  (10.4.2.16). The same argument shows that  $\text{rank } G - \dim Z(G) = \text{rank}(G^{\text{ad}})$ .

3:  $[G, G]$  is semisimple by (10.4.2.21), and the same argument as in item 2 shows that  $\text{rank}([G, G]) = \text{rank}(G^{\text{ad}})$ .

□

**Def. (10.4.2.24) [Anisotropic Reductive Groups].**  $G \in \text{RedGrp}$  is called an **isotropic reductive group** if it contains a non-trivial split torus; otherwise, it is called an **anisotropic reductive group**.

┘

**Def. (10.4.2.25) [Elliptic Maximal Tori].** For  $G \in \mathcal{R}\text{edGrp}/k$ , an **elliptic maximal torus** of  $G$  is a maximal torus that is anisotropic modulo  $Z(G)$  (10.4.2.15)(10.4.2.24). ┘

**Prop. (10.4.2.26) [Elliptic Elements].** For  $G \in \mathcal{R}\text{edGrp}/k$ ,  $x \in G(k)$  is called an **elliptic element** if  $x$  is not contained in any elliptic torus of  $G$  (10.4.2.25). ┘

If  $G$  admits an elliptic maximal torus (e.g.  $k \in p\text{-LField}$  (20.7.7.4) or  $k \in \mathbf{NField}$  ? or  $k = \mathbb{R}$  and  $G$  has Shimura variety ?), then  $x \in G(k)$  is elliptic iff  $Z_G(x)^0/Z_G^0$  is anisotropic. ┘

*Proof:* ? □

**Prop. (10.4.2.27) [Maximal Tori and Isogenies].** If  $G \in \mathcal{R}\text{edGrp}/k$  and  $T \leq G$  is a maximal torus, then ┘

- The intersection  $T_{\text{der}} = T \cap [G, G]$  is a maximal torus of  $[G, G]$ , and  $T = T_{\text{der}}R(G)$ . This induces a bijection between the set of maximal tori of  $[G, G]$  and the set of maximal tori of  $G$ .
- The preimage  $T_{\text{sim.cntd}} \leq G_{\text{sim.cntd}}$  is a maximal torus.

• The image  $T_{\text{ad}} \leq G_{\text{ad}}$  is a maximal torus.

If  $S \leq G$  is a maximal split torus, then

- The intersection  $S \cap [G, G]$  contains a unique maximal split torus  $S_{\text{der}}$  of  $G_{\text{der}}$ , and  $S = S_{\text{der}}A_G$  where  $A_G \leq G$  is the maximal split central torus. This induces a bijection between the set of maximal split tori of  $[G, G]$  and the set of maximal split tori of  $G$ .
- The preimage  $S_{\text{der}}$  in  $G_{\text{sim.cntd}}$  contains a unique maximal split torus  $S_{\text{sim.cntd}}$  of  $G_{\text{sim.cntd}}$ .
- The image  $S_{\text{ad}} \leq G_{\text{ad}}$  is a maximal split torus. ┘

*Proof:* ? 1: □

For 2, 3, notice that any maximal torus contains the center (10.4.2.15).

1:

2:

3: □

### Parabolic Subgroups of Reductive Groups

**Prop. (10.4.2.28) [Levi-Decompositions].** For  $P \in \mathcal{G}\text{rpVar}/k$ , a **Levi-factor** of  $P$  is a subgroup variety  $L \leq P$  s.t.  $L_{\bar{k}} \rightarrow P_{\bar{k}}/R_u(P_{\bar{k}})$  is an isomorphism. In other words,  $L$  is a reductive subgroup of  $P$  (10.4.2.11) s.t.  $P_{\bar{k}} = R_u(P_{\bar{k}}) \rtimes L_{\bar{k}}$ . If  $U \leq G$  is a subgroup s.t.  $U_{\bar{k}} = R_u(P_{\bar{k}})$  (e.g. when  $k = k^{\text{perf}}$  take  $U = R_u(P)$ ), then  $P = U \rtimes L$ , which is called a **Levi-decomposition** of  $P$ . ┘

**Def. (10.4.2.29) [Levi-Subgroups].** For  $G \in \mathcal{R}\text{edGrp}/k$ , a **levi-subgroup** is a Levi-factor (10.4.2.28) of a parabolic subgroup of  $G$ . ┘

**Thm. (10.4.2.30) [Mostov, [Mos56]].** If  $\text{char } k = 0$  and  $G \in \text{linAlgGrp}/k$  is arbitrary, then there exists  $M \leq G$  s.t.  $M^0$  is reductive and  $G = M.R_u(G)$ . Any such  $M$  is conjugate under conjugation by  $R_u(G)(k)$ . ┘

*Proof:* □

**Prop. (10.4.2.31) [Parabolic Subgroups are Rationally Conjugate].** If  $G \in \mathcal{RedGrp}/k$  and  $P, P' \leq G$  are two parabolic subgroups that are conjugate over  $\bar{k}$ , then they are conjugate.  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (10.4.2.32) [Parabolic Subgroups are Concentrator Subgroups].** For  $G \in \mathcal{RedGrp}/k$ , any parabolic subgroup  $P \leq G$  is of the form  $P_G(\lambda)$  (10.4.4.3) for some cocharacter  $\lambda \in X_*(G)$ . In particular, its unipotent radical is split (10.2.4.10).  $\lrcorner$

*Proof:* Cf. [CGP15]P67.  $\square$

**Cor. (10.4.2.33) [Parabolic Subgroups have Levi-Decompositions].** Let  $G \in \mathcal{RedGrp}/k$  and  $P \leq G$  be a parabolic subgroup, then  $P$  has Levi-decompositions (10.4.2.28), and any two Levi subgroups of  $P$  are conjugate by a unique element of  $R_u(P)(k)$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P559.  $\square$

**Cor. (10.4.2.34).** For  $G \in \mathcal{RedGrp}/k$ ,  $G$  has a properly-contained parabolic subgroup iff it has a non-central split torus.  $\lrcorner$

**Prop. (10.4.2.35) [Parabolic Subgroups and Reductive Quotients].** For  $\text{pr} : G \rightarrow G/R_u(G)$ , the maps

$$P \mapsto \text{pr}(P), \bar{P} \rightarrow f^{-1}(\bar{P})$$

are mutually-inverse  $\text{Gal}_k$ -equivariant  $G(k)$ -equivariant inclusion-preserving bijections between sets of parabolic subgroups of  $G$  and  $G/R_u(G)$ . In particular,  $G$  is quasi-split iff  $G/R_u(G)$  is quasi-split.  $\lrcorner$

*Proof:* Cf. [K-P23]P84.  $\square$

**Cor. (10.4.2.36).** If  $k = k^{\text{perf}}$ , then  $G$  has a properly-contained parabolic subgroup iff  $G/R_u(G)$  has a non-central split torus, by (10.4.2.34) and (10.4.2.11).  $\lrcorner$

### 3 Geometric Aspects

**Def. (10.4.3.1) [Simply-Connected Groups].**  $G$  is called a **simply-connected group variety** if every multiplicative isogeny (10.1.5.18) from another group variety  $G' \rightarrow G \in \mathcal{GrpVar}/k$  is an isomorphism.  $\lrcorner$

**Def. (10.4.3.2) [Isogenies of Reductive Pairs].** An **isogeny of reductive pairs**  $\varphi : (G, T) \rightarrow (G', T')$  is an isogeny  $\varphi : G \rightarrow G' \in \mathcal{RedGrp}/k$  s.t.  $\varphi(T) \subset T'$ .  $\lrcorner$

**Prop. (10.4.3.3) [Liftings].** If  $G$  is simply-connected (10.4.3.1) and  $\varphi : G' \rightarrow G$  is a multiplicative isogeny, then  $\varphi$  admits a section if  $k = k^{\text{perf}}$  or  $G$  is a perfect group.  $\lrcorner$

*Proof:* Cf. [Mil17]P388.  $\square$

**Def. (10.4.3.4) [Universal Covering].** A **universal covering** of  $G$  is a multiplicative isogeny  $\pi : G' \rightarrow G$  where  $G' \in \mathcal{GrpVar}/k$  is simply-connected.  $\lrcorner$

**Prop. (10.4.3.5) [Galois Theory].** Let  $\text{pr} : \tilde{G} \rightarrow G \in \mathcal{GrpVar}$  be a universal covering (10.4.3.4). If  $k = k^{\text{perf}}$  or  $G$  is perfect, then  $\text{pr}$  factors uniquely through any other multiplicative isogeny  $G' \rightarrow G \in \mathcal{GrpVar}/k$ .  $\lrcorner$



*Proof:* As  $\tilde{G}$  is a group variety, it admits no finite quotient(10.1.5.16), thus if  $[G, G] = G$ , then  $[\overline{G}, \overline{G}] \ker \pi = \overline{G}$ , thus  $G/[G, G]$  is finite, thus trivial, so  $\tilde{G}$  is also perfect.

The map  $G' \times_G \tilde{G}$  is surjective with finite kernel of multiplicative type, thus it has a section by(10.4.3.3), and the composite of this section with projection to  $G'$  is the desired lifting. If  $\alpha, \beta$  are two liftings, then  $\alpha/\beta$  is a map from  $\tilde{G}$  to  $\ker(\varphi)$ , which is finite, thus  $\alpha = \beta$ .  $\square$

**Prop.(10.4.3.6)[Simply-Connected Coverings].** Every  $G \in \mathcal{R}\text{edGrp}^{\text{ss}}/k$  admits an essentially unique universal covering(10.4.2.23).

In particular, for any  $G \in \mathcal{R}\text{edGrp}/k$ , there exists a universal covering  $G_{\text{sim.cntd}}$  of  $[G, G]$ , called the **simply-connected covering** of  $G$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop.(10.4.3.7).** If  $G$  is semisimple, then  $G$  is simply-connected(10.4.3.1) iff every central isogeny  $G' \rightarrow G$  with  $G'$  semisimple is an isomorphism.  $\lrcorner$

*Proof:* If  $G' \rightarrow G$  is a multiplicative isogeny then  $G'$  is semisimple by(10.4.2.20). If  $G' \rightarrow G$  is a central isogeny with  $G'$  semisimple, then it is a multiplicative isogeny by(10.4.2.17).  $\square$

## 4 Root Data

**Notation(10.4.4.1).**

- Let  $G \in \mathcal{R}\text{edGrp}/k$ .
- Let  $S \leq G$  be a maximal split torus.
- Let  $T \leq G$  be a maximal torus containing  $S$ .
- Let  $\Phi(T, G)$ (resp.  $\Phi(S, G)$ ) be the set of roots of the conjugation action of  $T$ (resp.  $S$ ) on  $G$ (10.2.4.13).
- Let  $\mathfrak{g}$  be the Lie algebra of  $G$ (10.2.2.2).
- Let  $\mathfrak{t}$  be the Lie algebra of  $T$ .

$\lrcorner$

**Def.(10.4.4.2)[Split Reductive Groups].** A **split reductive pair** is a pair  $(G, T)$  where  $G \in \mathcal{R}\text{edGrp}/k$  and  $T$  is a split maximal torus(10.1.1.4) of  $G$ . A reductive group is called a **split reductive group** if it is contained in some reductive pair. By(10.2.3.6), any reductive group is split reductive after a finite separable field extension.

This definition is compatible with that of(10.4.1.1).  $\lrcorner$

*Proof:* If  $G$  has a split maximal torus, then it has a split Borel subgroup by(10.4.4.12). Conversely, if  $G$  has a split Borel subgroup, then  $B$  contains a split maximal torus by(10.2.7.13), which is also maximal in  $G$  by base change to  $\bar{k}$  and use(10.4.1.7).  $\square$

**Thm.(10.4.4.3)[Parabolic Subgroups].** If  $G \in \mathcal{R}\text{edGrp}/R$ , then for any  $\lambda \in X_*(G)$ ,  $P_G(\lambda)$  is a parabolic subgroup scheme with unipotent radical  $U_G(\lambda)$ , and any ?  $\lrcorner$

*Proof:* ? Cf.[CGP15]  $\square$

**Prop.(10.4.4.4)[Action of Weyl Groups].** If  $\alpha \in \Phi(G, S)$ (10.4.4.1) and  $s \in W(G, S)(k)$ (10.4.1.23), then  $s\alpha \in \Phi(G, S)$ , and  $\mathfrak{g}_{s\alpha} = s\mathfrak{g}_\alpha$ , where  $s\mathfrak{g}_\alpha$  means  $n\mathfrak{g}_\alpha$  for any  $n \in N_G(S)(k)$  representing  $s$ , which is well-defined.  $\lrcorner$



*Proof:* Let  $n \in N_G(T)(k)$  represent  $s$ , and let  $t \in T(R)$  for some  $R \in \mathcal{C}\text{Ring}/k$ , then  $(s\alpha)(t) = \alpha(n^{-1}tn)$ . For any  $x \in \mathfrak{g}_\alpha$ , we have  $t(nx) = t(nx) = n(n^{-1}tn)x = \alpha(n^{-1}tn)n(x) = ((s\alpha)(t))(nx)$ . So  $\mathfrak{g}_{s\alpha} = s\mathfrak{g}_\alpha$ .  $\square$

**Prop. (10.4.4.5).** The natural map  $\text{Hom}(X^\bullet(G), \mathbb{Q})^{\text{Gal}_k} \rightarrow \text{Hom}(X^*(G), \mathbb{Q})$  is an isomorphism.  $\lrcorner$

*Proof:* The exact sequence  $0 \rightarrow X^\bullet(G) \rightarrow X^\bullet(G) \rightarrow X^\bullet(G)/X^*(G) \rightarrow 0$  remains exact after applying  $\text{Hom}(-, \mathbb{Q})^{\text{Gal}_k}$ , but

$$\text{Hom}(X^\bullet(G)/X^*(G), \mathbb{Q})^{\text{Gal}_k} = \text{Hom}((X^\bullet(G)/X^*(G))_{\text{Gal}_k}, \mathbb{Q})$$

vanishes because  $(X^\bullet(G)/X^*(G))_{\text{Gal}_k}$  is finite, because  $(X^\bullet(G)/X^*(G))^{\text{Gal}_k} = 0$ ?  $\square$

**Prop. (10.4.4.6) [Characters of the Center].** Let  $Z\Phi$  be the  $\mathbb{Z}$ -submodule of  $X^\bullet(T)$  generated by  $\Phi = \Phi(G, T)$ , then  $Z(G)$  and  $T/Z(G)$  are diagonalizable groups with

$$X^\bullet(Z(G)) = X^\bullet(T)/Z\Phi, \quad X^\bullet(T/Z(G)) = Z\Phi.$$

$\lrcorner$

*Proof:* By (10.4.2.15), there is an exact sequence

$$0 \rightarrow Z(G) \rightarrow T \xrightarrow{t \mapsto (\alpha(t))_\alpha} \prod_{\alpha \in \Phi(G, T)} \mathbb{G}_m,$$

so the assertions follow from (10.2.3.4).  $\square$

**Prop. (10.4.4.7).** If  $A_G$  is the maximal central split torus of  $G$ , then the natural map  $X^*(G) \rightarrow X^*(A_G)$  is injective with finite cokernel.  $\lrcorner$

*Proof:* It follows from (10.4.2.16) that  $X^*(G) = X^*(G/[G, G])$  and  $G/[G, G]$  is a connected quotient of  $Z(G)$  with finite kernel. Then the assertion follows from the fact that

$$X^\bullet(Z(G)_{\text{lf}})^{\text{Gal}_K} \rightarrow (X^\bullet(Z(G))_{\text{Gal}_K})_{\text{lf}}$$

is injective with finite cokernel.  $\square$

**Thm. (10.4.4.8) [Root Datum of  $G$ ].** Notation as in (10.2.4.20), if  $S$  is split, then

- $(G_\alpha, S)$  is a split reductive group of semisimple rank 1.
- $\text{Lie}(G_\alpha) = \mathfrak{t} \oplus \mathfrak{g}_{-\alpha} \oplus \mathfrak{g}_\alpha$ , and  $\dim \mathfrak{g}_\alpha = \dim \mathfrak{g}_{-\alpha} = 1$ . In particular,  $\mathbb{Q} \cdot \alpha \cap \Phi(G, S) = \{-\alpha, 0, \alpha\}$ .
- $U_\alpha$  is normalized by  $S$  and is isomorphic to  $\mathbb{G}_a$ , with Lie algebra  $\mathfrak{g}_\alpha$ . Any other  $T$ -normalized subgroup variety of  $G$  contains  $S_\alpha$  iff its Lie algebra contains  $\mathfrak{g}_\alpha$ .
- $W(G_\alpha, S) = \{1, s_\alpha\}$ , where  $s_\alpha$  can be represented by some  $n_\alpha \in N_{G_\alpha}(T)(k)$ .
- There exists a unique  $\alpha^\vee \in X_\bullet(S)$  s.t.  $s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha$  for any  $x \in X^\bullet(T)$ . Moreover,  $\langle \alpha, \alpha^\vee \rangle = 2$ .

And  $G$  is generated by  $T$  and the root groups  $\{U_\alpha\}_{\alpha \in \Phi(G, S)}$ .

The same hold in the non-split case, except that  $U_\alpha$  is not isomorphic to  $\mathbb{G}_a$  but split unipotent. In fact, if  $2\alpha \notin \Phi(G, T)$ , it is a vector group, and when  $2\alpha \in \Phi(G, T)$ ,  $U_\alpha/U_{2\alpha}$  is a vector space.  $\lrcorner$

*Proof:* Cf.[Mil17]P428.?

□

**Cor.(10.4.4.9).** Situation as in(10.4.4.8),  $U_\alpha$  is the unique subgroup of  $G$  that is normalized by  $T$ , isomorphic to  $\mathbb{G}_a$ , and  $T$  acts on it through  $\alpha$ .  $\lrcorner$

*Proof:* Cf.[Mil17]P431?.

□

**Cor.(10.4.4.10) [The Cocharacter  $\alpha^\vee$ ].** Situation as in(10.4.4.8), for  $\alpha \in \Phi(G, S) \in X^\bullet(S)$ , let  $C_\alpha = \ker(\alpha) \subset S$ , then  $S/C_\alpha \cong \mathbb{G}_m$ , and the homomorphism

$$S \rightarrow S : s \mapsto s/s_\alpha(s)$$

factors through  $C_\alpha$  and induces a map  $\mathbb{G}_m \rightarrow S/C_\alpha \rightarrow S$ . Then this map is exactly  $\alpha^\vee \in X_\bullet(S)$ (10.4.4.8).  $\lrcorner$

*Proof:* Cf.[K-P23]P94.

□

**Def.(10.4.4.11) [Regular Cocharacters].** Situation as in(10.4.4.8),  $\lambda \in X_\bullet(S)$  is called a **regular cocharacter** if it is contained in the interior of some Weyl chamber(3.9.5.8), i.e.  $\langle \alpha, \lambda \rangle \neq 0$  for any  $\alpha \in \Phi(G, S)$ .  $\lrcorner$

**Prop.(10.4.4.12) [Borel Subgroups Containing  $T$  are Split].** Let  $\lambda \in X_\bullet(T)$  be a regular cocharacter(10.4.4.11), then  $P(\lambda)$ ?? is the unique Borel subgroup of  $G$  containing  $T$  with Lie algebra

$$\mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi(G, T), \langle \alpha, \lambda \rangle > 0} \mathfrak{g}_\alpha.$$

In this way, we get a bijection between the set of Weyl chambers to the set of Borel subgroups containing  $T$ . In particular, all the Borel subgroups of  $G_{\bar{k}}$  containing  $T_{\bar{k}}$  are defined over  $k$ , and they are split solvable.  $\lrcorner$

*Proof:* Cf.[Mil17]P433, 434, 435?.

□

**Cor.(10.4.4.13) [Root Data for  $G$ ].** If  $T$  is split,  $(X^\bullet(T), \Phi(G, T), X_\bullet(T), \Phi(G, T)^\vee, W(G, T)) \in \text{RootDat}_{\text{red}}$  is a reduced root datum(3.9.5.40). Moreover, its isomorphism type doesn't depend on  $T$ , and is functorial under isomorphisms of  $G$ . And its Weyl group is exactly  $W(G, T)$ (10.4.1.23).  $\lrcorner$

*Proof:* To check  $(X^\bullet(T), \Phi(G, T), W(G, T)(k)) \in \text{RootSys}_{\text{red}}$ ,  $s_\alpha$  exists by(10.4.4.8), and  $s_\alpha(\Phi) = \Phi$  by(10.4.4.4). These  $s_\alpha$  generate the Weyl group  $W(G, T)$  because  $G$  is generated by  $T$  and the root groups  $\{U_\alpha\}_{\alpha \in \Phi}$ .

Choose a Borel subgroup  $B \leq G$  containing  $T$ (10.4.4.12). Suppose  $(B', T')$  is another split Borel pair, then there exists  $g \in G(k)$  s.t.  $(B', T') = g(B, T)g^{-1}$  by(10.4.1.7). This  $g$  defines an isomorphism between  $\Phi(G, T')$  and  $\Phi(G, T)$ . If  $(B', T') = g'(B, T)(g')^{-1}$  for some other  $g' \in G(k)$ , then  $g' = gh$  for some  $h \in N_G(T)(k)$ , but then  $h \in T$  by(10.4.4.12) and(3.9.5.16). Thus this inner isomorphism is unique.  $\lrcorner$

□

**Thm.(10.4.4.14) [Relative Root Datum for  $G$ ].**  $(X^\bullet(S), \Phi(G, S), X_\bullet(S), \Phi(G, S)^\vee, W(G, S)) \in \text{RootDat}$  is a root datum(3.9.5.40), and its Weyl group is exactly  $W(G, S)$ (10.4.1.23). This root system is called the **relative root datum** of  $G$  w.r.t.  $S$ .  $\lrcorner$

*Proof:* ? Cf.[Borel91]Thm21.6.

□

**Cor.(10.4.4.15)[Absolute Root Datum for  $G$ ].** By(10.4.4.14),

$$(X^\bullet(T), \Phi(G_{k^{\text{sep}}}, T_{k^{\text{sep}}}), X_\bullet(T), \Phi(G_{k^{\text{sep}}}, T_{k^{\text{sep}}})^\vee, W(G_{k^{\text{sep}}}, T_{k^{\text{sep}}}))$$

is a another root datum, called the **absolute root datum** of  $G$  w.r.t.  $T$ . We denote  $\Phi(G, T) \triangleq \Phi(G_{k^{\text{sep}}}, T_{k^{\text{sep}}})$ .

Then  $\Phi(G, S)$  is contained in the image of  $\Phi(G, T)$  under the map  $X^\bullet(T) \rightarrow X^\bullet(S)$  ┘

*Proof:* ? Cf.[Borel91]Thm21.8. □

**Prop.(10.4.4.16)[Root Systems and Isogenies].** For a maximal torus  $S \leq G$ , the natural maps

$$\Phi(G_{\text{sim.cntd}}, S_{\text{sim.cntd}}) \rightarrow \Phi(G_{\text{der}}, S_{\text{der}}) \rightarrow \Phi(G, S) \rightarrow \Phi(G_{\text{ad}}, S_{\text{ad}}) \quad (10.4.2.27)$$

are bijective. And for  $\alpha \in \Phi(G, S)$ , these maps induce isomorphism on the corresponding root groups  $U_\alpha$ . ┘

*Proof:* ? □

**Def.(10.4.4.17)[Positive System of Roots].** Let  $B \leq G$  be a minimal parabolic subgroup containing  $T$  with Lie algebra  $\mathfrak{b}$ , define the system of **positive roots attached to  $B$** :

$$\Phi^+(B) = \{\alpha \in \Phi | \mathfrak{g}_\alpha \subset \mathfrak{b}\},$$

and the **dominant Weyl-chamber**

$$C_+(B) = \{\lambda \in X_\bullet(T)_\mathbb{R} | \langle \alpha, \lambda \rangle > 0, \quad \forall \alpha \in \Phi^+(B)\}.$$

┘

**Prop.(10.4.4.18)[Positive System of Roots and Minimal Parabolics].** Construction(10.4.4.17) gives a bijection between set of positive system of roots of  $\Phi$  and the set of minimal parabolic subgroups containing  $T$ . ┘

*Proof:* ? □

**Cor.(10.4.4.19)[Quasi-Splitness and Centralizers of Maximal Split Tori].**  $G$  is quasi-split iff the centralizer of a maximal split torus is a maximal torus(i.e. solvable). ┘

*Proof:* The centralizer of a maximal split torus is the Levi of a minimal parabolic subgroup, thus it is a maximal torus iff the minimal parabolic subgroup is solvable. □

**Prop.(10.4.4.20).**  $G$  is semisimple iff its root datum(10.4.4.13) is semisimple(3.9.5.1). ┘

*Proof:*  $G$  is semisimple iff it has finite center(10.4.2.18), which is equivalent to  $\mathbb{Z}\Phi$  having finite index in  $X^\bullet(T)$  by(10.4.4.6). □

**Prop.(10.4.4.21).**

- If  $G$  is semisimple, then  $G$  is generated by the root groups  $U_\alpha$ (10.2.4.20) for each  $\alpha \in \Phi(G, T)$ . Thus it is also generated by the subgroups  $G^\alpha$  for each  $\alpha \in \Phi(G, T)$  by(10.2.4.22).
- If  $\Delta \subset \Phi(G, T)$  is a base, then  $[G, G]$  is generated by the subgroups  $G^\alpha$ (10.2.4.20) for each  $\alpha \in \Delta$ , and  $G$  is generated by the subgroups  $G_\alpha$  for each  $\alpha \in \Delta$ , thus also by  $T$  and  $U_\alpha$  for each  $\alpha \in \Delta$  by(10.2.4.22).

┘

*Proof:* Cf.[Mil17]P439, P443. □

**Prop. (10.4.4.22).** For any  $\alpha \in \Phi(G, S)$  and any  $u \in U_\alpha(k) \setminus \{1\}$ , there exists unique  $u', u'' \in U_{-\alpha}(k)$  s.t.  $m(u) = u'uu'' \in N_G(S)(k)$ . Moreover we have

- $u' \neq 1, u'' \neq 1$ .
- $m(u)$  acts on  $S$  as  $s_\alpha$ (10.4.4.8).
- $m(u) = m(u') = m(u'')$ .
- $m(u)^{-1} = m(u^{-1})$ .

┘

*Proof:* Cf.[K-P23]Prop C.2.24.

3 follows from the identities

$$m(u) = (m(u)u''m(u)^{-1})u'u = uu''(m(u)^{-1}u'm(u))$$

and the fact that  $m(u)u''m(u)^{-1}, m(u)^{-1}u'm(u) \in U_{-\alpha}(k)$  by item2.

4 is easy. □

**Def. (10.4.4.23) [Generic Unipotent Characters].** Let  $B = LU \leq G$  be a Borel-subgroup with associated base  $\Delta \subset \Phi(G, T)$ , then a character  $\psi : U(k) \rightarrow \mathbb{C}^\times$  is called a **generic unipotent character** on if  $\psi|_{U_\alpha(k)}$ (10.2.4.20) is non-trivial for each  $\alpha \in \Delta$ . Equivalently, the stabilizer of  $\psi$  in the  $k$ -points  $T = B/U$  is  $Z(k)$ .

A **Whittaker-datum** is a  $G(F)$ -conjugacy class of pairs  $(B, \psi)$  where  $B$  is a Borel subgroup of  $G$  and  $\psi$  is a generic unipotent character on  $B$ . ┘

**Prop. (10.4.4.24).** There are only f.m. Whittaker-data for  $G$ . And if  $G$  is an adjoint group, there exists a unique Whittaker-datum(10.4.4.23). ┘

*Proof:* ? □

### Reductive Groups of Semisimple Rank $\leq 1$

**Prop. (10.4.4.25) [Classifying Reductive Groups of Semisimple Rank  $\leq 1$ ].** Any  $G \in \mathcal{RedGrp}/k$  of semisimple rank 1(10.4.1.18) is isomorphic to exactly one of the groups

$$\mathbb{G}_m^r \times \mathrm{SL}(2), \quad \mathbb{G}_m^r \times \mathrm{PGL}(2), \quad \mathbb{G}_m^r \times \mathrm{PGL}(2), \quad r \in \mathbb{N}.$$

┘

*Proof:* Cf.[Mil17]P419. □

**Prop. (10.4.4.26).** Any  $G \in \mathcal{GrpVar}^1/k$  is commutative. ┘

*Proof:* Cf.[Mil17]P286. □

### Bruhat Decompositions

**Thm. (10.4.4.27)[Bruhat-Decompositions].** Let  $(G, B, T, S)$  be a split reductive group, denote  $N = N_G(T)$  and  $R = \{s_\alpha | \alpha \in S\}$ , then  $(B, N)$  is a Tits system (20.6.1.1).

In particular, we have a Bruhat decomposition

$$G(K) = B(K)WB(K).$$

┘

*Proof:* ? Cf. [Mil17]P448.

□

**Cor. (10.4.4.28)[Bruhat Decompositions for  $\mathrm{GL}(n)$ ].** For  $K \in \mathbf{Field}$ ,

$$\mathrm{GL}(n; K) = B(K)WB(K) = B(K)W \cdot \mathrm{Unip}(n; K)$$

where  $W$  is the set of permutation matrices,  $B$  is the invertible upper triangular matrices, and the decomposition is a disjoint union w.r.t.  $W$ .

Moreover, if  $K$  is a topological field, there is a lexicographical stratification  $0 = W_0 \subset W_1 \subset \dots \subset W_n$  of  $\mathrm{GL}(n; K)$  that  $W_k$  is closed in  $W_{k+1}$  and each  $W_{k+1} \setminus W_k$  is a double coset  $BwB$ .

For  $g \in \mathrm{GL}(n; K)$ , in order to find  $w \in W$  s.t.  $g \in BwB$ , it suffices to manipulate  $g$  by the following two operations:

- Add a row to rows before it.
- Add a column to columns after it.
- Multiply a row by non-zero scalars.

┘

*Proof:* For any matrix  $M \in \mathrm{GL}(n; K)$ , consider the first column, then there is a lowest term  $a_{i_1 1}$  that are not zero, then we can left multiply an upper triangular matrix  $b_1$  s.t. the first column of  $b_1 M$  has only one nonzero entry  $a_{i_1 1} = 1$ , then consider the second column but ignore the  $k$ -th row, we can find a lowest term  $a_{i_2 2}$  that is non-zero, then left multiply an upper triangular matrix  $b_2$  that the second column of  $b_2 b_1 M$  has only one non-zero entry  $a_{i_2 2} = 1$ . Now continuing this way, we find a permutation  $\sigma$  that only the entries  $a_{ij}$  that  $j \geq \sigma^{-1}(i)$  are non-zero. Then we can find an upper triangular matrix  $c$  that  $b_n b_{n-1} \dots b_1 M c$  is a permutation matrix  $M_{\sigma^{-1}}$ .

So we proved  $BWB = \mathrm{GL}(n; K)$ . Now it suffices to show if  $M_{\sigma_1}^{-1} b M_{\sigma_2} \in B$  for some  $b \in B$ , then  $\sigma_1 = \sigma_2$ : Because  $M_\sigma = \sum_i e_{\sigma(i)i} = \sum_i e_{i\sigma^{-1}(i)}$ ,

$$M_{\sigma_1}^{-1} b M_{\sigma_2} = \sum_{ij} b_{\sigma_1(i)\sigma_2(j)} e_{ij}.$$

This is an element in  $B$ , both its  $(\sigma_1^{-1}(k), \sigma_2^{-1}(k))$ -entry is  $b_{kk} \neq 0$ , thus  $\sigma_1^{-1}(k) \leq \sigma_2^{-1}(k)$ , which implies  $\sigma_1 = \sigma_2$ . □

**Cor. (10.4.4.29).** For  $K \in \mathbf{Field}$ ,  $\mathrm{SL}(2; K)$  is generated by  $\mathrm{Unip}(K)$  and any element in  $\mathrm{SL}(2; K) \setminus B(K)$ . ┘

## 5 Isogeny and Existence Theorems

**Prop. (10.4.5.1) [Isogenies of Root Data].** If  $\varphi : (G', T') \rightarrow (G, T)$  is an isogeny of split reductive groups (10.4.3.2), then  $X^\bullet(\varphi) : X^\bullet(T) \rightarrow X^\bullet(T')$  is an isogeny of root data (3.9.5.50), where  $\alpha \in \Phi$  and  $\alpha' \in \Phi'$  correspond iff  $\varphi(U_\alpha) = U_{\alpha'}(10.2.4.20)$ , in which case

$$\varphi(u_\alpha(a)) = u_{\alpha'}(c_\alpha a^{q(\alpha)}),$$

where  $a \in k^\times$  and  $q(\alpha) \in \max(\text{char } k, 1)^\mathbb{N}$  s.t.  $f(\alpha') = q(\alpha)\alpha$ .

$\varphi$  is central/isomorphism/Frobenius iff  $X^\bullet(\varphi)$  is (3.9.5.51). ┘

*Proof:* Cf. [Mil17]P485. □

**Prop. (10.4.5.2).** If two isogenies of split reductive groups (10.4.3.2)  $\varphi_1, \varphi_2 : (G', T') \rightarrow (G, T)$  induce the same map on root data (10.4.5.1), then they differ by an inner automorphism by a unique element  $t \in (T/Z(G))(k)$ . ┘

*Proof:* Cf. [Mil17]P486. □

**Thm. (10.4.5.3) [Isogeny Theorem, Chevalley-Demazure].** Let  $\varphi : (G, T) \rightarrow (G', T')$  be an isogeny of split reductive pairs over  $k$  (10.4.3.2), then  $X^\bullet(\varphi) : X^\bullet(T') \rightarrow X^\bullet(T)$  induces an isogeny of root data  $\Phi(G', T') \rightarrow \Phi(G, T)$  (3.9.5.50), and every isogeny of root data  $\Phi(G', T') \rightarrow \Phi(G, T)$  arises in this way. Moreover,  $\varphi$  is uniquely determined by  $X^\bullet(T)$  up to an inner automorphism defined by an element of  $T'/Z(G')(k)$ .

The statement also holds for “isogeny” replaced by “central isogeny”, “Frobenius-isogeny” or “isomorphism”. ┘

*Proof:* This follows from (10.4.5.1)(10.4.5.2) and [Mil17]P487. ? □

**Cor. (10.4.5.4).** If  $(G, T)$  and  $(G', T')$  are split reductive groups over  $k$  s.t.  $(G, T)_{\bar{k}} \cong (G', T')_{\bar{k}}$ , then  $(G, T) \cong (G', T')$ . ┘

**Cor. (10.4.5.5).** If  $(G, T)$  and  $(G', T')$  are reductive groups over  $k$  s.t.  $(G, T)_{\bar{k}} \cong (G', T')_{\bar{k}}$ , then  $(G, T)_{k'} \cong (G', T')_{k'}$  for some  $k'/k$  separable. ┘

**Def. (10.4.5.6) [Pinnings].** A **pinning on split reductive group** on  $G$  is a pair  $(\Delta, (e_\alpha)_{\alpha \in \Delta})$ , where  $\Delta$  is a base of  $\Phi(G)$ , and  $e_\alpha$  is a non-zero element of  $\mathfrak{g}_\alpha$  for each  $\alpha \in \Delta$ . A **pinned split reductive group** is a split reductive group with a pinning. ┘

**Def. (10.4.5.7) [Isogenies of Pinned Groups].** An **isogeny of pinned groups**  $\varphi : (G, T, \Delta, (e_\alpha)_{\alpha \in \Delta}) \rightarrow (G', T', \Delta', (e'_\alpha)_{\alpha \in \Delta'})$  s.t.

- Under the correspondence  $\Phi \rightarrow \Phi' : \alpha \mapsto \alpha'$  (10.4.5.1),  $\Delta$  corresponds to  $\Delta'$ .
  - $\text{Lie}(\varphi)(e_\alpha) = e_{\alpha'}$ .
- ┘

**Thm. (10.4.5.8) [Isogeny of Pinned Split Reductive Groups].** Let  $(G, T, \Delta, (e_\alpha)_{\alpha \in \Delta}), (G', T', \Delta', (e'_\alpha)_{\alpha \in \Delta'}) \in \text{RedGrp}$  be pinned split reductive groups over  $k$  and let  $f : (\Phi(G'), \Delta') \rightarrow (\Phi(G), \Delta)$  be an isogeny of based root data (i.e. an isogeny of root data (3.9.5.50) mapping  $\Delta'$  to  $\Delta$ ), then there exists a unique isogeny of pinned groups  $\varphi : (G, T, \Delta, (e_\alpha)_{\alpha \in \Delta}) \rightarrow (G', T', \Delta', (e'_\alpha)_{\alpha \in \Delta'})$  s.t.  $X^\bullet(\varphi) = f$ .

In particular, we have

$$\text{Aut}(\Phi(G), \Delta) \cong \text{Aut}((G, T, \Delta, (e_\alpha)_{\alpha \in \Delta})) \cong \text{OAut}(G).$$
┘

*Proof:* This follows from isogeny theorem(10.4.5.3) and? Cf.[Mil17]P496.  $\square$

**Thm.(10.4.5.9)[Existence Theorem, Chevalley-Demazure].** Every root datum arises from a split reductive group over  $k$  as in(10.4.4.8).  $\lrcorner$

*Proof:* Cf.[Mil17]P501.  $\square$

**Cor.(10.4.5.10) [Split  $k$ -Forms].** For arbitrary  $G \in \mathcal{R}\text{edGrp}/k$ , there exists a unique split reductive group  $G_0 \in \mathcal{R}\text{edGrp}/k$  s.t.  $(G_0)_{k^{\text{sep}}} \cong G_{k^{\text{sep}}}$ , called the **split  $k$ -form** of  $G$ , and  $G$  is called an **inner reductive group** if  $G_0$  is an inner form of  $G$ ??, and an **outer reductive group** otherwise.  $\lrcorner$

*Proof:* Take a maximal torus  $T \leq G$ (10.4.1.17), then  $T_{k^{\text{sep}}}$  is split, then we can take  $(G_0, T_0) \in \mathcal{R}\text{edGrp}/k$  split with root system equal to that of  $(G, T)_{k^{\text{sep}}}$  by(10.4.5.9). Then  $(G_0)_{k^{\text{sep}}} \cong G_{k^{\text{sep}}}$  by(10.4.5.3).  $G_0$  is unique by isomorphism theorem(10.4.5.3).  $\square$

**Thm.(10.4.5.11) [Classifying Split Semisimple Groups].** Associating root diagrams to split semisimple groups(10.4.4.20) is bijection between  $\mathcal{R}\text{edGrp}^{\text{ss}}/k$  to the set of isomorphism classes of rational root diagrams(3.9.5.31).

A split semisimple group  $G \in \mathcal{R}\text{edGrp}^{\text{ss}}/k$  is simply-connected iff its root diagram is simply-connected??(i.e.  $X = P(\Phi(G))$ ).  $\lrcorner$

*Proof:* The first assertion follows from(3.9.5.44) and the isomorphism and existence theorem(10.4.5.3)(10.4.5.9).

For the last assertion, for any central isogeny  $\varphi : G' \rightarrow G$ ,  $f = X^\bullet(\varphi)$  is central(10.4.5.1), so

$$f(X) \subset X' \subset P(\Phi') = f_{\mathbb{Q}}(P(\Phi)).$$

If  $X = P(\Phi)$ , then  $f(X) = X'$  is an isomorphism, thus  $\varphi$  is an isomorphism by isogeny theorem(10.4.5.3) and(10.4.3.7).

Conversely, if  $X \neq P(\Phi)$ , then  $(X, \Phi, \Phi^\vee) \hookrightarrow (P(\Phi), \Phi, \Phi^\vee)$  is a central isogeny, which corresponds to a non-isomorphic central isogeny by existence and isogeny theorem(10.4.5.9)(10.4.5.3)  $\square$

**Def.(10.4.5.12) [Strictly Isogenous Semisimple Groups].** Two  $G, G' \in \mathcal{R}\text{edGrp}^{\text{ss}}/k$  are called **strictly isogenous semisimple groups** if there exist central isogenies  $H \rightarrow G, H \rightarrow G' \in \mathcal{R}\text{edGrp}/k$ .  $\lrcorner$

**Prop.(10.4.5.13) [Classifying Strictly Isogenous Split Semisimple Groups].** Strict-isogeny classes(10.4.5.12) of split semisimple groups over  $k$  are in bijection with the set of Dynkin-diagrams(3.9.5.29). Such a group is (geometrically)almost-simple iff its Dynkin-diagram is connected.  $\lrcorner$

*Proof:* Simply-connected semisimple groups over  $k$  are classified by their root systems(10.4.5.11), which are classified by Dynkin diagrams by(3.9.5.29). The last assertion follows from(10.4.5.18).  $\square$

### Inner Forms

**Notation(10.4.5.14).**

- Use notations defined in Cohomologies and Extensions.

$\lrcorner$

**Prop. (10.4.5.15) [Quasi-Split Forms].** Every element of  $H^1(k, \text{OAut}(G_{k^{\text{sep}}}))$  is the class of a quasi-split  $k$ -form of  $G$ , and two quasi-split  $k$ -forms of  $G$  are isomorphic iff they have the same class.  
 $\lrcorner$

*Proof:* Cf. [Mil17]P499.  $\square$

**Cor. (10.4.5.16) [Quasi-Split Inner Forms].** Any  $G \in \text{RedGrp}/k$  is an inner form of a quasi-split reductive group  $H \in \text{RedGrp}/k$ , and  $H$  is uniquely determined up to isomorphism.  $\lrcorner$

*Proof:* Cf. [Mil17]P500.  $\square$

### More on Reductive Groups

**Prop. (10.4.5.17) [Semisimple Groups are Perfect].** If  $G \in \text{RedGrp}^{\text{ss}}/k$ , then  $G$  is a perfect group, i.e.  $G = [G, G]$ .  $\lrcorner$

*Proof:* It suffices to prove for  $k = \bar{k}$ . Choose a maximal torus  $T \leq G$ , Then notice that each  $G^\alpha$  for  $\alpha \in \Phi(G, T)$  is split by [Mil17]Prop20.24  $\color{red}{?}$ , and they generate  $G$  by (10.4.4.21).  $\square$

**Prop. (10.4.5.18) [Almost-Simple Factors].** Any  $G \in \text{RedGrp}^{\text{ss}}/k$  has only f.m. almost-simple normal subgroup varieties  $G_1, \dots, G_r$  (10.4.2.6), and is an almost-direct product of them (10.1.5.9). These are called the **almost-simple factors** of  $G$ .

Every normal subgroup variety of  $G$  is an almost-direct product of the almost-simple factors it contains, and is centralized by the remaining ones.  $\lrcorner$

*Proof:* Cf. [Mil17]P440.  $\square$

**Prop. (10.4.5.19).** An isogeny of geometrically almost-simple split algebraic groups over  $k$  is an isomorphism if it restricts to an isomorphism  $T \rightarrow T'$ , except for the isogenies  $\text{SO}(2n+1) \rightarrow \text{Sp}(2n)$  when  $\text{char } k = 2$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P494.  $\square$

**Prop. (10.4.5.20) [Deconstructing Semisimple Groups].** If  $G$  is simply-connected and semisimple, then

$$G \cong \text{Res}_{k_1/k}(G_1) \times \dots \times \text{Res}_{k_m/k}(G_s),$$

where  $k \leq k_i \leq k^{\text{sep}}$ ,  $G_i \in \text{RedGrp}^{\text{ss}}/k$  is geometrically almost-simple for each  $i$ . And each  $\text{Res}_{k_i/k}(G_i)$  is almost-simple.

If  $G$  is an adjoint group (10.4.2.1), then

$$G \cong \text{Res}_{k_1/k}(G_1) \times \dots \times \text{Res}_{k_m/k}(G_s),$$

where  $k \leq k_i \leq k^{\text{sep}}$ ,  $G_i \in \text{RedGrp}^{\text{ss}}/k$  is geometrically simple. And each  $\text{Res}_{k_i/k}(G_i)$  is simple.  $\lrcorner$

*Proof:* Consider  $G_{k^{\text{sep}}}$  and the Galois action on the decomposition (10.4.5.18).  $\square$

**Prop. (10.4.5.21) [Reductive Groups and Semisimple Groups].** The following are equivalent:

- $G$  is reductive.
- $R(G_{\bar{k}})$  is a torus.
- $G$  is an almost-product of a torus and a semisimple group, (in fact,  $G = R(G) \cdot [G, G]$ ) (10.4.2.23).



And in this case,  $G$  has a semisimple representation with finite kernel. The converse holds if  $k = k^{\text{perf}}$ .  
 $\perp$

*Proof:* Cf. [Mil17]P442<sup>?</sup>.  $\square$

**Prop. (10.4.5.22) [Quasi-Split Twists of Reductive Groups].** Given a connected reductive group  $G/F$  and its dual group  $\widehat{G}$  (10.4.6.2), there exists a split reductive group that is isomorphic to  $G$  over  $\overline{F}$ <sup>?</sup>. Choose an isomorphism  $\psi$  over  $\overline{F}$  from  $G$  to such a group, then the map

$$\sigma \mapsto \sigma(\psi^{-1})\psi, \quad \sigma \in \text{Gal}_F$$

induces a map  $\text{Gal}_F \rightarrow \text{OAut}(G)$ .

Then to any such homomorphism, there corresponds to a unique quasi-split reductive group over  $F$ <sup>?</sup>.  $\perp$

*Proof:*  $\square$

**Prop. (10.4.5.23).**  $\perp$

### Algebraic Fundamental Groups

**Prop. (10.4.5.24) [Algebraic Fundamental Groups, Borovoi].** For  $G \in \text{RedGrp}/k$  and  $T \leq G$  a maximal torus,  $\text{Gal}_k$  acts on the quotient  $X_{\bullet}(T)/Q(\Phi^{\vee}(G, T))$  (3.9.5.30). If  $T' \leq G$  is another maximal torus, then there exists some  $g \in G(k)$  s.t.  $T' = gTg^{-1}$ , and we get an isomorphism  $\text{Ad}(g) : X_{\bullet}(T)/Q^{\vee}(\Phi(G)) \cong X_{\bullet}(T')/Q^{\vee}(\Phi(G, T'))$ . Because  $W(G, T)$  acts on  $X_{\bullet}(T)$  by adding co-roots, it acts trivially on  $P(\Phi(G, T))$ , and the isomorphism  $X_{\bullet}(T)/Q^{\vee}(\Phi(G, T)) \cong X_{\bullet}(T')/Q^{\vee}(\Phi(G, T'))$  doesn't depend on  $g$  and is canonical. In particular, it is  $\text{Gal}_k$ -equivariant. Then we define

$$\pi_1(G) = \lim_{T \leq G \text{ maximal tori}} X_{\bullet}(T)/Q(\Phi^{\vee}(G, T)),$$

called the **algebraic fundamental group** of  $G$ .

This assignment  $G \mapsto \pi_1(G)$  induces a functor  $\pi_1 : \text{RedGrp}/k \rightarrow \text{Mod}_{\text{Gal}_k}^{\text{fg}}$ .  $\perp$

*Proof:* Any  $f : H \rightarrow G \in \text{RedGrp}/k$  lifts to a map  $f_{\text{sim.cntd}} : H_{\text{sim.cntd}} \rightarrow G_{\text{sim.cntd}}$ <sup>?</sup>, and for any maximal torus  $T \leq H$ , we can take a maximal torus  $T' \leq G$  containing  $f(T)$ .  $\square$

**Cor. (10.4.5.25).** For  $G \in \text{RedGrp}/k$ ,

- If  $G = T$  is a torus, then  $\pi_1(G) = X_{\bullet}(T)$ .
  - $G$  is semisimple and simply-connected iff  $\pi_1(G) = \mathbf{1}$ .
  - If  $k = \mathbb{C}$ , then  $\pi_1(G) \cong \pi_1(G^{\text{an}})$ .
  - If  $\widehat{G} \in \text{RedGrp}/\overline{\mathbb{Q}}$  is the Langlands dual group of  $G$  (10.4.6.2), then there is an canonical isomorphism  $\pi_1(G) \cong X^{\bullet}(Z(\widehat{G}))$ .
- $\perp$

*Proof:* 1 is trivial.

2: This follows from (10.4.5.11).

3: This follows from item2 and the fact that  $G$  is simply-connected iff  $G^{\text{an}}$  is simply-connected<sup>?</sup>.

4: <sup>?</sup>  $\square$

**Prop. (10.4.5.26)** [ $\pi_1$  is Exact].  $\pi_1 : \text{RedGrp}/k \rightarrow \text{Ab}_{\text{Gal}_K}^{\text{fg}}$  is an exact functor.  $\lrcorner$

*Proof:* ? Cf. [Borovoi]Prop1.5.  $\square$

**Prop. (10.4.5.27)** [Inner Forms]. If  $G' \in \text{RedGrp}/k$  is an inner form of  $G \in \text{RedGrp}/k$ , then there is a canonical isomorphism

$$\pi_1(G) \cong \pi_1(G').$$

$\lrcorner$

*Proof:* ?  $\square$

### Z-Extensions

**Def. (10.4.5.28)** [Z-Extensions, Ono[Ono65]/Langlands[Lan79b]]. For  $G \in \text{RedGrp}/k$ , an **z-extension** of  $G$  is a central extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  s.t.

- $Z$  is an induced torus (10.2.3.11).
- $[\tilde{G}, \tilde{G}]$  is simply-connected.

$\lrcorner$

**Prop. (10.4.5.29)** [Properties of Z-Extensions, Langlands[Lan79b]/Kottwitz[Kot84]]. For  $G \in \text{RedGrp}/k$ ,

- If  $L/F \in \text{GalField}^{\text{fin}}$  s.t.  $G_L$  is split, there is a z-extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  s.t.  $Z$  is a product of the form  $\text{Res}_{L/F} \mathbb{G}_m$ .
- If  $\tilde{G}_1 \rightarrow G, \tilde{G}_2 \rightarrow G$  are two z-extensions, so is their fiber product.
- If  $f : G \rightarrow H \in \text{RedGrp}/k$  and  $\tilde{G} \rightarrow G, \tilde{H} \rightarrow H$  are z-extensions (10.4.5.28), then there exists a z-extension  $\tilde{G}_1 \rightarrow G$  factoring through  $\tilde{G}$  and a commutative diagram

$$\begin{array}{ccc} \tilde{G}_1 & \longrightarrow & \tilde{H} \\ \downarrow & & \downarrow \\ G & \longrightarrow & H \end{array}$$

$\lrcorner$

*Proof:* 1:  $Z(G)^0 \times G_{\text{sim.cntd}} \rightarrow G$  is surjective because  $[G, G] \rightarrow G_{\text{ad}}$  does, and it has finite kernel by (10.4.2.16). Denote the kernel by  $\mu$ , then  $\mu$  splits over  $L$  and the  $\text{Gal}(L/F)$ -module  $X^\bullet(\mu)$  is a quotient of a free  $\text{Gal}(L/K)$ -module  $M$ . Let  $Z = \mathbf{D}(M)$ , then  $Z$  is an induced torus defined over  $F$  with an injection  $\mu \hookrightarrow Z$ . Define  $\tilde{G} = ((Z(G)^0 \times G_{\text{sim.cntd}}) \times Z) / \mu$ , then  $\tilde{G}$  is a central extension of  $G$  by  $Z$  (10.4.2.13). Thus  $\tilde{G}$  is reductive.  $G_{\text{sim.cntd}}$  embeds into  $G$  as a normal subgroup with Abelian quotient, so  $[G, G] = G_{\text{sim.cntd}}$  is simply-connected.

2: The kernel  $Z$  of  $\tilde{G}_3 = \tilde{G}_1 \times_G \tilde{G}_2 \rightarrow G$  is the product of kernels of  $\tilde{G}_1 \rightarrow G$  and  $\tilde{G}_2 \rightarrow G$ , thus induced.  $\tilde{G}_3$  is reductive because  $\tilde{G}_3$  is a central extension of  $G$  by  $Z$  (10.4.2.13). And  $[\tilde{G}, \tilde{G}]$  is simply-connected because  $\tilde{G} \rightarrow \tilde{G}_1$  is a (surjective) central extension.

3: Take  $\tilde{G}_1 = \tilde{G} \times_H \tilde{H}$ , then  $\tilde{G}_1 \rightarrow G$  is surjective with kernel the product of kernels of  $\tilde{G} \rightarrow G$  and  $\tilde{H} \rightarrow H$ , so  $\tilde{G}_1$  is reductive and  $[\tilde{G}_1, \tilde{G}_1]$  is simply-connected for the same reason as in item2.  $\square$

**Prop. (10.4.5.30) [Z-Extensions and Inner Twists].** For  $G \in \mathcal{R}\text{edGrp}/k$  and an inner twisting  $\psi : G_{k^{\text{sep}}} \cong H_{k^{\text{sep}}}$  s.t.  $H \in \mathcal{R}\text{edGrp}/k$  is quasi-split. If  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  is a z-extension, then there exists a unique z-extension  $1 \rightarrow \tilde{Z} \rightarrow \tilde{H} \rightarrow H \rightarrow 1$  and a unique inner twisting  $\tilde{G}_{k^{\text{sep}}} \cong \tilde{H}_{k^{\text{sep}}}$  s.t.  $\tilde{H}$  is quasi-split, and the following diagram is commutative:

$$\begin{array}{ccc} \tilde{G} & \xrightarrow{\psi'} & \tilde{H} \\ \downarrow & & \downarrow \\ G & \xrightarrow{\psi} & H \end{array}$$

Moreover,  $\tilde{\psi} : Z \cong Z'$  is defined over  $k$ . ┘

*Proof:* Cf. [Kot82]P799. ? □

### Relative Dynkin-Diagrams

Cf. [K-P23]P98.

**Def. (10.4.5.31) [Tits-indices].** For  $G \in \mathcal{R}\text{edGrp}^{\text{ss}}/k$ , a **Tits-index** is a tuple  $(\Delta, \Delta_0, \rho)$  where  $\Delta \subset \Phi(G, T)$  is a set of absolute simple roots,  $\Delta_0 \subset \Delta$  is a subset consisting of the simple roots of the anisotropic kernel  $G^1$  ?, and  $\rho$  is a continuous action of  $\text{Gal}_k$  on  $\Delta$  preserving  $\Delta_0$ . ┘

## 6 Langlands L-Groups

References are [Bor79].

**Notation (10.4.6.1).**

- Let  $G \in \mathcal{R}\text{edGrp}/k$ . ┘

**Def. (10.4.6.2) [Langlands Dual Groups].** For  $G \in \mathcal{R}\text{edGrp}/k$ , there is a **Langlands dual group**  $\hat{G} \in \mathcal{R}\text{edGrp}/\overline{\mathbb{Q}}$  s.t.

- Any maximal torus  $\hat{T}$  of  $\hat{G}$  is isomorphic to the complex dual torus  $T^*(T)_{\mathbb{C}^\times} = \text{Hom}(X_*(T), \mathbb{C}^\times)$ .

- The Coxeter-Dynkin diagram of  $\hat{G}$  is dual to that of  $G$ . ?  
Then there is a canonical isomorphism  $\text{OAut}(\hat{G}) \cong \text{OAut}(G)$ .

Such a  $\hat{G}$  is only determined up to inner homomorphisms. ┘

*Proof:* Cf. [Bor79], [Kot84]Section1 and [B-G14] ?. □

**Def. (10.4.6.3) [Langland L-Groups].** The **Langlands L-group**  ${}^L G$  is a (possibly non-connected) reductive group over  $\overline{\mathbb{Q}}$  defined as a semi-direct product

$${}^L G = \hat{G} \rtimes \text{Gal}_k,$$

where  $\text{Gal}_k$  acts on  $\hat{G}$  via a homomorphism to  $\text{OAut}(\hat{G})$  (10.4.5.22) and a section of  $\text{Aut}(\hat{G}) \rightarrow \text{OAut}(\hat{G})$  via pinning ?.

The action factors through a finite quotient group  $\text{Gal}(K/k)$  for some  $K/k \in \text{GalField}^{\text{fin}}$ , so we can consider the **reduced Langlands L-group**  ${}^L G = \hat{G} \rtimes \text{Gal}(K/k)$ .

$$\mu_G : \text{Gal}_k \rightarrow \text{Aut}(\Phi(G_{\bar{k}}, B, T)) \cong \text{Aut}(\Phi(\widehat{G}, \widehat{B}, \widehat{T})) \rightarrow \text{Aut}(\widehat{G}).$$

And for each  $v \in \Sigma_F$ , a complex analytic group  ${}^L G_{F_v}$ , and complex analytic homomorphisms  ${}^L G_{F_v} \rightarrow {}^L G_F$  defined up to conjugacy.  $\lrcorner$

**Prop. (10.4.6.4) [Langlands L-Groups of Inner Forms].** An inner twisting  $G_{k^{\text{sep}}} \xrightarrow{\xi} G'_{k^{\text{sep}}}$  induces a  $\text{Gal}_K$ -equivariant isomorphism of Langlands dual groups  $\widehat{G} \cong \widehat{G}'$ , thus an isomorphism of Langlands L-groups  ${}^L G \cong {}^L G'$ .  $\lrcorner$

*Proof:* ?  $\square$

**Example (10.4.6.5) [Langlands Dual Groups].**

- $\widehat{\text{GL}(n)} = \text{GL}(n)$ .
- $\widehat{\text{SL}(n)} = \text{PGL}(n)$  and  $\widehat{\text{PGL}(n)} = \text{SL}(n)$ .
- $\widehat{\text{SO}(2n+1)} = \text{Sp}(2n)$  and  $\widehat{\text{Sp}(2n)} = \text{SO}(2n+1)$ .
- $\widehat{\text{SO}(2n)} = \text{SO}(2n)$ .
- $\widehat{\text{GSp}(2n)} = \text{GSpin}_{2n+1}(\mathbb{C})$ .
- $\widehat{\text{G2}} = \text{G2}$ ,  $\widehat{\text{F4}} = \text{F4}$ ,  $\widehat{\text{E6}} = \text{E6}$ ,  $\widehat{\text{E7}} = \text{E7}$ ,  $\widehat{\text{E8}} = \text{E8}$ .
- If  $G$  is semi-simple and simply-connected, then  $\widehat{G}$  is an adjoint group (10.4.2.1).
- If  $G$  is an adjoint group, then  $\widehat{G}$  is semi-simple and simply-connected.  $\lrcorner$

*Proof:* ? Cf. [Bor79]P30.  $\square$

**Example (10.4.6.6) [Langlands L-Groups for  $\text{U}(m)$ ].** If  $G = \text{U}(m)$  is the standard unitary group attached to a CM extension  $E/F$  (10.5.3.19), then the reduced Langlands L-group of  $\text{U}(m)$  is  ${}^L \text{U}(m) = \text{GL}(m; \mathbb{C}) \rtimes \text{Gal}(E/F)$  where  $c \in \text{Gal}(E/F)$  acts on  $\text{GL}(m; \mathbb{C})$  via

$$cMc = \varphi_m M^{-t} \varphi_m^{-1}, \quad \varphi_m = \begin{bmatrix} & & & (-1)^m \\ & & \cdots & \\ & -1 & & \\ 1 & & & \end{bmatrix}$$

$\lrcorner$

### Parabolic Subgroups of Langlands L-Groups

References are [Bor79].

**Notation (10.4.6.7).**

- Let  $G \in \text{RedGrp}/k$ .  $\lrcorner$

**Def. (10.4.6.8) [Parabolic Subgroups of  ${}^L G$ ].** A parabolic subgroup of  ${}^L G_{\mathbb{C}}$  (10.4.6.3) is a closed subgroup  $P \leq {}^L G_{\mathbb{C}}$  that surjects onto  $\text{Gal}_k$  and is a normalizer of a parabolic subgroup of  $\widehat{G}_{\mathbb{C}}$ .

${}^L B = \widehat{B} \rtimes \text{Gal}_k$  is a parabolic subgroup of  ${}^L G_{\mathbb{C}}$ . A parabolic subgroup  $P \leq {}^L G_{\mathbb{C}}$  is called a **standard parabolic subgroup** if it contains  ${}^L B$ .

Any parabolic subgroup of  ${}^L G_{\mathbb{C}}$  is conjugate to a unique standard parabolic subgroup ?.

Let  $\text{Par}({}^L G_{\mathbb{C}})$  be the conjugacy classes of parabolic subgroups of  ${}^L G(\mathbb{C})$ . ┘

**Def. (10.4.6.9) [Relevant Parabolic Subgroups of  ${}^L G$ ].** There is a bijection  $\text{Par}(G)_F \cong \text{Par}({}^L G_{\mathbb{C}})$  (10.4.6.8), and  $P \in \text{Par}({}^L G_{\mathbb{C}})$  is called a **relevant parabolic subgroup** if it corresponds to a class in  $p(G/F)$  under this bijection.

If  $G$  is quasi-split, then any parabolic subgroup of  ${}^L G(\mathbb{C})$  is relevant.

Let  ${}^L \text{Par}({}^L G(\mathbb{C}))$  denote the set of relevant parabolic subgroups of  ${}^L G(\mathbb{C})$ . ┘

*Proof:* ? Cf. [Bor79]P32. □

**Prop. (10.4.6.10).** If  $G' \in \text{RedGrp}/F$  is a quasi-split inner form of  $G$ , then

$${}^L p({}^L G_{\mathbb{C}}) \subset {}^L p({}^L G'_{\mathbb{C}}) = p({}^L G_{\mathbb{C}}).$$

┘

*Proof:* ? Cf. [Bor79]P32. □

## 10.5 Group Theory IV: More on Reductive Groups

*“Before Genesis, God gave the devil free reign to construct much of creation, but dealt with certain matters himself, including semisimple groups.”*

– C. Chevalley.

References are [B-T65, D-G70, D-G70a, D-G70b, Hum75, Bor91, P-R94, Spr98, Mil17, Con14].

### Notation(10.5.0.1).

- Let  $k \in \mathbf{Field}$ .
- Let  $R \in \mathbf{CRing}$ .
- Let  $\emptyset \neq S \in \mathbf{Sch}$ .
- Use notations on reductive groups defined in [Group Theory III: Reductive Groups](#).
- Let  $G, G' \in \mathbf{RedGrp}/k$ .

┘

### 1 Conjugacy Classes

References are [Kot82] and [Conjugacy classes, Springer-Steinberg].

### Notation(10.5.1.1).

- Suppose  $k = k^{\text{perf}}$ .

┘

### Stable Conjugacy Classes

References are [Kot82].

**Thm.(10.5.1.2)**[Steinberg]. Suppose  $T \leq G$  be a maximal torus and let

$$T/W(G, T) \triangleq \text{Spec} \left( \Gamma(T)^{W(G, T)} \right),$$

then there exists a  $G$ -invariant morphism  $G \rightarrow T/W(G, T)$ . If  $k = k^{\text{sep}}$ , then this map induces a bijection between the set of semi-simple conjugacy classes of  $G(k)$  and  $(T/W(G, T))(k)$ . ┘

*Proof:* ?

□

**Def.(10.5.1.3)**[Stable Conjugacy Classes, Kottwitz[Kot82]].  $x, y \in G(k)$  are called **stably conjugate** if there exists  $g \in G(k^{\text{sep}})$  s.t.  $gxg^{-1} = y$  and  $g^{-1}\sigma(g) \in Z_G(x_s)^0$ , where  $x_s$  is the semisimple part of  $x$  in the abstract Jordan-decomposition(10.3.3.1).

In this case, applying  $\sigma \in \text{Gal}_k$  to the equation  $gxg^{-1} = y$  implies that  $g^{-1}\sigma(g) \in Z_G(x)$ . Thus  $g^{-1}\sigma(g) \in Z_G(x) \cap Z_G(x_s)^0$ . Denote  $G_x^* \triangleq Z_G(x) \cap Z_G(x_s)^0$ . Notice that when  $Z_G(x_s)$  is connected(e.g.  $\pi_1([G, G]) = 1$  or  $x_s$  is strongly regular semi-simple(10.5.1.17)), we have  $G_x^* = Z_G(x)$  by(10.3.3.5), and the notion of stable conjugacy is the same as the notion of conjugacy under  $G(k^{\text{sep}})$ . ┘

**Cor.(10.5.1.4)**[ $G_{\text{ad}}$ -Conjugacy Implies Stable Conjugacy]. By(10.1.1.26), if  $x, y \in G(k)$  are conjugate under  $G_{\text{ad}}(k)$ , then  $x, y$  are stably conjugate(10.5.1.3), by(10.1.5.6). ┘

**Cor. (10.5.1.5).** If  $x, y \in G(k)$  and  $g \in G(k^{\text{sep}})$  s.t.  $\text{Ad}(g)$  is a stable conjugacy from  $x$  to  $y$ , then  $\text{Ad}(g)$  induces an inner twisting  $(G_x^*)_{k^{\text{sep}}} \cong (G_y^*)_{k^{\text{sep}}}$ .  $\lrcorner$

**Prop. (10.5.1.6) [Conjugacy Classes in Stable-Conjugacy Classes].** Suppose  $x, y \in G(k)$ . If  $g \in G(k^{\text{sep}})$  s.t.  $\text{Ad}(g)$  is a stable conjugacy from  $x$  to  $y$  (10.5.1.3), then  $(g^{-1}g^\sigma)_{\sigma \in \text{Gal}_k}$  is a 1-cycle of  $\text{Gal}_k$  in  $G_x^*$  whose cohomology class  $\text{inv}(x, y)$  lies in the kernel of

$$H^1(k, G_x^*) \rightarrow H^1(k, G).$$

Similarly, if  $g x g^{-1} = y$ , then  $(g^{-1}g^\sigma)_{\sigma \in \text{Gal}_k}$  defines a 1-cycle of  $\text{Gal}_k$  in  $G_x$  whose cohomology class lies in the kernel of

$$H^1(k, G_x) \rightarrow H^1(k, G).$$

In this way, we get

- An isomorphism between the set of conjugacy classes of  $x$  within the set of  $k^{\text{sep}}$ -conjugacy classes of  $x$  and  $\ker(H^1(k, G_x) \rightarrow H^1(k, G))$ .
- An isomorphism between the set of conjugacy classes of  $x$  within the set of stable-conjugacy classes of  $x$  and

$$\text{Im}\left(\ker\left(H^1(k, G_x^*) \rightarrow H^1(F, G)\right) \rightarrow \ker\left(H^1(k, G_x) \rightarrow H^1(F, G)\right)\right).$$

In particular, if  $G_x$  is connected (e.g.  $\pi_1([G, G]) = 1$  or  $x_s$  is strongly-regular semi-simple), then this set is just  $\ker(H^1(k, G_x) \rightarrow H^1(F, G))$ .  $\lrcorner$

*Proof:* This is easy from the definition (9.8.4.1).  $\square$

**Prop. (10.5.1.7) [Stable Conjugacy Classes and Z-Extensions].** Let  $1 \rightarrow Z \rightarrow G \xrightarrow{\alpha} H \rightarrow 1 \in \text{RedGrp}/k$  be a z-extension (10.4.5.28), then

- If  $y \in G(k)$  and  $x = \alpha(y) \in H(k)$ , then  $1 \rightarrow Z \rightarrow Z_H(y) \xrightarrow{\alpha} G_x^* \rightarrow 1$  (10.5.1.3) is an exact sequence. In particular,  $G_x^* \in \text{RedGrp}/k$ .
- $x, x' \in H(k)$  are stably conjugate iff there exists stably conjugate elements  $y, y' \in G(k)$  s.t.  $\alpha(y) = x, \alpha(y') = x'$ .  $\lrcorner$

*Proof:* Cf. [Kot82]P789.  $\square$

**Prop. (10.5.1.8).** If  $x, y \in G(k)$  are stably conjugate (10.5.1.3), then  $G_x^*$  and  $G_y^*$  are inner forms of each other. And if  $G_x = G_x^*$ , then there is a choice of inner twisting  $(G_x)_{k^{\text{sep}}} \cong (G_y)_{k^{\text{sep}}}$  that is canonical up to inner automorphisms of  $G_x$ .  $\lrcorner$

*Proof:* Cf. [Kot82]P789.  $\square$

**Prop. (10.5.1.9).** If  $G$  is quasi-split and  $x \in G(k)$  is semi-simple, then there exists  $y \in G(k)$  stably-conjugate to  $x$  (10.5.1.3) s.t.  $Z_G(y)^0$  is quasi-split.  $\lrcorner$

*Proof:* Cf. [Kot82]P790.  $\square$

### Rational Conjugacy Classes

**Def. (10.5.1.10) [c-complete Reductive Groups].**  $G$  is called a **c-complete reductive group** if every semi-simple conjugacy class  $C$  in  $G(k^{\text{sep}})$  that is defined over  $k$  (i.e.  $\sigma(C) = C$  for all  $\sigma \in \text{Gal}_k$ ) contains an element of  $G(k)$ .  $\lrcorner$

**Thm. (10.5.1.11) [Kottwitz[Kot82]].** If  $G$  is quasi-split and  $\pi_1([G, G]) = 1$ , then  $G$  is c-complete(10.5.1.10).  $\lrcorner$

*Proof:* Cf.[Kot82]P791.  $\square$

**Thm. (10.5.1.12) [Rational Unipotent Conjugacy Classes, Kottwitz[Kot82]].** If  $\text{char } k = 0$  and  $G$  is quasi-split, then every unipotent conjugacy class in  $G(k^{\text{sep}})$  that is defined over  $k$  contains an element of  $G(k)$ .  $\lrcorner$

In fact the same holds for  $\text{char } k$  large enough relative to the Dynkin-diagram of  $G$ .?  $\lrcorner$

*Proof:* Cf.[Kot82]P793.  $\square$

**Cor. (10.5.1.13).** if  $\text{char } k = 0$ ,  $G$  is quasi-split and  $\pi_1([G, G]) = 1$ , then every conjugacy class in  $G(k^{\text{sep}})$  that is defined over  $k$  contains an element of  $G(k)$ .  $\lrcorner$

In fact the same holds for  $\text{char } k$  large enough relative to the Dynkin-diagram of  $G$ .  $\lrcorner$

*Proof:* Let  $x \in G$  s.t. the conjugacy class of  $x$  is defined over  $k$ . Let  $x = us = su$  be the abstract Jordan-decomposition of  $x$  in  $G$ (10.3.3.1). For any  $\sigma \in \text{Gal}_k$ , there exists  $g \in G(k)$  s.t.  $gxg^{-1} = \sigma(x)$ , and it follows from the uniqueness of Jordan-decomposition that  $gs\sigma(s)^{-1} = \sigma(s)$ . Thus the conjugacy class of  $s$  is also defined over  $k$ . Thus it follows from(10.5.1.11) that if we replace  $x$  by a conjugate, then we may assume that  $s \in G(k)$ . And it follows from(10.5.1.9) that by replacing  $x$  by a stable conjugate, we may assume that  $Z_G(s)$  is quasi-split (and still  $s \in G(k)$ ).

Similarly,  $gu\sigma(u)^{-1} = \sigma(u)$ , and the conjugacy class of  $u$  is also defined over  $k$ , and it follows from(10.5.1.12) that there exists  $h \in Z_G(s)$  s.t.  $huh^{-1} \in Z_G(s)(k)$ . And then  $huh^{-1} = s(huh^{-1}) \in G(k)$ .  $\square$

### Cyclic Norm Maps

**Def. (10.5.1.14) [Cyclic Norm Maps, Kottwitz[Kot82]].** Suppose  $\text{char } k = 0$  and  $L/k \in \text{GalField}^{\text{fin}}$  is a cyclic extension of degree  $l$  with  $\text{Gal}(L/k) = \langle \sigma \rangle$ . Choose an inner twisting  $\psi : G_{\bar{k}} \rightarrow H_{\bar{k}}$  where  $H \in \text{RedGrp}/k$  is quasi-split(10.4.5.16). Then we can define a map  $\mathfrak{N}$  from  $G(L)$  to the set of stable-conjugacy classes of  $H(k)$  as follows:

- If  $[G, G]$  is simply-connected and  $x \in G(L)$ , define  $\text{Nm}_{L/k}(x) = \sigma^{l-1}(x)\sigma^{l-2}(x)\dots\sigma(x)x$ , then the conjugacy class of  $\psi(\text{Nm}_{L/k}(x))$  is defined over  $k$ , thus by(10.5.1.13) defines a  $k^{\text{sep}}$ -conjugacy class in  $H(k)$  (which is the same as stable-conjugacy class by(10.5.1.3)).
- In general, take a  $z$ -extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  and a  $z$ -extension  $1 \rightarrow Z' \rightarrow \tilde{H} \rightarrow H \rightarrow 1$  where  $H'$  is quasi-split and inner twisting  $\tilde{\psi} : (\tilde{G})_{k^{\text{sep}}} \cong (\tilde{H})_{k^{\text{sep}}}$  as in(10.4.5.30). Then there exists a unique map  $\mathfrak{N}$  from  $G(L)$  to the set of stable conjugacy classes of  $H(k)$  making the following commutative diagram commutative:

$$\begin{array}{ccc} \tilde{G}(L) & \xrightarrow{\mathfrak{N}_{\sigma, \tilde{\psi}}} & \{\text{stable-conjugacy classes of } \tilde{H}(k)\} \\ \downarrow & & \downarrow \\ G(L) & \xrightarrow{\mathfrak{N}_{\sigma, \psi}} & \{\text{stable-conjugacy classes of } H(k)\} \end{array} .$$



This  $\mathfrak{N}_{\sigma,\psi}$  doesn't depends on  $\tilde{G}$  chosen. And if  $H = G, \psi = \text{id}_G$ , then denote  $\mathfrak{N}_\sigma$ . ┘

*Proof:* 1:  $\sigma(\text{Nm}(x)) = x \text{Nm}(x) x^{-1}$ , so the conjugacy class of  $\text{Nm}(x) \in G(k^{\text{sep}})$  is defined over  $k$ . And the same holds for  $\psi(\text{Nm}_{L/k}(x))$  because  $\psi$  is an inner twisting.

2: Uniqueness of  $\mathfrak{N}_{\sigma,\psi}$  follows from the fact that  $\tilde{G}(L) \rightarrow G(L)$  is surjective by (10.2.3.12). To show existence, notice that if  $x, y \in \tilde{G}(L)$  s.t.  $y = xz$  for some  $z \in Z(k)$ , then we have  $\text{Nm}_{L/k}(y) = \text{Nm}_{L/k}(x) \text{Nm}_{L/k}(z)$ , thus  $\mathfrak{N}_{\sigma,\psi}(x) = \mathfrak{N}_{\sigma,\psi}(y)$  modulo  $Z'$ . The independence of  $\psi$  follows from (10.4.5.29) and the definition in item1. □

**Prop. (10.5.1.15)[Conjugacy and Cyclic Extensions].** Situation as in (10.5.1.14), let  $I = \text{Res}_{L/k} G_L$ , then ?

In particular, for  $x, y \in G(L)$ ,  $\text{Nm}_{L/k}(x), \text{Nm}_{L/k}(y) \in G(k^{\text{sep}})$  are conjugate iff  $x, y \in I(k^{\text{sep}})$  are  $\sigma$ -conjugate. ┘

*Proof:* Cf. [Kot82]P801 ? □

**Prop. (10.5.1.16)[Stable Conjugacy and Cyclic Extensions].** Situation as in (10.5.1.14), for  $x, y \in G(L)$ ,  $x, y$  are stably conjugate iff  $\mathfrak{N}(x) = \mathfrak{N}(y)$ . ┘

*Proof:* Cf. [Kot82]P804. □

### Strongly Regular Semi-Simple Elements

**Def. (10.5.1.17) [Regular Semi-Simple Elements].** A semi-simple element  $\gamma \in G(k)$  is called a **regular semi-simple element** if  $Z_G(\gamma)^0 \in \mathcal{T}_{\text{ori}}/k$ , and it is called a **strongly regular semi-simple element** if  $Z_G(\gamma) \in \mathcal{T}_{\text{ori}}/k$ . ┘

**Prop. (10.5.1.18).** If  $\text{char } k = 0, \gamma, \gamma' \in G(k)$  are strongly regular semi-simple elements and  $g \in G(k^{\text{sep}})$  s.t.  $g\gamma g^{-1} = \gamma', \text{Ad}(g) : Z_G(\gamma) \rightarrow Z_G(\gamma')$  doesn't depends on  $g$  chosen because  $Z_G(\gamma)$  is a torus, and we denote it by  $\varphi_{\delta, \delta'}$ . ┘

## 2 Cohomology of Reductive Groups

References are [Mil17]Chap25.g and [Ser97a] and [Number theory and Algebraic groups].

**Prop. (10.5.2.1)[Cohomology of Parabolic Subgroups].** Let  $G \in \text{RedGrp}/k$  and  $P \leq G$  a parabolic subgroup with Levi-subgroup  $L$  (10.4.2.33), then the natural map  $H^1(k, L) \rightarrow H^1(k, P)$  is bijective and  $H^1(k, P) \rightarrow H^1(k, G)$  is injective. ┘

*Proof:* □

### Cohomology over Local Fields

References are [Kne65].

**Thm. (10.5.2.2) [Tate-Nakayama Theorem].** For  $K \in \text{LField}$  and  $T \in \mathcal{T}_{\text{ori}}/K$  with splitting field  $L/K$ , then for any  $n \in \mathbb{Z}_+$ , we have

$$H_{\text{Tate}}^n(L/K, T) \cong H_{\text{Tate}}^{2-i}(L/K, X^\bullet(T)) \cong H_{\text{Tate}}^{i-2}(L/K, X_\bullet(T)) \quad (9.8.1.5).$$

In particular,  $H^1(L/K, T) \cong H^1(L/K, X^\bullet(T)) = (X^\bullet(T)_{\text{Gal}_K})_{\text{tor}} = \pi_0(\hat{T}^{\text{Gal}_K})^\vee$ . ┘

**Thm. (10.5.2.3)**[Steinberg's Theorem, Lang[Lan56]/Serre[Ser62]/Springer[Spr62]/Steinberg[Ste65]].

If  $k$  is a cd-1 field(9.8.3.18)(e.g. a finite field?? or a Henselian field with alg.closed residue field??) and  $G \in \mathcal{R}edGrp/k$ , then  $H^1(k, G) = \mathbf{1}$  and  $G$  is quasi-split.

Moreover, if  $k = k^{\text{perf}}$ , then the above assertion holds for any  $G \in \mathcal{A}lgGrp_{\text{c}ntd}/K$ .  $\lrcorner$

*Proof:* Cf.[Ser97a]Section2.2, 2.3. This statement is right!! [P-R94] has a proof of Lang's theorem.  $\square$

**Prop. (10.5.2.4).** Suppose  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  is Henselian and  $\kappa = \bar{\kappa}$ . If  $G \in \mathcal{G}rpVar/K$ , then  $H^1(K, G) \rightarrow H^1(\bar{K}, G)$  is a bijection. In particular, if  $G$  is reductive, then  $H^1(K, G) = \mathbf{1}$ .  $\lrcorner$

*Proof:* Cf.[Fibres principaux sur les corps values henseliens]Prop3.5.3.(2). or [K-P23]P81?  $\square$

The last assertion follows from(10.5.2.3).  $\square$

**Prop. (10.5.2.5).** If  $k$  is a cd-1 field(9.8.3.18) and  $T/k$  is a torus, then  $H^i(k, T) = 0$  for any  $i \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* Take an exact sequence  $1 \rightarrow R_1 \rightarrow R_0 \rightarrow T \rightarrow 1$  where  $R_1$  is a torus and  $R_0$  is an induced torus(10.2.3.13), then  $H^i(k, R_0)$  is a finite product of  $H^i(k', \mathbb{G}_m)$  for  $k'/k$  finite separable.  $H^1(k', \mathbb{G}_m)$  by Hilbert's multiplicative Satz90(9.8.3.2), and  $H^2(k', \mathbb{G}_m) = \text{Br}(k') = 0$  by(9.8.3.18). And also  $H^i(k', \mathbb{G}_m) = 0$  for  $i \geq 3$  by??. Now  $H^{i+2}(k, R_1) = 0$  for any  $i \in \mathbb{Z}_+$  by(9.8.3.18), so  $H^{i+1}(k, T) = 0$ , and then also  $H^{i+1}(k, R_1) = 0$  for all  $i \in \mathbb{Z}_+$ , so  $H^i(k, T) = 0$  for any  $i \in \mathbb{Z}_+$ .  $\square$

**Thm. (10.5.2.6).** Suppose  $(\mathcal{O}_K, K) \in \text{DVR}$  is Henselian s.t.  $\kappa$  is a perfect cd-1 field(9.8.3.18). If  $C \subset \mathcal{B}(G_{K^{\text{sh}}})$  is a  $\text{Gal}_\kappa$ -stable chamber(20.5.2.31) with corresponding Iwahori-subgroup  $\mathcal{I} \subset G(K^{\text{sh}})$ , and  $\mathcal{N} = N_{G(K^{\text{sh}})}(\mathcal{I}) = G(K^{\text{sh}})_C$  the normalizer, then the natural maps

$$H^1(\text{Gal}_\kappa, \mathcal{N}/\mathcal{I}) \leftarrow H^1(\text{Gal}_\kappa, \mathcal{N}) \rightarrow H^1(\text{Gal}_\kappa, G(K^{\text{sh}})) \rightarrow H^1(K, G)$$

are bijective.  $\lrcorner$

*Proof:* ? Cf.[K-P23].  $\square$

**Cor. (10.5.2.7)** [Cohomology of Simply-Connected Semisimple Groups, Kneser/Bruhat-Tits[B-T87]]. Suppose  $(\mathcal{O}_K, K) \in \text{DVR}$  is Henselian s.t.  $\kappa$  is a perfect cd-1 field(9.8.3.18). If  $G$  is semisimple and simply-connected, then  $H^1(K, G) = \mathbf{1}$ .  $\lrcorner$

*Proof:* In this case  $G(K)^0 = G(K)$ (20.5.2.18), thus  $\mathcal{N} = \mathcal{I}$  by(20.6.4.14) and(20.6.1.5), so the assertion follows from(10.5.2.6).  $\square$

**Prop. (10.5.2.8)**[Anisotropic Tori]. If  $K \in p\text{-LField}$  and  $T \in \mathcal{T}ori/K$  is anisotropic, then  $H^2(K, T) = 0$ .  $\lrcorner$

*Proof:* Cf.[K-P23]P400.  $\square$

**Thm. (10.5.2.9)**[Finiteness Theorems]. If  $K \in \text{LField}$  and  $G \in \text{linAlgGrp}/K$ , then  $\#H^1(K, G) < \infty$ .  $\lrcorner$

*Proof:* Cf.[P-R94]P316.  $\square$

### Cohomology over Global Fields

**Notation (10.5.2.10).**

- Let  $F \in \mathbf{GField}$ .
- Let  $G \in \mathbf{RedGrp}/F$ .

┘

**Thm. (10.5.2.11) [Finiteness Theorems].** If  $F \in \mathbf{NField}$  and  $G \in \mathbf{linAlgGrp}/F$ , then

$$\# \ker \text{III}(G/F) < \infty \text{ (9.8.3.28)}.$$

┘

*Proof:* Cf. [P-R94]P316. □

**Thm. (10.5.2.12) [Hasse-Principle for Simply-Connected Semisimple Groups, Kneser/Harder/Chernousov].** If  $F \in \mathbf{NField}$  and  $G$  is semi-simple and simply-connected, then

$$H^1(F, G) \rightarrow \prod_{v \in \Sigma_\infty(F)} H^1(F_v, G)$$

is a bijection. In particular,  $\text{III}(G/F) = 1$ . ┘

*Proof:* Cf. [P-R94]P286. ? □

**Cor. (10.5.2.13) [Hasse-Principle for Adjoint Groups].** If  $F \in \mathbf{NField}$  and  $G$  is an adjoint group, then

$$\text{III}(G/F) = 1.$$

┘

*Proof:* Cf. [P-R94]P336. ? □

**Prop. (10.5.2.14) [Cohomology of Semi-Simple Groups].** If  $K \in p\text{-LField} \cup \mathbf{NField}$  and  $G \in \mathbf{RedGrp}^{\text{ss}}/K$ , then the connecting map  $H^1(K, G) \xrightarrow{\delta} H^2(K, \pi_1(G))$  associated to the exact sequence  $1 \rightarrow \pi_1(G) \rightarrow G_{\text{sim.cntd}} \rightarrow G \rightarrow 1$  is surjective. And if  $K \in p\text{-LField}$ , it is moreover bijective. ┘

*Proof:* Cf. [R-P94]P326. ? □

**Thm. (10.5.2.15) [Kneser [Kne65]].** If  $F \in \mathbf{NField}$ ,  $T \in \mathbf{Tori}/F$  and  $T_{F_v}$  is anisotropic for some  $v \in \Sigma_F$ , then

$$H^2(F, T) \rightarrow \prod_{v \in \Sigma_F} H^2(F_v, T)$$

is injective. ┘

*Proof:* Cf. [P-R94]P315. ? □

### 3 Classical Groups

**Def. (10.5.3.1) [Classical Groups and Exceptional Groups].**  $G \in \mathcal{R}ed\mathcal{G}rp/K$  is called a **classical group** if it is semisimple and the simply-connected covering group  $\tilde{G}_{\bar{k}}$  of  $G_{\bar{k}}$  has associated Dynkin diagrams with connected components of type  $A_n, B_n, C_n, D_n$  excluding the type  ${}^3D_4$  or  ${}^6D_4$  (Galois action).

$G$  is called an **exceptional group** if it is semisimple and the simply-connected covering group  $\tilde{G}_{\bar{k}}$  of  $G_{\bar{k}}$  with associated Dynkin diagrams has connected components of type  $E_6, E_7, E_8, F_4, G_2$ .

$G$  is called a **trialitarian group** if it is semisimple and the simply-connected covering group  $\tilde{G}_{\bar{k}}$  of  $G_{\bar{k}}$  has associated Dynkin diagrams with connected components of type  $D_4$  and the action of  $\text{Gal}_k$  on them permutes with three outer vertices.  $\lrcorner$

#### Split Classical Groups

**Def. (10.5.3.2) [Examples of Classical Groups over  $\mathbb{Z}$ ].** For  $n \in \mathbb{Z}_+$ ,

- the **general linear group**  $\text{GL}(n) = \mathbb{Z}[T_{ij}][1/\det]$  representing the group functor  $\mathcal{C}Alg_{\mathbb{Z}} \rightarrow \mathcal{G}rp : R \mapsto \text{GL}(n; R)$ .
- the **multiplicative group**  $\mathbb{G}_m = \text{Spec } \mathbb{Z}[T, T^{-1}]$ , which is just  $\text{GL}(1)$ .
- the **special linear group**  $\text{SL}(n)$  is the algebraic subgroup scheme of  $\text{GL}(n)$  defined by the ideal  $(\det - 1)$ , representing the group functor  $\mathcal{C}Alg_{\mathbb{Z}} \rightarrow \mathcal{G}rp : R \mapsto \text{SL}(n, R)$ .
- the (standard)**orthogonal group**  $\text{O}(n)$  is the algebraic subgroup of  $\text{GL}(n)$  generated by the entries of the equation  $(T_{ij})^t C_n (T_{ij}) = C_n$ , where  $C_n = \begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1}_k \\ \mathbf{0} & \mathbf{1}_k & \mathbf{0} \end{bmatrix}$  if  $n = 2k+1$  and  $\begin{bmatrix} \mathbf{0} & \mathbf{1}_k \\ \mathbf{1}_k & \mathbf{0} \end{bmatrix}$  if  $n = 2k$ , representing the functor  $\mathcal{C}Alg_{\mathbb{Z}} \rightarrow \mathcal{G}rp : R \mapsto \{g \in \text{GL}(n, R) \mid g^t C_n g = C_n\}$ .
- the (standard)**general orthogonal group**  $\text{GO}(n)$  is the algebraic subgroup of  $\text{GL}(n) \times \mathbb{G}_m \subset \text{GL}(n+1)$  generated by the entries of the equation  $(T_{ij})^t C_n (T_{ij}) = T C_n$ , representing the functor

$$\mathcal{C}Ring_{\mathbb{Z}} \rightarrow \mathcal{G}rp : R \mapsto \{g \in \text{GL}(n, R) \times \mathbb{R}^\times \mid g^t C_n g = r C_n\}.$$

There is an exact sequence

$$1 \rightarrow \text{O}(n) \rightarrow \text{GO}(n) \xrightarrow{\nu} \mathbb{G}_m \rightarrow 1,$$

where  $\nu : (g, r) \mapsto r$  is called the **amplitude character**.

- the **special orthogonal groups**  $\text{SO}(n) = \text{O}(n) \cap \text{SL}(n)$  (when  $\text{char } k \neq 2$ ).
- the (standard)**symplectic group**  $\text{Sp}(n)$  is the algebraic subgroup of  $\text{GL}(2n)$  defined by the ideal generated by the entries of the equation  $(T_{ij})^t J_n (T_{ij}) = J_n$ , where  $J_n = \begin{bmatrix} \mathbf{0} & \mathbf{1}_n \\ -\mathbf{1}_n & \mathbf{0} \end{bmatrix}$ , representing the functor

$$\mathcal{C}Ring_{\mathbb{Z}} \rightarrow \mathcal{G}rp : R \mapsto \{g \in \text{GL}(2n; R) \mid g^t J_n g = J_n\} = \left\{ \begin{bmatrix} A & B \\ C & D \end{bmatrix} \mid A^t C = C^t A, A^t D - C^t B = I, B^t D = D^t B \right\}.$$

- the (standard)**general symplectic group**  $\text{GSp}(n)$  is the algebraic subgroup of  $\text{GL}(2n) \times \mathbb{G}_m \subset \text{GL}(2n+1)$  generated by the entries of the equation  $(T_{ij})^t J_n (T_{ij}) = T J_n$ , representing the functor

$$\mathcal{C}Ring_{\mathbb{Z}} \rightarrow \mathcal{G}rp : R \mapsto \{(g, r) \in \text{GL}(2n; R) \times R^* \mid g^t J_n g = r J_n\}.$$

There is an exact sequence

$$1 \rightarrow \mathrm{Sp}(2n) \rightarrow \mathrm{GSp}(2n) \xrightarrow{\nu} \mathbb{G}_m \rightarrow 1,$$

where  $\nu : (g, r) \mapsto r$  is called the **amplitude character**.

- $\mathrm{PGL}(n)$  is the quotient group of  $\mathrm{GL}(n)$  by its center.
- $\mathrm{PSL}(n)$  is the quotient group of  $\mathrm{SL}(n)$  by its center.
- $\mathrm{PSO}(n)$  is the quotient group of  $\mathrm{SO}(n)$  by its center.
- the **standard Borel subgroup**  $B(n)$  is the algebraic subgroup of  $\mathrm{GL}(n)$  representing the upper-triangular matrices.
- the **standard unipotent subgroup**  $\mathrm{Unip}(n)$  is the algebraic subgroup of  $B(n)$  representing the unipotent matrices.
- the **diagonal group**  $T(n)$  is the unipotent subgroup of  $B(n)$  representing diagonal matrices.

┘

*Proof:* By (10.1.1.3), it suffices to show  $\mathrm{Hom}(-, G)$  is a group functor when restricted to affine schemes.  $\square$

**Prop. (10.5.3.3) [Linear Groups].** For any  $n \in \mathbb{Z}_+$ ,  $\mathrm{PGL}(n; k) = \mathrm{GL}(n; k)/k^\times$  because  $H^1(k, \mathbb{G}_m) = 1$ ?

There is an exact sequence  $1 \rightarrow \mu_n \rightarrow \mathrm{SL}(n) \rightarrow \mathrm{PGL}(n) \rightarrow 1$  because  $\mathrm{SL}(n, \bar{k}) \rightarrow \mathrm{PGL}(n, \bar{k})$  is surjective (10.1.5.6).  $\square$

**Def. (10.5.3.4) [Spin Algebraic Groups].** There are algebraic subgroups of  $\mathrm{Cl}^+(L)_{\mathbb{Q}}$  generating the spin group construction (3.11.5.6):

$$\mathrm{Spin}(L)(R) = \{g \in \mathrm{Cl}^+(L)_R : gg' = 1, gLg^{-1} = L\}$$

$$\mathrm{GSpin}(L)(R) = \{g \in \mathrm{Cl}^+(L)_R : gg' = \lambda(g) \in R^*, gLg^{-1} = L\}.$$

Called the **spin algebraic group** and the **GSpin algebraic group**.  $\mathrm{Spin}(L)$  and  $\mathrm{GSpin}$  acts on  $L$  via conjugation.  $\square$

**Prop. (10.5.3.5).** There is a diagram with exact rows and columns:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mu_2 & \longrightarrow & \mathrm{Spin}(L) & \longrightarrow & \mathrm{SO}(L) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \parallel \\ 1 & \longrightarrow & \mathbb{G}_m & \longrightarrow & \mathrm{GSpin}(L) & \longrightarrow & \mathrm{SO}(L) \longrightarrow 1 \\ & & \downarrow 2 & & \downarrow \lambda & & \\ & & \mathbb{G}_m & \xlongequal{\quad} & \mathbb{G}_m & & \end{array}$$

┘

*Proof:*

$\square$

**Prop. (10.5.3.6) [Simply-Connectedness].**  $\mathrm{Spin}(n)$  are simply-connected.  $\square$

### Parabolic Subgroups

**Prop. (10.5.3.7)[Borel Subgroups of  $\mathrm{GL}(n)$ ].** If  $k = \bar{k}$  and  $G = \mathrm{GL}(n)$ , then by Lie-Kolchin(10.2.7.7), the Borel subgroups of  $G$  are exactly the conjugates by  $G(k)$  of  $B(n)$ .  $\lrcorner$

*Proof:* Cf.[Mil17]P354.  $\square$

**Prop. (10.5.3.8).**  $B(n)$  is split solvable and  $\mathrm{Unip}(n)$  is split nilpotent.

Moreover,  $\mathrm{Unip}(n)$  has no non-zero subtorus, by successively using(10.1.1.8).  $\lrcorner$

*Proof:* Cf.[Mil17]P137.  $\square$

**Prop. (10.5.3.9) [Maximal Borel Subgroups].** Let  $G = \mathrm{SO}(2n), \mathrm{SO}(2n+1)$  or  $\mathrm{Sp}(2n)$ , then the maximal Borel subgroup of  $G$  are the stabilizers of the maximal totally anisotropic flags in  $G$ (of length  $n$ ).  $\lrcorner$

*Proof:* Cf.[Mil17]P354.  $\square$

**Prop. (10.5.3.10)[ $\mathrm{Sp}(2n)$  and  $\mathrm{SL}(2n)$ ].**  $\mathrm{Sp}(2n) \subset \mathrm{SL}(2n)$ , and  $\mathrm{Sp}(2) = \mathrm{SL}(2)$ .  $\lrcorner$

*Proof:* The first inclusion follows from(3.5.11.28), and the second follows from the fact that the volume form the definition(10.5.3.2).  $\square$

### Almost-Simple Semisimple Groups

#### Semisimplicity

**Prop. (10.5.3.11)[Reductive and Semisimple Groups].**  $\mathrm{GL}(n), \mathrm{SL}(n+1), \mathrm{SO}(n), \mathrm{Sp}(2n)$  are reductive by(10.3.5.1), as they are connected, and their standard representation is geo.simple.

$\mathrm{SL}(n), \mathrm{SO}(n), \mathrm{Sp}(2n)$  are semisimple, as they have finite centers(10.4.2.18).  $\lrcorner$

#### Root Systems

**Prop. (10.5.3.12) [Maximal Tori of  $\mathrm{GL}(n)$ ].**  $T(n)$  is a maximal torus in  $\mathrm{GL}(n)$  because  $C_{\mathrm{GL}(n)}(T(n)) = T(n)$ . And for  $n \in \mathbb{Z}_+$ , the conjugacy classes of maximal tori in  $\mathrm{GL}(n)$  is in bijection with the isomorphism classes of étale  $k$ -algebras of rank  $n$ . In particular,  $\lrcorner$

*Proof:* If  $T \leq \mathrm{GL}(n)$  is a maximal torus, then  $V = k^n$  decomposes as a direct sum of simple  $T$ -modules(10.2.3.23)  $V = \bigotimes_i V_i$ , and each  $k_i = \mathrm{End}_{T_i}(V_i)$  is a separable field extension of  $k$  s.t.  $\dim_{k_1} V_i = 1$  by(10.2.3.23), and  $T \leq \prod_i \mathrm{Res}_{k_i/k} \mathbb{G}_m$ . By comparing ranks, we see  $T = \prod_i \mathrm{Res}_{k_i/k} \mathbb{G}_m$ , and  $\prod_i k_i$  is an étale  $k$ -algebra of rank  $n$ . Conversely, for any étale  $k$ -algebra  $\prod_i k_i$  of rank  $n$ ,  $T = \prod_i \mathrm{Res}_{k_i/k} \mathbb{G}_m$  is a maximal torus contained in  $\mathrm{GL}(\prod_i k_i) \cong \mathrm{GL}(n)$ (maximal because it has rank  $n$ , which equals the diagonal torus). Clearly this establishes a bijection.  $\square$

**Prop. (10.5.3.13)[Classical Semisimple Groups].**

- For the split reductive pair  $(\mathrm{GL}(n), T(n))$  where  $n \in \mathbb{Z}_{\geq 2}$  and  $T$  is the diagonal torus,  $\Phi(\mathrm{GL}(n), T(n)) = \{e_i - e_j\}_{i \neq j \in [n]_+}$ , where  $e_i \in X^\bullet(T(n))$  where

$$e_i(\mathrm{diag}(t_1, \dots, t_n)) = t_i, \quad s_{i,j}(t_1, \dots, t_n) = (t_1, \dots, t_{i-1}, t_j, t_{i+1}, \dots, t_{j-1}, t_i, t_{j+1}, \dots, t_n),$$

$$(e_i - e_j)^\vee = e_i^\vee - e_j^\vee : t \mapsto (1, \dots, [t]_i, \dots, [t^{-1}]_j, \dots, 1).$$

$A_n$  : For the split reductive pair  $(\mathrm{SL}(n+1), T)$ , where  $n \in \mathbb{Z}_{\geq 2}$  and  $T$  is the diagonal torus,  $\Phi(\mathrm{SL}(n+1), T) = \{\pm\alpha_{ij}\}$ , where  $\alpha \in X^\bullet(T)$  s.t.

$$\alpha_{ij}(\mathrm{diag}(t_1, \dots, t_{n+1})) = t_i/t_j, \quad s_{i,j}(t_1, \dots, t_{n+1}) = (t_1, \dots, t_{i-1}, t_j, t_{i+1}, \dots, t_{j-1}, t_i, t_{j+1}, \dots, t_{n+1}),$$

$$\alpha_{ij}^\vee(t) = (1, \dots, [t]_i, \dots, [t^{-1}]_j, \dots, 1).$$

and the base corresponding to the standard Borel is  $\Delta = \{\alpha_{12}, \alpha_{23}, \dots, \alpha_{n,n+1}\}$ . Thus it is of type  $A_n$  (3.9.6.3). It is simply-connected and geometrically almost-simple because  $X = P(\Phi)$  (10.4.5.11) (10.4.5.13), and has center  $D(P(\Phi)/Q(\Phi)) \cong \mu_{n+1}$  (10.4.4.6). The unique outer automorphism reversing the Dynkin diagram is given by  $X \mapsto X^{-t}$ .

- For the split reductive pair  $(\mathrm{PGL}(n), \overline{T}(n))$ ,  $\Phi(\mathrm{PGL}(n), \overline{T}(n)) = \{\alpha_{ij}\}_{i \neq j \in [n]_+}$ , where  $\alpha_{ij} \in X^\bullet(T(n))$  s.t.  $\alpha_{ij}(\mathrm{diag}(t_1, \dots, t_n)) = t_i/t_j$ .

$B_n$  : The split reductive pair  $(\mathrm{SO}(2n+1), T)$  where  $T = \{\mathrm{diag}(1, t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})\}$ ,

$$\Phi(\mathrm{SO}(2n+1), T) = \{\pm(e_i + e_j), \pm(e_i - e_j), \pm e_i\}_{i \neq j \in [n]_+},$$

where

$$e_i(\mathrm{diag}(1, t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})) = t_i, \quad (e_i - e_j)^\vee = e_i^\vee - e_j^\vee : t \mapsto (1, \dots, [t]_i, \dots, [t^{-1}]_j, \dots, 1),$$

$$s_{i,j}(\mathrm{diag}(1, t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})) = \mathrm{diag}(1, t_1, \dots, t_{i-1}, t_j, t_{i+1}, \dots, t_{j-1}, t_i, t_{j+1}, \dots, t_n, \\ t_1^{-1}, \dots, t_{i-1}^{-1}, t_j^{-1}, t_{i+1}^{-1}, \dots, t_{j-1}^{-1}, t_i^{-1}, t_{j+1}^{-1}, \dots, t_n^{-1}).$$

The eigenvectors in  $\mathfrak{so}(2n+1)$  is given in [Mil17]P458. And the base corresponding to the standard Borel is  $\Delta = \{e_1 - e_2, e_2 - e_3, \dots, e_{n-1} - e_n, e_n\}$ .

Thus it is of type  $B_n$  (3.9.6.3). It is geometrically almost-simple of adjoint-type because  $X = Q(\Phi)$  (10.4.4.6) (10.4.5.13). Its simply-connected covering group is  $\mathrm{Spin}(2n+1) \rightarrow \mathrm{SO}(2n+1)$  with kernel  $D(P(\Phi)/Q(\Phi)) \cong \mu_2$  (10.4.5.3), which is the center of  $\mathrm{Spin}(n)$ .

$C_n$  : For the split reductive pair  $(\mathrm{Sp}(n), T)$  where  $n \in \mathbb{Z}_{\geq 2}$  and  $T$  is the diagonal torus,

$$\Phi(\mathrm{Sp}(n), T) = \{\pm 2e_i, \pm e_i \pm e_j\}_{i \neq j \in [n]_+},$$

where

$$e_i(\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})) = t_i, \quad (e_i - e_j)^\vee = e_i^\vee - e_j^\vee : t \mapsto (1, \dots, [t]_i, \dots, [t^{-1}]_j, \dots, 1),$$

$$s_{i,j}(\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})) = \mathrm{diag}(t_1, \dots, t_{i-1}, t_j, t_{i+1}, \dots, t_{j-1}, t_i, t_{j+1}, \dots, t_n, \\ t_1^{-1}, \dots, t_{i-1}^{-1}, t_j^{-1}, t_{i+1}^{-1}, \dots, t_{j-1}^{-1}, t_i^{-1}, t_{j+1}^{-1}, \dots, t_n^{-1}).$$

The eigenvectors in  $\mathfrak{sp}(n)$  is given in [Mil17]P459. And the base corresponding to the standard Borel is  $\Delta = \{e_1 - e_2, e_2 - e_3, \dots, e_{n-1} - e_n, 2e_n\}$ .

Thus it is of type  $D_n$  (3.9.6.3). It is geometrically almost-simple and simply-connected because  $X = P(\Phi)$  (10.4.5.11) (10.4.5.13), and has center  $D(P(\Phi)/Q(\Phi)) \cong \mu_2$ .

$D_n$  : For the split reductive pair  $(\mathrm{SO}(2n), T)$  where  $n \in \mathbb{Z}_{\geq 2}$  and  $T = \{\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})\}$ ,

- $\Phi(\mathrm{SO}(2n), T) = \{\pm e_i \pm e_j\}_{i \neq j \in [n]_+}$ , where
 
$$e_i(\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})) = t_i, \quad (e_i - e_j)^\vee = e_i^\vee - e_j^\vee : t \mapsto (1, \dots, [t]_i, \dots, [t^{-1}]_j, \dots, 1),$$
- $s_{i,j}(\mathrm{diag}(1, t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})) = \mathrm{diag}(1, t_1, \dots, t_{i-1}, t_j, t_{i+1}, \dots, t_{j-1}, t_i, t_{j+1}, \dots, t_n, t_1^{-1}, \dots, t_{i-1}^{-1}, t_j^{-1}, t_{i+1}^{-1}, \dots, t_{j-1}^{-1}, t_i^{-1}, t_{j+1}^{-1}, \dots, t_n^{-1})$ .
- The eigenvectors in  $\mathfrak{so}(2n)$  is given in [Mil17]P460.
- The base corresponding to the standard Borel is  $\Delta = \{e_1 - e_2, e_2 - e_3, \dots, e_{n-1} - e_n, e_{n-1} + e_n\}$ .
- Thus it is of type  $D_n$  (3.9.6.3). It is geometrically almost-simple by (10.4.5.13). Its non-simply-connected and non-adjoint-type with center  $D(X/Q(\Phi)) \cong \mu_2$ . It simply-connected covering group is  $\mathrm{Spin}(2n) \rightarrow \mathrm{SO}(2n)$  with kernel  $D(P(\Phi)/Q(\Phi)) \cong \begin{cases} \mu_2 \times \mu_2 & , 2|n \\ \mu_4 & , 2 \nmid n \end{cases}$  (10.4.5.3).

⌋

**Prop. (10.5.3.14)**[Root Systems of Semisimple Rank 1]. There are only three reduced root systems of semisimple rank 1, given by  $(\mathbb{Z}^r, \{\pm\alpha\}, \{\pm\alpha^\vee\})$  with

$$\begin{cases} \alpha = 2e_1 \\ \alpha^\vee = e_1^\vee \end{cases}, \quad \begin{cases} \alpha = e_1 \\ \alpha^\vee = 2e_1^\vee \end{cases}, \quad \begin{cases} \alpha = e_1 - e_2 \\ \alpha^\vee = e_1^\vee - e_2^\vee \end{cases},$$

corresponding resp. to the reductive groups

$$\mathrm{SL}(2) \times \mathbb{G}_m^{r-1}, \quad \mathrm{PGL}(2) \times \mathbb{G}_m^{r-1}, \quad \mathrm{GL}(2) \times \mathbb{G}_m^{r-2}.$$

⌋

*Proof:* ?

□

### Forms of Classical Groups

**Prop. (10.5.3.15)**[Pure Inner Forms]. Suppose  $\mathrm{char} k = 2$ .

- For  $K/k \in \mathrm{Field}$  of degree 2 and  $V \in \mathrm{Herm}_n(K/k)$ , the inner forms of  $\mathrm{U}(V)$  are exactly the groups  $\mathrm{U}(V')$  where  $V' \in \mathrm{Herm}_n(K/k)$ .
- For  $V \in \mathrm{SymSpa}_n(k)$ , the inner forms of  $\mathrm{SO}(V)$  are exactly the groups  $\mathrm{SO}(V')$  where  $V' \in \mathrm{SymSpa}_n(k)$  and  $\mathrm{disc}(V') = \mathrm{disc}(V)$ .

⌋

*Proof:* This follows from (10.2.5.6).

□

**Thm. (10.5.3.16)**. If  $\mathrm{char} k \neq 2$ , then the geometrically almost-simple simply-connected classical groups over  $k$  are the following:

$A_n$  : The groups  $\mathrm{SL}(m; D)$  for  $D \in \mathrm{Az}_k$ , which are inner forms of  $\mathrm{SL}(n)$ . The special unitary groups attached to a Hermitian form for a division ring of simple unitary type (3.6.4.33), which are outer forms for  $\mathrm{SL}(n)$ . There are no pure inner forms, by (10.2.5.13).

$C_n(n \geq 3)$  : The unitary groups attached to a  $\epsilon$ -skew Hermitian form for a central division ring with an involution of  $\epsilon$ -symplectic type (3.6.4.31) over  $k$ , which are inner forms for  $\mathrm{Sp}(2n)$ .



$B_{n \geq 2}, D_{n \geq 4}$  : The unitary groups attached to a  $\epsilon$ -skew Hermitian form for a central division ring with an involution of  $\epsilon$ -orthogonal type(3.6.4.31) over  $k$ , which are  $k$ -forms for  $\text{Spin}(n)$ .  $\lrcorner$

*Proof:* 1: ? Cf.[Mil17]P524, 525.

2: ? Cf.[Mil17]P526.

3: ? Cf.[Mil17]P534.  $\square$

**Cor.(10.5.3.17).** If  $k$  is a special field(11.2.2.10) and  $\text{char } k \neq 2$ , then the geometrically almost-simple simply-connected classical groups over  $k$  are the following:

$A_n$  : The groups  $\text{SL}(m; D)$  for  $D \in \text{Az}_k$ , which are inner forms of  $\text{SL}(n)$ . The special unitary groups attached to a Hermitian form for a division ring of simple unitary type(3.6.4.33), which are outer forms for  $\text{SL}(n)$ .

$C_n(n \geq 3)$  : The unitary groups attached to a  $\epsilon$ -skew Hermitian form for a quaternion field with an involution of  $\epsilon$ -symplectic type(3.6.4.31) over  $k$ , which are inner forms for  $\text{Sp}(2n)$ .

$B_{n \geq 2}, D_{n \geq 4}$  : The unitary groups attached to a  $\epsilon$ -skew Hermitian form for a central division ring with an involution of  $\epsilon$ -orthogonal type(3.6.4.31) over  $k$ , which are  $k$ -forms for  $\text{Spin}(n)$ .  $\lrcorner$

### Symplectic Groups

**Def.(10.5.3.18) [Siegel Parabolic Subgroups].** Situation as in(10.5.3.19), the **Siegel parabolic subgroup** of  $\text{Sp}(2n)$  is defined to be the parabolic subgroup  $P_{\text{Sie}} = LU$  where

$$L = \{\text{diag}(A, A^{-t}) : A \in \text{GL}(n)\} \cong \text{GL}(n).$$

$$U = \left\{ \begin{bmatrix} \mathbf{1}_n & B \\ \mathbf{0} & \mathbf{1}_n \end{bmatrix} : B \in \text{Sym}_n(F) \right\} \cong \text{Sym}_n.$$

By(12.10.1.21), the modular quasi-character of  $P_{\text{Sie}}$  is

$$\delta_L(g) = \det(\text{Ad}(g)|\mathfrak{U}) = |\det(A(g))|^{n+1}.$$

$\lrcorner$

### Unitary Groups

**Def.(10.5.3.19) [Standard Unitary Groups].** Let  $k \in \mathbf{Field}$  and  $E/k$  an étale  $k$ -algebra of degree 2 with a non-trivial involution  $\mathfrak{c}$ , then  $\text{U}(n; n)_F$  be defined to be algebraic subgroup of  $\text{Res}_{E/F} \text{GL}(2n)$  representing the functor

$$\mathfrak{c}\text{Ring}_k \rightarrow \mathfrak{Grp} : R \mapsto \left\{ g \in \text{GL}(2n; R \otimes_k E) : g^t \begin{bmatrix} & & 1 \\ & \ddots & \\ 1 & & \end{bmatrix} g^{\mathfrak{c}} = \begin{bmatrix} & & 1 \\ & \ddots & \\ 1 & & \end{bmatrix} \right\}.$$

$\lrcorner$

**Def. (10.5.3.20) [Siegel Parabolic Subgroups].** Situation as in (10.5.3.19), the **Siegel parabolic subgroup** of  $U(n, n)_F$  is defined to be the parabolic subgroup  $P_{\text{Sie}} = LU$  where

$$L = \{\text{diag}(A, c(A)^{-t}) : A \in \text{Res}_{E/F} \text{GL}(n)\} \cong \text{Res}_{E/F} \text{GL}(n).$$

$$U = \left\{ \begin{bmatrix} \mathbf{1}_n & b \\ \mathbf{0}_n & \mathbf{1}_n \end{bmatrix} : b \in \text{Herm}_n(E/F) \right\} \cong \text{Herm}_n(E/F).$$

The modular character of  $L$  is

$$\det(\text{Ad}_{\mathfrak{u}}(g)) = |\det(A(g))|^n.$$

┘

**Def. (10.5.3.21) [Unitary Groups associated to \*-Algebras].** Let  $k \in \text{Field}$  and  $E/k$  an étale  $k$ -algebra of degree 2 with a non-trivial involution  $c$ . Let  $D$  be a  $k$ -algebra of rank  $2m^2$  with center  $E$  that is simple over  $E$ , equipped with an involution  $*$  of second kind (3.2.1.10). Let  $G \in \text{AlgGrp}/k$  be defined as follows:

$$G(R) = \{x \in (D \otimes_k R)^\times \mid xx^* = 1\}.$$

There are two cases:

- If  $E \cong k \times k$ , then  $D \cong D_1 \times D_2$  where  $D_i$  are central simple  $k$ -algebras with  $*$  :  $D_1 \cong D_2^{\text{op}}$ . Then  $G \cong \Delta_i^*$ .
- If  $E \in \text{Field}$ , then  $G$  is called a **unitary group** attached to  $E/k$ . Moreover, if  $D \cong \text{Mat}(m; k)$ , then  $*$  must be an adjunction w.r.t. a non-degenerate  $c$ -Hermitian form  $f$  on  $E^m$ . Moreover, if  $f$  is the **standard anti-diagonal form**

$$f(xe_i, ye_j) = c(x)y\delta_{i, m+1-i},$$

then  $G$  is called the **quasi-split unitary group** attached to  $E/k$ .

┘

### Langlands Dual Groups

**Prop. (10.5.3.22).** The Langlands dual group of  $\text{GL}(n)$  is  $\text{GL}(n)$ . The Langlands dual group of  $\text{SL}(n)$  is  $\text{PGL}(n)$ .

┘

## 4 Exceptional Groups

See [Springer, Linear Algebraic Groups] and [Book of Involutions].

## 5 Real Reductive Groups

References are [Gaitsgory, Real Reductive Groups] and [Mil17].

**Def. (10.5.5.1) [Real Forms].** Let  $G$  be a connected complex Lie group, then a **real form** of  $G$  is a connected real Lie subgroup  $K \subset G$  that  $\mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{g}$ .

┘

**Prop. (10.5.5.2) [Real Lie Groups and Algebraic Groups].** Let  $(H, \mathfrak{h})$  be a connected Lie group in  $\text{LieGrp}/\mathbb{C}$ , then there needs not be an algebraic group  $G$  over  $\mathbb{R}$  that  $G(\mathbb{R})^0 = H$ . For example, the topological fundamental group of  $\text{SL}(2; \mathbb{R})$  is  $\mathbb{Z}$ , so it has many coverings of finite degree, none of which is algebraic, because  $\text{SL}(2)$  as an algebraic group is simply connected **?**.

However, if  $H$  admits a f.d. representation  $H \hookrightarrow \text{GL}(V)$ , then there exists an algebraic group  $G \subset \text{GL}(V)$  that  $\text{Lie}(G) = [\mathfrak{h}, \mathfrak{h}]$ . So if  $H$  is semisimple, then there exists an algebraic group  $G \subset \text{GL}(V)$  s.t.  $\text{Lie}(G) = [\mathfrak{h}, \mathfrak{h}]$ . When  $H$  is semisimple, this means  $G^0 = H$ .

┘

### Compact Real Algebraic Groups

**Def. (10.5.5.3) [Compact Real Algebraic Groups].** A **compact real algebraic group** is an algebraic group  $G \in \text{AlgGrp}/\mathbb{R}$  s.t.  $G(\mathbb{R})$  is a compact Lie group.  $\lrcorner$

**Prop. (10.5.5.4) [Compactness and Representations].** Let  $G \in \text{AlgGrp}_{\text{cnd}}/\mathbb{R}$ . If  $G$  is compact, then every f.d. real representation  $\rho : G \rightarrow \text{GL}(V)$  carries a  $G(\mathbb{R})$ -invariant inner form. Conversely, if a faithful f.d. real representation carries such a form, then  $G$  is compact.  $\lrcorner$

*Proof:* If  $G$  is compact, then  $H = \rho_{\mathbb{R}}(G(\mathbb{R}))$  is compact, so we can take an arbitrary inner form on  $V$  and take average on  $H$ . The converse is easy.  $\square$

**Def. (10.5.5.5) [Relevant Groups].** A compact real algebraic group is called **relevant group** iff the map  $\pi_0(G(\mathbb{R})) \rightarrow \pi_0(G)(\mathbb{R})$  is surjective.  $\lrcorner$

**Lemma (10.5.5.6).** Let  $Z$  be an affine variety over  $\mathbb{R}$ , let  $X$  be a subset of  $Z(\mathbb{R})$ , and let  $I_X$  be the ideal of regular functions on  $Z$  that vanishes at  $X$ , then  $X' = V(I_X)$  satisfies  $X \subset X'(\mathbb{R})$ . Also by construction,  $X'$  is relevant and  $X$  intersects real points of every connected components of  $X'$ .

Now if  $Z$  is acted on by a compact Lie group  $K$  and  $X$  is a single  $K$ -orbit, then  $X \cong X'(\mathbb{R})$ .  $\lrcorner$

*Proof:* Cf. [Gaitsgory P17].  $\square$

**Prop. (10.5.5.7) [Compact Relevant Groups and Compact Real Lie Groups].** The functor  $G \mapsto G(\mathbb{R})$  is an equivalence of categories from the category of relevant real compact algebraic groups to the category of compact real Lie groups.  $\lrcorner$

*Proof:* For the fully faithfulness: given a map  $\varphi : G_1(\mathbb{R}) \rightarrow G_2(\mathbb{R})$ , we need to show it comes from a unique algebraic group homomorphism. Let  $K \subset G_1(\mathbb{R}) \times G_2(\mathbb{R})$  be the graph of  $\varphi$ , then let  $\Gamma$  be the subgroup of  $G_1 \times G_2$  corresponding to  $K$  in (10.5.5.6), then it suffices to prove that the map  $\Gamma \rightarrow G_1$  is an isomorphism. It is an isomorphism after passing to real points, so isomorphism on the level of Lie algebras. And then it is an isomorphism, because both groups are relevant?  $\square$

**Cor. (10.5.5.8).** The proof actually works if only  $G_1$  is relevant compact real group. So if we choose  $G_2 = \text{Res}_{\mathbb{C}/\mathbb{R}} \text{GL}(n)$ , then by adjointness there is a bijection

$$\text{Hom}_{\text{AlgGrp}}(G_{\mathbb{C}}, \text{GL}(n)) \cong \text{Hom}_{\text{LieGrp}}(G(\mathbb{R}), \text{GL}(n; \mathbb{C})).$$

That is, their complex representations correspond.  $\lrcorner$

### Complex Reductive Algebraic Groups

**Prop. (10.5.5.9).** If  $G$  is a real reductive group, then its complexification  $G_{\mathbb{C}}$  is complex reductive.  $\lrcorner$

*Proof:* It suffices to show that  $\text{Rep}(G(\mathbb{C}))$  is semisimple. For this, notice  $\text{Rep}(G)$  is semisimple by definition, so it suffices to show for any representation  $V$  of  $G_{\mathbb{C}}$ , if  $W$  is a  $G$ -invariant subspace, then  $W$  is also  $W$  is  $G_{\mathbb{C}}$ -invariant. But the invariance condition is a vanishing of some matrix coefficients, they vanish on  $G$  so also vanish on  $G_{\mathbb{C}}$ .  $\square$

**Def. (10.5.5.10) [Real Form].** A **real form** on a complex reductive algebraic group is an anti-linear group isomorphism  $\sigma : G \rightarrow G$  that  $\sigma^2 = 1$ . It is called **compact** iff  $G^{\sigma}$  is compact real, and it is called **relevant** iff  $G^{\sigma}$  is relevant compact (10.5.5.5).  $\lrcorner$

**Prop. (10.5.5.11)[Polar Decomposition].** If  $G \in \text{AlgGrp}/\mathbb{C}$  and  $K \in G(\mathbb{C})$  is a compact Lie subgroup s.t.

- $\mathfrak{g} \cong \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C}$ .

- $K$  intersects non-trivially every connected components of  $G(\mathbb{C})$ .

then the group  $G$  contains a unique real structure  $\sigma$  that  $K = G(\mathbb{C})^\sigma$ . And if  $\mathfrak{p} \subset \mathfrak{g}$  be the subspace  $\{\xi \in \mathfrak{g} | \sigma(\xi) = -\xi\}$ , then the map

$$k \times \mathfrak{p} \rightarrow G(\mathbb{C}) : (k, p) \mapsto k \cdot \exp(p)$$

is a diffeomorphism. ┘

*Proof:* Cf.[Gaitsgory P18]. □

**Cor. (10.5.5.12).** If we denote  $P = \exp(\mathfrak{p})$ , then  $P \subset \tilde{P} = \{g \in G(\mathbb{C}) | \sigma(g) = g^{-1}\}$ , and there is a diffeomorphism:

$$\coprod_{k \in K, k^2=1} \{k\} \times P \cong \tilde{P}.$$

┘

**Cor. (10.5.5.13).** In the situation of(10.5.5.11),  $G$  is reductive, by(10.5.5.9). ┘

**Cor. (10.5.5.14).**  $K \rightarrow G(\mathbb{C})$  is a homotopy equivalence. ┘

### Cartan Involutions

**Def. (10.5.5.15)[Cartan Involutions].** Let  $G \in \text{AlgGrp}_{\text{cntd}}/\mathbb{R}$ , an **involution on  $G$**  is an isomorphism  $\theta : G \cong G \in \text{AlgGrp}/\mathbb{R}$  s.t.  $\theta^2 = \text{id}$ .

A **Cartan involution** is an involution s.t.  $G^{(\theta)}(\mathbb{R}) = \{g \in G(\mathbb{C}) | g = \theta(\bar{g})\}$  is compact. ┘

**Prop. (10.5.5.16).** Cartan involution and compactness. ┘

**Example (10.5.5.17).** Let  $G = \text{SL}(2)_{\mathbb{R}}$  and  $\theta = \text{ad}\left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right)$ , then  $\text{SL}(2)^{(\theta)}(\mathbb{R}) = \text{SU}(2)$  is compact, so this is a Cartan involution on  $G$ . ┘

**Thm. (10.5.5.18) [Satake].** Let  $G \in \text{AlgGrp}_{\text{cntd}}/\mathbb{R}$ , then  $G$  admits a Cartan involution(10.5.5.15) iff  $G$  is reductive. And in this case, any two Cartan involutions differ by a conjugation by elements in  $G(\mathbb{R})$ , i.e.  $(\tau = \text{ad}(g_0)^{-1} \circ \theta \circ \text{ad}(g_0))$ . ┘

*Proof:* Cf.[Satake, Algebraic structures of symmetric domains, volume 4 of Kano Memorial Lectures, 1980]I.4.3. □

**Cor. (10.5.5.19) [Satake].**

- $G$  is a connected algebraic group, then  $G$  is compact iff  $\text{id}$  is a Cartan involution on  $G$ . And in this case, this is the only Cartan involution of  $G$ . In particular, a compact connected algebraic group is reductive.
- If  $V$  is a f.d. real vector space and  $G = \text{GL}(V)$ , then the choice of a basis for  $V$  determines a Cartan involution  $M \mapsto M^t$ , and by(10.5.5.18), any Cartan involution is of this form.
- If  $G \subset \text{GL}(V)$ , then  $G$  is reductive iff  $G$  is stable under a Cartan involution of  $\text{GL}(V)$ . And any Cartan involutions of  $G$  is of this form.

┘

*Proof:* Cf.[Satake, Algebraic structures of symmetric domains, volume 4 of Kano Memorial Lectures, 1980]I.4.4. □

**Prop. (10.5.5.20)[*c*-Polarizations].** Let  $G \in \text{AlgGrp}/\mathbb{R}$  and  $c \in G(\mathbb{R})$  (equivalently  $c^2 \in Z(G(\mathbb{R}))$ ), and  $\text{ad}(c)^{-1} = \text{ad}(c^{-1})$ . Then for a real representation  $V$  of  $G$ , a ***c*-polarization** is a  $G(\mathbb{R})$ -invariant bilinear form  $\varphi$  s.t. the form  $(u, v) \mapsto \varphi(u, cv)$  is symmetric and positive-definite.

Then for such  $c$ , if  $\text{ad}(c)$  is a Cartan involution, then any f.d. real representation of  $G$  admits a  $c$ -polarization. Conversely, if a faithful real representation of  $G$  admits a  $c$ -involution, then  $\text{ad}(c)$  is a Cartan involution. ┘

*Proof:* Cf.[?Mil17b]P16. □

### Maximal Compact Subgroup

**Thm. (10.5.5.21) [Existence and Uniqueness of Maximal Compact Subgroups].** For  $G \in \text{linAlgGrp}/\mathbb{R}$ ,  $G(\mathbb{K})$  has a maximal compact subgroup  $\mathcal{K}$  that is unique up to conjugation.

For any compact subgroup  $K' \subset G(\mathbb{R})$ , there exists an element  $g \in P^\tau$  s.t.  $\text{Ad}_g(K') \in K^\tau$ . ┘

*Proof:* Cf.[Getz]P114 ? and [Gaitsgory P25]. □

**Example (10.5.5.22)[ $\text{GL}(n)$ ].** Fix the following maximal compact subgroup of  $\text{GL}(n; F \otimes \mathbb{R})$ :

$$\mathcal{K}_\infty = \prod \mathcal{K}_v, \quad \mathcal{K}_v = \begin{cases} \text{O}(n) & v \text{ real} \\ \text{U}(n) & v \text{ complex} \end{cases}.$$

Define  $\mathfrak{gl}_{n,\infty} = \prod_{v \in S_\infty} \mathfrak{gl}(n, F_v)$ , then we can define  $(\mathfrak{gl}_{n,\infty}, \mathcal{K}_\infty)$ -modules. ┘

### Complex Reductive Lie Groups

**Def. (10.5.5.23)[Compact Part].** If  $G = ((\mathbb{C}^\times)^r \times G_{ss})/Z$  is a connected complex reductive Lie group, then  $Z \subset (S^1)^r \times G_{ss}^c$  (the compact part), then we denote  $G^c = (S^1)^r \times G_{ss}^c/Z$  the compact subgroup of  $G$ . Then the restriction of f.d. representations of  $G$  to  $G^c$  is an equivalence by (16.5.6.3). ┘

**Example (10.5.5.24).**  $\text{GL}(n; \mathbb{C}) = (\mathbb{C}^* \times \text{SL}(n; \mathbb{C}))/\mu_n$  is a complex reductive Lie group. ┘

## 6 Reductive Group Schemes

**Def. (10.5.6.1)[Transporter Functors].** For  $G \in \text{GrpSch}/S$  and  $Y, Y' \rightarrow G \in \text{Sch}/S$ , the **transporter functor**  $\text{bxTransp}_G(Y, Y')$  is the functor on  $S$ -schemes:

$$\text{Transp}_G(Y, Y') : S' \in \text{Sch}/S \mapsto \{g \in G(S') \mid gY_{S'}g^{-1} \subset Y'_{S'}\}.$$

And when  $Y = Y'$  is f.p. over  $S$ , this is denoted by  $N_G(Y)$ . (Cf. [Con14]P56). ┘

**Prop. (10.5.6.2) [Smoothness of Transporter Subgroups].** For  $G \in \text{GrpSch}^{\text{pf}}/S$  and f.p. closed subschemes  $Y, Y' \subset G \in \text{Sch}^{\text{pf}}/S$ . If  $Y$  is either a multiplicative type subgroup or finite flat over  $S$ , then  $\text{Transp}_G(Y, Y')$  (10.5.6.1) is represented by a f.p. closed subscheme of  $G$ .

And if  $G/S$  is smooth and  $Y, Y'$  are both of multiplicative type over  $S$ , then  $\text{Transp}_G(Y, Y')$  is smooth over  $S$ . ┘

*Proof:* Cf.[Con14]P56. □

**Prop. (10.5.6.3) [Smoothness of Centralizers].** For  $G \in \mathcal{G}rpSch^{pf}/S$  and f.p. closed subschemes  $Y, Y' \subset G \in \mathcal{S}ch^{pf}/S$ . If  $Y$  satisfies one of the following:

- $Y$  is a multiplicative type subgroup.
- $Y$  is smooth over  $S$  with connected non-empty fibers.
- $Y$  is finite flat over  $S$ .

Then  $C_G(Y)$  is representable by a f.p. closed subgroup of  $G$ .

And  $C_G(Y)$  is smooth when  $G$  is smooth and  $Y$  is a multiplicative type subgroup, in which case  $\mathrm{Lie}(C_G(Y)) = \mathrm{Lie}(G)^Y$ . ┘

**Def. (10.5.6.4) [ $S$ -Tori].**  $T \in \mathcal{G}rpSch^{aff}/S$  is called an  **$S$ -torus** if it is of multiplicative type with smooth connected fibers. ┘

**Def. (10.5.6.5) [Semisimple and Reductive Schemes].**  $G \in \mathcal{G}rpSch^{aff}/S$  is called a **reductive scheme** if it has reductive geometric fibers. It is called a **semisimple scheme** if it has semisimple geometric fibers. ┘

**Prop. (10.5.6.6) [Non-Connected Reductive Groups].** If  $G \in \mathcal{G}rpSch^{sm,sep,pf}/S$  has reductive geometric fibers, then  $G^0$  is a clopen reductive  $S$ -scheme, and  $G/G^0$  is a separated étale  $S$ -group of finite presentation. ┘

*Proof:* Cf.[Con14]P82. □

**Prop. (10.5.6.7) [Spreading Out Subgroups].** For  $G \in \mathcal{G}rpSch^{aff,sm}/S$ ,  $s \in S$  and  $H_0 \leq G_s$  a subgroup of multiplicative type, then there exists an étale nbhf  $(S', s')$  of  $(S, s)$  with  $\kappa(s') = \kappa(s)$  and a multiplicative type group  $H' \leq G_{S'}$  s.t.  $H'_{s'} = H_0$  inside  $(G_{S'})_{s'} = G_s$ . ┘

*Proof:* Cf.[Con14]P83. □

**Prop. (10.5.6.8) [Locality of Reductive Group Schemes and Maximal Tori].** Let  $G \in \mathcal{G}rpSch^{aff,sm}/S$  and  $s \in S$  s.t.  $G_s^0$  is reductive, then

- There exists an open subscheme  $s \in U \subset S$  s.t.  $G_u^0$  is reductive for any  $u \in U$ .
  - If  $T \leq G$  is an  $S$ -torus (10.5.6.4) s.t.  $T_s \leq G_s^0$  is maximal, then there exists an open subscheme  $s \in V \subset S$  s.t.  $C_G(T)_V^0 = T_V$ , and  $T_v$  is a maximal torus in  $G_v^0$  for any  $v \in V$ .
- ┘

*Proof:* Cf.[Con14]P85. □

**Def. (10.5.6.9) [Maximal Tori].** For  $G \in \mathcal{G}rpSch^{aff,sm}/S$ , an  $S$ -torus (10.5.6.4)  $T \leq G$  is called a **maximal  $S$ -torus** if  $T_s \leq G_s$  is a maximal torus for any  $s \in S$ . ┘

**Prop. (10.5.6.10).** Let  $G \in \mathcal{G}rpSch^{aff,sm}/S$  and  $S' \in \mathcal{S}ch/S$ . If  $T \leq G$  is a maximal  $S$ -torus (10.5.6.9), then  $T_{S'} \leq G_{S'}$  is a maximal  $S'$ -torus, by (10.4.1.17).

And if  $T \leq G$  is an  $S$ -torus s.t.  $T_s \leq G_s$  is a maximal torus for each  $s \in S$ , then  $T$  is a maximal torus of  $G$ , by (10.5.6.8). ┘

**Prop. (10.5.6.11).** Let  $G \in \mathcal{G}rpSch^{aff,sm}/S$  s.t.  $G_s^0$  is reductive for any  $s \in S$ , then the locus of  $s \in S$  s.t.  $G_s$  is connected is closed in  $G$ . ┘

*Proof:* Cf.[Con14]P89. □

**Thm. (10.5.6.12) [Moduli of Maximal Tori].** Let  $G \in \text{GrpSch}^{\text{aff}, \text{sm}}/S$  s.t.  $G_s^0$  is reductive for any  $s \in S$ . Then the functor on  $S$ -schemes:

$$\underline{\text{Tor}}_{G/S} : S' \in \text{Sch}/S \mapsto \{\text{maximal tori in } G_{S'}\}$$

is represented by a smooth quasi-affine  $S$ -scheme  $\text{Tor}_{G/S}$ , and  $\text{Tor}_{G/S} \rightarrow S$  is surjective.

If  $T \leq G$  is a maximal  $S$ -torus, then the map  $G/N_G(T) \rightarrow \text{Tor}_{G/S}$  defined by  $G$ -conjugation on  $T$  is an isomorphism. In particular, any two maximal tori of  $G$  are conjugate étale-locally on  $S$ .  $\lrcorner$

*Proof:* Cf.[Con14]P94.  $\square$

**Cor. (10.5.6.13) [Maximal Tori Exist].** Any  $G \in \text{RedGrp}/S$  admits a maximal torus étale-locally on  $S$ .  $\lrcorner$

*Proof:* This is because a smooth surjective admits a section étale-locally. ?  $\square$

**Prop. (10.5.6.14) [Finite Weyl Groups].** For  $G \in \text{RedGrp}/S$  and a maximal  $S$ -torus  $T \leq G$ , the Weyl group  $W_G(T) = N_G(T)/C_G(T)$  is finite étale over  $S$ .  $\lrcorner$

*Proof:* Cf.[Con14]P96.  $\square$

**Prop. (10.5.6.15).** For  $G \in \text{RedGrp}/S$ ,  $Z(G) = \ker(\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}))$ .  $\lrcorner$

*Proof:* Cf.[Con14]P102.  $\square$

**Def. (10.5.6.16) [Isogenies of Reductive Schemes].**  $f : G \rightarrow G' \in \text{GrpSch}^{\text{sm}, \text{aff}}/S$  is called an **isogeny** if it is a finite flat surjective. And it is called a **central isogeny** if its kernel is central in  $G$ .  $\lrcorner$

**Prop. (10.5.6.17) [Fibral Criterion of Central Isogenies].** If  $f : G \rightarrow G' \in \text{RedGrp}/S$  is surjective, then  $f$  is a central isogeny iff  $f_{\bar{s}}$  has finite central kernel for all  $s \in S$ .  $\lrcorner$

*Proof:* Cf.[Con14]P103.  $\square$

### Split Reductive Schemes

**Def. (10.5.6.18) [Split Reductive Schemes].**  $G \in \text{RedGrp}/S$  is called a **split reductive scheme** if it has a maximal torus  $T \leq G$  with an isomorphism  $T \cong D_S(M)$  for some finite free  $\mathbb{Z}$ -module  $M$  s.t.

- Every root  $\alpha : T \rightarrow \mathbb{G}_m$  (characters of  $T$  appearing in  $\mathfrak{g}$ ) arises from elements of  $M$ .
- Every root space  $\mathfrak{g}_\alpha$  is free of rank 1 over  $\mathcal{O}_S$ .
- Every coroot  $\alpha^\vee : \mathbb{G}_m \rightarrow T$  arises from elements of  $M^\vee$ .

$\lrcorner$

**Prop. (10.5.6.19).** Every  $G \in \text{RedGrp}/S$  is split(10.5.6.18) étale-locally on  $S$ .  $\lrcorner$

*Proof:* Cf.[Con14]P136.  $\square$

**Thm. (10.5.6.20) [Split Reductive Schemes of Semisimple Rank 1].** Let  $(G, T, M) \in \text{RedGrp}/S$  be split reductive with geometric fibers of semisimple rank 1, then Zariski-locally on  $S$ ,  $(G, T)$  is isomorphic to product of a split central torus with one of the following:

- $(\text{SL}(2), D)$  with  $D$  the diagonal torus.



- $(\mathrm{PGL}(2), \overline{D})$  with  $\overline{D}$  the diagonal torus.
- $(\mathrm{GL}(2), T)$  with  $T$  the diagonal torus.

⌋

*Proof:* Cf.[Con14]P139.

□

**Def. (10.5.6.21) [Parabolic Subgroup Schemes].** For  $G \in \mathrm{RedGrp}/S$ , a **parabolic subgroup scheme**  $P \leq G$  is a smooth  $S$ -affine subgroup scheme  $P \in \mathrm{GrpSch}^{\mathrm{aff}, \mathrm{sm}}/S$  equipped with a monic homomorphism  $P \leq G$  that embeds  $P_{\overline{s}}$  as a parabolic subgroup of  $G_{\overline{s}}$  for any  $s \in S$ .

⌋

**Prop. (10.5.6.22).** Suppose  $G \in \mathrm{RedGrp}/S$ . Any parabolic subgroup scheme  $P \leq G$  (10.5.6.21) is a closed subscheme of  $G$ , and étale-locally there exists a maximal torus  $T \leq G$  s.t.  $T \leq P$ .

Conversely, if  $G$  admits a maximal torus  $T$  s.t.  $T \leq P$  and  $(G, T)$  is split, then Zariski-locally on  $S$ , there exists  $\lambda : \mathbb{G}_m \rightarrow T$  s.t.  $P = P_G(\lambda)$ ?

⌋

*Proof:* Cf.[Con14]P152.?

□

**Prop. (10.5.6.23) [Moduli of Parabolic Subgroups].** Let  $G \in \mathrm{RedGrp}/S$ , then the functor on  $S$ -schemes:

$$\underline{\mathrm{Par}}_{G/S} : S' \in \mathrm{Sch}/S \mapsto \{\text{parabolic subgroups of } G_{S'}\}$$

is represented by a smooth proper  $S$ -scheme  $\mathrm{Par}_{G/S}$  represented by the canonical  $S$ -ample line bundle  $\det(\mathrm{Lie}(\mathcal{P}))^*$ , where  $\mathcal{P} \subset G \times \mathrm{Par}_{G/S}$  is the universal parabolic subgroup of  $G$ .

⌋

*Proof:* Cf.[Con14]P156.

□

**Def. (10.5.6.24) [Borel Subgroup Schemes].** For  $G \in \mathrm{RedGrp}/S$ , a **Borel subgroup scheme**  $P \leq G$  is a parabolic subgroup scheme  $P \leq G$  (10.5.6.21) s.t.  $P_{\overline{s}}$  is a Borel subgroup of  $G_{\overline{s}}$  for any  $s \in S$ .

$G$  is called a **quasi-split reductive group scheme** over  $S$  if it admits a Borel subgroup scheme.

⌋

**Prop. (10.5.6.25) [Moduli of Borel Subgroups].** For  $G \in \mathrm{RedGrp}/S$ ,

- If  $P \leq G$  is a parabolic subgroup scheme, then the locus of  $s \in S$  s.t.  $P_{\overline{s}}$  is Borel in  $G_{\overline{s}}$  is a clopen subset of  $S$ .
- Any two Borel subgroups of  $G$  are conjugate étale-locally on  $S$ .
- the functor on  $S$ -schemes:

$$\underline{\mathrm{Bor}}_{G/S} : S' \in \mathrm{Sch}/S \mapsto \{\text{Bor subgroups of } G_{S'}\}$$

is represented by a smooth proper  $S$ -scheme  $\mathrm{Bor}_{G/S}$  represented by the canonical  $S$ -ample line bundle  $\det(\mathrm{Lie}(\mathcal{B}))^*$ , where  $\mathcal{B} \subset G \times \mathrm{Bor}_{G/S}$  is the universal Borel subgroup of  $G$ .

⌋

*Proof:* Cf.[Con14]P158.

□

**Cor. (10.5.6.26) [Reductive Groups over henselian Local Rings are Quasi-Split].** If  $R \in \mathcal{CRing}$  is a Henselian local ring with finite residue field and  $G \in \mathrm{RedGrp}/R$ , then  $G$  is quasi-split, and becomes split over a finite étale extension of  $R$ .

⌋

*Proof:* Cf.[Con14]P159.

□



### Isogeny and Existence Theorems

**Def. (10.5.6.27) [Compatibility with Splittings].** Let  $(G, T, M), (G', T', M') \in \mathcal{R}\text{edGrp}/S$  be split, then a quasi-finite surjection  $f : (G', T') \rightarrow (G, T)$  is said to be **compatible with splittings** if there is a homomorphism  $h : M \rightarrow M'$ , a bijection  $d : \Phi' \cong \Phi$  and a function  $q : \Phi' \rightarrow \mathbb{Z}_+$  valued in prime powers s.t.

The induced map  $X^\bullet(f) : M_S \rightarrow M'_S$  arises from  $h$ .

- For all  $\alpha' \in \Phi'$ ,  $h(d(\alpha')) = q(\alpha')\alpha'$  and  $h^\vee(\alpha'^\vee) = q(\alpha')d(\alpha')^\vee$ .
- If  $q(\alpha') \in p(\alpha')^{\mathbb{Z}}$  for some  $p(\alpha') \in \text{Prime}$ , then  $S \in \text{Sch}/\mathbb{F}_{p(\alpha')}$ .

┘

**Prop. (10.5.6.28).** Let  $(G, T, M), (G', T', M') \in \mathcal{R}\text{edGrp}/S$  be split, then any quasi-finite surjection  $f : (G', T') \rightarrow (G, T)$  is Zariski-locally compatible with splitting. In particular, if  $S$  is connected, then  $f$  is compatible with the splittings.

┘

*Proof:* Cf. [Con14]P190.

□

**Thm. (10.5.6.29) [Isogeny Theorem].** For split reductive groups  $(G', T', M'), (G, T, M) \in \mathcal{R}\text{edGrp}/S$ , any  $p(S)$ -morphism  $\Phi(G', T', M') \rightarrow \Phi(G, T, M)$  is induced by an isogeny  $f : (G', T', M') \rightarrow (G, T, M)$  compatible with the splittings, and such  $f$  is unique up to an action of  $(T/Z(G))(S)$  on  $G$ .

┘

*Proof:* Cf. [Con14]P196.

□

**Thm. (10.5.6.30) [Existence Theorem].** Every root datum arises from a split reductive scheme over  $S$ .

┘

*Proof:* Cf. [Con14]P196.

□

### Dynamical Methods

#### Chevalley Groups

References are [Groups over  $\mathbb{Z}$ , Gross].

**Def. (10.5.6.31) [Chevalley Group Schemes].** A **Chevalley group scheme** is a split reductive group scheme over  $\mathbb{Z}$  that admits maximal tori (10.4.1.16).

┘

**Prop. (10.5.6.32) [Classifying Chevalley Groups].** Chevalley group schemes are determined up to isomorphisms by its reduced root datum, and each reduced root data arises in this way. In particular, two Chevalley groups are isomorphic iff they are isomorphic over  $\mathbb{C}$ .

┘

*Proof:* Cf. [Con14]P230.

□

## 10.6 Topics in Group Schemes

**Notation(10.6.0.1).**

- Use notations on group schemes defined in [Group Schemes I: Structure Theory](#). ┘

**Thm.(10.6.0.2) [Lang-Steinberg].** Let  $k \in \mathbf{Field}$ ,  $G \in \mathbf{AlgGrp}_{\text{cntd}}/k$  and  $F : G_{\bar{k}} \rightarrow G_{\bar{k}} \in \mathbf{AlgGrp}/\bar{k}$  s.t. the fixed point of  $F$  is finite, then the **Lang map**

$$\mathcal{L} : G(\bar{k}) \rightarrow G(\bar{k}) : g \mapsto g^{-1}F(g)$$

is surjective. ┘

*Proof:* Cf.[Steinberg, Endomorphisms of Linear Algebraic Groups, P67]. ?

We only prove this when  $\#k < \infty$  and  $F$  is a Steinberg endomorphism(10.6.1.2).

By(10.1.4.4) and(10.1.4.2), we can assume  $G$  is smooth.  $G$  acts on itself by  $g(x) = gxF(g)^{-1}$ . By(10.2.1.15), it has a closed orbit  $\Omega$ . Thus it suffices to show that  $\dim \Omega = \dim G$ . Take  $x \in \Omega(k)$ , by(10.2.1.11) and(10.1.1.27), it suffices to show there exists only f.m.  $g \in G(k)$  s.t.  $gxF(g)^{-1} = x$ . Let  $F^m(x) = x$ ,  $f : G \rightarrow G : f(g) = xF(g)x^{-1}$ . Notice  $xF(x) \cdot F^2(x) \cdot \dots \cdot F^{m-1}(x)$  is contained in  $\ker(F^m)$ , so it is of finite order, let's say  $r$ . Then  $f^{mr} = F^{mr}$ , which has only f.m. solutions, thus there are only f.m.  $g$  s.t.  $gxF(g)^{-1} = x$ . □

### 1 over Finite Fields

Main references are [Representations of Finite Groups of Lie Type, Digne and Michel].

**Notation(10.6.1.1).**

- Let  $p \in \mathbf{Prime}$ ,  $r \in \mathbb{Z}_+$ ,  $q = p^r$ .
- Let  $k \in \mathbf{Field}^p$ ,  $\#k = q$ . ┘

**Def.(10.6.1.2) [Steinberg-Endomorphisms].** Let  $k \in \mathbf{Field}^p$  and  $G \in \mathbf{GrpVar}/k$ , a **Steinberg-endomorphism** of  $G$  is an endomorphism  $F : G_{\bar{k}} \rightarrow G_{\bar{k}} \in \mathbf{AlgGrp}/\bar{k}$  s.t. some power of  $F$  equals some power of the geometric Frobenius  $\text{Fr}_X$ (7.2.10.1). ┘

**Prop.(10.6.1.3) [Orbits Contains a Rational Point].** Let  $V \in \mathbf{Sch}^{\text{ft}}/k$ ,  $G \in \mathbf{AlgGrp}_{\text{cntd}}/k$ ,  $G$  acts on  $V$ , then any  $G$ -orbits  $O$  contains a rational point. ┘

*Proof:* Let  $v \in O(\bar{k})$ , then  $F(v) = gv$  for some  $g \in G(\bar{k})$ . Then by Lang's theorem(10.6.0.2),  $g = F(h)^{-1}h$  for some  $h \in G(\bar{k})$ . Then  $F(h(v)) = h(v)$ , so  $h(v) \in O$  is a rational point. □

**Cor.(10.6.1.4).** If  $G \in \mathbf{AlgGrp}/k$ ,  $H \leq G$  is a connected subgroup, then  $G(k)/H(k) = G/H(k)$ . ┘

**Prop.(10.6.1.5) [Any Group Variety over Finite Fields is Quasi-Split].** Any group variety  $G/k$  is quasi-split(10.4.1.1). ┘

*Proof:* Let  $B \leq G_{\bar{k}}$  be a Borel subgroup(10.4.1.4), then  $\text{Frob}^{-1}(B(\bar{k})) = hB(\bar{k})h^{-1}$  for some  $h \in G(\bar{k})$  by(10.4.1.7). By(10.6.0.2),  $h = g^{-1}\text{Frob}(g)$  for some  $g \in G(\bar{k})$ . Then  $g\text{Frob}^{-1}(B(\bar{k}))g^{-1}$  is the  $\bar{k}$ -points of a Borel subgroup of  $G$  and is fixed by  $\text{Frob}$ , so it descends to a Borel subgroup of  $G$ . □

## 10.7 Formal Group Schemes

References are [Zin84].

**Notation (10.7.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $R \in \mathbb{C}\text{Ring}$ .
- Let  $S \in \text{Sch}$ .

┘

### 1 Formal Power Series

**Def. (10.7.1.1) [Notations].** Let  $\mathbf{X} = \{X_1, \dots, X_n\}$ ,  $\mathbf{Y} = \{Y_1, \dots, Y_n\}$ .

Let  $R$  be a commutative unital ring,  $R[[\mathbf{X}]]$  be the power series ring. It is a local ring with the maximal ideal  $(\mathbf{X})$ .

Let  $i_1 : R[[\mathbf{X}]] \rightarrow R[[\mathbf{X}, \mathbf{Y}]]$  and  $i_2 : R[[\mathbf{Y}]] \rightarrow R[[\mathbf{X}, \mathbf{Y}]]$  be the natural embeddings.

Let  $\Omega_{R[[\mathbf{X}]]/R}^1$  be the free  $R[[X]]$ -module with basis given by  $dX_1, \dots, dX_n$ . And there is a universal derivative  $D : R[[\mathbf{X}]] \rightarrow \Omega_{R[[\mathbf{X}]]/R}^1$  given by  $D(f) = \sum_i \frac{\partial f}{\partial X_i} dX_i$ . It satisfies the usual universal property of Kähler differentials, but with morphisms changed to continuous morphisms. ┘

**Prop. (10.7.1.2) [Formal Sums and Products].**

- Let  $f_n \in R[[\mathbf{X}]]$  for any  $n \in \mathbb{N}$  s.t. for any  $m \in \mathbb{N}$ , there exists a  $N(m) \in \mathbb{N}$  s.t.  $f_n \in (\mathbf{X})^m$  for any  $n \geq N(m)$ , then define the **formal sum**

$$\sum_n f_n = \varinjlim_{n \rightarrow \infty} \sum_{k=0}^n f_k.$$

In this case, we say  $\sum_n f_n$  is well-defined.

- Let  $g_n \in R[[\mathbf{X}]]$  for any  $n \in \mathbb{N}$  s.t. for any  $m \in \mathbb{N}$ , there exists a  $N(m) \in \mathbb{N}$  s.t.  $g_n \in 1 + (\mathbf{X})^m$  for any  $n \geq N(m)$ , then define the **formal product**

$$\prod_n g_n = \varinjlim_{n \rightarrow \infty} \prod_{k=0}^n g_k.$$

In this case, we say  $\prod_n g_n$  is well-defined. ┘

**Prop. (10.7.1.3) [Automorphisms].** If  $F_i$  are power series without constant terms that the matrix degree 1 terms of  $(F_i)$  (the Jacobi matrix) is invertible in  $R$ , then there are unique power series  $G_i$  without constant terms that  $G \circ F = \text{id}$  and  $F \circ G = \text{id}$ . ┘

*Proof:* It  $F_i$  induces a map  $F : R[[X_1, \dots, X_n]] \rightarrow R[[X_1, \dots, X_n]]$  which in turn induces a graded map  $K[X_1, \dots, X_n] \rightarrow K[X_1, \dots, X_n]$ . It is clear that  $(\frac{\partial F_i}{\partial X_j})_{ij}$  is invertible iff the induced graded ring map is an isomorphism, and because  $K[[X_1, \dots, X_n]]$ , a map is an isomorphism iff its induced graded map is an isomorphism.  $\square$

**Prop. (10.7.1.4).** If  $R$  be a torsion-free algebra, and let

$$f(T) = \sum_{n \geq 1} \frac{a_n}{n!} T^n \in (R \otimes \mathbb{Q})[[T]], \quad a_1 \in R^*.$$

Then the unique  $g(T)$  s.t.  $f(g(T)) = T$  is of the form

$$g(T) = \sum_{n \geq 1} \frac{b_n}{n!} T^n \in (R \otimes \mathbb{Q})[[T]], \quad b_1 \in R^*.$$

⌋

*Proof:* Repeatedly differentiating the equation  $f(g(T)) = T$ , we get that  $f'(g(T))g^{(n)}(T)$  can be expressed as integral polynomials in the variables

$$f^{(i)}(g(T)), 1 \leq i \leq n, \quad g^{(j)}(T), 1 \leq j \leq n-1.$$

Then evaluating at  $T = 0$ , we get that  $b_n \in R$  and  $b_1 \in R^*$ , as  $a_1 \in R^*$ . □

### 1-Dimensional Formal Power Series

**Def. (10.7.1.5) [Formal Exponential and Logarithm].** The **formal exponential** and **formal logarithm** is defined to be

$$\exp(X) = \sum_{n \geq 0} \frac{X^n}{n!}, \quad \log(1 + X) = - \sum_{n > 0} \frac{(-X)^n}{n} \in \mathbb{Q}[[X]].$$

They satisfies  $\exp(\log(1 + X)) = 1 + X$ ,  $\log(\exp(X)) = X$ . ⌋

**Remark (10.7.1.6).** WARNING: It should be made clear that  $\log(1 + f)$  is defined only for  $f \in xK[[x]]$ , and  $\log(1 + x)$  is a symbol for the function  $\text{Log}(x) = \log(1 + x)$ .  $\log(x)$  is not defined. ⌋

*Proof:* It suffices to prove  $\text{Exp}(x) = \exp(x) - 1$  and  $\text{Log}(x) = \log(1 + x)$  are inverse to each other. It suffices to show  $\log(\exp(x)) = x$ , because then by (10.7.1.3) the inverse of  $\text{Log}$  must be just  $\text{Exp}$  by (10.7.1.3).

We notice  $\text{Exp}$  are the unique formal power series without constant term that satisfied  $d(\text{Exp}) = \text{Exp} + 1$ , and  $\text{Log}$  is the unique formal power series that satisfies  $d(\text{Log}(x)) = \frac{1}{1+x}$ . Thus

$$d(\log(\exp(x))) = \frac{\exp(x)}{\exp(x)} = 1,$$

so  $\log(\exp(x)) = x$ , because it has no constant term. □

**Prop. (10.7.1.7) [Multiplicative Properties].** Suppose  $\mathbb{Q} \subset R$ , then

- for any  $f, g \in R[[\mathbf{X}]]$ ,  $\exp(f + g) = \exp(f) \exp(g)$ .
- If  $\sum_n f_n$  is well-defined (10.7.1.2), then  $\prod_n \exp(f_n)$  is well-defined and

$$\exp\left(\sum_n f_n\right) = \prod_n \exp(f_n).$$

- for any  $f, g \in 1 + (\mathbf{X})$ ,  $\log(f) + \log(g) = \log(fg)$ .

- If  $g_n \in 1 + (\mathbf{X})$  and  $\prod_n g_n$  is well-defined (10.7.1.2), then  $\sum_n \log(f_n)$  is well-defined and

$$\log\left(\sum_n f_n\right) = \prod_n \log(f_n).$$

┘

*Proof:* Brutal force calculation. ?

□

**Def. (10.7.1.8) [Formal Powers].** Let  $f \in 1 + xR[[x]]$ ,  $g \in R[[x]]$ ,  $f^g$  is defined to be

$$\exp(g \log(f)).$$

In particular,  $\log(f^g) = g \log(f)$  by (10.7.1.5).

┘

**Prop. (10.7.1.9).** If  $p \in \text{Prime}$ ,  $m, n \in \mathbb{Z}_+$ ,  $(n, p) = 1$ , and  $f(x) \in 1 + T\mathbb{Z}_{(p)}[[T]]$ , then  $f(T)^{m/n} \in 1 + T\mathbb{Z}_{(p)}[[T]]$ .

┘

*Proof:* If  $g(T) = f(T)^{m/n}$ , it is easy to see  $g(T)^n = f(T)^m$  and  $g(T) \in 1 + T\mathbb{Q}[[T]]$ . Let

$$f(T)^m = 1 + \sum_{m \geq 1} a_m T^m, \quad g(T) = 1 + \sum_{m \geq 1} b_m T^m,$$

Then it is easy to use induction to show  $b_m \in \mathbb{Z}_{(p)}$  for any  $m \in \mathbb{Z}_+$ .

□

**Prop. (10.7.1.10).** In  $\mathbb{Q}[[x]]$ ,

$$\exp(x) = \prod_{d>0} \left(\frac{1}{1-x^d}\right)^{\frac{\mu(d)}{d}} = \prod_{d>0} B_{-\mu(d)/d}(-X^d).$$

┘

*Proof:* Taking log, we prove its convergence and equality at once:

$$\sum_{d>0} \log\left(\left(\frac{1}{1-x^d}\right)^{\frac{\mu(d)}{d}}\right) = \sum_{d>0} \frac{\mu(d)}{d} \log\left(\frac{1}{1-x^d}\right) = \sum_{d>0} \frac{\mu(d)}{d} \sum_{d'|d} \frac{x^{dd'}}{d'} = \sum_{n>0} \frac{x^n}{n} \sum_{d|n} \mu(d) = x \text{ (2.6.1.26)}.$$

□

**Prop. (10.7.1.11) [Artin-Hasse Exponential].** For  $p \in \text{Prime}$ , the **Artin-Hasse exponential** is the power series  $\text{hexp}(x)$  defined to be

$$\text{hexp}(x) = \exp\left(x + \frac{x^p}{p} + \frac{x^{p^2}}{p^2} + \dots\right) \in \mathbb{Q}[[x]].$$

Then it satisfies

$$\text{hexp}(x) = \prod_{d \in \mathbb{Z}_+ \setminus (p)} \left(\frac{1}{1-x^d}\right)^{\frac{\mu(d)}{d}} = \prod_{d \in \mathbb{Z}_+ \setminus (p)} B_{-\mu(d)/d}(-X^d).$$

and in fact  $\text{hexp}(x) \in \mathbb{Z}_{(p)}[[x]]$ . (Compare (10.7.1.10)).

┘

*Proof:* Taking log, we see

$$\begin{aligned}
 \sum_{d \in \mathbb{Z}_+ \setminus (p)} \log \left( \left( \frac{1}{1-x^d} \right)^{\frac{\mu(d)}{d}} \right) &= \sum_{d \in \mathbb{Z}_+ \setminus (p)} \frac{\mu(d)}{d} \log \left( \frac{1}{1-x^d} \right) \\
 &= \sum_{d \in \mathbb{Z}_+ \setminus (p)} \frac{\mu(d)}{d} \sum_{d' > 0} \frac{x^{dd'}}{d'} \\
 &= \sum_{n \in \mathbb{Z}_+} \frac{x^n}{n} \sum_{d|n, p \nmid d} \mu(d) \\
 &= \sum_{k \in \mathbb{N}} \frac{x^{p^k}}{p^k} \text{ (2.6.1.26)}.
 \end{aligned}$$

Then the last assertion follows from (10.7.1.9).  $\square$

**Def. (10.7.1.12) [Bernoulli-Numbers].** For  $k \in \mathbb{N}$ , the **Bernoulli-numbers**  $B_k$  are defined by the following generating function:

$$\frac{T}{e^T - 1} = \sum_{k \in \mathbb{N}} B_k \frac{T^k}{k!} \in \mathbb{Q}[[T]].$$

Then  $B_k \in \mathbb{Q}$  for all  $k$ , and

$$B_0 = 1, B_1 = -\frac{1}{2}, B_{2k+1} = 0 (k \geq 1), \quad B_2 = \frac{1}{6}, B_4 = -\frac{1}{30}, B_6 = \frac{1}{42}, B_8 = -\frac{1}{30}.$$

More numbers can be found at [https://oeis.org/wiki/Bernoulli\\_numbers](https://oeis.org/wiki/Bernoulli_numbers).

For simplicity, for  $k \in \mathbb{Z}_{<0}$ , denote  $B_k = 0$ .  $\lrcorner$

*Proof:* To show that  $B_{2k+1} = 0$  for  $k \geq 1$ , notice

$$\frac{T}{e^T - 1} - \frac{-T}{e^{-T} - 1} = -T.$$

$\square$

**Def. (10.7.1.13) [Bernoulli-Polynomials].** For  $k \in \mathbb{N}$ , the **Bernoulli-polynomials**  $B_k(X) \in \mathbb{Q}[X]$  are defined by the following generating function:

$$\frac{T e^{TX}}{e^T - 1} = \left( \sum_{k \in \mathbb{N}} B_k \frac{T^k}{k!} \right) \left( \sum_{k \in \mathbb{N}} \frac{(XT)^k}{k!} \right) = \sum_{k \in \mathbb{N}} B_k(X) \frac{T^k}{k!} \text{ (10.7.1.12)}.$$

In particular,

$$B_k(X) = \sum_{i=0}^k \binom{k}{i} B_i X^{k-i},$$

and

$$B_0(X) = 1, \quad B_1(X) = X - \frac{1}{2}, \quad B_2(X) = X^2 - X + \frac{1}{6}, \quad B_3(X) = X^3 - \frac{3}{2}X^2 + \frac{1}{2}X.$$

$\lrcorner$

**Prop. (10.7.1.14) [Properties of Bernoulli-Polynomials].** For any  $k \in \mathbb{N}$ , we have

- $\frac{\partial}{\partial X} B_{k+1}(X) = (k+1)B_k(X)$ .
- $B_k(1-X) = (-1)^k B_k(X)$ .
- $B_k(X+1) - B_k(X) = kX^{k-1}$ .
- For any  $N \in \mathbb{Z}_+$ , we have

$$\sum_{a \in [N-1]} B_k(X + \frac{a}{N}) = N^{1-k} B_k(NX).$$

┘

*Proof:* These all follow from the functional equation, for example,

$$\frac{\partial}{\partial T} \left( \frac{T e^{TX}}{e^T - 1} \right) = T \cdot \frac{T e^{TX}}{e^T - 1}.$$

□

**Cor. (10.7.1.15).**

┘

**Prop. (10.7.1.16) [Generalized Bernoulli-Polynomials].** For  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}(N)$ , define the **generalized Bernoulli-polynomials**  $B_{k,\chi}(X) \in \mathbb{C}[X]$  by the following generating function:

$$\sum_{a \in [N]_+} \chi(a) \frac{T e^{aT} e^{TX}}{e^{NT} - 1} = \sum_{n \in \mathbb{N}} B_{k,\chi}(X) \frac{T^n}{n!}.$$

And denote

$$B_{k,\chi} = B_{k,\chi}(0).$$

Then for each  $M \in \mathbb{Z}_+$  s.t.  $N|M$  and each  $k \in \mathbb{Z}_+$ , we have

$$B_{k,\chi} = M^{k-1} \sum_{a \in [M]_+} \chi(a) B_k\left(\frac{a}{M}\right).$$

┘

**Def. (10.7.1.17) [Hankel-Determinants].** Let  $K$  be a field and  $f = \sum a_i T^i \in K[[T]]$ , then for  $k, M > 0$ , define the **Hankel-determinants** to be

$$H_k = \det(a_{i+j+k}), 1 \leq i, j \leq M$$

┘

**Prop. (10.7.1.18) [Characterizing Rational Functions].** Let  $f(T) = \sum_{i \in \mathbb{N}} a_i T^i \in K[[T]]$ , then  $f \in K[T]$  iff for some  $m \in \mathbb{Z}_+$ , the **Hankel determinants**  $H_{k,m} = \det(a_{k-2+i+j})_{1 \leq i \leq j \leq m} = 0$  for all  $k$  large.

┘

*Proof:* Suppose  $f(T) = P(T)/Q(T)$  where  $P(T) = \sum_{i=0}^m b_i T^i, Q(T) = \sum_{i=0}^n c_i T^i$ , then  $P(T) = F(T)Q(T)$ . So for  $i$  large,

$$\sum_{j=0}^n a_{i-n+j} c_{n-j} = 0.$$

And these will imply that  $H_{k,m} = 0$  for  $k \geq \max(m - n + 1, 1)$ .

Conversely, if  $H_{k,m} = 0$  for a minimal  $m$ , denote  $S \in \mathbb{Z}_+$  s.t.  $H_{k,m} = 0$  for  $k \geq S$ . Then we claim that  $H_{k,m-1} \neq 0$  for any  $k \geq S$ : If  $H_{s,m-1} = 0$  for some  $s \geq S$ , then in  $(a_{k-2+i+j})_{1 \leq i \leq j \leq m}$ , the upper-left  $(m-1) \times (m-1)$ -matrix has determinant 0. So we can use row linear combination to make some row zero except possibly the last entry. Then we can argue in each case, the lower-left  $(m-1) \times (m-1)$  matrix has determinant 0, which is just  $H_{s+1,m-1}$ . So by induction,  $H_{k,m-1} = 0$  for any  $k \geq s$ , contradicting the minimality of  $m$ .

Thus for any  $s \geq S$ ,  $H_{s,m} = 0$  and  $H_{s,m-1} \neq 0$ . Thus the last row of  $(a_{k-2+i+j})_{1 \leq i \leq j \leq m}$  is a linear combination of the first  $m-1$  rows. This implies that if we take any  $u_1, \dots, u_m \in K$  s.t.

$$a_S u_{m-1} + a_{S+1} u_{m-1} + \dots + a_{S+m-1} u_0 = 0,$$

$$a_{S+1} u_{m-1} + a_{S+2} u_{m-1} + \dots + a_{S+m} u_0 = 0,$$

.....

$$a_{S+m-2} u_{m-1} + a_{S+m-1} u_{m-1} + \dots + a_{S+2m-3} u_0 = 0,$$

also satisfies

$$a_{S+m-1} u_{m-1} + a_{S+m} u_{m-1} + \dots + a_{S+2m-2} u_0 = 0.$$

Thus by induction, for any  $s \geq S$ ,

$$a_s u_{m-1} + a_{s+1} u_{m-1} + \dots + a_{s+m-1} u_0 = 0.$$

This implies that

$$\left( \sum_{i=0}^{m-1} u_i T^i \right) \left( \sum_{i \in \mathbb{N}} a_i T^i \right) \in K[T].$$

□

**Cor. (10.7.1.19).** If  $K \subset L \in \mathbf{Field}$ , then  $K[[T]] \cap L[T] = K[T]$ . ┘

**Prop. (10.7.1.20).** Let  $P, Q \in \mathbb{Q}[T]$  be prime to each other with constant coefficient 1. if  $P/Q = Z \in \mathbb{Z}[[T]]$ , then we have  $P, Q \in \mathbb{Z}[T]$ . ┘

*Proof:* Let  $\lambda$  be a root of  $Q(T)$ , we prove that  $|\lambda^{-1}|_p \leq 1$  for any  $p \in \mathbf{P}$ : If  $|\lambda|_p < 1$ , then  $Z(\lambda)$  converges in  $\mathbb{Q}_p$  because it has integral coefficients, and then

$$P(\lambda) = Q(\lambda)Z(\lambda) = 0.$$

This contradicts the fact that  $P, Q$  are coprime. So  $\lambda^{-1} \in \mathbb{Z}$ , and  $Q \in \mathbb{Z}[T]$ . Consequently,  $P(T) = Q(T)Z(T) \in \mathbb{Z}[[T]] \cap \mathbb{Q}[T] = \mathbb{Z}[T]$ . □

## 2-Dimensional Formal Power Series

**Def. (10.7.1.21)[Binomial Power Series].** The **binomial power series**  $B_X(Y) = (1+Y)^X$  is defined to be

$$B_X(Y) = \sum_{n \in \mathbb{N}} \frac{X(X-1) \dots (X-n+1)}{n!} Y^n \in \mathbb{Q}[[X, Y]].$$

And for any  $a \in \mathbb{Q}$ , define

$$B_a(X) = B(a, X) \in \mathbb{Q}[[X]].$$

┘



### Geometric Objects

**Def. (10.7.1.22) [Differential Operators].** A continuous  $R$ -linear mapping  $D : R[[\mathbf{X}]] \rightarrow R[[\mathbf{X}]]$  is called a **formal differential operator** of order  $N \geq -1$  iff

$$L_D : R[[\mathbf{X}, \underline{Z}]] \rightarrow R[[\mathbf{X}]] : \sum p_\alpha(\mathbf{X})Z^\alpha \rightarrow \sum p_\alpha(\mathbf{X})D(X^\alpha)$$

vanish on  $J^{N+1}$ , where  $J = (X_i - Z_i)$ .

Then  $D$  is an operator of order  $N$  if  $[f, D] = fD - Df$  has order  $N - 1$  for any  $f \in R[[X]]$ .  $\lrcorner$

*Proof:* Let  $D$  be a differential operator of order  $N$ , since  $f(\mathbf{X}) - f(\underline{Z}) \in J$ , for all  $g(\mathbf{X}, \underline{Z}) \in J^N$ , we have  $L_D((f(\mathbf{X}) - f(\underline{Z}))g(\mathbf{X}, \underline{Z})) = 0$ , which is equivalent to  $L_{D \circ f - f \circ D}(g) = 0$ , so  $D \circ f - f \circ D$  is an operator of order  $N - 1$ . Conversely, if  $D \circ f - f \circ D$  is an operator of degree  $N - 1$ , then  $L_D((f(\mathbf{X}) - f(\underline{Z}))g) = 0$  for all  $g \in J^N$ , then  $D$  is an operator of order  $N$ .  $\square$

**Cor. (10.7.1.23).** A differential operator  $D : R[[\mathbf{X}]] \rightarrow R[[\mathbf{X}]]$  of order 1 is a linear map that satisfies  $D(fg) = D(f)g + fD(g)$ , which is just a derivative on  $R[[\mathbf{X}]]$ . Equivalently,  $D = \sum_i u_i(\underline{Z}) \frac{\partial}{\partial Z_i}$ .  $\lrcorner$

**Prop. (10.7.1.24) [Graded Module of Differential Operators].** Let  $D_1, D_2$  be differential forms of order  $N_1, N_2$ , then  $D_1 \circ D_2$  is a differential form of order  $N_1 + N_2$ , and  $[D_1, D_2]$  is a differential form of order  $N_1 + N_2 - 1$ . The  $R$ -algebra of differential operators on  $R[[\mathbf{X}]]$  is denoted by  $\mathcal{DO}$ .

In particular, the graded module of differential operators

$$\text{gr } \mathcal{DO} = \oplus \mathcal{DO}_N / \mathcal{DO}_{N-1}$$

is a commutative graded ring.  $\lrcorner$

*Proof:*

$\square$

**Prop. (10.7.1.25) [Basis of Differential Operators].** There is a representation  $g(X + Y) = \sum_\alpha D_\alpha g(X)Y^\alpha$  for any  $g \in R[[\mathbf{X}, \mathbf{Y}]]$ , where  $D_\alpha$  is a differential operator of degree  $|\alpha|$ . And  $\{D_\alpha\}$  form a free  $K[[\mathbf{X}]]$ -basis for the module of differential operators (10.7.1.24).

In fact,  $D_\alpha$  is just mimicking  $\frac{\partial^\alpha}{\alpha!}$  in all fields.  $\lrcorner$

*Proof:*  $g(\underline{Z}) = \sum_\alpha D_\alpha g(\mathbf{X})(\underline{Z} - \mathbf{X})^\alpha$ , thus if  $D$  is a differential operator of order  $N$ , then

$$D(g)(\mathbf{X}) = L_D(g(\underline{Z})) = \sum_{|\alpha| \leq N} L((\underline{Z} - \mathbf{X})^\alpha) D_\alpha g(\mathbf{X}),$$

which means  $D = \sum_{|\alpha| \leq N} a_\alpha(\mathbf{X}) D_\alpha$ , where  $a_\alpha = L((\underline{Z} - \mathbf{X})^\alpha)$ .  $\square$

**Def. (10.7.1.26) [Tangent Space].** The tangent space of  $K[[\mathbf{X}]]$  is the  $K$ -module  $\text{Hom}_K((\mathbf{X})/(\mathbf{X})^2, K)$ .  $\lrcorner$

**Def. (10.7.1.27) [Formal Curves].** A **formal curve** in  $K[[\mathbf{X}]]$  is an  $n$ -tuple  $(\gamma_1(T), \dots, \gamma_n(T))$  of elements in  $K[[T]]$ . The **tangent space of formal curve** is the map  $\text{Hom}_K((\mathbf{X})/(\mathbf{X})^2, K)$  given by  $X_i \mapsto \gamma_i(T) \in (T)/(T^2) \cong K$ .  $\lrcorner$

**Prop. (10.7.1.28) [Integral Curves].** For any  $\lrcorner$

## 2 Formal Identities

**Prop. (10.7.2.1).** In  $\mathbb{Z}[[T]]$ ,

$$\frac{1}{(1-T)^2} = \sum_{n \geq 0} (n+1)z^n$$

┘

*Proof:*

$$(1-T)^2 \sum_{n \geq 0} (n+1)z^n = (1-T) \sum_{n \geq 0} z^n = 1.$$

□

**Prop. (10.7.2.2).** In  $\mathbb{Z}[[T]]$ ,

$$\sum_{n \geq 1} \frac{nT^n}{1-T^n} = \sum_{n \geq 1} \frac{T^n}{(1-T^n)^2}$$

┘

*Proof:*

□

**Prop. (10.7.2.3).** In  $R[[z]]$ , if

$$\sum_{r=0}^{\infty} A(r)z^r = \frac{1}{(1-\alpha_1 z)(1-\alpha_2 z)}, \quad \sum_{r=0}^{\infty} B(r)z^r = \frac{1}{(1-\beta_1 z)(1-\beta_2 z)},$$

then

$$\sum_{r=0}^{\infty} A(r)B(r)z^r = (1-\alpha_1\alpha_2\beta_1\beta_2 z^2) \prod_{i=1}^2 \prod_{j=1}^2 \frac{1}{(1-\alpha_i\beta_j z)}.$$

┘

*Proof:*  $A(n)B(n) = \sum_{0 \leq k, j \leq n} \alpha_i^k \alpha_2^{n-k} \beta_1^j \beta_2^{n-j}$ , and the coefficients in the  $z^n$  term of  $\prod_{i=1}^2 \prod_{j=1}^2 \frac{1}{(1-\alpha_i\beta_j z)}$  is

$$\sum_{0 \leq k, j \leq n} \#\{(r_1, r_2, r_3, r_4) | 0 \leq r_i \leq n, r_1 + r_2 = a, r_1 + r_3 = b, r_1 + r_2 + r_3 + r_4 = n\} \alpha_i^k \alpha_2^{n-k} \beta_1^j \beta_2^{n-j}.$$

Notice

$$\#\{(r_1, r_2, r_3, r_4) | 0 \leq r_i \leq n, r_1 + r_2 = a, r_1 + r_3 = b, r_1 + r_2 + r_3 + r_4 = n\} = \min(a, b) - (a + b - n).$$

Thus the effect of multiplying  $(1-\alpha_1\alpha_2\beta_1\beta_2 z^2)$  reduces the coefficients of  $\alpha_i^k \alpha_2^{n-k} \beta_1^j \beta_2^{n-j} z^n$  to 1. So the assertion follows. □

**Prop. (10.7.2.4) [Euler Identities].**

- In  $\mathbb{Z}[x][[q]]$ ,

$$\prod_{n \geq 0} (1 + xq^n) = \sum_{n=0}^{\infty} \frac{q^{n(n-1)/2} x^n}{(1-q) \dots (1-q^n)}.$$

- In  $\mathbb{Z}[[x, q]]$ ,

$$\prod_{n \geq 0} (1 - xq^n)^{-1} = \sum_{n=0}^{\infty} \frac{x^n}{(1-q) \dots (1-q^n)}.$$

┘

*Proof:* These follow from interesting combinatorial identities:

1: The coefficient of  $q^m x^n$  on both sides are the number of ways to add  $n$  different non-negative integers to  $m$ .

2: The coefficient of  $q^m x^n$  on both sides are the number of ways to add  $n$  non-negative integers to  $m$ .  $\square$

**Prop.(10.7.2.5) [Jacobi's Triple Product Formula, Andrew[And65]].** in  $\mathbb{Z}[x][[x^{-1}q]]$  (notice  $\mathbb{Z}[x][[x^{-1}q]] \subset \mathbb{Z}[x, x^{-1}][[q]]$ ),

$$\sum_{n=-\infty}^{\infty} x^n q^{n^2} = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + xq^{2n-1})(1 + x^{-1}q^{2n+1}).$$

┘

*Proof:* By substituting  $q = q^2, x = xq$  in Euler identities (10.7.2.4), in  $\mathbb{Z}[x][[x^{-1}q]]$ ,

$$\begin{aligned} \prod_{n \geq 0} (1 + xq^{2n+1}) &= \sum_{n=0}^{\infty} \frac{q^{n^2} x^n}{(1 - q^2)(1 - q^4) \dots (1 - q^{2n})} \\ &= \left( \prod_{j \geq 0} (1 - q^{2j+2}) \right)^{-1} \sum_{n=0}^{\infty} q^{n^2} x^n \prod_{j=0}^{\infty} (1 - q^{2n+2+2j}) \\ &= \left( \prod_{j \geq 0} (1 - q^{2j+2}) \right)^{-1} \sum_{n=-\infty}^{\infty} q^{n^2} x^n \prod_{j=0}^{\infty} (1 - q^{2n+2+2j}) \\ (q = q^2, x = q^{2n+2} \text{ in (10.7.2.4)}) &= \left( \prod_{j \geq 0} (1 - q^{2j+2}) \right)^{-1} \sum_{n=-\infty}^{\infty} q^{n^2} x^n \sum_{m=0}^{\infty} \frac{(-1)^m q^{m^2+m+2nm}}{(1 - q^2)(1 - q^4) \dots (1 - q^{2n})} \\ &= \left( \prod_{j \geq 0} (1 - q^{2j+2}) \right)^{-1} \sum_{m=0}^{\infty} \frac{(-x^{-1}q)^m}{(1 - q^2) \dots (1 - q^{2n})} \sum_{n=-\infty}^{\infty} q^{(m+n)^2} x^{m+n} \\ (q = q^2, x = -x^{-1}q \text{ in (10.7.2.4)}) &= \left( \prod_{j \geq 0} (1 - q^{2j+2}) \right)^{-1} \prod_{j=0}^{\infty} (1 + x^{-1}q^{2j+1})^{-1} \sum_{n=-\infty}^{\infty} q^{n^2} x^n. \end{aligned}$$

 $\square$ 

**Cor.(10.7.2.6) [Dedekind Eta Function].** Substitute  $q = q^{3/2}$  and  $x = q^{-1/2}$ , we get:

$$\sum_{n=-\infty}^{\infty} (-1)^n q^{(3n^2+n)/2} = \prod_{n=1}^{\infty} (1 - q^{3n})(1 - q^{3n-1})(1 - q^{3n-2}) = \prod_{n=1}^{\infty} (1 - q^n) \in \mathbb{Z}[[q]].$$

By completing the square,

$$\sum_{n=-\infty}^{\infty} (-1)^n q^{(6n+1)^2/24} = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n) \in \mathbb{Z}[[q^{1/24}]].$$

the last term is known as the **Dedekind eta function**  $\eta(q) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n) \in \mathbb{Z}[[q^{1/24}]]$ .  $\square$

### 3 Formal Group Laws

In this subsection, the local structure of an algebraic group scheme at the origin is studied.

**Def. (10.7.3.1) [Formal Group Laws].** A **formal group law**  $\mathcal{G}$  of dimension  $n$  over  $R$  is a continuous local map  $\mu_{\mathcal{G}} : R[[\mathbf{X}]] \rightarrow R[[\mathbf{X}, \mathbf{Y}]]$  given by an  $n$ -tuple of power series  $G = (G_1, \dots, G_n)$  in  $R[[\mathbf{X}, \mathbf{Y}]]$  that

$$G(X, Y) \equiv X + Y \pmod{(\mathbf{X}, \mathbf{Y})^2}, \quad G(G(\mathbf{X}, \mathbf{Y}), Z) = G(X, G(\mathbf{Y}, Z)).$$

A **formal  $R$ -module** is a formal group law  $\mathcal{G}$  over  $R$  together with a ring homomorphism  $R \rightarrow \text{End}_R(G)$  that  $[a](X) = aX + \dots$

A morphism of formal groups laws  $\mathcal{G} \rightarrow \mathcal{H}$  is a continuous local map  $\varphi^* : R[[\mathbf{X}']] \rightarrow R[[\mathbf{X}]]$  given by an  $n'$ -tuple of power series  $\varphi = (\varphi_1, \dots, \varphi_{n'})$  in  $R[[\mathbf{X}]]$  that satisfies

$$\mu_{\mathcal{G}} \circ \varphi^* = (\varphi^* \otimes \varphi^*) \circ \mu_{\mathcal{H}}.$$

Or equivalently,  $\varphi(G(\mathbf{X}, \mathbf{Y})) = H(\varphi(\mathbf{X}), \varphi(\mathbf{Y}))$ . ┘

**Prop. (10.7.3.2).** For a formal group law  $G$  over  $R$  (10.7.3.1),

- $G(0, \mathbf{Y}) = \mathbf{Y}, G(\mathbf{X}, 0) = \mathbf{X}$ .
  - There exists a unique inverse  $\text{inv}(\mathbf{X})$  that  $G(\mathbf{X}, \text{inv}(\mathbf{X})) = 0$ . And this  $\text{inv}(\mathbf{X})$  satisfies  $G(\text{inv}(\mathbf{X}), \mathbf{X}) = 0$ , and  $\text{inv}^2 = \text{id}$ .
- ┘

*Proof:* ? □

**Remark (10.7.3.3).** WARNING: A formal group law of dimension  $n$  is not equivalent to a group scheme structure on  $\text{Spec } \mathbb{Z}[[X_1, \dots, X_n]]$  (10.1.1.1), as here we are taking a completion. But see (10.7.4.4). ┘

**Def. (10.7.3.4) [Multiplication Map].** Let  $G$  be a commutative formal group law over  $R$  and  $m \in \mathbb{Z}$ , then we can define a group homomorphism  $[m] : G \rightarrow G$  inductively as follows:

$$[0](\mathbf{X}) = 0, \quad [m+1](\mathbf{X}) = G([m](\mathbf{X}), \mathbf{X}), \quad [m-1](\mathbf{X}) = G([m](\mathbf{X}), i(\mathbf{X})).$$

Then it can be verified that this is well-defined and  $[m]$  is a group homomorphism. ┘

**Prop. (10.7.3.5) [Multiplications as Automorphisms].** Let  $\mathcal{G}$  be a commutative formal group law over  $R$  and  $m \in \mathbb{Z}$ , then  $[m](\mathbf{X}) = m\mathbf{X} + o(\mathbf{X})$ . In particular, if  $m$  is invertible in  $R$ , then  $[m]$  is an automorphism of  $G$ , by (10.7.1.3). ┘

**Example (10.7.3.6) [Additive and Multiplicative Groups].**  $\mathbb{G}_a$  is the one-dimensional formal group with  $\mathbb{G}_a(\mathbf{X}, \mathbf{Y}) = X + Y$ ,  $\mathbb{G}_m$  is the one-dimensional formal group with  $\mathbb{G}_m(\mathbf{X}, \mathbf{Y}) = X + Y + XY$ .

Over a  $\mathbb{Q}$ , there is an isomorphism between  $\mathbb{G}_a$  and  $\mathbb{G}_m$  giving by  $X \rightarrow \exp(X) - 1$ . ┘

**Def. (10.7.3.7) [Representing Functors of Formal Modules].** For  $\mathcal{O} \in \mathcal{CRing}$ , the functor

$$\mathcal{CRing}_{\mathcal{O}} \rightarrow \text{Set} : B \mapsto \{\text{Formal Groups over } B\}$$

is representable by an  $\mathcal{O}$ -algebra  $\Lambda_{\mathcal{O}}$ . ┘

**Def. (10.7.3.8) [Invariant Differential Forms].** An **invariant differential form** on a formal group law  $G$  over  $R$  is an element  $\omega = \sum a_i(\mathbf{X})dX_i \in \Omega_{R[[X]]/R}^1$  that satisfies  $(\mathbf{Y} \mapsto G(\mathbf{X}, \mathbf{Y}))_*\omega = \omega$ , i.e.

$$\sum_j a_j(\mathbf{Y})dY_j = \sum_i a_i(G(\mathbf{X}, \mathbf{Y})) \frac{\partial G_i}{\partial Y_j}(\mathbf{X}, \mathbf{Y})dY_j.$$

If  $G$  is commutative, then this is equivalent to  $\mu_*\omega = i_{1*}\omega + i_{2*}\omega$ . ┘

**Prop. (10.7.3.9).** The mapping  $\omega \mapsto (u_1(0), \dots, u_n(0))$  is an isomorphism of the  $R$ -module of invariant differential forms and  $R^n$ . ┘

*Proof:* Cf. [Zin84]P14. □

**Prop. (10.7.3.10) [Pullback of Invariant Differential Forms].** Let  $\varphi : G \rightarrow H$  be a homomorphism of formal group laws, then for any invariant differential form  $\omega = \sum a_i(\mathbf{X})dX_i$  on  $H$ ,  $\varphi^*\omega$  is a differential form on  $G$ , given by expanding  $\sum a_i(\varphi(\mathbf{X}))d(\varphi_i(\mathbf{X}))$ . ┘

*Proof:*

$$\begin{aligned} (\mathbf{Y} \mapsto G(\mathbf{X}, \mathbf{Y}))^*\varphi^*\omega &= \sum a_i(\varphi(G(\mathbf{X}, \mathbf{Y})))d(\varphi(G(\mathbf{X}, \mathbf{Y}))) \\ &= \sum a_i(H(\varphi(\mathbf{X}), \varphi(\mathbf{Y})))d(H(\varphi(\mathbf{X}), \varphi(\mathbf{Y}))) \\ &= \sum_i a_i(\varphi(\mathbf{Y}))d(\varphi(\mathbf{Y})) \\ &= \varphi^*\omega \end{aligned}$$

□

**Def. (10.7.3.11) [Invariant Differential Operators].** An **invariant differential operator** on a formal group  $G$  of dimension  $n$  is a differential operator  $D$  s.t.

$$\mu \circ D = (1 \otimes D) \circ \mu.$$

Equivalently, if  $D = \sum_i u_i(\underline{Z}) \frac{\partial}{\partial Z_i}$ , it satisfies

$$\sum_j u_j(\mathbf{Y}) \frac{\partial G_i}{\partial Y_j} = u_i(G(\mathbf{X}, \mathbf{Y})).$$

The space of invariant operators is stable under composition. The  $R$ -algebra of invariant differential operators on  $G$  is denoted by  $\mathcal{DO}_G$ . ┘

**Prop. (10.7.3.12).** The space of invariant differential operators on  $G$  of order  $\leq N$  are in bijection with the space of  $K$ -linear maps  $l : K[[\mathbf{X}]] \rightarrow K$  s.t.  $l((\mathbf{X})^{N+1}) = 0$  via

$$Df(\mathbf{X}) = (1 \otimes l)f(G(\mathbf{X}, \mathbf{Y})).$$

┘

*Proof:* Cf. [Zin84]P22. □

**Cor.(10.7.3.13) [Basis of Invariant Differential Operators].** For any  $f \in K[[\mathbf{X}]]$ , write  $f(G(\mathbf{X}, \mathbf{Y})) = \sum_{\alpha} (H_{\alpha}f)(\mathbf{X})Y^{\alpha}$ .

Then  $H_{\alpha}$  are differential operators of order  $|\alpha|$ , and the space of invariant differential operators of order  $N$  form a free  $R$ -module with basis  $H_{\alpha}$ . In particular, the space of invariant derivatives is isomorphic to  $R^n$ .  $\lrcorner$

*Proof:* By (10.7.3.12),

$$Df(\mathbf{X}) = (1 \otimes l)f(G(\mathbf{X}, \mathbf{Y})) = (1 \otimes l)\left(\sum_{\alpha} (H_{\alpha}f)(\mathbf{X})Y^{\alpha}\right) = \sum_{\alpha} l(\mathbf{Y}^{\alpha})(H_{\alpha}f)(\mathbf{X}).$$

□

**Cor.(10.7.3.14) [Composition of Invariant Differential Operators].** Let  $D_1, D_2 \in \mathcal{DO}_G$ , then

$$(D_1 \circ D_2)f(\underline{Z}) = (1 \otimes l_1 \otimes l_2)f(G(\underline{Z}, G(\mathbf{X}, \mathbf{Y}))),$$

in particular,

$$l_{D_1 \circ D_2}(f) = (l_1 \otimes l_2)f(G(\mathbf{X}, \mathbf{Y})).$$

□

*Proof:*

$$(D_1 \circ D_2)f(\underline{Z}) = D_1(1 \otimes l_1)f(G(\underline{Z}, \mathbf{Y})) = (1 \otimes l_2 \otimes 1)(1 \otimes 1 \otimes l_1)f(G(G(\underline{Z}, \mathbf{X}), \mathbf{Y})) = (1 \otimes l_1 \otimes l_2)f(G(\underline{Z}, G(\mathbf{X}, \mathbf{Y}))).$$

□

**Prop.(10.7.3.15) [Pushforward of Invariant Differential Operators].** A homomorphism  $\varphi$  of formal groups  $G \rightarrow H$  induces a map from the linear maps  $l : K[[\mathbf{X}]] \rightarrow K$  to linear maps  $l \circ \varphi^* : K[[\mathbf{X}']] \rightarrow K$ , which induces a  $K$ -homomorphism of algebras  $\varphi_* : \mathcal{DO}_G \rightarrow \mathcal{DO}_H$ .  $\lrcorner$

*Proof:* To show preserves algebra structure, notice for any  $f \in R[[\mathbf{X}']]$ , by (10.7.3.14),

$$\begin{aligned} \varphi_*(D) \circ \varphi_*(D')f(0) &= (l \circ \varphi^*) \otimes (l' \circ \varphi^*)f(H(\mathbf{X}', \mathbf{Y}')) \\ &= (l \otimes l')f(H(\varphi(\mathbf{X}), \varphi(\mathbf{Y}))) \\ &= (l \otimes l')f(\varphi(G(\mathbf{X}, \mathbf{Y}))) \\ &= [(l \otimes l') \circ \varphi^*]f(G(\mathbf{X}, \mathbf{Y})) \\ &= \varphi_*(D \circ D')f(0). \end{aligned}$$

Thus  $\varphi_*(D) \circ \varphi_*(D') = \varphi_*(D \circ D')$ , by (10.7.3.12).  $\lrcorner$

□

**Prop.(10.7.3.16) [Q-Theorem].** Any commutative connected formal group over a  $\mathbb{Q}$ -algebra  $R$  is a direct sum of  $\widehat{\mathbf{G}}_a$ .  $\lrcorner$

□

*Proof:* Cf. [Zin84] P19.  $\lrcorner$

□

**Thm.(10.7.3.17) [Lazard 1955].** If  $S \rightarrow R \in \mathcal{CRing}$  is surjective, then any formal group law over  $R$  comes from a formal group law over  $S$ .  $\lrcorner$

□

*Proof:* Cf. [Zin84] ?  $\lrcorner$

□

### 1-dimensional Formal Group Laws

**Def. (10.7.3.18) [Normalized Invariant Differential Form].** For a 1-dimensional formal group  $G$  over  $R[[X]]$ , the module of invariant differentials is isomorphic to  $R$  (10.7.3.11). An invariant differential form  $\omega = P(T)dT$  is called **normalized** if  $P(0) = 1$ .

Then the unique normalized invariant differential form on  $G$  is given by  $\omega_G = G_X(0, T)^{-1}dT$ .  $\lrcorner$

*Proof:* We need to check  $G_X(0, G(T, S))^{-1}G_X(T, S) = G_X(0, T)^{-1}$ , and this is just  $G(U, G(T, S)) = G(G(U, T), S)$  differentiated at  $U$  and let  $U = 0$ .  $\square$

**Prop. (10.7.3.19).** For a morphism  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  of 1-dimensional formal groups laws over  $R$ ,  $\varphi^*\omega_{\mathcal{G}} = \varphi'(0)\omega_{\mathcal{F}}$ .  $\lrcorner$

*Proof:* By (10.7.3.10),  $\varphi^*\omega_{\mathcal{G}}$  is an invariant differential form, and compare their constant coefficients.  $\square$

**Cor. (10.7.3.20).** Let  $p$  be a prime, then for any formal group law of dimension 1,

$$[p](T) = pf(T) + g(T^p)$$

for some  $f(0) = g(0) = 0$ .  $\lrcorner$

*Proof:* It suffices to show that  $[p]'(T) \in pR[[T]]$ . But by (10.7.3.19) and (10.7.3.5),

$$p\omega(T) = [p]^*\omega(T) = G_X(0, [p](T))^{-1}[p]'(T)dT$$

so this is true.  $\square$

**Def. (10.7.3.21) [Formal Logarithm].** When  $R$  is torsion-free, the **formal logarithm**  $\log_{\mathcal{F}}$  for a 1-dimensional formal group is the integration of invariant differential

$$\int_0^T \omega_{\mathcal{F}} = T + \frac{c_1}{2}T^2 + \cdots \in (R \otimes \mathbb{Q})[[T]].$$

Then the **formal power exponential** is the unique power series  $\exp_{\mathcal{F}}$  that is the inverse of  $\log_{\mathcal{F}}$ . It exists uniquely by (10.7.1.3), and by (10.7.1.4), it is of the form

$$\exp_{\mathcal{F}} = \sum_{n \geq 1} \frac{b_n}{n!} T^n \in (R \otimes \mathbb{Q})[[T]], b_n \in R.$$

$\lrcorner$

**Prop. (10.7.3.22).** For  $R$  is torsion-free and an 1 dimensional formal group  $\mathcal{F}$  over  $R$ ,

$$\log_{\mathcal{F}} : \mathcal{F} \rightarrow \mathbb{G}_a$$

is an isomorphism of formal groups laws over  $R \otimes_{\mathbb{Z}} \mathbb{Q}$ .

And if  $\mathcal{F}$  is a formal  $R$ -module, then it is an isomorphism of  $R$ -modules, because from (10.7.3.19) that  $\omega_{\mathcal{F}} \circ [a] = a\omega_{\mathcal{F}}$ , thus  $\log_{\mathcal{F}} \circ [a] = a \cdot \log_{\mathcal{F}}$ .  $\lrcorner$

*Proof:* From  $\omega_{\mathcal{F}}(F(T, S)) = \omega_{\mathcal{F}}(T)$ , we get that  $\log_{\mathcal{F}}(F(T, S)) = \log_{\mathcal{F}}(S) + \log_{\mathcal{F}}(T)$ . So it is a homomorphism. Now the inverse  $\exp_{\mathcal{F}}$  is already given, so it is an isomorphism.  $\square$

**Cor. (10.7.3.23) [1-Dimensional Formal Group Law is Commutative].** Any 1-dimensional formal group over a ring  $R$  that has no torsion nilpotents is commutative.  $\lrcorner$

*Proof:* We only prove for  $R$  torsion free  $\color{red}{?}$ .  $F(T, S) = \exp_{\mathcal{F}}(\log_{\mathcal{F}}(T) + \log_{\mathcal{F}}(S))$ .  $\square$

### Lubin-Tate Formal Group Law

**Notation(10.7.3.24).**

- Let  $(K, \mathcal{O}_K, \mathfrak{m}_K, \varpi_K, k) \in p\text{-LField}$ .

┘

**Def.(10.7.3.25)[Lubin-Tate Formal Group Law].** A **Lubin-Tate power series** for  $\varpi$  is a power series  $\varphi(X) \in \mathcal{O}_K[[X]]$  s.t.

$$\varphi(X) \equiv \varpi X \pmod{X^2}, \quad \varphi(X) \equiv X^q \pmod{\mathfrak{p}_v}.$$

A **Lubin-Tate module**  $G$  over  $\mathcal{O}_K$  is a formal  $\mathcal{O}_K$ -module(10.7.3.1) s.t.  $[\varpi_K](X)$  is a Lubin-Tate power series.

┘

**Prop.(10.7.3.26).** Given a  $p$ -adic number field  $K$  with residue field  $\mathbb{F}_q$ , we consider the set  $\xi_\pi$  of all Lubin-Tate power series for  $\pi$ .

If  $f, g \in \xi_\pi$  and  $L(\mathbf{X}) = \sum a_i X_i$  be a linear form, then there exists a unique power series  $F(\mathbf{X})$  that  $F(X) \equiv L(X) \pmod{(\mathbf{X})^2}$  and  $f(F(\mathbf{X})) = F(g(\mathbf{X}))$ .

┘

*Proof:* Choose  $F$  consecutively, if  $F_{r+1} = F_r + \Delta_r$ , then must

$$\Delta \equiv \frac{f(F_r(X)) - F_r(g(X))}{\pi^{r+1} - \pi} \pmod{\text{degree}(r+2)}.$$

This has coefficient in  $\mathcal{O}$  because  $f \equiv g \equiv Z^q \pmod{\pi}$ .

□

**Cor.(10.7.3.27).** If we let  $f = g, L = X + Y$  to get  $F_f$  and  $f, g, L = aX$  to get  $a_{f,g}$ , then

- $F_f(X, Y) = F_f(Y, X)$ .
- $F_f(F_f(X, Y), Z) = F_f(X, F_f(Y, Z))$ .
- $a_{f,g}(F_g(X, Y)) = F_f(a_{f,g}(X), a_{f,g}(Y))$ .
- $a_f b_f(Z) = (ab)_f(Z)$ .
- $(a + b)_f(Z) = F_f(a_f(Z), b_f(Z))$ .
- $\pi_f(Z) = f(Z)$ .

all follow from the unicity of the last proposition.

┘

**Cor.(10.7.3.28)[Existence of Lubin-Tate Modules].** We get a commutative formal  $\mathcal{O}$ -module  $F_f$  for every  $f$ . And this group can act on  $\mathfrak{p}_L$  for an alg.ext  $L/K$ . The set of zeros  $\Lambda_{f,n}$  of  $f^n$  in  $L$ , as the elements annihilated by  $\pi^n$ , is a submodule of  $\mathfrak{p}_L^{(f)}$ .

And  $u_{g,f}$  for any unit  $u \in \mathcal{O}$  defines an isomorphism between  $F_f$  and  $F_g$ , thus this formal group only depends on  $\pi$ , called  $F_\pi$ . Hence  $L_{f,n} = K(\Lambda_{f,n})$  only depends on  $\pi$ , with Galois group  $G_{\pi,n}$ .

┘

**Prop.(10.7.3.29)[Different Uniformizers].** For two uniformizers  $\varpi, \varpi'$ , it is proven that  $F_\varpi$  and  $F_{\varpi'}$  are isomorphic, but isomorphic over  $\mathcal{O}_{\check{K}}$ .

Thus  $L_{\varpi,n}$  and  $L_{\varpi',n}$  may not be isomorphic, but  $K^{\text{ur}}.L_{\pi,n} = K^{\text{ur}}.L_{\pi',n}$  since  $\check{K}.L_{\pi,n} = \check{K}.L_{\varpi',n}$  and both of them is the algebraic closure of  $K$  in it.

┘

*Proof:* Cf.[Neukirch CFT P105].?

□



**Lemma (10.7.3.30).** The Newton polygon of  $[\varpi_K^n]/\varpi_K^n$  has vertices

$$(1, 0), (q, -1/e_K), (q^2, -2/e_K), \dots$$

┘

*Proof:* Notice  $[\varpi_K^n]$  has no infinite edge of negative slope because all its coefficient are in  $\mathcal{O}_K$ . Now look at its roots, it has a root 0, and  $q - 1$  roots of valuation  $v_p(\varpi_K)/(q - 1)$ ,  $q(q - 1)$  roots of valuation  $v_p(\varpi_K)/q(q - 1)$ , and so on. So by factor out these roots,  $[\varpi_K^n]/\varpi_K^n$  is left with a power series whose Newton polygon is a single line, which shows the desired result.  $\square$

**Prop. (10.7.3.31).** The formal logarithm (10.7.3.21) of the Lubin-Tate formal group  $F_\pi$  satisfies:

$$\log_{\mathcal{F}_\varpi}(T) = \lim_{n \rightarrow \infty} [\varpi_{\mathcal{F}}^n]/\varpi_{\mathcal{F}}^n.$$

┘

*Proof:* By (10.7.3.22) we have

$$\log_{\mathcal{F}}(T) = \log_{\mathcal{F}}([\varpi_{\mathcal{F}}^n]/\varpi_{\mathcal{F}}^n) = ([\varpi_K^n] + a_2/2[\varpi_K^n]^2 + \dots)/\varpi_K^n$$

and for any degree  $n$ , the coefficient of  $[\varpi_K^{2n}]/\varpi_K^{2n}$  is bounded below by a  $c(n)$ , so  $[\varpi_K^{2n}]/\varpi_K^{2n}$  converges to 0, thus the result.  $\square$

**Cor. (10.7.3.32).** The Newton polygon of  $\log_{\mathcal{F}}(T)$  has vertices  $(1, 0), (q, -1/e_K), (q^2, -2/e_K), \dots$  ┘

The discussion is continued at 2.

### Canonical and Quasi-Canonical Liftings

**Def. (10.7.3.33).** Let  $(K, \mathcal{O}_K, \varpi, \kappa) \in p\text{-Field}$ . ┘

**Prop. (10.7.3.34) [Canonical Liftings].** Let  $L/K$  be a quadratic extension with a uniformizer  $t$  and residue field  $k_L$ , and  $\check{L}$  the maximal unramified extension of  $L$  with residue field  $k$ . Let  $G$  be the unique formal  $\mathcal{O}_K$ -module of height 2 over  $k$  (10.7.4.19), and  $R = \text{End}_k(G)$  is isomorphic to the maximal order in the quaternion division algebra  $B$  over  $F$ .

Then there exist inclusions  $\alpha : \mathcal{O}_L \hookrightarrow B$  whose action on  $k = \text{Lie}_k(G)$  is compatible with the reduction map ?. Then this makes  $G$  an  $\mathcal{O}_L$ -module of height 1.

Then  $G$  can be lifted uniquely to a formal  $\mathcal{O}_L$ -module  $\underline{G}$  over  $\mathcal{O}_{\check{L}}$ , called the **canonical lifting** of the pair  $(G, \alpha)$ . ┘

*Proof:* The Lubin-Tate formal group gives a formal  $\mathcal{O}_L$ -module over  $\mathcal{O}_L$  with  $[t](X) \equiv tX + X^{\#k_L}$ , so its of height 1 modulo  $t$ , so isomorphic to  $G$ . This together with the fact that height 1 formal groups has trivial deformations spaces (Cf. [Lubin-Tate, Formal moduli for one-parameter formal Lie groups]1. Prop4.2 ?) implies that there are canonical lifting to  $\mathcal{O}_{\check{L}}$ .  $\square$

**Prop. (10.7.3.35) [Gross].** Situation as in (10.7.3.34), for  $n \in \mathbb{Z}_+$ , denote  $R_n = \text{End}_{\mathcal{O}_{\check{L}}/(t^n)}(\underline{G})$ , then there are inclusions

$$\dots \subset R_n \subset \dots \subset R_1 \subset R_0 = R.$$

And by completeness, we have  $\cap_n R_n = \mathcal{O}_L$ .

Then in fact  $R_n = \mathcal{O}_L + t^n R$ . ┘

*Proof:* Cf. [Gross, Canonical and QuasiCanonical Liftings]P322. ┘

## 4 Formal Groups

**Notation(10.7.4.1).**

- Let  $\text{Nil}_R$  be the exact category of nilpotent commutative (possibly non-unital)  $R$ -algebras.  $\lrcorner$

**Def.(10.7.4.2) [Formal Groups].** A (commutative) **formal group** is an exact functors  $\text{Nil}_R \rightarrow \mathcal{A}b$  that commutes with infinite direct sums.  $\lrcorner$

**Remark(10.7.4.3).** Don't confuse formal groups with formal schemes, they are totally different notions.  $\lrcorner$

**Prop.(10.7.4.4) [Formal Group Law as Formal Groups].** For  $n \in \mathbb{Z}_+$ , a commutative formal group law of dimension  $n$  (10.7.3.1) is equivalent to an Abelian functor on  $\text{Nil}_R$  whose underlying set-theoretic functor if  $N \mapsto N^{\oplus n}$ , in particular, it is a formal group over  $R$  (10.7.4.2).  $\lrcorner$

*Proof:* Each commutative formal group law  $G$  of dimension  $n$  defines a functor

$$\text{Nil}_R \rightarrow \mathcal{A}b : N \mapsto (N^n, G)$$

Conversely, if  $\tilde{G} : \text{Nil}_R \rightarrow \mathcal{A}b$  is a functor s.t. the underlying set-theoretic map is  $N \mapsto N^n$ , then  $\tilde{G}((\mathbf{X}, \mathbf{Y})/(\mathbf{X}, \mathbf{Y})^k) \cong [(\mathbf{X}, \mathbf{Y})/(\mathbf{X}, \mathbf{Y})^k]^n$  as sets. Suppose

$$(X_1, \dots, X_n) + (Y_1, \dots, Y_n) = (G_1^k(\mathbf{X}, \mathbf{Y}), \dots, G_n^k(\mathbf{X}, \mathbf{Y}))$$

in  $\tilde{G}((\mathbf{X}, \mathbf{Y})/(\mathbf{X}, \mathbf{Y})^k)$ , then  $G_i^{k+1} \equiv G_i^k \pmod{(\mathbf{X}, \mathbf{Y})^k}$ , and their limit defines a commutative formal group law of dimension  $n$ . Because for any nilpotent algebra  $N$  and  $(a_1, \dots, a_n), (b_1, \dots, b_n) \in N^n$ , there is a surjective map  $(\mathbf{X}, \mathbf{Y})/(\mathbf{X}, \mathbf{Y})^k \rightarrow N$  for some  $k$  that maps  $X_i$  to  $a_i$ ,  $Y_i$  to  $b_i$ , thus by functoriality,  $\tilde{G}(N) \cong G(N)$ .  $\square$

**Prop.(10.7.4.5) [Formal Groups are Left Exact].** If  $G$  is a formal group over  $R$ , then  $G(0) = 0$ , and  $G$  commutes with fiber products.  $\lrcorner$

*Proof:* ?  $\square$

**Prop.(10.7.4.6) [Formal Completion of Group Schemes].** Let  $G \in \text{GrpSch}/R$  be commutative and smooth, then

$$\hat{G} : \text{Nil}_R \rightarrow \mathcal{A}b : N \mapsto \ker(G(R \oplus N) \rightarrow G(R))$$

is a formal group over  $R$ , where  $R \oplus N$  is endowed with the commutative  $R$ -ring structure  $(r_1, n_1)(r_2, n_2) = (r_1 r_2, r_1 n_2 + r_2 n_1 + n_1 n_2)$  and  $R \oplus N \rightarrow R$  is the map  $\text{id} \oplus 0$ .  $\lrcorner$

*Proof:* Cf.  $[p]$ -divisible groups, lecture, Zink]Prop1.5. ?  $\square$

**Example(10.7.4.7).** For  $S \in \text{Nil}_R$ , the functor  $\text{Res}_{S/R}(G_m)$  is a commutative formal group.  $\lrcorner$

**Prop.(10.7.4.8) [Formal Group Laws over Complete Local Rings].** Let  $G$  be a commutative formal group law defined over a complete local ring  $(R, \mathfrak{m})$ , then

- For each  $n$ ,  $G(\mathfrak{m}^n)/G(\mathfrak{m}^{n+1}) \cong \mathfrak{m}^n/\mathfrak{m}^{n+1}$  as groups.
- Let  $p$  be the characteristic of the residue field  $k$  ( $p$  may be 0), then  $\mathcal{F}(\mathfrak{m})_{\text{tor}} = \mathcal{F}(\mathfrak{m})[p^\infty]$ .  $\lrcorner$

*Proof:* 1: This is because  $F(x, y) \equiv x + y \pmod{\mathfrak{m}^n}$ .

2: This is because  $[m]$  is an automorphism by (10.7.3.5) as  $m$  is invertible in  $R$ .  $\square$

**Prop. (10.7.4.9) [Tangent Functors].** For a formal group  $G$  over  $R$  and  $M \in \text{Mod}_R$ , we can regard  $M$  as an element of  $\text{Nil}_R$  by requiring that  $M^2 = 0$ . Then there is a natural  $R$ -module structure on  $G(M)$ , and we get a functor

$$t_G : \text{Mod}_R \rightarrow \text{Mod}_R.$$

┘

*Proof:* Cf. [ $p$ -Divisible Groups, lecture, Zink]Prop1.9.  $\square$

**Prop. (10.7.4.10).** Suppose  $G$  is a formal group over  $R$ . For  $M \in \text{Mod}_R$  and  $m \in M$ , define  $c_m : R \rightarrow M : a \mapsto am$ , then the map

$$t_G(R) \otimes_R M \rightarrow t_H(M) : \xi \otimes m \mapsto t_H(c_m)(\xi)$$

is a map of  $R$ -modules, and it is an isomorphism. In particular,  $t_H(R)$  is flat over  $R$ .  $\square$

*Proof:* Cf. [ $p$ -Divisible Groups, lecture, Zink]Prop1.11.  $\square$

**Prop. (10.7.4.11) [Characterizing Formal Group Laws].** If  $G$  is a formal group over  $R$  s.t.  $t_G(R) \cong R^{\oplus n}$  for some  $n \in \mathbb{Z}_+$ , then  $G$  comes from a formal group law of dimension  $n$  over  $R$  as in (10.7.4.4).  $\square$

*Proof:* Cf. [ $p$ -divisible groups, lecture, Zink]Prop1.14.  $\square$

**Prop. (10.7.4.12) [Base Change of Formal Groups].** If  $f : R \rightarrow S \in \mathcal{C}\text{Ring}$  and  $G$  is a formal group over  $R$ , then the forgetful functor  $\text{Nil}_R \rightarrow \text{Nil}_S$  makes  $G$  a formal group over  $S$ , denoted by  $f_*G$ .  $\square$

**Thm. (10.7.4.13) [Lazard].**  $\square$

### Grothendieck-Messing Exponentials

**Def. (10.7.4.14) [Lie Algebras of Formal Groups].** For a formal group  $G$  over  $R$ , define

$$\text{Lie}(G) : \text{Nil}_R \rightarrow \text{Set} : N \mapsto G(N'),$$

where  $N' \in \text{Nil}_R$  is the module  $N$  with a different algebra structure  $N^2 = 0$ . Then  $\text{Lie}(G)$  naturally prolongs to a map  $\text{Lie}(G) : \text{Nil}_R \rightarrow \mathcal{A}\text{b}$  because  $G(N' \times_R N') = G(N') \times G(N')$  by (10.7.4.5).  $\square$

**Prop. (10.7.4.15) [Grothendieck-Messing Exponentials].** If  $N \in \text{Nil}_R$  has a pd-structure  $\{\gamma_n : N \rightarrow N\}$ , and  $G$  is a strictly pro-representable commutative formal group over  $R$  (10.7.4.2), then there is an isomorphism

$$\exp_N : \text{Lie}(G)(N) \cong G(N) \in \mathcal{A}\text{b} \text{ (10.7.4.14)}$$

functorial in  $(N, \{\gamma_n\}_{n \in \mathbb{Z}_+})$  and in  $G$ , called the **Grothendieck-Messing exponential map**.

In particular, if  $\mathbb{Q} \subset R$ , then for any strictly pro-representable commutative formal group  $G$  over  $R$ , there is an isomorphism

$$\exp_G : \text{Lie}(G) \cong G$$

functorial in  $G$ .  $\square$

*Proof:* Cf. [ $p$ -divisible Groups, lecture, Zink]Prop1.38, 1.47.  $\square$

### 1-Dimensional Formal $\mathcal{O}$ -Modules over Fields of Finite Characteristic

**Def. (10.7.4.16).** Let  $(K, \mathcal{O}_K, \varpi, \kappa) \in p\text{-Field}$ ,  $A \in \mathcal{CRing}/\mathcal{O}_K$  and  $G$  a formal  $\mathcal{O}_K$ -module over  $A$ , then  $\quad \lrcorner$

**Prop. (10.7.4.17) [Heights of Homomorphisms].** Let  $E$  be a field over  $\kappa$ , then any homomorphism  $\varphi$  between formal  $\mathcal{O}$ -modules over  $E$  is of the form  $\varphi(x) = \psi(x^{p^k})$  where  $\psi^\lambda \neq 0$ .

Then from the fact  $\varphi$  commutes with  $\mathcal{O}$ -actions, we can deduce that  $\log_p \# \kappa | k$  ?. The number  $k / \log_p \# \kappa$  is called the **height of  $\varphi$** . The height of a formal  $\mathcal{O}$ -module is the height of the multiplication by  $\varpi$ .  $\quad \lrcorner$

**Prop. (10.7.4.18).**

- There exists  $\mathcal{O}$ -modules of arbitrary heights.
- Non-zero homomorphisms exists only between  $\mathcal{O}$ -modules of the same height.
- Any formal module of height  $h$  is isomorphic to  $G_a$  modulo  $q^h$ .  $\quad \lrcorner$

*Proof:* Cf. [Elliptic Modules, Drinfeld] P566. ?

2. This is because  $\text{ht}(g \circ f) = \text{ht}(g) + \text{ht}(f)$ .  $\quad \square$

**Prop. (10.7.4.19) [ $\mathcal{O}$ -Modules over Separably Closed Fields].** If  $E/\kappa$  is a field extension and  $E^s = E$ , then

- For  $h \in \mathbb{Z}_+$ , all  $\mathcal{O}$ -modules over  $E$  of height  $h$  are isomorphic.
- The ring of endomorphisms of such an  $\mathcal{O}$ -module is isomorphic to the maximal order of a central division algebra with invariant  $1/h$  over  $K$ .  $\quad \lrcorner$

*Proof:* Cf. [Elliptic Modules, Drinfeld] P566. ?  $\quad \square$

### 1-Dimensional Commutative Formal Groups over DVR

**Prop. (10.7.4.20).** Let  $(K, \mathfrak{m}_K, \kappa)$  be a CDVR with mixed characteristic  $(0, p)$ ,  $\mathcal{F}$  a formal group law over  $R$ . Then  $\mathcal{F}(\mathfrak{m})_{\text{tor}} = \mathcal{F}[p^\infty]$ , and if  $x \in \mathcal{F}(\mathfrak{m})$  has exact order  $p^n$ , then  $v(x) \leq \frac{v(p)}{p^n - p^{n-1}}$ .  $\quad \lrcorner$

*Proof:* If  $(m, p) = 1$ , then  $\mathcal{F}(\mathfrak{m})[m] = 0$  by (10.7.3.5).

By (10.7.3.20),  $[p](x) = pf(x) + g(x^p)$ , where  $f(0) = 1$ . Thus it is possible only if  $pv(x) \geq v(p^x)$ , which means  $v(x) \leq v(p)/(p-1)$ .

Now if this is true for  $n \geq 1$ , let  $x \in \mathcal{F}(\mathfrak{m})$  with order  $p^{n+1}$ , then  $v([p](x)) = v(pf(x) + g(x^p)) \geq \min(v(px), pv(x))$ . But  $[p](x)$  has order  $p^n$ , thus  $\frac{v(p)}{p^n - p^{n-1}} \geq \min(v(px), pv(x))$ . But  $n \geq 1$  and  $v(x) > 0$ , it is impossible that  $\frac{v(p)}{p^n - p^{n-1}} \geq v(px)$ , thus we have  $\frac{v(p)}{p^n - p^{n-1}} \geq pv(x)$ .  $\quad \square$

**Prop. (10.7.4.21).** Let  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa)$  be a CDVR of mixed characteristic  $(0, p)$ ,  $\mathcal{F}$  a formal group law over  $K$ , then

- the formal logarithm (10.7.3.21) induces an homomorphism  $\log_{\mathcal{F}} : \mathcal{F}(\mathfrak{m}) \rightarrow K$ .
- For  $r > \frac{v(p)}{p-1}$ ,  $\log_{\mathcal{F}}$  is an isomorphism  $\log_{\mathcal{F}} : \mathcal{F}(\mathfrak{m}^r) \cong \mathfrak{m}^r$ .  $\quad \lrcorner$

*Proof:* This follows from determining the convergence of  $\log_{\mathcal{F}}$  and  $\exp_{\mathcal{F}}$  by (19.1.7.3), after which it is a homomorphism by (10.7.3.22).  $\quad \square$

**Cor. (10.7.4.22) [Group Structure of CDVRs].** Take  $F(X, Y) = (1 + X)(1 + Y) - 1$ , then  $\log_{\mathcal{F}}$  is given by  $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ , and it induces an isomorphism  $(1 + \mathfrak{m}^r)^{\times} \cong \mathfrak{m}^r$  for  $r > \frac{v(p)}{p-1}$ .  
 $\perp$

**Prop. (10.7.4.23) [ $\mathbb{Z}_p$ -Multiplication].** Let  $(R, \mathfrak{m})$  be a CDVR with residue field  $k$  of characteristic  $p$ ,  $F$  a formal group law over  $R$ , then for any  $x \in \mathcal{F}(\mathfrak{m})$ ,  $\lim_{n \rightarrow \infty} [p^n](x) \rightarrow 0$ .  
 In particular, we can define  $[\alpha](x)$  for any  $\alpha \in \mathbb{Z}_p$ .  $\perp$

*Proof:* By (10.7.3.20),  $v([p](x)) = v(pf(x) + g(x^p)) \geq \min(v(x) + v(p), pv(x))$ .  $\square$

## 5 Cartier-Theory

References are [Zin84].

## 6 Isogenies of Formal Groups

References are [Zin84].

**Def. (10.7.6.1) [Isogenies of Formal Groups].** A homomorphism  $\varphi : G_1 \rightarrow G_2$  of formal groups of the same dimension is called an **isogeny of formal groups** if the kernel of  $\varphi$  is representable.

If  $G_i = \mathrm{Spf} R_i$  and  $\mathfrak{a}_i$  is the augmentation ideal of  $G_i$ , then  $\varphi$  is an isogeny iff  $\mathfrak{a}_i$  is nilpotent in  $R_1/\varphi^*(\mathfrak{a}_2)R_1$ .  $\perp$

*Proof:* Cf. [Zin84]P87.  $\square$

**Def. (10.7.6.2) [Heights in Positive Characteristic].** For  $R = k \in \mathbf{Field}^p$  and  $\varphi : F \rightarrow G$  an isogeny of formal groups over  $k$ , then  $\# \ker(\varphi) = p^h$  for some  $h \in \mathbb{N}$  by [Zin84]P86, then  $\varphi$  is said to be of height  $h$ . In general, if  $R \in \mathbf{Field}^p$  and  $\varphi : F \rightarrow G$  is an isogeny of formal groups over  $R$ , then  $\varphi$  is said to have **height of isogeny**  $h$  if  $\varphi_{\kappa(\mathfrak{p})}$  is of height  $h$  for each  $\mathfrak{p} \in \mathrm{Spec}(R)$ .  $\perp$

**Prop. (10.7.6.3) [Composition of Isogenies].** If  $\varphi_1 : G_1 \rightarrow G_2, \varphi_2 : G_2 \rightarrow G_3$  are morphisms of formal groups of the same dimension, then  $\varphi_2 \circ \varphi_1$  is an isogeny iff  $\varphi_1, \varphi_2$  are both isogenies. And in this case we have

$$\mathrm{ht}(\varphi_1) + \mathrm{ht}(\varphi_2) = \mathrm{ht}(\varphi_2 \circ \varphi_1)$$

if two of the heights are defined (10.7.6.2).  $\perp$

*Proof:* Cf. [Zin84]P90.  $\square$

**Def. (10.7.6.4) [Heights of Isogenies between 1-Dimensional Formal Groups].** For  $R \in \mathbf{CRing}^p$  and  $\varphi : F \rightarrow G$  be a homomorphism of formal group laws over  $R$  of dimension 1, the **height of homomorphism**  $\mathrm{ht}(\varphi)$  is the largest integer  $h$  s.t.  $\varphi(T) = g(T^{p^h})$  for some  $h$ .

For a formal group law  $F$  over  $R$ , the **height of formal group law**  $\mathrm{ht}(F)$  is  $\mathrm{ht}([p]_F)$ . By (10.7.3.20), the height is always positive.  $\perp$

**Prop. (10.7.6.5).** Let  $R \in \mathbf{CRing}^p$  and  $f : F \rightarrow G$  be a homomorphism of formal group laws over  $R$  of dimension 1, then

- If  $f^\wedge(0) \neq 0$ , then  $\mathrm{ht}(f) > 0$ .
- If  $f = g(T^{p^h})$  with  $h = \mathrm{ht}(f)$ , then  $g'(0) \neq 0$ .

In particular, the first non-zero term of  $f(T)$  is  $T^{p^h}$ , where  $h$  is the height of  $f$ .  $\perp$

*Proof:* 1: Let  $\omega_F, \omega_G$  be the normalized invariant differential forms of  $F$  and  $G$ , then

$$0 = f'(0)\omega_F(T) = \omega_G(f(T)) = G_X(0, T)^{-1}f'(T)dT,$$

thus  $f'(T) = 0$ , which means  $\text{ht}(f) > 0$ .

2: Let  $q = p^h$  and  $F^{(q)}(T) = F(X^{1/q}, Y^{1/q})^q$ . Then it is easy to see  $F^{(q)}$  is another formal group law, and  $g$  is a homomorphism from  $F^{(q)}$  to  $F$ :

$$g(F^{(q)}(X^q, Y^q)) = g(F(X, Y)^q) = f(F(X, Y)) = F(f(X), f(Y)) = F(g(X^q), g(Y^q)).$$

Which means  $g(F^{(q)}(X, Y)) = F(g(X), g(Y))$ . Thus if  $g'(T) = 0$ , by item1,  $g(T) = g_1(T^p)$ , contradicting  $h = \text{ht}(f)$ .  $\square$

**Thm.(10.7.6.6) [Fiber Criterion of Isogenies].** Suppose  $R \in \mathcal{CRing}/\mathbb{Z}_{(p)}$  and  $\varphi : G_1 \rightarrow G_2$  is a morphism of formal groups of the same dimension over  $R$ . If  $\varphi_{\kappa(\mathfrak{p})}$  is an isogeny for each  $\mathfrak{p} \in \text{Spec}(R/(p))$ , then  $\text{ht}(\mathfrak{p}) \geq \text{ht}(\mathfrak{q})$  for each  $\mathfrak{q} \subset \mathfrak{p}$ . And if  $\text{ht}(\mathfrak{p})$  is independent of  $p$  and  $R$  has only f.m. minimal non-trivial prime ideals, then  $\varphi$  is an isogeny.  $\lrcorner$

*Proof:* Cf. [Zin84](#)P91 ?  $\square$

## 10.8 Algebraic Stacks

Basic references are [Knu71, L-MB00, Vis08, Ols16, Sta] and [Fibered Category to Algebraic Stacks, Lamb].

**Notation (10.8.0.1).**

- Let  $S \in \mathbf{Sch}$ .
- Use notations on sites, sheaves and stacks defined as in [Sites, Sheaves, Topoi and Stacks](#).
- A fibered category/stack over  $S$  means a fibered category/stack over  $\mathbf{Sch}_{\text{fppf}}/S$  (obsolete).

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### 1 Schemes as Functors

**Prop. (10.8.1.1) [Strong Yoneda Lemma].** For any  $S \in \mathbf{Sch}$ , the functor

$$\mathbf{Sch}/S \rightarrow \mathbf{Sh}_{\text{fppf}}/S \rightarrow \mathbf{Sh}^{\text{set}}(\mathbf{Aff}_{\text{fppf}}/S) \rightarrow \mathbf{PSh}^{\text{set}}(\mathbf{Aff}/S) : X \mapsto (\downarrow_X : \text{Spec } R \mapsto \text{Mor}_S(\text{Spec } R, X))$$

is a fully faithful embedding of the categories.

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*Proof:* (7.1.2.25) applied to  $\mathbf{Aff}_{\text{fppf}}/S \rightarrow \mathbf{Sch}_{\text{fppf}}/S$  implies the restriction map  $\mathbf{Sh}_{\text{fppf}}/S \rightarrow \mathbf{Sh}(\mathbf{Aff}_{\text{fppf}}/S)$  is an equivalence, thus the assertion follows from Yoneda's lemma and the fact  $\mathbf{Sh}^{\text{set}}(\mathbf{Aff}_{\text{fppf}}/S) \rightarrow \mathbf{PSh}^{\text{set}}(\mathbf{Aff}/S)$  is fully faithful.  $\square$

**Def. (10.8.1.2) [Closed Subfunctors].** For  $X \in \mathbf{PSh}^{\text{set}}(\mathbf{Aff}_S)$ , a subfunctor  $Z$  of  $\downarrow_X \in \mathbf{PSh}^{\text{set}}(\mathbf{Aff}_S)$  is called a **closed subfunctor** of  $X$  if for any  $T \in \mathbf{Aff}/S$  and  $f \in \text{Mor}(\downarrow_T, X)$ , the functor  $Z \otimes_{X, f} \downarrow_T$  is represented by a closed subscheme of  $T$ .

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**Prop. (10.8.1.3) [Closed Subfunctor of Schemes].** For  $X \in \mathbf{Sch}/S$ , a closed subfunctors of  $\downarrow_X$  (10.8.1.2) are exactly of the form  $\downarrow_Z$  for a closed subscheme  $Z \subset X$ .

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*Proof:* If  $Z$  is a closed subscheme of  $X$ , then for any  $f : h^A \rightarrow X$ ,  $f^{-1}(Z)$  is the pullback of  $Z$  along  $\text{Spec } A \rightarrow X$ , so it is a closed subscheme of  $\text{Spec } A$ . Conversely, if  $Z$  is a closed subfunctor of  $X$ , then for each affine open subset  $U$  of  $X$ ,  $Z \cap h_U$  is represented by a quotient of  $\mathcal{O}(U)$  by some ideal  $\mathcal{I}(U)$ . Because of the uniqueness,  $\mathcal{I}(U)$  and  $\mathcal{I}(U')$  coincides on the intersection  $U \cap U'$ , thus  $U \mapsto \mathcal{I}(U)$  defines a sheaf of ideals  $\mathcal{I}$  on  $X$ .

Now  $Z = h_{Z'}$ , where  $Z'$  is the closed subscheme of  $X$  defined by  $\mathcal{I}$ , because for any  $\text{Spec } R \rightarrow X$ , the pullback of  $Z$  and  $Z'$  to  $R$  are the same, because they are all closed subschemes of  $\text{Spec } R$  and they are equal on an open covering of  $\text{Spec } R$  (The pullback of the open coverings of  $X$ ). Now if  $\text{Spec } R \rightarrow X$  is represented by an element  $\alpha \in X(R)$ ,  $Z \times_X h^R(R)$  is the set  $\{\varphi \in \text{Hom}(R, R) \mid X(\varphi)(\alpha) \in Z(R)\}$ . So  $\text{id}_R \in Z \times_X h^R(R) \iff \alpha \in Z(R)$ . From this we see that  $Z(R) = Z'(R)$  for any  $R$ .  $\square$

**Def. (10.8.1.4) [Open Subfunctors].** Let  $Z$  be a subfunctor of  $X \in \mathbf{PSh}^{\text{set}}(\mathbf{Aff}/S)$ .  $Z$  is called an **open subfunctor** of  $X$  if for any  $T \in \mathbf{Aff}_S$ ,  $f \in \text{Mor}(\downarrow(T), X)$ , the functor  $U \otimes_X h_T$  is represented by an open subscheme of  $T$ .

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**Prop. (10.8.1.5) [Open Subfunctors of Schemes].** For  $X \in \mathbf{Sch}_S$ , the open subfunctors of  $\downarrow_X$  (10.8.1.4) are exactly of the form  $\downarrow_U$  for an open subscheme  $U \subset X$ .

┘

*Proof:* The proof is exactly the same as that of (10.8.1.3).  $\square$

**Def. (10.8.1.6) [Open Coverings by Functors].** An **open covering of a functor**  $X \in \mathcal{PSh}^{\text{Set}}(\text{Sch}_S)$  is a family of open subfunctors  $\{U_i\}$  that for any  $T \in \text{Aff}/S$ ,  $h_T \times_X U_i$  is an open covering of  $T$ .  $\lrcorner$

**Prop. (10.8.1.7) [Open Coverings of Schemes].** For  $X \in \text{Sch}_S$ , the open coverings of the functor  $\lrcorner_X$  are exactly open coverings of  $X$ .  $\lrcorner$

*Proof:* The proof is exactly the same as that of (10.8.1.3).  $\square$

**Prop. (10.8.1.8).** The pullback of a closed subfunctor is also a closed subfunctor. The intersection of closed subfunctors is a closed subfunctor.  $\lrcorner$

**Lemma (10.8.1.9).** Let  $k$  be a field and  $B \in \mathcal{CAlg}_k$ ,  $X \in \mathcal{PSh}^{\text{Set}}(\mathcal{CAlg}_B)$ , define

$$X_* \in \mathcal{PSh}^{\text{Set}}(\mathcal{CAlg}_k) : X_*(R) = X(R \otimes_k B).$$

Then if  $Z$  is a closed subfunctor of  $X$ ,  $Z_*$  is also a closed subfunctor of  $X_*$ .  $\lrcorner$

*Proof:* Let  $A$  be a  $k$ -algebra, and  $\alpha \in X_*(A)$ . To prove  $Z_*$  is closed in  $X_*$ , we need to show there exists an ideal  $\mathfrak{a} \subset A$  that for any homomorphism  $\varphi : A \rightarrow R$ ,

$$X_*(\varphi)(\alpha) \in Z_*(R) \iff \varphi(\mathfrak{a}) = 0.$$

Because  $Z$  is closed in  $X$ , there exists an ideal  $\mathfrak{b}$  of  $A \otimes_k B$  that for any  $\varphi : A \rightarrow R$ ,

$$X(\varphi \otimes B)(\alpha) \in Z_*(R) \iff (\varphi \otimes B)(\mathfrak{b}) = 0.$$

Now by (5.1.1.29), there is an ideal  $\mathfrak{a} \subset A$  that an ideal  $I$  of  $A$  satisfies  $\mathfrak{b} \subset I \otimes B \iff \mathfrak{a} \subset I$ , thus we are done.  $\square$

**Prop. (10.8.1.10).** Let  $S \in \text{Sch}$ ,  $X \in \mathcal{PSh}^{\text{Set}}(\text{Aff}/S)$  and  $Z$  a closed subfunctor of  $X$ . If  $Y \in \text{Sch}_R$ , then  $\text{Mor}(\lrcorner_Y, Z)$  is a closed subfunctor of  $\text{Mor}(\lrcorner_Y, X)$ .  $\lrcorner$

*Proof:* If  $Y = h^B$ , then  $\text{Mor}(Y, X)(R) = X(B \otimes R)$ , thus the conclusion follows from (10.8.1.9). For a general  $Y$ , let  $Y_i$  be an affine open covering of  $Y$ , then there are maps  $\rho_i : \text{Hom}(Y, X) \rightarrow \text{Hom}(Y_i, X)$ . Now  $\text{Hom}(Y_i, Z)$  is closed subfunctor of  $\text{Hom}(Y_i, X)$ , thus we are done if we can show that  $\text{Hom}(Y, Z) = \cap_i \rho_i^{-1}(\text{Hom}(Y_i, Z))$ . But this is equivalent to any map  $Y_R \rightarrow X_R$  that maps  $(Y_i)_R$  into  $Z_R$  maps  $Y_R$  into  $Z_R$ , which is clear.  $\square$

**Def. (10.8.1.11) [Fat Subfunctors].** Let  $S \in \text{Sch}$  and  $\mathcal{F} \in \text{Sh}^{\text{Set}}(\text{Aff}_{\text{fppf}}/S)$ , then a subfunctor  $D$  of  $\mathcal{F}$  is called a **fat subfunctor** if the shification of  $D$  w.r.t. the fppf topology is just  $\mathcal{F}$ .  $\lrcorner$

**Prop. (10.8.1.12) [Extending Group Structures].** Let  $S \in \text{Sch}$ ,  $\mathcal{F} \in \text{Sh}^{\text{grp}}(\text{Aff}_{\text{fppf}}/S)$  and  $D$  a fat subfunctor of  $\mathcal{F}$ , then every group structure on  $D$  extends uniquely to a group structure on  $\mathcal{F}$  by shification.  $\lrcorner$

## 2 Algebraic Spaces

**Def. (10.8.2.1) [Schematic Morphism].** A **schematic morphism** of fibered categories  $\mathcal{X} \rightarrow \mathcal{Y} \in \text{FibCat}^{\text{grp}}(\text{Sch}_{\text{fppf}}/S)$  over  $S$  is a representable morphism of fibered categories over  $\text{Sch}_{\text{fppf}}/S$  (4.2.3.34).  $\lrcorner$

**Prop. (10.8.2.2).** Suppose  $\mathcal{F} \rightarrow \mathcal{G} \in \mathcal{PSh}_{\text{fppf}}^{\text{Set}}/S$  be schematic (10.8.2.1) and  $\mathcal{G} \in \text{Sh}_{\text{fppf}}^{\text{Set}}/S$ , then  $\mathcal{F} \in \text{Sh}_{\text{fppf}}^{\text{Set}}/S$ .  $\lrcorner$



*Proof:* Cf.[Sta]02WD.  $\square$

**Prop. (10.8.2.3).** Suppose  $\mathcal{F} \rightarrow \mathcal{G} \in \mathcal{PSh}_{\text{fppf}}^{\text{Set}}/S$  is schematic(10.8.2.1), then  $\Delta_{\mathcal{F}/\mathcal{G}} : \mathcal{F} \rightarrow \mathcal{F} \times_{\mathcal{G}} \mathcal{F}$  is also schematic.  $\lrcorner$

*Proof:* Cf.[Sta]05L9.  $\square$

**Def. (10.8.2.4) [Properties of Schematic Morphisms].** For a property  $\mathcal{P}$  of maps of schemes that is fppf-local on the target and stable under base change, we say that  $\mathcal{P}$  holds for a schematic map(10.8.2.1)  $\mathcal{X} \rightarrow \mathcal{Y} \in \text{FibCat}(\text{Sch}_{\text{fppf}}/S)$  if  $\mathcal{P}$  holds for the map  $T \times_{\mathcal{Y}} \mathcal{X} \rightarrow T$  for any  $T \in \text{Sch}/S$ .

It is easy to show that this definition is compatible with that of schemes when  $\mathcal{X}, \mathcal{Y} \in \text{Sch}_{\text{fppf}}/S$ .  $\lrcorner$

**Prop. (10.8.2.5) [Descent for Properties].** Let  $f : \mathcal{X} \rightarrow \mathcal{Y} \in \mathcal{PSh}^{\text{Set}}(\text{Sch}_{\text{fppf}}/S)$  be a schematic morphism and  $g : \mathcal{Z} \rightarrow \mathcal{Y} \in \mathcal{PSh}^{\text{Set}}(\text{Sch}_{\text{fppf}}/S)$ . If  $g^{\#}$  is an epimorphism, then  $f$  has  $\mathcal{P}$  iff  $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \rightarrow \mathcal{Z}$  has  $\mathcal{P}$ .  $\lrcorner$

*Proof:* Cf.[Sta]03KD.  $\square$

**Prop. (10.8.2.6) [Properties of Diagonals].** For  $\mathcal{F} \in \mathcal{PSh}^{\text{Set}}(\text{Sch}_{\text{fppf}}/S)$ , the following are equivalent:

- $\Delta_{\mathcal{F}}$  is schematic(10.8.2.1).
- For any  $U \in \text{Sch}_{\text{fppf}}/S$  and  $a \in \mathcal{F}(U)$ , the map  $a : \mathcal{Y}_U \rightarrow \mathcal{F}$  is representable.
- For any  $U, V \in \text{Sch}_{\text{fppf}}/S$  and  $a \in \mathcal{F}(U), b \in \mathcal{F}(V)$ , the fiber product  $\mathcal{Y}_U \times_{a, \mathcal{F}, b} \mathcal{Y}_V$  is representable.

Moreover,  $\Delta_{\mathcal{F}}$  has a property  $\mathcal{P}$  iff for any  $U, V \in \text{Sch}_{\text{fppf}}/S$  and  $a \in \mathcal{F}(U), b \in \mathcal{F}(V)$ , the map  $\mathcal{Y}_U \times_{a, \mathcal{F}, b} \mathcal{Y}_V \rightarrow U \times V$  has property  $\mathcal{P}$ .  $\lrcorner$

*Proof:* Cf.[Sta]025W, 0CBT.  $\square$

**Prop. (10.8.2.7).** If a schematic morphism  $f : \mathcal{X} \rightarrow \mathcal{Y} \in \mathcal{PSh}^{\text{Set}}(\text{Sch}_{\text{fppf}}/S)$  is flat, loc.p. and surjective, then  $f$  is an epimorphism.  $\lrcorner$

*Proof:* Cf.[Sta]05VM.  $\square$

### Algebraic Spaces

**Def. (10.8.2.8) [Algebraic Space].** An **algebraic space** is a sheaf  $\mathcal{X} \in \text{Sh}_{\text{fppf}}^{\text{Set}}/S$  s.t.

- The diagonal map  $\Delta_{\mathcal{X}}$  is schematic(10.8.2.1),
- There exists some scheme  $U \in \text{Sch}/S$  with an étale surjective map  $\mathcal{Y}_U \rightarrow \mathcal{X}$  in  $\text{Sch}_{\text{fppf}}/S$ , called an **étale atlas for the algebraic space  $\mathcal{X}$** . (Notice  $\mathcal{Y}_U \rightarrow \mathcal{X}$  is schematic by(10.8.2.6)).

and

The category of algebraic spaces is the full subcategory of  $\text{Sh}_{\text{fppf}}^{\text{Set}}/S$ , denoted by  $\text{AlgSp}/S$ .  $\lrcorner$

**Prop. (10.8.2.9) [Schemes as Algebraic Spaces].**  $\mathcal{Y} : \text{Sch}/S \rightarrow \text{AlgSp}/S$ (10.8.2.8) is a fully faithful embedding.  $\lrcorner$

*Proof:* For  $X \in \text{Sch}/S$ ,  $h_X$  is a sheaf because fppf site is subcanonical by(7.1.4.35), its diagonal is representable, and the identity  $\text{id}_X : X \rightarrow X$  is surjective and étale.  $\square$

**Remark (10.8.2.10)[Kollar].** In general, the quotient of a scheme by a finite group action is an algebraic space that is not a scheme. Naively one can think of algebraic spaces as quotients of schemes by finite groups.  $\lrcorner$

**Def. (10.8.2.11) [Sites over Algebraic Spaces].** The Zariski/étale/smooth/fppf/... sites over an algebraic space is defined verbatim as that of over algebraic schemes.  $\lrcorner$

**Prop. (10.8.2.12) [Algebraic Spaces and Étale Equivalence Relations].** Cf. [Sta]02WW.  $\lrcorner$

*Proof:*  $\square$

**Prop. (10.8.2.13).** For  $\mathcal{X} \in \mathcal{A}lgSp/S$  (10.8.2.8), the diagonal map  $\Delta_{\mathcal{X}/S}$  is locally of finite type, locally quasi-finite, separated and is a monomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]02X4.  $\square$

**Def. (10.8.2.14) [Underlying Space of Algebraic Spaces].** For  $\mathcal{X} \in \mathcal{A}lgSp/S$ , the underlying space  $|\mathcal{X}|$  is a topological space whose points are

$$\lim_{K \in \mathbf{Field}} \text{Mor}_{\mathbf{Sh}_{fppf}^{\text{set}}/S}(\text{Spec } K, \mathcal{X}),$$

and the topology is defined in [Sta]03BX<sup>?</sup>.

Then  $X \mapsto |X|$  induces a functor  $|-| : \mathcal{A}lgSp/S \rightarrow \mathcal{T}op/|S|$ .  $\lrcorner$

**Prop. (10.8.2.15).** Let  $\mathcal{X} \in \mathcal{A}lgSp/S$ ,  $T \in \mathcal{S}ch/S$ ,  $f : T \rightarrow \mathcal{X} \in \mathcal{A}lgSp/S$ , then  $f$  is surjective iff  $|f| : |T| \rightarrow |X|$  (10.8.2.14) is surjective.  $\lrcorner$

*Proof:*  $\square$

### 3 Properties of Algebraic Spaces

**Def. (10.8.3.1) [Properties of Algebraic Spaces].** Let  $X \in \mathcal{A}lgSp/S$ . For a property  $\mathcal{P}$  of schemes that is étale-local, the following are equivalent:

- There exists some  $U \in \mathcal{S}ch$  with a surjective étale morphism  $U \rightarrow \mathcal{X} \in \mathcal{A}lgSp/S$  (10.8.2.4) s.t.  $U$  has property  $\mathcal{P}$ .
- For any  $U \in \mathcal{S}ch$  with an étale morphism  $U \rightarrow \mathcal{X} \in \mathcal{A}lgSp/S$ ,  $U$  has property  $\mathcal{P}$ .

In which case, we say  $X$  has property  $\mathcal{P}$ . This is compatible with the definition in the scheme case when  $X \in \mathcal{S}ch/S$ .  $\lrcorner$

*Proof:*  $2 \rightarrow 1$  is clear. For  $1 \rightarrow 2$ : If  $U \in \mathcal{S}ch$  with a surjective étale morphism  $U \rightarrow \mathcal{X} \in \mathcal{A}lgSp/S$  s.t.  $U$  has property  $\mathcal{P}$ , and  $V \in \mathcal{S}ch$  with an étale morphism  $V \rightarrow \mathcal{X} \in \mathcal{A}lgSp/S$ , then  $U \times_X V \rightarrow U$  is étale, and  $U \times_X V \rightarrow V$  is an étale covering. Thus  $U \times_X V \rightarrow U$  has property  $\mathcal{P}$  because  $U$  does, and  $V$  has property  $\mathcal{P}$  because  $U \times_X V$  does.  $\square$

**Def. (10.8.3.2) [Dimensions of Algebraic Stacks].** For  $X \in \mathcal{A}lgSp/S$  and  $x \in |X|$ , define the local dimension

$$\dim_x(X) \triangleq \dim_u(U)$$

for any étale morphism  $f : U \rightarrow X$  with  $u \in U$  s.t.  $f(u) = x$ . Notice this is well-defined by (10.8.2.8) and (5.4.7.28). And we define

$$\dim(X) = \sup_{x \in |X|} \dim_x(X),$$

which is compatible with the definition in the scheme case when  $X \in \mathcal{S}ch/S$ , by (7.4.1.31).  $\lrcorner$

**Remark (10.8.3.3).** WARNING: unlike the case of schemes (7.4.1.31), for  $X \in \mathcal{A}lgSp/S$ , it may not be true that  $\dim(X) = \dim(|X|)$ , Cf. [Sta]02Z8.  $\lrcorner$

**Thm. (10.8.3.4) [Algebraic Spaces are Fpqc Shaves, Gabber].**  $\mathcal{A}lgSp/S \subset \mathcal{S}h^{\text{Set}}(\mathcal{A}ff_{\text{fpqc}}/S)$ .  $\lrcorner$

*Proof:* Cf. [Sta]0APL.  $\square$

## 4 Morphisms between Algebraic Spaces

**Remark (10.8.4.1).** We need to define properties for not necessarily schematic morphisms, and the definition should be compatible with (10.8.2.4).  $\lrcorner$

### Open Morphisms

**Def. (10.8.4.2) [Open Morphisms].** For a representable morphism  $f : X \rightarrow Y \in \mathcal{A}lgSp/S$  (4.2.3.34), the following are equivalent:

- $f$  is universally open (10.8.2.4).
- For any  $Z \rightarrow Y \in \mathcal{A}lgSp/S$ ,  $|Z \times_Y X| \rightarrow |Z|$  is open.

Thus we can define an **open morphism of algebraic spaces** to be a morphism  $f : X \rightarrow Y \in \mathcal{A}lgSp/S$  s.t.  $|f| : |X| \rightarrow |Y|$  is open. This is compatible with the definition in (10.8.2.4).  $\lrcorner$

*Proof:* Cf. [Sta]03Z1.  $\square$

## 5 Algebraic Spaces over Fields

## 6 Algebraic Stacks

References are [Sta].

**Def. (10.8.6.1) [Representable by Algebraic Spaces].** A morphism  $f : \mathcal{X} \rightarrow \mathcal{Y} \in \mathcal{F}ibCat^{\text{grpd}}_{\text{fppf}}/S$  is called **representable by algebraic spaces** if for any  $T \in \mathcal{S}ch_{\text{fppf}}/S$  and a morphism  $(\mathcal{S}ch_{\text{fppf}}/U) \rightarrow \mathcal{Y}$ , the fibered category  $(\mathcal{S}ch_{\text{fppf}}/T) \times_{\mathcal{Y}} \mathcal{X}$  is equivalent to an algebraic space over  $T$  (10.8.2.8). (Notice it is a fibered category by (4.2.3.13) and (4.2.3.14))  $\lrcorner$

**Def. (10.8.6.2) [Properties of Morphisms Representable by Algebraic Spaces].** For a property  $\mathcal{P}$  of maps of schemes that is fppf-local and stable under base change, we say that  $\mathcal{P}$  holds for a morphism  $f : \mathcal{X} \rightarrow \mathcal{Y} \in \mathcal{F}ibCat^{\text{grpd}}(\mathcal{S}ch_{\text{fppf}}/S)$  representable by algebraic spaces (10.8.6.1) iff for any  $U \in \mathcal{S}ch_{\text{fppf}}/S$ ,  $\mathcal{P}$  holds for the map of algebraic spaces  $U \times_{\mathcal{Y}} \mathcal{X} \rightarrow U \in \mathcal{A}lgSp/S$  (in the sense of Morphisms between Algebraic Spaces).  $\lrcorner$

**Prop. (10.8.6.3) [Properties Under Equivalences].** Having properties are stable under equivalences. Cf. [Sta]0459.  $\textcolor{red}{?}$   $\lrcorner$

**Def. (10.8.6.4) [2-Category of Algebraic Stacks].** An **algebraic stack** over  $S$  is a stack fibered in groupoids  $\mathcal{X} \in \mathcal{S}ta^{\text{grpd}}_{\text{fppf}}/S$  s.t. the diagonal is representable by an algebraic space (10.8.6.1), and there exists some scheme  $U \in \mathcal{S}ch/S$  with a smooth surjective map  $\mathcal{S}ch_{\text{fppf}}/U \rightarrow \mathcal{X}$  in  $\mathcal{S}ta_{\text{fppf}}/S$ , called an **atlas for the algebraic stack**  $\mathcal{X}$ .

It is called a **Deligne-Mumford stack** if moreover there exists some  $U \in \mathcal{S}ch_{\text{fppf}}/S$  with a smooth étale map  $\mathcal{S}ch_{\text{fppf}}/U \rightarrow \mathcal{X}$  in  $\mathcal{S}ta_{\text{fppf}}/S$ .

The 2-category of algebraic (resp. Deligne-Mumford) stacks is the full sub-2-category of  $\mathcal{S}ta_{\text{fppf}}/S$ , denoted by  $\mathcal{A}lgSta/S$  (resp.  $\mathcal{A}lgSta^{\text{DM}}/S$ ).  $\lrcorner$

**Prop. (10.8.6.5).** If  $X, Y \in \mathcal{C}at(\mathcal{S}ch/S)$  are equivalent, then  $\mathcal{X} \in \mathcal{A}lg\mathcal{S}p/S$  (resp.  $\mathcal{A}lg\mathcal{S}ta^{DM}/S$ ) iff so is  $\mathcal{Y}$ . ┘

*Proof:* By (7.1.3.8),  $\mathcal{Y}$  is also a stack in groupoid. And  $\Delta_Y$  is equivalent to  $\Delta_X$ , and notice that having property  $\mathcal{P}$  is stable under equivalences (10.8.6.3). □

**Prop. (10.8.6.6) [2-Fibered Products of Algebraic Stacks].** The 2-category  $\mathcal{A}lg\mathcal{S}p/S$  has 2-fibered products. ┘

*Proof:* Cf. [Sta]04TF. □

**Prop. (10.8.6.7) [Algebraic Stacks and Étale Groupoid Object].** There is a bijection of étale groupoid objects on a scheme with the category of algebraic stacks. ┘

*Proof:* Cf. [Lamb, P39]. □

**Prop. (10.8.6.8) [Algebraic Spaces and Étale Equivalence Relations].** There is a bijection of étale equivalence relations on a scheme with the category of algebraic spaces. ┘

*Proof:* This is a corollary of (10.8.6.7). □

## 7 Sheaves on Algebraic Stacks

## 8 Representability

## 9 Artin's Axioms

## 10 Quot and Hilbert Stacks

## 11 Properties of Algebraic Stacks

## 12 Morphisms of Algebraic Stacks

## 13 Limits of Algebraic Stacks

## 14 Cohomology of Algebraic Stacks

## 15 Derived Categories of Stacks

## 10.9 Moduli Problems

### 1 Moduli problems

*“...the theory of moduli spaces has the wonderful feature of having many doors, many techniques by which this theory can be developed. ...This is another part of its charm. ...one fact stands out: the story is not finished.”*

– D. Mumford

. . . the theory of moduli spaces has the wonderful feature of having many doors, many techniques by which this theory can be developed. ...This is another part of its charm. . . . one fact stands out: the story is not finished. —David Mumford

References are [Sta] and [Stacks and Moduli, Alper].

### Moduli Fields

**Def. (10.9.1.1) [Fields of Moduli].** For  $k \in \mathbf{Field}$ ,  $k = \bar{k}$  and  $X \in \mathbf{Var}/\bar{k}$ , the **moduli field** of  $X$  is the fixed field of  $\{\sigma \in \mathrm{Gal}(\bar{k}/k) \mid X^\sigma \cong X\}$ .  $\lrcorner$

**Prop. (10.9.1.2) [Hammer-Herrlich [H-H03]].** If  $k \in \mathbf{Field}$ ,  $k = \bar{k}$  and  $C \in \mathbf{SmProjVar}^1/k$ , then  $C$  is defined on a finite extension of the moduli field  $M$ . Sometimes,  $X$  cannot be defined on the moduli field.

In fact, this works nearly for any kind of smooth projective varieties with a coarse moduli space.

$\lrcorner$

*Proof:* Cf. [H-H03].

There exists a f.g.  $M$ -algebra  $A \subset k$  s.t.  $C$  has model  $\mathcal{C}$  over  $A$ . By localizing  $A$ , we may assume that  $\mathcal{C}/A$  is smooth and projective. Let  $G$  be the subgroup of  $\mathrm{Aut}(C)$  consisting of  $\sigma$  s.t.  $C^\sigma \cong C$ . Consider  $\mathrm{Spec}(A \otimes_M k)$ , then there is a natural  $k$ -point  $p$ , and for any  $\sigma \in G$ , the fiber of  $\mathcal{C}$  over  $\sigma(p)$  is  $C^\sigma$  which is isomorphic to  $C$ .

Then we prove that  $\{\sigma(p)\}_{\sigma \in G}$  is dense in  $\mathrm{Spec}(A \otimes_M k)$ ? After this, we see that the family  $\mathcal{C}$  corresponds to a map from  $\mathrm{Spec}(A \otimes_M k)$  to the moduli space of smooth genus  $g$  curves that is constant on a dense subset, thus constant. But  $\mathrm{Spec}(A \otimes_M k)$  contains  $\bar{M}$  points, so  $C$  is defined over  $\bar{M}$ .  $\square$

**Prop. (10.9.1.3).** For  $k \in \mathbf{Field}^0$ ,  $k = \bar{k}$  and  $C \in \mathbf{SmProjVar}^1/k$ ,

- If  $C$  has genus  $\leq 1$ , then  $C$  is defined over the moduli field of  $C$ .
- $C/\mathrm{Aut}(C)$  is defined over the moduli field of  $C$ .

$\lrcorner$

*Proof:* 1: If  $g(C) = 0$ , then  $C \cong \mathbb{P}^1$ , which is defined over  $\mathbb{Q}$ . If  $g(C) = 1$ , then the moduli field and the field of definition are both  $\mathbb{Q}(j(C))$ .

2: Cf. [Koc04].?  $\square$

### Representability

**Prop. (10.9.1.4) [A Representability Criterion].** Let  $\mathcal{F} \in \mathbf{Sh}^{\mathrm{Set}}(\mathbf{Sch}_{\mathrm{Zar}})$  and there is a covering  $F = \cup F_i$  by open subfunctors (10.8.1.4)(10.8.1.6) that are representable by schemes, then  $F$  is representable by a scheme.  $\lrcorner$

*Proof:* Let  $(X_i, \xi_i)$  represents  $F_i$ , where  $\xi \in F_i(X_i)$ . Because  $F_j \subset F$  is representable by open immersion, there are open subsets  $U_{ij} \subset X_i$  that  $T \rightarrow X_i$  factors through  $U_{ij}$  iff  $\xi|_T \in F_j(T)$ . In particular,  $\xi_i|_{U_{ij}} \in F_j(U_{ij})$ , and therefore there is a canonical map  $\varphi_{ij} : U_{ij} \rightarrow X_j$  that  $\varphi_{ij}^* \xi_j = \xi_i|_{U_{ij}}$ . By definition of  $U_{ji}$  this map factors through  $U_{ji}$ .

For the rest, Cf. [Sta]01JJ. ?

□

**Cor. (10.9.1.5) [Representing Group Functors].** Let  $S \in \text{Sch}$  and  $\mathcal{G} \in \text{Sh}^{\text{grp}}(\text{Aff}_{\text{Zar}}/S)$  and there is an open subfunctor  $\mathcal{F} \subset \mathcal{G}$  s.t. for any  $\text{Spec } K \in \text{Aff}_S$  where  $K$  is a field and  $g \in \mathcal{G}(K)$ , there exists a  $g' \in \mathcal{F}(K)$  s.t.  $gg' \in \mathcal{F}(K)$ , then  $\mathcal{G}$  is representable.  $\lrcorner$

*Proof:* For any  $g \in \mathcal{G}(S)$ , let  $F_g = \tau_g^* F$ , then  $F_g \cong F$  and are also open subfunctors of  $\mathcal{G}$ . then  $\{F_g \rightarrow G\}$  is an open covering because for any  $T \in \text{Sch}$  and  $\xi \in \text{Mor}(h_T, \mathcal{G})$ ,  $F_g \times_{\mathcal{G}} h_T$  is represented by open subschemes of  $T$ , and it must covers  $T$  because otherwise there are some map  $\text{Spec } k(x) \rightarrow T$  s.t.  $F_g \times_{\mathcal{G}} h_{k(x)}$  are all empty, contradicting the hypothesis. Thus the assertion follows from (10.9.1.4).  $\square$

**Prop. (10.9.1.6) [Hom( $\mathcal{E}, \mathcal{F}$ ) for Coherent Sheaves].** Let  $S$  be a locally Noetherian scheme and  $X$  a projective scheme over  $S$ . Let  $\mathcal{E}, \mathcal{F}$  be coherent sheaves on  $X$  and  $\mathcal{F}$  is flat, then the functor

$$\underline{\text{Hom}}(\mathcal{E}, \mathcal{F}) : \text{Sch}_S \rightarrow \text{Set} : T \mapsto \text{Hom}_T(\mathcal{E}_T, \mathcal{F}_T)$$

is representable by a vector bundle  $\mathbf{V}$  over  $S$  (7.5.2.21).  $\lrcorner$

*Proof:* Cf. [Kle05]P16.  $\square$

**Cor. (10.9.1.7).** For any  $f \in \text{Hom}(\mathcal{E}, \mathcal{F})$ ,  $f$  corresponds to a morphism  $X \rightarrow \mathbf{V}$ , then the inverse image of  $\mathbf{V}_0 \subset \mathbf{V}$  in  $X$  is the closed subscheme  $X'$  with the universal property that for any  $\varphi : T \rightarrow X$ ,  $\varphi^* f = 0$  iff  $\varphi$  factors through  $X'$ .  $\lrcorner$

### Coarse Moduli Schemes

**Def. (10.9.1.8) [Coarse Moduli Schemes].** For  $k \in \text{Field}$  and  $\mathcal{F} : \text{Sch}/k \rightarrow \text{Set}$ , a **coarse moduli scheme** for  $\mathcal{F}$  is a scheme  $Z \in \text{Sch}/k$  with a morphism  $\varphi : \mathcal{F} \rightarrow \mathcal{Y}_Z$  satisfying the following universal property: for any  $T \in \text{Sch}/k$ ,

$$\text{Hom}(\mathcal{F}, \mathcal{Y}_T) = \text{Hom}(Z, T).$$

And  $Z$  is called a **fine moduli scheme** for  $\mathcal{F}$  if  $\varphi$  is an isomorphism.  $\lrcorner$

## 2 Hilbert & Quot Schemes

References are [Nit05].

For simplicity, we restrict to the category of locally Noetherian schemes over  $S$ , and assume  $X/S$  is separated.

**Def. (10.9.2.1) [Hilbert Functors].** For  $X \in \text{NSch}/S$ , the **Hilbert functor** is the functor

$$\underline{\text{Hilb}}_{X/S} : \text{Sch}_S \rightarrow \text{Set} : T \mapsto \{\text{closed subschemes of } X_T \text{ that is flat over } T\}.$$

□

**Def. (10.9.2.2) [Quot Functors].** Let  $S \in \text{NSch}$ ,  $f : X \rightarrow S \in \text{Sch}^{\text{ft}}/S$  and  $\mathcal{E} \in \text{Coh}(X)$ , then for any  $T \in \text{Sch}_S$ , define a **family of quotients of  $\mathcal{E}$  parametrized by  $T$**  to be a pair  $(\mathcal{F}, q)$  where

- $\mathcal{F} \in \text{Coh}(X_T)$  s.t.  $\text{Supp}(\mathcal{F})$  is proper over  $T$ .
- $q : \mathcal{E}_T \rightarrow \mathcal{F}$  is a surjective homomorphism of sheaves.

Then the **Quot functor** is the functor

$$\underline{\text{Quot}}_{\mathcal{E}/X/S} : \text{Sch}_S \rightarrow \text{Set} : T \mapsto \{\text{isomorphism classes of family of quotients of } \mathcal{E} \text{ parametrized by } T\}.$$

┘

**Cor. (10.9.2.3) [Hilbert Functors as Quot Functors].**  $\underline{\text{Hilb}}_{X/S} = \underline{\text{Quot}}_{\mathcal{O}_X/X/S}$ .

┘

**Prop. (10.9.2.4) [ $\underline{\text{Quot}}_{\mathcal{F}/X/S}$  are Fpqc Sheaves].**  $\underline{\text{Quot}}_{\mathcal{F}/X/S}$  is a fpqc sheaf over  $\text{Sch}$ .

┘

*Proof:* This follows from the fact  $\mathcal{QCoh}/\text{Sch}$  is a fpqc sheaf (7.1.5.12). □

**Prop. (10.9.2.5) [Stratification by Hilbert Polynomials].** Let  $\mathcal{L} \in \text{Pic}(X_T)$ , then for any  $t \in T$ , the Hilbert polynomial of  $\mathcal{F}_t$  w.r.t. the line bundle  $\mathcal{L}_t$  on  $X_t$  is locally constant on  $T$ ?, and is stable under extension of residue fields, thus we can define  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  to represent the subfunctor of  $\underline{\text{Quot}}_{\mathcal{E}/X/S}$  consisting of pairs  $(\mathcal{F}, q)$  s.t.  $\mathcal{F}_t$  has Hilbert polynomial  $\Phi$  w.r.t.  $\mathcal{L}_t$  for all  $t \in T$ .

Thus if each  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  is representable,  $\underline{\text{Quot}}_{\mathcal{E}/X/S}$  is also representable, and there is a decomposition

$$\underline{\text{Quot}}_{\mathcal{E}/X/S} = \coprod_{\Phi \in \mathbb{Q}[\lambda]} \underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}.$$

┘

**Prop. (10.9.2.6) [Valuation Criterion].** Let  $X \rightarrow S$  be proper, then the morphism of functors  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}} \rightarrow h_S$  satisfies the discrete valuation criterion for properness.

┘

*Proof:* The valuation criterion says if  $R$  is a valuation ring with fraction field  $K$  with a morphism  $\text{Spec } R \rightarrow S$ , then any pairs  $(\mathcal{F}_K, q_K)$  where  $\mathcal{F}_K \in \text{Coh}(X_K)$  and  $q_K : \mathcal{E}_K \rightarrow \mathcal{F}_K$  a surjection can extend uniquely to a surjective map  $q_R : \mathcal{E}_R \rightarrow \mathcal{F}_R \in \text{Coh}(X_R)$  s.t.  $\mathcal{F}_R$  is flat over  $R$ .

For this, let  $j : X_K \rightarrow X_R$  be the open immersion, take  $\mathcal{F}_R$  to be the image of the map  $\mathcal{E}_R \rightarrow i_* \mathcal{E}_K \xrightarrow{i_* q_K} i_* \mathcal{F}_K$ , then  $i^* \mathcal{F}_R = \mathcal{F}_K$ , and  $\mathcal{F}_R$  is flat over  $R$  because it is torsion-free (5.4.1.12). Notice the properness of  $X/S$  is used to show that  $\text{Supp}(\mathcal{F}) \subset X_R$  is proper over  $R$ . □

**Lemma (10.9.2.7) [Preliminary Reductions].** Situation as in (10.9.2.2), then

- Let  $\nu \in \mathbb{Z}$ , then tensoring  $\mathcal{L}^\nu$  gives an isomorphism of functors  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}} \cong \underline{\text{Quot}}_{\mathcal{E}(\nu)/X/S}^{\Psi, \mathcal{L}}$ , where  $\Psi(\lambda) = \Phi(\lambda + \nu)$ .
- Let  $\varphi : \mathcal{E} \rightarrow \mathcal{G}$  be a surjective homomorphism in  $\text{Coh}(X)$ , then the corresponding natural transformation  $\underline{\text{Quot}}_{\mathcal{G}/X/S}^{\Phi, \mathcal{L}} \rightarrow \underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  is a closed subfunctor.
- If  $X/S$  is proper, let  $i : U \rightarrow X$  be an open subscheme, then  $\underline{\text{Quot}}_{i_* \mathcal{E}/U/S}^{\Phi, \mathcal{L}}$  is an open subfunctor of  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$ .

┘

*Proof:* 1 is obvious.

2: It suffices to show for any locally Noetherian scheme  $T \in \text{Sch}_S$  and a pair  $(\mathcal{F}, q) \in \underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}(T)$ , there exists a closed subscheme  $T'$  of  $T$  s.t. for any locally Noetherian scheme  $U$  and  $f : U \rightarrow T$ ,  $q_U : \mathcal{E}_U \rightarrow \mathcal{F}_U$  factors through  $\mathcal{G}_U$  iff  $f$  factors through  $T'$ . For this, just take  $T'$  to be the vanishing scheme of  $\ker(\varphi_T) \rightarrow \mathcal{E}_T \rightarrow \mathcal{F}_T$  (10.9.1.7) (or by direct verification).



3: Firstly the natural transformation  $\underline{\text{Quot}}_{i^*\mathcal{E}/U/S}^{\Phi, \mathcal{L}} \rightarrow \underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  is given by

$$\eta_T : (\mathcal{F}, i^*\mathcal{E}_T \rightarrow \mathcal{F}) \mapsto (\text{Im}(\mathcal{E}_T \rightarrow i_*\mathcal{F}), \text{adjunction}).$$

Notice restriction is a left inverse to this transformation. Then by inspection, this is an open subfunctor iff for any locally Noetherian scheme  $T \in \text{Sch}_S$  and a pair  $(\mathcal{F}, q) \in \underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}(T)$ , for any locally Noetherian scheme  $Q$  and  $f : Q \rightarrow T$ , the restriction of  $(\mathcal{F}, q)$  to  $Q$  is in  $\underline{\text{Quot}}_{i^*\mathcal{E}/U/S}^{\Phi, \mathcal{L}}(Q)$  iff  $\mathcal{F}_Q \rightarrow i_{Q*}i_Q^*\mathcal{F}_Q$  is injective. ?  $\square$

**Thm. (10.9.2.8) [Grothendieck].** Let  $S \in \text{NSch}_{\text{qc}}$  and  $f : X \rightarrow S$  a (quasi-/strongly)projective morphism,  $\mathcal{L}$  an  $f$ -ample line bundle on  $X$ , then for any  $\mathcal{E} \in \text{Coh}(X)$  and  $\Phi \in \mathbb{Q}[\lambda]$ , the functor  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  (10.9.2.5) is representable by a (quasi-/strongly)projective  $S$ -scheme.  $\square$

*Proof:* Cf. [Nit05]P24.

For the quasi-projective case, use (10.9.2.7) and the fact any coherent sheaf can be extended.  $\square$

**Remark (10.9.2.9).** Cf. [Hartshorne Appendix B, 3.4.1] has an example of a  $X \in \text{SmPrpr}^3/\mathbb{C}$  with a free  $G = \mathbb{Z}/(2)$ -action for which the quotient  $X/G$  is not a scheme. Which means that  $\underline{\text{Hilb}}_{X/\mathbb{C}}$  is not representable by a scheme.  $\square$

**Cor. (10.9.2.10).** Let  $S$  be a locally Noetherian scheme and  $f : X \rightarrow S$  be  $H$ -projective,  $\mathcal{L} = \mathcal{O}_{\mathbb{P}_S(V)}(1)|_X$ ,  $\mathcal{E}$  a coherent quotient sheaf of  $\mathcal{O}_X^{\oplus p}(\nu)$  where  $p > 0, \nu \in \mathbb{Z}$ , and  $\Phi \in \mathbb{Q}[\lambda]$ , then the functor  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  is representable by an  $H$ -projective  $S$ -scheme.  $\square$

**Lemma (10.9.2.11) [Altman-Kleinman].** Let  $S \in \text{NSch}$  and  $f : X \rightarrow S$  be a closed subscheme of some  $\mathbb{P}_S(V)$  where  $V$  is a finite locally free sheaf on  $S$ ,  $\mathcal{L} = \mathcal{O}_{\mathbb{P}_S(V)}(1)|_X$ ,  $\mathcal{E}$  a coherent quotient sheaf of  $f^*(W)(\nu)$  where  $W$  is a finite locally free sheaf on  $S$  and  $\nu \in \mathbb{Z}$ , and  $\Phi \in \mathbb{Q}[\lambda]$ , then the functor  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  is representable by a projective  $S$ -scheme that can be embedded in  $\mathbb{P}_S(F)$  for some finite locally free sheaf  $F$  on  $S$  that is an exterior power of  $W$  with a symmetric power of  $V$ .  $\square$

*Proof:* Cf. [Nit05]P24.  $\square$

**Lemma (10.9.2.12).** Let  $S \in \text{NSch}$  and  $f : X = \mathbb{P}_S(V) \rightarrow S$  where  $V$  is a finite locally free sheaf on  $S$ ,  $\mathcal{L} = \mathcal{O}_{\mathbb{P}_S(V)}(1)$ ,  $\mathcal{E} = f^*(W)(\nu)$  where  $W$  is a finite locally free sheaf on  $S$  and  $\nu \in \mathbb{Z}$ , and  $\Phi \in \mathbb{Q}[\lambda]$ , then the functor  $\underline{\text{Quot}}_{\mathcal{E}/X/S}^{\Phi, \mathcal{L}}$  is representable by a projective  $S$ -scheme that can be embedded in  $\mathbb{P}_S(F)$  for some finite locally free sheaf  $F$  on  $S$  that is an exterior power of  $W$  with a symmetric power of  $V$ .  $\square$

*Proof:* Cf. [Nit05]P24.  $\square$

### Examples of Quot Schemes

**Prop. (10.9.2.13) [Projective Spaces].**  $\underline{\text{Quot}}_{\mathcal{O}_{\mathbb{P}^n}^{\oplus(n+1)}/\mathbb{Z}/\mathbb{Z}}^{1, \mathcal{O}_{\mathbb{P}^n}} \cong \mathbb{P}_{\mathbb{Z}}^n$ , with the universal element the tautological quotient  $\mathcal{O}_{\mathbb{P}^n}^{\oplus(n+1)} \rightarrow \mathcal{O}_{\mathbb{P}^n}(1)$ .

Moreover, if  $\mathcal{E} \in \text{Coh}^{\text{free}}(S)$ , then  $\underline{\text{Quot}}_{\mathcal{E}/S/S}^{1, \mathcal{O}_S} \cong \mathbb{P}(\mathcal{E})$ , with the universal element the tautological quotient  $\pi^*(\mathcal{E}) \rightarrow \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ .  $\square$

*Proof:* For any surjective map  $\mathcal{E}_T \rightarrow \mathcal{L}$  where  $\mathcal{L}$  is a line bundle, locally  $H^0(\mathcal{L}) \cong H^0(\mathcal{E}_T)$  is of dimension  $n$  and thus  $\mathcal{L}$  is basepoint-free. Now the associated map  $\varphi_{\mathcal{L}} : T \rightarrow \mathbb{P}(\mathcal{E})$  ? pulls  $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$  to  $\mathcal{L}$ .  $\square$



**Def. (10.9.2.14) [Relative Grassmannians].** Let  $\mathcal{E} \in \text{Coh}(S)$ , define the **relative Grassmannian of**  $\mathcal{E}$  to be  $\text{Gr}_S(\mathcal{E}, k) = \underline{\text{Quot}}_{\mathcal{E}/S/S}^{k, \mathcal{O}_S}$ .

Then if  $\mathcal{E}$  is finite locally free, it is a quotient of the group scheme  $\text{GL}(n)_S$ , and when  $\mathcal{E} \cong \mathcal{O}_S^n$  and  $S = \mathbb{Z}$ , this is just the Grassmannian variety  $\text{Gr}(n, k)$  defined in (10.2.1.25).

In particular,  $\text{Gr}(n+1, 1) \cong \mathbb{P}_{\mathbb{Z}}^n$  by (10.9.2.13).  $\lrcorner$

*Proof:* First we prove for  $\mathcal{E}$  finite locally free: There is a natural action of  $\text{GL}(n)$  on  $\underline{\text{Quot}}_{\mathcal{E}/S/S}^{k, \mathcal{O}_S}$ , it is clearly a quotient map, and the kernel is a closed subgroup of  $\text{GL}(n)_S$  by (10.3.1.4). Thus  $\text{Gr}(\mathcal{E}, k)$  is a quotient of  $\text{GL}(n)$  by definition (10.1.1.19).

In general,  $\mathcal{E}$  is a locally a quotient of a finite locally free sheaf, thus  $\text{Gr}(\mathcal{E}, k)$  is locally projective by (10.9.2.7).  $\square$

**Prop. (10.9.2.15) [Quotient by Flat Projective Equivalence Relations].**  $\lrcorner$

*Proof:* Cf. [Nit05]P31.  $\square$

**Prop. (10.9.2.16) [Grassmannian Varieties are projective].** The Grassmannian variety  $\text{Gr}(\mathcal{E}, k)$  is locally projective, and when  $\mathcal{E}$  is locally free, it is projective, and if  $\mathcal{E}$  is trivial, it is  $H$ -projective.  $\lrcorner$

*Proof:* It injects into  $\mathbb{P}(\wedge^k \mathcal{E})$  by the natural transformation

$$T_S : (\mathcal{O}_T^n \rightarrow \mathcal{Q}) \mapsto (\wedge^k (\mathcal{O}_T^n) \rightarrow \wedge^k \mathcal{Q}).$$

In particular, the corresponding very ample line bundle is just  $\wedge^k \mathcal{F}$ , where  $f^* \mathcal{E} \rightarrow \mathcal{F}$  is the universal element and  $f : \text{Gr}(\mathcal{E}, k) \rightarrow S$ , by (10.9.2.13).  $\square$

**Cor. (10.9.2.17) [Grassmannian Varieties and Projective Space].** There is a canonical isomorphism  $G(n, n-1) \cong \mathbb{P}^{n-1}$  identifying sections with surjections from  $\mathcal{O}_X^n$ .  $\lrcorner$

**Prop. (10.9.2.18) [Flag Varieties].** The flag variety (10.2.1.27) represents the functor  $T \mapsto \{ \text{ the isomorphism classes of flags in } \mathcal{O}_T^n \text{ of the given dimensions} \}$ .  $\lrcorner$

**Cor. (10.9.2.19) [Flag Varieties are Projective].** The flag varieties are projective as they are closed subsets of a product of Grassmannian varieties.  $\lrcorner$

### 3 Picard Schemes

References are [Kle05].

**Notation (10.9.3.1).**

- For simplicity, we assume  $S \in \text{NSch}$  and restrict to  $\text{NSch}^{\text{sep}}/S$ .
- Recall a morphism  $X \in \text{NSch}/S$  is said to have integral geometric fibers if for any alg.closed field  $k$  and a morphism  $\text{Spec } k \rightarrow S$ ,  $X_k$  is integral.

$\lrcorner$

#### Picard Functors

**Def. (10.9.3.2) [Picard Functors].** Let  $X \in \text{Sch}/S$ , the **Picard-functor** is the functor

$$\underline{\text{Pic}}_X : \text{Sch}_S \rightarrow \text{Grp} : T \mapsto \text{Pic}(X_T) = H^1(X_T, \mathcal{O}_{X_T}^*).$$

The problem of this functor is that it is not a Zariski presheaf.

Define the **relative Picard functor**  $\widetilde{\text{Pic}}_{X/S} : \text{Sch}_S \rightarrow \text{Grp} : T \mapsto \text{Pic}(X_T)/\text{pr}_T^*(\text{Pic}(T))$ . Let the shification of this ring in the Zariski/étale/fppf topology denoted by  $\underline{\text{Pic}}_{X/S, \text{Zar}}$ ,  $\underline{\text{Pic}}_{X/S, \text{ét}}$ ,  $\underline{\text{Pic}}_{X/S}$ . Then there are maps of presheaves:

$$\underline{\text{Pic}}_X \rightarrow \widetilde{\text{Pic}}_{X/S} \rightarrow \underline{\text{Pic}}_{X/S, \text{Zar}} \rightarrow \underline{\text{Pic}}_{X/S, \text{ét}} \rightarrow \underline{\text{Pic}}_{X/S}.$$

┘

**Remark (10.9.3.3).** In the following we will frequently use a technical condition s.t.  $\mathcal{O}_S \rightarrow f_*(\mathcal{O}_X)$  is an isomorphism. This is true when  $X$  is a proper variety over  $S$ , by (7.12.5.2). ┘

**Prop. (10.9.3.4).** Assume  $\mathcal{O}_S \cong f_*\mathcal{O}_X$ , then the functor  $\mathcal{N} \rightarrow f^*\mathcal{N}$  is fully faithful from the category of finite locally free sheaves on  $S$  to the category of finite locally free sheaves on  $S$  s.t.  $f_*$  is a left partial inverse. ┘

*Proof:* For any finite locally free sheaf  $\mathcal{N}$ , there is an isomorphism  $\mathcal{N} \cong f_*f^*\mathcal{N}$  by checking locally, and for any other finite locally free sheaf  $\mathcal{N}'$ ,  $\text{Hom}(\mathcal{N}, \mathcal{N}')$  is also finite locally free, thus the natural map  $f^*\text{Hom}(\mathcal{N}, \mathcal{N}') \rightarrow \text{Hom}(f^*\mathcal{N}, f^*\mathcal{N}')$  is an isomorphism by checking locally. Thus  $\text{Hom}(\mathcal{N}, \mathcal{N}') \cong \text{Hom}(f^*\mathcal{N}, f^*\mathcal{N}')$ .  $\square$

**Def. (10.9.3.5) [Rigidification].** Assume  $f : X \rightarrow S \in \text{NSch}$  has a section  $g : S \rightarrow X$ , then for any  $T \in \text{Sch}/S$  and  $\mathcal{L} \in \text{Pic}(X_T)$ , a  **$g$ -rigidification of  $\mathcal{L}$**  is an isomorphism  $u : \mathcal{O}_T \cong g_T^*\mathcal{L}$ . ┘

**Prop. (10.9.3.6).** Let  $f : X \rightarrow S$  be a morphism of schemes with a section  $g$ , then for any  $T \in \text{Sch}/S$ , the group of isomorphism classes of pairs  $(\mathcal{L}, u)$  where  $\mathcal{L} \in \text{Pic}(X_T)$  and  $u$  is a  $g$ -rigidification of  $\mathcal{L}$ , is isomorphism to  $\underline{\text{Pic}}_{X/S}(T)$ . ┘

*Proof:* For any  $\mathcal{M} \in \text{Pic}(X_T)$ , let  $\mathcal{L} = \mathcal{M} \otimes (f_T^*g_T^*\mathcal{M})^{-1}$ , then  $\mathcal{L} = \mathcal{M} \in \text{Pic}(X_T)/\text{pr}_T^*\text{Pic}(T)$ , and there is a canonical isomorphism  $g_T^*\mathcal{L} \cong \mathcal{O}_T$ .

Conversely, if  $u : \mathcal{O}_T \cong g_T^*\mathcal{L}$  and there exists some  $\mathcal{N} \in \text{Pic}(T)$  s.t.  $v : \mathcal{L} \cong f_T^*\mathcal{N}$ , let  $w = g_T^* \circ u : \mathcal{O}_T \cong g_T^*\mathcal{L} \cong \mathcal{N}$ , then there are isomorphism of pairs:  $v : (\mathcal{L}, u) \cong (f_T^*\mathcal{N}, w)$ ,  $f_T^*w : (\mathcal{O}_{X_T}, \text{can}) \cong (f_T^*\mathcal{N}, w \circ \text{can})$ , thus  $(\mathcal{L}, u) = e$ .  $\square$

**Prop. (10.9.3.7) [No Automorphisms after Rigidification].** Let  $f : X \rightarrow S \in \text{NSch}$ . If for any  $T \in \text{Sch}/S$ ,  $f_T : X_T \rightarrow T$  satisfies  $\mathcal{O}_T \cong (f_T)_*\mathcal{O}_{X_T}$ , and  $f$  has a section  $g$ , then for any  $(\mathcal{L}, u)$  where  $u$  is a  $g$ -rigidification of  $\mathcal{L}$ ,  $\text{Aut}(\mathcal{L}, u)$  is trivial. ┘

*Proof:* For such an automorphism  $v : \mathcal{L} \rightarrow \mathcal{L}$ ,  $g_T^*(v) = \text{id}$ . Also notice  $\text{Hom}(\mathcal{L}, \mathcal{L}) = H^0(\text{Hom}(\mathcal{L}, \mathcal{L})) = H^0(\mathcal{O}_{X_T}) \cong H^0(\mathcal{O}_T)$ , thus  $v$  is tensoring a line bundle  $\mathcal{L}$  from  $\mathcal{O}_T$ . Thus  $g^*v$  is multiplying  $\mathcal{L}$ . So  $\mathcal{L} \cong \mathcal{O}_T$  and  $v$  is trivial.  $\square$

**Prop. (10.9.3.8) [Comparison Theorems].** Let  $f : X \rightarrow S \in \text{NSch}$ . If for any  $T \in \text{Sch}_S$ ,  $f_T : X_T \rightarrow T$  satisfies  $\mathcal{O}_T \cong (f_T)_*\mathcal{O}_{X_T}$ , then

- The morphisms  $\underline{\text{Pic}}_{X/S} \hookrightarrow \underline{\text{Pic}}_{X/S, \text{Zar}} \hookrightarrow \underline{\text{Pic}}_{X/S, \text{ét}} \hookrightarrow \underline{\text{Pic}}_{X/S}$  are injections. In particular,  $\underline{\text{Pic}}_{X/S}$  is separated in the fppf topology.
- If  $f$  has a section, then  $\widetilde{\text{Pic}}_{X/S} \cong \underline{\text{Pic}}_{X/S, \text{Zar}} \cong \underline{\text{Pic}}_{X/S, \text{ét}} \cong \underline{\text{Pic}}_{X/S}$ .
- If  $f$  has a section Zariski locally, then  $\underline{\text{Pic}}_{X/S, \text{Zar}} \cong \underline{\text{Pic}}_{X/S, \text{ét}} \cong \underline{\text{Pic}}_{X/S}$ .
- if  $f$  has a section étale locally, then  $\underline{\text{Pic}}_{X/S, \text{ét}} \cong \underline{\text{Pic}}_{X/S}$ .

┘

*Proof:* 1: It suffices to show that  $\widetilde{\text{Pic}}_{X/S} \rightarrow \text{Pic}_{X/S}$  are injective. For this, notice first  $\text{Pic}_{X/S}$  is also the fppf-sheaf associated to  $\text{Pic}_X$ , as any line bundle on  $T$  is Zariski-locally trivial. Then if  $\mathcal{L} \in \text{Pic}(X_T)$  is fppf-locally trivial, then for some fppf covering  $\pi : T' \rightarrow T$ ,  $(f_{T'})_* \pi_X^* \mathcal{L} \cong \mathcal{O}_{T'}$  by hypothesis. And this equals  $\pi^*(f_T)_* \mathcal{L}$  by flat base change. So  $(f_T)_* \mathcal{L}$  is an invertible sheaf (by hypothesis used on  $f_T$ ) and  $(f_T)_*(f_T)^* \mathcal{L} \cong \mathcal{L}$  because it is true after f.f. base change to  $T'$  (flat base change used).

2: By item 1, it suffices to show  $\widetilde{\text{Pic}}_{X/S}$  satisfies the fppf-descent, but this follows from the facts that  $\text{Pic}_{X/S}$  is a rigid fibered category (7.1.3.6),  $\mathcal{QCoh}/\text{Sch}$  is a stack (7.1.5.12) and  $\text{Pic}$  satisfy the fpqc descent (7.5.3.1).

3, 4 follow from 2 by base change. □

**Prop. (10.9.3.9) [Comparison of Points].** Let  $\text{Spec } A \in \text{Sch}/S$  where  $A$  is a local ring, then

- the natural maps  $\text{Pic}_X(A) \rightarrow \text{Pic}_{X/S}(A) \rightarrow \text{Pic}_{X/S, \text{Zar}}(A)$  are isomorphisms.
- If  $A$  is Artin local with alg. closed residue field, then  $\text{Pic}_X(A) \cong \text{Pic}_{X/S, \text{ét}}(A)$ .
- If  $A = k \in \mathbf{Field}$  and  $k = \bar{k}$ , then  $\text{Pic}_X(k) \cong \text{Pic}_{X/S}(k)$

┘

*Proof:* 1: Since  $A$  is local,  $\text{Pic}(\text{Spec } A) = 0$ , and it has a unique minimal point, thus 1 is clear.

2: Every étale  $A$ -algebra of f.t. is a direct product of copies of  $A^?$ , thus it is clear any fppf local scheme is a scheme.

3: In this case, any fppf covering of  $\text{Spec } k$  has a section, thus this follows from (10.9.3.8). □

**Prop. (10.9.3.10).** If  $X/S$  is proper of f.p., then an element in  $\text{Pic}_{X/S}(S)$  is trivial iff it is trivial Zariski-locally on  $S$ . ┘

*Proof:* Cf. [Neron Model, P202]. □

### Relative Effective Divisors

**Prop. (10.9.3.11) [Divisor Functor].** The divisor functor  $\text{Div}_{X/S} : \text{Sch}/S \rightarrow \text{Grp}$  is given by  $T \mapsto \{\text{relative effective divisors on } X_T/T\}$ . This is truly a functor by (7.8.1.14). ┘

**Thm. (10.9.3.12) [ $\text{Div}_{X/S}$  is Representable].** If  $S$  is a locally Noetherian scheme and  $X \rightarrow S$  is projective and flat, then  $\text{Div}_{X/S}$  (10.9.3.11) is representable by an open subscheme of the Hilbert scheme  $\text{Hilb}_{X/S}$ . ┘

*Proof:* Let  $H = \text{Hilb}_{X/S}$  and  $W \subset X \times H$  the universal closed subscheme. Then  $\text{pr}_H$  is projective and flat. Let  $V$  be the open loci of points of  $W$  s.t.  $W$  is an effective divisor,  $U = H \setminus q(W \setminus V)$ , then  $U$  is open in  $H$  and  $q^{-1}U$  is an effective divisor on  $X \times U/U$ . By definition of Hilbert scheme, for any effective divisor  $D$  on some  $X_T/T$  is the pullback of  $W$  via  $g$ . Now we show for any  $t \in D$ ,  $g(t) \in U$ . This follows from (7.8.1.12) and f.f. descent of regularness (7.1.5.28). Thus  $g$  factors through  $U$ , as  $U$  is open. □

**Def. (10.9.3.13) [Relative Abel Map].** The relative Abel map is the functor

$$A_{X/S}(T) : \mapsto \text{Div}_{X/S}(T) \rightarrow \widetilde{\text{Pic}}_{X/S}(T) : D \mapsto \mathcal{L}_{X_T}(D).$$

┘

**Def. (10.9.3.14) [Linear System Functor].** Let  $X \in \text{Sch}_S$  and  $\mathcal{L} \in \text{Pic}(X)$ , the **linear system functor**  $\underline{\text{LinSys}}_{\mathcal{L}/X/S} : \text{Sch}_S \rightarrow \text{Set}$  is defined to be the inverse image of  $\mathcal{L} \in \widetilde{\text{Pic}}_{X/S}$  in  $\underline{\text{Div}}_{X/S}$  via the Abel map(10.9.3.13).  $\lrcorner$

**Thm. (10.9.3.15) [ $\underline{\text{LinSys}}_{\mathcal{L}/X/S}$  is Representable].** Let  $X$  be a proper variety over  $S$  and  $\mathcal{L} \in \text{Pic}(X)$ , then  $\underline{\text{LinSys}}_{\mathcal{L}/X/S}$  is represented by some projective space  $\mathbb{P}(\mathcal{Q})$  over  $S$ .  $\lrcorner$

*Proof:* Cf. [Kle05]P25. ?  $\square$

**Prop. (10.9.3.16).** Assume  $X/S$  is proper and  $\mathcal{F} \in \text{Coh}(X)$ , then there exists a unique  $\mathcal{Q} \in \text{Coh}(S)$  with functorial isomorphism

$$q : \text{Hom}(\mathcal{Q}, \mathcal{N}) \cong f_*(\mathcal{F} \otimes f^*\mathcal{N})$$

for any  $\mathcal{N} \in \mathcal{QCoh}(S)$ .

And the formation of  $\mathcal{Q}$  commutes with base change.  $\lrcorner$

*Proof:* Cf. [Kle05]P24. ?  $\square$

**Prop. (10.9.3.17).** Situation as in(10.9.3.16), if  $S$  is a local ring with closed point  $s$ , the following are equivalent:

- The  $\mathcal{O}_S$ -module  $\mathcal{Q}$  is locally free.
- For all  $\mathcal{N} \in \mathcal{QCoh}(\mathcal{O}_S)$ , the functor  $\mathcal{N} \mapsto f_*(\mathcal{F} \otimes f^*\mathcal{N})$  is right exact.
- For all  $\mathcal{N} \in \mathcal{QCoh}(\mathcal{O}_S)$ , the natural map  $f_*(\mathcal{F}) \otimes \mathcal{N} \rightarrow f_*(\mathcal{F} \otimes f^*\mathcal{N})$  is an isomorphism.
- The natural map  $H^0(X, \mathcal{F}) \otimes k(s) \rightarrow H^0(X_s, \mathcal{F}_s)$  is a surjection.

$\lrcorner$

*Proof:* Cf. [Kle05]P24. ?  $\square$

**Prop. (10.9.3.18) [ $\underline{\text{Div}}_{X/S}$  and  $\underline{\text{LinSys}}_{\mathcal{P}/X_{\underline{\text{Pic}}_{X/S}/\underline{\text{Pic}}_{X/S}}$ ].** If  $X$  is a proper variety over  $S$  and  $\widetilde{\text{Pic}}_{X/S}$  is representable with the universal sheaf  $\mathcal{P}$ , then  $\underline{\text{Div}}_{X/S}$  with the relative Abel map(10.9.3.13) is isomorphic to  $\underline{\text{LinSys}}_{\mathcal{P}/X_{\underline{\text{Pic}}_{X/S}/\underline{\text{Pic}}_{X/S}}$  and also  $\mathbb{P}(\mathcal{Q})$  over  $\underline{\text{Pic}}_{X/S}$ , where  $\mathcal{Q}$  is the coherent sheaf associated to  $\mathcal{P}$ (10.9.3.16).  $\lrcorner$

*Proof:* It follows from definitions(10.9.3.14)(10.9.3.11) and(10.9.3.13) and the definition of a universal sheaf. The last assertion follows from(10.9.3.15).  $\square$

**Prop. (10.9.3.19) [ $\underline{\text{Div}}_{X/S}$  is Proper over  $\underline{\text{Pic}}_{X/S}$ ].** Let  $X/S$  be locally projective flat with integral geometric fibers, then

- $A_{X/S} : \underline{\text{Div}}_{X/S} \rightarrow \underline{\text{Pic}}_{X/S}(T)$  is proper.
- If  $\widetilde{\text{Pic}}_{X/T} \cong \text{Pic}_{X/T}$ , then  $A_{X/S} : \underline{\text{Div}}_{X/S} \rightarrow \underline{\text{Pic}}_{X/S}(T)$  is projective.

$\lrcorner$

*Proof:* If  $\widetilde{\text{Pic}}_{X/T} \cong \text{Pic}_{X/T}$ , then by(10.9.3.18), it is projective. For general case, notice proper satisfies fpqc descent, so we can base change to  $X \times_S X \rightarrow X$ , then it has a section, i.e. the diagonal section. Thus by(10.9.3.8),  $\widetilde{\text{Pic}}_{X/T} \cong \text{Pic}_{X/T}$  and  $A_{X \times_S X/X}$  is projective, thus  $A_{X/S}$  is proper.  $\square$

### Picard Schemes

**Def. (10.9.3.20) [Picard Scheme].** For the 5 different presheaves on  $\text{Sch}_S$  in (10.9.3.2), if one of them is representable, then all the sheaves after it are all isomorphic to it and representable, because fpqc sites are subcanonical. So it will make no confusion to call the representing scheme the **Picard scheme**.  $\lrcorner$

**Prop. (10.9.3.21).** If  $\underline{\text{Pic}}_{X/S}$  is representable, then it is locally of f.t. over  $S$ .  $\lrcorner$

*Proof:* Because  $S$  is locally Noetherian, by (7.8.4.2), it suffices to show that for any directed inverse system  $(T_i, f_{ii'}) \in \text{Aff}_S$ ,

$$\varinjlim_I \underline{\text{Pic}}_{X/S}(T_i) \cong \underline{\text{Pic}}_{X/S}(\varinjlim_I T_i).$$

For the rest, Cf. [Kle05]P33.  $\color{red}?$   $\square$

**Prop. (10.9.3.22) [Points of  $\underline{\text{Pic}}_{X/S}$ ].** If  $\underline{\text{Pic}}_{X/S}$  exists, then schematic points of  $\underline{\text{Pic}}_{X/S}$  corresponds to line bundles on the geometric fibers of  $X/S$ .  $\lrcorner$

*Proof:* This follows from the definition of schematic points and (10.9.3.9) item 3.  $\square$

**Thm. (10.9.3.23) [Grothendieck].** Let  $f : X \rightarrow S$  be locally projective, flat with integral geometric fibers, then

- $\underline{\text{Pic}}_{X/S, \text{ét}}$  is representable by a scheme separated and locally of f.p. over  $S$ .
- If moreover  $S$  is Noetherian and  $X/S$  is projective, then  $\underline{\text{Pic}}_{X/S, \text{ét}}$  is a disjoint union of open subschemes, each an increasing union of open quasi-projective  $S$ -schemes.

Moreover, by (10.9.3.8), if  $f$  has a section  $e$ , then there is a **Poincaré-class**  $p_{X/S} \in \text{Pic}(X \times \underline{\text{Pic}}_{X/S, \text{ét}})$  with a rigidification along  $e$ .  $\lrcorner$

*Proof:* Cf. [Kle05]P27.  $\color{red}?$   $\square$

**Cor. (10.9.3.24).** If  $S$  is Noetherian and  $X/S$  is projective flat with integral geometric fibers, then any qc locally closed subscheme of  $\underline{\text{Pic}}_{X/S}$  is quasi-projective.  $\lrcorner$

**Prop. (10.9.3.25) [Tangent Space of  $\underline{\text{Pic}}_{X/k}$ ].** Let  $k \in \text{Field}$  and  $X \in \text{Sch}/k$ . Assume  $\underline{\text{Pic}}_{X/k}$  is representable by a scheme and equals  $\underline{\text{Pic}}_{X/k, \text{ét}}$ , then  $T_e(\underline{\text{Pic}}_{X/k}) \cong H^1(X, \mathcal{O}_X)$ .  $\lrcorner$

*Proof:* By (10.2.2.2), the tangent space equals  $\ker(\underline{\text{Pic}}_{X/k}(k[\varepsilon]) \rightarrow \underline{\text{Pic}}_{X/k}(k))$ .

Take the first order thickening  $X_k \subset X_{k[\varepsilon]} = X_\varepsilon$ , there is an exact sequence  $0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{X_\varepsilon}^* \rightarrow \mathcal{O}_X^* \rightarrow 1$ , where  $\mathcal{O}_X \rightarrow \mathcal{O}_{X_\varepsilon}^* : a \mapsto 1 + a\varepsilon$ . And it is split. Thus we have a split exact sequence

$$0 \rightarrow H^1(\mathcal{O}_X) \rightarrow H^1(\mathcal{O}_{X_\varepsilon}^*) \rightarrow H^1(\mathcal{O}_X^*) \rightarrow 0.$$

Now  $\underline{\text{Pic}}_{X/S, \text{ét}}$  is also the shification of  $\widetilde{\underline{\text{Pic}}}_{X/k} : T \mapsto H^1(X_T, \mathcal{O}_{X_T}^*)$ , thus there is a natural map

$$H^1(\mathcal{O}_X) \rightarrow \ker(\underline{\text{Pic}}_{X/k}(k[\varepsilon]) \rightarrow \underline{\text{Pic}}_{X/k}(k)) = T_e(\underline{\text{Pic}}_{X/k}).$$

And it can be shown that this is a  $k$ -homomorphism.

To show this map is an isomorphism, by flat base change we can assume  $k$  is alg.closed. Then by (10.9.3.9), the maps  $H^1(\mathcal{O}_X^*) \rightarrow \underline{\text{Pic}}_{X/k}(k)$  and  $H^1(\mathcal{O}_{X_\varepsilon}^*) \rightarrow \underline{\text{Pic}}_{X/k}(k[\varepsilon])$  are isomorphisms, thus  $H^1(\mathcal{O}_X) \rightarrow T_e(\underline{\text{Pic}}_{X/k})$  is also an isomorphism by five lemma.  $\square$

**Thm. (10.9.3.26)[Grothendieck].** Assume  $X$  is proper over  $S$  integral, then there exists a non-empty open subscheme  $V \subset S$  s.t.  $\underline{\text{Pic}}_{X_V/V}$  is representable, and is a disjoint union of quasi-projective subschemes.  $\lrcorner$

*Proof:* Cf.[Kle05]P34.  $\square$

**Cor. (10.9.3.27)[Murre-Oort/Artin].** Let  $X$  be a proper scheme over a field  $k$ , then  $\underline{\text{Pic}}_{X/k}$  is representable by a disjoint union of quasi-projective subschemes.  $\lrcorner$

**Prop. (10.9.3.28).** Let  $X \rightarrow Y$  be a surjective morphism of proper schemes over a field  $k$ , then the dual map  $\underline{\text{Pic}}_{Y/k} \rightarrow \underline{\text{Pic}}_{X/k}$  is affine.  $\lrcorner$

*Proof:*  $\square$

**Prop. (10.9.3.29).** If  $X/S$  is projective and flat with integral geometric fibers, then any connected component of  $\underline{\text{Pic}}_{X/S}$  is clopen of f.t..  $\lrcorner$

*Proof:* By(10.9.3.23),  $\underline{\text{Pic}}_{X/S}$  is locally Noetherian, so each connected components are clopen(7.4.1.23), and they are of f.t. as  $\underline{\text{Pic}}_{X/S}^\Phi$  are(10.9.3.41).  $\square$

**Prop. (10.9.3.30) [Projectiveness of Subschemes].** Let  $S$  be Noetherian and  $X \rightarrow S$  be smooth projective with irreducible geometric fibers, then every quasi-compact closed subscheme of  $\underline{\text{Pic}}_{X/S}$  over  $S$  is projective.  $\lrcorner$

*Proof:* By(10.9.3.24), it suffices to show it is universally closed. As  $X \rightarrow S$  is both proper and flat, it suffices to show for  $X \rightarrow S$  f.f.. Then we can base change via the f.f. covering  $X \rightarrow S$ , thus we can assume it has a section, so  $\widetilde{\underline{\text{Pic}}}_{X/S} \cong \underline{\text{Pic}}_{X/S}$ . Then we use valuation criterion: for any valuation ring  $A$  with fraction field  $K$ , it suffices to extend a line bundle on  $X_K$  to a line bundle on  $X_A$ , because  $Z$  is closed and  $A$  is reduced. By replacing  $\mathcal{L}$  with  $\mathcal{L}(n)$ , we can assume  $\mathcal{L}$  has a global section, which implies  $\mathcal{L}$  is an effective Cartier divisor  $D$ , as  $X_K$  is integral. Notice  $A$  is regular and  $X_T/T$  is smooth, so  $X_T$  is regular?? and thus locally factorial. Then the closure of  $D$  in  $X_T$  is also a divisor?.  $\square$

**Cor. (10.9.3.31).** If  $k \in \text{Field}$  and  $X \in \text{SmProj}/k$ , then all connected components of  $\underline{\text{Pic}}_{X/k}$  are proper.  $\lrcorner$

*Proof:* This follows from(10.9.3.30) and(10.1.4.18).  $\square$

**Prop. (10.9.3.32)[Product Varieties].** If  $k \in \text{Field}$  and  $X, Y$  are two complete varieties over  $k$  with rational points, then  $\underline{\text{Pic}}_{X \times Y/k} \cong \underline{\text{Pic}}_{X/k} \times \underline{\text{Pic}}_{Y/k}$ , with  $p_{X \times Y} = \text{pr}_1^* p_X \otimes \text{pr}_2^* p_Y$ .  $\lrcorner$

*Proof:* This follows from see-saw principal(7.12.1.22), the theorem of the cube(7.12.1.23) and(10.9.3.6).  $\square$

$\underline{\text{Pic}}_{X/S}^0$

**Def. (10.9.3.33)[ $\underline{\text{Pic}}_{X/S}^0$ ].** If  $\underline{\text{Pic}}_{X/S}$  exists, then its identity component is a subgroup scheme(10.1.4.12), denoted by  $\underline{\text{Pic}}_{X/S}^0$ .  $\lrcorner$

**Prop. (10.9.3.34)[Projectiveness of  $\underline{\text{Pic}}_{X/k}^0$ ].** Let  $X$  be a projective variety over a field  $k$ , then  $\underline{\text{Pic}}_{X/k}^0$  is representable(10.9.3.23) and is quasi-projective. And it is projective if  $X/k$  is geo.normal.  $\lrcorner$

*Proof:* Cf. [Kle05]P37. ?

□

**Cor. (10.9.3.35) [Subfamily and Morphisms Relations].** The Poincaré class  $p_X$  in  $\text{Pic}(X \times \underline{\text{Pic}}^0(X))$  is the unique line bundle that satisfies  $p_b = b$  for a point  $b \in \underline{\text{Pic}}^0(X)$ , and  $p_0$  is trivial.

For a subfamily  $c$  of  $\text{Pic}^0(X)$  parametrized by an scheme  $T$  over  $k$ , there is a morphism

$$T \rightarrow \underline{\text{Pic}}^0(X) : t \mapsto c_t \in \underline{\text{Pic}}^0(X)(\kappa(t))$$

over  $k$ .

┘

*Proof:* A point  $b \in \underline{\text{Pic}}^0(X)$  is a morphism  $k(b) \rightarrow \underline{\text{Pic}}^0(X)$ , and the restriction of  $p$  to  $b$  is just  $b$ , by (10.9.3.23).

The second assertion follows from the first, because this subfamily corresponds to a morphism  $T \rightarrow \underline{\text{Pic}}^0(X)$ , and the restriction of  $p$  at the image of  $t$  in  $\underline{\text{Pic}}^0(X)$  is just the subfamily restricted at  $t$ , which is  $c_t$ . □

**Prop. (10.9.3.36) [ $\underline{\text{Pic}}_{X/k, \text{red}}^0$  is an Abelian Variety].** If  $k \in \text{Field}$  and  $X \in \text{SmProjVar}/k$ , then  $\underline{\text{Pic}}_{X/k, \text{red}}^0$  is an Abelian variety. ┘

*Proof:*  $\underline{\text{Pic}}_{X/S, \text{red}}^0$  is also a smooth proper connected algebraic group by (10.9.3.31)(10.9.3.29), thus it is an Abelian variety. □

**Prop. (10.9.3.37) [Dual Picard-Maps].** By functoriality, if  $X/S, X'/S$  are connected schemes with sections s.t.  $\underline{\text{Pic}}_{X/S}$  and  $\underline{\text{Pic}}_{X'/S}$  are representable, then the pullback along  $\varphi$  induces a dual homomorphism of group schemes:

$$\varphi^\vee : \underline{\text{Pic}}_{X'/S, \text{red}}^0 \rightarrow \underline{\text{Pic}}_{X/S, \text{red}}^0.$$

In other words, it is the unique morphism  $\underline{\text{Pic}}_{X'/S, \text{red}}^0 \rightarrow \underline{\text{Pic}}_{X/S, \text{red}}^0$  s.t.

$$(\varphi \times \text{id}_{\underline{\text{Pic}}_{X'/S, \text{red}}^0})^* p_{X'/S} = (\text{id}_X \times \varphi^\vee)^* p_{X/S}.$$

┘

**Prop. (10.9.3.38) [Double Picard-Map].** Let  $X/S$  be a connected scheme s.t.  $\widetilde{\underline{\text{Pic}}}_{X/S, \text{red}}$  is representable with a Poincaré class  $p_X \in \text{Pic}(X \times \underline{\text{Pic}}_{X/S, \text{red}}^0)$ , and  $(\underline{\text{Pic}}_{X/S, \text{red}}^0)^\vee$  is representable, then  $p_X$  satisfies  $(p_X)_0 = 0$ , thus inducing a **double Picard-map**  $\kappa_X : X \rightarrow (\underline{\text{Pic}}_{X/S, \text{red}}^0)^\vee$ . ┘

**Def. (10.9.3.39) [Néron-Severi Group Scheme].** Let  $X$  be a complete  $k$ -variety, the **Néron-Severi group**  $\text{NS}(X_{\bar{k}})$  of  $X_{\bar{k}}$  is defined to be

$$\text{NS}(X_{\bar{k}}) = \text{Pic}(X_{\bar{k}}) / \underline{\text{Pic}}^0(X_{\bar{k}}) = \pi_0(\underline{\text{Pic}}_{\bar{X}/\bar{k}})(\bar{k}) \text{ (9.1.11.4) (9.3.1.2),}$$

which in fact equals  $\pi_0(\underline{\text{Pic}}_{X_{k^s}/k^s})(k^s) = \text{Pic}(X_{k^s}) / \underline{\text{Pic}}^0(X_{k^s}) = \text{NS}(X_{k^s})$ .

Also denote

$$N^1(X_{\bar{k}}) = \text{Pic}(X_{\bar{k}}) / \text{Pic}^\tau(X_{\bar{k}}) = \text{NS}(X_{\bar{k}}) / \text{NS}(X_{\bar{k}})_{\text{tor}} \text{ (9.1.11.25) } = N^1(X_{k^s}).$$

┘

**Prop. (10.9.3.40) [Theorem of the Base].**  $\text{NS}(X_{\bar{k}})$  is f.g. In particular,  $N^1(X_{\bar{k}})$  is a finite free  $\mathbb{Z}$ -module, and its rank  $\rho(X)$  is called the **Picard number** of  $X$ . ┘



*Proof:* Cf. [Kleiman, Toward a numerical theory of ampleness, P334].  $\square$

**Prop. (10.9.3.41)** [ $\underline{\text{Pic}}_{X/S}^\Phi$ ]. If  $X/S$  is locally projective, flat with integral geometric fibers,  $\Phi \in \mathbb{Q}[\lambda]$ , let  $\underline{\text{Pic}}_{X/S}^\Phi$  denote the subfunctor of  $\underline{\text{Pic}}_{X/S}$  representing invertible sheaves  $\mathcal{L}$  with  $\chi(\mathcal{L}(n)) = \Phi(n)$ , then  $\underline{\text{Pic}}_{X/S}^\Phi$  are clopen subschemes of  $\underline{\text{Pic}}_{X/S}$  of f.t. and form a disjoint cover of it, and forming it commutes with base change. Moreover, if  $X/S$  is projective and  $S$  is Noetherian, then each  $\underline{\text{Pic}}_{X/S}^\Phi$  is quasi-projective over  $S$ .  $\sqsubset$

*Proof:* Cf. [Kle05]P60. ?

The first assertion follows from the fact Euler character is locally constant so  $\underline{\text{Pic}}_{X/S}^\Phi$  form a clopen disjoint covering of  $\underline{\text{Pic}}_{X/S}$  and use (10.8.1.5) and (10.8.1.7).  $\square$

**Prop. (10.9.3.42)** [Quasi-Coherence and Boundedness]. If  $\underline{\text{Pic}}_{X/S}$  is representable and  $\Lambda \subset \underline{\text{Pic}}_{X/S}$  be a subset with corresponding set of line bundles  $L$ , then  $\Lambda$  is qc iff  $L$  is bounded (9.1.11.24).  $\sqsubset$

*Proof:* If  $L$  is bounded by  $T \in \text{Sch}^{\text{ft}}/S$ , then there is a map  $\theta : T \rightarrow \underline{\text{Pic}}_{X/S}$  that  $\Lambda \subset \theta(T)$ . Notice  $T$  is a Noetherian space, thus so is  $\theta(T)$ , and  $\Lambda$  is qc.

Conversely, if  $\Lambda$  is qc, then it is contained in an open subscheme  $U \subset \underline{\text{Pic}}_{X/S}$  of f.t. over  $S$ , then  $U$  gives out a line bundle on  $X_T$  for some fppf covering  $T$  of  $U$ . Such  $T$  can be chosen to be f.t. over  $S$  as  $U$  is f.t., so  $L$  is bounded by  $T$ .  $\square$

**Cor. (10.9.3.43)** [Quasi-Coherence and Hilbert-Polynomials]. If  $S$  is Noetherian and  $X/S$  is projective and flat, and  $\underline{\text{Pic}}_{X/S}$  is representable, let  $\Lambda \subset \underline{\text{Pic}}_{X/S}$  be any subset and  $\Pi$  the corresponding set of Hilbert polynomials, then  $\#\Pi < \infty$  if  $\Lambda$  is qc, and  $\#\Pi = 1$  if  $\Lambda$  is connected.  $\sqsubset$

*Proof:* If  $\Lambda$  is qc, then  $L$  is bounded by some  $T \in \text{Sch}^{\text{ft}}/S$ , and then by locally constancy of Hilbert polynomials,  $\#\Pi < \infty$ .

If  $\Lambda$  is connected, then so is the induced reduced structure on its closure  $\overline{\Lambda}$ .  $\overline{\Lambda}$  gives out a line bundle on  $X_T$  for some fppf covering  $T$  of  $\overline{\Lambda}$ . Notice  $T \rightarrow \overline{\Lambda}$  is open and Hilbert polynomial is locally constant on  $T$ , so there is only one Hilbert polynomial type.  $\square$

### Torsion Components

**Prop. (10.9.3.44)**. If  $X/S$  is projective, flat with integral geometric fibers, then for  $n \in \mathbb{Z}_+$ ,  $[n] : \underline{\text{Pic}}_{X/S} \rightarrow \underline{\text{Pic}}_{X/S}$  is of f.t.  $\sqsubset$

*Proof:* Cf. [Kle61].  $\square$

**Prop. (10.9.3.45)**. If  $X/S$  is proper and  $\underline{\text{Pic}}_{X/S}$  exists, then  $\underline{\text{Pic}}_{X/S}^\tau$  is an open subgroup of f.t..  $\sqsubset$

*Proof:* Cf. [Kle61]P59.  $\square$

**Prop. (10.9.3.46)**. If  $X$  is a  $k$ -scheme s.t.  $\underline{\text{Pic}}_{X/k}$  is representable, then  $\underline{\text{Pic}}_{X/S}^\tau$  is an open subgroup, and forming it commutes with change of fields. Moreover, if  $X/S$  is projective, then  $\underline{\text{Pic}}_{X/S}^\tau$  is clopen of f.t..  $\sqsubset$

*Proof:* Using (10.1.4.19) and (10.9.3.21), it suffices to prove  $\underline{\text{Pic}}_{X/S}^\tau$  is of f.t. when  $X$  is projective over  $S$ . For this, it suffices to prove for  $k = \overline{k}$ . In this case, by (9.1.11.25) and (9.1.11.26), there is an algebraic  $k$ -scheme  $T$  and  $\mathcal{M} \in \mathcal{T}$  s.t. the corresponding map  $\theta : T \rightarrow \underline{\text{Pic}}_{X/k}$  satisfies  $\underline{\text{Pic}}_{X/S}^\tau \subset \theta(T)$ . Notice  $T$  is a Noetherian space, thus so is  $\theta(T)$ , and  $\Lambda$  is qc.  $\square$



**Cor. (10.9.3.47).** Let  $X$  is a projective variety over a field  $k$ , then  $\underline{\text{Pic}}_{X/k}^\tau$  is quasi-projective. And if  $X$  is geo.normal, then it is projective.  $\lrcorner$

*Proof:* By (10.9.3.46)(10.9.3.24),  $\underline{\text{Pic}}_{X/S}$  is quasi-projective. To show it is proper, it suffices to show for  $k = \bar{k}$ . In this case,  $\underline{\text{Pic}}_{X/k}^\tau$  is covered by f.m. translates of  $\text{Pic}_{X/k}^0$  as it is qc, and  $\underline{\text{Pic}}_{X/k}^0$  is projective by (10.9.3.34), so  $\underline{\text{Pic}}_{X/k}^\tau$  is also proper, as it is closed so is the scheme-theoretic image of these copies  $\underline{\text{Pic}}_{X/k}^0$ .  $\square$

**Prop. (10.9.3.48).** Let  $X/S$  be locally projective flat with integral geometric fibers, then  $\underline{\text{Pic}}_{X/S}^\tau$  is a clopen subgroup of  $\underline{\text{Pic}}_{X/S}$  of f.t., and forming it is compatible with base change. Moreover, if  $X/S$  is projective with  $S$  Noetherian, then  $\underline{\text{Pic}}_{X/S}^\tau$  is quasi-projective.  $\lrcorner$

*Proof:* Cf. [Kle05]P58.  $\square$

**Cor. (10.9.3.49) [Torsion Components are Projective].** If  $X/S$  is smooth projective with integral geometric fibers and  $S$  is Noetherian, then  $\underline{\text{Pic}}_{X/S}^\tau$  is projective.  $\lrcorner$

*Proof:* This follows from (10.9.3.48) and (10.9.3.30).  $\square$

**Prop. (10.9.3.50).** If  $S$  is Noetherian and  $X/S$  is locally projective flat with integral geometric fibers, then for any  $s \in S$  with residue field  $k(s)$  and an alg.closure  $\bar{k}(s)$ , the group

$$\underline{\text{Pic}}_{X_{\bar{k}(s)}/\bar{k}(s)}^\tau(\bar{k}(s)) / \underline{\text{Pic}}_{X_{\bar{k}(s)}/\bar{k}(s)}(\bar{k}(s))$$

is a finite group, and its order is uniformly bounded.  $\lrcorner$

*Proof:* It is finite by (10.9.3.48), and the number of connected components of  $\underline{\text{Pic}}_{X_{\bar{k}(s)}/\bar{k}(s)}^\tau$  is constant for a non-empty open subset of  $S$  by [EGA 4.3, 9.7.9], so it is bounded by Noetherian induction.  $\square$

## 4 Picard Spaces

**Thm. (10.9.4.1) [Representability by Algebraic Spaces, Artin].** If  $X \rightarrow S$  is proper flat and f.p. morphism of algebraic spaces s.t. forming  $f_*\mathcal{O}_X$  commutes with base change, then  $\underline{\text{Pic}}_{X/S}$  is representable by an algebraic space, and is locally of f.p. over  $S$ .  $\lrcorner$

*Proof:* Cf. [Artin, Algebraization of formal moduli: 1]Thm7.3.  $\square$

## 5 Moduli of Curves

**Def. (10.9.5.1) [Smooth Curves of Genus  $g$ ].** Given a scheme  $S$ , there is a category  $\mathcal{M}_g$  fibered in sets over  $\text{Sch}/S$  where  $\mathcal{M}_g(T)$  is the set of smooth and proper morphisms of schemes  $C \rightarrow T$  that the fibered are all geometrically connected curves of genus  $g$ .

Similarly there is a category  $Z_g$  fibered in sets over  $\text{Sch}/S$  of smooth pointed curves of genus  $g$   $\lrcorner$

## 10.10 Geometric Invariant Theory

## 10.11 Group Algebraic Spaces

Main references are [Sta]Chap76.

**Notation (10.11.0.1).**

- Use notations defined in Algebraic Stacks 10.8.
- Fix  $S \in \text{Sch}$  and  $B \in \text{AlgSp}/S$ .  $\text{AlgSp}/B = (\text{AlgSp}/S)/B$ .

┘

**Def. (10.11.0.2) [Group Algebraic Spaces].** The category  $\text{GrpSp}/B$  of **group algebraic space** over  $B$  is defined to be

$$\text{GrpSp}/B = \text{Sh}_{\text{fppf}}^{\text{grp}}/B \cap \text{AlgSp}/B \subset \text{Sh}_{\text{fppf}}/B.$$

┘

### 1 Quotient of Groupoids

Main references are [Quotient Spaces Modulo Algebraic Groups, Kollar, 1997].

**Def. (10.11.1.1) [Quotients of Groupoids].** Let  $U \in \text{AlgSp}/B$ ,  $j : R \rightarrow U \times_B U$  be a pre-relation on  $U$  over  $B$ ,  $\varphi : U \rightarrow X \in \text{AlgSp}/B$ , then

- For  $u \in |U|$ , the  **$R$ -orbit** of  $u$  is the equivalent class of  $u \in |U|$  generated by  $|R| \subset |U| \times |U|$ .
- For  $T \subset |U|$ ,  $T$  is called  **$R$ -invariant** if  $s^{-1}(T) = t^{-1}(T) \subset |R|$ .
- $\varphi$  is said to be  **$R$ -invariant** if for it equalizes the two maps  $s, t : R \rightarrow U$ .
- $\varphi$  is said to be **set-theoretically  $R$ -invariant** if for any  $\text{Spec } k \in \text{Sch}/B, k = \bar{k}$ ,  $\varphi(k)$  equalizes the two maps  $s, t : R(k) \rightarrow U(k)$ .
- $\varphi$  is said to **separate  $R$ -orbits** if it is set-theoretically  $R$ -invariant and for any  $\text{Spec } k \in \text{Sch}/B, k = \bar{k}$  and  $u, u' \in U(k)$ ,  $\varphi(u) = \varphi(u') \in X(k)$  implies  $u, u'$  are in the same  $R$ -orbits.
- $\varphi$  is said to be an **orbit space for  $R$**  if it is  $R$ -invariant and surjective, and separates  $R$ -orbits.
- $\varphi$  is said to be a **course quotient for  $R$**  if it is a categorical quotient and it is an orbit space for  $R$ .
- If  $S = B, U, R, X \in \text{Sch}$ , then  $\varphi$  is said to be a **course quotient in schemes for  $R$**  if it is a categorical quotient in schemes and it is an orbit space for  $R$ .
- If  $\varphi$  is  $R$ -invariant, The **sheaf of  $R$ -invariant functions**  $(\varphi_* \mathcal{O}_U)^R$  as the étale  $\mathcal{O}_X$ -subalgebra of  $\varphi_* \mathcal{O}_U$  which is the equalizer of two maps induced from  $s, t : R \rightarrow U$ .

┘

**Prop. (10.11.1.2).** Situation as in (10.11.1.1), if  $R, U$  are locally of f.t. over  $B$ ,  $\varphi$  is an orbit space for  $R$  iff it is  $R$ -invariant and for any  $\text{Spec } k \in \text{Sch}/B, k = \bar{k}$ ,

$$U(k)/(\text{equivalent relations generated by } j(R(k))) \rightarrow X(k)$$

is bijective.

┘

*Proof:* Cf. [Sta]04A0.

□

**Def. (10.11.1.3) [Good Quotients].** Situation as in (10.11.1.1),  $\varphi$  is said to be a **good quotient** if

- $\varphi$  is affine, surjective,  $R$ -invariant.
- For any base change  $\varphi'$  of  $\varphi$ ,  $|\varphi'|$  is a closed map, and  $|\varphi|(Z_1 \cap Z_2) = |\varphi|(Z_1) \cap |\varphi|(Z_2) \subset |X|$  for any closed subsets  $Z_1, Z_2 \subset |U|$ .
- $\mathcal{O}_X \rightarrow (\varphi_* \mathcal{O}_U)^R$  is an isomorphism.

┘

**Def.(10.11.1.4) [Geometric Quotients].** Situation as in (10.11.1.1),  $\varphi$  is said to be a **geometric quotient** if

- $\varphi$  is an orbit space for  $R$ ,
- $\varphi$  is universally submersive,
- $\mathcal{O}_X \rightarrow (\varphi_* \mathcal{O}_U)^R$  is an isomorphism.

┘

**Thm.(10.11.1.5) [Kollar].** Let  $S \in \text{Sch}$  be excellent,  $G \in \text{AlgGrp}_{\text{Aff}}/S$ ,  $X \in \text{AlgSp}^{\text{sep,ft}}/S$ . Let  $m : G \times X \rightarrow X$  be a proper  $G$ -action on  $X$ , and one of the following holds:

- $G$  is a reductive group scheme over  $S$ ,
- $S = \text{Spec } k$  where  $k \in \text{Field}$ ,  $\text{char } k > 0$ .

┘

*Proof:* then a geometric quotient  $\varphi : X \rightarrow X/G$  exist (10.11.1.4), and  $X/G \in \text{AlgSp}^{\text{sep,ft}}/S$ . Cf.[Quotient Schemes modulo Algebraic Groups, Kollar]P35.  $\square$

## 2 Quotients of Schemes

**Prop.(10.11.2.1).** Let  $u_0, u_1 : X_1 \rightarrow X_0$  be an equivalence relation on the algebraic scheme  $X_0$  over  $R_0$ . Assume that

- $u_0 : X_1 \rightarrow X_0$  is locally free of rank  $r$ .
- For all  $x_0 \in X$ ,  $u_0(u_1^{-1}(x))$  is contained in an open affine subscheme of  $X_0$ .

Then a quotient  $u : X_0 \rightarrow X$  exists. Moreover,  $u$  is locally free of rank  $r$ .

┘

*Proof:* Cf[?Mil17b]P597.  $\square$

## 10.12 Higher Dimensional Geometry

1 Bend and Break

2 Cone Theorem

3 Homological Methods

Prop. (10.12.3.1) [Birkar-Cascini-Hacon-Mkernan].

└

4 Minimal Model Program

### 10.13 Toric Varieties

# 11 | Algebraic $K$ -Theory of Schemes

## 11.1 Algebraic $K$ -Theory

### 1 Introduction

Algebraic  $K$ -theory is about natural constructions of cohomology theories/spectra from algebraic data such as commutative rings, symmetric monoidal categories and various homotopy theoretic refinements of these.

When applied to the stack of vector bundles then algebraic  $K$ -theory subsumes topological  $K$ -theory and also differential  $K$ -theory.

### 2 $K$ -Theory of Categories

#### $K$ -Groups of Triangulated Categories

**Def. (11.1.2.1) [ $K$ -Groups of Triangulated Categories].** Let  $\mathcal{D}$  be a triangulated categories, then the  $K$ -group of  $\mathcal{D}$  is defined to be the quotient

$$\bigoplus_{X \rightarrow Y \rightarrow Z \text{ distinguished}} 1 \rightarrow \bigoplus_{X \in \mathcal{D}} \rightarrow K_0(\mathcal{D}) \rightarrow 0$$

where  $e_{X \rightarrow Y \rightarrow Z}$  is mapped to  $e_Y - e_X - e_Z$ . ┘

**Prop. (11.1.2.2) [Naturality].**

- An exact functor between triangulated categories induce a map of their corresponding  $K$ -groups,
- A bi-exact bifunctor  $F : \mathcal{D} \times \mathcal{D}' \rightarrow \mathcal{E}$  (4.8.4.9) induces a bilinear map  $K_0(\mathcal{D}) \times K_0(\mathcal{D}') \rightarrow K_0(\mathcal{E})$  sending  $([X], [X'])$  to  $F(X, X')$ .
- A  $\delta$ -functor from an Abelian category to a triangulated category induces a map of their corresponding  $K$ -groups (Notice these two  $K$ -groups are defined differently).
- A functor from an exact category to a triangulated category that sends exact sequences to distinguished triangles induces a map of their corresponding  $K$ -groups. ┘

#### $K$ -Theory of quasi-exact categories

**Def. (11.1.2.3) [Quasi-Exact Categories].** A **quasi-exact category** is a pair  $(\mathcal{C}, \mathcal{E})$  where  $\mathcal{C}$  is a pointed balanced small additive category and  $\mathcal{E}$  is a set of short sequences

$$0 \rightarrow X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z \rightarrow 0$$

(where we call  $\varphi$  an **admissible monomorphism** and  $\psi$  an **admissible epimorphism**) satisfying the following:

- **Ex1:** For any complex  $0 \rightarrow X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z \rightarrow 0$  in  $\mathcal{E}$ ,  $\varphi$  is the kernel of  $\psi$  and  $\psi$  is the cokernel of  $\varphi$ .
- For any  $X, Y \in \mathcal{C}$ ,  $0 \rightarrow X \rightarrow X \oplus Y \rightarrow Y \rightarrow 0$  is in  $\mathcal{E}$ .
- $\mathcal{E}$  is closed under isomorphisms.
- Admissible monomorphisms are stable under compositions and base change by admissible epimorphisms.
- Admissible epimorphisms are stable under compositions and base change by admissible monomorphisms.

┘

**Def. (11.1.2.4).** For any quasi-exact category  $(\mathcal{C}, \mathcal{E})$ , we can define the category  $Q(\mathcal{C})$  as the category s.t.

- $\text{Ob}(Q(\mathcal{C})) = \text{Ob}(\mathcal{C})$ ,
- For  $X, Y \in \text{Ob}(Q(\mathcal{C}))$ ,  $\text{Map}_{Q(\mathcal{C})}(X, Y)$  consists of isomorphism classes of roofs  $(X \xleftarrow{\varphi} S, S \xrightarrow{f} Y)$  s.t.  $\varphi$  is an admissible epimorphism and  $f$  is an admissible monomorphism.
- The compositions are given by base change.

The conditions in (11.1.2.3) ensures that  $Q(\mathcal{C})$  is truly a category.

┘

**Def. (11.1.2.5) [ $K$ -Groups of Quasi-Exact Categories].** For any quasi-exact category  $(\mathcal{C}, \mathcal{E})$ ,  $Q(\mathcal{C})$  is path-connected as  $\mathcal{C}$  is pointed, so we can define the  **$K$ -group of quasi-separated categories** as

$$K_i(\mathcal{C}, \mathcal{E}) = \pi_{i+1}(\mathbb{B}(Q(\mathcal{C}))).$$

┘

**Prop. (11.1.2.6) [ $K_0$ -Groups].** For any quasi-exact category  $(\mathcal{C}, \mathcal{E})$ ,  $K_0(\mathcal{C}, \mathcal{E})$  can be described as the kernel:

$$\bigoplus_{[0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0] \in \mathcal{E}} \mathbb{Z} \rightarrow \bigoplus_{A \in \mathcal{C}} \mathbb{Z} \rightarrow K_0(\mathcal{A}) \rightarrow 0,$$

where  $e_{0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0}$  is mapped to  $e_B - e_A - e_C$ .

┘

*Proof:*

□

### 3 $K$ -Theory of Rings

**Def. (11.1.3.1) [ $K$ -Theories of Rings].** For  $R \in \mathcal{C}\text{Ring}$ , Let  $\text{Proj}_R^{\text{fg}}$  be the 1-category of f.g. projective  $R$ -modules, and let  $\iota \text{Proj}_R^{\text{fg}}$  be the maximal subgroupoid, then it is a  $E_\infty$ -space. Let

$$K(R) = (\iota \text{Proj}_R^{\text{fg}})^{\infty\text{-ab}} \text{ (4.9.2.7)}.$$

And let

$$K_0(R) = \pi_0(K(R)), \quad K_j(R) = \pi_j(K(R), 0).$$

┘



**Prop. (11.1.3.2) [Ring Structure].** There is an  $\mathbb{E}_\infty$ -Ring structure on  $K_{\geq 0}(R)$ , so

$$K_*(R) = \bigoplus_{j \geq 0} K_j(R)$$

has a graded ring structure. ┘

**Prop. (11.1.3.3) [Matsumoto].** For  $k \in \mathbf{Field}$ , there are natural isomorphisms

$$K_i^{\text{Mil}}(k) \cong K_i(k)$$

for  $i = 0, 1, 2$ , thus inducing a map

$$K_*^{\text{Mil}}(k) \rightarrow K_*(k)$$

that is not necessarily an isomorphism. ┘

*Proof:* ? □

#### 4 K-Theory of Schemes

**Thm. (11.1.4.1) [Thomason, Quillen].** Let  $X$  be an affine regular and separated scheme, and  $X = U \cup V$  is an affine open cover, then there is a Cartesian square of spectra

$$\begin{array}{ccc} K(X) & \longrightarrow & K(U) \\ \downarrow & & \downarrow \\ K(V) & \longrightarrow & K(U \cap V) \end{array}$$

*Proof:* ┘

**Cor. (11.1.4.2) [Descent Spectral Sequence].** There is a spectral sequence

$$H^p(X; \pi_q(K)) \Rightarrow K_{p-q}(X).$$

*Proof:* ? ┘

**Def. (11.1.4.3) [Graded Determinant Map].** For  $X \in \mathbf{Sch}$  affine or regular, let  $\mathbf{Pic}^{\mathbb{Z}}(X) = \mathbf{Pic}(X) \rtimes H^0(X; \mathbb{Z})$ , where  $H^0(X; \mathbb{Z})$  acts on  $\mathbf{Pic}(X)$  locally by  $n \cdot \mathcal{L} = \mathcal{L}^{(-1)^n}$ . Then there is a graded determinant map

$$(\det, \text{rank}) : \iota \text{Proj}^{\text{fg}}(X) \rightarrow \mathbf{Pic}^{\mathbb{Z}}$$

that is a morphism of  $\mathbb{E}_\infty$ -rings:

$$\begin{array}{ccc} \det(M \oplus N) & \xrightarrow{\cong} & \det(M) \otimes \det(N) \\ \downarrow \cong, (-1)^{\text{rank}(M) \text{rank}(N)} & & \downarrow \cong \\ \det(N \oplus M) & \xrightarrow{\cong} & \det(N) \otimes \det(M) \end{array}$$

is commutative. Thus by universal property of group completion, we get a map

$$K(X) \rightarrow \mathbf{Pic}^{\mathbb{Z}}(X).$$

Thus we have maps

$$K_j(X) \xrightarrow{(\det, \text{rank})} \pi_j(\mathbf{Pic}^{\mathbb{Z}}(X)),$$

and the kernel is denoted by  $SK_j(X)$ . ┘

**Prop. (11.1.4.4).** The graded determinant map (11.1.4.3) induces a surjective ring homomorphism

$$K_0(X) \rightarrow \mathbf{Pic}^{\mathbb{Z}}(X).$$

┘

*Proof:*  $\mathcal{L} \otimes \mathcal{O}_X^{n-1}$  is mapped to  $(\mathcal{L}, n)$ , and these generate  $\mathbf{Pic}^{\mathbb{Z}}(X)$ . □

**Prop. (11.1.4.5).** If  $R$  is a local ring, then there is an isomorphism

$$K_1(R) \xrightarrow{(\det, \text{rank})} \pi_1(\mathbf{Pic}^{\mathbb{Z}}(R)) \cong R^*.$$

┘

**Prop. (11.1.4.6).** Let  $X$  be a an irreducible regular Noetherian scheme of dimension 1, then there is an isomorphism

$$K_0(X) \xrightarrow[\cong]{(\det, \text{rank})} \mathbf{Pic}^{\mathbb{Z}}(X).$$

┘

*Proof:* [Motives at p]L7P4. ? □

**Prop. (11.1.4.7) [Properties of K-Groups].** For  $j \in \mathbb{Z}$ ,  $K_j : \text{Sch}^{\text{op}} \rightarrow \mathcal{A}b$  are functors satisfying the following properties:

- there is a rank map  $\text{rank} : K_0(X) \rightarrow H^0(X, \mathbb{Z})$ , which is an isomorphism if  $X$  is a local scheme.
- If  $X$  is qcqs and  $X = U \cup V$  is an open cover, there is a long exact sequence

$$\cdots \rightarrow K_j(X) \rightarrow K_j(U) \oplus K_j(V) \rightarrow K_j(U \cap V) \rightarrow K_{j-1}(X) \rightarrow \cdots.$$

- If  $X$  is qcqs and  $Y \rightarrow X$  is a closed immersion s.t.  $Y$  is regular, then there are natural maps

$$i_* : K_j(Y) \rightarrow K_j(X)$$

that fits into a long exact sequence

$$\cdots \rightarrow K_j(Y) \xrightarrow{i_*} K_j(X) \rightarrow K_j(X \setminus Y) \rightarrow K_{j-1}(Y) \rightarrow \cdots.$$

- If  $X$  is regular, then the projection  $X \times \mathbb{A}^1 \rightarrow X$  induces isomorphisms

$$K_j(X) \cong K_j(X \times \mathbb{A}^1).$$

- If  $X$  is qcqs, there is a natural isomorphism

$$K_j(X)\{0\} \oplus K_j(X)\{0(1) - 0\} \cong K_j(\mathbb{P}_X^1). ?$$

┘

*Proof:* ? □

**Def. (11.1.4.8) [K-Groups of Schemes].** For  $X \in \text{Sch}_{\text{qcqs}}$ , its algebraic K-theory is defined to be the spectrum

$$K(X) = \varprojlim_{\text{Spec } R \rightarrow X} K(R).$$

It follows from (11.1.4.1) that this defines a ┘

**Prop. (11.1.4.9)[Cup Products, Push and Pull].** For any regular separated Noetherian scheme  $X$ , there is a cup product map

$$\cup : K_i(X) \times K_j(X) \rightarrow K_{i+j}(X).$$

And if  $f : X' \rightarrow X$  is a morphism of regular separated Noetherian schemes, there is a graded ring homomorphism  $f^* : K_*X \rightarrow K_*X'$ . And if  $f$  is proper, there are pushforward maps  $f_* : K_i(X') \rightarrow K_i(X)$ . And in this case, they satisfy the projection formula

$$f_*(f^*a \cup b) = a \cup f_*b.$$

There is a canonical inclusion  $\mathcal{O}^*(X) \subset K_1(X)$ , and the restriction of  $f^*(\mathcal{O}^*(X)) \subset \mathcal{O}^*(X')$ , which is just the pullback map. And if  $f$  is finite flat, then the image of  $f_*(\mathcal{O}^*(X')) \subset \mathcal{O}^*(X)$ , which is just the norm map.

The cup product map restricted to  $\mathcal{O}^*(X) \times \mathcal{O}^*(X) \rightarrow K_2(X)$  is called the **universal norm map**, denoted by  $\{u, v\}$ . ┘

*Proof:* ? □

**Conj. (11.1.4.10)[Beilinson-Parshin].** For  $k \in \mathbf{Field}$ ,  $\#k < \infty$ ,  $X \in \mathbf{SmProj}/k$ ,  $i > 0$ ,  $K_i(X) \otimes \mathbb{Q} = 0$ . ┘

*Proof:* □

**Remark (11.1.4.11).** In fact, when  $\dim X = 0$  this is done by Quillen's computation of K-groups of fields, Cf.[On the Cohomology and K-theory of the General Linear Groups over a Finite Field, Quillen, 1972].

when  $\dim X = 1$ ,  $K_i(X)$  are in fact finite, by [Finite Generation of K-Groups of a Curve over a Finite Field, Don 1982]. ┘

### Comparison Maps

**Prop. (11.1.4.12)[Logarithm Differential Maps].** For  $S \in \mathcal{C}\mathbf{Ring}$ ,  $R \in \mathcal{C}\mathbf{Ring}/S$ ,  $X = \mathbf{Spec} R$  and  $q \in \mathbb{N}$ , there are homomorphisms

$$\mathrm{dlog} : K_q(R) \rightarrow \Omega_{R/S}^q,$$

satisfying

- $\mathrm{dlog}(a \cup b) = \mathrm{dlog} a \cup \mathrm{dlog} b$ .
- If  $a \in R^* \subset K_1(R)$ , then  $\mathrm{dlog} a = a^{-1}da \in \Omega_{R/S}^1$ .
- On  $K_0(R)$ ,  $\mathrm{dlog}$  is the degree map.
- If  $R'/R$  is a finite flat extension of regular rings, then

$$\mathrm{tr}_{R'/R}^\Omega \circ \mathrm{dlog}_{R'} = \mathrm{dlog}_R \mathrm{tr}_{R'/R}^K.$$

*Proof:* □

**Prop. (11.1.4.13).** If  $S \in \mathcal{C}\mathbf{Ring}$ , and  $Y = \mathbf{Spec} R$  is an smooth affine precurve over  $S$ , with a smooth completion  $X$  over  $S$ , and  $Z = X \setminus Y$ , then in fact the image of  $\mathrm{dlog}$  is contained in

$$H^0(X, \Omega_{X/S}^q(\log Z)).$$

*Proof:* Cf.[Scholl, P411] ?. □

$K_2$  of Curves

**Prop. (11.1.4.14) [Faddeev's Exact Sequence].** Let  $X$  be a non-singular complete curve over a field  $k$ . Then for any point  $P \in X$ , consider the tame symbol  $\{-, -\}_P$  at  $P$  (3.10.1.5), then there is an exact sequence

$$0 \rightarrow K_2(X) \rightarrow K_2(R(X)) \xrightarrow{\prod_{P \in X} \{-, -\}_P} \kappa(P)^\times.$$

┘

*Proof:* ?

□

**Prop. (11.1.4.15) [Étale Abel-Jacobi Maps].** If  $Y$  is a smooth variety over a number field  $F$ ,  $p \in \mathbf{P}$ ,  $n, j \in \mathbb{Z}_+$ , and  $H^{j+1}(Y_{\mathbb{Q}}, \mathbb{Q}_p(n))^{\text{Gal}_F} = 0$ , then there is an **Abel-Jacobi map**

$$\text{AJ}_{n,j} : K_{2n-j-1}(Y) \rightarrow H^1(\text{Gal}_F, H^j(Y_{\mathbb{Q}}, \mathbb{Q}_p)(n)).$$

In case  $Y$  is a curve,  $j = 1$  and  $n = 2$ , then the Abel-Jacobi map is defined integrally:

$$\text{AJ}_2 : K_2(Y) \rightarrow H^1(\text{Gal}_F, H^1(Y_{\mathbb{Q}}, \mathbb{Z}_p)(2)).$$

It is defined as follows: There is a Chern class map  $c_2 : K_2(Y) \rightarrow H^2(Y, \mathbb{Z}_p(2))$ , and by torsion-freeness (9.4.7.3) and weight reason ?, the first column in the Hochschild-Serre spectral sequence

$$E_2^{i,j} = H^i(\text{Gal}_F, H^j(Y_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(n))) \Rightarrow H^{i+j}(Y, \mathbb{Z}_p(n))$$

vanishes. So there is a low degree exact sequence

$$0 \rightarrow H^2(\text{Gal}_F, H^0(Y_{\mathbb{Q}}, \mathbb{Z}_p(2))) \rightarrow H^2(Y, \mathbb{Z}_p(2)) \xrightarrow{\text{H-S}_2} H^1(\text{Gal}_F, H^1(Y_{\mathbb{Q}}, \mathbb{Z}_p(2))) \rightarrow H^3(\text{Gal}_F, H^0(Y_{\mathbb{Q}}, \mathbb{Z}_p(2))).$$

Then define  $\text{AJ}_2 = -\text{H-S}_2 \circ c_2$ .

┘

*Proof:*

□

## 11.2 Brauer-Grothendieck Groups

References are [Central Simple Algebras and Galois Cohomology, Gille and Szamuely] and [The Brauer-Grothendieck Group].

### 1 Brauer Groups

**Def. (11.2.1.1) [Brauer Groups].** For  $X \in \text{Sch}$ , the **Brauer group**  $\text{Br}(X)$  is defined to be  $\text{Br}(X) = H_{\text{et}}^2(X, \mathbb{G}_m) = H_{\text{fppf}}^2(X, \mathbb{G}_m)$  (9.4.6.37). And for  $R \in \mathbb{C}\text{Ring}$ ,  $\text{Br}(R)$  is defined to be  $\text{Br}(R) = \text{Br}(\text{Spec } R)$ .  $\lrcorner$

**Prop. (11.2.1.2).** If  $X$  is a regular Noetherian scheme that is qc or integral,  $\text{Br}(X) \rightarrow \text{Br}(R(X))$  is injective, and  $\text{Br}(X)$  is a torsion Abelian group.  $\lrcorner$

*Proof:* The qc case reduces to the integral case, and this follows from [Brauer-Grothendieck Groups]Chap3.5.

The second assertion follows from the first and the fact  $\text{Br}(R(X))$  is torsion (11.2.2.4).  $\square$

**Cor. (11.2.1.3).** If  $X \rightarrow Y$  is a birational morphism of integral regular schemes, then  $\text{Br}(Y) \rightarrow \text{Br}(X)$  is injective.  $\lrcorner$

*Proof:* This follows from the fact  $\text{Br}(Y) \rightarrow \text{Br}(X) \rightarrow \text{Br}(R(X)) = \text{Br}(R(Y))$  is injective.  $\square$

### Azumaya Brauer Groups

**Prop. (11.2.1.4) [Azumaya Algebras of Schemes].** For  $X \in \text{Sch}$ , an **Azumaya algebra** over  $X$  is a coherent  $\mathcal{O}_X$ -algebra  $\mathcal{A}$  s.t.  $\mathcal{A}_x \neq 0$  for any  $x \in X$ , and satisfies the following equivalent definitions:

- There exists an étale covering  $\{U_i \rightarrow X\}$  s.t. for each  $i$ , there exists  $r_i \in \mathbb{Z}_+$  s.t.  $\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{O}_{U_i} \cong \text{End}(\mathcal{O}_{U_i}^{\otimes r_i})$ .
- There exists an fppf covering  $\{U_i \rightarrow X\}$  s.t. for each  $i$ , there exists  $r_i \in \mathbb{Z}_+$  s.t.  $\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{O}_{U_i} \cong \text{End}(\mathcal{O}_{U_i}^{\otimes r_i})$ .
- $\mathcal{A} \in \text{Vect}_X$ , and for any arithmetic point  $x \in X$ ,  $\mathcal{A}_x$  is isomorphic to  $\text{Mat}(r_x, \kappa(x))$  for some  $r_x \in \mathbb{Z}_+$ . And this  $r_x$  is called the **degree of  $\mathcal{A}$  at  $x$** .
- $\mathcal{A} \in \text{Vect}_X$ , and the canonical homomorphism  $\mathcal{A} \otimes \mathcal{A}^{\text{op}} \rightarrow \text{End}_{\mathcal{O}_X}(\mathcal{A})$  is an isomorphism.

In particular, this definition is compatible with the definition of Azumaya algebras over fields in (3.6.4.5), by (3.6.4.26).

The category of Azumaya algebras over  $X$  is denoted by  $\text{Az}_X$ . For  $n \in \mathbb{Z}_+$ , the subset of Azumaya algebras over  $X$  of constant degree  $n$  is denoted by  $\text{Az}^n(X)$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$  is trivial by (3.6.4.26).  $4 \rightarrow 1$ : Cf. [Milne80 Etale Cohomologies]P141.  $\square$

**Def. (11.2.1.5) [Azumaya Brauer Groups].** Let  $X \in \text{Sch}$ , the **Azumaya Brauer group**  $\text{Br}_{\text{Az}}(X)$  is defined to be the set of equivalence class of Azumaya algebras over  $X$  under the equivalence relation that  $\mathcal{A} \sim \mathcal{A}'$  iff there exists  $\mathcal{E}, \mathcal{E}' \in \text{Vect}(X)$  with positive ranks at each point s.t.

$$\mathcal{A} \otimes_{\mathcal{O}_X} \text{End}(\mathcal{E}) \cong \mathcal{A}' \otimes_{\mathcal{O}_X} \text{End}(\mathcal{E}').$$

Moreover,  $\text{Br}_{\text{Az}}(X)$  is a group under tensor product and the inverse is given by  $\mathcal{A} \rightarrow \mathcal{A}^{\text{op}}$ .  $\lrcorner$

*Proof:* To show this is an equivalence and the group operation is well-defined, use the fact that for  $\mathcal{E}, \mathcal{E}' \in \text{Vect}(X)$ ,

$$\mathbf{End}(\mathcal{E}) \otimes_{\mathcal{O}_X} \mathbf{End}(\mathcal{E}') \cong \mathbf{End}(\mathcal{E} \otimes \mathcal{E}').$$

And the inverse is of the form given, by (11.2.1.4).  $\square$

**Lemma (11.2.1.6).** Let  $(\mathcal{C}, \mathcal{O})$  be a ringed site and  $\mathcal{F}, \mathcal{G} \in \text{Vect}(\mathcal{O})$  s.t.  $\mathbf{End}(\mathcal{F}) \cong \mathbf{End}(\mathcal{G})$ , then there exists an invertible sheaf  $\mathcal{L}$  s.t.  $\mathcal{F} \otimes \mathcal{L} \cong \mathcal{G}$ .  $\lrcorner$

*Proof:* Let  $\mathcal{L} \subset \text{Hom}(\mathcal{F}, \mathcal{G})$  be generated as an  $\mathcal{O}_X$ -module by local isomorphisms  $\varphi : \mathcal{F} \cong \mathcal{G}$  s.t. the conjugation by  $\varphi$  coincides with the given isomorphism, then local computation and the fact all automorphisms of  $\text{Mat}(n)$  is inner shows that  $\mathcal{L}$  is invertible, and the evaluation map

$$\mathcal{F} \otimes \mathcal{L} \rightarrow \mathcal{G}$$

is an isomorphism.  $\square$

**Prop. (11.2.1.7).** If  $\mathcal{A} \in \text{Az}_X$  has constant degree  $d$ , then  $[\mathcal{A}] \in \text{Br}_{\text{Az}}(X)$  is annihilated by  $d$ .  $\lrcorner$

*Proof:* Choose an étale covering  $\{U_i \rightarrow X\}$  and isomorphisms  $\mathcal{A}|_{U_i} \cong \text{Hom}(\mathcal{F}_i, \mathcal{F}_i)$ , where  $\mathcal{F}_i \in \text{Vect}^d(X)$ , then

$$\mathcal{A}^{\otimes d}|_{U_i} \cong \mathbf{End}(\mathcal{F}_i^{\oplus d}).$$

Consider the maps

$$p_i : \mathcal{F}_i^{\oplus d} \rightarrow \wedge^d \mathcal{F}_i \subset \mathcal{F}_i^{\oplus d},$$

then  $p_i^2 = d!p_i$  and  $\text{rank}(p_i) = 1$ . We show now that these  $p_i$  glue together to get a global section  $p$  of  $\mathcal{A}^{\otimes d}$ : by (11.2.1.6), there exist compatible invertible sheaves  $\mathcal{L}_{ij}$  on  $U_i \cap U_j$  s.t.

$$\mathcal{F}_i|_{U_{ij}} \otimes \mathcal{L}_{ij} \cong \mathcal{F}_j|_{U_{ij}}.$$

These isomorphisms can clearly generate isomorphisms to glue  $\{p_i\}$  together.

Then consider  $\mathcal{H} = \mathcal{A}^{\otimes d} \circ p \subset \mathcal{A}^{\otimes d}$ , we claim that

- $\dim \mathcal{H} = d^d$ ,
- left multiplication by  $\mathcal{A}^{\otimes d}$  induces an isomorphism  $\mathcal{A}^{\otimes d} \cong \mathbf{End}(\mathcal{H})$ .

Which will imply that  $d[\mathcal{A}] = 0 \in \text{Br}_{\text{Az}}(X)$ . To show these claims, it suffices to do local calculations:

If  $\mathcal{F} \cong \mathcal{O}_X e_1 \oplus \dots \oplus \mathcal{O}_X e_d$ , then  $\mathcal{A} \cong \mathbf{End}(\mathcal{O}_X^d)$ , and

$$p : e_{\sigma(1)} \otimes \dots \otimes e_{\sigma(d)} \mapsto \text{sgn}(\sigma) e_1 \otimes \dots \otimes e_d, \sigma \in \mathcal{S}_d, \quad 0 \text{ on other basis vectors}$$

So

$$\mathcal{H} = \{f : \mathcal{F}^{\otimes d} \rightarrow \mathcal{F}^{\otimes d} : e_{\sigma(1)} \otimes \dots \otimes e_{\sigma(d)} \mapsto \text{sgn}(\sigma) v_1 \otimes \dots \otimes v_d, \quad 0 \text{ on other basis vectors}\}$$

Then  $\mathcal{A} \cong \mathbf{End}(\mathcal{O}_X^d)$  is clear.  $\square$

**Cor. (11.2.1.8)[Azumaya Brauer Groups are Torsion].** If  $X \in \text{Sch}$  is qc or connected, then  $\text{Br}_{\text{Az}}(X)$  is a torsion Abelian group.  $\lrcorner$

### Comparison of Two Brauer Groups

**Prop. (11.2.1.9) [Azumaya Brauer Groups via Cohomology].** Let  $X \in \text{Sch}$ , there exists an isomorphism of pointed sets

$$\text{Az}_X^n \cong \check{H}_{\text{ét}}^1(X, \text{PGL}(n)).$$

┘

*Proof:* As  $\text{PGL}(n) = \text{Aut}(\text{Mat}(n))$  and any Azumaya algebra is a twist-form for  $\text{Mat}(n)_X$ , any  $\mathcal{A} \in \text{Az}_X^n$  defines a 1-cocycle for  $\text{PGL}(n)$ . It is clear that if this cocycle is a coboundary, then  $\mathcal{A} \cong \text{Mat}(n)_X$ . Its left to show that any 1-cocycle comes from these: As  $\text{PGL}(n) \subset \text{GL}(n^2)$ , and 1-cocycle is a 1-cocycle for  $\text{GL}(n^2)$ , thus by (7.1.6.1) corresponds to a vector bundle  $\mathcal{E}$  of rank  $n^2$ . By taking a refinement, this means there is a covering  $\{U_i \rightarrow X\}$  and isomorphisms  $\varphi_i : \text{Mat}(n, \mathcal{O}_{U_i}) \cong \mathcal{E}|_{U_i}$  as modules such that the transformation maps  $\varphi_i^{-1} \circ \varphi_j \in \text{PGL}(n, \mathcal{O}_{U_{ij}}) \subset \text{GL}(n^2, \mathcal{O}_{U_{ij}})$ , which means exactly  $\mathcal{E} \in \text{Az}_X^n$ .  $\square$

**Prop. (11.2.1.10) [Br(X) and  $\text{Br}_{\text{Az}}(X)$ ].** Assume  $X \in \text{Sch}$  is qc and every finite subset of  $X$  is contained in an affine scheme, by (7.3.2.19), the exact sequence

$$1 \rightarrow \mathbb{G}_m \rightarrow \text{GL}(n) \rightarrow \text{PGL}(n) \rightarrow 1$$

of algebraic groups gives a map

$$\check{H}_{\text{ét}}^1(X, \text{GL}(n)) \rightarrow \text{Az}^n(X) = \check{H}_{\text{ét}}^1(X, \text{PGL}(n)) \xrightarrow{\delta_n} \check{H}_{\text{ét}}^2(X, \mathbb{G}_m) \hookrightarrow H_{\text{ét}}^2(X, \mathbb{G}_m) = \text{Br}(X).$$

by (11.2.1.9) and (7.3.2.14). So the kernel of  $\delta : \text{Az}_X^n \rightarrow \text{Br}(X)$  is given by Azumaya algebras of the form  $\mathbf{End}(\mathcal{E})$  for  $\mathcal{E} \in \text{Vect}^n(X)$ . These maps for various  $n$  give a map

$$\text{Az}(X) \rightarrow \text{Br}(X).$$

Then

- This map is an injective homomorphism, so  $\text{Br}_{\text{Az}}(X) \subset \text{Br}(X)$ .
- If  $X$  has an ample invertible sheaf, then the map induces an isomorphism

$$\text{Br}_{\text{Az}}(X) \cong \text{Br}(X)_{\text{tor}}.$$

In particular, if  $X$  is a regular quasi-projective scheme over some  $A \in \mathbb{C}\text{Ring}$ , then by (11.2.1.2),

$$\text{Br}_{\text{Az}}(X) \cong \text{Br}(X)_{\text{tor}} = \text{Br}(X).$$

For general  $X$ , this is also doable, by Gabber and de Jong? ?

┘

*Proof:* 1: By taking disjoint union, it suffices to show that for  $\mathcal{A} \in \text{Az}_X^n, \mathcal{B} \in \text{Az}_X^m$ ,

$$\delta_{mn}(\mathcal{A} \otimes \mathcal{B}) = \delta_n(\mathcal{A}) \cdot \delta_m(\mathcal{B}).$$

But this is clear from the description of  $\delta_n$ : by the hypothesis and [Milne80 Etale Cohomology, Prop4.2.19] ? and (7.3.2.19), we may take an étale refinement  $\{U_i\}$  and assume the cocycle corresponding to  $\mathcal{A}$  is given by a cocycle  $c_{ij}$  s.t.  $c_{ij} \in \Gamma(U_{ij}, \text{GL}(n))$ . Then  $\delta(\mathcal{A})$  is represented by the 2-cocycle with  $a_{ijk} = c_{jk}c_{ik}^{-1}c_{ij} \in \Gamma(U_{ijk}, \mathbb{G}_m)$ .

The injectivity follows from the presentation

$$\mathrm{Br}_{\mathrm{Az}}(X) \cong \varinjlim \mathrm{Az}_X^n / \sim = \varinjlim \check{H}_{\mathrm{et}}^1(X, \mathrm{PGL}(n)) / \check{H}_{\mathrm{et}}^1(X, \mathrm{GL}(n)).$$

and the fact the image of  $\check{H}_{\mathrm{et}}^1(X, \mathrm{GL}(n))$  in  $\check{H}_{\mathrm{et}}^1(X, \mathrm{PGL}(n))$  is the class of **End**( $n, E$ ): a  $\mathbf{a} \in \check{H}_{\mathrm{et}}^1(X, \mathrm{GL}(n))$  corresponds to a vector bundle  $\mathcal{E}$  of rank  $n$  with trivializations  $\varphi_i : \mathcal{O}_{U_i}^n \cong E|_{U_i}$  s.t.  $\mathbf{a}$  is represented by  $(\varphi_i^{-1}\varphi_j)$ . Then  $\mathcal{A} = \mathbf{End}(n, \mathcal{E}) \in \mathrm{Az}_X^n$  has trivializations  $\psi_i : \mathrm{Mat}(n, \mathcal{O}_{U_i}) \cong \mathbf{End}(n, \mathcal{E})|_{U_i}$ , and  $\mathcal{A}$  corresponds to the 1-cocycle

$$(\psi_i^{-1}\psi_j) = (\alpha_{ij}), \quad \alpha_{ij}(a) = \varphi_j^{-1}\varphi_i a \varphi_i^{-1}\varphi_j, a \in \mathrm{Mat}(n, \mathcal{O}_{U_{ij}})$$

which is exactly the image of the class  $\mathbf{a}$  under the map  $\mathrm{GL}(n) \rightarrow \mathrm{PGL}(n) : u \mapsto \mathrm{conj}(u)$ .

2: Cf.[Brauer-Grothendieck Group]Chap4.?

□

**Prop.(11.2.1.11)[Kummer Exact Sequences].** For  $X \in \mathrm{Sch}$ ,  $n \in \mathbb{Z}_+$ , the Kummer exact sequence of algebraic groups

$$1 \rightarrow \mu_n \rightarrow \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \rightarrow 1$$

gives exact sequences

$$0 \rightarrow \mathrm{Pic}(X)/n \mathrm{Pic}(X) \rightarrow H_{\mathrm{fppf}}^2(X, \mu_n) \rightarrow \mathrm{Br}(X)[n] \rightarrow 0,$$

$$0 \rightarrow \mathrm{Br}(X)/n \mathrm{Br}(X) \rightarrow H_{\mathrm{fppf}}^3(X, \mu_n) \rightarrow H_{\mathrm{et}}^3(X, \mathbb{G}_m)[n] \rightarrow 0.$$

by(9.4.6.37).

┘

**Prop.(11.2.1.12)[Mayer-Vietoris Exact Sequence].** Let  $X \in \mathrm{Sch}$  and  $X = U \cup V$  be a Zariski covering with  $U \cap V = W$ , then by(11.2.1.2) there is an exact sequence

$$\begin{aligned} 0 \rightarrow \Gamma(X, \mathcal{O}_X^\times) &\rightarrow \Gamma(V, \mathcal{O}_X^\times) \oplus \Gamma(V, \mathcal{O}_X^\times) \rightarrow \Gamma(W, \mathcal{O}_X^\times) \\ &\rightarrow \mathrm{Pic}(X) \rightarrow \mathrm{Pic}(U) \oplus \mathrm{Pic}(V) \rightarrow \mathrm{Pic}(W) \\ &\rightarrow \mathrm{Br}(X) \rightarrow \mathrm{Br}(U) \oplus \mathrm{Br}(V) \rightarrow \mathrm{Br}(W) \rightarrow H_{\mathrm{et}}^3(X, \mathbb{G}_m) \rightarrow \dots \end{aligned}$$

And when  $U$  is locally factorial, then  $\mathrm{Pic}(U) \rightarrow \mathrm{Pic}(W)$  is surjective?, so there is an exact sequence

$$0 \rightarrow \mathrm{Br}(X) \rightarrow \mathrm{Br}(U) \oplus \mathrm{Br}(V) \rightarrow \mathrm{Br}(W) \rightarrow H_{\mathrm{et}}^3(X, \mathbb{G}_m).$$

┘

**Prop.(11.2.1.13)[Hochschild Spectral Sequence].**

┘

### Residue Homomorphism

**Prop.(11.2.1.14)[Faddeev].** For  $k \in \mathrm{Field}$ , there is an exact sequence

$$0 \rightarrow \mathrm{Br}(k) \rightarrow \mathrm{Br}(\mathbb{P}_k^1) \xrightarrow{\mathrm{res}} \bigoplus_{x \in |\mathbb{P}_k^1|_0} H^2(\kappa(x), \mathbb{Q}/\mathbb{Z}) \rightarrow H^1(k, \mathbb{Q}/\mathbb{Z}) \rightarrow 0.$$

┘

*Proof:* Cf. [GS06, 6.4.5]?

□



**Artin Conjecture**

**Conj. (11.2.1.15)**[Artin]. For  $X \in \text{Sch}/\mathbb{Z}$  proper,  $\# \text{Br}(X) < \infty$ . ┘

*Proof:* Cf.[Central Simple Algebras and Galois Cohomology, 6.4.5.]. □

**Prop. (11.2.1.16)**[Artin & Tate Conjecture]. For a proper surface  $X/\mathbb{Z}$ , the Artin conjecture for  $X$  (11.2.1.15) is equivalent to Tate conjecture for divisors of  $X$ . ┘

*Proof:* □

**Prop. (11.2.1.17)**[Artin vs. Tate-Šafarevič Conjecture]. For  $F \in \mathbf{NField}$ , for a regular integral scheme  $X$  of dimension 2 that is flat proper over  $\mathcal{O}_F$  that has a section, the Artin conjecture for  $X$  is equivalent to the finiteness of  $\text{III}(\text{Jac}(X_F))$ . ┘

*Proof:* □

**2 Field case**

**Notation (11.2.2.1).**

- Let  $k \in \mathbf{Field}$ .
- ┘

**Prop. (11.2.2.2)**[Indices of Brauer Classes]. Any Brauer class in  $\text{Br}_{\text{Az}}(k)$  is represented by a unique central division ring.

So we can define the **index of the Brauer class**  $D$  as the  $\text{ind}([D]) = \sqrt{[D : k]} \in \mathbb{Z}_+$  (3.6.4.16). And the **period of the Brauer class**  $D$  to be the exponent of  $[D] \in \text{Br}_{\text{Az}}(k)$ . ┘

*Proof:* The existence of this division ring follows from Wedderburn theorem (3.6.2.22), and to show uniqueness, notice if

$$\text{Mat}(n; D) \cong \text{Mat}(m; D') = A$$

when we can recover  $D$  as  $D = D' = \text{End}_A(V)$ , where  $V$  is the unique simple module, by (3.6.2.21). □

**Prop. (11.2.2.3)**[Brauer Groups and Galois Cohomology]. The **Brauer group**  $\text{Br}(k)$  is defined as the profinite cohomology  $H^2(\text{Gal}(k^{\text{sep}}/K), (k^{\text{sep}})^{\times})$ . For a Galois extension  $L/k$ ,  $\text{Br}(L/k)$  is defined as  $H^2(\text{Gal}(L/k), L^{\times})$ . Then by (9.8.2.3) we have

$$\varinjlim \text{Br}(L/k) = \text{Br}(k).$$

And by Hochschild-Serre spectral sequence and by Hilbert's multiplicative theorem90:  $H^1(H, k_s^*) = 0$ , we have the low term:

$$0 \rightarrow \text{Br}(L/k) \xrightarrow{\text{inf}} \text{Br}(k) \xrightarrow{\text{res}} \text{Br}(L)^{\text{Gal}(L/k)} \rightarrow H^3(\text{Gal}(L/k), L^{\times}) \rightarrow H^3(k, (k^{\text{sep}})^{\times}).$$

So  $\text{Br}(L/k)$  is the kernel of  $\text{Br}(k) \rightarrow \text{Br}(L)$ . ┘

*Proof:* Cf.[Neukirch Cohomology of Number Fields Chap6.3]. ? □

**Cor. (11.2.2.4)**[Brauer Groups are Torsion].  $\text{Br}(k) \in \mathcal{A}b_{\text{tor}}$ , by (9.8.2.4). In particular,  $\text{Br}_{\text{Az}}(k) \cong \text{Br}(k)$  by (11.2.1.10). ┘

**Cor. (11.2.2.5).** If  $k \in \mathbf{Field}^{\text{fin}}$ , we have  $\text{Br}(k) = 0$ , ┘

**Prop. (11.2.2.6)[Involutions].** If  $D \in \text{Az}_k$  admits an involution of the first kind, then period of  $[D]$  is 1 or 2. In particular, periods of a quaternion ring over  $k$  is 1 or 2. ┘

*Proof:* An involution of first kind defines an isomorphism  $A \cong A^{\text{op}}$ , which implies that  $A \otimes A \cong \text{End}(A)$ . Thus  $2[A] = 0 \in \text{Br}_{\text{Az}}(k)$ . □

**Prop. (11.2.2.7)[Finite Fields, Wedderburn].** If  $k \in \mathbf{Field}^{\text{fin}}$ , then for any  $n \in \mathbb{N}$ ,  $\check{H}^1(k, \text{PGL}(n)) = 0$ , and  $\text{Br}(k) = 0$ . ┘

*Proof:* This follows from (11.2.2.2) and the fact any finite division ring is commutative (3.6.1.1). Alternatively, because the finite Galois extensions are cyclic and unramified. □

**Prop. (11.2.2.8)[Valued Fields].** For  $(R, K, k) \in \text{CDVR}$ , there exists a split exact sequence

$$0 \rightarrow \text{Br}(k) \rightarrow \text{Br}_{\text{tame}}(K) \rightarrow H^1(k, \mathbb{Q}/\mathbb{Z}) \rightarrow 0.$$

┘

*Proof:* Cf. [Brauer-Grothendieck Group, P32]. □

**Prop. (11.2.2.9).** For any  $D \in \text{Az}_k$  with index  $n$ , there is a map from the conjugacy classes of  $D^\times$  to the conjugacy classes of  $\text{GL}(n; k)$ . Cf. [Wed08]P293? ┘

**Def. (11.2.2.10)[Special Fields].**  $k$  is called a **special field** if every element of order 2 is represented by a quaternion ring over  $k$  (11.2.2.6). ┘

**Prop. (11.2.2.11).** Separably-closed fields, finite fields, local fields and global fields are special. ┘

*Proof:* ? Their Brauer group has descriptions. □

### 3 Local Field case

**Notation (11.2.3.1).**

- Let  $K \in \mathbf{LField}$ .

┘

**Thm. (11.2.3.2)[Brauer Groups of Local Fields].**

- the invariant map (19.7.2.6) is a canonical injection

$$\text{inv}_K : \text{Br}(K) \hookrightarrow \mathbb{Q}/\mathbb{Z}$$

whose image is

$$\begin{cases} \mathbb{Q}/\mathbb{Z} & , K \in p\text{-LField} \\ \frac{1}{2}\mathbb{Z}/\mathbb{Z} & , K = \mathbb{R} \\ 0 & , K = \mathbb{C} \end{cases}$$

- Every Azumaya algebra over  $K$  is cyclic?.
- Every element of  $\text{Br}(K)$  has period equal to index (11.2.2.2).

┘

*Proof:* 1: This follows from the definition of a class formation (19.7.1.1).

The rest follows from [Poonen, P25]. ?

□

**Cor. (11.2.3.3).** For  $K \neq \mathbb{C} \in \mathbf{LField}$ , there exists a unique nontrivial quaternion algebra over  $K$ , and its the only division ring with exponent 2. ┘

**Prop. (11.2.3.4).** For any  $D \in \mathbf{Az}_K$  of dimension  $\dim_K D = r^2$ , we can define an additive norm on  $D$  by

$$v_D(\delta) = \frac{1}{r} v_K(\text{Nmrd}(\delta)).$$

Then  $D$  is complete w.r.t. this norm. Define

$$\mathcal{O}_D = \{\delta \in D : v_D(\delta) \geq 0\}, \quad \mathfrak{m}_D = \{\delta \in D : v_D(\delta) > 0\}.$$

Then  $\mathfrak{m}_D$  is a two-sided ideal of  $\mathcal{O}_D$  and  $\kappa_D = \mathcal{O}_D/\mathfrak{m}_D$  is division ring that is finite over  $k$  thus a field.

Then  $L = W(\kappa_D)[\frac{1}{p}] \subset D$ , and  $f(D) \triangleq [\kappa_D : \kappa] = [L : K] \leq r$ . But also  $e(D) \leq r$ , and  $ef = r^2$ , so  $e = f = r$ .

By the Skolem-Noether theorem, there exists  $\delta_D \in D^\times$  s.t.  $\delta_K(x) = \delta x \delta^{-1}$  for any  $x \in L$ . Then we have  $\text{inv}_K(D) = v_D(\delta) \in \mathbb{Q}/\mathbb{Z}$ . ? ┘

**Prop. (11.2.3.5) [Description of Azumaya Algebras].** For  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-}\mathbf{LField}$  and  $r \in \mathbb{Z}_+, s \in [0, r]$  s.t.  $(r, s) = 1$ , denote  $\lambda = r/s$ , then the unique  $D_{K, \lambda} \in \mathbf{Az}_k$  with  $\text{inv}_K(D) = \lambda$  is defined as follows:

$$D_{K, \lambda} = K_r[\Pi]$$

where  $K_r$  is the unramified extension of  $K$  of degree  $r$ , and  $\Pi$  is an indeterminate satisfying the relations

$$\Pi^r = \varpi_K^r, \quad \Pi.a = \sigma_K(a).\Pi.$$

Then  $\dim D_\lambda = r^2$ , and  $\text{ind}(D_\lambda) = r$ . ┘

*Proof:* ?

□

## 4 Global Field case

**Thm. (11.2.4.1) [Brauer-Hasse-Noether].** For  $F \in \mathbf{NField}$ ,

- there is a canonical exact sequence

$$1 \rightarrow \text{Br}(F) \rightarrow \bigoplus_{\mathfrak{p} \in \Sigma_F} \text{Br}(F_{\mathfrak{p}}) \xrightarrow{\text{inv}_F} \mathbb{Q}/\mathbb{Z} \rightarrow 0,$$

and in the characterization via Azumaya algebras (11.2.1.5)(11.2.1.10), the first map sends  $[D]$  to the family  $([D \otimes_F F_v])_v$ .

- Every Azumaya algebra over  $F$  is cyclic ?.
- Every element of  $\text{Br}(F)$  has exponent equal to index.

┘

*Proof:* 1: Choose for each  $v \in \Sigma_F$  a fixed valuation of  $K^{\text{sep}}$  extending  $v$ , and for each  $L/K \in \text{GalField}^{\text{fin}}$ , consider the homomorphisms

$$1 \rightarrow H^2(\text{Gal}(L/F), L^\times) \rightarrow H^2(\text{Gal}(L/F), \mathbf{I}_F) \cong \otimes_{v \in \Sigma_F} H^2(\text{Gal}(L_v/K_v), L_v^\times) \xrightarrow{\text{inv}_{L/K}} \frac{1}{[L:F]} \mathbb{Z}/\mathbb{Z} \rightarrow 1 \quad (19.7.3.3),$$

by taking direct limit, we get the homomorphism

$$\text{Br}(F) \rightarrow \bigoplus_{\mathfrak{p} \in \Sigma_F} \text{Br}(F_{\mathfrak{p}}) \xrightarrow{\text{inv}_F} \mathbb{Q}/\mathbb{Z}.$$

But by (19.7.3.15), these groups are unions over only the cyclic extensions  $L/F$ . But for cyclic extensions the above sequence is exact (19.7.3.13), so the limit is also exact.

2: [Poonen, Rational Points, P26]. ?

3: It follows from item1 that for any division ring  $D/F$ ,  $[D] \in \text{Br}(F)$  has exponent  $n = \text{lcm}(\text{ord}(\text{inv}_v([D])))$ . So it follows from (11.2.3.2) that there  $n_v | \text{ind}([D])$  for any  $v$ , so  $n | \text{ind}([D])$ .

And it follows from Grünwald-Wang (19.7.4.32) and the local-global compatibility of  $\text{inv}_v$  that there exists a cyclic extension  $L/K$  of degree  $n$  that splits  $D$ . Then it follows from (3.6.4.18) (11.2.2.2) that there exists  $B = \text{Mat}(r; D)$  s.t.  $L \subset B$  and  $[B:F] = [L:K]^2$ . Thus  $[D:F] | [L:K]^2$ , so  $\text{ind}(D) = n$ .  $\square$

**Prop. (11.2.4.2) [Norm Groups of Division Rings, Eichler].** If  $F \in \text{NField}$  and  $D/F$  is a division ring of index  $n$ , then  $\text{Nm}_{D/K}(D^\times) \subset F^\times$  is the set of elements that is positive under all  $v \in \Sigma_F^{\mathbb{R}}$  s.t.  $D_v \not\cong \text{Mat}(n; F_v)$ .  $\lrcorner$

*Proof:* Cf. [P-R94] P38.  $\square$

**Prop. (11.2.4.3) [Wang].** If  $F \in \text{NField}$  and  $D/F$  is a division ring of index  $n$ , then  $\text{SL}(1; D) = [D^\times, D^\times]$ .  $\lrcorner$

**Prop. (11.2.4.4).**  $\lrcorner$

## 5 Function Field case

**Conj. (11.2.5.1) [Period-Index Conjecture, Colliot-Thelene(01)].** If  $k = \bar{k}$  and  $X \in \text{SmProj}/k$ , if  $\alpha \in \text{Br}(R(X))$ , then  $\text{ind}(\alpha) | \text{per}(\alpha)^{\dim X - 1}$  (11.2.2.2).  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj. Cor. (11.2.5.2).** If  $k = \bar{k}$  and  $X \in \text{SmProj}^1/k$ , then  $\text{Br}(R(X)) = 1$ . In particular, the period-index conjecture is true in this case.

If  $k = \bar{k}$  and  $X \in \text{SmProj}^1/k$ , then the period-index conjecture is true by deJong-Lieblich.  $\lrcorner$

## 6 Severi-Brauer Varieties

References are [Severi – Brauer varieties: a geometric treatment, Kollar] and [Central Simple Algebras and Galois Cohomology].

**Def. (11.2.6.1) [Severi-Brauer Varieties].** For  $k \in \text{Field}$ , a **Severi-Brauer variety** over  $k$  is a  $k$ -variety  $X$  that is a  $k$ -form for  $\mathbb{P}_k^n$  for some  $n \in \mathbb{N}$ .

If  $K/k$  is a field extension and  $X_K \cong \mathbb{P}_K^n$ , then  $K$  is called a **splitting field** for  $X$ .  $\lrcorner$

**Prop. (11.2.6.2).** Let  $k \in \mathbf{Field}$  and  $X$  a Severi-Brauer variety over  $k$ , then a **twisted-linear subvariety** of  $X$  is a closed subvariety of  $X$  s.t. the inclusion  $Y_{\bar{k}} \subset X_{\bar{k}} \cong \mathbb{P}_{\bar{k}}^n$  embeds  $Y_{\bar{k}}$  as a linear subvariety of  $\mathbb{P}_{\bar{k}}^n$ .  $\lrcorner$

**Prop. (11.2.6.3) [Châtelet].** For  $k \in \mathbf{Field}$  and an  $n$ -dimensional Severi-Brauer variety  $X$  over  $k$ , the following are equivalent:

- $X \cong \mathbb{P}_k^n$ .
- $X \sim \mathbb{P}_k^n$ .
- $X(k) \neq \emptyset$ .
- $X$  contains a twisted-linear subvariety of codimension 1.

 $\lrcorner$ 

*Proof:*  $1 \rightarrow 2$  is trivial. If 2 holds, then to show item 3, if  $\#k = \infty$ , then any Zariski open subset of  $\mathbb{P}_k^n$  contains a  $k$ -point, and if  $\#k < \infty$ , clearly any Severi-Brauer variety over  $k$  splits over a finite Galois field extension of  $k$ , so by the proof of (11.2.6.6)  $\color{red}?$  shows that  $SB_n(K) \cong H^1(k, \mathrm{PGL}(n)) = 1$  (11.2.2.7). Thus  $X \cong \mathbb{P}_k^n$  has a rational point.

$4 \rightarrow 1$ : The twisted-linear subvariety is a divisor of  $X$   $\color{red}?$ , so it defines a rational map  $\varphi_D$  into some projective variety, and when base changed to  $\bar{k}$ , it is the isomorphism of  $X$  with  $\mathbb{P}_{\bar{k}}^n$ . Thus it must be an isomorphism.

$3 \rightarrow 4$ : Cf. [Central Simple Algebras] P341.  $\color{red}?$

 $\square$ 

**Cor. (11.2.6.4) [Galois Splitting Fields].** Let  $k \in \mathbf{Field}$  and  $X$  a Severi-Brauer variety over  $k$ , then  $X$  has a finite Galois splitting field.  $\lrcorner$

*Proof:* It suffices to show that  $k^s$  is a splitting field for  $X$ . But for this, by the theorem, it suffices to notice  $X_{k^s}$  always has a rational point (7.12.1.10).  $\square$

**Cor. (11.2.6.5).** Severi-Brauer groups satisfy the local-global property.  $\lrcorner$

*Proof:* Cf. [Poonen] P108.  $\square$

**Prop. (11.2.6.6) [Severi-Brauer Varieties and Galois Cohomologies].** For  $k \in \mathbf{Field}$ , there is an isomorphism of pointed sets  $H^1(k, \mathrm{PGL}(n)) \cong SB_n(L/K)$ , where  $SB_n(L/K)$  is the isomorphism classes of Severi-Brauer varieties of dimension  $n - 1$  that splits in  $L$ .  $\lrcorner$

*Proof:* Cf. [Neukirch Cohomology of Number Fields P348].  $\square$

## 7 Brauer-Manin Obstruction

**Prop. (11.2.7.1).** For  $F \in \mathbf{GField}$  and  $X \in \mathbf{Sch}^{\mathrm{sm}}/F$ , there is a natural continuous right-linear pairing

$$\gamma : X(\mathbf{A}_F) \times \mathrm{Br}(X) \rightarrow \mathbb{Q}/\mathbb{Z}$$

s.t. the restriction to  $X(F) \times \mathrm{Br}(X)$  is trivial.

Let  $X(\mathbf{A}_F)^{\mathrm{Br}}$  denote the left kernel of  $\gamma$ , then we say the **Brauer-Manin obstruction** is sufficient for  $X$  if  $X(F) \subset X(\mathbf{A}_F)^{\mathrm{Br}}$  is dense.  $\lrcorner$

*Proof:*  $\color{red}?$

 $\square$ 

**Prop. (11.2.7.2).** If  $F \in \mathbf{GField}$  and  $X \in \mathbf{SmProj}/F$ , then if  $X$  satisfies the weak approximation, then Brauer-Manin obstruction is sufficient.  $\lrcorner$

*Proof:*

 $\square$

## 8 Norm Residue Isomorphism Theorem

See Voevodsky's work.

## Part III

# Analysis and Geometry





# 12 | Condensed Mathematics and Analysis

## 12.1 Condensed Mathematics(Scholze)

Main references are [Condensed Mathematics, Scholze]. [Lectures on Condensed Mathematics, Clausen-Scholze], [Lectures on Analytic Geometry, Clausen-Scholze].

**Notation(12.1.0.1).**

- Use notations defined in [Profinite Cohomology](#).

┘

### 1 Introduction

**Remark(12.1.1.1).** Delete this subsection. ?

┘

Condensed mathematics is invented to overcome the subtleties when dealing with algebra together with topology. For example, the category of topological Abelian groups doesn't form an Abelian category.

### 2 Condensed Objects

**Def.(12.1.2.1) [ $\kappa$ -Condensed Objects].** The pro-étale site  $*_{pro\acute{e}t}$  of a point is isomorphic to  $\mathcal{P}rof$  in the standard topology. For any uncountable strong limit cardinal  $\kappa$ , the site  $\mathcal{P}rof_\kappa$  is the category of  $\kappa$ -small profinite sets  $S$  in the standard topology.

Then for any  $\mathcal{C} \in \mathcal{C}at$ , we can define the category  $\mathcal{C}ond_\kappa(\mathcal{C})$  of  $\kappa$ -**condensed objects** in  $\mathcal{C}$  to be the category  $\mathcal{F}unc(\mathcal{P}rof_\kappa, \mathcal{C})$ .

┘

**Prop.(12.1.2.2)[Adjointness].** Given an uncountable strong limit cardinal  $\kappa$ , the functors

$$X \mapsto \underline{X} : \mathcal{P}rof / \mathcal{G}rp / \mathcal{C}Alg \rightarrow \mathcal{C}ond(\mathcal{S}et) / \mathcal{C}ond(\mathcal{G}rp) / \mathcal{C}ond(\mathcal{C}Alg)$$

are faithful, and fully faithful when restricted to the category of  $\kappa$ -small objects.

And there is an adjunction

$$X \mapsto \underline{X} : \mathcal{P}rof \rightleftarrows \mathcal{C}ond(\mathcal{S}et) : T \mapsto T(*)_{\text{top}} = \varinjlim_{S \in \mathcal{P}rof_\kappa / T} S.$$

In particular,  $\underline{X}(*)_{\text{top}} \cong X_{\kappa\text{-cg}}$ .

┘

*Proof:* Firstly,  $\underline{X}$  is truly a condensed set: for the sheaf condition, let  $S' \rightarrow S$  be surjective morphism of profinite spaces, then the sheaf condition is true set-theoretically, and for any map  $S \rightarrow T$ , if  $S' \rightarrow S \rightarrow T$  is continuous, then  $S \rightarrow T$  is continuous, because  $S' \rightarrow S$  is a closed map.

It suffices to check the set case, and the faithfulness and fully faithfulness follows from the adjointness

$$\mathrm{Hom}_{\kappa\text{-Cond}(\mathrm{Set})}(T, \underline{X}) = \mathrm{Hom}(T(*)_{\mathrm{top}}, X).$$

Notice that any  $s \in T(S)$  induces a map of sets  $\bar{s} : S \rightarrow T(*) : x \mapsto T(* \xrightarrow{x} S)(s)$ , inducing the topology of  $T(*)$ . And by definition, a morphism  $T \rightarrow \underline{X}$  is just a morphism  $T(*) \rightarrow X$  that for any  $S \rightarrow T$ , the composition  $S \rightarrow T(*) \rightarrow X$  is continuous. And by definition, this is just a morphism from  $T(*)_{\mathrm{top}} \rightarrow X$ .

The last assertion follows from the fact any compact Hausdorff space  $S$  is a quotient space of its Stone-Ćech compactification.  $\square$

**Prop. (12.1.2.3) [Free Condensed Abelian Groups].** By the adjoint functor theorem (4.1.1.35), the forgetful functor from  $\mathrm{Cond}(\mathcal{A}b)$  to  $\mathrm{Cond}(\mathrm{Set})$  has a left adjoint  $T \mapsto \mathbb{Z}[T]$ . Concretely,  $\mathbb{Z}[T]$  is the sheafification of the functor that sends a compact Hausdorff space  $S$  to the free Abelian group  $\mathbb{Z}[T(S)]$ . In particular, by Yoneda lemma, for any compact Hausdorff space  $S$ , there is a condensed Abelian group  $\mathbb{Z}[S]$  that for any condensed Abelian group  $M$ ,  $\mathrm{Hom}(\mathbb{Z}[S], M) = M(S)$ .  $\lrcorner$

**Lemma (12.1.2.4).** Consider the site of all  $\kappa$ -small compact Hausdorff topological spaces, then the category of sheaves on this site is equivalent to that of  $\mathrm{Prof}_{\kappa}$  (via restriction).  $\lrcorner$

*Proof:* Use (7.1.2.25), because any  $\kappa$ -small compact Hausdorff space is a quotient space of a  $\kappa$ -small profinite space  $(\beta S_0)$  (4.4.2.15), noticing that  $|\beta S| \leq 2^{2^{|S|}} < \kappa$ .  $\square$

**Lemma (12.1.2.5).** Consider the site of all  $\kappa$ -small extremally disconnected spaces, then the category of sheaves on this site is equivalent to that of  $\kappa$ -small condensed sets, via restriction.  $\lrcorner$

*Proof:* Because any  $\kappa$ -small compactly generated space is a quotient space of a  $\kappa$ -small extremally disconnected space  $(\beta S)$  (4.4.2.17), noticing that  $|\beta S| \leq 2^{2^{|S|}} < \kappa$ .  $\square$

**Cor. (12.1.2.6) [Cond( $\mathcal{A}b$ ) and Extremally Disconnected Spaces].** The category of  $\kappa$ -condensed Abelian groups is equivalent to the category of presheaves  $\mathcal{F}$  on the category of  $\kappa$ -extremally disconnected spaces that  $\mathcal{F}(\emptyset) = 0$  and  $\mathcal{F}(S_1 \coprod S_2) = \mathcal{F}(S_1) \times \mathcal{F}(S_2)$ .  $\lrcorner$

*Proof:* It suffices to show that the second sheaf condition is automatic: For a surjective map of extremally disconnected spaces  $f : \tilde{S} \rightarrow S$ , there is an isomorphism

$$\mathcal{F}(S) \xrightarrow{\mathcal{F}(f)} \{g \in \mathcal{F}(\tilde{S}) \mid \mathcal{F}(p_1)(g) = \mathcal{F}(p_2)(g)\}.$$

By (4.4.1.32), there is a section  $\sigma : S \rightarrow \tilde{S}$  that  $f \circ \sigma = \mathrm{id}_S$ , thus  $\mathcal{F}(\sigma) \circ \mathcal{F}(f) = \mathrm{id}$ , thus  $\mathcal{F}(f)$  is injective. For the surjectivity, suppose  $\mathcal{F}(p_1)(g) = \mathcal{F}(p_2)(g)$ , then

$$\mathcal{F}(p_2)(\mathcal{F}(f)\mathcal{F}(\sigma)(g)) = \mathcal{F}((\sigma \circ f) \times_S \mathrm{id}_{\tilde{S}})\mathcal{F}(p_1)(g) = \mathcal{F}((\sigma \circ f) \times_S \mathrm{id}_{\tilde{S}})\mathcal{F}(p_2)(g) = \mathcal{F}(p_2)(g)$$

And similarly  $\mathcal{F}(p_2)$  is injective, thus  $g = \mathcal{F}(f)\mathcal{F}(\sigma)(g)$  is in the image.  $\square$

**Prop. (12.1.2.7) [Category of Condensed Abelian Groups].** The category of  $\kappa$ -condensed Abelian groups satisfies Grothendieck's Axiom  $AB3, AB4, AB5, AB6, AB3^*, AB4^*$ . And also it is generated by compact projective objects.  $\lrcorner$

*Proof:* We use (12.1.2.6). Because all limits and colimits of Abelian groups commutes with finite products, the limits and colimits in the category of condensed Abelian groups are just the pointwise limits and colimits, thus the axioms follow from that of the category  $\mathcal{A}b$ .

By (12.1.2.3), the condensed Abelian group  $\mathbb{Z}[S]$  for  $S$   $\kappa$ -extremally disconnected satisfies  $\text{Hom}(\mathbb{Z}[S], M) = M(S)$ , and by arguments above,  $M \rightarrow M(S)$  commutes with all limits and colimits, thus  $\mathbb{Z}[S]$  is compact and projective. And we show every  $M$  admits a surjection from some direct sum of  $\mathbb{Z}[S]$ : use Zorn's lemma, choose the maximal object  $M'$  that admits a surjection, if  $M/M' \neq 0$ , then find a nonzero map  $\mathbb{Z}[S] \rightarrow M/M'$  (because  $M(S) = 0$  for any  $S$  implies  $M = 0$ ), then it lifts to a nonzero map  $\mathbb{Z}[S] \rightarrow M$  by projectivity, contradiction.  $\square$

**Cor. (12.1.2.8).**

- We can define the tensor of two condensed Abelian groups  $M, N$  as the shiffication of the presheaf  $S \mapsto M(S) \otimes N(S)$ .
- We can define an internal Hom, which is right adjoint to tensor operator. In particular, for any compact Hausdorff space  $S$ ,  $\underline{\text{Hom}}(M, N)(S) = \text{Hom}(\mathbb{Z}[S] \otimes M, N)$ .
- The derived category  $D(\text{Cond}(\mathcal{A}b))$  is also compactly generated, and we can define Rtensor and RHom as in 2.

*Proof:* Cf.[Condensed Mathematics, P13].?  $\square$

**Prop. (12.1.2.9).** Let  $\kappa' > \kappa$  be uncountable strongly limit cardinals, then there is a natural functor from the set of  $\kappa$ -condensed sets to the category of  $\kappa'$ -condensed sets by pulling back along the morphism of sites  $\text{Prof}_{\kappa'} \rightarrow \text{Prof}_{\kappa}$ . Then this functor is fully faithful and commutes with all colimits and  $\lambda$ -small limits, where  $\lambda$  is the cofinality of  $\kappa$ .  $\lrcorner$

*Proof:* This should have something to do with (7.1.2.25), thus it is left adjoint to the restriction functor and  $i^s i_s \cong \text{id}$ , thus it is fully faithful and commutes with all colimits. For the limits, cf.[Condensed Mathematics, P14].?  $\square$

**Def. (12.1.2.10) [Condensed Objects].** For any  $\mathcal{C} \in \text{Cat}$ , define the category  $\text{Cond}(\mathcal{C})$  of **condensed objects** in  $\mathcal{C}$  is defined to be the filtered colimits of the category of  $\kappa$ -condensed sets along the filtered poset of all uncountable limit cardinals  $\kappa$ .  $\lrcorner$

**Prop. (12.1.2.11).** If  $X$  is a  $T_1$  topological space, then  $\underline{X}$  is a condensed set that all maps from points are quasicompact. Conversely, if  $T$  is a condensed set that all maps from points are quasicompact, then  $T(*)_{\text{top}}$  is a compactly generated  $T_1$  space.  $\lrcorner$

*Proof:* Cf.[Condensed Mathematics, P16].?  $\square$

**Prop. (12.1.2.12) [ $\mathcal{T}_{\text{op}}$  and  $\text{Cond}(\text{Set})$ ].**

- The functor  $X \mapsto \underline{X}$  induces an equivalence between compact Hausdorff space and qcqs condensed sets.
- A compactly generated space  $X$  is weak Hausdorff iff  $\underline{X}$  is quasi-separated. For any quasi-separated condensed set  $T$ , the space  $T(*)_{\text{top}}$  is compactly generated weak Hausdorff.  $\lrcorner$

*Proof:* Cf.[Condensed Mathematics, P16].?  $\square$

**Prop. (12.1.2.13).** The example  $(R, \text{disc}) \rightarrow (R, \text{Nat})$ .?

In particular, enlarging topological abelian groups into an abelian category precisely forces us to include non-quasi-separated objects.  $\lrcorner$

### 3 Cohomologies

**Prop. (12.1.3.1).** For any set  $I$ , there is an isomorphism

$$\check{H}^i(\prod_I S^1, \mathbb{Z}) \cong \bigwedge^i (\oplus_I \mathbb{Z}).$$

preserving the natural cup product.  $\lrcorner$

*Proof:* If  $I$  is finite, this follows from the classical calculation of the cohomology of the tori. If  $I$  is infinite, then we should use lemma (12.1.3.2) below.  $\square$

**Lemma (12.1.3.2).** If  $S_j, j \in J$  is a filtered system of compact Hausdorff spaces and  $S = \varprojlim_{j \in J} S_j$ , then there are natural maps

$$\varprojlim_j \check{H}^i(S_j, \mathbb{Z}) \rightarrow \check{H}^i(S, \mathbb{Z})$$

are isomorphisms.  $\lrcorner$

*Proof:*  $\square$

**Prop. (12.1.3.3) [Z-Cohomology].** Let  $S$  be a compact Hausdorff space, then there are natural functorial isomorphisms

$$H^i(S, \mathbb{Z}) \cong H_{\text{Cond}}^i(S, \mathbb{Z}).$$

*Proof:* Because the Čech and sheaf cohomology of  $S$  are equal by (7.3.5.13), it suffices to calculate for Čech cohomology.  $\lrcorner$

Firstly, if  $S$  is a profinite set, let  $S = \varprojlim_j S_j$  where  $S_j$  are finite, then  $\square$

**Prop. (12.1.3.4) [R-Cohomologies Vanish].** For any compact Hausdorff space  $S$ ,

$$H_{\text{Cond}}^i(S, \mathbb{R}) = 0$$

for  $i > 0$ , and  $H^0(S, \mathbb{R}) = C(S, \mathbb{R})$ .  $\lrcorner$

*Proof:* Cf. [Condensed Mathematics, P21].  $\square$

### 4 LCAs

### 5 Solid Abelian Groups

**Def. (12.1.5.1) [Solid Abelian Group].** For a profinite set  $S = \varprojlim S_i$ , define the condensed Abelian group

$$\mathbb{Z}[S]^{\blacksquare} = \varprojlim \mathbb{Z}[S_i].$$

There is a natural map  $S \rightarrow \mathbb{Z}[S]^{\blacksquare}$ , inducing a map  $\mathbb{Z}[S] \rightarrow \mathbb{Z}[S]^{\blacksquare}$ .

Then a **solid Abelian group** is a condensed Abelian group  $A$  that for any profinite set  $S$  and a morphism of Abelian groups  $\mathbb{Z}[S] \rightarrow A$  extends to a morphism  $\mathbb{Z}[S]^{\blacksquare} \rightarrow A$ .

A complex  $C \subset D(\text{Cond}(\mathcal{A}b))$  is called a **solid complex** if for all profinite set  $S$ , the natural map

$$R\text{Hom}(\mathbb{Z}[S]^{\blacksquare}, C) \rightarrow R\text{Hom}(\mathbb{Z}[S], C)$$

is an isomorphism.  $\lrcorner$

**Prop. (12.1.5.2)[Free Solid Abelian Group].** For a profinite set  $S$ ,

- Consider ?

$$\mathbb{Z}[S]^{\blacksquare} = \varprojlim \mathbb{Z}[S_i] = \varprojlim \underline{\mathrm{Hom}}(C(S_i, \mathbb{Z}), \mathbb{Z}) = \underline{\mathrm{Hom}}(C(S, \mathbb{Z}), \mathbb{Z}).$$

This means that the underlying Abelian group of  $\mathbb{Z}[S]^{\blacksquare}$  is the  $\mathbb{Z}$ -valued measures on  $S$ .

- There is some set  $|I| \leq 2^{|S|}$ , that there is an isomorphism  $\mathbb{Z}[S]^{\blacksquare} \cong \prod_I \mathbb{Z}$ .
- $\mathbb{Z}[S]^{\blacksquare}$  is solid both as a module and a complex.

┘

*Proof:* 2: Take an isomorphism  $C(S, \mathbb{Z}) \cong \oplus_I \mathbb{Z}$ ?, then

$$\mathbb{Z}[S]^{\blacksquare} = \underline{\mathrm{Hom}}(C(S, \mathbb{Z}), \mathbb{Z}) \cong \underline{\mathrm{Hom}}(\bigoplus_I \mathbb{Z}, \mathbb{Z}) \cong \prod_I \mathbb{Z}.$$

3: We need to show the extension property, but by 2, it suffices to show for  $\mathbb{Z}[S]^{\blacksquare} = \mathbb{Z}$ ?. □

**Prop. (12.1.5.3)[Category of Solid Abelian Groups].** The category  $\mathrm{Solid} \subset \mathrm{Cond}(\mathcal{A}\mathrm{b})$  is an Abelian subcategory that is stable under all limits, colimits and extensions. The objects  $\prod_I \mathbb{Z}$ , where  $I$  is any set, form a family of compact projective generators. The inclusion  $\mathrm{Solid} \subset \mathrm{Cond}(\mathcal{A}\mathrm{b})$  admits a left adjoint  $M \mapsto M^{\blacksquare}$  which is the unique colimit-preserving extension of  $\mathbb{Z}[S] \mapsto \mathbb{Z}[S]^{\blacksquare}$ .

The functor  $D(\mathrm{Solid}) \rightarrow D(\mathrm{Cond}(\mathcal{A}\mathrm{b}))$  is fully faithful and its essential image is precisely the solid Abelian groups, and the inclusion admits a left adjoint  $C \rightarrow C^L$  which is left derived functor of  $M \rightarrow M^{\blacksquare}$ . ┘

## 6 Analytic Geometry

## 7 Complex Geometry

## 12.2 Topological Commutative Algebra(Adic Rings)

References are [Hub93, Sch12, Bos15, B-S19, Mor19].

**Notation(12.2.0.1).**

- Use notations on topological groups defined in [Topology II](#).
- Use notations on  $p$ -adic commutative algebra defined in [p-adic Commutative Algebras](#).

┘

### 1 Topological Abelian Groups and Rings

**Def.(12.2.1.1) [Topological Rings].** A **topological Abelian group** is an Abelian group with a topology structure that the addition and inversion are all continuous.

A **topological (commutative)ring** is a commutative ring  $R \in \mathcal{CRing}$  endowed with a topology structure that the addition, multiplication and inversion are all continuous. The category of topological rings is denoted by  $\mathcal{TopCRing}$ .

Similarly we can define a **topological module** over a topological ring. ┘

**Def.(12.2.1.2) [Topologically Nilpotent Element].** For  $A \in \mathcal{TopCRing}$ ,  $x \in A$  is called **topologically nilpotent element** iff  $x^n \rightarrow 0$  when  $n \rightarrow \infty$ . The set of topologically nilpotent elements is denoted by  $A^{00} \subset A$ . ┘

**Def.(12.2.1.3) [Bounded Sets].** For  $A \in \mathcal{TopCRing}$ , a subset  $S \subset A$  is called a **bounded subset** iff for all open nbhd  $U$  of 0, there exists an open nbhd  $V$  of 0 that  $VS \subset U$  ┘

**Def.(12.2.1.4) [Power-Bounded Elements].** For  $A \in \mathcal{TopCRing}$ ,  $x \in A$  is called a **power-bounded element** if  $\{x^n | n \in \mathbb{Z}_+\}$  is a bounded subset of  $A$  (12.2.1.3). The set of power-bounded elements are denoted by  $A^0$ . ┘

**Def.(12.2.1.5) [Uniform Rings].**  $A \in \mathcal{TopCRing}$  is called a **uniform ring** if  $A^{00}$  is bounded in  $A$  (12.2.1.2)(12.2.1.3). ┘

**Def.(12.2.1.6) [Strict Morphism].** A **strict morphism** of topological rings is a continuous morphism that the quotient topology and the subspace topology coincides on the image. ┘

### Completion of Topological Abelian Groups

**Prop.(12.2.1.7) [Completion].** There exists a completion functor left adjoint to the forgetful functor from the category of complete topological Abelian groups to the category of topological Abelian groups, given by Cauchy filters. ┘

*Proof:* ┘

**Prop.(12.2.1.8) [Subgroups and Completion].** Let  $A \in \mathcal{TopAb}$ , then the completion  $i : A \rightarrow \hat{A}$  induces a bijection between the set of open subgroups of  $A$  and the open subgroups of  $\hat{A}$ , given by  $G \mapsto \overline{i(G)} = \hat{G}$ . ┘

*Proof:* Cf.[Mor19]P74. ┘

**Def.(12.2.1.9) [Restricted Power Series].** For  $R \in \mathcal{TopCRing}$ , the **restricted power series** over  $R$  is defined to be

$$R\langle X_1, \dots, X_n \rangle \triangleq \left\{ \sum a_v X^v \in R[[X_1, \dots, X_n]] : \lim_{v \rightarrow \infty} a_v \rightarrow 0 \right\}.$$

┘

### Adic Rings

**Def. (12.2.1.10) [Adic Rings].** An **adic ring** is a topological ring  $R \in \text{TopCRing}$  s.t. the topology coincides with the  $\mathfrak{a}$ -adic topology for some ideal  $\mathfrak{a} \in \text{Ideal}(R)$ , and any such  $\mathfrak{a}$  is called a **ideal of definition**.

Write  $(R, \mathfrak{a}) \in \text{AdicCRing}$  to denote that  $R$  is an adic ring with an ideal of definition  $\mathfrak{a}$ .  $\lrcorner$

**Prop. (12.2.1.11).** If  $(R, \mathfrak{a}) \in \text{AdicCRing}$  (12.2.1.10) and  $R$  is Noetherian, then  $R$  has a maximal ideal of definition.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (12.2.1.12) [Topologically Nilpotent Elements].** For  $A \in \text{AdicCRing}$  and  $x \in A$ , the following are equivalent:

- $x$  is topologically nilpotent.
- There exists an ideal of definition  $I$  that the image of  $x$  in  $A/I$  is nilpotent.
- There exists an ideal of definition  $I$  that  $x \in I$ .

In particular, the  $A^{00}$  is an open radical ideal of  $A$  (12.2.1.2), and it is the union of all ideals of definitions in  $A$  (12.2.1.10).  $\lrcorner$

*Proof:*  $3 \rightarrow 1 \rightarrow 2$  is trivial.  $2 \rightarrow 3$ : if  $x$  is nilpotent in  $A/I$ , let  $J = I + xA$ , then  $J$  is an open ideal, and  $J^n \in I$ , so  $I$ -adic and  $J$ -adic topologies on  $A$  coincide, so  $J$  is an ideal of definition.

The rest is easy.  $\square$

**Prop. (12.2.1.13) [Adic Localization].** For  $(R, \mathfrak{a}) \in \text{AdicCRing}$ , the restricted power series  $R\langle \xi \rangle$  is complete w.r.t. the  $\mathfrak{a}$ -adic topology, and in fact

$$R\langle \xi \rangle \cong \varprojlim_n R/\mathfrak{a}^n[\xi].$$

For  $f \in R$ , we can define the **adic completion** of  $R$  at  $f$  to be the

$$R\langle f^{-1} \rangle = \varprojlim_n (R/\mathfrak{a}^n[\frac{1}{f}]).$$

The natural map  $R\langle \xi \rangle \rightarrow R\langle f^{-1} \rangle$  induces an isomorphism

$$R\langle \xi \rangle / (1 - f\xi) \cong R\langle f^{-1} \rangle.$$

$\lrcorner$

*Proof:* There is an isomorphism  $R[f^{-1}]/(\mathfrak{a}^n) \cong (R/\mathfrak{a}^n)[f^{-1}]$  because localization is flat, so we are done.  $\square$

**Remark (12.2.1.14).** There is another canonical morphism  $R[f^{-1}] \rightarrow R\langle f^{-1} \rangle$  which exhibits  $R\langle f^{-1} \rangle$  as the completion of  $R[f^{-1}]$  w.r.t. the ideal  $\mathfrak{a}R[f^{-1}]$ .

Then  $R\langle f^{-1} \rangle$  is endowed with an  $\mathfrak{a}$ -adic topology, and if  $\mathfrak{a}$  is f.g., then  $R\langle f^{-1} \rangle$  is complete w.r.t. to the  $\mathfrak{a}R\langle f^{-1} \rangle$ -adic topology. Also we see  $R\langle f^{-1} \rangle$  doesn't depend on the choice of ideal of definition of  $\mathfrak{a}$ .  $\lrcorner$

*Proof:* Consider the projective system of exact sequences:

$$0 \rightarrow (1 - f\xi)R/\mathfrak{a}^n[\xi] \rightarrow R/\mathfrak{a}^n[\xi] \rightarrow R/\mathfrak{a}^n[f^{-1}] \rightarrow 0.$$

which is a surjective system so by Mittag-Leffler(4.1.1.45),  $\varprojlim$  is exact(5.9.3.2), and there is an exact sequence

$$0 \rightarrow \varprojlim_n (1 - f\xi)R[\xi] \rightarrow \varprojlim_n R[\xi] \rightarrow \varprojlim_n R[f^{-1}] \rightarrow 0.$$

Now  $\varprojlim_n (1 - f\xi)R[\xi] \cong (1 - f\xi) \cong (1 - f\xi)R\langle\xi\rangle$ , because  $(1 - f\xi)$  is not a zero-divisor in  $R/\mathfrak{a}^n$ , and we get the desired isomorphism.  $\square$

**Def. (12.2.1.15) [Completed Tensor Product].** Let  $(A, \mathfrak{a}), (B, \mathfrak{b}) \in \text{AdicCRing}$  be complete, then we can define a **complete tensor product** ring  $(A \widehat{\otimes} B, (\mathfrak{a} \otimes B + A \otimes \mathfrak{b})) \in \text{AdicCRing}$  as

$$A \widehat{\otimes} B = \varprojlim_n (A/\mathfrak{a}^n \otimes B/\mathfrak{b}^n)$$

this is just the  $(\mathfrak{a} \otimes B + A \otimes \mathfrak{b})$ -adic completion of the tensor product  $A \otimes B$ , because  $(\mathfrak{a} \otimes B + A \otimes \mathfrak{b})^n$  and  $(\mathfrak{a}^n \otimes B + A \otimes \mathfrak{b}^n)$  are cofinal.  $\lrcorner$

### Properties of $R$ -Algebras

**Def. (12.2.1.16) [Topologically of Finite Presentation].** For  $(R, I) \in \text{AdicCRing}$ ,  $(A, I_A) \in \text{AdicCRing}/R$  is called

- **of topologically finite type** if  $A \cong R\langle\zeta_1, \dots, \zeta_n\rangle/\mathfrak{a}$  and  $I_A = IA$ , where  $\mathfrak{a} \in \text{Ideal}(R\langle\zeta_1, \dots, \zeta_n\rangle)$ .
- **of topologically finite presentation** if moreover the ideal  $\mathfrak{a}$  is f.g..
- **admissible** if it is of topologically finite presentation and  $A[I] = 0$ .

$\lrcorner$

**Prop. (12.2.1.17) [Raynaud-Gruson].** For  $(R, I) \in \text{AdicCRing}$ , if  $A \in \text{AdicCRing}$  is an  $R$ -algebra of topologically finite type(12.2.1.16) and  $M \in \text{Mod}_A$  is flat over  $R$ . Then  $M$  is an  $A$ -module of f.p..  $\lrcorner$

*Proof:* Cf.[Bos15]P165.  $\square$

**Cor. (12.2.1.18).** For  $(R, I) \in \text{AdicCRing}$ , if  $A \in \text{AdicCRing}$  is an  $R$ -algebra of topologically finite type(12.2.1.16) s.t.  $A[I] = 0$ , then  $A$  is of topologically finite presentation, in particular admissible(12.2.1.16).  $\lrcorner$

*Proof:* Cf.[Bos15]P166.  $\square$

**Prop. (12.2.1.19) [Topologically of Finite Presentation is Local].** Let  $A$  be an  $R$ -algebra that is  $I$ -adically complete and separated,  $f_1, \dots, f_r$  be a set of elements generating the unit ideal, then  $A$  is of topologically finite type/topologically finite presentation/admissible iff each  $A\langle f_i^{-1} \rangle$  does.  $\lrcorner$

*Proof:* Cf.[Bos15]P169.  $\square$



## 2 Valuation Rings

**Def.(12.2.2.1) [Valuation Ring].** A **valuation ring** is the maximum elements in the dominating ordering of local rings in a field  $K$ , where  $B$  **dominates**  $A$  iff  $A \subset B$  and  $\mathfrak{m}_B \cap A = \mathfrak{m}_A$ .

A valuation ring in  $K$  is called **absolutely algebraically closed** if  $K$  is alg.closed. ┘

**Prop.(12.2.2.2) [Valuation is Maximal].** Any local ring  $A$  in a field  $K$  is dominated by a valuation ring with fractional field  $K$ . ┘

*Proof:* Note that the dominating relation satisfies the condition of the Zorn's lemma, so it suffices to prove that  $A$  is not maximal if its fractional field is not  $K$ . Let  $t \notin K_0 = \text{frac}A$ . If  $t$  is transcendental over  $K_0$ , then  $A[t]$  with the maximal ideal  $(\mathfrak{m}, t)$  dominate  $A$ . If  $t$  is algebraic over  $K_0$ , then there is a  $a$  that  $at$  is integral over  $A$ , hence by (5.2.1.5) there is a maximal ideal of  $A[at]$  above  $A$ , which proves the lemma. □

**Prop.(12.2.2.3) [Valuation Ring Criterion].**  $A$  is a valuation ring with field of fraction  $K$  iff for any  $x \in K$ ,  $x$  or  $x^{-1}$  is in  $A$ . ┘

*Proof:* If  $A$  is a valuation ring, then for  $x \notin A$ , we know that  $A[x]$  is a local ring, hence there is no prime over  $\mathfrak{m}$  otherwise  $A[x]_{\mathfrak{p}}$  is a bigger local ring, so we see  $\mathfrak{m}A[x] = A[x]$ , i.e.  $1 = \sum t_i x^i$ , so  $x^{-1}$  is integral over  $A$ . Now  $A[x^{-1}]$  has a  $\mathfrak{m}'$  over  $\mathfrak{m}$ , so  $A = A[x^{-1}]_{\mathfrak{m}'}$ , which shows  $x^{-1} \in A$ .

Conversely, if for any  $x \in K$ ,  $x$  or  $x^{-1}$  is in  $A$ , we assume  $A$  is not  $K$ , so  $A$  is not field by the condition. Then it has a non-zero maximal ideal, but only one, otherwise we can choose  $x, y$  that  $x/y, y/x \notin A$ . And  $A$  is maximal because if there is a  $A \subset A'$ , and a  $x \in A'$ , then if  $x \notin A$ , then  $x^{-1} \in A$ , hence also in  $\mathfrak{m}_A$ , so it is in  $\mathfrak{m}_{A'}$ , but now  $x^{-1}$  cannot be in  $A'$ , contradiction. □

**Cor.(12.2.2.4).** For  $K \subset L$  subfield, if  $A$  is a valuation ring of  $L$ , then  $A \cap K$  is a valuation ring of  $K$ . And if  $L/K$  is algebraic and  $A$  is not a field, then  $A \cap K$  is not a field. (This is because the primes of  $A$  are all over 0 so cannot contain each other (5.2.1.5) so  $A$  is a field). ┘

**Cor.(12.2.2.5).** The quotient  $A/p$  at a prime is a valuation ring, and any localization of valuation ring is a valuation ring, by this criterion. ┘

**Prop.(12.2.2.6) [Valuation Rings are Normal].** Valuation rings are normal rings. ┘

*Proof:* If  $A$  is a valuation ring and  $x \in \text{Frac}(A)$  is algebraic over  $A$ , then either  $x \in A$ , or  $x$  is a combination of  $x^{-1}$  thus in  $A$ , by (12.2.2.3). □

**Cor.(12.2.2.7) [Integral Closure and Valuation Ring].** The integral closure of a subring in a field  $K$  is the intersection of valuation rings containing  $A$ . ┘

*Proof:* Valuation ring is integrally closed, so it suffices to prove if  $x$  is not algebraic over  $A$ , then there is a valuation ring of  $A$  not containing  $x$ . This is because  $x \notin B = A[x^{-1}]$  otherwise  $x$  is integral over  $A$ . Now  $x^{-1}$  is not a unit in  $B$ , hence  $x \in p \in B$ , hence  $B_p$  is dominated by some valuation ring  $V$ , and  $x \notin V$  because  $x^{-1} \in \mathfrak{m}_V$ . □

**Prop.(12.2.2.8) [Bezout Domain and Valuation Ring].** A valuation ring is equivalent to a Bezout local domain. ┘

*Proof:* One way is because the element of minimum valuation generate the ideal. Conversely, for  $f, g \in A$ ,  $(f, g) = (h)$ , so  $f = ah, g = bh$ , and  $h = cf + dg$ , then  $ab + cd = 1$ , hence  $a$  or  $b$  is a unit, so  $f/g \in A$  or  $g/f \in A$ . By (12.2.2.3),  $A$  is a valuation ring. □

**Prop. (12.2.2.9).** A valuation ring is Noetherian iff it is discrete valuation iff it is PID.  $\lrcorner$

*Proof:* Only need to prove Noetherian then  $\Gamma = \mathbb{Z}$ . we know ideals of  $\Gamma$  of the form  $\{x | x \geq \gamma\}$ , where  $\gamma > 0$  has a maximal element, so there is a minimal element bigger than 0, so  $\Gamma \cong \mathbb{Z}$ .  $\square$

**Prop. (12.2.2.10).** In a fixed field, any inclusion relation of two valuation ring is given by localization.  $\lrcorner$

*Proof:* Just localize at the image of the maximal ideal  $\mathfrak{m}_B \cap A$ , then they are valuation rings (12.2.2.5) that dominate each other, thus they are the same by definition (12.2.2.1).  $\square$

**Def. (12.2.2.11) [Extension of Valuation Rings].** An injective local homomorphism of valuation rings is called an **extension of valuation rings**. By (5.4.1.28), it is equivalent to a f.f. morphism of valuation rings.  $\lrcorner$

### 3 Valuations on Fields

**Notation (12.2.3.1).**

- Let  $K \in \text{Field}$ .

**Def. (12.2.3.2) [Valuations on Fields].** For  $K \in \text{Field}$ , a **field valuation** on  $K$  is a surjective homomorphism  $v : K^\times \rightarrow \Gamma \in \text{Ab}$  s.t.  $v(x + y) \geq \min\{v(x), v(y)\}$  for any  $x \neq -y \in K^\times$ , where  $\Gamma$  is an ordered Abelian group (3.2.9.2). This  $\Gamma$  is called the **value group** of  $v$ .

The **rank of a valuation on field** is defined as the height of its value group (3.2.9.5).

$v$  is called **discrete valuation on field** if  $\Gamma \cong (\mathbb{Z}, < 0)$ .  $\lrcorner$

**Prop. (12.2.3.3) [Valuation Ring and Valuation].** For  $K \in \text{Field}$ , given a valuation ring  $A \leq K$  (12.2.2.1), let  $K^\times \rightarrow \Gamma = K^\times / A^*$  be the natural valuation (12.2.3.2). Conversely, given a valuation  $v$  on  $K$ ,  $A = v^{-1}(\{x \geq 0\}) \cup \{0\}$  is a valuation ring.

Thus valuation rings of  $K$  is equivalent to valuations on  $K$ .  $\lrcorner$

*Proof:* These are definitely valuation rings, and if  $A$  is a valuation ring by (12.2.2.3), then we set  $\Gamma = K^*/A^*$ , where  $A^*$  is the invertible elements of  $A$  and  $x \leq y$  iff  $y/x \in (A - \{0\})/A^*$ . This is totally ordered by (12.2.2.3).  $\square$

**Cor. (12.2.3.4) [Rank and Dimension].** A valuation ring of rank  $n$  (12.2.3.2) has Krull dimension  $n$ , because the set of convex subgroups of  $K^\times / A^*$  (3.2.9.4) is in bijection with the set of ideals of  $A$ .  $\lrcorner$

**Prop. (12.2.3.5) [DVRs].** For a Noetherian local domain  $(A, \mathfrak{m}, \kappa)$  of dimension 1 that is not a field, the following are equivalent:

1.  $A$  is a valuation ring corresponding to a discrete valuation in (12.2.3.3).
2.  $A$  is normal.
3.  $\mathfrak{m}$  is a principal ideal.
4.  $A$  is regular.
5. Every nonzero ideal is a power of  $\mathfrak{m}$ .
6. There exists a  $x \in A$  that every nonzero ideal is of the form  $(x^k)$ .

Such a ring is called a **discrete valuation ring** or a DVR. The category of discrete valued rings are denoted by DVR.  $\lrcorner$

*Proof:*  $1 \rightarrow 2$  : Valuation ring is integrally closed, by(12.2.2.6).

$2 \rightarrow 3$  : As the radical of any non-trivial ideal  $(a)$  is  $\mathfrak{m}$ , and  $A$  is Noetherian, so there is an  $n$  that  $\mathfrak{m}^n \subset a$  and  $\mathfrak{m}^{n-1} \not\subset a$ . Then choose  $b \in a - \mathfrak{m}^{n-1}$ ,  $x = a/b \in K$ , then  $x^{-1} \notin A$ , then it is not integral over  $A$ . So  $x^{-1}\mathfrak{m} \not\subset \mathfrak{m}$ , but  $x^{-1}\mathfrak{m} \subset A$ , so it equals  $A$ , which means  $\mathfrak{m} = (x)$ .

$3 \rightarrow 4$  : Clear.

$4 \rightarrow 5$  : For any ideal  $a$ , its radical is  $\mathfrak{m}$  and  $A$  is Noetherian, so  $\mathfrak{m}^n \subset a$ . Now  $A/\mathfrak{m}^n$  is Artinian by(5.1.3.4), so by(5.1.3.6)  $a$  is a power of  $\mathfrak{m}$ .

$5 \rightarrow 6$  : And  $x \in \mathfrak{m} - \mathfrak{m}^2$  will do.

$6 \rightarrow 1$  : Define  $v(a) = k$  if  $(a) = (x^k)$ . □

**Cor.(12.2.3.6) [Noetherian Valuation Rings are DVRs].** A Noetherian valuation rings are automatically DVRs. ┘

*Proof:* This is implicit in the proof above. □

**Prop.(12.2.3.7).** Let  $R$  be a Noetherian local domain with fraction field  $K$  and  $R \neq K$ , then there exists a Noetherian local ring  $R'$  of dimension 1 that dominates  $R$  s.t.  $R \rightarrow R'$  is essentially of f.t.. ┘

*Proof:* Cf.[Sta]00P8. □

**Cor.(12.2.3.8) [Dominance by DVRs].** Let  $R$  be a Noetherian local domain with fraction field  $K$  and  $R \neq K$ ,  $L/K$  a f.g. filed extension, then there exists a a DVR  $A$  with fraction field  $L$  which dominants  $R$ . ┘

*Proof:* First we can reduce to the case  $L/K$  is finite: If it is not finite, choose a transcendence basis  $x_1, \dots, x_r$ , and replace  $R$  by  $R[x_1, \dots, x_r]_{\mathfrak{m}_{R,x_1,\dots,x_r}}$ .

In the finite case, first we can reduce to the case  $\dim R = 1$  by(12.2.3.7), and let  $A$  be the integral closure of  $R$  in  $L$ , then by(5.1.1.53),  $A$  is Noetherian, and  $R \rightarrow A$  is integral so there exists a maximal prime  $\mathfrak{n} \subset A$  that  $\mathfrak{n} \cap R = \mathfrak{m}_R$ . Thus  $A_{\mathfrak{n}}$  is a DVR by(12.2.3.6). □

### Valuations of Rank 1

In this subsubsection, all valuations are of rank 1(12.2.3.2).

**Remark(12.2.3.9) [Multiplicative Valuations].** As any ordered Abelian group of height 1 can be embedded into  $\mathbb{R}$ (3.2.9.6), a field valuation  $v$  on a field  $K$  of rank 1 is equivalent to a real-valued ultrametric valuation.

In this case, define  $|\cdot| : K \rightarrow \mathbb{R}_+ : |a| = \begin{cases} \exp(-v(a)) & , a \neq 0 \\ 0 & , a = 0 \end{cases}$ . ┘

**Def.(12.2.3.10) [Valuations].** For  $R \in \mathcal{CRing}$ , a **valuation** on  $R$  is a function  $|\cdot| : R \rightarrow \mathbb{R}_{\geq 0}$  s.t.

- $|1| = 1$ .
- $|x + y| \leq |x| + |y|$  for all  $x, y \in R$ .
- $|ab| = |a| \cdot |b|$  for any  $a, b \in R$ .

And a commutative ring with a valuation is called a **valued ring**.

A valuation is called a **non-Archimedean valuation** iff  $|x + y| \leq \max\{|x|, |y|\}$  for any  $x, y$ , i.e.  $|\cdot|$  is a ultrametric valuation(19.1.1.3). It is called an **Archimedean valuation** iff it is not non-Archimedean.

A **non-Archimedean field**(resp. Archimedean field) is a field with a non-Archimedean(resp. Archimedean) valuation. ┘

**Prop. (12.2.3.11).** A valuation on  $K$  is non-Archimedean(12.2.3.10) iff  $\{|n| : n \in \mathbb{N}\} \subset \mathbb{R}_{\geq 0}$  is bounded.  
 $\lrcorner$

*Proof:* If it is non-archimedean, then clearly by induction and  $n = 1 + (n - 1)$   $|n| \leq 1$ . Conversely, if  $|n| \leq M$ , then consider  $|x + y|^n = |(x + y)^n| \leq \sum |C_n^k| x^k y^{n-k} \leq M \max\{|x|, |y|\}$ , so letting  $n$  be large, clearly  $|x + y| \leq \max\{|x|, |y|\}$ .  $\square$

**Cor. (12.2.3.12).** If  $p \in \text{Prime}$  and  $R \in \mathcal{CRing}/\mathbb{F}_p$ , then any valuation on  $R$  is non-Archimedean(12.2.3.10).  
 $\lrcorner$

**Def. (12.2.3.13) [Equivalent Valuations].** For  $R \in \mathcal{CRing}$ , two valuations on  $R$ (12.2.3.10) are called **equivalent** if they induce the same topology on  $K$ .  
 $\lrcorner$

**Prop. (12.2.3.14) [Equivalent Valuations].** Two valuations on a field is equivalent(12.2.3.13) iff  $|x|_1 < 1 \implies |x|_2 < 1$ . And in this case,  $|x|_1 = |x|_2^s$  for some  $s \in \mathbb{R}_+$ .  
 $\lrcorner$

*Proof:* if two valuation are equivalent, then  $x^n \rightarrow 0$  in  $\tau_1$  iff  $x^n \rightarrow 0$  in  $\tau_2$ , so  $|x|_1 < 1 \implies |x|_2 < 1$ .  
 If  $|x|_1 < 1 \implies |x|_2 < 1$ , then let  $y$  be an element that  $|y|_1 > 1$ , then any element  $|x| = |y|^\alpha$  for some  $\alpha \in \mathbb{R}$ . Let  $\frac{n_i}{m_i}$  converges to  $\alpha$  from above, then  $|\frac{x^{n_i}}{y^{m_i}}|_1 < 1$ , so  $|\frac{x^{n_i}}{y^{m_i}}|_2 < 1$ , so  $|x|_2 \leq |y|_2^\alpha$ . Similarly,  $|x|_2 \geq |y|_2^\alpha$ , so  $|x|_2 = |y|_2^\alpha$ . So  $|x|_1 = |x|_2^s$  for some  $s > 0$ .  
 If  $|x|_1 = |x|_2^s$  for some  $s > 0$ , then these two valuations are clearly equivalent.  $\square$

**Cor. (12.2.3.15) [Weak Approximation].** If  $|\cdot|_1, \dots, |\cdot|_n$  be pairwise inequivalent valuations on  $K$ , then for any  $a_1, \dots, a_n \in K$  and  $\varepsilon > 0$ , there is an  $x \in K$  that  $|x - a_i|_i < \varepsilon$ .  
 $\lrcorner$

*Proof:* As  $|\cdot|_1, |\cdot|_n$  are inequivalent, there are  $\alpha, \beta$  that  $|\alpha|_1 < 1, |\alpha|_n \geq 1, |\beta|_n < 1, |\beta|_1 \geq 1$  by(12.2.3.14), so let  $y = \beta/\alpha$ , then  $|y|_1 > 1, |y|_n < 1$ .  
 Now we prove by induction that there is an  $\alpha$  that  $|\alpha|_1 > 1, |\alpha|_i < 1$  for  $i = 2, \dots, n$ . the case  $n = 2$  is done, for general  $n$ , if the  $z$  for  $n - 1$  satisfies  $|z|_n \leq 1$ , then  $z^m y$  will do, for  $m$  large. if  $|z| > 1$ , then the sequence  $|t_m|_i = |\frac{z^m}{1+z^m}|_i$  converges to 1 for  $i = 1, n$  and converges to 0 for  $i = 2, \dots, n - 1$ , so  $t^m y$  will do, for  $m$  large.  $\square$

**Prop. (12.2.3.16) [Gelfand].** Any field  $K$  with an Archimedean place is a subfield of  $\mathbb{C}$ .  
 $\lrcorner$

*Proof:* We consider its completion. when it contains  $\mathbb{C}$ , this is a corollary of??, otherwise, we consider  $K \otimes \mathbb{C}$ , then it is a finite dimensional module over  $K$  thus also complete.  $\square$

**Remark (12.2.3.17).** Because of this, we usually don't consider only non-Archimedean valuations, and refer to all valuations as **places**, cf.(19.3.1.5).  
 $\lrcorner$

**Thm. (12.2.3.18) [Ostrowski].**

1.  $\Sigma_{\mathbb{Q}} = \text{Prime} \cup \{\infty\}$ . Thus any complete Archimedean field is isomorphic to  $\mathbb{R}$  or  $\mathbb{C}$  by(12.2.3.16).
2. Any non-trivial valuation on  $\mathbb{F}_q(t)$  is of the form  $|\cdot|_p$  or  $|\cdot|_\infty$ , where  $p$  is an irreducible polynomial in  $\mathbb{F}_q[t]$ .

$\lrcorner$

*Proof:* 1: if it is non-Archimedean, then  $|n| \leq 1$ , and it is not trivial, so there is a minimal  $p$  that  $|p| < 1$ . Then  $p$  is easily seen to be a prime. Then for any  $(a, p) = 1$ ,  $a = dp + r$ , so  $|r| = 1$ , so  $|a| = 1$ .

And if it is Archimedean, then we prove that in  $\mathbb{N}$ ,  $|m| = m^\lambda$  for some  $\lambda$ : Let  $F(n) = |n|$  and  $f = \log_2 F$ , then  $f(m+n) \leq \max\{f(m), f(n)\} + 1$ , and if  $m = \sum_{i=1}^r d_i n^i$ , then  $f(m) \leq r(1+f(n)) + a_n$ , where  $a_n = \sup\{f(k) | k < n\}$ . And  $r \leq \log m / \log n$ , so

$$\frac{f(m)}{\log m} \leq \frac{a + f(n)}{\log n} + \frac{b}{\log n}$$

then letting  $m \rightarrow m^k, k \rightarrow \infty$ , and then let  $n \rightarrow n^k, k \rightarrow \infty$ , we get  $\frac{f(m)}{\log m} \leq \frac{f(n)}{\log n}$ , for any  $m, n$ .

2: Any valuation on  $\mathbb{F}_q(t)$  is non-Archimedean(12.2.3.12), and  $|n| = 1$  if  $(n, p) = 1$ , because  $n^{p-1} = 1$ . Similarly, if there is a minimal hence irreducible  $P$  that  $|P| < 1$ , then use induction and  $Q = sP + r$  for some  $s, r$  of degree  $< \deg Q$ , so  $|Q| = 1$  for all  $(Q, P) = 1$ . Otherwise,  $|P| \leq 1$  for all  $P$ , then  $|t| > 1$ , otherwise  $|\cdot|$  is trivial, so it is easy by induction that  $|F(t)| = |t|^{\deg F}$ .  $\square$

**Lemma(12.2.3.19) [Continuity of Roots].** If  $(K, |\cdot|)$  is a valued field and  $K = \overline{K}$ , then for a separable polynomial  $f$  over a valued alg.closed field  $\overline{K}$ , there is a  $\varepsilon$  that every polynomial  $g$  that are closed enough to  $f$ , the roots of  $g$  is closed to roots of  $f$  respectively.  $\lrcorner$

*Proof:* This is easy to see by decomposition as each root of  $f$  is close to a root of  $g$ .  $f, g$  have the same degree so the roots correspond to each other.  $\square$

**Prop.(12.2.3.20) [Fundamental Inequality].** if  $(K, v)$  is a valued field and  $L/K$  be a field extension of degree  $n$ , if  $w_i$  are the valuations of  $L$  above  $v$ , then

$$\sum_{w_i|v} e(w_i/v) f(w_i/v) \leq [L : K].$$

The equality holds when  $v$  is discrete and  $L/K$  is separable.  $\lrcorner$

*Proof:* Cf.[Clark note Theorem4].?  $\square$

**Def.(12.2.3.21) [Spherically Complete Valued Field].** A valued field  $K$  is called **spherically complete** iff each descending chain of metric balls has a nonempty intersection.  $\lrcorner$

### Microbial Valuations

**Prop.(12.2.3.22) [Microbial Valuation].** For a valuation ring  $(R, K)$  with the corresponding valuation on  $K$ . The following are equivalent:

- The topology on  $K$  coincides with a rank 1 topology.
- $K^{00} \neq 0$ .
- $R$  has a prime ideal of height 1.

If this is the case, then the valuation defined by  $A$  is called **microbial valuations**.

And in this case, for any topologically nilpotent element  $\varpi$ ,  $K = R[\varpi^{-1}]$ , and  $\varpi^r \in R$  for some  $r$ , and the topology on  $R$  is  $\varpi^r$ -adic. And if  $\mathfrak{p}$  is a prime ideal of rank 1, then the valuation ring  $R_{\mathfrak{p}}$  is of rank 1, and defines the same topology on  $R$ .  $\lrcorner$

*Proof:* 1  $\rightarrow$  2: if there is a rank 1 valuation  $|\cdot|'$  that defines the same topology as  $R$ , then any  $|x|' < 1$  will be a topological nilpotent element by(12.2.3.23).

2  $\rightarrow$  3: if  $\varpi$  is a topological nilpotent element, then  $\mathfrak{p} = \sqrt{(\varpi)}$  is a prime ideal, and it is minimal, because if there is another  $\mathfrak{q} \subsetneq \mathfrak{p}$ , then  $\varpi \notin \mathfrak{q}$ , but  $\mathfrak{p} \subset (\varpi^n)$  by induction: because  $(\varpi) \not\subseteq \mathfrak{q}$ ,  $\mathfrak{q} \subset (\varpi)$ ,

and if  $x \in \mathfrak{q}$ , then  $x = \varpi^n y$ , and  $\varpi \notin \mathfrak{q}$ , so  $y \in \mathfrak{q} \subset (\varpi^n)$ , so  $x \in (\varpi^{n+1})$ . Now  $\mathfrak{q} = 0$  because  $\varpi$  is topological nilpotent.

$3 \rightarrow 1$ : It suffices to prove that the valuation defined by  $R_{\mathfrak{p}}$  is the same as the topology of  $R$ . But this is true in general, just notice that the valuation topology of a nontrivial valuation is also defined by  $B(a, \gamma]$ .

The final remark is clear as  $x\varpi^n \in R \iff |\varpi^n| \leq |x^{-1}|$ .  $\square$

**Lemma (12.2.3.23).** Let  $R$  be a valuation ring, if  $x \in R^*$  is topologically nilpotent, then  $|x| < 1$ , and the converse is also true if  $R$  has rank 1.  $\lrcorner$

*Proof:* if  $|x| \geq 1$ , then  $x^{\mathbb{N}} \not\subset B(0, 1)$ , so it is not topologically nilpotent. And if  $R$  has rank 1,  $|x| < 1$ , then for any  $\delta \neq 0$ , there is some  $n$  that  $|\delta^{-1}| < |x^{-n}|$  (3.2.9.6), so  $|x^m| < |\delta|$  for  $m$  large, thus  $x$  is topologically nilpotent.  $\square$

**Prop. (12.2.3.24) [Constructing Microbial Valuations].** If  $A$  is a valuation ring and  $f \in A$  is a non-zero non-unit, then the  $f$ -adic Hausdorffization  $\overline{A} = A / \cap_n (f^n)$  and the completion  $\hat{A}$  are all microbial.  $\lrcorner$

*Proof:* Easy, Cf. [Perfectoid Spaces, Bhatt]P63.  $\square$

## 4 Affinoid Algebras

References are [BGR84]Part B.

**Notation (12.2.4.1).**

- Let  $(K, |\cdot|_K, \mathcal{O}_K, \kappa)$  be a complete non-Archimedean field.

$\lrcorner$

### Tate-Algebras

**Def. (12.2.4.2) [Tate-Algebras].** The **Tate- $K$ -algebra**  $T_n = K \langle x_1, \dots, x_n \rangle$  is defined to be the restricted power series (12.2.1.9) consists of elements  $\sum_{\alpha} a_{\alpha} x^{\alpha}$  s.t.  $a_{\alpha} \in K$  and  $\lim_{|\alpha| \rightarrow \infty} |a_{\alpha}| = 0$ . It is endowed with the norm  $|f| = \max |a_v|$ .

The norm satisfies  $|fg| = |f||g|$  and  $|f + g| \leq |f| + |g|$ .

There is a **reduction map** from  $T_n$  to  $\kappa[x_1, \dots, x_n]$ , which is surjective.  $\lrcorner$

*Proof:*  $T_n$  is an algebra because the values of coefficients of  $f$  is bounded.  $|fg| \leq |f| \cdot |g|$  is easy, to show  $|fg| \geq |f||g|$ , we assume  $|f| = |g| = 1$ , then their reduction in  $K[x_1, \dots, x_n]$  is non-zero, thus  $\overline{fg}$  is non-zero, which shows  $|fg| \geq 1$ .  $\square$

**Prop. (12.2.4.3) [Maximum Principle].** A formal power series  $f \in K[[\mathbf{X}]]$  converges in  $B^n(\overline{K})$  iff it is in  $T_n$ .

And when it is in  $T_n$ ,  $|f(x)|$  attains its maximum value  $|f|$  in  $B^n(\overline{K})$ .  $\lrcorner$

*Proof:* If it converges at  $(1, \dots, 1)$ , then  $\lim_{|v| \rightarrow \infty} |a_v| = 0$  by (19.1.2.10). Conversely, for any point in  $B^n(\overline{K})$ , it can be considered in a finite extension field of  $K$ , thus complete, hence we can apply (19.1.2.10) again.

For the second assertion, we assume  $|f| = 1$ , then consider its reduction to  $k[x_1, \dots, x_n]$ , then there is a  $\bar{x}$  in the alg. closure of  $k$  that  $\bar{f}(\bar{x}) \neq 0$ . Now  $\bar{k}$  can be seen as the residue field of  $\overline{K}$ . Then the lifting of  $\bar{x}$  to a  $x \in \overline{K}$  has valuation 1 and  $|f(x)| = 1$ .  $\square$

**Prop. (12.2.4.4).**  $T_n$  is a Banach-algebra(12.7.4.1). ┘

**Cor. (12.2.4.5).** An element  $f$  of norm 1 of  $T_n$  is invertible in  $T_n$  iff its reduction in  $k[x_1, \dots, x_n]$  is a unit. Elements of other norms can be reduced to the case of norm 1. ┘

*Proof:* One direction is trivial, the other is because  $|f - f(0)| < 1$ , hence  $f = f(0)(1 + g)$ , this is invertible by power expansion as  $T_n$  is complete. □

**Def. (12.2.4.6) [Distinguished Power Series].** A restricted power series  $g = \sum g_v X_n^v \in T_n \in T_{n-1} \langle X_n \rangle$  is called  $X_n$ -**distinguished of order  $s$**  iff  $g_s$  is a unit in  $T_{n-1}$ ,  $|g_s| = |g|$  and  $|g_s| > |g_v|$  for all  $v > s$ . ┘

**Lemma (12.2.4.7).** For any f.m. elements  $f_i \in T_n$ , there is a continuous automorphism of  $T_n$  that maps  $T_n \rightarrow T_n, T_i \rightarrow T_i + T_n^{\alpha_i}$  that maps  $f_i$  to  $X_n$ -distinguished elements. ┘

*Proof:* Cf.[Rigid and Formal Geometry P16]. □

**Prop. (12.2.4.8) [Weierstrass Division].** If  $g \in T_n$  is  $X_n$ -distinguished of order  $s$ , for any  $f \in T_n$ , there is a unique form  $f = qg + r$ , where  $q \in T_n$  and  $r \in T_{n-1}[X_n]$  of degree  $r < s$ . Moreover,  $|f| = \max\{|q||g|, |r|\}$ . ┘

*Proof:* Cf.[Rigid and Formal Geometry P17]. □

**Cor. (12.2.4.9) [Weierstrass Preparation].** If  $g \in T_n$  is  $X_n$ -distinguished of order  $s$ (12.2.4.6), then there exists uniquely a  $r \in T_{n-1}[X_n]$  of degree  $s$  and  $g = re$ , where  $e$  is a unit in  $T_n$ . ┘

*Proof:* By(12.2.4.8) applied to  $X_n^s = qg + r$  with  $|r| \leq 1$ . Then  $\omega = X_n^s - r$  is  $X_n$  is the desired polynomial, it suffice to show  $q$  is a unit. Let  $g$  be normalized that  $|g| = 1$ , then  $|q| = 1$ , by reduction to polynomials, we see  $\tilde{\omega} = \tilde{q}\tilde{g}$ , and  $\tilde{\omega}, \tilde{g}$  are both polynomials of degree  $s$ , so  $\tilde{q} \in k^*$ , so  $q$  is a unit by(12.2.4.5).

Uniqueness: if  $g = e\omega$ , then  $X_n^s = e^{-1}g + (X_n^s - \omega)$ , so uniqueness follows from that of Weierstrass division. □

**Prop. (12.2.4.10) [Noether Normalization].** For any proper ideal  $\mathfrak{a} \in \text{Ideal}(T_n)$ , there exists some  $d \in \mathbb{N}$  with a finite injection  $T_d \hookrightarrow T_n/\mathfrak{a}$ . ┘

*Proof:* We may assume  $\alpha \neq 0$ , thus choose a  $g \in \alpha \neq 0$ , then using(12.2.4.7), we may assume  $g$  is  $X_n$ -distinguished. Now the Weierstrass division theorem(12.2.4.8) says that  $T_{n-1} \rightarrow T_n/(g)$  is finite. Hence  $T_{n-1} \rightarrow T_n/(g) \rightarrow T_n/\mathfrak{a}$  is finite. Now we can use induction to find a  $T_d \rightarrow T_{n-1}/T_{n-1} \cap \mathfrak{a}$  finite, thus also  $T_d \rightarrow T_n/\mathfrak{a}$  is finite. □

**Cor. (12.2.4.11) [Finite Residue Fields].** The residue field of a maximal ideal  $\mathfrak{m} \in |\text{Spec } T_n|_0$  is a finite extension field of  $K$  ┘

*Proof:* Because finite injection  $T_d \rightarrow T_n/\mathfrak{m}$  shows  $T_n$  is a field(5.2.1.3), thus we must have  $d = 0$ . because  $T_n/\mathfrak{m}$  has dimension 0, thus  $K \rightarrow T_n/\mathfrak{m}$  is a finite injection. □

**Cor. (12.2.4.12).** The map from  $B^n(\overline{K})$  to the set of maximal ideals of  $T_n$  are surjective. ┘

*Proof:* Evaluation map defines a map  $T_n \rightarrow K(x_1, \dots, x_n)$  that is surjective, thus the kernel is a maximal ideal. Conversely, for any maximal ideal  $\mathfrak{m} \subset T_n$ ,  $K' = T_n/\mathfrak{m}$  is finite over  $K$ , so we may assume  $K' \subset \overline{K}$ .



We show that this map  $\varphi : T_n \rightarrow \overline{K}$  is contractive, otherwise there is a  $|a| = 1, |\alpha = \varphi(a)| > 1$ . Consider the minimal polynomial  $p$  of  $|\alpha|$ , all the conjugates of  $\alpha$  has the same valuation as  $K$ , as  $K$  is Henselian, thus  $p$  has ascending Newton polygon, thus by (12.2.4.5) it is invertible in  $T_n$ . But  $\varphi(p(a)) = 0$ , contradiction.

So  $|\varphi(x)| \leq |x|$ , then it is continuous, and  $(\varphi(T_1), \dots, \varphi(T_n)) \subset B^n(K^n)$ , so we are done.  $\square$

**Cor. (12.2.4.13) [Ring Properties of  $T_n$ ].**  $T_n$  is Noetherian, UFD, Jacobson of Krull dimension  $n$ .  $\lrcorner$

*Proof:* Noetherian: Use induction, as in the proof of (12.2.4.10),  $T_{n-1} \rightarrow T_n/(g)$  is finite for some  $g \in \mathfrak{a}$ , then also  $T_n/\mathfrak{a}$  is finite over  $T_{n-1}$ , thus Noetherian as a  $T_{n-1}$  module, thus Noetherian as a ring.

UFD: Cf. [Rigid and Formal Geometry P20].

Jacobson: We need to show that any prime ideal  $\mathfrak{p}$  is an intersection of maximal ideals. The case of  $\mathfrak{p}$  is by (12.2.4.3). For  $\mathfrak{p} \neq 0$ , by Noetherian normalization (12.2.4.10), there is a  $T_d \subset T_n/\mathfrak{p}$  finite. Then use induction and generalized Nullstellensatz (5.2.6.10),  $T_n/\mathfrak{p}$  is Jacobson, thus  $\mathfrak{p} = \text{rad}(T_n/\mathfrak{p})$ .

Dimension  $n$ : Cf. [Formal and Rigid Geometry P22].  $\square$

**Prop. (12.2.4.14).** For an ideal  $\mathfrak{a} \in T_n$ , there are  $a_1, \dots, a_r$  which generate  $\mathfrak{a}$  that  $|a_i| = 1$ , and any elements in  $f$  has a representation of the form  $\sum f_i a_i$  with  $|f_i| \leq |f|$ .

The same assertion holds for submodules of  $T_n^k$ .  $\lrcorner$

*Proof:* Cf. [Formal and Rigid Geometry P27,29].  $\square$

**Cor. (12.2.4.15) [Affinoid Ideals are Closed].** Each ideal of  $T_n$  is closed hence complete in  $T_n$ . This follows immediately from (12.2.4.13) and (19.1.6.13).  $\lrcorner$

**Cor. (12.2.4.16).** For any ideal  $\mathfrak{a}$  of  $T_n$ , the distance from an element to  $\mathfrak{a}$  attains minimum.  $\lrcorner$

*Proof:* Cf. [Bos15] P28.  $\square$

### Affinoid Algebras

**Def. (12.2.4.17) [Affinoid Tate  $K$ -Algebras].** A normed algebras of the form  $A = T_n/\mathfrak{a}$  are called **affinoid  $K$ -algebras**, so it is Noetherian and Jacobson by (12.2.4.13). An affinoid algebra has a natural semi-norm by  $|f|_{\text{sup}} = \sup |f|_{\mathfrak{m}}$  in  $A/\mathfrak{m}$  for a maximal ideal  $\mathfrak{m}$  of  $A$  by (12.2.4.11).  $\lrcorner$

*Proof:* We need to show the sup is finite, for this, notice  $|f| = |g|$  for some  $g$  in the residue norm (12.2.4.18), so for any maximal ideal  $\mathfrak{m}$  of  $A$ , the inverse is a maximal ideal  $\mathfrak{n}$  in  $T_n$  by finiteness, thus  $|f|_{\mathfrak{m}} = |g|_{\mathfrak{n}} \leq |g|_{\text{sup}} = |g| = |f|$ , so  $|f|_{\text{sup}} \leq |f|$ .

For the second-last equality, notice on  $T_n$ ,  $|\cdot|_{\text{sup}}$  and  $|\cdot|$  equal, by (12.2.4.3) and (12.2.4.12).  $\square$

**Def. (12.2.4.18) [Residue Norm].** For an affinoid  $K$ -algebra  $A = T_n/\mathfrak{a}$ , there is a natural residue norm on it. This is a complete  $K$ -algebra norm on  $A$  s.t.  $T_n \rightarrow A$  is continuous and open. For any  $f \in A$ , the residue norm is obtained at an element of  $T_n$ .

Any residue norm is bigger than the sup-norm, by the proof of (12.2.4.17).  $\lrcorner$

*Proof:* It is a  $K$ -algebra norm is easily verified, should notice  $|f| = 0$  iff  $f = 0$ , because  $\mathfrak{a}$  is closed (12.2.4.15). The last assertion follows from (12.2.4.16).  $\square$

**Remark (12.2.4.19).** The sup norm may not even be a norm, if  $\mathfrak{a}$  is not radical, but the fact that sup norm is smaller than any residue norm, together with (12.2.4.23), is enough for use.  $\lrcorner$



**Prop. (12.2.4.20).** For  $T_d \rightarrow A$  a finite injection, assume  $A$  is a torsion-free  $T_d$ -module, then for any  $f \in A$ , there is a unique minimal monic polynomial  $P$  of  $f$  over  $T_d$ .

In this case,  $|f|_{\sup} = \sup |a_i|_{\sup}^{1/i}$  where  $a_i$  are coefficients of  $P$ .  $\lrcorner$

*Proof:* Because  $A$  is torsion-free, we reduce to the quotient field of  $T_n$ , then  $f$  has a minimal monic polynomial, and  $T_n$  is UFD, hence Gauss lemma shows that this polynomial has coefficients in  $T_d$ . Hence  $T_n[f] = T_n[X]/(p)$ .

For the second, notice first for finite extension the Spec map is surjective, thus we may assume  $A = T_n[f] = T_n[X]/(p)$ , and for a maximal ideal  $\mathfrak{m}$  of  $T_n$ , let  $T_n/\mathfrak{m} = k$ , then  $A/(\mathfrak{m}) = k[X]/(\bar{p})$ , then maximal ideals of  $A/(\mathfrak{m})$  corresponds to roots  $\alpha_i$  of  $\bar{p}$  in  $\bar{k}$ , so

$$\sup_{\varphi^{-1}(\mathfrak{n})=\mathfrak{m}} |f|_{\mathfrak{n}} = \sup |\alpha_i| = \max |a_i|_{\mathfrak{m}}^{1/i},$$

so the result follows.  $\square$

**Cor. (12.2.4.21).**  $|f|_{\sup} \in \sqrt[N]{|K|}$  for some  $N$  and all  $f \in A$ , because the minimal polynomial has coefficients in  $T_n$ , and sup norm and Gauss norm coincide on  $T_n$  by the proof of (12.2.4.17).  $\lrcorner$

**Cor. (12.2.4.22) [Maximum Principle].** For any  $f \in A$ ,  $\|f\|_{\sup} = \|f\|_{\mathfrak{m}}$  for some maximal ideal  $\mathfrak{m}$ .  $\lrcorner$

*Proof:* Since  $A$  is Noetherian (12.2.4.13), it has f.m. minimal primes, hence  $|f|_{\sup} = |f|_{\sup}$  in  $A/p_i$  for some minimal prime of  $A$ . Hence we reduce to the case of (12.2.4.20), hence the conclusion follows from (12.2.4.3) and the proof of (12.2.4.20).  $\square$

**Prop. (12.2.4.23) [Residue Norms Equivalent].** Any morphism from a Noetherian  $k$ -algebra to an affinoid algebras  $A$  is continuous w.r.t any residue norms. In particular, any  $k$ -Banach algebra topology on  $A$  coincides with the  $k$ -affinoid topology on  $A$ , and all residue norms on an affinoid algebra are equivalent.

Moreover, for any morphism of  $k$ -affinoid algebras  $B \rightarrow A$ , the norm on  $A$  can be replaced by an equivalent one that makes  $A$  into a normed  $B$ -algebra.  $\lrcorner$

*Proof:* Use (19.1.6.10), it suffices to show the condition holds, for  $\mathfrak{B} = \{\mathfrak{m}^v\}$  where  $\mathfrak{m}$  are maximal ideals of  $A$ : The residue field is finite by (12.2.4.11), their intersections is  $(0)$  because if  $f \in \cap_{\mathfrak{m}} \cap_n \mathfrak{m}^n$ , Krull's theorem (5.2.2.15) (use localization) says for each maximal ideal  $\mathfrak{m}$  there is a  $m \in \mathfrak{m}$  that  $(1 - m)f = 0$ , so  $\text{Ann}(f) = (1)$ , so  $f = 0$ .

For the second assertion, see [non-Archimedean Analysis P229].  $\square$

**Cor. (12.2.4.24).** The notion of power-boundedness and topological nilpotence is independent of residue norm chosen.  $\lrcorner$

**Cor. (12.2.4.25) [Restricted Power Series].** For an affinoid algebra  $A$ , the restricted power series in  $A$ :

$$A\langle X_i \rangle = \left\{ \sum a_v X^v \mid \lim_{|v| \rightarrow \infty} a_v = 0 \right\}$$

is an affinoid algebra, and is independent of the residue norm chosen.  $\lrcorner$

**Def. (12.2.4.26) [Strongly Noetherian].**  $A$  is called **strongly Noetherian** if  $A\langle T_1, \dots, T_n \rangle$  are Noetherian for any  $n \geq 0$ .  $\lrcorner$

**Lemma (12.2.4.27).** The image  $A$  is dense in  $A\langle X \rangle / (X - f)$  (in the residue norm, and thus in all other norms, by (12.2.4.23)(12.2.4.18)), this is because a restricted power series can be truncated by a finite part and a part with small norm, and the finite part is in the image of  $A$ .  $\lrcorner$

**Def. (12.2.4.28) [Affinoid Generators].** For a morphism of affinoid algebras  $A \rightarrow A'$ , a set of elements  $h_i$  in  $A'$  is called a set of **affinoid generator** iff there is a surjection

$$A\langle X_1, \dots, X_n \rangle \rightarrow A', \quad X_i \mapsto h_i$$

Of course  $h_i$  is power-bounded, by the residue norm given.  $\lrcorner$

**Lemma (12.2.4.29).** If  $\pi' : A\langle X_1, \dots, X_n \rangle \rightarrow A' : X_i \mapsto h'_i$  is a surjective morphism of affinoid algebras that  $A\langle X_1, \dots, X_n \rangle$  is endowed with the Gauss norm and  $A'$  is endowed with the residue norm, then any set of elements  $h = (h_1, \dots, h_n)$  that  $|h_i - h'_i| < 1$  is a set of affinoid generators.  $\lrcorner$

*Proof:* By non-Archimedean property,  $|h_i| \leq 1$  thus also  $|h'_i| \leq 1$ , and Let  $\varepsilon = \max\{|h_i - h'_i|\} < 1$ . The strategy is simple, if for each  $g$  in  $A'$ , we can find a  $f$  that  $|f| = |g|, |\pi(f) - g| \leq \varepsilon|g|$ , then by iteration, there is a  $f$  that  $\pi(f) = g$ . But by (12.2.4.18) and (12.2.4.16), if we choose a  $f$  that  $\pi'(f) = g$  and  $|f| = |g|$ , then

$$|\pi(f) - g| = \left| \sum a_v h^v - \sum a_v h'^v \right| = \left| \sum a_v (h^v - h'^v) \right| \leq \varepsilon |f| = \varepsilon |g|.$$

□

**Def. (12.2.4.30) [Distinguished Element].** For an affinoid algebra  $A$  and an element  $x \in \text{Sp } A$  (27.2.1.1), a element  $f \in A\langle X_1, \dots, X_n \rangle$  is called  **$X_n$ -distinguished of order  $s$  at  $x$**  iff it is distinguished in  $A/\mathfrak{m}_x\langle X_1, \dots, X_n \rangle$  is distinguished of order  $s$  in the sense of (12.2.4.6) (notice  $A/\mathfrak{m}_x$  is a complete valued field by (12.2.4.11)).  $\lrcorner$

**Prop. (12.2.4.31) [Fibered Pushouts].** When  $R, A_1, A_2$  are all affinoid algebras, the amalgamated sum is also an affinoid algebra. In other words, the category of affinoid algebras admits amalgamated sums (fibered pushouts by (19.1.2.6)).  $\lrcorner$

*Proof:* Cf. [Formal and Rigid Geometry P245].  $\lrcorner$

**Prop. (12.2.4.32).**  $T_n \hat{\otimes} T_m \cong T_{m+n}$ .  $K' \hat{\otimes} T_{n,K} = T_{n,K'}$ .  $\lrcorner$

**Prop. (12.2.4.33).** For affinoid algebras  $R, A_1, A_2$  and ideals  $\mathfrak{a}_1 \subset A_1, \mathfrak{a}_2 \subset A_2$ , there is an isomorphism:

$$(A_1 \hat{\otimes}_R A_2) / (\mathfrak{a}_1, \mathfrak{a}_2) \cong (A_1 / \mathfrak{a}_1) \hat{\otimes}_R (A_2 / \mathfrak{a}_2)$$

□

*Proof:* Cf. [Rigid and Formal Geometry P248].  $\lrcorner$

**Prop. (12.2.4.34) [Finite Extension of Affinoid Algebras].** If  $B$  is an affinoid  $K$ -algebra and  $\varphi : B \rightarrow A$  is a finite ring map, then  $A$  can be provided a topology to make it an affinoid  $K$ -algebra, and  $\varphi$  is continuous.  $\lrcorner$

*Proof:* We can associate to  $A$  a Banach algebra topology induced from  $B^n \rightarrow A \rightarrow 0$  that is continuous. Now it is an affinoid  $K$ -algebra: we may assume  $B = T_n$ , then  $A = \sum T_n a_i$ , and we may assume  $|a_i| < 1$  then clearly there is a continuous extension  $T_n\langle X_i \rangle \rightarrow A$  extending this map, so  $A$  is affinoid.  $\lrcorner$

### Construction of Affinoid Tate Algebras

**Def. (12.2.4.35) [Affinoid Localizations].** Let  $A$  be an affinoid Tate algebra (12.2.4.17), then for a finite set of elements  $\{f_i, g_j\} \subset A$ , we can define the localization

$$A\langle f_i, g_j^{-1} \rangle = A\langle \zeta_1, \dots, \zeta_i, \xi_1, \dots, \xi_j \rangle / (\zeta_i - f_i, 1 - \xi_j g_j),$$

which is also an affinoid Tate algebra.  $\lrcorner$

## 5 Huber-Rings

**Def. (12.2.5.1) [Huber-Rings].** A **Huber-ring** is a topological ring  $A \in \text{TopCRing}$  s.t. there is an open subring  $A_0 \subset A \in \text{CRing}$  s.t. the induced topology on  $A_0$  is  $I$ -adic for some f.g. ideal  $I \in \text{Ideal}(A_0)$ . Such an  $A_0$  is called a **ring of definition** of  $A$ , and  $I$  is called the **ideal of definition** of  $A$ .

The category  $\text{HubCRing}$  is defined to be the subcategory of topological rings consisting of Huber rings. We write  $(A, A_0, I) \in \text{HubCRing}$  to denote that  $A$  is a Huber-ring with a ring of definition  $A_0$  and an ideal of definition  $I \in \text{Ideal}(A_0)$ .  $\lrcorner$

**Prop. (12.2.5.2) [Boundedness and Rings of Definition].** For  $A \in \text{HubCRing}$ ,  $A_0 \subset A \in \text{CRing}$  is a ring of definition (12.2.5.1) iff  $A_0$  is open and bounded in  $A$ .  $\lrcorner$

*Proof:* Clearly a ring of definition is open and bounded, for the converse, let  $(A'_0, I)$  be a couple of definition, and  $A_0$  is an open and bounded subset of  $A$ , then  $I^k \subset A_0$  for some  $n$ , and set  $J = I^k A_0$ . As  $A_0$  is bounded, for any open nbhd  $U$  of 0, there exists  $m > 0$  that  $I^{km} A_0 \subset U$ , thus  $J^m \subset U$ . This shows  $J$  is a fundamental system of nbhd of 0, thus  $A_0$  is  $J$ -adic and is a ring of definition.  $\square$

**Cor. (12.2.5.3).** For  $A \in \text{HubCRing}$ .

- If  $A_0, A_1$  are two rings of definition of  $A$ , then so does  $A_0 \cap A_1$  and  $A_0 A_1$ .
- Every open subring  $B$  of  $A$  is a Huber ring.
- If  $B \leq C$  are subrings of  $A$  and  $B$  is bounded,  $C$  is open, then there is a ring of definition  $A_0$  of  $A$  s.t.  $B \leq A_0 \leq C$ .  $\lrcorner$

*Proof:* 1: By (12.2.5.2).

2: Let  $I^n \subset B$ , then  $(B \cap A_0, I^n)$  is a couple of definition of  $B$ .

3: By 2,  $C$  is Huber, take a ring of definition  $A_0$  of  $C$ , then  $A_0 B$  is open and bounded in  $C$ , thus a ring of definition.  $\square$

**Lemma (12.2.5.4).** If  $A$  is a Huber ring and  $T \subset A$  is a subset that generates an open ideal of  $A$ , then for any open nbhd  $U$  of  $A$ , the subgroup  $T^n U$  is open.  $\lrcorner$

*Proof:* Let  $(A_0, I)$  be a couple of definition. By assumption the ideal  $J$  generated by  $T$  is open, thus  $J^n$  is also open, and contains some  $I^m$ . Now we can change  $I^m$  to  $I$ . Now  $I$  is f.g., so there is a finite subset  $M \subset A$  that  $I \subset T^n M$ . Notice  $M$  is bounded because it is finite, so there is an integer  $r$  that  $I^r M \subset U$ , thus  $I^{r+1} \subset T^n U$ , and  $T^n U$  is open.  $\square$

**Def. (12.2.5.5) [Tate-Huber Rings].** A **Tate-Huber ring** is a Huber-ring  $A \in \text{HubCRing}$  s.t. there exists an open subring  $A_0$  that the induced topology on  $A_0$  is  $t$ -adic for some  $t \in A_0 \cap A^*$ . Such a  $t$  is called a **pseudo-uniformizer** of  $A$ .  $\lrcorner$

**Prop. (12.2.5.6) [Examples of Tate-Huber-Rings].**

- If  $K$  is a complete non-Archimedean field and  $R$  is a  $K$ -Banach algebra, then  $R$  is Tate with a ring of definition by  $(R_{\leq 1}, t)$ , where  $t$  is a pseudo-uniformizer of  $K$ .
- If  $A_0 \in \mathcal{CRing}$  and  $g \in A_0$  is a nonzero-divisor, then  $(A_0[g^{-1}], (g))$  is an Tate-Huber ring.

⌋

**Prop. (12.2.5.7) [Properties of Tate-Huber-Rings].** If  $(A, A_0) \in \text{HubCRing}$  is Tate with a pseudo-uniformizer  $g$  (12.2.5.5), then there exists  $n \in \mathbb{Z}_+$  s.t.  $g^n \subset A_0$ . And in this case,  $A_0$  is  $g^n$ -adic and  $A = A_0[(g^n)^{-1}]$ .

In this case, a subset  $S \subset A$  is bounded iff  $S \subset g^{-n}A_0$  for some  $n$ .

⌋

*Proof:* Because  $A_0$  is open in  $g$ , there is some  $n$  that  $g^n \subset A_0$ , and then with  $n$  even larger we can assume  $g \in I$ , because  $g$  is topologically nilpotent, and  $gA_0$  is also open in  $A_0$ , thus it contains  $I^m$  for some  $m$ . So now  $g^{mn}A_0 \subset I^m \subset g^nA_0$ , which means  $A_0$  is  $g^n$ -adic.

To show  $A = A_0[(g^n)^{-1}]$ , it suffices to notice  $g^{kn}x \rightarrow 0$  as  $k \rightarrow \infty$  for any  $x \in A$ , so for  $k$  large,  $g^{kn}x \in A_0$ .

The last assertion is easy, as open subsets of  $A$  and  $g^{kn}A_0$  are cofinal.

□

**Prop. (12.2.5.8) [Power-Bounded Elements and Topologically Nilpotent Elements].** The subset  $A^0$  of power-bounded elements in  $A$  is a subring, and it is the filtered colimit of all the ring of definition in  $A$ , thus open. It is also integrally closed in  $A$ .

$A^{00}$  (12.2.1.2) is a radical ideal of  $A^0$ . But it is in general not an ideal of  $A$ .

Recall  $A$  is called uniform if  $A^0$  is bounded (12.2.1.5), or equivalently  $A^0$  is a ring of definition, by (12.2.5.2).

⌋

*Proof:* By (12.2.5.3), every power-bounded element is contained in a ring of definition, and any ring of definition is bounded, so  $A^0$  is the union of all rings of definitions of  $A$ , and this is filtered by (12.2.5.3).

To show  $A^0$  is integrally closed, notice by what we already proved, if  $a$  is integral over  $A^0$ , then it is integral over a ring of definition  $A_0$ , but then  $\{a^n\} \subset A_0[a]$  is bounded, so  $a \in A^0$ .

Showing  $A^{00}$  is a radical ideal of  $A^0$  is easy and omitted.

□

**Cor. (12.2.5.9).** If  $A \in \text{HubCRing}$  is separated, Tate (12.2.5.5) and uniform (12.2.1.5), then  $A$  is reduced.

⌋

*Proof:* Assume that  $A_0$  the set of power-bounded elements is a ring of ideal and  $g \in A_0$  is a pseudo-uniformizer. If  $x$  is nilpotent, then  $g^{-n}$  is nilpotent for any  $x$ , so power-bounded and  $g^{-n}x \in A_0$ , which means  $x \in g^nA_0$  for any  $n$ . But  $A_0$  is separated, so  $x = 0$ .

□

**Cor. (12.2.5.10).** For  $A \in \text{HubCRing}$ , an ideal  $J \in \text{Ideal}(A)$  is open iff  $A^{00} \in \sqrt{J}$ .

⌋

*Proof:* If  $J$  is open, then clearly  $A^{00} \subset \sqrt{J}$ . Conversely, if  $A^{00} \subset \sqrt{J}$  and  $(A_0, I)$  is a couple of definition, then  $I \subset A^{00}$  by (12.2.1.12), so  $I \subset \sqrt{J}$ . And  $I$  is f.g., so  $I^N \subset J$  for some  $J$ , thus  $J$  is open.

□

**Prop. (12.2.5.11).** If  $K$  is a complete non-Archimedean field, then any Banach  $K$ -algebra  $R$  is a complete Tate ring, and if  $K, R$  are perfectoids, then  $R^{00} = K^{00}R^0$ .

⌋

*Proof:* In the perfectoid case, first  $K^{00}R^0 \subset R^{00}$ , and for any topological nilpotent  $\alpha$ ,  $\alpha^n \subset tR^{00}$  for a pseudo-uniformizer  $t$ . Thus  $R^{00}$  and  $K^{00}R^0$  has the same radical, it suffices to show  $K^{00}R^0$  is radical, but the quotient  $R^0/K^{00}R^0$  is a perfect  $K^0/K^{00}$ -algebra by perfectoidness, thus it must be radical.

□

**Prop.(12.2.5.12) [Complete Perfect Tate ring are Uniform, André].** If  $A \in \text{HubCRing}$  is a complete Tate ring s.t.  $\text{char } A = p \in \text{Prime}$  and  $A = A^{\text{perf}}$ , then  $A$  is uniform(12.2.1.5).  $\lrcorner$

*Proof:* Let  $(A_0, t)$  be a ring of definition, let  $A_n = A_0^{\frac{1}{p^n}}$ , then  $A_\infty = \text{colim } A_n = (A_0)_{\text{perf}}$ . We check  $t^{\frac{1}{p^n}} A^0 \subset A_\infty \subset t^{-1} A_0$ , which shows  $A^0$  is bounded.

If  $f \in A^0$ , then  $t^a f^{\mathbb{N}} \subset A \subset A_\infty$ , and  $A_\infty$  is perfect, so  $t^{\frac{a}{p^n}} f \in A_\infty$  for all  $n$ . Notice the Frobenius is a continuous bijection of Banach spaces, so it is open by Banach theorem(12.7.2.5), so  $A_0^p \supset t^{mp} A_0$ , thus  $t^m A_1 \subset A_0$ , and then  $t^{\frac{m}{p^n}} A_{n+1} \subset A_n$ . So  $t^{\sum_n m/p^n} A_n \subset A_1$ . So  $t^c A_\infty \subset A_0$ , for  $c$  large.  $\square$

### Huber-Pairs

**Def.(12.2.5.13) [Huber-Pairs].** For  $A \in \text{HubCRing}$ , a **ring of integral elements** is an open and integrally closed subring of  $A$  contained in  $A^0$ (12.2.5.8)(e.g.  $A^0$  itself, by(12.2.5.8)).

A **Huber-pair** is a pair  $(A, A^+)$  where  $A \in \text{HubCRing}$  and  $A^+ \leq A$  is a ring of integral elements. A morphism of Huber pairs should preserve the ring of integers.  $\lrcorner$

**Prop.(12.2.5.14).** For  $(A, A^+) \in \text{HubPair}$ , we have  $A^{00} \in \text{Ideal}(A^+)$ . In particular,  $A^+$  contains any pseudo-uniformizer, and the set of rings of integral elements is in bijection with integrally closed subrings of  $A/A^{00}$ .

Also,  $A^+$  is a filtered colimits of rings of definitions.  $\lrcorner$

*Proof:*  $t \in A^{00}$  is topologically nilpotent hence  $t^n \subset A^+$  for some  $n$  as it is open, and then  $t \in A^+$  as it is integrally closed. It is an ideal because it is an ideal of  $A^0$ (12.2.5.8).

For the last assertion, notice that  $A^0$  is the filtered colimits of rings of definitions(12.2.5.8), and the intersection of a ring of definition with  $A^+$  is also a ring of definition, because it is open and bounded(12.2.5.2), the result follows.  $\square$

**Def.(12.2.5.15) [Zariski, Henselian, Complete Huber Pairs].**  $(A, A^+) \in \text{HubPair}$  is called

- **complete Huber-pair** iff  $A$  is complete.
- **Henselian Huber-pair** iff  $(A^+, A^{00})$  is a Henselian pair.
- **Zariski Huber-pair** iff  $(A^+, A^{00})$  is a Zariski pair.

$\lrcorner$

**Prop.(12.2.5.16).** An affinoid Tate ring  $(A, A^+) \in \text{HubCRing}$  with a ring of definition  $(A_0, I)$  that  $A_0 \subset A^+$  is

- Zariski iff  $I$  is in the Jacobson radical of  $A_0$ .
- Henselian iff the pair  $(A_0, I)$  is Henselian.
- Complete then it is Henselian.
- Henselian then it is Zariski.

$\lrcorner$

*Proof:* 1: We prove that if  $t \in \text{rad}(A_0)$ , then for any other  $B_0 \supset A_0$ ,  $t \in \text{rad}(B_0)$ . If this is true, then as  $A^+$  is a filtered colimits of rings of definitions(because  $A^0$  does), it is clear that  $t$  lies in the maximal ideal (check  $1 + at$  is unit). For this, if  $\mathfrak{m} \subset B_0$  is maximal and  $t \notin \mathfrak{m}$ , choose  $n$  that  $t^n B_0 \subset A_0$ , and an element  $b \in B_0$  that maps to  $t^{-n-1}$  modulo  $\mathfrak{m}$ , then  $a = t^n b \in A_0$  is mapped to  $t^{-1}$ . Thus the composition  $A_0 \rightarrow B_0 \rightarrow B_0/\mathfrak{m}$  is surjective:  $\bar{b}$  is the image of  $a^n(t^n b) \in A$ . So  $t$  is not in a maximal ideal of  $A_0$ , contradiction.

Conversely, Cf.[Bhatt Perfectoid Space P57].

2: Cf.[Bhatt Perfectoid Spaces P57].

3:  $A$  is complete then  $A_0$  is complete, hence  $(A_0, I)$  is Henselian by (5.3.10.6), so it is Henselian by item2. 4: Trivial.  $\square$

### Adic Morphisms

**Def. (12.2.5.17) [Adic Morphisms].**  $f : A \rightarrow B \in \text{HubCRing}$  is called an **adic morphism** if there are rings of definitions  $A_0, B_0$  and an ideal of definition  $I$  of  $A$  that  $f(A_0) \subset B_0$ , and  $f(I)B_0$  is an ideal of definition of  $B_0$ .

A morphism  $(A, A^+) \rightarrow (B, B^+)$  of Huber pairs is called an **adic morphism** if  $A \rightarrow B$  is.  $\lrcorner$

**Prop. (12.2.5.18).** Let  $f : A \rightarrow B \in \text{HubCRing}$  be an adic morphism (12.2.5.17), then:

1.  $f$  is continuous and open.
2. If  $A_0, B_0$  are rings of definition s.t.  $f(A_0) \leq B_0$ , then for any ideal of definition  $I \subset A_0$ ,  $f(I)B_0$  is an ideal of definition in  $B_0$ .
3.  $f$  maps bounded sets to bounded sets.

$\lrcorner$

*Proof:* Let  $f(A_0) \subset B_0$  and  $J = f(I)B_0$ . Then  $I^n \subset f^{-1}(J^n)$ , so  $f$  is continuous.

If  $E$  is bounded in  $A$ , for any  $n$ , let  $m$  be that  $I^m E \subset I^n$ , then  $f(E)f(I)^m = f(EI^m) \subset f(I^n) \subset J^n$ , thus  $f(E)J^m \subset J^n$ , so  $f(E)$  is bounded.  $\square$

### Construction of Huber Rings

References are [Mor19].

**Prop. (12.2.5.19) [Quotient].** Let  $(A, A^+) \in \text{HubPair}$  and  $\mathfrak{a} \in \text{Ideal}(A)$ , then the quotient pair  $(A/\mathfrak{a}, (A/\mathfrak{a})^+)$  where  $A/\mathfrak{a}$  is the integral closure of  $A^+/\mathfrak{a}$  in  $A/\mathfrak{a}$ .  $\lrcorner$

**Prop. (12.2.5.20) [Completion of Huber Rings].** For  $(A, A_0, I) \in \text{HubCRing}$ , define  $\widehat{A} = \varprojlim_n A/I^n$  (as an Abelian group), then:

1. The canonical map  $\widehat{A_0} \rightarrow \widehat{A}$  is injective and  $\widehat{A_0} \cap A = A_0$ .
2. If we put the unique topology on  $\widehat{A}$  that  $\widehat{A_0}$  is an open subgroup, then  $\widehat{A}$  is complete.
3. There is a unique topological ring structure on  $\widehat{A}$  that  $A \rightarrow \widehat{A}$  is continuous.
4.  $\widehat{A}$  is Huber with a couple of definition  $(\widehat{A_0}, I\widehat{A_0})$ , and the canonical map  $A \rightarrow \widehat{A}$  is adic.
5.  $\widehat{A_0} \otimes_{A_0} A \rightarrow \widehat{A}$  is an isomorphism.

$\lrcorner$

*Proof:* Cf.[Mor19]P72.  $\square$

**Prop. (12.2.5.21).** Let  $A$  be a Huber ring and  $i : A \rightarrow \widehat{A}$  be the completion, then under the bijection of (12.2.1.8),

- $\widehat{A^0} = \widehat{A^0}$ ,  $\widehat{A^{00}} = \widehat{A^{00}}$ .
- $G \subset A$  is a ring of definition iff  $\widehat{G} \subset \widehat{A}$  is a ring of definition.
- the map  $\text{Cont}(\widehat{A}) \rightarrow \text{Cont}(A)$  is a bijection.

┘

*Proof:* Cf[[Mor19](#)] P75. □

**Prop. (12.2.5.22).** For  $A \in \text{HubCRing}$ , under the bijection of [\(12.2.1.8\)](#), an open ring  $A_0$  is a ring of integral elements of  $A$  iff  $A_0^\wedge$  is a ring of integral elements of  $A^\wedge$ . ┘

*Proof:* It is easy to show  $A_0$  is a ring iff  $A_0^\wedge$  is a ring, and by [\(12.2.5.21\)](#),  $A_0 \subset A^0$  iff  $A_0^\wedge \subset (A^\wedge)^0$ .

It suffices to prove that if  $A_0$  is open and integrally closed, then  $A_0^\wedge$  is integrally closed in  $A^\wedge$ . Let  $x \in A^\wedge$  satisfy  $x^d + a_{d-1}x^{d-1} + \dots + a_0 = 0$ , where  $a_i \in A_0^\wedge$ , because  $A_0^\wedge$  is open, we can find  $x' \in A$  and  $a'_i \in A_0$  that  $(x')^d + a'_{d-1}(x')^{d-1} + \dots + a'_0 \in A'_0$ , but then  $x' \in A_0$  because  $A_0$  is integrally closed, and thus  $x = (x - x') + x' \in A'_0$ . □

**Cor. (12.2.5.23) [Completion of Huber Pairs].** The forgetful functor from the category of complete Huber-pairs to  $\text{HubPair}$  has a left adjoint called completion, where  $(A, A^+)^\wedge = (A^\wedge, (A^\wedge)^+)$ , where  $A^\wedge$  is defined in [\(12.2.5.30\)](#), and  $(A^\wedge)^+$  is the closure of the image of  $A^+$  in  $A^\wedge$ . ┘

**Prop. (12.2.5.24) [Completion, Henselization, Zariski Localization].** There are left adjoint to the forgetful functors from the category of complete/Henselian/Zariski pairs to the category of pairs, called the **completion/Henselization/Zariski localization** of pairs. And there are natural maps

$$(A, A^+) \rightarrow (A, A^+)_{\text{Zar}} \rightarrow (A, A^+)_{\text{Hens}} \rightarrow (A, A^+)^\wedge.$$

┘

*Proof:* □

**Prop. (12.2.5.25) [Tensor Products of Adic Maps of Huber Rings].**

- If  $(A, A^+) \rightarrow (B, B^+) \in \text{HubPair}$  is adic [\(12.2.5.17\)](#), then pullback along the associated map of topological spaces  $\text{Spa}(B, B^+) \rightarrow \text{Spa}(A, A^+)$  preserves rational subsets.
- Let  $(B, B^+) \leftarrow (A, A^+) \rightarrow (C, C^+) \in \text{HubPair}$  be adic morphisms. Let  $A_0, B_0, C_0$  be rings of definition compatible with the morphisms, and  $I \subset A_0$  be an ideal of definition. Let  $D = B \otimes_A C$  and let  $D_0$  be the image of  $B_0 \times_{A_0} C_0$  in  $D$ . Make  $D$  into a Huber ring by declaring  $D_0$  to be a ring of definition with  $ID_0$  as its ideal of definition and  $D^+$  be the integral closure of the image of  $B^+ \otimes_{A^+} C^+$  in  $D$ . Then  $(D, D^+)$  is a Huber pair, and it is the pushout of the diagram in the category of Huber pairs. ┘

*Proof:* For 1, it suffices to show that if  $T$  is a finite set of  $A$  that  $TA$  is open, then  $TB$  is open in  $B$ :  $I \subset TA$  for some ideal of definition  $I \subset A_0$ , in which case  $IB_0 \subset B_0$  is also an ideal of definition by [\(12.2.5.18\)](#), thus open, and so  $TB$  is also open as  $IB \subset TB$ . □

2 just follows from the definition. □

**Remark (12.2.5.26) [Non-Adic Morphisms].** Pushouts may not exist for non-adic morphisms of Huber rings. For example,  $\mathbb{Z}_p \rightarrow \mathbb{Z}_p[[T]]$  is not adic [\(12.2.5.18\)](#), and the diagram

$$(\mathbb{Z}_p[[T]], \mathbb{Z}_p[[T]]) \leftarrow (\mathbb{Z}_p, \mathbb{Z}_p) \rightarrow (\mathbb{Q}_p, \mathbb{Z}_p)$$

has no pushout in the category of Huber pairs: If there is a pushout  $(D, D^+)$ , then we will have a morphism

$$(D, D^+) \rightarrow (\mathbb{Q}_p \langle T, \frac{T^n}{p} \rangle, \mathbb{Z}_p \langle T, \frac{T^n}{p} \rangle)$$



for each  $n \geq 1$ . But notice that  $T$  is nilpotent in  $D$ , and  $1/p \in D <$  so  $T^n/p \rightarrow 0 \in D$  as  $n \rightarrow \infty$ . So  $T^n/p \in D^+$  for some  $n$ , but then  $T^n/p \in \mathbb{Z}_p\langle T, \frac{T^{n+1}}{p} \rangle$ , which is impossible.  $\perp$

**Def. (12.2.5.27)[Topological Polynomial Functions].** Let  $A$  be a non-Archimedean topological ring, and  $\{X_i\}_{i \in I}$  be a family of indeterminates,  $\{T_i\}_{i \in I}$  be a family of subsets of  $A$  that satisfies  $T_i^n U$  is open for any  $n > 0, i$  and open nbhd  $U$  of  $A$ .

Let  $\mathbb{N}^{(I)}$  be the set of functions  $I \rightarrow \mathbb{N}$  with finite support, then for any  $\nu \in \mathbb{N}^{(I)}$ , let  $T^\nu = \prod T_i^{\nu(i)}$ . For any nbhd  $U$  of  $A$ , we set

$$U_{[X,T]} = \left\{ \sum_{\nu \in \mathbb{N}^{(I)}} a_\nu X^\nu \mid a_\nu \in T^\nu U \right\}.$$

Then there exists a unique topological structure on  $A[X]$  that  $U_{[X,T]}$  form a fundamental system of nbhds of 0, and denote it by  $A[X]_T$ . It satisfies:

- the canonical inclusion  $\iota : A \rightarrow A[X]_T$  is continuous and the set  $\{T_i X_i\}_{i \in I}$  is power-bounded.
- $\iota$  satisfies the universal property that any continuous map  $f : A \rightarrow B$  to another non-Archimedean topological ring  $B$  that  $\{f(T_i)X_i\}_{i \in I}$  is power-bounded factors through  $A[X]_T$ .  $\perp$

*Proof:* Just notice that  $(U \cap V)_{[X,T]} \subset U_{[X,T]} \cap V_{[X,T]}$  and  $U_{[X,T]} \cdot V_{[X,T]} = (UV)_{[X,T]}$ , so they form a topological basis because  $A$  is topological.

The first properties are easily verified. For the second, the extension  $f' : A[X] \rightarrow B$  exists abstractly, and it suffices to show it is continuous. If we let  $E \subset B$  be the subring generated by  $\{f(T_i)X_i\}_{i \in I}$ , then  $E$  is bounded, so for any open subgroup  $H \subset B$ , there is some open subgroup  $G \subset B$  that  $EG \subset H$ , and thus  $f^{-1}(G)$  is open and contains some nbhd  $U$ , then  $U_{[X,T]} \subset (f')^{-1}(G)$ , so  $f'$  is continuous.  $\square$

**Def. (12.2.5.28)[Topological Power Series].** Let  $A, X, T$  as in (12.2.5.27), then the set

$$A\langle X \rangle_T = \left\{ \sum_{\nu \in \mathbb{N}^{(I)}} a_\nu X^\nu \in A[[X]] \mid a_\nu \in T^\nu U \text{ a.e.} \right\},$$

is a subring of  $A[[X]]$ , and there is a unique topological structure on  $A[[X]]$  that

$$U_{\langle X, T \rangle} = \left\{ \sum_{\nu \in \mathbb{N}^{(I)}} a_\nu X^\nu \in A\langle X \rangle_T \mid a_\nu \in T^\nu U \right\},$$

form a fundamental system of nbhds of  $A\langle X \rangle_T$ .  $\perp$

*Proof:* The proof is not hard and similar to that of (12.2.5.27) so omitted.  $\square$

**Prop. (12.2.5.29).** Let  $A, X, T$  as in (12.2.5.27), then

- $A[X]_T$  is dense in  $A\langle X \rangle_T$  and the topology coincide.
- If  $A$  is Hausdorff and  $T_i$  is bounded for any  $i \in I$ , then  $A[X]_T$  and  $A\langle X \rangle_T$  are all Hausdorff.
- If  $A$  is complete and  $T_i$  is bounded for any  $i \in I$ , then  $A\langle X \rangle_T$  is complete, so it is the completion of  $A[X]_T$ .  $\perp$



*Proof:* Only the completeness needs proof, Cf.[Mor19]P80.  $\square$

**Prop. (12.2.5.30)[Power Series is Huber].** Let  $A$  be a Huber ring with a couple of definition  $(A_0, I)$ , and  $X = \{X_\lambda\}$  be a finite set of indeterminates,  $T_\lambda$  is a family of subsets of  $A$  that  $T_\lambda A$  is open in  $A$ , then

- $A[X]_T$  is a Huber ring with a couple of definition  $((A_0)_{[X,T]}, I_{[X,T]})$ , and there is a canonical map  $A \rightarrow A[X]_T$  which is adic.
- $A\langle X \rangle_T$  is Huber with a couple of definition  $((A_0)_{\langle X,T \rangle}, I_{\langle X,T \rangle})$ , and there is a canonical map  $A \rightarrow A\langle X \rangle_T$  which is adic.

┘

*Proof:* It suffices to prove for any ideal of definition  $J$ ,  $J_{[X,T]} = J \cdot (A_0)_{[X,T]}$  and  $J_{\langle X,T \rangle} = J \cdot (A_0)_{\langle X,T \rangle}$ . The first is clear. For the second, use the fact  $J$  is f.g. and  $\{J^n\}$  is a fundamental basis of nbhds of 0.  $\square$

**Prop. (12.2.5.31)[Example].** Let  $A = \mathbb{Z}_p$  and  $T = p$ , then there is a Huber ring

$$\mathbb{Z}_p\langle X \rangle_T = \left\{ \sum_{n \geq 0} a_n X^n \in \mathbb{Z}_p[[X]] \mid l^{-n} a_n \rightarrow 0 \right\}$$

with a ring of definition

$$(Z_p)_{\langle X,T \rangle} = \left\{ \sum_{n \geq 0} a_n X^n \in \mathbb{Z}_p\langle X \rangle_T \mid a_n \in p^n \mathbb{Z}_p \right\}.$$

Notice that  $\mathbb{Z}_p\langle X \rangle_T$  is not adic although  $\mathbb{Z}_p$  is (because  $p$  is nilpotent but  $pX$  is not).  $\square$

**Def. (12.2.5.32)[Localizations].** Let  $A$  be a non-Archimedean topological ring and  $\{T_i\}$  is a family of subsets of  $A$  satisfying  $T_i^n U$  is open for any  $n > 0, i$  and open nbhd  $U$  of  $A$ , and  $\{s_i\}$  is a family of elements of  $A$ , which generates a multiplicative subset  $R \subset A$ .

Then there is a unique non-Archimedean ring structure on  $R^{-1}A$ , denoted by  $A(\frac{T}{S})$ , that the canonical map  $\varphi : A \rightarrow A(\frac{T}{S})$  is continuous and the set  $\{\frac{\varphi(t)}{\varphi(s_i)}\}$  is power-bounded, and it is the initial map for all maps  $A \rightarrow B$  satisfying this property.  $\square$

*Proof:* Cf.[Mor19]P83.  $\square$

**Cor. (12.2.5.33).** Let  $J$  be the ideal of  $A[X]_T$  generated by  $\{1 - s_i X_i\}$ , then  $A[X]_T/J$  with the quotient topology satisfies the same universal property as  $A(\frac{T}{S})$ , so there is a canonical isomorphism

$$A[X]_T/J \cong A(\frac{T}{S}).$$

In particular,  $A(\frac{T}{S})$  is a Huber ring, and the canonical map  $A \rightarrow A(\frac{T}{S})$  is adic. Explicitly,  $B_0$  is the  $(A_0)_{[X,T]}$ -subalgebra of  $B$  generated by the elements  $\frac{T_i}{s_i}$ .  $\square$

**Def. (12.2.5.34).** If  $A$  is Huber ring, then we denote the completion of  $A(\frac{T}{S})$  by  $A\langle \frac{T}{S} \rangle$ , which is also a Huber ring, and the canonical map  $A \rightarrow A\langle \frac{T}{S} \rangle$  is adic, by (12.2.5.30) and (12.2.5.33). It satisfies the natural universal property.  $\square$

**Cor. (12.2.5.35).** If  $A$  is complete, then we can also regard  $A\langle \frac{T}{S} \rangle$  as the quotient of  $A\langle X \rangle_T$  by the closure of the ideal generated by  $\{1 - s_i X_i\}$ .  $\square$

**Prop. (12.2.5.36)[Example].** Let  $A = \mathbb{Z}_p[[T]]$  with the  $(p, T)$ -adic topology, then

$$A\left(\frac{p, T}{T}\right) = \mathbb{Z}_p[[T]][T^{-1}]$$

with a ring of definition  $A[\frac{p}{T}]$ , and

$$A\left(\frac{p, T}{p}\right) = \mathbb{Z}_p[[T]][p^{-1}]$$

with a ring of definition  $A[\frac{T}{p}]$ .

In  $A\langle\frac{p, T}{T}\rangle$ , a ring of definition is  $A_{\langle X, T \rangle}/(1 - pX)$ , which is isomorphic to

$$A\left\langle\frac{T}{p}\right\rangle = \left\{ \sum_{n \geq 0} a_n \left(\frac{T}{p}\right)^n \mid a_n \in A, a_n \rightarrow 0 \right\}$$

by (12.2.5.33). ┘

## 6 Analytic Points and Analytic Huber Pairs

**Def. (12.2.6.1)[Analytic Huber Rings].** A Huber ring  $A \in \text{HubCRing}$  is called an **analytic Huber-ring** if  $AA^{00} = A$  (12.2.1.2). Any Tate-Huber ring (12.2.5.5) is analytic. ┘

**Prop. (12.2.6.2)[Equivalent Definitions of Analytic Rings].** For  $A \in \text{HubCRing}$ , the following are equivalent:

1.  $A$  is analytic (12.2.6.1).
2. Any ideal of definition in any ring of definition of  $A$  generates the unit ideal of  $A$ .
3. Any open ideal of  $A$  is trivial.
4. For any non-trivial ideal  $I$  of  $A$ , the quotient topology on  $A/I$  is not discrete.
5. The only discrete  $A$ -module is the 0-module.
6. The set  $\text{Spa}(A, A^+)$  contains no point with induced topology on the residue field trivial. ┘

*Proof:* Cf. [Ked19] P4. ? □

**Prop. (12.2.6.3)[Analytic Open Mapping Theorem].** If  $A$  is an analytic Huber ring, and  $M, N$  are complete Banach  $A$ -modules, then any continuous surjective map  $M \rightarrow N$  is open. ┘

*Proof:* [Hub94] L2.4(i). ? □

### Analytic Points

**Def. (12.2.6.4)[Analytic Points].** For  $A \in \text{HubCRing}$ , a point  $x \in \text{Cont}(A)$  is called an **analytic point** if  $\mathfrak{p}_x$  is not open in  $A$ . The set of analytic points of  $\text{Cont}(A)$  is denoted by  $\text{Cont}(A)_{\text{an}}$ .

If  $A$  is Tate, then  $\text{Cont}(A)_{\text{an}} = \text{Cont}(A)$ , because the only open ideal of  $A$  is  $A$  itself. ┘

**Prop. (12.2.6.5)[Characterizations of Analytic Points].** Let  $A$  be a Huber ring, then for a point  $x \in \text{Cont}(A)$ , the following is equivalent:

1.  $x$  is analytic.
2.  $|A^{00}|_x \neq 0$ .
3. For any ring of definition and ideal of definition  $(A_0, I)$  of  $A$ ,  $|I|_x \neq 0$ .

┘

*Proof:* 1  $\rightarrow$  2:  $\mathfrak{p}_x$  is non-open, so it cannot contain the open subset  $A^{00}$  of  $A$ ??.

2  $\rightarrow$  3: trivial because any ring of definition contains  $A^{00}$ (12.2.5.8). □

**Cor.(12.2.6.6).** Let  $A$  be a Huber ring and  $I$  an ideal of definition with a set of generators  $(f_1, \dots, f_n)$ , then

$$\text{Cont}(A)_{an} = \cup_{i=1}^n U\left(\frac{f_1, \dots, f_n}{f_i}\right).$$

┘

**Prop.(12.2.6.7)[Analytic Valuation Microbial].** Let  $A$  be a Huber ring and  $x \in \text{Cont}(A)_{an}$ , then  $x$  has  $\text{rank} \geq 1$ , and the valuation  $|\cdot|_x$  on  $k(x)$  is microbial(12.2.3.22). ┘

*Proof:* If  $x$  has rank 1, then  $\Gamma_x = 1$ , and  $\mathfrak{p}_x = \{a \in A \mid |a|_x < 1\}$  is open.

If  $x$  is analytic, then there are some  $a \in A^{00}$  that  $|a|_x \neq 0$ (12.2.3.22), thus the image of  $a$  in  $k(x)$  is non-zero and topologically nilpotent, thus  $k(x)$  is microbial by(12.2.3.22). □

**Prop.(12.2.6.8)[Adic Morphism and Analytic Points].** Let  $\varphi : (A, A^+) \rightarrow (B, B^+)$  be a morphism of Huber pairs, then:

- If  $x \in \text{Spa}(B, B^+)$  is not analytic, then  $\text{Spa}(\varphi)(x)$  is not analytic.
- If  $\varphi$  is adic, then  $\text{Spa}(\varphi) : \text{Spa}(B, B^+) \rightarrow \text{Spa}(A, A^+)$  carries analytic points to analytic points.
- If  $B$  is complete and  $\text{Spa}(\varphi) : \text{Spa}(B, B^+) \rightarrow \text{Spa}(A, A^+)$  carries analytic points to analytic points, then  $\varphi$  is adic.
- If  $\varphi$  is adic, then the  $\text{Spa}(\varphi)$  maps rational subsets to rational subsets. In particular,  $\text{Spa}(\varphi)$  is spectral.

┘

*Proof:* 1: Trivial.

2: If  $f(x)$  is not analytic, then  $I \subset \varphi^{-1}(\mathfrak{p}_x)$ , so  $f(I) \subset \mathfrak{p}_x$ , which means  $\mathfrak{p}_x$  is not analytic because  $f(I)A_0$  is open.

3: Cf. Morel P96?.

4: Only notice that  $(f(T))$  is open if  $(T)$  is open. □

**Cor.(12.2.6.9).** If  $A$  is an analytic Huber ring, then any continuous morphism  $f : A \rightarrow B$  is adic, by(27.3.4.27). ┘

## 7 Huber and Banach Rings

Cf.[Ked19]1.5.

**Prop.(12.2.7.1).** Let  $A$  be a uniform Huber ring, then for  $x = \sum_{n=0}^{\infty} x_n T^n \in A\langle T \rangle$  such that the coefficients  $x_n$  generate the unit ideal of  $A$ , then multiplication by  $x$  defines a strict inclusion  $A\langle T \rangle \rightarrow A\langle T \rangle$ , i.e.  $|xg| \geq |g|$ . ┘

*Proof:* Cf.[Ked19]P25. □

## 8 Perfectoid Fields

**Notation(12.2.8.1).** Let  $(K, \mathcal{O}_K, \mathfrak{m}_K, k)$  be a non-Archimedean complete valued field.  $\lrcorner$

**Prop.(12.2.8.2).** The valuation can in fact be constructed from  $K^0$  as  $|x| = \sup\{\frac{n}{k} | x^k \in t^n K^0\}$  by(3.2.9.6), as it is a rank 1 valuation.  $\lrcorner$

**Def.(12.2.8.3)[Perfectoid Field].** If  $\text{char } K = p \in \text{Prime}$ , then  $K$  is called a **perfectoid field** iff:

- The value group  $|K^\times| \subset \mathbb{R}_+^\times$  is not discrete.
- $\mathcal{O}_K/(p)$  is semi-perfect.

$\lrcorner$

**Prop.(12.2.8.4)[Examples of Perfectoid Fields].**

- $K = \mathbb{Q}_p(p^{\frac{1}{p^\infty}})^\wedge$ .  $\mathcal{O}_K = \mathbb{Z}_p(p^{\frac{1}{p^\infty}})^\wedge$ , and  $\mathcal{O}_K/(p) \cong [\mathbb{F}_p(t^{\frac{1}{p^\infty}})/(t)]^\wedge$ , which is clearly semi-perfect. And its value group is  $\mathbb{Z}[p^{-1}]$ .
- $K = \mathbb{C}_p$ , its value group is  $\mathbb{Q}$ , and  $K = \overline{K}$ , so  $\mathcal{O}_K/(p)$  is clearly perfect.
- if  $\text{char } K = p$ , then  $K$  is a perfectoid field iff  $K$  is perfect: if  $K$  is perfect, then it is clearly perfectoid, and the semi-perfectness of  $\mathcal{O}_K$  implies its perfectness, so also  $K$  is perfect(multiply by a  $p$ -power of an element in  $\mathfrak{m}_K$ ).
- If  $K \in \text{PerfdField}$  and  $|p| \leq |\varpi| < 1$  is a pseudo-uniformizer, then  $K/(\varpi)$  is perfect hence perfectoid.

$\lrcorner$

**Prop.(12.2.8.5)[Perfectoid Field and Integral Perfectoid Rings].** The ring of integers  $\mathcal{O}_K$  for a perfectoid field  $K$  is an integral perfectoid ring(27.6.3.1).  $\lrcorner$

*Proof:* We assume that  $K$  is of char0, then we check conditions in(27.6.3.6). It is clear that  $\mathcal{O}_K$  is  $p$ -adically complete,  $p$ -normal. To find  $\varpi^p = pu$ , as  $|K^*|$  is not discrete, we find  $x$  that  $x^p$  divides  $p$ , and then there exists some  $y$  that  $y^p \equiv x^p/p \pmod{p}$ , thus  $(xy)^p \equiv 1 \pmod{p}$ , thus  $\varpi = xy$  satisfies the condition.  $\square$

**Prop.(12.2.8.6).** If  $K \in \text{Perfd}$ , then

- $|K^\times|$  is a  $p$ -divisible Abelian group.
- $\mathfrak{m}_K^2 = \mathfrak{m}_K$ , and  $\mathfrak{m}_K$  is flat ?.
- $\mathcal{O}_K$  is not Noetherian.

$\lrcorner$

*Proof:* 1 : First if  $|p| < |x| \leq 1$ , we show  $|x|$  is  $p$ -divisible: there is a  $y, z \in K^0$  that  $y^p = x + pz$ , so  $|y|^p = |x|$ . Now because  $|K^*|$  is not discrete, so there is a  $|x| \notin |p|^\mathbb{Z}$ , by rescaling, we may assume  $|p| < |x| \leq 1$ , thus  $p = xy$  for some  $y$ , and  $|p| < |y| \leq 1$ , too. So  $|p|$  is also divisible by  $p$ , so it is clear now  $|K^*|$  is divisible by  $p$ .

2 follows from(27.6.3.5)(12.2.8.5).

2  $\rightarrow$  3 by Nakayama's lemma, because otherwise  $K^{00} = 0$ .  $\square$

**Cor.(12.2.8.7).** The proof of 1 also shows that  $|K^\times|$  is generated by  $|x|$  that  $|p| < |x| < 1$ .  $\lrcorner$

**Lemma(12.2.8.8).** If  $C^\flat$  is a perfectoid space of residue characteristic  $p$ , then  $1 + \mathfrak{m}_{C^\flat}$  is a  $\mathbb{Q}_p$ -algebra.  $\lrcorner$

*Proof:* Both  $\varphi$  and exponentiation of  $\mathbb{Z}_p^*$  is definable, so  $p^n t \cdot (1 + x) = (\varphi^n(1 + x))^k$ .  $\square$

### Tilting

**Prop. (12.2.8.9) [Pseudo-Uniformizers].** Fix a **pseudo-uniformizer**  $|p| \leq |\varpi| < 1$ , consider the tilting(5.5.1.9)  $\mathcal{O}_K^\flat$ , then by(5.5.1.12), this topological group doesn't depends on  $\pi$  chosen.  $\lrcorner$

**Remark (12.2.8.10).** There are diagrams:

$$\begin{array}{ccc} \lim_{x \rightarrow x^p} \mathcal{O}_K & \xrightarrow{\quad} & \mathcal{O}_K \\ \downarrow \cong & \nearrow \# & \downarrow \\ \mathcal{O}_K^\flat = \lim_{\varphi} K^0/(\varpi) & \longrightarrow & \mathcal{O}_K/(\varpi) \end{array} . \quad \lrcorner$$

**Prop. (12.2.8.11) [Tilting of  $\mathcal{O}_K$ ].** There is an element  $t \in \mathcal{O}_K^\flat$  that  $|t^\sharp| = |\varpi|$ ,  $t$  maps into  $(\pi)$  and gives an isomorphism  $\mathcal{O}_K^\flat/(t) \cong \mathcal{O}_K/(\varpi)$ .

Moreover, the  $t$ -adic topology on  $\mathcal{O}_K^\flat$  is complete, and coincides with the topology of  $\mathcal{O}_K^\flat$  given as in(5.5.1.9).  $\lrcorner$

*Proof:* There are canonical surjective map  $K^{0b} \rightarrow K^0/p \rightarrow K^0/\pi$ , and by  $p$ -divisibility of the value group(12.2.8.6), there is a  $f \in K^0$  that  $|f|^p = |\pi|$ , so in particular  $|f| > |\pi|$ , thus  $f \neq 0 \in K^0/\pi$ . and choose a  $g \in K^{0b}$  lifting  $f \bmod \pi$ , then  $g^\sharp \equiv f \bmod \pi$ , see diagram(12.2.8.10). so  $|g^\sharp| = |f|$  as  $|f| > |\pi|$ . Now let  $t = g^p$ , then  $|t^\sharp| = |f|^p = |\pi|$ .

Now clearly  $t$  maps into  $(\pi)$ , and if  $g$  maps to 0 in  $K^0/\pi$ , then by the diagram again,  $g^\sharp \in (\pi)$ , and  $(t^\sharp) = (\pi)$ , so  $g^\sharp = at^\sharp$  for some  $a \in K^0$ . so  $t|g$  in  $K^{0b}$ , as by(5.5.1.17),  $R^{0b}$  is a valuation ring in the valuation  $|\cdot| \circ \sharp$ .

For the last assertion, just use the commutative diagrams:

$$\begin{array}{ccc} K^{0b}/(t^{p^n}) & \longrightarrow & K^{0b}/(t^{p^{n-1}}) \\ \downarrow & & \downarrow \\ K^0/(\pi) & \xrightarrow{\varphi} & K^0/(\pi) \end{array} , \text{ where}$$

the vertical are isomorphisms, and compute their inverse limits.  $\square$

**Cor. (12.2.8.12) [Tilting of Perfectoid Fields].**

- $\mathcal{O}_K^\flat$  is a valuation ring of rank 1, with the field of fraction  $K^\flat = \mathcal{O}_K^\flat[t^{-1}] \in \text{Perfd}$ .
- Its maximal ideal is  $(t^{\frac{1}{p^\infty}})$ , with Krull dimension 1.
- The value group and residue field of  $K$  and  $K^\flat$  is canonical isomorphic.

$\lrcorner$

*Proof:*  $K^{0b}$  has rank no more than  $K^0$  which is 1(12.2.8.3), and it is non-trivial because  $|t| = |\pi|$ , so the rank is 1, and it is perfect by definition, so  $K$  is perfectoid by(12.2.8.4).

For the maximal ideal, the maximal ideal of  $K^{0b}/t$  is its nilradical, as it is a valuation ring of rank 1(3.2.9.6), which is clearly  $(t^{\frac{1}{p^\infty}})$ . For the dimension, by(12.2.3.4), the Krull dimension equal the rank, which is 1.

For the residue field, use the isomorphism(12.2.8.11),  $K^{0b}/t = K^0/\pi$  and the second item just proved, and for the value group, the same lemma(12.2.8.11) gives any  $|p| \leq |\pi| < 1$  are in the value group of  $K^\flat$ , and  $|K^\times|$  is generated by these values by(12.2.8.7).  $\square$

**Prop. (12.2.8.13) [Tilting Continuous Valuations].** If  $K \in \text{Perfd}$ , for any continuous valuation on  $K$  of any rank, the function  $|\cdot|^\flat = |\cdot| \circ \sharp$  is a continuous valuation on  $\mathcal{O}_K^\flat$ , and every continuous valuation of  $\mathcal{O}_K^\flat$  comes from this way.  $\lrcorner$

**Remark(12.2.8.18).** Notice that  $K = \mathbb{Q}_p(\mu_{p^\infty})^\wedge$  and  $K = \mathbb{Q}_p(p^{\frac{1}{p^\infty}})^\wedge$  have the same tiltings, so the tilting functor is not faithful. this is due to the fact that  $\mathbb{Q}_p$  is not perfectoid. This will not happen over a perfectoid base field, see(12.2.9.13).  $\square$

## 9 Perfectoid Algebras

**Def. (12.2.9.1) [Perfectoid Algebra].** For  $K$  a perfectoid field with tilt  $K^\flat$ , let  $t \in K^\flat$  be a pseudo-uniformizer (12.2.5.5) with  $\varpi = t^\sharp$ , so it has a compatible collection of  $p^n$ -th roots  $(t^{\frac{1}{p^n}})^\sharp$ . Now:

- A **perfectoid algebra** over  $K$  is a uniform Banach  $K$ -algebra  $R$  that  $R^0/\pi$  is semi-perfect.
- A **perfectoid algebra** over  $K^{0a}$  is a  $K^{0a}$ -algebra  $A$  that is  $t$ -adically complete and flat over  $K^{0a}$  (or  $A_*$  over  $K^0$ , by (5.7.3.3)), and  $K^{0a}/\pi \rightarrow A/\pi$  is relative perfect, i.e. the Frobenius induces an isomorphism  $A/\pi^{\frac{1}{p}} \cong A/\pi$ .
- A **perfectoid algebra** over  $K^{0a}/\pi$  is a  $K^{0a}/\pi$ -algebra  $A$  that is flat over  $K^{0a}/\pi$  (or  $A_*$  over  $K^0/\pi$ , by (5.7.3.3)), and the map  $K^{0a}/\pi \rightarrow A$  is relatively perfect, i.e. the Frobenius induces an isomorphism  $A/\pi^{\frac{1}{p}} \cong A$ .

┘

**Remark (12.2.9.2).** notice the definition regarding the relative perfectness doesn't depends on  $\pi$  chosen, by the power lifting theorem (2.6.1.10). ┘

**Prop. (12.2.9.3) [Faithfully flatness of Perfectoids].** Nonzero flat  $\mathcal{O}_K^a/(\varpi)$ -algebras are faithfully flat, so does  $t$ -adically complete flat  $K^{0a}$ -algebras. In particular,  $\text{Perfd}_{\mathcal{O}_K^a/(\varpi)}$  and  $\text{Perfd}_{\mathcal{O}_K^a}$  are all faithfully flat modules. ┘

*Proof:* If  $K^{0a}/\pi \rightarrow A$  is not faithfully flat, then there is an ideal  $J \subset K^{0a}/\pi$  that  $K^0/J \neq 0$  but  $A/J = 0$ . Now this implies  $J \subsetneq I$ , so there is a  $\varpi \in I - J$  hence  $J \subset (\varpi)$ . Hence  $A/\varpi = 0$  as well. Now there are exact sequences  $0 \rightarrow K^{0a}/\varpi^n \xrightarrow{\varpi} K^{0a}/\varpi^{n+1} \rightarrow K^{0a}/\varpi \rightarrow 0$ , so tensoring with  $A$  and induct, we get  $K^{0a}/\varpi^n \otimes A = 0$ , but  $|\varpi^n| < |\pi|$  for some  $n$ , so  $A = 0$ .

The other case is similar, now  $A/\varpi = 0$ , so use (5.7.3.3),  $A_*/\varpi \subset (A/\varpi)_* = 0$ , but  $A_*$  is also  $t$ -adically complete, so  $A_* = 0$ , and  $A = (A_*)^a = 0$ . ┘

**Prop. (12.2.9.4) [Examples of Perfectoid Algebras].**

- If  $\text{char } K = p$ , then a  $K$ -Banach algebra is perfectoid iff it is uniform and perfect. Likewise, a  $\pi$ -adically complete and  $\pi$ -torsion free  $K^{0a}$ -algebra is perfectoid iff it is perfect.
- Let  $A = \mathcal{O}_K[x_1^{\frac{1}{p^\infty}}, \dots, x_n^{\frac{1}{p^\infty}}]^\wedge$ , then  $A^a \in \text{Perfd}_{\mathcal{O}_K^a}$ , and  $R = A[\pi^{-1}] \in \text{Perfd}_K$  in the Banach metric as in (19.1.6.8).

┘

*Proof:* 1: a perfectoid algebra of char  $p$  is perfect, because by semi-perfectness,  $x = x_1^p + \pi z_1 = x_1^p + \pi x_2^p + \pi^2 z_2 = \dots$ , so  $x = (x_1 + \pi^{\frac{1}{p}} x_2 + \pi^{\frac{2}{p}} x_3 + \dots)^p$ . In fact, uniformity is automatically implied by perfectness, by (12.2.5.12). The case of  $\mathcal{O}_K^a$ -algebra is similar.

2:  $A^a$  is  $K^{0a}$ -flat because  $A_*$  does, because it is a colimit of completions of polynomial algebras over  $I$  and  $I$  is flat over  $K^0$  (12.2.8.6). and  $R$  is perfectoid by (19.1.6.8) because  $A$  is totally integrally closed in  $R$ , because  $K^0[x_1^{\frac{1}{p^\infty}}, \dots, x_n^{\frac{1}{p^\infty}}]$  does (trivially), and use (5.7.2.10). ┘

### Tilting Equivalence

**Prop. (12.2.9.5) [Tilting Equivalence].** There are canonical isomorphisms of categories:

$$\text{Perfd}_K \cong \text{Perfd}_{K^{0a}} \cong \text{Perfd}_{K^{0a}/\pi},$$

where the first map is by  $R \mapsto R^{0a}$  and  $A \rightarrow A_*[t^{-1}]$  just as in (19.1.6.8). The second map is reduction by  $\pi$ .

In particular, using tilting (12.2.8.11), there are canonical isomorphisms of categories:

$$\mathrm{Perfd}_K \cong \mathrm{Perfd}_{K^{0a}} \cong \mathrm{Perfd}_{K^{0a}/\pi} = \mathrm{Perfd}_{K^{b0a}/t} \cong \mathrm{Perfd}_{K^{b0a}} \cong \mathrm{Perfd}_{K^b}.$$

If  $R \in \mathrm{Perfd}_K$  corresponds to  $S \in \mathrm{Perfd}_{K^b}$ , then we call  $S = R^b$  the **tilting** of  $R$  and  $R = S^\sharp$  the **untilting** of  $S$ .  $\square$

*Proof:*  $[\mathrm{Perfd}_K \cong \mathrm{Perfd}_{K^{0a}}]$

Firstly, if  $R \in \mathrm{Perfd}_K$ , then  $A = R^{0a} \in \mathrm{Perfd}_{K^{0a}}$ :  $R^0/\pi^{\frac{1}{p}} \rightarrow R^0/\pi$  is surjective by definition, for injectivity, if  $x^p/\pi \in R^0$ , then  $x/\pi^{\frac{1}{p}}$  is also power bounded, thus in  $R^0$ . And by (19.1.6.8),  $A$  is  $\pi$ -adically complete and  $\pi$ -torsion-free, hence  $R$ -flat by (5.4.1.12).

Next we show if  $A \in \mathrm{Perfd}_{K^{0a}}$ , then  $A_*$  is  $\pi$ -adically complete,  $t$ -torsion free and  $p$ -root closed in  $A[\pi^{-1}]$ , hence is a left inverse to the mapping  $R \rightarrow R^{0a}$ , by (5.7.3.5). It is complete by (5.7.3.3)2,3.

For  $p$ -root closedness, by (5.7.3.3),  $A_*/\pi^{\frac{1}{p}} \subset (A/\pi^{\frac{1}{p}})_* \hookrightarrow (A/\pi)_*$  by Frobenius, and then so does  $A_*/\pi^{\frac{1}{p}} \rightarrow A_*/\pi$ . Now if  $x \in A_*[\pi^{-1}]$  satisfies  $x^p \in A_*$ , then  $y = \pi^{\frac{k}{p}}x \in A_*$  for some  $k$ , and we want to lower  $k$  by 1 inductively, thus showing  $x \in A_*$ : As  $y^p \in \pi A_*$ ,  $y \in \pi^{\frac{1}{p}}A_*$  by what we have proved, thus  $\pi^{\frac{k-1}{p}}x \in A_*$ .

For surjectivity of Frob:  $A_*/\pi \rightarrow A_*/\pi$ , notice first it is almost surjective, because  $(A_* \rightarrow A_*/\pi)^a = A \rightarrow (A_*/\pi)^a \subset (A/\pi)_*^a = A/\pi$  is surjective by hypothesis, then by (5.7.2.3), it suffices to show that Frob is surjective on  $A/IA$ . For some  $x \in A^*$ , choose  $0 < 1 < c$ , almost surjectivity shows that  $\pi^c x \equiv y^p \pmod{\pi A_*}$ , so  $(y/p^{\frac{c}{p}})^p \in A_*$ , thus  $y \in p^{\frac{c}{p}}A_*$ , thus  $x \equiv (y/p^{\frac{c}{p}})^p \pmod{\pi^{1-c}A_*} \subset IA$ , so we are done.

Finally, this is also a right inverse, because we know that  $A_* \cong R^0$  by (5.7.3.5), thus  $A \cong R^{0a}$  in  $\mathrm{Mod}_R^a$ .  $\square$

*Proof:*  $[\mathrm{Perfd}_{K^{0a}} \cong \mathrm{Perfd}_{K^{0a}/\pi}]$

Firstly the reduction is a perfectoid  $K^{0a}/\pi$ -algebra: it is flat because flatness is stable under base change, and the rest are trivial. To construct a converse is a problem of deformation theory, we need to lift from  $K^{0a}/\pi$ -algebra via  $K^{0a}/\pi^n$ -algebras to a  $K^{0a}$ -algebra, suppose each lifting is unique up to isomorphism and the lift  $A_n$  is flat over  $K^{0a}/\pi^n$ , then we can form their inverse limit, which is flat, because it is  $\pi$ -torsion-free: if  $\pi(x_n) = 0$ , then by  $0 \rightarrow \pi^n K^{0a}/\pi^{n+1} \rightarrow K^{0a}/\pi^{n+1} \xrightarrow{\pi} K^{0a}/\pi^n \rightarrow 0$  and the flatness of  $A_{n+1}$ ,  $x_{n+1} \in \pi^n A_{n+1}$ , thus  $x_n = 0$ , and  $x = 0$ .

Now  $0 \neq A \in \mathrm{Perfd}_{K^{0a}/\pi}$ , then  $A$  is faithfully flat by (12.2.9.3), then by (5.7.1.8),  $A_{!!}$  is faithfully flat, and  $(-)_!!$  preserves all colimits and also Frobenius, so  $A_{!!}$  is relatively perfect. Then we use the above argument, and (8.1.4.1) to show that there is a  $\tilde{A} \in \mathcal{C}$  which is  $\pi$ -adically complete and  $K^0$ -flat, then  $\tilde{A} = (\tilde{A}_{!!})^a$  is also  $p$ -adically complete and  $K^{0a}$ -flat, by (5.7.3.3).

And we check  $\tilde{A}/\pi = (\tilde{A}_{!!}/\pi)^a = (A_{!!}/\pi)^a = A$  as  $(-)_!!$  commutes with colimits, and conversely, if  $A \in \mathrm{Perfd}_{K^{0a}}$ , we need to show  $A = \tilde{A}/\pi$ , notice by hypothesis,  $A_{!!}$  is faithfully flat  $K^0$ -algebra that is relatively flat over  $K^0/\pi$ , now it is also complete, because  $A_{!!} \rightarrow A_*$  is an injection (because  $(-)^a$  is exact) and almost isomorphism, so the cokernel is  $\pi$ -torsion, and  $A_*$  is complete, so does  $A_{!!}$ , by (5.2.3.9). Now  $A_{!!}/\pi = (A_{!!}/\pi)$  as  $(-)_!!$  commutes with colimits, so  $A_{!!}$  is just the lift, and  $(\tilde{A}_{!!}/\pi)^a = A_{!!}^a = A$ .  $\square$

**Cor. (12.2.9.6) [Tilting via Foutain's Functors].** The tilt  $R^b$  is just the Foutain's tilting, i.e.  $R^b = R^{0b}[t^{-1}]$ , and  $R^b = \lim_{x \mapsto x^p} R$ ,  $R^{0b} = (R^b)^0$ .  $\square$



*Proof:* Consider the diagram

$$\begin{array}{ccccc}
 K^{b0}/t^{p^n} & \xrightarrow{\varphi^{-n}} & K^{b0}/t \cong K^0/\pi & \longrightarrow & R^0/t \\
 & \searrow & \downarrow \varphi^n & & \downarrow \varphi^n \\
 & & K^{b0}/t \cong K^0/\pi & \longrightarrow & R^0/t
 \end{array}$$

Then the upper row is just the unique flat and relative perfect lifting along  $K^{b0}/t^{p^n} \rightarrow K^{b0}/t$ . Taking inverse limit, we get the structure map  $K^{b0} \rightarrow R^{b0}$ , so after almostification, this is just the lifting we are looking for, because it is unique. So  $(R^{0b})^a = (R^{b0})^a$ , and  $R^b = R^{0b}[t^{-1}]$  unwinding the tilting equivalence.

For  $R^b$ , notice there is a map

$$R^b \cong (\lim_{x \mapsto x^p} R^0)[t^{-1}] \rightarrow \lim_{x \mapsto x^p} (R^0[\pi^{-1}]) \cong \lim_{x \mapsto x^p} R$$

Now injectivity is clear as  $t$  is non-zero-divisor, and if  $(f_n) \in \lim_{x \mapsto x^p} R$ , then  $\pi^c f_n \in R^0$  for some  $c$ , then  $\pi^{\frac{c}{p^n}} f_n \in R^0$  because  $R^0$  is  $p$ -root closed(5.7.3.5), so  $t^c(f_n) \subset R^{0b}$ .

For the last assertion, it is true if  $R^{0b}$  is totally integrally closed in  $R^b$ , by(5.7.3.5). For this, if  $t^c f^{\mathbb{N}} \subset R^{0b}$ , then  $\pi^c(f^{\sharp})^{\mathbb{N}} \subset R^0$ , thus  $f^{\sharp} \in R^0$ . And by  $p$ -root closedness,  $p^n$ -th roots of  $f^{\sharp}$  are all in  $R^0$ , so  $f = (f_n) \in \lim_{x \mapsto x^p} R$  is in  $R^{0b}$ .  $\square$

**Prop. (12.2.9.7)[Fountain's Functor  $\theta$ ].** Given a perfectoid field  $K$ , the kernel of the Fontaine's map  $\theta : A_{\text{inf}}(K) \rightarrow K^0$ (5.5.1.15) is generated by a non-zero-divisor, in fact, if  $\text{char} K = 0$ , the generator can be chosen to be any element that maps to a generator of  $\ker \bar{\theta}$  and if  $\text{char} K = p$  this diagram is trivial. In particular, the diagram is a pushout.  $\lrcorner$

*Proof:* See the proof of(27.6.3.6) in the  $p$ -torsionfree case.  $\square$

**Prop. (12.2.9.8)[Untilting via  $A_{\text{inf}}$ ].** For any perfect  $\mathcal{O}_K^b$ -algebra  $A$ , by deformation theory(or in fact Witt theory) there is a unique lifting  $W(A)$  lifting it to  $A_{\text{inf}}(K^{0b})$ . And then pushout  $W(A) \otimes_{A_{\text{inf}}(K^0)} K^0$  is just the lifting of  $A/\pi$ , because the diagram above is pushout. This is in fact the method of [Kedlaya-Liu] used to prove the tilting-equivalence without the use of almost mathematics and deformation theory.  $\lrcorner$

**Cor. (12.2.9.9) [Limits and Colimits].** Any of the categories in(12.2.9.5) has arbitrary limits and colimits.  $\lrcorner$

*Proof:* We construct for  $\text{Perfd}_{K^{0ba}}$ : The limits is just the limits of topological rings, as the properties of  $t$ -adically complete,  $t$ -torsion free and perfect is preserved by limits(5.2.3.18). For the colimit, just use the  $t$ -adic completion of the left perfection of the colimits in the category of  $K^{0ba}$ -algebras, its  $t$ -torsion is almost zero because of perfectness, thus it is almost flat(5.7.3.3).  $\square$

**Remark (12.2.9.10).** Note also for further reference that in the category  $\text{Perfd}_{K^{0a}}$ , a filtered colimits is just the  $\pi$ -adically completion of the filtered limits as rings, because perfectness and flatness is preserved(5.4.1.6).  $\lrcorner$

**Prop. (12.2.9.11)[Tilting Equivalence Identifies Fields].**  $R \in \text{Perfd}_K$  is a perfectoid field iff its tilt  $R^b$  is a perfectoid field.  $\lrcorner$

*Proof:* It is proven that if  $R$  is a perfectoid field, then  $R^b$  is a perfectoid field. Conversely,  $R$  is a perfectoid field if the spectral norm given by  $\|x\| = \inf\{|t|^{-1} | t \in R^*, tx \in R^0\}$  is the Banach valuation of  $R$  and  $R$  is a field.

For the multiplicativeness of  $\|\cdot\|_R$ , notice that  $R^b$  is a perfectoid field, so its non-Archimedean valuation coincides with the spectral norm of  $\|\cdot\|_{R^b}$ , and this equals  $\|\cdot\|_R \circ \sharp$ , because  $R^{0b} = R^{b0}$ , an element  $f \in R^{b0}$  iff  $f^\sharp \in R^0$ . Now the norm extends that of  $K$  and commutes with scalar multiplication, so for any  $f, g$ , we may assume  $f, g \in R^0 - 0\pi^{\frac{1}{p}}R^0$ , now choose  $a, b \in R^b$  that  $a^\sharp - f, b^\sharp - g \in \pi R^0$ , this can be done because  $R^{b0} = R^{0b} \rightarrow R^0/\pi$  is surjective, then  $a, b, ab \notin \pi R^{b0}$  because  $R^{0b} = R^{b0}$ . Then clearly  $\|f\|_R = \|a\|_{R^b}, \|g\|_R = \|b\|_{R^b}, \|fg\|_R = \|ab\|_{R^b}$ , so it is multiplicative by the multiplicativeness of  $R^b$ .

To show  $R$  is a field, consider and  $f \in R - \pi^{\frac{1}{p}}R$ , choose  $a \in R^b$  that  $f = a^\sharp + \pi g$ , then as  $R^b$  is a field, there is a  $b$  that  $ab = 1$ . Now  $\|\pi\|_R < \|\pi^{\frac{1}{p}}\|_R \subset \|f\|_R = \|a\|_{R^b} \leq 1$ , so we get  $\|\pi b^\sharp g\| < 1$ , then

$$f^{-1} = \frac{1}{a^\sharp + \pi g} = \frac{b^\sharp}{1 + \pi b^\sharp g} = b^\sharp \left( \sum (-\pi b^\sharp g)^k \right)$$

can be constructed in  $R$ . □

### Perfectoid Affinoid Algebra

**Def. (12.2.9.12) [Perfectoid Affinoid  $K$ -algebras].** An **affinoid  $K$ -algebra**  $(R, R^+)$  is just an affinoid Tate ring over  $(K, K^0)$ . It is called a **perfectoid affinoid  $K$ -algebra** iff  $R$  is a perfectoid algebra. ┘

**Prop. (12.2.9.13) [Affinoid Tilting Equivalence].** The categories of perfectoid affinoid algebras (12.2.9.12) over  $K$  and  $K^b$  are equivalent, where  $(R, R^+)$  is identified with  $(R^b, R^{b+})$  iff  $R^b$  is the tilting of  $R$  and

$$\begin{array}{ccc} R^+/\mathfrak{m}R^0 & \xrightarrow{\cong} & R^{b+}/\mathfrak{m}^b R^{b0} \\ \downarrow & & \downarrow \\ R^0/\mathfrak{m}R^0 & \xrightarrow{\cong} & R^{b0}/\mathfrak{m}^b R^{b0} \end{array}.$$

Moreover, in this case,  $R^+/\pi$  is semi-perfect, and  $R^{+b} \cong R^{b+}$  as a subring of  $R^{0b} \cong R^{b0}$ . ┘

*Proof:* The case  $R^+ = R^0$  is already known by tilting equivalence (12.2.9.5) and (12.2.9.6).

By (12.2.5.11) and (12.2.5.14),  $\mathfrak{m}R^0 = R^{00} \subset R^+ \subset R^0$ , thus  $R^+ \rightarrow R^0$  is an almost isomorphism and  $R^+$  is determined by its image  $\overline{R^+} \subset R^0/\mathfrak{m}R^0$ , which is integrally closed if  $R^+$  does, so the identification is clear.

For the semi-perfectness: as  $R^+/\mathfrak{m}R^0$  is integrally closed, it is perfect. Now  $R^+ \rightarrow R^0$  is an almost isomorphism, so Frob on  $R^+/\pi$  is almost surjective because it does on  $R^0/\pi$  by definition, and now we know Frob is surjective on  $R^+/\pi$  by (5.7.2.3).

To show  $R^{+b} \cong R^{b+}$ , we show there is a Cartesian diagram

$$\begin{array}{ccc} R^{+b} & \longrightarrow & R^{b+}/\mathfrak{m}^b R^{b0} \\ \downarrow & & \downarrow \\ R^b & \longrightarrow & R^b/\mathfrak{m}^b R^{b0} \end{array}, \text{ but this is the}$$

Cartesian diagram

$$\begin{array}{ccc} R^+/\pi & \longrightarrow & R^+/\mathfrak{m}R^0 \cong R^{b+}/\mathfrak{m}^b R^{b0} \\ \downarrow & & \downarrow \\ R^0/\pi & \longrightarrow & R^0/\mathfrak{m}R^0 \end{array}$$

applied the functor  $(-)^{\text{perf}}$ , which preserves

limits(5.5.1.9). (Notice that  $R^+/\mathfrak{m}R^0 \cong R^{b+}/\mathfrak{m}^b R^{b0}$  is already perfect).  $\square$

**Cor.(12.2.9.14).** Notice that the proof also shows that  $R^+ \rightarrow R^0$  is an almost isomorphism, thus if  $R$  is a perfectoid  $K$ -algebra, then  $R^+$  is automatically a perfectoid  $K^{0a}$ -algebra by(12.2.9.1).  $\lrcorner$

**Cor.(12.2.9.15) [Perfectoid Affinoid Field].** A perfectoid affinoid  $K$ -algebra  $(R, R^+)$  is called a **perfectoid affinoid field** iff  $R$  is a perfectoid field and  $R^+$  is an open valuation ring.

Notice this is equivalent to  $R^+/\mathfrak{m}R^0$  is a valuation ring in  $R^0/\mathfrak{m}R^0$ . In particular, combining with(12.2.9.11), affinoid perfectoid fields are preserved under tilting and untilting.  $\lrcorner$

**Cor.(12.2.9.16).** The tilting equivalence also shows that for any perfectoid affinoid  $K$ -algebra  $(R, R^+)$ , the tilting induces an equivalence of categories  $\mathcal{P}\text{erf}_R \cong$   $\lrcorner$

**Prop.(12.2.9.17) [Filtered Colimits of Perfectoid Affinoid  $K$ -Algebras].** The category of perfectoid affinoid  $K$ -algebras has filtered colimits, and it is just the colimits in the category of complete uniform affinoid Tate rings(27.3.2.21). In particular, the filtered colimits of  $(A_i, A_i^+)$  is  $(\text{colim}_i A_i, \text{colim}_i A_i^+)$ .  $\lrcorner$

*Proof:* The colimit is perfectoid because the filtered colimits is exact.  $\square$

## 10 Almost Purity Theorem

**Thm.(12.2.10.1) [Almost Purity Theorem, Fontaine-Wintemberger[F-W79]/Faltings[Fal02]/Scholze[Sch12]].** For  $R \in \mathcal{P}\text{erf}_K$  with tilt  $S$ (12.2.9.5),

- Almost purity in characteristic  $p$ : (take  $(-)_*$  and) Inverting  $t$  gives an equivalence  $S_{\text{afét}}^0 \cong S_{\text{fét}}$ .
- Almost purity in characteristic 0: Inverting  $\pi$  gives an equivalence  $R_{\text{afét}}^0 \cong R_{\text{fét}}$ .
- Tilting and untilting functors induce equivalences  $R_{\text{afét}}^0 \cong S_{\text{afét}}^0$ .

In particular, there are equivalences

$$S_{\text{fét}} \xleftarrow{a} S_{\text{afét}}^{0a} \xrightarrow{b} (S^{0a}/t)_{\text{afét}} \cong (R^{0a}/\pi)_{\text{afét}} \xleftarrow{c} R_{\text{afét}}^{0a} \xrightarrow{d} R_{\text{fét}},$$

$\lrcorner$

*Proof:* The map  $a$  is already given in(5.7.2.14) by passing the power bounded-elements(equivalently,  $S_*$ ) and inverting  $t$ . And it is an isomorphism.

The equivalence of  $b$  and  $c$  follows from[Almost Ring theory, Thm5.3.27] **?**.

The functor  $d$  is given by  $A \rightarrow A_*[t^{-1}]$ . Firstly,  $A$  is a perfectoid  $K^{0a}$ -algebra. This is because it is almost finite projective thus almost flat, and  $R^{0a}/\pi \rightarrow A/\pi$  is weakly relative perfect by(5.7.2.15), so does  $K^{0a}/\pi \rightarrow A/\pi$  because relative perfect is stable under composition. And it is finite projective thus almost direct summand of a finite free module.

So now the tilting equivalence(12.2.9.5) shows that  $A_*[t^{-1}] \in R_{\text{fét}}$ : it is finite etale because the  $A_*$  is finite projective by the right adjointness of  $(-)_*$ , and unramified is defined in terms of  $A_*$ . The converse of  $d$  is supposed to be the functor that extract from  $A_*$  from  $A_*[t^{-1}]$  the total integral closure  $A_{\text{tic}}$  of  $R^0$ , which is functorial. We already know that  $A_*$  is totally integrally closed in  $A_*[t^{-1}]$  by(12.2.9.5), so  $A_{\text{tic}} \subset A_*$ . Conversely, as  $A$  is almost finitely generated over  $R^0$ , for  $f \in A_*$ ,  $\pi f^{\mathbb{N}}$  lies in a f.g.  $R^0$ -submodule of  $A_*$ , so  $f^{\mathbb{N}}$  is totally integral over  $R^0$ , so  $A_* = A_{\text{tic}}$ .

It's left to show that  $d$  is essentially surjective, but this uses perfectoid spaces. For now, we only check that this is true for  $R$  being a perfectoid field(of char 0).For this, we show directly that the untilting functor  $\sharp : K_{\text{fét}}^b \rightarrow K_{\text{fét}}$  is essentially surjective. Now  $\sharp$  is an equivalence of categories

$\text{Perfd}_{K^\flat} \rightarrow \text{Perfd}_K$ , and it preserves degree, at least for field extensions, so it preserves Galois extensions. Now that finite étale algebra over fields are just disjoint of finite separable extensions(5.4.7.23), so it suffices to show that any finite extension of  $K$  is contained in some  $L^\sharp$ .

Consider  $M = \widehat{K^\flat}$ , it is alg.closed of char  $p$  so clearly a perfectoid field, and by(12.2.8.16)  $M^\sharp$  is alg.closed.  $M^\sharp$  is just the colimit in the category of uniform Banach  $K$ -algebras, so its valuation ring is just the completion of the valuation ring of  $L^\sharp$  for  $L/K^\sharp$  finite Galois. Then if  $N = \cup L^\sharp$ , then  $N$  is dense in  $M^\sharp$ , and  $N/K$  is clearly algebraic and in particular Hensel. So  $N \subset \overline{N} \subset M^\sharp$  is dense, so by Krasner's lemma(19.1.2.29),  $N = \overline{N}$ . Now  $N = \cup L^\sharp$  is an alg.closure of  $K$ , so every finite extension of  $K$  is contained in some  $L^\sharp$ .

The proof of the general case of the essentially surjectivity of  $d$  is continued at(27.4.3.10). □

## 12.3 Measure Theory and Functions on $\mathbb{R}^n$

References are [Lan93, Fol99, T-W06, Zhou08], [Measure theory and fine properties of functions, Evans] and [Real Analysis and Probability, Dudley], [Real and Complex Analysis, Rudin] and [Stein, Real Analysis].

### Notation (12.3.0.1).

- Use notations on topology defined in [Topology I](#).
- Use notations on formal power series defined in [Formal Power Series](#).

┘

## 1 Measures

**Def. (12.3.1.1) [ $\sigma$ -Algebras]**. For  $X \in \text{Set}$ , an **algebra of subsets** of  $X$  is a subset of  $\mathcal{P}(X)$  that is closed under taking complements and finite unions. A  **$\sigma$ -algebra** on  $X$  is an algebra of subsets of  $X$  that is closed under countable unions. ┘

**Def. (12.3.1.2) [Measurable Space]**. A **measurable space** is a tuple  $(X, \mathcal{M})$  where  $X \in \text{Set}$  and  $\mathcal{M}$  is a  $\sigma$ -algebra on  $X$  (12.3.1.1). ┘

**Prop. (12.3.1.3) [Constructing  $\sigma$ -Algebras]**.

- Suppose  $X \in \text{Set}$  and  $X_\alpha \subset X$  are subsets s.t.  $X = \bigcup_{\alpha \in I} X_\alpha$ , and  $\mathcal{M}_\alpha$  are  $\sigma$ -algebras of  $X_\alpha$ , then we can construct a  $\sigma$ -algebra  $\mathcal{M}$  on  $X$  consisting of  $A \in \mathcal{P}(X)$  s.t.  $A \cap X_i \in \mathcal{M}_i$ , called the **colimit  $\sigma$ -algebra** on  $X$ .
- Suppose  $\text{pr} : Y \rightarrow X \in \text{Set}$  is a surjection and  $\mathcal{M}$  is a  $\sigma$ -algebra on  $Y$ , then we can construct a  $\sigma$ -algebra on  $X$  consisting of elements  $A \in \mathcal{P}(X)$  s.t.  $\text{pr}^{-1}(A) \in \mathcal{M}$ , called the **quotient  $\sigma$ -algebra** on  $X$ .

┘

**Example (12.3.1.4) [Borel  $\sigma$ -Algebras]**. For any  $X \in \text{Top}$ , the  $\sigma$ -algebra generated by the open subsets of  $X$  is called the **Borel  $\sigma$ -Algebra** on  $X$ . The elements in this  $\sigma$ -algebra are called **Borel subsets** of  $X$ . ┘

**Def. (12.3.1.5) [Separated Measurable Spaces]**. A measurable space  $(X, \mathcal{M})$  is called a **separable measurable space** if  $\{x\} \in \mathcal{M}$  for each  $x \in X$ .

And  $(X, \mathcal{M})$  is called a **countably-separated measurable space** if there is a countable family  $\{E_j\}_{j \in \mathbb{Z}_+} \in \mathcal{M}$  s.t. for any  $x \in X$ ,  $\{x\} = \bigcap_{j \in \mathbb{Z}_+} E_j$ .

$(X, \mathcal{M})$  is called a **Polish space** if  $X$  is a separable completely-metrizable topological space, and  $\mathcal{M}$  is the Borel  $\sigma$ -algebra on  $X$  (12.3.1.4).

$(X, \mathcal{M})$  is called a **standard measurable space** if it is isomorphic to a Borel subset (12.3.1.4) of a Polish space. It is clear that a standard measurable space is countably-separated. ┘

**Lemma (12.3.1.6) [Measurable Schröder-Bernstein Theorem]**. If  $(X, \mathcal{A}), (Y, \mathcal{B})$  are measurable spaces and there exists an injective bimeasurable map  $f : (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$  and also an injective bimeasurable map  $g : (Y, \mathcal{B}) \rightarrow (X, \mathcal{A})$ , then there exists an isomorphism of measurable spaces  $h : (X, \mathcal{A}) \cong (Y, \mathcal{B})$ . ┘

**Thm. (12.3.1.7) [Borel Isomorphism Theorem]**. Any two standard measurable spaces (12.3.1.5) are isomorphic iff they have the same cardinality.

In particular, any uncountable standard Borel set is isomorphic to  $\mathbb{C}$ , or equivalently  $[0, 1]$ . ┘

*Proof:* Cf. [ELEMENTARY PROOF OF THE BOREL ISOMORPHISM THEOREM]. □

### Measures

**Def. (12.3.1.8) [Measures and Measure Spaces].** Let  $(X, \mathcal{M})$  be a measurable space (12.3.1.2), then a **measure** on  $(X, \mathcal{M})$  is a function  $\mu : \mathcal{M} \rightarrow [0, \infty]$  that

- $\mu(\emptyset) = 0$ .
- (Countable additivity) If  $E_i$  is a countable family of disjoint sets in  $\mathcal{M}$ , then  $\mu(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mu(E_i) \in [0, \infty]$ .

A **measure space** is a measurable space together with a measure  $\mu$ . And  $\mu$  is called a

- **finite measure** if  $\mu(X) < \infty$ .
- **probabilistic measure** if  $\mu(X) = 1$ .
- **$\sigma$ -finite measure** if there exists  $(E_i)_{i \in \mathbb{Z}_+} \in \mathcal{M}$  s.t.  $X = \cup_i E_i$  and  $\mu(E_i) < \infty$  for each  $i$ .

┘

**Prop. (12.3.1.9).** Let  $(X, \mathcal{M}, \mu)$  be a measure space (12.3.1.8), then

- For any  $E, F \in \mathcal{M}$  s.t.  $E \subset F$ , we have  $\mu(E) \leq \mu(F)$ .
- For any  $(E_i)_{i \in \mathbb{Z}} \in \mathcal{M}$ , we have  $\mu(\cup_i E_i) \leq \sum_{i \in \mathbb{Z}_+} \mu(E_i)$ .
- For any  $(E_i)_{i \in \mathbb{Z}} \in \mathcal{M}$  s.t.  $E_1 \subset E_2 \subset \dots$ , we have  $\mu(\cup_i E_i) = \lim_{n \rightarrow \infty} \mu(E_n)$ .
- For any  $(E_i)_{i \in \mathbb{Z}} \in \mathcal{M}$  s.t.  $\mu(E_1) < \infty$ , we have  $\mu(\cap_i E_i) = \lim_{i \rightarrow \infty} \mu(E_i)$ .

┘

*Proof:* Cf. [Fol99]P25. ?

□

**Def. (12.3.1.10) [Complete Measures].** A measure space  $(X, \mathcal{M}, \mu)$  is called a **complete measure space** if the following is satisfied: For any  $E \in \mathcal{M}$  s.t.  $\mu(E) = 0$  and  $F \subset E$ , we have  $F \in \mathcal{M}$ .

Denote  $\mathcal{N} = \{N \in \mathcal{M} | \mu(N) = 0\}$  and denote

$$\overline{\mathcal{M}} = \{X \in \mathcal{P}(X) | X = E \cup F, E \in \mathcal{M}, F \in \mathcal{N} \text{ for some } E, F\}.$$

Then  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra on  $X$  (12.3.1.1), and there is a unique extension of  $\mu$  to a measure  $\bar{\mu}$  on  $\overline{\mathcal{M}}$ . This  $\bar{\mu}$  is called the **completion measure** of  $\mu$ .

┘

*Proof:* Cf. [Fol99]P27. ?

□

**Def. (12.3.1.11) [Signed Measures].** Let  $(X, \mathcal{M})$  be a measurable space (12.3.1.2), then a **signed measure** on  $X$  is a function  $\mu : \mathcal{M} \rightarrow [-\infty, \infty]$  that

- $\mu(\emptyset) = 0$ .
- $\mu$  assumes at most one of the values  $\pm\infty$ .
- If  $E_i$  is a countable family of disjoint sets in  $\mathcal{M}$ , then  $\mu(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mu(E_i)$ , where the sum converges absolutely if  $\mu(\cup_{i=1}^{\infty} E_i) \in (-\infty, \infty)$ .

$\mu$  is called a **finite signed measure** if  $\text{Im}(\mu) \subset (-\infty, \infty)$ .

┘

**Prop. (12.3.1.12).** Let  $X \in \text{Set}$  and  $\mathcal{M}$  be an algebra on  $X$ , then a finitely-additive real-valued function  $\mu$  on  $\mathcal{M}$  is countably-additive iff  $\lim_n \mu(A_n) = 0$  for any sequence  $(A_n)_{n \in \mathbb{N}} \subset \mathcal{M}$  s.t.  $A_0 \supset A_1 \supset \dots \supset A_n$  and  $\cap_n A_n = \emptyset$ .

┘

*Proof:* [Dudley, P86].

□

**Def. (12.3.1.13) [Outer Measures].** Let  $X \in \text{Set}$ , an **outer measure** on  $X$  is a map

$$\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$$

s.t.

- $\mu^*(\emptyset) = 0$ .
- (Countable sub-additivity)  $\mu^*(A) \leq \sum_{i \in \mathbb{Z}_+} \mu^*(A_i)$  if  $A \subset \bigcup_{i \in \mathbb{Z}_+} A_i$ .
- $\mu^*(\bigcup_{i \in \mathbb{Z}_+} A_i) \leq \sum_{i \in \mathbb{Z}_+} \mu^*(A_i)$  for any  $(A_i)_{i \in \mathbb{Z}_+} \subset X$ .

┘

**Prop. (12.3.1.14) [Constructing Outer Measures].** Suppose  $X \in \text{Set}$  and  $\mathcal{E} \subset \mathcal{P}(X)$  be a family of subsets of  $X$  s.t.  $\emptyset \in \mathcal{E}$ ,  $X \in \mathcal{E}$ , and  $\rho : \mathcal{E} \rightarrow [0, \infty]$  is a map. For any  $A \subset X$ , define

$$\mu_\rho^*(A) = \inf \left\{ \sum_{i \in \mathbb{Z}_+} \rho(E_i) \mid E_i \in \mathcal{E}, A \subset \bigcup_{i \in \mathbb{Z}_+} E_i \right\},$$

then  $\mu^*$  is an outer measure on  $X$  (12.3.1.13). ┘

*Proof:* Only the countable sub-additivity property needs a proof: For any  $(A_i)_{i \in \mathbb{Z}_+} \subset X$  and  $\varepsilon \in \mathbb{R}_+$ , choose  $(E_i^k)_{i \in \mathbb{Z}_+} \subset X$  s.t.  $\sum_{k \in \mathbb{Z}_+} \rho(E_i^k) \leq \mu_\rho^*(A) + \varepsilon 2^{-i}$ . Then  $\{E_i^k\}_{i,k \in \mathbb{Z}_+}$  covers  $\bigcup_{i \in \mathbb{Z}_+} A_i$ , and thus

$$\mu_\rho^*\left(\bigcup_{i \in \mathbb{Z}_+} A_i\right) \leq \sum_{i,k \in \mathbb{Z}_+} \rho(E_i^k) \leq \sum_{i \in \mathbb{Z}_+} \mu_\rho^*(A_i) + \varepsilon.$$

Since  $\varepsilon$  is arbitrary, the assertion follows.  $\square$

**Thm. (12.3.1.15) [Carathéodory's Theorem].** Let  $X \in \text{Set}$  and  $\mu^*$  be an outer measure on  $X$ . A subset  $A \subset X$  is called a  $\mu^*$ -**measurable set** if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for any  $E \subset X$ . Let  $\mathcal{M} \subset \mathcal{P}(X)$  be the set of  $\mu^*$ -measurable sets, then  $\mathcal{M}$  is a  $\sigma$ -algebra, and the restriction of  $\mu^*$  to  $\mathcal{M}$  is a complete measure (12.3.1.10). ┘

*Proof:* Cf. [Fol99] P29. ?  $\square$

**Prop. (12.3.1.16) [Hahn Decomposition Theorem].** Let  $(X, \mathcal{M})$  be a measurable space and  $\mu$  a signed measure on  $X$ , a set  $E \in \mathcal{M}$  is called **positive (resp. negative/null) for  $\mu$**  if  $\mu(F) \geq 0$  (resp.  $\mu(F) \leq 0 / \mu(F) = 0$ ) for any  $F \in \mathcal{M} \cap \mathcal{P}(E)$ .

Then there exists a disjoint decomposition  $X = P \sqcup N$ ,  $P, N \in \mathcal{M}$ , s.t.  $P$  is positive for  $\mu$  and  $N$  is negative for  $\mu$ . Such a decomposition is unique up to a null subset of  $\mu$ . ┘

*Proof:* Cf. [Follan, P86]. ?  $\square$

**Def. (12.3.1.17) [Orthogonal Measures].** Let  $(X, \mathcal{M})$  be a measurable space, then two signed measures  $\mu, \nu$  on  $X$  is called **orthogonal measures** if there is a decomposition  $X = E \sqcup F$ ,  $E, F \in \mathcal{M}$  s.t.  $E$  is null for  $\mu$  and  $F$  is null for  $\nu$ . Orthogonal measures are denoted by  $\mu \perp \nu$ . ┘

**Prop. (12.3.1.18) [Jordan-Decompositions].** Let  $(X, \mathcal{M})$  be a measurable space and  $\mu$  a signed measure on  $X$ , then there exist unique positive measures  $\mu^+$  and  $\mu^-$  on  $X$  s.t.  $\mu = \mu^+ - \mu^-$  and  $\mu^+ \perp \mu^-$  (12.3.1.17). Such a decomposition is called the **Jordan-decomposition of measures**.

$|\mu| = \mu^+ + \mu^-$  is called the **total variation** of  $\mu$ . And  $\mu$  is called a **finite signed measure** /  **$\sigma$ -finite signed measure** iff  $|\mu|$  is finite /  $\sigma$ -finite. ┘

*Proof:* This follows immediately from (12.3.1.16).  $\square$

**Def. (12.3.1.19) [Complex Measures].** A **complex measure** on a measurable space  $(X, \mathcal{M})$  is a function  $\nu : \mathcal{M} \rightarrow \mathbb{C}$  s.t.

- $\nu(\emptyset) = 0$ .
- If  $E_i$  is a countable family of disjoint sets in  $\mathcal{M}$ , then  $\mu(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mu(E_i)$ , where the series converges absolutely.

┘

**Def. (12.3.1.20) [Lebesgue Space].** A one-point subset with positive measure is called an **atomic measure space**. A **Lebesgue space** is a finite measure space that is isomorphic to a finite union of intervals and countably many atomic measure spaces.

┘

### Borel Measures

**Notation (12.3.1.21).**

- Let  $X \in \text{LCH}$ .

┘

**Def. (12.3.1.22) [Radon Measure].** A **Borel measure** on  $X$  is a measure defined on the Borel  $\sigma$ -algebra of  $X$  (12.3.1.4)

A Borel measure  $\mu$  is called **inner regular** on a Borel set  $E$  iff  $\mu(E) = \inf\{\mu(K) \mid K \subset E \text{ compact}\}$  for every Borel set  $E$ . It is called **outer regular** on a Borel set  $E$  iff  $\mu(E) = \sup\{\mu(U) \mid E \subset U \text{ open}\}$ .

$\mu$  is called a **regular measure** if it is finite on compact sets and outer regular on Borel sets.

A **Radon measure** is a Borel measure that is finite on compact set, outer regular on Borel sets, inner regular on open sets.

┘

**Def. (12.3.1.23) [Discrete Measures and Continuous Measures].** A complex or signed Borel measure  $\mu$  on Hausdorff space  $X$  is called a **discrete measure** if there exists a countable set  $\{x_i\} \subset X$  and  $c_j \in \mathbb{C}$  s.t.  $\sum |c_j| < \infty$ , and  $\mu = \sum c_j \delta_{x_i}$ . And  $\mu$  is called a **continuous measure** if  $\mu(\{x\}) = 0$  for any  $x \in X$ .

Any complex or  $\sigma$ -finite signed measure  $\mu$  on  $X$  can be written uniquely as  $\mu = \mu_d + \mu_c$ , where  $\mu_d$  is a discrete measure and  $\mu_c$  is a continuous measure.

┘

*Proof:* It is easy to see that  $E = \{x \in X : \mu(\{x\}) > 0\}$  is countable. Thus we can define  $\mu_d(A) = \mu(A \cap E)$  and  $\mu_c(A) = \mu(A \setminus E)$ .  $\square$

**Prop. (12.3.1.24).** Every Radon measure is inner regular on all of its  $\sigma$ -finite sets.

┘

*Proof:* Cf. [Folland, Real Analysis, P216].  $\square$

**Cor. (12.3.1.25).** Every  $\sigma$ -finite Radon measure is regular. In particular, if  $X$  is  $\sigma$ -compact, then every Radon measure is regular.

┘

### Measurable Maps

**Def. (12.3.1.26) [Measurable Map].** A **measurable map** or a measurable function  $f : (X, \mathcal{M}) \rightarrow (Y, \mathcal{N})$  between two measurable spaces is a map  $f : X \rightarrow Y$  that  $f^{-1}(E) \in \mathcal{M}$  for any  $E \in \mathcal{N}$ .

┘

**Def. (12.3.1.27) [Non-Singular Maps].** A **non-singular measurable map** is a measurable map of measure spaces that the preimage of every set of measure 0 has measure 0.

┘



**Prop. (12.3.1.28)[convergences].** There are three different kinds of convergences:

- **almost everywhere convergence** iff  $f_n(x) \rightarrow f(x)$  a.e.
- **almost uniform convergence** iff for any  $\delta > 0$ , there is a measurable subset  $E_\delta$  that  $f_n$  convergent to  $f$  uniformly on  $E - E_\delta$ .
- **convergence in measure** iff  $\lim_{k \rightarrow \infty} m(\{x \in E \mid |f_n(x) - f(x)| > \varepsilon\}) = 0$ .

┘

**Prop. (12.3.1.29)[Relations between Convergences].**

- (Egoroff) If  $m(E) < \infty$  and  $f_k$  converges to  $f$  a.e. then  $f_k$  converges to  $f$  almost uniformly.
- If  $f_k$  converges to  $f$  almost uniformly, then  $f_k$  converges to  $f$  in measure.
- (Riesz) If  $f_k$  converges to  $f$  in measure, then there is a subsequence  $f_{n_k}$  that converges to  $f$  a.e..

┘

*Proof:* 1: Cf.[实变函数周明强 P113].

2: Trivial.

3: Cf.[实变函数周明强 P118].

□

## 2 Integrations

**Def. (12.3.2.1)[Simple Functions].**

┘

**Def. (12.3.2.2)[Integrable Functions].** Let  $(X, \mathcal{M})$  be a measurable space. If  $\mu$  is a measure, ?

If  $\mu$  is a signed measure, then define  $L^1(\mu) = L^1(|\mu|) = L^1(\mu^+) \cap L^1(\mu^-)$  (12.3.1.18). And  $f$  is called an **extended integrable function** if either  $f^+$  or  $f^-$  is integrable.

┘

**Def. (12.3.2.3)[Locally Integrable Functions].** A measurable function  $f : \mathbb{R}^n \rightarrow \mathbb{C}$  is called **locally integrable** if  $\int_K |f(x)| dx < \infty$  for every bounded measurable set  $K$  of  $\mathbb{R}$ . The set of locally integrable function is denoted by  $L^1_{\text{loc}}(\mathbb{R}^n)$ .

┘

**Prop. (12.3.2.4).** A function  $f$  is real analytic on an open set of  $\mathbb{R}$  iff there is a extension to a complex analytic function to an open set of  $\mathbb{C}$ . And this is equivalent to: For every compact subset, there is a constant  $C$  that for every positive integer  $k$ ,  $|\frac{d^k f}{dx^k}(x)| \leq C^{k+1} k!$ .

┘

*Proof:* Use Lagrange residue(中值定理) to show that it will converge to  $f$ .

□

**Prop. (12.3.2.5)[Monotone-convergence-theorem].**

┘

**Prop. (12.3.2.6)[Dominant Convergence Theorem].**

┘

**Prop. (12.3.2.7)[Fatou's Lemma].**

┘

*Proof:* ?

□

**Prop. (12.3.2.8).** The set  $E$  of nowhere differentiable functions are of second category in  $C[0, 1]$ , and its complement set is of first category.

┘

*Proof:* let  $A_n$  be the sets of functions  $f$  that there exists a  $s$  that for any  $|h| \leq 1/n$ ,  $|\frac{f(s+h)-f(s)}{h}| \leq n$ . It is easy to see that  $C[0, 1] - E \subset \cup_n A_n$ , so it suffices to show each  $A_n$  is of first category.

Firstly  $A_n$  is closed, because if  $s \notin A_n$ , then for any  $s$ , there is a  $|h_s| \leq 1/n$  that  $|f(s+h_s)-f(s)| > n|h_s|$ . So by continuity, there is a  $\varepsilon_s > 0$  and some nbhd  $J_s$  of  $s$  that  $|f(\sigma-h_s)-f(\sigma)| > n|h_s| + 2\varepsilon_s$  for all  $\sigma \in J_s$ . Then there are f.m.  $J_{s_i}$  that covers  $[0, 1]$ , so let  $\varepsilon = \min\{\varepsilon_i\}$ , then if  $\|g-f\| < \varepsilon$ , then  $g \notin A_n$ .

And  $A_n$  has no interior point, because for any  $f \in A_n$ ,  $f$  can be approximated by a polynomial  $g$ , by Stone-Weierstrass theorem(12.3.8.2), and by Mean-value theorem, there is a  $M$  that  $|g(s+h)-g(s)| \leq M|h|$  for all  $s$  and  $|h| < 1/n$ . So if  $p$  is a pairwise-linear function that  $\|p\|$  is small and the slopes of  $p$  are bigger than  $M+n$ , then  $g+p$  is near  $f$  but  $g+p \notin A_n$ .

Finally,  $E$  is of second category by Baire theorem(4.4.9.2).  $\square$

**Prop. (12.3.2.9) [Fubini-Tonelli].** For two  $\sigma$ -finite measure spaces  $X, Y$ ,

- If  $f \in L^+(X \times Y)$ , then  $f_x \in L^+(Y)$  and  $f^y \in L^+(X)$ , and

$$\int_{X \times Y} f dx dy = \int_Y \int_X f dx dy = \int_X \int_Y f dy dx.$$

- If  $f \in L^1(X \times Y)$ , then  $f_x \in L^1(Y)$  and  $f^y \in L^1(X)$ , a.e. and the product formula is definable and holds.  $\lrcorner$

*Proof:* Cf.[Folland P67].  $\square$

**Prop. (12.3.2.10).** Let  $(X, \mathcal{M})$  be a measurable space with a positive measure  $\mu$ , and  $f$  is an extended  $\mu$ -integrable function(12.3.2.2), then we can define a singed measure  $\nu$  on  $X$  by

$$\nu(E) = \int_E f d\mu, \quad E \in \mathcal{M}.$$

This relation is denoted by  $d\nu = f d\mu$ . It is clear that  $\nu \ll \mu$ .  $\lrcorner$

**Cor. (12.3.2.11).** Let  $(X, \mathcal{M})$  be a measurable space with a positive measure  $\mu$ , and  $f \in L^1(\mu)$ , then it follows from(12.3.3.2) that: For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\delta \in \mathbb{R}_+$  s.t.  $|\int_E f d\mu| < \varepsilon$  whenever  $\mu(E) < \delta$ .  $\lrcorner$

### Miscellaneous

**Lemma (12.3.2.12) [Smooth Test Functions].** If  $[t_0, t_1] \subset \mathbb{R}$  is an interval and  $f \in C([t_0, t_1])$  s.t.  $\int_{t_0}^{t_1} f(t)h(t)dt = 0$  for any  $h(t) \in C^\infty([t_0, t_1])$  that  $h(t_0) = h(t_1) = 0$ , then  $f = 0$ .  $\lrcorner$

*Proof:* Use bump functions as a test function.  $\square$

### Henstock-Kurzweil integrations

**Prop. (12.3.2.13).** For  $f : \mathbb{R} \rightarrow \mathbb{C}$ ,  $f \in L^1(m)$  iff both  $f^+$  and  $f^-$  are Henstock-Kurzweil integrable.  $\lrcorner$

*Proof:*  $\square$

**Thm. (12.3.2.14).** If  $a < b \in \mathbb{R}$  and  $f : [a, b] \rightarrow \mathbb{R}$  is Henstock-Kurzweil integrable and  $F(x) = {}^{HK}\int_a^x f(x)dx$ , then

$$F^\Delta(x) = f(x).$$

$\lrcorner$

*Proof:*

□

**Thm. (12.3.2.15).** If  $a < b \in \mathbb{R}$  and  $F : [a, b] \rightarrow \mathbb{R}$  is differentiable, then

$$F(b) - F(a) = {}^{HK} \int_a^b F'(x) dx.$$

┘

### 3 Differentiations

#### Absolute Continuity

**Def. (12.3.3.1) [Absolutely Continuous].** Let  $(X, \mathcal{M})$  be a measurable space with a positive measure  $\mu$ . A signed measure  $\nu$  on  $X$  is called **absolutely continuous** w.r.t.  $\mu$  (denoted by  $\nu \ll \mu$ ) if for any  $E \in \mathcal{M}$ ,  $\mu(E) = 0 \implies \nu(E) = 0$ . ┘

**Prop. (12.3.3.2).** Let  $(X, \mathcal{M})$  be a measurable space with a positive measure  $\mu$ , and  $\nu$  a finite signed measure on  $X$ , then  $\nu \ll \mu$  iff for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\delta \in \mathbb{R}_+$  s.t.  $|\nu(E)| < \varepsilon$  whenever  $\mu(E) < \delta$ . ┘

*Proof:* Cf.[Folland]P89.

□

**Lemma (12.3.3.3).** Let  $(X, \mathcal{M})$  be a measurable space and  $\mu, \nu$  are positive measures on  $X$ , then either  $\nu \perp \mu$ , or there exists  $\varepsilon \in \mathbb{R}_+$  and  $E \in \mathcal{M}$  s.t.  $\mu(E) > 0$ , and  $E$  is a positive set for  $\nu - \varepsilon\mu$ . ┘

*Proof:* Cf.[Folland]P90.

□

**Thm. (12.3.3.4) [Lebesgue-Radon-Nikodym].** Let  $(X, \mathcal{M})$  be a measurable space and  $\mu$  is positive  $\sigma$ -finite measures on  $X$ . Then for any  $\sigma$ -finite signed measures  $\nu$  on  $X$ , there exists unique  $\sigma$ -finite signed measures  $\lambda, \rho$  on  $X$  s.t.

$$\lambda \perp \mu, \quad \rho \ll \mu \text{ (12.3.3.1)}, \quad \nu = \lambda + \rho.$$

Moreover, there exists an extended  $\mu$ -integrable function  $f : X \rightarrow \mathbb{R}$  s.t.  $d\rho = f d\mu$ , and  $f$  is determined a.e. w.r.t.  $\mu$ . Such  $f$  is called the **Radon-Nikodym derivative** of  $\rho$  w.r.t.  $\mu$ , denoted by  $d\rho/d\mu$ .

The same holds true for a complex measure  $\nu$  (12.3.1.19), by applying to the real and complex parts. ┘

*Proof:* This is a special case of the **Freudenthal Spectral Theorem** (12.9.3.18). Cf.[Folland]P90. □

**Cor. (12.3.3.5) [Lebesgue Decomposition Theorem].** By (12.3.1.23), any (complex or  $\sigma$ -finite signed) Borel measure  $\nu$  on  $X$  can be written uniquely as a sum of Borel measures

$$\nu = \nu_d + \nu_{ac} + \nu_{sc},$$

where  $\nu_d$  is discrete,  $\mu_{ac} \ll dm$ , and  $\nu_{sc} \perp dm$  but continuous. ┘

**Def. (12.3.3.6) [Total Variation for Complex Measures].** For a complex measure  $\nu$  on a measurable space  $(X, \mathcal{M})$ , we can define its total variation to be the positive measure on  $\mathcal{M}$  defined as follows: Write  $d\mu = f d(|\mu_r| + |\mu_i|)$ , where  $\mu_r$  and  $\mu_i$  are the real and complex parts of  $\mu$ . Then define  $d|\mu| = |f| d(|\mu_r| + |\mu_i|)$ . This is compatible with the definition for signed measures (12.3.1.18). ┘

**Prop. (12.3.3.7).** For a complex measure  $\nu$  on a measurable space  $(X, \mathcal{M})$ ,

- $|\nu(E)| \leq |\nu|(|E|)$  for any  $E \in \mathcal{M}$ .
- $\nu \ll |\nu|$ , and  $d\nu/d|\nu|$  has absolute value 1  $|\nu|$ -a.e.
- $L^1(\nu) = L^1(|\nu|)$ , and if  $f \in L^1(\nu)$ , then  $|\int f d\nu| \leq \int |f| d|\nu|$ .

┘

*Proof:* Cf. [Fol99]P94.

□

### Differentiation on Euclidean Spaces

**Lemma (12.3.3.8) [Vitali Covering Theorem].** Let  $\mathcal{C}$  be a collection of balls in  $\mathbb{R}^n$ , and let  $U = \cup_{B \in \mathcal{C}} B$ . Then if  $c > m(U)$ , then there exists disjoint  $B_1, \dots, B_k \in \mathcal{C}$  that  $\sum_{i=1}^k m(B_k) > 3^{-n}c$ .

┘

*Proof:*

□

**Lemma (12.3.3.9).** If  $f \in L^1_{\text{loc}}$  and  $A_r f(x) = \frac{1}{\text{Vol}(B(r,x))} \int_{B(r,x)} f(y) dy$ , then  $A_r f$  is continuous in both  $r$  and  $x$ .

┘

*Proof:* Cf. [Folland P96].

□

**Prop. (12.3.3.10).** If  $f \in L^1_{\text{loc}}$ , then  $\lim_{r \rightarrow 0} A_r f(x) = f(x)$  for a.e.  $x \in \mathbb{R}^n$ .

┘

*Proof:* Cf. [Folland P97].

□

**Prop. (12.3.3.11) [Fermat].** Let  $x_0 \in \mathbb{R}$ ,  $\delta \in \mathbb{R}_+$ ,  $f$  is a function on  $U(x_0, \delta)$ . If  $x_0$  is an extreme point of  $f$ , and  $f'(x_0)$  exists, then  $f'(x_0) = 0$ .

┘

*Proof:* By changing  $f$  to  $-f$  if necessary, we can assume  $x_0$  is a supremum point. Then for  $0 < h < \delta$ ,  $\frac{f(x_0+h)-f(x_0)}{h} \leq 0$ , and for  $-\delta < h < 0$ ,  $\frac{f(x_0+h)-f(x_0)}{h} \geq 0$ , so  $f'(x_0) = 0$ .

□

**Lemma (12.3.3.12) [Rolle's Mean Value Theorem].** If  $a < b \in \mathbb{R}$ ,  $f \in C([a, b])$  is differentiable on  $[a, b]$ , and  $f(a) = f(b)$ , then there exists some  $\xi \in (a, b)$  s.t.  $f'(\xi) = 0$ .

┘

*Proof:* As  $f$  is continuous on  $[a, b]$ , which is compact,  $f$  has minimum  $m$  and maximum  $M$  on  $[a, b]$ . If  $m = M$ , then  $f$  is constant, and any  $\xi \in (a, b)$  will do. If  $M > m$ , then  $M \neq f(a)$  or  $m \neq f(a)$ . Suppose WLOG the first case happens, then if  $f(\xi) = M$ ,  $\xi \in (a, b)$ , then  $f'(\xi) = 0$  by Fermat's theorem (12.3.3.11).

□

**Thm. (12.3.3.13) [Lagrange's Mean Value Theorem].** If  $a < b \in \mathbb{R}$ ,  $f \in C([a, b])$  is differentiable on  $[a, b]$ , then there exists some  $\xi \in (a, b)$  s.t.  $f'(\xi) = \frac{f(b)-f(a)}{b-a}$ .

┘

*Proof:* This follows from Rolle's mean value theorem by considering the function

$$F(x) = f(x) - \left[ f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right].$$

□

**Cor. (12.3.3.14) [Cauchy's Mean Value Theorem].** If  $a < b \in \mathbb{R}$ ,  $f, g \in C([a, b])$  is differentiable on  $[a, b]$ , and  $g' \neq 0$ . then there exists some  $\xi \in (a, b)$  s.t.  $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(\xi)}{g'(\xi)}$ .

┘

*Proof:* This follows from Rolle's mean value theorem by considering the function

$$G(x) = f(x) - \left[ f(a) + \frac{f(b) - f(a)}{g(b) - g(a)} (g(x) - g(a)) \right].$$

□

**Cor.(12.3.3.15) [Integral Mean Value Theorem].** For  $a < b \in \mathbb{R}$ ,  $f, g \in C([a, b])$ , there exists  $\xi \in (a, b)$  s.t.

$$\int_a^b f(t)g(t)dt = f(\xi) \int_a^b g(t)dt.$$

┘

*Proof:* Consider

$$H(x) = \int_a^x f(t)g(t)dt, \quad G(x) = \int_a^x g(t)dt,$$

then it follows from Cauchy's mean value theorem(12.3.3.14) that there exists  $\xi \in (a, b)$  s.t.

$$\frac{H(b) - H(a)}{G(b) - G(a)} = \frac{\int_a^b f(t)g(t)dt}{\int_a^b g(t)dt} = \frac{H'(\xi)}{G'(\xi)} = f(\xi).$$

□

**Prop.(12.3.3.16) [Commutativity of Integration].** Let  $n \in \mathbb{Z}_{\geq 2}$ ,  $\Omega \subset \mathbb{R}^n$  be open and  $f \in C(\Omega)$ . If  $1 \leq j < k \leq n$  and  $\frac{\partial}{\partial x_j} \frac{\partial}{\partial x_k} f, \frac{\partial}{\partial x_k} \frac{\partial}{\partial x_j} f$  exist and continuous on  $\Omega$ , then

$$\frac{\partial}{\partial x_j} \frac{\partial}{\partial x_k} f = \frac{\partial}{\partial x_k} \frac{\partial}{\partial x_j} f.$$

In particular, this holds for  $f \in C^2(\Omega)$ .

┘

*Proof:* For simplicity we prove for  $n = 2, i = 1, j = 2$ , and the general case is verbatim. For  $\mathbf{x} = (x, y) \in \Omega$ , if  $\Delta x, \Delta y$  is sufficiently small, define

$$I(\Delta x, \Delta y) = \frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0)}{\Delta x \Delta y} - \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta x \Delta y}$$

and

$$g(x) = f(x, y_0 + \Delta y) - f(x, y_0), \quad h(y) = f(x_0 + \Delta x, y) - f(x_0, y).$$

Then by mean value theorem(12.3.3.13),

$$\begin{aligned} I(\Delta x, \Delta y) &= \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x \Delta y} \\ &= \frac{g'(x_0 + \theta_1 \Delta x)}{\Delta y} \\ &= \frac{\frac{\partial}{\partial x} f(x_0 + \theta_1 \Delta x, y_0 + \Delta y) - \frac{\partial}{\partial x} f(x_0 + \theta_1 \Delta x, y_0)}{\Delta y} \\ &= \frac{\partial}{\partial y} \frac{\partial}{\partial x} f(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y) \end{aligned}$$

where  $\theta_1, \theta_2 \in [0, 1]$ , and

$$\begin{aligned} I(\Delta x, \Delta y) &= \frac{h(y_0 + \Delta y) - h(y_0)}{\Delta x \Delta y} \\ &= \frac{h(y_0 + \theta_3 \Delta y)}{\Delta x} \\ &= \frac{\frac{\partial}{\partial y} f(x_0 + \Delta x, y_0 + \theta_3 \Delta y) - \frac{\partial}{\partial x} f(x_0, y_0 + \theta_3 \Delta y)}{\Delta x} \\ &= \frac{\partial}{\partial x} \frac{\partial}{\partial y} f(x_0 + \theta_4 \Delta x, y_0 + \theta_3 \Delta y) \end{aligned}$$

where  $\theta_3, \theta_4 \in [0, 1]$ . Then we get

$$\frac{\partial}{\partial y} \frac{\partial}{\partial x} f(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y) = \frac{\partial}{\partial x} \frac{\partial}{\partial y} f(x_0 + \theta_4 \Delta x, y_0 + \theta_3 \Delta y).$$

Now let  $\Delta x, \Delta y \rightarrow 0$  and use the fact  $\frac{\partial}{\partial x_j} \frac{\partial}{\partial x_k} f, \frac{\partial}{\partial x_k} \frac{\partial}{\partial x_j} f$  are continuous to finish the proof.  $\square$

**Thm. (12.3.3.17) [Taylor's Expansion with Integral Remainder].** Let  $n \in \mathbb{Z}_+, k \in \mathbb{N}, f \in C^{k+1}(\mathbb{R}^n)$ , then for  $\underline{x} \in \mathbb{R}^n$ ,

$$f(\underline{x}) = \sum_{j=0}^k \frac{1}{j!} \left( \left( \sum_i x_i D^i \right)^j f \right)(0) + \int_0^1 \frac{(1-t)^k}{k!} \left( \left( \sum_i x_i D^i \right)^{k+1} f \right)(t\underline{x}) d\underline{x}$$

Equivalently,

$$f(\underline{x}) = \sum_{|\alpha| \leq k} D^\alpha f(0) \frac{x^\alpha}{\alpha!} + (k+1) \sum_{|\alpha|=k+1} \left( \int_0^1 (1-t)^k D^\alpha f(t\underline{x}) dt \right) \frac{x^\alpha}{\alpha!}.$$

This is called the a **Taylor's expansion with integral remainder**.  $\lrcorner$

*Proof:* Consider the function  $\varphi(u) = f(u\underline{x}), u \in \mathbb{R}$ , it is reduced to the  $n = 1$  case. Thus it suffices to prove that

$$f(x) - \sum_{j=0}^k f^{(j)}(0) \frac{x^j}{j!} = \int_0^1 (1-t)^k \frac{x^{k+1}}{k!} f^{(k+1)}(tx) dt = \int_0^x \frac{(x-t)^k}{k!} f^{(k+1)}(t) dt.$$

Use induction on  $k$ : For  $k = 0$ , this is just the fundamental theorem of calculus. In general, it suffices to notice that if  $f \in C^{k+2}(\mathbb{R}^n)$ ,

$$\begin{aligned} \int_0^x \frac{(x-t)^k}{k!} f^{(k+1)}(t) dt &= \left( -\frac{(x-t)^{k+1}}{(k+1)!} f^{(k+1)}(t) \right) \Big|_0^x + \int_0^x \frac{(x-t)^{k+1}}{(k+1)!} f^{(k+2)}(t) dt \\ &= f^{(k+1)}(0) \frac{x^{k+1}}{(k+1)!} + \int_0^x \frac{(x-t)^{k+1}}{(k+1)!} f^{(k+2)}(t) dt. \end{aligned}$$

$\square$

**Cor. (12.3.3.18) [Taylor's Expansion with Lagrangian Remainder].** Let  $n \in \mathbb{Z}_+, k \in \mathbb{N}, f \in C^{k+1}(\mathbb{R}^n)$ , then for  $\underline{x} \in \mathbb{R}^n$ ,

$$f(\underline{x}) = \sum_{j=0}^k \left( \left( \sum_i x_i D^i \right)^j f \right)(0) \frac{x^j}{j!} + \frac{1}{(k+1)!} \left( \left( \sum_i x_i D^i \right)^{k+1} f \right)(\theta \underline{x})$$

for some  $\theta \in (0, 1)$ . This is called a **Taylor's expansion with Lagrangian remainder**.  $\lrcorner$

*Proof:* It follows from (12.3.3.17) and the integral mean value theorem (12.3.3.15) that there exists some  $\theta \in (0, 1)$  s.t.

$$\begin{aligned} f(\underline{x}) - \sum_{|\alpha| \leq k} D^\alpha f(0) \frac{\underline{x}^\alpha}{\alpha!} &= \int_0^1 \frac{(1-t)^k}{k!} \left( \left( \sum_i x_i D^i \right)^{k+1} f \right) (t\underline{x}) dt \\ &= \left( \left( \sum_i x_i D^i \right)^{k+1} f \right) (\theta \underline{x}) \int_0^1 \frac{(1-t)^k}{k!} dt \\ &= \frac{1}{(k+1)!} \left( \left( \sum_i x_i D^i \right)^{k+1} f \right) (\theta \underline{x}) \end{aligned}$$

□

**Cor. (12.3.3.19) [Taylor's Expansion with Peano Remainder].** Let  $n \in \mathbb{Z}_+$ ,  $k \in \mathbb{N}$ ,  $f \in C^{k+1}(\mathbb{R}^n)$ , then for  $\underline{x} \in \mathbb{R}^n$ ,

$$f(\underline{x}) = \sum_{j=0}^k \left( \left( \sum_i x_i D^i \right)^j f \right) (0) \frac{\underline{x}^j}{j!} + o(|\underline{x}|^k)$$

for some  $\theta \in (0, 1)$ . This is called a **Taylor's expansion with Peano remainder**. ┘

*Proof:* This follows from (12.3.3.18) and the continuity (thus local boundedness) of the derivatives of  $f$ . □

**Prop. (12.3.3.20) [Derivatives of Reciprocals].** For any  $k \in \mathbb{N}$ ,

$$\left( \frac{1}{f} \right)^{(k)} = \frac{1}{f} \sum_{a_1, a_2, \dots \in \mathbb{N}, a_1 + 2a_2 + \dots = k} \binom{a_1 + a_2 + \dots}{a_1, a_2, \dots} \left( \frac{-f'}{1!f} \right)^{a_1} \cdot \left( \frac{-f^{(2)}}{2!f} \right)^{a_2} \cdots$$

┘

*Proof:* ? □

**Prop. (12.3.3.21) [Derivatives of Composite Functions, Faà di Bruno].** For  $m \in \mathbb{Z}_+$  and  $g, f \in C^m(\mathbb{R})$ ,

$$\begin{aligned} (g \circ f)^{(m)}(t) &= \sum_{b_1 + 2b_2 + \dots + mb_m = m} \frac{m!}{b_1! \dots b_m!} g^{(b_1 + \dots + b_m)}(f(t)) \left( \frac{f'(t)}{1!} \right)^{b_1} \left( \frac{f^{(2)}(t)}{2!} \right)^{b_2} \cdots \left( \frac{f^{(m)}(t)}{m!} \right)^{b_m} \\ &= \sum g^{(k)}(f(t)) \left( \frac{f'(t)}{1!} \right)^{b_1} \left( \frac{f^{(2)}(t)}{2!} \right)^{b_2} \cdots \left( \frac{f^{(m)}(t)}{m!} \right)^{b_m} \end{aligned}$$

where in the last expression, the summation is over all partitions of  $[m]_+$ , and for each partition,  $k$  is the number of blocks, and  $b_i$  is the number of blocks with exactly  $i$  elements. ┘

*Proof:* Firstly it is easy to prove by induction that the expression of  $(g \circ f)^{(m)}$  is of the form

$$(g \circ f)^{(m)}(t) = \sum_{k \in \mathbb{Z}_+} g^{(k)}(f(t)) B_k(f'(t), \dots, f^{(m)}(t))$$

where  $B_k$  are polynomials. If we take  $g(t) = e^{pt}$  and  $f$  analytic, then we can express

$$(g \circ f)(x+h) = e^{pf(x+h)} = e^{pf(x)} e^{pf'(x)h} e^{pf^{(2)}(x)\frac{h^2}{2!}} \cdots$$

$$= \sum_{m \in \mathbb{N}} h^m \sum_{k \in \mathbb{N}} p^k e^{pf(x)} \sum \frac{1}{(b_1)! \dots (b_m)!} \left( \frac{f^{(1)}(t)}{1!} \right)^{b_1} \left( \frac{f^{(2)}(t)}{2!} \right)^{b_2} \dots \left( \frac{f^{(m)}(t)}{m!} \right)^{b_m}$$

for any  $h$ , where the summation is over all  $b_1, \dots, b_m \in \mathbb{N}$  s.t.

$$b_1 + \dots + b_m = k, \quad b_1 + 2b_2 + \dots + mb_m = m.$$

Now by Taylor's formula, we get

$$(g \circ f)^{(m)}(x) = \sum_{k \in \mathbb{N}} p^k e^{pf(x)} \sum \frac{m!}{(b_1)! \dots (b_m)!} \left( \frac{f^{(1)}(t)}{1!} \right)^{b_1} \left( \frac{f^{(2)}(t)}{2!} \right)^{b_2} \dots \left( \frac{f^{(m)}(t)}{m!} \right)^{b_m}.$$

And because  $p$  is arbitrary and the Taylor expansion of  $f$  at  $x$  is arbitrary, the desired formula follows.  $\square$

### Functions of Bounded Variation

**Prop. (12.3.3.22) [Bounded Variations].** For a function  $F : \mathbb{R} \rightarrow \mathbb{C}$ , define the **total variation function** of  $F$ :

$$T_F : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0} : T_F(x) = \sup \left\{ \sum_{j=1}^n |F(x_j) - F(x_{j-1})| : n \in \mathbb{N}, -\infty < x_0 < x_1 < \dots < x_n = x \right\}.$$

Then  $T_F$  is a non-decreasing function, and for  $a < b \in \mathbb{R}$ ,

$$T_F(b) - T_F(a) = \sup \left\{ \sum_{j=1}^n |F(x_j) - F(x_{j-1})| : n \in \mathbb{N}, a = x_0 < x_1 < \dots < x_n = b \right\}.$$

Then  $F$  is called a **function of bounded variation** if  $T_F(\infty) = \lim_{x \rightarrow \infty} T_F(x) < \infty$ . The space of functions of bounded variation is denoted by  $BV(\mathbb{R})$ .

Similarly we can define the totally variation function for a function  $F : [a, b] \rightarrow \mathbb{C}$ , where  $a < b \in \mathbb{R}$ , by regarding it as a function on  $\mathbb{R}$  s.t.  $F(x) = F(a)$  if  $x < a$  and  $F(x) = F(b)$  if  $x > b$ . So we can define  $BV([a, b])$ .  $\lrcorner$

**Prop. (12.3.3.23) [Properties of Bounded Variations].**

- $F \in BV(\mathbb{R})$  iff  $\operatorname{Re} F, \operatorname{Im} F \in BV(\mathbb{R})$ .
- For  $F : \mathbb{R} \rightarrow \mathbb{R}$ ,  $F \in BV$  iff  $F$  is the difference of two bounded non-decreasing functions. And if this is the case, then these functions can be chosen to be  $\frac{1}{2}(T_F + F)$  and  $\frac{1}{2}(T_F - F)$ , which is called the **Jordan decomposition** of  $F$ .
- If  $F \in BV(\mathbb{R})$ , then  $F(x-), F(x+)$  exists for any  $x \in \mathbb{R}$ . And also  $F(\pm\infty)$  exist.
- If  $F \in BV(\mathbb{R})$ , then  $F$  has at most countably many discontinuous points.
- If  $F \in BV(\mathbb{R})$  and  $G(x) = F(x+)$ , then  $F^\wedge$  and  $G^\wedge$  exist, and they are equal.

$\lrcorner$

*Proof:* Cf. [Folland]P103.  $\square$

**Def. (12.3.3.24) [Normalized Bounded Variations].** The space NBV of **normalized bounded variation functions** consists of functions  $F \in BV(\mathbb{R})$  (12.3.3.22) s.t.  $F(-\infty) = 0$  and  $F$  is right-continuous.  $\lrcorner$



**Prop. (12.3.3.25)[Bounded Variations and Measures].** If  $\mu$  is a complex Borel measure on  $\mathbb{R}$ , and  $F(x) = \mu((-\infty, x])$  for any  $x \in \mathbb{R}$ , then  $F \in \text{NBV}$ . Conversely, if  $F \in \text{NBV}$ , then there exists a unique complex Borel measure  $\mu_F$  on  $\mathbb{R}$  s.t.  $F(x) = \mu_F((-\infty, x])$ . Moreover,  $|\mu_F| = \mu_{T_F}$ .  $\lrcorner$

*Proof:* Cf.[Folland]P104.  $\square$

**Def. (12.3.3.26) [Lebesgue-Stieltjes Integration].** For  $F \in \text{NBV}$  and any function  $G : X \rightarrow \mathbb{C}$ , denote

$$\int g dF \triangleq \int g d\mu_F \text{ (12.3.3.25)}$$

when it is integrable, which is called the **Lebesgue-Stieltjes integral**.  $\lrcorner$

**Thm. (12.3.3.27)[Integration by Parts].** If  $F, G \in \text{NBV}$  and at least one of them is continuous, then for  $a < b \in \mathbb{R}$ ,

$$\int_{(a,b]} F dG + \int_{(a,b]} G dF = F(b)G(b) - F(a)G(a).$$

*Proof:* Cf.[Fol99]P107.  $\square$

**Def. (12.3.3.28) [Absolute Continuity].** For  $a < b \in [-\infty, \infty]$ ,  $f \in C([a, b])$  is called **absolutely continuous** on  $[a, b]$  if for any  $\varepsilon > 0$ , there exists  $\delta < 0$  s.t. for any f.m. disjoint intervals  $(a_i, b_i), i \leq N$ ,

$$\sum_{i=1}^N (b_i - a_i) < \delta \implies \sum_{i=1}^N |F(b_i) - F(a_i)| < \varepsilon.$$

**Prop. (12.3.3.29).** For  $F \in \text{NBV}$ ,  $F$  is absolutely continuous iff  $\mu_F \ll m$  (12.3.3.25).  $\lrcorner$

*Proof:* Cf.[Folland]P105.  $\square$

**Prop. (12.3.3.30).** For  $a < b \in [-\infty, \infty]$ ,  $f \in C([a, b])$  is differentiable and  $F^\Delta$  is bounded, then  $F$  is absolutely continuous.  $\lrcorner$

*Proof:* This follows from the fact  $|F(b_i) - F(a_i)| \leq (\sup |F^\Delta|)|b_i - a_i|$  by mean value theorem (12.3.3.13).  $\square$

**Thm. (12.3.3.31)[Fundamental Theorem of Calculus for Lebesgue Measure].** If  $a < b \in \mathbb{R}$  and  $F \in C([a, b])$ , then the following are equivalent:

- $F$  is absolutely continuous on  $[a, b]$ .
- $F(x) - F(a) = \int_a^x f(t) dt$  for some  $f \in L^1([a, b], m)$ .
- $F$  is differentiable a.e. on  $[a, b]$ ,  $F^\Delta \in L^1([a, b], m)$ , and  $F(x) - F(a) = \int_a^x F^\Delta(t) dt$ .

In particular, this is the case for  $F$  everywhere differentiable and  $F^\Delta$  bounded, by (12.3.3.30).  $\lrcorner$

*Proof:* Cf.[Fol99]P106.  $\square$

## 4 Smooth Functions

**Prop. (12.3.4.1).** For  $n \in \mathbb{Z}_+$  and  $f \in C^\infty(\mathbb{R}^n)$ , if  $f(0) = 0$ , then there exists  $g_1, \dots, g_n \in C^\infty(\mathbb{R}^n)$  s.t.

$$f(\mathbf{x}) = x_1 g_1(\mathbf{x}) + \dots + x_n g_n(\mathbf{x}).$$

┘

*Proof:*

$$f(\mathbf{x}) = \int_0^1 f(t\mathbf{x}) dt = \int_0^1 \frac{\partial}{\partial x_i} f(t\mathbf{x}) x_i dt.$$

So we can take  $g_i(\mathbf{x}) = \int_0^1 \frac{\partial}{\partial x_i} f(t\mathbf{x}) dt$ . □

**Thm. (12.3.4.2) [Implicit Function Theorem].** ┘

## 5 Series

**Lemma (12.3.5.1).** For a sequence  $(a_n)_{n \in \mathbb{N}}$  in  $\mathbb{R}_+$ ,

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} \leq \lim_{n \rightarrow \infty} a_n^{1/n} \leq \overline{\lim}_{n \rightarrow \infty} a_n^{1/n} \leq \overline{\lim}_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$$

┘

*Proof:* Let  $\lim_{n \rightarrow \infty} a_n^{1/n} = R$ . If  $R > 0$ , then by definition, for any  $\varepsilon > 0$  and  $M \in \mathbb{Z}_+$ , there exists some  $m \geq M$  s.t.  $a_m^{1/m} < R + \varepsilon$ . Then for such  $M$ ,  $a_m/a_1 \leq (R + \varepsilon)^m/a_1$ . So for some  $k \leq m$ ,  $a_k/a_{k-1} \leq ((R + \varepsilon)^m/a_1)^{m-1}$ . Then if  $M$  is very large,  $a_k/a_{k-1} < R + 2\varepsilon$ . Thus  $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} \leq R$ .

Then  $\overline{\lim}_{n \rightarrow \infty} a_n^{1/n} \leq \overline{\lim}_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$  follows from this by considering the sequence  $b_n = a_n^{-1}$ . The other inequality is trivial. □

**Prop. (12.3.5.2).** If  $(a_n)_{n \geq n_0, n \in \mathbb{N}}$  is a series of real numbers and  $C \in \mathbb{R}$  s.t.  $a_{m+n} \geq a_m + a_n + C$  for any  $m, n \geq n_0 \in \mathbb{N}$ , then  $a_m/m$  converges to  $\overline{\lim} a_n/n \in (-\infty, \infty]$ .

Notice by taking  $b_n = -a_n$ , the  $\leq$  case is also true. In particular, if  $|a_{m+n} - a_m - a_n| \leq C$  for any  $m, n \geq n_0 \in \mathbb{N}$ , then  $a_m/m$  converges in  $\mathbb{R}$ . ┘

*Proof:* Let  $\lambda = \overline{\lim} a_n/n$ . By induction, for any  $N \geq n_0 \in \mathbb{N}$ ,  $a_{2^k N} \geq 2^k a_N + (2^k - 1)C$  for any  $k \in \mathbb{N}$ , so  $a_{2^k N}/2^k N \geq a_N/N + \min\{C, 0\}$ , so  $\lambda > -\infty$ .

If  $\lambda = +\infty$ , then for any  $M \in \mathbb{R}$ , there exists  $n > |C|/\varepsilon$  s.t.  $a_n/n \geq M$ . For any  $N$  large,  $N = kn + m$  for some  $k, m$  and  $n_0 \leq m < n_0 + n$ , so  $a_N \geq ka_n + a_m + (k+1)C$ , so  $\underline{\lim} a_N/N \geq a_n/n - \varepsilon \geq M - \varepsilon$  for  $N$  large, so  $\underline{\lim} a_n/n = \overline{\lim} a_n/n$ .

If  $\lambda < \infty$ , for any  $\varepsilon > 0$ , there exists  $n > |C|/\varepsilon$  s.t.  $\lambda - a_n/n < \varepsilon$ . For any  $N$  large,  $N = kn + m$  for some  $k, m$  and  $n_0 \leq m < n_0 + n$ , so  $a_N \geq ka_n + a_m + (k+1)C$ , so  $\underline{\lim} a_N/N \geq a_n/n \geq \lambda - 2\varepsilon$  for  $N$  large, so  $\underline{\lim} a_n/n = \overline{\lim} a_n/n$ . □

## Power Series

**Prop. (12.3.5.3) [Cauchy-Hadamard].** For any power series  $a_0 + a_1 X + \dots + a_n X^n + \dots \in \mathbb{R}[[X]]$ , take  $1/R = \overline{\lim} |a_n|^{1/n}$ , where we assume  $1/0 = \infty$  and  $1/\infty = 0$ , then

- The series converges absolutely for every  $|x| < R$ , and if  $\rho < R$ , then the convergence is uniform for  $|x| \leq \rho$ . In particular, the series converges to a continuous function in  $\mathbb{D}(0, R)$  by (4.4.8.21).

- If  $|x| > R$ , the terms are unbounded, and the series diverges.

And  $R$  is called the **radius of convergence** of the sequence.  $\lrcorner$

*Proof:* 1: For  $0 < \rho < R$ , take  $\rho_1 \in (\rho, R)$ , then the definition of  $R$  implies that for some  $N \in \mathbb{Z}_+$  and any  $n \geq N$ ,  $|a_n| \leq R^{-n} < \rho_1^{-n}$ , so  $|a_n x^n| \leq (\rho/\rho_1)^n$ , so  $\sum_{n \in \mathbb{N}} a_n x^n$  converges absolutely and uniformly for  $|x| \leq \rho$ .

2: For  $R < |x|$ , the definition of  $R$  implies that for any  $N \in \mathbb{Z}_+$ , there exists  $n \geq N$  s.t.  $|a_n| \geq R^{-n}$ , so  $|a_n x^n| \geq |x/R|^n > 1$ , so this sequence cannot be convergent.  $\square$

**Prop. (12.3.5.4).** For any power series  $a_0 + a_1 x + \dots + a_n x^n + \dots$  in  $\mathbb{R}$ , by (12.3.5.1), if  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho \in [0, \infty]$ , then the radius of convergence (12.4.3.1)  $R = 1/\rho$ , where we assume  $1/0 = \infty$  and  $1/\infty = 0$ .  $\lrcorner$

### Abel Summation

**Prop. (12.3.5.5) [Abel Summation].** For  $(a_n)_{n \in \mathbb{Z}_+}, (b_n)_{n \in \mathbb{Z}_+} \in \mathbb{C}^{\mathbb{Z}_+}$  and  $N \in \mathbb{N}, k \in \mathbb{Z}_+$ , if we denote  $A(n) = \sum_{N < m \leq n} a_m$ , then

$$\sum_{N < n \leq N+k} a_n b_n = A_{N+k} b_{N+k+1} + \sum_{N < n \leq N+k} A_n (b_n - b_{n+1}).$$

$\lrcorner$

**Cor. (12.3.5.6) [Abel's Criterion of Convergence].** If  $(a_n)_{n \in \mathbb{N}} \in \mathbb{C}^{\mathbb{N}}$  is a sequence and  $(b_n) \in \mathbb{R}_{\geq 0}^{\mathbb{N}}$  is a non-increasing sequence, and there exists  $A \in \mathbb{R}_+$  s.t.

$$\lim_{n \rightarrow \infty} b_n = 0, \quad \sup_{N \in \mathbb{N}} \left| \sum_{n \in [N]} a_n \right| \leq A,$$

then the series  $\sum_{n \in \mathbb{N}} a_n b_n$  converges, and for each  $N \in \mathbb{N}$ , we have

$$\left| \sum_{n > N} a_n b_n \right| \leq 2A b_{N+1}.$$

$\lrcorner$

**Cor. (12.3.5.7) [Abel Integrations].** If  $(a_n)_{n \in \mathbb{Z}_+} \in \mathbb{C}^{\mathbb{Z}_+}$  and we denote  $A(X) = \sum_{n \in [X]_+} a_n$ , then for any  $X \in \mathbb{R}_{\geq 1}$  and  $b(X) \in C^1([1, X])$ , we have

$$\sum_{n \in [X]_+} a_n b(n) = A(X) b(X) - \int_1^X A(t) b'(t) dt.$$

$\lrcorner$

### Double Series

**Def. (12.3.5.8) [Pringsheim Convergence, Pringsheim [Pri00]].** A double sequence  $x = (x_{ij})_{i,j \in \mathbb{Z}_+} \in \mathbb{C}^{\mathbb{Z}_+ \times \mathbb{Z}_+}$  is said to be **convergent in Pringsheim's sense** if there exists  $L \in \mathbb{C}$  s.t. for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\mathbb{N} \in \mathbb{Z}_+$  s.t.  $|x_{jk} - L| \leq \varepsilon$  for any  $j, k \in \mathbb{Z}_{\geq \mathbb{N}}$ .

And it is said to be a **Cauchy sequence in Pringsheim's sense** if for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\mathbb{N} \in \mathbb{Z}_+$  s.t.  $|x_{jk} - x_{pq}| \leq \varepsilon$  for any  $p, q, j, k \in \mathbb{Z}_{\geq \mathbb{N}}$ .  $\lrcorner$

### Statistical Convergence

References are [Fri85].

**Def. (12.3.5.9) [Statistical Convergent Sequences].** A sequence  $x = (x_i)_{i \in \mathbb{Z}_+} \in \mathbb{C}^{\mathbb{Z}_+}$  is called a **statistically convergent** if there exists  $L \in \mathbb{C}$  s.t. for any  $\varepsilon \in \mathbb{R}_+$ , we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \# \{k \in [n]_+ : |x_k - L| \leq \varepsilon\} = 0.$$

And it is said to be a **statistically Cauchy sequence** if for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $N \in \mathbb{Z}_+$  sut

$$\lim_{n \rightarrow \infty} \frac{1}{n} \# \{k \in [n]_+ : |x_k - x_N| \leq \varepsilon\} = 0.$$

┘

**Def. (12.3.5.10) [Statistical Convergent Double Sequences].** A double sequence  $x = (x_{i,j})_{i,j \in \mathbb{Z}_+} \in \mathbb{C}^{\mathbb{Z}_+ \times \mathbb{Z}_+}$  is called a **statistical convergent double sequence** if there exists  $L \in \mathbb{C}$  s.t. for any  $\varepsilon \in \mathbb{R}_+$ , the set

$$\{(j, k) \in \mathbb{Z}_+ \times \mathbb{Z}_+ : |x_{jk} - L| \geq \varepsilon\}$$

has natural double density 0.

A double sequence  $x = (x_{i,j})_{i,j \in \mathbb{Z}_+} \in \mathbb{C}^{\mathbb{Z}_+ \times \mathbb{Z}_+}$  is called a **statistical Cauchy double sequence** if for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $M, N \in \mathbb{Z}_+$  s.t.

$$\{(j, k) : |x_{jk} - x_{MN}| \geq \varepsilon\}$$

has natural double density 0.

Then a complex double sequence  $x = (x_{i,j})_{i,j \in \mathbb{Z}_+}$  is statistical Cauchy iff it is statistically convergent. ┘

*Proof:* ?

□

**Prop. (12.3.5.11).** Convergent double sequences are statistically convergent, but not conversely. ┘

*Proof:* Cf. [M-O03].

□

**Prop. (12.3.5.12).** The set of bounded statistically convergent double sequences is a closed subspace of the space of bounded sequences, but it is nowhere dense. ┘

### Cesáro-Summable(convergent) Sequences

**Def. (12.3.5.13) [Cesáro-Summable Sequences].** A sequence  $x = (x_i)_{i \in \mathbb{Z}_+}$  is called **Cesáro-summable** if the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i$$

exists.

A double sequence  $x = (x_{i,j})_{i,j \in \mathbb{Z}_+}$  is called **Cesáro-summable** if the limit

$$\lim_{n,m \rightarrow \infty} \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n x_{i,j}$$

exists. ┘

**Prop. (12.3.5.14).** Let  $x$  be a double sequence and  $p \in \mathbb{R}_+$ , then

- If  $x$  is bounded, then  $x$  is strongly Ceáro- $p$ -summable iff it is strongly Ceáro-summable.
- If  $x$  is bounded and convergent in Pringsheim's sense (12.3.5.8), then it is strongly Ceáro- $p$ -summable.
- If  $x$  is strongly Ceáro- $p$ -summable, then it is statistically convergent. And the converse is true if  $x$  is bounded.

┘

**Cor. (12.3.5.15).** If  $x$  is a sequence or a double sequence that is bounded and statistically convergent (12.3.5.10), then it is also Cesáro-summable (12.3.5.13).

┘

### Borel-Summable Series

**Prop. (12.3.5.16) [Borel Summation].** Suppose  $\sum_{n \in \mathbb{N}} a_n x^n$  is a power series with zero radius of convergent, and suppose that the **exponential generating function**

$$g(x) = \sum_{n \in \mathbb{N}} a_n \frac{x^n}{n!}$$

is convergent on some nbhd of 0, and analytically extends to a smooth function on  $[0, \infty)$ , and satisfies that  $g(x) = O(x^N)$ .

Then we can define a **Borel summation** of this power series as

$$I(x) = \int_0^\infty g(xu) e^{-u} du = x^{-1} \int_0^\infty g(u) e^{-\frac{u}{x}} du = x^{-1} (\mathcal{L}g)(x^{-1}) \quad (12.5.4.4).$$

Then  $I(x)$  is convergent on  $[0, \frac{1}{N})$ , and has the Taylor expansion  $\sum_{n \in \mathbb{N}} a_n x^n$  at the origin.

┘

*Proof:* Take derivatives, and notice that for  $n \in \mathbb{N}$ ,  $\int_0^\infty x^n e^{-x} dx = n!$ .

□

**Example (12.3.5.17).** Consider the power series

$$\sum_{n \in \mathbb{N}} (-1)^n n! x^n,$$

then  $g(x) = \sum_{n \in \mathbb{N}} (-1)^n x^n = \frac{1}{1+x}$ , so

$$I(x) = \int_0^\infty \frac{e^{-u}}{1+xu} du = x^{-1} e^{1/x} E_1\left(\frac{1}{x}\right),$$

where  $E_1(x) = \int_x^\infty \frac{e^{-u}}{u} du$  is the integral exponential.

┘

## 6 $L^p$ -spaces

**Lemma (12.3.6.1) [Hölder].** if  $\sum x_i = 1, x_i \geq 0$ , then for any  $a_i \geq 0$

$$\prod a_i^{x_i} \leq \sum a_i x_i.$$

┘

*Proof:* ?

□

**Thm. (12.3.6.2) [Hölder-Cauchy-Bunyakovsky-Schwarz Inequality].** Let  $X$  be a measure space, and  $1 \leq p, q \leq \infty$  satisfies  $1/p + 1/q = 1$ , then

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

More generally, if  $\sum_{i=1}^n 1/p_i = 1$ , then

$$\left\| \prod f_i \right\|_1 \leq \prod \|f_i\|_{p_i}.$$

┘

*Proof:* The both sides are homogenous for  $f_i$ , so we may assume  $\|f_i\|_{p_i} = 1$ , then use Hölder's Lemma(12.3.6.1) for  $x_i = q/p_i$ .  $\square$

**Cor. (12.3.6.3) [Generalized Cauchy-Schwartz Inequality].** Let  $f : I \rightarrow J \in \text{finSet}$  and  $\{a_j\}_{j \in J} \subset \mathbb{C}$ , then

$$\sum_{i \in I} |a_{f(i)}| \leq \left( \sum_{j \in J} |a_j|^2 \right)^{1/2} \left( \sum_{i \in I} \#f^{-1}(f(i)) \right)^{1/2}.$$

┘

**Prop. (12.3.6.4) [Dual of  $L^p(\mu)$ ].** For a  $\sigma$ -finite measurable space  $(X, \mathcal{M}, \mu)$ , for  $1 \leq p < \infty$

$$L^p(X, \Omega, \mu)^* = L^q(X, \Omega, \mu).$$

┘

*Proof:* Firstly, Hölder inequality(12.3.6.2) shows that a  $g \in L^q(X, \Omega, \mu)$  defines a functional by  $f \mapsto \int f g d\mu$ . Conversely, if given a functional  $F$ , define a measure  $v(E) = F(\chi_E)$  for all measurable set  $E \in \Omega$ . It is countably additive: first it is finitely additive, and if  $E_n$  is a descending sequence of measurable sets that  $\cap E_n = \emptyset$ , then

$$v(E_n) \leq \|F\| \|\chi_{E_n}\|_{L^p} = \|F\| \mu(E_n)^{1/p} \rightarrow 0.$$

(where we used the fact  $p < \infty$ ). And it is clearly absolutely continuous w.r.t.  $\mu$ .

So by Radon-Nikodym(12.3.3.4), there is a measurable function  $g$  that  $v(E) = \int_E g d\mu$ . So for all simple function  $f$ ,  $F(f) = \int f g d\mu$ . Next we want to prove  $\|g\|_q \leq \|F\|$ , because any measurable function  $f$  can be approximated by simple functions  $f_i$  in  $L^p$  norm(12.3.8.6), so

$$\left| \int (f(x) - f_i(x)) g(x) d\mu \right| \leq \|f - f_i\|_p \cdot \|g\|_q \leq \|f - f_i\|_p \|g\|_q.$$

So  $F(f) = \lim F(f_i) = \lim \int f_i g d\mu = \int f g d\mu$ .

To prove this, if  $1 < p$ , then let  $E_t = \{x : |g(x)| \leq t\}$ , and  $f = \chi_{E_t} |g|^{q-2} g$ , then

$$\int_{E_t} |g|^q d\mu = \int f g d\mu = F(f) \leq \|F\| \|f\|_{L^p} = \|F\| \left( \int_{E_t} |g|^q d\mu \right)^{1/p}$$

which is equivalent to  $\|g \chi_{E_t}\|_{L^q} \leq \|F\|$ . Let  $t \rightarrow \infty$ , then the monotone convergence theorem(12.3.2.5) gives us the result.

If  $p = 1$ , then  $q = \infty$ . For any  $\varepsilon > 0$ , let  $A = \{x : |g(x)| > \|F\| + \varepsilon\}$ ,  $E_t = \{x : |g(x)| \leq t\}$ , and let  $f = \chi_{E_t \cap A} \text{sgn}(g)$ , then  $\|f\|_{L^1} = \mu(E_t \cap A)$ , and

$$\mu(E_t \cap A) (\|F\| + \varepsilon) \leq \int_{A \cap E_t} |g| d\mu = \int f g d\mu \leq \|F\| \mu(E_t \cap A)$$

If  $\mu(A) \neq 0$ , then let  $t \rightarrow \infty$ , this is a contradiction. So  $\|g\|_\infty \leq \|F\|$ .  $\square$

**Prop. (12.3.6.5) [Hilbert Basis for Products].** For two  $\sigma$ -finite measure spaces  $M, N$  and Hilbert basis (12.7.4.8)  $\{e_i\}$  of  $L^2(M)$  and  $\{f_j\}$  of  $L^2(N)$ ,  $\{e_i \otimes f_j\}$  gives a Hilbert basis for  $L^2(M \times N)$ . (Use Fubini).  $\lrcorner$

*Proof:* It is easily verified that  $e_i \otimes f_j$  are mutually orthogonal, and if some  $f \in L^2(M \times N)$  satisfies  $(f, e_i \otimes f_j) = 0$  for all  $i, j$ , then

$$\int_M e_i(x) \int_N f(x, y) f_j(y) dy = 0$$

for all  $i$ . But  $\int_N f(x, y) f_j(y) dy \in L^2(M)$  for a.e.  $x$  ( $f(x, y) \in L^2(N)$  a.e.  $x$  by Fubini-Tonelli), thus it vanishes. So  $\int_N f(x, y) f_j(y) dy = 0 \in L^2(M)$  for a.e.  $x$ , so by Fubini-Tonelli again,  $\|f\|_{L^2(M \times N)} = 0$ , thus  $f = 0$ .  $\square$

**Prop. (12.3.6.6) [Minkowski's Inequality].** Let  $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$  be  $\sigma$ -finite measure spaces, and  $f$  a  $(\mathcal{M} \otimes \mathcal{N})$ -measurable function on  $X \times Y$ .

- If  $f \geq 0$  and  $1 \leq p < \infty$ , then

$$\left[ \int \left( \int f(x, y) d\nu(y) \right)^p d\mu(x) \right]^{1/p} \leq \int \left[ \int f(x, y)^p d\mu(x) \right]^{1/p} d\nu(y).$$

- If  $1 \leq p \leq \infty$ ,  $f(\cdot, y) \in L^p(\mu)$  for a.e.  $y$ , and the function  $y \mapsto \|f(\cdot, y)\|_p$  is in  $L^1(\nu)$ , then  $f(x, \cdot) \in L^1(\nu)$  for a.e.  $x$ , and the function  $x \mapsto \int f(x, y) d\nu(y)$  is in  $L^p(\mu)$ , and

$$\left\| \int f(\cdot, y) d\nu(y) \right\|_p \leq \int \|f(\cdot, y)\|_p d\nu(y).$$

$\lrcorner$

*Proof:* 1: If  $p = 1$ , then this is just Tonelli's theorem (12.3.2.9), and when  $1 < p < \infty$ , let  $q^{-1} + p^{-1} = 1$ , and  $g \in L^q(\mu)$ , then by Tonelli's theorem and Holder's inequality (12.3.6.2),

$$\int \left[ \int f(x, y) d\nu(y) \right] |g(x)| d\mu(x) = \int \int f(x, y) |g(x)| d\mu(x) d\nu(y) \leq \|g\|_q \int [f(x, y)^p d\nu(y)]^{1/p} d\mu(x).$$

So we finish by (12.3.6.4).

2 follows from item 1 and Fubini's theorem. And when  $p = \infty$ , this is trivial.  $\square$

### $L^2$ -Space

**Def. (12.3.6.7).** Let  $X$  be a measure space, then there is a map

$$\star : L^2(X \times X) \times L^2(X \times X) \rightarrow L^2(X \times X) : (K_1 \star K_2)(u, v) = \int_X K_1(u, x) K_2(x, v) dx.$$

By Schwarz inequality,  $\|K_1 \star K_2\|_2 \leq \|K_1\|_2 \star \|K_2\|_2$ , thus  $\star$  makes  $L^2(G \times G)$  into a Banach algebra (without a unit).

This Banach algebra can left and right act on  $L^2(X)$ , also denoted by  $\star$ , then for  $K \in L^2(X \times X)$ ,  $P, Q \in L^2(X)$ ,

$$(P \otimes Q) \star K = P \otimes (Q \star K), \quad K \star (P \otimes Q) = (K \star P) \otimes Q.$$

And also for  $S, T \in L^2(X)$ ,

$$(P \otimes Q) \star (\bar{S} \otimes T) = (Q, S)_{L^2} P \otimes T.$$

$\lrcorner$

**Prop. (12.3.6.8).**

- An element  $K \in L^2(X \times X)$  has the form  $P \otimes Q$  iff  $K \star K' \star K$  is proportional to  $K$  for all  $K' \in L^2(X \times X)$ .
- Let  $K_1 = P_1 \otimes Q_1$  and  $K_2 = P_2 \otimes Q_2$ , then  $P_1$  and  $P_2$  are proportional iff  $K_1 \star K$  and  $K_2 \star K$  are proportional for all pure tensors  $K \in L^2(X \times X)$ . Similarly,  $Q_1$  and  $Q_2$  are proportional iff  $K \star K_1$  and  $K \star K_2$  are proportional for all pure tensor  $K$ .
- For any uniform transformation  $s : L^2(X \times X) \rightarrow L^2(X \times X)$  respecting  $\star$ , there exists a unitary transformation  $s_0 : L^2(X) \rightarrow L^2(X)$  s.t.  $s(P \otimes \overline{Q}) = s_0(P) \otimes \overline{s_0(Q)}$ . And it can be chosen to be invertible iff  $s$  is.

┘

*Proof:* Cf.[Bump, P527].?

□

 **$L^\infty$ -Spaces**

**Def. (12.3.6.9) [Slowly Oscillating Functions].** A **slowly oscillating function** on  $\mathbb{R}^n$  is a function  $f \in L^\infty(\mathbb{R}^n)$  s.t. for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $A, \delta \in \mathbb{R}_+$  s.t.  $|f(x) - f(y)| < \varepsilon$  whenever  $|x| > A, |y| > A, |x - y| < \delta$ .

┘

**7 Estimations**

**Prop. (12.3.7.1).**  $\lim_{n \rightarrow \infty} n^{1/n} = 1$ .

┘

*Proof:* Let  $n^{1/n} = 1 + \delta_n, \delta_n > 0$ , then  $n = (1 + \delta_n)^n \geq 1 + \frac{n(n-1)}{2} \delta_n^2$ , so  $\delta_n \leq \sqrt{2/n}$ , and  $\lim_{n \rightarrow \infty} \delta_n = 0$ .

□

**Prop. (12.3.7.2).** For any  $x \in (0, \frac{\pi}{2})$ ,  $\frac{2}{\pi}x < \sin x < x$ .

┘

*Proof:* it suffices to show that  $\frac{\sin x}{x}$  is decreasing on  $x \in (0, \frac{\pi}{2})$ :

$$\left(\frac{\sin x}{x}\right)' = \frac{x \cos x - \sin x}{x^2},$$

and  $x \cos x - \sin x < 0$  because

$$(x \cos x - \sin x)' = -x \sin x < 0.$$

□

**8 Approximations**

**Notation (12.3.8.1).**

- Let  $X \in \text{LCH}$ .
- Let  $\mu$  be a Radon measure on  $X$  (12.3.1.22).

┘

**Prop. (12.3.8.2) [Stone-Weierstrass Approximation].** If  $X$  is compact and  $A$  is a unital  $*$ -subalgebra of  $C(X)$  that separates points, then  $A$  is dense in  $C(X)$ .

┘

*Proof:* This is a consequence of Bishop theorem (12.8.3.18) because in this case the real functions in  $A$  separate points, so all  $A$ -antisymmetric sets consists of one point.

□



**Cor. (12.3.8.3).** The set of Polynomial functions are dense in  $C([0, 1]^n)$  for any  $n \in \mathbb{Z}_+$ . ┘

**Cor. (12.3.8.4).** The set of trigonometric polynomials are dense in  $\mathbb{C}([0, 1]^n)$  for any  $n \in \mathbb{Z}_+$ . ┘

**Prop. (12.3.8.5) [Simple Function Approximation].** Let  $E$  be a measure space,

- If  $f(x)$  is a non-negative measurable function on  $E$ , then there is an ascending sequence of simple functions  $(\varphi_n(x))$  that converges to  $f$  point-wise.
- If  $f(x)$  is a measurable function on  $E$ , then there is a sequence of simple functions  $\varphi_n$  that  $|\varphi_k(x)| \leq |f(x)|$ , and converges to  $f$  pointwise.
- If  $f(x)$  is bounded, then the convergence can be chosen to be uniform.

┘

*Proof:* Cf. [实变函数周明强 P110]. □

### $L^p$ -Approximation

**Prop. (12.3.8.6) [Simple Function Approximation].** Any function in  $L^p$  can be approximated by compactly supported simple functions in  $L^p$  norm. ┘

*Proof:* □

**Prop. (12.3.8.7) [Approximation by Compactly Supported Functions].** for  $1 \leq p < +\infty$ ,  $C_c(X)$  are dense in  $L^p(X, \mu)$ , but not for  $p = \infty$ . ┘

*Proof:* Use compactly supported simple function approximation (12.3.8.6) and then use outer regular approximation (12.3.1.22) and then Tietz extension. □

**Prop. (12.3.8.8) [Lusin].** If  $f$  is almost everywhere finite on  $E$ , then for any  $\delta > 0$ , there is a closed subset  $F \subset E$  that  $f$  is continuous function on  $F$ . ┘

*Proof:* First if  $f$  is a simple function  $f = \sum_{i=1}^n c_i \chi_{E_i}$ , then for each  $E_i$ , choose a closed subset  $F_i \subset E_i$  that  $m(E_i - F_i) < \frac{\delta}{n}$ , and then  $\cup F_i$  satisfies the required condition.

Now if  $f$  is arbitrary, let  $g(x) = \frac{f(x)}{1+|f(x)|}$  to make it bounded, then by (12.3.8.5), there is a sequence of simple functions  $\varphi_k$  converging to  $f$ , and for each  $k$ , we choose a closed subset  $F_k$  that  $m(E - F_k) < \frac{\delta}{2^k}$ , so if we let  $F = \cap F_k$ , then  $\varphi_k$  are all continuous on  $F$ , so by the uniform convergence,  $f$  is also continuous on  $F$ . □

**Cor. (12.3.8.9).** If  $f$  is measurable function on  $E$  that is a.e. finite, then for any  $\delta$ , there is a continuous function  $g$  that  $m(\{x \in E | f(x) \neq g(x)\}) < \delta$ . And if  $E$  is bounded,  $g$  can be chosen to be compactly supported. ┘

*Proof:* Now that there is a closed subset  $F$  that  $m(E - F) < \delta$  and  $f$  is continuous on  $F$ , we can use Tietze extension (4.4.6.4), there is a function  $g$  that equals  $f$  on  $F$ .

If  $E \subset B(0, R)$ , then we can choose a bump function to multiply with  $g$ . □

**Prop. (12.3.8.10).** for  $1 \leq p < +\infty$ , trigonometric polynomials are dense in  $L^p(\mathbb{T})$  and  $C(\mathbb{T})$ , but not for  $p = \infty$ . So  $e^{2\pi i n x}$  forms an orthogonal basis in  $L^2(\mathbb{T})$ . ┘

Thus, the Parseval's identity holds. ┘

*Proof:* Just use the fact that Fejer kernels are an approximate identity. □

**Prop. (12.3.8.11).** For a integrable function  $u$  that has compact support,  $u_\delta = j_\delta * u$  is a smooth function of compact support that  $\|u_\delta - u\|_{C^k} \rightarrow 0$  when  $u \in C^k$ . Where  $j_\delta$  is the scaling of a smooth function of compact support. So Smooth function of compact support are dense in  $C_0^k$ .  $\lrcorner$

**Prop. (12.3.8.12).**  $D(\mathbb{R}^n)$  is dense in  $W^{m,p}(\mathbb{R}^n)$ .  $\lrcorner$

*Proof:* Use the fact that  $C_0$  are dense in  $L^p$  by (12.3.8.11). And  $f_\delta \rightarrow f$  in  $L^p$  norm for  $f \in C_0$ . So we can use the three-part argument applied to  $D_\alpha u$  to get  $D_\alpha(u_\delta) \rightarrow D_\alpha u$  in  $L^p$  norm for  $|\alpha| \leq m$ . Thus the result.  $\square$

## 9 Example of Calculations

**Prop. (12.3.9.1).** For  $s \in \mathbb{C}, \operatorname{Re}(s) > 1/2$ , then

$$\int_{-\infty}^{\infty} (1+x^2)^{-s} e^{ik \arctan(-\frac{1}{x})} dx = (-i)^k \sqrt{\pi} \frac{\Gamma(s)\Gamma(s-\frac{1}{2})}{\Gamma(s+\frac{k}{2})\Gamma(s-\frac{k}{2})}.$$

$\lrcorner$

*Proof:* Cf.[Bump, Automorphic Forms and Representations, P230].  $\square$

## 10 Hausdorff Measures and Hausdorff Dimensions

References are [Fol99].

**Def. (12.3.10.1) [Metric Outer Measure].** For a metric space  $(X, d)$ , an outer measure  $\mu^*$  on  $X$  is called a **metric outer measure** if

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$$

for any  $A, B \subset X$  s.t.  $\operatorname{Dist}(A, B) > 0$ .  $\lrcorner$

**Prop. (12.3.10.2) [Metric Outer Measure is a Borel Measure].** If  $(X, d)$  is a metric space and  $\mu^*$  is an outer measure, then every Borel subset of  $X$  is  $\mu^*$ -measurable (12.3.1.15).  $\lrcorner$

*Proof:* Because closed subsets of  $X$  generates the  $\sigma$ -algebra of Borel subsets of  $X$ , by (12.3.1.15), it suffices to show that any closed subset  $A$  of  $X$  is  $\mu^*$ -measurable, i.e.  $\mu^*(A) = \mu^*(A \cap F) + \mu^*(A \setminus F)$  for any  $F \subset X$ . We may assume  $\mu^*(A) < \infty$ , and for each  $n \in \mathbb{Z}_+$ , denote  $B_n = \{x \in A \setminus F \mid \operatorname{Dist}(x, F) \geq n^{-1}\}$ , then  $B_n$  is an increasing sequence of sets with union  $A \setminus F$ , and  $\operatorname{Dist}(B_n, F) \geq n^{-1}$ . Thus

$$\mu^*(A) \geq \mu^*((A \cap F) \cup B_n) = \mu^*(A \cap F) + \mu^*(B_n).$$

Thus it suffices to show that  $\lim_n \mu^*(B_n) = \mu^*(A \setminus F)$ .

Denote  $C_j = B_{j+1} \setminus B_j$ , then  $\operatorname{Dist}(C_{j+1}, B_j) > \frac{1}{j(j+1)}$ , thus

$$\mu^*(B_{2k}) \geq \sum_{j=1}^k \mu^*(C_{2j-1}), \quad \mu^*(B_{2k+1}) \geq \sum_{j=1}^k \mu^*(C_{2j}).$$

In particular,  $\sum_{j \in \mathbb{Z}_+} \mu^*(C_j)$  converges. Then it follows from the inequality

$$\mu^*(A \setminus F) \leq \mu^*(B_n) + \sum_{i=n+1}^{\infty} \mu^*(C_i)$$

that  $\limsup \mu^*(B_n) \leq \mu^*(A \setminus F)$ .  $\square$

**Def. (12.3.10.3) [Hausdorff Measures].** Suppose  $(X, d)$  is a metric space, then for  $p \in \mathbb{R}_{\geq 0}, \delta \in \mathbb{R}_+$  and  $A \subset X$ , define

$$H_{p,\delta}(A) = \inf \left\{ \sum_{j=1}^{\infty} \text{diam}(B_j)^p \mid B_j \subset X, A \subset \cup_j B_j, \text{diam } B_j \leq \delta \right\}.$$

with the convention  $\inf \emptyset = \infty$ , and the  $p$ -dimensional **Hausdorff outer measure**

$$H_p(A) = \lim_{\delta \rightarrow 0} H_{p,\delta}(A).$$

Then  $H_p$  is a metric outer measure (12.3.10.1). In particular, it restricts to a Borel measure on  $X$  by (12.3.10.1).  $\lrcorner$

*Proof:*  $H_{p,\delta}$  is an outer measure by (12.3.1.14), and it is easy to show that the limit  $H_p$  is also an outer measure. It is a metric outer measure because for any  $A, B \subset X$ , when  $\delta < \text{Dist}(A, B)$ , any subset  $B_i$  of  $X$  with  $\text{diam}(B_i) < \delta$  can only intersect either  $A$  or  $B$ .  $\square$

**Prop. (12.3.10.4).** For  $p \in \mathbb{R}_{\geq 0}$ , the  $p$ -dimensional Hausdorff measure is invariant under isometries, and if  $f : (X, d) \rightarrow (Y, d')$  is a map and there exists  $C \in \mathbb{R}_+$  s.t.  $d'(f(x), f(y)) \leq Cd(x, y)$  for any  $x, y \in X$ , then  $H_p(f(A)) \leq CH_p(A)$  for any  $A \subset X$ .  $\lrcorner$

**Def. (12.3.10.5) [Hausdorff Dimensions].** Suppose  $(X, d)$  is a metric space and  $p \in \mathbb{R}_{\geq 0}, A \subset X$ . If  $H_p(A) < \infty$ , then  $H_q(A) = \infty$  for any  $q \in \mathbb{R}_{>p}$ . And if  $H_p(A) > 0$ , then  $H_q(A) = \infty$  for any  $q \in \mathbb{R}_{<p}$ . In particular,  $\inf\{p \in \mathbb{R}_{\geq 0} : H_p(A) = 0\} = \sup\{p \in \mathbb{R}_{\geq 0} : H_p(A) = \infty\}$ , which is called the **Hausdorff dimension** of  $A$ , denoted by  $\dim^H(A)$ .  $\lrcorner$

*Proof:* If  $H_p(A) < \infty$ , then for any  $\delta \in \mathbb{R}_+$ , there exists  $(B_i)_{i \in \mathbb{Z}_+} \subset X$  s.t.  $A \subset \cup_{i \in \mathbb{Z}_+} B_i$ ,  $\text{diam}(B_i) < \delta$  and  $\sum_i \text{diam}(B_i)^p < H_p(A) + 1$ . Then for any  $q \in \mathbb{R}_{>p}$ , we have

$$H_{q,\delta}(A) \leq \sum_i \text{diam}(B_i)^q \leq \delta^{q-p} \sum_i \text{diam}(B_i)^p \leq \delta^{q-p}(H_p(A) + 1).$$

Taking limit  $\delta \rightarrow 0$ , we see  $H_q(A) = 0$ .

The second assertion follows by contraposition.  $\square$

**Prop. (12.3.10.6).** Suppose  $(X, d)$  is a metric space and  $(A_i)_{i \in \mathbb{Z}_+} \subset X$ , then  $\dim^H(\cup_i A_i) = \sup_i \dim^H(A_i)$ .  $\lrcorner$

**Prop. (12.3.10.7).** For  $n \in \mathbb{Z}_+$ , there exists  $\gamma_n \in \mathbb{R}_+$  s.t.  $\gamma_n H_n$  is the Lebesgue measure on  $\mathbb{R}^n$ .  $\lrcorner$

*Proof:* Cf. [Fol99]P351.  $\color{red}{?}$   $\square$

**Prop. (12.3.10.8) [Hausdorff Measure of  $C^1$ -Manifolds].** Let  $M$  be a  $k$ -dimensional  $C^1$ -submanifold of  $\mathbb{R}^n$  parametrized by  $f : V \rightarrow \mathbb{R}^n$  for some open subset  $U \subset \mathbb{R}^k$ . If  $A \subset U$  is a Borel subset, then  $f(A)$  is also a Borel subset, and

$$H_k(f(A)) = \int_A J(df_x) dH_k(x).$$

Moreover, if  $\varphi$  is any Borel-measurable function on  $M$  that is either non-negative or in  $L^1(M, H_k)$ , then

$$\int_M \varphi(y) dH_k(y) = \int_U \varphi(f(x)) J(df_x) dH_k(x).$$

*Proof:*  $\color{red}{?}$   $\square$

### Self-Similarity and Hausdorff Dimensions

**Prop. (12.3.10.9)[Semilitudes and Invariant Measures].** Let  $r \in (0, 1)$  and  $\mathcal{S} = (S_1, \dots, S_m)$  is a family of semilitudes of  $\mathbb{R}^n$  with scaling factor  $r$ , and  $X \subset \mathbb{R}^n$  is a non-empty compact set that is invariant under  $\mathcal{S}$ . For  $x \in \mathbb{R}^n, E \subset \mathbb{R}^n, \mu \in \text{Meas}(\mathbb{R}^n)$  and  $i_1, \dots, i_k \in [m]_+$ , denote

$$x_{i_1, \dots, i_k} = S_{i_1} \circ \dots \circ S_{i_k}(x), \quad E_{i_1, \dots, i_k} = S_{i_1} \circ \dots \circ S_{i_k}(E), \quad \mu_{i_1, \dots, i_k}(E) = \mu((S_{i_1} \circ \dots \circ S_{i_k})^{-1}(E)).$$

Then there is a Borel probability measure on  $X$  s.t. for any  $k \in \mathbb{Z}_+$ ,

$$\mu = \frac{1}{m^k} \sum_{i_1, \dots, i_k \in [m]_+} \mu_{i_1, \dots, i_k}.$$

┘

*Proof:* Choose  $x \in X$  and for each  $k \in \mathbb{Z}_+$ , denote

$$\mu^k = \frac{1}{m^k} \sum_{i_1, \dots, i_k \in [m]_+} [\delta_x]_{i_1, \dots, i_k}.$$

Then it is easy to prove by uniform continuity that  $\{\mu^k(f)\}$  is a Cauchy sequence for each  $f \in C(X)$  and the limits define a linear functional on  $C(X)$ . Let  $\mu$  be the associated Radon measure on  $X$ .

Then  $\mu$  is clearly a probability measure. And  $\text{Supp}(\mu) = X$  because the points  $x_{i_1, \dots, i_k}$  are dense in  $X$ . And the invariance properties of  $\mu$  follows because convergence of measures commutes with similitudes.  $\square$

**Def. (12.3.10.10)[Separating Sets].** Let  $r \in (0, 1)$  and  $\mathcal{S} = (S_1, \dots, S_m)$  is a family of semilitudes of  $\mathbb{R}^n$  with scaling factor  $r$ , then a **separating set** for  $\mathcal{S}$  is a non-empty bounded open set  $U \subset \mathbb{R}^n$  s.t.

$$S_i(U) \subset U, \quad S_i(U) \cap S_j(U) = \emptyset, \forall i \neq j \in [m]_+.$$

┘

**Prop. (12.3.10.11).** Suppose  $r \in (0, 1)$  and  $\mathcal{S} = (S_1, \dots, S_m)$  is a family of semilitudes of  $\mathbb{R}^n$  with scaling factor  $r$  with a separating set  $U \subset \mathbb{R}^n$  (12.3.10.10), then

$$X = \bigcap_{n \in \mathbb{Z}_+} \left( \bigcup_{i=1}^m S_i \right)^{\circ n} (\bar{U})$$

is the unique non-empty compact subset of  $\mathbb{R}^n$  that is invariant under  $S = \bigcup_{i=1}^m S_i$ .  $\square$

*Proof:* Clearly  $X$  is a non-empty compact invariant subset for  $S$ . And if  $Y$  is another non-empty compact subset of  $\mathbb{R}^n$  that is invariant under  $S$ , because  $S(Y) = Y, S(X) = X$ , we have

$$d_X^{\text{sup}}(Y) = \max_i d_X^{\text{sup}}(S_i(Y)) \leq r d_X^{\text{sup}}(Y).$$

thus  $d_X^{\text{max}}(Y) = 0$ , and thus  $Y \subset X$ . The same argument shows that  $X \subset Y$ , thus  $X = Y$ .  $\square$

**Prop. (12.3.10.12)[Hausdorff Dimension of Invariant Subsets with Similitudes].** Suppose  $r \in (0, 1)$  and  $\mathcal{S} = (S_1, \dots, S_m)$  is a family of semilitudes of  $\mathbb{R}^n$  with scaling factor  $r$  with a separating set  $U \subset \mathbb{R}^n$  (12.3.10.10), and  $X \subset \mathbb{R}^n$  is the unique non-empty invariant compact subset. Denote  $p = \log_{1/r}(m)$ , then

- $0 < \dim_p^H(X) < \infty$ . In particular,  $X$  has Hausdorff dimension  $p$ .
- $\dim_p^H(S_i(X) \cap S_j(X)) = 0$  for  $i \neq j$ .

┘

*Proof:* •  $\square$

## 11 Area and Coarea Formulas

## 12 Inequalities

References are [HLP88].

### Limits

**Prop. (12.3.12.1)**[e]. For  $x \in \mathbb{R}_+$ ,

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

┘

*Proof:*

□

### Convex Functions

**Def. (12.3.12.2)**[Convex Functions]. Let  $X \subset \mathbb{R}^n$  be a convex set, then a **convex function** on  $X$  is a function  $f : X \rightarrow \mathbb{R}$  s.t. for any  $x, y \in X$  and  $t \in [0, 1]$ ,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

┘

**Prop. (12.3.12.3)**. A convex function on  $\mathbb{R}$  is absolutely continuous.

┘

*Proof:* This is because it is absolutely continuous.

□

**Prop. (12.3.12.4)**. Let  $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R}$  be a non-decreasing function s.t.  $\varphi(0) = 0$ , and  $\varphi(e^\xi)$  is a convex function of  $\xi$ . Suppose  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \in \mathbb{R}_+$ ,  $\kappa_i \in \mathbb{R}$  satisfy

$$\lambda_1 \leq \kappa_1, \quad \lambda_1 \lambda_2 \leq \kappa_1 \kappa_2, \quad \dots \quad \lambda_1 \dots \lambda_m \leq \kappa_1 \dots \kappa_m,$$

then

$$\sum_i \varphi(\lambda_i) \leq \sum_i \varphi(\kappa_i).$$

┘

*Proof:* For  $z \in \mathbb{R}_{\geq 0}$ , define

$$f(z) = \prod_i \max(1, \kappa_i z), \quad g(z) = \prod_i \max(1, \lambda_i z),$$

then by argue case by case,  $g(z) \leq f(z)$ .

Notice if there exists an increasing function  $\psi : [0, \infty) \rightarrow \mathbb{R}$  s.t. for any  $\lambda \in \mathbb{R}_+$ ,

$$\varphi(\lambda) = \int_{\lambda^{-1}}^{\infty} \log(\lambda z) d\psi(z),$$

where the RHS is the Lebesgue-Stieltjes integral(12.3.3.26), then

$$\sum_i \varphi(\lambda_i) = \int_0^\infty \log g(z) d\psi(z) \leq \int_0^\infty \log f(z) d\psi(z) = \sum_i \varphi(\kappa_i).$$

Denote  $G(\xi) = \varphi(e^\xi)$ , then we show that  $\psi(z) = -G^\lambda(-\log z)$  suffices: If  $\xi = \log \lambda$ , then

$$\int_{\lambda^{-1}}^{\infty} \log(\lambda z) d\psi(z) = \int_{-\xi}^{\infty} (\xi + t) d(-G^\lambda(-t)) = G(\xi) = \varphi(\lambda).$$

Where we used integration by parts?? and the fact  $\lim_{t \rightarrow -\infty} tG^\lambda(t) = 0$ , which is because  $\lim_{t \rightarrow -\infty} G(t) = 0$  and

$$tG^\lambda(t) \leq 2 \int_t^{t/2} G^\lambda(t) dt = 2[G(\frac{t}{2}) - G(t)].$$

□

### Hardy Inequality

**Thm. (12.3.12.5) [Hardy Inequality].** For  $p \in \mathbb{R}_{>1}$  and a measurable function  $f(x)$  on  $(0, \infty)$  s.t.  $f(x) \geq 0$ ,

$$\int_0^\infty \left( \frac{1}{x} \int_0^x f(t) dt \right)^p dx \leq \left( \frac{p}{p-1} \right)^p \int_0^\infty f(x)^p dx.$$

┘

*Proof:* Let  $F(x) = \int_0^x f(t) dt$ , then  $\frac{\partial}{\partial x} F(x)^p = pF(x)^{p-1} f(x)$ . So by integration by part, for any  $\alpha < A \in \mathbb{R}_+$ ,

$$\begin{aligned} \int_\alpha^A \left( \frac{F(x) - F(\alpha)}{x} \right)^p dx &= -\frac{A^{1-p}}{p-1} (F(A) - F(\alpha))^p + \frac{p}{p-1} \int_\alpha^A \left( \frac{F(x) - F(\alpha)}{x} \right)^{p-1} f(x) dx \\ (12.3.6.2) \leq \frac{p}{p-1} \left( \int_\alpha^A \left( \frac{F(x) - F(\alpha)}{x} \right)^p dx \right)^{(p-1)/p} \left( \int_\alpha^A f(x)^p dx \right)^{1/p}, \end{aligned}$$

which implies that

$$\int_\alpha^A \left( \frac{F(x) - F(\alpha)}{x} \right)^p dx \leq \left( \frac{p}{p-1} \right)^p \int_\alpha^A f(x)^p dx.$$

Then the result follows by taking limit  $A \rightarrow \infty$  and then take limit  $\alpha \rightarrow 0$ . □

**Cor. (12.3.12.6) [Hardy-Landau Inequality].** If  $p \in \mathbb{R}_{>1}$  and  $(a_k)_{k \in \mathbb{Z}_+} \subset \mathbb{R}_+$  is a sequence of positive real numbers, then

$$\sum_{n=1}^{\infty} \left( \frac{1}{n} \sum_{k=1}^n a_k \right)^p \leq \left( \frac{p}{p-1} \right)^p \sum_{n=1}^{\infty} a_n^p.$$

┘

*Proof:* We may assume that  $a_1 > a_2 > \dots$ , and  $\sum_{n=1}^{\infty} a_n^p < \infty$ . Define  $f : (0, \infty) \rightarrow \mathbb{R}_+ : x \mapsto a_{[x]}$ , then  $f \in L^p(0, \infty)$ , and

$$\frac{1}{x} \int_0^x f(t) dt = \frac{\sum_{k \leq x} a_k + a_{[x]}(x - [x])}{x},$$

which is decreasing in  $x$ .

Then

$$\sum_{n=1}^{\infty} \left( \frac{1}{n} \sum_{k=1}^n a_k \right)^p \leq \sum_{n=1}^{\infty} \left( \int_{n-1}^n \frac{1}{x} \int_0^x f(t) dt \right)^p dx \leq \left( \frac{p}{p-1} \right)^p \int_0^\infty f(x)^p dx = \left( \frac{p}{p-1} \right)^p \sum_{n=1}^{\infty} a_n^p$$

□

**Cor. (12.3.12.7) [Carleman Inequality].** If  $(a_k)_{k \in \mathbb{Z}_+} \subset \mathbb{R}_+$  is a sequence of positive real numbers, then

$$\sum_{n=1}^{\infty} (a_1 \dots a_n)^{1/n} \leq e \sum_{n=1}^{\infty} a_n.$$

┘

*Proof:* By [Hardy-Landau Inequality](#), for any  $p \in \mathbb{R}_{>1}$ ,

$$\left(\frac{p}{p-1}\right)^p \sum_{n=1}^{\infty} a_n \geq \sum_{n=1}^{\infty} \left(\frac{1}{n} \sum_{k=1}^n a_k^{1/p}\right)^p \geq \sum_{n=1}^{\infty} (a_1 \dots a_n)^{1/n}.$$

Then we use the fact  $\lim_{p \rightarrow \infty} \left(\frac{p}{p-1}\right)^p = e$  [\(12.3.12.1\)](#). □

**Cor. (12.3.12.8) [Pólya-Knopp Inequality].** For any measurable function  $f(x)$  on  $(0, \infty)$  s.t.  $f(x) > 0$ ,

$$\int_0^{\infty} \exp\left(\frac{1}{x} \int_0^x \log f(t) dt\right) dx \leq e \int_0^{\infty} f(x) dx.$$

┘

*Proof:* The proof is similar in essence to that of [Carleman Inequality](#). □

**Thm. (12.3.12.9) [Copson Inequality].** If  $p \in \mathbb{R}_{>1}$  and  $(a_k)_{k \in \mathbb{Z}_+} \subset \mathbb{R}_+$  is a sequence of positive real numbers, then

$$\sum_{n=1}^{\infty} \left(\sum_{k=n}^{\infty} \frac{a_k}{k}\right)^p \leq p^p \sum_{n=1}^{\infty} a_n^p.$$

┘

*Proof:* □

## 12.4 Complex Analysis I

References are [Ahl78, S-S03, Li04, T-W06].

**Notation (12.4.0.1).**

- Use notations defined in [Measure Theory and Functions on  \$\mathbb{R}^n\$](#) .
- Let  $\Omega$  be a region [\(12.4.1.1\)](#).

┘

### 1 Basics

**Def. (12.4.1.1) [Regions].** A **region** is a nonempty connected open subset of  $\mathbb{C}$ .

┘

**Def. (12.4.1.2) [Conjugations].** The non-zero element in  $\text{Gal}(\mathbb{C}/\mathbb{R})$  is denoted by  $c$ . And for  $z \in \mathbb{C}$ ,  $c(z)$  is also denoted by  $\bar{z}$ .

For  $z \in \mathbb{C}$ , denote  $\text{Re}(z) = \frac{z+\bar{z}}{2}$ ,  $\text{Im}(z) = \frac{z-\bar{z}}{2i}$ , called the **real part** and **complex part** of  $z$  resp..  
So

$$z = \text{Re}(z) + i \text{Im}(z), \quad \bar{z} = \text{Re}(z) - i \text{Im}(z).$$

┘

**Prop. (12.4.1.3) [C is Complete].** The natural extended value from  $\mathbb{R}$  to  $\mathbb{C}$  is of the form  $|x + iy| = \sqrt{x^2 + y^2}$ . In particular, it is easy to prove that  $\mathbb{C}$  is a complete valued field.

┘

**Def. (12.4.1.4) [Derivatives].** We introduce the following notations:

$$\frac{\partial}{\partial z} = \frac{1}{2} \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right), \quad dz = dx + i dy \quad d\bar{z} = dx - i dy.$$

Then  $dz$  is dual to  $\frac{\partial}{\partial \bar{z}}$  and  $d\bar{z}$  is dual to  $\frac{\partial}{\partial z}$ . And for any function  $f \in C^1(\mathbb{C})$ ,

$$df = \frac{\partial}{\partial x} f dx + \frac{\partial}{\partial y} f dy = \frac{\partial}{\partial z} f dz + \frac{\partial}{\partial \bar{z}} f d\bar{z}.$$

And also,

$$dz d\bar{z} = 2 dx dy = 2 r dr d\theta$$

┘

**Prop. (12.4.1.5) [Orientations].** For  $f \in C^1(\mathbb{C})$ ,

$$\text{Jac}_f(z) = \frac{\partial}{\partial x} f \frac{\partial}{\partial y} f = \left| \frac{\partial}{\partial z} f \right|^2 - \left| \frac{\partial}{\partial \bar{z}} f \right|^2.$$

Thus  $f$  preserves orientations at  $z$  iff  $\left| \frac{\partial}{\partial z} f \right| > \left| \frac{\partial}{\partial \bar{z}} f \right|$ .

┘

**Def. (12.4.1.6) [Cross Ratios].** For any three pts  $z_2, z_3, z_4 \in \overline{\mathbb{C}}$ , there is a unique linear transformation that maps them to  $1, 0, \infty$ . In fact, the linear transformation is just  $Sz = \frac{z-z_3}{z-z_4} / \frac{z_2-z_3}{z_2-z_4}$ .

Then for any point  $z_1$ , the **cross ratio**  $(z_1, z_2, z_3, z_4)$  is the image of  $z_1$  under the linear transformation that carries  $z_2, z_3, z_4$  to  $1, 0, \infty$ .

┘

**Prop. (12.4.1.7).** The cross ratio is invariant under linear transformation, and it is real iff  $z_1, z_2, z_3, z_4$  are colinear or cocycle.

┘



*Proof:* The first is because there is only one linear transformation that maps  $z_2, z_3, z_4$  to  $1, 0, \infty$ .

For the second, notice by (12.4.1.6),  $\arg(z_1, z_2, z_3, z_4) = \arg \frac{z_1 - z_3}{z_1 - z_4} - \arg \frac{z_2 - z_3}{z_2 - z_4}$ , and this is real iff  $\angle z_4 z_2 z_3 = \angle z_4 z_1 z_3$  or  $\pi - \angle z_4 z_1 z_3$ , which is equivalent to cocycle. For other degenerate cases, we need some other argument.  $\square$

**Cor. (12.4.1.8).** A linear transformation maps colinear/cocycle points to colinear/cocycle points.  $\lrcorner$

**Lemma (12.4.1.9) [Invariant Factor].** If  $a, b, c, d \in \mathbb{R}$ , then

$$\operatorname{Im}\left(\frac{az + b}{cz + d}\right) = \frac{ad - bc}{|cz + d|^2} \operatorname{Im}(z).$$

$\lrcorner$

### Analytic Functions

**Def. (12.4.1.10) [Analytic Functions].** For an open subset  $\Omega \subset \mathbb{C}$ , a complex-valued function  $f$  on  $\Omega$  is called **analytic** or **holomorphic** if  $\frac{\partial}{\partial \bar{z}} f(z) = 0$  for any  $z \in \Omega$  (12.4.1.4). Equivalently,

$$\frac{\partial}{\partial x} f = -i \frac{\partial}{\partial y} f, \quad \text{i.e.} \quad \begin{cases} \frac{\partial}{\partial x} \operatorname{Re}(f) = \frac{\partial}{\partial y} \operatorname{Im}(f) \\ \frac{\partial}{\partial x} \operatorname{Im}(f) = -\frac{\partial}{\partial y} \operatorname{Re}(f) \end{cases}$$

i.e.  $f$  has the same derivative vertically and horizontally, hence in every direction.

The space of analytic functions on  $\Omega$  is denoted by  $\mathcal{O}(\Omega)$ . More generally, if  $\Omega \subset \mathbb{C}$  is any subspace,  $\mathcal{O}(\Omega) = C(\Omega) \cap \mathcal{O}(\Omega^\circ)$ .

For  $f \in \mathcal{O}(\Omega)$ , denote

$$f^\wedge(z) = \frac{\partial}{\partial z} f(z) = \frac{\partial}{\partial x} f(z) = -i \frac{\partial}{\partial y} f(z).$$

and for  $n \in \mathbb{N}$ , denote inductively

$$f^{(0)} = f, \quad f^{(1)} = f^\wedge, \quad f^{(n+1)} = (f^{(n)})^\wedge.$$

$\lrcorner$

**Lemma (12.4.1.11).** Let  $\Omega \subset \mathbb{C}$  be a region and  $f \in \mathcal{O}(\Omega)$  s.t.  $f^\wedge = 0$ , then  $f$  is a constant function.  $\lrcorner$

*Proof:* This follows from the fact any two points in  $\Omega$  can be connected by a path consisting of vertical or horizontal segments, and use the fundamental theorem of calculus (12.3.3.31).  $\square$

**Prop. (12.4.1.12) [ $\bar{\partial}$ -Equation].** Let  $\Omega \subset \mathbb{C}$  be a region and  $f \in C^k(\Omega)$ ,  $k \geq 1$ , then locally near every point, there exists a  $C^k$ -function  $g$  s.t.

$$\frac{\partial}{\partial \bar{z}} g = f.$$

And such a function is defined up to an analytic function.  $\lrcorner$

*Proof:* Taking a bump function, we may assume  $f \in C_c^k(\Omega)$ . Then define

$$g(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{f(\zeta)}{\zeta - z} d\zeta \wedge d\bar{\zeta} = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{f(\zeta + z)}{\zeta} d\zeta \wedge d\bar{\zeta}.$$

This singular integration is convergent because  $h(\zeta) = \frac{1}{\zeta - z}$  is locally  $L^1$  (this follows from (12.4.1.4)). And the integration is uniformly convergent in  $z$ . Then the differentiation commutes with integration and shows  $g$  is  $C^k$ . Moreover,

$$\frac{\partial}{\partial \bar{z}} g(z) = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi i} \int_{\mathbb{C} \setminus \mathbb{D}(0, \varepsilon)} \frac{\partial}{\partial \bar{\zeta}} f(\zeta) \frac{d\zeta \wedge d\bar{\zeta}}{\zeta - z} = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi i} \int_{\partial \mathbb{D}(0, \varepsilon)} f(\zeta) \frac{d\zeta}{\zeta - z} = f(z)$$

by Stoke's formula and continuity.  $\square$

## 2 Complex Integration

**Lemma (12.4.2.1).** If  $f$  is analytic on a rectangle  $R$  minus f.m. points  $\zeta_i$  and if  $\lim_{z \rightarrow \zeta_i} (z - \zeta_i)f(z) = 0$ , then  $\int_{\partial R} f(z)dz = 0$ .  $\square$

*Proof:* We consider first the case that no points are omitted. Cut the rectangle  $R$  into 4 rectangles  $R^1, R^2, R^3, R^4$  that is similar to  $R$ , then

$$\int_{\partial R} f(z)dz = \sum_{i=1}^4 \int_{\partial R^i} f(z)dz.$$

Let  $|\int_{\partial R} f(z)dz| = T$ , then there exists some  $R^i$  that  $|\int_{\partial R^i} f(z)dz| \geq \frac{1}{4}T$ . Denote this  $R^i$  by  $R_1$ . Then we can do the same for  $R_1$  to find an  $R_2$  s.t.  $|\int_{\partial R_2} f(z)dz| \geq \frac{1}{4^2}T$ . Continuing this process, we find a sequence  $R_1 \supset R_2 \supset \dots \supset R_n \supset$ , and their intersection is a single point  $z_0$  as  $R$  is compact. Now as  $f$  is analytic, for any  $\varepsilon > 0$ , for  $n$  sufficiently large, for any  $z \in \partial R_n$ ,

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \leq \varepsilon.$$

Notice also by direct calculating that

$$\int_{\partial R_n} dz = 0, \quad \int_{\partial R_n} z dz = 0,$$

we have

$$\frac{T}{4^n} \leq \left| \int_{\partial R_n} f(z)dz \right| = \left| \int_{\partial R_n} [f(z) - f(z_0) - f'(z_0)(z - z_0)]dz \right| \leq \varepsilon \int_{\partial R_n} |z - z_0| |dz| \leq \varepsilon \frac{dL}{4^n}$$

where  $d, L$  are the length of the diagonal and the perimeter of  $R$ . So  $T \leq \varepsilon$ . As  $\varepsilon$  is arbitrary, this means  $T = 0$ .

In general, by cutting into several rectangles, it suffices to prove for the case that only one point is omitted, and in this case, we can use what we have proved and the hypothesis to reduce the rectangle to any small enough rectangle  $R_0$  around  $\zeta_1$  s.t. for any  $z \in \partial R_0$ ,

$$f(z) \leq \frac{\varepsilon}{|z - \zeta_1|}.$$

And then

$$\left| \int_{\partial R_0} f(z)dz \right| \leq \varepsilon \int_{\partial R_0} \frac{|dz|}{|z - \zeta_1|} \leq 8\varepsilon$$

by elementary estimation. As  $\varepsilon$  is arbitrary, this means  $\int_{\partial R_0} f(z)dz = 0$ .  $\square$

**Thm. (12.4.2.2) [Cauchy].** If  $\Omega \subset C$  is a simply-connected region and  $\Omega'$  is the region obtained from  $\Omega$  by omitting f.m. points  $\zeta_i$ . Suppose  $f \in \mathcal{O}(\Omega')$ , and  $\lim_{z \rightarrow \zeta_i} (z - \zeta_i)f(z) = 0$ , then  $\int_{\gamma} f(z)dz = 0$  for any closed piecewise  $C^1$  curve  $\gamma \subset \Omega'$ .

Moreover, if  $f \in \mathcal{O}(\overline{\Omega'})$ , then  $\int_{\gamma} f(z)dz = 0$  for any closed piecewise  $C^1$  curve  $\gamma \subset \overline{\Omega'} \setminus \{\zeta_1, \dots, \zeta_i\}$ .  $\square$

*Proof:* Fix a  $z_0 \in \Omega$ , then for any  $z \in \Omega$ , choose a path  $\gamma$  from  $z_0$  to  $z$  consisting of vertical or horizontal segments, and let  $F(z) = \int_{\gamma} f(z)dz$ . Then this  $F$  is well-defined: if there are two paths  $\gamma, \gamma'$ ,  $\gamma - \gamma'$  is a sum oriented boundary of rectangles, and these rectangles must be contained in  $\Omega$

because  $\Omega$  is simply-connected. Thus the above lemme(12.4.2.1) shows that  $F(z)$  is independent of the path chosen, and it is clear that  $F(z)$  has the same derivative in both directions so analytic by definition(12.4.1.10). So clearly  $\int_{\gamma} f(z)dz = F(z)\Big|_{\gamma(0)}^{\gamma(1)} = 0$ .

For the last assertion, any such a closed piecewise  $C^1$  curve  $\gamma \subset \overline{\Omega'} \setminus \{\zeta_1, \dots, \zeta_i\}$  can be uniformly approximated by a path in  $\Omega'$ .  $\square$

**Cor. (12.4.2.3) [Existence of Primitive].** If  $\Omega \subset \mathbb{C}$  is a simply connected region and  $f \in \mathcal{O}(\Omega)$ , then there exists a function  $F \in \mathcal{O}(\Omega)$  s.t.  $F' = f$ .  $\lrcorner$

*Proof:* Take  $z_0 \in \Omega$  and take  $F(z) = \int_{\gamma} f(z)dz$  for any path  $\gamma$  from  $z_0$  to  $z$ .  $F$  is well-defined by the Cauchy theorem(12.4.2.2).  $\square$

**Prop. (12.4.2.4) [Generating Analytic Functions].** If  $\varphi(\zeta)$  is continuous on an arc  $\gamma$ , then the function

$$F_n(\zeta) = \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z)^n} d\zeta$$

is analytic on each connected component of  $\mathbb{C} \setminus \gamma$ , and its derivative is  $F'_n(z) = nF_{n+1}(z)$ .  $\lrcorner$

*Proof:* Use induction on  $n$ : for  $n = 1$ , firstly we prove  $F_1$  is continuous: for  $z_0 \notin \gamma$ , choose  $\delta > 0$  s.t.  $U(z_0, \delta) \cap \gamma = \emptyset$ , then for  $z \in U(z_0, \delta)$ ,  $d(z, \gamma) > \delta/2$ , and

$$|F_1(z) - F_1(z_0)| = |z - z_0| \left| \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z)(\zeta - z_0)} d\zeta \right| \leq |z - z_0| \frac{2}{\delta^2} \int_{\gamma} |\varphi| |d\zeta|,$$

thus  $F_1$  is continuous. Moreover, the above argument applied to the function  $\Phi(\zeta) = \varphi(\zeta)/(\zeta - z_0)$  implies that

$$\lim_{z \rightarrow z_0} \frac{F_1(z) - F_1(z_0)}{z - z_0} = \lim_{z \rightarrow z_0} \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z)(\zeta - z_0)} d\zeta = \lim_{z \rightarrow z_0} \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z)} d\zeta = \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z_0)} d\zeta = F_2(z_0),$$

hence  $F_1' = F_2$ .

For  $n > 1$ , suppose we have shown  $F_{n-1}' = (n-1)F_n$ , then with notations as above, we have

$$\begin{aligned} F_n(z) - F_n(z_0) &= \left[ \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z)^{n-1}(\zeta - z_0)} d\zeta - \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^n} d\zeta \right] + (z - z_0) \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z)^n(\zeta - z_0)} d\zeta \\ &= \left[ \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z)^{n-1}} d\zeta - \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z_0)^{n-1}} d\zeta \right] + (z - z_0) \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z)^n} d\zeta \end{aligned}$$

Then we see by induction hypothesis that  $\lim_{z \rightarrow z_0} F_n(z) - F_n(z_0) = 0$  and  $F_n$  is continuous for any  $\varphi$ . Moreover,

$$\begin{aligned} \lim_{z \rightarrow z_0} \frac{F_n(z) - F_n(z_0)}{(z - z_0)} &= \lim_{z \rightarrow z_0} \frac{1}{(z - z_0)} \left[ \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z)^{n-1}} d\zeta - \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z_0)^{n-1}} d\zeta \right] + \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z)^n} d\zeta \\ &= n \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z_0)^n} d\zeta + \int_{\gamma} \frac{\Phi(\zeta)}{(\zeta - z_0)^n} d\zeta \\ &= (n+1)F_{n+1}(z_0) \end{aligned}$$

$\square$

**Prop. (12.4.2.5) [Index of a Point w.r.t a Curve].** If  $\gamma$  is a piecewise  $C^1$  curve that doesn't pass a point  $a$ , then  $\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z-a}$  is an integer  $n(\gamma, a)$ , called the **index of  $a$  w.r.t  $\gamma$** , or the **winding number**.

And this index function is constant on each connected component of  $\mathbb{C} \setminus \gamma$ , and 0 on the unbounded component. In particular, if  $\gamma$  is a circle and  $a$  is contained in the interior of this circle, then  $n(\gamma, a) = 1$ .  $\lrcorner$

*Proof:* Cf. [Ahl78]P115. ?

this index function is constant on each connected component of  $\mathbb{C} \setminus \gamma$ , and 0 on the unbounded component by the continuity of the integral by (12.4.2.4).

For the last assertion, it suffices to show for  $\gamma = \partial \mathbb{D}(0, R)$  and  $a = 0$ , by what we just said. And in this case,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z-a} = \frac{1}{2\pi i} \int_0^{2\pi} \frac{iRe^{i\theta}}{Re^{i\theta}} d\theta = 1$$

$\square$

**Cor. (12.4.2.6) [Cauchy Integral Formula, Cauchy1825].** if  $\Omega \subset \mathbb{C}$  is a simply-connected region,  $f \in \mathcal{O}(\Omega)$ , then for any piecewise  $C^1$  closed curve  $\gamma \subset \Omega$  and  $a \notin \gamma$ ,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(\xi)}{\xi-a} d\xi = n(\gamma, a)f(a) \text{ (12.4.2.5).}$$

In particular, if  $\gamma$  is the boundary of a disk  $D$  contained in  $\mathbb{C}$ , then for any  $a \in D$ ,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(\xi)}{\xi-a} d\xi = f(a).$$

Moreover, if  $f \in \mathcal{O}(\overline{\Omega})$ , then this is true for any piecewise  $C^1$  closed curve  $\gamma \subset \Omega$  and  $a \notin \gamma$ .  $\lrcorner$

*Proof:* Consider the function  $F(z) = \frac{f(z)-f(a)}{z-a}$ , then it is analytic for  $z \neq a$ , and at  $a$  it satisfies the condition of Cauchy theorem (12.4.2.2), so  $\int_{\gamma} F(z)dz = 0$  which is  $\int_{\gamma} \frac{f(z)dz}{z-a} = f(a) \int_{\gamma} \frac{dz}{z-a}$ , and use (12.4.2.5).  $\square$

**Cor. (12.4.2.7) [Higher Derivations].** For any region  $\Omega$   $f \in \mathcal{O}(\Omega)$ , if  $a \in \Omega$  and  $\gamma$  is a small piecewise-smooth curve  $\gamma$  in  $\Omega$  centered at  $a$ ,

$$f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta-a} d\zeta$$

by Cauchy integral theorem (12.4.2.6). So derivatives of  $f$  are all analytic, and satisfy:

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta-z)^{n+1}} d\zeta.$$

In particular,  $\mathcal{O}(\Omega) \subset C^\infty(\Omega)$ .  $\lrcorner$

**Cor. (12.4.2.8) [Morera].** If  $f$  is continuous on a region  $\Omega$  and if  $\int_{\gamma} f dz = 0$  for any piecewise closed curve  $\gamma \subset \Omega$  consisting of vertical or horizontal segments, then  $f(z) \in \mathcal{O}(\Omega)$ .  $\lrcorner$

*Proof:* There is an analytic function  $F$  that  $F' = f$ , by the same method of the proof of (12.4.2.2), so  $f$  is analytic, by (12.4.2.7).  $\square$

**Cor. (12.4.2.9) [Cauchy Estimate].** If  $f \in \mathcal{O}(\overline{\mathbb{D}(a, r)})$ , and  $|f| \leq M$  on the boundary, then  $|f^{(n)}(a)| \leq Mn!r^{-n}$  for any  $a \in \mathbb{D}$ .  $\lrcorner$

**Cor. (12.4.2.10) [Liouville].** Any bounded holomorphic function on  $\mathbb{C}$  is constant.  $\lrcorner$

*Proof:* If  $|f(z)| \leq M$ , then the Cauchy estimate (12.4.2.9) shows that  $|f'(a)| \leq Mr^{-1}$ , letting  $r$  tends to  $\infty$ , then  $f'(a) = 0$  for all  $a$ , thus  $f$  is constant.  $\square$

**Cor. (12.4.2.11) [Mean Value Property].** If  $f \in \mathcal{O}(\mathbb{D})$ , then  $|f(0)| \leq \int_{\mathbb{D}} |f(z)| dx dy$ .  $\lrcorner$

*Proof:*  $|f(0)| \leq \frac{1}{2\pi} \int f(re^{i\theta}) d\theta$ , so if multiplied by  $rdr$  and integrate, then

$$|f(0)| \leq \int \int f(re^{i\theta}) r dr d\theta = \int \int f(z) dx dy.$$

$\square$

**Prop. (12.4.2.12).** For any  $f \in \mathcal{O}(\overline{\mathbb{D}(0, R)})$ , for  $z \in \mathbb{D}(0, R)$ :

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} \operatorname{Re}\left(\frac{\zeta+z}{\zeta-z}\right) f(\zeta) \frac{d\zeta}{\zeta}.$$

$\lrcorner$

*Proof:* Let  $F(\zeta) = \frac{f(\zeta)}{\zeta - R^2/\bar{z}}$ , then  $F \in \mathcal{O}(\overline{\mathbb{D}(0, R)})$ , and by Cauchy's theorem (12.4.2.2),

$$\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - R^2/\bar{z}} d\zeta = 0.$$

And by Cauchy's integral formula,

$$\frac{1}{2\pi i} \int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - z} d\zeta = f(z).$$

so

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} \left[ \frac{f(\zeta)}{\zeta - z} - \frac{f(\zeta)}{\zeta - R^2/\bar{z}} \right] d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=R} \left[ \frac{\zeta}{\zeta - z} + \frac{\bar{z}}{\zeta - \bar{z}} \right] f(\zeta) \frac{d\zeta}{\zeta} = \frac{1}{2\pi i} \int_{|\zeta|=R} \operatorname{Re}\left(\frac{\zeta+z}{\zeta-z}\right) f(\zeta) \frac{d\zeta}{\zeta}$$

$\square$

**Prop. (12.4.2.13) [Invariance of Winding Numbers].** Let  $S$  be a smooth Jordan curve in  $\mathbb{C}$ , and let  $(C_s)_{s \in S}$  be a smoothly varying family of smooth Jordan curves (i.e. there exists a diffeomorphism  $J : S \times \partial\mathbb{D}(0, 1) \cong \{(s, z) : s \in S, z \in C_s\}$ ), with  $D_s = \operatorname{int} C_s$ . Let  $a, b, f : S \rightarrow \mathbb{C}$  be smooth functions s.t.  $a(s), b(s) \in D_s$ , and  $f(s) \in C_s$ , then

$$\int_S \frac{1}{f(s) - a(s)} ds = \int_S \frac{1}{f(s) - b(s)} ds$$

$\lrcorner$

*Proof:* The use of the smoothly varying property is that: by isotopy extension theorem (16.1.7.3), we can extend  $J$  to a diffeomorphism

$$J : S \times \overline{\mathbb{D}(0, 1)} \cong \{(s, z) : s \in S, z \in \overline{D_s}\},$$

so we can define a homotopy from  $a(s)$  to  $b(s)$  via

$$e_t(s) = J_s((1-t)J_s^{-1}(a(s)) + tJ_s^{-1}(b(s))).$$

$\square$

### Constructing Analytic Functions

**Prop. (12.4.2.14) [Holomorphic Function Defined by Integrations].** Let  $(X, \mathcal{M}, \mu)$  be a  $\sigma$ -finite measure space and  $\Omega \subset \mathbb{C}$  be a region. For any  $F \in L^1(\Omega \times X)$ , if

- $F(z, x)$  is analytic in  $z$  for any  $x \in X$ .
- $z \mapsto \int_X |F(z, x)| dx$  is uniformly convergent on compact subsets.

then  $f(z) = \int_X F(z, x) dx$  is an analytic function on  $\Omega$ .

In particular, item2 holds if  $F \in C(\Omega \times X)$  and  $X$  is compact.  $\lrcorner$

*Proof:* This follows from Morera's theorem(12.4.2.8) and Fubini-Tonelli theorem(12.3.2.9).  $\square$

**Cor. (12.4.2.15) [Weierstrass].** If  $(f_n)_{n \in \mathbb{Z}_+}$  is a sequence of holomorphic functions on  $\Omega$  that converges uniformly to  $f^\wedge$  on every compact subset, then  $f$  is holomorphic on  $\Omega$ . And for any  $k \in \mathbb{N}$ ,  $(f_n^{(k)})_{n \in \mathbb{Z}_+}$  converges uniformly on  $f^\wedge$  on every compact subset.  $\lrcorner$

*Proof:* For the last assertion, use Cauchy's integral formula(12.4.2.6).  $\square$

### Local Properties of Analytic Functions

**Prop. (12.4.2.16) [Taylor Expansions].** Let  $\Omega \subset \mathbb{C}$  be a region and  $f \in \mathcal{O}(\Omega)$ , then if  $D \subset \Omega$  be a closed disk with center  $z_0$ , then for any  $z \in D$ ,

$$f(z) = \sum_{n \in \mathbb{N}} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

$\lrcorner$

*Proof:* By Cauchy's integral formula(12.4.2.6), if  $C$  is the boundary of  $D$ , then

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi - z} d\xi$$

But

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \frac{1}{1 - \left(\frac{\xi - z}{\xi - z_0}\right)} = \sum_{n \in \mathbb{N}} \frac{(\xi - z)^n}{(\xi - z_0)^{n+1}}$$

and the convergence is uniform on  $C$ , so by(12.4.2.7),

$$f(z) = \int_C \left( \frac{1}{2\pi i} \int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}} \right) (\xi - z)^n = \sum_{n \in \mathbb{N}} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

$\square$

**Cor. (12.4.2.17).** For any region  $\Omega \subset \mathbb{C}$  and  $f \in \mathcal{O}(\Omega)$ . If  $z_0 \in \Omega$  and  $f^{(n)}(z_0) = 0$  for any  $n \in \mathbb{N}$ , then  $f = 0 \in \mathcal{O}(\Omega)$ .  $\lrcorner$

*Proof:* The subset  $E = \{z_0 \in \Omega | f^{(n)}(z_0) = 0, \forall n \in \mathbb{N}\}$  is an open subset by the Taylor expansion(12.4.2.16), and the complement is also an open subset as all the derivatives are continuous. So  $E = \Omega$ , and  $\Omega$  is connected.  $\square$

**Prop. (12.4.2.18) [Removable Singularities].** Let  $\Omega \subset \mathbb{C}$  and  $\Omega'$  is the region obtained from  $\Omega$  by omitting f.m. points  $\zeta_i$ ,  $f \in \mathcal{O}(\Omega')$ , then  $f$  can be extended to an analytic function  $f \in \mathcal{O}(\Omega)$  iff  $\lim_{z \rightarrow \zeta_i} (z - \zeta_i) f(z) = 0$  for each  $i$ .  $\lrcorner$

*Proof:* The necessary is clear. For the other direction, choose for each  $\zeta_i$  a disk  $D(\zeta_i, \delta)$  contained in  $\Omega$  with boundary  $\gamma$ , then for any  $z_0 \in D(\zeta_i, \delta)$ , by Cauchy theorem(12.4.2.2) applied to the analytic function  $F(z) = \frac{f(z)-f(z_0)}{z-z_0}$  on  $D(z_0, \delta) \setminus \{z_0, \zeta_i\}$ , we see

$$\int_{\gamma} \frac{f(\xi)}{\xi - z_0} d\xi = \int_{\gamma} \frac{f(z_0)}{\xi - z_0} = f(z_0) \quad (12.4.2.5).$$

But by(12.4.2.4),  $z \mapsto \int_{\gamma} \frac{f(\xi)}{\xi - z} d\xi$  is a holomorphic function on  $D(\zeta_i, \delta)$ , and it extends  $f(z_0)$ . So  $f$  can be extended to an analytic function  $f \in \mathcal{O}(\Omega)$ .  $\square$

**Def. (12.4.2.19)[Orders of Vanishing].** For any region  $\Omega \subset \mathbb{C}$  and  $f \neq 0 \in \mathcal{O}(\Omega)$ , then for any  $z_0 \in \Omega$ , there exists a smallest  $n \in \mathbb{N}$  s.t.  $f^{(n)}(z_0) \neq 0$  by(12.4.2.17). So by repeatedly using(12.4.2.18) on  $\frac{f(z)}{z-z_0}$ , we get

$$f(z) = (z - z_0)^n f_n(z)$$

on a nbhd of  $z_0$ , where  $f_n \in \mathcal{O}(\Omega')$  and  $f_n(z_0) \neq 0$  for some nbhd  $\Omega'$  of  $z_0 \in \Omega$ . Such an  $n$  is called the **order of vanishing** of  $f$  at  $z_0$ .

In particular, if  $n > 0$ ,  $f_n(z) \neq 0$  on a nbhd of  $z_0$ , so  $f(z) \neq 0$  on nbhd of  $z_0$ , thus the vanishing point of  $f$  is isolated in  $\Omega$ .  $\lrcorner$

**Cor. (12.4.2.20)[Uniqueness].** If the zeros of a holomorphic function  $f$  has a convergent point in the domain of definition, then  $f = 0$ . In particular, if  $f, g \in \mathcal{O}(\Omega)$  satisfies  $f(z) = g(z)$  for any  $z \in E$ , where  $E \subset \Omega$  is a subset with convergence points in  $\Omega$ , then  $f = g \in \mathcal{O}(\Omega)$ .  $\lrcorner$

**Prop. (12.4.2.21)[Singularities].** If  $\Omega \subset \mathbb{C}$  is region and  $\Omega'$  is the region obtained from  $\Omega \setminus E$  where  $E = \{\zeta_1, \dots, \zeta_n, \dots\} \subset \Omega$  is a discrete subset. Suppose  $f \in \mathcal{O}(\Omega')$ , then  $f$  is said to have a **singularity** at  $\zeta_i$ . For each  $i$ , let  $\zeta = \zeta_i$ , the following are all the cases:

- $\lim_{z \rightarrow \zeta} |z - \zeta|^\alpha |f(z)| = 0$  for any  $\alpha \in \mathbb{R}$ . In this case,  $f = 0 \in \mathcal{O}(\Omega)$ .
- there exists  $h \in \mathbb{R}$  s.t.  $\lim_{z \rightarrow \zeta} |z - \zeta|^\alpha |f(z)| = \begin{cases} 0 & , \alpha < h \\ \infty & , \alpha > h \end{cases}$ . In this case,  $h \in \mathbb{Z}$ , and  $z_0$  is called a **zero** of  $f$  if  $h > 0$  and **pole** of  $f$  if  $h < 0$ . And if  $h = 0$ , it is called a **removable singularity**, which is discussed in(12.4.2.18).
- Neither of the above holds. In this case,  $z_0$  is called a **essential singularity** of  $f$ .  $\lrcorner$

*Proof:* In case 1,  $f$  can be extended to a function on  $\Omega' \cup \{\zeta\}$  with  $f(\zeta) = 0$  by(12.4.2.18), and at this point, all the derivatives of  $f$  vanish by(12.4.2.19), so  $f = 0 \in \mathcal{O}(\Omega)$  by(12.4.2.20).

In case 2,  $|z - \zeta|^m |f(z)| = 0$  for some  $m \in \mathbb{Z}$  larger than  $h$ , so  $(z - \zeta)^m f(z)$  can be extended to a function on  $\Omega' \cup \{\zeta\}$ . As  $f$  is not identically zero,  $(z - \zeta)^m f(z)$  has a finite order of vanishing at  $\zeta$ , so  $(z - \zeta)^m f(z) = (z - \zeta)^k f_k(z)$  for some  $k \in \mathbb{Z}$ , and  $f_k$  is analytic on a nbhd of  $\zeta$ , by(12.4.2.19). Thus  $f(z) = (z - \zeta)^{k-m} f_k(z)$ , and clearly  $h$  is an integer.  $\square$

**Def. (12.4.2.22)[Meromorphic Functions].** Situation as in(12.4.2.21), if all the singularities of  $f$  on  $\Omega$  are zeros or poles,  $f$  is called a **meromorphic function** on  $\Omega$ . Notice the poles and zeros of  $f$  are discrete. The space of meromorphic functions on  $\Omega$  is denoted by  $\mathcal{M}(\Omega)$ .  $\lrcorner$

**Prop. (12.4.2.23).**  $\mathcal{M}(\Omega) \in \text{Field}$ , and for any  $f \in \mathcal{M}(\Omega)^\times$  and  $z_0 \in \Omega$ , either  $f(z_0) \neq 0$ , or  $f(z) = (z - z_0)^{n_{z_0}} g(z)$  for some  $n_{z_0} \in \mathbb{Z}$  and  $g(z) \in \mathcal{M}(\Omega)$  s.t.  $g(z_0) \neq 0$ .  $\lrcorner$

*Proof:* This follows from (12.4.2.21) and (12.4.2.18).  $\square$

**Prop. (12.4.2.24) [Maximum Principle].** Let  $f(z)$  be analytic and non-constant on a region  $\Omega$ , then its absolute value attains no minimum or maximum on  $\Omega$ .  $\lrcorner$

*Proof:* This follows easily from Cauchy's integral formula (12.4.2.6). Notice if it attains an extremal at  $z_0 \in \Omega$ , then the Cauchy integral implies that  $f$  is constant on a small circle surrounding  $z_0$ , so it is constant by uniqueness theorem (12.4.2.20).  $\square$

**Prop. (12.4.2.25) [Analytic Functions on Annulus].** If  $0 \leq r < R \leq \infty$  and  $f$  is holomorphic on  $\{z \in \mathbb{C} | r \leq |z| \leq R\}$ , show that  $f$  has a series expansion

$$f = \sum_{n \in \mathbb{Z}} a_n z^n$$

$\lrcorner$

*Proof:* For  $\zeta \in C_R$ ,

$$\frac{1}{\zeta - z} = \frac{1}{\zeta} \cdot \frac{1}{1 - \frac{z}{\zeta}} = \sum_{n \in \mathbb{N}} \frac{z^n}{\zeta^{n+1}},$$

and the convergence is uniform on  $C_R$ , so

$$\frac{1}{2\pi i} \int_{C_R} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n \in \mathbb{N}} \underbrace{\left( \frac{1}{2\pi i} \int_{C_R} \frac{f(\zeta)}{\zeta^{n+1}} d\zeta \right)}_{a_n} z^n.$$

Similarly, for  $\zeta \in C_r$ ,

$$\frac{1}{\zeta - z} = \frac{1}{z} \cdot \frac{1}{1 - \frac{\zeta}{z}} = \sum_{n \in \mathbb{N}} \frac{\zeta^n}{z^{n+1}},$$

and the convergence is uniform on  $C_r$ , so

$$\frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n \in \mathbb{Z}^+} \underbrace{\left( \frac{1}{2\pi i} \int_{C_r} f(\zeta) \zeta^{n-1} d\zeta \right)}_{a_{-n}} z^{-n}.$$

Thus it follows from the Cauchy integral formula (12.4.2.6) that

$$f(z) = \frac{1}{2\pi i} \int_{C_R} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n \in \mathbb{Z}} a_n z^n,$$

and the convergence is uniform on the annulus.  $\square$

## Residues

**Def. (12.4.2.26) [Residues].** If  $\Omega \subset \mathbb{C}$  is a simply-connected region and  $f$  is a function on  $\Omega$  analytic except for isolated singularities  $\{a_j\}$ , then for each  $i$ , if  $\mathbb{D}(a_i, \delta)$  is a nbhd of  $a_i$  contained in  $\Omega$  and contains no other singularities of  $f$ , then define

$$\text{res}_{z=a_i} f = \frac{1}{2\pi i} \int_{\gamma} f(z) dz,$$

where  $\gamma = \partial \mathbb{D}(a_i, \delta)$ . This quantity is invariant of the nbhd chosen by Cauchy's theorem (12.4.2.2), and is called the **residue** of  $f$  at  $a_i$ .  $\lrcorner$



**Thm. (12.4.2.27) [Residue theorem].** If  $\Omega \subset \mathbb{C}$  is a simply-connected region and  $f$  is a function on  $\Omega$  analytic except for isolated singularities  $\{a_j\}$ , then for any piecewise  $C^1$  closed curve  $\gamma \subset \Omega$  not passing through the singularities,

$$\frac{1}{2\pi i} \int_{\gamma} f(z) dz = \sum_i n(\gamma, a_i) \operatorname{res}_{z=a_i} f,$$

where the RHS is a finite sum. ┘

*Proof:* This is because the interior of  $\gamma$  contains only f.m. singularity points, and  $\gamma$  is homologous to a linear combination of cycles around each singularity with multiplicity  $n(\gamma, a_i)$ . □

**Prop. (12.4.2.28).** For  $n \in \mathbb{Z}$ ,

$$\operatorname{res}_{z=0} z^n = \begin{cases} 1 & , n = -1 \\ 0 & , n \neq -1 \end{cases}.$$

In particular, if locally near  $z = 0$ ,  $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$ , then

$$\operatorname{res}_{z=0} f(z) = a_{-1}.$$

┘

*Proof:*

$$\begin{aligned} \int_{|z|=r} z^n dz &= \int_0^{2\pi} z^n(t) z'(t) dt = \int_0^{2\pi} i r^{n+1} e^{i(n+1)t} dt \\ &= \begin{cases} \frac{r^{n+1}}{n+1} e^{i(n+1)t} \Big|_{t=0}^{2\pi} = 0 & , n \neq -1 \\ 2\pi i & , n = -1 \end{cases} \end{aligned}$$

□

**Prop. (12.4.2.29).** If  $\Omega \subset \mathbb{C}$  is a region and  $f$  is a function on  $\Omega$  analytic with singularities. If  $f$  is analytic around  $z_0 \in \Omega$ , then for  $n \in \mathbb{Z}_+$ , by (12.4.2.7),

$$\operatorname{res}_{z=z_0} \frac{f(z)}{(z-z_0)^n} = \frac{f^{(n-1)}(z_0)}{(n-1)!}.$$

┘

**Prop. (12.4.2.30) [Generalized Argument Principle].** If  $\Omega \subset \mathbb{C}$  is a simply-connected region and  $f \in \mathcal{M}(\Omega)$  with zeros  $\{a_i\}$  and poles  $\{b_i\}$ , and  $g(z) \in \mathcal{O}(\Omega)$ , then for any piecewise  $C^1$  closed curve  $\gamma \subset \Omega$  not passing through zeros or poles of  $f$ ,

$$\frac{1}{2\pi i} \int_{\gamma} g(z) \frac{f'(z)}{f(z)} dz = \sum_i n(\gamma, a_i) g(a_i) - \sum_j n(\gamma, b_j) g(b_j)$$

where the RHS is a finite sum, and counted with multiplicity by the order.

Moreover, notice if  $\Gamma$  is the closed curve  $f \circ \gamma$ , then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \frac{1}{2\pi i} \int_{\Gamma} \frac{1}{z} dz = n(\Gamma, 0).$$

┘

*Proof:* Notice if  $z_0 \in \Omega$  and  $f(z) = (z - z_0)^h f_h(z)$  near  $z_0$  where  $h \in \mathbb{Z}$ ,  $f_h(z)$  is analytic on a nbhd of  $z_0$  and  $f_h(z_0) = 0$ , then

$$g(z) \frac{f'(z)}{f(z)} = \frac{hg(z)}{z - z_0} + f'_h(z)g(z),$$

so  $\text{res}_{z=z_0} [g(z) \frac{f'(z)}{f(z)}] = hg(z_0)$  by (12.4.2.29) and (12.4.2.5). Thus the assertion follows from the residue formula (12.4.2.27) applied to the meromorphic function  $F = gf'/f \in \mathcal{M}(\Omega)$ .  $\square$

**Cor. (12.4.2.31) [Rouché].** If  $\Omega \subset \mathbb{C}$  is a region and  $\gamma \subset \Omega$  is a piecewise  $C^1$  closed curve that is a boundary of a subset of  $\Omega$  homeomorphic to a  $\mathbb{D}$ . Suppose  $f, g \in \mathcal{O}(\Omega)$  satisfies  $|f(z) - g(z)| < |f(z)|$  for  $z \in \gamma$ , then  $f(z)$  and  $g(z)$  have the same number of zeros or poles on the interior of  $\gamma$ .  $\lrcorner$

*Proof:* This follows from the argument principle (12.4.2.30) applied to the meromorphic function  $F = f/g \in \mathcal{M}(\Omega)$ . Notice that  $n(\gamma, a) = 1$  for any  $a$  in the interior of  $\gamma$ , and  $n(F \circ \gamma, 0) = 0$  because

$$|F(z) - 1| < 1$$

For  $z \in \gamma$  by hypothesis.  $\square$

### Logarithm

**Def. (12.4.2.32) [Branch of Logarithm].** Let  $\Omega \subset \mathbb{C}^\times$  be a region, a branch of **logarithm** is an analytic function  $\ell(z) \in \mathcal{O}(\Omega)$  s.t.  $e^{\ell(z)} = z$  (12.4.3.6).

By connectedness, If no confusion is made, we will denote any branch of logarithm by  $\log(z)$ .  $\lrcorner$

**Prop. (12.4.2.33).** If  $\Omega \subset \mathbb{C}$  is a simply-connected region, and  $f \in \mathcal{O}(\Omega)$  is non-vanishing, then there exists some  $g \in \mathcal{O}(\Omega)$  s.t.  $f(z) = e^{g(z)}$ .  $\lrcorner$

*Proof:* By (12.4.2.3), there exists  $\ell \in \mathcal{O}(\Omega)$  s.t.  $g' = \frac{f'}{f}$  and we can assume  $e^{g(z_0)} = f(z_0)$  for some  $z_0 \in \Omega$  because  $\exp : \mathbb{C} \rightarrow \mathbb{C}^\times$  is clearly surjective. Then notice  $[f(z) \exp(-g(z))]' = \exp(-g(z))(f'(z) - f(z)g'(z)) = 0$ , so  $f(z) \exp(-g(z)) = 0$ .  $\square$

**Cor. (12.4.2.34) [Existence of Logarithm].** If  $\Omega \subset \mathbb{C}^\times$  be simply-connected region, then there exists a branch of logarithm on  $\Omega$ , and also a branch of  $\sqrt[n]{f}$  on  $\Omega$ .  $\lrcorner$

## 3 Series and Product Developments

*“The letter e may now no longer be used to denote anything other than this positive universal constant.”*

—E. Landau

**Prop. (12.4.3.1) [Abel-Hadamard].** For any power series  $a_0 + a_1z + \dots + a_nz^n + \dots \in \mathbb{C}[[z]]$ , take  $1/R = \overline{\lim} |a_n|^{1/n}$ , where we assume  $1/0 = \infty$  and  $1/\infty = 0$ , then

- The series converges absolutely for every  $|z| < R$ , and if  $\rho < R$ , then the convergence is uniform for  $|z| \leq \rho$ .
- If  $|z| > R$ , the terms are unbounded, and the series diverges.

And  $R$  is called the **radius of convergence** of the sequence. Notice the radius of convergence is the same as that of  $|a_0| + |a_1|z + \dots + |a_n|z^n + \dots$  in  $\mathbb{R}$  (12.3.5.3).  $\lrcorner$

*Proof:* 1: For  $0 < \rho < R$ , take  $\rho_1 \in (\rho, R)$ , then the definition of  $R$  implies that for some  $N \in \mathbb{Z}_+$  and any  $n \geq N$ ,  $|a_n| \leq R^{-n} < \rho_1^{-n}$ , so  $|a_n z^n| \leq (\rho/\rho_1)^n$ , so  $\sum_{n \in \mathbb{N}} a_n z^n$  converges absolutely and uniformly for  $|z| \leq \rho$ .

2: For  $R < |z|$ , the definition of  $R$  implies that for any  $N \in \mathbb{Z}_+$ , there exists  $n \geq N$  s.t.  $|a_n| \geq R^{-n}$ , so  $|a_n z^n| \geq |z/R|^n > 1$ , so this sequence cannot be convergent.  $\square$

**Cor. (12.4.3.2).** By (12.3.5.4) and (12.4.3.1), for any power series  $a_0 + a_1 x + \dots + a_n x^n + \dots$  in  $\mathbb{R}$ , if  $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = \rho \in [0, \infty]$ , then the radius of convergence (12.4.3.1)  $R = 1/\rho$ , where we assume  $1/0 = \infty$  and  $1/\infty = 0$ .  $\lrcorner$

**Prop. (12.4.3.3).** The power series  $a_0 + a_1 z + \dots + a_n z^n + \dots$  defines an analytic function  $f(z)$  on its disk of convergence. And also

$$f'(z) = \sum_{n \in \mathbb{N}} n a_n z^{n-1}.$$

In particular,  $f(z)$  is infinite differentiable in its disk of convergence.  $\lrcorner$

*Proof:* The partial sum converges to  $f$  uniformly on compact subset of the disk of convergence (12.4.3.1), so this follows from (12.4.2.15).  $\square$

**Prop. (12.4.3.4).** Any holomorphic function  $f$  defined on the punctured disk  $0 < |z| < 1$  is of the form

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$$

where  $\overline{\lim}_{n \rightarrow \infty} |a_n|^{1/n} \leq 1$ , and  $\lim_{n \rightarrow \infty} |a_{-n}|^{1/n} = 0$ .  $\lrcorner$

*Proof:*

$\square$

**Prop. (12.4.3.5).** If  $f \in C((0, 1])$  has an expansion

$$f(x) = \sum_{\nu \in \Sigma} a(\nu) x^\nu$$

near  $x = 0$ , where  $\Sigma \subset \mathbb{R}$  is a discrete subset bounded from below, then

$$\int_0^1 f(x) x^s \frac{dx}{x}$$

has meromorphic continuation to all  $s \in \mathbb{C}$ , and has only simple poles at  $s \in -\Sigma$ , and the residue at  $s = -\nu$  is  $a(\nu)$ . Moreover, for any  $N \in \mathbb{R}$ ,  $f$  is essentially bounded on  $\text{Re}(s) > -N$ .  $\lrcorner$

*Proof:* For any  $N \in \mathbb{R}$ , let

$$f(x) = \sum_{\nu \in \Sigma, \nu < N} a(\nu) x^\nu + R(x), \quad R(x) = O(x^N),$$

then

$$\int_0^1 f(x) x^s \frac{dx}{x} = \sum_{\nu \in \Sigma, \nu < N} \frac{a(\nu)}{s + \nu} + \int_0^1 R(x) x^s \frac{dx}{x}.$$

$\square$

**Prop. (12.4.3.6) [Trigonometric Functions].** The exponential function

$$\exp(z) = \sum_{n \in \mathbb{N}} \frac{z^n}{n!} \quad (10.7.1.5)$$

is convergent and analytic on  $\mathbb{C}$ , by (12.4.3.1), and it is also denoted by  $e^z$ . We can also define the **trigonometric functions**

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} = \sum_{n \in \mathbb{N}} (-1)^n \frac{z^{2n}}{(2n)!}, \quad \sin(z) = \frac{e^{iz} - e^{-iz}}{2i} = \sum_{n \in \mathbb{N}} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$

which are analytic on  $\mathbb{C}$ . ┘

**Prop. (12.4.3.7) [Eulerian Identity].** The smallest positive real number  $\rho$  that  $e^{i\rho} + 1 = 0$  is  $\pi$ . In particular, the Eulerian identity  $e^{i\pi} + 1 = 0$  holds. ┘

*Proof:* □

**Prop. (12.4.3.8) [Infinite Products].** For a sequence  $(b_n) \in \mathbb{C} \setminus \{-1\}$ , the infinite product  $\prod_{n \in \mathbb{Z}_+} (1 + b_n)$  is said to **converge** if the sequence

$$\Pi_m = \prod_{n=1}^m (1 + b_n)$$

converges. It is said to **converge absolutely** if  $\prod_{n \in \mathbb{Z}_+} \log(1 + b_n)$  converges absolutely. Then:

- If  $\prod_{n \in \mathbb{Z}_+} (1 + b_n)$  converges absolutely, then  $\prod_{n \in \mathbb{Z}_+} (1 + b_n)$  converges to a non-zero limit.
- $\prod_{n \in \mathbb{Z}_+} (1 + b_n)$  converges absolutely iff  $\sum_{n \in \mathbb{Z}_+} b_n$  converges absolutely.
- If  $(a_n) \in \mathbb{C}$  and  $\sum_{n \in \mathbb{Z}_+} a_n$  converges absolutely, then  $\prod_{n \in \mathbb{Z}_+} (1 + a_n) = 0$  iff  $a_k = -1$  for some  $k$ . ┘

*Proof:* 1:  $\prod_{n \in \mathbb{Z}_+} (1 + b_n) = \exp(\sum_n \log(1 + b_n))$ .

2: It follows from the fact  $\lim_{z \rightarrow 0} \frac{\log(1+z)}{z} = 1$  that there exists a  $0 < \varepsilon < 1/2$  that

$$(1 - \varepsilon)|a_n| < |\log(1 + a_n)| \leq (1 + \varepsilon)|a_n|$$

for  $n$  sufficiently large.

3: By omitting f.m. terms, this follows from item2. □

**Prop. (12.4.3.9).** If  $(F_n)$  is a sequence of holomorphic functions on a region  $\Omega$ , and there exists constants  $c_n > 0$  s.t.

$$\sum c_n < \infty, |F_n(z) - 1| \leq c_n, \forall z \in \Omega,$$

then

- The product  $\prod_n F_n(z)$  converges uniformly on  $\Omega$  to a holomorphic function  $F(z)$ .
- If  $F_n \neq 0$  for any  $n$ , then

$$\frac{F'(z)}{F(z)} = \sum_n \frac{F'_n(z)}{F_n(z)}.$$

- If  $F_n \neq 0$  for any  $n$ , then the zeros of  $F(z)$  are exactly zeros of  $F_n(z)$  (counting multiplicity).

┘

*Proof:* 1: The proof is the same as that of(12.4.3.8), and notice that the convergence is uniform so the resulting function is holomorphic.

2: we can omit f.m. terms, so we may assume that each  $F_n$  is non-vanishing. Let  $G_N(z) = \prod_{k=1}^N F_n(z)$ , then  $G_N(z) \rightarrow F(z)$  uniformly on compact subsets. So by(12.4.2.15),  $G_N^\lambda$  converges to  $F^\lambda(N)$  uniformly on compact subset, and

$$\frac{F^\lambda(z)}{F(z)} = \lim_{N \rightarrow \infty} \frac{G_N^\lambda(z)}{G_N(z)} = \sum_n \frac{F_n^\lambda(z)}{F_n(z)}.$$

3: This is because we can omit f.m. terms and assume that each  $F_n$  is non-vanishing. Then the resulting □

### Partial Fractions

**Def. (12.4.3.10).** ┘

**Prop. (12.4.3.11).**

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n \neq 0 \in \mathbb{Z}} \left( \frac{1}{z-n} + \frac{1}{n} \right) = \frac{1}{z} + \sum_{n \in \mathbb{Z}_+} \frac{2z}{z^2 - n^2} = \frac{1}{z} - 2 \sum_{k \in \mathbb{Z}_+} \zeta(2k) z^{2k-1}.$$

┘

*Proof:* This follows from taking logarithmic derivative of the Hadamard product of  $\sin(\pi z)$ (12.4.3.24). □

**Cor. (12.4.3.12).**

$$\frac{\pi}{\sin(\pi z)} = \frac{1}{z} + \sum_{n \in \mathbb{Z}_+} (-1)^n \left( \frac{1}{z-n} + \frac{1}{z+n} \right).$$

┘

*Proof:* Use the formula  $\frac{1}{\sin(2z)} = \cot(z) - \cot(2z)$ . □

**Cor. (12.4.3.13).**

$$\frac{\pi^2}{\sin^2(\pi z)} = \sum_{n \in \mathbb{Z}} \frac{1}{(z-n)^2}.$$

┘

*Proof:* This follows from taking derivative of(12.4.3.11). □

### Entire Functions

**Thm. (12.4.3.14) [Product Development, Weierstrass].**

- If  $f$  is an entire function, then the zeros of  $f$  is at most countable(counting multiplicity), and if they can be ordered by their modules,  $|a_1| \leq |a_2| \leq \dots \leq |a_n| \leq \dots$  with  $\lim_{n \rightarrow \infty} |a_n| = \infty$ .

- If  $S \subset \mathbb{Z}_+$ , and  $(a_n)_{n \in S} \subset \mathbb{C}$  be a sequence of complex numbers, which satisfies  $\lim_{n \rightarrow \infty} a_n = \infty$  when  $\#S = \infty$ , then every entire function with  $(a_n)$  as zeros(counting multiplicity) can be written in the form

$$f(z) = z^m e^{g(z)} \prod_{n \in S} \left[ \left(1 - \frac{z}{a_n}\right) \exp\left(\frac{z}{a_n} + \frac{1}{2}\left(\frac{z}{a_n}\right)^2 + \dots + \frac{1}{m_n}\left(\frac{z}{a_n}\right)^{m_n}\right) \right]$$

for some sequence  $(m_n)_{n \in S}$  of positive integers. ┘

*Proof:* Firstly, for any such sequence, the RHS converges to an entire function: for any  $z \in \mathbb{C}$ , we can discard f.m. terms s.t.  $|a_n| \leq |z|$ , and for  $|a_n| > |z|$ ,

$$\left(1 - \frac{z}{a_n}\right) \exp\left(\frac{z}{a_n} + \frac{1}{2}\left(\frac{z}{a_n}\right)^2 + \dots + \frac{1}{m_n}\left(\frac{z}{a_n}\right)^{m_n}\right) = \exp\left(-\frac{1}{m_n+1}\left(\frac{z}{a_n}\right)^{m_n+1} - \frac{1}{m_n+2}\left(\frac{z}{a_n}\right)^{m_n+2} - \dots\right)$$

and the module of the exponent is bounded by

$$\frac{1}{m_n+1} \left(\frac{|z|}{|a_n|}\right)^{m_n+1} \left(1 - \frac{|z|}{|a_n|}\right)^{-1}$$

So if we choose  $m_n = n$ , the RHS converges in a nbhd of  $z$ , and it is entire.

Then any such function with the desired zeros differ by the RHS by an entire non-vanishing function, which must be of the form  $\exp(g(z))$  by (12.4.2.34). □

**Cor. (12.4.3.15).** Any entire function on  $\mathbb{C}$  is a quotient of two meromorphic functions. ┘

**Cor. (12.4.3.16).** If  $S \subset \mathbb{Z}_+$ , and  $(a_n)_{n \in S}, (A_n)_{n \in S} \subset \mathbb{C}$  be two sequence of complex numbers s.t.  $a_m \neq a_n$  for  $m \neq n$ , which satisfies  $\lim_{n \rightarrow \infty} a_n = \infty$  when  $\#S = \infty$ , then there exists an entire function  $f(z)$  which satisfies  $f(a_n) = A_n$ . ┘

*Proof:* Let  $g(z)$  be an entire function satisfying  $g(z)$  has simple zeros at  $a_n$ , then consider

$$f(z) = \sum_{n \in S} g(z) \frac{e^{\gamma_n(z-a_n)} A_n}{z - a_n g'(a_n)}.$$

□

**Def. (12.4.3.17) [Genus of Entire Functions and Canonical Products].** The **genus of an entire function**  $f$  is the smallest  $h \in \mathbb{N}$  s.t.  $f$  can be written in the form

$$f(z) = z^m e^{g(z)} \prod_{n \in S} \left[ \left(1 - \frac{z}{a_n}\right) \exp\left(\frac{z}{a_n} + \frac{1}{2}\left(\frac{z}{a_n}\right)^2 + \dots + \frac{1}{h}\left(\frac{z}{a_n}\right)^h\right) \right]$$

where  $S \subset \mathbb{N}, m \in \mathbb{N}, g(z)$  is a polynomial of degree  $\leq h$ . if no such  $h$  exists,  $f$  is said to have genus  $\infty$ .

In view of the Weierstrass theorem (12.4.3.14), this  $h$  is equal to the minimal non-negative integer s.t.

$$\sum_{n \in \mathbb{Z}_+} |a_n|^{-h-1} < \infty.$$

If  $f$  has finite genus, then the canonical form of  $f$  is unique. ┘

**Def. (12.4.3.18) [Order of Growth].** The **order of growth** of an entire function  $f$  is defined to be

$$\lambda = \overline{\lim}_{r \rightarrow \infty} \frac{\log \log \max_{|z|=r} |f(z)|}{\log r},$$

or equivalently the smallest number  $\lambda \in \mathbb{R} \cup \{\infty\}$  s.t.

$$\max_{|z|=r} |f(z)| \leq e^{r^{\lambda+\varepsilon}}$$

for any  $\varepsilon > 0$ .

More generally,  $f \in \mathcal{M}(\mathbb{C})$  is said to have order of growth  $\leq \lambda$  iff it can be written as a quotient  $f = g/h$  where  $g, h \in \mathcal{O}(\mathbb{C})$  both have order of growth  $\leq \lambda$ .  $\lrcorner$

**Lemma (12.4.3.19) [Order of Growth and Roots].** Let  $f$  be an entire function with order of growth  $\rho$ .

- For  $r \in \mathbb{R}_+$ , let  $N(r)$  be the number of zeros of  $f$  in  $\mathbb{D}(0, r)$ , then for any  $\varepsilon > 0$ , there exists  $C > 0$  s.t.  $N(r) \leq Cr^{\rho+\varepsilon}$  for  $r$  sufficiently large.
- Let  $\{a_n\}_{n \in \mathbb{Z}_+}$  be the zeros of  $f$ , then for any  $\varepsilon > 0$ ,  $\sum_{k \in \mathbb{N}} |a_k|^{-\rho-\varepsilon} < \infty$ .

$\lrcorner$

*Proof:* 1: dividing  $f(z)$  by a power of  $z$ , we may assume  $f(0) \neq 0$ . Then it follows from the Jensen formula (12.5.2.11) on the disk  $\mathbb{D}(0, 2\rho)$  that

$$N(r) \log 2 \leq \frac{1}{2\pi} \int_0^{2\pi} \log |f(2re^{i\theta})| d\theta - \log |f(0)| \leq (2r)^{\rho+\varepsilon} - \log |f(0)|.$$

2: By item 1, for  $n$  large,  $n \leq N(|a_n|) \leq |a_n|^{\rho+\varepsilon/2}$ , so

$$\sum |a_n|^{-\rho-\varepsilon} \leq \sum_{n \in \mathbb{Z}_+} |n|^{-\frac{\rho+\varepsilon}{\rho+\varepsilon/2}} < \infty.$$

$\square$

**Lemma (12.4.3.20).** If  $g$  is an entire function on  $\mathbb{C}$ ,  $\rho \in \mathbb{R}_+$  and there is a sequence of positive real numbers  $(r_n)$  that extends to infinity, and  $u = \operatorname{Re}(g)$  satisfies

$$\max_{|z|=r_n} u(z) \leq Cr_n^\rho$$

for each  $n$ , then  $g$  is a polynomial of degree  $\leq \rho$ .  $\lrcorner$

*Proof:* Let  $g(z) = \sum_{n \in \mathbb{N}} a_n z^n$ , then for any  $k \in \mathbb{Z}_+$ , let  $r = r_k$ , by Cauchy's formula

$$\frac{1}{2\pi} \int_0^{2\pi} g(re^{i\theta}) e^{-im\theta} d\theta = \begin{cases} a_m r^m & , m \in \mathbb{N} \\ 0 & , n \in \mathbb{Z}_- \end{cases},$$

so by taking conjugation and adding,

$$\frac{1}{\pi} \int_0^{2\pi} \operatorname{Re}(g)(re^{i\theta}) e^{-in\theta} d\theta = \begin{cases} a_n r^n & , n \in \mathbb{Z}_+ \\ \operatorname{Re}(a_0) & , n = 0 \end{cases}.$$

Then for  $n \in \mathbb{Z}_+$ ,

$$|a_n| = \frac{1}{\pi r^n} \left| \int_0^{2\pi} [u(re^{i\theta}) - Cr^\rho] e^{-in\theta} d\theta \right| \leq \frac{1}{\pi r^n} \int_0^{2\pi} [Cr^\rho - u(re^{i\theta})] d\theta = 2Cr^{\rho-n} - 2\operatorname{Re}(a_0)r^{-n}.$$

Then letting  $r = r_k \rightarrow \infty$  finishes the proof.  $\square$

**Thm. (12.4.3.21)[Hadamard].** The genus  $h$  and the order of growth  $\lambda$  (12.4.3.18) of an entire function  $f$  satisfies

$$0 \leq h \leq \lambda \leq h + 1 \leq \infty.$$

(Notice if  $\lambda \in \mathbb{Z}$ , it might not be able to determine  $h$  from  $\lambda$ ). ┘

*Proof:* It follows from the proof of Weierstrass theorem (12.4.3.14) that  $f$  is of genus  $h < \infty$ , then  $\lambda \leq h + 1$  (taking  $m_n = h$  for each  $n$ ). Conversely, if  $f$  has order of growth  $\lambda < \infty$ , then we need to show that  $h \leq \lambda$ . Take  $h = \lfloor \lambda \rfloor$ , and let  $(a_n)$  are zeros of  $f$  (counting multiplicity and ordered by modulus), then firstly, we show that the product

$$\prod_{n \in S} \left[ \left(1 - \frac{z}{a_n}\right) \exp\left(\frac{z}{a_n} + \frac{1}{2}\left(\frac{z}{a_n}\right)^2 + \dots + \frac{1}{h}\left(\frac{z}{a_n}\right)^h\right) \right]$$

converges: For this, by the proof of Weierstrass theorem (12.4.3.14), it suffices to show that  $\sum_{n \in S} |a_n|^{-h-1}$  converges, and this follows from (12.4.3.19).

Then it's left to show that the entire function  $g(z)$  as in (12.4.3.14) is a polynomial of degree  $\leq h$ . For this, firstly we prove that if  $\varepsilon > 0$  is small that  $\rho + \varepsilon < h + 1$ , and  $|z - a_n| \geq |a_n|^{h+1}$  for any  $n \in S$ , then there exists  $C \in \mathbb{R}_+$  s.t.

$$\left| \prod_{n \in S} \left[ \left(1 - \frac{z}{a_n}\right) \exp\left(\frac{z}{a_n} + \frac{1}{2}\left(\frac{z}{a_n}\right)^2 + \dots + \frac{1}{h}\left(\frac{z}{a_n}\right)^h\right) \right] \right| \geq e^{-C|z|^{\rho+\varepsilon}}.$$

Deonte  $E_h(s) = (1 - s) \exp\left(s + \frac{1}{2}s^2 + \dots + \frac{1}{h}s^h\right)$ , we use a lemma:

**Lemma (12.4.3.22).** there exists constant  $C$  s.t.

- If  $|s| \leq 1/2$ ,  $|E_h(s)| \geq e^{-C|z|^{h+1}}$ .
- If  $|s| \geq 1/2$ ,  $|E_h(s)| \geq |1 - s|e^{-C|z|^h}$ .

┘

*Proof:* For  $|s| \geq 1/2$ ,

$$|E_h(s)| \geq |(1 - s)|e^{-|s| - \frac{|s|^2}{2} - \dots - \frac{|s|^h}{h}} \geq |1 - s|e^{-C|z|^h}.$$

And for  $|s| \leq 1/2$ ,

$$|E_h(s)| \geq |(1 - s)|e^{-\frac{|s|^{h+1}}{h+1} - \frac{|s|^{h+2}}{h+2} - \dots} \geq e^{-C|z|^{h+1}}.$$

□

Then if  $|a_n| \geq 2|z|$ ,

$$\left| E_h\left(\frac{z}{a_n}\right) \right| \geq e^{-C|z/a_n|^{h+1}} \geq e^{-C|z/a_n|^{\rho+\varepsilon}},$$

and if  $|a_n| \leq 2|z|$ , by the hypothesis,

$$\left| E_h\left(\frac{z}{a_n}\right) \right| \geq \left| \left(1 - \frac{z}{a_n}\right) \right| e^{-C|z/a_n|^h} \geq |a_n|^{-h-2} e^{-C|z/a_n|^{\rho+\varepsilon}}.$$

Thus by (12.4.3.19),

$$\prod_{n \in S} \left| E_h\left(\frac{z}{a_n}\right) \right| \geq e^{-C'z} \prod_{|a_n| \leq 2|z|} |a_n|^{-h-2} \geq e^{-C'z} |2z|^{-(h+2)N(2|z|)} \geq e^{-C''|z|^{\rho+\varepsilon}'}$$



for any  $\varepsilon' > \varepsilon$ .

Finally, we get

$$e^{\operatorname{Re}(g(z))} = |e^{g(z)}| = \left| \frac{f(z)}{\prod_{n \in S} |E_h(\frac{z}{a_n})|} \right| \leq C e^{\rho + \varepsilon}$$

whenever  $|z - a_n| \geq |a_n|^{h+1}$  for any  $n \in S$ . Then as  $\sum_n |a_n|^{h+1} < \infty$ , we can apply (12.4.3.20) to show that  $g$  is a polynomial of degree  $\leq \rho + \varepsilon < h + 1$ , thus we are done.  $\square$

**Cor. (12.4.3.23).** For an entire function  $f$  with order  $\lambda \in \mathbb{R}_+ \setminus \mathbb{Z}_+$ , then  $\#f^{-1}(a) = \infty$  for any  $a \in \mathbb{C}$ .  $\perp$

*Proof:* As  $f$  and  $f - a$  has the same order, it suffices to show that such an  $f$  has i.m. zeros. Suppose it has only f.m. zeros, then we can divide a polynomial  $P(z)$  and see that  $F(z) = f(z)/P(z)$  also has the same order but no zero. Thus  $F(z) = e^{g(z)}$  for some entire  $g$ . But then  $g$  is a polynomial of degree  $\lambda$ , which is impossible.  $\square$

**Example (12.4.3.24) [Canonical Forms].**

$$\sin(\pi z) = \pi z \prod_{n \in \mathbb{Z}^\times} \left[ \left(1 - \frac{z}{n}\right) e^{z/n} \right] = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

is of genus 1 and order of growth 1.  $\perp$

*Proof:*  $|\sin(\pi z)| \leq e^{\pi|z|}$ , so it has order of growth  $\leq 1$ , and it has a simple zero at  $z = 0$ , so by Hadamard's theorem (12.4.3.21),

$$\sin(\pi z) = \pi z e^{Az+B} \prod_{n \neq 0 \in \mathbb{Z}} \left[ \left(1 - \frac{z}{n}\right) e^{z/n} \right] = \pi z e^{Az+B} \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

for some  $A, B \in \mathbb{C}$ . Letting  $z \rightarrow 0$  implies that  $B = 0$ . And letting  $z \rightarrow 1$  implies that

$$e^A = 2 \prod_{n=1}^{\infty} \left(1 - \frac{1}{n^2}\right) = 1,$$

so  $A = 0$ .  $\square$

## 4 Analytic Continuations

**Lemma (12.4.4.1) [Symmetry Principle].** If  $\Omega \subset \mathbb{C}$  is a region s.t.  $c(\Omega) = \Omega$ , denote  $\Omega^+ = \Omega \cap \mathcal{H}$ ,  $\Omega^- = c(\Omega^+)$ ,  $I = \Omega \cap \mathbb{R}$ . Suppose  $f^+ \in \mathcal{O}(\overline{\Omega^+})$ ,  $f^- \in \mathcal{O}(\overline{\Omega^-})$ , and  $f^+(z) = f^-(z)$  for  $z \in \mathbb{R}$ , then the function

$$f : \Omega \rightarrow \mathbb{C} : z \mapsto \begin{cases} f^+(z) & , z \in \overline{\Omega^+} \\ f^-(z) & , z \in \Omega^- \end{cases}$$

is analytic on  $\Omega$ .  $\perp$

*Proof:*  $f$  is clearly continuous. Thus it is easily seen to be analytic by Morera's theorem (12.4.2.8).  $\square$

**Prop. (12.4.4.2) [Schwarz Reflection Principle].** If  $\Omega \subset \mathbb{C}$  is a region s.t.  $c(\Omega) = \Omega$ , denote  $\Omega^+ = \Omega \cap \mathcal{H}$ ,  $\Omega^- = c(\Omega^+)$ ,  $I = \Omega \cap \mathbb{R}$ . Suppose  $f^+ \in \mathcal{O}(\overline{\Omega^+})$  satisfies  $f(I) \subset \mathbb{R}$ , then  $f^+$  can be analytically extended to an analytic function on  $\Omega$ .  $\perp$

*Proof:* For  $z \in \Omega^-$ , define  $f(z) = \overline{f(\bar{z})}$ , then  $f$  is a continuous function on  $\Omega$ , because for  $z \in I$ ,  $f(z) = \overline{f(\bar{z})}$ . To show that  $f$  is analytic, by (12.4.4.1), it suffices to show that  $f$  is analytic on  $\Omega^-$ : For any  $z_0 \in \Omega^-$ , if  $z$  is close to  $z_0$ , then  $\bar{z}, \bar{z}_0 \in \Omega^+$ . By Taylor expansion of  $f$  around  $\bar{z}_0$  (12.4.2.16)

$$f(z) = \overline{f(\bar{z})} = \overline{\sum_{n \in \mathbb{N}} a_n(\bar{z} - \bar{z}_0)^n} = \sum_{n \in \mathbb{N}} \bar{a}_n(z - z_0)^n.$$

And this is a power series around  $z_0$  with the same radius of convergence as that of  $\bar{z}_0$ , thus  $f$  is analytic around  $z_0$ . Because  $z_0$  is arbitrary,  $f$  is analytic on  $\Omega^-$ .  $\square$

**Cor. (12.4.4.3).** If  $\Omega \subset \mathbb{C}^\times$  is a region that is stable under the involution  $\iota : z \mapsto \bar{z}^{-1}$ , denote  $\Omega^+ = \Omega \cap \mathcal{H}$ ,  $\Omega^- = \iota(\Omega^+)$ ,  $I = \Omega \cap \partial\mathbb{D}$ . Suppose  $f^+ \in \mathcal{O}(\Omega^+)$  satisfies  $f(I) \subset \mathbb{R}$ , then  $f^+$  can be analytically extended to an analytic function on  $\Omega$ .  $\lrcorner$

*Proof:* For  $z \in \Omega^-$ , define  $f(z) = \overline{f(\iota(z))}$ , then  $f$  is a continuous function on  $\Omega$ , because for  $z \in I$ ,  $f(z) = \overline{f(\iota(z))}$ . To show that  $f$  is analytic, by a similar lemma as (12.4.4.1), it suffices to show that  $f$  is analytic on  $\Omega^-$ : For any  $z_0 \in \Omega^-$ , if  $z$  is close to  $z_0$ , then  $\iota(z), \iota(z_0) \in \Omega^+$ . By Taylor expansion of  $f$  around  $\iota(z_0)$  (12.4.2.16)

$$f(z) = \overline{f(\iota(z))} = \overline{\sum_{n \in \mathbb{N}} a_n(\iota(z) - \iota(z_0))^n} = \sum_{n \in \mathbb{N}} \frac{\bar{a}_n}{z_0} \frac{(z - z_0)^n}{z^n} = g\left(\frac{1}{z}\right),$$

where  $g(z) = \sum_{n \in \mathbb{N}} (-1)^n \bar{a}_n (z - \frac{1}{z_0})^n$  is a power series around  $z_0$  with the same radius of convergence as that of  $\bar{z}_0$ , so analytic around  $z_0^{-1}$ . Thus  $f$  is analytic around  $z_0$ . Because  $z_0$  is arbitrary,  $f$  is analytic on  $\Omega^-$ .  $\square$

## 5 Theorems

**Prop. (12.4.5.1) [Runge's Theorem].** Let  $K$  be a compact subset of  $\bar{\mathbb{C}}$  and let  $f$  be a function which is holomorphic on an open set containing  $K$ . If  $A$  is a set containing at least one complex number from every bounded connected component of  $\bar{\mathbb{C}} \setminus K$ , then there exists a sequence of rational functions which converges uniformly to  $f$  on  $K$  and all the poles of the functions are in  $A$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (12.4.5.2) [Mergelyan].** If  $K$  is compact in  $\mathbb{C}$  and  $f$  is a continuous function on  $K$  that is holomorphic in  $\text{int}(K)$ , then  $f$  can be uniformly approximated by polynomials.  $\lrcorner$

**Prop. (12.4.5.3) [Weierstrass].** For a ascending sequence of regions  $\Omega_1 \subset \Omega_2 \subset \dots$ ,  $\cup_n \Omega_n = \Omega$ , and  $f_n$  is analytic on  $\Omega_n$ , and  $f_n(z)$  converges to a function  $f(z)$  in the compact-open topology, then  $f(z)$  is also analytic, and moreover,  $f'_n(z)$  converges to  $f'(z)$  in the compact-open topology.  $\lrcorner$

*Proof:* The analyticity follows from Morera's theorem (12.4.2.8) as the integration on a closed curve commutes with uniform convergence, the same argument applied to the limit of equations

$$f'_n(z) = \frac{1}{2\pi i} \int_{\partial\mathbb{D}(0,r)} \frac{f_n(\zeta) d\zeta}{(\zeta - z)^2}$$

shows that the derivative also converges, and uniformly on  $\overline{\mathbb{D}(0, \rho)}$  for  $\rho < r$ .  $\square$

**Cor. (12.4.5.4) [Hurwitz].** Cf. [Ahlfors P178].  $\lrcorner$

**Thm. (12.4.5.5) [Montel].** If  $\Omega \subset \mathbb{C}$  is region and  $\mathcal{S} = \{f_\alpha\}$  is a set of holomorphic functions on  $\Omega$  bounded in the topology of  $H(\Omega)$ , i.e. inter convex uniform convergence, then  $\mathcal{S}$  is sequentially compact in  $H(\Omega)$ . Equivalently,  $\mathcal{O}(\Omega)$  has the Heine-Borel property.  $\lrcorner$

*Proof:* By Arzela-Ascoli(4.4.8.28) that it suffices to show that  $\mathcal{S}$  is uniformly bounded on any compact subset  $K \subset \Omega$ . Choose  $\delta$  small s.t.

$$K_0 = \{z \in \mathbb{C} | d(z, K) \leq 2\delta\} \subset \Omega,$$

then  $|f_\alpha(z)| \leq M$  for any  $f_\alpha \in \mathcal{S}$  and  $z \in K_0$  for some  $M$ . So by Cauchy's formula,  $|f'_\alpha(z)| \leq \frac{M}{\delta}$  for any  $z \in K$ ,  $f_\alpha \in \mathcal{S}$ . Thus it is clear  $\mathcal{S}$  are equicontinuous on  $K$ .  $\square$

**Thm. (12.4.5.6) [Little Picard Theorem].** The image of a non-constant entire function  $f : \mathbb{C} \rightarrow \mathbb{C}$  is either  $\mathbb{C}$  or  $\mathbb{C}$  with one point omitted.  $\lrcorner$

*Proof:* The modular curve  $Y(2)$  is the sphere minus three points(28.3.3.4)(26.3.3.14)(26.3.3.16), and the linear transformations of  $S^1$  is triply transitive, thus we can assume  $f$  is a analytic map  $\mathbb{C} \rightarrow Y(2)$ . As  $\mathbb{C}$  is simply connected, we can lift this map to the covering of  $Y(2)$ , which is  $\mathcal{H}$ . But  $\mathcal{H}$  is biholomorphic to the open disk, but a bounded entire function is constant(12.4.2.10), so  $f$  is constant.  $\square$

**Thm. (12.4.5.7) [Phragmén-Lindelöf].** Let  $f$  be a function that is holomorphic on  $\{\sigma + it \in \mathbb{C} : \sigma_1 \leq \sigma \leq \sigma_2, t \geq c\}$ , and  $f(\sigma + it) = O(e^{t^\alpha})$  for some  $\alpha > 0$  uniformly for any  $\sigma_1 \leq \sigma \leq \sigma_2$ .

Suppose further that there exist  $M_a, M_b \in \mathbb{R}_+$  s.t.

$$f(a + it) \leq M_a(1 + |t|)^\alpha, \quad f(b + it) \leq M_b(1 + |t|)^\beta,$$

then

$$f(\sigma + it) \leq M_a^{\ell(\sigma)} M_b^{1-\ell(\sigma)} (1 + |t|)^{\alpha\ell(\sigma) + \beta(1-\ell(\sigma))},$$

where  $\ell$  is the linear function on  $\mathbb{R}$  s.t.  $\ell(a) = 1$  and  $\ell(b) = 0$ .  $\lrcorner$

*Proof:* First assume that  $M_a = M_b = M$  and  $\alpha = \beta = 0$ , then there exists  $M$  that  $|\varphi(s)| \leq M$  for  $\text{Re}(s) = \sigma_i$ . Now let  $m \equiv 2 \pmod{4}$  and  $m > \alpha$ , then  $\text{Re}((\sigma + i\tau)^m)$  is a polynomial of  $\sigma$  and  $\tau$  with highest term of  $\tau$  being  $-\tau^m$ , so we have

$$\text{Re}(s^m) = -\tau^m + O(|\tau|^{m-1}), \quad |\tau| \rightarrow \infty$$

uniformly on the strip. So  $\text{Re}(s^m)$  has an upper bound  $N$  on the strip. Thus for any  $\varepsilon > 0$ ,

$$|f(s)e^{\varepsilon s^m}| \leq M e^{\varepsilon N}$$

on the boundary of the strip and

$$|f(s)e^{\varepsilon s^m}| = O(e^{|t|^\alpha - \varepsilon|t|^m})$$

uniformly on the strip, thus converges uniformly to 0 as  $|\text{Im}(s)| \rightarrow \infty$ .

Then we can use maximum principle to see that

$$|f(s)e^{\varepsilon s^m}| \leq M e^{\varepsilon N}$$

on the strip. Let  $\varepsilon \rightarrow 0$ , we get  $|\varepsilon(s)| \leq M$ , thus the theorem.

In general, define  $\psi(s) = (s - \sigma_1 + 1)^b$ ?, then  $|\psi(s)| = |s - \sigma_1 + 1|^b \sim |\tau|^b$  when  $|\tau| \rightarrow \infty$ . Thus  $f_1(s) = f(s)/\psi(s)$  satisfies the same condition as  $\varphi$  with  $b = 0$ , so  $f(s)/\psi(s)$  is bounded on the strip, and thus  $|f(s)| = O(|t|^b)$ .  $\square$

**Remark (12.4.5.8).** Some condition on the growth rate of  $|f(\sigma + it)|$  is necessary, otherwise we can consider  $e^{iz}$  on the strip  $-\frac{\pi}{2} \leq \text{Re}(z) \leq \frac{\pi}{2}$ , then it is bounded for  $\text{Re}(z) = \pm \frac{\pi}{2}$ , but not bounded for  $-\frac{\pi}{2} < \text{Re}(z) < \frac{\pi}{2}$ .  $\lrcorner$

## 6 Calculating Definite Integrals

### 1

**Prop. (12.4.6.1).** If  $f$  has a primitive  $F$ , then for any arc  $\gamma$ ,

$$\int_{\gamma} f(z) dz = F(\gamma(1)) - F(\gamma(0)).$$

┘

**Example (12.4.6.2).** For  $a > 0$ , evaluate the integrals

$$\int_0^{\infty} e^{-ax} \cos(bx) dx, \quad \int_0^{\infty} e^{-ax} \sin(bx) dx.$$

┘

*Proof:*

$$\int_0^{\infty} e^{-(a+bi)z} dz = \lim_{R \rightarrow \infty} \left. \frac{-1}{a+bi} e^{-(a+bi)z} \right|_0^R = \frac{1}{a+bi} - \lim_{R \rightarrow \infty} \frac{1}{a+bi} e^{-(a+bi)R}.$$

As  $|\frac{1}{a+bi} e^{-(a+bi)R}| = \frac{1}{\sqrt{a^2+b^2}} e^{-aR} \rightarrow 0$  as  $R \rightarrow \infty$ ,

$$\int_0^{\infty} e^{-(a+bi)z} dz = \frac{1}{a+bi}.$$

Then taking the real and imaginary part, we see that

$$\int_0^{\infty} e^{-ax} \cos(bx) dx = \frac{a}{a^2 + b^2}, \quad \int_0^{\infty} e^{-ax} \sin(bx) dx = \frac{b}{a^2 + b^2}.$$

□

### From Real to Complex

**Prop. (12.4.6.3).** For any  $\xi \in \mathbb{C}$ ,

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{2\pi i x \xi} dx.$$

┘

*Proof:* Firstly we show that both sides are holomorphic functions in  $\xi$ : The LHS is clear. For the RHS, by (12.4.2.14), it suffices to show that

$$\xi \mapsto \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi x \operatorname{Im}(\xi)} dx$$

is a uniformly convergent on compact subsets. But because  $\operatorname{Im}(z)$  is bounded on compact subsets, it suffices to show that  $\int_{-\infty}^{\infty} e^{-\pi x^2} e^{Ax} dx$  is convergent for any  $A \in \mathbb{R}$ . And by a change of variable, it suffices to notice that  $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$ , by [Stein]P42.

It has been proven in Stein's book Page42? that the assertion is true for  $\xi \in \mathbb{R}$ . Then as both sides are holomorphic functions in  $\xi$ , this is true for all  $\xi \in \mathbb{C}$ , by the uniqueness theorem. □

**Cor. (12.4.6.4).** For any  $a \in \mathbb{C}$ ,  $\operatorname{Re} a > 0$  and  $\xi \in \mathbb{C}$ ,

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\xi x - \frac{1}{2}ax^2} dx = \frac{1}{\sqrt{a}} e^{-\frac{1}{2a}\xi^2}.$$

┘

*Proof:* Firstly we show that both sides are holomorphic functions in  $a$ : The RHS is clear. For the LHS, by (12.4.2.14), it suffices to show that

$$a \mapsto \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\xi x - \frac{1}{2}ax^2} dx$$

is a uniformly convergent on compact subsets. But because  $\operatorname{Re}(a) > 0$  is bounded below on compact subsets, it suffices to show that  $\int_{-\infty}^{\infty} e^{-Ax^2 - Bx} dx$  is convergent for any  $A \in \mathbb{R}_+$ . And by a change of variable, it suffices to notice that  $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$ , by [Stein]P42.

It follows from (12.4.6.3) and a change of variable that the assertion is true for  $a \in \mathbb{R}_+$ . Then as both sides are holomorphic functions in  $\xi$ , this is true for all  $a$ , by the uniqueness theorem.  $\square$

**Prop. (12.4.6.5).** For  $\alpha \in \mathbb{C}$  s.t.  $\operatorname{Re}(\alpha^2) \geq 0$ ,

$$\int_0^{\infty} e^{-\alpha^2 x^2} dx = \frac{\sqrt{\pi}}{2\alpha}.$$

┘

*Proof:* By a change of variable, it suffices to show for  $\alpha = e^{i\theta}$ . But in this case,

$$\alpha \int_0^R e^{-\alpha^2 x^2} dx - \int_0^R e^{-x^2} dx = \int_{\gamma} e^{-z^2} dz$$

where  $\gamma : t \in [0, \theta] \mapsto Re^{it}$ . But the RHS is bounded by

$$R \int_0^{\pi/4} e^{-R^2 \sin(2\theta)} d\theta \leq R(\delta + \frac{\pi}{4} e^{-R^2 \sin(2\delta)})$$

for any  $\delta \in (0, \pi/4)$ . If we take  $\delta = R^{-3/2}$ , then for  $R$  large, the RHS is bounded by

$$\frac{1}{\sqrt{R}} + \frac{\pi R}{2} e^{-\sqrt{R}}$$

which tends to 0. Thus the integral

$$\lim_{R \rightarrow \infty} \alpha \int_0^R e^{-\alpha^2 x^2} dx = \lim_{R \rightarrow \infty} \alpha \int_0^R e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \text{ (12.4.6.3)}.$$

$\square$

### Rational Functions

**Prop. (12.4.6.6).** Let  $f, g$  be real polynomials, and  $\deg(g) \geq \deg(f) + 2$ , and  $g(z)$  has no zeros on the real line. Then

$$\int_{-\infty}^{\infty} \frac{f(x)}{g(x)} dx$$

can be calculated. ┘

*Proof:* As the integral is absolutely convergent, we can take  $R \in \mathbb{R}$ , and  $\gamma$  to be the loop which consists of the line from  $-R$  to  $R$  and the hemisphere with origin 0 and radius  $R$  from  $R$  to  $-R$ , then when  $R$  is large, we can assume all the zeros of  $g(z)$  in  $\mathcal{H}$  is enclosed by  $\gamma$ . Then the integral equals  $2\pi i$  times the sum of residues of  $\frac{f(z)}{g(z)}$  at the zeros of  $g(z)$  in  $\mathcal{H}$ . □

**Example (12.4.6.7).** Evaluate the integrals

$$\int_{-\infty}^{\infty} \frac{1}{1+x^4} dx.$$

What are the poles of  $1/(1+z^4)$ ? ┘

*Proof:*  $g(z) = 1 + z^4$  has two zeros  $e^{\frac{\pi}{4}i}$  and  $e^{-\frac{3\pi}{4}i}$  on the upper half plane, so the integral equals

$$\int_{\gamma_1} \frac{(x - e^{\frac{\pi}{4}i})^{-1}(x - e^{\frac{5\pi}{4}i})^{-1}(x - e^{\frac{7\pi}{4}i})^{-1}}{x - e^{\frac{3\pi}{4}i}} + \int_{\gamma_2} \frac{(x - e^{\frac{3\pi}{4}i})^{-1}(x + e^{\frac{5\pi}{4}i})^{-1}(x - e^{\frac{7\pi}{4}i})^{-1}}{x - e^{\frac{\pi}{4}i}}$$

which equals  $2\pi i$  times

$$\left( \frac{x^4 + 1}{x - e^{\frac{\pi}{4}i}} \right)^{-1} \Big|_{x=e^{\frac{\pi}{4}i}} + \left( \frac{x^4 + 1}{x - e^{\frac{3\pi}{4}i}} \right)^{-1} \Big|_{x=e^{\frac{3\pi}{4}i}}$$

so it equals

$$(2\pi i) \left[ \frac{1}{4e^{\frac{3\pi}{4}i}} + \frac{1}{4e^{\frac{9\pi}{4}i}} \right] = \frac{\pi}{\sqrt{2}}$$

□

**Prop. (12.4.6.8)[Fractional Powers].** Let  $f, g$  be real polynomials, and  $\deg(g) \geq \deg(f) + 2$ , and  $g(z)$  has at most one simple zero at the origin and no other zeros, then the integral

$$\int_0^{\infty} x^{\alpha} \frac{f(x)}{g(x)} dx, 0 < \alpha < 1$$

can be evaluated. ┘

*Proof:* The integral equals  $2 \int_0^{\infty} t^{2\alpha+1} \frac{f(t^2)}{g(t^2)} dt$ , and we can choose a branch of  $z^{\alpha}$  that is defined on  $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$ . Then we can integrate along the loop in  $\mathcal{H}$  that consists of two hemisphere of radius  $\varepsilon, R$  centered at the origin,  $[\varepsilon, R] \cup [-R, -\varepsilon]$ . Then when  $R \rightarrow \infty$  and  $\varepsilon \rightarrow 0$ , the integration on the hemisphere tends to 0, and the integration on  $[\varepsilon, R] \cup [-R, -\varepsilon]$  tends to

$$\int_{-\infty}^{\infty} z^{2\alpha+1} \frac{f(z^2)}{g(z^2)} dz = (1 - e^{2\pi i \alpha}) \int_0^{\infty} t^{2\alpha+1} \frac{f(t^2)}{g(t^2)} dt.$$

□

4

**Prop. (12.4.6.9).** Let  $f, g$  be real polynomials, and  $\deg(g) \geq \deg(f) + 1$ , and  $g(z)$  has no zeros on the real line, then

$$\int_{-\infty}^{\infty} \frac{f(x)}{g(x)} e^{ix} dx, \quad \int_{-\infty}^{\infty} \frac{f(x)}{g(x)} \cos(x) dx, \quad \int_{-\infty}^{\infty} \frac{f(x)}{g(x)} \sin(x) dx$$

can be calculated. ┘

*Proof:* If  $\deg(g) \geq \deg(f) + 2$ , then as before we can use the same hemisphere methods to evaluate, because on the hemisphere, we can bound

$$\left| \int_{\gamma} \frac{f(Re^{i\theta})}{g(Re^{i\theta})} iRe^{iz} d\theta \right| \leq C \int_{\gamma} \frac{e^{-y}}{R} d\theta \leq C \int_{\gamma} \frac{1}{R} d\theta \leq \frac{2\pi C}{R}$$

which converges 0 as  $R \rightarrow \infty$ .

But if  $\deg(g) = \deg(f) + 1$ , this hemisphere integral is no longer converging to 0, so we try another method: Integrate around the square with vertices

$$\{(X_1, 0), (X_1, X_1 + X_2), (-X_2, X_1 + X_2), (-X_2, 0)\}.$$

The integral on the vertical lines are bounded by

$$C \int_0^{X_1+X_2} e^{-y} \frac{dy}{|z|} \leq C \frac{1}{X_2} \int_0^{X_1+X_2} e^{-y} dy \leq \frac{C}{X_2}$$

And the same is true for the left vertical lines. And the horizontal line is bounded by

$$C \int_{-X_1}^{X_2} \frac{e^{-(X_1+X_2)}}{X_1 + X_2} dt = Ce^{-(X_1+X_2)}$$

and all these converge to 0 as  $X_1, X_2 \rightarrow \infty$ . □

**Prop. (12.4.6.10).** Show that

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

┘

*Proof:* To do this one, we need a variation of the technique from above. Choose the semi-contour, then it equals the half of the integral

$$\int_{\gamma} \frac{e^{iz}}{z} dz = 2\pi i$$

minus the integral

$$\int_{\gamma} \frac{e^{iz}}{z} dz$$

where  $\gamma : [0, \pi] \rightarrow \mathbb{C} : t \mapsto \varepsilon e^{it}$ . So the integral equals

$$\int_0^{\pi} e^{iz} i dz$$

which converges to  $\pi i$ . □

## 7 Biholomorphisms

**Def. (12.4.7.1) [Biholomorphisms].** For regions  $U, V \subset \mathbb{C}$ , a **biholomorphism** from  $U$  to  $V$  is a holomorphic map  $f : U \rightarrow V$  s.t. there is an inverse  $g : V \rightarrow U$  which is also holomorphic, and  $f \circ g = \text{id}_V, g \circ f = \text{id}_U$ .  $\lrcorner$

**Prop. (12.4.7.2) [Analytic Functions are Open].** Let  $\Omega \subset \mathbb{C}$  be a region, then any non-constant  $f \in \mathcal{O}(\Omega)$  defines an open map  $\Omega \rightarrow f(\Omega)$ .  $\lrcorner$

*Proof:* Let  $z_0 \in \Omega, w_0 = f(z_0)$ . by (12.4.2.20), we can find  $\varepsilon > 0$  s.t.  $\mathbb{D}(z_0, 2\varepsilon) \subset U$ , and  $z_0$  is the only zero of  $f - w_0$  in  $\mathbb{D}(z_0, 2\varepsilon)$ . Let  $\gamma$  be the circle  $|z - z_0| = \varepsilon$ , and  $\Gamma = f(\gamma)$ . Then because  $w_0 \notin \Gamma$ , there exists  $\delta$  s.t.  $\mathbb{D}(w_0, \delta) \cap \Gamma = \emptyset$ . Then by Rouché's theorem (12.4.2.31), for any  $w \in \mathbb{D}(w_0, \delta)$ ,  $f - w$  has a zero  $z \in \mathbb{D}(z_0, \varepsilon)$ . Thus  $f(\mathbb{D}(z_0, \varepsilon)) \supset \mathbb{D}(w_0, \delta)$ , and this implies  $f$  is open.  $\square$

**Prop. (12.4.7.3) [Local Biholomorphisms].** A holomorphic map  $f : U \rightarrow V \subset \mathbb{C}$  is a local bijection iff  $f'(z) \neq 0$  for any  $z \in U$ .  $\lrcorner$

*Proof:* For  $z_0 \in U$ , let  $F(z) = f(z) - f(z_0)$ , then by discreteness of the zeros, there exists  $\delta > 0$  s.t.  $\mathbb{D}(z_0, \delta) \subset U$  and  $F(z) \neq 0$  for  $0 < |z - z_0| < \delta$ . Because  $\{F(z) : |z - z_0| = \delta\}$  is compact, there exists some  $\varepsilon > 0$  s.t.  $\mathbb{D}(f(z_0), \varepsilon) \subset V$  and  $|F(z)| \geq \varepsilon$  for any  $|z - z_0| = \delta$ .

If  $F(z)$  has zero of order exactly 1 at  $z_0$ , then for  $\xi \in \mathbb{D}(0, \varepsilon)$ , by Rouché's theorem (12.4.2.31),  $F(z) - \xi$  has exactly 1 zeros in  $\mathbb{D}(z_0, \delta)$ , which implies  $F^{-1}(\mathbb{D}(0, \varepsilon)) \xrightarrow{F} \mathbb{D}(0, \varepsilon)$  is a bijection, and  $z_0 \in F^{-1}(\mathbb{D}(0, \varepsilon))$ . Thus  $F$  and also  $f$  is a local bijection at  $z_0$ .

If  $F(z)$  has zero of order  $d \geq 1$  at  $z_0$ , then for  $\xi \in \mathbb{D}(0, \varepsilon)$ , by Rouché's theorem (12.4.2.31),  $F(z) - \xi$  has exactly  $d$  zeros in  $\mathbb{D}(z_0, \delta)$ , which implies  $F^{-1}(\mathbb{D}(0, \varepsilon')) \xrightarrow{F} \mathbb{D}(0, \varepsilon')$  is a  $d$ -fold covering for any  $\varepsilon' < \varepsilon$ , so  $F$  thus also  $f$  can never be a local bijection at  $z_0$ .

Thus  $f$  is a local bijection at  $z_0$  iff  $f'(z_0) \neq 0$  has zero of order exactly 1 at  $z_0$ , which is also clearly equivalent to  $f'(z_0) \neq 0$ . So the assertion is true.  $\square$

**Cor. (12.4.7.4).** If  $U, V$  are regions of  $\mathbb{C}$  and  $f : U \rightarrow V$  is holomorphic and bijective, then  $f'(z) \neq 0$  for any  $z \in U$ , and the inverse of  $f$  is also holomorphic.  $\lrcorner$

*Proof:* To show the inverse is holomorphic, notice that for  $w_0 = f(z_0) \in V$ , if  $w = f(z)$  is close to  $w_0$ , then

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w - w_0}{g(w) - g(w_0)}} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}.$$

Because when  $z \rightarrow z_0, w \rightarrow w_0$ , so we conclude that  $g$  is holomorphic at  $w_0$ , and its derivative is the reciprocal of the derivative of  $f$ .  $\square$

**Cor. (12.4.7.5).** For regions  $U, V \subset \mathbb{C}$ , any holomorphic bijection from  $U$  to  $V$  is a biholomorphism.  $\lrcorner$

**Def. (12.4.7.6) [Univalent Functions].** A **univalent function** is a holomorphic function that is injective.  $\lrcorner$

**Prop. (12.4.7.7).** Any  $C^1$  conformal map in  $\mathbb{C}$  is holomorphic or anti-holomorphic. In higher dimension, conformal is equivalent to  $\langle df_p(v_1), df_p(v_2) \rangle = \lambda^2(p) \langle v_1, v_2 \rangle$ .  $\lrcorner$

*Proof:* Cf. [Ahlfors P74].  $\square$

**Prop. (12.4.7.8) [Automorphism Groups].**



- The only holomorphic automorphisms of  $\mathbb{D}$  fixing the origin are the rotations.
- $\text{Aut}(\mathbb{D}) = \{e^{i\theta} \frac{\alpha - z}{1 - \bar{\alpha}z}, \alpha \in \mathbb{D}\}$ . Moreover, we denote by  $\psi_\alpha$  the automorphism  $\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}$ , then  $\psi_\alpha^2 = \text{id}$ .
- $\text{Aut}(\mathcal{H}) = \text{PSL}(2, \mathbb{R})$ .
- $\text{Aut}(\bar{\mathbb{C}}) = \text{PSL}(2, \mathbb{C})$ .
- $\text{Aut}(\mathbb{C}) = \mathbb{C} \ltimes \mathbb{C}^\times$ .

□

*Proof:* 1:  $g, g^{-1}$  are both automorphisms of  $\mathbb{D}$  that fixes the origin, so by Schwartz's lemma(12.5.1.3),  $|g(z)| \leq |z|, |g^{-1}(z)| \leq |z|$ . Thus  $|g(z)| = |z|$  for any  $z \in \mathbb{D}$ , and by Schwartz's lemma(12.5.1.3),  $g$  is a rotation.

2: For any  $f \in \text{Aut}(\mathbb{D})$ , there exists some  $\alpha \in \mathbb{D}$  s.t.  $f(\alpha) = 0$ . Then  $g = f \circ \psi_\alpha$  maps 0 to 0. Then by item1,  $g(z) = e^{i\theta}z$ , and  $f(z) = e^{-i\theta}\psi_\alpha(z)$ .

3: Firstly  $SL(2, \mathbb{R})$  acts transitively on  $\mathcal{H}$  it suffices to show that any  $z \in \mathcal{H}$  can be mapped to  $i$ :  $\begin{bmatrix} & -c^{-1} \\ c & \end{bmatrix} (z) = \frac{1}{c^2 z}$ , so we may take  $c \in \mathbb{R}$  s.t.  $M_1 = \begin{bmatrix} & -c^{-1} \\ c & \end{bmatrix}$  maps  $z$  to  $z_1$  with  $\text{Im}(z_1) = 1$ , and then we can take some  $b \in \mathbb{R}$  s.t.  $M_2 = \begin{bmatrix} 1 & b \\ & 1 \end{bmatrix}$  maps  $z_1$  to  $i$ .

Then for any  $f \in \text{Aut}(\mathcal{H})$ , there exists some  $\beta \in \mathcal{H}$  s.t.  $f(\beta) = i$ . take some matrix  $\gamma$  s.t.  $\gamma(\beta) = i$ , then  $g = f \circ \gamma^{-1}$  preserves  $i$ . Let  $F : \mathcal{H} \cong \mathbb{D} : z \mapsto \frac{z-i}{z+i}$  be a biholomorphism, then it can be checked that

$$F \circ \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \circ F^{-1} = e^{i\theta} : \mathbb{D} \rightarrow \mathbb{D}.$$

Thus by Schwartz lemma(12.5.1.3), there exists some  $\theta \in \mathbb{R}$  s.t.  $g = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \kappa_\theta$ , and then  $f = \kappa_\theta \circ \gamma \in \text{SL}(2; \mathbb{R})$ .

4: For any distinct points  $z_1, z_2, z_3 \in \mathbb{C} \cup \{\infty\}$ ,

$$F_{z_1, z_2, z_3} : \mathbb{C} \cup \{\infty\} \rightarrow \mathbb{C} \cup \{\infty\} : z \mapsto \frac{z - z_2}{z - z_3} / \frac{z_1 - z_2}{z_1 - z_3}$$

satisfies  $F_{z_1, z_2, z_3}(z_1) = 1, F_{z_1, z_2, z_3}(z_2) = 0, F_{z_1, z_2, z_3}(z_3) = \infty$ , and

$$G_{z_1, z_2, z_3} : \mathbb{C} \cup \{\infty\} \rightarrow \mathbb{C} \cup \{\infty\} : w \mapsto \frac{w z_3 \frac{z_1 - z_2}{z_1 - z_3} - z_2}{w \frac{z_1 - z_2}{z_1 - z_3} - 1}$$

is an inverse of  $f_{z_1, z_2, z_3}$ , so  $f_{z_1, z_2, z_3}$  is a bijection.

Now consider  $h = F_{f(1), f(0), f(\infty)} \circ f$ , then  $h$  is also a bijection and  $h(1) = 1, h(0) = 0, h(\infty) = \infty$ . Thus  $h$  is entire, and by(12.4.2.23)  $h(z) = zL(z)$  for some  $L(z) \in \mathcal{M}(\mathbb{C})$  entire and  $L(1) = 1, L(\infty) = 0$ , so by Liouville's theorem(12.4.2.10),  $L = 1$ , and  $h(z) = z$ .

Thus  $f(z) = G_{f(1), f(0), f(\infty)}$  is of the form  $\frac{az+b}{cz+d}$ . To show that  $ad - bc \neq 0$ , notice if  $ad - bc = 0$ , then  $f(z)$  is constant, contradiction.

5: Notice  $g(z) = (f(z) - f(0))/(f(1) - f(0))$  is also injective and entire, and  $g(0) = 0, g(1) = 1$ . But by Picard's great theorem,  $g(z)$  is meromorphic at  $\infty$ . So  $g(\infty) = \infty$ , otherwise  $g$  is entire and bounded, contradicting Liouville's theorem(12.4.2.10). So  $g(z) = z$ , by item4. Thus  $f(z) = (f(1) - f(0))z + f(0)$  is linear. □

### Riemann Mapping Theorem

**Lemma (12.4.7.9) [Limits of Univalent Functions are Univalent].** A limit of a sequence of univalent holomorphic functions in the compact-open topology is univalent or constant.  $\lrcorner$

*Proof:* If  $f = \lim_{n \rightarrow \infty} f_n$  is non-constant and  $f(z_1) = f(z_2)$ , we can take a Jordan curve  $\gamma$  surrounding  $z_1$  and  $z_2$  s.t.  $0 \notin f(\gamma)$ . Then  $f_n$  converges uniformly to  $f$  on  $\gamma$ , and then it follows from Rouché's theorem (12.4.2.31) that when  $n$  is large,  $f_n - f(z_1)$  has two zeros inside  $\gamma$ , contradiction.  $\square$

**Thm. (12.4.7.10) [Riemann Mapping Theorem, Koebe].** Let  $\Omega \subset \mathbb{C}$  be a simply-connected region and  $\Omega \neq \mathbb{C}$ , then for any  $z_0 \in \Omega$ , there exists a unique biholomorphism  $F : \Omega \rightarrow \mathbb{D}$  s.t.  $F(z_0) = 0$  and  $F'(z_0) \in \mathbb{R}_+$ .  $\lrcorner$

**Remark (12.4.7.11).** This theorem can be generalized to other simply-connected Riemann surfaces, see (16.9.9.2).  $\lrcorner$

*Proof:* For the uniqueness, if there are two such maps  $\varphi$  and  $\varphi'$ , then  $h = \varphi' \circ \varphi^{-1}$  is an automorphism of  $\mathbb{D}$  s.t.  $h(0) = 0$  and  $h'(0) \in \mathbb{R}_+$ . Then it follows from (12.4.7.8) that  $h = \text{id}$ .

For the existence, let  $\mathcal{F}$  be the space of univalent holomorphic functions on  $\Omega$  with values in  $\mathbb{D}$ . Then  $\mathcal{F} \neq \emptyset$ : Let  $a \neq b \in \partial\Omega$ , then by the map  $g(z) = \frac{z-a}{z-b}$ , we may assume that  $\{0, \infty\} \subset \partial\Omega$ . Then  $\sqrt{\cdot}$  has two branches on  $\Omega$ , denoted by  $h_+$  and  $h_-$ . Suppose  $w_0 \in h_-(\Omega)$  with  $\mathbb{D}(w_0, \delta) \subset h_-(\Omega)$ , then the function

$$f_0 : z \mapsto \frac{\delta}{h_+(z) - w_0} \in \mathcal{F}.$$

Let

$$\alpha = \sup_{f \in \mathcal{F}} |f'(z_0)|.$$

If  $\mathbb{D}(z_0, \delta_1) \subset \Omega$ , then by Cauchy's formula,  $0 < |f'(z_0)| \leq \delta_1^{-1}$ , so  $\alpha \in (0, \infty)$ , and there exists  $f_n \in \mathcal{F}$  s.t.  $|f'_n(z_0)| \geq \alpha - \frac{1}{n}$ . Then it follows from Montel's theorem (12.4.5.5) that there is a subsequence of  $\{f_n\}$  that converges to a holomorphic function  $f$  in compact-open topology, and  $|f'(z_0)| = \alpha > 0$  by Weierstrass theorem (12.4.5.3). Notice  $f$  is also univalent by (12.4.7.9), so  $f \in \mathcal{F}$ .

It remains to show that  $f(z_0) = 0$  and  $f_0(\Omega) = \Delta$ , as a rotation of  $f$  will satisfy the requirement. If  $f(z_0) = \beta \neq 0$ , then take

$$f'(z) = \psi_\beta \circ f$$

then  $f' \in \mathcal{F}$ , and

$$|(f')'(z_0)| = \left| \frac{f'(z_0)}{1 - |\beta|^2} \right| > |f'(z_0)|,$$

contradicting the maximality of  $|f'(0)|$ .

If  $w \in \mathbb{D} \setminus f(\Omega)$ , then  $\psi(z) = \psi_\beta \circ f$  is non-vanishing on  $\Omega$ , thus  $\sqrt{\cdot}$  has a branch  $h$  on  $\psi(\Omega)$ . Now suppose

$$f'(z) = \psi_{h(z_0)} \circ h,$$

then

$$f = \psi_w^{-1} \circ (z \mapsto z^2) \circ \psi_{h(w)}^{-1} \circ f',$$

where  $\Phi = \psi_w^{-1} \circ (z \mapsto z^2) \circ \psi_{h(w)}^{-1}$  is an automorphism of  $\mathbb{D}$  that fixes 0, so  $|\Phi'(0)| < 1$  by Schwartz's lemma (12.5.1.3), and

$$|f'(0)| = |\Phi'(0)| |(f')'| \leq |(f')'|,$$

contradicting the maximality of  $|f'(0)|$ .  $\square$

**Cor. (12.4.7.12).** Let  $\Omega \subset \overline{\mathbb{C}}$  be a simply-connected region, then  $D$  is biholomorphic to exactly one of the following:

- $\mathbb{D} \cong \mathcal{H}$ , in which case  $\#\partial\Omega > 1$ , and  $\Omega$  is said to be of **hyperbolic type**.
- $\mathbb{C}$ , in which case  $\#\partial\Omega = 1$ .
- $\overline{\mathbb{C}}$ , in which case  $\#\partial\Omega = 0$ .

┘

*Proof:* If  $\#\partial\Omega > 1$ , by a fractional transformation, we can assume  $\Omega \subset \mathbb{C}$ , and  $\Omega \neq \mathbb{C}$ , so  $\Omega \cong \mathbb{D}$  by (12.4.7.10). If  $\#\partial\Omega = 1$ , then clearly  $\Omega = \overline{\mathbb{C}} \setminus \{z_0\} \cong \mathbb{C}$ . □

**Cor. (12.4.7.13).** Any two simply-connected regions  $\Omega, \Omega' \subset \mathbb{C}$  are homeomorphic. ┘

### Injective Holomorphic Maps

**Thm. (12.4.7.14)**[Bieberbach Conjecture1916, de Branges1984]. Let  $f(z) = z + a_2 z^2 + \dots \in \mathcal{O}(\mathbb{D})$  is an injective map, then  $|a_n| \leq n$ . And the equation holds iff  $f$  is a rotation composed with the Koebe function. ┘

*Proof:* Cf.[李忠]. □

## 12.5 Complex Analysis II

References are [Ahl78, S-S03, Li04, T-W06].

### Notation(12.5.0.1).

- Use notations defined in [Measure Theory and Functions on  \$\mathbb{R}^n\$](#) .
- Let  $\Omega$  be a region(12.4.1.1).

┘

### 1 Poincaré Metric

**Def.(12.5.1.1) [Hyperbolic Metric].** For any region  $\Omega \subset \mathbb{C}$  of hyperbolic type??, there exists a covering  $\mathbb{D} \rightarrow \Omega$ , and different coverings differ by an automorphism of  $\mathbb{D}$ . Thus we can define a **Poincaré metric** on  $\Omega$  by pushforward of the Poincaré metric  $d^P s = \rho_\Omega(s)|dz|$  on  $\mathbb{D}$ . Then this is well-defined and independent of the covering map. ┘

*Proof:* It is well-defined because for any two local inverse map  $\Omega \rightarrow \mathbb{D}$  differ by a Deck transformation which is an automorphism of  $\mathbb{D}$  thus preserves the Poincaré metric?. And any two covering  $\mathbb{D} \rightarrow \Omega$  differ by a covering map which is also an automorphism of  $\mathbb{D}$ , thus preserves the Poincaré metric. □

**Prop.(12.5.1.2) [Hyperbolic Metrics].**

- $\rho_{\mathcal{H}}(z) = 1/\text{Im}(z)$ .
- $\rho_{\mathbb{D}(0,(0,r))}(z) = 1/(|z| \log \frac{r}{|z|})$ .

┘

*Proof:* 1: ?.

2: There is a covering map  $\mathcal{H} \rightarrow \mathbb{D}(0,(0,r)) : z \mapsto re^{iz}$ , so we can plug in the inverse map  $w = i \log z$  into  $|dw|/\text{Im } w$  to get the desired formula. □

**Thm.(12.5.1.3) [Schwartz Lemma].** If  $f \in \mathcal{O}(\mathbb{D})$  and satisfies  $|f(z)| \leq 1$ ,  $f(0) = 0$ , then  $|f(z)| \leq |z|$ , and  $|f'(0)| \leq 1$ . Moreover, if  $|f(z)| = |z|$  for some  $z$  or  $|f'(0)| = 1$ , then  $f(z) = cz$  for some  $|c| = 1$ .

┘

*Proof:* The function  $g(z) = f(z)/z$  is analytic on  $0 < |z| < 1$  with a removable singularity at 0(12.4.2.18) and extends to an analytic function on  $|z| < 1$  with  $g(0) = f'(0)$ , and  $\lim_{|z| \rightarrow 1} |g(z)| \leq 1$ , thus by maximal principle(12.4.2.24),  $|g(z)| \leq 1$  for any  $|z| < 1$ , thus we are done. The last assertion also follows from(12.4.2.24). □

**Cor.(12.5.1.4).** If  $g \in \mathcal{O}(\mathbb{D}(0,r))$  satisfies  $|g(z)| \leq B$  for any  $z \in \mathbb{D}(0,r)$ , then for any  $z, \zeta \in \mathbb{D}(0,r)$ , we have

$$|g(z) - g(\zeta)| \leq 2B \left| \frac{r(z - \zeta)}{r^2 - \bar{\zeta}z} \right|.$$

┘

**Lemma(12.5.1.5) [Schwartz-Pick].** Let  $f \in \mathcal{O}(\mathbb{D})$ , and  $f(\mathbb{D}) \subset \mathbb{D}$ ,  $f(0) = 0$ , then in the Poincaré metric, for any  $z_1, z_2 \in \mathbb{D}$ ,

$$d^P(f(z_1), f(z_2)) \leq d^P(z_1, z_2).$$

And if the equality holds for some  $z_1 \neq z_2$ , then  $f \in \text{Aut}(\mathbb{D})$ . ┘

*Proof:* Cf.[李忠]P29. □

**Prop. (12.5.1.6) [Generalized Schwartz Lemma].** Let  $\Omega_1, \Omega_2$  be regions of hyperbolic type with Poincaré metrics  $ds_1 = \sigma_1(z)|dz|$  and  $ds_2 = \sigma_2(z)|dz|$ , then for any holomorphic map  $f : \Omega_1 \rightarrow \Omega_2$ ,

$$\sigma_2(f(z))|f'(z)| \leq \sigma_1(z).$$

In particular, if  $\Omega_1 \subset \Omega_2$ , then  $\sigma_2(z) \leq \sigma_1(z)$ . ┘

*Proof:* Let  $h_1 : \mathbb{D} \rightarrow \Omega_1, h_2 : \mathbb{D} \rightarrow \Omega_2$  be covering maps, then  $f : \Omega_1 \rightarrow \Omega_2$  lifts to a holomorphic map  $F : \mathbb{D} \rightarrow \mathbb{D}$ . Thus Schwartz-Pick lemma implies that  $\sigma_{\mathbb{D}}(F(z))|F'(z)| \leq \sigma_{\mathbb{D}}(z)$ . And this implies that

$$\sigma_1(z) \geq \sigma_{\mathbb{D}}(F \circ \psi_1(z))|(F \circ \psi_1)'(z)| = \sigma_{\mathbb{D}}(\psi_2 \circ f(z))|(\psi_2 \circ f)'(z)| = \sigma_2(f(z))|f'(z)|.$$

□

### Montel-Normal Family

**Def. (12.5.1.7) [Montel-Normal Family].** If  $\Omega \subset \mathbb{C}$  is a region, then a set  $\mathcal{S} = \{f_\alpha\}$  of holomorphic functions on  $\Omega$  is called a **Montel-normal family** if for any sequence of functions in  $\mathcal{S}$ , there exists a subsequence that converges uniformly in  $\Omega$  to  $\infty$  or to a holomorphic function. Or equivalently, for any compact subset  $D \subset \Omega$ , there exists a subsequence that converges uniformly in the compact-open topology of  $C(D, \mathbb{P}^1(\mathbb{C}))$ .

$\mathcal{S}$  is called normal at a point  $z \in \mathbb{C}$  if there exists a region  $z \in \Omega$  s.t.  $\mathcal{S}|_\Omega$  is a Montel-normal family. ┘

**Prop. (12.5.1.8).** If  $\Omega, \Omega' \subset \mathbb{C}$  is a region, and  $\{f_\alpha\}$  is a Montel-normal family of holomorphic maps from  $\Omega$  to  $\Omega'$ . Suppose  $g \in \mathcal{O}(\Omega')$ , then  $\{g \circ f_\alpha\}$  is also Montel-normal on  $\Omega$ . ┘

**Prop. (12.5.1.9) [Montel].** If  $\Omega \subset \mathbb{C}$  is a region, and  $\mathcal{S} = \{f_\alpha\}$  is a set of holomorphic functions on  $\Omega$  s.t.

$$\mathcal{S}' = \left\{ \frac{|f'_\alpha(z)|}{1 + |f_\alpha(z)|^2} \right\}$$

is a set of functions on  $\Omega$  bounded in the compact-open topology, then  $\mathcal{S}$  is a Montel-normal family. ┘

*Proof:* □

**Thm. (12.5.1.10) [Montel].** If  $\Omega \subset \mathbb{C}$  is a region, and  $\mathcal{S} = \{f_\alpha\}$  is a set of holomorphic functions on  $\Omega$  s.t. there exists  $a, b \in \mathbb{C}$  s.t.  $f(z) \neq a, b$  for any  $f \in \mathcal{S}, z \in \Omega$ , then  $\mathcal{S}$  is a Montel-normal family (12.5.1.7). ┘

*Proof:* By Riemann-mapping theorem, there is a covering map  $\rho : \mathbb{D} \rightarrow \mathbb{C} \setminus \{0, 1\}$ . Then each map  $f_\alpha$  lifts via  $\rho$  to a map  $\tilde{f}_\alpha : \Omega \rightarrow \mathbb{D}$ . Then  $\tilde{f}$  is bounded, and it follows from the formula  $f'(z) = \frac{1}{2\pi} \int_\gamma \frac{f(\zeta)}{(\zeta-z)^2} d\zeta$  that the derivatives of  $\tilde{f}_\alpha$  are also uniformly bounded. Thus  $\tilde{f}_\alpha$  is equicontinuous and uniformly bounded, so Montel-normal by Arzela-Ascoli. Finally, by (12.5.1.8),  $\{f_\alpha\}$  is also Montel-normal. □

**Prop. (12.5.1.11).** If  $\Omega \subset \mathbb{C}$  is a region and  $f \in \mathcal{M}(\Omega)$ , then for  $k \in \mathbb{Z}_+$ , the family  $\{f^{(n)}\}_{n \in \mathbb{Z}_+}$  is Montel-normal on  $\Omega$  iff the family  $\{f^{(kn)}\}_{n \in \mathbb{Z}_+}$  is Montel-normal on  $\Omega$ . ┘

$$\rho_{0,1}(z)$$

**Prop. (12.5.1.12)**  $[\rho_{0,1}]$ . Suppose the Poincaré metric on  $\mathbb{D} \setminus \{0, 1\}$  is  $\rho_{0,1}(z)|dz|$ , then

- $\rho_{0,1}(z) = \rho_{0,1}(1 - z)$ .
- $\rho_{0,1}(1/z) = \rho_{0,1}(z)/z^2$ .
- $\rho_{0,1}(z) = \rho_{0,1}(1 - z) > \frac{1}{4|z|}$  when  $|z|$  is small.
- $\rho_{0,1}(z) \leq \frac{1}{|z| \log |\frac{1}{z}|}$ .

┘

*Proof:* 1, 2 follows from the uniqueness of the Poincaré metric.

3 follows from [李忠]P46. ?

4: This follows from (12.5.1.2) and generalized Schwartz lemma (12.5.1.6).

□

**Prop. (12.5.1.13)**. Let  $\eta = \min\{\rho_{0,1}(z) : |z| = 1\}$ , then  $\eta = \rho_{0,1}(-1) = 4\pi^2/\Gamma^4(\frac{1}{4}) = 0.2284733$ .

┘

*Proof:*

□

**Prop. (12.5.1.14)**. For  $0 < |z| \leq 1$ ,

$$\rho_{0,1}(z) \geq \frac{1}{|z|(\eta^{-1} - \log |z|)},$$

where  $\eta = \min\{\rho_{0,1}(z), |z| = 1\} > 0$  by (12.5.1.12).

┘

*Proof:*

□

**Cor. (12.5.1.15)**. When  $z \rightarrow 0$ ,  $\rho_{0,1}(z) \sim \frac{1}{|z| \log |\frac{1}{z}|}$ .

┘

**Thm. (12.5.1.16)** [Landau]. If  $f \in \mathcal{O}(\mathbb{D})$  doesn't take the values 0, 1, then

$$|f'(0)| \leq 2|f(0)|(\eta^{-1} + |\log |f(0)||) \quad (12.5.1.13).$$

And the equality can be achieved.

┘

*Proof:* Cf. [李忠]P41.

□

**Thm. (12.5.1.17)** [Great Picard Theorem]. If an analytic function  $f$  has an essential singularity at a point  $w$ , then on any punctured nbhd of  $w$ ,  $f$  takes any value infinitely often, except for at most one single exception.

┘

*Proof:* We may assume  $f : \mathbb{D} \setminus \{0\} \rightarrow \mathbb{C} \setminus \{0, 1\}$ , then by generalized Schwartz lemma (12.5.1.6) and (12.5.1.2),

$$\rho_{0,1}(f(z))|f'(z)| \leq \frac{1}{|z| \log \frac{1}{|z|}}.$$

We want to prove that for  $z \in \mathbb{D}(0, e)$ , there exists some  $C$  s.t.

$$-\log |f(z)| < C \log \frac{1}{|z|}.$$

When  $f(z) \geq 1$ , this is clearly true, so we may assume  $f(z) < 1$ . Let  $m = \max\{|f(z)| : |z| = \rho\}$ , and let  $z = re^{i\theta}$ . Consider two cases:

If  $f(te^{i\theta}) \in \mathbb{D}$  for any  $r \leq t \leq e$ , then by (12.5.1.14),

$$\frac{|f'(z)|}{|f(z)|} [\eta^{-1} - \log |f(te^{i\theta})|]^{-1} \leq \frac{1}{|z| \log \frac{1}{|z|}}.$$

Integration on  $t \in [r, \rho]$  implies that

$$\log[\eta^{-1} - \log |f(z)|] \leq \log \log \frac{1}{|z|} - \log \log \frac{1}{\rho} + \log[\eta^{-1} - \log |f(\rho e^{i\theta})|] \leq \log \log \frac{1}{|z|} + \log \log \frac{e^{\eta^{-1}}}{m},$$

which implies that

$$-\log |f(z)| < C \log \frac{1}{|z|}.$$

If  $r_0 \in (r, e]$  is the smallest number s.t.  $|f(r_0 e^{i\theta})| = 1$ , then integration on  $t$  from  $r$  to  $r_0$  implies that

$$\log[\eta^{-1} - \log |f(z)|] \leq \log \log \frac{1}{|z|} - \log \log \frac{1}{r_0} + \log(\eta^{-1}) \leq \log \log \frac{1}{|z|},$$

which also implies that

$$-\log |f(z)| < \eta^{-1} \log \frac{1}{|z|}.$$

Now notice we can do all above for  $f$  replaced by  $1/f$ , so it implies that

$$\log |f(z)| < C' \log \frac{1}{|z|},$$

and then  $f$  is meromorphic at  $z = 0$ . □

**Thm. (12.5.1.18) [Schottky].** If  $f \in \mathcal{O}(\mathbb{D})$  doesn't take the values 0, 1, then

$$\log |f(z)| \leq [\eta^{-1} + \max\{\log |f(0)|, 0\}] \frac{1 + |z|}{1 - |z|} - \eta^{-1},$$

and the equality can be achieved. ┘

*Proof:* It follows from generalized Schwartz lemma (12.5.1.6) that

$$\rho_{0,1}(f(z)) |f'(z)| \leq \frac{2}{1 - |z|^2}.$$

For  $z = re^{i\theta} \in \mathbb{D}$ , if  $f(te^{i\theta}) \in \mathbb{D}$  for any  $t \in [0, r]$ , then it follows from (12.5.1.14) that

$$\frac{|f'(z)|}{|f(z)|} [\eta^{-1} - \log |f(te^{i\theta})|]^{-1} \leq \frac{2}{1 - |z|^2}.$$

Integration on  $t \in [0, r]$  implies that

$$\frac{\eta^{-1} - \log |f(z)|}{\eta^{-1} - \log |f(0)|} \leq \frac{1 + |z|}{1 - |z|}.$$

If  $r_0 \in (0, r]$  is the largest number s.t.  $|f(r_0 e^{i\theta})| = 1$ , then integration on  $t$  from  $r_0$  to  $r$  implies that

$$\eta(\eta^{-1} - \log |f(z)|) \leq \frac{1 + |z|}{1 - |z|} \frac{1 - r_0}{1 + r_0} < \frac{1 + |z|}{1 - |z|}.$$

So in any case we get

$$\eta^{-1} - \log |f(z)| \leq [\eta^{-1} + \max\{\log |f(0)|, 0\}] \frac{1 + |z|}{1 - |z|}.$$

And all the above applies to  $f$  replaced by  $1/f$ , so we get the desired formula.

To show the equality can be achieved, Cf.[李忠]P45. □

## 2 Harmonic Functions

**Def. (12.5.2.1) [Harmonic Functions].** A real-valued function on a region  $\Omega \subset \mathbb{C}$  is called **harmonic** iff it is  $C^1$  and has second order derivatives and

$$\Delta u = \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} f = 0.$$

This is an elliptic differential operator, so  $u$  is automatically smooth, by (13.1.8.4).

The vector space of Harmonic functions on  $\Omega$  is denoted by  $\mathcal{H}(\Omega) \subset C^\infty(\Omega)$ . ┘

**Prop. (12.5.2.2) [Schwarz's Theorem].** Cf.[Ahlfors P169]. ┘

**Def. (12.5.2.3) [Mean-Value Property].** A real valued function  $u$  on a region  $\Omega$  is said to have the **mean-value property** iff

$$u(z_0) = \frac{1}{2\pi} \int_0^{1\pi} u(z_0 + r e^{i\theta}) d\theta$$

whenever  $\mathbb{D}(z_0, r) \subset \Omega$ . ┘

**Lemma (12.5.2.4) [Harmonic Mean Value].** If  $u$  is a Harmonic function between two concentric circles, then the arithmetic mean of it over circles  $|z| = r$  is a linear function of  $\log r$ :

$$\frac{1}{2\pi} \int_{|z|=r} u d\theta = \alpha \log r + \beta.$$

In particular, if  $u$  is harmonic in the disk, then by continuity,  $\alpha = 0$ , and the mean value is a constant. ┘

*Proof:* Cf.[Ahlfors P165]. □

**Prop. (12.5.2.5) [Harmonicity and Mean-Value Property].** A harmonic function satisfies the mean-value property, and conversely, and continuous function satisfying the mean-value property is harmonic. ┘

*Proof:* Harmonic function satisfies the mean-value property by (12.5.2.4). Conversely, for any  $z_0$ , by Schwarz's theorem (12.5.2.2), there is a harmonic function  $v(z)$  that is harmonic in  $\mathbb{D}(z_0, \rho)$  and equals  $u(z)$  on  $\partial B(z_0, \rho)$ . Now the maximal and minimal principles apply to  $u - v$ , thus  $u = v$  is harmonic. □



**Cor. (12.5.2.6) [Maximum Principle].** If  $u$  is a harmonic function, then it attains neither maximum nor minimum at its region of definition.  $\lrcorner$

**Prop. (12.5.2.7) [Real Part of an Analytic Function is Harmonic].** On  $\mathbb{C}$ , formally,

$$4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \Delta.$$

Thus for any open subset  $\Omega \subset \mathbb{C}$ , if  $f \in \mathcal{O}(\Omega)$ , then  $\operatorname{Im} f, \operatorname{Re} f \in \mathcal{H}(\Omega)$ .  $\lrcorner$

*Proof:* By (12.4.1.4),

$$4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

$$4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

If  $f \in \mathcal{O}(\Omega)$ , then  $f \in C^\infty(\Omega)$  by (12.4.2.7), then by (12.3.3.16) the formal calculation above applies to  $f$ , and by (12.4.1.10)

$$\Delta \operatorname{Re} f + i \Delta \operatorname{Im} f = \Delta f = \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} f = 0,$$

so  $\Delta \operatorname{Re} f = \Delta \operatorname{Im} f = 0$ , and  $\operatorname{Im} f, \operatorname{Re} f \in \mathcal{H}(\Omega)$ .  $\square$

**Cor. (12.5.2.8).** If  $\Omega \subset \mathbb{C}$  is a region and  $z \in \Omega$ ,  $f \in \mathcal{O}(\Omega)$  is non-vanishing, then  $\log |f| \in \mathcal{H}(\Omega)$ .  $\lrcorner$

*Proof:* Being harmonic is local, so we may look locally and assume  $\Omega$  is a disk. Then by (12.4.2.33), there exists a  $g \in \mathcal{O}(\Omega)$  s.t.

$$f(z) = \exp(g(z)).$$

Then it follows  $\operatorname{Re}(g(z)) = \log |f(z)|$ , which is harmonic by (12.5.2.7).  $\square$

**Prop. (12.5.2.9) [Harmonic Functions as the Real Parts of Analytic Functions].** Let  $\Omega \subset \mathbb{C}$  be a simply-connected region,  $u \in \mathcal{H}(\Omega)$ , then there exists  $f \in \mathcal{O}(\Omega)$  s.t.  $\operatorname{Re} f = u$ . And any two such  $f$  differ by a purely imaginary constant.  $\lrcorner$

*Proof:* Let  $g(z) = 2 \frac{\partial}{\partial \bar{z}} u$ , then  $\frac{\partial}{\partial \bar{z}} g = 2 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial \bar{z}} u = \frac{1}{2} \Delta u = 0$ , so  $g \in \mathcal{O}(\Omega)$ . Then by (12.4.2.3), there exists  $f \in \mathcal{O}(\Omega)$  s.t.  $\frac{\partial}{\partial \bar{z}} f = g = 2 \frac{\partial}{\partial \bar{z}} u$ . So

$$2 \frac{\partial}{\partial x} u = \frac{\partial}{\partial x} \operatorname{Re} f + \frac{\partial}{\partial y} \operatorname{Im} f = 2 \frac{\partial}{\partial x} \operatorname{Re} f, \quad 2 \frac{\partial}{\partial y} u = \frac{\partial}{\partial y} \operatorname{Re} f - \frac{\partial}{\partial x} \operatorname{Im} f = 2 \frac{\partial}{\partial y} \operatorname{Re} f.$$

So  $u - \operatorname{Re} f$  is a constant function. And we may modify  $f$  to make  $\operatorname{Re} f = u$ . For the last assertion, use the Cauchy-Riemann equations to show that  $\frac{\partial}{\partial x} (\operatorname{Im} f - \operatorname{Im} f') = \frac{\partial}{\partial y} (\operatorname{Im} f - \operatorname{Im} f') = 0$ , so  $\operatorname{Im} f - \operatorname{Im} f'$  is a constant function.  $\square$

## Properties

**Prop. (12.5.2.10) [Poisson Formula].** For  $u \in \mathcal{H}(\overline{\mathbb{D}(0, R)})$ ,

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 - |z|^2}{|Re^{i\theta} - z|^2} u(Re^{i\theta}) d\theta = \operatorname{Re} \left[ \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{\zeta + z}{\zeta - z} u(\zeta) \frac{d\zeta}{\zeta} \right].$$

for  $z \in \mathbb{D}(0, R)$ .

In particular, for any  $f \in \mathcal{O}(\overline{\mathbb{D}(0, R)})$ , there is a constant  $C \in \mathbb{R}$  s.t. for any  $z \in \mathbb{D}(0, R)$ :

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=\rho} \frac{\zeta + z}{\zeta - z} \operatorname{Re}(f(\zeta)) \frac{d\zeta}{\zeta} + iC.$$

┘

*Proof:* This follows from (12.5.2.9) and (12.4.2.12). □

**Cor. (12.5.2.11) [Poisson-Jensen Formula].** For  $f \in \mathcal{O}(\overline{\mathbb{D}(0, R)})$  with no zeros on the boundary and zeros  $a_1, \dots, a_n$  inside (counting multiplicity), then for  $z \in \mathbb{D}(0, R)$  s.t.  $f(z) \neq 0$ ,

$$\log |f(z)| = - \sum_i \log \left| \frac{R^2 - \bar{a}_i z}{R(z - a_i)} \right| + \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 - |z|^2}{|Re^{i\theta} - z|^2} \log |f(Re^{i\theta})| d\theta.$$

In particular, if  $f(0) \neq 0$ , then

$$\log |f(0)| = - \sum_i \log \left| \frac{R}{a_i} \right| + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta.$$

┘

*Proof:* Firstly, this is true when  $f$  is non-vanishing on  $\mathbb{D}(0, \rho)$ , because in this case  $\log |f|$  is harmonic by (12.5.2.8) and we can use mean value theorem (12.5.2.5). In general, consider the function

$$F(z) = f(z) \prod_{i=1}^n \frac{R^2 - \bar{a}_i z}{R(z - a_i)},$$

then it satisfies  $F(z) = f(z)$  for  $|z| = R$ , and has no zeros on  $\mathbb{D}(0, R)$ , so

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(\rho e^{i\theta})| d\theta = \log |F(0)| = \log |f(0)| + \sum_i \log \left| \frac{R^2 - \bar{a}_i z}{R(z - a_i)} \right|.$$

□

**Prop. (12.5.2.12) [Hadamard's Three Circle Theorem].** Let  $f(z)$  be analytic in the annulus  $r_1 < |z| < r_2$ , and continuous on the boundary, if  $M(r)$  denotes the maximum of  $|f(z)|$  for  $|z| = r$ , then:

$$M(r) \leq M(r_1)^\alpha M(r_2)^{1-\alpha}$$

where  $\alpha = \log(r_2/r) : \log(r_2/r_1)$ . ┘

*Proof:* Apply the maximum principle (12.5.2.6) for

$$g(z) = \log |f(z)| - \log(M(r_1))(\log(r_2/|z|) : \log(r_2/r_1)) - \log(M(r_2))(1 - \log(r_2/|z|) : \log(r_2/r_1)),$$

it is harmonic by (12.5.2.8), then  $g(z) \leq 0$  on  $|z| = r_1$  and  $|z| = r_2$ , so  $g(z) \leq 0$  on all the annulus. □

**Cor. (12.5.2.13).** Let  $f(z)$  be analytic in the annulus  $r_1 < |z| < r_2$ , then the function

$$s \mapsto \max_{z=e^s} |f(z)|$$

is convex on the interval  $[\log(r_1), \log(r_2)]$ . ┘

**Prop. (12.5.2.14) [Reflection Principle].** ┘

*Proof:* □

**Prop. (12.5.2.15) [Harnack's Inequality].** For a positive harmonic function  $u$  on  $B(0, \rho)$ ,

$$\frac{\rho - |z|}{\rho + |z|} u(0) \leq u(z) \leq \frac{\rho + |z|}{\rho - |z|} u(0).$$

┘

*Proof:* By Poisson formula,

$$u(z) = \frac{1}{2\pi} \int \frac{\rho^2 - |z|^2}{|\rho e^{i\theta} - z|^2} u(\rho e^{i\theta}) d\theta$$

for  $|z| < \rho$ , so the conclusion follows from the obvious inequality

$$\frac{\rho - |z|}{\rho + |z|} \leq \frac{\rho^2 - |z|^2}{|\rho e^{i\theta} - z|^2} \leq \frac{\rho + |z|}{\rho - |z|}$$

and Mean-value property (12.5.2.5). □

**Cor. (12.5.2.16).** If  $E$  is a compact subset of a region  $\Omega$ , there is a constant  $M$ , depending on  $E, \Omega$  that for any positive harmonic function  $u(z)$  on  $\Omega$ ,  $u(z_2) \leq Mu(z_1)$  for any  $z_1, z_2 \in E$ . ┘

*Proof:* This is an easy consequence of Harnack inequality and the compactness of  $E$ . □

**Cor. (12.5.2.17) [Harnack's Principle].** Consider a sequence of functions  $u_n(z)$ , each harmonic in a region  $\Omega_n$ , and there is a region  $\Omega$  that every point has a nbhd that is contained in all but f.m.  $\Omega_n$ , and in this nbhd  $u_n(z) \leq u_{n+1}(z)$  for  $n$  large, then either  $u_n(z)$  tends to  $+\infty$  in the compact open topology, or they tends to a harmonic function in compact open topology. ┘

*Proof:* The uniform continuity follows easily from Harnack's inequality, and for the harmonicity of the limit function  $u(z)$  is a consequence of the Poisson formula. □

### Bounds on Locations of Zeros

**Prop. (12.5.2.18).** For  $f \in \mathcal{O}(\mathbb{D})$  and  $r \in (0, 1)$ , denote

$$A(r) = \frac{1}{2r} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta.$$

Then if  $f$  has no zeros on  $\partial\mathbb{D}(0, r)$ ,

$$\frac{\partial}{\partial r} A(r) = \frac{1}{r} \#\{\text{zeros of } f \text{ in } \mathbb{D}(0, r)\}.$$

┘

*Proof:* We may divide  $f$  by  $z^k$  to make  $f(0) \neq 0$ , then the assertion follows from (12.5.2.11). □

**Prop. (12.5.2.19).** Let  $f \in \mathcal{O}(\overline{\mathbb{D}})$  s.t.  $\max_{z \in \overline{\mathbb{D}}} |f| = K|f(0)|$ . Then for any  $\delta \in (0, 1)$ ,

- $\#\text{Zero}(f) \cap \mathbb{D}(0, 1 - \delta) \leq \frac{\log K}{\delta}$ .

- There exists  $C(\delta) \in \mathbb{R}_+$  s.t. for any  $s \in \mathbb{D}(0, 1 - \delta)$ ,

$$\left| \frac{f^n}{f}(s) + \sum_{\text{Zero}(f) \cap \mathbb{D}(0, 1 - \delta)} \frac{1}{s - \rho} \right| \leq C(\delta) \log K.$$

┘

*Proof:* 1: By (12.5.2.18) and fundamental theorem of calculus,

$$\#\text{Zero}(f) \cap \mathbb{D}(0, 1 - \delta) \leq \frac{A(1) - A(1 - \delta)}{\delta} \leq \frac{\max_{\partial \mathbb{D}} \log |f| - \log |f(0)|}{\delta} = \frac{\log K}{\delta}.$$

2: Cf. [Larry Guth]P83.

□

### Dirichlet Problem

### 3 Subharmonic Functions

**Def. (12.5.3.1) [Subharmonic Functions].**

┘

**Prop. (12.5.3.2) [Hartog's Lemma].** If  $(u_i)_{i \in \mathbb{Z}_+}$  is a sequence of real-valued subharmonic functions (12.5.3.1) on an open subset  $U \subset \mathbb{C}$  that is uniformly bounded from above, and there exists  $C \in \mathbb{R}$  s.t.

$$\overline{\lim}_i u_i(z) \leq C$$

for each  $z \in U$ . Then for any  $\varepsilon \in \mathbb{R}_+$  and any compact subset  $K \subset U$ , there exists  $N \in \mathbb{Z}_+$  s.t. for any  $k \in \mathbb{Z}_{\geq N}$ , we have

$$u_k(z) \leq C + \varepsilon.$$

┘

*Proof:* By shrinking  $U$  (but still contains  $K$ ), we may assume that  $(u_i)$  are uniformly bounded on  $U$ . We may assume that  $u_i \leq 0$ . Let  $r = d(K, U^c)/4$ , then for any  $z \in K$ ,

$$\overline{\lim}_k \int_{\mathbb{D}(z, r)} u_k(\zeta) d\zeta \leq 2\pi r^2 C$$

by Fatou's lemma (12.3.2.7). Thus for  $k$  sufficiently large,

$$\int_{\mathbb{D}(z, r)} u_k(\zeta) d\zeta \leq 2\pi r^2 (C + \varepsilon/2).$$

Now for any  $0 < \delta < r$  and  $w \in D(z, \delta)$ , we have

$$\pi(r + \delta)^2 u_k(w) \leq \int_{D(w, r + \delta)} u_k(\zeta) d\zeta \leq \int_{\mathbb{D}(z, r)} u_k(\zeta) d\zeta \leq 2\pi r^2 (C + \varepsilon)$$

Thus for suitable  $\delta$ , we have  $u_k(w) \leq C + \varepsilon$ . Since  $K$  is compact, the assertion follows.

□

#### 4 Mellin-Transform

**Def. (12.5.4.1) [Mellin-Types].** For  $-\infty \leq \alpha < \beta \leq \infty$ , a function  $f : \mathbb{R}_+ \rightarrow \mathbb{C}$  is called of **Mellin-type**  $(\alpha, \beta)$  if

$$f(y)y^{s-1} \in L^1(\mathbb{R})$$

for each  $s \in \mathbb{C}$  s.t.  $\operatorname{Re}(s) \in (\alpha, \beta)$ . ┘

**Prop. (12.5.4.2) [Mellin Transform and Inversion].** Given a function  $f : \mathbb{R}_+ \rightarrow \mathbb{C}$  of Mellin-type  $(\alpha, \beta)$ , its **Mellin-transformation** is defined to be

$$(\mathcal{M}f)(s) = \int_0^\infty f(t)t^s \frac{dt}{t}.$$

where  $s \in \mathbb{C}$  is in the domain s.t. this integral is absolutely convergent (this is the case when  $\operatorname{Re}(s) \in (\alpha, \beta)$ ).

Notice if  $\int_0^1 f(t)t^s \frac{dt}{t}$  is convergent for some  $s_0 \in \mathbb{C}$ , then it converges for any  $s \in \mathbb{C}$  with  $\operatorname{Re}(s) > \operatorname{Re}(s_0)$ , and if  $\int_1^\infty f(t)t^s \frac{dt}{t}$  converges for some  $s$ , then it converges for any smaller  $s$ . So the domain of  $\mathcal{M}(f)$  is a vertical strip  $\sigma_1 < \operatorname{Re}(s) < \sigma_2$  for  $\sigma_1, \sigma_2 \in [-\infty, \infty]$ , if nonempty. And in this case,  $\mathcal{M}(f)$  is an analytic function on it, by (12.4.2.14).

Then  $f$  can be recovered from  $\mathcal{M}f$ : for any  $\sigma_1 < \sigma < \sigma_2$ , we have

$$f(x) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} (\mathcal{M}f)(s)x^{-s} ds.$$

┘

*Proof:* Using the isomorphism of groups  $t = e^x : \mathbb{R}^+ \rightarrow \mathbb{R}$ , this is just the usual Fourier-transformation on  $\mathbb{R}$  (12.11.3.34). □

**Prop. (12.5.4.3) [L-Functions].** Let  $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{C}$  be a smooth function that decays exponentially at infinity, then the **L-function**

$$L(f; s) = \frac{1}{\Gamma(s)} \int_0^\infty f(t)t^s \frac{dt}{t} \quad (12.5.4.2)$$

converges absolutely to a holomorphic function for  $\operatorname{Re}(s) > 0$ . Moreover,

$$L(f; s) = -L(f^\vee, s+1).$$

In particular,  $L(f; s)$  extends to a holomorphic function on  $\mathbb{C}$ . And for  $n \in \mathbb{N}$ ,

$$L(f; -n) = (-1)^n f^{(n)}(0).$$

┘

*Proof:*  $\mathcal{M}h(s)$  is analytic for  $\operatorname{Re}(s) > 0$  by (12.5.4.2). The equation  $L(f; s) = -L(f^\vee, s+1)$  follows by integration by part and the fact  $\Gamma(s) = (s-1)\Gamma(s-1)$  (12.6.1.2). Thus

$$L(f; -n) = (-1)^{n+1} L(f^{(n+1)}; 0) = (-1)^{n+1} \int_0^\infty f(t) dt = (-1)^n f^{(n)}(0).$$

□

**Def. (12.5.4.4) [Laplace-Transformation].** Let  $h : \mathbb{R}_+ \rightarrow \mathbb{R}$  be a piecewise-continuous function, the **Laplace-transformation** of  $h$  is the function

$$(\mathcal{L}h)(s) = \int_0^\infty e^{-st} h(t) dt, s \in \mathbb{C}$$

whenever it is convergent. It is a holomorphic function for  $\operatorname{Re}(s) > c$  if  $h(t) \leq \kappa e^{ct}$  for some  $\kappa \in \mathbb{R}_+$ .  
 $\lrcorner$

*Proof:* The last assertion follows from (12.4.2.14).  $\square$

**Thm. (12.5.4.5).** Let  $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$  be a bounded piecewise-continuous function that its Laplace-transform  $(\mathcal{L}f)(s)$  (12.5.4.4) extends to a holomorphic function on  $\operatorname{Re}(s) \geq 0$ , then the integral  $\int_0^\infty f(t) dt$  converges and equals  $\mathcal{L}f(0)$ .  $\lrcorner$

*Proof:* Cf. [Suthurland, Number Theory1, L16]. ?  $\square$

### Calculations

References are [Table of integrals, series, and products].

**Example (12.5.4.6).** For  $s = \sigma + i\tau \in \mathbb{C}$ ,

- $\mathcal{M}(e^{-y})(s) = \Gamma(s)$  if  $\sigma > 0$ .
- $\mathcal{M}(\cos y)(s) = \Gamma(s) \cos\left(\frac{\pi s}{2}\right)$  if  $0 < \sigma < 1$ .
- $\mathcal{M}(\sin y)(s) = \Gamma(s) \sin\left(\frac{\pi s}{2}\right)$  if  $-1 < \sigma < 1$ .
- $\mathcal{M}\left(\frac{1}{1+y}\right)(s) = \frac{\pi}{\sin(\pi s)}$  if  $0 < \sigma < 1$ .
- $\mathcal{M}(\log(1+y))(s) = \frac{\pi}{s \sin(\pi s)}$  if  $-1 < \sigma < 0$ .

$\lrcorner$

*Proof:* Cf. [Table of integrals, series, and products] Chap3.  $\square$

**Example (12.5.4.7) [Integral Representation of  $\log^+ x$ ].** Because

$$\mathcal{M}(\log^+ x)(s) = s^{-2}, \quad \operatorname{Re}(s) < 0,$$

we have

$$\log^+ x = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} x^s s^{-2} ds, \quad \sigma \in \mathbb{R}_+.$$

$\lrcorner$

## 5 Miscellaneous

**Def. (12.5.5.1) [ $L^2(\mathbb{D}, \nu_k)$  Unit Disc].** Define a density  $\nu_k = \frac{4(1-|w|^2)^{k-2} du dv}{|1-w|^{2k}} (k \geq 2)$  on  $\mathbb{D}$  and define  $L^2(\mathbb{D}, \nu_k)$  to be the space of holomorphic functions on  $\mathbb{D}$  that is  $L^2(\nu_k)$  bounded. Then the space  $L^2(\mathbb{D}, \nu_k)$  is complete.  $\lrcorner$

*Proof:* By (12.4.2.11), the uniform norm is bounded by the local  $L^1$ -norm hence also the local  $L^2$ -norm. Hence for a compact subset  $K$ , the uniform norm is also bounded by  $L^2(\nu_k)$ -norm. Thus any Cauchy sequence in  $L^2(\mathbb{D}, \nu_k)$  converges to a holomorphic function by Weierstrass theorem (12.4.5.3).  $\square$

**Prop. (12.5.5.2).** The space  $L^2(\mathbb{D}, \nu_k)$  has an orthogonal basis consisting of holomorphic functions

$$\{\psi_n = w^n(1-w)^k\}_{n \geq 0}.$$

┘

*Proof:* Firstly,  $\psi_n$  is convergent: In the polar coordinate,  $\nu_k = \frac{4(1-r^2)^{k-2}rdrd\theta}{|1-w|^{2k}}$ , so

$$\|\psi_n\|^2 = 4 \int_0^{2\pi} \int_0^1 r^{2n}(1-r^2)^{k-2} dr d\theta < \infty.$$

And if  $m \neq n$ ,

$$\int_{\mathbb{D}} \psi_m(w) \overline{\psi_n(w)} dw = 4 \int_{\mathbb{D}} w^m (\overline{w})^n (1-r^2)^{k-2} dr d\theta = 4 \int_0^1 r^{m+n} (1-r^2)^{k-2} \int_0^{2\pi} e^{i(m-n)\theta} d\theta = 0.$$

□

**Def. (12.5.5.3) [Upper Half Plane  $\mathcal{H}_k$ ].** On the upper half plane  $\mathcal{H}$ , we can define a density  $\mu_k = y^k \frac{dx dy}{y^2}$  ( $k \geq 2$ ), and define  $L^2(\mathcal{H}, \mu_k)$  to be the space of holomorphic functions on  $\mathcal{H}$  that is  $L^2(\mu_k)$  bounded. Then under the Cayley map  $z \mapsto w = \frac{z-i}{z+i}$ , this density is mapped to the density  $\nu_k$  of  $\mathbb{D}$  and induces an isomorphism of spaces

$$L^2(\mathcal{H}, \mu_k) \cong L^2(\mathbb{D}, \nu_k).$$

In particular,  $L^2(\mathcal{H}, \mu_k)$  is complete, and it has an orthogonal basis

$$\{\varphi_n = \left(\frac{z-i}{z+i}\right)^n \frac{(2i)^k}{(z+i)^k}\}.$$

by (12.5.5.1) and (12.5.5.2).

┘

## 6 Elliptic Functions

References are [Ahl78]Chap7, [Sil99], [Sil16]. [Elliptic Functions according to Eisenstein and Kronecker, Weil].

### Trigonometric Functions

#### Elliptic Functions(Eisenstein)

**Def. (12.5.6.1) [Doubly Periodic Function].** Let  $\Lambda$  be a lattice of  $\mathbb{C}$ , then a meromorphic function  $f$  on  $\mathbb{C}$  is called **doubly periodic w.r.t.**  $\Lambda$  if it is invariant under  $\Lambda$ .

┘

**Prop. (12.5.6.2).** Let  $f$  be a periodic function for  $\Lambda$ , not identically zero, and let  $D$  be a fundamental parallelogram for  $\Lambda$  that  $f$  has no zeros or poles on the boundary of  $D$ . Then

- $\sum_{P \in D} \text{ord}_P(f) = 0$ .
- $\sum_{P \in D} \text{res}_P(f) = 0$ .
- $\sum_{P \in D} \text{ord}_P(f) \cdot P = 0$ .

┘

*Proof:*  $f$  can be realized as meromorphic functions on the Riemann surface  $\mathbb{C}/\Lambda$ , so 1, 2 are direct consequences of (16.9.9.9) 1, 2. 3 is 2 applied to the meromorphic function  $zf'(z)/f(z)$ .  $\square$

**Def. (12.5.6.3)[Order].** The order of an elliptic function  $f$  is defined to be the number of poles of  $f$  in  $D$ . Equivalently, it is the number of zeros of  $f$  in  $D$  (12.5.6.2).  $\lrcorner$

**Prop. (12.5.6.4).** The total number of orders of a non-constant elliptic function  $\geq 2$ .  $\lrcorner$

*Proof:* If its order is 0, then it is bounded on  $D$  and also on  $\mathbb{C}$  thus constant by Liouville's theorem. If its order is 1, then it has a simple pole  $z_0$ , but then  $\text{Res}_{z_0}(f) \neq 0$ , contradicting (12.5.6.2).  $\square$

**Def. (12.5.6.5)[Weierstrass  $\wp$ -Function].** For a lattice  $\Lambda \subset \mathbb{C}$ , consider the function

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left( \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right)$$

Then it is a doubly periodic meromorphic function that has a double pole at 0, and so does its derivative  $\wp'$ , having a triple pole at 0. Hence they descend to a rational function on  $\mathbb{C}/\Lambda$ .

If  $\Lambda = \Lambda_\tau$ , then the function  $\wp(z; \tau) = \wp(z)$  is called the **Weierstrass  $\wp$ -function**.  $\lrcorner$

**Prop. (12.5.6.6).** Let  $G_k(\Lambda)$  be given by (26.3.4.5),

$$\wp(z) = \frac{1}{z^2} + \sum_{k \geq 1} (2k + 1) G_{2k+2}(\Lambda) z^{2k}.$$

$\lrcorner$

*Proof:* This follows from (10.7.2.1).  $\square$

**Prop. (12.5.6.7)[Fields of Elliptic Functions].** The field of doubly periodic functions for  $\Lambda$  is just  $\mathbb{C}(\wp, \wp')$ . And  $\wp, \wp'$  also satisfies the following equation:

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$$

where  $g_2 = 60G_4(\Lambda)$ ,  $g_3 = 140G_6(\Lambda)$  (26.3.4.5).  $\lrcorner$

*Proof:* For the equational function, just notice we can calculate directly that the difference of the two sides is a holomorphic function in  $z$  without constant terms, and they are both doubly periodic, thus is zero.

For the first statement, notice that any function is a sum of an odd function and an even function.  $\wp'$  is odd, thus an odd function is  $\wp'$  times an even function. Thus it reduces to show that any even doubly periodic function is a rational function of  $\wp$ .

For any even periodic function  $f$ ,  $f$  has the same order at  $z_0$  and  $-z_0$ , also if  $z_0 \equiv -z_0 \pmod{\Lambda}$ , the order of  $f$  at  $z_0$  must be odd. Now consider  $\wp(z) - \wp(z_0)$ , then it is a function with two poles at 0, so it has two zeros. If  $z_0 \equiv -z_0$ , then  $\wp(z) - \wp(z_0)$  has a zero of order 2 at  $z_0$ , and otherwise it has simple zeros at  $z_0$  and  $-z_0$ .

So for any even doubly periodic function  $f$ , we can use the product  $\prod (\wp(z) - \wp(z_0))^{m_i}$  to get a function with the same order of poles and zeros as  $f$ , which implies it equals  $f$ .  $\square$

**Prop. (12.5.6.8)[Uniformization Theorem].** For any complex numbers  $A, B \in \mathbb{C}$  that  $A^3 - 27B^2 \neq 0$ , there exists a unique lattice  $\Lambda \in \mathbb{C}$  that  $g_2(\Lambda) = A$ ,  $g_3(\Lambda) = B$  (12.5.6.7).  $\lrcorner$



*Proof:* The  $j$ -function (26.3.4.22) is surjective, thus there is a  $z \in \mathbb{C}$  s.t.

$$1728 \frac{1}{1 - \frac{E_6^2}{E_4^3}} = j(z) = 1728 \frac{A^3}{A^3 - 27B^2}$$

And if we assume  $A, B \neq 0$ , then by (26.3.4.5) (31.3.5.3) and (10.7.1.12), assume this implies

$$\left(\frac{B}{g_3(\tau)}\right)^2 \left(\frac{g_2(\tau)}{A}\right)^3 = 1.$$

Then we can scale  $\Lambda$  by  $\Lambda' = \alpha\Lambda$  s.t.  $g_3(\Lambda') = B$ ,  $g_2(\Lambda) = A$ .

The case that  $A = 0$  or  $B = 0$  is similar.

The uniqueness follows from (20.9.8.10). □

## 7 Quasi-Conformal Mappings

### Modulus

**Def. (12.5.7.1) [Conformal Annulus].** A **conformal annulus** is an open subset  $A$  of  $\mathbb{P}^1(\mathbb{C})$  that can be mapped conformally to an annulus  $\mathbb{D}(0, (r, s))$  for  $r < s \in \mathbb{R}_{\geq 0}$ . And in this case, the **module of**  $A$  is defined to be  $\frac{1}{2\pi} \log(\frac{s}{r})$ . And this is an invariance of  $A$ . ┘

*Proof:* □

**Def. (12.5.7.2) [Uniformly Perfect Subsets].** A compact subset  $E \subset \mathbb{P}^1(\mathbb{C})$  is called **uniformly perfect** if  $\#E \geq 2$ , and the modules of the conformal annuli in  $\mathbb{P}^1(\mathbb{C}) \setminus E$  that separate  $E$  are bounded. ┘

### Beltrami Equations

**Def. (12.5.7.3) [Beltrami Coefficients].** For  $f \in C^1(\mathbb{C})$  and  $z \in \mathbb{C}$  s.t.  $df$  is non-singular at  $z$ , then the **Beltrami coefficient** of  $f$  at  $z$  is defined to be the number

$$\mu = \frac{\frac{\partial}{\partial \bar{z}} f}{\frac{\partial}{\partial z} f}.$$

By (12.4.1.5),  $f$  preserves orientation iff  $|\mu| < 1$  at any point.

If  $f^{(-1)}$  is the inverse of  $f$  near  $f(z)$ , then

$$|\mu_{f^{-1}}(f(z))| = |\mu_f(z)|,$$

because

$$\frac{\partial}{\partial z} f^{-1} \frac{\partial}{\partial z} f + \frac{\partial}{\partial \bar{z}} f^{-1} \overline{\frac{\partial}{\partial \bar{z}} f} = 0$$

┘

**Def. (12.5.7.4) [Beltrami Equations].** For  $\mu \in C(\mathbb{C})$ , the equation  $\frac{\partial}{\partial z} f = \mu \frac{\partial}{\partial \bar{z}} f$  is called a **Beltrami equation**. ┘

**Prop. (12.5.7.5).** If  $f \in \mathbb{C}^1(\mathbb{C})$  is non-singular and  $\varphi$  is conformal, then

$$\begin{aligned}\mu_{\varphi \circ f}(z) &= \mu_f(z), \\ \mu_{f \circ \varphi}(z) &= \mu_f(\varphi(z)) \frac{\overline{\varphi'(z)}}{\varphi'(z)}.\end{aligned}$$

┘

*Proof:* These follow from the fact that  $\frac{\partial}{\partial \bar{z}}\varphi = 0$ . □

**Def. (12.5.7.6) [Quasiconformal Maps].** For a region  $\Omega \subset \mathbb{C}$  and  $k \in [0, 1)$ , a  **$k$ -quasiconformal map**  $f$  on  $\Omega$  is a function  $f^\infty(\Omega)$  s.t.  $|\mu| \leq k < 1$  at each point  $z \in \Omega$ . ┘

**Def. (12.5.7.7) [Quasi-Circles].** A **quasi-circle** is a subset of  $\mathbb{P}^1(\mathbb{C})$  that can be mapped via a quasiconformal homeomorphism of the sphere to  $\partial\mathbb{D}(0, 1) \in \mathbb{P}^1(\mathbb{C})$ . ┘

## 8 Multi-Variable case

### Basics

Should cover the part from [Complex Analytic and Differential Geometry Demailly], [Principle of Algebraic Geometry Griffith/Harris] and [Huy05].

**Def. (12.5.8.1) [Polydiscs].** For  $z \in \mathbb{C}^n$  and  $\varepsilon \in (\mathbb{R}_+)^n$ , the **polydisc**  $B(z, \varepsilon) \subset \mathbb{C}^n$  is defined to be the set  $\{z : |z_i - a_i| < \varepsilon_i\}$ . ┘

**Def. (12.5.8.2) [Holomorphic Functions in Several Variables].** Let  $\Omega \subset \mathbb{C}^n$  be an open subset, then  $f : \Omega \rightarrow \mathbb{C}$  is called a **holomorphic function in several variables** iff it is holomorphic for each coordinate separately. ┘

**Thm. (12.5.8.3) [Hartog's Theroem].** Let  $\Omega \subset \mathbb{C}^n$  be an open subset and  $f : \Omega \rightarrow \mathbb{C}$  be a holomorphic function (12.5.8.2), then for any  $z \in \Omega$ ,  $f$  is a convergent power series in some nbhd of  $z$ . ┘

*Proof:* Firstly, if  $f$  is continuous on some closed polydisc  $\overline{\mathbb{D}}(z, \varepsilon)$  around  $z$  (12.5.8.1), then

$$\begin{aligned}f(z) &= \frac{1}{2\pi i} \int_{|\zeta_1 - z_1| = \varepsilon_1} \cdots \int_{|\zeta_n - z_n| = \varepsilon_n} \frac{f(\zeta)}{(\zeta_1 - z_1) \cdots (\zeta_n - z_n)} d\zeta_1 \cdots d\zeta_n \\ &= \sum_{a_i \in \mathbb{N}} z_1^{a_1} \cdots z_n^{a_n} \cdot \int_{|\zeta_1 - z_1| = \varepsilon_1} \cdots \int_{|\zeta_n - z_n| = \varepsilon_n} \frac{f(\zeta)}{\zeta_1^{a_1+1} \cdots \zeta_n^{a_n+1}} d\zeta_1 \cdots d\zeta_n.\end{aligned}$$

is a power series around  $z$ .

$f$  is continuous on some closed polydisc around  $z$  if it is bounded on this closed polydisc, by Schwarz lemma (12.5.1.4) applied to each coordinate.

Now we prove the assertion by induction on  $n$ .  $n = 1$  if trivial, for  $n \geq 2$ , let  $\overline{\mathbb{D}} = \prod_{i=1}^n \overline{\mathbb{D}}(z_i, 2r)$  be a polydisc around  $z$ . For  $B \in \mathbb{Z}_+$ , let

$$\Omega_B = \{z' \in \prod_{1 \leq j \leq n-1} \overline{\mathbb{D}}(z_j, r) : |f(z', z_n)| \leq B, \forall z_n \in \overline{\mathbb{D}}(z_n, r)\},$$

then  $\Omega_B$  is closed because for each  $z_n \in \overline{\mathbb{D}}(z_n, r)$ ,  $f(z', z_n)$  is analytic in  $z'$  by induction hypothesis. But for fixed  $z' \in \prod_{1 \leq j \leq n-1} \overline{\mathbb{D}}(z_j, r)$ ,  $f(z', z_n)$  is continuous in  $z_n$ , thus bounded on  $\overline{\mathbb{D}}(z_n, r)$ . So we have

$$\bigcup_{B \in \mathbb{Z}_+} \Omega_B = \prod_{1 \leq j \leq n-1} \overline{\mathbb{D}}(z_j, r).$$

But then by Baire category theorem, some  $\Omega_B$  must have a non-empty interior, thus contains a closed polydisc  $\prod_{1 \leq j \leq n-1} \overline{\mathbb{D}}(w_j, \varepsilon) \times \overline{\mathbb{D}}(z_n, r)$ . Thus  $f$  is analytic on the interior of this polydisc by the argument above.

Now take the Taylor expansion

$$f(\zeta', \zeta_n) = \sum_{\alpha} c_{\alpha}(\zeta_n)(\zeta' - w')^{\alpha},$$

where each  $c_{\alpha}(\zeta_n)$  is holomorphic in  $\zeta_n$  and satisfies  $|c_{\alpha}(\zeta_n)| \leq M/\varepsilon^{|\alpha|}$  for some  $M \in \mathbb{R}_+$  uniformly in  $\zeta_n$  and  $\alpha$  by Cauchy's integral formula.

For any  $0 < r_1 < r_2 < r$  and  $\zeta_n \in \mathbb{D}(z_n, r)$ , we have  $\lim_{|\alpha| \rightarrow \infty} |c_{\alpha}(\zeta_n)| r_2^{|\alpha|} = 0$  because of the convergence of power series for  $\zeta' \in \prod_{1 \leq j \leq n-1} \mathbb{D}(w_j, r)$ . Thus the functions

$$\{u_{\alpha}(\zeta_n) = \frac{1}{|\alpha|} \log |c_{\alpha}(\zeta_n)|\}_{|\alpha| \geq 1}$$

are uniformly bounded on  $\mathbb{D}(z_n, r)$ , and for any fixed  $\zeta_n$ ,

$$\overline{\lim}_{|\alpha| \rightarrow \infty} u_{\alpha}(\zeta_n) \leq \log(1/r_2).$$

Thus by Hartog's lemma (12.5.3.2), for all  $\alpha$  sufficiently large, we have  $u_{\alpha}(\zeta_n) \leq \log(1/r_1)$  uniformly in  $\zeta_n \in \mathbb{D}(z_n, r_1)$ . Thus

$$f(\zeta', \zeta_n) = \sum_{\alpha} c_{\alpha}(\zeta_n)(\zeta' - w')^{\alpha}$$

converges uniformly to an analytic function on the polydisc  $\prod_{1 \leq j \leq n-1} \mathbb{D}(w_j, r_1) \times \mathbb{D}(z_n, r)$ . Since  $r_1 < r_2$  are arbitrary, we see  $f$  is analytic on the polydisc  $\prod_{1 \leq j \leq n-1} \mathbb{D}(w_j, r) \times \mathbb{D}(z_n, r)$ , which contains  $z$ , so  $f$  is analytic at  $z$ .  $\square$

**Thm. (12.5.8.4) [Hartog's Extension Theorem].** If  $n \in \mathbb{Z}_{\geq 2}$  and  $K$  is a compact subset in an open domain  $\Omega \subset \mathbb{C}^n$  s.t.  $\Omega \setminus K$  is connected, then any holomorphic function on  $\Omega \setminus K$  extends to a holomorphic function on  $\Omega$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (12.5.8.5).** Let  $\varepsilon = (\delta, \dots, \delta)$  and  $f$  be a holomorphic function on the polydisc  $\overline{B_{\varepsilon}(0)}$ . Then if  $f$  vanishes of order  $k$  at the origin and  $|f(z)| \leq C$ , then

$$f(z) \leq C \left( \frac{|z|}{\delta} \right)^k$$

for all  $z \in \overline{B_{\varepsilon}(0)}$ .  $\lrcorner$

*Proof:* Fix  $z \in \overline{B_{\varepsilon}(0)} \neq 0$ , consider the one-variable function  $g_z(w) = w^{-k} f(w \cdot \frac{z}{|z|})$ , then  $g_z$  is holomorphic and  $|g_z(w)| \leq \delta^{-k} C$  for  $|w| = \delta$ . So maximal principle implies that  $g_z(w) \leq \delta^{-k} C$  for all  $|w| \leq \delta$ . Hence  $|z|^{-k} |f(z)| = |g_z(z)| \leq \delta^{-k} C$ .  $\square$

## 12.6 Special Functions

References are [Special Functions].

### 1 Gamma Function

**Def. (12.6.1.1) [Gamma Function, Euler(1729)].** For  $s \in \mathbb{C}$  s.t.  $\operatorname{Re}(s) > 0$ , the **Gamma function** is defined to be

$$\Gamma(s) = \int_0^\infty e^{-t} t^s \frac{dt}{t} = \mathcal{M}(e^{-t})(s) \text{ (12.5.4.2).}$$

which is convergent for any  $\operatorname{Re}(s) > 0$  thus an analytic function for  $\operatorname{Re}(s) > 0$  by (12.4.2.14).  $\lrcorner$

**Prop. (12.6.1.2).** For  $s \in \mathbb{C}$  that  $\operatorname{Re}(s) > 0$ ,  $\Gamma(s+1) = s\Gamma(s)$ .  $\lrcorner$

*Proof:* For any  $\varepsilon \in \mathbb{R}_+$ ,

$$\int_\varepsilon^{1/\varepsilon} \frac{\partial}{\partial t} (e^{-t} t^s) dt = - \int_\varepsilon^{1/\varepsilon} e^{-t} t^s dt + s \int_\varepsilon^{1/\varepsilon} e^{-t} t^{s-1} dt.$$

Then letting  $\varepsilon \rightarrow 0$  gives the desired formula.  $\square$

**Cor. (12.6.1.3).**  $\Gamma(s)$  extends to a meromorphic function for all  $s \in \mathbb{C}$ , with simple poles at  $s \in \mathbb{Z}_{\leq 0}$ . And for  $n \in \mathbb{N}$ , the Laurent series of  $\Gamma(s)$  at  $s = -n$  is given by

$$\Gamma(s) = \frac{(-1)^n}{n!} \left( \frac{1}{s+n} + \left(1 + \frac{1}{2} + \dots + \frac{1}{n} - \gamma_0\right) + \dots \right),$$

where  $\gamma_0$  is the Euler's constant (35.1.2.8).  $\lrcorner$

*Proof:* ? We only prove that  $\operatorname{res}_{s=-n} \Gamma(s) = \frac{(-1)^n}{n!}$ .

In fact, for  $m \in \mathbb{Z}_+$ ,

$$F_m(s) = \frac{\Gamma(s+m)}{s(s+1)\dots(s+m-1)}$$

extends  $\Gamma(s)$  meromorphically to  $\operatorname{Re}(s) > -m$  with simple poles at  $s = 0, -1, \dots, -m+1$ , and

$$\operatorname{res}_{s=-m+1} F_m(s) = \frac{\Gamma(1)}{(-1)(-2)\dots(-m+1)} = \frac{(-1)^{m-1}}{(m-1)!}.$$

$\square$

**Lemma (12.6.1.4) [Gamma Function is Logarithmically Convex].**  $\log \Gamma(x)$  is convex on  $\mathbb{R}_+$ .  $\lrcorner$

*Proof:* It suffices to show that  $\Gamma''(x)\Gamma(x) \geq \Gamma'(x)^2$ . But notice

$$\Gamma'(x)^2 = \left( \int_0^\infty e^{-t} t^{x-1} \log t dt \right)^2 \leq \Gamma(x) \Gamma''(x)$$

by the Cauchy-Schwarz inequality.  $\square$

**Prop. (12.6.1.5) [Characterizing Gamma Function, Artin].** Let  $\Phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  be differential s.t.  $\log \Phi$  is convex and  $x\Phi(x) = \Phi(x+1)$  for all  $x \in \mathbb{R}_+$ , then we have  $\Phi(x) = \Gamma(1)\Gamma(x)$  for all  $x \in \mathbb{R}_+$ .  $\lrcorner$

*Proof:* Let  $H = \Phi/\Gamma$ , then  $H$  is 1-periodic, and  $H(1) = \Phi(1)$ . Since  $\Phi'/\Phi$  and  $\Gamma'/\Gamma$  are non-decreasing(12.6.1.4), we have

$$\frac{\Phi'(n)}{\Phi(n)} - \frac{\Gamma'(n+1)}{\Gamma(n+1)} \leq \frac{H'(n+x)}{H(n+x)} = \frac{\Phi'(n+x)}{\Phi(n+x)} - \frac{\Gamma'(n+x)}{\Gamma(n+x)} \leq \frac{\Phi'(n+1)}{\Phi(n+1)} - \frac{\Gamma'(n)}{\Gamma(n)}$$

for each  $n \in \mathbb{Z}_+$  and  $x \in \mathbb{R}_{\geq 0}$ . However, both  $\Gamma$  and  $\Phi$  satisfy the functional equation

$$\frac{f'(x+1)}{f(x+1)} - \frac{f'(x)}{f(x)} = \frac{1}{x},$$

so for each  $x \in \mathbb{R}_+$ , we get

$$\frac{H'(1)}{H(1)} - \frac{1}{n} \leq \frac{H'(x)}{H(x)} \leq \frac{H'(1)}{H(1)} + \frac{1}{n}.$$

Notice  $n \in \mathbb{Z}_+$  is arbitrary, so we see  $H'(x) = \kappa H(x)$  for some  $\kappa \in \mathbb{R}$ . Thus  $H(x) = H(1)e^{\kappa x}$ . Because  $H$  is 1-periodic, we see  $\kappa = 0$ .  $\square$

**Prop. (12.6.1.6).** For  $t \in \mathbb{R}_+$ ,  $\Gamma(t)$  is decreasing for  $t \leq 1$ , and increasing for  $t \geq 1$ .  $\lrcorner$

*Proof:*

$$\frac{\partial}{\partial t} \Gamma(t) = \int_0^\infty \log t e^{-t} t^s \frac{dt}{t}.$$

$\square$

**Prop. (12.6.1.7) [Euler's Reflection Formula].**

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}.$$

$\lrcorner$

*Proof:* By uniqueness theorem, it suffices to prove for  $0 < s < 1$ . Then

$$\begin{aligned} \Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t} t^{s-1} \Gamma(1-s) dt = \int_0^\infty e^{-t} t^{s-1} \left( t \int_0^\infty e^{-vt} (vt)^{-s} dv \right) dt \\ &= \int_0^\infty \int_0^\infty e^{-t(v+1)} v^{-s} dv dt = \int_0^\infty \frac{v^{1-s}}{v(v+1)} dv \\ (12.4.6.8) &= \frac{\pi}{\sin(\pi s)} \end{aligned}$$

$\square$

**Prop. (12.6.1.8) [Special Values].**

$$\Gamma(n+1) = n!, \quad \forall n \in \mathbb{N}, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

$\lrcorner$

*Proof:* For the first assertion, use induction:  $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$ , and the induction process follows from(12.6.1.2). The second assertion follows from(12.6.1.7).  $\square$

**Cor. (12.6.1.9).** For  $k \in \mathbb{N}$ ,

$$\Gamma\left(\frac{1-2k}{2}\right) = \frac{\sqrt{\pi}(-1)^k k! 2^{2k}}{(2k)!}.$$

┘

**Thm. (12.6.1.10) [Hadamard Product for  $\Gamma^{-1}(s)$ ].** For  $s \in \mathbb{C}$ ,

$$\frac{1}{\Gamma(s)} = e^{\gamma_0 s} \prod_{n \in \mathbb{Z}_+} \left[ \left(1 + \frac{s}{n}\right) e^{-s/n} \right].$$

where the product is absolutely convergent.

┘

*Proof:* By (12.6.1.21)(12.6.1.3) and (12.4.3.21),

$$\frac{1}{\Gamma(s)} = e^{As+B} s \prod_{n \in \mathbb{Z}_+} \left[ \left(1 + \frac{s}{n}\right) e^{-s/n} \right]$$

for some  $A, B \in \mathbb{C}$ . Taking  $s \rightarrow 0$ , we get  $B = 0$  by (12.6.1.3). And  $\Gamma(1) = 1$  (12.6.1.8) implies that

$$e^{-A} = \prod_{n \in \mathbb{Z}_+} \left[ \left(1 + \frac{1}{n}\right) e^{-1/n} \right] = \lim_{N \rightarrow \infty} e^{\log(N+1) - \sum_{n=1}^N 1/n} = e^{-\gamma} \quad (35.1.2.8)$$

□

**Cor. (12.6.1.11) [Legendre Duplication Formula].**

$$\Gamma(2s-1) = \frac{4^{s-1}}{\sqrt{\pi}} \Gamma\left(s - \frac{1}{2}\right) \Gamma(s),$$

or equivalently,

$$2^{1-2s} \sqrt{\pi} \Gamma(2s) = \Gamma(s) \Gamma\left(s + \frac{1}{2}\right).$$

┘

*Proof:*

□

**Cor. (12.6.1.12) [Euler's Formula].**

$$\frac{1}{\Gamma(s)} = \lim_{n \rightarrow \infty} \frac{s(s+1) \cdots (s+n)}{n^n n!}$$

┘

*Proof:* This follows from the definition of the infinite product and the definition of Euler's constant (12.6.1.7). □

**Cor. (12.6.1.13).** Taking logarithmic derivative, we get

$$-\frac{\Gamma'(s)}{\Gamma(s)} = \gamma_0 + \frac{1}{s} + \sum_{n \in \mathbb{Z}_+} \left( \frac{1}{s+n} - \frac{1}{n} \right).$$

In particular, by (12.6.1.8),

$$\Gamma'(1) = -\gamma_0 = \int_0^\infty e^{-t} \log t dt.$$

┘

**Prop. (12.6.1.14)**[Gauss-Legendre Multiplication Formula]. For  $s \in \mathbb{C}$ ,

$$\frac{\prod_{h=0}^{m-1} \Gamma_p(\frac{s+h}{m})}{\Gamma_p(s) \prod_{h=1}^{m-1} \Gamma_p(\frac{h}{m})} = m^{1-s}.$$

┘

*Proof:*

□

**Prop. (12.6.1.15)**[Mellin-Inversion Formula]. By (12.5.4.2) applied to  $f(x) = e^{-x}$ , for any  $c \in \mathbb{R}_+$ ,

$$e^{-x} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) x^{-s} ds, x > 0.$$

┘

**Thm. (12.6.1.16)**[Stirling's Formula]. For any  $\varepsilon \in (0, \pi)$ , we have

$$\Gamma(s) = \left(\frac{2\pi}{s}\right)^{\frac{1}{2}} \left(\frac{s}{e}\right)^s \left(1 + O_\varepsilon\left(\frac{1}{|s|}\right)\right)$$

for any  $s \in \mathbb{C}^\times$  with  $|\arg(s)| < \pi - \varepsilon$ . In particular, if  $\sigma \in \mathbb{R}$  is fixed, we have

$$\Gamma(\sigma + it) = \sqrt{2\pi}(it)^{\sigma-\frac{1}{2}} e^{-\frac{\pi}{2}|t|} \left(\frac{|t|}{e}\right)^{it} \left(1 + O_\varepsilon\left(\frac{1}{|t|}\right)\right).$$

┘

*Proof:* ? Cf. [Ten15]P175.

□

**Cor. (12.6.1.17)**[Stirling's Inequality]. For any  $n \in \mathbb{Z}_+$ , we have

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$

┘

*Proof:* ?

□

**Cor. (12.6.1.18)**. For any  $n \in \mathbb{Z}_+$ , we have

$$n \log n - n + 1 \leq \log(n!) \leq n \log n - n + 1 + \log n.$$

┘

**Def. (12.6.1.19)**[Archimedean  $L$ -Factors]. Define

$$L_{\mathbb{R}}(s) = \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right), \quad L_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s) = L_{\mathbb{R}}(s) L_{\mathbb{R}}(s+1).$$

Notice  $L_{\mathbb{R}}(1) = 1, L_{\mathbb{C}}(1) = \pi^{-1}$ , by (12.6.1.8).

┘

### Estimations

**Prop. (12.6.1.20).** For  $n \in \mathbb{Z}_{\geq 2}$ ,

$$\binom{2n}{n} \geq 2^n, \quad \binom{2n+1}{n} > 2^{n+1}.$$

*Proof:* It is true for  $n = 2$ , and for  $n \geq 3$ , we have

$$\binom{2n+1}{n} \geq \frac{2^{2n+1} - 2}{2n} \geq 2^{n+1}, \quad \binom{2n}{n} \geq \frac{2^{2n} - 2}{2n - 1} \geq 2^n$$

┘

□

**Prop. (12.6.1.21).** There exists  $c_1, c_2 \in \mathbb{R}_+$  s.t. for any  $s \in \mathbb{C}$ ,

$$\left| \frac{1}{\Gamma(s)} \right| \leq c_1 e^{c_2 |s| \log |s|}.$$

Thus  $1/\Gamma(s)$  has order of growth 1.

┘

*Proof:* For  $\operatorname{Re}(s) > 0$

$$\Gamma(s) = \int_0^1 e^{-t} t^{s-1} dt + \int_1^\infty e^{-t} t^{s-1} dt = \sum_{n \in \mathbb{N}} \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t} t^{s-1} dt,$$

and this equality holds for all  $s \in \mathbb{C}$ . Thus by (12.6.1.7),

$$\left| \frac{1}{\Gamma(s)} \right| = \left| \frac{\sin(\pi s) \Gamma(1-s)}{\pi} \right| \leq \left| \sum_{n \in \mathbb{N}} \frac{(-1)^n}{n!(n+1-s)} \right| \frac{|\sin(\pi s)|}{\pi} + \left| \int_1^\infty e^{-t} t^{-s} dt \right|.$$

Suppose  $k = \lfloor \operatorname{Re}(s) + \frac{1}{2} \rfloor$ , then

$$\left| \sum_{n \in \mathbb{N}} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin(\pi s)}{\pi} \right| \leq \sum_{n \in \mathbb{N}, n \neq k} \left| \frac{1}{n!(n+1-s)} \frac{\sin(\pi s)}{\pi} \right| + \left| \frac{\sin(\pi s)}{(k-1)!(s-k)\pi} \right|$$

and the first term is bounded by  $e^{\pi|s|}$ , and the second term is bounded by a constant independent of  $k$ , because  $\sin(\pi s)$  has a zero at  $s = k$ .

Also,

$$\left| \int_1^\infty e^{-t} t^{-s} dt \right| \leq \int_1^\infty e^{-t} t^{k+1} dt = (k+1)! \leq e^{(k+1) \log(k+1)}.$$

So the assertion follows.

□

## 2 Bessel Function

**Def. (12.6.2.1) [Bessel Function].** The **Bessel function** is defined to be

$$B(r, s) = \int_0^1 (1-y)^{r-1} y^{s-1} dy.$$

┘

**Prop. (12.6.2.2).**

$$B(x, y) = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)}.$$

┘

*Proof:*

□



### 3 K-Bessel Function

**Def. (12.6.3.1).** The **K-Bessel function** is defined to be

$$K_s(y) = \frac{1}{2} \int_0^\infty e^{-y(t+\frac{1}{t})/2} t^s \frac{dt}{t}, \quad y > 0.$$

it satisfies  $K_{-s}(y) = K_s(y)$ . ┘

**Prop. (12.6.3.2).**  $|K_s(y)| \leq e^{-y/2} K_{\operatorname{Re} s}(2)$  when  $y > 4$ . ┘

*Proof:* This is because  $e^{-ab} < e^{-a}e^{-b}$  when  $a, b > 2$ . □

**Prop. (12.6.3.3).**

$$\frac{\partial}{\partial y} K_s(y) = \frac{1}{2} \int_0^\infty \left(-\left(t + \frac{1}{t}\right)/2\right) e^{-y(t+\frac{1}{t})/2} t^s \frac{dt}{t} = -\frac{1}{2} (K_{s+1}(y) + K_{s-1}(y))$$
┘

**Prop. (12.6.3.4).** If  $\operatorname{Re}(s) > 1/2$  and  $r \in \mathbb{R}$ ,

$$\left(\frac{y}{\pi}\right)^s \Gamma(s) \int_{-\infty}^\infty (x^2 + y^2)^{-s} e^{2\pi i r x} dx = \begin{cases} \pi^{1/2-s} \Gamma(s - \frac{1}{2}) y^{1-s} & r = 0 \\ 2|r|^{s-1/2} \sqrt{y} K_{s-1/2}(2\pi|r|y) & r \neq 0 \end{cases}$$
┘

*Proof:* By (12.6.1.1), the LHS equals

$$\int_{-\infty}^\infty \int_0^\infty e^{-t} \left(\frac{ty}{\pi(x^2 + y^2)}\right)^s e^{2\pi i r x} \frac{dt}{t} dx = \int_0^\infty \int_{-\infty}^\infty e^{-\pi t(x^2 + y^2)/y} t^s e^{2\pi i r x} dx \frac{dt}{t}$$

where we interchanged the order of integration because it is absolutely convergent, and made a change of variable. Also notice

$$\int_{-\infty}^\infty e^{-t\pi x^2/y} e^{2\pi i r x} dx = \sqrt{\frac{y}{t}} e^{-y\pi r^2/t}$$

by Fourier inversion (12.11.3.3), so we get the final answer. □

**Prop. (12.6.3.5).** If  $\operatorname{Re}(s) + k/2 > 1/2$  and  $r$  is real,

$$\left(\frac{y}{\pi}\right)^s \Gamma(s) \int_{-\infty}^\infty \frac{1}{(x + iy)^k (x^2 + y^2)^s} e^{-2\pi i r x} dx = \begin{cases} \pi^{1/2-s} \Gamma(s - \frac{1}{2}) y^{1-s} & r = 0 \\ 2|r|^{s-1/2} \sqrt{y} K_{s-1/2}(2\pi|r|y) & r \neq 0 \end{cases}$$
┘

*Proof:* By (12.6.1.1), the LHS equals

$$\int_{-\infty}^\infty \int_0^\infty e^{-t} \left(\frac{ty}{\pi(x^2 + y^2)}\right)^s e^{2\pi i r x} \frac{dt}{t} dx = \int_0^\infty \int_{-\infty}^\infty e^{-\pi t(x^2 + y^2)/y} t^s e^{2\pi i r x} dx \frac{dt}{t}$$

where we interchanged the order of integration because it is absolutely convergent, and made a change of variable. Also notice

$$\int_{-\infty}^\infty e^{-t\pi x^2/y} e^{2\pi i r x} dx = \sqrt{\frac{y}{t}} e^{-y\pi r^2/t}$$

by Fourier inversion (12.11.3.3), so we get the final answer. □

### 4 Hypergeometric Functions

### 5 Confluent Hypergeometric Functions

### 6 Dilogarithm

## 12.7 General Functional Analysis

Basic references are [Rudin Functional Analysis],[Nonarchimedean Functional Analysis].

This section only contains theorems that are applicable to both Archimedean and non-Archimedean valuations. For theorems specialized to non-Archimedean valuations, See [p-adic Analysis](#), for theorems specialized to Archimedean valuations, See [Archimedean Functional Analysis](#). Many propositions in Functional Analysis can be transplanted in the general case, but I haven't finish yet.

The major problem is that convexity is not definable, so Hahn-Banach fail, causing many to fail.

### Notation(12.7.0.1).

- Fix a complete valued field  $K$ (12.2.3.10).
- If  $\text{char } K \neq 0$ , then we fix a sequence of elements  $\{a_n\}$  that  $\lim |a_n| = \infty$ . This will be applied for example in the proof of Banach-Steinhaus theorem, but we will just write  $n$  instead of  $a_n$ .  $\lrcorner$

### 1 Topological Vector Space

**Def.(12.7.1.1) [Topological Vector Spaces].** A **topological vector space**(TVS) over  $\mathbb{K}$  is a  $\mathbb{K}$ -vector space with a topological structure s.t. the addition and scalar multiplications are continuous maps.  $\lrcorner$

**Def.(12.7.1.2)[Cauchy Sequences in Topological Spaces and Completeness].** For  $X \in \text{TVS}/\mathbb{K}$ , a **Cauchy-sequence(in TVS)** in  $X$  is a sequence  $(x_i)_{i \in \mathbb{Z}_+} \in X$  s.t. for any nbhd  $V$  of  $0 \in X$ , there exists some  $N \in \mathbb{Z}_+$  s.t.  $x_m - x_n \in V$  for any  $m, n \in \mathbb{Z}_{\geq N}$ . This is compatible with the notion of Cauchy-sequences in metric spaces when  $X$  is metrizable.

And  $X$  is called a **complete topological vector space** if every Cauchy-sequences in  $X$  converges.  $\lrcorner$

**Prop.(12.7.1.3).** For subsets  $K, C$  of a TVS  $X$  that  $K$  is compact and  $C$  is closed, there is a nbhd  $V$  that  $(K + V) \cap (C + V) = \emptyset$ .  $\lrcorner$

*Proof:* For each  $x \in K$ , there are symmetric nbhd  $V_x$  that  $(x + V_x + V_x + V_x) \cap C = \emptyset$ . Then  $(x + V_x + V_x) \cap (C + V_x) = \emptyset$ . Because  $K$  is compact, there are f.m.  $x_i$  that  $K \subset \cup(x_i + V_{x_i})$ , so let  $V = \cap V_{x_i}$ , then  $(K + V) \cap (C + V) = \emptyset$ .  $\square$

**Cor.(12.7.1.4)[Closed Subbasis].** Every nbhd of  $0$  in a TVS contains a closure of another nbhd of  $0$ . (Apply the above proposition for  $K = \{0\}$ ).  $\lrcorner$

**Def.(12.7.1.5)[Balanced Subsets].** A subset containing  $0$  is called **balanced subset** iff  $kU = U$  for each  $k \in K$  s.t.  $|k| = 1$ .  $\lrcorner$

**Prop.(12.7.1.6)[Balanced Subbasis].** Every nbhd  $U$  of  $0$  in a TVS contains a balanced nbhd of  $0$ . By(12.7.1.4), we can even assume  $\overline{V} \subset U$ .  $\lrcorner$

*Proof:* Since scalar multiplication is continuous, there is a  $\delta > 0$  and a nbhd  $V$  that  $\alpha V \subset U$ , for each  $|\alpha| < \delta$ . Then let  $W = \cup_{|\alpha| < \delta} \alpha V$ .  $\square$

**Def.(12.7.1.7)[F-Spaces and Fréchet spaces].** A space is called an **F-space** if its topology is induced by a complete invariant metric. F-space is of second Baire-category by(4.4.9.2)

A locally convex F-space is called a **Fréchet space**.  $\lrcorner$

**Def. (12.7.1.8) [(Semi-)Norms].** For  $X \in \text{Vect}_K$ , a **semi-norm** on a vector space  $X$  is a real-valued function  $p$  that  $p(x+y) \leq p(x) + p(y)$ , and  $p(\alpha x) = |\alpha|p(x)$  for  $\alpha \in k$ .

It is called a **norm** if moreover  $p(x) = 0 \iff x = 0$ , in which case it is called a **normed  $K$ -space**.

A family of seminorms  $\{p_i\}$  on  $X$  is called **separating** iff for each  $x$ , at least one  $p_i$  satisfies  $p_i(x) \neq 0$ .  $\lrcorner$

**Prop. (12.7.1.9).** A TVS is metrizable by a translation-invariant metric iff it has a countable basis.  $\lrcorner$

*Proof:* One direction is trivial, for the other, Cf.[Rudin P18].  $\square$

**Prop. (12.7.1.10).** If a subspace  $Y$  of a TVS  $X$  is a F-space, then it is closed in it.  $\lrcorner$

*Proof:* Choose an invariant metric  $d$  compatible with its topology, Let  $U_n$  be a nbhd of  $X$  that  $U_n \cap Y = B(0, 1/n)$ , and choose a symmetric nbhd  $V_n$  of  $X$  that  $V_n + V_n \subset U_n$ , and  $V_{n+1} \subset V_n$ .

If  $y \in \overline{Y}$ , then for any  $y_n \in Y \cap (y + V_n) = E_n$ , then  $y_n - y_m \in U_{\min(m,n)} \cap Y = B(0, 1/n)$ , so it is a Cauchy sequence in  $Y$ , hence all  $E_n$  has a unique element  $y_0$  in common. Now if we intersect each  $V_n$  by a nbhd  $W$  of  $X$ , the same argument shows that there is a unique element  $y_W$  in  $Y \cap (y + W \cap V_n)$ , and this must be just  $y_0$ , but  $y - y_W \in W$ , so we must have  $y = y_0 \in Y$ .  $\square$

**Def. (12.7.1.11) [Totally Bounded Sets].** A set  $E$  in a TVS is called **totally bounded subset** if for every nbhd  $V$  of 0, there is a finite set  $F$  that  $E \subset F + V$ .  $\lrcorner$

## 2 Completeness

**Thm. (12.7.2.1) [Banach-Steinhaus].** Let  $X, Y \in \text{TVS}$  and  $\Gamma \subset \text{Hom}(X, Y)$ . If the set

$$B = \{x \in X \mid \Gamma(x) \text{ is bounded}\}$$

is a second Baire-category set in  $X$ , then  $B = X$  and  $\Gamma$  is equicontinuous, (thus maps bounded sets to bounded sets).  $\lrcorner$

*Proof:* For an open balanced nbhd  $W$  of 0, choose a balanced nbhd  $U$  s.t.  $\overline{U} + \overline{U} \subset W$  (12.7.1.6), set  $E = \bigcap_{\Lambda \in \Gamma} \Lambda^{-1}(\overline{U})$ , then  $B \subset \bigcup_{i=1}^{\infty} nE$ , so by Baire theorem (4.4.9.2),  $E$  has a interior point thus has a nbhd  $V$  s.t.  $\Gamma(V) \subset \overline{U} + \overline{U} \subset W$ . Thus we are done.  $\square$

**Cor. (12.7.2.2) [Uniform Boundedness Theorem].** If  $X, Y \in \text{TVS}$  and  $X$  is an F-space (12.7.1.7). Suppose  $\Gamma \subset \text{Hom}(X, Y)$  satisfies  $\Gamma(x)$  is bounded for any  $x \in X$ , then  $\Gamma$  is equicontinuous.  $\lrcorner$

**Cor. (12.7.2.3).** Let  $X, Y \in \text{TVS}$  and  $X$  is an F-space. If  $(A_n) \in \text{Hom}(X, Y)^{\mathbb{Z}_+}$  is a sequence of continuous linear mapping, then  $\varinjlim A_n = A$  iff  $\{\|A_n\|\}$  is bounded and  $\varinjlim A_n x = Ax$  for  $x$  in a dense subset of  $X$ .  $\lrcorner$

*Proof:* One direction is immediate from Banach-Steinhaus and (12.7.1.7), the converse is an easy  $\varepsilon/3$ -technique.  $\square$

**Cor. (12.7.2.4) [Strong operator convergence].** Let  $X, Y \in \text{TVS} / \mathbb{K}$  and  $X$  is an F-space (12.7.1.7). If  $(A_n) \in \text{Hom}(X, Y)^{\mathbb{Z}_+}$  is a sequence of continuous linear mapping s.t.  $\varinjlim A_n x = Ax$  for any  $x \in X$ , then  $\varinjlim A_n = A$ .  $\lrcorner$

**Thm. (12.7.2.5) [Open Mapping theorem].** Let  $X, Y \in \text{TVS}/\mathbb{K}$  and  $X$  is an F-space (12.7.1.7). Suppose  $T \in \text{Hom}(X, Y)$  and  $\text{Im}(T)$  is of second Baire-category in  $Y$ , then it is a surjective open mapping and  $Y$  is an F-space.

In fact, we only need  $T$  be defined on a subspace  $D(T) \subset X$  and it is a **closed operator** in the sense that its graph is closed.  $\perp$

*Proof:* For any  $n \in \mathbb{N}$ ,  $V_n = T(B(0, \frac{1}{2^n}))$  is of second category by the definition (4.4.9.1) and the fact  $\cup_{m \in \mathbb{Z}_+} mV_n = R(T)$ . In particular  $\overline{V_n}$  has an interior. Then also it contains a nbhd of  $0 \in V$  because  $\overline{V_{n+1}} - \overline{V_{n+1}} \subset \overline{V_n}$ .

Now we show  $\overline{V_{n+1}} \subset V_n$ , which will imply that  $T$  is open.

For any  $y \in \overline{V_{n+1}}$ , since  $\overline{V_{n+1}}$  contains a nbhd of 0, we can consecutively choose  $x_m \in B(0, \frac{1}{2^{n+m}})$  s.t.

$$y - \sum_{i=0}^m T(x_i) \in \overline{V_{n+m+1}}.$$

for any  $m \in \mathbb{Z}_+$ . Then because  $X$  is complete and  $T$  is closed, we see  $\sum x_i$  converges to some  $x \in D(T) \cap B(0, \frac{1}{2^{n-1}})$ , and  $Tx = y \in V_{n-1}$ .

Any open linear mapping must be surjective, hence  $Y \cong X/\ker(T)$ , so  $Y$  is also an F-space.  $\square$

**Cor. (12.7.2.6) [Banach Theorem].** If a continuous map  $T$  between F-spaces is a bijection, then it has a continuous inverse.  $\perp$

**Cor. (12.7.2.7).** If a F-space is complete w.r.t two different topologies and one is stronger than the other, then they are equivalent.  $\perp$

**Cor. (12.7.2.8).** For every operator between F-spaces that has closed image, we have  $X/N(T) \cong R(T)$ .  $\perp$

**Cor. (12.7.2.9) [Closed Graph Theorem].** If  $T : X \rightarrow Y$  is a closed linear mapping between two F-spaces, i.e. the graph of it is closed, then it is continuous.

Notice the graph being closed is equivalent to if  $x_i \rightarrow x$  and  $Tx_i \rightarrow y$ , then  $y = Tx$ . This is very useful when proving some map is continuous.  $\perp$

*Proof:* Because the metric induced by the graph is stronger than the original one, we can use Banach Theorem.  $\square$

**Cor. (12.7.2.10).** If  $A, B, C$  are F-spaces, and  $f : A \rightarrow B, g : B \rightarrow C$ , if  $gf, g$  is continuous and  $g$  is injective, then  $f$  is continuous.  $\perp$

*Proof:* Use closed graph theorem, if  $x_i \rightarrow x, f(x_i) \rightarrow z$ , then  $gf(x_i) \rightarrow g(z)$ , so  $gf(x) = g(z)$ , so  $f(x) = z$ .  $\square$

**Cor. (12.7.2.11) [Finite Codimensional Image].** If the image of a continuous linear mapping  $T$  between F-spaces has finite codimensional image, then the image is closed and complemented.  $\perp$

*Proof:* It has finite codimension, so we can construct  $K^n \oplus X/N(T) \rightarrow Y$ , by Banach theorem (12.7.2.6) it is a homeomorphism, and the image of  $X/N(T)$  corresponds to the image, so the image is closed.  $\square$

**Prop. (12.7.2.12) [Separate Continuous].** If a bilinear map  $B : X \times Y \rightarrow Z$  where  $X$  is a F-space is closed in each coordinate, then  $B(x_n, y_n)$  converges to  $B(x_0, y_0)$ .  $\perp$

*Proof:* Use Banach-Steinhaus to prove  $B_{y_n}(x)$  is equicontinuous, then use  $B(x_n - x_0, y_n) + B(x_0, y_n - y_0)$ . ?  $\square$

### 3 Dual Space

**Prop. (12.7.3.1) [Operator Spaces].** If  $X, Y$  are normed spaces then  $L(X, Y)$  is also a normed space with the metric  $\|\Lambda\| = \sup\{\|\Lambda x\| : \|x\| \leq 1\}$ , called the **operator space** from  $X$  to  $Y$ . And if  $Y$  is Banach, then  $L(X, Y)$  is also Banach. The proof is easy.

In particular,  $X^*$  is a Banach space.  $\lrcorner$

**Prop. (12.7.3.2).** For a bounded operator  $T$ ,

$$\overline{R(T)} = N(T^*)^\perp, \quad \overline{R(T^*)} = N(T)^\perp.$$

In particular, using Hahn Banach,  $R(T)$  is dense in  $Y$  iff  $T^*$  is injective,  $T$  is injective iff  $T^*$  is weak\*-dense in  $X^*$ .  $\lrcorner$

**Prop. (12.7.3.3).** Let  $\Lambda_1, \dots, \Lambda_n, \Lambda$  are linear functionals on a vector space  $X$ , let  $N = \cap \ker f_i$ , the following are equivalent:

1.  $\Lambda = \sum \alpha_i \Lambda_i$ .
2.  $|\Lambda x| \leq \gamma |\Lambda_i x|$ .
3.  $\ker \Lambda \subset N$ .

$\lrcorner$

*Proof:* Only need to show  $3 \rightarrow 1$ : define  $\pi : X \rightarrow k^n : x \mapsto (\Lambda_1 x, \dots, \Lambda_n x)$ , then by hypothesis  $f(\pi_i(x)) = \Lambda(x)$  defined a linear functional on  $\pi(X)$ . This can be extended to a functional on  $k^n : F(u_1, \dots, u_n) = \sum \alpha_i u_i$ , so  $\Lambda$  is a linear combination of  $\Lambda_i$ .  $\square$

### Weak Convergence

**Def. (12.7.3.4) [Operator Topologies].** There are three topologies on  $L(X)$  for a normed space  $X$ :

- norm topology:  $\|T_i - T\| \rightarrow 0$ .
- strong topology:  $\|(T_i - T)x\| \rightarrow 0$  for any  $x \in X$ .
- weak topology:  $\lim f(T_n x) = f(Tx)$  for any  $x \in X$  and  $f \in X^*$ .

$\lrcorner$

**Prop. (12.7.3.5) [Weak Convergence and Boundedness].** In a normed space  $X$ , if  $x_n \rightarrow x$  weakly iff  $\{x_n\}$  is bounded and  $\lim f(x_n) = f(x)$  for  $f$  in a dense subset a dense subset  $M \subset X^*$ .  $\lrcorner$

*Proof:* This follows from (12.7.2.3), as  $X^*$  is a Banach space, by (12.7.3.1).  $\square$

### 4 Banach Spaces

**Def. (12.7.4.1) [Banach Spaces and Banach Algebras].** A completely normed (valued)  $K$ -vector space (12.7.1.8) is called a **Banach space**.

A completely normed  $K$ -algebra  $A \in \mathcal{Alg}/K$  s.t.  $\|xy\| \leq \|x\| \|y\|$  for all  $x, y \in A$  is called a **Banach-algebra**.

Denote by  $\text{BanSpa}_K$  the category of Banach spaces over  $K$ , and denote  $\text{BanAlg}_K$  the category of Banach algebras over  $K$ .  $\lrcorner$

**Prop. (12.7.4.2).** The dual space of a Banach space is a Banach space. (Immediate from (12.7.3.1)).  $\lrcorner$

**Prop. (12.7.4.3).** If  $A$  is a Banach space as well as a topological group, then there is a norm on  $A$  which induce the same topology and makes  $A$  into a Banach algebra.  $\lrcorner$

*Proof:* embed  $A$  into  $L(A)$  by left multiplication, which is injective, and  $\|x\| = \|xe\| = \|M_x e\| \leq \|M_x\| \|e\|$ , so its inverse is continuous. Now if we show the image  $\tilde{A}$  is closed in  $L(A)$ , then the open mapping theorem(12.7.2.5) shows that  $A \cong \hat{A}$ , and  $\hat{A}$  is clearly a Banach algebra.

To show it is closed, if  $T = \lim T_i$ , notice  $T_i(y) = T_i(e)y$ , so take a limit,  $T(y) = T(e)y = M_{T(e)}y$ .

□

**Cor. (12.7.4.4).** Any  $A \in \text{BanSpa}_K^{\text{fd}}$  is isomorphic to  $\text{Mat}(n; K)$  for some  $n \in \mathbb{N}$ . In particular, if  $xy = e$ , then  $yx = e$ .  $\lrcorner$

*Proof:* Embed  $A$  into  $L(A)$ .  $\square$

**Remark (12.7.4.5)[Inequivalent Banach Norms].** There exist two complete norm on a vector space that are inequivalent. For this, just choose a Banach space  $X$ , and notice if we can choose a discontinuous bijection  $X \rightarrow X$ , then the induced metric is also complete, and it cannot be equivalent by Banach theorem(12.7.2.6). For this, choose an infinite dimensional Banach space over  $\mathbb{C}$ , and choose a  $\mathbb{C}$ -basis  $x_i$  for it, and choose a sequence  $x_n$  and maps  $x_n$  to  $nx_n$ , the rest are invariant, then this is not continuous.  $\lrcorner$

### Hilbert Spaces

**Def. (12.7.4.6).** For any pre-Hilbert space  $\mathcal{H}$ , there are different topologies on  $B(\mathcal{H})$ :

- The **norm operator topology** is defined by the operator norm  $\|T\|$ .
- The **ultra-strong topology** is defined by the separating seminorms(12.7.1.8)

$$\{T \mapsto \sum_{n \in \mathbb{Z}} \|Tv_n\|^2 : v_i \in \mathcal{H}, \sum_i \|v_i\|^2 < \infty\}.$$

- The **strong operator topology** is defined by the separating seminorms

$$\{T \mapsto \|Tu\| : u \in \mathcal{H}\}.$$

- The **weak operator topology**: defined by the separating seminorms

$$\{T \mapsto (Tu, v) : u, v \in \mathcal{H}\}.$$

By default,  $B(\mathcal{H})$  is equipped with the norm operator topology.  $\lrcorner$

**Prop. (12.7.4.7).** For any pre-Hilbert space  $\mathcal{H}$ , the strong and weak operator topologies(12.7.4.6) coincide on the set  $U(\mathcal{H})$  of unitary operators on  $\mathcal{H}$ .  $\lrcorner$

*Proof:* If  $T_n$  converges to  $T$  in the weak operator topology, then

$$\|(T_n - T)u\|^2 = \|Tu\|^2 + \|T_n u\|^2 - 2\text{Re}(T_n u, Tu).$$

The right hand side is clearly bounded by the weak seminorms, so the two topologies coincide.  $\square$

**Prop. (12.7.4.8)[Hilbert Bases].** For  $(\mathcal{H}, S = \{e_\alpha\}) \in \mathcal{HilbSpa}$ , the following are equivalent:

1. For any  $x$ ,  $x = \sum(x, e_\alpha)e_\alpha$ , (notice the sum are in fact infinite sum).

2. There is a no nonzero element  $x$  that is orthogonal to all  $e_\alpha$ .

3. **Parseval equality** holds:  $\|x\|^2 = \sum |(x, e_\alpha)|^2$ .

If these are true, then  $S$  is called a **Hilbert basis** of  $H$ , a Hilbert basis always exists, by Zorn's lemma. We use the notation  $(\mathcal{H}, S = \{e_\alpha\}) \in \mathcal{HilbSpa}$  to denote that  $\mathcal{H}$  is a Hilbert space and  $S$  is a Hilbert basis of  $\mathcal{H}$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 2$  : if  $(x, e_\alpha) = 0$  for all  $e_\alpha$ , then by 1,  $x = \sum (x, e_\alpha)e_\alpha = 0$ .

$2 \rightarrow 3$  : Notice  $y = x - \sum (x, e_\alpha)e_\alpha$  is orthogonal to all  $e_\alpha$ , and

$$\|y\| = \|x\|^2 - \sum |(x, e_\alpha)|^2.$$

In particular, the Parseval equality holds.

$3 \rightarrow 1$  :  $\|x - \sum (x, e_\alpha)e_\alpha\| = 0$ .  $\square$

**Prop. (12.7.4.9)[Symmetric Operators are Continuous, Hellinger-Toeplitz].** For  $\mathcal{H} \in \mathcal{HilbSpa}$ , any symmetric operator on  $\mathcal{H}$  is continuous.  $\lrcorner$

*Proof:* Use the [Closed Graph Theorem](#). Suppose  $x_i \rightarrow x$  and  $Tx_i \rightarrow y$ , then for any  $z \in \mathcal{H}$ ,  $(y, z) = \lim_i (Tx_i, z) = \lim_i (x_i, Tz) = (x, Tz) = (Tx, z)$ . Thus  $y = Tx$ . Then by [Closed Graph Theorem](#),  $T$  is continuous.  $\square$

### Ultrarnormed Banach Spaces

The ultrarnormed Banach spaces are defined in [\(19.1.6.5\)](#).

### Nuclear Maps and Spaces

## 12.8 Archimedean Functional Analysis

References are [Rud91]. Chap11,13 needs to be revisited.

This section studies functional analysis over Archimedean local fields.

**Notation(12.8.0.1).**

- Let  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ .
- Let  $\Omega \subset \mathbb{K}^n$  be a non-empty open subset.
- Use notation on general Functional Analysis defined in [General Functional Analysis](#).
- Let  $X \in \text{TVS}/\mathbb{K}$ .

┘

### 1 Topological Vector Space

**Def.(12.8.1.1)[Seminorms].** A **sublinear functional** on  $X$  is a function  $p : X \rightarrow \mathbb{R}$  s.t.

- $p(x + y) \leq p(x) + p(y)$  for any  $x, y \in X$ .
- $p(\lambda x) = \lambda p(x)$  for any  $x \in X$  and  $\lambda \in \mathbb{R}_+$ .

And  $p$  is called a **seminorm** if moreover  $p$  is a non-negative, and  $p(\alpha x) = |\alpha|p(x)$  for all  $\alpha \in \mathbb{K}$ .

┘

**Def.(12.8.1.2)[Absorbing Sets].** A **absorbing set** is a convex set  $A \subset X$  s.t.  $\cup_{k>0} kA = X$ . A convex nbhd of 0 is clearly absorbing.

┘

**Def.(12.8.1.3)[Minkowski Functional].** For an absorbing set  $A$ , the **Minkowski functional**  $\mu_A$  is defined to be  $\mu_A(x) = \inf\{t > 0, x/t \in A\}$ . It satisfies:

- $\mu_A(x + y) \leq \mu_A(x) + \mu_A(y)$ .
- $\mu_A(kx) = k\mu_A(x)$  if  $k > 0$ .
- $\mu_A$  is a seminorm if  $A$  is balanced.
- If  $B = \{x | \mu(x) < 1\}$ ,  $C = \{x | \mu(x) \leq 1\}$ , then  $B \subset A \subset C$  and  $\mu_A = \mu_B = \mu_C$ .

┘

*Proof:* Cf.[Rudin P27].

□

**Cor.(12.8.1.4)[Seminorm and Absorbing set].** A seminorm on  $X$  is exactly the Minkowski functional of a balanced absorbing set  $W$ , but the set may not be unique. and it is uniformly continuous iff 0 is an interior point.

┘

*Proof:* If  $p$  is a seminorm, then  $\{x | p(x) < 1\}$  is convex, balanced and absorbing by definition(12.7.1.8). The converse is by(12.8.1.3). The last assertion is easy.

□

**Prop.(12.8.1.5)[Minkowski Functional and Separating Seminorms].** If  $\mathfrak{B}$  is a convex local base in a TVS  $X$ , then the Minkowski functionals of elements of  $\mathfrak{B}$  forms a separating family of seminorms.

Conversely, a separation family  $\mathfrak{P}$  of seminorms on a vector space defines a convex balanced local base for a topology  $\tau$  that is locally convex. And in this topology, a sequence converges iff  $p(x_i - x) \rightarrow 0$  for  $p \in \mathfrak{P}$ , a set is bounded if each  $p$  is bounded on it.

┘



*Proof:* For any  $V \in \mathfrak{B}$ ,  $V = \{x \in X \mid \mu_V(x) < 1\}$ , because  $V$  is open and convex. (12.8.1.3) shows each  $\mu_V$  is a seminorm, and it is continuous because it is bounded on  $V$ . And they are separating because  $\mathfrak{B}$  is a local base.

Defined  $V(p, n) = \{x \in X \mid p(x) < 1/n\}$ , and let these be a local subbasis at 0, and make it a topology by translation. This is checked to be a locally convex TVS. For the last assertion, if  $E$  is bounded, then  $E \subset kV(p, 1)$  for  $k$  large, so  $p$  is bounded on  $E$ , and conversely, for each nbhd  $U$ , there are  $p_i$  and  $M_i$  and  $\cap V(p_i, M_i) \subset U$ , so  $E \subset kU$  for  $n$  large.  $\square$

**Prop. (12.8.1.6).** If  $\mathfrak{P}$  is a family of countable separating family of semi-norms on  $X$ , then the topology defined in (12.8.1.5) is in fact metrizable, by a metric  $d(x, y) = \sum \frac{1}{2^k} \frac{p_i(x-y)}{1+p_i(x-y)}$ .  $\lrcorner$

### Finite Dimensional Subspace

**Prop. (12.8.1.7) [Finite Dimensional and Locally Compact].** If  $\dim_{\mathbb{K}} X < \infty$ , then there is only one topological vector space structure on  $X$ , and  $X$  is complete in this topology.

$X$  is locally compact iff it is f.d.  $\lrcorner$

*Proof:* Cf. [Rudin P17].

For the second assertion, if  $X$  is locally compact, then 0 has a nbhd  $V$  that is precompact, so bounded, hence  $2^{-n}V$  forms a local basis. the compactness of  $\overline{V}$  shows there are f.m.  $x_i$  that  $\overline{V} \subset \cup(x_i + \frac{1}{2}V)$ . Let  $Y$  be the subspace spanned by  $x_i$ , then it is of f.d, thus closed. Now  $V \subset Y + \frac{1}{2}V$ , so  $\frac{1}{2}V \subset Y + \frac{1}{4}V$ , hence continuing this way,  $V \subset \cap(Y + 2^{-n}V)$ , so  $V \subset \overline{Y} = Y$ . But then  $Y = X$ .  $\square$

**Cor. (12.8.1.8) [Finite Subspace Closed].** A f.d subspace in a TVS over  $\mathbb{C}$  is closed, because it must be a  $F$ -space, hence it is closed by (12.7.1.10).  $\lrcorner$

**Prop. (12.8.1.9) [Finite Subspace in Banach Space].** For a finite dimensional space  $V$  in an Archimedean Banach space, there is a continuous projection onto it. In particular, any finite dimensional space in an Archimedean Banach space is complemented.

Also finite codimensional subspace in any Banach space is complemented by (12.7.2.11).  $\lrcorner$

*Proof:* Choose a basis  $e_i$  for  $V$ , consider the dual basis  $f_i$ . Because a finite dimensional space only has one topology (12.8.1.7), these  $f_i$  are bounded on  $V$ . Extend them to bounded functional on  $X$ , then consider  $p(x) = \sum f_i(x)e_i$ , then this is a continuous projection onto  $V$ .  $\square$

## 2 Various Spaces and Duality

**Example (12.8.2.1) [Various Spaces and Duality].**

- For any  $m \in \mathbb{Z}_+$ , the **Sobolev Space**  $W^{m,p}(\Omega)$  is the completion of a subspace of  $C^\infty(\Omega)$  with the norm?

$$\|u\|_{m,p} = \left( \sum_{|\alpha| \leq m} \int_{\Omega} |\partial^\alpha u(x)|^p dx \right)^{1/p}.$$

And we denote  $H^m(\Omega) \triangleq W^{m,2}(\Omega)$ . It is also a subspace of  $L^p(\Omega)$  that satisfies this, without completion (12.11.4.1).

- $C_0^\infty(\Omega)$  is the subspace of  $C^\infty(\Omega)$  that have compact support in  $\Omega$ . Its completion  $W_0^{m,p}(\Omega)$  is a closed subspace of  $W^{m,p}(\Omega)$ . And we denote  $W_0^{m,2}(\Omega)$  by  $H_0^m(\Omega)$  and the dual space of  $H_0^m(\Omega)$  by  $H^{-m}(\Omega)$ .

- For a locally convex space  $X$ ,  $C(X)$  in the topology of compact convergence is a Fréchet space(12.7.1.7).
- $\mathcal{O}(\Omega) \subset C(\Omega)$  the space of holomorphic functions in  $\Omega$  is a closed subspace of  $C(\Omega)$  by(12.4.2.15), thus it is a Fréchet space. By Hausdorff theorem(4.4.8.27), Montel's theorem says exactly that  $\mathcal{O}(\Omega)$  has the Heine-Borel property(12.7.1.7).
- $\mathcal{O}^2(\Omega)$  the space of holomorphic functions on  $\Omega$  that is also  $L^2$ . It has the  $L^2$ -norm.
- $C^\infty(\Omega)$  in the topology defined by seminorms  $p_N(f) = \max\{|D^\alpha f(x)| : x \in K_N, |\alpha| \leq N\}$ , is a Fréchet-space thus locally convex and it has the Heine-Borel property by Arzela-Ascoli.
- For  $K \subset \Omega$  compact,  $D(K)$  is the closed subspace of  $C^\infty(\Omega)$  consisting of functions  $f$  s.t.  $\text{Supp}(f) \subset K$ . It is a Fréchet-space with Heine-Borel property.
- The set of **test functions**  $\mathcal{T}(\Omega)$  is the space of smooth functions on  $\Omega$  with support contained in  $\Omega$ . It has topology defined in(12.11.2.2). s.t.  $\text{Supp}(f) \subset \Omega$  making  $\mathcal{T}(\Omega)$  into a locally convex TVS. It is complete and has Heine-Borel property, by(12.11.2.3).
- The space of **Schwartz-functions**  $\mathcal{S}(\mathbb{R}^n)$  is defined as smooth functions on  $\mathbb{R}^n$  s.t.

$$\sup_{|\alpha| \leq N, x \in \mathbb{R}^n} (1 + |x|^2)^N |(D_\alpha f)(x)| < \infty$$

for any  $N \in \mathbb{Z}_+$ . And it is a Fréchet-space defined by these seminorms. ┘

### Dual Spaces

**Prop.(12.8.2.2).**

- For a  $\sigma$ -finite measure space  $(X, \Omega, \mu)$  and  $1 \leq p < \infty$ , by(12.3.6.4),

$$L^p(X, \Omega, \mu)^* = L^q(X, \Omega, \mu).$$

- $C[0, 1]^* = BV[0, 1]$  and  $C[X]^* = \text{Meas}(X)$ , the space of complex measure on compact  $X$  with the norm of total variance, by Riesz-representation theorem(12.10.1.9). ┘

### 3 Convexity

**Prop.(12.8.3.1).** Every convex nbhd of 0 contains a balanced convex nbhd of 0. By(12.7.1.4), we can even assume  $\bar{V} \subset U$ . ┘

*Proof:* If  $U$  is convex, choose  $W$  as in(12.7.1.6), then  $W \subset A = \bigcap_{|\alpha|=1} \alpha U$  because it is balanced. Then  $W \subset A^\circ$  and  $A^\circ$  is open and balanced, satisfying the requirement. □

**Prop.(12.8.3.2).** For a compact convex set  $K$  in a TVS  $X$ , if a set  $\Gamma$  of continuous linear mapping is bounded for every  $x \in K$ , then  $\Gamma$  is equicontinuous on  $K$ . ┘

*Proof:* The proof is similar to that of Banach-Steinhaus(12.7.2.1). For  $K$  compact convex, the same argument shows that there is a nbhd  $V$  that  $K \cap (x_0 + V) \subset nE$ , fix  $p > 1$  that  $K \subset x_0 + pV$ , then for any  $x \in K$ , consider  $z = (1 - p^{-1})x_0 + p^{-1}x$ , then  $z \in K$  as  $K$  is convex and  $z - x_0 = p^{-1}(x - x_0) \in V$ , so  $z \in nE$ , and since  $x = pz - (p - 1)x_0$ ,  $\Lambda x \in pnW$  for each  $\Lambda \in \Gamma$ , so  $\Gamma$  is equicontinuous. □

**Hahn-Banach**

**Prop. (12.8.3.3) [Real Hahn].** For a sublinear functional  $p$  on a real linear space  $X$  and a subspace  $X_0$ , if a functional  $f$  satisfies  $f(x) \leq p(x)$  on  $X_0$ , then it can be extended to a functional  $\Lambda$  on  $X$  that  $|\Lambda(x)| \leq |p(x)|$ .  $\lrcorner$

*Proof:* Use Zorn's lemma, if the maximum extension is not on the whole space but on  $M$ , choose  $x_1 \in X - M$ , we want to define  $f(x_1)$ . Now let  $M_1 = \{x + tx_1 | x \in M\}$ . Since for  $x, y \in M$ ,  $f(x) + f(y) = f(x + y) \leq p(x + y) \leq p(x - y) + p(x_1 + y)$ , so

$$f(x) - p(x - x_1) \leq p(y + x_1) - f(y)$$

for  $x, y \in M$ . Let the maximum of the left side be  $\alpha$ , and define  $f(x_1) = \alpha$ , then it is clear  $f(z) \leq p(z)$  still.  $\square$

**Prop. (12.8.3.4) [Complex Hahn].** For a seminorm  $p$  (i.e. it can attain 0) on a complex linear space  $X$  and a subspace  $X_0$ , if a functional  $f$  satisfies  $|f(x)| \leq p(x)$  on  $X_0$ , then it can be extended to a functional on  $X$  with the same condition.  $\lrcorner$

*Proof:* Let  $g(x) = \operatorname{Re} f(x)$  and extend it by Hahn and set  $f(x) = g(x) - ig(ix)$ , then  $f$  is complex linear, and for any  $x$ , for some  $|\alpha| = 1$ ,  $|f(x)| = f(\alpha x) = g(\alpha x) \leq p(\alpha x) = p(x)$ .  $\square$

**Cor. (12.8.3.5) [Hahn].** In a normed space  $X$ , a bounded linear functional on a subspace  $X_0$  can be extended to a bounded functional on  $X$  with the same norm.  $\lrcorner$

**Cor. (12.8.3.6) [Extending Functional Preserving Norm].** If  $X$  is a normed space and  $N$  is a closed subspace, if  $x_0$  satisfies  $d = d(x_0, N) > 0$ , then here is a continuous functional  $f$  that  $f(x) = 0$  and  $f(x_0) = d$  and  $\|f\| = 1$ .  $\lrcorner$

*Proof:* Define  $f(m + \alpha x) = |\alpha|d$  on  $\operatorname{span}\{N, x\}$ , then  $f(m + \alpha x) = |\alpha|d = |\alpha|d(x_0, N) \leq |\alpha|(\|\frac{x'}{\alpha} + x_0\|) = \|\frac{x'}{\alpha} + x_0\| = \|x' + \alpha x\| = \|x\|$ . So  $\|f\| \leq 1$ , so we can use Hahn-Banach to extend it to a functional on  $X$ .  $\square$

**Prop. (12.8.3.7) [Geometric Hahn].**

- If  $E_1$  and  $E_2$  are two convex set that  $E_1 \cap E_2 = \emptyset$  and  $E_1$  has interior point, then there is a continuous linear functional that separate them, i.e.  $\operatorname{Re} f(E_1) < \operatorname{Re} f(E_2)$ . (The interior point is here to assure  $f$  is continuous).
- In a locally convex TVS, if  $E_1$  is convex compact and  $E_2$  is convex closed, then there is a real functional that separate them, i.e.  $\operatorname{Re} f(E_1) < \gamma_1 < \gamma_2 < \operatorname{Re} f(E_2)$ . Thus for a set  $E$  and a point  $x$ ,  $x \in \overline{\operatorname{span} E} \iff$  for all  $f$  that  $f(E) = 0$ ,  $f(x) = 0$ .  $\lrcorner$

*Proof:* The complex case follows from the real case, so assume it is real. Consider  $a_0 \in E_1, b_0 \in E_2$ , let  $x_0 = a_0 - b_0$  and let  $C = E_1 - E_2 + x_0$ , then  $C$  is a convex nbhd of 0. Let  $p$  be the Minkowski functional of  $C$ , then  $p$  is sublinear by (12.8.1.3) and  $p(x_0) \geq 1$ . Let  $f(tx_0) = t$  on the subspace  $M$  generated by  $x$ , then it extends to a functional  $\Lambda$  that  $\leq 1$  on  $C$ , thus it is bounded by 1 on  $C \cap (-C)$ , hence continuous. For any  $a \in E_1, b \in E_2$ , because  $\lambda(x_0) = 1$  and  $a - b + x_0 \in C$  open,  $\Lambda a < \Lambda b$ .

For the second, There is a convex nbhd  $V$  of 0 that  $E_1 + V \cap B = \varphi$ , so the above argument applied with  $E_1 + V$  and  $B$  shows that there is a  $f$  that separate them. And  $f(E_1 + V)$  is open and  $f(E_1)$  is compact, so the conclusion follows.  $\square$

**Cor. (12.8.3.8)[Banach-Saks].** The weak closure of a convex set in a locally convex metric space equals the original closure.

Thus if a sequence  $\{x_n\}$  weakly converges to  $x$  in a metrizable locally convex space, then a convex combination of  $\{x_n\}$  strongly converge to  $x$ , i.e.  $x \in \overline{\text{co}}(\{x_n\})$ , because metric space is first countable.  $\perp$

*Proof:* A weak closed set is closed, and to show the closure is weakly closed, use(12.8.3.7).  $\square$

**Prop. (12.8.3.9).** If  $A_i$  are compact convex sets in a TVS  $X$ , then  $\text{co}(A_1 \cup \dots \cup A_n)$  is compact.  $\perp$

*Proof:* Firstly, the image  $K$  of  $S \times A_1 \times \dots \times A_n$ ,  $(s_1, \dots, s_n) \times (a_1, \dots, a_n) \mapsto \sum s_i a_i$  is closed, where  $S = \{0 \leq x_i, \sum x_i = 1\}$ . And we show  $K$  is just the convex closure: it contains all  $A_i$ , and it is convex because each  $A_i$  is.  $\square$

**Prop. (12.8.3.10).** In  $F$ -space, a closed subset is compact iff it is totally bounded by(4.4.8.27).  $\perp$

**Prop. (12.8.3.11).** In a locally convex space, if  $E$  is totally bounded, then  $\text{co}(E)$  is totally bounded. Thus in a Fréchet space, if  $K$  is compact, then  $\overline{\text{co}}(K)$  is compact.  $\perp$

*Proof:* For a nbhd  $U$  of 0, choose a convex nbhd  $V$  that  $V + V \subset U$ , then  $E \subset F + V$  for some finite set  $F$ , hence  $\text{co}(E) \subset \text{co}(F) + V$ . But  $\text{co}(F)$  is compact by(12.8.3.9). So  $\text{co}(F) \subset F_1 + V$  for some finite set  $F_1$ , then  $\text{co}(E) \subset F_1 + U$ .

If  $K$  is compact, then it is totally bounded, and then  $\text{co}(K)$  is totally bounded and  $\overline{\text{co}(K)}$  is totally bounded by(4.4.8.25), so it is compact by(12.8.3.10).  $\square$

**Prop. (12.8.3.12)[Weakly Bounded and Locally Convex].** If  $X \in \text{TVS}$  is locally convex, then a subset  $U \subset X$  is bounded iff it is weakly bounded.  $\perp$

*Proof:* One direction is trivial, for the other, suppose  $E$  is weakly bounded and  $U$  is a closed nbhd of 0. Because  $X$  is locally convex, there is a convex, balanced nbhd of 0 that  $\overline{V} \subset U$ (12.8.3.1). Now  $\overline{V} = V^{**}$  the polar(12.8.4.1) by(12.8.3.8).

Now  $V^*$  is weak\*-compact and  $|\Lambda(x)| \leq \gamma(\Lambda)$  for each  $\Lambda \in X^*$  for some  $\gamma(\Lambda)$  because  $E$  is weakly bounded. So we can use(12.8.3.2) to show that  $|\Lambda x| \leq \gamma$  for some  $\gamma$  and all  $\Lambda \in V^*$ . So we have  $\gamma^{-1}E \subset \overline{V} \subset U$ . This proves that  $E$  is bounded.  $\square$

**Prop. (12.8.3.13)[Markov-Kakutani Fixed Point Theorem].** For a commuting family  $\mathcal{F}$  of continuous affine maps from  $K$  to  $K$  where  $K$  is a compact convex set in a TVS, then there is a fixed point in  $K$  for all maps in  $\mathcal{F}$ .  $\perp$

*Proof:* Consider the semigroup  $\mathcal{F}^*$  generated by these maps together with their average, it is also commutative because they are all affine. For any  $f, g \in \mathcal{F}^*$ ,  $f(K) \cap g(K) \supset f \circ g(K)$ , so by finite intersection property, there is a point in  $p \in K$  in all  $f(K)$ .

For this  $p$ , consider  $p = \frac{1}{n}(I + T + T^2 + \dots + T^{n-1})(x)$ , then  $p - Tp = \frac{1}{n}(x - T^n x) \in \frac{1}{n}(K - K)$ . as  $K$  is bounded and  $n$  is arbitrary, this means that  $p = Tp$  for all  $T$ .  $\square$

**Cor. (12.8.3.14)[Invariant Hahn].** For a commuting family  $\Gamma$  of operators on a normed space and  $Y$  an invariant space, then for any  $\Gamma$ -invariant continuous functional  $f$  on  $Y$ , it has a  $\Gamma$ -invariant Hahn extension.  $\perp$

*Proof:* We may assume  $\|f\| = 1$ . Let  $K$  be all extensions of  $f$  that has norm  $\leq 1$ .  $K$  is obviously convex, and it is weak\*-compact by Banach-Alaoglu(12.8.4.1). The adjoint action of  $T$  is checked to be continuous in the weak\*-topology, so by(12.8.3.13), there is some  $F \in K$  that is invariant under  $\Gamma$ .  $\square$

**Krein-Milman theorem**

**Thm. (12.8.3.15) [Krein-Milman].** For a compact convex set  $K$  in a TVS  $X$  that is weakly-Hausdorff ( $X^*$  separate points),  $K = \overline{\text{co}}(\text{Extreme}(K))$ .

If  $K$  is a compact set in a locally convex space, then  $K \subset \overline{\text{co}}(E(K)) = \overline{\text{co}}(K)$ .  $\lrcorner$

*Proof:* First show that every compact extreme set  $S$  of  $K$  contains an extreme point. Notice arbitrary intersection of compact extreme sets of  $K$  is compact extreme, because compact is closed, because  $X$  is Hausdorff. And for any functional  $\Lambda \in X^*$ , the maximal value point in  $K$  is compact extreme. Now we use Zorn's lemma to find a minimal compact extreme set in  $S$ , then it must be a point because  $X^*$  separate points.

Now use the weak topology Hahn(12.8.4.2), if  $\overline{\text{co}}(E(K)) \subset K$  is not  $K$ , then it is compact, then we can find a functional that separate  $\overline{\text{co}}(E(K))$  and some point of  $K$ . This is a contradiction because the extreme value point for any functional on  $K$  is an extreme set.

In the locally convex case, the convexity of  $K$  is not needed, and we can show using geometric Hahn(12.8.3.7) instead that,  $K \subset \overline{\text{co}}(K)$ .  $\square$

**Prop. (12.8.3.16) [Milman's Theorem].** If  $K$  is a compact set in a locally convex space  $X$  and if  $\overline{\text{co}}(K)$  is also compact (e.g. in a Fréchet space(12.8.3.11)), then every extreme point of  $\overline{\text{co}}(K)$  lies in  $K$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (12.8.3.17).** For a compact Hausdorff space  $S$  and an algebra  $A \subset C(S)$ , a subset  $E$  is called  **$A$ -antisymmetric** iff every  $f \in A$  that is real on  $E$  is constant on  $E$ . There are in fact maximal  $A$ -antisymmetric subsets of  $S$ .  $\lrcorner$

**Prop. (12.8.3.18) [Bishop Theorem].** If  $A$  is a closed subalgebra of  $C(S)$ . If  $g \in C(S)$  satisfies  $g|_E \in A|_E$  for every maximal  $A$ -antisymmetric set  $E$ , then  $g \in A$ . This theorem is a generalization of Stone-Weierstrass approximation.  $\lrcorner$

*Proof:* The annihilator  $A^\perp$  of  $A$  consists of all regular complex Borel measure  $\mu$  on  $S$  that  $\int f d\mu = 0$  for all  $f \in A$  by Riesz representation(12.10.1.9). ? Cf.[Rudin P122].  $\square$

**Prop. (12.8.3.19) [Schauder Fixed Point Theorem].** If  $C$  is a closed convex subset in a metrizable TVS and continuous  $T : C \rightarrow C$  has sequentially compact image (e.g.  $C$  is compact and  $X$  is locally convex hence  $X^*$  separate points), then  $T$  has a fixed point.  $\lrcorner$

*Proof:* As  $T(C)$  is sequentially compact, for each  $n$ , there is a  $1/n$ -net  $N_n = \{y_i\} \subset T(C)$ , let  $E_n = \text{span}\{N_n\}$ .

Define a map  $T(C) \rightarrow \text{co}(N_n) : I_n(y) = \sum y_i \lambda_i(y)$ , where  $\lambda_i(y) = \frac{m_i(y)}{\sum m_i(y)}$ , and  $m_i(y) = 1 - n\|y - y_i\|$  if  $y \in B(y_i, 1/n)$ , and 0 otherwise.

Now  $\|I_n(y) - y\| = \|\sum (y_i - y) \lambda_i(y)\| \leq \sum \|y_i - y\| \lambda_i(y) \leq \frac{1}{n}$  for each  $y \in T(C)$ . As  $C$  is convex,  $\text{co}(N_n) \subset C$ , if we let  $T_n = I_n \circ T$ , then  $T_n$  has a fixed pt  $x_n$  in  $\text{co}(N_n)$  by Brouwer Fixed-Point Theorem.

As  $T(C)$  is sequentially compact and  $C$  is closed, there is a subsequence  $Tx_{n_k}$  that converges to  $x \in C$ . And then

$$\|x_{n_k} - x\| = \|I_n Tx_{n_k} - x\| \leq \|I_n Tx_{n_k} - Tx_{n_k}\| + \|Tx_{n_k} - x\| < \frac{1}{n} + \|Tx_{n_k} - x\|$$

so  $x_{n_k} \rightarrow x$ , and then by continuity,  $Tx = x$ .  $\square$

### Vector-valued Integrations

**Def. (12.8.3.20) [Vector-Valued Integrations].** Given a measure space  $(Q, \mu)$  and  $X$  is an Archimedean TVS on which  $X^*$  separate points (e.g. Fréchet space). If  $f$  is a function from  $M$  to  $X$  that  $\Lambda \circ f$  are integrable w.r.t  $\mu$  for any  $\Lambda \in X^*$ . The **vector-valued integration**  $\int_M f d\mu$  of  $f$  w.r.t  $Q$  is an element  $y$  that

$$\Lambda y = \int_Q (\Lambda f) d\mu$$

for any  $\Lambda \in X^*$ . ┘

**Prop. (12.8.3.21) [Vector-Valued Integrations of Compactly-Supported Functions].** If  $X \in \text{TVS}/\mathbb{R}$  on which  $X^*$  separate points (e.g. Fréchet space),  $Q \in \text{LCH}$  with a compactly supported Radon measure  $\mu$ , and  $f$  is continuous s.t.  $\overline{\text{Conv}(f(Q))}$  is compact (e.g. when  $X$  is Fréchet (12.8.3.11)), then the integral  $y = \int_Q f d\mu$  exists, and belongs to the closed linear span of the range of  $f$ . Moreover if  $\mu$  is positive and  $\mu(Q) = 1$ , then  $y \in \overline{\text{Conv}(f(Q))}$ . ┘

*Proof:* Cf.[Rudin P78]. □

**Cor. (12.8.3.22).** If  $Q$  is Hausdorff,  $d\mu$  is compactly supported,  $X$  is Archimedean Banach and  $f : Q \rightarrow X$  is continuous, then

$$\left\| \int_Q f d\mu \right\| \leq \int_Q \|f\| d\mu.$$

┘

*Proof:* Let  $y = \int_Q f d\mu$ . By (12.8.3.6), there is a  $\|\Lambda\| \leq 1$  that  $\Lambda y = \|y\|$ , so

$$\Lambda y = \|y\| = \int_Q \Lambda f d\mu = \left| \int_Q \Lambda f d\mu \right| \leq \int_Q |\Lambda f| d\mu \leq \int_Q \|f\| d\mu.$$

□

**Prop. (12.8.3.23).** If  $X \in \text{TVS}/\mathbb{R}$  on which  $X^*$  separate points,  $Q$  is a compact subset of  $X$ , and  $\overline{\text{co}}(Q)$  is compact, then  $y \in \overline{\text{co}}(Q)$  iff there is a regular Borel measure  $\mu$  on  $Q$  that  $y = \int_Q x d\mu(x)$ . ┘

*Proof:* Cf.[Rudin P79]. □

**Prop. (12.8.3.24) [Action of Measure].** For a fixed map  $f : Q \rightarrow X$ , assume  $X$  is a Fréchet space, then the integration functor in (12.8.3.21) induces a continuous map

$$\text{Meas}_c(Q) \rightarrow X : \mu \mapsto \int_\mu f$$

that maps  $\delta_x$  to  $f(x)$ . ┘

*Proof:* It suffices to verify continuity: for any seminorm  $\rho$ , by convexity,

$$\rho\left(\int_\mu f\right) \leq (\mu, \rho(f)),$$

thus for  $\mu \in U$  satisfying  $(\mu, \rho(f)) < 1$ ,  $\rho(\int_\mu f) < 1$ . This proves continuity. □

**Prop. (12.8.3.25) [Vector Valued Integration of  $L^1$ -Functions, Folland].** If  $V \in \text{BanSpa}/\mathbb{C}$  and  $X \in \text{LCH}$  with a Radon measure  $\mu$ . If  $g \in L^1(\mu)$  and  $f : X \rightarrow V$  is continuous with bounded image, then the vector-valued integration  $\int gf d\mu$  (12.8.3.20) exists and belongs to the closed linear span of the range of  $f$ . Moreover,

$$\left\| \int gf(x) d\mu(x) \right\| \leq \sup_{x \in X} \|f(x)\| \int |g(x)| d\mu(x).$$

□

*Proof:* Clearly  $\Lambda(gH) \in L^1(\mu)$  for all  $\Lambda \in V^*$ . And  $\mu$  is Radon, so there is a sequence  $\{g_n\} \in C_c(X)$  that converges to  $g$  in  $L^1$ , so  $\int g_n f d\mu$  is integrable by (12.8.3.21), and

$$\int \|g_n(x)f(x) - g_m(x)f(x)\| d\mu(x) \leq \int |g_n(x) - g_m(x)| d\mu(x) \rightarrow 0,$$

thus this is a Cauchy sequence, converging to some  $y$ . Now for any  $\Lambda \in V^*$ ,

$$\Lambda(y) = \lim \Lambda \left( \int g_n f d\mu(x) \right) = \lim \int \Lambda \circ (g_n f) d\mu(x) = \int \varphi \circ (gf) d\mu(x).$$

The last equality uses boundedness again.

Moreover, each  $\int g_n f d\mu(x)$  belongs to the closed range of  $f$  by (12.8.3.21), hence so does  $\int gf d\mu(x)$ . And last assertion also follows from (12.8.3.21). □

### Differentiable Vector-Valued Functions



**Def. (12.8.3.26) [Differentiable Vector-Valued Functions].** Let  $U \subset \mathbb{R}$  be open and  $X \in \text{TVS}/\mathbb{R}$  be a Fréchet space, then a function  $f : U \rightarrow X$  is called a

- **weakly smooth vector-valued function** if  $\Lambda \circ f$  is smooth for any  $\Lambda \in X^*$ .
- **strongly smooth vector-valued function** if  $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$  exists for each  $x_0 \in U$  and is a continuous function  $f' : U \rightarrow X$ , and similarly  $f^{(2)}$  exists and continuous and so on.

Similarly we can define weakly- $C^k$  vector-valued functions and strongly  $C^k$ -vector valued functions  $f : U \rightarrow X$  for each  $k \in \mathbb{Z}_+$  and open subsets  $U \subset \mathbb{R}^n$ .  $\lrcorner$

**Prop. (12.8.3.27) [Weakly- $C^k$  Implies Strongly- $C^{k-1}$ ].** Let  $X \in \text{TVS}/\mathbb{R}$  be a Fréchet space (12.7.1.7) and  $f : (0, 1) \rightarrow X$  is weakly- $C^k$  for  $k \in \mathbb{Z}_+$ , then  $f$  is strongly  $C^{k-1}$ .

In particular  $f$  is weakly-smooth iff it is smooth.  $\lrcorner$

*Proof:* For  $k = 1$ , this follows from (12.8.3.28). So we may assume  $k \geq 1$ .

Suppose  $k = 2$ . For any  $b \in (0, 1)$ , consider  $g(x) = \frac{f(x) - f(b)}{x - b}$  for  $x \in [0, b) \cup (b, 1]$ , then by hypothesis,  $\lambda \circ g$  extends to a  $C^1$ -function on  $(-1, 1)$  (by calculation using L'Hospital theorem). Thus  $g$  extends to a continuous function on  $(0, 1)$  by (12.8.3.28), and  $f'$  exists. Moreover,  $f'(x)$  is weakly- $C^1$  by hypothesis, so it is continuous by (12.8.3.28). Thus  $f$  is strongly- $C^1$ .

Now use induction on  $k \geq 2$ . Let  $g(x) = f'(x)$  (which exists by the above), then  $g(x)$  is weakly- $C^{k-1}$ , and then by induction hypothesis,  $g(x)$  is strongly- $C^{k-2}$ , which means  $f$  is strongly- $C^{k-1}$ .  $\square$

**Lemma (12.8.3.28).** Let  $X \in \text{TVS}/\mathbb{R}$  be a Fréchet space (12.7.1.7) and  $f : [-1, 0] \cup [0, 1] \rightarrow X$  satisfies: for each  $\Lambda \in X^*$ ,  $\Lambda \circ f$  extends to a  $C^1$ -function on  $[-1, 1]$ , then  $f(b) \in X$  can be chosen s.t.  $f$  extends to a continuous function  $[-1, 1] \rightarrow X$ .

In particular, if  $f : (-1, 1) \rightarrow X$  is weakly- $C^1$ , then it is continuous.  $\lrcorner$

*Proof:* For  $\Lambda \in X^*$ , let  $F_\Lambda$  be the  $C^1$ -extension of  $\Lambda \circ f$ . Because  $F_\Lambda$  is  $C^1$ , by mean value theorem,  $\{\frac{F_\Lambda(x) - F_\Lambda(y)}{x - y} : x \neq y \in [-1, 1]\}$  is bounded. Thus  $\{\frac{f(x) - f(y)}{x - y} : x \neq y \in [-1, b) \cup (b, 1]\} \subset X$  is weakly-bounded. Thus by (12.8.3.12), it is bounded. Thus  $\{f(x)\}_{x \rightarrow b}$  is a Cauchy net, and  $f(b)$  exists because  $X$  is complete.  $\square$

### Holomorphic Vector-Valued Functions

**Def. (12.8.3.29) [Holomorphic Vector-Valued Functions].** Let  $\Omega \subset \mathbb{C}$  be open, and  $X \in \text{TVS}/\mathbb{C}$ , then a function  $f : \Omega \rightarrow X$  is called a

- **weakly holomorphic vector-valued function** if  $\Lambda \circ f$  is holomorphic for any  $\Lambda \in X^*$ .
- **strongly holomorphic vector-valued function** if  $\lim_{w \rightarrow z} \frac{f(w) - f(z)}{w - z}$  exists for every  $z \in \Omega$ .

A strongly holomorphic function is clearly weakly holomorphic, and the converse is true when  $X$  is Fréchet space, by the following proposition (12.8.3.30).  $\lrcorner$

**Prop. (12.8.3.30) [Weak and Strong Holomorphy].** Let  $\Omega \subset \mathbb{C}$  be open, and  $X \in \text{TVS}/\mathbb{C}$  be a Fréchet space. Suppose  $f : \Omega \rightarrow X$  is weakly holomorphic (12.8.3.29), then  $f$  is strongly continuous, and the Cauchy integral formula (12.4.2.6) holds for  $f$ , and  $f$  is strongly holomorphic.  $\lrcorner$

*Proof:* We may assume  $0 \in \Omega$ , then Let  $B(0, 2r) \subset \Omega$  and  $\Gamma$  the boundary of  $B(0, 2r)$ , since  $\Lambda f$  is holomorphic,

$$\frac{(\Lambda f)(z) - (\Lambda f)(0)}{z} = \frac{1}{2\pi i} \int_{\Gamma} \frac{(\Lambda f)(\zeta)}{(\zeta - z)\zeta} d\zeta \quad 0 < |z| < 2r.$$



Therefore  $\{\frac{f(z)-f(0)}{z} | 0 < |z| \leq r\}$  is weakly bounded, so it is also bounded by (12.8.3.12), so  $f$  is strongly continuous.

The integral exists by (12.8.3.21), so  $f$  satisfies Cauchy integral formula because it satisfies this when acting with any functional  $\Lambda$ , and  $X^*$  separate points.

For the last assertion, Cf.[Rudin P84].  $\square$

**Prop. (12.8.3.31)[Liouville's Theorem].** If  $X \in \text{TVS}/\mathbb{C}$  on which  $X^*$  separate points and  $f : \mathbb{C} \rightarrow X$  is weakly holomorphic and  $f(\mathbb{C})$  is weakly bounded, then  $f$  is constant.  $\lrcorner$

*Proof:* Immediate from Liouville's theorem (12.4.2.10).  $\square$

## 4 Duality Theory

**Prop. (12.8.4.1)[Banach-Alaoglu].** For  $X \in \text{TVS}$  and a nbhd  $V$  of  $0 \in X$ , the set

$$K = \{f : |f(x)| \leq 1, \forall x \in V\}$$

is weak\*-compact in  $X^*$ , which is called the **polar**  $V^*$  of  $V$ .  $\lrcorner$

*Proof:* Consider the Minkowski function  $\gamma$  of  $V$ , then for each  $\Lambda \in K$ ,  $|\Lambda x| \leq \gamma(x)$ . If we consider the space  $P = \prod_{x \in X} [-\gamma(x), \gamma(x)]$ , then  $P$  is compact by Tychonoff (4.4.2.5).

The point is that the weak\*-topology coincides with the pointwise convergence topology on  $K$ , because they have the same generating subbasis. If we show  $K$  is a closed subspace of  $P$ , this will finish the proof that  $K$  is weak\*-compact. For this, consider any  $f_0$  in its closure, then for each  $x, y \in X$ ,  $\alpha, \beta \in K$ , there is a  $f \in K$  that is close to  $f_0$  at  $x, y$  and  $\alpha x + \beta y$ . So  $f_0$  is linear. Similarly we can show  $|f_0(x)| \leq 1$  for  $x \in V$ , so  $f_0 \in K$ .  $\square$

**Prop. (12.8.4.2).** If  $X$  is a TVS that  $X^*$  separate points (e.g. locally convex), then the weak topology  $X_w$  is a locally convex space, and  $(X_w)^* = X^*$ .  $\lrcorner$

*Proof:* If  $\Lambda$  is a functional that is continuous in  $X_w$ -topology, then  $|\lambda x| < 1$  for some set defined by elements in  $X^*$ , so by (12.7.3.3),  $\Lambda = \sum \alpha_i \Lambda_i$  which is continuous w.r.t the original topology.  $\square$

**Prop. (12.8.4.3)[Hahn Weak Topology case].** If  $X$  is a TVS that  $X^*$  separate points, then if  $A, B$  are disjoint nonempty, compact convex sets in  $X$ , then there is a  $\Lambda \in X^*$  that separate  $A$  and  $B$ , i.e.  $\text{Re } f(E_1) < \gamma_1 < \gamma_2 < \text{Re } f(E_2)$ .  $\lrcorner$

*Proof:* Let  $X_w$  be  $X$  with the weak topology, then the sets  $A$  and  $B$  are compact in  $X_w$  as it's weaker. And they are also closed because  $X_w$  is Hausdorff.  $X_w$  is convex, so we can use geometric Hahn (12.8.3.7). Now  $(X_w)^* = X^*$ , so the chosen functional is also continuous in the original topology.  $\square$

**Prop. (12.8.4.4)[Dual Banach Space].** For a normed space  $X$ ,  $x \in X$  can be seen as functional on  $X^*$ , of norm exactly  $\|x\|$ . And the closed ball  $B^*$  of the dual space  $X^*$  is weak\*-compact.  $\lrcorner$

*Proof:* The first assertion is because of (12.8.3.6), the last assertion is because of Banach-Alaoglu (12.8.4.1).  $\square$

**Prop. (12.8.4.5)[Adjoint Norm].** For  $X, Y$  normed, the adjoint of  $T : X \rightarrow Y$  satisfies  $\|T^*\| = \|T\|$ .  $\lrcorner$

*Proof:* Use (12.8.4.4),  $\|T\| = \sup\{|\langle Tx, y^* \rangle| : \|x\| \leq 1, \|y^*\| \leq 1\} = \|T^*\|$ .  $\square$

**Prop. (12.8.4.6)[Closed Range Theorem].** Let  $T$  be continuous mapping between Banach spaces  $X$  and  $Y$ , let  $U, V$  be open balls of  $X, Y$  particularly. then the following are equivalent:

1.  $\|T^*y^*\| \geq \delta \|y^*\|$  for some  $\delta$ .
2.  $\delta V \subset \overline{T(U)}$ .
3.  $\delta V \subset T(U)$ , i.e.  $T^{-1}$  is continuous.
4.  $T(X) = Y$ .
5.  $T^*$  is one-to-one and  $R(T^*)$  is closed in  $X$ .

┘

*Proof:* 1  $\rightarrow$  2: If  $\|T^*y^*\| \geq \delta \|y^*\|$ , first prove  $\delta V \subset \overline{T(U)}$ . If  $y_0 \notin \overline{T(U)}$ , since  $\overline{T(U)}$  is convex closed and balanced, geometric Hahn shows that there is a  $y^*$  that  $|y^*(y)| \leq 1$  for every  $y \in T(U)$ , and  $|y^*(y_0)| > 1$ . Then it follows  $\|T^*y^*\| \leq 1$ . So

$$\delta < \delta |y^*(y_0)| \leq \delta \|y_0\| \|y^*\| \leq \|y_0\| \|T^*y^*\| \leq \|y_0\|$$

This shows  $\delta V \subset \overline{T(U)}$ .

2  $\rightarrow$  3: may assume  $\delta = 1$ . Then  $\overline{V} \subset \overline{T(U)}$ . Then for every  $y \in Y$  and every  $\varepsilon > 0$ , there is a  $x$  that  $\|x\| \leq \|y\|$  and  $\|y - Tx\| < \varepsilon$ . For any  $y_1 \in V$ , pick  $\varepsilon_n > 0$  that  $\sum \varepsilon_n < 1 - \|y_1\|$ , then choose  $\|x_n\| \leq \|y_n\|$  that  $\|y_n - Tx_n\| < \varepsilon_n$ , and let  $y_{n+1} = y_n - Tx_n$ . Then is verified that  $x = \sum x_n \in U$  and  $Tx$ .

3  $\rightarrow$  1:  $\|T^*y^*\| = \sup\{|\langle x, T^*y^* \rangle| : x \in U\} \geq \sup\{|\langle y, y^* \rangle| : y \in V\} = \delta \|y^*\|$ .

3  $\iff$  4: follows from open mapping theorem.

4  $\rightarrow$  5:  $T^*$  is injective by (12.7.3.2). By open mapping theorem,  $T^*$  is a multiple of a dilation, so  $R(T^*)$  is closed by (4.4.8.30).

5  $\rightarrow$  4:  $R(T)$  is dense in  $Y$  by (12.7.3.2), and it is closed by the proposition (12.8.4.7) below.  $\square$

**Prop. (12.8.4.7)[Closed Range Theorem].** If  $X, Y$  are Banach spaces and  $T \in L(X, Y)$ , the following are equivalent:

1.  $R(T)$  is closed in  $Y$ .
2.  $R(T^*)$  is weak\*-closed in  $X^*$ .
3.  $R(T^*)$  is closed in  $X$ .

┘

*Proof:* 1  $\rightarrow$  2: As  $N(T)^\perp$  is the weak\*-closure of  $R(T^*)$ , it suffices to prove  $N(T)^\perp \subset R(T^*)$ . As  $R(T)$  is complete, the open mapping theorem applies to  $X \rightarrow R(T)$ , showing that each  $y \in R(T)$  corresponds to an element  $x \in X$  that  $Tx = y$  and  $\|x\| \leq K\|y\|$ .

For  $x^* \in N(T)^*$ , define a functional  $\Lambda$  on  $R(T)$  by  $\Lambda Tx = \langle x, x^* \rangle$ , this is well-defined, and  $|\Lambda y| = |\Lambda Tx| \leq K\|y\| \|x^*\|$ . So it is continuous and by Hahn-Banach some continuous functional  $y^* \in Y^*$  extends  $\Lambda$ . Then  $\langle Tx, y^* \rangle = \Lambda Tx = \langle x, x^* \rangle$ , so  $x^* = T^*y^*$  is in the image of  $T^*$ , so we are done.

3  $\rightarrow$  1: let  $Z = \overline{R(T)}$ .  $RT$  is dense in  $Z$ , so (12.7.3.2) shows  $T^* : Z^* \rightarrow X^*$  is injective. And for each  $z^* \in Z^*$ , there is an extension  $y^*$  by Hahn-Banach, and then  $\langle x, T^*y^* \rangle = \langle Tx, y^* \rangle = \langle Tx, z^* \rangle = \langle x, T^*z^* \rangle$ , so  $T^*(Y^*) = T^*(Z^*)$ , which is closed by hypothesis.

Now use open mapping theorem for  $Z^* \rightarrow R(T^*)$ , then there is a  $c$  that  $c\|z^*\| \leq \|T^*z^*\|$ . So  $T : X \rightarrow Z$  is surjective, by (12.8.4.6). So  $R(T) = Z$  is closed.  $\square$

**Prop. (12.8.4.8).** In a normed space, iff  $x_n \rightarrow x$  weakly, then  $\lim \|x_n\| \geq \|x\|$ .  $\square$

*Proof:* Choose a functional that  $\|f\| = 1$  and  $|f(x)| = 1$  by (12.8.3.6), then use the definition of weak convergence.  $\square$

**Prop. (12.8.4.9) [Eberlein-Smulian].** For a set  $A$  in a Banach space  $X$ ,  $A$  is weak\*-sequentially compact iff its weak precompact.  $\lrcorner$

*Proof:* ?

We prove here that the case that the closed unit ball of a reflexive Banach space is weakly\*-self sequentially compact.

To prove this, first we show that a bounded sequence has a subsequence that is weak\*-convergent in  $X$ . Let  $X_0 = \overline{\text{span}\{x_n\}}$ , then  $X_0$  is reflexive by (12.8.4.13), and it is separable, so  $X_0^*$  is separable by (12.8.4.11). Then the result follows from (12.8.4.14).

Finally, the weak limit  $x$  is in the closed unit ball, by (12.8.4.8).  $\square$

### Reflexive and Separable

**Def. (12.8.4.10) [Reflective Banach Space].** If  $X$  is a Banach space, there is an isometric immersion of  $X$  onto a closed subspace of  $X^{**}$  (closed because  $X$  is complete).  $X$  is called **reflexive** iff  $X \cong X^{**}$ .  $\lrcorner$

**Prop. (12.8.4.11) [Separability Banach].** For a normed space  $X$ , if  $X^*$  is separable, then  $X$  is separable.  $\lrcorner$

*Proof:* Choose a countable dense set in  $X^*$ , then their projection to the unit sphere  $S^* \{g_n\}$  are dense in  $S^*$  (easily checked), and choose for each of them a  $x_n$  that  $\|x_n\| = 1$  and  $g_n(x_n) > \frac{1}{2}$ .

Now I claim  $x_n$  are dense in  $X$ , i.e.  $X_0 = \overline{\text{span}\{x_n\}} = X$ . If this is not the case, then there is a  $\|x\| = 1$  not in  $X_0$ , so by (12.8.3.6), there is a  $f$  that  $f(X_0) = 0$  and  $\|f\| = 1$  and  $f(x) = 1$ . Then  $\|g_n - f\| = \sup_{\|x\|=1} \{|g_n(x) - f(x)|\} \geq |g_n(x_n) - f(x_n)| = |g_n(x_n)| \geq 1/2$ , contradicting the fact  $g_n$  is dense in  $S^*$ .  $\square$

**Prop. (12.8.4.12) [Duality Exact].** If  $X$  is a closed subspace of a normed space  $Y$ , and  $Y/X$  is the quotient field, then  $(Y/X)^*$  is a closed subspace of  $Y^*$ , and  $X^*$  is the quotient.  $\lrcorner$

*Proof:*  $(Y/X)^* \rightarrow Y^*$  is clearly injective, and the  $X^*$  are all functionals on  $Y$  modulo the functionals that vanish on  $X$ .  $\square$

**Cor. (12.8.4.13) [Pettis].** Closed subspace and quotient space of a reflexive normed space is reflexive.  $\lrcorner$

*Proof:* Use the fact that  $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$  induces an exact sequence  $0 \rightarrow X^{**} \rightarrow Y^{**} \rightarrow Z^{**} \rightarrow 0$ , and there is a map  $X \rightarrow X^{**}$ , so we can use snake lemma (as modules).  $\square$

**Prop. (12.8.4.14) [Separable Ball Weak\*-Sequentially Compact].** If a normed space  $X$  is separable, then the closed unit ball of  $X^*$  is weak\*-sequentially compact.  $\lrcorner$

*Proof:* Let  $x_n$  be a countable dense subset of  $X$ , then by diagonal method, for each bounded sequence of  $f_n \in X^*$ , there is a subsequence  $f_{n_k}$  that  $f_{n_k}(x_m)$  converges for each  $x_m$ . Then by (12.7.2.3),  $f_{n_k}$  converges to some  $f \in X^*$ . So the theorem is finished.  $\square$

**Prop. (12.8.4.15) [Reflexive Ball Weak\*-Sequentially Compact].** In a reflexive Banach space  $X$ , then a set in  $X$  is bounded iff it is weak\*-sequentially compact.  $\lrcorner$

*Proof:* If it is reflexive, then the unit ball is weak\*-compact by Alaoglu, so it is weak\*-sequentially compact by Eberlein(12.8.4.9). Conversely, if it is weak\*-sequentially compact, then its closure is weak\*-compact, thus bounded.  $\square$

**Prop.(12.8.4.16).** A closed convex set of a reflexive Banach space attains minimal norm.  $\lrcorner$

*Proof:* By Hahn, a closed convex set is weakly closed. let  $d = \inf\{\|x\|\}$ , then if  $d \leq \|x_n\| < d + q/n$ , then  $\{x_n\}$  is bounded, so by(12.8.4.15) it is weak-sequentially compact(12.8.4.9), thus some  $x_n \rightarrow x$  weakly. and use(12.8.4.8),  $x$  must attain minimal norm  $d$ .  $\square$

## 5 Compact Operators & Fredholm Operators

**Def.(12.8.5.1) [Compact Operators].** A **compact operator** is an operator between Banach spaces that maps bounded set to sequentially compact(equivalently precompact or totally bounded(4.4.8.27)) set. It is necessarily continuous because the norm function is continuous thus  $\|Tx\|$  is bounded on the unit ball. The set of compact operators between  $X, Y$  is denoted by  $\mathfrak{C}(X, Y)$ .  $\lrcorner$

**Prop.(12.8.5.2) [Examples of Compact Operators].**

- Let  $X$  be a compact measure space and  $Lu(x) = \int_X K(x, y)u(y)dy$  for  $K \in C(X \times X)$ . This is a compact operator on  $C(X)$  by [Arzela-Ascoli](#).
- Let  $\Omega$  be a  $\sigma$ -finite measure space and let  $\mathcal{L}u(x) = \int_\Omega K(x, y)u(y)dy$  for  $K(x, y) \in L^2(\Omega \times \Omega)$ . This is a compact operator on  $L^2(\Omega)$ , as it is Hilbert-Schmidt(12.9.4.4)(12.9.4.3).  $\lrcorner$

**Prop.(12.8.5.3) [Properties of Compact Operators].**

1. For a continuous operator, it has f.d. image iff it is compact and the image is closed.
2. The space of compact operator is a closed subspace of  $L(X, Y)$ . Thus the limit of f.d. operators is compact.
3. If one of  $A$  or  $B$  is compact and the other is continuous, then  $AB$  is compact, because continuous maps bounded to bounded and compact to compact.  $\lrcorner$

*Proof:*

1. A finite dimensional space is closed by(12.8.1.8), and a finite dimensional space is Heine-Borel(4.4.5.2), so it maps closed ball to precompact set, as it is continuous. Conversely, if it is compact and the image is closed, then it is an open map to its image, by open mapping theorem, and the image is locally compact because  $T$  is compact, so it has finite dimension(12.8.1.7).
2.  $S+T$  is continuous because sum of precompact set is precompact. To show it is closed subspace, Use totally bounded definition, for  $T$  is the closure, let  $\|S - T\| < r$ , then if  $S(x_i)$  is a  $r$ -net for  $S(B(0, 1))$ , then  $T(x_i)$  is a  $3r$ -net for  $T(B(0, 1))$ .
3. Because continuous function preserves both boundedness and (pre)compactness.  $\square$

**Prop.(12.8.5.4) [Compact Operators and Totally Convergence].** Let  $x_n \rightarrow x$  weakly, if  $T$  is compact, then  $Tx_n \rightarrow Tx$  strongly. The converse is true when  $X$  is reflexive. In particular, this applies to Hilbert space.  $\lrcorner$

*Proof:* Assume the contrary, if  $Tx_n$  doesn't converge to  $Tx$ , there is a subsequence  $x_{n_k}$  that  $\|Tx_{n_k} - Tx\| \geq \varepsilon_0$ . Now  $\{x_n\}$  is bounded by (12.7.3.5), so by  $T$  compact, there is a subsubsequence  $Tx_{n_k} \rightarrow z$  strongly. But because  $x_{n_i} \rightarrow x$  weakly,  $Tx_{n_i} \rightarrow Tx$  weakly because  $T$  is continuous, and thus  $z = Tx$ .

The converse is by Eberlein (12.8.4.9), because the bounded  $x_n$  has a weak convergent subsequence, and it is mapped to convergent sequence by  $T$ .  $\square$

**Prop. (12.8.5.5).**  $T$  is compact  $\iff T^*$  is compact.  $\lrcorner$

*Proof:* We need only to show that  $T^*y_n^*$  has a uniformly convergent subsequence on the unit sphere, but for this it suffice to prove  $y_n^*$  is sequentially compact in  $C(\overline{T(B(0,1))})$ . And we use Arzela-Ascoli because  $\overline{T(B(0,1))}$  is compact. For the other half, use the double dual space.  $\square$

**Lemma (12.8.5.6).** If there is a chain of closed subspaces  $M_1 \subset M_2 \subset \dots$  that  $T(M_n) = M_n$  and  $(T - \lambda_n I)M_n \subset M_{n-1}$  for some  $\lambda_n \in \sigma(A) - B(0, r)$ , then  $T$  is not compact.  $\lrcorner$

*Proof:* There are  $y_n \in M_n$  that  $\|y_n\| \leq 1$  and  $\|y_n - x_n\| \geq 1/2$  for  $x \in M_{n-1}$ , so if  $m < n$ ,  $\|Ty_m - Ty_n\| = \|\lambda y_n - (Ty_m - (T - \lambda_n)y_n)\| \geq \frac{|\lambda_n|}{2} \geq \frac{r}{2}$ , so  $Ty_n$  has no convergent subsequence.  $\square$

**Lemma (12.8.5.7).** If  $A$  is compact and  $T = 1 - A$ , then if  $T$  is not injective, then it is not surjective. And for any  $r > 0$ ,  $\sigma_p(A) - B(0, r)$  is a finite set.  $\lrcorner$

*Proof:* We use (12.8.5.6). If  $R(T) = X$ , then let  $M_n = N(T^n)$ , then  $M_0 \neq 0$  because there is a  $Tx_0 = 0$ , and  $M_n \subset M_{n+1}$  because there is a  $T^n x_{n+1} = x_0$ , so  $x_{n+1} \in M_{n+1} - M_n$ .

If  $\sigma_p(A) - B(0, r)$  is infinite, then choose  $M_n$  to be generated by  $n$  eigenvectors, then it is clear that a chain like above will be found.  $\square$

**Lemma (12.8.5.8).** If  $A$  is compact and  $T = 1 - A$ , then  $R(T)$  is closed.  $\lrcorner$

*Proof:* it suffices to show  $T^{-1} : R(T) \rightarrow X/N(T)$  is continuous, if this is not the case, then there is a sequence  $\|x_n\| = 1$  but  $Tx_n \rightarrow 0$ . But  $A$  is compact, so there is a subsequence that  $Ax_{n_k} \rightarrow z$ , so  $x_{n_k} \rightarrow z$ . So  $Tz = 0$  so  $z = 0$ , but then  $x_{n_k} \rightarrow 0$ , contradiction.  $\square$

**Prop. (12.8.5.9) [Riesz-Fredholm].** For a compact operator  $A \in L(X)$ , let  $T = I - A$ . Then:

1.  $0 \in \sigma(A)$  if  $X$  is not f.d.
2.  $T$  is Fredholm of index 0 (12.8.5.17). Equivalently,  $\sigma(A) \setminus \{0\} = \sigma_p(A) \setminus \{0\}$  (because either  $T$  not injective or  $T$  is surjective).
3.  $\sigma(A)$  has at most one convergent point 0 (it must attain 0 if  $X$  is a infinite-dimensional). Hence it has at most countable spectrum.  $\lrcorner$

*Proof:* 1: If 0 is a regular value, then  $T$  is invertible, thus  $T^{-1}T = \text{id}$  is compact, thus  $X$  has f.d. (12.8.1.7).

2: Firstly  $\dim N(T) < \infty$ . This is because  $T|_{N(T)} = \text{id}_{N(T)}$ , so it is compact iff it is f.d. (12.8.5.3). by [Rudin P108] ?.

3: By (12.8.5.7).  $\square$

**Prop. (12.8.5.10) [Lomonosov's Invariant Subspace Theorem].** If  $X$  is an infinite dimensional complex Banach space, and  $T \neq 0$  is a compact operator in  $L(X)$ , then there is a proper closed subspace  $M$  of  $X$  that is invariant under  $S$  for any  $S$  that commutes with  $T$ .

In particular, if  $S$  commutes with some compact operator  $T$ , then  $S$  has an invariant closed subspace.  $\lrcorner$

*Proof:* If  $\Gamma$  is the subspace of all operators that commutes with  $T$ , then it is a subalgebra of  $L(X)$ , and for each  $y \in X$ , let  $\Gamma(y) = \{Sy | S \in \Gamma\}$ , then  $S(\Gamma(y)) \subset \Gamma(y)$ , then so does the closure of  $\Gamma(y)$ . So if the proposition is false, then  $\Gamma(y)$  is dense in  $X$  for each  $y$ .

Pick  $x_0$  that  $Tx_0 \neq 0$ , then there is an open ball  $B$  of  $x_0$  that  $\|Tx\| \geq \frac{1}{2}\|Tx_0\|$  and  $\|x\| \geq \frac{1}{2}\|x_0\|$  for  $x \in B$ . Now our assumption shows that for every  $y \neq 0$ , there is a nbhd  $W$  of it that maps into  $B$  by some  $S \in \Gamma$  (notice  $\Gamma$  is a subspace).

Now  $K = \overline{T(B)}$  is compact because  $T$  is compact, so there are f.m. open sets  $W_i$  whose union cover  $K$ , and  $S_i(W_i) \in B$ , where  $S_i \in \Gamma$ . Now let  $\mu = \max\{\|S_i\|\}$ . Consider  $Tx_0 \in K$ , so there is a  $S_{i_1}Tx_0 \in B$ , then  $TS_{i_1}Tx_0 \in K$ , so there is a  $S_{i_2}TS_{i_1}Tx_0 \in B$ . Continuing this way, we get

$$\frac{1}{2}\|x_0\| \leq \|x_N\| \leq \mu^N \|T^N\| \|x_0\|,$$

so by Gelfand theorem (12.9.1.8),  $\rho(T) > 0$ , so there is a eigenvalue  $\lambda$  of  $T$  (by (12.8.5.9)) that  $N(T - \lambda I)$  is finite dimensional, so not equal to  $X$ , and it is clearly invariant under  $\Gamma$ .  $\square$

**Prop. (12.8.5.11) [Jordan Decomposition for Compact Operators].** For a compact operator  $A$  and all the non-zero eigenvalues  $\lambda_i$ , we can find a subspace

$$\bigoplus_{i=1}^{\infty} N((\lambda_i - A)^{p_i}), \quad \lambda_i \neq 0$$

on which  $A$  has a Jordan form.  $\lrcorner$

*Proof:* Let  $T = 1 - A$ , By (3.2.4.5), we only have to prove there are some  $m, n$  that There is a  $p$  that  $N(T^p) = N(T^{p+1})$  and a  $q$  that  $R(T^q) = R(T^{q+1})$ , because then we have a decomposition  $X = N((T - \lambda I)^p) \oplus R((T - \lambda I)^p)$ , and all these  $N((T - \lambda I)^p)$  are pairwise disjoint.

Now  $q < \infty$ , because if  $R(T) \supset R(T^2) \supset \dots$ , because  $T^k$  is of the form  $1 + \text{compact operator}$ ,  $R(T^k)$  are all closed by (12.8.5.8), so by (12.8.5.6), this is impossible.

For  $p$ , use Riesz-Fredholm (12.8.5.9),

$$\dim N(T^q) = \text{codim} R(T^q) = \text{codim} R(T^{q+1}) = \dim N(T^{q+1}) < \infty$$

So  $p \leq q < \infty$ .  $\square$

**Prop. (12.8.5.12).** If  $X, Y$  are Banach spaces and  $T, K \in L(X, Y)$ ,  $K$  is compact and  $R(A) \subset R(K)$ , then  $A$  is compact.  $\lrcorner$

*Proof:* Use (12.7.2.10), then we can lift the function to a map  $\tilde{T} : X \rightarrow X/N(K)$ , which is also continuous, so  $T = \tilde{K} \rightarrow \tilde{T}$  is also compact.  $\square$

### Schauder Basis

**Def. (12.8.5.13) [Schauder Basis].** Let  $X$  be a Banach space, a sequence  $e_n$  is called a **Schauder basis** iff for any  $x \in X$ , there is a unique sequence  $C_n(x)$  that  $x = \lim \sum_{n=1}^N C_n(x)e_n$ . Notice in this case  $X$  is automatically separable.  $\lrcorner$

**Prop. (12.8.5.14).** If  $X$  has a Schauder basis, then  $C_n(x)$  are continuous functional on  $X$ .  $\lrcorner$

*Proof:* Consider the module  $\|x\|_1 = \sup \|S_N x\|$ , Firstly, it is complete, because  $\|x\| = \lim \|S_N x\| \leq \|x\|_1$ , so if there is a Cauchy sequence  $\{x_i\}$  in  $|\cdot|_1$ , then it is a Cauchy sequence in  $|\cdot|$ , then it converges to some  $x$ . Now  $C_N(x) = S_N(x_i) - S_{N-1}(x_i)$  are all Cauchy sequence, uniform in  $N$ , so they converges to some sequence  $c_N$ .

It is left to verify that  $s_N = \sum_{i=1}^N c_n e_n$  converges to  $x$ , because then it is easy to verify that  $\lim \|x_i - x\|_1 = 0$ . For this, choose  $N_1$  large that  $\|x_n - x\| \leq \varepsilon$  for  $n \geq N_1$ , and choose  $N_2$  large that  $\|S_k(x_n) - s_k\| \leq \varepsilon$  for all  $k$  and  $n \geq N_2$ . Then for  $x_{N_1+N_2}$ , there is a  $N_3$  that  $\|S_n x_{N_1+N_2} - x_{N_1+N_2}\| \leq \varepsilon$ , so  $\|x - s_k\| \leq \|x - x_{N_1+N_2}\| + \|S_k x_{N_1+N_2} - x_{N_1+N_2}\| + \|S_k(x_{N_1+N_2}) - s_k\| \leq 3\varepsilon$  for  $k$  large.

Now by Banach(12.7.2.7),  $\|x\|_1 \leq M\|x\|$  for some  $M$ , so  $|C_n(x)e_n| \leq 2M\|x\|$  and  $C_n$  is continuous.  $\square$

**Prop. (12.8.5.15).** If  $X$  has a Schauder basis, then any compact operator is a limit of operators of f.d. range.  $\lrcorner$

*Proof:* Let  $S_N(x) = \sum_{n=1}^N C_n(x)e_n$ , it is continuous by(12.8.5.14). And it converges, so  $\|S_N\| \leq M$ , by Banach-Steinhaus(12.7.2.1).

For any compact operator, we want to find f.d. range operator  $T_i$  that  $T_i \rightarrow T$ . For this, given any  $\varepsilon > 0$ , because  $\overline{T(B(0,1))}$  is compact, there are operators that is a  $\varepsilon/M^2$ -net  $y_i$ , then choose  $N$  large enough that  $|S_N y_i - y_i| \leq \varepsilon/M^2$ , then for any  $x$ , there is a  $y_i$  that  $|Tx_i - y_i| < \varepsilon/M^2$ , so  $|S_N T x_i - S_N y_i| < \varepsilon/M$ , and then  $|S_N T x - T x_i| < \varepsilon$ , and notice  $S_N T$  has f.d. range.  $\square$

**Prop. (12.8.5.16)[Compact Operator as Limits of F.D. Operators].** Any compact operator on a Hilbert space is a limit of f.d. operators.  $\lrcorner$

*Proof:* Cf.[Trace Classes and Hilbert-Schmidt Operators, Thm10].  $\square$

### Fredholm Operator

**Def. (12.8.5.17)[Fredholm Operator].** A bounded operator between Banach space is called a **Fredholm operator** if  $\dim N(T) < \infty$  and  $\text{codim} R(T) < \infty$ . It necessarily has closed image by(12.7.2.11), so  $R(T) = N(T^*)^\perp$ (12.7.3.2).

The **index** of a Fredholm operator is defined as  $\text{ind}(T) = \dim N(T) - \text{codim} R(T)$ , thus for a compact operator  $A$ ,  $I - A$  has index 0, by(12.8.5.9).  $\lrcorner$

**Prop. (12.8.5.18).** For a Fredholm operator between Banach space, we have

$$X = N(T) \oplus R(T) \quad Y = Y/R(T) \oplus R(T)$$

and  $X/N(T) \cong R(T)$ .  $\lrcorner$

*Proof:* Because  $R(T)$  and  $N(T)$  are finite/cofinite hence closed and complemented by(12.8.1.9). If  $X = N(T) \oplus M_1$  and  $Y = R(T) \oplus M_2$ , then  $M_1 \cong X/N(T)$ ,  $X/N(T) \cong R(T)$  and  $M_2 \cong Y/R(T)$  by Banach theorem.  $\square$

**Prop. (12.8.5.19)[Characterizing Fredholm Operator].**  $T$  is Fredholm from  $X$  to  $Y$  iff there exist a bounded  $S_1, S_2$  from  $Y$  to  $X$  that  $S_1 T = I - A_1$ ,  $T S_2 = I - A_2$ , where  $A_1, A_2$  is compact. If this is the case,  $S_1$  and  $S_2$  can be chosen the same as  $S$ , then  $S$  is called the **regulator** of  $T$ , and  $S$  is Fredholm as well.

So the Fredholm operator is the set of operators invertible 'modulo compact ones'.  $\lrcorner$



*Proof:* By (12.8.5.18)  $T : X/N(T) \cong R(T)$ , and there is a projection of  $\pi : Y$  onto  $R(T)$ . Thus we composed them to get a  $S = T^{-1} \circ \pi : Y \rightarrow X$ . And  $ST$  and  $TS$  are both 1 minus a projection with f.d. image, hence compact (12.8.5.3).

For the converse,  $R(T) \supset R(1 - A_2)$  is of finite codimension because  $1 - A_2$  is Fredholm, and  $N(T) \subset N(1 - A_1)$  is of finite dimension because  $1 - A_1$  is Fredholm.  $\square$

**Cor. (12.8.5.20).** The set of Fredholm operators is closed under composition. Index is a locally constant function on it, and  $\text{ind}(T_1 T_2) = \text{ind}(T_1) + \text{Ind}(T_2)$ .  $\lrcorner$

*Proof:* There is a long exact sequence (use (4.8.8.4) in the category of vector spaces)

$$0 \rightarrow \ker T_2 \rightarrow \ker T_1 T_2 \rightarrow \ker T_2 \rightarrow \text{Coker } T_2 \rightarrow \text{Coker } T_1 T_2 \rightarrow \text{Coker } T_1 \rightarrow 0.$$

which shows the composition and index is additive.

For the openness and locally constancy, use (12.8.5.19), when adding a small  $R$ ,  $S(T + R) = 1 - A_1 + SR$ , and if when  $\|R\| < \|S\|^{-1}$ ,  $E_1 = (I + RS)^{-1}$  is bounded, so  $E_1 S(T + R) = I - E_1 A_1$ , and similarly does  $(T + R)SE_2$ , so  $T + R$  is Fredholm. And  $\text{ind } E_1 + \text{ind } S + \text{ind}(T + R) = \text{ind}(1 - E_1 A_1) = 0$ , and  $\text{ind } E_1 = 0$  because it is invertible, and  $\text{ind } S + \text{ind } T = \text{ind}(1 - A_1) = 0$ , so  $\text{ind } T = \text{ind}(T + R)$ .  $\square$

**Cor. (12.8.5.21).** If  $T$  is Fredholm and  $A$  is compact, then  $T + A$  is also Fredholm and  $\text{ind}(T + A) = \text{ind}(T)$ . In particular,  $\text{ind}$  is in fact defined on the quotient of  $L(X, Y)$  by compact operators.  $\lrcorner$

*Proof:* It is Fredholm by (12.8.5.19), and we notice  $S(T + A)$  and  $ST$  are both 1 minus compact operators, thus (12.8.5.20) and (12.8.5.9) gives the result.  $\square$

**Cor. (12.8.5.22).** If  $T$  is Fredholm, then  $T^*$  is Fredholm, and  $\text{ind}(T^*) = -\text{ind}(T)$ .  $\lrcorner$

*Proof:* The first follows from (12.8.5.19) and (12.8.5.5). For the second, use the fact  $R(T)$  and  $N(T)$  are all closed.  $\square$

## 6 Unbounded Operators



## 12.9 Archimedean Banach Algebra

References are [Rud91].

**Notation (12.9.0.1).**

- Use notations on functional analysis defined in [Archimedean Functional Analysis](#).
- Every Banach space in this section is assumed to be defined over  $\mathbb{C}$ .

┘

### 1 Banach Algebras

#### Spectra of Bounded Operators

**Def. (12.9.1.1) [Spectrums].** Suppose  $X$  is Banach space and  $A \in L(X)$ . Then  $\lambda \in \mathbb{C}$  is called a:

- **point spectrum** if  $\lambda I - A$  is not injective;
- **continuous spectrum** if it is not a point spectrum and  $R(\lambda I - A) \neq X$  but  $\overline{R(\lambda I - A)} = X$ .
- **residue point** if it is not a point spectrum and  $\overline{R(\lambda I - A)} \neq X$ .
- **regular point** if  $\lambda I - A$  is injective and  $R(\lambda I - A) = X$ , in which case  $(\lambda I - X)^{-1}$  is continuous, by Banach.

denote  $\sigma(A) = \mathbb{C} \setminus \{\text{regular points of } A\}$  the **spectrum** of  $A$ , and  $\rho(A) = \sup\{|\lambda| \mid \lambda \in \sigma(A)\}$  is called the **spectral radius** of  $A$ . ┘

**Prop. (12.9.1.2).** If  $A \in \text{BanAlg}$  and  $x$  is invertible in  $A$ , and  $h \in A$  satisfies  $\|h\| < \frac{1}{2} \|x^{-1}\|^{-1}$ , then  $x + h$  is also invertible and

$$\|(x + h)^{-1} - x^{-1} + x^{-1}hx^{-1}\| \leq 2 \|x^{-1}\|^3 \|h\|^2$$

┘

*Proof:*  $x + h = x(e + x^{-1}h)$  and  $\|x^{-1}h\| < \frac{1}{2}$ , so  $x + h$  is invertible by (12.9.1.14), and

$$\|(x + h)^{-1} - x^{-1} + x^{-1}hx^{-1}\| = \left\| \left[ (e + x^{-1}h)^{-1} - e + x^{-1}h \right] x^{-1} \right\| \leq 2 \|x^{-1}\|^3 \|h\|^2$$

also by (12.9.1.14). ┘

**Cor. (12.9.1.3).** If  $A \in \text{BanSpa}$  then the invertible elements  $G(A)$  is an open subset of  $A$ , and the mapping  $x \mapsto x^{-1}$  is a homeomorphism of  $G(A)$  onto  $G(A)$ . ┘

**Prop. (12.9.1.4).** For  $X \in \text{BanSpa}$  and  $T \in L(X)$ ,  $\mathbb{C} \setminus \sigma(T)$  is an open set and  $\lambda \rightarrow (\lambda I - T)^{-1}$  is a operator-valued holomorphic function on  $\mathbb{C} \setminus \sigma(T)$ .

In particular,  $\sigma(T)$  is not empty. ┘

*Proof:* The first assertion is by (12.9.1.3), for the second, let  $f(\lambda) = (\lambda e - x)^{-1}$  is defined on  $\Omega = \mathbb{C} - \sigma(x)$  and (12.9.1.2) shows

$$\|f(\mu) - f(\lambda) + (\mu - \lambda)f^2(\lambda)\| \leq 2 \|f(\lambda)\|^3 \cdot |\mu - \lambda|^2.$$

So  $\lim_{\mu \rightarrow \lambda} \frac{f(\mu) - f(\lambda)}{\mu - \lambda} = -f^2(\lambda)$ , which means that  $f$  is strongly holomorphic in  $\Omega$ .

Now if  $|\lambda| > \|x\|$ , then  $|f(\lambda)| = |\lambda^{-1}e + \lambda^{-2}x + \dots| \leq \frac{1}{|\lambda| - \|x\|}$ , so  $\sigma(x)$  cannot be empty by Liouville theorem (12.8.3.31). ┘

**Cor. (12.9.1.5) [Gelfand-Mazur].** If  $A$  is a Banach algebra over  $\mathbb{C}$  and  $A^\times = A^*$ , then  $A = \mathbb{C}$ .  $\lrcorner$

*Proof:* Any nonzero element  $x$  has a nonempty spectrum, so there is a  $\lambda(x)$  that  $x - \lambda(x)e$  is not invertible, so it must be 0. That is, the mapping  $\mathbb{C} \rightarrow A : \lambda \mapsto \lambda e$  is bijective, so is isomorphism by Banach.  $\square$

**Prop. (12.9.1.6).** Notice  $(1-T)$  is invertible for  $\|T\| < 1$  and the inverse can be calculated by definition.

In particular, for a Banach algebra  $A$  and any  $x \in A$ , when  $\lambda > \|x\|$ ,  $e - \lambda^{-1}x$  is invertible, so the spectrum of  $x$  is bounded. Now that its complement is open as the inverse image of  $G(A)$  by  $\lambda \mapsto \lambda e - x$ , so the spectrum of  $x$  is compact.  $\lrcorner$

**Cor. (12.9.1.7) [Spectrum is Continuous].** The spectrum of an element of a Banach algebra is continuous, i.e. if  $\sigma(x) \subset \Omega$  for some open subset  $\Omega \subset \mathbb{C}$ , then there is a  $\delta > 0$  that  $\sigma(x+y) \subset \Omega$  for  $\|y\| < \delta$ .  $\lrcorner$

*Proof:* As  $\|(\lambda e - x)^{-1}\|$  is a continuous function of  $\lambda$  in the complement of  $\sigma$ , and since the norm tends to 0 as  $\lambda \rightarrow \infty$ , there is a  $M$  that  $\|(\lambda e - x)^{-1}\| < M$  for all  $\lambda \notin \Omega$ . Now if  $\|y\| < 1/M$  and  $\lambda \notin \Omega$ , then  $\lambda e - (x+y) = (\lambda e - x)[e - (\lambda e - x)^{-1}y]$  is invertible.  $\square$

**Prop. (12.9.1.8) [Gelfand].** The spectrum radius  $\rho(A) = \lim_{n \rightarrow \infty} \|A^n\|^{\frac{1}{n}} = \inf \|A^n\|^{\frac{1}{n}} \leq \|A\|$ .

This formula is remarkable, as the LHS depends only on the algebraic structure, and the RHS depends on the metric structure.  $\lrcorner$

*Proof:* For  $r > \rho(x)$

$$x^n = \frac{1}{2\pi i} \int_{\Gamma_r} \lambda^n f(\lambda) d\lambda.$$

Let  $M(r) = \max \|f(re^{i\theta})\|$ , then  $\|x^n\| \leq r^{n+1}M(r)$ , hence  $\limsup \|x^n\|^{1/n} \leq r$ , so  $\limsup \|x^n\|^{1/n} \leq \rho(x)$ .

For the converse, if  $\lambda \in \sigma(x)$ , then  $\lambda^n \in \sigma(x^n)$ , because  $\lambda^n e - x^n = (\lambda e - x)(\lambda^{n-1}e + \dots + x^{n-1})$ , and this two commutes. So  $|\lambda^n| \leq \|x^n\|$ , so  $\rho(x) \leq \inf \|x^n\|^{1/n}$ .  $\square$

**Prop. (12.9.1.9).**  $\sigma(A) = \sigma(A^*)$ .  $\lrcorner$

*Proof:* It suffices to show if  $T$  is invertible iff  $T^*$  is invertible. If  $T$  is invertible, then  $T^*$  is invertible with inverse  $(T^{-1})^*$ . Conversely, if  $T^*$  is invertible, then  $T^{**}$  is invertible, so, as the restriction of  $T^{**}$ ,  $T$  is injective and image is closed. If the image is not  $X$ , then there is a  $f$  that vanish on the image, so  $T^*f = 0$ , but then  $f = 0$ .  $\square$

**Prop. (12.9.1.10).** In a Banach algebra  $A$ ,  $e - xy$  is invertible iff  $e - yx$  is invertible, thus  $\sigma(xy) \cup \{0\} = \sigma(yx) \cup \{0\}$ .  $\lrcorner$

*Proof:* Let  $z = (e - xy)^{-1}$ , then we claim  $e + yzx$  is just the inverse of  $e - yx$ :  $(e - yx)(e + yzx) = e - yx + yzx - yxyzx = e$  and  $(e + yzx)(e - yx) = e + yzx - yx - yzxyx = e$ .  $\square$

**Lemma (12.9.1.11).** If  $A$  is a Banach algebra and  $x_n \in G(A)$  converges to  $x \notin G(A)$ , then  $\|x_n^{-1}\| \rightarrow \infty$ .  $\lrcorner$

*Proof:* If  $\|x_n^{-1}\| < M$ , choose  $n$  that  $\|x_n - x\| < 1/M$ , then  $\|e - x_n^{-1}x\| = \|x_n^{-1}(x_n - x)\| < 1$ , so  $x_n^{-1}x$  is invertible, so  $x$  is invertible.  $\square$

**Prop. (12.9.1.12).** For Banach algebra  $B$  and its closed subalgebra  $A$ ,  $\sigma_A(x)$  is obtained from  $\sigma_B(x)$  by filling some holes. So when  $\sigma_B(x)$  doesn't separate  $\overline{\mathbb{C}}$  or  $\sigma(A)$  has empty interior, then  $\sigma_A(x) = \sigma_B(x)$ .  
 $\lrcorner$

*Proof:* Cf.[Rudin P256].  $\square$

**Prop. (12.9.1.13).** if  $A$  is a Banach algebra over  $\mathbb{C}$  that  $\|x\|\|y\| \leq M\|xy\|$  for some fixed  $M$ , then  $A$  is isomorphic to  $\mathbb{C}$ .  
 $\lrcorner$

*Proof:* If  $y$  is a boundary pt of  $G(A)$ ,  $y = \lim y_n$ , then  $\|y_n^{-1}\| \rightarrow \infty$ . But  $\|y_n\|\|y_n^{-1}\| \leq M\|e\|$ , so  $y_n \rightarrow 0$ , so  $y = 0$ .

But any boundary point of  $\sigma(x)$  gives a boundary point  $\lambda e - x$  of  $G(A)$ , so  $x = \lambda e$ , so  $A \cong \mathbb{C}$ .  $\square$

### Complex Homomorphism

**Prop. (12.9.1.14).** Suppose  $A$  is a Banach algebra over  $\mathbb{C}$ ,  $x \in A$  satisfies  $\|x\| < 1$ , then  $\|(e - x)^{-1} - e - x\| \leq \frac{\|x\|^2}{1 - \|x\|}$ , and  $|\varphi(x)| < 1$  for any complex homomorphism  $\varphi$  on  $A$ . In particular, any complex homomorphism is continuous.  
 $\lrcorner$

*Proof:*  $\|(e - x)^{-1} - e - x\| = \|x^2 + x^3 + \dots\| \leq \sum_{n=2}^{\infty} \|x\|^n = \frac{\|x\|^2}{1 - \|x\|}$ .

For the second, notice  $e - \lambda^{-1}x$  is invertible for each  $|\lambda| \geq 1$ , so  $1 - \lambda\varphi(x) \neq 0$ , so  $\varphi(x) \neq \lambda$ .  $\square$

**Prop. (12.9.1.15)[Gleason-Kahane-Zelazko].** If  $\varphi$  is a linear functional on a Banach algebra  $A$  over  $\mathbb{C}$ , if  $\varphi(e) = 1$  and  $\varphi(x) \neq 0$  for every invertible element  $x \in A$ , then  $\varphi$  is a complex homomorphism.  
 $\lrcorner$

*Proof:* Cf.[Rudin P251].  $\square$

### Symbolic Calculus

**Prop. (12.9.1.16)[Symbolic Calculus].** For a Banach algebra  $A$ . For a domain  $\Omega$  in  $\mathbb{C}$ , define  $A_\Omega$  as the set of  $x$  that  $\sigma(x) \in \Omega$ , it is an open set by (12.9.1.7), then:

$$f \mapsto \tilde{f}(x) = \frac{1}{2\pi i} \int_{\Gamma} f(\lambda)(\lambda e - x)^{-1} d\lambda$$

for any contour  $\Gamma$  that surrounds  $\sigma(x)$ , is a continuous algebra isomorphism of  $H(\Omega)$  into the set of  $A$ -valued functions on  $A_\Omega$  with the compact-open topology.

We have  $\widetilde{g \circ f} = \tilde{g} \circ \tilde{f}$ .  $\lrcorner$

*Proof:* The nontrivial part is that this map is multiplicative, but for this we can use Runge's theorem to approximate any function on  $\sigma(x)$ .  $\square$

This theorem makes it possible to implant complex analysis to the study of Banach Algebra.

**Cor. (12.9.1.17).**  $\exp(x)$  is defined on  $A$  and is continuous. If  $\sigma(x)$  doesn't separate 0 from  $\infty$ , then  $\log(x)$  is defined but might not be continuous.  $\lrcorner$

**Prop. (12.9.1.18)[Spectral Mapping Theorem].**  $\tilde{f}(x)$  is invertible in  $A$  iff  $f(\lambda) \neq 0$  on  $\sigma(x)$ . Thus we have  $\sigma(\tilde{f}(x)) = f(\sigma(x))$ .  $\lrcorner$

**Prop. (12.9.1.19).** If  $f$  doesn't vanish identically on any component of  $\Omega$ , then  $f(\sigma_p(T)) = \sigma_p(\tilde{f}(T))$ .  
 Cf.[Rudin P266].  $\lrcorner$

### Commutative Banach Algebra

**Lemma (12.9.1.20).** For  $A$  a commutative Banach algebra, the set of maximal ideals has codimension 1 corresponds to kernels of complex homomorphisms to  $\mathbb{C}$ . (Consider the quotient space and use Gelfand-Mazur). Note that a complex homomorphism is all continuous because  $\lambda e - x$  maps to nonzero.

$\lambda \in \sigma(x)$  iff there is a complex homomorphism  $h$  s.t.  $h(x) = \lambda$ . (Because  $x$  is invertible iff it is not contained in any proper ideal of  $A$ .  $\lrcorner$ )

*Proof:*

$\square$

**Prop. (12.9.1.21)[Gelfand Transform].** The **spectrum**  $\Delta_A$  of a unital commutative Banach algebra  $A$  is defined to be the set  $\Delta$  of maximal ideals of  $A$ . It is a locally compact Hausdorff space w.r.t to the weak\*-topology and the Gelfand transform:  $x \mapsto \hat{x}(h) = h(x)$  is a continuous map of  $A$  into  $C(\Delta)$ . And the range of  $\hat{x}$  equals  $\sigma(x)$ , so  $\|\hat{x}\| = \rho(x) \leq \|x\|$ .  $\lrcorner$

*Proof:* First we prove it is compact Hausdorff: As  $\sigma(A) = \{h \in \text{closed ball of } A^* | h(e) = 1, h(xy) = h(x)h(y)\}$  which is a closed subset of the closed ball of  $A^*$ , so it is compact Hausdorff. The rest is clear and follows from (12.9.1.20).  $\square$

**Prop. (12.9.1.22).** For  $A = C(X)$  where  $X$  is compact Hausdorff,  $\Delta$  is homeomorphic to  $X$ . (otherwise it has finite  $f_i \neq 0$ , then  $\sum |f_i|^2$  is positive thus invertible but maps to 0). In fact, for a space  $X$ ,  $\Delta(C(X))$  is the stone-Čech compactification of  $X$ .  $\lrcorner$

**Prop. (12.9.1.23).** For  $A = L^\infty(m)$ , the spectrum of  $f$  is just the essential range of  $f$ .  $\lrcorner$

**Lemma (12.9.1.24).** If  $\hat{A} \subset C(\Delta)$  with a chosen topology that makes it compact, and  $A$  separate points, then the topology of it is the same of the weak\*-topology. (Compact to Hausdorff).  $\lrcorner$

**Prop. (12.9.1.25).** The algebra  $L^1(\mathbb{R}^n) \oplus \delta$  with the multiplication by convolution has the spectrum  $\mathbb{R}^n \cap \{\infty\}$ . (Use  $(L^p)^* = L^q$  and see when will it be homomorphism).  $\lrcorner$

## 2 Hilbert spaces

**Prop. (12.9.2.1)[Optimal Approximation].** A closed convex subset in a Hilbert space has a unique element that attains the minimum norm.  $\lrcorner$

*Proof:* Assume  $0 \notin C$ , so let  $d = \inf_{z \in C} \|z\| > 0$ , then there are  $x_n$  that  $d \leq \|x_n\| \leq d + 1/n$ . It suffices to show that  $x_n$  is a Cauchy sequence, because then it has a convergent point in  $C$ . Now

$$\|x_n - x_m\|^2 = 2(\|x_n\|^2 + \|x_m\|^2) - 4 \left\| \frac{x_n + x_m}{2} \right\|^2 \leq 2[(d + 1/n)^2 + (d + 1/m)^2] - 4d^2 \rightarrow 0.$$

For the unicity, if  $\|x_1\| = \|x_2\| = d$ , then

$$\|x_1 - x_2\|^2 = 2(\|x_1\|^2 + \|x_2\|^2) - 4 \left\| \frac{x_1 + x_2}{2} \right\|^2 \leq 4d^2 - 4d^2 = 0.$$

$\square$

**Cor. (12.9.2.2)[Orthogonal Decomposition].** The orthogonal complement of a closed subspace of a Hilbert space exists. and the projection on to a closed subspace exists. This is a good trait of Hilbert space.  $\lrcorner$

*Proof:* For any element  $x$ , let  $y$  be the optimal approximation(12.9.2.1) of  $x$ , then  $z = x - y$  is orthogonal to  $y$ .  $\square$

**Thm. (12.9.2.3) [Riesz].** Linear functionals on a Hilbert space over  $\mathbb{C}$  are all of the form  $x \mapsto (x, z)$  (Choose an orthogonal of the kernel). In other words, Hilbert spaces are reflexive.  $\lrcorner$

*Proof:* Choose a  $x_0$  orthogonal to  $N(f)$  by(12.9.2.2) and  $\|x_0\| = 1$ , then any  $x = \alpha x_0 + y$  where  $y \in N(f)$ . Inner product with  $x_0$ , we get  $\alpha = (x, x_0)$ , so  $f(x) = \alpha f(x_0) = (x, \overline{f(x_0)x_0})$ .  $\square$

**Cor. (12.9.2.4).** For Hilbert spaces  $\mathcal{H}, \mathcal{H}'$  and  $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ ,  $\|T\| = \sup\{(Tx, y) \mid \|x\| \leq 1, \|y\| \leq 1\}$ .  $\lrcorner$

*Proof:* Use(12.8.3.6) to find for each  $x$  a functional  $f$  of norm 1 that  $|f(Tx)| = \|Tx\|$ , then use Riesz theorem. In particular, if we define  $\square$

**Cor. (12.9.2.5)[Reproducing Kernel].** For a Hilbert space  $H$ , if elements of  $H$  are all complex valued functions on a set  $S$ , and  $J_x : f \mapsto f(x)$  is continuous functional for  $H$ , then there is a function  $K(x, y)$  on  $S \times S$  that  $K_y(x) = K(x, y) \in H$ , and  $f(y) = (f, K_y)_H$ , called the **reproducing kernel**.

And if  $e_\alpha$  is a basis for  $H$ , then  $K(x, y) = \sum e_\alpha(x) \overline{e_\alpha(y)}$ .  $\lrcorner$

*Proof:* For any  $y$ , there is a  $K_y \in H$  that  $f(y) = (f, K_y)_H$  by Riesz representation. If we let  $K(x, y) = (K_y, K_x) = K_y(x)$ , then this is the desired kernel.

If  $e_\alpha$  is a basis, then  $K_x = (K_x, e_n)e_n = \overline{e_n(x)}e_n$ , so by Parseval equality,  $K(x, y) = \sum e_\alpha(x) \overline{e_\alpha(y)}$ .  $\square$

**Prop. (12.9.2.6).** Let  $\mathcal{H}$  be a Hilbert space, then a sequence  $x_n$  converges to  $x$  iff  $x_n$  converges to  $x$  weakly and  $\|x\|_n \rightarrow \|x\|$ .  $\lrcorner$

*Proof:* One direction is trivial, for the other, notice that  $\|x_n - x\|^2 = \|x_n\|^2 + \|x\|^2 - 2\operatorname{Re}(x, x_n)$  which converges to 0.  $\square$

**Thm. (12.9.2.7) [Lax-Milgram].** If  $a(x, y)$  is a sesquilinear form on a Hilbert space  $H$  over  $\mathbb{C}$  that  $|a(x, y)| \leq M\|x\|\|y\|$ , then there is a unique continuous operator  $A \in L(H)$  that  $a(x, y) = (x, Ay)$ . If moreover  $|a(x, x)| \geq \delta\|x\|^2$ , then  $A$  is bijective and  $\|A^{-1}\| \leq \frac{1}{\delta}$ .  $\lrcorner$

*Proof:* For any  $y$ ,  $x \mapsto a(x, y)$  is a continuous functional, so by(12.9.2.3), there is an element  $Ay$  that  $a(x, y) = (x, Ay)$ .

Now  $Ay$  depends linearly on  $y$ , and  $\|Ay\| = \sup |a(x, y)|/\|x\| \leq M\|y\|$ .

If  $|a(x, x)| \geq \delta\|x\|^2$ , then  $A$  is clearly injective, and  $R(A)$  is closed, because for any  $z = \lim Av_n$ , it is easily verified that  $v_n$  is a Cauchy sequence. And  $R(A)^\perp = 0$ , because if  $(w, Av) = 0$  for any  $v \in H$ , then  $\delta\|w\|^2 \leq |a(w, w)| = 0$ .  $A^{-1}$  exists by Banach theorem(12.7.2.6), and  $\delta\|x\|^2 \leq |a(x, x)| = (x, Ax) \leq \|x\|\|Ax\|$ , so  $\delta\|x\| \leq \|Ax\|$ .  $\square$

**Cor. (12.9.2.8) [Variational Inequality].** If  $H$  is a Hilbert space that  $a(x, y)$  is an anti-symmetric bilinear function that  $\delta\|x\|^2 \leq a(x, x) \leq M\|x\|^2$ , then if  $u_0 \in H$ , and  $C$  is a closed convex subset of  $X$ , the function

$$f : x \mapsto a(x, x) - \operatorname{Re}(u_0, x)$$

attains minimum at  $C$ .  $\lrcorner$

*Proof:* Similar to the proof of (12.9.2.1).  $f(x) \geq \delta \|x\|^2 - \|u_0\| \|x\|$  is bounded below on  $C$ . if  $x_n$  is a sequence that converge to the infimum  $d$ , then

$$\begin{aligned} a(x_n - x_m, x_n - x_m) &= 2(a(x_n, x_n) + a(x_m, x_m)) - 4a\left(\frac{x_n + x_m}{2}, \frac{x_n + x_m}{2}\right) \\ &= 2(f(x_n) + f(x_m)) - 4f\left(\frac{x_n + x_m}{2}\right) \leq 2[(d + 1/n)^2 + (d + 1/m)^2] - 4d^2 \rightarrow 0. \end{aligned}$$

So  $x_n$  is a Cauchy sequence by the condition, and it contains a unique minimum.  $\square$

**Cor. (12.9.2.9) [Involutions].** For a Hilbert space  $\mathcal{H}$  over  $\mathbb{C}$ , for any  $T \in L(H)$ , there is an operator  $T^* \in L(H)$  that  $(Tx, y) = (x, T^*y)$ , which is called the **formal adjoint** or involution of  $T$ . Notice it is defined on  $H$ , not on  $H^*$ .

Moreover,  $\|T\| = \|T^*\| = \|T^*T\|^{1/2}$ .  $\lrcorner$

*Proof:* Use Lax-Milgram (12.9.2.7) for  $a(x, y) = (Tx, y)$ . For the last assertion,  $\|T\| = \|T^*\|$  by (12.9.2.4). And we notice

$$\|Tx\|^2 = (Tx, Tx) = (T^*Tx, x) \leq \|T^*T\| \|x\|^2,$$

so  $\|T\| \leq \|T^*T\|^{1/2}$ .  $\square$

**Remark (12.9.2.10) [Examples].** The dual operator of the integral operators (12.8.5.2) with kernel  $K(x, y)$  is also an integral operator with kernel  $K^*(x, y) = \overline{K(y, x)}$ , by Fubini-Tonelli theorem.  $\lrcorner$

### 3 Spectral Theory on Hilbert Spaces

#### Resolution of Identity

**Def. (12.9.3.1) [Resolution of Identity on a Hilbert Spaces].** A resolution of identity on a Hilbert space  $\mathcal{H} \in \text{HilbSpa}$  for a  $\sigma$ -algebra on a set  $\Omega$  is a  $E$  that:

1.  $E(\emptyset) = 0, E(\Omega) = 1$ .
2.  $E(\omega)$  is self-adjoint projection.
3.  $E(\omega' \cap \omega) = E(\omega')E(\omega)$ .
4.  $E(\omega \cup \omega') = E(\omega) + E(\omega')$  for disjoint  $\omega, \omega'$ .
5.  $E_{x,y}(\omega) = (E(\omega)x, y)$  is a complex measure on  $E$ .

Thus for any  $x, \omega \rightarrow E(\omega)x$  is a countably additive  $\mathcal{H}$ -valued measure.

This will generate an isometric\*-isomorphism  $\Psi$  of the Banach algebra  $L^\infty(E)$  onto a closed normal subalgebra  $A$  of  $B(H)$ . (Define on simple function first).

$$\Psi(f) = \int_{\Omega} f dE, \quad (\Psi(f)x, y) = \int_{\Omega} f dE_{x,y}.$$

$\lrcorner$

*Proof:* Cf. [Rudin P319].  $\square$

**Prop. (12.9.3.2) [Spectral Decomposition for Normal Algebra].** For any closed normal algebra  $A$  of  $B(H)$ , there is a unique resolution of identity  $E$  (12.9.3.1) on the Borel subsets of  $\Delta_A$  that the inverse of Gelfand transform extends to an isometric \*-isomorphism  $\Phi$  of the algebra  $L^\infty(E)$  to a closed subalgebra  $B$  containing  $A$ .

In fact,  $B = \Gamma(\Gamma(A))$  is normal by Fuglede theorem (12.9.3.10).  $\lrcorner$

*Proof:* Cf.[Rudin P322].  $\square$

**Cor.(12.9.3.3)[Generalized Symbolic Calculus for Normal Operator].** For a normal operator  $T$  and the minimal closed commutative  $B^*$ -algebra  $A$  it generates, the inverse of Gelfand transform gets us a map  $\Psi : C(\sigma(x)) \rightarrow A$  that  $\Psi(z) = x$ ,  $\Psi(\bar{z}) = x^*$ , by(14.1.1.4).

Then the above proposition says there is a resolution of identity on the Borel set of  $\sigma(T)$  that  $\Psi$  extends to a function that maps  $L^\infty(m)$  to  $B(H)$  and  $\|\Psi(f)\| = \|f\|_\infty$ .  $\lrcorner$

**Cor.(12.9.3.4)[Normal and Invariant Subspace].** Any closed normal algebra  $A$  has many invariant subspaces, just choose a decomposition of Borel sets  $\Delta_A = \omega \amalg \omega'$ , then  $R(E(\omega)) \oplus R(E(\omega')) = H$ .

In particular, any normal operator has an invariant subspace.  $\lrcorner$

### Normal Operators on Hilbert Space

**Lemma(12.9.3.5).** For a Hilbert space  $H$  and  $T \in L(H)$ ,  $T$  is defined by values  $(Tx, x)$ .  $\lrcorner$

*Proof:* If  $(Tx, x) = 0$ , then  $(Tx, y) + (Ty, x) = 0$ , so  $-i(Tx, y) + i(Ty, x) = 0$ , solving  $(Tx, y) = 0$  for all  $x, y$ , so  $T = 0$ .  $\square$

**Prop.(12.9.3.6)[Normal Operators].**

1. An operator is normal iff  $\|Tx\| = \|T^*x\|$ . So  $N(T) = N(T^*)$  thus  $\sigma_p(T^*) = \overline{\sigma_p(T)}$ , and  $R(T)$  is dense iff  $T$  is injective. And different eigenspaces are orthogonal.
2. An operator is unitary iff  $R(U) = H$  and  $\|Ux\| = \|x\|$  for every  $x$ . (Because an operator is defined by its value  $(Tx, y)$ ).

$\lrcorner$

*Proof:*  $\|Tx\|^2 = (T^*Tx, x)$ ,  $\|T^*x\|^2 = (TT^*x, x)$ , and they are equal iff  $T, T^*$  commutes by(12.9.3.5). In particular, for different eigenvectors,  $\alpha(x, y) = (Tx, y) = (xT^*y) = (x, \bar{\beta}y) = \beta(x, y)$ .

For unitary, one way is obvious, for the other, if  $\|Ux\| = \|x\|$ , then  $(U^*Ux, x) = (x, x)$ , so  $U^*U = id$  by(12.9.3.5), and  $U$  is a bijection. So it is invertible.  $\square$

**Cor.(12.9.3.7).** For a normal operator  $T$  on a Hilbert space  $T$  is invertible iff there is a  $\delta$  that  $\|Tx\| = \|T^*x\| \geq \delta\|x\|$ .  $\lrcorner$

*Proof:*  $T$  is injective iff  $R(T)$  is dense, and if  $\|Tx\| = \|T^*x\| \geq \delta\|x\|$ , then  $R(T)$  is closed by(4.4.8.30), so it is invertible by Banach theorem.  $\square$

**Prop.(12.9.3.8).** If  $T$  is normal, then

1.  $\|T\| = \sup\{|(Tx, x)| \mid \|x\| = 1\}$ .
2.  $T$  is self-adjoint iff  $\sigma(T)$  is real.
3.  $T$  is unitary iff  $|\sigma(T)| = 1$ .

$\lrcorner$

*Proof:* For 1,  $\|T\| = \rho(T) = \|z_0\|$  for some  $z_0 \in \rho(T)$  by Naimark(14.1.1.3), then Urysohn lemma to show  $E(U) \neq 0$  for a open  $U$  near  $x$ (because otherwise there is a continuous function supported in  $U$  that are mapped to 0), then there are  $\|x_0\| = 1$  that  $E(U)x_0 = x_0$ .

Consider now  $f = (z - z_0)i_U(z)$ , then  $f(T)(x_0) = Tx_0 - \lambda_0x_0$ , so

$$(Tx_0, x_0) - \lambda_0 = |(f(T)x_0, x_0)| \leq \|f(T)\| = \|f\| \leq \varepsilon$$

This shows that  $\|T\| = \sup\{|(Tx, x)| \mid \|x\| = 1\}$ .

For 2, 3, by generalized symbolic calculus(12.9.3.3),  $\hat{T} = \lambda$  on  $\sigma$  and  $\hat{T}^* = \bar{\lambda}$  on  $\sigma$ , so they are equal iff  $\sigma(T)$  is real, and  $TT^* = I$  iff  $\lambda\bar{\lambda} = 1$  on  $\sigma(T)$ .  $\square$

**Prop. (12.9.3.9) [Decomposition of Operators].** For  $\mathcal{H} \in \mathcal{HilbSpa}$ , every operator  $S \in B(\mathcal{H})$  is a linear combination of two self-adjoint operator and a linear combination of four unitary operator.  $\lrcorner$

*Proof:* The first assertion is easy as  $S = (S + S^*)/2 + (S - S^*)/2$ . Now any self-adjoint operator is a multiple of a self-adjoint operator of norm  $\|S\| \leq 1$ , so  $1 - S^2$  is positive, and we have  $S = \frac{1}{2}(f_+(S) + f_-(S))$ , where  $f_{\pm}(s) = s \pm i\sqrt{1 - s^2}$ .  $\square$

**Prop. (12.9.3.10) [Fuglede].** If  $N_1$  and  $N_2$  are normal operators and  $A$  is a bounded linear operator on a Hilbert space such that  $N_1 A = A N_2$ , then  $N_1^* A = A N_2^*$ .  $\lrcorner$

*Proof:* For any  $S \in B(H)$ ,  $\exp(S - S^*)$  is unitary thus  $\|\exp(S - S^*)\| = 1$ ,  $\exp(N_1)A = A\exp(N_2)$ . Because  $\exp(M)T = T\exp(N)$ , if we let  $U_1 = \exp(M^* - M)$ ,  $U_2 = \exp(N - N^*)$ , then

$$\|\exp(N_1^*)T\exp(-N_2^*)\| = \|U_1 T U_2\| \leq \|T\|,$$

because  $\lambda N_i$  is normal. Now

$$\|\exp(\lambda N_1^*)T\exp(-\lambda N_2^*)\| = \|U_1 T U_2\| \leq \|T\|$$

also holds, thus by Liouville,  $\exp(\lambda N_1^*)T\exp(-\lambda N_2^*) = T$ . Compare the coefficients of  $\lambda$ , we get the result.  $\square$

**Prop. (12.9.3.11).** An operator  $T \in B(H)$  has the same spectrum w.r.t all the closed  $B^*$ -algebras of  $B(H)$  containing it.  $\lrcorner$

*Proof:* If  $T$  is invertible, because  $TT^*$  is self-adjoint thus has real spectrum (12.9.3.8) so doesn't separate  $\mathbb{C}$  thus it is invertible in any closed  $B^*$ -algebra of  $B(H)$  (12.9.1.12). so does  $T^{-1} = T^*(TT^*)^{-1}$ .  $\square$

**Prop. (12.9.3.12).** For  $T$  normal and  $E$  its spectral decomposition, then if  $f \in C(\sigma(T))$  and  $\omega_0 = f^{-1}(0)$ , then  $N(f(T)) = R(E(\omega_0))$ .  $\lrcorner$

*Proof:*  $\chi_{\omega_0} f = 0$ , so  $f(T)R(E(\omega_0)) = 0$ , and if we let  $\omega_n = f^{-1}([1/(n-1), 1/n])$ , and let  $f_n(\lambda) = 1/f(\lambda)\chi_{\omega_n}$ , then  $f_n(T)f(T) = E(\omega_n)$ , so if  $f(T) = 0$ , then  $E(\omega_n)x = 0$ , so countable additivity shows that  $E(\sigma \setminus \omega_0)x = 0$ , so  $E(\omega_0)x = x$ . These shows the desired result.  $\square$

**Cor. (12.9.3.13).**

1.  $N(T - \lambda I) = R(\{\lambda\})$ .
2. every isolated point of  $\sigma(T)$  is point spectrum, because this point is open thus is  $E(\{x\}) \neq 0$  by Urysohn lemma.
3. if  $\sigma(T)$  is countable, then every  $x \in H$  has a unique orthogonal decomposition  $x = \sum E(\lambda_i)x$  and  $T(E(\lambda_i)x) = \lambda_i E(\lambda_i)x$ .

$\lrcorner$

**Prop. (12.9.3.14) [Normal Compact Operator].** A normal operator  $T \in B(H)$  is compact iff  $\sigma(T)$  has no limit point except possibly 0 and  $\dim N(T - \lambda I) < \infty$  for  $\lambda \neq 0$ .

In particular, a normal compact operator is a limit of f.d. operators  $\lrcorner$

*Proof:* One direction is general, by (12.8.5.9), for the other, it is a limit of operators of finite dimensional range by general symbolic calculus (12.9.3.3).  $\square$



**Cor. (12.9.3.15) [Spectral Theorem].** A compact normal operator (in particular a normal operator on a f.d linear space) is unitarily diagonalizable.  $\lrcorner$

*Proof:* it suffices to find a basis of eigenvectors, but this is easy, just by (12.9.3.13).  $\square$

**Cor. (12.9.3.16) [Hilbert-Schmidt].** For a self-adjoint compact operator  $A$  on a Hilbert space  $H$ , there is a set of orthonormal basis that  $A$  is diagonal on it. And of course, its eigenvalues are real and can only converges to 0 (12.9.3.8).  $\lrcorner$

**Prop. (12.9.3.17).** For a normal compact operator  $T \in L(H)$ , then:

1.  $T$  has an eigenvalue  $|\lambda|$  that  $|\lambda| = \|T\|$ .
2.  $f(T)$  is compact if  $f \in C(\sigma(T))$  and  $f(0) = 0$ .
3.  $f(T)$  is not compact if  $f \in C(\sigma(T))$  and  $f(0) \neq 0$  and  $\dim H = \infty$ .

$\lrcorner$

*Proof:* 1: The spectrum of maximal norm is isolated (12.8.5.9) hence a point spectrum by (12.9.3.13). And  $|\lambda| = \|T\|$  by symbolic calculus (12.9.3.3).

2: Cf. [Rudin P330].

3: The 2 sill show that  $f(0)I - f$  is compact, If  $f$  is compact, then  $f(0)I$  is compact, so  $\dim H < \infty$  (12.8.5.3).  $\square$

**Prop. (12.9.3.18) [Freudenthal Spectral Theorem].**  $\lrcorner$

**Prop. (12.9.3.19) [Positive Equivalent Definition].** A  $T \in L(H)$  is positive, i.e.  $T = T^*$  and  $\sigma(T) \subset \mathbb{R}_{\geq 0}$  iff  $(Tx, x) \geq 0$ .  $\lrcorner$

*Proof:* If  $(Tx, x) \geq 0$ , then  $(Tx, x) = (x, Tx) = (T^*x, x)$ , so  $T = T^*$  by (12.9.3.5), so  $\sigma(T)$  is real (12.9.3.8), and for  $\lambda > 0$ ,

$$\lambda \|x\|^2 = (\lambda x, x) \leq ((T + \lambda I)x, x) \leq \|(T + \lambda I)x\| \|x\|,$$

so  $T + \lambda I$  is invertible by (12.9.3.7), so  $\sigma(T) \subset \mathbb{R}_{\geq 0}$ .

Conversely, if  $T$  is positive, then it is normal, so  $(Tx, x) = \int_{\sigma(T)} \lambda dE_{x,x} \geq 0$ .  $\square$

**Prop. (12.9.3.20) [Polar Decomposition].**

1. Every positive operator  $T$  has a positive square root, which is invertible if  $T$  is.
2. Polar decomposition exists in  $B(H)$ : Any  $T \in L(H)$  invertible has a unique decomposition  $T = UP$  where  $U$  is unitary and  $P$  is positive. And  $\|Px\| = \|Tx\|$  for all  $x$ .
3. Any normal operator has commuting decomposition  $UP$ , where  $U, P, T$  commutes.

$\lrcorner$

*Proof:* 1: Use general symbolic calculus, then  $S = \sqrt{\lambda}(T)$  is the square root of  $T$ . If  $T$  is invertible, then  $S^{-1} = T^{-1}S$ .

2:  $(T^*Tx, x) = (Tx, Tx) \geq 0$ , so  $T^*T$  is positive (12.9.3.19), so let  $P = \sqrt{T^*T}$ , then it is also invertible, and  $U = TP^{-1}$  is unitary.

3: Use general symbolic calculus, let  $p(\lambda) = |\lambda|$ ,  $u(\lambda) = \lambda/|\lambda|$  if  $\lambda \neq 0$ , and  $u(0) = 0$ . Then  $T = UP$ , and they are commutative.  $\square$

**Cor. (12.9.3.21) [Similar Normal Operator].** Similar normal operators are unitarily equivalent.  $\lrcorner$

*Proof:* It suffices to show that if  $M = TNT^{-1}$ , and  $T = UP$  is the polar decomposition, then  $M = UNU^{-1}$ . Fuglede (12.9.3.10) shows  $M^*T = TN^*$ , so  $NP^2 = NT^*T = T^*MT = T^*TN = P^2N$ , so  $N$  commutes with any functions  $f(P)$ , in particular  $P$ . Hence  $M = (UP)N(UP)^{-1} = UNU^{-1}$ .  $\square$

#### 4 Hilbert-Schmidt Operators & Trace Classes

References are [Trace Classes and Hilbert-Schmidt Operators].

In this subsection, all Hilbert spaces are separable.

**Def. (12.9.4.1) [Hilbert-Schmidt Operator].** Let  $\mathcal{H}, \mathcal{K}$  be Hilbert spaces and  $T \in L(\mathcal{H}, \mathcal{K})$ . Then for any orthonormal basis  $\{e_k\}$  of  $\mathcal{H}$  and  $\{f_k\}$  of  $\mathcal{K}$ ,

$$\sum_j \|Te_j\|_{\mathcal{K}}^2 = \sum_j \|T^*f_j\|_{\mathcal{H}}^2.$$

In particular, this value is independent of the basis  $\{e_k\}$  chosen.

$T$  is called a **Hilbert-Schmidt** iff

$$\|T\|_{\text{H-S}} = \left( \sum_j \|Te_j\|_{\mathcal{K}}^2 \right)^{1/2} < \infty$$

for some/all basis  $e_j$  of  $\mathcal{H}$ . The space of all Hilbert-Schmidt maps from  $\mathcal{H}$  to  $\mathcal{K}$  is denoted by  $S_2(\mathcal{H}, \mathcal{K})$ .  $\lrcorner$

*Proof:*

$$\sum_i \|Te_i\|^2 = \sum_i \sum_j |(Te_i, f_j)|^2 = \sum_i \sum_j |(T^*f_j, e_i)|^2 = \sum_j \|T^*f_j\|^2.$$

□

**Cor. (12.9.4.2) [Properties of  $S_2(\mathcal{H}, \mathcal{K})$ ].**

- If  $A \in S_2(\mathcal{H}, \mathcal{K})$  then  $A^* \in S_2(\mathcal{K}, \mathcal{H})$  with the same HS-norm.
- For  $A \in S_2(\mathcal{H}, \mathcal{K})$ ,  $\|A\| \leq \|A\|_{\text{H-S}}$ .
- $S_2(\mathcal{H}, \mathcal{K})$  is a complete in the HS-norm. And it is moreover Hilbert space because we can define

$$(A, B)_{\text{H-S}} = \sum_{\alpha} (A^* B e_{\alpha}, e_{\alpha}).$$

- If  $\mathcal{H}_1, \mathcal{K}_1$  are separable Hilbert spaces and  $T \in S_2(\mathcal{H}, \mathcal{K})$ ,  $A \in \mathcal{L}(\mathcal{H}_1, \mathcal{H})$ ,  $B \in \mathcal{L}(\mathcal{K}, \mathcal{K}_1)$ , then  $BT A \in S_2(\mathcal{H}_1, \mathcal{K}_1)$ .  $\lrcorner$

*Proof:* 1 follows from (12.9.4.1). 2 is because we can extend  $u$  to a basis of  $\mathcal{H}$ .

3:  $\|-\|_{\text{H-S}}$  is clearly a semi-norm (12.7.1.8), and it is a norm by item 2. To show the completeness, if  $A_j$  is an HS-Cauchy sequence, then it is a Cauchy sequence in the operator norm, thus converges to an operator  $A$ . Then for any  $\varepsilon$ , there is an  $N$  that for any  $j, k \geq N$ ,  $\|A_j - A_k\|_{\text{H-S}} \leq \varepsilon$ . This implies

$$\sum_{\alpha \in S} \|(A_k - A_j)e_{\alpha}\|_{\mathcal{K}}^2 \leq \varepsilon^2$$

for any finite subset  $S \subset I$ . Then letting  $k \rightarrow \infty$  and then letting  $S$  be any subset, we get  $\|A - A_j\|_{\text{H-S}} \leq \varepsilon$ . Thus  $A$  is Hilbert-Schmidt and  $A_j \rightarrow A$  in HS-norm.

4: it is clear that  $\|BT\|_{\text{H-S}} \leq \|B\| \|T\|_{\text{H-S}}$ , and use the transpose invariance of HS-norms and operator norms (12.9.2.9). □

**Prop. (12.9.4.3) [Hilbert-Schmidt Operators are Compact].** If  $A \in S_2(\mathcal{H}, \mathcal{K})$  and  $\{f_k\}$  is an orthonormal basis of  $\mathcal{K}$ , and we denote  $\text{pr}_n$  as the projection of  $\mathcal{K}$  onto the span of  $\{f_1, \dots, f_n\}$ , then

$$\|\text{pr}_n A - A\|_{\text{H-S}} \rightarrow 0.$$

In particular,  $A$  is compact, by (12.9.4.2) and (12.8.5.3).  $\lrcorner$

*Proof:* By (12.9.4.2), it suffices to show that  $\|A^* \pi_n - A^*\|_{\text{H-S}} \rightarrow 0$ . But this norm is just  $\sum_{k>N} \|A^* f_k\|_{\mathcal{H}}^2$ , which converges to 0.  $\square$

**Prop. (12.9.4.4) [Integral Operator is Hilbert-Schmidt].** Let  $\Omega$  be a  $\sigma$ -finite measure space and  $K(x, y) \in L^2(\Omega \times \Omega)$ , then the operator  $Lu(x) = \int_{\Omega} K(x, y)u(y)dy$  defined in (12.8.5.2) is a Hilbert-Schmidt operator on  $L^2(\Omega)$ . In fact,  $\|L\|_2 = \|K\|_{L^2}$ .  $\lrcorner$

*Proof:* Let  $\mathcal{E}$  be an Hilbert basis of  $L^2(\Omega)$ , then we have

$$\|L\|_2^2 = \sum_{f_1, f_2 \in \mathcal{E}} |(Lf_1, f_2)|^2 = \sum_{f_1, f_2 \in \mathcal{E}} \left| \int_X \int_X \overline{f_2(y)} K(x, y) f_1(x) dx dy \right|^2 = \sum_{f_1, f_2 \in \mathcal{E}} (K, f_1 \otimes f_2)^2.$$

But by (12.3.6.5),  $\{f_i \otimes f_j\}$  form a Hilbert Basis for  $L^2(\Omega \times \Omega)$ , so the equation equals  $\|K\|_2^2$ .  $\square$

### Trace Classes

**Def. (12.9.4.5) [Trace Classes].** Let  $\mathcal{H}$  be a Hilbert space,  $\{e_i\}, \{f_i\}$  are two orthonormal basis,  $A \in B(\mathcal{H})$ . Let  $|A| = (A^*A)^{1/2}$  which is positive, then

$$\sum_i (|A|e_i, e_i) = \sum_i (|A|^{1/2}e_i, |A|^{1/2}e_i) = \sum_i \| |A|^{1/2}f_i \|^2 = \sum_i (|A|f_i, f_i)$$

by (12.9.4.1), thus we can define  $\|A\|_1 = \sum_i (|A|e_i, e_i)$ , and say  $A$  is a **trace class** if  $\|A\|_1 < \infty$ . The space of trace classes is denoted by  $S_1(\mathcal{H})$ .

A trace-class  $A$  is clearly compact as  $|A|$  is a limit of f.d. range operators. (Use diagonalization, then there are only countably many eigenvectors of  $|A|$ ).  $\lrcorner$

**Prop. (12.9.4.6).** If  $A \in \mathcal{B}(\mathcal{H})$ , then the following are equivalent:

- $A \in S_1(\mathcal{H})$ .
- $|A|^{1/2} \in S_2(\mathcal{H})$ .
- $|A|$  is a product of two elements in  $S_2(\mathcal{H})$ .
- $A$  is a product of two elements in  $S_2(\mathcal{H})$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2 \rightarrow 3$  is clear, for  $3 \rightarrow 4$ : if  $|A| = TS$ , then by polar decomposition  $A = U|A| = (UT)S$ , and  $UT \in S_2(\mathcal{H})$  by (12.9.4.2), so  $A$  is a product of two elements in  $S_2(\mathcal{H})$ .

$4 \rightarrow 3$  is similar to  $3 \rightarrow 4$ .

$3 \rightarrow 1$ : if  $A = BC$  where  $B, C \in S_2(\mathcal{H})$ , then  $B^* \in S_2(\mathcal{H})$  also by (12.9.4.2), and

$$\|A\|_1 = \sum_i (Ae_i, e_i) = \sum_i (Ce_i, B^*e_i) \leq \sum_i \|Ce_i\| \|B^*e_i\| \leq \|C\|_2 \|B^*\|_2 < \infty$$

$\square$

**Lemma (12.9.4.7).** if  $T$  is a positive trace-class and  $S \in \mathcal{L}(\mathcal{H})$ , then if  $x_i$  is an orthonormal basis of  $\mathcal{H}$ , then

$$\sum_i (STx_i, x_i) \leq \|S\| \|T\|_1$$

is absolutely convergent, and is independent of the basis chosen.  $\lrcorner$

*Proof:* Let  $e_i$  be the basis of eigenvectors of  $T$  of eigenvalues  $\lambda_i > 0$ , then  $\sum_i \lambda_i < \infty$  by (12.9.4.6), and

$$(STx_i, x_i) = \sum_j (x_i, e_j)(STe_j, x_i) = \sum_j \lambda_j (x_i, e_j)(Se_j, x_i)$$

And

$$\sum_i \sum_j \lambda_j |(x_i, e_j)(Se_j, x_i)| \leq \sum_j \lambda_j \|e_j\| \|Se_j\| \leq \|S\| \sum \lambda_\alpha < \infty.$$

Moreover:

$$\sum_i (STx_i, x_i) = \sum_i \sum_j ((x_i, e_j)STe_j, x_i) = \sum_i \sum_j (STe_j, (e_j, x_i)x_i) = \sum_j (STe_j, e_j).$$

□

**Prop. (12.9.4.8) [Singular Trace].** If  $\mathcal{H} \in \mathcal{HilbSpa}$  with an orthonormal Hilbert basis  $\{x_\alpha\}$  and  $T \in S_1(\mathcal{H})$ , then  $\sum (Tx_\alpha, x_\alpha)$  absolutely converges, and is independent of the basis chosen, called the **singular trace**  $\text{tr}T$  of  $T$ . The singular trace is a positive definite linear functional on  $S_1(\mathcal{H})$ .  $\lrcorner$

*Proof:* Use polar decomposition  $T = U|T|$  (12.9.3.20) and notice  $|T|$  is a positive trace class (12.9.4.6) and then use (12.9.4.7).  $\square$

**Prop. (12.9.4.9) [Trace of Integral Operators].** Let  $A, B$  be  $L^2$  integral operators on a  $\sigma$ -finite measure space  $\Omega$  with kernel  $K_1(x, y), K_2(x, y) \in L^2(\Omega \times \Omega)$ , then  $AB$  is also an integral operator with kernel

$$\int K_1(x, z)K_2(z, y)dz,$$

and

$$\text{tr}(AB) = \int \int K_1(x, y)K_2(y, x)dxdy.$$

$\lrcorner$

*Proof:* The formula for integral kernel is an immediate consequence of Fubini-Tonelli theorem. For the trace, observe that  $A^*$  is the integral operator with kernel  $\overline{K_1(y, x)}$  (12.9.2.10), thus

$$\begin{aligned} \text{tr}(AB) &= \sum_i (ABe_i, e_i) = \sum_i (Be_i, A^*e_i) \\ &= \sum_i \sum_k (e_k, A^*e_i)(Be_i, e_k) \\ &= \sum_i \sum_k (e_k \otimes \overline{e_i}, K_1^*)(K_2, e_k \otimes \overline{e_i}) \\ &= (K_2, K_1^*) = \int \int K_1(x, y)K_2(y, x)dxdy \end{aligned}$$

as  $\{e_i \otimes \overline{e_k}\}$  is a Hilbert basis for  $\Omega \times \Omega$  (12.3.6.5).  $\square$

**Prop. (12.9.4.10) [Properties of Trace Classes].**

1.  $S_1(\mathcal{H})$  is a two-sided  $*$ -ideal of  $\mathcal{L}(\mathcal{H})$ .
2.  $\|\cdot\|_1$  is a norm on  $S_1(\mathcal{H})$ .
3. If  $T \in S_1(\mathcal{H})$ , then  $\text{tr}(T^*) = \overline{\text{tr}(T)}$ .
4. For any  $T \in S_1(\mathcal{H})$  and  $S \in L(\mathcal{H})$ ,  $\text{tr}(ST) = \text{tr}(TS)$ , and  $|\text{tr}(ST)| \leq \|S\| \|T\|_1$ . In particular the singular trace is a bounded linear functional on  $S_1(\mathcal{H})$ .

□

*Proof:* Let  $S, T \in S_1(\mathcal{H})$ ,  $S = V|S|$ ,  $T = W|T|$ ,  $S + T = X|S + T|$  where  $V, W, X$  are unitary, then  $|S + T| = X^*(S + T)$  is positive compact, so it has an orthonormal eigenbasis  $e_n$  by (12.9.3.16), so

$$\sum (|S + T|x_i, x_i) = \sum (X^*V|S|x_i, x_i) + \sum (X^*W|T|x_i, x_i) \leq \|S\|_1 + \|T\|_1$$

by (12.9.4.7). So  $S + T$  is a trace class, and  $\|S + T\|_1 \leq \|S\|_1 + \|T\|_1$ .

Now if  $U$  is unitary and  $T \in S_1(\mathcal{H})$ , then  $(UT)^*UT = T^*T$ , so  $UT$  is a trace class, and  $(TU)^*TU = U^{-1}T^*U$  has  $|TU| = U^{-1}|T|U$ , so  $TU$  is also a trace class. Moreover,  $\text{tr}(TU) = \sum (TUx_i, x_i) = \sum (UTUx_i, Ux_i) = \text{tr}(UT)$ .

Then notice very  $S \in L(\mathcal{H})$  is a linear combination of four unitary operator (12.9.3.9), so the proposition is true, and if  $T$  is a trace class, then  $T = V|T|$ , and  $T^* = |T|V^*$  is also a trace class.

4:  $|\text{tr}(ST)| \leq \|SV\| \|T\|_1 = \|S\| \|T\|_1$  by (12.9.4.7). □

**Prop. (12.9.4.11) [Trace Classes form a Banach Space].** Let  $S_0(\mathcal{H})$  be the space of operators of f.d. range, then the map

$$\rho : S_1(\mathcal{H}) \rightarrow S_0(\mathcal{H})^* : \rho(A) : C \mapsto \text{tr}(CA)$$

is an isometric isomorphism. In particular,  $S_1(\mathcal{H})$  is a Banach space, by (12.7.3.1). □

*Proof:* Clearly  $\rho$  is a linear map as singular trace is. For  $T \in S_1(\mathcal{H})$ ,  $\|\rho(T)\| \leq \|T\|_1$  by (12.9.4.10).

If  $\Phi \in S_0(\mathcal{H})^*$ ,  $g, h \in \mathcal{H}$ , consider  $g \otimes h^* \in S_0(\mathcal{H})$  that  $g \otimes h^*(v) = (v, h)g$ , then  $B(g, h) = \Phi(g \otimes h^*)$  is a sesquilinear form on  $H$  that is bounded by  $\|\Phi\|$ . Thus by Lax-Milgram (12.9.2.7), there is a unique  $T \in \mathcal{B}(\mathcal{H})$  that  $B(g, h) = (g, Th)$ .

Now let  $A = T^*$  and let  $A = U|A|$  be the polar decomposition,  $\mathcal{E}$  an orthonormal basis of  $\mathcal{H}$  and  $S \subset \mathcal{E}$  a finite subset, define

$$C_S = \left( \sum_{e \in S} e \otimes e^* \right) U^* = \sum_{e \in S} e \otimes (Ue)^*.$$

Then  $C_S$  is of f.d. and  $\|C_S\| \leq 1$ . And:

$$\sum_{e \in S} (|A|e, e) = \sum_{e \in S} (U^*Ae, e) = \sum_{e \in S} (e, T Ue) = \sum_{e \in S} B(e, Ue) = \sum_{e \in S} \Phi(e \otimes (Ue)^*) = \Phi(C_S).$$

So  $\|A\|_1 \leq \|\Phi\|$ .

If  $C$  is any operator of f.d. range that  $C = \oplus g_k \otimes h_k^*$ , then

$$\Phi(C) = \sum \Phi(g_k \otimes h_k^*) = \sum B(g_k, h_k) = \sum (Ag_k, h_k) = \sum \text{tr}(A(g_k \otimes h_k^*)) = \text{tr}(AC) = \rho(A)(C)$$

so the image is  $A$  is just  $\Phi$ . This shows that  $\rho$  is surjective and moreover  $\|A\|_1 = \|\rho(A)\|$ , so we are done. □

### Tensor Product of Hilbert Spaces

References are [Fol16].

**Def. (12.9.4.12) [Hilbert-Schwartz Tensor Product].** For  $\mathcal{H}_1, \mathcal{H}_2 \in \mathcal{HilbSpa}$ , the **Hilbert space dual**  $\mathcal{H}_1^*$  is defined to be  $\mathcal{H}_1^\vee = S_2(\mathcal{H}_1, \mathbb{C}) \in \mathcal{HilbSpa}$  (12.9.4.2), and the **tensor product of Hilbert spaces** is defined to be

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = S_2(\mathcal{H}_2^\vee, \mathcal{H}_1) \in \mathcal{HilbSpa} \text{ (12.9.4.2)}.$$

For  $u \in \mathcal{H}_1, v \in \mathcal{H}_2$ , we can define  $u \otimes v \in \mathcal{H}_1 \otimes \mathcal{H}_2$  as

$$(u \otimes v)(\lambda) = \lambda(u)v.$$

┘

**Prop. (12.9.4.13).** If  $\{u_\alpha\}_{\alpha \in I}$  is an orthonormal basis of  $\mathcal{H}_1$  and  $\{v_\beta\}_{\beta \in J}$  is an orthonormal basis of  $\mathcal{H}_2$ , then

$$\{u_\alpha \otimes v_\beta\}_{\alpha \in I, \beta \in J}$$

is an orthonormal basis of  $\mathcal{H}_1 \otimes \mathcal{H}_2$ .

┘

*Proof:* Cf. [Fol16]P281. □

**Prop. (12.9.4.14) [Non-Existence of Tensor Product of Hilbert Spaces].** If  $\mathcal{H}, \mathcal{K} \in \mathcal{HilbSpa}$  are two infinite dimensional separable Hilbert spaces, then there doesn't exist a tensor product of  $\mathcal{H}, \mathcal{K} \in \mathcal{HilbSpa}$  satisfying the universal property. ┘

*Proof:* We firstly prove the Hilbert-Schwartz tensor product  $\mathcal{H} \otimes \mathcal{K}$  (12.9.4.12) doesn't satisfy the universal property:

We may identify  $\mathcal{K}$  with  $\mathcal{H}^*$  and identify  $\mathcal{H} \otimes \mathcal{K}$  with  $S_2(\mathcal{H})$  (12.9.4.12). Then there is a bilinear map  $\mathcal{H} \otimes \mathcal{H}^\vee \rightarrow \mathbb{C} : (u, \lambda) \mapsto \lambda(u)$ . This map induces a map  $\mathcal{H} \otimes \mathcal{H}^\vee \rightarrow \mathbb{C}$  with is the trace map. But there are clearly elements in  $\mathcal{H} \otimes \mathcal{H}^\vee$  that the map cannot be defined continuously: Let  $\{e_i\}$  be a complete orthonormal basis of  $\mathcal{H}$ , then the trace of the element

$$\sum_{n \in \mathbb{Z}_+} \frac{1}{n} e_i \otimes e_i^\vee \in \mathcal{H} \otimes \mathcal{H}^\vee$$

cannot be defined.

Next we prove that no other Hilbert spaces  $X$  can satisfy the universal property. If  $X$  does, then there is a unique map  $T : X \rightarrow \mathcal{H} \otimes \mathcal{H}^\vee$  by universal property, and thus  $\mathcal{H} \otimes \mathcal{H}^* \rightarrow X$  is injective. And clearly  $\mathcal{H} \otimes \mathcal{H}^*$  is dense in  $X$ .

Let  $T^* : X \rightarrow \mathcal{H} \hat{\otimes} \mathcal{H}^*$  be the adjoint, then for any  $v, w \in \mathcal{H}$  and  $\lambda, \mu \in \mathcal{H}^*$ , we have

$$\langle v \otimes \lambda, T^*(w \otimes \mu) \rangle_X = \langle T(v \otimes \lambda), w \otimes \mu \rangle = \langle v \otimes \lambda, w \otimes \mu \rangle.$$

But  $v$  and  $\lambda$  are arbitrary, thus  $T^*(w \otimes \mu)$  is a multiple of  $w \otimes \mu$ . But  $w$  and  $\mu$  are arbitrary, and we can take  $w \otimes \mu = (v + w) \otimes (\mu + \lambda)$ , so  $T^*$  acts as a constant on  $V \otimes V^*$ . Thus the norm on  $X$  and  $\mathcal{H} \hat{\otimes} \mathcal{H}^*$  differ by a constant, thus  $X$  cannot have the tensor product property. □

**Prop. (12.9.4.15) [Tensor Product of Operators].** Let  $\mathcal{H}_1, \mathcal{H}_2 \in \mathcal{HilbSpa}$  and  $S, S' \in B(\mathcal{H}_1), T, T' \in B(\mathcal{H}_2)$ , then we can define

$$(S \otimes T)f = TfS^\vee, \text{ where } f \in S_2(\mathcal{H}_2^\vee, \mathcal{H}_1).$$

- $S \otimes T \in B(\mathcal{H}_1 \otimes \mathcal{H}_2)$  and  $\|S \otimes T\| = \|S\| \|T\|$ .
- $(S \otimes T)(u \otimes v) = Su \otimes Tv$  for any  $u \in \mathcal{H}_1, v \in \mathcal{H}_2$ .
- $(S \otimes T)(S' \otimes T') = (SS') \otimes (TT')$ .
- $(S \otimes T)^* = S^* \otimes T^*$ .
- If  $S$  and  $T$  are unitary, then so is  $S \otimes T$ .

┘

## 12.10 Harmonic Analysis on Locally Compact Groups

*“Harmonic analysis, in the broadest sense refers to a general principle in mathematics that links geometric objects with spectral objects.”*

–Arthur

References are [Bum97, Fol16].

All representations in this section are assumed to be over  $\mathbb{C}$ .

**Notation (12.10.0.1).**

- Use notations on Topology defined in [Topology I](#).
- Use notations on Representations defined in [Smooth Representations](#).
- Let  $G \in \text{LCG}$ .
- Let  $X \in \text{LCH}$ .

┘

### 1 Locally Compact Groups

**Def. (12.10.1.1) [Left and Right Regular Actions].** The **left regular action** and **right regular action** of  $G$  on  $C_c(G)$  are defined as follows:  $L_y f(x) = f(y^{-1}x)$ ,  $R_y f(x) = f(xy)$ .

More generally, if  $G$  acts on  $X$  on the left and  $X$  has a left- $G$ -invariant Radon measure, then  $G$  acts on  $C_c(X)$  by  $L_g(f)(x) = f(g^{-1}x)$ , and  $R_g(f)(x) = f(xg)$ . ┘

**Prop. (12.10.1.2) [Translation is Continuous].** Situation as in (12.10.1.1), then any  $f \in C_c(X)$  is left and right uniformly continuous. Equivalently,  $G \rightarrow C_c(G) : y \mapsto R_y(f)$  and  $y \mapsto L_y(f)$  are continuous group homomorphisms from  $G$  to  $C_c(G)$ . ┘

*Proof:* Cf. [Folland Abstract Harmonic Analysis P38]. □

**Def. (12.10.1.3) [Involution].** For  $f \in C_c(G)$  or  $f \in L^p(G)$  for some  $p$ , let  $\tilde{f}(x) = \overline{f(x^{-1})}$ . ┘

**Prop. (12.10.1.4).**  $G$  is a normal topological space. ┘

*Proof:* Notice that by choosing a precompact symmetric open neighbourhood  $U$  of identity, there exists a  $\sigma$ -compact clopen subgroup  $H$ . So  $H$  can  $\sigma$ -locally refine every open cover, thus  $G$  can, too. So by (4.4.7.2)  $G$  is paracompact. As a topological group,  $G$  is regular, thus  $G$  is normal by (4.4.7.6). □

**Prop. (12.10.1.5).**  $G$  has a subgroup  $G_0$  that is clopen and  $\sigma$ -compact. ┘

*Proof:* Let  $U$  be a symmetric precompact nbhd of 1 in  $G$ , then let  $U_n = U^n$ , then  $\overline{U_n} \subset U_{n+1}$ , so let  $G_0 = \cup_n U_n = \cup \overline{U_n}$ , then it is open because each  $U_n$  does, and compact because each  $\overline{U_n}$  does. □

**Prop. (12.10.1.6).** If  $H \leq G$  is a locally compact subgroup, then  $H$  is closed in  $G$ . ┘

*Proof:* By hypothesis there exists an open nbhd  $U$  of  $e \in G$  that  $U \cap H$  has compact closure  $K \subset H$ . But then  $K$  is also compact in  $G$  thus closed. So  $K$  is the closure of  $U \cap H$  in  $G$ . Choose a symmetric open nbhd  $V$  of  $e \in G$  that  $VV \subset U$ , and suppose  $x \in \overline{H}$ , then  $x^{-1} \in \overline{H}$  and  $Vx^{-1} \cap H \neq \emptyset$ . Let  $y \in Vx^{-1} \cap H$ . For any nbhd  $U_i$  of  $e \in G$ , choose  $x' \in xU_i \cap H$ , then  $yx' = yx(x^{-1}x') \in yxU_i$  and also  $yx' \in H$ ,  $yx' \in Vx^{-1}xV \subset U$ . By arbitrariness of  $U_i$ , this means  $yx \in \overline{U \cap H} = K \subset H$ , thus  $x \in H$ , and  $H$  is closed. □



### Integration on Locally Compact Groups

**Def. (12.10.1.7) [Positive Linear Functional].** A **positive linear functional** on  $X$  is a linear functional  $I$  on  $C_c(X)$  that  $I(f) \geq 0$  whenever  $f \geq 0$ . How is this definition compatible with that of (14.1.1.10)?  $\lrcorner$

**Lemma (12.10.1.8) [Positive Linear Functional is Continuous].** A positive linear functional (12.10.1.7)  $I$  on  $C_c(X)$  is automatically continuous, where  $C_c(X)$  is given compact convergence topology as in (12.8.2.1).  $\lrcorner$

*Proof:* We need to prove that for any compact subset  $K$  of  $X$ , there is a constant  $C_K$  that for any  $f \in C(G)$  with support in  $K$ , we have  $|I(f)| \leq C_K \|f\|_\infty$ .

Given any  $K$ , choose by Urysohn lemma a  $\varphi \in C_c(X, [0, 1])$  that  $\varphi = 1$  on  $K$ , so if  $\text{Supp } f \subset K$ , then  $|f| \leq \|f\|_\infty \varphi$ , thus the positivity of  $I$  shows that  $|I(f)| \leq I(\varphi) \|f\|_\infty$ .  $\square$

**Prop. (12.10.1.9) [Riesz-Markov-Kakutani Representation Theorem].**

- If  $I$  is a positive linear functional (12.10.1.7) on  $C_c(X)$ , there is a unique Radon measure  $\mu$  on  $X$  such that  $I(f) = \int f d\mu$ . Moreover,

$$\mu(U) = \sup\{I(f) : f < U\}, \text{ where } U \subset X \text{ open,}$$

$$\mu(K) = \inf\{I(f) : f > \chi_K\}, \text{ where } K \subset X \text{ compact.}$$

- If  $I$  is a continuous linear functional on  $C_0(X)$ , there is a unique regular complex Borel measure  $\mu$  on  $X$  that  $I(f) = \int f d\mu$ .

In particular the space  $\text{Meas}(X)$  of Radon measures on  $X$  is the “positive part” of the the dual space of  $C_c(X)$ .  $\lrcorner$

*Proof:* Cf. [Real Analysis Folland P212].  $\square$

**Thm. (12.10.1.10) [Haar Measure].** A left (resp. right) **Haar measure** on an arbitrary  $G \in \text{TopGrp}$  is a non-zero Radon measure (12.3.1.22)  $\mu$  on  $G$  that satisfies  $\mu(xE) = \mu(E)$  (resp.  $\mu(Ex) = \mu(E)$ ). A Radon measure  $\mu$  is a Haar measure iff it satisfies  $\int L_y f d\mu = \int f d\mu$  for any  $f \in C_c^+(G)$  and  $y \in G$  (12.10.1.2).  $\lrcorner$

Then  $G$  possesses a left Haar measure  $\mu_G$  if  $G \in \text{LCG}$ , unique up to a scalar.  $\lrcorner$

*Proof:* If  $\mu$  is a Haar measure, then  $\int L_y f d\mu = \int f d\mu$  by approximation by simple functions (12.3.8.5). Conversely, if  $\int L_y f d\mu = \int f d\mu$  for any  $f \in C_c(G)$ , then it holds for all  $f \in C_c(G)$ , hence  $\mu(xE) = \mu(E)$  by Riesz-Markov-Kakutani representation theorem (12.10.1.9).

For  $f, \varphi \in C_c^+(G)$ , define  $(f : \varphi)$  to be the infimum of all finite sums  $\sum_{i=1}^n c_i$  that  $f \leq \sum_{i=1}^n c_i L_{x_i} \varphi$  for some  $x_1, \dots, x_n \in G$ . This makes sense because  $f$  has finite support so can be covered by f.m. translation of any open set. This quantity satisfies the following properties:

- $(f : \varphi) = (L_y f : \varphi)$  for any  $y \in G$ .
- $(f_1 + f_2 : \varphi) \leq (f_1 : \varphi) + (f_2 : \varphi)$ .
- $(cf : \varphi) = c(f : \varphi)$ .
- $(f_1 : \varphi) \leq (f_2 : \varphi)$  for  $f_1 \leq f_2$ .
- $(f : \varphi) \geq \|f\|_{\text{sup}} / \|\varphi\|_{\text{sup}}$ .
- $(f : \varphi) \leq (f : \psi)(\psi : \varphi)$  for any  $\psi \in C^+(G)$ .

Now choose a  $f_0 \in C_c^+(G)$  and define  $I_\varphi(f) = \frac{(f:\varphi)}{(f_0:\varphi)}$  for  $f, \varphi \in C_c^+(G)$ . Then  $I_\varphi$  is left-invariant, sub-additive, homogeneous of degree 1, and monotone. Moreover, it satisfies  $(f_0 : f)^{-1} \leq I_\varphi(f) \leq (f : f_0)$ .

Let  $X_f$  be the interval  $[(f_0 : f)^{-1}, (f : f_0)]$ , and let  $X = \prod_{f \in C_c^+(G)} X_f$ , then for each nbhd  $V$  of  $1 \in G$ , let  $K(V)$  be the closure in  $X$  of  $\{I_\varphi | \text{Supp}(\varphi) \subset V\}$ , then these sets satisfy finite intersection property. So by compactness, there is an  $I$  contained in every  $K(V)$ . Which means for any nbhd  $V$  of  $1 \in G$  and  $\varepsilon > 0$  and  $f_1, \dots, f_n \in C_c^+(G)$ , there exists  $\varphi \in C_c^+(V)$  that  $|I(f_i) - I_\varphi(f_i)| < \varepsilon$ . Then by some argument,  $I$  commutes with left translation, addition and multiplication by positive scalars. Now extend  $I$  to a positive linear functional on  $C_c(G)$ , and then use Riesz-Markov-Kakutani representation theorem(12.10.1.9) to finish.  $\square$

**Lemma (12.10.1.11).** For any  $f_1, f_2 \in C_c^+(G)$  and  $\varepsilon > 0$ , there is a neighborhood  $V$  of  $1 \in G$  that in the notation in the proof of(12.10.1.10),  $I_\varphi(f_1) + I_\varphi(f_2) \leq I_\varphi(f_1 + f_2) + \varepsilon$  whenever  $\text{Supp}(\varphi) \subset V$ .  $\lrcorner$

*Proof:* Let  $g \in C_c^+(G)$  that  $g = 1$  on  $\text{Supp}(f_1 + f_2)$  and  $\delta > 0$ , let  $h_i = f_i/(f_1 + f_2 + \delta g)$ , then  $h_i \in C_c^+(G)$  and there is a nbhd  $V$  of  $1 \in G$  s.t.  $|h_i(x) - h_i(y)| < \delta$  for  $i = 1, 2$  and  $y^{-1}x \in V$ . Take  $\varphi \in C_c^+(G)$  with  $\text{Supp}(\varphi) \subset V$ . If  $h \leq \sum c_i L_{x_i} \varphi$ , then

$$f_i(x) = h(x)h_i(x) = \sum c_j \varphi(x_j^{-1}x)h_i(x) \leq \sum c_j \varphi(x_j^{-1}x)[h_i(x_j) + \delta]$$

because whenever  $\varphi(x_j^{-1}x) \neq 0$ ,  $|h_i(x) - h_i(x_j)| < \delta$ . As  $h_1 + h_2 < 1$ ,

$$(f_1 : \varphi) + (f_2 : \varphi) \leq \sum c_j [h_1(x_j) + \delta] + \sum c_j [h_2(x_j) + \delta] \leq \sum c_j [1 + 2\delta]$$

which implies

$$I_\varphi(f_1) + I_\varphi(f_2) \leq (1 + 2\delta)I_\varphi(h) \leq (1 + 2\delta)[I_\varphi(f_1 + f_2) + \delta I_\varphi(g)].$$

Notice  $\delta$  is arbitrary, thus we can choose  $\delta$  small enough that the assertion is true.  $\square$

**Prop. (12.10.1.12).** If  $\mu$  is a left Haar measure on  $G$ , then for any open subset  $U$  of  $G$ ,  $\mu(U) > 0$ .  $\lrcorner$

*Proof:* By inner regularity,  $\mu(K) > 0$  for some compact subset  $K$ . Suppose  $\mu(U) = 0$  for an open subset  $U$ , then f.m. translates of  $U$  covers  $K$ , contradiction.  $\square$

**Cor. (12.10.1.13).** Let  $\mu$  be a left Haar measure on  $G$ , then

- $G$  is discrete iff  $\mu(\{e\}) > 0$ .
- $G$  is compact iff  $\mu(G) < \infty$ .

$\lrcorner$

**Prop. (12.10.1.14).** Integration of a nontrivial character of  $G$  w.r.t. the Haar measure  $\mu_G$  is 0.  $\lrcorner$

*Proof:*  $\int f(x)d\mu(x) = \int f(yx)d\mu(yx) = f(y) \int f(x)d\mu(x)$ . Now choose a  $y$  that  $f(y) \neq 1$ .  $\square$

**Def. (12.10.1.15) [Modular Quasi-Characters].** For a left Haar measure  $\mu$  on  $G$ ,  $\mu_x(E) = \mu(Ex)$  is also a left Haar measure, so there is a  $\Delta_G(x) \in \mathbb{R}_+$  s.t.  $\mu_x = \Delta_G(x)\mu$ . Then the function  $\Delta_G$  is a group homomorphism from  $G$  to  $\mathbb{R}^+$ , which is called the **modular quasi-character** of  $G$ .

$G$  is called **unimodular** iff  $\Delta_G = 1$ , i.e. a left Haar measure is also a right Haar measure. Obviously, a locally compact Abelian group is unimodular.

Sometimes we use right Haar measures, so we denote  $\delta_G(g) = \Delta(g)^{-1}$ .  $\lrcorner$

**Prop. (12.10.1.16).**  $\Delta$  is a continuous group homomorphism from  $G$  to  $\mathbb{R}_+$ , and

$$\int R_y f d\mu = \Delta(y^{-1}) \int f d\mu.$$

equivalently,  $d\mu(xy_0) = \Delta(y_0)d\mu(x)$ .  $\lrcorner$

*Proof:* For the continuity of  $\Delta$ , because  $y \mapsto R_y(f)$  is continuous for each  $f$  (12.10.1.2), so  $y \mapsto \int R_y f d\lambda$  is continuous, as  $\mu$  is Radon measure, so by the equation just proved,  $\Delta$  is continuous.

Now for any measurable function  $E$ ,  $\chi_E(xy) = \chi_{Ey^{-1}}(x)$ , thus

$$\int \chi_E(xy) d\mu = \mu(Ey^{-1}) = \Delta(y^{-1})\mu(E) = \Delta(y^{-1}) \int \chi_E(x) d\mu(x),$$

which proves the equation for  $f = \chi_E$ . Then the general case follows from approximating  $f$  by simple functions (12.3.8.6).  $\square$

**Prop. (12.10.1.17)[Involution Measure].** If  $\mu$  is a left Haar measure on  $G$  and  $\rho$  is defined by  $\rho(E) = \mu(E^{-1})$ , then  $\rho$  is a right Haar measure on  $G$ , and  $d\rho(x) = d\mu(x^{-1}) = \Delta(x^{-1})d\mu(x)$ .  $\lrcorner$

*Proof:* Notice

$$\int R_y(f)(x) \Delta(x^{-1}) d\mu(x) = \Delta(y) \int f(xy) \Delta((xy)^{-1}) d\mu(x) = \int f(x) \Delta(x^{-1}) d\mu(x),$$

so  $\Delta(x^{-1})d\mu(x)$  is a right Haar measure, hence  $c d\mu(x^{-1})$  for some  $c$ . If  $c \neq 1$ , we let  $U$  be a precompact symmetric nbhd  $U$  of 1 that  $|\Delta(x^{-1}) - 1| \leq \frac{1}{2}|c - 1|$  on  $U$ . But then  $|c - 1|\mu(U) = |\int_U (\Delta(x^{-1}) - 1) d\mu(x)| \leq \frac{1}{2}|c - 1|\mu(U)$ , contradiction.  $\square$

**Prop. (12.10.1.18) [Unimodular Groups].** For a compact group  $K$  of  $G$ ,  $\Delta$  is trivial on  $K$ . So compact group is unimodular, and if  $G/[G, G]$  is compact, then it is also unimodular.

A discrete group is unimodular.  $\lrcorner$

*Proof:* These all follow from (12.10.1.16) and the fact that a compact subgroup of  $\mathbb{R}^+$  is  $\{1\}$ , and  $\mathbb{R}$  is Abelian.

A discrete group is unimodular because  $\mu(\{e\}) = \mu(\{x\}) = \Delta(x)\mu(\{e\}) > 0$  (12.10.1.13).  $\square$

**Prop. (12.10.1.19) [Analytic Variety case].** Suppose  $G$  is an open subset of  $K^N$  where  $K$  is a local field, and the left translation is given by

$$xy = A(x)y + b(x)$$

then the Haar measure of  $G$  is given by  $|\det A(x)|^{-1} dx$ , where  $dx$  is the Haar measure on  $K^N$ .

Also if we want to calculate the right Haar measure, consider the right action.

More generally, if  $G$  is an  $n$ -dimensional analytic variety over  $K$  and  $\omega$  is a non-zero left-invariant differential form on  $G$  (which is in bijection with  $\wedge^n \mathfrak{g} \setminus \{0\}$ ), then we get a left Haar measure  $|\omega|$  on  $G$  through the positive linear functional

$$f \mapsto \int_{G(F_v)} f(g) |\omega|_v(g) \text{ where } f \in C_c(G(F_v))$$

by Riesz representation theorem (12.10.1.9). It is defined locally as follows: If on a coordinate map near a point  $x_0 \in G$ ,  $\omega(x) = f(x) dx_1 \wedge \dots \wedge dx_n$ , then  $|\omega|(x) = |f(x)|_v \cdot |dx_1| \times \dots \times |dx_n|$ . (Cf. [Kna02] Section 8.2.)  $\lrcorner$

*Proof:* Use change of variable formula, because  $A(xy) = A(x)A(y)$ , and

$$|\det A(ax)|^{-1}d(ax) = |\det A(ax)|^{-1}d(A(a)x + b(x)) = |\det A(x)|^{-1}dx.$$

□

**Cor. (12.10.1.20) [Examples of Analytic Variety Measures].**

- $dx/|x|$  is the Haar measure on  $\mathbb{R}^*$ .
- $dx dy/(x^2 + y^2)$  is the Haar measure on  $\mathbb{C}^\times$ .
- $x_{11}x_{22}^2 \dots x_{nn}^n \prod_{i < j} dx_{ij}$  (resp.  $x_{11}^n x_{22}^{n-1} \dots x_{nn} \prod_{i < j} dx_{ij}$ ) are the left (resp. right) Haar measure on the group  $B(n; \mathbb{R})$  of upper-triangular matrixes in  $GL(n; \mathbb{R})$ .
- $\prod_{i < j} dx_{ij}$  is the left and right Haar measure on  $Unip(n; \mathbb{R})$ .
- $|\det T|^{-n} dT$  is the left and right Haar measure on the group  $GL(n; \mathbb{R})$ , where  $dT$  is the Lebesgue measure on  $Mat(n; \mathbb{R}) = \mathbb{R}^{n^2}$ .
- The  $ax + b$  group  $G$  of all affine (invertible) translations of  $\mathbb{R}$  has left measure  $dadb/a^2$  and right Haar measure  $dadb/a$ .

┘

*Proof:* Clear.

□

**Prop. (12.10.1.21) [Modular Quasi-Characters of Analytic Varieties].** If  $K \in \mathbf{LField}$ ,  $G/K$  is an analytic group and  $\text{Ad}$  is the adjoint action of  $G$  on  $\mathfrak{g}$ , then  $\Delta_G(x) = |\det \text{Ad}(x^{-1})|$  (12.10.1.15). ┘

*Proof:* Let  $n = \dim G$ , then the Haar measure on  $G$  is given by the absolute value of a left-invariant  $m$ -form  $\omega$  by (12.10.1.19). Now for any  $X \in \mathfrak{g}$  corresponding to a left invariant vector space  $L_X$ ,

$$d(R_g)_p((L_X)_p) = (L_{\text{Ad}(g^{-1})X})_{pg}$$

by (16.5.3.5), so  $R_g^* \omega = \det(\text{Ad}(g^{-1})) \omega$ . So  $\Delta(g) \omega = R_g^* |\omega| = |\det(\text{Ad}(g^{-1}))| |\omega|$ .

□

**Cor. (12.10.1.22) [Unimodular Algebraic Groups].** For  $K \in \mathbf{LField}$  and  $G \in \mathbf{AlgGrp}/K$ , then  $G(K)$  is unimodular if one of the following holds:

- $G$  is Abelian,
- $G(K)$  is compact.
- $G^0 \in \mathbf{RedGrp}/K$ .
- $G$  is nilpotent.

┘

*Proof:* The Abelian case is trivial.

The compact case follows from (12.10.1.18).

For the reductive case, it suffices to show that  $\Delta(G^0) = 1$ , because then  $\Delta$  factors through  $G/G^0$ , which is a finite group, thus trivial. Now suppose  $G$  is connected, we prove that in fact  $\det \text{Ad}(G) = 1$ .  $\det \text{Ad}$  is trivial on  $[G, G]$  and also on the center  $Z(G)$ . Thus by (10.4.5.21), it is trivial on  $G$ .

The nilpotent case follows directly from (12.10.1.21), as  $\det \text{Ad}(x) = \exp \text{tr}(\text{ad}(x))$  and  $\text{ad}(x)$  is nilpotent.

□

## 2 Convolutions

**Def. (12.10.2.1) [Convolution of Measures].** If  $\mu, \nu$  are two complex (hence finite) Radon measures on  $G$ , the map

$$I(\varphi) = \int \int \varphi(xy) d\mu(x) d\nu(y)$$

is clearly a linear functional on  $C_c(G)$  that satisfies  $|I(\varphi)| \leq \|\varphi\|_\infty \|\mu\| \|\nu\|$ , so it defines a measure on  $G$  by Riesz representation (12.10.1.9), called the **convolution** of  $\mu$  and  $\nu$ , denoted by  $\mu * \nu$ , that  $\|\mu * \nu\| \leq \|\mu\| \|\nu\|$ .  $\square$

**Prop. (12.10.2.2) [Measure Algebra].**

- The convolution of measure is associative.
- $\delta_x * \delta_y = \delta_{xy}$ .
- The convolution of measure is commutative iff  $G$  is commutative.
- The convolution makes the set of complex measures on  $G$  into a unital Banach algebra, called the **measure algebra** of  $G$ , denoted by  $\text{Meas}(G)$ .

$\square$

*Proof:* 1: If  $\varphi \in C_c(G)$ , then

$$\begin{aligned} \int_G \varphi d[\mu * (\nu * \sigma)] &= \int \int \varphi(xy) d\mu(x) d(\nu * \sigma)(y) \\ &= \int \int \int \varphi(xyz) d\mu(x) d\nu(y) d\sigma(z) \\ &= \int \int \varphi(yz) d(\mu * \nu)(y) d\sigma(z) \\ &= \int \varphi d[(\mu * \nu) * \sigma] \end{aligned}$$

by Fubini theorem, which shows  $\mu * (\nu * \sigma) = (\mu * \nu) * \sigma$ .

2:

$$\int \int \varphi d(\delta_x * \delta_y) = \int \int \varphi(uv) d\delta_x(u) d\delta_y(v) = \varphi(xy) = \int \varphi d\delta_{xy}.$$

3: If  $G$  is commutative, then  $\varphi(xy) = \varphi(yx)$ , then the commutativity follows from Fubini theorem. The converse follows from item 2.

4 It is a Banach algebra because  $\|\mu * \nu\| \leq \|\mu\| \|\nu\|$  (12.10.2.1). And the point measure  $\delta_1$  is the unit:

$$\int \varphi d(\delta * \mu) = \int \int \varphi(xy) d\delta(x) d\mu(y) = \int \varphi(y) d\mu(y)$$

shows  $\delta * \mu = \mu$  for any  $\mu$ , and similarly  $\mu * \delta = \mu$ , so  $\delta$  is the identity.  $\square$

**Prop. (12.10.2.3) [Involution of Measure].**  $\text{Meas}(G)$  has a canonical involution that preserves total measure:

$$\mu \mapsto \mu^* : \mu^*(E) = \overline{\mu(E^{-1})}.$$

$\square$

*Proof:*  $\mu^*$  clearly satisfies  $\|\mu^*\| = \mu^*(G) = \mu(G) = \|\mu\|$ . And for any  $\varphi \in C_c(G)$ ,

$$\varphi d(\mu * \nu)^* = \int \varphi(x^{-1}) d(\overline{\mu * \nu})(x) = \int \varphi((xy)^{-1}) d\overline{\mu}(x) d\overline{\nu}(y) = \int \int \varphi(yx) d\mu^*(x) d\nu^*(y) = \int \varphi d(\nu^* * \mu^*),$$

which shows  $(\mu * \nu)^* = \nu^* * \mu^*$ .  $\square$

**Def. (12.10.2.4)** [ $L^1(G)$  and  $\text{Meas}(G)$ ]. Fix a Haar measure  $d\mu$  on  $G$ ,  $L^1(G)$  embeds into  $\text{Meas}(G)$  (12.10.2.2) by identifying  $f$  with the measure  $f(x)d\mu(x)$ , and this is in fact an isometry.

So the convolution and involution can be defined on  $L^1(G)$ , and the outcome turns out to be a.e. defined and in  $L^1(G)$  too:

$$f * g(x) = \int f(y)g(y^{-1}x)d\mu(y).$$

by Fubini-Tonelli theorem, and the involution

$$f^*(x) = \Delta(x^{-1})\overline{f(x^{-1})}.$$

Equivalently,

$$f * g = \int f(y)L_y(g)d\mu(y) = \int g(y^{-1})R_y(f)d\mu(y) \quad (12.8.3.25)$$

as vector-valued integrations.  $\lrcorner$

*Proof:*

$$\begin{aligned} (f * g)(\varphi) &= \int \int \varphi(xy)f(x)g(y)dx dy = \int \int \varphi(y)f(x)g(x^{-1}y)dx dy = \int \varphi(y) \left( \int f(x)g(x^{-1}y)dx \right) dy. \\ (f d\mu)^* &= \overline{(f d\mu)(x^{-1})} = \overline{f(x^{-1})}\Delta(x^{-1})d\mu(x) \quad (12.10.1.17). \end{aligned}$$

For the last assertion, Cf. [Fol16]P287.  $\color{red}?$   $\square$

**Prop. (12.10.2.5).** The convolution  $f * g$  can be calculated in multiple ways by left invariance and (12.10.1.17):

$$f * g(x) = \int f(y)g(y^{-1}x)dy = \int f(xy)g(y^{-1})dy = \int f(y^{-1})g(yx)\Delta(y^{-1})dy = \int f(xy^{-1})g(y)\Delta(y^{-1})dy.$$

In particular, if  $G$  is unimodular, then it can be calculated anyway you want.  $\lrcorner$

**Prop. (12.10.2.6).** For  $1 \leq p < \infty$ , the left and right translations of  $G$  on  $L^p(G)$  are all continuous.  $\lrcorner$

*Proof:* Cf. [Fol15]P58.  $\square$

**Prop. (12.10.2.7)** [ $L^p$ -Estimate]. The convolution on  $L^1(G)$  defined in (12.10.2.4) can be extend to  $L^p$ :

If  $1 \leq p < \infty$ ,  $f \in L^1(G)$  and  $g \in L^p(G)$ , then

- $f * g \in L^p(G)$ , and  $\|f * g\|_p \leq \|f\|_1 \|g\|_p$ .
- If  $G$  is unimodular or  $f$  has compact support, then the same as above holds for  $g * f$ .

More generally, for  $\mu \in \text{Meas}(G)$  and  $f \in L^p(G)$ , we can define the convolution  $\mu * f$  as

$$\mu * f(x) = \int f(y^{-1}x)d\mu(y).$$

Then  $\|\mu * f\|_p \leq \|\mu\| \|f\|_p$ . And when  $G$  is unimodular of  $p = 1$ , we can define  $f * \mu$  similarly. In particular,  $L^1(G)$  is a two-sided ideal in  $\text{Meas}(G)$ .  $\lrcorner$

*Proof:* 1: By Minkowski's inequality (12.3.6.6),

$$\|f * g\|_p = \left\| \int f(y)L_y g dy \right\|_p \leq \int |f(y)| \|L_y g\|_p dy = \|f\|_1 \|g\|_p.$$

2: This is similar, using (12.10.2.5).  $\square$

**Prop. (12.10.2.8) [Young's Inequality for Convolutions].** The convolution in (12.10.2.7) can be further extended:

- Suppose  $G$  is unimodular and  $f \in L^p(G), g \in L^q(G)$  with  $1/p + 1/q = 1, 1 < p < \infty$ , then  $f * g \in C_0(G)$ , and  $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$ .
- Suppose  $f \in L^1(G), g \in L^\infty(G)$ , then  $f * g$  is left uniformly continuous, and  $g * f$  is right uniformly continuous.

More generally, by Riesz-Thorin interpolation theorem<sup>?</sup>, if  $G$  is unimodular and  $f \in L^p(G), g \in L^q(G)$  with  $1/p + 1/q = 1/r + 1, 1 < p, q, r < \infty$ , then  $f * g \in C_0(G)$ , and  $\|f * g\|_r \leq \|f\|_p \|g\|_q$ .  $\lrcorner$

*Proof:* Cf. [Fo15]P57, P58.<sup>?</sup>

By Riesz representation (12.10.1.9), it suffices to show that: for  $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 2$ ,

$$\int \int f(x)g(y-x)h(y) \leq \|f\|_p \|g\|_q \|h\|_r.$$

write the LHS as

$$\int \int (f^p(x)g(y-x)^q)^{1-\frac{1}{r}} (f^p(x)h^r(y))^{1-\frac{1}{q}} (g^q(y-x)h^r(y))^{1-\frac{1}{p}}$$

and use Holder inequality for three functions (12.3.6.2).  $\square$

### Dirac Sequences

**Prop. (12.10.2.9) [Approximate Identities].** Let  $\mathcal{U}$  be a neighborhood base of  $1 \in G$ . A family of functions  $\{\varphi_U\}_{U \in \mathcal{U}} \in C_c(G)$  are called an **approximate identity** if:

1.  $\text{Supp}(\Phi_U)$  is compact and contained in  $\subset U$ .
2.  $\Phi_U(x) \geq 0, \int_0^1 \Phi_U(x) dx = 1$ .
3.  $\Phi_U(x^{-1}) = \Phi_U(x)$ .

Approximation identities exist.  $\lrcorner$

*Proof:* To show existence, we can take the neighbourhood bases of  $1 \in G$  to be symmetric precompact and take  $\Phi_U(x) = \mu(U)^{-1} \mathbf{1}_U$ .  $\square$

**Prop. (12.10.2.10).** For any approximate identity, if  $1 \leq p < \infty$  and  $f \in L^p(G)$ , or  $p = \infty$  and  $f$  is left (resp. right) uniformly continuous, then  $\Phi_U * f \rightarrow f \in L^p(G)$  (resp.  $f * \Phi_U \rightarrow f \in L^p(G)$ ).  $\lrcorner$

*Proof:* Cf. [Fo15]P58.<sup>?</sup>  $\square$

**Prop. (12.10.2.11) [Dirac sequences].** A **Dirac sequence** on  $G$  is a sequence  $f_n \in C_c(G)$  that  $f_n \mu_{\text{Haar}} \rightarrow \delta_1$  in the weak topology of  $\text{Meas}_c(G)$ .

If  $G$  is second-countable, then any approximate identity (12.10.2.9) gives a Dirac sequence by (12.10.2.10).  $\lrcorner$

**Cor. (12.10.2.12).**  $C_c(G)$  is dense in  $\text{Meas}_c(G)$ .  $\lrcorner$

*Proof:* By translations of Dirac sequences, we can show that  $C_c(G)$  is dense in the space  $\text{Meas}_c^0(G)$  consisting of point measures. Then notice also that  $\text{Meas}_c^0(G)$  is dense in  $\text{Meas}_c(G)$ , because otherwise by Hahn-Banach, there is a linear functional on  $\text{Meas}_c(G)$  vanishing on  $\text{Meas}_c^0(G)$ . Clearly any linear functional on  $\text{Meas}_c^0(G)$  is evaluating on a function  $f \in C_c(G)$ . But then clearly  $f = 0$ , contradiction.  $\square$

**Prop. (12.10.2.13).** A closed subspace  $I \subset L^1(G)$  is a left ideal(resp. right ideal) iff it is invariant under left translations(resp. right translations).  $\lrcorner$

*Proof:* Suppose  $I$  is a left ideal, then for any  $f \in I$  and  $g \in G$ , let  $\{\Phi_U\}$  be an approximate identity, then  $L_x(\Phi_U * f) = L_x(\Phi_U) * f \in I$ , thus  $L_x(f) = \lim L_x(\Phi_U * f) \in I$  by (12.10.2.10). Conversely, suppose  $I$  is invariant under left translations, then for  $f \in I$  and  $g \in L^1(G)$ ,  $g * f = \int g(y)L_y(f)d\mu(y)$  (12.10.2.4) is in the closed linear span of the functions  $L_y(f)$  by (12.10.2.4) and (12.8.3.20), thus in  $I$ .  $\square$

### 3 Homogenous Spaces

**Def. (12.10.3.1)[Notations].** If  $\mu$  is a left Haar measure on  $G$  and  $H \leq G$  is a closed subgroup with a left Haar measure  $d\mu(\xi)$ , let  $q : G \rightarrow G/H$  be the quotient map.  $\lrcorner$

**Prop. (12.10.3.2).** Let  $G$  is  $\sigma$ -compact and  $S \in \text{LCH}$  is a transitive  $G$ -space. If  $s_0 \in S$  and  $\text{Stab}_G(s_0) = H$ , then  $G/H \cong S$  as  $G$ -spaces.  $\lrcorner$

*Proof:* Cf.[Folland P60].  $\square$

**Lemma (12.10.3.3).** If  $E \subset G/H$  is compact, then there is a compact  $K \subset G$  that  $q(K) = E$ .  $\lrcorner$

*Proof:* Choose a precompact nbhd  $V$  of 1 in  $G$ , since  $q$  is open, the set  $q(xV)$  is an open cover of  $E$ , so there are f.m.  $x_i$  that  $E \subset \cup q(x_iV)$ . Then let  $K = q^{-1}(E) \cap (\cup x_i\bar{V})$ , this will suffice.  $\square$

**Def. (12.10.3.4)[Fundamental Domain].** A **fundamental domain** for a group  $\Gamma$  acting discontinuously on a locally compact second countable Hausdorff space  $X$  is an Borel subset  $F \in X$  that:

- $\cup_{\gamma \in \Gamma} \gamma F = \mathcal{H}$ .
- if  $\gamma \neq 1 \in \Gamma$ , then  $\gamma F \cap F = \emptyset$ .

Then fundamental domains exist.  $\lrcorner$

*Proof:* The quotient space  $G \backslash \Gamma$  is locally compact second countable, thus there is a countable set of precompact open basis  $\{B_i\}$  for  $G \backslash \Gamma$ , and a countable set of precompact open basis  $\{C_i\}$  for  $X$ . For each  $\bar{x} \in G \backslash \Gamma$ , choose  $\bar{x} \in B_{i(\bar{x})}$  and choose a preimage  $x \in X$  and a nbhd  $C_{i(x)}$  that  $C_{i(x)} \cap \gamma(C_{i(x)}) = \emptyset$  for any  $\gamma \neq 1$ , then there is a nbhd  $B_{j(x)}$  contained in the image of  $C_{i(x)}$ . Then we can take a precompact preimages of  $B_{j(\bar{x})}$  in  $C_{i(x)}$  (by (12.10.3.3)), labeled by  $U_i$ . Then  $U_i$  maps isomorphically to their images in  $G \backslash \Gamma$  and covers  $G \backslash \Gamma$ .

Now take  $V_1 = W_1 = U_1$ , and  $W_{n+1} = U_{n+1} \setminus \cup_{\gamma \in \Gamma} \gamma(V_n)$ ,  $V_{n+1} = V_n \cup W_{n+1}$ . Then  $\cup V_i$  is a fundamental domain.  $\square$

**Prop. (12.10.3.5)[Projection of Functions].** There is a map  $P : C_c(G) \rightarrow C_c(G/H) : Pf(xH) = \int_H f(x\xi)d\mu(\xi)$ .  $Pf$  is continuous and this map is well-defined and surjective.

And  $\text{Supp}(Pf) \subset q(\text{Supp } f)$ , and if  $\varphi \in C_c(G/H)$ , then  $P((\varphi \circ q)f) = \varphi Pf$ .  $\lrcorner$

*Proof:*  $Pf$  is continuous because  $f$  is left uniformly continuous (12.10.1.2). By (12.10.3.6), there is a  $g \geq 0 \in C_c(G)$  that  $Pg = 1$  on  $\text{Supp } \varphi$ . Let  $f = (\varphi \circ f)g$ , then  $Pf = \varphi(Pg) = \varphi$ .  $\square$

**Lemma (12.10.3.6).** If  $F \in G/H$  is compact, then there is a  $f \in C_c(G)$  that  $f \geq 0$  and  $Pf = 1$  on  $F$ .  $\lrcorner$



*Proof:* Let  $E$  be a precompact nbhd of  $F$  in  $G/H$ , choose a compact  $K \subset G$  that  $q(K) = \overline{E}$  by (12.10.3.3). Choose  $g \in C_c(G) \geq 0$  that is positive on  $K$  and  $\varphi \in C_c(G/H)$  that is 1 on  $F$  and vanish outside  $E$ , then set

$$f = \frac{\varphi \circ q}{Pg \circ q}g$$

then  $f \geq 0$  and  $Pf = (\varphi/Pg)Pg = \varphi$ . □

**Prop. (12.10.3.7) [Quotient Measures in Regular cases].** If  $H \leq G$  is a closed subgroup and  $d\mu(g), d\mu(h)$  are Haar measures on  $G$  and  $H$  resp., then there is a  $G$ -invariant positive Radon measure  $\mu$  on  $G/H$  iff  $\Delta_G|_H = \Delta_H$ . If this is the case, then this measure is unique up to constant, and when suitably chose, it satisfies:

$$\int_G f(x)dx = \int_{G/H} Pf d\mu(xH) = \int_{G/H} \left( \int_H f(x\xi) d\xi \right) d\mu(xH) \quad (12.10.3.7).$$

for any  $f \in C_c(G)$ . This measure is denoted by  $\frac{d\mu(g)}{d\mu(h)}$ . ┘

*Proof:* Cf. [Fol16]P62? □

**Cor. (12.10.3.8) [Product Decomposition of Measure].** If  $G$  is a unimodular and  $P, K$  are closed subgroups s.t.  $P \cap K$  is compact and  $G = PK$ . Let  $\mu_P, \rho_K$  be the left and right Haar measure on  $P, K$  respectively, then a Haar measure on  $G$  can be given s.t.

$$\int_G f(g) d\mu_G(g) = \int_K \int_P f(pk) d\mu_P(p) d\rho_K(k), \text{ where } f \in C_c(G).$$

┘

*Proof:* Consider  $H = P \times K$  and  $M = P \cap K$  embedded diagonally in  $H$ , then there is a homeomorphism  $H/M \cong G$  given by  $(p, k) \mapsto pk^{-1}$ . Then we can verify both side are  $H$ -invariant measure on  $G \cong H/M$  (compactness of  $P \cap K$  is used to guarantee that the RHS is integrable?). Thus by uniqueness in (12.10.3.7), the equation is true. □

**Prop. (12.10.3.9) [Triple Product Decomposition of Measures].** If  $G$  is unimodular and  $H, A, U \leq G$  be closed subgroups s.t.

- The product morphism  $H \times A \times U \rightarrow G$  is a homeomorphism.
- $A$  normalizes  $U$ .
- $A, H$  are unimodular.

Then there is a Haar measure on  $G$  s.t.  $d\mu_G(hau) = \rho(a) d\mu_H(h) d\mu_A(a) d\mu_U(u)$ . ┘

*Proof:* ? □

**Def. (12.10.3.10) [Fundamental Domains].** ┘

### Non-regular Quotients

**Def. (12.10.3.11) [Rho-Functions].** Let  $H \leq G$  be a closed subgroup and  $\Delta = \Delta_G/\Delta_H$ . Let  $\mathcal{S}(G, \Delta)$  be the space of continuous functions on  $G$  that satisfies

- for any  $h \in H$ ,  $f(hg) = \Delta(h)f(g)$ .

- $f$  is compactly supported in  $H \backslash G$ .

┘

**Lemma (12.10.3.12).** Let  $H \leq G$  a closed subgroup, then there is a continuous function  $f_0 : G \rightarrow [0, \infty)$  that

- $f_0^{-1}((0, \infty)) \cap Hx \neq \emptyset$  for any  $x \in G$ .
- $\text{Supp } f_0 \cap HK$  is compact for any compact group  $K$  of  $G$ .

┘

*Proof:* Cf.[Folland, P64].

□

**Lemma (12.10.3.13).** If  $f \in C_c(G)$  and  $Pf = 0$ , then  $\int f\rho = 0$  for any rho-function  $\rho$ . In fact, this is true if  $\rho$  is allowed to take value 0.

┘

*Proof:* Cf.[Folland P65].

□

**Lemma (12.10.3.14).** There is an operator  $p : C_c(G) \rightarrow C_c(G/H)$  given by

$$p(f)(g) = \int_H f(hg) \Delta_G^{-1}(h) d\mu_H(h).$$

Then  $p$  is right  $G$ -invariant, and  $p(ff') = fp(f')$  for any  $f' \in C_c(G)$  and  $f \in C(G)$  that is  $H$ -invariant.

- The image of  $p$  is  $\mathcal{S}(G, \Delta)$ , and if  $s \geq 0 \in \mathcal{S}(G, \Delta)$ , then there is a non-negative  $f \geq 0 \in C_c(G)$  that  $p(f) = s$ .
- If  $p(f) = 0$ , then  $\int_G f(x) d\nu_G(x) = 0$ , where  $d\nu_G$  is a right Haar measure on  $G$ .

┘

*Proof:* 1: It is clearly that  $p(f)(gh) = \Delta(h)p(f)(g)$ , and let  $f_0$  be defined as in (12.10.3.12), then  $s_0 = p(f_0)$  is positive-valued. Now for any  $s \in \mathcal{S}(G, \Delta)$ ,  $p(ss_0^{-1}f_0) = ss_0^{-1}p(f_0) = s$ , and  $sf_0 \in C_c(G)$  by hypothesis.

2: If  $p(f) = 0$ , then

$$\begin{aligned} \int_G \int_H s_0^{-1}(g) f_0(g) f(hg) \Delta_G^{-1}(h) d\mu_H(h) d\nu_G(g) &= \int_G \int_H \Delta(h) s_0^{-1}(g) f_0(h^{-1}g) f(g) d\mu_H(h) d\nu_G(g) \\ &= \int_G \left[ \int_H f_0(hg) \Delta_G(h)^{-1} d\mu_H(h) \right] s_0^{-1}(g) f(g) d\nu_G(g) \\ &= \int_G f(g) d\nu_G(g) \end{aligned}$$

□

**Prop. (12.10.3.15) [Haar Measure on Rho-Functions].** There exists a unique continuous positive functorial  $\nu_{H \backslash G}$  on  $\mathcal{S}(G, \Delta)$  that for any  $f \in C_c(G)$ ,

$$\int_G f(x) d\nu_G(x) = \int_{H \backslash G} p(f)(y) d\nu_{H \backslash G}(y) = \int_{H \backslash G} \int_H f(hg) \Delta_G^{-1}(h) d\mu_H(h) d\nu_{H \backslash G}.$$

and it is invariant under the right action of  $G$ . We denote  $\nu_{H \backslash G}(s) = \int_{H \backslash G} s(g) d\nu_{H \backslash G}(g)$  for  $s \in \mathcal{S}(G, \Delta)$ .

┘

*Proof:* This follows from (12.10.3.14).

□

**Cor. (12.10.3.16).** If  $G$  is a unimodular and  $H, K$  are closed subgroups s.t.  $H \cap K$  is compact and  $G = HK$ , then the the quotient measure  $\mu_{H \backslash G}$  is given by

$$f \mapsto \int_K f(k) d\nu_K(k).$$

where  $d\nu_K(k)$  is a right Haar measure on  $K$ , by comparison with (12.10.3.8).  $\lrcorner$

**Def. (12.10.3.17) [Class Functions].** A measurable function on  $G$  is called a **class function** iff  $f(y^{-1}xy) = f(x)$  a.e.  $(x, y) \in G \times G$ . Denote  $ZL^p(G)$  the space of class functions in  $L^p(G)$  and  $ZC(G)$  the space of continuous class functions on  $G$ .  $\lrcorner$

### Restricted Direct Products

**Def. (12.10.3.18) [Restricted Direct Products].** Let  $S \in \text{Set}, \#S \leq \aleph_0$  be a set of indices and  $\{X_\lambda\}_{\lambda \in \Lambda}$  is a family of locally compact spaces, and for a.e.  $\lambda \in \Lambda$  an open compact subset  $K_\lambda \subset X_\lambda$ . Then the **restricted direct product** is defined to be

$$X = \prod' (X_\lambda, K_\lambda) \triangleq \varinjlim_{\emptyset \subset \text{fin } S \in \Lambda} \left( \prod_{\lambda \in S} X_\lambda \times \prod_{\lambda \in \Lambda \setminus S} K_\lambda \right)$$

endowed with the colimit space topology. And we denote

$$X'_S \triangleq \prod_{\lambda \in S} X_\lambda \times \prod_{\lambda \in \Lambda \setminus S} K_\lambda, \quad X^S \triangleq \prod_{\lambda \in \Lambda \setminus S} K_\lambda.$$

This topology is stronger than the product topology of  $\prod_\lambda X_\lambda$ . It has an open basis  $N = \prod_\lambda N_\lambda$ , where  $N_\lambda$  is open in  $X_\lambda$  and  $N_\lambda = K_\lambda$  for a.e.  $\lambda \in \Lambda$ . It is locally compact because each  $X'_S$  does.  $\lrcorner$

**Prop. (12.10.3.19).** Every compact subset  $N$  of  $X$  is contained in some  $\prod_\lambda M_\lambda$ , where  $M_\lambda \subset X_\lambda$  is compact and  $M_\lambda = H_\lambda$  for a.e.  $\lambda$ .  $\lrcorner$

*Proof:* This is because  $X_S$  is an open covering of  $X$ , and the union of f.m.  $X_{S_i}$  is also of the form  $X_S$ . So  $N$  is contained in some  $X_S$ , thus its projection in the  $S$ -coordinates is compact.  $\square$

**Prop. (12.10.3.20) [Tensor Product Groups].** Situation as in (12.10.3.18), if all  $X_\lambda, K_\lambda$  are topological groups, then  $X$  has a natural topological group structure, making it a locally compact group.  $\lrcorner$

**Situation (12.10.3.21).** In the following, we assume that  $X_\lambda = G_\lambda, K_\lambda = H_\lambda$  are all Abelian topological groups.  $\lrcorner$

**Prop. (12.10.3.22) [Quasi-Characters on  $G$ ].** Quasi-characters on  $G$  are of the form  $\otimes_\lambda c_\lambda$  that  $c_\lambda \in G_\lambda^\vee$  is trivial on  $H_\lambda$  for a.e.  $\lambda \in \Lambda$ .  $\lrcorner$

*Proof:* Let  $c$  be a quasi-character, choose a nbhd of  $1 \in U \subset \mathbb{C}$  that contains no subgroup, then  $c^{-1}(U)$  has contains an open basis  $\prod_{\lambda \in S} N_\lambda \times G^S$ , where  $N_\lambda$  are open nbhds of 1, so  $c(G^S) = 1$ . Thus  $c(\mathfrak{a}) = \prod_\lambda c_\lambda(a_\lambda)$  is true for any  $\mathfrak{a} \in G$ .

Conversely, clearly  $\otimes_\lambda c_\lambda$  is a quasi-character on  $G$ , it is continuous.  $\square$

**Prop. (12.10.3.23) [Dual of  $G$ ].** In each  $G_\lambda^\vee$ , by (12.11.1.8)  $H_\lambda$  is compact, so  $H_\lambda^\vee = G_\lambda^\vee / H_\lambda^\perp$  is discrete, so  $H_\lambda^\perp$  is open;  $H_\lambda$  are open, so  $H_\lambda^\perp = (G_\lambda / H_\lambda)^\vee$  are compact. So we can define the space  $\prod'(G_\lambda^\vee, H_\lambda^\perp)$ . Then the dual group  $G^\vee \cong \prod'(G_\lambda^\vee, H_\lambda^\perp)$  as a topological group.  $\lrcorner$

*Proof:* (12.10.3.22) shows that this is an algebraic isomorphism, so it suffices to prove this is a topological homeomorphism(12.11.1.7):

For any compact  $B \in G_1 = \prod_{\mathbf{p} \in S} N_\lambda \times \prod_{\mathbf{p} \notin S} H_\lambda$ , for any  $\varepsilon > 0$ , if  $c \in \prod_{\mathbf{p} \in S} N'_\lambda \times \prod_{\mathbf{p} \notin S} H_\lambda^\perp$ , where  $N'_\lambda = \{c_\lambda | |c_\lambda(N_\lambda - 1)| < \varepsilon/|S|\}$ , then  $|c(B) - 1| < \varepsilon$ .

Conversely, if  $\varepsilon$  is small enough, then if  $c(\prod_{\mathbf{p} \in S} N_\lambda \times \prod_{\mathbf{p} \notin S} H_\lambda) - 1| < \varepsilon$ , then  $c \in \prod_{\mathbf{p} \in S} N'_\lambda \times \prod_{\mathbf{p} \notin S} H_\lambda^\perp$ , where  $N'_\lambda = \{c_\lambda | |c_\lambda(N_\lambda - 1)| < \varepsilon\}$   $\square$

**Prop. (12.10.3.24)[Restricted Product Measure].** Let measures  $\mu_\lambda$  be given on  $X_\lambda$  that  $\mu_\lambda(K_\lambda) = 1$  for a.e.  $\mathbf{p}$ , define a Haar measure on  $G$  as follows:

On  $X'_S$ ,  $d\mu_S = (\prod_{\lambda \in S} d\mu_\lambda) \times d\mu^S$ , where  $\mu^S$  is the product measure on  $X^S$ .

Then these can define a positive left-invariant functional  $I$  on  $C_c(X)$  s.t.  $|I(f)| \leq \|f\|$  for any  $f$  that depends only on f.m. coordinates  $\lambda \in S$ . Then Stone-Weierstrass theorem shows these functions are dense in  $C_c(G)$ , thus  $I$  can be uniquely extended to a functional on  $C_c(G)$ , and this defines a Haar measure on  $G$  by Riesz representation(12.10.1.9), denoted by  $\mu = \prod'_\lambda \mu_\lambda$ , called the **restricted product measure**.  $\lrcorner$

**Prop. (12.10.3.25).** For a function  $f$  on  $G = \prod'(G_\lambda, H_\lambda)$  measurable, if either  $f \geq 0$  or  $f \in L^1(G)$ , then

$$\int_G f(\mathbf{a}) d\mathbf{a} = \lim_{\vec{S}} \int_{G_S} f(\mathbf{a}) d\mathbf{a}$$

as a net limit.  $\lrcorner$

*Proof:* The second case follows from the first case, as  $\int_G f = \lim_{\vec{B} \text{ compact}} \int_B f$  by monotone convergence theorem, and any  $B$  compact is contained in some  $G_S$ (12.10.3.19).  $\square$

**Cor. (12.10.3.26).** If  $f(\mathbf{a}) = \prod_\lambda f_\lambda(\mathbf{a}_\lambda)$ , where  $f_\lambda \in L_1(G_\lambda)$  and  $f_\lambda = \chi_{H_\lambda}$  a.e.  $\mathbf{p}$ , then if

$$\prod_\lambda \int_{G_\lambda} |f_\lambda(\mathbf{a}_\lambda)| d\mathbf{a}_\lambda < \infty,$$

then  $f \in L^1(G)$ , and

$$\int_G f(\mathbf{a}) d\mathbf{a} = \prod_\lambda \left( \int f_\lambda(\mathbf{a}_\lambda) d\mathbf{a}_\lambda \right).$$

$\lrcorner$

**Def. (12.10.3.27)[Dual Measure].** Notice if  $f_\lambda = \chi_{H_\lambda}$ , then

$$f_\lambda^\vee(c_\lambda) = \int_{H_\lambda} \overline{c_\lambda(\alpha_\lambda)} d\mathbf{a}_\lambda = d\mathbf{a}_\lambda(H_\lambda) \chi_{H_\lambda^\perp}(c_\lambda).$$

So by Fourier transform(12.11.1.24), if  $dc_\lambda$  is the dual measure on  $G_\lambda^\vee$ , then  $\chi_{H_\lambda^\perp} = dc_\lambda(H_\lambda^\perp) d\mathbf{a}_\lambda(H) \chi_{H_\lambda^\perp}$ , which means  $dc_\lambda(H_\lambda^\perp) = 1, a.e. \mathbf{p}$ , thus we can define a measure on  $G^\vee$  as  $dc = \prod'_\lambda dc_\lambda$ .

Then  $dc$  is the measure on  $\hat{G}$  dual to  $d\mathbf{a}$  on  $G$ .  $\lrcorner$

*Proof:* The duality is by the lemma below(12.10.3.28), applied to both  $f$  and  $\hat{f}$ .  $\square$

**Lemma (12.10.3.28)[Fourier Transform on Product].** if  $f_\lambda \in B_1(G_\lambda)$  and  $f_\lambda = \chi_{H_\lambda}$  a.e.  $\lambda$ , then  $f(\mathbf{a}) = \prod f_\lambda(\mathbf{a}_\lambda) \in B_1(G)$ , and  $f^\vee(c) = \prod f_\lambda^\vee(c_\lambda)$ .  $\lrcorner$

*Proof:* For any character  $c$ , because

$$f(\mathfrak{a})\bar{c}(\mathfrak{a}) = \prod f_\lambda(\mathfrak{a}_\lambda)\bar{c}_\lambda(\mathfrak{a}_\lambda)$$

and every  $f_\lambda(\mathfrak{a}_\lambda)\bar{c}_\lambda(\mathfrak{a}_\lambda) \in L^1(G_\lambda)$ . So (12.10.3.26) applies and shows the equations. Similarly, because  $\widehat{f}_\lambda = \chi_{H_\lambda^\perp}$  a.e.  $\mathfrak{p}$ , we have  $\widehat{f} \in L_1(\widehat{G})$ , so  $f(\mathfrak{a}) \in B_1(G)$ .  $\square$

### Compactly Generated Groups

**Def. (12.10.3.29) [Compactly Generated Groups].**  $\Gamma \in \mathcal{T}\text{op}\mathcal{G}\text{rp}$  is called a **compactly generated group** if there exists a compact subset  $C \subset \Gamma$  s.t.  $C$  is a defining set for  $\Gamma$  (3.1.5.8).  $\lrcorner$

**Prop. (12.10.3.30) [Criterion for Compactly Generatedness].** Let  $\Gamma \in \mathcal{T}\text{op}\mathcal{G}\text{rp}$  with a map  $|\cdot| : \Gamma \rightarrow \mathbb{N}$  s.t.

- $|1| \neq 0$ .
- $|g_1g_2| \leq |g_1| + |g_2|$  for any  $g_1, g_2 \in \Gamma$ .
- $|g^{-1}| = |g|$  for  $g \in \Gamma$ .
- For any  $n \in \mathbb{N}$ ,  $\Gamma_n = \{g \in \Gamma : |g| \leq n\}$  is compact.

Moreover, assume there exists  $c, b, d \in \mathbb{Z}_+$  s.t.

- If  $g \in \Gamma$  and  $|g| > c$ ; then there exists  $g_1, g_2 \in \Gamma$  s.t.  $g = g_1g_2$ ,  $|g_1| < C$  and  $|g_2| < |g| - d$ .
- If  $f, g, h \in \Gamma$  s.t.  $fgh = 1$ , then there exist

$$- t \in \mathbb{Z}_+$$

$$- g_1, \dots, g_t \in \Gamma$$

s.t.

$$- t \leq |g| + b.$$

$$- |g_i| \leq c.$$

$$- g = g_1 \dots g_t.$$

$$- |fg_1 \dots g_j| \leq \max(|f|, |h|) + d \text{ for any } j \in [t].$$

Then  $\Gamma$  is compactly generated.  $\lrcorner$

*Proof:* Cf. [P-R94]P157.  $\square$

**Prop. (12.10.3.31).** If  $H \trianglelefteq G$  is an open normal subgroup of finite index and  $H$  is compactly generated, then  $G$  is also compactly generated.  $\lrcorner$

*Proof:* Cf. [P-R94]P152.  $\square$

### Maximal Compact Subgroups

**Def. (12.10.3.32) [Maximal Compact Subgroup].** A **maximal compact subgroup** of a locally compact group  $G$  is a maximal object in the set of all compact subgroup of  $G$ .  $\lrcorner$

**Prop. (12.10.3.33) [Cartan-Iwasawa-Malcev theorem].** Maximal compact subgroups for  $G$  exist.  $\lrcorner$

*Proof:*  $\square$

#### 4 Locally Profinite Groups

**Def. (12.10.4.1) [Locally Profinite Group].** A **locally profinite group** is a locally profinite topological group. A profinite group is locally profinite, and any compact open subgroup of a locally profinite group is profinite.  $\lrcorner$

**Cor. (12.10.4.2).** A closed subgroup of a locally profinite group is locally profinite, and a quotient group is locally profinite.  $\lrcorner$

*Proof:* The proof is very similar to that of (3.1.12.4), as the result of (4.4.1.25) remains true, because any connected nbhd of  $e$  is contained in any compact open subgroup.  $\square$

**Prop. (12.10.4.3) [Compact Open Subgroups Form a Basis].** If  $G \in \text{locProfGrp}$ , then the set of compact open subgroups of  $G$  form a basis of the nbhd of 1.  $\lrcorner$

*Proof:* For any nbhd  $U$  of 1, choose a precompact nbhd  $V$  of 1 contained in  $U$ , then there is another compact open subgroup contained in  $V$ , by (4.12.1.24).  $\square$

**Prop. (12.10.4.4) [Quotient of Locally Profinite Group].** A quotient subspace of a locally profinite group is locally profinite.  $\lrcorner$

*Proof:* Consider the  $H$  action on  $G$ , then it is regular, because the graph is the preimage of  $H$  in the map  $G \times G \rightarrow G : (g_1, g_2) \mapsto g_1^{-1}g_2$ . So by (4.12.1.10)  $G/H$  is Hausdorff. But clearly  $G \rightarrow G/H$  is open and  $G/H$  is locally profinite as it has a basis of locally compact subsets.  $\square$

**Lemma (12.10.4.5).** Let  $G \in \text{locProfGrp}$  and  $H \leq G$  be a closed subgroup, then for any open compact subspace  $V \subset G/H$ , there is an open compact subspace  $U \subset G$  that  $p(U) = V$ .  $\lrcorner$

*Proof:* The preimage  $p^{-1}(V)$  is open, so there is a covering of  $p^{-1}(V)$  by open compact subsets  $U_i$ . Then  $p(U_i)$  are open and covers  $V$ , thus there are f.m.  $U_i$  that  $p(\cup U_i) = V$ .  $\square$

**Prop. (12.10.4.6) [Homogeneous Group].** Let  $G$  is a locally profinite group that is  $\sigma$ -compact. If  $G$  acts transitively on a locally profinite space  $X$ , let  $x_0 \in X$  and  $\text{Stab}(x_0) = H$ , then  $G/H \rightarrow X$  is a homeomorphism. ? Is this true for locally compact groups?  $\lrcorner$

*Proof:* Let  $N$  be a compact open subset of  $N$ , and  $g_i$  be the left coset representatives for  $N$ , which is countable. Now  $X = \cup_i \gamma(g_i N)x_0$ . Because a locally profinite space is a Baire space, some  $\gamma(g_i N)x_0$  contains a nbhd of  $\gamma(g_i n)x$ . Now left acts  $(g_i n)^{-1}$ , we see  $x_0$  is an interior point of  $\gamma(N)x_0$ . Now  $N$  is arbitrary, thus (12.10.4.3) shows  $g \mapsto \gamma(g)x$  is open, thus  $G/H \rightarrow X$  is open. It is clearly continuous, thus  $G/H \cong X$ .  $\square$

## 12.11 Harmonic Analysis on Locally Compact Abelian Groups

### 1 Locally Compact Abelian Group

**Notation(12.11.1.1).**

- In this subsection, let  $G \in \text{LCA}$ .
- Use notations on Functional Analysis defined in [Archimedean Functional Analysis](#).
- Use notations on Harmonic Analysis defined in [Harmonic Analysis on Locally Compact Groups](#).

┘

**Thm.(12.11.1.2)[Pontryagin-van.Kampen Fundamental Structure Theorem].**  $G$  is isomorphic to  $E \times \mathbb{R}^n$  for some  $n \in \mathbb{N}$  and some  $E \in \text{LCA}$  which has a compact open subgroup.

┘

*Proof:*

□

### Dual Group

**Def.(12.11.1.3)[Dual Group].** Because  $G$  is Abelian, the irreducible unitary representations of  $G$  in  $\widehat{G}$  are all 1-dimensional by (24.7.1.9). An element of  $G^\vee$  is called a **character of  $G$** . And a continuous homomorphism from  $G$  to  $\mathbb{C}^\times$  is called a **quasi-character**.

The topologies on  $G^\vee$  that makes it into a LCA group is given in (12.11.1.7).

┘

**Remark(12.11.1.4).**  $\widehat{\mathbb{R}} \cong \mathbb{R}$ , and the quasi-characters of  $\mathbb{R}$  are all of the form  $x \rightarrow e^{sx}$  for  $s \in \mathbb{C}$ .

┘

*Proof:* If  $\varphi \in \widehat{\mathbb{R}}$ , then  $\varphi(0) = 1$ , and there is an  $a > 0$  that  $\int_0^a \varphi(t)dt \neq 0 = A$ . Now  $A\varphi(x) = \int_x^{x+a} \varphi(t)dt$ , so taking derivative,

$$\varphi'(x) = \frac{\varphi(x+a) - \varphi(x)}{A} = \frac{\varphi(a) - 1}{A} \varphi(x),$$

which shows  $\varphi(x) = e^{sx}$  for some  $s \in \mathbb{C}$ .

□

**Prop.(12.11.1.5)[Dual Group as Spectrum of  $L^1(G)$ ].** The dual group  $\widehat{G}$  can be regarded as the spectrum of  $L^1(G)$ , i.e. multiplicative homomorphism of  $L^1(G)$ :

$$\xi \mapsto \left( \xi(f) = \int \overline{(x, \xi)} f(x) dx \right).$$

┘

*Proof:* First,  $\xi$  is multiplicative because

$$\xi(f * g) = \int \int f(y)g(y^{-1}x)(x, \xi) dy dx = \int \int f(y)g(x)(xy, \xi) dy dx = \xi(f)\xi(g).$$

Conversely, any continuous functional on  $L^1$  is like  $\varphi(f) = \int f(x)\varphi(x)dx$  for some  $\varphi \in L^\infty$ , and it is multiplicative, so

$$\varphi(f) \int \varphi(x)g(x) = \varphi(f)\varphi(g) = \varphi(f * g) = \int \int \varphi(y)f(yx^{-1})g(x) dx dy = \int \varphi(L_x(f))g(x) dx$$

So  $\varphi(x) = \frac{\varphi(L_x(f))}{\varphi(f)}$ , a.e., for any  $f$ . so  $\varphi(x)$  can be chosen to be continuous, as  $x \rightarrow L_x(f)$  is continuous (12.10.1.2). And clearly  $\varphi$  is multiplicative.

□

**Cor. (12.11.1.6).**  $G^\vee \subset P_1(G)$ , because  $\int (f^* * f) \varphi d\mu = |\Phi(f)|^2 \geq 0$ .  $\lrcorner$

**Cor. (12.11.1.7) [Dual Group as a LCA Group].** Now we can give  $\widehat{G}$  the compact-open topology, then the group operation is clearly continuous, and the topology coincides with that inherited by the weak\*-topology of the  $L^\infty$  by (24.7.1.27), so  $\widehat{G} \cup \{0\}$  is a compact Hausdorff space because  $\widehat{G} \subset P_1(G)$  and it is the subset of  $L^\infty$  that  $\{h(xy) = h(x)h(y)\}$  which is weak\*-closed hence weak\*-compact. In particular,  $G^\wedge$  is a locally compact topological group.  $\lrcorner$

**Prop. (12.11.1.8) [Duality between Discrete Groups and Compact Groups].** if  $G$  is discrete, then  $\widehat{G}$  is compact, if  $G$  is compact, then  $\widehat{G}$  is discrete.  $\lrcorner$

*Proof:* if  $G$  is discrete, then there is a unit  $\delta$  in  $L^1(G)$ , which is 1 on  $e$  and 0 otherwise. So the spectrum of  $L^1(G)$  is compact by (12.9.1.21).

If  $G$  is compact, then  $1 \in L^1$ , so  $U = \{f \in L^\infty(G) : |f| > \frac{1}{2}\}$  is weak\*-open, but  $U \cap G^\vee = \{1\}$  by (12.10.1.14), so  $\widehat{G}$  is discrete.  $\square$

### Fourier Transform

**Prop. (12.11.1.9) [Fourier Transforms].** The **Fourier transform** on  $G$  is defined as in (12.11.1.5) to be the map

$$L^1(G) \rightarrow C(G^\vee) : f \mapsto \mathcal{F}f(\xi) = f^\vee(\xi) = \int f(x) \overline{\chi(x, \xi)}.$$

It is a norm-decreasing \*-homomorphism from  $L^1(G)$  to  $C_0(G^\vee)$ , and its range is a dense subspace of  $C_0(G^\vee)$ .

Equivalently, the Fourier transform is just the Gelfand transform of  $L^1(G)$  (12.9.1.21) composed with an inverse map.  $\lrcorner$

*Proof:* Cf. [Fol16] P102.  $\square$

**Prop. (12.11.1.10).** There is another map from  $\text{Meas}(G^\vee)$  to bounded continuous functions on  $G$ :

$$\mu \mapsto \left( \varphi_\mu : x \mapsto \int (x, \xi) d\mu(\xi) \right).$$

This is a norm decreasing injection from  $\text{Meas}(G^\vee)$  to  $L^\infty(G)$ , and if  $\mu$  is positive, then  $\varphi_\mu$  is a function of positive type.  $\lrcorner$

*Proof:* It suffices to prove injectivity, but if  $\varphi_\mu = 0$ , then  $0 = \int \int f(x) (x, \xi) d\mu(\xi) dx = \int \widehat{f}(\xi^{-1}) d\mu(\xi)$  for all  $f \in L^1(G)$ , so but this shows  $\mu = 0$  because of (12.11.1.9) and Riesz representation.

For the positive type, notice that

$$\int \int f(x) \overline{f(y)} \varphi_\mu(y^{-1}x) dx dy = \int \int \int f(x) \overline{f(y)} \overline{(y, \xi)} (x, \xi) d\mu(\xi) dx dy = \int |\widehat{f}(\xi)|^2 d\mu(\xi) \geq 0.$$

$\square$

**Prop. (12.11.1.11) [Bochner's Theorem].** For  $\varphi \in \text{Pos}(G)$  (24.7.1.16), there exists a unique positive  $\mu \in \text{Meas}(G^\vee)$  s.t.  $\varphi = \varphi_\mu$ .  $\lrcorner$



*Proof:* We have the map defined in (12.11.1.10) injects  $M(\widehat{G})$  into  $P(G)$  (norm-decreasing), so it suffices to prove the existence. For this, we may assume  $\varphi \in P_0(G)$ . Let  $M_0$  be the set of positive measure  $\mu \in M(\widehat{G})$  that  $\mu(\widehat{G}) \leq 1$ , then  $M_0$  is weak\*-compact in  $M(\widehat{G})$ . Now

$$\int f(x)\varphi_\mu(x)dx = \int \int f(x)d\mu(\xi)dx = \int \widehat{f}(\xi^{-1})\mu(x),$$

so the mapping  $\mu \rightarrow P_0$  must be continuous w.r.t their weak\*-topologies, so the image is a compact convex subset of  $P_0$ . But the image contains all characters and 0 (by taking the point mass), which are the extreme points of  $P_0$ , by (24.7.1.25), so it contains all the  $P_0$ , by Krein-Milman (12.8.3.15).  $\square$

**Cor. (12.11.1.12).**  $\{\varphi_\mu\} = B(G)$  (24.7.1.28), by (12.11.1.11) and (12.11.1.10). Thus the inverse  $B(G) \rightarrow M(G^\vee)$  is denoted by  $f \mapsto d\mu_f$ .  $\square$

**Cor. (12.11.1.13) [Herglotz].** A sequence  $(a_n)_{n \in \mathbb{Z}_+} \in \mathbb{C}$  is positive-definite iff there is a positive measure  $\mu \in \text{Meas}(\mathbb{T}^1)$  s.t.  $a_n = \varphi_\mu(n)$ .  $\square$

**Prop. (12.11.1.14).** The set of regular Borel probability measures on a compact  $X$  is weak\*-compact in  $C(X)^*$ . (Use Alaoglu).  $\square$

**Prop. (12.11.1.15) [Fourier Inversion Formula].** (special case of (12.11.1.24)) If  $f \in B^1(G)$  (24.7.1.28), then  $\widehat{f} \in L^1(G^\vee)$ , and if the Haar measure  $d\xi$  of  $\widehat{G}$  is suitably normalized w.r.t. the Haar measure of  $G$ , then  $d\mu_f(\xi) = \widehat{f}(\xi)d\xi$  (12.11.1.12), i.e.  $f(x) = \int (x, \xi)\widehat{f}(\xi)d\xi$ . This measure  $d\xi$  is called the **dual measure** of  $dx$ .  $\square$

*Proof:* Cf. [Folland Abstract Harmonic Analysis P105].  $\square$

**Cor. (12.11.1.16).** If  $f \in L^1(G) \cap \text{Pos}(G)$ , then  $\widehat{f} \geq 0$ , as  $d\mu_f(\xi) = \widehat{f}(\xi)d\xi$  and  $\mu_f$  is positive, by Bochner's theorem (12.11.1.11).  $\square$

**Prop. (12.11.1.17) [Dual Measure of Discrete Group].** If  $\mu$  is the counting measure on a discrete group, then its dual measure satisfies  $|\widehat{G}| = 1$ , and if  $G$  is compact and  $|G| = 1$ , then the dual measure is the counting measure on  $\widehat{G}$ .  $\square$

*Proof:* First (12.11.1.8) should be noticed. If  $G$  is compact and  $|G| = 1$ , then if  $g = 1$ , then  $\widehat{g} = \chi_{\{1\}}$ , so  $g(x) = \sum (x, \xi)\widehat{g}(\xi)$ , which shows the dual measure is counting measure by definition (12.11.1.15).  $\square$

**Prop. (12.11.1.18) [Plancherel Theorem].** The Fourier transform on  $L^1(G) \cap L^2(G)$  extends uniquely to an isomorphism from  $L^2(G)$  to  $L^2(\widehat{G})$  that satisfies Fourier inversion formula.  $\square$

*Proof:* Cf. [Folland P108].  $\square$

**Cor. (12.11.1.19) [Hausdorff-Young Inequality].** Let  $1 \leq p \leq 2$  and  $p^{-1} + q^{-1} = 1$ . If  $f \in L^p(G)$ , then  $\widehat{f} \in L^q(G)$ , and  $\|\widehat{f}\|_q \leq \|f\|_p$ .  $\square$

*Proof:* Cf. [Folland, P109].  $\square$

### Schrödinger Representations

**Def. (12.11.1.20)**  $[A(G)]$ . Denote  $A(G) = G^* \times G \times \mathbb{T}^1$  with the group law

$$(v_1^*, v_1, t_1)(v_2^*, v_2, t_2) = (v_1^* + v_2^*, v_1 + v_2, t_1 t_2 \langle v_1, v_2^* \rangle).$$

Also we denote for  $w, w' \in G^* \times G$ ,  $[w, w'] = \langle v_1, v_2^* \rangle$ .

Notice the commutator

$$(v_1^*, v_1, 1)(v_2^*, v_2, 1)(v_1^*, v_1, 1)^{-1}(v_2^*, v_2, 1)^{-1} = (0, 0, \langle v_1, v_2^* \rangle \langle v_2, v_1^* \rangle^{-1}),$$

thus  $\langle v_1, v_2^* \rangle \langle v_2, v_1^* \rangle^{-1}$  defines a multiplicative skew-symmetric, bilinear and perfect pairing  $[-, -]$  of  $G^* \times G$  with itself.  $\perp$

**Thm. (12.11.1.21) [Segal-Shale-Weil].**

- There is an unitary representation of  $A(G)$  on  $L^2(G)$  given by

$$(\rho(v^*, v, t)\Phi)(u) = t \langle u, v^* \rangle \cdot \Phi(u + v),$$

called the **Schrödinger representation**. In fact, it is the induced representation  $\text{Ind}_{G \times \mathbb{T}^1}^{A(G)} \chi$ , where  $\chi(g, t) = t$ .

- This Schrödinger representation is irreducible.
- Any irreducible unitary representation of  $A(G)$  is isomorphic to the Schrödinger representation, and the isomorphism is unique up to a scalar.  $\perp$

**Remark (12.11.1.22).** This should be a direct consequence of the Stone-Von-Neumann theorem. Cf. [History of Stone Von-Neumann Theorem] ?.  $\perp$

*Proof:* Cf. [EASY PROOF OF THE STONE-VON NEUMANN-MACKEY THEOREM].

This is clearly a unitary representation.  $\rho$  induces an action of  $G^* \times G$  on  $L^2(G)$  via  $t = 1$ , thus an action of  $C_c(G^* \times G)$  on  $L^2(G)$ :

$$(\rho(\varphi)\Phi)(u) = \int_{G^* \times G} \varphi(w)(\rho(w, 1)\Phi)(u)dw = \int_G K_\varphi(u, v)\Phi(v)dv$$

where  $K_\varphi(u, u + v) = \int_{G^*} \varphi(v^*, v) \langle u, v^* \rangle dv^*$  is the Fourier transform of  $\varphi$  w.r.t.  $G^*$ . As Fourier transform is  $L^2$ -isometry,  $\varphi \mapsto K_\varphi$  extends to an isometry  $\lambda : L^2(G^* \times G) \cong L^2(G \times G)$ , whose inverse is given by

$$\varphi(v^*, v) = \int_G K_\varphi(u, u + v) \langle -u, v^* \rangle du,$$

and also the action  $\rho$  extends to all  $\varphi \in L^2(G^* \times G)$ .

It can be verified that  $\rho(\varphi_1) \circ \rho(\varphi_2) = \rho(\varphi_1 \star \varphi_2)$ , where

$$(\varphi_1 \star \varphi_2)(w) = \int_{G^* \times G} \varphi_1(w_1)\varphi_2(w - w_1)[w_1, w - w_1]dw_1.$$

Then by comparison with the equation above,  $K_{\varphi_1 \star \varphi_2} = K_{\varphi_1} \star K_{\varphi_2}$ , where  $\star$  is defined in (12.3.6.7).

$\sigma \in B_0(G)$  is of the form  $\sigma(w, t) = (s(w), f(w)t)$ , where  $w \in G^* \times G$ ,  $f : G^* \times G \rightarrow \mathbb{T}$  is a map satisfying

$$f(w_1 + w_2) = f(w_1)f(w_2)[s(w_1), s(w_2)][w_1, w_2]^{-1}.$$

Notice  $s$  preserves Haar measure on  $G^* \times G$ : it is preserved the pairing  $[\cdot, \cdot]$  as it is a commutator and  $\sigma$  fixes  $Z(A(G))$ . Thus  $s$  preserves Haar measure of  $G^* \times G$ , by (12.11.1.35).

Now we define a unitary transformation  $\Sigma$  of  $L^2(G^* \times G)$  by  $(\Sigma\varphi)(w) = f(w)^{-1}\varphi(s(w))$ , then it can be checked? by using the equations above that

$$\Sigma(\varphi_1 \star \varphi_2) = \Sigma(\varphi_1) \star \Sigma(\varphi_2).$$

Thus by the isomorphism  $\lambda : L^2(G^* \times G) \cong L^2(G \times G)$ ,  $\Sigma$  induces a unitary transformation on  $L^2(G \times G)$ , also denoted by  $\Sigma$  and it preserves  $\star$ . Then by (12.3.6.8), there is a unitary map  $\omega : L^2(G) \rightarrow L^2(G)$  that

$$\Sigma(P \otimes \overline{Q}) = \omega^{-1}(P) \otimes \overline{\omega^{-1}Q}.$$

Next, notice for any  $(w, t) \in A(G)$ ,

$$\bar{t}\lambda^{-1}(P \otimes Q)(w) = \int_G P(u)\overline{tQ(u+v)} \langle u, v^* \rangle du = (P, \rho(w, t)Q)_{L^2},$$

thus

$$\begin{aligned} (P, \rho(\sigma(w, t)))_{L^2} &= (P, \rho(s(w), f(t))Q)_{L^2} = \overline{tf(w)}\lambda^{-1}(P \otimes \overline{Q})(s(w)) \\ &= \bar{t}(\Sigma\lambda^{-1}(P \otimes \overline{Q}))(w) = \bar{t}\lambda^{-1}(\omega^{-1}(P) \otimes \overline{\omega^{-1}Q})(w) \\ &= (\omega^{-1}(P), \rho(w, t)\omega^{-1}(Q))_{L^2} \\ &= ((P), \omega\rho(w, t)\omega^{-1}(Q))_{L^2} \end{aligned}$$

Because  $P, Q$  are arbitrary, this means  $\rho(\sigma(w, t)) = \omega(\sigma) \circ \rho(w, t) \circ \omega(\sigma)^{-1}$ .

It remains to show that  $\rho$  is irreducible: For any endomorphism of  $L^2(G)$  commuting with  $\rho$ , it commutes with  $\rho(\varphi)$  for any  $\varphi \in L^2(G^* \times G)$ . Take  $\varphi = \lambda^{-1}(P \otimes \overline{Q})$ , where  $P, Q \in L^2(G)$ , then  $\rho(\varphi)\Phi = (\Phi, Q)_{L^2}P$ , so  $(T\Phi, Q)_{L^2}P = (\Phi, Q)_{L^2}TP$ . Then  $T$  is a scalar.  $\square$

### Pontryagin Duality

**Thm. (12.11.1.23) [Pontryagin Duality].** The map

$$G \rightarrow (G^\vee)^\vee : x \mapsto (\chi \mapsto \chi(x))$$

is an isomorphism of topological groups.  $\lrcorner$

*Proof:* Cf. [Folland Abstract Harmonic Analysis P110].  $\square$

**Cor. (12.11.1.24) [Fourier Inversion Theorem].** If  $f \in L^1(G)$  and  $\hat{f} \in L^1(G^\vee)$  and the measure are dual to each other (12.11.1.15), then  $f(x) = f^{\vee\vee}(x^{-1})$ , i.e.  $f(x) = \int (x, \xi) \hat{f}(\xi) d\xi$  a.e..  $\lrcorner$

*Proof:* As

$$\hat{f}(\xi) = \int \overline{(x, \xi)} f(x) dx = \int (x^{-1}, \xi) f(x) dx = \int (x, \xi) f(x^{-1}) dx,$$

so by definition  $\hat{f} \in B^1(\hat{G})$ , and  $d\mu_{\hat{f}}(x) = f(x^{-1})dx$ . Then by (12.11.1.15),  $f(x^{-1}) = f^{\wedge\wedge}(x)$ .  $\square$

**Cor. (12.11.1.25) [Fourier's Uniqueness Theorem].** If  $\mu, \nu \in M(G)$  satisfy  $\hat{\mu} = \hat{\nu}$ , then  $\mu = \nu$ . In particular, if  $f, g \in L^1(G)$  and  $\hat{f} = \hat{g}$ , then  $f = g$ .  $\lrcorner$

*Proof:* By (12.11.1.10) (norm decreasing),  $\mu$  is uniquely determined by  $\varphi_\mu(\xi) = \widehat{\mu}(\xi^{-1})$  by Fourier inversion.  $\square$

**Lemma (12.11.1.26).** If  $\varphi, \psi \in C_c(\widehat{G})$ , then  $\varphi * \psi = \widehat{h}$  where  $h \in B^1(G)$ . In particular,  $\mathcal{F}(B^1(G))$  is dense in  $L^p(\widehat{G})$  for  $p < \infty$ .  $\lrcorner$

*Proof:* Cf. [Folland, P109].  $\square$

**Prop. (12.11.1.27).**  $(fg)^\wedge = \widehat{f} * \widehat{g}$  is satisfied for  $f, g \in L^2(G)$  also.  $\lrcorner$

*Proof:* Cf. [Folland, P112].  $\square$

**Prop. (12.11.1.28) [Duality of Subgroups].**  $(H^\perp)^\perp = H$  for closed subgroup  $H$  of a locally compact Abelian group  $G$ .  $\lrcorner$

*Proof:* Suffices to prove  $(H^\perp)^\perp \subset H$ . If  $x_0 \notin H$ , then Gelfand-Raikov shows that there is a character  $\eta$  on  $G/H$  that  $\eta(q(x_0)) \neq 1$ , so  $x_0 \notin (H^\perp)^\perp$ .  $\square$

**Prop. (12.11.1.29).** If  $H$  is a closed subgroup of  $G$ , then there are natural isomorphisms of LCA groups:

$$\Phi : \widehat{(G/H)} \cong H^\perp, \quad \Psi : \widehat{G}/H^\perp \cong \widehat{H}$$

$\lrcorner$

*Proof:*  $\Phi$  is clearly algebraic isomorphism. If  $|\eta(q(K)) - 1| < \varepsilon$ , then  $|\eta(K) - 1| < \varepsilon$ , so  $\Phi$  is continuous in the compact-open topology. Similarly, to show  $\Phi$  is open, it suffices to show a compact subset of  $G/H$  has a compact inverse image in  $G$ , but this is just (12.10.3.3).

Now for  $\Psi$ , notice  $\widehat{G}/H^\perp \cong (H^\perp)^\perp \cong H$  by (12.11.1.28), so by Pontryagin duality theorem,  $\widehat{G}/H^\perp \cong \widehat{H}$ .  $\square$

**Cor. (12.11.1.30) [Hahn-Banach for LCA Groups].** By the surjectivity of  $\Psi$ , any character of  $\widehat{H}$  extends to a character of  $G$ .  $\lrcorner$

**Prop. (12.11.1.31) [Poisson Summation Formula].** For a closed subgroup  $H \leq G$  and  $f \in C_c(G)$ , define  $F(xH) = \int_H f(xy)dy$  on  $G/H$ , then  $F \in C(G/H)$  by (12.10.3.7), and:

- $\widehat{F} = \widehat{f}|_{H^\perp}$ , where  $\widehat{G}/H^\perp$  is identified with  $H^\perp$  by (12.11.1.29).
- If  $\widehat{f}|_{H^\perp} \in L^1(H^\perp)$ , then with the dual measure of  $G/H$  on  $H^\perp$  (12.11.1.15), we have

$$\int_H f(xy)dy = \int_{H^\perp} \widehat{f}(\xi)(x, \xi)d\xi.$$

In particular, take  $x = e$ , then

$$\int_H f(y)dy = \int_{H^\perp} \widehat{f}(\xi)d\xi.$$

$\lrcorner$

**Remark (12.11.1.32).** The condition that  $f \in C_c(G)$  can be loosen to  $f \in C(G) \cap L^1(G)$  if it can be proven that  $F \in C(G/H)$ . For example this is case if  $H$  is discrete and the summation w.r.t.  $H$  is locally uniformly continuous.  $\lrcorner$

*Proof:* Notice for  $\xi \in H^\perp$ ,

$$\widehat{F}(\xi) = \int_{G/H} F(xH) \overline{(x, \xi)} d(xH) = \int_{G/H} \int_H f(xy) \overline{(xy, \xi)} dy d(xH) = \int_G f(x) \overline{(x, \xi)} dx = \widehat{f}(\xi)$$

by (12.10.3.7). And 2 is just (12.11.1.24) applied to  $F(xH)$  on  $G/H$ .  $\square$

**Cor. (12.11.1.33).** In the situation of (12.11.1.31), if  $H$  is discrete in  $G$  and  $G/H$  is compact, then both  $H, H^\perp$  are discrete. By considering the dual measure using (12.11.1.17), the Poisson' summation formula reads:

$$\sum_H f(xy) = \frac{1}{\text{Vol}(G/H)} \sum_{H^\perp} \widehat{f}(\xi)(x, \xi).$$

┘

### Perfect Pairings

**Def. (12.11.1.34) [Self-Adjoint Haar Measures].** If there is an isomorphism  $G \cong G^\vee$ , or a perfect bilinear pairing  $G \times G \rightarrow \mathbb{C}^*$ , then by Fourier inversion (12.11.1.15), a Haar measure  $d\mu$  on  $G$  corresponds to a Haar measure  $d\alpha$  on  $\widehat{G}$ , but via the isomorphism,  $d\alpha$  corresponds to a measure  $\widehat{d\alpha}$  on  $G$ . Now anyway, there is a unique  $d\mu$  that  $d\mu = \widehat{d\alpha}$ , and this is called the **self-dual Haar measure** on  $G$ .

Via this pairing, we define the Fourier transform as an isomorphism

$$L^2(G) \cong L^2(G) : \Phi(f)(x) = \int_G f(y) \overline{\langle x, y \rangle} d\mu(y).$$

Then  $d\mu$  is a self-dual measure is equivalent to  $\Phi(\Phi(f))(x) = f(-x)$ , or equivalently  $f(y) = \int_G \widehat{f}(y) \langle x, y \rangle d\mu(y)$ .  $\square$

**Prop. (12.11.1.35).** If  $G$  is a locally compact Abelian group, and there is a perfect bilinear pairing  $G \times G \rightarrow \mathbb{C}^*$  and  $\sigma : G \rightarrow G$  is a group automorphism of  $G$  that preserves this pairing, then  $\sigma$  preserves the Haar measure on  $G$ .  $\square$

*Proof:* It is clear that  $\sigma^* d\mu = |\sigma| d\mu$  for some real constant  $|\sigma| > 1$ . Consider the Fourier transform w.r.t. this pairing:  $\mathcal{F}\varphi(x) = \int_G \varphi(y) \langle x, y \rangle dy$ , then  $\mathcal{F}(\varphi \circ \sigma) = |\sigma|^{-1} \mathcal{F}(x)$ . But  $\mathcal{F}$  is an isomorphism  $L^2(G) \cong L^2(G)$ , so  $|\sigma|^{-1} = |\sigma|$ , thus  $|\sigma| = 1$ .  $\square$

**Prop. (12.11.1.36) [Self Duality of Topological Fields].** Let  $K$  be a locally compact topological field. If  $X$  is a non-trivial character on the additive group  $K^+$ , then for any  $\eta \in K^+$ ,  $\xi \mapsto X(\eta\xi)$  is also a character, and

$$F_X : \eta \mapsto (\xi \mapsto X(\eta\xi))$$

is an isomorphism of topological groups of  $K^+$  and  $(K^+)^{\vee}$ .

Then by (12.11.1.34), we find a self-dual Haar measure  $dx$  on  $K^+$  w.r.t.  $X$ .

In fact, such a character  $X$  does exist, by Gelfand-Raikov (24.7.1.29).  $\square$

*Proof:* First this is clearly a homomorphism of groups, and it is injective, because if  $X(\eta\xi) = 1$  for all  $\xi$ , then  $\eta K^+ \neq K^+$  (because  $X$  is nontrivial), so  $\eta = 0$ .

Now the image of  $F$  is dense, because if  $X(\eta\xi) = 1$  for all  $\eta$ , then  $\xi = 0$ , so  $\overline{\text{Im}(F)}^\perp = 1$ . Now  $(H^\perp)^\perp = H$  for  $H$  closed (12.11.1.28) and use Pontryagin duality (12.11.1.23), so  $\text{Im } f$  is dense in  $\widehat{G}$ .

Now  $F$  is open and continuous, because: for any  $B \in G$  compact, there is a nbhd  $V$  of 0 that  $|X(V) - 1| < \varepsilon$ , so there is a nbhd  $V'$  that  $V'B \subset V$ , so if  $\eta \in V$ ,  $|X(\eta B) - 1| < \varepsilon$ , so  $F$  is continuous ( $\widehat{K^+}$  has the compact-open topology). And if we choose  $\xi_0$  that  $X(\xi_0) \neq 1$ , then choose  $B = \overline{B(0, \frac{|\varepsilon_0|}{\varepsilon})}$  compact, if  $|X(\eta B) - 1| < |X(\xi_0) - 1|$ , then  $\xi_0 \notin \eta B$ , which means that  $|\eta| < \varepsilon$ . This means  $F(\overline{B(0, \varepsilon)})$  contains  $V(B, |X(\xi_0) - 1|)$ , so  $F$  is open.

So the image of  $F$  is a locally compact subgroup of  $\widehat{G}$ , so by (12.10.1.6) it is closed, hence equals  $G$  as it is dense, so  $F$  is surjective, and is an isomorphism.  $\square$

## 2 Distributions on $\mathbb{R}^n$

Reference are [Rud91], [泛函分析张恭庆] and [Harmonic Analysis: Real Variable Methods, Orthogonality, and Oscillatory Integrals, Stein]. Notice much should be rewritten in greater generality of [Harmonic Analysis on Locally Compact Groups](#).

**Notation (12.11.2.1).**

- Let  $n \in \mathbb{Z}_+$ .
- Let  $\Omega \subset \mathbb{R}^n$  be an open subset.

┘

**Def. (12.11.2.2) [Test Functions].** The vector space

$$\mathcal{T}(\Omega) = \varinjlim_{K \subset \Omega \text{ compact}} D(K) \quad (12.8.2.1)$$

with the colimit topology is called the space of **test functions** on  $\Omega$ . Then  $(\mathcal{T}(\Omega), \tau) \in \text{TVS}(12.7.1.1)$ .

┘

**Prop. (12.11.2.3) [Test Functions].**

- $\mathcal{T}(\Omega)$  is locally convex.
- If  $E \subset \mathcal{T}(\Omega)$  is bounded, then  $E \subset D(K)$  for some compact subset  $K \subset \Omega$ , and  $\|E\|_n(12.8.2.1)$  is bounded for each  $n \in \mathbb{Z}_+$ .
- $\mathcal{T}(\Omega)$  has the Heine-Borel property.
- $\mathcal{T}(\Omega)$  is complete (12.7.1.2).

┘

*Proof:* Cf. [Rud91]P153?.

$\square$

**Prop. (12.11.2.4).** If  $\Lambda : \mathcal{T}(\Omega) \rightarrow Y \in \text{TVS}$  and  $Y$  is locally convex, then  $\Lambda$  is continuous iff it is bounded iff it is sequentially-continuous iff its restriction to each  $D(K)$  for compact subset  $K \subset \Omega$  is continuous.

┘

*Proof:* Cf. [Rud91]P155?.

$\square$

**Cor. (12.11.2.5) [Differential Operators].** For each multi-index  $\alpha$ ,  $D^\alpha$  defines a continuous map  $\mathcal{T}(\Omega) \rightarrow \mathcal{T}(\Omega)$ .

┘

**Def. (12.11.2.6) [Distributions].** The space of continuous linear functionals of  $\mathcal{T}(\Omega)$  (12.11.2.2) is called the space of **distributions** on  $\Omega$ , denoted by  $\text{Dist}(\Omega)$ . It is equivalence to the restriction to every  $D(K)$  is continuous, by (12.11.2.4).

┘

**Def. (12.11.2.7) [Order of Distributions].** For  $\Lambda \in \text{Dist}(\Omega)$  (12.11.2.6), the **order (of distribution)** of  $\Lambda$  is the infimum of all  $N \in \mathbb{N}$  s.t. for any compact subset  $K \subset \Omega$ , there exists  $C_K \in \mathbb{R}_+$  that  $|\Lambda\varphi| \leq C_K \|\varphi\|_N$  for every  $\varphi \in D(K)$ . The order might be  $\infty$ .  $\lrcorner$

**Def. (12.11.2.8) [Dirac Measure].** The **Dirac measure** at  $x \in \mathbb{R}^n$  is defined to be the distribution  $\delta_x \in D'(\Omega) : \varphi \mapsto \varphi(x)$ .  $\delta_0$  is also denoted by  $\delta$ .  $\lrcorner$

**Def. (12.11.2.9) [Differentiation of Distributions].** The **differentiation of a distribution**  $\Lambda$  is defined as  $D^\alpha \Lambda(\varphi) = (-1)^{|\alpha|} \Lambda(D^\alpha \varphi)$ .

Multiplication of a distribution by a smooth function  $f$  is defined by  $(f\Lambda)(\varphi) = \Lambda(f\varphi)$ .

The following relation holds:

$$D^\alpha(f\Lambda) = \sum_{\beta \leq \alpha} C_{\alpha\beta}(D^{\alpha-\beta}f)(D^\beta\Lambda).$$

$\lrcorner$

### Support of Distributions

**Def. (12.11.2.10) [Support of Distributions].** The **support of a distribution** is the complement  $\text{Supp}(\Lambda)$  of the open sets  $U$  that  $\Lambda(f) = 0$  for any  $f$  with support in  $U$ .

If  $\text{Supp}(\Lambda)$  is compact, then  $\Lambda$  has finite order and  $|\Lambda\phi| \leq C \|\phi\|_N$  for some  $N$ , and  $\Lambda$  extends uniquely to a continuous linear functional on  $C^\infty(\Omega)$ .  $\lrcorner$

*Proof:* This is because its support is compact so we can choose a smooth  $\psi$  that  $\equiv 1$  on  $\text{Supp} \varphi$  and has support in  $W \subset \Omega$ . Then by (12.11.2.2), there is a  $C$  that  $|\Lambda(\psi\varphi)| < C \|\psi\varphi\|_N$ , and Leibniz rule will give us the result.  $\square$

**Prop. (12.11.2.11).** If the support of a  $\Lambda$  is a pt  $p$  (thus has finite order  $m$ ), then it is a linear combination of  $D^\alpha \delta_p, |\alpha| \leq m$ . (use approximate identity and show the kernel of  $\Lambda$  is contained in the kernel of  $D^\alpha \delta_p$ ).  $\lrcorner$

*Proof:* Cf. [Rudin P165].  $\square$

**Prop. (12.11.2.12).** For any distribution  $\Lambda$ , there exist continuous functions  $g_\alpha$  in  $C^\infty(\Omega)$  that each compact  $K$  intersects support of f.m  $g_\alpha$  and  $\Lambda = \sum D^\alpha g_\alpha$ . When  $\Lambda$  has finite order, we can use only f.m  $g_\alpha$ .  $\lrcorner$

*Proof:* use partition of unity. Then for a compact  $K$ , find a compact-open  $W$ , then find a bump function between  $K \subset W$ , thus reduce to the case of  $D_{\overline{W}}$ . For the rest, Cf. [Rudin P169].  $\square$

### Convolution on $\mathbb{R}^n$

**Notation (12.11.2.13).**

- Denote  $\mathcal{D} = \mathcal{D}(\mathbb{R}^n)$  and  $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$ .  $\lrcorner$

**Def. (12.11.2.14) [Translations and Convolutions of Distributions].** The **translation** of a distribution  $u$  is defined as  $(\tau_x u)(\varphi) = u(\tau_{-x} \varphi)$ , where  $\tau_x \varphi(y) = \varphi(y - x)$ .

The **convolution** of a test function with a distribution  $u$  is defined as  $(u * \varphi)(x) = u(\tau_x \check{\varphi})$ , where  $\check{\varphi}(y) = \varphi(-y)$ .  $\lrcorner$

**Prop. (12.11.2.15)**[Special Case of (12.11.2.18)]. For  $u \in D', \varphi \in D, \psi \in D$ ,

- $\tau_x(u * \varphi) = (\tau_x u) * \varphi = u * (\tau_x \varphi)$ .
- $u * \varphi \in C^\infty$  and  $D^\alpha(u * \varphi) = (D^\alpha u) * \varphi = u * (D^\alpha \varphi)$ .
- $u * (\varphi * \psi) = (u * \varphi) * \psi$ .

If  $u$  has compact support, then (12.11.2.10) shows that  $u$  can extend to  $C^\infty$ , thus convolution is defined for  $\varphi \in C^\infty$  and the first two formulae still hold, and when  $\psi \in D$ ,

$$u * \psi \in D, \quad u * (\varphi * \psi) = (u * \varphi) * \psi = (u * \psi) * \varphi$$

┘

*Proof:* Cf.[Rudin P171], [Rudin P174].

□

**Cor. (12.11.2.16).**  $L : \varphi \mapsto u * \varphi$  is a continuous linear map into  $C^\infty$  that commutes with  $\tau_x$ . And any these map comes from a  $u$ : let  $u = (L\check{\varphi})(0)$ .

┘

*Proof:* It is continuous because of of closed graph theorem (12.7.2.9),  $\lim(u * \varphi_i)(x) = \lim u(\tau_x \check{\varphi}) = u(\tau_x \check{\varphi})$ .

□

**Cor. (12.11.2.17).** When  $u, v \in D'$  and one of them has compact support, then similar to (12.11.2.16),  $L\varphi = u * (v * \varphi)$  is a continuous linear map that commutes with  $\tau_x$ , so there is a unique **convolution distribution**  $u * v$  that  $(u * v) * \varphi = u * (v * \varphi)$ . This convolution is compatible with the previous one when  $v \in D$ .

┘

**Prop. (12.11.2.18)**[Convolution of Distributions]. For  $u, v, w \in D'$ ,

- if one of  $u, v$  has compact support, then  $u * v = v * u$ , and  $\text{Supp}(u * v) \subset \text{Supp}(u) + \text{Supp}(v)$ .
- if two of three of  $u, v, w$  has compact support, then  $(u * v) * w = u * (v * w)$ .
- $D^\alpha u = (D^\alpha \delta) * u$ .
- if one of  $u, v$  has compact support, then  $D^\alpha(u * v) = (D^\alpha u) * v = u * (D^\alpha v)$ .

┘

*Proof:* Cf.[Rudin P177].

□

**Def. (12.11.2.19).** A **approximate identity** here is a  $h \in D$  that  $h_k(x) = k^n h(kx)$ . Then we will have  $\lim \varphi * h_j = \varphi$  for  $\varphi \in D$ ,  $\lim u * h_j = u$  in  $D'$ .

┘

### 3 Fourier Analysis on $\mathbb{R}^n$

Reference are [Red91], [泛函分析张恭庆] and [Harmonic Analysis: Real Variable Methods, Orthogonality, and Oscillatory Integrals, Stein]. Notice much should be rewritten in greater generality of [Harmonic Analysis on Locally Compact Groups](#).

**Def. (12.11.3.1)**[Normalized Differentials]. Denote the normalized notation  $\mathbb{E}^n$  as  $dm = (\sqrt{2\pi})^{-n} dx$  and

$$D_\alpha = \frac{1}{i^{|\alpha|}} D^\alpha = \frac{1}{i^{|\alpha|}} \frac{\partial^\alpha}{\partial x^\alpha},$$

which will simplify many notations compared to  $D^\alpha$ . The **Fourier transform** here of a function  $f \in L^1(\mathbb{R}^n)$  is the function  $\mathcal{F}(f) \in C(\mathbb{R}^n)$  s.t.

$$\mathcal{F}f(t) = \int_{\mathbb{R}^n} f e_{-t} dm_n = (f * e_t)(0).$$

See (12.11.3.12) for general Fourier transform.

┘



**Prop. (12.11.3.2)[Invariant Properties].** For  $f \in L^1(\mathbb{R})$ ,

$$\mathcal{F}(\tau_x f) = e_{-x} \mathcal{F} f, \quad \mathcal{F}(e_{-x} f) = \tau_x \mathcal{F} f,$$

$$\widehat{f * g} = \widehat{f} \cdot \widehat{g}, \quad \widehat{f(x/\lambda)}(t) = \lambda^n \widehat{f}(\lambda t).$$

(Note  $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ ). ┘

**Lemma (12.11.3.3).** Let  $f = e^{-1/2|x|^2}$ , then  $f \in \mathcal{S}$ ,  $\widehat{f} = f$  and  $f(0) = \int \widehat{f}$ . ┘

*Proof:* Reduce to the 1 dimensional case, in which case,  $f^\lambda + xy = 0$ , and  $\mathcal{F} f$  also satisfies this. □

**Lemma (12.11.3.4).** For  $f, g \in L^1$ , Fubini theorem shows  $\int \widehat{f} g = \int f \widehat{g}$ . ┘

**Prop. (12.11.3.5)[Classical Fourier-Transform].**

- $\mathcal{S}(\mathbb{R}^n)$  is a Fréchet space in the topology defined by these norms.
- multiplication by  $g \in \mathcal{S}(\mathbb{R}^n)$  and derivations are continuous linear map from  $\mathcal{S}$  to  $\mathcal{S}$  (direct calculation).
- $\widehat{P(D)f}(t) = P(t)\widehat{f}(t)$  and  $\widehat{Pf} = P(-D)\widehat{f}$ .
- The Fourier-transform is a continuous linear bijection of  $\mathcal{S}$ , and  $\mathcal{F}^2(f)(t) = f(-t)$ . ┘

*Proof:* 1:

2:

3: use (12.11.2.18) for the first one, and for the second one, should use definition of derivative and dominated convergence.

4:  $\Psi f \in \mathcal{S}$  by 3, and it is continuous by closed graph theorem. By (12.11.3.4) and (12.11.3.2),  $\int \widehat{f}(t)g(t/\lambda) = \int f(t/\lambda)\widehat{g}(y)$ . If  $\widehat{f}, \widehat{g} \in L^1$ , dominant convergence shows  $g(0) \int \widehat{f} = f(0) \int \widehat{g}$ . So we only need one  $f$  that  $f(0) = \int \widehat{f}$ ,  $f = e^{-1/2|x|^2}$  will suffice (12.11.3.3). Hence  $g(0) = \int \widehat{g}$  for every such  $g$ , and the conclusion follows by translation (12.11.3.2), and (12.11.1.15) also follows. □

**Cor. (12.11.3.6).** If  $f \in L^1(\mathbb{R}^n)$ , then  $\mathcal{F} f \in C_0(\mathbb{R}^n)$ , and  $\|\mathcal{F} f\|_\infty \leq \|f\|_1$ , because  $\mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$  is dense. ┘

**Prop. (12.11.3.7)[Fourier-Inversion Theorem].** If  $f \in L^1(\mathbb{R}^n)$  and  $\widehat{f} \in L^1(\mathbb{R}^n)$ , then  $\check{f} = \Psi^2 f$  a.e. ┘

*Proof:* In (12.11.3.4), let  $g \in \mathcal{S}$  and substitute  $g = \Psi g$  and use Fubini, we get  $\check{f} - \Psi^2 f$  is orthogonal to every  $\mathcal{S}$ , then every continuous function with compact support by (12.3.8.11). Thus they equal a.e. □

**Cor. (12.11.3.8).** For  $f, g \in \mathcal{S}$ , we have  $\mathcal{F}(fg) = \mathcal{F}(f) * \mathcal{F}(g)$  (apply Fourier one time and use (12.11.3.2)), and thus  $f * g \in \mathcal{S}$ . ┘

**Prop. (12.11.3.9)[Fourier-Plancherel].** If  $f, g \in \mathcal{S}$ , then

$$\int f \bar{g} = \int \bar{g}(x) \widehat{f}(t) e^{ixt} = \int \widehat{f}(t) \int \bar{g}(x) e^{ixt} = \int \widehat{f} \bar{\widehat{g}}$$

by inversion formula. And  $\mathcal{S}$  is dense in  $L^2$ , thus it extends to a linear isometry of  $L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ . This coincides with the Fourier transform on  $L^1 \cap L^2$ . ┘

**Prop. (12.11.3.10) [Tempered Distributions].**  $D$  injects into  $\mathcal{S}$  and is dense. (Notice they both are complete, but the subspace topology are different) (Use scaling, Cf. [Rudin Functional Analysis P189]). So we call a distribution **tempered** iff it comes from a continuous functional of  $\mathcal{S}$ .

From (12.11.2.10), any distribution with compact support is tempered. By Hölder, every  $f \in L^p(\mathbb{R}^n), p \geq 1$  is tempered distribution, and every polynomial or functions of polynomial growth are tempered distribution.

$$D \subset \mathcal{S} \subset L^2 = (L^2)^\vee \subset \mathcal{S}' \subset D'.$$

$\mathcal{S}, \mathcal{S}'$  is complete (12.7.4.2).  $\lrcorner$

**Prop. (12.11.3.11).**  $f \in \mathcal{S}'$  iff  $f = \sum_{|\alpha| \leq m} D_\alpha(u_\alpha(1 + |x|^2)^{m/2})$  for some  $m \in \mathbb{Z}_+$ , where  $u_\alpha \in L^2(\mathbb{R}^n)$ .  $\lrcorner$

*Proof:* In fact,

$$\|\varphi\|'_m = \left( \sum_{|\alpha| \leq m} \int (1 + |x|^2)^m |D_\alpha \varphi|^2 dx \right)^{1/2}$$

is an equivalent set of norms of  $\mathcal{S}'$ , Cf. [泛函分析张恭庆 P182]. And each of them defines a Hilbert space. So by Riesz we get the result.  $\square$

**Prop. (12.11.3.12) [Fourier Transform of Tempered Distributions].** Define the **Fourier transformation** of a tempered distribution  $u \in \mathcal{S}'(\mathbb{R}^n)$  as the tempered distribution  $\mathcal{F}u(\varphi) = u(\mathcal{F}\varphi)$ .

It is easily verified this definition is compatible with previously defined Fourier transform when seen as tempered distributions by?? In particular, this is defined for compactly supported distribution,  $L^p(\mathbb{R}^n), p \geq 1$  and smooth functions of polynomial growth (12.11.3.10).  $\lrcorner$

**Prop. (12.11.3.13) [ $\mathcal{S}' = \mathcal{F}(\mathcal{S}')$ ].** For  $u \in \mathcal{S}'(\mathbb{R}^n)$ , we have

$$\mathcal{F}(P(D)u) = P \mathcal{F}u, \quad \mathcal{F}(Pu) = P(-D) \mathcal{F}u.$$

And the Fourier transformation is a continuous linear isometry of  $\mathcal{S}'$  in the weak\* topology.  $\lrcorner$

**Cor. (12.11.3.14).**  $\hat{1} = \delta$ , thus for any  $P \in \mathbb{R}[X_1, \dots, X_n]$ , we have

$$\mathcal{F}P = P(-D)\delta, \quad \mathcal{F}(P(D)\delta) = P.$$

Now (12.11.2.11) tells us a distribution is the Fourier transform of a polynomial iff it has support in the origin.  $\lrcorner$

**Prop. (12.11.3.15) [Convolution of Tempered Distributions].** For  $u \in \mathcal{S}'(\mathbb{R}^n)$  and  $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$ ,

- $u * \varphi$  is smooth of polynomial growth and  $D^\alpha(u * \varphi) = (D^\alpha u) * \varphi = u * (D^\alpha \varphi)$ .
- $u * (\varphi * \psi) = (u * \varphi) * \psi$ .
- If  $P \in \mathbb{R}[X_1, \dots, X_n]$ , then  $D^\alpha u, Pu$  and  $\varphi u$  are all tempered.
- $\mathcal{F}(u * \varphi) = \mathcal{F}\varphi \cdot \mathcal{F}u, \mathcal{F}u * \mathcal{F}\varphi = \mathcal{F}(\varphi u)$ .

$\lrcorner$

*Proof:* Cf. [Rudin Functional Analysis P195] for the first 3.  $\square$

**Poisson's Summation Formula**

**Def. (12.11.3.16) [Poisson's Summation Formula for Lattices].** Let  $V \in \text{Vect}^n/\mathbb{R}$  with an Haar measure  $\mu$ ,  $\Gamma \subset V$  a complete lattice, and  $\Gamma' \subset V^\vee$  be its  $\mathbb{Z}$ -dual, then for any  $f \in \mathcal{S}(V)$ , we have

$$\sum_{x \in \Gamma} f(x) = \frac{1}{\text{Vol}(V/\Gamma)} \sum_{y \in \Gamma'} \mathcal{F}f(y).$$

In fact it suffices to assume that  $f(x) = O((1 + |x|)^{-d-\delta})$ ,  $\text{Fou}f(x) = O((1 + |x|)^{-d-\delta})$  for some  $\delta \in \mathbb{R}_+$ . These functions are called **admissible functions**.  $\lrcorner$

*Proof:* Use (12.11.1.33). It suffices to show that the function  $F(x) = \sum_{\gamma \in \Gamma} f(x + \gamma)$  is locally uniformly continuous, which is clear.  $\square$

**Cor. (12.11.3.17) [1-Dimensional Poisson's Summation Formula].** Suppose  $f \in \mathcal{S}(\mathbb{R})$ , then we have

$$\sum_{m \in \mathbb{Z}} f(m) = \sum_{n \in \mathbb{Z}} \mathcal{F}f(n).$$

And by changing  $f(x)$  into  $f(ux + v)$  where  $u \in \mathbb{R}_+$  and  $v \in \mathbb{R}$ , we have

$$\sum_{m \in \mathbb{Z}} f(vm + u) = \frac{1}{v} \sum_{n \in \mathbb{Z}} (\mathcal{F}f)\left(\frac{n}{v}\right) e^{2\pi i \frac{un}{v}},$$

or equivalently

$$\sum_{m \in \mathbb{Z}} f\left(\frac{n}{v}\right) e^{2\pi i \frac{un}{v}} = v \sum_{m \in \mathbb{Z}} (\mathcal{F}f)(vm + u).$$

More generally, for any  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}(N)$ , we have

$$\sum_{m \in \mathbb{Z}} f(m) \chi(m) = \frac{\tau(\chi)}{N} \sum_{n \in \mathbb{Z}} (\mathcal{F}f)\left(\frac{n}{N}\right) \bar{\chi}(n) \quad (35.3.6.8).$$

$\lrcorner$

*Proof:* The last assertion follows from (35.3.6.11).  $\square$

**Cor. (12.11.3.18).** For  $y \in \mathbb{R}_+$ , we have

$$\sum_{n \in \mathbb{Z}} e^{2\pi i n x} e^{-\pi n^2/y} = \sqrt{y} \sum_{m \in \mathbb{Z}} e^{-\pi(m+x)^2 y}.$$

In particular, we have

$$\sum_{n \in \mathbb{Z}} e^{-\pi n^2/y} = \sqrt{y} \sum_{m \in \mathbb{Z}} e^{-\pi m^2 y}.$$

$\lrcorner$

**Prop. (12.11.3.19) [Fourier-Transform of Radial Functions].** Let  $d \in \mathbb{Z}_+$  and let  $f \in L^1(\mathbb{R}^d)$  be a radial function, then w.r.t, the Euclidean measure,

$$\mathcal{F}(f)(t) = 2\pi |t|^{1-\frac{d}{2}} \int_0^\infty f(r) J_{d/2-1}(2\pi r|t|) r^{d/2} dr,$$

where  $J_{d/2-1}$  is the Bessel function of order  $d/2 - 1$  ?. In particular,  $\mathcal{F}(f)$  is also radial.  $\lrcorner$

*Proof:* Cf.[Andrews, R. Askey, and R. Roy, Special Functions]9.10.3? . □

**Cor. (12.11.3.20).** Let  $k \in \mathbb{Z}_+$  and let  $f = g(|x|^2)|x|^{1-k/2} \in C_c^\infty(\mathbb{R}^d)$  where  $g \in C_c^\infty(\mathbb{R}_+)$ , then

$$\mathcal{F}(f)(y) = h(|y|^2)|y|^{1-k/2},$$

where

$$h(y) = \pi \int_0^\infty J_{k/2-1}(2\pi\sqrt{xy})g(x)dx.$$

┘

*Proof:* Cf.[I-K04]P90. □

**Def. (12.11.3.21) [Spherical Harmonics].** Let  $k \in \mathbb{Z}_+$  and  $L = \frac{\partial^2}{\partial x_1^2} + \dots \frac{\partial^2}{\partial x_k^2}$  is the Laplacian-operator on  $\mathbb{R}^k$ , then a homogenous polynomial  $P \in \mathbb{C}[x_1, \dots, x_k]$  is called a **spherical harmonic polynomial** if it satisfies  $L(P) = 0$ .

If  $P$  is a spherical harmonic polynomial of degree  $d$ , then we have

$$\mathcal{F}(P(x)e^{-\pi x^2})(y) = i^d P(y)e^{-\pi y^2}.$$

┘

**Prop. (12.11.3.22) [Spherical Harmonics and Fourier-Transforms of Radial Functions].** If  $P$  is a spherical harmonic polynomial of degree  $d$ (12.11.3.21), then for any  $g \in C_c^\infty(\mathbb{R}_+)$  we have

$$\mathcal{F}\left(P(y)g(|x|^2)|x|^{1-k/2}\right) = P(y)h(|y|^2)|y|^{1-k/2},$$

where

$$h(y) = \pi \int_0^\infty J_{k/2+d-1}(2\pi\sqrt{xy})g(x)dx \text{ (12.11.3.20).}$$

┘

*Proof:* ? Compare with(12.11.3.20). □

### Paley-Wiener Theory

**Thm. (12.11.3.23) [Paley-Wiener Theorem].** For  $\varphi \in D(\mathbb{R}^n)$  with support in  $rB$ , the You-Know-How defined  $\widehat{\varphi}(z)$  is an entire function of several variable and satisfies:

$$|\varphi^\vee(z)| \leq \gamma_N(1 + |z|)^{-N} e^{r|\operatorname{Im} z|}.$$

For  $N \geq 0$ . Conversely, any such function correspond to a  $\varphi \in D(\mathbb{R}^n)$  that has support in  $rB$ . ┘

*Proof:* Cf.[Rudin P198]. □

**Prop. (12.11.3.24) [Fourier-Laplace transformation].** For  $u \in D'(\mathbb{R}^n)$  of order  $N$  that has support in  $rB$ , the  $\widehat{u}(z) = u(e_{-z})$  is an entire function of several variable and satisfies:

$$|f(z)| \leq \gamma(1 + |z|)^N e^{r|\operatorname{Im} z|}.$$

Conversely, any such function correspond to a  $u \in D'(\mathbb{R}^n)$  that has support in  $rB$ . ┘

*Proof:* Cf.[Rudin P199]. □

### Gaussian Integrals

**Notation (12.11.3.25).** Let  $V \in \text{Vect}^d/\mathbb{R}$  and  $M(V)$  be the set of non-degenerate complex symmetric bilinear forms  $B$  on  $V$  with  $\text{Re } B \geq 0$ . And let  $M^0(V) \subset M(V)$  be the open dense subset consisting of forms  $B$  with  $\text{Re } B > 0$ .

Suppose  $V$  is endowed with an Haar measure  $dx$ , so we can define  $\det B$  for  $B \in M(V)$ . As  $M(V)$  is star-shaped thus simply-connected, we can also define the principle square  $(\det B)^{1/2}$ .

Any such  $B$  induces an isomorphism  $\varphi_B : V \rightarrow V^* : x \mapsto B(x, -)$ , and thus induces an inner product on the dual space:

$$B^{-1}(l, l') = B(\varphi_B^{-1}l, \varphi_B^{-1}l') = l(\varphi_B^{-1}l').$$

And there are isomorphisms:

$$B \mapsto B^{-1} : M(V) \cong M(V^*), \quad M^0(V) \cong M^0(V^*).$$

┘

**Lemma (12.11.3.26) [Gaussian Integral].** For  $B \in M(V)$ ,  $e^{-\frac{1}{2}B(x,x)} \in \mathcal{S}(V)$ , and its Fourier transform is

$$\mathcal{F}(e^{-\frac{1}{2}B(x,x)})(p) = (\det B)^{-1/2} e^{-\frac{1}{2}B^{-1}(p,p)}, \quad p \in V^*.$$

┘

*Proof:* By continuity ?, it suffices to prove this for  $B \in M^0(V)$ . In this case  $B$  is diagonalizable and the statement is reduced to the case  $\dim V = 1$ . In this case, it suffices to show that for any  $a \in \mathbb{C}, \text{Re } a > 0$  and  $p \in \mathbb{R}$ ,

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ipx - \frac{1}{2}ax^2} dx = \frac{1}{\sqrt{a}} e^{-\frac{1}{2a}p^2}.$$

And this follows from??.

□

**Prop. (12.11.3.27).** For  $g \in \mathcal{S}(V)$ ,  $B \in M(V)$ , consider the integral

$$I_g(\hbar) = \int_V g(\hbar x) e^{-B(x,x)} dx, \quad \hbar \geq 0,$$

then  $I_g \in C^\infty([0, \infty))$  with

$$I_g'(\hbar) = I_{\frac{1}{2}\Delta_B(g)}(\hbar) \text{ (13.1.3.2).}$$

And also

$$I_g(0) = (2\pi)^{d/2} \frac{g(0)}{\sqrt{\det B}}.$$

┘

*Proof:* The last assertion follows from (12.11.3.26) where  $p = 0$ .

Then we prove  $I_g$  is continuous: Only the continuity at  $\hbar = 0$  is non-trivial. By Plancherel formula,

$$I_g(\hbar) = \left( g(\hbar^{\frac{1}{2}}x), e^{-\frac{1}{2}B(x,x)} \right) = \hbar^{-\frac{d}{2}} (\det B)^{-\frac{1}{2}} (g^\vee(\hbar^{-\frac{1}{2}}p), e^{-\frac{1}{2}B^{-1}(p,p)}) = (\det B)^{-\frac{1}{2}} (g^\vee(p), e^{-\frac{\hbar}{2}B^{-1}(p,p)}).$$

But  $e^{-\frac{\hbar}{2}B^{-1}(p,p)} \rightarrow 1$  in  $\mathcal{S}'(V^*)$  as  $\hbar \rightarrow 0$ , so

$$\lim_{\hbar \rightarrow 0} I_g(\hbar) = (\det B)^{-\frac{1}{2}} (g^\vee(p), 1) = (2\pi)^{\frac{d}{2}} (\det B)^{-\frac{1}{2}} g(0) = I_g(0).$$

To prove smoothness: If  $\hbar > 0$ , then by differentiation,

$$I_g'(h) = \frac{1}{2}h^{-1}I_{Eg}(\hbar),$$

where  $E = \sum_i e_i^* \partial_{e_i}$  is the Euler vector field on  $V$  (13.1.3.1). So by (12.11.3.28),

$$I_g'(\hbar) = I_{\frac{1}{2}\Delta_B g}(\hbar).$$

Then it suffices to prove that  $I_g \in C^1([0, \infty))$  and use induction. For this, notice first that if  $C$  is a positive definite real form on  $V$ , then by (12.11.3.26),

$$I_{e^{-\frac{1}{2}C(x,x)}}(\hbar) = \int_V e^{-\frac{1}{2}(B+\hbar C)(x,x)} dx = (2\pi)^{\frac{d}{2}} \det(B + \hbar C)^{-\frac{1}{2}}$$

is smooth. Thus by linearity, we may assume that  $g(0) = 0$ . Then it follows from Taylor expansion (12.3.3.17) that  $g$  is a linear combination of functions of the form  $lf$ , where  $l \in V^*$  and  $f \in \mathcal{S}(V)$ . Thus we may assume  $g = lf$ , and then it follows from (12.11.3.28) that

$$\lim_{\hbar \rightarrow 0} \frac{I_g(\hbar) - I_g(0)}{\hbar} = I_{\varphi_B^{-1}(l)f}(\hbar),$$

which is just  $I_{\frac{1}{2}\Delta_B g}(\hbar)$ , because

$$\frac{1}{2}\Delta_B g(0) = \sum_j l(e_j) \partial_{\varphi_B^{-1}e_j^*} f(0) = \partial_{\varphi^{-1}(l)} f(0).$$

This proves that the assertion. □

**Lemma (12.11.3.28).** For  $f \in \mathcal{S}(V)$  and  $l \in V^*$ ,

$$I_{lf}(\hbar) = \hbar I_{\varphi_B^{-1}(l)f}(\hbar).$$

┘

*Proof:*

$$\begin{aligned} I_{lf}(\hbar) &= \hbar^{\frac{1}{2}}(l(x)f(\hbar^{\frac{1}{2}}x), e^{-\frac{1}{2}B(x,x)}) = \hbar^{\frac{1}{2}}(f(\hbar^{\frac{1}{2}}x), l(x)e^{-\frac{1}{2}B(x,x)}) \\ &= -\hbar^{\frac{1}{2}}(f(\hbar^{\frac{1}{2}}x), \partial_{\varphi_B^{-1}(l)} e^{-\frac{1}{2}B(x,x)}) = \hbar^{\frac{1}{2}}(\partial_{\varphi_B^{-1}(l)} f(\hbar^{\frac{1}{2}}x), e^{-\frac{1}{2}B(x,x)}) \\ &= \hbar((\partial_{\varphi_B^{-1}(l)} f)(\hbar^{\frac{1}{2}}x), e^{-\frac{1}{2}B(x,x)}) = \hbar I_{\varphi_B^{-1}(l)f}(\hbar) \end{aligned}$$

□

**Prop. (12.11.3.29) [Riemann's Lemma].** Let  $D \subset \mathbb{R}^n$  be a compact region with smooth boundary, and  $f, g \in C^\infty(D)$  s.t.  $g^{(n)}$  vanishes on the boundary for any  $n \in \mathbb{N}$ , and  $df \neq 0$  on  $\text{Supp}(g)$ , then the function

$$I(\hbar) = \int_D g(x) e^{\frac{if(x)}{\hbar}} dx$$

extends to a smooth function on  $[0, \infty)$ , and has rapidly decaying derivatives of all orders as  $\hbar \rightarrow 0$ .

┘

*Proof:* As  $\text{Supp } g$  is compact and  $df \neq 0$  on  $\text{Supp } g$ , we can cover  $D$  by f.m. charts s.t.  $f(x) = x_1$ . Then by using smooth partition of unity, we may reduce to the case that  $f(x) = x_1$ . Then we are reduced to the 1-dimensional case and  $f(x) = x$ . Then in this case, it suffices to notice that

$$\int_a^b g(x) e^{i\frac{x}{\hbar}} dx = i\hbar \int_a^b g'(x) e^{i\frac{x}{\hbar}} dx,$$

and use induction. □

### Steepest Descent and Stationary Phase Formulae

**Thm. (12.11.3.30) [Steepest Descent Formula].** Let  $D \subset \mathbb{R}^n$  be a compact region with smooth boundary and  $f, g \in C^\infty(D)$ . Assume that  $f$  attains a global minimal at a unique non-degenerate critical point  $c \in D$  with  $\text{Hess } f(c)$  positive definite, then

$$\int_D g(x) e^{-f(x)/\hbar} dx = \hbar^{\frac{d}{2}} e^{-f(c)/\hbar} I(\hbar)$$

where  $I(x)$  extends to a smooth function on  $[0, \infty)$  with

$$I(0) = (2\pi)^{\frac{d}{2}} \frac{g(c)}{\sqrt{\det \text{Hess } f(c)}}.$$

Moreover, this generalizes to the case that  $f$  attains global minimums at multiple non-degenerate critical points.  $\square$

*Proof:* WLOG, we may assume that  $c$  is the origin and  $f(c) = 0$ . By Morse lemma, there is a smooth coordinate near the origin that  $f(x) = \frac{1}{2} \text{Hess}(f)(x, x)$  on a nbhd of the origin. Then we can use smooth partition of function to reduce to the case  $g$  has support on a small nbhd of 0 or the case that  $g$  vanishes near 0. For the last case, it is clear by direct differentiation that  $I(\hbar)$  is rapidly decaying as  $\hbar \rightarrow 0$ . And for the first case, by a change of variable,

$$I(\hbar) = \int_{\mathbb{R}^n} g'(\hbar y) e^{-\frac{1}{2} \text{Hess } f y^2} dy.$$

Then we finish by (12.11.3.27).  $\square$

**Thm. (12.11.3.31) [Stationary Phase Formula].** Let  $D \subset \mathbb{R}^n$  be a compact region with smooth boundary, and  $f \in C^\infty(D), g \in C^\infty(D)$ . Assume that  $f$  has a unique critical point  $c \in D$  and  $f$  is non-degenerate at  $c$ , and  $g^{(n)}$  vanishes on the boundary for any  $n \in \mathbb{N}$ , then

$$\int_D g(x) e^{if(x)/\hbar} dx = \hbar^{\frac{d}{2}} e^{if(c)/\hbar} I(\hbar)$$

where  $I(x)$  extends to a smooth function on  $[0, \infty)$  with

$$I(0) = (2\pi)^{\frac{d}{2}} e^{\sigma\pi i/4} \frac{g(c)}{\sqrt{|\det \text{Hess } f(c)|}},$$

where  $\sigma$  is the signature of  $\text{Hess } f(c)$ .

Moreover, this generalizes to the case that  $f$  has multiple non-degenerate critical points.  $\square$

*Proof:* The proof is similar to that of (12.11.3.30). In particular, using the partition function to separate the near to 0 part of  $g$ . And the away-from-0 part of  $g$  is smooth and rapidly decaying by Riemann lemma (12.11.3.29). Then by Morse lemma, it reduces to the case that  $f$  is quadratic. Then the assertion follows from (12.11.3.27).  $\square$

### Fourier Analysis on $\mathbb{R}$

**Prop. (12.11.3.32).** For  $f \in L^2(\mathbb{R})$ , the Fourier-transform of  $f$  is given by

$$\mathcal{F}(f) \in C(\mathbb{R}) : y \mapsto \int_{\mathbb{R}} f(x) e^{-2\pi i xy} dx.$$

Then  $\mathcal{F}(f)$  is uniformly continuous, and  $\lim_{|y| \rightarrow \infty} f(y) = 0$  (Riemann-Lebesgue lemma).

And the map  $f \mapsto \mathcal{F}(f)$  is injective, by Fourier's uniqueness theorem (12.11.1.25).  $\square$

**Example (12.11.3.33) [Fourier-Transforms].**

•

$$\mathcal{F}(e^{-2\pi|x|})(y) = \frac{1}{\pi(1+y^2)}.$$

•

$$\mathcal{F}(e^{-\pi x^2})(y) = e^{-\pi y^2}.$$

•

$$\mathcal{F}\left(\frac{1}{\text{ch}(\pi x)}\right)(y) = \frac{1}{\text{ch}(\pi y)}.$$

- If  $Y \in \mathbb{R}_+$  and  $\rho_Y(x) = \max(1 - \frac{|x|}{Y}, 0)$ , then

$$\mathcal{F}(\rho_Y)(x) = \left( \frac{\sin(\pi Y x)}{\pi Y x} \right)^2,$$

called the **Fejér-Kernel**. ┘

**Prop. (12.11.3.34) [Fourier-Inversion Formula].** ┘

#### 4 Sobolev-Spaces

**Def. (12.11.4.1) [Sobolev-Spaces].** For  $1 \leq p < \infty$ , the **Sobolev-space**  $W^{m,p}(\Omega)$  is the space of functions  $u$  that  $D^\alpha u \in L^p(\Omega)$  for every  $|\alpha| \leq m$ , with the norm  $\|u\| = \sum_{|\alpha| \leq m} \int_\Omega |D^\alpha u(x)|^p dx$ . The **Sobolev space**  $W_0^{m,p}(\Omega)$  is the completion of the subspace  $C_0^\infty(\Omega)$ . ┘

**Prop. (12.11.4.2) [Meyers-Serrin].** The Sobolev space  $W^{m,p}(\Omega)$  is the completion of  $u \in C^\infty(\Omega)$  that  $D^\alpha u \in L^p(\Omega)$  for every  $|\alpha| \leq m$ . ┘

*Proof:* Choose a countable partition of unity  $\psi_k$ , then as in the proof of (12.3.8.12), we can choose  $\delta_k$  small enough and  $\|\psi u - (\psi u)_{\delta_k}\| < \varepsilon/2^k$  and  $\varphi = \sum (\psi u)_{\delta_k}$  is definable. □

**Prop. (12.11.4.3).** We denote  $H^m(\Omega) = W^{m,2}(\Omega)$  and  $H_0^m(\Omega) = W_0^{m,2}(\Omega)$  and  $H^{-m}(\Omega) = (H_0^m(\Omega))^*$  when  $m$  is an integer. Notice derivative is not applicable for  $H^{-m}(\Omega)$  unless  $\Omega = \mathbb{R}^n$ .

When  $\Omega = \mathbb{R}^n$ ,  $D(\mathbb{R}^n)$  is dense in  $W^{m,p}(\mathbb{R}^n)$ , thus  $W_0^{m,p} = W^{m,p}$ . Define the **Sobolev-space**

$$H^s = \{u | (1 + |y|^2)^{s/2} |\hat{u}| \in L^2\}$$

$H^s$  is a Hilbert space and  $H^s \subset \mathcal{S}'$  for every  $s$  (use Holder to show  $\hat{u} \in \mathcal{S}'$ ).  $H^m$  coincides with previously defined  $H^m$  when  $m$  is a positive integer thus also negative-integer. A linear operator on  $H = \cup H^s$  is said to have **order**  $t$  if it maps every  $H^s$  continuously into  $H^{s-t}$ . ┘

*Proof:* By Plancherel, we have

$$\|\varphi\|'_m = \left( \sum_{|\alpha| \leq m} \|D_\alpha u\|_2^2 \right)^{1/2} \quad \text{and} \quad \left( \int (1 + |x|^2)^m |\hat{u}|^2 \right)^{1/2}$$

are equivalence norms on  $H^m$ . □

**Lemma (12.11.4.4) [Poincare Inequality].** For  $\Omega$  bounded, on  $C_0^m(\Omega)$  the  $W^{m,p}$  norm is controlled by  $L^p$  norms of its  $m$ th order derivatives. ┘



*Proof:* We may assume  $\Omega \subset \prod_{i=1}^n [0, a]$ , then for any  $u \in W^{m,p}$ ,  $u(x) = \int_0^{x_1} D^1 u(t, x_2, \dots, x_n) dt$ , so by Holder inequality,

$$|u(x)| \leq a^{1/q} \left( \int_0^a |D^1 u|^p dx_1 \right)^{1/p}.$$

so

$$\int_{\Omega} |u(x)|^p dx \leq a^q \int_{\Omega} |D^1 u|^p dx_1.$$

Doing the same for all other derivatives, we can see the norm is controlled by the highest( $m$ -th) order norms.  $\square$

**Prop. (12.11.4.5).** When  $t < s$ ,  $H^s \subset H^t$ . And  $H^s$  are isometric to  $H^t$  by  $\hat{v} = (1 + |y|^2)^{t/2} \hat{u}$  and is of order  $t$ .  $D^\alpha$  is of order  $|\alpha|$ . If  $f \in \mathcal{S}$ , then  $u \rightarrow fu$  is an operator of order 0.

Every distribution of compact support is in some  $H^s$  (12.11.2.10), in particular  $D(\Omega)$ .  $\lrcorner$

*Proof:* Cf. [Rudin P217].  $\square$

**Prop. (12.11.4.6) [Sobolev's Embedding Theorem].** On a manifold of dimension  $n$  which is compact with Lipschitz boundary or complete of positive injective radius and bounded sectional curvature,

- if  $k > l$  be integers and

$$\frac{1}{p} - \frac{k}{n} = \frac{1}{q} - \frac{l}{n}$$

then  $W^{k,p}(\text{int}(M)) \subset W^{l,q}(M)$  continuously.

- if

$$\frac{1}{p} - \frac{k}{n} = -\frac{r+\alpha}{n}$$

then  $W^{k,p}(\text{int}(M)) \subset C^{r,\alpha}(M)$  continuously.  $\lrcorner$

*Proof:* Cf. [Evans P290].  $\square$

**Cor. (12.11.4.7) [Gagliardo–Nirenberg–Sobolev].** On a manifold of dimension  $n$  which is compact with Lipschitz boundary or a complete of positive injective radius and bounded sectional curvature, if  $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$  (Sobolev conjugate), then  $W^{1,p}(\text{int}(M)) \subset L^{p^*}(M)$  continuously.  $\lrcorner$

**Cor. (12.11.4.8).** On a manifold of dimension  $n$  which is compact with Lipschitz boundary or a complete of positive injective radius and bounded sectional curvature, if  $m > n/2$ , then  $W^{m,2}(\text{int}(M)) \subset C(\overline{\Omega})(M)$  continuously. And the functions in  $W_0^{m,2}$  are continuous and vanish at the boundary, by  $C_0$  approximation.  $\lrcorner$

*Proof:* The  $\mathbb{R}^n$  case can be directly proved: because we have the equivalent norm (12.11.4.3),  $\hat{u} \in L^2$  thus  $u \in L^2$ , and

$$\int |\hat{u}| \leq \left( \int (1 + |x|^2)^m |\hat{u}|^2 \right)^{1/2} \cdot \left( \int 1/(1 + |x|^2)^m \right)^{1/2}.$$

We have  $\hat{u} \in L^1$ , thus inversion formula applies that  $u$  is continuous and  $\|u\|_\infty \leq \|\hat{u}\|_1 \leq C\|u\|_{H^m}$ .  $\square$

**Cor. (12.11.4.9).**  $\cap_s H^s = C^\infty(M)$ .  $\lrcorner$

**Prop. (12.11.4.10)[Rellich-Kondrechov].** On a compact manifold with  $C^1$  boundary of dimension  $n$ , if  $k > l$  and

$$\frac{1}{p} - \frac{k}{n} < \frac{1}{q} - \frac{l}{n}$$

then  $W^{k,p} \subset W^{l,q}$  completely continuously.  $\lrcorner$

*Proof:* Cf.[Distributions and Operators P199], [Evans P290].  $\square$

**Cor. (12.11.4.11).** On a bounded extension domain of  $\mathbb{R}^n$ ,  $W^{1,p} \subset L^p$  completely continuously.  $\lrcorner$

*Proof:* We prove the  $p = 2$  case. For a sequence  $u_m$  in  $W^{1,2}$ , we have  $\|u_m - u_p\|_2 = \|U_m - U_p\|_2 = \|\hat{U}_m - \hat{U}_p\|_2$ . By (12.8.4.9), there is a subsequence that  $\hat{U}_m$  pointwise converge. Notice they are uniformly bounded, Now apply two region argument, for  $|x| < r$ , use Lebesgue dominant convergence, and for  $|x| > r$ , use  $\int (1 + |x|^2)|\hat{U}_m - \hat{U}_p|^2$  is bounded to conclude  $\|u_m - u_p\|_2 \rightarrow 0$ .  $\square$

**Prop. (12.11.4.12).**  $u \in D'(\Omega)$  is a locally  $H^s \iff \psi u \in H^s$  for every  $\psi \in D(\Omega) \iff D_\alpha u$  is locally  $L^2$  for every  $|\alpha| \leq s$ .

Thus every smooth function is locally  $H^s$  for every  $s$ .  $\lrcorner$

*Proof:*  $1 \rightarrow 2$  use partition of unity,  $2 \rightarrow 1$  easy, and 2, 3 are all equivalent to  $D_\alpha(\psi u) \in L^2$  for every  $\psi \in D(\Omega)$ . by Leibniz+Plancherel or (12.11.4.5).  $\square$

**Prop. (12.11.4.13).** If  $r > p + n/2$ , then if a function  $f$  on  $\Omega$  has all the distribution derivative  $D_i^k f$  locally  $L^2$ ,  $= g_{is}$ , for  $0 \leq k \leq r$ , then  $f \in C^p(\Omega)$  a.e.  $\lrcorner$

**Cor. (12.11.4.14).** If  $u \in D'(\Omega)$  is locally  $H^s$ , then  $u \in C^{s-n/2}(\Omega)$ . Thus  $\cap \text{locally } H^s = C^\infty(\Omega)$ .  $\lrcorner$

### Holder Space

**Def. (12.11.4.15).** Holder space  $C^{k,\alpha}(\Omega)$  is the subspace of  $C^k(\Omega)$  with the norm

$$\|f\|_{C^{k,\alpha}} = \|f\|_{C^k} + \max_{|\beta|=k} \sup_{x \neq y \in \Omega} \frac{|D^\beta f(x) - D^\beta f(y)|}{|x - y|^\alpha}.$$

$\lrcorner$

## 5 Fourier Analysis on Tori

**Notation (12.11.5.1).**

- Notice  $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ .

$\lrcorner$

**Prop. (12.11.5.2)[Basic Properties of Fourier Analysis].**

- For  $f \in L^1(\mathbb{T})$ , we have the **Fourier-coefficients**

$$c_n(f) \triangleq \mathcal{F}(f)(n) = \int_{\mathbb{T}} f(x) e^{-2\pi i n x} dx.$$

for each  $n \in \mathbb{Z}$ . And

$$\sum_{n \in \mathbb{Z}} c_n(f) e^{2\pi i n x}$$

is called the **Fourier-series** of  $f$ .  $f$  is determined by its Fourier-coefficients, by Fourier's uniqueness theorem (12.11.1.25).

- If  $f \in L^1(\mathbb{T})$  is absolutely continuous, then  $c_n(f') = 2\pi i n \cdot c_n(f)$  for each  $n \in \mathbb{Z}$ , by (12.11.3.5).
- (Fourier-kernels) If  $f \in L^1(\mathbb{T}^n)$  with Fourier series  $\sum_{n \in \mathbb{Z}} c_n(f) e^{2\pi i n x}$ , and

$$P_N = c_n(P_N) \sum_{|n| \leq N} e^{2\pi i n x}$$

is an trigonometric polynomial, then by (12.11.3.2), the Fourier series of  $f * P_N$  is

$$\sum_{|n| \leq N} c_n(f) c_n(P_N) e^{2\pi i n x}.$$

- For  $f, g \in L^2(\mathbb{T}) \subset L^1(\mathbb{T})$  with Fourier-coefficients  $c_n(f), c_n(g)$ , we have

$$(f, g) = \sum_{n \in \mathbb{Z}} c_n(f) c_n(g) \in \mathbb{R}.$$

┘

**Prop. (12.11.5.3) [Pointwise Convergence].** If  $f$  is a periodic function on  $\mathbb{R}$  with period  $2\pi$  and of bounded variance (e.g. piecewise-differentiable), then the Fourier series of  $f$  converges to  $\frac{1}{2}(f(x+) + f(x-))$  pointwise. ┘

*Proof:* [武胜健 2, P264]. □

**Prop. (12.11.5.4) [Uniform Convergence].** If  $f$  is a periodic differentiable function on  $\mathbb{R}$  with period  $2\pi$  and  $f'$  is integrable at  $[-\pi, \pi]$ , then the Fourier series of  $f$  converges to  $f$  uniformly. ┘

*Proof:* [武胜健 2, P281]. □

## **12.12 Non-Commutative Harmonic Analysis(Reductive Groups)**

### **1 Non-Commutative Plancherel Formula(Harish-Chandra)**

## 12.13 Harmonic Analysis on Finite Groups

References are [Fourier analysis on Finite Groups and applications, Terras].

### 1 Finite Abelian Groups

**Def. (12.13.1.1) [Finite Fourier Transform].** Let  $G \in \mathcal{Ab}^{\text{fin}}$  and  $e : G \times G \rightarrow S^1$  is a perfect pairing, then for any  $f : G \rightarrow \mathbb{S}^1$ , we can define the Fourier transform of  $f$  as

$$\mathcal{F}f(\xi) = \frac{1}{\#G} \sum_{x \in G} f(x) \overline{e(x, \xi)}.$$

The matrix  $(e(x, \xi))_{x, \xi \in G}$  is called the **Fourier matrix**. ┘

**Thm. (12.13.1.2) [Uncertainty Principle].** Let  $G \in \mathcal{Ab}^{\text{fin}}$  and  $e : G \times G \rightarrow S^1$  is a perfect pairing, then for any non-zero function  $f : G \rightarrow \mathbb{C}$ , we have:

$$\# \text{Supp}(f) \cdot \# \text{Supp}(\mathcal{F}f) \geq \#G.$$

┘

*Proof:*

$$\begin{aligned} \sup_{\xi \in G} |\mathcal{F}f(\xi)| &\leq \frac{1}{\#G} \sum_{x \in G} |f(x)| \\ &\leq \sqrt{\frac{\# \text{Supp}(f)}{\#G}} \left( \frac{1}{\#G} \sum_{x \in G} |f(x)| \right) \\ &= \sqrt{\frac{\# \text{Supp}(f)}{\#G}} \left( \sum_{x \in G} |f(x)| \right) \\ &\leq \sqrt{\frac{\# \text{Supp}(f) \cdot \# \text{Supp}(f^\vee)}{\#G}} \sup_{\xi \in G} |f^\vee(\xi)| \end{aligned}$$

Thus the assertion follows. □

### Finite Fields

**Lemma (12.13.1.3).** Suppose  $p \in \text{Prime}$ ,  $n \in \mathbb{Z}_+$ ,  $P(\underline{X}) \in \mathbb{Z}[X_1, \dots, X_n]$  and  $\omega_i \in \mu_p$  s.t.  $P(\omega_1, \dots, \omega_n) = 0$ , then  $p | P(1, \dots, 1)$ . ┘

*Proof:* Suppose  $\omega_i = e^{2\pi i k_i / p}$ , and consider

$$Q(z) = P(z^{k_1}, \dots, z^{k_n}) \pmod{z^n - 1},$$

which satisfies  $Q(e^{2\pi i / p}) = 0$  and  $\deg(Q) \leq p - 1$ . Thus  $Q$  is a multiple of the polynomial  $1 + X + \dots + X^{p-1}$ . Thus  $p | P(1, \dots, 1)$  holds. □

**Thm. (12.13.1.4)**[Fourier Matrix Invertible for Prime Fields, Chebotarev(37)]. The standard pairing  $\mathbb{F}_p \times \mathbb{F}_p \rightarrow S^1 : (x, \xi) \mapsto e^{2\pi i x \xi / p}$  induces the Fourier matrix  $(e^{2\pi i x \xi / p})_{x, \xi \in \mathbb{F}_p}$  (12.13.1.1). Then all its principal minors are non-degenerate.

In particular, for any  $A, A' \subset \mathbb{F}_p$  with the same cardinality,

$$\ell^2(A) \rightarrow \ell^2(A') : f \mapsto \mathcal{F}f|_{A'}$$

is invertible. ┘

*Proof:* For any  $n \in [p]_+$  and  $\{x_1, \dots, x_n\} \subset [p-1]$ ,  $\{\xi_1, \dots, \xi_n\} \subset [p-1]$ , denote

$$D(X_1, \dots, X_n) = \det \left( X_i^{\xi_j} \right)_{i, j \in [n]_+} \in \mathbb{Z}[X_1, \dots, X_n].$$

then  $X_i - X_j | D(X_1, \dots, X_n)$  for any  $i < j \in [n]_+$  and we denote

$$D(X_1, \dots, X_n) = P(X_1, \dots, X_n) \prod_{i < j \in [n]_+} (X_i - X_j), \text{ where } P(X_1, \dots, X_n) \in \mathbb{Z}[X_1, \dots, X_n].$$

Then we have

$$\left( Z_1 \frac{\partial}{\partial Z_1} \right) \left( Z_2 \frac{\partial}{\partial Z_2} \right)^2 \cdots \left( Z_n \frac{\partial}{\partial Z_n} \right)^n D(X_1, \dots, X_n) \Big|_{(X_1, \dots, X_n) = (1, \dots, 1)} = (n-1)! \cdots 2! \cdot 1! P(1, \dots, 1).$$

And the LHS equals

$$\det \text{Van}(\xi_1, \dots, \xi_n) \neq 0 \text{ (3.5.9.18)}.$$

Thus  $P(1, \dots, 1) \neq 0$ , and by (12.13.1.3), we have  $P(e^{2\pi i x_1/p}, \dots, e^{2\pi i x_n/p}) \neq 0$ . Thus  $D(x_1, \dots, x_n) \neq 0$ , and the assertion follows. □

**Cor. (12.13.1.5)**[Tao[Tao05]]. If  $p \in \text{Prime}$  and  $f : \mathbb{F}_p \rightarrow \mathbb{C}$  is a non-zero function, then

$$\# \text{Supp}(f) + \# \text{Supp}(\mathcal{F}f) \geq p + 1.$$

Conversely, if  $A, B \subset \mathbb{Z}/(p)$  are two non-empty subsets s.t.  $\#A + \#B \geq p + 1$ , then there exists a non-zero function  $f : \mathbb{Z}/(p) \rightarrow \mathbb{C}$  s.t.  $\text{Supp}(f) = A$  and  $\text{Supp}(f^\vee) = B$ . ┘

*Proof:* If  $\# \text{Supp}(f) + \# \text{Supp}(f^\vee) \leq p$ , then there exists a subset  $A' \subset \mathbb{F}_p$  disjoint with  $\text{Supp}(\mathcal{F}f)$ . This contradicts (12.13.1.4).

Conversely, it suffices to prove that if  $A, B \subset \mathbb{Z}/(p)$  are two non-empty subsets s.t.  $\#A + \#B = p + 1$ , then there exists a non-zero function  $f : \mathbb{Z}/(p) \rightarrow \mathbb{C}$  s.t.  $\text{Supp}(f) = A$  and  $\text{Supp}(\mathcal{F}f) = B$  (because the  $\#A + \#B \geq p + 1$  case follows from taking linear combinations).

Then we choose  $A' \subset \mathbb{F}_p$  s.t.  $A' \cap B = \{\xi\}$ . Then by (12.13.1.4) there exists  $f : \mathbb{F}_p \rightarrow \mathbb{C}$  with  $\text{Supp}(f) \subset A$  and  $\mathcal{F}f$  vanishes on  $A' \setminus \{\xi\}$  and  $\mathcal{F}(f)(\xi) \neq 0$ . Then  $\text{Supp}(f) = A$  and  $\text{Supp}(\mathcal{F}f) = B$ , otherwise the uncertainty principle just proven is contradicted. □

**Cor. (12.13.1.6).** If  $p \in \text{Prime}$ ,  $k \in [p-1]$  and  $f(X) = \sum_{j=0}^k a_j X^{n_j} \in \mathbb{C}[X]^\times$  and  $0 \leq n_0 < n_1 < \dots < n_k < p$ , then  $f$  has at most  $k$ -zeros (without multiplicity) in  $\mu_p$ . ┘

*Proof:* This is because the restriction of  $f$  on  $\mu_p$  is basically the Fourier transform of a function on  $\mu_p$  whose support has cardinality  $\leq k + 1$ . □

## 2 Finite Non-Abelian Groups

# 13 | Differential Equations and Dynamical Systems

## 13.1 Differential Equations

### 1 ODE-Fundamentals

**Prop. (13.1.1.1).**

$$x^{(2)} = f(x)$$

It can be solved. ┘

*Proof:*

$$\begin{aligned} x'x^{(2)} &= f(x)x' \\ \frac{1}{2}(x')^2 &= \int^x f(t)dt \end{aligned}$$

□

**Prop. (13.1.1.2) [Wronsky].** ┘

### 2 ODE-Theorems

**Prop. (13.1.2.1) [Existence and Uniqueness of ODE of Lipschitz Type].** If  $F(t, x)$  defined on  $[-h, .h] \times [\eta - \delta, \eta + \delta]$  is a function that is locally Lipschitz: that is,  $\exists \delta, L$ , s.t. if  $|t| \leq h, |x_i - \eta| \leq \delta$ , then

$$|F(t, x_1) - F(t, x_2)| \leq L|x_1 - x_2|.$$

Then the initial value problem:

$$(Tx)(t) = \eta + \int_0^t F(\tau, x(\tau))d\tau$$

has a unique solution on the interval  $[-h, h]$  if  $h < \min\{\delta/M, 1/L\}$ , where  $M$  is the maximum of  $F$  on  $[-h, .h] \times [\eta - \delta, \eta + \delta]$ . Because  $T$  is a contraction. ┘

**Prop. (13.1.2.2) [Existence of ODE of Continuous Type (Caratheodory)].** If  $F(t, x)$  defined on  $[-h, .h] \times [\eta - \delta, \eta + \delta]$  is a continuous function, then

$$(Tx)(t) = \eta + \int_0^t F(\tau, x(\tau))d\tau$$

has a unique solution on the interval  $[-h, h]$  if  $h < \delta/M$ , where  $M$  is the maximum of  $F$  on  $[-h, .h] \times [\eta - \delta, \eta + \delta]$ . (Use Schauder fixed point theorem and Arzela-Ascoli). ┘

**Thm. (13.1.2.3) [Existence Theorem for Complex Differential Equations].** Let  $\Omega \subset \mathbb{C}^{n+1}$  be a domain and  $f(z, \mathbf{w}) : \Omega \rightarrow \mathbb{C}^n$ , then the initial value problem

$$\mathbf{w}' = f(z, \mathbf{w}), \quad \mathbf{w}(z_0) = \mathbf{w}_0$$

has exactly one holomorphic solution locally (Thus on a simply connected domain).  $\lrcorner$

**Cor. (13.1.2.4).** So a holomorphic high-order ODE for a complex variable can be solved. And luckily it can be solved even  $\bar{z}$  appears (just regard it as a constant).  $\lrcorner$

*Proof:* Cf. [Ordinary Differential Equations, P110].  $\square$

**Prop. (13.1.2.5).** For the equation:

$$\frac{d\mathbf{y}}{d\mathbf{x}} = \mathbf{A}\mathbf{y},$$

One solution basis is:

$$\begin{cases} e^{\lambda_1 x} \mathbf{P}_1^{(1)}(x), \dots, e^{\lambda_1 x} \mathbf{P}_{n_1}^{(1)}(x); \\ \dots\dots\dots \\ e^{\lambda_s x} \mathbf{P}_1^{(d)}(x), \dots, e^{\lambda_s x} \mathbf{P}_{n_s}^{(1)}(x); \end{cases}$$

Where

$$\mathbf{P}_j^{(i)}(x) = \mathbf{r}_{j0}^{(i)} + \frac{x}{1!} \mathbf{r}_{j1}^{(i)} + \dots,$$

where  $\mathbf{r}_{j0}^{(i)}$  is a basis of solution of  $(\mathbf{A} - \lambda_i I)^n \mathbf{x} = 0$ , and  $\mathbf{r}_{k+1}^{(i)} = (\mathbf{A} - \lambda_i I) \mathbf{r}_k^{(i)}$ .  $\lrcorner$

*Proof:* Cf. [常微分方程丁同仁定理 6.6].  $\square$

**Cor. (13.1.2.6).** For the equation:

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = 0,$$

If the characteristic equation has  $s$  different roots  $\lambda_1, \dots, \lambda_s$  and corresponding multiplicities  $n_1, \dots, n_s$ , then:

$$\begin{cases} e^{\lambda_1 x}, x e^{\lambda_1 x}, \dots, x^{n_1-1} e^{\lambda_1 x}; \\ \dots\dots\dots \\ e^{\lambda_s x}, x e^{\lambda_s x}, \dots, x^{n_s-1} e^{\lambda_s x}; \end{cases}$$

is a solution basis.  $\lrcorner$

*Proof:* Cf. [常微分方程丁同仁 P198].  $\square$

**Prop. (13.1.2.7) [Lyapunov].** Consider the Lyapunov stability of an autonomous system of the form:

$$\frac{\partial}{\partial t} x = Ax + o(|x|),$$

Then:

1. If  $A$  has a eigenvalue whose real part is positive, then the trivial solution is not weak stable.
2. If all eigenvalues of  $A$  has negative real part, then the trivial solution is strong stable.

$\lrcorner$



Stum-Liouville

**Prop. (13.1.2.8) [Stum-Liouville].** The eigenvalue BVP problem of L-S equation:

$$Lu = (pu')' + qu = \lambda u, \quad a_1 u(a) + a_2 u'(a) = 0, b_1 u(b) + b_2 u'(b) = 0, \sigma(x) > 0.$$

can be solved by the method of Green's function. For the function:

$$G(x, s) = \begin{cases} Cu_1(x)u_2(s), & x < s \\ Cu_2(x)u_1(s), & x > s \end{cases}$$

for some  $C$ , where  $u_1$  is a solution of the L-S equation with boundary value at  $a$ , and  $u_2$  with boundary value at  $b$  that are linear independent (This happens when the homogenous equation has no solution). It satisfies:  $LG(x, s) = \delta(x - s)$  and satisfies the boundary conditions.

Because  $L$  is self-adjoint, we have:

$$Gf(x) = \int f(s)G(x, s)ds, LG = \text{id}, GL = \text{id}$$

thus the eigenvalues of  $L$  is the reciprocal of the eigenvalues of  $G$ , and  $G$  is a compact self-adjoint operator on  $L^2(\sigma, \mathbb{R})$ , so by spectral theorem, the eigenvectors are countable and form an orthonormal basis.

And when the homogenous problem do have a solution  $\phi$ , then we have:  $Lu = f$  has a solution iff  $(f, \phi) = 0$ . one way is simple and the other way is because we solve the initial problem of ODE and find that it automatically satisfies the boundary condition. Cf. [Stum Liouville Theory].  $\lrcorner$

**Prop. (13.1.2.9).** More generally, if there the boundary is mixed of  $u(a), U'(a), u(b), u'(b)$ , the solution of

$$Lu = (pu')' + qu = 0, B_1(u) = \alpha, B_2(u) = \beta.$$

has a unique solution for any  $\alpha, \beta$  iff the homogenous equation has only non-trivial solution. (Because the solution space is of 2 dimensional.  $\lrcorner$

**Prop. (13.1.2.10) [Stum Seperation Theorem].**  $\lrcorner$

*Proof:*  $\square$

**Prop. (13.1.2.11) [Stum Comparison Theorem].** If  $y'' + K_i(x)y = 0$  are equations. If  $y_i(0) = 0$  and  $|y_1'(0)| = |y_2'(0)|$ , then if  $K_1(x) \geq K_2(x)$ , then  $y_1(x) \geq y_2(x)$  until  $y_2(x)$  is zero. (directly from (16.3.5.10)).  $\lrcorner$

*Proof:*  $\square$

### 3 Linear PDE

#### Linear Operators

**Def. (13.1.3.1) [Euler Differential Operator].** For  $V \in \text{Vect}/\mathbb{R}$ , the **Euler differential** is the first order linear differential operator

$$E = \sum_i e_i^* \partial_{e_i},$$

where  $\{e_i\}$  is any basis of  $V$ .  $\lrcorner$

*Proof:* □

**Def. (13.1.3.2) [Laplacians].** For  $V \in \text{Vect}/\mathbb{R}$  and a non-degenerate bilinear form  $B$  on  $V$ , the **Laplacian operator** of  $B$  is defined to be the second order linear differential operator

$$\Delta_B = \sum_i \partial_{\varphi_B^{-1} e_i^*} \circ \partial_{e_i} : \mathcal{S}(V) \rightarrow \mathcal{S}(V)$$

┘

*Proof:* □

### Existence of Solutions

**Def. (13.1.3.3) [Fundamental Solutions].** For a linear PDE with constant coefficients  $P(D)u = v$ , the **fundamental solution** is a distribution  $E \in D'(\mathbb{R}^n)$  that  $P(D)E = \delta$ . This is important because if  $v$  is a distribution with compact support,  $P(D)(E * v) = (P(D)E) * v = \delta * v = v$  (12.11.2.18), so  $u = E * v$  is a distribution solution. ┘

**Prop. (13.1.3.4).** When  $v \in D'(\mathbb{R}^n)$  has compact support,  $P(D)u = v$  has a solution  $u$  with compact support iff  $Pg = \hat{v}$  has a solution  $g$  entire. In this case,  $g = \hat{u}$  for some distribution  $u$ , and  $u$  has support in the convex hull of the support of  $v$ . ┘

*Proof:* Use (12.11.3.24), and some bound relation between  $g$  and  $Pg$ . Cf. [Rudin Functional Analysis P212]. □

**Prop. (13.1.3.5).** The fundamental solution always exist when for PDE of constant coefficients. ┘

*Proof:* For a  $\varphi \in D(\mathbb{R}^n)$ , there is at most one  $\psi$  that  $\psi = P(D)\varphi$  because  $\hat{\psi} = P\hat{\varphi}$  and they are entire function. Thus the task is to verify the functional  $u : P(D)\varphi \rightarrow \varphi(0)$  is continuous and extend to a distribution  $u \in D'(\mathbb{R}^n)$ . Cf. [Rudin Functional Analysis P215]. □

## 4 Differential Operators on Manifolds

**Def. (13.1.4.1) [Differential Operators].** For  $X \in \text{Diff}$ , define  $D_{-1,X} = 0$ , and for any  $n \in \mathbb{N}$ , define  $D_{n,X}$  to be the subbundle of  $\text{End}(\mathcal{O}_X)$  s.t. for any  $f \in \mathcal{O}_X$ ,  $[L, f] \in D_{n-1}(X)$ . Then  $D_X = \cup_n D_{n,X}$  is called the algebra of **differential operators** on  $X$ . ┘

**Prop. (13.1.4.2).** For  $X \in \text{Diff}$ ,

- $D_{1,X} \cong \mathcal{O}_X \rtimes T_X$ .
- $\text{gr } D_X = \oplus_{i \in \mathbb{N}} \text{Sym}^i T_X = \mathcal{P}\text{oly}(T_X^*)$ . In particular,  $D_X$  is left and right Noetherian.
- $\Gamma(X, D_X)$  is generated over  $\Gamma(X, \mathcal{O}_X)$  by  $L_v$  for  $v \in \text{Vect}(X)$  subject to the following relations:

$$[f, g] = 0, \quad [L_v, f] = v(f), \quad L_{fv} = fL_v, \quad [L_v, L_w] = L_{[v, w]}, \quad f, g \in \Gamma(X, \mathcal{O}_X), v, w \in \text{Vect}(X).$$

┘

**Def. (13.1.4.3) [D-Modules].** Let  $X$  be a differential manifold, then a left/right  **$D$ -module(differential)** on  $X$  is a left/right  $D_X$ -modules. ┘

**Def. (13.1.4.4) [Index Theorem P109].** has a nice definition of symbol of a differential operator on a manifold as a map from  $\text{Sym}^m T^*M \otimes \mathbb{C} \rightarrow \text{Hom}(E, F)$ . ┘

*Proof:* □

## 5 Pseudo-Differential Operator

**Def. (13.1.5.1) [Japanese Brackets].** Denote the **Japanese bracket**  $[x] = (1 + |x|^2)^{1/2} = \Theta(1 + |x|)$ .

Motivated by the formula  $(P \cdot f^\vee)^\vee = P(D)f$  for  $f \in \mathcal{S}$  and polynomial  $P$  of  $\xi$  with coefficients smooth functions of  $x$ ?? we define the **symbol class**  $S^{\mu, \beta}$  as the space of smooth functions  $a : \mathbb{R}^{2n} \rightarrow \mathbb{C}$  that

$$|D_{x, \alpha} D_{\xi, \beta} a(x, \xi)| \leq C_{\alpha, \beta} [x]^\mu [\xi]^{m-|\beta|}$$

and denote  $S^m = S^{0, m}$ .

We denote the **symbol class**  $\mathcal{A}^v$  as the space of smooth functions  $a : \mathbb{R}^{2n} \rightarrow \mathbb{C}$  that  $|D_\alpha a| \leq C_\alpha [x + \xi]^v$  for any  $\alpha$ . So  $S^{\mu, m} \subset \mathcal{A}^{|\mu|+|m|}$

And we define the **pseudo-differential operator of symbol**  $a$ :

$$(a(x, D)u)(x) = \int_{\xi} e^{ix\xi} a(x, \xi) \hat{u}$$

Moreover, we can define the **amplitude function**  $p(x, y, \xi)$  and define

$$Pu(x) = \int e^{i(x-y)\xi} p(x, y, \xi) u(y) dy.$$

┘

**Def. (13.1.5.2) [Polyhomogenous Symbols].** We define the space  $S^d$  of **polyhomogenous symbols of degree**  $d$  as the set of all symbols in  $S_{0,1}^d$  that there exists a set of  $p_{d-l}$  homogenous in  $\xi$  of degree  $d-l$  that  $p = \sum p_{d-l}$  modulo an operator in  $S^{-\infty}$ . Note that when  $p_{d-l}$  is homogenous of degree  $d-l$ , then it is automatically in  $S_{0,1}^{d-l}$ .

┘

**Def. (13.1.5.3) [Elliptic Operators].** A pseudo operator  $a$  is called **elliptic operator** if  $\sigma(a) \in S^m$  and  $\sigma(a) \geq [\xi]^{-m}$  for  $\xi$  big enough.

┘

**Prop. (13.1.5.4) [Peetre's Inequality].** For all  $v \in \mathbb{R}$ , there is a constant  $C$  that

$$[X + Y]^v < C[X]^v[Y]^v.$$

┘

*Proof:* For  $v > 0$ , just as normal. For  $v < 0$ , use  $X = (X + Y) + (-Y)$  applied to  $-v$ . □

**Prop. (13.1.5.5).** The mapping  $a(x, \xi) \times u(x) \mapsto a(x, D)u$  is continuous from  $\mathcal{A}^v \times \mathcal{S} \rightarrow \mathcal{S}$ , thus also continuous from  $S^{\mu, m} \times \mathcal{S} \rightarrow \mathcal{S}$ . Cf.[Pseudo Differential Operator P28].

┘

**Lemma (13.1.5.6) [Schur Test].** For a function  $K$  on  $\mathbb{R}^{2n}$  and  $u \in L^p(\mathbb{R}^n)$ , let  $\|K\|_1 = \sup_x \int |K(x, y)| dy$  and  $\|K(x, y)\|_2 = \sup_y \int |K(x, y)| dx$ . Let  $Au(x) = \int K(x, y) u(y) dy$ , then

$$\|Au\|_{l^p} \leq \|K\|_1^{1-1/p} \|K\|_2^{1/p} \|u\|_{L^p}.$$

by Holder. ┘

**Prop. (13.1.5.7) [Calderón-Vaillancourt].** There is a constant  $C, N_{CV}$  that for  $u \in \mathcal{A}^0$  and  $\varphi \in \mathcal{S}$ ,

$$\|Op(u)\varphi\|_{L^2} \leq C \max_{|\alpha|+|\beta| \leq N_{CV}} \|\partial_x^\alpha D_{\beta, \xi} u\|_{L^\infty} \|\varphi\|_{L^2}.$$

This in particular applies to  $u \in S^0$ . ┘

*Proof:* Cf.[Calderon-Vaillancourt]. □

**Cor. (13.1.5.8).**  $S^m$  maps  $H^s$  to  $H^{s-m}$ . Because by symbolic calculus(13.1.5.10), we have

$$Op([\xi]^{s-m})Op(u)Op([\xi]^{-s}) = Op(b) \in S^0,$$

thus  $Op(u) = Op([\xi]^{m-s})Op(b)Op([\xi]^s)$  maps  $H^s$  into  $H^{s-m}$ . ┘

### Symbolic Calculus

**Def. (13.1.5.9) [Semiclassical Operator].** For  $a \in S^{\mu,m}$  and  $h \in (0, 1]$ , we denote  $a_h(x, \xi) = a(x, h\xi)$ , it is also in  $S^{\mu,m}$ .  $\lrcorner$

**Prop. (13.1.5.10) [Composition].** If  $a \in S^{\mu_1, m_1}$  and  $b \in S^{\mu_2, m_2}$ , there is a pseudo-differential operator  $(a \# b)(h) \in S^{\mu_1 + \mu_2, m_1 + m_2}$  for every  $h \in (0, 1]$  that

$$Op(a_h)Op(b_h) = Op((a \# b)(h)_h)$$

and for all  $J > 0$ ,  $(a \# b)(h)$  can be written as

$$a \# b(h) = \sum_{j < J} h^j \left( \sum_{|\alpha|=j} \frac{1}{\alpha!} \partial_\xi^\alpha a D_x^\alpha b \right) + h^J r_J^\#(a, b, h)$$

where  $r_J^\#(a, b, h) \in S^{\mu_1 + \mu_2, m_1 + m_2 - J}$  and it is bilinear of  $a, b$  and equicontinuous independently of  $h$ .  $\lrcorner$

*Proof:* Cf. [Pseudo Differential Operator P36].  $\square$

**Prop. (13.1.5.11) [Adjoint].** If  $a \in S^{\mu,m}$  and  $u, v \in \mathcal{S}$ , there is a pseudo-differential operator  $a^*(h)$  for every  $h \in (0, 1]$  that

$$(u, Op(a_h)v) = (Op(a^*(h)_h)u, v)$$

in the  $L^2$  norm and for all  $J > 0$ ,  $a^*(h)$  can be written as

$$a^*(h) = \sum_{j < J} h^j \left( \sum_{|\alpha|=j} \frac{1}{\alpha!} \partial_\xi^\alpha D_x^\alpha \bar{a} \right) + h^J r_J^*(a, h)$$

where  $r_J^*(a, h) \in S^{\mu, m-J}$  and it is anti-linear of  $a$  and equicontinuous independently of  $h$ .  $\lrcorner$

*Proof:* Cf. [Pseudo Differential Operator P30].  $\square$

**Def. (13.1.5.12).** For  $u \in \mathcal{S}'$ , we define the action of  $a(x, \xi)$  on  $u$  by

$$(Op(a_h)u)(\bar{\varphi}) = u(\overline{Op(a^*(h)_h)\varphi}).$$

This is compatible with the definition on  $\mathcal{S}$ .  $\lrcorner$

## 6 General PDE

Cf. [Gilbarg, David; Trudinger, Neil S. Elliptic partial differential equations of second order. Reprint of the 1998 edition. Classics in Mathematics. Springer-Verlag, Berlin, 2001. xiv+517 pp. ISBN: 3-540-41160-7].

### Direct Solutions

**Prop. (13.1.6.1) [Characteristic Line].** Consider a 1-dimensional parabolic equation:

$$p_t + c(p, x, t)p_x = r(p, x, t)$$

Let  $P(t) = p(X(t), t)$ , this equation is equivalent to

$$P_t = r(X(t), t, P(t)), \quad X_t = c(X(t), t).$$

an ODE equation.  $\lrcorner$

**Prop. (13.1.6.2).** A set of equations:

$$\frac{\partial}{\partial x^i} \mu = A_i \mu$$

where  $\mu$  is a  $n$ -vector. It has a solution iff

$$[A_i, A_j] = \frac{\partial}{\partial x^i} A_j - \frac{\partial}{\partial x^j} A_i.$$

┘

*Proof:*

□

**Cor. (13.1.6.3).** This seems to be able to derive Frobenius integrability theorem, but I cannot figure it out. ┘

## 7 Analysis on Manifolds

**Prop. (13.1.7.1)[Peetre's Theorem].** For a linear operator from  $C^\infty(M)$  to  $C^\infty(M)$  that  $\text{Supp}(Lu) \subset \text{Supp}(u)$  where  $M$  is a compact manifold, then on every compact subset of a coordinate chart  $L$  looks like a differential operator of finite order. ┘

*Proof:* The first thing is to prove on a chart  $\Omega$ ,  $L$  is continuous on  $C_0^\infty(\Omega)$ . In fact, it suffice to show it is continuous from  $C_0^\infty(\Omega)$  to  $C_0^0(\Omega)$  because we can apply to  $D_\alpha L$ . For this, Cf.[Pseudo Differential Operator P86].

Then we have  $|Lu| \leq C \max_{|\alpha| \leq m} \sup_K |D_\alpha \varphi|$  for every  $\varphi \in C_0(K)$ . And the functional  $\varphi \rightarrow (L\varphi)(x)$  is a distribution supported on  $x$ , thus by (12.11.2.11), it is of the form

$$Lu(x) = \sum_{|\alpha| \leq m} a_\alpha(x) D_\alpha \varphi(x).$$

We need to show  $a_\alpha$  is smooth, which we choose a bump function  $\chi$  to show  $a_0$  is smooth and then choose  $x_i \chi$  applied to  $L\varphi - a_0 \varphi$  to show  $a_i$  is smooth, etc. □

**Prop. (13.1.7.2).** The property of  $\psi$ do of order  $d$  is preserved under diffeomorphism, Cf.[Distributions and Operators P195], giving us the possibility to define  $\psi$ do differential operator on manifolds, and the principal symbol variate in this way that it forms a function from the cotangent bundle to the  $M_n(\mathbb{C})$ . And the Sobolev space is defined by the property that all of its restrictions on a atlas are Sobolev, using the partition of unity. ┘

**Prop. (13.1.7.3).** All the norms of different are commensurable up to constant factor w.r.t. each other, so it doesn't quite matter with different norms. ┘

**Prop. (13.1.7.4).** The parametrix exists for an elliptic operator on manifolds. Cf.[Distributions and Operators P207]. ┘

## 8 Elliptic Operators

**Prop. (13.1.8.1).** Elliptic operator is a Fredholm operator. And the kernel and cokernel are smooth functions, so it is also a Fredholm operator on  $C^\infty(\Omega)$ . ┘

*Proof:* It suffice to find a left and right inverse modulo compact operators, and in fact we find it module  $S^{-\infty}$ . Since  $S^{-\infty}$  are all compact operators, i.e. it has a parametrix. Cf.[Distributions and Operators, P184]. □

**Prop. (13.1.8.2)[Garding Inequality].** For an elliptic operator of order  $d$  on  $\Gamma(E)$ ,

$$\|f\|_{H^s} \leq C(\|f\|_{H^{s-d}} + \|Pf\|_{H^{s-d}})$$

┘

*Proof:*

□

**Cor. (13.1.8.3) [Elliptic Regularity Theorem].** The inverse image of a smooth function under an elliptic operator is a smooth function, because the intersection of  $H^s(E)$  is  $C^\infty(E)$ . ┘

**Cor. (13.1.8.4) [Elliptic Regularity Theorem].** For  $L = \sum_{|\alpha| \leq N} f_\alpha D_\alpha$ , where  $f_\alpha \in C^\infty(\Omega)$  and the equation  $Lu = v$  for distributions  $u$  and  $v \in D'(\Omega)$ , when  $v$  is locally  $H^s$ ,  $u$  is locally  $H^{s+N}$ . Thus if  $v \in C^\infty(\Omega)$ , then  $u \in C^\infty(\Omega)$  by (12.11.4.12)(12.11.4.14). ┘

*Proof:* We prove the case when  $L$  has leading coefficients constant. For every  $\varphi \in D(\Omega)$  that is 1 on some open ball  $B$ ,  $\varphi u$  has compact support thus in some  $H^t$  and then we use a sublemma that says if  $\psi$  is 1 on the support of  $\varphi$ , then if  $\psi u$  is in  $H^t$ , where  $t \leq s + N - 1$ , then  $\varphi u \in H^{t+1}$ . In this way, we can shrink the nbhd to reach  $H^{s+N}$ . The proof of the lemma is in [Rudin Functional Analysis P220]. □

**Prop. (13.1.8.5)[Analytic Ellipticity theorem].** Suppose  $L$  is an analytic elliptic differential operator on a domain  $M \subset \mathbb{R}^n$ , then every solution to  $L\varphi = 0$  is analytic. ┘

*Proof:*

□

**Prop. (13.1.8.6).** The formal adjoint of an elliptic operator is an elliptic operator. ┘

*Proof:*

□

**Cor. (13.1.8.7).** The index of an elliptic operator, regarded as an operator form  $L_s \rightarrow L_{s-d}$  doesn't depend on  $s$ , because all the kernel of  $P$  and  $P^*$  are smooth. ┘

**Prop. (13.1.8.8).** For an elliptic operator, It has a inverse, the Green function which is a compact operator, so it has countable eigenfunctions consisting of smooth functions on  $L^2$  with eigenvalues converging to  $\infty$ . Moreover, the eigenvalues satisfy  $|\lambda_n| \geq Cn^\delta$  for some  $\delta, C$ . ┘

*Proof:* We prove for  $P$  self-adjoint. Use (13.1.8.1),  $\ker P$  is all smooth, so there is a map  $P(H^{-2d}) \rightarrow P(H^{-d})$  which is bijective thus an isomorphism by Banach. So the inverse of this isomorphism composed with the Sobolev embedding  $H^{-d} \rightarrow L^2$  is a compact operator  $G$ . we notice that this map has the same eigenfunctions as  $P$ , thus the result from that of compact operators.

For the second assertion, it suffice to prove  $\dim N(\lambda) \leq C\lambda^M$ . Using Garding inequality and Sobolev embedding, we have for  $f \in N(\lambda)$ ,  $\|f\|_{C^0} \leq C(1 + \lambda^l)\|f\|_{L^2}$  for large  $l$ . So if we choose an orthonormal basis  $f_i$ , then  $|a_i f_i(x)| \leq C(1 + \lambda^l)\sqrt{\sum |a_i|^2}$ . Let  $a_i = f_i(x)$  and integrate over  $M$ , we get the desired result. □

**Cor. (13.1.8.9).** For a self-adjoint elliptic operator  $P$  which is not a constant,  $L^2(E)$  has a basis consisting of eigenfunctions of  $P$ . ┘

**Cor. (13.1.8.10)[Sturm-Liouville].** This can be used to solve for example eigenvalue problem for Liouville's equation:

$$(pu')' + qu = \lambda \sigma u.$$

where  $p$  and  $\sigma$  are positive. Cf. (13.1.2.8). ┘

**Cor. (13.1.8.11).** The Hermite functions  $C_n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2}$ , as the eigenvector of  $\hat{H} = x^2 - \frac{d^2}{dx^2}$ , forms a complete basis for  $L^2(\mathbb{R})$ . Because it is  $e^{-x^2}$  times the solution of the operator  $(e^{-x^2} F')' - e^{-x^2} F$ .  $\lrcorner$

**Prop. (13.1.8.12).** For a formally self-adjoint elliptic operator  $P$  of degree  $d$  on  $E$ ,  $\Gamma(E) = \text{Im } P \oplus P(\Gamma(E))$ .  $\lrcorner$

*Proof:* We know that  $L^2(E) = P(H^d E) \oplus \ker P$ , and  $\ker P$  are all smooth by (13.1.8.3), so  $\Gamma(E) = \ker P \oplus P(H^d E) \cap \Gamma(E)$ . Now use Garding's inequality (13.1.8.2), the intersection is just  $P(\Gamma(E))$ , thus the result.  $\square$

**Prop. (13.1.8.13) [Asymptotic Heat Equation].** In this case we have the series

$$h_t(A^*A) = \sum_{\lambda} e^{-\lambda t} \dim \Gamma_{\lambda}(E)$$

converges and  $h_t$  has an asymptotic expansion

$$h_t = \sum_{k \geq -n} t^{k/2m} U_k(A^*A)$$

where  $n = \dim M$  and  $U_k = \int_M \mu_k$  for a differential form on  $M$ . Cf. [Heat Equation and the Index Theorem P297].

By the proposition above, the eigenspaces of eigenvalue non-zero neutralize, so  $\text{Ind } A = h_t(A^*A) - h_t(AA^*)$ , so

$$\text{Ind } A = U_0(A^*A) - U_0(AA^*) = \int_M \mu_0(A^*A) - \mu_0(AA^*).$$

The proof consists of the following propositions,  $\lrcorner$

**Prop. (13.1.8.14).** Using the fact that an elliptic operator has a countable basis, for an elliptic operator  $P$ , when  $t > 0$ , we let  $K(t, x, y) = \sum_n e^{-t\lambda_n} \Phi_n(x) \bar{\Phi}_n(y)$ , then

$$e^{-tP} f(x) = \int K(t, x, y) f(y) dy.$$

$K(t, x, y)$  is smooth. and the trace of  $e^{-tA^*A}$  is exactly  $h_t(A^*A)$  as in the last proposition. And the trace is just  $\int_M K(t, x, x)$ , as can be easily seen.  $\lrcorner$

*Proof:* Use Garding inequality and (13.1.8.8), we can show  $\|K\|_{C^k}$  is bounded.  $\square$

## 13.2 Calculus of Variations

### 1 Euclidean case

**Def. (13.2.1.1) [Functionals on Function Spaces].** A **functional** on the space of finite curves in  $\mathbb{R}^N$  is a real-valued function on the space of smooth curves in  $\mathbb{R}^N$ .  $\lrcorner$

**Def. (13.2.1.2) [Differentiable Functionals].** A **differentiable functional** on the space of finite curves in  $\mathbb{R}^N$  is a functional on the space of curves s.t. for any  $\gamma : I \rightarrow \mathbb{R}^N$  and  $h \in C^\infty(I, \mathbb{R}^N)$ ,

$$\Phi(\gamma + h) - \Phi(\gamma) = F(h) + O(\|h\|^2 + \|h'\|^2).$$

when  $\|h\|^2 + \|h'\|^2$  is small, where  $F$  is a linear functional  $F : C^\infty(I, \mathbb{R}^N) \rightarrow \mathbb{R}$ . Such a function  $F$  (necessarily unique) is called the **variation of the functional**  $\Phi$ .  $\lrcorner$

**Def. (13.2.1.3) [Extremal Curves of a Functionals].** An **extremal curve** for a differentiable functional (13.2.1.2) is a finite curve  $\gamma : [t_0, t_1] \rightarrow \mathbb{R}^N$  s.t. the variation  $F(h) = 0$  for any  $h \in C^\infty([t_0, t_1], \mathbb{R}^N)$  s.t.  $h(t_0) = h(t_1) = 0$ .  $\lrcorner$

**Prop. (13.2.1.4).** If  $L = L(a, b, c) \in C^\infty(\mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R})$ , then, the functional

$$\Phi(\gamma) = \int_{t_0}^{t_1} L(\gamma(t), \dot{\gamma}(t), t) dt, \quad \gamma : [t_0, t_1] \rightarrow \mathbb{R}^N :$$

is differentiable, and its variation is given by

$$F(h) = \int_{t_0}^{t_1} \left[ \frac{\partial}{\partial \gamma} L - \frac{\partial}{\partial t} \frac{\partial}{\partial \dot{\gamma}} L \right] \cdot h dt + \left( h \cdot \frac{\partial}{\partial \dot{\gamma}} L \right) \Big|_{t_0}^{t_1}.$$

$\lrcorner$

*Proof:*

$$\begin{aligned} \Phi(\gamma + h) - \Phi(\gamma) &= \int_{t_0}^{t_1} [L(\gamma + h, \dot{\gamma} + h', t) - L(\gamma, \dot{\gamma}, t)] dt \\ &= \int_{t_0}^{t_1} \left[ \frac{\partial}{\partial \gamma} L \cdot h + \frac{\partial}{\partial \dot{\gamma}} L \cdot h' \right] dt + O(\|h\|^2 + \|h'\|^2) \\ &= \int_{t_0}^{t_1} \left[ \frac{\partial}{\partial \gamma} L - \frac{\partial}{\partial t} \frac{\partial}{\partial \dot{\gamma}} L \right] \cdot h dt + \left( h \cdot \frac{\partial}{\partial \dot{\gamma}} L \right) \Big|_{t_0}^{t_1} + O(\|h\|^2 + \|h'\|^2) \end{aligned}$$

The last assertion follows easily from (12.3.2.12).  $\square$

**Cor. (13.2.1.5) [Euler-Lagrangian Equation].**  $\gamma$  is an extremal curve (13.2.1.3) for this functional (13.2.1.4) if

$$\frac{\partial}{\partial \gamma} L - \frac{\partial}{\partial t} \frac{\partial}{\partial \dot{\gamma}} L = \mathbf{0}$$

along the curve  $\gamma$ . This equation is called the **Euler-Lagrangian equation** for this functional.

More concretely, we can write

$$\frac{\partial}{\partial \gamma} \Phi(t) = \lim_{\varepsilon \rightarrow 0} \frac{\Phi[f(t') + \varepsilon \delta(t' - t)] - \Phi[f(t')]}{\varepsilon} = \frac{\partial}{\partial \gamma} L(t) - \frac{\partial}{\partial t} \frac{\partial}{\partial \dot{\gamma}} L(t).$$

$\lrcorner$

**Cor. (13.2.1.6).** If  $\Phi(\gamma) = \int_{\gamma} \|\nabla \gamma\|^2$ , then  $\frac{\partial}{\partial \gamma} \Phi = -2\Delta \gamma$ .  $\lrcorner$

### 2 Riemannian Manifold case



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### 13.3 Micro-local Analysis

## 13.4 Scattering Theory

## 13.5 Differential Equations on Varieties

References are [Del70] and [Katz, Nilpotent connections and the monodromy theorem].

**Notation (13.5.0.1).**

- Use notations on complex geometry defined in [Complex Geometry](#).
- We assume that all analytic spaces(16.9.3.4) are  $\sigma$ -compact.

┘

### Analytic Spaces

**Def. (13.5.0.2) [Principle of Analytic Continuation].** For a non-empty connected locally path-connected and locally simply path-connected pointed topological space  $X \in \mathcal{Top}$  and  $\mathcal{F} \in \mathcal{Sh}(X)$ ,  $\mathcal{F}$  is said to satisfy the **principle of analytic continuation** if for any two local sections of  $\mathcal{F}$ , the set of points on  $X$  where these two sections agree is clopen in  $X$ . ┘

**Prop. (13.5.0.3).** Situation as in(13.5.0.2), if  $\mathcal{F}$  satisfies the principle of analytic continuation, then the submodule of  $\mathcal{F}$  generated by the global sections of  $\mathcal{F}$  is a constant sheaf on  $X$ . ┘

**Def. (13.5.0.4) [Multi-Valued Sections].** For a non-empty connected locally path-connected and locally simply path-connected pointed topological space  $(X, x_0) \in \mathcal{Top}$ , let  $\text{pr} : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$  be the universal covering space. Then for each  $\mathcal{F} \in \mathcal{Sh}(X)$ , a **multi-valued section** of  $\mathcal{F}$  is an element of  $\Gamma(\tilde{X}, \text{pr}^* \mathcal{F})$ . ┘

**Def. (13.5.0.5) [First-Order Infinitesimal Neighbors].** For  $x, y : S \rightarrow X \in \text{AnSpa}/\mathbb{C}$ ,  $x, y$  are called **first-order infinitesimal neighbors** if the map  $(x, y) : S \rightarrow X \times X$  factors through the first-order infinitesimal neighbourhood of  $\Delta_X \subset X \times X$ . ┘

**Def. (13.5.0.6) [Holomorphic Connections].** For  $X \in \mathbb{C}\text{-Mani}$  and  $\mathcal{V} \in \mathcal{Coh}_{\text{lf}}(X)$ , a **holomorphic connection** on  $\mathcal{V}$  is a function that assigns to each tuple  $(S, x, y)$  where  $S \in \text{AnSpa}/\mathbb{C}$  and first-order infinitesimal neighbors  $x, y : S \rightarrow X \in \text{AnSpa}/\mathbb{C}$  an isomorphism  $\gamma_{x,y} : x^* \mathcal{V} \cong y^* \mathcal{V}$  s.t.

- For any such tuple  $(S, x, y)$  and  $f : T \rightarrow S \in \text{AnSpa}/\mathbb{C}$ , we have  $f^*(\gamma_{x,y}) = \gamma_{x \circ f, y \circ f}$ .

For each such tuple  $(S, x, y)$  s.t.  $x = y$ , we have  $\gamma_{x,x} = \text{id}_{x^* \mathcal{V}}$ . ┘

**Prop. (13.5.0.7) [Equivalent Definition of Connections, Koszul].** For  $X \in \mathbb{C}\text{-Mani}$  and  $\mathcal{V} \in \mathcal{Coh}_{\text{lf}}(X)$ , a holomorphic connection on  $\mathcal{V}$ (13.5.0.6) is equivalent to a map

$$\nabla : \mathcal{V} \rightarrow \Omega_X^1 \otimes \mathcal{V} \in \mathcal{Sh}(\mathbb{C}_X)$$

s.t. the **Leibniz-rule** holds for any local section  $f$  of  $\mathcal{O}_X$  and local section  $s$  of  $\mathcal{V}$ :

$$\nabla(fs) = df \otimes s + f \nabla(s).$$

The set of connections on  $\mathcal{V}$  is denoted by  $\text{Conn}(\mathcal{V})$ , and the category of pairs  $(\mathcal{V}, \nabla)$  where  $\mathcal{V} \in \mathcal{Coh}_{\text{lf}}(X)$  and  $\nabla$  is a connection of  $\mathcal{V}$  is denoted by  $\text{Conn}(X)$ . ┘

*Proof:* ?

□

**Example (13.5.0.8) [Connections].** Let  $X \in \mathbb{C}\text{-Mani}$ ,

- The **canonical connection** on  $\mathcal{O}_X$  is given by  $\nabla(f) = f$ .
- 

┘

**Def. (13.5.0.9) [Horizontal Sections].** For  $X \in \mathbb{C}\text{-Mani}$  and  $(\mathcal{V}, \nabla) \in \text{Conn}(X)$ , a **horizontal section** on  $\mathcal{V}$  is a local section  $s$  of  $\mathcal{V}$  s.t.  $\nabla(s) = 0$ . ┘

## 1 Regular Connections

### Regularity in Dimension 1

**Def. (13.5.1.1) [Regular Singularities].** Let  $U \subset \mathbb{C}$  is an domain containing 0 and

$$E : \frac{\partial^n}{\partial z^n} y + \sum_{i \in [n-1]} a_i(z) \frac{\partial^i}{\partial z^i} y = 0,$$

where  $a_i(z) \in \mathcal{O}(U \setminus \{0\})$ . Then this equation  $E$  is said to have a **regular singularity** at  $z = 0$  if  $z^{n-i} a_i(z) \in \mathcal{O}(U)$  for each  $i \in [n-1]$ . If this holds, then we can write  $E$  as

$$\left( z \frac{\partial}{\partial z} \right)^n y + \sum_{i \in [n-1]} b_i(x) \left( z \frac{\partial}{\partial z} \right)^i y = 0$$

for some  $b_i(z) \in \mathcal{O}(U)$ . ┘

### Logarithmic Poles

**Def. (13.5.1.2) [Logarithmic deRham-Complexes].** For  $X \in \text{Sch}^{\text{ft,sm}}/\mathbb{C}$  and a normal-crossing divisor  $Y \subset X$ , denote  $X^* = X \setminus Y$  and  $j : X^* \hookrightarrow X$ , the **logarithmic deRham-complex** of  $X$  along  $Y$  is the smallest subcomplex

$$\Omega_X^\bullet \langle Y \rangle \subset j_* \Omega_{X^*}^\bullet$$

s.t.

- $\Omega_X^\bullet \langle Y \rangle$  is stable under exterior product.
- $\Omega_X^\bullet \langle Y \rangle$  contains  $\Omega_{X^*}^\bullet$ .
- For any local section  $f$  of  $\mathcal{O}_{X^*}$  s.t.  $f$  is meromorphic along  $Y$ , we have  $df/f \in \Omega_X^1 \langle Y \rangle$ .

And local sections of  $\Omega_X^\bullet \langle Y \rangle$  are said to present a logarithmic pole along  $Y$ .

Then we have

- For a local section  $\alpha$  of  $j_* \Omega_{X^*}^p$ ,  $\alpha \in \Omega_X^\bullet \langle Y \rangle$  iff  $\alpha$  and  $d\alpha$  have at worst simple poles along  $Y$ .
  - $\Omega_X^1 \langle Y \rangle$  is locally free, and  $\Omega_X^p \langle Y \rangle = \wedge^p (\Omega_X^1 \langle Y \rangle)$  for each  $p \in \mathbb{Z}_+$ .
  -
- ┘

### Regularity in Higher Dimensions

**Thm. (13.5.1.3) [Regularity along Closed Analytic Subspaces, Deligne[Del70]].** Suppose

- $X \in \text{AnSpa}/\mathbb{C}$  and  $Y \subset X \in \text{AnSpa}/\mathbb{C}$  is a closed subspace s.t.  $X \setminus Y$  is smooth.
- $X' \rightarrow X$  is the normalization of  $X$  and  $Y'$  is the inverse image of  $Y$  in  $X'$ .
- Let  $(\mathcal{V}, \nabla) \in \text{Conn}(X)$  be meromorphic along  $Y$  ?.

Then the following conditions are equivalent:

- 
- For each map  $\varphi : \mathbb{D}^1 \rightarrow X \in \text{AnSpa}$  with  $\varphi^{-1}(Y) = \{0\}$ , the inverse image  $\varphi^*(\mathcal{V}, \nabla)$  is regular on  $\mathbb{D}^1$  ?.
- The multi-valued horizontal sections of  $(\mathcal{V}, \nabla)$  are of moderate growth along  $Y$  ?.

└

*Proof:* Cf. [Del70]Thm4.1.?

□

## 13.6 D-Modules

References are [HTT08].

**Notation (13.6.0.1).**

- Let  $k \in \mathbf{Field}, k = \bar{k}$ .
- Let  $X \in \mathcal{V}\mathbf{ar}^{\text{sm}}/k$ .

┘

### 1 Basics

**Def. (13.6.1.1) [Differential Operators].** Define  $D_{0,X} = \mathcal{O}_X$ , and for any  $n \in \mathbb{Z}_+$ , define  $D_{n,X}$  to be the subbundle of  $\underline{\text{Hom}}(\mathcal{O}_X, \mathcal{O}_X)$  s.t. for any local section  $f \in \mathcal{O}_X$ ,  $[L, f] \in D_{n-1}(X)$ . Then  $D_X = \cup_n D_{n,X}$  is a sheaf of algebras, called the algebra of **differential operators** on  $X$ . ┘

**Prop. (13.6.1.2).**

- $D_{1,X} \cong \mathcal{O}_X \rtimes T_X$ .
- $\text{gr } D_X = \oplus_{i \in \mathbb{N}} \text{Sym}^i T_X = \mathcal{P}\text{oly}(T_X^*)$ . In particular,  $D_X$  is left and right Noetherian.
- $\Gamma(X, D_X)$  is generated over  $\Gamma(X, \mathcal{O}_X)$  by  $L_v$  for  $v \in \mathcal{V}\text{ect}(X)$  subject to the following relations:

$$[f, g] = 0, \quad [L_v, f] = v(f), \quad L_{fv} = fL_v, \quad [L_v, L_w] = L_{[v,w]}, \quad f, g \in \Gamma(X, \mathcal{O}_X), v, w \in \mathcal{V}\text{ect}(X).$$

┘

**Def. (13.6.1.3) [D-Modules].** A left/right **D-module(algebraic)** on  $X$  is a left/right qco  $D_X$ -modules. The category of left D-modules on  $X$  is denoted by  $\text{Mod}_D(X)$ . The category of left quasi-coherent D-modules on  $X$  is denoted by  $\mathcal{Q}\text{Coh}_D(X)$ . ┘

**Def. (13.6.1.4) [Connections].** For  $M \in \text{Mod}(\mathcal{O}_X)$ , a **connection** on  $M$  is a  $k$ -linear morphism of sheaves  $\nabla : M \rightarrow M \otimes_{\mathcal{O}_X} \Omega_X^1$  s.t.

$$\nabla(fm) = f\nabla(m) + m \otimes df$$

for any local sections  $f \in \mathcal{O}_X$  and  $m \in M$ . ┘

**Prop. (13.6.1.5) [Curvatures].** Given  $M \in \text{Mod}(\mathcal{O}_X)$  and a **connection**  $\nabla$  on  $M$ , there is a  $\mathcal{O}_X$ -linear map

$$\nabla^2 : M \rightarrow M \otimes_{\mathcal{O}_X} \Omega_X^2$$

s.t. for local sections  $m \in M, v, w \in T_X$ , it is given by  $\nabla^2(m)(v, w) = ([\nabla_v, \nabla_w] - \nabla_{[v,w]})m$ , where  $\nabla_v : M \rightarrow M$  is given by  $\nabla_v(m) = \nabla(m)(v)$ . This map  $\nabla^2$  is called the **curvature** of  $\nabla$ . And  $\nabla$  is called a **flat connection** if  $\nabla^2 = 0$ . ┘

*Proof:* ?

□

**Prop. (13.6.1.6) [D-Modules and Connections].** A left  $D_X$ -module is the same thing as an  $\mathcal{O}_X$ -module with a flat connection. ┘

*Proof:* If  $M$  is a left  $D_X$ -module, then there is a  $D_X$ -action locally given by  $L_v.m = \nabla(m)(v)$ . This is an action by (13.6.1.2). Conversely, the same formula can be used to define a connection on  $M$  that is flat. ┘

**Def. (13.6.1.7)[Integrable Connections].** An **integrable connection** over  $X$  is a vector bundle with a flat connection. The category of integrable connections on  $X$  is denoted by  $\text{Conn}(X)$ .  $\lrcorner$

**Prop. (13.6.1.8)[Coherence].**

- $D_X$  is a coherent sheaf of rings.
- $M \in \text{Mod}_D(X)$  is coherent over  $D_X$  iff it is Qco and locally f.g. over  $\mathcal{O}_X$ .
- $M \in \text{Mod}_D(X)$  is coherent over  $X$  iff it is an integral connection(13.6.1.7).

$\lrcorner$

*Proof:* Cf.[D-modules, perverse sheaves, and representation theory]P27.?

$\square$

**Prop. (13.6.1.9).** The support of an irreducible  $D$ -module  $M \in \text{Mod}_D(X)$  is irreducible.  $\lrcorner$

*Proof:* If  $\text{Supp}(M) = Z_1 \cup Z_2$ , consider  $i : Z_1 \hookrightarrow X$  and  $j : X \setminus Z \hookrightarrow X$ , then there is an exact sequence

$$0 \rightarrow j_! j^* M \rightarrow M \rightarrow i_* i^* M \rightarrow 0.$$

which shows that  $M$  is reducible, contradiction.  $\square$

**Prop. (13.6.1.10) [Direct/ Inverse Images and Intermediate Extensions].** For  $\pi : X \rightarrow Y \in \mathcal{V}\text{ar}^{\text{sm}}/k$ , there are inverse image and direct image functors of  $D$ -modules as the general formalism of(7.2.2.9). the module  $D_{X/Y} = f^* D_Y$  is a left  $D_X$ -module by defining

$$\nabla_v(f \otimes L) = v(f) \otimes L + f \otimes \pi_*(v)L, \quad f \in \mathcal{O}_X, L \in D_Y.$$

In particular, for any  $M \in \text{Mod}_D(Y)$ ,  $f^*(M) = D_{X/Y} \otimes f^{-1}M$  is a left  $D_X$ -module, called the **inverse image D-module**.

And if  $\pi$  is affine, then for any right  $D_X$ -module  $M$ ,  $\pi_*(M) \otimes D_Y = \pi_*(M \otimes D_{X/Y})$  is a right  $D_Y$ -module, called the **direct image D-module**. There are derived versions in the non-affine situation?

If  $j : U \hookrightarrow X$  is an inclusion of open subschemes, then there is also a **intermediate extension D-module**  $j_! M$  for any  $D$ -module  $M \in \text{Mod}_D(X)$ (7.2.2.9).  $\lrcorner$

### Categories of D-Modules

**Def. (13.6.1.11) [D-Affine Varieties].** For  $X \in \mathcal{V}\text{ar}^{\text{sm}}/k$ , then  $X$  is called a **D-affine variety** if the functor

$$\Gamma : \mathcal{Q}\text{Coh}_D(X) \rightarrow \text{Mod}(D(X))$$

is an equivalence of categories.  $\lrcorner$

**Kashiwara's Equivalence****2 Coherent D-Modules****3 Holonomic D-Modules****Regular Holonomic D-Modules****4 Analytic D-Modules and the deRham Functor****5 Meromorphic Connections****6 Analytic D-Modules and the deRham Functor****7 Beilinson-Bernstein Localization Theorem**

**Prop. (13.6.7.1).** Partial flag varieties of semisimple groups are D-affine varieties. ┘

*Proof:* Cf.[Representation of Lie groups, Etingof]P108. □

**Thm. (13.6.7.2)**[Beilinson-Bernstein Localization Theorem]. ┘

**8 Riemann–Hilbert Correspondence**



## 13.7 Dynamical Systems

References are [Dynamical System, Katok] and [B-S02].

### 1 Topological Dynamical Systems

**Def. (13.7.1.1) [Topological Dynamic Systems].** A **topological dynamic system** is a topological space  $X$  and either a continuous map  $f : X \rightarrow X$  or a continuous (semi)flow  $f^t$  on  $X$ .  $\lrcorner$

**Def. (13.7.1.2) [Conjugacies].** Let  $(X, f), (Y, g)$  be topological dynamical systems, then a **semiconjugacy of dynamical systems** from  $(X, f)$  to  $(Y, g)$  is a surjection  $\text{pr} : X \rightarrow Y$  s.t.  $g \circ \text{pr} = \text{pr} \circ f$ . An invertible semiconjugacy is called a **conjugacy of dynamical systems**.  $\lrcorner$

**Def. (13.7.1.3) [Dynamical Notions].** Let  $f : X \rightarrow X$  be a discrete topological dynamic system (13.7.1.1), then

- $x \in X$  is called a **periodic point** for  $f$  if  $f^{(p)}(x) = x$  for some  $p \in \mathbb{Z}_+$ . And the minimal  $p \in \mathbb{Z}_+$  s.t.  $f^{(p)}(x) = x$  is called the **period** of  $f$  at  $x$ .  $x \in X$  is called a **pre-periodic point** for  $f$  if  $f^{(m)}(x) = f^{(n)}(x)$  for some  $0 < m < n \in \mathbb{Z}_+$ . The set of pre-periodic points of  $f$  is denoted by  $\text{PrePer}(f)$ .
- A **periodic cycle** is a finite set  $S \subset X$  s.t.  $f|_S$  is a transitive permutation.
- $x \in X$  is called **almost periodic** if for any nbhd  $U$  of  $x$ ,  $\{i | f^i(x) \in U\}$  is relatively dense in  $\mathbb{N}$ , i.e. appear in every  $k$  consecutive integers for some  $k$ .
- A periodic point  $x \in X$  for  $f$  with periodicity  $k$  is called **attracting** if there exists a precompact nbhd  $U$  of  $x \in X$  s.t.  $f(\overline{U}) \subset U$ , and  $\cap_{n \in \mathbb{N}} f^{(n)}(U) = \{x\}$ . Similarly,  $x$  is called **repelling** if there exists a nbhd  $U$  of  $x \in X$  s.t.  $\overline{U} \subset f(U)$ , and  $\cap_{n \in \mathbb{N}} f^{(-n)}(U) = \{x\}$ .
- For an attracting periodic point  $x \in X$  for  $f$ , the **basin of attraction** of  $x$  is defined to be the set

$$BA(x) = \{y \in X | \exists j \in \mathbb{N}, \lim_{n \rightarrow \infty} f^{(n)}(y) = f^j(x)\}.$$

If  $U$  is a nbhd of  $x \in X$  s.t.  $f(\overline{U}) \subset U$ , and  $\cap_{n \in \mathbb{N}} f^{(n)}(U) = \{x\}$ , then  $BA(x) = \cup_{n \in \mathbb{Z}_+} f^{(-n)}(x)$ , so it is open. The connected components  $BA_0(x)$  of  $BA(x)$  containing some  $f^{(j)}(x)$  for  $j \in \mathbb{N}$  is called the **intermediate basin of attraction** of  $x$ .

- For  $x \in X$ , its  **$\omega$ -limit points** are defined to be

$$\omega(x) = \bigcap_{n \in \mathbb{N}} \overline{\bigcup_{i \geq n} f^i(x)}$$

- If  $f$  is invertible, for  $x \in X$ , its  **$\alpha$ -limit points** are defined to be

$$\alpha(x) = \bigcap_{n \in \mathbb{N}} \overline{\bigcup_{i \leq -n} f^i(x)}$$

- The set of **(positively)recurrent** points for  $f$  is defined to be  $\mathcal{R}(f) = \{x \in X | x \in \omega(x)\}$ .
- The set  $NW(f)$  of **non-wandering points** are the points  $x$  that for any nbhd  $U$  of  $x$ , there is an  $n > 0$  that  $f^n(U) \cap U \neq \emptyset$ .
- Denote the orbit sets  $\mathcal{O}(x) = \bigcup_{n \in \mathbb{Z}} f^n(x)$ ,  $\mathcal{O}^+(x) = \bigcup_{n \geq 0} f^n(x)$ .

- Let  $X$  be compact, then a closed non-empty forward  $f$ -invariant  $Y \subset X$  is called a **minimal set** for  $f$  if there is no smaller such set. ┘

**Prop. (13.7.1.4).**

- $NW(f)$  is closed,  $f$ -invariant, and contains  $\alpha(x), \omega(x)$  for all  $x$ .
- Every recurrent point is non-wandering, thus  $\overline{\mathcal{R}(f)} \subset NW(f)$ .
- Let  $X$  be compact Hausdorff, then  $\overline{\mathcal{O}^+(x)}$  is minimal for  $f$  iff  $x$  is almost periodic. ┘

*Proof:* item1 is easy, 2 follows from 1 as  $x \in \omega(x)$ .

For 3: suppose  $x$  is almost periodic and  $y \in \overline{\mathcal{O}^+(x)}$ , we need to show that  $x \in \overline{\mathcal{O}^+(y)}$ . For any nbhd  $x \in U$ , there is a small nbhd  $x \in U' \subset U$  and a nbhd  $\Delta \subset V \subset X \times X$ , that if  $x_1 \in U'$  and  $(x_1, x_2) \in V$ , then  $x_2 \in U$ . Since  $x$  is periodic, there is a  $K$  that for any  $j \geq 0$ , there is a  $f^{j+k}(x) \in U'$  for some  $0 \leq k \leq K$ . Let  $V' = \cap_{i=0}^K f^{-1}(V)$ , then  $V'$  is open and contains the diagonal. Thus there is a nbhd  $W$  of  $y$  that  $W \times W \subset V'$ . Now choose  $f^n(x) \in W$  by almost periodicity, and  $f^{n+k}(x) \in U'$  for some  $0 \leq k \leq K$ , then we have  $(f^{n+k}(x), f^k(x)) \in V$  by the definition of  $V'$  and  $W$ , and hence  $f^k(x) \in U$ . This shows  $x \in \overline{\mathcal{O}^+(y)}$ .

Conversely, if  $x$  is not almost periodic, then there is a nbhd  $U$  of  $x$ , that there is a sequence  $\{a_i\} \in \mathbb{N}$ , that  $f^{a_i+j}(x) \notin U$  for  $j \leq i$ . By convergence theorem and passing to a subsequence, we assume  $y$  is the limit of  $f^{a_i}(x)$ , and  $f^j(y) \notin U$  for any  $j > 0$ , thus  $x \notin \overline{\mathcal{O}^+(y)}$ , showing  $\overline{\mathcal{O}^+(x)}$  is not minimal. □

**Prop. (13.7.1.5)[Minimal Set Exists on Compact Dynamic System].** If  $f$  is a topological dynamic system on a compact space, then there exists a minimal set. In particular, there exists an almost periodic point, in priori positively recurrent point, by (13.7.1.4). ┘

*Proof:* Use Zorn's lemma and the finite intersection property. □

**Def. (13.7.1.6)[Topologically Transitive].** A topological dynamic system  $f : X \rightarrow X$  is called **topologically transitive** if there is a point  $x$  that  $\overline{\mathcal{O}(x)} = X$ . ┘

**Prop. (13.7.1.7).** Let  $f$  be a continuous map of a locally compact Hausdorff second countable space  $X$ . Suppose that for any two non-empty open set  $U, V$ , there is  $n > 0$  that  $f^n(U) \cap V \neq \emptyset$ , then  $f$  is topologically transitive. ┘

*Proof:* For any open subset  $V$ , the hypothesis says  $\cup_{n>0} f^{-n}(V)$  is dense in  $X$ . Let  $V_i$  be a countable basis for the topology of  $X$ , and  $Y = \cap_i \cup_{n>0} f^{-n}(V_i)$ , then it is non-empty, by Baire category theorem (4.4.9.2). Now the orbit of any  $y \in Y$  enters every  $V_i$ , thus its orbit is dense in  $X$ . □

**Prop. (13.7.1.8).** Let  $f : X \rightarrow X$  be a homeomorphism of a compact metric space, and suppose  $X$  has no isolated points, then if there is a dense orbit  $\mathcal{O}(x)$ , there will be a dense orbit  $\mathcal{O}^+(x)$ . ┘

*Proof:* The hypothesis that  $X$  has no isolated points shows that  $\mathcal{O}(x)$  meets every open subset  $U$  infinitely many times, thus we can choose  $f^{n_k}(x) \in B(x, 1/k)$  that  $|n_k| \rightarrow \infty$ . Thus  $f^{n_k+l}(x) \rightarrow f^l(x)$  for any  $l$ .

If there are infinitely many  $n_k > 0$ , then we have  $\mathcal{O}(x) \subset \overline{\mathcal{O}^+(x)}$ , hence  $\mathcal{O}^+(x)$  is dense. If there are infinitely many  $n_k < 0$ , then  $\mathcal{O}^-(x)$  is dense. Then for any open subset  $U, V$ , we can find  $i < j < 0$  that  $f^i(x) \in U, f^j(x) \in V$ , thus  $f^{j-i}(U) \cap V \neq \emptyset$ . Hence we use (13.7.1.7) to conclude  $f$  is topologically transitive. □

**Def. (13.7.1.9) [Topological Mixing].** A topological dynamic system  $f : X \rightarrow X$  is called **topologically mixing** if for any two non-empty open subsets  $U, V$ , there is  $N > 0$  that  $f^n(U) \cap V \neq \emptyset$  for any  $n \geq N$ .  $\lrcorner$

**Def. (13.7.1.10).** A homeomorphism  $f : X \rightarrow X$  is called **expansive** if there is a  $\delta > 0$  that for any two distinct points  $x, y$ , there is some  $n \in \mathbb{Z}$  that  $d(f^n(x), f^n(y)) \geq \delta$ . Similarly, we can define **positively expansive** for a non-invertible continuous map  $f : X \rightarrow X$ . This constant  $\delta$  is called a **expansiveness constant** of  $f$ .  $\lrcorner$

**Prop. (13.7.1.11) [Compact Metric Space not positively expansive].** If  $f$  be a continuous map of an infinite compact metric space  $X$ , then it is not positively expansive.  $\lrcorner$

*Proof:* First assume  $f$  is invertible. Fix  $\varepsilon > 0$ , consider all  $m$  that there are points  $x \neq y$  that

$$d(f^n(x), f^n(y)) < \varepsilon, \quad 0 < n \leq m, \quad d(x, y) \geq \varepsilon.$$

If these  $m$  are infinite, then we can use convergence point theorem to find a point  $x, y$  that  $d(f^n(x), f^n(y)) < \varepsilon$  for any  $n > 0$ , thus  $f$  is not expansive.

If these  $m$  are finite, let  $M$  be a maximal, then by absolute convergence, there is a  $\delta$  that if  $d(x, y) \leq \delta$ , then  $d(f^n(x), f^n(y)) < \varepsilon$  for any  $0 \leq n \leq m$ . Then by definition of  $M$ ,  $d(f^{-1}(x), f^{-1}(y)) < \varepsilon$ , and similarly  $d(f^{-n}(x), f^{-n}(y)) < \varepsilon$  for any  $n < 0$ .

Now choose a finite  $\delta/2$ -net  $\{x_i\}$  of  $M$ , then for each  $j \in \mathbb{Z}$ , there are two  $f^j(x_s), f^j(x_t)$  in the same  $B(\delta/2, x_j)$ , thus  $d(f^n(x_s), f^n(x_t)) < \varepsilon$  for  $n \leq j$ . Now there are only f.m. pairs of elements in  $M$ , there are some pair  $(x_\alpha, x_\beta)$  appeared infinitely many times for different  $j > 0$ , thus we have  $d(f^n(x_\alpha), f^n(x_\beta)) < \varepsilon$  for any  $n \in \mathbb{Z}$ .

For the non-invertible case, the proof is the same, noticing that  $f^{-1}$  is chosen wisely when it can be defined.  $\square$

**Cor. (13.7.1.12).** Let  $f$  be an expansive homeomorphism of an infinite compact metric space  $X$ , then there are distinct points  $x_0, y_0$  that  $\lim_{n \rightarrow \infty} d(f^n(x_0), f^n(y_0)) = 0$ .  $\lrcorner$

*Proof:* Let  $\delta$  be an expansive constant of  $f$ , then by (13.7.1.11), there are  $x_0 \neq y_0 \in X$  that  $d(f^n(x_0), f^n(y_0)) \leq \delta$  for all  $n > 0$ . Suppose  $d(f^n(x), f^n(y)) \not\rightarrow 0$ , then by compactness in  $X \times X$ , there is a subsequence  $\{n_k\} \in \mathbb{N}$  that  $f^{n_k}(x) \rightarrow x', f^{n_k}(y) \rightarrow y'$  with  $x' \neq y'$ . Then this will show that  $d(f^m(x), f^m(y)) < \delta$  for any  $m \in \mathbb{Z}$ , contradicting expansiveness of  $f$ .  $\square$

**Def. (13.7.1.13).** Let  $f : X \rightarrow X$  be a homeomorphism of compact Hausdorff space, then

- two points  $x, y$  are called **proximal** if the closure of orbits  $\overline{\mathcal{O}((x, y))}$  under  $f \times f$  intersect the diagonal  $\Delta \subset X \times X$ . It is called **distal** otherwise.  $f$  is called **distal** if every two distinct points  $x, y$  are distal.
- $f$  is called **equicontinuous** if the family  $\{f^n\}_{n \in \mathbb{Z}}$  is equicontinuous.  $\lrcorner$

**Prop. (13.7.1.14).** A equicontinuous homeomorphism  $f$  of a compact metric space are distal.  $\lrcorner$

*Proof:* Suppose  $f$  is not distal, then there is a proximal pair  $(x, y)$ , thus there is a sequence  $\{n_k\} \in \mathbb{Z}$  that  $d(f^{n_k}(x), f^{n_k}(y)) \rightarrow 0$ . let  $d(x, y) = \varepsilon$ , thus for any  $\delta > 0$ , there is some  $d(f^{n_k}(x), f^{n_k}(y)) < \varepsilon$ , thus contradicting the equicontinuity of  $f^{-n}$ .  $\square$

**Def. (13.7.1.15) [Almost Periodic Set].** For a subset  $A \subset X$  and a homeomorphism  $f : X \rightarrow X$ , denote by  $f_A$  the action of  $f$  on  $\text{Map}(A, X)$ . Then  $A$  is called **almost periodic** if every  $z \in \text{Map}(A, A)$  is almost periodic for  $f_A$ . Equivalently, for any finite set of points  $\{x_i\} \in A$ , and nbhds  $x_i \in U_i$ , the set  $\{k \in \mathbb{Z} \mid f^k(x_i) \in U_i\}$  is relatively dense in  $\mathbb{Z}$ . This is compatible with the previous definition of almost periodic point (13.7.1.3).  $\lrcorner$

**Def. (13.7.1.16).** A homeomorphism of a compact Hausdorff space is called **pointwise almost periodic** if every point  $x$  is almost periodic. By (13.7.1.4), we can see that this is equivalent to  $X$  is a union of minimal sets.  $\lrcorner$

**Lemma (13.7.1.17).** Every almost periodic set  $A$  is contained in a maximal almost periodic set in  $X$ .  $\lrcorner$

*Proof:* It is because of the second definition of almost periodic set (13.7.1.15) that the sum of an ordered family of almost periodic sets is also almost periodic.  $\square$

**Prop. (13.7.1.18).** Let  $f$  be a homeomorphism of a compact Hausdorff space  $X$ , then every point  $x \in X$  is proximal to an almost periodic point.  $\lrcorner$

*Proof:* If  $x$  is almost periodic, then we are done. If not, consider a maximal almost periodic set  $A \subset X$ , then  $x \notin A$ . Then for  $z \in X^A$  with range  $A$ , consider  $(x, z) \in X \times X^A$ , and find an almost periodic point  $(x_0, z_0)$  of  $f \times f_A$  in  $\overline{\mathcal{O}(x, z)}$ , by (13.7.1.5). Because  $z$  is almost periodic,  $z \in \overline{\mathcal{O}(z_0)}$ , by (13.7.1.4). Thus we see  $(x', z) \in \overline{\mathcal{O}(x_0, z_0)}$  for some  $x'$ , by the compactness of  $X$ . Then  $(x', z)$  is also almost periodic, and we can forget about  $x_0, z_0$ .

Therefore,  $\{x'\} \cup \text{Im}(z) = \{x'\} \cup A$  is almost periodic for  $f$ , and since  $A$  is maximal,  $x' \in A = \text{Im}(z)$ . This shows  $(x', x') \in \overline{\mathcal{O}(x, x')}$ , showing  $x$  is proximal to  $x'$ .  $\square$

**Cor. (13.7.1.19) [Distal Homeomorphism is Pointwise Almost Periodic].** If  $f$  is a distal homeomorphism of a compact Hausdorff space, then  $f$  is pointwise almost periodic.  $\lrcorner$

**Prop. (13.7.1.20).**

- A homeomorphism of a compact Hausdorff space is distal iff the product system  $(X \times X, f \times f)$  is pointwise almost periodic.
  - A factor system of a pointwise almost periodic homeomorphism is pointwise almost periodic.
  - A factor system of a distal homeomorphism is distal.
- $\lrcorner$

*Proof:* 1: If  $f$  is distal, then so is  $f \times f$ , hence it is pointwise almost periodic, by (13.7.1.19). Conversely, if  $f \times f$  is pointwise almost periodic, then if  $x, y$  is proximal, then  $(z, z) \in \overline{\mathcal{O}(x, y)}$ , but since  $\overline{\mathcal{O}(x, y)}$  is minimal, by (13.7.1.4),  $(x, y) \subset \overline{\mathcal{O}(z, z)} \subset \Delta$ , hence  $x = y$ .

2: This is easy.

3: if  $f$  is a factor of  $g$ , then  $f \times f$  is a factor of  $g \times g$ , but the latter is pointwise locally periodic, thus the first is also locally periodic, by 2, and then  $f$  is distal, by 1.  $\square$

### Topological Entropy

**Def. (13.7.1.21) [Definitions].** Let  $(X, d)$  be a compact metric space, and  $f$  a continuous map  $X \rightarrow X$ , we define

- for  $x, y \in X$ ,  $d_n(x, y) = \max_{0 \leq k \leq n-1} d(f^k(x), f^k(y))$ .

- a subset  $A \subset X$  is called  $(n, \varepsilon)$ -**spanning** if it is a  $\varepsilon$ -net in  $(X, d_n)$ . Similarly, we can define  $(n, \varepsilon)$ -**separated** and  $\text{cov}(n, \varepsilon, f)$  the minimal number of covering of  $X$  of  $d_n$ -diameter  $< \varepsilon$ .  $\lrcorner$

**Lemma (13.7.1.22).**  $\text{cov}(n, 2\varepsilon, f) \leq \text{span}(n, \varepsilon, f) \leq \text{sep}(n, \varepsilon, f) \leq \text{cov}(n, \varepsilon, f)$ .  $\lrcorner$

*Proof:* Easy.  $\square$

**Prop. (13.7.1.23)[Topological Entropy].** The number

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log(\text{cov}(n, \varepsilon, f)) = h_\varepsilon(f)$$

is well-defined and finite. It is increasing as  $\varepsilon$  decreases, so  $h(f) = \lim_{\varepsilon \rightarrow 0^+} h_\varepsilon(f)$  exists, which lies in  $[0, \infty]$ . It is called the **topological entropy** of  $f$ .

Notice the entropy can be calculated by either  $\text{cov}$ ,  $\text{span}$  or  $\text{sep}$ , by (13.7.1.22).  $\lrcorner$

*Proof:* For this, we need to notice if  $U$  has  $d_n$ -diameter  $< \varepsilon$  and  $V$  has  $d_m$ -diameter  $< \varepsilon$ , then  $U \cap f^{-m}(V)$  has  $d_{m+n}$ -diameter  $< \varepsilon$ . Hence

$$\text{cov}(m+n, \varepsilon, f) \leq \text{cov}(m, \varepsilon, f) \cdot \text{cov}(n, \varepsilon, f).$$

Thus we can use (12.3.5.2) to conclude.  $\square$

**Prop. (13.7.1.24).** The topological entropy doesn't depend on the metric generating the topology.  $\lrcorner$

*Proof:* This is because  $d'$  is a continuous function on  $(X \times X, d \times d)$ , thus it is uniformly continuous as  $X$  is compact, so

$$\text{cov}(n, \varepsilon, f) \leq \text{cov}(n, \delta(\varepsilon), f)$$

where  $\delta(\varepsilon) \rightarrow 0$  when  $\varepsilon \rightarrow 0$ . This shows the topological entropies are the same.  $\square$

**Cor. (13.7.1.25).** Two conjugate dynamic systems have the same topological entropy.  $\lrcorner$

**Prop. (13.7.1.26)[Properties of Entropy].** Let  $f : X \rightarrow X$  be a continuous map of a compact metric space  $X$ ,  $g : Y \rightarrow Y$  be a continuous map of a compact metric space  $Y$ , then

- $h(f^m) = m \cdot h(f)$  for  $m > 0$ .
- If  $f$  is invertible, then  $h(f^{-1}) = h(f)$ .
- Let  $A_i$  be a finite family closed forward  $f$ -invariant subsets of  $X$  whose union is  $X$ , then

$$h(f) = \max h(f|_{A_i}).$$

- $h(f \times g) = h(f) + h(g)$ , and if  $f$  is an extension of  $g$ , then  $h(f) \geq h(g)$ .  $\lrcorner$

*Proof:* 1: Use two inequalities:

$$\text{span}(n, \varepsilon, f^m) \leq \text{span}(mn, \varepsilon, f), \quad \text{span}(n, \delta(\varepsilon), f^m) \geq \text{span}(mn, \varepsilon, f)$$

where  $\delta(\varepsilon) \rightarrow 0$  if  $\varepsilon \rightarrow 0$ , and (13.7.1.22).

- 2: Because  $(n, \varepsilon)$ -separated sets for  $f$  and  $(n, \varepsilon)$ -separated sets for  $f^{-1}$  corresponds via  $f^n$ .

3: Use two inequalities:

$$\text{span}(n, \varepsilon, f) \leq \sum_{i=1}^k \text{span}_i(n, \varepsilon, f) \leq k \cdot \max \text{span}_i(n, \varepsilon, f),$$

$$\text{sep}(n, \varepsilon, f) \geq \max \text{sep}(n, \varepsilon, f).$$

4: Noticing two inequalities:

$$\text{cov}(n, \varepsilon, f \times g) \geq \text{cov}(n, \varepsilon, f) \cdot \text{cov}(n, \varepsilon, g),$$

$$\text{sep}(n, \varepsilon, f \times g) \geq \text{sep}(n, \varepsilon, f) \cdot \text{sep}(n, \varepsilon, g).$$

and the proof of the last assertion is similar to that of (13.7.1.24).  $\square$

**Prop. (13.7.1.27).** Let  $f : X \rightarrow X$  be an expansive homeomorphism of a compact metric space of expansiveness constant  $\delta$ , then  $h(f) = h_\varepsilon(f)$  for  $\varepsilon < \delta$ .  $\lrcorner$

*Proof:* For  $0 < \gamma < \varepsilon < \delta$ , we show that  $h_{2\gamma}(f) = h_\varepsilon(f)$ . For this, it suffices to prove  $\leq$ . By expansiveness, if  $x \neq y$ , then there is some  $i \in \mathbb{Z}$  that  $d(f^i(x), f^i(y)) \geq \delta > \varepsilon$ . Since the set  $\{(x, y) \in X \times X \mid d(x, y) \geq \gamma\}$  is compact, there is a  $k = k(\gamma, \varepsilon)$  that if  $d(x, y) \geq \gamma$ , then  $d(f^i(x), f^i(y)) > \varepsilon$  for some  $|i| \leq k$ . Thus if  $A$  is a  $(n, \gamma)$ -separated set, then  $f^{-k}(A)$  is a  $(n + 2k, \varepsilon)$ -separated set. Hence by (13.7.1.22),  $h_{2\gamma}(f) \leq h_\varepsilon(f)$ .  $\square$

### Application to Ramsey Theory

**Def. (13.7.1.28)[IP-System].** Let  $\mathcal{F}$  be the collection of all finite non-empty subset of  $\mathbb{N}$ . For  $\alpha, \beta \in \mathcal{F}$ , write  $\alpha < \beta$  if every element of  $\alpha$  is smaller than that of  $\beta$ .

For a commutative group  $G$ , an **IP-system** in  $G$  is a map  $T : \mathcal{F} \rightarrow G$  that  $T_{i_1, \dots, i_k} = T_{i_1} \cdot \dots \cdot T_{i_k}$ .  $\lrcorner$

**Prop. (13.7.1.29)[Furstenberg-Weiss].** Let  $G$  be a commutative group acting minimally on a compact topological space  $X$ , then for any open subset  $U$  of  $X$ ,  $n > 0$  and  $\alpha \in \mathcal{F}$ , and any IP-systems  $T^1, \dots, T^n$  on  $G$  (13.7.1.28), there is a  $\beta \in \mathcal{F}$  that  $\alpha < \beta$ , and

$$U \cap T_\beta^1(U) \cap \dots \cap T_\beta^n(U) \neq \emptyset.$$

$\lrcorner$

*Proof:* Cf. [Dynamic System, P49].  $\square$

**Cor. (13.7.1.30).** Let  $G$  be a commutative group acting homeomorphically on a compact metric space  $X$  and  $T^1, \dots, T^n$  be IP-systems on  $G$ , then for any  $\alpha \in \mathcal{F}$  and  $\varepsilon > 0$ , there are  $x \in X$  and  $\beta > \alpha$  that  $d(x, T_\beta^i(x)) < \varepsilon$  for any  $i$ .  $\lrcorner$

*Proof:* Just need to find a minimal closed subset of  $X$  for  $G$ , but this is easy by finite intersection theorem.  $\square$

**Cor. (13.7.1.31)[Multiple Recurrence Property].** Let  $T$  be a homeomorphism of a compact metric space  $X$ , then for any  $\varepsilon > 0$  and  $q > 0$ , there are  $p > 0$  and  $x \in X$  that  $d(T^{jp}(x), x) < \varepsilon$  for  $0 \leq j \leq q$ .  $\lrcorner$

*Proof:* This is a special case of (13.7.1.30), by taking  $G = \{T^k\}$ , and  $T_\alpha^i = T^{i|\alpha|}$ , where  $|\alpha|$  is the sum of elements in  $\alpha$ .  $\square$

**Cor. (13.7.1.32) [Generalized van der Waerden Theorem].** For each finite partition  $\mathbb{Z} = \cup_{i=1}^m S_k$ , one of the the subset  $S_k$  contains arbitrarily long arithmetic progressions.

More generally, let  $A$  be a finite subset of  $\mathbb{Z}^d$ , then for each partition  $\mathbb{Z}^d = \cup_{i=1}^m S_k$ , there are some  $k$ ,  $z_0 \in \mathbb{Z}^d$  and  $n > 0$  that  $z_0 + na \in S_k$  for any  $a \in A$ .  $\lrcorner$

*Proof:* Consider the product space  $\Sigma_m = \{1, 2, \dots, m\}^{\mathbb{Z}}$  with the 2-adic metric with shifting  $\sigma$ . Then a partition of  $\mathbb{Z}$  can be viewed as an element  $\xi$  in  $\{1, 2, \dots, m\}^{\mathbb{Z}}$ , with  $\xi_i = k$  if  $i \in S_k$ . Let  $X = \overline{\cup_{i=-\infty}^{\infty} \sigma^i(\xi)}$ . Let  $A_k = \{\omega \in X | \omega_0 = k\}$ , then if  $x \in A_k, y \in X$  with  $d(x, y) < 1$ , then  $y \in A_k$  also. Thus for any  $q > 0$ , we can use (13.7.1.31) to show that there is an  $\omega \in X$  that  $d(\sigma^{ip}(\omega), \omega) < 1$  for  $0 \leq i \leq q$ , thus there is an  $r \in \mathbb{Z}$  that  $\xi_j = \omega_0$  for  $j = r, r + k, \dots, r + pq$ . And this proves the theorem.

The proof of the general case is similar.  $\square$

## 2 Symbolic Dynamics

### Shifts

**Def. (13.7.2.1) [Subshifts].** A **subshift** is a closed subset  $X \subset \Sigma_m$  that is invariant under the shift  $\sigma$  and  $\sigma^{-1}$ . A map between to subshifts of  $\Sigma_m$  is called a **code** if it commutes with  $\sigma$ .  $\lrcorner$

**Prop. (13.7.2.2).** Let  $X$  be a subshift of  $\Sigma_m$  (13.7.2.1), let  $W_n(X)$  be the set of words of length  $n$  that occurs in  $X$ ,  $\sigma|_X$  is clearly expansive of constant 1, thus we have

$$h(\sigma|_X) = \lim_{n \rightarrow \infty} \frac{1}{n} \log |W_n(X)|.$$

$\lrcorner$

*Proof:* This is because every element in  $W_n$  appears in the first  $n$  term of some  $\omega \in X$ , and elements in a set of  $d_n$ -diameter  $< 1$  has first  $n$  entries the same. For the other direction, notice a  $(n, 1)$ -separated set has the first  $n$ -entries not the same, thus contribute to different elements in  $W_n(X)$ , thus  $\text{sep}(n, 1, \sigma|_X) < W_n(X)$ .  $\square$

**Def. (13.7.2.3) [block code].** Let  $X$  be a subshift,  $k, l \geq 0, n = k + l + 1$ , and  $\alpha$  be a map from  $W_n(X)$  to  $\{0, 1, \dots, m' - 1\}$ , then a  $(k, l)$ -block code is a morphism from  $X$  to  $\Sigma_{m'}$  that maps  $x$  to the sequence  $a_\alpha(x)$  that  $a_\alpha(x)_i = \alpha((x_{i-k}, \dots, x_{i+l}))$ .

When  $\{0, 1, \dots, m' - 1\}$  is chosen to be  $W_n(X)$  itself, then this is called a **higher block presentation** of  $X$ .  $\lrcorner$

**Prop. (13.7.2.4) [Every Code is a block code, Curtis-Lyndon-Hedlund].** Every code  $c : X \rightarrow Y$  is a block code.  $\lrcorner$

*Proof:* Let  $\mathcal{A}$  be the symbol set of  $Y$ , define  $\hat{c} : X \rightarrow \mathcal{A} : x \mapsto c(x)_0$ . This is a continuous map, and  $X$  is compact thus it is uniformly continuous, thus there is a  $\delta > 0$  that  $\hat{c}(x) = \hat{c}(y)$  if  $d(x, y) < \delta$ . Thus there is a large  $k$  that  $\hat{c}$  only depends on the first  $2k + 1$  term, and it commutes with shifting, thus it is a block code.  $\square$

**Def. (13.7.2.5) [SFT].** For  $k \in \mathbb{Z}_+$ , a **subshift of finite type** or SFT of  $k$ -step is a subshift  $X$  that are defined to be the elements in  $\Sigma_m$  that doesn't contain some set of words of length  $k + 1$ . When  $k = 1$ , this is also called a **topological Markov chain**.  $\lrcorner$



**Prop. (13.7.2.6).** Every SFT is isomorphic to a vertex shift.  $\lrcorner$

*Proof:* For this SFT  $X$  of step  $k$ , construct a graph, whose vertices are  $W_k(X)$ , and two element of  $W_k(X)$  are connected by an edge if they adjoint to an element in  $W_{k+1}(X)$ .  $\square$

**Cor. (13.7.2.7).** Every SFT is isomorphic to an edge shift. This is because every vertex shift can be 2-block isomorphic to its edge shift.  $\lrcorner$

**Prop. (13.7.2.8) [Perron].**  $\lrcorner$

### Sofic Shifts and Data Transmission

**Def. (13.7.2.9) [Sofic Shifts].** A subshift  $X \subset \Sigma_m$  is called **sofic** if it is a factor of a subshift of finite type.  $\lrcorner$

**Prop. (13.7.2.10).** A subshift  $X \subset \Sigma_m$  is sofic iff it is isomorphic to an infinite path shift for some directed graph  $\Gamma$  (Notice that different edge in  $\Gamma$  can be labeled the same).  $\lrcorner$

*Proof:* Clearly such a path shift is a factor of the edge shift of  $\Gamma$ , thus it is sofic. Conversely, A sofic shift is a factor of some edge shift  $c : \Sigma_A^e \rightarrow X$ , by (13.7.2.7), and  $c$  is a block code, by (13.7.2.3). Composing with the higher block code presentation, we may assume  $c$  is a  $(0, 1)$ -block code,  $\square$

## 3 Ergodic Theory

**Prop. (13.7.3.1) [Poincaré's Recurrence Theorem].** Let  $T$  be a measure-preserving transformation of a finite measure space  $X$ , if  $A$  is a measurable set, then for a.e.  $x \in A$ , there is some  $n > 0$  that  $T^n(x) \in A$ .  $\lrcorner$

*Proof:* Let  $B$  be the set of points contradicting this property,  $B = A \setminus \bigcup_{k>0} T^{-k}A$ , thus all the preimages  $T^{-k}B$  are disjoint, measurable and have the same measure as  $B$ , thus it must has measure 0 as  $X$  has finite measure.  $\square$

**Cor. (13.7.3.2) [Derivative Transformation].** Given a finite measurable space and a measure-preserving map  $T : X \rightarrow X$  and a measurable subset  $A$  of finite measure, then the **derivative transformation**  $T_A : A \rightarrow A : x \mapsto T^k(x)$ , where  $k > 0$  is the first integer that  $T^k(x) \in A$ . By Poincaré's Recurrence theorem, the derivative transformation is defined on a subset of full measure.  $\lrcorner$

**Prop. (13.7.3.3).** Let  $X$  be a second countable metric space and  $\mu$  a Borel probability measure on  $X$ , and  $f : X \rightarrow X$  is a measure preserving continuous map, then a.e. point  $x \in X$  is recurrent, i.e.  $\text{Supp } \mu \subset \mathcal{R}(f)$ .  $\lrcorner$

*Proof:* There is a countable family of basis  $\{U_i\}$  of nbhds of  $X$ , and for each  $U_i$ , elements in  $U_i$  returns to  $U_i$  except for a set  $X_i$  of measure 0. Then  $\mathcal{R}(f) = X \setminus \bigcup_i X_i$  has full measure.  $\square$

**Def. (13.7.3.4) [Ergodicity].** A measure-preserving transformation  $T$  of a measure space  $X$  is called **ergodic** if any essentially  $T$ -invariant measurable subset has measure 0 or full measure.  $\lrcorner$

**Prop. (13.7.3.5).** Let  $T$  be a measure-preserving transformation on a finite measure space  $X$  and  $p \in (0, \infty]$ , then  $T$  is ergodic iff every essentially  $T$ -invariant function  $f \in L^p(X, \mu)$  is constant.  $\lrcorner$

*Proof:*

$\square$



**Prop. (13.7.3.6).** Let  $X$  be a measure space and  $f$  is an essentially invariant function for a measure-preserving map or flow  $T$  on  $X$ , then there is a strictly invariant measurable function  $\hat{f}$  that  $f = \hat{f}$  a.e.. ┘

*Proof:* [Dynamic System P74]. □

**Def. (13.7.3.7).** A measure-preserving transformation or flow on a probability space  $X$  is called **mixing** if

$$\lim_{t \rightarrow \infty} \mu(T^{-t}A \cap B) = \mu(A)\mu(B).$$

It is called **weak mixing** if

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n |\mu(T^{-i}A \cap B) - \mu(A)\mu(B)| = 0.$$

or

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t |\mu(T^{-t}A \cap B) - \mu(A)\mu(B)| = 0.$$

┘

**Prop. (13.7.3.8).** Mixing transformation is weak mixing, and weak mixing is ergodic. ┘

*Proof:* For a weak mixing transformation, if  $A$  is essentially invariant, then  $\mu(A) = \mu(A)^2$ , thus  $\mu(A) = 1$  or  $0$ . □

**Prop. (13.7.3.9) [Mixing and Topological Mixing].** Let  $X$  be a compact metric space,  $T : X \rightarrow X$  be a continuous map and  $\mu$  a  $T$ -invariant Borel measure on  $X$ , then

- If  $T$  is ergodic, then the orbit of a.e. point  $x \in X$  is dense in  $\text{Supp } \mu$ .
- If  $T$  is mixing, then  $T$  is topologically mixing on  $\text{Supp } \mu$ .

┘

*Proof:* 1: Let  $U$  be an open subset intersecting  $\text{Supp } \mu$ , then the  $T$ -invariant subset  $\cup_{k>0} T^{-k}U$  has full measure, thus the forward orbit of a.e.  $x$  intersect  $U$ . Then take a countable open basis of  $X$ , thus the forward orbit of a.e.  $x$  is dense in  $\text{Supp } \mu$ .

2: Because  $\lim_{t \rightarrow \infty} T^{-t}(A) \cap B \rightarrow \mu(A)\mu(B) > 0$ , so does  $\lim_{t \rightarrow \infty} A \cap T^t(B)$ . □

### Ergodic Theorems

**Prop. (13.7.3.10) [von Neumann Ergodic Theorem].** If  $U \in L(H)$  is unitary and  $x \in H$ , then the average  $A_n x = \frac{1}{n}(x + Ux + \dots + U^{n-1}x)$  converges to some  $Px$ , where  $P$  is the projection to the fixed space of  $U$ . ┘

*Proof:* Define  $a_n = \frac{1}{n}(1 + \lambda + \dots + \lambda^{n-1})$  on the unit circle, and  $b(1) = \chi_{\{1\}}$ , then if  $y = b(U)x = Px$ , then  $\|y - A_n x\|^2 = \int_{\sigma(U)} |b - a_n|^2 dE_{x,x}(\lambda)$ . But this integral converges to 0 by dominated convergence theorem. □

**Prop. (13.7.3.11) [Birkhoff Ergodic Theorem].** If  $T$  is a measure-preserving transformation of a finite measure space  $(X, \mu)$ , and let  $f \in L^1(X, \mu)$ , then the limit

$$\bar{f}(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(T^i(x))$$

exists for a.e.  $x$  and is  $\mu$ -integrable and  $T$ -invariant, and satisfies

$$\int_X \bar{f}(x) d\mu = \int_X f(x) d\mu$$

In addition, if  $f \in L^2(X, \mu)$ , then  $\bar{f}$  is just the projection of  $f$  to the subspace of  $T$ -invariant measures.

The same thing is true for a measure-preserving flow.  $\lrcorner$

*Proof:* Let

$$A = \{x \in X | f(x) + f(T(x)) + \dots + f(T^k(x)) \geq 0, \quad \exists k \geq 0\}.$$

Then firstly we have  $\int_A f(x) d\mu \geq 0$ : Cf.[Dynamic System, P82].  $\color{red}?$   $\square$

**Cor.(13.7.3.12).** A measure-preserving map  $T : X \rightarrow X$  in a finite measure space  $(X, \mu)$  is ergodic iff for each  $f \in L^1(X, \mu)$ ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(T^i(x)) = \frac{1}{\mu(X)} \int_X f(x) d\mu, \text{ a.e. } x.$$

I.e., the time average equals the space average for any  $L^1$ -function.  $\lrcorner$

*Proof:* If  $T$  is ergodic, then the function  $\bar{f}$  defined in (13.7.3.11) is a constant, thus the equation. Conversely, if this equation holds, then if  $f$  is  $T$ -invariant, the RHS is constant.  $\square$

**Cor.(13.7.3.13).** Taking a dense subset of  $L^2(X, \mu)$  in the above corollary, we see that a measure-preserving map  $T : X \rightarrow X$  is ergodic iff for any measurable subset  $A$  and a.e.  $x \in X$ ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \chi_A(T^i(x)) = \frac{\mu(A)}{\mu(X)}.$$

$\lrcorner$

## Invariant Measure for a Transformation

### 4 Chaos

#### Li-Yorke Chaos

**Prop.(13.7.4.1)** [Li-Yorke[L-Y75]]. Let  $J \subset \mathbb{R}$  be an interval,  $F : J \rightarrow J$  a continuous map and there is a point  $a \in J$  s.t.

$$F^{\circ 3}(a) \leq a < F(a) < F^{\circ 2}(a)$$

or

$$F^{\circ 3}(a) \geq a > F(a) > F^{\circ 2}(a).$$

Then

- For any  $k \in \mathbb{Z}_+$ , there exists a periodic point in  $J$  with period  $k$ .
- There is an uncountable subset  $S \subset J$  that satisfies:
  - $S$  contains no periodic points.
  - For any  $p \neq q \in S$ ,  $\limsup_n |F^{\circ n}(p) - F^{\circ n}(q)| > 0$  and  $\liminf_n |F^{\circ n}(p) - F^{\circ n}(q)| = 0$ .
  - For any  $p \in S$  and periodic point  $q \in J$ ,  $\limsup_n |F^{\circ n}(p) - F^{\circ n}(q)| > 0$ .

┘

**Remark (13.7.4.2).** Notice the hypothesis is satisfied when  $F$  has a period point with period 3. ┘

*Proof:* 1: We may assume that  $F^{\circ 3}(a) \leq a < F(a) < F^{\circ 2}(a)$  and let  $K = [a, F(a)]$ ,  $L = [F(a), F^{\circ 2}(a)]$ . For  $k \in \mathbb{Z}_+$ , define a sequence of intervals  $(I_n)_{n \in \mathbb{N}}$  s.t.

- If  $k = 1$ , define  $I_n = L$ .
- if  $k \geq 1$ , define  $I_0 = I_1 = \dots = I_{k-1} = L$ ,  $I_k = K$ , and  $I_{n+k} = I_n$  for any  $n \in \mathbb{N}$ .

Then the hypothesis of (13.7.4.3) is satisfied and we get a nested sequence of intervals  $(Q_n)_{n \in \mathbb{N}}$ . Now  $Q_k \subset Q_0 \subset I_0 = F^{(k)}(Q_k)$ , so there exists  $p_k \in Q_k \subset L$  s.t.  $F^{(k)}(p_k) = p_k$ . Then the hypothesis implies  $p_k$  has period  $k$ , because otherwise  $k \geq 2$  and  $F^{(k-1)}(p_k) = F(a)$ , thus  $p_k = F^{\circ 2}(a)$ , contradicting the fact that  $F(p_k) \subset L$ .

2: Cf. [L-Y75]P990. ?

□

**Lemma (13.7.4.3).** Let  $J \subset \mathbb{R}$  be an interval,  $F : J \rightarrow J$  a continuous map and  $(I_n)_{n \in \mathbb{N}}$  be a sequence of compact intervals contained in  $J$  s.t.  $I_{n+1} \subset F(I_n)$ . Then there is a sequence of intervals  $(Q_n)_{n \in \mathbb{N}}$  with  $Q_{n+1} \subset Q_n \subset I_0$ , and  $F^{(n)}(Q_n) = I_n$ . ┘

*Proof:* Define  $Q_0$  and define inductively  $Q_n$ . If  $Q_n$  is defined, then  $I_{n+1} \subset F^{(n+1)}(Q_n) = F(I_n)$ , so it is easy to find  $Q_{n+1} \subset Q_n$  s.t.  $F^{(n+1)}(Q_{n+1}) = I_{n+1}$ . □

**Remark (13.7.4.4).** ? Is it true that the existence of a stable periodic point implies that almost every point is asymptotically periodic? ┘

## 5 Hyperbolic Dynamics

## 13.8 Complex Dynamical Systems

See [Milnor, John Dynamics in one complex variable], [Complex Dynamics and Renormalization, McMullen] and [C-G93].

### 1 Local Behaviors

**Notation (13.8.1.1).**

- For  $d \in \mathbb{Z}_{\geq 2}$ , denote  $\text{Rat}_d$  the set of non-constant rational maps  $f : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}^1(\mathbb{C})$  of degree  $d$ . And denote  $\text{Rat} = \bigcup_{d \in \mathbb{Z}_{\geq 2}} \text{Rat}_d$ . ┘

**Def. (13.8.1.2)[Attracting and Repelling].** For a Riemann surface  $\Omega$  and  $f : \Omega \rightarrow \Omega$ , suppose  $x \in \Omega$  is a periodic point for  $f$  of periodicity  $k$ , the **Multiplier** of  $f$  at  $x$  is defined to be  $\lambda_f(x) = (f^{(k)})'(x) \in \mathbb{C}$ . Then  $x$  is called

- an **anattracting point** for  $f$  iff  $|\lambda_f(x)| < 1$ .
- a **repelling pint** for  $f$  iff  $|\lambda_f(x)| > 1$ .
- a **superattracting point** if  $\lambda_f(x) = 0$ .
- called a **rationally neutral point** if  $\lambda_f(x) \in \mu(\mathbb{C})$ .
- called an **irrationally neutral point** if  $|\lambda_f(x)| = 1$  but  $\lambda_f(x) \notin \mu(\mathbb{C})$ .

Hyperbolic points and irrationally neutral points are all called **neutral points** of  $f$ . ┘

**Def. (13.8.1.3)[Stable Points].** For a Riemann surface  $\Omega$  and  $f : \Omega \rightarrow \Omega$ , suppose  $x \in \Omega$  is a fixed point of  $f$ , then  $x$  is called a **stable point** if there exists  $0 \leq r_0, r \leq R$  s.t.  $f^{on}(z) \in \mathbb{D}(z_0, r)$  for any  $z \in \mathbb{D}(z_0, r_0)$ ,  $n \in \mathbb{Z}_+$ .

Later it will be clear that a fixed point  $z_0$  is stable iff  $z_0 \in \text{Fat}(f)$  (13.8.3.1). ┘

**Prop. (13.8.1.4)[Attracting Fixed Points, Koenigs].** Suppose  $f \in \mathcal{O}(\mathbb{C})$  has an attracting fixed point  $z_0 \in \mathbb{C}$  with  $\lambda = f'(z_0) \in B(0, (0, 1))$ . Then there is a unique conformal mapping  $\varphi$  from a nbhd of  $z_0$  to a nbhd of 0 that conjugates  $f$  to the linear function  $g(\zeta) = \lambda\zeta$ , and  $\varphi'(z_0) = 1$ . ┘

*Proof:* The uniqueness is clear. For the existence, suppose  $z_0 = 0$ . For  $n \in \mathbb{Z}_+$ , let  $\varphi_n = \lambda^{-n} f^{(n)}$ , then  $\varphi_n$  satisfies  $\varphi_n \circ f = \lambda \varphi_{n+1}$ . If there exists  $\varphi \in \mathcal{O}(\mathbb{C})$  s.t.  $\varphi_n \rightarrow \varphi$  pointwise, then  $\varphi \circ f = \lambda \varphi$ .

Notice for  $\delta \in \mathbb{R}_+$  small,  $|f(z) - \lambda z| \leq C|z|^2$  for any  $|z| \leq \delta$ . So  $|f(z)| \leq |\lambda z| + C|z|^2 \leq (|\lambda| + C\delta)|z|$ . So when  $\delta$  is small s.t.  $|\lambda| + C\delta < 1$ ,

$$|f^{(n)}(z)| \leq (|\lambda| + C\delta)^n |z|, \quad \forall |z| \leq \delta.$$

If  $\delta$  is small s.t.  $\rho = (|\lambda| + C\delta)^2 / |\lambda| < 1$ , then

$$|\varphi_{n+1}(z) - \varphi_n(z)| = \left| \frac{f(f^{(n)}(z)) - \lambda f^{(n)}(z)}{\lambda^{n+1}} \right| \leq \frac{C|f^{(n)}(z)|^2}{|\lambda|^{n+1}} \leq \frac{\rho^n C|z|^2}{|\lambda|}.$$

So  $\varphi_n$  converges uniformly for  $|z| \leq \delta$ , to some holomorphic function  $\varphi$ . □

**Cor. (13.8.1.5)[Repelling Fixed Points].** The same applies to repelling fixed points, because we can consider  $f^{(-1)}$  instead. The corresponding Schröder's equation for  $h = \varphi^{(-1)}$  is  $h(\lambda\zeta) = f(h(\zeta))$ . Then it can be verified that  $h$  can be extended to an entire function of finite order. ┘

**Prop. (13.8.1.6) [Superattracting Fixed Points, Boettcher].** Suppose  $f \in \mathcal{O}(\mathbb{C})$  has a superattracting fixed point  $z_0 \in \mathbb{C}$  of order  $p \geq 2$  s.t.  $f(z) = z_0 + a_p(z - z_0)^p + \dots$ , then there exists a conformal mapping  $\varphi$  of some nbhd of  $z_0$  to a nbhd of 0 that conjugates  $f$  to the mapping  $g(\zeta) = \zeta^p$ . And the conjugation is unique up to multiplication by a  $(p-1)$ -th roots of unity.  $\lrcorner$

*Proof:* Suppose  $z_0 = 0$ . If we change variables  $w = cz$  with  $c^{p-1} = 1/a_p$ , then  $f(w) = w^p + \dots$ . Let  $\varphi_n = (f^{(n)})^{p^{-n}}$ , then  $\varphi_{n-1} \circ f = (f^{(n-1)} \circ f)^{p^{-n+1}} = \varphi_n^p$ . If there exists  $\varphi \in \Omega(\mathbb{C})$  s.t.  $\varphi_n \rightarrow \varphi$  pointwise, then  $\varphi \circ f = \varphi^p$ .

Notice for  $\delta \in \mathbb{R}_+$  small,  $|f(z)| \leq C|z|^p$  for some  $C \in \mathbb{R}_{\geq 1}$ . So  $|f^{(n)}(z)| \leq (C|z|)^{p^n}$ . Thus

$$\frac{\varphi_{n+1}}{\varphi_n} = \left( \frac{\varphi_1 \circ f^{(n)}}{f^{(n)}} \right)^{p^{-n}} = (1 + O(|f^{(n)}|))^{p^{-n}} = 1 + O(p^{-n}(C|z|)^{p^n}) = 1 + O(p^{-n}),$$

when  $|z| \leq 1/C$ . So  $\prod_n \frac{\varphi_{n+1}}{\varphi_n}$  is uniformly bounded for  $|z| \leq 1/C$ , so  $\varphi_{n+1}$  converges to some holomorphic function  $\varphi$ .  $\square$

**Prop. (13.8.1.7) [Neutral Fixed Points].** Suppose  $f \in \mathcal{O}(\mathbb{C})$  has a neutral fixed point  $z_0 \in \mathbb{C}$  with  $f'(z_0) = \lambda$ . We may find a local conformally mapping  $h$  from a nbhd of  $z_0$  to a nbhd of 0 that conjugates  $f$  to a linear function  $g(\zeta) = \lambda\zeta$ . If  $h = \varphi^{(-1)}$ , then  $h$  must satisfy the Schröder's equation

$$h(\lambda\zeta) = f(h(\zeta)).$$

Then this equation is solvable iff  $z_0$  is a stable points for  $f$  (13.8.1.3)  $\lrcorner$

*Proof:* Suppose  $z_0$  is stable, let  $\Omega = \cup_{n \in \mathbb{Z}_+} f(\mathbb{D}(z_0, r_0))$ , then  $\Omega \subset \mathbb{D}(0, r)$  and is connected. Suppose  $\varphi : \mathbb{D} \rightarrow \Omega$  is a covering map that  $\varphi(0) = 0$  (12.4.7.10), then the map  $f : \Omega \rightarrow \Omega$  lifts to a map  $F : \mathbb{D} \rightarrow \mathbb{D}$ , thus by Schwartz lemma (12.5.1.3),  $F(z) = \lambda z$ , and the Schröder's equation has a solution  $\varphi$ .

Conversely, it is clear if the Schröder's equation has a solution  $\varphi$ , then  $z_0$  is stable.  $\square$

**Cor. (13.8.1.8).** Situation as in (13.8.1.7),  $f$  is topologically conjugate to  $g(\zeta) = \lambda\zeta$  near  $z_0$  iff  $f$  is conformally conjugate to  $g(\zeta) = \lambda\zeta$  near  $z_0$ .  $\lrcorner$

**Def. (13.8.1.9) [Siegel Disks].** A **Siegel disk** is a simply-connected  $f$ -fixed region  $\Omega \subset \mathbb{C}$  s.t. the action of  $f$  on it is conjugate to a rotation.

By (13.8.1.7), if  $f$  has a fixed point  $z_0$ , then a nbhd of  $z_0$  that is a Siegel disk iff  $z_0$  is a stable point for  $f$ .  $\lrcorner$

**Thm. (13.8.1.10) [Siegel].** Situation as in (13.8.1.7),

- If  $\lambda^n = 1$  for some  $n \in \mathbb{Z}_+$ , then  $z_0$  is stable for  $f$  iff  $f^{(m)}(z_0) = z_0$  for some  $m \in \mathbb{Z}_+$ . And if this is the case, then  $f^{(n)} = 1$ .
- If  $\lambda = e^{2\pi i \omega}$  where  $\omega$  is a Diophantine number (20.2.4.6), then  $z_0$  is a stable point for  $f(z)$ .
- There exists a case  $\lambda = e^{2\pi i \theta}$  s.t. the Schröder's equation is unsolvable.  $\lrcorner$

*Proof:* 1: If  $z = 0$  is stable, then  $f^{(n)}(z) = z$  by (13.8.1.7). Conversely, if  $f^{(n)}(z) = z$ , then clearly  $z = 0$  is stable.

3: Take  $f(z) = \lambda z + \dots + z^d$ , and suppose there is a solution  $h$  to the Schröder's equation that covers a nbhd  $\mathbb{D}(z_0, \delta)$  of  $z_0$ , then  $f^{(n)}(z) = z$  has only one zero in  $\mathbb{D}(0, \delta)$ . But

$$f^{(n)}(z) - z = z^{d^n} + \dots + (\lambda^n - 1)z,$$

so all the zeros except for  $z =$  are outside  $\mathbb{D}(0, \delta)$ . Thus

$$\delta^{d^n} \leq \prod |z_j| = |\lambda^n - 1|.$$

But if we take  $\lambda = e^{2\pi i \theta}$  and  $\theta = \sum_{k \in \mathbb{Z}} 2^{-q_k}$  where  $q_1 < q_2 < \dots$  is an sequences of integers increasing rapidly, then the above inequality is not possible.

2:(See also [C-G93]P43?) Use(13.8.1.7), the Schröder functional equation writes:

$$\sum_{k=2}^{\infty} c_k (a_1^k - a_1) \zeta^k = \sum_{l=2}^{\infty} a_l (\zeta + \sum_{r=2}^{\infty} c_r \zeta^r)^l.$$

Thus it is clear that  $c_2 = a_2/a_1(a_1 - 1)$ , and  $c_k$  is determined by  $c_2$  inductively by

$$c_k = \frac{1}{a_1^k - a_1} \sum_{l_1 + \dots + l_r = k} c_{l_1} \cdots c_{l_r}.$$

So  $\varphi$  is formally determined by  $f$ . Moreover, the modulus of  $c_k$  are bounded by the solution of the functional equation

$$\sum_{k=2}^{\infty} c_k |a_1^{k-1} - 1| \zeta^k = \sum_{l=2}^{\infty} |a_l| (\zeta + \sum_{r=2}^{\infty} c_r \zeta^r)^l,$$

and the bound depends positively on  $|a_l|$  and negatively on  $|a_1^{k-1} - 1|$ .

We have  $\overline{\lim} |a_n|^{\frac{1}{n}} = 1/R$ , so we have  $|a_n| \geq a^{n-1}$  for some  $a \in \mathbb{R}_+$  and any  $n \geq 2$ . Notice that the Schröder functional remains true under the transformation  $f(z) \mapsto af(z/a)$ ,  $\varphi(\zeta) \mapsto a\varphi(\zeta/a)$ , so we may assume that  $|a_n| \leq 1$  for  $n \geq 2$ . Then it suffices to show the solution  $\varphi(\zeta) = \zeta + \sum_{k=2}^{\infty} \tau_k \zeta^k$  of the functional equation

$$\sum_{k=2}^{\infty} c_k |a_1^{k-1} - 1| \zeta^k = \sum_{l=2}^{\infty} (\zeta + \sum_{r=2}^{\infty} c_r \zeta^r)^l$$

has positive radius of convergence. We also denote the solution of the functional equation

$$\sum_{k=2}^{\infty} c_k \zeta^k = \sum_{l=2}^{\infty} (\zeta + \sum_{r=2}^{\infty} c_r \zeta^r)^l$$

by  $\psi(\zeta) = \zeta + \sum_{k=2}^{\infty} \sigma_k \zeta^k$ . Then

$$\psi(\zeta) - \zeta = \psi(\zeta)^2 / (1 - \psi(\zeta)),$$

thus

$$4\psi(\zeta) = \zeta + 1 - \sqrt{1 - 6\zeta + \zeta^2}$$

has radius of convergence  $R = 2 - 2\sqrt{2}$ . And we can prove inductively that

$$\sigma_k \leq \delta_k \tau_k,$$

where  $\delta_k$  are defined as follows:

$$\delta_1 = 1, \quad \delta_k = |a_1^{k-1} - 1| \cdot \max_{l_1 + \dots + l_r = k} \delta_{l_1} \cdots \delta_{l_r}.$$

But then by(13.8.1.11),  $\sigma_k < k^{-2\nu} 2^{(5\nu+1)(k-1)} \tau_k$ . So  $\varphi(\zeta)$  has radius of convergence  $\geq (3-2\sqrt{2})2^{-5\nu-1}$ .

□

**Lemma (13.8.1.11).** Situation as in (13.8.1.10), suppose  $|a_1^n - 1| \leq (2n)^\nu$  for  $\nu \in \mathbb{R}_+$  and any  $n \in \mathbb{Z}_+$ , then

$$\delta_k \leq k^{-2\nu} 2^{(5\nu+1)(k-1)}.$$

┘

*Proof:* Cf. [Sie42] Lemma 3. ?

□

**Cor. (13.8.1.12).** Let  $f(z) = z + \sum_{k \geq 2} a_k z^k$  be a power series with positive radius of convergence, then  $z = 0$  is a stable point iff  $f(z) = z$ .

┘

**Prop. (13.8.1.13) [Rationally Neutral Fixed Points].** Suppose  $f \in \mathcal{O}(\mathbb{C})$  has a neutral fixed point  $z_0 \in \mathbb{C}$  with  $f(z) = \lambda z + az^{p+1} + \dots$ ,  $a \in \mathbb{C}^\times$ ,  $p \in \mathbb{Z}_+$ , then there is a conformally mapping  $\varphi$  from a “half nbhd” of  $z_0$  to a nbhd of  $\infty \in \mathcal{H} \cup \{\infty\}$  with  $g(\zeta) = \zeta + 1$ .

┘

*Proof:*

□

## 2 Rational Dynamics

**Prop. (13.8.2.1).** Any  $f \in \text{Rat}$  has infinitely many periodic points.

┘

*Proof:* Notice the degree of  $f^{(n)}(z) - z$  tends to infinity as  $n \rightarrow \infty$ , so if there are only f.m. periodic points  $\{x_1, \dots, x_r\}$ , the multiplicity of one of these points as a period point of  $f^{(n)}(z) - z$  must be large when  $n$  is large. But suppose  $f^{(n)}(\xi) - \xi = 0$  and  $\xi$  has multiplicity  $m \geq 2$  for  $f^{(n)}$ , then near  $\xi$ ,

$$f^{(n)}(z) = z + (z - \xi)^m + \dots,$$

thus for  $k \in \mathbb{Z}_+$ , by induction,

$$f^{(nk)}(z) = z + k(z - \xi)^m + \dots,$$

so  $\xi$  has multiplicity  $m$  for  $f^{(nk)}$ . Thus we can argue that for all sufficiently divisible  $N$ ,  $f^{(N)}(z) - z$  has only f.m. solutions (counting multiplicity), contradiction. □

**Thm. (13.8.2.2) [Shishikuka87].** For  $f \in \text{Rat}$ , there are at most  $2d - 2$  attracting or neutral periodic points for  $f$ .

┘

*Proof:* ?

□

**Cor. (13.8.2.3).** Any  $f \in \text{Rat}$  contains infinitely many repelling periodic points, by (13.8.2.1).

┘

## 3 Julia Sets

**Def. (13.8.3.1) [Fatou Sets and Julia Sets].** For any  $f \in \text{Rat}$ , the **Fatou set**  $\text{Fat}(f)$  is defined to be the set of points  $z \in \mathbb{P}^1(\mathbb{C})$  s.t. the family  $\{f^{(n)}\}_{n \in \mathbb{Z}_+}$  is Montel-normal at  $z$ . And the **Julia set**  $\text{Jul}(f)$  is defined to be  $\text{Jul}(f) = \mathbb{P}^1(\mathbb{C}) \setminus \text{Fat}(f)$ . Each component of  $\text{Fat}(f)$  is called a **Fatou component** of  $f$ . By (13.8.2.3) and (13.8.3.6),  $\text{Jul}(f) \neq \emptyset$ .

┘

**Prop. (13.8.3.2).** For  $f \in \text{Rat}$  and  $k \in \mathbb{Z}_+$ ,  $\text{Jul}(f) = \text{Jul}(f^{(k)})$ , by (12.5.1.11).

┘

**Thm. (13.8.3.3) [Complete Invariance of Julia Sets].** For any  $f \in \text{Rat}$ ,  $\text{Jul}(f)$  and  $\text{Fat}(f)$  are both completely invariant under  $f$ , i.e.  $f(\text{Jul}(f)) = f^{-1}(\text{Jul}(f)) = \text{Jul}(f)$ , and  $f(\text{Fat}(f)) = f^{-1}(\text{Fat}(f)) = \text{Fat}(f)$ .

┘

*Proof:* If  $\{f^{(n)}\}$  is Montel-normal at  $f(x)$ , then it is Montel-normal at  $x$  because  $f$  is continuous. Conversely, if  $\{f^{(n)}\}$  is Montel-normal at  $x$ , suppose  $f$  is Montel-normal on a nbhd  $U$  of  $x \in X$ , and let  $V$  be a nbhd of  $x$  s.t.  $x \in V \subset \bar{V} \subset U$ . Then because  $f$  is open, we can show that  $\{f^{(n)}\}$  is Montel-normal on  $f(V)$ .  $\square$

**Prop. (13.8.3.4).** For  $f \in \text{Rat}$ ,  $\text{Jul}(f)$  is either  $\mathbb{P}^1(\mathbb{C})$  or has no interior.  $\lrcorner$

*Proof:* Suppose  $U$  is open and  $U \subset \text{Jul}(f)$ , then  $\{f^{(n)}\}$  is not Montel-normal on  $U$ . Then by (12.5.1.10),  $\cup_{n \in \mathbb{N}} f^{(n)}(U) \subset \mathbb{P}^1(\mathbb{C})$  omits at most 2 points. As  $\text{Jul}(f)$  is  $f$ -invariant and closed, this implies  $\text{Jul}(f) = \mathbb{P}^1(\mathbb{C})$ .  $\square$

**Prop. (13.8.3.5)[Julia Sets and Periodic Points].** For  $f \in \text{Rat}$ ,  $\text{Jul}(f)$  contains all repelling periodic points and unstable neutral periodic points(not corresponding to Siegel disks); and  $\text{Fat}(f)$  contains all attracting periodic points and all stable neutral periodic points(corresponding to Siegel disks).  $\lrcorner$

*Proof:* This follows from (13.8.1.9) and the definition of stable points(13.8.1.3).  $\square$

**Prop. (13.8.3.6).** For  $f \in \text{Rat}$ , if  $x$  is an attracting point for  $f$ , then  $BA(x) \subset \text{Fat}(f)$ . And if  $x$  is a repelling periodic point, then  $x \in \text{Jul}(f)$ .  $\lrcorner$

*Proof:* If  $x$  is attracting, then  $\{f^{(n)}\}$  is eventually uniformly bounded on each compact subset of  $BA(x)$ , thus Montel-normal. If  $x$  is repelling, then  $|f'(z)| > 1$ , so  $|(f^{(n)})'(z)|$  tends to infinity as  $n \rightarrow \infty$ . Then it is clear that  $\{f^{(n)}\}$  is not Montel-normal at  $x$ .  $\square$

**Prop. (13.8.3.7)[Exceptional Sets].** For  $f \in \text{Rat}$  and  $z \in \text{Jul}(f)$ , the **exceptional set**  $E_z$  of  $f$  at  $z$  is defined to be

$$E_z = \cup_{z \in U} \mathbb{P}^1(\mathbb{C}) \setminus \left( \cup_{n \in \mathbb{Z}_+} f^{(n)}(U) \right).$$

Then  $\#E_z \leq 2$ , and it is in fact independent of  $z \in \text{Jul}(f)$ .

- If  $\#E = 1$ , then it can be conjugated to  $\infty$ , and then  $f$  is a polynomial.
- If  $\#E = 2$ , then it can be conjugated to  $\{0, \infty\}$ , and then  $f(z) = Cz^d, d \in \mathbb{Z} \setminus \{0, \pm 1\}$ .

In particular,  $E \subset \text{Fat}(f)$ .  $\lrcorner$

*Proof:*  $\#E_z \leq 2$  by (12.5.1.10). By definition,  $f^{-1}(E_z) \subset E_z$ , so if  $\#E_z = 2$  for some  $z$ , it can be conjugated to the form as above, and then it can be verified that  $E_z = \{0, \infty\}$  for any  $z \in \text{Jul}(f)$ . If  $\#E_z = 1$  for some  $z$ , then it can be conjugated to the form as above, and  $\#E_z \geq 1$  for any  $z \in \text{Jul}(f)$ . But we also know from the above argument that  $\#E_{z'} = 2$  cannot happen for any  $z' \in \text{Jul}(f)$ , so  $E_z = \{\infty\}$  for any  $z \in \text{Jul}(f)$ .  $\square$

**Cor. (13.8.3.8).** Suppose  $f \in \text{Rat}$ , and  $z \in \text{Jul}(f)$ , then  $\cup_{n \in \mathbb{Z}_+} f^{(-n)}(z)$  is dense in  $\text{Jul}(f)$ . In particular, any completely invariant subset of  $\text{Jul}(f)$  is dense in  $\text{Jul}(f)$ .  $\lrcorner$

*Proof:* By the definition of  $E$ , if  $z \notin E$ , then for any open subset  $U \subset \mathbb{P}^1(\mathbb{C})$  s.t.  $U \cap \text{Jul}(f) \neq \emptyset$ , there exists  $n \in \mathbb{Z}_+$  s.t.  $z \in f^{(n)}(U)$ . In particular,  $U \cap \left( \cup_{n \in \mathbb{Z}_+} f^{(-n)}(z) \right) \neq \emptyset$ . As  $E \subset \text{Fat}(f)$ , any  $z \in \text{Jul}(f)$  suffices.  $\square$

**Thm. (13.8.3.9)[Mañé-Rocha].** For  $f \in \text{Rat}$ ,  $\text{Jul}(f)$  is perfect and uniformly perfect(12.5.7.2) subset.  $\lrcorner$

*Proof:* Cf. [C-G93]P57, 64.  $\square$



**Prop. (13.8.3.10) [Repelling Periodic Points are Dense].** For  $f \in \text{Rat}$ ,  $\text{Jul}(f)$  is the closure of repelling periodic points of  $f$ .  $\square$

*Proof:* As  $f$  has only f.m. attracting or neutral periodic points (13.8.2.2), and  $\text{Jul}(f)$  is perfect (13.8.3.9), so if the assertion is wrong, we will get a non-empty open subset  $U \subset \mathbb{C}$  that intersect  $\text{Jul}(f)$ , and contains no periodic points, poles and critical points of  $f$ . Since  $\deg(f) > 1$ , there are two different branches  $h_1, h_2$  of  $f^{(-1)}$  on  $U$ , and  $h_1(z) \neq f^{(n)}(z), h_2(z) \neq f^{(n)}(z)$  for any  $z \in U$  and  $n \in \mathbb{N}$ . But then the family of functions

$$h_n(z) = \frac{f^{(n)}(z) - h_1(z)}{f^{(n)}(z) - h_2(z)} \cdot \frac{z - h_2(z)}{z - h_1(z)}$$

doesn't take values  $0, 1, \infty$ , thus is a Montel-normal family (12.5.1.10). And then also  $\{f^{(n)}\}$  is a Montel-normal family on  $U$ , contradicting the definition of  $\text{Jul}(f)$ .  $\square$

**Prop. (13.8.3.11) [Self-Similarity of Julia Sets].** For  $f \in \text{Rat}$ , if  $U$  is an open subset of  $\mathbb{P}^1(\mathbb{C})$  s.t.  $U \cap \text{Jul}(f) \neq \emptyset$  then there exists some  $N \in \mathbb{Z}_+$  s.t.  $f^N(U \cap \text{Jul}(f)) = \text{Jul}(f)$ .  $\square$

*Proof:* By (13.8.3.10), choose  $z \in U \cap \text{Jul}(f)$  a repelling periodic point with period  $n$ . Then choose a nbhd  $V$  of  $z \in \mathbb{P}^1(\mathbb{C})$  s.t.  $V \subset f^{(n)}(V)$ , so  $f^{(n(j-1))}(V) \subset f^{(nj)}(V)$  for any  $j \in \mathbb{Z}_+$ . By (13.8.3.7),

$$\text{Jul}(f) \subset \mathbb{P}^1(\mathbb{C}) \setminus E \subset \bigcup_{j \in \mathbb{Z}_+} f^{(nj)}(V),$$

so by compactness of  $\text{Jul}(f)$ ,  $\text{Jul}(f) \subset f^{(nj)}(V)$  for some  $j$ .  $\square$

## 4 Fatou Components

**Prop. (13.8.4.1).** For  $f \in \text{Rat}$  and a Fatou component  $U$  of  $f$ ,  $f(U)$  is also a Fatou component of  $f$ , and the map  $f : U \rightarrow f(U)$  is proper of degree  $\leq \deg(f)$ . And  $f^{-1}(U)$  is a disjoint union of at most  $d$  Fatou components.  $\square$

*Proof:* By (13.8.3.3), this is true for any analytic covering map.  $\square$

**Prop. (13.8.4.2).** For  $f \in \text{Rat}$ , if  $U$  is a completely invariant Fatou component for  $f$ , then  $\partial U = \text{Jul}(f)$ , and every other Fatou components is simply-connected. There are at most two completely invariant Fatou components for  $f$ .  $\square$

*Proof:* Cf. [C-G93]P70. ?  
 $\partial U = \text{Jul}(f)$  by (13.8.3.8).  $\square$

**Prop. (13.8.4.3).** For  $f \in \text{Rat}$ , there can only be 0, 1, 2 or infinitely many Fatou components.  $\square$

*Proof:* Suppose there are only f.m. Fatou components of  $f$ . Let  $U$  be a Fatou component, and suppose  $U_0 = U$ , and  $f(U_{k+1}) = U_k$  for  $k \in \mathbb{N}$ . then there are some  $U_m = U_n$  for  $n \neq m$ . In this case, we see that  $U$  is periodic. And then there exists  $N \in \mathbb{Z}_+$  s.t.  $f^{(N)}(U) = U$  for any Fatou component  $U$  of  $f$ . Then each component  $U$  is totally invariant under  $f^{(N)}$ , which implies that there are at most 2 Fatou components by (13.8.4.2).  $\square$

**Thm. (13.8.4.4) [No-Wandering-Domain Theorem, Sullivan1985].** For  $f \in \text{Rat}$ , every Fatou component  $U$  of  $f$  is eventually periodic under permutation by  $f$ , i.e. there exists  $0 < n < m \in \mathbb{Z}_+$  s.t.  $f^{(n)}(U) = f^{(m)}(U)$ .  $\square$

*Proof:* If there is a contradictory Fatou component  $U$ , call it a wandering domain. Then we have:

**Lemma (13.8.4.5) [Baker].** If  $U$  is a wandering domain, then  $f^{(n)}(U)$  is simply-connected for  $n$  sufficiently large. ┘

*Proof:* □

**Lemma (13.8.4.6).**  $B(U)$  contains a infinitely-dimensional subspace  $N(U)$  of compactly-supported Beltrami differentials with the following property: If  $\mu \in N(U)$  satisfies  $\mu = \bar{\partial}v$  for some quasi-conformal vector field  $v$  and  $v|_{\partial U} = 0$ , then  $\mu = 0$ . ┘

*Proof:* □

?

□

**Thm. (13.8.4.7).** If  $f \in \text{Rat}$ , and  $A \subset \text{Jul}(f)$  is a stable point that intersect any equivalent class of  $f$  (where  $x \sim y \iff f^{(n)}(x) = f^{(n)}(y)$  for some  $n \in \mathbb{N}$ ) in at most one point, then  $\mu(A) = 0$ . ┘

*Proof:* □

**Def. (13.8.4.8) [Parabolic Components].** For  $f \in \text{Rat}$ , a **parabolic component** for  $f$  is a Fatou component  $U$  of  $f$  of period  $n$  with a neutral fixed point  $\zeta$  for  $f^{(n)}$  on its boundary with multiplier 1, and that all points of  $U$  converges to  $\zeta$  under iterations by  $f^{(n)}$ . ┘

**Def. (13.8.4.9) [Herman Rings].** For  $f \in \text{Rat}$ , a **Herman ring** for  $f$  is a doubly-connected Fatou component of  $f$  of period  $n$  s.t. the action of  $f^{(n)}$  is conjugate to either a isometry of an annulus. ┘

**Thm. (13.8.4.10) [Classifying Periodic Fatou Components].** For  $f \in \text{Rat}$ , each Fatou component  $U$  for  $f$  is exactly one of the following:

- $U$  contains an attracting periodic point
  - $U$  is parabolic(13.8.4.8).
  - $U$  is a Sigel disk(13.8.1.9).
  - $U$  is a Herman ring(13.8.4.9).
- ┘

*Proof:* Cf.[C-G93]P74. □

**Thm. (13.8.4.11) [Sullivan].** For  $f \in \text{Rat}$ , if  $\text{Fat}(f)$  has two components and  $f$  is hyperbolic on  $\text{Jul}(f)$ , then  $\text{Jul}(f)$  is a quasi-circle(12.5.7.7). ┘

*Proof:* Cf.[C-G93]P102. □

**Prop. (13.8.4.12) [Woff-Denjoy].** ┘

## 5 Conjectures

**Conj. (13.8.5.1) [Density of Hyperbolicity].** The set of hyperbolic rational maps is dense and open in the set  $\text{Rat}_d$  of rational maps on  $\mathbb{P}^1(\mathbb{C})$  of dimension  $d$ . ┘

*Proof:* □

**Conj. (13.8.5.2) [No Invariant Line Fields].** For  $f \in C(z)$ , there are no invariant fields on  $\text{Jul}(f)$ , unless  $f$  is double covered by an integral torus endomorphism. ┘

## 6 Polynomial Iterates

**Prop. (13.8.6.1) [Julia Sets and Critical Points].** For  $P \in \text{Poly}(z)$ ,

- $\text{Jul}(P)$  is connected iff there are no finite critical points in  $AB(\infty)$ , i.e. the forward orbits of any finite critical point of  $f$  is bounded.
- $\text{Jul}(P)$  is totally disconnected if for any finite critical point  $q$ ,  $P^{(n)}(q) \rightarrow \infty$  as  $n \rightarrow \infty$ .

┘

*Proof:* Cf. [C-G93]P66, 67. □

## 7 Quadratic Maps

**Prop. (13.8.7.1).** Since any polynomial  $f \in \text{Poly}(z)$  of degree 2 can be conjugated via maps  $z \mapsto z + A$  to a map of the form  $z \mapsto z^2 + c$ , to study quadratic polynomial maps, it suffices to study maps of the form  $f(z) = z^2 + c$ .

By (13.8.6.1),  $\text{Jul}(f)$  is either connected or totally disconnected (Notice if  $\{f^{(n)}(0)\}$  is unbounded, then it tends to  $\infty$ ). ┘

**Def. (13.8.7.2) [Mandelbrot Set].** The **Mandelbrot set**  $\mathbf{Man}$  is define to be the set of  $c \in \mathbb{C}$  s.t.  $\{f_c^{(n)}(0)\}$  is bounded, where  $f_c(z) = z^2 + c$ . ┘

**Prop. (13.8.7.3).**  $\mathbf{Man} \subset \overline{\mathbb{D}}(0, 2)$ . ┘

*Proof:* This is because we can show by induction that  $|f^{(n)}(0)| \geq |a|(|a| - 1)^{n-1}$  for  $n \in \mathbb{Z}$ . □

**Prop. (13.8.7.4) [Douady – Hubbard1982].** The Mandelbrot set  $\mathbf{Man} = \{a \in \mathbb{C} \mid (z^2 + a)^{(n)}(0) \leq 2, \forall n \in \mathbb{N}\}$ , and it is compact, connected and  $\mathbb{P}^1(\mathbb{C}) \setminus \mathbf{Man}$  is connected. ┘

*Proof:* Let  $f_a(z) = z^2 + a$ . If  $|a| > 2$ , then  $a \notin \mathbf{Man}$ . If  $|a| \leq 2$ ,  $f_a^{(n)}(0) = 2 + \alpha$  for some  $\alpha \in \mathbb{R}_+$ , then it can be shown by induction that  $f_a^{(n+k)}(0) \geq 2 + 4^k \alpha$  for any  $k \in \mathbb{N}$ , so  $a \notin M$ . This implies  $\mathbf{Man} = \{a \in \mathbb{C} \mid (z^2 + a)^{(n)}(0) \leq 2, \forall n \in \mathbb{N}\}$ . In particular, it is closed.

For the last assertion, suppose  $\Omega$  is a connected component of  $\mathbb{C} \setminus M$ . Then  $\max_{a \in \overline{\Omega}} |f_a^{(n)}(0)| > 2$ . If  $\Omega$  is bounded, then by maximum principle,  $|f_a^{(n)}(0)| \geq 2$  for some  $a \in \partial\Omega \subset \mathbf{Man}$ , which contradict the first assertion. So any connected component of  $\mathbb{C} \setminus \mathbf{Man}$  is unbounded, thus  $\mathbb{P}^1(\mathbb{C}) \setminus \mathbf{Man}$  is connected.

To show that  $\mathbf{Man}$  is connected, denote  $P_n(c) = f_c^n(0)$ , let  $R \in (2, \infty)$ , then

$$\mathbf{Man} = \bigcap_{n \in \mathbb{Z}_+} P_n^{-1}(\overline{\mathbb{D}}(0, R))$$

is a nested intersection of compact sets. We may choose  $R$  s.t.  $\partial\mathbb{D}(0, R)$  contains no critical points of any  $P_n$ . Thus by (4.4.6.2), it suffices to show that each  $P_n^{-1}(\overline{\mathbb{D}}(0, R))$  is connected. Suppose  $U$  is a connected component of  $P_n^{-1}(\overline{\mathbb{D}}(0, R))$  s.t.  $0 \notin \overline{U}$ , and this  $n$  is minimum. Then we show that  $0 \notin P_k(\overline{U})$  for all  $1 \leq k \leq n$ . To show this, use induction on  $k$ :  $k = 1$  case follows from the hypothesis, suppose that this is true for  $k \leq k_0 < n$ , then for  $k = k_0$ ,

$$f_c^{k+1}(0) \neq c, \quad c \in \overline{U}.$$

Notice for  $c \in \partial U$ ,

- $f_c^{k+1-n}(\partial D(0, R))$  is a Jordan curve, which varies smoothly in  $c$ : By the minimality of  $n$ ,  $P_{n-k-1}^{-1}(\mathbb{D}(0, R))$  is connected, so for any  $c \in \overline{U}$ ,  $f_c^{k+1-n}(\partial \mathbb{D}(0, R))$  is a smooth Jordan curve (because  $f_c^{k+1-n}$  has a critical point if  $|f_c^m(0)| = R$  for some  $m < n$ , which contradicts the fact  $|f_c^n(0)| \leq R$ ), and

$$F : \{(c, z) : c \in \overline{U}, f_c^m(z) \in \partial \mathbb{D}(0, R)\} \rightarrow U \times \partial \mathbb{D}(0, R)$$

is a smooth covering map. Notice  $U$  is simply-connected so there is a map  $\partial U \times I \rightarrow \overline{U}$ . Thus the pullback of  $F$  just induces a smooth isomorphism of this family of Jordan curves with the trivial family, so it depends smoothly on  $c$ .

- $f_c^{k+1}(0) \in f_c^{k+1-n}(C_R)$ , because  $|f_c^{n-1}(c)| = R$

- $0, c \in f_c^{k+1-n}(\mathbb{D}(0, R))$ , because  $|f_c^{n-1}(c)| = R$ .

So  $n(\partial U, c \mapsto f_c^{(k+1)}(0) - c) = n(\partial U, c \mapsto f_c^{(k+1)}(0) - 0)$  by??. Thus

$$f_c^{k+1}(0) \neq 0, \quad c \in \overline{U},$$

by argument principle. Then the induction process is finished.

Finally, notice  $P_n : U \rightarrow D(0, R)$  is a branched covering, so  $0 \in D(0, R) = P_n(U)$ , contradiction.

For another proof, Cf.[Mandelbrot set is connected].?

□

**Conj. (13.8.7.5) [Locally-Connectedness].** The Mandelbrot set **Man** is locally connected. ┘

*Proof:*

□

**Prop. (13.8.7.6) [Main Cardioid].** **Man** contains the **main cardioid** which is given by

$$\{a \in \mathbb{C} : |1 + \sqrt{1 - 4a}| < 1\} \cup \{a \in \mathbb{C} : |1 - \sqrt{1 - 4a}| < 1\} = \left\{ \frac{\mu(2 - \mu)}{4} \mid \mu \in \mathbb{D} \right\}.$$

┘

*Proof:* The main cardioid is just the set  $a \in \mathbb{C}$  s.t.  $f_a(z) = z^2 + a$  has a attracting fixed point.

And in this case, it can be shown that  $\mathbb{D}(0, \frac{1}{2}) \subset BA(\frac{1+\sqrt{1-4a}}{2})$ . Thus  $a \in \mathbf{Man}$ . □

**Prop. (13.8.7.7).**  $\mathbf{Man} \cap \mathbb{R} = [-2, \frac{1}{4}]$ . ┘

*Proof:* For  $a \in \mathbb{R}$ , if  $a > 1/4$ , then  $z^2 + a > z + (a - \frac{1}{4})$ , so  $a \notin \mathbf{Man}$ . If  $a < -2$ , then  $a \notin \mathbf{Man}$  by (13.8.7.3).

And for  $a \in [0, 1/4]$ ,  $f^{(n)}(0) \in [0, 1/2]$ , so  $a \in \mathbf{Man}$ . Finally, for  $a \in [-2, 0]$ ,  $|f^{(n)}(0)| \leq a$ , so  $a \in \mathbf{Man}$ . □

**Prop. (13.8.7.8).** Any point in  $\partial \mathbf{Man}$  is an accumulation point of the set of values  $a \in \mathbb{C}$  s.t.  $f_a(z) = z^2 + a$  has a super-attracting point other than  $\infty$  (i.e. 0 is periodic for  $f_a$ ). ┘

*Proof:* Let  $U$  be an small open set of  $\mathbb{C}$  intersecting  $\partial \mathbf{Man}$  and  $0 \notin U$ . Suppose  $f_a(z)$  doesn't has a super-attracting point other than  $\infty$ , then  $f_a^{(n)}(0) \neq 0$  for any  $n \in \mathbb{Z}_+$ . In particular,  $f_a^{(n)}(0) \neq \sqrt{a}$ , where  $\sqrt{a}$  is a branch of square root. Consider the family of functions  $\{g_n(a) = f_a^{(n)}(z)/\sqrt{a}\}$  on  $a$ , then  $g_n(a) \neq \{0, 1, \infty\}$  for any  $a \in D$ . So by (12.5.1.10),  $\{g_n\}$  is normal on  $D$ . As  $D \cap \mathbf{Man} \neq \emptyset$ , there exists  $a \in D$  s.t.  $\{g_n(a)\}$  is bounded. But  $D \setminus \mathbf{Man} \neq \emptyset$ , so there exists  $\xi \in D$  and a sequence  $(g_n^{(n_i)}(\xi))$  that tends to infinity. So  $\{g_n\}$  is normal on  $D$  cannot be normal on  $D$ , contradiction. □

**Conj. (13.8.7.9)[Hyperbolicity].** Every component of the interior of the Mandelbrot set is hyperbolic. Equivalently, the set of  $c \in \mathbb{C}$  s.t.  $f_c(z) = z^2 + c$  is hyperbolic is an open dense subset of  $\mathbb{C}$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (13.8.7.10)[Yoccoz].** If  $f \in \mathbb{C}[z]$  is a quadratic polynomial, and there is an invariant line field on  $\text{Jul}(f)$ , then  $f$  is infinitely renormalizable.  $\lrcorner$

*Proof:*  $\square$

**Thm. (13.8.7.11)[Robust Rigidity].** If  $f \in \mathbb{C}[z]$  is a robust infinitely renormalizable quadratic polynomial, then there are no invariant line fields on  $\text{Jul}(f)$ .  $\lrcorner$

*Proof:* Cf.[McMullen]  $\square$

**Cor. (13.8.7.12).** If  $f \in \mathbb{C}[z]$  is a real quadratic polynomial, then there are no invariant line fields on  $\text{Jul}(f)$ .  $\lrcorner$

*Proof:*  $\square$



# 14 | Operator Algebras

## 14.1 Operator Algebras

**Notation(14.1.0.1).**

- Use notations on Archimedean Banach spaces defined in [Archimedean Banach Algebra](#).

┘

Cf.[Princeton Companion], [\[Fol16\]](#), [Non-commutative Geometry, Connes]Chap5.

The theory of operator algebras is the non-commutative analogue of measure theory.

### 1 $C^*$ -algebras

**Def. (14.1.1.1)[ $C^*$ -Algebras, Rickart(1946)/Segal(1947)].** A  $B^*$ -algebra(or  $C^*$ -algebra) is a Banach algebra  $A$  over  $\mathbb{C}$  with an involution  $*$  :  $A \rightarrow A$  s.t.  $\|xx^*\| = \|x\|^2$  for any  $x \in A$ .

┘

**Prop. (14.1.1.2).** For  $H \in \mathcal{H}ilbSpa$ , the adjoint operation serves as an involution and makes  $B(\mathcal{H})$  into a  $B^*$ -algebra, by [\(12.9.2.9\)](#).

┘

**Prop. (14.1.1.3)[Gelfand-Naimark].** For a commutative  $C^*$ -algebra, the Gelfand transform  $x \mapsto \hat{x}$  is an isomorphism from  $A$  to  $C(\Delta)$  with  $\|x\| = \|\hat{x}\|_\infty$  and  $\widehat{x^*} = \widehat{\hat{x}}$ .

┘

*Proof:* First use  $\|xx^*\| = \|x\|^2$  to prove that a Hermitian element is mapped to real function, and use Stone-Weierstrass to show that the image is dense, then let  $y = xx^*$  and  $\|y^{2^m}\| = \|y\|^{2^m}$  to prove  $\|\hat{x}\| = \|x\|$ , so its image is closed.  $\square$

**Cor. (14.1.1.4).** If  $A$  is a commutative  $B^*$ -algebra that contains an element  $x$  s.t polynomials of  $x, x^*$  are dense in  $A$ , then  $\hat{x}$  is an isomorphism from  $\Delta_A$  to  $\sigma(x)$ , in particular, the Gelfand transform [\(12.9.1.21\)](#) is an isomorphism from  $C(\sigma(x))$  to  $A$ .

┘

*Proof:* Cf.[Rudin P290].  $\square$

Now we want to apply commutative algebra methods in the non-commutative case, there are two ways.

**Prop. (14.1.1.5).** For a commutative set of elements  $S$  in  $A$ , its bicommutant [\(14.1.2.2\)](#)  $B = \Gamma(\Gamma(S))$  is commutative, closed and contains  $S$ . And  $\sigma_B(x) = \sigma_A(x)$  for  $x \in B$ .

┘

*Proof:* Because  $S \subset \Gamma(S)$ ,  $\Gamma(\Gamma(S)) \subset \Gamma(S)$ , thus  $\Gamma(\Gamma(S))$  is commutative. And if  $xy = yx$ , then  $x^{-1}y = yx^{-1}$ , so the inverse, if exists, are in  $B$ .  $\square$

**Cor. (14.1.1.6).** In a Banach algebra, if  $x, y$  commutes, then

$$\sigma(x + y) \subset \sigma(x) + \sigma(y), \quad \sigma(xy) \subset \sigma(x)\sigma(y).$$

┘

*Proof:* Because  $\sigma(x)$  is just the range of  $\hat{x}$  on  $\Delta_A$  where  $A = \Gamma(\Gamma(\{x, y\}))$  (14.1.1.5)(12.9.1.21).  $\square$

The second method applies to normal elements:

**Def. (14.1.1.7) [Normal].** In a Banach algebra with an involution, a set  $S$  is called **normal** if it is commutative and  $S^* = S$ . An element  $x$  is called:

- **normal** iff  $x$  commutes with  $x^*$ .
- **unitary** iff  $x^* = x^{-1}$ .
- **Hermitian** iff  $x^* = x$ .
- **positive** iff  $x = x^*$  and  $\sigma(x) \subset [0, \infty)$ .

┘

**Prop. (14.1.1.8).** A maximal normal set  $B$  in  $A$  is a closed subalgebra and  $\sigma_B(x) = \sigma_A(x)$  for  $x \in B$ .

┘

*Proof:* Cf.[Rudin P294].  $\square$

**Cor. (14.1.1.9) [Normalness and Spectra].** In a  $B^*$ -algebra  $A$ ,

- Hermitian elements have real spectra.
- If  $x$  is normal, then  $\rho(x) = \|x\|$ .
- If  $u, v \geq 0$ , then  $u + v \geq 0$ .
- $yy^* \geq 0$ . Thus  $e + yy^*$  is invertible.

┘

*Proof:* Cf.[Rudin P295].  $\square$

**Def. (14.1.1.10) [Positive Functional].** Let  $(A, *)$  be a Banach algebra with an involution, a **positive functional** on  $A$  is a function  $F : A \rightarrow \mathbb{C}$  such that  $F(xx^*) \geq 0$ . It has the following properties.

- $F(x^*) = \overline{F(x)}$  and  $|F(xy^*)|^2 \leq F(xx^*)F(yy^*)$ . (Use Swartz like trick).
- $|F(x)|^2 \leq F(e)F(xx^*) \leq F(e)^2\rho(xx^*)$ , because  $e = ee^*$ . Thus  $|F(x)| \leq F(e)\rho(x)$  for every normal  $x$  by (14.1.1.9), so  $\|F\| = F(e)$  if  $A$  is commutative.

┘

*Proof:* Cf.[Rudin P297].  $\square$

**Prop. (14.1.1.11) [Positive Functionals and Measures].** If  $A$  is a commutative Banach algebra with an involution that  $h(x^*) = \overline{h(x)}$ , then The map

$$\mu \rightarrow F(x) = \int_{\Delta} \hat{x} d\mu$$

is a one-to-one correspondence between the convex set of measures  $\mu$  that  $\mu(\Delta) \leq 1$  to the convex set  $K$  of positive functionals on  $A$  of norm  $\leq 1$ , i.e.  $F(e) \leq 1$ , so maps the extreme points, i.e. the point mass to extremes points, thus the extreme points of  $K$  is exactly  $\Delta$ . This can be used to prove Bochner's theorem (12.11.1.11).  $\square$



*Proof:* Use the last prop to show that there is a functional on  $C(\Delta)$  and use Riesz representation. It is positive and by Stone-Weierstrass, it is unique.  $\square$

**Thm. (14.1.1.12) [Gelfand-Naimark-Segal].** Any  $C^*$ -algebra  $A$  is isomorphic to a closed subspace of  $B(\mathcal{H})$  for some  $\mathcal{H} \in \mathcal{HilbSpa}$  as  $*$ -algebras.

Such a representation  $A \rightarrow B(\mathcal{H})$  is called a  **$*$ -representation**. It is called a **non-degenerate  $*$ -representation** if there are no  $u \in \mathcal{H}$  s.t.  $Au = 0$ .  $\lrcorner$

*Proof:* Cf. [Rudin].  $\square$

## 2 von-Neumann-Algebras

**Def. (14.1.2.1) [von-Neumann-Algebras].** A **von Neumann Algebra** is a  $C^*$ -subalgebra of  $B(\mathcal{H})$  (14.1.1.1)(14.1.1.2) that contains the identity and is closed in the weak operator topology (12.7.3.4).  $\lrcorner$

**Def. (14.1.2.2) [Bicommutant].** For  $\mathcal{S} \subset L(\mathcal{H})$ , define the **commutator**  $\Gamma(\mathcal{S})$  be the algebra of operators that commutes with any  $S \in \mathcal{S}$ , then  $\mathcal{S} \subset \Gamma(\Gamma(\mathcal{S}))$ .  $\Gamma(\Gamma(\mathcal{S}))$  is called the **bicommutant** of  $\mathcal{S}$ .

If  $\mathcal{A}$  is a  $*$ -algebra in  $L(\mathcal{H})$ , then  $\Gamma(\Gamma(\mathcal{A}))$  is a von-Neumann algebra.  $\lrcorner$

*Proof:* If  $\mathcal{A}$  is a  $*$ -algebra, then  $\Gamma(\Gamma(\mathcal{A}))$  is clearly a  $*$ -algebra, and it is weakly-closed because if  $SV_i = V_iS$  and  $V_i x \rightarrow Vx$  for any  $x$ , then  $SVx = VSx$  for any  $x$ .  $\square$

**Thm. (14.1.2.3) [von-Neumann Density Theorem].** Let  $\mathcal{A}$  be a  $*$ -subalgebra of  $L(\mathcal{H})$ , then  $\mathcal{A}$  is dense in  $\Gamma(\Gamma(\mathcal{A}))$  in the ultra-strong operator topology (12.7.4.6). i.e.  $\Gamma(\Gamma(\mathcal{A}))$  is the von-Neumann algebra generated by  $\mathcal{A}$  (14.1.2.1).  $\lrcorner$

*Proof:* Firstly we prove that  $\mathcal{A}$  is dense in  $\Gamma(\Gamma(\mathcal{A}))$  in the strong operator topology. For any  $S \in \Gamma(\Gamma(\mathcal{A}))$  and any  $x_1, \dots, x_N \in \mathcal{H}$ ,  $\varepsilon > 0$ , we need to prove there exists  $A \in \mathcal{A}$  that  $\sum \|Sx_i - Ax_i\| \leq \varepsilon^2$ .

For the case  $N = 1$  and  $x_1 = x$ , consider the closure  $\mathcal{X}$  of  $\{Ax\}$ , then the orthogonal projection  $P$  onto  $\mathcal{X}$  is an operator in  $\Gamma(\mathcal{A})$ . This implies  $A(1 - P)x = (1 - P)Ax = 0$ , thus  $x = Px$  because of non-degeneracy. Thus  $S$  commutes with  $P$ , and  $Sx = SPx = PSx \in \mathcal{X}$ , thus there exists an  $A \in \mathcal{A}$  that  $Ax$  is close to  $Sx$ . For  $N > 1$ , we can just apply the result to  $\mathcal{H}^N$ .

For the ultra-strong operator topology, consider the Hilbert space  $\mathcal{W} \in \mathcal{HilbSpa}$  consisting of sequence  $(v_n)_{n \in \mathbb{Z}_+}$  in  $\mathcal{H}$  s.t.  $\sum_n \|v_i\|^2 < \infty$ . Then  $\mathcal{A}$  acts on  $\mathcal{W}$  diagonally.

For  $S \in \Gamma(\Gamma(\mathcal{A}))$ , we prove that the diagonal action of  $S$  on  $\mathcal{W}$  commutes with the action of  $\mathcal{A}$ . Let  $\text{pr}_m : \mathcal{W} \rightarrow \mathcal{H}$  be the projection to the  $m$ -th component and  $i_n : \mathcal{H} \rightarrow \mathcal{W}$  the inclusion to the  $n$ -th component, then it suffices to show that for any  $T \in \Gamma(\mathcal{A}^\Delta)$ , we have

$$\text{pr}_m(S^\Delta T - TS^\Delta)i_n = 0$$

for any  $m, n \in \mathbb{Z}_+$ . Notice that  $\text{pr}_m Ti_n \in \Gamma(\mathcal{A})$  because for any  $x \in \mathcal{A}$  and  $v \in \mathcal{H}$ , we have

$$\text{pr}_m Ti_n Av = \text{pr}_m TA^\Delta i_n v = \text{pr}_m A^\Delta Ti_n v = A \text{pr}_m Ti_n v.$$

Thus

$$\text{pr}_m TS^\Delta i_n = \text{pr}_m Ti_n S = S \text{pr}_m Ti_n = \text{pr}_m S^\Delta Ti_n.$$

And the proved strong-operator density for  $\mathcal{W}$  implies the ultra-strong operator density of  $\mathcal{A}$  in  $\Gamma(\Gamma(\mathcal{A}))$ .  $\square$

**Def. (14.1.2.4) [Factors].** A **factor** is a non-Neumann-algebra (14.1.2.1) with center  $\mathbb{C}$ .  $\lrcorner$

### 3 Type-I Factors

**Remark (14.1.3.1).** Factors (14.1.2.4) are defined as type I, II, III depending on their lattices of orthogonal projections.  $\lrcorner$

**Def. (14.1.3.2) [Type-I Factors].** A **type-I factor** is a factor (14.1.2.4) that is isomorphic to  $B(\mathcal{H})$  for some  $\mathcal{H} \in \mathcal{HilbSpa}$ .  $\lrcorner$

**Thm. (14.1.3.3).** If  $A \leq B(\mathcal{H})$  is a type-I factor on  $\mathcal{H} \in \mathcal{HilbSpa}$  (14.1.3.2), isomorphic to  $B(\mathcal{H}_1)$  for  $\mathcal{H}_1 \in \mathcal{HilbSpa}$ , then there exists  $\mathcal{H}_2 \in \mathcal{HilbSpa}$  and unitary isomorphism  $\mathcal{H} \cong \mathcal{H}_1 \otimes \mathcal{H}_2$  s.t.

$$UAU^{-1} = B(\mathcal{H}_1) \otimes \text{id}_{\mathcal{H}_2}, \quad U\Gamma(A)U^{-1} = \text{id}_{\mathcal{H}_1} \otimes B(\mathcal{H}_2).$$

$\lrcorner$

*Proof:* Cf. [Dixmier, Von Neumann Algebras] 1.8.2, Cor3.  $\square$

**Cor. (14.1.3.4).** If  $A \leq B(\mathcal{H})$  is a type-I factor (14.1.3.2), then  $\Gamma(A)$  (14.1.2.2) is also a type I factor.  $\lrcorner$

# 15 | Geometric Topology

## 15.1 Low Dimensional Topology

### 1 Thurston's Geometrization Conjecture(3-Manifolds)

**Prop.(15.1.1.1) [Prime Decomposition Theorem, Kneser1929/Milnor[Mil62]].** Any compact manifold pos. with boundary is a connected sum of f.m. compact manifolds s.t. they are not connected sum of non-trivial manifolds, called **prime manifolds**.  $\lrcorner$

*Proof:* ? Cf.[Kneser] for proof that it can be written as a connected sum of f.m. compact manifolds.  $\square$

**Def.(15.1.1.2) [Incompressible Surfaces].** A surface  $S \in \text{Diff}_\partial$  in a 3-manifold  $M \in \text{Diff}_\partial^3$  is called a **two-sided surface** if  $\partial S \subset \partial M$  and the normal bundle of  $S$  is oriented. A **incompressible surface** is a two-sided surface  $S \subset M$  s.t. any simple curve on  $S$  that bounds a disk on  $M$  with interior disjoint from  $N$  also bounds a disk in  $M$ .  $\lrcorner$

**Def.(15.1.1.3) [Atoroidal Manifolds].**  $M \in \text{Diff}_\partial^3$  is called **geometrically atoroidal** if every incompressible torus in  $M$ (15.1.1.2) is isotopic to a boundary component.  $M$  is called **homotopically atoroidal** if every map  $\mathbb{T}^2 \rightarrow M$  that is injective on  $\pi_1$  is homotopic to a boundary component.

Homotopically atoroidal manifolds are geometrically atoroidal. ?  $\lrcorner$

**Example(15.1.1.4) [Homotopically Atoroidal Manifolds].** The following two manifolds are homotopically atoroidal(15.1.1.3):

- $M = \mathbb{T}^2 \times \mathbb{I}$ .
- $M = (\mathbb{T}^2 \times \mathbb{I}) / \{\pm 1\}$ .

$\lrcorner$

*Proof:* ?  $\square$

### Haken Manifolds

**Def.(15.1.1.5) [Haken Manifolds].** A **Haken manifold** is a 3-dimensional prime manifold(15.1.1.1) that is homeomorphic to  $\mathbb{S}^3$  or contains a 2-sided incompressible surface(15.1.1.2) not homeomorphic to  $\mathbb{S}^2$ .  $\lrcorner$

**Prop.(15.1.1.6) [Haken Manifolds].** Any 3-dimensional prime manifold with boundary(15.1.1.1) is Haken(15.1.1.5). Any 3-dimensional closed manifold  $M$  with  $H^1(M; \mathbb{Q}) \neq 0$  is Haken.  $\lrcorner$

*Proof:* ?  $\square$

### Model Geometries

**Def.(15.1.1.7) [Model Geometries].** A **model geometry** is a simply-connected smooth manifold  $X \in \mathcal{D}\text{iff}$  together with a transitive action of a Lie group  $G$ (16.5.1.1) on  $X$  with compact stabilizers, and maximal among groups acting smoothly and transitively on  $X$  with compact stabilizer  $\mathcal{K}$ .  $\lrcorner$

**Def.(15.1.1.8)[Geometric Structures].** A **geometric structure** on a manifold pos. with boundary  $M$  is a diffeomorphism from  $M$  to  $X/\Gamma$  for some model geometry  $X = G/\mathcal{K}$ (15.1.1.7), where  $\Gamma$  is a discrete subgroup of  $G$  acting freely on  $X$ .  $\lrcorner$

**Thm.(15.1.1.9) [Thurston Geometries].** There are exactly 8 3-dimensional model geometries(15.1.1.7) s.t. there exists a compact manifold(pos. with boundary) with this geometric structure(15.1.1.8):

- (Spherical geometry:) $X = \mathbb{S}^3 = \text{SO}(4; \mathbb{R})/\text{SO}(3; \mathbb{R})$ . Closed manifolds modeled on this geometry are exactly those with finite fundamental groups, these are called **spherical manifolds**. They are classified.
- (Euclidean geometry:) $X = \mathbb{R}^3 = (\text{O}(3; \mathbb{R}) \ltimes \mathbb{R}^3) / \text{O}(3; \mathbb{R})$ . There are exactly 10 finite-volume closed 3-manifolds with this geometry, 6 orientable and 4 non-orientable.
- (Hyperbolic geometry:) $X = \mathcal{H}^3 = \text{O}^+(1, 3; \mathbb{R})/\text{O}(3; \mathbb{R})$ .
- ( $X = \mathbb{S}^2 \times \mathbb{R} = ((\text{O}(3; \mathbb{R}) \times \mathbb{R}) / \{\pm 1\}) / (\text{O}(2; \mathbb{R}) \times \{\pm 1\})$ :)
- ( $\mathcal{H}^2 \times \mathbb{R}$ :)
- ( $\widetilde{\text{SL}(2; \mathbb{R})}$ :) $X = \left( \mathbb{Z} \setminus \left( \mathbb{R} \times \widetilde{\text{SL}(2; \mathbb{R})} \right) \right) / \text{SO}(2; \mathbb{R})$ .
- (Nilgeometry:) $X$  is the quotient of the Heisenberg group quotient by  $\text{O}(2; \mathbb{R})$ .
- (Solgeometry:)

$\lrcorner$

*Proof:* ?

$\square$

**Thm.(15.1.1.10) [Thurston's Geometrization Conjecture].** The interior of any compact 3-manifold pos. with boundary has a canonical decomposition into pieces with geometric structures(15.1.1.8).  $\lrcorner$

*Proof:* ?

$\square$

**Lemma(15.1.1.11)[Hyperbolic Manifolds with Boundaries].** The interior of a compact 3-manifold with boundary  $M \in \mathcal{D}\text{iff}_\partial$  has a hyperbolic structure(15.1.1.9) iff  $M$  is prime(15.1.1.1), homotopically atoroidal and not homeomorphic to  $(\mathbb{T}^2 \times \mathbb{I}) / \{\pm 1\}$ (15.1.1.4). In particular, the Thurston's geometrization conjecture holds for all compact 3-dimensional prime manifolds with boundary.

And if  $M$  has a hyperbolic structure, then the interior has finite volume iff  $\partial M$  consists of tori and not homeomorphic to  $\mathbb{T}^2 \times \mathbb{I}$ (15.1.1.4).  $\lrcorner$

*Proof:* Cf.[Hyperbolic structures on 3-manifolds, 1: deformations of acylindrical manifold, Thurston].  $\square$

**Lemma(15.1.1.12) [Haken Manifolds].** The Thurston's geometrization conjecture holds for any Haken manifold(15.1.1.5). A closed Haken manifold has a hyperbolic structure iff it is homotopically atoroidal(15.1.1.3).  $\lrcorner$

## 2 Floer Homology

**Def. (15.1.2.1) [Witten Complex].** Let  $M \in \text{Diff}_{\text{cpet}}$  and  $f \in C(M)$  is a Morse function, then at each critical point  $P$  of  $f$ , the Hessian  $H_P(f)$  is a non-degenerate quadratic form with signature  $n_P^+, n_P^-$ .

We define the **Witten complex**  $C$  as follows:

- For  $0 \leq q \leq \dim M$ ,  $C_q$  is the free group generated by the critical points with  $n_P^- = q$ .
- Choose a metric on  $M$ , define the flow generated by  $\text{grad} f$ . Then for critical points  $P, Q$  with  $n_P^- = q, n_Q^- = q - 1$ , the number of trajectories under  $\text{grad} f$  from  $P$  to  $Q$  is finite, and this gives the boundary map coefficients  $\partial_{P,Q}$ .

┘

References are [Supersymmetry and Morse Theory, Witten].

### Casson Invariants

**Def. (15.1.2.2) [Casson Invariants].** Let  $Y$  be a homological 3-sphere, the **Casson invariant**  $\lambda(Y)$  of  $Y$  is defined to be the half of the number of isomorphism classes of irreducible representation  $\pi_1(Y) \rightarrow SU(2)$ .

┘

**Prop. (15.1.2.3).** For a homological 3-sphere  $Y$ , let

- $\mathcal{A}$  be the space of  $SU(2)$ -connections for the trivial bundle on  $Y$ ,
- $\mathcal{G}$  be the group of gauge transformations  $Y \rightarrow SU(2)$ .
- $\mathcal{C} = \mathcal{A}/\mathcal{G}$ .

Then  $\mathcal{C}$  is an infinite-dimensional manifold, and the connection map  $F : \mathcal{A} \mapsto F_A$  defines a 1-form  $F$  on  $\mathcal{C}$ .

┘

*Proof:*

□

**Prop. (15.1.2.4) [Taubes].** Situation as in (15.1.2.3), the zeros of  $F$ , i.e. the set of flat connections, corresponds to irreducible representations  $\pi_1(Y) \rightarrow SU(2)$ . And we can use Fredholm perturbation to calculate the number, i.e.  $2\lambda(Y)$ .

┘

*Proof:* ?

□

### Floer Homology Groups

**Remark (15.1.2.5) [Relative Morse Indices].** The difficulty to construct Morse theory for  $Y$  lies in the fact that the Hessian of  $f : \mathcal{C} \rightarrow \mathbb{R}/\mathbb{Z}$  at critical points has both Morse indices  $n^+, n^-$  infinite. The way around is to notice for critical points  $P, Q$ , the **relative Morse index**  $n_{P,Q}^- = n_P^- - n_Q^-$  is finite.

┘

**Prop. (15.1.2.6) [Floer Homology Theory].** Using the flow generated by  $\text{grad} f$ , we can get construct a chain complex with (mod 8)-grading, which is a finite complex with the critical points of  $f$  as simplexes. The corresponding homology groups are called the **Floer homology** of  $Y$ , denoted by  $HF_q(Y)$ , where  $q \in \mathbb{Z}/(8)$ .

Notice reversing the orientation of  $Y$  induces an action on  $HF_*(Y)$  corresponding to Poincaré duality.

┘

**Prop. (15.1.2.7) [Floer Homologies and Casson Invariants].** It is clear now that  $2\lambda(M) = \sum_{q=0}^7 \dim HF(Y)$ . Thus the Floer homology groups form a refinement of  $\lambda(Z)$ .

┘

### 3 Gauge Theory

### 4 Donaldson-Floer Theory

Main references are [Geometry of 4-Manifolds, Donaldson, ICM1987].

**Prop. (15.1.4.1) [Hodge].** For an algebraic surface  $S/\mathbb{C}$ , the signature  $b_2^+$  of the intersection form on  $H_2(S(\mathbb{C}))$  satisfies

$$b_2^+ = 1 + 2p_g(S).$$

┘

**Def. (15.1.4.2) [Donaldson Invariants].** Let  $Z$  be an oriented simply-connected differentiable 4-manifold, let  $b_2^+, b_2^-$  be the signature of the intersection form on  $H_2(Z)$ . Assume  $b_2^+ > 1$  is odd, the **Donaldson invariants** are a sequence of integral polynomials  $\varphi_k$  on  $H_2(Z)$  for  $k$  sufficiently large, and  $\deg(\varphi_k) = 4k = 3\frac{b_2^++1}{2}$ . ?

┘

**Thm. (15.1.4.3).** If  $Z = Z_1 \# Z_2$  is a connected sum with  $b_2^+(Z_i) \neq 0$ , then  $\varphi_k(Z) = 0$  for all  $k$ .

┘

*Proof:*

□

**Thm. (15.1.4.4).** If  $Z$  is an algebraic surface, then for  $k$  sufficiently large,  $\varphi_k(Z(\mathbb{C})) \neq 0$ . In particular,  $Z$  is essentially indecomposable.

┘

*Proof:*

□

**Prop. (15.1.4.5) [Λ-Splitting].** Suppose the intersection form on  $H_2(Z)$  decomposes as  $H_2(Z) = A_1 \oplus A_2$ , where  $A_i^+ > 0$ . If  $\varphi_k(Z) \neq 0$  for some  $k$ , then by (15.1.4.3),  $Z$  cannot be decomposed to a connected sum  $Z_1 \# Z_2$  having intersection form  $A_i$ . However, there exists such a decomposition  $Z = Z_1 \coprod_Y Z_2$  where  $Y$  is a homotopy 3-sphere.

┘

*Proof:* Cf. [Freeman and Taylor, Λ-Splitting 4-Manifolds].

□

**Prop. (15.1.4.6).** For a Λ-splitting  $Z = Z_1 \coprod_Y Z_2$ , the Donaldson invariant polynomials are related to the Floer homology groups of  $Y$ . ?

┘

## 15.2 Other Geometries

### 1 Hyperbolic Geometry

**Prop. (15.2.1.1).** Isometries of hyperbolic ball are all given by Mobius transformations, because the distance to three non-colinear point can localize a point. Cf.[双曲几何 刘毅].  $\lrcorner$

**Def. (15.2.1.2) [Hyperbolic Disk].** The **hyperbolic disk** is a Riemannian manifold homeomorphic to  $\mathbb{D}$  endowed with the **Poincaré metric** or **hyperbolic metric**

$$d^P s = \frac{|dz|}{1 - |z|^2} = \sigma_{\mathbb{D}}(z)|dz|.$$

$\lrcorner$

**Prop. (15.2.1.3).** The Poincaré metric on  $\mathbb{D}$  is preserved by  $\text{Aut}(\mathbb{D})$ .  $\lrcorner$

*Proof:* Cf.[李忠, P26].  $\square$

**Prop. (15.2.1.4).** For any  $z_1, z_2 \in \mathbb{D}$ ,

$$d^P(z_1, z_2) = \log \frac{|1 - \bar{z}_1 z_2| + |z_1 - z_2|}{|1 - \bar{z}_1 z_2| - |z_1 - z_2|}.$$

$\lrcorner$

*Proof:* Cf.[李忠, P27].  $\square$

### Mostow Rigidity Theorem

**Conj. (15.2.1.5) [Rationality of Hyperbolic Volumes].** The volumes of hyperbolic 3-manifolds are not all rationally related.  $\lrcorner$

*Proof:*  $\square$

**Thm. (15.2.1.6) [Mostow Rigidity Theorem].**  $\lrcorner$

### Klein Groups

References are [Three-dimensional manifolds, Kleinian groups and hyperbolic geometry, Thurston].

**Def. (15.2.1.7) [Klein Groups].** A **Klein group** is a discrete subgroup of  $\text{PSL}(2; \mathbb{C})$ .

A Klein group is called a **Fuchsian group** if its limit set in  $\mathbb{C}$  is a circle. It is called a **quasi-Fuchsian group** if its limit set is a Jordan curve  $\lrcorner$

## 2 Metric Geometry(Gromov)

**Notation(15.2.2.1).**

- Use notations on metric spaces defined in [Metric Spaces](#).
- Let  $(X, \text{dist}) \in \text{MetSpa}(4.4.8.3)$ .

┘

**Def.(15.2.2.2)[Hausdorff dimension].**  $\dim^H(X)$ .

┘

**Def.(15.2.2.3)[Hausdorff Distances].** The **Hausdorff distance** for two subset  $Y_1, Y_2 \subset X$  is defined to be

$$d_X^H(Y_1, Y_2) = \inf\{r \in \mathbb{R}_+ | Y_2 \subset B(Y_1, r), Y_1 \subset B(Y_2, r)\}.$$

The **Gromov-Hausdorff metric** for two metric space is

$$d^{GH}(X_1, X_2) = \inf_Z \{d_Z^H(i_1(X_1), i_2(X_2))\}$$

where  $i_1, i_2$  are isometry of  $X_1, X_2$  into som metric space  $Z$ .

This metric makes the set of all compact metric space into a complete Hausdorff space  $\mathcal{MET}$ . ┘

**Def.(15.2.2.4).** A map from  $X$  to  $Y$  is called a  $\varepsilon$ -**approximation** iff  $B(f(X), \varepsilon) = Y$  and  $|d(x, y) - d(f(x), f(y))| \leq \varepsilon$ .

We have: if there is a  $\varepsilon$  approximation, then  $d^{GH}(X, Y) \leq 3\varepsilon$ , and if  $d^{GH}(X, Y) \leq \varepsilon$ , there is a  $3\varepsilon$  approximation. ┘

**Prop.(15.2.2.5).** Fix a function  $N : (0, 1) \rightarrow \mathbb{N}$ , the space  $\mathcal{MET}(D, N)$  of complete metric space of diameter bounded by  $D$  and for every  $\varepsilon$ , there is a  $\varepsilon$ -net with no more then  $N(\varepsilon)$  points. Then it is a compact subspace of  $\mathcal{MET}$ . ┘

*Proof:* We show it is totally bounded and closed. It is totally bounded because the space of discrete space of no more than  $N(\varepsilon)$  is compact and it  $\varepsilon$  approximate  $\mathcal{MET}(D, N)$  by definition. Thus we have it is totally bounded. And  $\square$

**Prop.(15.2.2.6).** The set of isometries of ┘

**Prop.(15.2.2.7)[Gromov Compactness Theorem].** Denote the space  $\mathcal{RIC}_{*, -1}^D(n)$  of manifold with Ricci curvature bounded below by  $-1$  and diameter bounded above by  $D$ , then it is a precompact subset of  $\mathcal{MET}$ . ┘

*Proof:* By Bishop-Gromov(16.3.6.18), there is a  $N(\varepsilon)$  that  $M$  can only have  $N(\varepsilon)$  many balls of radius  $\varepsilon$ , because  $M$  has bounded diameter (Packing argument). So  $\mathcal{RIC}_{*, -1}^D(n) \subset \mathcal{MET}(D, 2N)$  is precompact.  $\square$

**Prop.(15.2.2.8).** Any metric space  $X$  in the closure of  $\mathcal{RIC}_{*, -1}^D(n)$  has Hausdorff dimension  $\dim^H(X) \leq n$ . ┘

**Prop.(15.2.2.9)[Gromov].** If a sequence of manifold  $\{M_i\}$  in  $\mathcal{M}_{V, -k}^{D, k}(n)$ , then they has a limit point  $X \in \mathcal{MET}$ . Then  $X$  is a  $C^\infty$  manifold and there is a  $C^{1, \alpha}$ -metric for every  $\alpha < 1$ . And  $M_i$  are all diffeomorphic to  $X$  for large  $X$ .

In particular, this implies that there are only finitely many diffeomorphic classes. ┘

**Prop.(15.2.2.10)[Peterson].**  $\mathcal{M}_{*, v, k}^D(n)$  has only finitely many homotopy classes. ┘



### 3 Nilgeometry

**Def. (15.2.3.1) [Nilmanifolds].** A **nilmanifold** is a differential manifold of the form  $X = G/\Gamma$  where  $G \in \mathcal{L}\text{ieGrp}$  is a nilpotent Lie group and  $\Gamma \subset G$  is a cocompact discrete subgroup.  $\lrcorner$

### 4 Spectral Geometry

**Def. (15.2.4.1) [Spectra of Manifolds].** Let  $M$  be a manifold(possibly with boundary), then  $M$  has a Laplacian operator  $\Delta$ , defined as

$$\Delta(f) = -\operatorname{div}(\operatorname{grad} f).$$

The **spectrum of manifold** of  $M$  is defined to be the sequence of eigenvalues of  $\Delta$ .

Two manifolds are called **isospectral** if they have the same spectrum.  $\lrcorner$

**Def. (15.2.4.2) [Almost-Conjugate Subgroups].** Let  $G \in \mathcal{G}\text{rp}^{\text{fin}}$  and  $H, K$  be subgroups of  $G$ , then the following are equivalent:

- There is a bijection of sets  $f : H \rightarrow K$  s.t.  $h$  and  $f(h)$  are conjugate in  $G$  for any  $h \in H$ .
- The two Hilbert spaces  $L^2(H \backslash G)$  and  $L^2(K \backslash G)$  are isomorphic.

$\lrcorner$

**Thm. (15.2.4.3) [Isospectral Quotient Manifolds, Sunada-Berard].** Let  $M$  be a compact Riemannian manifold(possibly with boundary) and  $G \in \mathcal{G}\text{rp}^{\text{fin}}$  s.t.  $G$  acts on  $M$  by isometries. Suppose  $H, K$  are almost-conjugate(15.2.4.2) subgroups of  $G$  that acts freely on  $M$ , then the quotient orbifolds  $M_1 = H \backslash M$  and  $M_2 = K \backslash M$  are isospectral. (If they have boundary, they are isospectral both in the Dirichlet-sense and Neumann-sence.)  $\lrcorner$

*Proof:*

$\square$

### 15.3 Knots and Links

References are [Ada04], [Knot Theory, Lickorish].

**Def. (15.3.0.1) [Knots and Links].** For  $M \in \text{Mani}^3$ , a **knot**  $K$  in  $M$  is an embedding  $K : \mathbb{S}^1 \hookrightarrow M$ , and a **link** is a collection of disjoint unions knots in  $M$ . If  $M$  is piecewise-smooth or piecewise-linear,  $K$  is also assumed to be smooth or piecewise-linear.

If  $M$  is not specified, then it is assumed to be  $\mathbb{R}^3$ . If we talk about two knots, then it is assumed that they are disjoint.  $\lrcorner$

**Def. (15.3.0.2) [Unknots].** An **unknot** (or a **trivial knot**) is a knot  $K \subset M$  that is (piecewise-smoothly/linearly) null-homotopic. Two knots/links are called **equivalent knots/links** if they are isotopic.

If we consider the knots in  $\mathbb{R}^3$ , we tacitly assume that the links are piecewise-linear.  $\lrcorner$

**Def. (15.3.0.3) [Knot Diagrams].** For a (piecewise-linear) link  $L \subset \mathbb{R}^3$ , a **link diagram** of  $K$  is given by a linear projection  $\text{pr} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  s.t.

- $\#p^{-1}(x) \leq 2$  for any  $x \in \mathbb{R}^2$ .
- $\#p^{-1}(x) = 2$  for only f.m.  $x \in \mathbb{R}^2$ .
- For any vertex  $x$  of  $L$ ,  $\text{pr}^{-1}(\text{pr}(x)) = \{x\}$ .

together with a labeling of overcrossing and undercrossing at each pair  $(x, y) \in K$  s.t.  $p(x) = p(y)$ , according to their height w.r.t. the projection.  $\lrcorner$

**Def. (15.3.0.4) [Crossing Numbers].** The **crossing number**  $\text{Cr}(L)$  of a link  $L$  is the minimal number of crossings for any link diagram of  $L$ .  $\lrcorner$

**Def. (15.3.0.5) [Unknot Numbers].** The **unknot number** of a knot  $K$  is the minimal number of crossings needed to change on some knot diagram of it to make it an unknot. Why does such an unknot always exists?  $\lrcorner$

**Def. (15.3.0.6) [Sign of Crossings].** There are three kinds of crossing relation between two oriented strands:  $L_+$ ,  $L_-$ ,  $L_0$ . Cf.[Lickorish]P11Figures.  $\lrcorner$

**Def. (15.3.0.7) [Reidemeister Moves].** There are three **Reidemeister moves** on a knot diagram:

1. A local untwist/twist (i.e.  $\nleftrightarrow$  changed to  $\rightarrow$ ).
2. A local strand passing over or under a strand.
3. A local strand passing over or under a crossing.

$\lrcorner$

**Thm. (15.3.0.8).** Given two knots  $K, K'$  and two projections  $D, D'$  of them, then  $K \sim K'$  iff  $D$  and  $D'$  are connected by a finite series of Reidemeister moves.  $\lrcorner$

*Proof:* ?  $\square$

**Def. (15.3.0.9) [Connected Sums of Knots].** Given two knots  $K, K' \in \mathbb{S}^3$ , we can consider their connected sum. And a **prime knot** is a non-trivial knot that is not a connected sum of two non-trivial knots.  $\lrcorner$

**Def. (15.3.0.10) [Knot Invariants].** A **knot invariant** is a function on the set of knots that has the same value on equivalent knots. By (15.3.0.8), a knot invariant is equivalent to a function on knot diagrams that is invariant under Reidemeister moves.  $\lrcorner$

**Def. (15.3.0.11) [Knot Groups].** For any knot  $K \subset \mathbb{R}^3$ , the **knot group**  $\Pi(K)$  is defined to be  $\Pi(K) = \pi_1(\mathbb{R}^3 \setminus K)$ . This is a knot invariant.  $\lrcorner$

**Def. (15.3.0.12) [Tricolorable Knots].** A **tricoloring of a knot diagram** is a tricoloring of each stand of the diagram s.t.

- Every color is used.
  - At each crossing, the three strands are either of the same color or pairwise different colors.
- having a tricoloring is a property that is invariant under Reidemeister moves, so it is a knot invariant. A **tricolorable knot** is a knot that has a tricolorable knot diagram.  $\lrcorner$

**Def. (15.3.0.13) [Alternating Knots].** An **alternating knot** is a knot that has a diagram s.t. if one travels through the knot, the ‘up’ and ‘down’ nature of the crossing alternates.  $\lrcorner$

### Knots and Braids

**Def. (15.3.0.14) [Braids].** For  $n \in \mathbb{Z}_+$ , let  $A$  be the set of  $n$  points in  $\mathbb{R}^3$  with coordinates  $y = 0, z = 1, x = 1, 2, \dots, n$ , and  $B$  the set of  $n$  points in  $\mathbb{R}^3$  with coordinates  $y = 0 = z, x = 1, 2, \dots, n$ . Then an  $n$ -strand braid is a set of  $n$  piecewise-smooth non-intersecting paths with descending  $z$ -coordinates connecting the points in  $A$  bijectively to points in  $B$ . Two braids are called **equivalent** if there is an isotopy of  $\mathbb{R}^3$  taking one into another with the endpoints fixed.

The  $n$ -strand **braid group** is the set of equivalent classes of  $n$ -strand braids with the group structure such that: For any two braids  $X, Y$ ,  $X \circ Y$  is the braid with the bottom of  $X$  connected to the top of  $Y$ . It is easy to check that the braids form a group under this multiplication.  $\lrcorner$

**Def. (15.3.0.15) [Closure of Braids].** Given any braid  $B$ , the **closure of braid** of  $B$  is the oriented link  $L(B)$  generated by connecting each  $(k, 0, 0)$  with  $(k, 0, 1)$  by a smooth path contained in the plane  $x = k$  and non-intersecting with  $B$ , where the orientation of  $L(B)$  is given by going through the braid from the top to the bottom.  $\lrcorner$

**Prop. (15.3.0.16).** For  $n \in \mathbb{Z}_+$ , the following three groups are equivalent:

- The  $n$ -strand braid group.
- The fundamental group of the unordered  $n$ -configuration space for  $\mathbb{R}^2$ .
- The free group generated by words  $\sigma_1, \dots, \sigma_{n-1}$  under the relations:

$$\sigma_i \sigma_j = \sigma_j \sigma_i, \forall 1 \leq i \leq j-2 \leq n-1,$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \forall 1 \leq i \leq n-2.$$

$\lrcorner$

*Proof:* 1 and 3: Any braid can be projected along the  $y$ -direction, and we can perturb it s.t. this is a link diagram, and for each  $z_0$ , there is at most one crossing with  $z = z_0$ . In this way, we can decompose any braid into a composite of braids s.t. each braid is a swap of two consecutive vertices  $i$  and  $i+1$ . Then there is a map from the free group  $\langle \sigma_1, \dots, \sigma_{n-1} \rangle$  to the  $n$ -strand braid group mapping  $\sigma_i$  to the swap of vertices  $i$  and  $i+1$  with  $i$  at front, and  $\sigma^{-1}$  mapped to the swap of vertices

$i$  and  $i + 1$  with  $i + 1$  at front. Then it can be checked that it is truly a map, and it factors through the relations

$$\begin{aligned}\sigma_i \sigma_j &= \sigma_j \sigma_i, \forall 1 \leq i \leq j - 2 \leq n - 1, \\ \sigma_i \sigma_{i+1} \sigma_i &= \sigma_{i+1} \sigma_i \sigma_{i+1}, \forall 1 \leq i \leq n - 2.\end{aligned}$$

So it remains to prove that any two equivalent braids are connected by a sequence of moves according to these relations. This proof is similar to that of Reidemeister moves (15.3.0.8) ?.

1 and 2: Connecting the top and bottom of each braid gives a loop in the  $n$ -configuration space of  $\mathbb{R}^2$ .  $\square$

**Def. (15.3.0.17) [Markov Moves].** There are three kinds of **Markov moves** on a braid:

1. Conjugation by  $\sigma_i$  for some  $i$ .
2. Make an  $n$ -strand braid to an  $n + 1$ -strand braid by adding a straight strand at the end and multiply by  $\sigma_n$  or  $\sigma_n^{-1}$  on the right.

┘

**Prop. (15.3.0.18) [Markov].** For two braids  $B, B'$  (may have different number of strands), their closure represent equivalent links iff they are connected by a finite number of Markov moves (15.3.0.17).  $\square$

*Proof:* Cf. [A study of Braids] P148. ?  $\square$

**Thm. (15.3.0.19) [Alexander].** Any oriented link  $L$  is equivalent to the closure of a braid.  $\square$

*Proof:* We only prove for piecewise-linear links. Take a link diagram  $D$  of  $L$  and take a point  $P \in \mathbb{R}^2 \setminus D$  that is not on the lines generated by any segment of  $D$ . Then we can observe each segment is clockwise or counter-clockwise w.r.t.  $P$ . We can make them all clockwise: For counter-clockwise segments we draw a general triangle that include  $P$ , and then use Reidemeister moves to make a new diagram for  $L$ . Then in this way, projection clockwise from  $P$  gives us a braid whose closure is equivalent to  $L$ .  $\square$

**Cor. (15.3.0.20) [Braid Indices].** The **braid index** of an oriented link  $L$  is the minimal  $n \in \mathbb{Z}_+$  s.t.  $L$  is equivalent to the closure of an  $n$ -strand braid.  $\square$

### Braid Groups

**Cor. (15.3.0.21) [Pure braids].** Due to the covering map  $\mathbb{C}^n \setminus \cup \{z_i = z_j\} \rightarrow \mathbb{C}^n \setminus \cup \{z_i = z_j\} / S^n$  with fiber  $S^n$ , there is a map  $B_n \rightarrow S_n$  which is easily seen to be surjective and with kernel  $P_n = \pi_1(\mathbb{C}^n \setminus \cup \{z_i = z_j\})$ , called the group of **pure braids**.  $\square$

**Prop. (15.3.0.22).** There is a group homomorphism  $B_3 \rightarrow \text{PSL}(2; \mathbb{Z})$  that maps  $a = \sigma_1 \sigma_2 \sigma_1$  to  $S$  and  $b = \sigma_1 \sigma_2$  to  $T$ . The kernel of this map is the group generated by  $c = a^2 = b^3$ .  $\square$

### Coding of Knots

**Def. (15.3.0.23) [Chessboard Coloring of Knots].** Given a knot diagram  $K$ , we can chessboard 2-color the components of the complements of  $K$ , s.t. two adjacent components have different colors. (This is because each vertex has order 4).  $\square$

**Def. (15.3.0.24) [Dowker-Thistlethwaite Notations].** Given a knot diagram, the **Dowker-Thistlethwaite notation** is given as follows: ?  $\square$

**Def. (15.3.0.25)**[Conway Notations]. ┘

**Def. (15.3.0.26)**[Gauss Codes]. Cf.[Kau99]P667. ? ┘

**Prop. (15.3.0.27)**[Gauss Codes are Evenly Intersticed]. Each Gauss code of a knot satisfies that:  
There is an even number of labels in between the two appearances of any label. ┘

*Proof:* This follows from the Jordan curve theorem in the plane. □

### Stick Numbers

**Def. (15.3.0.28)**[Stick Numbers]. For a knot  $K \in \mathbb{R}^3$ , the **stick number**  $s(K)$  is the minimal number  $n \in \mathbb{Z}_+$  s.t.  $K$  is equivalent to a piecewise-linear knot with  $n$  vertices. ┘

**Prop. (15.3.0.29)**. For any non-trivial knot  $K$ ,  $s(K) \geq 6$ . ┘

*Proof:* Use the fact that any knot diagram with  $\leq 2$  crossings is an unknot(15.3.5.1). □

**Prop. (15.3.0.30)**. For any non-trivial knot  $K$ ,

$$\frac{5 + \sqrt{8 \operatorname{Cr}(K) + 9}}{2} \leq s(K) \leq \frac{3}{2}(\operatorname{Cr}(K) + 1).$$

┘

*Proof:* Cf.[https://en.wikipedia.org/wiki/Stick\\_number](https://en.wikipedia.org/wiki/Stick_number). □

**Prop. (15.3.0.31)**[Jin]. If  $K_1, K_2$  are knots, then

$$s(K_1 \# K_2) \leq s(K_1) + s(K_2) - 3.$$

Is it possible to make it -4(which would be optimal in the case  $K_1, K_2$  being trefoil) ?. ┘

*Proof:* Cf.[Jin, Polygon Indices and Superbridge Indices of Torus Knots and links]. □

### Bridge Numbers

**Def. (15.3.0.32)**[Bridge Numbers]. The **bridge number**  $b(K)$  of a knot  $K$  is the minimal number of strands that overpass at least one other strands in some knot diagram of it. Clearly,  $b(K) \leq \operatorname{Cr}(K)$ . ┘

**Prop. (15.3.0.33)**. For any non-trivial knots,  $b(K) \geq 2$ . ┘

*Proof:* ? □

**Prop. (15.3.0.34)**[Schburt]. If  $K_1, K_2$  are knots, then

$$b(K_1 \# K_2) \leq s(K_1) + s(K_2) - 1.$$

┘

*Proof:* ?[Schburt1954]. □

**Def. (15.3.0.35)**[Two-Bridged Knots]. A **two-bridged knot** is a knot  $K \in \mathbb{R}^3$  with  $b(K) = 2$ . In particular, by(15.3.0.34), two-bridged knots are all prime knots. ┘

**Prop. (15.3.0.36)**. Two-bridged knots are just rational knots. ┘

*Proof:* ? □

### Slice Knots

**Def. (15.3.0.37) [Slice Knots].** A **slice knot** is a knot  $K \in \mathbb{S}^3 = \partial\mathbb{D}^4$  that is the boundary of an embedding  $M : \mathbb{D}^2 \hookrightarrow \mathbb{D}^4$ . ┘

**Def. (15.3.0.38) [Ribbon Knots].** A **ribbon knot** is a slice knot  $K \in \mathbb{S}^3$  which is a boundary of an embedding  $M : \mathbb{D}^2 \hookrightarrow \mathbb{D}^4$  s.t. the function  $\|x\|$  on  $\mathbb{D}^4$  restricted to  $M$  has no interior local maximum on  $M$ . ┘

## 1 Seifert Surfaces

**Def. (15.3.1.1) [Siefert Surfaces].** A **Siefert surface** of an oriented link  $L \subset \mathbb{S}^3$  is a connected compact oriented surface with boundary in  $\mathbb{S}^3$  that has  $L$  as its oriented boundary. ┘

**Thm. (15.3.1.2).** Any oriented link  $L \subset \mathbb{S}^3$  has a Siefert surface. ┘

*Proof:* Cf.[Lickorish]P16. □

**Cor. (15.3.1.3) [Genus of Knots].** For any knot  $K$ , the **genus of knot**  $g(K)$  of  $K$  is defined to be the minimal genus of a Siefert surface of  $K$ . ┘

**Prop. (15.3.1.4).** For any two disjoint knots  $K_1, K_2$ ,  $g(K_1 + K_2) = g(K_1) + g(K_2)$ . ┘

*Proof:* Cf.[Lickorish]P17. □

**Thm. (15.3.1.5) [Schönflies Theorem].** For any piecewise-linear embedding  $e : \mathbb{S}^2 \rightarrow \mathbb{S}^3$ ,  $\mathbb{S}^3 \setminus e(\mathbb{S}^2)$  has two connected components, and the closure of each of them is a piecewise-linear  $\mathbb{D}^2$ . ┘

*Proof:* Cf.[Lickorish. The irreducibility of the three-sphere]. □

## 2 Polynomial Invariants

### HOMFLY Polynomials

**Thm. (15.3.2.1) [HOMFLY].** There exists a unique knot invariant  $P$  on the equivalent classes of tame oriented links to the set of homogenous Laurent polynomials of degree 0 in  $x, y, z$ , s.t.

- $P_{\bigcirc} = 1$ .
- $xP_{L_+}(x, y, z) + yP_{L_-}(x, y, z) + zP_{L_0}(x, y, z) = 0$ , which means a local surgery of the crossing according to (15.3.0.6). ┘

**Prop. (15.3.2.2).** Given two oriented links  $L, L'$ ,

$$P_{L \# L'}(x, y, z) = P_L(x, y, z) \cdot P_{L'}(x, y, z).$$
┘

**Example (15.3.2.3).**  $P_{\coprod^n \bigcirc}(x, y, z) = \left(-\frac{x+y}{z}\right)^{n-1}$ . ┘

Kauffman PolynomialsAlexander-Conway Polynomials

**Def. (15.3.2.4)** [Alexander-Conway Polynomials]. **Alexander-Conway polynomial.** ┘

**Prop. (15.3.2.5).** The Alexander-Conway polynomial is just

$$\Delta_L(t) = P_L(1, -1, t^{1/2} - t^{-1/2}) \text{ (15.3.2.1).}$$

┘

*Proof:*

□

Jones Polynomials

Cf. [STATE MODELS AND THE JONES POLYNOMIAL, Kauffman].

**Def. (15.3.2.6)** [Kauffman Brackets]. The **Kauffman bracket** is a function from unoriented link diagrams in  $\mathbb{S}^2$  to  $\mathbb{Z}[T][\frac{1}{T}]$  defined inductively by the following rules:

- $\langle \bigcirc \rangle = 1$ .
- $\langle D \amalg \bigcirc \rangle = -(T^{-2} + T^2) \langle D \rangle$ .
- $\langle L_+ \rangle = T \cdot \langle \rangle + T^{-1} \cdot \langle \succ \rangle$ , which means a local surgery of the crossing according to (15.3.0.6).

The Kauffman bracket of a diagram is invariant under type2,3 Reidemeister moves, but changed by type1 Reidemeister moves. ┘

**Def. (15.3.2.7)** [Writhe]. The **Writhe of a link diagram**  $w(D)$  of an oriented link diagram  $D$  is the sum of sign of crossings of  $D$  (15.3.0.6). The writhe of a diagram is invariant under type2,3-Reidemeister moves, but changed by type1 Reidemeister moves.

The writhe is invariant under change of orientations of  $D$ , so it is an invariant of un-oriented link diagrams. ┘

**Thm. (15.3.2.8)** [Jones Polynomials]. Let  $D$  be a diagram of a link  $L$ , then the expression

$$X_L(T) = (-T)^{3w(D)} \langle D \rangle \text{ (15.3.2.7)(15.3.2.6)}$$

is a link invariant (15.3.0.10) of the oriented link  $L$ . The **Jone polynomial** of  $L$  is defined to be

$$V_L(t) = X_L(t^{-1/4}) \in \mathbb{Z}[t^{1/2}, t^{-1/2}].$$

┘

*Proof:* Cf. [Lickorish]P26.

It suffices to show that this  $X_L(T)$  is invariant under Reidemeister moves, and this is easy to verify (Cf. [Ada04]). ┘

**Remark (15.3.2.9).** Rewrite the definition to fit in HOMFLY's framework. Cf. [Kau99]P677?. ┘

**Prop. (15.3.2.10).** The Jones polynomial is just

$$V_L(t) = P_L(t, -t^{-1}, t^{1/2} - t^{-1/2}) \text{ (15.3.2.1).}$$

┘

*Proof:*

□

**Prop. (15.3.2.11) [Properties of Jones Polynomials].**

- If  $K$  is a knot, then  $V_K(\omega) = 1$ .
- $V_L(1) = (-2)^{\pi_0(L)-1}$ .
- if  $K$  is a knot, then  $\frac{\partial}{\partial t} V_K(1) = 0$ .
- If  $K$  is a knot, then the Arf invariant of  $K$  is just  $\text{Arf} = (1 - V_K(i))/2$ .
- $\Delta_K(-1) \equiv 1, 5 \pmod{8}$ , according to  $\text{Arf} = 0$  or  $1$ .

┘

*Proof:* Cf. [Jones].

□

**Prop. (15.3.2.12).** If  $K$  is a knot and  $|\Delta_K(i)| > 3$ , then  $K$  cannot be represented as a closed 3-braid.

If  $K$  is a knot and  $\Delta_K(e^{2\pi i/5}) > 6.5$ , then  $K$  cannot be represented as a closed 4-braid.

┘

*Proof:* Cf. [Jones].

□

**Conj. (15.3.2.13).** If  $K$  is a non-trivial knot, then  $V_K(T) \neq 1$ .

┘

*Proof:*

□

### Arf Invariants

**Def. (15.3.2.14) [Pass Moves].** A **Pass move** on an oriented knot diagram is a local strand move by passing two strands of the opposite orientation across two strands of the opposite orientation at the same time. Notice that the orientation of  $K$  doesn't influence the definition of this pass move. Two knots are called **pass-equivalent** if they can be connected by a sequence of pass moves.

┘

**Def. (15.3.2.15) [Arf Invariants].** Any knot is pass-equivalent to either the unknot or a trefoil knot (Notice the two trefoil knots are pass-equivalent), and a trefoil knot is not pass-equivalent to the unknot.

The **Arf invariant**  $\text{Arf}(K) \in \mathbb{Z}/(2)$  of a knot  $K$  is defined to 0 if it is pass-equivalent to the unknot, and 1 otherwise.

┘

*Proof:* Cf. [Ada04]P223. ?

□

**Prop. (15.3.2.16).** On a local crossing, if  $\langle L_+ \rangle$  and  $\langle L_- \rangle$  are all knots and  $\langle \rangle \langle \rangle$  is a link with two components, then

$$\text{Arf}(\langle L_+ \rangle) = \text{Arf}(\langle L_- \rangle) + \text{lk}(\langle \rangle \langle \rangle).$$

Notice the orientation on  $\langle \rangle \langle \rangle$  doesn't matter to the parity.

┘

*Proof:* This may be done using invariant polynomials. ?

□



### 3 Chirality

**Def. (15.3.3.1) [Chirality].** A knot  $K \subset \mathbb{R}^3$  is called **chiral** if it is not equivalent to its mirror image. It is called **achiral** or **amphicheiral** otherwise. ┘

**Prop. (15.3.3.2).** Any alternating knot diagram with an odd number of crossings is chiral. ┘

*Proof:* □

### 4 Linkage

**Def. (15.3.4.1) [Linking Numbers].** For an oriented link  $L \subset \mathbb{R}^3$  with two component, let  $D$  be a link diagram of it, then the **linking number** of  $L$  is defined to be half of the sum of crossing signs (15.3.0.6) between two components of  $L$ , where ‘up’ is +1 and ‘down’ is -1. Then this number is an integer, and it is independent of the diagram chosen by checking Reidemeister moves (15.3.0.7), so it is a link invariant of  $L$ .

If  $L = K \amalg K'$ , this number is denoted by  $\text{lk}(K, K')$ . ┘

**Prop. (15.3.4.2).** Suppose  $K$  be an oriented knot in  $\mathbb{S}^3$ . Let  $N$  be a tubular nbhd of  $K$ , and  $X = \mathbb{S}^3 \setminus X$ . Then  $H_1(X)$  is isomorphic to  $\mathbb{Z}$  by Alexander duality, and it is generated by a class represented by a simply closed curve  $\mu \subset \partial N$  that bounds a disk in  $N$  meeting  $K$  at a single point. Let  $C$  be another oriented simple closed curve in  $X$ , then  $\text{lk}(C, K)$  equals  $[C] = \text{lk}(C, K) \cdot \mu \in H_1(X)$ . ┘

*Proof:* Cf. [Lickorish]P12. □

### 5 Tables of Knots

For a table of knots, Cf. [Ada04]P280.

**Prop. (15.3.5.1) [2 Crossings].** Any knot diagram with  $\leq 2$  crossings determine an unknot. ┘

*Proof:* ? The only way I can think is to enumerate. □

**Prop. (15.3.5.2).** There are no knot diagrams with exact two crossings. ┘

*Proof:* □

### 6 Quandles

**Def. (15.3.6.1) [Quandles on Sets].** For  $Q \in \text{Set}$ , a **quandle** on  $Q$  is a binary map  $*$  :  $Q \times Q \rightarrow Q$  that satisfies:

- For any  $a \in Q$ ,  $a * a = a$ .
- For any  $a, b \in Q$ , there exists a unique  $c \in Q$  s.t.  $c * b = a$ .
- For any  $a, b, c \in Q$ ,  $(a * b) * c = (a * c) * (b * c)$ .

┘

**Def. (15.3.6.2) [Quandles of Links].** ┘

## 7 Virtual Knots and Links

References are [Kau99], [CKS02], [FJSK04] and [Man10], [Kau12].

**Def. (15.3.7.1) [Virtual Link Diagrams].** A **virtual link diagram**  $D$  is a 4-regular planer graph  $D$  with decorations on the nodes which is either  $L_+, L_-$  (15.3.0.6) or virtual crossing ( $\otimes$ ). ┘

**Def. (15.3.7.2) [Virtual Reidemeister Moves].** ┘

**Def. (15.3.7.3) [Virtual Links].** The set VL of **Virtual Links** is defined to be the set of equivalence classes of virtual link diagrams where two diagrams are called equivalent if they are connected by a finite sequence of virtual Reidemeister moves (15.3.7.2). ┘

*Proof:* Cf. [Kau99]. □

### Gauss Codes

**Def. (15.3.7.4) [Gauss Codes].** The **Gauss codes** of a virtual link is defined the same as the Gauss code of a planer link (15.3.0.26) by omitting the virtual crossings. ┘

**Prop. (15.3.7.5).** There is a bijection between VL and the Reidemeister equivalence classes of Gauss codes. ┘

*Proof:* Cf. [Kau99].? □

**Prop. (15.3.7.6) [Virtual Knots and Gauss Codes].** If  $K$  is a virtual knot whose underlying Gauss code is planar and whose sign sequence is standard?, then  $K$  is equivalent to a classical knot. ┘

*Proof:* Cf. [Kau99]P669.?  
□

### Abstract Link Diagrams

References are [K-K00].

**Def. (15.3.7.7) [Abstract Link Diagrams].** ┘

**Def. (15.3.7.8) [Abstract Moves of Abstract Link Diagrams].** ┘

**Def. (15.3.7.9) [Abstract Links].** The set AL of **abstract links** is defined to be the set of equivalence classes of abstract link diagrams, where two diagrams are called equivalent if they are connected by a finite sequence of abstract moves (15.3.7.8). ┘

**Prop. (15.3.7.10) [Virtual Links and Abstract Links].** There is a bijection of sets  $\varphi : \text{VL} \cong \text{AL}$ , called the **skimming process**. ┘

*Proof:* Cf. [K-K00].?  
□

### Stable Equivalence of Link Diagrams

**Def. (15.3.7.11) [Stable Equivalence of Links on Thickened Surfaces].** Let  $\mathcal{D}$  be the set of pairs  $(\Sigma, D)$  where  $\Sigma$  is a compact oriented surface (possibly with boundaries) and  $D$  is a link diagram on  $\Sigma$ . Two objects  $(\Sigma_1, D_1), (\Sigma_2, D_2) \in \mathcal{D}$  are said to be connected by a **stable Reidemeister move** if there exists a closed oriented surface  $\Sigma_3$  and orientation-preserving embeddings  $f_1 : \Sigma_1 \rightarrow \Sigma_3, f_2 : \Sigma_2 \rightarrow \Sigma_3$  s.t. the diagrams  $f_1(D_1)$  and  $f_2(D_2)$  are related by Reidemeister moves on  $\Sigma_3$ .

Then two objects in  $\mathcal{D}$  are called **stably equivalent diagrams** if they are connected by a finite sequence of stable Reidemeister moves.  $\lrcorner$

**Prop. (15.3.7.12) [Virtual Links and Links on Thickened Surfaces].** The natural inclusion map  $\text{AL} \hookrightarrow \mathcal{D} / \sim$  (15.3.7.11) is a bijection. In particular, by (15.3.7.10), there are bijections

$$\varphi : \text{VL} \cong \text{AL} \cong \mathcal{D} / \sim .$$

$\lrcorner$

*Proof:* This follows by taking regular nbhd of a link diagram in a surface.  $\square$

**Prop. (15.3.7.13) [Kuperberg].** Every equivalence class in  $\mathcal{D} / \sim$  has a unique irreducible representative.  $\lrcorner$

*Proof:* Cf. [Kup03].  $\square$

### Fundamental Group, Crystals, Racks and Quandles

**Prop. (15.3.7.14).** If  $K$  and  $K'$  are classical knot diagrams that are equivalent under extended virtual Reidemeister moves, then  $K$  and  $K'$  are equivalent under classical Reidemeister moves.  $\lrcorner$

*Proof:* Cf. [Kau99]P675.  $\square$

**Def. (15.3.7.15) [Jones Polynomials].** The Jones polynomial is defined the same way as that of Planer links (15.3.2.8) by omitting the virtual crossings.  $\lrcorner$

**Prop. (15.3.7.16) [Trivial Jones Polynomials].** To each non-trivial classical knot diagram  $K$  there is a corresponding non-trivial virtual knot diagram  $\text{Virt}(K)$  with trivial Jones polynomial.  $\lrcorner$

*Proof:* Cf. [Kau12]Thm6.1.  $\square$

## 8 Virtual Concordance and Slices

References are [CKS02].

## 9 Links and Graphs

**Def. (15.3.9.1) [Intrinsically Linked Graphs].** An **intrinsically linked graph** is a graph  $K$  s.t. any embedding of  $K$  in  $\mathbb{R}^3$  contains a non-trivial link.  $\lrcorner$

**Def. (15.3.9.2) [Intrinsically Knotted Graphs].** An **intrinsically knotted graph** is a graph  $K$  s.t. any embedding of  $K$  in  $\mathbb{R}^3$  contains a knotted cycle.  $\lrcorner$

**Prop. (15.3.9.3) [Conway-Gordon].** Any embedding of the graph  $K_6$  in  $\mathbb{R}^3$  contains two triangles with odd linking number. In particular,  $K_6$  is intrinsically linked.  $\lrcorner$

*Proof:* In the embedding shown as in [Ada04]P216Figure8.2, it is easy to see that sum of linking numbers between the 10 pairs of 3-cycles in  $K_6$  is odd. Then the assertion follows from (15.3.9.6).  $\square$

**Lemma (15.3.9.4).** The parity of the sum of linking numbers of all the 10 pairs of 3-cycles in  $K_6$  under any embedding is invariant under a crossing change.  $\lrcorner$

*Proof:* Clearly a crossing of an edge with itself or a crossing of two adjacent edges doesn't change the sum. And a crossing of two non-adjacent edges changes two linking number by 1, so the parity of the sum is unchanged.

For a crossing of two adjacent edges,  $\square$

**Prop. (15.3.9.5)[Conway-Gordon].** Any embedding of the graph  $K_7$  in  $\mathbb{R}^3$  contains a Hamilton cycle with non-zero Arf invariant.  $\lrcorner$

*Proof:* In the embedding shown as in [Ada04]P230Figure, all Hamilton cycles except one is unknotted, and the rest one is a trefoil, so the total sum of Arf invariants of all 360 Hamilton cycles in  $K_7$  under this embedding is 1. Then the assertion follows from (15.3.9.6).  $\square$

**Lemma (15.3.9.6).** The sum of Arf invariants of all 360 Hamilton cycles in  $K_7$  under any embedding is invariant under a crossing change.  $\lrcorner$

*Proof:* Cf. [C-G83]P448.  $\square$

**Prop. (15.3.9.7).**  $K_{5,5}$  is intrinsically knotted.  $\lrcorner$

*Proof:* Cf. [Miki Shimabara1988].  $\square$

**Prop. (15.3.9.8).** Intrinsically linked graphs are intrinsically knotted.  $\lrcorner$

*Proof:* Cf. [RST93].  $\square$

**Def. (15.3.9.9)[Peterson Graphs].** A **Peterson graph** is a graph connected to  $K_6$  through  $Y - \Delta$  transforms. There are exactly 7 Peterson graphs, Cf. [Ada04]P222.  $\lrcorner$

**Thm. (15.3.9.10)[Robertson-Seymour-Thomas].** A graph is intrinsically knotted iff it contains a Peterson graph (15.3.9.9).  $\lrcorner$

*Proof:* One direction is trivial, for the other, Cf. [RST93]  $\lrcorner$ .  $\square$

**Conj. (15.3.9.11).** Any intrinsically knotted graph contains a subgroup that is an extension (splitting of vertices) of  $K_7$  or  $K_{5,5}$ .  $\lrcorner$

*Proof:*  $\square$

## 15.4 Geometric and Combinatorial Group Theory

Cf.[Princeton Companion].

### Mednykh's Formula

**Thm. (15.4.0.1).** For  $G \in \mathcal{Ab}^{\text{fin}}$ , and  $\Sigma_g$  the oriented closed surface of genus  $g$ , then

$$\# \text{Hom}(\pi_1(\Sigma_g), G) = \#G \sum_i \left( \frac{\#G}{\dim V_i} \right)^{2g-2}.$$

┘

**Prop. (15.4.0.2).** For any oriented triangulated surface  $\Sigma$  and  $G \in \mathcal{Ab}^{\text{fin}}$ , we can associate a rational number in the following way:

1. Let  $V = \mathbb{Q}[G]$  which is a non-commutative  $\mathbb{Q}$ -algebra.
2. Let  $a = \frac{1}{\#G^2} \sum_{g,h,k \in G, ghk=1} g \otimes h \otimes k \in V^{\otimes 3}$ .
3. Let  $N \in (V^{\otimes 2})^*$  be given by  $N(g, h) = \#G \delta_{gh,1}$ .
4. Label the edges  $e_1, \dots, e_E$  and given each of them an arbitrary orientation, and faces  $f_1, \dots, f_F$  with orientation given by the orientation of  $\Sigma$ .
5. For each face  $f_i$ , let the orientation on the boundary be given by the ordered pair  $(f_i(1), f_i(2), f_i(3))$  (which subjects to a 3-cycle permutation).
6. Consider the element

$$\otimes_i a \in \otimes_i (V^{\otimes 3}),$$

and for each edge  $e_j = f_x(a) = f_y(b)$ , use the element  $N \in (V^{\otimes 2})^*$  to contract the  $(x, a)$ -th and  $(y, b)$ -th component of  $\otimes_i (V^{\otimes 3})$ . Then the image of  $\otimes_i a$  under this contraction is an element in  $\mathbb{Q}$ .

Then this element is invariant under isogenies of  $\Sigma$  and Pachner moves of the triangulations. In particular, this element only depends on  $\Sigma$ , by (4.4.11.11). ┘

*Proof:* ?

□

**Prop. (15.4.0.3).** Situation as in (15.4.0.2), the number we get is just

$$(\#G)^{\chi(G)-1} \cdot \# \text{Hom}(\pi_1(\Sigma), G).$$

┘

*Proof:* ?

□

### 1 Space Symmetry Groups

**Prop. (15.4.1.1) [Finite Group of Symmetries of Euclidean Spaces].** For any  $n \in \mathbb{Z}_+$ , any finite group  $G$  of congruences of  $\mathbb{R}^n$  fixes a point. In particular, when consider finite group of symmetries, we can assume that the origin is fixed. ┘

*Proof:* Let  $P_1, \dots, P_{\#G}$  be the images of the origin, then  $\frac{1}{N}(P_1 + \dots + P_N)$  is fixed. □

**Prop. (15.4.1.2) [Spherical Area Formula].** For any  $k \in \mathbb{Z}_{\geq 3}$  and a convex  $k$ -gon  $X = v_1 \dots v_k$  on  $\mathbb{S}^2$  with  $\angle v_{i-1}v_i v_{i+1} = A_i$ , then

$$\sum_i A_i = \text{Vol}(X) + (k-2)\pi.$$

┘

*Proof:* By dividing into triangles, it suffices to prove for  $k = 3$ . In this case, the proof is easy by drawing great lines.  $\square$

## 2D Euclidean Groups

**Def. (15.4.1.3) [Orbifold Notations, Thurston-Conway].** ?

┘

**Thm. (15.4.1.4) [Magic Theorem, Conway].** There are exactly 17 discrete rigid symmetry groups of  $\mathbb{R}^2$ .

┘

*Proof:* Cf. [C-S03]P19. ?

┘

## Finite 3D Symmetry Groups

**Prop. (15.4.1.5) [Spherical Rotation Relations].** The product of rotations about the vertices of a spherical triangle through twice the angles of that triangle, is identity.

┘

*Proof:* Let  $P, Q, R$  be the reflections along the sides of the triangle, then the rotations are exactly  $PQ, QR, RP$ . So  $(PQ)(QR)(RP) = \text{id}$ .  $\square$

**Cor. (15.4.1.6) [Rotation Groups].** The 3D symmetry group generated by rotations consists all of rotations. Such groups are called **3D rotation groups**.

┘

**Thm. (15.4.1.7) [Finite 3D Reflection Groups].** Any 3D finite reflection group  $G$  is generated by the reflections along the sides of a convex spherical polygon whose angles are of the form  $\pi/n_i$  where  $n_i \in \mathbb{Z}_{\geq 2}$ . Such a reflection group is denoted by  $*n_1 \dots n_k$ , called the **orbifold notation**.

Then every 3D finite reflection group is in one of the following classes:

- $*532 = \pm I = \pm I_{60} \cong \text{Alt}(5) \times \mathbb{Z}/(2)$ .
- $*432 = \pm O = \pm O_{24} \cong \text{Sym}(4) \times \mathbb{Z}/(2)$ .
- $*332 = \pm T = \pm T_{12} \cong \text{Alt}(4) \times \mathbb{Z}/(2)$ .
- $*22n = \pm D = \pm D_{2n} \cong D_{2n} \times \mathbb{Z}/(2)$ , where  $n \in \mathbb{Z}_+$ .
- $*nn = \pm C = \pm C_n \cong C_n \times \mathbb{Z}/(2)$ , where  $n \in \mathbb{Z}_+$ .

The first three are called **polyhedral reflection groups**, and the last two classes are called **axial reflection groups**.  $\square$

*Proof:* For the first assertion, let the plane of reflections divide the sphere into convex polygons, then any two adjacent polygons are isomorphic by a reflection. This implies that  $G$  is generated by the reflections along the sides of any one of these polygons.

If  $G$  is generated by the reflection group of a convex  $k$ -gon where  $k \in \mathbb{Z}_{\geq 2}$ , then by (15.4.1.2),  $k/2\pi > (k-2)\pi$ , which means  $k = 2$  or  $k = 3$ .

If  $k = 3$ , then  $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1 + \text{Vol}(X)/\pi$  has solutions  $(a, b, c) \in \{(2, 2, n), (2, 3, 3), (2, 3, 4), (2, 3, 5)\}$ . If  $k = 2$ , the convex 2-gon consists of two semicircles, and the group is isomorphic to  $*nn$ .

For the description of the groups, only the icosahedral group needs a proof. ?

┘

**Thm. (15.4.1.8) [Finite 3D Rotation Groups].** Every finite 3D rotation group is generated by two rotations, and it consists of rotations in a reflection group  $*n_1 \dots n_k$ , so it is denoted by  $n_1 \dots n_k$ .

Then every 3D finite rotation group is in one of the following classes:

- $532 = I = I_{60} \cong \text{Alt}(5)$ , called the **icosahedral group**.
- $432 = O = O_{24} \cong \text{Sym}(4)$ , called the **octahedral group**.
- $332 = T = T_{12} \cong \text{Alt}(4)$ , called the **tetrahedral group**.
- $22n = D = D_{2n} \cong D_{2n}$ , where  $n \in \mathbb{Z}_+$ , called the **dihedral groups**.
- $nn = C = C_n \cong C_n$ , where  $n \in \mathbb{N}$ , called the **cyclic groups**.

The first three are called **polyhedral rotation groups**, and the last two classes are called **axial rotation groups**. ┘

*Proof:* Firstly we prove that a symmetry group generated by two rotations is contained in a reflection group with index 2: Let  $x, y$  be the rotation centers with rotations  $\rho_x, \rho_y$ , and let  $M$  be the reflection along the plane  $\langle x, y, O \rangle$ . Then  $M$  commutes with  $\rho_x, \rho_y$ , and  $\rho_x M, \rho_y M$  are all reflections. So  $\langle M, \rho_x, \rho_y \rangle$  is a finite reflection group, which contains  $G = \langle x, y \rangle$  as a subgroup of index 2.

Now any two rotation of  $G$  generates a subgroup that is isomorphic to one of the above. So we can consider the distances between the rotation centers, to show that  $G$  is in fact generated by some two elements (Cf. [C-S03]P30). □

**Def. (15.4.1.9) [Diploid and Haploid Symmetry Groups].** A subgroup of  $O(n; \mathbb{R})$  is called a **diploid symmetry group** if it contains  $-1$ . And it is called a **haploid symmetry group** otherwise. It is called a **hybrid symmetry group** if it is haploid but not contained in  $\text{SO}(n; \mathbb{R})$ . ┘

**Prop. (15.4.1.10) [Finite 3D Hybrid Symmetry Groups].** By considering the projective image, every 3D finite hybrid symmetry group is in one of the following classes:

- $CC_{2n} = +C_n - C_{2n}, n \in \mathbb{Z}_+$ , called the **cyclo-cyclic group**.
- $CD_{2n} = +C_n - D_{2n}, n \in \mathbb{Z}_+$ , called the **cyclo-dihedral group**.
- $DD_{4n} = +D_{2n} - D_{4n}, n \in \mathbb{Z}_+$ , called the **dihedral-dihedral group**.
- $TO_{24} = +T_{12} - D_{24}$ , called the **tetra-octahedral group**. ┘

*Proof:* The point is to figure out the index-2 subgroups of  $\text{Sym}(4)$  and  $D_{2n}$ . □

**Prop. (15.4.1.11) [Tabulations].** There is a tabulate of all 3D symmetry groups of polyhedrons and their orbifold notations in [C-S03]P25. ? ┘





# 16 | Differential Geometry

## 16.1 Smooth Manifolds I: Generalities

*“A line is length without breadth.”*

*–D. Euclid*

References are [G-P74, Mil97, Lee13] and [Geometric Analysis, Jost].

**Notation(16.1.0.1).**

- Use notations on topology defined in [Topology I](#).
- Use notations on real analysis defined in [Measure Theory and Functions on  \$\mathbb{R}^n\$](#) .
- Use notations on algebraic topology defined in [Algebraic Topology II: Homologies and Cohomologies](#).
- All smooth manifolds(16.1.1.1) in this section is assumed to be smooth over  $\mathbb{R}$ (16.1.1.1) or analytic over  $\mathbb{C}$ .
- Let  $M, N \in \text{Diff}_\partial$ (16.1.1.1).

┘

### 1 Local Properties

**Def.(16.1.1.1) [Smooth manifolds].** For  $n \in \mathbb{N}$ , a **smooth manifold**(resp. a **smooth manifold possibly with boundaries**) of dimension  $n$  is a topological manifold(resp. topological manifold possibly with boundaries) of dimension  $n$ (4.4.11.3) with an open covering  $\{U_\alpha\}$  together with an embedding of topological spaces  $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$ (resp.  $\varphi_\alpha : U_\alpha \rightarrow H^n$ ) s.t. for any  $\alpha, \beta$ ,

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is smooth.

A morphism of smooth manifolds  $f : (M, \{U_\alpha, \varphi_\alpha\}) \rightarrow (N, \{V_{\alpha'}, \varphi_{\alpha'}\})$  is a map of manifolds s.t. for any  $\alpha, \alpha'$ , the map

$$\varphi_{\alpha'} \circ f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap f^{-1}(V_{\alpha'})) \rightarrow \varphi_{\alpha'}(U_{\alpha'})$$

is smooth.

The category of smooth manifolds possibly with boundary is denoted by  $\text{Diff}_\partial$ . The subcategory of smooth manifolds is denoted by  $\text{Diff}$ .

An isomorphism in  $\text{Diff}_\partial$  is called a **diffeomorphism**.

┘

**Remark(16.1.1.2).** There must be a more categorical way to define a manifold.

┘

**Def. (16.1.1.3) [Tangent Vectors].** For  $M \in \text{Diff}_\partial(16.1.1.1)$ , a **tangent vector** at  $p \in M$  is defined to be a linear map  $v : C^\infty(M) \rightarrow \mathbb{R}$  satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

The vector space of all tangent vectors of  $M$  at  $p$  is denoted by  $T_p M$ . ┘

**Def. (16.1.1.4) [Differential of Smooth Maps].** For  $F : M \rightarrow N \in \text{Diff}_\partial$  and  $p \in M$ , there is a map

$$dF_p : T_p M \rightarrow T_{F(p)} N : dF_p(v)(f) = v(f \circ F), \text{ where } v \in T_p M, f \in C^\infty(N).$$

Then:

- $dF_p : T_p M \rightarrow T_{F(p)} N$  is linear,
  - (Chain rule) Taking differentials commute with compositions of maps.
  - $d(\text{id}) = \text{id}_{T_p M}$ .
- ┘

**Prop. (16.1.1.5).** For  $M \in \text{Diff}^n$  and any  $p \in M$ ,  $\dim T_p M = n$ . ┘

**Prop. (16.1.1.6).** For  $M_1, \dots, M_k \in \text{Diff}$  and  $p = (p_1, \dots, p_k) \in M = M_1 \times \dots \times M_k$ , there is a natural isomorphism  $T_p M \cong T_{p_1}(M_1) \times \dots \times T_{p_k}(M_k)$ . ┘

**Prop. (16.1.1.7) [Collar Neighborhood Theorem].** If  $X$  is a smooth paracompact manifold with boundary, then there is a nbhd of  $\partial X$  in  $X$  which is diffeomorphic to the product  $\partial X \times [0, 1]$ . ┘

*Proof:*

□

**Prop. (16.1.1.8) [Rank Theorem].** Suppose  $r \in \mathbb{N}$ ,  $f : M \rightarrow N \in \text{Diff}$  s.t.  $\text{rank } df_p = r$  for any  $p \in M$ , then for any  $p \in M$ , there exists smooth charts centered at  $p, F(p)$  s.t. the coordinate representation of  $F$  is

$$F(x_1, \dots, x_r, x_{r+1}, \dots, x_m) \mapsto (x_1, \dots, x_r, 0, \dots, 0).$$

┘

*Proof:* Cf. [Lee13]P81. □

**Def. (16.1.1.9) [Immersion].**  $f : M \rightarrow N \in \text{Diff}$  is called a **smooth immersion** of manifolds if  $df(x) : T_x(M) \rightarrow T_{f(x)}(N)$  is injective for any  $x \in M$ .

$f$  is called a **smooth embedding** if it is both a smooth immersion and a topological embedding.

$f$  is called a **smooth submersion** if it is a smooth map that  $df(p) : T_p(M) \rightarrow T_{f(p)}(N)$  is surjective for any  $p \in M$ . ┘

**Prop. (16.1.1.10).** Suppose  $f : M \rightarrow N \in \text{Diff}$  and  $p \in M$ . If  $df_p$  is surjective, then there is a nbhd  $p \in U \subset M$  s.t.  $f : U \rightarrow N$  is a submersion. And if  $df_p$  is injective, then there is a nbhd  $p \in U \subset M$  s.t.  $f : U \rightarrow N$  is an immersion. ┘

**Def. (16.1.1.11) [Local Diffeomorphism].**  $f : M \rightarrow N \in \text{Diff}_\partial$  is called a **local diffeomorphism** if for any  $p \in M$ , there exists an open subset  $p \in U \subset M$  s.t.  $U \rightarrow f(U)$  is a diffeomorphism. ┘

**Prop. (16.1.1.12) [Inverse Function Theorem for Manifolds].** Suppose  $f : M \rightarrow N \in \text{Diff}_\partial$  and  $p \in M$  s.t.  $df_p$  is invertible, then there exist  $p \in U \subset M$  s.t.  $f : U \rightarrow f(U)$  is a diffeomorphism (16.1.1.1). ┘

In particular,  $f$  is a local diffeomorphism (16.1.1.11) iff  $df_p$  is invertible for any  $p \in M$ . ┘

*Proof:* Cf. [Lee13]P79. □

**Prop. (16.1.1.13) [Local Section Theorem].**  $\text{pr} : M \rightarrow N \in \mathcal{D}\text{iff}$  is a smooth submersion iff each point of  $M$  is in the image of a local section of  $\text{pr}$ . ┘

*Proof:* If each point of  $M$  is in the image of a local section of  $F$ , the differential is surjective at every point. Conversely, use the rank theorem (16.1.1.8). □

**Prop. (16.1.1.14).** A smooth submersion  $F : M \rightarrow N \in \mathcal{D}\text{iff}$  is an open map, and a surjective smooth submersion is a quotient map. ┘

*Proof:* Let  $W \subset M$ , and  $q = \pi(p)$  where  $p \in W$ , then there is a local section  $\sigma : U \rightarrow M$  that  $\sigma(q) = p$  (16.1.1.13), thus  $\sigma^{-1}(W)$  is open in  $N$ . But for any  $y \in \sigma^{-1}(W)$ ,  $y = \pi(\sigma(y)) \in \pi(W)$ . So  $p \in \sigma^{-1}(W) \subset \pi(U)$ , which means  $\pi(W)$  is open, and  $\pi$  is an open map.  $\dagger$  The last assertion follows as any open surjective map is a quotient map (4.4.1.10). □

**Prop. (16.1.1.15) [Characteristic Property of Surjective Smooth Submersions].** Let  $\pi : M \rightarrow N$  be a smooth submersion of manifolds,  $P$  another smooth manifold, then

- Any map  $F : N \rightarrow P$  is smooth iff  $F \circ \pi$  is smooth.
  - Any smooth map  $\tilde{F} : M \rightarrow P$  that is constant on the fibers of  $\pi$  induces a smooth map  $F : N \rightarrow P$  that  $\tilde{F} = F \circ \pi$ .
- ┘

*Proof:* 1: Use local section theorem (16.1.1.13).

2: There is a constant map  $F : N \rightarrow P$  that  $\tilde{F} = F \circ \pi$  by (16.1.1.14) and universal property of quotient maps, and it is smooth by item 1. □

**Def. (16.1.1.16) [Smooth Covering Space].** A **smooth covering space** of a smooth manifold  $X$  is a space  $\tilde{X}$  together with a smooth map  $\pi : \tilde{X} \rightarrow X$  that there is a covering  $U_\alpha$  of  $X$  that for each  $\alpha$ ,  $\pi^{-1}(U_\alpha)$  is a disjoint union of open subsets of  $\tilde{X}$ , each of which is mapped diffeomorphically onto  $U_\alpha$ . ┘

**Prop. (16.1.1.17) [Proper Free Action].** Let  $\pi : E \rightarrow M$  be a smooth covering map, then with the discrete topology,  $\text{Aut}_\pi(E)$  is a discrete Lie group acting smoothly, freely and properly on  $E$ .

Conversely, suppose  $M$  is a smooth manifold and  $\Gamma$  is discrete group acting smoothly, freely and properly on a manifold, then the quotient space  $M/\Gamma$  is a topological manifold, and it has a unique smooth structure that the quotient map  $\pi : M \rightarrow M/\Gamma$  is a smooth normal covering map. ┘

*Proof:* If  $\pi : E \rightarrow M$  is a smooth covering map, then the action is continuously, freely and properly by (4.15.1.28). Smoothness can be seen by applying (16.1.1.15).  $\text{Aut}_\pi(E)$  is a Lie group because it is countable: Let  $q \in M$ , and  $U$  an evenly covered nbhd of  $q$ , then  $\pi^{-1}(U)$  is a union of open subsets each containing one element of  $\pi^{-1}(q)$ . so  $|\pi^{-1}(q)|$  is countable, and because  $\text{Aut}_\pi(E)$  acts freely, it is also countable.

Because this action is a covering map action by (4.12.1.19), (4.15.1.31) shows this is a normal covering map. The quotient space is locally Euclidean, and also Hausdorff by (4.12.1.12)(4.12.1.10), so it is a topological manifold. The smooth structure is clear. □

**Def. (16.1.1.18) [Smooth Embedding].** A **smooth embedding** of manifolds  $f : M \rightarrow N$  is a smooth immersion that is also a homeomorphism onto its image. ┘

**Prop. (16.1.1.19) [Global Rank Theorem].** Let  $F : M \rightarrow N$  be a smooth map of manifolds of constant rank, then:

- if it is an injection, then it is a submersion.
- if it is a surjection, then it is a submersion.
- if it is a bijection, then it is a diffeomorphism.

┘

*Proof:* Cf. [Lee Smooth Manifold P83].

□

**Prop. (16.1.1.20) [Local Embedding Theorem].** If  $F : M \rightarrow N$  is a smooth morphism of manifolds, then it is a smooth immersion if it is locally a smooth embedding on the source. ┘

*Proof:* Let  $p \in M$ , then there exists a nbhd  $U_1$  of  $p \in M$  that  $F$  is injective. Now choose another precompact nbhd  $U$  of  $p$  that  $\overline{U} \subset V$ , then  $F|_{\overline{U}}$  is an injective map with compact domain, so it is a topological embedding by (4.4.2.11). Thus  $F|_U$  is a smooth embedding. □

### Submanifolds

**Def. (16.1.1.21) [Submanifolds].** For a smooth manifold  $M$ , an **embedded submanifold**  $S \subset M$  is a subset  $S$  that is a manifold in the induced topology together with a smooth structure that the inclusion is a smooth embedding of manifolds (16.1.1.18).

An **immersed submanifold**  $S \subset M$  is a subset endowed with a topology and a smooth manifold structure that the inclusion is a smooth immersion (16.1.1.9).

An **weakly embedded submanifold**  $S \subset M$  is an immersed submanifold that any smooth map  $F : N \rightarrow M$  from some space  $N$  that has image in  $S$  is smooth as a map from  $N$  to  $S$ . ┘

**Remark (16.1.1.22).** Examples of immersed submanifolds that is not an embedded submanifolds are the 8-figure and the dense curve in a torus. However, an immersed submanifold is locally embedded on the source, by (16.1.1.20). ┘

**Def. (16.1.1.23) [Slice Charts].** Let  $U \subset \mathbb{R}^n$ , then a  $k$ -slice of  $U$  is the set  $S = \{(x^1, \dots, x^n) \in U \mid x^{k+1} = \dots = x^n = 0\}$ .

Let  $M$  be a manifold, a **slice chart** of a subset  $S \subset M$  is a smooth char  $(U, \varphi)$  that  $\varphi(U \cap S)$  is a  $k$ -slice of  $\varphi(U)$  for some  $k$ . ┘

**Prop. (16.1.1.24) [Local Slice Criterion for Embedded Submanifolds].** Let  $S \in \text{Diff}_0^k$  be an embedded submanifold of  $M$ , then each point of  $S$  is contained in the domain of a slice chart (16.1.1.23). Conversely, if  $S \subset M$  is a subset that each point of  $S$  is contained in the domain of a slice chart, then the induced topology makes  $S$  a topological manifold, and there is a smooth structure on  $S$  that makes it an embedded submanifold of  $M$ . ┘

*Proof:* Suppose  $S$  is an embedded submanifold, then the rank theorem (16.1.1.8) shows there exists coordinates that the image of  $i(S)$  is contained in a  $k$ -slice of  $M$ . Shrinking the open subset a little bit, we get a slice chart of  $S$ .

Conversely, if each point of  $S$  is contained in the domain of a slice chart, then we can use these smooth charts to get an atlas for  $S$ , which makes  $S$  an embedded topological submanifold of  $M$ . The transition maps are also smooth because they are restrictions of the corresponding transition map of  $M$ , so  $S$  is an embedded submanifold of  $M$ . □

**Cor. (16.1.1.25)[Boundary Submanifolds].** Suppose  $\partial M \neq \emptyset$ , then  $\partial M$  is an embedded submanifold of  $M$  of codimension 1.  $\lrcorner$

**Prop. (16.1.1.26).** If  $M$  is a compact manifold, then any injective immersion  $f : M \hookrightarrow M$  is an embedding of submanifolds.  $\lrcorner$

*Proof:* The topology of  $M$  is equivalent to the induced topology by (4.4.2.10).  $\square$

**Prop. (16.1.1.27) [Immersed Submanifolds are Locally Embedded].** If  $S \subset M$  is an immersed submanifold, then for any  $p \in S$ , there is a nbhd  $U$  of  $p \in S$  that  $U \subset M$  is an embedded submanifold.  $\lrcorner$

*Proof:* This follows immediately from (16.1.1.20). Notice that the topology on  $U$  must be the induced topology, because it is a smooth embedding thus a homeomorphism onto its image (16.1.1.18).  $\square$

**Lemma (16.1.1.28).** If  $i : S \hookrightarrow M$  is an immersed submanifold, if  $F : N \rightarrow M$  is a smooth morphism of manifolds that has image in  $S$ , if  $F : N \rightarrow S$  is continuous, then  $N \rightarrow S$  is smooth.  $\lrcorner$

*Proof:* Let  $p \in N$  mapping to  $q = F(p)$ . By (16.1.1.27), there is a nbhd  $V$  of  $q \in S$  that  $i|_V$  is an smooth embedding. Thus there is a slice chart (16.1.1.24)  $(W, \psi)$  of  $M$  that  $(V_0, \tilde{\psi})$  is a smooth chart for  $V$ , where  $V_0 = W \cap V$  and  $\tilde{\psi} = \pi \circ \psi$ , where  $\pi$  is the projection  $\mathbb{R}^n \rightarrow \mathbb{R}^k$  onto the first  $k$  coordinates, and also a smooth chart for  $S$ .

Let  $U = F^{-1}(V_0)$  be open in  $F$ , then there is a smooth chart of  $N$  contained in  $U$ . Now the coordinate representation of  $F$  in the slice chart of  $S$  is just the representation of  $F : N \rightarrow M$  composed with the projection  $\pi$ , so it is smooth.  $\square$

**Remark (16.1.1.29).**  $N \rightarrow S$  being continuous is a necessary condition, otherwise consider the figure 8.  $\lrcorner$

**Prop. (16.1.1.30) [Restricting Codomain of Smooth Morphism].** If  $S \subset M$  is an embedded submanifold and  $N \rightarrow M$  is a smooth that has image contained in  $S$ , then  $N \rightarrow S$  is smooth.  $\lrcorner$

*Proof:* Because in this case,  $S$  has the induced topology, so it is easily seen that  $N \rightarrow S$  is continuous. Then the assertion follows from (16.1.1.28).  $\square$

**Prop. (16.1.1.31) [Generalized Inverse Function Theorem].** Suppose  $f : M \rightarrow N \in \mathcal{D}\text{iff}_\partial$  and  $X \subset M$  is an embedded submanifold s.t.  $df_x$  is invertible for any  $x \in X$  and  $f$  maps  $X$  diffeomorphically onto  $f(X)$  (16.1.1.1), then there exists a nbhd  $U$  of  $X$  in  $M$  s.t.  $f : U \rightarrow f(U)$  is a diffeomorphism.

In particular,  $f$  is a local diffeomorphism (16.1.1.11) iff  $df_p$  is invertible for any  $p \in M$ .  $\lrcorner$

*Proof:* Take a local inverse on a nbhd of every  $x \in X$  using (16.1.1.12), then take a locally finite subcovering and glue them together.  $\square$

### Sard's Theorem

**Def. (16.1.1.32) [Measure-Zero Sets in Manifolds].** For  $(M, \{\varphi_\alpha\}) \in \mathcal{D}\text{iff}_\alpha$ , a subset  $A \subset M$  is called a **measure zero set** if for  $\varphi_\alpha(U_\alpha \cap A) \subset \mathbb{R}^{n_\alpha}$  has measure zero.  $\lrcorner$

**Lemma (16.1.1.33) [Invariance of Measure-Zero Sets].** If  $m, n \in \mathbb{Z}_+$  and  $S \subset \mathbb{R}^n$  has measure zero, then for any smooth map  $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ,  $g(S)$  also has measure zero.  $\lrcorner$

**Lemma (16.1.1.34).** Cf. [Lee13]P127. ┘

**Prop. (16.1.1.35) [Measure Zero Sets].** For  $M \in \text{Diff}_\alpha$ ,

- A subset of a measure-zero set is a measure-zero set.
- A countable union of measure-zero sets is a measure-zero set.
- A measure-zero set is nowhere-dense in  $M$ .
- If  $f : M \rightarrow N \in \text{Diff}_\partial$  and  $A \subset M$  is a measure-zero set, then  $f(A) \subset N$  is also a measure-zero set, by (16.1.1.33). ┘

**Def. (16.1.1.36) [Regular Values and Critical Values].** For  $f : X \rightarrow Y \in \text{Diff}$ , a point  $x \in X$  is called a **critical point** iff  $df_x$  is not surjective, otherwise it is called a **regular point**.  $y \in Y$  is called a **critical value** if  $f^{-1}(y)$  contains a critical point, otherwise it is called a **regular value**. ┘

**Thm. (16.1.1.37) [Regular Level Set Theorem].** If  $y \in N$  is a regular value for a map  $f : M \rightarrow N \in \text{Diff}(16.1.1.36)$ , then  $f^{-1}(y) \subset M$  has a natural closed embedded submanifold structure of codimension  $\dim N$ . ┘

*Proof:* The set  $U$  of points  $p \in M$  s.t.  $\text{rank } df_p = n$  is open in  $M$  by (16.1.1.10), so  $f : U \rightarrow N$  is submersion. Then we can use rank theorem (16.1.1.8) on  $U$  and slice-chart criterion to show that  $f^{-1}(y)$  is a closed embedded submanifold of  $U$ . Thus it is also a closed embedded submanifold of  $M$ . □

**Remark (16.1.1.38).** This theorem is useful in proving some subset is a smooth manifold. ┘

**Prop. (16.1.1.39) [Stack of Records Theorem].** If  $y$  is regular value of a map  $f : X \rightarrow Y$ , where  $X$  is compact and  $\dim X = \dim Y$ , then  $f$  is a covering map locally on  $f^{-1}(U)$  for some nbhd  $U$  of  $y$ . ┘

*Proof:* □

**Thm. (16.1.1.40) [Sard Theorem].** For  $f : M \rightarrow N \in \text{Diff}_\partial$ , the set of critical values in  $N$  is a measure-zero set. In particular, the regular points are dense in  $N$  (16.1.1.35). ┘

*Proof:* Cf. [Lee13]P131. □

**Cor. (16.1.1.41).** ┘

### Whitney Theorems

**Def. (16.1.1.42) [Tubular Neighborhoods].** For an embedded submanifold  $M \subset \mathbb{R}^n$ , a **tubular neighbourhood** of  $M$  is a nbhd  $U$  of  $M \subset \mathbb{R}^n$  that is the diffeomorphic image of an open subset  $V \subset NM$  of the form

$$V = \{(x, v) \in NM : |v| < \delta(x)\}$$

where  $\delta : M \rightarrow \mathbb{R}_+$  is some continuous function. ┘

**Prop. (16.1.1.43) [TM and  $N(\Delta_M, M \times M)$ ].** For any embedded submanifold  $M \subset \mathbb{R}^n$ , there is an isomorphism  $TM \cong N(\Delta_M, M \times M)$  define by sending  $(x, v)$  to  $((x, x), (-v, v))$ . ┘

**Thm. (16.1.1.44) [Tubular Neighbourhood Theorem].** Every embedded submanifold  $M \subset \mathbb{R}^n$  admits a tubular nbhd (16.1.1.42). ┘

*Proof:* Cf.[Lee13]P139.?

□

**Cor.(16.1.1.45)[Tubular Neighbourhood of Submanifolds].** For any embedded submanifold  $M \subset \mathbb{R}^n$  and an embedded submanifold  $N \subset M$ , there is a nbhd  $U$  of  $N \subset M$  that is a diffeomorphic image of an open subset  $V \subset N_M(N)$ . ┘

*Proof:* Take a tubular nbhd  $Y_\varepsilon$  of  $M \subset \mathbb{R}^n$ , and define  $h : N_M(N) \rightarrow \mathbb{R}^n : (z, v) \mapsto z + v$ . Let  $W = h^{-1}(Y_\varepsilon)$ , then  $W \rightarrow Y_\varepsilon \rightarrow Y$  is the identity on  $Z$ . Then we can use(16.1.1.31). □

**Prop.(16.1.1.46)[A Neighbourhood of  $M$  in  $TM$ ].** For any  $M \in \text{Diff}_\partial$ , there is a diffeomorphism of a nbhd of  $M \subset TM$  with a nbhd of  $\Delta_M \subset M \times M$ , extending the diffeomorphism  $\Delta_M : M \cong \Delta_M$ . And when  $M$  is oriented, this diffeomorphism preserves the orientation given by

$$(x^1, \dots, x^n, \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x^n}), \quad (x^1, \dots, x^n, x^1, \dots, x^n)$$

resp. ┘

*Proof:* By Whitney embedding(16.1.1.47), we may assume that  $M \subset \mathbb{R}^{2 \dim M + 1}$ . Then the assertion follows from the tubular nbhd theorem(16.1.1.44) and(16.1.1.43). □

**Thm.(16.1.1.47)[Whitney Embedding Theorem].** Any  $M \in \text{Diff}_\partial$  can be embedded as a closed embedded submanifold of  $\mathbb{R}^{2 \dim M + 1}$ . ┘

*Proof:* Cf.[Pollack P51]. □

**Cor.(16.1.1.48)[Whitney Immersion Theorem].** Any  $M \in \text{Diff}_\partial$  admits a smooth immersion into  $\mathbb{R}^{2 \dim M}$ . ┘

*Proof:* □

### Transversality

**Def.(16.1.1.49)[Transversality].** Let  $Z \subset Y \in \text{Diff}$  be an embedded submanifold, then  $f : X \rightarrow Y \in \text{Diff}_\partial$  is said to be **transversal** to  $Z$  if for any  $x \in f^{-1}(Z)$ , we have  $\text{Im}(df_x) + T_{f(x)}(Z) = T_{f(x)}(Y)$ . We use  $f \bar{\cap} Z$  to denote that  $f$  is transversal to  $Z$ .

More generally, if  $X, Z, Y \in \text{Diff}_\partial$  and  $Z$  is compact,  $f : X \rightarrow Y, g : Z \rightarrow Y \in \text{Diff}_\partial$  are said to be **transversal** if for any  $x \in X$  and  $z \in Z$  s.t.  $f(x) = g(z)$ , we have  $\text{Im}(df_x) + \text{Im}(dg_z) = T_{f(x)}(Y)$ . We use  $f \bar{\cap} g$  to denote that  $f$  is transversal to  $g$ . ┘

**Prop.(16.1.1.50)[Transversal Stable under Perturbations].** The property of transversality for a map  $f : X \rightarrow Y$  for a compact manifold  $X$  to a fixed submanifold  $Z \subset Y$  is stable under smooth deformation. ┘

*Proof:* We can assume the submanifold is defined by a slice, so the transversality is in fact equivalent to locally submersion in the vertical direction. Thus it is clearly stable under deformation. □

**Prop.(16.1.1.51)[Transversal Preimage are Submanifolds].** If  $Z \subset Y \in \text{Diff}$  and  $f : X \rightarrow Y \in \text{Diff}_\partial$  s.t.  $f \bar{\cap} Z, \partial f \bar{\cap} Z$ , then the preimage  $f^{-1}(Z) \subset X$  is an embedded submanifolds possibly with boundaries of codimension  $\text{codim}_Y(Z)$  in  $X$ . ┘



*Proof:* For any  $x \in f^{-1}(Z)$  with  $f(x) = z$ , we can choose a slice chart around  $z \in Z$  (16.1.1.23) by (16.1.1.24) s.t.

$$Z = \{(x^1, \dots, x^n) | x^{k+1} = \dots = x^n = 0\} \subset Y.$$

So the transversal hypothesis (16.1.1.49) exactly means that the origin is a regular value for  $f$  composed with the local projection  $\text{pr}_{(x^{k+1}, \dots, x^n)}$ . Thus  $f^{-1}(Z) = (\text{pr}_{(x^{k+1}, \dots, x^n)} \circ f)^{-1}(0)$  is an embedded submanifold of codimension  $\text{codim}_Y(Z)$  in  $X$  by (16.1.1.37).

If  $\partial X \neq \emptyset$ , Cf.[Pollack]P60. ?

□

**Cor. (16.1.1.52) [Transversal Intersection are Submanifolds].** If  $Z \subset Y \in \text{Diff}$  is an embedded submanifold and  $X \subset Y$  is a submanifolds s.t.  $X \bar{\cap} Z$ , then  $X \cap Z$  is an embedded submanifold of  $X$ , and  $\text{codim}_X(X \cap Z) = \text{codim}_Y(Z)$ . ┘

**Prop. (16.1.1.53) [Parametric Transversality Theorem].** Suppose  $N$  and  $M$  are smooth manifolds,  $X \subset M$  is an embedded submanifold, and  $F_s$  is a smooth family of maps from  $N$  to  $M$ . If the map  $F : N \times S \rightarrow M$  is transverse to  $X$ , then for a.e.  $s$ , the map  $F_s : N \rightarrow M$  is transverse to  $X$ . ┘

*Proof:* Cf.[Smooth Manifold Lee T6.35].

□

**Thm. (16.1.1.54) [Transversality Homotopy Theorem].** Suppose  $M, N \in \text{Diff}$  and  $X \subset M$  is an embedded submanifold, then every  $f : N \rightarrow M \in \text{Diff}$  is homotopic to a smooth map  $g : N \rightarrow M$  that is transverse to  $X$ . ┘

*Proof:* Embed  $M$  into an  $R^k$  and take a tubular neighbourhood, then we can construct a  $N \times D^k$  transversal to  $M$ . Cf.[Smooth Manifold Lee T6.36]. ?

□

**Prop. (16.1.1.55) [Transversality Extension Theorem].** Suppose  $X \in \text{Diff}_\partial, Y \in \text{Diff}$  and  $C \in \text{Diff}_\partial$  is a closed submanifold of  $X$ ,  $Z \in \text{Diff}$  is a closed submanifold of  $Y$ . Let  $f : X \rightarrow Y \in \text{Diff}_\partial$  s.t.  $f|_C \bar{\cap} Z$  and  $\partial f|_{C \cap \partial X} \bar{\cap} Z$ , then there exists  $g : X \rightarrow Y \in \text{Diff}_\partial$  that is smoothly homotopic to  $f$ , and  $g = f$  on a nbhd of  $C$  s.t.  $g \bar{\cap} Z$  and  $\partial g \bar{\cap} Z$ . ┘

*Proof:* Cf.[Pollack P72]. ?

□

## 2 Smooth Vector Bundles

**Def. (16.1.2.1) [Smooth Vector Bundle].** A smooth vector bundle over a  $M$  is a vector bundle over  $M$  with an atlas of smooth trivialization maps. ┘

**Def. (16.1.2.2) [Smooth Fiber Bundle].** ┘

### Tangent and Cotangent Bundles

**Lemma (16.1.2.3) [Differential in Coordinates].** Let  $F : U \rightarrow V$  be a smooth map where  $U \subset \mathbb{R}^n$  and  $V \subset \mathbb{R}^n$ , with corresponding coordinates  $(x^i)$  and  $(y^i)$ , then

$$dF_p\left(\frac{\partial}{\partial x^i}\right)_p = \sum_j \frac{\partial F^j}{\partial x^i}(p) \frac{\partial}{\partial y^j} \Big|_{F(p)}.$$

┘



*Proof:* for any smooth function  $f$ ,

$$dF_p\left(\frac{\partial}{\partial x^i}\bigg|_p\right)(f) = \frac{\partial}{\partial x^i}\bigg|_p(f \circ F) = \sum \frac{\partial f}{\partial y^j}(F(p)) \frac{\partial F^j}{\partial x^i}(p) = \left(\sum_j \frac{\partial F^j}{\partial x^i}(p) \frac{\partial}{\partial y^j}\bigg|_{F(p)}\right)(f).$$

□

**Lemma (16.1.2.4)[Change of Coordinates].** Suppose  $(U, \varphi), (V, \psi)$  be two smooth charts of a smooth manifold, and the transition function on  $U \cap V$  is denoted by

$$\psi \circ \varphi^{-1}(x) = (\tilde{x}^1(x), \dots, \tilde{x}^n(x)),$$

then (16.1.2.3) shows

$$\begin{aligned} \frac{\partial}{\partial x^i}\bigg|_p &= d(\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right) = d(\psi^{-1})_{\psi(p)} \cdot d(\psi \cdot \varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right) \\ &= d(\psi^{-1})_{\psi(p)} \left( \sum_j \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j}\bigg|_{\psi(p)} \right) = \sum_j \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j}\bigg|_p \end{aligned}$$

┘

**Def. (16.1.2.5)[Tangent Vectors].** Let  $M$  be a smooth manifold, then a **tangent vector** at a point  $p$  is a linear map  $v : C^\infty(M) \rightarrow \mathbb{R}$  satisfying:

$$v(fg) = f(p)v(g) + g(p)v(f).$$

The set of tangent vectors at  $p$  is a vector space, denoted by  $T_p M$ .

┘

**Def. (16.1.2.6)[Tangent Bundle].** Let  $n = \dim M$ , the tangent bundle of  $M$  is defined to be the set  $TM = \coprod T_p M$ . It has a smooth manifold structure that makes it into a  $2n$ -dimensional manifold, and the projection  $\pi : TM \rightarrow M$  is smooth. And it is a  $n$ -dimensional vector bundle over  $M$ .

┘

*Proof:* Let  $(U, \varphi)$  be a smooth chart of  $M$ , with coordinate functions  $\varphi^1, \dots, \varphi^n$ , then we define a map  $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$  by

$$\tilde{\varphi}\left(\sum v^i \frac{\partial}{\partial x^i}\bigg|_p\right) = (x^1(p), \dots, x^n(p), v^1, \dots, v^n).$$

Then for two different open subset  $U, V$ , the transition map is

$$\tilde{\psi} \cdot \tilde{\varphi}^{-1}(x^1, \dots, x^n, v^1, \dots, v^n) = (\tilde{x}^1(x), \dots, \tilde{x}^n(x), \sum_j \frac{\partial \tilde{x}^1}{\partial x^j}(x)v^j, \dots, \sum_j \frac{\partial \tilde{x}^n}{\partial x^j}(x)v^j)$$

which is clearly smooth. So this defines a smooth vector bundle over  $M$ , called the **tangent bundle** of  $M$ .

□

**Def. (16.1.2.7)[Cotangent Bundle].** The **cotangent bundle**  $T^*M$  of a smooth manifold  $M$  is the dual of the tangent bundle  $TM$ .

┘

**Def. (16.1.2.8)[Parallelizable manifold].**  $M$  is called **parallelizable** iff  $TM$  is trivial.

┘

### Vector Fields

**Def. (16.1.2.9) [Smooth Vector Field].** A **smooth vector field** on  $M$  is a smooth global section of the tangent bundle  $TM \rightarrow M$  (16.1.2.6). The space of smooth vector fields on  $M$  is denoted by  $\mathfrak{X}(M)$ .  $\lrcorner$

**Prop. (16.1.2.10) [Check Smoothness].** Let  $X$  be a section of the vector bundle  $TM \rightarrow M$ , then  $X$  is a smooth vector field iff  $Xf \in C^\infty(M)$  for any  $f \in C^\infty(M)$ .  $\lrcorner$

*Proof:* If for any  $f \in C^\infty(M)$ ,  $Xf \in C^\infty(M)$ , let  $U$  be a trivializing nbhd of  $M$  with coordinate functions  $x^i$ , then near any point  $p \in U$ , we can use bump function to extend  $x^i$  to a smooth function on  $M$ . Then  $X(x^i) = X^i$  near  $p$ , thus the coordinates of  $X$  in this trivialization are all smooth, so  $X$  is smooth near  $p$ , thus smooth everywhere.

Conversely, for any  $f \in C^\infty(M)$ , in a trivializing nbhd  $U$  of  $M$ ,  $Xf(x) = (\sum X^i(x) \frac{\partial}{\partial x^i} |_x)(f) = \sum X^i(x) \frac{\partial f}{\partial x^i}(x)$  is smooth.  $\square$

**Def. (16.1.2.11) [Coordinate Vector Fields].** For any smooth chart  $(U, (x^i))$  of  $M$  and  $i \in [\dim M]_+$ , the map

$$p \mapsto \frac{\partial}{\partial x^i} |_p, \text{ where } p \in U$$

defines a smooth vector field on  $U$  by (16.1.2.10), called the  $i$ -th **coordinate vector field**, denoted by  $\frac{\partial}{\partial x^i}$ .  $\lrcorner$

**Def. (16.1.2.12) [Smooth Frames].** An ordered  $k$ -tuple of vector fields defined on a subset  $A \subset M$  is called **linearly independent** if  $(X_{1,p}, \dots, X_{k,p})$  is a linearly independent  $k$ -tuple in  $T_p M$ .

An ordered  $k$ -tuple of vector fields defined on an open subset  $U \subset M$  is called a **smooth local frame** if  $(X_{1,p}, \dots, X_{k,p})$  is linearly independent and spans  $T_p M$  for each  $p \in U$ . It is called a **smooth global frame** if  $U = M$ .

$M$  is called **smoothly parallelizable** if it admits a smooth global frame.  $\lrcorner$

**Prop. (16.1.2.13) [Vector Fields Simplification].** For every vector field  $X$  and every point  $X(p) \neq 0$ , there exists a coordinate nbhd  $(x_1, \dots, x_{n-1}, t)$  such that  $X = \frac{\partial}{\partial t}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Def. (16.1.2.14) [Pushforward of Vector Fields].** Let  $F : M \rightarrow N \in \text{Diff}$  is a diffeomorphism, then for any  $X \in \mathfrak{X}(M)$ , there exists a  $Y \in \mathfrak{X}(N)$  that  $dF_p(X_p) = Y_{F(p)}$ .  $\lrcorner$

*Proof:* We just define  $Y_q = dF_{F^{-1}(q)}(X_{F^{-1}(q)})$ , it suffices to show this a smooth vector field. But  $Y : N \rightarrow TN$  is the composition

$$N \xrightarrow{F^{-1}} M \xrightarrow{X} TM \xrightarrow{TF} TN,$$

so it is smooth.  $\square$

**Def. (16.1.2.15) [Lie Brackets].** The **Lie bracket** of two vector fields  $X, Y \in \mathfrak{X}(M)$  is defined to be the vector field  $[X, Y] \in \mathfrak{X}(M)$  s.t.  $[X, Y](f) = (XY - YX)f$  for any  $f \in C^\infty(M)$ .

If  $X = \sum a_i \frac{\partial}{\partial x_i}$ ,  $Y = \sum b_i \frac{\partial}{\partial x_i}$ , then  $[X, Y] = \sum (X(b_i) - Y(a_i)) \frac{\partial}{\partial x_i}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (16.1.2.16)[Jacobi Identity].**  $\mathfrak{X}(M)$  is a Lie algebra w.r.t. Lie Bracket defined in (16.1.2.15).  $\lrcorner$

*Proof:* Direct verification by the definition (16.1.2.15).  $\square$

**Prop. (16.1.2.17).**  $[fu, v] = f[u, v] - df(u)v$ .  $\lrcorner$

*Proof:* Direct from the definition (16.1.2.15).  $\square$

**Prop. (16.1.2.18)[Pushforward of Lie Bracket].** Let  $F : M \rightarrow N$  be a diffeomorphism and  $X_1, X_2 \in \mathfrak{X}(M)$ , then  $F_*[X_1, X_2] = [F_*X_1, F_*X_2]$  (16.1.2.14).  $\lrcorner$

*Proof:* For any  $f \in C^\infty(N)$ ,  $F_*X_i = Y_i$ , then

$$[X_1, X_2](f \circ F) = X_1X_2(f \circ F) - X_2X_1(f \circ F) = X_1(Y_2(f) \circ F) - X_2(Y_1(f) \circ F) = (Y_1Y_2(f) - Y_2Y_1(f)) \circ F,$$

which means exactly  $F_*[X_1, X_2] = [Y_1, Y_2]$ .  $\square$

### Tensor Fields

**Def. (16.1.2.19)[Lie Derivatives of Tensor Fields].** Cf. [Lee13]P321.  $\lrcorner$

## 3 Differential Forms

**Prop. (16.1.3.1)[Frobenius Theorem].** If  $X$  is an involutive distribution on  $M$ , then there is a unique maximal integration manifold passing through it. Where a distribution is involutive if it is closed under Lie bracket.  $\lrcorner$

*Proof:* The key to the proof is to prove that involutive is equivalent to integrable, i.e. flat locally as  $\{\frac{\partial}{\partial x_i}\}$  for some local coordinate. Cf. [李群讲义 项武义 P226]  $\square$

**Cor. (16.1.3.2).**  $X, Y \in \text{Lie}(M)$  commute iff their corresponding vector fields commute.  $\lrcorner$

### Interior and Exterior Derivatives

#### Lie Derivatives

**Prop. (16.1.3.3).**  $[X, Y] = \frac{\partial}{\partial t}(d(\varphi_{-t})Y)|_{t=0}$ , where  $\varphi$  is the flow of  $X$ .  $\lrcorner$

*Proof:* For any function  $f$ , set  $g(t, q) = \frac{f(\phi_t(q)) - f(q)}{t}$ ,  $g(0, q) = Xf(q)$ . Then  $g$  is differentiable (because  $g(t, q) = \int_0^1 Xf(\phi_{ts}(p))ds$ , and:

$$\begin{aligned} \lim_{t \rightarrow 0} d(\phi_{-t})Yf(p) &= \lim_{t \rightarrow 0} \frac{Yf(p) - Y(f\phi_{-t})(\phi_t(p))}{t} \\ &= \lim_{t \rightarrow 0} \frac{Yf(p) - Yf(\phi_t p) - Y(tg(-t, \phi_t(p)))}{t} \\ &= ((XY - YX)f)(p) \\ &= [X, Y]f(p) \end{aligned}$$

$\square$

**Prop. (16.1.3.4) [Derivative formula].**

$$d\omega(X, Y) = X\omega(Y) - Y\omega(X) - \omega([X, Y]).$$

┘

*Proof:*

□

**Prop. (16.1.3.5) [Cartan's magic formula].**

$$L_X\omega = \iota_X(d\omega) + d(\iota_X\omega)$$

$$\iota([X, Y]) = [L_X, \iota_Y]$$

┘

*Proof:* Notice that four of them are derivatives (check because  $\iota_X(w \wedge v) = \iota_X w \wedge v + (-1)^{|w|} w \wedge \iota_X v$ ). So by induction, we only has to verify them on dimension 0 and 1. □

**Prop. (16.1.3.6) [Stoke's theorem].**

$$\oint_{\Omega} d\omega = \oint_{\partial\Omega} i^* \omega.$$

In a 3-dimensional Riemannian manifold, If we set:

$$df = \omega^1_{\text{grad} f}, \quad d\omega^1_A = \omega^2_{\text{curl} A}, \quad d\omega^2_A = (\nabla A)\omega^3,$$

Then:

$$f(y) - f(x) = \int_l \text{grad} f \cdot dl.$$

$$\int_l A \cdot dl = \oint_S \text{curl} A \cdot dn.$$

$$\oint_U \nabla \cdot F dV = \oint_{\partial U} F \cdot ndS.$$

┘

*Proof:*

□

### Hodge Star

**Def. (16.1.3.7) [Hodge Star Operator].** given a volume-form  $\omega$  on a vector space, the Hodge star operator  $*$  is an operator from  $\bigwedge^k V \rightarrow \bigwedge^{n-k} V$  such that:

$$\alpha \wedge (*\beta) = \langle \alpha, \beta \rangle \omega.$$

On a closed oriented Riemannian manifold, given a volume form  $\omega$ , the star operator satisfies:

$$(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle \omega = \int_M \alpha \wedge *\beta.$$

And  $** = (-1)^{p(n-p)}$  on  $\Omega^p M$ . ┘

**Def. (16.1.3.8).** For a operator  $d$  on  $\Omega^* M$ , we define the adjoint  $d^* = (-1)^{n(p+1)+1} * d *$  on  $\Omega^p$ , which satisfies the adjoint property by calculation:

$$(d^* \alpha, \beta) = (\alpha, d\beta).$$

The laplacian  $\Delta = d^* d + d d^*$ . It can be verified that  $\Delta$  commutes with  $*$  and  $d$ . ┘

### Orientations

**Def. (16.1.3.9) [Continuous Orientations].** A **continuous orientation** on  $M$  is a choice of orientation on  $T_p M$  for each  $p \in M$  s.t. for each  $p \in M$ , there exists a nbhd  $U \subset M$  and a local frame  $(E^1, \dots, E^n)$  on  $U$  s.t. for each  $p \in U$ , the orientation on  $T_p M$  is induced from the ordered tuple  $(E_{1,p}, \dots, E_{n,p})$ .

$M$  is called an **orientable smooth manifold** if it admits a continuous orientation, and called a **non-orientable smooth manifold** if not. A **oriented smooth manifold** is an orientable smooth manifold together with an orientation on it.  $\lrcorner$

**Prop. (16.1.3.10) [Orientation Forms].** For  $M \in \text{Diff}_\partial^n$ , a continuous orientation on  $M$  determines and is determined by a non-vanishing  $n$ -form on  $M$ .

In particular, any parallelizable smooth manifold is orientable.  $\lrcorner$

*Proof:* Cf. [Lee13]P381.  $\square$

**Cor. (16.1.3.11).** If  $M$  is connected and oriented, then it has exactly two orientations. If two orientations on  $M$  agree on one point, then they are equal.  $\lrcorner$

**Def. (16.1.3.12) [Orientation-Preserving Maps].** A local diffeomorphism (16.1.1.11)  $f : M \rightarrow N \in \text{Diff}_\partial$  is called **orientation preserving** if for any  $p \in M$ ,  $df_p$  takes oriented bases of  $T_p M$  to oriented bases of  $T_{f(p)} N$ . And it is called **orientation-reversing** if for any  $p \in M$ ,  $df_p$  takes oriented bases of  $T_p M$  to negatively-oriented bases of  $T_{f(p)} N$ .  $\lrcorner$

**Prop. (16.1.3.13) [Orientation of Hypersurfaces].** Suppose  $M \in \text{Diff}_\partial^n$  is oriented,  $S \subset M \in \text{Diff}_\partial$  is an immersed hypersurface, and  $n$  is a vector field along  $S$  that is nowhere tangent to  $S$ , then there is a unique orientation on  $S$  s.t. for each  $p \in S$ ,  $(E^1, \dots, E^{n-1})$  is an oriented basis for  $T_p S$  iff  $(n, E^1, \dots, E^{n-1})$  is an oriented basis for  $T_p M$ . And if  $\omega$  is an orientation form for  $M$ , then  $\iota_S^*(N \lrcorner \omega)$  is an orientation form for the induced orientation on  $S$ .  $\lrcorner$

*Proof:* Cf. [Lee13]P385.  $\square$

**Cor. (16.1.3.14) [Boundary Orientations].** Suppose  $\partial M \neq \emptyset$ , then  $\partial M \subset M$  is an embedded hypersurface by (16.1.1.25), thus orientable by (16.1.3.13), with the induced orientation defined by any outward-pointing vector fields along  $\partial M$ , called the **induced orientation** on  $\partial M$ .  $\lrcorner$

*Proof:* Cf. [Lee13]P386.  $\square$

**Prop. (16.1.3.15).** If  $M \in \text{Diff}$  is oriented, then as oriented manifolds, we have

$$\partial(I \times M) = M \times \{1\} - M \times \{0\}.$$

$\lrcorner$

## 4 Differential Topology

References are [Mil97].

### Intersection Numbers Modulo 2

**Def. (16.1.4.1) [Intersection Numbers Modulo 2].** Suppose  $X, Y \in \mathcal{D}\text{iff}$ ,  $X$  compact and  $Z \subset Y$  a closed submanifold s.t.  $\dim X + \dim Z = \dim Y$ . For any  $f : X \rightarrow Y \in \mathcal{D}\text{iff}$  s.t.  $f \bar{\cap} Z$ , define  $I_2(f, Z)$  as the number of points of  $f^{-1}(Z)$  modulo 2, called the **intersection number modulo 2** of  $f$  and  $Z$ .  $\lrcorner$

**Lemma (16.1.4.2) [Boundary Theorem].** Situation as in (16.1.4.10), if  $X = \partial W$  for some  $W \in \mathcal{D}\text{iff}_\partial$  compact and  $f : X \rightarrow Y \in \mathcal{D}\text{iff}$  extends to some  $g : W \rightarrow Y \in \mathcal{D}\text{iff}_\partial$ , then  $I_2(f, Z) = 0$ .  $\lrcorner$

*Proof:* Use extension theorem (16.1.1.55), (16.1.1.51) and (16.1.12.3).  $\square$

**Prop. (16.1.4.3) [Homotopy Invariance].** Situation as in (16.1.4.10), for any  $f, f' : X \rightarrow Y$  s.t.  $f \bar{\cap} Z, f' \bar{\cap} Z$ , if  $f$  is smoothly homotopic to  $f'$ , then  $I_2(f, Z) = I_2(f', Z)$ .  $\lrcorner$

*Proof:* Immediate from boundary theorem (16.1.4.2).  $\square$

**Def. (16.1.4.4) [General Intersection Numbers Modulo 2].** Suppose  $X, Y \in \mathcal{D}\text{iff}$ ,  $X$  compact and  $Z \subset Y$  a closed submanifold s.t.  $\dim X + \dim Z = \dim Y$ , then for any  $f : X \rightarrow Y \in \mathcal{D}\text{iff}$ , we can choose  $f' : X \rightarrow Y \in \mathcal{D}\text{iff}$  smoothly homotopic to  $f$  s.t.  $f' \bar{\cap} Z$  by (16.1.1.54), and define  $I_2(f, Z) = I_2(f', Z)$ . This definition is well-defined and is compatible with the definition in (16.1.4.10) by (16.1.4.3).  $\lrcorner$

**Def. (16.1.4.5) [Intersection Number modulo 2 of Two Maps].** Suppose  $X, Z \in \mathcal{D}\text{iff}_{\text{cpct}}, Y \in \mathcal{D}\text{iff}$ ,  $f : X \rightarrow Y, g : Z \rightarrow Y$  s.t.  $\dim X + \dim Z = \dim Y$ , then we can choose  $f' : X \rightarrow Y$  s.t.  $f' \sim_{\text{sm}} f$  and  $g' : Z \rightarrow Y$  s.t.  $g \sim_{\text{sm}} g'$  ?. Then  $(f')^{-1}(g'(Z))$  is a finite set of points, so we can define the **intersection number modulo 2 of two maps**  $I_2(f, g) \in \mathbb{Z}/(2)$  to be the parity of  $\#(f')^{-1}(g'(Z))$ , called the **intersection number modulo 2 of two maps**. Then:

- This is well-defined.
- $f, g$  are transversal iff  $f \times g : X \times Z \rightarrow Y \times Y$  are transversal to  $\Delta_Y$ .
- $I_2(f, g) = (-1)^{\dim Z} I_2(f \times g, \Delta_Y)$ .
- If  $g : Z \rightarrow Y$  is an immersed submanifold, then  $I_2(f, g) = I_2(f, g(Z))$ .

$\lrcorner$

*Proof:* This a simple local tangent vector calculation, Cf. [Pollack]P113, 115.  $\square$

**Prop. (16.1.4.6) [Properties of Intersection Numbers of Two Maps].** Situation as in (16.1.4.5),

- If  $f' \sim_{\text{sm}} f, g' \sim_{\text{sm}} g$ , then  $I_2(f, g) = I_2(f', g')$ , because  $f \times g \sim_{\text{sm}} f' \times g'$ .

$\lrcorner$

**Prop. (16.1.4.7) [Mod 2 Brouwer Degree of Maps].** If  $f : X, Y \in \mathcal{D}\text{iff}$  s.t.  $\dim X = \dim Y$ ,  $X$  is compact and  $Y$  is connected, then  $I_2(f, \{y\})$  (16.1.4.4) is the same for each  $y \in Y$ , called the **mod 2 degree of  $f$** . If  $f$  is smoothly homotopic to  $f'$ , then  $I(f, Z) = I(f', Z)$ , and  $I(f, Z) = 0$  if  $f$  is a boundary map, by (16.1.4.3)(16.1.4.2).  $\lrcorner$

*Proof:* Cf. [Pollack P80]. ?  $\square$

### Orientable Intersection Numbers

**Prop. (16.1.4.8) [Transversal Preimage Orientation].** Let  $Y \in \text{Mani}$  be oriented and  $Z \subset Y$  an oriented closed submanifold. If  $X \in \text{Diff}_\partial$  is oriented and  $f : X \rightarrow Y \in \text{Diff}_\partial$  satisfies  $f \pitchfork Z, \partial f \pitchfork Z$ , then the orientation of  $Z, Y, Z$  defines canonically an orientation on  $S = f^{-1}(Z)$  s.t. for any  $x \in f^{-1}(Z)$  and any subspace  $H \subset T_x(X)$  complementary to  $T_x(S)$ , the orientation on  $S$  is defined by the following decompositions:

$$H \oplus T_x(S) = T_x(X), \quad df_x(H) \oplus T_{f(x)}(Z) = T_{f(x)}(Y).$$

called the **preimage orientation** of  $f^{-1}(Z)$ . ┘

*Proof:* Cf.[Pollack]P100. ? □

**Prop. (16.1.4.9).** Situation as in (16.1.4.8), as oriented manifolds,

$$\partial[f^{-1}(Z)] = (-1)^{\text{codim } Z} (\partial f)^{-1}(Z).$$

*Proof:* Cf.[Pollack]P101. □

**Def. (16.1.4.10) [Intersection Numbers].** Suppose  $X, Y \in \text{Diff}$ ,  $X$  compact and  $Z \subset Y$  a closed submanifold s.t.  $\dim X + \dim Z = \dim Y$ . For any  $f : X \rightarrow Y \in \text{Diff}$  s.t.  $f \pitchfork Z, f^{-1}(Z)$  is a finite set of points with orientations give in (16.1.4.8). So we can define  $I(f, Z)$  to be the sum of the orientations or  $f^{-1}(Z)$ , called the **intersection number** of  $f$  and  $Z$ . ┘

**Lemma (16.1.4.11) [Boundary Theorem].** If  $W \in \text{Diff}_\partial$ ,  $Y \in \text{Diff}$  is oriented and  $Z \subset Y$  is an oriented closed subscheme s.t.  $\dim \partial W + \dim Z = \dim Y$ . If  $f : W \rightarrow Y$  is a map of smooth manifolds s.t.  $\partial f \pitchfork Z$ , then  $I(\partial f, Z) = 0$ . ┘

*Proof:* The same as the proof of (16.1.4.2). □

**Prop. (16.1.4.12) [Homotopy Invariance].** Homotopic transversal maps always have the same intersection number w.r.t  $Z$ . ┘

**Def. (16.1.4.13) [General Intersection Numbers].** Suppose  $X, Y \in \text{Diff}$ ,  $X$  compact and  $Z \subset Y$  a closed submanifold s.t.  $\dim X + \dim Z = \dim Y$ , then for any  $f : X \rightarrow Y \in \text{Diff}$ , we can choose  $f' : X \rightarrow Y \in \text{Diff}$  smoothly homotopic to  $f$  s.t.  $f' \pitchfork Z$  by (16.1.1.54), and define  $I(f, Z) = I(f', Z)$ . This definition is well-defined and is compatible with the definition in (16.1.4.10) by (16.1.4.12). ┘

**Prop. (16.1.4.14) [Intersection Numbers of Two Maps].** Suppose  $X, Y, Z \in \text{Diff}_{\text{or mtd}}$  s.t.  $\dim X + \dim Z = \dim Y$  and  $X, Z$  are compact. Let  $f : X \rightarrow Y, g : Z \rightarrow Y$  be transversal maps. Then we can choose  $f' : X \rightarrow Y$  s.t.  $f' \sim_{\text{sm}} f$  and  $g' : Z \rightarrow Y$  s.t.  $g \sim_{\text{sm}} g'$  ?. Then  $f^{-1}(g(Z))$  is a finite set of points with orientations given in (16.1.4.8). So we can define the **intersection number**  $I(f, g)$  to be the sum of the orientations or  $f^{-1}(Z)$ . Then:

- This is well-defined.
- $f, g$  are transversal iff  $f \times g : X \times Z \rightarrow Y \times Y$  are transversal to  $\Delta_Y$ .
- $I(f, g) = (-1)^{\dim Z} I(f \times g, \Delta_Y)$ .
- If  $g : Z \rightarrow Y$  is an immersed submanifold, then  $I(f, g) = I(f, g(Z))$ .

┘

*Proof:* This is a simple local tangent vector calculation, Cf.[Pollack]P113, 115. □

**Prop. (16.1.4.15)[Properties of Intersection Numbers of Two Maps].** Situation as in (16.1.4.14),

- If  $f' \sim_{\text{sm}} f, g' \sim_{\text{sm}} g$ , then  $I(f, g) = I(f', g')$ , because  $f \times g \sim_{\text{sm}} f' \times g'$ .
- $I(f, g) = (-1)^{\dim X \cdot \dim Z} I(g, f)$ , by definition (16.1.4.8).

┘

*Proof:* ? □

**Cor. (16.1.4.16)[Self-Intersections].** If  $X \subset Y \in \text{Diff}$  and  $X$  is compact with  $\dim X \in 2\mathbb{N} + 1$  and  $\dim Y = 2 \dim X$ , then  $I(X, X) = 0$ . If this fails, then  $Y$  is not orientable, for example the Möbius band with the central circle. ┘

### Degrees and Indices of Vector Fields

**Prop. (16.1.4.17)[Brouwer Degrees].** If  $X, Y \in \text{Diff}_{\text{orntd}}$  s.t.  $\dim X = \dim Y$  and  $X$  is compact, and  $f : X \rightarrow Y \in \text{Diff}$ , then  $I(f, \{y\})$  (16.1.4.13) is the same for each  $y \in Y$  by (16.1.4.15), called the **Brouwer degree** of  $f$ . If  $f$  is smoothly homotopic to  $f'$ , then  $I(f, Z) = I(f', Z)$ , and  $I(f, Z) = 0$  if  $f$  is a boundary map, by (16.1.4.11)(16.1.4.12). ┘

*Proof:* The proof is the same as that of (16.1.4.7). □

**Prop. (16.1.4.18)[Compositions].** If  $X, Y, Z \in \text{Diff}_{\text{orntd}}$  s.t.  $\dim X = \dim Y = \dim Z$  and  $f : X \rightarrow Y, g : Y \rightarrow Z \in \text{Diff}$ , then  $\deg(g \circ f) = \deg(f) \deg(g)$ . ┘

**Def. (16.1.4.19)[Indices].** For  $V \in \mathfrak{X}(M)$  and an isolated zero  $x \in M$  of  $V$ , choose a local chart  $\varphi$  of  $U \subset M$  centered at  $x$  containing no other zeros of  $V$ , and choose a chart sphere  $S_\varepsilon^{\dim M - 1}$  in this chart. By identifying the tangent vectors of points in  $U$  with those in  $\varphi(U)$ , we have a map

$$S_\varepsilon^{\dim M - 1} \rightarrow \mathbb{S}^{\dim M - 1} : z \mapsto v(z) / \|v(z)\|.$$

Then we can define  $\text{ind}_x(V)$  to be the degree of this map.

Notice that this is independent of this  $\varepsilon$  chosen, because it is continuous in  $\varepsilon$  and the degree is an integer. Also it is independent of the local charts chosen, because a local transition map between charts maps a sufficiently small chart sphere to a topological sphere homotopic to a chart sphere preserving orientation. So the degrees are the same because it is preserved under homotopy (16.1.4.17). ┘

**Prop. (16.1.4.20)[Non-degenerate Vanishing Points].** Situation as in (16.1.4.19), for each vanishing point  $x$  of  $V$ , consider  $V$  as a map  $M \rightarrow TM$ , then at a vanishing point  $x \in X$  of  $V$ , the image of  $dV_x : T_x M \rightarrow T_x(TM)$  is contained in  $TM$ , because otherwise  $x$  is not an isolated zero (by looking locally). Then  $x$  is called a **non-generate zero** if  $dV_x : T_x M \rightarrow T_x M$  is non-degenerate.

For a non-degenerate zero  $x \in M$  of  $V$ , by looking at the definition (16.1.4.19), we see  $\text{ind}_x(V) = 1$  if  $dV_x$  preserves orientation and  $\text{ind}_x(V) = -1$  if  $dV_x$  reverses orientation. ┘

**Prop. (16.1.4.21)[Vector Fields with Isolated Zeros].** If  $M$  is compact, then there exists a vector field  $M$  with only isolated zeros. And a vector bundle over a  $k$  dimensional manifold splits to components of dimension no bigger than  $k$ , determined by its Chern class. ┘



*Proof:* This follows from (4.15.2.21).  $\square$

**Thm. (16.1.4.22) [Poincare-Hopf Index theorem].** For  $M \in \text{Diff}_{\text{cpct}}$  and  $V \in \mathfrak{X}(M)$  with only f.m. zeros, the sum of indices of  $V$  (16.1.4.19) equals  $I(\Delta_M, \Delta_M)$ .  $\lrcorner$

*Proof:* We can smoothly perturb  $V$  s.t. it has only non-degenerate vanishing points (16.1.4.20) as follows: It suffices to prove for a small nbhd of a vanishing point  $x$ , so we may assume  $x \in U \subset \mathbb{R}^n$  for some  $n \in \mathbb{Z}_+$ . Choose a bump function supported on  $U$  and has value 1 on a nbhd of  $x$ , and define

$$V_a(z) = V(z) - \rho(z)\mathbf{a}, \text{ where } z \in U.$$

Because  $x$  is an isolated zero, when  $\|\mathbf{a}\|$  is small,  $V_a(z)$  can have zero only when  $\rho(z) = 1$ . Thus for  $\mathbf{a}$  sufficiently small. Thus if we choose  $\mathbf{a}$  small s.t.  $\mathbf{a}$  is a regular value of  $V$ , then  $V_a$  has only non-degenerate zeros.

By (16.1.1.46), there is an oriented diffeomorphism of a nbhd of  $M \subset TM$  with  $\Delta_M \subset M \times M$  identifying  $M$  with  $\Delta_M$ . Thus it is clear from the definition that

$$I_{M \times M}(\Delta_M, \Delta_M) = I_{TM}(M, V(M)),$$

and the latter is a sum of local orientations at the vanishing points of  $V$ . By homotopy invariance, we may assume  $V$  has only non-degenerate zeros, then it is easy to see  $M \overline{\cap} V(M)$ , and the local intersection numbers (16.1.4.10) are just indices by (16.1.4.20). Thus  $I_{M \times M}(\Delta_M, \Delta_M)$  equals the sum of indices of  $V$ .  $\square$

**Prop. (16.1.4.23) [Euler Characteristics and Self-Intersection of Diagonals].** The Euler character of an orientable compact manifold  $Y$  equals the intersection of the diagonals  $I(\Delta, \Delta)$ .  $\lrcorner$

*Proof:* For this, we use the Poincare-Hopf theorem (16.1.4.22). It is clear that on a triangulation, we can place a source on the center of each face/edge/..., thus producing a smooth vector fields, thus it is clear the sum of their indices equals both the combinatorial Euler character and the defined character.

Alternatively, we can use Morse theory to construct a non-degenerate vector field?  $\square$

**Cor. (16.1.4.24).** The Euler character of an odd dimensional compact smooth manifold  $Y$  is 0, by (16.1.4.16).  $\lrcorner$

### Lefschetz Fixed Point Theorems

**Def. (16.1.4.25) [Lefschetz Numbers].** For  $X \in \text{Diff}_{\text{cpct, orntd}}$  and  $f : X \rightarrow X \in \text{Diff}$ , the (global) smooth Lefschetz number of  $f$  is defined to be

$$L(f) \triangleq I(\text{Graph}(f), \Delta_X).$$

$L(f)$  is invariant under homotopy of  $f$  by (16.1.4.14).  $\lrcorner$

**Thm. (16.1.4.26) [Lefschetz Fixed Point Theorem].** Situation as in (16.1.4.25), if  $L(f) \neq 0$ , then  $f$  has a fixed point on  $X$ .  $\lrcorner$

*Proof:* If  $f$  has no fixed point on  $X$ , then  $I(\text{Graph}(f), \Delta_X) = 0$ .

There are topological proofs.?  $\square$

**Cor. (16.1.4.27) [Euler Characteristics and Lefschetz Numbers].** Situation as in (16.1.4.25), if  $f$  is smoothly homotopic to  $\text{id}_X$ , then  $L(f) = \chi(X)$ . In particular, if  $f$  has no fixed points on  $X$ , then  $\chi(X) = 0$ .  $\lrcorner$

*Proof:*  $L(f) = L(\text{id}_X) = I(\Delta_X, \Delta_X) = \chi(X)$  (16.1.4.23).  $\square$

**Remark (16.1.4.28).** This is useful for calculating Euler characteristics.  $\lrcorner$

**Prop. (16.1.4.29) [Lefschetz Maps].** For  $X \in \text{Diff}_{\text{cpct, orntd}}$  and  $f : X \rightarrow X \in \text{Diff}$ , there exists  $f' : X \rightarrow X$  s.t.  $f \sim_{\text{sm}} f'$  and  $\text{Graph}(f) \cap \Delta_X$ . Such an  $f'$  is called a **Lefschetz map**.  $\lrcorner$

*Proof:* Cf. [Pollack]P120. ?  $\square$

**Def. (16.1.4.30) [Local Lefschetz Numbers].**  $\lrcorner$

**Prop. (16.1.4.31) [Local and Global Lefschetz Numbers].** Situation as in (16.1.4.25), if  $f$  has only f.m. fixed points, then

$$L(f) = \sum_{x \in X, f(x)=x} L_x(f) \text{ (16.1.4.30).}$$

$\lrcorner$

*Proof:* Cf. [Pollack]P130. ?  $\square$

**Prop. (16.1.4.32).** The only finite group  $G \in \text{Grp}^{\text{fin}}$  that can act freely on  $S^{2n}$  is  $\mathbb{Z}/(2)$  or 1.  $\lrcorner$

*Proof:* Consider the degree map, then it is a homomorphism from  $G$  to  $\mathbb{Z}$ , thus the image is just  $\pm 1$ . Now it is by Lefschetz fixed point theorem that  $\deg(g) = -1$  for  $g \neq 1$ , thus it is injective to  $\pm$ .  $\square$

### Framed Cobordism

**Def. (16.1.4.33) [Smooth Cobordisms].** For  $M \in \text{Diff}$ , two smooth submanifolds  $N, N' \subset M \in \text{Diff}$  is called **smoothly cobordant** if the subset

$$N \times [0, \frac{1}{4}) \cup N' \times (\frac{3}{4}, 1]$$

can be extended to a compact smooth manifold

$$X \subset M \times [0, 1]$$

s.t.  $\partial X = N \times \{0\} \cup N' \times \{1\}$ .  $\lrcorner$

**Def. (16.1.4.34) [Framings].** For  $N \subset M \in \text{Diff}$ , a **framing** of  $N$  in  $M$  is a smooth function that assigns to each  $x \in N$  a basis of  $N_M(N)_x$ . A submanifold of  $M$  with a framing is called a **framed manifold**.  $\lrcorner$

**Def. (16.1.4.35) [Framed Cobordisms].** For  $M \in \text{Diff}$ , two smooth embedded submanifolds  $N, N' \subset M \in \text{Diff}$  is called **smoothly framed cobordant** if the subset

$$N \times [0, \frac{1}{4}) \cup N' \times (\frac{3}{4}, 1]$$

can be extended to a compact smooth manifold

$$X \subset M \times [0, 1]$$

s.t.  $\partial X = N \times \{0\} \cup N' \times \{1\}$ , and there exists a framing on  $X$  that restricts to the framing on  $N$  and  $N'$  resp..  $\lrcorner$

**Prop. (16.1.4.36) [Pontryagin Framed Manifolds].** For  $f : M \in \mathbb{S}^n \in \text{Diff}_{\text{cpct}}$ , take a regular value  $y \in \mathbb{S}^n$  for  $f$  (16.1.1.40), then  $S = f^{-1}(y)$  is a smooth manifold by (16.1.1.37), and for each  $x \in S$ ,  $N_M(S)_x \cong T_x M / T_x S$  is naturally isomorphic to  $T_y(\mathbb{S}^n)$ , so any basis of  $T_y(\mathbb{S}^n)$  gives a framing on  $f^{-1}(y)$ . Then:

- For a different regular value  $y' \in \mathbb{S}^n$ , the framed manifolds  $f^{-1}(y)$  and  $f^{-1}(y')$  are smoothly framed cobordant. Any of these framed manifolds is called a **Pontryagin framed manifold** associated to  $f$ .
- For  $f, g : M \rightarrow \mathbb{S}^n \in \text{Diff}_{\text{cpct}}$ ,  $f$  and  $g$  are smoothly homotopic iff their associated Pontryagin framed manifolds (16.1.4.36) are framed cobordant.
- Any compact framed submanifold  $N$  of codimension  $r$  in  $M$  occurs as a Pontryagin manifold for some smooth  $f : M \rightarrow \mathbb{S}^r$ .

In particular, the set of smooth homotopy classes of maps from  $M$  to  $\mathbb{S}^n$  are in bijection with the set of framed cobordism classes of framed submanifolds of  $M$  of codimension  $r$ .  $\lrcorner$

*Proof:* Firstly it is clear that different bases of  $T_y(\mathbb{S}^n)$  give the same cobordant classes, because  $\text{GL}(n; \mathbb{R})^+$  is connected.

If  $f, g$  are smoothly homotopic by a homotopy  $F : M \times [0, 1] \rightarrow \mathbb{S}^n$ , then their Pontryagin framed manifolds  $f^{-1}(y)$  and  $g^{-1}(y)$  are cobordant by choosing a common regular value  $y'$  for both  $f$  and  $g$  that is close enough to  $y$  by Sard's theorem (16.1.1.40) and (16.1.4.37).

Now for any rotation  $\tau$  of  $\mathbb{S}^n$ ,  $\tau \circ f$  is homotopic to  $\tau$ , so  $f^{-1}(y)$  is framed cobordant to  $f^{-1}(z)$  for any regular values  $y, z \in \mathbb{S}^n$  for  $f$ .

For any compact framed submanifold  $N$  of codimension  $r$  in  $M$ , we may embed  $M$  into  $\mathbb{R}^n$ , and take the tubular nbhd of  $U$  in  $M$ . Then by a smooth  $\text{GL}(n)$ -action, we can assume that the map

$$V \subset N \times \mathbb{R}^r \rightarrow U \subset M \rightarrow N_M(N)$$

maps the canonical basis of  $\mathbb{R}^r$  to the chosen frame. Then we can take the smooth map  $M \rightarrow \mathbb{S}^r$  by compactifying the map  $U \rightarrow N \times \mathbb{R}^r \rightarrow \mathbb{R}^r$  and map  $M \setminus U$  to the north pole. Then the Pontryagin framed manifold of this map is just the framed manifold  $N$ .

Finally, if the associated Pontryagin framed manifolds for  $f$  and  $g$  are framed cobordant, then there is a framed cobordism  $X \subset M \times [0, 1]$  between  $f^{-1}(y)$  and  $g^{-1}(y)$ , then the tubular nbhd argument as above shows that there is a homotopy  $F : M \times [0, 1] \rightarrow \mathbb{S}^n$  whose Pontryagin manifold  $F^{-1}(y)$  is exactly the framed manifold  $X \subset M \times [0, 1]$ . Then because  $F_0^{-1}(y) = f^{-1}(y)$  and  $F_1^{-1}(y) = g^{-1}(y)$ , we have  $F_0 \sim_{\text{sm}} f, F_1 \sim_{\text{sm}} g$  by (16.1.4.38), and thus  $f \sim_{\text{sm}} g$ .  $\square$

**Lemma (16.1.4.37).** For  $f : M \rightarrow \mathbb{S}^n \subset \text{Diff}_{\text{cpct}}$ , if  $y \in \mathbb{S}^n$  is a regular value for  $f$ , then for every regular value  $z$  for  $f$  in a nbhd of  $y$ ,  $f^{-1}(z)$  and  $f^{-1}(y)$  are framed cobordant.  $\lrcorner$

*Proof:* Because  $M$  is compact, the set of regular values of  $\mathbb{S}^n$  for  $f$  is compact, so there exists a nbhd  $U$  of  $y$  consisting of regular values. Then for any  $z \in U$ , we can take a one-parameter family of rotations  $r_t$  of  $\mathbb{S}^n$  taking  $y$  to  $z$ , and let

$$F : M \times [0, 1] \rightarrow \mathbb{S}^n : (x, t) \mapsto r_t f(x),$$

then  $F^{-1}(y)$  is a framed cobordism between  $f^{-1}(y)$  and  $f^{-1}(z)$ .  $\square$

**Lemma (16.1.4.38).** For  $f, g : M \rightarrow \mathbb{S}^n \subset \text{Diff}_{\text{cpct}}$ , if  $y \in \mathbb{S}^n$  is a common regular value for  $f$  and  $Y$ , and  $f^{-1}(y)$  is equal to  $g^{-1}(y)$  as framed manifolds, then  $f \sim_{\text{sm}} g$ .  $\lrcorner$

*Proof:* Denote  $N = f^{-1}(y)$ . Firstly, if  $f$  coincides with  $g$  on a nbhd of  $N$ , then we may linearly deform  $f$  to  $g$  in  $\mathbb{S}^n \setminus \{y\}$ . Take a tubular nbhd  $V$  of  $N$  in  $M$  with a product representation

$$N \times \mathbb{R}^p \cong V$$

s.t.  $V$  is small enough that neither  $f(V)$  or  $g(V)$  contains the antipode of  $y$ . Identifying  $\mathbb{S}^n \setminus \{y\}$  with  $\mathbb{R}^n$ , we get maps

$$F, G : N \times \mathbb{R}^n \rightarrow \mathbb{R}^n$$

s.t.  $F^{-1}(0) = G^{-1}(0) = N \times \{0\}$  and  $dF_{(x,0)} = dG_{(x,0)} = \text{pr}_2$  for any  $x \in N$ .

Then we can find  $c \in \mathbb{R}_+$  s.t.  $F(x, u) \cdot u > 0, G(x, u) \cdot u > 0$  for any  $u \in \mathbb{R}^n$  s.t.  $0 < \|u\| < c$ . So the line between  $F(x, u)$  and  $G(x, u)$  doesn't contain 0.

Now take a smooth function  $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}$  s.t.  $\lambda(u) = 1$  if  $\|u\| \leq c/2$  and  $\lambda(u) = 0$  for  $\|u\| \geq c$ , then the homotopy

$$F_t(x, u) = [1 - \lambda(u)t]F(x, u) + \lambda(u)tG(x, u)$$

deforms  $F = F_0$  into a mapping  $F_1$  that coincides with  $G$  in a nbhd of  $N$ , and  $F^{-1}(0) = N$ . Then  $F_1$  corresponds to a map  $f_1 : M \rightarrow \mathbb{S}^n$ , and we can use the initial method to deform  $f_1$  into  $g$ . Thus  $f \sim_{\text{sm}} g$ .  $\square$

**Cor. (16.1.4.39)[Smooth Hopf Theorem].** For  $M \in \text{Diff}_{\text{cpct, orntd}}^n$ , two maps  $f, g : M \rightarrow \mathbb{S}^n \in \text{Diff}_{\text{cpct}}^n$  are smooth homotopic iff they have the same Brouwer degree.

For  $M \in \text{Diff}_{\text{cpct, non-orntd}}^n$ , two maps  $f, g : M \rightarrow \mathbb{S}^n \in \text{Diff}_{\text{cpct}}^n$  are smooth homotopic iff they have the same mod 2 Brouwer degree(16.1.4.7).  $\lrcorner$

*Proof:* For the orient case, this is because it is clear that two framed 0-dimensional submanifolds of  $M$  are cobordant iff they have the same signature.

For the non-orient case, this is because any framed point of  $M$  is cobordant to its opposite framed point along some loop in  $M$ , because otherwise  $M$  is orientable.  $\square$

## 5 Flow

**Def. (16.1.5.1)[Integral Curves].** Let  $V$  be a vector field over a smooth manifold  $M$ , then an **integral curve** of  $V$  is a smooth curve  $\gamma : J \rightarrow M$  that  $\gamma'(t) = V_{\gamma(t)}$  for any  $t \in J$ .  $\lrcorner$

**Def. (16.1.5.2)[Flow].** Let  $M$  be a manifold, then a **flow** on  $M$  is a continuous map  $\theta : D \rightarrow M$ , where

- $D \in \mathbb{R} \times M$  is an open subset.
- for any  $p \in M, D^p = \{t | (t, p) \in D\}$  is an open interval containing 0.
- When it is defined,  $\theta(s, \theta(t, p)) = \theta(s + t, p)$ .

If  $\theta$  is a flow, we define  $\theta_t(p) = \theta^{(p)}(t) = \theta(t, p)$ .

If  $\theta$  is smooth, then we can define the **infinitesimal generator** of  $\theta$  to be the vector field  $V_p = \theta^{(p)'}(0)$ .  $\lrcorner$

**Def. (16.1.5.3)[Complete Vector Fields].** A complete vector field on a smooth manifold is a vector field that generates a global flow.  $\lrcorner$

**Prop. (16.1.5.4).** If  $\theta : D \rightarrow M$  is a smooth flow, then the infinitesimal generator  $V$  of  $\theta$  is a smooth vector field, and each  $\theta^{(p)}$  is an integral curve of  $V$ .  $\lrcorner$

*Proof:* Cf. [Lee13]P212.  $\square$

**Prop. (16.1.5.5) [Isotopy Extension Theorem].** Let  $A \subset M$  be a compact subset, then any isotopy  $F : A \times \mathbb{I} \rightarrow M$  can be extended to an diffeotopy of  $M$ .  $\lrcorner$

*Proof:* Consider  $F(A \times I) \subset M \times I$  is a compact set, and  $TM \times I \rightarrow M \times I$  is a vector bundle. The time lines generate a section  $F(A \times I) \rightarrow TM \times I$ , so ?? guarantees an extension  $M \times I \rightarrow TM \times I$ , and because manifolds are locally compact, this section can be chosen to be compactly supported, then the flow it generates is a diffeotopy.  $\square$

## 6 Distributions and Foliations

Cf. [Lee13] Chap19.

## 7 Smooth Isotopies

**Def. (16.1.7.1) [Smooth Isotopies].** For  $V, M \in \mathcal{D}\text{iff}$ , a **smooth isotopy** is an isotopy (4.13.2.1)  $F : V \times I \rightarrow M$  that is smooth.

If  $V = M$ ,  $F_0 = \text{id}_M$ , and each  $F_t$  is a diffeomorphism of  $M$ , then this is called a **diffeotopy**.  $\lrcorner$

**Prop. (16.1.7.2).** Any orientation preserving self-diffeomorphism  $f$  of  $\mathbb{R}^n$  is diffeotopic to  $\text{id}_{\mathbb{R}^n}$ .  $\lrcorner$

*Proof:* We may assume  $f(0) = 0$ . And we can define an isotopy

$$F : \mathbb{R}^n \times [0, 1] \rightarrow \mathbb{R}^n : (x, t) \mapsto \begin{cases} f(tx)/t & t \neq 0 \\ df_0(x) & , t = 0 \end{cases},$$

then clearly each  $F_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is a diffeomorphism, and  $F$  is smooth because by (12.3.4.1) we can write

$$f(x) = x_1 g_1(x) + \dots + x_n g_n(x), \text{ where } g_i(x) \in C^\infty(\mathbb{R}^n),$$

so  $f(tx)/t = x_1 g_1(tx) + \dots + x_n g_n(tx)$  is smooth.

So  $F$  is diffeotopic to a non-degenerate linear map, and then clearly diffeotopic to  $\text{id}_{\mathbb{R}^n}$ .  $\square$

**Thm. (16.1.7.3) [Isotopy Extension Theorem].** Let  $M \in \mathcal{D}\text{iff}_\partial$  and  $V \subset M$  a compact submanifold. If  $F : V \times I \rightarrow M$  is a smooth isotopy of  $V$  (16.1.7.1), then  $F$  extends to a diffeotopy of  $M$  with compact support.  $\lrcorner$

*Proof:* [Hirsch, Differential Topology] P180.  $\square$

**Cor. (16.1.7.4) [Smooth Homogeneity Lemma].** If  $M \in \mathcal{D}\text{iff}_{\text{cntd}, \partial}$  and  $x, y \in \text{Int}(M)$ , then there is a self-diffeomorphism of  $M$  that is diffeotopic to  $\text{id}_M$  and takes  $x$  to  $y$ .  $\lrcorner$

**Thm. (16.1.7.5) [Isotopy Extension Theorem].** let  $M \in \mathcal{D}\text{iff}$ ,  $U \subset M$  be an open subset and  $A \subset U$  a compact set. Let  $F : U \times I \rightarrow M$  be an isotopy s.t.  $\hat{F}(U \times I) \subset M \times I$  is open, then there is a diffeotopy of  $M$  with compact support, which agrees with  $F$  on a nbhd of  $A \times I$ .  $\lrcorner$

*Proof:* [Hirsch, Differential Topology] P180.  $\square$

## 8 DeRham Theory

References are [Bott-Tu].

## 9 Čech-DeRahm Complexes

References are [Bott-Tu].

## 10 Spectral Sequence

References are [Bott-Tu].

## 11 Characteristic Classes

References are [Bott-Tu].

In this subsection, we study characteristic classes of manifolds, especially of tangent bundles.

**Prop.(16.1.11.1)[Antipodal Map of  $\mathbb{S}^n$ ].** The antipodal map of  $\mathbb{S}^n$  has degree  $(-1)^{n+1}$ , because it is a composition of  $(n+1)$  reflections. In particular, if  $n$  is even, then this antipodal map is not homotopic to  $\text{id}_{\mathbb{S}^n}$ .  $\lrcorner$

**Prop.(16.1.11.2)[Non-Vanishing Vector Fields].**

- If  $n \in 2\mathbb{Z}_+ + 1$ , then  $\mathbb{RP}^n$  and  $\mathbb{S}^n$  admit a non-vanishing vector field.
- If  $n \in 2\mathbb{Z}_+$ , then  $\mathbb{S}^n$  and  $\mathbb{RP}^n$  admit no non-vanishing vector field.

$\lrcorner$

*Proof:* If  $n = 2k + 1$ , consider the non-vanishing vector field on  $\mathbb{S}^{2k+1} : (x_1, \dots, x_{2k+2}) \mapsto (x_2, -x_1, x_4, -x_3, \dots, x_{2k+2}, -x_{2k+1})$ . Then This descends to a non-vanishing vector field on  $\mathbb{RP}^{2k+1}$ .

Conversely, if  $n = 2k$ , then a non-vanishing vector field defines an homotopy between the antipodal map to  $\text{id}_{\mathbb{S}^n}$ . But the former has degree  $-1$  by (16.1.11.1) and the latter has degree 1.  $\square$

## 12 Classifying Smooth Manifolds

**Prop.(16.1.12.1)[0-Dimensional Manifolds].** Any 0-dimensional connected manifold is smooth and diffeomorphic to pt.  $\lrcorner$

**Prop.(16.1.12.2)[1-Dimensional Manifolds].** Any 1-dimensional connected smooth manifold possibly with boundaries is diffeomorphic to one of the following:

$$\mathbb{S}^1, [0, 1], (0, 1], (0, 1).$$

$\lrcorner$

*Proof:* For a finite interval  $I \subset \mathbb{R}$ , we call a map  $f : I \rightarrow M$  a **parametrization** if it maps  $I$  diffeomorphically onto an open subset of  $M$ , and in any smooth chart  $U \subset M$ , the velocity vector of the map  $f : f^{-1}(U) \rightarrow U$  has length bounded below by some positive constant  $c \in \mathbb{R}_+$  depending on  $U$ .

Take any parametrization by arc length  $f : I \rightarrow M$  that is maximal in the sense it cannot be extended to a parametrization on a larger interval  $I'$  containing  $I$  (which exists because  $\mathbb{R}$  is well-ordered).

If  $f$  is onto, then  $M \cong I$ . If  $f$  is not onto, then there exists a limit point  $x \in M \setminus f(I)$  of  $f(I)$ . There exists another parametrization  $g : J = (-1, 1) \rightarrow M$  with  $g(0) = x$ .

Consider the graph  $\Gamma \subset I \times J$  consisting of all  $(s, t)$  s.t.  $f(s) = g(t)$ , then the graph is smooth and it is always from upper-left to lower-right or from lower-left to upper-right. So it is clear that it

has at most two components. (To be rigorous, we can normalize the domain of  $f$  or  $g$  s.t.  $f^{-1} \circ g$  has constant velocity).

it is clear that if it has two components, then we can glue  $f$  and  $g$  to get a diffeomorphism  $\mathbb{S}^1 \rightarrow M$ . And if it has only one component, then we can extend  $f$  using  $g$ , contradicting the hypothesis that  $f$  is maximal. So the assertion is proved.  $\square$

**Cor. (16.1.12.3).** The boundary of any smooth manifold of dimension 1 consists of even number of points.  $\lrcorner$

**Thm. (16.1.12.4) [2-Dimensional Manifolds].** Every compact oriented manifold is diffeomorphic to  $\Sigma_g$  for a unique  $g \in \mathbb{N}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (16.1.12.5) [3-Dimensional Manifolds].**  $\lrcorner$

## 13 Others

### Real Algebraic Geometry

**Prop. (16.1.13.1).** Any compact smooth manifolds in  $\mathbb{R}^n$  can be approximated by a real algebraic variety.  $\lrcorner$

*Proof:* Cf. [Nash's work on Algebraic Geometry].  $\square$

**Prop. (16.1.13.2).** Let  $Y$  be a projective variety over  $\mathbb{R}$  and  $Z \subset Y$  a closed subvariety, then there exists a triangulation of the pair  $(Y(\mathbb{R}), Z(\mathbb{R}))$ .  $\lrcorner$

*Proof:* Cf. [Hironaka1975, Triangulations of Algebraic Sets] and [Lojasiewicz1964, Triangulation of semi-analytic sets].  $\square$

## 16.2 Locally Symmetric Spaces

### 1 Symmetric Spaces

Main references are [Hel78], [Mil17b] and [Lan20].

**Def. (16.2.1.1) [Symmetric Spaces].** A Riemannian manifold is called **locally symmetric** at  $p$  if  $\nabla R(p) = 0$ . Locally symmetric is equivalent to the fact that every local reversing map is an isometry.

A **symmetric space** is a Riemannian manifold that  $\nabla R = 0$  everywhere.

A symmetric space is complete because two folding is an extension of geodesics. In particular, a symmetric space is homogenous, and to check symmetrically, it suffices to show it is homogenous and locally symmetric at a point.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.2.1.2).** A Lie group with a bi-invariant metric is a symmetric space.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.2.1.3).** The conjugate points in a symmetric space is easy to calculate, they are  $\exp(\frac{\pi k}{\sqrt{e_i}} V)$ , counting multiplicity, where  $e_i$  is the eigenvalue of the self-adjoint operator  $K_V(W) = R(V, W)V$  at  $p$ .  $\lrcorner$

**Prop. (16.2.1.4) [Lie Group of Isometries].** Let  $M$  be a symmetric space, then the group of isometries  $\text{Isom}(M^\infty, g)$  of  $M$  has a natural structure of a Lie group.  $\lrcorner$

*Proof:* Cf. [Helgason, homogenous Spaces, 4.3.2].  $\square$

**Prop. (16.2.1.5) [Symmetric Space is a Homogenous Space].** Let  $(M, g)$  be a symmetric space, and  $p \in M$ , then the subgroup  $K_p \subset \text{Aut}(M, g)^0$  fixing  $p$  is compact, and the natural map

$$\text{Isom}(M, g)^0 / K_p \rightarrow M^\infty$$

is an isomorphism of smooth manifolds, where  $\text{Isom}(M, g)^0 / K_p$  is given the homogenous space structure (16.2.1.4). In particular,  $\text{Isom}(M, g)^0$  acts transitively on  $M$ .  $\lrcorner$

*Proof:* Cf. [Mil17c] P12.  $\square$

### 2 Decompositions of Symmetric Spaces

#### 3 Non-Compact Type

#### 4 Compact Type

#### 5 Hermitian Symmetric Spaces

**Def. (16.2.5.1) [Hermitian Symmetric Spaces].** A **Hermitian symmetric space** is a Hermitian manifold that is a symmetric space (16.2.1.1) and that the local symmetries are all holomorphic.  $\lrcorner$

**Prop. (16.2.5.2) [Lie Group of Isometries].** For a Hermitian symmetric space, the group of holomorphic symmetries  $\text{Aut}(M, g)$  is closed in the group of isometries  $\text{Aut}(M^\infty, g)$ , which is a Lie group by (16.2.1.4), so it is also a Lie group.  $\lrcorner$



**Prop. (16.2.5.3) [Basic Hermitian Symmetric Spaces].** There are three different families of Hermitian symmetric spaces (but not complete):

- (Non-compact Type): These spaces are non-compact and simply connected, with negative curvatures, and  $\text{Isom}(M, g)^0$  is adjoint and non-compact.
- (Compact Type): These spaces are compact and simply-connected, with positive curvatures, and  $\text{Isom}(M, g)^0$  is adjoint and compact.
- (Euclidean Type): These spaces have constant curvature 0.

┘

*Proof:* Cf. Helgason 1978, Chap8.

□

**Prop. (16.2.5.4) [Decomposition].** Any Hermitian symmetric space  $M$  decomposes into a product  $M^0 \times M^+ \times M^-$  of Hermitian symmetric spaces with  $M^0$  Euclidean,  $M^-$  of non-compact type and  $M^+$  of compact type.

┘

*Proof:*

□

**Prop. (16.2.5.5).** Any Hermitian symmetric space of Euclidean type is a quotient of  $\mathbb{C}^g$  by a discrete subgroup of translations.

┘

*Proof:*

□

**Example (16.2.5.6).** The projective space  $\mathbb{P}^n(\mathbb{C})$  with the Fubini-Study metric is a Hermitian symmetric space. For any  $p$ , the (descent of) the rotation through  $\pi$  about the axis through  $p$  and its polar opposite is the geodesic isomorphism at  $p$ .

┘

*Proof:* See (16.10.1.6).

□

## 6 Hermitian Symmetric Domains

**Def. (16.2.6.1) [Hermitian Symmetric Domain].** A Hermitian symmetric space (16.2.5.1) of non-compact type (16.2.5.3) is called a **Hermitian symmetric domain**. In particular, a Hermitian symmetric domain is simply-connected. A **bounded symmetric domain** is a bounded open connected symmetric subset of  $\mathbb{C}^n$ .

┘

**Prop. (16.2.6.2) [Hermitian Symmetric Domains are Bounded].** Every Hermitian symmetric domain can be embedded into  $\mathbb{C}^n$  for some  $n$ , and the image is bounded.

┘

*Proof:*

□

**Prop. (16.2.6.3) [Bergman Metric].** Every bounded symmetric domain has a canonical Hermitian metric called the **Bergman metric**, it is invariant under holomorphic automorphisms, and it has negative curvatures.

┘

*Proof:* Cf. [Mil17]P11 and [Hel78]8.3.3.

□

**Cor. (16.2.6.4).** Every Hermitian symmetric domain  $D$  has a unique Hermitian metric that maps to the Bergman metric under any isomorphism of  $D$  onto a bounded symmetric domain.

┘

**Cor. (16.2.6.5).** A complex manifold is a symmetric Hermitian domain iff it is biholomorphic to a bounded symmetric domain.

┘

*Proof:*

□

**Def. (16.2.6.6) [Siegel Upper Half Space  $\mathcal{H}_g$ ].** For  $g \in \mathbb{Z}_+$ , the **Siegel upper half space**  $\mathcal{H}_g$  is defined to be

$$\mathcal{H}_g = \{Z \in \text{Mat}(n, \mathbb{C}) \mid Z^t = Z, \text{Im}(Z) \in \text{Pos}(g, \mathbb{C})\}.$$

It is identified with an open subset of  $\mathbb{C}^{g(g+1)/2}$ . We also define  $\mathcal{H}_g^\pm = \mathcal{H}_g \cup \overline{\mathcal{H}_g}$ .

The  $\text{Sp}(2g; \mathbb{R})$  (16.6.1.1) acts transitively on  $\mathcal{H}_g$  via

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} Z = (AZ + B)(CZ + D)^{-1}$$

and induces isomorphisms

$$\text{Sp}(2g; \mathbb{R})/\text{U}(n) \cong \mathcal{H}_g, \quad \text{GSp}(2g; \mathbb{R})/\text{U}(n) \cong \mathcal{H}_g^\pm.$$

The matrix  $\begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}$  acts as an involution on  $\mathcal{H}_g$ , and has  $i\mathbf{1}_g$  as its fixed point, so  $\mathcal{H}_g$  is homogeneous and symmetric.

The injection into  $\mathbb{C}^g$  is not holomorphic, so we cannot see from this that  $\mathcal{H}_g$  is holomorphic, but we can see from ?

┘

*Proof:* ? Cf. [Lan20]P15.

□

**Cor. (16.2.6.7) [Upper Half Plane].** By (16.6.1.10), the group  $\text{GL}(2; \mathbb{R})$  acts continuously on  $\mathbb{C}$  by

$$\gamma(z) = \frac{az+b}{cz+d} \text{ where } \gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The groups  $\text{GL}(2; \mathbb{R})$  preserves the upper plane  $\mathcal{H}$ , by (12.4.1.9). The action of  $\text{SL}(2; \mathbb{R})$  on  $\mathcal{H}$  is transitive and the stabilizer of  $i$  is  $\text{SO}(2, \mathbb{R})$ , thus we have  $\mathcal{H} \cong \text{SL}(2; \mathbb{R})/\text{SO}(2; \mathbb{R})$ , and  $\text{PSL}(2; \mathbb{R})$  is the group of holomorphic automorphisms of  $\mathcal{H}$  by (12.4.7.8).

Also, the Riemannian metric  $\frac{dx dy}{y^2}$  on  $\mathcal{H}$  is fixed by the action of  $\text{GL}(2; \mathbb{R})$ .

┘

**Def. (16.2.6.8) [Siegal Unit Disk  $\mathbb{D}_g$ ].** Let  $\mathbb{D}_g$  be the set of symmetric complex matrixes that  $I - Z^*Z$  is positive definite, then it is identified with an open subset of  $\mathbb{C}^{g(g+1)/2}$ , this is a holomorphic embedding.

There is an isomorphism from  $\mathcal{H}_g$  (16.2.6.6) to  $\mathbb{D}_g$ :

$$Z \mapsto (Z - iI_g)(Z + iI_g)^{-1}.$$

This is an isomorphism, so  $\mathbb{D}_g$  is symmetric, and  $\mathcal{H}_g$  has an invariant metric, so they are both Hermitian symmetric domains ?

┘

*Proof:* ? [Lan20]P15.

□

**Prop. (16.2.6.9).** Let  $(M, g)$  be a Hermitian symmetric domain, then the inclusions

$$\text{Aut}(M^\infty, g) \supset \text{Aut}(M, g) \subset \text{Aut}^{\text{Hol}}(M)$$

induce equalities

$$\text{Aut}(M^\infty, g)^0 = \text{Aut}(M, g)^0 = \text{Hol}(M)^0$$

Then  $\text{Hol}(M)^0$  acts transitively on  $M$ , the stabilizer  $K_p$  of  $p$  in  $\text{Aut}^{\text{Hol}}(M)^0$  is compact, and

$$\text{Hol}(M)^0/K_p \cong M^\infty.$$

┘

*Proof:* Cf.[?Mil17b]P12. □

**Prop. (16.2.6.10).** The Lie group  $\text{Aut}^{\text{Hol}}(M)$  in (16.2.6.9) is a real semisimple Lie group with only f.m. connected components and trivial center. ?

If  $G$  is a connected simple real algebraic group with trivial center s.t.  $D = G(\mathbb{R})^0/K$  for some maximal compact subgroup  $K \subset G(\mathbb{R})^0$ , then  $\text{Aut}(D) \cap G(\mathbb{R}) = G(\mathbb{R})^0$ , and  $G(\mathbb{R})$  has one or two connected components. ┘

*Proof:* ? □

**Prop. (16.2.6.11) [Rotation at a Point].** Let  $D$  be a Hermitian symmetric domain and  $p \in D$ , then there is a unique homomorphism  $u_p : U_1 \rightarrow \text{Hol}(D)$  that  $u_p(z)$  fixes  $p$  and acts on  $T_p(D)$  as multiplication by  $z$ . ┘

**Prop. (16.2.6.12) [Classification of Hermitian Symmetric Domains].** The isomorphism classes of irreducible Hermitian symmetric domains are classified by the special nodes on connected Dynkin diagrams. ┘

*Proof:* Cf.[Mil17]P20. □

## 7 Locally Symmetric Varieties

**Prop. (16.2.7.1) [ $D(\Gamma)$ ].** Let  $D$  be a Hermitian symmetric domain, and let  $\Gamma$  be a discrete subgroup of  $\text{Aut}^{\text{Hol}}(D)^0$ . If  $\Gamma$  is torsion-free, then  $\Gamma$  acts freely on  $D$ , and there is a unique complex manifold structure on  $\Gamma \backslash D$  that the quotient map  $\pi : D \rightarrow \Gamma \backslash D$  is a holomorphic covering space.

In this case, denote  $D(\Gamma) = \Gamma \backslash D$ , and  $D$  is a universal covering of  $D(\Gamma)$ , by (16.2.5.3) and (16.2.6.1). ┘

*Proof:* Cf.[?Mil17b]P32. □

**Prop. (16.2.7.2).** Let  $D$  be a Hermitian symmetric domain and  $\Gamma \subset \text{Hol}(D)^0$  is a discrete subgroup, then by (16.2.6.9),  $\Gamma$  has finite covolume in  $\text{Hol}(D)^0$  iff  $\Gamma \backslash D$  has finite covolume, ┘

*Proof:* □

**Prop. (16.2.7.3).**  $D(\Gamma)$  has only f.m. automorphisms, as a complex manifold. ┘

*Proof:* Cf.[?Mil17b]P41. □

**Thm. (16.2.7.4) [Satake-Baily-Borel Compactifications].** Let  $D$  be a locally symmetric Hermitian space and  $\Gamma \subset \text{Hol}(D)^0$  an arithmetic subgroup (20.4.4.1), then  $D(\Gamma)$  can be realized as an open subset of a projective variety  $\overline{D(\Gamma)}$  over  $\mathbb{C}$ , and it is a normal algebraic variety. If moreover  $\Gamma$  is torsion-free, then it is smooth, by (16.2.7.1).

Such a projective variety  $D(\Gamma)$  is called a **locally symmetric variety**. ┘

*Proof:* Cf. [BAILY-BOREL, Compactification of arithmetic quotients of bounded symmetric domains. 84:442–528. 1966] and [CASSELMAN, Geometric rationality of Satake compactifications, pp. 81–103.1997].

Cf.[?Mil17b]P38 for a history. □

**Thm. (16.2.7.5) [Borel].** Let  $D(\Gamma)$  be a quotient variety (16.2.7.4) of a Hermitian symmetric domain by a torsion-free arithmetic subgroup  $\Gamma$  in  $\text{Aut}^{\text{Hol}}(D)^0$ , then for any other smooth quasi-projective variety  $V$  over  $\mathbb{C}$ , any holomorphism map  $V^{\text{an}} \rightarrow D(\Gamma)^{\text{an}}$  comes from a morphism. ┘

*Proof:*

□

**Cor. (16.2.7.6)[Algebraic Structure is Unique].** The algebraic variety structure on  $D(\Gamma)$  is unique. Any variety of the form is called a **locally symmetric variety**. ┘

*Proof:* If there is another structure, then by GAGA it is a smooth variety, then the holomorphic map extends to a bijective morphism, which must be an isomorphism, by [Milne, Algebraic Geometry, P188] ?. □

**Cor. (16.2.7.7)[Borel].** The Satake-Baily-Borel compactification (16.2.7.4)  $\overline{D(\Gamma)}$  is minimal in the sense that it factors through any compactification  $D(\Gamma) \subset V$  s.t.  $V \setminus D(\Gamma)$  is a divisor with only normal crossings as singularities. ┘

*Proof:* Cf.[Borel 1972] ?

□

## 16.3 Riemannian Geometry

References are [Riemannian Geometry Do Carmo], [Geometric Analysis Jost] and [Differential Geometry Loring Tu].

**Notation (16.3.0.1).**

- Use notations on smooth manifolds defined in [Smooth Manifolds I: Generalities](#).
- Let  $M \in \mathcal{D}\text{iff}_{\text{cntd}}$ .

┘

### 1 Euclidean Geometry

**Thm. (16.3.1.1) [Ptolemy].** Let  $A, B, C, D \in \mathbb{R}^2$ , then

$$d(A, C) \cdot d(B, D) \leq d(A, B) \cdot d(C, D) + d(A, D) \cdot d(B, C).$$

And the equality holds if  $A, B, C, D$  are consequence points on a circle.

┘

*Proof:* Use complex numbers. there is an identity

$$(A - B)(C - D) + (A - D)(B - C) = (A - C)(B - D),$$

thus by triangle inequality,

$$|(A - C)(B - D)| \leq |(A - B)(C - D)| + |(A - D)(B - C)|,$$

where the equality holds if  $A, B, C, D$  are consequence points on a circle.

□

**Thm. (16.3.1.2) [Euclid].** There are only five Platonic solids: tetrahedron, hexahedron, octahedron, dodecahedron and the icosahedron.

┘

*Proof:*

□

**Thm. (16.3.1.3) [Pappus Hexagon Theorem].** Let 1, 2, 3, 4, 5 be five points in  $\mathbb{TP}^2$ , and define the cross products:

$$a = 1 \otimes 4, b = 2 \otimes 4, c = 3 \otimes 4, a' = 1 \otimes 5, b' = 2 \otimes 5, c' = 3 \otimes 5,$$

$$6 = b \otimes c', 7 = a' \otimes c, 8 = a \otimes b', a'' = 1 \otimes 6, b'' = 2 \otimes 7, c'' = 3 \otimes 8.$$

Then the three tropical lines  $a'', b'', c''$  are concurrent.

┘

*Proof:*

□

### Higher Dimensions

**Remark (16.3.1.4).** Instead of using time to visualize higher dimensions, we can also use color to visualize higher dimensions. In fact, human eyes are even 3-color based, which means theoretically we can visualize  $\mathbb{R}^3$  simply using colors.

┘

## 2 Basics

**Def. (16.3.2.1) [Riemannian Manifolds].** A **Riemannian metric** on  $M$  is a smooth symmetric covariant 2-tensor field on  $M$  that is positive-definite at each point. A **Riemannian manifold** (possibly with boundaries) is a pair  $(M, g)$  where  $M \in \text{Diff}_\partial$  and  $g$  is a Riemannian metric on  $M$ .

The category of Riemannian manifolds is denoted by  $\text{RieMani}$ .  $\lrcorner$

**Prop. (16.3.2.2) [Existence of Riemannian Metrics].** Every smooth manifold  $M \in \text{Diff}_\partial$  admits a Riemannian metric.  $\lrcorner$

*Proof:* Use a Euclidean open covering and partition of unity (4.4.7.8).  $\square$

**Def. (16.3.2.3).** For  $(M, g) \in \text{RieMani}$ , in any local coordinate  $(x^i)$ ,  $g$  can be written as a symmetric tensor  $g = \sum g_{ij} x^i \otimes x^j$ .

For  $p \in M$  and  $v, w \in T_p M$ ,

- The **length** of  $v$  is defined to be

$$|v|_g^2 \triangleq \langle v, v \rangle_g^{1/2} \triangleq g_p(v, v)^{1/2}.$$

- The **angle** of  $v, w$  is defined to be the unique  $\theta \in [0, \pi]$  s.t.

$$\cos \theta = \frac{\langle v, w \rangle_g}{|v|_g |w|_g}.$$

- $v, w$  is said to be **orthogonal** if  $\langle v, w \rangle_g = 0$ .  $\lrcorner$

**Def. (16.3.2.4) [Orthonormal Frames].** For  $(M, g) \in \text{RieMani}$  and  $U \subset M$ , an **smooth orthonormal frame** is a local frame  $(E_1, \dots, E_n)$  on  $U$  s.t.  $(E_1|_p, \dots, E_n|_p)$  is an orthonormal frame for  $T_p M$  at each  $p \in U$ .  $\lrcorner$

**Prop. (16.3.2.5) [Constructing Orthonormal Frames].** For  $(M, g) \in \text{RieMani}$  and  $U \subset M$  with a smooth local frame  $(X_1, \dots, X_n)$  on  $U$ , there exists a smooth orthonormal frame  $(E_1, \dots, E_n)$  on  $U$  s.t.  $\text{span}(E_{1,p}, \dots, E_{j,p}) = \text{span}(X_{1,p}, \dots, X_{j,p})$  for each  $j \in [n]_+$  and  $p \in U$ .  $\lrcorner$

**Prop. (16.3.2.6).** If the metric tensor on the tangent space is  $g$  in a coordinate, then it is  $g^{-1}$  in the cotangent space. (Follows from??).  $\lrcorner$

**Def. (16.3.2.7) [Riemannian Volume Forms].** Suppose  $(M, g)$  is an oriented Riemannian manifold possibly with boundaries, then there is a unique smooth form  $\omega_g \in \Omega^n(M)$  s.t.

$$\omega_g(E_1, \dots, E_n) = 1$$

for each oriented local orthonormal from  $(E^1, \dots, E^n)$  on  $M$  (16.3.2.4), called the **Riemannian volume form**.

In any oriented smooth local chart  $(x^i)$ , this Riemannian volume form is given by

$$\omega_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n.$$

$\lrcorner$

*Proof:* Cf. [Lee13] P389.  $\square$

### Pseudo-Riemannian Manifolds

**Def. (16.3.2.8) [Pseudo-Riemannian Metrics].** A **pseudo-Riemannian metric** on  $M$  is a smooth symmetric covariant 2-tensor field on  $M$  that is non-degenerate at each point. A **pseudo-Riemannian manifold** (possibly with boundaries) is a pair  $(M, g)$  where  $M \in \text{Diff}_\partial$  and  $g$  is a pseudo-Riemannian metric on  $M$ .

A pseudo-Riemannian metric of signature  $(1, n-1)$  at each point is called a **Lorentz metric**. A **Lorentz manifold** (possibly with boundaries) is a pair  $(M, g)$  where  $M \in \text{Diff}_\partial$  and  $g$  is a Lorentz metric on  $M$ .  $\lrcorner$

## 3 $\mathbb{R}^3$ -Geometry

### Different Coordinates

**Prop. (16.3.3.1).** In a polar coordinate,

$$g_{11} = 1, g_{12} = 0, g_{22} = \left| \frac{\partial}{\partial \theta} f \right|^2, \quad K = -\frac{(\sqrt{g_{22}})_{\rho\rho}}{\sqrt{g_{22}}}$$

And  $\sqrt{g_{22}} \sim \rho$ . (Use the formula relating Jacobi Field with curvature)  $\lrcorner$

### Moving Frame Method

**Thm. (16.3.3.2) [Theorema Egregium, Gauss 1827].**

$$R_{1212} = K(g_{11}g_{22} - g_{12}^2)$$

Which is a special case of the definition of curvature.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.3.3.3) [Gauss-Bonnet].** Let  $M$  be a compact oriented 2-dimensional Riemannian manifold, then

$$\chi(M) = \frac{1}{2\pi} \int_M K \text{ Vol.}$$

*Proof:* Should be an direct corollary of (16.7.5.6).  $\square$

### Topology and Geometry

**Prop. (16.3.3.4).** Every compact orientable surface of genus  $p > 1$  can be provided with a metric of constant negative curvature.  $\lrcorner$

*Proof:*  $\square$

**Remark (16.3.3.5) [Hilbert Theorem].** There exist complete surfaces with  $K \leq 0$  in  $\mathbb{R}^3$ , but the hyperbolic surface cannot be immersed into  $\mathbb{R}^3$ .  $\lrcorner$

#### 4 Connections

**Def. (16.3.4.1) [Affine Connection].** An **affine connection** on a vector bundle  $E$  is a map  $D : \Gamma(E) \rightarrow \Gamma(E) \otimes \Gamma(T^*M)$  that satisfies differential-like properties, it can be written as  $D = d + \omega$ , with  $\omega \in \Omega^1(\text{End}(E))$ .  $\lrcorner$

**Prop. (16.3.4.2) [Transformation Law].** In two coordinates  $\bar{e} = ea$  for  $a : U \rightarrow GL(r, \mathbb{R})$ ,  $d_A = d + \omega, d + \bar{\omega}$ , then  $\bar{\omega} = a^{-1}\omega a + a^{-1}da$ .

Moreover, giving any locally compatible  $d + \omega, \omega \in \Omega^1(\mathfrak{g})$  in the sense above, then for any  $G$ -associated bundle  $E$ , where  $G$  has lie algebra  $\mathfrak{g}$ , there is a connection that locally looks like  $d + \omega$ , (where  $\mathfrak{g}$  embeds into  $\mathfrak{gl}(E)$ ).  $\lrcorner$

**Cor. (16.3.4.3) [Local Nature of Connection].** From the description of connection given above, it's easy to say if there is a local connection that satisfies these transformation laws, then it generate a global connection. So by partition of unity (4.4.7.8), connection exists in any vector bundle over a manifold.  $\lrcorner$

**Cor. (16.3.4.4) [Simplification].**  $d_{gA}(s) = g d_A(g^{-1}(s))$ , So for any connection  $d_A$  and any point  $x_0$ , there is a gauge transformation that makes  $d_A = d$  at  $x_0$ .  $\lrcorner$

*Proof:* Just need to have  $s(x_0) = \text{id}$ ,  $ds(x_0) = -A(x_0)$ . this is possible because  $A \in \Omega^1(\text{Ad}E)$  which is the fiber of the frame bundle, use  $\exp$ .  $\square$

**Prop. (16.3.4.5) [Induced connections].** The connection action  $d_A = d + \omega$  on a vector bundle  $E$  induces connection on many relevant bundles. the action on dual bundle is by

$$d_A(s^*) = ds^* - \omega^t(s^*) = ds^* - s^* \circ \omega.$$

And the connection on  $\text{End } E$  by

$$d_A(\alpha) = d\alpha + [\omega, \alpha] = [\nabla, \alpha]$$

And they act on  $\Omega^*(E)$  by Leibniz rule thus the formula looks the same. (Note that the convention is section write on the left of the differential forms, so for example,  $[\omega, \omega] = 2\omega \wedge \omega$ ).  $\lrcorner$

*Proof:* Cf. [Jost P110].  $\square$

**Cor. (16.3.4.6).** For a line bundle  $L$ , for a connection on it with curvature  $\Omega$ , the induced on the dual line bundle  $L^*$  has connection  $-\Omega$ . (because  $\Omega = d\omega$  and  $\omega' = -\omega$ ).  $\lrcorner$

**Prop. (16.3.4.7) [Second Bianchi's Identity].** A affine connection on  $E$  looks locally like  $d_A = d + \omega$ , where  $\omega \in \Omega^1(\text{End } E)$ . And  $F_A = d_A \circ d_A \in \Omega^2(\text{End}(E))$  satisfies

$$d_A F_A = dF_A + [\omega, F_A] = 0.$$

$\lrcorner$

*Proof:* Notice  $dF_A = dd\omega + d(\omega \wedge \omega) = d\omega \wedge \omega - \omega \wedge d\omega$ , and  $\omega(d\omega + \omega \wedge \omega) - (d\omega + \omega \wedge \omega)\omega = \omega \wedge d\omega - d\omega \wedge \omega$ .  $\square$

**Def. (16.3.4.8) [Christoffel Symbol].** The **Christoffel symbol** of a connection is defined by the equations:  $\nabla_{X_i} X_j = \sum_k \Gamma_{ij}^k X_k$ .

The **geodesic equations** is  $\frac{D}{dt}(\frac{d\gamma}{dt}) = \ddot{x}_k + \sum_{i,j} \Gamma_{ij}^k \dot{x}_i \dot{x}_j = 0 \quad \forall k$ .  $\lrcorner$



**Def. (16.3.4.9).** The **torsion tensor** of a connection  $\nabla$  on  $TM$  is defined as  $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$ . The connection is called **torsion-free** if  $T = 0$ . This is equivalent to  $\Gamma_{i,j}^k = \Gamma_{j,i}^k$ .

A connection is called **metric** if it preserves metric. i.e.  $\nabla g = 0$ .  $\lrcorner$

*Proof:*  $T$  is a tensor because it is skew-symmetric, and

$$T(fX, Y) = f\nabla_X Y - f\nabla_Y X - df(Y)X - (f[X, Y] - df(Y)X) = fT(X, Y),$$

where (16.1.2.17) is used.  $\square$

**Prop. (16.3.4.10).** If  $\nabla$  is torsion-free connection on  $TM$ , then its induced connection on  $T^*M$  satisfies

$$(d\alpha)(v_1, \dots, v_k) = \sum (-1)^i (D_{v_i} \alpha)(v_1, \dots, \widehat{v_i}, \dots, v_k).$$

$\lrcorner$

*Proof:*  $\square$

**Def. (16.3.4.11) [Curvature Tensor].** The **curvature tensor** of a (affine) connection  $d_A$  is  $F_A = d_A \circ d_A \in \Omega^2(\text{End}(E))$ . The curvature tensor it induced is

$$F_A(Z)(X, Y) = R(X, Y)Z = D_Y D_X Z - D_X D_Y Z + D_{[X, Y]} Z.$$

In particular, the curvature depends only on the point, and locally  $F_A = d\omega + \omega \wedge \omega$

In two coordinates  $\bar{e} = ea$  for  $a : U \rightarrow GL(r, \mathbb{R})$ ,  $\overline{F_A} = a^{-1} F_A a$ .

The connection is called **flat** if  $F_A = 0$ .  $\lrcorner$

*Proof:* To verify the equation, check first the left side is pointwise, and the third component of the right side assures it is pointwise, too, thus we can check for a local coordinate vector field  $([X_i, X_j] = 0)$ , then because  $\nabla s = \sum_i \nabla_i s dx_i$ ,

$$\nabla^2 s = \nabla \left( \sum_i \nabla_i s dx_i \right) = \sum_{ij} \nabla_j \nabla_i s dx_j dx_i = \sum_{i < j} (\nabla_i \nabla_j - \nabla_j \nabla_i) s dx_i \wedge dx_j$$

$\square$

**Prop. (16.3.4.12) [Flat coordinate].** A connection on  $TM$  assumes near every point a flat coordinate, i.e.  $\nabla(\partial/\partial x^i) = 0$ , iff it is flat and torsion-free.  $\lrcorner$

*Proof:* One side is easy because its Christoffels vanish. On the other side, use integrability theorems (13.1.6.2). Cf. [Jost P115].  $\square$

**Prop. (16.3.4.13).**

$$\Delta \langle \varphi, \varphi \rangle = 2(\langle D^* D \varphi, \varphi \rangle - \langle D \varphi, D \varphi \rangle).$$

$\lrcorner$

*Proof:* Cf. [Jost P118].  $\square$

**Prop. (16.3.4.14).** For a flat connection, there is a bundle isomorphism (Gauge transform) that transforms  $d_A$  into natural  $d$ .  $\lrcorner$

*Proof:* Because  $d_{gA}(s) = gd_A(g^{-1}(s))$ ,  $d_{gA} = d - dg \cdot g^{-1} + g \cdot \omega \cdot g^{-1}$ . Solve this PDE directly. (Cf.[Topics in Geometry Xie Yi week3]).  $\square$

**Cor. (16.3.4.15).** For a flat connection, by (16.3.4.14), the parallel transportation only depends on the homotopy type of the loop, thus gives an action of  $\pi(X)$  on  $SO(T_p(X))$  (or  $SU(T_p(X))$ ). (because it is locally constant).

In this way, connections module gauge equivalence (preserving matrix) equals representation of  $\pi(X)$  module conjugations. The reverse map is giving by principal bundle.  $\lrcorner$

*Proof:*  $\square$

### Levi-Civita Connection

**Def. (16.3.4.16) [Levi-Civita Connection].** The Levi-Civita connection is the unique connection on  $M$  that is metric and torsion-free:

$$X\langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle, \quad \nabla_X Y - \nabla_Y X = [X, Y].$$

It satisfies:

$$\langle Z, \nabla_Y X \rangle = 1/2 \{ X\langle Y, Z \rangle + Y\langle Z, X \rangle - Z\langle X, Y \rangle - \langle [X, Z], Y \rangle - \langle [Y, Z], X \rangle - \langle [X, Y], Z \rangle \}.$$

Then

$$\Gamma_{ij}^m = 1/2 \sum_k \{ g_{jk,i} + g_{ki,j} - g_{ij,k} \} g^{km}$$

Thus geodesic is a solution that only depends on the metric (16.3.4.8), so a local isometry preserves geodesics.  $\lrcorner$

**Prop. (16.3.4.17).** Now the Lie derivative has the form:

$$L_X(S)(Y_1, \dots, Y_p) = \nabla_X(S)(Y_1, \dots, Y_p) + \sum_{i=1}^p S(Y_1, \dots, Y_{i-1}, \nabla_{Y_i} X, \dots, Y_p).$$

The exterior derivative  $d$  and its adjoint  $d^*$  has the form:

$$d\omega(Y_i) = \sum (-1)^p \nabla_{Y_i} \omega(\check{Y}_p), \quad d^* \omega(Y_i) = - \sum \nabla_{e_j} \omega(e_j, Y_i)$$

where  $e_i$  is an orthonormal basis. Cf.[Jost P140].  $\lrcorner$

**Prop. (16.3.4.18) [Covariant Differential Symmetry].** For a parametrized surface:  $s : (u, v) \rightarrow M$ ,

$$\frac{D}{\partial u} \frac{\partial s}{\partial v} = \frac{D}{\partial v} \frac{\partial s}{\partial u}.$$

$\lrcorner$

*Proof:*

$$\frac{D}{\partial u} \frac{\partial s}{\partial v} = \frac{D}{\partial u} \left( \sum \frac{\partial s_i}{\partial v} X_i \right) = \frac{\partial^2 s_i}{\partial u \partial v} + \sum \frac{\partial s_i}{\partial v} \left( \sum \frac{\partial s_j}{\partial u} \nabla_j X_i \right) = \frac{\partial^2 s_i}{\partial u \partial v} + \sum_{ij} \frac{\partial s_i}{\partial v} \frac{\partial s_j}{\partial u} \nabla_j X_i.$$

But now the Levi-Civita connection is symmetric, thus  $\nabla_j X_i = \nabla_i X_j$ , showing the symmetry in  $u$  and  $v$ .  $\square$

**Lemma (16.3.4.19) [Gauss].** Let  $p \in M$  and  $v \in T_p M$  s.t.  $\exp_p v$  is defined,  $w \in T_p M$ , then

$$\langle (d\exp_p)_v(v), (d\exp_p)_v(w) \rangle = \langle v, w \rangle.$$

┘

*Proof:* Cf.[Do Carmo P69].

□

**Prop. (16.3.4.20) [Geodesic Locally Minimizing].** In a normal nbhd of  $p$ , the geodesic starting at  $p$  is the minimal line. ┘

*Proof:* And curve  $c(t)$  can be written as  $\exp_p(r(t)v(t)) = f(r(t), t)$ , where  $f(s, t) = \exp_p(sv(t))$ , so by Gauss lemma,  $\langle \frac{\partial f}{\partial r}, \frac{\partial f}{\partial t} \rangle = 0$ . Now  $dc/dt = \partial f / \partial r r'(t) + \partial f / \partial t$ , so

$$|dc/dt|^2 = |r'(t)|^2 + |\partial f / \partial t|^2 \geq |r'(t)|^2.$$

Integrate this will give us the desired result. □

**Prop. (16.3.4.21) [Totally normal nbhd].** For any point  $p$ , there exists a nbhd  $W$  and a number  $\delta > 0$  s.t. for every  $q \in W$ ,  $\exp_q$  is a diffeomorphism on  $B_\delta(0)$  and  $\exp_q(B_\delta(0)) \supset W$ . Thus, fine cover exists in every smooth manifold, because Riemannian metric exists on these manifolds. ┘

*Proof:* Cf.[Do Carmo P72].

□

- **(Geodesic Frame)** In a neighborhood of every point  $p$ , there exists  $n$  vector fields, orthonormal at each point, and  $\nabla_{E_i} E_j(p) = 0$ . (Choose normal nbhd and parallel a orthonormal basis to every point. (WARNING: this is not a flat coordinate, it only helps when dealing with point-wise properties).)

**Def. (16.3.4.22) [Killing Fields].** A **Killing field** is a vector field which generates an infinitesimal isometry.  $X$  is killing  $\iff L_X(g) = 0 \iff \langle \nabla_Y X, Z \rangle + \langle \nabla_Z X, Y \rangle = 0$  for all  $Y, Z$ , which is called the **Killing equation**. ┘

*Proof:* Use Lie formula,

$$L_X(g)(Y, Z) = X\langle Y, Z \rangle - \langle [X, Y], Z \rangle - \langle Y, [X, Z] \rangle$$

. and Levi-Civita connection is torsion-free. □

**Prop. (16.3.4.23).** Let  $M$  be a compact Riemannian manifold of even dimension with positive sectional curvatures, then every Killing field on  $M$  has a singularity. ┘

*Proof:* Cf.[Do Carmo P104].

□

**Def. (16.3.4.24) [Geometric Differential Notations].** For  $f \in C^\infty(M)$ ,  $X \in \Gamma(M, TM)$ ,

- The **gradient** of  $f$  is defined to be the tangent field  $\text{grad} f \in \Gamma(M, TM)$  s.t. for any  $X \in \Gamma(M, TM)$ ,  $\langle \text{grad} f, X \rangle_p = X(f)(p)$ .
- The **divergence** of  $X$  is defined to be a function  $\text{div} X \in C^\infty(M)$  s.t. for any  $p \in M$ ,  $\text{div} X(p) = \text{trace of the linear map } Y(p) \rightarrow \nabla_Y X(p) = \sum_i \langle \nabla_{E_i} X, E_i \rangle$ . It measures the variation of the volume and it depends only on the point.
- The **Hessian** of  $f$  is defined to be a symmetric form  $\text{Hess} f \in \Gamma(M, T^*M^{\otimes 2}) \cong \text{Hom}(TM, TM)$  s.t. for any  $Y, Z \in \Gamma(M, TM)$ ,  $(\text{Hess} f)(Y, Z) = \langle (\text{Hess} f)Y, Z \rangle$  where  $(\text{Hess} f)Y = \nabla_Y \text{grad} f$ .

- The **Laplacian** of  $f$  is defined to be the function  $\Delta f = \operatorname{div} \operatorname{grad} f = \operatorname{tr}(\operatorname{Hess} f) \in C^\infty(M)$ .

┘

**Prop. (16.3.4.25).** In a geodesic frame,

$$\operatorname{grad} f(p) = \sum_{i=1}^n (E_i(f)) E_i(p)$$

$$\operatorname{div} X(p) = \sum_{i=1}^n E_i(f_i)(p), \text{ where } X = \sum_i f_i E_i.$$

$$\Delta f = \sum_i E_i(E_i(f))(p).$$

┘

**Cor. (16.3.4.26).**

$$\Delta(f \cdot g) = f \Delta g + g \Delta f + 2\langle \operatorname{grad} f, \operatorname{grad} g \rangle,$$

because these only depends on the point.

┘

**Prop. (16.3.4.27).**  $d(i(X)m) = (\operatorname{div} X)m$ . where  $m$  is the volume form.

┘

*Proof:* Choose a geodesic frame  $E_i$ ,  $\theta_i$  is a dual form of  $E_i$ , let  $X = \sum f_i E_i$ , then  $\iota(X)m = \sum_i (-1)^{i+1} f_i \theta_i$ , so

$$d(\iota(X)m) = \sum (-1)^{i+1} df_i \wedge \theta_i + \sum (-1)^{i+1} f_i \wedge d\theta_i = \left( \sum E_i(f_i) \right) m + \sum (-1)^{i+1} f_i \wedge d\theta_i.$$

Notice that  $d\theta_i = 0$ , because  $d\theta_k(E_i, E_j) = E_i \theta_k(E_j) - E_j \theta_k(E_i) - \theta_k([E_i, E_j]) = 0$  (16.1.3.4), as it is a geodesic frame. And  $\sum E_i(f_i) = \operatorname{div}(X)$  (16.3.4.25).  $\square$

**Prop. (16.3.4.28) [Hopf theorem].** If  $f$  is a differentiable function on a compact orientable manifold with  $\Delta f \geq 0$ , then  $f$  is constant.

┘

*Proof:* Let  $\operatorname{grad}(f) = X$ , then

$$\int_M \Delta f dm = \int_M \operatorname{div}(X) dm = \int_M d(\iota(X)m) = 0.$$

So  $\Delta f = 0$ . Now

$$0 = \int_M \Delta(f^2/2) dm = \int_M f \Delta f dm + \int_M |\operatorname{grad}(f)|^2 dm$$

by (16.3.4.26), thus  $\operatorname{grad}(f) = 0$ , so  $f$  is constant.  $\square$

**Def. (16.3.4.29) [Riemannian Curvatures].**

- The **sectional curvature**  $K(X, Y) = \frac{\langle R(X, Y)X, Y \rangle}{|X \wedge Y|^2}$ .
- The **Ricci curvature**  $\operatorname{Ric}(x) = \operatorname{Ric}(x, x)$ , where  $\operatorname{Ric}(x, y)$  is the symmetric form of  $\frac{1}{n}$  of trace of the map  $z \rightarrow R(x, z)y$ .  
Thus  $\operatorname{Ric}_p(x) = \frac{1}{n-1} \sum \langle R(x, z_i)x, z_i \rangle$ , for  $x$  a unit vector, where  $z_i$  is an orthonormal basis orthogonal to  $x$ .
- The **scalar curvature**  $K(p) = 1/n \sum \operatorname{Ric}_p(z_i)$ , where  $z_i$  is an orthonormal basis.

The curvatures only depends on the point (16.3.4.11).  $\square$

**Lemma (16.3.4.30).**

$$\frac{D}{\partial t} \frac{D}{\partial s} V - \frac{D}{\partial s} \frac{D}{\partial t} V = R\left(\frac{\partial f}{\partial s}, \frac{\partial f}{\partial t}\right)V.$$

┘

*Proof:* Obvious because  $\frac{\partial}{\partial s}, \frac{\partial}{\partial t}$  commutes. ?

□

**Prop. (16.3.4.31) [Sectional Curvature Define Curvature].** The curvature tensor is determined by its sectional curvature.

In particular, if  $M$  is isotropic at a point  $p$  (The sectional curvature depends only on the point), then

$$R(X, Y, W, Z) = K_0(\langle X, W \rangle \langle Y, Z \rangle - \langle X, Z \rangle \langle Y, W \rangle).$$

┘

*Proof:* Cf.[Do Carmo P95], should use the cyclicity of the first three terms.

□

**Prop. (16.3.4.32) [Bianchi Identities].** Recall the covariant differential

$$\nabla R(Y_i, Z) = Z(R(Y_i)) - \sum_i R(\nabla_Z Y_i, Y_j) \quad (16.3.4.5).$$

We have

- (Bianchi Identity)  $\sum_{(X,Y,Z)} R(X, Y)Z = 0$ .
- (Second Bianchi Identity)  $\sum_{(Z,W,T)} \nabla R(X, Y, Z, W, T) = 0$ .

┘

*Proof:* 1: Cf.[Do Carmo P91], should reduce to Jacobi identity.

2:

□

**Prop. (16.3.4.33) [Schur's Theorem].** Suppose  $\dim M \geq 3$  and the sectional curvature only depends on  $p$ , then  $M$  has constant curvature.

┘

*Proof:* Use the second Bianchi Identity and geodesic frame and (16.3.4.31). Cf.[Do Carmo P106].

□

**Def. (16.3.4.34) [Eisenstein Curvature].**  $M$  is called a **Eisenstein manifold** iff its Ricci curvature  $\lambda(p)$  only depends on the point. Then

- If  $M$  is connected and Eisenstein of dimension  $\geq 3$ , then it has constant Ricci curvatures everywhere every direction.
- If  $M$  is connected and Eisenstein of dimension 3, then it has constant sectional curvatures.

┘

*Proof:* 1: Cf.[Do Carmo P108].

2: Now it has constant Ricci curvature, then

$$R_{1212} + R_{1313} = \lambda = R_{1212} + R_{2323} = R_{1313} + R_{2323}.$$

So we can solve these curvatures out.

□

**Prop. (16.3.4.35) [Riemannian Curvature Identities].**

•

- $R(X, Y, Z, W) = R(Z, W, X, Y)$ ,  $R(X, Y, Z, W) = R(X, Y, W, Z)$ .

┘

*Proof:* Cf.[DO Carmo P91].

□

- $B(X, Y) = \bar{\nabla}_X \bar{Y} - \nabla_X Y$ . It is bilinear and symmetric.
- $H_\eta(x, y) = \langle B(x, y), \eta \rangle$ . Thus  $B(x, y) = \sum H_i(x, y) E_i$  for an orthonormal frame  $E_i$  in  $\mathfrak{X}(U)^\perp$ .
- $S_\eta(x) = -(\bar{\nabla}_x \eta)^T$ . It satisfies:  $\langle S_\eta(x), y \rangle = H_\eta(x, y) = \langle B(x, y), \eta \rangle$ . It is self-adjoint. When codimension 1, it is the derivative of the Gauss mapping.
- (**Gauss Formula**): let  $x, y$  be orthonormal tangent vector. Then:

$$K(x, y) - \bar{K}(x, y) = \langle B(x, x), B(y, y) \rangle - |B(x, y)|^2.$$

- An immersion is called **geodesic** at  $p$  if the second fundamental form  $S_\eta$  is zero for all  $\eta$ , (which means  $\nabla_X Y$  has no normal component). It is called **minimal** if the trace of  $S_\eta$  is zero.
- An immersion is called umbilic if there exists a normal unit field  $\eta$  s.t.  $\langle B(X, Y), \eta \rangle(p) = \lambda(p) \langle X, Y \rangle$ .
- If the ambient space has constant sectional curvature and the immersed manifold is totally umbilic, then  $\lambda$  is constant.
- mean curvature tensor of immersion  $f = 1/n \sum_i (\text{tr } S_i) E_i = 1/n \text{tr } B$ . It is zero if  $f$  is minimal.
- normal connection  $\nabla_X^\perp \eta = (\bar{\nabla}_X \eta)^N = \bar{\nabla}_X \eta + S_\eta(X)$ .

**Prop.(16.3.4.36).** • (Gauss equation)

$$\langle \bar{R}(X, Y)Z, T \rangle = \langle R(X, Y)Z, T \rangle - \langle B(Y, T), B(X, Z) \rangle + \langle B(X, T), B(Y, Z) \rangle.$$

- (Ricci equation)

$$\langle \bar{R}(X, Y)\eta, \zeta \rangle - \langle R^\perp(X, Y)\eta, \zeta \rangle = \langle [S_\eta, S_\zeta]X, Y \rangle.$$

- (Codazzo equation)

$$\langle \bar{R}(X, Y)Z, \eta \rangle = (\bar{\nabla}_Y B)(X, Z, \eta) - (\bar{\nabla}_X B)(Y, Z, \eta). \quad (\text{Lie bracket})$$

┘

### Parallel Transportations

**Def.(16.3.4.37)[Parallel Transportation].**

┘

**Def.(16.3.4.38)[Holonomy Group].** The **holonomy group**  $Hol_x(g)$  of a Riemannian manifold  $M$  w.r.t to the Levi-Civita connection is defined to be the subgroup of  $O(T_x(M))$  induced by the parallel transportation along a loop. If  $M$  is connected, For different points, holonomy groups are conjugate, so holonomy group is defined up to conjugation.

┘

**Prop.(16.3.4.39)[Trivial Holonomy Group].** If  $M$  is a Riemannian manifold and the holonomy group is trivial, then for any  $X, Y, Z \in X(M)$ ,  $R(X, Y)Z = 0$ .

┘

*Proof:* Cf.[Do Carmo P105].

□

**Prop. (16.3.4.40) [Berger].** In fact, the groups that can be realized as a holonomy group of a simply connected complete Riemannian manifold can be classified.  $\lrcorner$

*Proof:* Cf. [Complex geometry Daniel P214].  $\square$

**Def. (16.3.4.41).** The **Geodesic flow** for a connection on  $TM$  is the flow on  $TM$  whose trajectories are  $t \mapsto (\gamma(t), \gamma'(t))$ , where  $\gamma$  is a geodesic on  $M$ .  $\lrcorner$

**Prop. (16.3.4.42) [The smoothness of geodesics].** For every point  $p$ , there exists a nbhd  $V$  and a  $C^\infty$  mapping

$$\gamma : (-\delta, \delta) \times V \times B(0, \epsilon) \rightarrow M,$$

s.t.  $\gamma(t, q, v)$  is the geodesic passing through  $p$  with velocity  $v$ .  $\lrcorner$

**Prop. (16.3.4.43) [Curvature and Metric, Cartan].** Let  $M, \widetilde{M}$  be two Riemannian manifold of dimension  $n$  and let  $p \in M, \tilde{p} \in \widetilde{M}$ . Choose a linear isometry  $i : T_p(M) \cong T_{\tilde{p}}(\widetilde{M})$ . Let  $V$  be a normal neighbourhood of  $p$  that  $\exp_p$  is defined on  $i \circ \exp_p^{-1}(V)$ . Define a mapping  $f : V \rightarrow \widetilde{M}$  by  $f(q) = \exp_{\tilde{p}} \circ i \circ \exp_p^{-1}(q)$ .

For any  $q \in V$ , there is a unique normalized geodesic  $\gamma : [0, t] \rightarrow M$  that  $\gamma(0) = p, \gamma(t) = q$ . Denote by  $P_t$  the parallel transportation along  $\gamma$ , and the map  $\varphi_t : T_q(M) \rightarrow T_{f(q)}(\widetilde{M})$  by  $\varphi_t(v) = \tilde{P}_t \circ i \circ P_t^{-1}(v)$ .

If for all  $q \in V$  and all  $x, y, u, v \in T_q(M)$ , we have

$$\langle R(x, y)u, v \rangle = \langle \tilde{R}(\varphi_t(x), \varphi_t(y)\varphi_t(u), \varphi_t(v)) \rangle,$$

then  $f : V \rightarrow f(V) \subset \widetilde{M}$  is an isometry and  $df_p = i$ .  $\lrcorner$

*Proof:* Cf. [Do Carmo P157]. Use Jacobi fields. The point is that the hypothesis implies that the map of a Jacobi field is also a Jacobi field.  $\square$

**Cor. (16.3.4.44).** Let  $M, \widetilde{M}$  be Riemannian manifolds with the same dimension  $n$  in which parallel transportation preserves sectional curvature. Let  $p \in M, \tilde{p} \in \widetilde{M}$ . If there is a linear isometry  $i : T_p(M) \cong T_{\tilde{p}}(\widetilde{M})$  s.t.  $K(p, E) = K(\tilde{p}, i(E))$  for any 2-dimensional subspace  $E \subset T_p(M)$ , then there exist nbhd  $V$  of  $p$ , nbhd  $\tilde{V}$  of  $\tilde{p}$  and an isometry  $f : V \rightarrow \tilde{V}$  that  $df_p = i$ .  $\lrcorner$

**Cor. (16.3.4.45).** If in situation (16.3.4.44),  $M$  and  $\widetilde{M}$  are moreover complete and simply connected, then there is a unique isometry  $f : M \rightarrow \widetilde{M}$  s.t.  $f(p) = \tilde{p}$  and  $df_p = i$ .  $\lrcorner$

### Complete manifold

**Thm. (16.3.4.46) [Hopf-Rinow theorem].** The following is equivalent definition of **completeness** for  $M \in \text{Diff}_{\text{cnd}}$ :

1.  $\exp_p$  is defined for all of  $T_p(M)$  and all  $p \in M$ .
2. The closed and bounded sets of  $M$  are compact.
3.  $M$  is complete as a metric space.
4.  $M$  is  $\sigma$ -compact and if  $q_n \notin K_n$ ,  $d(p, q_n) \rightarrow \infty$ .
5. The length of any divergent (compact escaping) curve is unbounded.

and if  $M$  is complete, then for any  $p, q \in M$ , there exists a minimizing geodesic between  $p, q$ . In particular, any compact submanifold of a complete manifold is complete.  $\lrcorner$

*Proof:* Cf.[Do Carmo P147]. □

- For any two manifold of the same constant curvature and any two orthogonal basis, there is a local isometry (It is locally isotropic).
- Any complete manifold with a sectional curvature is like  $\tilde{M}/\Gamma$ , where  $\tilde{M}$  is  $\mathbf{H}^n$ ,  $\mathbf{R}^n$  or  $\mathbf{S}^n$ .

**Prop.(16.3.4.47) [Cartan].** in any nontrivial homotopy class in a compact manifold , there exists a closed geodesic. ┘

*Proof:* □

## 5 Jacobi Field and Comparison Theorems

**Def.(16.3.5.1) [Jacobi Field].** The **Jacobi field equation** along a normalized geodesic  $\gamma$  is defined to be

$$D^2 J(t) + R(\dot{\gamma}(t), J(t))\dot{\gamma}(t) = 0.$$

It is defined by its initial condition  $J(0)$  and  $J'(0)$ . It can be used to detect the sectional curvature, the critical point of  $\exp_p$  and calculate variation of energy. ┘

**Prop.(16.3.5.2) [Constant Curvature Case].** On a manifold with constant curvature  $K$ , the Jacobi field equation for a vector field  $J$  normal to  $\gamma$  is equivalent to

$$D^2 J(t) + KJ(t) = 0.$$

┘

*Proof:* Use(16.3.4.31), we have

$$\langle R(\gamma', J)\gamma', T \rangle = K\{\langle \gamma', \gamma' \rangle \langle J, T \rangle - \langle \gamma', T \rangle \langle J, \gamma' \rangle\} = K \langle J, T \rangle$$

So  $R(\gamma', J)\gamma' = KJ$ . □

**Prop.(16.3.5.3).** The Jacobi field along a point with initial velocity 0 all has the form

$$J(t) = (d\exp_p)_{t\dot{\gamma}(0)}(tJ'(0)).$$

┘

*Proof:* Cf.[Do Carmo P113]. Should use uniqueness theorem of ODE. □

**Cor.(16.3.5.4) [Conjugate Points].** If two points  $p, q$  are connected by a geodesic  $\gamma$ , and  $q = \exp_p(v_0)$ , then  $p, q$  are called **conjugate** along  $\gamma$ , if there is a non-trivial Jacobi field on  $\gamma$  that  $J(p) = J(q) = 0$ .

Then  $q$  is conjugate to  $p$  iff  $v_0$  is the critical point of  $\exp_p$ , and the multiplicity of conjugacy is equal to the kernel of  $(\exp_p)_{v_0}$ . ┘

**Prop.(16.3.5.5).** For a Jacobi field  $J$  along  $\gamma$ ,  $\langle J(t), \gamma'(t) \rangle$  is linear in  $t$ . ┘

*Proof:* Take second derivatives. □

- If  $J$  is a Jacobi field  $J(t) = (d\exp_p)_{tv}(tw)$ ,  $|v| = |w| = 1$ , then

$$|J(t)| = t - \frac{1}{6}K_p(v, w)t^3 + o(t^3).$$



**Prop. (16.3.5.6).** There are no conjugate points on a Riemannian manifold of non-positive curvature.  $\perp$

*Proof:* Cf.[Do Carmo P119].  $\square$

**Prop. (16.3.5.7)[Killing Field is everywhere Jacobi].** A Killing field is a Jacobi field along geodesics. And if  $X(p) = 0$ , then  $X$  is tangent to the geodesic sphere near  $p$ , because  $X$  preserves length.  $\perp$

*Proof:*  $\square$

### Energy Analysis

**Def. (16.3.5.8)[Energy].** The **energy** of a geodesic  $\gamma : [0, a] \rightarrow M$  is defined to be

$$E(s) = \int_0^a \left| \frac{\partial}{\partial t} \gamma(s, t) \right|^2 dt.$$

$\perp$

**Prop. (16.3.5.9).** A minimizing geodesic must minimize energy.  $\perp$

*Proof:*  $\square$

- **(First Variation of Energy)**

$$1/2E'(0) = - \int_0^a \langle V(t), D\dot{c}(t) \rangle dt + \langle V(a), \dot{c}(a) \rangle - \langle V(0), \dot{c}(0) \rangle.$$

A piecewise differentiable curve is a geodesic iff every proper variation has first derivative 0.

- **(Second Variation of Energy)** If  $\gamma$  is a geodesic,

$$1/2E''(0) = \int_0^a \{ \langle DV(t), DV(t) \rangle - \langle R(\dot{\gamma}, V)\dot{\gamma}, V \rangle \} dt + \langle D_s V(a), \dot{\gamma}(a) \rangle - \langle D_s V(0), \dot{\gamma}(0) \rangle.$$

- a variation is equivalent to a vector field along the curve, and a variation that  $f_s(t)$  are all piecewise geodesics corresponds to a piecewise Jacobi field(Choose a normal partition).

**Prop. (16.3.5.10)[Rauch Comparison theorem].** Let  $M$  and  $\tilde{M}$  be manifolds,  $\dim \tilde{M} \geq \dim M$ .

If  $J$  and  $\tilde{J}$  be two normal Jacobi fields along geodesics  $\gamma$  and  $\tilde{\gamma}$  that  $|J(0)| = |J'(0)| = 0$  and  $|J'(0)| = |\tilde{J}'(0)|$ . If  $\tilde{\gamma}$  has no conjugate point or focal point free and  $\tilde{K}(\tilde{x}, \dot{\tilde{\gamma}}(t)) \geq K(x, \dot{\gamma})$  for any vector  $x, \tilde{x}$ , then  $|\tilde{J}| \leq |J|$ .  $\perp$

**Cor. (16.3.5.11)[Injectivity Radius Estimate].** If the sectional curvature of  $M$  satisfies:  $0 < L \leq K \leq H$ , then the distance between any two conjugate points satisfies:  $\frac{\pi}{\sqrt{H}} \leq d \leq \frac{\pi}{\sqrt{L}}$ .  $\perp$

**Prop. (16.3.5.12).** If two manifold  $M$  and  $M'$  satisfy  $K \leq K'$ , then in a normal nbhd of a point  $p$  in  $M$  and a nbhd of  $p'$  that exp is nonsingular, the transformation of a curve  $c$  shortens length.

Note that this is not Toponogov theorem, because if you try to map from a large curvature manifold to a small curvature, then you cannot guarantee that the mapped curve is the shortest.  $\perp$

**Cor. (16.3.5.13).** In a complete simply connected manifold of non-positive curvature,

$$A^2 + B^2 - 2AB \cos \gamma \leq C^2$$

thus  $\alpha + \beta + \gamma \leq \pi$ .  $\perp$

**Prop. (16.3.5.14) [Moore theorem].** Let  $\bar{M}$  be a complete simply connected manifold of sectional curvature  $\bar{K} \leq -b \leq 0$ ,  $M$  a compact manifold of sectional curvature satisfying  $K - \bar{K} \leq b$ . If  $\dim \bar{M} < \dim M$ ,  $M$  cannot be immersed into  $\bar{M}$ . (use Hadamard theorem to choose the furthest geodesic and calculate the second variation of energy and use Gauss formula).  $\lrcorner$

**Cor. (16.3.5.15).** Let  $\bar{M}$  be a complete simply connected manifold of sectional curvature  $\bar{K} \leq 0$ ,  $M$  a compact manifold of sectional curvature satisfying  $K \leq \bar{K}$ . If  $\dim \bar{M} < \dim M$ ,  $M$  cannot immerse into  $\bar{M}$ .  $\lrcorner$

**Lemma (16.3.5.16) [Klingenberg Lemma].** Let  $M$  be a complete manifold of sectional curvature  $K \geq K_0$ , let  $\gamma_0, \gamma_1$  be two homotopic geodesics from  $p$  to  $q$ , then there exists a middle curve  $\gamma_s$  s.t.

$$l(\gamma_0) + l(\gamma_s) \geq \frac{2\pi}{\sqrt{K_0}}.$$

$\lrcorner$

*Proof:* Assume  $l(\gamma_0) < \frac{2\pi}{\sqrt{K_0}}$ , otherwise we are done. Then by Rauch comparison (16.3.5.10), the  $\exp_p : T_p M \rightarrow M$  has no critical point in the open ball  $B$  of radius  $\pi/\sqrt{K_0}$ . Now we want to lift  $\gamma_s$  to  $T_p M$ . It is clear that we cannot lift  $\gamma_1$ , because otherwise it is not a curve. Hence for every  $\varepsilon > 0$ , there is a curve  $\alpha_{t(\varepsilon)}$  that can be lifted and has a point with distance smaller than  $\varepsilon$  to the boundary  $\partial B$ , otherwise the  $s$  that can be lifted will be open and closed in  $[0, 1]$ , thus containing 1.

So now if we choose a sequence of lifts curves  $\gamma_s$  converging to the boundary, then  $s$  has a convergent point, then we have  $l(\gamma_0) + l(\gamma_{t_0}) \geq \frac{\pi}{\sqrt{K_0}}$ .  $\square$

**Prop. (16.3.5.17) [Klingenberg].** Let  $M$  be a simply connected compact manifold of dimension  $n \geq 3$  such that  $\frac{1}{4} < K \leq 1$ , then  $i(M)$  (The infimum of distance to the cut locus)  $\geq \pi$ .  $\lrcorner$

**Cor. (16.3.5.18).** If  $M$  is a compact orientable manifold of even dimension satisfying  $0 < K \leq 1$ , then  $i(M) \geq \pi$ .  $\lrcorner$

**Prop. (16.3.5.19) [1/4-pinch Sphere Theorem].** Let  $M$  be a compact simply connected manifold satisfying  $0 < 1/4K_{\max} < K \leq K_{\max}$ , then  $M$  is homeomorphic to a sphere.

(Use Klingenberg Theorem, this is a special case of diameter geodesic sphere theorem). Cf. (16.3.5.29).

It can be shown that in this case, this sphere is even diffeomorphic to  $S^n$  using Ricci flow.  $\lrcorner$

**Remark (16.3.5.20).**  $0 < 1/4K_{\max} < K$  cannot be changed to  $\geq$ . In fact, the Funibi-Study metric on  $CP^n$  has sectional curvature  $1 \geq K \geq 4$ . Cf. ??  $\lrcorner$

$\text{Hess}\rho(X, Y)$  where  $\rho$  is the distance to a fixed point, is important.

**Prop. (16.3.5.21).**  $\text{Hess}\rho(X, Y)$  is positive definite on the tangent space of the geodesic sphere within the injective radius, and its principal value is  $|\frac{J'}{J}|$  for a Jacobi field in that direction. And it is zero on the normal direction.

So there would be a Riccati comparison theorem on the eigenvalue of  $\Pi_2 : \lambda' \leq -K - \lambda^2, \text{Hess}(\rho)$  is bounded.  $\lrcorner$

*Proof:* Notice that

$$\text{Hess}\rho(X, Y) = (\nabla_X \text{grad}\rho, Y) = XY\rho - (\nabla_X Y)\rho$$

so if choose a normal geodesic  $\gamma$  of initial vector  $X$ , then

$$\begin{aligned} \text{Hess}\rho(X, X) &= X\langle \dot{\gamma}, d\rho \rangle - (\nabla_X \dot{\gamma})\rho = X\langle \dot{\gamma}, d\rho \rangle = \langle \dot{\gamma}, d\langle \dot{\gamma}, d\rho \rangle \rangle = E''(0) \\ &= I_q(X, X) = ((\nabla_{\dot{\gamma}} X)(q), X(q)) = \frac{\langle J', J \rangle}{|J|^2} \end{aligned}$$

□

**Prop. (16.3.5.22) [Toponogov].** Let  $M$  be a complete manifold with  $K \geq H$ .

If a hinge satisfies  $\gamma_1$  is minimal and  $\gamma_2 \geq \frac{\pi}{\sqrt{H}}$  if  $H > 0$ ., then on  $M^H$  the same hinge has smaller distance of endpoints than this hinge

┘

*Proof:* Cf. [Cheeger Comparison Theorems in Riemannian Geometry P42]. And there is another triangle version: For a minimal geodesic triangle, the comparison triangle has smaller angles. NOTE this theorem cannot be derived from Rauch Comparison Theorem. □

### Critical Point for Distance Function

**Prop. (16.3.5.23).** The critical point for distance function on a complete manifold is that for every direction  $v$ , there is a minimal geodesic  $\gamma$  s.t.  $\langle \gamma'(l), v \rangle \leq \frac{\pi}{2}$ .

The set of regular point is open and there exists a smooth gradient like vector field (i.e. acute angle with every minimal geodesic) on this open subset .

┘

**Prop. (16.3.5.24) [Berger's Lemma].** A maximal point for the distance function is a critical point. ┘

*Proof:* If not, choose a convergent point  $v$  of the minimal geodesics with endpoint in a curve of that direction, then exp near  $v$  will generate a Jacobi field with endpoint Jacobi is the sam of that direction. So the distance will increase by  $\cos \theta$  along that direction, contradiction. □

**Prop. (16.3.5.25) [Soul Lemma].** Let  $M$  is a Riemannian manifold and  $A$  is a closed submanifold. If  $\text{dist}(A, -)$  has no critical point on  $D(A, R) \setminus A$ , then  $B(A, R)$  is diffeomorphic to the normal bundle of  $A \rightarrow M$ .

┘

*Proof:*  $A$  has a normal exp radius  $\epsilon$ , and we can vary the gradient-like vector field to be identical to the normal vector near  $A$ , and use Morse lemma (the flow) to get a diffeomorphism. □

**Cor. (16.3.5.26) [Disk Theorem].** If  $A$  is a point then  $M$  is diffeomorphic to a disk. ┘

**Lemma (16.3.5.27) [Generalized Schoenflies Theorem].** Easy to do, just use the fact that exp is continuous to find a boundary sphere depending continuously on the direction (both  $p$  and  $q$ ). ┘

**Prop. (16.3.5.28) [Sphere Theorem].** If  $M$  is a closed manifold and has a distance function with only one critical point (the furthest one), then  $M$  is homeomorphic to a twisted ball. ┘

*Proof:* There exists a  $\epsilon$  and  $r$  that  $B(q, \epsilon)$  and  $B(p, r)$  covering  $M$ , (Use the convergent point argument). Then use the generalized Schoenflies theorem. □

**Prop. (16.3.5.29) [Diameter Sphere Theorem].** If a closed manifold  $M$  satisfies  $\text{sec } M \geq K > 0$ , and  $\text{diam}(M) > \frac{\pi}{2\sqrt{K}}$ , then  $M$  is homeomorphic to  $S^n$ . ┘

*Proof:* First, if there are two maximal distance point, then use Toponogov to show contradiction. Second, at other points  $x$ ,

$$\angle pqx > \frac{\pi}{2}$$

(Regular domain) because of Toponogov and The formula

$$\cos \tilde{\alpha} = \frac{\cos l - \cos l_1 \cos l_2}{\sin l_1 \sin l_2}.$$

So the geodesic direction  $\vec{xq}$  will serve as a geodesic-like vector field (might need paracompactness).  
□

**Prop. (16.3.5.30) [Critical Principle].** In a complete manifold  $M$  of sectional curvature  $> K$ , if  $q$  is a critical point of  $p$ , then for any point  $x$  with  $d(p, x) > d(p, q)$  and any minimal geodesic from  $p$  to  $x$ , the  $\angle xpq$  is smaller than the  $\cosh_K^{-1}(\frac{d(p, x)}{d(p, q)})$ . ┘

*Proof:* Use Toponogov for the hinge  $xpq$ . Then notice that there is a different minimal geodesic from  $p \rightarrow q$  that makes the  $\angle pqx < \pi/2$  by the definition of critical point, thus there is another Toponogov inequality, this two inequality contradicts. □

**Cor. (16.3.5.31).** For a complete open manifold whose  $K$  are lower bounded, then it is homeomorphic to the interior of a manifold with boundary. (Use Soul lemma, otherwise there will be a sequence of critical point whose angles are big). ┘

**Prop. (16.3.5.32).** ray construction and Line construction ? ┘

**Prop. (16.3.5.33) [Soul Theorem].** If  $M$  is an open manifold with  $K \geq 0$ , then there is a totally geodesic submanifold  $S$  that  $M$  is diffeomorphic to the normal bundle over  $S$ . ┘

*Proof:* Use the ray construction to get a totally convex compact subset, hence it is a manifold or with boundary, if it has boundary, then find to set of maximal distance to the distance to boundary, the distance to the boundary is a convex function, so it is a smaller totally geodesic manifold. So a  $S$  without boundary must exist and this constitutes a stratification, all the level set is strongly convex. Thus all point outside  $S$  is not critical, hence the soul lemma applies. Cf.[GeJian Comparison theorems in Riemannian Geometry Lecture7]. □

**Prop. (16.3.5.34) [Perelman].** There is a distance non-increasing contraction unto the soul, and it must be just the projection along the normal bundle. Moreover, for any geodesic on the soul and a parallel vector field in the normal bundle along it, it spans a flat surface (by Rauch comparison). ┘

**Cor. (16.3.5.35) [Soul Conjecture].** For an open(non-compact) complete manifold  $M$  with  $K \geq 0$ , if it has a point  $p$  s.t. sectional curvature at  $p$  are all positive, then  $M$  is diffeomorphic to  $\mathbb{R}^n$ . (It's enough to show that its soul is a point, otherwise for any point, it must has a direction that is flat,  $K = 0$ ). ┘

## 6 Curvature Inequalities and Topology

### Sectional Curvature

**Prop. (16.3.6.1) [Hadamard theorem].**  $M$  a complete simply connected Riemann manifold of sectional curvature  $\leq 0$ , then  $\exp_p : T_p M \rightarrow M$  is an isomorphism of  $M$  to  $\mathbb{R}^n$ . (negative sectional curvature to show  $\exp$  is a local isomorphism, complete to show it is a covering map) ┘

**Prop. (16.3.6.2) [Liouville Theorem].** Any conformal mapping for an open subset of  $\mathbb{R}^n, n > 2$  is restriction of a composition of isometry, dilations and/or inversions, at most once.  $\lrcorner$

**Prop. (16.3.6.3) [Positive Curved, Closed Geodesic not Minimal].** If  $M$  is an even dimensional orientable Riemannian manifold with positive sectional curvature, let  $\sigma : [0, 1] \rightarrow M$  be a closed geodesic curve, then there exists an  $\varepsilon > 0$  that parametrized closed curves  $F; [0, 1] \times (-\varepsilon, \varepsilon) \rightarrow M$  near  $\sigma$  with lengths less than that of  $\sigma$ .  $\lrcorner$

*Proof:* Cf. [Solution to Yau Test Geometry Individual 2013 Prob5].  $\square$

**Prop. (16.3.6.4) [Synge].**  $f$  is an isometry of a compact oriented manifold  $M^n$  of positive sectional curvature,  $f$  alter orientation by  $(-1)^n$ , then  $f$  has a fixed pt.  $\lrcorner$

*Proof:* Cf. [Do Carmo P203].  $\square$

**Cor. (16.3.6.5).**  $M$  a compact manifold of positive sectional curvature, then

1. If  $M$  is orientable and  $n$  is even, then  $M$  is simply connected. So If  $M$  is compact and even dimension, then  $\pi(M) = 1$  or  $\mathbb{Z}_2$ .
2. If  $n$  is odd, then  $M$  is orientable.

(Use the universal cover and covering transformation.)  $\lrcorner$

**Conj. (16.3.6.6) [Hopf Conjecture].** If  $M$  is a compact Riemannian manifold of even dimension that  $K > 0$ , then it has positive Euler characteristic.  $\lrcorner$

### Morse Index

**Prop. (16.3.6.7) [Index Lemma].** Among the piecewise differentiable vector fields along a geodesic without conjugate point or without focal point, with initial value 0 and fixed end value, the Jacobi field attain minimum of the index form:

$$I_a(V, V) = \int_0^a \{ \langle DV(t), DV(t) \rangle - \langle R(\dot{\gamma}, V)\dot{\gamma}, V \rangle \} dt.$$

$\lrcorner$

**Cor. (16.3.6.8).**  $I_l(J, J) = \langle J, J' \rangle(l)$  for a Jacobi field.  $\lrcorner$

**Prop. (16.3.6.9).** a focal point is a critical value of  $\exp^\perp$ . For an embedded manifold, the focal point equals  $x + 1/t\eta$ , where  $\eta$  is a vertical vector and  $t$  is a principal value of  $S_e ta$ .  $\lrcorner$

**Prop. (16.3.6.10) [Morse Index theorem].** The index of the the index form  $I_a(V, W)$  on the space of vector fields 0 at the endpoints, equal to the number of points conjugate to  $\gamma(0)$  in  $[0, a)$ .  $\lrcorner$

**Cor. (16.3.6.11).** If  $\gamma$  is minimizing,  $\gamma$  has no conjugate points on  $(0, a)$ ,  $\gamma$  has a conjugate point, it is not minimizing.  $\lrcorner$

**Prop. (16.3.6.12) [Morse].** If  $M$  is complete with non-negative sectional curvature, then  $\pi_1(M)$  have no finite non-trivial cyclic group and  $\pi_k(M) = 0$ .  $\lrcorner$

*Proof:* because universal cover of  $M$  is contractible, so the higher homotopy group vanish and  $H^k(M) = H^k(\pi_1(M))$ , so if a subgroup is finite cyclic, its homology is periodic, contradiction.  $\square$

**Prop. (16.3.6.13) [Preissman].** For a compact manifold with  $K < 0$ , any nontrivial abelian subgroup of  $\pi_1$  is infinite cyclic.  $\lrcorner$

**Prop. (16.3.6.14).** If  $M$  is compact and  $K < 0$ ,  $\pi_1(M)$  is not abelian.  $\lrcorner$

Assuming  $M$  complete,

- The cut point of  $p$  along  $\gamma$  is the maximum  $\gamma(t)$  s.t.  $d(p, \gamma(t)) = t$ . It is either the first conjugate point of  $p$  or the intersection of two minimizing geodesics.
- Conversely, if a point is a conjugate point of  $p$  or is intersection of two geodesics of equal length, then there is a cut point before it. So, if intersection of two minimizing geodesics happens, it must happen before the occurrence of conjugate point.
- thus the cut point relation is reflexive, and if  $q \in M \setminus C_m(p)$ , then there exists a unique minimizing geodesic joining  $p$  and  $q$ .
- $M \setminus C_m(p)$  is homeomorphic to an open ball through  $\exp$ .
- the distance of  $p$  to the cut locus is continuous, thus  $C_m(p)$  is closed.
- If  $M$  is complete and there is a  $p$  which has a cut point for every geodesic, then  $M$  is compact.
- for  $q$  the closest of  $C_m(p)$  to  $p$ , either there exists a minimizing geodesic and  $q$  is conjugate to  $p$  or there is to minimizing geodesic connecting at  $q$ .

**Prop. (16.3.6.15).** The index of a geodesic will decrease when transferred to a manifold of smaller sectional curvature  $K$ .  $\lrcorner$

**Prop. (16.3.6.16).** In a complete manifold, if there is a sequence of points  $\{p_i\}$  converging to a point  $p$ , choose for each point a minimal geodesic, then a subsequence of them will converge to a minimal geodesic to  $p$ .  $\lrcorner$

*Proof:* The convergence is by smoothness and of  $\exp$  and Hadamard. The minimality is by comparing distance.  $\square$

### Ricci Curvature

**Prop. (16.3.6.17) [Ricci Comparison].** Volume comparison, Laplacian Comparison, Mean Curvature comparison. Cf. [葛健 Week13].  $\lrcorner$

**Prop. (16.3.6.18) [Bishop-Gromov].** Let  $M$  be an open manifold with  $\text{Ric} \geq H$ , let  $\tilde{M}(H)$  be a complete simply connected manifold of constant sectional curvature  $H$ , then

$$\text{Vol}(B_r(x)) \leq \text{Vol}(B_r(\tilde{p})), \quad \frac{\text{Vol}(B_R(x))}{\text{Vol}(B_r(x))} \leq \frac{\text{Vol}(B_R(\tilde{p}))}{\text{Vol}(B_r(\tilde{p}))}.$$

Cf. [葛健 Week13].  $\lrcorner$

**Prop. (16.3.6.19) [Bonnet-Myer].**  $M$  a complete manifold of Ricci curvature  $\text{Ric}_p(v) \geq \frac{1}{r^2}$ , Then  $M$  is compact and have diameter  $\leq \pi r$ .

And if the identity is achieved,  $M \cong \mathbb{S}^n$ .  $\lrcorner$

*Proof:* Use Laplacian comparison  $\Delta r \leq (n-1) \cot r$ . Cf. [葛健 week13].  $\square$

**Cor. (16.3.6.20) [Positive Ricci Finite Fundamental Groups].**  $M$  is a complete manifold of Ricci curvature  $\geq \delta > 0$ , then the universal cover is compact thus  $\pi_1(M)$  is finite. This can be seen as an obstruction for a compact manifold to have positive Ricci curvature.  $\lrcorner$

**Cor. (16.3.6.21) [Calabi-Yau].** For an open manifold with non-negative Ricci curvature, for any point,  $\text{Vol}(B(p, r)) \geq c_p r$ .  $\lrcorner$

**Prop. (16.3.6.22) [Milnor].** Let  $M$  be an open manifold of non-negative Ricci curvature of dimension  $n$ , then any f.g. subgroup of  $\pi_1(M)$  has polynomial growth  $\leq n$ . Milnor conjectured that  $\pi_1(M)$  in fact is f.g..  $\lrcorner$

**Prop. (16.3.6.23) [First Betti Number Theorem].** There is a number  $f(n, \lambda, D)$ ,  $f(n, 0, D) = n$ ,  $f(n, \lambda, D) = 0$  for  $\lambda > 0$  that for a manifold of diameter  $\leq D$  and Ricci curvature  $\geq \lambda$ ,  $b_1(M) \leq f(n, \lambda, D)$ .  $\lrcorner$

**Cor. (16.3.6.24) [Splitting Theorem].** The universal cover of a compact Riemannian manifold with non-negative Ricci curvature splits isometrically as a product  $\widetilde{M} = N \times \mathbb{R}^k$  where  $N$  is a compact manifold manifold.  $\lrcorner$

### Scalar Curvature

## 16.4 Smooth Manifolds II: Differential Topology

This section uses the tool of differential forms to analyze algebraic topology.

References are [Differential Forms in Algebraic Geometry Bott-Tu] and [G-P74].

**Notation(16.4.0.1).**

- Use notations on smooth manifolds defined in [Smooth Manifolds I: Generalities](#).

┘

### 1 Basics

**Prop.(16.4.1.1)[Cohomological Generator of Sphere].** Let  $v : x \mapsto x/|x|$  be the outward-pointing vector field on  $\mathbb{R}^n \setminus \{0\}$ , then the differential form

$$v \lrcorner dm = \frac{1}{|x|} \sum x_i (-1)^{i-1} dx_1 \wedge dx_2 \wedge \dots \widehat{dx_i} \wedge \dots dx_n$$

restricts to a differential form on  $S^{n-1}$  that is a generator of  $H^{n-1}(\mathbb{S}^{n-1})$ .

┘

*Proof:* First we calculate  $d(\iota(v)(dm)) = \operatorname{div}(v)dm = \frac{n-1}{|r|}dm$  (16.3.4.27). Consider using Stoke's formula:

$$\int_{\partial B_1} \iota(v)(dm) - \int_{\partial B_\varepsilon} \iota(v)(dm) = \int_{B(0,1) - B(0,\varepsilon)} \frac{n-1}{|r|} dm = 0.$$

Letting  $\varepsilon \rightarrow 0$ ,  $\int_{\partial B_\varepsilon} \iota(v)(dm)$  converges to 0 as  $|\iota(v)(dm)|$  is bounded and  $V(B_\varepsilon)$  converges to 0. And using the polar coordinate  $dm = r^{n-1}drd\omega$ , the right hand side is just

$$\int_{\mathbb{S}^{n-1}} \int_0^1 (n-1)r^{n-2}drd\omega = \operatorname{Vol}(\mathbb{S}^{n-1}).$$

□

**Prop.(16.4.1.2)[Degree Formula].** For any  $f : X \rightarrow Y \in \mathcal{M}\text{ani}_{\text{cpct,orntd}}^k$  and any  $k$ -form  $\omega \in \Omega^k(Y)$ , we have

$$\int_X f^* \omega = \deg(f) \int_Y \omega.$$

┘

*Proof:*  $\deg$  is defined in (16.1.4.17). Cf.[Pollack P188] ?

□

**Prop.(16.4.1.3)[Hopf Invariant].** For  $n \in \mathbb{Z}_{\geq 2}$  and any map  $\mathbb{S}^{2n-1} \rightarrow \mathbb{S}^n$ , let  $\alpha$  be a generator of  $H^n(\mathbb{S}^n)$ , then  $f^* \alpha = d\omega$  on  $\mathbb{S}^{2n-1}$  for some  $\omega$ . Define the **Hopf invariant** of  $f$  to be  $\operatorname{Hopf}(f) = \int_{\mathbb{S}^{2n-1}} \omega \wedge d\omega$ , then:

- The definition of Hopf invariant is independent of  $\omega$  chosen.
- For  $n \in 2\mathbb{Z} + 1$ , the Hopf invariant is 0.
- Homotopic maps  $f, g$  have the same Hopf invariant.

┘

*Proof:* 1: If  $d\omega = d\omega'$ , then

$$\int_{\mathbb{S}^{2n-1}} \omega' \wedge d\omega' - \int_{\mathbb{S}^{2n-1}} \omega \wedge d\omega = \int_{\mathbb{S}^{2n-1}} (\omega' - \omega) \wedge d\omega = \pm \int_{\mathbb{S}^{2n-1}} d((\omega - \omega') \wedge \omega) = 0.$$



2: If  $n$  is odd, then  $\omega$  is of even dimensional, thus  $\omega \wedge d\omega = \frac{1}{2}d(\omega \wedge \omega)$ , so  $H(f) = 0$  by Stokes.

3: If  $F : S^{2n-1} \times I \rightarrow S^n$  is a homotopy of  $f, g$ , then  $F^*\alpha = d\omega$  for some  $\omega$  on  $S^{2n-1} \times I$ . Thus consider

$$H(f) - H(g) = \int_{S^{2n-1}} \omega_1 \wedge d\omega_1 - \int_{S^{2n-1}} \omega_0 \wedge d\omega_0 = \int_{\partial(S^{2n-1} \times I)} \omega \wedge d\omega = \int_{S^{2n-1} \times I} d\omega \wedge d\omega,$$

But  $d\omega \wedge d\omega = F^*(\alpha \wedge \alpha)$ , and  $\alpha \wedge \alpha = 0$ . □

## 2 Morse Theory(Milnor)

References are [Mil63] and [Supersymmetry and Morse Theory, Witten].

**Def. (16.4.2.1) [Non-Degenerate Critical Point].** For  $X \in \text{Diff}$ ,  $f \in C^\infty(X)$ , a critical point of  $f$  is called a **non-degenerate critical point** iff the Hessian matrix is non-singular at  $x$ .

The notion of non-degenerate critical is independent of the coordinate chosen. ┘

*Proof:* Cf.[Pollack P42]. □

**Prop. (16.4.2.2) [Non-Degeneracy is Generic].** Non-degenerate critical points are the general situation in the following sense: For a manifold  $M \subset \mathbb{R}^n$ , for any smooth function  $f$  on  $M$ , consider the functions  $f_a = f + \sum a_i x_i$ , then for almost all  $(a_i)$ , all critical points of  $f_a$  is non-degenerate. ┘

*Proof:* Cf.[Pollack P43] □

**Prop. (16.4.2.3) [Morse Lemma].** In a non-degenerate critical point of  $f$ , there is a coordinate that

$$f = f(p) + x_1^2 + \cdots + x_{n-\lambda}^2 - y_1^2 - \cdots - y_\lambda^2.$$

┘

*Proof:* By induction on  $n$ , it suffices to show the following: If  $f$  is a smooth function on  $\mathbb{R}^n$  s.t.  $f(0) = 0$  and the origin is a non-degenerate critical point of  $f$ , then there is a local coordinate system near 0 s.t.  $f(x_1, \dots, x_n) = \pm x_1^2 + f(0, x_2, \dots, x_n)$ .

By linear algebra, after a linear change of variable, we may assume that the quadratic part of  $f$  is of the form  $\pm x_1^2 + Q(x_2, \dots, x_n)$ . Then consider the function  $g(x) = \frac{\partial}{\partial x_1} f(x)$ , then the linear part of  $g$  at the origin is  $\pm 2x_1$ . So by implicit function theorem(12.3.4.2),  $\{g(x) = 0\}$  is a smooth hypersurface near the origin with the tangent plane at the origin given by the equation  $x_1 = 0$ . Then by Taylor's expansion, for any  $y = (x_2, \dots, x_n)$ ,

$$f(u, y) = f(0, y) + \frac{\partial}{\partial u} f(0, y) x_1 + h(x_1, y) x_1^2 = f(0, y) + h(x_1, y) x_1^2,$$

where  $h$  is a smooth function near the origin with  $h(0, 0) = \pm 1$ . Thus with a change of variable  $\tilde{x}_1 = \sqrt{|h(x_1, y)|} x_1$  and  $y$  unchanged, we may assume that  $h = \pm 1$ . So  $f(x_1, y) = \pm u^2 + f(0, y)$ , as desired. □

**Prop. (16.4.2.4).** If  $M \in \text{Diff}$ ,  $f \in C^\infty(M)$  s.t.  $f^{-1}([a, b])$  is compact and with no critical points of  $f$ , then  $F^{-1}(a)$  is a deformation retracts of  $f^{-1}(b)$  using the flow  $\text{grad} f / |\text{grad} f|^2$ . ┘

*Proof:* □

**Prop. (16.4.2.5) [Morse Main Lemma].** If  $f$  is a smooth function with  $p$  a non-degenerate critical point and  $\lambda$  downward pointing direction. If for some  $f^{-1}([c - \epsilon, c + \epsilon])$  is compact, then  $M^{c+\epsilon}$  is homotopic to  $M^{c-\epsilon}$  gluing a  $\lambda$  dimensional cell.  $\lrcorner$

*Proof:* Cf.[Milnor Prop3.2].  $\square$

**Prop. (16.4.2.6).** For an embedded manifold and almost all point  $p$ , the distance to  $p$  is a morse function. (Use Sard theorem and degenerate  $\iff p$  is a focal point.  $\lrcorner$

**Cor. (16.4.2.7).** smooth manifold has CW type; on a compact manifold any vector field with discrete singular points has its index sum equal to  $\chi(M)$  (Hopf-Rinow), and there exists one.  $\lrcorner$

**Prop. (16.4.2.8).** for  $\Omega(p, q)^c$  the path space of energy  $< c$ , the piecewise geodesic path space  $B$  (piece fixed), the energy function is smooth and  $B^a$  is compact and is the deformation contraction of  $\text{int}\Omega^a$  for  $a < c$ .  $E$  has the same critical point and same index and nullity on  $B$  and  $\Omega^c$ . (Just geodesicize any path in  $\Omega$ .

So for two point not conjugate in  $B^a$ ,  $\Omega^a$  has a finite CW complex type and a  $\lambda$ -dimensional cell for every geodesic of index  $\lambda$  in  $B^a$ .  $\lrcorner$

**Prop. (16.4.2.9)[Morse Main Theorem].** If  $p$  and  $q$  are not conjugate along any geodesic, then  $\Omega(p, q)$  has a countable CW complex type and has a  $\lambda$ -cell for every geodesic of index  $\lambda$ .

If  $M$  has nonnegative Ricci curvature, then  $M$  has only finite cell for every dimension.  $\lrcorner$

*Proof:* Cf.[Mil63]Prop17.3.  $\square$

**Cor. (16.4.2.10).** The path space homotopy type only depend on the homotopy type of  $M$  (use the two homotopy to id to get a composition of homotopy of the two path space), so one can get the information of path space of  $M$  by looking at the homotopy type of  $M$ .  $\lrcorner$

**Prop. (16.4.2.11) [Minimal Geodesics].** If  $p, q$  in a complete manifold  $M$  has distance  $\sqrt{d}$  and the minimal geodesics form a topological manifold, and if all non-minimal geodesic has index  $\geq \lambda$ , then for  $0 \leq i < \lambda$ ,  $\pi_i(\Omega, \Omega^d) = 0$ .  $\lrcorner$

**Lemma (16.4.2.12).** In  $SU(2m)$ , the minimal geodesic from  $I$  to  $-I$  is homeomorphic to Grassmannian  $G_m(\mathbb{C}^{2m})$  and non-minimal geodesic has index  $\geq 2m + 2$ .

Similarly, The space of minimal geodesic from  $I$  to  $-I$  in  $O(2m)$  is homeomorphic to the space of complex structures in  $\mathbb{R}^{2m}$ , and any non-minimal geodesic has index  $\geq 2m - 2$ .  $\lrcorner$

*Proof:* Cf.[Milnor Morse Theory Lemma23.1 Lemma24.4].  $\square$

**Lemma (16.4.2.13).**  $\Omega_{k+1}$  is homotopic to the space of minimal geodesics in  $\Omega_k$  from  $J$  to  $-J$ . (The same way, calculate the index of geodesics from  $J$  to  $-J$  and use (16.4.2.11)). Cf.[Milnor Morse Theory Prop24.5] for definition of  $\Omega_{k+1}$ .  $\lrcorner$

## 16.5 Lie Groups

*“In these days the angel of topology and the devil of abstract algebra fight for the soul of every individual discipline of mathematics.”*

–H. Weyl

References are [?Kna96, Lee13, ?Eti21]. For classical groups, see [Classical Groups](#).

**Notation(16.5.0.1).**

- Let  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ .
- Let  $G \in \mathcal{LieGrp}/\mathbb{K}$  with Lie algebra  $\mathfrak{g}$ (16.5.1.1).

┘

### 1 Basics

**Def.(16.5.1.1)[Lie Groups].** A **Lie group** is a group object in the category of smooth  $\mathbb{K}$ -manifolds. Notice it suffices to check that multiplication is smooth. The left and right translations  $L_g, R_g$  are all smooth morphisms hence diffeomorphisms.

A **homomorphism of Lie groups** is a smooth morphism that is also a group homomorphism. By translation invariance, a group homomorphism always has constant ranks, so a homomorphism of Lie groups that is a bijection is an isomorphism by global rank theorem(16.1.1.19).

The tangent space of  $G$  at  $e$  is denoted by  $\mathfrak{g}$ .  $\mathfrak{g}$  is naturally an Lie algebra?.

┘

*Proof:* We show if a topological group  $G$  that is a smooth manifold satisfies  $m : G \times G \rightarrow G$  is smooth, then  $G$  is a smooth manifold: consider the map  $F : G \times G \rightarrow G \times G : (g, h) \mapsto (g, gh)$ , it is a smooth map that is bijective.

The tangent map of  $F$  at  $(g, h)$  is  $(X, Y) \mapsto (X, (dR_h)_g(X) + (dL_g)_h(Y))$ , which is surjective because  $L_g$  is a diffeomorphism by(16.1.1.19). Then  $F^{-1} : G \times G \rightarrow G \times G : (g, h) \mapsto (g, g^{-1}h)$  is smooth, and  $g \mapsto g^{-1}$  is smooth.  $\square$

**Prop.(16.5.1.2).** A connected Lie group is automatically second countable.

┘

*Proof:* This follows from the fact that a connected Lie group is a manifold hence locally second-countable and it is a union of products of a nbhd of  $e$ (4.12.1.3).  $\square$

**Prop.(16.5.1.3).** Any homomorphism of smooth manifolds has constant rank.

┘

*Proof:*  $F$  being a homomorphism means that  $F \circ L_g = L_{F(g)} \circ F$ . Taking derivative and noticing the fact  $L_g, L_{F(g)}$  are diffeomorphisms shows  $dF_{g_0}$  and  $dF_e$  have the same rank for any  $g$ .  $\square$

**Prop.(16.5.1.4)[Discrete Subgroups].** Any discrete subgroup  $\Gamma \leq G$  is a closed Lie subgroup of dimension 0.

┘

*Proof:* Firstly  $\Gamma$  is countable: Let  $U$  be a nbhd of  $e$  containing no other points, choose another nbhd of  $e$  that  $VV \subset U$ , then  $\{gV\}_{g \in \Gamma}$  is a family of disjoint open subsets of  $G$ , so there are countably many because  $G$  is second countable. Secondly  $\Gamma$  is closed in  $G$ , because

$\Gamma$  is closed in  $G$ : Let  $U$  be a nbhd of  $e$  containing no other points, choose another nbhd of  $e$  that  $VV \subset U$ , then  $\{gV\}_{g \in \Gamma}$  is a family of disjoint open subsets of  $G$  each containing an element of  $\Gamma$ . Then it is clear  $\Gamma$  is closed. Then(16.5.1.22) shows  $\Gamma$  is a closed Lie subgroup of dimension 0.  $\square$

**Def. (16.5.1.5) [Adjoint Representation].** The conjugation map  $C_g : G \rightarrow G : h \mapsto ghg^{-1}$  is a Lie group homomorphism. Let  $\text{Ad}(g) : \mathfrak{g} \rightarrow \mathfrak{g}$  denote its derivative, then  $\text{Ad} : g \mapsto \text{Ad}(g)$  is an action of  $G$  on  $\mathfrak{g}$ . Because  $C_{g_1g_2} = C_{g_1}C_{g_2}$ ,  $\text{Ad}(g_1g_2) = \text{Ad}(g_1)\text{Ad}(g_2)$ . This action is called the **adjoint representation** of  $G$ .  $\lrcorner$

**Prop. (16.5.1.6).** Suppose  $G$  is connected and  $\Gamma \subset G$  is a discrete normal subgroup, then  $\Gamma$  is in the center of  $G$ .  $\lrcorner$

*Proof:* For  $\gamma \in \Gamma$ , consider the map  $G \rightarrow G : g \mapsto g\gamma g^{-1}$ , then it is a map with images in  $\Gamma$ . But  $\Gamma$  is discrete, so its image must be a single point, which is  $\gamma$  because  $e\gamma e^{-1} = \gamma$ . This means  $\gamma$  is in the center of  $G$ .  $\square$

**Prop. (16.5.1.7) [Aut( $\mathfrak{g}$ )].** For  $\mathfrak{g} \in \text{LieAlg}^{\text{fin}}$ ,  $\text{Aut}(\mathfrak{g})$  is a closed Lie subgroup of  $\text{GL}(\mathfrak{g})$ , and its Lie algebra is  $\text{Der}(\mathfrak{g})$ . Denote  $\text{Int}(\mathfrak{g}) = \text{Der}(\mathfrak{g})^0$ .  $\lrcorner$

*Proof:* For the Lie algebra, it suffices to show for  $A \in \text{End}(\mathfrak{g})$ ,  $[e^{tA}X, e^{tA}Y] = e^{tA}[X, Y]$  iff  $A([X, Y]) = [A(X), Y] + [X, A(Y)]$ .

One direction is by taking derivative w.r.t.  $t$ , for the other, we can show  $y_1(t) = [e^{tA}X, e^{tA}Y]$  and  $y_2(t) = e^{tA}[X, Y]$  both satisfy the ordinary differential equation  $y_i'(t) = Ay_i(t)$ .  $\square$

### Exponential Map

**Def. (16.5.1.8) [One-Parameter Subgroup].** A **one-parameter subgroup** of  $G$  is a Lie group homomorphism  $\mathbb{K} \rightarrow G$ .  $\lrcorner$

**Prop. (16.5.1.9).** Let  $X \in \mathfrak{g}$ , then there exists a unique morphism of Lie groups  $\gamma : \mathbb{K} \rightarrow G$  that  $\gamma'(0) = X$ .  $\lrcorner$

*Proof:* If  $\gamma$  is such a group homomorphism, then  $\gamma(s+t) = \gamma(s)\gamma(t)$ . Differentiating for  $s$ , we get

$$\gamma'(t) = d(L_{\gamma(s)})_0(X).$$

Thus  $\gamma$  is an integral curve of the left-invariant vector field on  $G$  corresponding to  $X$  (16.5.3.3), which is unique.

Now to construct such a homomorphism, we use ODE theorems to construct a  $\gamma$  satisfying this for  $|t| < \varepsilon$ , and then check  $\gamma(s+t) = \gamma(s)\gamma(t)$  because they are both integral curves for  $t$  starting at  $\gamma(s)$ . In particular,  $d(L_{\gamma(t)})(X) = d(R_{\gamma(t)})(X)$ . Now we can extend this  $\gamma$  to whole of  $\mathbb{K}$  by defining  $\gamma(2^n s) = \gamma(s)^{2^n}$ , then we check by induction on  $n$  that

$$\gamma'(t) = \frac{1}{2} (d(R_{\gamma(\frac{t}{2})})\gamma'(\frac{t}{2}) + d(L_{\gamma(\frac{t}{2})})\gamma'(\frac{t}{2}))(X) = d(R_{\gamma(\frac{t}{2})})d(R_{\gamma(\frac{t}{2})})(X) = d(R_{\gamma(t)})(X).$$

$\square$

**Cor. (16.5.1.10) [One-Parameter Subgroup and Lie Algebras].** One-parameter subgroups of  $G$  correspond to maximal integral curves of left invariant vector fields starting at  $e$ . In particular, the one-parameter subgroups of  $G$  corresponds to  $\mathfrak{g}$  and also  $T_e(G)$ .

Also, the flow of the right-invariant vector field  $R_X$  is given by  $(g, t) \mapsto \exp(tX)g$ , and the flow of the left-invariant vector field  $L_X$  is given by  $(g, t) \mapsto g\exp(tX)$ .  $\lrcorner$

**Def. (16.5.1.11) [Exponential Map].** Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ , then we can define an **exponential map**  $\exp : \mathfrak{g} \rightarrow G$  that for any  $X \in \mathfrak{g}$ ,  $\exp(X) = \gamma_X(1)$ , where  $\gamma_X$  is the one-parameter subgroup of  $G$  generated by  $X$  (16.5.1.10). It can be shown that  $\gamma(sX)$  is the one-parameter subgroup of  $G$  generated by  $X$ .  $\lrcorner$

**Prop. (16.5.1.12) [Properties of Exponential Maps].**

1.  $\exp : \mathfrak{g} \rightarrow G$  is a smooth map which is a local diffeomorphism near  $e$  that  $\exp(0) = e$ ,  $\exp_* = \text{id}_{\mathfrak{g}}$ .
2.  $\exp(s + t) = \exp(s)\exp(t)$  for  $s, t \in \mathbb{K}$ .
3. For any group homomorphism  $\varphi : G \rightarrow H$  and  $X \in \mathfrak{g}$ ,  $\varphi(\exp(X)) = \exp(\varphi_*(X))$ .
4.  $g \exp(tX) g^{-1} = \exp(\text{Ad}(g)X)$ . Also,  $\text{Ad}_* = \text{ad}$ , or equivalently by item3,  $\text{Ad}(\exp(X)) = \exp(\text{ad}(X))$  as operators.
5. If we identify  $\mathfrak{gl}(n; \mathbb{K})$  with  $\text{GL}(n; \mathbb{K})$ , then we can check directly that the exponential map of  $\text{GL}_n(\mathbb{K})$  is

$$\exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k.$$

$\lrcorner$

*Proof:* 1: The smoothness follows from the smoothness of the solution of ODE. The smooth inverse theorem shows it is a local diffeomorphism at identity.

2: trivial.

3: Both  $\varphi(\exp(tX))$  and  $\exp(t\varphi_*(X))$  are integral curves of the vector field  $L_{\varphi_*(X)}$  and have the same initial point.

4: The first assertion is just item3 applied to the conjugate action  $C_g$ . For the Lie algebra homomorphism,

$$d(\text{Ad})(X)Y = \frac{\partial}{\partial s} \frac{\partial}{\partial t} \exp(sX) \exp(tY) \exp(-sX) = [X, Y].$$

by (16.5.1.14).  $\square$

**Prop. (16.5.1.13).** Let  $\mathfrak{g} = A \oplus B$  a decomposition into subspaces, then the map

$$F : A \oplus B \rightarrow G : (X, Y) \mapsto \exp(X) \exp(Y)$$

is a local diffeomorphism at  $0 \in \mathfrak{g}$ .  $\lrcorner$

*Proof:* Identify  $\mathfrak{g}$  and the tangent space of  $G$  at  $e$ , then the differential  $F$  is identity, so it is a local diffeomorphism.  $\square$

**Thm. (16.5.1.14) [Baker-Campbell-Hausdorff].** For  $X, Y \in \mathfrak{g}$ ,

$$\exp(tX) \exp(tY) = \exp\left(\sum_{n=1}^{\infty} \frac{t^n}{n!} \mu_n(X, Y)\right).$$

where  $\mu_n(X, Y)$  can be written as  $\mathbb{Q}$ -Lie polynomials of  $X$  and  $Y$  that is invariant of  $G$ .

In particular,  $\mu_1(X, Y) = X + Y$ ,  $\mu_2(X, Y) = \frac{1}{2}t^3([X, [X, Y]] + [Y, [Y, X]])$  and so on.  $\lrcorner$

*Proof:* By the Lie correspondence(16.5.3.15), we can first assume that  $G$  is simply-connected, then there is a mapping of  $G$  onto some subgroup of  $GL(n, \mathbb{K})$  with discrete kernel. If we can prove the formula for  $G = GL(n, \mathbb{K})$ , then

$$\exp\left(\sum_{n=1}^{\infty} \frac{t^n}{n!} \mu_n(X, Y)\right) \exp(tY)^{-1} \exp(tX)^{-1}$$

is contained the kernel, but it is a smooth function in  $t$ , and its value is 1 for  $t = 0$ , thus it holds for any  $t$ .

Now let  $T\mathbb{K}^2 = \mathbb{K}\langle x, y \rangle$  be the free non-commutative algebra in variables  $x, y$ , the series  $\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  can be viewed as an element in  $\widehat{\mathbb{K}\langle x, y \rangle}$ . Then we can define

$$\mu = \log(\exp(x) \exp(y)) \in \widehat{\mathbb{K}\langle x, y \rangle},$$

where  $\log(A) = -\sum_{n=1}^{\infty} \frac{(1-A)^n}{n!}$ . Then  $\mu = \sum_{n=1}^{\infty} \frac{\mu_n}{n!}$ , where  $\mu_n \in \mathbb{K}\langle x, y \rangle$  are polynomials in  $x, y$  of degree  $n$  with coefficients in  $\mathbb{Q}$ .

Then it remains to show that  $\mu_n$  can be written as Lie polynomials in  $x, y$ . Notice  $\Delta(x) = x \otimes 1 + 1 \otimes x$ , thus  $\Delta(\exp(x)) = \exp(x) \otimes \exp(x)$ , thus  $\Delta(\exp(x) \exp(y)) = \exp(x) \exp(y) \otimes \exp(x) \exp(y)$ , and

$$\begin{aligned} \Delta(\log(\exp(x) \exp(y))) &= \log(\Delta(\exp(x) \exp(y))) = \log((\exp(x) \exp(y) \otimes 1)(1 \otimes \exp(x) \exp(y))) \\ &= \log(\exp(x) \exp(y)) \otimes 1 + 1 \otimes \log(\exp(x) \exp(y)). \end{aligned}$$

then by separating degrees, each  $\mu_n$  is primitive(3.11.2.4), thus they are contained in the free Lie-algebra generated by  $x, y$ , by(3.7.8.16) and(3.7.8.19).

The calculation is invariant of  $n$  in  $GL(n; \mathbb{R})$ , thus it is invariant of  $G$ . □

**Cor. (16.5.1.15).** Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ ,  $X, Y \in \mathfrak{g}$ , then

$$\lim_{n \rightarrow \infty} \left( \exp\left(\frac{1}{n}X\right) \exp\left(\frac{1}{n}Y\right) \right)^n = \exp(X + Y).$$

┘

*Proof:*

$$\left( \exp\left(\frac{1}{n}X\right) \exp\left(\frac{1}{n}Y\right) \right)^n = \left( \exp\left(\frac{1}{n}(X + Y) + O\left(\frac{1}{n^2}\right)\right) \right)^n = \exp\left(X + Y + O\left(\frac{1}{n}\right)\right).$$

Taking  $n \rightarrow \infty$ , we get the desired result. □

**Prop. (16.5.1.16).** Let  $G, H \in \mathcal{LieGrp}/\mathbb{K}$ , then

- A continuous homomorphism  $\gamma : \mathbb{K} \rightarrow H$  is smooth.
- A continuous homomorphism between smooth Lie groups  $F : G \rightarrow H$  is smooth.
- There is at most one smooth structure on  $G$  that makes it a Lie group.

┘

*Proof:* 1: Let  $V$  be a nbhd of  $0 \in \mathfrak{h}$  that  $\exp$  is a diffeomorphism on  $2V$ (16.5.1.12). Choose  $t_0$  small that  $\gamma(t) \in \exp(V)$  for any  $|t| \leq t_0$ , and let  $X \in V$  that  $\exp(X) = \gamma(t_0)$ , then we can show  $\gamma(t) = \exp(tX)$  for any  $t = \frac{m}{2^n}$ , so this holds for any  $t$  by continuity/anlyticity, and  $\gamma$  is smooth.

2: By the proof of 1, we can construct a map(not necessary continuous)  $F_* : \mathfrak{g} \rightarrow \mathfrak{h}$  with commutative diagram

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{F_*} & \mathfrak{h} \\ \downarrow \exp & & \downarrow \exp \\ G & \xrightarrow{F} & H \end{array}$$

Now using (16.5.1.15) and the continuity of  $F$ , we can show  $F_*$  is linear:

$$\begin{aligned} \exp(F_*(X + Y)) &= F(\exp(X + Y)) = F\left(\lim_{n \rightarrow \infty} \exp\left(\frac{1}{n}X\right) \exp\left(\frac{1}{n}Y\right)\right)^n \\ &= \lim_{n \rightarrow \infty} \left(\exp\left(\frac{1}{n}F_*X\right) \exp\left(\frac{1}{n}F_*Y\right)\right)^n = \exp(F_*(X + Y)) \end{aligned}$$

Thus  $F$  is smooth at a nbhd of  $G$ , and smooth everywhere by translation.  $\square$

### Group Aspects

**Def. (16.5.1.17) [Lie Subgroup].** A Lie subgroup of  $G$  is a subgroup that is also an immersed submanifold (16.1.1.21).

An **embedded Lie subgroup** of  $G$  is a subgroup that is also an immerse submanifold (16.1.1.21).

┘

**Prop. (16.5.1.18) [Lie subgroup is Weakly Embedded].** Any Lie subgroup of  $G$  is weakly embedded.

┘

*Proof:* Cf. [Lee13] P506.  $\square$

**Lemma (16.5.1.19).** Let  $H \leq G$  be a subgroup that is also an embedded submanifold, then  $H$  is an embedded Lie subgroup.  $\square$

*Proof:* We need to show the multiplication and inverse on  $H$  is smooth:  $H \times H \rightarrow G$  is smooth and has image in  $H$ , thus  $H \times H \rightarrow H$  is also smooth, by (16.1.1.30).  $\square$

**Lemma (16.5.1.20).** Let  $H \leq G$  be a Lie subgroup that is also an embedded submanifold, then  $H$  is closed in  $G$ .  $\square$

*Proof:* Assume  $H$  is an embedded submanifold of  $G$ , then it is locally compact in the induced topology, so (12.10.1.6) shows  $H$  is closed in  $G$ .  $\square$

**Lemma (16.5.1.21).** Let  $H \leq G$  that is also a closed subset of  $G$ , then  $H$  is an embedded Lie subgroup.  $\square$

*Proof:* By (16.5.1.20), it suffices to show that  $H$  is an embedded submanifold of  $G$ . Let  $\mathfrak{g}$  be the Lie algebra of  $G$ , and define a subspace  $\mathfrak{h} \subset \mathfrak{g}$  that  $\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H, \forall t \in \mathbb{R}\}$ . By (16.5.1.15) and the fact  $H$  is closed in  $G$ ,  $\mathfrak{h}$  is a linear subspace of  $\mathfrak{g}$ .

Next we show that there exists a nbhd  $U$  of  $0 \in \mathfrak{g}$  that  $\exp$  is a diffeomorphism and also  $\exp(U \cap \mathfrak{h}) = \exp(U) \cap H$ : Let  $U$  be any open nbhd of  $0 \in \mathfrak{g}$  that  $\exp$  is a diffeomorphism  $U \rightarrow \exp(U)$ , then  $\exp(U \cap \mathfrak{h}) \subset \exp(U) \cap H$  by definition.

Let  $\mathfrak{b} \subset \mathfrak{g}$  be chosen s.t.  $\mathfrak{h} \oplus \mathfrak{b} = \mathfrak{g}$ , then  $F : \mathfrak{h} \oplus \mathfrak{b} \rightarrow G : (X, Y) \mapsto \exp(X) \exp(Y)$  is a local diffeomorphism. Choose nbhd  $U$  of  $0 \in \mathfrak{g}$  and  $\tilde{U}$  of  $0 \in \mathfrak{h} \oplus \mathfrak{b}$  that both  $\exp|_U$  and  $F|_{\tilde{U}}$  are diffeomorphisms, and choose a countable nbhd basis  $\{U_i\}$  of  $0 \in \mathfrak{g}$ . Denote  $V_i = \exp(U_i)$  and  $\tilde{U}_i = F^{-1}(V_i)$ , then  $V_i$  is a nbhd basis of  $e \in G$  and  $\tilde{U}_i$  is a nbhd basis of  $0 \in \mathfrak{h} \oplus \mathfrak{b}$ . We may assume  $U_i \subset U$  and  $\tilde{U}_i \subset \tilde{U}$ .

If  $\exp(U_i \cap \mathfrak{h}) \subset \exp(\tilde{U}_i) \cap H$  for any  $i$ , then we can choose  $h_i = \exp(Z_i) \in H$  that  $h_i \notin \exp(U_i \cap \mathfrak{h})$ . Because  $\exp(U_i) = F(\tilde{U}_i)$ , set  $h_i = \exp(X_i) \exp(Y_i)$ , where  $(X_i, Y_i) \in \tilde{U}_i$ . Now  $Y_i \neq 0$ , otherwise  $\exp(Z_i) = \exp(X_i)$ , which implies  $Z_i = X_i$  and  $h \in \exp(U_i \cap \mathfrak{h})$ . Notice  $\exp(Y_i) = \exp(X_i)^{-1} h_i \in H$ .

Because  $\tilde{U}_i$  is a basis of  $\mathfrak{h} \oplus \mathfrak{b}$ ,  $Y_i \rightarrow 0$ . Choose an inner product on  $\mathfrak{b}$ , and let  $c_i = |Y_i|$ , then  $c_i^{-1} Y_i$  lies on the unit sphere of  $\mathfrak{b}$ . Replacing by a subsequence, we can assume  $c_i^{-1} Y_i \rightarrow Y$  for some  $Y \in \mathfrak{b}$ . Then  $|Y| = 1$  by continuity.

For any  $t \in \mathbb{R}$ , let  $n_i = \lfloor \frac{t}{c_i} \rfloor$ , then  $|nc_i - t| \leq c_i \rightarrow 0$ , which means  $n_i Y_i \rightarrow tY$ , so  $\exp(n_i Y_i) \rightarrow \exp(tY)$ . But  $\exp(n_i Y_i) = \exp(Y_i)^{n_i} \in H$ , so  $\exp(tY) \in H$  because  $H$  is closed. Thus  $Y \in \mathfrak{h}$ , contradiction.

Thus in this way we can construct a slice chart  $\varphi$  of  $H$  at  $e$ , and for any  $h \in H$ , because

$$L_h((\exp(U) \cap H)) = L_h(\exp(U)) \cap H,$$

$\varphi \circ L_{h^{-1}}$  is a slice chart of  $H$  at  $h$ . Thus  $H$  is an embedded submanifold of  $G$  by (16.1.1.24).  $\square$

**Prop. (16.5.1.22) [Closed Subgroup Theorem].** Let  $H \leq G$  be a subgroup, then the following are equivalent:

- $H$  is closed in  $G$ .
- $H$  is an embedded submanifold.
- $H$  is an embedded Lie subgroup.

┘

*Proof:*  $3 \rightarrow 2$  is trivial,  $3 \rightarrow 1$  is (16.5.1.20).  $1 \rightarrow 3$  is (16.5.1.21),  $2 \rightarrow 3$  is (16.5.1.19).  $\square$

**Remark (16.5.1.23).** A dense line of the torus is a Lie subgroup that is not a closed Lie subgroup.  $\perp$

**Lemma (16.5.1.24) [One-Parameter subgroup of Subgroups].** Let  $H \leq G$  be a Lie subgroup, then the one-parameter subgroups of  $H$  are exactly those of  $G$  with initial velocity in  $T_e(H)$ .  $\perp$

*Proof:* This is because a one-parameter subgroup of  $H$  is naturally a one-parameter subgroup of  $G$ , and two one-parameter subgroups with the same initial velocity is identical.  $\square$

**Prop. (16.5.1.25).** Let  $H \leq G$  be a Lie subgroup, then the exponential map of  $H$  is the exponential map of  $G$  restricted to  $\mathfrak{h}$ , and

$$\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H, \forall t \in \mathbb{R}\}.$$

┘

*Proof:* The first assertion is an immediate corollary of (16.5.1.24). Now if  $X \in \mathfrak{h}$ , then the first assertion shows  $\exp(tX) \in H$  for all  $t$ . Conversely, if  $\exp(tX) \in H$  for all  $t$ , then  $\exp(tX)$  is a smooth map to  $H$ , by (16.5.1.18), thus its derivative at  $e$  is in  $\mathfrak{h}$ , which means  $X \in \mathfrak{h}$ .  $\square$

**Prop. (16.5.1.26) [Semidirect Product].** Let  $G, H \in \text{LieGrp}/\mathbb{K}$  and  $\tau$  is an action of  $G$  on  $H$ , then the Lie algebra of  $G \ltimes_{\tau} H$  is  $\mathfrak{g} \ltimes_{d\tau} \mathfrak{h}$ .  $\perp$

*Proof:* Notice the differential of the action  $\tau(g)$  on  $H$  defines a map  $G \rightarrow GL(\mathfrak{h})$ , and then the differential of this map gives a map  $d\tau : \mathfrak{g} \rightarrow \text{Der}(\mathfrak{h})$ , so we can form  $\mathfrak{g} \ltimes_{d\tau} \mathfrak{h}$  (3.7.1.6).

For the rest, Cf. [Knapp, P60].  $\square$



**Prop. (16.5.1.27).** Let  $G, H$  be simply-connected Lie groups with Lie algebra  $\mathfrak{g}, \mathfrak{h}$ , and let  $\pi : \mathfrak{g} \rightarrow \text{Der}(\mathfrak{h})$  be a Lie algebra homomorphism, then there is an action  $\tau$  of  $G$  on  $H$  by automorphisms that  $d\tau = \pi$ , and  $G \ltimes_{\tau} H$  is a simply-connected Lie group with Lie algebra  $\mathfrak{g} \ltimes_{\pi} \mathfrak{h}$ .  $\lrcorner$

*Proof:* Cf.[Knapp, P60].  $\square$

**Prop. (16.5.1.28)[Quotient Theorem for Lie Groups].** Let  $H \leq G$  be a normal closed Lie subgroup, then the quotient  $G/H$  is a Lie group, and the quotient map  $\pi$  is a surjective Lie group homomorphism with kernel  $H$ .  $\lrcorner$

*Proof:* By(16.5.2.6),  $G/H$  is a smooth manifold that the quotient map is surjective, smooth, and is a group homomorphism with kernel  $H$ . It suffices to show the multiplication of  $G/H$  is smooth, which is easy by(16.1.1.15).  $\square$

**Prop. (16.5.1.29)[First Isomorphism Theorem for Lie Groups].** Let  $\varphi : G \rightarrow H \in \text{LieGrp}$ , then there kernel of  $F$  is an closed Lie subgroup of  $G$  with Lie algebra  $\ker(\varphi_*)$ . The image of  $\varphi$  has a unique smooth structure making it a Lie subgroup of  $H$  that  $G/\ker(F) \rightarrow \text{Im}(\varphi)$  is a diffeomorphism, and it is a closed Lie subgroup when it is embedded in  $H$ , e.g. when  $\varphi$  induces a proper action(16.5.2.3).  $\lrcorner$

*Proof:* This follows from(16.5.2.2) and(16.5.1.22).  $\square$

**Def. (16.5.1.30)[Adjoint Group].** The center  $Z(G)$  of  $G$  is a closed Lie subgroup because it the kernel of  $\text{Ad}$ (16.5.1.29). We call the group  $G/Z(G)$  the **adjoint group** of  $G$ , which is an immersed subgroup of  $GL(\mathfrak{g})$  by(16.5.1.29).  $\lrcorner$

### Lie Groups and Analytic Groups

**Prop. (16.5.1.31).** What condition makes a Lie group a complex Lie group?  $\lrcorner$

**Prop. (16.5.1.32).** Any connected Lie group has a compact subgroup as deformation retraction.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.5.1.33) [Gleason; Montgomery-Zippin].** A real topological group  $G$  admits a (unique) Lie group structure iff the underlying topological space  $G$  is a topological manifold.  $\lrcorner$

*Proof:* Cf.<https://terrytao.wordpress.com/2011/06/17/hilberts-fifth-problem-and-gleason-metrics/>,  $\square$

**Prop. (16.5.1.34)[Lie Groups and Analytic Groups].** A real Lie group admits a unique real analytic structure, So it is not important to distinguish between a Lie group and an analytic Lie group, and call it **analytic group** if it is a connected Lie group. The uniqueness follows from(16.5.1.16).  $\lrcorner$

*Proof:* Use(16.5.1.14) to show that in a local exp char  $(U, \varphi)$  of 1,  $U \times U \rightarrow U$  is analytic, then we can choose an analytic atlas on  $G$  given by  $\{(gU, \varphi \circ L_{g^{-1}})\}$ . This is an analytic atlas, because the transition function is  $\varphi L_{g^{-1}h} \varphi^{-1}$  on  $U \cap h^{-1}gU$ . Because  $hU \cap gU \neq \emptyset$ , let  $x = hu_1 = gu_2$ , then  $L_{g^{-1}h} = L_{u_2u_1^{-1}}$ , which is analytic.

To show multiplication is analytic w.r.t. this atlas: ?  $\square$

**Cor. (16.5.1.35).** The proof above can be used to show that any  $C^k$ -group manifold be upgraded uniquely to a Lie group structure. So basically the study of  $C^0$ -Group manifold and analytic groups are the same.  $\lrcorner$

## 2 Homogeneous Spaces

### Actions of Lie Groups

**Prop. (16.5.2.1) [Fundamental Theorem on Lie Group Actions].** Let  $\theta$  be a right smooth action of  $G$  on a smooth manifold  $M$ , then we can define a complete Lie algebra homomorphism

$$\tilde{\theta} : \mathfrak{g} \rightarrow \mathfrak{X}(M) : \theta(X)_p = \frac{d}{dt}\bigg|_{t=0} p \cdot \exp(tX) = d(\theta^{(p)})_e(X_e),$$

called the **infinitesimal generator** of  $\theta$ . where a Lie homomorphism  $\tilde{\theta} : \mathfrak{g} \rightarrow \mathfrak{X}(M)$  is called complete iff for any  $X \in \mathfrak{g}$ ,  $\tilde{\theta}(X)$  is a complete vector field (16.1.5.3).

Conversely, if  $G$  is simply-connected and  $\tilde{\theta} : \mathfrak{g} \rightarrow \mathfrak{X}(M)$  is a complete Lie algebra homomorphism, then there exists a unique smooth right action  $\theta$  of  $G$  on  $M$  with infinitesimal generator  $\tilde{\theta}$ .  $\lrcorner$

*Proof:*  $\theta(X)$  is smooth because it is the infinitesimal generator of the smooth flow  $\mathbb{R} \times M \rightarrow M : (t, p) \mapsto p \exp(tX)$  (16.1.5.4).  $\tilde{\theta}$  is a Lie algebra homomorphism by (16.1.2.18).?  $\square$

**Prop. (16.5.2.2) [Isotropy Group and Orbits].** Let  $\theta$  be a smooth action of  $G$  on a smooth manifold  $M \in \text{Diff}$ , let  $\theta_x : \mathfrak{g} \rightarrow T_x M$  be given by  $\tilde{\theta}_x(X) = \tilde{\theta}(X)_x$ . Then

- The stabilizer  $G_x$  is a closed subgroup of  $G$  with Lie subalgebra  $\mathfrak{g}_x = \ker(\tilde{\theta}_x)$ .
- $Gx$  has a unique smooth structure making it an immersed subgroup of  $M$  that  $G/G_x \rightarrow Gx$  is a diffeomorphism, and  $T_x(Gx) \cong \text{Im}(\tilde{\theta}_x) \cong \mathfrak{g}/\mathfrak{g}_x$ .

$\lrcorner$

*Proof:* Cf. [?Eti21]P48.  $\square$

**Prop. (16.5.2.3).** If  $G$  acts smoothly and properly on a smooth manifold  $M \in \text{Diff}$ , then each orbit is a closed submanifold of  $M$ , and each isotropy group is compact, by (4.12.1.15) and (4.12.1.17).  $\lrcorner$

**Prop. (16.5.2.4) [Quotient Map Theorem].** Let  $G$  act smoothly, freely and properly on a smooth manifold  $M \in \text{Diff}$ , then the quotient space  $M/G$  is a topological manifold with dimension  $\dim M - \dim G$ , and it has a unique smooth structure that  $M \rightarrow M/G$  is a smooth submersion.  $\lrcorner$

*Proof:* Cf. [Lee13]P544.  $\square$

### Homogeneous Spaces and Fiber Bundles

**Def. (16.5.2.5) [Homogeneous Spaces].** A **homogeneous space** for  $G$  is a smooth manifold  $M \in \text{Diff}$  with a smooth transitive  $G$ -action.  $\lrcorner$

**Prop. (16.5.2.6) [Characterizing Homogeneous Spaces].**

- If  $H$  is a closed subgroup of  $G$ , then the left coset space  $G/H$  is a topological manifold of dimension  $\dim G - \dim H$ , and has a unique smooth structure that the quotient map  $G \rightarrow G/H$  is a smooth submersion. With this smooth structure, the left action of  $G$  on  $G/H$  turns it into a homogeneous space.
- If  $M$  is a homogeneous  $G$ -space, and  $p \in M$ , then the isotropy group  $G_p$  of  $M$  is a closed subgroup of  $G$ , and  $G/G_p \rightarrow M$  is a diffeomorphism of  $G$ -spaces.

$\lrcorner$

*Proof:* Cf. [Lee13]P551, P552.  $\square$

**Cor.(16.5.2.7) [Homogeneous Space Structure on Sets].** Suppose  $X$  is a set with a transitive action of  $G$  that for some point  $p \in X$ , the isotropy group  $G_p$  is closed in  $G$ . Then  $X$  has a unique smooth manifold structure with respect to which the given action is smooth. With this structure,  $\dim X = \dim G - \dim G_p$ .  $\lrcorner$

*Proof:* This is because  $G/G_p$  is a smooth manifold that is  $G$ -equivariantly isomorphism to  $X$ , and the uniqueness also follows from the proposition.  $\square$

**Prop.(16.5.2.8) [Quotients of Lie Groups by Discrete Subgroups].** Let  $\Gamma \leq G$  be a discrete subgroup, then  $G/\Gamma \in \text{Diff}$ , and the quotient map  $G \rightarrow G/\Gamma$  is a smooth normal covering.  $\lrcorner$

*Proof:* (16.5.1.4) and the proof of (16.5.2.6) shows  $\Gamma$  acts smoothly, freely, and properly on  $G$  on the right. Then the theorem is a consequence of (16.1.1.17).  $\square$

**Prop.(16.5.2.9) [Contractible Homogeneous Space].** If  $X$  is a homogeneous  $G$ -manifold that is contractible,  $x \in X$ , then  $G$  is diffeomorphic to  $G_x \times X$ .  $\lrcorner$

*Proof:*  $\square$

**Prop.(16.5.2.10) [Orientability of Homogenous Spaces].** Let  $X$  be a homogeneous manifold of  $G$ . Let  $x \in X$  and  $H = \text{Stab}_G(x)$ .

1.  $X$  is orientable iff values of  $\text{Ad}(h_i)$  are of the same sign for any two element  $h_1, h_2 \in H$  lying in a same connected component of  $G$ .
2. Show that when  $H$  is connected,  $X$  is orientable.
3. There exist a  $G$ -invariant volume form on  $X$  iff  $\text{Ad}(H) \subset \text{SL}(\mathfrak{g}/\mathfrak{h})$ .

$\lrcorner$

**Prop.(16.5.2.11) [Fiber Bundle of Homogenous Spaces].** Let  $G \in \text{LieGrp}_{\text{cpct}}$ ,  $K \leq H \leq G$  be closed subgroups, then the map  $G/K \rightarrow G/H$  is a  $G$ -locally trivial fiber bundle.  $\lrcorner$

*Proof:* ?  $\square$

**Prop.(16.5.2.12) [Examples of Homogeneous Spaces].** Let  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ .

- The **Grassmannian manifold**  $\text{Gr}(k, \mathbb{K}^n)$  is defined to be the set of  $k$ -dimensional spaces in  $\mathbb{K}^n$ .  $U(n, \mathbb{K})$  acts transitively on  $\text{Gr}(k, \mathbb{K}^n)$ , and the stabilizer is  $U(k, \mathbb{K}) \times U(n-k, \mathbb{K})$ . Thus  $\text{Gr}(k, \mathbb{K}^n)$  is a homogeneous  $U(n, \mathbb{K})$ -space by (16.5.2.7), and  $U(n, \mathbb{K}) \rightarrow \text{Gr}(k, \mathbb{K}^n)$  is a fiber bundle with fiber  $U(k, \mathbb{K}) \times U(n-k, \mathbb{K})$  by (16.5.2.11).
- The **Stiefel manifold**  $V_k(\mathbb{K}^n)$  of orthonormal  $k$ -frames in  $\mathbb{K}^n$  is defined to be the set of tuples  $(v_1, \dots, v_n)$  in  $\mathbb{K}^n$  that  $(v_i, v_j) = \delta_{ij}$ .  $U(n, \mathbb{K})$  acts transitively on  $V_k(\mathbb{K}^n)$  with stabilizer  $U(n-k, \mathbb{K})$ , Thus  $V_k(\mathbb{K}^n)$  is a homogeneous  $U(n, \mathbb{K})$ -space by (16.5.2.7), and  $U(n, \mathbb{K}) \rightarrow V_k(\mathbb{K}^n)$  is a fiber bundle with fiber  $U(n-k, \mathbb{K})$ , and  $V_k(\mathbb{K}^n) \rightarrow \text{Gr}(k, \mathbb{K}^n)$  is a fiber bundle with fiber  $U(k, \mathbb{K})$ .
- The **flag variety**. ?

$\lrcorner$

*Proof:* 1: Let  $\dim X = n$ .  $X$  is orientable iff there is a non-vanishing  $n$ -form on  $X$ . Because  $o_x : G \rightarrow X$  is submersive, choose  $n$  left invariant vector fields  $X_1, \dots, X_n$  that  $(do_x)_e(v_i)$  generate  $T_x(X)$ , then by homogeneity,  $(do_x)_g(X_{ig})$  generate  $T_g(X)$  for any  $g \in G$ . Thus  $X$  is orientable iff

there is a  $n$ -form  $\omega$  on  $G$  that satisfies  $r_h^* \omega = \omega$ ,  $\omega_g = \Delta(g)l_{g^{-1}}^* \omega_e$  for some  $\Delta(g) \in \mathbb{R}^*$ , and for any  $h \in H$ , and  $\omega(X_1, \dots, X_n) \neq 0$ .

If such a form exists, then  $\Delta(gh)l_{(gh)^{-1}}^* \omega_e = r_h^* l_{g^{-1}}^* \omega_e$ , which is equivalent to  $\Delta(g)\text{Ad}(h)^* \omega_e = \Delta(gh)\omega_e$  for any  $h \in H$ , thus values of  $\text{Ad}(h_i)$  are of the same sign for any two element  $h_1, h_2 \in H$  lying in a same connected component of  $G$ .

Conversely, if values of  $\text{Ad}(h_i)$  are of the same sign for any two element  $h_1, h_2 \in H$  lying in a same connected component of  $G$ , then we can define  $\omega$  as above, where  $\Delta(h) = \det(\text{Ad}(h)|(\mathfrak{g}/\mathfrak{h}))$ , and extend  $\Delta$  to  $G$  that satisfy  $\Delta(gh) = \Delta(g)\Delta(h)$ . This can be because  $G$  is a fiber bundle over  $G/H$ .

2: When  $H$  is connected, clearly the values of  $\text{Ad}(H)$  are of the same sign.

3: The proof is the same as that of item1. In this case, all  $\Delta(g) = 1$ , thus the existence of  $G$ -invariant volume form on  $X$  is equivalent to  $\text{Ad}(H) \subset SL(\mathfrak{g}/\mathfrak{h})$ .  $\square$

### 3 Lie Theory

#### Lie Algebras of Lie Groups

**Def.(16.5.3.1) [Invariant Vector Fields].** A smooth vector field  $X \in \mathfrak{X}(G)$  is called left-invariant if  $d(L_g)_{g'}(X_{g'}) = X_{gg'}$  for any  $g, g' \in G$ . The set of left-invariant vector fields on  $G$  is denoted by  $\text{Lie}(G)$ .  $\lrcorner$

**Prop.(16.5.3.2).** If  $X, Y$  are left-invariant vector fields over  $G$ , then  $[X, Y]$  is also left-invariant, by (16.1.2.18).  $\lrcorner$

**Prop.(16.5.3.3) [Invariant Vector Fields and Tangent Spaces].** Let  $G$  be a Lie group, then the evaluation map  $\text{Lie}(G) \rightarrow \mathfrak{g} : X \mapsto X_e$  is a vector space isomorphism.  $\lrcorner$

*Proof:* The inverse map is given by  $X \mapsto (L_X)_g = d(L_g)_e(X)$ . This is clearly a left-invariant vector field. It suffices to show that this  $\tilde{X}$  is smooth. By (16.1.2.10), it suffices to show  $L_X(f)$  is smooth for any smooth function  $f$ .

$$L_X(f)(g) = d(L_g)_e(X)(f)(g) = X(L_g f)(0) = \left( \frac{d}{dt} f(g \exp(tX)) \right) \Big|_{t=0}$$

which is a differential of a smooth map  $\mathbb{R} \times G \rightarrow \mathbb{R} : (t, g) \mapsto f(g \exp(tX))$ , so it is smooth in  $g$ .  $\square$

**Cor.(16.5.3.4) [Lie Group is Parallelizable].**  $G$  admits a left-invariant smooth global frame, thus in particular parallelizable.  $\lrcorner$

**Prop.(16.5.3.5).** If  $X \in \mathfrak{g}$  corresponds to a left-invariant vector field  $L_X$ , then for any  $g \in G$ ,

$$d(R_g)_p((L_X)_p) = (L_{\text{Ad}(g^{-1})X})_{pg}.$$

$\lrcorner$

*Proof:*

$$\begin{aligned} d(R_g)_p((L_X)_p)(f)(pg) &= (L_X)_p(R_g f) = d(L_p)_e(X)(R_g f) \\ &= X(L_p R_g(f)) = \left( \frac{d}{dt} (f(p \exp(tX)g)) \right) \Big|_{t=0} \end{aligned}$$

$$\begin{aligned}
&= \left( \frac{d}{dt} f \left( p \exp(t \operatorname{Ad}(g^{-1})X) \right) \right) \Big|_{t=0} \\
&= \operatorname{Ad}(g^{-1}X)(L_p(f)) \\
&= (L_{\operatorname{Ad}(g^{-1})X})_{pg}(f)(pg)
\end{aligned}$$

□

**Def. (16.5.3.6) [Lie Algebra of a Lie Group].**  $\operatorname{Lie}(G) \in \operatorname{LieAlg}$  w.r.t. the Lie bracket (16.1.2.15). It is called the **Lie algebra associated to  $G$** . ┘

*Proof:* This is clear from the definition  $[X, Y](f) = XY(f) - YX(f)$ . □

**Prop. (16.5.3.7) [Induced Map of Lie Algebras].** Any homomorphism  $f : G \rightarrow H$  induces a homomorphism of their Lie algebras via the tangent space. ┘

*Proof:* For any  $X \in \operatorname{Lie}(G)$ , define  $f_*(X)_g = (dL_g)_e(df_e(X_e))$ , then this is a left-invariant vector field on  $H$ , and it clearly corresponds to the tangent map via isomorphism in (16.5.3.3). This map is a Lie algebra map by a variant of (16.1.2.18). □

**Cor. (16.5.3.8).** If  $H \leq G$  is a Lie subgroup, then there is a natural isomorphism

$$\mathfrak{h} \cong \{X \in \operatorname{Lie}(G) \mid X_e \in T_e H\}.$$

In particular, the tangent space  $\mathfrak{h}$  of  $H$  is a Lie subalgebra of  $\mathfrak{g}$ . ┘

*Proof:* There is a commutative diagram □

$$\begin{array}{ccc}
\operatorname{Lie}(H) & \xrightarrow{X \mapsto X_e} & \mathfrak{h} \\
\downarrow \iota_* & & \downarrow (\iota_*)_e \\
\operatorname{Lie}(G) & \xrightarrow{X \mapsto X_e} & \mathfrak{g}
\end{array}$$

**Prop. (16.5.3.9) [Covering of Lie Groups].** For  $F : G \rightarrow H \in \operatorname{LieGrp}_{\text{cnd}}$ , the following are equivalent:

- $F$  is surjective with discrete kernel.
- $F$  is a smooth covering map.
- $F$  is a local diffeomorphism.
- The induced homomorphism  $F_* : \mathfrak{g} \rightarrow \mathfrak{h}$  is an isomorphism.

┘

*Proof:* 1  $\rightarrow$  2:  $F$  is surjective thus  $H$  is a homogeneous  $G$ -space, so (16.5.2.6) shows  $H \cong G/\ker(F)$ . And  $G \rightarrow G/\ker(F)$  is a smooth covering map by (16.5.2.8).

2  $\rightarrow$  3 is trivial.

3  $\rightarrow$  1: If  $F$  is a local diffeomorphism, then  $\ker(F)$  is discrete, and  $F$  is open. Thus  $F(G)$  is an open subgroup of  $H$ , thus all of  $H$  because  $H$  is connected (4.12.1.3).

3  $\rightarrow$  4 is trivial. 4  $\rightarrow$  3 is by inverse function theorem. □

**Cor. (16.5.3.10) [Homomorphism with Discrete Kernel].** Let  $G, H \in \operatorname{LieGrp}$  with  $\dim G = \dim H$ ,  $H$  is connected, and  $f : G \rightarrow H \in \operatorname{LieGrp}$  with discrete kernel, then  $f$  is a covering space map and  $\ker(f) \subset Z(G)$ , by (16.5.1.6). ┘

*Proof:* This homomorphism is locally injective at 1, so it has rank  $\dim G = \dim H$ , so it is local diffeomorphism. In particular, the image contains a nbhd of 1 in  $H$ , thus it contains all  $H$  by (4.12.1.3).  $\square$

**Prop. (16.5.3.11) [Universal Covering Lie Group].** If  $G \in \text{LieGrp}_{\text{cntd}}$ , then its universal covering space  $\tilde{G}$  can be given a Lie group structure that the covering map is a group homomorphism. Moreover, the group structure is unique up to isomorphism over  $G$ . Moreover, the kernel of  $\tilde{G} \rightarrow G$  is a discrete central subgroup of  $\tilde{G}$ .  $\lrcorner$

*Proof:* Because  $\tilde{G}$  is simply connected, so does  $\tilde{G} \times \tilde{G}$ , let  $\tilde{e}$  be an element over  $e$ , we can lift the map  $\tilde{G} \times \tilde{G} \rightarrow G \times G \xrightarrow{m} G$  to a map  $\tilde{m} : \tilde{G} \times \tilde{G} \rightarrow \tilde{G}$  that  $\tilde{m}(\tilde{e}, \tilde{e}) = \tilde{e}$ . Similar we can lift an inverse map  $\tilde{i}$  that  $\tilde{i}(\tilde{e}) = \tilde{e}$ . These maps are smooth because  $\pi : \tilde{G} \rightarrow G$  is a local diffeomorphism.

It's left to show that  $(\tilde{m}, \tilde{i})$  makes  $\tilde{G}$  into a Lie group: For example, the map  $L_{\tilde{e}} : \tilde{G} \rightarrow \tilde{G}$  is a lift of  $\text{id}_G$  and it coincides with  $\text{id}_{\tilde{G}}$  on a point  $\tilde{e}$ , thus it is just  $\text{id}_{\tilde{G}}$ , which means  $\tilde{e}$  is a left identity. The rest is easy.

For the uniqueness: By the universal property of covering, if there are two coverings, we can lift it to a map connecting them that maps identity to identity, then show it is a group homomorphism.

The last assertion follows from (16.5.3.10).  $\square$

**Prop. (16.5.3.12) [Lie Subalgebras and Subgroups].** For any lie subalgebra  $\mathfrak{h} \subset \mathfrak{g}$ , there exists uniquely a connected Lie subgroup  $H \leq G$  s.t.  $\mathfrak{h}$  is the lie algebra of  $H$ .  $\lrcorner$

*Proof:* By (16.1.3.1), there is a maximal connected manifold  $H$  corresponding to  $\mathfrak{h}$ , we only need to show that it is a group. But the left invariance of  $\mathfrak{h}$  shows that  $HH \subset H$  because  $H$  is maximal.

Cf. [Lee13]P506.  $\textcolor{red}{?}$   $\square$

**Cor. (16.5.3.13).** If  $G_1$  is a simply connected Lie group and  $G_2$  is a connected Lie group, then any Lie algebra homomorphism  $\tilde{h} : \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$  can be lifted to a unique Lie group homomorphism.  $\lrcorner$

*Proof:* Consider the image of  $\tilde{h} : \Gamma(\tilde{h}) \subset \mathfrak{g}_1 \times \mathfrak{g}_2$ , which is a Lie subalgebra. First notice a Lie group homomorphism  $h$  is equivalent to a Lie subgroup  $G_h$  of  $G_1 \times G_2$  that  $\pi_1|_{G_h}$  is a diffeomorphism onto  $G_1$ . And this Lie homomorphism induces the desired Lie algebra homomorphism iff the Lie algebra of  $G_h$  is just  $\tilde{h}$ .

(16.5.3.12) shows that there exists a unique Lie group  $G$  in  $G_1 \times G_2$  with Lie subalgebra  $\Gamma(\tilde{h})$ . The projection  $\pi_1|_G$  is a diffeomorphism onto  $G_1$ , because the tangent map at  $e$  is an isomorphism, thus a local diffeomorphism and by (16.5.3.10) a covering map, so it must be an isomorphism because  $G_1$  is simply connected and  $G$  is connected.  $\square$

**Cor. (16.5.3.14) [Representations of Simply-Connected Lie Groups].** The category of representations of simply-connected Lie groups is equivalent to the category of representations of its Lie algebra.  $\lrcorner$

**Thm. (16.5.3.15) [The Lie Correspondence].**

- The category of finite dimensional Lie algebras is equivalent to the category of simply connected Lie groups.
- For a f.d. Lie algebra  $\mathfrak{g}$ , the connected Lie groups with Lie algebras isomorphic to  $\mathfrak{g}$  corresponds to  $G/\Gamma$ , where  $G$  is a simply connected subgroup with Lie algebra  $\mathfrak{g}$ , and  $\Gamma$  is a discrete central subgroup of  $G$ .

$\lrcorner$

*Proof:* 1: By (16.5.3.12)(16.5.3.13) together with Ado's theorem??.

2: (16.5.3.11) and (16.5.3.9) shows any Lie group is a quotient of its universal covering Lie group by a discrete central subgroup. Conversely, for any discrete central subgroup,  $G/\Gamma$  is a Lie subgroup and  $\pi : G \rightarrow G/\Gamma$  is a homomorphism with kernel  $\Gamma$  by (16.5.1.4) and (16.5.1.28), thus  $\text{Lie}(G) = \text{Lie}(G/\Gamma) \cong \mathfrak{g}$  (16.5.1.28).  $\square$

**Prop. (16.5.3.16) [Ideals and Normal Subgroups].** Let  $G$  be a connected Lie group and  $H$  a connected Lie subgroup, then  $H$  is a normal subgroup of  $G$  iff  $\mathfrak{h}$  is an ideal of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Because  $G, H$  are both connected, (4.12.1.3) shows  $H$  is normal in  $G$  iff for any  $X \in \mathfrak{g}, Y \in \mathfrak{h}$ ,  $\exp(X)\exp(Y)\exp(X)^{-1} \in H$ . Taking derivative w.r.t.  $Y$ , this is equivalent to  $d(\text{Ad}(\exp(X)))(Y) = \text{ad}(X)Y \in \mathfrak{h}$  (16.5.1.12), by (16.5.1.25), which is equivalent to  $\mathfrak{h}$  being an ideal of  $\mathfrak{g}$ .  $\square$

**Prop. (16.5.3.17) [Center].** Let  $G$  be a connected subgroup with Lie algebra  $\mathfrak{g}$  and  $Z$  the center of  $G$ ,  $\mathfrak{z}$  the center of  $\mathfrak{g}$ , then  $Z$  is a closed Lie subgroup of  $G$  with Lie algebra  $\mathfrak{z}$ .  $\lrcorner$

*Proof:* Because  $G$  is connected,  $g \in Z$  iff  $g$  commutes with all  $\exp(tX), X \in \mathfrak{g}$ . Thus  $Z = \ker \text{Ad}$ . Now the assertion follows from (16.5.1.29).  $\square$

**Prop. (16.5.3.18) [Chevalley's Theorem].** Let  $G$  be a complex connected Lie group and  $\mathfrak{g}$  the Lie algebra of  $G$ ,  $\mathfrak{h}$  the Cartan subalgebra of  $\mathfrak{g}$ , and  $W$  the Weyl group, then the restriction of functions induces a graded algebra isomorphism

$$\mathbb{C}[\mathfrak{g}]^G \cong \mathbb{C}[\mathfrak{h}]^W.$$

$\lrcorner$

*Proof:* ?  $\square$

### Classifications

**Prop. (16.5.3.19) [Simply-Connected Compact Lie Groups].** Any simply connected compact Lie groups is a product of the following types:

- $\text{Spin}(n)$  for  $n \geq 3$ .
- $\text{SU}(2)$  for  $n \geq 2$ .
- $\text{Sp}(n)$  for  $n \geq 1$ .
- $E_6, E_7, E_8, F_4, G_2$ .

$\lrcorner$

*Proof:* ?  $\square$

## 4 Compact Lie Groups and Representations

**Prop. (16.5.4.1).** Any compact connected complex Lie group is Abelian. And it is a complex tori. So we only consider only compact real Lie groups.  $\lrcorner$

*Proof:* Mimic the proof that Abelian variety is commutative (20.9.1.5), using a similar rigidity lemma.  $\square$

**Cor. (16.5.4.2).**  $U(n)$  is not a complex Lie group, in particular not a complex algebraic variety.  $\lrcorner$



**Prop. (16.5.4.3).** Let  $G$  be a compact connected Lie group with Lie algebra  $\mathfrak{g}$  and center  $Z$ . Let  $G_{ss}$  be the connected subgroup of  $G$  corresponding to the Lie subalgebra  $[\mathfrak{g}, \mathfrak{g}]$  (16.5.3.12), then  $G_{ss}$  has finite center, and  $Z^0, G_{ss}$  are closed in  $G$ , and  $G = Z^0 G_{ss}$ .  $\lrcorner$

*Proof:* Cf. [Kna96] P198.  $\square$

**Prop. (16.5.4.4) [Compact Lie Group and Representations].** A compact topological group  $G$  is a real Lie group iff it has a faithful real f.d. representation. And in this case, it is a closed subgroup of  $U(n)$  for some  $n$ .  $\lrcorner$

*Proof:* If it has a faithful f.d. representation, then  $G \subset GL(n, \mathbb{R})$  compact hence closed, thus a Lie subgroup by (16.5.1.22).

Conversely, if  $G$  is a Lie group, then we can choose a small nbhd  $U$  of  $e \in G$  that contains no non-trivial subgroup of  $G$  (choose  $\exp(\frac{1}{2}V)$  where  $\exp$  is an diffeomorphism on  $V$ ). Consider kernel  $K_\pi$  for irreducible representations  $\pi$  of  $G$ , then  $\cap_\pi K_\pi = \emptyset$  by Gelfand-Raikov (24.7.1.29), in particular  $\cap_\pi (K_\pi - U) = \emptyset$ . But  $G - U$  is compact, hence there are f.m.  $\pi_i$  that  $\cap_i K_{\pi_i} \in U$ , but by definition of  $U$ ,  $\cap_i K_{\pi_i} = \{e\}$ , which gives a f.d. faithful representation of  $G$ , by (24.7.3.5).  $\square$

**Prop. (16.5.4.5) [Reduction to Faithful Representations].** Let  $V$  be a faithful f.d. representation of a compact Lie group  $G$  and  $Y$  is an irreducible f.d. representation of  $G$ , then  $Y$  is a direct summand of  $V^{\otimes n} \otimes V^{*\otimes m}$  for some  $m, n \geq 0$ . Moreover, if  $G$  is unimodular, we can take  $m = 0$ .  $\lrcorner$

*Proof:* Cf. [Etingof, P175].  $\square$

**Remark (16.5.4.6).** Theory of representations of compact Lie groups are a special case of abstract harmonic analysis 12.10.  $\lrcorner$

### Maximal Tori

**Prop. (16.5.4.7) [Tori].** Any connected compact Abelian real Lie group is a torus.  $\lrcorner$

*Proof:* By (16.6.4.1), the exponential map realizes  $\mathfrak{g}$  as the universal cover of  $G$ , and the kernel is then a complete lattice  $\Lambda$  in  $\mathbb{R}^n$ , so  $G \cong \mathbb{R}^n / \Lambda \cong (S^1)^n$ .  $\square$

**Prop. (16.5.4.8).** The maximal tori in a compact Lie group  $G$  corresponds to the maximal Abelian subalgebras of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* Let  $T$  be a maximal tori, then  $\mathfrak{t}$  is maximal Abelian by fundamental theorem of Lie (16.5.3.12). Conversely, if  $\mathfrak{t}$  is a maximal tori, then the corresponding Lie subgroup  $T$  is Abelian, and  $\overline{T}$  is also Abelian. The Lie algebra of  $T, \overline{T}$  are the same by maximality of  $\mathfrak{t}$ , so  $T = \overline{T}$ , and it is a torus by (16.5.4.7), and it is clearly maximal.  $\square$

**Prop. (16.5.4.9).** Let  $G$  be a compact connected Lie group with Lie algebra  $\mathfrak{g}$ , then any two maximal Abelian subalgebra of  $\mathfrak{g}$  are conjugate via  $\text{Ad}(G)$ .  $\lrcorner$

*Proof:* Let  $\mathfrak{t}, \mathfrak{t}'$  be two maximal Abelian subalgebra of  $\mathfrak{g}$ . As  $\mathfrak{g}$  is reductive (3.7.5.3), if we choose  $X \in \mathfrak{t}, X' \in \mathfrak{t}'$  that are not zero for any roots, then  $Z_{\mathfrak{g}}(X) = \mathfrak{t}, Z_{\mathfrak{g}}(X') = \mathfrak{t}'$ . Let  $(\cdot, \cdot)$  be a invariant inner product on  $\mathfrak{g}$  defined in (16.5.7.2), choose a  $g_0 \in G$  that  $(\text{Ad}(g_0)X, X')$  is maximal, then  $0 = ([Z, \text{Ad}(g_0)X, X'] = (Z, [\text{Ad}(g_0)X, X']$  for any  $Z \in \mathfrak{g}$ , so  $\text{Ad}(g_0)X \in Z_{\mathfrak{g}}(X') = \mathfrak{t}'$ . Thus  $\mathfrak{t}' \subset Z_{\mathfrak{g}}(\text{Ad}(g_0)X) = \text{Ad}(g_0)Z_{\mathfrak{g}}(X) = \text{Ad}(g_0)\mathfrak{t}$ , and the equality holds as  $\mathfrak{t}'$  is maximal.  $\square$



**Cor. (16.5.4.10) [Maximal Tori are Conjugate].** Let  $G$  be a compact connected Lie group, then any two maximal tori of  $G$  are conjugate via  $\text{Ad}(G)$ .  $\lrcorner$

**Prop. (16.5.4.11).** Let  $G$  be a compact connected Lie group, then any element of  $G$  is connected in a maximal torus.  $\lrcorner$

*Proof:* By (16.6.4.1), this element is contained in a 1-parameter subgroup  $T$  of  $G$ , and  $\bar{T}$  is also Abelian, so is a torus by (16.5.4.7). Then choose a maximal torus containing  $\bar{T}$ .  $\square$

**Cor. (16.5.4.12).**  $Z(G)$  is contained in any maximal torus, by (16.5.4.10).  $\lrcorner$

**Prop. (16.5.4.13).** Let  $S$  be a torus in a compact Lie group  $G$  and  $g \in G$  that commutes with elements in  $S$ , then there is a torus containing both  $S$  and  $g$ .  $\lrcorner$

*Proof:* Let  $A$  be the closure of  $\cup_{i \in \mathbb{Z}} g^i S$ , and  $A_0$  the identity component of  $A$ , then as  $A$  is compact,  $A_0$  is open in  $A$  and compact, so  $\cup_{i \in \mathbb{Z}} g^i A_0 = A$ , and also  $A/A_0$  is a cyclic group. As  $A_0$  is a torus, we can find  $a \in A$  that the closure of  $\{a^n | n \in \mathbb{Z}\}$  is  $A$ . By (16.6.4.1), let  $a = \exp_G(X)$ , then the closure of the 1-parameter group generated by  $X$  is a torus containing both  $S$  and  $g$ .  $\square$

**Cor. (16.5.4.14).** In a compact connected Lie group  $G$ , then centralizer of a torus  $T$  is connected. In fact, it is the union of maximal tori containing  $T$ . In particular, a maximal torus is self-centralizing.  $\lrcorner$

### Highest Weight Theory

Cf. [Compact Lie Groups, Sepanski]. Chap 7.

### Examples of Representations

**Prop. (16.5.4.15) [Representations of  $\text{SU}(2)$ ].**  $\text{SU}(2) \cong \mathbb{S}^3$  is simply connected with Lie algebra  $\mathfrak{sl}_2(\mathbb{R})$  (16.6.1.6), thus we can use (16.5.3.14) and (24.4.1.14) to see that the representations of  $\text{SU}(2)$  are all of the form  $W_n$ , where  $W_n$  is the representation of  $\text{SU}(2)$  on the space of homogenous polynomials of degree  $n$  in two variables  $x, y$  induced by  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$ .  $\lrcorner$

*Proof:* Check the central character of the induced representations of Lie algebra.  $\square$

**Prop. (16.5.4.16).** Let  $G = \text{SO}(n)$  for  $n \not\equiv 2 \pmod{4}$ . Show that:

1. For any element  $g \in G$ ,  $g$  and  $g^{-1}$  are conjugate in  $\text{SO}(n)$ .
2. Any  $(\rho, V) \in \text{Irr}(G)$  is isomorphic to its dual.

$\lrcorner$

*Proof:* 1: By (3.5.11.19), if  $n = 2m + 1$ ,  $g$  is orthogonally diagonal to a matrix of the form  $\text{diag}\{\text{SO}(2; \mathbb{R}), \dots, \text{SO}(2; \mathbb{R}), 1\}$ . Notice  $\text{SO}(2; \mathbb{R}) = \{k_\theta, \theta \in (0, 2\pi]\}$ , where  $k_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ , and  $k_\theta^{-1} = k_{-\theta} = k_\theta^t$ . If we take  $W = \text{diag}\left\{\begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \dots, \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, (-1)^m\right\}$ , then  $W \in \text{SO}(n, \mathbb{R})$  and  $WgW^{-1} = g^{-1}$ .

If  $n = 4k$ , then  $g$  is orthogonally diagonal to a matrix of the form  $\text{diag}\{\text{SO}(2; \mathbb{R}), \dots, \text{SO}(2; \mathbb{R})\}$ . If we take  $W = \text{diag}\left\{\begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \dots, \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}\right\}$ , then  $W \in \text{SO}(n; \mathbb{R})$  and  $WgW^{-1} = g^{-1}$ .

2: Because  $\chi_{V^*}(g) = \chi_V(g^{-1}) = \chi_V(g)$  as  $g, g'$  are in the same conjugacy class,  $V$  and  $V^*$  are isomorphic, because f.d. representations are determined by characters(3.6.2.13)(applied to the group algebra  $\mathbb{C}[G]$ ).  $\square$

### Maximal Compact Subgroup

For maximal compact subgroup of general locally compact subgroups, see [Maximal Compact Subgroups](#).

**Prop.(16.5.4.17) [Uniqueness for Semisimple Lie group].** Maximal compact subgroup exists by(12.10.3.33), and for a semisimple Lie group  $G$ , the maximal compact subgroup is unique up to conjugation.  $\lrcorner$

*Proof:* Cf.[Wiki].  $\square$

**Prop.(16.5.4.18)[Examples of Maximal Compact Subgroups].**

- $O(n)$  is the maximal compact subgroup of  $GL(n, \mathbb{R})$ .
- $SO(n)$  is the maximal compact subgroup of  $SL(n, \mathbb{R})$ .
- $SO(m, n)$  is the maximal compact subgroup of  $S(O(m) \times O(n))$ .
- $SO(n)$  is the maximal compact subgroup of  $SL(n, \mathbb{R})$ .
- $SU(n)$  is the maximal compact subgroup of  $SL(n, \mathbb{C})$ .
- $SU(m, n)$  is the maximal compact subgroup of  $S(U(m) \times U(n))$ .
- $SO(n)$  is the maximal compact subgroup of  $SO(n, \mathbb{C})$ .
- $U(n)$  is the maximal compact subgroup of  $GL(n, \mathbb{C})$ .
- $SU(n)$  is the maximal compact subgroup of  $SL(n, \mathbb{C})$ .
- $Sp(n)$  is the maximal compact subgroup of  $Sp(n, \mathbb{C})$ , by(16.6.1.15).
- $Sp(m, n)$  is the maximal compact subgroup of  $Sp(m) \times Sp(n)$ .
- $Sp(n)$  is the maximal compact subgroup of  $SL(n, \mathbb{H})$ , by(16.6.1.15).
- $U(n)$  is the maximal compact subgroup of  $O^*(n)$ , by(16.6.1.15).

And the above maximal compact subgroups are also deformation retractions.  $\lrcorner$

*Proof:* This follows from polar decomposition(16.5.5.2), notice that the projection of a compact group is a compact group in  $\mathbb{R}^n$ , so it is trivial.  $\square$

## 5 Decompositions

**Prop.(16.5.5.1) [Cartan Decomposition].** Let  $G = GL(n; \mathbb{R})$ ,  $K = O(n)$  or  $G = GL(n, \mathbb{R})^+$ ,  $K = SO(n)$ , then every double coset  $K \backslash G / K$  has a unique representation of diagonal matrix  $D$  with decreasing positive entries.  $\lrcorner$

*Proof:* For the existence, given  $g$ , consider  $S = g^t g = k_1^{-1} \text{diag}(\lambda_1, \dots, \lambda_n) k_1$ , where  $k_1 \in SO(n)$ (3.5.11.9). Then consider

$$k_2 = g k_1^{-1} \text{diag}(\lambda_1^{-1/2}, \dots, \lambda_n^{-1/2})$$

it is orthogonal and  $g = k_2 \operatorname{diag}(\lambda_1^{1/2}, \dots, \lambda_n^{1/2}) k_1$ .

For the uniqueness, consider  $g k_1^{-1} \operatorname{diag}(\lambda_1^{-1/2}, \dots, \lambda_n^{-1/2})$  is orthogonal, thus  $k_1 S k_1^{-1}$  is diagonal with decreasing positive entries, thus uniquely defined.  $\square$

**Prop. (16.5.5.2) [Polar Decomposition for Linear Groups].** Let  $G \subset \operatorname{GL}(n; \mathbb{C})$  be a closed linear group that is defined by the a family of real valued polynomials in the real and imaginary parts of entries of  $G$ , and  $G$  is closed under conjugation. Let  $K = G \cap U(n)$ , and let  $\mathfrak{p}$  be the subspace of Hermitian matrices in  $\mathfrak{g}$ , then the map  $K \times \mathfrak{p} \rightarrow G : (k, X) \mapsto k \exp(X)$  is a homeomorphism. In particular,  $G$  has  $K$  as deformation retracts.  $\lrcorner$

*Proof:* The  $\operatorname{GL}(n; \mathbb{C})$  case follows from (12.9.3.20) and (16.6.4.1), and for the general case, it suffices to show if  $g \in G$  and  $g = k \exp(X)$ , then  $X \in \mathfrak{g}$ : By hypothesis,  $g^* g = e^{2X} \in G$ . Take a basis that  $2X = \operatorname{diag}(a_1, \dots, a_n)$  with  $a_i$  real, then  $e^{2kX} = \operatorname{diag}(e^{ka_1}, \dots, e^{ka_n})$  are all in  $G$ . Then it can be easily shown any polynomial that vanishes at all such  $e^{2kX}$  vanishes on all  $e^{2tX}$ , thus  $X \in \mathfrak{g}$ .  $\square$

**Cor. (16.5.5.3).**  $\operatorname{GL}(n, \mathbb{R}) \cong P \cdot \operatorname{O}(n)$ , where  $P$  is the set of positive symmetric matrix, by (16.6.4.1).  $\lrcorner$

**Prop. (16.5.5.4) [QR-Decompositions].**

- Any real matrix  $A$  has the form  $A = QR$  where  $Q$  is orthogonal and  $R$  is upper triangular with positive diagonal entries. Moreover, if  $A$  is invertible, then the decomposition is unique.
- Any complex matrix  $A$  has the form  $A = QR$  where  $Q$  is unitary and  $R$  is upper triangular with positive diagonal entries. Moreover, if  $A$  is invertible, then the decomposition is unique.
- Any Quaternion matrix  $A$  has the form  $A = QR$  where  $Q \in \operatorname{U}(n; \mathbb{H})$  (16.6.1.15) and  $R$  is upper triangular with positive diagonal entries. Moreover, if  $A$  is invertible, then the decomposition is unique.

$\lrcorner$

*Proof:* We only prove for  $\operatorname{GL}(n; \mathbb{R})$ , the rest is similar. Use Gram-Schmidt orthogonalization: choose a basis  $v = \{v_1, \dots, v_n\}$  of  $V$  and  $A$  acts on  $V$ , then  $A$  maps  $\{v_1, \dots, v_n\}$  to a set  $\{w_1, \dots, w_n\}$ . Then we can define an orthonormal basis  $\{e_1, \dots, e_n\}$  that  $v_k \in \operatorname{span}\{e_1, \dots, e_k\}$ . Now let  $\{w_1, \dots, w_n\} = \{e_1, \dots, e_n\}R$ , then  $R$  is upper triangular, and  $Q = [e_1, \dots, e_n]_v$  is orthogonal, and  $A = QR$ . We can make diagonal entries of  $R$  positive by left multiplying a diagonal orthogonal matrix.

To show the uniqueness when  $A$  is invertible, let  $A = Q_1 R_1 = Q_2 R_2$ , then  $A^t A = R_1^t R_1 = R_2^t R_2$ . Then  $(R_2)^{-t} R_1^t = R_2 R_1^{-1}$ , where the LHS is lower-triangular and the RHS is upper-triangular, which means both of them are diagonal. Then if  $\alpha_i$  are the diagonal entries of  $R_1$  and  $\beta_i$  are the diagonal entries of  $R_2$ , then  $\alpha_i / \beta_i = \beta_i = \alpha_i$ , which means  $\alpha_i = \beta_i$ , and  $R_2 R_1^{-1} = 1$ .  $\square$

**Prop. (16.5.5.5) [Smith Normal Form].** Let  $(\mathcal{O}_K, K)$  be a PID. Choose a representative  $\mathcal{P}$  for associativity classes of any prime in  $R$  (to eliminate the distraction of units), then there is complete set of representatives for the double cosets of  $\operatorname{GL}(n; \mathcal{O}_K) \backslash \operatorname{GL}(n, K) / \operatorname{GL}(n; \mathcal{O}_K)$  consisting of diagonal matrices  $\operatorname{diag}(f_1, \dots, f_n)$ , where  $f_i \in K$  are products of elements in  $\mathcal{P}$ , and  $f_k$  divides  $f_{k+1}$ . Notice  $\operatorname{GL}(n, R)$  is the matrices with unit determinants in  $\mathcal{O}_K$ .  $\lrcorner$

*Proof:* For the uniqueness, clearly the row operators doesn't change the greatest common divisor of  $k \times k$  minors of  $M$  for any  $k$  (change by a scalar but the diagonal entries are monic), thus the entries are determined by the minors of  $M$ .

For the existence, for any  $g \in \operatorname{GL}(n; K)$  there is an  $r \in \mathcal{O}_K$ , that  $rg$  has coefficients in  $R$ , and also  $r$  is a product of elements in  $\mathcal{P}$ . let  $M$  be the submodule of  $R^n$  generated by the rows of  $rg$ , then

by the elementary divisor theorem(3.2.4.24), there exists a basis  $\xi_i$  for  $R^n$  and  $d_i \in R$  that  $d_i | d_{i+1}$ , and  $\{d_i \xi_i\}$  form a basis of  $M$ . we may assume  $d_i$  are products of elements in  $\mathcal{P}$ . Then the matrix  $\xi$  with rows  $\xi_i$  are in  $\text{GL}(n; \mathcal{O}_K)$ , and the rows of the matrix  $\text{diag}(d_1, \dots, d_n)\xi$  span the same module as  $rg$ . Then

$$K_1 rg = \text{diag}(d_1, \dots, d_n)\xi$$

for some  $K_1 \in \text{GL}(n; \mathcal{O}_K)$ , so  $g$  are in the same double coset as  $\text{diag}(r^{-1}d_1, \dots, r^{-1}d_n)$ .  $\square$

**Def.(16.5.5.6)[Generalized Upper Half Plane].** The generalized **upper half plane**  $\mathfrak{h}^n$  associated to  $\text{GL}(n; \mathbb{R})$  is defined to be the subset of  $\text{GL}(n; \mathbb{R})$  of the form  $z = xy$  where

$$x = \begin{bmatrix} 1 & x_{1,2} & x_{1,3} & \dots & x_{1,n} \\ & 1 & x_{2,3} & \dots & x_{2,n} \\ & & \ddots & & \dots \\ & & & 1 & x_{n-1,n} \\ & & & & 1 \end{bmatrix}, \quad y = \text{diag}(y_1 y_2 \dots y_{n-1}, y_1 y_2 \dots y_{n-2}, \dots, y_2 y_1, y_1, 1).$$

where  $x_{i,j} \in \mathbb{R}$  and  $y_i \in \mathbb{R}_+$ .  $\lrcorner$

**Thm.(16.5.5.7)[Iwasawa Decomposition].** For  $n \geq 2$ , each matrix  $g \in \text{GL}(n; \mathbb{R})$  has a factorization of the form  $g = \tilde{g} \cdot d \cdot k$  where  $\tilde{g} \in \mathfrak{h}^n$ ,  $k \in \text{O}(n; \mathbb{R})$  and  $d$  is diagonal. Moreover,  $\tilde{g}$  is uniquely determined and  $(d, k)$  is uniquely determined up to  $\{\pm 1\}$ .  $\lrcorner$

*Proof:* Cf.[Gol06]Prop1.2.6. ?  $\square$

**Def.(16.5.5.8)[Action of  $\text{GL}(n; \mathbb{R})$  on  $\mathfrak{h}^n$ ].** There is an action of  $\text{GL}(n; \mathbb{R})$  on  $\mathfrak{h}^n$ (16.5.5.6) defined by

$$g.z = \widetilde{g}.z(16.5.5.7).$$

$\lrcorner$

*Proof:* ?  $\square$

## 6 Semisimple Lie Groups

**Def.(16.5.6.1)[Semisimple Lie Group].** A **semisimple/solvable/nilpotent/simple Lie group** is a Lie group with a semisimple/solvable/nilpotent/simple Lie algebra.  $\lrcorner$

**Def.(16.5.6.2)[Adjoint Lie Group].** An **adjoint Lie group** is a semisimple real Lie group(16.5.6.1) with trivial center.  $\lrcorner$

**Prop.(16.5.6.3).** Let  $G$  be a semisimple complex Lie group, then the center  $Z$  of  $G$  is contained in  $G^c$ , thus the restriction of f.d. representations of  $G$  to  $G^c$  is an equivalence of categories.  $\lrcorner$

*Proof:* Cf.[Etingof, P209]  $\square$

### Maximal Tori

**Prop.(16.5.6.4).** Let  $G$  be a complex semisimple Lie group with Lie algebra  $\mathfrak{g}$ , compact form  $\mathfrak{g}^c$  and compact part  $G^c$ , then  $\lrcorner$

## 7 Analysis

**Lemma (16.5.7.1).** Let  $H$  be a Lie subgroup of  $G$  and  $g \notin H$ , then there is a smooth function  $\Phi$  on  $G$  that  $\Phi(xh) = \Phi(x)$  for any  $x \in G, h \in H$ , and  $\Phi(H) = 0$ , yet  $\Phi(g) \neq 0$ .  $\lrcorner$

*Proof:* Because  $H$  is closed, there is a nbhd  $U$  of  $g$  disjoint from  $H$  and a function  $\varphi$  supported in  $U$ . Then  $\Phi(x) = \int_H \varphi(xh)dh$  satisfies the desired condition.  $\square$

### Bi-Invariant Metric

**Lemma (16.5.7.2).** Bi-invariant metric exists in a compact Lie group.  $\lrcorner$

*Proof:* Because the Haar measure on a compact metric is bi-invariant. Choose a Riemann metric and set

$$\langle V, W \rangle = \int_{G \times G} \langle L_{\sigma^*} R_{\tau^*}(V), L_{\sigma^*} R_{\tau^*}(W) \rangle d\mu(\sigma) d\mu(\tau).$$

Note that  $L_*$  and  $R_*$  commute.  $\square$

**Prop. (16.5.7.3).** For a left-invariant metric on a connected Lie group  $G$ , if it is bi-invariant, then the inner product at the origin  $e$  is invariant under  $\mathfrak{g}$ (3.7.1.13), and the converse is also true if  $G$  is connected.  $\lrcorner$

*Proof:* If the metric is invariant, then for any  $X, Y, Z \in \mathfrak{g}$ ,  $\langle \text{Ad}(\exp(tX))Y, \text{Ad}(\exp(tX))Z \rangle = \langle X, Y \rangle$ , so we take derivation w.r.t.  $t$  to get

$$\langle \text{ad}(X)Y, Z \rangle + \langle Y, \text{ad}(X)Z \rangle = 0$$

by(16.5.1.12).

Conversely, if this is invariant, then using  $\exp(tX) = \exp((t - t_0)X) \exp(t_0X)$ , we get

$$\frac{\partial}{\partial t} \langle \text{Ad}(\exp(tX))Y, \text{Ad}(\exp(tX))Z \rangle = 0$$

for all  $t$ , thus  $\langle \text{Ad}(\exp(tX))Y, \text{Ad}(\exp(tX))Z \rangle = \langle X, Y \rangle$ , and it is invariant under right actions by  $\exp(tX)$  also. As  $G$  is generated by the elements  $\exp(X)$ (4.12.1.3), it is right-invariant under  $G$ .  $\square$

**Prop. (16.5.7.4).** If  $G$  has a bi-invariant metric, then

$$\nabla_X Y = 1/2[X, Y], \quad R(X, Y)Z = 1/4[[X, Y], Z], \quad K(\sigma) = 1/4|[X, Y]|^2.$$

So it has non-positive sectional curvature, and its curvature is non-negative, and all 1-parameter subgroups are geodesics from  $e$ .  $\lrcorner$

*Proof:* It suffices to show that  $\langle Z, \nabla_X Y \rangle = 1/2 \langle Z, [X, Y] \rangle$  for any  $Z$ , and this follows from(16.3.4.16). The second follows from the first and the definition(16.3.4.11).  $\square$

**Cor. (16.5.7.5).** A bi-invariant Lie group with  $Z(\mathfrak{g}) = 1$  is compact and  $\#\pi_1(G) < \infty$ .  $\lrcorner$

*Proof:* The Ricci curvature has a positive lower bound, otherwise for some  $X$ ,  $[X, Y] = 0$  for all  $Y$ , thus  $X$  is in the center. Hence we use Myer theorem(16.3.6.20).  $\square$

**Cor. (16.5.7.6).** If  $G$  is connected and has a bi-invariant metric, then  $\exp : \mathfrak{g} \rightarrow G$  is surjective. In particular, this holds if  $G$  is compact, by(16.5.7.2).  $\lrcorner$

*Proof:* Because  $\exp$  is defined for any  $X \in \mathfrak{g}$ , and for any  $g \in G$  and  $v \in T_g(G)$ ,  $\exp_p(v) = L_g(\exp(dL_{g^{-1}})_g(v))$  because  $L_g$  is an isomorphism of Riemann manifolds. So by Hopf-Rinow(16.3.4.46),  $G$  is complete. Thus for any  $q$ , there is a geodesic connecting  $e, q$ , which is a 1-parameter subgroup, thus  $q = \exp(X)$  for some  $X \in \mathfrak{g}$ .  $\square$

**Prop. (16.5.7.7) [Structure of bi-invariant Lie groups].** A simply-connected Lie group with a bi-invariant metric is equal to  $G' \times R^k$ ,  $G'$  compact.  $\lrcorner$

*Proof:* Because the orthogonal complement of the center of  $\mathfrak{g}$  is a Lie algebra,  $G$  is like  $G' \times R^k$ , and a simply connected abelian Lie group is  $R^k$  ?.  $\square$

## 16.6 Classical Groups

In this subsection, classical groups(10.5.3.2) over Archimedean fields and related Lie groups are studied.

References are [Symmetry, Representations and Invariants].

For more classical groups, Cf.[Classical Groups Baker].

**Notation(16.6.0.1).**

- Use notations on Lie groups defined in [Lie Groups](#).

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### 1 Basics

**Def.(16.6.1.1)[Examples of Classical Groups].** Let  $\mathbb{K}$  be either  $\mathbb{R}$  or  $\mathbb{C}$ ,

- For any associative algebra  $\mathbb{K}$  over a field, the **general linear group**  $GL(n; \mathbb{K})$  is the subgroup of  $Mat(n; \mathbb{K})$  consisting of invertible matrices.
- The **unitary group**  $U(n; \mathbb{C})$  is the subgroup of  $GL(n; \mathbb{C})$  consisting of matrices fixing a non-degenerate Hermitian form.
- The **special unitary group**  $SU(n; \mathbb{C}) = U(n) \cap SL_n(\mathbb{C})$ .
- The **pseudo-unitary groups**  $U(p, q, \mathbb{K})$ : If  $\mathbb{K} = \mathbb{R}$ , it is the subgroup of  $GL(n; \mathbb{R})$  consisting of matrices preserving a bilinear form of signature  $(p, q)$ . If  $\mathbb{K} = \mathbb{C}$ , it is the subgroup of  $GL(n; \mathbb{C})$  consisting of matrices preserving a Hermitian form of signature  $(p, q)$ . If  $\mathbb{K} = \mathbb{H}$ , it is the subgroup of  $GL(n; \mathbb{H})$  consisting of matrices preserving a non-degenerate quaternionic Hermitian form of signature  $(p, q)$ (3.5.11.1).
- The **quaternionic orthogonal group**  $O^*(2n)$  is the subgroup of  $GL(n; \mathbb{H})$  consisting of matrices preserving a non-degenerate quaternionic skew-Hermitian form(3.5.11.1).  $SO^*(2n)$  is the subgroup of  $O^*(2n)$  consisting of elements of determinant 1.
- $PU(n; \mathbb{C})$  is the quotient group of  $SU(n)$ (or  $U(n; \mathbb{C})$ ) by scalar matrices.

┘

*Proof:*  $GL(n; \mathbb{K})$  has a natural smooth structure as an open subset of  $\mathbb{K}^{n^2}$ , and the multiplication map is clearly smooth, so it is a Lie group by(16.5.1.1). The other groups are closed subgroups of  $GL(n; \mathbb{K})$ , so they have unique smooth manifold structure making them Lie subgroups of  $GL(n; \mathbb{K})$ , by(16.5.1.22)(16.5.1.16). Finally the quotient group by normal closed subgroups have natural Lie group structure by(16.5.1.28).  $\square$

**Prop.(16.6.1.2)[Classification of  $2 \times 2$  Matrices].** Let  $\mathbf{1} \neq \alpha \in SL(2; \mathbb{R})$ , then by Jordan decomposition,  $\alpha$  is conjugate to a matrix of one the two following types:

$$\begin{bmatrix} \lambda & 1 \\ & \lambda \end{bmatrix}, \quad \begin{bmatrix} \lambda & \\ & \mu \end{bmatrix}, \lambda \neq \mu,$$

according as it has repeated eigenvalues or distinct eigenvalues. In the first case,  $\alpha$  is called **parabolic transform**, and in the second case, if  $\mu = \bar{\lambda} \in \mathbb{S}^1$ , it is called **elliptic transform**, If  $\lambda/\mu$  is real and positive, it is called **hypobolic transform**.

Consider the action of  $SL(2; \mathbb{R})$  on  $\mathcal{H} \cup \mathbb{P}^1(\mathbb{R})$ , then:

- If  $\alpha \in \mathrm{SL}(2; \mathbb{R})$  is parabolic, then it has a unique real eigenvector, which means it has a unique fixed point in  $\mathbb{P}^1(\mathbb{R})$ .
- If  $\alpha \in \mathrm{SL}(2; \mathbb{R})$  is elliptic, then  $\alpha$  has two conjugate eigenvectors, which means it has exactly one fixed point  $z$  in  $\mathcal{H}$ , and a second fixed point, namely  $\bar{z}$ , in the lower half plane.
- If  $\alpha \in \mathrm{SL}(2; \mathbb{R})$  is hyperbolic, then it has two real eigenvectors, which means it has two distinct fixed points in  $\mathbb{P}^1(\mathbb{R})$ .

┘

**Prop. (16.6.1.3) [Iwasawa Decomposition].** Any element of  $\mathrm{GL}(2; \mathbb{R})^+$  has a unique representation of the form (16.5.5.4):

$$g = \begin{bmatrix} u & \\ & u \end{bmatrix} \begin{bmatrix} y^{1/2} & xy^{-1/2} \\ & y^{-1/2} \end{bmatrix} k_\theta$$

where  $k_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ . So by (12.10.3.8) and (12.10.1.19) the Haar measure is calculated to be  $dg = \frac{1}{2\pi} \frac{du}{u} \frac{dxdy}{y^2} d\theta$ , and it is unimodular by (12.10.1.22). (Notice the upper triangular matrix group  $B \subset \mathrm{GL}(2; \mathbb{R})^+$  is not unimodular).

┘

*Proof:* For the calculation of Haar measure, notice that it suffices to calculate for  $u, x, y$  and it is

$$\frac{d(uy^{1/2})d(uy^{-1/2})d(uxy^{-1/2})}{(uy^{1/2})^2uy^{-1/2}} = \frac{dxd(uy^{1/2})d(uy^{-1/2})}{u^2y} = \frac{dxdydu}{uy^2}$$

□

**Cor. (16.6.1.4).** Any element of  $\mathrm{SL}(2; \mathbb{R})$  has a unique representation of the form

$$g = \begin{bmatrix} e^{\frac{u}{2}} & 0 \\ 0 & e^{-\frac{u}{2}} \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} k_\theta.$$

And the Haar measure is given by  $dg = \frac{1}{2\pi} du dx d\theta$  by similar calculation, which is unimodular. ┘

**Prop. (16.6.1.5).**  $\mathcal{H}$  is a homogenous space for  $\mathrm{PGL}(2; \mathbb{R})$ , and  $\mathrm{Stab}_{\mathrm{PGL}(2; \mathbb{R})}(\mathbf{i}) = \mathrm{SO}(2; \mathbb{R})$ . Also it fixes the hyperbolic metric  $ds^2 = y^{-2}dxdy$ . ┘

*Proof:* This follows from the Iwasawa decomposition (16.6.1.3). □

**Prop. (16.6.1.6) [SU(2)].**

$$\mathrm{SU}(2) = \{A_{\alpha, \beta} = \begin{bmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{bmatrix}, \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1\}$$

is isomorphic to the group of unit quaternions and diffeomorphic to  $S^3$ . The Lie algebra of  $\mathrm{SU}(2)$  is isomorphic to  $\mathfrak{sl}_2(\mathbb{R})$ . ┘

*Proof:* Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SU}(2)$ , then

$$a\bar{a} + b\bar{b} = 1, \quad a\bar{c} + b\bar{d} = 0, \quad c\bar{c} + d\bar{d} = 1, \quad ac - bd = 1.$$

So  $(c, d) = \lambda \cdot (-\bar{b}, \bar{a})$ , and we can calculate  $\lambda = 1$ . So the first assertion follows.

By (3.5.11.34),  $\mathrm{SU}(2)$  is isomorphic to the group of unit quaternions, which is clearly isomorphic to  $S^3$ . □



**Prop. (16.6.1.7)**[Actions of  $SU(2)$ ].

- There is a double covering of Lie groups  $SU(2) \rightarrow SO(3)$  with kernel  $\{\pm 1\}$ .
- There is a double covering of Lie groups  $SU(2) \times SU(2) \rightarrow SO(4)$  with kernel  $\{\pm 1\}$ .

⌋

*Proof:* 1: Regard  $SU(2)$  as the unit quaternions and  $SO(3)$  the transformation group of pure unit quaternions, then  $SU(2) \rightarrow SO(3)$  is given by

$$u \mapsto (v \mapsto uv\bar{u}).$$

Because  $u \cdot \bar{u}$  preserves orthogonality relations, it preserves the space of pure quaternions, so it has image in  $O(3)$ . But it has image in  $SO(3)$  because  $SU(2) \cong S^3$  is connected. Its kernel is the center of  $\mathbb{H}$ , which is  $\pm 1$ . Now (16.5.3.10) shows this is a covering space map.

2: Consider the action  $SU(2) \times SU(2)$  on the section of unit vectors of  $\mathbb{H}$ :  $(u, v)(z) = uzv^{-1}$ . if  $(u, v)$  is in the kernel of this map, then  $uv^{-1} = 1$ , so  $u = v$ , and  $u$  is in the center of  $\mathbb{H}$ , so  $u = v = \pm 1$ . Because  $\dim SU(2) = 3$  and  $\dim SO(4) = 6$ , (16.5.3.10) shows this is a double cover.  $\square$

**Prop. (16.6.1.8).**

- There is a double cover  $SU(4) \rightarrow SO(6; \mathbb{R})$  with kernel  $\{\pm 1\}$ .
- There is a double cover  $SL(4; \mathbb{C}) \rightarrow SO(6; \mathbb{C})$  with kernel  $\{\pm 1\}$ .
- There is a double cover  $Sp(4; \mathbb{C}) \rightarrow SO(5; \mathbb{C})$  with kernel  $\{\pm 1\}$ .

⌋

*Proof:* 1:  $SU(4)$  acts on a 4-dimensional Hermitian space  $V$ . Then it acts on the Hermitian space  $\wedge^2 V$ . Consider the Hodge star operator  $*$ :  $\wedge^2 V \rightarrow \wedge^2 V$  (16.1.3.7), it is an anti-linear operator,  $*^2 = \text{id}$  and  $*$  commutes with  $SU(4)$  action. Then  $W = \ker(* - \text{id})$  is a real vector space of dimension 6 that  $SU(4)$  acts on. This kernel of this action is the same as the kernel of the action  $\wedge^2 V$ , which is  $\{\pm 1\}$ .

The Hermitian form induces a symmetric form on  $W$ : Take a conjugation on  $V$ , because  $V = W \oplus iW$ ,  $W = \text{Im}(* + \text{id})V$ . So for  $a, b \in W$ , let  $a = *c + c, b = *d + d$ , then  $(a, b)\omega = ((*a, *b) + (*a, b) + (a, *b) + (a, b))\omega = *a \wedge b + \overline{*a} \wedge \overline{b} + a \wedge b + *a \wedge \overline{b}$  is a real form. Then this representation induces a map  $SU(4) \rightarrow SO(6)$  (because  $SU(4)$  is connected). Because  $\dim SU(4) = \dim SO(6) = 6$ , it is a double cover by (16.5.3.10).

2:  $SL(4, \mathbb{C})$  acts on a 4-dimensional complex vector space  $V$ , then it acts on the space  $W = \wedge^2 V^*$ . We construct a non-degenerate bilinear form on  $W$  given by  $\wedge : W \times W \rightarrow \wedge^4 V^* \cong \mathbb{C}$ . Then  $SL(4, \mathbb{C})$  preserves this bilinear form, thus induces a map  $SL(4, \mathbb{C}) \rightarrow SO(6, \mathbb{C})$ . The kernel of this map is  $\{\pm 1\}$ , because if  $A$  preserves all  $v_1^* \wedge v_2^*$ , then  $Av_1^* \wedge Av_2^* = v_1^* \wedge v_2^*$ , so  $Av_1^* \wedge Av_2^* \wedge v_i^* = 0$ , so  $Av_1^* \in \{v_1^*, v_2^*\}$ , or  $Av_2^* \in \{Av_1^*, v_1^*\} \cap \{Av_1^*, v_2^*\} = Av_1^*$ , then  $Av_1^* \wedge Av_2^* = 0$ , contradiction. So  $Av_1^* \in \{v_1^*, v_2^*\}$ .  $v_1, v_2$  is arbitrary, thus  $A$  is diagonal, and it is clear  $A = \pm 1$ . Finally,  $SL(4, \mathbb{C}) \rightarrow SO(6, \mathbb{C})$  is a double cover by (16.5.3.10).

3: The representation of  $SL(4, \mathbb{C})$  on  $W = \wedge^2 V^*$  restricts to a representation of  $Sp(4, \mathbb{C})$ , and it fixes the vector  $\sigma = v_1^* \wedge v_2^* + v_3^* \wedge v_4^*$  by definition, so it also fixes the orthogonal  $\{\sigma\}^\perp$ , thus inducing a map  $Sp(4, \mathbb{C}) \rightarrow SO(5, \mathbb{C})$  with kernel  $\{\pm 1\}$ , which is a double cover by (16.5.3.10).  $\square$

**Prop. (16.6.1.9)**[ $SU(1, 1)$  and  $SL(2; \mathbb{R})$ ]. Let  $SL(2; \mathbb{C})$  act continuously on  $\mathbb{P}^1(\mathbb{C})$  by  $\gamma(z) = \frac{az+b}{cz+d}$  where

$$\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then}$$

- The stabilizer of the unit disk  $\mathbb{D}$  is

$$SU(1, 1) = \left\{ \begin{bmatrix} a & b \\ \bar{b} & \bar{a} \end{bmatrix} \mid |a|^2 - |b|^2 = 1 \right\}.$$

- $SU(1, 1)$  is conjugate to  $SL(2, \mathbb{R})$  in  $SL(2, \mathbb{C})$ .
- The subgroup of  $SU(1, 1)$  fixing  $0 \in \mathbb{D}$  is the subgroup of rotations  $\{\text{diag}(e^{i\theta/2}, e^{-i\theta/2})\}$ .

┘

*Proof:* 3: This follows from Schwartz lemma(12.5.1.3).

1, 2: We first describe  $SU(1, 1)$ :

$$SU(1, 1) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : |a|^2 - |c|^2 = 1, \quad a\bar{b} - c\bar{d} = 0, \quad |b|^2 - |d|^2 = -1, \quad ad - bc = 1 \right\}$$

which means  $(b, d) = \lambda(\bar{c}, \bar{a})$ , and  $\lambda = 1$ .

Let  $C = \frac{1}{\sqrt{2}e^{2\pi i/4}} \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix}$ , then  $C$  induces an isomorphism between  $\mathcal{H}$  and  $\mathbb{D}$ , because  $\frac{z-i}{z+i} \in \mathbb{D}$  iff  $|z-i| < |z+i|$  iff  $z$  is in the upper half plane. By this isomorphism and item 3, the element of  $SL(2, \mathbb{C})$  stabilizing  $\mathbb{D}$  and fixing  $i$  is of the group  $SO(2; \mathbb{R}) \subset SL(2; \mathbb{R})$ .

Notice that  $SL(2; \mathbb{R})$  preserves  $\mathcal{H}$  by(12.4.1.9), and it acts transitively on it because  $\sqrt{y} \begin{bmatrix} y & x \\ 1 & 1 \end{bmatrix}$  maps  $i$  to  $x + iy \in \mathcal{H}$ . Now if  $\gamma \in SL(2; \mathbb{C})$  stabilizes  $\mathcal{H}$ , then  $\gamma g$  fixes  $i$  for some  $g \in SL(2, \mathbb{R})$ , thus  $\gamma g \in SL(2; \mathbb{R})$ , thus  $\gamma \in SL(2, \mathbb{R})$ . Thus we have shown the stabilizer of  $\mathcal{H}$  is  $SL(2, \mathbb{R})$ . Then the stabilizer of  $\mathbb{D}$  is

$$\begin{aligned} C \cdot SL(2, \mathbb{R})C^{-1} &= \left\{ \frac{1}{2i} \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} i & i \\ -1 & 1 \end{bmatrix} \mid a, b, c, d \in \mathbb{R}, ad - bc = 1 \right\} \\ &= \left\{ \frac{1}{2} \begin{bmatrix} a+d+(b-c)i & a-d-(b+c)i \\ a-d+(b+c)i & a+d-(b-c)i \end{bmatrix} \mid a, b, c, d \in \mathbb{R}, ad - bc = 1 \right\} \end{aligned}$$

which is exactly  $SU(1, 1)$ . So we get 1 and 2. □

**Cor. (16.6.1.10)[Upper Half Plane  $\mathcal{H}$ ].** By(12.4.7.8), the groups  $GL(2; \mathbb{R})$  preserves the upper plane  $\mathcal{H}$ , by(12.4.1.9). The action of  $SL(2; \mathbb{R})$  on  $\mathcal{H}$  is transitive and the stabilizer of  $i$  is  $SO(2; \mathbb{R})$ , thus we have  $\mathcal{H} \cong SL(2; \mathbb{R})/SO(2; \mathbb{R})$ . ┘

**Prop. (16.6.1.11)[ $Sp(2; \mathbb{K})$  and  $SL(2; \mathbb{K})$ ].** For any field  $\mathbb{K}$ ,  $Sp(2; \mathbb{K}) \cong SL(2; \mathbb{K})$ : They both consists of linear maps preserving the differential form  $dx \wedge dy$ . ┘

**Prop. (16.6.1.12)[Center of  $SU(p, q)$ ].**  $Z_{U(p, q)}(SU(p, q))$  are all scalar matrices. ┘

*Proof:* Firstly we consider  $Z_{U(2)}(SU(2))$ , if  $A \in Z(SU(2))$ , then it commutes with  $\text{diag}(i, -i)$ , so  $A$  is a scalar. Similarly, If  $A \in Z_{U(1, 1)}(SU(1, 1))$ , then  $A$  commutes with  $\text{diag}(i, -i)$ , thus  $A$  is a scalar.

If  $X \in Z(SU(p, q))$  is not a scalar matrix, then it has two eigenvectors  $s, t$  with different eigenvalues. Consider the space  $V$  generated by  $s, t$  and its orthogonal complement, then  $X$  restricted to this space is in the center of  $U(V)$ . Now  $SU(V) \cong SU(1, 1)$  or  $SU(2)$ , both have the set of scalar matrices as center, so  $X$  cannot have different eigenvalue, contradiction. Thus  $X$  is a scalar matrix. □

**Prop. (16.6.1.13)**[PGL(2;  $\mathbb{C}$ )  $\cong$  SO(3)].  $V \cong \mathbb{C}^2$  has a natural symmetric form  $(x, y) \mapsto x \wedge y \cong \mathbb{C}$ , and this form is preserved by  $\text{SL}(2; \mathbb{C})$  by definition. The kernel of this map is  $\{\pm 1\}$ . Thus  $\text{SL}(2; \mathbb{C})$  acts on  $\text{Sym}^2(V)$ , thus induces a map  $\text{PGL}(2; \mathbb{C}) \rightarrow \text{SO}(3)$ , which an isomorphism by (16.5.3.10).  $\lrcorner$

**Prop. (16.6.1.14)**[Real and Complex Matrices].  $\text{GL}(n; \mathbb{C})$  can be embedded into  $\text{GL}(2n; \mathbb{R})$ , with determinant  $|\det|^2$ . And in this way,  $U(n)$  is mapped into  $O(2n; \mathbb{R})$ . Also,  $O(n; \mathbb{R})$  embeds into  $U(n)$  diagonally.  $\lrcorner$

*Proof:*

$$X + iY \mapsto \begin{bmatrix} X & -Y \\ Y & X \end{bmatrix} \sim \begin{bmatrix} X & -Y \\ iX + Y & X - iY \end{bmatrix} \sim \begin{bmatrix} X + iY & Y \\ 0 & X - iY \end{bmatrix}$$

$\square$

**Prop. (16.6.1.15)**[Symplectic Groups].

- $U(p, q; \mathbb{H}) = \text{Sp}(2n; \mathbb{C}) \cap U(2p, 2q, \mathbb{C})$ .  $U(n, \mathbb{H}) = U(n, 0, \mathbb{H})$  is also denoted by  $\text{Sp}(n)$ , called the **compact symplectic group**.
- $O^*(2n) = U(n, n) \cap O(2n; \mathbb{C})$ .
- $\text{Sp}(2n; \mathbb{C}) = \text{SL}(n; \mathbb{H})$ .
- $\text{Sp}(2n, \mathbb{R}) \cap O(2n; \mathbb{R}) = \text{Sp}(2n, \mathbb{R}) \cap \text{GL}(n; \mathbb{C}) = O(2n, \mathbb{R}) \cap \text{GL}(n; \mathbb{C}) = U(n) = \left\{ \begin{bmatrix} X & -Y \\ Y & X \end{bmatrix}, X + iY \in U(n) \right\}$ .

$\lrcorner$

*Proof:* 1: By (3.5.11.38), notice that any  $\mathbb{C}$ -linear automorphism preserving  $B_1$  and  $B_2$  is  $\mathbb{H}$ -linear.

2: By (3.5.11.38), notice that any  $\mathbb{C}$ -linear automorphism preserving  $B_1$  and  $B_2$  is  $\mathbb{H}$ -linear.

3: If  $A \in \text{Sp}(2n; \mathbb{R}) \cap O(2n; \mathbb{R})$ , then  $AA^t = A^t A = 1$ ,  $A^t J A = 1$ , then  $AJ = JA$ . The rest and the other identities are easy by (16.6.1.14).  $\square$

## 2 Spin Groups

**Notation (16.6.2.1).**

- Use notations on Clifford algebras defined in [Clifford Algebras](#).

$\lrcorner$

**Prop. (16.6.2.2).** For  $n$  even,  $\mathbb{C}(V)$  is naturally isomorphic to  $\text{End}_{\mathbb{C}}(\wedge^* W)$ , where  $W = \{\frac{1}{\sqrt{2}}(e_{2i-1} - ie_{2i})\}$ . This isomorphism is not obvious and restrict to a Spinor representation of  $\text{Spin}(n)$  and  $\rho(\Gamma)^2 = 1$  induce two representations of  $Cl(n)^0$ , in particular  $\text{Spin}(n)$ , called the **(half Spinor representations)**. This has a unique extension to representation of  $\text{Spin}^c$ .  $\wedge^* W$  comes with a Hermitian metric which is preserved by the action of  $\text{Pin}(n)$  (check). So the image is  $\text{SO}(n)$  is in  $\text{SO}(\wedge^* W)$ . Cf. [Jost Geometric analysis P72].  $\lrcorner$

**Def. (16.6.2.3)**[Spin Groups  $\text{Spin}(n)$ ]. Denote by  $\text{Pin}(n)$  as the group in  $\text{Cl}_n(\mathbb{R})$  generated by vectors  $v_i$  of norm 1. Because  $v_i \cdot v_i = -1$ , it is a group. And denote  $\text{Spin}(n)$  as the subgroup of  $\text{Pin}(n)$  generated by even numbers of  $v_i$ s.  $\lrcorner$

**Prop. (16.6.2.4)**[Action of  $\text{Spin}(n)$ ]. The conjugation action  $\text{Ad}(v) = v(-)v = \text{reflection w.r.t } v$ , maps  $\text{Pin}(n)$  to  $O(n)$  and  $\text{Spin}(n)$  to  $\text{SO}(n)$ . The kernel of this mapping is  $\{\pm 1\}$  when  $n$  is even. This is a double covering of  $\text{SO}(n)$  and  $O(n)$ , it is nontrivial because  $1, -1$  is connected by  $(\cos te_1 + \sin te_2)(\cos te_1 - \sin te_2)$ .

In particular,  $\text{Spin}(n)$  is a universal covering of  $\text{SO}(n)$  and thus simply-connected for  $n \geq 3$ .  $\lrcorner$

*Proof:* Let  $\alpha = e_i\beta + \gamma$ , then  $\beta, \gamma \in Cl^0$  and so  $\alpha = ce_1 \dots e_n + d$ , and  $c$  can happen only when  $n$  is odd.  $\square$

$$\text{Prop. (16.6.2.5) [Center of Spin}(n)\text{]}. Z(\text{Spin}(n)) = \begin{cases} S^1 & n = 2 \\ \mathbb{Z}/2\mathbb{Z} & n = 2k + 1, \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & n = 4k, \\ \mathbb{Z}/4\mathbb{Z} & n = 4k + 2, \end{cases} \quad \lrcorner$$

*Proof:*  $\square$

**Prop. (16.6.2.6) [Low Dimensional Accidental Isomorphisms].**

- $\text{Spin}(2) \cong \text{SO}(2) \cong \text{U}(1)$ .
- $\text{Spin}(3) \cong \text{SU}(2) \cong \text{Sp}(1)$
- $\text{Spin}(4) \cong \text{SU}(2) \times \text{SU}(2)$  because they are both universal coverings of  $\text{SO}(4)$ , by (16.6.3.1) and (16.6.2.4).
- $\text{Spin}(5) \cong \text{Sp}(2)$ .
- $\text{Spin}(6) \cong \text{SU}(4)$ .

$\lrcorner$

*Proof:* 2: because  $\text{Spin}(3), \text{SU}(2)$  are both universal covering of  $\text{SO}(3)$ , by (16.6.3.1) and (16.6.2.4), and  $\text{Sp}(1)$  acts transitively on the set of unit vectors in  $\mathbb{H}$  with trivial kernel.  $\square$

### 3 Fundamental Groups

**Prop. (16.6.3.1).**

- $\text{SU}(n-1) \rightarrow \text{SU}(n) \rightarrow \mathbb{S}^{2n-1}$  shows  $\text{SU}(n)$  are connected and simply connected. Also  $\pi_2(\text{SU}(n)) \cong \pi_2(\text{SU}(2)) \cong \pi_2(\mathbb{S}^3) = 0$ .
- $\text{SO}(n-1) \rightarrow \text{SO}(n) \rightarrow S^{n-1}$  shows  $\text{SO}(n)$  are connected and  $\pi_1(\text{SO}(n)) \cong \pi_1(\text{SO}(3)) = \mathbb{Z}/(2)$  for  $n \geq 3$  by (16.6.1.7) and (16.6.1.6). And  $\pi_1(\text{SO}(2)) = \mathbb{Z}$ .
- $\text{U}(n-1) \rightarrow \text{U}(n) \rightarrow S^{2n-1}$  shows  $\text{U}(n)$  are connected and  $\pi_1(\text{U}(n)) \cong \mathbb{Z}$ .
- $\text{SO}(n; \mathbb{R})$  is a deformation retraction of  $\text{SL}(n, \mathbb{R})$ , and  $\text{SU}(n)$  is a deformation retraction of  $\text{SL}(n; \mathbb{C})$ .

$$\bullet \pi_1(\text{PSO}(n)) = \begin{cases} \mathbb{Z} & n = 2 \\ \mathbb{Z}/2\mathbb{Z} & n = 2k + 1, \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & n = 4k, \\ \mathbb{Z}/4\mathbb{Z} & n = 4k + 2, \end{cases} \text{ . Because for } n \geq 3, \text{ its universal covering is } \text{Spin}(n), \text{ so } \pi_1(\text{PS}(n)) = Z(\text{Spin}(n)) \text{ (16.6.2.5).}$$

- $\pi_1(\text{PU}(n)) = \mathbb{Z}/(n)$ .
- $\text{Sp}(n-1) \rightarrow \text{Sp}(n) \rightarrow \mathbb{S}^{4n-1}$  shows  $\text{Sp}(n)$  are connected and simply connected.
- $\pi_1(\text{Sp}(2n; \mathbb{R})) = \pi_1(\text{U}(n)) = \mathbb{Z}$  and the determinant induces an isomorphism onto  $\pi_1(S^1)$ . In fact, this is used to define the Maslov index.

$\lrcorner$

## 4 Generals

**Prop. (16.6.4.1)[Image of Exponential Maps].**

1. The exponential map for  $\mathrm{GL}(n; \mathbb{C})$  is surjective.
2. The image of the exponential map for  $\mathrm{GL}(n; \mathbb{R})$  is  $\mathrm{GL}(n; \mathbb{R})^2$ .
3. The image of exponential map for  $B_+$  (the subgroup of  $\mathrm{GL}(n; \mathbb{R})$  consisting of upper-triangular matrices with positive entries), is surjective.
4. The exponential defines a diffeomorphism from the space of Hermitian (symmetric) matrices to positive definite Hermitian (resp. symmetric) matrices in  $\mathrm{GL}(n; \mathbb{C})$  (resp.  $\mathrm{GL}(n; \mathbb{R})$ ).
5. The exponential for a compact connected Lie group is surjective, by (16.5.7.6). In particular, this applies to  $G \in \{\mathrm{SO}(n), \mathrm{U}(n), \mathrm{SU}(n)\}$ .
6. The exponential for  $\mathrm{SL}(2; \mathbb{R})$  is not surjective. ?

┘

*Proof:* 1: Use Jordan forms. Notice the logarithm of  $(cI + N)$ ,  $c \neq 0$  is definable for  $N$  nilpotent, and it is a polynomial function of the matrix itself.

2: It is clear the image is contained in  $\mathrm{GL}(n; \mathbb{R})^2$ , conversely, we see from the complex case that any  $B \in \mathrm{GL}(n; \mathbb{R})$  is of the form  $\exp(P(B))$  for some polynomial  $P \in \mathbb{C}[X]$ , then  $T = B^2 = \exp(P(B) + \overline{P}(B)) \in \exp(\mathrm{GL}(n; \mathbb{R}))$ .

3: Use Jordan forms. Notice the logarithm of  $(cI + N)$ ,  $c \neq 0$  is definable for  $N$  nilpotent and  $c > 0$ , and it is a polynomial function of the matrix itself, so it is also upper-triangular.

4: By (3.5.11.9) it is clearly surjective. For injectivity, consider  $\exp(X) = \exp(Y)$ , then at least  $X, Y$  are both unitarily conjugate to the same diagonal matrix  $\mathrm{diag}(d_1, \dots, d_n)$ , and we may assume  $Y = \mathrm{diag}(d_1, \dots, d_n)$ , then

$$X = \tau^{-1} Y \tau, \quad \mathrm{diag}(D_1, \dots, D_n) = \tau^{-1} \mathrm{diag}(D_1, \dots, D_n) \tau,$$

where  $D_i = e^{d_i}$ . Consequently,  $\mathrm{diag}(D_1^k, \dots, D_n^k) = \tau^{-1} \mathrm{diag}(D_1^k, \dots, D_n^k) \tau$ , and we can choose  $c_i$  that  $\sum c_i D_j^i = d_j$  for any  $j$ , because the Vandermonde matrix is nonsingular. Hence  $\mathrm{diag}(d_1, \dots, d_n) = \tau^{-1} \mathrm{diag}(d_1, \dots, d_n) \tau$ , so  $X = Y$ .

5:

□

**Prop. (16.6.4.2)[Injectivity of Exponential Maps].** There is a characterization of when  $\exp : \mathfrak{g} \rightarrow G$  is injective:

<https://math.stackexchange.com/questions/475385/under-what-conditions-is-the-exponential-map-injective>

┘

## 16.7 Geometric Analysis II

References are [L-W89, Jos11].

### 1 Spin Structures

**Notation(16.7.1.1).**

- Use notations on real Clifford algebras defined in [Clifford Algebras](#).
- Use notations on spin groups defined in [Spin Groups](#).
- Use notations on fiber bundles defined in [Fiber Bundles & Topological K-Theory](#).
- Use notations on Riemannian geometry defined in [Riemannian Geometry](#).
- Let  $M \in \text{RieMani}_{\text{cpct}, \text{cntd}, \partial}$ .

┘

#### Spin Structures

**Def.(16.7.1.2) [Spin Structures].** For  $X \in \text{Mani}$ ,  $n \geq 2$  and  $E \in \text{Vect}_{\mathbb{R}, \text{orntd}}^n(X)$ , let  $P_{\text{SO}}(E)$  be the principle  $\text{SO}(n)$ -bundle of orthonormal frames in  $E$ , then a **spin structure** on  $E$  is a principle  $\text{Spin}(n)$ -bundle  $P_{\text{Spin}}(E)$  over  $M$  with a 2-sheeted covering:

$$\xi : P_{\text{Spin}}(E) \rightarrow P_{\text{SO}}(E)$$

s.t.  $\xi(pg) = \xi(p)\xi_0(g)$  for any  $p \in P_{\text{Spin}}(E)$  and  $g \in \text{Spin}(n)$ . Where if  $n = 2$ , we assume  $\text{Spin}(2) = \text{SO}(2) \xrightarrow{\xi_0 = \times 2} \text{SO}(2)$ .

It is clear that the spin structures on  $E$  are in bijection with double coverings of  $P_{\text{SO}}(E)$  that are non-trivial on each fiber of  $E$ . ┘

**Cor.(16.7.1.3) [Invariance of the Metric].** The set of spin structures on  $E$  are independent of the metric on  $E$ , because  $P_{\text{SO}}(E) \subset P_{\text{GL}^+}(E)$  is a homotopy equivalence. ┘

**Prop.(16.7.1.4) [Spin Structure Obstruction].** Situation as in (16.7.1.2), then by (4.15.4.6),  $E$  has a spin structure iff  $w_2(E) = 0$ , and in this case, the spin structures on  $E$  are in bijection with  $H^1(M, \mathbb{Z}/(2))$ . ┘

**Cor.(16.7.1.5).** For  $X \in \text{Mani}$ , if  $E, E' \in \text{Vect}_{\mathbb{R}}^n(X)$  with  $E = E_1 \oplus E_2$ , then a choice of orientations on two of  $\{E, E_1, E_2\}$  will determine a choice of orientation on the third.

Similarly, a choice of spin structures on two of  $\{E, E_1, E_2\}$  will determine a choice of spin structure on the third. ┘

*Proof:* It is easy to see that two of  $\{E, E_1, E_2\}$  is spin implies the rest is also spin. For the determination of the rest spin structure, Cf. [L-W89]P85. ? ┘

**Def.(16.7.1.6) [Spinor Bundles].** Let  $X \in \text{Mani}$  and  $E/X \in \text{Vect}_{\text{Riem}, \partial}^n(X)$  with a spin structure  $\xi : P_{\text{Spin}}(E) \rightarrow P_{\text{SO}}(E)$ , then for any real (resp. complex) left  $\text{Cl}(\mathbb{R}^n)$ -module  $M$ , we get an associate bundle (4.15.3.4) which is a real (resp. complex) **spinor bundle**  $S_\mu(E) = P_{\text{Spin}}(E) \times M$ . ┘

**Example(16.7.1.7) [Spinor Bundles].** Let  $E/X \in \text{Vect}_{\text{Riem}, \partial}^n(X)$  with a spin structure  $\xi : P_{\text{Spin}}(E) \rightarrow P_{\text{SO}}(E)$ ,

- When  $M = \text{Cl}(\mathbb{R}^n)$  with the adjoint action of  $\text{Cl}(\mathbb{R}^n)$ , the the associated spinor bundle is denoted by

$$\text{Cl}(E) \triangleq P_{\text{Spin}}(E) \times_{\text{Ad}} \text{Cl}(\mathbb{R}^n) = P_{\text{SO}}(E) \times_{\rho_n} \text{Cl}(\mathbb{R}^n)$$

is a principle  $\text{Cl}(\mathbb{R}^n)$ -bundle, called the **Clifford bundle** of  $E$ . And  $\text{Cl}(E)$  is locally just defined by the Clifford construction.

- When  $M = \text{Cl}(\mathbb{R}^n)$  with the left action of  $\text{Cl}(\mathbb{R}^n)$ , then the associated spinor bundle is denoted by

$$\text{Cl}_{\text{Spin}}(E) \triangleq P_{\text{Spin}}(E) \times_L \text{Cl}(\mathbb{R}^n)$$

is a principle  $\text{Cl}(\mathbb{R}^n)$ -bundle, called the **canonical spinor bundle** of  $E$ . ┘

**Prop. (16.7.1.8).** Situation as in (16.7.1.6), any spinor bundle  $S(E)$  is a bundle of left modules over the bundle of algebras  $\text{Cl}(E)$ . ┘

*Proof:* The action is just  $P_{\text{Spin}}(E) \times_L \text{Cl}(\mathbb{R}^n) \times M \rightarrow P_{\text{Spin}}(E) \times M$ . □

**Prop. (16.7.1.9) [Spin<sup>c</sup>-structure].** The group  $\text{Spin}^c$  is the covering space of  $\text{SO}(n) \times S^1$  ( $n > 2$ ) that corresponds to the group of elements mod 0 mod 2 in  $\mathbb{Z}_2 \times \mathbb{Z}$ , i.e.  $\text{Spin}(n) \times S^1 / \{\pm 1\}$ .

For example,  $\text{Spin}^c(4) = \{(A_1, A_2) \in U(2) \times U(2) \mid \det A_1 = \det A_2\}$ , and  $\text{Spin}^C(3) = U(2)$ .

Then a  $\text{SO}(n)$  bundle can be lift to be a  $\text{Spin}^c$ -bundle if the line bundle determined by  $S^1$  is determine the same  $w_2$  as it, i.e.  $w_2 = c_1(L) \bmod 2$ , This is equivalent to  $w_2$  is in the image of  $H^2(X, \mathbb{Z})$ , and this is equivalent to the Bockstein image of it is zero.

Use a variant of Wu's formula:  $w_2(TM)[\alpha] = \alpha \cdot \alpha \bmod 2$  for  $M$  orientable of dimension 4, we have any orientable manifold of dimension 4 has a  $\text{Spin}^c$ -structure. Cf.[XieYi 几何学专题 Homework3]. ┘

There is a connection on the Clifford bundle and on the Spinor bundle induced by the Levi-Civita connection of  $M$  (16.3.4.2). This is compatible with the Clifford action. and it is also metric because the connection 1-form is in  $\mathfrak{so}(n)$  because the action of  $\text{SO}(n)$  preserves metric.

### Spin Manifolds

**Def. (16.7.1.10) [Spin Manifolds].**  $M$  is called a **spin manifold** if  $M$  is oriented and  $TM$  has a spin structure (16.7.1.2).

By (16.7.1.4),  $M$  has a spin structure if  $w_2(M) = 0$ , and if  $w_2(M) = 0$ , then the spin structures on  $M$  are in bijection with  $H^1(M, \mathbb{Z}/(2))$ .

By (16.7.1.3), the spin structure doesn't depend on the Riemannian metric of  $M$ , and it is even homotopy-invariant by (4.15.4.10). For this reason, we define a **topological spin manifold** to be a topological compact manifold with  $w_1 = w_2 = 0$  (which doesn't depend on the smooth structure). ┘

**Cor. (16.7.1.11).** If  $M$  is a complex manifold, then  $M$  is spin iff  $c_1(M) \equiv 1 \pmod{2}$ , by (4.15.3.31). ┘

**Cor. (16.7.1.12).** If  $M$  is 2-connected, then there exists a unique spin-structure on  $M$ . ┘

**Prop. (16.7.1.13).** If  $M, M'$  are spin manifolds, then  $M \times M'$  is also a spin manifold.

If  $M$  is a spin manifold and  $N \subset M$  is a submanifold with a spin structure on  $\mathcal{N}_{N/M}$ , then  $N$  is a spin manifold. In particular, if  $M$  is a spin manifold and  $\partial M \neq \emptyset$ , then  $\partial M$  is also a spin manifold. ┘

**Prop. (16.7.1.14) [Signature of Topological Spin Manifolds].** If  $X$  is a topological spin manifold (16.7.1.10), then the intersection form on  $H^2(X, \mathbb{Z})$  is even. In particular,  $\text{sgn}(X) \in 8\mathbb{Z}$  by (19.5.8.1).

Moreover, by [The topology of 4-manifolds, Freedman], there exists a compact topological spin 4-manifold  $X$  of signature 8.  $\lrcorner$

*Proof:* Cf. [L-W89]P89?  $\square$

## Spin Cobordisms

### Dirac Operators

**Def. (16.7.1.15) [Dirac Operators].**  $\lrcorner$

## 2 $\text{Spin}^c$ -Structures

**Def. (16.7.2.1) [ $\text{Spin}^c$ -Structures].** Let  $X \in \text{Mani}$  and  $P_{\text{SO}(n)}$  be a principal  $\text{SO}(n)$ -bundle over  $X$ , then  $\lrcorner$

**Prop. (16.7.2.2) [Spin Structure Obstructions].** A principle  $\text{SO}(n)$ -bundle  $P$  carries a  $\text{Spin}^c$ -structure iff  $w_2(P)$  is the mod2 reduction of an integral class, i.e.  $W_3(P) = 0$  (Cf. [L-W89]Appendix B.13?).  $\lrcorner$

*Proof:* ?  $\square$

**Def. (16.7.2.3) [ $\text{Spin}^c$ -Manifolds].**  $M$  is called a  $\text{Spin}^c$ -manifold if  $M$  is oriented and  $TM$  has a  $\text{Spin}^c$ -structure??.

By??,  $M$  has a spin structure if  $W_3(M) = 0$ . and if  $W_2(M) = 0$ , then the spin structures on  $M$  are in bijection with  $2H^1(M; \mathbb{Z}) \oplus H^1(M, \mathbb{Z}/(2))$ .

? By (16.7.1.3), the  $\text{Spin}^c$ -structure doesn't depend on the Riemannian metric of  $M$ , and it is even homotopy-invariant by (4.15.4.10). For this reason, we define a **topological spin manifold** to be a topological compact manifold with  $w_1 = w_2 = 0$  (which doesn't depend on the smooth structure).  $\lrcorner$

**Thm. (16.7.2.4).** If  $M$  is a compact  $\text{Spin}^c$ -manifold of even dimension and  $D$  is the Dirac operator of the fundamental complex spinor bundle, then

$$\text{ind}(D^+) = \{e^{\frac{c}{2}} \hat{A}(X)\}[X],$$

where  $c$  is the canonical class of the  $\text{Spin}^c$ -structure (16.7.2.1).  $\lrcorner$

*Proof:* ?  $\square$

## 3 Yang-Mills Equation & Seiberg-Witten Equation

[Atiyah, M. F.; Bott, R. The Yang-Mills equations over Riemann surfaces. Philos. Trans. Roy. Soc. London Ser. A 308 (1983), no. 1505, 523–615].

**Def. (16.7.3.1) [Young-Mills].** The Young-Mills functional on connections  $A$  on a bundle  $E$  on a compact oriented space:

$$YM(A)^2 = \|F_A\|^2 = - \int_X \text{tr}(F_A \wedge *F_A)$$

it is a critical point when  $d_A \star F_A = 0$  and  $d_A F_A = 0$ .  $\lrcorner$



**Prop. (16.7.3.2) [2-dim Case].**  $\star F \in \Omega^0(\mathfrak{su}(E))$  is parallel thus its characteristic spaces is orthogonal and a stable under parallel transport. So an irreducible YM  $SU(2)$ -connection must be flat, thus correspond to irreducible  $SU(2)$  representation of  $\pi_1(X)$ .  $\lrcorner$

**Prop. (16.7.3.3) [4-dim Case].**  $** = (-1)^{2*2} = \text{id}$  on  $\Omega^2(E)$  on  $E$  a  $SU(n)$ -bundle, so  $\Omega^2(E) = \Omega^+ \oplus \Omega^-$ . We have

$$\|F_A^+\|^2 + \|F_A^-\|^2 \geq \|F_A^-\|^2 - \|F_A^+\|^2 = \int_X \text{tr}(F_A \wedge F_A) = 8\pi^2 c_2(E)$$

Cf. [谢毅 Lecture5]. So it attains minimum at the connection that  $\star F_A = \pm F_A$  and  $d_A F_A = 0$  ((Anti)self-dual((anti)instanton)) depending on the sign of  $c_2(E)$ .  $\lrcorner$

**Prop. (16.7.3.4) [Anti-Instanton Connection on Complex Line Bundle].** For a  $U(1)$ -bundle,  $d_A F_A = dF_A$ , so  $F_A$  is harmonic, thus  $c_1(L) = [\frac{-1}{2\pi i} F_A] \in H^2(X, \mathbb{Z}) \cap \mathcal{H}_-^2(X, \mathbb{R})$ . In fact, this is equivalent to the existence of a anti-self-dual connection on this bundle.

If this is the case, then we have the ASD-connections module Gauge equivalence is isomorphic to  $H^1(X, \mathbb{R})/H^1(X, \mathbb{Z}) = T^{b_1(X)}$ .  $\lrcorner$

*Proof:* Because a gauge is just a  $X \rightarrow S^1$ , and its connected component thus equals  $[X, S^1] = H^1(X, \mathbb{Z})$  (MacLane space), and its identity is just the map that is homotopic to id. and  $d(gA) = dA - g^{-1}dg = dA - idu$ , for  $g = \exp(iu)$ , so  $\Omega^1/\mathcal{G} = H^1(X, \mathbb{R}/H^1(X, \mathbb{Z}) = T^{b_1(X)}$ .  $\square$

**Lemma (16.7.3.5) [Weizenbock Formula].** On a Riemannian manifold  $M$ , the Laplace operator has the form:

$$\Delta = -\nabla_{e_i e_i}^2 - \xi^i \wedge \iota(e_j) R(e_i, e_j)$$

where  $\nabla_{X,Y}^2 = \nabla_X \nabla_Y - \nabla_{\nabla_X Y}$ .

$$\int |\mathcal{D}_A \varphi|^2 = \int |\nabla_A \varphi|^2 + \frac{1}{4} R |\varphi|^2 + \frac{1}{2} \langle F_A^+ \varphi, \varphi \rangle.$$

If  $M$  is a spin manifold, then the Dirac operator  $D$  satisfies:

$$D^2 = -\nabla_{e_i e_i}^2 + \frac{1}{4} R$$

where  $R$  is the scalar curvature form on  $M$ . If  $M$  is a  $Spin^c$  manifold with a  $Spin^c$  connection  $\nabla_A$ , then the Dirac operator satisfies

$$D_A^2 = -\nabla_{A, e_i e_i}^2 + \frac{1}{4} R + \frac{1}{2} F_A$$

Cf. [Geometric Analysis Jost P143,153].  $\lrcorner$

**Prop. (16.7.3.6) [Seiberg-Witten].** The Seiberg-Witten equation functional for a unitary connection  $A$  on the determinant bundle of a  $Spin^c$  structure of  $M$  and a section of  $\mathcal{S}^+$  is:

$$\begin{aligned} SW(\varphi, A) &= \int \left( |\nabla_A \varphi|^2 + |F_A^+|^2 + \frac{R}{4} |\varphi|^2 + \frac{1}{8} |\varphi|^2 \right) Vol. \\ &= \int \left( |\mathcal{D}_A \varphi|^2 + |F_A^+ - \frac{1}{4} \langle e_j e_k \varphi, \varphi \rangle e^j \wedge e^k|^2 \right) Vol \end{aligned}$$

So the Seiberg-Witten equation is the lowest topological possible value of the Seiberg-Witten functional. It writes:

$$\mathcal{D}_A \varphi = 0, \quad F_A^+ = \frac{1}{4} \langle e_j e_k \varphi, \varphi \rangle e^j \wedge e^k.$$

Cf. [Jost Chapter 7].  $\lrcorner$

**Cor. (16.7.3.7).** If a compact oriented  $\text{Spin}^c$  manifold  $M$  has nonnegative scalar curvature, then the only possible solution is  $\varphi = F_A^+ = 0$ . (See from the equivalence of forms of Seiberg-Witten functional.)  
 $\lrcorner$

#### 4 Index Theorems(Atiyah-Singer)

References are [Heat equation and the Index Theorem Atiyah] and [Index Theorem].

**Prop. (16.7.4.1) [Gilkey].** For a natural transformation  $\omega$  from the functor  $p : M \rightarrow$  the Riemannian structure on  $M$  to the functor  $q : M \rightarrow k$ -forms on  $M$ , if it is homogenous of weight 0 w.r.t to metric  $g$  (i.e.  $\omega(\lambda^2 g) = \omega(g)$ ) and in local coordinates it has the coefficients of  $\omega(g)$  generated by  $g_{ij}$  and  $\det g^{-1}$  and their derivatives, then is a polynomial of Pontryagin classes of the given dimension. (not only up to homology).  
 $\lrcorner$

*Proof:* Cf.[Heat equation and the Index Theorem Atiyah P284].  $\square$

**Prop. (16.7.4.2) [Gilkey Generalized].** For a natural transformation  $\omega$  from the functor  $p : M \rightarrow$  Riemannian structures on  $M$  with a Hermitian bundle  $E$  with a Hermitian connection and the functor  $q : M \rightarrow k$ -forms on  $M$ , if it is homogenous of weight  $(0, 0)$  w.r.t to metric  $g, h$  and the Hermitian structure (i.e.  $\omega(\lambda^2 g, \mu^2 \xi) = \omega(g, \xi)$ ) and in local coordinates it has the coefficients of  $\omega(g, \xi)$  generated by  $g_{ij}, h_{ij}, \det h^{-1}, \det g^{-1}$  and  $\Gamma_k^{ij}$  (the connection form) and their derivatives, then is a polynomial of Pontryagin classes and Chern classes of  $E$  of the given dimension. (not only up to homology).  
 $\lrcorner$

*Proof:* Cf.[Heat equation and the Index Theorem Atiyah P290].  $\square$

**Cor. (16.7.4.3).** For a natural transformation  $\omega$  from the functor  $p : M \rightarrow$  Hermitian bundle  $E$  on  $M$  with a Hermitian connection and the functor  $q : M \rightarrow k$ -forms on  $M$ , if it is homogenous of weight 0 w.r.t to metric  $h$  and the Hermitian structure (i.e.  $\omega(\mu^2 \xi) = \omega(\xi)$ ) and in local coordinates it has the form  $\omega(g, \xi)$  generated by  $h_{ij}, \det h^{-1}$  and  $\Gamma_k^{ij}$  (the connection form) and their derivatives, then is a polynomial of Chern classes of  $E$  of the given dimension. Because when composed with the forgetful functor, it gives a transformation as above. And it is obviously independent of  $g$ .  
 $\lrcorner$

**Prop. (16.7.4.4) [Hodge].** For any differential operator  $A$  from a vector bundle  $E$  to a vector bundle  $F$ , we form two operators  $AA^*$  and  $A^*A$ , then they are both self adjoint elliptic operators, let these corresponding eigenspace be  $\Gamma_\lambda(E)$  and  $\Gamma_\lambda(F)$ , then  $A$  and  $A^*$  define an isomorphism between  $\Gamma_\lambda(E)$  and  $\Gamma_\lambda(F)$ .  
 $\lrcorner$

*Proof:*  $\square$

**Prop. (16.7.4.5) [Hirzebruch Signature Formula].** On a  $4n$ -dimensional orientable manifold  $M$ , the Poincare duality defines a bilinear pairing  $H^{2n}(M) \times H^{2n}(M) \rightarrow \mathbb{R}$ , its signature  $\sigma(M)$  is given by:

$$\sigma(M) = \int_M L_n(p_1, \dots, p_n).$$

Where  $L_n$  is the degree  $n$  part of the Taylor expansion of  $\prod_{i=1}^r \frac{\sqrt{x_i}}{\tanh \sqrt{x_i}}$  in terms of the symmetric polynomial.  
 $\lrcorner$

*Proof:* We consider the operator  $\tau : \alpha \mapsto i^{l+p(p-1)} * \alpha$ ,  $\tau^2 = 1$ , thus  $\Gamma^*$  is decomposed into two eigenspaces of  $\tau$ . We define the **signature operator**  $A$  as the restriction of  $\Delta = d - \tau d \tau$  to  $\Gamma_+$ .  $\Delta$  anti commutes with  $\tau$  thus maps  $\Omega_+$  to  $\Omega_-$ , then we have  $\ker A = \ker \Delta \cap \Omega_+$ , which is the positive harmonic forms  $H_+$ . So

$$\text{Ind} A = \dim H_+ - \dim H_-.$$

And we notice the positive and negative harmonic forms neutralize each other unless on the  $2n$ -forms, so only need to consider them. In fact, if we consider  $4n+2$  manifolds, then  $\tau$  is pure imaginary and the conjugation neutralize even the  $2n+1$  forms, so there are no signature.

Now the inner product  $\alpha \rightarrow \int \alpha \wedge * \alpha$  is positive definite for a real form  $\alpha$ , so this index of  $A$  is just the signature of the intersection form defined by cup product.  $\square$

**Cor. (16.7.4.6).** For a  $4n$ -dimensional  $M$  which is a boundary of a manifold, its signature is 0.  $\lrcorner$

*Proof:* By Stokes theorem, if  $M$  is a boundary of a manifold, then all its Pontryagin numbers, i.e.  $\int_M \prod p_i^{n_i}, \sum n_i = n$ , vanish.  $\square$

**Prop. (16.7.4.7) [Generalized Hirzebruch Signature Formula].** Let  $M$  be a  $2l$  dimensional smooth manifold and  $E$  be a Hermitian bundle over  $M$ , then The index of the generalized signature operator is giving by

$$\text{Ind} A_\eta = 2^l \cdot \text{ch}(E) L(p_1, \dots, p_l).$$

where  $L(M)(p_i) = \prod \frac{x_i/2}{\tanh x_i/2}$ .  $\lrcorner$

**Prop. (16.7.4.8) [Hirzebruch-Riemann-Roch].** For a  $n$ -dimensional complex line bundle  $L$  over a compact Kähler manifold  $M$ ,

$$\chi(M, L) = \int_M [\text{ch}(E) \text{td}(T^{1,0} M)]_n.$$

Where  $\chi(M, L) = \sum_{q=0}^n (-1)^q \dim H^q(M, E)$ ,  $\text{ch}$  is the Chern character (16.7.5.5) and  $\text{td}(T^{1,0} M)$  is the Todd polynomial, i.e. Taylor expansion of  $\prod_{i=1}^r \frac{t_i}{1 - e^{-t_i}}$  in terms of the symmetric polynomial, applied to  $c_i(T^{1,0} M)$ .  $\lrcorner$

**Cor. (16.7.4.9) [Riemann-Roch].** For a  $n$ -dimensional complex vector bundle  $E$  over a Riemann Surface  $M$ , let  $\deg E = \int_M c_1(E)$ , then

$$\chi(M, E) = H^0(M, E) - \dim H^1(M, E) = \deg E + \text{rk}(E)(1 - g).$$

Cf. [Index Theorem P115].  $\lrcorner$

### Hodge Theory

**Prop. (16.7.4.10) [Hodge].** By (13.1.8.12), if we investigate the Laplace operator  $\Delta_d$  on a compact orientable Riemannian manifold, we get that

$$\Omega^i = \mathcal{H}^i \oplus \text{Im } \Delta_d = \mathcal{H}^i \oplus \text{Im } d \oplus \text{Im } d^*.$$

Thus  $H^i$  can be uniquely represented by elements of  $\mathcal{H}^i$ .  $\lrcorner$

*Proof:* It suffice to prove  $\Delta_d$  is self-adjoint elliptic.

$\text{Im } \Delta_d \subset \text{Im } d \oplus \text{Im } d^*$ , and the result follows if we show  $\mathcal{H}^i, \text{Im } d, \text{Im } d^*$  are orthogonal. In fact, let  $\omega$  be harmonic, then  $(\omega, d^* \xi) = (d\omega, \xi) = 0$ ,  $(\omega, d\eta) = (d^* \omega, \eta) = 0$ ,  $(d\eta, d^* \xi) = (dd\eta, \xi) = 0$ .  $\square$

**Cor. (16.7.4.11) [Poincare Duality for deRham Cohomology].** If  $M$  is a  $n$ -dimensional compact orientable Riemannian manifold, then

$$H_{dR}^p(M) \cong H_{dR}^{n-p}(M)$$

Induced by  $*$ , because  $** = \pm 1$  and  $*$  commutes with  $\Delta_d$  (16.1.3.8), so it induce an isomorphism  $\mathcal{H}^p \cong \mathcal{H}^{n-p}$ .

Moreover,  $*$  in fact induces a perfect pairing:

$$H_{dR}^k(M) \times H^{n-k}(M) \rightarrow \mathbb{R}$$

induced by the map

$$*: \mathcal{H}^k(M) \times \mathcal{H}^{n-k}(M) \rightarrow \mathbb{R} : (\alpha, \beta) \mapsto \int_M \alpha \wedge * \beta$$

As  $\int_M \alpha \wedge * \alpha = \|\alpha\|^2 \neq 0$ .  $\lrcorner$

**Prop. (16.7.4.12).** On a compact complex manifold, the formal adjoint of  $\bar{\partial}$  is  $*\bar{\partial}*$ . (By direct calculation). Also  $d^* = (-1)^{n(p+1)+1} * d * = - * d *$ .  $\lrcorner$

**Prop. (16.7.4.13) [Hodge].** Given a compact Hermitian complex manifold  $(X, J, g)$  and a holomorphic line bundle  $E$  over it, there is a Hermitian metric on  $A^{p,q}E$ , and an operator  $\bar{\partial}$  on it. Then  $\bar{\partial}$  has a formal adjoint  $\bar{\partial}^*$ , and  $\Delta_{\bar{\partial}_E}$  can be defined. Let  $\mathcal{H}_E^{p,q}$  be the kernel of  $\Delta_{\bar{\partial}_E}$  on  $A^{p,q}E$ , called the  **$E$ -valued  $(p, q)$ -forms**, then there is a orthonormal decomposition

$$A^{p,q}E = \mathcal{H}_E^{p,q} \oplus \text{Im } \Delta_{\bar{\partial}_E} = \mathcal{H}_E^{p,q} \oplus \text{Im } \bar{\partial}_E \oplus \text{Im } \bar{\partial}_E^*$$

And thus  $\mathcal{H}^{p,q}(X, E) \cong H_{\bar{\partial}}^{p,q}(X, E)$ .  $\lrcorner$

*Proof:* It suffice to prove  $\Delta_{\bar{\partial}_E}$  is self-adjoint elliptic. The rest is verbatim as the proof of (16.7.4.10).  $\square$

**Cor. (16.7.4.14) [Hodge].** In case  $E = \mathcal{O}_X$ ,  $\mathcal{H}^{p,q}(X) \cong H_{\bar{\partial}}^{p,q}(X)$ .  $\lrcorner$

**Cor. (16.7.4.15) [Kodaira-Serre Duality].** For a Hermitian line bundle over a compact Hermitian complex manifold  $X$ , from Hodge theorem and (16.9.5.2), we get

$$H^p(X, \Omega^q(E)) \cong H^{n-p}(X, \Omega^{n-q}(E^*))$$

induced by  $\bar{*}_E$  and  $\bar{*}_{E^*}$ .  $\lrcorner$

## 5 Characteristic Classes of Manifolds

### Chern-Weil Theory

**Prop. (16.7.5.1) [Chern-Weil].** An **Invariant polynomial** of the entries of  $M_n(k)$  is one that is invariant under the conjugation action (3.3.4.5).

For any connection on  $E$ , the **Chern-Weil** map  $CW$  from invariant polynomial ring to  $H^*(X) : P \mapsto [P(\Omega)]$  is a ring homomorphism independent on the connection  $A$ .

There are relations between  $c_i$  and  $\text{tr}(\Omega^k)$ , they can be derived formally by considering diagonal elements.  $\lrcorner$

*Proof:* To prove  $P(\Omega)$  is closed, notice by (3.3.4.5), it suffice to show  $\text{tr}(\Omega^k)$  is closed. By (16.3.4.7),  $d\text{tr}(\Omega^k) = \text{tr}(\omega \wedge \Omega^k - \Omega^k \wedge \omega) = 0$ , which is zero because  $\Omega$  is of even dimension.

For the independence of connections, use (4.14.3.29). For two connection  $\nabla_i$ ,  $\nabla = t\nabla_0 + (1-t)\nabla_1$  (you can smooth it) is a connection on the vector bundle  $\pi^*E$  on  $M \times I$ , and the section 0 and 1 induces the connection  $\nabla_0$  and  $\nabla_1$ . Thus  $s_0^*$  and  $s_1^*$  are the same map, thus  $CW_M(p) = s_i^*CW_{M \times I}(p)$  are all the same map.  $\square$

**Cor. (16.7.5.2).** For a complex line bundle of degree  $r$  over a complex manifold,

$$\det\left(1 - \frac{1}{2\pi i}F_A\right) = 1 + c_1 + \dots + c_r$$

gives out the **Chern class**, because it satisfies the axioms of Chern class (4.15.4.17). In other words,  $c_k = \text{tr}((- \frac{1}{2\pi i}F_A)^k)$ .

For a real line bundle of degree  $r$ ,

$$\det\left(1 - \frac{1}{2\pi i}F_A\right) = 1 + p_1 + \dots + p_{\lfloor \frac{r}{2} \rfloor}$$

gives out the **Pontryagin class**, where  $p_k \in H^{4k}(X)$ . (Notice the  $\omega$  thus  $\Omega$  can be chosen to be skew-symmetric, thus for odd  $k$  the classes  $\text{tr}(\Omega^k) \in H^{2k}(X)$  vanish).

For an oriented real bundle of degree  $2r$ , the  $\omega$  and thus  $\Omega$  can be chosen to be skew-symmetric and the transformation matrix in  $SO(2r)$ , then

$$\text{Pf}\left(\frac{1}{2\pi}\Omega\right) \in H^{2r}(X)$$

is well-defined and closed and gives the **Euler class**  $e(E)$  (recall  $e(E)^2 = p_r(E)$ ). (Use  $\text{Pf}^2 = \det$  to get that  $[\frac{\partial \text{Pf}}{\partial \Omega_{ij}}]^t$  commutes with  $\Omega$ , then calculate  $d\text{Pf}(\Omega) = 0$ ).  $\lrcorner$

*Proof:* In fact, the construction is natural w.r.t the connection because connection can be pulled back and summed. Then the only task is the normality, which is direct calculation on  $\mathbb{CP}^1$ .  $\square$

**Cor. (16.7.5.3).**

$$c_1(E) = c_1(\wedge^{\dim E} E).$$

Direct from the formula.  $\lrcorner$

**Cor. (16.7.5.4) [Whitney Product Formula].**

$$c(E \oplus F) = c(E)c(F), \quad p(E \oplus F) = p(E)p(F)$$

Directly from the product connection on  $E \oplus F$ .  $\lrcorner$

**Prop. (16.7.5.5) [Chern Character].** The Chern character

$$ch(E) = [\text{tr} \exp(\frac{i}{2\pi}F_A)]$$

satisfies  $ch(E \oplus F) = ch(E) + ch(F)$  and  $ch(E \otimes F) = ch(E)ch(F)$  by simple calculation. So it defines a ring homomorphism from  $K(X)$  to  $H^*(X)$ .  $\lrcorner$

**Prop. (16.7.5.6) [Chern-Gauss-Bonnet].** For a  $2n$ -dimensional orientable manifold  $M$ ,

$$\int_M e(TM) = \chi(M).$$

┘

**Prop. (16.7.5.7).** For a vector bundle and a flat connection  $d_A$  on a manifold, i.e.  $d_A^2 = 0$ , we have a deRham like cohomology, and there is a sheaf of flat sections.

$$H^*(X, A) = H^*(X, E).$$

┘

### Integrality of Characteristic Numbers

**Thm. (16.7.5.8) [Atiyah-Hirzebruch].** If  $k \in \mathbb{Z}_+$  and  $M$  is a compact spin manifold of dimension  $4k$ , then  $\hat{A}(M) \in \mathbb{Z}$ . Moreover, if  $k \in 2\mathbb{Z} + 1$ , then  $\hat{A}(M) \in 2\mathbb{Z}$ .

┘

**Cor. (16.7.5.9) [Rochlin].** If  $M$  is a compact spin 4-manifold, then the signature of  $M$  is a multiple of 16. Notice the smoothness is necessary by (16.7.1.14).

┘

*Proof:* This is because for 4-manifolds, the signature is always 8 times the  $\hat{A}$ -genus, Cf. [L-W89]2.6.17?.

□

**Thm. (16.7.5.10) [ $\hat{A}$ -Genus of  $\text{Spin}^c$ -manifolds are Integral].** Let  $X$  be a compact orientable manifold and  $c \in H^2(X, \mathbb{Z})$  s.t.  $c \equiv w_2(X) \pmod{2}$  (hence  $X$  is a  $\text{Spin}^c$ -manifold (16.7.2.3)), then

$$\{e^{\frac{c}{2}} \hat{A}(X)\}[X] \in \mathbb{Z}.$$

┘

*Proof:* Cf. [L-W89]P400?.

□

**Thm. (16.7.5.11) [Bott].** If  $E \in \mathcal{V}\text{ect}^{\text{fin}}_{\mathbb{C}}(\mathbb{S}^{2n})$ , then

$$\frac{1}{(n-1)!} c_n(E)[\mathbb{S}^{2n}] \in \mathbb{Z}.$$

┘

*Proof:* Cf. [L-W89]P281?.

□

### Immersions of Manifolds and the Vector Field Problem

#### 6 Positive Curvatures

#### Group Actions on Manifolds with Positive Scalar Curvature

#### Kähler Geometry, Pure Spinors and Twistor Geometry

#### Theory of Calibrations

#### Riemannian Metrics with Reduced Holonomy

## 16.8 Symplectic Geometry

Cf. [Arn89]Chap8, [Lee13]Chap22, [辛几何讲义范辉军].

**Notation (16.8.0.1).**

- Use notations on smooth manifolds defined in [Smooth Manifolds I: Generalities](#).
- Let  $n \in \mathbb{Z}_+$ .
- Let  $M \in \text{Diff}^{2n}$  be a symplectic manifold (16.8.1.1).

┘

### 1 Basics

#### Symplectic Forms

**Def. (16.8.1.1) [Symplectic Forms and Symplectic Manifolds].** For  $X \in \text{Diff}$ , a **symplectic form**  $\omega$  on  $X$  is a closed 2-form on  $X$  that is non-degenerate on any point. A smooth manifold with a symplectic form is called a **symplectic manifold**. A symplectic manifold must be even dimensional and orientable, by [Darboux's Theorem](#).

┘

**Prop. (16.8.1.2) [Cotangent Space is a Symplectic Manifold].** For any  $X \in \text{Diff}$ , a coordinate chart  $\varphi = (p_1, \dots, p_n)$  of  $X$  gives a **Lagrangian coordinate chart**  $(p_1, \dots, p_n, dp_1, \dots, dp_n)$  of  $T^*X$ . We can denote  $q_i = dp_i$ , then there is a symplectic form on  $T^*X$  given in this coordinate chart by

$$\omega^2 = dp \wedge dq.$$

┘

*Proof:* Consider the 1-form  $\omega^1$  on  $T^*M$  s.t. for any  $p \in T^*M$  with  $x = \text{pr}(p) \in M$  and  $\xi \in T(T^*M)$ ,  $\omega^1(\xi) = \langle p, \text{pr}_*(\xi) \rangle$ . Then it can be verified that in the coordinate chart  $(x_1, \dots, x_n, dx_1, \dots, dx_n)$  it is just  $\omega^1 = p dq$ . Thus  $d\omega^1 = \omega^2$  is closed.

Finally, it is clearly non-degenerate. □

**Prop. (16.8.1.3).** A Hamiltonian phase flow preserves the symplectic form.  $g^{t*}\omega = \omega$ . ┘

*Proof:* by Cartan's magic formula,

$$\frac{d}{dt}(g^t)^*\omega = L_X\omega = \iota_X(d\omega) + d(\iota_X\omega) = d(\iota_X\omega)$$

because  $\omega$  is closed. And by definition,  $d(\iota_X\omega)(\eta) = \omega(JdH, \eta) = \langle dH, \eta \rangle$ , so  $d(\iota_X\omega) = dH$ , Thus the theorem. □

For the following Cf. [辛几何讲义范辉军 lecture3].

**Prop. (16.8.1.4) [Moser's Stability].** If  $\omega_t$  is a smooth family of cohomologous forms on a closed manifold  $M$ , then there exists an isotopy  $\Psi_t$  s.t.

$$\Psi_t^*(\omega_t) = \omega_0.$$

┘

**Prop. (16.8.1.5) [Relative Moser Stability].** If  $M$  is a closed manifold and  $S$  is a compact submanifold, then if two closed 2-form equals on  $S$ , then there is an open neighborhood  $N_0, N_1$  of  $S$  and a diffeomorphism  $\Psi : N_0 \rightarrow N_1$  that

$$\Psi|_S = \text{id}, \quad \Psi^*\omega_1 = \omega_0.$$

┘

**Cor. (16.8.1.6) [Darboux's Theorem].** Every symplectic form  $\omega$  on  $M$  is locally diffeomorphic to the standard form  $\omega_0$  on  $\mathbb{R}^{2n}$ .

┘

*Proof:* Choose  $S = \text{pt}$  and uses relative Moser stability.

□

**Prop. (16.8.1.7).** For a compact symplectic manifold  $M$ , its even dimensional cohomology groups doesn't vanish, because  $\omega^k$  are nontrivial.

┘

*Proof:* This is because  $\omega^n$  is a volume form on  $M$  that never vanish, so it gives  $M$  an orientation and  $\int_M \omega^n \neq 0$ . If  $\omega^k$  is exact, then  $\omega^n$  is exact, so  $\int_M \omega^n = 0$  by Stokes', contradiction.

□

## 2 Poisson Structure

**Prop. (16.8.2.1) [Legendre Transform and Hamiltonian Systems].** [https://en.wikipedia.org/wiki/Legendre\\_transformation#Legendre\\_transformation\\_on\\_manifolds](https://en.wikipedia.org/wiki/Legendre_transformation#Legendre_transformation_on_manifolds)

┘

*Proof:* ?

□

**Def. (16.8.2.2) [Hamiltonian Vector Fields].** A **Hamiltonian function** is a smooth function on  $M$ . For any Hamiltonian function  $H$ , the symplectic form  $\omega$  maps  $dH$  to a vector field  $L_H \in \mathfrak{X}(M)$  s.t.  $\xi(H) = \omega(\xi, L_H)$  for any  $\xi \in \mathfrak{X}(M)$ , called the **Hamiltonian vector field** of  $H$ .

┘

**Def. (16.8.2.3) [Poisson Brackets].** For any

In particular, for  $X \in \text{Diff}$  and any Lagrangian coordinate chart  $(\mathbf{p}, \mathbf{q})$ , the Poisson bracket of  $f, g \in C^\infty(T^*X)$  is given by

$$\{f, g\}_{\text{Pois}} = \frac{\partial}{\partial \mathbf{q}} f \frac{\partial}{\partial \mathbf{p}} g - \frac{\partial}{\partial \mathbf{p}} f \frac{\partial}{\partial \mathbf{q}} g.$$

┘



## 16.9 Complex Geometry

References are [Voi02, Huy05], [Principle of Algebraic Geometry Griffith/Harris], more advanced stuffs to add from [Kähler Geometry]. Even more advanced is [Demailly Complex Analytic and Differential Geometry].

### 1 Complex Manifolds

**Def. (16.9.1.1) [Complex Manifold].** A **complex manifold** is an even dimensional manifold that the transformation matrices are holomorphic. The category of complex manifolds are denoted by  $\mathbb{C}\text{-Mani}$ .  $\lrcorner$

**Prop. (16.9.1.2) [Andreotti-Franzcel].** Let  $M^n \subset \mathbb{C}^n \in \mathbb{C}\text{-Mani}^n$ , then  $M$  is homotopic to a CW complex of real dimension  $\leq n$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.9.1.3) [Adjunction Formula].** The normal sheaf of a submanifold  $Y \subset X$  is defined the same as the case of nonsingular varieties (7.12.1.17), then the same is true of the adjunction formula:

$$\mathcal{K}_Y \cong \mathcal{K}_X \otimes \det \mathcal{N}_{Y/X}$$

In case  $Y$  is of codimension 1,  $\mathcal{N}_{Y/X} \cong \mathcal{L}(Y)|_Y = \mathcal{O}_Y(Y)$ .  $\lrcorner$

**Prop. (16.9.1.4) [Remmert].** A non-compact complex manifold admits a proper holomorphic embedding into  $\mathbb{C}^N$  for some  $N$  iff it is a Stein manifold.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (16.9.1.5) [Siegel].** Let  $X \in \mathbb{C}\text{-Mani}^n$ , then the field  $R(X)$  of meromorphic functions on  $X$  has transcendence degree  $\leq n$  over  $\mathbb{C}$ . And in case  $\text{tr.deg}[R(X) : \mathbb{C}] = n$ , it is a f.g. field extension of  $\mathbb{C}$ . Then we define the **algebraic dimension** of a compact connected complex manifold  $X$  to be  $\dim^{\text{alg}}(X) = \text{tr.deg } R(X)$ .  $\lrcorner$

*Proof:* It suffices to show that given any meromorphic functions  $f_1, \dots, f_{n+1}$ , there is an algebraic relation between them.

Now for each  $x$ , there is a nbhd  $U_x$  that any  $f_i$  writes as the quotient of two holomorphic functions  $\frac{g_{i,x}}{h_{i,x}}$ . And assume  $W_x \subset V_x \subset \overline{V}_x \subset U_x$  are the metric balls  $B(x, \frac{1}{2}) \subset B(x, 1)$ . As  $X$  is compact, there are  $N$   $x_k$  that  $X = \cup W_{x_k}$ .

As on the intersections,  $\frac{g_{i,k}}{h_{i,k}} = \frac{g_{i,l}}{h_{i,l}}$ , any we can assume they are all prime, so  $\frac{g_{i,k}}{g_{i,l}} = \varphi_{i,kl}$  is a unit. Let  $\varphi_{kl} = \prod_i \varphi_{i,kl}$ , as  $X$  is compact, let  $C = \max_{k,l} \varphi_{kl} \geq 1$ .

For any homogenous polynomial  $F \in \mathbb{C}[X_1, \dots, X_{n+1}]$  of deg  $m$ , let  $G_k = F(\frac{g_{1,k}}{h_{1,k}}, \dots, \frac{g_{n,k}}{h_{n,k}})(\prod_i h_{ik})^m$ . Then  $G_k$  are holomorphic and  $G_k = \varphi_{lk}^m G_l$  on the intersection. Now I claim for any  $M > 0$ , there is a  $F$  that  $G_k$  vanishes up to at least order  $M$  at  $x_k$ .

For this, just consider the dimension of all homogenous polynomials of degree  $m$  is  $C_{m+n+1}^m$ , and the number of desired equations of elements needs to be vanish is  $N \cdot C_{m'-1+n}^{m'-1}$ , so this always can be achieved when  $m$  is sufficiently large.

By Schwartz lemma (12.5.8.5),  $|G_k(x)| \leq (\frac{1}{2})^{m'} C'$ , where  $C' = \max\{|G_k(x)| | k = 1, \dots, n, x \in \overline{V}_k\}$ .

If  $C' = |G_k(x)|$ , and  $x \in W_l$ , then  $C' = |G_l(x)| |\varphi_{lk}^m(x)| \leq \frac{C'}{2^{m'}} \cdot C^m$ . If for some  $m, m'$ ,  $C^m < 2^{m'}$ , then this shows  $C' = 0$  which will finish the proof.

Look back at the condition of  $m, m'$ ,  $C_{m+n+1}^m > N \cdot C_{m'-1+n}^{m'-1}$  can be achieved together with  $m < \lambda m'$  for any  $\lambda$ , because the left hand is degree  $n+1$  in  $m$  and the right hand is degree  $n$  in  $m'$ .  
 $\square$

### Almost Complex Structure

**Def. (16.9.1.6) [Almost Complex Structures].** For  $M$  a real orientable manifold of dimension  $2n$ , an **almost complex structure** is a real bundle map  $J : T_{\mathbb{C}}M \rightarrow T_{\mathbb{C}}M$  satisfying  $J^2 = -1$ . A manifold with an almost complex structure is called an **almost complex manifold**.

A complex manifold has an almost complex structure, just define

$$J\left(\frac{\partial}{\partial x_i}\right) = \frac{\partial}{\partial y_i}, \quad J\left(\frac{\partial}{\partial y_i}\right) = -\frac{\partial}{\partial x_i}.$$

┘

**Def. (16.9.1.7) [Complex Differentials].** Situation as in (16.9.1.6),  $J$  will define a bundle map  $J : T^*M \rightarrow T^*M$ , and it has two eigenvalues  $\pm i$ , denoted by  $T^{*1,0}M$  and  $T^{*0,1}M$ . The **formal differential forms**  $\wedge^k T^*M \cong \bigoplus_{0 \leq p \leq k} \wedge^{p,k-p} T^*M$ .

Define  $\partial = \text{pr}_{p+1,q} \circ d$  on  $\wedge^{p,q} T^*M$ , and  $\bar{\partial} = \text{pr}_{p,q+1} \circ d$  on  $\wedge^{p,q} T^*M$ .

┘

**Def. (16.9.1.8) [Integrability].** An almost complex structure  $(M, J)$  is called **integrable** iff it satisfies the following equivalent conditions:

- $d\alpha = \partial\alpha + \bar{\partial}\alpha$ .
- $d\alpha = \partial\alpha + \bar{\partial}\alpha$  is true for  $\alpha \in \Omega^{1,0}(M)$
- $[T^{0,1}X, T^{0,1}X] \subset T^{0,1}X$ .
- $\bar{\partial}^2 f = 0$  for functions  $f$ .

┘

*Proof:* 1  $\iff$  3 is because by (16.1.3.4), if  $u, v \in T^{0,1}X$ ,

$$d\alpha(u, v) = u(\alpha(v)) - v(\alpha(u)) - \alpha([u, v]) = -\alpha([u, v]).$$

3  $\iff$  4 is because by (16.1.3.4), if  $\alpha = \bar{\partial}f$  and  $u, v \in T^{0,1}X$ , then

$$\bar{\partial}^2 f(u, v) = u(\bar{\partial}f(v)) - v(\bar{\partial}f(u)) - \bar{\partial}f([u, v]) = u(df(v)) - v(df(u)) - \bar{\partial}f([u, v]) = \partial f([u, v])$$

$\square$

**Thm. (16.9.1.9) [Nirenberg-Newlander].** Given an almost complex manifold  $(M, J)$ , it is integrable iff it comes from a complex structure.

┘

*Proof:* Cf. [Foundation of Differential Geometry Kobayashi Chap9.2].

$\square$

**Cor. (16.9.1.10).** For  $M \in \mathbb{C}\text{-Mani}$ ,

$$\bar{\partial}^2 = 0, \quad \partial\bar{\partial} + \bar{\partial}\partial = 0, \quad \partial^2 = 0.$$

┘

## 2 Deformation of Complex Structures

Cf.[Kähler Geometry] and [Complex Geometry, Daniel Chap6], should be completed as soon as possible.

## 3 Analytic Spaces and Coherence Sheaves

Cf.[Demailly] and [GAGA Serre].

### Analytic Subvarieties

**Def. (16.9.3.1)[Analytic subvariety].** An **analytic subvariety** is a closed subset of a complex manifold that is locally defined by f.m. holomorphic functions. The **regular points** of an analytic subvariety locally defined by  $k$  functions is the points that  $\text{rank}((\frac{\partial}{\partial z_j} f_i)_{i \leq k, j \leq n}) = k$ .  $\lrcorner$

**Prop. (16.9.3.2) [Proper Mapping Theorem].** If  $U, M$  are complex manifolds and  $M \subset U$  is an analytic subvariety, then if  $f : U \rightarrow N$  is a holomorphic mapping whose restriction on  $M$  is proper, then  $f(M)$  is an analytic subvariety of  $N$ .  $\lrcorner$

*Proof:* Cf.[Griffith/harris P395].  $\square$

**Def. (16.9.3.3) [Local Analytic Spaces].** An **analytic subspace of  $\mathbb{C}^n$**  is an analytic subvariety of  $\mathbb{C}^n$ . On an analytic space, there is a sheaf of holomorphic functions  $\mathcal{O}_U$ . So we can define **holomorphic map  $\varphi$**  as continuous functions that maps holomorphic germs to holomorphic germs, which is equivalent to the coordinates of  $\varphi$  are all holomorphic.  $\lrcorner$

**Def. (16.9.3.4)[Analytic Spaces].** An **analytic space** is a Hausdorff local-ringed space  $(X, \mathcal{O}_X)$  that is locally isomorphic to an analytic subspace of  $\mathbb{C}^n$  (16.9.3.3). Morphisms are continuous maps that are locally holomorphic. Sub-analytic spaces are defined as usual.

The products of analytic spaces can be defined, and it has the product topology, unlike the case of schemes.  $\lrcorner$

**Prop. (16.9.3.5)[Analytic Modules].** Let  $(X, \mathcal{O}_X)$  be an analytic space (16.9.3.4), an **analytic module** over  $X$  is just an  $\mathcal{O}_X$ -module. For a sub-analytic space  $Y$ , we have a sheaf of ideals  $\mathcal{I}_Y$  which is the sheaf of germs vanishing at  $Y$ , and  $\mathcal{O}_X/\mathcal{I}_Y$  is a sheaf on  $X$  that is zero outside  $Y$ , and we identify it with  $\mathcal{O}_Y$ .  $\lrcorner$

**Def. (16.9.3.6)[Coherent Analytic Sheaves].**  $\textcolor{red}{?}$   $\lrcorner$

**Prop. (16.9.3.7)[Coherence of Structure Sheaf].** If  $(X, \mathcal{O}_X)$  is an analytic space, then  $\mathcal{O}_X$  is coherent  $\textcolor{red}{?}$ , and the sheaf of ideals  $\mathcal{I}_Y$  of any sub-analytic-space  $Y$  is also coherent.  $\lrcorner$

*Proof:* First prove for  $X$  is an open subset of  $\mathbb{C}^n$ , Cf.[GAGA Serre P4]. And by definition  $\mathcal{O}_X$  is a  $\mathcal{O}_X$ -module of f.t., and it is also coherent  $\textcolor{red}{?}$ , so  $\mathcal{O}_X$  is coherent.  $\mathcal{I}_Y$  is coherent because it is a kernel of  $\mathcal{H}_X \rightarrow \mathcal{H}_Y$ .  $\square$

## 4 Positive Current

## 5 Hermitian Vector Bundles

**Def. (16.9.5.1)[Holomorphic and Hermitian Vector Bundles].** H A **holomorphic vector bundle** is a vector bundle on a complex manifold that the transition functions are holomorphic. A **Hermitian vector bundle** is a holomorphic vector bundle endowed with a Hermitian metric<sup>?</sup>. Any holomorphic vector bundle has a Hermitian structure, by partition of unity method.  $\lrcorner$

**Prop. (16.9.5.2) [Hodge Star for Hermitian bundles].** If  $\mathcal{E}$  is a Hermitian vector bundle over a compact complex manifold  $X$  of complex dimension  $n$ , we define a conjugate-linear operator  $\bar{*} : A^{p,q}(\mathcal{E}) \rightarrow A^{n-p,n-q}(\mathcal{E}) : \eta \mapsto \bar{*}\eta$ , and a conjugate-linear functor  $\tau E \rightarrow E^*$  induced by the Hermitian metric on  $E$ .

Then we can define  $\bar{*}_E : A^{p,q}(E) \rightarrow A^{n-p,n-q}E : \eta \otimes s \mapsto \bar{*}(\eta) \otimes \tau(s)$ . It can be checked that

$$(\alpha, \beta) * 1 = \alpha \wedge *_E \beta,$$

$$\bar{\partial}_E^* = -\bar{*}_{E^*} \bar{\partial}_{E^*} \bar{*}_E, \quad \bar{*}_E \Delta_{\bar{\partial}_E} = \Delta_{\bar{\partial}_{E^*}} \bar{*}_{E^*}, \quad \bar{*}_{E^*} \bar{*}_E = (-1)^{p+1} \text{ on } \Omega^{p,q}(E).$$

$\lrcorner$

### Hermitian Manifold

**Def. (16.9.5.3) [Holomorphic Tangent Bundle].** Let  $M$  be a complex manifold, the complexified tangent bundle  $T_{\mathbb{C}}M$  is defined as  $TM \otimes_{\mathbb{R}} \mathbb{C}$ , the **holomorphic tangent bundle**  $T^{1,0}M$  and anti-holomorphic bundle  $T^{0,1}M$  are defined to be the vectors generated resp. by  $\frac{\partial}{\partial z_i}$  and  $\frac{\partial}{\partial \bar{z}_i}$ . The **holomorphic cotangent bundle** and anti-holomorphic cotangent bundle is defined to be the covectors generated by  $dz_i$  and  $d\bar{z}_i$ .  $\lrcorner$

**Def. (16.9.5.4) [Hermitian Metric].** Let  $M$  be an almost complex manifold, a **Hermitian metric** on  $T_{\mathbb{C}}M$  is a metric that is  $J$ -invariant, that is  $g(Ju, Jv) = g(u, v)$ . Notice(3.5.11.31) shows a Hermitian metric is equivalent to a non-degenerate Hermitian form on  $T_{\mathbb{C}}M$ , where  $g$  appears as the real part of the Hermitian form.  $\lrcorner$

**Def. (16.9.5.5) [Hermitian Manifolds].** A complex manifold with a Hermitian metric is called a **Hermitian manifold**.  $\lrcorner$

**Def. (16.9.5.6) [the Kähler Form].** Given a Hermitian manifold  $M$ , define the **Kähler form**  $\omega_g$  as  $\omega_g(u, v) = g(Ju, v)$ . Then it is a real 2-form on  $M$ .

Notice  $g(u, v) = \omega_g(u, Jv)$ , so  $g$  can be constructed by  $\omega_g$ , iff  $\omega_g$  is positive(16.10.6.1).  $\lrcorner$

### Analytic Picard Groups

**Def. (16.9.5.7) [Analytic Picard Groups].** The group of isomorphisms of holomorphic line bundles on a complex manifold  $X$  is denoted by  $\text{Pic}_{\mathbb{C}}(X)$ , called the **analytic Picard group** of  $X$ .  $\lrcorner$

**Prop. (16.9.5.8) [First Chern Class].** For a connected space  $X$ , there is an exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_X^* \rightarrow 0,$$

and it induces a map  $\text{Pic}_{\mathbb{C}}(X) = H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z})$ , which is a just the first Chern class(same proof as in(4.15.4.20)).

**WARNING:** in this case it is not necessarily isomorphism, not as in the case of topological line bundles.

in particular, The image of the first Chern class is trivial in  $H^2(X, \mathcal{O}_X)$ .  $\lrcorner$

**Def. (16.9.5.9).** The dual of the universal line bundle on  $\mathbb{CP}^n$  is called the **hyperplane line bundle**, denoted by  $H$  or  $\mathcal{O}(1)$ .  $\lrcorner$

**Prop. (16.9.5.10).**  $\text{Pic}_{\mathbb{C}}(\mathbb{CP}^n) \cong \mathbb{Z}$ , with  $\mathcal{O}(1)$  as a generator.  $\lrcorner$

*Proof:* As  $\mathbb{CP}^n$  is Kähler, use (16.11.5.5), then  $H^{0,k}(X, \mathbb{C}) \cong H^k(X, \mathcal{O}_X) = H^k(X, \mathcal{K}_X \otimes \mathcal{O}(2)) = 0$  for  $k \geq 1$  by Kodaira vanishing (16.10.7.3), and then  $\text{NS}(X) = H^{1,1}(X) = H^2(X, \mathbb{Z}) = \mathbb{Z}$  by Lefschetz (1,1)-form theorem (16.11.5.6). It remains to prove  $c_1(\mathcal{O}(1))$  is the generator, for this, Cf. [Demailly P280].  $\square$

**Prop. (16.9.5.11).** Let  $S_d$  be the set of homogenous polynomials of degree  $d$ , then

$$H^0(\mathbb{CP}^n, \mathcal{O}(d)) = \begin{cases} S_d & d \geq 0 \\ 0 & d < 0 \end{cases}$$

$\lrcorner$

*Proof:* This is because it is sections that satisfy  $f_{\alpha}([z]) = (\frac{z_{\beta}}{z_{\alpha}})^k f_{\beta}([z])$ , which says  $f_{\alpha}$  glue together to give a holomorphic function homogenous of degree  $k$  on  $\mathbb{C}^n \setminus \{0\}$ , which extends to a function on  $\mathbb{C}^n$  by (12.5.8.4), then it is easy to see it is a homogenous polynomial using the power series expansion.  $\square$

**Def. (16.9.5.12)[Complex Neron-Severi Group].** For a compact complex manifold  $X$ , the **complex Néron-Severi group**  $\text{NS}_{\mathbb{C}}(X)$  is the image of  $\text{Pic}_{\mathbb{C}}(X) \rightarrow H^2(X, \mathbb{Z})$ .  $\text{rank}(\text{NS}(X))$  is called the **Picard number** of  $X$ .

There is a good description of  $\text{NS}(X)$  in case  $X$  is Kähler, See Lefschetz theorem (16.11.5.6).

This is compatible with the algebraic Néron-Severi group (10.9.3.39), i.e.  $\text{NS}_{\mathbb{C}}(X) = \text{Pic}_{\mathbb{C}}(X) / \text{Pic}_{\mathbb{C}}^0(X)$  Cf. [G-H94]P457, [Vak17]P484.  $\lrcorner$

### Chern Connection

**Prop. (16.9.5.13)[Chern Connection].** Given a Hermitian holomorphic bundle  $E \rightarrow M$  on a complex manifold, there is a unique **Chern connection**  $\nabla$  on  $E$ , that  $\nabla$  is holomorphic (i.e. the connection matrix is holomorphic w.r.t a holomorphic frame), and it is compatible with the Hermitian metric.  $\lrcorner$

*Proof:* Write out the requirement: if  $H = h_{ij}$  is the matrix of the Hermitian metric, so  $H$  is Hermitian, and we need  $dh_{ij} = (\nabla e_i, e_j) + (e_i, \nabla e_j) = \sum_k \omega_{ik} h_{kj} + \sum_k \overline{\omega_{jk}} h_{ik} \omega$  is holomorphic, so must

$$\partial H = \theta H, \quad \bar{\partial} H = H \bar{\theta}^t.$$

But  $H^t = \bar{H}$  so these two equations are equivalent and  $\theta = \partial H H^{-1}$ .  $\square$

**Cor. (16.9.5.14).** The curvature of the Chern connection is  $\Omega = \bar{\partial}(\partial(h)h^{-1})$ . In particular, it is a skew-symmetric matrix of  $(1,1)$ -forms. If it is of dimension 1, then  $\Omega = \bar{\partial}\partial \log h$ .  $\lrcorner$

*Proof:*  $\Omega$  is locally  $d\omega + \omega \wedge \omega$ , so if we choose a unitary basis, then  $\omega$  is skew-symmetric by definition and  $\omega \wedge \omega$  is also skew-symmetric, so  $\Omega$  is skew-symmetric. The calculation is direct calculation.  $\square$

**Prop. (16.9.5.15).** The transformation matrix of a complex manifold is holomorphic, so it is possible to define globally  $\bar{\partial}$  operator. And locally on a nbhd,  $\partial$  is defined as  $d - \bar{\partial}$ .  $\lrcorner$

**Prop. (16.9.5.16) [Normal Coordinate].** For a Hermitian vector bundle  $E$  over a complex manifold  $X$ , given any coordinate frame  $(z_j)$ , there exists a holomorphic frame  $(e_\lambda)$  that

$$\langle e_\lambda, e_\mu \rangle = \delta_{\lambda,\mu} - \sum c_{jk\lambda\mu} z_j \bar{z}_k + O(|z|^3)$$

where  $c_{ij\lambda\mu}$  is the coefficient of the Chern connection  $\Omega$ . Such a coordinate is called the **normal coordinate frame** of  $E$  at  $x$ .  $\lrcorner$

*Proof:* Cf. [Demailly P270].  $\square$

## 6 Cohomology

**Lemma (16.9.6.1) [Dolbeault Complex,  $\bar{\partial}$ -Poincaré Lemma].** If  $X$  is a complex manifold of dimension  $n$ , and  $\mathcal{E}$  a holomorphic vector bundle, then there is an exact sequence:

$$0 \rightarrow \Omega^{p,0}(\mathcal{E}) \xrightarrow{\bar{\partial}_\mathcal{E}} \Omega^{p,1}(\mathcal{E}) \xrightarrow{\bar{\partial}_\mathcal{E}} \dots \xrightarrow{\bar{\partial}_\mathcal{E}} \Omega^{p,n-p}(\mathcal{E}) \rightarrow 0$$

If  $p = 0$ , it is called the **Dolbeault complex** of  $\mathcal{E}$ .  $\lrcorner$

*Proof:* Because tensoring  $\mathcal{E}$  is exact, it suffices to show for  $\mathcal{E} = \mathbb{C}_X$ . As  $\bar{\partial}^2 = 0$  (16.9.1.10), and the sequence is clearly exact at  $\Omega^{p,0}(\mathcal{E})$ , it suffices to show that if  $\alpha \in \Omega^{p,q}(X)$ ,  $q > 0$  satisfies  $\bar{\partial}\alpha = 0$ , then  $\alpha = \bar{\partial}\beta$  for some  $\beta \in \Omega^{p,q-1}(X)$ . Let

$$\alpha = \sum_I \sum_J \alpha_{I,J} dz_I \wedge d\bar{z}_J = \sum_I dz_I \left( \sum_J \alpha_{I,J} d\bar{z}_J \right),$$

then it is clear it suffices to prove for  $(\sum_J \alpha_{I,J} d\bar{z}_J)$ , and the question is reduced to the case  $p = 0$ .

Suppose  $\alpha = \sum_J \alpha_J \wedge d\bar{z}_J$ , then we use induction on  $k$  which equals the maximal number s.t. there is a  $J$  with  $k \in J$  and  $\alpha_J \neq 0$ . Notice  $k \geq q$ . If  $k = q$ , then

$$\alpha = f d\bar{z}_1 \wedge \dots \wedge d\bar{z}_q,$$

and  $f$  is holomorphic in the variables  $z_l, l > q$ . Then the proof of (12.4.1.12) gives us a smooth function  $g$  s.t.  $\frac{\partial}{\partial \bar{z}_q} g = f$  that is holomorphic in the variables  $z_l, l > q$ . So

$$\bar{\partial}((-1)^{q-1} g d\bar{z}_1 \wedge \dots \wedge d\bar{z}_{q-1}) = f d\bar{z}_1 \wedge \dots \wedge d\bar{z}_q.$$

So the case  $k = q$  is proved. Suppose the assertion is true for  $k - 1 \geq q$ , suppose

$$\alpha = \alpha_1 + \alpha_2 \wedge d\bar{z}_k,$$

where only the coordinates of index  $< k$  appear in  $\alpha_i$ . Then  $\alpha_2$  is holomorphic in the same argument as above shows that for each  $J$ ,  $\alpha_{2,J} = \frac{\partial}{\partial \bar{z}_k} \beta_{2,J}$  that each  $\beta_{2,J}$  is holomorphic in the variables  $z_l, l > q$ . Then

$$\bar{\partial} \left( \sum_J \beta_{2,J} d\bar{z}_J \right) = (-1)^{q-1} \alpha_2 \wedge d\bar{z}_k + \alpha'_1$$

where  $\alpha'_1$  involves only the coordinates  $z_l$  for  $l < k$ . Then we eliminated  $d\bar{z}_k$  and can use induction hypothesis from now.  $\square$

**Def. (16.9.6.2)[Dolbeault Cohomology].** The **Dolbeault cohomology group**  $H_{\bar{\partial}}^{p,q}(X, \mathcal{E})$  of a holomorphic vector bundle  $\mathcal{E}$  over a complex manifold  $X$  is defined to be the  $q$ -th cohomology group of the complex

$$0 \rightarrow \Omega^{p,0}(\mathcal{E}) \xrightarrow{\bar{\partial}_{\mathcal{E}}} \Omega^{p,1}(\mathcal{E}) \xrightarrow{\bar{\partial}_{\mathcal{E}}} \dots \xrightarrow{\bar{\partial}_{\mathcal{E}}} \Omega^{p,n-p}(\mathcal{E}) \rightarrow 0$$

and  $H_{\bar{\partial}}^{p,q}(X)$  is defined to be  $H_{\bar{\partial}}^{p,q}(X, \underline{\mathbb{C}}_X)$ . By (16.9.6.5),  $H_{\bar{\partial}}^{p,q}(X, \mathcal{E}) \cong H_{\bar{\partial}}^q(M, \Omega_{\text{hol}}^p \otimes_{\mathcal{O}_X} \mathcal{E})$ .  $\lrcorner$

**Cor. (16.9.6.3).** If  $X$  is a complex manifold of dimension  $n$ , there are exact sequences:

$$0 \rightarrow \underline{\mathbb{C}}_X \xrightarrow{\partial} \mathcal{O}_X \xrightarrow{\partial} \Omega_{\text{hol}}^1 \xrightarrow{\partial} \dots \xrightarrow{\partial} \Omega_{\text{hol}}^n \rightarrow 0$$

$\lrcorner$

*Proof:* This follows from the Poincaré lemma (7.3.5.10) and  $\bar{\partial}$ -Poincaré lemma (16.9.6.1), by applying the spectral sequence (I mean in the category of sheaves).  $\square$

**Prop. (16.9.6.4)[Holomorphic Cohomology].** For  $X \in \mathbb{C}\text{-Mani}$ ,

$$H^p(X, \underline{\mathbb{C}}) = H^p(X, \Omega_{\text{hol}}^{\bullet}).$$

$\lrcorner$

**Prop. (16.9.6.5)[Dolbeault].** For  $X \in \mathbb{C}\text{-Mani}$  and  $\mathcal{E}$  a holomorphic bundle,

$$H_{\bar{\partial}}^{p,q}(X, \mathcal{E}) \cong H^q(M, \Omega_{\text{hol}}^p \otimes_{\mathcal{O}_X} \mathcal{E}),$$

where the left is Dolbeault cohomology and the right is sheaf cohomology. (Use the fact that smooth sheaf is fine so adapted to sheaf cohomology (7.3.5.4), and  $\bar{\partial}$ -Poincaré lemma (16.9.6.1)).

Moreover, there is a spectral sequence of

$$E_1^{p,q} = H_{\bar{\partial}}^{p,q}(X) \Rightarrow E^n = H_{dR}^n(X, \mathbb{R}) \times_{\mathbb{R}} \mathbb{C}.$$

$\lrcorner$

**Prop. (16.9.6.6)[Cartan].** The class of Coh-Acyclic subsets of an analytic space is exactly the class of Stein manifolds.  $\lrcorner$

*Proof:*  $\square$

### De.Rham Cohomology

**Prop. (16.9.6.7).** The Frölicher spectral sequence of a compact Kähler manifold generates at  $E_1$ .  $\lrcorner$

*Proof:*  $\square$

## 7 GAGA

Main references are [Ser55].

### Analytification of Algebraic Varieties and Sheaves

**Prop. (16.9.7.1)[Analytification].** For any variety over  $\mathbb{C}$ , any open affine subset is isomorphic to an analytic space of  $\mathbb{C}^n$ , hence can be given an analytic structure  $X^{\text{an}}$  called the **analytification** of  $X$ . It is a locally ringed space, together with a map  $X^{\text{an}} \rightarrow X$  of local ringed spaces.

This construction extends to a functor

$$\text{Sch}^{\text{loc.ft,sep}}/\mathbb{C} \rightarrow \text{AnSpa}/\mathbb{C}.$$

An **algebraic analytic space** is an analytic space that is in the essential image of this functor.

Notice  $X^{\text{an}}$  and  $X$  have in fact the same underlying sets.  $\lrcorner$

**Remark (16.9.7.2).** There is in fact a more general analytification for any scheme locally of finite type over  $\mathbb{C}$ . That is, we define it as the right adjoint to the forgetful functor from analytic spaces to local ringed spaces. Where an analytic space is a local ringed space that locally has immersions into  $\mathbb{C}^n$ . Should consult [Grothendieck EGA1-7].  $\lrcorner$

*Proof:* Notice the schemes that have an analytification is stable under open subscheme, closed subscheme and products, and we can make a glue a large space from open subschemes by the unicity. So we only need to consider  $\text{Spec } \mathbb{C}[T]$ , whose analytification is  $\mathbb{C}$ .  $\square$

**Def. (16.9.7.3)[Betti Cohomologies].** For any variety defined over  $\mathbb{C}$ , and  $\mathbb{R}$  a ring, let  $H_{\text{Betti}}^i(X, R) = H^i(X^{\text{an}}, R)$  be the **Betti cohomology** of  $X$  with coefficients in  $R$ .  $\lrcorner$

**Prop. (16.9.7.4)[Transfer of Properties].**

- $X^{\text{an}}$  is locally compact and  $\sigma$ -compact.
- $X^{\text{an}}$  is Hausdorff
- A morphism  $f : X \rightarrow Y$  is smooth/étale iff  $f^{\text{an}}$  is smooth/a local isomorphism.  $X$  is smooth over  $\mathbb{C}$  iff  $X^{\text{an}}$  is a complex manifold.
- A morphism  $f : X \rightarrow Y$  is proper iff  $X^{\text{an}} \rightarrow Y^{\text{an}}$  is proper. In particular,  $X$  is complete(proper) iff  $X^{\text{an}}$  is compact.
- If  $X$  is projective and connected, then  $X^{\text{an}}$  is connected iff  $X$  is connected.  $\lrcorner$

*Proof:* 1:  $X$  is qc hence covered by f.m. affine subsets hence second-countable and use(4.4.2.23).  $X^{\text{an}}/X$  is flat because completion of Noetherian rings are flat(5.2.3.24).

2: Because analytification preserves products and morphisms, and separatedness of  $X$  shows that  $\Delta(X)$  is closed in  $X \times X$ , hence it is also closed in the analytification.

3: This follows from the Jacobian criterion(5.4.5.24).

4: Cf.[GAGA Serre P8].

5: This follows from(16.9.7.16), as  $H^0(X, \mathcal{O}_X) = H^0(X^{\text{an}}, \mathcal{O}_X)$ .  $\square$

**Prop. (16.9.7.5).** There is a natural map from  $\mathcal{O}_x$  to  $\mathcal{H}_x$  that maps  $\mathfrak{m}_x$  to  $\mathfrak{m}_x \mathcal{H}_x$ , thus inducing a map  $\hat{\theta} : \hat{\mathcal{O}}_x \rightarrow \hat{\mathcal{H}}_x$ . This is an isomorphism. In particular,  $\theta : \Omega_x \rightarrow \mathcal{H}_x$  is injective.

Moreover, if  $Y$  is a locally closed subscheme of  $X$ , then the local ideal of functions vanishing at  $Y$  maps to  $\mathcal{A}_x(Y)$ , and  $\mathcal{A}_x(Y)$  is generated by  $\theta(\mathcal{I}_x(Y))$ . Moreover,  $\mathcal{H}_{x,Y} = \mathcal{H}_x/\mathcal{I}_x(Y)$ .  $\lrcorner$

*Proof:* [GAGA Serre P6].  $\square$



**Cor. (16.9.7.6).** The inclusion  $\mathcal{O}_x \subset \mathcal{H}_x$  is flat ring extension, by (5.4.1.19) and the fact  $\hat{A}/A$  is flat. And  $\dim \mathcal{O}_x = \dim \mathcal{H}_x$  because  $\dim A = \dim \hat{A}$  (5.2.4.16).  $\lrcorner$

**Cor. (16.9.7.7).** Given an open and dense subscheme  $U$  of an algebraic variety  $X$  over  $\mathbb{C}$ ,  $U^{\text{an}}$  is dense in  $X^{\text{an}}$ .  $\lrcorner$

*Proof:* Consider the complement  $Y$ , if  $U^{\text{an}}$  is not dense in  $X^{\text{an}}$ , then there exists a  $x$  that  $\mathcal{A}_x(Y) = 0$ , so by (16.9.7.5),  $\mathcal{I}_x(Y) = 0$ , so  $Y$  is not dense near  $x$ , contradicton.  $\square$

**Cor. (16.9.7.8).** For a morphism  $f$  of algebraic varieties over  $\mathbb{C}$ ,  $\overline{f(X)^{\text{an}}} = \overline{f(X)}^{\text{an}}$ .  $\lrcorner$

*Proof:* By Chevalley theorem (7.6.1.6), there is a open dense subscheme  $U$  of  $\overline{f(X)}$  that is contained in  $f(X)$ , then (16.9.7.7) shows  $U^{\text{an}}$  is dense in  $\overline{f(X)}^{\text{an}}$ , so  $\overline{f(X)}^{\text{an}} \subset \overline{f(X)^{\text{an}}}$ . The converse is obvious.  $\square$

**Def. (16.9.7.9) [Analytification of Sheaves].** Denote for a sheaf  $\mathcal{F}$  over  $X$   $\mathcal{F}'$  the inverse image sheaf over  $X^{\text{an}}$  pulled back along  $X^{\text{an}} \rightarrow X$ . Define  $\mathcal{F}^{\text{an}}$  the **analytification of  $\mathcal{F}$**  as the sheaf  $\mathcal{F}' \otimes_{\mathcal{O}_X} \mathcal{H}_X$ .  $\lrcorner$

**Prop. (16.9.7.10).**  $\mathcal{F} \rightarrow \mathcal{F}^{\text{an}}$  is exact from the category of sheaves on  $X$  to the category of analytic sheaves on  $X^{\text{an}}$ ,  $\mathcal{F}' \rightarrow \mathcal{F}^{\text{an}}$  is injective, and it maps coherent sheaves to coherent analytic sheaves.  $\lrcorner$

*Proof:* The first two follows from the fact that  $\mathcal{H}_X$  is flat over  $X^{\text{an}} \rightarrow X$  (16.9.7.6). For the last assertion, notice if  $\mathcal{O}_X^p \rightarrow \mathcal{O}_X^q \rightarrow \mathcal{F} \rightarrow 0$ , then  $\mathcal{H}_X^p \rightarrow \mathcal{H}_X^q \rightarrow \mathcal{F}^{\text{an}} \rightarrow 0$ , so it is coherent because  $\mathcal{H}_X$  is coherent (16.9.3.7) (7.2.2.28).  $\square$

**Prop. (16.9.7.11).** Let  $i : Y \rightarrow X$  be a closed subscheme, then for a coherent sheaf  $\mathcal{F}$  on  $Y$ ,  $(i^{\text{an}})_* \mathcal{F}^{\text{an}} \cong (i_* \mathcal{F})^{\text{an}}$ .  $\lrcorner$

*Proof:* These two sheaves are both 0 outside  $Y^{\text{an}}$ , consider a point of  $Y$ , their stalks are respectively  $\mathcal{F}_x \otimes_{\mathcal{O}_{x,X}} \mathcal{H}_{x,X}$  and  $\mathcal{F}_x \otimes_{\mathcal{O}_{x,Y}} \mathcal{H}_{x,Y}$ . By (16.9.7.5) we notice

$$\mathcal{H}_{x,Y} = \mathcal{H}_{x,X} / \mathcal{I}_x(Y) \mathcal{H}_{x,X} = \mathcal{H}_{x,X} \otimes_{\mathcal{O}_{x,X}} \mathcal{O}_{x,Y}.$$

So this two are equal by associativity of tensor product.  $\square$

**Prop. (16.9.7.12).** By Leray Spectral sequence (7.3.1.9), for an analytic sheaf  $\mathcal{G}$ , there is a boundary map  $H^k(X, \mathcal{G}) \rightarrow H^k(X^{\text{an}}, \mathcal{G})$ . So for a sheaf  $\mathcal{F}$  on  $X$ , there is a map

$$\varepsilon : H^k(X, \mathcal{F}) \rightarrow H^k(X, \text{an}_* \mathcal{F}^{\text{an}}) \rightarrow H^k(X^{\text{an}}, \mathcal{F}^{\text{an}})$$

$\lrcorner$

### Equivalence between Algebraic Variety and Analytic Spaces

**Remark (16.9.7.13) [GAGA Principle].** In principal, any complete analytic object in  $\mathbb{CP}^n$  is algebraic.  $\lrcorner$

**Prop. (16.9.7.14).** Let  $X, Y$  be algebraic varieties over  $\mathbb{C}$  and  $f : X \rightarrow Y$  is morphism that is bijective, if  $f^{\text{an}}$  is an analytic isomorphism, then  $f$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [GAGA Serre P9].  $\square$

**Cor. (16.9.7.15).** Let  $X, Y$  be algebraic varieties over  $\mathbb{C}$ , iff  $f : X^{\text{an}} \rightarrow Y^{\text{an}}$  is holomorphic map and the image of  $f$  in  $X^{\text{an}} \times Y^{\text{an}} = (X \times Y)^{\text{an}}$  comes from an algebraic subscheme, then  $f$  comes from an algebraic morphism. (Because  $X^{\text{an}} \rightarrow \Gamma(X)$  is an analytic isomorphism).  $\lrcorner$

**Prop. (16.9.7.16) [GAGA on Coh(X)].** Let  $X$  be a proper scheme over  $\mathbb{C}$ , then  $\mathcal{F} \rightarrow \mathcal{F}^{\text{an}}$  defines an equivalence of categories between  $\text{Coh}(X)$  and  $\text{Coh}^{\text{an}}(X^{\text{an}})$ .  $\lrcorner$

*Proof:* Cf. [GAGA Serre P13], [SGA1, Chap12]..  $\square$

**Prop. (16.9.7.17) [GAGA on Projective Varieties].**

- (Chow) Any analytic subvariety of  $\mathbb{CP}^n$  is algebraic.
- Any meromorphic function on an algebraic variety  $V \subset \mathbb{CP}^n$  is rational.
- Any meromorphic differential form on a smooth variety is algebraic.
- Any holomorphic map between smooth varieties can be given by rational maps.
- Any holomorphic vector bundle on a smooth variety is algebraic, i.e. transition function can be made rational.

Cf. [Griffith/Harris P168,170]. ?  $\lrcorner$

**Cor. (16.9.7.18).** If the analytification of a variety  $X$  is a compact complex manifold, i.e.  $X$  is smooth (16.9.7.4), then  $K(X) = K(X^{\text{an}})$ , as they are both morphism to  $\mathbb{P}^1$ .  $\lrcorner$

### Applications

**Prop. (16.9.7.19) [Generalized Riemann Existence Theorem].** Let  $X$  be a normal scheme of finite type over  $\mathbb{C}$ . Given any finite morphism of analytic spaces (i.e. proper and has finite fibers) form a normal complex analytic space  $f : X' \rightarrow X^{\text{an}}$ , then there is a unique normal scheme  $X'$  and a finite morphism  $g : X' \rightarrow X$  that  $g^{\text{an}} = f$ .  $\lrcorner$

*Proof:* Cf. [SGA1, Chap12], using resolution of singularities.  $\square$

**Cor. (16.9.7.20) [Algebraic Fundamental Group].**  $\lrcorner$

## 8 Algebraic Compact Complex Manifolds

**Def. (16.9.8.1) [Moishezon Manifolds].** A compact complex manifold is called a **Moishezon manifold** iff  $\text{tr. deg. } K(X) = \dim X$ , by (16.9.1.5) this is the highest degree it can have. When  $X$  is an analytification of an algebraic variety  $X^{\text{an}}$ ,  $K(X^{\text{an}}) = K(X)$  by (16.9.7.18), so Moishezon is a necessary condition for a compact complex manifold to be algebraic.  $\lrcorner$

**Prop. (16.9.8.2) [Chow-Kodaira].** Any Moishezon manifold of dimension 2 is algebraic and projective.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.9.8.3) [Moishezon].** Any Moishezon manifold becomes algebraic and projective after a finite number of monoidal transformations with non-singular centers.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.9.8.4) [Artin].** The category of smooth proper algebraic spaces over  $\mathbb{C}$  is equivalent to the category of Moishezon manifolds.

In particular, there are non-algebraic Moishezon manifolds in dimension  $\geq 3$ . Examples are given in [Har77]P444.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.9.8.5) [Moishezon].** Every Moishezon manifold that is Kähler is projective and algebraic.  $\lrcorner$

*Proof:* Cf. [Moishezon On  $n$ -dimensional compact varieties with  $n$  algebraically independent meromorphic functions].  $\square$

## 9 Riemann Surfaces

**Prop. (16.9.9.1) [Genus].**  $\lrcorner$

**Thm. (16.9.9.2) [Uniformization Theorem, Poincaré-Klein-Koebe].** Any simply-connected Riemann surface is analytically isomorphic to one of the following:

$$\mathcal{H} \cong \mathbb{D}, \mathbb{C}, \mathbb{P}^1(\mathbb{C}).$$

$\lrcorner$

*Proof:*  $\square$

**Cor. (16.9.9.3) [Classifying Riemann Surfaces].** For any Riemann surface  $S$  with universal covering space  $\tilde{S}$ ,

- If  $\tilde{S} \cong \overline{\mathbb{C}}$ , then  $S$  is compact, so by (16.9.9.6) and Riemann-Hurewitz (7.13.1.33),  $\tilde{S} \rightarrow S$  is an isomorphism. So in this case  $S \cong \overline{\mathbb{C}}$ .
- If  $\tilde{S} \cong \mathbb{C}$ , then  $\pi_1(S) \subset \text{Aut}(\tilde{S}) = \mathbb{C} \ltimes \mathbb{C}^\times$  by (12.4.7.8). But  $\pi_1(S)$  can have no fixed point, so they are all of the form  $\tau : z \mapsto z + b$ . So it is a lattice of  $\mathbb{C}$ , thus isomorphic to  $\mathbb{1}, \mathbb{Z}$  or  $\mathbb{Z}^2$ . The corresponding  $S$  is  $\mathbb{C}, \mathbb{C}^\times$  or  $\mathbb{C}/\Lambda$  a torus.
- If  $\tilde{S} \cong \mathcal{H}$ , then  $S = \mathcal{H}/\Gamma$  where  $\Gamma \subset \text{PSL}(2, \mathbb{R})$  is a Fuchs group ?. Such  $S$  are called a **hyperbolic Riemann surface**.  $\lrcorner$

**Cor. (16.9.9.4).** If  $S$  is a Riemann surface with a non-constant meromorphic function omitting at most 3 points, then  $S$  is of hyperbolic type.  $\lrcorner$

*Proof:* This is because  $\mathbb{C}$  or  $\mathbb{P}^1(\mathbb{C})$  cannot be mapped into  $\mathbb{D}$ .  $\square$

**Prop. (16.9.9.5) [Metrics on Riemann Surfaces].** For a Riemann surface  $S$ ,

- If  $S = \overline{\mathbb{C}}$ , then  $S$  has a natural Fubini-Study metric

$$ds^2 = \frac{4|dz|^2}{(1 + |z|^2)^2}.$$

with curvature 1.

- If  $\tilde{S} = \mathbb{C}$ , then  $\pi_1(S) \subset \text{Aut}(\mathbb{C})$  are all translations so preserves the Euclidean metric  $ds^2 = |dz|^2$ , thus  $S$  has a flat metric.
- If  $\tilde{S} = \mathcal{H}$ , then  $\pi_1(S) \subset \text{Aut}(\mathbb{C}) \cong \text{PSL}(\mathbb{R})$  preserves the hyperbolic metric  $ds^2 = y^{-2}dx dy$  by (16.6.1.5), thus inducing a hyperbolic metric on  $S$  with curvature  $-1$ .  $\lrcorner$

### Compact Riemann Surfaces

References are [Li04].

**Prop. (16.9.9.6) [Riemann Existence Theorem].** Any compact Riemann Surface is a Hodge manifold, thus projective algebraic, by (16.10.8.6) and Chow's lemma??. And the category of compact Riemann surfaces is equivalent to the category of compact algebraic curves, by GAGA (16.9.7.17).  $\lrcorner$

*Proof:* Because  $H^{1,1}(X) = H^2(X, \mathbb{Z})$ , so it clearly contains integral classes. And it is positive because there is a basis generated by any Hermitian metric on  $X$ . So the theorem follows from (16.10.8.6).  $\square$

**Remark (16.9.9.7).** In fact the same argument shows that any Kähler manifold with  $H^{0,2}(X) = 0$  is projective.  $\lrcorner$

**Cor. (16.9.9.8).** Any compact Riemann surface of genus 0 is analytically isomorphic to  $\overline{\mathbb{C}}$ .  $\lrcorner$

**Prop. (16.9.9.9).**

- For any meromorphic function  $f$  on a compact Riemann surface,  $(f) = (f)_0 - (f)_\infty$  has degree 0.
- Let  $\omega$  be a differential form on a compact Riemann surface, then the sum of residues of  $\omega$  at its poles is zero.

$\lrcorner$

*Proof:* 2: Choose a triangularization of the Riemann surface, then use the fact for any simple region  $\Omega$  be boundary  $C$ ,

$$\int_C \omega = 2\pi i \left( \sum_{\text{poles}} \text{res}_p \omega \right).$$

And the integrals cancel out.

- 1: This is a direct consequence of 2 applied to the differential form  $\omega = df/f$ .  $\square$

### Abel's Theorem and Reciprocity Law

**Prop. (16.9.9.10) [Reciprocity Law I].** Cf. [Griffith/Harris P230].  $\lrcorner$

**Prop. (16.9.9.11) [Weil].**  $f, g$  are meromorphic functions on a compact Riemann surface that  $(f), (g)$  are disjoint, then

$$\prod f(p)^{v_p(g)} = \prod g(p)^{v_p(f)}.$$

$\lrcorner$

*Proof:* Cf. [Griffith/Harris, P242].  $\square$

**Prop. (16.9.9.12) [Differentials on Plane Curves].**  $\lrcorner$

## 10 Conformal Geometry

## 16.10 Kähler Geometry

Basic References are [Voi02], [Principle of Algebraic Geometry Griffith/Harris], more advanced stuffs to add from [Kähler Geometry] and [Huy05]. Even more advanced is [Demailly Complex Analytic and Differential Geometry].

**Notation(16.10.0.1).**

- Use notations as in [Complex Geometry](#).

┘

### 1 Kähler Metric

**Def.(16.10.1.1)[Kähler Manifolds].** A metric  $g$  on a manifold  $M$  is called a **Kähler metric** if the metric form  $\omega_g$  is closed. In which case, it is called the **Kähler class** or Kähler form of  $g$  in  $H_{\text{dR}}^2(M)$ . A complex manifold with a Kähler metric is called a **Kähler manifold**.

If  $g_{ij} = g(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j})$ , then  $\omega_g = \sum_{ij} g_{ij} dz_i \wedge d\bar{z}_j$ . Then the condition of  $\omega_g$  being closed can in fact be written in derivatives of  $g$ .

┘

**Prop.(16.10.1.2).** If  $g$  is Hermitian, then  $\omega_g$  is real, non-degenerate and  $\frac{1}{n!}\omega^n$  is a volume form on  $M$ . In particular, if  $\omega$  is Kähler, then it is a symplectic form.

┘

*Proof:* If  $g = \sum \varphi_i \otimes \bar{\varphi}_i$ , then  $\omega = i \sum \varphi_i \wedge \bar{\varphi}_i$ , so it is clear that  $\bar{\omega} = \omega$ .  $\omega$  is non-degenerate as  $g$  is. The last assertion follows from (3.5.11.26).  $\square$

**Cor.(16.10.1.3).** If  $M$  is a compact Kähler manifold, then its even dimensional cohomology group doesn't vanish (16.8.1.7).

┘

**Remark(16.10.1.4).** Notice there are notions like almost Hermitian and almost Kähler, similar to the definition of Hermitian and Kähler, but they are just defined using an almost complex structure on  $M$ . And an almost Kähler structure is Kähler iff  $\nabla J = 0$ , Cf.[Foundation of Differential Geometry Kobayashi].

┘

**Example(16.10.1.5)[Kähler Manifolds].**

- If  $M = \mathbb{R}^{2n}$ ,  $g = \sum dx_i \wedge dx_i + \sum dy_i \wedge dy_i$ , then  $\omega_g = \frac{i}{2} \sum dz_i \wedge d\bar{z}_i$  is Kähler.
- The metric  $\omega_g = \sum dz_i \wedge d\bar{z}_i$  on a complex tori  $\mathbb{C}^n/\Lambda$  is Kähler.
- Any Riemann surface is Kähler, because  $d\omega$  is a 3-form so vanish.
- if  $M = B(0, 1) \subset \mathbb{C}^n$  and  $\omega_g = i\partial\bar{\partial} \log \frac{1}{1-|z|^2}$ , then it is Kähler.
- The product metric on the product space  $M \times N$  of two Kähler manifold is Kähler.
- A submanifold of a Kähler manifold is Kähler, as the Kähler form is the pullback of the Kähler form of the large manifold.

┘

**Prop.(16.10.1.6)[Fubini-Study Metric].** The **Fubini-Study metric** form on  $\mathbb{CP}^n$  is defined locally to be  $i\partial\bar{\partial}|s|^2$ , for any local lifting  $s$  of the projection  $\mathbb{C}^n \setminus \{0\} \rightarrow \mathbb{CP}^n$ . This doesn't depend on the lifting, as  $\partial\bar{\partial}(\log f + \log \bar{f}) = 0$ , so they glue together to be a global form on  $\mathbb{CP}^n$ . It can be checked,  $\omega$  is translation invariant and on the coordinate  $(1, w_1, \dots, w_n) \rightarrow (w_1, \dots, w_n)$ ,  $\omega|_{(0, \dots, 0)} = \sum dw_i \wedge d\bar{w}_i$ , so it is positive definite.

┘

**Cor. (16.10.1.7).** Any projective manifold is Kähler. ┘

**Prop. (16.10.1.8).** the Fubini-Study metric on  $\mathbb{CP}^n$  has sectional curvature  $1 \leq K \leq 4$ . ┘

*Proof:* Cf.[Do Carmo P188]. ? □

**Prop. (16.10.1.9)[Kähler Normal Coordinate].** For a Hermitian metric  $g$  on  $M$ ,  $g$  is Kähler iff for any point  $p$  of  $M$ , there is a holomorphic coordinate centered at  $p$ ,  $\omega_g = \sum g_{ij} dz_i \wedge d\bar{z}_j$  satisfying  $g_{ij}(p) = 0$  and  $dg_{ij}(p) = 0$ . This coordinate is called **Kähler normal coordinate**. (Notice this is different from Darboux theorem, because this coordinate should be holomorphic). ┘

*Proof:* Cf.[Complex Geometry P210]. □

## 2 Geometry of Kähler Manifolds

**Prop. (16.10.2.1).** Let  $(M, J, g)$  be a Kähler manifold, then the complexification of the Levi-Civita connection of  $g$  restricts to the Chern connection on  $T^{1,0}M$ . ┘

*Proof:* Cf.[Complex Geometry note 石亚龙 48] and [Complex geometry Daniel Chap4.A]. □

**Prop. (16.10.2.2).** For a Kähler manifold,  $\nabla J = 0$ . ┘

*Proof:* The problem depends only on first derivative, so choosing a Kähler normal nbhd(16.10.1.9), we may choose  $J$  to be constant, so obviously  $\nabla J(p) = 0$ ,  $P$  is arbitrary, so  $\nabla J = 0$ . □

**Cor. (16.10.2.3).**  $\nabla(JX) = J\nabla X$ , so  $R(X, Y)JZ = JR(X, Y)Z$ , thus

$$\langle R(JX, JY)Z, W \rangle = \langle R(Z, W)JX, JY \rangle = \langle R(Z, W)X, Y \rangle = \langle R(X, Y)Z, W \rangle,$$

so  $R(JX, JY)Z = R(X, Y)Z$ . ┘

**Prop. (16.10.2.4).** The curvature tensor of the complexified Levi-Civita connection on a Kähler manifold can be calculated in terms of  $\partial_i, \bar{\partial}_j$ , Cf.[Complex Geometry note 石亚龙 50]. ┘

## 3 Kähler Identities

**Notation (16.10.3.1).**

- Let  $X$  be a compact complex Kähler manifold. ┘

**Def. (16.10.3.2).** Introduce some operators:

- $d^c = i(\bar{\partial} - \partial)$ , then  $dd^c = 2i\partial\bar{\partial}$ .
- The **Lefschetz operator**  $L(\eta) = \omega \wedge \eta$ .  $\Lambda$  is defined as the formal adjoint of  $L$  as  $A^{p,q}$  is an inner space. In fact,  $\Lambda = \pm * L*$ .
- $h = (k - n)$  on  $\mathcal{A}^k(X)$ . ┘

**Prop. (16.10.3.3).**  $[L, \Lambda] = p + q - n$  on  $(p, q)$ -forms. ┘

*Proof:* The problem doesn't depends on the derivatives, so using the Kähler normal coordinate(16.10.1.9), it suffice to prove for  $\mathbb{C}^n$ , for this, Cf.[Griffith/Harris P120] or [Complex Geometry P34]. □

**Prop. (16.10.3.4)**[Kähler Identities].

$$[\Lambda, \partial] = i\bar{\partial}^*, \quad [\Lambda, \bar{\partial}] = -i\partial^*.$$

┘

*Proof:* The second one follows from the first because  $\omega$  is a real form. For the first, notice only first derivation are involved, so by using the Kähler normal coordinate, it suffice to prove for  $\mathbb{C}^n$ , and this is by [Complex Geometry 石亚龙 P61].  $\square$

**Cor. (16.10.3.5).**

$$[\Lambda, d^c] = d^*, \quad [\Lambda, d] = -d^{c*}.$$

┘

**Prop. (16.10.3.6).**  $\Delta_d$  commutes with both  $L$  and  $\Lambda$ .

┘

*Proof:*  $L$  commutes with  $d$  because  $\omega$  is closed, so taking adjoints,  $\Lambda$  commutes with  $d^*$ . Now by Kähler identities,

$$\Lambda\Delta_d = \Lambda(dd^* + d^*d) = -d^{c*}d^* + dd^*\Lambda - dd^{*c} + d^*d\Lambda = \Delta_d\Lambda.$$

So taking adjoints,  $\Delta_d$  also commutes with  $L$ .  $\square$

**Prop. (16.10.3.7).** In the Kähler case,  $\Delta_\partial = \Delta_{\bar{\partial}} = \frac{1}{2}\Delta_d$ .

┘

*Proof:*

$$\Delta_d = (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) + (\partial^* + \bar{\partial}^*)(\partial + \bar{\partial}) = \Delta_\partial + \Delta_{\bar{\partial}} + (\partial\bar{\partial}^* + \bar{\partial}^*\partial) + (\bar{\partial}\partial^* + \partial^*\bar{\partial})$$

So it suffice to prove  $\partial\bar{\partial}^* + \bar{\partial}^*\partial = 0$  (so  $\bar{\partial}\partial^* + \partial^*\bar{\partial} = 0$  by conjugation), and  $\Delta_\partial = \Delta_{\bar{\partial}}$ . For the first, use Kähler identities, then

$$i(\partial\bar{\partial}^* + \bar{\partial}^*\partial) = \partial[\Lambda, \partial] + [\Lambda, \partial]\partial = 0$$

For the second, using Kähler identities,

$$i\Delta_{\bar{\partial}} = \bar{\partial}[\Lambda, \partial] + [\Lambda, \partial]\bar{\partial} = \bar{\partial}\Lambda\partial + \partial\bar{\partial} - \Lambda\bar{\partial}\partial - \partial\Lambda\bar{\partial}$$

and the same is miraculous true for  $\Delta_\partial$ , so the result is true.  $\square$

## 4 Hodge Theory

**Thm. (16.10.4.1)**[Hodge Decomposition for compact Kähler-Manifolds]. For a compact Kähler manifold  $X$ ,

$$H_{\text{dR}}^r(X, \mathbb{C}) = \bigoplus_{p+q=r} H_{\bar{\partial}}^{p,q}(X) \cong \bigoplus_{p+q=r} H^q(X, \Omega^p)$$

and  $\overline{H_{\bar{\partial}}^{p,q}(X)} \cong H_{\bar{\partial}}^{q,p}(X)$ . Moreover, this decomposition doesn't depends on the Kähler metric.  $\square$

*Proof:* (16.10.3.7) shows that  $\Delta_d$  maps  $A^{p,q}$  to  $A^{p,q}$ , so  $\mathcal{H}_d^{p+q} \cap A^{p,q} = H_{\bar{\partial}}^{p,q}(X)$ . The last assertion is seen using the  $\Delta_d$  definition.

If chosen two different Kähler metric  $g, g'$ , there  $\mathcal{H}^{p,q}(X, g) \cong H^{p,q}(X) \cong H^{p,q}(X, g')$ . If  $\alpha, \alpha'$  be  $g, g'$   $\bar{\partial}$ -harmonic respectively, so by definition  $\alpha - \alpha' = \bar{\partial}\gamma$  for some  $\gamma$ , and they are both  $d$ -harmonic, so  $d\bar{\partial}\gamma = 0$ , and  $\bar{\partial}\gamma$  is  $g$ -orthogonal to  $\mathcal{H}^k(X, g)$  by Hodge decomposition for  $\bar{\partial}$  with metric  $g$ , so by Hodge theorem for  $d$  with metric  $g$ ,  $\partial\gamma$  is  $d$ -exact, so  $[\alpha] = [\alpha']$ .  $\square$

**Cor. (16.10.4.2).** Betti number  $b_r = \sum_{p+q=r} h^{p,q}$ ,  $h^{p,q} = h^{q,p}$ . In particular,  $b_{2k+1}$  is always even.  $\lrcorner$

**Cor. (16.10.4.3)[Holomorphic Forms on Kähler Manifolds are Closed].**  $\mathcal{H}_{\bar{\partial}}^{p,0}(X) = H^0(X, \Omega^p)$ .

Now a  $(p, 0)$ -form is automatically  $\bar{\partial}^*$ -closed, so it is  $\bar{\partial}$ -harmonic iff it is holomorphic. So we conclude any holomorphic  $p$ -form on a Kähler manifold is  $d$ -closed, even  $d$ -harmonic.  $\lrcorner$

**Lemma (16.10.4.4) [ $\partial\bar{\partial}$ -lemma].** A closed differential form  $\eta$  on a compact Kähler manifold  $M$  is  $d$ -exact iff it is  $\partial$ -exact iff it is  $\bar{\partial}$ -exact iff it is  $\partial\bar{\partial}$ -exact.  $\lrcorner$

*Proof:* Now  $\Delta_d, \Delta_{\bar{\partial}}, \Delta_{\partial}$  are all the same, By Hodge theorem, it suffice to prove, if a form is orthogonal to  $\mathcal{H}^{p,q}(X)$ , then it is  $\partial\bar{\partial}$ -exact (this implies other exactness).

Noe  $\eta$  is  $d$ -closed hence  $\partial$  and  $\bar{\partial}$ -closed, then  $\eta = \partial\gamma$  for some  $\gamma$ , and then  $\gamma = \bar{\partial}\beta + \bar{\partial}^*\beta' + \beta''$  for  $\beta''$  harmonic. So  $\eta = \partial\bar{\partial}\beta + \partial\bar{\partial}^*\beta'$ , and then  $\bar{\partial}\eta = \bar{\partial}\partial\bar{\partial}^*\beta = 0$ , but then inner product with  $\partial\beta$  shows  $\bar{\partial}^*\partial\beta = 0$ , so  $\eta = \partial\bar{\partial}\beta$ .  $\square$

**Cor. (16.10.4.5)[Kodaira-Serre Duality].** By (16.7.4.15), For a Hermitian line bundle over a compact Hermitian complex manifold  $X$ , from Hodge theorem and (16.9.5.2), we get

$$H^p(X, \Omega^q(E)) \cong H^{n-p}(X, \Omega^{n-q}(E^*))$$

induced by  $\bar{*}_E$  and  $\bar{*}_{E^*}$ . Moreover, there is a perfect pairing

$$H^p(X, \Omega^q(E)) \times H^{n-p}(X, \Omega^{n-q}(E^*)) \rightarrow \mathbb{C}$$

induced by

$$\mathcal{H}^{p,q}(X, E) \times \mathcal{H}^{n-p, n-q}(X, E^*) \rightarrow \mathbb{C} : (\alpha, \beta) \mapsto \int_X \alpha \wedge \bar{*}_E \beta$$

In fact,  $\int_X \alpha \wedge \bar{*}_E \alpha = \|\alpha\|^2 \neq 0$ .  $\lrcorner$

**Prop. (16.10.4.6).** Holomorphic 1-forms on a compact complex surface is closed. ?  $\lrcorner$

**Prop. (16.10.4.7)[Hard Lefschetz Theorem].** For a compact Kähler manifold  $M$ , the map

$$L^k : H^{n-k}(M) \rightarrow H^{n+k}(M) \quad (16.10.3.2)$$

is an isomorphism, (notice it is defined because  $L$  commutes with  $\Delta_d$  (16.10.3.6)).

Define the **primitive cohomology class**  $H_{\text{prm}}^{n-k}(M) = \ker(L^{k+1}|_{H^{n-k}})$ , then

$$H^m(M) = \oplus_k L^k H_{\text{prm}}^{m-2k}(M).$$

$\lrcorner$

*Proof:* Cf. [Griffith/Harris P122], using representation theory of  $\mathfrak{sl}_2$ .  $\square$

**Thm. (16.10.4.8)[Hodge-Riemann Bilinear Relation].** Let  $(X, \omega)$  be a Kähler manifold of dimension  $n$ , then for  $k \leq n$ , we can define a Hermitian form

$$H(\alpha, \beta) = i^k \int_X \omega^{n-k} \wedge \alpha \wedge \beta = i^k \langle L^{n-k} \alpha, \beta \rangle$$

on  $H^k(X; \mathbb{R})$ . Then it satisfies:

- The Hodge decomposition (16.10.4.1) is orthogonal for  $H$ .



- If  $\alpha \neq 0 \in H_{\text{prim}}^{p,q}(X)$ , then

$$i^{p-q-k}(-1)^{\frac{k(k-1)}{2}}H(\alpha) > 0$$

┘

*Proof:* Cf.[Griffith/Harris] or [Complex Geometry Daniel P138] or[Hodge Theory, Chap6.3.2].  $\square$

**Cor. (16.10.4.9).** For a compact Kähler manifold of complex dimension  $2m$ ,

$$\text{sgn}(X) = \sum_{p,q=0}^m (-1)^p h^{p,q}(m)$$

┘

*Proof:* Cf.[Complex Geometry Daniel P140].  $\square$

**Prop. (16.10.4.10) [Hirzebruch-Riemann-Roch].** By(16.7.4.8), for a  $n$ -dimensional complex line bundle  $L$  over a compact Kähler manifold  $M$ ,

$$\chi(M, L) = \int_M [\text{ch}(E)\text{td}(T^{1,0}M)]_n.$$

Where  $\chi(M, L) = \sum_{q=0}^n (-1)^q \dim H^q(M, E)$ ,  $\text{ch}$  is the Chern character(16.7.5.5) and  $\text{td}(T^{1,0}M)$  is the Todd polynomial, i.e. Taylor expansion of  $\prod_{i=1}^r \frac{t_i}{1 - e^{-t_i}}$  in terms of the symmetric polynomial, applied to  $c_i(T^{1,0}M)$ .  $\square$

**Cor. (16.10.4.11) [Riemann-Roch].** By(16.7.4.9), for a complex vector bundle  $E$  over a Riemann surface  $M$ , let  $\deg E = \int_M c_1(E)$ , then

$$\chi(M, E) = H^0(M, E) - \dim H^1(M, E) = \deg E + \text{rk}(E)(1 - g).$$

┘

**Cor. (16.10.4.12).** For other examples of corollaries of Hirzebruch-Riemann-Roch theorem, Cf.[Complex Geometry P232].  $\square$

## 5 Complex Tori

**Def. (16.10.5.1) [Complex Tori].** A **complex torus** is a pointed complex manifold isomorphic to the complex manifold  $V/\Lambda$ , where  $V$  is a f.d. vector space over  $\mathbb{C}$  and  $\Lambda$  is a complete lattice in  $V$ .  $\square$

**Prop. (16.10.5.2).** Let  $X = V/\Lambda, X' = V'/\Lambda'$  be complex tori, then any  $\mathbb{C}$ -linear map  $\alpha : V \rightarrow V'$  s.t.  $\alpha(\Lambda) \subset \Lambda'$  defines a holomorphic map  $X \rightarrow X'$  sending 0 to 0. And any holomorphic map  $X \rightarrow X'$  sending 0 to 0 is of this form.  $\square$

*Proof:* Any  $\varphi : X \rightarrow X'$  lifts to a continuous map  $\psi : V \rightarrow V'$ , and it is holomorphic because the morphisms  $V \rightarrow X, V' \rightarrow X'$  are locally biholomorphic,  $\psi$  is also holomorphic. For any  $\omega \in \Lambda$ ,  $\psi(z + \omega) - \psi(z)$  has value in  $\Lambda'$  for any  $z$ , so  $\psi(z + \omega) = \psi(z) + a(\omega)$ . From here it is easy to see that  $\psi$  is linear.  $\square$

**Def. (16.10.5.3) [Riemann Pairs].** A **Riemann pair** is a pair  $(\Lambda, J)$  where  $\Lambda$  is a finite  $\mathbb{Z}$ -module of finite rank, and  $J$  is a complex structure on  $\Lambda \otimes \mathbb{R}$ . A homomorphism of Riemann pairs is a group homomorphism  $\Lambda \rightarrow \Lambda'$  that preserves the complex structure.  $\lrcorner$

**Prop. (16.10.5.4) [Riemann Pairs and Abelian Varieties].** There is an equivalence of the category of Riemann pairs with  $\mathcal{AbVar}/\mathbb{C}$  by

$$(\Lambda, J) \mapsto (\Lambda \otimes \mathbb{R})/\Lambda.$$

 $\lrcorner$ 

*Proof:* It is fully faithful by (16.10.5.2), and it is clearly essentially surjective.  $\square$

**Prop. (16.10.5.5) [Cohomology of Complex Tori].** For any complex torus  $X = V/\Lambda$ ,  $H_1(X, \mathbb{Z}) \cong \Lambda$ , so  $H^1(X, \mathbb{Z}) \cong \text{Hom}(\Lambda, \mathbb{Z})$ , and by Künneth formula,

$$H^n(X, \mathbb{Z}) \cong \wedge^n H^1(X, \mathbb{Z}) \cong \wedge^n \text{Hom}(\Lambda, \mathbb{Z}).$$

Then

$$H^n(X, \mathbb{C}) \cong \wedge^n \text{Hom}_{\mathbb{R}}(V, \mathbb{C}) \cong \wedge^n (\text{Hom}_{\mathbb{C}}(V, \mathbb{C}) \oplus \text{Hom}_{\mathbb{C}}(\bar{V}, \mathbb{C})) = \bigoplus_{p+q=n} \wedge^p V^* \otimes \wedge^q \bar{V}^*$$

and

$$H_1(X, \mathbb{C}) \cong \Lambda \otimes_{\mathbb{Z}} \mathbb{C} \cong V \otimes_{\mathbb{R}} \mathbb{C} \cong V \otimes \bar{V} = \text{Lie}(X) \oplus \overline{\text{Lie}(X)}.$$

 $\lrcorner$ 

**Cor. (16.10.5.6) [Complex Tori are Kähler].** A complex torus (16.10.5.1)  $X = V/\Lambda$  is a Kähler manifold, and  $H^2(X, \mathbb{R}) \cong \wedge^2 \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ , by (16.10.5.5).  $\lrcorner$

**Prop. (16.10.5.7) [Hodge-Theory of Complex Tori].** Let  $X = V/\Lambda$  be a complex torus, as  $X$  is Kähler (16.10.5.6), there is a canonical decomposition

$$H_{\text{dR}}^n(X, \mathbb{C}) \cong \bigoplus_{p+q=n} H^q(X, \Omega^p)$$

by (16.10.4.1). Then this decomposition corresponds to the decomposition in (16.10.5.5) via the isomorphism  $H_{\text{dR}}^n(X, \mathbb{C}) \cong H^n(X, \mathbb{C})$ . In particular, there is a canonical isomorphism

$$H^q(X, \Omega^p) \cong \wedge^p V^* \otimes \wedge^q \bar{V}^*.$$

In particular,

$$H^1(X, \mathcal{O}_X) \cong \bar{V}^*.$$

 $\lrcorner$ 

*Proof:*  $\square$

**Def. (16.10.5.8) [Dual Complex Tori].** Let  $X = V/\Lambda$  be a complex torus with a Riemann form, then we can define the **dual complex torus** as

$$X^\vee = V^*/\Lambda^*$$

where  $V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$  and  $\Lambda^* = \{f \in V^* \mid f(\Lambda) \subset \mathbb{Z}\}$ .  $\lrcorner$

**Riemann Forms**

**Prop. (16.10.5.9) [Riemann Forms].** For a complex torus  $V/\Lambda$ , it is projective iff there exists a **Riemann form** on  $V$ , which is an alternating bilinear form  $\omega : V \times V \rightarrow \mathbb{R}$  that:

- $\omega(iu, iv) = \omega(u, v)$ .
- $\omega(v, iv) > 0$  for  $v \neq 0$ .
- $\omega(u, v) \in \mathbb{Z}$  for  $u, v \in \Lambda$ .

Notice  $\omega$  is clearly non-degenerate. ┘

*Proof:* Use (16.10.5.6). The conditions are just equivalent to  $\omega$  being an integral positive Kähler form (16.10.8.6). ? How to interpret? □

**Cor. (16.10.5.10).** Notice these conditions are equivalent to the fact the Hermitian form

$$\lambda(x, y) = \omega(ix, y) + i\omega(x, y)$$

is Hermitian and positive definite. ┘

*Proof:* □

**Cor. (16.10.5.11) [Riemann Relations].** Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Choose a complex basis  $e_1, \dots, e_g$  of  $V$  and a  $\mathbb{Z}$ -basis  $u_1, \dots, u_{2g}$  of  $\Lambda$ . Let  $\Pi \in \text{Mat}(n \times 2n, \mathbb{C})$  be the basis of  $(u_1, \dots, u_{2g})$  in  $(e_1, \dots, e_g)$ , so  $X \cong \mathbb{C}^g / \Pi \mathbb{Z}^{2g}$ . Then  $X \in \mathcal{AbVar}/\mathbb{C}$  iff there exists a non-degenerate skew-symmetric matrix  $A \in \text{GL}(2g, \mathbb{Z})$  satisfying the **Riemann relations**

$$\Pi A^{-1} \Pi^t = 0, \quad i\Pi A^{-1} \bar{\Pi}^t \in \text{Pos}(g, \mathbb{C}).$$

┘

*Proof:* By (16.10.5.9), it suffices to show that if  $\omega$  is the alternating form on  $\Lambda$  corresponding to  $A$ , then these two conditions are equivalent to the condition that  $\omega$  is a Riemann form.

Notice that  $E(\Pi x, \Pi y) = x^t A y$  for any  $x, y \in \mathbb{R}^{2g}$ , and if

$$I = \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix}^{-1} \begin{bmatrix} i\mathbf{1} & \\ & -i\mathbf{1} \end{bmatrix}^{-1} \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix},$$

then

$$\begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix} I = i \begin{bmatrix} \Pi \\ -\bar{\Pi} \end{bmatrix},$$

so  $i\Pi = \Pi I$ . Then  $\omega(iu, iv) = \omega(u, v)$  is equivalent to  $I^t A I = A$ , or equivalently

$$\begin{bmatrix} i\mathbf{1} & \\ & -i\mathbf{1} \end{bmatrix} \left( \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix} A^{-1} \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix}^t \right)^{-1} \begin{bmatrix} i\mathbf{1} & \\ & -i\mathbf{1} \end{bmatrix} = \left( \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix} A^{-1} \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix}^t \right)^{-1}.$$

By comparing blocks, this is equivalent to  $\Pi A^{-1} \Pi^t = 0$ .

And if  $x, y \in \mathbb{R}^{2g}$  and  $u = \Pi x, v = \Pi y$ , using the fact  $\Pi A^{-1} \Pi^t = 0$ ,

$$\omega(iu, u) = x^t I^t A x = \begin{bmatrix} u \\ \bar{u} \end{bmatrix}^t \begin{bmatrix} i\mathbf{1} & \\ & -i\mathbf{1} \end{bmatrix} \left( \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix} A^{-1} \begin{bmatrix} \Pi \\ \bar{\Pi} \end{bmatrix}^t \right)^{-1} \begin{bmatrix} v \\ \bar{v} \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} u \\ \bar{u} \end{bmatrix}^t \begin{bmatrix} 0 & i(\bar{\Pi}A^{-1}\Pi^t)^{-1} \\ i(\Pi A^{-1}\bar{\Pi}^t)^{-1} & 0 \end{bmatrix} \begin{bmatrix} v \\ \bar{v} \end{bmatrix} \\
&= iu^t(\bar{\Pi}A^{-1}\Pi^t)^{-1}\bar{v} - i\bar{u}^t(\Pi A^{-1}\bar{\Pi}^t)^{-1}v
\end{aligned}$$

And because  $\omega$  is alternating,

$$u^t(\bar{\Pi}A^{-1}\Pi^t)^{-1}\bar{u} + \bar{u}^t(\Pi A^{-1}\bar{\Pi}^t)^{-1}u = 0,$$

so

$$\omega(iu, u) = 2\bar{u}i(\bar{\Pi}A^{-1}\Pi^t)^{-1}\bar{u},$$

and it is positive definite iff  $i\Pi A^{-1}\bar{\Pi}^t \in \text{Pos}(n, \mathbb{C})$ .  $\square$

## 6 Positivity

**Def. (16.10.6.1) [Positive Line Bundle].** A 2-form  $\omega$  on a Hermitian complex manifold  $M$  is called **positive** iff  $\omega(u, Ju) \geq 0$  for  $u \neq 0 \in TM$ , which is equivalent to  $-i\omega(v, \bar{v}) > 0$  for all  $v \in T^{1,0}X$ .

A holomorphic vector bundle is called **(Griffith-)positive** iff there exists a Hermitian metric on it that the curvature form  $\Omega$  for the Chern connection (16.9.5.13) satisfies  $h(\Omega(s), s)(v, \bar{v}) > 0$  for all  $s \in E$  and  $v \in T^{1,0}X$ .

The pullback of a positive line bundle along an immersion is positive.  $\lrcorner$

**Prop. (16.10.6.2) [Positivity on Kähler Manifolds].** On a compact Kähler manifold, being positive is a topological property for line bundles. It is equivalent to the first Chern class of  $L$  can be represented by a positive form in  $H_{dR}^2(M)$ .  $\lrcorner$

*Proof:*  $c_1(L) = [\frac{i}{2\pi}\Omega]$ , so one direction is trivial, and if  $c_1(L) = [\frac{i}{2\pi}\theta]$ , choose an arbitrary Hermitian metric  $h$  on  $L$ , then by  $\partial\bar{\partial}$ -lemma (16.10.4.4),  $\theta = \Omega + \bar{\partial}\partial\rho$  for some smooth function  $\rho$ . Then  $e^\rho h$  has  $\Omega = \theta$  by formula (16.9.5.14).  $\square$

**Cor. (16.10.6.3).** On a compact Kähler manifold, if  $L$  is positive, then for any other Hermitian line bundle  $L'$ ,  $kL + L'$  is positive.  $\lrcorner$

**Prop. (16.10.6.4).** The hyperplane line bundle  $\mathcal{O}(1)$  (16.9.5.9) is positive.  $\lrcorner$

*Proof:* The hyperplane line bundle is dual to the tautological line bundle. The metric on the tautological line bundle is given by locally  $g_i = \frac{1}{|z_i|^2} \sum |z_i|^2$ . It is compatible with the transition map, and then by (16.9.5.14), the Chern curvature is

$$\bar{\partial}\partial\left(\frac{1}{|z_i|^2} \sum |z_i|^2\right) = \bar{\partial}\partial\left(\sum |z_i|^2\right).$$

So by (16.3.4.6) the curvature of the hyperplane line bundle times  $i$  is just the Fubini-Study metric form (16.10.1.6), so it is positive.  $\square$

**Prop. (16.10.6.5).** For  $\tilde{X} \rightarrow X$  the blowing-up of  $X$  at a point  $x$ , If  $L$  is a positive line bundle on  $X$ , then for any integer  $n$ , there exists a  $k > 0$  that  $\pi^*L^k - nE$  is a positive line bundle on  $\tilde{X}$ , where  $E$  is the exceptional divisor.  $\lrcorner$

*Proof:* Involves explicit metric calculation, Cf. [Kodaira Embedding Theorem P11] and [Complex Geometry P249]..  $\square$

## 7 Kodaira Vanishing Theorem

**Prop. (16.10.7.1) [Nakano Identities].** For a holomorphic vector bundle over a compact Kähler manifold  $(M, \omega)$  with Hermitian metric  $h$ , introduce operators  $L$  and  $\Lambda$  as before. If we denote the  $(1, 0)$  and  $(0, 1)$ -part of the Chern connection on  $E$  by  $D'$  and  $D'' = \bar{\partial}$ , then

$$[\Lambda, \bar{\partial}] = -iD'^*, \quad [\Lambda, D'] = i\bar{\partial}^*$$

┘

*Proof:* The question is local, choose normal coordinate frame at  $x$  (16.9.5.16), then by the formula of Chern connection (16.9.5.14),  $\nabla_E = d + A$ ,  $A(x) = 0$ , and  $\nabla_{E^*} = d + B$ ,  $B(x) = 0$ . so

$$[\Lambda, \bar{\partial}_E] + iD'^* = [\Lambda, \partial] + i\partial^* + [\Lambda, A^{0,1}] + iB^{0,1}$$

where the usual Kähler identities (16.10.3.4) are used. Then it is zero when evaluated at  $x$ , Cf. [Demailly Complex Analytic and Differential Geometry P329].  $\square$

**Cor. (16.10.7.2) [Bochner-Kodaira-Nakano Identity].**

$$\Delta_{\bar{\partial}, E} - \Delta_{D', E} = i[\Omega, \Lambda]$$

┘

*Proof:*

$$-i\Delta_{D', E} = D'[\Lambda, \bar{\partial}] + [\Lambda, \bar{\partial}]D' = D'\Lambda\bar{\partial} - D'\bar{\partial}\Lambda + \Lambda\bar{\partial}D' - \bar{\partial}\Lambda D'$$

and similar calculation for  $i\Delta_{\bar{\partial}, E}$ , so

$$i\Delta_{\bar{\partial}, E} - i\Delta_{D', E} = \Lambda(\bar{\partial}D' + D'\bar{\partial}) - (\bar{\partial}D' + D'\bar{\partial})\Lambda = -[\Omega, \Lambda].$$

$\square$

**Prop. (16.10.7.3) [Kodaira-Akizuki-Nakano Vanishing Theorem].** If  $\mathcal{L}$  is a positive line bundle on a compact Kähler manifold  $M$ , then

$$H^p(M, \Omega^q(\mathcal{L})) = 0$$

for  $p + q > n$ . In particular,  $H^p(M, \mathcal{K}_M \otimes \mathcal{L}) = 0$  for  $p > 0$ .  $\square$

*Proof:* By Hodge theorem (16.7.4.13), it suffice to prove there are no harmonic  $(p, q)$ -forms  $\in \mathcal{H}^{p,q}(X, L)$  on  $L$ .

As  $i\Omega = \omega$  is positive, we may endow  $M$  with the metric  $\omega$ , then by (16.10.7.2) and (16.10.3.3),  $\Delta_{\bar{\partial}} - \Delta_{D'} = [L, \Lambda] = p + q - n$  on  $A^{p,q}$ .

So if  $s \in \mathcal{H}^{p,q}(X, L)$ , then  $(\Delta_{\bar{\partial}}s - \Delta_{D'}s, s) = (p + q - n)||s||^2 \geq 0$ , but  $(\Delta_{\bar{\partial}}s - \Delta_{D'}s, s) = -(\Delta_{D'}s, s) = -||D's||^2 - ||D'^*s||^2 \leq 0$ , so  $s = 0$ .  $\square$

**Cor. (16.10.7.4) [Serre's Theorem].** Let  $\mathcal{L}$  be a positive line bundle on a compact complex Kähler manifold  $M$ , then for any holomorphic vector bundle  $\mathcal{E}$ , for  $m$  large,  $H^q(M, \mathcal{L}^m \otimes \mathcal{E}) = 0$ .  $\square$

*Proof:* Same notation as in the proof of (16.10.7.3), choose Hermitian structure on  $E$  and  $L$  and their Chern connections by  $\nabla_E, \nabla_L$ , the corresponding Chern connection on  $E \otimes L^m$  is denoted by  $\nabla$ , and make sure  $\frac{i}{2\pi}F_{\nabla_L}$  is the Kähler form  $\omega$ , then for any harmonic form  $\alpha \in \mathcal{H}^{p,q}(X, E \otimes L^m)$ , by (16.10.7.2),  $\frac{i}{2\pi}([\Lambda, F_{\nabla}](\alpha), \alpha) \geq 0$ , but  $\frac{i}{2\pi}F_{\nabla} = \frac{i}{2\pi}F_{\nabla_E} + m\omega$ , so

$$0 \leq \frac{i}{2\pi}([\Lambda, F_{\nabla_E}](\alpha), \alpha) + m(n - p - q)\|\alpha\|^2$$

Notice  $|([\Lambda, F_{\nabla_E}](\alpha), \alpha)|$  has a bound by Schwartz inequality, then if  $p + q > n$  and  $m$  sufficiently large,  $\alpha$  must be 0. In this case  $\mathcal{H}^{p,q}(X, E \otimes L^m) = 0$ , but  $\mathcal{H}^{0,q}(X, \mathcal{K}_X \otimes E \otimes L^m) \subset \mathcal{H}^{n,q}(X, E \otimes L^m)$ , so it is 0. Now we've proved  $H^q(X, \mathcal{K}_X \otimes E \otimes L^m) = 0$  for any  $E$  if  $m$  is large. But  $E$  is arbitrary, so the conclusion is true.  $\square$

**Cor. (16.10.7.5) [Grothendieck's Lemma].** Every holomorphic vector bundle  $E$  over  $\mathbb{CP}^1$  is uniquely isomorphic to a finite direct sum of  $\mathcal{O}(a_i)$ .  $\lrcorner$

*Proof:* If  $E$  has rank 1, this is the content of (16.9.5.10), so use induction on rank of  $E$ . Choose a maximal  $a$  that  $\text{Hom}(\mathcal{O}(a), E) = H^0(\mathbb{CP}^1, E(-a)) \neq 0$ . This  $a$  exists because Serre's Theorem (16.10.7.4) shows that  $H^1(\mathbb{CP}^1, E(-a)) = 0$  for  $a$  sufficiently small, and Riemann-Roch (16.7.4.9) shows that  $\chi(\mathbb{CP}^1, E(-a)) = \deg E + \text{rk}(E)(1 - a)$  is positive for  $a$  sufficiently small, so  $H^0(\mathbb{CP}^1, E(-a)) \neq 0$ . Conversely, if  $a$  is sufficiently large, then  $H^0(\mathbb{CP}^1, E(-a)) \cong H^1(\mathbb{CP}^1, E^*(a - 2)) = 0$  (Notice  $\mathcal{K}_{\mathbb{CP}^n} = \mathcal{O}(-n - 1)$ ).

So now there is an exact sequence of sheaves

$$0 \rightarrow \mathcal{O}(a) \xrightarrow{s} E \rightarrow E_1 \rightarrow 0$$

I claim  $E_1$  is also a vector bundle, because  $s$  never vanishes, otherwise if it vanishes at some  $x$ , then we can divide by a linear factor  $s_x \in H^0(\mathbb{CP}^1, \mathcal{O}(1))$  to get a map  $\mathcal{O}(a + 1) \rightarrow E$ , contradicting the maximality. So by induction  $E_1 = \oplus \mathcal{O}(a_i)$ , then I claim  $a_i \leq a$ , because otherwise  $H^0(\mathbb{CP}^1, E_1(-a - 1)) \neq 0$ , and by the exact sequence  $0 \rightarrow \mathcal{O}(-1) \rightarrow E(-a - 1) \rightarrow E_1(-a - 1) \rightarrow 0$ ,  $H^0(\mathbb{CP}^1, E(-a - 1)) \neq 0$ , contradiction.

Then we want to show the above sequence splits, this is equivalent to

$$0 \rightarrow E_1^*(a) \rightarrow E^*(a) \rightarrow \mathcal{O} \rightarrow 0$$

splits, and this follows from the fact  $H^1(\mathbb{CP}^1, E_1^*(a)) = H^1(\mathbb{CP}^1, \oplus \mathcal{O}(a - a_i)) = 0$ , by Serre duality. So there is a section lifting  $\mathcal{O} \rightarrow E^*(a)$ , which splits the sequence.  $\square$

**Prop. (16.10.7.6) [Weak Lefschetz Theorem].** Let  $X$  be a compact Kähler manifold and  $Y$  be a submanifold that the line bundle  $\mathcal{L}(Y)$  is positive, then the canonical restriction map  $H^k(X, \mathbb{C}) \rightarrow H^k(Y, \mathbb{C})$  is isomorphism for  $k \leq n - 2$  and injective for  $k = n - 1$ .  $\lrcorner$

*Proof:* In fact, using Hodge decomposition, it suffices to prove on the level of  $H^q(X, \Omega_X^p)$ . Tensoring the exact sequence

$$0 \rightarrow \mathcal{O}_X(-Y) \rightarrow \mathcal{O}_X \rightarrow i_*i_Y^*\mathcal{O}_X \rightarrow 0$$

with  $\Omega_X^p$  and taking the cohomology. By Serre duality and Kodaira vanishing (16.10.7.3), the map  $H^q(X, \Omega_X^p) \rightarrow H^q(X, \Omega^p i_*i_Y^*\mathcal{O}_X)$  is isomorphism for  $p + q < n - 1$  and injection for  $p + q = n - 1$ .

Next consider the exact sequence  $0 \rightarrow TY \rightarrow TX \rightarrow \mathcal{N}_{Y/X} \rightarrow 0$ . By (7.5.1.26) there is an exact sequence

$$0 \rightarrow \wedge^p TY \rightarrow \wedge^p TX|_Y \rightarrow \wedge^{q-1} TY \mathcal{N}_{Y/X} \rightarrow 0$$

Taking dual and applying adjunction formula(16.9.1.3), it becomes:

$$0 \rightarrow \Omega_Y^{q-1} \otimes \mathcal{O}(-N) \rightarrow \Omega_X^q|_Y \rightarrow \Omega_Y^q \rightarrow 0$$

Taking cohomology and use Serre duality and Kodaira vanishing as before, the result follows, and the composition is also true.  $\square$

**Remark(16.10.7.7).** There is a topological proof of weak Lefschetz theorem in [Bott, On a Theorem of Lefschetz].  $\lrcorner$

## 8 Kodaira Embedding

**Prop.(16.10.8.1)[Kodaira map].** For a holomorphic line bundle  $L$  on a compact complex manifold  $M$ , if  $s_0, \dots, s_n$  be a basis of  $H^0(X, L)$ , we try to define a map from  $M$  to  $\mathbb{CP}^n : x \rightarrow [s_0(x), \dots, s_n(x)]$ . This is independent of the change of coordinates because  $g_{\alpha\beta}$  is invertible, and it is definable iff  $L$  is basepoint-free. This map is holomorphic where it is definable.  $\lrcorner$

**Def.(16.10.8.2)[Ample Holomorphic Line Bundles].** A holomorphic line bundle  $\mathcal{L}$  on a compact complex manifold  $X$  is called a

- **semi-ample holomorphic line bundle** iff for  $m$  large,  $\mathcal{L}^m$  is basepoint-free.
  - **very ample holomorphic line bundle** iff  $L$  is basepoint-free and the Kodaira map  $\iota_L : X \rightarrow \mathbb{CP}^N$  is a holomorphic embedding.
  - **ample holomorphism line bundle** iff for  $m$  large,  $L^m$  is very ample.
- $\lrcorner$

**Lemma(16.10.8.3)[Cohomological Method for Very Ampleness].** For the above Kodaira map to be a holomorphic embedding, it suffice to show that the map is definable, injective and surjective on cotangent space. For these, it is equivalent to  $H^0(X, L) \rightarrow L_x$  surjective,  $H^0(X, L) \rightarrow L_x \oplus L_y$  surjective, and  $L \otimes \mathcal{I}_x \rightarrow L_x \otimes T^{1,0*}(X)_x$  surjective. And they are true if

$$H^1(X, L \otimes \mathcal{I}_x) = 0, \quad H^1(X, L \otimes \mathcal{I}_{x,y}) = 0, \quad H^1(X, L \otimes \mathcal{I}_x^2) = 0.$$

respectively.  $\lrcorner$

*Proof:* Basepoint-free at  $x$  is easily seen to be equivalent to  $H^0(X, L) \rightarrow L_x$  surjective. And there is an exact sequence of sheaves:

$$0 \rightarrow L \otimes \mathcal{I}_x \rightarrow L \rightarrow L_x \rightarrow 0$$

where  $L_x$  means the skyscraper sheaf. So  $H^1(X, L \otimes \mathcal{I}_x^2) = 0$  induces the result.

Injective is easily seen to be equivalent to  $H^0(X, L) \rightarrow L_x \oplus L_y$  surjective. And there is an exact sequence of sheaves:

$$0 \rightarrow L \otimes \mathcal{I}_{x,y} \rightarrow L \rightarrow L_x \oplus L_y \rightarrow 0$$

where  $\mathcal{I}_{x,y}$  is the sheaf of functions vanishing at  $x$  and  $y$ , and  $L_x \oplus L_y$  means the skyscraper sheaf. So  $H^1(X, L \otimes \mathcal{I}_{x,y}) = 0$  induces the result.

For the surjection on cotangent spaces, given any point  $x$ , choose a basis  $s_1, \dots, s_n$  of sections in  $H^0(X, L)$  vanishing at  $x$ , and by basepoint-free, there is a  $s_0$  not vanishing at  $x$ , then on a coordinate, the Kodaira map is given by  $x \rightarrow (s_1/s_0, \dots, s_n/s_0)$ , then it need to be checked  $d_x(s_i/s_0) = d_x(x_i)/s_0$  span  $T^{1,0*}(X)_x$ . But there are exact sequences of sheaves:

$$0 \rightarrow L \otimes \mathcal{I}_x^2 \rightarrow L \otimes \mathcal{I}_x \xrightarrow{d_x} L_x \otimes T_x^{1,0*} \rightarrow 0$$

where  $d_x$  is given by  $d_x(s \otimes f) = s(x) \otimes d_x(f)$  (by the universal property of skyscraper sheaf), it suffice to give a map  $(L \otimes \mathcal{I}_x \rightarrow L_x \otimes T_x^{1,0*})$ , notice this is independent of the coordinate because  $d_x(s_\alpha) = d_x(g_{\alpha\beta}s_\beta) = g_{\alpha\beta}d_x(s_\beta)$ , as  $s_\alpha$  vanishes at  $x$ , so this is truly a sheaf map, and its kernel is  $L \otimes \mathcal{I}_x^2$ . So  $H^1(X, L \otimes \mathcal{I}_x^2) = 0$  induces the result.  $\square$

**Prop. (16.10.8.4) [Ampleness and Positivity].** A holomorphic line bundle  $\mathcal{L}$  on a compact Kähler manifold is ample iff it is positive.  $\lrcorner$

*Proof:* If  $L$  is ample, then  $L^m$  is the pullback of the hyperplane bundle by the Kodaira map. The hyperplane line bundle is positive by (16.10.6.4), so  $L^m$  is positive with the induced metric, so  $L$  is also positive given the  $m$ -th roots of the induced metric (notice the metric of line bundle is just locally a number compatible with transition map).

Conversely, using (16.10.8.3), we want to find a  $L^k$  that  $H^1(X, L^k \otimes \mathcal{I}_x) = 0$ ,  $H^1(X, L^k \otimes \mathcal{I}_{x,y}) = 0$ ,  $H^1(X, L^k \otimes \mathcal{I}_x^2) = 0$ . First notice it suffice to prove for single points when  $k$  is sufficiently large, because the holomorphic embedding is an open property and  $X$  is compact so a sufficiently large  $k$  will suffice.

Consider the blowing-up  $\tilde{X}$  at a point  $x$ , there is a commutative diagram

$$\begin{array}{ccc} H^0(X, L^k) & \longrightarrow & L_x^k \\ \downarrow \pi^* & & \downarrow \cong \\ H^0(\tilde{X}, \pi^* L^k \otimes L_x^k) & \longrightarrow & H^0(E, \mathcal{O}_E) \otimes L_x^k \end{array}$$

The right vertical map is isomorphism as  $E \cong \mathbb{CP}^n$ , so  $H^0(E, \mathcal{O}_E) = \mathbb{C}$ . The left exact sequence is also isomorphism: it is injective because  $\pi$  is surjective, and it is surjective because: if  $\dim X = 1$ , then  $\pi = \text{id}$  so trivially true, and if  $\dim X \geq 2$ , then because  $\pi : \tilde{X} - E \cong X - \{x\}$ , any holomorphic function on  $\tilde{X}$  induces a holomorphic function on  $X - \{x\}$  and by Hartog's theorem (12.5.8.4), it comes from a holomorphic function on  $X$ .

Now the second horizontal line is part of the cohomology exact sequence of (7.5.3.15)

$$0 \rightarrow \pi^* L^k \otimes \mathcal{O}_{\tilde{X}}(-E) \rightarrow \pi^* L^k \rightarrow \pi^* L^k|_E \rightarrow 0$$

So it is reduced to prove  $H^1(\tilde{X}, \pi^* L^k \otimes \mathcal{O}(-E)) = 0$ , but by (7.8.2.8),  $\pi^* L^k - E = \pi^* L^k - E + \mathcal{K}_{\tilde{X}} - \pi^* \mathcal{K}_X - (n-1)E = \mathcal{K}_{\tilde{X}} + (\pi^* L^k - E) + \pi^*(L^k - \mathcal{K}_X)$ , and by (16.10.6.5)(16.10.6.3) the last two are positive when  $k$  is large, so the conclusion follows from Kodaira vanishing (16.10.7.3).

The proof of  $H^1(X, L^k \otimes \mathcal{I}_{x,y}) = 0$  is verbatim, just use blowing-up at two different points.

To prove  $H^1(X, L^k \otimes \mathcal{I}_x^2) = 0$ , consider the blowing-up  $\tilde{X}$  at a point  $x$ , notice there is a commutative diagram

$$\begin{array}{ccc} H^0(X, L^k \otimes \mathcal{I}_x) & \xrightarrow{d_x} & L_x^k \otimes T^{1,0*} X_x \\ \downarrow \pi^* & & \downarrow \cong \\ H^0(\tilde{X}, \pi^* L^k - E) & \longrightarrow & L_x^k \otimes H^0(E, -E) \end{array}$$

In fact this comes from the two commuting exact sequences twisted with  $\pi^* L^k$ :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi^* \mathcal{I}_x^2 & \longrightarrow & \pi^* \mathcal{I}_x & \xrightarrow{d_x} & \pi^* T^{1,0*} X_x \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{O}(-2E) & \longrightarrow & \mathcal{O}(-E) & \longrightarrow & \mathcal{O}_E(-E) \longrightarrow 0 \end{array}$$



The second line is (7.5.3.15) and the fact a section vanishing at  $x$  lifts to a section vanishing at  $E$  thus equivalent to a section in the twisted sheaf  $- \otimes \mathcal{O}(-E)$ . These two exact sequences commutes because

Back to the commutative diagram, the above argument also shows that the first vertical map is isomorphism. To show the second vertical map is isomorphism, notice by (7.8.2.7)  $\mathcal{O}(-E)$  is just the hyperplane line bundle on  $E$ , so  $H^0(E, -E) \cong T^{1,0*}X_x$ , we need to know the vertical map is the natural map  $V^* \rightarrow H^0(\mathbb{P}(V), \mathcal{O}(1))$ . This in fact need some careful calculation using coordinates in (7.8.2.7).?

Now the map  $d_x$  is surjective iff the second horizontal map is surjective, with is part of the cohomology exact sequence of

$$0 \rightarrow \pi^* L^k \otimes \mathcal{O}_{\tilde{X}}(-2E) \rightarrow \pi^* L^k \mathcal{O}_{\tilde{X}}(-E) \rightarrow \pi^* L^k|_E \rightarrow 0$$

So it is reduced to prove  $H^1(\tilde{X}, \pi^* L^k \otimes \mathcal{O}(-wE)) = 0$ , which is by Kodaira vanishing theorem the same reason as before.  $\square$

**Cor. (16.10.8.5) [Kodaira Embedding Theorem].** If a compact complex manifold  $M$  has a positive line bundle, then it is projective.  $\lrcorner$

**Def. (16.10.8.6) [Hodge Manifolds].** A compact Kähler manifold  $X$  is projective iff it has a closed positive  $(1, 1)$ -form  $\omega$  whose cohomology class  $[\omega]$  is rational (resp. integral) (i.e. in  $H^2(X, \mathbb{Q})$  (resp.  $H^2(X, \mathbb{Z})$ )). In fact, a compact Kähler manifold with a Hodge metric is called a **Hodge manifold**. A pair  $(X, [\omega])$  where  $X$  is a compact Kähler manifold and  $[\omega] \in H^2(X, \mathbb{Z})$  is called a **polarized manifold**.

So compact Hodge manifolds are just those compact Kähler manifolds together with an ample line bundle class.  $\lrcorner$

*Proof:* if  $\omega$  is rational, then a multiple of it is integral, then there is a  $L$  that  $c_1(L) = k[\omega]$  by Lefschetz theorem on  $(1, 1)$ -forms (16.11.5.6), so  $L$  is positive by (16.10.6.2), which is equivalent to ampleness by (16.10.8.4), so  $X$  is projective. Conversely, the Chern class of the pullback of the hyperplane line bundle is positive and rational (16.10.6.4) (16.11.5.6).  $\square$

**Cor. (16.10.8.7) [Compact Riemann Surfaces are Hodge Manifolds].** A **Riemann surface** is a complex variety of dimension 1. Any compact Riemann surface is a compact Hodge manifold.  $\lrcorner$

*Proof:* This is because  $H^2(X, \mathbb{C}) = H^2(X, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}$  is generated by the metric form  $\omega$ , so there must be a multiple of  $\omega$  that is integral.  $\square$

**Cor. (16.10.8.8).** if  $\tilde{X}$  is the blowing-up of a Kähler manifold  $X$  at a point  $x$ , then if  $X$  is projective, then  $\tilde{X}$  is also projective, because by (16.10.6.5)  $\pi^* L^k \setminus E$  is positive for  $k$  large.  $\lrcorner$

**Cor. (16.10.8.9).** For a finite unbranched cover of compact Kähler manifolds  $\tilde{X} \rightarrow X$ ,  $\tilde{X}$  is projective iff  $X$  is projective.  $\lrcorner$

*Proof:* A positive rational closed  $(1, 1)$ -form on  $X$  pull backs to a positive rational closed  $(1, 1)$ -form on  $\tilde{X}$ , and it can even be pulled forward:  $\omega' = \sum_{y \in \pi^{-1}(x)} (\pi^{-1})^* \omega(y)$ , then it is also positive closed.

It is rational because  $\int_X \omega' \wedge \eta = \frac{1}{d} \int_{\tilde{X}} \omega \wedge \pi^* \eta$ , where  $\tilde{X} \rightarrow X$  is branched of degree  $d$ .  $\square$

**Def. (16.10.8.10) [the Kähler Cone].** For a Kähler manifold  $X$ , the **Kähler cone**  $K_X$  is defined to be the set of closed real positive  $(1, 1)$ -forms. Then  $K_X$  is an open convex cone in  $H^{1,1}(X) \cap H^2(X, \mathbb{R})$ . Then (16.10.8.6) says  $X$  is projective iff  $K_X \cap H^2(X, \mathbb{Z}) \neq 0$ .  $\lrcorner$

## 9 Fujiki Manifolds

## 16.11 Hodge Theory

### 1 Hodge Structures

Cf. [Huy05] Chap3.A, [Del71], [Hodge 2, 3, Deligne] and [Shimura Varieties, Milne].

**Def. (16.11.1.1) [Mixed Hodge Structures].** An integral **mixed Hodge structure** is given by a f.g. Abelian group  $\Lambda$  together with

- An increasing **weight filtration**  $W_\bullet$  of  $\Lambda_{\mathbb{Q}}$ .
- A decreasing **Hodge filtration**  $F^\bullet$  of  $\Lambda_{\mathbb{C}}$ .
- A bigradation  $V^{\bullet,\bullet}$  of  $\Lambda_{\mathbb{C}}$ .

such that

- For any  $k \in \mathbb{Z}$ ,  $W_k \otimes \mathbb{C} = \bigoplus_{p+q \leq k} V^{p,q}$ .
- For any  $p \in \mathbb{Z}$ ,  $F^p = \bigoplus_{r \geq p} V^{r,s}$ .
- $\overline{V^{p,q}} = V^{q,p}$ .

It is called a **pure Hodge structure** if the weight filtration  $W_\bullet$  splits over  $\mathbb{Q}$ .  $\lrcorner$

**Prop. (16.11.1.2).** For any subring  $Q \subset \mathbb{C}$ , we can naturally define morphisms between  $Q$ -mixed Hodge structures. Then integral Hodge structures form an Abelian category, denoted by  $m\text{Hdg}_{\mathbb{Q}}$ .

And it is easy to define tensor products of mixed integral Hodge structures.  $\lrcorner$

*Proof:* Cf. [Theorie de Hodge, Deligne].  $\square$

**Def. (16.11.1.3) [Integral Hodge Structures].** For  $k \in \mathbb{Z}$ , an integral **Hodge structure** of weight  $k$  is a mixed Hodge structure pure of weight  $k$ .

Equivalently, it is a f.g. free Abelian group  $\Lambda$  together with a **Hodge decomposition** of graded  $\mathbb{R}$ -vector spaces

$$\Lambda_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$$

satisfying  $\overline{V^{p,q}} = V^{q,p}$ .

This is because given an integral Hodge structure of weight  $k$ , we can associate a **Hodge filtration**  $F^\bullet V : F^p V = \bigoplus_{r \geq p} V^{r,k-r}$ . It satisfies

$$V_{\mathbb{C}} = F^p \Lambda \oplus \overline{F^{k-p+1} \Lambda}$$

for any  $p$ , and for any  $p+q=k$ ,

$$V^{p,q} = F^p \Lambda \cap \overline{F^q \Lambda}.$$

Thus giving the Hodge decomposition is equivalent to giving the Hodge filtration.

The category of pure  $Q$ -Hodge structures of weight  $k$  is denoted by  $\text{Hdg}_Q^k$ .  $\lrcorner$

**Prop. (16.11.1.4) [Hodge Structure as Representations of Deligne Torus].** Giving a real representation  $h : \mathbb{S} \rightarrow \text{GL}(V)_{\mathbb{R}}$  (7.1.5.33) is equivalent to giving a bigradation  $V^{p,q}$  of  $V_{\mathbb{C}}$  s.t.  $\overline{V^{p,q}} = V^{q,p}$  by (10.2.3.23), thus equivalent to a Hodge structure.

And in this correspondence,  $V$  is pure of weight  $n$  iff  $h \circ w : \mathbb{G}_m \rightarrow \text{GL}(V)$  acts as character  $z \mapsto z^n$ .

Then for any subring  $Q \subset \mathbb{R}$ , this defines an equivalence of categories:  $\text{Rep}_Q(\mathbb{S}) \cong \text{Hdg}_Q$ .  $\lrcorner$

**Def. (16.11.1.5) [Weil Operator].** Given a Hodge structure  $(V, h)$ , the **Weil operator** is defined to be the real homomorphism  $C = h(i)$  (16.11.1.4). Equivalently,  $C$  acts by  $i^{q-p}$  on  $V^{p,q}$ .  $\perp$

**Def. (16.11.1.6) [Tate Hodge Structure].** A **Tate Hodge Structure** is the Hodge structure  $\mathbb{Z}(1)$  pure of type  $(-1, -1)$  s.t.

$$\mathbb{Z}(1) = 2\pi i\mathbb{Z} \subset \mathbb{Z}(1)_{\mathbb{C}} \cong \mathbb{C}.$$

Also denote  $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$ . In the correspondence (16.11.1.4),  $\mathbb{Z}(n)$  correspond to the representation

$$\mathbb{S} \rightarrow \mathbb{G}_m : z \mapsto (z\bar{z})^n.$$

In particular, the Weil operator is identity on a Tate Hodge structure.  $\perp$

**Def. (16.11.1.7) [Hodge Group].** For  $V \in \text{Hdg}_Q$  with the corresponding map  $\mathbb{S} \rightarrow \text{GL}(V)_{\mathbb{R}}$ , the **Hodge group**  $\text{Hdg}(V)$  is the smallest subgroup  $H \subset \text{GL}(V)$  s.t. the map  $h|_{\mathbb{S}^1}$  factors through  $H_{\mathbb{R}}$ .  $\perp$

**Prop. (16.11.1.8) [Hodge Structures of Type  $(-1, 0), (0, -1)$ ].** A Hodge structure of type  $(-1, 0), (0, -1)$  is equivalent to a complex vector space. And the Weil operator is just acting by  $i$ .  $\perp$

*Proof:* This follows from (3.5.11.30).  $\square$

**Prop. (16.11.1.9) [Mixed Hodge Structure from Algebraic Geometry].** For any f.g. field  $Q \subset \mathbb{C}$ , there is a functor  $\text{Var}/K \rightarrow m\text{Hdg}_Q$ .  $\perp$

*Proof:* Cf. [Deligne, Hodge Theory]  $\color{red}{?}$   $\square$

### Polarizations

**Def. (16.11.1.10) [Hodge Tensors].** For  $(V, h) \in \text{Hdg}_Q^n$ , an  $r$ -**Hodge tensor** is a map of  $Q$ -Hodge-structures

$$T : V^{\otimes r} \rightarrow Q(-\frac{nr}{2}).$$

In particular,  $T(C_V v_1, \dots, C_V v_r) = T(v_1, \dots, v_r)$ .  $\perp$

**Def. (16.11.1.11) [Polarizations].** For  $(V, h) \in \text{Hdg}_Q^n$ , a **Hodge polarization** is a Hodge tensor  $\psi : V \times V \rightarrow Q(-n)$  s.t.

$$\psi_C(u, v) = (2\pi i)^n \psi(u, Cv)$$

is symmetric and positive-definite. In particular,

$$\psi(u, v) = \psi(Cu, Cv) = (-1)^n \psi(v, u).$$

A isomorphism of Hodge structures preserving their polarization is called a **Hodge isometry**.  $\perp$

**Def. (16.11.1.12) [Polarizations].**  $\color{red}{?}$  An integral **polarized Hodge structure** of weight  $k \in \mathbb{Z}$  is given by an integral Hodge structure  $(\Lambda, F^p \Lambda)$  of weight  $k$  (16.11.1.3) together with a Hermitian form  $H$  on  $\Lambda_{\mathbb{C}}$  s.t.

- The Hodge decomposition (16.11.1.3) is orthogonal for  $H$ .
- If  $\alpha \neq 0 \in V^{p,q}(X)$ , then

$$i^{p-q-k} (-1)^{\frac{k(k-1)}{2}} H(\alpha) > 0$$

┘

**Example (16.11.1.13) [Smooth Projective Varieties].** If  $X \in \text{SmProjVar}/\mathbb{C}$ , then any embedding  $X \hookrightarrow \mathbb{P}_{\mathbb{C}}^n$  induces a polarized rational Hodge structure on the primitive part of  $H^n(X, \mathbb{Q})$  for each  $n$ , by the Hodge index theorem. ┘

*Proof:* Cf.[Voison]Thm6.3.2..? ┘

**Example (16.11.1.14) [Complex Abelian Varieties].** [Frobenius-Lefschetz-Poincaré-Riemann-Weierstrass] By Riemann theorem(16.10.5.9) and(16.11.1.12), for any  $n \in \mathbb{Z}_+$ , there is an equivalence of categories:

$$\text{AbVar}^{\text{polar}}/\mathbb{C} \cong \text{int-Hdg}^{-1, \text{polar}}.$$

┘

**Def. (16.11.1.15) [ $K$ -Hodge-deRham Structures].** For a subfield  $\iota : K \subset \mathbb{C}$ , a  **$K$ -Hodge-deRham structure** is a quadruple  $V = (V_K, \{V^{p,q}\}_{p,q \in \mathbb{Z}}, V_{\mathbb{Q}}, u)$  where

- $V_{\mathbb{Q}} \in \text{Vect}_{\mathbb{Q}}$ ,
- $(V_K, \{V^{p,q}\}_{p,q \in \mathbb{Z}})$  is a  $K$ -mixed Hodge structure(16.11.1.2).
- $u$  is an isomorphism

$$u : V_K \otimes_K \mathbb{C} \cong V_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C},$$

endowing  $V_K \otimes_K \mathbb{C}$  with a real structure. ┘

**Def. (16.11.1.16) [Primitive Parts].** ┘

### Hodge-Classes

**Def. (16.11.1.17) [Hodge-Classes].** For  $\Lambda \in \text{int-Hdg}^{2p}$ , define the space of **Hodge-classes** of degree  $2p$  to be

$$\text{Hdg}^{2p}(\Lambda) \triangleq \Lambda \cap V^{p,p} = \ker(\Lambda \rightarrow \Lambda_{\mathbb{C}}/F^p V).$$

And if  $X$  is a compact Kähler manifold and  $k \in \mathbb{N}$ , the **Hodge-classes** of  $X$  of degree  $2k$  is defined to be  $\text{Hdg}^{2k}(X) = \text{Hdg}^{2k}(H^{2k}(X, \mathbb{Z})) = H^{2k}(X, \mathbb{Z}) \cap H_{\text{Betti}}^{k,k}(X)$ . ┘

**Prop. (16.11.1.18) [Lelong1957].** If  $X$  is a compact Kähler manifold, then the cohomology classes of analytic subsets of  $X$  and the Chern classes of holomorphic vector bundles on  $X$  are Hodge classes(16.11.1.17). And these two subsets are equal. ┘

*Proof:* Cf.[Voisin, Hodge]P273, 280.? ┘

**Prop. (16.11.1.19) [Lelong].** If  $X$  is a projective complex manifold and  $\mathcal{L}$  is a holomorphic line bundle on  $X$  and  $\sigma$  a non-zero section of  $\mathcal{L}$ , then the cohomology class of  $\text{Div}(\sigma)$  is equal to  $c_1(\mathcal{L})$  in  $H^2(X, \mathbb{Z})$ .

In particular, the Chern classes of line bundles and the cohomology class of divisors on  $X$  are the same. ┘

*Proof:* Cf.[Voisin, Hodge]P281.? ┘

For the last assertion, use the fact that any line bundle has a meromorphic section(using the ample line bundle). ┘

**Def. (16.11.1.20) [Künneth Hodge Classes].** If  $X$  is a compact Kähler manifold of dimension  $n$ , then the Hodge decomposition and the Künneth decomposition on  $H^*(X \times X)$  are compatible. In particular, for  $a, b \in \mathbb{N}$ ,  $a+b = n$ , the Künneth components  $\text{cl}(\Delta)_{a,b} \in H^a(X) \otimes H^b(X) \subset H^{2n}(X \times X)$  is a Hodge class by (16.11.1.18).  $\lrcorner$

**Example (16.11.1.21) [Künneth Hodge Classes].** If  $X$  is a compact Kähler manifold of dimension  $n$ , then the Hodge decomposition and the Künneth decomposition on  $H^*(X \times X)$  are compatible. In particular, for  $a, b \in \mathbb{N}$ ,  $a+b = n$ , the Künneth components  $\text{cl}(\Delta)_{a,b} \in H^a(X) \otimes H^b(X) \subset H^{2n}(X \times X)$  is a Hodge class by (16.11.1.18).  $\lrcorner$

**Example (16.11.1.22) [Inverse Hyperplane Section Hodge Classes].** If  $X \in \text{SmProjVar}^n/\mathbb{C}$ , and  $\eta \in H^2(X, \mathbb{Z})$  is the class of a hyperplane section, then by hard lefschetz, for any  $p \in \mathbb{N}$ ,  $\cup \eta^p : H^{n-p}(X, \mathbb{C}) \rightarrow H^{n+p}(X, \mathbb{C})$  is an isomorphism. Let  $\mathfrak{z} \in H^{n-p}(X, \mathbb{C}) \times H^{n+p}(X, \mathbb{C}) \subset H^{2n}(X \times X)$  s.t.

$$(\eta^p)^{-1}(c) = \text{pr}_{1!}(\mathfrak{z} \cup \text{pr}_2^* c),$$

then  $\mathfrak{z}$  is a Hodge class.  $\lrcorner$

### Variations of Hodge Structures

**Thm. (16.11.1.23).** Let  $S \in \text{Var}/\mathbb{C}$  and  $X \in \text{SmProjVar}/S$ . Given an integral class  $c$  on a typical fiber  $X_0$ , then the locus where  $c$  is  $\lrcorner$

## 2 Mumford-Tate Groups

**Def. (16.11.2.1) [Mumford-Tate Groups].** For  $V \in \text{Hdg}_{\mathbb{Q}}$  with the corresponding homomorphism  $\$ \rightarrow \text{GL}(V)_{\mathbb{R}}$ , The **Mumford-Tate group**  $\text{MT}(V)$  is defined to be the minimal algebraic subgroup  $M \leq \text{GL}(V)$  s.t.  $h$  factors through  $M_{\mathbb{R}}$ .

More generally, the **big Mumford-Tate group** is defined to be  $\text{MT}^{\sharp}(V) = \text{MT}(V \oplus \mathbb{Q}(1))$ .

For  $X \in \text{SmProj}/\mathbb{C}$ , define the Mumford-Tate group of  $X$  to be  $\text{MT}^r(X) = \text{MT}(H_{\text{Betti}}^{2r}(X, \mathbb{Q}(r)))$ .  $\lrcorner$

**Prop. (16.11.2.2).** For  $V \in \text{Hdg}_{\mathbb{Q}}$ ,

- $\text{MT}(V)$  is connected.
- Suppose  $V$  is pure of weight  $m$ . Then if  $m = 0$ ,  $\text{MT}(V) \subset \text{SL}(V)$ , and if  $m \neq 0$ ,  $\text{MT}(V)$  contained  $\mathbb{G}_m$ .
- Suppose  $V$  is pure of weight  $m$ . Then if  $m = 0$ ,  $\text{MT}^{\sharp}(V) \cong \text{MT}(V) \times \mathbb{G}_m$ , and if  $m \neq 0$ , then  $\text{MT}^{\sharp}(V) \rightarrow \text{MT}(V)$  is an isogeny. Moreover, if  $m = 1$ , then this isogeny is an isomorphism.  $\lrcorner$

*Proof:* ? Cf.[Mumford-Tate Groups, Moonen]P9.  $\square$

**Prop. (16.11.2.3).** The Mumford-Tate group of a polarizable Hodge structure is a reductive group.  $\lrcorner$

*Proof:* Cf.[Hodge Cycles on Abelian Varieties, Deligne-Milne]P24?.

Use the fact that in characteristic 0, an algebraic group is reductive iff it has a completely reducible faithful representation?. Then use the fact that  $\text{Hdg}_{\mathbb{Q}}^{\text{polar}}$  is a semisimple category?. And the fact that sub-Hodge structure of  $V$  are exactly  $\text{MT}(V)$ -submodules.  $\square$

**Prop. (16.11.2.4).** If  $m \in \mathbb{Z}, n \in \mathbb{Z}_+$  and  $V \in \text{Hdg}_{\mathbb{Q}}^m$ , then  $\text{MT}(V^n) = \text{MT}(V)$ , where  $\text{MT}(V)$  is the diagonal embedded in  $\text{GL}(V^n)$ . ┘

*Proof:* ? □

**Prop. (16.11.2.5) [Mumford-Tate Group of Abelian Varieties].** For  $A \in \text{AbVar}/\mathbb{C}$ ,  $A$  has CM iff  $\text{MT}(A)$  is a torus. ┘

*Proof:* Cf. [Mumford-Tate Groups, Moonen] P11. ? □

### 3 Hodge Conjecture

References are [Hodge cycles on abelian varieties, Deligne] and [Voisin].

**Conj. (16.11.3.1) [Hodge].** For  $X \in \text{SmProj}/\mathbb{C}$ , then  $\text{Hdg}^{2k}(X) \otimes \mathbb{Q}$  is generated by cohomology class of an algebraic cycle of  $X$  (16.11.1.18). ┘

WARNING: The hodge conjecture is wrong without the hypothesis that  $X$  is projective. ┘

*Proof:* □

**Thm& Conj. Cor. (16.11.3.2) [Lefschetz (1,1)-Form].** The Hodge conjecture is true in codimension 1 by Lefschetz (1, 1)-form theorem (16.11.5.6). ┘

*Proof:* □

**Thm& Conj. Cor. (16.11.3.3) [Simple Abelian Varieties, Tankeev].** If  $A \in \text{AbVar}/\mathbb{C}$  is a simple Abelian variety s.t.  $\dim(A) \in \text{Prime}$ , then the Hodge conjecture (16.11.3.1) holds for any powers of  $A$ . ┘

*Proof:* □

**Conj. (16.11.3.4) [Generalized Hodge Conjecture, Grothendieck].** Let  $X \in \text{SmProj}/\mathbb{C}$ , and  $L \subset H^{2k+l}(X, \mathbb{Q})$  is a  $\mathbb{Q}$ -sub-Hodge structure contained in  $F^k H^{2k+l}(X)$ , then there exist subvarieties  $Y_i \subset X$  of codimension  $k$  s.t.  $L$  is contained in  $\sum_i j_{i*} H_{2n-l}(Y_i, \mathbb{Q})$ . ┘

*Proof:* Cf. [Hodge Theory, Voisin] P285. ? □

### Arithmetic Theory

**Conj. (16.11.3.5) [Deligne].** ┘

*Proof:* □

**Thm& Conj. Cor. (16.11.3.6) [Deligne].** Let  $k \in \text{Field}^0, k = \bar{k}, A \in \text{AbVar}/k$ , and let  $t \in H_{\mathbb{A}}^{2p}(A)(p)$ . Then if  $t$  is a Hodge cycle w.r.t. one embedding  $\sigma : k \rightarrow \mathbb{C}$ , then it is absolutely Hodge. ┘

*Proof:* □

**Def. (16.11.3.7) [deRham-Hodge Classes].** Let  $k \in \text{Field}^0, k = \bar{k}$ , and  $X \in \text{SmProj}/k$ , then for any embedding  $\sigma : k \hookrightarrow \mathbb{C}$ ,  $H_{\text{dR}}^n(X/k)$  satisfies

$$H_{\text{dR}}^n(X/k) \otimes_{k, \sigma} \mathbb{C} \cong H_{\text{dR}}^n(X, \mathbb{C}) \cong H_{\text{Betti}}^n(X, \mathbb{C}). ??$$

Thus we can define the space  $\text{Hdg}_{\text{dR}}^{2k}(X)$  of **deRham-Hodge classes** on  $X$  of degree  $2k$  w.r.t.  $\sigma$  as the subspace of elements in  $H_{\text{dR}}^{2k}(X/k)$  that are mapped to  $(2\pi i)^k$  times a Hodge class of  $\sigma X$  of degree  $2k$  (16.11.1.17). ┘

This notion is independent of the embedding  $\sigma$ . ┘

*Proof:* ? □

**Prop. (16.11.3.8)**[Tate+Deligne  $\implies$  Hodge Conjectures]. If the Tate conjecture holds for  $X$ , then every absolutely Hodge class of  $X$  is algebraic. In particular, if the Deligne conjecture is true (e.g. for Abelian varieties), the Tate conjecture implies the Hodge conjecture. ┘

*Proof:* Cf.[Tate conjecture over Finite fields, Milne]P11. ? □

## 4 Absolute Hodge Cycles

**Thm. (16.11.4.1)** [Principle A, Deligne]. Let  $(X_\alpha/\mathbb{C})$  be a family of varieties, and let  $\mathcal{T}$  be the category of the spaces obtained from tensors and duals of  $H_{\text{Betti}}^*(X_\alpha, \mathbb{Q})$  and  $\mathbb{Q}(1)$ . Let  $t_i \in T_i \in \mathcal{T}, i = 1, \dots, N$  be absolute Hodge cycles and let

$$G \subset \prod_{\alpha, n_\alpha} \text{GL}(H_{\text{Betti}}^{n_\alpha}(X_\alpha, \mathbb{Q})) \times \mathbb{G}_m$$

be the subgroup fixing  $t_i$ . Then any  $t \in T \in \mathcal{T}$  that is fixed by  $G$  is an absolute Hodge cycle. ┘

*Proof:* Cf.[Absolute Hodge Cycles on Abelian Varieties, Deligne-Milne]P26. ? □

## 5 Abel-Jacobi Maps

### Intermediate Jacobians

**Def. (16.11.5.1)**[Intermediate Jacobians]. For any integral Hodge structure of weight  $2k-1$ , we can define

$$J^{2k-1}(\Lambda) = \Lambda_{\mathbb{C}} / (F^k \Lambda \oplus \overline{F^k \Lambda}).$$

This is a complex torus. ┘

*Proof:* We have a decomposition

$$\Lambda_{\mathbb{C}} = F^k \Lambda \oplus \overline{F^k \Lambda},$$

so  $F^k \Lambda \cap \Lambda_{\mathbb{R}} = \{0\}$ , thus the map

$$\Lambda_{\mathbb{R}} \rightarrow \Lambda_{\mathbb{C}} / F^k \Lambda$$

is an isomorphism of  $\mathbb{R}$ -vector spaces. □

**Prop. (16.11.5.2)**. If  $(\Lambda, F^\bullet \Lambda) \rightarrow (\Lambda', F^\bullet \Lambda')$  is a morphism of Hodge-structure of bidegree  $(r, r)$ , then it induces a morphism of complex tori

$$J^{2p-1}(\Lambda) \rightarrow J^{2(p+r)-1}(\Lambda').$$

┘

**Def. (16.11.5.3)**[Jacobian and Albanese Varieties]. If  $X$  is a compact Kähler manifold of dimension  $n$ , then  $J^1(X)$  is also denoted by  $\text{Jac}(X)$ , called the **Jacobian variety of  $X$** , and  $J^{2n-1}(X)$  is denoted by  $\text{Alb}(X)$ , called the **Albanese variety of  $X$** . ┘

**Prop. (16.11.5.4)[Jacobian].** The **Jacobian**  $\text{Jac}(X)$  of a compact Kähler manifold  $X$  is defined to be  $H^1(X, \mathcal{O}_X)/H^1(X, \mathbb{Z})$ , so it is a complex torus of dimension  $b_1(X)$  by (16.11.5.1), it is also the kernel of the first Chern class map by the long exact sequence (16.9.5.8), i.e.

$$0 \rightarrow \text{Jac}(X) \rightarrow \text{Pic}(X) \xrightarrow{c_1} \text{NS}(X) \rightarrow 0$$

┘

**Lemma (16.11.5.5).** if  $X$  is compact Kähler, then the natural map  $H^k(X, \mathbb{C}) \rightarrow H^k(X, \mathcal{O}_X)$  is just the projection onto the  $(0, k)$ -part. In particular, the image is in  $H^{0,k}(X)$ . ┘

*Proof:* This follows from Hodge decomposition, the definition of Dolbeault cohomology and the following commutative diagram:

$$\begin{array}{ccccccc} \mathbb{C} & \longrightarrow & \mathcal{A}^0(X) & \xrightarrow{d} & \mathcal{A}^1(X) & \xrightarrow{d} & \mathcal{A}^2(X) \dots \\ \downarrow & & \downarrow = & & \downarrow \pi_{0,1} & & \downarrow \pi_{0,2} \\ \mathcal{O}_X & \longrightarrow & \mathcal{A}^0(X) & \xrightarrow{\bar{\partial}} & \mathcal{A}^1(X) & \xrightarrow{\bar{\partial}} & \mathcal{A}^2(X) \dots \end{array}$$

□

**Thm. (16.11.5.6) [Lefschetz theorem on  $(1, 1)$ -forms].** By (16.9.5.8), the image of  $\text{Pic}_{\mathbb{C}}(X) \rightarrow H^2(X, \mathbb{Z})$  is trivial in  $H^2(X, \mathcal{O}_X)$ . And if  $X$  is compact Kähler, there is Hodge decomposition (16.10.4.1)  $H^2(X, \mathcal{O}_X) = H^{2,0}(X) \oplus H^{1,1}(X) \oplus H^{0,2}(X)$ .

So the image of  $\text{Pic}_{\mathbb{C}}(X)$  is contained in  $H^{1,1}(X, \mathbb{Z}) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X)$  by (16.11.5.5) and duality. Then it is also surjective, this is to say,  $\text{NS}(X) = H^{1,1}(X, \mathbb{Z})$  (16.9.5.12). ┘

*Proof:* Because by the long exact sequence of (16.9.5.8) and (16.11.5.5) again, an  $\alpha \in H^2(X, \mathbb{C})$  is in  $H^{1,1}(X, \mathbb{Z})$  iff  $\alpha$  is in the image of  $\text{Pic}_{\mathbb{C}}(X) \rightarrow H^{1,1}(X, \mathbb{Z})$ . □

### Abel-Jacobi Maps

**Def. (16.11.5.7)[Abel-Jacobi Map].** For any  $k \in \mathbb{Z}_+$ , there is an **Abel-Jacobi map**

$$\Phi_X^k : Z^k(X)_{\text{hom}} \rightarrow J^{2k-1}(X).$$

┘

*Proof:* Cf.[Voison, P292]. ? □

**Thm. (16.11.5.8) [Griffiths1968].** Let  $X$  is a compact Kähler manifold,  $Y$  be a connected complex manifold and  $y_0 \in Y$ ,  $Z \subset Y \times X$  a cycle of codimension  $k$  s.t. each component  $Z_i$  of  $Z$  is smooth and the projection  $Z_i \rightarrow Y$  is a submersion. Then the map

$$Y \rightarrow J^{2k-1}(X) : y \mapsto \Phi_X^k(Z_y - Z_{y_0})$$

is holomorphic. ┘

*Proof:* Cf.[Voison, P294]. ? □



**Def. (16.11.5.9) [Albanese Maps].** Fix a base point  $x_0$  of  $X$ , by Griffith's theorem (16.11.5.8) applied to  $\Delta \subset X \times X$ , we get an **Albanese map**

$$\text{Alb}_X : X \rightarrow \text{Alb}(X) : x \mapsto \Phi_X^{2n-1}(x - x_0)$$

that is holomorphic and functorial in  $(X, x_0)$ .  $\lrcorner$

**Prop. (16.11.5.10).** if  $T$  is a torus, then the Albanese map  $\text{Alb}_T : T \rightarrow \text{Alb}(T)$  is an isomorphism.  $\lrcorner$

*Proof:*  $\square$

**Prop. (16.11.5.11) [Universal Properties].** The Albanese map satisfies the following universal property: Any complex torus  $T$  and a map  $f : X \rightarrow T$  s.t.  $f(x_0) = 0$  factors through the Albanese map.  $\lrcorner$

*Proof:*  $\square$

**Lemma (16.11.5.12).** For  $k \in \mathbb{Z}_+$  sufficiently large,

$$\text{Alb}^k : X^k \rightarrow \text{Alb}(X)$$

is surjective.  $\lrcorner$

*Proof:* Cf. [Voison, P298].  $\square$

**Prop. (16.11.5.13).** If  $X \in \text{SmProj}/\mathbb{C}$ , then  $\text{Alb}(X) \in \text{AbVar}/\mathbb{C}$ .  $\lrcorner$

*Proof:* Cf. [Voison, P300].  $\square$  The factorized map is

$$g : \text{Alb}(X) = H^0(X, \Omega_X)^*/H_1(X, \mathbb{Z}) \rightarrow T = H^0(T, \Omega_T)^*/H_1(T, \mathbb{Z}).$$

$\square$

## 6 Deligne Cohomology

Cf. [Voisin1, 2] and [Esnault-Viehweg, Deligne-Beilinson Cohomology].

**Def. (16.11.6.1) [Deligne Complex].** Let  $X$  be a complex manifold and  $p \in \mathbb{Z}_+$ , the (real) **Deligne complex**  $\mathbb{R}_{\text{Del}}(p)$  is the complex

$$0 \rightarrow \mathbb{R}[0] \xrightarrow{(2\pi i)^p} \mathcal{O}_X \xrightarrow{d} \Omega_X \xrightarrow{d} \dots \xrightarrow{d} \Omega_X^{p-1} \rightarrow 0.$$

Similarly we can define the Deligne complex with coefficients in  $A$  for any ring  $A \subset \mathbb{R}$ .  $\lrcorner$

**Def. (16.11.6.2) [Deligne Cohomologies].** Let  $X$  be a complex manifold and  $p \in \mathbb{Z}, k \in \mathbb{N}$ , the **Deligne cohomology**  $H_{\text{Del}}^k(X, \mathbb{R}(p))$  is defined to be the hypercohomology  $\mathbb{H}^k(X, \mathbb{R}_{\text{Del}}(p))$  (16.11.6.1). Similarly we can define the Deligne cohomology with coefficients in  $A$  for any ring  $A \subset \mathbb{R}$ .  $\lrcorner$

**Example (16.11.6.3).** For  $p = 1$ ,  $\mathbb{Z}_{\text{Del}}(1)$  is quasi-isomorphic to  $\mathcal{O}_X^*[-1]$ , so  $H_{\text{Del}}^{k+1}(X, \mathbb{Z}(1)) \cong H^k(X, \mathcal{O}_X^*)$ .

For  $p = 2$ , there is a quasi-isomorphism

$$\begin{array}{ccccc} (2\pi i)^2 \mathbb{Z} & \longrightarrow & \mathcal{O}_X & \longrightarrow & \Omega_X^1 \\ & & \downarrow \exp((2\pi i)^{-1}(-)) & & \downarrow (2\pi i)^{-1} \\ & & \mathcal{O}_X^* & \xrightarrow{f \mapsto \text{dlog}(f)} & \Omega_X^1 \end{array}$$

so  $H_{\text{Del}}^1(X, \mathbb{Z}(2)) = \mathbb{C}^\times$ , and  $H_{\text{Del}}^2(X, \mathbb{Z}(2))$  corresponds to holomorphic line bundles with a holomorphic connection.  $\lrcorner$

**Prop. (16.11.6.4).** If  $X$  is a compact Kähler manifold and  $A \subset \mathbb{R}$ , then there is a long exact sequence

$$\dots \rightarrow H_{\text{Del}}^k(X, A(p)) \rightarrow H^k(X, A) \rightarrow H^k(X, \mathbb{C}) / \text{Fil}^p H^k(X, \mathbb{C}) \rightarrow H_{\text{Del}}^{k+1}(X, A(p)) \rightarrow \dots$$

$\lrcorner$

*Proof:* There is an exact sequence of complexes

$$0 \rightarrow \Omega_X^{\leq p-1}[-1] \rightarrow A_{\text{Del}}(p) \rightarrow A(p) \rightarrow 0,$$

which induces a long exact sequence

$$\dots \rightarrow H_{\text{Del}}^k(X, A(p)) \rightarrow H^k(X, A(p)) \rightarrow \mathbb{H}^k(X, \Omega_X^{\leq p-1}) \rightarrow H_{\text{Del}}^{k+1}(X, A(p)) \rightarrow \dots$$

And the exact sequence of complexes

$$0 \rightarrow \Omega_X^{\geq p} \rightarrow \Omega_X^\bullet \rightarrow \Omega_X^{\leq p-1} \rightarrow 0$$

and the fact  $\mathbb{H}^k(X, \Omega_X^{\geq p}) \cong \text{Fil}^p H^k(X, \mathbb{C}) \subset H^k(X, \mathbb{C})$  ? implies that  $\mathbb{H}^k(X, \Omega_X^{\leq p-1}) \cong H^k(X, \mathbb{C}) / F^p H^k(X, \mathbb{C})$ .  $\square$

**Cor. (16.11.6.5).** There is an exact sequence

$$0 \rightarrow J^{2p-1}(X) \rightarrow H_{\text{Del}}^{2p}(X, \mathbb{Z}(p)) \rightarrow \text{Hdg}^{2p}(X, \mathbb{Z}) \rightarrow 0. \text{(16.11.5.1)(16.11.1.17)}$$

$\lrcorner$

**Prop. (16.11.6.6).** Let  $X$  be compact Kähler of dimension  $p$  and  $i < 2p$ , then

$$H^i(X, \mathbb{R}(i)) \hookrightarrow H^k(X, \mathbb{C}) / \text{Fil}^p H^k(X, \mathbb{C}).$$

So there are exact sequences

$$0 \rightarrow H^{i-1}(X, \mathbb{R}(p)) \rightarrow H^{i-1}(X, \mathbb{C}) / \text{Fil}^p H^{i-1}(X, \mathbb{C}) \rightarrow H_{\text{Del}}^i(X, \mathbb{R}(p)) \rightarrow 0.$$

And because  $\mathbb{C} = \mathbb{R}(p) \oplus \mathbb{R}(p-1)$ , there are exact sequences

$$0 \rightarrow \text{Fil}^p H^{i-1}(X, \mathbb{C}) \rightarrow H^{i-1}(X, \mathbb{R}(p-1)) \rightarrow H_{\text{Del}}^i(X, \mathbb{R}(p)) \rightarrow 0$$

is injective.  $\lrcorner$

*Proof:* This is because  $\mathbf{c}$  acts as multiplication by  $(-1)^p$  on the LHS, and  $\text{Fil}^p H^k(X, \mathbb{C}) \cap \overline{\text{Fil}^p H^k(X, \mathbb{C})} = \oplus_{r,s \geq p} H^{r,s}(X)$ , so the kernel is  $\{0\}$ .  $\square$

**Prop. (16.11.6.7) [Comparison of Complex Conjugations].** For  $X \in \text{SmProj}/\mathbb{R}$ , the canonical isomorphism

$$H_{\text{Betti}}(X, \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} = H_{\text{Betti}}(X, \mathbb{C}) \xrightarrow{I_{\text{dR}}} H_{\text{dR}}^i(X^{\text{an}}) \xrightarrow{GAGA} H_{\text{dR}}^i(X) \otimes_{\mathbb{R}} \mathbb{C}$$

is an isomorphism that identifies the complex conjugation  $\mathbf{c}^* \otimes \mathbf{c}$  on the LHS with  $\mathbf{c}$  on the RHS, so  $\perp$

*Proof:* Deligne, Prop1.4.  $\square$

**Cor. (16.11.6.8).** By taking the complex conjugation fixed part, we get an exact sequence

$$0 \rightarrow \text{Fil}^p H_{\text{Betti}}^{i-1}(X, \mathbb{R}) \rightarrow H_{\text{Betti}}^{i-1}(X, \mathbb{R}(p-1))^{\mathbf{c}^* = (-1)^{p-1}} \rightarrow H_{\text{Del}}^i(X/\mathbb{R}, \mathbb{R}(p)) \rightarrow 0$$

$\perp$

### Differential Characters

**Def. (16.11.6.9) [Differential Characters].** Let  $X$  be a differential manifold and  $Z_l^\infty$  be the subgroup of closed singular differentiable chains, and  $\Xi_\infty^l(X) \subset \text{Hom}(Z_l^\infty, \mathbb{R}/\mathbb{Z})$  consisting of characters  $\chi$  s.t. there exists  $\omega \in \Omega^{l+1}(X)$  s.t.

$$\chi(\partial\varphi) = \int_{\Delta_{l+1}} \varphi^* \omega \in \mathbb{R}/\mathbb{Z}, \quad \forall \varphi \in C_{l+1}^\infty(X).$$

Such an  $\omega$  is clearly uniquely determined by  $\chi$ .  $\perp$

**Prop. (16.11.6.10).** For any  $\chi \in \Xi_\infty^l(X)$ ,  $\omega_\chi$  is an integral closed form.  $\perp$

*Proof:* Cf. [Voisin, P306].  $\textcolor{red}{?}$   $\square$

**Prop. (16.11.6.11).** If  $X$  is a complex manifold, and  $\mu \in \Omega_{\mathbb{R}}^{l-1}(X)$ , then for any closed chain  $\varphi \in C_l^\infty(X)$ ,

$$\overline{\int_{\Delta_l} \varphi^*(i\partial\mu)} = \int_{\Delta_l} \varphi^*(-i\bar{\partial}\mu) = \int_{\Delta_l} \varphi^*(-i d\mu + i\partial\mu) = \int_{\Delta_l} \varphi^*(i\partial u),$$

so  $\int_{\Delta_l} \varphi^*(i\partial u) \in \mathbb{R}$ , and we can define a differential character

$$\int i\partial\mu \in \Xi_\infty^l(X) : \varphi \mapsto \int_{\Delta_l} \varphi^*(i\partial u).$$

$\perp$

**Prop. (16.11.6.12).** For a compact Kähler manifold  $X$  and  $p \in \mathbb{N}$ , consider the subgroup  $\Xi_\infty^{2p-1}(X)^{p,p} \subset \Xi_\infty^{2p}(X)$  consisting of differential characters  $\chi$  s.t.  $\omega_\chi$  is of the type  $(p, p)$ . Then

$$H_{\text{Del}}^{2p-1}(X)^{p,p} \cong K_\infty^{2p-1}(X) = \Xi_\infty^{2p-1}(X)^{p,p} / \left\{ \int i\partial\mu \mid \mu \in \Omega^{p-1,p-1}(X) \right\}.$$

$\perp$

*Proof:* Cf. [Voisin, P307].  $\textcolor{red}{?}$   $\square$

### Properties of the Deligne Cohomology

**Prop. (16.11.6.13) [Cup Products].** There are cup products

$$H_{\text{Del}}^p(X, \mathbb{Z}(q)) \times H_{\text{Del}}^{p'}(X, \mathbb{Z}(q')) \rightarrow H_{\text{Del}}^{p+p'}(X, \mathbb{Z}(q+q')).$$

┘

*Proof:*

□

**Prop. (16.11.6.14) [Cycle Class Maps].** For  $X \in \text{SmProj}/\mathbb{C}$  and  $p \in \mathbb{N}$ , there are class maps

$$\text{CH}^p(X) \rightarrow H_{\text{Del}}^{2p}(X, \mathbb{Z})$$

that lifts the class  $[Z] \in H^{2p}(X, \mathbb{Z})$  (16.11.6.4), and commutes with products and cup products. ┘

*Proof:* Cf. [Voisin, P311]. ?

□

### Modified Hodge Conjecture (Clausen)

#### 7 Coniveau

**Def. (16.11.7.1) [Coniveau].** The **coniveau** of  $\alpha \in H_{\text{Betti}}^\bullet(X, \mathbb{Q})$  is the smallest number  $c$  s.t. there is a closed algebraic subscheme  $Y \subset X$  s.t.  $\alpha|_{X \setminus Y} = 0 \in H_{\text{Betti}}^\bullet(X \setminus Y, \mathbb{Q})$ . ┘

**Thm. (16.11.7.2) [Deligne].** If  $\alpha \in H_{\text{Betti}}^\bullet(X, \mathbb{Q})$  is mapped to  $0 \in H_{\text{Betti}}^\bullet(X \setminus Y, \mathbb{Q})$  where  $Y$  is pure of codimension  $c$ , then  $\alpha = j_*\beta$ , where  $j : \tilde{Y} \rightarrow Y \rightarrow X$  is a resolution of singularities of  $Y$  and  $\beta \in H_{\text{Betti}}^{\bullet-2c}(\tilde{Y}, \mathbb{Q})$ . ┘

*Proof:*

□

## 16.12 Compactifications of Locally Symmetric Spaces

References are [Smooth compactifications of locally symmetric varieties, Rapoport-Scholze](contains a lot of references), [Toroidal Compactification of Siegel Spaces (1980).pdf], [Goresky, M. 2005. Compactifications and cohomology of modular varieties, pp. 551–582. In Harmonic analysis, the trace formula, and Shimura varieties] and [SATAKE, 2001. Compactifications, old and new].

**Def. (16.12.0.1)[Toroidal Compactifications].** Cf. [BOREL, A. AND JI, L. 2006. Compactifications of symmetric and locally symmetric spaces].  $\square$

For Siegel modular varieties, there is a short note in [Conrad, Mordell conjecture seminar].

### 16.13 Calabi-Yau Manifolds

# 17 | Non-Commutative Geometry(Connes)

## 17.1 Non-Commutative Geometry

References are [Connes, Geometry from the spectral point of view], [Connes, Noncommutative geometry], [Bost-Connes, Hecke Algebras, Type 3 factors and phase transitions with spontaneous symmetry breaking in number theory].

### 1 Spectral Triples

**Def. (17.1.1.1) [Spectral Triples].** A **spectral triple** is a triple  $(\mathcal{A}, \mathcal{H}, D)$  where  $\mathcal{A}$  is a  $*$ -algebra,  $\rho : \mathcal{A} \rightarrow B(\mathcal{H})$  is a representation on a Hilbert space  $\mathcal{H}$ , and  $D \in L(\mathcal{H})$  (not necessarily bounded), s.t.

- $D$  is self-adjoint.
- For each  $\lambda \in \mathbb{C} \setminus \mathbb{R}$ ,  $(D - \lambda)^{-1}$  is compact on  $\mathcal{H}$ .
- For each  $a \in \mathcal{A}$ , the commutator  $[a, D] \in B(\mathcal{H})$ .

┘

## 17.2 Supersymmetry

**Def. (17.2.0.1) [Supermanifolds].** A **supermanifold** of dimension  $(n, m)$  is a smooth manifold  $|M|$  together with a sheaf  $\mathcal{O}_M$  of  $\mathbb{Z}/(2)$ -graded algebras that is locally isomorphic to  $\mathcal{O}_{|M|} \otimes \Lambda[\xi_1, \dots, \xi_m]$ . The odd functions on  $M$  generate a nilpotent ideal  $\mathcal{J} \subset \mathcal{O}_M$ , with  $(|M|, \mathcal{O}_M/\mathcal{J})$  a manifold, called the underlying manifold of  $M$ , denoted by  $M_{\text{red}}$ .

A map of supermanifolds  $\varphi : M \rightarrow N$  is a map of manifolds  $\varphi_0 : M_0 \rightarrow N_0$  together with a map of supervectors spaces  $\varphi_0^* \mathcal{O}_N \rightarrow \mathcal{O}_M$ .  $\lrcorner$

**Remark (17.2.0.2).** In a similar fashion, we can define complex supermanifold, analytic supermanifolds or superschemes.  $\lrcorner$

**Prop. (17.2.0.3) [Morphisms between Supermanifolds].** Let  $M$  be a supermanifold and  $U \subset \mathbb{R}^{p|q}$  an open submanifold, then a map of supermanifolds  $f : M \rightarrow U$  is equivalent to a tuple  $(f_1, \dots, f_p, \varphi_1, \dots, \varphi_q)$  s.t. for any  $x \in |M|$ ,  $(f_1(x), \dots, f_p(x)) \in |U|$ .

In fact, giving such a tuple, suppose  $f_i = f_i^0 + r_i$  where  $f_i^0 \in \mathcal{O}_{|M|}$  and  $r_i \in \mathcal{J}$ , then in local coordinates, the pullback of  $F = \sum_I F_I \theta^I$  is

$$F(f_1, \dots, f_p, \varphi_1, \dots, \varphi_q) = \sum_I \left( \sum \partial^\alpha F(f_1^0, \dots, f_p^0, \varphi_1, \dots, \varphi_q) \frac{r^\alpha}{\alpha!} \right) \varphi^I.$$

$\lrcorner$

**Def. (17.2.0.4) [Super Vector Bundles].** A **super-vector bundle** over a supermanifold  $M$  is an  $\mathcal{O}_M$ -sheaf  $\mathcal{E}$ , locally free of dimension  $p|q$ .  $\lrcorner$

### 1 Real Structures

### 2 Super Lie Groups

**Def. (17.2.2.1) [Super Classical Groups].**

- $\text{GL}(p|q)$ .
- There is a Berezinian map  $\text{Ber} : \text{GL}(p|q) \rightarrow \text{GL}(1|0)$ , and the kernel is denoted by  $\text{SL}(p|q)$ .
- $\text{OSp}(n|2m)$  is the group fixing an even supersymmetric non-degenerate bilinear form  $\Phi$ .
- $\pi \text{Sp}(n|n)$  is the group fixing an odd super-anti-symmetric non-degenerate bilinear form  $\Phi$ .
- The kernel of  $\text{Ber} : \pi \text{Sp}(n|n) \rightarrow \text{GL}(1|0)$  is called the group of P-series.
- Let  $D = \mathbb{R}[\eta]$  be the super division algebra with  $\eta$  odd and  $\eta^2 = -1$ , then there is a odd determinant map

$$\text{odet} : \text{GL}(n; D) \rightarrow \mathbb{R}^{0|1},$$

the kernel of  $\text{odet}$  is called the group of Q-series.  $\lrcorner$

*Proof:* Cf.[Deligne, Supersymmetry]P70.  $\square$



# 18 | Theoretical Physics

## 18.1 Hamiltonian Mechanics

References are [Arn89] and [Notes on Quantum Field Theory, Etingof].

**Notation(18.1.0.1).**

- Use notations on Riemannian geometry defined in [Riemannian Geometry](#).
- Use notations on symplectic geometry defined in [Symplectic Geometry](#).

┘

### 1 Hamiltonian Mechanics

#### Newtonian Mechanics

**Physical Law(18.1.1.1)[Space and Time in Newtonian Mechanics].** The **Galilean space-time structure** is a 4-dimensional real vector space. Its points are called world points or events.

The **time** is a projection from the space to  $\mathbb{R}$ :  $t : \mathbb{R}^N \rightarrow \mathbb{R}$ . Two events  $a, b$  are called **simultaneous** if  $t(a) = t(b)$ .

The set of events simultaneous to a given event forms a 3-dimensional affine subspace of  $\mathbb{R}^3$ , called a **space of simultaneous events**.

┘

**Def.(18.1.1.2) [Galilean Transformations].** By(18.1.1.1), we can assume that the Galilean space-time is isomorphic to  $\mathbb{R} \times \mathbb{R}^3$ , where the first coordinate is time. Any such isomorphism determines a **Galilean coordinate system** on the space-time structure. Thus on the Galilean space-time, we can define the Eisenstein metric  $ds^2 = dx_1^2 + dx_2^2 + dx_3^2 - dt^2$ .

The **Galilean group** is the group of automorphisms of the Galilean space preserving the Eisenstein metric. The transformations of a Galilean coordinate system is also the class of **inertia coordinate systems** w.r.t. this coordinate system.

┘

**Prop.(18.1.1.3).** Any Galilean transformation  $g$  can be written as a product  $g_1 \circ g_2 \circ g_3$  where

- $g_1$  is a **uniform motion** with velocity  $v$ :  $g_1(t, x) = (t, x + tv)$ .
- $g_2$  is a **translation of the origin**:  $g_2(t, x) = (t + s, x + x')$ .
- $g_3$  is a **rotation of coordinate axis**:  $g_3(t, x) = g_3(t, Gx)$ , where  $G \in O(3, \mathbb{R})$ .

┘

*Proof:*

□

**Def.(18.1.1.4)[Motion, Velocity and Acceleration].** A **motion** in  $\mathbb{R}^N$  is a smooth curve  $x : I \rightarrow \mathbb{R}^N$ , where  $I \subset \mathbb{R}$  is an interval.

For  $t_0 \in I$ , the **velocity of a motion**  $x : I \rightarrow \mathbb{R}^N$  is defined to be the derivative  $\dot{x} = \frac{\partial}{\partial t}x|_{t=t_0}$ . And the **acceleration vector** of  $t_0$  is defined to be the the second derivative  $\ddot{x} = \frac{\partial^2}{\partial t^2}x|_{t=t_0}$ .  $\lrcorner$

**Physical Law (18.1.1.5)[Newtonian Mechanics and Newton's Principle of Determinacy]**. Any motion of a newtonian mechanical system uniquely determined by its initial positions(i.e.  $x(t_0) \in \mathbb{R}^N$ ) and its initial states(i.e.  $\dot{x}(t_0) \in \mathbb{R}^N$ ).

The study of motions in a newtonian system is called a **Newtonian mechanics**(of particles).  $\lrcorner$

**Def. (18.1.1.6)[Newton's Equation]**. In a given Galilean coordinate system, it follows from the newton's principle of determinacy(18.1.1.5) that there is a function  $F : \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}$  s.t. for any motion  $x$  in  $\mathbb{R}^N$ ,

$$\ddot{x}(t) = F(x, \dot{x}, t).$$

This is called the **Newton's equation**.  $\lrcorner$

**Axiom (18.1.1.7)[Galileo's Principle of Relativity]**. The Newton's equations(18.1.1.6) are invariant under the action of Galilean group(18.1.1.2). Equivalently,

$$F(x, \dot{x}, t) = F(x, \dot{x}, t + s),$$

$$F(x + tv, \dot{x} + v, t) = F(x + tv, \dot{x} + v, t),$$

$$GF(x, \dot{x}, t) = F(Gx, G\dot{x}, t), \quad G \in O(3, \mathbb{R}).$$

$\lrcorner$

**Def. (18.1.1.8)[Forces]**. A **force** is a vector field on the configuration space.

A **conservative force** is a force  $F$  s.t. there exists a potential energy  $U$  s.t.  $\text{grad}U = F$ .  $\lrcorner$

**Def. (18.1.1.9)[Work of a Force]**. The **work** of a force  $F$  on a path  $\gamma$  is defined to be

$$A = (F, \gamma) = \int_{\gamma} (F, d\gamma).$$

$\lrcorner$

**Prop. (18.1.1.10)**. If the configuration space is simply-connected(e.g.  $\mathbb{R}^N$ ), then a force is conservative iff its work along any path only depends on the endpoints.  $\lrcorner$

### Hamiltonian Mechanics

**Def. (18.1.1.11)[Lagrangian Dynamical System]**. A **Lagrangian (dynamical) system** of particles consists of the following:

- A differentiable manifold  $M \in \text{Mani}$ .
- A differentiable function  $\mathcal{L} \in C^\infty(TM \times \mathbb{R})$  called the **Lagrangian of the system**.

If  $\mathcal{L}$  factors through the projection to  $TM$ , then it is called an **autonomous Lagrangian system**, otherwise it is called a **non-autonomous Lagrangian system**.

In particular, a Lagrangian system is a classical field theory(18.6.0.2).  $\lrcorner$

**Def. (18.1.1.12)[Generalized Positions and Momentums]**. Given a Lagrangian system  $(M, \mathcal{L})$ , any smooth chart  $\varphi_\alpha : U_\alpha \subset M \cong V_\alpha \subset \mathbb{R}^n$  defines a **generalized position coordinate**  $q_i = (\varphi_\alpha)_i$  on  $U_\alpha$ . And  $p_i \triangleq \frac{\partial}{\partial \dot{q}_i} \mathcal{L}$  are called the **generalized momentum coordinate**.  $\lrcorner$

**Def. (18.1.1.13) [Lagrangian Mechanics and the Principle of Least Action].** For any Lagrangian dynamical system (18.1.1.11) and any generalized coordinate  $\mathbf{q}$ , then a **motion** in this system is a path  $\gamma : [t_0, t_1] \rightarrow TM$  s.t.  $\gamma$  is a extremal curve (13.2.1.3) of the functional

$$\Phi(\gamma) = \int_{t_0}^{t_1} \mathcal{L}(\mathbf{q}(\gamma), \dot{\mathbf{q}}(\gamma), t) dt.$$

Equivalently, it satisfies the Euler-Lagrangian equation (13.2.1.5)

$$\frac{\partial}{\partial t} \frac{\partial}{\partial \dot{\mathbf{q}}} \mathcal{L} = \frac{\partial}{\partial \mathbf{q}} \mathcal{L}$$

along  $\gamma$ .

The study of motions satisfying this equation in a Lagrangian system is called a **Lagrangian mechanics**. ┘

**Def. (18.1.1.14) [Natural Lagrangians].** For a Lagrangian system with phase space  $M$  (18.1.1.11), a Riemannian structure on  $M$  will give a **kinetic energy**

$$T = \frac{1}{2} \langle v, v \rangle_x, \quad v \in T_x M,$$

and a **potential energy** is a function  $U \in C^\infty(M)$  that is viewed as a function on  $TM$  via projection.

Then the Lagrangian  $\mathcal{L}$  is called a **natural Lagrangian** if it is of the form  $\mathcal{L} = T - U$  for some kinetic energy  $T$  and some potential energy  $U$ . ┘

**Thm. (18.1.1.15) [Hamilton's Equations and Hamiltonian Systems].** Situation as in (18.1.1.13), we define its **Hamiltonian** to be

$$\mathcal{H}(\mathbf{p}, \mathbf{q}, t) = \mathbf{p} \cdot \dot{\mathbf{q}} - \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t).$$

and the Euler-Lagrangian equation (18.1.1.13) rewrites as

$$\dot{\mathbf{q}} = \frac{\partial}{\partial \mathbf{p}} \mathcal{H}, \quad \dot{\mathbf{p}} = -\frac{\partial}{\partial \mathbf{q}} \mathcal{H}.$$

(This is because by Legendre transform (16.8.2.1),  $(\mathbf{p}, \mathbf{q})$  is a new coordinate for  $TM$ ).

This implies that we can just start with a Hamiltonian  $\mathcal{H}$ , and consider a **Hamiltonian mechanics** where the motion in a Hamiltonian system is a path  $\gamma : I \rightarrow T^*M$  that is a extremal curve of the functional

$$S(\gamma) = \int_I (\mathbf{p} \cdot \dot{\mathbf{q}} - \mathcal{H}(\mathbf{p}, \mathbf{q}, t)) dt.$$

Then a Hamilton's system is dual to a Lagrangian system. ┘

**Cor. (18.1.1.16) [Hamiltonian Fixed in Autonomous Systems].** In a Lagrangian system, if the Lagrangian  $\mathcal{L}$  is independent of the time, then the Hamiltonian is fixed in a Hamiltonian mechanical system. ┘

*Proof:*

$$\dot{\mathcal{H}} = \frac{\partial}{\partial \mathbf{p}} \mathcal{H} \cdot \dot{\mathbf{p}} + \frac{\partial}{\partial \mathbf{q}} \mathcal{H} \cdot \dot{\mathbf{q}} = \dot{\mathbf{q}} \cdot \dot{\mathbf{p}} - \dot{\mathbf{p}} \cdot \dot{\mathbf{p}} = 0.$$

□

### Poisson Brackets

**Prop. (18.1.1.17) [Preserved Scalar Fields].** For a Hamiltonian system, if  $F(\mathbf{p}, \mathbf{q}) \in C^\infty(TM)$  doesn't depend on  $t$ , then

$$\dot{F} = \frac{\partial}{\partial \mathbf{p}} F \cdot \dot{\mathbf{p}} + \frac{\partial}{\partial \mathbf{q}} F \cdot \dot{\mathbf{p}} = \{F, \mathcal{H}\}_{\text{Pois}} \quad (18.1.1.15)(16.8.2.3).$$

In particular, if  $\{F, \mathcal{H}\} = 0$ , then  $F$  is preserved in any Hamiltonian motion.  $\lrcorner$

### Examples

**Def. (18.1.1.18) [Energy-Momentum Coordinates for  $\mathbb{R}^n$ ].** When considering particles in  $\mathbb{R}^n$ , we can use another coordinate other than the usual coordinate  $(\mathbf{x}, t)$ : Consider its Fourier transform:

$$\tilde{f}(\omega, \mathbf{k}) = \int_{\mathbb{R}^n \times \mathbb{R}} e^{i(\omega t - \mathbf{k} \cdot \mathbf{x})} f(\mathbf{x}, t) d\mathbf{x} dt.$$

where  $\omega$  is the **temporal frequency coordinate** and  $\mathbf{k}$  is the **spatial frequency coordinate**. Also  $E = \hbar\omega$  is called the **energy coordinate** and  $\mathbf{p} = \hbar\mathbf{k}$  is called the **momentum coordinate**.

Then we have the Fourier inversion:

$$f(\mathbf{x}, t) = \frac{1}{(2\pi)^{\frac{n+1}{2}}} \int_{\mathbb{R}^n \times \mathbb{R}} e^{i(\mathbf{x} \cdot \mathbf{k} - t\omega)} \tilde{f}(\omega, \mathbf{k}) d\mathbf{k} d\omega.$$

or the partial Fourier inversion:

$$f(\mathbf{x}, t) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} e^{i\mathbf{x} \cdot \mathbf{k}} \tilde{f}(t, \mathbf{k}) d\mathbf{k}.$$

$\lrcorner$

**Example (18.1.1.19) [Free Particles].** In the system of a free mass point of weight  $m$  in  $\mathbb{R}^3$ , if  $\mathbf{q}$  is the usual position coordinate, then

$$\mathcal{L} = T = \frac{m \|\dot{\mathbf{q}}\|^2}{2}.$$

Thus the generalized momenta are  $\mathbf{p} = m\dot{\mathbf{q}}$ ,  $\mathcal{H} = \frac{\|\mathbf{p}\|^2}{2m}$ , and the Hamiltonian equation is

$$\dot{\mathbf{q}} = \frac{1}{m} \mathbf{p}, \quad \dot{\mathbf{p}} = 0.$$

$\lrcorner$

**Example (18.1.1.20) [Central Potential Fields].** In the system of a free mass point in  $\mathbb{R}^2$  with a central potential field  $U = U(r)$  in the polar coordinate  $(r, \varphi)$ . Then in this coordinate  $q_1 = r, q_2 = \varphi$ ,

$$\mathcal{L}(q, \dot{q}) = T(q, \dot{q}) - U(q) = \frac{m}{2} (\dot{q}_1^2 + q_1^2 \dot{q}_2^2) - U(q_1).$$

Then the generalized momentum coordinates are

$$p_1 = m\dot{r}, \quad p_2 = mr^2\dot{\varphi}$$

and the Hamiltonian equations are

$$\dot{p}_1 = q_1 \dot{q}_2^2 - \frac{1}{m} \frac{\partial}{\partial r} U, \quad \dot{p}_2 = 0.$$

Notice the second equation is called the **law of conservation of angular momentum**. This is a special case of (18.1.1.22).  $\lrcorner$

**Example (18.1.1.21) [String Waves].** For a small perturbation of a string of mass  $m$  and length  $l$  and tension  $k$  with coordinate given by  $\varphi(x, t)$ , the kinetic energy and potential energy is given by

$$T = \frac{m}{2l} \int_0^l \frac{\partial}{\partial t} \psi(x, t) dx, \quad U = \frac{k}{2} \int_0^l \frac{\partial}{\partial x} \psi(x, t).$$

Where the potential is calculated by noticing the small distortion and Hooke's formula ?. Then the Euler-Lagrangian equation (18.1.1.13) is given by

$$\frac{\partial^2}{\partial x^2} \psi = \frac{1}{v^2} \frac{\partial^2}{\partial t^2} \psi, \quad v = \sqrt{kl/m}.$$

┘

**Prop. (18.1.1.22) [Generalized Law of Conservation of Angular Momentum].** A coordinate  $q_i$  not appearing in the Lagrangian (i.e.  $\frac{\partial}{\partial q_i} \mathcal{L} = 0$ ) is called a **cyclic coordinate**. Then it follows from ?? that the generalized momentum corresponding to a cyclic coordinate remains constant. ┘

### Symmetries

**Thm. (18.1.1.23) [Noether's Theorem].** ┘

**Thm. (18.1.1.24) [Continuity Equation].** ┘

**Prop. (18.1.1.25) [Dimensional Analysis].** In an equation raising from a physical problem, we can normalize all the indeterminants to get a non-dimensional one, giving the equation some kind of characteristic length. ┘

## 2 Symplectic Structure

### 3 Unit System

**Fact (18.1.3.1) [International System of Quantities].** The **international system of quantities** use the following 7 fundamental measurements of reality: time(T), length(L), mass(M), electric current(I), absolute temperature( $\Theta$ ), amount of substance(N), and luminous intensity (J).

We can define a unit system by solving the equation asserting the values of 7 constants (with linearly independent measure dimensions). (So basically the base units can be expressed in these units). ┘

**Def. (18.1.3.2) [Base of International System of Units(SI)].** The **international system of units** is a unit system defined by choose the base units

- (mole)mol : N.
- (second)s : T.
- (metre)m : L.
- (kilogram)kg : M.
- (ampere)A : I.
- (Kelvin)K :  $\Theta$ .
- (candela)cd: J.

s.t.

- The **Avogadro constant**:  $N_A = 6.02214076 \cdot 10^{23} \text{mol}^{-1} : \text{N}^{-1}$ .
- The **hyperfine transition frequency** of Cs:  $\Delta\nu_{Cs} = 9192631770 \text{s}^{-1} : \text{T}^{-1}$ .
- The **speed of light in vacuum**:  $c = 299792458 \text{m} \cdot \text{s}^{-1} : \text{LT}^{-1}$ .
- The **plank constant**:  $h = 6.62607015 \cdot 10^{-34} \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-1} : \text{ML}^2 \text{T}^{-1}$ .
- The **elementary particle**:  $e = 1.602176634 \cdot 10^{-19} \text{A} \cdot \text{s} : \text{IT}$ .
- The **Boltzmann constant**:  $k_B = 1.380649 \cdot 10^{-23} \text{K}^{-1} \cdot \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} : \Theta^{-1} \text{ML}^2 \text{T}^{-2}$ .
- The luminous efficacy of 540 THz radiation:  $K_{\text{cd}} = 683 \text{lm/W} : \text{JM}^{-1} \text{L}^{-2} \text{T}^3$ .

┘

**Def. (18.1.3.3) [Derived SI Units].**

- (Herz)  $\text{Hz} = \text{s}^{-1} : \text{T}^{-1}$ .
- (Newton)  $N = \text{kg} \cdot \text{m} \cdot \text{s}^{-2} : \text{MLT}^{-2}$ .
- (Pascal)  $\text{Pa} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2} : \text{ML}^{-1} \text{T}^{-2}$ .
- (joule)  $J = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} : \text{ML}^2 \text{T}^{-2}$ .
- (Watt)  $W = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} : \text{ML}^2 \text{T}^{-3}$ .
- (Coulomb)  $C = \text{A} \cdot \text{s} : \text{IT}$ .
- (Tesla)  $T = \text{A}^{-1} \cdot \text{kg} \cdot \text{s}^{-2} : \text{I}^{-1} \text{MT}^{-2}$ .
- (volt)  $V = \text{A}^{-1} \cdot \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} : \text{I}^{-1} \text{ML}^2 \text{T}^{-3}$ .
- (farad)  $F = \text{A}^2 \cdot \text{kg}^{-1} \cdot \text{m}^{-2} \cdot \text{s}^4 : \text{I}^2 \text{M}^{-1} \text{L}^{-2} \text{T}^4$ .
- (Ohm)  $\Omega = \text{A}^{-2} \cdot \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} : \text{I}^2 \text{ML}^2 \text{T}^{-3}$ .
- (lumen)  $\text{lm} = \text{cd} \cdot \text{sr} : \text{J}$ .

┘

**Fact (18.1.3.4) [Physical Constants].**

- The **reduced plank constant**:  $\hbar = h/2\pi : \text{ML}^2 \text{T}^{-1}$ .
- The **Newtonian constant of gravitation**:  $G = 6.67430(15) \cdot 10^{-11} \text{kg}^{-1} \cdot \text{m}^3 \cdot \text{s}^{-2} : \text{M}^{-1} \text{L}^3 \text{T}^{-2}$ .
- The **vacuum magnetic permeability**(or the **magnetic constant**):

$$\mu_0 = 1.25663706212(19) \cdot 10^{-6} \text{A}^{-2} \cdot \text{kg} \cdot \text{m} \cdot \text{s}^{-2} \approx 4\pi \cdot 10^{-7} \text{A}^{-2} \cdot \text{kg} \cdot \text{m} \cdot \text{s}^{-2} : \text{I}^{-2} \text{MLT}^{-2}.$$

- The **vacuum permittivity constant**:

$$\varepsilon_0 = \frac{1}{\mu_0 c^2} \approx \frac{625000}{22468879468420441\pi} \text{A}^2 \cdot \text{kg}^{-1} \cdot \text{m}^{-3} \cdot \text{s}^4 : \text{I}^2 \text{M}^{-1} \text{L}^{-3} \text{T}^4.$$

┘

**Fact (18.1.3.5) [Dimensionless Physical Constants].**

- The **fine structure constant**

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137.035999084} \approx 0.0072973525693.$$

There is no exhaustive list of such constants but it does make sense to ask about the minimal number of fundamental constants necessary to determine a given physical theory. Thus, the Standard Model requires 25 physical constants, about half of them are the masses of fundamental particles (which become dimensionless when expressed relative to the Planck mass or, alternatively, as coupling strength with the Higgs field along with the gravitational constant).

Fundamental physical constants cannot be derived and have to be measured. Developments in physics may lead to either a reduction or an extension of their number: discovery of new particles, or new relationships between physical phenomena, would introduce new constants, while the development of a more fundamental theory might allow the derivation of several constants from a more fundamental constant.

A long-sought goal of theoretical physics is to find first principles (theory of everything) from which all of the fundamental dimensionless constants can be calculated and compared to the measured values.

The large number of fundamental constants required in the Standard Model has been regarded as unsatisfactory since the theory's formulation in the 1970s. The desire for a theory that would allow the calculation of particle masses is a core motivation for the search for "Physics beyond the Standard Model".

┘

## 18.2 Gauge Theory



## 18.3 General Relativity(Gravity Fields)

### 1 Basics

**Prop.(18.3.1.1)[Maxwell's Equation].** Normal Maxwell's equation reads:

$$\begin{cases} \operatorname{div} E = q & (\text{Coulomb's law}) \\ \operatorname{div} H = 0 & (\text{Gaussian law}) \\ \operatorname{curl} E = -\frac{1}{c} \frac{\partial H}{\partial t} & (\text{Faraday's law}) \\ \operatorname{curl} H = j + \frac{1}{c} \frac{\partial E}{\partial t} & (\text{Ampère-Maxwell law}) \end{cases}$$

where  $E$  is the magnetic field,  $H$  is the electric field,  $q$  the charge density,  $j$  the electric current.

In Minkowski space, we define the electromagnetic 2-form

$$F = \frac{1}{2} F_{\alpha\beta} dx^\alpha \wedge dx^\beta,$$

where  $F_{i0} = E_i$ ,  $F_{ij} = H_k$ , and electric current  $J$ ,  $J^i = -j^i$ ,  $J^0 = q$ .

Maxwell's equation can be re-written as:

$$d^*F = J \quad dF = 0.$$

Where  $d^* = *d*$ . ┘

*Proof:* The Minkowski space is flat, the equivalence can be seen by direct calculation. □

## 18.4 Quantum Mechanics

References are [B-L14, Nap].

### 1 Basics

**Remark (18.4.1.1).** Quantum mechanics are all about **probability amplitudes** or the **propagator**. Any physical observable  $X$  in quantum mechanics can be written as an operator  $\lrcorner$

**Meta Def. (18.4.1.2) [Equivalent Quantum Pictures].** A **quantum picture** is a way to explain reality and allows calculations to making predictions about reality that can be experimentally verified.

Two quantum pictures are called **experimentally equivalent** if they give the same predictions for reality that can be experimentally verified.  $\lrcorner$

### Evolution of Quantum Systems

**Prop. (18.4.1.3).** There are several ways to put time in a quantum system:

- The Schrödinger picture, where all the time dependence is located in the operators.
- The Heisenberg picture, where the wave functions carry the time dependence.
- The Diract picture(interaction representation), where both the operators and wave functions carry the time dependence.  $\lrcorner$

**Physical Law (18.4.1.4) [Schrödinger Picture of Quantum Mechanics].** The Schrödinger equation can be derived from the Dirac-von.Neumann axioms:

The **states of particles** is a countable dimensional Hilbert space, and

- The **observables** of a quantum system are defined to be the (possibly unbounded) Hermitian operators  $A$  on  $\mathbb{H}$ . Then any continuous observable is unitarily diagonalizable, with real eigenvalues by Hilbert-Schmidt(12.9.3.16). The eigenvalues of the matrix corresponding to an observable is called the **observations**.
- The **state**  $\varphi$  of the quantum system is a unit vector of  $\mathbb{H}$ , up to scalar multiples.
- The expectation value of an observable  $A$  for a system in a state  $\varphi$  is given by the inner product  $(\varphi, A\varphi)$ .
- (Unitarity) The time evolution of a quantum state according to the Schrodinger equation is mathematically represented by a unitary operator  $U(t)$  (depends only on the state an relative time)(one-parameter subgroup).

Now that  $\varphi(t) = \hat{U}(t)\varphi(t_0)$ , so  $\hat{U}(t)\varphi(t_0) = e^{-i\hat{\mathcal{H}}t}$ , where  $\hat{\mathcal{H}}$  is Hermitian, called the **quantum Hamiltonian**.

So now take derivative w.r.t  $t$ , we get  $i\frac{\partial}{\partial t}\varphi = \hat{\mathcal{H}}\varphi$ . By **quantum correspondence principle**, it is possible to derive the expression of  $\hat{\mathcal{H}}$  by classical methods.  $\lrcorner$

**Physical Law (18.4.1.5) [Heisenberg Picture of Quantum Mechanics].** ? For any quantum observable  $\hat{\mathcal{O}}$ , the **Heisenberg equation of motion**:

$$\dot{\hat{\mathcal{O}}}(t) = -i[\hat{\mathcal{O}}, \hat{\mathcal{H}}].$$

$\lrcorner$

**Def. (18.4.1.6) [Qubits].** A **qubit** is a state that is complex combination of  $|0\rangle$  and  $|1\rangle$ , i.e.  $|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$ .  $\lrcorner$

**Def. (18.4.1.7) [Quantum Entanglement and Singlets].** Any tensor in a tensor space of two Hilbert space that is not a pure tensor is called a **quantum entanglement state**. The simplest quantum entanglement is the **singlet state**

$$|\Psi_{-}\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle \in (\mathbb{C}\{|0\rangle, |1\rangle\})^{\otimes 2} \cong \mathbb{C}^4.$$

$\lrcorner$

**Def. (18.4.1.8) [Pauli Observables].** The observables on a two dimensional Hilbert space, i.e. a qubit state space, are all combinations of **Pauli observables** plus **1**:

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Their corresponding eigenvalues are denoted by

$$\uparrow = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \downarrow = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \rightarrow = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \leftarrow = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \otimes = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} \quad \odot = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix}.$$

The Pauli differentials are special because  $\{i\sigma_x, i\sigma_y, i\sigma_z\} = \mathfrak{su}(2; \mathbb{C})$  gives the standard representation of  $SU(2)$ .  $\lrcorner$

**Remark (18.4.1.9).** The solution of a Schrodinger equation for a non Relativistic particle is assumed to be a Schwartz function (Vanish fast enough at infinity). The coefficients is assumed smooth enough to guarantee at least uniqueness and existence locally.  $\lrcorner$

**Prop. (18.4.1.10).** The wave function on the  $(p, t)$  coordinates is the Fourier Transform of the wave function on the  $(x, t)$  coordinates, because the eigenstate of the  $p$ -operator  $i\hbar \frac{\partial}{\partial x}$  is  $e^{ikx}$ , the coefficients of which is the value (probability) of the wave function of the  $(p, t)$  coordinates.  $\lrcorner$

**Prop. (18.4.1.11) [Schrödinger Uncertainty Principle].** Set  $\sigma_A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}$ , then:

$$\sigma_A^2 \sigma_B^2 \geq \left| \frac{1}{2} \langle \hat{A}\hat{B} + \hat{B}\hat{A} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle \right|^2 + \left| \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right|^2.$$

$\lrcorner$

*Proof:* Derived from definition and Schwarz inequality, Cf.[Wiki].  $\square$

**Cor. (18.4.1.12) [Heisenberg Uncertainty Principle].**  $\sigma_x \sigma_p \geq \frac{\hbar}{2}$ .  $\lrcorner$

*Proof:*

$$[x, i\hbar \frac{\partial}{\partial x}] = i\hbar.$$

$\square$

**Prop. (18.4.1.13) [Spectral Decomposition].** In Quantum physics, one need to use spectral decomposition of the Hamiltonian operator. But at most cases, there are only countably many eigenstate and the eigenvalue has a lower bound and tends to infinity. In this case,  $(\hat{H} + A)^{-1}$  is a compact operator thus by spectral theorem(12.9.3.15) the eigenstate of  $\hat{H}$  forms a set of complete basis.  $\lrcorner$

**Prop. (18.4.1.14) [No-Cloning Theorem]. ?**  $\lrcorner$

### Quantizations

**Def. (18.4.1.15) [First Quantizations].** A **first quantization** of a Hamiltonian system (18.1.1.13) is a Hilbert space  $\mathcal{H} \in \mathcal{HilbSpa}$  with an map  $f \mapsto \hat{f}$  from all classical dynamical variables to linear Hermitian operators on  $\mathcal{H}$  s.t.

- $(\alpha f + \beta g)^\wedge = \alpha \hat{f} + \beta \hat{g}$ .
- The dynamical variable 1 that is unity everywhere is mapped to the identity operator  $\mathbf{1}_{\mathcal{H}}$ .
- $[\hat{f}, \hat{g}] = i\hbar\{f, g\}^\wedge + O(\hbar^2)$ .

┘

**Remark (18.4.1.16).** ? The intuition comes from the comparison of the equations:

$$\dot{F} = \{F, \mathcal{H}\}_{\text{Pois}}, \quad \{q_j, p_k\} = \delta_{jk}$$

in classical Hamiltonian system and the equations:

$$\langle \dot{\hat{F}} \rangle = \frac{1}{i} \langle [\hat{F}, \hat{\mathcal{H}}] \rangle, \quad \{\hat{q}_j, \hat{p}_k\} = i\delta_{jk}.$$

┘

**Remark (18.4.1.17) [Canonical Quantizations].** Canonical quantizations of a Hamiltonian system (18.1.1.15) work as follows:

- Write down the classical Lagrangian density in terms of fields.
- Calculate the momentum density and work out the Hamiltonian density in terms of fields.
- Treat the fields and momentum density as operators. Impose commutation relations on them to make them quantum mechanical.
- Expand the fields in terms of creation/annihilation operators. This will allow us to use occupation numbers and stay sane.

?

┘

**Example (18.4.1.18).** We can quantize the energy functor  $E$  as  $\hat{E} = i\hbar \frac{\partial}{\partial t}$ . Then the Hamilton's equation quantize to the Schrödinger equation

$$i\hbar |\dot{\psi}(t)\rangle = \hat{H} |\psi(t)\rangle.$$

And we have

$$[\hat{q}^i, \hat{q}^j] = 0, \quad [\hat{p}_i, \hat{p}_j] = 0, \quad [\hat{q}^i, \hat{p}_j] = i\hbar \delta_{i,j}.$$

$$(\dot{p}_i)^\wedge = -\frac{\partial}{\partial \hat{q}^i} \hat{H}, \quad (\dot{q}^i)^\wedge = -\frac{\partial}{\partial \hat{p}_i} \hat{H}.$$

┘

**Example (18.4.1.19) [State-Space for Non-Relativistic 3-dimensional Quantum Particles].**

Firstly, the Hilbert space  $\mathcal{H}_x$  for a single 1-dimensional quantum particle can be represented by  $L^2$ -functions on  $\mathbb{R}$ :  $|\psi_x\rangle \in \mathcal{H}_x$  is of the form

$$|\psi\rangle = \int_{\mathbb{R}} \psi(x) |x\rangle dx, \quad \int_{\mathbb{R}} |\psi(x)|^2 dx < \infty,$$

where  $\psi(x) = \langle x|\psi\rangle$ . And the position and momentum Hermitian operators  $\widehat{x}, \widehat{p}_x$  acts on  $|\psi\rangle \in \mathcal{H}_x$  via the integral representation

$$\widehat{x}|\psi\rangle = \int_{\mathbb{R}} \widehat{x}\psi(x)|x\rangle dx, \quad \widehat{p}|\psi\rangle = \int_{\mathbb{R}} -i\hbar \frac{\partial}{\partial x}\psi(x)|x\rangle dx.$$

And the Hilbert space  $\mathcal{H}_1$  for a single 3-dimensional quantum particle can be represented by

$$\mathcal{H}_1 = \mathcal{H}_x \otimes \mathcal{H}_y \otimes \mathcal{H}_z.$$

with position and momentum Hermitian operators  $\widehat{\mathbf{r}} = \widehat{x} \otimes \widehat{y} \otimes \widehat{z}$ ,  $\widehat{\mathbf{p}} = \widehat{p}_x \otimes \widehat{p}_y \otimes \widehat{p}_z$ .

Similarly, for  $n \in \mathbb{Z}_+$ , the Hilbert space  $\mathcal{H}_1$  for a single 3-dimensional quantum particles can be defined to be  $\mathcal{H}_n = (\mathcal{H}_1)^{\otimes n}$ .  $\lrcorner$

### Second Quantizations: Particle-Like Waves

**Def. (18.4.1.20) [Phonons].** Particle-like waves modes are called **phonons**.  $\lrcorner$

**Def. (18.4.1.21) [Quantum Harmonic Oscillators].** The **quantum harmonic oscillator** is a quantization of the usual 1-dimensional oscillator ?: A usual oscillator with mass  $m$  and spring constant  $K$  is give by

$$\frac{1}{2m}p^2 + \frac{1}{2}Kx^2 = E.$$

in the  $x$ -coordinate where  $E$  is the fixed energy. Then the quantization is given by the Schrödinger equation

$$\left(-\frac{1}{2m}\widehat{p}^2 + \frac{1}{2}Kx^2\right)\psi = \left(-\frac{1}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}Kx^2\right)\psi = E\psi$$

of the waveform  $\psi$ .

If we denote  $K = m\omega^2$ , then the quantum Hamiltonian is given by  $\widehat{H} = \frac{1}{2m}\widehat{p}^2 + \frac{1}{2}m\omega^2\widehat{x}^2$ .

If we define the annihilation and creation operators:

$$\widehat{a} = \sqrt{\frac{m\omega}{2}}\left(\widehat{x} + \frac{i}{m\omega}\widehat{p}\right), \quad \widehat{a}^* = \sqrt{\frac{m\omega}{2}}\left(\widehat{x} - \frac{i}{m\omega}\widehat{p}\right),$$

then

$$[\widehat{a}, \widehat{a}^*] = 1, \quad \widehat{H} = \omega\left(\widehat{a}\widehat{a}^* + \frac{1}{2}\right).$$

Then by [B-L14]P23, there are unique eigenstate  $E_n = |n\rangle$  with eigenvalue  $n + 1/2$  for any  $n \in \mathbb{N}$ , and

$$|n\rangle = \frac{(\widehat{a}^*)^n}{n!} |0\rangle.$$

$\lrcorner$

**Example (18.4.1.22) [Phonons in Coupled Quantum Harmonic Oscillators].** Consider a linear chain of  $N$  atoms each of mass  $m$  connected by springs of unstretched length  $a$  and which have spring constant  $K$ . The Hamiltonian for this system is

$$\widehat{H} = \frac{1}{2m} \sum_j \widehat{p}_j^2 + \sum_j \frac{K}{2} (\widehat{x}_{j+1} - \widehat{x}_j)^2.$$

If we take discrete Fourier transform

$$\hat{x}'_k = \frac{1}{\sqrt{N}} \sum_j x_j e^{-2\pi i k j / N}, \quad \hat{p}'_k = \frac{1}{\sqrt{N}} \sum_j p_j e^{-2\pi i k j / N},$$

then

$$\hat{p}'_k = \hat{p}'_{-k}, \quad \hat{x}'_k = \hat{x}'_{-k}, \quad [\hat{x}'_k, \hat{p}'_{k'}] = i\delta_{k, -k'},$$

and

$$\hat{H} = \sum_k \left[ \frac{1}{2m} \hat{p}'_k \hat{p}'_{-k} + \frac{m\omega_k^2}{2} \hat{x}'_k \hat{x}'_{-k} \right], \quad \omega_k = \frac{4K}{m} \sin^2\left(\frac{2\pi k}{2N}\right).$$

Define the creation and annihilation operators as

$$\hat{a}_k = \frac{\sqrt{m\omega_k}}{2} \left( \hat{x}'_k + \frac{i}{m\omega_k} \hat{p}'_k \right), \quad \hat{a}_k^* = \frac{\sqrt{m\omega_k}}{2} \left( \hat{x}'_{-k} - \frac{i}{m\omega_k} \hat{p}'_{-k} \right),$$

then

$$[\hat{a}_k, \hat{a}_{k'}] = 0, \quad [\hat{a}_k^*, \hat{a}_{k'}^*] = 0, \quad [\hat{a}_k, \hat{a}_{k'}^*] = \delta_{k, k'},$$

and

$$\hat{H} = \sum_k \frac{\omega_k}{2} (\hat{a}_k \hat{a}_k^* + \hat{a}_{-k} \hat{a}_{-k}^*) = \sum_{k=1}^N \omega_k \left( \hat{a}_k \hat{a}_k^* + \frac{1}{2} \right).$$

So in this Fourier coordinate, the motion is just like  $N$  uncoupled quantum harmonic oscillators (18.4.1.21). Thus the phonons in this case are just these  $\hat{x}'_k$ .  $\lrcorner$

### Calculations

**Prop. (18.4.1.23)[Virial Theorem].** For a system that  $V(r) \sim r^n$ , the average kinetic energy and the average potential energy has the relation :

$$2 \langle T \rangle = n \langle V \rangle.$$

$\lrcorner$

*Proof:* ?

$\square$

### Spin

## 2 Relativistic Quantum Mechanics

### Klein-Gordon Equation

## 3 Quantum Computations

### Classical Logic Gates

**Def. (18.4.3.1)[Boolean Functions].** A **Boolean function** is a function  $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ .  $\lrcorner$

**Def. (18.4.3.2)[Classical Logic Gates].** There are four classical logic gates, if we let  $0, 1 \in \mathbb{F}_2$ , then:

- **AND gate:**  $(a, b) \mapsto ab$ .
- **OR gate:**  $(a, b) \mapsto ab + a + b$ .

- **NOT gate:**  $a \mapsto a + 1$ .
- **COPY gate:**  $a \mapsto (a, a)$ .

┘

**Def. (18.4.3.3) [Reversible Gates].** A gate is called **reversible** iff it is a bijection from  $\mathbb{F}_2^n$  to  $\mathbb{F}_2^n$ . ┘

**Def. (18.4.3.4) [Simulations].** A set of gates is said to be able to **simulate** a boolean function  $f$  iff there is a composition of these gates that maps

$$(x_1, \dots, x_{m+n}) \rightarrow (g_1(x_1, \dots, x_{m+n}), \dots, g_k(x_1, \dots, x_{m+n}))$$

such that if we let  $x_{i_1} = a_1, x_{i_2} = a_2, \dots, x_{i_m} = a_m$  be fixed, where  $\{0, 1, \dots, m+n\} - \{i_1, \dots, i_m\} = \{j_1, \dots, j_n\}$  (in some order), then  $g_1(x_1, \dots, x_{m+n}) = f(x_{j_1}, \dots, x_{j_n})$ .

A set of gates is called **universal** iff they can simulate all Boolean function  $f : \mathbb{F}_2^n \mapsto \mathbb{F}_2$ . ┘

**Prop. (18.4.3.5) [Classical Gates are Universal].** The four classical gates are universal. In fact,  $\text{And}(x, y) = \text{OR}(\text{NOT}(x), \text{NOT}(y))$ , so even AND is disposable, and COPY is not used as well. ┘

*Proof:* Just use OR gate to juxtapose all possible combinations that are mapped to 1. □

**Example (18.4.3.6) [Reversible Gates].**

- The **CNOT gate** is defined to be  $\text{CNOT} : (a, b) \mapsto (a, b + a)$ .
- The **Toffoli gate** is defined to be  $\text{CCNOT} : (a, b, c) \mapsto (a, b, c + ab)$ .
- The **Fredkin gate** or CSWAP gate is a three-bit gate defined as the quantization of the gate given by:  $(a, b, c) \mapsto (a, (1 - a)c + ab, (1 - a)b + ac)$ .

┘

**Prop. (18.4.3.7).** CNOT gate cannot simulate AND. In particular, CNOT is not universal. ┘

*Proof:* It can be shown that any Boolean function that can be simulated by CNOT gate is of the form  $(x_1, \dots, x_n) \mapsto \sum a_i x_i + b$ . But AND is of the form  $(a, b) \mapsto ab$ , which is not of the form, so it is not simulated by CNOT. □

**Prop. (18.4.3.8).** The Toffoli gate (18.4.3.6) is universal (18.4.3.4). ┘

*Proof:* By (18.4.3.5), it suffices to show that it can simulate AND and NOT, then it can simulate OR because  $\text{OR}(x, y) = \text{NOT}(\text{NOT}(x), \text{NOT}(y))$ .

AND is outputted in the third bit with  $c = 0, a = x, b = y$ , NOT is outputted in the third bit with  $a = 1 = c, b = x$ . □

**Prop. (18.4.3.9).** The Fredkin gate (18.4.3.6) is universal (18.4.3.4). ┘

*Proof:* If we set the third input to be 0, then the second output of  $\text{CSWAP}(x, y, 0)$  is  $\text{And}(x, y)$ , and if we set the second input to be 0 and the third input to be 1, then the second output of  $\text{CSWAP}(x, 0, 1)$  is  $\text{NOT}(x)$ . Thus the Fredkin gate can simulate both AND and NOT gates, thus universal, by (18.4.3.5). □

### Quantum Logic Gates

Our model of quantum computation is quantum logic gates.

**Def. (18.4.3.10) [Quantum Logic Gates].** A **quantum logic gate** is a unitary matrix. In particular a quantum logic gate is always reversible.  $\lrcorner$

**Prop. (18.4.3.11) [Examples of Quantum Gates].**

- The **Hadamard gate**  $H$  is a rotation on one single qubit given by the matrix  $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$ .
- If a classical gate is reversible and its matrix is unitary, then the same matrix will give out a quantum gate with all entries 0 and 1, called the **quantization of classical gates**.  $\lrcorner$

## 4 Quantum Algorithms

**Algo. (18.4.4.1) [Deutsch-Jozsa Algorithm].** The **Deutsch-Jozsa problem** is that: given a function  $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ , which is either a constant function or a function that takes half value 0 and half value 1,

Consider the gate  $U_f : (x_1, \dots, x_n, x) \mapsto (x_1, \dots, x_n, x + f(x_1, \dots, x_n))$ , then there is a **Deutsch-Jozsa algorithm** that can determine if  $f$  is a constant function just using the gate  $U_f$  once.  $\lrcorner$

*Proof:* Make the state

$$H^{\otimes n+1} \circ U_f \circ H^{\otimes(n+1)}(0, 0, \dots, 0, 1) = \left( \frac{1}{2^n} \sum_{(a_1, \dots, a_n) \in \mathbb{F}_2^n} \left( \sum_{(x_1, \dots, x_n) \in \mathbb{F}_2^n} (-1)^{f(a_1 \dots a_n) + \sum_i a_i x_i} \right) |a_1 \dots a_n\rangle \right) \otimes |0\rangle,$$

and then measure all the first  $n$  bits in the  $|0\rangle/|1\rangle$ -basis. Then the result is  $(0, 0, \dots, 0)$  iff  $f$  is constant.  $\square$

### Making Classical Algorithms Reversible

Because quantum computers can only deal with reversible algorithms, we need to spend either time or space to make classical algorithms revertible. This is because by von-Neumann-Landauer limit, it takes energy to erase a bit of information.

For example, The modular exponentiation algorithm(2.10.2.1) can be made revertible as in [Sho97].

### Quantum Fourier Transform

**Def. (18.4.4.2) [Quantum Fourier Transform].** The (inverse)quantum Fourier transform is an algorithm represented by the matrix

$$U_{\text{QFT}} = \frac{1}{\sqrt{N}} \text{Van}(1, \zeta_N, \zeta_N^2, \dots, \zeta_N^{N-1}).$$

*Proof:* It suffices to prove this matrix is unitary, which is easy.  $\square$



**Lemma (18.4.4.3) [Tensor Representation].** The trick of the quantum Fourier transform lies in its connection with the 2-adic decimal representation:

$$\begin{aligned} U_{\text{QFT}}(|x_n x_{n-1} \dots x_1\rangle) &= \frac{1}{\sqrt{N}} (|0\rangle + \exp(2\pi i \cdot 0.x_1)|1\rangle \\ &\quad \otimes (|0\rangle + \exp(2\pi i \cdot 0.x_2 x_1)|1\rangle \\ &\quad \otimes \dots \\ &\quad \otimes (|0\rangle + \exp(2\pi i \cdot 0.x_n x_{n-1} \dots x_1)|1\rangle) \end{aligned}$$

┘

*Proof:* Direct calculation. ?

□

**Algo. (18.4.4.4) [Quantum Fourier Transform Algorithm, Shor].** Because of the tensor representation of quantum Fourier transform, it is anticipated that the transformation can be represented using Hadamard gate and phase change gates. The circuit given in [https://en.wikipedia.org/wiki/Quantum\\_Fourier\\_transform](https://en.wikipedia.org/wiki/Quantum_Fourier_transform) shows that for  $n \in \mathbb{Z}_+$ ,  $U_{\text{QFT}}$  on  $N = 2^n$  can be represented by a circuit with  $n(n+1)/2$  gates up to a swap. And there are  $O(n^2)$  qbit swaps to be performed, which shows that the quantum Fourier transform has complexity  $O(n^2)$ . ┘

### Shor's Algorithms

References are [Sho94], [Sho97]

**Def. (18.4.4.5).** For a number  $M = pq$ , where  $p, q$  are different odd prime numbers,  $x \in (\mathbb{Z}/(M))^*$  is called **good** iff  $r = \text{ord}(x)$  is even, and neither of  $x^{r/2} \pm 1$  is divisible by  $M$ .

Then at least half of  $(\mathbb{Z}/(M))^*$  is good. ┘

*Proof:* It suffices to consider a fixed order  $2a$ , and this is additive in  $\mathbb{Z}/(2a, p-1)\mathbb{Z} \times \mathbb{Z}/(2a, q-1)\mathbb{Z}$ . ?

□

**Algo. (18.4.4.6) [Shor's Algorithm for Computing Orders].** If we find a good  $x$  for  $M$ , then  $x^{r/2} \pm 1$  contains separately a prime  $p$  or  $q$ , so we can use Euclidean algorithm to extract a prime of  $M$ . This is just the idea of Shor's algorithm (18.4.4.7). ┘

**Algo. (18.4.4.7) [Shor's Prime Factorization Algorithm].** For  $M = pq$ , we can factor  $p, q$  out with high probability in  $O((\log M)^2)$  time. ┘

*Proof:* Cf. [Napkin P274].

□

**Algo. (18.4.4.8) [Shor's Discrete Logarithm Algorithm].** ┘

### Grover's Algorithm

**Prop. (18.4.4.9).** ┘

**Prop. (18.4.4.10) [Grover's Selection Algorithm].** If there are  $n$  items labeled  $\{0, \dots, n-1\}$ , and there is a marked item  $w$ . then there is a quantum algorithm that find  $w$  in  $O(\sqrt{n})$  times. ┘

*Proof:* Cf. [Quantum Algorithm MIT P33,35].

□

## 5 Quantum Informations

### Quantum Cryptography

x1dddd

## 18.5 Statistical Mechanics

### 1 Basics

**Def. (18.5.1.1) [Statistical Ensembles, Gibbs].** A **statistical ensemble** is a probability distribution over the microstates of particles used in statistical mechanics to describe a single system. It formalizes the notion that an experimenter repeating an experiment again and again under the same macroscopic conditions.  $\lrcorner$

**Def. (18.5.1.2) [Statistical Equilibriums].** A **statistical equilibrium** is a system that doesn't change over time.

A **thermodynamic ensemble** is a statistical ensemble (18.5.1.1) that each microstate of particles is in statistical equilibrium.  $\lrcorner$

**Def. (18.5.1.3) [Gibbs Distribution].** In a system with Hamiltonian (classical or quantum)  $\mathcal{H}_0$  that has eigenstates  $|\lambda\rangle$  with energy eigenvalues  $E_\lambda$  for  $\lambda \in M$ . Suppose this system is parametrized by a parameter  $\beta$  (usually  $\beta = \frac{1}{k_B T}$ .)

Then the **Gibbs distribution** is a function on  $M$  whose value at  $|\lambda\rangle$  is given by

$$p_\lambda = \frac{e^{-\beta E_\lambda}}{Z}, \quad Z(\beta) = \int_M e^{-\beta E_\lambda} = \text{tr}(e^{-\beta \mathcal{H}_0}).$$

This  $Z(\beta)$  is called the **partition function** of this system. And  $\rho = e^{-\beta \mathcal{H}_0}/Z$  is called the **probability density** (operator).  $\lrcorner$

**Prop. (18.5.1.4).** The Hammersley-Clifford theorem implies that any probability measure that satisfies a Markov property is a Gibbs measure for an appropriate choice of (locally defined) energy function. Therefore, the Gibbs measure applies to widespread problems outside of physics, such as Hopfield networks, Markov networks, Markov logic networks, and boundedly rational potential games in game theory and economics. A Gibbs measure in a system with local (finite-range) interactions maximizes the entropy density for a given expected energy density; or, equivalently, it minimizes the free energy density.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (18.5.1.5) [Principle of Maximum Entropy].** Gibbs measure serves to maximize the entropy of a system, subject to a set of constraints.  $\lrcorner$

*Proof:* ?  $\square$

**Def. (18.5.1.6) [Generating Functions].** For any  $n$  observables (operators)  $\varphi_i$ , we want to calculate their correlations functions on a statistical system  $G_{ij} = \mathbf{E}(\varphi_i \varphi_j)$ . This can be done by writing  $\mathcal{H} = \mathcal{H}_0 - \frac{1}{\beta} \sum_k J_k \varphi_k$ . and consider

$$Z(\mathbf{J}) = \text{tr}(e^{-\beta \mathcal{H}_0 + \sum_k J_k \varphi_k}).$$

In this way,

$$G_{ij} = \frac{1}{Z(\mathbf{0})} \frac{\partial^2}{\partial J_i \partial J_j} Z(\mathbf{J}).$$

And similarly for higher correlation functions.  $\lrcorner$

## 2 Evolution of Statistical Systems

### Liouville's Equation

## 3 Ising Models and Potts Models

Cf.[The Knot Book, Adams].

**Prop.(18.5.3.1) [Potts Models and Tutte Polynomials].** Let  $G$  be a planer graph and  $q \in \mathbb{Z}_+$ , then the Potts model on  $G$  with  $q$ -state and interaction energy  $E(S_i, S_j) = \delta_{i,j}$  has partition function

$$Z = T_G(q, e^{-\frac{1}{kT}} - 1),$$

where  $T_G(q, t)$  is the Tutte polynomial of  $G$ . ┘

*Proof:* Cf.[Adams, P240]. ? □

**Prop.(18.5.3.2) [Potts Models and Jones Polynomials].** Any knot diagram can be turned into a signed planer graph  $G$ . Then for  $q \in \mathbb{Z}_+$ , there is aa Potts model on  $G$  with  $q$ -state and suitable choice of signed interaction energies  $E_{\pm}(S_i, S_j)$  s.t. the partition function is just

$$Z = V_G(t)$$

where  $V_G(t)$  is the Jones polynomial, and  $t \in \mathbb{C}$  satisfies  $q = 2 + t + t^{-1}$ . ┘

*Proof:* Cf.[Adams, P213]. ? □

## 4 Statistical Quantum Mechanics

## 5 Thermodynamical Systems

## 18.6 Field Theory

Field theory are continuum limits of Hamiltonian mechanics 18.1, which are discrete systems.

**Notation (18.6.0.1).**

- Use notations on smooth manifolds defined in [Smooth Manifolds I: Generalities](#).
- Choose the Heaviside-Lorentz system of units s.t.  $c = \hbar = \mu_0 = 1$  (18.1.3.4)(18.1.3.2), which can be done because their measure dimensions are independent. And then  $\varepsilon_0 = \frac{1}{\mu_0 c^2} = 1$ .
- We use the notation  $\dot{X} = \frac{\partial}{\partial t} X$  for any function  $X$  depending on the time  $t$ .

┘

**Def. (18.6.0.2) [Field Theories].** A **field theory** consists of the following:

- A differentiable manifold  $M \in \text{Diff}$  called the **space-time**.
- A differential vector bundle  $E/M$  called the **physical field**.
- A **Lagrangian**  $\mathcal{L}$  which is a differential functional on  $C^\infty(M, E)$  (13.2.1.2).

Moreover, it is called a **field theory of order**  $\leq k$  if  $\mathcal{L} = \mathcal{L}(\{\partial_I \varphi\}_{|I| \leq k})$  is smooth function on the  $k$ -th jet space of  $M$  valued in  $E$ .

A field theory of order  $\leq 1$  is also called a **classical field theory**. A field theory of order  $\leq 2$  is called a **semi-classical field theory**.

┘

### 1 Classical Field Theory

**Example (18.6.1.1) [Gravity Fields, Kepler/Newton].**

┘

**Example (18.6.1.2) [Electromagnetic Fields, Faraday/Maxwell].** The Maxwell equations are

$$\begin{cases} \nabla \cdot \mathbf{E} = \rho \\ \nabla \times \mathbf{E} = -\dot{\mathbf{B}} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{B} = \mathbf{J} + \dot{\mathbf{E}} \end{cases}$$

┘

**Example (18.6.1.3) [Velocity Fields, Euler].**

┘

### 2 Semi-Classical Field Theory

TBA

**Prop. (18.6.2.1).** Yang-Mills Fields.

┘

### 3 Perturbative Field Theory

**Remark (18.6.3.1) [Path Integrals].**

┘

## 4 Conformal Field Theory

## 5 Symmetry Breaking

**Remark (18.6.5.1).** When symmetry breaks, there are four features one should look out for:

**Phase transition:** The transition separates two distinct states of different symmetry. The low-temperature phase has lost some symmetry, more precisely it is missing a symmetry element.

**New excitations:** New particles known as Goldstone modes can emerge upon symmetry breaking.

**Rigidity:** Any attempt to deform the field in the broken symmetry state results in new forces emerging. Examples of rigidity include phase stiffness in superconductors, spin stiffness in magnets and the mechanical strength of crystalline solids.

**Defects:**

└

**Example (18.6.5.2) [Landau's Theory of Magnets].**

└

## 6 Statistical Field Theory

### Generating Functionals

One of the marvelous things about field theory is that even when we can't solve a theory exactly, the generating functional  $Z[J]$  and the Green's functions  $G(n)(x_1, \dots, x_n)$  can be expressed in terms of Feynman diagrams.

## 7 Higgs Fields

## 8 Dualities

**Def. (18.6.8.1).** Two field theory are called **equivalent field theories** if expectation values in both theories equal. In particular, the partition function should equal. ?

└

## 18.7 Quantum Field Theory

*“The mathematical problems that have been solved or techniques that have arisen out of physics in the past have been the lifeblood of mathematics.”*

–*M. Atiyah*

References are [Witten, Super-symmetry and Morse Theory]. [Witten, Topological Quantum Field Theory, 1988]. [Witten, Elliptic Genera and Quantum Field Theory], [Quantum Field Theory and the Jones Polynomial, Witten], [Supersymmetry and Morse Theory, Witten], [Etingof’s notes], [Skinner’s notes, <https://www.damtp.cam.ac.uk/user/dbs26/AQFT.html>]. [Undergraduate Lecture Notes in Topological Quantum Field Theory].

**Def. (18.7.0.1) [Quantum Field Theory].** A quantum field theory consists of the following:

- Two Riemannian manifold  $(M, g), (N, g_1) \in \text{RieMani}_\partial$ .
- A space  $\mathcal{C}$  of maps from  $M$  to  $N$ , called the **configuration space** on  $M$ . Each  $\varphi \in \mathcal{C}$  is called a **quantum field**.

┘

### Photons

#### 1 Path Integrals(Feynmann1942)

**Def. (18.7.1.1) [Actions of Quantum Field Theory].** An action of a quantum field theory(18.7.0.1) is a map  $S : \mathcal{C} \rightarrow \mathbb{R}$ . The critical points of  $S$  corresponds to classical Euler-Lagrangian equations. Usually  $S$  is assumed to be local, i.e. can be written as functions of  $\varphi$  and its derivatives.

The main tool to study QFT is the path integral

$$\int_{\mathcal{C}} g(\varphi) e^{-S(\varphi)/\hbar} d\varphi.$$

If  $g = \mathbf{1}$ , this is called the path integral  $Z(M, g, S)$ , and if  $\mathcal{O}_i$  are some operators(functions on  $\mathcal{C}$ ), the integral

$$\left\langle \prod_i \mathcal{O}_i(\varphi) \right\rangle = \int_{\mathcal{C}} \prod_i \mathcal{O}_i(\varphi) e^{-S(\varphi)/\hbar} d\varphi$$

is called the **correlation function** between the  $\mathcal{O}_i$ s.

For example, analogous as in [Brownian Motion](#) or [Generating Functions](#), the motion  $q = q(t)$  of a quantum particle in  $\mathbb{R}^N$  (1-dimensional QFT) is determined by the correlation functions

$$\text{Cor}(q_{j_1}(t_1), \dots, q_{j_n}(t_n)) = \int_{P_{a,b}} q_{j_1}(t_1) \cdot \dots \cdot q_{j_n}(t_n) e^{-S(q)/\hbar} Dq, \quad S(q) = \int_a^b \left( \frac{1}{2} \dot{q}^2 - U(q) \right) dt.$$

┘

**Def. (18.7.1.2) [Path Integral on Manifolds with Boundaries].** When  $\partial M \neq \emptyset$ , the possible fields on each boundary component  $B_i$  form a Hilbert space  $\mathcal{H}_i$ . Thus path integral defines a map

$$\otimes \mathcal{H}_i \rightarrow \mathbb{C} : \otimes \psi_i \mapsto \int_{\varphi, \varphi|_{B_i} = \psi_i} e^{-S(\varphi)/\hbar} d\varphi.$$

In particular if  $M = N \times \mathbb{I}$ , then by duality it defines a map

$$U : \mathcal{H} \rightarrow \mathcal{H} : (U\psi_0, \psi_1) = \int_{\varphi, \varphi|_{N \times \{0\}} = \psi_0, \varphi|_{N \times \{1\}} = \psi_1} e^{-S(\varphi)/\hbar} d\varphi.$$

┘

**Def. (18.7.1.3) [Feynman's Action-Amplitudes Formalism].** Feynman's amplitudes is a space-time version of the Schrödinger picture of quantum mechanics(18.4.1.4). ┘

### Instantons and Tunnelling

**Def. (18.7.1.4) [Instantons].** Instantons are constant energy solutions to quantum mechanical equations of motion with a characteristic, stable structure localized in time associated with critical points in the action. ┘

## 2 Perturbative QFT

### 0-dimensional QFT, Feynman Calculus

**Notation (18.7.2.1).**

- Consider the 0-dimensional quantum field theory for  $M = \text{pt}$  and  $N = \mathbb{R}^n$ .

┘

**Remark (18.7.2.2) [Steepest Descent Approximations].** By(12.11.3.30) with  $f(x) = S(x)$ , we get a smooth function  $I(x) \in C^\infty([0, \infty))$  s.t.

$$I(\hbar) = \hbar^{-\frac{d}{2}} e^{\frac{S(c)}{\hbar}} \int_D g(x) e^{-\frac{S(x)}{\hbar}} dx,$$

and by(12.11.3.27), the expansion of  $I(\hbar)$  at  $\hbar = 0$  only depends on derivatives of  $g$  at  $c$ . Thus to understand the expansion of  $I(\hbar)$ , it suffices to understand the **correlation functions**:

$$\langle l_1, \dots, l_N \rangle = \hbar^{-\frac{d}{2}} \int_D l_1(x) \dots l_N(x) e^{-\frac{S(x)}{\hbar}} dx$$

for any  $N \in \mathbb{N}$  and  $l_1, \dots, l_N \in V^*$ . When  $N = 0$ , this is just the partition function(18.5.1.3).

Notice in view of the expansion of  $\hbar$ , we may replace  $D$  with  $V$ (by reason as in the proof of(12.11.3.30)), and WLOG, we may assume that  $c = 0$ , and the Taylor expansion of  $S(x)$  at  $x = 0$  is

$$S(x) = S(0) + \frac{B(x, x)}{2} - \sum_{i \geq 3} \frac{B_i(x, \dots, x)}{i!}, B_i = \partial^i f(0),$$

where  $B^{-1}$  is called the **propagator**, and  $B_i/i!$  are called the **interaction terms**. ┘

**Def. (18.7.2.3) [Feynman Graphs].** A **Feynman graph** is a finite multi-graph s.t. there are no vertices of valence 2. For  $N \in \mathbb{N}$ , let  $\text{Feyn}_{\geq 3}(N)$  be the isomorphism classes of Feynman graphs with  $N$  fixed vertices of valence 1 with a labeling of them.

And if  $\underline{n} = (n_0, \dots, n_k, \dots)$ , let  $\text{Feyn}(\underline{n})$  be the isomorphism classes of finite multi-graphs s.t. for any  $k \in \mathbb{N}$ , there are exactly  $n_k$ -many vertices of valence  $k$ . ┘

**Def. (18.7.2.4) [Feynman Amplitudes].** Situation as in (12.11.3.25), given

- $B_k \in \text{Sym}^k(V^*)$  for each  $k \in \mathbb{Z}_{\geq 3}$ ,
- $N \in \mathbb{N}$ , and  $l_1, \dots, l_N \in V^*$ ,
- a Feynman graph  $\Gamma \in \text{Feyn}(N)$ ,

let the **Feynman amplitude**  $\text{Feyn}_\Gamma(l_1, \dots, l_N)$  to be defined by the following procedure:

- For  $i \in [N]_+$ , put  $l_i$  on the  $i$ -th labeled vertex of  $\Gamma$  of valence 1,
- For  $k \in \mathbb{Z}_+$ , put  $B_k$  on each unlabeled vertex of  $\Gamma$  of valence  $k$ .
- Put  $B^{-1}$  on each edge of  $\Gamma$ .
- Tensor the tensors on the vertices together, and contract them using the  $B^{-1}$  on the edges. The resulting tensor is a real number.

┘

**Lemma (18.7.2.5) [Wick's Theorem].** Situation as in (12.11.3.25), if  $N \in \mathbb{Z}_+$  and  $l_1, \dots, l_N \in V^*$ , then

$$\int_V l_1(x) \dots l_N(x) e^{-\frac{B(x,x)}{2}} dx = \begin{cases} \frac{(2\pi)^{d/2}}{\det B} \sum_{\sigma \in \Pi_{N/2}} \prod_{i \in \{1, \dots, n\}/\sigma} B^{-1}(l_i, l_{\sigma(i)}) & 2|N \\ 0 & 2 \nmid N \end{cases},$$

where  $\Pi_{N/2}$  is the set of matching on  $[N/2]_+$ .

┘

*Proof:* If  $2 \nmid N$ , this is obvious because the integrand is an odd function. Suppose  $n = 2k$ , then both sides are symmetric and multilinear in  $(l_1, \dots, l_N)$ , so it suffices to show for  $l_1 = l_2 = \dots = l_n = l$ . And by change of variable, we may assume that there is a basis  $\{x_1, \dots, x_n\}$  of  $V$  s.t.  $l(x) = x_1$ , and  $B(x_1, \dots, x_n) = x_1^2 + \dots + x_n^2$ . Then by integrating out  $x_2, \dots, x_n$  using (12.11.3.27), it suffices to show that

$$\int_{-\infty}^{\infty} x^{2k} e^{-x^2/2} dx = \sqrt{2\pi} (2k-1)!!,$$

which is true by integration by part and ??.

□

**Thm. (18.7.2.6) [Feynman Calculus, Variational Form].** Suppose

$$S(x) = \frac{B(x,x)}{2} + \tilde{S}(x),$$

where  $\tilde{S}(x) = -\sum_{i \in \mathbb{N}} g_i B_i(x, \dots, x)/i!$ ,  $g_i$  being formal parameters ( $g_0$  is called the **vacuum energy**,  $g_1$  is called the **tadpole**,  $g_2$  is called the **mass**), and consider the partition function

$$Z = \hbar^{-\frac{d}{2}} \int_V e^{-S(x)/\hbar}.$$

Then

$$Z_S = \frac{(2\pi)^{\frac{d}{2}}}{\sqrt{\det B}} \sum_{\underline{n} \in (\mathbb{N})^{\mathbb{N}}} \left( \prod_i g_i^{n_i} \right) \sum_{\Gamma \in \text{Feyn}(\underline{n})} \frac{\hbar^{b(\Gamma)}}{\# \text{Aut}(\Gamma)} \text{Feyn}_\Gamma,$$

where  $b(\Gamma) = \#E(\Gamma) + N - \#V(\Gamma)$ .

And by decomposing  $\Gamma$  into connected components, there is also a logarithmic form:

$$\log \frac{Z_S}{Z_0} = \sum_{\underline{n} \in (\mathbb{N})^{\mathbb{N}}} \left( \prod_i g_i^{n_i} \right) \sum_{\Gamma \in \text{Feyn}_{\text{cnd}}(\underline{n})} \frac{\hbar^{b(\Gamma)}}{\# \text{Aut}(\Gamma)} \text{Feyn}_\Gamma, \quad Z_0 = \frac{(2\pi)^{\frac{d}{2}}}{\sqrt{\det B}}.$$

This is called the **loop expansion**, because for  $\Gamma$  connected,  $b(\Gamma) + 1$  is the number of loops in  $\Gamma$ . ┘



**Cor.(18.7.2.7) [Feynman Calculus, Taylor Expansion Form].** Situation as in (18.7.2.2), for any  $N \in \mathbb{N}$  and  $l_1, \dots, l_N \in V^*$ , we have

$$\langle l_1, \dots, l_N \rangle = \frac{(2\pi)^{\frac{d}{2}}}{\sqrt{\det B}} \sum_{\Gamma \in \text{Feyn}(N)} \frac{\hbar^{b(\Gamma)}}{\# \text{Aut}(\Gamma)} \text{Feyn}_{\Gamma}(l_1, \dots, l_N)$$

as a formal power series in  $\hbar$ , . ┘

*Proof:* As both sides are symmetric and multilinear in  $(l_1, \dots, l_N)$ , it suffices to show for  $l_1 = \dots = l_n = l$ . Then it suffices to calculate the Taylor expansion of

$$\langle e^{l(x)}, \dots, e^{l(x)} \rangle = \hbar^{-\frac{d}{2}} \int_V e^{l(x) - \frac{S(x)}{\hbar}} dx$$

which is exactly calculated by (18.7.2.6). And the conclusions match, noticing that one is labeled and one is unlabeled so there is a multiple by  $N!$ . □

### Matrix Integrals

**Remark(18.7.2.8) [Situations].** For  $N \in \mathbb{Z}_+$ , let  $\text{Herm}^N$  be the set of Hermitian matrices of size  $N$ , with the inner product form  $(A_1, A_2) = \text{tr}(A_1 A_2)$ , with a Lebesgue measure  $d\mu$  normalized s.t.

$$\int_{\text{Herm}^N} e^{-\frac{\text{tr}(A^2)}{2}} d\mu A = 1.$$

Consider the path integral

$$Z_N = \hbar^{-N^2/2} \int_{\text{Herm}_N} e^{-\frac{S(A)}{\hbar}} d\mu A,$$

where the action functional is

$$S(A) = \frac{\text{tr}(A^2)}{2} - \sum_{m \in \mathbb{N}} g_m \frac{\text{tr}(A^m)}{m}.$$

We are interested in the behavior of the coefficients of the expansion of  $Z_N$  in  $g_i$  for large  $N$ . ┘

## 3 Propagators and perturbations

### Scattering Theory

**Def.(18.7.3.1) [Scattering Matrices, Wheeler].** ┘

## 4 Relativistic QFT

### Klein-Gordon Equation

## 5 Fermions

References are [Supersymmetry and Morse Theory, Witten].

## Fermionic Integrals

### 6 Topological QFT

References are [Ati88], [On the Classification of Topological Field Theories, Lurie].

**Remark (18.7.6.1).** It now seems clear that the way to investigate the subtleties of low-dimensional manifolds is to associate to them suitable infinite-dimensional manifolds (e.g. spaces of connections) and to study these by standard linear methods (homology, etc.). In other words we use quantum field theory as a refined tool to study low-dimensional manifolds. —Atiyah.  $\lrcorner$

**Def. (18.7.6.2) [Topological Quantum Field Theory].** A **topological quantum field theory** (or TQFT for short) in dimension  $d$  defined over a ground ring  $\Lambda \in \mathcal{CAlg}$  consists of the following data:

- For any  $\Sigma \in \text{Diff}_{\text{orntd}, \text{cpct}}^d$ , there is a finite  $\Lambda$ -module  $Z(\Sigma)$ , called the **states of particles of  $\Sigma$** .
- For any  $M \in \text{Diff}_{\text{orntd}, \partial}^{d+1}$ , there is an element  $Z(M) \in Z(\partial M)$ , called the **vacuum state defined by  $M$** .

s.t.

- $Z$  are functorial in orientation-preserving diffeomorphisms of  $\Sigma$  or  $M$ .
- $Z$  are involutory, i.e.  $Z(\Sigma^*) = Z(\Sigma)^*$ , where  $\Sigma^*$  is  $\Sigma$  with the dual orientation and  $Z(\Sigma)^*$  is the dual module of  $Z(\Sigma)$ .
- $Z$  is multiplicative, i.e.  $Z(\Sigma_1 \amalg \Sigma_2) = Z(\Sigma_1) \otimes Z(\Sigma_2)$ , and if  $\partial M_1 = \Sigma_1 \amalg \Sigma_3$ ,  $\partial M_2 = \Sigma_3 \amalg \Sigma_2$ ,  $M = M_1 \amalg_{\Sigma_3} M_2$ , then

$$Z(M) = \langle Z(M_1), Z(M_2) \rangle$$

where the pairing is the natural pairing  $Z(\Sigma_3) \times Z(\Sigma_3)^* \rightarrow \Lambda$ .

- $Z(\emptyset^d) = \Lambda$ , and  $Z(\emptyset^{d+1}) = 1 \subset \Lambda = Z(\emptyset^d)$ .

$\lrcorner$

**Prop. (18.7.6.3) [Cobordisms].** If For  $M \in \text{Diff}_{\text{orntd}, \partial}^{d+1}$ , if  $\partial M = \Sigma_1^* \amalg \Sigma_2$ , then  $Z(M) \in Z(\Sigma_1)^* \otimes Z(\Sigma_2)$  can be regarded as a homomorphism  $Z(M) : Z(\Sigma_1) \rightarrow Z(\Sigma_2)$ , and this is compatible with composition of cobordisms. In particular, if  $\Sigma \in \text{Diff}_{\text{orntd}, \text{cpct}}^d$ ,  $M = \Sigma \times [0, 1]$ , then  $Z(M) = \text{id} \in \text{End}(Z(\Sigma))$ .

In particular, the states of particles and the vacuum states (18.7.6.2) can be calculated by cut-and-paste.  $\lrcorner$

**Def. (18.7.6.4) [Vacuum-Vacuum Expectation Values].** For  $M \in \text{Diff}_{\text{orntd}, \text{cpct}}^{d+1}$ ,  $Z(M) \in Z(\partial M = \emptyset) = \Lambda$  is an element, called the **vacuum-vacuum expectation value of  $M$** .  $\lrcorner$

**Prop. (18.7.6.5) [Homotopy Invariance].** Let  $\Sigma, \Sigma' \in \text{Diff}_{\text{orntd}, \text{cpct}}^d$ ,  $M = \Sigma \times [0, 1]$ ,  $M' = \Sigma' \times [0, 1]$ , and  $F : M \rightarrow M'$  is a homotopy of morphisms between  $f_1, f_2 : \Sigma \rightarrow \Sigma'$ , then  $Z(f) = Z(f') : Z(\Sigma') \rightarrow Z(\Sigma)$ , by (18.7.6.3).  $\lrcorner$

## Examples of Topological QFT

**Remark (18.7.6.6).** For  $d = 1$ , there are Floer/Gromov theory and holomorphic conformal field theories.  $\lrcorner$

**Remark (18.7.6.7).** For  $d = 2$ , there are Jones/Witten theory, Casson theory, Johnson theory, and “Thurston” theory.  $\lrcorner$

**Remark (18.7.6.8).** For  $d = 3$ , there are Floer/Donaldson theory.  $\lrcorner$

## 7 Vector Operator Algebras and Haag-Kastler Axioms

Cf.[Princeton Companion].

**Def.(18.7.7.1)[Haag-Kastler Axioms].** An **algebraic quantum field theory** consists of a Hilbert space  $\mathcal{H} \in \mathcal{HilbSpa}$  and for each open bounded subset  $\mathcal{O}$  of the Minkowski space a von.Neumann algebra  $\mathcal{A}(\mathcal{O})$  s.t. the following holds:

- (Isotony) If  $\mathcal{O}_1 \subset \mathcal{O}_2$ , then  $\mathcal{A}(\mathcal{O}_1) \subset \mathcal{A}(\mathcal{O}_2)$ .
- (Causality) If  $\mathcal{O}_1$  is space-like separated from  $\mathcal{O}_2$ , then  $[\mathcal{A}(\mathcal{O}_1), \mathcal{A}(\mathcal{O}_2)] = 0$ .
- (Poincaré invariance) There is a unitary representation of the Poincaré group  $\mathcal{P}$  (affine automorphism group of  $\mathbb{R}^4$ ) on  $\mathcal{H}$  s.t.  $\mathcal{A}(g\mathcal{O}) = g\mathcal{A}(\mathcal{O})g^*$  for any  $g \in \mathcal{P}$ .
- (Spectrum Condition) The joint spectrum  $\text{Spec}(P)$  of the energy-momentum operator  $P$  (i.e. the generator of space-time translations) is contained in the closed forward lightcone.
- (Existence of a vacuum vector) A cyclic Poincaré-invariant vector  $\Omega \in \mathcal{H}$  exists.

For such an algebraic quantum field theory,  $\mathcal{A} = \overline{\cup_{\mathcal{O}} \mathcal{A}(\mathcal{O})}$  is called the **quasi-local algebra**. ┘

## 8 Wightman QFT

**Def.(18.7.8.1)[Double Covering of the Poincaré Group].** The Poincaré group  $\mathbf{P}$  of the affine translations of  $\mathbb{R}^3$  has a double covering  $\tilde{\mathbf{P}}$ . ┘

**Def.(18.7.8.2)[Positive Energy Representations].**  $(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}}(\tilde{\mathbf{P}})$  is called a **positive energy representation** if the spectrum of the Hermitian (pos.unbounded) operator  $\hat{H}_\pi = it|_{t=0}\pi(t)$  is bounded below. ┘

## 9 Osterwalder-Schrader Axioms

## 18.8 Renormalization

## 18.9 Mirror Symmetry

Cf.[Princeton Companions].

## 18.10 Standard Model

### Non-Commutative Geometry Approach

References are [Gravity and the standard model with neutrino mixing, Connes].

starting with a certain noncommutative manifold, described by an algebra, one can derive the full Standard Model of high-energy physics, including the Higgs field, via a so-called spectral action principle applied to a generalized noncommutative Laplacian on this manifold. The commutative counterpart of this is the derivation of the Einstein equations (for positive-definite signature of the metric) from the spectrum of the ordinary Laplacian on a Riemannian manifold.

#### 1 Higgs Fields

#### 2 Beyond Standard Model

## 18.11 String Theory

### 1 Basics

**Def. (18.11.1.1)[B-Models].** For a Calabi-Yau manifold  $X$  with  $\dim_{\mathbb{C}} X \in 2\mathbb{Z} + 1$ , a **B-model string theory** is a ?

┘

**Def. (18.11.1.2)[Branes].** For any complex submanifold  $Y \subset X$  of dimension  $k$ , there is a brane ?.

┘

**Def. (18.11.1.3)[Open-String States].** ?

┘

**Example (18.11.1.4).** The open-string states connecting  $\mathbb{C}^k \subset \mathbb{C}^d$  with itself is the polynomial ring over the Dolbault complex

$$\Omega^{0,*}(\mathbb{C}^k)[\varepsilon_1, \dots, \varepsilon_{d-k}].$$

This is a differential graded algebra with differential  $\bar{\partial}$ .

Moreover, if there are  $N$  subcomplexes, the open-string states are

$$\mathcal{A} = \mathrm{GL}(N; \Omega^{0,*}(\mathbb{C}^k)[\varepsilon_1, \dots, \varepsilon_{d-k}]),$$

and we have a trace map

$$\mathrm{tr} : \mathcal{A} \rightarrow \mathbb{C} : A \mapsto \int_{\mathbb{C}^{k|d-k}} \mathrm{tr} A dz_1 \dots dz_k d\varepsilon_1 \dots d\varepsilon_{d-k}.$$

┘

**Def. (18.11.1.5)[Open-String Field Theory].** The **open-string field theory** has

- Fields:  $\prod \mathcal{A}$ ,
- Action:  $S(\alpha) = \frac{1}{2} \mathrm{tr}(\alpha \bar{\partial} \alpha) + \frac{1}{3} \mathrm{tr} \alpha^3$ .

┘





## Part IV

# Number Theory and Arithmetic Geometry



## 19 | Algebraic Number Theory

## 19.1 $p$ -adic Analysis

References are [Non-Archimedean Analysis Part A] and [Kob84].

This section should only contain theorems that are only applicable to non-Archimedean valuations. Theorems that are applicable to both Archimedean and non-Archimedean valuations should be put into 12.7.

Many properties proved in Functional Analysis independent of complex analysis has counterparts in the non-Archimedean setting, and in fact the goal of this section is to build an analytic theory parallel to complex analysis.

**Notation (19.1.0.1).**

- Let  $p \in \text{Prime}$ .
- Use notations on topological commutative algebra defined in [Topological Commutative Algebra \(Adic Rings\)](#).
- In this section, all norms are non-Archimedean.

┘

### 1 Ultr norms and (Non-Archimedean) Valuations

#### Ultr norms Groups

**Def. (19.1.1.1) [(Semi-)Normed Abelian Groups].** A **semi-normed (Abelian) group** is an Abelian group  $G \in \mathcal{Ab}$  with a function  $- : G \rightarrow \mathbb{R}_{\geq 0}$  s.t.

- $|0| = 0$ .
- $|x - y| \leq \max\{|x|, |y|\}$  for all  $x, y \in R$ .

Such a function is called an **ultrametric valuation**. We can similarly define additive versions s.t.  $v : G \rightarrow \mathbb{R} \cup \{\infty\}$  is an **additive ultrametric valuation** if  $\exp \circ (-v)$  is an ultrametric valuation.

And a semi-normed group  $(G, |-|)$  is called a **normed group** if  $\ker(|-|) = 0$ .

┘

**Cor. (19.1.1.2) [Topological Structures].** For a semi-normed group  $G$  (19.1.1.1),  $G$  has a natural topological structure given by the inverse topology via  $|-|$ . Then  $G$  is a topological Abelian group (12.2.1.1) in this topology. If  $G$  is a normed group, then it is totally disconnected, because open balls at 0 are subgroups hence closed.

$G$  is called a **complete semi-normed group** if  $G$  is a complete metric space.

┘

#### Ultr norms Rings

**Def. (19.1.1.3) [(Semi-)ultr normed Ring].** A **(semi-)ultr normed ring** is a commutative ring  $A \in \mathcal{CRing}$  with a function  $|-| : A \rightarrow \mathbb{R}_{\geq 0}$  s.t.

- $(A^+, |-|)$  is a (semi-)normed Abelian group.
- $|1| = 1$  or  $|-|$  is identically zero.
- $|ab| \leq |a| \cdot |b|$  for any  $a, b \in A$ .

Such a function is called an **ultrametric valuation**. We can similarly define additive versions s.t.  $v : G \rightarrow \mathbb{R} \cup \{\infty\}$  is an **additive ultrametric valuation** if  $\exp \circ (-v)$  is an ultrametric valuation.

An **(ultra)valued ring** is a ultr normed ring with  $|ab| = |a| \cdot |b|$  for any  $a, b \in A$ . Clearly an ultravalued ring is just a ring with a non-Archimedean valuation (12.2.3.10).

It is called **degenerate (semi-)ultr normed ring** if all non-zero valuation value  $\geq 1$ .

A (non-Archimedean) **Banach-ring** is a normed ring that is complete under the valuation.  $\lrcorner$

**Prop. (19.1.1.4).** In an ultranormed ring (19.1.1.3), every triangle is an acute isosceles triangle. (This is because the biggest is smaller than the maximal of the other two, thus the biggest two are equal). Hence we have, for a circle  $B(O, r)$ , any interior point  $P$  is a center of circle, because  $OP < r$ .  $\lrcorner$

**Def. (19.1.1.5) [Bald Rings].** A normed ring  $R$  is call a **B-ring** if elements of valuation 1 is invertible, it is called **bald ring** if there is a  $\varepsilon$  that no elements has valuation in  $(1 - \varepsilon, 1)$ .  $\lrcorner$

**Prop. (19.1.1.6).** If  $K$  is a normed field with valuation ring  $R$ , the smallest subring containing a zero sequence  $a_0, a_1, \dots$  is bald (19.1.1.5).  $\lrcorner$

*Proof:* Cf. [Formal and Rigid Geometry P25].  $\square$

**Def. (19.1.1.7) [Topologically Nilpotent Elements].** An element  $a$  in a normed ring  $A$  is called **topologically nilpotent** iff  $\lim a^n = 0$ . The set of all topological nilpotent elements in  $A$  are denoted by  $A^0$ .  $\lrcorner$

**Prop. (19.1.1.8).**  $A^0$  is a subgroup of  $A^+$ , which is multiplicatively closed, and  $A^0$  is clopen in  $A$ . In particular,  $A^0$  is complete if  $A$  is complete.  $\lrcorner$

*Proof:* Cf. [Non-Archimedean analysis P27].  $\square$

**Prop. (19.1.1.9) [Nakayama's Lemma].** If  $A$  is complete normed ring and  $M$  is a  $A$ -module, if there are f.m. elements  $x_i$  of  $M$  that  $M = N + \sum x_i M$ , then  $M = N$ .  $\lrcorner$

*Proof:* The proof is verbatim as the proof of the usual Nakayama lemma.  $\square$

## 2 Normed Modules

**Def. (19.1.2.1) [Ultranormed Module].** For a normed ring  $(A, |-\|) \in \mathcal{CRing}(19.1.1.1)$ ,  $M \in \text{Mod}_A$  with an ultrametric valuation  $\|-\| : M \rightarrow \mathbb{R}_{\geq 0}$  is called **normed module** if

- $(M, \|-\|)$  is a normed Abelian group (19.1.1.1).
- $\|ax\| \leq |a| \|x\|$  for any  $a \in A, m \in M$ .

Such a function is called an **ultrametric valuation** on  $M$ . We can similarly define additive versions s.t.  $v : G \rightarrow \mathbb{R} \cup \{\infty\}$  is an **additive ultrametric valuation** if  $\exp \circ (-v)$  is an ultrametric valuation.

If  $A$  is a valued ring and the equality always holds, we call it a **faithfully ultranormed module** or a **valued module**.

If  $A$  is a valued field, any normed module is valued.  $\lrcorner$

**Remark (19.1.2.2).** This is the non-Archimedean analogue of (12.7.1.8).  $\lrcorner$

**Prop. (19.1.2.3) [Ultranormed Algebra].** A normed algebra is an  $A$  algebra  $B$  with  $A \rightarrow B$  bounded of norm 1.  $\lrcorner$

**Prop. (19.1.2.4).** For two valued module over  $A$ , if  $A$  is non-degenerate, a morphisms is bounded iff it is continuous. This is because we can multiply by elements of  $A$  to reduce to a nbhd of 0.

This applies to the case when  $A$  contains a field where the valuation is non-trivial, because we can use (19.1.2.1).  $\lrcorner$

**Def. (19.1.2.5)[Completed Tensor Product].** For two normed modules over a normed ring  $R$ , there is a complete normed  $R$ -module  $M \hat{\otimes} N$  called the **completed tensor product**, satisfying the following universal properties:  $M \times N \rightarrow M \hat{\otimes} N$  is bounded by 1, and for any complete normed  $R$ -module  $T$  and a  $R$ -map  $M \times N \rightarrow T$  bounded by  $a$ , then it factor through a  $R$ -map  $M \hat{\otimes} N \rightarrow T$  bounded by  $a$ .

It satisfies many universal properties as you can imagine.  $\lrcorner$

*Proof:* Cf.[Formal and Rigid Geometry P238].  $\square$

**Cor. (19.1.2.6).** By (19.1.2.4), when  $A$  is non-degenerate, then the amalgamated sum is just the fibered pushout when restricted to the category of complete valued module over  $A$  with continuous maps as morphisms, because it satisfies the universal property.  $\lrcorner$

**Prop. (19.1.2.7)[Amalgamated Sum].** For two normed  $R$ -algebras there is an operation of **amalgamated sum** which satisfies universal properties similar to (19.1.2.5). In fact, it is just the completed tensor product when seen as modules.  $\lrcorner$

*Proof:* Cf.[Formal and Rigid Geometry P242].  $\square$

### Weakly Cartesian Space

**Def. (19.1.2.8)[Weakly Cartesian Vector Spaces].** A normed  $K$ -vector space over a valued field  $K$  is called **weakly Cartesian** if?  $\lrcorner$

**Prop. (19.1.2.9).** If  $K$  is a complete valued field, then each normed  $K$ -vector space  $V$  is weakly Cartesian.  $\lrcorner$

*Proof:* Cf.[Non-Archimedean Analysis P92].  $\square$

### Completeness

**Prop. (19.1.2.10)[Cauchy Sequence of Non-Archimedean field].** For a sequence  $\sum a_i$  in a non-Archimedean field, it is a Cauchy sequence iff  $\lim |a_i| = 0$ .

In particular, convergent sequence are all absolutely convergent and for a Cauchy sequence not converging to 0, the valuations of the terms stabilize.  $\lrcorner$

*Proof:* One way is easy, the other way, notice  $|\sum_{v=i}^j a_i| \leq \max_{i,i+1,\dots,j} |a_v| < \varepsilon$ .  $\square$

**Prop. (19.1.2.11)[Completion of a Field].** The completion of a non-Archimedean field is preferred to choose the definition of Cauchy sequence, so we see by (19.1.2.10) that  $v(\widehat{K}) = v(K)$ .  $\lrcorner$

**Prop. (19.1.2.12).** For a complete field  $K$  and any finite vector space  $L$ ,  $L$  has only one norm up to equivalence and it is complete.  $\lrcorner$

*Proof:* Cf.[Formal and Rigid Geometry P230].  $\square$

**Prop. (19.1.2.13).** A complete valuation on a field can extend uniquely to a valuation on its alg.closure. And in the finite case, it is  $|\alpha| = |\text{Nm}(\alpha)|^{\frac{1}{d}}$ . This is an immediate consequence of (19.1.2.24) and (19.1.2.21), and  $|\alpha| \leq 1$  iff it is integral over valuation ring  $R$  of  $K$ .  $\lrcorner$

**Prop. (19.1.2.14).** Any infinite separable algebraic extension of a complete field is never complete.  $\lrcorner$

*Proof:* We use Krasner's lemma(19.1.2.26). By Ostrowski theorem(12.2.3.18), we can assume it is non-Archimedean, otherwise it cannot be infinite dimensional. Choose an infinite linearly independent basis of decreasing value rapidly enough, then we can see the field generated by the limit contains all the partial sums, contradiction.  $\square$

**Prop. (19.1.2.15)**  $[\mathbb{C}_K]$ . If  $K = \overline{K}$ , then  $\widehat{K} = \overline{\widehat{K}}$ , denoted by  $\mathbb{C}_K$ .  $\lrcorner$

*Proof:* Let  $L = \overline{\widehat{K}}$ , then we can extend to a valuation on  $L$ , now let  $f$  be a monic polynomial with coefficients in  $\widehat{K}$ , we show its root  $\alpha \in L$  can be approximated by elements in  $K$ , now let  $g$  monic in  $K[X]$  be an approximation of  $f$  that  $|g(\alpha)| \leq \varepsilon^n$ , then there is a root  $\beta$  of  $g$  that  $|\alpha - \beta| < \varepsilon$ , and  $\beta \in K$  by alg.closedness.  $\square$

**Prop. (19.1.2.16)**. If  $K$  is a complete, then  $K^{\text{sep}}$  is dense in  $\overline{K}$ .  $\lrcorner$

*Proof:* Assume  $F$  is non-Archimedean, then for  $y \in F^{\text{alg}}$ , there is a  $n$  that  $y^{p^n} = \alpha \in F^{\text{sep}}$ . We may assume  $|\alpha| \leq 1$ , then let  $\pi$  be an element that  $|\pi| < 1$ , then if  $y_i$  is a root of the separable polynomial  $Y^{p^n} - \pi^i Y - \alpha = 0$ , then  $(y - y_i)^{p^n} = \pi^i y_i$ . So  $|y - y_i| \rightarrow 0$ .  $\square$

**Def. (19.1.2.17)**  $[\mathbb{Q}_p, \text{Hensel1897}]$ .  $\mathbb{Q}_p$  is defined to be the  $p$ -adic completion of  $\mathbb{Q}$ , called the **field of  $p$ -adic numbers**. Its ring of integer is  $\mathbb{Z}_p$ , which equals the  $p$ -adic completion of  $\mathbb{Z}$ , called the **ring of  $p$ -adic integers**.  $\lrcorner$

**Prop. (19.1.2.18)**. There is a non-canonical isomorphism  $\overline{\mathbb{Q}_p} \cong \mathbb{C}$ , not compatible the topology.  $\lrcorner$

*Proof:* This follows from(3.2.7.5) and the fact they both have the same cardinality  $\aleph_1$ .  $\square$

**Cor. (19.1.2.19)**. The  $p$ -adic valuation on  $\mathbb{Q}$  can be extended to  $\mathbb{C}$  non-canonically.  $\lrcorner$

### Henselian Valued Fields

**Def. (19.1.2.20)** **[Henselian Valued Field]**. A **Henselian valued field** is a valued field  $K$  that the valuation ring  $\mathcal{O}_K$  is a Henselian local ring(5.3.10.1).  $\lrcorner$

**Prop. (19.1.2.21)**. A valued field  $K$  is Henselian(19.1.2.20) iff the valuation of  $K$  has a unique extension to any finite extension  $L/K$ .  $\lrcorner$

*Proof:* Cf.[Algebraic Number Theory Neukirch P144].  $\square$

**Def. (19.1.2.22)** **[Ramification Degrees]**. If  $L/K$  is a finite extension of valued field of degree  $n$ , then  $v$  extends uniquely to  $w(\alpha) = \frac{1}{n}v(N_{L/K}(\alpha))$ , now we define the **ramification degree** as  $[w(L^*) : v(K^*)]$ , and the **inertia degree** as the degree of the residue field extension.  $\lrcorner$

Thus for a normal extension,  $x$  and  $\sigma(x)$  has the same valuation. Hence any polynomial in  $K[X]$  has a decomposition into polynomials where all their roots has the same valuation.  $\lrcorner$

**Prop. (19.1.2.23)** **[Generalized Hensel's Lemma]**. Let  $(K, \mathcal{O}_K)$  be a complete valued non-Archimedean field. If  $P, Q, R \in \mathcal{O}_K[X]$  and  $0 \leq \lambda < 1$  that  $\deg P = m + n$ ,  $\deg Q = n$ ,  $\deg R = m$ , and

$$\deg(P - QR) \leq m + n - 1, \quad \|P - QR\|_G \leq \lambda |\text{res}(Q, R)|^2$$

where  $\|-\|_G$  is the Gauss norm on  $K[X]$ . Then there exist polynomials  $U, V$  that

$$\bullet \quad \|U\|_G, \|V\|_G \leq \lambda |\text{res}(Q, R)|,$$

- $\deg U \leq n - 1, \quad \deg V \leq m - 1.$
- $P = (Q + U)(R + V).$

┘

*Proof:* If  $\rho = |\text{res}(Q, R)| = 0$ , then  $P = QR$ . Otherwise, the map  $\theta_{Q,R} : W_m \oplus W_n \rightarrow W_{m+n}$  is invertible (3.3.1.7) where  $W_t$  is the free  $K$ -vector space of polynomials of degree  $< t$ , and we can define  $\varphi(U, V) = \theta_{Q,R}^{-1}(P - QR - UV)$ .

If  $U, V \in B(0, \lambda\rho)$ , then  $|\varphi(U, V)|_G \leq \lambda\rho$ . And it can be proved  $\varphi$  is a contraction map from  $B(0, \lambda\rho)^2$  to itself with contraction factor  $\lambda$ , so it has a fixed point  $(U, V)$  by (4.4.8.29). Then  $QU + RV = P - QR - UV$ , and we are done.  $\square$

**Cor. (19.1.2.24) [Hensel's Lemma].** Let  $(K, \mathcal{O}_K)$  be a complete valued non-Archimedean field,  $P(X) \in \mathcal{O}_K[X]$  and  $\alpha_0 \in \mathcal{O}_K$  s.t.  $|P(\alpha_0)| < |P'(\alpha_0)|^2$ , then there exists a  $\alpha \in \mathcal{O}_K$  that  $P(\alpha) = 0$  and  $|\alpha - \alpha_0| \leq |P(\alpha_0)/P'(\alpha_0)|$ .

In particular, this holds when  $|P'(\alpha_0)| = 1$  and  $|P(\alpha_0)| < 1$ , in which case we can pass to the residue field. In particular, a complete valued field is Henselian (19.1.2.20).  $\square$

*Proof:* Let  $\lambda = \|P(\alpha_0)\| / \|P'(\alpha_0)\|^2$ . Notice if we let  $P(X) = Q(X)(X - \alpha_0) + P(\alpha_0)$  where  $Q(X) \in \mathcal{O}_K[X]$ , then  $\text{res}(Q(X), X - \alpha_0) = Q(\alpha_0) = P'(\alpha_0)$  (3.3.1.9). So we can use the generalized Hensel's lemma (19.1.2.23) to conclude.  $\square$

**Cor. (19.1.2.25).** Let  $(K, \mathcal{O}_K)$  be a complete valued non-Archimedean field. If  $Q, R \in \mathcal{O}_K[X]$  are relatively prime in  $\kappa[X]$ , then there exists  $U, V \in \mathcal{O}_K[X]$  s.t.

- $\deg U \leq m - 1, \deg V \leq n - 1.$
- $QU + RV = 1.$

┘

*Proof:* Use the fact  $\theta_{Q,R} : W_n \times W_m \rightarrow W_{m+n}$  is invertible where  $W_t$  is free  $\mathcal{O}_K$ -module of polynomials of degree  $< t$ .  $\square$

**Thm. (19.1.2.26) [Krasner's Lemma].** Let  $(\mathcal{O}_K, K) \in \text{DVR}$  be a Henselian and  $\alpha, \beta \in \overline{K}$  s.t.  $|\beta - \alpha| < |\alpha - \sigma(\alpha)|$  for all  $\sigma$ , then  $K(\alpha, \beta)/K(\beta)$  is purely inseparable. In particular, if  $\alpha$  is separable over  $K$ , we have  $K(\alpha) \subset K(\beta)$ .  $\square$

*Proof:* It suffice to prove  $\tau(\alpha) = \alpha$  for all field morphism  $\tau : K(\alpha, \beta) \rightarrow \overline{K}$  fixing  $K(\beta)$ . This is because  $|\tau(\alpha) - \beta| = |\alpha - \beta| < |\alpha - \sigma(\alpha)|$ , thus  $|\tau(\alpha) - \alpha| \leq \max\{|\tau(\alpha) - \beta|, |\beta - \alpha|\} < |\alpha - \sigma(\alpha)|$ .  $\square$

**Cor. (19.1.2.27).** If  $f$  is a separable irreducible polynomial and  $\alpha$  is a root, then for  $g$  closed enough to  $f$ , there is a root  $\beta$  of  $g$  that  $K(\beta) = K(\alpha)$ . (Immediate consequence of (12.2.3.19)).  $\square$

**Cor. (19.1.2.28).** Any finite separable extension  $\mathcal{L}/\widehat{K}$  is of the form  $L_0\widehat{K}$  for some finite separable extension  $L_0/K$ , by primitive element theorem (3.2.6.18).  $\square$

**Cor. (19.1.2.29).** If  $K \subset K^{\text{sep}}$  is dense, then  $K = K^{\text{sep}}$ .  $\square$

### 3 Extensions of Norms and Valuations

See [BGR84] Chap3.



## 4 Extensions of Henselian Valued Fields

**Notation (19.1.4.1).**

- Let  $(K, v, \mathcal{O}_K, \mathfrak{m}_v, \varpi, k)$  be a Henselian non-Archimedean valued field of residue characteristic  $p \in \text{Prime}$ . If  $\mathcal{O}_K$  is DVR, assume  $(\varpi) = \mathfrak{m}_v k m_v$ .

┘

**Lemma (19.1.4.2) [Extension is Monogenous].** For a finite extension of CDVR, if the residue field extension  $\lambda/k$  is separable, then there exists a  $x \in \mathcal{O}_L$  that  $\mathcal{O}_K[x] = \mathcal{O}_L$ .

┘

*Proof:* If  $\bar{x}$  is an element of  $\lambda$  that generate  $\lambda$  over  $k$ , by primitive element theorem, then let  $\bar{f}$  be the minipoly of  $\bar{x}$ , then let  $f, x$  be lifting of them, then  $f(x)$  is a uniformizer, otherwise  $f'(x)$  has valuation 0, so  $f(x + \varpi_L)$  is a uniformizer. Now we see that  $x^i f(x)^j$  is a basis of  $\mathcal{O}_L$  over  $\mathcal{O}_K$ ,  $\square$

**Prop. (19.1.4.3).** If  $K \subset_{\text{fin}} L$  is a finite separable extension and if  $I \in \text{Ideal}(\mathcal{O}_L)$ , then  $v_K(\text{tr}_{L/K}(I)) = [v_K(I \cdot \mathfrak{D}_{L/K})]$ .

┘

*Proof:* By definition,  $\text{tr}_{L/K}(x\mathcal{O}_L) \subset \mathcal{O}_K$  iff  $x \in \mathcal{D}_{L/K}^{-1}$ , thus  $\text{tr}_{L/K}(I) \subset J$  iff  $I \subset \mathcal{D}_{L/K}^{-1}J$ , i.e.  $\text{tr}_{L/K}(I)$  is the smallest ideal  $J$  of  $\mathcal{O}_K$  that contains  $I \cdot \mathcal{D}_{L/K}$ , thus the result.  $\square$

**Prop. (19.1.4.4) [Galois Group of Completions].** If  $K$  is a DVR, then for any  $L \subset K^{\text{sep}}$ , we have  $\widehat{L} = L \otimes_K \widehat{K}$ , and this induces an equivalence between finite separable extensions of  $K$  and finite separable extensions of  $\widehat{K}$ . In particular,  $\text{Gal}_K \cong \text{Gal}_{\widehat{K}}$ .

┘

*Proof:* Cf. [Ber93]P45. ?

┘

### Unramified Extensions

**Def. (19.1.4.5) [Unramified Extensions].** A finite extension  $L/K$  is called **unramified extension** if the residue field extension  $\lambda/k$  is separable and  $[L : K] = [\lambda : k]$ . Any algebraic extension is called **unramified** iff any finite extension is unramified.

This is compatible because unramified extensions form a distinguished class. So we can talk about the **maximal unramified extension**  $T$  of  $K$ , and a field extension  $L/K$  is called **unramified** if all finite subextensions are unramified.

┘

*Proof:* It is faithfully transitive because the field extension degree is transitive, and for base change, as the residue field is separable, we let  $\lambda = k[\bar{\alpha}]$ , and choose a lift  $\alpha \in \mathcal{O}_L$ , the minipoly of  $\alpha$  is  $f(X) \in \mathcal{O}_K[X]$ . Then we have

$$[\lambda : k] \leq \deg \bar{f} = \deg f = [K(\alpha) : K] \leq [L : K] = [\lambda : k]$$

So  $L = K(\alpha)$  and  $\bar{f}$  is the minipoly of  $\bar{\alpha}$ . Then  $L' = K'(\alpha)$ , and let  $g(X)$  be the minipoly of  $\alpha$  over  $K'$ , then  $\bar{g}$  is a factor of  $\bar{f}$  so separable, hence irreducible by Hensel's lemma. Noe:

$$[\lambda' : k'] \leq [L' : K'] = \deg g = \deg \bar{g} = [k'(\alpha) : k'] \leq [\lambda' : k].$$

So  $[\lambda' : k'] = [L' : K']$ .

┘

**Prop. (19.1.4.6) [Maximal Unramified Extension].** The residue field of the maximal unramified extension  $K^{\text{ur}}/K$  is  $\bar{k}$ , and the value group is the same as  $K$ .

┘

*Proof:* The first assertion is because for any separable polynomial, it has a lift which is irreducible has a root lifting  $\bar{\alpha}$ , contradicting the maximality. For the second, look at finite subextensions, then it results from the fundamental inequality (12.2.3.20).  $\square$

### Tamely Ramified Extensions

**Def. (19.1.4.7) [Tamely Ramified Extension].** For  $K$  a Henselian non-Archimedean valued field,  $K \subset_{\text{fin}} L \in \text{Field}$  is called a **tamely ramified extension** if the residue field extension is separable and  $(e_{L/K}, p) = 1$ .  $\lrcorner$

**Prop. (19.1.4.8).** Tamely unramified extensions form a distinguished class, so we can talk about the maximal tamely unramified extensions, and a field extension  $L/K$  is called **tamely ramified** if all finite subextensions are tamely unramified.  $\lrcorner$

*Proof:* Cf.[Algebraic Number Theory Neukirch P156].  $\square$

**Prop. (19.1.4.9).**  $L/K^{\text{ur}} \subset K^{\text{sep}}/K^{\text{ur}}$  is tamely ramified iff the extension is generated by radicals over the maximal unramified extension:  $L = K^{\text{ur}}(\sqrt[p]{a_i}), (m, p) = 1$ , where  $a_i \in K^{\text{ur}}$ , (WARNING: make sure if  $a_i \in K$  or not?).  $\lrcorner$

*Proof:* Cf.[Algebraic Number Theory Neukirch P155].  $\square$

**Prop. (19.1.4.10).** The value field of tamely ramified extensions. Cf.[Neukirch P157].  $\lrcorner$

### Totally Ramified Extensions

**Def. (19.1.4.11) [Eisenstein Polynomial].** For  $(K, \mathcal{O}_K) \in \text{DVR}$ , an **Eisenstein polynomial** in  $\mathcal{O}_K[T]$  is a polynomial of the form

$$f(T) = T^n + a_{n-1}T^{n-1} + \dots + a_1T + a_0$$

where  $a_i \in \mathfrak{m}$  for any  $i$  and  $a_0 \in \mathfrak{m} \setminus \mathfrak{m}^2$ .  $\lrcorner$

**Prop. (19.1.4.12) [Totally Ramified Extensions via Eisenstein Polynomials].** If  $e(T)$  is an Eisenstein polynomial in  $\mathcal{O}_K[T]$  and  $\Pi$  is a root of  $e(T)$  in  $\overline{K}$ , then  $L = K(\Pi)$  is a totally ramified extension of  $K$  with uniformizer  $\Pi$ . Conversely, if  $L/K$  is a totally ramified and  $\Pi \in L$  is a uniformizer, then  $L = K(\Pi)$  and the minimal polynomial of  $\Pi$  is an Eisenstein polynomial.  $\lrcorner$

*Proof:*  $\square$

**Remark (19.1.4.13).** More about totally ramified extensions are discussed in [Totally Ramified Extensions](#).  $\lrcorner$

### Ramification Groups

**Def. (19.1.4.14) [Ramification Groups].** For a Galois extension  $L/K$  of CVDRs, denote  $\lambda/k$  residue fields extension of  $w|v$ , and denote  $\mathfrak{p}_w, \mathfrak{p}_v$  by  $\mathfrak{P}, \mathfrak{p}$ . Define:

The **inertia group** is  $I(L/K) = \{\sigma \in \text{Gal}(L/K) | \sigma(x) \equiv x \pmod{\mathfrak{P}}\}$ .

The **ramification group** is  $R(L/K) = \{\sigma \in \text{Gal}(L/K) | \sigma(x)/x \equiv 1 \pmod{\mathfrak{P}}\}$ .  $\lrcorner$

**Prop. (19.1.4.15).** For local fields, the ramification degree  $e$  equals the order of inertia group  $|I_{L/K}|$ .  $\lrcorner$

**Prop. (19.1.4.16).** The residue field extension  $\lambda/k$  is normal and there is an exact sequence

$$1 \rightarrow I(L/K) \rightarrow \text{Gal}(L/K) \rightarrow \text{Gal}(\lambda/k) \rightarrow 1.$$

$\lrcorner$

*Proof:* Cf.[Neukirch P172].  $\square$

**Prop. (19.1.4.17).**  $R(L/K)$  is the unique pro- $p$ -Sylow subgroup of  $\text{Gal}(L/K)$  (3.1.12.11).  $\lrcorner$

*Proof:* Cf.[Neukirch P174].  $\square$

**Prop. (19.1.4.18).** There is an exact sequence

$$1 \rightarrow R(L/K) \rightarrow I(L/K) \rightarrow \chi(L/K) \rightarrow 0$$

where  $\chi_v(L/K) = \text{Hom}(\Delta/\Gamma, \lambda^\times)$  and  $\Delta = w(L^\times), \Gamma = v(K^\times)$ . Moreover, in case  $L/K$  is a finite extension, this exact sequence splits.  $\lrcorner$

*Proof:* For any  $\sigma \in I_w$ , define the map  $\chi_\sigma : \Delta/\Gamma \rightarrow \lambda^*$  as follows: for any  $\delta \in \Delta/\Gamma$ , let  $\delta = w(x)$  for some  $x \in L^*$ , let  $\chi_\sigma(\delta) = \frac{\sigma(x)}{x} \bmod \mathfrak{P}_w$ . This is independent of  $x$  chosen, because if  $w(x) \equiv w(x') \bmod \Gamma$ , then  $w(x) = w(ax)$  for some  $a \in K^*$ , thus  $x = axu$  for some  $u \in \mathcal{O}_w^*$ . Now  $\frac{\sigma(u)}{u} \equiv 1 \bmod \mathfrak{P}$  as  $\sigma \in I_w$ , so  $\frac{\sigma(x)}{x} = \frac{\sigma(x')}{x'} \in \lambda^*$ . And the kernel of this map is  $R_w$  by definition.  $\square$

The sequence is exact on the right by [Neukirch P175].  $\textcolor{red}{?}$

### Higher Ramification Groups

**Notation (19.1.4.19).**

- Let  $(L, \mathcal{O}_L)/(K, \mathcal{O}_K) \in \text{CDVR}$  be a finite Galois extension.

$\lrcorner$

**Def. (19.1.4.20) [Higher Ramification Groups].** For  $s \in \mathbb{R}_+$ , define the  $s$ -th ramification group  $G_s(L/K) = \{\sigma \in G \mid v_L(\sigma(x) - x) \geq s + 1 \text{ for all } a \in \mathcal{O}_L\}$ .

Then we have  $G = G_{-1} \supset G_0 \supset G_1 \supset \dots$ . And  $G_0$  is the inertia group.  $\lrcorner$

**Prop. (19.1.4.21).** When  $K$  has finite residue field,  $G_1$  is the ramification group  $R_w$  (19.3.1.22). In this case, we have

$$G_s(L/K) = \{\sigma \in G_0 \mid \frac{\sigma(\varpi_L)}{\varpi_L} \in U_L^s\}, \text{ for } s \geq 0.$$

So there are injective morphism  $G_s/G_{s+1} \rightarrow U_L^s/U_L^{s+1} : \sigma \mapsto \sigma(\varpi_L)/\varpi_L$  for  $s \geq 0$ . (This is independent of  $\varpi_L$  chosen because units are mapped mod  $U_L^{s+1}$ ).  $\lrcorner$

*Proof:*  $G_1 = R_w$ : one direction is trivial, for the other, we use Teichmüller representatives, then  $R_w$  preserves all them, and  $\sigma(x) - x \equiv 0 \bmod \mathfrak{P}^2$  is true for  $\pi$ , so it is true for all.  $\square$

**Prop. (19.1.4.22).** For local fields  $L/K$ , if  $\sigma$  is in the inertia group, then

$$v_L\left(\frac{\sigma(x)}{x} - 1\right) \geq v_L\left(\frac{\sigma(\varpi_L)}{\varpi_L} - 1\right) + \delta_{v_L(x), 0}$$

for any  $x \in \mathcal{O}_L$  and a uniformizer  $\varpi_L$ . Equality holds when  $v_L(x) = 1$ .  $\lrcorner$

*Proof:* if  $L$  has residue field  $\mathbb{F}_q$ , then any element of  $\mathcal{L}$  can be written as  $\sum \xi_n \varpi_L^n$ , where  $\xi_n$  are all  $q - 1$ -th roots of unity. And because  $\sigma$  is inertia group, all  $q - 1$ -th roots of unity are preserved, so  $\sigma(\xi_n \varpi_L^n) - \xi_n \varpi_L^n = \xi_n \varpi_L \left(\frac{\sigma(\varpi_L)}{\varpi_L} - 1\right) (\sigma(\varpi_L)^{n-1} + \sigma(\varpi_L)^{n-2} \varpi_L + \dots + \varpi_L^{n-1})$  has valuation  $\geq v\left(\frac{\sigma(\varpi_L)}{\varpi_L} - 1\right) + n$ . Thus the result.  $\square$

In the sequel, we assume that the residue field extension is separable, as to use the proposition (19.1.4.2).

**Lemma (19.1.4.23).** We define  $i_{L/K}(\sigma) = v_L(\sigma x - x)$ , where  $x$  is the generator of  $\mathcal{O}_L/\mathcal{O}_K$ . If  $L/L'/K$  are Galois extensions that  $e$  is the ramification index of  $L/L'$ . Then

$$i_{L'/K}(\sigma') = \frac{1}{e} \sum_{\sigma|_{L'}=\sigma'} i_{L/K}(\sigma).$$

┘

*Proof:* Cf.[Neukirch Algebraic Number Theory P178].

□

**Def. (19.1.4.24) [Upper Numbering].** We define the **Herbrand function**  $\varphi_{L/K}(u) = \int_0^u \frac{dx}{(G_0:G_x)}$ . It maps  $\{x \geq 1\}$  to itself and is strictly increasing.

If  $m \leq s < m+1$ , then it is just  $\varphi_{L/K}(s) = \frac{1}{g_0}(g_1 + g_2 + \dots + g_m + (s-m)g_{m+1})$ , where  $g_i = |G_i|$ . By a double counting, it is

$$\varphi_{L/K}(s) = \frac{1}{g_0} \sum_{\sigma \in G} \min\{i_{L/K}(\sigma), s+1\} - 1.$$

The derivative of  $\varphi_{L/K}$  is  $\varphi'_{L/K}(s) = \frac{|G_s|}{g_0}$ .

Let  $\psi_{L/K}$  be the inverse function of  $\varphi_{L/K}$ . We define  $G^t = G_{\psi_{L/K}(t)}$ , this is called the **upper numbering**.

┘

**Lemma (19.1.4.25).** For  $L/L'/K$  Galois extensions, one has  $G_s(L/K)H/H = G_t(L'/K)$ , where  $t = \varphi_{L/L'}(s)$ . Equivalently,  $G_s/H_s = (G/H)_{\varphi_{L/L'}(s)}$ . ┘

*Proof:* For  $\sigma' \in G(L'/K)$ , we choose a inverse image  $\sigma \in G(L/K)$  of maximal  $i_{L/K}(\sigma)$ , then  $i_{L'/K}(\sigma') - 1 = \varphi_{L/L'}(i_{L/K}(\sigma) - 1)$ . To prove this, let  $i_{L/K}(\sigma) = m$ , then we see  $i_{L/K}(\sigma\tau) = \min\{i_{L/K}(\tau), m\}$ , so by (19.1.4.23),  $i_{L'/K}(\sigma') = \frac{1}{e} \sum_{\tau \in H} \min\{i_{L/K}(\tau), m\}$ . And  $e = |H_0|$  by (19.1.4.15). So the assertion follows from (19.1.4.24).

Now  $\sigma'$  is in the image of  $G_s$  is equivalent to  $i_{L/K}(\sigma) - 1 \geq s \iff \varphi_{L/L'}(i_{L/K}(\sigma) - 1) \geq \varphi_{L/L'}(s)$ , which by what proved is equivalent to  $\sigma' \in G_t(L'/K)$ . ┘

**Cor. (19.1.4.26).** For  $L/L'/K$  Galois extensions,  $\varphi_{L/K} = \varphi_{L'/K} \circ \varphi_{L/L'}$ , hence similar formula holds for  $\psi$ . ┘

*Proof:* By the proposition and multiplicity of ramification index  $e$ , we get

$$\frac{1}{e_{L/K}} |G_s| = \frac{1}{e_{L'/K}} |(G/H)_t| \frac{1}{e_{L/L'}} |H_s|.$$

where  $t = \varphi_{L/L'}(s)$ , which is equivalent to the derivative

$$\varphi'_{L/K}(s) = \varphi'_{L'/K}(t) \varphi'_{L/L'}(s) = (\varphi_{L'/K} \circ \varphi_{L/L'})'(s),$$

and they are equal at  $s = 0$ , so the conclusion follows. ┘

**Prop. (19.1.4.27) [Herbrand's Theorem].** For  $L/L'/K \in \mathbf{Field}$  s.t.  $L/K, L'/K \in \mathbf{GalField}$  and any  $t \in \mathbb{R}$ ,  $G^t(L'/K)$  is the image of  $G^t(L/K)$  under the quotient. ┘

*Proof:* Let  $r = \varphi_{L'/K}(t)$ , by the above lemme and corollary,

$$G^t H/H = G_{\varphi_{L/K}(t)} H/H = G'_{\varphi_{L/L'}(\psi_{L/K}(t))} = G'_{\varphi_{L/L'}(\psi_{L/L'}(r))} = G_r(L'/K) = G^t(L'/K)$$

□

**Prop. (19.1.4.28) [Hasse-Arf].** For an Abelian extension of CDVRs  $L/K$  that the residue field extension is separable, the jump in the upper numbering of higher ramification group  $G^v$  must happen at integers. (Note: The proof in the case where  $K$  is a local field is much easier by Lubin-Tate group, See(19.7.2.33)).

┘

*Proof:* The theorem is just saying that if  $G_s \neq G_{s+1}$  for  $s$  integer, then  $\varphi_{L/K}(s)$  is an integer.

This follows from the following lemma, because if  $G$  is not totally ramified, then we can change it to the Galois field of  $G_0$ , this didn't change anything by the definition of(19.1.4.24), and the fact  $\varphi(0) = 0$ . And when  $G^v \neq G^{v+}$ , then we consider splitting  $G/G^{v+}$  into product of cyclic groups, thus there is one cyclic group  $H$  that the projection of  $G^v$  into  $H$  is not trivial. Now  $H$  is a Galois group of some  $L'/K$ , and Herbrand's theorem shows that  $H^v \neq H^{v+}$ , hence  $v$  is an integer by the following lemma.

□

**Lemma (19.1.4.29).** For a cyclic totally ramified extension  $L/K \in \text{CDVR}$  s.t. the residue field extension is separable, if  $\mu$  is the maximal integer that  $G_\mu \neq 1$ , then  $\varphi_{L/K}(G_\mu)$  is an integer.

┘

*Proof:* Cf.[Serre Local Fields P94].

□

**Example (19.1.4.30).** If  $K_n = \mathbb{Q}_p(\mu_{p^n})$ , then

$$\text{Gal}_s(K_n/\mathbb{Q}_p) = \text{Gal}(F_n/F_t) \quad \text{for } p^t - 1 \leq s < p^{t+1} - 2.$$

Thus  $\text{Gal}^i(K_n/\mathbb{Q}_p) = \text{Gal}(K_n/K_i)$ .

┘

*Proof:* This is because  $\zeta_{p^n} - 1$  is a uniformizer of  $K_n$ (19.1.5.28).

□

## 5 Local Fields

### $p$ -adic Fields

**Def. (19.1.5.1) [ $p$ -adic Fields].** A  $p$ -adic field is a CDVR  $(K, v, \mathcal{O}_K, \mathfrak{p}_v, \varpi_K, \kappa)$  s.t.  $\text{char } K = 0$ ,  $\text{char } \kappa = p$ ,  $\kappa = \kappa^{\text{perf}}$ . A  $p$ -adic local field(19.1.5.10) is a  $p$ -adic field(19.1.5.1). The category of  $p$ -adic fields is denoted by  $p\text{-Field}$ .

┘

**Def. (19.1.5.2) [Topological Completion].** For  $K \in p\text{-Field}$ (19.1.5.1), define  $\mathbb{C}_K = \widehat{\overline{K}}$ , which is an alg.closed complete valued field, by(19.1.2.24)(19.1.2.21) and(19.1.2.15). Also denote  $\mathbb{C}_p = \mathbb{C}_{\mathbb{Q}_p}$ .

┘

**Lemma (19.1.5.3).** If  $K \in p\text{-Field}$  and  $P(X) \in \overline{K}[X]$  is a monic polynomial of degree  $n$ , and all of its roots satisfied  $v_p(\alpha) \geq c$  for some constant  $c$ . Let  $q = p^k$  if  $n = p^k d$ ,  $d \neq 1$  or  $n = p^{k+1}$ .

Then the derivative  $P^{(q)}(X)$  has a root  $\beta$  with  $v_p(\beta) \geq c$  or in case  $n = p^{k+1}$ ,  $v_p(\beta) \geq c - \frac{1}{p^k(p-1)}$ .

┘

*Proof:* Let  $P = X^n + a_{n-1}X^{n-1} + \dots + a_0$ , then  $v_p(a_i) \geq (n-i)c$ . And

$$1/q! \cdot P^{(q)}(X) = \sum_{i=0}^{n-1} C_{n-i}^q a_{n-i} X^{-i-q}.$$

So at least one root  $\beta$  satisfies

$$v_p(\beta) \geq \frac{1}{n-q}((n-q)c - v_p(C_n^q)) = c - \frac{1}{p^k(p-1)}.$$

□

**Lemma (19.1.5.4).** If  $K$  is a  $p$ -adic field and  $\alpha \in \overline{K}$ , let  $\Delta_K(\alpha) = \inf_{g \in \text{Gal}_K} v_p(g(\alpha) - \alpha)$ , then there exists a  $\delta \in K$  that  $v_p(\alpha - \delta) \geq \Delta_K(\alpha) - p/(p-1)^2$ . ┘

*Proof:* We strengthen the assertion and use induction on  $n = [K(\alpha) : K]$  to prove that there is a  $\delta$  that  $v_p(\alpha - \delta) \geq \Delta_K(\alpha) - \sum_{k=0}^m \frac{1}{p^k(p-1)}$ , where  $p^{m+1}$  is the largest power of  $p$  that  $\leq n$ .

$n = 1$  is sure, let the minipoly of  $\alpha$  over  $K$  be  $P(X)$ . By lemma (19.1.5.3), there is a root  $\beta$  of  $P^{(q)}$  that  $v_p(\beta - \alpha) \geq v_p(\alpha)$  or minus a factor when  $n = p^{k+1}$ . Then for any  $\sigma$ ,  $v_p(\sigma(\beta) - \beta) \geq v_p(\sigma(\alpha) - \alpha)$  or minus a factor. Then  $\Delta(\beta) \geq \Delta(\alpha)$  or minus a factor. Now  $[K(\beta) : K] < [K(\alpha) : K] = n$ , so we can use induction hypothesis to get the result. □

**Remark (19.1.5.5).** The constant  $p/(p-1)^2$  can be replaced by  $1/(p-1)$ , and it is optimal: this is a theorem of Borgne in [Bor10]. ┘

**Thm. (19.1.5.6) [Ax-Sen-Tate].** If  $K \in p\text{-Field}$  and  $K \subset L \subset K^{\text{sep}}$ , then  $(\mathbb{C}_K)^{\text{Gal}_L} = \widehat{L}$ . ┘

*Proof:* Any  $\alpha \in \widehat{K}$  can be written as  $\sum \alpha_n$  with  $\alpha_n \in \overline{K}$ . Then  $\Delta_L(\alpha_n) \rightarrow \infty$ , and  $\alpha_n$  can be approximated by  $\delta_n \in L$  by lemma (19.1.5.4), thus  $\alpha \in \widehat{L}$ . □

**Prop. (19.1.5.7).** For any  $K \in p\text{-Field}$ ,  $\mathcal{O}_K^*$  thus also  $K^\times$  are locally compact. ┘

*Proof:* For  $n$  large,  $U^n \cong \mathfrak{m}_K^n$  by the  $p$ -adic logarithm map (19.1.5.16), so it is compact. □

### Local Fields

**Notation (19.1.5.8).**

- Let  $p \in \text{Prime}$  and  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa) \in p\text{-NField}$  (19.1.5.12). ┘

**Def. (19.1.5.9) [Local Fields].** A **local field** is a locally compact valued field. A local field is clearly complete. The category of local fields is denoted by **LField**. ┘

**Prop. (19.1.5.10) [Complete Valued Fields].** A complete Archimedean valued field must be  $\mathbb{R}$  or  $\mathbb{C}$ , by (12.2.3.18).

Any complete non-Archimedean valued field is discretely valued, and has finite residue fields of characteristic  $p \in \text{Prime}$ . Such a field is called a  **$p$ -adic local fields**. The category of  $p$ -adic local fields is denoted by  $p\text{-LField}$ . ┘

*Proof:* Cf. [Sutherland, L9]. ? □

**Remark (19.1.5.11).** The valuation ring of a  $p$ -adic local field is a DVR, thus the theory of Dedekind domains<sup>7</sup> applies to this case. ┘

**Prop. (19.1.5.12) [ $p$ -adic Local Fields].**  $p$ -adic local fields are precisely the finite extensions of the field  $\mathbb{Q}_p$  or  $\mathbb{F}_p((t))$ , called  **$p$ -adic number field** and  **$p$ -adic function field** respectively.

The category of  $p$ -adic number fields are denoted by  $p\text{-NField}$ , and the category of  $p$ -adic function fields are denoted by  $p\text{-FField}$ . ┘

*Proof:* Cf. [Neukirch Algebraic Number Theory P135]. □

### Group Structures

**Prop. (19.1.5.13).** For  $m > 0$ , there is an isomorphism  $(-)^m : U^n \cong U^{n+v(m)}$  when  $n$  is sufficiently large.  $\lrcorner$

*Proof:* Let  $m = u\pi^{v(m)}$ . For surjectivity, we need to find  $x$ , that  $1 + a\pi^{n+v(m)} = -(1 + x\pi^n)^m$ . i.e.

$$-a + ux + \pi^{n-v(m)}f(x) = 0.$$

This has a solution  $x$  by Hensel's lemma.  $\square$

**Cor. (19.1.5.14).**  $(K^*)^m$  is an open subgroup of  $K^*$ , and  $\bigcap_m (K^*)^m = 1$ . (Because if  $a \in \bigcap_m (K^*)^m = 1$ , then  $a$  is a unit, thus  $a \in \bigcap_m (U)^m = 1$ , thus  $a \in U^n$  for every  $n$  thus  $a = 1$ ).  $\lrcorner$

**Prop. (19.1.5.15).**  $[K^\times : (K^\times)^m] = m \cdot |m|_{\mathfrak{p}}^{-1} \cdot |\mu_m(K)|$ .  $\lrcorner$

*Proof:* Use the multiplicative Herbrand quotient (4.8.8.7),  $(K^* : (K^*)^m) = q_{0,m}(K^*)|\mu_m(K)|$ .  $q_{0,m}$  is additive, thus

$$q_{0,m}(K^*) = q_{0,m}(K/U)q_{0,m}(U/U^n)q_{0,m}(U^n).$$

$q_{0,m}(K/U) = m$ ,  $q_{0,m}(U/U^n) = 1$  as  $U/U^n$  is finite (4.8.8.9). It For  $q_{0,m}(U^n)$ , when  $n$  is large, it equals  $(U^n : U^{n+v(m)})$  by (19.1.5.13), which is  $|m|_{\mathfrak{p}}^{-1}$ .  $\square$

**Prop. (19.1.5.16) [ $p$ -adic Logarithm].** There is a unique multiplicative  $p$ -adic logarithm function  $\log : K^\times \rightarrow K$  that  $\log(\varpi_K) = 0$ , and for  $x \in \mathfrak{m}_K$ ,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

Moreover, for  $t > \frac{e_K}{p-1}$ , there is a map  $\exp : \mathfrak{p}^t \rightarrow U^t$  : which is an inverse to  $\log$  on  $U^t$ , inducing an isomorphism  $U^t \cong \mathfrak{p}^t$ .

Moreover, because of the uniqueness and continuity, this map in fact can be defined to be a locally analytic from  $\mathbb{C}_p^\times$  to  $\mathbb{C}_p$  with derivative  $1/x$ .  $\lrcorner$

*Proof:* This follows from (10.7.4.22). Notice that any such function must vanish on the Teichmüller lifts of elements in  $\kappa$ .  $\square$

**Prop. (19.1.5.17) [Multiplicative Group Structures].** For  $K \in p\text{-LField}$ ,

- If  $\text{char } K = 0$ , then  $\mathcal{O}_K^* \cong \mathbb{Z}/(p-1) \oplus \mathbb{Z}/(p^a) \oplus \mathbb{Z}_p^d$ , where  $d = [K : \mathbb{Q}_p]$  and  $a \in \mathbb{N}$ .
- If  $\text{char } K = p$ , then  $\mathcal{O}_K^* \cong \mathbb{Z}/(p-1) \oplus \mathbb{Z}_p^{\mathbb{N}}$ .

$\lrcorner$

*Proof:* Cf. [Neukirch P140].  $\square$

**Prop. (19.1.5.18) [Multiplicative Group Structures].** For  $p \in \mathbf{P}$ ,

$$\mathbb{Q}_p^\times \cong \begin{cases} \mathbb{Z} \times \mathbb{Z}_p \times \mathbb{Z}/(p-1) & , p \neq 2 \\ \mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}/(2) & , p = 2 \end{cases}$$

$\lrcorner$

**Cor. (19.1.5.19).** For  $p \neq \ell \in \mathbf{P}$ ,  $\mathbb{Q}_p \not\cong \mathbb{Q}_\ell$ .  $\lrcorner$

*Proof:* Because  $\mathbb{Q}_p^\times \not\cong \mathbb{Q}_\ell^\times$ : Firstly look at the torsion groups, then also look at  $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2 \not\cong \mathbb{Q}_\ell^\times/(\mathbb{Q}_\ell^\times)^2$ .  $\square$

**Prop. (19.1.5.20).** Any automorphism of  $\mathbb{R}$  or a  $p$ -adic number field is identity.  $\lrcorner$

*Proof:* It suffices to show that an automorphism is continuous. For  $\mathbb{R}$ , this is because  $a > 0 \iff a = b^2 \iff \sigma(a) = \sigma(b)^2 \iff \sigma(a) > 0$ , and  $\mathbb{Q}$  is dense in  $\mathbb{R}$ .

For a local field, we prove that  $\sigma(\mathcal{O}_K^*) \subset \mathcal{O}_K^*$ .  $\mathcal{O}_K^*$  is characterized by the property that  $\{n|y^n = x\}$  are infinite. This is because  $x^p = a$  has a root for  $a \in \mathcal{O}_K^*$  for  $p$  large prime, by Henselian lemma.  $\square$

**Def. (19.1.5.21) [Norm Groups].** For any extension of local fields  $L/K$ ,  $\text{Nm}_{L/K}$  is open, and  $\mathcal{N}_{L/K} = \text{Nm}_{L/K} L^\times \subset K^\times$  is called the **norm group** of  $L/K$ .  $\lrcorner$

### Extension Fields

**Prop. (19.1.5.22).**  $[\mathbb{Q}_p^{\text{ab,tame}} : \mathbb{Q}_p^{\text{ab,ur}}] < \infty$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (19.1.5.23).** The maximal unramified extension of  $\mathbb{F}_p((t))$  is  $T = \overline{\mathbb{F}_p}((t))$ , and the maximal tamely unramified extension of  $\mathbb{F}_p((t))$  is  $T(\sqrt[m]{t} | m \geq 1, (m, p) = 1)$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (19.1.5.24).** Any finite quotient group of  $\text{Gal}_K$  is solvable.  $\lrcorner$

*Proof:* This follows from (19.1.4.17)(19.1.4.16)(19.1.4.18) and (3.1.12.12) and (3.1.6.2).  $\square$

**Def. (19.1.5.25) [Tame Characters].** There is an isomorphism

$$\hat{t} : I_K/R_K \cong \text{Gal}(K^{\text{tame}}/K^{\text{ur}}) \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$$

and the projection to  $\mathbb{Z}_\ell$  is called the  $\ell$ -adic (additive) **tame character** of  $I_K$ .

Equivalently, if  $\ell \in \mathbf{P} \setminus \{p\}$ , for any compatible system of  $\ell^\infty$ -th roots  $\{\varpi_{\ell^n}\}$  of  $\varpi$  in  $\overline{K}$ , it is the character  $t_\ell$  of  $\text{Gal}_K$  that  $\sigma(\varpi_{\ell^n}) = \zeta_{p^n}^{t_\ell(\sigma)} \varpi_{\ell^n}$ .  $\lrcorner$

### Ramifications of Cyclotomic Fields

References are [Tat67].

**Prop. (19.1.5.26) [Unramified cases].** Let  $\#\kappa = q$  and  $n \in \mathbb{Z}_+ \setminus (p)$ , consider  $L = K(\zeta_n)$ . Then  $L/K$  is unramified of inertia degree  $f$  where  $f$  is the minimal number that  $q^f \equiv 1 \pmod n$ . And  $\mathcal{O}_L = \mathcal{O}_K[\zeta_n]$ .  $\lrcorner$

*Proof:*  $\zeta_n$  is a root of  $\Psi_n(X)|X^n - 1$ , which is separable over  $\kappa$ , so  $\Psi_n$  and  $\overline{\Psi_n}$  are both irreducible of the same degree by Hensel's lemma. So it is unramified, and  $\lambda$  is the minimal extension of  $\mathbb{F}_q$  that contains the  $n$ -th roots and are generated by it, thus the result by the theory of finite fields.

For the last assertion, notice it is unramified so  $\mathcal{O}_L = \mathcal{O}_K[\zeta_n] + p\mathcal{O}_L$  hence the result follows from Nakayama's lemma.  $\square$



**Cor. (19.1.5.27).** The maximal unramified extension of  $K$  is generated by adjoining all  $n$ -th roots where  $(n, p) = 1$ . This is because there is an inclusion relation and their residue field  $\overline{\mathbb{F}}_p$  is already generated by roots of unity.  $\lrcorner$

**Prop. (19.1.5.28)[Totally Ramified cases].** Consider  $\mathbb{Q}_p$  (other local fields behave different), we have the  $\mathbb{Q}(\zeta_{p^n})/\mathbb{Q}$  is totally ramified of degree  $\varphi(p^n)$  the Galois group is  $(\mathbb{Z}/(p^n))^*$ . The ring of valuation of  $\mathbb{Q}_p(\zeta_{p^n})$  is  $\mathbb{Z}_p[\zeta_{p^n}]$  and  $1 - \zeta_{p^n}$  is a uniformizer.  $\lrcorner$

*Proof:* Notice

$$\Psi_{p^n}(X) = \frac{X^{p^n} - 1}{X^{p^{n-1}} - 1} \equiv (X - 1)^{p^{n-1}(p-1)} \pmod{p}$$

and  $\Phi(1) = p$ . So  $\text{Nm}(1 - \zeta_{p^n}) = \prod(1 - \sigma(\zeta_{p^n})) = \Phi(1) = p$ . Thus  $\mathbb{Q}(\zeta_{p^n})/\mathbb{Q}$  is totally ramified of degree  $p^{n-1}(p-1)$  and  $1 - \zeta_{p^n}$  is a uniformizer. The ring of integer is generated by a uniformizer by (12.2.3.20) as the extension is totally ramified.  $\square$

**Prop. (19.1.5.29)[Infinite Cyclotomic Field].** Let  $K_n = K(\zeta_{p^n})$  and  $K_\infty = \cup K_n$ . Let  $\chi_{\text{cycl}}$  be the cyclotomic character, then  $\chi(\text{Gal}_K)$  is an open subgroup of  $\mathbb{Z}_p^*$ , thus contains a  $U_n$  for some  $n$ . Thus there is an isomorphism of groups:  $\chi^{-1}(U_n) \cap \text{Gal}_K / \chi^{-1}(U_{n+1}) \cap \text{Gal}_K \cong U_n / U_{n+1}$  which has order  $p$ , for  $n$  large.

So  $K_{n+1}/K_n$  is totally ramified of degree  $p$ , because  $K_n = K(\mu_{p^n})$ , and its value group extension is of degree  $p$ , too.

And  $[K_n : \mathbb{Q}(\mu_{p^n})]$  is decreasing and eventually equals to  $[K_\infty : \mathbb{Q}(\mu_{p^\infty})]$ . This is because its order equals  $\chi^{-1}(U_n)/\chi^{-1}(U_n) \cap \text{Gal}_K \cong \chi^{-1}(U_n) \text{Gal}_K / \text{Gal}_K$ , which is eventually  $\ker(\chi) \text{Gal}_K / \text{Gal}_K$ , because  $U_n \subset \chi(\text{Gal}_K)$ .  $\lrcorner$

**Cor. (19.1.5.30).** For  $n$  large, if  $x_i$  is a set of basis of  $\mathcal{O}_{K_n}$  over  $\mathcal{O}_{\mathbb{Q}(\mu_{p^n})}$ , then they form a basis of  $K_N$  over  $F_N$  for all  $N \geq n$ . This is because it generate  $K_N$  over  $F_N$  and  $[K_N : F_N] = [K_n : F_n]$ .  $\lrcorner$

**Prop. (19.1.5.31).**  $p^n v_p(\mathfrak{D}_{K_n/F_n})$  is bounded and eventually constant. In particular  $v_p(\mathfrak{D}_{K_n/F_n})$  converges to 0.  $\lrcorner$

*Proof:* Cf.[Galois representation Berger P20].  $\square$

**Cor. (19.1.5.32).** If  $L/K$  is a finite extension, then  $\text{tr}_{L_\infty/K_\infty}(\mathfrak{m}_{L_\infty}) = \mathfrak{m}_{K_\infty}$ .  $\lrcorner$

*Proof:* By (19.1.4.3) and the fact  $\text{Gal}(L_\infty/K_\infty) \cong \text{Gal}(L_n/K_n)$  for  $n$  large by (19.1.5.29), we have  $\text{tr}_{L_\infty/K_\infty}(\mathfrak{m}_{L_n}) = \mathfrak{m}_{K_n}^{c_n}$ , where  $c_n = [v_{K_n}(\mathfrak{m}_{L_n} \mathcal{D}_{L_n/K_n})]$ . By the above proposition,  $c_n$  is bounded by a  $c$ . But if  $x \in \mathfrak{m}_{K_\infty}$ ,  $x \in \mathfrak{m}_{K_n}^c$  for  $n$  large, so  $x \in \text{tr}_{L_\infty/K_\infty}(\mathfrak{m}_{L_\infty})$ .  $\square$

**Lemma (19.1.5.33).** For any  $\delta > 0$ , when  $n$  is large, if  $x \in \mathcal{O}_{K_{n+1}}$  and  $g \in \text{Gal}(K_{n+1}/K_n)$ ,  $v_p(g(x) - x) \geq \frac{1}{p-1} - \delta$ . In particular,  $v(N_{K_{n+1}/K_n}(x) - x^p) \geq \frac{1}{p-1} - \delta$ .  $\lrcorner$

*Proof:* Choose a basis  $e_i$  of  $\mathcal{O}_{K_n}/\mathcal{O}_{F_n}$ , then  $e_i^*$  is a basis for  $\mathcal{D}_{K_n/F_n}$ , and if  $x_i = \text{tr}_{K_{n+1}/F_{n+1}}(xe_i)$ , then  $x_i \in \mathcal{O}_{F_{n+1}}$  and  $x = \sum x_i e_i$ , by (19.1.5.30), and we have by (19.1.4.30),  $v(g(x_i) - x_i) \geq 1/(p-1)$ , so when  $n$  is large, by (19.1.5.31),  $v(x_i) \geq -\delta$ , so the require is satisfied.  $\square$

**Prop. (19.1.5.34).** if  $\delta > 0$  and  $I$  is the ideal of elements of valuation  $\geq 1/(p-1) - \delta$ , then for  $n$  large, there is a map  $x \mapsto x^p : \mathcal{O}_{K_{n+1}}/I \cap \mathcal{O}_{K_{n+1}} \rightarrow \mathcal{O}_{K_n}/I \cap \mathcal{O}_{K_n}$ , and it is surjective.  $\lrcorner$

*Proof:* For  $n$  large, choose a uniformizer  $\pi_{n+1}$  of  $K_{n+1}$ , then  $\pi_n = N_{K_{n+1}/K_n}(\pi_{n+1})$  is the uniformizer of  $K_n$  because it is totally ramified (19.1.5.29), so any element  $x \in \mathcal{O}_{K_{n+1}}$  can be written as  $\sum \pi_{n+1}^i [x_i]$ , where  $x_i \in k_{K_{n+1}} = k_{K_\infty}$ . Then  $x^p \equiv \sum \pi_{n+1}^{pi} [x_i]^p \equiv \sum \pi_n^i [x_i^p] \pmod{I}$  by the above proposition. And the surjection is verbatim.  $\square$

**Def. (19.1.5.35) [Tate's Normalized Trace, [Tat67]].** The function  $R_n(x) = p^{-k} \text{tr}_{\mathbb{Q}(\mu_{p^{n+k}})/\mathbb{Q}(\mu_{p^n})}(x)$  is compatible with varying  $k$  and defines a  $\mathbb{Q}(\mu_{p^n})$ -linear projection from  $\mathbb{Q}(\mu_{p^\infty})$  to  $\mathbb{Q}(\mu_{p^n})$ , and it commutes with the  $\text{Gal}_{\mathbb{Q}}$ -action, called the **Tate's normalized trace**.

From (19.1.5.36) it's easily verified that  $R_n(\mathcal{O}_{\mathbb{Q}(\mu_{p^{n+k}})}) \subset \mathcal{O}_{\mathbb{Q}(\mu_{p^n})}$ , thus  $R_n(\pi_n^j \mathcal{O}_{\mathbb{Q}(\mu_{p^{n+k}})}) \subset \pi_n^j \mathcal{O}_{\mathbb{Q}(\mu_{p^n})}$ . So we have  $v(R_n(x)) \geq v(x) - v(\pi_n)$ . So  $R_n$  extends by continuity to a map  $R_n : \widehat{\mathbb{Q}(\mu_{p^\infty})} \rightarrow \widehat{\mathbb{Q}(\mu_{p^n})}$ . If  $x \in \mathbb{Q}(\mu_{p^\infty})$ , then  $R_n(x) = x$  for  $n$  large, thus  $R_n(x) \rightarrow x$  for any  $x \in \widehat{\mathbb{Q}(\mu_{p^\infty})}$ .

Now for  $K \in p\text{-NField}$ , when  $n$  is large, if  $e_i$  is a set of basis of  $\mathcal{O}_{K_n}/\mathcal{O}_{\mathbb{Q}(\mu_{p^n})}$ , then for any  $x \in \mathcal{O}_{K_{n+k}}$ ,  $x = \sum x_i e_i^*$ , where  $x_i = \text{tr}_{K_{n+k}/F_{n+k}}(x e_i) \in \mathcal{O}_{F_{n+k}}$ , as in the proof of (19.1.5.33). So we can define  $R_n(x) = \sum R_n(x_i) e_i^*$ . Notice this is defined only for  $n$  large, and is independent of  $e_i$  chosen, and by the following lemma, it is continuous and extends to a  $K_n$ -linear projection  $R_n : \widehat{K_\infty}$  to  $K_n$ .  $\lrcorner$

**Lemma (19.1.5.36).** Let  $k \geq 0$  and  $n \geq 1$ , then  $R_n(\zeta_{p^{n+k}}^j) = 1$  for  $j = 0$  and vanishes otherwise.  $\lrcorner$

*Proof:* This is clear from the fact  $\text{tr}_{\mathbb{Q}(\mu_{p^{n+k}})/\mathbb{Q}(\mu_{p^n})}(\zeta_{p^{n+k}}^j) = \zeta_{p^{n+k}}^j \sum_{\eta^{p^k}=1} \eta^j$ .  $\square$

**Lemma (19.1.5.37).** for any  $\delta \in \mathbb{R}_+$ , when  $n$  is large,  $v(R_n(x)) \geq v(x) - \delta$ .  $\lrcorner$

*Proof:* We have  $v(x_i) \geq v(x) - v(\pi_{n+k})$  by  $\mathbb{Q}(\mu_{p^{n+k}})$ -linearity, and  $v(R_n(x_i)) \geq v(x_i) - v(\pi_n)$  as in (19.1.5.35), and  $v(e_i^*) \geq -\delta$  when  $n$  is large, by (19.1.5.31). Thus the result.  $\square$

**Prop. (19.1.5.38) [Refinement of Hilbert's Theorem90].** There is a decomposition of  $\widehat{K_\infty} = K_n \oplus X_n$ , where  $X_n = \ker R_n$ . If  $\delta > 0$ , then for  $n$  large,  $\alpha \in \mathbb{Z}_p^*$  and  $\gamma_n$  that  $\chi(\gamma_n)$  is a topological generator of  $\Gamma_{F_n}$ ,  $1 - \alpha\gamma_n : X_n \rightarrow X_n$  (because  $\gamma_n$  commutes with  $R_n$ ) is invertible and

$$v_p((1 - \alpha\gamma_n)^{-1}x) \geq v_p(x) - 1/(p-1) - \delta,$$

unless  $\alpha = -1$  and  $p = 2$ , in which case it is only invertible on  $X_{n+1}$ .  $\lrcorner$

*Proof:* As usual,  $x_i$  is a basis of  $\mathcal{O}_{K_n}/F_n$ , then  $x = \sum x_i e_i^*$ ,  $x_i = \text{tr}_{K_\infty/F_\infty}(x e_i) \in \widehat{F_\infty}$ , and  $R_n(x) = 0$ . Then  $(1 - \alpha\gamma_n)$  acts on  $x_i$ , so it reduce to the case  $K = \mathbb{Q}_p$ .

**Injectivity:** If  $\alpha = 1$ , this is Ax-Sen-Tate theorem. In other situations,  $(1 - \alpha\gamma_n)(R_{n+k}(x)) = 0$  for all  $k \geq 0$ , so  $R_{n+k}(x) = \alpha^{p^k} \gamma_n^{p^k}(R_{n+k}(x)) = \alpha^{p^k} R_{n+k}(x)$ , so  $R_{n+k}(x) = 0$ , hence  $x = 0$  by continuity.

**Surjectivity:** Let  $F_{n+k}^* = \bigoplus_{j=1, p \nmid j}^{p^k-1} F_n \zeta_{p^{n+k}}^j$ , then  $F_{n+k} = F_n^* \oplus F_{n+1}^* \oplus \dots \oplus F_{n+k}^*$ , and  $F_{n+k} \cap X_n = F_{n+1}^* \oplus \dots \oplus F_{n+k}^*$ . Now if  $x = \sum_{j=1, p \nmid j}^{p^k-1} x_j \zeta_{p^{n+k}}^j$  with  $x_j \in \mathcal{O}_{F_n}$ , then

$$x = (1 - \alpha^{p^{k-1}} \gamma_n^{p^{k-1}}) \sum_{j=1, p \nmid j}^{p^{k-1}-1} x_j \frac{\zeta_{p^{n+k}}^j}{1 - \alpha^{p^{k-1}} \zeta_p^j}.$$

Now  $v_p(1 - \alpha^{p^{k-1}} \zeta_p^j) \leq 1/(p-1)$ , and

$$(1 - \alpha\gamma_n)^{-1} = \frac{1 - \alpha^{p^{k-1}} \gamma_n^{p^{k-1}}}{1 - \alpha\gamma_n} (1 - \alpha^{p^{k-1}} \gamma_n^{p^{k-1}})^{-1},$$

so  $\alpha_n : 1 - \alpha\gamma_n : F_{n+k}^* \rightarrow F_{n+k}^*$  is invertible and

$$v_p((1 - \alpha\gamma_n)^{-1}x) \geq v_p(x) - 1/(p-1) - v_p(\zeta_{p^n} - 1)$$

holds. And the assertion holds by uniform continuity.  $\square$

### Miscellaneous

**Def. (19.1.5.39) [Lattices].** If  $K \in \mathbf{LField}$  and  $V \in \mathbf{Vect}^{\text{fd}}/K$ , then a **lattice in**  $V$  is

- a compact open  $\mathcal{O}_K$ -submodule of  $V$  if  $K$  is non-Archimedean. Equivalently, it is a f.g.  $\mathcal{O}_K$ -submodule that contains a  $K$ -basis of  $V$ .
- a discrete subgroup  $\Lambda$  of  $V$  s.t.  $V/\Lambda$  is compact if  $K$  is Archimedean.

┘

**Prop. (19.1.5.40).** If  $K \in p\text{-Field}$ ,  $V \in \mathbf{Vect}_K^{\text{fd}}$  and  $\Lambda$  be an  $\mathcal{O}_K$ -submodule of  $V$ , then  $\Lambda$  is a lattice in  $V$  iff it is a finite  $\mathcal{O}_K$ -module and generate  $V$  as a  $K$ -vector space. ┘

*Proof:* If  $\Lambda$  is an  $\mathcal{O}_K$ -submodule, then it clearly generates  $V$  as a  $K$ -vector space, and the f.g.  $\mathcal{O}_K$ -submodules of  $\Lambda$  is a cover of  $\Lambda$ , which has a finite subcover as  $\Lambda$  is compact open, thus  $\Lambda$  is f.g. over  $\mathcal{O}_K$ .

Conversely, if  $\Lambda$  is a f.g.  $\mathcal{O}_K$ -submodule that generate  $V$  as a  $K$ -vector space, then it is a quotient of  $\mathcal{O}_K^n$  for some  $n$ , thus compact. And let  $S$  be a  $K$ -basis of  $V$  contained in  $\Lambda$ , then  $\mathcal{O}_K S$  is an open nbhd of  $0 \subset \Lambda$ , which means it is open. ┘

**Prop. (19.1.5.41).** Let  $V \in \mathbf{Vect}_{\mathbb{R}}^{\text{fd}}$  and  $\Lambda < V$ , then the following are equivalent:

- $\Lambda$  is a lattice in  $V$ .
- $\Lambda$  is discrete and contains an  $\mathbb{R}$ -basis of  $V$ .
- $\Lambda$  has a  $\mathbb{Z}$ -basis that is also an  $\mathbb{R}$ -basis of  $V$ .

┘

**Remark (19.1.5.42).** WARNING: These are not equivalent to  $\Lambda$  is a f.g.  $\mathbb{Z}$ -module and generated  $V$  as an  $\mathbb{R}$ -vector space: Consider  $\mathbb{Z}\{1, \alpha\} \subset \mathbb{R}$ . ┘

*Proof:*  $1 \rightarrow 2$ : Let  $W$  be a complementary subspace of  $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} \subset V$ , then  $W \cong \overline{W}$  is closed thus compact in  $V/\Lambda$ , which implies  $W = 0$ .

$2 \rightarrow 3$ : We may assume  $V = \mathbb{R}^n$  and  $\Lambda$  contains the canonical basis  $e_1, \dots, e_n$ , then  $\Lambda$  is generated by  $S = \{\lambda \in \Lambda \mid \max_i |\lambda_i| \leq 1\}$ . Because  $\Lambda$  is discrete thus closed in  $V$ ,  $S$  is finite. Thus  $\Lambda$  is a f.g.  $\mathbb{Z}$ -module, and has no torsion, thus a free Abelian group. But  $\Lambda/\mathbb{Z}\{e_i\}$  is finite, thus  $\Lambda \cong \mathbb{Z}^n$ . Hence a basis of  $\Lambda$  must also be a basis of  $V$ .

$3 \rightarrow 1$  is clear. ┘

**Def. (19.1.5.43) [Dual Lattice].** If  $K \in p\text{-Field}$  and  $V \in \mathbf{Vect}_K^{\text{fd}}$ ,  $B$  is a non-degenerate bilinear form on  $V$  and  $\psi$  is a non-trivial character of  $K$ , then for any lattice  $L \subset V$ , the **dual lattice**  $L' = \{u \in V \mid \psi(2(B(u, v))) = 1, \forall v \in L\}$  is also a lattice in  $V$ . ┘

*Proof:* It is an open group by no-small-subgroup argument, as  $\psi, B$  are continuous. it is compact because take any basis  $\{v_1, \dots, v_n\}$  of  $V$  that  $v_i \in L$ ,  $\{u \in V \mid \psi(2(B(u, v_i))) = 1, \forall i\}$  is compact. ┘

## 6 Ultrarnormed Banach Spaces

**Notation (19.1.6.1).**

- Let  $(K, v, \mathcal{O}_K, \mathfrak{p}_v, \varpi, k)$  be a complete non-Archimedean valued field (of rank 1),  $K^0 = \mathfrak{p}_v$ , and  $\varpi \in K^\times, |\varpi| < 1$  a uniformizer. If  $\mathcal{O}_K$  is DVR, assume  $(\varpi) = \mathfrak{p}_v$ .

┘

### Ultrarnormed $K$ -modules

**Prop. (19.1.6.2).** Any normed  $K$ -module is weakly-Cartesian.  $\lrcorner$

*Proof:* Cf.[Non-Archimedean Analysis P92].  $\square$

**Cor. (19.1.6.3).** Any two valuation on a finite  $K$ -vector space are equivalent.  $\lrcorner$

*Proof:* Cf.[Non-Archimedean Analysis P93].  $\square$

**Prop. (19.1.6.4).** If  $V$  is a normed  $\mathbb{Q}_p$  vector space and  $V_0 = \{x \in V \mid |x| \leq 1\}$ , then  $V^\wedge \cong (V_0)_p^\wedge[p^{-1}]$ .  $\lrcorner$

### Ultrarnormed Banach Spaces

**Def. (19.1.6.5) [Ultrarnormed Banach Spaces].** In the non-Archimedean case, an **ultrarnormed Banach algebra** is defined as in (12.7.4.1), but additionally  $|a + b| \leq \max\{|a|, |b|\}$ .  $\lrcorner$

**Def. (19.1.6.6) [Uniform Banach Space].** For a complete non-Archimedean field  $K$  and a Banach algebra  $R$ , define  $R^0$  to be the ring of **power bounded elements**. Then it is a subring, and it is open, as it contains the closed ball  $\overline{D}(0, 1)$ .

Recall  $R$  is called uniform if  $R^0$  is itself bounded in  $R$  (12.2.1.5). Notice a uniform Banach space must be reduced, because a nilpotent element is clearly power bounded, and any scalar multiple of it is nilpotent.  $\lrcorner$

**Lemma (19.1.6.7).** Fix a uniformizer  $t$  in a non-Archimedean complete field  $K$ , if  $|K^*|$  is discrete, then if  $A$  is a  $t$ -adically complete and  $t$ -torsion-free  $K^0$ -algebra, let  $R = A[t^{-1}]$ , then the norm

$$|f| = \inf\{|t|^n \mid f \in t^n A\},$$

then this makes  $R$  into a  $K$ -Banach space that the  $t$ -adic topology of  $A$  is the same as the metric topology of  $A$ , so  $A \subset R_{\leq 1} \subset R^0$ .

Notice if  $|K^*|$  is not discrete but there is a pseudo-uniformizer  $t$  that has a compatible system of  $p^n$ -th roots, if  $A$  is a  $t$ -adically complete and  $t$ -torsion-free  $K^0$ -algebra, let  $R = A[t^{-1}]$ , then the norm

$$|f| = \inf\{|t|^{\frac{n}{p^k}} \mid f \in t^{\frac{n}{p^k}} A\},$$

then this makes  $R$  into a  $K$ -Banach space that the  $t$ -adic topology of  $A$  is the same as the metric topology of  $A$ , so  $A \subset R_{\leq 1} \subset R^0$ , and in this case  $R^0 = A_* = \text{Hom}((t^{\frac{1}{p^\infty}}), A)$  (5.7.2.2).  $\lrcorner$

**Prop. (19.1.6.8) [Uniform  $K$ -Banach Space and  $K^0$ -Algebra].** Fix a pseudo-uniformizer  $t$  in a non-Archimedean complete field  $K$ , the following category are equivalent:

- The category  $\mathcal{C}$  of uniform Banach  $K$ -algebras  $R$ .
- The category  $\mathcal{D}_{tic}$  of  $t$ -adically complete and  $t$ -torsionfree  $K^0$ -algebras  $A$  with  $A$  totally integrally closed (5.2.1.1) in  $A[t^{-1}]$ .

$\lrcorner$

*Proof:* The functor  $F : \mathcal{C} \rightarrow \mathcal{D}_{tic} : \text{if } R \text{ is uniform Banach space, then } F(R) = R^0$ :  $R^0$  is open subring by (19.1.6.6), and  $R^0 \in B(0, r]$  for some  $r > 0$  by uniformity. As  $R$  is  $K$ -Banach,  $\cap t^n B(0, r] = 0$ , so  $R^0$  is  $t$ -adically separated, and also it is complete. If  $f^\mathbb{N} \in t^{-k} R^0 \subset t^{-k} B(0, r]$ ,

then clearly  $f$  is power bounded thus  $f \in R^0$ , so  $R^0$  is totally integrally closed in  $R$ .  $R \rightarrow R^0$  is preserved by continuous mappings, so  $F$  is truly a functor.

Conversely, lemma above(19.1.6.7) shows  $R = A[t^{-1}]$  is a  $K$ -Banach algebra, this is a functor  $G : \mathcal{D}_{tic} \rightarrow \mathcal{C}$ , and  $A \subset R^0$ . We show  $A = R^0$ , as this is equivalent to  $FG \cong \text{id}$ : as the  $t$ -adic topology and metric topology are the same(19.1.6.7), if  $t^c f^{\mathbb{N}} \subset A$  for some  $c$ , thus  $f$  is totally integral over  $A$ , thus  $f \in A$  by tic.

Finally, we need to show  $GF \cong \text{id}$ , which in fact that the given Banach algebra norm on  $R$  is equivalent to the norm  $|\cdot|'$  given in(19.1.6.7) w.r.t  $R^0$ .  $R'_{<1} \subset R^0 \subset R_{\leq c}$  by uniformity, and conversely,  $R_{\leq 1} \subset R^0 \subset R'_{\leq 1}$ , thus this two norms are equivalent.  $\square$

**Prop.(19.1.6.9).** Let  $\varphi : A \rightarrow B$  be a  $k$ -homomorphism between  $k$ -Banach algebras that there is a family  $\mathfrak{B}$  of ideals of  $B$  that for each  $b \in \mathfrak{B}$ :

- $B$  is closed and  $\varphi^{-1}(b)$  is closed in  $A$ .
- $\dim_k B/b < \infty$ .
- $\cap_{b \in \mathfrak{B}} b = (0)$ .

Then  $\varphi$  is continuous.  $\lrcorner$

*Proof:* Consider the map  $A/\varphi^{-1}(b) \rightarrow B/b$  with the residue norms, Cf.[non-Archimedean analysis P167].  $\square$

**Cor.(19.1.6.10).** Let  $\varphi : A \rightarrow B$  be a  $k$ -homomorphism between Noetherian  $k$ -Banach algebras that there is a family  $\mathfrak{B}$  of ideals of  $B$  that for each  $b \in \mathfrak{B}$ ,  $\dim_k B/b < \infty$  and  $\cap_{b \in \mathfrak{B}} b = (0)$ , then  $\varphi$  is continuous. (Because the closedness condition is automatic by(19.1.6.13)).  $\lrcorner$

**Cor.(19.1.6.11).** All complete  $k$ -algebra norms on a Noetherian  $k$ -algebra  $B$  satisfying the condition of(19.1.6.10) are equivalent.  $\lrcorner$

### Modules over $K$ -Banach Spaces

**Prop.(19.1.6.12).** If  $M$  is a normed module over a  $k$ -Banach algebra  $A$ , if the completion of  $M$  is a finite  $A$ -module, then  $M$  is complete.  $\lrcorner$

*Proof:* There are morphism  $\pi : A^n \rightarrow \widehat{M}$  that are surjective continuous, so by open mapping theorem(12.7.2.5), this map is open, so  $\sum \check{A}x_i = \pi(A^n)$  is a nbhd of 0 in  $\widehat{M}$ , because  $\check{A}$  is open(19.1.1.8) and then  $\widehat{M} = M + \sum \check{A}x_i$ , because  $M$  is dense in  $\check{M}$ , then we are done by(19.1.1.9).  $\square$

**Cor.(19.1.6.13) [Noetherian and Submodule Closed].** For a complete normed module over a  $k$ -Banach algebra  $A$ ,  $M$  is Noetherian iff all submodules of  $M$  are closed. In particular,  $A$  is Noetherian iff all ideals of  $A$  are closed.  $\lrcorner$

*Proof:* If  $M$  is Noetherian, then the completion of any submodule is finite over  $A$ , so it is complete hence closed by(19.1.6.12). Conversely, if any ideal of  $M$  is closed, then for a chain of ideals of  $M : \cup M_i = M'$ ,  $M'$  is complete hence Baire space by(4.4.9.2), so some  $M_i$  must contain a nbhd of  $M'$ , because it is an ideal, but then  $M_i = M'$ .  $\square$

## 7 $p$ -adic Power Series

References are [ $p$ -adic Analysis, Robert] and [Kob84].

**Notation(19.1.7.1).**

- Let  $p \in \text{Prime}$ .
- Let  $K \in p\text{-Field}$ (19.1.5.1).

┘

### Power Series

References are [Dwo60, Kob84, Gou20].

**Prop. (19.1.7.2) [ $p$ -adic Cauchy-Hadamard].** Let  $(R, \mathfrak{m})$  be a CDVR of mixed characteristic  $(0, p)$ , then for any power series  $a_0 + a_1x + \dots + a_nx^n + \dots$  in  $R$ , take  $1/r = \overline{\lim} |a_n|^{1/n}$ , where we assume  $1/0 = \infty$  and  $1/\infty = 0$ , then

- The series converges absolutely for every  $|x| < r$ , and if  $\rho < r$ , then the convergence is uniform for  $|x| \leq \rho$ . In particular, the series converges to a continuous function on  $\mathbb{D}(0, r)$  by (4.4.8.21).
- If  $|x| \geq r$ , the terms are unbounded, and the series diverges.

By analogy with (19.1.7.2),  $r$  is called the **radius of convergence** of the sequence. ┘

*Proof:* The proof is the same as that of (19.1.7.2). □

**Cor. (19.1.7.3) [Convergence of Power Series].** Let  $(R, \mathfrak{m})$  be a CDVR of mixed characteristic  $(0, p)$ , then

- If  $f(T) = \sum_{n \geq 1} \frac{a_n}{n} T^n \in (R \otimes \mathbb{Q})[[T]]$  with  $a_i \in R$ , then  $r(f) \geq 1$ .
- If  $g(T) = \sum_{n \geq 1} \frac{b_n}{n!} T^n \in (R \otimes \mathbb{Q})[[T]]$  with  $b_i \in R$ , then  $r(f) \geq |p|^{1/(p-1)}$ .

┘

*Proof:* 1:  $v(a_n x^n / n) \geq nv(x) - v(n) \geq nv(x) - \log_p(n)v(p)$  converges to  $\infty$  for  $n \rightarrow \infty$  when  $v(x) > 0$ .

2:  $v(b_n x^n / n!) \geq nv(x) - v(n!) \geq nv(x) - (n-1)v(p)/(p-1)$  (2.6.1.33) converges to  $\infty$  for  $n \rightarrow \infty$  when  $v(x) > v(p)/(p-1)$  □

**Prop. (19.1.7.4) [Taylor Expansions give no Analytic Extensions].** If  $f(T) \in \mathbb{C}_K[[T]]$  converges for some  $\alpha \in \mathbb{C}_K$ , then its Taylor expansion at  $T = \alpha$

$$g(T) = \sum_{m \in \mathbb{N}} b_m (T - \alpha)^m \in \mathbb{C}_K[[T]] \text{ where } b_m = \sum_{n \geq m} \binom{n}{m} a_n \alpha^{n-m}$$

has the same region of convergence as  $f$ , and they are equal on this region. ┘

*Proof:* Use the non-Archimedean inequality to prove that  $g$  converges whenever  $f$  converges, and then the assertion follows by switching  $f$  and  $g$ . □

**Prop. (19.1.7.5) [Euclidean Division].** Let  $f, g \in \mathcal{O}_K[[T]]$  and assume  $f = a_0 + a_1T + \dots$  where  $a_i \in \mathfrak{m}_K$  for  $0 \leq i \leq n-1$ , and  $a_n \in \mathcal{O}_K^*$ , then we can uniquely write

$$g = qf + r$$

where  $q \in \mathcal{O}_K[[T]]$  and  $r \in \mathcal{O}_K[T]$  is a polynomial of degree  $\leq n-1$ . ┘

*Proof:* Cf. [Mazur Control Theorem, P4]. □

**Def. (19.1.7.6) [Distinguished Polynomials].**  $P(T) \in \mathcal{O}_K[T]$  is called a **distinguished polynomial** if  $P(T) = T^n + a_{n-1}T^{n-1} + \dots + a_0$ , where  $a_i \in \mathfrak{m}_K$  for  $0 \leq i \leq n-1$ . ┘

**Prop. (19.1.7.7) [ $p$ -adic Weierstrass Preparation].** Every power series  $f(T) \in \mathcal{O}_K[[T]]^\times$  can be written uniquely as  $f(T) = \varpi_K^\mu P(T)U(T)$ , where  $\mu \in \mathbb{N}$ ,  $P(T)$  is a distinguished polynomial (19.1.7.6), and  $U(T) \in \mathcal{O}_K[[T]]^*$ .  $\lrcorner$

*Proof:* Cf. [Mazur Control Theorem, P5].  $\square$

**Cor. (19.1.7.8).** If  $f(T) \in \mathcal{O}_K[[T]]^\times$ , there are only f.m.  $x \in \mathcal{O}_{\mathbb{C}_K}$  with  $f(x) = 0$ .  $\lrcorner$

**Cor. (19.1.7.9).** If  $P(T) \in \mathcal{O}_K[T]$  is a distinguished polynomial, and  $g(T) \in \mathcal{O}_K[T]$ . Suppose  $g(T)/P(T) \in \mathcal{O}_K[[T]]$ , then  $g(T)/P(T) \in \mathcal{O}_K[T]$ .  $\lrcorner$

*Proof:* Let  $g(T) = f(T)P(T) \in \mathcal{O}_K[[T]]$ . Suppose  $x \in \mathbb{C}_K$  is a root of  $P(T)$ , then  $|x| < 1$ , thus  $f(x)$  converges, and  $g(x) = 0$ . Then we can divide both  $g(T)$  and  $P(T)$  by  $(T - x)$ , and by induction (and change  $K$  by a larger field), we can get  $P(T)$  divides  $g(T)$  as polynomials.  $\square$

**Prop. (19.1.7.10).**  $\mathcal{O}_K[[T]]$  is a UFD.  $\lrcorner$

*Proof:* It follows from  $p$ -adic Weierstrass preparation theorem (19.1.7.7) that each element can be written as products of  $\pi_K$  and irreducible distinguished polynomials. And irreducible distinguished polynomials are all primes: if  $P(T)$  is irreducible distinguished and  $a, b \in \mathcal{O}_K[[T]]$  s.t.  $P(T) \mid ab$ , then we can use (19.1.7.5) to assume  $a, b$  are polynomials, and then by (19.1.7.9),  $P_n(T) \mid ab$  as polynomials. Then  $P(T) \mid a$  or  $P(T) \mid b$  because  $P(T)$  is irreducible and  $\mathcal{O}_K[T]$  is a UFD (3.2.3.8). Then  $\mathcal{O}_K[[T]]$  is a UFD, by (3.2.3.6).  $\square$

**Cor. (19.1.7.11).** If  $f \in \mathcal{O}_K[[T]]$  is not invertible, then  $\mathcal{O}_K[[T]]/(f)$  is infinite.  $\lrcorner$

**Lemma (19.1.7.12).** Let  $f, g \in \mathcal{O}_K[[T]]$  be relatively prime (3.2.3.4), then the ideal  $(f, g)$  is of finite index in  $\mathcal{O}_K[[T]]$ .  $\lrcorner$

*Proof:* Cf. [Mazur Control Theorem, P8].  $\square$

**Prop. (19.1.7.13)** [ $\text{Spec}(\mathcal{O}_K[[T]])$ ]. The prime ideals of  $\mathcal{O}_K[[T]]$  are  $0, (\varpi_K, T), (\varpi_K)$ , and the ideal  $(P(T))$  where  $P(T)$  is distinguished and irreducible. And the ideal  $(p, T)$  is the unique maximal ideal.  $\lrcorner$

*Proof:* The above ideals are all primes by (19.1.7.10). And if  $\mathfrak{p}$  is a non-zero prime, suppose  $f$  is a non-zero polynomial in  $\mathfrak{p}$  of minimal degree, which exists by  $p$ -adic Weierstrass preparation theorem (19.1.7.7). Then because  $\mathfrak{p}$  is a prime ideal, by Weierstrass preparation, we can assume that  $f = p$  or  $f$  is a distinguished irreducible polynomial. Suppose  $(f) \neq \mathfrak{p}$ , then there exists  $g \in \mathfrak{p} \setminus (f)$ . Then  $f, g$  are relatively prime, then  $\mathcal{O}_K[[T]]/(f, g)$  is finite (19.1.7.12). Thus  $\mathfrak{p}$  is also of finite index in  $\mathcal{O}_K[[T]]$ . Then  $p^n \in \mathfrak{p}$  for some  $n$ , and by primality,  $p \in \mathfrak{p}$ . And there are two  $T^i, T^j$  that are equal in  $\mathcal{O}_K[[T]]/\mathfrak{p}$ , so  $(1 - T^{j-i})T^i \in \mathfrak{p}$ . But  $(1 - T^{j-i})$  is invertible, so  $T \in \mathfrak{p}$ . Then  $(p, T) \subset \mathfrak{p}$ , and  $\mathfrak{p} = (p, T)$ .  $\square$

For the last assertion, notice if  $P(T)$  is distinguished, then  $P(T) \in (p, T)$ .  $\square$

**Prop. (19.1.7.14) [Characterizing Integral Power Series, Dwork [Dwo60]].** Let  $R \in \mathbb{C}\text{Ring}_{\mathbb{Z}_p}$  be integral domain,  $\varphi : R \rightarrow R$  be a lift of the Frobenius of  $R/(p)$ , and  $R_0 = R[\frac{1}{p}]$ . Let  $F(\underline{X}) \in 1 + (\underline{X})R_0[[\underline{X}]]$ , then  $F(\underline{X}) \in 1 + (\underline{X})R[[\underline{X}]]$  iff

$$\frac{F^\varphi(X_1^p, \dots, X_n^p)}{(F(\underline{X}))^p} \in 1 + p(\underline{X})R[[\underline{X}]].$$

$\lrcorner$

*Proof:* If  $F(\underline{X}) \in R[[\underline{X}]]$ , then  $F^\varphi(X_1^p, \dots, X_n^p) \equiv F(\underline{X})^p \pmod{p}$ . For the other direction, denote  $F(\underline{X}) = \sum_{\alpha} a_{\alpha} X^{\alpha}$ , then

$$\sum_{\alpha} \varphi(a_{\alpha}) X^{p\alpha} = \left( \sum_{\alpha} a_{\alpha} X^{\alpha} \right)^p (1 + pT \sum_{\alpha} b_{\alpha} X^{\alpha})$$

where  $b_{\alpha} \in R$ . We use induction on  $k$  to show that  $a_{\alpha} \in R$  for any  $|\alpha| \leq n$ . The case  $k = 0$  is true, and if the cases  $k < n$  are all true, for  $k = n$  consider the above equation modulo  $(p, \{X^{\alpha}\}_{|\alpha| \geq n+1})$ , then we get

$$\sum_{|\alpha| < n} \varphi(a_{\alpha}) X^{p\alpha} \equiv \left( \sum_{|\alpha| \leq n} a_{\alpha} X^{\alpha} \right)^p \equiv \sum_{|\alpha| < n} a_{\alpha}^p T^{p\alpha} + p \sum_{|\alpha| = n} a_{\alpha} X^{\alpha}.$$

Thus for any  $|\alpha| = n$ , we get  $a_{\alpha} \in R$ . The induction process is completed.  $\square$

### Examples of Power Series

For power series with  $\mathbb{Q}$  coefficients, see [Formal Power Series](#).

**Def. (19.1.7.15)**  $[F(X, Y)]$ . Define

$$\begin{aligned} F(X, Y) &= B_X(Y) B_{(X^p - X)/p}(Y^p) \cdots B_{(X^{p^2} - X^p)/p^2}(Y^{p^2}) \cdots \quad (10.7.1.21) \\ &= (1 + Y)^X (1 + Y^p)^{\frac{X^p - X}{p}} (1 + Y^{p^2})^{\frac{X^{p^2} - X^p}{p^2}} \cdots \in \mathbb{Q}_p[[X, Y]]. \end{aligned}$$

Then  $F(X, Y)$  is well-defined and  $F(X, Y) \in 1 + (X, Y)\mathbb{Z}_p[[X, Y]]$ . Moreover,

$$F(X, Y) = \sum_{n \in \mathbb{N}} \left( X^n \sum_{m=n}^{\infty} a_{m,n} Y^m \right), \quad a_{m,n} \in \mathbb{Z}_p.$$

┘

*Proof:* It is clear that  $F(X, Y)$  is well-defined. to show that  $F(X, Y) \in \mathbb{Z}_p[[X, Y]]$ , use [\(19.1.7.14\)](#), it suffices to show that

$$\frac{F(X^p, Y^p)}{(F(X, Y))^p} = \frac{(1 + Y^p)^X}{(1 + Y)^{pX}} \in 1 + p(X, Y)\mathbb{Z}_p[[X, Y]].$$

Because  $1 + Y \in 1 + Y\mathbb{Z}_p[[Y]]$ , by [\(19.1.7.14\)](#),  $(1 + Y^p)/(1 + Y)^p \in 1 + pY\mathbb{Z}_p[[Y]]$ . Thus it is also clear to see that

$$\frac{(1 + Y^p)^X}{(1 + Y)^{pX}} = \left( \frac{1 + Y^p}{(1 + Y)^p} \right)^X \in 1 + p(X, Y)\mathbb{Z}_p[[X, Y]].$$

For the last assertion, notice that in each term of the definition, the degree of  $X$  is no more than the degree of  $Y$ .  $\square$

**Prop. (19.1.7.16) [Dwork Splitting Function].** For  $s \in \mathbb{Z}_+$ ,  $q = p^s$ , there exists a **Dwork splitting function**  $\Theta(T) = \sum a_n T^n \in \mathbb{Q}_p(\zeta_p)[[T]]$  s.t.  $v_p(a_n) \geq n/(p-1)$ , and for any  $a \in \mathbb{F}_q$ ,

$$\Theta([a])\Theta([a^p]) \cdots \Theta([a^{p^{s-1}}]) = \zeta_p^{\text{tra}},$$

where  $[a]$  is the Teichmüller lift of  $a$ .  $\square$



*Proof:* Let  $t = [a]$ , Set  $\Theta(T) = F(T, \zeta_p - 1) = \sum_{n \in \mathbb{N}} a_n T^n$ , where  $a_n = \sum_{m=n}^{\infty} (\zeta_p - 1)^m \in \mathbb{Q}_p(\zeta_p)$ . Thus clearly  $v_p(a_n) \geq n/(p-1)$  and  $\Theta(T)$  converges for  $T \in \mathbb{D}(0; p^{1/(p-1)})$ .

It is easy to verify by using the fact  $t^{p^s} = t$  and the definition of (19.1.7.15) that

$$(1 + Y)^{t+t^p+\dots+t^{p^{s-1}}} = F(t, Y)F(t^p, Y) \dots F(t^{p^{s-1}}, Y) \in \overline{\mathbb{Q}_p}[[Y]].$$

In particular, take  $Y = \zeta_p - 1$ , then we get

$$F(t, \zeta_p - 1)F(t^p, \zeta_p - 1) \dots F(t^{p^{s-1}}, \zeta_p - 1) = \zeta_p^{t+t^p+\dots+t^{p^{s-1}}},$$

which implies the desired identity.  $\square$

### Laurent Series

References are [Kob84].

#### Notation (19.1.7.17).

- Denote by  $\mathcal{L}_K$  the set of (possibly infinite) Laurent series with coefficients in  $K$ .

┘

**Def. (19.1.7.18) [Convergences].** For  $f = \sum_{n \in \mathbb{Z}} a_n T^n \in \mathcal{L}_K$  and  $r \in \mathbb{R}$ ,  $f$  is called convergent on the radius  $r$  if for any  $a \in \mathbb{C}_K$  with  $v_p(a) = r$ ,  $f(a)$  converge. And  $f$  is said to converge on  $\infty$  (resp.  $-\infty$ ) if  $a_n = 0$  for  $n$  sufficiently large (resp. sufficiently small). Denote  $\text{Conv}(f) \subset [-\infty, \infty]$  to be the set of radius that  $f$  is convergent on. Then  $\text{Conv}(f)$  is an interval of  $[-\infty, +\infty]$ .

Let  $\mathcal{A}(I)$  denote the set of elements in  $K$  of valuation in  $I$ . And if  $f$  is bounded at  $r_1$  and  $r_2$ , then it is convergent on  $(r_1, r_2)$ .  $\perp$

**Def. (19.1.7.19) [Valuations].** For  $f \in \mathcal{L}_K$  and  $r \in [-\infty, \infty]$ , define  $v^{(r)}(f) = \inf_{n \in \mathbb{Z}} \{v(a_n) + nr\} \in [-\infty, \infty]$ .

And if  $r \in \text{Conv}(f) \cap \mathbb{R}$ , denote  $n(f, r)$  and  $N(f, r)$  to be the smallest and largest integer s.t.  $v^{(r)}(f) = v_p(a_i(f)) + ir$ .  $\perp$

**Def. (19.1.7.20) [Boundedness].**  $f \in \mathcal{L}_K$  is called bounded on the radius  $r \in \mathbb{R}$  if  $v^{(r)}(f) > -\infty$ . If  $f$  is convergent on radius  $r$ , then it is bounded on radius  $r$ .  $\perp$

**Def. (19.1.7.21).** Denote  $\mathcal{L}_K[r_1, r_2] = \{f | f \text{ is convergent on } [r_1, r_2]\}$ .

$$\mathcal{L}_K(r_1, r_2) = \{f | f \text{ is convergent on } (r_1, r_2)\}.$$

$$\mathcal{L}_K]r_1, r_2] = \{f | f \text{ is convergent on } (r_1, r_2] \text{ and bounded at } r_1\}.$$

$$\mathcal{L}_K\{r\} = \mathcal{L}_K[r, r].$$

$\mathcal{B}_L(I)$  is the subset of bounded functions. These are all rings under addition and multiplication.

┘

*Proof:* Cf. [Foundations of Theory of  $(\varphi, \Gamma)$ -modules over the Robba Ring P31]. ?  $\square$

**Prop. (19.1.7.22).** For  $I = [r_1, r_2]$  or  $(r_1, r_2]$ ,  $\mathcal{L}_K I$  is an integral domain. Moreover, for  $r \in I$  and  $f, g \in \mathcal{L}_K I^\times$ ,

$$v^{(r)}(fg) = v^{(r)}(f) + v^{(r)}(g),$$

$$n(fg, r) = n(f, r) + n(g, r),$$

$$N(fg, r) = N(f, r) + N(g, r).$$

┘

*Proof:* Cf.[Foundations of Theory of  $(\varphi, \Gamma)$ -modules over the Robba Ring]P32. ? □

**Prop. (19.1.7.23).** If  $f, g \in \mathcal{L}_K$  is bounded at  $r \in \mathbb{R}$ , then for  $f, g$  bounded on radius  $r$ ,

- $v^{(r)}(f) = \infty \iff f = 0$ .
- $v^{(r)}(f + g) \geq \min(v^{(r)}(f), v^{(r)}(g))$ .
- For  $\lambda \in L$ ,  $v^{(r)}(\lambda f) = v_p(\lambda) + v^{(r)}(f)$ .

┘

**Cor. (19.1.7.24).** By (19.1.7.22) and (19.1.7.23), if  $I = [r_1, r_2]$  or  $(r_1, r_2]$  and  $r \in I$ , then  $v^{(r)}$  is an additive valuation on  $\mathcal{L}_K I$ . ┘

**Def. (19.1.7.25).** If we set for  $\mathcal{L}_K[r_1, r_2]$  the the valuation  $v^{[r_1, r_2]}(f) = \min\{v^{(r_1)}(f), v^{(r_2)}(f)\}$ , then this is a valuation on it. ┘

**Prop. (19.1.7.26).**  $\mathcal{L}_K(\{r\})$  is complete under valuation  $v^{(r)}$ . Similarly the valuation  $v^{[r_1, r_2]}(f)$  makes  $\mathcal{L}_K[r_1, r_2]$  a Banach space unless  $r_1 = r_2 = \infty$ . ┘

*Proof:* We let  $r = 0$ . For a Cauchy sequence of Laurent series, we see that each coefficient is a Cauchy sequence, hence converge to some element in  $L$ , so it converge term-wise to a Laurent series  $f$ , so it converge to  $f$  in  $v^{(r)}$ . □

**Cor. (19.1.7.27)[Fréchet Topology].** We consider  $\mathcal{L}_K(0, r]$ , then it has a countable sequence of norms  $v^{1/n, r}$ , which makes it a locally convex space, and the last proposition shows that these valuations are complete, and a Cauchy sequence must converge to the term-wise limit, so  $\mathcal{L}_K(0, r]$  is a complete Fréchet space in the Fréchet topology. ┘

**Cor. (19.1.7.28).** The same method shows that  $\mathcal{L}_K(I)$  is a Fréchet space for any interval  $I$ . ┘

**Def. (19.1.7.29) [Newton Polygon].** For  $P(X) = a_0 + a_1X + \dots + a_dX^d + \dots \in K[[X]]$ , its **Newton polygon** is defined to be the lower convex hull of the set of points  $\{(0, v(a_0)), (1, v(a_1)), \dots, (d, v(a_d)), \dots\}$ . ┘

**Prop. (19.1.7.30) [Roots and Newton Polygons].** For  $P(T) \in K[[T]]$ , if the newton polygon of  $P$  has a segment of slope  $-\lambda$  and finite length  $N$ , then there are  $N$  roots of  $P$  in  $\mathbb{C}_K$  with valuation  $\lambda$ . ┘

*Proof:* Firstly if  $P$  is a polynomial, we may assume  $P$  is monic, then its coefficients are elementary polynomials of roots of  $P$ . Thus the conclusion follows as  $K$  is non-Archimedean.

Next, we can use  $p$ -adic Weierstrass preparation (19.1.7.7) to get the desired results. ? □

**Prop. (19.1.7.31).** If  $I = ]0, +\infty]$  and  $f(X) \in \mathcal{H}(I)$ , then the number of zeros of  $f(X)$  in  $\mathcal{A}(I)$  equals the length of the segment of  $NP(f)$  whose slope is  $-s$ , and these roots gives a  $P_s(X) \in K[X]$  that  $f(X) = P_s(X)G(X)$ ,  $G(X) \in \mathcal{H}(I)$ . ┘

*Proof:* Cf.[Zeros of Power Series over complete Valued Field Lazard]. □

**Cor. (19.1.7.32).** If  $f(X) \in \mathcal{H}(I)$ , then  $f(X) \in \mathcal{B}_L(I)$  iff it has f.m. zeros in  $\mathcal{A}(I)$ . ┘

*Proof:* Let  $r = \inf I$  and  $s = \sup I$ . First notice that  $f \in \mathcal{L}_K(I)$  is in  $\mathcal{B}(I)$  iff  $v(a_n) + nr$  is bounded from below as  $n \rightarrow +\infty$  and  $v(a_n) + ns$  is bounded below as  $n \rightarrow -\infty$ . And from the graph of  $NP(f)$ , this is equivalent to  $f$  has f.m. zeros in  $\mathcal{A}(I)$ . □

**Prop. (19.1.7.33).**  $\mathcal{H}(I)$  is a Bezout domain. ┘

*Proof:* [Lazard]. □

**Robba Ring**

**Def. (19.1.7.34)** [Robba Ring and Overconvergent Elements]. Define

$$\mathcal{E}_K = \left\{ f = \sum_{n \in \mathbb{Z}} a_n T^n \in \mathcal{L}_K[0[: \lim_{n \rightarrow -\infty} v(a_n) = \infty \right\} \quad (19.1.7.21),$$

and also the **overconvergent elements**  $\mathcal{E}^\dagger$  and **Robba ring**  $\mathcal{R}$  as

$$\mathcal{E}_K^\dagger = \bigcup_{r>0} \mathcal{L}_K]0, r], \quad \mathcal{R}_K = \bigcup_{r \in (0,1)} \mathcal{L}_K(r, 1]$$

and equip them with the final topology w.r.t. the Fréchet topologies on  $\mathcal{L}_K(0, r]$  (19.1.7.27). Then  $\mathcal{E}_K^\dagger = \mathcal{R}_K \cap \mathcal{E}_K$ . Also denote

$$\mathcal{E}_K^\pm = \mathcal{E}_K^\dagger \cap K[[T]], \quad \mathcal{R}_K^\pm = \mathcal{R}_K \cap K[[T]].$$

┘

**Prop. (19.1.7.35)** [Lazard].

- $\mathcal{R}_K$  is a Bezout-domain.
- Any finite submodule of  $\mathcal{R}_K^n$  admits elementary divisors.

┘

*Proof:* Cf. [Ber02].

□

**Prop. (19.1.7.36)**. There is a Frobenius action  $\varphi$  and an action of  $\mathbb{Z}_p^*$  on  $\mathcal{R}_K$  defined by

$$\varphi(f)(T) = f((T+1)^p - 1), \quad \gamma(f)(T) = f((T+1)^\gamma - 1).$$

Also denote  $t = \log(T+1) = \sum_{n \geq 1} (-1)^{n+1} \frac{T^n}{n} \in \mathcal{R}_K$ , then  $\varphi(t) = pt, \gamma(t) = \gamma t$ .

┘

*Proof:*

□

**Prop. (19.1.7.37)**. The only principle ideals of  $\mathcal{R}_K$  stable under the action of  $\varphi$  and  $\mathbb{Z}_p^*$  are  $t^i \mathcal{R}_K$ ,  $i \in \mathbb{N}$ .

┘

*Proof:* Cf. [L. Berger, Equations différentielles p-adiques et  $(\phi, \Gamma)$ -modules filtrés].

□

For more properties of Robba ring, See [Foundations of Theory of  $(\varphi, \Gamma)$ -modules over the Robba Ring]Chap4.

## 19.2 $p$ -adic Lie Groups

References are [Schneider].

## 19.3 Global Fields

References are [Sen80, Neu99, R-V99, Cox13], <https://math.mit.edu/classes/18.785/2015fa/lectures.html>, <http://www.math.columbia.edu/~chaoli/docs/ClassFieldTheory.html#thm:CFTlocalnorm>, should also consult notes of P. L. Clark.

### Notation(19.3.0.1).

- Use notations on Dedekind domains defined in [Commutative Algebra II](#).
- Use notations on valuations defined in [Valuations of Rank 1](#), and all valuations are assumed to be of rank 1.
- Use notations on non-commutative algebras defined in [More on \(Non-Commutative\)Algebras](#).
- Use notations on  $p$ -adic analysis defined in [p-adic Analysis](#).
- Use notations on restricted tensor products defined in [Harmonic Analysis on Locally Compact Groups](#).
- Let  $(K, \mathcal{O}_K, \mathfrak{m}_K, \varpi) \in \mathbf{LField}$ .
- Let  $L/K \in \mathbf{LField}$  be a finite separable extension.
- Let  $(F, \mathcal{O}_F) \in \mathbf{GField}(19.3.1.1)(19.3.1.2)$ .
- Let  $E/F \in \mathbf{GField}$  be a finite separable extension.

┘

## 1 Global Fields

**Def.(19.3.1.1)[Global Fields].** A **global field** is a finite extension of  $\mathbb{Q}$  or  $\mathbb{F}_p((t))$ , without a valuation. The former is called a **number field** and the latter a **function field**. The set of global fields is denoted by  $\mathbf{GField}$ . The partially ordered set of number fields is denoted by  $\mathbf{NField}$ . The partially ordered set of function fields is denoted by  $\mathbf{FField}$ .

┘

**Def.(19.3.1.2)[Ring of Integers].**  $\mathbb{Z} \subset \mathbb{Q}$  and  $\mathbb{F}_p[t] \subset \mathbb{F}_p((t))$  are PIDs, thus Dedekind domains, thus for a global field  $F(19.3.1.1)$ , we can define the **ring of integers**  $\mathcal{O}_F$  of  $F$  to be the integral closure of  $\mathbb{Z}$  or  $\mathbb{F}_p[t]$  in  $F$ , which is an Dedekind domain, by(5.2.7.12) and(3.2.3.17).

┘

**Remark(19.3.1.3).** The section [Dedekind-Domains](#) applies to the ring of integers  $\mathcal{O}_F$  for  $F \in \mathbf{GField}$ .

┘

**Def.(19.3.1.4)[Roots of Unity].** For  $F \in \mathbf{GField}$ ,  $\#F_{\text{tor}}^\times < \infty$ .

┘

*Proof:* ?

□

**Def.(19.3.1.5)[Places].**

- $\Sigma_F^{\text{fin}}$  is the equivalent classes of (non-Archimedean)valuations of  $F$ , called the **finite places** of  $F$ .
- $\Sigma_F^\infty$  is the equivalent classes of (Archimedean valuations) $|\cdot|_v = -\log|\tau(\cdot)|$  of  $F$  corresponding to embeddings  $\tau: K \rightarrow \mathbb{C}(12.2.3.16)$ , called the **infinite places** of  $F$ .
- $\Sigma_F^\mathbb{R}$  is the set of infinite places  $v$  of  $F$  corresponding to embeddings  $F \rightarrow \mathbb{R}$ , called the set of **real places** of  $F$ . Denote  $r_1 = \#\Sigma_F^\mathbb{R}$  if it is finite.

- $\Sigma_F^{\mathbb{C}}$  is the set of infinite places  $v$  of  $F$  that is non-real, called the set of **complex places** of  $F$ . Notice two embeddings corresponds to the same place iff they are conjugate. Denote  $r_2 = \#\Sigma_F^{\mathbb{C}}$  if it is finite.
- For  $\mathfrak{p} \in \text{Spec}(\mathcal{O}_F)$ ,  $v_{\mathfrak{p}}$  is the valuation of  $F$  corresponding to  $\mathfrak{p}$ . Most of the time we will not distinguish between a maximal prime  $\mathfrak{p}$  and its corresponding valuation.
- For a finite extension  $L/F$  and  $v \in \Sigma_F$ ,  $\Sigma_L^v$  is the set of finite places over  $L$  over  $v$ .
- For  $v \in \Sigma_F$ , denote  $F_v$  the completion of  $F$  w.r.t.  $v$ .
- For any non-zero ideal  $\mathfrak{a} \neq 0 \in \text{Ideal}(\mathcal{O}_F)$ , denote  $\|\mathfrak{a}\| = \#\mathcal{O}_F/\mathfrak{a}$ , called the **norm of ideal  $\mathfrak{a}$** .

┘

**Def. (19.3.1.6) [Constant Fields].**

┘

**Prop. (19.3.1.7) [Valuations].** For  $F \in \mathbf{GField}$ ,  $v \in \Sigma_F$ , let  $\mathfrak{p} \in \mathbf{Prime} \cup \{\infty\}$  s.t.  $v|\mathfrak{p}$ , then define the **inertia degree**

$$f_v = \begin{cases} [\kappa(v) : \kappa(\mathfrak{p})] & , v \in \Sigma_F^{\text{fin}} \\ [F_v : F_{\mathfrak{p}}] & , v \in \Sigma_F^{\infty} \end{cases}, \quad \|v\| = \begin{cases} \mathfrak{p}^{f_v} & , v \in \Sigma_F^{\text{fin}} \\ e^{f_v} & , v \in \Sigma_F^{\infty} \end{cases}, \quad |\cdot|_v = \|\cdot\|^{-v(a)}$$

This is compatible with the definition in (5.2.7.20) when  $v \in \Sigma_F^{\text{fin}}$ .

┘

**Thm. (19.3.1.8) [Product Formula].** Let  $F \in \mathbf{GField}$  and  $a \in F^{\times}$ , then  $|a|_w = 1$  for a.e.  $w \in \Sigma_F$ , and

$$\prod_{w \in \Sigma_F} |a|_w = 1.$$

┘

*Proof:* Let  $F_0 = \mathbb{Q}$  or  $\mathbb{F}_p(t)$  be the constant field of  $F$  (19.3.1.6), then the assertion is easy to verify for  $F_0$ , and

$$\prod_{w \in \Sigma_F} |a|_w = \prod_{v \in F_0} \prod_{w|v} |a|_w = \prod_{v \in F_0} |\text{Nm}_{F/F_0}(a)|_v = 1.$$

□

**Prop. (19.3.1.9) [Artin-Whaples].** Let  $F \in \mathbf{Field}$  and  $\Sigma_F$  the set of places of  $F$ , then  $F \in \mathbf{GField}$  iff

- There exists representatives  $|\cdot|_v$  for  $v \in \Sigma_F$  s.t. the product formula  $\prod_{v \in \Sigma_F} |a|_v = 1$  for any  $a \in F^{\times}$ .
- $F_v \in \mathbf{LField}$  (19.1.5.9) for any  $v \in \Sigma_F$ .

┘

*Proof:*

□

**Prop. (19.3.1.10) [Local to Global Fields].** Let  $F \in \mathbf{GField}$ ,  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$  and  $k \in \mathbb{Z}_+$ . Suppose for each  $v \in S$ , there is given a separable field extension  $F'_v/F_v$  of degree  $k$ , then there exists  $L/F \in \mathbf{GField}$  of degree  $k$  s.t. for each  $v \in S$  s.t.  $L_v/F_v$  is inert, and  $L_v \cong F'_v$ .

┘

*Proof:* This follows from weak approximation, Krane's lemma (19.1.2.26) and continuity of roots. Notice Krasner's lemma also holds for  $v \in \Sigma_F^{\infty}$ .

□

**Def. (19.3.1.11) [Algebraic Numbers and Algebraic Integers].** An element  $\alpha \in \overline{\mathbb{Q}}$  is called an **algebraic number**. It is called an **algebraic integer** iff it satisfies a monic polynomial equation  $p(X) \in \mathbb{Z}[T]$ . Notice the set of algebraic integers equals

$$\mathcal{O}_{\overline{\mathbb{Q}}} = \bigcup_{F \subset \mathbb{C}} \mathcal{O}_F.$$

where the union is taken over all number fields  $F$ . ┘

**Prop. (19.3.1.12) [Characterizing Roots of Unity].** If  $\alpha$  is an algebraic integer s.t. any of its conjugate has absolute value 1, then  $\alpha$  is a root of unity. ┘

*Proof:* For any power  $\alpha^r$  of  $\alpha$ , its minimal polynomial over  $\mathbb{Q}$  has bounded degree and bounded coefficients independent of  $r$ , so there are only f.m. such polynomial and f.m. such roots. Thus  $\alpha^i = \alpha^j$  for some  $i \neq j \in \mathbb{Z}_+$ , and  $\alpha \in \mu(\mathbb{Q})$ . □

**Prop. (19.3.1.13) [Discriminants].** For  $F \in \mathbf{NField}$ , there is a discriminant  $\mathfrak{d}_{F/\mathbb{Q}} \in \text{Ideal}(\mathbb{Z})$  (5.2.7.36), and if  $\alpha_1, \dots, \alpha_n$  is a  $\mathbb{Z}$ -basis of  $\mathcal{O}_F$ , then  $\mathfrak{d}_{F/\mathbb{Q}} = (d(\alpha_1, \dots, \alpha_n))$ . And this  $\text{disc}(F) = d(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}^\times$  is invariant of the basis chosen. So it is also called the **discriminant of  $F$** .

Then  $\text{disc}(F) \equiv 0, 1 \pmod{4}$ . ┘

*Proof:* □

**Prop. (19.3.1.14) [Sign of Discriminants].** For  $F \in \mathbf{NField}$ , we have  $\text{disc}(F) \in (-1)^{r_2} \mathbb{Z}_+$ . ┘

*Proof:*  $d(F) \neq 0$  by (3.2.6.34), and by definition,  $d(F) = \det((\sigma\alpha_i)_{\sigma,i})^2$ , where  $\sigma$  runs through embeddings  $F \hookrightarrow \mathbb{C}$ , and  $\alpha_1, \dots, \alpha_n$  is a  $\mathbb{Z}$ -basis of  $\mathcal{O}_F$ . Then notice

$$\overline{\det((\sigma\alpha_i)_{\sigma,i})} = (-1)^{r_2} \det((\sigma\alpha_i)_{\sigma,i}),$$

so if  $r_2$  is odd, then  $\det((\sigma\alpha_i)_{\sigma,i})$  is purely imaginary, and if  $r_2$  is even,  $\det((\sigma\alpha_i)_{\sigma,i})$  is real. Thus the assertion follows. □

### Special Algebraic Numbers

**Def. (19.3.1.15) [Pisot and Salem Numbers].** A **Pisot number** is an algebraic number  $\alpha \in \overline{\mathbb{Z}} \cap \mathbb{R}_{>1}$  s.t. all the Galois conjugates  $\alpha_i$  of  $\alpha$  except  $\alpha$  itself satisfy  $|\alpha_i| < 1$ .

A **Salem number** is algebraic number  $\alpha \in \overline{\mathbb{Z}} \cap \mathbb{R}_{>1}$  s.t. all the Galois conjugates  $\alpha_i$  of  $\alpha$  except  $\alpha$  itself satisfy  $|\alpha_i| \leq 1$ , and  $|\alpha_i| = 1$  for some  $i$ . By (3.3.1.6), this implies the minimal polynomial of  $\alpha$  is reciprocal, and also the conjugates of  $\alpha$  is  $\{\alpha, \alpha^{-1}, \alpha_1, \dots, \alpha_r\}$  where  $|\alpha_i| = 1$ . ┘

**Def. (19.3.1.16) [Weil-Numbers].** For  $p \in \mathbf{Prime}$ ,  $q \in p^{\mathbb{Z}}$ , a  **$q$ -Weil-integer** of weight  $w$  is an algebraic integer  $\alpha \in \mathcal{O}_{\overline{\mathbb{Q}}}$  that for any embedding  $\iota : \mathbb{Q}(\alpha) \rightarrow \mathbb{C}$ ,  $|\iota(\alpha)| = q^{w/2}$ . The set of Weil  $q$ -numbers of weight  $w$  is denoted by  $\text{Weil}(q^{w/2})$ .

A  **$q$ -Weil-number** of weight  $w$  is an algebraic number  $\alpha \in \overline{\mathbb{Q}}$  s.t. for  $|\alpha|_v = 0$  for each  $v \in \Sigma_{\mathbb{Q}(\alpha)}^{\text{fin}} \setminus S_{\mathbb{Q}(\alpha)}(p)$ , and  $|\iota(\alpha)| = q^{w/2}$  for each embedding  $\iota : \mathbb{Q}(\alpha) \rightarrow \mathbb{C}$ . ┘

**Prop. (19.3.1.17) [Smyth].** Let  $\alpha \in \overline{\mathbb{Z}}$  with absolute conjugate  $\alpha = \alpha_1, \alpha_2, \dots, \alpha_d$ , and  $P(X)$  is the minimal polynomial of  $\alpha$ . If

$$\prod_i \min(1, |\alpha_i|) < \theta_0,$$

where  $\theta_0 \approx 1.32472$  is the real root of the polynomial  $\theta^3 - \theta - 1 = 0$ , then  $P(X)$  is a reciprocal polynomial. ┘

*Proof:* Cf.[Product of the Conjugates Outside the Unit Circle of an Algebraic Integer, Smyth].  $\square$

**Cor.(19.3.1.18).**  $\theta_0$ (19.3.1.17) is smallest Pisot number(19.3.1.15).  $\lrcorner$

*Proof:* If  $\alpha$  is a smaller Pisot number, then by(19.3.1.17), the minimal polynomial of  $\alpha$  is reciprocal, so it must have degree 2, and of the form  $X^2 - aX + 1$ . Then it is easy to verify that  $\alpha > \theta_0$ , contradiction.  $\square$

**Def.(19.3.1.19)[Lehmer's Number].** Consider the **Lehmer's polynomial**

$$f(X) = X^{10} + X^9 - X^7 - X^6 - X^5 - X^4 - X^3 + X + 1 \in \mathbb{Z}[X],$$

then its largest real root  $\lambda \approx 1.17628 \in \overline{\mathbb{Z}}$  is called the **Lehmer's number**.  $\lrcorner$

**Def.(19.3.1.20)[Mahler measures].** For a monic polynomial  $f(X) \in \mathbb{Z}[X]^\times$ , the **Mahler measure**  $M(f)$  is defined to be the absolute value of the product of roots of  $f$  with norm greater than one.  $\lrcorner$

**Conj.(19.3.1.21)[Lehmer].** For any monic polynomial  $f(X) \in \mathbb{Z}[X]^\times$ ,  $M(f) \geq \lambda$  where  $\lambda$  is the Lehmer's number(19.3.1.19), and the equality holds exactly when  $f$  is the Lehmer's polynomial(19.3.1.19).  $\lrcorner$

*Proof:*  $\square$

### Galois Theory of Extensions

**Def.(19.3.1.22)[Ramification Groups].** For a Galois extension of global fields  $L/K \in \mathbf{GField}$  and a valuation extension  $w|v$ , denote  $\lambda/k$  residue fields extension of  $w|v$ , and denote  $\mathfrak{p}_w, \mathfrak{p}_v$  by  $\mathfrak{P}, \mathfrak{p}$ .

Then the **decomposition group** is  $G_w(L/K) = \{\sigma \in G(L/K) | w \circ \sigma = w\}$ . The **decomposition field**  $Z_w$  is the fixed field of  $G_w$ .

When  $w$  is non-Archimedean, we further define:

The **inertia group** is  $I_w(L/K) = \{\sigma \in G_w(L/K) | \sigma(x) \equiv x \pmod{\mathfrak{P}}\}$ . The **inertia field**  $T_w$  is the fixed field of  $I_w$ .

The **ramification group** is  $R_w(L/K) = \{\sigma \in G_w(L/K) | \sigma(x)/x \equiv 1 \pmod{\mathfrak{P}}\}$ . The **ramification field**  $V_w$  is the fixed field of  $R_w$ .

Similarly we can define for higher ramification groups  $G_{w,s}(L/F)$ .  $\lrcorner$

**Prop.(19.3.1.23)[Local and Global Ramification Groups].** Let  $L/F$  be a Galois extension of global fields,  $v \in \Sigma_F, w \in \Sigma_L^v$ , then any embedding  $L \rightarrow \overline{F}_v$  induces isomorphisms

$$G_{w,s}(L/F) \cong G_s(L_w/F_v)$$

by natural restriction.  $\lrcorner$

*Proof:* Cf.[ANT, Neukirch].  $\square$

**Cor.(19.3.1.24)[Local and Global Galois Groups].** There is an embedding  $\mathrm{Gal}(L_w/F_v) \hookrightarrow \mathrm{Gal}(L/F)$  that is determined up to conjugation. In particular, there is an embedding

$$\mathrm{Gal}_{F_v} \hookrightarrow \mathrm{Gal}_F$$

determined up to conjugation.  $\lrcorner$



*Proof:* It is determined up to conjugation because  $\text{Gal}(L/F)$  acts transitively on the places over  $v$ : For finite places this follows from (5.2.7.22), and for infinite places this means  $\text{Gal}(L/F)$  acts transitively on the extension of embedding to  $\mathbb{C}$ , which is clearly true.  $\square$

**Prop. (19.3.1.25)**  $[Z_w]$ .

- The restriction  $w_Z$  of  $w$  to  $Z_w$  extends uniquely to  $L$ .
- If  $v$  is non-Archimedean,  $w_Z$  has the same residue field and value group as  $v$ . In particular,  $w_Z/v$  is unramified.
- $Z_w = L \cap K_v \subset L_w$ .

┘

*Proof:* Cf. [Neukirch P171].  $\square$

**Prop. (19.3.1.26)**.  $T_w/Z_w$  is the maximal unramified subextension of  $L/Z_w$ .  $\text{┘}$

*Proof:* Cf. [Neukirch P173].  $\square$

**Prop. (19.3.1.27)**.  $V_w/Z_w$  is the maximal tamely ramified subextension of  $L/Z_w$ .  $\text{┘}$

*Proof:* Cf. [Neukirch P175].  $\square$

### Discriminants

**Def. (19.3.1.28) [Lattices]**. Let  $(\mathcal{O}_K, K)$  be a Dedekind domain and  $V \in \text{Vect}^{\text{fin}}/K$ , then an  $\mathcal{O}_K$ -lattice in  $V$  is a f.g.  $\mathcal{O}_K$ -module  $\Lambda$  s.t.  $\Lambda \otimes_{\mathcal{O}_K} K = V$ .

Any  $\mathcal{O}_F$ -lattice in  $F^n$  is of the form  $\Lambda = \mathcal{O}_F x_1 \oplus \mathcal{O}_F x_2 \oplus \dots \oplus \mathcal{O}_F x_{n-1} \oplus \mathfrak{a} x_n$  where  $\mathfrak{a} \in \text{Ideal}(\mathcal{O}_F)$  (Cf. [P-R94]P42, may have to do with (5.2.7.15)?), and it is called a **free lattice** if  $L \cong \mathcal{O}_F^n$ . Notice this is always the case if  $\text{cl}(\mathcal{O}_F) = 1$ , e.g. when  $F = \mathbb{Q}$  or  $F \in p\text{-LField}$ .  $\text{┘}$

**Prop. (19.3.1.29) [Local to Global Compatibility for Lattices]**. Let  $F \in \text{NField}$ , then

1. If  $\Lambda$  is an  $\mathcal{O}_F$ -lattice in  $F^n$ , then for any  $v \in \Sigma_F$ , its completion in  $F_v$  is an  $\mathcal{O}_{F_v}$ -lattice in  $F_v^n$ , and for a.e. place  $v$ ,  $\Lambda_v = \mathcal{O}_{F_v}^n$ .
2. Conversely, if for any  $v \in \Sigma_F^f$ ,  $\Lambda_v$  is an  $\mathcal{O}_{F_v}$ -lattice in  $F_v^n$ , and  $\Lambda_v = \mathcal{O}_{F_v}^n$  for a.e.  $v$ , then there is a unique  $F$ -lattice  $\Lambda \in F^n$  s.t.  $\Lambda_v$  is the closure of  $\Lambda$  in  $F_v$ . In fact,  $\Lambda = \bigcap_{v \in \Sigma_F^f} (F^n \cap \Lambda_v)$ .
3.  $\Lambda$  is determined by  $\Lambda_v$  for each  $v \in \Sigma_F^f$ .

┘

*Proof:* 1 is easy. For 2, notice that  $\mathcal{O} = \bigcap_{v \in \Sigma_F^f} (\mathcal{O}_{F_v})$ , so  $\Lambda$  defined as above is commensurable with  $\mathcal{O}_F^n$ , so it is clearly a lattice, because  $\mathcal{O}_F$  is Noetherian. 3 follows from 2. (I think this argument also holds for global function fields?)  $\square$

**Prop. (19.3.1.30) [Lower Bounds for Discriminant]**. If  $F \in \text{NField}$  and  $n = [F : \mathbb{Q}]$ , then

$$|\text{disc}(F)| \geq \frac{n^{2n}}{(n!)^2} \left(\frac{\pi}{4}\right)^{2r_2} \geq \frac{n^{2n}}{(n!)^2} \left(\frac{\pi}{4}\right)^n \geq \frac{1}{2\pi e^{\frac{1}{6}n}} \left(\frac{\pi e^2}{4}\right)^n \geq \frac{0.1347}{n} \cdot (5.8)^n \text{ (12.6.1.17)}.$$

┘

*Proof:* Cf. [Neukirch, P204].?  $\square$

**Cor. (19.3.1.31) [Hermite's Theorem].** There are only f.m. number fields with a given discriminant.  $\lrcorner$

*Proof:* (19.3.1.30) shows the degree is controlled by the discriminant, thus the theorem follows from (20.8.1.8).  $\square$

**Cor. (19.3.1.32) [Minkowski's Theorem].** For  $\mathbb{Q} \neq F \in \mathbf{NField}$ , we have  $\text{disc}(F) \neq \pm 1$ .  $\lrcorner$

**Cor. (19.3.1.33).**  $\mathbb{Q}$  doesn't have any non-trivial unramified extensions, by (5.2.7.39).  $\lrcorner$

### Orders

References are [Neu99]Chap1.12, [P-R94]P42 and [Vig80].

**Def. (19.3.1.34) [Orders].** An **order** is a Noetherian integral domain  $\mathcal{O}$  of dimension 1 whose conductor (5.2.7.17) is non-zero, or equivalently its integral closure in the fraction field is finite over  $\mathcal{O}$  by (5.2.7.18). A Dedekind domain is an order.  $\lrcorner$

**Def. (19.3.1.35) [A-Ideals and A-Orders].** If  $(A, K)$  is Dedekind domain and  $B \in \mathbf{Ring}^{\text{fd}}/K$ , then

- An **A-ideal** in  $B$  is a complete  $A$ -lattice in  $B$  (19.3.1.28).
- An **A-order** in  $B$  is an  $A$ -ideal in  $B$  that is also a subring.
- An **Eichler A-order** in  $B$  is the intersection of two distinct maximal  $A$ -orders in  $B$ .

$\lrcorner$

**Prop. (19.3.1.36) [Orders in Number Fields].** Situation as in (5.2.7.13), if  $L/K$  is separable, then a subring  $\mathcal{O}$  of  $L$  is an  $\mathcal{O}_K$ -order iff it is an order in  $L$  (19.3.1.34) with integral closure  $\mathcal{O}_L$ . in particular,  $\mathcal{O}_L$  is the maximal  $\mathcal{O}_K$ -order in  $L$ .

In particular, for  $F \in \mathbf{NField}$ , the  $\mathbb{Z}$ -orders in  $F$  are exactly subrings of  $\mathcal{O}_F$  that contains a basis of  $F$ , and they all have finite indexes in  $\mathcal{O}_F$ .  $\lrcorner$

*Proof:* If  $\mathcal{O}$  is an  $\mathcal{O}_K$ -order, then every element of  $\mathcal{O}$  is integral over  $\mathcal{O}_K$ , by acting on  $\mathcal{O}$ .  $\mathcal{O}_K$  is contained in  $\mathcal{O}$ , and  $\mathcal{O} \otimes_{\mathcal{O}_K} K = L$ , so the fraction field of  $\mathcal{O}$  is  $L$ . Thus the integral closure of  $\mathcal{O}$  in  $L$  is  $\mathcal{O}_L$ . Then  $\dim \mathcal{O} = 1$  as  $\dim \mathcal{O}_L = 1$  (5.2.4.14), and it is Noetherian because it is f.g. over  $A$  (5.1.1.43). And  $\mathcal{O}_L$  is finite over  $\mathcal{O}_K$  by (5.2.7.21), thus is also f.g. over  $\mathcal{O}$ . Thus  $\mathcal{O}$  is an order.

If  $\mathcal{O}$  is an order with integral closure  $\mathcal{O}_L$ , then it is f.g. over  $\mathcal{O}_K$  as it is contained in the f.g.  $\mathcal{O}_K$ -module  $\mathcal{O}_L$ . And it contains a  $K$ -basis of  $L$  because  $L = K\mathcal{O}_L$  and  $a\mathcal{O}_L \subset \mathcal{O}$  for some  $a \in \mathcal{O}^\times$  because  $\mathcal{O}$  is an order. Thus  $\mathcal{O}$  is an  $A$ -lattice in  $L$  that is a ring, so it is an  $A$ -order.

For the last assertion, notice that one direction is clear, and if  $\mathcal{O}_F$  contains a basis of  $F$  and is contained in  $\mathcal{O}_F$ , then it must be a  $\mathbb{Z}$ -lattice of  $F$ , as  $\mathcal{O}_F$  is free because it is finite over  $\mathbb{Z}$  (5.2.7.21). Then it is clear  $\mathcal{O}$  is an order with integral closure  $\mathcal{O}_F$ .  $\square$

**Cor. (19.3.1.37).** If  $F \in \mathbf{NField}$  and  $L = F(\alpha_i)$  where  $\alpha_i \in \mathcal{O}_L$ , then  $\mathcal{O} = \mathcal{O}_F[\alpha_i]$  is an  $\mathcal{O}_K$ -order in  $L$ .  $\lrcorner$

**Prop. (19.3.1.38) [Multiplier Orders].** Let  $(A, K)$  be a Dedekind domain,  $B \in \mathbf{Ring}^{\text{fd}}/K$  and  $I$  be an  $A$ -ideal of  $B$ , then the ring of left and right multipliers

$$\mathcal{O}_l = \{\alpha \in B \mid \alpha I \subset I\}, \quad \mathcal{O}_r = \{\alpha \in B \mid I\alpha \subset I\}$$

are  $A$ -orders in  $B$ . And  $I$  is called

- a **normal ideal** if  $\mathcal{O}_l$  and  $\mathcal{O}_r$  are maximal orders,
- an **integral ideal** if  $I \subset \mathcal{O}_l, \Lambda \subset \mathcal{O}_r$ .
- A **principal ideal** if there exists  $a \in A^\times$  s.t.  $\Lambda = \mathcal{O}_l a = a \mathcal{O}_r$ .

The **inverse ideal** of  $I$  is the ideal  $I^{-1} = \{b \in B : IbI \subset I\}$ .  $\lrcorner$

*Proof:* For any  $\alpha \in B$ , we have  $\alpha I \subset B = I \otimes K$ , so there exists  $b \in A$  s.t.  $b\alpha \in \mathcal{O}_l$ . So  $\mathcal{O}_l \otimes K = B$ .

Let  $0 \neq a \in A \cap I$ , then  $a\mathcal{O}_l \in I$ , so  $\mathcal{O}_l \leq a^{-1}I$  is a lattice in  $B$  because  $A$  is Noetherian.  $\square$

**Prop. (19.3.1.39) [Properties of Orders].** Let  $(A, K)$  be a Dedekind domain,  $B \in \text{Ring}^{\text{fd}}/K$ , then

- An  $A$ -order in  $B$  is equivalent to a subring  $\mathcal{O}$  of  $B$  consisting of elements integral over  $A$ , containing  $A$ , and satisfying  $K\mathcal{O} = B$ .
- There exist  $A$ -orders in  $B$ .
- Every  $A$ -order in  $B$  is contained in a maximal  $A$ -order.  $\lrcorner$

*Proof:* •  $\square$

**Prop. (19.3.1.40) [Local Properties of Maximal Orders].** Let  $F \in \text{NField}$  and  $B \in \text{Ring}^{\text{fd}}/F$ , then an order  $\mathcal{O}$  in  $B$  is maximal iff for each  $v \in \Sigma_F^f$ ,  $\mathcal{O}_v \subset B_{F_v}$  is maximal. Similar arguments work for  $\mathcal{O}_r$ .  $\lrcorner$

*Proof:* This is clear from (19.3.1.29) and the definition (19.3.1.35).  $\square$

**Def. (19.3.1.41) [Discriminants of Orders].** Compatible with (19.3.1.13), if  $F \in \text{NField}$  and  $\mathcal{O} \subset \mathcal{O}_F$  is an order, then we define the **discriminant of the order**  $\mathcal{O}$  to be the  $d(\mathcal{O}) = d(\alpha_1, \dots, \alpha_n)$  where  $\alpha_1, \dots, \alpha_n$  is a  $\mathbb{Z}$ -basis of  $\mathcal{O}$ . Notice this is invariant of the basis chosen.  $\lrcorner$

**Lemma (19.3.1.42).** For  $K \in p\text{-LField}$ ,

- $\text{Mat}(n; \mathcal{O}_K)$  is a maximal order in  $\text{Mat}(n; K)$ .
- Any two maximal orders in  $\text{Mat}(n; K)$  are conjugate.
- Any order  $\mathcal{O} \subset \text{Mat}(n; K)$  is contained in some maximal order. And it is contained in f.m. maximal orders.  $\lrcorner$

*Proof:* 1: If another order contains  $\text{Mat}(n; \mathcal{O}_K)$  and also an element  $x$ , then we can clearly see that it contains an element of the form  $A = \text{diag}(x_1, \dots, x_n)$  s.t.  $x_1 \notin \mathcal{O}_K$ . Then  $\{A^{\mathbb{Z}}\}$  is not bounded.

3: As  $\mathcal{O}$  is compact and  $\text{Mat}(n; \mathcal{O}_K)$  is open, there exists f.m.  $x_i$  s.t.  $B \subset \cup_i (x_i + \text{Mat}(n; \mathcal{O}_K))$ . Then it follows that  $\Lambda = \oplus \mathcal{O}_K^n x_i$  is a lattice stable under  $\mathcal{O}$ , thus  $\mathcal{O} \subset \text{Stab}(\Lambda)$  is conjugate to  $\text{Mat}(n; \mathcal{O}_K)$  thus maximal.

Next we show  $\mathcal{O}$  is contained in f.m. maximal orders: Suppose  $\mathcal{O} \subset C = \text{Stab}(\Lambda)$  and  $\mathcal{O} \subset C' = \text{Stab}(\Lambda')$ , then there exists some  $\alpha$  that  $\varpi^\alpha \text{Stab}(\Lambda) \subset \mathcal{O} \subset \text{Stab}(\Lambda')$ . Replacing  $\Lambda$  by a constant doesn't change stabilizer, so we may assume  $\Lambda \subset \Lambda'$ , and assume  $\Lambda = \mathcal{O}_K \{e_1, \dots, e_n\}$  and  $\Lambda' = \mathcal{O}_K \{e_1, \varpi^{\alpha_2} e_2, \dots, \varpi^{\alpha_n} e_n\}$ . Since  $\varpi^\alpha \text{Stab}(\Lambda) \subset \text{Stab}(\Lambda')$ ,  $\text{Stab}(\Lambda)(\Lambda') \subset \varpi^{-\alpha} \Lambda'$ . Also by using the matrices permuting  $e_1$  and  $e_i$ , we see  $\alpha_i < \alpha$  for each  $i$ , so  $\varpi^\alpha \Lambda \subset \Lambda'$ , and then  $\varpi^n \text{Stab}(\Lambda')(\Lambda) \subset \text{Stab}(\Lambda')(\Lambda') = \Lambda' \subset L$ . This implies  $\varpi^\alpha \text{Stab}(\Lambda') \subset \text{Stab}(\Lambda)$ . From this it is clear that there are only f.m. maximal orders containing  $\mathcal{O}$ .

2: It follows from the proof of item3 that any order is contained in a stabilizer, then it suffices to show that if  $\text{Stab}(\Lambda) = \text{Stab}(\Lambda')$ , then  $\Lambda = \Lambda'$ . But this is already implicit in the proof of item2: In this case,  $\alpha = 0$ .  $\square$

**Thm. (19.3.1.43)[Maximal Orders in Semisimple Algebras over Local Fields].** If  $K \in p\text{-LField}$  and  $B$  is a semisimple algebra over  $K$ , then any  $\mathcal{O}_K$ -order  $\mathcal{O} \subset B$  is contained in a maximal order. And there are only f.m. maximal orders containing it.  $\lrcorner$

*Proof:* Writing  $B$  as a product of simple algebras, it suffices to prove for  $B$  simple, and by base change to the center  $L$  of  $B$ , it suffices to show for  $B$  central simple: This is because if two orders are isomorphic after extension by  $\mathcal{O}_L$ , then their transformation matrix has coefficients in  $\mathcal{O}_L \cap K = \mathcal{O}_K$ . Then we need to show that there are only f.m. orders containing  $\mathcal{O}$ . Choose a splitting field  $L/K$  for  $B$ , it reduces to prove for  $B = \text{Mat}(n; K)$ : This reason is the same as above. Then the assertion follows from (19.3.1.42).  $\square$

**Thm. (19.3.1.44)[Maximal Orders in Semisimple Algebras over Global Fields].** If  $F \in \text{NField}$  and  $B$  is a semisimple algebra over  $F$ , then any order  $\mathcal{O} \subset B$  is contained in some maximal order. And there are only f.m. maximal orders containing it.  $\lrcorner$

*Proof:* Writing  $B$  as a product of simple algebras, it suffices to prove for  $B$  simple, and by base change to the center  $L$  of  $B$ , it suffices to show for  $B$  central simple: This is because if two orders are isomorphic after extension by  $\mathcal{O}_L$ , then their transformation matrix has coefficients in  $\mathcal{O}_L \cap F = \mathcal{O}_F$ . Then we need to show that there are only f.m. orders containing  $\mathcal{O}$ . Choose a splitting field  $L/F$  for  $B$ , it reduces to prove for  $B = \text{Mat}(n; F)$ : This reason is the same as above.

And for  $B = \text{Mat}(n; F)$ , by (19.3.1.42) and (19.3.1.29),  $\mathcal{O}$  is already maximal at a.e. place, so the assertion follows from (19.3.1.29) and the local case (19.3.1.43).  $\square$

### Quadratic Fields

**Def. (19.3.1.45)[ $\mathcal{O}_D$ ].** For  $D \in \mathbb{Z}^\times$  s.t.  $D \equiv 0, 1 \pmod{4}$ , denote  $\mathcal{O}_D = \mathbb{Z}[\frac{D+\sqrt{D}}{2}]$ , which is an order in  $\mathbb{Q}(\sqrt{D})$ .  $\lrcorner$

**Prop. (19.3.1.46)[Ring of Integers in Quadratic Fields].** Let  $n \in \mathbb{Z}^\times$  be a square-free, and let  $\mathcal{K} = \mathbb{Q}(\sqrt{n})$ .

- If  $n \equiv 2, 3 \pmod{4}$ , then  $\mathcal{O}_{\mathcal{K}} = \mathbb{Z}[\sqrt{n}]$ , and  $\text{disc}(\mathcal{K}) = 4n$ .
- If  $n \equiv 1 \pmod{4}$ , then  $\mathcal{O}_{\mathcal{K}} = \mathbb{Z}[\frac{1+\sqrt{n}}{2}]$ , and  $\text{disc}(\mathcal{K}) = n$ .

In particular,  $\mathcal{O}_{\mathcal{K}} = \mathcal{O}_{\text{disc}(\mathcal{K})} = \mathbb{Z}[\frac{\text{disc}(\mathcal{K})+\sqrt{\text{disc}(\mathcal{K})}}{2}]$ .  $\lrcorner$

*Proof:* 1: the minimal polynomial of  $\sqrt{n}$  is  $X^2 - n$ , whose discriminant is  $4n$ , which doesn't have a proper divisor  $\beta$  that  $4n/\beta$  is a square and  $\beta \equiv 0, 1 \pmod{4}$ , so by (19.3.1.13)  $\mathbb{Z}[\sqrt{n}]$  is the ring of integers.

2: the minimal polynomial of  $\frac{1+\sqrt{n}}{2}$  is  $X^2 - X + \frac{1-n}{4}$ , whose discriminant is  $n$ , which doesn't have a proper divisor  $\beta$  that  $4n/\beta$  is a square, so by (19.3.1.13)  $\mathbb{Z}[\frac{1+\sqrt{n}}{2}]$  is the ring of integers.

For the last assertion, given the basis for the ring of integers, we can easily calculate the discriminant as in (3.2.6.33). It equals  $4n$  in the first case and  $n$  in the second case. Thus the assertion follows.  $\square$

**Def. (19.3.1.47)[Fundamental Discriminants].** A **fundamental discriminant** is an element  $d \in \mathbb{Z}^\times$  s.t.  $d$  is the discriminant of quadratic field  $\mathcal{K} \in \text{NField}$ .

Then any fundamental discriminant is a product of distinct elements in  $\{-4, 8, -8\} \cup \{(\frac{-1}{p})p\}_{p \in \text{Prime}_{\geq 3}}$ , by (19.3.1.46).  $\lrcorner$

**Def. (19.3.1.48) [Imaginary Quadratic Orders].**

- $\mathcal{O}_{-1} = \mathbb{Z}[i]$  is called the ring of **Gaussian integers**.
- $\mathcal{O}_{-3} = \mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$  is called the ring of **Eisenstein integers**.
- $\mathcal{O}_{-7} = \mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$  is called the ring of **Kleinian Integers**.

These are all PIDs. ┘

**Thm. (19.3.1.49) [Primes in Quadratic Fields].** Let  $\mathcal{K}$  be a quadratic field with discriminant  $\text{disc}(\mathcal{K})$ .

Then for  $p \in \text{Prime}$ ,

- If  $p \mid \text{disc}(\mathcal{K})$ , then  $p\mathcal{O}_{\mathcal{K}} = \mathfrak{p}^2$ , where  $\mathfrak{p}$  is a prime in  $\mathcal{O}_{\mathcal{K}}$ , and  $\mathfrak{p} = (p, \sqrt{\text{disc}(\mathcal{K})})$  if  $p$  is odd.
- If  $\left(\frac{\text{disc}(\mathcal{K})}{p}\right) = 1$ , then  $p\mathcal{O}_{\mathcal{K}} = \mathfrak{p}\mathfrak{p}'$ , where  $p, \mathfrak{p} \neq \mathfrak{p}'$  are primes in  $\mathcal{O}_{\mathcal{K}}$ .
- If  $\left(\frac{\text{disc}(\mathcal{K})}{p}\right) = -1$ , then  $p\mathcal{O}_{\mathcal{K}}$  is a prime in  $\mathcal{O}_{\mathcal{K}}$ .

And every maximal prime in  $\mathcal{O}_{\mathcal{K}}$  are of the form.

In particular,  $p$  ramifies in  $\mathcal{K}$  iff  $p \mid \text{disc}(\mathcal{K})$ , and  $p$  splits completely in  $\mathcal{K}$  iff  $\left(\frac{\text{disc}(\mathcal{K})}{p}\right) = 1$ . ┘

*Proof:* 1 is easy.

2, 3: Cf. [Cox13]P92.

To show that every maximal prime  $\mathfrak{p} \subset \mathcal{O}_{\mathcal{K}}$  is of this form, notice that  $\mathfrak{p} \mid \text{Nm}(\mathfrak{p})$  which is an ideal in  $\mathbb{Z}$ , so its prime factors are all of the form above. □

**Cor. (19.3.1.50).** For  $p \in \text{Prime}_{\geq 3}$ ,  $\mathbb{Q}\left(\sqrt{\left(\frac{-1}{p}\right)p}\right)$  is the unique quadratic extension of  $\mathbb{Q}$  unramified outside  $p$ . ┘

**Prop. (19.3.1.51) [Differents].** For a quadratic field  $\mathcal{K}$ , by (5.2.7.32) and (19.3.1.46), we have:

$$\mathfrak{d}_{\mathcal{K}} = (\sqrt{\text{disc}(\mathcal{K})}), \quad \|\mathfrak{d}_{\mathcal{K}}\| = |\text{disc}(\mathcal{K})|.$$

┘

**Prop. (19.3.1.52) [Primitive Ideals].** For a quadratic field  $\mathcal{K}$  with  $D = \text{disc}(\mathcal{K})$ , an ideal  $\mathfrak{a} \in \text{Ideal}^{\times}(\mathcal{O}_{\mathcal{K}})$  is called a **primitive ideal** if  $\mathfrak{a} \mid (m)$  implies  $m = 1$  for any  $m \in \mathbb{Z}_+$ . Then every primitive ideal is of the form

$$\mathfrak{a} = \mathbb{Z}\left\{a, \frac{b + \sqrt{D}}{2}\right\},$$

where  $a = \|\mathfrak{a}\|$  and  $b \in \mathbb{Z}_+$  is determined by  $-a < b \leq a, b^2 \equiv D \pmod{4a}$ . ┘

*Proof:* Clearly  $a = \|\mathfrak{a}\|$  is contained in  $\mathfrak{a}$ , and it is the smallest positive integer contained in  $\mathfrak{a}$ . Thus there must be another element  $\frac{b + \sqrt{D}}{2}$  s.t.  $\left\{a, \frac{b + \sqrt{D}}{2}\right\}$  is a basis of  $\mathfrak{a}$ . But then  $b^2 \equiv D \pmod{4a}$ , and we may assume  $-a < b \leq a$ . And conversely, if  $(b')^2 \equiv D \pmod{4a}$ , then it is easy to verify  $\mathbb{Z}\left\{a, \frac{b + \sqrt{D}}{2}\right\}$  is a primitive ideal with norm  $\mathfrak{a}$ . Which implies that  $b' \equiv b \pmod{2a}$ . So the solution of  $b$  is unique. □

### Orders in Imaginary Quadratic Fields

**Prop. (19.3.1.53) [Orders in Quadratic Fields].** Let  $\mathcal{K}$  be a quadratic field, then for any  $f \in \mathbb{Z}_+$ , there exists a unique order in  $\mathcal{O}_{\mathcal{K}}$  of index  $f$ , which is

$$\mathcal{O}_{\mathcal{K},f} = \mathbb{Z}[1, f \frac{\text{disc}(\mathcal{K}) + \sqrt{\text{disc}(\mathcal{K})}}{2}].$$

And  $\text{disc}(\mathcal{O}_{\mathcal{K},f}) = f^2 \text{disc}(\mathcal{K})$  (19.3.1.41).  $\lrcorner$

*Proof:* Let  $\mathcal{O}$  be an order of index  $f$  in  $\mathcal{O}_{\mathcal{K}}$ , then it is clear that  $f\mathcal{O}_{\mathcal{K}} + \mathbb{Z} = \mathcal{O}_{\mathcal{K},f} \subset \mathcal{O}$ . And this one has index  $f$ . The last assertion is clear from the definition of discriminant.  $\square$

**Prop. (19.3.1.54) [Proper Fractional Ideals].** Let  $F \in \mathbf{NField}$  and  $\mathcal{O} \subset \mathcal{O}_F$  an order. A fractional ideal  $\mathfrak{a}$  of  $\mathcal{O}$  is called a **proper fractional ideal** if

$$\mathcal{O} = \{\beta \in \mathcal{K} | \beta \mathfrak{a} \subset \mathfrak{a}\}.$$

Then any invertible fraction ideal is proper.

Moreover, if  $F$  is a quadratic field, then a fractional ideal is invertible iff it is proper.  $\lrcorner$

*Proof:* Cf. [Cox13]P122.  $\square$

**Lemma (19.3.1.55) [ $\mathcal{O}$ -Ideals Prime to Conductor].** Let  $\mathcal{K}$  be a quadratic field,  $f \in \mathbb{Z}_+$  and  $\mathcal{O} = \mathcal{O}_{\mathcal{K},f}$ , then an  $\mathcal{O}$ -ideal  $\mathfrak{a}$  is prime to  $f$  iff  $(\|\mathfrak{a}\|, f) = 1$ . And any such  $\mathcal{O}$ -ideal  $\mathfrak{a}$  is proper (19.3.1.54).  $\lrcorner$

*Proof:* If  $\beta \in \mathcal{K}$  and  $\beta \mathfrak{a} = \mathfrak{a}$ , then  $\beta \in \mathcal{O}_{\mathcal{K}}$ , and  $\beta \mathcal{O} = \beta(\mathfrak{a} + f\mathcal{O}) = \beta \mathfrak{a} + \beta f\mathcal{O} \subset \mathfrak{a} + f\mathcal{O}_{\mathcal{K}} \subset \mathfrak{D}$ . Thus  $\beta \in \mathcal{O}$ . And then  $\mathfrak{a}$  is proper.  $\square$

**Lemma (19.3.1.56).** Let  $\mathcal{K}$  be an imaginary quadratic field,  $f \in \mathbb{Z}_+$  and  $\mathcal{O} = \mathcal{O}_{\mathcal{K},f}$ , then any ideal class of  $\text{Cl}(\mathcal{O})$  contains a proper  $\mathcal{O}$ -ideal that is relatively prime to  $f$ .  $\lrcorner$

*Proof:* Cf. [Cox13]P128. ?  $\square$

**Prop. (19.3.1.57) [Class Groups of Imaginary Quadratic Orders].** Let  $\mathcal{K}$  be an imaginary quadratic field and  $f \in \mathbb{Z}_+$ ,  $\mathcal{O} = \mathcal{O}_{\mathcal{K},f}$ . Denote

- $I(\mathcal{O}, f)$  the subgroup of  $I(\mathcal{O})$  generated by  $\mathcal{O}$ -ideals  $\alpha \subset \mathcal{O}$  that is prime to  $f$ .
- $P(\mathcal{O}, f)$  the group of  $P(\mathcal{O})$  generated by  $\alpha \in \mathcal{O}$  s.t.  $(\text{Nm}(\alpha), f) = 1$ .
- $I_{\mathcal{K}}(f)$  the subgroup of  $I(\mathcal{O}_{\mathcal{K}})$  generated by  $\mathcal{O}_{\mathcal{K}}$ -ideals  $\mathfrak{a} \in \text{Ideal}(\mathcal{O}_{\mathcal{K}})$  that is prime to  $f$ .
- $P_{\mathcal{K},\mathbb{Z}}(f)$  the subgroup of  $P(\mathcal{K})$  generated by  $\alpha \in \mathcal{O}_{\mathcal{K}}$  s.t.  $\alpha \equiv a \pmod{f\mathcal{O}_{\mathcal{K}}}$  for some  $a \in \mathbb{Z} \setminus (f)$ .

Then  $\mathfrak{a} \mapsto \mathfrak{a} \cap \mathcal{O}$  induces an isomorphism  $I_{\mathcal{K}}(f) \cong I(\mathcal{O}, f)$  with inverse given by  $\mathfrak{a} \mapsto \mathfrak{a}\mathcal{O}_{\mathcal{K}}$ , and  $\|\mathfrak{a}\| = \|\mathfrak{a} \cap \mathcal{O}\|$ . Moreover, there is an isomorphism

$$\text{Cl}(\mathcal{O}) \cong I(\mathcal{O}, f) / P(\mathcal{O}, f) \cong I_{\mathcal{K}}(f) / P_{\mathcal{K},\mathbb{Z}}(f).$$

$\lrcorner$

*Proof:* The map  $I(\mathcal{O}, f) \rightarrow I(\mathcal{O})$  is surjective by (19.3.1.56), and the kernel of  $I(\mathcal{O}, f) \rightarrow \text{Cl}(\mathcal{O})$  is  $P(\mathcal{O}) \cap I(\mathcal{O}, f)$ . It is easy to show that this group is just  $P(\mathcal{O}, f)$ .

If  $\mathfrak{a} \in \text{Ideal}(\mathcal{O}_K)$  is prime to  $f$ , then  $\mathfrak{a} \cap \mathcal{O}$  is also prime to  $f$  because there is a natural injection  $\mathcal{O} \cap (\mathfrak{a} \cap \mathcal{O}) \hookrightarrow \mathcal{O}_K/\mathfrak{a}$ . And this map is also a surjective because multiplying by  $f$  is a bijection on  $\mathcal{O}_K/\mathfrak{a}$  and  $f\mathcal{O}_K \subset \mathcal{O}$ . And also  $\mathfrak{a}\mathcal{O}_K \cap \mathcal{O} = \mathfrak{a}$  because

$$\mathfrak{a}\mathcal{O}_K \cap \mathcal{O} = (\mathfrak{a}\mathcal{O}_K \cap \mathcal{O})\mathcal{O} = (\mathfrak{a}\mathcal{O}_K \cap \mathcal{O})(\mathfrak{a} + f\mathcal{O}) \subset \mathfrak{a} + \mathfrak{a} \cdot f\mathcal{O}_K \subset \mathfrak{a}.$$

If  $\mathfrak{a} \in \text{Ideal}(\mathcal{O})$  is prime to  $f$ , then  $\mathfrak{a}\mathcal{O}_K + f\mathcal{O}_K = (\mathfrak{a} + f\mathcal{O})\mathcal{O}_K = \mathcal{O}\mathcal{O}_K = \mathcal{O}_K$ . So  $\mathfrak{a}\mathcal{O}_K$  is also prime to  $f$ . And also  $(\mathfrak{a} \cap \mathcal{O})\mathcal{O}_K = \mathfrak{a}$  because  $\mathfrak{a} = \mathfrak{a}\mathcal{O} = \mathfrak{a}(\mathfrak{a} \cap \mathcal{O} + f\mathcal{O}) \subset (\mathfrak{a} \cap \mathcal{O})\mathcal{O}_K + f\mathfrak{a} \subset (\mathfrak{a} \cap \mathcal{O})\mathcal{O}_K$ .

The bijection given above is clearly multiplicative, thus induces a homomorphism  $I(\mathcal{O}, f) \rightarrow I_K(f)$ . Then it suffices to show that subgroup  $P(\mathcal{O}, f)$  is mapped onto  $P_{K, \mathbb{Z}}(f)$ : This follows from the fact that: For  $\alpha \in \mathcal{O}_K$ ,  $\alpha \equiv a \pmod{f\mathcal{O}_K}$  for some  $a \in \mathbb{Z} \setminus (f)$  iff  $\alpha \in \mathcal{O}$  and  $(\text{Nm}(\alpha), f) = 1$ .  $\square$

## 2 Cyclotomic Fields

**Def. (19.3.2.1) [Cyclotomic Units].** A compatible system of roots of unity is a system  $\{\zeta_n, n \in \mathbb{Z}_+\} \subset \overline{\mathbb{Q}}$  s.t.

- $\zeta_m \neq 1$  for  $m > 1$ .
- For  $k, n \in \mathbb{Z}_+$ ,  $\zeta_{kn}^k = \zeta_n$ .
- $\zeta_4 = i$ .

We fix a choice of compatible system of unity throughout this book.  $\lrcorner$

**Prop. (19.3.2.2).**  $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/(n))^*$ .  $\lrcorner$

*Proof:* We choose a prime  $p$  prime to  $n$  and show that  $\zeta_n^p$  is conjugate to  $\zeta_n$ .

Let  $X^n - 1 = f(X)h(X)$  with  $f(X)$  minimal polynomial of  $\zeta_n$ . If  $f(\zeta_n^p) \neq 0$ , then  $h(\zeta_n^p) = 0$ , thus  $h(X^p) = f(X)g(X)$ . So modulo  $p$ ,  $X^n - 1$  has a multi root, which is impossible.  $\square$

**Lemma (19.3.2.3).** For  $p \in \mathbf{P}$ ,  $r \in \mathbf{Z}_+$ , consider  $\mathbb{Q}(\zeta_{p^r})$ , then  $(p) = (1 - \zeta_{p^r})^{p^{r-1}(p-1)}$ , and

$$\text{disc}(1, \zeta_{p^r}, \dots, \zeta_{p^r}^{p^{r-1}(p-1)-1}) = \pm p^{p^{r-1}(r(p-1)-1)},$$

where the sign is positive if  $p^r = 4$  or  $p \equiv 3 \pmod{4}$ .  $\lrcorner$

*Proof:* As  $\Psi_{p^r}(X) = X^{p^{r-1}(p-1)} + X^{p^{r-1}(p-2)} + \dots + X^{p^{r-1}} + 1$ , by taking  $X = 1$ , we get

$$p = \prod_{g \in (\mathbb{Z}/(n))^*} (1 - \zeta_{p^r}^g).$$

But it is easy to see that for any  $g, g' \in (\mathbb{Z}/(n))^*$ ,  $1 - \zeta_{p^r}^g$  and  $1 - \zeta_{p^r}^{g'}$  differ by a unit, so by (3.2.6.35), if  $\zeta_i$  are the conjugates of  $\zeta_{p^r}$ , then

$$d(1, \dots, \theta^{p^{r-1}(p-1)-1}) = \prod_{i < j} (\zeta_i - \zeta_j)^2 = \pm \prod_i \Psi_{p^r}^\vee(\zeta_i) = \pm \text{Nm}_{\mathbb{Q}(\zeta_{p^r})/\mathbb{Q}}(\Psi_{p^r}^\vee(\zeta_{p^r})).$$

Differentiating the equation

$$(X^{p^{r-1}} - 1)\Psi_{p^r}(X) = X^{p^r} - 1,$$

we get

$$(\zeta_p - 1)\Psi_{p^r}^\vee(\zeta_{p^r}) = p^r \zeta_{p^r}^{-1}.$$

Then notice as  $p$  is totally ramified in  $\mathbb{Q}(\zeta_p)$  (19.1.5.28),

$$\mathrm{Nm}_{\mathbb{Q}(\zeta_{p^r})/\mathbb{Q}}(\zeta_p - 1) = (\mathrm{Nm}_{\mathbb{Q}(\zeta_p)/\mathbb{Q}}(\zeta_p - 1))^{p^{r-1}} = \pm p^{p^{r-1}},$$

and the assertion follows. The last assertion follows from (19.3.1.14).  $\square$

**Prop. (19.3.2.4).** For  $n \in \mathbb{Z}_+$ ,  $\mathcal{O}_{\mathbb{Q}(\zeta_n)} = \mathbb{Z}[\zeta_n]$ .  $\lrcorner$

*Proof:* First consider the case  $n = p^r$  a prime power. By (19.3.2.3),  $d(1, \zeta_{p^r}, \dots, \zeta_{p^r}^{p^{r-1}(p-1)-1}) = \pm p^s$  for some  $s \in \mathbb{Z}_+$ , so  $p^s \mathcal{O} \subset \mathbb{Z}[\zeta_{p^r}] \subset \mathcal{O}$ . Because  $p$  totally ramifies by (19.1.5.28),  $\mathcal{O} = \mathbb{Z}[\zeta_{p^r}] + (1 - \zeta_{p^r})\mathcal{O}$ , thus  $\mathcal{O} = \mathbb{Z}[\zeta]$  by Nakayama.

In general, if  $n = \prod_i p_i^{r_i}$ , for different  $p_i$ , and  $\mathbb{Q}(\zeta_n) = \prod \mathbb{Q}(\zeta_{p_i^{r_i}})$  by Chinese remainder theorem, and the fields  $\mathbb{Q}(\zeta_{p_i^{r_i}})$  are disjoint and the discriminant are pairwise coprime, thus by (5.2.7.16), the products of the integral basis form an integral basis.  $\square$

**Cor. (19.3.2.5).**  $\mathcal{O}_{\mathbb{Q}(\zeta_n + \zeta_n^{-1})} = \mathbb{Z}[\zeta_n + \zeta_n^{-1}]$ .  $\lrcorner$

**Prop. (19.3.2.6).** By (35.3.6.16), for  $p \geq 3 \in \mathbf{P}$ ,  $\mathbb{Q}(\sqrt{(-1)^{\frac{p-1}{2}} p}) \subset \mathbb{Q}(\mu_p)$ .  $\lrcorner$

**Prop. (19.3.2.7).** For  $p \in \mathbf{P}$ , if  $\varepsilon \in (\mathbb{Z}[\zeta_p])^*$ , then there exists  $\varepsilon_1 \in \mathbb{Q}(\zeta_p + \zeta_p^{-1})$  and  $r \in \mathbb{Z}$  s.t.  $\varepsilon = \zeta_p^r \varepsilon_1$ .  $\lrcorner$

*Proof:* The case  $p = 2$  is clear. Assume  $p \geq 3$ , and  $\alpha = \bar{\varepsilon}/\varepsilon$ , then any conjugate of  $\alpha$  has absolute value 1. Thus  $\alpha$  is a root of unity (19.3.1.12), Assume  $\alpha = \pm \zeta_p^a$  ?. Let  $\varepsilon = b_0 + b_1 \zeta_p + \dots + b_{p-2} \zeta_p^{p-2}$ , if  $\alpha = -\zeta_p^a$ , then

$$\bar{\varepsilon} = b_0 + b_1 \zeta_p^{-1} + \dots \equiv b_0 + b_1 + \dots \equiv \varepsilon = -\zeta_p^a \bar{\varepsilon} \equiv \bar{\varepsilon} \pmod{1 - \zeta_p}.$$

Then  $2\bar{\varepsilon} \in (1 - \zeta_p)$ , which is not possible.

Thus  $\alpha = \zeta_p^a$ . Assume  $a \equiv 2r \pmod{p}$ , and  $\varepsilon_1 = \zeta^{-r} \varepsilon$ , then  $\bar{\varepsilon}_1 = \varepsilon_1$ , and  $\varepsilon = \varepsilon_1 \zeta_p^r$ .  $\square$

**Prop. (19.3.2.8).**  $p \in \mathbf{P}$  splits in  $\mathbb{Q}(\zeta_n)$  iff  $p \equiv 1 \pmod{n}$ .  $\lrcorner$

*Proof:* First, if it splits, then  $f_p = 1$ . Because the ring of integers is  $\mathbb{Z}[\zeta_n]$ , so  $X^n - 1$  splits in  $\mathbb{F}_p$  (19.3.2.4), thus  $p \equiv 1 \pmod{n}$ . And if  $p \equiv 1 \pmod{n}$ , it is unramified and  $X^n - 1$  splits in  $\mathbb{F}_p$ , so  $f_p = 1$ .  $\square$

**Prop. (19.3.2.9) [Ramifications].**  $p \in \mathbf{Prime}$  is ramified in  $\mathbb{Q}(\mu_n)$  iff  $p|n$ .  $\lrcorner$

*Proof:* This follows from (19.1.5.26) and (19.1.5.28).  $\square$

**Cor. (19.3.2.10).** If  $m, n \in \mathbb{Z}_+$  are coprime, then  $\mathbb{Q}(\mu_m) \cap \mathbb{Q}(\mu_n) = \mathbb{Q}$ .  $\lrcorner$

*Proof:* This follows from Minkowski's theorem (19.3.1.32).  $\square$

**Prop. (19.3.2.11).** If  $n \in \mathbb{Z}_+$  has at least two prime divisors, then  $1 - \zeta_n$  is a unit in  $\mathbb{Q}(\mu_n)$ .  $\lrcorner$

*Proof:* By (3.3.2.11),  $\mathrm{Nm}(1 - \zeta_n) = \Psi_n(1) = 1$ .  $\square$

**Prop. (19.3.2.12).**

- If  $n \in \mathbb{Z}_+$  is not of the form  $p^r$  or  $2p^r$  for  $p \in \mathbf{P}$ , then  $\mathbb{Q}(\zeta_n)/\mathbb{Q}(\zeta_n + \zeta_n^{-1})$  is unramified at finite places.



- If  $p \in \mathbf{P}$  and  $n \in p^{\mathbb{Z}_+}$  or  $n \in 2p^{\mathbb{Z}_+}$ , then  $\mathbb{Q}(\zeta_n)/\mathbb{Q}(\zeta_n + \zeta_n^{-1})$  is ramified at any places above  $p$ , and unramified at other finite places.  $\lrcorner$

*Proof:* For the first case, let  $p, q$  are two prime divisors of  $n$  (or  $q = 2$ , in which case take  $q = 4$ ). Then  $\zeta_p, \zeta_q$  are not in  $\mathbb{Q}(\zeta_n + \zeta_n^{-1})$ , so  $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_n + \zeta_n^{-1}, \zeta_p) = \mathbb{Q}(\zeta_n + \zeta_n^{-1}, \zeta_q)$ . But then  $\mathbb{Q}(\zeta_n)/\mathbb{Q}(\zeta_n + \zeta_n^{-1})$  can only ramify at finite places that are both over  $p$  and over  $q$ , so it is unramified at all finite places.

For the second case, use the fact that  $\mathbb{Q}(\zeta_n)/\mathbb{Q}$  is totally ramified at  $p$  and unramified at other places (19.1.5.26) and (19.1.5.28).  $\square$

**Prop. (19.3.2.13) [No Unramified Extensions].** Let  $m, n \in \mathbb{Z}_+$  and  $\mathbb{Q}(\zeta_m) \subset F \subset \mathbb{Q}(\zeta_n)$  s.t.  $F/\mathbb{Q}(\zeta_m)$  is unramified, then  $F = \mathbb{Q}(\zeta_m)$ .  $\lrcorner$

*Proof:* This follows from class field theory.  $\square$

### $\mathbb{Z}_p$ -Extensions

**Def. (19.3.2.14) [ $\text{cycl}_p(K)$ ].** For  $p \in \mathbf{P}$  and  $K \in p\text{-Field}$  or  $K \in \mathbf{GField}$ , Denote  $K_{p^\infty} = K(\zeta_{p^\infty})$ . Then by (19.3.2.2) and (19.1.5.17),  $\text{Gal}(K_{p^\infty}/K) \subset \mathbb{Z}_p^* \cong \mathbb{Z}_p \oplus G$  is a subgroup of finite index, where  $G$  is a finite group. So there is a unique subextension  $\text{cycl}_p(K) \subset K(\mu_{p^\infty})$  with  $\text{Gal}(\text{cycl}_p(K)/K) \cong \mathbb{Z}_p$ , called the **cyclotomic  $\mathbb{Z}_p$ -extension** of  $K$ .

Also denote  $\text{cycl}_{p,n}(K) \subset \text{cycl}_p(K)$  the subextension of degree  $p^n$ .  $\lrcorner$

**Prop. (19.3.2.15).** If  $\ell$  is a  $p$  is totally ramified in  $\text{cycl}_p(\mathbb{Q})$ , and any  $\ell \in \mathbf{P} \setminus \{p\}$  is unramified in  $\text{cycl}_p(\mathbb{Q})$ .  $\lrcorner$

*Proof:* Cf. [Washington, P265].  $\square$

### Others

**Prop. (19.3.2.16) [Sum of Primitive Roots of Unity].** For  $n \in \mathbb{Z}_+$ , the sum of primitive  $n$ -th roots of unity equals  $\mu(n)$  (2.6.1.25).  $\lrcorner$

*Proof:* The assertion is multiplicative, so it suffices to prove for prime powers. Let  $p \in \mathbf{Prime}, k \in \mathbb{Z}_+$ . If  $k = 1$ , then clearly the sum is -1. If  $k \geq 2$ , then the sum equals

$$\sum_{i=0}^{p^k-1} \zeta_{p^k}^i - \sum_{i=0}^{p^{k-1}-1} \zeta_{p^{k-1}}^i = 0.$$

$\square$

**Prop. (19.3.2.17).** Find all  $n \in \mathbb{Z}_+$  and primitive  $n$ -th roots of unity  $\zeta_1, \dots, \zeta_4$  s.t.

$$\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4 = 1.$$

$\lrcorner$

*Proof:* By (19.3.2.16),

$$4\mu(n) = \sum_{k \in (\mathbb{Z}/(n))^*} \sum_{i=1}^4 \zeta_i^k = \phi(n).$$

Thus  $\varphi(n) = 4$  and  $\mu(n) = 1$ . Then  $n$  can only be 5, 8, 10, 12, and  $n$  can only be 10.

Now  $\Psi_{10}(X) = X^4 - X^3 + X^2 - X + 1$ , so the identity  $\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4 = 1$  must be essentially identical to this one. So the only possibility is

$$\{\zeta_1, \zeta_2, \zeta_3, \zeta_4\} = \{e^{2\pi i \frac{1}{10}}, e^{2\pi i \frac{3}{10}}, e^{2\pi i \frac{7}{10}}, e^{2\pi i \frac{9}{10}}\}.$$

□

### 3 Class Numbers

**Def. (19.3.3.1) [Class Groups and Class Numbers].** The **class group of  $F$**  is defined to be the class group  $\text{Cl}(\mathcal{O}_F)$  of  $\mathcal{O}_F$ . It is denoted by  $\text{Cl}(F)$ . The cardinality of  $\text{Cl}(F)$  is called the **class number of  $F$** , denoted by  $\text{cl}(F)$ .

$$\text{Also denote } h(D) = \begin{cases} \text{cl}(\mathcal{O}_D) & , D \equiv 0, 1 \pmod{4} \\ 0, & \text{otherwise} \end{cases} \text{ and } \text{Cl}(D) = \begin{cases} \text{Cl}(\mathcal{O}_D) & , D \equiv 0, 1 \pmod{4} \\ 0, & \text{otherwise} \end{cases} . \quad \lrcorner$$

**Lemma (19.3.3.2).** For  $\mathfrak{a} \neq 0 \in \text{Ideal}^\times(\mathcal{O}_F)$ , there exists  $a \in \mathfrak{a}^\times$  s.t.

$$|\text{Nm}_{F/\mathbb{Q}}(a)| \leq \left(\frac{2}{\pi}\right)^s \sqrt{|\text{disc}(F)|} \cdot \|\mathfrak{a}\| .$$

□

*Proof:* This follows immediately from (20.8.1.4). □

**Thm. (19.3.3.3) [Class Numbers are Finite].**  $\text{cl}(F) = \# \text{Cl}(F) < \infty$ . □

*Proof:* It is clear that there are only f.m. ideals  $\mathfrak{a} \in \text{Ideal}(\mathcal{O}_F)$  of bounded norm. Thus it suffices to prove that each ideal class is represented by an integral ideal  $\mathfrak{b} \in \text{Ideal}(\mathcal{O}_F)$  with  $\|\mathfrak{b}\| \leq \left(\frac{2}{\pi}\right)^s$ . For this, let the ideal class be represented by  $\mathfrak{a}$ , and let  $\gamma \in \mathcal{O}_F^\times$  s.t.  $\mathfrak{b} = \gamma \mathfrak{a}^{-1} \subset \mathcal{O}_F$ . Then by (19.3.3.2), there exists  $\alpha \in \mathfrak{b}^\times$  s.t.  $|\text{Nm}_{F/\mathbb{Q}}(\alpha)| \leq \left(\frac{2}{\pi}\right)^s \sqrt{|d_F|} \|\mathfrak{b}\|$ . Then  $\mathfrak{a} = \alpha \mathfrak{b}^{-1} = \alpha \gamma^{-1} \mathfrak{a}$  is contained in  $\mathcal{O}_F$ , and

$$\|\mathfrak{a}\| = |\text{Nm}_{F/\mathbb{Q}}(\alpha)| / \|\mathfrak{b}\| \leq \left(\frac{2}{\pi}\right)^s \sqrt{|d_F|} .$$

□

**Conj. (19.3.3.4).** There are infinitely many real quadratic field with class number 1. □

*Proof:*

□

**Conj. Cor. (19.3.3.5).** There are infinitely many number fields with class number 1. □

*Proof:*

□

### Class Numbers of Imaginary Quadratic Orders

References are [Gol85], [Sta67], [Cox13].

**Thm. (19.3.3.6)** [Dirichlet's Class Number Formula]. Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{K}) = -\frac{\#\mu(\mathcal{K})}{2|d_{\mathcal{K}}|} \sum_{n=1}^{|d_{\mathcal{K}}|-1} \left(\frac{d_{\mathcal{K}}}{n}\right)_n.$$

┘

*Proof:* Cf. [Cox13]P134. □

**Thm. (19.3.3.7).** Let  $\mathcal{K}$  be an imaginary quadratic field and  $f \in \mathbb{Z}_+$ , then  $\text{cl}(\mathcal{O}_{\mathcal{K}}) \mid \text{cl}(\mathcal{O}_{\mathcal{K},f})$ , and

$$\text{cl}(\mathcal{O}_{\mathcal{K},f}) = \frac{\text{cl}(\mathcal{O}_{\mathcal{K}})f}{[\mu(\mathcal{O}_{\mathcal{K}}) : \mu(\mathcal{O}_{\mathcal{K},f})]} \prod_{p \mid f} \left(1 - \left(\frac{d_{\mathcal{K}}}{p}\right)\frac{1}{p}\right).$$

┘

*Proof:* Cf. [Cox13]P132. □

**Prop. (19.3.3.8).** Let  $p \equiv 3 \pmod{4} \in \text{Prime}$ , then for  $D = -p$  or  $-4p$ ,  $\text{cl}(\mathcal{O}_D)$  (19.3.1.45) is odd. ┘

*Proof:* Cf. [Cox, Prop3.11 and Thm7.7(ii)]. ? □

**Thm. (19.3.3.9)** [Siegel]. Let  $\mathcal{K}$  denote an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\lim_{d_{\mathcal{K}} \rightarrow -\infty} \frac{\text{cl}(\mathcal{K})}{\log |d_{\mathcal{K}}|} = \frac{1}{2}.$$

┘

*Proof:* Cf. [Sie35]. □

**Thm. (19.3.3.10)** [Gauss Class Number Problem1801, Siegel [Sie35]/Goldfeld(1976)/Gross-Zagier(1983)].

For any imaginary quadratic field  $\mathbb{Q}(\sqrt{D})$  with discriminant  $D < 0$ , we have

$$\text{cl}(\mathcal{O}_D) \geq \frac{1}{55} \log(|D|) \prod_{p \in S(D)} \left(1 - \frac{[2\sqrt{p}]}{p+1}\right).$$

┘

*Proof:* This follows from (35.5.3.4) and (35.4.4.10). See also [Goldfeld, The class numbers of quadratic fields and the conjectures of Birch and Swinnerton-Dyer]. □

**Def. (19.3.3.11)** [Hurwitz's Class Numbers]. Let  $\mathcal{K}$  be an imaginary quadratic field and  $\mathcal{O} \subset \mathcal{O}_{\mathcal{K}}$  be an imaginary quadratic order, then define the **Hurwitz's class number**

$$H(\mathcal{O}) = \sum_{\mathcal{O}' \subset \mathcal{O}_{\mathcal{K}}} \frac{2 \text{cl}(\mathcal{O}')}{\#\mathcal{O}'^*}.$$

And if  $\mathcal{O}_D$  has discriminant  $D$ , then also denote

$$H(D) = H(\mathcal{O}_D) = \sum_{f^2 \mid D} \hat{h}(D/f^2)$$

where  $\hat{h}(\Delta) = 2h(\Delta)/\#\mathcal{O}_{\Delta}^*$  (19.3.3.1). In particular, if  $D \neq -3, -4$  is a fundamental discriminant, then  $H(D) = h(D)$ . ┘

### Quadratic Forms and Class Numbers

**Def. (19.3.3.12) [Primitive Binary Quadratic Forms].** A **primitive binary quadratic form** is a binary quadratic form of the form  $\varphi(x, y) = ax^2 + bxy + cy^2$  s.t.  $a, b, c \in \mathbb{Z}$  and  $(a, b, c) = 1$ .  $\lrcorner$

**Def. (19.3.3.13) [Representing Numbers].** For a binary quadratic form  $\varphi(x, y) = ax^2 + bxy + cy^2$ , an integer  $m \in \mathbb{Z}$  is said to be **represented by the form**  $\varphi$  if  $m = \varphi(x, y)$  for some  $x, y \in \mathbb{Z}$ . And  $m$  is said to be **properly represented by the form**  $\varphi$  if  $m = \varphi(x, y)$  for some  $x, y \in \mathbb{Z}$  s.t.  $(x, y) = 1$ .  $\lrcorner$

**Prop. (19.3.3.14).** A binary quadratic form  $\varphi(x, y) = ax^2 + bxy + cy^2$  properly represents an integer  $m \in \mathbb{Z}$  iff  $\varphi$  is equivalent to a form  $\phi(x, y) = mx^2 + Bxy + Cy^2$  for some  $B, C \in \mathbb{Z}$ .  $\lrcorner$

*Proof:* One direction is clear, for the other, if  $f(p, q) = m$  and  $(p, q) = 1$ , then we can find  $r, s \in \mathbb{Z}$  s.t.  $ps - qr = 1$ , then we have

$$\varphi(px + ry, qx + sy) = f(p, q)x^2 + (2apr + bps + brq + 2cqs)xy + f(r, s)y^2.$$

□

**Def. (19.3.3.15) [Equivalence between Binary Quadratic Forms].** Two binary quadratic forms  $f(x, y), g(x, y)$  are called **equivalent** if there exists  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{GL}(2; \mathbb{Z})$  s.t.  $f(x, y) = g(ax + by, cx + dy)$ .

And  $f, g$  are called **properly equivalent** if there exists  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{SL}(2; \mathbb{Z})$  s.t.  $f(x, y) = g(ax + by, cx + dy)$ .  $\lrcorner$

**Def. (19.3.3.16) [Reduced Primitive Binary Quadratic Forms].** A **reduced primitive binary quadratic form** is a primitive binary quadratic form  $\varphi(x, y) = ax^2 + bxy + cy^2$  (19.3.3.15) s.t.  $|b| \leq a \leq c$ , and  $b \geq 0$  if either  $|b| = a$  or  $a = c$ .  $\lrcorner$

**Prop. (19.3.3.17).** Every primitive binary quadratic form  $\varphi(x, y) = ax^2 + bxy + cy^2$  is properly equivalent to a unique reduced primitive binary quadratic form.  $\lrcorner$

*Proof:* Cf. [Cox13]P25?  $\lrcorner$

**Prop. (19.3.3.18).** For  $D \in \mathbb{Z}_-$ , there are only f.m. reduced primitive binary quadratic forms  $\varphi(x, y) = ax^2 + bxy + cy^2$  (19.3.3.16) with discriminant  $b^2 - 4ac = D$ . Let  $h(D)$  be this number.  $\lrcorner$

*Proof:* Because  $\varphi(x, y)$  is reduced, we have  $-D = 4ac - b^2 \geq 4a^2 - a^2 = 3a^2$ , so  $a \leq \sqrt{-D/3}$ , so there are only f.m. choices of  $a$ , and then for each  $a$ , because  $|b| \leq a$ , there are only f.m. choices of  $b$ , and we notice that  $c$  is determined by  $a$  and  $b$ .  $\lrcorner$

**Exercise (19.3.3.19).** If  $p \in \text{Prime}$  is represented by two binary quadratic forms  $f(x, y)$  and  $g(x, y)$  of the same discriminant, show that  $f$  and  $g$  are equivalent. And if  $f(x, y) = x^2 + ny^2$  for some  $n \in \mathbb{Z}_+$  and  $g$  is reduced, show that  $f = g$ .  $\lrcorner$

**Exercise (19.3.3.20).** Given a primitive binary quadratic form  $\varphi(x, y) = ax^2 + bxy + cy^2$ , show that  $-\varphi = [\tilde{\varphi}]$  in the class group, where  $\tilde{\varphi}(x, y) = ax^2 - bxy + cy^2$ .  $\lrcorner$

**Thm. (19.3.3.21)[Class Groups of Binary Quadratic Forms].** Let  $D \in \mathbb{Z}_-$  s.t.  $D \equiv 0, 1 \pmod{4}$ , and  $C(D)$  the set of equivalence classes of primitive positive definite forms of discriminant  $D$ , then the Dirichlet composition is a well-defined binary operation on  $C(D)$  which makes  $C(D)$  into a finite Abelian group, whose identity element is given by

$$\begin{cases} x^2 - \frac{D}{4}y^2 & , D \equiv 0 \pmod{4} \\ x^2 + xy + \frac{1-D}{4}y^2 & , D \equiv 1 \pmod{4} \end{cases}.$$

And the inverse of the class containing  $ax^2 + bxy + cy^2$  is the class containing  $ax^2 - bxy + cy^2$ .  $\lrcorner$

*Proof:* Cf.[Cox13]P45.  $\square$

**Prop. (19.3.3.22)[ $\#C(D)[2]$ ].** Situation as in (19.3.3.21), let  $r = \#S(D)$ , then there are exact  $(2^\mu - 1)$ -many classes in  $C(D)$  of order  $\leq 2$ , where

$$\mu = r + \begin{cases} 0 & , D \equiv 1 \pmod{4} \\ 0 & , D = -4n \ \& \ n \equiv 3 \pmod{4} \\ 1 & , D = -4n \ \& \ n \equiv 1, 2 \pmod{4} \\ 1 & , D = -4n \ \& \ n \equiv 4 \pmod{8} \\ 1 & , D = -4n \ \& \ n \equiv 0 \pmod{8} \end{cases}.$$

$\lrcorner$

*Proof:* Cf.[Cox13]P47.  $\square$

**Prop. (19.3.3.23)[Siegel's Mass Formula].** Let  $D \in \mathbb{Z}_-$  s.t.  $D \equiv 0, 1 \pmod{4}$  and let  $m \in \mathbb{Z}_+ \setminus (D)$ , and denote  $R_D(m)$  the number of representations of  $m$  by a complete system of inequivalent forms of discriminant  $D$ , then we have

$$R_D(m) = w(D) \sum_{d \in \mathbb{Z}_+, d|m} \left( \frac{D}{d} \right), \text{ where } w(D) = \begin{cases} 6 & , D = -3 \\ 4 & , D = -4 \\ 2 & , D < -4 \end{cases}.$$

$\lrcorner$

*Proof:*  $\color{red}{?}$  Cf.[Cox13]Lemma3.25.  $\square$

**Conj. (19.3.3.24)[Euler's Convenient Number Conjecture].** If  $D = -4N$  where  $N \in \mathbb{Z}_+$  satisfies  $h_1(D) = 1$ , i.e.  $C(D)$ (19.3.3.21) has one class in each genus  $\color{red}{?}$ , then one of the cases happen:

$$\begin{cases} N \in \{1, 2, 3, 4, 7\} & , h(D) = 1 \\ N \in \{5, 6, 8, 9, 10(\text{and } 10 \text{ more})\} & , h(D) = 2 \\ N \in \{21, 24, 30, 33(\text{and } 20 \text{ more})\} & , h(D) = 4 \\ N \in \{105, 120(\text{and } 15 \text{ more})\} & , h(D) = 8 \\ N \in \{840, 1320, 1365, 1848\} & , h(D) = 16 \end{cases}.$$

$\lrcorner$

### List of Imaginary Quadratic Fields by Class Number

**Thm. (19.3.3.25)**[Imaginary Quadratic Fields with Class Number 1, Baker-Heegner-Stark].

- Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 1 \iff d_{\mathcal{K}} \in \{-3, -4, -7, -8, -11, -19, -43, -67, -163\}.$$

These numbers are called **Heegner-numbers**.

- Let  $D \in \mathbb{Z}_{<0}$  and  $D \equiv 0, 1 \pmod{4}$ , then

$$\text{cl}(\mathcal{O}_D) = 1 \iff D \in \{-3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163\}.$$

┘

*Proof:* 1: Cf.[Cox, P247] ?

2: Let  $\mathcal{K} = \mathbb{Q}(\sqrt{D})$  and  $D = f^2 d_{\mathcal{K}}$ ,  $f \in \mathbb{Z}_+$ , then by (19.3.3.7),  $\text{cl}(\mathcal{O}_{\mathcal{K}}) = 0$ , thus

$$d_{\mathcal{K}} \in \{-3, -4, -7, -8, -11, -19, -43, -67, -163\}.$$

If  $\mu(\mathcal{K}) = \{\pm 1\}$ , then by (19.3.3.7) again,  $f \leq 2$ . And it can be checked directly that  $f = 2$  only happens when  $d_{\mathcal{K}} \equiv \pm 1 \pmod{8}$ , so  $d_{\mathcal{K}} = -7$ .

If  $|\mu(\mathcal{K})| > 2$ , then  $d_{\mathcal{K}} = -3$  or  $-4$ . For  $d_{\mathcal{K}} = -4$ ,  $f \leq 2$ , and it can be checked that  $f = 2$  works, and  $D = -16$ . If  $d_{\mathcal{K}} = -3$ , then  $f \leq 3$ , and it can be checked that  $f = 2$  or  $f = 3$  works.  $\square$

**Prop. (19.3.3.26)**[Imaginary Quadratic Fields with Class Number 2]. For  $n \in \mathbb{Z}_+$  that is square-free,

$$\text{cl}(\mathbb{Q}(\sqrt{-n})) = 2 \iff n \in \{5, 6, 10, 13, 15, 22, 35, 37, 51, 58, 91, 115, 123, 187, 235, 267, 403, 427\}.$$

┘

*Proof:* Cf.[Ribenoim, Classical Theory of Algebraic Numbers].  $\square$

**Prop. (19.3.3.27)**[Imaginary Quadratic Fields with Class Number 3]. Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 3 \iff d_{\mathcal{K}} \in \left\{ \begin{array}{l} -23, -31, -59, -83, -107, -139, -211, -283, \\ -307, -331, -379, -499, -547, -643, -883, -907 \end{array} \right\}.$$

┘

*Proof:* Cf.[Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop. (19.3.3.28)**[Imaginary Quadratic Fields with Class Number 4, Arno]. For  $n \in \mathbb{Z}_+$  that is square-free,

$$\text{cl}(\mathbb{Q}(\sqrt{-n})) = 4 \iff n \in \left\{ \begin{array}{l} 14, 17, 21, 30, 33, 34, 39, 42, 46, 55, 57, 70, 73, 78, 82, 85, 93, \\ 97, 102, 130, 133, 142, 155, 177, 190, 193, 195, 203, 219, 253, \\ 259, 291, 323, 355, 435, 483, 555, 595, 627, 667, 715, 723, 763, \\ 795, 955, 1003, 1027, 1227, 1243, 1387, 1411, 1435, 1507, 1555 \end{array} \right\}.$$

┘

*Proof:* Cf.[Arno, The imaginary quadratic fields of class number 4].  $\square$

**Prop.(19.3.3.29) [Imaginary Quadratic Fields with Class Number 5].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 5 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 47, 79, 103, 127, 131, 179, 227, 347, 443, 523, 571, 619, 683, 691, \\ 739, 787, 947, 1051, 1123, 1723, 1747, 1867, 2203, 2347, 2683 \end{array} \right\}.$$

┘

*Proof:* Cf.[Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.30) [Imaginary Quadratic Fields with Class Number 6].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 6 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 87, 104, 116, 152, 212, 244, 247, 339, 411, 424, 436, 451, 472, 515, \\ 628, 707, 771, 808, 835, 843, 856, 1048, 1059, 1099, 1108, 1147, 1192, \\ 1203, 1219, 1267, 1315, 1347, 1363, 1432, 1563, 1588, 1603, 1843, 1915, \\ 1963, 2227, 2283, 2443, 2515, 2563, 2787, 2923, 3235, 3427, 3523, 3763. \end{array} \right\}.$$

┘

*Proof:* Cf.[Wagner, Class number 5,6 and 7].  $\square$

**Prop.(19.3.3.31) [Imaginary Quadratic Fields with Class Number 7].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 7 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 71, 151, 223, 251, 463, 467, 487, 587, 811, 827, 859, 1163, \\ 1171, 1483, 1523, 1627, 1787, 1987, 2011, 2083, 2179, 2251, \\ 2467, 2707, 3019, 3067, 3187, 3907, 4603, 5107, 5923 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.32) [Imaginary Quadratic Fields with Class Number 9, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 9 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 199, 367, 419, 491, 563, 823, 1087, 1187, 1291, 1423, 1579, 2003, \\ 2803, 3163, 3259, 3307, 3547, 3643, 4027, 4243, 4363, 4483, 4723, \\ 4987, 5443, 6043, 6427, 6763, 6883, 7723, 8563, 8803, 9067, 10627 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.33) [Imaginary Quadratic Fields with Class Number 11, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 11 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 167, 271, 659, 967, 1283, 1303, 1307, 1459, 1531, 1699, 2027, 2267, \\ 2539, 2731, 2851, 2971, 3203, 3347, 3499, 3739, 3931, 4051, \\ 5179, 5683, 6163, 6547, 7027, 7507, 7603, 7867, 8443, 9283, \\ 9403, 9643, 9787, 10987, 13003, 13267, 14107, 14683, 15667 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.34) [Imaginary Quadratic Fields with Class Number 13, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 13 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 191, 263, 607, 631, 727, 1019, 1451, 1499, 1667, 1907, 2131, \\ 2143, 2371, 2659, 2963, 3083, 3691, 4003, 4507, 4643, \\ 5347, 5419, 5779, 6619, 7243, 7963, 9547, 9739, 11467, \\ 11587, 11827, 11923, 12043, 14347, 15787, 16963, 20563 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.35) [Imaginary Quadratic Fields with Class Number 15, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 15 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 239, 439, 751, 971, 1259, 1327, 1427, 1567, 1619, 2243, 2647, 2699, 2843, \\ 3331, 3571, 3803, 4099, 4219, 5003, 5227, 5323, 5563, 5827, 5987, 6067, 6091, \\ 6211, 6571, 7219, 7459, 7547, 8467, 8707, 8779, 9043, 9907, 10243, 10267, \\ 10459, 10651, 10723, 11083, 11971, 12163, 12763, 13147, 13963, 14323, \\ 14827, 14851, 15187, 15643, 15907, 16603, 16843, 17467, 17923, 18043, \\ 18523, 19387, 19867, 20707, 22003, 26203, 27883, 29947, 32323, 34483 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.36) [Imaginary Quadratic Fields with Class Number 17, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 17 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 383, 991, 1091, 1571, 1663, 1783, 2531, 3323, 3947, 4339, \\ 4447, 4547, 4651, 5483, 6203, 6379, 6451, 6827, 6907, 7883, \\ 8539, 8731, 9883, 11251, 11443, 12907, 13627, 14083, 14779, \\ 14947, 16699, 17827, 18307, 19963, 21067, 23563, 24907, \\ 25243, 26083, 26107, 27763, 31627, 33427, 36523, 37123 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop.(19.3.3.37) [Imaginary Quadratic Fields with Class Number 19, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 19 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 311, 359, 919, 1063, 1543, 1831, 2099, 2339, 2459, 3343, 3463, \\ 3467, 3607, 4019, 4139, 4327, 5059, 5147, 5527, 5659, 6803, \\ 8419, 8923, 8971, 9619, 10891, 11299, 15091, 15331, 16363, \\ 16747, 17011, 17299, 17539, 17683, 19507, 21187, 21211, \\ 21283, 23203, 24763, 26227, 27043, 29803, 31123, 37507, 38707 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$



**Prop. (19.3.3.38) [Imaginary Quadratic Fields with Class Number 21, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 21 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 431, 503, 743, 863, 1931, 2503, 2579, 2767, 2819, 3011, 3371, \\ 4283, 4523, 4691, 5011, 5647, 5851, 5867, 6323, 6691, 7907, \\ 8059, 8123, 8171, 8243, 8387, 8627, 8747, 9091, 9187, 9811, 9859, \\ 10067, 10771, 11731, 12107, 12547, 13171, 13291, 13339, 13723, \\ 14419, 14563, 15427, 16339, 16987, 17107, 17707, 17971, 18427, \\ 18979, 19483, 19531, 19819, 20947, 21379, 22027, 22483, 22963, \\ 23227, 23827, 25603, 26683, 27427, 28387, 28723, 28867, 31963, \\ 32803, 34147, 34963, 35323, 36067, 36187, 39043, 40483, \\ 46027, 49603, 51283, 52627, 55603, 58963, 59467, 61483 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

**Prop. (19.3.3.39) [Imaginary Quadratic Fields with Class Number 23, Arno].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $d_{\mathcal{K}}$ , then

$$\text{cl}(\mathcal{O}_{\mathcal{K}}) = 23 \iff -d_{\mathcal{K}} \in \left\{ \begin{array}{l} 647, 1039, 1103, 1279, 1447, 1471, 1811, 1979, 2411, 2671, 3491, \\ 3539, 3847, 3923, 4211, 4783, 5387, 5507, 5531, 6563, 6659, 6703, \\ 7043, 9587, 9931, 10867, 10883, 12203, 12739, 13099, 13187, 21019, \\ 15307, 15451, 16267, 17203, 17851, 18379, 20323, 20443, 20899, \\ 21163, 22171, 22531, 24043, 25147, 25579, 25939, 26251, 26947, \\ 27283, 28843, 30187, 31147, 31267, 32467, 34843, 35107, 37003, \\ 40627, 40867, 41203, 42667, 43003, 45427, 45523, 47947, 90787 \end{array} \right\}.$$

┘

*Proof:* [Arno-Robinson-Wheeler, Imaginary quadratic fields with small odd class number].  $\square$

### Class Number of Cyclotomic Fields

**Prop. (19.3.3.40) [Regular Primes, Kummer].** For  $p \geq 3 \in \text{Prime}$ ,  $p \mid \text{cl}(\mathbb{Q}(\zeta_p))$  iff there exists some  $k \in \mathbb{Z}_+$  s.t.  $k \leq \frac{p-3}{2}$  and  $p$  divides the denominator of  $B_{2k}$  (10.7.1.12). Such primes  $p$  are called **regular primes**.  $\square$

*Proof:* ?  $\square$

**Prop. (19.3.3.41) [Montgomery/Uchida].** For  $p \in \text{Prime}$ ,  $\text{cl}(\mathbb{Q}(\zeta_p)) = 1$  iff  $p \leq 19$ .  $\square$

*Proof:*  $\square$

**Prop. (19.3.3.42) [Iwasawa [Iwa56]].** Let  $E/F \in \text{NField}$  be Galois, then

- If  $E/F$  is totally ramified at some place  $v \in \Sigma_F^{\text{fin}}$ , then  $\text{cl}(F) \mid \text{cl}(E)$ .
- If moreover  $p \in \text{Prime}$  and  $E/F$  is a cyclic extension of  $p$ -power degree and is ramified only at  $v$ , then  $p \nmid \text{cl}(F) \implies p \nmid \text{cl}(E)$ .

┘

*Proof:* 1: Let  $H_1/F, H_2/E$  be the Hilbert class fields of  $F, E$  resp.. Then  $H_1 \cap E = F$  because  $v$  is totally ramified in  $E$ . Thus  $\text{Gal}(H_1E/E) \cong \text{Gal}(H_1/F)$  is Abelian, and  $H_1E/E$  is also unramified by?? So  $H_1E \subset H_2$ , which implies that  $\text{cl}(F) = [H_1 : F][H_2 : E] = \text{cl}(E)$ .

2: Let  $H_1/F, H_2/E$  be the Hilbert class fields of  $F, E$  resp., and let  $E \subset H' \subset H_2$  s.t.  $[H_2 : H']$  is prime to  $p$  and  $[H' : E]$  is a  $p$ -power. If  $p \mid \text{cl}(E)$ , then  $H' \neq E$ . Let  $N = \text{Gal}(H'/E)$  and  $G = \text{Gal}(H'/F)$ . Because  $G$  is a  $p$ -group thus nilpotent(3.1.8.11), there is a normal subgroup  $M$  of  $G$  contained in  $N$  s.t.  $[N : M] = p$ . Then  $N/M$  is contained in the center of  $G/M$ . Now  $G/N$  is cyclic, we get that  $G/M$  is Abelian by(3.1.7.1).

Let  $E \subset E' \subset H'$  corresponds to  $M$ , then  $E'/F$  is Abelian, and all the places of  $E'$  over  $v$  have the same inertia field(19.3.1.22)  $F'$ . Because  $v$  is totally ramified in  $E$  and  $E'/E$  is unramified, we have  $[F' : F] = p$ . No places of  $F$  other than  $v$  is ramified in  $F'/F$ , so  $F'/F$  is cyclic unramified of degree  $p$  and is contained in  $H_1$ . Thus  $\text{cl}(F) = [H_1 : F]$  is therefore divisible by  $p = [F' : F]$ .  $\square$

**Cor. (19.3.3.43).**

- If  $p \in \text{Prime}$  is a regular prime(19.3.3.40), and  $F/\mathbb{Q}$  is an Abelian extension with conductor  $p^e$ , then  $p \nmid \text{cl}(F)$ .
- If  $F/\mathbb{Q}$  is a cyclic extension of degree  $p^{e-1}$  and conductor  $p^e$ , then  $p \nmid \text{cl}(F)$ .

┘

*Proof:* These are just the proposition applied to the extensions  $\mathbb{Q}(\zeta_{p^e})/\mathbb{Q}(\zeta_p)$ ,  $\mathbb{Q}(\zeta_{p^e})/F$  or  $F/\mathbb{Q}$ .  $\square$

## 19.4 Adeles and Ideles

*“Adeles make life possible.”*

– J. Arthur.

**Notation (19.4.0.1).**

- Use notations defined in [Global Fields](#).

┘

### 1 Adeles

**Notation (19.4.1.1) [Adeles, Artin-Whaples [A-W45]].** For  $S \in \Sigma_F$ ,  $\#S < \infty$ ,

- The **adele group** (adele=additive element) of  $F$  is defined to be  $\mathbf{A}_F = \prod'_v (F_v, \mathcal{O}_v)$  [\(12.10.3.18\)](#).
- $\mathbf{A}_{F,S} = \prod_{v \in S} F_v$ .
- $\mathbf{A}_F^S = \prod'_{v \notin S} (F_v, \mathcal{O}_v)$ , called the group of  $S$ -**adeles** of  $F$ .
- $\mathbf{A}'_{F,S} = \mathbf{A}_{F,S} \times (\prod_{v \notin S} \mathcal{O}_{F_v})$ .
- The **finite adeles**  $\mathbf{A}_{F,f} = \mathbf{A}_F^{S_\infty} = \prod'_{v \notin S_\infty} (F_v, \mathcal{O}_v)$ .
- The **infinite adeles**  $\mathbf{A}_{F,\infty} = \mathbf{A}_{F,S_\infty} = \prod_{v \in S_\infty} F_v$ .
- For  $v \in \Sigma_F$ ,  $x \in F_v$ , let  $[x]_{\mathfrak{p}}^\oplus$  be the image of  $x$  under the map  $F_{\mathfrak{p}} \rightarrow \mathbf{A}_F$ .

┘

**Prop. (19.4.1.2) [Extension of Adeles].** If  $L/F$  is separable, then

$$\mathbf{A}_F \otimes_F L \cong \mathbf{A}_L$$

via diagonal embedding.

┘

*Proof:*  $\mathbf{A}_F \otimes_F L \cong \prod'_v (F_v, \mathcal{O}_v \otimes_{\mathcal{O}_F} \mathcal{O}_L)$ , because for any element  $x \in L$ ,  $|\text{Nm}_{L/F}(x)|_v \neq 0$  for only f.m.  $v$ . And there are isomorphisms by diagonal maps.

$$F_v \otimes_F L \cong \prod_{w|v} L_w, \quad \mathcal{O}_v \otimes_{\mathcal{O}_F} \mathcal{O}_L \cong \prod_{w|v} \mathcal{O}_w \text{ (5.2.7.26)}.$$

□

**Prop. (19.4.1.3) [ $F$  Cocompact in  $\mathbf{A}_F$ ].**  $F$  is discrete in  $\mathbf{A}_F$  and  $F \backslash \mathbf{A}_F$  is compact.

┘

*Proof:* Let  $\infty$  be any prime, consider  $U = \{\mathfrak{a} \in \mathbf{A}_F \mid |a|_\infty < 1, |a|_v \leq 1\}$ , then  $U \cap K = 0$  by [\(19.4.1.5\)](#).

Now we show  $\mathbf{A}_F/F$  is compact. By [\(19.4.1.2\)](#), it suffices to prove for  $F = \mathbb{Q}$  or  $\mathbb{F}_p(t)$ . Let  $U_\infty = \{x \in F_\infty \mid |x|_\infty \leq 1\}$ , and Cf. [MIT notes, 22.12]. ?

□

### Ideles

**Notation (19.4.1.4) [Ideles, Chevalley [Che34]].** For  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ .

- The **idele group** (idele=ideal element) of  $F$  is defined to be  $\mathbf{I}_F = \prod'_v (F_v^\times, \mathcal{O}_{F_v}^*)$  [\(12.10.3.18\)](#), which is set-theoretically just  $\mathbf{A}_F^\times$ . Notice the topology on  $\mathbf{I}_K$  is stronger than the subspace topology induced from  $\mathbf{A}_K$ .

- The **idele class group**  $C_F = \mathbf{I}_F / F^\times$ .
- $\mathbf{I}_F$  is naturally a valuation ring with valuation  $|\cdot|$ , called the **idelic norm**.
- $\mathbf{I}_F^1 \subset \mathbf{I}_F$  is the subgroup consisting of elements of idelic norm 1, called the set of **unit ideles**.
- $\mathbf{I}_F^S = \prod'_{v \notin S} (F_v^\times, \mathcal{O}_v^*)$ , called the group of  **$S$ -ideles** of  $F$ .
- $\mathbf{I}_{F,S} = \prod_{v \in S} F_v^\times$ .
- $\mathbf{I}'_{F,S} = \mathbf{I}_{F,S} \times \left( \prod_{v \notin S} \mathcal{O}_{F_v}^* \right)$ .
- $F^S = \mathcal{O}_{F,S}^* = F^\times \cap \mathbf{I}'_{F,S}$  is the set of **S-units** of  $F$  (5.2.7.10).
- The **finite ideles**  $\mathbf{I}_{F,f} = \mathbf{I}_F^{\Sigma_F^\infty} = \prod'_{v \notin \Sigma_\infty} (F_v^\times, \mathcal{O}_v^*)$ .
- The **infinite ideles**  $\mathbf{I}_{F,\infty} = \mathbf{I}_{F,S_\infty} = \prod_{v \in S_\infty} F_v^\times$ .
- For  $v \in \Sigma_F$ ,  $x \in F_v^\times$ , let  $[x]_v$  be the image of  $x$ , and similarly for the embedding  $F_v \hookrightarrow \mathbf{A}_F$ .

┘

**Prop. (19.4.1.5) [Product Formula].** If  $\mathfrak{a} \in F^\times \subset \mathbf{I}_F$ , then  $|\mathfrak{a}|_F = 1$ . In other words,  $F^\times \subset \mathbf{I}_F^1$ .

┘

*Proof:* Consider the restricted product measure  $d\mu$  on  $\mathbf{A}_F$ , then clearly  $d\mu(\mathfrak{a}x) = \|\mathfrak{a}\| d\mu(x)$ , and multiplying by  $\mathfrak{a}$  induces an isomorphism of  $\mathbf{A}_F$ , but preserves the counting measure on  $F$ . But  $\mathbf{A}_F$  is compact (19.4.1.3) thus has finite volume, so  $\|\mathfrak{a}\| = 1$ .  $\square$

**Prop. (19.4.1.6).** If  $F \in \mathbf{FField}$ , the norm group  $|\mathbf{I}_F|$  of the Adelic norm is  $|\mathbf{I}_F| = q^\mathbb{Z}$ , where  $q = \#F_0$  and  $F_0$  is the maximal finite field contained in  $F$ .

┘

*Proof:* This is because in this case,  $F$  corresponds to smooth curve over the prime field  $F_0$ . Then by Weil conjecture (31.1.3.4), there exists a point of  $F_0$  with residue order  $q^n$  for any  $n$  large. Then their quotient gives  $q$ .  $\square$

**Prop. (19.4.1.7) [Splitting of  $\mathbf{I}_F$ ].**

- If  $F \in \mathbf{NField}$ , take  $v \in \Sigma_F^\infty$ , then the exact sequence  $1 \rightarrow \mathbf{I}_F^1 \rightarrow \mathbf{I}_F \xrightarrow{|\cdot|_{\mathbf{A}}} \mathbb{R}_+ \rightarrow 1$  splits, so there are a non-canonical isomorphisms

$$\mathbf{I}_F \cong \mathbf{I}_F^1 \times \mathbb{R}_+, \quad F^\times \backslash \mathbf{I}_F \cong (F^\times \backslash \mathbf{I}_F^1) \times \mathbb{R}_+.$$

- If  $F \in \mathbf{FField}$ , take  $v \in \Sigma_F$ , then for some  $q \in \|v\|^{\mathbb{Z}_+}$ , the exact sequence  $1 \rightarrow \mathbf{I}_F^1 \rightarrow \mathbf{I}_F \xrightarrow{|\cdot|_{\mathbf{A}}} q^\mathbb{Z} \rightarrow 1$  splits, so there are a non-canonical isomorphisms

$$\mathbf{I}_F \cong \mathbf{I}_F^1 \times q^\mathbb{Z}, \quad F^\times \backslash \mathbf{I}_F \cong (F^\times \backslash \mathbf{I}_F^1) \times q^\mathbb{Z}.$$

In particular, any Hecke character of  $F$  is of the form  $\chi = \chi_0 \cdot \|\cdot\|_{\mathbf{A}}^{\lambda_i}$  for some  $\lambda \in \mathbb{R}$ .

┘

**Lemma (19.4.1.8).**  $\mathbf{I}_F^1$  is closed in  $\mathbf{A}_F$ , and the subspace topologies from  $\mathbf{A}_F$  and  $\mathbf{I}_F$  are the same on  $\mathbf{I}_F^1$ .

┘

*Proof:* ?  $\square$

**Lemma (19.4.1.9) [Blichfeldt-Minkowski].** There exists  $\kappa_F > 0$  s.t.: For any  $x \in \mathbf{I}_F$  s.t.  $|x|_F \geq \kappa_F$ , if

$$W(x) = \{y \in \mathbf{I}_F : |y_v|_v \leq |x_v|_v, \forall v \in \Sigma_F\}.$$

then  $W(x) \cap F^\times \neq \emptyset$ .

┘

*Proof:* ?

□

**Prop. (19.4.1.10) [Fujisaki's Lemma].**  $F^\times$  is discrete and cocompact in  $\mathbf{I}_F$ . Thus  $C_F = F^\times \backslash \mathbf{I}_F$  is Hausdorff and locally compact. ┘

*Proof:*  $F^\times$  is discrete in  $\mathbf{I}_F$  because it is already discrete in  $\mathbf{A}_F$  (19.4.1.3). By (19.4.1.8), using (19.4.1.9), it suffices to show that for  $\|x\|_F \geq C$ ,  $W(x) \cap \mathbf{I}_F^1 \rightarrow \mathbf{I}_F^1/F^\times$  is surjective: If  $y \in \mathbf{I}_F^1$ , then  $\|x/y\| = \|x\| \geq C$ , thus there exists  $z \in K^\times \cap W(x/y)$ , thus  $zy \in W(x)$ , and  $y$  is in the image of  $W(x) \cap \mathbf{I}_F^1$ . □

**Cor. (19.4.1.11) [Shimura Varieties for  $G_m$ ].**

- If  $F \in \mathbf{NField}$ , then  $F^\times \backslash \mathbf{I}_F / \mathbf{I}_{F,\infty}$  is compact.
- For any open compact subgroup  $\mathcal{K} \subset \mathbf{I}_{F,f}$ , the set

$$X_{\mathcal{K}} = F^\times \backslash \mathbf{I}_F / \mathcal{K} \mathbf{I}_{F,\infty}$$

is a finite set. ┘

*Proof:* 1 follows from (19.4.1.7)(19.4.1.10) and the fact  $\text{val}(I_{F,\infty}) = \mathbb{R}_+$ .

2: Notice that  $\mathcal{K} \mathbf{I}_{F,\infty}$  is open in  $\mathbf{I}_F$ , thus  $\mathbf{I}_F / \mathcal{K} \mathbf{I}_{F,\infty}$  is discrete. If  $F \in \mathbf{NField}$ , then by 1,  $X_{\mathcal{K}}$  is both discrete and compact, thus finite. If  $F \in \mathbf{FField}$ , then notice first that there is an exact sequence

$$1 \rightarrow F^\times \backslash \mathbf{I}_F^1 / \mathcal{K}^1 \rightarrow F^\times \backslash \mathbf{I}_F / \mathcal{K} \xrightarrow{\text{val}_F} \text{val}_F(\mathbf{I}_F) / \text{val}_F(\mathcal{K}) \rightarrow 1,$$

and then use (19.4.1.10) and the fact  $\# \text{val}_F(\mathbf{I}_F) / \text{val}_F(\mathcal{K}) < \infty$ . □

**Prop. (19.4.1.12).** If  $L/F$  is Galois, then  $\mathbf{I}_F \subset \mathbf{I}_L$ , and  $\mathbf{I}_L^{\text{Gal}(L/F)} = \mathbf{I}_F$ , this is be the diagonal inclusion to all the primes above a given prime, and the action is by  $(\sigma \mathfrak{a})_{\mathfrak{p}} = \sigma \mathfrak{a}_{\sigma^{-1}\mathfrak{p}}$ . This induces an inclusion  $\mathbf{C}_F \subset \mathbf{C}_L$  and  $\mathbf{C}_L^{\text{Gal}(L/F)} = \mathbf{C}_F$ . The last assertion uses long exact sequence and  $H^1(\text{Gal}(L/F), L^\times) = 0$ . ┘

**Lemma (19.4.1.13).** The map  $\mathbf{I}_F \rightarrow \mathbf{A}_F \times \mathbf{A}_F : x \mapsto (x, x^{-1})$  is a homeomorphism of  $\mathbf{I}_F$  onto a closed subspace of  $\mathbf{A}_F \times \mathbf{A}_F$ . ┘

*Proof:* Compare their topological basis. □

**Prop. (19.4.1.14) [Idele Groups and Extensions].** For an extension of global fields  $L/F$ ,

- the diagonal embedding  $\mathbf{A}_F \rightarrow \mathbf{A}_L$  induces a closed embedding, and also a closed embedding  $\mathbf{I}_F \rightarrow \mathbf{I}_L$  by (19.4.1.13), thus also a closed embedding  $\mathbf{I}_F / F^\times \rightarrow \mathbf{I}_L / L^\times$ .
- $\mathbf{A}_L$  is a finite  $\mathbf{A}_F$ -module, thus there is a norm map  $\text{Nm}_{L/K} : \mathbf{A}_L \rightarrow \mathbf{A}_F$ , which restricts to  $\text{Nm}_{L/K} : \mathbf{I}_L \rightarrow \mathbf{I}_F$ , which is compatible with  $\text{Nm}_{L/K} : L^\times \rightarrow K^\times$ , thus inducing a map  $\text{Nm}_{L/K} : \mathbf{C}_L \rightarrow \mathbf{C}_F$ . Then this map is continuous, open and proper.
- $[n] : \mathbf{C}_F \rightarrow \mathbf{C}_F$  is continuous and proper. ┘

*Proof:* 2: It is continuous because it is compatible with the local norms  $\text{Nm}_{L_{\mathfrak{p}}/F_{\mathfrak{p}}}$ , and  $\text{Nm}_{L_{\mathfrak{p}}/F_{\mathfrak{p}}}^{-1}(\mathcal{O}_{F,\mathfrak{p}}^\times) = \mathcal{O}_{L,\mathfrak{p}}^\times$ . To show it is open, use the fact the local norms are open and for unramified places  $\text{Nm}_{L_{\mathfrak{p}}/F_{\mathfrak{p}}} \mathcal{O}_{L,\mathfrak{p}} = \mathcal{O}_{F,\mathfrak{p}}$  (19.1.5.21)(19.7.2.21). To show it is proper, use the splitting  $\mathbf{C}_F \cong F^\times \backslash \mathbf{I}_F^1 \times q^{\mathbb{Z}}$  (19.4.1.7) and the fact  $F^\times \backslash \mathbf{I}_F^1$  is proper (19.4.1.10). □

3: Use the splitting (19.4.1.7).

$\mathbf{C}_F$

**Def. (19.4.1.15) [Hecke Character, Hecke[[Hec20](#)]].** A **Hecke character** of  $F$  is a character of  $\mathbf{C}_F$ .

┘

**Def. (19.4.1.16) [Unramified Places].** Let  $\chi$  be a Hecke character of  $\mathbf{C}_F$ , then for a.e.  $v \in \Sigma_F^{\text{fin}}$ , the conductor of  $\chi_v$  is  $\mathcal{O}_v^*$ . For these  $v$ ,  $\chi$  is said to be **unramified** at  $v$  (which is compatible with the definition in [\(19.4.2.2\)](#)).

┘

**Prop. (19.4.1.17).** Let  $p \in \text{Prime}$ ,  $F \in p\text{-FField}$ , and  $\ell \in \text{Prime} \setminus \{p\}$ ,  $E \in \ell\text{-NField}$ , then any continuous homomorphism  $\chi : \mathbf{C}_F \rightarrow \mathcal{O}_E^*$  that is unramified outside a finite set of places is of the form  $\chi = \chi_1 \cdot c^{\deg(\cdot)}$ , where  $\chi_1 : \mathbf{C}_F \rightarrow \mathcal{O}_E^*$  is of finite order and  $c \in \mathcal{O}_E^*$ .

┘

*Proof:* Let  $\sigma \in \mathbf{C}_F^{\text{ab}}$  s.t.  $\deg(\sigma) = 1$ , let  $c = \chi(\sigma)$ , and  $\chi_1 = \chi \cdot c^{-\deg(\cdot)}$ , then  $\chi_1(\mathbf{C}_F) = \chi_1(\mathbf{I}_F)$ . But as  $\#\text{Cl}(F) < \infty$ , by [\(19.7.4.8\)](#), it suffices to show that  $\chi_1(F^\times U^1/F^\times)$  is finite. Then it suffices to show that  $\#\chi(U^1) < \infty$ . But if  $\chi$  is unramified outside a finite set  $S$  of places, then it suffices to show that  $\#\chi(\prod_{v \in S} \mathcal{O}_v^*) < \infty$ . But this is because  $\prod_{v \in S} \mathcal{O}_v^*$  has a pro- $p$ -group of finite index and  $\mathcal{O}_E^*$  has a pro- $\ell$ -group of finite index.  $\square$

**Def. (19.4.1.18) [Norm Groups].** Let  $\mathcal{N}_{E/F} = \text{Nm}_{E/F} \mathbf{C}_E$ , called the **norm group** of  $E/F$ .

┘

**Prop. (19.4.1.19) [Connected Component of identity of  $\mathbf{C}_F$ ].** If  $F \in \text{NField}$ , let  $\mathbf{I}_{F,\infty}^0 \cong \mathbb{R}_+^{r_1} \times (\mathbb{C}^\times)^{r_2}$  be the connected component of identity of  $\mathbf{I}_{F,\infty}$ ,  $D_F \subset \mathbf{C}_F$  the closure of  $\mathbf{I}_{F,\infty}^0 \subset \mathbf{C}_F$ , then

- $D_F$  is the connected component of identity of  $\mathbf{C}_F$ .
- $D_F = \cap_{n \in \mathbb{Z}_+} \mathbf{C}_F^n$  is the group of divisible elements of  $\mathbf{C}_F$ .
- $\mathbf{C}_F/D_F$  is a profinite group. (which will be isomorphic to  $\text{Gal}_F^{\text{ab}}$ , as we will see in [\(19.7.3.29\)](#)).

┘

*Proof:*  $\mathbf{I}_{F,\infty}^0$  is divisible, thus so does its image in  $\mathbf{C}_F$ . Then  $D_F$  is also divisible as  $[n] : \mathbf{C}_F \rightarrow \mathbf{C}_F$  is continuous and proper [\(19.4.1.14\)](#).

Consider the map  $\prod_{v \in \Sigma_F^{\text{fin}}} \mathcal{O}_{F,v}^* \rightarrow \mathbf{C}_F/D_F$ , its cokernel is finite because  $\mathbf{C}_F/\prod_{v \in \Sigma_F^{\text{fin}}} \mathbf{I}_{F,\infty} \cong \text{Cl}(\mathcal{O}_F)$  is finite and  $\mathbf{I}_{F,\infty}/\mathbf{I}_{F,\infty}^0$  is finite. But  $\prod_{v \in \Sigma_F^{\text{fin}}} \mathcal{O}_{F,v}^*$  is a profinite group, thus  $\mathbf{C}_F/D_F$  is also a profinite group.

To show  $D_F = \mathbf{C}_F^0$ , firstly  $\mathbf{C}_F^0 \subset D_F$  by the fact  $\mathbf{C}_F/D_F$  is connected thus totally disconnected, and the reverse is true because  $D_F$  is connected.

To show  $D_F = \cap_{n \in \mathbb{Z}_+} \mathbf{C}_F^n$ , notice any divisible element is in  $D_F$  as a profinite group  $\mathbf{C}_F/D_F$  doesn't contain non-trivial divisible elements.  $\square$

**Thm. (19.4.1.20) [Class Numbers and Unit Theorems].** For  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  s.t.  $S \neq \emptyset$ ,

1. The  $S$ -class group  $\text{Cl}(\mathcal{O}_{F,S}) \cong F^\times \backslash \mathbf{I}_F / \mathbf{I}_{F,S}'$  is finite. In particular,  $\mathbf{I}_\mathbb{Q} = \mathbb{Q}^\times \times \mathbb{R}_+^\times \times \prod_{p \in \text{Prime}} \mathbb{Z}_p^\times$ .
2. If  $x \in F^\times$  and  $|x|_v = 1$  for any  $v \in \Sigma_F$ , then  $x \in F_{\text{tor}}^\times$ .
3. The  $S$ -unit group [\(5.2.7.10\)](#)  $\mathcal{O}_{F,S}^* \cong F_{\text{tor}}^\times \times \mathbb{Z}^{\#S-1}$ .

┘

*Proof:* 1 follows from [\(19.4.1.11\)](#).

2, 3: Consider the regulator map

$$\text{Reg}^S : \prod_{v \in S} F_v^\times \rightarrow \mathbb{R}^{\#S} : (x_v) \mapsto (-\log |x_v|_v).$$

Then this map restricted to  $F^S$  has image a discrete subset of the hyperplane  $H = \{\sum x_i = 0\}$ . In fact the image is a full lattice in  $H$ : let  $n_1$  be the number of infinite places in  $S$  and  $n_2$  the number of finite places in  $S$ , then  $\text{Reg}^S((\prod_{v \in S} F_v^\times)^1) \cong \mathbb{R}^{n_1-1} \times \mathbb{Z}^{n_2}$  if  $n_1 > 0$  and  $\mathbb{Z}^{\#S-1}$  if  $n_1 = 0$ . Then notice  $(\prod_{v \in S} F_v)^1 / F^S$  is finite, which is true as  $(\mathbf{I}_F^S)^1 / F^S$  is compact, so the image of  $F^S$  is a full lattice of  $H$ .

Also notice if  $x \in F^S$  is in the kernel of  $\text{Reg}^S$ , then  $|x|_v = 1$  for any  $v \in \Sigma_F$ , so  $\{x^n | n \in \mathbb{Z}\}$  is a bounded subset in  $\mathbf{I}'_{F,S}$ , but it is also contained in the discrete subset  $F^S$  (19.4.1.3), thus it is a finite subset, and  $x$  is a root of unity.  $\square$

**Cor. (19.4.1.21).** If  $S \subset \Sigma_F$  is sufficiently large, then  $\mathbf{I}_F = \mathbf{I}_F^S \cdot F^\times$ , and  $\mathbf{C}_F = \mathbf{I}_F^S \cdot F^\times / F^\times$ .  $\lrcorner$

*Proof:* The ideal class group is finite, hence we can find a finite set of representative for it. Only finite set of primes are involved in it, thus we let  $S$  contain all these primes and infinite primes, then for any  $\mathfrak{a}$ ,  $\prod_{p \nmid \infty} a_p = A_i \cdot (x)$ , and  $A_i \in \mathbf{I}_F^S$ , hence  $\mathfrak{a} \in \mathbf{I}_F^S \cdot F^\times$ .  $\square$

**Prop. (19.4.1.22) [Hecke Characters as Dirichlet Characters].** For an finite modulus  $\mathfrak{m} \in \text{Ideal}^\times(\mathcal{O}_F)$ , we may define a **Dirichlet character** as a character

$$\chi_0 : (\mathcal{O}_F / \mathfrak{m})^* \rightarrow \mathbb{C}^\times.$$

There is an exact sequence

$$1 \rightarrow \mathcal{O}_F^* \rightarrow \prod_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^* \times F_\infty^\times \rightarrow \mathbf{C}_F \rightarrow \text{Cl}(F) \rightarrow 1.$$

For any Hecke character  $\chi$  of  $F$ ,  $\chi_f$  factors through  $\prod_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^* \rightarrow (\mathcal{O}_F / \mathfrak{m})^*$  for some finite modulus  $\mathfrak{m} = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_k^{e_k}$  by continuity constraint, thus inducing a Dirichlet character  $\chi_0$  modulo  $\mathfrak{m}$ , called the **Dirichlet component of  $\chi$** .

However, not every naive Dirichlet character  $\chi_0$  and an infinite character  $\chi_\infty$  can form a Hecke character. For example, they must satisfy  $\chi_0(\varepsilon)\chi_\infty(\varepsilon) = 1$  for any unit  $\varepsilon \in \mathcal{O}_F^*$ . And if  $\chi_0(\varepsilon)\chi_\infty(\varepsilon) = 1$ , then there exists exactly  $\text{cl}(F)$ -many Hecke character  $\chi$  extending  $\chi_0$  and  $\chi_\infty$ , because the surjection

$$F^\times \backslash \mathbf{I}_F / \prod_{\mathfrak{p}} (1 + \mathfrak{p}^{e_{\mathfrak{p}}}) \rightarrow \text{Cl}(F) \rightarrow 1$$

has sections (take case of the infinite places).  $\lrcorner$

**Def. (19.4.1.23) [Hecke Characters and Ideal Characters].** Any Hecke character  $\chi$  of  $F$  (19.4.1.15) is of the form  $\chi = \chi_f \times \chi_\infty$ , where  $\chi_f : \mathbf{I}_{F,f} \rightarrow \mathbb{C}$  and  $\chi_\infty : (F \otimes \mathbb{R})^\times \rightarrow \mathbb{C}$ . Then by continuity, there exists a smallest modulus  $\mathfrak{m} = \prod_{v \in \Sigma_F^{\text{fin}}} \mathfrak{p}_v^{e_v}$  s.t.  $\chi_f$  is trivial on

$$\prod_{v \in \Sigma_F^{\text{fin}}} (1 + \mathfrak{p}_v^{e_v}).$$

Thus the restriction of  $\chi$  to  $\mathbf{I}^{\text{m}\infty}$  gives us a ideal class map  $J^{\text{m}} \rightarrow \mathbb{C}$  which we also denote by  $\chi$ . And the restriction of  $\chi$  to  $\prod_{v \in \Sigma_F^{\text{fin}}} \mathcal{O}_{F_v}^*$  gives a Dirichlet character  $\chi_0^{-1} : (\mathcal{O}_F / \mathfrak{m})^* \rightarrow \mathbb{C}$ .

Then we have  $\chi((\alpha))\chi_\infty(\alpha) = \chi_0(\alpha)$  for any  $\alpha \in \mathcal{O}$  that is prime to  $\mathfrak{m}$ .  $\lrcorner$

**Example (19.4.1.24) [Dirichlet Characters and Hecke Characters for  $\mathbb{Q}$ ].** Hecke characters of  $\mathbb{Q}$  are exactly of the form  $\chi(x) = \chi_1(x)|x|^\lambda$  for some  $\lambda \in i\mathbb{R}$ , where  $\chi_1$  a Hecke character of finite order.

And a Hecke character of finite order with conductor  $N$  corresponding to a primitive Dirichlet character  $\chi_0 \in \text{Diri}'(N)$  via (19.4.1.20):

$$\chi : \mathbf{I}_{\mathbb{Q}} \rightarrow \prod_p \mathbb{Z}_p^* \rightarrow (\mathbb{Z}/(N))^* \xrightarrow{\chi_0^{-1}} \mathbb{C}^\times.$$

In this way,  $\chi([p]_p) = \chi_0(p)$  for any  $p \in \text{Prime}$  s.t.  $(p, N) = 1$ . ┘

### Algebraic Hecke-Characters

References are [CCO14], [Serre-Tate, Good reduction of abelian varieties], [Serre, Abelian l-adic representations and elliptic curves], [Milne, Canonical models of (mixed) Shimura varieties and automorphic vector bundles], [Milne-Shih, Langlands' construction of the Taniyama group].

**Def. (19.4.1.25) [Algebraic Characters].** For  $K'/K, F/K \in \text{Field}$  s.t.  $\#K = \infty$  and  $K'/K$  is separable, a homomorphism  $\chi : (K')^\times \rightarrow F^\times$  is called an **algebraic character**(w.r.t.  $K$ ) if one of the following equivalent conditions holds:

- $\chi$  is induced by a homomorphism  $\varphi : (\text{Res}_{K'/K}(\mathbb{G}_m))_F \rightarrow \mathbb{G}_m \in \text{AlgGrp}/F$ , which is equivalent to a homomorphism  $\varphi' : \text{Res}_{K'/K}(\mathbb{G}_m) \rightarrow \text{Res}_{F/K}(\mathbb{G}_m)$  when  $F/K$  is finite.
- If  $e_1, \dots, e_n$  is a  $K$ -basis of  $K'$ , then there exists  $f(X_1, \dots, X_n) \in F(\mathbf{X})$  defined on each non-zero element of  $K^n$ , s.t.  $\chi(\sum x_i e_i) = f(x_1, \dots, x_n)$ .
- There exists a  $\text{Gal}_F$ -invariant function  $m : \text{Hom}_K(K', F^{\text{sep}}) \rightarrow \mathbb{Z}$  s.t.

$$\chi(x) = \prod_{\tau \in \text{Hom}_K(K', F^{\text{sep}})} \tau(x)^{m(\tau)}, \text{ where } x \in (K')^\times.$$

┘

*Proof:* Cf. [CCO14]P117? □

**Remark (19.4.1.26).** Situation as in (19.4.1.25), if  $K \subset F \subset K'$ , then the only algebraic character  $(K')^\times \rightarrow F^\times$  is  $\text{Nm}_{K'/F}$ . ┘

**Def. (19.4.1.27) [Algebraic Idele-Characters].** For  $F \in \text{NField}$ , an **algebraic idele-character** is a continuous homomorphism  $\epsilon : \mathbf{I}_F \rightarrow L^\times$  where  $L \in \text{Field}^0$  is endowed with discrete topology, s.t. its restriction to  $F^\times \subset \mathbf{I}_F$  is an algebraic character  $\epsilon_{\text{alg}}$  w.r.t.  $\mathbb{Q}$  (19.4.1.25), called the **algebraic part of the algebraic idele-character**.

And for each  $\mathfrak{m} \in \text{Ideal}(\mathcal{O}_F)^\times$ , a homomorphism  $\chi : I_{\mathfrak{m}} \rightarrow L^\times$  for some field  $L \in \text{Field}^0$  is called an **algebraic idele-character with conductor  $\mathfrak{m}$**  if there exists an algebraic character  $\chi_{\text{alg}} : F^\times \rightarrow L^\times$  (19.4.1.25) s.t.

$$\chi((x)) = \chi_{\text{alg}}(x)$$

for each  $x \in F^\times$  s.t.  $x \in 1 + \mathfrak{m}$  and is positive at all real places of  $F$ . Such  $\chi_{\text{alg}}$  is called the **algebraic part of the algebraic idele-character**. ┘

**Def. (19.4.1.28) [Serre-Tori].** ┘

**Prop. (19.4.1.29) [Algebraic Idele-Characters and Serre-Tori, Serre].** Let  $F, L \in \text{NField}$  and  $\mathfrak{m} \in \text{Ideal}(\mathcal{O}_F)^\times$ .



- For each  $\rho : S_F \rightarrow \text{Res}_{L/\mathbb{Q}}(\mathbb{G}_m) \in \text{GrpSch}/\mathbb{Q}$ , the composition

$$\mathbf{I}_F \rightarrow S_F(\mathbb{Q}) \xrightarrow{\rho} L^\times$$

is an algebraic Hecke-character (19.4.1.27), and every algebraic Hecke-character  $\mathbf{I}_F \rightarrow L^\times$  is of this form.

- For each  $\rho_m : S_m \rightarrow \text{Res}_{L/\mathbb{Q}}(\mathbb{G}_m) \in \text{GrpSch}/\mathbb{Q}$ , the composition

$$\mathbf{I}_m = \mathbf{I}_F^{S(m^\infty)} / \prod_{v \in \Sigma_F^{\text{fin}} \setminus S(m)} \mathcal{O}_{F_v}^* \rightarrow \mathbf{I}_F/U_{m^\infty} \rightarrow S_m(\mathbb{Q}) \xrightarrow{\rho_m} L^\times$$

is an algebraic Hecke-character with conductor  $\mathfrak{m}$ , and every algebraic Hecke-character  $\mathbf{I}_m \rightarrow L^\times$  is of this form.

┘

*Proof:* Cf. [Abelian  $l$ -adic representations and elliptic curves]2.2.6. □

**Def. (19.4.1.30) [Algebraic Hecke-Characters].** For  $F \in \mathbf{NField}$ , an (complex-valued) **algebraic Hecke-character** of  $F$  is a Hecke-character  $\chi : \mathbf{C}_F \rightarrow \mathbb{C}^\times$  s.t. there exists a map  $a : \text{Hom}(F, \mathbb{C}) \rightarrow \mathbb{Z}$  that

$$\chi|_{(\mathbf{I}_{F,\infty})^0}(\mathbf{x}) = \prod_{\tau \in \text{Hom}(F, \mathbb{C})} x_\tau^{a_\tau}.$$

┘

**Prop. (19.4.1.31) [Global Class Field Theory for Algebraic Hecke-Characters].** Suppose  $F \in \mathbb{C}$ ,  $\chi$  is an algebraic Hecke-character of  $F$ , then

- For each  $\ell \in \mathbf{Prime}$  and an isomorphism  $\iota_\ell : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$ , there is a unique continuous character

$$r_{\ell, \iota_\ell}(\chi) : \text{Gal}_F \rightarrow \mathbb{C}^\times$$

s.t. for each  $v \in \Sigma_F^{\text{fin}} \setminus S(\ell)$ , we have

$$\iota \circ r_{\ell, \iota_\ell}(\chi)|_{W_{F_v}} \circ \text{Art}_{F_v} = \chi_v.$$

- There exists  $\text{wt}(\chi) \in \mathbb{Z}$  s.t.  $|\chi| = \|\cdot\|_F^{-\text{wt}(\chi)/2} : \mathbf{C}_F \rightarrow \mathbb{C}^\times$ .

And every geometric  $\ell$ -adic characters of  $\text{Gal}_F$  (30.1.2.1) arises in this way. ┘

*Proof:* Cf. [Potential automorphy and change of weight, Gee-Geraghty-Taylor]Appendix A.2. □

## 2 Fourier Analysis on Adeles

References are [Tat67b, R-V99, Poo15].

### Local Notations

**Def. (19.4.2.1) [Normalized Valuations].** Let  $d\mu$  a Haar measure on  $K$ , let the valuation on  $K$  be given by  $d\mu(\alpha\xi) = |\alpha|d\mu(\xi)$ . Then for  $k \in K$ ,

$$|k| = \begin{cases} |k| & K = \mathbb{R} \\ |k|^2 & K = \mathbb{C} \\ \frac{1}{\|\mathfrak{p}\|^{v(k)}} & K \in p\text{-LField} \end{cases}.$$

┘

*Proof:* If  $K = \mathbb{R}, \mathbb{C}$ , this is routine calculation. If  $K$  is non-Archimedean, then by the translation invariance of  $\mu$ ,  $\mu(\alpha\mathcal{O}) = \frac{\mu(\mathcal{O})}{N(\alpha)} = |\alpha|\mu(\mathcal{O})$ .  $\square$

**Def. (19.4.2.2)[Unramified Quasi-Character].** The multiplicative group  $K^\times$  is also a locally compact group. For a quasi-character  $\chi$  of  $K^\times$ , it is called **unramified(quasi-character)** iff  $\chi(\mathcal{O}_K^*) = 1$ .

An unramified quasi-character on  $K^\times$  is all of the form  $|\cdot|^s$  for  $s \in \mathbb{C}$ .  $\lrcorner$

*Proof:* An unramified quasi-character is equivalent to a continuous group homomorphism from  $\text{val}(K^\times) \rightarrow \mathbb{Z}$ . But  $\text{val}(K^\times)$  must be isomorphic to  $\mathbb{Z}$  or  $\mathbb{R}$ , so the assertion follows from (12.11.1.4).  $\square$

**Lemma (19.4.2.3)[Canonical Character of Local Fields].** Consider  $k$  the closure of the base field of  $K$ , which is  $\mathbb{R}, \mathbb{Q}_p$  or  $\mathbb{F}_p((t))$  by Ostrowski (12.2.3.18). Now let

$$\lambda(x) = \begin{cases} x \pmod{1} & k = \mathbb{R} \\ \text{a rational number } \lambda(x) \text{ that } \lambda(x) - x \in \mathbb{Z}_p \text{ in } \mathbb{Q}/\mathbb{Z} & k = \mathbb{Q}_p \\ a_{-1}/p = \text{res}(x)/p & k = \mathbb{F}_p((t)) \end{cases}$$

Then  $\lambda$  is a continuous additive function on  $k$ . Now let

$$\Lambda(x) = \begin{cases} \lambda(\text{tr}_{K/k}(x)) & \text{number field case} \\ \lambda(\text{tr}_{K_v/k_v}(x\omega_v)) & \text{function field case, where } \omega \text{ is a chosen global meromorphic form on } X. \end{cases}$$

And  $X(x) = e^{2\pi i \Lambda(x)}$ . Notice that this is just a rigorous definition of the character  $e^{2\pi i \text{tr}_{K/k}(x)}$ .  $\lrcorner$

**Cor. (19.4.2.4).**  $F(\eta) = e^{2\pi i \Lambda(\eta\xi)}$  is trivial on  $\mathcal{O}_K$  is equivalent to  $\xi \in \mathfrak{d}^{-1}$ , where  $\mathfrak{d} = \mathfrak{d}_{K/k}$ . In other words, adopting the isomorphism of (12.11.1.36),  $\mathcal{O}^\perp = \mathfrak{d}^{-1}$ ,  $(\mathfrak{d}^{-1})^\perp = \mathcal{O}$ . And  $\mathfrak{c}(\Lambda) = \mathfrak{d}^{-1}$ .  $\lrcorner$

*Proof:* Because  $\Lambda(\eta\mathcal{O}) = 0$  iff  $\text{tr}_{K/k}(\eta\mathcal{O}) \subset \mathcal{O}_k$ , which is equivalent to  $\eta \in \mathfrak{d}^{-1}$ .  $\square$

**Prop. (19.4.2.5)[Canonical Self-Adjoint Haar Measures].** We can calculate the self-adjoint Haar measure w.r.t. the canonical character on  $K^+$  (19.4.2.3) as follows:

$$d\mu = \begin{cases} d\mathbf{m} & , K = \mathbb{R} \\ 2 d\mathbf{m} & , K = \mathbb{C} \\ \text{the measure that } \mu(\mathcal{O}_K) = \|\mathfrak{d}\|^{-1/2} & , \text{otherwise} \end{cases}$$

$\lrcorner$

*Proof:* We only calculate for the  $p$ -adic fields **?**.

Let  $f = \mathbf{1}_{\mathcal{O}}$ , then

$$\widehat{f}(\eta) = \int_{\mathcal{O}} e^{-2\pi i \Lambda(\eta\xi)} d\mu(\xi) = \begin{cases} \mu(\mathcal{O}) & \eta \in \mathfrak{d}^{-1} \\ 0 & \text{otherwise} \end{cases} = \mu(\mathcal{O}) \mathbf{1}_{\mathfrak{d}^{-1}}(\eta).$$

By (19.4.2.4) and (12.10.1.14). So

$$I_{\mathcal{O}}(\xi) = \int_G \widehat{f}(F(\eta))(\xi, F(\eta)) d\mu(\eta) = \int_{\mathfrak{d}^{-1}} \mu(\mathcal{O}) e^{2\pi i \Lambda(\eta\xi)} d\mu(\eta) = \mu(\mathcal{O}) \mu(\mathfrak{d}^{-1}) I_{\mathcal{O}}(\eta).$$

So  $\mu(\mathcal{O}) \mu(\mathfrak{d}^{-1}) = N(\mathfrak{d}) \mu(\mathcal{O})^2 = 1$ , which shows the desired result.  $\square$

**Remark (19.4.2.6).** In fact, if we use other characters  $\psi$  of  $K^+$ , then  $\mu(\mathcal{O}_K) = \|\mathfrak{c}_\psi\|^{1/2}$ .  $\lrcorner$

**Cor. (19.4.2.7) [Quasi-Character of  $K^\times$ ].** There is a continuous morphism from  $K^\times \rightarrow \mathcal{O}_K^*$ :  $\tilde{\alpha} = \alpha/\pi^{v(\alpha)}$  when  $\alpha$  is non-Archimedean or  $\tilde{\alpha} = \alpha/|\alpha|$  if  $K$  is Archimedean. So any quasi-character  $c$  is of the form  $c(\alpha) = c(\tilde{\alpha})$  times an unramified quasi-character, which is of the form  $|\cdot|^s$ , where  $\operatorname{Re}(s)$  is called the **exponent** of  $c$ . Now  $\mathcal{O}_K^*$  is a compact group, so continuous quasi-characters  $\tilde{c}$  on it must be a character.  $\lrcorner$

**Def. (19.4.2.8) [Haar Measure on  $K^\times$ ].** Notice that if  $g(\alpha) \in C_c(K^\times)$ , then  $\frac{g(\alpha)}{|\alpha|} \in C_c(K^+ \setminus \{0\})$ , so if we define  $\Phi(g) = \int_{K^+ \setminus \{0\}} g(\xi) |\xi|^{-1} d\xi$ , then

$$\Phi(ag) = \int_{K^+ \setminus \{0\}} g(a\xi) |a\xi|^{-1} |a| d\xi = \int_{K^+ \setminus \{0\}} g(a\xi) |a\xi|^{-1} da\xi = \Phi(g).$$

By (19.4.2.1). So By Riesz representation, there is a Haar measure  $d_1^\times \mu$  on  $K^\times$  that  $\int_{K^\times} g(\alpha) d_1^\times \mu(\alpha) = \int_{K^+ \setminus \{0\}} g(\xi) |\xi|^{-1} d\mu(\xi)$ , for any  $g \in C_c(K^\times)$ .

But when  $K$  is non-Archimedean, renormalize  $d^\times \mu = (1 - \frac{1}{\|\mathfrak{p}\|})^{-1} d_1^\times \mu$ .  $\lrcorner$

**Remark (19.4.2.9).** The reason behind this normalization is when  $d\mu$  is the canonical measure (19.4.2.5), we want to make  $\mu^\times(\mathcal{O}^*) = \|\mathfrak{d}\|^{-1/2}$ :

$$\int_{\mathcal{O} \setminus \{0\}} d\xi = \sum_{k=0}^{\infty} \int_{\pi^k \mathcal{O}^*} d\xi = (1 + \frac{1}{N\mathfrak{p}} + \frac{1}{N\mathfrak{p}^2} + \dots) \int_{\mathcal{O}^*} d\xi = \frac{1}{1 - \frac{1}{N\mathfrak{p}}} \int_{\mathcal{O}^*} d\xi$$

so

$$\int_{\mathcal{O}^*} d_1^\times \alpha = \int_{\mathcal{O}^*} |\xi|^{-1} d\xi = \frac{\|\mathfrak{p}\| - 1}{\|\mathfrak{p}\|} \int_{\mathcal{O} \setminus \{0\}} d\xi = \frac{\|\mathfrak{p}\| - 1}{\|\mathfrak{p}\|} \|\mathfrak{d}\|^{-1/2} \quad (19.4.2.5)$$

$\lrcorner$

**Def. (19.4.2.10) [Schwartz-Bruhat Function].** Define the set  $\mathcal{S}(K)$  of **Schwartz-Bruhat function** on  $K$  as Schwartz functions on  $K$  if  $K \cong \mathbb{R}, \mathbb{C}$  (12.8.2.1) and locally constant functions with compact support if  $K \in p\text{-LField}$ .  $\lrcorner$

**Prop. (19.4.2.11) [Local Schwarz Functions].** The space  $\mathcal{S}(K)$  of Schwartz functions satisfy the following properties: If  $f \in \mathcal{S}(K)$ ,

- $f, f^\vee \in L^1(K^+)$ .
- $f(\alpha) |\alpha|^\sigma, f^\vee(\alpha) |\alpha|^\sigma \in L^1(K^\times)$  for  $\sigma > 0$ .
- $f^\vee \in \mathcal{S}(K)$  too.

$\lrcorner$

*Proof:* 3:  $p$ -adic case: Let  $\mathfrak{p}^n$  be the conductor of  $\psi$ , then the Fourier transform of  $\chi_{\mathfrak{p}^{-k}}$  is  $V(\mathfrak{p}^{-k}) \chi_{\mathfrak{p}^{n+k}}$ . Then for  $k$  large,  $\chi_{\mathfrak{p}^{-k}} v = V(\mathfrak{p}^{-k}) V(\mathfrak{p}^{n+k}) v$ , thus  $v \in C_c^\infty(F)$ .

Archimedean case: (12.11.3.5).  $\square$

### Global notations

**Notation (19.4.2.12).**

- There are natural Haar measures  $d\mu_K, d^\times \mu_K$  on  $\mathbf{A}_F$  and  $\mathbf{I}_F$  defined by (12.10.3.24) and (19.4.2.5) (19.4.2.8). They satisfy  $d^\times \mu_K = \frac{1}{|\cdot|_K} d\mu_K$ .

- Let  $\mathfrak{d} = \mathfrak{d}_{F/\mathbb{Q}}$  when  $F \in \mathbf{NField}$ , or  $\mathfrak{d} = \{x | \text{tr}(\text{res}(x\mathcal{O})) = 0\}^{-1}$  when  $F \in \mathbf{FField}$ .
- Fix  $\psi = \prod_v \psi_v$  an additive character of  $\mathbf{A}_F/F$ .

┘

**Def. (19.4.2.13)[Global Fourier Transform].** (12.11.1.36) shows  $\psi$  induces a canonical isomorphism

$$\mathbf{A}_F \cong \mathbf{A}_F^\vee : \eta \mapsto (\xi \rightarrow \psi(\eta\xi)),$$

and we choose the corresponding self-dual Haar measure  $d\mu$ , then the Fourier transform and inversion formula on  $\mathbf{A}_F$  is then written as:

$$f^\vee(y) = \int_{\mathbf{A}_F} f(x) \overline{\psi(xy)} dx, \quad f(x) = \int_{\mathbf{A}_F} f^\vee(y) \psi(xy) dy.$$

In fact  $dx = \prod dx_{\mathfrak{p}}$  is the restricted product measure on  $\mathbf{A}_F$  (12.10.3.24), where  $dx_{\mathfrak{p}}$  is the self-dual measure w.r.t  $\psi_{\mathfrak{p}}$  in (19.4.2.5), then  $dx$  is the self-dual Haar measure w.r.t the global canonical character  $\psi$  by (12.10.3.27), called the **Tamagawa measure** on  $\mathbf{A}_F$ .

┘

**Def. (19.4.2.14)[Global Schwartz-Bruhat Functions].** For a global field  $F$ , the set  $\mathcal{S}(F)$  of global Schwartz-Bruhat functions is defined to be

$$\mathcal{S}(F) = \bigotimes_{v \in \Sigma_F}^{\prime} \mathcal{S}(F_v) \text{ (3.6.5.13)}.$$

┘

**Prop. (19.4.2.15)[Schwartz-Bruhat Functions].** A Schwartz-Bruhat function  $f \in \mathcal{S}(\mathbf{A}_F)$  (19.4.2.14) satisfies:

- $f(x), f^\vee \in L^1(\mathbf{A}_F)$  and they are continuous.
- $\sum_{\xi \in K} f(\mathfrak{a}(x + \xi))$  and  $\sum_{\xi \in K} \widehat{f}(\mathfrak{a}(x + \xi))$  converges uniformly absolutely on compact sets of  $\mathbf{A}_F$ .
- $f(\mathfrak{a})|\mathfrak{a}|^\sigma, \widehat{f}(\mathfrak{a})|\mathfrak{a}|^\sigma \in L^1(\mathbf{I}_F)$  for  $\sigma \in \mathbb{R}_{>1}$ .

┘

*Proof:* 1, 2 is the same as in (19.4.2.23), for 3:  $\int_{\mathbf{I}_F} |f||\mathfrak{a}|^\sigma d\mathfrak{a} = \prod_{\mathfrak{p}} \int_{F_{\mathfrak{p}}^\times} |f_{\mathfrak{p}}||\mathfrak{a}|^\sigma d\mathfrak{a}_{\mathfrak{p}}$ , and for a.e.  $\mathfrak{p}$ ,

$$\int_{F_{\mathfrak{p}}^\times} |f_{\mathfrak{p}}||\mathfrak{a}|^\sigma d\mathfrak{a}_{\mathfrak{p}} = \int_{\mathcal{O}_{\mathfrak{p}}^\times} |\mathfrak{a}_{\mathfrak{p}}|_{\mathfrak{p}}^\sigma d\mathfrak{a}_{\mathfrak{p}} = \frac{1}{1 - \frac{1}{\|\mathfrak{p}\|^\sigma}} \int_{\mathcal{O}_{\mathfrak{p}}^*} d\mathfrak{a}_{\mathfrak{p}} = \frac{1}{1 - \frac{1}{\|\mathfrak{p}\|^\sigma}}.$$

Thus the global integral converges by comparison with the Dedekind Zeta function (29.5.2.1). □

**Def. (19.4.2.16)[Global Standard Character].** Define the **global standard character**

$$X : \mathbf{A}_F \rightarrow \mathbb{C}^\times : x \mapsto e^{2\pi i \Lambda(x)}, \quad \Lambda(x) = \sum_{\mathfrak{p} \in \Sigma_F} \Lambda_{\mathfrak{p}}(x_{\mathfrak{p}}) \text{ (19.4.2.3)}.$$

Notice this is definable because  $x_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}$  a.e.  $\mathfrak{p}$ , thus  $\Lambda_{\mathfrak{p}}(x_{\mathfrak{p}}) = 1$  a.e..

Then  $X(F) = 1$ .

┘

*Proof:* In the number field case,

$$\Lambda(\xi) = \sum_p \sum_{\mathfrak{p}|p} \lambda_p(\mathrm{tr}_{\mathfrak{p}/p}(\xi)) = \sum_p \lambda_p(\mathrm{tr}_{K/\mathbb{Q}}(\xi))$$

so to show  $\lambda$  is an integer, it suffices to show  $\Lambda(a)$  is a  $q$ -adic integer for any  $q$  and any  $a \in \mathbb{Q}$ , but for this, notice

$$\sum \lambda_{\mathfrak{p}}(x) = \sum_{p \neq q, \infty} \lambda_p(x) + \lambda_q(x) - x$$

is a  $q$ -adic integer, by definition(19.4.2.3).

In the function field case, this follows from the fact that the sum of residues of a meromorphic 1-form is 0? .  $\square$

**Prop. (19.4.2.17).**  $F^\perp = F$ , i.e.  $X(xy) = 0, \forall y \in F \iff x \in F$ .  $\lrcorner$

*Proof:* Because  $F^\perp \cong \widehat{\mathbf{A}/F}$  and  $\mathbf{A}_F/F$  is compact(19.4.1.3),  $F^\perp$  is discrete(12.11.1.8) and contains  $F$ . So  $F^\perp/F$  is discrete hence finite in  $\mathbf{A}/F$ . But  $F^\perp$  is clearly a vector space over  $F$ , thus  $F^\perp = F$  must be true, because  $\#F = \infty$ .  $\square$

**Cor. (19.4.2.18).** By(12.11.1.36) and(19.4.2.17), any non-trivial character on  $\mathbf{A}_F/F$  is of the form  $\mathfrak{a} \mapsto X(k\mathfrak{a})$  for some  $k \in K$ .

In particular, for any such character  $\psi$ ,  $\psi_v$  is non-trivial.  $\lrcorner$

**Def. (19.4.2.19)[Unramified Places].** Let  $\chi$  be a Hecke character of  $F$ , then  $v \in \Sigma_F$  is called **unramified** if  $v \in \Sigma_F^{\mathrm{fin}}$ ,  $\mathfrak{d}_v = 1$ ,  $\mathfrak{c}(\psi_v) = \mathcal{O}_v$ ,  $\chi_v$  is unramified(19.4.2.2). Notice a.e. place  $v$  is unramified.  $\lrcorner$

*Proof:* To show that for any unramified character  $\psi$ , the conductor of  $\varphi_v$  is  $\mathcal{O}_v$ , consider the canonical character  $X$  defined(19.4.2.16), it can be verified the conductor of  $X_v$  is  $\mathcal{O}_v$  for a.e.  $v$ , and  $\psi$  must be of the form  $\psi(x) = X(ax)$  for some  $a \in K$  by(19.4.2.13) and(19.4.2.17), thus this is also true for  $\psi$ .  $\square$

**Prop. (19.4.2.20).**

$$\mathrm{Vol}(\mathbf{I}_F^1/F^\times) = \frac{2^{r_1}(2\pi)^{r_2}}{w_F \sqrt{|\mathrm{disc}(F)|}} \mathrm{cl}(F) \mathrm{Reg}(F),$$

where  $w_F = \#F_{\mathrm{tor}}^\times$ ,  $\mathrm{cl}(F)$  is the class number, and  $\mathrm{Reg}(F)$  is the regulator(20.8.1.7).

In particular, for  $F = \mathbb{Q}$ , this equals 1.  $\lrcorner$

*Proof:* Cf.[Tate Thesis, P337] or[GTM186, P281].?  $\square$

**Lemma (19.4.2.21)[Poisson Formula].** If

- $f \in C(\mathbf{A}_F) \cap L^1(\mathbf{A}_F)$ ,
- $F(x) = \sum_{\xi \in F} f(x + \xi)$  is locally absolutely continuous,
- $\sum_{\xi \in F} |f^\vee(\xi)| < \infty$ .

Then w.r.t. the self-dual Haar measure,

$$\sum_{\xi \in F} f^\vee(\xi) = \sum_{\xi \in F} f(\xi).$$

and  $V(\mathbf{A}_F/F) = 1$ .  $\lrcorner$

*Proof:* By (19.4.2.17) and (19.4.1.3), this is a special case of (12.11.1.31). In fact, we know it is true for a constant  $V(\mathbf{A}_F/F)$ . But it is symmetric, so  $V(\mathbf{A}_F/F)^2 = 1$ .  $\square$

**Prop. (19.4.2.22) [Riemann-Roch].** If for any idele  $\mathfrak{a} \in \mathbf{I}_F$ ,

- $f(\mathfrak{a}x) \in C(\mathbf{A}_F) \cap L^1(\mathbf{A}_F)$ .
- $F(x) = \sum_{\xi \in F} f(x + \xi)$  is locally absolutely continuous,
- $\sum_{\xi \in F} |f^\vee(\mathfrak{a}\xi)| < \infty$ .

Then for any  $\mathfrak{a} \in \mathbf{I}_F$ ,

$$\frac{1}{|\mathfrak{a}|} \sum_{\xi \in F} f^\vee\left(\frac{\xi}{\mathfrak{a}}\right) = \sum_{\xi \in F} f(\mathfrak{a}\xi).$$

┘

*Proof:* Consider  $g(x) = f(\mathfrak{a}x)$ , then

$$g^\vee(x) = \int_{\mathbf{A}_F} f(\mathfrak{a}\eta) e^{-2\pi i \lambda(x\eta)} d\eta = \frac{1}{|\mathfrak{a}|} \int_{\mathbf{A}_F} f(\eta) e^{-2\pi i \lambda(x\eta/\mathfrak{a})} d\eta = \frac{1}{|\mathfrak{a}|} f^\vee(x/\mathfrak{a}).$$

Then apply Poisson formula (19.4.2.21) to  $g$ .  $\square$

**Prop. (19.4.2.23).** Schwartz-Bruhat functions (19.4.2.14) satisfy the condition in (19.4.2.22).  $\square$

*Proof:*  $\int_{\mathbf{A}_F} |f| = \prod_v \int_{F_v} |f_v| dx_v < \infty$ , noticing (19.4.2.5) and  $N(\mathfrak{d}_v) = 1$  for a.e.  $v$ . And for any Schwartz function  $f$  and any  $x$ , the set of  $k$  that  $f_v(x_v + k_v) \neq 0$  is  $v$ -bounded for  $v$  non-Archimedean and  $|k|_v \leq 1$  a.e., so in the function case, these  $k$  are finite because  $F$  is discrete in  $\mathbf{A}_F$ .

And in the number field case, these  $k$  is contained in some fractional ideal  $I$ , but  $I$  is then a lattice in  $F_\infty$  by Minkowski theory, so

$$\sum_{\xi \in K} |f(x + \xi)| \leq C \sum_{x \in K} \prod_{\mathfrak{p} \in S_\infty} |f_{\mathfrak{p}}(x + \xi)|$$

but  $f_{\mathfrak{p}}$  is an Archimedean Schwartz function, thus this is absolutely convergent.

Now we also showed that  $\int_{\xi \in F} f^\vee(\xi) < \infty$ , because  $f^\vee \in \mathcal{S}(\mathbf{A}_F)$ , by (19.4.2.11).  $\square$

**Prop. (19.4.2.24) [True Riemann-Roch For Function Fields].** Cf. [Fourier Analysis on Number Fields, P267]. ?  $\square$

## 19.5 Quadratic Forms

References are [Lam05], [Quadratic Forms Clark] and [Algebraic and Geometric Theory of Quadratic Forms].

**Notation(19.5.0.1).**

- Let  $K \in \text{Field}$  s.t.  $\text{char } K \neq 2$ .
- Use notations on quadratic forms defined in [Quadratic Forms on Modules](#).

┘

### 1 Quadratic Forms

This subsection is a continuation of [Sesqui-linear Forms](#). In fact, most materials in this subsection are trivial facts.

**Def. (19.5.1.1)[Discriminants].** For a quadratic form  $Q$  on a finite  $K$ -module  $V$  valued in  $K$  [\(3.2.4.30\)](#), the **discriminant of forms**  $\text{disc}(Q)$  is defined to be the determinant of the Gram matrix [\(3.5.11.3\)](#) of this quadratic form w.r.t. any basis of  $V$ . It is a function from the set of quadratic forms to  $K^\times / (K^\times)^2$  that is invariant under congruences. ┘

**Def. (19.5.1.2)[Quadratically Closed].**  $K$  is called a **quadratically closed field** if  $(K^\times)^2 = K^\times$ , or equivalently  $K$  has no quadratic field extensions. ┘

**Def. (19.5.1.3)[Quadratic Spaces].** The category of **quadratic spaces** is a category with objects as finite dimensional spaces with a quadratic form, and its morphisms are isometric embeddings. ┘

**Def. (19.5.1.4)[Universal Quadratic Form].** For a quadratic form  $A$ , let  $D_F(A)$  be the set of elements representable by  $A$ . A **universal quadratic form** is a quadratic form that represents every element of  $K^*$ . ┘

**Prop. (19.5.1.5).** If every binary quadratic form over  $K$  is universal, then any two non-degenerate quadratic forms over  $K$  are isomorphic iff they have the same rank and discriminant [\(19.5.1.1\)](#). ┘

*Proof:* This is because  $\langle a, b \rangle \cong \langle 1, ab \rangle$  and then use induction. □

#### Non-Degeneracy

**Def. (19.5.1.6) [Non-degeneracy].** A quadratic space is called **non-degenerate** if  $v \mapsto B(v, \cdot)$  is an isomorphism from  $V$  to  $V^*$ . Notice if  $\dim V = \infty$ , this cannot happen, because  $\dim V^* > \dim V$  [\(3.5.3.9\)](#). And in case  $\dim V < \infty$ ,  $\dim V = \dim V^*$ , so it suffices to show  $v \mapsto B(v, \cdot)$  is injective, i.e. if  $v \neq 0$ , then there is a  $w$  that  $B(v, w) \neq 0$ . ┘

**Prop. (19.5.1.7)[Radical Splitting].** The **radical** of a quadratic space is defined to be  $\text{rad}(V) = V^\perp$ . Then for any quadratic form  $V$ , there is an orthogonal decomposition  $V = \text{rad}(V) \oplus W$ , where  $W$  is a non-degenerate form. ┘

*Proof:* In fact, by the definition, any complement space of  $\text{rad}(V)$  in  $V$  can be chosen as the orthogonal complement  $W$ . □

**Prop. (19.5.1.8).** If  $W$  is a non-degenerate sub-quadratic space of  $V$ , then  $W \oplus W^\perp = V$ . ┘

*Proof:* Since  $W$  is non-degenerate,  $W \cap W^\perp = 0$ . and for any  $v \in V$ ,  $B(v, \cdot) \in W^*$ , so by degeneracy, there is a  $w \in W$  that  $B(v, \cdot) = B(w, \cdot)$ , then  $z = v - w \in W^\perp$  and  $v = w + z$ .  $\square$

**Prop.(19.5.1.9) [Orthogonal Complement and Non-Degeneracy].** If  $V$  is a non-degenerate quadratic space, then for any non-degenerate subspace  $W$ ,  $\dim W + \dim W^\perp = \dim V$ , and  $(W^\perp)^\perp = W$ .  $\lrcorner$

*Proof:* The first is immediate from the fact  $\dim \ker + \dim \text{Coker} = \dim V$ . The second is by dimensional reason.  $\square$

**Cor.(19.5.1.10).** A subspace  $W$  of a non-degenerate quadratic space  $V$  is a non-degenerate quadratic space iff  $W \cap W^\perp = 0$ .  $\lrcorner$

**Remark(19.5.1.11).** We basically only care about non-degenerate forms, so from now on, we only care about non-degenerate forms.  $\lrcorner$

### Diagonalizability

**Prop.(19.5.1.12) [Quadratic Form Representable].** Any quadratic forms  $Q$  over  $K$  of  $\text{char} \neq 2$  is diagonalizable, and if  $\alpha \in K^\times$  is represented by  $Q$ , then it is diagonalizable to a matrix with first entry  $\alpha$ .

We will use the notation  $\langle \alpha_1, \dots, \alpha_n \rangle$  for the diagonal quadratic form  $\sum \alpha_i x_i^2$ .  $\lrcorner$

*Proof:* Use(3.5.11.11), since in this case, a quadratic form is equivalent to a symmetric form. And if  $\alpha = B(v, v)$ , then we can choose  $v$  in the first place in the proof of(3.5.11.11).  $\square$

**Cor.(19.5.1.13).** If  $K$  is quadratically-closed, any non-degenerate quadratic form is congruent to  $x_1^2 + \dots + x_n^2$ .  $\lrcorner$

*Proof:* Because in this case, we can make  $\sum a_i x_i^2$  into  $\sum (\sqrt{a_i} x_i)^2$ .  $\square$

**Prop.(19.5.1.14) [Pfister].** For any  $n \in \mathbb{Z}_+$ , the product of two element of  $k$  that are sums of  $2^n$  squares is another such number.  $\lrcorner$

*Proof:* Cf.[C-S03]P78.  $\square$

**Cor.(19.5.1.15).** If  $N \in \mathbb{Z}_+$  is a minimal number s.t.  $-1$  can be expressed as a sum of  $N$  squares in  $k$ , then  $N \in 2^{\mathbb{Z}_+}$ .  $\lrcorner$

*Proof:* Suppose  $N = 2^n + k$  where  $n, k \in \mathbb{Z}_+, k < 2^n$ , and

$$-1 = x_1^2 + \dots + x_{2^n}^2 + y_1^2 + \dots + y_k^2,$$

then  $s = x_1^2 + \dots + x_{2^n}^2 \neq 0$ . So

$$-s = y_1^2 + \dots + y_k^2 = y_1^2 + \dots + y_{2^n}^2.$$

But then

$$-s^2 = (x_1^2 + \dots + x_{2^n}^2)(y_1^2 + \dots + y_{2^n}^2) = z_1^2 + \dots + z_{2^n}^2$$

by(19.5.1.14), contradiction.  $\square$



### Isotropic and Hyperbolic Spaces

**Def. (19.5.1.16) [Isotropic].** Given a non-degenerate quadratic space  $V$ , a vector  $v$  is called **isotropic** if  $B(v, v) = 0$ .  $V$  is itself called **isotropic** if it is non-degenerate and there exists an isotropic vector, otherwise it is called **anisotropic**.  $\lrcorner$

**Def. (19.5.1.17) [Hyperbolic Plane].** The **hyperbolic plane**  $\mathbb{H}$  is the 2-dimensional space with quadratic form  $H(x, y) = xy$ , which is congruent to  $\frac{1}{2}(x^2 - y^2)$ .

A quadratic space is called **hyperbolic** if it is isomorphic to a direct sum of hyperbolic planes.  $\lrcorner$

**Lemma (19.5.1.18).** If  $V$  is a non-degenerate isotropic space, then there is an isometric imbedding of the hyperbolic plane into  $V$ .  $\lrcorner$

*Proof:* There is a  $u \in V$  that  $B(u, u) = 0$ . By non-degeneracy, there is a  $w$  that  $B(u, w) \neq 0$ . We may assume  $B(u, w) = 1$ . Now I claim there is an  $\alpha$  that  $q(\alpha u + w) = 0$ : in fact,  $q(\alpha u + w) = 2\alpha B(u, w) + q(w)$ , so take  $\alpha = -q(w)/2$ . Let  $v = \alpha u + w$ , then  $q(u) = q(v) = 0$ , and  $B(u, v) = 1$ , so it is isomorphic to  $\mathbb{H}$ .  $\square$

**Prop. (19.5.1.19) [Isotropic Complement].** If  $V$  is a non-degenerate quadratic space, and  $W \subset V$  is an isotropic space with basis  $u_1, \dots, u_m$ , then there is another isotropic space  $W'$  with basis  $v_1, \dots, v_m$  that  $B(u_i, v_j) = \delta_{ij}$ .  $\lrcorner$

*Proof:* Use induction on  $m$ . The  $m = 1$  case is lemma(19.5.1.18) above. If this is true for  $n < m$ , let  $W = \{u_2, \dots, u_m\}$ , then if  $W^\perp \subset \{u_1\}^\perp$ , then  $u_1 \in W$  by(19.5.1.9), contradiction, so there is a  $v \in W^\perp$  that  $B(u_1, v) \neq 0$ , so by the same proof as(19.5.1.18), there is a  $\alpha u_1 + v$  that is isotropic, and a  $\mathbb{H} \subset W^\perp$ , so by(19.5.1.9),  $W \subset \mathbb{H}^\perp$ , so by induction, we can find in  $\mathbb{H}^\perp$  elements  $v_2, \dots, v_m$  that satisfies the requirement.  $\square$

**Cor. (19.5.1.20).**  $\langle a, -a \rangle \cong \mathbb{H}$ , because it is isotropic, and it has dimension 2.  $\lrcorner$

**Cor. (19.5.1.21) [Isotropic Forms are Universal].** A non-degenerate isotropic space is universal, because hyperbolic plane does.  $\lrcorner$

**Cor. (19.5.1.22).** A maximal totally isotropic space in a non-degenerate quadratic space  $V$  has dimension at most  $\frac{1}{2} \dim V$ , and equality holds if  $V$  is hyperbolic.  $\lrcorner$

**Prop. (19.5.1.23) [First Representation Theorem].** If  $q$  is a non-degenerate quadratic form, then  $q$  represents  $\alpha \in K^\times$  iff  $q \oplus \langle -\alpha \rangle$  is isotropic.  $\lrcorner$

*Proof:* If  $q$  represent  $\alpha$ , then by(19.5.1.12) shows that  $q$  is equivalent to  $\langle \alpha, \alpha_1, \dots, \alpha_n \rangle$ , so  $q \oplus \langle -\alpha \rangle$  contains a  $\langle \alpha, -\alpha \rangle$  which is isomorphic to  $\mathbb{H}$  by(19.5.1.18).

Conversely, if  $q \oplus \langle -\alpha \rangle$  is isotropic, then there is a  $-\alpha x_0^2 + \sum \alpha_i x_i^2 = 0$ . If  $x_0 \neq 0$ , then  $q$  represent  $\alpha$ , and if  $x_0 = 0$ , then  $q$  is isotropic, thus represent any element(19.5.1.21).  $\square$

**Cor. (19.5.1.24).** The following are equivalent:

- Any  $n$ -quadratic form over  $K$  is universal.
- Any  $(n + 1)$ -quadratic form over  $K$  is isotropic.

$\lrcorner$

**Cor. (19.5.1.25) [Transform of Binary Forms].** For any  $a, b \in F^*$  that  $a + b \in F^*$ ,  $\langle a, b \rangle \cong \langle a + b, (a + b)ab \rangle$ .  $\lrcorner$

**Prop. (19.5.1.26) [Isotropy Criterion].** For two non-degenerate forms  $f, g$  over  $K$ ,  $h = \langle f, -g \rangle$  is isotropic iff there is an  $\alpha \in K^*$  that is represented by both  $f$  and  $g$ .  $\lrcorner$

*Proof:* Easy, notice to use isotropic form is universal (19.5.1.21).  $\square$

## 2 Witt Theory

**Prop. (19.5.2.1) [Witt Cancellation Theorem].** If  $U_1, U_2, V_1, V_2$  are quadratic spaces and  $V_1 \cong V_2$ ,  $V_1 \oplus U_1 \cong V_2 \oplus U_2$ , then  $U_1 \cong U_2$ .  $\lrcorner$

*Proof:* We may identify  $V_1 = V_2 = V$ , and  $W = U_1 \oplus V = U_2 \oplus V$ .

First if  $V$  is totally isotropic and  $U_1$  is non-degenerate, then there is a matrix  $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$  that

$$M^t \begin{bmatrix} 0 & 0 \\ 0 & B_2 \end{bmatrix} M = \begin{bmatrix} 0 & 0 \\ 0 & B_1 \end{bmatrix}$$

So  $B_1 = D^t B_2 D$ . As  $B_1$  is non-singular, so is  $D$ , thus  $U_1 \cong U_2$ .

Now if  $V$  is isotropic but  $U_1, U_2$  are not non-degenerate, then we may assume in their diagonalization,  $U_1$  has less 0s, it has  $r$  0s, then we can extract from both  $U_i$  a zero part, thus reducing to the above case.

Now if  $\dim V = 1$ ,  $V = \langle a \rangle$ , if  $a = 0$ , then we are done by the above argument, and if  $a \neq 0$ , then find  $q(x) = a$ , then by (19.5.2.12), we can find a  $\tau \in O(W)$  that  $\tau(V_1) = V_2$ , so now  $U_1, U_2$  as the orthogonal complement of  $V_1, V_2$ , they are isometric under the map  $\tau$ .

So now in general, we can cancel  $V$  out by moving its diagonal part once a time.  $\square$

**Cor. (19.5.2.2).** If  $X$  is a quadratic space and  $V_1, V_2$  are non-degenerate subspaces of  $X$ , then any isometry  $V_1 \cong V_2$  extends to an isometry of  $X$ .  $\lrcorner$

*Proof:*  $V_i \oplus V_i^\perp = X$  by (19.5.1.8).  $\square$

**Cor. (19.5.2.3) [Witt's Extension Theorem].** If  $X$  is a non-degenerate quadratic space and  $f : W_1 \rightarrow W_2$  is an isometry of two subspaces of  $X$ , then  $f$  extends to an isometry of  $X$ .

Notice this also holds for symplectic spaces  $X$ , by the same method of proof.  $\lrcorner$

*Proof:* If  $W_1$  is non-degenerate, then so does  $W_2$ , and we can use (19.5.2.2). By (19.5.1.7), we can write  $W_i = U_i \oplus V_i$  where  $U_i$  is totally isotropic and  $V_i$  is non-degenerate. Now  $X, V_i$  are non-degenerate,  $V_i^\perp$  is non-degenerate also, so there is an isotropic complement  $U'_i \subset V_i^\perp$  (19.5.1.19). Let  $T_i = \langle U_i, U'_i \rangle V_i$ , then  $T_i$  is non-degenerate and  $W \subset T$ . As  $U_i$  is the radical of  $W_i$ ,  $U_1 \cong U_2$ , and then  $\langle U_1, U'_1 \rangle \cong \langle U_2, U'_2 \rangle$ . By Witt cancellation,  $W_1 \cong W_2$ . So we reduced to the non-degenerate case.  $\square$

**Cor. (19.5.2.4).** If  $V$  is a non-degenerate quadratic space, then the group of isometries of  $X$  acts transitively on the set of all totally isotropic subspaces of a fixed dimension  $d$ .  $\lrcorner$

**Thm. (19.5.2.5) [Witt's Decomposition Theorem].** For any quadratic space  $V$ , there is an orthogonal decomposition

$$V \cong \text{rad}(V) \oplus \mathbb{H}^{\oplus k} \oplus V'$$

where  $V'$  is anisotropic (19.5.1.16). Moreover the number  $k = \text{ind}(V)$  which is called the **Witt index** of  $V$  and the isometry class of  $V' = w(V)$  which is called the **non-isotropic kernel** is independent of the decomposition.  $\lrcorner$

*Proof:* The existence of the decomposition follows from (19.5.1.7) and an easy induction using (19.5.1.18). The uniqueness is an easy corollary of (19.5.1.18) and Witt's cancellation theorem.  $\square$

**Cor. (19.5.2.6).** The Witt index equals the maximal dimension of a maximal totally isotropic subspace of  $W$ , by (19.5.1.19).  $\lrcorner$

**Remark (19.5.2.7).** This is a good reason that we will only consider non-degenerate quadratic forms from now on.  $\lrcorner$

**Cor. (19.5.2.8) [Sylvester's Law of Signature].** Let  $q_{r,s} = [r]\langle 1 \rangle \oplus [s]\langle 1 \rangle$ , then any non-degenerate quadratic form  $q$  over  $\mathbb{R}$  is congruent to exactly one of  $q_{r,s}$ , and  $r - s$  is called the **signature** of  $q$ .  $\lrcorner$

**Prop. (19.5.2.9) [Witt's Chain Equivalence Theorem].** Two quadratic forms  $q_1 = \langle a_1, \dots, a_n \rangle$  and  $q_2 = \langle b_1, \dots, b_n \rangle$  are called **simply equivalent** iff there are two indices that  $\langle a_i, a_j \rangle \cong \langle b_i, b_j \rangle$ . Two quadratic forms are called **chain equivalent** iff there is a chain of simply equivalence between them.

Then two diagonal quadratic forms over  $K$  are equivalent iff they are chain equivalent.  $\lrcorner$

*Proof:* Chain equivalent is clearly equivalent, Conversely, by Witt's decomposition theorem, it is easy to reduce to the non-degenerate case.

Now if  $q = \langle \alpha_1, \dots, \alpha_n \rangle \cong q' = \langle \beta_1, \dots, \beta_n \rangle$ , any form  $q = \langle \gamma_1, \dots, \gamma_n \rangle$  that is chain equivalent to  $q$  is equivalent to  $q'$ , so  $\beta_1$  is represented by it, choose a form that there is a minimal  $l$  that  $\beta_1$  is represented by  $\langle \gamma_1, \dots, \gamma_l \rangle$ , we prove that  $l = 1$ :

if the minimal  $l$  is not 1, then  $d = \gamma_1 a_1^2 + \gamma_2 a_2^2 \neq 0$  (otherwise  $l$  can be smaller), so  $\langle \gamma_1, \gamma_2 \rangle \cong \langle d, \gamma_1 \gamma_2 d \rangle$  by (19.5.1.12) and invariance of  $\det$ . so  $q \cong \langle d, \gamma_3, \dots, \gamma_n, d \gamma_1 \gamma_2 \rangle$  (notice permutation is chain equivalence), and this is smaller, contradiction.

Now  $l = 1$ , so we may assume  $\alpha_1 = \beta_1$ , and then Witt's cancellation (19.5.2.1) shows that  $\langle \alpha_2, \dots, \alpha_n \rangle \cong \langle \beta_2, \dots, \beta_n \rangle$ , so we win by induction.  $\square$

## Orthogonal Group

**Def. (19.5.2.10) [Orthogonal Groups].** The **orthogonal group** of a quadratic space  $(V, Q) \in \text{QuadSpa}/K$  is the set of matrixes  $M$  that  $g(Mx) = q(x)$ . And it is clear  $\det M = \pm 1$ , so we can also define  $O^+(V)$  and  $O^-(V)$ .  $\lrcorner$

**Def. (19.5.2.11).** A **hyperplane reflection** for a non-isotropic vector  $v$  is defines by  $x \mapsto x - \frac{2B(x,v)}{q(v)}v$ , it is an element in  $O(V)$ .  $\lrcorner$

**Prop. (19.5.2.12).** If  $x, y$  are two non-isotropic vectors that  $q(x) = q(y)$ , then there is a  $\tau \in O(V)$  that  $\tau(x) = y$ .  $\lrcorner$

*Proof:* First notice  $q(x+y) + q(x-y) = 2q(x) + 2q(y) = 4q(x) \neq 0$ , so one of  $x+y, x-y$  is non-isotropic. And it can be easily calculated that  $\tau_{x-y}(x) = y$  or  $-\tau_{x+y}(x) = y$ .  $\square$

**Prop. (19.5.2.13) [Cartan-Dieudonné].** Let  $V$  be a non-degenerate quadratic form of dimension  $n$ , then every element of the orthogonal group  $O(V)$  can be represented as a product of  $n$  reflections.  $\lrcorner$

*Proof:* Cf. [Quadratic Forms Clark P22].  $\square$

### 3 Witt Rings

References are [Quadratic Forms 2, Clark].

**Def. (19.5.3.1) [Witt Ring].** The **Witt ring**  $W(K)$  of  $K$  is a free commutative ring over  $\mathbb{Z}$  generated by equivalent classes of anisotropic(19.5.1.16) quadratic forms over  $K$ , modulo the relations  $[q_1] + [q_2] = [q_1 \oplus q_2]$  and  $[q_1] \cdot [q_2] = [q_1 \otimes q_2]$ .

There is another ring, the **Grothendieck-Witt ring**  $\widehat{W}(K)$  which is defined as the ring generated by all non-degenerate quadratic forms over  $K$ .  $\lrcorner$

**Cor. (19.5.3.2) [Rank Functor].** There are rank functors from  $\widehat{W}(K) \rightarrow \mathbb{Z}$ , which is a ring homomorphism. In particular, two elements of the same rank are equal in  $\widehat{W}(K)$  iff they are equal in  $W(K)$ , by Witt's cancellation theorem(19.5.2.1).  $\lrcorner$

**Prop. (19.5.3.3).** The rank functor is an isomorphism  $\widehat{W}(K) \rightarrow \mathbb{Z}$  iff  $K$  is quadratically closed. In this case,  $W(K) \cong \mathbb{Z}/2\mathbb{Z}$ .  $\lrcorner$

*Proof:* This is equivalent to  $\langle a \rangle \cong \langle 1 \rangle$  for any  $a \in K^*$ , which means  $K$  is quadratically closed.  $\square$

**Prop. (19.5.3.4).** The subgroup  $[\mathbb{H}]$  generated by the hyperbolic plane is an ideal of  $\widehat{W}(K)$ . And  $\widehat{W}(K)/\mathbb{Z}[\mathbb{H}] \cong W(K)$ .  $\lrcorner$

*Proof:*  $\mathbb{Z}[\mathbb{H}]$  is an ideal because  $[\mathbb{H}] \cdot [\langle a_1, \dots, a_n \rangle] \cong \sum \langle a_i, -a_i \rangle$  which is a multiple of  $[\mathbb{H}]$ , by(19.5.1.20). The last assertion follows from Witt's decomposition(19.5.2.5).  $\square$

**Cor. (19.5.3.5).**  $[\langle a_1, \dots, a_n \rangle] + [\langle -a_1, \dots, -a_n \rangle] = 0 \in W(K)$ , by(19.5.1.20).  $\lrcorner$

**Prop. (19.5.3.6) [Presentation of the Witt Ring].**  $\widehat{W}(K)$  is isomorphic to the quotient of the free commutative ring generated by  $\{[a] \mid a \in K^\times\}$  module the following relations:

- $[1] - 1$ .
- $[ab] - [a] - [b]$ .
- $[a] + [b] - [a + b](1 + [ab])$ ,  $a, b, a + b \in K^\times$ .

$\lrcorner$

*Proof:* Cf.[Lam, P39].  $\square$

### 4 over Finite Fields

**Notation (19.5.4.1).**

- Let  $k \in \text{Field}^{\text{fin}}$  with  $\text{char } k \neq 2$ .
- Let  $1 \neq \psi \in k^\vee$ .

$\lrcorner$

**Notation (19.5.4.2).**

- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-LField}$  with  $\text{char } \kappa \neq 2$ .
- Let  $1 \neq \psi \in K^\vee$ .

$\lrcorner$

**Prop. (19.5.4.3)  $[W(\mathbb{F}_q)]$ .**

- If  $q \equiv 1 \pmod{4}$ , then  $W(\mathbb{F}_q)$  is isomorphic to  $\mathbb{Z}/(2)[F^\times/(F^\times)^2] \cong \mathbb{Z}/2\mathbb{Z}[\mathbb{F}_2]$  as rings.

- If  $q \equiv 3 \pmod{4}$ , then  $W(\mathbb{F}_q)$  is isomorphic to  $\mathbb{Z}/(4)$  as rings.  $\lrcorner$

*Proof:* 1: In this case,  $-1$  is a square, let  $s$  be a non-square, then there are only three anisotropic forms:  $\langle 1 \rangle, \langle s \rangle, \langle 1, s \rangle$ . Thus the assertion is clear.

2: In this case, we can take  $s = -1$ , thus any anisotropic form must be of the form  $\langle 1, \dots, 1 \rangle$  or  $\langle -1, \dots, -1 \rangle$ . As  $\langle 1, 1 \rangle$  is universal by (37.6.1.8),  $\langle 1, 1, 1 \rangle \cong \langle -1 \rangle$  by representation theorem (19.5.1.23), thus  $4[\langle 1 \rangle] = 0$ .  $\square$

**Prop. (19.5.4.4) [Binary Quadratic Forms over Finite Fields are Universal].** Every non-degenerate binary quadratic form over a finite field  $\mathbb{F}_q$  is universal.  $\lrcorner$

*Proof:* By (19.5.4.3), there are essentially three quadratic forms,  $\langle 1, 1 \rangle, \langle s, s \rangle, \langle 1, s \rangle$ , where  $s$  is a non-square. By (37.6.1.8), these are all universal.  $\square$

**Cor. (19.5.4.5) [Quadratic Forms over Finite Fields].** Any non-degenerate quadratic form over a finite field  $k$  is equivalent to the form  $Q(x_1, \dots, x_r) = x_1^2 + \dots + x_{n-1}^2 + sx_n^2$  for some  $s \in \mathbb{F}_q$ . And two non-degenerate quadratic forms over  $\mathbb{F}_q$  are equivalent iff they have the same rank and discriminant.  $\lrcorner$

*Proof:* This follows from (19.5.4.4) and (19.5.1.12).  $\square$

**Cor. (19.5.4.6).** Any quaternion algebra over a finite field splits, by Hilbert's criterion (19.6.1.16).  $\lrcorner$

## 5 over Local Fields

**Prop. (19.5.5.1) [ $W(\mathbb{R})$ ].**

- There exists two anisotropic forms at each rank  $n$ :  $\langle 1, \dots, 1 \rangle$  and  $\langle -1, \dots, -1 \rangle$ .
- $W(\mathbb{R}) \cong \mathbb{Z}$ .
- Two non-degenerate forms over  $\mathbb{R}$  are equivalent iff they have the same rank and signature, by (19.5.2.8).
- $\widehat{W}(\mathbb{R}) \cong \mathbb{Z}[\mathbb{F}_2]$  as a ring.  $\lrcorner$

*Proof:* Only 4 need a proof: It suffices to show  $[\langle 1 \rangle]$  and  $[\langle -1 \rangle]$  are linearly independent in  $\widehat{W}(\mathbb{R})$ , because they generate all other rings. But if  $a[\langle 1 \rangle] + b[\langle -1 \rangle] = 0$ , then  $a + b = 0$  by rank reason, but when pass to  $W(F)$ ,  $a - b = 0$ , thus  $a = b = 0$ .  $\square$

**Prop. (19.5.5.2).** Let  $\langle a_1, \dots, a_n \rangle_K$  be a quadratic space over  $K \in p\text{-LField}$ , we may assume  $a_i \in \mathcal{O}_F$ . If  $\langle a_1, \dots, a_n \rangle_{\mathcal{O}_K/\mathfrak{p}^n}$  is isotropic for all  $n$ , then  $\langle a_1, \dots, a_n \rangle_K$  is isotropic.  $\lrcorner$

*Proof:* This is because we can find non-zero elements  $v_n \in (\mathcal{O}_F/\mathfrak{p}^n)^d$  that  $q(v_n) = 0 \in \mathcal{O}_F/\mathfrak{p}^n$ , thus we can find elements  $v \in \mathcal{O}_F^d$  that  $q(v) = 0$ .  $\square$

**Prop. (19.5.5.3) [Unique Quaternion Algebra].** If  $\mathbb{C} \not\cong K \in \text{LField}$ , then there is a unique non-split quadratic algebra over  $F$ , which is of the form  $Q_K(a, b)$  where  $a, b \in K^\times$  with  $(a, b)_K = -1$ .  $\lrcorner$

*Proof:* Cf. [Lam, P156].  $\square$

**Cor. (19.5.5.4).** For this unique quaternion algebra  $A$  in (19.5.5.3), the valuation  $|N(x)|$  is a valuation of  $A$ , i.e. defines the topology.  $\lrcorner$

*Proof:* This is because  $|N(-)|$  is bounded on the compact subset  $\{x \mid |x| = 1\}$ .  $\square$

**Prop. (19.5.5.5) [Hilbert Symbol].** The Hilbert symbol (19.6.1.20)  $(-, -)_K : K^\times / (K^\times)^2 \times K^\times / (K^\times)^2 \rightarrow \{\pm 1\}$  satisfies

- It is symmetric and bi-multiplicative for both terms.
- It is non-degenerate: if  $F \not\cong \mathbb{C}$  is a local field, for any  $y \in F^* \setminus (F^*)^2$ , then there exists a  $z \in F^*$  that  $(y, z)_F = -1$ .
- $(a, -a) = 1$ .
- If  $a(a-1) \neq 0$ ,  $(a, 1-a) = 1$ .

┘

*Proof:* 1: To show it is multiplicative, by (19.6.1.16),  $(a, b)_F = 1$  iff  $a \in \text{Nm}_{E/F}(E^\times)$ , where  $E = F(\sqrt{b})$ . Notice  $\text{Nm}_{E/F}(E^\times)$  has index 1 or 2 in  $F^\times$  by class field theory (19.7.2.9), thus  $a \mapsto (a, b)_F$  is a character of  $a$  in any case.

2: Cf. [Lam, P160]. ?

3: By (19.6.1.17).  $\square$

**Def. (19.5.5.6) [Hasse-Invariants].** If  $(V, Q)$  is a quadratic space over  $K$  that is equivalent to the diagonal form  $a_1x_1^2 + \dots + a_nx_n^2$ , then the **Hasse invariant** or the  **$\epsilon$ -invariant** of  $V$  is defined to be

$$\epsilon(V) = \prod_{i < j} (a_i, a_j)_K \in \{\pm 1\}.$$

Then this  $\epsilon(V)$  is well-defined and is an invariant of  $V$ .  $\square$

*Proof:* Cf. [Ser73]P35. ?  $\square$

**Prop. (19.5.5.7).** If  $(V, Q)$  is a quadratic space over  $K$  of rank  $n$ , then  $Q$  represents 0 iff

- $n = 2$  and  $\text{disc}(V) = -1$ .
- $n = 3$  and  $(-1, -d)_K = \epsilon$ .
- $n = 4$  and either  $d \neq 1$  or  $d = 1$  and  $\epsilon(V) = (-1, -1)_K$ .
- $n \geq 5$  and  $K \in p\text{-LField}$ .

┘

*Proof:* Cf. [Ser73]P36. ?

If  $n = 2$ , then  $Q$  represents 0 iff  $-a_1/a_2 = -\text{disc}(Q) = 1 \in K^\times / (K^\times)^2$ .  $\square$

**Cor. (19.5.5.8).** If  $(V, Q) \in \text{QuadSpa}^n / K$ , then  $Q$  represents  $a \in K^\times / (K^\times)^2$  iff

- $n = 1$  and  $a = \text{disc}(V)$ .
- $n = 2$  and  $(a, -\text{disc}(V))_K = \epsilon(V)$ .
- $n = 3$  and either  $a \neq -\text{disc}(V)$  or  $a = -\text{disc}(V)$  and  $(-1, -d)_K = \epsilon(V)$ .
- $n \geq 4$  and  $K \in p\text{-LField}$ .

┘

*Proof:* This follows from the theorem because for  $a \in K^\times / (K^\times)^2$ , the form  $Q_a = Q - aZ^2$  has a zero iff  $a$  is representable by  $Q$ , and  $\text{disc}(Q_a) = -a\text{disc}(Q)$ ,  $\epsilon(Q_a) = (-a, \text{disc}(Q))_K \epsilon(Q)$ .  $\square$

**Prop. (19.5.5.9) [Classifying Quadratic Forms over  $\mathbb{Q}_p$ ].** Two quadratic spaces over  $\mathbb{Q}_p$  are equivalent iff they have the same rank, discriminant and  $\epsilon$ -invariants.

And two quadratic forms over  $\mathbb{R}$  are equivalent iff they have the same rank  $\leq 3$ , discriminant and  $\epsilon$ -invariants.  $\lrcorner$

*Proof:* It follows from (19.5.5.8) that two quadratic forms over  $K$  has the same rank, discriminant and  $\epsilon$ -invariants represents the same set of elements in  $K^\times / (K^\times)^2$ , thus we can use (19.5.1.12) to do an induction on the rank.  $\square$

**Cor. (19.5.5.10).** There exists a unique anisotropic quadratic space of rank 4 over  $\mathbb{Q}_p$ , which has discriminant 1 and  $\epsilon$ -invariant  $= -(-1, -1)_{\mathbb{Q}_p}$ . It is of the form  $Q = z^2 - ax^2 - by^2 + abt^2$  for some  $a, b \in K^\times$  with  $(a, b)_K = -1$ . This is the reduced form of the unique quaternion algebra over  $K$  (19.5.5.3).  $\lrcorner$

**Prop. (19.5.5.11) [Classifying Hermitian Forms over  $K$ ].** If  $K/\mathbb{Q}_p$  is a field extension of degree 2 and  $Q$  is a non-degenerate  $c$ -Hermitian form of rank  $m$  over  $K/\mathbb{Q}_p$ , then

- If  $m = 2k + 1$ , then  $Q$  is equivalent to

$$\sum_{i=1}^k q_0(z_{2i-1}, z_{2i}) + (-1)^k \text{disc}(Q) \text{Nm}(z_{2k+1}).$$

- If  $m = 2k$ , then  $Q$  is equivalent to

$$\sum_{i=1}^{k-1} q_0(z_{2i-1}, z_{2i}) + \text{Nm}(z_{2k-1}) + (-1)^{k-1} \text{disc}(Q) \text{Nm}(z_{2k}).$$

where  $q_0$  is the hyperbolic form  $q_0(x, y) = (xc(y) + yc(x))/2$ .  $\lrcorner$

*Proof:* Because any quadratic form over  $\mathbb{Q}_p$  of rank  $\geq 5$  represents 0 by (19.5.5.7), so if  $m \geq 3$ ,  $Q$  contains a  $q_0$ . So it suffices to prove for  $m \leq 2$ . The case  $m = 1$  is trivial, and for  $m = 2$ , apply the above argument for the Hermitian form  $Q((z_1, z_2)) - \text{Nm}(z_3)$ , we get that  $Q$  represents 1, and then the theorem follows, by the same reasoning as in the proof of (19.5.1.12).  $\square$

### $\gamma$ -Invariants

**Prop. (19.5.5.12) [Weil].** Let  $(V, B) \in \text{QuadSpa}^d/K$ , then  $V$  be identified with  $V^*$  via  $(v, v^*) \mapsto \psi(-2B(v, v^*))$ . Denote the Fourier transform  $\mathcal{F}$  to be compatible with this identification.

Let  $F_B(v) = \psi(B(v, v))$ , then for any  $a \in K^\times$ , then  $\varphi * F_{aB} \in \mathcal{S}(V)$  for any  $\varphi \in \mathcal{S}(V)$ , and there is a complex number  $\gamma(aB)$  with norm 1 that for any  $\varphi \in \mathcal{S}(V)$ ,

$$\mathcal{F}(\varphi * F_{aB}) = |a|^{-d/2} \gamma(aB) \mathcal{F}(\varphi) F_{-a^{-1}B},$$

which means  $|a|^{-d/2} \gamma(aB) F_{-a^{-1}B}$  is formally the Fourier transform of  $F_{aB}$ . This number  $\gamma(B)$  is called the  $\gamma$ -invariant of  $V$ .  $\lrcorner$

*Proof:* We only prove for  $a = 1$ :

$$(\Phi * F_B)(v) = \int_V \Phi(u) \psi(B(v - u, v - u)) du = F_B(v) \mathcal{F}(\Phi * F_B)(-v)$$

is an element of  $\mathcal{S}(V)$ , as  $\mathcal{F}(\mathcal{S}(V)) = \mathcal{S}(V)$  (19.4.2.11). Now use the last equation of (3.1.5.9) for  $z = 1$  and the projective unitary representation  $\omega_1$  of  $\mathrm{SL}(2; F)$  on  $L^2(V)$  (26.8.1.5),

$$\omega_1(w_1)\omega_1(n(1))\omega_1(w_1)\omega_1(n(1))\omega_1(-1)\Phi = \gamma(B)\omega_1(n(-1))\omega_1(w_1)\Phi$$

for some  $|\gamma(B)| = 1$ . Then it can be computed that the LHS equals  $\widehat{\Phi * F_B}$ , and the RHS equals  $F_B \cdot \widehat{\Phi}$ , thus we are done.  $\square$

**Cor. (19.5.5.13) [Analytic Interpretation of Isotropy].** For  $(V_1, B_1), (V_2, B_2) \in \mathrm{QuadSpa}_K$ ,

- $\gamma(B_1 \oplus B_2) = \gamma(B_1)\gamma(B_2)$ .
- $\gamma(-B) = \gamma(B)^{-1}$ .
- If  $(V_1, B_1)$  is hyperbolic (19.5.1.17), then  $\gamma(B_1) = 1$ .
- $\gamma$  is a group homomorphism  $W(F) \rightarrow \mathbb{C}^*$  (19.5.3.1).

┘

*Proof:* 1, 2 follow from taking test functions  $\varphi$  to be  $\varphi_1 \otimes \varphi_2$  or  $\bar{\varphi}$ . 3 follows from the fact  $\gamma(\langle -1, 1 \rangle) = \gamma(\langle 1 \rangle)\gamma(\langle -1 \rangle) = 1$ . 4 follows from 1 and 3.  $\square$

**Prop. (19.5.5.14).** Situation as in (19.5.5.12),

- For any  $\Phi \in \mathcal{S}(V)$ ,  $\int_V (\Phi * F_B)(v) dv = \gamma(B) \int_V \Phi(v) dv$ .
- If  $K \in p\text{-LField}$ , then for any lattice in  $V$  (19.1.5.39) sufficiently large,  $\gamma(B) = \int_L F_B(v) dv$ .
- If  $\#K < \infty$ , then  $\gamma(B) = \int_V F_B(v) dv$ .

┘

*Proof:* 1 follows from (19.5.5.12) evaluated at  $a = 1$  and  $v = 0$ .

2: Consider the dual lattice  $L^\vee$  (19.1.5.43), if  $L$  is sufficiently large, then  $L'$  is sufficiently small, s.t.  $F_B(u) = 1$  for all  $u \in L'$ . Then

$$(\chi_{L'} * F_B)(v) = \int_{L'} F_B(v - u) du = \psi(B(v, v)) \int_{L'} \psi(-2B(u, v)) du,$$

which equals  $V(L')\chi_L F_B$ . Then let  $\Phi = \chi_{L'}$  in item 1, we get the desired assertion.

3: This is the same as 2, take  $\Phi = 1$ .  $\square$

**Prop. (19.5.5.15) [Weil's Characterization of Hilbert Symbol].** Let  $a, b \in K^\times$  and  $q : Q_K(a, b) \rightarrow K$  be the norm form, then  $\gamma(q) = (a, b)_K$  (19.6.1.20).

In particular,  $(a, b)_F = \gamma(\langle 1, -a, -b, ab \rangle)$ .  $\square$

*Proof:* By (19.5.5.13) and (19.5.5.5), it suffices to show for a non-split  $Q_F(a, b)$ ,  $\gamma(\langle 1, -a, -b, ab \rangle) = -1$ .

If  $K$  is Archimedean or finite, then the existence of a non-split  $Q_F(a, b)$  over  $A$  shows  $F = \mathbb{R}$ , and  $a = b = -1$ , and  $\gamma(\langle 1, 1, 1, 1 \rangle) = (\gamma(\langle 1 \rangle))^4$ . Let  $\psi : \mathbb{R} \rightarrow \mathbb{C} : x \mapsto e^{2\pi i x}$  as in (19.4.2.3), and  $\Phi(x) = e^{-\pi x^2}$ , then it can be calculated using (19.5.5.12) that  $\gamma(B)$  is an 8-th roots of unity, thus we are done.

If  $K \in p\text{-LField}$ , denote the Haar measure on  $A = Q_K(a, b)$  by  $dz$ , and then by (19.6.1.12) and (12.10.1.19), the Haar measure on  $A^*$  is of the form  $d^\times z = |N(z)|^{-2} dz$ . By (19.5.5.4), for large  $n$ ,  $L = N^{-1}(\mathfrak{p}^{-n})$  is a sufficiently large lattice in  $V$ , thus we can use (19.5.5.14) to evaluate the sign of

$$\gamma(N) = \int_{L \setminus \{0\}} \psi(N(z)) |N(z)|^2 d^\times z.$$



But this integrand factors through  $N : L \setminus 0 \rightarrow \mathfrak{p}^{-n}$ , thus it suffices to evaluate the sign of

$$\int_{\mathfrak{p}^{-n} \setminus 0} \psi(x) |x|^2 d^\times x = \int_{\mathfrak{p}^{-n}} \psi(x) |x| dx,$$

which is  $-q^{1-2r}(1-q^{-1})^{-1}$ , where the conductor of  $\psi$  is  $\mathfrak{p}^r$ . So  $\gamma(N) = -1$ .  $\square$

**Cor. (19.5.5.16).** Let  $r_i \in F^*$ , then  $\gamma(\langle ar_1, \dots, ar_{2n} \rangle) = ((-1)^n r_1 \dots r_{2n}, a)_F \gamma(\langle r_1, \dots, r_{2n} \rangle)$ .  $\lrcorner$

*Proof:* By (19.5.5.15), it suffices to show for  $n = 1$ .  $\gamma(\langle r_1, r_2 \rangle) = (r_1, r_2)_F \gamma(\langle 1, r_1 r_2 \rangle)$ , thus also  $\gamma(\langle ar_1, ar_2 \rangle) = (ar_1, ar_2)_F \gamma(\langle 1, a^2 r_1 r_2 \rangle)$ , so

$$\gamma(\langle ar_1, ar_2 \rangle) / \gamma(\langle r_1, r_2 \rangle) = (ar_1, ar_2)_F / (r_1, r_2)_F = (a, -r_1 r_2)$$

by (19.5.5.5).  $\square$

**Cor. (19.5.5.17).** Let  $(V, Q)$  has dimension  $2n$ , then  $\gamma(Q)^2 = ((-1)^n \text{disc}(Q), -1)_F$ .  $\lrcorner$

*Proof:* It reduces to  $n = 1$ , in which case, it suffices to show that  $\gamma(\langle r_1, r_2 \rangle)^2 = (-1, -r_1 r_2)_F$ . But  $(-1, -r_1 r_2)_F = \gamma(\langle 1, 1, r_1 r_2, r_1 r_2 \rangle) = \gamma(\langle 1, r_1 r_2 \rangle)^2$ . By (19.5.5.15),  $\gamma(\langle 1, -r_1, -r_2, r_1 r_2 \rangle)^2 = 1$ , thus  $\gamma(\langle 1, r_1 r_2 \rangle)^2 = \gamma(\langle r_1, r_2 \rangle)^2$ .  $\square$

## 6 over Global Fields

References are [Quadratic Forms over Global Fields, Clark].

**Prop. (19.5.6.1).** Suppose  $k = \bar{k}$  and  $F = k(t)$ , then any binary quadratic form  $Q$  over  $F$  is universal.  $\lrcorner$

*Proof:* We may assume  $Q = \langle 1, f \rangle$ , where  $f \in F^\times$ . We can assume  $f \notin -(F^\times)^2$ , otherwise this is clearly universal. It is easy to see that the  $\mathbb{F}_2$  vector space  $F^* / (F^*)^2$  has a basis  $\{t - a | a \in k\}$ . Notice  $D_F(Q)$  is a subgroup of  $F^*$  as it is the norm group of  $F(\sqrt{-f})$ , it suffices to show that  $t - a \in D_F(Q)$ . After a change of variable, it suffices to show that  $t \in D_F(Q)$ , or  $\langle 1, f, -t \rangle$  is isotropic, or equivalently  $-f \in \langle 1, -t \rangle$  for any  $f$ . The same argument show that it suffices to show that  $\langle 1, -t, t - a \rangle$  is isotropic. But  $t - a + (-t) + (\sqrt{a})^2 = 0$ .  $\square$

**Cor. (19.5.6.2).** Any quaternion algebra over  $\bar{k}(t)$  splits, by Hilbert's criterion (19.6.1.16).  $\lrcorner$

**Prop. (19.5.6.3) [Examples].**

- $Q_{\mathbb{Q}}(-1, -1) \cong Q_{\mathbb{Q}}(-2, -3)$ .
- $Q_{\mathbb{Q}}(-1, -1) \not\cong Q_{\mathbb{Q}}(-2, -5)$ .
- Let  $p$  be a prime number,  $Q_{\mathbb{Q}}(-1, p)$  splits iff  $p = 2$  or  $p \equiv 1 \pmod{4}$ .
- Let  $p$  be a prime number,  $Q_{\mathbb{Q}}(-2, p)$  splits iff  $p = 2$  or  $p \equiv 1, 3 \pmod{8}$ .
- $Q_{\mathbb{Q}}(-3, 5)$  is a division ring, but splits over  $K = \mathbb{Q}(\sqrt{17})$ .

$\lrcorner$

*Proof:* 1: By (19.5.1.25),  $Q_{\mathbb{Q}}(1, 1, 1, 1) \cong (1, 1, 2, 2) \cong (1, 3, 6, 2)$ , thus  $Q_{\mathbb{Q}}(-1, -1) \cong Q_{\mathbb{Q}}(-2, -3)$  by (19.6.1.16).

2: if they are isomorphic, then  $\langle 2, 5, 10 \rangle \cong \langle 1, 1, 1 \rangle \cong \langle 2, 2, 1 \rangle$ , thus  $\langle 5, 10 \rangle \cong 1, 2$ , which is impossible as 1 is not representable by  $\langle 5, 10 \rangle$ .

3: By Hilbert's criterion(19.5.1.25), if  $Q_{\mathbb{Q}}(-1, p)$  splits, then  $-x^2 + py^2 = z^2$  for some  $x, y, z \in \mathbb{Z}$  that  $(x, y, z) = 1$ , thus  $-1$  is a square in  $\mathbb{F}_p$ , which means  $p \equiv 1 \pmod{4}$  or  $p = 2$ . Conversely, if  $p \equiv 1 \pmod{4}$ , then by(35.5.1.8),  $p = x^2 + y^2$  for some  $x, y \in \mathbb{Z}$ , thus  $\langle -1, p \rangle$  represents 1, so  $Q_{\mathbb{Q}}(-1, p)$  splits. if  $p = 2$ , then  $\langle -1, 2 \rangle$  represents 1, thus it splits.

4: By Hilbert's criterion(19.5.1.25), if  $Q_{\mathbb{Q}}(-1, p)$  splits, then  $-2x^2 + py^2 = z^2$  for some  $x, y, z \in \mathbb{Z}$  that  $(x, y, z) = 1$ , thus  $-2$  is a square in  $\mathbb{F}_p$ , which means  $p \equiv 1, 3 \pmod{8}$  or  $p = 2$ . Conversely, if  $p \equiv 1 \pmod{4}$ , then by(35.5.1.8),  $p = x^2 + y^2$  for some  $x, y \in \mathbb{Z}$ , thus  $\langle -1, p \rangle$  represents 1, so  $Q_{\mathbb{Q}}(-1, p)$  splits. If  $p = 2$ , then it splits as above.

5: If it splits, then  $-3x^2 + 5y^2 = z^2$  for some  $x, y, z \in \mathbb{Z}$  that  $(x, y, z) = 1$ . This is a contradiction modulo 3. In  $K = \mathbb{Q}(\sqrt{17})$ , however,  $5 \cdot 2^2 - 3 = 17$ , thus  $Q_{\mathbb{Q}(\sqrt{17})}(-3, 5)$  splits.  $\square$

**Prop. (19.5.6.4)[Hasse-Minkowski Principle].** If  $F \in \mathbf{GFie}ld$ ,  $\text{char } F \neq 2$ , and  $Q$  is a quadratic form over  $F$ , then  $Q$  is isotropic over  $F$  iff  $Q$  is isotropic over  $F_{\mathfrak{p}}$  for all place  $\mathfrak{p}$ .  $\lrcorner$

*Proof:* Cf.[Lam, P170].? Also [Vigneras]P57 has an analytic proof.  $\square$

## 7 Algebraic Extensions

## 8 Integral Quadratic Forms and Hilbert's 11th Problem

References are [Cogdell, sums of three squares].

**Prop. (19.5.8.1)[Signatures of Even Bilinear Forms, van.der.Blij].** The signature of a unimodular even integral bilinear form is divisible by 8.  $\lrcorner$

*Proof:* Cf.[Symmetric Bilinear forms, Milnor-Husemoller]P242?  $\square$

### Counting Representations by Sums of Square

**Prop. (19.5.8.2)**  $[r_2(n)]$ . For  $n \in \mathbb{Z}_+$ ,

$$r_2(n) = 4 \sum_{d \in \mathbb{Z}_+, d|n} \chi_{-4}(d) = 4 \sum_{d \in 2\mathbb{Z}_+ - 1, d|n} (-1)^{(d-1)/2} \text{(35.3.6.4)}.$$

$\lrcorner$

*Proof:* It is clear that

$$\sum_{n \in \mathbb{Z}_+} r_2(n) n^{-s} = \zeta_{\mathbb{Q}(i)}(s) = 4\zeta(s)L(\chi_{-4}; s)$$

$\square$

**Prop. (19.5.8.3)**  $[r_3(n), \text{Legendre/Gauss}]$ . For  $n \in \mathbb{Z}_+$ ,

$$r_3(n) = \begin{cases} 24H(-n) & , n \equiv 3 \pmod{8} \\ 12H(-4n) & , n \equiv 1, 2 \pmod{4} \\ 0 & , n \equiv 7 \pmod{8} \\ r_3(n/4) & , n \equiv 0 \pmod{4} \end{cases}.$$

where  $H(D)$  is the Hurwitz class number(19.3.3.11). In particular,  $r_3(n) > 0$  iff  $n$  is not of the form  $n = 4^a(8k + 7)$ .  $\lrcorner$

*Proof:*

□

**Prop. (19.5.8.4)** [ $r_4(n)$ , **Lagrange**]. For  $n \in \mathbb{Z}_+$ ,

$$r_4(n) = 8(2 + (-1)^n) \sum_{d|n, 2 \nmid d} d > 0.$$

┘

*Proof:*

□

**Prop. (19.5.8.5)** [ $r_6(n)$ ]. For  $n \in \mathbb{Z}_+$ ,

$$r_6(n) = 4 \sum_{d|n} (4\chi_{-4}\left(\frac{n}{d}\right) - \chi_{-4}(d))d^2,$$

where  $\chi_{-4}$  is the Kronecker character (35.3.6.4).

┘

*Proof:*

□

**Prop. (19.5.8.6)** [ $r_8(n)$ , **Jacobi**]. For  $n \in \mathbb{Z}_+$ ,

$$r_8(n) = 16 \sum_{d|n} (-1)^{n+d} d^3,$$

where  $\chi_{-4}$  is the Kronecker character (35.3.6.4).

┘

*Proof:*

□

**Prop. (19.5.8.7).** There are closed form formulas for  $r_{4s^2}(n)$  and  $r_{4s(s+1)}(n)$ , Cf. [Infinite Families of Exact Sums of Squares Formulas, Jacobi Elliptic Functions, Continued Fractions, and Schur Functions].

┘

*Proof:*

□

**Prop. (19.5.8.8)** [Estimating Sums of  $r_k(n)$ ]. For  $X \in \mathbb{R}_+$  and  $k \in \mathbb{Z}_{\geq 4}$ , we have

$$\sum_{n \in [X^2]_+} r_k(n) = \frac{\pi^{k/2}}{\Gamma(k/2 + 1)} X^k + O(X^{k-2} \log X).$$

┘

*Proof:* For  $k = 4$ , use the formula (19.5.8.4), then

$$\begin{aligned} \sum_{n \in [X^2]_+} r_k(n) &= 8(2 + (-1)^m) \sum_{dm \leq X, 2 \nmid d} \left( \frac{X^4}{4m^2} + O\left(\frac{X}{m}\right) \right) \\ &= 2X^4 \sum_{m \in \mathbb{Z}_+} (2 + (-1)^m) m^{-2} + O(X^2 \log X) \\ &= 3\zeta(2)X^4 + O(X^2 \log X) \\ &= \frac{1}{2}\pi^2 X^4 + O(X^2 \log X). \end{aligned}$$

For general  $k$ ,  $r_k = r_4 * r_{k-4}$ , so we approximate the summation on the other  $k - 4$  terms via integration. ?

□

### Hilbert's 11th Problem

Hilbert's 11th problem is to classify quadratic forms over integral rings of number fields

## 19.6 Arithmetic of Quaternions and Octonions

References are [Vig80, C-S03].

**Notation (19.6.0.1).**

- Use notations on global fields defined in [Global Fields](#).
- Let  $K \in \text{Field}$ .

┘

### 1 Quaternion Algebras

References are [Quaternion Algebras].

**Notation (19.6.1.1).**

- In this subsection,

┘

**Def. (19.6.1.2) [Quaternion Algebras].** A **quaternion algebra** over  $K$  is a central simple algebra over  $K$  of dimension 4.

┘

**Prop. (19.6.1.3) [ $L\{\sqrt{\theta}\}$ ].** For a quaternion algebra  $Q/K$  (19.6.1.2), if  $L/K$  is a separable quadratic algebra contained in  $Q$ , then there exists  $u \in Q$  s.t.  $u^2 = \theta \in K^\times$ , and  $um = \overline{m}u$  for any  $m \in L$ , where  $\overline{(-)}$  is the unique non-trivial  $K$ -automorphism of  $L$ . Denote this quaternion algebra by  $L\{\sqrt{\theta}\}$ . In particular, by (3.6.4.20), any quaternion algebra over  $K$  has a representation of this type.

┘

*Proof:* If  $L \cong K \times K$ , then by Wedderburn theorem,  $Q \cong \text{Mat}(2; K)$ , and we can take  $u = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix}$  with  $u^2 = 1$ . If  $L \in \text{Field}$ , then by Skolem-Noether (3.6.4.11), then non-trivial  $K$ -automorphism of  $L$  is induced by some inner isomorphism of  $Q$ , which is  $h \mapsto uhu^{-1}$ . Then  $u^2 \in C_Q(L) = L$  (3.6.4.15). Thus  $u^2 - \theta = 0 \in Q$  for some  $\theta \in L^\times$ . But  $u$  also satisfies  $u^2 - \text{trrd}(u)u + \text{Nmrd}(u) = 0 \in Q$ , which implies that  $\text{trrd}(u) = 0$ , otherwise  $u \in L$ , which is impossible, and thus  $\theta \in K^\times$ .  $\square$

**Def. (19.6.1.4) [Bar Involution].** There is an anti-involution of  $L\{\sqrt{\theta}\}$  given s.t.  $\overline{\{m_1 + m_2u\}} = \overline{m_1} - m_2u$  for  $m_1, m_2 \in L$ , called the **bar involution**.

Thus if we define the **reduced norm map** and **trace map**

$$\text{Nmrd}(x) = x\overline{x}, \quad \text{trrd}(x) = x + \overline{x},$$

then they have images in  $K$ . More explicitly,

$$\text{Nmrd}(m_1 + m_2u) = \text{Nm}(m_1) + \theta \text{Nm}(m_2), \quad \text{trrd}(m_1 + m_2u) = \text{tr}(m_1).$$

In particular, for  $Q = \text{Mat}(2; K)$ ,  $\overline{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ , and  $\text{trrd}, \text{Nmrd}$  are usual trace and determinant, resp.

┘

**Cor. (19.6.1.5) [Invariance of Bar Involutions].** Notice the bar involution can be defined intrinsically, by defining pure vectors, which are vectors  $x$  s.t.  $x^2 \in F$  but  $x \notin F$ . In particular, the bar involution, trace and determinants are independent of the form given.

It is clear that if  $\alpha \in Q_K(a, b) \setminus K$ , then the minimal polynomial of  $\alpha$  over  $K$  is  $T^2 - \text{tr}(\alpha)T + \text{Nmrd}(\alpha) = 0$ .

┘

**Prop. (19.6.1.6)[Characterizing  $\text{Mat}(2; K)$ ].** For a quadratic algebra  $Q = L\{\sqrt{\theta}\}/K$ ,  $Q \cong \text{Mat}(2; K)$  iff  $L$  is not a field or  $\theta \in \text{Nm}(L^\times)$ .  $\lrcorner$

*Proof:* If  $L \cong K \times K$ , then by Wedderburn theorem,  $Q \cong \text{Mat}(2; K)$ . Suppose  $L \in \mathbf{Field}$ . If  $\theta \in \text{Nm}(L^\times)$ , then there exists some  $m \in L$  s.t.  $(mu)^2 = 1$  but  $\mu \neq \pm 1$ , so  $Q \cong \text{Mat}(2; K)$ . Conversely, if  $Q \cong \text{Mat}(2; K)$ , then  $\text{Nmrd}(m_1 + m_2u) = \text{Nm}(m_1) + \text{Nm}(m_2)\theta$ , thus the norm map is injective. But the norm map of  $\text{Mat}(2; K)$  is clearly not injective, thus  $Q \not\cong \text{Mat}(2; K)$ .  $\square$

$Q_K(a, b)$

**Notation (19.6.1.7).**

- In this subsubsection, we assume  $\text{char } K \neq 2$ .  $\lrcorner$

**Prop. (19.6.1.8).** Let  $a, b \in K^\times$ , define a  $K$ -algebra  $Q(a, b)$  as

$$Q_K(a, b) = Q \langle X, Y \rangle / (X^2 - a, Y^2 - b, XY + YX).$$

Denote  $\overline{X} = i, \overline{Y} = j, \overline{XY} = k$ . Then:

- $\dim_K Q_K(a, b) = 4$ , thus any element of  $Q_K(a, b)$  is of the form  $x + yi + zj + wk$ .
- $Q_K(ax^2, by^2) \cong Q_K(a, b)$  for  $x, y \in K^\times$ .
- $Q_K(-1, 1) \cong \text{Mat}(2; K)$ .
- $Q_K(a, b)$  is a quaternion algebra over  $F$ .  $\lrcorner$

*Proof:* 1:  $Q_K(a, b) \otimes_F \overline{F} = Q_{\overline{F}}(a, b) \cong Q_{\overline{F}}(-1, 1) \cong M_2(\overline{F})$  has dimension 4 over  $\overline{F}$ , by item2 and3.

2: trivial.

3: The map is given by  $\varphi : Q_K(-1, 1) \rightarrow \text{Mat}(2; F) : i \mapsto \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}, j \mapsto \begin{bmatrix} & \\ 1 & \end{bmatrix}$ . The verification is straightforward.

4: Consider  $Q_K(a, b) \otimes_K \overline{K} \cong \text{Mat}(2, \overline{K})$  by item2, which simple with center  $\overline{K}$ , the center of  $Q_K(a, b)$  has to be  $K$ , and also  $Q_K(a, b)$  has to be simple.  $\square$

**Def. (19.6.1.9)[Pure Tensors].** A quaternion element of the form  $yi + zj + wk$  is called a **pure tensor**. The space of pure tensor is denoted by  $A_0$ .

Let  $v \neq 0 \in Q_K(a, b)$ , then  $v \in A_0$  iff  $v \notin F$  but  $v^2 \in F$ .  $\lrcorner$

*Proof:* By direct calculation,

$$(x + yi + zj + wk)^2 = (x^2 + ay^2 + bz^2 - abw) + 2x(yi + zj + wk).$$

$\square$

**Def. (19.6.1.10)[Bar Involution].** There is an anti-involution of  $Q_K(a, b) = Q \langle X, Y \rangle / (X^2 - a, Y^2 - b, XY + YX)$  given by  $X \mapsto -X, Y \mapsto -Y$  (thus  $XY \mapsto (-Y)(-X) = YX$ ), called the **bar involution**.

Thus if we define the **reduced norm map** and **trace map**

$$\text{Nmrd}(x) = x\bar{x}, \quad \text{trrd}(x) = x + \bar{x},$$

then they have images in  $K$ . If  $\alpha = x + yi + zj + wk$ , then  $\text{tr}(\alpha) = 2x, \text{Nmrd}(\alpha) = x^2 - ay^2 - bz^2 + abw^2$ .

$\lrcorner$

**Cor. (19.6.1.11) [Invariance of Bar Involutions].** Notice the bar involution can be defined intrinsically, by defining pure vectors, which are vectors  $x$  s.t.  $x^2 \in F$  but  $x \notin F$ . In particular, the bar involution, trace and determinants are invariant under isomorphisms of quaternion algebras.

It is clear that if  $\alpha \in Q_K(a, b) \setminus K$ , then the minimal polynomial of  $\alpha$  over  $K$  is  $T^2 - \text{tr}(\alpha)T + \text{Nmrd}(\alpha) = 0$ .  $\lrcorner$

**Cor. (19.6.1.12).** For  $x \in Q_K(a, b)$ , if we let  $T(x)$  be the left(or right) multiplication by  $x$  on  $Q_K(a, b)$ , then the determinant of  $T(x)$  equals  $\text{Nmrd}(x)^2$ .  $\lrcorner$

*Proof:* After base change, all the values won't change, thus it suffices to prove for  $\text{Mat}(2; K)$ , in which case, it can be verified by (19.6.1.8) that  $\det(x) = \text{Nmrd}(x)$ , and the  $T(x)$  has determinant  $\det(x)^2$ .  $\square$

**Prop. (19.6.1.13) [Norm Forms].** If  $\text{char } k \neq 2$ , define a quadratic form on  $Q_K(a, b)$ :  $B(x, y) = \text{trrd}(x\bar{y})/2$ . Its norm form is just  $\text{Nmrd}(x)$ .

Thus this quadratic space has  $\{1, i, j, k\}$  as an orthonormal basis. It is non-degenerate and isomorphic to

$$\langle 1, -a, -b, ab \rangle \cong \langle 1, -a \rangle \otimes \langle 1, -b \rangle.$$

$\lrcorner$

**Prop. (19.6.1.14).** Let  $A, A'$  be two quaternion algebras over  $K$ , then the following are equivalent:

- $A, A'$  are isomorphic as quaternion algebras.
- $A, A'$  are isomorphic as quadratic spaces (19.6.1.13).
- $A_0, A'_0$  (19.6.1.9) are isomorphic as quadratic spaces.

$\lrcorner$

*Proof:* Cf. [Lam, P57].  $\square$

**Cor. (19.6.1.15).**  $Q_K(a, -1) \cong Q_K(a, a)$ .  $\lrcorner$

*Proof:* These algebras have norm forms  $\langle 1, -a, -1, -a \rangle$  and  $\langle 1, -a, -a, a^2 \rangle$ , which are clearly isometric quadratic spaces.  $\square$

**Prop. (19.6.1.16) [Split or Division Ring].** Let  $A = Q_K(a, b)$ , then the following are equivalent:

1.  $A \cong \text{Mat}(2; F)$ .
2.  $A$  is not a division algebra.
3.  $A$  is isotropic as a quadratic space.
4.  $A$  is hyperbolic as a quadratic space.
5.  $A_0$  is isotropic as a quadratic space.
6.  $(\langle a \rangle - 1)(\langle b \rangle - 1) = 0$  in  $W(F)$  (or  $\widehat{W}(F)$ , by (19.5.3.2)).
7. (Hilbert's Criterion) The binary form  $\langle a, b \rangle$  represents 1.
8.  $a \in \text{Nm}_{E/F}(E^*)$ , where  $E = F(\sqrt{b})$ .

$\lrcorner$

*Proof:*  $4 \rightarrow 5 \rightarrow 3$  is clear. As  $Q_K(-1, 1) \cong M(2, F)$ , 1, 4, 6, 7 are equivalent by (19.5.1.12) and (19.6.1.14). Also  $3 \iff 4$  as the the norm form of  $A$  has determinant in  $(F^*)^2$ . Also  $1 \rightarrow 2 \rightarrow 3$  is clear, as  $x \in A$  is invertible iff  $\text{Nmrd}(x) \neq 0$ . Thus 1 to 7 are all equivalent.

Now we show  $7 \iff 8$ : Firstly assume  $b \notin (F^\times)^2$ , otherwise both are true. Let  $E = F(\sqrt{b})$ , then  $\text{Nm}_{E/F}(x + y\sqrt{b}) = x^2 - by^2$ , thus 8 says  $a \in D_F(\langle 1, -b \rangle)$ , which is equivalent to  $\langle 1, -b, -a \rangle$  isotropic, which is equivalent to  $\langle a, b \rangle$  represents 1, by representation theorem (19.5.1.23).  $\square$

**Cor. (19.6.1.17).** Let  $a \in F^\times$ ,

- $Q_K(1, -a) \cong Q_K(a, -a) \cong \text{Mat}(2; F)$ .
- If  $a \neq 1$ , then  $Q_K(a, 1 - a)$  splits.
- $Q_K(-1, a)$  splits iff  $a$  is a sum of two squares in  $F$ .

┘

*Proof:* These follows from Hilbert's criterion (19.6.1.16) and representation theorem (19.5.1.23).  $\square$

**Prop. (19.6.1.18) [Characterizing Quaternion Algebras].**

- If  $A \neq K$  is a central simple algebra over  $K$  of dimension  $\leq 4$ , then  $A$  is a quaternion algebra and is isomorphic to  $Q_K(a, b)$  for some  $a, b \in K^\times$ .
- If  $B \neq K$  is a f.d. central simple  $K$ -algebra equipped with an  $K$ -algebra involution  $x \mapsto \bar{x}$  s.t.  $x + \bar{x} \in K, x\bar{x} \in K$  for any  $x \in B$ , then  $B$  is isomorphic to a quaternion algebra over  $K$ .

┘

*Proof:* 1: By Wedderburn's theorem, then  $A \cong \text{Mat}(n; D)$  where  $D$  is a central division ring over  $K$ . Then  $A = M(2, F) \cong Q_K(1, -1)$  or  $D$ . If  $A = D$ , choose  $i \notin F$ , and  $K = F(i)$  a subfield of  $D$ . Then  $F \subset K \subset D$ , thus  $\dim K = 2$ , thus we may modify  $i$  that  $i^2 = a \in F$ . The conjugation of  $i$  on  $D$  satisfies  $(\text{ad}(i))^2 = \text{id}$ , thus  $A = A^+ \oplus A^-$  where  $\text{ad}(i)$  acts by 1 and  $-1$  resp.. Let  $j \neq 0 \in A^-$ , then  $ij = -ji, j^2i = ij^2$ , thus  $j^2 \in A^+$ . Now  $K(j)$  is also a quadratic field over  $F$ , thus  $j^2 + cj - b = 0$  for  $c, b \in F$ . Then  $cj = b - j^2 \in A^+ \cap A^- = 0$  shows  $c = 0$ . Then  $A = F \oplus Fi \oplus Fj \oplus Fij$  is the quaternion algebra  $Q_K(a, b)$ .

2: By Wedderburn's theorem,  $A \cong \text{Mat}(n, D)$  where  $D$  is a central division ring over  $K$ . The hypothesis shows every element of  $B$  satisfies a degree 2 equation  $x^2 - (x + \bar{x})x + x\bar{x} = 0$ , thus  $n \leq 2$ . If  $n = 2$ , consider the diagonal elements, then we get  $D = F$ . If  $B = D$ , then the same argument as above shows we can find  $i, j$  that  $ij = -ji$  and  $i^2 = a, j^2 = b, a, b \in F^*$ . Then  $j$  induces an isomorphism  $A^+ \cong A^-$ . If there are some  $l \in A^+ \setminus F(i)$ , then  $\bar{l} = \bar{i}l = il$ , thus  $il + \bar{l} = 2il \in F$ , contradiction. Thus  $A^+ = F(i)$ , and  $D = Q_K(a, b)$ .  $\square$

**Prop. (19.6.1.19) [Classifying Binary Forms].** Let  $q = \langle a, b \rangle, q' = \langle a', b' \rangle$  be non-singular binary forms, then they are isomorphic iff  $\det(q) = \det(q')$  and  $Q_K(a, b) \cong Q_K(a', b')$ .  $\square$

*Proof:* If  $q \cong q'$ , then  $ab = a'b' \in F^\times / (F^\times)^2$ , thus  $\langle 1, -a, -b, ab \rangle \cong \langle 1, -a', -b', a'b' \rangle$ , thus  $Q_K(a, b) \cong Q_K(a', b')$  by (19.6.1.16). The converse follows the same way by cancellation theorem (19.5.2.1).  $\square$

**Def. (19.6.1.20) [Hilbert Symbol].** For  $a, b \in K^\times$ , define the **Hilbert symbol**

$$\{a, b\}_K = \begin{cases} 1 & Q_K(a, b) \cong \text{Mat}(2; K) \\ -1 & \text{otherwise} \end{cases}.$$

┘

**Prop. (19.6.1.21) [Computing Hilbert Symbols].** Cf. [Vigneras]P27.

For  $\mathbb{Q}_2$ , we can calculate using the reciprocity formula (19.6.5.4).  $\square$

### Splitting Fields

**Def. (19.6.1.22) [Splitting Fields].** A quaternion algebra  $A$  over  $F$  is said to **split** in a field extension  $K/F \in \mathbf{Field}$  if  $A \otimes_F K \cong \mathrm{Mat}(2, K)$ .  $K$  is said to be a **splitting field** for  $A$ .  $\lrcorner$

**Prop. (19.6.1.23).** Let  $A = Q_K(a, b)$  and  $K = F(\sqrt{c})$ ,  $c \in F^\times$ , then the following are equivalent:

- $A$  splits over  $K$ .
- $A \cong Q_K(c, d)$  for some  $d \in F^\times$ .
- $K$  is a subalgebra of  $A$  over  $F$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2$ : Consider the pure tensor space  $A_0 \cong \langle -a, -b, ab \rangle$ . If it is isotropic, then  $A_0 \cong \langle -c, -d, cd \rangle$  for some  $d \in F^\times$ . If it is anisotropic, then because  $A_K \cong \mathrm{Mat}(2; K)$ , by (19.6.1.16), there exists  $x_i, y_i$  not all zero s.t.

$$-a(x_1 + \sqrt{a}y_1)^2 - b(x_2 + \sqrt{a}y_2)^2 + ab(x_3 + \sqrt{a}y_3)^2 = 0$$

Thus  $\mathbf{x}$  is orthogonal to  $\mathbf{y}$  and  $N(\mathbf{x}) + cN(\mathbf{y}) = 0$ .  $\mathbf{y} \neq 0$ , because otherwise  $N(\mathbf{x}) = 0$  and  $\mathbf{x} = 0$ . Thus there is an orthogonal basis  $\{x, y, z\}$ , and

$$A_0 \cong \langle N(z), N(y), -cN(y) \rangle,$$

and because  $\det(A_0) = 1$ ,  $N(z) \in c(K^\times)^2$ , thus item2 holds.

$2 \rightarrow 3$ : In fact, if  $i^2 = c \in A$ , then  $F(i) \cong K$ .

$3 \rightarrow 1$ : This follows from (3.6.4.18).  $\square$

**Prop. (19.6.1.24).**  $\lrcorner$

### Orders and Ideals

**Notation (19.6.1.25).**

- Let  $(\mathcal{O}_K, K)$  be a Dedekind domain.
- Let  $Q/K$  be a quadratic algebra.

$\lrcorner$

**Def. (19.6.1.26) [Ideals and Orders].** An element  $x \in Q$  is called a **integral element** over  $\mathcal{O}_K$  if  $\mathbb{Z}[x]$  is a  $\mathcal{O}_K$ -lattice. Equivalently, the reduced trace and reduced norm of  $x$  lies in  $\mathcal{O}_K$ .  $\lrcorner$

## 2 Real Quaternions and Octonions

References are [C-S03].

**Prop. (19.6.2.1).**  $\mathrm{Aut}(\mathbb{H}) = \mathrm{SO}(3)$ .  $\lrcorner$

**Prop. (19.6.2.2).** For a coordinate-wise description of the octonion ring, see [C-S03]P75  $\lrcorner$

*Proof:*

$\square$

**Cor. (19.6.2.3).**  $\mathrm{Aut}(\mathbb{O}) = G_2$ .  $\lrcorner$



*Proof:* see[C-S03]P76. □

**Cor. (19.6.2.4).** Any subalgebra of  $\mathbb{O}$  generated by two elements is associative. ┘

*Proof:* This is because we can always use automorphism to assume that the two generators are of the form  $a + i_1x$  and  $c + i_1d + i_2f$ . And then we can use the fact  $\langle 1, i_1, i_2, i_4 \rangle \cong \mathbb{H}$ . □

**Prop. (19.6.2.5) [Finite Subgroups of  $\mathbb{H}$ ].** the finite groups of  $\mathbb{H}^\times$  are just

$$2I, \quad 2O, \quad 2T, \quad 2D_{2n}, \quad 2C_n, \quad C_{2k+1}$$

where  $n, k \in \mathbb{N}$ . ┘

*Proof:* By considering the covering map  $\mathbb{H}^\times \rightarrow \mathrm{SO}(3)$ , the image of the finite group can be only  $I, O, T, D_{2n}, C_n$  where  $n \in \mathbb{Z}$  (15.4.1.8). Then notice if the image contains an order-2 rotation  $g$ , then if  $[q] = g$ ,  $q^2 = 1$ , and so  $q = -1$ . So the finite group is just the inverse image of  $I, O, T, D_{2n}, C_n$  except for  $C_{2k+1}$  where  $k \in \mathbb{1}$ . □

### 3 Real Division Algebras

**Thm. (19.6.3.1) [Hopf/Bott-Milnor/Kervaire].** Every f.d. possibly non-associative division rings over  $\mathbb{R}$  has dimension  $\in \{1, 2, 4, 8\}$ . ┘

*Proof:* □

**Thm. (19.6.3.2) [Fobenius1878/Zorn1931].** Any alternative (3.6.9.1) division algebra  $D$  over  $\mathbb{R}$  is isomorphic to  $\mathbb{R}, \mathbb{C}, \mathbb{H}$  or  $\mathbb{O}$ . And  $\mathbb{O}$  is not associative. ┘

*Proof:* We use (3.6.9.2). For any  $x \in D$ , the map  $R[T] \rightarrow D : T \mapsto x$  is an algebraic homomorphism, so by finite dimensionality of  $D$ ,  $x$  is algebraic over  $\mathbb{R}$ , thus  $x^2 \in \mathbb{R} + \mathbb{R}x$ . But then if  $D \neq \mathbb{R}$ , we can find some  $i \in D$  with  $i^2 = -1$ . And  $C = \mathbb{R} + \mathbb{R}i$  is isomorphic to  $\mathbb{C}$ .

Then also  $C = \{x \in D | xi = ix\}$ . This is because if  $x \in D$  and  $xi = ix$ , we may assume  $x \notin \mathbb{R}$ , so  $(x - a)^2 = (ic)^2$  for some  $a, c \in \mathbb{R}$ . But then  $(x - a - ci)(x - a + ci) = 0$ , which shows  $x \in C$ .

Now suppose  $D \neq C$ . Let  $C^- = \{x \in D | xi = -ix\}$ , then  $D = C \oplus C^-$ , because we have

$$x = \frac{1}{2}(x - ixi) + \frac{1}{2}(x + ixi), \quad x \in D.$$

Let  $x \in C^- \setminus \{0\}$ , then  $x^2 \in C \cap (\mathbb{R} + \mathbb{R}x) = \mathbb{R}$ . And also  $x^2 \in \mathbb{R}_{<0}$ . So we find some  $j \in C^-$  s.t.  $j^2 = -1$ . Then  $ij = -ji$ , and so  $\langle 1, i, j \rangle \cong \mathbb{H}$ . Denote  $ij = k$ .

Define  $H = \{x \in D | xk = (xi)j\}$ , then

**Lemma (19.6.3.3).**  $H = C + Cj$ . ┘

*Proof:* Firstly  $C + Cj \subset H$ . And  $H = C \oplus (C^- \cap H)$ , because of the identity  $x = \frac{1}{2}(x - ixi) + \frac{1}{2}(x + ixi)$  above and the fact  $x + ixi \in H$  as

$$(x + ixi)k = xk - i(xk)i = xk - (ix)(ki) = xk - (ix)j = ((x + ixi)i)j.$$

Notice the linear transformation  $T : x \mapsto xj$  preserves  $H$ : If  $x \in H$ , then  $(xj)k = -(xk)j = -((xi)j)j = xi$ , and  $((xj)i)j = -((xj)j)i = xi$ .

Also notice that if  $x \in H$ , then  $0 = \text{Asso}(x, i, j) = -\text{Asso}(i, x, j) = -i(xj) + (ix)j$ , so if  $x \in C^- \cap H$ , then

$$(xj)i = -(xi)j = (ix)j = i(xj),$$

so  $xj \in H$ . In this way we get  $\dim H \leq 4$ , so  $H = C + Cj$ .  $\square$

Now suppose  $D \neq H$ , and let  $H^- = \{x \in D \mid xk = -(xi)j\}$ , then  $D = H \oplus H^-$ , because we have

$$x = \frac{1}{2}(x - ((xi)j)k) + \frac{1}{2}(x + ((xi)j)k), \quad x \in D,$$

and also

$$(((xi)j)k)i)j = (((xii)j)k)j = -((xj)k)j = -xk$$

by (3.6.9.2) item4 and Moufang law.

**Lemma (19.6.3.4).** Elements in  $H^-$  anti-commute with  $i, j, k$ .  $\lrcorner$

*Proof:* If  $x \in H^-$ , then  $xk = -(xi)j$ , so  $x = [(xi)j]k$ , and

$$kx = [k(xi)](jk) = [(ij)(xi)]i = (i(jx)i)i = -i(jx).$$

So  $kx = -\frac{1}{2}\text{Asso}(i, j, x) = -\frac{1}{2}\text{Asso}(x, i, j) = -xk$ .

The proof of  $jx = -jx$  is similar using the fact  $x = -[(xi)k]j$ .

The proof of  $ix = -ix$  is similar.  $\square$

Then  $H^- \subset C^-$ , and there by the same argument as above, there exists  $h \in H^-$  s.t.  $h^2 = -1$ . Now we prove that  $H^- = Hh$ : By (3.6.9.2) item4, if  $x \in H$ , then

$$(xh)k = -(xk)h = -((xi)j)h = -((xh)i)j,$$

so  $xh \in H^-$ . Conversely, if  $x \in H^-$ , we can prove that  $xh \in H$ .

So  $D = \mathbb{R}\{1, i, j, k, h, hi, hj, hk\}$ , and by Moufang equality,  $(ih)(kh) = -(hi)(kh) = -h(ik)h = hjh = -(jh)h = j$ . Similarly we can calculate the multiplication of any two elements in  $D$ . And in this way, it is isomorphic to  $\mathbb{O}$ .  $\square$

**Def. (19.6.3.5) [Composite Rings].** A (possibly non-associative) composite ring over  $\mathbb{R}$  is defined to be a unital ring  $D \in \text{naAlg}/\mathbb{R}$  with a (non-degenerate and positive-definite) scalar product  $\langle -, - \rangle$  s.t. the norm  $\text{Nm}(x) = \langle x, x \rangle$  is multiplicative from  $D$  to  $\mathbb{R}$ .

For any such composite ring  $D$ , every element  $x \in D$  has a decomposition  $x = a + x'$  where  $a \in \mathbb{R}$  and  $\langle x', 1 \rangle = 0$ . Then we can define the **involution** of  $D$ :  $\bar{x} = a - x'$ .  $\lrcorner$

**Prop. (19.6.3.6).** For any possibly non-associative composite ring  $D$  over  $\mathbb{R}$  and  $x, u, v \in D$ , we have

$$\langle xu, v \rangle = \langle u, \bar{x}v \rangle, \quad \langle xu, xv \rangle = \text{Nm}(x) \langle u, v \rangle$$

And also  $\text{Nm}(x) = \bar{x}x = x\bar{x}$ .  $\lrcorner$

*Proof:* It follows from the relation  $\langle x, y \rangle = \frac{1}{2}(\text{Nm}(x+y) - \text{Nm}(x) - \text{Nm}(y))$  and the multiplicativity of  $\text{Nm}$  that

$$\langle xy, xz \rangle = \text{Nm}(x) \langle y, z \rangle.$$

So if  $\langle x, 1 \rangle = 0$ , then  $\langle x'y, y \rangle = 0$  for any  $y$ . This implies that  $\langle x'u, v \rangle = -\langle x'v, u \rangle$  for any  $u, v$ . Now any element  $x \in D$  is of the form  $x = a + x'$  where  $a \in \mathbb{R}$  and  $\langle x', 1 \rangle = 0$ , then the assertion follows.

For the last assertion, notice

$$\langle \bar{x}x, y \rangle = \langle x, xy \rangle = \langle \text{Nm}(x), y \rangle,$$

so  $\bar{x}x = \text{Nm}(x)$ .  $\square$

**Prop. (19.6.3.7)[Composite Algebras, Hurwitz1898].** Any f.d. possibly non-associative composite ring  $D$  over  $\mathbb{R}$  is alternative and isomorphic to  $\mathbb{R}, \mathbb{C}, \mathbb{H}$  or  $\mathbb{O}$  by (19.6.3.2).  $\lrcorner$

*Proof:* It follows from the fact  $\text{Nm}(x) = \bar{x}x$  that  $D$  is a division ring. to show alternativity, notice

$$\langle z, (\bar{x}x)y \rangle = \text{Nm}(x) \langle z, y \rangle \stackrel{(19.6.3.6)}{=} \langle xz, xy \rangle = \langle z, \bar{x}(xy) \rangle.$$

Thus  $(\bar{x}x)y = \bar{x}(xy)$ . But because  $x + \bar{x} \in \mathbb{R}$ , we get  $x^2y = x(xy)$ . The right-alternativity is similar.  $\square$

**Thm. (19.6.3.8)[Benkart-Britten-Osborn].** Classify real flexible division algebras.  $\lrcorner$

*Proof:*  $\square$

## 4 Quaternions over Local Fields

**Notation (19.6.4.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K) \in p\text{-LField}$ .

$\lrcorner$

**Prop. (19.6.4.2)[Hasse Invariants].** If  $K \not\cong \mathbb{C}$ , then there exist a unique quaternion field over  $K$ .

In particular, for a quaternion algebra  $Q/K$ , we can define its **Hasse invariant of quaternion algebra** to be  $\epsilon(Q) = 1$  if  $Q$  is a quaternion field, and  $-1$  otherwise.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (19.6.4.3)[Properties of the Quaternion Algebra].** Let  $Q/K$  be the unique quadratic field, then

- The map  $|h| = |\text{Nmrd}(h)|^{1/2}$  is a discrete valuation on  $K_{\text{ur}}\{\sqrt{\pi}\}$ . Denote its valuation ring by  $\mathcal{O}$ , then it is the maximal order of  $K_{\text{ur}}\{\sqrt{\varpi}\}$ .
- The quaternion field is isomorphic to  $K_{\text{ur}}\{\sqrt{\varpi}\}$ , where  $K_{\text{ur}}$  is the unramified extension of  $K$ , and its prime ideal  $\mathfrak{P} + \mathcal{O}\sqrt{\varpi}$  satisfies  $\mathfrak{P}^2 = (\varpi)$ .
- 

$\lrcorner$

*Proof:* 1: To show that  $w$  is a discrete valuation, but this follows from considering the subfield  $K(h) \subset K_{\text{ur}}\{\sqrt{\pi}\}$ .

2: Firstly, we show that the unramified quadratic extension  $K_{\text{ur}}$  is contained in  $Q$ : Cf.[Vigneras]P25 ?  $\square$

**Prop. (19.6.4.4).** If  $Q/K$  is the unique quaternion field, then a finite separable extension  $L/K$  is a splitting field for  $Q$  iff  $[F : K]$  is even.  $\lrcorner$

*Proof:* If  $[L : K]$  is even, then either  $e_{L/K}$  is even or  $f_{L/K}$  is even. In the former case,  $\varpi_K$  is a square in  $L$ , so  $Q \otimes F \cong \text{Mat}(2; F)$  by??

If  $[L : K]$  is odd, then  $LK_{\text{ur}}$  is the unramified quadratic extension of  $L$ , and  $\varpi_K$  is not a uniformizer of  $L$ , so  $\square$

## Maximal Orders

### 5 Quaternions over Global Fields

**Notation (19.6.5.1).**

- Let  $F \in \mathbf{GField}$ .

┘

**Def. (19.6.5.2) [Ramified Places].** Let  $Q/F$  be a division algebra of dimension  $n^2$ , then  $v \in \Sigma_F$  is said to be a **ramified place** for  $Q$  if  $Q \otimes_F F_v$  is not isomorphic to  $\text{Mat}(n; F_v)$ .

The set of ramified places of  $Q$  is denoted by  $\text{Ram}_F(Q)$ . Denote  $\text{disc}(H) = \prod_{v \in \text{Ram}_F(Q) \cap \Sigma_F^{\text{fin}}} \mathfrak{p}_v$ , called the **reduced discriminant** of  $Q$ .

┘

**Thm. (19.6.5.3).** For a quaternion algebra  $Q/F$ ,  $\text{Ram}_F(Q)$  is finite, and  $\# \text{Ram}_F(Q)$  is even.

Conversely, for any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ , there exists a unique quaternion algebra  $Q/F$  s.t.  $S = \text{Ram}_F(Q)$ .

┘

*Proof:* The first assertion follows from (11.2.4.1) and (11.2.2.3). To show the existence, by the multiplicative structure of  $\text{Br}(F)[2]$ , it suffices to construct for  $S = \{v, w\}$ : Take a quadratic extension  $L/F$  that  $v, w$  are inert in  $L$  (which exists by (19.3.1.10)), then  $[\mathbf{I}_F : F^\times \text{Nm}_{L/F} \mathbf{I}_L] = 2$ . Take  $a_v \in F_v \setminus \text{Nm}(L_v), a_w \in F_w \setminus \text{Nm}(F_w)$ , then

$$[a_v]_v [a_w]_w \subset F^\times \text{Nm}_{L/F} \mathbf{I}_L,$$

which means there exists  $\theta \in F^\times$  s.t.  $\theta$  is not a norm in  $L_v$  or  $L_w$ , but for any  $u \in \Sigma_F \setminus \{v, w\}$ , either  $L_u$  is not a field or  $\theta$  is a norm in  $L_u$ . Thus  $L\{\sqrt{\theta}\}$  is ramified exactly at  $v, w$ .  $\square$

**Cor. (19.6.5.4) [Reciprocity Formula for Hilbert Symbols].** If  $\text{char } F \neq 2$ , then for any  $a, b \in F^\times$ , we have

$$\prod_{v \in \Sigma_F} \{a, b\}_v = 1 \text{ (19.6.1.20).}$$

┘

**Prop. (19.6.5.5) [Hasse-Weil Principle].** For a quaternion algebra  $Q/F$ ,  $Q$  is split iff  $Q \otimes F_v$  is split for each  $v \in \Sigma_F$ .

┘

*Proof:* Here is an analytic proof: By (29.5.4.17) and (29.5.4.16), if  $Q$  is non-split, then  $\text{ord}_{s=1/2} \Lambda(Q; s) \geq 0$ , which means  $\# \text{Ram}_F(Q) \geq 2$ .  $\square$

**Cor. (19.6.5.6).** For a quaternion algebra  $Q/F$  and  $L/F \in \mathbf{Field}$ ,  $L$  is a splitting field for  $Q$  iff  $L_v$  is a splitting field for  $Q \otimes F_v$  for each  $v \in \Sigma_F$ .

┘

**Thm. (19.6.5.7) [Norm Theorem].** Let  $Q/F$  be a quaternion algebra, then  $\text{Nm}(Q)$  is exact the set of elements of  $F$  which are positive at all the ramified real places of  $F$ .

┘

*Proof:* ?

$\square$

**Cor. (19.6.5.8).** Let  $Q/F$  be a quaternion algebra, then  $\text{Nm}(\mathcal{O}_Q) = \text{Nm}(Q) \cap \mathcal{O}_F$ .

┘

*Proof:* ?

$\square$

**Thm. (19.6.5.9) [Strong Approximation].** Let  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$  s.t.  $S$  contains at least one infinite place of  $F$  if  $F \in \text{NField}$ . Suppose  $Q/F$  is a quaternion algebra s.t.  $Q_S^1$  is not compact, then  $Q_K^1 Q_S^1 = Q_A^1$ .  $\lrcorner$

*Proof:* ?  $\square$

**Def. (19.6.5.10) [Eichler Condition].** For a quaternion algebra  $Q/F$ ,  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$  is said to satisfy the **Eichler condition** for  $Q$  if  $S$  contains at least one place of  $F$  that is unramified in  $Q$ .  $\lrcorner$

### Hurwitz and Lipschitz Integers

**Def. (19.6.5.11) [Hurwitz and Lipschitz Integers].** The **Lipschitz integers** is defined to be  $\mathbb{H}(\mathbb{Z})$ . And the **Hurwitz integers**  $\mathbb{H}$  is defined to be

$$\mathbb{H} = \{a + bi + cj + dk \in \mathbb{H} \mid a, b, c, d \in \mathbb{Z} \text{ or } a, b, c, d \in \frac{1}{2} + \mathbb{Z}\},$$

which contains  $\mathbb{H}(\mathbb{Z})$  as an index-2 sub-lattice.  $\lrcorner$

**Prop. (19.6.5.12).** Let  $D_4$  be the set of Lipschitz integers of even norm (19.6.5.11), then we have a coset decomposition

$$\mathbb{H}(\mathbb{Z}) = D_4 \cup (1 + D_4), \quad \mathbb{H} = D_4 \cup (1 + D_4) \cup (\omega + D_4) \cup (\omega^2 + D_4)$$

where  $\omega = \frac{1+i+j+k}{2}$ . And the multiplications between the cosets are compatible.  $\lrcorner$

**Prop. (19.6.5.13).** Every left or right ideal of  $\mathbb{H}$  is principle.  $\lrcorner$

*Proof:* This is because the Euclid division holds in  $\mathbb{H}$ .  $\square$

**Prop. (19.6.5.14) [Unique Factorization of Primitive Hurwitz Quaternions].** A **primitive Hurwitz integer** is a Hurwitz integer  $Q \in \mathbb{H}$  that is not divisible by a rational  $n \in \mathbb{Z}_{\geq 2}$ . Then for any primitive integer  $Q \in \mathbb{H}$  and a prime factorization  $\text{Nm}(Q) = p_1 \dots p_k$  where  $p_i \in \text{Prime}$ , up to units there is a unique decomposition

$$Q = P_1 \dots P_k, \quad \text{Nm}(P_i) = p_i.$$

$\lrcorner$

*Proof:* By (19.6.5.13), the ideal  $p_1 H + QH$  is a principle ideal  $P_1 H$ . Then  $\text{Nm}(P_1) \mid \text{Nm}(p_1) = p_1^2$ . So  $\text{Nm}(P_1) = 1, p_1$  or  $p_1^2$ . But it cannot be 1 because any element  $p_1 a + Qb$  has norm  $p_1^2 \text{Nm}(a) + p_1 \dots p_k \text{Nm}(b) + (2p_1 a, Qb)$  which divides  $p_1$ . But it can be  $p_1^2$  neither, because if this is the case, then  $p_1 = P_1 U$  where  $U$  is a unit, so  $p_1 \mid U$ , contradicting the fact  $U$  is primitive.

So we get  $P_1$ . Now by induction, we get  $P_2, \dots, P_k$ .  $\square$

**Prop. (19.6.5.15) [Factorization of Lipschitz Integers].** Any factorization of a Lipschitz integer into Hurwitz integers is equivalent up to units to a factorization into Lipschitz integers.  $\lrcorner$

*Proof:* In fact, by the coset decomposition as in (19.6.5.12), any such factorization can be made to a factorization into Lipschitz integers just by multiplying by  $\omega$ .  $\square$

## 6 Applications to Arithmetic Groups

### 7 Without Eichler Condition

## 19.7 Cohomology of Arithmetic Fields

References are [Neu15, Mil20], [Cohomology of Number Fields, Neukirch], <http://www.math.columbia.edu/~chaoli/docs/ClassFieldTheory.html#sec28> and [Arithmetic Duality Theory, Milne].

**Notation (19.7.0.1).**

- Use notations on profinite groups defined in [Profinite Groups](#).
- Use notations on cohomologies on profinite groups defined in [Cohomology of Profinite Groups](#).
- Use notations on number fields defined in [Global Fields](#).

┘

### 1 Abstract Class Field Theory

**Def. (19.7.1.1) [Class Formations].** A **profinite formation** consists of a profinite group  $G \in \text{ProfGrp}$  regarded as a Galois group of a field  $\text{Gal}_K$  and  $A \in T_G$ . It is called a **field formation** iff for any normal extension  $L/K$ ,  $H^1(L/K, A^L) = 0$ .

For a field extension, by (9.8.1.14),  $\text{inf}$  is an injection on  $H^2$ . We denote  $H^2(K)$  as the profinite cohomology group  $H^2(G, A) = \text{Br}(K)$ . Inflation should be thought of as inclusions.

It is called a **class formation** if moreover for every normal extension  $L/K$ , there is an canonical isomorphism

$$\text{inv}_{L/K} : H^2(L/K) \rightarrow \frac{1}{[L : K]} \mathbb{Z}/\mathbb{Z}$$

that is compatible with inflation and restriction in the sense that:

- If  $N/L/K$  with  $N/K$  and  $L/K$  normal, then  $\text{inv}_{L/K} = \text{inv}_{N/K}|_{H^2(L/K)}$  via inflation.
- If  $N/L/K$  with  $N/L$  and  $N/K$  normal, then  $\text{inv}_{N/L} \circ \text{res}_L = [L : K] \cdot \text{inv}_{N/K}$ .

$\text{inv}_{L/K}^{-1}(\frac{1}{[L:K]} + \mathbb{Z}) \in H^2(L/K)$  is called the **fundamental class**  $u_{L/K}$ .

┘

**Prop. (19.7.1.2).**  $\text{inv}$  also commutes with  $\text{cor}$  and conjugation:

$$\text{inv}_{N/K}(\text{cor}_K c) = \text{inv}_{N/L} c, \quad \text{inv}_{\sigma N/\sigma K}(\sigma^* c) = \text{inv}(c).$$

The first is because  $\text{inv}$  commutes with  $\text{res}$  thus  $\text{res}$  is surjective, thus there is a  $c'$  that  $c = \text{res } c'$ . Because of  $\text{cor } \text{res} = [L : k]$ , we have  $\text{cor}_K(c) = c'^{[L:K]}$ . Thus  $\text{inv}_{N/K}(\text{cor}_K c) = [L : K] \text{inv}_{N/K}(c') = \text{inv}_{N/L}(\text{res}_L c') = \text{inv}_{N/L}(c)$ .

For the conjugation, Cf. [Neu15]P69?

┘

**Cor. (19.7.1.3).** From this we easily get that

$$u_{L/K} = (u_{N/K})^{[N:L]}, \quad \text{res}_L(u_{N/K}) = u_{L/K}$$

$$\text{cor}_K(u_{N/L}) = (u_{N/K})^{[L:K]}, \quad \sigma^*(u_{N/K}) = u_{\sigma N/\sigma K}.$$

┘

**Prop. (19.7.1.4) [Class Formation Main Theorem].** Tate's theorem (9.8.1.25) tells us for a class formation, for  $L/K$  normal extension, there is a **Nakayama isomorphism**

$$\theta_{L/K} = u_{L/K} \cup - : H^q(\text{Gal}(L/K), \mathbb{Z}) \cong H^{q+2}(L/K).$$

Especially, for  $q = -2$ , there is a canonical isomorphism  $\text{Gal}^{\text{ab}}(L/K) \cong A_K / \text{Nm}_{L/K} \mathbf{A}_L$ . Its inverse is called the **Artin reciprocity isomorphism** and induces a **norm residue symbol** map  $(-, L/K)$

$$1 \rightarrow \text{Nm}_{L/K} A_L \rightarrow A_K \xrightarrow{(-, L/K)} \text{Gal}^{\text{ab}}(L/K) \rightarrow 1.$$

This norm residue symbols induce an **Artin map**

$$\text{Art}_K = (-, K) : A_K \rightarrow \text{Gal}_K^{\text{ab}} = \varprojlim_{L/K} \text{Gal}^{\text{ab}}(L/K)$$

that has dense image. ┘

**Lemma (19.7.1.5).** Let  $L/K$  be a normal extension,  $a \in A_K$  and  $\chi \in \text{Gal}^{\text{ab}}(L/K)^\vee = H^1(G(L/K), \mathbb{Q}/\mathbb{Z})$  is a character, then

$$\chi((a, L/K)) = \text{inv}_{L/K}(a \smile \delta\chi) \in \frac{1}{[L : K]} \mathbb{Z}/\mathbb{Z}.$$

┘

*Proof:* Cf.[Neukirch CFT P71]. □

**Prop. (19.7.1.6) [Properties of Norm Residue Symbols].** For formal field extensions  $N/L/K$  with  $N/K$  normal, there are commutative diagrams:

$$\begin{array}{ccccc} A_K & \xrightarrow{(-, N/K)} & \text{Gal}(N/K)^{\text{ab}} & & A_K & \xrightarrow{(-, N/K)} & \text{Gal}(N/K)^{\text{ab}} & & A_L & \xrightarrow{(-, N/L)} & \text{Gal}(N/L)^{\text{ab}} \\ \downarrow \text{id} & & \downarrow \text{pr} & & \downarrow \text{Nm}_{L/K} & & \downarrow i & & \downarrow \sigma & & \downarrow \sigma^* \\ A_K & \xrightarrow{(-, L/K)} & \text{Gal}(L/K)^{\text{ab}} & & A_L & \xrightarrow{(-, N/L)} & \text{Gal}(N/L)^{\text{ab}} & & A_{\sigma L} & \xrightarrow{(-, \sigma N / \sigma L)} & \text{Gal}(\sigma N / \sigma L)^{\text{ab}} \end{array}$$

Where Ver is the transfer map defined in?? ┘

*Proof:* Cf.[Neukirch CFT P72]. ? □

**Remark (19.7.1.7) [Non-Abelian Problems].** For a finite normal extension  $L/K$ ,  $\text{Nm}_{L/K} A_L = \text{Nm}_{L^{\text{ab}}/K} A_{L^{\text{ab}}}$ . This is because the quotient both correspond to  $\text{Gal}(L/K)^{\text{ab}}$ . So class field theory doesn't tell about non-Abelian extension. ┘

**Prop. (19.7.1.8) [Norm Groups and Abelian Extension].** The map  $L \mapsto \mathcal{N}_{L/K} = \text{Nm}_{L/K} A_L$  defines a inclusion reversing isomorphism between the lattice of Abelian extension  $L$  of  $K$  and the lattice of norm groups of  $A_K$ , i.e.:

$$\mathcal{N}_{L_1 L_2 / K} = \mathcal{N}_{L_1 / K} \cap \mathcal{N}_{L_2 / K}, \quad \mathcal{N}_{L_1 \cap L_2 / K} = \mathcal{N}_{L_1 / K} \cdot \mathcal{N}_{L_2 / K}.$$

And any group that contains a norm group is a norm group. ┘

*Proof:* By the first commutative diagram of inv, if  $(a, L_i/K) = 0$ , then  $(a, L_1 L_2 / K)$  is trivial on  $\text{Gal}(L_i/K)$ , thus trivial on  $\text{Gal}(L_1 L_2 / K)$ , thus  $a \in \mathcal{N}_{L_1 L_2}$ . so  $\mathcal{N}_{L_1} \cap \mathcal{N}_{L_2} \subset \mathcal{N}_{L_1 L_2}$ , the other side is easy. the second is because  $|\mathcal{N}_{L_1 \cap L_2} / \mathcal{N}_{L_1}| = |G_{L_1 / L_1 \cap L_2}| = |G_{L_1 L_2 / L_2}| = |I_{L_1} I_{L_2} / I_{L_1}|$ . Also we deduce  $I_{L_1} \subset I_{L_2} \iff L_2 \subset L_1$ , thus by canonical isomorphism, groups containing  $\mathcal{N}_{L/K}$  are in bijection with middle fields of  $L/K$  by counting numbers. □

**Remark (19.7.1.9).** This shows the philosophy of CFT, i.e. the property of Abelian extensions of a field is can be read from its multiplicative group structure. Of course, determining and characterizing these norm groups requires some work. ┘

### Abstract Weil Groups

References are [A-T09] and [Tat79].

**Def. (19.7.1.10)** [Abstract Weil Groups]. ┘

**Def. (19.7.1.11)** [Essentially Inner Automorphisms]. ┘

## 2 Local Class Field Theory

**Notation (19.7.2.1).**

- Let  $p \in \text{Prime}$ ,
  - Let  $(K, \mathcal{O}_K, \mathfrak{p}_K, \varpi_K) \in p\text{-LField}$ ,
  - Let  $L/K$  be a finite separable extension.
- ┘

### Unramified Extensions

**Lemma (19.7.2.2).** If  $L/K$  is unramified, then  $H^q(\text{Gal}(L/K), \mathcal{O}_L^*) = 0$  for all  $q \in \mathbb{Z}$ . ┘

*Proof:* Cf. [Neukirch P83]. □

**Prop. (19.7.2.3)** [Unramified Class Formations]. The unramified extensions of  $K$  form a class formation. ┘

*Proof:* We first define the inv map: use the exact sequence  $1 \rightarrow \mathcal{O}_L^* \rightarrow L^\times \xrightarrow{v_L} \mathbb{Z} \rightarrow 0$ , using the lemma (19.7.2.2), we have an isomorphism

$$H^2(\text{Gal}(L/K), L^\times) \xrightarrow{v_L} H^2(\text{Gal}(L/K), \mathbb{Z}) \xrightarrow{\delta^{-1}} H^1(\text{Gal}(L/K), \mathbb{Q}/\mathbb{Z}) = \text{Gal}(L/K)^\vee.$$

And there is an isomorphism

$$\varphi : \text{Gal}(L/K)^\vee \cong \frac{1}{[L; K]} \mathbb{Z}/\mathbb{Z}, \quad \varphi(\chi) = \chi(\sigma_{L/K})$$

where  $\sigma_{L/K}$  is the geometric Frobenius which generates  $\text{Gal}(L/K)$ . Then define

$$\text{inv}_{L/K} = \varphi \circ \delta^{-1} \circ v_L : H^2(\text{Gal}(L/K), L^\times) \cong \frac{1}{[L : K]} \mathbb{Z}/\mathbb{Z}$$

To verify this is a class formation, we should verify (19.7.1.1), Cf. [Neukirch P85] ?. □

**Prop. (19.7.2.4)** [Local Norm Symbol is given by Geometric Frobenius]. If  $L/K$  is unramified, then  $(a, L/K) = \sigma_{L/K}^{v_K(a)}$ . The same holds for  $L$  replaces by  $K^{\text{ur}}$ , in which case

$$1 \rightarrow \mathcal{O}_K^* \rightarrow K^\times \rightarrow \text{Gal}(K^{\text{ur}}/K) \rightarrow 0$$

is exact. ┘

*Proof:* We use (19.7.1.5). For any  $\chi$ ,

$$\chi(a, L/K) = \text{inv}_{L/K}(\bar{a} \cup \delta \chi) = \varphi \circ \delta^{-1} \circ v_K(\bar{A} \cup \delta \chi) = \varphi(\delta^{-1}(v_K(a) \delta \chi)) = \varphi(v_K(a) \chi) = v_K(a) \chi(\sigma_{L/K}) = \chi(\sigma_{L/K}^{v_K(a)}),$$

The second assertion follows from the last prop (19.7.2.21). □



### Ramified Extensions

**Lemma (19.7.2.5).** If  $L/K$  is normal, then  $\#H^2(L/K) \mid [L : K]$ . ┘

*Proof:* Cf.[Neukirch CFT P89]. Should use the fact that  $G_{L/K}$  is solvable and Herbrand quotient. □

**Lemma (19.7.2.6) [Invariant Maps].** If  $L/K$  is normal and  $L'/K$  is another unramified extension of the same degree, then  $H^2(L/K) = H^2(L'/K) \subset \text{Br}(K)$ . In particular, we can define an invariant map  $\text{inv} : \text{Br}(K) \rightarrow \mathbb{Q}/\mathbb{Z}$ . ┘

*Proof:* In view of (19.7.2.5) and (19.7.2.3), we only need to prove  $H^2(L'/K) \subset H^2(L/K)$ . For this, we let  $N = LL'$ , then there is an exact sequence (9.8.1.14)

$$1 \rightarrow H^2(L/K) \rightarrow H^2(N/K) \xrightarrow{\text{res}_L} H^2(N/L)$$

then we only need to prove  $\text{res}_L(c) = 0$ , and this follows from  $\text{inv}_{N/L}(\text{res}_L c) = 0$ . This will follow, if we have

$$\text{inv}_{N/L}(\text{res}_L c) = [L : K] \cdot \text{inv}_{L'/K} c.$$

This follows from (19.7.2.7). □

**Lemma (19.7.2.7).** For two subextensions  $L/K, L'/K$  in  $M/L$  normal with  $L'/K$  unramified extension,  $N = LL'$ , for  $c \in H^2(L'/K)$ ,

$$\text{inv}_{N/L}(\text{res}_L c) = [L : K] \cdot \text{inv}_{L'/K} c.$$

┘

*Proof:* Cf.[Neukirch CFT P90]. □

**Lemma (19.7.2.8).**  $(\text{Gal}_K, (K^{\text{sep}})^{\times}, \text{inv}_K)$  forms a class formation. ┘

*Proof:* This almost follows from that of unramified extensions (19.7.2.3). We verify axioms (19.7.1.1) that  $\text{inf}$  is natural and commutes with  $\text{res}$ . It is natural because it is natural on unramified extensions, it commutes with  $\text{res}$  because we can assume  $c \in H^2(L'/K)$  unramified and use (19.7.2.7). □

**Cor. (19.7.2.9) [Local Artin's Reciprocity Law].** If  $L/K$  is normal, then the homomorphism

$$u_{L/K} \cup - : H^q(\text{Gal}(L/K), \mathbb{Z}) \cong H^{q+2}(L/K)$$

is an isomorphism. ┘

**Cor. (19.7.2.10).**  $H^3(L/K) = 1, H^4(L/K) = \text{Gal}^{\text{ab}}(L/K)^{\vee}$ , by (9.8.1.7). ┘

**Thm. (19.7.2.11) [Artin's Reciprocity Law].** By (19.7.1.4), cup product with the fundamental class in  $H^2(L/F)$  defines an isomorphism

$$\theta_{L/K} : \text{Gal}^{\text{ab}}(L/K) \cong H^{-2}(\text{Gal}(L/K), \mathbb{Z}) \rightarrow H^0(L/K) = K^{\times} / \text{Nm}_{L/K} L^{\times},$$

called the **Nakayama map**. And the reverse map is called the **norm residue symbol**  $(-, L/K)$

$$1 \rightarrow \text{Nm}_{L/K} L^{\times} \rightarrow K^{\times} \xrightarrow{(-, L/K)} \text{Gal}^{\text{ab}}(L/K) \rightarrow 1.$$

This norm residue symbols induce an **Artin map**

$$\text{Art}_K = (-, K) : K^{\times} \rightarrow \text{Gal}_K^{\text{ab}}$$

with dense image. ┘

**Cor. (19.7.2.12).** By (19.7.1.6), there are commutative diagrams

$$\begin{array}{ccccc}
 K^\times & \xrightarrow{(-, N/K)} & \text{Gal}(N/K)^{\text{ab}} & & L^\times & \xrightarrow{(-, N/L)} & \text{Gal}(N/L)^{\text{ab}} \\
 \downarrow \text{id} & & \downarrow \text{pr} & \text{Nm}_{L/K} \updownarrow i & \downarrow \sigma & & \downarrow \sigma^* \\
 K^\times & \xrightarrow{(-, L/K)} & \text{Gal}(L/K)^{\text{ab}} & & \sigma L^\times & \xrightarrow{(-, \sigma N/\sigma L)} & \text{Gal}(\sigma N/\sigma L)^{\text{ab}}
 \end{array}$$

Where Ver is the transfer map defined in (3.1.9.6).  $\lrcorner$

**Cor. (19.7.2.13)[Geometric Frobenius].** By (19.7.2.4),  $(\varpi_K, K)$  is the image of an element  $\sigma_K \in \text{Gal}_K$  which is a lift of  $\text{Frob}_K^{-1}$ , i.e. a **geometric Frobenius element**, which is unique up to  $I_K$ .  $\lrcorner$

**Cor. (19.7.2.14)[Quadratic Character].** If  $L/K$  is a quadratic extension, then  $\mathcal{N}_{L/K}$  is a subgroup of  $K^\times$  of order 2. Let  $\chi$  be the non-trivial character of  $K^\times$  that is trivial on  $\mathcal{N}_{L/K}$ , called the **quadratic character** of  $K^\times$  attached to  $L/K$ .  $\lrcorner$

**Prop. (19.7.2.15)[Higher Ramification Groups].** If  $L/K$  is Abelian, the higher principal units  $U_K^n$  are mapped under the higher ramification groups of  $\text{Gal}(L/K)$  under the upper numbering. **?**  $\lrcorner$

*Proof:*  $\square$

**Def. (19.7.2.16)[Conductors].** If  $L/K$  is Abelian, define the **conductor(local Abelian extension)**  $f_{L/K} \in \mathbb{Z}_+$  to be the smallest  $n \in \mathbb{Z}_+$  s.t.  $U_K^n \subset \mathcal{N}_{L/K}$ .  $\lrcorner$

**Cor. (19.7.2.17).** For an Abelian extension  $L/K$ ,  $f_{L/K} = 1$  iff  $L/K$  is unramified, by (19.7.2.21).  $\lrcorner$

### Characterize the Norm Groups of $K^\times$

**Prop. (19.7.2.18)[Norm Group and Abelian Extension].** The map  $L \mapsto \mathcal{N}_{L/K}$  defines an inclusion reversing isomorphism between the lattice of Abelian extension  $L$  of  $K$  and the lattice of norm groups of  $K^\times$ , i.e.:

$$\mathcal{N}_{L_1 L_2/K} = \mathcal{N}_{L_1/K} \cap \mathcal{N}_{L_2/K}, \quad \mathcal{N}_{L_1 \cap L_2/K} = \mathcal{N}_{L_1/K} \cdot \mathcal{N}_{L_2/K}.$$

And any group that contains a norm group is a norm group.  $\lrcorner$

*Proof:* This follows from (19.7.1.8) and (19.7.2.8).  $\square$

**Prop. (19.7.2.19).** The norm groups are precisely the open(closed) subgroups of finite index in  $K^\times$ . In fact finite index are itself open because it contains  $(K^\times)^n$  which is open.  $\lrcorner$

*Proof:* One part follows from (19.7.2.18) and the fact that  $(K^\times)^m$  is open (19.1.5.14). For the converse, we only need to prove  $(K^\times)^m$  is a norm group. This uses Kummer theory and Cf.[Neukirch CFT P96].  $\square$

**Prop. (19.7.2.20)[Norm Groups of Local Fields].** The norm groups of  $K^\times$  are exactly the groups containing  $U_K^n \times (\varpi_K^f)$  for some  $n, f$ .  $\lrcorner$

*Proof:*  $U_K^n \times (\varpi_K^f)$  is a norm group because it is closed of finite index. Conversely, any norm group contains some  $U_K^n$  because it is open and contains some  $(\varpi_K^f)$  because it is of finite index.  $\square$

**Cor. (19.7.2.21)[Norm Group of Unramified Extensions].** The norm group of an unramified extension of degree  $f$  is

$$\mathcal{O}_K^* \times (\varpi_K^f)^{\mathbb{Z}}.$$

In particular,  $L/K$  is unramified iff  $\mathcal{O}_K^* \subset \mathcal{N}_{L/K}$ .  $\lrcorner$

*Proof:* This follows from (19.7.2.4) and (19.7.2.18).  $\square$

### Local Weil Groups

**Prop. (19.7.2.22)** [Local Weil Group, Weil[Wei51]]. The Weil group defined in (19.7.1.10) is a topological group that fits into the exact sequence

$$1 \rightarrow I_K \rightarrow W_K \rightarrow \mathbb{Z} \rightarrow 0.$$

And the relative Weil group  $W_{L/K}$  a topological group that fits into the exact sequence

$$1 \rightarrow \text{Gal}(\bar{L}/K^{\text{ur}}) \rightarrow W_{L/K} \rightarrow \mathbb{Z} \rightarrow 0.$$

In particular, there is a natural isomorphism

$$W_K \cong \varprojlim_{L/K} W_{L/K}.$$

┘

*Proof:*

□

**Prop. (19.7.2.23).**  $W_L = \text{Gal}_L \cap W_K$ .

┘

*Proof:* The group  $W_K \cap \text{Gal}_L \subset \text{Gal}_K$  consists of elements that fixes  $L$  and induces a residual morphism  $\varphi_\kappa^n$  for some  $n \in \mathbb{Z}$ . Thus  $[\kappa_L : \kappa] | n$ , and it induces a residual morphism  $\varphi_{\kappa_L/\kappa}^{n/[\kappa_L:\kappa_K]}$ , which means that is in  $W_L$ . □

**Prop. (19.7.2.24)** [Artin Maps and Weil Groups]. The Artin map  $\text{Art}_K : K^\times \rightarrow \text{Gal}_K^{\text{ab}}$  is injective because  $(K^\times)^n$  are all norm groups by (19.7.2.19), so the kernel is contained in their intersection with 1 by (19.1.5.14). Its image is just  $W_K^{\text{ab}}$ . ┘

*Proof:*

□

**Def. (19.7.2.25)** [Norm on Weil Groups]. The Artin map (19.7.2.24) gives a norm  $|x| = |\text{Art}^{-1}(x)|$  on  $W_K^{\text{ab}}$ , which maps a lift of the geometric Frobenius in  $\text{Gal}_\kappa$  to  $q^{-1}$ . ┘

### Totally Ramified Extensions

References are [L-T65].

**Notation (19.7.2.26).**

- Use notations defined in Lubin-Tate Formal Group Law.

┘

**Prop. (19.7.2.27)** [Tate Modules]. There is an isomorphism of  $\mathcal{O}$ -modules  $\Lambda_{f,n} \cong \mathcal{O}_K/(\varpi^n)$ , Cf.[Neukirch CFT P101]. Thus the automorphism of  $\Lambda_{f,n}$  is all of the form  $u_f$  for units, isomorphic to  $U_K/U_K^n$ .

So we can define a **Tate module**  $TG = \varprojlim \ker[\varpi_K^n]$ , it is a free  $\mathcal{O}_K$ -module of rank 1. ┘

**Def. (19.7.2.28)** [Lubin-Tate Character]. As  $TG$  is a free  $\mathcal{O}_G$ -module of dimension 1, and  $\text{Gal}_K$  acts on  $TG$ , there can be attached a **Lubin-Tate character**  $\chi_K : \text{Gal}_K \rightarrow \mathcal{O}_K^*$  by  $g(\alpha) = [\chi_K(g)](\alpha)$ , this depends on  $\pi_K$ , but its restriction on  $I_K$  doesn't depend on  $\pi_K$ , and is just the local CFT isomorphism composed with  $x \rightarrow x^{-1}$ . ┘

*Proof:*  $[\chi_K(g)]$  is, by definition, the morphism that is  $\text{id}$  on  $K^{\text{ur}}$  and  $g$  on  $L_\pi$ . So it equals  $g$  on all  $K^{\text{ab}}$  iff  $g$  is  $\text{id}$  on  $K^{\text{ur}}$ , that is,  $g \in I_K$ . So if  $g \in I_K$ , by local CFT,  $(\chi(g))^{-1}$  corresponds to  $g$ , uniquely.  $\square$

**Prop. (19.7.2.29).**  $G_{\pi,n} \cong \mathcal{O}_K^*/U_K^n$ , thus we have  $G_\pi \cong \mathcal{O}_K^*$ .  $L_{\pi,n}/K$  (10.7.3.28) is Abelian totally ramified of degree  $p^{n-1}(p-1)$  generated by a Eisenstein polynomial with constant coefficient  $\varpi$ . In particular,  $\varpi$  is in the norm group.  $\lrcorner$

*Proof:* For this, first note Galois action induce an isomorphism on  $\Lambda_{f,n}$ , thus correspond to an element of  $U_K/U_K^n$  by (19.7.2.27), this is an injection because  $\Lambda_{f,n}$  generate  $L_{\pi,n}$ . Then we use the canonical polynomial  $f(Z) = \pi Z + Z^q$ ,  $f^n = f^{n-1}\varphi(n)$ , where  $\varphi(n)$  is a Eisenstein polynomial, thus  $L_{\pi,n}/K$  is totally ramifies with  $|G_{\pi,n}| = q^{n-1}(q-1) = |U_K/U_K^n|$ , thus the result.  $\square$

**Prop. (19.7.2.30)[Explicit Local Norm Residue Symbol].** Now we can write the universal residue symbol little bit more explicitly. For  $a = u\varpi^m$ ,  $(a, K)$  acts as  $\varphi^m$  on  $K^{\text{ur}}$ , and on  $L_{\pi,n}$ , its action is generated by the action  $(u^{-1})_f$  on the generating set  $\Lambda_{f,n}$ .

Thus the norm group of  $L_{\pi,n}$  is just  $U_K^n$  by (19.7.2.29).  $\lrcorner$

*Proof:* Cf. [Neukirch CFT P106].  $\square$

**Cor. (19.7.2.31)[Norm Groups of Totally Ramified Extensions].** The norm groups of the totally ramified Abelian extension are precisely the groups that contains some  $U_K^n \times \varpi^{\mathbb{Z}}$  for some uniformizer  $\pi$ . And every totally ramified Abelian extension  $L/K$  is contained in some  $L_{\varpi,n}$ .  $\lrcorner$

*Proof:* For any totally ramified extension, choose a uniformizer, then its norm is a uniformizer  $\varpi$  of  $K$ . And  $\text{Nm}_{L/K}$  is open, thus it contains some  $U^n$ . The rest follows from local CFT (19.7.2.18).  $\square$

**Cor. (19.7.2.32)[Maximal Abelian Extension of  $p$ -adic Local Fields].** Let  $L_\pi = \cup L_{\pi,n} = K(\Lambda_f)$ , where  $\Lambda_f = \cup \Lambda_{f,n}$ , then  $T \cdot L_\pi$  is the maximal Abelian extension of  $K$ . Hence  $G_K^{\text{ab}} = G_{T,K} \times G_\pi$ .  $\lrcorner$

*Proof:* This follows immediately from (19.7.2.20).  $\square$

**Cor. (19.7.2.33)[Hasse-Arf].** We can prove Hasse-Arf in the case where  $K$  is a local field. This is because we already know the maximal Abelian extension, and  $\text{Gal}(K^{\text{ab}}/T) \cong \text{Gal}(L_\pi/K) \cong \mathbb{Z}_p$  for which we know the Galois action well (19.7.2.27)(19.7.2.29), so  $i(\sigma) = v(\sigma(\alpha_n) - \alpha_n) = v([\sigma - 1](\alpha))$ , which jumps at  $U_K^n$  (the same pattern as  $K = \mathbb{Q}_p$  (19.1.4.30)), thus the result.  $\lrcorner$

**Example (19.7.2.34)[Cyclotomic Fields].** When  $K = \mathbb{Q}_p$ , we can choose  $f(T) = (1+T)^p - 1$ , thus  $L_{\varpi,n}$  is just  $\mathbb{Q}_p(\zeta_{p^n})$ . And we have  $r_f = (1+T)^r - 1$ , thus we have

$$(a, \mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p)(\zeta_{p^n}) = \zeta_{p^n}^r$$

where  $a = up^m$ , and  $r \equiv u^{-1} \pmod{p^n}$ .  $\lrcorner$

### 3 Global Class Field Theory

**Notation (19.7.3.1).**

- Let  $L/F \in \text{GField}$ .

$\lrcorner$

**Prop. (19.7.3.2).** Let  $\mathfrak{P}$  be a prime of  $L$  lying over  $\mathfrak{p}$ , then  $H^q(G, I_L^{\mathfrak{p}}) \cong H^q(G_{\mathfrak{P}}, L_{\mathfrak{P}}^*)$ . If  $\mathfrak{p}$  is a finite unramified prime of  $L$ , then  $H^q(G, U_L^{\mathfrak{p}}) = 1$  for all  $q$ .  $\lrcorner$

*Proof:* Notice  $I_L^{\mathfrak{p}} = \prod_{\sigma \in G/G_{\mathfrak{P}}} L_{\sigma\mathfrak{P}}^* = \prod_{\sigma \in G/G_{\mathfrak{P}}} \sigma L_{\mathfrak{P}}^*$  is an induced module, so by (9.8.2.8), we have  $H^q(G_{\mathfrak{P}}, L_{\mathfrak{P}}^*)$ , and similarly for  $U_{\mathfrak{p}}$ , which vanish by (19.7.2.2).  $\square$

**Cor. (19.7.3.3) [Cohomology of Ideles].**

$$H^q(G, \mathbf{I}_L^S) = \bigoplus_{p \in S} H^q(G_{\mathfrak{P}/\mathfrak{p}}, L_{\mathfrak{P}}^*), \quad H^q(G, \mathbf{I}_L) = \bigoplus_p H^q(G_{\mathfrak{P}}, L_{\mathfrak{P}}^*).$$

And the isomorphism is natural, by restriction to components.  $\lrcorner$

*Proof:* For this, just notice  $\mathbf{I}_L = \cup_S \mathbf{I}_L^S$ , then use the last proposition, notice group cohomology commutes with colimits (9.8.2.3).  $\square$

**Cor. (19.7.3.4).**  $H^1(G, \mathbf{I}_L) = H^3(G, \mathbf{I}_L) = 0$ , by (19.7.2.10).  $\lrcorner$

**Cor. (19.7.3.5).** An idele  $\mathbf{a} \in \mathbf{I}_F$  is the norm of an idele  $\mathbf{b}$  in  $\mathbf{I}_L$  if each component  $\mathbf{a}_{\mathfrak{p}}$  is the norm of an element  $b_{\mathfrak{p}} \in L_{\mathfrak{P}}^*$ .  $\lrcorner$

**Prop. (19.7.3.6).** The decomposition commutes with inf, res and cor. Cf. [Neukirch CFT P125].  $\lrcorner$

### Cyclic Class Formations

**Lemma (19.7.3.7).** For a cyclic extension  $L/F$  of order  $p$ ,  $\mathbf{C}_L$  is a Herbrand module with Herbrand quotient  $h(\mathbf{C}_L) = p$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (19.7.3.8) [First Fundamental Inequality].**  $[\mathbf{C}_F : \text{Nm}_{L/F} \mathbf{C}_L] \geq p$   $\lrcorner$

*Proof:*  $\square$

**Prop. (19.7.3.9) [Second Fundamental Inequality, Chevalley [Che40]].** If  $\mathbb{Q}(\zeta_p) \subset F$  and  $L/F$  is a cyclic extension of order  $p$ , then  $[\mathbf{C}_F : \text{Nm}_{L/F} \mathbf{C}_L] \leq p$ .  $\lrcorner$

*Proof:*  $\square$

**Cor. (19.7.3.10).** If  $L/F$  is a cyclic extension of order  $p \in \text{Prime}$ , then  $[\mathbf{C}_F : \text{Nm}_{L/F} \mathbf{C}_L] = p$ .  $\lrcorner$

*Proof:*  $\square$

**Cor. (19.7.3.11) [Hasse Principle for Norms for Cyclic Extensions].** If  $L/F$  is a cyclic extension and  $a \in F^\times$ , then  $a \in \text{Nm}_{L/F} L^\times$  iff  $a \in \text{Nm}_{L/F}(\mathbf{I}_L)$ .  $\lrcorner$

*Proof:* Use the long exact sequence for  $1 \rightarrow L^\times \rightarrow \mathbf{I}_L \rightarrow \mathbf{C}_L \rightarrow 1$ , we see that  $H^0(G, L^\times) \rightarrow H^0(G, \mathbf{I}_L)$  is an injection, which is

$$F^\times / \text{Nm}_{L/F} L^\times \hookrightarrow \bigoplus_p F_{\mathfrak{p}}^\times / \text{Nm}_{L_{\mathfrak{P}}/F_{\mathfrak{p}}} L_{\mathfrak{P}}^\times.$$

In fact, by (19.7.3.4), we say that this is equivalent to  $H^1(\text{Gal}(L/K), \mathbf{C}_L) = 1$ , which is equivalent to second fundamental inequality.  $\square$

**Remark (19.7.3.12).** WARNING: the Hasse norm principle is not true for non-cyclic Galois extensions, for examples  $\mathbb{Q}(\sqrt{13}, \sqrt{17})/\mathbb{Q}$ .  $\lrcorner$

**Prop. (19.7.3.13).** If  $L/F$  is cyclic, then there is an exact sequence

$$1 \rightarrow H^2(\text{Gal}(L/F), L^\times) \rightarrow H^2(\text{Gal}(L/F), \mathbf{I}_L) \xrightarrow{\text{inv}_{L/F}} \frac{1}{[L:F]} \mathbb{Z}/\mathbb{Z} \rightarrow 0.$$

 $\lrcorner$ 

*Proof:* Cf.[Neukirch]P143.  $\square$

### General Class Formations

**Prop. (19.7.3.14).** For  $L/F$  normal,  $\#H^2(G, \mathbf{C}_L) | [L:F]$ .  $\lrcorner$

*Proof:* Cf.[Neukirch P137].  $\square$

**Prop. (19.7.3.15)** [Reduction to Cyclic cases].

$$\text{Br}(F) = \bigcup_{L/F \text{ cyclic}} H^2(\text{Gal}(L/F), L^\times), \quad H^2(\text{Gal}_K, \mathbf{I}_{\bar{K}}) = \bigcup_{L/K \text{ cyclic}} H^2(\text{Gal}(L/K), \mathbf{I}_L).$$

 $\lrcorner$ 

*Proof:* Cf.[Neukirch P127].  $\square$

**Def. (19.7.3.16)** [Invariant Maps]. For  $c = (c_p) \in H^2(\text{Gal}(L/K), \mathbf{I}_L)$ , define the **invariant map**

$$\text{inv}_{L/K} c = \sum_{\mathfrak{p}} \text{inv}_{L_{\mathfrak{p}}/K_{\mathfrak{p}}} c_{\mathfrak{p}} \text{ (19.7.2.8)}.$$

 $\lrcorner$ 

**Prop. (19.7.3.17).** If  $c \in H^2(\text{Gal}(L/F), L^\times)$ , then  $\text{inv}_{L/F} c = 0$ .  $\lrcorner$

*Proof:* Cf.[Neukirch P141].  $\square$

**Cor. (19.7.3.18).** Now we can define the inv map for  $H^2(\text{Gal}(L/K), \mathbf{C}_L)$ . From the exact sequence  $1 \rightarrow L^\times \rightarrow \mathbf{I}_L \rightarrow \mathbf{C}_L \rightarrow 1$ , we get an exact sequence:

$$1 \rightarrow H^2(\text{Gal}(L/K), L^\times) \rightarrow H^2(\text{Gal}(L/K), \mathbf{I}_L) \rightarrow H^2(\text{Gal}(L/K), \mathbf{C}_L) \rightarrow H^3(G_{L/K}, L^\times).$$

The last one is 1 if  $L/F$  is cyclic, thus by this proposition, inv is defined for  $H^2(G_{L/K}, \mathbf{C}_L)$ .  $\lrcorner$

**Prop. (19.7.3.19).** The homomorphism

$$\text{inv}_F : H^2(\text{Gal}(\Omega/F), \mathbf{I}_\Omega) \rightarrow \mathbb{Q}/\mathbb{Z}$$

is surjective.  $\lrcorner$

*Proof:* Cf.[Neukirch]P145.  $\square$

**Prop. (19.7.3.20)** [Reduce to Cyclic Case]. If  $L/K$  is normal and  $L'/K$  is cyclic with the same degree, then  $H^2(L'/K) = H^2(L/K) \subset H^2(\bar{K}/K)$ .  $\lrcorner$

*Proof:*

□

**Cor. (19.7.3.21).**  $H^2(\overline{K}/K) = \cup_{L/K \text{ cyclic}} H^2(L/K)$ , thus the homomorphism  $H^2(\text{Gal}_K, I_{\overline{K}}) \rightarrow H^2(\overline{K}/K)$  is surjective by (19.7.3.18).  $\lrcorner$

*Proof:* Why can always find such a cyclic extension?  $\color{red}?$

□

**Cor. (19.7.3.22).** The inv map is defined for  $H^2(\overline{K}/K)$ , and  $\text{inv}_{L/K} : H^2(L/K) \rightarrow \frac{1}{[L:K]} \mathbb{Z}/\mathbb{Z}$  is an isomorphism for every normal extension  $L/K$ .  $\lrcorner$

**Lemma (19.7.3.23) [Main Lemma].** The formation  $(\text{Gal}_F, \mathbf{C}_{F^{\text{sep}}}, \text{inv}_F)$  is a class formation (19.7.1.1).  $\lrcorner$

*Proof:*

□

**Thm. (19.7.3.24) [Artin's Reciprocity Law, Artin1927].** By (19.7.1.4), cup product with fundamental classes  $u_{L/F} \in H^2(L/F)$  define an isomorphism

$$\theta_{L/F} : \text{Gal}^{\text{ab}}(L/F) \cong H^{-2}(\text{Gal}(L/F), \mathbb{Z}) \rightarrow H^0(L/F) = \mathbf{C}_F / \text{Nm}_{L/F} \mathbf{C}_L,$$

called the **Nakayama isomorphism**. And the reverse map is called the **norm residue symbol** map  $(-, L/F)$

$$1 \rightarrow \text{Nm}_{L/F} \mathbf{C}_L \rightarrow \mathbf{C}_F \xrightarrow{(-, L/F)} \text{Gal}^{\text{ab}}(L/F) \rightarrow 1.$$

These norm residue symbols induce an **Artin map**

$$\text{Art}_F = (-, F) : \mathbf{C}_F \rightarrow \text{Gal}_F^{\text{ab}}$$

□

**Thm. (19.7.3.25) [Local-Global Compatibility].** For  $L/F$  Abelian and  $\mathfrak{a} \in \mathbf{I}_F$ :

$$(\mathfrak{a}, L/F) = \prod_{\mathfrak{p}} (a_{\mathfrak{p}}, L_{\mathfrak{p}}/K_{\mathfrak{p}}) \in \text{Gal}^{\text{ab}}(L/F) \text{ (19.7.2.11)}.$$

In particular, if  $v \in \Sigma_F^{\text{fin}}$  is unramified in  $L$ , then

$$([\varpi_{\mathfrak{p}}]_{\mathfrak{p}}, L/F) = \text{Frob}_{L_{\mathfrak{p}}/F_{\mathfrak{p}}} \in \text{Gal}_{L_{\mathfrak{p}}/F_{\mathfrak{p}}} \subset \text{Gal}_F.$$

□

*Proof:* Cf. [Neukirch CFT P154].

□

**Prop. (19.7.3.26).** There are commutative diagrams:

$$\begin{array}{ccccc} \mathbf{C}_F & \xrightarrow{(-, N/F)} & \text{Gal}(N/F)^{\text{ab}} & & \mathbf{C}_F & \xrightarrow{(-, N/F)} & \text{Gal}(N/F)^{\text{ab}} & & \mathbf{C}_L & \xrightarrow{(-, N/L)} & \text{Gal}(N/L)^{\text{ab}} \\ \downarrow \text{id} & & \downarrow \text{pr} & & \uparrow \text{Nm}_{L/F} \begin{array}{c} \uparrow i \\ \downarrow i \end{array} & & \uparrow \text{Ver} & & \downarrow \sigma & & \downarrow \sigma^* \\ \mathbf{C}_F & \xrightarrow{(-, L/F)} & \text{Gal}(L/F)^{\text{ab}} & & \mathbf{C}_L & \xrightarrow{(-, N/L)} & \text{Gal}(N/L)^{\text{ab}} & & \mathbf{C}_{\sigma L} & \xrightarrow{(-, \sigma N / \sigma L)} & \text{Gal}(\sigma N / \sigma L)^{\text{ab}} \end{array}$$

Where Ver is the transfer map defined in??.

□

*Proof:*

□

**Cor. (19.7.3.27).** By (19.7.1.8), the map  $L \mapsto \mathcal{N}_{L/F} = \text{Nm}_{L/F} \mathbf{C}_L$  defines a inclusion reversing isomorphism between the lattice of Abelian extension  $L/K$  and the lattice of norm groups of  $\mathbf{C}_F$ , i.e.:

$$\mathcal{N}_{L_1 L_2/F} = \mathcal{N}_{L_1/F} \cap \mathcal{N}_{L_2/F}, \quad \mathcal{N}_{L_1 \cap L_2/F} = \mathcal{N}_{L_1/F} \cdot \mathcal{N}_{L_2/F}.$$

And any group that contains a norm group is a norm group.

┘

**Prop. (19.7.3.28) [Existence Theorem].** The norm groups of  $\mathbf{C}_F$  are precisely the (open)closed subgroups of finite index.

┘

*Proof:* Cf. [Neukirch P162] or notes taken by Chao Li. ?

□

**Cor. (19.7.3.29) [The Kernel of  $\text{Art}_F$ ].**

- If  $F \in \mathbf{NField}$ , the kernel  $\cap_{L/F} \mathcal{N}_{L/F}$  of  $\text{Art}_F$  is exactly the connected component  $(\mathbf{C}_F)^0$  of  $1 \in \mathbf{C}_F$ , which is the group of divisible elements in  $\mathbf{C}_F$  (19.4.1.19). Moreover,  $\text{Art}_F : \mathbf{C}_F / (\mathbf{C}_F)^0 \rightarrow \text{Gal}_F^{\text{ab}}$  is an isomorphism.
- If  $F \in \mathbf{FField}$ , then  $\text{Art}_F$  is injective but not surjective. In fact, there is an exact sequence (by etale fundamental group)

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbf{I}_F^1 / F^\times & \longrightarrow & \mathbf{I}_F / F^\times & \longrightarrow & q^\mathbb{Z} \longrightarrow 0 \\ & & \downarrow \cong & & \downarrow (-, F) & & \downarrow \\ 0 & \longrightarrow & \mathbf{I}_F^{\text{ab}} & \longrightarrow & \text{Gal}_F^{\text{ab}} & \longrightarrow & \text{Gal}_k \cong \widehat{\mathbb{Z}} \longrightarrow 0 \end{array}$$

thus  $\mathbf{C}_F$  can be regarded as the Weil group of  $F$ .

┘

*Proof:* 1: As  $\text{Gal}_F^{\text{ab}}$  is totally disconnected, it must factors through  $D_F$ . But  $\mathbf{C}_F / (\mathbf{C}_F)^0$  is a profinite group, thus  $(\mathbf{C}_F)^0$  is an intersection of open subgroups of finite index, thus it is in the kernel, by (19.7.3.28).

2: The map is injective because by splitting it is a product of  $q^\mathbb{Z}$  and a profinite group, thus the intersection of open subgroups of finite index is trivial (19.7.3.28). To show the diagram is commutative, it suffices to show that  $(x, F)$  acts as  $\text{Frob}_k^{|x|_F}$  on  $\bar{k}$ : This is because on any finite unramified Abelian extension  $L/F$ ,  $(x, F)$  acts via  $\varphi_{L_w/F_v}^{|x_v|_v}$  by definition (19.7.2.4), and these add up to  $|x|_v$  as  $L/F$  is cyclic.

□

## 4 Decomposition Law

**Notation (19.7.4.1).**

- Let  $L/F$  be an Abelian extension of Global fields.

┘

**Prop. (19.7.4.2).**  $\mathcal{N}_{E/K} \cap F_{\mathfrak{p}}^\times = \mathcal{N}_{L_{\mathfrak{p}}/F_{\mathfrak{p}}}.$

┘

*Proof:* For the non-trivial part, notice if  $\mathfrak{a} \in N_{\mathfrak{p}} L_{\mathfrak{p}}^*$  is a norm times a  $a \in K^*$ , then it is a norm at all primes except  $\mathfrak{p}$ , thus it is also norm at  $\mathfrak{p}$  by the multiplicative definition of the inv map (19.7.3.16).

□



**Cor. (19.7.4.3) [Ramifications].** Let  $\mathcal{N} = \text{Nm}_{L/K} \mathbf{C}_L$  be the norm group, then

- $\mathfrak{p}$  is unramified in  $L$  iff  $\mathcal{O}_{F,\mathfrak{p}}^* \subset \mathcal{N}_{L/F}$ .
- $\mathfrak{p}$  splits completely in  $L$  iff  $F_{\mathfrak{p}}^\times \subset \mathcal{N}_{L/F}$ .

┘

*Proof:*

□

**Prop. (19.7.4.4) [Decomposition Law].** Let  $n = [L : F]$  and  $\mathfrak{p} \in \text{Unr}(F)$  and  $\varpi_{\mathfrak{p}}$  is a uniformizer, then if  $f$  is the smallest positive integer s.t.  $[\varpi_{\mathfrak{p}}]_{\mathfrak{p}}^f \in \mathcal{N}_{L/F}$ , then  $\mathfrak{p}$  factors in the extension  $L$  into  $r = n/f$  distinct primes of degree  $f$ .

Notice to determine whether a place is ramified or not, we can use criterion (19.7.4.17). ┘

*Proof:* The degree the extension of  $\mathfrak{p}$  is just the order of the Frobenius automorphism of  $G_{\mathfrak{p}/\mathfrak{p}}$ , which is just the order in  $\text{Gal}(L/F) \cong \mathbf{C}_F / \text{Nm}_{L/F} \mathbf{C}_L$ . The Frobenius of  $\mathfrak{p}$  correspond exactly to  $(\dots, 1, \pi, 1, \dots)$  by (19.7.3.25) and (19.7.2.4), so the result follows. □

**Prop. (19.7.4.5) [Unramified Kummer Extensions are Rare].** Let  $F$  be a global field or the function field of a smooth curve over an alg.closed field  $k$ , and  $S \subset \Sigma_F$  be a finite set of places, and  $m \in \mathbb{Z} \cap F^*$ , then there are only f.m. Kummer extensions over  $F$  that have Galois group of exponent  $m$  and are unramified at all finite places outside  $S$ . ┘

*Proof:* We may add all the  $m$ -th roots of unity to  $F$ . Then by Kummer theory (3.2.8.15), Kummer extensions of exponent  $m$  over  $F$  corresponds to  $F(\sqrt[m]{a})$ , where  $a \in F$ . But we can only add those  $a$  s.t.  $\text{ord}_v(a) \equiv 0 \pmod{m}$  in order to get unramified extensions, so the desired Kummer extensions correspond to

$$T_S = \{a \in F^\times / (F^\times)^n \mid \text{ord}_v(a) \equiv 0 \pmod{m}, \forall v \in \Sigma_K^0 \setminus S\}.$$

If  $F \in \mathbf{GField}$ , after enlarging  $S$ , we may assume  $\mathcal{O}_{K,S}$  is PID, then this group is a quotient of  $\mathcal{O}_{K,S}^* / \mathcal{O}_{K,S}^m$ , which is finite by unit theorem (19.4.1.20). If  $F = K(C)$ , where  $C$  is a complete non-singular curve, then by Riemann-Roch there is an exact sequence

$$0 \rightarrow T_\emptyset \rightarrow T_S \rightarrow (\mathbb{Z}/(m))^{\#S} \rightarrow 0$$

And for  $f \in T_\emptyset$ ,  $\text{div}(f) = mD_f$  for some  $D_f \in \text{Pic}(C)[m]$ . Notice  $\#\text{Pic}(C)[m] < \infty$  by (7.13.2.18), thus the theorem follows. □

### Ray Class Fields

**Def. (19.7.4.6) [Notations].**

- A **modulus**  $\mathfrak{m}$  is a formal product  $\prod_{v \in \Sigma_F} \mathfrak{p}_v^{e_v}$  where  $e_v \geq 0$ , and  $e_v = 0$  if  $v \in \Sigma_F^{\mathbb{C}}$ , and  $e_v = 0, 1$  if  $v \in \Sigma_F^{\mathbb{R}}$ .
- For a modulus  $\mathfrak{m}$ , define

$$U^{\mathfrak{m}} = \prod_{v \in \Sigma_F^{\text{fin}}, e_v=0} F_{\mathfrak{p}} \times \prod_{v \in \Sigma_F^{\text{fin}}, e_v>0} (1 + \mathfrak{p}_v^{e_v}) \times \prod_{v \in \Sigma_F^{\infty}, e_v=0} F_v^\times \times \prod_{v \in \Sigma_F^{\mathbb{R}}, e_v=1} \mathbb{R}_+^\times \subset \mathbf{I}_F$$

and  $\mathbf{C}^{\mathfrak{m}}$  is the image of  $U^{\mathfrak{m}}$  in  $\mathbf{C}_F$ , which is an open subgroup of finite index. ┘

**Def. (19.7.4.7) [Ray Class Fields].** For a modulus  $\mathfrak{m}$ ,

- $F_{\mathfrak{m}}$  is defined to be the Abelian extension of  $F$  corresponding to  $\mathbf{C}^{\mathfrak{m}}$  via (19.7.3.28), called the **ray class field** modulo  $\mathfrak{m}$ . In particular,  $\text{Gal}(F_{\mathfrak{m}}/F) \cong \mathbf{C}_F/\mathbf{C}^{\mathfrak{m}}$  via Artin's map (19.7.3.24).
- $J^{\mathfrak{m}}$  is defined to be the multiplicative subgroup generated by ideals  $\mathfrak{a} \in \text{Ideal}^{\times}(\mathcal{O}_F)$  relatively prime to  $\mathfrak{m}$ .
- Define  $P^{\mathfrak{m}} \triangleq F^{\times} \cap U^{\mathfrak{m}}$ , called the **ray mod  $\mathfrak{m}$** .

⌋

**Def. (19.7.4.8) [Ray Class Groups].**

$$\text{Cl}^{\mathfrak{m}}(F) \triangleq J^{\mathfrak{m}}/P^{\mathfrak{m}} \cong \mathbf{C}_F/\mathbf{C}^{\mathfrak{m}} \cong \text{Gal}(F^{\mathfrak{m}}/F) \text{ (19.7.4.7)(19.7.4.6)}$$

is a finite group, called the **ray class group** modulo  $\mathfrak{m}$ .

⌋

*Proof:*  $\mathbf{C}_F/\mathbf{C}^{\mathfrak{m}} \cong F^{\times} \backslash \mathbf{I}_F/U^{\mathfrak{m}}$ , and by weak approximation,  $\mathbf{I}_F = \mathbf{I}^{\mathfrak{m}}U^{\mathfrak{m}}F^{\times}$ , so

$$\mathbf{C}_F/\mathbf{C}^{\mathfrak{m}} \cong F^{\times} \backslash \mathbf{I}^{\mathfrak{m}}U^{\mathfrak{m}}F^{\times}/U^{\mathfrak{m}} = \mathbf{I}^{\mathfrak{m}}/\mathbf{I}^{\mathfrak{m}} \cap (U^{\mathfrak{m}}F^{\times}) \xrightarrow{\cong} J^{\mathfrak{m}}/P^{\mathfrak{m}}.$$

where the final map is given by

$$\mathbf{I}^{\mathfrak{m}} \rightarrow J^{\mathfrak{m}} : (a_v)_{v \in \Sigma_F} \mapsto \prod_{v \in \Sigma_F^{\text{fin}}} \mathfrak{p}_v^{v(a_v)}.$$

$\text{Cl}^{\mathfrak{m}}(F)$  is finite because it is a quotient group of  $\mathbf{I}_F/U^{\mathfrak{m}}$ , which is finite.

□

**Cor. (19.7.4.9) [Cyclotomic Fields].** When  $F = \mathbb{Q}$ ,  $m \in \mathbb{Z}_+$  and  $\mathfrak{m} = m \cdot \infty$ ,  $\mathbf{Q}_{\mathfrak{m}} = \mathbb{Q}(\mu_m)$ .

In particular, we can think of the ray classes fields for general function fields as “generalized cyclotomic fields”.

⌋

*Proof:* For any  $p \in \text{Prime}$ , let  $m = p^{e_p} \times m'$ , then  $\mathbb{Q}(\zeta_m) = \mathbb{Q}(\zeta_{p^{e_p}})\mathbb{Q}(\zeta_{m'})$ . Notice both  $\mathcal{N}_{\mathbb{Q}(\zeta_{p^{e_p}})/\mathbb{Q}}$  and  $\mathcal{N}_{\mathbb{Q}(\zeta_{m'})/\mathbb{Q}}$  contains  $(1 + \mathfrak{p}^{e_p})$  by (19.7.2.34)(19.7.2.30) and (19.7.2.21). Thus by (19.7.3.27), the  $\mathcal{N}_{\mathbb{Q}(\zeta_m)/\mathbb{Q}}$  also contains  $(1 + \mathfrak{p}^{e_p})$ . Thus  $U^{\mathfrak{m}}$  is contained in  $\text{Nm}_{\mathbb{Q}(\zeta_m)/\mathbb{Q}} \mathbf{I}_{\mathbb{Q}(\zeta_m)}$ .

To show the reverse inclusion, we calculate  $(\mathbf{C}_{\mathbb{Q}} : \mathbf{C}^{\mathfrak{m}})$ . As  $(\mathbf{C}_{\mathbb{Q}} : \mathbf{C}_{\mathbb{Q}}^1) = \text{Cl}(\mathbb{Q}) = 1$ ,

$$[\mathbf{C}_{\mathbb{Q}} : \mathbf{C}^{\mathfrak{m}}] = [U_{\mathbb{Q}}^1 \mathbb{Q}^{\times} : U_{\mathbb{Q}}^{m\infty} \mathbb{Q}^{\times}] = [U_{\mathbb{Q}}^1 : U_{\mathbb{Q}}^{m\infty}] [U_{\mathbb{Q}}^1 \cap \mathbb{Q}^{\times} : U_{\mathbb{Q}}^{m\infty} \cap \mathbb{Q}^{\times}] = \varphi(m) = [\mathbb{Q}(\zeta_m) : \mathbb{Q}].$$

□

**Cor. (19.7.4.10) [Kronecker].** Every Abelian extension  $F/\mathbb{Q}$  is a subfield of  $\mathbb{Q}(\zeta_m)$  for some  $m \in \mathbb{Z}_+$ .

⌋

**Def. (19.7.4.11) [Hilbert Class Fields].** The ray class field of  $F$  mod 1 is called the **Hilbert class field** of  $F$ , denoted by  $F_1$  or  $H(F)$ . Its Galois group is isomorphic to  $\mathbf{C}_F/\mathbf{C}_F^1 \cong J_F/P_F = \text{Cl}(F)$  (19.7.4.8). In particular,  $[F_1 : F] = \text{cl}(F)$ .

⌋

## Conductors

**Def. (19.7.4.12) [Admissible Modulus].**

- An **admissible modulus** for  $L/F$  is a modulus  $\mathfrak{m}$  s.t.  $\mathbf{C}^{\mathfrak{m}} \subset \mathcal{N}_{L/F}$  or equivalently  $L \subset F_{\mathfrak{m}}$  (19.7.4.7).

- All subgroups of  $\text{Cl}_{\mathfrak{m}}(F)$  (19.7.4.8) are called **ideal groups defined mod  $\mathfrak{m}$** .
- If  $\mathfrak{m}$  is an admissible modulus for  $L/F$ , then  $H_{L/F}^{\mathfrak{m}} = \text{Nm}_{L/F} J_L^{\mathfrak{m}} \cdot P_F^{\mathfrak{m}}$  is called the **ideal group defined mod  $\mathfrak{m}$** . ┘

**Prop. (19.7.4.13).** A modulus  $\mathfrak{m}$  is admissible for  $L/F$  iff  $U^{\mathfrak{m}} \subset \text{Nm}_{L/F}(\mathbf{I}_L)$ . ┘

*Proof:* This is because  $\mathcal{N}_{L/F} \cap K_p = \mathcal{N}_{L_{\mathfrak{p}}/K_p}$  by (19.7.4.2). □

**Cor. (19.7.4.14).** The norm subgroups of  $\mathbf{C}_F$  are exactly those containing some  $C^{\mathfrak{m}}$ . ┘

*Proof:* This is because  $\mathcal{N}_{L/F} \mathbf{A}_L$  is open in  $\mathbf{A}_K$ , by (19.4.1.14), so it must contain some  $U^{\mathfrak{m}}$ . The converse is also clear. □

**Def. (19.7.4.15) [Conductors].** By (19.7.4.14), there is a minimal admissible modulus for  $L/F$  (19.7.4.12), called the **conductor(global Abelian extension)** of  $L/F$ , denoted by  $\mathfrak{f}_{L/F}$ . ┘

**Prop. (19.7.4.16) [Local and Global Conductors].**

$$\mathfrak{f}_{L/F} = \prod_{v \in \Sigma_F} \mathfrak{f}_{L_w/F_p} \quad (19.7.2.16).$$

┘

*Proof:* This follows from (19.4.1.14). □

**Cor. (19.7.4.17) [Conductor Detects Ramifications].**  $v \in \Sigma_F$  is ramified in  $L/F$  iff  $v | \mathfrak{f}_{L/F}$ , by (19.7.2.16). In particular,  $\text{Ram}(F_{\mathfrak{m}}/F) = S(\mathfrak{m})$ . ┘

**Prop. (19.7.4.18).** If  $S$  is a finite set of primes that contains all the infinite primes and all the primes lying above the primes dividing  $n$ , and  $\mathbf{I}_F = \mathbf{I}_F^S \cdot F^{\times}$ , then  $\mathbf{C}_F^n \cdot U_F^S$  is a norm group. If  $\mu_n \subset F$ , then it corresponds to the Kummer extension  $T = F(\sqrt[n]{F^S}/F)$ . ┘

### Hilbert Class Fields

**Prop. (19.7.4.19).** The Hilbert class field  $F_1$  (19.7.4.11) is the maximal unramified Abelian extension of  $F$ , by (19.7.4.17). ┘

**Def. (19.7.4.20) [Hilbert Class Field Tower].** Let  $F_1$  be the Hilbert class field of  $F$ , and for any  $n \in \mathbb{Z}_+$ , let  $F_{n+1}$  be the Hilbert class field of  $F_n$ , then such a series of extensions is called the **Hilbert class field tower** of  $F$ .

Similarly for  $p \in \text{Prime}$ , we can consider the maximal Abelian  $p$ -extensions of  $F$ , which gives us a series

$$F \subset F_1^{(p)} \subset F_2(p) \subset \dots \subset F_{\infty}^{(p)}.$$

┘

**Prop. (19.7.4.21).** Each  $F_n$  is Galois over  $F$ , and  $F_1$  is the largest Abelian subfield of  $F_2/F$ . ┘

*Proof:* For the last assertion, notice that that largest Abelian subfield of  $F_2/F$  is unramified Abelian over  $F$ , then use (19.7.4.19). □

**Prop. (19.7.4.22) [Principal Ideal Theorem].** In the Hilbert class field over  $F$ , every ideal  $\mathfrak{a}$  of  $F$  becomes a principal ideal. ┘

*Proof:* As  $J_F/P_F \cong \mathbf{C}_F/\mathbf{C}_F^1$ , It suffices to show that the map  $i : \mathbf{C}_F \rightarrow \mathbf{C}_{F_1}$  has image in  $\mathbf{C}_{F_1}^1$ . By the commutative diagram(19.7.3.26)

$$\begin{array}{ccccccc} & & \mathbf{C}_F & \xrightarrow{(-, F_2/F)} & \mathrm{Gal}(F_2/F)^{\mathrm{ab}} & & \\ & & \downarrow i & & \downarrow \mathrm{Ver} & & \\ 0 & \longrightarrow & \mathbf{C}_{F_1}^1 & \longrightarrow & \mathbf{C}_{F_1} & \xrightarrow{(-, F_2/F_1)} & \mathrm{Gal}(F_2/F_1)^{\mathrm{ab}} \longrightarrow 0 \end{array}$$

It suffices to show that  $\mathrm{Ver}$  is trivial. Then this follows from(19.7.4.21) and(3.1.9.3).  $\square$

**Thm. (19.7.4.23) [Brumer].** Let  $F \in \mathbf{NField}$ ,  $p \in \mathbf{Prime}$ , and  $t_F^{(p)}$  be the number of primes  $\ell \in \mathbf{Prime}$  s.t.  $p|e_{\mathfrak{L}/\ell}$  for any prime  $\mathfrak{L}$  above  $\ell$ , then

$$\dim_{\mathbb{F}_p} \mathrm{Cl}(F)/(p) \geq t_F^{(p)} - 2(d-1)$$

┘

*Proof:*

$\square$

**Prop. (19.7.4.24) [Golod-Šafarevič].** Let  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = d$  and  $p \in \mathbf{Prime}$ , if  $[F_{\infty}^{(p)} : F] < \infty$ , then

$$\dim_{\mathbb{F}_p} \mathrm{Cl}(F)/(p) \leq 1 + 2\sqrt{d+1}.$$

┘

*Proof:*

$\square$

**Cor. (19.7.4.25).** By(19.7.4.23), if  $t_F^{(p)} \geq 2d + 2\sqrt{d+1}$ , then  $[F_{\infty}^{(p)} : F] = \infty$ .

For example, if we take  $F = \mathbb{Q}(\sqrt{d})$  where  $d$  is square-free with at least 8 different prime factors, then  $[F_{\infty}^{(p)} : F] = \infty$ .  $\square$

### Classical Formulation

**Def. (19.7.4.26) [Artin Symbols].** There is a homomorphism  $J^{\mathfrak{m}} \rightarrow \mathrm{Gal}(L/K)$  called the **Artin symbol**  $\left(\frac{L/K}{-}\right)$ . On primes  $\mathfrak{p}$ , it maps a prime  $\mathfrak{p}$  which is unramified by(19.7.4.12) to its local Frobenius  $\mathrm{Frob}_{\mathfrak{p}} \in \mathrm{Gal}(\mathfrak{P}/\mathfrak{p}) \subset \mathrm{Gal}(L/K)$ . This is well-defined only up to conjugacy in  $\mathrm{Gal}(L/K)$ , and thus well-defined when  $L/K$  is Abelian.  $\square$

**Lemma (19.7.4.27).** If  $\mathfrak{m}$  is an admissible modulus for  $L/F$ , the restriction to finite part defines isomorphism  $\mathcal{N}_{L/F}/\mathbf{C}_F^{\mathfrak{m}} \cong H_{L/F}^{\mathfrak{m}}/P^{\mathfrak{m}}$ (19.7.4.12).  $\square$

*Proof:* Cf.[Neukirch CFT P176].  $\square$

**Prop. (19.7.4.28) [Classical Artin Reciprocity Law].** If  $\mathfrak{m}$  is an admissible modulus, then there is a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathcal{N}_{L/F} & \longrightarrow & \mathbf{C}_F & \xrightarrow{(-, L/F)} & G_{L/F} \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \parallel \\ 1 & \longrightarrow & H_{L/F}^{\mathfrak{m}}/P^{\mathfrak{m}} & \longrightarrow & J^{\mathfrak{m}}/P^{\mathfrak{m}} & \xrightarrow{\left(\frac{L/K}{-}\right)} & \mathrm{Gal}(L/F) \longrightarrow 1 \end{array}$$

Thus the second row is exact by(19.7.3.24), and  $\mathrm{Gal}(L/F) \cong J^{\mathfrak{m}}/H_{L/F}^{\mathfrak{m}}$ .  $\square$

*Proof:* Cf.[Neukirch CFT, P178]. □

**Prop. (19.7.4.29)[Decomposition Law].** Let  $n = [L : F]$  and  $\mathfrak{m}$  be an admissible modulus for  $L/F$  (e.g. the conductor) and  $\mathfrak{p} \nmid \mathfrak{m}$ . If  $f$  is the smallest number that  $\mathfrak{p}^f \in H_{L/F}^{\mathfrak{m}}$ , then  $\mathfrak{p}$  factors in the extension  $L$  into  $r = n/f$  distinct primes of degree  $f$ . ┘

*Proof:* The degree the extension of  $\mathfrak{p}$  is just the order of the Frobenius automorphism of  $G_{\mathfrak{p}/\mathfrak{p}}$ , which is just the order in  $\text{Gal}(L/F) \cong J^{\mathfrak{m}}/H_{L/F}^{\mathfrak{m}}$ . The Frobenius of  $\mathfrak{p}$  correspond exactly to  $\mathfrak{p}$  by (19.7.2.4), so the result follows. □

### $n$ -th Powers

Cf.[?A-T67]Chap9,10.

**Prop. (19.7.4.30) [Grünwald-Wang].** Let  $n = 2^t \cdot m \in \mathbb{Z}_+$ , and assume  $F \in \mathbf{FField}$  or  $F(\zeta_{2t})/F$  is cyclic. Then  $a \in F$  is in  $F^n$  iff  $a \in F_v^n$  for a.e.  $v \in \Sigma_F$ . ┘

*Proof:* Cf.[Chao Li's notes] or [Mil20]P229. □

**Thm. (19.7.4.31) [Grünwald-Wang].** For any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ , if for each  $v \in S$ ,  $\chi'_v$  is a continuous character of  $F_v^\times$  of finite order  $n_v$ , then there exists a Hecke character  $\chi$  of  $F$  with the  $\chi_v = \chi'_v$ . Moreover, if  $n = \text{lcm}(n_v)$ ,  $t = \text{ord}_2(n)$ , and assume  $F \in \mathbf{FField}$  or  $F(\zeta_{2t})/F$  is cyclic, then  $\chi$  can be taken to be of order  $n$ . ┘

*Proof:* Cf.[Chao Li's notes] or [Artin-Tate, Class Field Theory, Thm10.5]. ? □

**Cor. (19.7.4.32)[Local-Global Compatibility for Cyclic Extensions].** For each  $v \in \Sigma_F$ , a positive integer  $n_v$  is given s.t.

- For a.e.  $v$ ,  $n_v = 1$ .
  - For  $v \in \Sigma_F^{\mathbb{R}}$ ,  $n_v \in \{1, 2\}$ .
  - For  $v \in \Sigma_F^{\mathbb{C}}$ ,  $n_v = 1$ .
- ┘

*Proof:* The point is that we can choose  $\chi_v$  s.t. the exceptions in (19.7.4.31) can be avoided, and then get a Hecke character  $\chi$  of  $F$  of order  $n$  which corresponds to a cyclic extension via cyclic class formation. ? Cf.[Artin-Tate, Class Field Theory, Thm10.5, P105]. ? □

### Higher Reciprocity Law

References are [Mil20]Chap8 and [Cox13]P149. Contributors include: Kronecker/Weber/Hilbert/-Furtwangler/Takagi.

**Def. (19.7.4.33)[Higher Hilbert Symbols].** ┘

**Prop. (19.7.4.34)[Quadratic Reciprocity].** For any  $p \in \text{Prime}$  and  $n \in \mathbb{Z}$ , we have

$$\left( \frac{\left( \frac{-1}{p} \right) p}{n} \right) = \left( \frac{n}{p} \right) \text{ (2.6.1.27).}$$
┘

*Proof:* By multiplicity, it suffices to prove for  $n \in \{0, -1\} \cup \text{Prime}(2.6.1.27)$ . For  $n \in \{0, -1, 2\}$  or  $p = 2$  this is trivial. Assume  $p \geq 3$  and  $n = q \in \text{Prime}$ , then this follows from the Artin reciprocity for  $\mathbb{Q}(\sqrt[p]{\left(\frac{-1}{p}\right)})$  or from (35.3.5.4) and (35.3.5.2) by calculating  $G_{pq}(1; 2)$  twice.  $\square$

**Prop. (19.7.4.35)[Cubic Reciprocity].**  $\lrcorner$

**Prop. (19.7.4.36)[Quadric Reciprocity].**  $\lrcorner$

### Ring Class Fields for Imaginary Quadratic Fields

References are [Cox13]Chap9.

**Def. (19.7.4.37) [Ring Class Fields].** If  $\mathcal{K}$  is an imaginary quadratic field,  $f \in \mathbb{Z}_+$ , the **ring class field** of  $\mathcal{K}$  corresponding to  $\mathcal{O}_{\mathcal{K},f}$  (19.3.1.53) is an Abelian extension  $L/K$  corresponding to  $I_K(f)/P_{\mathcal{K},\mathbb{Z}}(f) \cong \text{Cl}(\mathcal{O}_{\mathcal{K},f})$  under the Artin map (19.3.1.57)(19.7.4.7).  $I_K(f)/P_{\mathcal{K},\mathbb{Z}}(f) \cong \text{Cl}(\mathcal{O}_{\mathcal{K},f})$  is a quotient group of  $\text{Cl}^f(\mathcal{K})$ , so  $L \subset \mathcal{K}_{\mathfrak{m}_f}$ , the ray class group modulo  $f$  (19.7.4.7).  $\lrcorner$

**Thm. (19.7.4.38)[Generalized Dihedral Extensions].** If  $\mathcal{K}$  is an imaginary quadratic field, then for any order  $\mathcal{O} \subset \mathcal{O}_{\mathcal{K}}$ , the ring class field  $L/K$  is Galois over  $\mathbb{Q}$ , and

$$\text{Gal}(L/\mathbb{Q}) \cong \text{Gal}(L/\mathcal{K}) \rtimes \text{Gal}(\mathcal{K}/\mathbb{Q}),$$

where  $\text{Gal}(\mathcal{K}/\mathbb{Q}) = \{1, c\}$  and  $c$  acts on  $\text{Gal}(L/\mathcal{K})$  via  $\sigma^c = \sigma^{-1}$ .

Such an  $L$  is said to be **generalized dihedral over  $\mathbb{Q}$** . Then conversely, any Abelian extension  $L/K$  s.t.  $L$  is generalized dihedral over  $\mathbb{Q}$  is contained in a ring class field of  $\mathcal{K}$ .  $\lrcorner$

*Proof:* Cf. [Cox13]P163 and P172. ?  $\square$

## 5 $L$ -Functions

**Notation (19.7.5.1).**

- Let  $L/F$  be an Abelian extension of Global fields.
- For two functions  $f(s), g(s) : (1, \infty) \rightarrow \mathbb{C}$ , write  $f \stackrel{1}{\sim} g$  iff  $f - g$  is bounded on  $(1, 1 + \varepsilon)$  for some  $\varepsilon > 0$ .

$\lrcorner$

**Thm. (19.7.5.2)[Effective Chebotarev].** For any subset  $C \subset \text{Gal}(L/F)$  that is stable under conjugation, for  $X > 0$ , let

$$\pi_C(X) = \{v \in \Sigma_F | e_v = 1, (\mathfrak{p}_v, L/F) \in C, \|v\| \leq X\},$$

then

$$\#\pi_C(X) = \frac{\#C}{\#\text{Gal}(L/F)} \frac{X}{\log X} + O\left(\frac{X^{\beta_0}}{\log X}\right).$$

$\lrcorner$

*Proof:* Cf. [Lagarias and Odlysko, Effective Chebotarev].  $\square$

**Cor. (19.7.5.3)[Chebotarev].** Let  $L/F \in \mathbf{NField}$  be a Galois extension, then for any subset  $C \subset \text{Gal}(L/F)$  that is stable under conjugation, let

$$T = \{\mathfrak{p} \in \text{Unr}(F) | (\mathfrak{p}, L/F) \in C\},$$

then  $T$  has Dirichlet density  $\delta(T) = \#C/\#\text{Gal}(L/F)$ .  $\lrcorner$

*Proof:* This follows from (19.7.5.2) and the prime ideal theorem (35.5.9.2).  $\square$

**Prop. (19.7.5.4) [Polar Chebotarev Density Theorem].** ?  $\lrcorner$

**Prop. (19.7.5.5) [Chebotarev for Function Fields].** Cf. [Sta].  $\lrcorner$

*Proof:*  $\square$

**Prop. (19.7.5.6) [Every Ideal Class Contains a Prime].** For  $F \in \mathbf{NField}$ , each ideal class of  $F$  contains a prime ideal. In fact, the primes contained in any fixed ideal class has natural density  $1/\text{cl}(F)$ .  $\lrcorner$

*Proof:* Consider the Hilbert class field  $H(F)$  of  $F$  (19.7.4.11), then  $\text{Cl}(F) \cong \text{Gal}(H(F)/F)$  via the norm residue map. Then the assertion follows from (19.7.5.2).  $\square$

### Splitting of Primes

**Def. (19.7.5.7) [Splitting Sets].** Let  $\text{Spl}(L/F) \subset \Sigma_F^{\text{fin}}$  be the set of finite places of  $F$  that splits completely in  $L$ . By (5.2.7.28), If  $N/F$  is the Galois closure of  $L/F$ , then  $\text{Spl}(N/F) = \text{Spl}(L/F)$ .  $\lrcorner$

**Prop. (19.7.5.8) [Splitting in Cyclotomic Fields].** Let  $L/\mathbb{Q}$  be Galois, then  $L \in \mathbb{Q}(\zeta_m)$  for some  $m \in \mathbb{Z}_+$  by (19.7.4.10), and let  $\Lambda = \mathcal{N}_{L/\mathbb{Q}}/\mathbb{C}^m \subset (\mathbb{Z}/(m))^*$  corresponds to  $L$ , then  $p \in \mathbf{Prime}$  splits in  $L$  iff  $p(\bmod m) \in \Lambda$ .  $\lrcorner$

**Cor. (19.7.5.9) [Dirichlet's Problem].** By (19.7.5.2) and (19.7.5.8), for any  $m \in \mathbb{Z}_+$  and  $[a] \in (\mathbb{Z}/(m))^\times$ , there exists i.m.  $p \in \mathbf{Prime}$  s.t.  $p \equiv a(\bmod m)$ .  $\lrcorner$

**Prop. (19.7.5.10) [Frobenius].**  $\text{Spl}(L/K)$  has Dirichlet density  $1/[L : K]$ .  $\lrcorner$

*Proof:* For  $v \in \Sigma_F^{\text{fin}}$  unramified, it splits in  $L$  iff  $(\mathfrak{p}_v, L/F) = \text{id}$ , by (19.7.3.25). So we can use Chebotarev theorem (19.7.5.3) for  $C = \{\text{id}\}$ .  $\square$

**Cor. (19.7.5.11).** If  $L/F, L/F'$  satisfies  $\text{Spl}(L/F) = \text{Spl}(L'/F)$ , then  $L = L'$ .  $\lrcorner$

*Proof:* The hypothesis implies that  $\text{Spl}(L/F) = \text{Spl}(LL'/F) = \text{Spl}(L'/F)$ , which then implies  $L = LL' = L'$ , by (19.7.5.10).  $\square$

**Cor. (19.7.5.12).** If  $L/K$  is a finite separable extension of global fields s.t.  $\text{Spl}(L/K)$  has Dirichlet density  $[L : K]$ , then  $L/K$  is Galois, by (19.7.5.7).  $\lrcorner$

**Cor. (19.7.5.13).** If  $L/K$  is a finite separable extension of global fields s.t.  $\text{Spl}(L/K)$  has Dirichlet density 1, then  $L = K$ , by (19.7.5.7).  $\lrcorner$

## 6 Explicit Construction of Class Fields

The explicit class field theory is the subject that tries to write the maximal Abelian extension of a field  $K$  as splitting field of polynomials.

**Prop. (19.7.6.1) [Known Cases].**

- $\mathbb{Q}^{\text{ab}} = \mathbb{Q}(\zeta_\infty)$  (19.7.4.10).
- If  $F \in \mathbf{FField}$ , the theory of Drinfeld modules gives most Abelian extensions of  $F$  ?.

- If  $K \in p\text{-NField}$ ,  $K^{\text{ab}}$  is given by adjoining all torsion points of Lubin-Tate formal groups over  $K$  (19.7.2.32).
- If  $F \in \text{NField}$  is a CM field, then by Kronecker's Jugendtraum (21.2.7.9), the Abelian extensions of  $F$  are given by  $j$ -invariants and torsion points of elliptic curves over  $F$  with CM.
- If  $F \in \text{NField}$  is totally real, the conjecture of Stark gives all the Abelian extensions. ?

┘

## 7 Explicit Reciprocity Law

Cf. [Iwasawa] [Kummer], [Chao Li] and [Kato].

## 8 Local Field cases

Poitou-Tate Duality

Artin-Verdier Duality

## 9 Global Field cases

Poitou-Tate Duality

Artin-Verdier Duality



## 19.8 Inverse Galois Problem

References are [\[Ser97\]](#).

**Prop. (19.8.0.1) [Specialization Methods].** Let  $K \in \mathbf{Field}$  be Hilbertian and  $G \in \mathbf{Grp}^{\text{fin}}$ . If there exists  $n \in \mathbb{N}$  and a Galois extension  $F/K(T_1, \dots, T_n)$  with Galois group  $G$ , then there is a Galois extension of  $K$  with Galois group  $G$ . Moreover, if  $K$  is alg.closed in  $F$ , then there exists infinitely many Galois extension of  $K$  with Galois group  $G$  that are linearly disjoint.  $\lrcorner$

*Proof:* Cf. [\[Ser97\]](#)P137. ?  $\square$

**Remark (19.8.0.2) [Noether's method].** If  $G \in \mathbf{Grp}^{\text{fin}}$  acts on a finite set  $E \in \mathbf{finSet}$ , then we can consider the action of  $G$  on  $\mathbb{Q}(\{T_{x_i}\}_{x_i \in E})$  via the action on the subscript. Then if the fixed field  $F$  is a purely transcendental extension of  $\mathbb{Q}$ , then we can use [\(19.8.0.1\)](#) to get a Galois extension of  $\mathbb{Q}$  with Galois group  $G$ .  $\lrcorner$

**Thm. (19.8.0.3) [Large Sieve Bounds on Galois Groups of Polynomials, van.der.Waerden(1936)/Gallagher[\[Gal73\]](#).** Let  $n \in \mathbb{Z}_+$ . For each  $N \in \mathbb{Z}_+$ , denote by  $D_N$  the set of monic polynomials of degree  $n$  with integer coefficients whose absolute values bounded by  $N$ . Then

- For each sequence  $r = (r_1, \dots, r_k)$  with  $r_1 + 2r_2 + \dots + kr_k = n$ , if we denote  $C_r \subset D_N$  the subset consisting of polynomials  $f$  s.t.  $f$  doesn't factor as a product of  $r_1$ -many linear factors,  $r_2$ -many irreducible quadratic factors,  $r_3$ -many cubic irreducible factors and so on. Then For each  $\varepsilon \in \mathbb{R}_+$ , there exists  $c_{r,\varepsilon} \in \mathbb{R}_+$  s.t. for each  $N \in \mathbb{Z}_+$  we have

$$\#C_r \leq c_{r,\varepsilon} N^{n-1/2} \log^{1-\delta_r+\varepsilon} N, \text{ where } \delta_r = \frac{1}{1^{r_1}(r_1)!2^{r_2}(r_2)! \dots k^{r_k}(r_k)!}.$$

- For each  $\varepsilon \in \mathbb{R}_+$ , there exists  $c_{n,\varepsilon} \in \mathbb{R}_+$  s.t. for each  $N \in \mathbb{Z}_+$  we have

$$\#\{f \in D_N : \text{the Galois-group of } f \text{ is not } \text{Sym}_n\} \leq c_{n,\varepsilon} N^{n-1/2} \log^{1-\frac{1}{n!}+\varepsilon} N.$$

$\lrcorner$

*Proof:* 1: We use the large sieve method[\(35.1.6.14\)](#). Notice that

$$\#C_r = \prod_{f \in \mathbb{Z}_+} \binom{\pi_p(f)}{r_f},$$

where  $\pi_p(f)$  is the number of degree- $f$  irreducible polynomials in  $\mathbb{F}_p[X]$ . By Möbius-inversion, we have

$$\pi_p(f) = \frac{1}{f} \sum_{d|f} \mu(d) p^{f/d},$$

which is a polynomial in  $p$  with leading term  $p^f/f$ . So the number  $\omega_p(r)$  of polynomials in  $\mathbb{F}_p[X]$  of type- $r$  is a polynomial in  $p$  with leading term

$$\prod_{f \in \mathbb{Z}_+} \frac{1}{r_f!} \left( \frac{p^f}{f} \right)^{r_f} = \delta_r p^n,$$

So there exists  $M_\varepsilon \in \mathbb{Z}_+$  s.t.  $\omega_p(r) \geq (\delta_r - \varepsilon) p^n$  for each  $p \in \mathbf{Prime}_{>M_\varepsilon}$ .

Then with notation as in (35.1.6.14), we have

$$L(Q) \geq L''(Q) \triangleq \sum_{n \in [Q]_+^b, (n, \prod_{p \in \text{Prime} \leq M_\varepsilon} p) = 1} \left( \frac{\delta_r}{1 - \delta_r} \right)^{\#S(n)}.$$

Then It follows from Tauberian theorem (35.1.1.5) as in the proof of (35.9.3.5) that

$$L''(Q) \sim \frac{1}{\Gamma\left(\frac{\delta_r}{1-\delta_r}\right)} \prod_{p \in [M_\varepsilon]_+} \left(1 - \frac{1}{p}\right)^{\delta_r/(1-\delta_r)} \prod_{p \in \text{Prime} > M_\varepsilon} \left(1 + \frac{\delta_r/(1-\delta_r)}{p}\right) \left(1 - \frac{1}{p}\right)^{\delta_r/(1-\delta_r)}$$

Taking  $Q = \sqrt{N}$ , the assertion follows from (35.1.6.14).

2: If the Galois-group of  $f$  is not  $\text{Sym}_n$ , then there exists a conjugacy class of  $\text{Sym}_n$  that is disjoint from the Galois-group of  $f$ , by (3.1.2.12). Thus there exists some  $r = (r_1, \dots, r_k)$  with  $r_1 + 2r_2 + \dots + kr_k = n$  s.t.  $f \in C_r$ . Thus the assertion follows from item1 and the fact  $\delta_r \geq 1/n!$ .  $\square$

## 1 $p$ -adic Case

**Prop. (19.8.1.1) [Abhyankar's conjecture, Raynaud].** A finite group  $\Gamma \in \mathcal{Ab}^{\text{fin}}$  is the Galois group of an unramified Galois covering of  $\mathbb{A}_{\mathbb{F}_p}^1$  iff it is generated by its  $p$ -Sylow subgroups.  $\lrcorner$

*Proof:*

$\square$

## 2 Solvable Groups

**Thm. (19.8.2.1) [Šafarevič].** For  $F \in \mathbf{GField}$  and  $G \in \mathcal{Ab}^{\text{fin}}$  is solvable, then there exists a Galois extension  $L/F$  s.t.  $\text{Gal}(L/F) \cong G$ .  $\lrcorner$

*Proof:* Cf. [Cohomology of Number Fields, Neukirch].

$\square$

## 19.9 Transcendental Number Theory

References are [Baker, Transcendental Number Theory].

### 1 Transcendental Numbers

**Def. (19.9.1.1) [Logarithm of Algebraic Numbers].** Define the set of **Logarithms of Algebraic numbers**

$$\mathbb{L} = \{\lambda \in \mathbb{C} | e^\lambda \in \overline{\mathbb{Q}}\}.$$

┘

**Conj. (19.9.1.2) [Schanuel].** Suppose  $z_1, \dots, z_n \in \mathbb{C}$  are linearly independent over  $\mathbb{Q}$ , then

$$\text{tr.deg}(\mathbb{Q}(z_1, \dots, z_n, e^{z_1}, \dots, e^{z_n})/\mathbb{Q}) \geq n.$$

┘

**Conj. Cor. (19.9.1.3).** If  $\lambda_1, \dots, \lambda_n \in \mathbb{L}$  are linearly independent over  $\mathbb{Q}$ , then they are alg.independent over  $\mathbb{Q}$ .

┘

**Thm. (19.9.1.4) [Hermite-Lindemann-Weierstrass].**

- If  $\alpha_1, \dots, \alpha_n \in \overline{\mathbb{Q}}$  are pairwise distinct, then  $e^{\alpha_1}, \dots, e^{\alpha_n}$  are linearly independent over  $\overline{\mathbb{Q}}$ .
- If  $\alpha_1, \dots, \alpha_n \in \overline{\mathbb{Q}}$  are linearly independent over  $\mathbb{Q}$ , then  $e^{\alpha_1}, \dots, e^{\alpha_n}$  are alg.independent over  $\mathbb{Q}$ . In particular, if  $\alpha \in \overline{\mathbb{Q}}$ , then  $e^\alpha \notin \overline{\mathbb{Q}}$ .

┘

*Proof:* Cf.[Baker, Thm1.4].

□

**Cor. (19.9.1.5).**  $\pi$  and  $e$  are not in  $\overline{\mathbb{Q}}$ .

┘

*Proof:*  $e^{2\pi i} = 1$ , and  $e^1 = e$ .

□

**Conj. (19.9.1.6) [p-adic Lindemann-Weierstrass Conjecture].** Suppose  $p \in \mathbf{P}$  and  $\alpha_1, \dots, \alpha_n \in \overline{\mathbb{Q}} \cap \mathbb{Q}_p$  s.t.  $|\alpha_i|_p < 1/p$  for any  $i$ . Then  $\exp_p(\alpha_1), \dots, \exp_p(\alpha_n) \in \mathbb{C}_p$  are alg.independent over  $\mathbb{Q}$ .

┘

*Proof:*

□

**Conj. (19.9.1.7) [Modular Conjecture, Bertrand].** Denote  $j(\tau) = J(e^{2\pi i \tau})$ . Suppose  $\tau_1, \dots, \tau_n \in \mathcal{H}$  s.t.

$$\{J(e^{2\pi i \tau_1}), J'(e^{2\pi i \tau_1}), J''(e^{2\pi i \tau_1}), \dots, J(e^{2\pi i \tau_n}), J'(e^{2\pi i \tau_n}), J''(e^{2\pi i \tau_n})\}$$

are alg.independent over  $\mathbb{Q}$ , then there exists two  $\tau_i, \tau_j$  that are linearly independent.

┘

*Proof:*

□

**Thm. (19.9.1.8) [Gelfond-Schneider].** If  $\alpha, \beta \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$  with  $\alpha \neq 0, 1$ , then  $\alpha^\beta \notin \overline{\mathbb{Q}}$ .

┘

*Proof:* Cf.[Sil16]P286.

□

**Cor. (19.9.1.9).**  $(\sqrt{2})^{\sqrt{2}} \notin \overline{\mathbb{Q}}$ .

┘

### Logarithmic Forms

**Thm. (19.9.1.10) [Baker-Wüstholz].** Suppose  $n \in \mathbb{Z}_+$ ,  $\alpha_1, \dots, \alpha_n \in \overline{\mathbb{Q}} \setminus \{0, 1\}$ ,  $K = \mathbb{Q}(\alpha_1, \dots, \alpha_n)$ , and  $d = [K : \mathbb{Q}]$ . Let  $b_1, \dots, b_n \in \mathbb{Z}$ , and  $L(z_1, \dots, z_n) = \sum b_i z_i$  be the linear form. If  $\log(\alpha_i)$  is any logarithmic root of  $\alpha_i$  for any  $i$ , and we define the modified heights

$$h'(\alpha_i) = \max(h^+(\alpha_i), \frac{1}{d}|\log \alpha_i|, \frac{1}{d}), \quad h'(L) = \max(h(L), \frac{1}{d}).$$

If  $L(\log(\alpha_1), \dots, \log(\alpha_n)) \neq 0$ , then we have

$$\log |L(\log(\alpha_1), \dots, \log(\alpha_n))| > -18(n+1)!n^{n+1}(32d)^{n+2} \log(2nd)h'(\alpha_1) \dots h'(\alpha_n)h'(L).$$

┘

*Proof:* Cf. [Logarithmic Forms and Group Varieties].

□

**Cor. (19.9.1.11).** Situation as above, if each  $\log(\alpha_i)$  is a principle value, and  $B = \max(|b_i|, e)$ ,  $A = \max(H(\alpha_i), e)$ , then

$$\log |L(\log(\alpha_1), \dots, \log(\alpha_n))| > -(16nd)^{2(n+2)}(\log A)^n \log B.$$

┘

*Proof:*

□

**Thm. (19.9.1.12) [Six Exponential Theorem].** If  $x_1, x_2, x_3$  are three complex numbers that are linearly independent over the rational numbers, and  $y_1, y_2$  are two complex numbers that are also linearly independent over the rational numbers, then at least one of the following numbers is transcendental:

$$e^{x_1 y_1}, e^{x_2 y_1}, e^{x_3 y_1}, e^{x_1 y_2}, e^{x_2 y_2}, e^{x_3 y_2}.$$

┘

*Proof:* ?

□

## 2 Periods

References are [Periods, Zagier].

**Def. (19.9.2.1) [Algebraic Functions].** Let  $F$  be a field contained in  $\mathbb{C}$ , then an **algebraic function** of degree  $d \in \mathbb{Z}_+$  in  $n$  variable over  $F$  is a function  $f(\underline{X}) \in C(\mathbb{R}^n)$  that is continuous in its domain and satisfies an equation  $P(\underline{X}, f(\underline{X})) = 0$  where  $P \in F[\underline{X}, T]$  that is of degree  $d$  in  $T$ .

┘

**Def. (19.9.2.2) [Periods].** A **period number** is a complex number that is an integral combinations of real numbers of the form  $\int_U f(x)dx$  where  $f$  is an algebraic function over  $\mathbb{Q}$  (19.9.2.1) and  $U$  is a precompact connected open domain of  $\mathbb{R}^n$  defined by polynomial inequalities with rational coefficients, for some  $n \in \mathbb{N}$ . The set of periods is denoted by  $\mathbb{P}$ . The **extended Period ring** is defined to be  $\widehat{\mathbb{P}} = \mathbb{P}[\frac{1}{2\pi i}]$ .

Clearly  $\overline{\mathbb{Q}} \subset \mathbb{P} \subset \mathbb{C}$ , and they form an algebra. Also,  $\#\mathbb{P} = \aleph_1$ .

┘

**Prop. (19.9.2.3).** In defining periods, we can even consider domains defined by inequalities by algebraic functions over  $\mathbb{Q}$  (19.9.2.1) defined on a larger open subset.

┘

*Proof:* If  $U$  satisfies  $f < a$  and  $f$  satisfies a polynomial function  $P(\underline{X}, f(\underline{X})) = 0$ , then  $U$  satisfies  $P(\underline{X}, a) > 0$  or  $P(\underline{X}, a) < 0$ . We can assume the first one holds, and we can isolate the part  $P(\underline{X}, a) > 0, f(\underline{X}) < a$  and the part  $P(\underline{X}, a) > 0, f(\underline{X}) > a$  by segments, so we are reduced to the polynomial case.  $\square$

**Prop. (19.9.2.4) [Algebraic Varieties].** If  $X$  is a smooth variety of dimension  $d$  over  $\mathbb{Q}$  and  $D \subset X$  a divisor with normal crossing,  $\omega \in \Omega^d(X)$  vanishing on  $D$ , and  $\gamma \in H_d(X(\mathbb{C}), D(\mathbb{C}), \mathbb{Q})$  is a singular  $n$ -chain with boundary in  $Y(\mathbb{C})$ , then the integral  $\int_C \omega \in \mathbb{P}$ .

In fact, any integration on a variety over  $\mathbb{Q}$  can be reduced to this case, by resolution of singularity and restriction of divisors, and any period number comes from these.  $\textcolor{red}{?}$   $\lrcorner$

*Proof:* Roughly because we can represent  $\gamma$  by a semi-algebraic chain.  $\textcolor{red}{?}$   $\square$

**Example (19.9.2.5).**

- $\sqrt{2} = \int_{2x^2 < 1} dx \in \mathbb{P}$ .
- $\pi = \int_{x^2 + y^2 < 1} dx dy \in \mathbb{P}$ .
- $\log 2 = \int_1^2 \frac{dx}{x} \in \mathbb{P}$ .

$\lrcorner$

**Prop. (19.9.2.6).** For  $n \in \mathbb{Z}_+, n \geq 2$ ,  $\zeta(n) \in \mathbb{P}$ .  $\lrcorner$

**Conj. (19.9.2.7) [Non-Examples].** There are no natural examples of numbers proven to be non-period.  $e$ ,  $1/\pi$  and Euler-Mascheroni constant  $\gamma$  are conjectured to be non-periods. Notice  $e$  is known to be transcendental but  $\gamma$  is not known to be rational or not yet!  $\lrcorner$

**Remark (19.9.2.8) [How to Distinguish Different Periods].** There may be numbers that can be written in different ways, such as

$$\sqrt{11 + 2\sqrt{29}} + \sqrt{16 - 2\sqrt{29} + 2\sqrt{55 - 10\sqrt{29}}} = \sqrt{5} + \sqrt{22 + 2\sqrt{5}}.$$

But as they are both algebraic numbers we can check this by brutal force:

1. Find polynomials satisfied by them and use Euclidean division to find their common divisor, and use inequalities to bound its roots.
2. Calculate these two numbers to sufficiently high precision and use the fact two algebraic numbers of bounded degree cannot be too close to each other by Diophantine geometry.

But how to do this for periods?  $\lrcorner$

*Proof:* They are both equal to the expression

$$\sqrt{11 + 2\sqrt{29}} + \sqrt{11 - 2\sqrt{29}} + \sqrt{5}$$

by using the equality  $\sqrt{a} + \sqrt{b} = \sqrt{a + b + 2\sqrt{ab}}$ .  $\square$

**Conj. (19.9.2.9) [Periods Conjecture].** Any two integral representations of a period can be transformed to each other by means of cut-and-paste, change of variables and Stoke's formula.  $\lrcorner$

*Proof:*  $\square$

**Remark (19.9.2.10).** A tip of thinking from this conjecture is that: Whenever you want to prove some two period numbers are equal, write them as integrals and then try to apply change of cut-and-paste, change of variables and Stoke's formula.  $\lrcorner$

**Example (19.9.2.11) [Calabi].** We give one indication of this conjecture by proving  $\zeta(2) = \pi^2/6$ :

$$(1 - \frac{1}{4})\zeta(2) = 1^{-2} + 3^{-2} + 5^{-2} + \dots = \int_{(0,1)^2} \frac{1}{(1-x^2y^2)} dx dy.$$

But the change of variables

$$(x, y) = (\frac{\sin u}{\cos v}, \frac{\sin v}{\cos u})$$

has Jacobian  $(1 - x^2y^2)$  and maps the triangle  $T = \{u, v > 0, u + v < \pi/2\}$  bijectively to  $(0, 1)^2$ , so

$$\int_{(0,1)^2} \frac{1}{(1-x^2y^2)} dx dy = \int_T du dv = \text{Vol}(T) = \pi^2/8$$

$\lrcorner$

**Def. (19.9.2.12) [Exponential Periods].** An **exponential period number** is a complex number that is an integral combinations of real numbers of the form  $\int_U f(x) \exp(g(x)) dx$  where  $f, g$  are algebraic functions over  $\mathbb{Q}$  (19.9.2.1) and  $U$  is a precompact connected open domain of  $\mathbb{R}^n$  defined by polynomial inequalities with rational coefficients, for some  $n \in \mathbb{N}$ . The set of periods is denoted by  $\mathbb{P}_{\text{exp}}$ .  $\lrcorner$

### 3 Periods and Differential Equations

### 4 Periods and $L$ -Functions

### 5 Periods and Motives

## 20 | Algebraic Geometry over Dedekind Schemes

### 20.1 Pseudo-Algebraically Closed Fields

References are [Field Arithmetic].

**Notation (20.1.0.1).**

- Let  $k \in \mathbf{Field}$ .

┘

#### 1 Pseudo-Algebraically Closed Fields

**Def. (20.1.1.1).**  $k \in \mathbf{Field}$  is called a **pseudo-algebraically closed field** or **PAC field** if every variety over  $k$  has a  $k$ -point.

┘

**Example (20.1.1.2) [PAC Fields].**

- Separably closed fields are PAC.
- Infinite algebraic extensions of finite fields are PAC.
- Ultraproducts of distinct finite fields are PAC.

┘

*Proof:*

□

#### 2 Hilbertian Fields

Cf. [Field Arithmetics] and [Ser97].

#### Thin Sets

**Def. (20.1.2.1) [Thin Sets].** For  $X \in \mathbf{Sch}_{\text{int}}^{\text{ft,sep}}/k$ , a subset  $\Omega \subset X(k)$  is called a **thin set** if there exists  $Y \in \mathbf{Sch}^{\text{ft,sep}}/k$  and a map  $f : Y \rightarrow X \in \mathbf{Sch}/k$  s.t.

- $\Omega \subset f(Y(k))$ .
- $f$  is generically quasi-finite and separable (7.4.5.6).
- $f$  has no rational sections.

┘

**Prop. (20.1.2.2).** For  $X \in \text{Sch}_{\text{int}}^{\text{ft,sep}}/k$  and  $U \subset X$  an non-empty open subscheme,  $\Omega \subset X(k)$  is a thin set in  $X$  iff  $\Omega \cap U(k)$  is a thin set in  $U$ .

A finite union of thin sets is a thin set. ┘

**Def. (20.1.2.3) [Two Types of Thin Sets].** For  $X \in \text{Sch}_{\text{int}}^{\text{ft,sep}}/k$ , there are two types of thin sets:

- A type 1 thin set is a set contained in a proper closed subscheme of  $X$  (because we can take the projection  $\text{id} : X \rightarrow X$ ).
- A type 2 thin set is a set of the form  $f(Y(k))$  where  $Y \in \text{Sch}_{\text{int}}^{\text{ft,sep}}/k$  and  $f : Y \rightarrow X$  is dominant finite separable of degree  $d \geq 2$  (7.4.5.6). We may assume that  $Y \in \text{Var}/k$ , because otherwise by (5.3.6.5)  $R(Y)$  contains a finite separable extension  $k'$  of  $k$ , and for any  $y \in Y(k)$ ,  $\mathcal{O}_{Y,y}$  doesn't contain  $k'$ , thus is non-normal (because  $k'$  is contained in its fraction field), and thus  $y$  is contained in the non-normal locus of  $Y$  (7.4.2.2), thus  $f(Y(k))$  is contained in a properly-contained closed subscheme.

Then every thin set of  $X(k)$  is contained in a finite union of thin sets of type1 or type2, by Zariski's main theorem (7.4.5.1). ┘

**Thm. (20.1.2.4).** Let  $X \in \text{Sch}/k$  be irreducible and  $f : X \rightarrow \mathbb{P}^n$  be a dominating morphism, then there exists a thin set  $\Omega \subset \mathbb{P}^n(k)$  for  $X$  (20.1.2.1) s.t. for all  $t \in \mathbb{P}^n(k) \setminus \Omega$ ,  $f^{-1}(t)$  is a finite set of points that are conjugate over  $k$ . ┘

*Proof:* Cf. [Ser97] P132. □

**Prop. (20.1.2.5) [Thin Sets and Linear Subspaces].** For  $k \in \text{Field}$  and  $1 \leq d \leq n \in \mathbb{Z}_+$  and any thin subset  $\Omega \subset \mathbb{P}^n(k)$ , there exists a non-empty open subset  $U \subset \text{Gr}(n; d)_k$  s.t.  $\Omega \cap L$  is a thin set in  $L$  for each  $L \in U(k)$  representing a rational linear subspace of  $\mathbb{P}^n$ . ┘

*Proof:* ? □

**Prop. (20.1.2.6) [Thin Sets and Field Extensions].** For a finite separable extension  $k'/k \in \text{Field}$ , if  $\Omega_1 \subset \mathbb{P}^n(k_1)$  is thin in  $\mathbb{P}^n/k_1$ , then  $\Omega = \Omega_1 \cap \mathbb{P}^n(k)$  is thin in  $\mathbb{P}^n/k$ . ┘

*Proof:* Cf. [Field Arithmetic, P223] or [Ser97]. □

**Remark (20.1.2.7).** The converse is false, Cf. [Field Arithmetic, 13.9.5]. ┘

### Hilbertian Fields

**Def. (20.1.2.8) [Hilbertian Fields].**  $k \in \text{Field}$  is called a **Hilbertian field** if  $\mathbb{P}^1(k)$  is not a thin set in the projective line  $\mathbb{P}^1/k$ . ┘

**Prop. (20.1.2.9).** For  $k \in \text{Field}$ ,

- If  $\mathbb{P}^1(k)$  is not a thin set in  $\mathbb{P}^1/k$ , then  $k$  is Hilbertian, by (20.1.2.5).
- If  $k$  is Hilbertian, then for each  $d \in \mathbb{Z}$  s.t.  $d \neq 0 \in k^\times$ , we have  $k^\times/(k^\times)^d$  is infinite. In particular, local fields are all non-Hilbertian.
- If  $k$  is Hilbertian and  $k'/k \in \text{Field}$  is a finite separable extension, then  $k_1$  is Hilbertian, by (20.1.2.6).
- If  $k$  is Hilbertian, then  $k(T)$  is also Hilbertian. (Cf. [Field Arithmetics] Chap13 ?). ┘



**Def. (20.1.2.10) [Hilbert Subsets].** For  $k \in \mathbf{Field}$  and a set of variables  $T_1, \dots, T_n$ , let  $f(\underline{T}, \underline{X}) \in k[\underline{T}, \underline{X}]$  be irreducible in  $k(\underline{T})[\underline{X}]$  and separable in  $\underline{X}$ , and let  $g \in k[\underline{T}]^\times$ . Then a subset of  $\mathbb{P}^n(k)$  the form

$$\{(t_1, \dots, t_r) \in k^r \mid g(\underline{t}) \neq 0, f(\underline{t}, \underline{X}) \text{ is irreducible}\}$$

is called a **separable Hilbert subset** of  $k^n$ . More generally, every finite intersection of such sets is also called a separable Hilbert subset of  $k^n$ .  $\perp$

**Prop. (20.1.2.11) [Hilbertian Fields and Separable Hilbert Subsets].** For  $k \in \mathbf{Field}$ ,  $k$  is a Hilbertian field iff every separable Hilbert subset of  $k^n$  is non-empty.  $\perp$

*Proof:* By (20.1.2.3), each separable Hilbertian subset of  $k^n$  is contained in the complement of a thin set in  $\mathbb{A}^n$ , thus  $k$  is a Hilbertian field if every separable Hilbert subset of  $k^n$  is non-empty.

Conversely, by (20.1.2.5), It suffices to show that every separable Hilbert subset of  $k^n$  contains a complement of a thin set in  $\mathbb{P}^n$ : Let  $f(\underline{T}, \underline{X})$  decompose as

$$f(\underline{T}, \underline{X}) = a_n(\underline{T}) \prod_{i=1}^n (X - \alpha_i(\underline{T}))$$

in  $\overline{k(\underline{T})}[\underline{X}]$ , then each  $(t_1, \dots, t_r)$  s.t.  $a_n(t_1, \dots, t_n) \neq 0$  and  $f(t_1, \dots, t_n, \underline{X})$  is reducible corresponds to a decomposition  $f(\underline{T}, \underline{X}) = g(\underline{T}, \underline{X})h(\underline{T}, \underline{X})$  in  $\overline{k(\underline{T})}[\underline{X}]$  s.t.  $g(t_1, \dots, t_n, \underline{X})$  has coefficients in  $k$ . But  $f$  is irreducible in  $k(\underline{T})[\underline{X}]$ , so  $g(\underline{T}, \underline{X})$  must have coefficients  $y \in \overline{k(\underline{T})} \setminus k(\underline{T})$ . Thus the curve  $\text{Spec}(k[x, y])$  which is separable over  $\mathbb{A}^n$  has a rational point mapping to  $(t_1, \dots, t_n)$ . Since there are only f.m. decompositions  $f(\underline{T}, \underline{X}) = g(\underline{T}, \underline{X})h(\underline{T}, \underline{X})$  in  $\overline{k(\underline{T})}[\underline{X}]$ , we know that this Hilbert subset contains the complement of a thin set, and the assertion follows.  $\square$

**Prop. (20.1.2.12) [Separable Extensions].** If  $L/K$  is a finite separable extension, then any separable Hilbert subset of  $L^r$  contains a separable Hilbert subset of  $K^r$ . In particular, if  $K$  is Hilbertian, then so is  $L$ .  $\perp$

### Examples of Hilbertian Fields

**Example (20.1.2.13) [Hilbertian Fields].**

- Global fields are Hilbertian.
- If  $k \in \mathbf{Field}$  and  $K$  is a f.g. transcendental extension of  $k$ , then  $K$  is Hilbertian.  $\perp$

*Proof:* Cf. [Field Arithmetic, Chap13].  $\square$

## 3 Haar Measures

## 4 Problems

## 5 Frobenius Fields

## 6 Undecidability

## 20.2 Heights and Diophantine Approximations

References are [Galois Cohomology, Serre]Chap2.4.5, [Wei50, B-G06], [Fundamentals of Diophantine Geometry, Lang].

### 1 Heights

**Def. (20.2.1.1) [Equivalent Height function].** Let  $X$  be a projective scheme, two height functions  $X(\overline{K}) \rightarrow \mathbb{R}$  are called equivalent iff they differ by a bounded function.  $\lrcorner$

**Prop. (20.2.1.2).** There is a way of constructing the Weil heights as a special case of the global heights, which are sum of local heights, Cf.[Diophantine Geometry, Chap2].  $\lrcorner$

### Heights on Projective Spaces

**Def. (20.2.1.3) [Canonical Heights on Projective Spaces].** For  $K \in \mathbf{LField}$  with base field  $K_0$  and  $x = (x_0, \dots, x_n) \in K^{n+1} \setminus \{0\}$ , suppose  $u$  is a normalized place on  $K^{\text{sep}}$  and  $v$  a normalized place of  $K$  s.t.  $u|v$ , denote  $\log(0) = -\infty$  and define

$$h_u((x_0, \dots, x_n)) = \max_i (\log |x_i|_u), \quad H_u((x_0, \dots, x_n)) = \exp(h_u((x_0, \dots, x_n))),$$

$$h_v((x_0, \dots, x_n)) = [K : K_0] h_u((x_0, \dots, x_n)), \quad H_v((x_0, \dots, x_n)) = \exp(h_v((x_0, \dots, x_n))).$$

And for  $F \in \mathbf{GField}$  over base field  $F_0$  and  $x \in \mathbb{P}^n(F)$ , define

$$h(x) = \frac{1}{[F : F_0]} \sum_{v \in \Sigma_F} h_v((x_0, \dots, x_n)), \quad H(x) = \exp(h(x)).$$

Equivalently,

$$H(x) = \|\mathbf{a}\|^{-1/[F:F_0]} \prod_{\tau \in \Sigma_F^\infty} \max_i |x_i|_\tau^{1/[F:F_0]}.$$

Then

- $h_u((x_0, \dots, x_n))$  is invariant under finite separable extensions of fields  $K'/K$ , thus this local height is defined on all  $(K^{\text{sep}})^{n+1}$ .
- $h(x)$  is well-defined.
- $h(x) \geq 0$ .
- $h(x)$  is invariant under extensions  $F'/F \in \mathbf{GField}$ , thus the height is defined on  $\mathbb{P}^n(F^{\text{sep}})$ .
- For  $\sigma \in \text{Gal}_F$ ,  $h(\sigma(x)) = h(x)$ .

These are called **canonical heights** and **multiplicative canonical heights** on  $\mathbb{P}^n$ .  $\lrcorner$

*Proof:* 1 is a consequence of the product formula(19.4.1.5). 2 follows from 1 because we can divide a constant to make a coordinate unit in  $K^*$ , thus clearly  $h(x) \geq 0$ . 3 follows from the fundamental identity.  $\square$

**Prop. (20.2.1.4).** For  $K \in \mathbf{LField}$  and  $P_1, \dots, P_r \in (K^{\text{sep}})^n$ ,

$$h_u(P_1 + \dots + P_r) \leq h_u(P_1) + \dots + h_u(P_r) + \varepsilon_u \log |r|_u,$$

where  $\varepsilon_u = 0$  if  $u$  is non-Archimedean and 1 otherwise.  $\lrcorner$

**Def. (20.2.1.5) [Height on Affine Spaces].** For  $F \in \mathbf{NField}$  and  $\underline{x} = (x_1, \dots, x_n) \in F^n$ , the **affine height** is defined to be:

$$h^+(\underline{x}) = h([1, \underline{x}]),$$

where  $[1, \underline{x}] \in \mathbb{P}^n(F)$ , and similarly we define  $H^+(\underline{x})$ .

In particular, if  $\alpha \in F$ , then the height of  $\alpha$  is

$$h^+(\alpha) = \frac{1}{[F : \mathbb{Q}]} \sum_{v \in \Sigma_F} \max(0, \log |\alpha|_v).$$

┘

**Lemma (20.2.1.6).** For  $\alpha \in \overline{\mathbb{Q}}$  and  $\lambda \in \mathbb{Q}$ , we have  $h(\alpha^\lambda) = |\lambda| h(\alpha)$ . ┘

*Proof:* For  $\lambda > 0$ , this is easy. So it suffices to consider  $\lambda = -1$ . Notice

$$\log |\alpha|_v = \max\{0, \log |\alpha|_v\} - \max\{0, \log |1/\alpha|_v\},$$

summing over all places  $v$  and use the product formula, we get the desired results.  $\square$

**Lemma (20.2.1.7) [Northcott].** Let  $F \in \mathbf{GField}$  and  $C > 0, d > 0$ , then  $\#\{x \in \mathbb{P}^n(\overline{F}) | h(x) \leq C, \deg(x) \leq d\} < \infty$ . ┘

*Proof:* We first reduce to the case  $k(x) = \mathbb{Q}$  or  $\mathbb{F}_p(t)$ : for a point  $x$  with  $[k(x) : \mathbb{Q}] \leq d$ , consider the point  $(X_0, \dots, X_m)$  in the projective space  $\mathbb{P}^m$  of forms of degree  $\deg(x)$  in  $n+1$  variables corresponding to  $\text{Nm}(\sum x_i T_i)$ . Notice the inverse image of any closed point in  $\mathbb{P}^m$  is a finite set in  $\mathbb{P}^n(\overline{K})$ , and the height of  $h((X_0, \dots, X_m)) \leq d!(n+1)h(x)$ , so it suffices to prove for  $\mathbb{P}^m(\mathbb{Q})$ . In this case, we normalize a point to a unique point with integral coordinates with no common divisors, then  $\max_i \log |x_i|_p = 0$  for any  $p$ , thus  $h(x) = h_\infty(x)$ , and clearly only f.m. points has bounded heights.  $\square$

**Prop. (20.2.1.8) [Change of Coordinates].** Let  $h_1, h_2$  be heights of  $\mathbb{P}(F^{\text{sep}})$  defined w.r.t. two coordinates systems, then  $h_1 - h_2 = O(1)$ . Thus we can consider heights in any particular coordinates that is convenient. ┘

*Proof:* The proof is straightforward.  $\square$

### Heights of Polynomials

**Def. (20.2.1.9) [Heights and Mahler Heights].** The height of a polynomial  $f(T) = x_0 + x_1 T + \dots + x_d T^d$  is defined to be  $h^+(f) = h^+((x_0, \dots, x_d))$ , and similarly for  $H^+(f)$ .

The **Mahler height** of  $f$  is defined to be? ┘

**Lemma (20.2.1.10) [Gelfond].** ┘

**Prop. (20.2.1.11).** Let  $f_1, \dots, f_m \in \overline{\mathbb{Q}}[X_1, \dots, X_n]$  of degree  $d$ , then

$$-d \log 2 + \sum_{j=1}^m h(f_j) \leq h(f) \leq d \log 2 + \sum_{j=1}^m h(f_j).$$

┘

*Proof:* Cf. [Bombieri, P28].?  $\square$

**Cor. (20.2.1.12).** Let  $d \in \mathbb{Z}_+$ , then there exists constants  $C_1, C_2 \in \mathbb{R}_+$  depending on  $d$  s.t. for any  $\alpha \in \overline{\mathbb{Q}}$  with  $\deg(\alpha) = d$ , if  $f_\alpha \in \mathbb{Z}[T]$  be a minimal polynomial of  $\alpha$  with coprime coefficients, then

$$dh^+(\alpha) - C_1 \leq h(f_\alpha) \leq dh^+(\alpha) + C_2.$$

┘

### House Heights

**Def. (20.2.1.13) [House Heights].** For  $F \in \mathbf{GField}$  and  $x = (x_1, \dots, x_n) \in \mathcal{O}_F^{\oplus n}$ , define

$$\overline{|x|} = \max_i \max_{\tau: F \rightarrow \mathbb{C}} |\tau(x)|.$$

┘

**Prop. (20.2.1.14) [House Heights and Projective Points].** For  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = d$  and  $n \in \mathbb{Z}_+$ , there exists  $c(F, n) \in \mathbb{R}_+$  s.t. any point  $x \in \mathbb{P}^n(F)$  is representable by  $x' = (x_0, \dots, x_n) \in \mathbb{A}^{n+1}(\mathcal{O}_F)$  s.t.

$$\overline{|x'|} \leq c(F, n) H(x) \text{ (20.2.1.13).}$$

┘

*Proof:* Let  $\mathfrak{c}_1, \dots, \mathfrak{c}_h \in \text{Ideal}(\mathcal{O}_F)^\times$  be a set of representative for the ideal class of  $F$ . Then we may choose  $x_0, \dots, x_n$  s.t.  $(x_0, \dots, x_n) = \mathfrak{c}_i$  for some  $i \in [h]_+$ . Then

$$H(x) = \|\mathfrak{c}_i\|^{-1/d} \prod_{\tau \in \Sigma_F^\infty} \max_j |x_j|_\tau^{1/d} \text{ (20.2.1.3).}$$

Now by property of the regulator map (20.8.1.6), there exists  $t \in \mathbb{R}_+$  and  $\varepsilon \in \mathcal{O}_F^\times$  s.t.

$$\log t + \log |\tau(\varepsilon)| = \max_j \log |\tau(x_j)| + O_F(1), \forall \tau : F \rightarrow \mathbb{C}.$$

Thus after multiplying each  $x_i$  by  $\varepsilon^{-1}$ , we get  $\max_j \log |x_j|_\tau = \log t + O_F(1)$  for each  $\tau$ . Then the assertion is clear.  $\square$

### Arakelov Heights

**Def. (20.2.1.15) [Arakelov Heights].** Let  $K \in \mathbf{LField}$  with base field  $K_0$  and  $x = (x_0, \dots, x_n) \in K^n$ , then for a normalized place  $u$  on  $K^{\text{sep}}$  and a normalized place  $v$  of  $K$  that  $u|v$ , define

$$h_u^{Ar}((x_0, \dots, x_n)) = \begin{cases} \log \max_i (|x_i|_u) & , K \in p\text{-Field} \\ \log \sqrt{\sum_{j=0}^n |x_j|_u^2} & , K \in \{\mathbb{R}, \mathbb{C}\} \end{cases}, \quad H_u^{Ar}((x_0, \dots, x_n)) = \exp(h_u^{Ar}((x_0, \dots, x_n)))$$

$$h_v^{Ar}((x_0, \dots, x_n)) = [K_v : K_0] h_u((x_0, \dots, x_n)), \quad H_v^{Ar}((x_0, \dots, x_n)) = \exp(h_v^{Ar}((x_0, \dots, x_n))).$$

And if  $F \in \mathbf{GField}$  over base field  $F_0$ , for  $x \in \mathbb{P}^n(F)$ , define

$$h^{Ar}(x) = \frac{1}{[F : F_0]} \sum_{v \in \Sigma_F} h_v(x), \quad H^{Ar}(x) = \exp(h^{Ar}(x)).$$

Then  $h^{Ar}(x), H^{Ar}(x)$  are well-defined, positive, and invariant under finite extensions of fields for the same reason as (20.2.1.3). Then they extends to the algebraic closure, called the **Arakelov heights** and **multiplicative Arakelov heights** on  $\mathbb{P}^n$ .  $\square$

**Cor. (20.2.1.16).** The Arakelov height is the height associated to the locally bounded metrized line bundle on  $\mathcal{O}_{\mathbb{P}^n}(1)$  with Fubini-Study metrics in the Archimedean places and standard metric in the non-Archimedean places, so it is equivalent to the canonical height.  $\square$

**Def. (20.2.1.17) [Arakelov Heights on Matrices].** The Arakelov heights induces a height function on the Grassmannians by the canonical embedding, and for any matrix  $A \in M_{n \times m}(\overline{F})$  of rank  $m$ , let the Arakelov heights of  $A$  be defined as the height of the point in the Grassmannian  $\text{Gr}_m(\overline{F})$  associated to the space spanned by the columns of  $A$ . And if  $A \in M_{m \times n}(F)$  of rank  $m$ , then its Arakelov height is defined to be  $H^{Ar}(A) = H^{Ar}(A^t)$ .

Equivalently, it is the height of the point in  $\mathbb{P}^{\binom{n}{m}-1}(F)$  represented by the  $m \times m$ -minors of  $A$ . And we can also define the local heights of  $A$  in this form.  $\perp$

**Prop. (20.2.1.18).** Let  $K = \mathbb{R}$  or  $\mathbb{C}$  and  $A \in M_{n \times m}(\overline{K})$  has rank  $m$ , then  $H_u(A) = \det(A^*A)^{1/2}$ , by Binet's formula (3.5.9.8).  $\perp$

**Prop. (20.2.1.19).** Let  $K$  be a local field and  $A = [B, C] \in M_{n \times m}(\overline{K})$  has rank  $m$ , then

$$H_u^{Ar}(A) \leq H_u^{Ar}(B)H_u^{Ar}(C).$$

$\perp$

**Prop. (20.2.1.20).** Let  $F \in \mathbf{GField}$  and  $W$  be a subspace of  $F^n$  and  $W^\perp \in (F^n)^* \cong F^n$  be the annihilator of  $W$ , then  $h^{Ar}([W]) = h^{Ar}([W^\perp])$ .  $\perp$

*Proof:* Cf. [B-G06]P68.  $\square$

**Prop. (20.2.1.21).** Let  $K$  be a local field and  $V, W$  be subspaces of  $\overline{K}^n$ , then

$$h_u^{Ar}([V + W]) + h_u^{Ar}([V \cap W]) \leq h_u^{Ar}([V]) + h_u^{Ar}([W]).$$

$\perp$

*Proof:* Cf. [B-G06]P69.  $\square$

**Prop. (20.2.1.22) [Metric on Projective Space].** Let  $K$  be a local field, for a normalized place  $u$  on  $\overline{K}$  and a normalized place  $v$  of  $K$  that  $u|v$ , for  $x, y \in (\overline{K})^n \setminus \{0\}$ , define

$$\delta_u(x, y) = \frac{H_u^{Ar}(x \wedge y)}{H_u^{Ar}(x)H_u^{Ar}(y)}, \quad \delta_v(x, y) = \delta_u(x, y)^{[K_v:(K_0)_{v_0}]},$$

then  $\delta(x, y) \in [0, 1]$  by (20.2.1.21), and defines a metric on  $\mathbb{P}^n(\overline{K})$ , called the **projective metric on  $\mathbb{P}^n$** . In the complex case, it is just the Fubini-Study metric.

And if  $F$  is a global field over base field  $F_0$ , for  $x, y \in \mathbb{P}^n(\overline{F})$ , define

$$\delta(x) = \prod_v \delta_v(x, y)^{\frac{1}{[F:F_0]}}.$$

$\perp$

*Proof:* Cf. [B-G06]P70.  $\square$

**Prop. (20.2.1.23).** Let  $F$  be a global field and  $x, y \in \mathbb{P}^1(\overline{F})$ , then

- If  $O = [0, 1]$ , then  $H^{Ar}(x) = \frac{1}{\delta(x, O)}$ .
- If  $x = [1, \alpha], y = [1, \beta]$ , then  $\delta(x, y) = \frac{1}{H^{Ar}(\alpha)H^{Ar}(\beta)}$ .

$\perp$

### Weil Heights

**Prop. (20.2.1.24).** Let  $X \in \mathcal{V}\text{ar}^{\text{Proj}}/k$  and  $\varphi : X \rightarrow \mathbb{P}^k, \psi : X \rightarrow \mathbb{P}^l \in \text{Sch}/k$ . If  $\varphi^*(\mathcal{O}_{\mathbb{P}^k}(1)) \cong \psi^*(\mathcal{O}_{\mathbb{P}^l}(1))$ , then the induced height function  $h_\varphi - h_\psi$  is bounded on  $X(k^{\text{sep}})$ .  $\lrcorner$

*Proof:* Let  $\varphi^*(\mathcal{O}_{\mathbb{P}^k}(1)) \cong \psi^*(\mathcal{O}_{\mathbb{P}^l}(1)) \cong L$ , then  $L$  is very ample. Consider a basis  $\{s_0, \dots, s_n\}$  of  $\Gamma(X, L)$ , then it induces a closed embedding  $\chi : X \rightarrow \mathbb{P}^n : x \mapsto [s_0(x), \dots, s_n(x)]$  and  $\chi^*(\mathcal{O}_{\mathbb{P}^n}(1)) \cong L$ . Then it suffices by symmetry to prove  $h_\varphi \sim h_\chi$ .

We may assume  $\varphi(X)$  is not contained in any proper linear subspace of  $X$ , so we can choose a basis  $T_0, \dots, T_k$  of  $\varphi^*(\Gamma(\mathbb{P}^k, \mathcal{O}_{\mathbb{P}^k}(1)))$ , and let  $s_0, \dots, s_k = T_0, \dots, T_k$ . Let  $x \in X(\overline{K})$ , let  $(x_0, \dots, x_n)$  be the coordinate of  $\chi(x)$  and  $(x_0, \dots, x_k)$  the coordinates of  $\varphi(x)$ . Then by the formula (20.2.1.3) clearly  $h_\varphi \leq h_\chi$ .

For the converse, let  $I$  be the homogenous ideal corresponding to  $X \subset \mathbb{P}^n$ , then  $X$  has coordinate ring  $R = K[T_0, \dots, T_n]/I$ . The  $T_0, \dots, T_k$  generates a radical ideal equal to  $(T_0, \dots, T_n)$ , because they have no common zero on  $X$ . Now there is an integer  $q$  and homogenous ideals  $F_{ij}$  that

$$T_{k+i}^q - \sum_{j=0}^k F_{ij}(T_0, \dots, T_n) T_j \in I$$

so

$$q \log |x_{k+i}|_v \leq (q-1) \max_{j \leq n} \log |x_j|_v + \max_{j \leq k} \log |x_j|_v + C_v$$

where  $C_v = 0$  unless  $v$  is Archimedean. Hence

$$\max_{j \leq n} \log |x_j|_v \leq \max_{j \leq k} \log |x_j|_v + C_v \Rightarrow h_\chi \leq h_\varphi + C.$$

$\square$

**Prop. (20.2.1.25) [Weil Height Functions].** Let  $F \in \mathbf{GField}$  and  $X \in \mathcal{V}\text{ar}^{\text{Proj}}/F$ , then for each element  $\mathcal{L} \in \text{Pic}(X)$ , we can assign a unique **Weil height**  $h_{\mathcal{L}} : X(F^{\text{sep}}) \rightarrow \mathbb{R}$ , determined up to bounded functions, s.t.

- $h_{\mathcal{L}_1 \otimes \mathcal{L}_2} - h_{\mathcal{L}_1} + h_{\mathcal{L}_2} = O(1)$ .
- If  $X = \mathbb{P}^n$ , then  $h_{\mathcal{O}(1)}$  is the height defined in (20.2.1.3).
- For any  $f : X \rightarrow Y \in \mathcal{V}\text{ar}^{\text{Proj}}/F$  and  $\mathcal{L} \in \text{Pic}(Y)$ , we have  $h_{f^*(\mathcal{L})} - h_{\mathcal{L}} \circ f = O(1)$ .

$\lrcorner$

*Proof:* The uniqueness follows from the fact that  $\text{Pic}(X)$  is generated by very ample line bundles because  $X$ . For the existence, we may take item2, 3 as definition of  $h_{\mathcal{L}}$  for  $\mathcal{L}$  very ample, and extend it to all  $\text{Pic}(X)$ . Then it suffices to verify item1.

Let  $L_1 = \varphi^*(\mathcal{O}(1)), L_2 = \psi^*(\mathcal{O}(1))$ , where  $\varphi : X \rightarrow \mathbb{P}^k, \psi : X \rightarrow \mathbb{P}^l$ . Denote  $\sigma : \mathbb{P}^k \times \mathbb{P}^l \rightarrow \mathbb{P}^{kl+k+l}$  the Segre embedding, then  $L_1 \otimes L_2 \cong \chi^*(\mathcal{O}_{\mathbb{P}^{kl+k+l}}(1))$  where  $\chi : X \xrightarrow{(\varphi, \psi)} \mathbb{P}^k \times \mathbb{P}^l \xrightarrow{\sigma} \mathbb{P}^{kl+k+l}$ . And we check  $h_\chi \sim h_\varphi + h_\psi$ .  $\square$

**Thm. (20.2.1.26) [Northcott].** Let  $F \in \mathbf{GField}$  and  $X \in \mathcal{V}\text{ar}^{\text{Proj}}/F$ , and let  $h_{\mathcal{L}}$  be a Weil height function associated to an ample class  $\mathcal{L} \in \text{Pic}(X)$  (20.2.1.25), then the set

$$\{P \in X(\overline{K}) | h_{\mathcal{L}}(P) \leq C, [K(P) : K] \leq d\}$$

is finite for any constant  $C, d$ .  $\lrcorner$

*Proof:* There is a  $m > 0$  that  $mc$  is very ample. Because  $mh_{\mathcal{L}} \sim h_{m\mathcal{L}}$  (20.2.1.25), we can assume that  $\mathcal{L}$  is very ample, thus reducing to  $X = \mathbb{P}^n$  and  $\mathcal{L} = \mathcal{O}(1)$  by (20.2.1.25). This follows from (20.2.1.7).  $\square$

**Prop. (20.2.1.27).** Let  $F \in \mathbf{NField}$ ,  $X \in \mathbf{SmProjVar}/F$  and  $\mathcal{L} \in \mathbf{Pic}(X)$  s.t.  $h_{\mathcal{L}} = O(1)$  on  $X(F^{\text{sep}})$ , then  $\mathcal{L} \in \mathbf{Pic}^0(X)$ .  $\lrcorner$

*Proof:* ? Cf. [Ser97]P46.  $\square$

**Prop. (20.2.1.28).** Let  $F \in \mathbf{GField}$ ,  $X \in \mathbf{SmProjVar}/F$  and  $c \in \mathbf{Pic}(X)$  an ample line bundle,  $c' \in \mathbf{Pic}^0(X)$ , then

$$h_{c'} = O(|h_c|^{1/2} + 1).$$

$\lrcorner$

*Proof:* By base change of fields, we may assume  $X$  has a rational point  $P_0$ . Consider the double Picard map  $\varphi : X \rightarrow \widehat{\mathbf{Pic}^0(X)} = \widehat{A}$ . Viewing  $c'$  as a rational point of  $A$ , consider  $c'' = (p_A)_{c'} \in \mathbf{Pic}(\widehat{A})$ , then  $\varphi^*(c'') = (\text{id} \times \varphi)^*(p_A)|_{c'} = (p_X)_{c'} = c'$ . Thus  $h_{c'} \sim h_{c''}$ .

Let  $\widehat{c}$  be an even ample line bundle on  $\widehat{A}$  (20.9.1.32), then for some  $n$  large,  $nc - \varphi^*(\widehat{c})$  is base-point-free, thus by (20.2.1.25),  $h_{\widehat{c} \circ \varphi} = O(|h_c| + 1)$ . Thus it suffices to prove for  $c', c$  changed to  $c'', \widehat{c}$ , which is true by (21.1.3.10).  $\square$

**Prop. (20.2.1.29)** [Weil Heights in Intersection Theory]. Cf. [B-G06]P44 ?  $\lrcorner$

### Bounded Sets

**Def. (20.2.1.30)** [Bounded Sets]. Let  $k \in \mathbf{Field}$  s.t.  $k = \overline{k}$  and  $|\cdot|$  be a valuation on  $k$ , Then:

- For  $X \in \mathbf{Aff}^{\text{ft}}/k$ , a subset  $E \subset X(\overline{k})$  is called **bounded** if for any  $f \in \mathcal{O}_X(X)$ ,  $|f|$  is bounded on  $E$ .
- For  $X \in \mathbf{Sch}^{\text{ft}}/k$ , a subset  $E \subset X(\overline{k})$  is called **bounded** if there exists a finite open affine covering  $U_i$  of  $X$  and sets  $E_i \subset U_i(\overline{k})$  that  $E_i$  is bounded in  $U_i$  and  $E = \cup E_i$ .

$\lrcorner$

**Prop. (20.2.1.31)** [Properties of Bounded Sets]. Let  $k \in \mathbf{Field}$ ,  $k = \overline{k}$  and  $|\cdot|$  valuation on  $k$ ,  $X \in \mathbf{Sch}^{\text{ft}}/k$  and  $E \subset X(\overline{k})$ , then:

- If  $k$  is locally compact, then  $E$  is bounded in  $X$  iff  $E \subset X(k)$  is relative compact in the  $k$ -topology.
- For a bounded set  $E$  in  $X$  and any finite open affine covering  $U_i$  of  $X$ , there is a division of  $E$ :

$$E = \cup E_i, \quad E_i \subset U_i(k)$$

and  $E_i$  being bounded in  $U_i$ .

- If  $E$  is bounded in  $X$  and  $Y$  is a closed subscheme of  $X$ , then  $E \cap Y(\overline{k})$  is bounded in  $Y$ .
- The image of a bounded set under a morphism is also bounded.
- $\mathbb{P}^n(\overline{k})$  is bounded in  $\mathbb{P}^n$ .
- The inverse image of bounded set under a proper morphism is bounded. In particular, if  $X/k$  is proper, then  $X(\overline{k})$  is bounded in  $X$ .

┘

*Proof:* 1: Because  $X$  is separated, it suffices to prove for  $X$  affine. And we can also take a refinement of the covering, thus assuming  $U_i = X_{h_i}$ . Suppose  $\sum g_i h_i = 1$ , and let  $E_i = \{P \in E \mid |h_i(P)| = \max_k |h_k(P)|\}$ , then  $E_i \subset U_i(\overline{K})$ . To show  $E_i$  is bounded in  $U_i$ , it suffices to show  $|1/h_i|$  is bounded in  $E_i$ . But this is clear, because  $|g_i|$  is bounded on  $E$ .

2: Use local coordinates.

4: Use the affine open covering  $X_i = \{x_i \neq 0\}$ , and let  $E_i = \{|x_i| = \max_{j=0,\dots,n} |x_j|\}$ , then clearly  $E_i$  is bounded in  $X_i$ .

5: Cf.[Diophantine Geometry, P55], may use Chow's lemma ?. □

**Prop. (20.2.1.32).** Let  $k \in \mathbf{Field}$  be a CDVR, and  $V \subset \mathbb{A}^n \in \mathbf{Sch}/k$  is a closed subscheme with no  $k$ -points at infinite, then  $V(k)$  is bounded in  $V$ . ┘

*Proof:* Let  $\overline{V}$  be the closure of  $V$  in  $\mathbb{P}^n$ , then we may assume that  $V$  is defined over  $\mathcal{O}_K$ . Then by Greenberg's theorem[Rational points in Henselian discrete valuation rings] ?, there exists  $n \in \mathbb{Z}_+$  s.t.  $(\overline{V} \setminus V)(\mathcal{O}_K/\mathfrak{m}_K^n) = \emptyset$ , which means that  $\min(v(x_i)) \leq n - 1$  for any  $x = (x_1, \dots, x_n) \in V(k)$ . □

**Prop. (20.2.1.33) [Σ-Bounded Sets].** Let  $F \in \mathbf{Field}$  and  $\Sigma$  is a set of places on  $K$  that any  $\alpha \in F^\times$  has only f.m. of nonzero valuations. Then  $(E^u)_{u \in \Sigma}$  is said to be **Σ-bounded** in an affine variety  $X$  if for any  $f \in K[X]$ ,

$$C_v(f) = \sup_{u \in M, u|v} \sup_{P \in E^u} |f(P)|_u$$

is finite for any  $v \in M_K$  and  $C_v(f) > 1$  for only f.m.  $v$ .

And we can also define  $\Sigma$ -bounded subsets of any variety  $X/K$  as in (20.2.1.30). Then similar properties as in (20.2.1.31) hold for  $M$ -bounded sets. ┘

**Def. (20.2.1.34) [Locally Bounded Functions].** Situation as in (20.2.1.33), a function  $f : X(\overline{K}) \rightarrow \mathbb{R}$  is called **locally bounded** if  $f(E)$  is bounded for every  $\Sigma$ -bounded set  $E$  in  $X$ . ┘

### Distance Functions

**Def. (20.2.1.35) [Distance Function on Curves].** Let  $C$  be a curve over a valued field  $K$ , let for  $P, Q \in C(K)$ , the  $v$ -**adic distance** function  $d_v(P; Q)$  be defined to be  $d_v(P; Q) = \min(|t_Q(P)|_v^{1/e}, 1)$ , where  $t_Q$  is a function with only one zero of order  $e$  at  $Q$ .

To understand this definition, it is just same as the pullback of the local height via  $t_Q$ . It is a special case of local heights. ┘

**Prop. (20.2.1.36).** Let  $C$  be a curve over a valued field  $K$  and  $F \in K(C)$ , then

$$\lim_{P \in C(K), d_v(P, Q) \rightarrow 0} \frac{\log |F(P)|_v}{\log d_v(P, Q)} = \text{ord}_Q(F).$$

┘

*Proof:* In taking limit, scaling  $t_Q$  by a constant doesn't matter, so If we change  $h_{\mathcal{O}(-Q)}$  to  $h_{\mathcal{O}(-kQ)}^{1/k}$ , the limit won't change, so we will do so and assume  $k$  is large enough s.t. there is a function that has only zeros at  $P$  of order  $k$ , so  $f^* \mathcal{O}(1) = \mathcal{O}(-kP)$ , and thus we may change  $\mathcal{O}(kP)$  to  $f^* \mathcal{O}(1)$  and then we see  $f^{\text{ord}_Q(F)} / F^k$  is regular and non-vanishing at  $Q$ , thus we can easily get the desired result. □



**Prop. (20.2.1.37).** If  $\varphi : C_1 \rightarrow C_2$  is a non-constant map of curves over a valued field  $K$ , then for  $Q \in C_1$ ,

$$\lim_{P \in C(K), d_v(P, Q) \rightarrow 0} \frac{\log d_v(\varphi(P), \varphi(Q))}{\log d_v(P, Q)} = e_Q(\varphi).$$

┘

*Proof:* Similar as that of (20.2.1.36). In fact (20.2.1.36) can be derived from this one.  $\square$

## Metricized Line Bundles

### Chevalley-Weil Theorem

References are [B-G06].

**Prop. (20.2.1.38) [Local Chevalley-Weil Theorem].** Let  $K$  be a non-Archimedean valued field,  $f : Y \rightarrow X \in \mathcal{V}\text{ar}/K$  be a finite unramified morphism of  $K$ -varieties and  $E$  a bounded set in  $X(\overline{K})$  (20.2.1.30). Then there is an  $\alpha \in \mathcal{O}_K^\times$  s.t.  $\alpha \in \widehat{\text{disc}}(P/Q)$  whenever  $P \in Y(\overline{K})$  and  $Q = f(P) \in E$ .  $\square$

*Proof:* An unramified map is locally of the form a closed embedding of a standard étale morphism<sup>?</sup>, thus there are f.m.  $U_i, V_i$  covering  $X, Y$  respectively that  $V_i \rightarrow U_i$  is closed embedding  $V_i \rightarrow W_i$  of a standard étale morphism  $W_i \rightarrow U_i$ . Because  $\varphi$  is finite hence proper,  $\varphi^{-1}(E)$  is bounded by (20.2.1.31), thus there is a decomposition of  $\varphi^{-1}(E)$  into bounded sets  $E'_i \subset V_i$ . Then it suffices to prove for standard étale morphisms, because we can then multiply them.

The image of  $E_i$  in  $W_i$  is also bounded. Let  $W_i \rightarrow U_i : \text{Spec}(A[t]/f) \rightarrow \text{Spec } A$ , where

$$f = t^d + a_1 t^{d-1} + \dots + a_d, \quad a_i \in A.$$

By boundedness, there is an  $a \neq 0 \in R$  that

$$\max_{i=1, \dots, d} \sup_{P \in E'} |a_i(\varphi(P))| \leq |a|^{-1}.$$

Then  $\xi = at_P$  is a root of the polynomial

$$g_Q(t) = t^d + aa_1(Q)t^{d-1} + \dots + a^d a_d(Q),$$

and it is easily verified that  $|\xi| \leq 1$  thus  $\xi \in R_P$ . Now let  $g_\xi$  be the minimal polynomial of  $\xi$  over  $\widehat{K(Q)}$ , then  $g_Q = g_\xi h$ ,  $h \in \widehat{K(Q)}[t]$ . By Gauss lemma in fact  $g_\xi, h \in \widehat{R_Q}[t]$ .

Now the elements  $1, \xi, \dots, \xi^{d-1}$  form a basis of  $\widehat{K(P)}/\widehat{K(Q)}$ , so the discriminant  $D_{g_\xi} \in \delta_{\widehat{K(P)}/\widehat{K(Q)}}$ .

By?? and (5.2.7.37),  $|D_{g_\xi}| = |N_{\widehat{K(P)}/\widehat{K(Q)}}(g'_\xi(\xi))| = |g'_\xi(\xi)|^{\widehat{d}}$ . But

$$g'_\xi(\xi)h(\xi) = g'_Q(\xi) = a^{d-1}f'_Q(\xi) = a^{d-1}f'_Q(t_P) = a^{d-1}f'(P).$$

Because  $f'$  is unit, we have  $|D_{g_\xi}^{-1}|$  is bounded on  $E'$ . So there is an  $\alpha \neq 0 \in R$  that  $|D_{g_\xi}| \geq |\alpha|$ , independent of  $P$ . Then we are done, as  $D_{g_\xi} \in \delta_{\widehat{K_P}/\widehat{K_Q}}$ .  $\square$

**Lemma (20.2.1.39) [Global Chevalley-Weil].** Let  $F \in \text{Field}$  and  $\Sigma$  be a set of discrete valuations of  $F$  s.t. any  $\alpha \in F^\times$  has only f.m. nonzero valuations. Suppose  $f : Y \rightarrow X \in \mathcal{V}\text{ar}^{\text{proper}}/K$  is an unramified finite map, and  $(E^u)_{u \in \Sigma}$  is a  $\Sigma$ -bounded family in  $X$  (20.2.1.33), then for every  $v \in \Sigma$  there exists  $\alpha_v \in \mathcal{O}_{F,v}^\times$  that  $\alpha_v \in \delta_{P/Q}^u$  whenever  $u|v$  and  $P \in Y(\overline{K})$  with  $\varphi(P) = Q \in E^u$ . Moreover,  $\alpha_v = 1$  for a.e.  $v \in \Sigma$ .  $\square$

*Proof:* The proof is exactly the same as that of (20.2.1.38), noticing that  $a, \alpha$  depend only on  $v$  but not  $u|v$ , and also  $a = \alpha = 1$  for a.e.  $v$  by (20.2.1.33).  $\square$

**Lemma (20.2.1.40) [Global Chevalley-Weil Theorem for Discrete Valuations].** Let  $F \in \text{Field}$  and  $\Sigma$  be a set of discrete valuations of a field  $F$  s.t. any  $\alpha \in F^\times$  has only f.m. nonzero valuations. Suppose  $f : Y \rightarrow X \in \text{Var}^{\text{proper}}/K$  is an unramified finite map, then there exists  $\emptyset \subset_{\text{fin}} S \subset \Sigma$  and for any  $v \in S$  a nonzero element  $\alpha_v \in \mathfrak{p}_v$  s.t. for any  $P \in Y(F^{\text{sep}})$  and  $Q = f(P)$ , and any place  $w_0|v$  of  $K(Q)$ ,  $\kappa(P)/\kappa(Q)$  is unramified outside  $S$  and if  $v \in S$  we have  $\alpha_v \in \delta_{P/Q}^{w_0}$ .  $\lrcorner$

*Proof:* We may assume  $f$  is surjective because it is closed (finite is proper). Notice  $X(\overline{K})$  is  $M$ -bounded in  $X$  (20.2.1.31), where  $M$  is the set of valuations of  $\overline{K}$  extending that of  $M_K$ . Then we can use global Chevalley-Weil theorem (20.2.1.39) to find elements  $\alpha_v$  for  $v \in M_K$ . Now notice

$$\text{disc}(P/Q)^{w_0} = \prod_{w|w_0} (\widehat{\text{disc}(P/Q)}^w \cap \mathcal{O}_{Q,v})$$

by (5.2.7.40), and the number of  $w \in M_{K(P)}, w|w_0$  is bounded by  $[K(P) : K(Q)]$ , which is further bounded by  $\deg(f)$  as in (20.2.1.38). Hence we can take  $\alpha_v = \alpha_v^{\deg(f)}$  to finish the proof.  $\square$

**Lemma (20.2.1.41).** Let  $F$  be a global field or the function field of a non-singular curve over a field  $k$ , and let  $f : Y \rightarrow X \in \text{Var}^{\text{proper}}/K$  be unramified and finite. Then there exists  $\alpha \in \mathcal{O}_F^\times$  s.t. for any  $P \in X(F^{\text{sep}})$  and  $Q = \varphi(P)$ , we have  $\alpha \in \text{disc}_{P/Q}$ .  $\lrcorner$

*Proof:* By (7.13.1.15), we can use the above lemma (20.2.1.40) to  $\Sigma = \Sigma_F$ . Notice that because

$$\text{disc}(P/Q) = \prod_{w_0 \in \Sigma_{\kappa(Q)}^{\text{fin}}} (\text{disc}(P/Q)^{w_0} \cap \mathcal{O}_{X,Q}),$$

we can assume  $\alpha_v \in \mathcal{O}_{X,Q}$ , and take  $\alpha = \prod_{v \in \Sigma} \alpha_v$ , which is a finite product by (20.2.1.40).  $\square$

**Thm. (20.2.1.42) [Chevalley-Weil].** Let  $F$  be a global field or the function field of a non-singular curve over a field  $k$ , and let  $f : Y \rightarrow X \in \text{Var}^{\text{proper}}/K$  be an unramified finite map. Then there exists  $L/K \in \text{GalField}^{\text{fin}}$  s.t. for any  $P \in Y(K^{\text{sep}})$  with  $f(P) \in X(K)$ , we have  $P \in Y(L)$ .  $\lrcorner$

*Proof:* By Chevalley-Weil theorem (20.2.1.41), there is an  $\alpha \in \mathcal{O}_F^\times$  that  $\alpha \in \text{disc}(P/Q)$  for any  $P \in Y(F^s)$  with  $f(P) \in X(K)$ . Thus  $\kappa(P)/\kappa(Q)$  is unramified outside  $S_F(\alpha)$ . But then Hemite's theorem (20.8.1.8) shows there are only f.m. possibilities of  $\kappa(P)$ , and the assertion follows.  $\square$

## 2 Canonical Heights

References are [Call-Silverman. Canonical heights on varieties with morphisms].

**Prop. (20.2.2.1) [Tate's Limiting argument].**  $\lrcorner$

*Proof:* Cf. [B-G06] P285. ?  $\square$

**Cor. (20.2.2.2).** Tate's limiting argument gives another way of constructing Néron-Tate heights. More generally, for any projective variety  $X$  over a global field  $K$ , if  $\varphi : X \rightarrow X$  a morphism over  $K$ ,  $\mathcal{L}$  a line bundle on  $X$  and  $k, l \in \mathbb{Z}, |k| > |l|$  that

$$l\varphi^*(\mathcal{L}) = k\mathcal{L},$$

then there is a unique function  $\widehat{h}_{\varphi, \mathcal{L}}$  in the equivalent class of  $h_{\mathcal{L}}$  that

$$l\widehat{h}_{\varphi, \mathcal{L}}(\varphi(x)) = k\widehat{h}_{\varphi, \mathcal{L}}(x).$$

In particular, the Néron-Tate heights on an Abelian variety for an even or odd line bundle is obtained by taking  $\varphi = [m]$  for some  $m \geq 2$ .  $\lrcorner$

*Proof:* Assume  $l \neq 0$ , consider the subgroup  $\mathcal{N} = \{\lambda^r | r \in \mathbb{N}\}$ , where  $\lambda = k/l$ , and  $\mathcal{N}$  acts on  $X(\overline{K})$  by  $\lambda^r \cdot x = \varphi^r(x)$ , then  $h_{\mathcal{L}}$  is quasi-homogenous of degree 1, so Tate's limiting argument (20.2.2.1) shows

$$\widehat{h}_{\varphi, \mathcal{L}}(x) = \lim_{r \rightarrow \infty} \lambda^{-r} h_{\mathcal{L}}(\varphi^r(x))$$

satisfies the requirement. And similarly, it is non-negative if  $c$  is ample or base-free.  $\square$

### 3 Approximations of Algebraic Numbers

#### Subspace Theorem

References are [Sch70, F-W94, Eve96, E-S02, B-G06].

**Lemma (20.2.3.1).**  $\lrcorner$

**Thm. (20.2.3.2) [Subspace Theorem, Schmidt/Vojta].** For  $F \in \mathbf{NField}$  and  $S \subset \Sigma_F$  a finite subset of places,  $n \in \mathbb{N}, \varepsilon \in \mathbb{R}_+$ . For  $v \in S$ , let  $\{L_{v0}, \dots, L_{vm_v}\}$  be a set of linear forms in  $\overline{F}_v[X_0, \dots, X_n]$  in general position. Then

- If  $\Sigma_F^\infty \subset S$ , then there are f.m. rational linear hyperspaces  $T_1, \dots, T_h$  of  $\mathbb{A}_F^{n+1}$  s.t.

$$\left\{ \underline{x} \in \mathcal{O}_{F,S}^{n+1} \setminus \{0\} \mid \prod_{v \in S} \prod_{i=0}^{m_v} |L_{vi}(\underline{x})|_v < H([\underline{x}])^{-\varepsilon} \right\} \subset T_1 \cup \dots \cup T_h.$$

- There exists f.m. rational hyperplanes  $T_1, \dots, T_h$  of  $\mathbb{P}_F^n$  s.t.

$$\left\{ [\underline{x}] \in \mathbb{P}^n(F) \mid \prod_{v \in S} \prod_{i=0}^{m_v} \frac{|L_{vi}(\underline{x})|_v}{H_v(\underline{x})} < H([\underline{x}])^{-n-1-\varepsilon} \right\} \subset T_1 \cup \dots \cup T_h.$$

$\lrcorner$

*Proof:* 2: Firstly, it suffices to prove for  $m_v = n$  for any  $v$ : We may assume  $m_v \geq n$  by adding coordinate functions. And after partitioning into f.m. cases and reordering, we may assume that

$$|L_{v0}(\underline{x})|_v \leq |L_{v1}(\underline{x})|_v \leq \dots \leq |L_{vm_v}(\underline{x})|_v.$$

As  $L_{v0}, \dots, L_{vn}$  are linearly independent, they form a basis for  $\mathcal{O}_{\mathbb{P}^n}(1)$ , so there are constants  $C_v$  s.t.

$$H_v(\underline{x}) \leq C_v \max_{0 \leq i \leq n} |L_{vi}(\underline{x})|_v = C_v |L_{vn}(\underline{x})|_v \leq C_v |L_{vk}(\underline{x})|_v, \forall k > n.$$

Thus

$$\prod_{v \in S} \prod_{i=0}^n \frac{|L_{vi}(\underline{x})|_v}{H_v(\underline{x})} \leq C \prod_{v \in S} \prod_{i=0}^{m_v} \frac{|L_{vi}(\underline{x})|_v}{H_v(\underline{x})},$$

and the assertion follows from the case  $m_v = n$ , as the constant  $C$  can be eliminated by applying Northcott's theorem (20.2.1.26).

Secondly, item2 implies item1: For  $\underline{x} \in \mathcal{O}_{F,S}^{n+1}$ ,

$$H([\underline{x}]) = \prod_{v \in \Sigma_F} H_v(\underline{x}) \leq \prod_{v \in S} H_v(\underline{x}),$$

so  $[\underline{x}]$  is in the set considered in item2. And any two  $\underline{x}, \underline{x}'$  with  $[\underline{x}] = [\underline{x}']$  are contained in the same hyperplane.

Conversely, item1 implies item2: We may assume that  $S$  is sufficiently large that  $\Sigma_F^\infty \subset S$  and  $\text{cl}(\mathcal{O}_{F,S}) < \infty$ , because if we add a place  $v$  and take  $L_{vi}(\underline{x}) = x_i$ , then  $|L_{vi}(\underline{x})|_v \leq H_v(\underline{x})$  for each  $i$ . Then for any  $[\underline{x}]$  satisfying the inequality of item2, take the fractional ideal

$$\mathfrak{X} = \sum x_i \mathcal{O}_{F,S},$$

then  $\mathfrak{X} = (\delta)$  for some  $\delta \in K^\times$ . And If  $v \notin S$ , we can see that  $|\delta|_v = \max_i |x_i|_v = H_v(\underline{x})$ . So we may change  $\underline{x}$  to  $\underline{x}' = \delta^{-1} \underline{x}$ , in which case,  $\underline{x}' \in \mathcal{O}_{F,S}^{n+1} \setminus \{0\}$ , and  $H_v(\underline{x}') = 1$  for  $v \notin S$ . So  $\prod_{v \in S} H_v(\underline{x}') = H([\underline{x}']) = H([\underline{x}])$ , and the inequality in item2 implies  $\underline{x}'$  is in the subset in item1. Thus we are done.

Thus it suffices to prove item1: For this, Cf. [B-G06]P197. ?

□

**Cor. (20.2.3.3) [Schmidt].** Let  $\alpha_0, \dots, \alpha_n \in \overline{\mathbb{Q}}$ , then for any  $\varepsilon \in \mathbb{R}_+$ , there are only f.m.  $\underline{x} \in \mathbb{Z}^{n+1}$  s.t.

$$0 < |\alpha_0 x_0 + \dots + \alpha_n x_n| \leq H(\underline{x})^{-n-\varepsilon}.$$

┘

*Proof:* Use induction on  $n$ . For  $n = 0$ , this is trivial. For  $n \geq 1$ , the subspace theorem (20.2.3.2) with

$$F = \mathbb{Q}, S = \{\infty\}, L_{v0} = \alpha_0 X_0 + \dots + \alpha_n X_n, \quad L_{vi} = X_i, 1 \leq i \leq n$$

implies that the desired set is contained in f.m. rational linear subspaces of  $\mathbb{A}_{\mathbb{Q}}^{n+1}$ . Thus we can assume that the desired set all satisfy a linear relation of the form  $\sum_j A_j X_j = 0$ , with  $A_n \neq 0$ . Then with  $\beta_n = \alpha_i - \alpha_n A_i / A_n$ ,

$$0 < |\alpha_0 x_0 + \dots + \alpha_n x_n| = |\beta_0 x_0 + \dots + \beta_{n-1} x_{n-1}| \leq H(x)^{-n-\varepsilon} \leq H(x)^{-n+1-\varepsilon}.$$

Thus the assertion follows from induction hypothesis.

□

**Cor. (20.2.3.4) [Schmidt].** Let  $\alpha \in \overline{\mathbb{Q}}$  and  $d \in \mathbb{Z}_+, \varepsilon \in \mathbb{R}_+$ , then there exists only f.m.  $\xi \in \overline{\mathbb{Q}}$  with  $\deg(\xi) = d$  and

$$|\alpha - \xi| \leq H^+(\xi)^{-d(d+1)-\varepsilon},$$

┘

*Proof:* For any such  $\xi$ ,  $|\xi - \alpha| \leq 1$ , and we may assume that  $\xi$  is not conjugate to  $\alpha$ , so  $f_\xi(\alpha) \neq 0$ . Suppose  $f_\xi = x_0 + x_1 T + \dots + x_d T^d \in \mathbb{Z}[T]$  with coprime coefficients, then  $H(f_\xi) = \Theta(H^+(\xi)^d)$  by (20.2.1.12). Thus by mean-value theorem,

$$0 < |f_\xi(\alpha)| = |x_0 + x_1 \alpha + \dots + x_d \alpha^d| \leq C_\alpha |\alpha - \xi| H(f_\xi) \leq H(f_\xi)^{-d-\frac{\varepsilon}{d}} = H(\underline{x})^{-d-\frac{\varepsilon}{d}}.$$

So the assertion follows from (20.2.3.3).

□

**Prop. (20.2.3.5) [Absolute Subspace Theorem, Evertse-Schlickewei].** [B-G06]P228. ?

┘

**Roth's Theorem****Prop. (20.2.3.6).** ┘**Lemma (20.2.3.7) [Roth's Lemma].** ┘**Thm. (20.2.3.8) [Roth].** Let  $F \in \mathbf{NField}$  and  $\Sigma_F^\infty \subset S \subset \Sigma_F$  be finite, and for each  $v \in S$  a number  $\alpha_v \in \overline{F}_v$ . Then for any  $\varepsilon > 0$ , there are only f.m.  $\beta \in F$  that

$$\prod_{v \in S} H_v(\beta - \alpha_v) = \prod_{v \in S} \min(1, |\beta - \alpha_v|_v) \leq H^+(\beta)^{-2-\varepsilon} \quad (20.2.1.5).$$

Moreover,  $\alpha_v$  can be  $\infty$  as well, in the sense that  $|\beta - \alpha_v|_v = |\beta|_v^{-1}$ . ┘*Proof:* It is clear that it suffices to show that there are only f.m.  $\beta \in F$  s.t.

$$\prod_{v \in S} |\beta - \alpha_v|_v \leq H^+(\beta)^{-2-\varepsilon}, \quad |\beta - \alpha_v|_v < 1.$$

Let  $x = [1, \beta] \in \mathbb{P}^1(K)$  and let  $L_{v0} = X_1 - \alpha_v X_0$  (If  $\alpha_v = \infty$ , take  $L_{v0} = X_0$ ), then for any such  $\beta$ ,

$$\prod_{v \in S} \frac{|L_{v0}(x)|_v}{H_v(\underline{x})} = \prod_{v \in S} |\beta - \alpha_v|_v \max(1, |\beta|_v)^{-1} \leq H^+(\beta)^{-2-\varepsilon}.$$

Thus the assertion follows from the subspace theorem (20.2.3.2). □**Remark (20.2.3.9).** The exponent in Roth's theorem is the best possible, by (20.2.4.4).This theorem is ineffective in the sense it doesn't give a maximal bound on  $H(\beta)$  for a solution  $\beta$ , but it is effective in the sense it gives a bound for the number of solutions. ┘**Remark (20.2.3.10).** This is not true for function fields, as if we choose  $F$  to be the splitting field of the separable polynomial  $x^q - x + t$  over  $\mathbb{F}_p(t)$ , then there is a valuation  $w$  over the valuation  $v$  corresponding to  $(t)$  s.t.  $F_w = \mathbb{F}_p((t))$ , and  $\alpha = t + t^q + t^{q^2} + \dots$ . Take  $\beta_k = t + t^q + \dots + t^{q^k}$ , then  $|\beta_k - \alpha|_v = c^{-q^{k+1}}$  with  $H(\beta_k) = c^{q^k} = H(\beta)^{-q}$ , thus cannot have a Roth's theorem. ┘*Proof:* □**Cor. (20.2.3.11) [Classical Roth's Theorem].** For any  $\alpha \in \overline{\mathbb{Q}}$  and  $\varepsilon > 0$ , there are only f.m.  $p/q \in \mathbb{Q}$  s.t.  $|p/q - \alpha| \leq |q|^{-2-\varepsilon}$ . ┘*Proof:* Take  $F = \mathbb{Q}$ ,  $S = \{\infty\}$  and  $\alpha_\infty = \alpha$ , then use Roth's theorem (20.2.3.8). □**Cor. (20.2.3.12).** For  $p \in \mathbf{Prime}$ ,  $\alpha \in \mathbb{Q}_p$  and  $\varepsilon > 0$ , there are only f.m.  $n \in \mathbb{Z}$  s.t.  $|n - \alpha|_p \leq |n|^{-1-\varepsilon}$ . ┘*Proof:* Take  $F = \mathbb{Q}$ ,  $S = \{\infty, p\}$  and  $\alpha_\infty = \infty$ ,  $\alpha_p = \alpha$ , then use Roth's theorem (20.2.3.8). □**Cor. (20.2.3.13) [Another Interpretation].** Let  $F \in \mathbf{GField}$  and  $v$  a valuation of  $\overline{F}$ ,  $C$  be a complete curve over  $F$  and  $Q \in C(\overline{F})$ , then for  $f \in K(C)^*$ ,

$$\lim_{P \in C(F), d_v(P, Q) \rightarrow 0} \frac{\log d_v(P; Q)}{\log H_F(f(P))} \geq -2.$$

┘

*Proof:* Replacing  $f$  by  $1/f$  if necessary, we can assume  $f(Q) \neq \infty$ . Let  $f - f(Q)$  has order  $e$  at  $Q$ , then by (20.2.1.36),

$$\lim_{P \in C(F), d_v(P, Q) \rightarrow 0} \frac{\log |f(P) - f(Q)|_v}{\log d_v(P, Q)} = e.$$

And Roth's theorem implies

$$H_K(f(P))^{2+\varepsilon} |f(P) - f(Q)|_v \geq 1$$

for a.e.  $P$ . Thus by taking limit and varying  $\varepsilon$ , the conclusion follows.  $\square$

**Cor. (20.2.3.14) [Thue's Equation].** Let  $F(x, y) \in \mathbb{Z}[x, y]$  be a homogenous polynomial that has at least three different linear factors in  $\mathbb{C}[x, y]$ , then for any  $m \in \mathbb{Z}^\times$ , there are only f.m. solutions of  $F(x, y) = m$  in  $\mathbb{Z}^2$ .  $\lrcorner$

*Proof:* Let  $F_1, \dots, F_r$  be non-isomorphic irreducible polynomials in  $\mathbb{Z}[x, y]$  dividing  $F$ , suppose there are i.m. solutions to  $F(x, y) = m$ , then there are inf.m. rational points  $x_n/y_n$  converging but not equal to a root of  $F_i$  for each  $i$ . Thus  $r = 1$ , and  $\deg F_1 \geq 3$  by hypothesis. And it is clear that for some root  $\alpha$  of  $F_1$  and some constant  $C > 0$ , there are i.m. rational points  $n$  s.t.  $|x_n/y_n - \alpha| \leq C|y_n|^{-3}$ , contradicting Roth's theorem (20.2.3.11).  $\square$

**Prop. (20.2.3.15) [Quantitative Bounds for Roth's Theorem].** Cf. [B-G06] Chap 6.5.  $\lrcorner$

**Prop. (20.2.3.16) [Liouville].** Let  $\alpha \in \overline{\mathbb{Q}}, d = \deg(\alpha) \geq 2$ , then there is a constant  $C > 0$  s.t. for all  $p/q \in \mathbb{Q}$ ,  $|p/q - \alpha| \geq C/q^d$ . Notice only the  $d = 2$  case is not covered by Roth's theorem (20.2.3.11).  $\lrcorner$

*Proof:* We can assume  $\alpha \in \mathbb{R}$ , because otherwise  $C = \text{Im } \alpha$  works. Let  $f(T) \in \mathbb{Z}[T]$  be a minimal polynomial of  $\alpha$  over  $\mathbb{Q}$ , then  $f$  has no roots in  $\mathbb{Q}$ . Suppose  $C_1$  is the maximum of  $|f'(t)|$  for  $t \in [\alpha - 1, \alpha + 1]$ . Suppose  $|p/q - \alpha| \leq 1$ , then  $|f(p/q)| = |f(p/q) - f(\alpha)| \leq C_1|p/q - \alpha|$ , and also  $q^d f(p/q) \in \mathbb{Z}$  thus  $|q^d f(p/q)| \geq 1$ . Thus we get  $|p/q - \alpha| \geq 1/C_1 q^d$ . Thus for general  $p/q$ ,  $C = \min(C_1^{-1}, 1)$  works.  $\square$

### Diophantine Approximation on Abelian Varieties

**Prop. (20.2.3.17) [Product Theorem, Faltings].** Suppose  $k \in \text{Field}^0, k = \overline{k}, m, n_1, \dots, n_m \in \mathbb{Z}_+$ ,  $P = \mathbb{P}_k^{n_1} \times \dots \times \mathbb{P}_k^{n_m}$ , for any  $f \in \Gamma(P, \mathcal{O}_P(d_1, \dots, d_m)) \setminus \{0\}$  where  $d_1 > d_2 > \dots > d_m$  are positive integers, denote

$$Z_\sigma(f) = \{x \in P(K) : \frac{\partial^\alpha}{\partial x^\alpha} f = 0, \quad \forall \alpha, \alpha/d < \sigma\}, \sigma \in \mathbb{R}_+.$$

Then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C > 0$  satisfying the following: If  $d_h/d_{h+1} \geq C$  for  $h = 1, 2, \dots, m-1$ , then for any  $\sigma \in \mathbb{R}_+$ , any irreducible component  $Z$  of  $Z_\sigma \cap Z_{\sigma+\varepsilon}$  is of the form  $Z = Z_1 \times \dots \times Z_m$ , where  $Z_i$  are closed subvarieties of  $\mathbb{P}_K^{n_i}$  with  $\deg(Z_i)$  bounded in terms of  $\varepsilon$  and  $n_1 + \dots + n_m$  only.  $\lrcorner$

*Proof:* ?  $\square$

## 4 Approximations by Algebraic Numbers

### Rational Approximations and Continued Fractions

**Def. (20.2.4.1) [Best Rational Approximations].** For  $\alpha \in \mathbb{R}$ ,  $\frac{a}{b} \in \mathbb{Q}$  is called a **best rational approximation** for  $\alpha$  if

$$|b\alpha - a| = ||b\alpha|| < ||d\alpha||$$

for any  $d \in \mathbb{Z}_+$ ,  $d < b$ . ┘

**Thm. (20.2.4.2) [Dirichlet].** For any  $\alpha \in \mathbb{R}$  and  $n \in \mathbb{Z}_+$ , there exists  $b \in \mathbb{Z}_+$ ,  $b \leq n$  s.t.

$$||b\alpha|| < \frac{1}{n+1}.$$

So there exists  $\frac{a}{b} \in \mathbb{Q}$  with  $1 \leq b \leq n$  s.t.

$$|\alpha - \frac{a}{b}| \leq \frac{1}{b(n+1)}.$$

┘

*Proof:* Consider the  $n+1$  numbers

$$0, \alpha, 2\alpha, \dots, n\alpha \in \mathbb{R}/\mathbb{Z},$$

then there are two of them with distance  $\leq \frac{1}{n+1}$ . Thus their difference gives such a  $b$ . □

**Cor. (20.2.4.3) [Characterizing Rational Numbers].** For  $\alpha \in \mathbb{R}$ , the following are equivalent:

- $\alpha \in \mathbb{Q}$ .
- There exists  $C \in \mathbb{R}_+$  s.t. for any  $q \in \mathbb{Z}_+$ ,  $\{q\alpha\} > 0 \implies \{q\alpha\} \geq C$ .
- There are only f.m.  $q \in \mathbb{Z}_+$  s.t.  $0 < q\{q\alpha\} < 1$ .

┘

**Thm. (20.2.4.4) [Dirichlet-Hurewitz].** For any  $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ , there are infinitely many  $\frac{p}{q} \in \mathbb{Q}$  s.t.  
 $|\frac{p}{q} - \alpha| \leq \frac{1}{\sqrt{5}q^2}$ . ┘

*Proof:* Cf. [G. H. Hardy and E. M. Wright. An introduction to the theory of numbers]Thm194<sup>?</sup>. We prove here for a weaker result:

For  $N > 0$ , consider the sets  $\{\{q\alpha\} | q = 0, \dots, N\}$ , where  $\{\beta\}$  are the fractional part of  $\beta$ . Then by pigeonhole principle gives two  $0 \leq q_1 < q_2 \leq N$  s.t.  $|(q_1 - q_2)\alpha - p| \leq 1/N \leq (q_2 - q_1)$ . Let  $q = q_2 - q_1$ , then  $|p/q - \alpha| \leq 1/q^2$ . □

**Def. (20.2.4.5) [Continued Fractions].** ┘

**Def. (20.2.4.6) [Diophantine Numbers].** For  $\mu \in \mathbb{R}_{\geq 2}$ , a  $(C, \mu)$ -**Diophantine number** is a number  $\theta \in \mathbb{R}$  s.t. there exists  $C \in \mathbb{R}_+$  s.t. for any  $p, q \in \mathbb{Z}_+$ ,  $q \neq 0$ ,

$$|\theta - \frac{p}{q}| \geq \frac{C}{q^\mu}.$$

And a **badly approximable number** is a  $(C, 2)$ -Diophantine number for some  $C \in \mathbb{R}_+$ . ┘

**Prop. (20.2.4.7).**  $\xi \in \mathbb{R}$  is badly approximable iff the fractional expansion  $\xi = [a_0; a_1, a_2, \dots]$  satisfies  $(a_n)_{n \in \mathbb{N}}$  is bounded.  $\lrcorner$

*Proof:* Cf. [Distributions Modulo 1]P245.  $\square$

**Prop. (20.2.4.8) [Approximation and Fractional Expansions].** Let  $M \in \mathbb{Z}_+$ ,  $\kappa, \mu \in [0, 1]$ ,  $A \in \mathbb{R}_+$ ,  $B \in \mathbb{R}_{>1}$ , and  $\frac{p}{q}$  is a fractional approximation of  $\kappa$  s.t.  $q > 6M$ . Let  $\varepsilon = ||q\mu|| - M||q\kappa||$ , then

- If  $\varepsilon > 0$ , then there is no solution of the inequality

$$|m\kappa - n + \mu| < AB^{-m}$$

for  $m, n \in \mathbb{Z}$  with  $\frac{\log(Aq/\varepsilon)}{\log B} \leq m \leq M$ .

- (This may apply when  $\kappa + \mu$  is close to 1) If  $\varepsilon < 1/3$  but  $\lfloor q\mu + 0.5 \rfloor = q - p$ , then there is no solution of the inequality

$$|m\kappa - n + \mu| < AB^{-m}$$

for  $m, n \in \mathbb{Z}$  with  $\max(\frac{\log(3Aq)}{\log B}, 1) < m \leq M$ .  $\lrcorner$

*Proof:* 1: Assume  $0 \leq m \leq M$ , then

$$|m(q\kappa - p) + (mp - nq) + q\mu| < qAB^{-m},$$

so

$$qAB^{-m} > |mp - nq + q\mu| - m\{\{q\kappa\}\} \geq \{\{q\mu\}\} - M\{\{q\kappa\}\} = \varepsilon.$$

2: Assume  $0 \leq m \leq M$ , then

$$|m(q\kappa - p) + (mp - nq + r) + (q\mu - r)| < qAB^{-m},$$

so

$$|mp - nq + r| < qAB^{-m} + ||q\mu|| + M||q\kappa|| < qAB^{-m} + 2/3.$$

If  $qAB^{-m} \leq 1/3$ , then  $mp - nq + r = 0$ . But then the hypothesis implies that  $m \equiv 1 \pmod{q}$ . As  $q > 6M$ , this means  $m = 1$ .  $\square$

### Higher Degree case

**Prop. (20.2.4.9) [Dirichlet-Hurewitz/Davenport-Schmidt].** For  $d \in \mathbb{Z}_+$ , let  $k_d \in [-1, \infty]$  be the supremum of numbers  $w \in \mathbb{R}_+$  s.t. for any  $\alpha \in \mathbb{R}$  not an algebraic number of degree  $\leq d$  and any  $\varepsilon \in \mathbb{R}_+$ , there exists a constant  $C = C(\alpha, \varepsilon)$  s.t. there are inf.m. real algebraic number  $\xi$  of degree  $d$  satisfying

$$|\alpha - \xi| \leq CH^+(\xi)^{-dk_d + \varepsilon}.$$

Then

- $k_1 = 2$  by (20.2.4.4).
- $k_2 = 3$ .
- $k_d \leq d + 1$ .

$\lrcorner$



*Proof:* 2: One direction is by (20.2.3.4), the other by [Davenport-Schmidt, Approximation to real number by quadratic irrationals] **?**.

3 follows from (20.2.3.4). □

**Prop. (20.2.4.10) [Dirichlet-Hurewitz/Davenport-Schmidt/Roy].** For  $d \in \mathbb{Z}_+$ , let  $k_d \in [-1, \infty]$  be the supremum of numbers  $w \in \mathbb{R}_+$  s.t. for any  $\alpha \in \mathbb{R}$  not an algebraic integer of degree  $\leq d$  and any  $\varepsilon \in \mathbb{R}_+$ , there exists a constant  $C = C(\alpha, \varepsilon)$  s.t. there are inf.m. real algebraic number  $\xi$  of degree  $d$  satisfying

$$|\alpha - \xi| \leq CH^+(\xi)^{-dk_d + \varepsilon}.$$

Then

- $k_1 = -1$  trivially.
  - $k_2 = 3$ .
  - $k_3 = \frac{3+\sqrt{5}}{2}$ .
  - $\lfloor \frac{d+1}{2} \rfloor \leq k_d \leq d+1$ .
- ┘

*Proof:* 2: One direction is Roth's theorem, the other is [Davenport-Schmidt, Approximation to real number by algebraic integers].

3: [Davenport-Schmidt, Approximation to real number by algebraic integers] and [Approximation by cubic algebraic integers, Roy].

4: (20.2.3.4) and [Davenport-Schmidt, Approximation to real number by algebraic integers] and [Approximation by cubic algebraic integers, Roy]. □

## 5 Dynamics on $\mathbb{T}^1$

References are [Bug12].

### Equidistributions

**Thm. (20.2.5.1) [Weyl's Criterion].** Given  $n \in \mathbb{Z}_+$  and a sequence  $(x_n) \in (\mathbb{R}^n)^{\mathbb{Z}_+}$ , the following are equivalent:

- $(x_n)$  is equidistributed modulo  $\mathbb{Z}^n$ .
- For any  $\mathbb{Z}^n$ -periodic function  $f \in C(\mathbb{R}; \mathbb{C})$ ,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(x_n) = \int_{[0,1]^n} f(x) dx.$$

- For any  $h \in \mathbb{Z}^n \setminus 0$ ,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{2\pi i \sum_{i=1}^n h_i x_i} = 0.$$
┘

*Proof:* 1 and 2 follows easily from the definition of Riemann integration.

2 and 3 follows from the Stone-Weierstrass theorem that the set of trigonometric functions is dense in  $\mathbb{C}_C([0, 1])$  (12.3.8.4). □

**Cor. (20.2.5.2).** For any  $m \in \mathbb{Z}_+$ , if  $\alpha_1, \dots, \alpha_m \in \mathbb{R}$  and  $\{1, \alpha_1, \dots, \alpha_m\}$  is linearly independent over  $\mathbb{Q}$ , then  $(n\alpha_1, \dots, n\alpha_m)_{n \in \mathbb{Z}_+}$  is equidistributed modulo  $\mathbb{Z}^m$ .  $\lrcorner$

*Proof:* The hypothesis implies that for any  $h \in \mathbb{Z}^n \setminus 0$ ,  $\sum h_i \alpha_i \notin \mathbb{Q}$ . Thus it is easy to see that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{2\pi i \sum_{i=1}^N h_i x_i} = 0.$$

□

**Cor. (20.2.5.3).** If  $(x_n) \in \mathbb{R}^{\mathbb{Z}_+}$  is a sequence that is equidistributed modulo 1, and  $(y_n) \in \mathbb{R}^{\mathbb{Z}_+}$  is a convergent sequence, then  $(x_n + y_n)_{n \in \mathbb{Z}_+}$  is also equidistributed modulo 1.  $\lrcorner$

**Thm. (20.2.5.4) [Weyl].** Let  $d \in \mathbb{Z}_+$  and  $P(X) = a_d X^d + \dots + a_1 X + a_0 \in \mathbb{R}[X]$  s.t. at least one of  $a_1, \dots, a_d$  is irrational, then the sequence  $(P(n))_{n \in \mathbb{Z}_+}$  is equidistributed modulo 1.  $\lrcorner$

*Proof:* Cf. [Bug12]P2. ?

□

**Prop. (20.2.5.5).** Given any sequence  $(x_n)_{n \in \mathbb{Z}_+} \in \mathbb{R}^{\mathbb{Z}_+}$  s.t.

$$\liminf_{n \rightarrow \infty} (x_{n+1} - x_n) > 0,$$

then for Lebesgue-a.e.  $\xi \in \mathbb{R}$ , the sequence  $(\xi x_n)_{n \in \mathbb{Z}_+}$  is equidistributed modulo 1.

In particular, this is true for  $x_n = \alpha^n$  where  $\alpha \in \mathbb{R}$ ,  $|\alpha| > 1$ .  $\lrcorner$

*Proof:* Cf. [Distribution modulo 1]P6. ?

□

**Prop. (20.2.5.6).** Given any  $\xi \in \mathbb{R}^\times$ , then for Lebesgue-a.e.  $\alpha \in \mathbb{R}$  with  $|\alpha| > 1$ , the sequence  $(\xi \alpha^n)_{n \in \mathbb{Z}_+}$  is equidistributed modulo 1.  $\lrcorner$

*Proof:* Cf. [Distribution modulo 1]P8. ?

□

### Discrepancy of Distributions

**Def. (20.2.5.7) [Discrepancy].** For  $N \in \mathbb{Z}_+$  and  $x_1, \dots, x_N \in \mathbb{R}$ , the **discrepancy** between  $x_1, \dots, x_N$  is defined to be

$$D_N(x_1, \dots, x_N) \triangleq \sup_{0 \leq u < v \leq 1} \left| \frac{\#\{k \in [N]_+ \mid \{x_k\} \in [u, v)\}}{N} - (v - u) \right|.$$

□

**Prop. (20.2.5.8).** A sequence  $(x_N) \in \mathbb{R}^{\mathbb{Z}_+}$  is equidistributed modulo 1 iff  $\lim_{N \rightarrow \infty} D_N(x_1, \dots, x_N) = 0$ .  $\lrcorner$

*Proof:* This follows from the uniform convergence theorem.  $\square$

**Prop. (20.2.5.9).** For any sequence  $(x_N) \in \mathbb{R}^{\mathbb{Z}_+}$ , there are infinitely many  $N \in \mathbb{Z}_+$  s.t.

$$D_N(x_1, \dots, x_N) \geq \frac{\log N}{25N}.$$

□

*Proof:* Cf. [Sequences, Discrepancies and Applications] or [Uniform distribution of sequences].  $\square$

### Multiplication by Rational Numbers

**Prop. (20.2.5.10).** If  $b \in \mathbb{Z}_{\geq 2}$  and  $\xi \in \mathbb{R} \setminus \mathbb{Q}$ , the numbers  $\{\xi b^n\}_{n \in \mathbb{Z}_+}$  cannot be contained in an interval of length strictly shorter than  $1/b$ .

And this bound is strict, by considering the Sturmian words(37.7.1.4).  $\lrcorner$

*Proof:* Let  $\{\xi\} = [0.a_1a_2\dots]_b$ , then we may assume that there exists some  $l \in [b-2]$  s.t.  $a_i \in \{l, l+1\}$  for any  $i \in \mathbb{Z}_+$ , otherwise it is easy to see that the assertion is true. In this case, by(37.7.1.3), for any  $m \in \mathbb{Z}_+$ , there exists a sequence  $W_m$  of  $m$ -blocks s.t.  $0 \amalg W_m$  and  $1 \amalg W_m$  both appear in the sequence  $a_1, a_2, \dots$ . Then this will imply that there exists two  $u_m, v_m \in \mathbb{Z}_+$  s.t.  $\{\xi b^{u_m}\} - \{\xi b^{v_m}\} > b^{-1} - b^{-m}$ . As  $m$  can be arbitrarily large, the assertion follows.  $\square$

**Prop. (20.2.5.11)** [Mahler[Mah57]]. For any  $\varepsilon \in \mathbb{R}_+$ ,  $p, q \in \mathbb{Z}, p > q \geq 2$ , and  $\gcd(p, q) = 1$ , there exists  $N \in \mathbb{Z}_+$  s.t.  $||(\frac{p}{q})^n|| \geq 2^{-\varepsilon n}$  for any  $n \in \mathbb{Z}_{\geq N}$ .  $\lrcorner$

*Proof:* Apply Roth's theorem(20.2.3.8) with  $K = \mathbb{Q}, S = \{\infty, 2, 3\}$  with  $\alpha_\infty = 1, \alpha_2 = \infty, \alpha_3 = 0$ , and  $\beta_k = 3^k/n2^k$  with  $n = \lceil (\frac{3}{2})^k \rceil$ . Then  $|\beta - \alpha_2|_2 \leq 2^{-k}, |\beta - \alpha_3|_3 \leq 3^{-k}|n|_3^{-1}$ , and  $H(\beta_k) = 3^k|n|_3$ . Thus Roth's theorem says for any  $\varepsilon > 0$ ,

$$|1 - 3^k/n2^k|_\infty \leq 2^k 3^k |n|_3 (3^k |n|_3)^{-2-\varepsilon}$$

holds for only f.m.  $k$ .  $\square$

**Def. (20.2.5.12)** [Mahler's **Z-Numbers**].  $\xi \in \mathbb{R}_+$  is called a **Mahler's Z-number** if  $0 < \{\xi(\frac{3}{2})^n\} < 1/2$  holds for any  $n \in \mathbb{Z}_+$ .  $\lrcorner$

### Multiplication by Algebraic Numbers

**Prop. (20.2.5.13)** [Pisot]. Let  $\alpha \in \mathbb{R}_{>1}, \xi \in \mathbb{R}^\times$  s.t.

$$\sum_{n \in \mathbb{Z}_+} ||\xi \alpha^n||^2$$

converges, then  $\alpha$  is a Pisot-number(19.3.1.15), and  $\xi \in \mathbb{Q}(\alpha)$ .  $\lrcorner$

*Proof:* Cf.[Distribution Modulo 1]P16. **?**  $\square$

**Thm. (20.2.5.14)** [Vijayaraghavan]. Let  $\alpha \in \mathbb{R}_{>1}, \xi \in \mathbb{R}^\times$  s.t.  $(\{\xi \alpha^n\})_{n \in \mathbb{Z}_+}$  has only f.m. limit points, then  $\alpha$  is a Pisot number(19.3.1.15), and  $\xi \in \mathbb{Q}(\alpha)$ .  $\lrcorner$

*Proof:*  $\square$

**Cor. (20.2.5.15).** If  $\xi \in \mathbb{R}^\times$  and  $p, q \in \mathbb{Z}, p > q \geq 2$ , then the sequence  $(\{\xi(\frac{p}{q})^n\})$  has infinitely many limit points.  $\lrcorner$

**Prop. (20.2.5.16).** If  $\alpha \in \mathbb{R}_{\geq 1} \cap \overline{\mathbb{Q}}$  with minimal polynomial  $q_d X^d + \dots + q_1 X + q_0 \in \mathbb{Z}[X]$ , and

$$\limsup_{n \rightarrow \infty} ||\xi \alpha^n|| < \frac{1}{|q_0| + \dots + |q_n|},$$

then  $\alpha$  is a Pisot or Salem number, and  $\xi \in \mathbb{Q}(\alpha)$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (20.2.5.17).** For any Salem number  $\alpha \in \mathbb{R}$ , the sequence  $(\alpha^n)_{n \in \mathbb{Z}_+}$  is dense modulo 1, but not equidistributed modulo 1. ┘

*Proof:* Cf.[Distributions Modulo 1]P57. □

**Prop. (20.2.5.18).** If  $\alpha \in \mathbb{R}$  is a Pisot or Salem number, then there are infinitely many composite numbers in the sequence  $(\lfloor \alpha^n \rfloor)_{n \in \mathbb{Z}_+}$ . ┘

*Proof:* Cf.[Integer parts of powers of Pisot and Salem numbers] ?. □

### Littlewood Conjecture

**Conj. (20.2.5.19) [Littlewood].** For any  $(\xi, \eta) \in \mathbb{R}^2$ ,  $\inf_{q \in \mathbb{Z}_+} q \cdot \{\{q\xi\}\} \cdot \{\{q\eta\}\} = 0$ . ┘

*Proof:* □

**Conj. (20.2.5.20) [Mixed Littlewood Conjecture, Mathan-Teulié].** For any  $p \in \text{Prime}$  and  $\xi \in \mathbb{R}$ ,  $\inf_{q \in \mathbb{Z}_+} q \cdot |q|_p \cdot \{\{q\xi\}\} = 0$ . ┘

## 20.3 Reductions of Schemes

References are [BLR90, Sta].

**Notation (20.3.0.1).**

- Let  $(\mathcal{O}_K, K)$  be a Dedekind domain.
- Let  $X, Y \in \text{Sch}^{\text{ft, sep}}/K$ .

┘

### 1 Schemes over DVR

**Def. (20.3.1.1) [Integral Models].** A **integral model** over  $\mathcal{O}_K$  for  $X$  is a scheme  $X \in \text{Sch}^{\text{ft, sep, flat}}/\mathcal{O}_K$  s.t.  $\mathcal{X} \times_{\mathcal{O}_K} K \cong X$ . It is called a **proper model** if  $\mathcal{X}/\mathcal{O}_K$  is proper, etc..

┘

**Prop. (20.3.1.2) [Models Exist].**  $X$  has a model over some open dense subset of  $\text{Spec } \mathcal{O}_K$ .

┘

*Proof:* Choose a finite covering  $X_i$ , and the transformation isomorphisms  $\varphi_{ij}$  are defined over a finite localization ring of  $\mathcal{O}_K$ . We can take a further localization to make it flat. It is clearly of f.t.

□

**Prop. (20.3.1.3) [Affine Integral Models].** If  $\mathcal{X} \in \text{Sch}/\mathcal{O}_K$  is an affine integral model for  $X \in \text{Aff}/K$ , then  $\Gamma(\mathcal{X}) \rightarrow \Gamma(X)$  is injective (because flatness is equivalent to torsion-free). Thus if  $\mathcal{Y} \in \text{Sch}/\mathcal{O}_K$  is an affine integral model of  $Y$ , then a morphism  $X \rightarrow Y$  extends to a morphism

┘

**Prop. (20.3.1.4) [Generalized Valuation Criterion].**

- If  $X \in \text{Sch}^{\text{sep}}/\mathcal{O}_K$ , then the natural map  $X(\mathcal{O}_K) \rightarrow X(K)$  is injective.
- If  $X \in \text{Sch}^{\text{proper}}/\mathcal{O}_K$ , then the natural map  $X(\mathcal{O}_K) \rightarrow X(K)$  is bijective.

┘

*Proof:* 1: This follows from (7.4.5.42) because  $\text{Spec } K \rightarrow \text{Spec } \mathcal{O}_K$  is dominant.

2: For the last assertion: for any  $f : \text{Spec } K \rightarrow X$ , we can spread out  $f$  into a map  $f : U \rightarrow X$  for some non-empty open subset  $U \subset \text{Spec } \mathcal{O}_K$ .  $\text{Spec } \mathcal{O}_K \setminus U$  is a finite set of closed points  $\mathfrak{p}_i$ . But for each  $\mathfrak{p}_i$ ,  $f$  can be extended to  $f : (\mathcal{O}_K)_{\mathfrak{p}_i} \rightarrow X$  by valuation criterion. This again spread out into a map  $f : V \rightarrow X$  for some non-empty open subset  $V \subset \text{Spec } \mathcal{O}_K$ . The restriction of these two maps to  $U \cap V$  are identical by item 1. So  $f$  extends to  $f : \text{Spec } \mathcal{O}_K \rightarrow X$ .

□

**Prop. (20.3.1.5) [Schematic Closures and Flat Models].** Let  $(\mathcal{O}_K, \mathfrak{m}_K, K) \in \text{CDVR}$  and  $X \in \text{Aff}/K$  with an affine integral model  $\mathcal{X} \in \text{Sch}/\mathcal{O}_K$  (20.3.1.1), then

- If  $Y \subset X$  is a closed subscheme, then the schematic closure  $\mathcal{Y}$  of  $Y$  in  $\mathcal{X}$  (7.4.5.13) is an affine integral model of  $Y$ .
- For any  $R \in \mathbb{C}\text{Ring}^{\text{flat}}/\mathcal{O}_K$ , we have  $\mathcal{Y}(R) = \mathcal{X}(R) \cap Y(R \otimes_{\mathcal{O}_K} K)$  inside  $X(R \otimes_{\mathcal{O}_K} K)$ .
- $(-)_K$  is a bijection between the set of closed flat subschemes of  $\mathcal{X}$  and the set of closed subschemes of  $Y$ , with inverse given by taking schematic closure.

┘

*Proof:* 1: The schematic closure is just  $\mathcal{Y} = \text{Spec}(\Gamma(X)/I(\mathcal{Y})) \text{Spec}(I(Y) \cap \Gamma(\mathcal{O}_X))$ , so  $\Gamma(\mathcal{Y})$  is a subring of  $\Gamma(Y)$ , thus  $\mathfrak{m}_K$ -torsion-free.

2:  $\mathcal{X}(R) \cap Y(R \otimes_{\mathcal{O}_K} K)$  is exactly the set of  $\mathcal{O}_K$ -homomorphism  $\varphi : \Gamma(X) \rightarrow \mathcal{O}_K$  s.t. its  $K$ -linear extension kills  $I(Y)$ , but then this is equivalent to  $\varphi(I(Y)) \text{Spec}(I(Y) \cap \Gamma(\mathcal{O}_X)) = 0$ , because  $R$  is  $\mathfrak{m}_K$ -torsion-free.

3: For a flat subscheme  $\mathcal{Y} \subset \mathcal{X} \in \text{Sch}/\mathcal{O}_K$ , there are exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & I(\mathcal{Y}) & \longrightarrow & \Gamma(\mathcal{X}) & \longrightarrow & \Gamma(\mathcal{Y}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & I(\mathcal{Y}) \otimes_{\mathcal{O}_K} K & \longrightarrow & \Gamma(X) & \longrightarrow & \Gamma(\mathcal{Y}_K) \longrightarrow 0 \end{array}$$

where the vertical maps are all injective.  $\square$

**Prop. (20.3.1.6).** Let  $(\mathcal{O}_K, K) \in \text{DVR}$  be Henselian and  $f : \mathcal{X} \rightarrow \mathcal{Y} \in \text{Sch}/\mathcal{O}_K$ . If  $f$  is of f.t.,  $f_K$  is smooth and  $\mathcal{X}(\widehat{\mathcal{O}_K}) \rightarrow \mathcal{Y}(\widehat{\mathcal{O}_K})$  is surjective, then  $\mathcal{X}(\mathcal{O}_K) \rightarrow \mathcal{Y}(\mathcal{O}_K)$  is also surjective.  $\lrcorner$

*Proof:* For  $y : \text{Spec}(\mathcal{O}_K) \rightarrow \mathcal{Y}$ , let  $\mathcal{X}_y \in \text{Sch}/\mathcal{O}_K$  be the base change of  $y$  along  $f$ . The generic fiber of  $\mathcal{X}_y$  is smooth by assumption, and it has a  $\widehat{\mathcal{O}_K}$ -valued point  $\hat{x}$ .  $\square$

## 2 Smooth Reductions

**Prop. (20.3.2.1) [Smoothness over DVRs].** For  $(\mathcal{O}_K, K, \varpi_K, \kappa) \in \text{DVR}$  and  $\mathcal{X} \in \text{Sch}^{\text{ft}}/\mathcal{O}_K$ ,

- If  $\mathcal{X}/\mathcal{O}_K$  is flat, then it is smooth iff both  $\mathcal{X}_K/K$  and  $\mathcal{X}_\kappa/\kappa$  are smooth.
- If  $\mathcal{X}$  is integral, then it is smooth iff  $\mathcal{X}_K \neq \emptyset$  and both  $\mathcal{X}_K/K$  and  $\mathcal{X}_\kappa/\kappa$  are smooth.

$\lrcorner$

*Proof:* This follows from (7.6.4.5). Notice that when  $\mathcal{X}$  is integral,  $\mathcal{X}/\mathcal{O}_K$  is flat iff  $\varpi_K$  is non-torsion on  $\mathcal{X}$  iff  $\varpi_K \neq 0$  on  $X$ , which means  $X_K \neq \emptyset$ .  $\square$

**Def. (20.3.2.2) [Good Reduction].** If  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$ ,  $X$  is said to have **good reduction** over  $\mathcal{O}_K$  if there is a smooth integral model  $\mathcal{X}/\mathcal{O}_K$  for  $X$  (20.3.1.1). If this is the case, then  $\tilde{X} = \mathcal{X}_k$  is a smooth scheme over  $k$ , called the **good reduction of  $X$** .

In this case, the reduction map  $X(\overline{K}) \cong \mathcal{X}(\overline{K}) \rightarrow \tilde{X}(\overline{k})$  is surjective, by valuation criterion and formal smoothness criterion.  $\lrcorner$

**Def. (20.3.2.3) [Relative Identity Component Groups].** For a Dedekind scheme  $(S, K)$  and  $\mathcal{G} \in \text{GrpSch}^{\text{sm}}/S$ , the **relative identity component**  $\mathcal{G}^0$  is defined to be the open subgroup scheme of  $\mathcal{G}$  deleting all non-identity components of  $\mathcal{G}_{\kappa(s)}$  for all  $s \in |S|_0$ .

If  $\mathcal{G}$  is affine, then so is  $\mathcal{G}^0$ , Cf. [Prasad-Yu, On quasi-reductive group schemes] Lemma 3.5. ?  $\lrcorner$

**Prop. (20.3.2.4) [Smooth Affine Integral Models].** Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  be Henselian and  $X \in \text{Aff}/K$ . If  $\mathcal{X}/\mathcal{O}_K$  is a smooth affine integral model for  $X$  (20.3.1.1), then

$$\Gamma(\mathcal{X}) = \{f \in \Gamma(X) : f(\mathcal{X}(\mathcal{O}_{\tilde{K}})) \subset \mathcal{O}_{\tilde{K}}\}.$$

$\lrcorner$

*Proof:* One direction is clear. For the other, suppose  $f \in \Gamma(X)$  s.t.  $f(\mathcal{X}(\mathcal{O}_{\tilde{K}})) \subset \mathcal{O}_{\tilde{K}}$  but  $f \notin \Gamma(\mathcal{X})$ , then there exists a smallest  $n \in \mathbb{Z}_+$  s.t.  $\varpi_K^n f \in \Gamma(\mathcal{X})$ . Then  $\varpi_K^n f$  reduces to a function  $g$  on  $\mathcal{X}_{\kappa^{\text{sep}}}$  s.t.  $g$  vanishes on the image of the map  $\mathcal{X}(\mathcal{O}_{\tilde{K}}) \rightarrow \mathcal{X}(\kappa^{\text{sep}})$ , by hypothesis on  $f$ . But by smoothness, this image is just  $\mathcal{X}(\kappa^{\text{sep}})$  (5.4.5.1), thus  $g$  vanishes, by (7.4.3.3).  $\square$

**Cor. (20.3.2.5).** Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  and  $\mathcal{X}, \mathcal{Y} \in \text{Sch}/\mathcal{O}_K$  be integral models of  $X = \mathcal{X}_K$  and  $Y = \mathcal{Y}_K$  resp. (20.3.1.1) s.t.  $\mathcal{X}$  is smooth and  $\mathcal{Y}$  is affine. Then a morphism  $\varphi : X \rightarrow Y$  (uniquely) extends to a morphism  $\varphi : \mathcal{X} \rightarrow \mathcal{Y}$  iff  $\varphi(X(\mathcal{O}_{\check{K}})) \subset Y(\mathcal{O}_{\check{K}})$  as subsets of  $Y(\check{K})$  (20.3.1.4).  $\lrcorner$

**Cor. (20.3.2.6).** The case  $X$  is affine follows from (20.3.2.4), and the general case follows by glueing and uniqueness.  $\lrcorner$

**Prop. (20.3.2.7) [Schematic Subsets].** Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  and  $X \in \text{Aff}/K$ . If  $\mathcal{U} \subset X(\check{K})$  is  $\text{Gal}(\check{K}/K)$ -invariant and there exists a smooth integral model  $\mathcal{X} \in \text{Sch}/\mathcal{O}_{\check{K}}$  of  $X_{\check{K}}$  s.t.  $\mathcal{U} = \mathcal{X}(\mathcal{O}_{\check{K}})$ , then  $\mathcal{X}$  is defined over  $\mathcal{O}_K$ .

Such a subset is called a **schematic subset** of  $X(\check{K})$ .  $\lrcorner$

*Proof:* By unramified Galois descent, Cf. [BLR90] Section 6.2. Example B.  $\square$

**Prop. (20.3.2.8).** Let  $(\mathcal{O}_K, K) \in \text{DVR}$  be Henselian.

- If  $X \in \text{Sch}^{\text{sm}}/K$ , then  $X(K)$  is dense in  $X(\widehat{K})$ .
- If  $\mathcal{X} \in \text{Sch}^{\text{sm}}/\mathcal{O}_K$ , then  $\mathcal{X}(\mathcal{O}_K)$  is dense in  $\mathcal{X}(\widehat{\mathcal{O}_K})$ .

$\lrcorner$

*Proof:* 1: By (7.6.6.7), we can assume that  $X$  is standard étale over  $\mathbb{A}_K^n$ , i.e. if  $R = K[T_1, \dots, T_n]$ , then  $X = \text{Spec}(R[Y]_{h,f}/f)$  where  $f(T, Y) \in R[Y]$  is monic in  $Y$  and  $h(T, Y) \in R[Y]$ . Thus a point of  $X(\widehat{K})$  is a tuple  $\mathbf{c} = (c_1, \dots, c_n) \in \widehat{K}^{\oplus n}$  and a simple root  $y \in \widehat{K}$  of  $f(\mathbf{c}, Y)$  with  $h(\mathbf{c}, y) \neq 0$ .

Take  $\mathbf{c}' \in K^{\oplus n}$  close enough to  $\mathbf{c}$ , then by continuity of separable roots,  $f(\mathbf{c}', Y)$  has a unique simple root  $y' \in (\widehat{K})^{\text{sep}}$  that is close to  $y$ . When it is close enough we have  $h(\mathbf{c}', y') \neq 0$ , and then the uniqueness of  $y'$  and (19.1.4.4) implies that  $y \in K$ .

2: This follows from item 1.  $\square$

### 3 Greenberg Functors

References are [Gre61, Gre63, B-GA18, K-P23].

**Notation (20.3.3.1).**

- Let  $(K, \mathcal{O}_k, \mathfrak{m}_K, \varpi_K, \kappa) \in p\text{-NField}$ .

$\lrcorner$

### 4 Néron Models

References are [Serre-Tate, Good Reduction of Abelian Varieties], [BLR90] and <http://virtualmath1.stanford.edu/~conrad/mordellsem/Notes/L11.pdf>.

**Def. (20.3.4.1) [Néron Models].** Let  $(S, K)$  be a Dedekind scheme and  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ , then a **Néron model** for  $X$  is a smooth scheme  $\mathcal{X} \in \text{Sch}^{\text{sm,sep,ft}}/S$  representing the functor

$$\text{Sch}^{\text{sm}}/S \rightarrow \text{Set} : \mathcal{Y} \mapsto \text{Hom}_K(\mathcal{Y} \otimes_S K, X).$$

In particular, the natural map  $\mathcal{X}(S) \rightarrow X(K)$  is a bijection.  $\lrcorner$

**Cor. (20.3.4.2) [Group Structures].** Let  $(S, K)$  be a Dedekind scheme and  $X \in \text{Grp}^{\text{sm,sep,ft}}/K$ , then any Néron model for  $X$  is naturally equipped with a group structure extending that of  $X$ .  $\lrcorner$

*Proof:* This follows from the universal properties.  $\square$

**Prop. (20.3.4.3) [Étale Valuation Criterion for Néron Models].** Let  $(R, K)$  be a DVR and  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ , then if  $\mathcal{X}/R$  is a Néron model for  $X$ , then  $\mathcal{X}$  satisfies valuation criterion for any  $R'$  that is the integral closure of  $R$  in an unramified field extension  $K'/K$ .

Conversely, if  $\mathcal{G} \in \text{GrpSch}^{\text{sm,ft}}/R$ , then  $\mathcal{G}$  is a Néron model of its generic fiber iff the natural map  $\mathcal{G}(R^{\text{sH}}) \rightarrow \mathcal{G}(K^{\text{sH}})$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [BLR90] Prop. 7.1.1.  $\square$

**Prop. (20.3.4.4) [Étale Base Change].** Let  $(R, K)$  be a Dedekind domain and  $\lrcorner$

*Proof:*  $\square$

**Prop. (20.3.4.5) [Unramified Base Change].** Let  $(R, K)$  be a DVR and  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ ,  $K'/K$  an unramified field extension,  $R'$  the integral closure of  $R$  in  $K'$ , then if  $\mathcal{X}$  is a Néron model for  $X$ , then  $\mathcal{X}_{R'}$  is a Néron-model for  $X_{K'}$ .  $\lrcorner$

*Proof:* Because  $R'/R$  is smooth (by (20.3.2.1) and the definition of unramifiedness (19.1.4.5)), for any  $\mathcal{Y}'/R'$  smooth,

$$\text{Mor}_{R'}(\mathcal{Y}', \mathcal{X}_{R'}) = \text{Mor}_R(\mathcal{Y}', \mathcal{X}) = \text{Mor}_K(\mathcal{Y}'_K, X) = \text{Mor}_{K'}(\mathcal{Y}'_{K'}, X_{K'}).$$

$\square$

**Prop. (20.3.4.6) [Étale Descent].** Let  $(R, K) \subset (R', K') \subset (R^{\text{sH}}, K^{\text{sH}}) \in \text{DVR}$ . If  $G \in \text{AlgGrp}^{\text{sm}}/K$  and  $G_{K'}$  has a Néron model  $\mathcal{G}'/R'$ , then  $\mathcal{G}'$  descends to a Néron model  $\mathcal{G}/R$  of  $G$ .  $\lrcorner$

*Proof:* Cf. [BLR90] Prop. 6.5.4.  $\square$

**Cor. (20.3.4.7).** Let  $(R, K) \subset (R', K') \in \text{DVR}$  be an unramified extension and  $G \in \text{Grp}^{\text{sm,ft}}/K$ , then  $G$  has a Néron model iff  $G_{K'}$  has a Néron model.  $\lrcorner$

*Proof:* Cf. [BLR90] Prop. 6.5.4.  $\square$

**Prop. (20.3.4.8) [Localness of Néron Property].** Let  $(S, K)$  be a Dedekind scheme and  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ .

- If  $(S_i)$  is an open cover of  $S$ , then  $X$  has a Néron model over  $S$  iff it has a Néron model over each  $S_i$ .
- $\mathcal{X} \in \text{Sch}^{\text{ft,sep}}/S$  is a Néron model for  $X$  iff for each closed point  $s \in S$ ,  $\mathcal{X} \times_S \text{Spec } \mathcal{O}_{S,s}$  is a Néron model for  $X$  over  $\mathcal{O}_{S,s}$ .  $\lrcorner$

*Proof:* 1: One direction follows from (20.3.4.4), for the other, notice they can be glued by universal properties.

2: Cf. [BLR, Prop 1.2.4]. If  $\mathcal{X}/S$  is a Néron model, then for any closed point  $s \in S$  and any  $\varphi : \mathcal{Y}_{(s)} \in \text{Sch}^{\text{sm}}/\mathcal{O}_{S,s}$  with a morphism  $Y_{(s),K} \rightarrow X$ , we need to show that it can be extended uniquely to a morphism over  $\mathcal{O}_{S,s}$ . We may assume that  $Y_{(s)}$  is f.p., and then it extends to a scheme  $Y/S'$  on a nbhd  $S'$  of  $s$ . Then because  $X/S_i$  is a Néron model by item 1, we can extend the  $\varphi$  over  $S'$ , which restricts to a morphism over  $\mathcal{O}_{S,s}$ . The uniqueness is clear.

The converse is similar. Just notice that smoothness is a stalkwise properties (7.6.4.5).  $\square$

**Prop. (20.3.4.9) [Extending Néron Models].** Let  $(S, K)$  be a Dedekind scheme and  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ , then  $X$  has a Néron model over  $S$  (20.3.4.1) iff there exists an open dense subscheme  $S' \subset S$  s.t.  $X$  has a Néron model over  $S'$ , and for any  $s \in S \setminus S'$ ,  $X$  has a Néron model over  $\mathcal{O}_{S,s}$ .  $\lrcorner$



*Proof:* One direction is obvious by (20.3.4.8), for the other, for each  $s_i \in S \setminus S'$ ,  $\mathcal{X}_i/\mathcal{O}_{S,s_i}$  extends to smooth scheme on a nbhd  $S_i$  of  $s_i$ , and since  $\mathcal{X}_i$  and  $\mathcal{X}/S'$  has isomorphic generic fibers, we can assume that they are isomorphic on  $S_i \cap S'$  after shrinking  $S_i$ . Then the glue together to a Néron model over  $\mathcal{O}_{S,s}$ . Then the assertion follows from (20.3.4.8).  $\square$

### Dilatations

**Thm. (20.3.4.10) [Dilatations].** Suppose  $(\mathcal{O}_K, K, \varpi_K, \kappa) \in \text{DVR}$  and  $\mathcal{X} \in \text{Aff}/\mathcal{O}_K$  is an affine integral model of  $X \in \text{Aff}/K$  (20.3.1.1). For a closed subscheme  $\mathbf{Y} \subset \mathcal{X}_\kappa$ , consider the **affine blow-up algebra**

$$A_{\mathbf{Y}} \triangleq A\left[\frac{I}{\varpi_K}\right] \triangleq \Gamma(\mathcal{X})\{\varpi^{-1}f : f \in I(\mathbf{Y})\} \subset \Gamma(X),$$

then  $\mathcal{X}_{\mathbf{Y}} \triangleq \text{Spec}(A_{\mathbf{Y}}) \in \text{Sch}/\mathcal{O}_K$  is an affine integral model of  $X$  with a natural map  $i_{\mathbf{Y}} : \mathcal{X}_{\mathbf{Y}} \rightarrow \mathcal{X}$ , and represents the functor

$$\mathbb{C}\text{Ring}^{\text{flat}}/\mathcal{O}_K \rightarrow \text{Set} : R \mapsto \{x \in \mathcal{X}(R) | x_\kappa \in Y(R \otimes_{\mathcal{O}_K} \kappa)\},$$

called the **dilatation** of  $\mathcal{X}$  w.r.t.  $\mathbf{Y}$ .  $\lrcorner$

*Proof:*  $A_{\mathbf{Y}}$  is  $\mathfrak{m}$ -torsion-free and f.g. over  $\mathcal{O}_K$ , so  $\mathcal{X}_{\mathbf{Y}}$  is flat and f.g. over  $\mathcal{O}_K$ , and  $A_{\mathbf{Y}}[\varpi_K^{-1}] = \Gamma(X)$ , so  $\mathcal{X}_{\mathbf{Y}}$  is an integral model of  $X$ .

For any  $R \in \mathbb{C}\text{Ring}^{\text{flat}}/\mathcal{O}_K$ ,  $\mathcal{X}_{\mathbf{Y}}(R) \rightarrow \mathcal{X}(R)$  is injective because its composition with  $\mathcal{X}(R) \rightarrow X(K)$  is injective. And it is clear that  $x : \Gamma(\mathcal{X}) \rightarrow R$  extends to a map  $A_{\mathbf{Y}} \rightarrow R$  iff  $x(I) \subset \mathfrak{m}R$ , or equivalently  $x_\kappa \in Y(R \otimes_{\mathcal{O}_K} K)$ .  $\square$

**Cor. (20.3.4.11) [Universal Property].** Situation as in (20.3.4.10), if  $\mathcal{Z} \in \text{Sch}^{\text{flat}}/\mathcal{O}_K$  and  $f : \mathcal{Z} \rightarrow \mathcal{X} \in \text{Sch}/\mathcal{O}_K$  satisfies  $f_\kappa$  factors through  $\mathbf{Y} \subset \mathcal{X}_\kappa$ , then  $f$  factors through  $\mathcal{X}_{\mathbf{Y}}$ .  $\lrcorner$

**Cor. (20.3.4.12) [Dilatations and Blowing-Ups].** Situation as in (20.3.4.10),  $\mathcal{X}_{\mathbf{Y}}$  is the open subscheme of  $\mathbf{Bl}_{\mathbf{Y}}(\mathcal{X})$  (7.8.2.1) consisting of the points where the exceptional invertible sheaf is generated by  $\varpi_K$ .  $\lrcorner$

**Prop. (20.3.4.13) [Smooth Dilatations].** Situation as in (20.3.4.10), if  $\mathcal{X}$  is smooth and  $\mathbf{Y}$  is smooth, then

- $\mathcal{X}_{\mathbf{Y}}/\mathcal{O}_K$  is an affine open subscheme of  $\mathbf{Bl}_{\mathbf{Y}}(\mathcal{X})$  (7.8.2.1), and is smooth.
- $(\mathcal{X}_{\mathbf{Y}})_\kappa \rightarrow \mathbf{Y}$  is an affine open sub-bundle of the projective bundle  $\mathbf{E}_{\mathbf{Y}}(X) \rightarrow \mathbf{Y}$ .
- If moreover the fibers of  $\mathcal{X}$  are connected,  $\mathbf{Y}$  is connected and  $\mathcal{X}_{\mathbf{Y}}(\mathcal{O}_K) \neq \emptyset$ , then the fibers of  $\mathcal{X}_{\mathbf{Y}}$  are connected.  $\lrcorner$

*Proof:*  $\mathcal{X}_{\mathbf{Y}}/\mathcal{O}_K$  is an affine open subscheme of  $\mathbf{Bl}_{\mathbf{Y}}(\mathcal{X})$  by definition, and the special fiber of the map  $i_{\mathbf{Y}} : \mathcal{X}_{\mathbf{Y}} \rightarrow \mathcal{X}$  is given by

$$\Gamma(\mathcal{X})_\kappa \subset \Gamma(\mathcal{X})_\kappa\{\varpi^{-1}f : f \in I(\mathbf{Y})\}/(f_1, \dots, f_n),$$

which clearly factors through  $\Gamma(\mathcal{X})/(f_1, \dots, f_n)$ , which means  $(i_{\mathbf{Y}})_\kappa$  factors through  $\mathbf{Y}$ .

The embedding  $\mathbf{Y} \rightarrow \mathcal{X}$  is a regular immersion (7.6.7.1), because both  $\mathbf{Y}$  and  $\mathcal{X}$  are smooth ?. Thus  $\mathbf{E}_{\mathbf{Y}}(\mathcal{X}) \rightarrow \mathbf{Y}$  is a projective bundle by (7.8.2.9), and  $(\mathcal{X}_{\mathbf{Y}})_\kappa \rightarrow \mathbf{Y}$  is an affine open sub-bundle.

Thus  $(\mathcal{X}_{\mathbf{Y}})_\kappa$  is smooth, and then  $\mathcal{X}_{\mathbf{Y}}$  is smooth by (20.3.2.1). And when  $\mathcal{X}_{\mathbf{Y}}(\mathcal{O}_K) \neq \emptyset$ ,  $(\mathcal{X}_{\mathbf{Y}})_\kappa$  is a non-empty open subset of  $\mathbf{E}_{\mathbf{Y}}(\mathcal{X})$ , thus connected.  $\square$

### Smoothing

**Def. (20.3.4.14) [Smoothing].** For  $(\mathcal{O}_K, K) \in \text{DVR}$  and  $\mathcal{X} \in \text{Sch}^{\text{ft}}/\mathcal{O}_K$  s.t.  $X = \mathcal{X}_K$  is smooth over  $K$ , a **smoothing** of  $\mathcal{X}$  is a map  $f : \mathcal{X}' \rightarrow \mathcal{X} \in \text{Sch}/\mathcal{O}_K$  s.t.

- $f$  is proper and  $f_K$  is an isomorphism.
- For each  $R' \in \mathbb{C}\text{Ring}/\mathcal{O}_K$  étale, the map  $\mathcal{X}'_{\text{sm}}(\mathcal{O}_K) \rightarrow \mathcal{X}(\mathcal{O}_K)$  is a bijection.

Notice that when  $\mathcal{X}'_{\text{sm}}/\mathcal{O}_K$  is affine and separated of f.t., then such  $\mathcal{X}'_{\text{sm}}$  is uniquely determined by (20.3.2.4), and denoted by  $\mathcal{X}^\infty \triangleq \mathcal{X}'_{\text{sm}}$ .  $\lrcorner$

**Thm. (20.3.4.15) [Smoothing Exists].** For  $(\mathcal{O}_K, K) \in \text{DVR}$  and  $\mathcal{X} \in \text{Sch}^{\text{ft}}/\mathcal{O}_K$  s.t.  $X = \mathcal{X}_K$  is smooth over  $K$ , a finite sequence of blowing-up centered in the special fibers gives a smoothing of  $\mathcal{X}$  (20.3.4.14). In particular, if  $X$  is quasi-projective, then so is  $\mathcal{X}'$ .

If  $\mathcal{X}$  is an affine group scheme, we construct inductively the a projective system of integral models  $\{\mathcal{X}^{(i)}\}$  as follows: Let  $\mathbf{Y}^{(i)}$  be the schematic closure of the image of the reduction map  $\mathcal{X}^{(i)}(\mathcal{O}_{K^{\text{sh}}}) \rightarrow \mathcal{X}^{(i)}(\kappa^{\text{sep}})$ , and let  $\mathcal{X}^{(i+1)}$  be the dilatation of  $\mathcal{X}^{(i)}$  w.r.t.  $\mathbf{Y}^{(i)}$  (20.3.4.10).

Notice that  $\mathbf{Y}^{(i)}$  is smooth  $\color{red}?$ , and  $\mathcal{X}^{(i+1)}(\mathcal{O}_{K^{\text{sh}}}) \rightarrow \mathcal{X}^{(i)}(\mathcal{O}_{K^{\text{sh}}})$  is bijective. Then there exists some  $i \in \mathbb{Z}_+$  s.t.  $\mathcal{X}^{(i)}$  is smooth, in particular, the system  $\{\mathcal{X}^{(i)}\}$  stabilizes.  $\lrcorner$

*Proof:*  $\color{red}?$  Cf. [BLR90] Prop 3.3.5. The crux is to show that after each dilatation, the Néron measure of the failure of smoothness at the origin, i.e. the valuation of the minors of the Jacobi matrix decreases.  $\square$

**Cor. (20.3.4.16) [Universal Property in the Affine case].** For  $(\mathcal{O}_K, K) \in \text{DVR}$  and  $\mathcal{X} \in \text{Sch}^{\text{ft}}/\mathcal{O}_K$  s.t.  $X = \mathcal{X}_K$  is smooth affine over  $K$ , the smoothing  $\mathcal{X}^\infty \rightarrow \mathcal{X}$  is the terminal element in the category of all maps from smooth  $\mathcal{O}_K$ -schemes to  $\mathcal{X}$ .  $\lrcorner$

*Proof:* By the formal smoothness property, any  $f : \mathcal{Z} \rightarrow \mathcal{X} \in \text{Sch}/\mathcal{O}_K$  with  $\mathcal{Z}$  smooth affine will lift inductively through the projective system  $\mathcal{X}^{(i)}$  by the universal property of dilatations (20.3.4.11).  $\square$

## 5 Semistable Reductions

## 6 Reductive of Groups

**Prop. (20.3.6.1) [Lie Algebras].** Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  and  $X \in \text{AlgGrp}/K$ . If  $\mathcal{X} \in \text{Grp}/\mathcal{O}_K$  is an affine integral model of  $X$  (20.3.1.1) compatible with the group structure, denote  $\mathcal{I}_e \subset \Gamma[\mathcal{X}]$  the kernel of the identity section  $\Gamma(\mathcal{X}) \rightarrow \mathcal{O}_K$ , then  $f \mapsto df \otimes 1$  is an isomorphism of  $\mathcal{O}_K$ -modules

$$\mathcal{I}_e/\mathcal{I}_e^2 \cong \Omega_{\mathcal{X}/\mathcal{O}_K}^1 \otimes_{\Gamma(\mathcal{X})} \mathcal{O}_K,$$

which is just the cotangent space  $\omega_{\mathcal{X},e}$  of  $\mathcal{X}$  at  $e$ .

Then  $\text{Lie}(\mathcal{X}) = \text{Hom}_{\mathcal{O}_K}(\omega_{\mathcal{X},e}, \mathcal{O}_K)$  and  $\text{Lie}(X) = \text{Hom}_{\mathcal{O}_K}(\omega_{\mathcal{X},e}, K)$ , so the natural map  $\text{Lie}(\mathcal{X}) \otimes_{\mathcal{O}_K} K \rightarrow \text{Lie}(X)$  is an isomorphism and  $\text{Lie}(\mathcal{X})$  is an  $\mathcal{O}_K$ -lattice of  $\text{Lie}(X)$ .

In particular,  $\mathcal{X}$  is smooth iff  $\omega_{\mathcal{X},e}$  is free iff  $\dim_F \text{Lie}(X) = \dim_\kappa \text{Lie}(\mathcal{X}_\kappa)$  iff  $\text{Lie}(\mathcal{X})_\kappa \rightarrow \text{Lie}(\mathcal{X}_\kappa)$  is surjective.  $\color{red}?$   $\lrcorner$

**Prop. (20.3.6.2) [Schematic Closures of Subgroups].** If  $K \in \text{CDVR}$  and  $\mathcal{G} \in \text{GrpSch}^{\text{aff,ft,flat}}/\mathcal{O}_K$  is an integral model of  $G = \mathcal{G}_K \in \text{AlgGrp}/K$ , and  $H \leq G \in \text{AlgGrp}/K$  is an algebraic subgroup, then the schematic closure  $\mathcal{H}$  of  $H$  in  $\mathcal{G}$  is an integral model of  $H$  and a subgroup scheme of  $\mathcal{G}$ , and  $\text{Lie}(\mathcal{H}) = \text{Lie}(H) \cap \text{Lie}(\mathcal{G})$  inside  $\text{Lie}(G)$ .  $\lrcorner$

*Proof:*  $\mathcal{H} \times_{\mathcal{O}_K} \mathcal{H}$  is an integral model of  $H \times H$ , and the multiplication map  $\text{mul.} : \mathcal{G} \times_{\mathcal{O}_K} \mathcal{G} \rightarrow \mathcal{G}$  is the composition of the isomorphism  $(x, y) \mapsto (xy, y)$  and  $\text{pr}_1$ , so it is flat, hence  $\text{mul.}^{-1}(\mathcal{H})$  is an integral model of  $\text{mul.}^{-1}(H)$ , thus  $\mathcal{H} \times_{\mathcal{O}_K} \mathcal{H} \subset \text{mul.}^{-1}(\mathcal{H})$  by (20.3.1.5). Similarly we can prove that the inverse of  $\mathcal{H}$  is in  $\mathcal{H}$  and the identity map factors through  $\mathcal{H}$ , thus proving that  $\mathcal{H}$  is a subgroup scheme of  $\mathcal{G}$ .

For the Lie algebra, we can apply (20.3.1.5) to  $\mathcal{O}_K[\varepsilon]/(\varepsilon^2)$  and notice  $\text{Lie}(\mathcal{H}) = \ker(\mathcal{H}(\mathcal{O}_K[\varepsilon]/(\varepsilon^2)) \rightarrow \mathcal{H}(\mathcal{O}_K))$ .  $\square$

**Prop. (20.3.6.3) [Dilatations of Groups].** Situation as in (20.3.4.10), if  $\mathcal{G} \in \text{GrpSch}/\mathcal{O}_K$  and  $\mathcal{U} \leq \mathcal{G}_\kappa$  is an algebraic subgroup, then  $\mathcal{G}_\mathcal{U} \in \text{GrpSch}/\mathcal{O}_K$  and  $i_\mathcal{U} : \mathcal{G}_\mathcal{U} \rightarrow \mathcal{G}$  is a group homomorphism.

In particular, the smoothing  $\mathcal{G}^\infty \rightarrow \mathcal{G}$  is a group homomorphism.  $\lrcorner$

*Proof:* Use the universal property (20.3.4.11).  $\square$

**Def. (20.3.6.4) [Congruence Group Schemes].** Situation as in (20.3.4.10), if  $\mathcal{G} \in \text{GrpSch}/\mathcal{O}_K$ ,

- For  $n \in \mathbb{N}$ , define  $\mathcal{G}(1 + \mathfrak{m}^n) = \ker(\mathcal{G}(\mathcal{O}_K) \rightarrow \mathcal{G}(\mathcal{O}_K/\mathfrak{m}_K^n))$ .
- Define  $\mathcal{G}_0 = \mathcal{G}$ , and define inductively  $\mathcal{G}_{i+1}$  the dilatation of  $\mathcal{G}_i$  w.r.t. the trivial subgroup of the special fiber. These  $\mathcal{G}_n$  are called **congruence group schemes** of  $\mathcal{G}$ .

$\lrcorner$

### Smooth Reductions

**Prop. (20.3.6.5) [Schematic Subgroups].** Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  and  $X \in \text{linAlgGrp}^{\text{sm}}/K$ . Then a  $\text{Gal}(\check{K}/K)$ -invariant subgroup  $\mathcal{U} \leq X(\check{K})$  is schematic (20.3.2.7) iff there exists an integral model  $\mathcal{X} \in \text{GrpSch}/\mathcal{O}_{\check{K}}$  of  $X_{\check{K}}$  preserving the group structure s.t.  $\mathcal{U} = \mathcal{X}(\mathcal{O}_{\check{K}})$ , because we can take the smoothing  $\mathcal{X}^\infty$  of  $\mathcal{X}$  (20.3.4.15)(20.3.6.3).  $\lrcorner$

**Cor. (20.3.6.6) [Intersection of Schematic Subgroups].** Situation as in (20.3.6.5), if  $\mathcal{U}_1, \mathcal{U}_2 \subset X(\check{K})$  are schematic, then so is  $\mathcal{U}_1 \cap \mathcal{U}_2$ .  $\lrcorner$

*Proof:* Let  $\mathcal{X}_i \in \text{GrpSch}/\mathcal{O}_K$  be smooth integral models of  $X$  s.t.  $\mathcal{X}_i(\mathcal{O}_{\check{K}}) = \mathcal{U}_i$ , which are naturally group schemes by (20.3.2.5), then we can take  $\mathcal{X}_3$  to be the schematic closure of the diagonal of  $X \times_K X$  in  $\mathcal{X}_1 \times_{\mathcal{O}_{\check{K}}} \mathcal{X}_2$ , which is an affine integral model of  $X$  with  $\mathcal{X}_3(\mathcal{O}_{\check{K}}) = \mathcal{U}_1 \cap \mathcal{U}_2$  by (20.3.1.5). And notice  $\mathcal{X}_3$  is a subgroup of  $\mathcal{X}_1 \times_{\mathcal{O}_K} \mathcal{X}_2$ , which can be checked at the generic fiber by the definition of schematic closure. Then we use (20.3.6.5).  $\square$

**Thm. (20.3.6.7) [Schematic Subgroups, Bruhat-Tits-Deligne].** Suppose

- $(\mathcal{O}_K, K) \in \text{DVR}$  is strictly-Henselian,
- $G \in \text{GrpSch}^{\text{aff,sm}}/K$  and  $\mathcal{U} \leq \mathcal{V} \leq G(K)$  are two subgroups,  $\mathcal{U}$  is normal in  $\mathcal{V}$ .
- $\mathcal{U}$  is schematic with smooth affine integral model  $\mathcal{U} \in \text{GrpSch}/\mathcal{O}_K$  (20.3.2.7).

Then

- There exists a smooth separated group scheme  $\mathcal{V} \in \text{GrpSch}/\mathcal{O}_K$  with generic fiber  $G$  s.t.  $\mathcal{V}(\mathcal{O}) = \mathcal{V}$ .
- There exists an open immersion  $\mathcal{U} \rightarrow \mathcal{V} \in \text{GrpSch}/\mathcal{O}_K$  that extends the inclusion  $\mathcal{U} \rightarrow \mathcal{V}$ , and induces isomorphism  $\mathcal{U}^0 \cong \mathcal{V}^0$ .
- If  $[\mathcal{V} : \mathcal{U}] < \infty$ , then  $\mathcal{V}$  is affine, thus is an integral model of  $H$  and  $\mathcal{V}$  is schematic.
- For any local ring extension  $R/\mathcal{O}_K$ , we have  $\mathcal{V}(R) = \mathcal{U}(R) \cdot \mathcal{V}$ .

- If  $(\mathcal{O}_{K'}, K') \in \text{DVR}$  is Henselian and  $(K')^{\text{sh}} = K$ ,  $G$  is defined over  $K$ , and  $\mathcal{V}$  is  $\text{Gal}(K/K')$ -invariant, then  $\mathcal{V}$  is defined over  $\mathcal{O}_{K'}$ . ┘

*Proof:* Cf. [K-P23]P647. □

**Prop. (20.3.6.8) [No  $\mathcal{G}$ -Torsors].** Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  be Henselian and  $\mathcal{G} \in \text{GrpSch}^{\text{aff}, \text{sm}}/\mathcal{O}_K$  with connected fibers, then the map  $H^1(\text{Gal}_\kappa, \mathcal{G}(\mathcal{O}_{\check{K}})) \rightarrow H^1(\text{Gal}_\kappa, \mathcal{G}(\kappa^{\text{sep}}))$  is injective. In particular, if  $\kappa$  is a cd-1 field (9.8.3.18), then  $H^1(\text{Gal}_\kappa, \mathcal{G}(\mathcal{O}_{\check{K}})) = \mathbf{1}$ . ┘

*Proof:* Cf. [K-P23]P285. □

## 7 Reduction of Curves

References are [Sta]Chap55 and [Qing Liu].

**Def. (20.3.7.1) [Semistable and Stable Unions of Curves].** Let  $k \in \text{Field}$ ,  $k$ ,  $C$  an algebraic union of curves over  $k$ , then  $C$  is called a **semistable union of curves** iff  $C_{\bar{k}}$  is reduced and the singular points of  $C_{\bar{k}}$  are all ordinary double points.  $C$  is called a **stable union of curves** if moreover it satisfies:

- $C_{\bar{k}}$  is connected and projective, with  $p_a(C) \geq 2$ .
- Let  $\Gamma$  be an integral component of  $C$  that is isomorphic to  $\mathbb{P}_k^1$ , then it meets other integral components at at least 3 points. ┘

### Minimal Models

**Def. (20.3.7.2) [Minimal Models].** Let  $R$  be a DVR with fraction field  $K$ . Let  $C$  be a smooth projective curve over  $K$ , then a **minimal model for  $C$**  is a regular proper model  $\mathcal{C}$  for  $C$  s.t. for any other such model  $\mathcal{C}'$ , any birational map of models  $\mathcal{C} \rightarrow \mathcal{C}'$  is an isomorphism.

Equivalently,  $\mathcal{C}$  is a regular proper model that doesn't contain an exceptional curve of the first kind, i.e. cannot be blown down? Cf. [Sta]0C21. ┘

**Prop. (20.3.7.3) [Resolution of Singularities for Arithmetic Surfaces].** Let  $(R, K)$  be a DVR,  $C$  a smooth projective curve over  $K$  of genus  $g$ . Then there exists a proper regular model for  $C$ . ┘

*Proof:* Cf. [Sil99] □

**Thm. (20.3.7.4) [Minimal Model for Curves].** If  $(\mathcal{O}_K, K) \in \text{DVR}$ , and  $C$  a smooth projective precurve over  $K$  of genus  $\geq 1$ , then there exists a unique minimal model  $\mathcal{C}$  for  $C$  (20.3.7.2).

In particular, for any other regular proper model  $\mathcal{C}'$ , there exists a unique domination map  $\mathcal{C}' \rightarrow \mathcal{C}$ . ┘

*Proof:* Cf. [Sil99]. □

**Prop. (20.3.7.5) [Semistable Reductions].** If  $(\mathcal{O}_K, K) \in \text{DVR}$ , for  $C \in \text{Cur}^{\text{sm}, \text{proper}}/K$ , the following are equivalent:

- There exists a proper model of  $C$  that is at-most-model of relative dimension 1 over  $\mathcal{O}_K$ .
- There exists a minimal model of  $C$  that is at-most-model of relative dimension 1 over  $\mathcal{O}_K$ .
- Any minimal model of  $C$  that is at-most-model of relative dimension 1 over  $\mathcal{O}_K$ .

If this is the case, then  $C$  is said to have **semistable reduction**. ┘

*Proof:* Cf.[Sta]0CDG. □

**Prop. (20.3.7.6) [Good Reductions].** Let  $R$  be a DVR with fraction field  $K$ , for a smooth projective curve  $C$  over  $K$ , the following are equivalent:

- $C$  has good reduction over  $R$ .
- There exists a minimal model of  $C$  that is smooth over  $R$ .
- Any minimal model of  $C$  is smooth over  $R$ .

┘

*Proof:* Cf.[Sta]0CDI. □

### Good Reduction for Curves

**Prop. (20.3.7.7) [Reduction of Morphisms].** For any morphism  $\varphi : C \rightarrow C'$  of smooth projective curves over  $\mathbb{Q}$  with good reduction at  $p$  and  $g(C') > 0$ , there exists a unique reduction morphism  $\tilde{\varphi} : \tilde{C} \rightarrow \tilde{C}'$  that commutes with the reduction map(20.3.2.2). This defines a functor from the category of smooth projective curves of positive genus with good reduction to the category of smooth curves over  $\mathbb{F}_p$ .

Moreover,  $\deg(\varphi) = \deg(\tilde{\varphi})$ , and by(20.3.2.2), if  $\varphi$  is surjective,  $\tilde{\varphi}$  is also surjective. And  $\varphi$  is an isomorphism iff  $\tilde{\varphi}$  is an isomorphism. ┘

*Proof:* Cf.[BLR90]Prop 9.5.1 ? □

**Cor. (20.3.7.8).** If  $\varphi : E \rightarrow E'$  is an isogeny of elliptic curves over  $\mathbb{Q}$  with good reduction at  $p$ , then the reduction morphism  $\tilde{\varphi} : \tilde{E} \rightarrow \tilde{E}'$  is also an isogeny. ┘

*Proof:* Because it maps  $\tilde{O}_E$  to  $\tilde{O}_{E'}$  and is surjective, this follows from(20.9.1.4). □

**Prop. (20.3.7.9) [Pushforward of Divisors].** Let  $C$  be a smooth projective curve over  $\mathbb{Q}$  with good reduction at  $p$ , then the reduction map  $X(\overline{K}) \rightarrow \tilde{X}(\overline{F}_p)$  induces a map

$$\mathrm{Div}^0(X) \rightarrow \mathrm{Div}^0(\tilde{X})$$

that maps principal divisors to principal divisors, so inducing a map  $\varphi_* : \mathrm{Pic}^0(C) \rightarrow \mathrm{Pic}^0(C')$ , which is compatible with pushforward of divisors along morphisms of curves: if  $h : C \rightarrow C'$  is a morphism of smooth projective curves over  $\mathbb{Q}$  with good reductions at  $p$ , then the following digram is commutative:

$$\begin{array}{ccc} \mathrm{Pic}^0(C) & \xrightarrow{h_*} & \mathrm{Pic}^0(C') \\ \downarrow & & \downarrow \\ \mathrm{Pic}^0(\tilde{C}) & \xrightarrow{\tilde{h}_*} & \mathrm{Pic}^0(\tilde{C}') \end{array}$$

┘

*Proof:* Cf.[BLR90]Prop 9.5.1 ? □

### Classification of Special Fibers of the Minimal Regular Models of Curves of Genus 2

Cf.[Y. Namikawa, K. Ueno, The complete classification of fibers in pencils of curves of genus two, Manuscripta Math. 9 (1973), 143–186.] and [A. P. Ogg, On pencils of curves of genus two, Topology 5 (1966), 355–362.].

## 20.4 Arithmetic Subgroups(Fundamental Domains)

References are [P-R94, Mor15].

**Notation (20.4.0.1).**

- Let  $F \in \mathbf{NField}$ .

┘

**Def. (20.4.0.2)[Arithmetic Subgroups].** For  $G \in \mathbf{linAlgGrp}/\mathbb{Q}$ , an **arithmetic subgroup**  $\Gamma \leq G(\mathbb{Q})$  is a subgroup that is commensurable with  $G(\mathbb{Q}) \cap \mathrm{GL}(n, \mathbb{Z})$  for some embedding  $G \hookrightarrow \mathrm{GL}(n)_{\mathbb{Q}}$ . This notion is independent of the embedding chosen.

If  $G \subset \mathrm{GL}(V)$  and  $L \subset V$  be a lattice, then the arithmetic subgroup  $G(\mathbb{Q}) \cap \mathrm{GL}(n, \mathbb{Z})$  is also denoted by  $G(\mathbb{Z}; L)$ , called the **group of units of  $G$** . If no confusion is make, it is also denoted by  $G(\mathbb{Z})$ .

┘

*Proof:* Cf.[P-R94]P171.

□

**Lemma (20.4.0.3) [Invariance of Arithmetic Subgroups].** For  $\varphi : G \rightarrow G' \in \mathbf{linAlgGrp}/\mathbb{Q}$ , the kernel of an arithmetic subgroup is an arithmetic subgroup.

┘

*Proof:*

□

**Cor. (20.4.0.4).** If  $G = H \ltimes N \subset \mathrm{GL}(V) \in \mathbf{linAlgGrp}/\mathbb{Q}$ , then  $H(\mathbb{Z}) \ltimes N(\mathbb{Z}) \subset G(\mathbb{Z})$  has finite index.

┘

*Proof:* Let  $\mathrm{pr} : G \rightarrow H$  be the projection, then  $H(\mathbb{Z}) \ltimes N(\mathbb{Z}) = \mathrm{pr}^{-1}(H(\mathbb{Z}))$ .

□

**Prop. (20.4.0.5).** If  $\varphi : G \rightarrow G' \in \mathbf{linAlgGrp}/\mathbb{Q}$  is surjective, then  $\varphi$  maps an arithmetic subgroup to an arithmetic subgroup.

┘

*Proof:* ?

□

**Prop. (20.4.0.6).** There exists arithmetic subgroups not of the form  $G(\mathbb{Z}; L)$ .

┘

*Proof:* ? By the proof of(20.4.0.3), any  $G_{\mathbb{Z}}^L$  must contain some congruence subgroup  $G_{\mathbb{Z}}(d)$ , but there are examples of subgroups of finite index of  $SL(2, \mathbb{Z})$  not of this form, Cf.[?R-P94]Chap9.5. □

**Prop. (20.4.0.7).** Let  $G \in \mathbf{linAlgGrp}/\mathbb{Q}$  and  $\Gamma$  be an arithmetic subgroup, then for any f.d. representation  $\rho \in \mathrm{Rep}_{\mathbb{Q}}^{\mathrm{fd}}(G)$ , there exists a  $\Gamma$ -invariant lattice.

┘

*Proof:* Cf.[?R-P94]P173.

□

### 1 Reduction Theory

References are [P-R94]Chap4.

**Thm. (20.4.1.1) [Borel Density Theorem].** Let  $G \in \mathbf{linAlgGrp}/\mathbb{Q}$  be semisimple of non-compact type, then any arithmetic subgroup  $\Gamma$ (20.4.0.2) is Zariski-dense in  $G$ .

┘

*Proof:* Cf.[?R-P94]P205. ?

□

### Fundamental Domains

**Thm. (20.4.1.2)[Open Fundamental Domains].** If  $G \in \mathcal{AlgGrp}/F$ , then there exists an open fundamental set  $\Omega \subset G(\mathbf{A}_{F,\infty})$  relative to  $G_{\mathcal{O}_F}$ , i.e.

- $\mathcal{K}\Omega = \Omega$  for suitable maximal compact subgroup  $\mathcal{K} \subset G(\mathbf{A}_{F,\infty})$ .
- $\Omega G_{\mathcal{O}_F} = G(\mathbf{A}_{F,\infty})$ .
- $\#\Omega^{-1}.\Omega \cap xG_{\mathcal{O}_F}y < \infty$  for any  $x, y \in G(K)$ .

┘

*Proof:* Cf.[P-R94]P277.

□

**Thm. (20.4.1.3)[Finite Presentedness].** If  $G \in \mathcal{AlgGrp}/F$ , then  $G_{\mathcal{O}_F}$  is f.p..

┘

*Proof:* Cf.[P-R94]P277.

□

### Compactness of $G(\mathbf{A}_{F,\infty})/G_{\mathcal{O}_F}$

**Prop. (20.4.1.4).** Let  $S \in \text{lin}\mathcal{AlgGrp}/\mathbb{Q}$  be a torus, then the following are equivalent:

- $S$  is  $\mathbb{Q}$ -anisotropic.
- $S(\mathbb{R})/S(\mathbb{Z})$  is compact.

┘

*Proof:* Cf.[?R-P94]P205.?

□

**Thm. (20.4.1.5)[Mostow-Tamagawa[M-T62]].** Let  $G \in \mathcal{AlgGrp}/F$ , then the following conditions are equivalent:

- $G(\mathbf{A}_{F,\infty})/G_{\mathcal{O}_F}$  is compact.
- The reductive part of  $G^0$  is anisotropic over  $F$ .

┘

*Proof:* Cf.[?R-P94]P210.?

□

**Prop. (20.4.1.6)[Mahler's Criterion].** A subset  $\Omega \subset GL(n; \mathbb{R})$  is compact modulo  $GL(n; \mathbb{Z})$  iff it satisfies:

- $\{\det(g) | g \in \Omega\}$  is bounded.
- $[\Omega.(\mathbb{Z}^n \setminus \{0\})] \cap U = \emptyset$  for some nbhd  $U$  of  $0 \in \mathbb{R}^n$ .

┘

*Proof:* Cf.[?R-P94]P211.?

□

### Finiteness of the Volume of $G(\mathbf{A}_{F,\infty})/G_{\mathcal{O}_F}$

**Thm. (20.4.1.7).**  $G(\mathbf{A}_{F,\infty})/G_{\mathcal{O}_F}$  has finite volume iff  $X(G^0) = 1$ , e.g.  $G \in \text{RedGrp}^{\text{ss}}/F$ .

┘

*Proof:* Cf.[?R-P94]P220.

□



### Finite Arithmetic Groups

**Prop. (20.4.1.8).** Let  $G \in \text{AlgGrp}/F$ , then there exists only f.m. conjugacy classes of finite subgroups of  $G_{\mathcal{O}_F}$ . ┘

*Proof:* ? □

**Conj. (20.4.1.9).** If  $G \leq \text{GL}(n)_{\mathbb{Q}} \in \text{AlgGrp}/\mathbb{Q}$  is of compact-type, then for any totally real extension  $F/\mathbb{Q}$ ,  $G(\mathcal{O}_F) = G(\mathbb{Z})$ . ┘

*Proof:* □

**Remark (20.4.1.10).** By (20.4.1.11), it suffices to prove this conjecture for  $\text{O}(V)$  for any inner space  $V/\mathbb{Q}$ . ┘

**Prop. (20.4.1.11).** If  $G \leq \text{GL}(n) \in \text{AlgGrp}/\mathbb{Q}$  satisfies  $G(\mathbb{R})$  is compact and Zariski dense in  $G$ , then  $G$  preserves a quadratic definite quadratic form  $f$ . ┘

*Proof:* Cf. [R-P94]P230. □

**Prop. (20.4.1.12) [Bartels-Kitaoka].** Let  $F/\mathbb{Q}$  be a totally real nilpotent Galois extension, and  $\Gamma \leq \text{GL}(n; \mathcal{O})$  is a finite  $\text{Gal}(F/\mathbb{Q})$ -invariant subgroup, then  $\Gamma \subset \text{GL}(n; \mathbb{Z})$ . ┘

*Proof:* Cf. [R-P94]P234. □

**Prop. (20.4.1.13).** Let  $V/\mathbb{Q}$  be an inner space and  $A = (a_{ij}) \in \text{GL}(n; \mathbb{Z})$  be an integral Gram matrix for  $V$ . If  $a_{ii} \leq 4\lambda$  for any  $i$ , where  $\lambda$  is the smallest eigenvalue of  $A$ , then for any totally real extension  $F/\mathbb{Q}$ ,  $\text{O}(V; \mathcal{O}_F) = \text{O}(V; \mathbb{Z})$ . ┘

*Proof:* Cf. [R-P94]P236. □

## 2 Congruence Subgroups

**Def. (20.4.2.1) [Congruence Subgroups].** Let  $G \in \text{AlgGrp}/\mathbb{Q}$  with an embedding  $G \hookrightarrow \text{GL}(n)_{\mathbb{Q}}$ , for  $N \in \mathbb{Z}_+$ , define

$$\Gamma(N) = G(\mathbb{Q}) \cap \ker(\text{GL}(n; \mathbb{Z}) \rightarrow \text{GL}(n; \mathbb{Z}/(N)))$$

and a **congruence subgroup** of  $G(\mathbb{Q})$  is any subgroup of  $G(\mathbb{Q})$  that contains some  $\Gamma(N)$  as a finite index subgroup.

The notion is compatible with that defined in (20.5.3.24). In particular, the set of congruence subgroups of  $G$  doesn't depend on the embedding ?. ┘

**Prop. (20.4.2.2) [Congruence Subgroup Problem].** Let  $G \in \text{RedGrp}/\mathbb{Q}$ .

- If  $G$  is split simply connected other than  $\text{SL}(2)$ , then every arithmetic subgroup of  $G$  is a congruence subgroup.
  - If  $G = \text{SL}(2)$  or non-simply connected, there are many arithmetic subgroups of  $G$  that is not congruence subgroups.
- ┘

*Proof:* Cf. [MATSUMOTO 1969. Sur les sous-groupes arithmetiques des groupes semi-simples deployes]. [SURY, B. 2003. The congruence subgroup problem, volume 24 of Texts and Readings in Mathematics. Hindustan Book Agency, New Delhi.] □



**Prop. (20.4.2.3).** The image of a congruence subgroup may not be a congruence subgroup.  $\lrcorner$

*Proof:*

□

**Prop. (20.4.2.4)[Minkowski].** The congruence subgroup  $1+p\mathrm{GL}(n;\mathbb{Z})$  is torsion-free for  $p \in \mathrm{Prime}_{\geq 3}$ .

$\lrcorner$

*Proof:* Cf.[P-R94]P232.

□

### 3 Adelic $S$ -Arithmetic Subgroups

References are [P-R94]Chap5.4.

### 4 Arithmetic Subgroup of Lie Groups

**Def. (20.4.4.1)[Arithmetic Subgroups].** Let  $H$  be a connected real Lie group, then an **arithmetic subgroup**  $\Gamma \subset H$  is a subgroup s.t. there exists an algebraic group  $G \in \mathrm{AlgGrp}/\mathbb{Q}$ , a surjective homomorphism  $G(\mathbb{R})^0 \rightarrow H$  with compact kernel, and  $\Gamma_0 \subset G(\mathbb{Q})$  an arithmetic subgroup(20.4.0.2) s.t.  $\Gamma_0 \cap G(\mathbb{R})^0$  is mapped to  $\Gamma$ .  $\lrcorner$

**Prop. (20.4.4.2).** Let  $H$  be a semisimple real Lie group that admits a faithful f.d. representation, then every arithmetic subgroup of  $H$  is discrete of finite covolume, and contains a torsion-free subgroup of finite index.  $\lrcorner$

*Proof:* Cf.[?Mil17b]P34.

□

**Prop. (20.4.4.3).** In any connected real Lie group there are only countably many arithmetic subgroups up to conjugacy.  $\lrcorner$

*Proof:* Cf.[Mor15]5.1.20.

□

**Thm. (20.4.4.4)[Margulis].** Every discrete subgroup of finite volume in a non-compact simple real Lie group  $H$  is arithmetic unless  $H$  is isogenous to  $SO(1, n)$  or  $SU(1, n)$ .

Note that  $SL(2, \mathbb{R})$  is isogenous to  $SO(1, 2)$ , so the theorem doesn't apply to it.  $\lrcorner$

*Proof:* Cf.[Mor15]5.2.?

□

### 5 Fuchsian Groups of $\mathrm{PGL}(2; \mathbb{R})$

References are [Miy06].

**Def. (20.4.5.1)[Limitin points].** For a subgroup  $\Gamma \leq \mathrm{PGL}(2; \mathbb{C})$ , denote  $\Lambda(\Gamma) \subset \mathbb{P}^1(\mathbb{C})$  the set of points  $z \in \mathbb{P}^1(\mathbb{C})$  s.t. there exists  $w \in \mathcal{H}$  and a infinite sequence  $(\gamma_n)_{n \in \mathbb{Z}_+}$  of distinct elements in  $\Gamma$  s.t.  $\lim_n \gamma_n w = z$ .

Then it is the smallest non-empty closed  $\Gamma$ -invariant subset of  $\mathbb{P}^1(\mathbb{C})$ . And we denote  $\Omega(\Gamma) = \mathbb{P}^1(\mathbb{C}) \setminus \Lambda(\Gamma)$ .  $\lrcorner$

*Proof:* ?

□

**Def. (20.4.5.2)[Fuchsian Groups].**  $\Gamma \leq \mathrm{PGL}(2; \mathbb{R})$  is called a **Fuchsian group** if?  $\lrcorner$

**Def. (20.4.5.3)[Geometrically Generated Fuchsian Groups].** A geometrically generated Fuchsian group  $\Gamma \leq \mathrm{SL}(2; \mathbb{R})$  is a Fuchsian group (20.4.5.2) s.t. a Dirichlet fundamental domain for  $\Gamma$  has f.m. edges.  $\lrcorner$

**Def. (20.4.5.4)[Elliptic Points].** A point  $z \in \mathcal{H}$  is called a **elliptic point** if it is the fixed point of an elliptic element  $\gamma$  of  $\Gamma$  (16.6.1.2).  $\lrcorner$

**Prop. (20.4.5.5).** Let  $\Gamma$  be a discrete subgroup of  $\mathrm{SL}(2, \mathbb{R})$  and  $z$  an elliptic point of  $\Gamma$ , then the stabilizer  $\Gamma_z$  of  $z$  in  $\Gamma$  is a finite cyclic subgroup.  $\lrcorner$

*Proof:* Because  $\mathrm{SL}(2, \mathbb{R})$  acts transitively on  $\mathcal{H}$ , by conjugacy we can assume the elliptic point is  $i$  for another  $\Gamma'$ . Then the stabilizer of  $i$  in  $\mathrm{SL}(2, \mathbb{R})$  is  $\mathrm{SO}(2, \mathbb{R}) \cong S^1$ , and  $\mathrm{SO}(2, \mathbb{R}) \cap \Gamma'$  is a compact and discrete subgroup, so it is finite cyclic.  $\square$

**Def. (20.4.5.6)[Fuchsian Groups of the First Kind].** For  $n \in \mathbb{Z}_+$ , a **Fuchsian group of the first kind** is a discrete subgroup  $\Gamma \subset \mathrm{SL}(n; \mathbb{R})$  s.t.  $\mathrm{Vol}(\mathrm{SL}(n; \mathbb{R})/\Gamma) < \infty$ .  $\lrcorner$

**Prop. (20.4.5.7)[Classifying Fuchsian Groups].** Suppose  $\Gamma \leq \mathrm{SL}(2; \mathbb{R})$  is a Fuchsian group (20.4.5.2), and let  $\Lambda(\Gamma) \subset \mathbb{P}^1(\mathbb{R})$  denote the limit points of  $\Gamma$ , then

- Any  $x \in \mathbb{P}^1(\mathbb{R})$  that is fixed by some element in  $\Gamma \setminus Z(\Gamma)$  lies in  $\Lambda(\Gamma)$ . In particular,  $\Lambda(\Gamma)$  contains all elliptic and cusp points of  $\Gamma$ .
- $\Gamma$  is countable because it is discrete in  $\mathrm{SL}(2; \mathbb{R})$ , thus there are uncountably many non-elliptic points in  $\mathcal{H}$ . For any non-elliptic point  $w \in \mathcal{H}$ ,  $\Lambda(\Gamma)$  equals the set of limit points of  $\Gamma.w$ . In particular,  $\Lambda(\Gamma)$  is  $\Gamma$ -invariant.
- If  $x \in \Lambda(\Gamma)$  is fixed by a hyperbolic  $\gamma \in \Gamma$  and  $y \in \Lambda(\Gamma)$  is not fixed by  $\gamma$ , then either  $\lim_{n \rightarrow \infty} \gamma^n y = x$  or  $\lim_{n \rightarrow \infty} \gamma^{-n} y = x$ . In particular,  $x$  is a limit point of  $\Lambda(\Gamma)$ .
- If  $x \in \Lambda(\Gamma)$  is not a hyperbolic point and  $\Gamma$  contains a hyperbolic  $\gamma$ , then  $x$  is a limit of hyperbolic points.
- If  $\#\Lambda(\Gamma) > 2$ , then  $\Gamma$  contains a hyperbolic element  $\gamma$  and  $\Lambda(\Gamma)$  is perfect, thus uncountable.
- If  $\#\Lambda(\Gamma) \neq \mathbb{P}^1(\mathbb{R})$  is perfect, then it is nowhere dense.

In particular, there are three kinds of Fuchsian groups:

- **elementary Fuchsian groups** are those with  $\#\Lambda(\Gamma) \leq 2$ .
- **Fuchsian groups of first kind** are those with  $\#\Lambda(\Gamma) = \mathbb{P}^1(\mathbb{R})$ .
- **Fuchsian groups of second kind** are those with  $\#\Lambda(\Gamma) \subset \mathbb{P}^1(\mathbb{R})$  perfect and nowhere dense.  $\lrcorner$

*Proof:* ?

4: Let  $y$  be a fixed point of  $\gamma$ . Choose  $w \in \mathcal{H}$  be non-elliptic for  $\Gamma$ , then there exists different  $\alpha_n \in \Gamma$  s.t.  $\lim_n \alpha_n w = y$ , and also different  $\beta_n \in \Gamma$  s.t.  $\lim_n \beta_n w = x$ , Then  $\lim_n \beta_n \alpha_n^{-1} y = x$ , and each  $\beta_n \alpha_n^{-1} y$  is a hyperbolic point.  $\square$

## Modular Groups

**Def. (20.4.5.8)[Principle Congruence Subgroups].** For  $n \in \mathbb{Z}_+$ , there are Fuchsian subgroups:

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) = \mathrm{SL}(n; \mathbb{Z})$$

where

$$\Gamma(N) = \ker(\mathrm{SL}(n; \mathbb{Z}) \rightarrow \mathrm{SL}(n; \mathbb{Z}/(n)),$$

$$\Gamma_1(N) = \left\{ \begin{bmatrix} A & B \\ C & d \end{bmatrix} \in \mathrm{GL}(n; \mathbb{Z}) \mid A \in \Gamma(N)_{n-1}, d \in \Gamma(N)_1, c \in N \cdot \mathrm{Mat}(1 \times n; \mathbb{Z}) \right\}.$$

$$\Gamma_0(N) = \left\{ \begin{bmatrix} A & B \\ C & d \end{bmatrix} \in \mathrm{GL}(n; \mathbb{Z}) \mid A \in \mathrm{Mat}(n; \mathbb{Z}), d \in \mathbb{Z}, c \in N \cdot \mathrm{Mat}(1 \times n; \mathbb{Z}) \right\}.$$

┘

**Def. (20.4.5.9) [Congruence Subgroups].** A **congruence subgroup** is a Fuchsian subgroup  $\Gamma \subset \mathrm{SL}(n; \mathbb{Z})$  that contains  $\Gamma(N)$  for some  $N \in \mathbb{Z}_+$ . In particular, a congruence subgroup has finite index in  $\Gamma(1)$ . ┘

**Def. (20.4.5.10) [Dirichlet Character for  $\Gamma_0(N)$ ].** For  $N \in \mathbb{Z}_+$  and  $\chi \in \mathrm{Diri}(N)$ , we can define a character of  $\Gamma_0(N)$ :

$$\chi_d \left( \begin{bmatrix} A & B \\ C & d \end{bmatrix} \right) = \chi(d).$$

┘

**Def. (20.4.5.11) [Fricke Involution].** For  $N \in \mathbb{Z}_+$ , denote

$$w_N = \begin{bmatrix} & -1 \\ N & \end{bmatrix}, \quad W_N = \frac{1}{\sqrt{N}} w_N.$$

Then  $W_N^2 = -1$ , and  $\Gamma_0(N)$  is stabilized by  $w_N$  and  $W_N$ . Denote

$$\Gamma_0^+(N) = \langle \Gamma_0, W_N \rangle \subset \mathrm{SL}(2, \mathbb{R}).$$

┘

*Proof:*

$$w_N \begin{bmatrix} a & b \\ c & d \end{bmatrix} w_N^{-1} = \frac{-1}{N} \begin{bmatrix} Nd & c \\ N^2b & Na \end{bmatrix} = \begin{bmatrix} -d & -c/N \\ -Nb & -a \end{bmatrix}.$$

□

**Prop. (20.4.5.12).**  $\Gamma_0^+(4) = \left\langle T = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}, W_4 \right\rangle$ . In fact, if  $\tilde{T} = W_4 T W_4^{-1} = \begin{bmatrix} 1 & \\ 4 & 1 \end{bmatrix}$ , then  $\overline{\Gamma_0(4)} = \overline{\langle T, \tilde{T} \rangle} \in \mathrm{PSL}(2; \mathbb{R})$ . ┘

*Proof:*

□

**Lemma (20.4.5.13) [Cartan Decomposition].** There is a complete set of coset representatives for

$$\mathrm{SL}(2, \mathbb{Z}) \backslash \mathrm{GL}(2, \mathbb{Q})^+ / \mathrm{SL}(2, \mathbb{Z})$$

consisting of the diagonal matrices  $\mathrm{diag}(d_1, d_2)$ , where  $d_1, d_2 \in \mathbb{Q}^\times$  and  $d_1/d_2$  is a positive integer.

More generally, for  $N > 0$ ,

$$\mathrm{GL}(2, \mathbb{Q})^+ = \coprod_{d_1, d_2 \in \mathbb{Q}^\times, d_1/d_2 \in \mathbb{Z}_+} \mathrm{SL}(2, \mathbb{Z}) \begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix} \Gamma_0(N).$$

Also, let  $\Sigma$  be the set of primes dividing  $N$ ,  $\mathbb{Z}_\Sigma$  be the localization of  $\mathbb{Z}$  at these primes, and  $G_0(N)$  be the subgroup of  $\mathrm{GL}(2, \mathbb{Z}_\Sigma)^+$  consisting of matrices  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  s.t.  $c \in N\mathbb{Z}_\Sigma$ . Then

$$G_0(N) = \coprod_{d_1, d_2 \in \mathbb{Z}_\Sigma, d_1/d_2 \in \mathbb{Z}_+} \Gamma_0(N) \begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix} \Gamma_0(N).$$

┘

*Proof:* The first assertion follows from (16.5.5.5), noticing the sign.

For the second assertion, for any  $g \in G_0(N)$ , let  $g = \gamma_1 \mathrm{diag}(d_1, d_2) \gamma_2$  where  $D = d_1/d_2 \in \mathbb{Z}$  and  $(D, N) = 1$ . Then for any  $\begin{bmatrix} a & b \\ Dc & d \end{bmatrix} \in \Gamma_0(N)$ , we can change  $(\gamma_1, \gamma_2)$  to  $(\gamma_1 \begin{bmatrix} a & Db \\ c & d \end{bmatrix}, \begin{bmatrix} a & b \\ Dc & d \end{bmatrix} \gamma_2)$ , in this way if  $\gamma_2 = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$ , we can change  $z$  to  $Dxc + dz$ . Now take  $k$  s.t.  $(kx + z, N) = 1$ , and let  $c$  be chosen s.t.  $N \mid D(kz + x)c + z$ , and  $d = kDc + 1$ , then  $(Dc, d) = 1$ , and we can find  $a, b$  s.t.  $ad - Dbc = 1$ . Then for this  $(a, b, c, d)$ ,  $\begin{bmatrix} a & b \\ Dc & d \end{bmatrix} \gamma_2 \in \Gamma_0(N)$ .

Finally, if  $g \in G_0(N)$  and  $\gamma_2 \in \Gamma_0(N)$ , then automatically  $\gamma_1 \in \Gamma_0(N)$ . □

**Prop. (20.4.5.14) [Conjugate of Congruence Subgroup].** If  $\Gamma$  is a congruence subgroup of level  $N$  and  $\alpha \in \mathrm{GL}(2, \mathbb{Q})^+$ , then  $\alpha^{-1}\Gamma\alpha$  is a congruence subgroup.

In particular,  $\alpha\Gamma\alpha^{-1} \cap \Gamma$  is also a congruence subgroup, and if  $\alpha \in \mathrm{SL}(2, \mathbb{Z})$  with  $\delta(\alpha) = D$ , then  $\alpha^{-1}\Gamma\alpha$  is a congruence subgroup of level  $DN$ . ┘

*Proof:* Let  $M_1, M_2$  be positive integers s.t.  $M_1\alpha, M_2\alpha^{-1} \in \mathrm{Mat}(2; \mathbb{Z})$ . (In the second case we can take  $M_1 = 1, M_2 = D$ ). If  $A$  is a matrix that  $A \equiv 1 \pmod{M_1M_2N}$ , then  $\alpha^{-1}(A - 1)\alpha \equiv 1 \pmod{N}$ . Thus  $\alpha^{-1}\Gamma\alpha$  contains  $\Gamma(M_1M_2N)$ . □

**Cor. (20.4.5.15) [Congruence Topology].** We can define a topology on the group  $\mathrm{GL}(2, \mathbb{Q})^+$  given by a basis at the origin given by the open subgroup  $\Gamma(N)$ . Then (20.4.5.14) can be used to show that this is truly a topology. In fact, its restriction on  $\mathrm{SL}(2, \mathbb{Z})$  is the same as the topology given by the inclusion  $\mathrm{SL}(2, \mathbb{Z}) \subset \mathrm{SL}(2, \widehat{\mathbb{Z}})$ .

But unfortunately,  $\mathrm{SL}(2, \mathbb{Z})$  is not compact in this topology, because otherwise  $\mathrm{SL}(2, \mathbb{Z})$  is closed in  $\mathrm{SL}(2, \widehat{\mathbb{Z}})$ , contradiction. ┘

**Lemma (20.4.5.16).** The action of  $\Gamma(1) = \mathrm{SL}(2, \mathbb{Z})$  on  $\mathcal{H}$  is properly discontinuous (4.12.1.13). ┘

*Proof:* Cf. [Bump P18]. □

**Prop. (20.4.5.17).** The subset  $F = \{z \in \mathcal{H} \mid |\mathrm{Re}(z)| < 1/2, |z| > 1\}$  is a fundamental domain for  $\Gamma(1) = \mathrm{SL}(2, \mathbb{Z})$ , and moreover, let  $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$  and  $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ , then  $S^2 = -1$ ,  $(ST)^3 = (TS)^3 = -1$ , and

(1) two elements  $z, z'$  of  $\overline{F}$  are equivalent under  $\Gamma(1)$  iff

- (a)  $\operatorname{Re}(z) = -1/2$  and  $z' = z + 1$ , then  $z' = T(z)$ .  
 (b)  $|z| = 1$  and  $z' = -\frac{1}{z}$ , then  $z' = S(z)$ .  
 (2) Let  $z \in \overline{F}$ , if the stabilizer of  $z$  is not  $\pm 1$ , then  
 (a)  $z = i$  and  $\operatorname{Stab}(i) = \langle S \rangle$ .  
 (b)  $z = \rho = e^{2\pi i/6}$  and  $\operatorname{Stab}(\rho) = \langle TS \rangle$ .  
 (c)  $z = \omega$ , and  $\operatorname{Stab}(\omega) = \langle ST \rangle$ .

┘

*Proof:* 1: Let  $\Gamma'$  be the subgroup of  $\Gamma(1)$  generated by  $S$  and  $T$ . By (12.4.1.9),

$$\operatorname{Im}(\gamma(z)) = \frac{\operatorname{Im}(z)}{|cz + d|^2}.$$

and there is a  $\gamma \in \Gamma'$  that  $|cz + d|$  attains the minimal value, then  $\operatorname{Im}(\gamma(z))$  attains a maximal value. Now there is a  $n$  that  $z' = \operatorname{Re}(T^n \gamma(z)) \in [-1/2, 1/2]$ . Now I claim that  $|z'| \geq 1$ , because otherwise

$$\operatorname{Im}(Sz') = \operatorname{Im}(-1/z') = \frac{\operatorname{Im}(z')}{|z'|^2} > \operatorname{Im}(z'),$$

contradiction.

2: Suppose  $z, z' \in \overline{F}$  are  $\Gamma$ -conjugate, we can assume  $\operatorname{Im}(z) \leq \operatorname{Im}(z')$ . Suppose  $z' = \gamma(z)$  and  $z = x + iy$ , then

$$(cx + d)^2 + (cy)^2 \leq 1.$$

Then  $|c| < 2$ . If  $c = 0$ , then  $d = \pm 1$ , and  $\gamma$  is a translation, thus we are in case 1.(a). If  $c = 1$ , then  $d = 0$ , unless  $z = \rho$  or  $\omega$ . If  $d = 0$ , then  $\gamma = \begin{bmatrix} a & 1 \\ -1 & \end{bmatrix}$ , thus  $\gamma(z) = a - \frac{1}{z}$ . If  $a = 0$ , then we are in case 1.(b), and if  $a \neq 0$ , then  $z = \rho$  or  $\omega$ . and the  $c = -1$  case is similar.  $\square$

**Prop. (20.4.5.18)[Cusps and Elliptic Points for  $\Gamma(1)$ ].**

- The cusps of  $\Gamma(1)$  are exactly  $\mathbb{P}^1(\mathbb{Q})$ , and each of them is  $\Gamma(1)$ -equivalent to  $\infty$ .
- The elliptic points of  $\Gamma(1)$  are exactly those that are  $\Gamma(1)$ -conjugate to  $i$  or  $\rho = e^{\pi i/3}$ , with corresponding stabilizer group cyclic of order 4 and 6.

┘

*Proof:* 1: Clearly  $\infty$  is fixed by the parabolic matrix  $T = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$ . Now for any  $m/n$  that  $m, n$  is coprime, there exists integers  $r, s$  that  $rm - sn = 1$ . Let  $\gamma = \begin{bmatrix} m & s \\ n & r \end{bmatrix}$ , then  $\gamma(\infty) = m/n$  thus  $m/n$  is also a cusp. Conversely, every parabolic matrix is conjugate to  $T$ , thus its fixed point is conjugate to  $\infty$ , which means the fixed point is in  $\mathbb{Q} \cup \{\infty\}$ .

2: For the elliptic points of  $\Gamma(1)$ , use (20.4.5.17).  $\square$

## 6 Schottky Groups of $\operatorname{PGL}(2; \mathbb{C})$

References are [Mumford, An analytic construction of degenerating curves over complete local fields].

**Def. (20.4.6.1)[Schottky Groups].** A **Schottky group** is a discrete f.g. free subgroup  $\Gamma \subset \mathrm{PSL}(2; \mathbb{C})$  that is purely loxodromic (i.e. has eigenvalues  $\lambda, \lambda^{-1}$  where  $|\lambda| \neq 1$ ).  $\lrcorner$

**Thm. (20.4.6.2)[Maskit].** If  $\Gamma \subset \mathrm{PSL}(2; \mathbb{C})$  is a Schottky group, then  $\mathcal{H}^3 \cup \Omega(\Gamma)/\Gamma$  (20.4.5.1) is homeomorphic to a split torus with  $n$ -handles with boundary  $X = \Omega(\Gamma)/\Gamma$  which is homeomorphic to a surface of genus  $n = \mathrm{rank}(\Gamma_{\mathrm{ab}})$ . This is called a **Schottky uniformization** of  $X$ .  $\lrcorner$

*Proof:* Cf.[Mumford, An analytic construction of degenerating curves over complete local fields]. ?  
 $\square$

## 20.5 Arithmetic of Algebraic Groups

References are [Tit79, P-R94, Mil17, K-P23].

### Notation(20.5.0.1).

- Use notations on group schemes defined in [Group Schemes I: Structure Theory](#).
- Use notations on reductive group schemes defined in [Group Theory III: Reductive Groups](#).
- Use notations on  $p$ -adic analysis defined in [p-adic Analysis](#).
- Use notations on harmonic analysis defined in [Harmonic Analysis on Locally Compact Groups](#).
- Use notations on adeles and ideles defined in [Global Fields](#).

┘

### 1 over Valued Fields

References are [Tit79].

### Notation(20.5.1.1).

- Let  $K \in \mathbf{Field}^0$  be a valued field.
- Let  $H \leq G \in \mathbf{AlgGrp}/K$ .
- $G(K)$  is a topological group with the topology coming from  $K$ .

┘

### Bounded Subgroups

**Def. (20.5.1.2)**  $[G(K)^1]$ . For  $G \in \mathbf{AlgGrp}/K$ , define

$$G(K)^1 = \bigcap_{\chi \in X^*(G)} \ker(|-| \circ \chi : G(K) \rightarrow \mathbb{R}_+).$$

┘

**Def. (20.5.1.3)** **[Bounded Subsets]**. For any  $X \in \mathbf{Aff}^{\text{sep}}/K$ , a subset  $B \subset X(K)$  is called a **bounded subset** if for any  $f \in \Gamma(X)$ , the set  $\{|f(b)| : b \in B\}$  is bounded.

Notice that if  $K \in \mathbf{LField}$ , then a subset  $B \subset X(K)$  is bounded iff it is pre-compact in the analytic topology.

┘

**Prop. (20.5.1.4)** **[Bounded Subsets and Finite Maps]**. For any  $X \rightarrow Y \in \mathbf{Aff}^{\text{ft}}/K$  is a finite map, then the inverse image of a bounded subset of  $Y(K)$  is a bounded subset of  $X(K)$ .

┘

*Proof:* We can write  $X$  as a closed subscheme of  $Y \times \mathbb{A}^n$  defined by  $n$  monic polynomials  $f_i(y)(T) \in \Gamma(Y)[T]$ . Then it suffices to notice that: The valuation of roots of any monic polynomial of bounded degree is dominated by the valuations of its coefficients.  $\square$

**Prop. (20.5.1.5)** **[Integral Points are Bounded]**. If  $\mathcal{G} \in \mathbf{Sch}^{\text{ft}}/\mathcal{O}_K$ , then  $\mathcal{G}(\mathcal{O}_K)$  is bounded in  $\mathcal{G}(K)$ .

┘

**Thm. (20.5.1.6)** **[Anisotropic Groups are Bounded]**. If  $K \in \mathbf{CDVR}$  and  $G \in \mathbf{RedGrp}/K$ ,  $G(K)$  is bounded(20.5.1.3) iff  $G$  is anisotropic(10.4.2.24).

┘

*Proof:* If  $G$  has a split torus, then it cannot be bounded by (20.5.1.4). Conversely, if  $G(K)$  is unbounded, then ? Cf. [K-P23]P73.  $\square$

**Cor. (20.5.1.7) [Largest Bounded Subgroups].** If  $G \in \text{RedGrp}/K$  and  $[G, G]$  is anisotropic (10.4.2.24), then  $G(K)^1$  (20.5.1.2) is the maximal bounded subgroup of  $G(K)$   $\lrcorner$

*Proof:* Let  $G_a$  be the inverse image in  $G$  of the maximal anisotropic subtorus of  $G_{\text{ab}}$ , then  $G_a$  is the maximal normal anisotropic subgroup variety of  $G$  (10.4.2.24). And if  $S$  is the maximal split central torus of  $G$ , then  $G = SG_a$  is an almost-direct product (10.1.5.9).  $\square$

### Topology and Analytic Structure

**Prop. (20.5.1.8) [Analytic Structures].** For any  $X \in \text{Sch}^{\text{ft}}/K$ ,  $X_{\text{sm}}(K)$  is an analytic variety over  $K$ . And for any map  $f : X \rightarrow Y \in \text{Sch}^{\text{ft}}/K$ ,  $f^{\text{an}} : X_{\text{sm}}(K) \cap f^{-1}(Y_{\text{sm}}(K)) \rightarrow Y_{\text{sm}}(K)$  is a map of analytic varieties over  $K$ .  $\lrcorner$

**Prop. (20.5.1.9) [Cartan-Serre].** Let  $K \in \{\mathbb{R}\} \cup \{\mathbb{Q}_p\}_{p \in \text{Prime}}$ , then closed subgroups of a Lie group over  $K$  is a Lie group, and every continuous maps between Lie groups over  $K$  is analytic.  $\lrcorner$

*Proof:* Cf. [Groupes algebriques et corps de classes, Serre]P260-263.  $\square$

**Prop. (20.5.1.10) [Normal Subgroups are Open].** If  $G$  is simple, then every normal subgroup of  $G(K)$  is open.  $\lrcorner$

**Prop. (20.5.1.11) [Reductive Groups are Unimodular].**  $G(K)$  is unimodular, by (12.10.1.22).  $\lrcorner$

**Thm. (20.5.1.12) [Compactness].**  $G(K)/H(K)$  is bounded iff  $H$  contains a maximal connected split solvable subgroup of  $G^0$ . In particular,  $G(K)$  is bounded iff  $G^0$  is reductive anisotropic (which also follows from (20.5.1.6)).  $\lrcorner$

*Proof:* Cf. [P-R94]P108. ?  $\square$

## 2 over Local Fields

References are [Tit79, K-P23]. For Bruhat-Tits theory, see ??.

### Valuation Homomorphisms

**Notation (20.5.2.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi, \kappa) \in p\text{-LField}$ .
- Let  $G \in \text{RedGrp}/K$ .

$\lrcorner$

**Prop. (20.5.2.2) [Valuation Homomorphism].** For  $G \in \text{RedGrp}/K$ , there is a homomorphism

$$\omega_G : G(K) \rightarrow \text{Hom}(X^\bullet(G)^{\text{Gal}_K}, \mathbb{Z}) : g \mapsto (\chi \mapsto -\text{ord}_K(\chi(g))).$$

The kernel is exactly  $G(K)^1$  (20.5.1.2). Let  $A_G \leq G$  be the maximal central split torus, then the restriction map  $X^\bullet(G)^{\text{Gal}_K} \rightarrow X^\bullet(A_G)$  is injective with finite cokernel by (10.4.4.7), thus composing  $\omega_G$  with the map

$$\text{Hom}(X^\bullet(G)^{\text{Gal}_K}, \mathbb{Z}) \hookrightarrow \text{Hom}(X^\bullet(G)^{\text{Gal}_K}, \mathbb{Q}) = \text{Hom}(X^\bullet(A_G), \mathbb{Q}) = X_\bullet(A_G) \otimes \mathbb{Q},$$



we get a map

$$\omega_G : G(K) \rightarrow X_\bullet(A_G) \otimes \mathbb{Q},$$

called the **valuation homomorphism**.  $\lrcorner$

**Cor. (20.5.2.3).** For a torus  $T/K$ , the image of the valuation homomorphism  $\omega_T$  (20.5.2.2) contains  $X_*(T)$  as a subgroup of finite index. And if  $T$  splits over  $\check{K}$ , then the image is exactly  $X_*(T)$ .  $\lrcorner$

*Proof:* If  $T$  is split, then the assertion is clear. And if  $S \leq T$  is a maximal split torus, then  $X_*(S) = X_*(T)$ , so the image of  $\omega_T$  contains  $X_*(S)$  by functoriality. If  $L/K$  is a finite Galois extension s.t.  $T_L$  is split, then by functoriality the image of  $\omega_T$  is exactly  $\frac{1}{e_{L/K}} X_*(T)$ .  $\square$

**Prop. (20.5.2.4) [Functoriality].** For  $f : G \rightarrow H \in \text{RedGrp}/k$ ,

- For any  $g \in G(K)$ ,  $\omega_H(f(g)) = f_* \omega_G(g)$ .
- $G(K)^1 = \{g \in G(K) : |\chi(g)| = 1 \forall \chi \in X^*(G)\} = \{g \in G(K) : |\chi(g)| = 1 \forall \chi \in X^\bullet(G)\}$ .

$\lrcorner$

*Proof:* 1:  $f$  induces a map of torus  $G/[G, G] \rightarrow H/[H, H]$ , so the functoriality is clear from the definition.

2: By item1 and the fact  $G/[G, G]$  is a torus, we are reduced to the case when  $G = T$  is a torus. Then notice that if  $L/K \in \text{GalField}^{\text{fin}}$  is a splitting field for  $T$ , then  $\omega_T = \omega_{T_L}|_{T(K)}$  by Galois invariance.  $\square$

**Prop. (20.5.2.5) [Maximal Bounded Subgroups of Tori].** For  $T \in \mathcal{Tori}/K$ ,

$$T(K)^1 = \ker(\omega_T) = \{t \in T(K) \mid \text{ord}_K(\chi(t)) = 0 \forall \chi \in X^\bullet(T)\}$$

is the unique maximal bounded subgroup of  $T(K)$ . In particular,  $T(K)$  is bounded iff  $T$  is anisotropic.  $\lrcorner$

*Proof:* The second equality follows from the fact . And  $T(K)^1$  is the unique maximal bounded subgroup by (20.5.1.7).  $\square$

**Cor. (20.5.2.6).** For  $G \in \text{RedGrp}/k$ ,  $?$   $\lrcorner$

**Prop. (20.5.2.7).** Let  $\kappa = \kappa^{\text{sep}}$  and  $f : T_1 \rightarrow T_2 \in \mathcal{Tori}/K$ , then  $f(T_1(K)^1) \subset T_2(K)^1$ . Moreover,

- If  $\ker(f)^0$  is anisotropic, then  $T_1(K)^1 = f^{-1}(T_2(K)^1)$ .
- If  $f$  is a quotient map and  $\ker(f)$  is connected, then  $f(T_1(K)^1)$  has finite index in  $T_2(K)^1$ .
- If  $f$  is a quotient map and  $\ker(f)$  is connected and anisotropic, then  $f(T_1(K)^1) = T_2(K)^1$ .

$\lrcorner$

*Proof:* 1: In this case,  $X^*(K)$  is finite by (10.4.4.5), There is an exact sequence

$$0 \rightarrow X^\bullet(\text{Coker}(f)) \rightarrow X^\bullet(T_2) \rightarrow X^\bullet(T_1) \rightarrow X^\bullet(\ker(f)) \rightarrow 0,$$

of  $\text{Gal}_K$ -modules. Suppose  $L \subset K^{\text{sep}}$  is a splitting field for both  $T_1$  and  $T_2$ , then the fact that  $\#X^*(K) < \infty$  and  $\#H^1(\text{Gal}(L/K), X^\bullet(\text{Coker}(f))) < \infty$  implies that  $X^*(T_2) \otimes \mathbb{Q} \rightarrow X^*(T_1) \otimes \mathbb{Q}$  is surjective, and  $X_*(T_1) \otimes \mathbb{Q} \rightarrow X_*(T_2) \otimes \mathbb{Q}$  is injective. So the assertion follows by functoriality of  $\omega_T$ .

3: It follows from (10.5.2.4) that  $T_1(K) \rightarrow T_2(K)$  is surjective, so this follows from item1.

2: By (20.5.2.3), there exists  $e \in \mathbb{Z}_+$  s.t.  $\omega_{T_i}(T_i(K)) \subset \frac{1}{e}X_*(T_i)$ , thus there is a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \ker(f)(K) & \longrightarrow & T_1(K) & \longrightarrow & T_2(K) \longrightarrow 0 \\ & & \downarrow \omega & & \downarrow \omega & & \downarrow \omega \\ 0 & \longrightarrow & \frac{1}{e}X_*(\ker(f)) & \longrightarrow & \frac{1}{e}X_*(T_1) & \longrightarrow & \frac{1}{e}X_*(T_2) \end{array}$$

Then it follows from the snake lemma?? and (20.5.2.3) that  $f(T_1(K)^1)$  has finite index in  $T_2(K)^1$ .  $\square$

**Prop. (20.5.2.8) [Finite Integral Subgroups].** Suppose  $n \in \mathbb{Z}_+$  and  $G \leq \mathrm{GL}(n)$ , then  $G \cap \mathrm{GL}(n; \mathcal{O}_K)$  has only f.m. conjugacy classes of finite subgroups.  $\lrcorner$

*Proof:* Cf. [P-R94]P117.  $\square$

**Thm. (20.5.2.9) [Sylow Pro- $p$ -Subgroups, Matsumoto].** If  $G \in \mathrm{RedGrp}^{\mathrm{ss}}/K$ , then any open subgroup  $H$  of  $G(K)$  contains a Sylow pro- $p$ -subgroup, unique up to conjugacy.  $\lrcorner$

*Proof:* Cf. [P-R94]P139. ?  $\square$

**Prop. (20.5.2.10) [An Algebraicity Theorem].**  $X \in \mathrm{Sch}^{\mathrm{ft}}/K$ , then  $X(K) \in \mathrm{locProf}$  in the  $K$ -topology (because varieties are closed, and use (4.4.4.6)). Let  $G \in \mathrm{linAlgGrp}/K$  and  $G \times X \rightarrow X$  is an action of  $G$  on  $X$ , then  $G(F) \times X(F) \rightarrow X(F)$  is a continuous constructible action (4.12.1.21).  $\lrcorner$

*Proof:* Cf. [Bernstein-Zelevinsky, Appendix].  $\square$

### Kottwitz-Maps ( $p$ -adic cases)

References are [Kot85, Kot86].

**Notation (20.5.2.11).**

- Let  $(K, \mathcal{O}_K, \kappa) \in \mathrm{DVR}$  s.t.  $K$  is Henselian and  $\kappa = \kappa^{\mathrm{perf}}$ .
- Let  $G \in \mathrm{RedGrp}/K$ .

$\lrcorner$

**Def. (20.5.2.12) [Kottwitz-Maps, Kottwitz [Kot85]].** There is a continuous homomorphism

$$\kappa_G : G(K) \rightarrow (\pi_1(G)_{I_K})^{\mathrm{Gal}_\kappa} \cong \left( X^\bullet(Z(\widehat{G}))_{I_K} \right)^{\mathrm{Gal}_\kappa} \quad (10.4.5.24)(10.4.5.25)$$

as follows:

1. If  $\kappa = \bar{\kappa}$  and  $G = T$  is a torus, there is a unique such  $\kappa_G$  s.t. for any  $L/K \in \mathrm{GalField}^{\mathrm{fin}}$  splitting  $T$ , the following diagram is commutative:

$$\begin{array}{ccc} T(L) & \xrightarrow{\omega_T} & X_\bullet(T) \\ \downarrow \mathrm{Nm}_{L/K} & & \downarrow \\ T(K) & \xrightarrow{\kappa_T} & X_\bullet(T)_{I_K} \end{array}$$

And this map is functional from the category of tori over  $K$  to  $\mathrm{Ab}_{\mathrm{Gal}_\kappa}^{\mathrm{fg}}$ .

2. If  $\kappa = \bar{\kappa}$  and  $[G, G]$  is simply-connected,  $G \rightarrow G_{\text{ab}} = G/[G, G]$  induces an isomorphism  $\pi_1(G) = \pi_1(G_{\text{ab}})$  by (10.4.5.26). Define  $\kappa_G$  to be the composite  $G(K) \rightarrow G_{\text{ab}}(K) \xrightarrow{\kappa_{G_{\text{ab}}}} \pi_1(G_{\text{ab}})_{I_K}$ .
3. If  $\kappa = \bar{\kappa}$ , choose a  $z$ -extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  (10.4.5.28), then  $[\tilde{G}, \tilde{G}]$  is simply-connected and  $\pi_1(\tilde{G}) \rightarrow \pi_1(G)$  is surjective by (10.4.5.26). Also  $\tilde{G}(K) \rightarrow G(K)$  is surjective by Steinberg's theorem (10.5.2.3). Then there is a unique map  $\kappa_G : G(K) \rightarrow \pi_1(G)_{I_K}$  s.t. the following diagram

$$\begin{array}{ccc} \tilde{G}(K) & \longrightarrow & G(K) \\ \downarrow \kappa_{\tilde{G}} & & \downarrow \kappa_G \\ \pi_1(\tilde{G})_{I_K} & \xrightarrow{=} & \pi_1(G)_{I_K} \end{array}$$

is commutative. And this  $\kappa_G$  doesn't depend on the choice of  $\tilde{G}$ .?

4. Define  $\kappa_G$  as the restriction of  $\kappa_{G_{K^{\text{sh}}}}$  to  $G(K)$ .

Then this  $\kappa_G$  induces a functor  $\mathcal{R}\text{edGrp}/K \rightarrow \mathcal{A}\text{b}^{\text{fg}}$ , called the **Kottwitz-map**.  $\square$

*Proof:* 1: The uniqueness follows from the fact that  $\text{Nm}_{L/K}(T(L)) = T(K)$ . For the existence, notice that it suffices to prove for a single Galois extension  $L/K$ , because if  $L'/K \in \mathcal{G}\text{alField}^{\text{fin}}$  and  $L \subset L'$ , then there is a commutative diagram

$$\begin{array}{ccc} T(L') & \xrightarrow{\omega_{T_{L'}}} & X_{\bullet}(T) \\ \downarrow \text{Nm}_{L'/L} & & \downarrow \\ T(L) & \xrightarrow{\omega_{T_L}} & X_{\bullet}(T) \end{array}$$

because  $L'/L$  is totally ramified. Now it suffices to prove that  $\text{Nm}_{L/K}$  induces an isomorphism  $T(L)_{I_K} \cong T(K)$ . For this, Cf. [K-P23]P421.?

3: Given a  $z$ -extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$ , there is an exact sequence  $1 \rightarrow \pi_1(Z) \rightarrow \pi_1(\tilde{G}) \rightarrow \pi_1(G) \rightarrow 1$  by (10.4.5.26), so there is a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 1 & \longrightarrow & Z(K) & \longrightarrow & \tilde{G}(K) & \longrightarrow & G(K) \longrightarrow 1 \\ & & \downarrow \kappa_Z & & \downarrow \kappa_{\tilde{G}} & & \\ & & \pi_1(Z)_{I_K} & \longrightarrow & \pi_1(\tilde{G})_{I_K} & \longrightarrow & \pi_1(G)_{I_K} \longrightarrow 1 \end{array}$$

where the first row is exact by Steinberg's theorem (10.5.2.3) and the square is commutative by the definition of  $\kappa_{\tilde{G}}$  and the functoriality of Kottwitz-maps for tori applied to the map  $Z \rightarrow \tilde{G}/[\tilde{G}, \tilde{G}]$ . Thus the uniqueness and existence of  $\kappa_G$  is clear.

For the independence of  $\tilde{G}$ , notice that fiber product of  $z$ -extensions are  $z$ -extensions (10.4.5.29), and the functoriality follows from (10.4.5.29).  $\square$

**Prop. (20.5.2.13) [Kottwitz-Maps and Valuation Maps].** There is a commutative diagram

$$\begin{array}{ccc} G(K) & \xrightarrow{\omega_G} & V(A_G) \\ \downarrow = & & \downarrow \cong \\ G(K) & \xrightarrow{\kappa_G \otimes \mathbb{Q}} & (X_{\bullet}(T)_{\mathbb{Q}}/\mathbb{R} \cdot \Phi^{\vee})_{I_K} \end{array}$$

In particular,  $G(K)^1 = \ker(\kappa_G \otimes \mathbb{Q})$ .  $\lrcorner$

*Proof:* By (20.5.2.12) and (20.5.2.4) it suffices to prove for  $G$  s.t.  $[G, G]$  is simply-connected. But in this case it suffices to prove for  $G/[G, G]$ , which is a torus, and the assertion follows by definition.  $\square$

**Prop. (20.5.2.14) [Kottwitz-Maps and Extension].** If  $L/K \in \mathbf{Field}$  is a finite separable extension, and  $T$  is a torus over  $K$ , then the following diagrams commute:

$$\begin{array}{ccc} T(L) & \xrightarrow{\kappa_{T_L}} & X_\bullet(T)_{I_L} \\ \downarrow \text{Nm}_{L/K} & & \downarrow \\ T(K) & \xrightarrow{\kappa_T} & X_\bullet(T)_{I_K} \end{array} \quad \begin{array}{ccc} T(L) & \xrightarrow{\kappa_{T_L}} & X_\bullet(T)_{I_L} \\ \uparrow i & & \uparrow \\ T(K) & \xrightarrow{\kappa_T} & X_\bullet(T)_{I_K} \end{array}$$

$\lrcorner$

*Proof:* By definition (20.5.2.12) it suffices to prove for  $\kappa = \bar{\kappa}$ . Choosing an exact sequence  $1 \rightarrow T_0 \rightarrow T_1 \rightarrow T \rightarrow 1 \in \mathbf{RedGrp}/k$  where  $T_0$  is a torus and  $T_1$  is an induced torus (10.2.3.13), we see that  $T_1(\check{K}) \rightarrow T(\check{K})$  is surjective by Steinberg's theorem (10.5.2.3). Then it suffices to prove for  $T_1$ , in which case  $X_\bullet(T)_{I_K}$  is torsion-free, and the assertion follows from (20.5.2.13) and (20.5.2.2).  $\square$

**Prop. (20.5.2.15) [Completions].** There is a commutative diagram

$$\begin{array}{ccc} G(K) & \xrightarrow{\kappa_G} & (\pi_1(G)_{I_K})^{\text{Gal}_\kappa} \\ \downarrow & & \downarrow \cong \\ G(\widehat{K}) & \longrightarrow & (\pi_1(G)_{I_{\widehat{K}}})^{\text{Gal}_\kappa} \end{array}$$

$\lrcorner$

*Proof:* Reduce to tori as in (20.5.2.13), in which case it follows from the uniqueness when defining  $\kappa_G$ .  $\square$

**Prop. (20.5.2.16) [Kottwitz-Maps and Restrictions].** If  $L/K \in \mathbf{Field}$  is a finite separable extension and  $G \in \mathbf{RedGrp}/L$ ,  $G' = \text{Res}_{L/K} G$ , then the following diagram are commutative:

$$\begin{array}{ccc} G'(K) & \xrightarrow{\kappa_{G'}} & \pi_1(G')_{I_K}^{\text{Gal}_\kappa} \\ \downarrow = & & \downarrow = \\ G(L) & \xrightarrow{\kappa_G} & \pi_1(G)_{I_L}^{\text{Gal}(\check{L}/L)} \end{array}$$

$\lrcorner$

*Proof:* Cf. [K-P23]P423.  $\square$

**Cor. (20.5.2.17).** If  $G = T$  is an induced torus (10.2.3.11), then  $\kappa_T$  is surjective.  $\lrcorner$

*Proof:* by (20.5.2.16) it suffices to prove for  $T = \mathbb{G}_m$ , which is trivial.  $\square$

**Def. (20.5.2.18) [Kottwitz-Kernel,  $G(K)^0$ , Kaletha-Prasad [K-P23]].** Define

$$G(K)^0 = \ker(\kappa_G) \text{ (20.5.2.12).}$$

Then  $G(K)^0 \leq G(K)^1$  (20.5.1.2) has finite index, and  $G(K)^0 \leq G(K)$  is normal and open. Moreover,

1. If  $G = T$  is a torus,  $\kappa = \bar{\kappa}$  and  $L/K \in \mathbf{GalField}^{\text{fin}}$  is a splitting field for  $T$ , then  $T(K)^0$  is the image of  $T(L)^1$  under the norm map  $\text{Nm}_{L/K}$ .
2. If  $1 \rightarrow T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow 1$  is an exact sequence of tori, then  $T_2(K)^0 \rightarrow T_3(K)^0$  is surjective. And if  $T_1$  is an induced torus, then  $1 \rightarrow T_1(K)^0 \rightarrow T_2(K)^0 \rightarrow T_3(K)^0 \rightarrow 1$  is exact.
3. If  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  is a z-extension(10.4.5.28), then the maps  $\tilde{G}(K) \rightarrow G(K), \tilde{G}(K)^0 \rightarrow G(K)^0$  are surjective.
4. If  $M_0 \leq G$  is a minimal Levi-subgroup, then  $G(K)^0 = G(K)^{\natural} M_0(K)^0$ , where  $G(K)^{\natural} = \text{Im}(G_{\text{sim.cntd}}(K) \rightarrow G(K))$ .

Using these, we can reconstruct  $G(K)^0$  as

$$G(K)^0 = G(K) \cap G(\check{K})^0 = G(K) \cap \left( G(\check{K})^{\natural} T(\check{K})^0 \right) = G(K) \cap \left( G(\check{K})^{\natural} \cdot \text{Nm}_{L/\check{K}} T(L)^1 \right),$$

where  $T \leq G_{\check{K}}$  is a maximal torus and  $L/\check{K} \in \mathbf{GalField}^{\text{fin}}$  is a splitting field for  $T$ .  $\lrcorner$

*Proof:* 1: By definition  $\text{Nm}_{L/K}(T(L)^0) \leq T(K)^0$ . For the converse, if  $t \in T(L)$  s.t.  $\text{Nm}_{L/K}(t) \in T(K)^0$ , then  $\kappa_{T_L}(t)$  is a sum of elements of the form  $y - \sigma(y)$  where  $y \in X_{\bullet}(T), \sigma \in I_K$ . But notice that  $\kappa_{T_L}(y(\varpi_L)/\sigma(y(\varpi_L))) = y - \sigma(y)$  and  $\text{Nm}_{L/K}(y(\varpi_L)/\sigma(y(\varpi_L))) = 1$ , so we can modify  $t$  to  $t'$  s.t.  $\text{Nm}_{L/K}(t) = \text{Nm}_{L/K}(t')$  and  $\kappa_{T_L}(t') = 0$ .

2: Cf.[K-P23]P91?.

3: The exact sequence  $1 \rightarrow Z(\check{K}) \rightarrow \tilde{G}(\check{K}) \rightarrow G(\check{K}) \rightarrow 1$  induces an exact sequence  $1 \rightarrow Z(K) \rightarrow \tilde{G}(K) \rightarrow G(K) \rightarrow 1$  by Szpiro's lemma. And in the exact sequence  $X_{\bullet}(Z)_{I_K} \rightarrow \pi_1(\tilde{G})_{I_K} \rightarrow \pi_1(G)_{I_K} \rightarrow 1$ , the first map is injective because  $X_{\bullet}(Z)_{I_K}$  is torsion-free(? Cf.[K-P23]Prop2.5.20). Now the  $\text{Gal}_{\kappa}$ -module  $X_{\bullet}(Z)_{I_K}$  is also induced, so we get an exact sequence  $(X_{\bullet}(Z)_{I_K})^{\text{Gal}_{\kappa}} \rightarrow (\pi_1(\tilde{G})_{I_K})^{\text{Gal}_{\kappa}} \rightarrow (\pi_1(G)_{I_K})^{\text{Gal}_{\kappa}} \rightarrow 1$ . Now the assertion follows from snake's lemma and(20.5.2.17).

4: We firstly prove for  $G$  quasi-split and  $[G, G]$  simply-connected:  $G(K)^{\natural} T(K)^0 \leq G(K)^0$  by functoriality. For the other direction, let  $T \leq G$  be a maximally split maximal torus and let  $T_{\text{sim.cntd}} = T \cap [G, G]$ , then  $1 \rightarrow T_{\text{sim.cntd}} \rightarrow T \rightarrow G/[G, G] \rightarrow 1$  is an exact sequence. Then  $T(K)^0 \rightarrow (G/[G, G])(K)^0$  is surjective by item2, and the assertion follows.

Next, if  $G$  is quasi-split, take a z-extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$ (10.4.5.28), then  $\tilde{G}$  is also quasi-split. Let  $T \leq G$  be a maximally split maximal torus,  $\tilde{T} \leq \tilde{G}$  the inverse image of  $T$ . Then  $\tilde{G}(K)^{\natural} = [\tilde{G}, \tilde{G}](K)$  is mapped into  $G(K)^{\natural}$ , and  $\tilde{T}(K)^0$  is mapped into  $T(K)^0$  by functoriality, and  $G(K)^{\natural} T(K)^0 \leq G(K)^0$  by functoriality. But we also have  $\tilde{G}(K)^0 = \tilde{G}(K)^{\natural} \tilde{T}(K)^0$  and  $\tilde{G}(K) \rightarrow G(K)$  is surjective by item3, so the assertion follows from the assertion for  $\tilde{G}$ , which is just proved.

Finally, for general  $G$ , it suffices to prove that if  $S \leq G$  is a maximal split torus and  $T \leq G$  is a special torus containing  $S$ (20.6.5.2), then

$$G(K)^{\natural} \left( T(\check{K})^0 \cdot Z_G(S)(\check{K})^{\natural} \right)^0 = G(K) \cap \left( G(\check{K})^{\natural} T(\check{K})^0 \right),$$

because then we are reduced to  $G_{\check{K}}$ , which is quasi-split by Steinberg's theorem. For this, Cf.[K-P23]Prop9.3.24?  $\square$

**Prop.(20.5.2.19) [Kottwitz-Maps are Surjective].** If  $\kappa$  is a cd-1 field, and  $C \subset \mathcal{B}(G_{K^{\text{sh}}})$  is a  $\text{Gal}_{\kappa}$ -invariant chamber(which exists by(20.5.2.31)), and  $G(K)_C$  the stablizer of  $C$  in  $G(K)$ , then  $\kappa_G(G(K)_C) = (\pi_1(G)_{I_K})^{\text{Gal}_{\kappa}}$ . In particular, by(20.5.2.13) it induces isomorphisms

$$G(K)_C / \left( G(K)_C \cap G(K)^0 \right) \cong (\pi_1(G)_{I_K})^{\text{Gal}_{\kappa}}, \quad \left( G(K)_C \cap G(K)^1 \right) / \left( G(K)_C \cap G(K)^0 \right) \cong ((\pi_1(G)_{I_K})_{\text{tor}})^{\text{Gal}_{\kappa}}.$$

$\lrcorner$

*Proof:* Firstly, if  $\kappa = \bar{\kappa}$ , then  $\kappa_G$  is surjective: By definition(20.5.2.12), this reduces to tori, in which case it is surjective by definition. Next notice that  $G(K)^{\natural} = \text{Im}(G_{\text{sim.cntd}}(K) \rightarrow G(K))$  acts transitively on chambers of  $\mathcal{B}(G_{K^{\text{SH}}})$ , and  $G(K)^{\natural} \leq G(K)^0$ , so  $\kappa_G : G(K)_C \rightarrow (\pi_1(G)_{I_K})^{\text{Gal}_{\kappa}}$  is also surjective.

Now for general  $\kappa$ , there is an exact sequence

$$1 \rightarrow G(K^{\text{SH}})_C^0 \rightarrow G(K^{\text{SH}})_C \rightarrow \pi_1(G)_{I_K} \rightarrow 1,$$

and  $G(K^{\text{SH}})_C^0 = \mathcal{G}_C^0(\mathcal{O}_{K^{\text{SH}}})$  where  $\mathcal{G}_C^0$  is the Bruhat-Tits group scheme(20.6.4.19), so it follows from(20.3.6.8) that  $G(K)_C \rightarrow (\pi_1(G)_{I_K})^{\text{Gal}_{\kappa}}$  is surjective.  $\square$

**Prop. (20.5.2.20).** For  $T \in \mathcal{T}\text{ori}/K$ , the Kottwitz-map  $\kappa_T$ (20.5.2.12) induces injective homomorphisms

$$T(K)^1/T(K)^0 \hookrightarrow ((X_{\bullet}(T)_{I_K})_{\text{tor}})^{\text{Gal}_{\kappa}}, \quad T(K)/T(K)^0 \hookrightarrow (X_{\bullet}(T)_{I_K})^{\text{Gal}_{\kappa}}.$$

which are isomorphism when  $\kappa$  is a cd-1 field, by(20.5.2.19).  $\perp$

*Proof:* Injectivity is clear from(20.5.2.13) and(20.5.2.18).  $\square$

**Prop. (20.5.2.21).** If  $\kappa$  is a cd-1 field and  $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1 \in \mathcal{T}\text{ori}/K$  is an exact sequence s.t.  $A$  has induced ramification?, then the sequence  $1 \rightarrow A(K)^0 \rightarrow B(K)^0 \rightarrow C(K)^0 \rightarrow 1$  is exact.  $\perp$

*Proof:* Cf.[K-P23]P434?.  $\square$

**Prop. (20.5.2.22) [Levi-Subgroups].** If  $M \leq G$  is a Levi subgroup, then  $M(K)^1 \cap G(K)^0 = M(K)^0$ .  $\perp$

*Proof:*  $M(K)^1 = M(K^{\text{SH}})^1 \cap M(K)$  by(20.5.2.2) and  $G(K)^0 = G(K^{\text{SH}})^0 \cap G(K)$  by(20.5.2.12), so it suffices to prove for  $K = K^{\text{SH}}$ . In particular,  $G, M$  are quasi-split by Steinberg's theorem(10.5.2.3).

One direction is clear. For the other, if  $m \in M(K)^1 \cap G(K)^0$ , then by(20.5.2.13)  $\kappa_M(m) \in (\pi_1(M)_{I_K})_{\text{tor}}$  has trivial image in  $\pi_1(G)_{I_K}$ . Notice there is an exact sequence

$$H_1(I_K, \pi_1(G)) \rightarrow (\ker(\pi_1(M) \rightarrow \pi_1(G)))_{I_K} \rightarrow \pi_1(M)_{I_K} \rightarrow \pi_1(G)_{I_K} \rightarrow 1.$$

So  $m$  lifts to a torsion element in  $(\ker(\pi_1(M) \rightarrow \pi_1(G)))_{I_K}$ . Then it suffices to prove that  $\ker(\pi_1(M) \rightarrow \pi_1(G))$  is a torsion-free induced module, then  $(\ker(\pi_1(M) \rightarrow \pi_1(G)))_{I_K}$  is non-torsion. For this, Cf.[K-P23]P432?.  $\square$

The map  $\pi_1(M) \rightarrow \pi_1(G)$  is clearly surjective, and

For more on Kottwitz-maps, see [B\(G\) and Isocrystals with Structures](#).

### Kottwitz-Maps for Archimedean Places

**Thm. (20.5.2.23).** For  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ , there is a unique extension of the Tate-Nakayama map to a functorial map

$$\kappa_G : \text{RedGrp}/\mathbb{K} \rightarrow (\pi_1(G)_{\text{Gal}_{\mathbb{K}}})_{\text{tor}}.$$

And if  $\mathbb{K} = \mathbb{R}$ , then

$$\ker(\kappa_G) = \text{Im} \left( H^1(\mathbb{R}, G_{\text{sim.cntd}}) \rightarrow H^1(\mathbb{R}, G) \right), \quad \text{Im}(\kappa_G) = \ker \left( \pi_1(G)_{\text{Gal}_{\mathbb{R}}} \xrightarrow{\text{Nm}_{\mathbb{C}/\mathbb{R}}} \pi_1(G) \right).$$

$\perp$

*Proof:* Cf.[Kot86]P368.?.  $\square$

### Unramified Groups

**Notation (20.5.2.24).**

- Let  $(K, \mathcal{O}_K, \kappa) \in \text{DVR}$  s.t.  $K$  is Henselian and  $\kappa = \kappa^{\text{perf}}$ .
- Let  $G \in \text{RedGrp}/K$ .

┘

**Def. (20.5.2.25) [Unramified Reductive Groups].**  $G$  is called an **unramified reductive group** over  $K$  if  $G$  is quasi-split and splits over  $K^{\text{sh}}$ . By (20.7.2.13),  $G$  is unramified reductive iff it has a reductive model  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_K$ .

┘

**Prop. (20.5.2.26) [Borel and Maximal Tori Exist].** For  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_K$ , there exists a Borel subgroup scheme  $\mathcal{B} \leq \mathcal{G}$  and a maximal  $\mathcal{O}$ -torus  $\mathcal{T} \leq \mathcal{B}$ .

┘

*Proof:* By (10.5.6.25), the moduli  $\text{Bor}_{\mathcal{G}/\mathcal{O}_K}$  of Borel subgroups of  $\mathcal{G}$  are smooth proper over  $\mathcal{O}_K$ , thus  $\text{Bor}_{\mathcal{G}/\mathcal{O}}(\mathcal{O}_K) = \text{Bor}_{\mathcal{G}/\mathcal{O}}(K) \neq \emptyset$  by valuation criterion of properness and (20.5.2.25). The maximal torus  $\mathcal{T} \leq G$  exists by ? Cf. [Getz, automorphic representations]P549.  $\square$

**Prop. (20.5.2.27) [Kottwitz Maps for Unramified Groups].** Let  $G$  be unramified with a maximal split torus  $S \leq G$  and  $x \in \mathcal{A}(T)$  a hyperspecial point with corresponding hyperspecial subgroup  $\mathcal{K}$ . Then for any  $b \in G(K)$ , there exists  $\mu \in X_{\bullet}(T)$  s.t.  $b \in \mathcal{K}\mu(\varpi_K)\mathcal{K}$  by Cartan-decomposition (20.7.2.14). Then  $\kappa_G(b)$  equals the image of  $\mu \in X_{\bullet}(T)$  in  $(\pi_1(G)_{I_K})^{\text{Gal}_{\kappa}}$  (20.5.2.12).

┘

*Proof:* This follows from the functoriality of Kottwitz-maps (20.5.2.12) and the fact that  $\mathcal{K} \leq G(K)^0$ .  $\square$

**Prop. (20.5.2.28) [Iwasawa-Decompositions].** For  $(\mathcal{G}, \mathcal{B}, \mathcal{T}) \in \text{RedGrp}/\mathcal{O}_K$  with  $(G, B, T) = (\mathcal{G}, \mathcal{B}, \mathcal{T})_K$ ,  $\mathcal{T}(\mathcal{O}_K)$  is a maximal compact subgroup of  $T(F)$ , and

$$G(K) = B(F)\mathcal{T}(\mathcal{O}_K), \quad B(\mathcal{O}_K) = \mathcal{T}(\mathcal{O}_K)\mathcal{N}(\mathcal{O}_K),$$

where  $\mathcal{N}(\mathcal{O}_K)$  is the smooth connected unipotent radical subscheme of  $\mathcal{B}$ .

┘

*Proof:* Cf. [Getz]P549.  $\square$

**Prop. (20.5.2.29) [Maximal Tori are Conjugate].** If  $\mathcal{H} \in \text{GrpSch}^{\text{ft}}/\mathcal{O}_K$  with connected special fiber, then any two maximal split  $\mathcal{O}_K$ -tori of  $\mathcal{H}$  are  $\mathcal{H}(\mathcal{O}_K)$ -conjugate.

┘

*Proof:* For maximal tori  $\mathcal{T}, \mathcal{T}' \leq \mathcal{G}$ , the transporter subgroup  $\text{Transp}_{\mathcal{G}}(\mathcal{T}, \mathcal{T}')$  is a smooth closed subgroup of  $\mathcal{H}$  (10.5.6.2), and  $\text{Transp}_{\mathcal{G}}(\mathcal{T}, \mathcal{T}')(\kappa) \neq 0$  by (10.4.1.7). Thus  $\text{Transp}_{\mathcal{G}}(\mathcal{T}, \mathcal{T}')(\mathcal{O}_K) \neq 0$  by valuation criterion of properness.  $\square$

### Residually Split and Residually Quasi-Split Groups

**Notation (20.5.2.30).**

- Let  $(K, \mathcal{O}_K, \kappa) \in \text{DVR}$  s.t.  $K$  is Henselian and  $\kappa = \kappa^{\text{perf}}$ .
- Let  $G \in \text{RedGrp}/K$ .

┘

**Prop. (20.5.2.31) [Residually Quasi-Split Groups].** The following are equivalent:



- There exists a  $\text{Gal}(K^{\text{sh}}/K)$ -stable chamber in the Bruhat-Tits building  $\mathcal{B}(G_{K^{\text{sh}}})$  (20.6.4.8).
- All maximal  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$  are chambers.
- For any maximal  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex  $C$ ,  $\bar{\mathcal{G}}_C^0$  is solvable.
- For any  $x \in \mathcal{B}(G)$ ,  $\bar{\mathcal{G}}_x^0$  is quasi-split.

If these conditions hold, then  $G$  is called a **residually quasi-split reductive group**. In particular, if  $\kappa$  is a cd-1 field, then  $G$  is always residually quasi-split, by Steinberg's theorem (10.5.2.3).  $\lrcorner$

*Proof:* 1  $\rightarrow$  2: This follows from (20.6.5.5).

2  $\rightarrow$  3: It follows from (20.6.4.20) and (10.4.2.34) that the absolute root system of the reductive quotient of  $\bar{\mathcal{G}}_C^0$  is empty, thus  $\bar{\mathcal{G}}_C^0$  is solvable.

3  $\rightarrow$  4: Choose a maximal  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex  $C$  containing  $x$ , then the image of  $\bar{\mathcal{G}}_C^0 \rightarrow \bar{\mathcal{G}}_x^0$  is a parabolic subgroup by (20.6.4.20), and also solvable, thus a Borel subgroup (10.4.1.1).

4  $\rightarrow$  1: By (20.6.4.20) and (10.4.2.35), for any  $x \in |\mathcal{B}(G)|$ , the set of  $\bar{\kappa}$ -Borel subgroups of  $\bar{\mathcal{G}}_x^0$  are in bijection with the set of chambers of  $\mathcal{B}(G_{\bar{\kappa}})$  containing  $x$ , and this map is  $\text{Gal}(K^{\text{sh}}/K)$ -equivariant ?. Thus item 4 implies that there exists a  $\text{Gal}(K^{\text{sh}}/K)$ -stable chamber in  $\mathcal{B}(G_{\bar{\kappa}})$ .  $\square$

**Prop. (20.5.2.32) [Quasi-Split Groups are Residually Quasi-Split].** If  $G$  is quasi-split, then it is residually quasi-split.  $\lrcorner$

*Proof:* For any special torus  $T \leq G$  containing a maximal split torus  $S \leq T$  (20.6.5.2), and  $x \in \mathcal{A}(S)$ , consider the tori  $\mathbf{S} \leq \mathbf{T} \leq \mathbf{G}_x$  corresponding to  $S \subset T$ , where  $\mathbf{S}$  is a maximal split torus and  $\mathbf{T}$  is a maximal torus (20.6.4.20). The assumption that  $G$  is quasi-split implies that the restriction of elements of  $\Phi(G, T)$  to  $S$  are non-trivial, in particular the restriction of elements of the sliced root system  $\Phi(G, T)_x$  to  $S$  is non-trivial. But  $\Phi(G, T)_x$  is canonical isomorphic to  $\Phi((\mathbf{G}_x)_{\bar{\kappa}}, \mathbf{T}_{\bar{\kappa}})$ , so the restriction of elements of  $\Phi((\mathbf{G}_x)_{\bar{\kappa}}, \mathbf{T}_{\bar{\kappa}})$  to  $\mathbf{S}$  is non-trivial, and  $\mathbf{G}_x$  is quasi-split (10.4.4.19).  $\square$

**Prop. (20.5.2.33).** Suppose  $G$  is residually quasi-split with  $\text{rank}_k(G) \geq 1$ , and

- $T \leq G$  is a special torus containing a maximal split torus  $S \leq T$ ,
- $\Psi \subset \mathcal{A}(T_{K^{\text{sh}}})^*$  the absolute affine root system and  $\Psi_K \subset \mathcal{A}(S)^*$  the relative affine root system,
- $C \subset \mathcal{A}(T_{K^{\text{sh}}})$  a  $\text{Gal}(K^{\text{sh}}/K)$ -stable chamber,  $\mathcal{C} = C^{\text{Gal}(K^{\text{sh}}/K)}$  the corresponding chamber of  $\mathcal{B}(G)$ ,
- $\Delta \subset \Psi$  the set of simple affine roots corresponding to  $C$ .

Then the set  $\Delta_K$  of the restrictions of elements of  $\Delta$  to  $\mathcal{A}(S)$  is the set of simple affine roots corresponding to  $\mathcal{C}$ . In particular, no elements of  $\Delta$  has constant restriction to  $\mathcal{A}(S)$ , and two elements has equal restriction iff they are  $\text{Gal}(K^{\text{sh}}/K)$ -conjugate.  $\lrcorner$

*Proof:* It suffices to prove for  $G$  almost simple ?.

Clearly two  $\text{Gal}(K^{\text{sh}}/K)$ -conjugate elements in  $\Delta$  has the same restriction to  $\mathcal{A}(S)$ , and each element of  $\Delta$  is non-negative on  $\mathcal{C}$ . As  $\text{rank}_k(G) \geq 1$ , each element of  $\Delta_K$  vanishes at some vertex of  $\mathcal{C}$ , thus are contained in the set of simple affine roots corresponding to  $\mathcal{C}$ .

If  $\text{Gal}(K^{\text{sh}}/K)$  has  $r$ -orbits on vertices of  $C$ , then  $\dim \mathcal{C} = r - 1$ , and each element of  $\Psi_K$  is a sum of elements of  $\Delta_K$ , thus  $\Delta_K$  is exactly the set of simple affine roots corresponding to  $\mathcal{C}$ , and two elements has equal restriction iff they are  $\text{Gal}(K^{\text{sh}}/K)$ -conjugate.  $\square$

**Prop. (20.5.2.34) [Characterizing Residually Split Groups].** The following are equivalent:



- The action of  $\text{Gal}(K^{\text{sh}}/K)$  on any special apartment of  $\mathcal{B}(G_{K^{\text{sh}}})$  is trivial, and the maximal central split torus of  $G_{K^{\text{sh}}}$  is defined over  $K$ .
- $\text{rank}_K(G) = \text{rank}_{K^{\text{sh}}}(G)$  (10.4.1.18).
- For any maximal  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex  $C \subset \mathcal{B}(G_{K^{\text{sh}}})$ ,  $\overline{\mathcal{G}}_C^0$  is split solvable.
- For any simplex  $\sigma \subset \mathcal{B}(G)$ ,  $\overline{\mathcal{G}}_\sigma^0$  is a split group (10.4.1.1).

If these conditions hold, then  $G$  is called a **residually split group**.  $\lrcorner$

*Proof:* 1  $\iff$  2: If the action of  $\text{Gal}(K^{\text{sh}}/K)$  on a special apartment  $\mathcal{A}(T_{K^{\text{sh}}})$  is trivial, then  $\text{rank}(T') = \text{rank}(S')$  where  $T'$  is the maximal  $K^{\text{sh}}$ -split torus contained in  $T \cap [G, G]$  and  $S'$  is the maximal split torus contained in  $S \cap [G, G]$ . Then  $\text{rank}_K(G) = \text{rank}_{K^{\text{sh}}}(G)$  by (10.4.2.27). The converse is also true by (10.4.2.27).

2  $\iff$  3: As  $\overline{\mathcal{G}}_\sigma^0$  is solvable by (20.6.4.20), and for any special torus  $T \leq G$  s.t.  $C \subset \mathcal{A}(T_{K^{\text{sh}}})$ ,  $\overline{\mathcal{T}}$  is a maximal torus of  $\overline{\mathcal{G}}_C^0$ , thus  $\overline{\mathcal{G}}_C^0$  is split solvable iff  $\overline{\mathcal{T}}$  is a maximal torus by (10.2.7.2), which is equivalent to  $T$  being split.

3  $\iff$  4: Let  $C$  be a  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex containing  $\sigma$ , then the image of  $\rho_{C,\sigma} : \overline{\mathcal{G}}_C^0 \rightarrow \overline{\mathcal{G}}_\sigma^0$  is a split Borel subgroup by (20.6.4.20) and (10.2.7.9).  $\square$

**Prop. (20.5.2.35) [Residually Split Groups are Quasi-Split].** Residually split groups are quasi-split, in particular residually quasi-split (20.5.2.32).  $\lrcorner$

*Proof:* This follows from (20.5.2.34) and (20.5.2.31).  $\square$

### Rigid Inner Forms

References are [Kal16].

**Notation (20.5.2.36).**

- Let  $K \in \text{LField}^0$ .
- Let  $G \in \text{linAlgGrp}/K$ .

$\lrcorner$

**Def. (20.5.2.37) [Canonical Galois Gerbes].** Define

$$P_K^{\text{rig}} \triangleq \varprojlim_{n \in \mathbb{Z}_+, L/K \in \text{GalField}^{\text{fin}}} \text{Res}_{L/K} \mu_n / \mu_n,$$

then there are canonical identifications:

$$H_{\text{cont}}^1(K, P_K^{\text{rig}}) = 1, \quad H_{\text{cont}}^2(K, P_K^{\text{rig}}) = \begin{cases} \widehat{\mathbb{Z}} & , K \in p\text{-NField} \\ \mathbb{Z}/(2) & , K = \mathbb{R} \\ 0 & , K = \mathbb{C} \end{cases}.$$

In particular, there is a unique isomorphism of continuous extensions

$$1 \rightarrow P_K^{\text{rig}} \rightarrow W_K^{\text{rig}} \rightarrow \text{Gal}_K \rightarrow 1,$$

and the group of automorphisms of this extension are exactly inner automorphisms coming from  $P_K^{\text{rig}}$ . And we can define  $H^1(W_K^{\text{rig}}, G) \triangleq H^1(W_K^{\text{rig}}, G(K^{\text{sep}}))$  independent of  $W$  chosen.  $\lrcorner$

*Proof:* Cf. [Kal16].  $\square$

**Def. (20.5.2.38) [Twisted Cohomology Groups].** Let  $\mathcal{A}_K$  be the category of injections  $Z \rightarrow G$  where  $G \in \text{linAlgGrp}/K$  and  $Z \leq Z(G)$  is a finite central subgroup. For  $[Z \rightarrow G] \in \mathcal{A}_K$ , let  $H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow G) \in \text{Set}$  be the cohomology classes of those 1-cocycles in  $Z^1(W_K^{\text{rig}}, G)$  (20.5.2.37) whose restriction to  $P_K^{\text{rig}}$  takes image in  $Z$ . Then there is a functorial map of sets

$$H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow G) \rightarrow H^1(K, G/Z).$$

And when  $Z(G)$  is finite, we denote  $H^1(W_K^{\text{rig}}, G)_{\text{bas}} \triangleq H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z(G) \rightarrow G)$ .  $\lrcorner$

**Thm. (20.5.2.39) [Properties of Twisted Cohomology Groups].** For each finite central subgroup  $Z \leq G$ , there is a commutative diagram

$$\begin{array}{ccccccc}
 1 & \longrightarrow & H^1(K, Z) & \xrightarrow{\text{inf}} & H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow Z) & \xrightarrow{\text{res}} & \text{Hom}(P_K^{\text{rig}}, Z) \\
 & & \downarrow & & \downarrow & & \downarrow = \\
 1 & \longrightarrow & H^1(K, G) & \xrightarrow{\text{inf}} & H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow G) & \xrightarrow{\text{res}} & \text{Hom}(P_K^{\text{rig}}, Z) \longrightarrow * \\
 & & \downarrow = & & \downarrow & & \downarrow b \\
 & & H^1(K, G) & \xrightarrow{\text{inf}} & H^1(K, G/Z) & \xrightarrow{\text{res}} & H^2(K, Z) \longrightarrow * \\
 & & \downarrow & \swarrow & \downarrow & & \downarrow \\
 & & H^1(K, G/Z(G)) & & 1 & & 1
 \end{array}
 \quad (20.5.2.38)$$

with exact rows and columns, where  $*$  is to be taken as  $H^2(K, G)$  if  $G$  is commutative and disregarded otherwise. In particular

- $\#\text{Hom}(P_K^{\text{rig}}, Z)^{\text{Gal}_K} < \infty$  and  $\#H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow Z) < \infty$ .
- Each fiber of the map  $H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow Z) \xrightarrow{\text{res}} \text{Hom}(P_K^{\text{rig}}, Z)$  is naturally identified with  $H^1(K, G^\dagger)$  for some inner form  $G^\dagger$  of  $G$ .

*Proof:* Cf. [Kal16](3.6).  $\square$

**Def. (20.5.2.40) [Rigid Inner Twists].** For  $G \in \text{RedGrp}/K$ , a **rigid inner twist** of  $G$  is pair  $(G', \xi, z)$  where  $G' \in \text{RedGrp}/K$  and  $\xi : G_{K^{\text{sep}}} \cong G'_{K^{\text{sep}}}$  is an inner twist, and  $z \in Z^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z(G) \rightarrow G)$  (20.5.2.38) s.t.  $\xi^{-1}\sigma(\xi) = \text{Ad}(\bar{z}(\sigma))$  for each  $\sigma \in \text{Gal}_K$ , where  $\bar{z}$  is the image of  $z$  in  $H^1(K, G_{\text{ad}})$ .

And an isomorphism of rigid inner twists  $(G'_1, \xi_1, z_1) \cong (G'_2, \xi_2, z_2)$  is a pair  $(f, \delta)$  where  $f : G'_1 \cong G'_2$  is an isomorphism and  $\delta \in G(K^{\text{sep}})$  s.t.  $\xi_2^{-1}f\xi_1 = \text{Ad}(\delta)$  and  $z_1(w) = \delta^{-1}z_2(w)\sigma_w(\delta)$  for any  $w \in W_K^{\text{rig}}$ , where  $\sigma_w$  is the image of  $w$  in  $\text{Gal}_K$  (20.5.2.37).  $\lrcorner$

**Prop. (20.5.2.41) [Rigid Inner Twists and Strong Rational Forms].** If  $K = \mathbb{R}$  and  $G \in \text{RedGrp}/K$ , then there is an equivalence between the groupoid of rigid inner twists of  $G$  and the groupoid of strong rational forms defined by [Adams-Barbasch-Vogan, The Langlands classification and irreducible characters for real reductive groups]  $\square$ .

*Proof:* Cf. [Kal16]P5.2. □

**Def. (20.5.2.42) [Twisted Cohomology Groups of Reductive Groups].** Let  $\mathcal{R}_K \subset \mathcal{A}_K$  (20.5.2.38) be the subcategory of injections  $Z \rightarrow G$  s.t.  $G \in \text{RedGrp}/K$ . For each  $[Z \rightarrow G] \in \mathcal{A}_K$ , denote  $\overline{G} = G/Z$ , then there is an isogeny  $\widehat{\overline{G}} \rightarrow \widehat{G}$ . Denote  $Z(\widehat{\overline{G}})^+$  to be the preimage of  $Z(\widehat{G})^{\text{Gal}_K}$  in  $\widehat{\overline{G}}$ , then we get a functor

$$\mathcal{R}_K \rightarrow \text{Set} : [Z \rightarrow G] \rightarrow \pi_0(Z(\widehat{\overline{G}}))^{\vee}.$$

┘

**Thm. (20.5.2.43) [Kaletha-Maps, Kaletha [Kal16]].** Situation as in (20.5.2.42),

- There exists a unique functorial map of sets

$$H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow G) \rightarrow \pi_0(Z(\widehat{\overline{G}}))^{\vee} \text{ (20.5.2.42),}$$

from  $\mathcal{R}_K$  to  $\text{Set}$  s.t.

- Its restriction to objects of the form  $[1 \rightarrow T]$  where  $T \in \text{Tori}/K$  is the Tate-Nakayama map (10.5.2.2).
- It satisfies the commutative diagram of functors

$$\begin{array}{ccc} H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow G) & \xrightarrow{\text{res}} & \text{Hom}(P_K^{\text{rig}}, Z) \\ & \searrow & \swarrow f \\ & \text{Hom}(P_K^{\text{rig}}, Z) & \end{array} \quad (20.5.2.39),$$

where  $f$  is the natural map ?

- The above map is an isomorphism when  $G \in \text{Tori}/K$ .
- The above map is an isomorphism when  $K \in p\text{-NField}$ . In particular,  $H^1(P_K^{\text{rig}} \rightarrow W_K^{\text{rig}}, Z \rightarrow Z)$  has a canonical Abelian group structure.
- The kernel and cokernel of the above map is can be explicitly described when  $K \in \{\mathbb{R}, \mathbb{C}\}$ .
- When  $Z = 1$ , the above map is the Kottwitz-map  $\kappa_G : H^1(K, G) \rightarrow \pi_0(Z(\widehat{G}))^{\vee}$  (28.6.1.30).

┘

*Proof:* Cf. [Kal16]Section4. ? □

### 3 over Adeles

References are [Weil and Grothendieck approaches to adelic points, Conrad] where he has more canonical way to define the topology, and [P-R94],

**Notation (20.5.3.1).**

- Let  $F \in \text{GField}$ .
- Let  $X, Y \in \text{Aff}^{\text{ft}}/F$ .
- Let  $G \in \text{linAlgGrp}/F$ .

┘

**Prop. (20.5.3.2)[Adelization of Affine Schemes].** Suppose  $X \subset \mathbb{A}^n$ , the f.m. defining equations has coefficients in  $\mathcal{O}_{F_v}$  for a.e.  $v \in \Sigma_F^{\text{fin}}$ , so it is clear to see that there is a bijection

$$X(\mathbf{A}_F) \cong \prod_{v \in \Sigma_F}^I (X(F_v), X(\mathcal{O}_{F_v})).$$

Then we endow the topology on  $X(\mathbf{A}_F)$  with that of the RHS.  $\square$

**Lemma (20.5.3.3).** If  $X \subset \mathbb{A}^m, Y \subset \mathbb{A}^n$  and  $f : X \rightarrow Y \in \text{Sch}$ , then  $f_{\mathbf{A}}(X(\mathbf{A}_F)) \subset Y(\mathbf{A}_F)$  (20.5.3.2), and this is a continuous map.  $\square$

*Proof:* The coordinates of  $f$  has coefficients in  $\mathcal{O}_{F_v}$  for a.e.  $v \in \Sigma_F$ , so  $f_{F_v}(X(\mathcal{O}_{F_v})) \subset Y(\mathcal{O}_{F_v})$  for a.e.  $v$ . This implies that  $f_{\mathbf{A}}$  is continuous.  $\square$

**Prop. (20.5.3.4)[Adelization of Algebraic Schemes].** For any  $X \in \text{Sch}^{\text{ft}}/F$ , choose a finite covering  $X = \cup_i X_i$  by affine open subschemes and choose embeddings  $f_i : X_i \rightarrow \mathbb{A}_F^{n_i}$ . Then define

$$X(\mathbf{A}_F) = \bigcup_i f_i^{-1}(\mathbf{A}_F^{n_i}) \quad (20.5.3.2)$$

as a topological space. Then this construction is independent of the choices of  $f_i$ , and any  $f : X \rightarrow Y \in \text{Sch}^{\text{ft}}/F$  induces a map of adele spaces  $f_{\mathbf{A}} : X(\mathbf{A}_F) \rightarrow Y(\mathbf{A}_F)$ .  $\square$

*Proof:* As in the proof of (20.5.3.3), it suffices to show that  $f_{F_v}(X(\mathcal{O}_{F_v})) \subset Y(\mathcal{O}_{F_v}) = \cup_j Y_j(\mathcal{O}_{F_v})$  for a.e.  $v \in \Sigma_F$ . By the definition of  $X(\mathbf{A}_F)$ , it suffices to assume that  $X \subset \mathbb{A}_F^n$  is affine.

Let the coordinates of the map  $X \rightarrow Y_j$  be given by  $y_1^j, \dots, y_{n_i}^j \in R(X)$ , and let  $I_j$  be the ideal of  $F[X_1, \dots, X_n]$  consisting of functions  $\psi$  s.t. each  $y_i^j$  can be written as  $y_i^j = \varphi_i^j/\psi$  where  $\varphi_i^j \in F[X_1, \dots, X_n]$  (such a representation may not be unique). Then  $\cap_j V(I_j) = \emptyset$ , because any  $x \in X$  is mapped to some  $Y_j$ , which means that some nbhd of  $x$  is mapped to  $Y_j$ , which means  $y_i^j$  can be represented by rational functions with denominator non-vanishing at  $x$ .

So there exists  $\psi_j \in I_j$  s.t.  $\sum_j \psi_j = 1$ . And let  $y_i^j = \varphi_i^j/\psi_j$  for each  $i \in [n_j]_+$  where  $\varphi_i^j \in F[X_1, \dots, X_n]$ . There are only f.m. polynomials, so for a.e.  $v \in \Sigma_F$ , all the coefficients of these functions are in  $\mathcal{O}_{F_v}$ .

If  $v$  is such a place, then  $f_{F_v}(X(\mathcal{O}_{F_v})) \subset \cup_j Y_j(\mathcal{O}_{F_v})$ : Because  $\psi_j(x) \in \mathcal{O}_{F_v}$  for each  $j$  and  $\sum_j \psi_j(x) = 1$ , there must be some  $j$  s.t.  $\psi_j(x) \in \mathcal{O}_{F_v}^*$ , and then  $y_i^j = \varphi_i^j/\psi_j \in \mathcal{O}_{F_v}$  for each  $i \in [n_j]_+$ .  $\square$

**Cor. (20.5.3.5).**

- If  $X \in \text{Sch}^{\text{ft}}/F$  and  $X = \cup X_i$  is a covering by open subschemes, then  $X(\mathbf{A}_F) = X_i(\mathbf{A}_F)$ .
- If  $X \rightarrow Y, Z \rightarrow Y \in \text{Sch}^{\text{ft}}/F$ , then  $(X \times_Y Z)(\mathbf{A}_F) = X(\mathbf{A}_F) \times_{Y(\mathbf{A}_F)} Z(\mathbf{A}_F)$ .
- If  $X \hookrightarrow Y \in \text{Sch}^{\text{ft}}/F$  is a closed embedding, then  $X(\mathbf{A}_F) \hookrightarrow Y(\mathbf{A}_F)$  is a closed embedding.
- If  $X \rightarrow Y \in \text{Sch}^{\text{ft}}/F$  is proper, then  $X(\mathbf{A}_F) \rightarrow Y(\mathbf{A}_F)$  is proper. ?
- If  $X \rightarrow Y \in \text{Sch}^{\text{ft}}/F$  is smooth surjective with geo.connected fibers, then  $X(\mathbf{A}_F) \rightarrow Y(\mathbf{A}_F)$  is open. ?
- Notice that for open immersions, the induced inclusion may not be immersions. For example, the embedding  $\mathbb{G}_m \subset \mathbb{A}^1$  induces  $\mathbf{I}_F \subset \mathbf{A}_F$ , which is not an immersion(it has finer topology).  $\square$

*Proof:* Cf.[Weil and Grothendieck approached to adelic points, Conrad].  $\square$

**Prop. (20.5.3.6) [Surjection of Adelic Points].** Let  $f : X \rightarrow Y \in \text{Sch}^{\text{ft}}/F$  be a morphism s.t. there exists a local section at each  $y \in Y$ , then  $f_{\mathbf{A}}$  is surjective.  $\lrcorner$

*Proof:* There exists a finite open covering  $Y_i$  of  $Y$  and subschemes  $X_i$  of  $X$  s.t.  $X_i \xrightarrow{f_i} Y_i$ , then by (20.5.3.4),  $Y(\mathbf{A}_F) = \cup_i Y_i(\mathbf{A}_F) = \cup_i f(X_i(\mathbf{A}_F)) \subset f(X(\mathbf{A}_F))$ .  $\square$

**Cor. (20.5.3.7).** If  $f : X \rightarrow Y \in \text{Sch}^{\text{ft}}/F$  is surjective on  $L$ -points for each  $L/F \in \mathbf{Field}$  and  $Y$  is irreducible, then  $f_{\mathbf{A}_F}$  is surjective.  $\lrcorner$

*Proof:* Apply the hypothesis to  $L = R(Y)$  gives that there exists a local section to  $f$  at each  $y \in Y$ .  $\square$

**Prop. (20.5.3.8).**  $G(\mathbf{A}_F)/G^0(\mathbf{A}_F)$  is compact.  $\lrcorner$

*Proof:* As  $G^0$  has finite index in  $G$ , for each  $v \in \Sigma_F$ , there exists a finite set  $C_v \in G(F_v)$  s.t.  $G(F_v) = C_v G^0(F_v)$ . It suffices to show that for a.e.  $v$ , such a  $C_v$  can be chosen to be contained in  $G(\mathcal{O}_{F_v})$ . For this, it suffices to show that  $G(F_v) = G(\mathcal{O}_{F_v})G^0(F_v)$ . Cf.[P-R94]P252?  $\square$

### Integral Models

**Prop. (20.5.3.9) [Smooth Group Models Exist for Affine Schemes].** For  $G \in \text{GrpSch}^{\text{aff,ft}}/F$ , there exists some  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and  $\mathcal{G} \in \text{GrpSch}^{\text{sm,ft}}/\mathcal{O}_{F,S}$  that is a smooth model for  $G$  (20.3.1.1). Any two such  $\mathcal{G}$  are isomorphic over  $\mathcal{O}_{F,S}$  for some  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$ .  $\lrcorner$

*Proof:* There exists  $\mathcal{G} \in \text{GrpSch}^{\text{ft}}/\mathcal{O}_{F,S}$  s.t.  $\mathcal{G}_F \cong G$  because the multiplication  $G \times G \rightarrow G$  is defined by f.m. equations. Then we can enlarge  $S$  to assume that  $G$  is smooth by (7.6.4.5). Any two model are isomorphic over  $F$ , and the isomorphism is given by f.m. equations, so they are isomorphic over  $\mathcal{O}_{F,S}$  for some  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$ .  $\square$

**Prop. (20.5.3.10) [Integral Models Exist and are a.e. Hyperspecial].** For  $G \in \text{RedGrp}/F$  and  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$ , if  $\mathcal{G} \in \text{GrpSch}/\mathcal{O}_{F,S}$  is a smooth model of  $G$  (20.3.1.1), then after enlarging  $S$ , we can assume  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_{F,S}$ . In particular,  $\mathcal{G}(\mathcal{O}_{F_v})$  is a hyperspecial subgroup of the reductive group  $G_{F_v} \in \text{RedGrp}/F_v$  for a.e.  $v \in \Sigma_F^{\text{fin}}$ .  $\lrcorner$

*Proof:* After enlarging  $S$ , we may assume  $\mathcal{G}/\mathcal{O}_{F,S}$  is smooth by (7.6.4.5). Then by (10.5.6.8) and (10.5.6.11),  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_{F,S}$ .  $\square$

**Cor. (20.5.3.11) [Compact Open Subsets are a.e. Hyperspecial].** For  $G \in \text{RedGrp}/F$ ,  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_{F,S}$  a model for  $G$  (which exists by (20.5.3.10)), if  $\mathcal{K}^S \leq G(\mathbf{A}_F^S)$  is a compact open subgroup, then there exists  $S \subset_{\text{fin}} S' \subset \Sigma_F$  s.t.  $\mathcal{K}^S = \mathcal{K}_{S'}^S \mathcal{K}^{S'}$ , where  $K_{S'}^S \leq G(\prod_{v \in S' \setminus S} F_v)$  and  $\mathcal{K}^{S'} \cong \mathcal{G}(\widehat{\mathcal{O}_F}^{S'})$ . In particular,  $\mathcal{K}_v \leq G(F_v)$  is a hyperspecial subgroup for  $v \in \Sigma_F \setminus S'$ .  $\lrcorner$

*Proof:* This follows from the definition of  $G(\mathbf{A}_F)$  (20.5.3.2).  $\square$

### Class Groups

**Thm. (20.5.3.12)** [Class Numbers are Finite, Borel/Conrad/Oesterle/Prasad]. For any  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and any compact open subgroup  $\mathcal{K}^S \leq G(\mathbf{A}_F^S)$ , we have

$$\#G(F) \backslash G(\mathbf{A}_F^S) / \mathcal{K}^S < \infty.$$

The **class number (of algebraic group)** of  $G$  is defined to be

$$\text{cl}(G) = \#G(K) \backslash G(\mathbf{A}_{F,f}) / G\left(\prod_{v \in \Sigma_F^{\text{fin}}} \mathcal{O}_v\right) < \infty.$$

┘

*Proof:* ? Cf. [P-R94]P251 and [Conrad, Finiteness theorems for algebraic groups over Function Fields].

By (20.5.3.8), we can assume that  $G$  is connected. And by restriction of scalars, we can assume that  $F = \mathbb{Q}$ .

? The double coset is finite because for any  $(g_v) \in G(\mathbf{A}_{F,f})$ , by strong approximation for  $\mathbb{G}_a$ , we can choose a  $g \in G(F)$  s.t.  $g^{-1}g_v \in K_v$  for any  $v \in \Sigma_F^{\text{fin}}$ , then the image is finite as  $U$  is compact open and  $\prod_v K_v$  is compact.  $\square$

**Lemma (20.5.3.13).** If  $G = N \rtimes H$  and  $N$  satisfies strong approximation for  $S_F(\infty)$  (20.5.3.15), then  $\text{cl}(G) \leq \text{cl}(H)$ .  $\square$

*Proof:* For any  $x \in G(\mathbf{A}_F)$ , because  $x^{-1}N'_{F,\infty}x$  is open in  $N(\mathbf{A}_F)$ , by strong approximation, we have  $N(\mathbf{A}_F) = (x^{-1}N'_{F,\infty}x)N(\mathbf{A}_{F,\infty})N(F) = x^{-1}N'_{F,\infty}xN(F)$ . In particular,  $N(\mathbf{A}_F) = N'_{F,\infty}N(F)$ , and

$$xN'_{F,\infty}N(F) = N'_{F,\infty}xN(F).$$

Thus

$$G(\mathbf{A}_F) = H(\mathbf{A}_F)N(\mathbf{A}_F) = H(\mathbf{A}_F)N'_{F,\infty}N(F) = N'_{F,\infty}H(\mathbf{A}_F)N(F).$$

Now if  $H(\mathbf{A}_F) = \cup_i H'_i x_i H(F)$ , then

$$G(\mathbf{A}_F) = \bigcup_i N'_{F,\infty} H'_i x_i H(F) N(F) = G'_{F,\infty} x_i G(F).$$

$\square$

**Def. (20.5.3.14)** [Hasse-Principle].  $G$  is said to satisfy the **Hasse-principle** if the map of sets:

$$H^1(F, G) \rightarrow \prod_{v \in \Sigma_F} H^1(F_v, G)$$

is injective.  $\square$

### Approximations

**Def. (20.5.3.15)** [Weak and Strong Approximations]. For  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ ,  $X \in \text{Sch}^{\text{sep}, \text{ft}}$  is said to satisfy

- The **strong approximation** w.r.t.  $S$  if the image of  $X(F) \rightarrow X(\mathbf{A}_F^S)$  is dense.

- The **weak approximation** w.r.t.  $S$  if the image of  $X(F) \rightarrow X(\mathbf{A}_{F,S})$  is dense.  
If  $X = G$  is an algebraic group, then the following are equivalent:
- The image of  $G(F)$  is dense in  $G(\mathbf{A}_F^S)$ ,
- $G(F)G(\mathbf{A}_{F,S})$  is dense in  $G(\mathbf{A}_F)$ .
- For any compact open subgroup  $U^S \subset G(\mathbf{A}^S)$ ,  $G(\mathbf{A}_F) = G(F)G(\mathbf{A}_{F,S})U^S$ .

In particular,

$$\Gamma \backslash G(\mathbf{A}_{F,S}) \cong G(F) \backslash G(\mathbf{A}_F) / U^S.$$

where  $\Gamma$  is the image of  $G(F) \cap (G(\mathbf{A}_{F,S}) \times U^S)$  in  $G(\mathbf{A}_{F,S})$ . ┘

**Prop. (20.5.3.16) [Strong Approximation for  $G_a$ ].** For any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ , strong approximation holds for  $S$  (20.5.3.15). In other words, the image of  $F$  is dense in  $\mathbf{A}_F^S$ , or equivalently,  $FF_v U = \mathbf{A}_F$  for any non-empty open subset  $U \subset \mathbf{A}_F^S$ . ┘

*Proof:* Cf. [MIT notes, 22.14]. ? □

**Cor. (20.5.3.17) [Strong Approximation for Unipotent Groups].** If  $U \in \text{AlgGrp}/F$  is unipotent, then  $U$  satisfies the strong approximation w.r.t. any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ . ┘

*Proof:*  $U$  is isomorphic to  $\text{Lie}(U)$  as schemes ?, so the assertion follows from (20.5.3.16). □

**Cor. (20.5.3.18) [Weak Approximation for Parabolic Subgroups].** If  $F \in \text{NField}$  and  $G$  is connected and has a Levi decomposition  $G = LU$  where  $U$  is the unipotent, then for any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ ,  $G$  admits weak (resp. strong) approximation w.r.t.  $S$  iff  $M$  does. ┘

*Proof:* Cf. [P-R94] Chap7. □

**Thm. (20.5.3.19) [Weak Approximations].** If  $F \in \text{NField}$  and  $G \in \text{RedGrp}^{\text{ss}}/F$  is simply-connected or adjoint, then  $G$  admits weak approximation (20.5.3.15) for any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ . ┘

*Proof:* Cf. [P-R94] Chap7.3. □

**Thm. (20.5.3.20) [Weak Approximations for Homogenous Spaces].** If  $F \in \text{NField}$  and  $G$  is connected s.t.  $G_{\text{s, st}}$  is simply-connected and  $G_{\text{tor}}$  is quasi-trivial (Cf. [Getz] P65 ?), and  $H \leq G$  is a connected subgroup s.t.  $H_{\text{tor}}$  splits over a cyclic extension of  $F$ , then  $G/H$  satisfies weak approximation w.r.t. any non-empty  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ . ┘

*Proof:* Cf. [Borovoi. The defect of weak approximation for homogeneous spaces] Cor3.14. □

**Thm. (20.5.3.21) [Strong Approximation].** If  $G$  is semisimple and simply-connected (10.4.3.1), and  $G(\mathbf{A}_{F,S})$  is non-compact for some  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$ , then  $G$  satisfies strong approximation for  $S$ . ┘

*Proof:* Cf. [Algebraic Groups and Number Theory, P427]. □

**Remark (20.5.3.22).** This is not true for non-semisimple or non-simply connected: for  $\mathbb{G}_m$ ,  $\mathbb{Q}^\times$  is not dense in  $\mathbf{I}_{\mathbb{Q},f}^\times$ .

For  $\text{PGL}(2)$ , the determinant of  $\text{PGL}(2; \mathbb{Q})$  is  $\mathbb{Q}^\times / (\mathbb{Q}^\times)^2$  and the determinant of  $\text{PGL}(2; \mathbf{A}_{\mathbb{Q},f})$  is  $\mathbf{I}_{\mathbb{Q},f} / \mathbf{I}_{\mathbb{Q},f}^2$ , which is not dense. ┘

**Cor. (20.5.3.23) [Strong Approximation for  $\text{SL}(n)$ ].**

- $\text{SL}(n)_F$  satisfies strong approximation for  $\Sigma_F^\infty$ .

- Let  $\mathcal{K}_0$  be an open compact subgroup of  $\mathrm{GL}(n; \mathbf{A}_{F,f})$ , then

$$\mathrm{GL}(n; F) \mathrm{GL}(n; A_\infty) \mathcal{K}_0 T_1(A) = \mathrm{GL}(n; A).$$

- Let  $\mathcal{K}_0$  be an open compact subgroup of  $\mathrm{GL}(n; \mathbf{A}_{F,f})$  that the image of  $\mathcal{K}_0$  under the determinant map is  $\prod_{v \in \Sigma_F^{\mathrm{fin}}} \mathcal{O}_v^*$ , then

$$\det : \mathrm{GL}(n; F) \mathrm{GL}(n; \mathbf{A}_\infty) \backslash \mathrm{GL}(n; \mathbf{A}_F) / \mathcal{K}_0 \rightarrow F^\times \mathbf{I}_{F,\infty} \backslash \mathbf{I}_F / \prod_{v \in \Sigma_F^{\mathrm{fin}}} \mathcal{O}_{F_v}^* = \mathrm{Cl}(F)$$

is an isomorphism. ┘

*Proof:* 1 is a direct consequence of (20.5.3.21).

2 is a direct consequence of 1.

3:  $\det$  is clearly surjective. To show it is injective, if  $\det(a) = \det(b) \in \mathrm{Cl}_1(F)$ , then  $\det(a) = \det(g_1 g_\infty b k_0)$  where  $g_1 \in \mathrm{GL}(n; F)$ ,  $g_\infty \in \mathrm{GL}(n; \mathbf{A}_{F,\infty})$ ,  $k_0 \in \mathrm{GL}(n; \mathbf{A}_{F,f})$  by hypothesis. Then  $g_1 g_\infty b k_0 a^{-1} \in \mathrm{SL}(n; \mathbf{A}_F)$ , and by item 1 it is of the form  $h_1 h_\infty k_0$  where  $h_1 \in \mathrm{SL}(n; F)$ ,  $h_\infty \in \mathrm{SL}(n; \mathbf{A}_{F,\infty})$ ,  $k_1 \in a(\mathcal{K}_0 \cap \mathrm{SL}(n; \mathbf{A}_{F,f}))a^{-1}$ . Then the assertion follows. □

**Prop. (20.5.3.24) [Passage From Archimedean to Adele via Congruence Subgroups].** For  $F \in \mathbf{NField}$  and  $\mathcal{K} \leq G(\mathbf{A}_{F,f})$  be a compact open subgroup, let  $\{g_i\} \subset G(\mathbf{A}_{F,f})$  be a finite set of double-coset representatives for  $G(F) \backslash G(\mathbf{A}_{F,f}) / \mathcal{K}$  (20.5.3.12), then

$$\coprod_i \Gamma_i \backslash G(F \otimes \mathbb{R}) \cong G(F) \backslash G(\mathbf{A}_F) / \mathcal{K}$$

where  $\Gamma_i$  is the image of  $(g_i \mathcal{K} g_i^{-1} \times G(F \otimes \mathbb{R})) \cap G(F)$  in  $G(F \otimes \mathbb{R})$ , which is an arithmetic subgroup of  $G(F)$  (20.4.0.2). ┘

*Proof:* The map  $\Gamma_i \backslash G(F \otimes \mathbb{R}) \rightarrow G(F) \backslash G(\mathbf{A}_F) / \mathcal{K}$  is given by

$$\Gamma_i g_\infty \mapsto G(F) g_\infty g_i \mathcal{K}.$$

□

**Thm. (20.5.3.25) [Strong Approximation for Homogenous Spaces].** If  $F \in \mathbf{NField}$ ,  $G \in \mathbf{RedGrp}^{\mathrm{ss}} / F$  is geometrically almost-simple (10.4.2.6) and  $H \leq G$  is a simply-connected semisimple subgroup, then for any  $\emptyset \subset_{\mathrm{fin}} S \subset \Sigma_F$ ,  $G/H$  satisfies strong approximation for  $S$  iff  $(G/H)(\mathbf{A}_{F,S})$  is non-compact. ┘

*Proof:* Cf. [Strong approximation for algebraic groups, Rapinchuk2014] Prop 2.4. □

## 4 Adelic Quotients

**Notation (20.5.4.1).**

- Let  $F \in \mathbf{GField}$ .
- Let  $X, Y \in \mathcal{A}^{\mathrm{ft}} / F$ .
- Let  $G \in \mathbf{linAlgGrp} / F$ .

┘



**Prop. (20.5.4.2).** For  $X \in \text{Sch}^{\text{ft}}/F$ ,  $X(F)$  is discrete and closed in  $X(\mathbf{A}_F)$ . ┘

*Proof:* It suffices to check for an affine covering so we can assume that  $X$  is affine, and the assertion reduces to  $\mathbb{A}^n$  and reduces to (19.4.1.3). □

**Def. (20.5.4.3) [Adelic Quotients].** Define

$$G(\mathbf{A}_F)^1 = \bigcap_{\chi \in X^*(G)} \ker(|-|_{\mathbf{A}} \circ \chi : G(\mathbf{A}_F) \rightarrow \mathbb{R}_+)$$

. Then  $G(F) \subset G(\mathbf{A}_F)$  is discrete and closed by product formula.

If  $F \in \mathbf{NField}$ , let  $A(G) \leq G(F_{\infty}) \leq G(\mathbf{A}_F)$  be the neutral component of the  $\mathbb{R}$ -point of the largest  $\mathbb{Q}$ -split torus  $A_G \leq \text{Res}_{F/\mathbb{Q}}(Z(G))$ . And for  $F \in \mathbf{FField}$  it is defined in [Getz]P68.

Define  $[G]_F = A(G)G(F) \backslash G(\mathbf{A}_F)$ , called the **adelic quotient** of  $G$ . ┘

**Prop. (20.5.4.4).** If  $G \in \text{RedGrp}/F$ , then  $A_G G(\mathbf{A}_F)^1 \leq G(\mathbf{A}_F)$  (20.5.4.3) is of finite index. And if  $F \in \mathbf{NField}$  we have  $A(G)G(\mathbf{A}_F)^1 = G(\mathbf{A}_F)$ .

In particular, if  $F \in \mathbf{NField}$  then there is a right  $G(\mathbf{A}_F)^1$ -equivariant homeomorphism

$$G(F) \backslash G(\mathbf{A}_F)^1 \cong [G] (20.5.4.3).$$

┘

*Proof:* ? □

### Tamagawa-Numbers

References are [Kottwitz. Tamagawa numbers], [Oesterlé. Nombres de Tamagawa et groupes unipotents en caractéristique  $p$ ], [P-R94]Chap5.3.

**Def. (20.5.4.5) [Tamagawa-Measures].** Suppose  $G$  is smooth and  $\dim G = n$ , take a  $F$ -rational differential  $n$ -form  $\omega \in \wedge^n \mathfrak{g}^*$ , then  $\omega$  induces a left-invariant measure  $\omega_v$  on  $G(F_v)$  for any  $v \in \Sigma_F$ . Thus it induces a Haar measure  $\mu_v$  on  $G(F_v)$  by (12.10.1.19).

Choose  $\lambda_v \in \mathbb{R}^+$  s.t.  $\prod_{v \in \Sigma_F} (\lambda_v \mu_v(G(\mathcal{O}_{F_v})))$  converges absolutely, then we get a **Tamagawa-measure**  $d\mu = \prod'_{v \in \Sigma_F} (\lambda_v d\mu_v)$  on  $G(\mathbf{A}_F)$ .

This Tamagawa-measure is independent of the choice of  $\omega$  by product formula (19.3.1.8), but depends on the normalizing factors  $\lambda_v$ . But usually there is a natural way to choose  $\lambda_v$  (Cf. [Oesterlé. Nombres de Tamagawa et groupes unipotents en caractéristique  $p$ ], [Weil, Adeles and Algebraic groups, Appendix 2] and [Kottwitz. Tamagawa numbers]). For example,  $G \in \text{RedGrp}^{\text{ss}}/F$ , then we can (and do) take  $\lambda_v = 1$  for each  $v \in \Sigma_F$ . ? ┘

**Prop. (20.5.4.6).** If  $G$  is smooth, then  $\Delta_{G(\mathbf{A}_F)}$  is trivial on  $G(\mathbf{A}_F)^1$  (and thus on  $G(F)$ ). In particular,  $G(\mathbf{A}_F)^1$  is unimodular by (12.10.3.7) because there is a  $G(\mathbf{A}_F)$ -invariant measure on  $G(\mathbf{A}_F)/G(\mathbf{A}_F)^1$  (which is isomorphic to  $\mathbb{R}_+^m$  or  $\mathbb{Z}_+^m$ ). And if  $G^0$  is reductive, then  $G(\mathbf{A}_F)$  is unimodular. ┘

*Proof:* Because the Haar measures on  $G(\mathbf{A}_F)$  are products of Haar measures on  $G(F_v)$  for  $v \in \Sigma_F$ , we have

$$\Delta_{G(\mathbf{A}_F)}(g) = \prod_{v \in \Sigma_F} \delta_{G(F_v)}(g_v) = |\det(\text{Ad}(g) : \mathfrak{g} \otimes_F \mathbf{A}_F \rightarrow \mathfrak{g} \otimes_F \mathbf{A}_F)| = 1.$$

Notice that  $\det \text{Ad} : G \rightarrow \mathbb{G}_m$  is a character, so by the formula as above,  $\Delta_{G(\mathbf{A}_F)}$  is trivial on  $G(\mathbf{A}_F)^1$ .

The last assertion follows from (12.10.1.22). □

**Def. (20.5.4.7) [Tamagawa-Measures].** For  $G \in \mathcal{G}\mathcal{r}\mathcal{p}\mathcal{V}\mathcal{a}\mathcal{r}/F$ , choose the right Tamagawa measure on  $G(\mathbf{A}_F)$  (20.5.4.5) and a suitably chosen measure on  $A_G$  ? and the discrete measure on  $G(F)$ , we get a canonical quotient measure on the adelic quotient  $[G]$  (20.5.4.3) by (20.5.4.6), called the **Tamagawa-measure** on  $[G]$ .  $\lrcorner$

**Thm. (20.5.4.8) [Tamagawa-Numbers, Borel/Oesterlé/Conrad12].** If  $G \in \mathcal{G}\mathcal{r}\mathcal{p}\mathcal{V}\mathcal{a}\mathcal{r}/F$ , then  $\tau(G) = \text{Vol}([G]_F) < \infty$  with the Tamagawa measure on  $[G]$  (20.5.4.7). This number is called the **Tamagawa-number** of  $G$ . And  $[G]$  is compact iff any  $F$ -split torus  $T \leq G$  satisfies  $T_{\overline{F}} \leq R(G_{\overline{F}})$ .  $\lrcorner$

*Proof:* Cf. [Finiteness theorems for algebraic groups over function fields, Conrad] Thm1.3.6. and ThmA.5.5.  $\square$

**Cor. (20.5.4.9).**

- $G(F) \backslash G(\mathbf{A}_F)$  is compact iff the reductive part of  $G^0$  is anisotropic (10.4.2.24) (e.g.  $G \in \text{Red}\mathcal{G}\mathcal{r}\mathcal{p}^{\text{ss}}/F$ ).
- $G(F) \backslash G(\mathbf{A}_F)$  has finite volume if  $X_*(G^0) = e$ .

$\lrcorner$

*Proof:* Cf. [P-R94] P260. ?  $\square$

### Tamagawa-Number Conjecture

**Prop. (20.5.4.10) [Tamagawa-Number of Inner Forms, Kottwitz[Kot88]].** If  $F \in \mathcal{N}\mathcal{F}\mathcal{i}\mathcal{e}\mathcal{l}\mathcal{d}$ ,  $G \in \text{Red}\mathcal{G}\mathcal{r}\mathcal{p}/F$  and  $G_0 \in \text{Red}\mathcal{G}\mathcal{r}\mathcal{p}/F$  is a quasi-split inner form of  $G$ , then  $\tau(G) = \tau(G_0)$ .  $\lrcorner$

*Proof:* Cf. [Kot88] ?  $\square$

**Lemma (20.5.4.11) [Ono/Kottwitz[Kot84]].** For  $G \in \text{Red}\mathcal{G}\mathcal{r}\mathcal{p}/F$ , we have

$$\tau(G) = \frac{\#\pi_0\left(Z(\widehat{G})^{\text{Gal}_F}\right)}{\#\text{III}(F, Z(\widehat{G}))} \quad (20.5.4.8).$$

$\lrcorner$

*Proof:* Cf. [Kottwitz, Trace formula: cuspidal tempered terms].  $\square$

**Thm. (20.5.4.12) [Tamagawa Number Conjecture, Weil; Demazure/Langlands/Lai/-Mars/Ono/Tamagawa/Weil/Kottwitz[Kot88]].** If  $G \in \text{Red}\mathcal{G}\mathcal{r}\mathcal{p}^{\text{ss}}/F$  is simply-connected, then  $\tau(G) = 1$  (20.5.4.8).  $\lrcorner$

*Proof:* ? [Langlands, The volume of the fundamental domain for some arithmetical subgroups of Chevalley group], [Lai, Tamagawa number of reductive algebraic group]. [Ono, Tamagawa numbers, in Algebraic Groups and Discontinuous Subgroups].  $\square$

### Reduction Theory of $[G]_F$ (Fundamental Domains)

Reduction theory allows us to control the adelic quotient  $[G]$  when it is non-compact.

**Notation (20.5.4.13).**

- Let  $G \in \text{Red}\mathcal{G}\mathcal{r}\mathcal{p}/F$ .

┘

**Def. (20.5.4.14) [Siegel Sets].** A Siegel set in  $G(\mathbf{A}_F)^1$  is a set of the form

$$\mathfrak{S}(t) = \Omega_N \Omega_M A_{T_0}(t) \mathcal{K}$$

where ? Cf. [Getz]P71.

┘

**Prop. (20.5.4.15) [Fundamental Domain].**  $\mathrm{GL}(n; F) \backslash \mathrm{GL}(n; \mathbf{A}_F)$  has a fundamental domain which can be covered by a sufficiently large Siegel sets  $\mathfrak{S}_{c,d}$ .

┘

*Proof:*

□

**Def. (20.5.4.16) [Global Siegel Sets].** For  $c, d > 0$ , then the global Siegel set  $\mathfrak{S}_{c,d} = K_f \times \mathfrak{S}_{c,d,\infty}$ ??  
And denote  $\overline{\mathfrak{S}_{c,d}}$  its image in  $Z(\mathbf{A}_F) \backslash \mathrm{GL}(2; \mathbf{A}_F)$ .

┘

**Prop. (20.5.4.17).** For  $c, d$  suitable chosen,  $\mathrm{GL}(2; \mathbf{A}_F) = \mathrm{GL}(2; F) \mathfrak{S}_{c,d}$ .

┘

*Proof:* We prove only for  $F = \mathbb{Q}$ ?: This is true for  $c \leq \sqrt{3}/2$  and  $d \geq 1$  because of the shape of the fundamental domain of  $\mathrm{GL}(2; \mathbb{R})$  for  $\mathrm{SL}(2; \mathbb{Z})$  (20.4.5.17).

□

## 5 over Number Fields

**Notation (20.5.5.1).**

- Let  $F \in \mathbf{GField}$ .

┘

**Def. (20.5.5.2) [Groups of Non-Compact Types].** Let  $G \in \mathbf{AlgGrp}/\mathbb{Q}$  be semisimple, then  $G$  is said to be a **group of compact type** if  $G(\mathbb{R})$  is compact, and said to be a **group of non-compact type** if it doesn't contain a non-trivial normal subgroup of compact type.

┘

**Prop. (20.5.5.3) [Real Approximations].** If  $G \in \mathbf{AlgGrp}/\mathbb{Q}$  satisfies each connected components of  $G$  contains a rational point, then  $G(\mathbb{Q})$  is dense in  $G(\mathbb{R})$ .

┘

*Proof:* Cf. [Mil17]P54.

□

### Global Kottwitz-Maps

**Cor. (20.5.5.4).** For  $F \in \mathbf{GField}$  and  $G \in \mathbf{RedGrp}/F$ , there is a natural exact sequence of pointed set

$$H^1(F, G) \rightarrow \bigoplus_{v \in \Sigma_F} H^1(F_v, G) \xrightarrow{\oplus_{v \in \Sigma_F} \kappa_{G_v}} \pi_0 \left( Z(\widehat{G})^{\mathrm{Gal}_F} \right)^\vee,$$

┘

*Proof:* Cf. [Kot86]P377?.

□

### Global Rigid Inner Twists

References are [Kal16, Kal16a] and [Kaletha, Global rigid inner forms and multiplicities of discrete automorphic representations].

**Def. (20.5.5.5) [Global Galois Gerbes].** Cf. [Kal16a]P31.?

┘

**Def. (20.5.5.6) [Global Rigid Inner Twists, Kal16a].** For  $G^* \in \mathbf{RedGrp}/F$ , a **rigid inner twist** of  $G^*$  is a triple  $(G, \xi, z)$  where  $\xi : G^* \rightarrow G$  is an inner twist and  $z \in Z^1(P^{\mathrm{rig}} \rightarrow \mathcal{E}^{\mathrm{rig}}, Z \rightarrow G^*)$  where  $Z \leq G$  is a finite central subgroup, s.t. the image of  $z$  in  $Z^1(F, G_{\mathrm{ad}}^*)$  equals  $z_{\mathrm{ad}}(\sigma) = \xi^{-1} \sigma(\xi)$ .

┘

## 20.6 Bruhat-Tits Theory

References are [I-M65, Tit79, B-T72, B-T84, B-T84a, B-T87, B-T87a, ?H-R08, Yu09, Yu15, K-P23], [Landvogt, A Compactification of the Bruhat-Tits Building], [the Bruhat-Tits Buildings of a  $p$ -adic Chevalley Group and an Application to Representation Theory, Rabinoff], [Buildings and Classical Groups, Garrett], [Period mappings and differential equations. From  $\mathbb{C}$  to  $\mathbb{C}_p$ , Andre],

### Notation (20.6.0.1).

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in \text{DVR}$  be Henselian with  $\kappa = \kappa^{\text{perf}}$  (if  $\# \kappa = \infty$ , use the notion of bounded subsets (20.5.1.3)).
- Let  $\Gamma_K = \{\text{ord}_K(x) | x \in K^\times\}$ .
- Use notations on reductive groups defined in [Group Theory III: Reductive Groups](#).
- Use notations on cohomology of reductive groups defined in [Group Theory III: Reductive Groups](#). Let  $G \in \text{RedGrp}/K$ .
- Let  $A_G \leq G$  be the maximal central split torus of  $G$ .
- Let  $H \leq G \in \text{AlgGrp}/K$ .
- $G(K)$  is a topological group with the topology coming from  $K$ .
- Use notations on (poly-)simplicial complexes defined in [Simplicial Complexes](#).

┘

### 1 Tits-Systems

References are [Bourbaki] and [K-P23].

**Def. (20.6.1.1) [Tits-Systems].** For  $G \in \text{Grp}$ , a **Tits-system** or a **BN-pair** in  $G$  is a tuple  $(B, N, S)$  where

- $B, N$  are subgroups of  $G$ ,  $S \subset N/(B \cap N)$ ,
- $G = \langle B \cup N \rangle$  and  $B \cap N \trianglelefteq N$ ,
- $S$  consists of elements of order 2, and generates  $W = N/(B \cap N)$ ,
- $BsBwB \subset BwB \cup BswB$  for any  $s \in S$  and  $w \in W$ ,
- $BsBsB \not\subseteq B$  for any  $s \in S$ ,

where for each  $w \in W$ , we denote  $Bw = Bn$  and  $wB = nB$  for any lift  $n \in N$  of  $w$ , which are well-defined cosets of  $B$  in  $G$ .

The group  $W = N/(B \cap N)$  is called the **Weyl-group of the Tits-system**. And  $B \cap N$  is usually denoted by  $T$ .

And  $(G, B, N, S)$  is called a **saturated Tits-system** if moreover we have

$$\bigcap_{n \in N} n^{-1}Bn = B \cap N.$$

┘

**Prop. (20.6.1.2)[Parabolic Subgroups].** In a Tits system  $(G, B, N, S)$ , any subgroup  $P \leq G$  containing  $B$  is of the form  $P = BW_X B$  where  $X \subset S$  and  $W_S$  is the subgroup of  $W$  generated by  $S$ . If  $P \neq G$ , then it is called a group is called a **standard parabolic subgroup** of  $G$  (w.r.t.  $B$ ). And any conjugate of a standard parabolic subgroup is called a **parabolic subgroup** of  $G$ .

Moreover, for  $X, X' \subset S$ ,  $BW_X B$  is conjugate to  $BW_{X'} B$  by  $g \in G$  iff  $X = X'$  and  $g \in BW_X B$ . Thus any parabolic subgroup  $P \leq G$  is conjugate to  $BW_X B$  for a unique subset  $X \leq S$ . This  $X$  is called the **type(parabolic subgroup)** of  $P$ .  $\lrcorner$

*Proof:* ? Cf.[Bourbaki, Lie Groups and Lie Algebras. Chapters 4-6, Elements of Mathematics]Cor4.2.5.  $\square$

**Cor. (20.6.1.3)[ $S$  is Uniquely Determined].** In a Tits system  $(G, B, N)$  (20.6.1.1),  $S \subset W$  consists exactly of elements  $s \in W$  s.t.  $B \cup BsB$  is a subgroup of  $G$ . In particular,  $S$  is determined by  $(B, N)$ .  $\lrcorner$

**Prop. (20.6.1.4)[Bruhat Decomposition].** Let  $(G, B, N)$  be a Tits system (20.6.1.1), then

$$G = \coprod_{w \in W} BwB.$$

$\lrcorner$

*Proof:* By (20.6.1.2), it suffices to show that these  $BwB$  are disjoint. ?  $\square$

**Prop. (20.6.1.5)[Parabolic Subgroups].** Let  $(G, B, N)$  be a Tits system, then

- Any parabolic subgroup of  $G$  (20.6.1.2) is equal to its normalizer.
- If  $Q \leq G$  is a parabolic subgroup containing two parabolic subgroups  $Q_1, Q_2 \leq G$ , then any  $g \in G$  s.t.  $gQ_1g^{-1} = Q_2$  satisfies  $g \in Q$ .

$\lrcorner$

*Proof:* ? Cf.[Bourbaki]Chap4.2.  $\square$

**Prop. (20.6.1.6)[Maps between Tits-Systems].** Let  $(G, B, N)$  be a Tits-system (20.6.1.1),

- If  $Z \leq T$  is a subgroup that is normal in  $G$ , set  $G' = G/Z, B' = B/Z, N' = N/Z$ , then  $(G', B', N')$  is a Tits-system with  $W' = W$ , and it is saturated iff  $(G, B, N)$  is (20.6.1.1).
- Suppose  $G \trianglelefteq G'$  and  $T' \leq G'$  normalizes  $B$  and  $N$ , and  $N$  normalizes  $T'$ , and  $G' = G.T'$  and  $T' \cap G = T$ . Set  $B' = B.T'$  and  $N' = N.T'$ , then  $(G', B', N')$  is a Tits-system with  $W' = W$ , and it is saturated iff  $(G, B, N)$  is.

$\lrcorner$

*Proof:* 1: This is clear as  $Z$  is normal in all groups under consideration.

2: Cf.[Kaletha-Prasad]P51.  $\square$

**Prop. (20.6.1.7).** Let  $(G, B, N, S), (G', B', N', S')$  be Tits-systems s.t.  $(G, B, N, S)$  is saturated, and  $f : G \rightarrow G'$  is a group homomorphism s.t.

- $\ker(f) \subset T$  (20.6.1.1).
- $f(G) \trianglelefteq G'$  and  $G'/f(G)$  is Abelian.
- $T'$  normalizes  $f(B)$ .
- $B' = f(B).T', N' = f(N).T'$ .

Then

- $G' = f(G).T'$ .
- $f^{-1}(T') = T$ .
- $T'$  normalizes  $f(BwB)$  for any  $w \in W$ .
- The map  $W \rightarrow W'$  induced by  $f$  (20.6.1.1) is an isomorphism that carries  $S$  bijectively onto  $S'$ .
- If  $P \leq G$  is parabolic (20.6.1.2), then so is  $P' = N_{G'}(f(P)) = f(P).T'$ , and  $P = f^{-1}(P')$ . This induces a type-preserving, order-preserving, conjugation-equivariant bijection between parabolic subgroups of  $G$  and  $G'$ .

┘

*Proof:* Cf. [Kaletha-Prasad]P52, 53. □

### Spherical Tits-Systems

**Prop. (20.6.1.8) [Tits-System from Reductive Groups].** Let  $k \in \mathbf{Field}$ ,  $G \in \mathbf{RedGrp}/k$  and  $P \leq G$  a minimal parabolic and  $N$  is the normalizer of a maximal  $k$ -split torus  $S \leq P$ , then  $(P(k), N(k))$  is a Tits-system in  $G(k)$  with finite Weyl group (20.6.1.1). And  $P \cap N = C_G(S)$ , which is a Levi  $k$ -subgroup of  $P$ . These Tits-systems are called **spherical Tits-systems**. ┘

*Proof:* Cf. [Borel, Linear Algebraic Groups]Prop21.15? □

**Prop. (20.6.1.9).** If  $k \in \mathbf{Field}$ ,  $\#k = \infty$ , then for  $G \in \mathbf{RedGrp}/k$ , any Tits-system in  $G(k)$  with finite Weyl group that satisfies a mild condition is a spherical Tits-system for some minimal parabolic  $P \leq G$  and some  $S$  (20.6.1.8). ┘

*Proof:* [Prasad, Weakly-split spherical Tits systems in quasi-reductive groups]Theorem B. □

### Decomposition into Irreducible Tits-Systems

**Prop. (20.6.1.10) [Coxeter-Systems].** For any Tits-system  $(G, B, N, S)$  (20.6.1.1),  $(W, S)$  is a Coxeter-system (3.9.1.1). And  $(G, B, N, S)$  is called an **irreducible Tits-system** if  $(W, S)$  is irreducible (3.9.1.2). ┘

## 2 Buildings

**Def. (20.6.2.1) [Buildings].** A **building** is a thick chamber complex  $X$  (4.13.4.2) with a family of chamber subcomplexes of  $X$  (4.13.4.2) called **apartments** of  $X$  s.t.

- Each apartment is a thin chamber complex (4.13.4.2) s.t. any two chambers of  $X$  is contained in some common apartment.
- For any two apartments  $a, a'$  containing a chamber  $C$  and a simplex  $\sigma$  (4.13.4.2), there is a simplicial isomorphism  $f : a \cong a'$  fixing vertices of  $C$  and  $\sigma$ .

More generally, we can define buildings for polysimplicial chamber complexes.

$X$  is called a **spherical building** if all apartments are spheres.  $X$  is called an **affine building** if all apartments are contractible. ┘

**Prop. (20.6.2.2) [Uniqueness Lemma].** Let  $X, Y$  be chamber complexes of the same dimension s.t. each codimension-1 simplex of  $Y$  is a subcomplex of at most 2 chambers. Let  $f, g : X \rightarrow Y$  be simplicial maps that sends non-stutter galleries (4.13.4.2) to non-stutter galleries, and agree on a chamber  $C$  of  $X$ , then  $f = g$ . ┘

*Proof:* Let  $D$  be any chamber of  $X$  and let  $(C = C_0, C_1, \dots, C_{r-1}, C_r = D)$  be a non-stutter gallery in  $X$ , then we can prove inductively that  $f$  and  $g$  agree on each  $C_i$ .  $\square$

**Cor. (20.6.2.3).** For any two apartments  $a, a'$  in a building  $X$  (20.6.2.1) containing a common chamber  $C$ , there exists a simplicial isomorphism  $f : a \cong a'$  fixing vertices of  $a \cap a'$ .

In fact, this is equivalent to the condition2 in the definition of buildings (20.6.2.1).  $\lrcorner$

*Proof:* For any simplex  $\sigma \subset a \cap a'$ , there is an isomorphism  $f_\sigma : a \rightarrow a'$  fixing vertices of  $C$  and  $D$  by the building axiom. But then by the uniqueness lemma these  $f_\sigma$  are all the same, thus inducing an isomorphism  $f : a \rightarrow a'$  fixing all vertices of  $a \cap a'$ .  $\square$

**Thm. (20.6.2.4) [Canonical Contractions].** Given an apartment  $\mathcal{A}$  in a building  $X$  (20.6.2.1) and a chamber  $C \subset a$ , there is a unique polysimplicial retraction  $\rho : X \rightarrow \mathcal{A}$  (4.13.4.1) sending non-stutter galleries starting at  $C$  to non-stutter galleries starting at  $C$ , called a **canonical contraction** from  $X$  to  $a$  centered at  $C$ . Moreover,

- This canonical contraction induces an isomorphism  $a' \rightarrow a$  for any apartment of  $X$  containing  $C$ .
- For any vertex  $x \in C$ , we have  $\rho^{-1}(x) = x$ .

$\lrcorner$

*Proof:* For any chamber  $D$  not contained in  $\mathcal{A}$ , let  $\mathcal{A}'$  be an apartment containing  $C, D$ , and let  $f : \mathcal{A}' \rightarrow \mathcal{A}$  be an isomorphism that fixes vertices of  $\mathcal{A}' \cap \mathcal{A}$  constructed in (20.6.2.3). Then  $\rho|_D$  doesn't depend on  $D$  chosen, by uniqueness lemma (20.6.2.2). It is clear that these maps for each  $D$  glue together to a contraction  $\rho : X \rightarrow \mathcal{A}$ .

For the last assertion, if  $\rho(y) = x$  is a vertex of  $C$ , let  $a'$  be an apartment containing  $y$  and  $C$ , then  $\rho$  is injective on  $\mathcal{A}'$ , thus we have  $y = x$ .  $\square$

**Cor. (20.6.2.5).** For two apartments  $C, D$  in a building  $X$  contained in a common apartment  $\mathcal{A}$ , a shortest gallery from  $C$  to  $D$  must be contained in  $\mathcal{A}$ .  $\lrcorner$

*Proof:* If  $C_0 = C, C_1, \dots, C_n = D$  is a shorted gallery from  $C$  to  $D$  that is not contained in  $a$ , then there exists  $j \in \mathbb{N}$  s.t.  $C_j \subset a$  but  $C_{j+1} \not\subset a$ . Let  $C'_{j+1}$  be the unique chamber in  $a$  different from  $C_{j+1}$  that  $C_{j+1} \cap C_j = C'_{j+1} \cap C_j$ , which exists because  $a$  is a thin chamber complex (20.6.2.1). Let  $\rho$  be the canonical retraction of  $X$  unto  $a$  with center  $C'_{j+1}$  (20.6.2.4), then  $\rho(C_{j+1}) = C_j$ , thus  $C_0 = C, \rho(C_1), \dots, \rho(C_{n-1}), C_n = D$  is a shorter path from  $C$  to  $D$ , contradiction.  $\square$

**Def. (20.6.2.6) [Foldings].** For a building  $X$ , a **folding** on an apartment  $a \in X$  is a simplicial map  $f : a \rightarrow a$  that is two-to-one and is a retraction onto its image. Two foldings  $f, f' : a \rightarrow a$  are called **opposite foldings** of each other if for any chamber  $C$  of  $X$  we have  $f^{-1}(f(C)) = f(C) \cup f'(C)$ .

The image of a folding is called a **half-apartment**.  $\lrcorner$

**Thm. (20.6.2.7) [Existence and Uniqueness of Foldings].** For a building  $X$  and two adjacent chambers  $C, D$  of  $X$  contained in a common apartment  $a$ , there exists mutually-opposite foldings (20.6.2.6)  $f, g : a \rightarrow a$  s.t.  $f(D) = C, g(C) = D$ .

Moreover, if we denote  $H = f(a), H' = g(a)$ , then  $f : H' \rightarrow H, g : H \rightarrow H'$  are isomorphisms,  $H \cup H' = a$ , and  $\dim(H \cap H') < \dim X$ . And we have

$$H = \bigcup_{E \text{ chambers of } a, \text{dist}(C,E) < \text{dist}(D,E)} E.$$

In particular,  $(f, g)$  are determined by  $(C, D)$ .  $\lrcorner$



*Proof:* Cf.[Buildings, Bruhat decompositions, unramified principal series, Garrett]P16.?

For the last assertion, for any chamber  $E \subset a$  s.t.  $f(E) = E$ , the image under  $f$  of any gallery  $(C_i)$  in  $a$  connecting  $D$  to  $E$  stutters: This is because there exists some  $i$  s.t.  $f(C_i) \neq C_i, f(C_{i+1}) = C_{i+1}$ , thus  $f(C_i) = f(C_{i+1}) = C_{i+1}$ . In particular,  $d(C, E) < d(D, E)$ . The converse inclusion follows from the same argument for  $g$ . Thus  $H$  is determined and  $f$  is identity on  $H$  and is an isomorphism on  $H'$ , thus  $f$  is determined by uniqueness lemma(20.6.2.2). Similarly  $g$  is determined.  $\square$

**Cor.(20.6.2.8)[Reflections].** Situation as in(20.6.2.7), there is a unique automorphic  $s : a \rightarrow a$  s.t.  $s$  fixes  $C \cap D$ ,  $s(C) = D$  and  $s(D) = C$ , called the **apartment reflection** along  $C \cap D$ . It follows that  $s^2 = \text{id}$ . In fact it is given by

$$s(x) = \begin{cases} f(x) & , x \in H' \\ g(x) & , x \in H \end{cases}.$$

┘

*Proof:*  $s$  is unique by the uniqueness lemma(20.6.2.2). The described function  $s$  satisfied the conditions because  $C \cap D \subset H \cap H'$ .  $\square$

### Metrics

**Prop.(20.6.2.9).** If  $X$  is an affine building and  $Y \subset X$  is bounded or has non-empty interior, then  $Y$  is contained in an apartment of  $X$  iff it is isometric to a subset of an apartment.  $\square$

*Proof:* [Buildings, Abramenko-Brown]Thm11.53.  $\square$

## 3 Groups Actioning on Buildings

**Def.(20.6.3.1)[Labeling of Chamber Complexes].** A **labeling of a chamber complex**  $X$  is a polysimplicial map  $\lambda : X \rightarrow \Delta^{\dim X}$ .  $\square$

**Def.(20.6.3.2)[Special Automorphisms].** A **special automorphism** of a chamber complex  $X$  is a polysimplicial isomorphism  $f : X \rightarrow X$  s.t. if  $f$  fixes a chamber, then it fixes all its vertices.  $\square$

**Def.(20.6.3.3)[Strongly-Transitive Group Actions].** If  $X$  is a building and  $G \in \text{Grp}$  acts on  $X$  by polysimplicial automorphisms, then the action is called a **strongly-transitive action on buildings** if given any two pairs  $(\mathcal{A}, C), (\mathcal{A}', C')$  where  $\mathcal{A}, \mathcal{A}' \subset X$  are apartments and  $C, C'$  are chambers in  $\mathcal{A}, \mathcal{A}'$  resp., there exists  $g \in G$  s.t.  $g(\mathcal{A}, C) = (\mathcal{A}', C')$ , and  $g$  fixes vertices of  $C \cap C'$ .  $\square$

**Prop.(20.6.3.4)[Labelings Exist for Buildings].** Any building  $X$  has a labeling(20.6.3.1).  $\square$

*Proof:* Let  $C \subset X$  be a chamber contained in an apartment  $a$ . Let  $r : X \rightarrow a$  be the canonical contraction(20.6.2.4). Let  $f_0, \dots, f_n$  be the foldings along the adjacent chambers  $D_i$  onto  $C$ (20.6.2.7), and let  $\varphi = f_0 \circ f_1 \dots f_n$ .

By(20.6.2.7), any chamber in  $a$  that is closer to  $D_i$  than to  $C$  is moved closer to  $C$  by  $f_i$ , and any chamber in  $a$  that is closer to  $C$  than to  $D$  is fixed by  $f$ . Thus we can define a labelings

$$\varphi^{\circ\infty} \circ r : X \rightarrow C.$$

$\square$



**Def. (20.6.3.5) [Parabolic Subgroups].** Let  $\lambda : X \rightarrow \Delta^{\dim X}$  be a building with a labeling, and let  $G$  be a group acting on  $X$  preserving labeling, s.t. the action is strongly transitive in the sense that it acts transitively on the pairs  $(C, a)$  where  $C$  is a chamber in an apartment  $a$ . Then a **parabolic subgroup** of  $G$  is a stabilizer of some simplex  $\sigma \subset X$ . A minimal parabolic subgroup (i.e. a stabilizer of a chamber) is called a **Iwahori subgroup** of  $G$ .

Iwahori subgroups of  $G$  are all conjugate by action of  $G$ .  $\lrcorner$

**Def. (20.6.3.6) [Weyl Groups].** Let  $\lambda : X \rightarrow \Delta^{\dim X}$  be a building with a labeling, and let  $G$  be a group acting on  $X$  preserving labeling. For an apartment  $a \in X$ , denote  $\mathcal{N}(a)$  the stabilizer of  $a$  in  $G$ ,  $A(a)$  the centralizer of vertices of  $a$  in  $G$ , and  $W(a) = \mathcal{N}(a)/A(a)$  the **Weyl group** of  $a$ .  $\lrcorner$

**Prop. (20.6.3.7) [Reflections Generate Weyl Groups].** Situation as in (20.6.3.6), for any chamber  $C \subset a$ , the set  $S$  of reflections along codimension-1 faces of  $C$  (20.6.2.8) generate  $W(a)$ .  $\lrcorner$

*Proof:* Cf. [Buildings, Bruhat decompositions, unramified principal series, Garrett]P15.  $\square$

**Def. (20.6.3.8) [Length Functions].** Situation as in (20.6.3.6), fix a chamber  $C$  contained in  $a$ , define the length of  $w \in W$  to be the distance from  $C$  to  $w(C)$  in  $a$ .  $\lrcorner$

**Prop. (20.6.3.9) [Lengths].** Situation as in (20.6.3.7), for any  $w \in W(a)$  and  $s \in S$ , we have

$$|\text{length}(ws) - \text{length}(w)| = |\text{length}(sw) - \text{length}(w)| = 1 \text{ (20.6.3.8)}.$$

$\lrcorner$

*Proof:* Cf. [Buildings, Bruhat decompositions, unramified principal series, Garrett]P20.  $\square$

### Tits System and $G$ -Buildings

**Thm. (20.6.3.10) [Tits-Buildings].** Let  $X = (G, B, N, S)$  be a Tits-system with  $\#S < \infty$ , and let  $\text{Par}(X)$  be the set of all maximal parabolic subgroups of the Tits-system (20.6.1.2), and let  $\mathcal{B}(X) \subset \mathcal{P}(\text{Par}(X))$  be the family of finite sets  $\{P_0, \dots, P_n\}$  of maximal parabolic subgroups s.t.  $\cap_i P_i$  is also a parabolic subgroup.  $G$  acts by conjugation on  $(\text{Par}(X), \mathcal{B}(X))$  on the left. Then

- $(\text{Par}(X), \mathcal{B}(X))$  is a simplicial complex (4.13.4.1).
- Let  $C \in \mathcal{B}(X)$  be the simplex corresponding to the set of all standard maximal parabolic subgroups (20.6.1.2), then it is a chamber of  $\mathcal{B}$ , called the **standard chamber** of  $\mathcal{B}$ .
- If  $g \in G$  fixes a simplex of  $\mathcal{B}$ , then it fixes all vertices of it. In particular,  $G$  acts by special automorphisms (20.6.3.2).
- Let  $\mathcal{A} \subset \mathcal{B}$  be the union of all  $N$ -conjugates of  $C$ , then  $\mathcal{B}$  is a building with set of apartments  $\{g \cdot \mathcal{A}\}_{g \in G}$ , called the **Tits-building** associated to  $X$ . The apartment  $\mathcal{A}$  is called the **standard apartment** of  $\mathcal{B}$ .
- If  $\sigma, \sigma'$  are faces of  $C$  and  $g \in G, n \in N$  s.t.  $\sigma \cup n\sigma' \subset g \cdot \mathcal{A}$ , then there exists  $g' \in G$  s.t.  $g \cdot a = g' \cdot a$  and  $g$  fixes vertices of  $\sigma \cup n\sigma'$ .
- (Parabolic Subgroups) For a simplex  $\sigma = \{P_0, \dots, P_n\} \in \mathcal{B}$ , denote  $P_\sigma = \cap_{i=0}^n P_i$ , then the map  $\sigma \mapsto P_\sigma$  induces a  $G$ -equivariant inclusion-reversing bijection from the set of simplexes of  $\mathcal{B}$  to the set of proper parabolic subgroups of  $G$  (20.6.1.2). In particular, the chambers corresponds to conjugates of  $B$ .
- $G$  acts strongly-transitively on  $\mathcal{B}(X)$  (20.6.3.3).

*Proof:* Cf.[Kaletha-Prasad]P56.?  
 $\mathcal{B}$  is clearly a simplicial complex.

┘

□

**Cor. (20.6.3.11) [Properties of Tits-Buildings].** Situation as in (20.6.3.10),

1.  $\mathcal{B}$  is spherical (20.6.2.1) iff it has finite Weyl group  $W$  (20.6.1.1).
2. Given any two apartments  $\mathcal{A}, \mathcal{A}'$  of  $\mathcal{B}$  containing a simplex  $\sigma \in \mathcal{B}$ , there exists  $g \in G$  s.t.  $g(\mathcal{A}) = \mathcal{A}'$  and fixes all vertices of  $\sigma$ . Equivalently,  $P_\sigma$  acts transitively on the set of apartments containing  $\sigma$ .
3. For any simplex  $\sigma \in \mathcal{B}$  and  $x$  in the interior of  $|\sigma|$ , we have  $P_x = P_\sigma$  (20.6.3.10), and the fixed point set of  $P_x$  is  $|\sigma|$ .
4. If  $\sigma, \tau$  are facets of the same chamber and  $g \in G$  s.t.  $g\sigma = \tau$ , then  $\sigma = \tau$  and  $g \in P_\sigma$ .
5. Two facets  $\sigma, \tau \in \mathcal{A}$  are conjugate under  $G$  iff they are conjugate under  $N$ .
6. If  $\mathcal{A} \subset \mathcal{B}$  is the standard apartment (20.6.3.10) and  $\sigma, \tau \in \mathcal{A}$  are simplexes, then  $G = P_\sigma N P_\tau$ .

┘

*Proof:* 1: ?

- 2: This follows from (20.6.3.10).
- 3: This follows from the bijection  $\sigma \mapsto P_\sigma$  (20.6.3.10) and the fact that any parabolic subgroup is self-normalizing (20.6.1.5).
- 4: This follows from the bijection  $\sigma \mapsto P_\sigma$  (20.6.3.10) and (20.6.1.2).
- 5: Choose a chamber  $C \subset \mathcal{A}$  containing  $\sigma$  and  $n \in N$  s.t.  $n^{-1}F \subset C$ , then by item 4 we get  $\sigma = n^{-1}\tau$ .
- 6: This follows easily from the Bruhat decomposition (20.6.1.4).

□

**Cor. (20.6.3.12) [Types of Simplexes].** Situation as in (20.6.3.10), via the bijection  $\sigma \mapsto P_\sigma$ , we can associate to each  $\sigma \in \mathcal{B}$  a **type** which is the type of  $P_\sigma$  (20.6.1.2).

┘

**Def. (20.6.3.13) [Restricted Tits-Buildings].** Situation as in (20.6.3.10), let  $S = S_1 \coprod \dots \coprod S_n$  with  $S_i$  irreducible and pairwise commuting, then we call  $X \subset S$  admissible if  $S_i \cap X \subsetneq S_i$  for each  $i$ . And we call a simplex  $\sigma \in \mathcal{B}$  admissible if the type of  $\sigma$  is an admissible subset of  $S$  (20.6.3.12). Then

- The subset  $\mathcal{B}' \subset \mathcal{B}$  consisting of admissible simplexes of  $\mathcal{B}$  form a (poly)building, with apartments given by intersection of apartments of  $\mathcal{B}$ .
- Every chamber of  $\mathcal{B}$  is admissible thus contained in  $\mathcal{B}'$ .
- $\mathcal{B}'$  is invariant under action of  $G$ , and the action is polysimplicial.
- The vertices of  $\mathcal{B}'$  are simplexes of  $\mathcal{B}$  of type  $X = X_1 \coprod \dots \coprod X_n$  where each  $X_i \subset S_i$  is a maximal properly-contained subset.
- The properties of  $\mathcal{B}$  (20.6.3.10) (20.6.3.11) all hold for  $\mathcal{B}'$ .

This  $\mathcal{B}'$  is called the **restricted Tits-building** of  $(G, B, N)$ .

┘

*Proof:* Cf.[Kaletha-Prasad]P62.

□

**Thm. (20.6.3.14) [BN-pair from a Building, Tits [Tit74]].** Let  $G$  be a group acting on a building  $\mathcal{B}$  (20.6.2.1) by special automorphisms (20.6.3.2) s.t. the action is strongly-transitive (20.6.3.3).

Choose a chamber  $C$  in an apartment  $\mathcal{A} \subset \mathcal{B}$ , and let  $B = \text{Stab}_G(C)$ ,  $N = \text{Stab}_G(\mathcal{A})$ , then  $(B, N)$  is a saturated Tits system in  $G$  (20.6.1.1).

┘

*Proof:* Cf.[Kaletha-Prasad]P54?.

□

**Cor. (20.6.3.15) [Contractuions are Induced by Group Actions].** Situation as in (20.6.3.14), if  $a \subset \mathcal{B}$  is an apartment and  $C \subset a$  is a chamber, let  $\rho : \mathcal{B} \rightarrow \mathcal{A}$  be the canonical contraction with center  $C$  (20.6.2.4), then for each apartment  $a'$  containing  $C$ , there exists a unique  $g \in G$  s.t.  $\rho(x) = gx$  for all  $x \in a'$ .  $\lrcorner$

*Proof:* By the hypothesis there exists a unique  $g \in G$  s.t.  $g$  fixes vertices of  $C$  and maps  $a'$  to  $a$ . The composition  $\rho \circ g^{-1}$  is a polysimplicial automorphism of  $a$  that fixes vertices of  $C$ , thus  $\rho \circ g^{-1}$  is trivial on  $a$  by uniqueness lemma (20.6.2.2). The uniqueness of  $g$  follows from the same argument.  $\square$

## 4 Bruhat-Tits Buildings

**Def. (20.6.4.1) [Maximal  $K$ -Split Tori].** For any maximal split torus  $S \leq G$ ,

- Let  $S' \leq S$  be the maximal split torus contained in  $[G, G]$ .
- Let  $V(S) = X_{\bullet}(S) \otimes \mathbb{R}, V(S') = X_{\bullet}(S') \otimes \mathbb{R}$ .
- $S' \rightarrow S \rightarrow S/A_G$  is an isogeny, so  $V(S') \rightarrow V(S) \rightarrow V(S/A_G)$  is an isomorphism, providing a canonical splitting  $V(S) = V(S') \oplus V(A_G)$ .?
- Let  $\Phi(G, S) \subset X^*(S)$  be the relative root datum of  $G$  w.r.t.  $S$  (10.4.4.14).
- Let  $W(G, S) = N_G(S)/Z_G(S)$  be the Weyl group of the relative root datum  $\Phi(G, S)$  (10.4.4.14).
- Let  $U_{\alpha}$  the unipotent root group w.r.t.  $S$ .

⌋

**Def. (20.6.4.2) [Spherical Buildings for Reductive Groups].** Let  $k \in \text{Field}, G \in \text{RedGrp}/k$  and  $P \leq G$  a minimal parabolic subgroup,  $S \leq P$  a maximal  $k$ -split torus, then the Tits-building (20.6.3.10) associated to the spherical Tits-system  $(P(k), N(k))$  (20.6.1.8) is called the **spherical building** of  $G/k$ . Its simplexes are in bijection with parabolic subgroups of  $G$  (20.6.4.13) by (20.6.3.10) and (10.4.1.6).  $\lrcorner$

**Prop. (20.6.4.3) [Tits-System for  $G(K)$ ].** Let  $S \leq G$  be a maximal  $K$ -split torus,  $N = N_G(S)(K)$ , and

$$H = \{x \in Z_G(S)(K) : \chi(x) \in \mathcal{O}_K^* \forall \chi \in X(Z_G(S))\}.$$

Then there exists an Iwahori subgroup  $B \subset G$  (20.6.4.13) s.t.  $B \cap N = H$  and  $(B, N)$  form a Tits system for  $G(K)$  (20.6.1.1).

Moreover, the Weyl group  $W$  for this Tits system is naturally an affine Weyl group of a root system?  $R_0$  of rank  $\dim S$ . In particular, there is a subset  $R \subset W$  consists of  $\dim S + 1$  elements  $r_1, \dots, r_{\dim S+1}$  s.t.  $W = \langle r_1, \dots, r_{\dim S+1} \rangle$ , and  $W_0 = \langle r_1, \dots, r_{\dim S} \rangle = N_G(S)/Z_G(S)$ .  $\lrcorner$

*Proof:* ?

□

**Cor. (20.6.4.4).** There are exactly  $\text{rank}_K G + 1$  many conjugacy classes of maximal parahoric proper subgroups of  $G$ .  $\lrcorner$

*Proof:* This follows from (20.6.6.20) (20.6.4.3) and (20.6.1.2).  $\square$

**Thm. (20.6.4.5) [Compactly-Generatedness, Behr [Beh67]].** For an arbitrary  $G \in \text{RedGrp}/K$ ,  $G(K)$  is compactly generated (12.10.3.29).  $\lrcorner$

*Proof:* Cf. [P-R94]P152. □

**Cor. (20.6.4.6).** For an arbitrary  $G \in \mathcal{R}\text{edGrp}/K$ , there are only f.m. conjugacy classes of maximal compact subgroups of  $G$ . ┘

*Proof:* This follows from (20.6.6.20)(20.6.4.3) and (20.6.1.2). □

**Thm. (20.6.4.7) [Cartan-Decompositions].** Situation as in (20.6.4.3), let  $\mathcal{K} = BW_0B$  be a maximal subgroup of  $G$ , then  $W = W_0 \ltimes T$ , where  $T$  is an Abelian group of rank  $\dim S$  generate by roots from  $R_0$ . Let  $T^+ \subset T$  be the set of positive roots. Let  $\nu : N \rightarrow W$  be the projection, then  $G(K) = \mathcal{K}\nu^{-1}(T^+)\mathcal{K}$ , and there is a bijection between  $T^+$  and  $[\mathcal{K} \backslash G(K) / \mathcal{K}]$  given by  $t \mapsto \mathcal{K}\nu^{-1}(t)\mathcal{K}$ . ┘

*Proof:* ? □

### Axiomatic Bruhat-Tits Buildings

**Axiom (20.6.4.8) [Bruhat-Tits Buildings].** The **Bruhat-Tits building**  $\mathcal{B}(G)$  is a building (20.6.2.1) s.t.

- $G(K)$  acts by polysimplicial automorphisms on  $\mathcal{B}(G)$  on the left.
- The set of apartments of  $\mathcal{B}(G)$  is  $G(K)$ -equivariantly bijective to the set of maximal split tori of  $G$ , and for a maximal split torus  $S \leq G$  corresponding to an apartment  $\mathcal{A}(S)$ , the stabilizer of  $\mathcal{A}(S)$  is  $N_G(S)(K)$ .

And a central isogeny  $G \rightarrow G' \in \mathcal{R}\text{edGrp}/K$  induces an equivariant isomorphism  $\mathcal{B}(G(K)) \cong \mathcal{B}(G'(K))$ . In particular, the buildings of  $G, G_{\text{ad}}, [G, G], G_{\text{sim.cntd}}$  are canonical identified, and the action of  $G(K)$  on  $\mathcal{B}(G)$  factors through  $G_{\text{ad}}(K)$ . ┘

*Proof:* This is Axiom4.1.1. of [Kaletha-Prasad]. If  $G$  is quasi-split, this follows from (20.6.4.28). In general it follows from Steinberg's theorem (10.5.2.3) and (20.6.5.7). □

**Axiom (20.6.4.9) [Functoriality].** Axiom (20.6.4.8) is satisfied, and  $\mathcal{B}(G)$  is functorial w.r.t. automorphisms of the field  $K$ . In particular, the  $G(K)$ -action on  $\mathcal{B}(G)$  extends to an action of  $\text{Aut}(K) \ltimes G(K)$ . ┘

*Proof:* This is proved in [K-P23]Prop7.9.5?. Basically it gives an intrinsic construction of the parabolic subgroups  $G(K)_{\Omega}^0$  (20.6.4.13). □

**Axiom (20.6.4.10) [Affine Structure on Apartments].** Axiom (20.6.4.8) holds, and for any maximal split torus  $S \leq G$ , notation as in (20.6.4.1), the apartment  $\mathcal{A}(S)$  corresponding to  $S$  satisfies

- $\mathcal{A}(S)$  is an affine  $V(S')$ -torsor.
- $N_G(S)(K)$  acts on  $\mathcal{A}(S)$  by affine automorphisms, whose derivative is induced by the Weyl group action on  $X_{\bullet}(S')$ . In particular,  $z \in Z_G(S)(K)$  acts on  $\mathcal{A}(S)$  by translation vector  $\nu(z) \in V(S')$ , where  $\nu(z)$  is the image of  $\omega_{Z_G(S)}(z)$  (20.5.2.2) under the projection  $V(S) \rightarrow V(S')$  (20.6.4.1).
- For any  $g \in G(K)$ , the isomorphism  $g : \mathcal{A}(S) \cong \mathcal{A}(gSg^{-1})$  is affine whose derivative  $dg : V(S) \rightarrow V(gS'g^{-1})$  is the natural isomorphism. ┘

*Proof:* ? This is Axiom4.1.4. of [Kaletha-Prasad].

If  $G$  is quasi-split, this follows from (20.6.4.28). □

**Cor. (20.6.4.11) [Uniqueness of  $\mathcal{A}(S)$ ].** For a maximal split torus  $S \leq G$ , notation as in (20.6.4.1), there is an exact sequence

$$1 \rightarrow Z_{[G,G]}(S')(K)/Z'(K)^1 \rightarrow N'(K)/Z'(K)^1 \rightarrow W(G, S) \rightarrow 1.$$

The valuation homomorphism  $\nu : Z(K) \rightarrow V(S')$  is  $W(G, S)$ -equivariant<sup>?</sup>, thus the pushout of the above sequence along  $\nu|_{Z_{[G,G]}(S')(K)}$  gives an exact sequence

$$1 \rightarrow V(S') \rightarrow \widetilde{W}_{\mathbb{R}} \rightarrow W \rightarrow 1. \quad (\star)$$

Then

- The set of splitting of the exact sequence $(\star)$  is an affine  $V(S')$ -torsor satisfying properties of axiom (20.6.4.10).
- If  $\mathcal{A}$  is an affine  $V(S')$ -torsor satisfying properties of  $\mathcal{A}(S)$  as in (20.6.4.10), then the following three affine  $V(S')$ -torsors are uniquely  $N_G(S)(K)$ -equivariantly isomorphic:
  - $\mathcal{A}$ .
  - The set of splittings of the the exact sequence $(\star)$ .
  - The set of maximal finite subgroups of the Lie group  $\widetilde{W}_{\mathbb{R}}$ .

In particular,  $\mathcal{A}(S)$  is uniquely determined by its properties listed in (20.6.4.10). ┘

*Proof:* Cf. [Kaletha-Prasad]P181.<sup>?</sup> □

**Axiom (20.6.4.12) [Stabilizers are Bounded].** The stabilizer  $G(K)_x^1 \triangleq \text{Stab}_{G(K)^1}(x)$  (20.5.1.2) for any  $x \in |\mathcal{B}(G)|$  is an open bounded (20.5.1.3) subgroup of  $G(K)$ . ┘

*Proof:* <sup>?</sup> This is Axiom 4.1.2 of [Kaletha-Prasad]. □

**Def. (20.6.4.13) [Parahoric Subgroups].** For any non-empty bounded (20.5.1.3) subcomplex of an apartment  $\mathcal{A} \subset \mathcal{B}(G)$  (20.6.4.28), let  $G(K)_{\Omega}^1$  denote the centralizer of  $\Omega$  in  $G(K)^1$ , and  $G(K)_{\Omega}^0$  the centralizer of  $\Omega$  in  $G(K)^0$ , which are open bounded subgroups of  $G(K)$  by (20.6.4.12) and (20.6.4.14).

For  $x \in |\mathcal{B}(G)|$ ,  $G(K)_x^0$  is called a **parahoric subgroups** of  $G(K)$  associated to  $x$ . Minimal parahoric subgroups (i.e. those associated to chambers) are called **Iwahori-subgroups** of  $G(K)$ . ┘

**Axiom (20.6.4.14) [Tits-Systems].** Axiom (20.6.4.8) holds, and for any maximal split torus  $S \leq G$  corresponding to an apartment  $\mathcal{A}(S) \subset \mathcal{B}(G)$  and  $C \subset \mathcal{A}(S)$  a chamber, then

$$(G(K)^0, G(K)_C^0, N_G(S)(K) \cap G(K)^0) \quad (20.6.4.13)$$

is a saturated Tits-system (20.6.1.1) with associated restricted Tits-building  $\mathcal{B}(G)$  (20.6.3.13).

In particular, the parahoric subgroups of  $G(K)$  (20.6.4.13) are exactly parabolic subgroups of this Tits-system (20.6.1.2), and for any simplex  $\sigma \in \mathcal{B}(G)$ ,  $G(K)_{\sigma}^0 = \text{Stab}_{G(K)^0}(\sigma)$ , and it follows from (20.6.3.11) (20.6.3.10) (20.6.3.13) that

- $G(K)^0$  acts by special automorphisms on  $\mathcal{B}(G)$  (20.6.3.2).
- $G(K)^0$  acts strongly-transitively on  $\mathcal{B}(G)$  (20.6.3.10).
- $N_G(S)(K)$  acts transitively on the set of chambers in  $\mathcal{A}(S)$ .
- For any maximal split torus  $S \leq G$ ,  $N_G(S)(K)$  acts transitively on the chambers of  $\mathcal{A}(S)$ .

- For any simplex  $\sigma \in \mathcal{B}$  and  $x$  in the interior of  $|\sigma|$ , we have  $P_x = P_\sigma$  (20.6.3.10), and the fixed point set of  $P_x$  is  $|\sigma|$ .

┘

*Proof:* ? This is Axiom 4.1.9. of [Kaletha-Prasad].

If  $G$  is quasi-split, this follows from (20.6.4.28). □

**Cor. (20.6.4.15).** If axiom (20.6.4.14) holds, then any  $G^0(K)$ -equivariant automorphism of  $|\mathcal{B}(G)|$  is identity. ┘

*Proof:* This follows immediately from the fact that the fixed point of  $P_x$  for a vertex  $x \in \mathcal{B}(G)$  is just  $x$ . □

**Axiom (20.6.4.16) [Affine Root Systems].** Axioms (20.6.4.8)(20.6.4.10) hold, and for any maximal torus  $S \leq G$ ,  $\alpha \in \Phi(G, S)$  and  $u \in U_\alpha(K) \setminus \{1\}$ , let  $u', u'' \in U_{-\alpha}(K) \setminus \{1\}$  and  $m(u) = u'uu'' \in N_G(S)(K)$  as in (10.4.4.22), then

- The action of  $m(u)$  on  $\mathcal{A}(S)$  fixes a hyperplane  $H_u$  that is an affine  $\ker(\alpha)$ -sub-torsor of  $\mathcal{A}(S)$ .
- Let  $\psi_\alpha^u \in \mathcal{A}(S)^*$  be the unique linear function with derivative  $\alpha$  and vanishes on  $H_u$ , then  $\Psi'(G, S) = \{\psi_\alpha^u | \alpha \in \Phi, u \in U_\alpha(K) \setminus \{1\}\}$  form an affine root system on  $\mathcal{A}(S)$  (3.9.8.2) with derived root system  $\Phi(G, S)$  (3.9.8.5).
- For  $\psi \in \mathcal{A}(S)^*$  with  $\dot{\psi} \in \Phi(G, S)$ , define  $\mathcal{U}_\psi = \{u \in U_\alpha(K) \setminus \{1\} | \psi_\alpha^u \geq \psi\}$  and  $\mathcal{U}_{\psi+} = \{u \in U_\alpha(K) \setminus \{1\} | \psi_\alpha^u > \psi\}$ . Then

$$\Psi(G, S) = \left\{ \psi \in \Psi'(G, S) \mid \mathcal{U}_\psi \subsetneq \mathcal{U}_{\psi+} \cdot U_{2\dot{\psi}}(K) \right\}$$

is an affine root system on  $\mathcal{A}(S)$ .

- $\Psi'(G, S) \subset \Psi(G, S) \cup \frac{1}{2}\Psi(G, S)$ .
- The image of the action map  $N_G(S)(K) \rightarrow \text{Aff}(\mathcal{A}(S))$  contains  $W(\Psi)$  (3.9.8.3), and is contained in  $W(\Psi(G, S))^{\text{ext}}$  (3.9.8.4). If  $G$  is simply-connected, then the image is just  $W(\Psi)$ , and if  $G$  is quasi-split and adjoint, the image is just  $W(\Psi(G, S))^{\text{ext}}$ .
- The polysimplicial structure on  $\mathcal{A}(S)$  induced by the affine root system  $\Psi(G, S)$  (3.9.8.11) is exactly the structure restricted from  $\mathcal{B}(G)$ .

┘

*Proof:* ? This is Axiom 4.1.6 of [Kaletha-Prasad]. □

**Axiom (20.6.4.17) [Action of  $\mathcal{U}_\psi$ ].** Axioms (20.6.4.8)(20.6.4.10)(20.6.4.16) hold, and for any maximal split torus  $S \leq G$  and any affine root  $\psi \in \Psi \subset \mathcal{A}^*$  (20.6.4.16),

- The action of  $\mathcal{U}_\psi$  on  $\mathcal{B}(G)$  fixes the half-apartment  $\mathcal{A}^{\psi \geq 0}$ .
- For any subcomplex  $\Omega \subset \mathcal{A}$ , the group  $G(K)_\Omega^0$  (20.6.4.13) is generated by  $Z_G(S)^0(K)$  and  $\mathcal{U}_\psi$  for all  $\psi \in \Psi$  with  $\psi(\Omega) \subset \mathbb{R}_{\geq 0}$ .
- For  $a \in \Phi$ ,  $U_a(K) \cap G(K)_\Omega^0 = \mathcal{U}_\psi$ , where  $\psi \in \Psi'$  is the smallest element with  $\dot{\psi} = a$  and  $\psi(\Omega) \subset \mathbb{R}_{\geq 0}$ .

┘

*Proof:* ? This is Axiom 4.1.8. of [Kaletha-Prasad]. □



**Cor. (20.6.4.18) [Uniqueness of  $\mathcal{B}(G)$ ].** Let  $\mathcal{B}, \mathcal{B}'$  be buildings satisfying Axioms (20.6.4.8) (20.6.4.10) (20.6.4.14) (20.6.4.16) (20.6.4.17), then there is a unique bijection  $F : \mathcal{B} \cong \mathcal{B}'$  that restricts to the canonical  $N_G(S)(k)$ -equivariant bijection  $f : \mathcal{A}(S) \cong \mathcal{A}'(S)$  for all maximal split torus  $S \leq G$  (20.6.4.11).  $\square$

*Proof:* Choose a maximal split torus  $S \leq G$ . Firstly we show that for any simplex  $\sigma \in \mathcal{A}(S)$ , the stabilizer of  $\sigma$  and  $f(\sigma)$  are equal.  $\color{red}?$  Cf. [Kaletha-Prasad]P184.

Now define  $F : \mathcal{B} \rightarrow \mathcal{B}'$  as follows: for any  $g \in G(K)$ ,  $F|_{g\mathcal{A}(S)} = g.f|_{\mathcal{A}(S)}$ . This map is well-defined by the claim we just proved, and it is surjective because  $G(K)$  acts transitively by (20.6.4.14). It is bijective because for  $g_i \in G(K)$  and simplexes  $\sigma_i \in \mathcal{A}(S)$ ,  $g_1\sigma_1 = g_2\sigma_2$  iff  $g_1.f(\sigma_1) = g_2.f(\sigma_2)$ : Write  $\sigma_2 = n\sigma_1$  for some  $n \in N_G(S)(K)$  (20.6.4.14), this assertion follows from what we just proved.

The uniqueness of the bijection is clear, because any automorphism of  $\mathcal{B}'$  that is  $G(k)$ -equivariant and fixes vertices of  $\mathcal{A}(S)$  automatically fixes  $\mathcal{B}'$  because  $G(K)$  acts transitively by (20.6.4.14).  $\square$

**Axiom (20.6.4.19) [Bruhat-Tits Group Schemes].** Axiom (20.6.4.8) holds, and for any non-empty bounded (20.5.1.3) subcomplex  $\Omega$  of an apartment  $\mathcal{A}(S)$  of  $\mathcal{B}(G)$ ,

- There exists  $\mathcal{G}_\Omega^1 \in \text{GrpSch}^{\text{sm,aff}}/\mathcal{O}_K$  s.t.  $(\mathcal{G}_\Omega^1)_K \cong G$  and  $\mathcal{G}_\Omega^1(\mathcal{O}_K) = G(K)_\Omega^1$  (20.6.4.13).
- Let  $\mathcal{G}_\Omega^0$  be the relative identity component of  $\mathcal{G}_\Omega^1$  (which is affine (20.3.2.3)), then  $\mathcal{G}_\Omega^0(\mathcal{O}_K) = G(K)_\Omega^0$  (20.6.4.13).
- If  $\mathcal{G}$  is semisimple and simply-connected, then  $\mathcal{G}_\Omega^1 = \mathcal{G}_\Omega^0$ .
- There is a closed  $\mathcal{O}_K$ -split torus  $\mathcal{S} \leq \mathcal{G}_\Omega^0$  with generic fiber  $S$ , whose special fiber is a maximal split torus of  $\overline{\mathcal{G}}_\Omega^0$ .

For  $x \in |\mathcal{B}(G)|$ ,  $\mathcal{G}_x^0 \in \text{GrpSch}/\mathcal{O}_K$  is called the **Bruhat-Tits parahoric subgroup** of  $G$  associated to  $x$ . Sometimes  $\mathcal{G}_x^0(\mathcal{O}_K)$  is also called the Bruhat-Tits parahoric subgroup of  $G$  associated to  $x$ .  $\square$

*Proof:*  $\color{red}?$  This is Axiom 4.1.20 of [Kaletha-Prasad].  $\square$

**Axiom (20.6.4.20) [Reductive Quotients of Special Fibers].** Axioms (20.6.4.8) (20.6.4.19) hold, and for any bounded sub-complexes  $\Omega \subset \Omega' \subset \mathcal{B}(G)$ , there is a unique homomorphism  $\rho_{\Omega, \Omega'} : \mathcal{G}_{\Omega'}^1 \rightarrow \mathcal{G}_\Omega^1 \in \text{GrpSch}/\mathcal{O}_K$  that induces identity on generic fiber, and on  $\mathcal{O}$ -points induces the natural inclusion  $G(K)_\Omega^1 \subset G(K)_{\Omega'}^1$ . Then we have:

- The kernel of the special fiber  $\overline{\rho}_{\Omega, \Omega'} : \overline{\mathcal{G}}_{\Omega'}^0 \rightarrow \overline{\mathcal{G}}_\Omega^0$  is a smooth connected unipotent subgroup of  $\overline{\mathcal{G}}_{\Omega'}^0$ .

And if  $\Omega = \sigma, \Omega' = \sigma'$  are simplexes of  $\mathcal{B}(G)$ , and let  $S \leq G$  is a maximal split torus s.t.  $\sigma' \subset \mathcal{A}(S)$  and  $\mathcal{S} \leq \mathcal{G}_\sigma^0$  the  $\mathcal{O}_K$ -split torus as in (20.6.4.19), then

1. The image  $p_\sigma(\sigma') = \overline{\rho}_{\sigma, \sigma'}(\overline{\mathcal{G}}_{\sigma'}^0)$  is a parabolic subgroup of  $\overline{\mathcal{G}}_\sigma^0$ , and the inverse image of  $p_\sigma(\sigma')(\kappa)$  under the the reduction map  $\mathcal{G}_\sigma^0(\mathcal{O}_K) \rightarrow \overline{\mathcal{G}}_\sigma^0(\kappa)$  is  $\mathcal{G}_{\sigma'}^0(\mathcal{O}_K)$ .
2. The special fiber  $\overline{\mathcal{S}}$  is a maximal split torus of  $\overline{\mathcal{G}}_{\sigma'}^0$  as well as  $\overline{\mathcal{G}}_\sigma^0$ , and its image in the reductive quotient  $\mathbf{G}_\sigma = \overline{\mathcal{G}}_\sigma^0/R_u(\overline{\mathcal{G}}_\sigma^0)$  is a maximal split torus  $\mathbf{S}$ .
3. Denote the image of  $p_\sigma(\sigma')$  in  $\mathbf{G}_\sigma$  by  $\mathbf{p}_\sigma(\sigma')$ . Then if  $x$  is in the interior of  $\sigma$  and  $v \in V(S)$  s.t.  $x + v$  is in the interior of  $\sigma'$ , then  $\Phi(\mathbf{p}_\sigma(\sigma'), \mathbf{S}) = \{\alpha \in \Phi(\mathbf{G}_\sigma, \mathbf{S}) : \langle v, \alpha \rangle \geq 0\}$ .
4. The map  $\sigma' \rightarrow \mathbf{p}_\sigma(\sigma')$  is an inclusion-reversing map from the poset of all simplexes properly containing  $\sigma$  to the poset of parabolic subgroups of  $\mathbf{G}_\sigma$ . In particular, the  $\kappa$ -rank of  $\mathbf{G}_\sigma$  equals the codimension of  $\sigma$  in  $X$ .

5. The root datum of  $G_\sigma$  w.r.t.  $S$  is given by  $(X^\bullet(S), \Phi_\sigma(G, S), X_\bullet(S), \Phi_\sigma^\vee)$ , where  $\Phi_\sigma(G, S) \subset \Phi(G, S) \subset X^\bullet(S)$  is the sliced root system(3.9.8.22). In particular, If  $C \subset \mathcal{B}(G)$  is a chamber containing  $\sigma$  and  $\Delta$  is the set of simples affine roots determined by  $C$ (3.9.8.14), then  $\Delta_\sigma = d(\Delta \cap \Psi_\sigma)$  is a basis of  $\Phi_\sigma$ , and the Dynkin diagram of  $G_\sigma$  is obtained by removing from the affine Dynkin diagram of  $\Psi$ (3.9.8.26) all vertices of  $\sigma$ , by(3.9.8.28).
6. Let  $G(K)_\sigma^{0+}$  be the inverse image of  $R_u(\bar{\mathcal{G}}_\sigma^0)(\kappa)$  under the the reduction map  $\mathcal{G}_\sigma^0(\mathcal{O}_K) \rightarrow \bar{\mathcal{G}}_\sigma^0(\kappa)$ , then  $G(K)_\sigma^{0+} \leq G(K)_{\sigma'}^{0+}$ .

┘

*Proof:* ? This is Axiom4.1.22 of [Kaletha-Prasad].

□

### Constructing Apartments

**Def. (20.6.4.21) [Valuation of Root Data].** If  $G$  is semisimple, for a maximal split torus  $S \leq G$ , a **valuation of root datum** of  $(G, S)$  is a collection  $\varphi = (\varphi_\alpha)_{\alpha \in \Phi(G, S)}$  of morphisms  $\varphi_\alpha : U_\alpha(K) \rightarrow \mathbb{R} \cup \{\infty\}$  s.t.

- For each  $\alpha \in \Phi(G, S)$ ,  $\#\text{Im}(\varphi_\alpha) \geq 3$ .
- For each  $\alpha \in \Phi(G, S)$  and  $r \in \mathbb{R}$ ,  $U_{\alpha, \varphi, r} = \varphi_\alpha^{-1}([r, \infty])$  is a subgroup of  $U_\alpha(K)$ , and  $U_{\alpha, \varphi, \infty} = \mathbf{1}$ .
- For each  $n \in N_G(S)(K)$  realizing  $s_\alpha$ , the function  $u \mapsto \varphi_\alpha(u) - \varphi_{-\alpha}(nun^{-1})$  is constant on  $U_\alpha(K) \setminus \{1\}$ .
- For  $\alpha, \beta \in \Phi(G, S)$  and  $r, s \in \mathbb{R}$  s.t.  $\beta \notin \mathbb{R}_{-}\alpha$ , the commutator subgroup  $[U_{\alpha, \varphi, r}, U_{\beta, \varphi, s}]$  is contained in the group generated by  $U_{p\alpha+q\beta, pr+qs}$  for all  $p, q \in \mathbb{Z}_+$  s.t.  $p\alpha + q\beta \in \Phi(G, S)$ .
- If  $\alpha, 2\alpha \in \Phi(G, S)$ , then  $\varphi_{2\alpha} = 2\varphi_\alpha|_{U_{2\alpha}(K)}$ .

If  $G$  is quasi-split, for a maximal split torus  $S \leq G$ , a **valuation of root datum** of  $(G, S)$  is defined to be a valuation of root datum of  $([G, G], S')$ .

Moreover, A valuation  $\varphi$  of root datum of  $(G, S)$  is said to be

- compatible with  $\text{ord}_K$  if for each  $\alpha \in \Phi(G, S)$  and  $z \in Z_G(S)(K)$ ,  $\varphi_\alpha(z^{-1}uz) = \varphi_\alpha(u) + \alpha(\nu(z))$ (20.6.4.10). In particular,  $\varphi_\alpha(s^{-1}us) = \varphi_\alpha(u) - \text{ord}_K(a(s))$  for  $s \in S(K)$ .
- compatible with topology on  $K$  if for each  $\alpha \in \Phi(G, S)$ , the subgroups  $\{U_{\alpha, \varphi, r}\}_{r \in \mathbb{R}}$  are bounded open subgroups of  $U_\alpha(K)$ , and form a nbhd base of  $\mathbf{1} \in U_\alpha(K)$ .

┘

**Def. (20.6.4.22) [Equipollence of Valuations of Root Data].** for a maximal split torus  $S \leq G$ ,  $V(S')$  acts on the set of valuations of root datum of  $(G, S)$ , by

$$(v.\varphi)_\alpha(u) = \varphi_\alpha(u) + \alpha(v), \text{ where } \alpha \in \Phi(G, S), u \in U_\alpha(K).$$

Cf. [K-P23]P200?.

Two valuations of root data  $\varphi, \varphi'$  are called **equipollent** if there exists  $v \in V(S')$  s.t.  $\varphi' = v.\varphi$ .

┘

**Def. (20.6.4.23) [Chevalley-Valuations].** Cf. [K-P23]6.1.20.?

┘

**Def. (20.6.4.24) [Apartments].** If  $G$  is semisimple, for a maximal split torus  $S \leq G$ , define the apartment  $\mathcal{A}(S)$  to be the affine  $V(S)$ -torsor whose elements are valuations of the root datum  $\Phi(G, S)$  that are equipollent to a Chevalley-valuation(20.6.4.23).

If  $G$  is quasi-split, for a maximal split torus  $S \leq G$ , define  $\mathcal{A}(S) = \mathcal{A}(S')$ .

┘



**Def. (20.6.4.25) [Apartments  $A(G(K), S)$ ].** Let  $\nu : S(K) \rightarrow X_*(S)_{\mathbb{R}}$  be the homomorphism defined by the following relation:

$$\langle \nu(z), \chi \rangle = -\text{ord}_K(\chi(z)), \text{ where } z \in Z_G(S)(K), \chi \in X^*(Z_G(S)).$$

Let  $\Lambda = \nu(Z_G(S)(K))$  which is a lattice in  $X_*(S)_{\mathbb{R}}$ , then  $\text{Hom}(X^*(S), \Gamma_K) \subset \Lambda \subset \text{Hom}(X^*(Z_G(S)), \Gamma_K)$  (20.7.0.1), and when  $G$  is unramified, then the first inclusion is an equality?, and if  $G$  is split both inclusions are equalities.  $\lrcorner$

**Def. (20.6.4.26) [Apartments  $A(G(K), S)$ ].** We can associate

- An affine space  $A(G(K), S)$ , which is a  $X_*(S)_{\mathbb{R}}$ -torsor on which  $N(K)$  acts on the right.
- A system  $\Phi_{\text{aff}}(G(K), S)$  of affine functions  $\alpha(x) = A(\alpha)X + v(\alpha)$  on  $A(G(K), S)$ .
- A map  $\alpha \mapsto X_\alpha$  mapping  $\alpha \in \Phi_{\text{aff}}(G(K), S)$  to a subgroup  $X_\alpha \leq G(K)$

s.t.

- $s^{-1}X_\alpha s = X_{\alpha^s}$  for any  $\alpha \in \Phi_{\text{aff}}(G(K), S)$  and  $s \in N(K)$ .
- For  $\alpha \in \Phi_{\text{aff}}(G(K), S)$ , the vector part  $v(\alpha)$  are elements of  $\Phi(G, S)$  (20.6.4.1).
- For  $a \in \Phi$ , the groups  $\{X_\alpha : v(\alpha) = a\}$  form a filtration of  $U_a(K)$ .

$\lrcorner$

**Def. (20.6.4.27) [Dynkin-Diagrams]. ?**

Since the chambers are permuted simply transitively by  $W$ , the isomorphism class of the Dynkin diagram doesn't depend on the choice of the chamber  $C$ .  $\lrcorner$

*Proof:* Cf. [Tit79]P36. ?  $\square$

### Constructing Buildings

**Def. (20.6.4.28) [Iwahori-Tits Systems and Bruhat-Tits Buildings].** The Tits-system  $(\mathcal{G}, \mathcal{I}, n, R)$  is called the **Iwahori-Tits system** of  $G$ . And define  $\mathcal{B}(G)$  to be the restricted Tits-building associated to this system (20.6.3.13).

In particular, axioms (20.6.4.8) (20.6.4.10) (20.6.4.14) hold for  $\mathcal{B}(G)$ .  $\lrcorner$

*Proof:* Cf. [K-P23]Chap7.5, 7.6. ?  $\square$

**Prop. (20.6.4.29) [Base Change].** If  $G$  is quasi-split, then for any finite field extension  $K'/K \in \text{Field}$ , there is a natural embedding  $|\mathcal{B}(G_K)| \hookrightarrow |\mathcal{B}(G_{K'})|$ .

We will see in (20.7.1.1) that the same holds when  $G$  is not quasi-split.  $\lrcorner$

*Proof:* Cf. [K-P23]P275. ?  $\square$

**Remark (20.6.4.30).** Notice that this map may not be polysimplicial.  $\lrcorner$

**Prop. (20.6.4.31) [Pass to Completions, Guy Rousseau].** If  $K \in \text{DVR}$  is a Henselian and  $G \in \text{RedGrp}/K$  is quasi-split, then the natural map  $|\mathcal{B}(G_K)| \rightarrow |\mathcal{B}(G_{\widehat{K}})|$  (20.6.4.29) is isomorphism of chamber complexes.  $\lrcorner$

*Proof:* Cf. [K-P23]P277.  $\square$

**Remark (20.6.4.32).** Notice however  $\mathcal{B}(G_{\widehat{K}})$  may have more apartments than  $\mathcal{B}(G)$  because  $G_{\widehat{K}}$  may have more maximal split tori than that of  $G$ .  $\lrcorner$

### Constructing Bruhat-Tits Group Schemes

**Notation (20.6.4.33).**

- In this subsection, notice that  $G_{K^{\text{SH}}}$  is quasi-split by Steinberg's theorem (10.5.2.3), so  $\mathcal{B}(G_{K^{\text{SH}}})$  (20.6.4.8) exists. ┘

**Thm. (20.6.4.34) [Bruhat-Tits Group Schemes].**

- For any non-empty bounded (20.5.1.3) subcomplex  $\Omega$  of an apartment  $\mathcal{A}$  of  $\mathcal{B}(G_{K^{\text{SH}}})$ , there exists  $\mathcal{G}_\Omega^1 \in \text{GrpSch}^{\text{sm}, \text{aff}}/\mathcal{O}_{K^{\text{SH}}}$  s.t.  $(\mathcal{G}_\Omega^1)_{K^{\text{SH}}} \cong G$  and  $\mathcal{G}_\Omega^1(\mathcal{O}_{K^{\text{SH}}}) = G(K^{\text{SH}})_\Omega^1$  (20.6.4.13).
- For any non-empty bounded (20.5.1.3) subset  $\Omega$  of an apartment  $\mathcal{A}$  of  $\mathcal{B}(G_{K^{\text{SH}}})$ , the groups  $G(K^{\text{SH}})_\Omega^1$  and  $G(K^{\text{SH}})_\Omega^\dagger$  are schematic and their smooth affine integral models share the same relative identity component group scheme. (20.6.4.13). ┘

*Proof:* Cf. [K-P23]P295, 296. ? □

**Prop. (20.6.4.35) [Pass to Completions].** For a bounded subset  $\Omega \subset |\mathcal{B}(G_{K^{\text{SH}}})|$  and its image  $\widehat{\Omega} \subset |\mathcal{B}(G_{\widehat{K}})|$ , the base change from  $\mathcal{O}_{K^{\text{SH}}}$  to  $\mathcal{O}_{\widehat{K}}$  of the Bruhat-Tits group schemes  $\mathcal{G}_\Omega^1, G_\Omega^0, G_\Omega^\dagger$  are the corresponding Bruhat-Tits group schemes exactly the Bruhat-Tits group schemes  $\mathcal{G}_{\widehat{\Omega}}^1, G_{\widehat{\Omega}}^0, G_{\widehat{\Omega}}^\dagger$ . ┘

*Proof:* Cf. [K-P23]P321 ? □

## 5 Unramified Descent

**Prop. (20.6.5.1) [Maximal  $K^{\text{SH}}$ -Split Tori].** Let  $\widetilde{Z}_G^0$  be the maximal split central torus of  $G_{K^{\text{SH}}}$ , then it is defined over  $K$  by Galois descent as  $\text{Gal}_{K^{\text{SH}}}$  is normal in  $\text{Gal}_K$ .

Then  $T' \mapsto T' \widetilde{Z}_G^0$  is a bijection between the set of maximal  $K^{\text{SH}}$ -split tori of  $[G, G]$  to the set of maximal  $K^{\text{SH}}$ -split tori of  $G$ . ┘

*Proof:* ? □

**Def. (20.6.5.2) [Special Tori].** A **special torus** is a torus  $T \leq G$  s.t.  $T$  contains a maximal split torus of  $G$  and  $T_{K^{\text{SH}}}$  is a maximal split torus of  $G_{K^{\text{SH}}}$ . Then

- Any maximal split torus  $S \leq G$  is contained in a special torus of  $G$ .
- $T' \mapsto T' \widetilde{Z}_G^0$  (20.6.5.1) is a bijection between the set of special tori of  $[G, G]$  to the set of special tori of  $G$ . ┘

*Proof:* Cf. [?K\_P23]Prop9.3.4. ? □

**Prop. (20.6.5.3) [Galois Actions].** As  $G_{K^{\text{SH}}}$  is quasi-split by Steinberg's theorem (10.5.2.3),  $\mathcal{B}' = \mathcal{B}(G_{K^{\text{SH}}})$  is defined by (20.6.4.8), and  $\text{Gal}(K^{\text{SH}}/K)$  acts on  $\mathcal{B}' = \mathcal{B}(G_{K^{\text{SH}}})$  by polysimplicial maps by (20.6.4.9). Then

1. If  $\mathcal{A} \subset \mathcal{B}'$  is an apartment and  $\gamma \in \text{Gal}(K^{\text{SH}}/K)$ , then  $\gamma(\mathcal{A})$  is also an apartment, and  $\gamma : \mathcal{A} \rightarrow \gamma(\mathcal{A})$  is an affine map.
2. If  $T \leq G_{K^{\text{SH}}}$  is a maximal torus and  $T$  is defined and splits over a subfield  $K' \subset K^{\text{SH}}$  containing  $K$ , then  $\mathcal{A}(T)$  is fixed by  $\text{Gal}(K^{\text{SH}}/K')$ . (? Cf. [K-P23]P325.)

3. For any  $x \in |\mathcal{B}(G)|$ ,  $\text{Gal}(K^{\text{sh}}/K).x$  is finite. (this follows easily from the last one). In particular,  $|\mathcal{B}^{\text{Gal}(K^{\text{sh}}/K)}| \neq \emptyset$  by Bruhat-Tits fixed-point theorem (20.6.6.12).

An apartment of  $\mathcal{B}'$  corresponding to a special torus of  $G$  (20.6.5.2) is called a **special apartment**.

⌋

*Proof:* ?

□

**Prop. (20.6.5.4) [Fixed Points  $|\mathcal{B}^{\text{Gal}(K^{\text{sh}}/K)}|$  and Special Apartments].** Situation as in (20.6.5.3), denote  $|\mathcal{B}| = |\mathcal{B}(G_{K^{\text{sh}}})|^{\text{Gal}(K^{\text{sh}}/K)}$ . Then

1.  $|\mathcal{B}|$  is a non-empty closed, convex subset of  $|\mathcal{B}(G_{K^{\text{sh}}})|$ , equipped with the restricted metric of that of  $|\mathcal{B}(G_{K^{\text{sh}}})|$  (20.6.6.5).
2. Every special apartment of  $\mathcal{B}(G_{K^{\text{sh}}})$  (20.6.5.3) contains a  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex of maximal dimension (among all  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes).
3. Any two simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$  is contained in a common special apartment.

⌋

*Proof:* 1: This follows from the fact  $\text{Gal}(K^{\text{sh}}/K)$ -action preserves metric and  $|\mathcal{B}(G_{K^{\text{sh}}})|$  is a unique-geodesic space. It is non-empty by (20.6.5.3).

2: Cf. [K-P23] Prop 9.3.6. ?

3: Cf. [K-P23] Prop 9.3.8. ?

□

**Prop. (20.6.5.5) [Polysimplicial Structure on  $|\mathcal{B}|$ ].** Situation as in (20.6.5.4),  $|\mathcal{B}|$  may not be the underlying space of a subcomplex of  $\mathcal{B}(G_{K^{\text{sh}}})$ . But:

1. If  $\sigma \neq \sigma'$  are  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$ , then  $|\sigma| \cap |\mathcal{B}| \neq |\sigma'| \cap |\mathcal{B}|$ .
2. There is a polysimplicial structure on  $|\mathcal{B}|$  s.t. its simplexes are exactly intersections of  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$  with  $|\mathcal{B}|$ . And if  $G$  is simple, then this is a simplicial structure. This polysimplicial structure is denoted by  $\mathcal{B}$ .
3.  $\mathcal{B}$  is a chamber complex (4.13.4.2).
4.  $G(K)$  acts by polysimplicial automorphisms on  $\mathcal{B}$ , and is continuous w.r.t. the metric (20.6.5.4).

⌋

*Proof:* Cf. [K-P23] P 9.3.10. ?

2: By (20.6.4.8), we may replace  $G$  with  $G_{\text{ad}}$ , and then a decomposition of  $G_{\text{ad}}$  into a product of simple groups (10.4.5.20) induces a product decomposition of  $\mathcal{B}(G_{K^{\text{sh}}})$  and  $|\mathcal{B}|$ , so it suffices to prove for  $G$  simple.

If  $\sigma$  is a  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$ , we prove that  $\sigma \cap \mathcal{B}$  is a single point: Let  $\mathcal{A}$  be a special apartment containing  $\sigma$  (20.6.5.4), then  $\text{Gal}(K^{\text{sh}}/K)$  stabilizes  $\mathcal{A}$  and acts on it by affine automorphism (20.6.5.3). The barycenter of  $\sigma$  is clearly fixed by  $\text{Gal}(K^{\text{sh}}/K)$ , and if  $x, y \in \sigma$  are two points fixed by  $\text{Gal}(K^{\text{sh}}/K)$ , then the whole line through  $x, y$  are fixed, thus meeting the boundary of  $\sigma$ , contradiction.

Also notice that any two  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of minimal dimension doesn't intersect. So for any  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex of  $\mathcal{B}(G_{K^{\text{sh}}})$ , its vertices are disjoint union of minimal  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes. By (10.4.5.20),  $G_{K^{\text{sh}}}$  is a product of copies of simple  $K^{\text{sh}}$ -groups permuted transitively by  $\text{Gal}(K^{\text{sh}}/K)$ , so any  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex of  $\mathcal{B}(G_{K^{\text{sh}}})$  is a product of copies of a single simplex permuted by  $\text{Gal}(K^{\text{sh}}/K)$ , so to consider the intersection with  $\mathcal{B}$  we are reduced to the case that  $G_{K^{\text{sh}}}$  is simple. In this case, it is easy to see that for any  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex  $\sigma$  of  $\mathcal{B}(G_{K^{\text{sh}}})$  and any set  $S$  of minimal  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of  $\sigma$ , the facet

of  $\sigma$  whose vertices are union of simplexes in  $S$ , is also a  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex because  $|\mathcal{B}|$  is convex. Thus we get a simplicial structure on  $|\mathcal{B}|$ .

3: Because any two simplexes of  $\mathcal{B}$  are contained in an apartment and  $|\mathcal{B}|$  is a convex subset (20.6.5.4), we see by Euclidean geometry that the interiors of simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$  intersecting  $\mathcal{B}$  are all of the same dimension, and are connected by a gallery.

4: This is easy, because  $G(K)$  preserves  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplexes of  $\mathcal{B}(G_{K^{\text{sh}}})$ .  $\square$

**Prop. (20.6.5.6) [Apartments].** Situation as in (20.6.5.5), for any maximal split torus  $S \leq G$ , denote  $\mathcal{B}(G_{K^{\text{sh}}})_S$  the set of  $|\mathcal{B}(G_{K^{\text{sh}}})|$  fixed by  $S(K)^1$ , which is a non-empty closed convex subset because  $S(K)^1$  is bounded (20.5.2.5)(20.6.6.19). And  $\mathcal{B}(G_{K^{\text{sh}}})_S$  is  $\text{Gal}(K^{\text{sh}}/K)$ -stable, so the fixed point  $\mathcal{A}(S) = \mathcal{B}(G_{K^{\text{sh}}})_S^{\text{Gal}(K^{\text{sh}}/K)}$  is a non-empty closed convex subset by (20.6.5.3). Then:

1. For any maximal split torus  $S \leq G$  and two points  $x, y \in \mathcal{A}(S)$ , there exists a special torus  $T \leq G$  (20.6.5.2) containing  $S$  s.t.  $x, y \in \mathcal{A}(T_{K^{\text{sh}}})$ .
2.  $\mathcal{B}(G_{K^{\text{sh}}})_S$  is the union of apartments of  $\mathcal{B}(G_{K^{\text{sh}}})$  corresponding to maximal split torus  $T \leq G_{K^{\text{sh}}}$  containing  $S_{K^{\text{sh}}}$ .
3. For any maximal split torus  $S \leq G$  contained in a special torus  $T \leq G$ , we have  $\mathcal{A}(S) = \mathcal{A}(T_{K^{\text{sh}}})^{\text{Gal}(K^{\text{sh}}/K)}$ , and the action of  $N_G(S)(K)$  and  $V(S')$  on  $\mathcal{A}(T_{K^{\text{sh}}})$  preserves  $\mathcal{A}(S)$ , making it an affine  $V(S')$ -torsor satisfying axiom (20.6.4.10).
4. For a maximal split torus  $S \leq G$ ,  $Z_G(S)(K)^1$  consists of all elements of  $N_G(S)(K) \cap G(K)^1$  that acts trivially on  $\mathcal{A}(S)$ .
5. The map  $S \mapsto \mathcal{A}(S)$  is a  $G(K)$ -equivariant bijection the set of maximal split torus of  $G$  and the set of special tori of  $G$  (20.6.5.2), and the set of special apartments of  $\mathcal{B}$  (20.6.5.3). In particular, the stablizer of  $\mathcal{A}(S)$  in  $G(K)$  is  $N_G(S)(K)$ .

┘

*Proof:* ?

- 1,2: Cf. [K-P23]Prop9.3.11.
- 3: Cf. [K-P23]Prop9.3.13.Prop9.3.16
- 4: Cf. [K-P23]Prop9.3.16
- 5: 9.3.17?  $\square$

**Thm. (20.6.5.7) [Unramified Descent for Buildings].** Situation as in (20.6.5.5),

1.  $\mathcal{B}$  is a building (20.6.2.1) with the set of apartments  $\mathcal{A}(S)$  (20.6.5.4) for all maximal split tori  $S \leq G$ .
2. Axiom (20.6.4.8) holds for  $\mathcal{B}$ .
3. Axiom (20.6.4.14) holds for  $\mathcal{B}$ . In particular,
4. Axiom (20.6.4.9) holds for  $\mathcal{B}$ .

┘

*Proof:* 1: Cf. [K-P23]Prop9.3.21. ?

2:

3: Cf. [K-P23]Prop.3.25?.

4: Axiom (20.6.4.9) holds because we can take the restriction of the action of  $\text{Aut}(K^{\text{sh}})$  on  $\mathcal{B}(G_{K^{\text{sh}}})$  to  $\mathcal{B}$ .  $\square$

**Prop. (20.6.5.8) [Unramified Descent of Affine Root Systems].** Situation as in (20.6.5.7), for any maximal split torus  $S \leq G$  contained in a special torus  $T \leq G$  (20.6.5.2), the restriction map from  $\mathcal{A}(T_{K^{\text{sh}}})$  to  $\mathcal{A}(S) = \mathcal{A}(T_{K^{\text{sh}}})^{\text{Gal}(K^{\text{sh}}/K)}$  (20.6.5.4) maps  $\Psi(G_{K^{\text{sh}}}, T_{K^{\text{sh}}})$  into  $\widehat{\Psi}(G, S)$ , and for any  $\psi \in \Psi(G, S) \subset \mathcal{A}(S)^*$  and  $\beta \in \Phi(G_{K^{\text{sh}}}, T_{K^{\text{sh}}})$  s.t.  $\beta$  restricts to  $\dot{\psi} \in V^*$ , there exists  $\psi_\beta \in \Psi(T_{K^{\text{sh}}}) \subset \mathcal{A}(T_{K^{\text{sh}}})^*$  with  $\dot{\psi} = \beta$  and restriction to  $\psi$  on  $\mathcal{A}(S)$ . Then:

- Axiom (20.6.4.16) holds for  $\mathcal{B}$ .

┘

*Proof:* Cf. [K-P23] 359. ?

□

**Prop. (20.6.5.9).**  $G(K)^1 = G(K^{\text{sh}})^1 \cap G(K)$ ,  $G(K)^0 = G(K^{\text{sh}})^0 \cap G(K)$ .

┘

*Proof:* Cf. [K-P23] P343. ?

□

**Thm. (20.6.5.10) [Unramified Descent of Bruhat-Tits Group Schemes].** Situation as in (20.6.5.7), for any non-empty bounded  $\text{Gal}(K^{\text{sh}}/K)$ -stable subset  $\Omega \subset |\mathcal{B}(G_{K^{\text{sh}}})|$ ,  $\mathcal{G}_\Omega^1$  is defined over  $\mathcal{O}_K$  by (20.3.2.4) and Galois-descent. So is the closed  $\mathcal{O}_K$ -split torus  $\mathcal{S} \leq \mathcal{G}_\Omega^0$  (20.6.4.19).

- $\mathcal{G}_\Omega^0(\mathcal{O}_K) = \left(G(K^{\text{sh}})_\Omega^0\right)^{\text{Gal}(K^{\text{sh}}/K)} = G(K)_\Omega^0$  by (20.6.5.9).
- Parahoric subgroups of  $G(K)$  (20.6.4.13) are exactly of the form  $P^{\text{Gal}(K^{\text{sh}}/K)}$  where  $P \leq G(K^{\text{sh}})$  is a  $\text{Gal}(K^{\text{sh}}/K)$ -stable parahoric subgroup. (because any  $\text{Gal}(K^{\text{sh}}/K)$ -stable parahoric subgroup of  $G(K^{\text{sh}})$  fixes a  $\text{Gal}(K^{\text{sh}}/K)$ -stable simplex).
- $G(K)_\Omega^0 = G(K^{\text{sh}})_\Omega^0 \cap G(K)$ ,  $G(K)_\Omega^1 = G(K^{\text{sh}})_\Omega^1 \cap G(K)$ .
- Axioms (20.6.4.19) (20.6.4.20) hold for  $\mathcal{B}$  and these  $\mathcal{G}_\Omega^0$ .

┘

*Proof:* Cf. [K-P23] Chap9.5. ?

Axiom (20.6.4.19) holds for  $\mathcal{B}$  because it holds for  $\mathcal{B}(G_{K^{\text{sh}}})$ .

□

## 6 Metrics and More Properties

**Notation (20.6.6.1).**

- Suppose that  $G$  satisfies all axioms in [Axiomatic Bruhat-Tits Buildings](#).

┘

**Def. (20.6.6.2) [Canonical Metrics].** By choosing a maximal split torus  $S \leq G$  and a  $W(G, S)$ -invariant metric on  $V(S')$  (20.6.4.1), we get a metric on  $V(S'_1)$  for each maximal split torus  $S_1 \leq G$ , s.t. conjugation by  $G(K)$  induces isometries between them. Such a choice is called a **canonical metric** on each  $V(S')$ .

┘

**Prop. (20.6.6.3).** Given a canonical metric on  $V(S')$  for each maximal split torus  $S \leq G$  (20.6.6.2), there exists a unique function  $\text{dist} : |\mathcal{B}(G)| \times |\mathcal{B}(G)| \rightarrow \mathbb{R}_{\geq 0}$  whose restriction on the apartment  $\mathcal{A}(S)$  for each maximal split torus  $S \leq G$  coincides with the metric induced by the  $V(S')$ -torsor structure (20.6.4.10). Moreover,  $\text{dist}$  is preserved by the action of  $G(K)$ .

┘

*Proof:* For any  $x, y \in |\mathcal{B}(G)|$ , there exists an apartment  $\mathcal{A}$  containing both  $x$  and  $y$  (20.6.2.1), so we can define the distance  $\text{dist}(x, y)$  as the distance in this apartment. This definition is independent of the apartment chosen, because for any other apartment  $\mathcal{A}'$  containing  $x$  and  $y$ , there exists  $g \in G(K)$  s.t.  $g\mathcal{A}' = \mathcal{A}$  and  $g$  preserves  $x$  and  $y$ , by (20.6.4.14), and this action preserves the canonical metric by (20.6.6.2). □

**Cor. (20.6.6.4) [Metric and Contractions].** For a chamber  $C$  contained in an apartment  $\mathcal{A} \subset \mathcal{B}(G)$ , let  $\rho : \mathcal{B}(G) \rightarrow \mathcal{A}$  be the canonical contraction centered at  $C$  (20.6.2.4), then

- The restriction of  $\rho$  to any apartment containing  $C$  is an isometry.
- If  $x, y \in |\mathcal{B}(G)|$  and either  $x \in C$  or  $x, y$  are contained in a common chamber, then  $\text{dist}(\rho(x), \rho(y)) = \text{dist}(x, y)$ .
- For any  $x, y \in |\mathcal{B}(G)|$ , we have  $\text{dist}(\rho(x), \rho(y)) \leq \text{dist}(x, y)$ .

┘

*Proof:* 1: This follows from the fact that the restriction of  $\rho$  on each apartment coincides with the action of some  $g \in G(K)$  by (20.6.3.15)(20.6.4.14), so it is an isometry by the constriction of  $\text{dist}$  (20.6.6.3).

2: This follows from item1 because in this case  $x, y$  is contained in an apartment containing  $C$ .

3: Let  $\mathcal{A}'$  be an apartment containing  $x, y$ , then the line segment  $[x, y]$  is divided into segments by the chambers contained in  $\mathcal{A}'$ . But  $\rho$  is an isometry on each segment by item2, so the assertion follows.  $\square$

**Cor. (20.6.6.5) [Metric and Line Segments].**  $\text{dist}$  (20.6.6.3) is a metric on  $|\mathcal{B}(G)|$ , and for any  $x, y \in |\mathcal{B}(G)|$ , the set

$$D_{x,y} = \{z \in |\mathcal{B}(G)| : \text{dist}(x, y) = \text{dist}(x, z) + \text{dist}(z, y)\}$$

lies in every apartment  $\mathcal{A}$  containing both  $x$  and  $y$ , and equals the line segment  $[x, y]$  there.  $\square$

*Proof:* To prove  $\text{dist}$  is a metric, it suffices to show that for any  $x, y, z \in |\mathcal{B}(G)|$ , we have  $\text{dist}(x, y) \leq \text{dist}(x, z) + \text{dist}(z, y)$ . For this, choose an apartment  $\mathcal{A} \subset \mathcal{B}(G)$  containing both  $x, y$ , and a chamber  $C \subset \mathcal{A}$ . Let  $\rho : \mathcal{B}(G) \rightarrow \mathcal{A}$  be the canonical contraction centered at  $C$  (20.6.2.4), then by (20.6.6.4), we have

$$\text{dist}(x, y) = \text{dist}(\rho(x), \rho(y)) \leq \text{dist}(\rho(x), \rho(z)) + \text{dist}(\rho(z), \rho(y)) \leq \text{dist}(x, z) + \text{dist}(z, y).$$

For the second assertion, for any apartment  $\mathcal{A}$  containing  $x, y$  and  $z \in D_{x,y}$ , take  $t$  in the line segment  $[x, y] \subset \mathcal{A}$  s.t.  $\text{dist}(x, t) = \text{dist}(x, z)$ ,  $\text{dist}(y, t) = \text{dist}(y, z)$ , and take an apartment  $C \subset \mathcal{A}$  containing  $t$ . Let  $\rho : \mathcal{B}(G) \rightarrow \mathcal{A}$  be the canonical contraction centered at  $C$ , then it follows from the contraction property (20.6.6.4) that  $\rho(z) = t$ . But then it follows from (20.6.2.4) that  $z = t$ .  $\square$

**Remark (20.6.6.6).** WARNING: The topology on  $|\mathcal{B}(G)|$  given by the metric may be weaker than the topology given by the CM-complex structure when  $\# \kappa = \infty$ .  $\square$

**Prop. (20.6.6.7).** The action  $G(K) \times |\mathcal{B}(G)| \rightarrow |\mathcal{B}(G)|$  is continuous w.r.t. the topology given by the metric (20.6.6.5)  $\square$

*Proof:* For any  $g, g' \in G(K)$  and  $x, x' \in \mathcal{B}(G)$ ,

$$\text{dist}(gx, g'x') \leq \text{dist}(gx, gx') + \text{dist}(gx', g'x') = \text{dist}(x, x') + \text{dist}(x', g^{-1}g'x'),$$

Now because  $G(K)_x^1$  is open (20.6.4.12), we see that this action is continuous.  $\square$

**Prop. (20.6.6.8) [ $|\mathcal{B}(G)|$  is Negatively-Curved].**  $|\mathcal{B}(G)|$  is a non-positively-curved metric space (4.4.8.13). In particular it is uniquely-geodesic (4.4.8.10).  $\square$

*Proof:* Cf. [K-P23]P170?.

Firstly  $|\mathcal{B}(G)|$  is uniquely-geodesic by (20.6.6.5) and the fact Euclidean spaces are uniquely-geodesic.

To show it is non-positively-curved, for any  $x, y \in |\mathcal{B}(G)|$ , let  $\mathcal{A}$  be an apartment containing  $x, y$ , and let  $m$  be the midpoint of  $[x, y]$ , then  $m \in |\mathcal{A}|$  (20.6.6.5). Choose an apartment  $C \subset \mathcal{A}$  containing  $m$ , and  $\rho : \mathcal{B}(G) \rightarrow \mathcal{A}$  the canonical contraction centered at  $C$  (20.6.2.4), then  $\text{dist}(m, z) = \text{dist}(m, \rho(z))$  for any  $z \in |\mathcal{B}(G)|$  by (20.6.6.4). Now it follows from Euclidean geometry that

$$\text{dist}(x, z)^2 + \text{dist}(y, z)^2 \geq 2\text{dist}(m, z)^2 + \frac{1}{2}\text{dist}(x, y)^2.$$

□

**Prop. (20.6.6.9) [ $|\mathcal{B}(G)|$  is Contractible, Solomon-Tits].** The map

$$\mathbb{I} \times |\mathcal{B}(G)| \times |\mathcal{B}(G)| \rightarrow |\mathcal{B}(G)| : (t, x, y) \mapsto tx + (1 - t)y$$

is continuous. In particular,  $|\mathcal{B}(G)|$  is a contractible topological space. ┘

*Proof:* The assertion will follow from the following claim: If  $x, y, z, z' \in |\mathcal{B}(G)|$  s.t.  $z \in [x, y]$  (20.6.6.5) and there exists  $\varepsilon \in \mathbb{R}_+$  s.t.

$$\text{dist}(x, z') \leq \text{dist}(x, z) + \varepsilon \text{dist}(x, y), \quad \text{dist}(y, z') \leq \text{dist}(y, z) + \varepsilon \text{dist}(x, y),$$

then we have

$$\text{dist}(z, z')^2 \leq 4\varepsilon \text{dist}(x, z) \text{dist}(y, z) + \varepsilon^2 \text{dist}(x, y)^2.$$

To prove this, choose an apartment  $\mathcal{A}$  containing  $x, y$  and a chamber  $C \subset \mathcal{A}$  containing  $z$ , then  $[x, y] \in \mathcal{A}$  by (20.6.6.5). Let  $\rho : \mathcal{B}(G) \rightarrow \mathcal{A}$  is canonical contraction centered at  $C$  (20.6.2.4), and let  $z'' = \rho(z')$ , then by (20.6.6.4) we have  $\text{dist}(z, z') = \text{dist}(z, z'')$ , and the same inequalities hold for  $z'$  replaced by  $z''$ . Thus we are reduced to Euclidean geometry, which is easy. □

**Thm. (20.6.6.10) [ $|\mathcal{B}(G)|$  is Complete].**  $|\mathcal{B}(G)|$  is a complete metric space. And any subcomplex is closed in  $|\mathcal{B}(G)|$ . ┘

*Proof:* We first prove that

**Lemma (20.6.6.11).** For any  $x \in |\mathcal{B}(G)|$ , there exists an open ball  $D = \mathbb{D}(x, \delta)$  for some  $\delta \in \mathbb{R}_+$  s.t.

- Every simplex intersecting  $D$  contains  $x$ .
- If  $g \in G(K)^0$  satisfies  $g.x \in D$ , then  $g$  fixes  $x$ .

┘

*Proof:* Let  $C$  be a chamber containing  $x$  and  $\mathcal{A}(S)$  be an apartment containing  $C$  corresponding to a maximal split torus  $S \leq G$ . By (20.6.4.16) and (3.9.8.2) and the fact that  $\#\Phi(G, S) < \infty$ , we can find a metric ball  $D(x, \delta)$  of  $x$  that only meets simplexes of  $\mathcal{A}(S)$  containing  $x$ .

Now if  $\sigma$  is a simplex of  $\mathcal{B}(G)$  that meets  $D$ , and  $\mathcal{A}'$  an apartment containing  $C$  and  $\sigma$ , then there exists  $g \in G(K)^0$  s.t.  $g\mathcal{A}' = \mathcal{A}$  and  $g$  fixes vertices of  $C$ , by (20.6.3.11). Thus  $g$  stabilizes  $D$ , and  $g\sigma$  intersects with  $D$ , which implies that  $x \in g\sigma$ , thus  $x \in \sigma$ .

And if  $g \in G(K)^0$  s.t.  $gx \in D$ , then the smallest simplex  $\sigma$  containing  $gx$  intersects with  $D$ , thus contains  $x$ . So  $g\sigma = \sigma$ , and then  $g$  fixes vertices of  $\sigma$  because  $G(K)^0$  acts by special automorphisms (20.6.4.14). So  $g$  fixes  $x$ . □



To prove the original assertion, it suffices to prove that any subcomplex  $M$  of  $\mathcal{B}(G)$  is complete. Let  $C$  be a chamber of  $\mathcal{B}(G)$ . For any Cauchy-sequence  $(x_n)_{n \in \mathbb{Z}_+}$  in  $M$ , let  $g_n \in G(K)^0$  s.t.  $y_n = g_n^{-1} \cdot x_n \in C$ . Passing to a subsequence, we may assume that  $y_n$  converges to some  $y \in C$ . Let  $D_y$  be the open ball satisfying (20.6.6.11). Because

$$\text{dist}(g_m y, g_n y) \leq \text{dist}(g_m y, g_m y_m) + \text{dist}(g_m y_m, g_n y_n) + \text{dist}(g_n y_n, g_n y) = \text{dist}(y, y_m) + \text{dist}(x_m, x_n) + \text{dist}(y, y_n),$$

we have  $g_m^{-1} g_n y \in D_y$  for  $m, n$  large, hence  $g_m y = g_n y$  for  $m, n$  large. Define this to be  $x$ , then clearly  $(x_n)$  converges to  $x$ . To show that  $x \in |M|$ , we can take an open ball  $D_x$  of  $x$  satisfying (20.6.6.11), then  $x_n \in D_x$  for  $n$  large, which means the simplex containing  $x_n$  contains  $x$ .  $\square$

**Cor. (20.6.6.12) [Bruhat-Tits Fixed-Point Lemma].**

- Any non-empty closed bounded (4.4.8.5) and convex (4.4.8.15) subset  $M \subset |\mathcal{B}(G)|$  has a unique barycenter (4.4.8.15).
- If  $M \subset X$  is a non-empty bounded subset and  $H \in \mathcal{G}rp$  acts on  $|\mathcal{B}(G)|$  by isometries and preserves  $M$ , then  $H$  has a fixed point in  $X$ .
- If  $H \in \mathcal{G}rp$  is a bounded group of isometric actions (4.4.8.8) on  $|\mathcal{B}(G)|$ , then  $H$  has a fixed point.  $\lrcorner$

*Proof:* This follows from (20.6.6.10) (20.6.6.8) and (4.4.8.31).  $\square$

**Prop. (20.6.6.13) [No Tubular Neighborhoods].** Let  $\mathcal{A}_1, \mathcal{A}_2$  be two apartments of  $\mathcal{B}(G)$  and there exists  $r \in \mathbb{R}_+$  s.t. for any  $x \in |\mathcal{A}_1|$  there exists  $x_2 \in |\mathcal{A}_2|$  s.t.  $\text{dist}(x_1, x_2) \leq r$ , then  $\mathcal{A}_1 = \mathcal{A}_2$ .  $\lrcorner$

*Proof:* Let  $\text{pr}_i : \mathcal{B}(G) \rightarrow \mathcal{A}_i$  be the closest-point map (4.4.8.34), then  $f = \text{pr}_2 \circ \pi_2 : \mathcal{A}_1 \rightarrow \mathcal{A}_1$  satisfies  $\text{dist}(x, f(x)) \leq 2r$ .

Choose an isomorphism of metric spaces  $\mathcal{A}_1 \cong \mathbb{R}^n$ , and for each  $y \in \mathbb{R}^n$ , define  $f_y : \mathbb{R}^n \rightarrow \mathbb{R}^n : f_y(x) = f(y - x) - (y - x)$ , then  $f_y$  maps  $\overline{\mathbb{D}}(0, 2r)$  into itself. Thus by Brouwer's fixed point theorem (4.14.4.2),  $f_y$  has a fixed point, which means  $f$  is surjective.

But this implies that  $\mathcal{A}_2$  contains  $\mathcal{A}_1$ , otherwise the closest-point map cannot map into the interiors of the chambers of  $\mathcal{A}_1$ . The same argument for  $\mathcal{A}_2$  implies that  $\mathcal{A}_1 = \mathcal{A}_2$ .  $\square$

**Prop. (20.6.6.14).** For any  $x \in |\mathcal{B}(G)|$ , the convex hull of  $G(K) \cdot x$  (4.4.8.17) equals  $|\mathcal{B}(G)|$ .  $\lrcorner$

*Proof:* Let  $S \leq G$  be a maximal split torus s.t.  $x \in |\mathcal{A}(S)|$ , and let  $C \subset \mathcal{A}(S)$  be a chamber containing  $x$ , then  $N_G(S)(K)$  acts transitively on the chambers of  $\mathcal{A}(S)$  (20.6.4.14). So it is clear that the convex hull of  $G(K) \cdot x$  contains  $\mathcal{A}(S)$ . But  $G(K)$  acts transitively on the set of apartments, so the assertion follows.  $\square$

**Prop. (20.6.6.15).** Let  $\mathcal{A}_1, \mathcal{A}_2$  be two apartments of  $\mathcal{B}(G)$ . There exists  $g \in G(K)^0$  s.t.  $g\mathcal{A}_1 = \mathcal{A}_2$  and fixes points of  $\mathcal{A}_1 \cap \mathcal{A}_2$ .  $\lrcorner$

*Proof:* Notice that  $\mathcal{A}_1 \cap \mathcal{A}_2$  is a closed convex subcomplex of  $\mathcal{B}(G)$ . If  $\sigma \subset \mathcal{A}_1 \cap \mathcal{A}_2$  is a simplex of maximal dimension, then by (20.6.3.11), there exists  $g \in G(K)^0$  s.t.  $g\mathcal{A}_1 = \mathcal{A}_2$  and  $g$  fixes vertices of  $\sigma$ .

But then this  $g$  in fact fixes every point of  $\mathcal{A}_1 \cap \mathcal{A}_2$ : If  $\sigma$  is a vertex, then  $\mathcal{A}_1 \cap \mathcal{A}_2$  is a vertex and there is nothing to prove, and if  $\dim \sigma > 0$ , we can take  $x_0$  in the interior of  $\sigma$ , then for any  $y \in \mathcal{A}_1 \cap \mathcal{A}_2$ , the geodesic segment  $[x_0, y]$  is fixed  $g$ .  $\square$

**Cor. (20.6.6.16).** For any maximal split torus  $S \leq G$  and  $g \in G(K)$ , there exists  $n \in N_G(S)(K)$  s.t.  $g \cdot x = n \cdot x$  for all  $x \in \mathcal{A}(S) \cap g^{-1} \mathcal{A}(S)$ .  $\lrcorner$



*Proof:* By (20.6.6.15) we can take  $h \in G(K)$  s.t.  $h.\mathcal{A}(S) = g^{-1}\mathcal{A}(S)$  and  $h$  fixes all vertices of  $\mathcal{A}(S) \cap g^{-1}\mathcal{A}(S)$ . Then  $n = gh$  stabilizes  $\mathcal{A}(S)$  thus belongs to  $N_G(S)(K)$  by (20.6.4.8).  $\square$

**Cor. (20.6.6.17).** If  $S_1, S_2 \leq G$  are maximal split tori,  $x \in \mathcal{A}(S_1) \cap \mathcal{A}(S_2)$  and  $C \leq S_1$  is a subtorus s.t.  $C(K).x \subset \mathcal{A}(S_2)$ , then  $C \leq S_2$ .  $\lrcorner$

*Proof:* Cf. [K-P23] Prop 4.2.26.  $\square$

**Prop. (20.6.6.18) [Bounded Subsets and Bounded Actions].** For a subset  $X \subset G(K)^1$ , the following are equivalent:

- $X$  is a bounded subset of  $G(K)$  (20.5.1.3).
- For some (hence any) pair  $(C, \mathcal{A})$  where  $\mathcal{A} \subset \mathcal{B}$  is an apartment and  $C \subset \mathcal{A}$  is a chamber, corresponding to Iwahori subgroup  $\mathcal{I}$  and maximal split torus  $S$  correspondingly (20.6.4.8) (20.6.4.13), there exists f.m.  $n_i \in N_G(S)(K)$  s.t.  $X \subset \cup_i \mathcal{I} n_i \mathcal{I}$ .
- $X$  is a bounded set of isometric actions of  $|\mathcal{B}(G)|$  (4.4.8.8).  $\lrcorner$

*Proof:* 1  $\iff$  2:  $\mathcal{I}$  is bounded by (20.6.4.12), so  $X$  is bounded iff  $\mathcal{I}X\mathcal{I}$  is bounded, which is equivalent to  $X \subset \mathcal{I}Y\mathcal{I}$  for some bounded subset  $Y \leq N(K)$ , by (20.7.1.4). Now because  $Z_G(S)$  has finite index in  $N_G(S)$ ,  $Z_G(S)(K)^1$  is bounded in  $Z_G(S)(K)$ , and  $\mathcal{I}$  contains  $Z_G(S)^0$ , which has finite index in  $Z_G(S)^1$ , we see that item1 and item2 are equivalent.

2  $\iff$  3: For any  $x \in |C|$ ,  $n_i \in N_G(S)(K)$  and  $g \in \mathcal{I} n_i \mathcal{I}$ , we have  $\text{dist}(gx_0, x_0) = \text{dist}(n_i x_0, x_0)$ , so  $\mathcal{I} n_i \mathcal{I} x_0$  is a bounded subset. So if  $X \subset \cup_i \mathcal{I} n_i \mathcal{I}$  for f.m.  $n_i \in N_G(S)(K)$ , we then  $X.x_0$  is bounded for any  $x_0 \in C$ . Conversely, if there are infinitely many  $n_i \in N_G(S)(K)$  s.t.  $\mathcal{I} n_i \mathcal{I} \cap G(K) \neq \emptyset$ , then  $G(K).x_0$  intersect infinitely many chambers  $n_i.C \subset \mathcal{A}$ , thus it is unbounded. This proves that 2  $\iff$  3, by (4.4.8.8).  $\square$

**Cor. (20.6.6.19) [Bounded Subgroups Has Fixed Points].** Any bounded subgroup  $\mathcal{G} \leq G(K)$  has a fixed point on  $|\mathcal{B}(G)|$ , and if  $\mathcal{G}$  preserves a non-empty closed convex subset, then it has a fixed point in that subset.  $\lrcorner$

*Proof:* This follows from (20.6.6.18) and (20.6.6.12).  $\square$

**Prop. (20.6.6.20) [Maximal Bounded Subgroups and Parahorics].**

- If  $x \in \mathcal{B}(G)$  is a vertex, then  $G(K)_x^1$  is a maximal bounded subgroup of  $G(K)$ .
- Any maximal bounded subgroup of  $G(K)$  is of the form  $G(K)_x^1$  for some  $x \in |\mathcal{B}(G)|$ . And if  $G(K)^0 = G(K)^1$ , then this  $x$  is a vertex.
- Maximal parahoric subgroups of  $G(K)$  (20.6.4.13) are exactly maximal bounded subgroups of  $G(K)^0$ .  $\lrcorner$

*Proof:* 1: If  $x \in |\mathcal{B}(G)|$  is a vertex, then  $G(K)_x^1$  is maximal because if  $\mathcal{K}$  is bounded and contains  $G(K)_x^1$ , then  $\mathcal{K}$  fixes some point  $y \in |\mathcal{B}(G)|$  by (20.6.6.19). But  $G(K)_x^0$  fixes only  $x$  by (20.6.4.14) (20.6.3.11), so  $y = x$  and  $\mathcal{K} = G(K)_x^1$ .

2: For any  $y \in |\mathcal{B}(G)|$ ,  $G(K)_y^1$  is open bounded by (20.6.4.12). If  $\mathcal{K} \leq G(K)$  is a bounded subgroup, then it fixes a point  $y \in |\mathcal{B}(G)|$  by (20.6.6.19). Thus  $\mathcal{K} = G(K)_y^1$  if  $\mathcal{K}$  is maximal. And if  $G(K)^0 = G(K)^1$ , then  $G(K)^1$  acts by special automorphisms on  $\mathcal{B}(G)$ , so  $G(K)^1$  fixes a vertex  $x$ , and then  $\mathcal{K} = G(K)_x^1$  by maximality.

3: The proof is verbatim as the proof of item1 and item2.  $\square$

**Remark (20.6.6.21).** When  $G(K)^0 \neq G(K)^1$ , there can exist maximal bounded subgroups of  $G(K)_x^1$  s.t.  $x$  is not a vertex of  $\mathcal{B}(G)$ . For example, when  $G = \mathrm{PGL}(2)$ , the **standard(extended) Iwahori-subgroup**, which is generated by the inverse image of the standard Borel subgroup of  $\mathrm{PGL}(2; \kappa)$  and  $\begin{bmatrix} & 1 \\ \varpi_K & \end{bmatrix}$ , stabilizes the midpoint of the standard chamber.  $\lrcorner$

**Prop. (20.6.6.22).** Suppose the Tits-system  $(G(K)^0, G(K)_C^0, N_G(S)(K) \cap G(K)^0)$  (20.6.4.14) is irreducible (20.6.1.10). If  $\mathcal{P} \neq \mathcal{P}'$  are two maximal parahoric subgroups of  $G(K)^0$ , then  $\mathcal{P} \cap \mathcal{P}'$  is a parahoric subgroup iff it is a maximal properly-contained subgroup of  $\mathcal{P}$ .  $\lrcorner$

*Proof:* If  $\mathcal{P} \cap \mathcal{P}'$  is a parahoric subgroup, then by (20.6.1.2) we can conjugate it into a subgroup containing  $\mathcal{G} = G(K)_C^0$  (i.e. a standard parahoric subgroup (20.6.1.2)). Since the Tits-system is irreducible, there exists  $r \neq r' \in S = N_G(S)(K) \cap G(K)^0$  s.t.  $\mathcal{P} = \mathcal{G}_{S \setminus \{r\}}, \mathcal{P}' = \mathcal{G}_{S \setminus \{r'\}}$ . Then  $\mathcal{P} \cap \mathcal{P}' = \mathcal{G}_{S \setminus \{r\}}, \mathcal{P}' = \mathcal{G}_{S \setminus \{r, r'\}}$  is a maximal properly-contained subgroup of  $\mathcal{P}$ .

Conversely, suppose  $\mathcal{P} \cap \mathcal{P}'$  is a maximal properly-contained subgroup of  $\mathcal{P}$ , and let  $x, x'$  be vertices of  $\mathcal{B}(G)$  fixed by  $\mathcal{P}, \mathcal{P}'$  resp. (20.6.4.13). Then  $\mathcal{P} \cap \mathcal{P}'$  fixes the simplex  $\sigma = [x, x']$ , and the parahoric subgroup corresponding to  $\sigma$ , then it contains  $\mathcal{P} \cap \mathcal{P}'$ , thus equals  $\mathcal{P} \cap \mathcal{P}'$  by maximality. So  $\mathcal{P} \cap \mathcal{P}'$  is parahoric.  $\square$

## 20.7 More Properties of Bruhat-Tits Buildings

References are [I-M65, Tit79, B-T72, B-T84, B-T84a, B-T87, B-T87a, P-R08, Yu09, Yu15, K-P23], [Landvogt, A Compactification of the Bruhat-Tits Building], [Buildings and Classical Groups, Garrett], [Period mappings and differential equations. From  $\mathbb{C}$  to  $\mathbb{C}_p$ , Andre].

### Notation (20.7.0.1).

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in \text{DVR}$  be Henselian with  $\kappa = \kappa^{\text{perf}}$  (if  $\#\kappa = \infty$ , use the notion of bounded subsets (20.5.1.3)).
- Let  $\Gamma_K = \{\text{ord}_K(x) | x \in K^\times\}$ .
- Use notations on Bruhat-Tits buildings defined in [Bruhat-Tits Theory](#).
- Let  $G \in \text{RedGrp}/K$ .
- Let  $A_G \leq G$  be the maximal central split torus of  $G$ .
- Let  $H \leq G \in \text{AlgGrp}/K$ .
- $G(K)$  is a topological group with the topology coming from  $K$ .

┘

### 1 Properties of Bruhat-Tits Buildings

**Prop. (20.7.1.1) [Finite Base Change].** If  $L/K \subset K^{\text{sep}}/K \in \text{Field}$  is a finite extension, the natural base change map  $\mathcal{B}(G_{K^{\text{SH}}}) \hookrightarrow \mathcal{B}(G_{L^{\text{SH}}})$  (20.6.4.29) induced a natural injection  $\mathcal{B}(G) \hookrightarrow \mathcal{B}(G_L)$ , because  $\text{Gal}(L^{\text{SH}}/L) \subset \text{Gal}(K^{\text{SH}}/K)$ . ┘

*Proof:* This is because the injection  $\mathcal{B}(G_{K^{\text{SH}}}) \hookrightarrow \mathcal{B}(G_{L^{\text{SH}}})$  is compatible with Galois action. ┐

**Prop. (20.7.1.2).**  $\mathcal{B}(G)$  (20.6.4.28) is locally-finite if  $\#\kappa < \infty$ . ┘

*Proof:* For each simplex  $\sigma \in \mathcal{B}(G)$  contained in a chamber  $C$ , the chambers containing  $\sigma$  are in bijection with  $G(K)_C^0/G(K)_\sigma^0$ , by (20.6.3.10)(20.6.3.11). Because  $\#\kappa < \infty$ , both  $\mathcal{P}_C$  and  $\mathcal{P}_\sigma$  are open compact (20.6.4.12), so this set is finite. ┐

**Prop. (20.7.1.3) [Parahoric Subgroups and pro- $p$ -Subgroups, Matsumoto].** When  $K \in p\text{-LField}$ , a Iwahori subgroup of  $G(K)$  (20.6.4.13) is a normalizer of a Sylow pro- $p$ -subgroup of  $G(K)$ . By (20.5.2.9), such a group is uniquely up to conjugacy. And a parahoric subgroup is a subgroup  $P \leq G(K)$  that is a finite union of double cosets of an Iwahori subgroup. ┘

*Proof:* Cf. [Kaletha-Prasad]P476 ? . ┐

### Iwahori-Decompositions

**Prop. (20.7.1.4) [Bruhat-Decompositions].** Assume axiom (20.6.4.14) holds. If  $C$  is a chamber in an apartment  $\mathcal{A} \subset \mathcal{B}(G)$ , and let  $\mathcal{I} \leq G(K)$  be the corresponding Iwahori subgroup (20.6.4.13) and  $S \leq G$  the maximal split torus (20.6.4.8), then

$$G(K) = \mathcal{I}.N_G(S)(k).\mathcal{I}.$$

┘

*Proof:* It follows from (20.6.4.14) and Bruhat-decomposition (20.6.1.4) that  $G(K)^0 = \mathcal{I}(N_G(S)(k) \cap G(K)^0) \mathcal{I}$ . Then the assertion follows from the fact that  $G(K) = Z_G(S)(K)G(K)^0$  (Cf. [KP23] 2.6.22).  $\square$

**Axiom (20.7.1.5) [Iwahori Decompositions].** Axiom (20.6.4.8)(20.6.4.10)(20.6.4.16) hold, and if  $S \leq G$  be a maximal split torus and  $C \subset \mathcal{A}(S)$  is a chamber with corresponding Iwahori subgroup  $\mathcal{I} = G(K)_C^0$  (20.6.4.13), then for any positive system of roots  $\Phi^+ \subset \Phi(G, S)$ ,

- The product map

$$(U^+(K) \cap \mathcal{I}) \times Z_G(S)(K)^0 \times (U^-(K) \cap \mathcal{I}) \rightarrow \mathcal{I}$$

is a bijection, where  $U^+$  (resp.  $U^-$ ) is the unipotent radical of the minimal parabolic subgroup containing  $S$  corresponding to  $\Phi^+$  ( $\Phi^-$ ) (10.4.4.18),

- The product map

$$\prod_{\alpha \in \Phi^{\pm, \text{short}}} u_{\psi_\alpha} \rightarrow U^\pm(K) \cap \mathcal{I} \quad (20.6.4.16)$$

is a bijection, where the product is taken in any order, and  $\psi_\alpha \in \Psi$  is the smallest affine functional on  $\mathcal{A}(S)$  with derivative  $\alpha$  that is non-negative on  $C$ .  $\lrcorner$

*Proof:*  $\color{red}{?}$  This is Axiom 4.1.16 of [Kaletha-Prasad].  $\square$

### Enlarged Buildings

**Def. (20.7.1.6) [Enlarged Bruhat-Tits Buildings].** The **enlarged Bruhat-Tits building**  $\tilde{\mathcal{B}}(G)$  is defined to be  $\mathcal{B}(G) \times V(A_G)$  (20.6.4.8) (where  $V(A_G) = X_*(A_G) \otimes \mathbb{R}$ ). And for a maximal split torus  $S \leq G$ , the **enlarged apartment** associated to  $S$  is defined to be  $\tilde{\mathcal{A}}(S) = \mathcal{A}(S) \times V(A_G)$ .

There is a canonical decomposition  $V(S) = V(S') \oplus V(A_G)$  (20.6.4.1), so  $\tilde{\mathcal{A}}(S)$  is naturally an affine  $V(S)$ -torsor. And choosing any Euclidean metric on  $V(A_G)$  makes  $|\mathcal{B}(G)|$  a complete non-positively-curved and uniquely-geodesic space, by the same arguments in [Metrics and More Properties](#).  $\lrcorner$

**Cor. (20.7.1.7).**  $Z(G)^0 \rightarrow G_{\text{ab}}$  is an isogeny, so  $V(A_G) \cong V(G_{\text{ab}})^{\text{Gal}_K}$ , and we get the evaluation map  $\nu : G(K) \rightarrow G_{\text{ab}}(K) \rightarrow V(G_{\text{ab}})^{\text{Gal}_K} \rightarrow V(A_G)$ . Then  $G(K)$  acts on  $V(A_G)$  (20.7.1.6) by translation by  $\nu(G)$ , in particular,

$$\langle g \cdot \lambda, \chi \rangle = \langle \lambda, \chi \rangle - \text{ord}(\chi(g)), \text{ where } \lambda \in V(A_G) \cong V(A_{\text{ab}})^{\text{Gal}_K}, \chi \in X^*(G).$$

Thus  $G(K)$  acts diagonally on  $\tilde{\mathcal{B}}(G)$ , and for any simplex  $\tilde{\sigma} \in \tilde{\mathcal{B}}(G)$  with image  $\sigma \in \mathcal{B}(G)$ , the stabilizer  $G(K)_{\tilde{\sigma}}$  equals  $G(K)_x^1$  (20.6.4.13).  $\lrcorner$

**Cor. (20.7.1.8).** Any bounded subgroup  $\mathcal{G} \subset G(K)$  has a fixed point in  $\tilde{\mathcal{B}}(G)$ , and if  $\mathcal{G}$  preserves a non-empty closed convex subset, then it has a fixed point in that subset.  $\lrcorner$

*Proof:* As  $\mathcal{G}$  is bounded, it is contained in  $G(K)^1$ , thus acts trivially on  $V(A_G)$ . So the assertions follow from (20.6.6.19).  $\square$

### Iwahori-Weyl Groups

See [P-R08, Ric16].

**Def. (20.7.1.9) [Iwahori-Weyl Group].** For a maximal split torus  $S \leq G$ , define the **Iwahori-Weyl group**  $\tilde{W}(G, S) = N_G(S)(K)/Z_G(S)(K)^0$ , with a natural surjection  $\tilde{W}(G, S) \rightarrow W(G, S)$ .  $\lrcorner$

## 2 Special Points

**Def. (20.7.2.1) [Special Points].** For  $x \in \mathcal{B}(G)$ , choose a maximal split torus  $S \leq G$  s.t.  $x \in \mathcal{A}(S)$ , then  $x$  is called a **special point** if  $x$  is a special point for the affine root system  $\Psi(G, S)$  (20.6.4.16)(3.9.8.23). By (20.6.4.10), this notion doesn't depend on  $S$  chosen. By (3.9.8.23), special points are vertices.

The parahoric subgroup corresponding to a special vertex (20.6.4.13) is called a **special maximal parahoric subgroup**.  $\lrcorner$

**Prop. (20.7.2.2).** Let  $S \leq G$  be a maximal split torus and  $x \in \mathcal{A}(S)$ , the following are equivalent:

- $x$  is special.
- The Weyl-group  $W(G_x)$  is canonically isomorphic to the Weyl-group  $W(G)$ .
- $N_G(S)(K) \cap G(K)_x^0$  maps onto  $W(G)$ .

$\lrcorner$

*Proof:* 1  $\iff$  2 follows from (3.9.8.24) and (20.6.4.20).  $\square$

**Prop. (20.7.2.3) [Special Point for  $\mathcal{B}(G_{K^{\text{sh}}})$ ].** If  $x \in \mathcal{B}(G)$  is special for  $\mathcal{B}(G_{K^{\text{sh}}})$ , then it is special for  $\mathcal{B}(G)$ .  $\lrcorner$

*Proof:* Let  $S \leq G$  be a maximal split torus s.t.  $x \in \mathcal{A}(S)$ . Let  $T \leq G$  be a special torus containing  $S$  (20.6.5.2). Because  $x$  is special, for any  $\beta \in \Phi(G, S)$ , there exists some  $\psi_\beta \in \Psi(G_{K^{\text{sh}}}, T_{K^{\text{sh}}})$  s.t.  $\psi_\beta(z) = 0$  and  $\psi_\beta$  is a non-constant multiple of  $\beta$ . Because  $\Phi(G_{K^{\text{sh}}}, T_{K^{\text{sh}}}) \rightarrow \Phi(G, S) \cup \{0\}$  is surjective, this implies that  $z$  is special for  $\mathcal{B}(G)$ .  $\square$

**Def. (20.7.2.4) [Super-special Points].** A point  $x \in \mathcal{B}(G)$  is called

- a **super-special point** if  $x$  is special (20.7.2.1) over any finite unramified extension  $K'/K \in \text{Field}$  (20.7.1.1). (By (20.7.2.3), this is equivalent to  $x$  special for  $\mathcal{B}(G_{K^{\text{sh}}})$ , Zhu called these very special points)
- an **absolutely special point** if  $x$  is special over any finite separable extension  $K'/K$ .
- a **hyperspecial point** if  $x$  is super-special and  $G$  splits over  $K^{\text{sh}}$ .

In particular, if  $G$  splits over  $K^{\text{sh}}$ , super-special, absolutely special and hyperspecial are equivalent.  $\textcolor{red}{?}$ .  $\lrcorner$

**Prop. (20.7.2.5) [Absolutely Special Points and Chevalley-Valuations].** Assume  $G$  is quasi-split. Let  $T \leq G$  is a maximally-split maximal torus (10.4.1.15) containing a maximal split torus  $S \leq G$  and  $\mathcal{A}$  the corresponding apartment, and  $o \in \mathcal{A}(S)$  is a Chevalley-valuation (20.6.4.23). Then for a point  $x = a + v \in \mathcal{A}(S)$ , the following are equivalent:

- $x$  is an absolutely-special point (20.7.2.4).
- $\alpha(v) \in v_\alpha \mathbb{Z}$  for all  $\alpha \in \Phi(G, T)$   $\textcolor{red}{?}$
- $x = t.o$  for some  $t \in T_{\text{ad}}(K)$ .
- $x$  is a Chevalley-valuation.

In particular, absolutely special points exist.  $\lrcorner$

*Proof:* Cf. [K-P23]P279.  $\square$

**Cor. (20.7.2.6) [Absolutely Special Points are Conjugate].** If  $G$  is quasi-split and adjoint (10.4.2.1), then any two absolutely special points are conjugate under action of  $G(K)$ .  $\lrcorner$

*Proof:* Cf. [K-P23]P280. ?

□

**Example (20.7.2.7) [Special Points of  $\mathcal{B}(\mathrm{SU}(3))$ ].** Cf. [K-P23]P280.

┘

**Prop. (20.7.2.8) [Super Special Points imply Quasi-Splitness].** Suppose  $\kappa$  is a cd-1 field (9.8.3.18). If  $x \in \mathcal{B}(G)$  is special in  $\mathcal{B}(G_{K^{\mathrm{sh}}})$ , then  $G$  is quasi-split. ┘

*Proof:* Let  $T \leq G$  be a special torus containing a maximal split torus  $S \leq G$  s.t.  $x \in \mathcal{A}(T) \subset \mathcal{B}(G_{K^{\mathrm{sh}}})$ . Let  $\mathbf{S} \leq \mathbf{T}$  be the tori of  $\mathbf{G}_x \in \mathrm{RedGrp}/\kappa$  corresponding to  $S \subset T$ , then  $\mathbf{S}$  is a maximal split torus and  $\mathbf{T}$  is a maximal torus by (20.6.4.20).

By Steinberg's theorem (10.5.2.3),  $\mathbf{G}_x$  is quasi-split, thus the restriction map  $\Phi((\mathbf{G}_x)_{\overline{\kappa}}, \mathbf{T}_{\overline{\kappa}}) \rightarrow \Phi(\mathbf{G}_x, \mathbf{S})$  is surjective. By the definition of specialness, for each  $\beta \in \Phi(G_{K^{\mathrm{sh}}}, T_{K^{\mathrm{sh}}})$ , some constant multiple of  $\beta$  belongs to  $\Phi((\mathbf{G}_x)_{\overline{\kappa}}, \mathbf{T}_{\overline{\kappa}})$ , thus  $\beta$  has a non-trivial restriction to  $S$ . This implies that  $Z_G(S) = Z_G(T)$ , which is a torus because  $G_{K^{\mathrm{sh}}}$  is quasi-split by Steinberg's theorem (10.5.2.3) and (10.4.4.19). So  $G$  is also quasi-split by (10.4.4.19). □

**Prop. (20.7.2.9) [Special Points and Absolutely-Special Points].** If  $G$  is quasi-split and  $\Phi(G)$  is reduced, then every special vertex is absolutely special. ┘

*Proof:* Cf. [K-P23]P280. ?

□

### Hyperspecial Points

**Thm. (20.7.2.10).** If  $\mathcal{G} \in \mathrm{RedGrp}/\mathcal{O}_K$  with  $\mathcal{G}_K \cong G$ , then

- $G$  splits over  $K^{\mathrm{sh}}$ .
- There exists a unique point  $x \in \mathcal{B}(G_{K^{\mathrm{sh}}})$  fixed by  $\mathcal{G}(\mathcal{O}_{K^{\mathrm{sh}}})$ , and it is a special point.
- There is a unique isomorphism  $\mathcal{G}_{\mathcal{O}_{K^{\mathrm{sh}}}} \rightarrow \mathcal{G}_x^0$  (20.6.4.34) extending the identity homomorphism on the generic fiber.

┘

*Proof:* Cf. [K-P23]P301. ?

□

**Cor. (20.7.2.11) [Reductive Integral Models and Hyperspecial Points].** For  $x \in \mathcal{B}(G_{K^{\mathrm{sh}}})$ ,  $\mathcal{G}_x^0 \in \mathrm{RedGrp}/\mathcal{O}_K$  iff  $x$  is a hyperspecial vertex. And any  $\mathcal{G} \in \mathrm{RedGrp}/\mathcal{O}_K$  with  $\mathcal{G}_K \cong G$  is of the form  $\mathcal{G}_x^0$  for some hyperspecial vertex  $x \in \mathcal{B}(G)$ . ┘

*Proof:* If  $\mathcal{G}_x^0$  is reductive, then  $(\mathcal{G}_x^0)_{\mathcal{O}_{K^{\mathrm{sh}}}}$  is also reductive, then  $G$  splits over  $K^{\mathrm{sh}}$  by (20.7.2.10), and,  $x$  is special for  $\mathcal{B}(G_{K^{\mathrm{sh}}})$  by (20.7.2.10), which implies  $x$  is hyperspecial for  $G$  by (20.7.2.3). Conversely, if  $x$  is hyperspecial, then this is [K-P23]Prop8.4.14. ?

And if  $\mathcal{G} \in \mathrm{RedGrp}/\mathcal{O}_K$  with  $\mathcal{G}_K \cong G$ , then by (20.7.2.10)  $G$  splits over  $K^{\mathrm{sh}}$ , and the unique point  $x \in \mathcal{B}(G_{K^{\mathrm{sh}}})$  fixed by  $\mathcal{G}(\mathcal{O}_{K^{\mathrm{sh}}})$  is contained in  $\mathcal{B}(G)$  by uniqueness, thus hyperspecial for  $G$  by (20.7.2.3). And the unique isomorphism  $\mathcal{G}_{\mathcal{O}_{K^{\mathrm{sh}}}} \rightarrow \mathcal{G}_x^0$  descends to an isomorphism  $\mathcal{G} \rightarrow \mathcal{G}_x^0$  by uniqueness. □

**Cor. (20.7.2.12) [Hyperspecial Parahorics are Conjugate].** Suppose  $\kappa$  is a cd-1 field (9.8.3.18). Hyperspecial points of  $\mathcal{B}(G)$  are conjugate to each other under action of  $G_{\mathrm{ad}}(K)$ , thus so are hyperspecial parahoric subgroups of  $G(K)$ . ┘

*Proof:* Notice  $\mathcal{B}(G) \rightarrow \mathcal{B}(G_{\mathrm{ad}})$  is an isomorphism inducing isomorphism of affine root systems (20.6.4.8), so the assertion follows from (20.7.2.4) (20.7.2.8) and (20.7.2.6). □

**Cor. (20.7.2.13)[Characterizing Hyperspecial Points].** If  $G$  is quasi-split and splits over  $K^{\text{sh}}$ , then  $G$  admits reductive integral models and hyperspecial points, by (20.7.2.4)(20.7.2.11) and (20.7.2.5). And the converse is true if  $\kappa$  is a cd-1 field, by (20.7.2.8), in which case all hyperspecial points are conjugate under  $G_{\text{ad}}(K)$  (20.7.2.12).  $\lrcorner$

### Cartan-Decompositions

**Thm. (20.7.2.14)[Cartan-Decomposition, Tits[Tit79]].** Let  $S \leq G$  be a maximal split torus,  $W = W(G, S)$ , and  $\Phi^+ \subset \Phi(G, S)$  a positive system of roots. Define

$$\mathcal{Z} = \{z \in Z_G(S)(K) \mid \alpha(\omega_{Z_G(S)}(z)) \geq 0, \forall \alpha \in \Phi^+\} \text{ (20.5.2.2),}$$

and let  $x \in \mathcal{A}(S)$  be a special vertex with corresponding parahoric subgroup  $\mathcal{P}_x$  (20.7.2.1), then

- $G(K) = \mathcal{P}_x \mathcal{Z} \mathcal{P}_x$ .
- For  $z, z' \in \mathcal{Z}$ ,  $\mathcal{P}_x z \mathcal{P}_x = \mathcal{P}_x z' \mathcal{P}_x$  iff  $z^{-1}z' \in Z(K)^0$ .

In particular,

$$\mathcal{H}(G(K), \mathcal{P}_x) \cong \mathbb{Z}[Z_G(S)(F)/Z_G(S)(F)^0]^W \xrightarrow{\kappa_{Z_G(S)}, \cong} \mathbb{Z} \left[ \left( X^\bullet \left( Z(\widehat{M}) \right)_{\text{Gal}_{\bar{K}}} \right)^{\text{Gal}_{\kappa}} \right].$$

In particular, if  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_K$ ,  $G = \mathcal{G}_K$ . If  $S \leq \mathcal{G}$  is a maximal split torus and  $B \leq G$  is a Borel subgroup containing  $S = \mathcal{S}_K$ , then

$$G(K) = \coprod_{\lambda \in P^+(\Phi(G, S))} \mathcal{G}(\mathcal{O}_K) \lambda(\varpi) \mathcal{G}(\mathcal{O}_K).$$

$\lrcorner$

*Proof:* 1: Let  $C \subset \mathcal{A}(S)$  be a chamber containing  $x$ , and  $\mathcal{I}$  the corresponding Iwahori subgroup. Because  $x$  is special,  $N_G(S)(K) \cap \mathcal{P}_x \rightarrow W(G, S)$  is surjective by (20.6.4.16) and (3.9.8.23) ?. Thus  $\mathcal{I} \mathcal{N}_x \subset \mathcal{P}_x Z_G(S)(K)$ , and the affine Bruhat-decomposition (20.7.1.4) implies that  $G(K) = \mathcal{P}_x Z_G(S)(K) \mathcal{I}$ . The fact that  $N_G(S)(K) \cap \mathcal{P}_x \rightarrow W(G, S)$  is surjective also implies that  $Z_G(S)(K) \subset \cup_{n \in N_G(S)(K) \cap \mathcal{P}_x} n \mathcal{Z} n^{-1}$ , so the assertion follows.

2: One direction is clear because  $Z_G(S)(K)^0 \subset \mathcal{P}_x$  (20.6.4.10). Conversely, if  $\mathcal{P}_x z \mathcal{P}_x = \mathcal{P}_x z' \mathcal{P}_x$ , then  $z^{-1}z' \in G(K)^0$ , and by (20.5.2.22), it suffices to show that  $z^{-1}z' \in Z(K)^1$ . By assumption there exists  $g, g' \in \mathcal{P}_x$  s.t.  $gz = z'g'$ . Thus  $x \in \mathcal{A}(S) \cap g\mathcal{A}(S)$ , and  $z'x = gzx \in \mathcal{A}(S) \cap g\mathcal{A}(S)$ . Thus it follows from (20.6.6.15) that there exists  $g_0 \in \mathcal{P}_x \cap z' \mathcal{P}_x (z')^{-1}$  s.t.  $g\mathcal{A}(S) = g_0\mathcal{A}(S)$ . So if we denote  $n_x = N_G(S)(K) \cap \mathcal{P}_x$ , we have  $g_0^{-1}g \in n_x$  by (20.6.4.8) and  $(z')^{-1}g_0z' \in \mathcal{P}_x$ . Thus

$$h = g_0^{-1}gz = g_0^{-1}z'g' = z'((z')^{-1}g_0^{-1}z')g' \in n_x z \cap z' \mathcal{P}_x = n_x z \cap z' n_x.$$

So we see there exists  $n_1, n_2 \in N_G(S)(K)$  s.t.  $z' = n_1 z n_2$ . Denote  $z'' = n_2^{-1} z n_2 \in Z_G(S)(K)$ , then  $z_0 = z'(z'')^{-1} = n_1 n_2 \in Z_G(S)(K) \cap n_x \leq Z_G(S)(K)^1$  (20.5.2.7). Thus

$$\omega_{Z_G(S)}(z), \omega_{Z_G(S)}(z'), \omega_{Z_G(S)}(z'') \in \{v \in X_\bullet(S) \otimes \mathbb{Q} \mid \alpha(v) \geq 0, \forall \alpha \in \Phi^+\}.$$

This is a fundamental domain for the action of  $W(G, S)$ , and  $z'' = n_2^{-1} z n_2$ , so there exists  $c \in Z_G(S)(K)^1$  s.t.  $z'' = cz$ , and  $z_0 = z'(z'')^{-1} = z'z^{-1}c^{-1}$ , and  $z'z^{-1} = z_0 c \in Z_G(S)(K)^1$ .

For the second assertion, Cf.[Satake Isomorphisms for Special Maximal Parahoric Hecke Algebras] ?



If  $G$  is unramified, then  $T = Z_G(S)$  is a maximal torus that splits over  $K^{\text{sh}}$ , so by (20.5.2.20),

$$\omega_G : T(K)/T(K)^0 \cong X_*(T), \quad \mathcal{Z}/T(K)^0 \cong P^+(\Phi(G, S)),$$

and a set-theoretical inverse of  $\omega_G$  is given by  $\lambda \mapsto \lambda(\varpi)$ .  $\square$

**Cor. (20.7.2.15) [ $G(K)$  is  $\sigma$ -Compact].**  $G(K)$  is  $\sigma$ -compact, by (20.7.2.5).  $\lrcorner$

### Iwasawa-Decompositions

**Thm. (20.7.2.16) [Iwasawa-Decomposition].** Let  $S \leq G$  be a maximal split torus, and  $\Phi^+ \subset \Phi(G, S)$  a system of positive roots with the corresponding unipotent radical  $U \leq G$ . Let  $C \subset \mathcal{A}(S)$  be a chamber and  $\mathcal{I}$  the corresponding Iwahori subgroup, then

$$G(K) = U(K) \cdot N_G(S)(K) \mathcal{I}.$$

$\lrcorner$

*Proof:* Cf. [K-P23] P189?  $\square$

**Thm. (20.7.2.17) [Iwasawa-Decomposition].** If  $x \in \mathcal{B}(G)$  is a special vertex with the corresponding parahoric subgroup  $\mathcal{P}_x \leq G(K)$ , and  $P \leq G$  is a parabolic subgroup, then  $G(K) = P(K) \mathcal{P}_x$ . In particular, when  $\#\kappa < \infty$ ,  $G(K)/P(K)$  is compact.  $\lrcorner$

*Proof:* Let  $C$  be a chamber containing  $x$  and let  $S \leq G$  be a maximal split torus s.t.  $C \subset \mathcal{A}(S)$ . Let  $\Phi^+ \subset \Phi(G, S)$  be a system of positive roots with the corresponding unipotent radical  $U \leq G$ .

Because  $x$  is special,  $N_G(S)(K) \cap \mathcal{P}_x \rightarrow W(G, S)$  is surjective by (20.6.4.16) and (3.9.8.23), thus  $Z_G(S)(K) \mathcal{P}_x = N_G(S)(K) \mathcal{P}_x$ . Then it follows from (20.7.2.16) that  $G(K) = U(K) N_G(S)(K) \mathcal{P}_C = U(K) Z_G(S)(K) \mathcal{P}_x$ .

As  $U Z_G(S)$  is a minimal parabolic subgroup of  $G$ , by (10.4.1.6) there exists  $g \in G(K)$  s.t.  $g^{-1} U Z_G(S) \subset P$ . Take  $h \in U(K) Z_G(K)$  s.t.  $hg \in \mathcal{P}_x$ , then

$$\begin{aligned} P(K) \mathcal{P}_x &\supset g^{-1} (U(K) Z_G(S)(K)) g \mathcal{P}_x \\ &= (hg)^{-1} (U(K) Z_G(S)(K)) (hg) \mathcal{P}_x \\ &= (hg)^{-1} (U(K) Z_G(S)(K)) \mathcal{P}_x (hg) \\ &= G(K). \end{aligned}$$

$\square$

**Cor. (20.7.2.18).** Let  $S \leq G$  be a maximal split torus, and  $\Phi^+ \subset \Phi(G, S)$  a system of positive roots with the corresponding unipotent radical  $U \leq G$ . If  $\mathcal{K} \subset G(K)^0$  is a bounded subgroup containing  $Z(K)^0$  and  $z, z' \in Z_G(S)(K)$  s.t.  $U(K) z \mathcal{K} = U(K) z' \mathcal{K}$ , then  $z' = Z_G(S)(K)^0 z$ .  $\lrcorner$

*Proof:* As  $Z_G(S)$  normalizes  $U$ , there exists  $u \in U(K)$  s.t.  $uz^{-1}z' \in \mathcal{K}$ . Then  $uz^{-1}z'$  lies in a bounded subgroup  $\mathcal{K} \cap (U \rtimes Z)(K)$ , and then its image  $z^{-1}z'$  under the natural map  $U \rtimes Z \rightarrow Z$  lies in  $Z(K)^1$ . But  $z^{-1}z' \in U \mathcal{P} \subset G(K)^0$ ?, hence  $z^{-1}z' \in Z(K)^1 \cap G(K)^0 = Z(K)^0$  (20.5.2.22).  $\square$

**Thm. (20.7.2.19) [Iwasawa-Decompositions].** Suppose  $F \in \mathbf{GField}$ ,  $G \in \mathbf{RedGrp}/F$ ,  $P \leq G$  is a parabolic subgroup with unipotent radical  $U$  and a Levi factor  $L$ , then there exists a maximal compact subgroup  $\mathcal{K} = \prod_{v \in \Sigma_F} \mathcal{K}_v \leq G(\mathbf{A}_F)$  s.t. for any  $v \in \Sigma_F$ , we have

$$G(F_v) = P(F_v) \mathcal{K}_v, \quad P(F_v) \cap \mathcal{K}_v = (L(F_v) \cap \mathcal{K}_v)(U(F_v) \cap \mathcal{K}_v).$$

$\lrcorner$

*Proof:* Cf. [Getz] P541.  $\square$



### Intersection of Cartan and Iwasawa Double Cosets

See [K-P23]Chap5.4.

## 3 Buildings for Classical Groups

References are [B-T84a, B-T87a].

$\mathrm{SL}(n)$

**Def. (20.7.3.1) [Splitting Bases].** Let  $X \in \mathcal{V}\mathrm{ect}^{\mathrm{fin}}/K$  and  $\alpha : X \rightarrow \mathbb{R} \cup \{\infty\}$  be an additive norm on  $X$  (19.1.2.1), then

- $\alpha$  is said to be **split** over a direct-sum decomposition  $X = X_1 \oplus \dots \oplus X_r$  if  $\alpha(\sum_{i=1}^r v_i) = \min\{\alpha(v_1), \dots, \alpha(v_r)\}$  for  $v_i \in X_i$ .
- A basis  $\{v_1, \dots, v_n\}$  of  $X$  is called a **splitting basis** for  $\alpha$  if  $\alpha$  splits over the direct-sum decomposition  $X = K.v_1 \oplus \dots \oplus K.v_n$ .
- $\alpha$  is called a **splitting additive norm** if it admits a splitting basis.

┘

**Prop. (20.7.3.2).** Situation as in (20.7.3.1), if  $\{v_1, \dots, v_r\}$  is a subset of  $X \setminus \{0\}$  s.t.  $\alpha(v_i) \nmid \alpha(v_j) \pmod{\mathbb{Z}}$ , then  $v_1, \dots, v_r$  are linearly independent, and  $v_1, \dots, v_r$  is splitting basis for  $\alpha|_{K\{v_1, \dots, v_r\}}$ .

In particular,  $\alpha(X \setminus \{0\})$  is a closed and discrete subset of  $\mathbb{R}$ .

┘

*Proof:* This follows from ultrametric inequality. □

**Prop. (20.7.3.3).** Situation as in (20.7.3.1),

- $\alpha$  is splitting iff  $L_{\alpha,0} = \{v \in X | \alpha(v) \geq 0\}$  is a lattice in  $X$ .
- If  $\alpha$  is splitting and  $Y \leq X$  is a subspace, then  $\alpha|_Y$  is also splitting, and there exists a complement  $Y'$  of  $Y$  in  $X$  s.t.  $\alpha$  splits over the direct-sum decomposition  $X = Y \oplus Y'$ .

┘

*Proof:* If  $\alpha$  is a splitting additive norm, then it is clear that  $L_{\alpha,0}$  is a lattice. We prove the converse of this and item2 together by inducting on  $\dim X$ .

If  $\dim X = 1$  this is clear. If  $\alpha$  is splitting, then  $L_{\alpha,0}$  is a lattice in  $X$ , so  $L_{\alpha,0} \cap Y$  is a lattice in  $Y$ , and  $\alpha|_Y$  is splitting by induction hypothesis.

Now if  $L_{\alpha,0}$  is a lattice and  $Y \subset X$  is a subspace, we find a complement  $Y'$  of  $Y$  in  $X$  s.t.  $\alpha$  splits over the direct-sum decomposition  $X = Y \oplus Y'$ . An easy induction shows it suffices to prove for  $\dim Y = \dim X - 1$ . In this case, take  $x_0 \in X \setminus Y$ , then there exists some  $n \in \mathbb{Z}_+$  s.t.  $(x_0 + \varpi^n L_{\alpha,0}) \cap Y = \emptyset$ . Thus  $\alpha(x_0 + Y) \subset \mathbb{R}$  is bounded above. But it is a closed discrete set by (20.7.3.2), so there exists a  $x \in x_0 + Y$  s.t.  $\alpha(x) \geq \alpha(x + y)$  for any  $y \in Y$ . Thus  $\alpha(x + y) = \min\{\alpha(x), \alpha(y)\}$  for any  $y \in Y$ . Then we can just take  $Y' = K.x$ .

So if  $L_{\alpha,0}$  is a lattice, we take a proper subspace  $Y \subset X$ , and find a complement  $Y'$  of  $Y$  in  $X$  s.t.  $\alpha$  splits over the direct-sum decomposition  $X = Y \oplus Y'$ , then  $L_{\alpha,0} \cap Y, L_{\alpha,0} \cap Y'$  are lattices in  $Y$  and  $Y'$  resp., so by hypothesis  $\alpha|_Y$  and  $\alpha|_{Y'}$  are splitting, so  $\alpha$  is also splitting. □

**Prop. (20.7.3.4).** Situation as in (20.7.3.1), if  $K = \widehat{K}$ , then  $\alpha$  is split. ┘

*Proof:* Cf. [K-P23]P516. □

**Def. (20.7.3.5) [Periodic Lattice Chains].** For  $X \in \mathcal{V}\text{ect}^{\text{fin}}/K$ , a **periodic lattice chain** in  $X$  is a totally-ordered set of lattices in  $X$  s.t. if  $L \in \mathcal{L}$ , then  $x.L \in \mathcal{L}$  for ally  $x \in K$ . Any period chain of lattices  $\mathcal{L}$  is of the form

$$\dots \subset L_0 \subset L_1 \subset \dots \subset L_r \subset \varpi_K^{-1}L_0 \subset \varpi_K^{-1}L_1 \subset \dots,$$

and the **rank** of  $\mathcal{L}$  is defined to be  $r + 1$ .

And if  $\mathcal{L}$  is a periodic chain, then a **grading** on  $\mathcal{L}$  is a inclusion-decreasing map  $c : \mathcal{L} \rightarrow \mathbb{R}$  s.t.  $c(\varpi_K L) = c(L) + 1$ . ┘

**Def. (20.7.3.6) [Adapted Bases].** For  $X \in \mathcal{V}\text{ect}^{\text{fin}}/K$  and  $\mathcal{L}$  a periodic chain of lattice in  $X$ , a basis  $(v_1, \dots, v_n)$  of  $X$  is said to be **adapted to  $\mathcal{L}$**  if for any  $L \in \mathcal{L}$ , there exists  $x_1, \dots, x_n \in K^\times$  s.t.  $(x_1 v_1, \dots, x_n v_n)$  is a basis of  $L$ . ┘

**Prop. (20.7.3.7).** For  $X \in \mathcal{V}\text{ect}^{\text{fin}}/K$ , every periodic lattice chain in  $X$  has a basis adapted to it. ┘

*Proof:* For a periodic lattice chain  $\mathcal{L}$ :

$$\dots \subset L_0 \subset L_1 \subset \dots \subset L_r \subset \varpi_K^{-1}L_0 \subset \varpi_K^{-1}L_1 \subset \dots,$$

denote  $\bar{L}_i = L_i/L_0$ , then there is a basis  $(\bar{v}_1, \dots, \bar{v}_n)$  adapted the flat  $\bar{L}_1 \subset \bar{L}_2 \subset \dots \subset \bar{L}_0$  in the sense that for each  $i \in [r]$  there exists  $k_i \in [n]$  s.t.  $\bar{L}_i = \kappa v_1 \oplus \dots \oplus \kappa v_{k_i}$ . Then any liftings of  $(v_1, \dots, v_n)$  is a basis adapted to  $L_0$  by Nakayama's lemma, and

$$L_i = \mathcal{O}_K v_1 \oplus \dots \oplus \mathcal{O}_K v_{k_i} \oplus \varpi_K^{-1} v_{k_{i+1}} \oplus \dots \oplus \varpi_K^{-1} v_n.$$

□

**Prop. (20.7.3.8) [Splitting Additive Norms and Periodic Lattice Chains].** Let  $X \in \mathcal{V}\text{ect}^{\text{fin}}/K$ . If  $\alpha$  is a splitting additive norm on  $X$ , for each  $r \in \mathbb{R}_+$ , define  $L_{\alpha,r} = \{v \in X | \alpha(v) \geq r\}$ , then  $L_{\alpha,r}$  is a lattice in  $X$  by (20.7.3.3), and  $\mathcal{L}_\alpha = \{L_{\alpha,r}\}_{r \in \mathbb{R}}$  is a periodic lattice chain (20.7.3.5), with grading  $c(L) = \inf_{v \in L} \alpha(v)$ . And if  $(\mathcal{L}, c)$  is a period lattice chain with a grading (20.7.3.5), then  $\alpha_{\mathcal{L}}(v) = \sup_{L \in \mathcal{L}, v \in L} c(L)$  is a splitting additive norm on  $X$  by (20.7.3.3).

The above construction gives mutually inverse bijections between the set of splitting additive norms (20.7.3.1) on  $X$  and the set of periodic lattice chains of  $X$  (20.7.3.5). And a basis of  $X$  is a splitting basis for an additive norm  $\alpha$  iff it is adapted to  $\mathcal{L}_\alpha$  (20.7.3.6). ┘

*Proof:* It is easy to see that these two maps are mutually inverse to each other. And the last assertion is also easy. □

**Prop. (20.7.3.9) [ $\mathcal{B}(\text{PGL}(n))$ ].** Let  $X \in \mathcal{V}\text{ect}^{\text{fin}}/K$ .

- If  $X = \ell_1 \oplus \dots \oplus \ell_n$  be a direct sum decomposition, define  $\tilde{\mathcal{A}}(\ell_1, \dots, \ell_n)$  to be the set of all additive norms that splits over this decomposition (20.7.3.1), then it is naturally an affine  $\mathbb{R}^n$ -torsor.
- Let  $\tilde{\mathcal{B}}(X)$  be the set of all splitting additive norms, then  $\mathbb{R}$ -acts on  $\tilde{\mathcal{B}}(X)$  by  $(\alpha + r)(x) = \alpha(x) + r$ . This action preserves all  $\tilde{\mathcal{A}}(\ell_1, \dots, \ell_n) \subset \tilde{\mathcal{B}}(X)$ , and we denote  $\mathcal{A}(\ell_1, \dots, \ell_n) = \tilde{\mathcal{A}}(\ell_1, \dots, \ell_n)/\mathbb{R}$  and  $\mathcal{B}(X) = \tilde{\mathcal{B}}(X)/\mathbb{R}$ .

- For each periodic lattice chain  $\mathcal{L}$  of  $X$ , define  $\Delta_{\mathcal{L}} \subset \mathcal{B}(X)$  the set of all additive norms  $\alpha \in \mathcal{B}(X)$  s.t.  $\mathcal{L}_{\alpha} = \mathcal{L}$  (20.7.3.8). For any splitting  $X = \ell_1 \oplus \dots \oplus \ell_n$  adapted to  $\mathcal{L}$  (20.7.3.6), which is the interior of a simplex of dimension  $\text{rank } \mathcal{L} - 1$  in each  $\mathcal{A}(\ell_1, \dots, \ell_n)$  s.t.  $\alpha_{\mathcal{L}}$  splits over the direct-sum decomposition  $X = \ell_1 \oplus \dots \oplus \ell_n$  (20.7.3.1). In this way, we defined a simplicial structure on  $\mathcal{B}(X)$  of dimension  $n - 1$ .
- In this way, each lattice in  $X$  corresponds to a vertex of  $\mathcal{B}(X)$ , and vertices corresponding to a set of lattices  $\{L_0, \dots, L_r\}$  constitute a simplex of dimension  $r$  iff up to permutation we have

$$L_0 \subset L_1 \subset \dots \subset L_r \subset \varpi_K^{-1} L_0.$$

- $\text{GL}(X)$  acts on  $\widetilde{\mathcal{B}(X)}$  by  $(g.\alpha)(x) = \alpha(g^{-1}x)$ . This induces an action of  $\text{PGL}(X)$  on  $\mathcal{B}(X)$  that preserves the simplicial structures.
- There is a canonical  $\text{PGL}(X)$ -equivariant isomorphism  $\mathcal{B}(X) \cong \mathcal{B}(\text{PGL}(X))$ , because we can compare stabilizers corresponding to each point. (Cf. [K-P23]P523?). And there is a non-canonical  $\text{GL}(X)$ -equivariant isomorphism  $\widetilde{\mathcal{B}}(X) \cong \widetilde{\mathcal{B}}(\text{GL}(X))$ .

┘

**Cor. (20.7.3.10).** If  $K \in \text{LField}$ , then

- $\text{GL}(n; \mathcal{O}_K)$  is the unique maximal compact subgroup of  $\text{GL}(n; K)$  up to conjugacy.
- $\text{SL}(n; K)$  has exactly  $n$ -many maximal compact subgroups up to conjugacy:

$$\text{SL}(2; \mathbb{Z}_p), \quad \begin{bmatrix} p & \\ & 1 \end{bmatrix}^{-1} \text{SL}(2; \mathbb{Z}_p) \begin{bmatrix} p & \\ & 1 \end{bmatrix}, \dots, \begin{bmatrix} p^{n-1} & \\ & 1 \end{bmatrix}^{-1} \text{SL}(2; \mathbb{Z}_p) \begin{bmatrix} p^{n-1} & \\ & 1 \end{bmatrix}$$

┘

*Proof:* This follows from (20.6.6.20)(20.7.2.13) and the fact these groups are clearly non-conjugate to each other.  $\square$

**Example (20.7.3.11) [Spherical Bruhat-Tits Buildings for  $\text{GL}(n+1; K)$ ].** For  $n \in \mathbb{Z}$ ,  $\mathcal{B}(\text{GL}(n+1; K))$  (20.6.4.28) is described as follows:

- The vertices of  $\mathcal{B}(\text{GL}(n+1; K))$  are proper subspaces of  $K^{\oplus(n+1)}$ .
- vertices  $x_1, \dots, x_r$  corresponding to subspaces  $V_1, \dots, V_r$  of  $K^{\oplus(n+1)}$  form a simplex of  $\mathcal{B}(\text{GL}(n+1; K))$  iff there exists a permutation  $\sigma \in \text{Sym}_r$  s.t.

$$V_{\sigma(1)} \subset V_{\sigma(2)} \subset \dots \subset V_{\sigma(r)}.$$

- For any unordered set  $S$  of 1-dimensional subspaces of  $K^{\oplus(n+1)}$  of cardinality  $n+1$ , there is an apartment of  $\mathcal{B}(\text{GL}(n+1; K))$  which is the chamber subcomplex spanned by vertices corresponding to a subspace  $V$  that is spanned by some subset of  $S$ .

Thus each apartment in  $\mathcal{B}(\text{GL}(n+1; K))$  is a simplicial  $n$ -sphere.  $\square$

*Proof:* Proof that  $\mathcal{B}(\text{GL}(n; K))$  is a building: it is clearly thick. And any apartment is clearly thin, and any two chambers are connected by a gallery, thus a chamber complex.

To prove that any two apartment is contained in a common apartment, Cf. [Garrett, Buildings, Bruhat decompositions, unramified principal series]P8.

To prove that for any two apartments  $a, a'$  in a building  $X$  (20.6.2.1) containing a common chamber  $C$ , there exists a simplicial isomorphism  $f : a \cong a'$  fixing vertices of  $a \cap a'$ , Cf. [Garrett, Buildings, Bruhat decompositions, unramified principal series]P9.  $\square$

**Example (20.7.3.12)**[Spherical Buildings for  $\mathrm{PGL}(n; K)$ ].  $\mathcal{B}(\mathrm{PGL}(n+1; K))$  (20.6.4.28) is described as follows:

- The vertices of  $\mathcal{B}(\mathrm{PGL}(n+1; K))$  are given by homothety classes of  $\mathcal{O}_K$ -lattices in  $K^{n+1}$  (19.3.1.28).
- A collection  $\{\Lambda_0, \dots, \Lambda_s\}$  is a  $s$ -simplex of  $\mathcal{B}(\mathrm{PGL}(n+1; K))$  if there exists  $M_i$  in the homothety class of  $\Lambda_i$  s.t. up to permutation, we have

$$M_0 \supsetneq M_1 \supsetneq \dots \supsetneq M_s \supsetneq \varpi_K M_0.$$

In particular, for any  $\mathcal{O}_K$ -lattice  $M$  in  $K^{n+1}$ , the set of  $s$ -simplexes of  $\mathcal{B}(\mathrm{PGL}(n+1; K))$  containing  $[M]$  is in bijection with the set of length- $(s+1)$ -flags in  $M/\varpi_K M$ .  $\lrcorner$

**Example (20.7.3.13)**[ $\mathrm{SL}(n; \mathbb{Q}_p)$ ]. The subgroup

$$\{(x_{ij}) \in \mathrm{SL}(n; \mathbb{Z}_p) : x_{ii} \equiv 1 \pmod{p}, x_{ij} \equiv 0 \pmod{p} \forall i < j\}$$

is a pro- $p$ -Sylow subgroup of  $\mathrm{SL}(n; \mathbb{Q}_p)$ , with corresponding Iwahori subgroup

$$B = \{(x_{ij}) \in \mathrm{SL}(n; \mathbb{Z}_p) : x_{ij} \equiv 0 \pmod{p} \forall i < j\}.$$

Let  $S \leq \mathrm{SL}(n; \mathbb{Q}_p)$  be the diagonal torus and  $N = N_G(S)$ , then  $(B, N)$  is a Tits system for  $\mathrm{SL}(n; \mathbb{Q}_p)$  (20.6.1.1).

A standard set  $R$  of generators of the Weyl group  $W$  is represented by the following matrices:

$$r_1 = \mathrm{diag}\left(\begin{bmatrix} & 1 \\ -1 & \end{bmatrix}, 1, \dots, 1\right), \quad r_2 = \mathrm{diag}\left(1, \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}, 1, \dots, 1\right), \dots, r_n = \begin{bmatrix} 0 & 0 & p^{-1} \\ 0 & \mathbf{1}_{n-1} & 0 \\ p & 0 & 0 \end{bmatrix}.$$

Let  $W_0 = \langle r_1, \dots, r_{n-1} \rangle$ , then  $P = BW_0B = \mathrm{SL}(n; \mathbb{Z}_p)$ . The other maximal compact subgroups  $P_i$  are equal to  $BW_iB$  where  $W_i = \langle r_1, \dots, \hat{r}_i, \dots, r_{n-1}, r_n \rangle$ , which is the stabilizer of the lattice  $\{e_1, \dots, e_{n-i}, pe_{n-i+1}, \dots, pe_n\}$ .

$W = W_0 \rtimes T$  where  $W_0 \cong S_n$  and  $T = S \cap \mathrm{SL}(n; p^{\mathbb{Z}})$ . Moreover,  $T^+ = \{(x_1, \dots, x_n) \in T : x_1 \leq x_2 \leq \dots \leq x_n\}$ .  $\lrcorner$

### $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$

**Notation (20.7.3.14).**

- Suppose  $\mathrm{char} \kappa \neq 2$ .
- Let  $\sigma$  be an involution on  $K$  with  $K_0 = K^\sigma$ .
- Let  $\epsilon \in \{\pm 1\}$ .
- Let  $(X, \langle -, - \rangle) \in \mathcal{V}\mathrm{ect}^{\mathrm{fin}}/K$  be a  $\epsilon$ -Hermitian space.
- $\langle -, - \rangle$  induces an isomorphism  $X^\sigma \cong X^* : x \mapsto \langle -, x \rangle$ .

$\lrcorner$

**Def. (20.7.3.15)**[Dual Splitting Additive Norms]. For  $X \in \mathcal{V}\mathrm{ect}^{\mathrm{fin}}/K$  and a splitting additive norm  $\alpha$  on  $X$  (20.7.3.1), define  $\alpha^* : X^* \setminus \{0\} \rightarrow \mathbb{R}$ :

$$\alpha^*(\lambda) = \inf_{x \in X \setminus \{0\}} (\mathrm{ord}_K(\lambda(x)) - \alpha(x)) \in \mathbb{R} \setminus \{-\infty\}.$$

Then

- $\alpha^*(\lambda) \in \mathbb{R}$  for any  $\lambda \in X^* \setminus \{0\}$ ,
- If  $\{v_1, \dots, v_n\}$  is a splitting basis for  $\alpha$ , then  $\{v_1^*, \dots, v_n^*\}$  is a splitting basis for  $\alpha^*$ , and  $\alpha^*(v_i^*) = -\alpha(v_i)$ .
- Under the identification  $X^{**} = X$ , the equality  $\alpha^{**} = \alpha$  holds.

┘

*Proof:* If  $\{v_1, \dots, v_n\}$  is a splitting basis for  $\alpha$ , then for  $x = \sum_i x_i v_i \in X$ , we have

$$\text{ord}_K(\lambda(x)) \geq \min_i (\text{ord}_K(x_i) + \text{ord}_K(\lambda(v_i))), \quad \alpha(x) = \min_i (\text{ord}_K(x_i) + \alpha(v_i)).$$

so

$$\text{ord}_K(\lambda(x)) - \alpha(x) \geq \min_i (\text{ord}_K(\lambda(v_i)) - \alpha(v_i)).$$

Thus  $\alpha^*(\lambda) = \min_i (\text{ord}_K(\lambda(v_i)) - \alpha(v_i))$ . Taking  $\lambda = v_i^*$  implies that  $\alpha^*(v_i^*) = -\alpha(v_i)$ , which implies that  $\alpha^*(\lambda) = \min_i (\text{ord}_K(\lambda(v_i)) + \alpha^*(v_i^*))$ , and  $\{v_1^*, \dots, v_n^*\}$  is a splitting basis for  $\alpha^*$ .  $\square$

**Def. (20.7.3.16) [Self-Dual Additive Norms].** For a splitting additive norm  $\alpha$  on  $X$  (20.7.3.1), let  $\alpha^\vee$  be the splitting additive norm on  $X^\sigma$  induced by  $\alpha^*$  (20.7.3.15) and the isomorphism  $X^\sigma \cong X^*$  (20.7.3.14). Then  $\alpha$  is called a **self-dual splitting additive norm** if  $\alpha^\sigma = \alpha^\vee$ , i.e. for any  $y \in X \setminus \{0\}$ , we have

$$\alpha(y) = \inf_{x \in X \setminus \{0\}} (\text{ord}_K(\langle x, y \rangle) - \alpha(x)).$$

┘

**Cor. (20.7.3.17).** If  $\alpha$  is a self-dual splitting additive norm on  $X$  (20.7.3.16), then  $\alpha(x) + \alpha(y) \leq \text{ord}_K(\langle x, y \rangle)$  for any  $x, y \in X$ .  $\square$

**Prop. (20.7.3.18).** If  $X$  is anisotropic, then there is a unique self-dual splitting additive norm  $\alpha$  on  $X$  (20.7.3.16), which is  $\alpha(x) = \frac{1}{2} \text{ord}_K(\langle x, x \rangle)$ .  $\square$

*Proof:* Firstly, we prove that  $2 \text{ord}_K(\langle x, y \rangle) \geq \text{ord}_K(\langle x, x \rangle) + \text{ord}_K(\langle y, y \rangle)$ : ?

Clearly  $\alpha(x) = \frac{1}{2} \text{ord}_K(\langle x, x \rangle)$  is an additive norm on  $X$  because

$$\alpha(x+y) = \frac{1}{2} (\text{ord}_K(\langle x, x \rangle) + \text{ord}_K(\langle x, y \rangle) + \text{ord}_K(\langle y, x \rangle) + \text{ord}_K(\langle y, y \rangle)) \geq \min\{\alpha(x), \alpha(y)\},$$

and it is self-dual by what we proved ?. And if  $\beta$  is another self-dual splitting additive norm on  $X$ , then we have  $\beta \leq \alpha$  by (20.7.3.17), which implies that  $\beta^\vee \geq \alpha^\vee$ , thus  $\beta = \alpha$ .  $\square$

**Prop. (20.7.3.19) [SU(3)].** Let  $L/K \in p\text{-LField}$  be a separable quadratic extension. There are two conjugacy classes of maximal compact subgroups in  $\text{SU}(3, L/K)$ :

$$\begin{bmatrix} \mathcal{O}_L & \mathcal{O}_L & \mathcal{O}_L \\ \mathcal{O}_L & \mathcal{O}_L & \mathcal{O}_L \\ \mathcal{O}_L & \mathcal{O}_L & \mathcal{O}_L \end{bmatrix}, \quad \begin{bmatrix} \mathcal{O}_L & \mathfrak{m}_L & \mathfrak{m}_L \\ \mathcal{O}_L & \mathcal{O}_L & \mathfrak{m}_L \\ \mathfrak{m}_L^{-1} & \mathcal{O}_L & \mathcal{O}_L \end{bmatrix}$$

These two subgroups are not conjugate under action of  $\text{PU}(3; L/K)$ .  $\square$

$\mathrm{SO}(n)$

### Non-Quasi-Split Groups

## 4 Finite Group Actions and Tamely-Ramified Descent

**Thm. (20.7.4.1) [Galois Descent].** For any  $K_1/K \in \mathrm{GalField}$  contained in  $K^{\mathrm{sep}}$ ,

- There is a unique  $G(K)$ -invariant injections  $\varphi : |\mathcal{B}(G)| \hookrightarrow |\mathcal{B}(G_{K_1})|$  (20.6.4.29) has image contained in  $|\mathcal{B}(G_{K_1})|^{\mathrm{Gal}(K_1/K)}$  (20.6.4.9), s.t. the restriction of  $\varphi$  to any apartment of  $\mathcal{B}(G)$  is an affine mapping into an apartment of  $\mathcal{B}(G_{K_1})$ .

┘

*Proof:* ? Cf. [K-P23]Chap7.9.

□

**Remark (20.7.4.2).** If  $K_1/K$  is not tamely-ramified, it may happen that  $\varphi : |\mathcal{B}(G)| \hookrightarrow |\mathcal{B}(G_{K_1})|^{\mathrm{Gal}(K_1/K)}$  is not surjective.

┘

## 5 Moy-Prasad Filtrations

## 6 Compactifications

References are [Landvogt, A Compactification of the Bruhat-Tits Building].

**Def. (20.7.6.1).**

┘

## 7 Classifying Groups over Local Fields

References are [Tit79], [K-P23]Chap10.7 and [Gross, Parahorics].

**Notation (20.7.7.1).**

- Assume that  $\kappa$  is a cd-1 field.

┘

### Anisotropic Groups

**Thm. (20.7.7.2) [Kneser/Bruhat-Tits [BT87]].** If  $G$  is absolutely almost-simple and anisotropic, then it splits over  $\check{K}$ , and is of type  $A_n$  for some  $n \in \mathbb{Z}_+$ .

And if  $K \in p\text{-LField}$ , then  $G$  is of inner type  $A_n$  for some  $n \in \mathbb{Z}_+$ . If  $G$  is moreover simply-connected, then  $G$  is isomorphic to  $\mathrm{SL}_{1,D}$  for some division algebra over  $K$ .

┘

*Proof:* Cf. [K-P23]P391 ?.

□

**Lemma (20.7.7.3) [Existence of Anisotropic Maximal Tori].** If every finite extension of  $\kappa$  admits cyclic Galois-extensions of every finite degree (e.g.  $\#\kappa < \infty$ ), then every quasi-split semisimple  $\kappa$ -group has an anisotropic maximal torus.

┘

*Proof:* Cf. [K-P23]P396.

□

**Thm. (20.7.7.4) [Existence of Elliptic Maximal Tori].** Suppose every finite extension of  $\kappa$  admits cyclic Galois-extensions of every finite degree (e.g.  $\#\kappa < \infty$ ). If  $G$  is semisimple, then  $G$  contains an anisotropic maximal torus  $T$  which contains a subtorus  $S \leq T$  s.t.  $S_{\check{K}}$  is a maximal  $\check{K}$ -split torus of  $G_{\check{K}}$ .

In particular, any  $G \in \mathrm{RedGrp}/K$  admits an elliptic maximal torus (10.4.2.25).

┘

*Proof:* Cf. [K-P23]P395.

□

List of Central-Isogeny Classes

**Thm.(20.7.7.5) [List of Central-Isogeny Classes of Absolutely Quasi-Simple Groups, Kneser65/ Bruhat-Tits[Tit79]].** Let  $K \in \mathbf{LField}$ . where

- A name of the shape  ${}^aX_n$  means  $X$  represents the absolute Dynkin diagram  $D(\Psi(G_{K^{\mathrm{sh}}}))$  (3.9.8.26) and  $a$  is the order of the automorphism group of  $\Delta_1$  induced by  $\mathrm{Gal}(K^{\mathrm{sh}}/K)$ . If  $a = 1$  (i.e. residually-split), it is omitted, and  $n = \mathrm{rank}_{K^{\mathrm{sh}}}(G)$ .
- 

┘

*Proof:* ?

□

- $C - B_n$  is the ramified even orthogonal group.
- $B - C_n$  and  $C - BC_n$  are ramified unitary groups.

## 20.8 Geometry of Numbers

References are [Cassels, Geometry of Numbers] and [Venkatesh's notes].

### 1 Minkowski Theory

Cf. [B-G06].

**Thm. (20.8.1.1) [Minkowski's Lattice Point Theorem].** Let  $V \in \text{Vect}_{\mathbb{R}}^n$  equipped with a Haar measure  $dx$  and  $\Gamma \subset V$  be a complete lattice. Suppose  $X \subset V$  is a centrally symmetric and convex measurable subset of  $V$  s.t.

$$\text{Vol}(X) > 2^n \|\Gamma\|,$$

then  $\#X \cap \Gamma > 1$ . ┘

*Proof:* Consider the lattice  $2\Gamma \subset \mathbb{R}^n$ , which has norm  $\leq \text{Vol}(X)$ . Thus the projection of  $X$  onto  $\mathbb{R}^n/2\Gamma$  must have two overlapping points, i.e. there exists  $z_1, z_2 \in X$  s.t.  $z_1 - z_2 \in 2\Gamma$ . Thus  $(z_1 - z_2)/2 \in X \cap \Gamma$ . □

**Def. (20.8.1.2) [Minkowski Spaces].** For  $F \in \text{NField}$ ,  $\mathbb{F}_{\mathbb{C}} \cong \prod_{\sigma: F \rightarrow \mathbb{C}} \mathbb{C}$ , and  $\text{Gal}_{\mathbb{R}}$  acts on  $\mathbb{F}_{\mathbb{C}}$  via permuting the coordinates. Then define

$$\mathbb{F}_{\mathbb{R}} = (\mathbb{F}_{\mathbb{C}})^{\text{Gal}_{\mathbb{R}}}.$$

And it is equipped with a symmetric inner space structure via restriction from  $\mathbb{F}_{\mathbb{C}} \cong \prod_{\sigma: F \rightarrow \mathbb{C}} \mathbb{C}$ . Such a space  $F_{\mathbb{R}}$  is called the **Minkowski space attached to  $F$** . ┘

**Prop. (20.8.1.3) [Embedding in Minkowski Spaces].** For  $F \in \text{NField}$ , there is a natural map  $j: \mathbb{F} \hookrightarrow \mathbb{F}_{\mathbb{R}}$ , and for  $\mathfrak{a} \in \text{Ideal}^{\times}(\mathcal{O}_F)$ ,  $j(\mathfrak{a})$  is a complete lattice in  $F_{\mathbb{R}}$  with

$$\|j(\mathfrak{a})\| = \sqrt{|\text{disc}(F)|} \|\mathfrak{a}\|.$$

┘

*Proof:* Cf. [Neukirch]P31. □

**Prop. (20.8.1.4).** Let  $F \in \text{NField}$  and  $\text{Ideal}^{\times}(\mathcal{O}_F)$ , and for each  $\tau \in \Sigma_F^{\infty}$ , let  $c_{\tau} \in \mathbb{R}_{+}$  s.t.  $c_{\tau} = c_{\bar{\tau}}$ , and

$$\prod_{\tau} c_{\tau} > \left(\frac{2}{\pi}\right)^s \sqrt{|\text{disc}(F)|} \|\mathfrak{a}\|,$$

then there exists  $a \in \mathfrak{a}^{\times}$  s.t. for any  $\tau \in \Sigma_F^{\infty}$ , we have

$$|\tau(a)| < c_{\tau}.$$

┘

*Proof:* Cf. [Neu99]P32. □

**Cor. (20.8.1.5).** For  $\mathfrak{a} \in \text{Ideal}^{\times}(\mathcal{O}_F)$ , there exists  $a \in \mathfrak{a}^{\times}$  s.t

$$|\tau(a)| < \left(\frac{2}{\pi}\right)^{\frac{s}{n}} |\text{disc}(F)|^{\frac{1}{2n}} \|\mathfrak{a}\|^{\frac{1}{n}}$$

for any  $\tau \in \Sigma_F^{\infty}$ . ┘



**Prop. (20.8.1.6) [Regulator Map].** There is a **regulator map**

$$\mathrm{Reg}_F : \mathcal{O}_F^* \rightarrow \left[ \prod_{\tau \in \Sigma_F^\infty} \mathbb{R} \right]^+ \cong \mathbb{R}^{r_1+r_2} : x \mapsto (\log |x|_\tau)_{\tau \in \Sigma_F^\infty}$$

such that the kernel is  $F_{\mathrm{tor}}^\times$ , and the image is a complete lattice in  $H = \{(x_i) \mid \sum x_i = 0\}$ .  $\lrcorner$

*Proof:* Cf.[Neukirch, P43].  $\square$

**Cor. (20.8.1.7) [Regulators].** Let  $\{\varepsilon_1, \dots, \varepsilon_t\}_{t=r_1+r_2-1}$  be a system of fundamental units in  $F$ , define  $\mathrm{Reg}(F)$  the **regular of  $F$**  to be the absolute value of any  $t \times t$  minor of the following matrix

$$\begin{bmatrix} \mathrm{Reg}^{(1)}(\varepsilon_1) & \dots & \mathrm{Reg}^{(1)}(\varepsilon_t) \\ \vdots & & \vdots \\ \mathrm{Reg}^{(t+1)}(\varepsilon_1) & \dots & \mathrm{Reg}^{(t+1)}(\varepsilon_t) \end{bmatrix},$$

then the covolume of  $\mathrm{Reg}(\mathcal{O}_F^*) \subset H$  (20.8.1.6) is

$$\mathrm{Vol}(\mathrm{Reg}(\mathcal{O}_F^*)) = \sqrt{r_1 + r_2} \mathrm{Reg}(F).$$

$\lrcorner$

*Proof:* Cf.[Neukirch, P43].  $\square$

**Thm. (20.8.1.8) [Bounded Ramifications are Rare, Hermite].** For  $n \in \mathbb{Z}_+$  and  $\Sigma_F^\infty \subset_{\mathrm{fin}} S \subset \Sigma_F$ , there are only f.m. field extensions  $L/F$  of degree  $n$  that are unramified outside  $S$ .  $\lrcorner$

*Proof:* The power of a prime  $\mathfrak{P}$  in the discriminant is controlled by  $n$  by (5.2.7.35). Together with (5.2.7.37), this shows the power of  $\mathfrak{p}$  in the discriminant of the extension is controlled by  $n$ , independent of the field. Also we can assume  $\sqrt{-1} \in F$ , because it changes the discriminant by a bounded factor, by (5.2.7.38). So it suffices to prove there are only f.m. field extension with fixed degree and discriminant. By (5.2.7.38), it suffices to prove for  $F = \mathbb{Q}$ .

For the rest, we use Minkowski's theorem (20.8.1.4), Cf.[Neukirch, P203] **?**.  $\square$

**Prop. (20.8.1.9) [Minkowski's Second Theorem].** Let  $F \in \mathrm{NField}$ ,  $\deg(F) = d$ ,  $n \in \mathbb{Z}_+$ ,  $V \in \mathrm{Vect}^n/F$ , and  $\Lambda$  a lattice in  $V$ . For any  $v \in \Sigma_F^\infty$ , let  $S_v$  be a non-empty open convex symmetric bounded subset  $S_v$  of  $V_v$ . For  $\lambda > 0 \in \mathbb{R}$ , denote  $\lambda S = \prod_{v \mid \infty} \lambda S_v \times \prod_{v \nmid \infty} \lambda \Lambda_v$ . For  $n \geq 1$ , denote the  **$n$ -th successive minimum** of  $S$  to be

$$\lambda_n = \inf\{t > 0 \mid tS \text{ contains } n \text{ linearly independent vectors of } \Lambda \text{ over } K\}.$$

Then

$$\left( \prod_{i=1}^n \lambda_i \right) \mathrm{Vol}(S)^{1/d} \leq 2^N,$$

where the volume is calculated w.r.t. the adelic measure given by (19.4.2.5).

Moreover, if  $S_v$  are totally symmetric, then

$$\frac{2^{dn} \pi^{sn}}{(n!)^r ((2n)!)^s} N(d_{K/\mathbb{Q}})^{n/2} \leq \left( \prod_i \lambda_i \right)^d \mathrm{Vol}(S).$$

$\lrcorner$

*Proof:* Cf.[B-G06]P611.  $\square$

## 20.9 Abelian Varieties

*“Think geometrically, prove algebraically.”*

–J. Tate

References are [Abelian Varieties notes Conrad], [BLR90, B-G06, Mil08, Sta], [Abelian Variety van der Geer]. See [Bhatt’s notes] for everything done in the relative setting. A history of Abelian varieties can be found in [Mil08]P125.

**Notation(20.9.0.1).**

- Let  $k \in \mathbf{Field}$ .
- Let  $S \in \mathbf{Sch}$ .

┘

### 1 Basics

**Def.(20.9.1.1) [Abelian Schemes].** For  $S \in \mathbf{Sch}$ , an **Abelian scheme** over  $S$  is a proper smooth group variety  $A \in \mathbf{GrpVar}/S$  (7.12.5.1). Denote  $\mathbf{AbVar}/S$  the subcategory of  $\mathbf{AlgGrp}/S$  consisting of Abelian schemes over  $S$ .

Being an Abelian scheme is stable under base change of fields, by (7.12.1.3). ┘

**Prop.(20.9.1.2).** For  $X \in \mathbf{AbVar}/S$ , any global tangent vector on a group variety is left invariant. ┘

*Proof:* Because  $\Gamma(X, \mathcal{O}_X) = k$  (7.12.1.12), so by (10.1.4.38),  $\Gamma(X, \mathcal{O}_X \otimes T_{X,e}) = T_{X,e}$  are all generated by left invariant vectors (left and right translation commutes). ┘

**Thm.(20.9.1.3) [Abelian Varieties are Projective, Weil].** Abelian varieties over  $k$  are projective, by (10.1.4.9) and (7.4.6.3). ┘

**Thm.(20.9.1.4) [Rigidity].** Let  $S \in \mathbf{Sch}$  and  $f : X \rightarrow Y \in \mathbf{AlgGrp}/S$  be a morphism from an Abelian varieties to a group variety, then it is a group homomorphism followed by a translation  $t_{f(e_X)}$ . ┘

*Proof:* Set  $y = i_Y(f(e_X))$  and consider  $h = t_y \circ f$ , then  $h(e_X) = e_Y$ , and consider the morphism:

$$g : X \times X \rightarrow Y : (x, x') \mapsto h(xx')(h(x)h(x'))^{-1}.$$

then  $g(e_X, X) = g(X, e_X) = e_Y$ , so the rigidity lemma (7.12.1.20) shows  $g$  is constant with value  $e_Y$ . Thus  $h \circ m_X = m_Y \circ (h \times h)$ , thus  $h$  is a group homomorphism. ┘

**Cor.(20.9.1.5) [Abelian Varieties are Commutative].**

- If  $A \in \mathbf{Var}/S$ , then the Abelian variety structure on  $A$  is determined by  $e_A$ .
- If  $A \in \mathbf{AbVar}/S$ , the group law on  $A$  is commutative, justifying the name.

┘

**Remark(20.9.1.6).** The completeness of  $A$  is essential for the proof. In fact, there are many non-commutative group varieties, like  $\mathrm{GL}(n)$ .

From now, use the additive notation for Abelian varieties. ┘

*Proof:* 1: If there are two Abelian variety structures  $(m, i), (n, j)$ , then consider  $X \times X \rightarrow X \rightarrow X : (x, y) \mapsto m(x, y)(n(x, y)^{-1})$ , then it is constant on  $e_X \times X$  and  $X \times e_X$ , thus it is constant, so  $m = n$ . And  $i = j$  is also clear by the associativity.

2: The inverse  $i$  is a group homomorphism by (20.9.1.4), thus it is commutative. ┘

**Prop. (20.9.1.7) [Rigidity of Morphism from Smooth Varieties].** If  $k \in \mathbf{Field}$  and  $X \in \mathcal{AbVar}/k$ , then any rational map from a smooth  $k$ -variety  $Y$  extends to a morphism  $Y \rightarrow X$ .  $\lrcorner$

*Proof:* By (7.4.6.15), a rational map is defined on a set whose complement has codimension  $\geq 2$ , but then (10.1.1.18) shows it must be defined on all of  $X$ .  $\square$

**Prop. (20.9.1.8).** For  $S \in \mathcal{Sch}$  and  $X, Y \in \mathcal{AbVar}/S$ . If  $f : X \rightarrow Y$  is a homomorphism over  $S$  s.t. for any arithmetic point  $s \in S$ ,  $f_s : X_s \rightarrow Y_s$  has finite kernel, then  $f$  is surjective, thus is an isogeny.  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (20.9.1.9) [Quotient by Finite Subgroups].** Quotients of an Abelian variety  $A$  are also an Abelian varieties.  $\lrcorner$

*Proof:* It is a group variety of dimension 1 with a rational point by (10.1.5.33). And it is complete by (7.4.6.3).  $\square$

**Remark (20.9.1.10).** WARNING: it is not true that the quotient of an Abelian variety by any finite group of automorphisms is an Abelian variety, as the quotient of an elliptic curve by the hyperelliptic involution (i.e.  $-1$ ) is just  $\mathbb{P}^1$ .  $\lrcorner$

**Cor. (20.9.1.11) [Even Functions].** For  $E \in \mathcal{EllCur}/k$  and  $f \in R(E)$ ,  $f$  is even iff  $f \in K(x)$ .  $\lrcorner$

*Proof:* Write  $f(x, y) = g(x) + h(x)y$ , then the condition says  $(2y + a_1x + a_3)h(x) = 0$  for any  $(x, y) \in E(\bar{K})$ . Thus  $h(x) = 0$ , otherwise  $E$  is degenerate, which is not possible.  $\square$

**Def. (20.9.1.12) [Isogenies of Abelian Schemes].** A homomorphism  $\varphi : A \rightarrow B \in \mathcal{AbVar}/S$  is called an **isogeny between Abelian schemes** if it is surjective with finite fibers. This is compatible with the definition of isogenies of algebraic groups over fields (10.1.5.12), by (10.1.5.16).  $\lrcorner$

**Prop. (20.9.1.13) [Weil-Restriction of Abelian Varieties].** if  $K'/K$  is a finite separable field extension and  $A' \in \mathcal{AbVar}/K'$ , then  $A = \text{Res}_{K'/K}(A')$  is also an Abelian variety over  $K$  s.t.  $\dim A = [K' : K] \dim A'$ .  $\lrcorner$

*Proof:* Cf. [CCO14]P22?  $\square$

### Pseudo-Abelian Varieties

**Def. (20.9.1.14) [Pseudo-Abelian varieties].** A **pseudo-Abelian variety** over  $k$  is a group variety over  $k$  with no non-trivial affine normal group subvarieties.  $\lrcorner$

**Prop. (20.9.1.15).** Abelian varieties over  $k$  are pseudo-Abelian varieties (20.9.1.14).  $\lrcorner$

*Proof:* Every normal group subvariety is closed thus complete, hence if it is affine, it equals  $e$ .  $\square$

**Prop. (20.9.1.16).** Being a pseudo-Abelian variety over  $k$  is stable under separable base change of fields.  $\lrcorner$

*Proof:* Cf. [Mil17]P149.  $\square$

**Prop. (20.9.1.17).** Let  $G \in \mathcal{GrpVar}/k$ , then there exists a unique affine normal group subvariety  $N$  s.t.  $G/N$  is a pseudo-Abelian variety. And this  $N$  is stable under base change.  $\lrcorner$

*Proof:* This follows from (10.1.4.37) and (10.1.5.34)(10.1.5.33).  $\square$

**Prop. (20.9.1.18) [Barsotti-Chevalley].** If  $k = k^{\text{perf}}$ , then any pseudo-Abelian variety over  $k$  is complete hence is an Abelian variety.  $\lrcorner$

*Proof:* Cf. [Mil17]P154.  $\square$

**Cor. (20.9.1.19).** If  $k = k^{\text{perf}}$  and  $G \in \text{GrpVar}/k$ , then there exists a unique affine normal group subvariety  $N \trianglelefteq G$  s.t.  $G/N \in \text{AbVar}/k$ , by (20.9.1.17).  $\lrcorner$

**Thm. (20.9.1.20) [Algebraic Groups are Extensions of Abelian Varieties, Chevalley].** For any  $G \in \text{AlgGrp}_{\text{cnd}}/k$ , there is a unique maximal connected affine normal algebraic subgroup  $N \trianglelefteq G$  s.t.  $G/N \in \text{AbVar}/k$ .  $\lrcorner$

*Proof:* Cf. [Mil17]P155.  $\square$

**Cor. (20.9.1.21).** Any pseudo-Abelian variety  $G$  is commutative.  $\lrcorner$

*Proof:* As  $G$  is smooth and connected, so is  $[G, G]$  (10.1.4.29), so it is a group variety (10.1.4.37). Let  $N$  be given by (20.9.1.20), then because  $G/N$  is an Abelian variety,  $[G, G] \subset N$ . Thus  $[G, G]$  is affine, which implies  $[G, G] = e$ .  $\square$

**Prop. (20.9.1.22) [Totaro(2013)].** Any pseudo-Abelian variety is an extension of a connected unipotent group variety  $U$  by an Abelian variety  $A$  in a unique way.  $\lrcorner$

*Proof:*  $\square$

### Semi-Abelian Varieties

**Def. (20.9.1.23) [Semi-Abelian Varieties].** For  $k \in \text{Field}$ , a **semi-Abelian variety** over  $k$  is a group variety  $A \in \text{GrpVar}/k$  that is an extension of an Abelian variety by a torus.  $\lrcorner$

**Def. (20.9.1.24) [Semi-Abelian Schemes].** For  $S \in \text{Sch}$ , a **semi-Abelian scheme**  $S$  is a commutative smooth group variety  $\mathcal{A} \in \text{GrpVar}/S$  s.t. each fiber is a semi-Abelian variety (20.9.1.23).  $\lrcorner$

### Line Bundles

**Remark (20.9.1.25).** As Abelian varieties over  $k$  are regular,  $\text{Cl}(X) \cong \text{Pic}(X)$ , by (7.5.3.15).  $\lrcorner$

**Prop. (20.9.1.26) [Theorem of the Cube].** If  $X$  is an Abelian variety, and  $L$  is a line bundle over  $X$ , then

$$\Theta(L) = \text{pr}_{123}^*(L) \otimes \text{pr}_{12}^*(L^{-1}) \otimes \text{pr}_{13}^*(L^{-1}) \otimes \text{pr}_{23}^*(L^{-1}) \otimes \text{pr}_1^*(L) \otimes \text{pr}_2^*(L) \otimes \text{pr}_3^*(L)$$

is trivial.  $\lrcorner$

*Proof:* This is trivial on  $0 \times X \times X, X \times 0 \times X, X \times X \times 0$ , so it is trivial, by (7.12.1.23).  $\square$

**Cor. (20.9.1.27).** There is a form of morphisms from a scheme to  $X$ , just by considering  $(f, g, h) : Y \times Y \times Y \rightarrow X \times X \times X$ , i.e.

$$(f + g + h)^*L \otimes (f + g)^*L^{-1} \otimes (f + h)^*L^{-1} \otimes (g + h)^*L^{-1} \otimes f^*L \otimes g^*L \otimes h^*L$$

is trivial.  $\lrcorner$

**Cor. (20.9.1.28) [Theorem of the Square].** Let  $A$  be an Abelian variety that  $\mathcal{L}$  is a line bundle, then for any  $x, y \in A$ ,

$$t_{x+y}^* L \otimes L \cong t_x^* L \otimes t_y^* L \in \text{Pic}(A_{k(x)k(y)}).$$

Notice this isomorphism is defined under the field generated by the residue fields of  $x$  and  $y$ .  $\lrcorner$

*Proof:* Apply the theorem of the cube (20.9.1.27) for  $f = \text{id}_X$  and  $g, h$  the function with constant value  $x, y$ .  $\square$

**Cor. (20.9.1.29).** For a line bundle  $L$  on an Abelian variety  $X$ , then the map  $\varphi_L : X \rightarrow \text{Pic}(X) : x \mapsto [t_x^* L \otimes L^{-1}]$  is a homomorphism.  $\lrcorner$

**Cor. (20.9.1.30).** For any  $\mathcal{L} \in \text{Pic}(A)$ ,

$$[n]^* L \cong L^{n(n+1)/2} \otimes [-1]^* L^{n(n-1)/2}.$$

$\lrcorner$

*Proof:* Use (20.9.1.27) in case  $f = [n], g = [1], h = [-1]$ , then we have:

$$n^* L^2 \otimes (n+1)^* L^{-1} \otimes (n-1)^* L^{-1} \cong (L \otimes [-1]^* L)^{-1}.$$

So we can use induction.  $\square$

**Def. (20.9.1.31) [Even Line Bundle].** Consider the involution  $[-1]$  of  $X$  and its action on the line bundles, then a **even/odd line bundle** is defined to be a line bundle  $L$  that  $[-1]^* \mathcal{L} \cong \mathcal{L}$  (resp.  $\mathcal{L}^{-1}$ ).  $\lrcorner$

**Prop. (20.9.1.32).** On an Abelian variety, there is an even very ample line bundle.  $\lrcorner$

*Proof:* Abelian variety is projective by (20.9.1.3), thus there is a very ample line bundle  $L$ , and  $[-1]^* L$  is also very ample, so  $L \otimes [-1]^* L$  is even and very ample, by (7.5.4.23).  $\square$

**Prop. (20.9.1.33).** For  $F \in \mathbf{NField}$ ,  $A \in \mathbf{AbVar}/F$  and  $c \in \text{Pic}(A)$ , there are an odd line bundle  $c^-$  and an even line bundle  $c^+$  that  $c = c_- + c_+$ .  $\lrcorner$

*Proof:* Consider  $2c = (c + [-1]^*(c)) + (c - [-1]^*(c))$ .  $c - [-1]^*(c)$  is odd thus is in  $\text{Pic}^0(X)$  by (20.9.4.12), thus by (20.9.6.18), there is a  $c^* \in \text{Pic}^0(X)$  that  $2c^* = c - [-1]^*(c)$ . Then  $c = (c - c^*) + c^*$  satisfies the requirement.  $\square$

**Lemma (20.9.1.34).** Let  $S \in \mathbf{Sch}$ ,  $A \in \mathbf{AbVar}/S$  and  $\text{pr}_i : A \times A \rightarrow A$  be the projections and  $m$  be the multiplication, then the following are equivalent:

- $m^*(c) \cong \text{pr}_1^*(c) + \text{pr}_2^*(c)$ .
- $\tau_a^*(c) \cong c$  for  $a \in A$ .

And if these are satisfied,  $c$  is even.  $\lrcorner$

*Proof:* The equivalence is a consequence of the formula  $(m^*(c) - \text{pr}_1^*(c) - \text{pr}_2^*(c))|_{A \times \{a\}} = \tau_a^*(c) - c$  and the see-saw principle (7.12.1.22). the last assertion is a consequence of the first equation pulled back via the morphism

$$A \rightarrow A \times A : a \mapsto (a, -a).$$

$\square$

**Prop. (20.9.1.35) [Projective Embeddings].** No Abelian varieties of dimension  $g$  can be embedded into  $\mathbb{P}^{2g-1}$ . No Abelian variety except for elliptic curves and Abelian surfaces of degree 10 in  $\mathbb{P}^4$  can be embedded into  $\mathbb{P}^{2g}$ .  $\lrcorner$

*Proof:* Cf. [Van de Geer P26], where algebraic topologies are used.  $\square$

## 2 Formal Groups

**Def. (20.9.2.1) [Formal Group Law of Abelian Varieties].** Given  $k \in \mathbf{Field}$ ,  $A \in \mathcal{AbVar}/k$ , the group structure on  $A$  induces a homomorphism  $\mathcal{O}_{A,e} \rightarrow (\mathcal{O}_{A,e} \times \mathcal{O}_{A,e})_{e \times e}$ , whose completion is a formal group law of dimension  $n$ .  $\lrcorner$

## 3 over Alg.Closed Fields

**Notation (20.9.3.1).**

- Let  $k \in \mathbf{Field}$ ,  $k = \bar{k}$ .

$\lrcorner$

**Prop. (20.9.3.2).** There is a closed pt 0 in  $X$  that corresponds to 0 in the group  $X(k)$ , if we denote  $\Omega_0$  the cotangent space at 0, it is the stalk of the differential  $\Omega_{X/k}$  at 0 (7.5.5.6).  $\lrcorner$

**Prop. (20.9.3.3).** Suppose  $k = \bar{k}$ , then for each Albert-algebra  $(D, *)$  (21.2.1.21), there exists  $A \in \mathcal{AbVar}^{\text{sim}}/k$  with a polarization  $\lambda$  s.t.  $(\text{End}^0(A), \text{Ros}_\lambda) \cong (D, *)$ .  $\lrcorner$

*Proof:* Cf. [CCO14]P28.  $\square$

**Prop. (20.9.3.4).** Suppose  $k = \bar{k}$  and  $A \in \mathcal{AbVar}^g/k$ , then  $\dim_k H^1(A, \mathcal{O}_A) \leq g$ . In fact, equality holds by (20.9.4.19).  $\lrcorner$

*Proof:* This follows from Borel's classification of Hopf-algebras (3.11.3.14). Cf. [Van de Geer P94] or [Conrad Notes, P45]. ?  $\square$

**Prop. (20.9.3.5).** Suppose  $k = \bar{k}$ ,  $A \in \mathcal{AbVar}/k$ , and  $X \subset A$  is an integral subvariety. Suppose  $\{a \in A(k) | a + X = X\} = 0$ , then for  $m$  large, the map

$$\alpha_m : X^m \rightarrow A^{m-1} : (x_1, \dots, x_m) \mapsto (x_1 - x_2, x_2 - x_3, \dots, x_{m-1} - x_m)$$

is quasi-finite with generic degree 1.  $\lrcorner$

*Proof:* Cf. [Equidistribution of Small points, Shou-wu]P163. ?  $\square$

## p-Divisible Groups

**Prop. (20.9.3.6).** For  $k \in \mathbf{Field}$ ,  $\text{char } k = p$ ,  $A \in \mathcal{AbVar}/k$ ,  $\ell \in \mathbf{P} \setminus \{p\}$ ,  $A(k^{\text{sep}})$  is an Abelian group and  $A[\ell^n] \cong (\mathbb{Z}/(\ell^n))^{2g}$  and  $A[p^n] \cong (\mathbb{Z}/(p^n))^r$  for some  $r \geq 1$ .  $\lrcorner$

**Prop. (20.9.3.7).** There is an isomorphism

$$H_{\text{ét}}^m(A_{k^{\text{sep}}}, \mathbb{Q}_\ell) \cong \bigwedge_{\mathbb{Q}_\ell}^m (V_\ell(A))^*.$$

$\lrcorner$

*Proof:* Cf. [Grothendieck Monodromy theorem].  $\square$

## 4 Dual Abelian Varieties

References are [CCO14]Section1.4.2.

**Def. (20.9.4.1) [Dual Abelian Varieties].** For  $S \in \text{Sch}$  and  $A \in \text{AbVar}/S$ , the **dual Abelian variety**  $A^\vee$  is defined to be its Picard-scheme  $\text{Pic}_{A/S}^0 \in \text{Sch}^{\text{sep, loc. pf}}/S$  (10.9.3.23), which represents the sheaf  $\widetilde{\text{Pic}}_{A/S}^0 \in \text{Sh}(\text{Sch}_{\text{fppf}}/S)$  (i.e. line bundles with  $e$ -rigidifications (10.9.3.6)). We will see it is an Abelian variety (i.e. smooth) in (20.9.4.19).

By definition, there is a universal line bundle  $p_A \in \text{Pic}(A \times A^\vee)$  with a rigidification along  $e$ .  $\lrcorner$

**Lemma (20.9.4.2).** For  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ ,  $\mathcal{L} \in \text{Pic}(A)$ , consider the double Picard-map (10.9.3.38)  $\kappa_A : A \rightarrow A^{\vee\vee}$ , then  $\lambda_{\mathcal{L}} = \lambda_{\mathcal{L}}^\vee \circ \kappa_A : A \rightarrow A^\vee$ .  $\lrcorner$

*Proof:* Cf. [van Der Geer, P100].  $\square$

**Lemma (20.9.4.3) [Double Duality Theorem].** For  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ , the double Picard-map (10.9.3.38)  $\kappa_A : A \rightarrow A^{\vee\vee}$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [van Der Geer, P101].  $\textcolor{red}{?}$   $\square$

**Prop. (20.9.4.4) [Dual Abelian Schemes].** If  $S \in \text{Sch}$  and  $A \in \text{AbVar}/S$ , then

- $\text{Pic}_{A/S}^0 = \text{Pic}_{A/S}^\tau$  is a projective Abelian scheme over  $S$ , called the **dual Abelian scheme** of  $A/S$ .
- the double Picard-map (10.9.3.38)  $A \rightarrow A^{\vee\vee}$  is an isomorphism.  $\lrcorner$

*Proof:* By reducing to the fiber, these reduces to the case that  $S$  is a field, and this case follows from (20.9.4.19) and (20.9.4.3).  $\square$

**Def. (20.9.4.5) [Symmetric Homomorphisms].** For  $S \in \text{Sch}$  and  $A \in \text{AbVar}/S$ , a homomorphism  $\alpha : A \rightarrow A^\vee$  over  $S$  is called a **symmetric homomorphism** if

$$\alpha = \alpha^\vee \circ \kappa_{A/S} : A \rightarrow (A^\vee)^\vee \rightarrow A^\vee \text{ (20.9.4.3).}$$

$\lrcorner$

### Ample Line Bundles

**Prop. (20.9.4.6) [ $A \rightarrow A^\vee$  Induced by Line Bundles].** Let  $S \in \text{Sch}$  and  $A \in \text{AbVar}/S$ . For any  $\mathcal{L} \in \text{Pic}_{A/S}(S)$  (line bundles with a rigidification (10.9.3.2)), there is a **Mumford line bundle** on  $A \times A$  given by

$$\Lambda(\mathcal{L}) = m^* \mathcal{L} - \text{pr}_1^* \mathcal{L} - \text{pr}_2^* \mathcal{L},$$

where  $m : A \times A \rightarrow A$  is the product. It is in  $\text{Pic}^0(A)$  because  $u : \Lambda(c)|_{e \times A} \cong \mathcal{O}_S$  and  $\Lambda(\mathcal{L})|_{A \times a} = \tau_a^* c - c$ . Then by definition, this line bundle corresponds to a morphism  $\lambda_{\mathcal{L}} : A \rightarrow A^\vee$  over  $S$ , and by (10.9.3.35)  $\lambda_c(a) = \tau_a^* c - c \in A^\vee(k(a))$ . This is also a homomorphism of Abelian schemes, by (20.9.1.4), as  $\varphi_c(e) = e$ .

It is clear that  $\mathcal{L} \mapsto \lambda_{\mathcal{L}}$  is a group homomorphism  $\text{Pic}_{A/S}(S) \mapsto \text{Hom}_S(A, A^\vee)$ . Also, denote  $K(\mathcal{L}) = \ker(\lambda_{\mathcal{L}})$ .  $\lrcorner$

**Cor. (20.9.4.7).** Situation as in (20.9.4.6),  $\lambda_{\mathcal{L}}$  is symmetric.  $\lrcorner$

*Proof:* This is because  $\Lambda(\mathcal{L})$  is symmetric(20.9.4.5).  $\square$

**Cor. (20.9.4.8).** For  $x \in A(k)$ ,  $\lambda_{\tau_x^* \mathcal{L}} = \lambda_{\mathcal{L}}$ .  $\lrcorner$

*Proof:* This follows from see-saw lemma, by observing that stalks of the line bundles in(20.9.4.6) are  $\tau_{a+x}^* \mathcal{L} - \tau_x^* \mathcal{L}$  and  $\tau_a^* \mathcal{L} - \mathcal{L}$  resp..  $\square$

**Cor. (20.9.4.9).**  $(\text{id}_A, \varphi_{\mathcal{L}})^* p_A = \mathcal{L} \otimes [-1]^* \mathcal{L}$ .  $\lrcorner$

*Proof:*

$$(\text{id}_A, \varphi_{\mathcal{L}})^* p_A = \Delta_A^*(\text{id} \times \lambda_{\mathcal{L}})^* p_A = \Delta_A^*(m^* \mathcal{L} - \text{pr}_1^* \mathcal{L} - \text{pr}_2^* \mathcal{L}) = [2]^* \mathcal{L} \otimes \mathcal{L}^{-2} \cong \mathcal{L} \otimes [-1]^* \mathcal{L} \text{ (20.9.1.30)}.$$

$\square$

**Prop. (20.9.4.10) [Poincaré-Classes are Even].** The Poincaré-class  $p_A \in \text{Pic}(A \times A^\vee)$  is even.  $\lrcorner$

*Proof:* Let  $b \in A^\vee$ , then

$$([-1]^*(p))|_{A \times \{b\}} = [-1]^*(p|_{A \times \{-b\}}) = [-1]^*(-b) = b.$$

and

$$([-1]^*(p))|_{\{0\} \times A^\vee} = [-1]^*(p|_{\{0\} \times \widehat{A}}) = 0.$$

Thus  $[-1]^* p \cong p$  by(10.9.3.35) and the see-saw principle(7.12.1.22).  $\square$

**Lemma (20.9.4.11).** If  $b \in \text{Pic}(A)$  and  $\lambda_b = 0$ , then for any ample  $c \in \text{Pic}(A)$ , there is some  $a \in A(K)$  that  $b = \tau_a^*(c) - c$ .  $\lrcorner$

*Proof:* [Mumford, P77].  $\square$

**Thm. (20.9.4.12) [Characterizing  $A^\vee$ ].** For  $\mathcal{L} \in \text{Pic}(A)$ ,  $[-1]^*(\mathcal{L}) \otimes \mathcal{L}^{-1} \in \text{Pic}^0(A)$ , and the following are equivalent:

1.  $\mathcal{L} \in \text{Pic}^0(A)$ .
2.  $\lambda_{\mathcal{L}} = 0$ .
3. For every ample line bundle  $c \in \text{Pic}(A)$ , there is an  $a \in A$  that  $\mathcal{L} \cong \tau_a^*(c) - c$ .
4. There is an ample line bundle  $c \in \text{Pic}(A)$  and an  $a \in A$  s.t.  $\mathcal{L} \cong \tau_a^*(c) - c$ .
5.  $\mathcal{L}$  is odd.
6. For any scheme  $X$ , the map  $\text{Mor}(X, A) \rightarrow \text{Pic}(X) : \varphi \mapsto \varphi^*(c)$  is a group homomorphism.

$\lrcorner$

*Proof:*  $2 \rightarrow 3$  is lemma(20.9.4.11).  $4 \rightarrow 1$  is by(20.9.4.6).  $3 \rightarrow 4$  is trivial.

$1 \rightarrow 2$ :  $\mathcal{L} \in \text{Pic}^0(A)$  corresponds to a section  $[\mathcal{L}] : k \rightarrow \underline{\text{Pic}}_A^0$ . Let

$$\Lambda(p_A) = (m \times \text{id}_{\underline{\text{Pic}}^0(A)})^*(p_A) - (\text{pr}_1 \times \text{id}_{\underline{\text{Pic}}^0(A)})^*(p_A) - (\text{pr}_2 \times \text{id}_{\underline{\text{Pic}}^0(A)})^*(p_A),$$

then it corresponds to a map  $\Lambda : A \times \underline{\text{Pic}}_A^0 \rightarrow \underline{\text{Pic}}_A^0$  s.t.

$$\lambda_{\mathcal{L}} = \Lambda \circ (i_1 \times [\mathcal{L}]) : A \rightarrow \underline{\text{Pic}}_A^0.$$

But  $\Lambda(A \times \{0\}) = 0$ , so by rigidity lemma(7.12.1.20) it factors through the projection  $\text{pr}_1$ . Thus  $\lambda_{\mathcal{L}} = 0$ .



$6 \rightarrow 5$  is trivial, and 6 is clearly equivalent to the assertion when  $X = A$ , and it is equivalent to 2 by (20.9.1.34).

We next prove  $[-1]^*(c) - c \in \text{Pic}^0(A)$ : because  $[-1]\tau_a = \tau_{-a}[-1]$ , we have

$$\tau_a^*([-1]^*(c)) - [-1]^*(c) = [-1](\tau_{-a}^*(c) - c).$$

since  $\tau_{-a}^*(c) - c \in \text{Pic}^0(X)$  by (20.9.4.6), the equation is equal to  $c - \tau_{-a}^*(c)$  by implication  $1 \rightarrow 2 \rightarrow 6$ . and this further equals  $\tau_a^*(c) - c$  by the theorem of square (20.9.1.28). Then

$$\tau_a^*([-1]^*(c) - c) - ([-1]^*(c) - c) = 0.$$

Hence  $[-1]^*(c) - c \in \text{Pic}^0(A)$  by the implication  $2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ .

$5 \rightarrow 1$ : Let  $c$  be an odd element, then  $-2c = [-1]^*(c) - c \in \text{Pic}^0(A)$  by what just proved, we have  $c \in \text{Pic}^0(A)$ , then  $\varphi_c$  has image in the kernel of  $[2]$  on  $\hat{A}$ , by the implication  $1 \rightarrow 2$ , thus it has trivial image, by (20.9.6.18).  $\square$

**Cor. (20.9.4.13).**  $\mathcal{L} \mapsto \lambda_{\mathcal{L}}$  induces an injection

$$\text{NS}(A_{\bar{k}}) = \text{Pic}(A_{\bar{k}}) / \text{Pic}^0(A_{\bar{k}}) \hookrightarrow \text{Hom}(A_{\bar{k}}, A_{\bar{k}}^{\vee}).$$

┘

**Cor. (20.9.4.14) [Abelian Varieties are Finite Torsion-Free].** For  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ ,  $\text{NS}(A_{\bar{k}})$  is a finite free  $\mathbb{Z}$ -module. (Although this is true for any complete variety?).  $\square$

*Proof:* This follows from (20.9.4.13) and (20.9.6.12).  $\square$

**Cor. (20.9.4.15).** Let  $S \in \text{Sch}$  and  $A \in \text{AbVar}/S$ . If  $b$  is a section of  $\underline{\text{Pic}}_{A/S}^0 \rightarrow S$ , and  $\mathcal{L} = b^*(p_A) \in \text{Pic}(A)$ , then  $\lambda_A = 0$ .  $\square$

*Proof:*  $\square$

**Cor. (20.9.4.16).** For  $\mathcal{L} \in \text{Pic}(A)$  and  $n \in \mathbb{Z}$ ,  $\lambda_{[n]\mathcal{L}} = n^2 \lambda_{\mathcal{L}}$ .  $\square$

*Proof:* By (20.9.1.30), it suffices to prove for  $n = -1$ . And this follows from (20.9.4.12) and the fact  $[-1]^*\mathcal{L} \otimes \mathcal{L}^{-1} \in \text{Pic}^0(A)$ .  $\square$

**Prop. (20.9.4.17).** Let  $k \in \text{Field}$ ,  $A \in \text{AbVar}/k$ . For  $\lambda \in \text{Hom}(A, A^{\vee})$ , consider  $\mathcal{M} = (\text{id}, \lambda)^* p_A$ , then  $\lambda_{\mathcal{M}} = \lambda + \lambda^{\vee}$ . In particular, if  $\lambda$  is symmetric, then  $\varphi_{\mathcal{M}} = 2\lambda$ .  $\square$

*Proof:* [Geer]P159.  $\square$

**Prop. (20.9.4.18) [Ample Implies Non-Degenerate].**  $\mathcal{L} \in \text{Pic}(A)$  is ample iff it is non-degenerate and  $H^0(A, \mathcal{L}) \neq 0$ .  $\square$

In particular, an effective line bundle on  $A$  is ample iff it is non-degenerate.  $\square$

*Proof:* Cf. [Diophantine Geometry, P253] or [Conrad notes].  $\square$

**Cor. (20.9.4.19) [Dual Abelian Variety].** Let  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ , then

$$A^{\vee} = \underline{\text{Pic}}_{X/S}^0 = \underline{\text{Pic}}_{X/S, \text{red}}^0 = \underline{\text{Pic}}_{X/S}^{\tau}$$

is also an Abelian variety, and  $\dim A = \dim A^{\vee} = \dim_k H^1(A, \mathcal{O}_A)$ .  $\square$

*Proof:* Take an ample line bundle  $\mathcal{L}$  on  $A$ , then  $\varphi_{\mathcal{L}} : A \rightarrow A^\vee$  has finite kernel, thus  $\dim \hat{A} \geq \dim A$ , but by (20.9.3.4) and (10.9.3.25),  $\dim_k H^1(A, \mathcal{O}_A) = T_e(A^\vee) \leq \dim A$ , thus it is regular by (5.3.5.18).

$\underline{\text{Pic}}_{X/S}^\tau = \underline{\text{Pic}}_{X/S}^0$  because we can pass to the  $k = \bar{k}$ , and use the fact Abelian varieties are torsion-free (20.9.4.14).  $\square$

**Def. (20.9.4.20) [Non-Degenerate Line Bundles].** For  $S \in \text{Sch}$  and  $A \in \text{AbVar}/S$ , a **non-degenerate line bundle** is a rigidified line bundle  $\mathcal{L} \in \underline{\text{Pic}}_{A/S}(S)$  s.t.  $K(\mathcal{L}) = \ker \varphi_{\mathcal{L}}$  is a finite flat group scheme, i.e.  $\varphi_{\mathcal{L}}$  is an isogeny.  $\lrcorner$

## 5 Cohomology on Abelian Varieties

**Thm. (20.9.5.1) [Cohomology of the Poincaré Class].** For  $k \in \text{Field}$  and  $A \in \text{AbVar}^g/k$ , Let  $\text{pr}_2 : A \times A^\vee \rightarrow A^\vee$  be the projection, then

$$R^n \text{pr}_{2*} p_A = e_*(k)^{\delta_{n,g}}, \quad H^n(A \times A^\vee, p_A) = k^{\delta_{n,g}}.$$

$\lrcorner$

*Proof:* Cf. [Conrad, P62] or [Geer, P126]  $\color{red}?$ .  $\square$

**Thm. (20.9.5.2) [Cohomology of Line Bundles].** Let  $k \in \text{Field}$  and  $A \in \text{AbVar}^g/k$ ,  $\mathcal{F} \in \text{Coh}(A)$ , then

1.  $\chi(\mathcal{F}) = \deg(\text{ch}_g(\mathcal{F}))$ .
2. If  $\mathcal{L} \in \text{Pic}(A)$ , then  $\chi(\mathcal{L}) = \deg(c_1(\mathcal{L})^g)/g!$ . In particular,  $\chi(\mathcal{L}^m) = m^g \chi(\mathcal{L})$ .
3.  $\deg(\lambda_{\mathcal{L}}) = \chi(\mathcal{L})^2$ . In particular,  $\deg(\lambda_{\mathcal{L}})$  is a square. Notice if  $\mathcal{L}$  is degenerate, then  $\deg(\lambda_{\mathcal{L}}) = 0$ .
4. If  $\mathcal{L} \in \text{Pic}(A)$  is a non-degenerate line bundle, then there is a unique integer  $0 \leq i \leq g$  s.t.  $H^i(X, \mathcal{L}) \neq 0$ . Such an  $i$  is called the **index of  $\mathcal{L}$** . And  $\text{ind}(\mathcal{L}^{-1}) + \text{ind}(\mathcal{L}) = g$ . Notice  $\text{ind}(\mathcal{L}) = 0$  is equivalent to it is effective or equivalent to it being ample (20.9.4.18).
5. Moreover, if  $k = \mathbb{C}$ , and  $A = \text{Lie}(A)/\Lambda$ ,  $\mathcal{L} \in \text{Pic}(A)$ , then  $c_1(\mathcal{L})$  corresponds to a Hermitian form  $H$   $\color{red}?$ , and  $\text{ind}(\mathcal{L})$  is just the number of negative eigenvalues of  $H$ .

$\lrcorner$

*Proof:* 1, 2: Because  $\mathcal{T}_{A/k}$  is trivial, the Todd class  $\text{Todd}(\mathcal{T}_{A/k}) = 1$ , so the assertions follows from [Hirzebruch-Riemann-Roch \(9.1.9.3\)](#).

3, 4. For the last assertion, Cf. [Van de Geer, P131]  $\color{red}?$

$$\dim_k H^n(X \times X, \Lambda(\mathcal{L})) = \deg(\lambda_{\mathcal{L}}) \delta_{n,g},$$

$$H^n(X \times X, \Lambda(\mathcal{L})) \cong H^n(X \times X, m^* \mathcal{L} \otimes \text{pr}_1^* \mathcal{L}^{-1}) \cong H^n(X \times X, \text{pr}_1^* \mathcal{L} \otimes \text{pr}_2^* \mathcal{L}^{-1}) \cong \bigoplus_{p+q=n} H^p(X, \mathcal{L}) \otimes H^q(X, \mathcal{L}^{-1}).$$

So item 4 follows from the two equations above.

5:  $\color{red}?$   $\square$

**Cor. (20.9.5.3).** Let  $k \in \text{Field}$  and  $f : X \rightarrow Y \in \text{IsogAbVar}_k$ , then  $\chi(f^* \mathcal{L}) = \deg(f) \chi(\mathcal{L})$ .  $\lrcorner$

*Proof:* We may base change to  $k = \bar{k}$ . By the proposition and the fact  $c_1(f^*(\mathcal{L})^g) = f^*(c_1(\mathcal{L})^g)$ , it suffices to show that  $\deg(f^*([P])) = \deg(f)$  for any point  $P \in Y(k)$ , which is trivial.  $\square$

**Prop. (20.9.5.4) [Properties of Indices].** For  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ , then

- The index is a locally constant function in algebraic families of non-degenerate line bundles. In particular, if  $\mathcal{L} \in \text{Pic}(A)$  is non-degenerate and  $\mathcal{L}' \in \text{Pic}^0(A)$ ,  $\text{ind}(\mathcal{L}) = \text{ind}(\mathcal{L} \otimes \mathcal{L}')$ .
- If  $f : B \rightarrow A \in \text{IsogAbVar}_k$ , and  $\mathcal{L} \in \text{Pic}(A)$  a non-degenerate line bundle, then  $\text{ind}(\mathcal{L}) = \text{ind}(f^*\mathcal{L})$ .
- If  $\mathcal{L}$  is non-degenerate, then for  $m \in \mathbb{Z}_+$ ,  $\mathcal{L}^{\otimes m}$  is non-degenerate too, and  $\text{ind}(\mathcal{L}^{\otimes m}) = \text{ind}(\mathcal{L})$ .
- If  $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_1 \otimes \mathcal{L}_2 \in \text{Pic}(A)$  are all non-degenerate, then  $\text{ind}(\mathcal{L}_1 \otimes \mathcal{L}_2) \leq \text{ind}(\mathcal{L}_1) + \text{ind}(\mathcal{L}_2)$ .
- If  $\mathcal{L}, \mathcal{L}' \in \text{Pic}(X)$  and  $\mathcal{L}, \mathcal{L} \otimes \mathcal{L}'$  are both non-degenerate, then  $\text{ind}(\mathcal{L} \otimes \mathcal{L}') \leq \text{ind}(\mathcal{L})$ .

⌋

*Proof:* Cf.[Geer, P134].?

□

**Prop. (20.9.5.5).** Let  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ . Suppose  $\mathcal{L}, \mathcal{H} \in \text{Pic}(A)$  and  $\mathcal{L}, \mathcal{L} \otimes \mathcal{H}$  are both non-degenerate, and that  $\text{ind}(\mathcal{L}) \neq \text{ind}(\mathcal{L} \otimes \mathcal{H})$ , then the polynomial  $P(t) = \deg((c_1(\mathcal{L}) + tc_1(\mathcal{H}))^g)$  has a root  $t \in [0, 1]$ .

⌋

*Proof:* Cf.[Geer, P136].?

□

**Cor. (20.9.5.6).** Let  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ . Then for any  $\mathcal{L} \in \text{Pic}(A)$  non-degenerate and  $m \in \mathbb{Z}_+$ ,  $\text{ind}(\mathcal{L}^m) = \text{ind}(\mathcal{L})$ .

⌋

*Proof:* ?

□

**Thm. (20.9.5.7) [Kempf-Mumford-Ramanujam].** Let  $\mathcal{L}$  be a line bundle on an Abelian variety  $A$ , and fix an ample line bundle  $H$  on  $A$ , let  $\Phi$  be the Hilbert polynomial of  $\mathcal{L}$  w.r.t.  $H$ , then

- The multiplicity of 0 in  $\Phi$  equals  $\dim K(\mathcal{L})$ .
- If  $\mathcal{L}$  is non-degenerate, then all roots of  $\Phi$  in  $\mathbb{C}$  are real, and the number of positive roots equals  $\text{ind}(\mathcal{L})$ .

⌋

*Proof:* Cf.[Van de Geer, P139,140].

□

**Prop. (20.9.5.8).** For any ample line bundle  $\mathcal{L}$  on  $A$ ,  $\mathcal{L}^{\otimes m}$  is basepoint-free for  $m \geq 2$  and very ample for  $m \geq 3$ .

⌋

*Proof:*

□

**Prop. (20.9.5.9).** Let  $S \in \text{Sch}$  is connected and  $X \in \text{AbVar}/S$ . If  $\mathcal{L} \in \text{Pic}(X)$  is non-degenerate, and there exists an arithmetic point  $s \in S$  s.t.  $\mathcal{L}_s$  is ample on  $X_s$ , then  $\mathcal{L}$  is relatively ample on  $X/S$ . ⌋

*Proof:*

□

## 6 Isogenies and Tate Modules

### Isogenies

**Prop. (20.9.6.1) [Isogenies of Abelian Varieties].** Let  $k \in \text{Field}$  and  $f : X \rightarrow Y \in \text{AbVar}/k$ , the following are equivalent:

- $f$  is an isogeny.

- $f$  is surjective and  $\dim X = \dim Y$ .
- $\ker f$  is a finite group scheme and  $\dim X = \dim Y$ .
- $f$  is finite.

The set of isogenies from  $X$  to  $Y$  is denoted by  $\text{IsogAbVar}(X, Y)$ .  $\text{IsogAbVar}(X, X) \cup \{0\}$  is denoted by  $\text{IsogAbVar}(X)$ .  $\lrcorner$

*Proof:* This follows from the theory of algebraic groups.  $\square$

**Prop. (20.9.6.2) [Separable Isogenies].** For  $f : X \rightarrow Y \in \text{IsogAbVar}_k$ ,

- For any  $Q \in Y(\bar{k})$ ,  $\#f^{-1}(Q)(\bar{k}) = \deg_s(f)$ , and for any  $P \in X$ ,  $e_P(f) = \deg_i(f)$ .
  - The map:  $X[f] \rightarrow \text{Gal}(K(X_{\bar{k}})/\varphi^*(K(Y_{\bar{k}}))) : P \mapsto \tau_P^*$  is an isomorphism of groups.
  - $f$  is unramified iff it is étale iff it is separable, and in this case,  $K(X_{\bar{k}})/K(Y_{\bar{k}})$  is an Abelian Galois extension of degree  $\deg(f)$ .
- $\lrcorner$

*Proof:* 1: By (7.12.3.5), this holds for  $Q$  in a dense open subset of  $Y$ , thus by homogeneity, it holds for all  $Q$ . The second assertion follows from the fact after shrinking  $f$  is finite locally free of rank  $\deg(f)$  by using homogeneity on  $X$ .

2: This is clearly a group homomorphism. By Galois theory, it suffices to show it is injective, which is clear.

3: This follows from (20.9.6.1) and generic unramifiedness (7.6.5.9).  $\square$

**Prop. (20.9.6.3) [Inverse Isogenies].** If  $k \in \text{Field}$ ,  $X, Y \in \text{AbVar}/k$  and  $f \in \text{IsogAbVar}(X, Y)$  with  $d = \deg(f)$ , then there exists  $g \in \text{IsogAbVar}(Y, X)$  s.t.  $g \circ f = [d]_X$ ,  $f \circ g = [d]_Y$ .

In particular, we can write  $X \sim Y$  to denote that  $X, Y$  are isogenous. This is an equivalence relation.  $\lrcorner$

*Proof:* As  $\ker(f)$  is a finite group scheme of rank  $d$ ,  $[d]\ker(f) = 0$  by Deligne's theorem (10.1.2.19), so  $[d]$  factors through  $f$  via  $g : Y \rightarrow X$ , so  $g \circ f = [d]_X$ . Then also  $[d]_Y \circ f = f \circ [d]_X = f \circ (g \circ f) = (f \circ g) \circ f$ , thus  $[d]_Y = f \circ g$  as  $f$  is a quotient map.  $\square$

**Def. (20.9.6.4) [Dual Isogenies].** Let  $\varphi : A \rightarrow B$  in  $\text{IsogAbVar}/S$ , we get a **dual homomorphism**  $\varphi^\vee \in \text{Hom}(B^\vee, A^\vee)$ . Then:

- $\widehat{\widehat{\varphi}} = \varphi$ .
  - $\deg(\varphi^\vee) = \deg(\varphi)$ .
  - $(\varphi + \psi)^\vee = \varphi^\vee + \psi^\vee$ .
  - $[m]_E^\vee = [m]_{E^\vee}$ .
- $\lrcorner$

*Proof:* 1: This is formal.

2: [Conrad, P66]. ?

3: This is clear from the modular description of  $\widehat{A}$ .

4: This follows from 3.  $\square$

**Thm. (20.9.6.5) [Kernels of Dual isogenies].** For  $f : A \rightarrow B \in \text{AbVar}/S$ , there is a canonical isomorphism

$$\ker(f)^D \cong \ker(f^\vee) \in \text{finAbSch}/k.$$

And double duality for Abelian varieties and Cartier-duality are compatible up to a sign: If we identify  $f$  with  $(f^\vee)^\vee$  using  $\kappa_{A/S}$  and  $\kappa_{B/S}$  (20.9.4.3), then the natural isomorphism

$$(\ker(f)^D)^D \cong \ker((f^\vee)^\vee) \cong \ker(f)$$

is the negative of the canonical isomorphism provided by Cartier-duality.

In particular, for each  $n \in \mathbb{Z}_+$ , by (20.9.6.4), there are canonical isomorphisms

$$A[n]^D \cong A^\vee[n].$$

┘

*Proof:* ? Cf. [Abelian Varieties, van der Geer]P100 for the case  $S \in \mathbf{Field}$ , and [Oda, The first de Rham cohomology group and Dieudonne modules]Thm. 1.1, Cor. 1.3 for the general case.  $\square$

**Prop. (20.9.6.6) [Being Isogeny is Clopen].** For  $\varphi : A \rightarrow B \in \mathcal{AbVar}/S$ , the set  $\{s \in S \mid \varphi_s \text{ is an isogeny}\}$  is clopen in  $S$ .  $\square$

*Proof:* Cf. [CC014]P146.  $\square$

Hom( $X, Y$ )

**Def. (20.9.6.7) [Simple Abelian Varieties].** A **simple Abelian variety** is an Abelian variety that has no non-trivial Abelian subvarieties.  $\square$

**Def. (20.9.6.8) [Isotypic Abelian-Varieties].**  $A \in \mathcal{AbVar}/k$  is called an **isotypic Abelian variety** if it is isogenous to  $C^e$  for some  $e \in \mathbb{Z}_+$  and  $C \in \mathcal{AbVar}^{\text{sim}}/k$  (20.9.6.7).  $\square$

**Prop. (20.9.6.9) [Hom( $A, B$ ) and Inseparable Extensions, Chow].** If  $K'/K \in \mathbf{Field}$  and  $K$  is separably closed in  $K'$ , then for any  $A, B \in \mathcal{AbVar}/K$ , the map  $\text{Hom}(A, B) \rightarrow \text{Hom}(A_{K'}, B_{K'})$  is an isomorphism.  $\square$

*Proof:* Cf. [Conrad, Chow's  $K/k$ -image and  $K/k$ -trace, and the Lang-Neron theorem].  $\square$

**Thm. (20.9.6.10) [Poincaré's Complete Reducibility Theorem].** Let  $B$  be an Abelian subvariety of  $A$ , then there exists an Abelian subvariety  $C$  of  $A$  that the addition gives an isogeny

$$B \times C \rightarrow A.$$

┘

*Proof:* Choose an ample line bundle  $c$  on  $A$ , let  $\iota : B \rightarrow A$  be the inclusion and  $\hat{\iota} : \hat{A} \rightarrow \hat{B}$  the dual map, then

$$(\hat{\iota} \circ \lambda_c)|_B = \lambda_{\iota^*(c)}.$$

Since  $\iota^*(c)$  is also ample,  $\varphi_{\iota^*(c)}$  is an isogeny, thus has finite kernel. Let  $C = \ker(\iota \circ \varphi_c)$ , then we have  $C \cap B$  is finite, whence  $B \times C \rightarrow A$  has finite kernel. The dimension theorem (10.1.4.39) applied to  $\iota \circ \varphi_c$  shows

$$\dim C + \dim B^\vee = \dim A^\vee.$$

and this together with (20.9.4.19) shows  $B \times C \rightarrow A$  is a surjection, because  $A$  is irreducible. Thus it is an isogeny.  $\square$

**Cor. (20.9.6.11).** For  $k \in \mathbf{Field}$ ,  $A \in \mathcal{AbVar}/k$ , there are simple Abelian subvarieties  $B_1, \dots, B_n$  of  $A$  that the inclusions give an isogeny

$$B_1 \times \dots \times B_n \rightarrow A.$$

┘

**Prop. (20.9.6.12)**[Hom( $A, A'$ ) is **Finite Free**]. Let  $k$  be a field and  $A, A' \in \mathcal{AbVar}/k$ , then  $\mathrm{Hom}(A_1, A_2)$  w.r.t. the addition is a finite free Abelian group of rank at most  $4g^2$ . ┘

*Proof:* Assume  $[m] \circ \varphi = 0$ , then  $\varphi \circ [m] = 0$ , and use the fact  $[m]$  is surjective (20.9.6.18). The last assertion follows from (21.1.4.3). □

**Def. (20.9.6.13)** [Quasi-Isogenies]. For  $A, A' \in \mathcal{AbVar}/k$ ,  $\mathrm{IsogAbVar}(A, A') \otimes \mathbb{Q}$  is called the set of **quasi-isogenies** from  $A$  to  $A'$ .

Define the category  $\mathrm{IsogAbVar}/k$  to be the category consisting of Abelian varieties over  $k$  with morphisms  $\mathrm{Mor}(A, A') = \mathrm{IsogAbVar}(A, A') \otimes \mathbb{Q}$ . Notice  $f : A \rightarrow A' \in \mathcal{AbVar}/k$  is an isogeny iff  $f$  is an isomorphism in  $\mathrm{IsogAbVar}/k$ , by (7.4.4.20).

Thus by (20.9.6.11),  $\mathrm{IsogAbVar}/k$  is a semisimple  $\mathbb{Q}$ -linear Abelian category. ┘

**Prop. (20.9.6.14)** [ $\mathrm{End}(A)_{\mathbb{Q}}$  is **Semi-Simple**]. For  $A, A' \in \mathcal{AbVar}/k$ , if  $A \sim A'$ , then  $\mathrm{End}(A)_{\mathbb{Q}} = \mathrm{End}(A')_{\mathbb{Q}}$ . Moreover,  $\mathrm{End}_{\mathbb{Q}}(A)$  is a semi-simple  $\mathbb{Q}$ -algebra, and it is simple iff  $A$  is isotypic (20.9.6.8). ┘

*Proof:* These follow from (20.9.6.3) and (20.9.6.11). □

**Def. (20.9.6.15)** [Abelian Varieties of  $\mathrm{GL}(2)$ -Type].  $A \in \mathcal{AbVar}/\mathbb{Q}$  is called an **Abelian variety of  $\mathrm{GL}(2)$ -type** if  $A$  is simple and  $\mathrm{End}(A)_{\mathbb{Q}}$  contains some  $E \in \mathbf{NField}$  with  $[E : \mathbb{Q}] = \dim A$ . ┘

**Thm. (20.9.6.16).**  $A \in \mathcal{AbVar}/\mathbb{Q}$  is of  $\mathrm{GL}(2)$ -type iff it is a simple quotient of  $J_1(N)/\mathbb{Q}$ . ┘

*Proof:* Cf. [Ribet, Abelian Varieties over  $\mathbb{Q}$  and Modular Forms]. And [Serre's Modularity Conjecture 1]P502. ? □

### Tate-Modules

**Prop. (20.9.6.17).** For  $S \in \mathbf{NSch}$ ,  $A \in \mathcal{AbVar}^g/S$ ,  $n \in \mathbb{Z}_+$  and  $n$  is invertible on  $S$ , then  $A[n]$  is a finite locally free  $S$ -group scheme of order  $n^{2g}$ . ┘

*Proof:* ? □

**Prop. (20.9.6.18)** [Multiplication Map, Weil]. Let  $k \in \mathbf{Field}$ ,  $A \in \mathcal{AbVar}/k$ , and  $n \in \mathbb{Z} \cap k^\times$ , then  $[n] : A \rightarrow A$  is finite faithfully flat of degree  $n^{2 \dim A}$ . In particular it is an isogeny (20.9.6.1), and

- If  $n \in \mathbb{Z} \setminus (\mathrm{char} k)$ , then  $[n]$  is finite étale and  $|A[n]| \cong (\mathbb{Z}/n\mathbb{Z})^{2 \dim A}$ .
- If  $n = 0 \in k$ , then  $[n]$  is not separable.
- If  $\mathrm{char} k = p \in \mathbf{P}$ , then  $|A(p)| \cong (\mathbb{Z}/p\mathbb{Z})^r \times \alpha_p^{2g-2r} \times \mu_p^r$  for some  $0 \leq r \leq g$ . In particular,  $A[p^e] = (\mathbb{Z}/(p^e))^r$  for any  $e \geq 0$ . This  $r$  is called the  **$p$ -rank** of  $A$ . It is invariant under isogenies.

┘

*Proof:* 1, 2: Let  $Z = \ker([n])$ , then it is proper. By (20.9.1.30), choose an ample line bundle  $\mathcal{L} = \mathcal{L}(D) \in \text{Pic}(A)$ , then  $\mathcal{O}_Z = \mathcal{L}_Z^{n(n+1)/2} \otimes [-1]^* \mathcal{L}_Z^{n(n-1)/2}$ . As  $\mathcal{L}_Z$  and  $[-1]^* \mathcal{L}_Z$  are both ample,  $\mathcal{O}_Z$  is ample. Now (7.4.4.20) shows  $Z$  is finite. Thus by dimension equation,  $[n]$  is finite and faithfully flat.

The action of  $[n]$  on the tangent space at the origin is given by multiplying by  $n$  by (10.2.2.2), thus it is unramified at the origin iff  $n \in \mathbb{Z} \cap k^*$  by (7.6.5.12), and also on any other closed points by homogeneity. Thus if  $n \in \mathbb{Z} \cap k^*$ ,  $[n]$  is finite étale, and otherwise  $[n]$  is not separable, by (7.6.5.12), as it is not generically unramified.

To calculate  $A[n]$  in the étale case, notice  $[n]^* D = n^2 D$ , and the intersection number formula

$$n^{2g}(D \cdot D \cdot \dots \cdot D) = ([n]^* D \cdot [n]^* D \cdot \dots \cdot [n]^* D) = \deg([n])(D \cdot D \cdot \dots \cdot D)$$

where the intersection is  $g$ -fold, thus  $(D \cdot D \cdot \dots \cdot D)$  equals the degree of the projection embedding of  $X$  via  $\mathcal{L}(D)$  thus positive, so we get  $\deg([n]) = n^{2g} = \#A[n]$  as  $[n]$  is finite étale. To see the structure of  $A[n]$ , notice this is true for any  $m|n$ .

3: ? Cf.[Conrad]. The last assertion follows from the fact the fact  $[p]$  is surjective and the structure theory of Abelian groups.  $\square$

**Def. (20.9.6.19) [Tate-Modules].** For  $A \in \text{AbVar}/k$  and  $\ell \in \text{Prime} \setminus \text{char } k$ , define  $T_\ell(A) \stackrel{\triangle}{=} \varprojlim_{n \geq 0} A[\ell^n]$ , called the **Tate module** of  $A$ . Then it is naturally a  $\mathbb{Z}_\ell$ -module and isomorphic to  $\mathbb{Z}_\ell^{2g}$  by (20.9.6.18).

Also define  $V_\ell(A) = T_\ell(A) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ . There are  $\text{Gal}_k$  actions on them, as  $[\ell^n]$  is étale and  $\text{Gal}_k$  preserves  $A[\ell^n]$  because  $O$  is  $\text{Gal}_k$ -invariant.

For  $\ell = \text{char } k = p > 0$ , we can also define  $T_p(A)$ , which is isomorphic to  $\mathbb{Z}_p^r$  for some  $0 \leq r \leq g$  by (20.9.6.18), but there is not Galois action on it.  $\lrcorner$

**Def. (20.9.6.20) [Adelic Tate Modules].** Situation as in (20.9.6.19), if  $\text{char } k = 0$ , define the **Adelic Tate module** to be

$$T_f(A) = \prod_{\ell \in \text{Prime}} T_\ell(A), \quad V_f(A) = \prod'_{\ell \in \text{Prime}} (V_\ell(A), T_\ell(A)).$$

$\lrcorner$

**Prop. (20.9.6.21) [Étale Cohomology and Tate Modules].** If  $k = k^{\text{sep}}$ ,  $A \in \text{AbVar}^g/k$ , and  $\ell \in \text{Prime} \setminus \text{char } k$ , then

- There are canonical isomorphisms  $H_{\text{ét}}^1(A, \mathbb{Z}_\ell) \cong \text{Hom}_{\mathbb{Z}_\ell}(T_\ell(A), \mathbb{Z}_\ell) \cong T_\ell(A)(1)$ . Thus  $H_{\text{ét}}^1(A, \mathbb{Z}_\ell) \cong \mathbb{Z}_\ell^{2g}$  by (20.9.6.18).
- The cup product defines isomorphisms  $\wedge^r H_{\text{ét}}^1(A, \mathbb{Z}_\ell) \cong H^r(A, \mathbb{Z}_\ell)$  for any  $r > 0$ .

$\lrcorner$

*Proof:* Cf.[Mil08]P55. ?  $\square$

**Cor. (20.9.6.22).** There is a natural  $\text{Gal}_k$ -invariant pairing  $\wedge^r T_\ell(A) \times H_{\text{ét}}^r(A, \mathbb{Z}_\ell) \rightarrow \mathbb{Z}_\ell$ .  $\lrcorner$

**Prop. (20.9.6.23).** For  $A \in \text{AbVar}/k$ ,  $\ell \in \text{Prime} \setminus \{\text{char } k\}$  and  $f \in \text{End}(A)_\mathbb{Q}$ , the representation  $\rho_\ell : \text{End}(A)_\mathbb{Q} \rightarrow \text{End}(V_\ell)$  satisfies:

$$\deg(f) = \det(\rho_\ell(f)).$$

In particular,  $\deg([n] - f) = \det(n - \rho_\ell(f))$  for any  $n \in \mathbb{Z}$ .  $\lrcorner$

*Proof:* ? □

**Thm. (20.9.6.24) [Semisimplicity].** Let  $F \in \mathbf{GField}$ ,  $\ell \in \mathbf{Prime} \setminus \text{char } F$ ,  $A \in \mathbf{AbVar}/F$ , then the action of  $\text{Gal}_F$  on  $V_\ell(A)$  is semisimple. ┘

*Proof:* □

## 7 Polarizations and Weil-Pairings

### Polarizations and Rosati-Involutions

**Def. (20.9.7.1) [Polarizations].** For  $A \in \mathbf{AbVar}/S$ , a **polarization of Abelian variety** of  $X/S$  is a symmetric isogeny  $\lambda : A \rightarrow A^\vee$  s.t. étale-locally  $\lambda$  is of the form  $\lambda_{\mathcal{L}}$  (20.9.4.6) for a relatively ample rigidified line bundle  $\mathcal{L} \in \text{Pic}(A/S)$ . Polarizations are symmetric, by (20.9.4.7).

A **principal polarization of an Abelian variety** is a polarization of degree 1.

The category of Abelian varieties over  $k$  of dimension  $g$  together with a polarization of degree  $d$  is denoted by  $\mathbf{AbVar}^{g, \text{polar}=d}/k$ . ┘

**Prop. (20.9.7.2) [Zarhin's Trick].** For any  $A \in \mathbf{AbVar}/k$ ,  $(A \times A^\vee)^4$  is canonically principally polarized. ┘

*Proof:* Cf. [Mil08] P60. □

**Thm. (20.9.7.3) [Finitely Many Polarizations].** For any  $A \in \mathbf{AbVar}/k$  and  $d \in \mathbb{Z}_+$ , there are only f.m. isomorphism classes of polarizations on  $A$  of degree  $d$ . ┘

*Proof:* Cf. [Mil08] P63. □

**Def. (20.9.7.4) [Rosati-Involutions].** For  $A \in \mathbf{AbVar}/k$  and  $\lambda : A \rightarrow A^\vee$  a polarization (20.9.7.1), the inverse  $\lambda^{-1} \in \text{Hom}(A^\vee, A)_{\mathbb{Q}}$  is definable by (20.9.6.3). Then the **Rosati-involution** corresponding to  $\lambda$  is defined to be the map

$$\text{End}(A)_{\mathbb{Q}} \rightarrow \text{End}(A)_{\mathbb{Q}} : \alpha \mapsto \lambda^{-1} \circ \alpha^\vee \circ \lambda.$$

Then it is an involution. ┘

**Prop. (20.9.7.5) [Rosati Involutions are Positive].** For  $A \in \mathbf{AbVar}/k$ , the bilinear form

$$(\alpha, \beta) \mapsto \text{tr}(\alpha \circ \beta^\dagger) : \text{End}(A)_{\mathbb{Q}}^2 \rightarrow \mathbb{Q}$$

is positive definite. ┘

*Proof:* □

**Prop. (20.9.7.6).** Let  $k \in \mathbf{Field}$  and  $(A, \lambda) \in \mathbf{AbVar}/k$ , then

- $\# \text{Aut}((A, \lambda)) < \infty$ .
  - for any  $n \geq 3$ , any automorphism of  $(A, \lambda)$  acting trivially on  $A[n]$  is the identity.
- ┘

*Proof:* Cf. [Milne, P62] ? □



**Weil-Pairings**

**Thm. (20.9.7.7)[Weil-Pairings].** Let  $S \in \text{Sch}$  and  $X, Y \in \text{AbVar}/S$ , then for any isogeny  $\alpha : X \rightarrow Y$ , there is a perfect pairing

$$e_\alpha : \ker(\alpha) \times \ker(\alpha^\vee) \rightarrow \mathbb{G}_m.$$

┘

*Proof:* Cf.[Mumford, P143] **?**.

□

**Def. (20.9.7.8)[Induced Pairings].** For  $k \in \text{Field}$ ,  $A \in \text{AbVar}$ , if  $e_m$  is any Weil pairing, and  $\lambda : A \rightarrow A^\vee \in \text{IsogAbVar}/k$ , denote

$$e_m^\lambda = e_m \circ (\text{id} \times \lambda) : A[m] \times A[m] \rightarrow \mu_m(1).$$

┘

**Def. (20.9.7.9).** Let  $k \in \text{Field}$ ,  $\ell \in \mathbb{P}$  and  $\text{char } k \neq 2, \ell$ . Then for  $A \in \text{AbVar}/k$ ,  $\lambda \in \text{Hom}(A, A^\vee)$  is

┘

**Prop. (20.9.7.10)[Weil-Pairings for Elliptic Curves].** Let  $E \in \text{EllCur}/k$ , then for any  $m \in \mathbb{Z} \cap k^*$ , there is a non-degenerate pairing

$$e_m : E[m] \times E[m] \rightarrow \mu_m$$

that is alternating, non-degenerate and  $\text{Gal}_k$ -invariant.

Moreover,  $e_{mm'}(S, T) = e_m([m']S, T)$  for  $S \in E[mm']$  and  $S \in T[m]$ . In particular, for  $\ell \in \mathbb{P} \setminus \text{char } k$ , we can define a Weil pairing on the Tate module

$$e : T_\ell(E) \times T_\ell(E) \rightarrow \mu_{\ell^\infty}.$$

┘

*Proof:* Let  $T \in E[m]$ , then by (21.3.1.23), there exists some  $f \in \overline{K}(E)^*$  s.t.  $\text{div}(f) = m[E] - m[O]$ . Let  $T' \in E[m^2]$  s.t.  $mT' = T$ , then there exists a function  $g \in \overline{K}(E)^*$  s.t.

$$\text{div}(g) = [m]^*(T) - [m]^*(O) = \sum_{R \in E[m]} ([T' + R] - [R]).$$

Notice  $f \circ [m]$  and  $g^m$  have the same divisor, so we may assume  $f \circ [m] = g^m$ . Then for any  $S \in E[m]$  and  $X \in E$ ,  $g(S + X)^m = f(mX) = g(X)^m$ , so  $E \rightarrow \mathbb{P}^1 : g(X + S)/g(X)$  is constant and has value in  $\mu_m$ , denote it by  $e(T, S)$ .

For bilinearity, Cf.[Sil16]P94 **?**.

For alternating, Cf.[Sil16]P94 **?**.

For non-degeneracy, if  $e_m(S, T) = 1$  for any  $T \in E[m]$ , then  $g$  is  $E[m]$ -invariant, which implies  $g = h \circ [m]$  for some  $h \in \overline{K}(E)^*$  by (20.9.6.2) and the fact  $[m]$  is separable. Then  $(h \circ [m])^m = f \circ [m]$ , so  $f = h^m$ , and  $\text{div}(h) = [T] - [O]$ , so  $T = O$ .

Galois invariance is clear.

The last assertion follows by taking  $g' = g \circ [m']$  and  $f' = f^{m'}$ .

□

**Cor. (20.9.7.11)[Primitive Pairing].** Let  $E \in \text{EllCur}/k$ , then for any  $m \in \mathbb{Z} \cap k^*$ ,  $\mu_m \subset K(E[m])$ , and there exists some  $S, T \in E[m]$  s.t.  $e_m(S, T)$  is a primitive  $m$ -th roots of unity.

┘

*Proof:* If the subgroup generated by all  $e_m(S, T)$  is  $\mu_d$ , then  $e_m([d]S, T) = 1$  for any  $S, T \in E[m]$ , so  $[d]S = 0$  by non-degeneracy, so  $d = m$  because  $\#E[m] = (\mathbb{Z}/(m))^2$ . Then the  $\text{Gal}_k$ -invariance of  $e$  shows  $\mu_m \subset K(E[m])$ .  $\square$

**Prop. (20.9.7.12) [Duality with Isogenies].** Let  $k \in \mathbf{Field}$  and  $\varphi : E_1 \rightarrow E_2 \in \text{IsogAbVar}_k^1$ ,  $m \in \mathbb{Z} \in k^*$ , then for  $S \in E_1[m], T \in E_2[m]$ ,

$$e_m(S, \varphi^\vee(T)) = e_m(\varphi(S), T).$$

In particular, for  $S \in T_\ell(E_1)$  and  $T \in T_\ell(E_2)$ ,  $e(S, \varphi^\vee(T)) = e(\varphi(S), T)$ .  $\lrcorner$

*Proof:* Cf. [Sil16]P97?.  $\square$

**Prop. (20.9.7.13).** Let  $E \in \text{EllCur}/k$  and  $\varphi \in \text{End}(E)$ ,  $\ell \in \mathbf{P} \setminus \text{char } k$ , then  $\det(\varphi_\ell) = \deg(\varphi)$ ,  $\text{tr}(\varphi_\ell) = 1 + \deg(\varphi) - \deg(1 - \varphi)$ .  $\lrcorner$

*Proof:* For any  $v, v' \in T_\ell(E)$ , use the Weil pairing:

$$e(v, v')^{\deg(\varphi)} = e([\deg(\varphi)]v, v') = e(\varphi_\ell v, \varphi_\ell v') = e(v, v')^{\det(\varphi_\ell)},$$

thus  $\deg(\varphi) = \det(\varphi_\ell)$ . The latter equality follows from the identity  $\text{tr}(A) = 1 + \det(A) - \det(1 - A)$  for any  $A \in M_2(R)$ .  $\square$

## 8 over Archimedean Local Fields

**Prop. (20.9.8.1) [Complex Abelian Varieties and Complex Tori].** By (16.10.5.9) and GAGA (16.9.7.17),  $\text{AbVar}/\mathbb{C}$  is the same as the category of complex tori  $V/\Lambda$  with a Riemann form  $\omega : \wedge^2 \Lambda \rightarrow \mathbb{Z}$  (16.10.5.9).

Such a complex tori  $V/\Lambda$  is called a **polarizable integral Hodge structure**.  $\lrcorner$

**Def. (20.9.8.2) [Types of Polarizations].** Given a complex tori  $X = V/\Lambda$  of dimension  $d$  with a Riemann form  $\omega$ , there exists a basis of  $\Lambda$  and  $d_1 | d_2 | \dots | d_n \in \mathbb{Z}_+$  s.t. the matrix  $\omega$  w.r.t. this basis is

$$\begin{bmatrix} & D \\ -D & \end{bmatrix}, \quad D = \text{diag}(d_1, \dots, d_n),$$

then  $X$  is said to have type  $D$ .  $\lrcorner$

**Prop. (20.9.8.3) [Abelian Varieties with Polarizations].** For  $n \in \mathbb{Z}_+$  and  $d_1 | d_2 | \dots | d_n \in \mathbb{Z}_+$ , the set of polarized Abelian varieties over  $\mathbb{C}$  of type  $D = \text{diag}(d_1, \dots, d_n)$  with a symplectic basis is in bijection with the set of  $\text{GL}(n, \mathbb{C})$  orbits of isomorphisms of real vector spaces  $\Pi_{\mathbb{R}} : U_D \otimes \mathbb{R} \rightarrow \mathbb{C}^g$  s.t. for any  $x, y \in V$ ,

$$(\Pi_{\mathbb{R}}^{-1}ix, \Pi_{\mathbb{R}}^{-1}iy)_D = (\Pi_{\mathbb{R}}^{-1}x, \Pi_{\mathbb{R}}^{-1}y)_D,$$

and the symmetric form

$$(x, y) \mapsto (\Pi_{\mathbb{R}}^{-1}ix, \Pi_{\mathbb{R}}^{-1}y)_D$$

is positive definite.

And it is moreover canonical in bijection to the Siegel upper half space  $\mathcal{H}_n$  (16.2.6.6).  $\lrcorner$

*Proof.* For any  $X = V/U \in \text{AbVar}^n/\mathbb{C}$  with a Riemann form  $\omega$  of type  $D$  and a basis  $\{u_1, \dots, u_n, v_1, \dots, v_n\}$  of  $U$  s.t. the matrix of  $\omega$  is  $\begin{bmatrix} & D \\ -D & \end{bmatrix}$ . Then by choosing any complex basis  $e_1, \dots, e_n$  of  $V$ , we get an isomorphism  $U \otimes \mathbb{R} \cong \mathbb{C}^g$ , and we can represent  $\{u_i\}$  in the basis  $\{e_i\}$  s.t.

$$(e_1, \dots, e_n, ie_1, \dots, ie_n) = \Pi_{\mathbb{R}}(u_1, \dots, u_n, v_1, \dots, v_n), \quad \Pi_{\mathbb{R}} = \begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{bmatrix} \in \text{Mat}(2n, \mathbb{R}).$$

Then

$$(e_1, \dots, e_n)(\Pi_{11} + i\Pi_{21}, \Pi_{21} + i\Pi_{22})(u_1, \dots, u_n),$$

and  $\Pi = (\Pi_{11} + i\Pi_{21}, \Pi_{21} + i\Pi_{22})$  as in (16.10.5.11). Then the first assertion follows from (16.10.5.9). Notice the  $\text{GL}(n, \mathbb{C})$ -orbit relation comes from the arbitrariness of the choice of basis  $e_1, \dots, e_n$ .

For the second assertion, notice that in each  $\text{GL}(n, \mathbb{C})$ -orbit, there is a canonical choice of basis  $\{e_1, \dots, e_n\}$  s.t.  $e_i = \frac{1}{d_i}v_i$ . Then in this basis,

$$\Pi_{\mathbb{R}} = \begin{bmatrix} \Pi_{11} & D \\ \Pi_{21} & 0 \end{bmatrix}, \quad \Pi = (\Pi_{11} + i\Pi_{12}, D).$$

So if  $Z = \Pi_{11} + i\Pi_{12}$ , the Riemann relations (16.10.5.11) are just equivalent to

$$\{Z^t = Z, \quad \text{Im}(Z) \in \text{Pos}(n, \mathbb{C})\}.$$

□

**Cor. (20.9.8.4).** For  $n \in \mathbb{Z}_+$  and  $d_1|d_2|\dots|d_n \in \mathbb{Z}_+$ , the set of polarized Abelian varieties over  $\mathbb{C}$  of type  $D = \text{diag}(d_1, \dots, d_n)$  with a symplectic basis is in bijection with the set of homomorphisms of real algebraic groups  $h_1 : S_1 \rightarrow \text{Sp}_{\mathbb{R}}(U_D)$  satisfying

- The complexification  $h_{1,\mathbb{C}} : \mathbb{C}^\times \rightarrow \text{Sp}_{\mathbb{R}}(U_D \otimes \mathbb{C})$  has weights  $\pm 1$  with multiplicities  $n$ .
- the symmetric form  $(x, y) \mapsto (h_1(i)x, y)_D$  is positive definite.

┘

**Cor. (20.9.8.5) [Classifying Abelian Varieties with Polarizations].** It follows from (20.9.8.3) that for  $n \in \mathbb{Z}_+$  and  $d_1|d_2|\dots|d_n \in \mathbb{Z}_+$ , the set of isomorphisms of Abelian varieties over  $\mathbb{C}$  of type  $D = \text{diag}(d_1, \dots, d_n)$  with a symplectic basis is bijection with the set

$$\text{Sp}_D(\mathbb{Z}) \backslash \mathcal{H}_n.$$

┘

**Prop. (20.9.8.6) [Dual Riemann Forms].** If  $X = V/\Lambda$  is a complex torus with a Riemann form  $\omega$ , then the dual complex torus (16.10.5.8)  $X^\vee$  also has a Riemann form defined as follows: any  $f \in V^*$  corresponds to an element  $v_f$  s.t.  $f(u) = \omega(u, v_f)$ . Thus we can define  $\omega'(f, g) = \omega(v_f, v_g)$ . And there is an isogeny (i.e. surjective isomorphism with finite kernel)

$$A \rightarrow A^\vee : \bar{a} \mapsto [x \mapsto \omega(x, a)]$$

which is compatible with the Riemann forms.

┘

**Prop. (20.9.8.7) [Rosati-Involutions are Positive].** Let  $X = V/\Lambda \in \mathcal{AbVar}/\mathbb{C}$ , then for any  $\alpha \in \text{End}_{\mathbb{Q}}(X)$ , there exists  $\alpha' \in \text{End}_{\mathbb{Q}}(X)$  s.t.

$$\omega(\alpha x, y) = \omega(x, \alpha' y).$$

And this determines an involution on  $\text{End}_{\mathbb{Q}}(X)$ , called the **Rosati-involution**. Notice this is compatible with that defined in (20.9.7.4), by using (20.9.8.6).

Then this is a positive involution on  $\text{End}_{\mathbb{Q}}(X)$ .  $\lrcorner$

*Proof:* Notice the form  $\omega_J : (x, y) \mapsto \omega(x, Jy)$  is symmetric and positive, and

$$\omega_J(\alpha x, y) = \omega(\alpha x, Jy) = \omega(x, \alpha' Jy) = \omega(x, J\alpha' y) = \omega_J(x, \alpha' y).$$

So this involution is positive by (21.2.1.20).  $\square$

### Elliptic Curve Case

**Prop. (20.9.8.8) [Complex Tori as Elliptic Curves].** Let  $\Lambda$  be a complete lattice of  $\mathbb{C}$ , then we can make  $\mathbb{C}/\Lambda$  into a Riemannian surface as the quotient space of  $\mathbb{C}$ . Then it is positive by??t is a smooth curve of genus 1 with the origin as the rational point, which means it is an elliptic curve.  $\lrcorner$

**Prop. (20.9.8.9) [Elliptic Curve as Complex Tori].** For an elliptic curve  $\mathbb{C}/\Lambda$  over  $\mathbb{C}$ , by (12.5.6.7), the Weierstrass functions (12.5.6.5)  $\wp, \wp'$  are just the rational functions in (21.3.1.7), so the map

$$z \mapsto [\wp(z), \wp'(z), 1]$$

is biholomorphic from  $\mathbb{C}/\Lambda$  to the elliptic curve defined by the equation  $y^2 = 4x^3 - g_2(\Lambda)x - g_3(\Lambda)$  that  $g_2^3 - 27g_3^2 \neq 0$ . And it also preserves the group structure.  $\lrcorner$

*Proof:* To show that the homomorphism preserves group structure, notice that if  $z_1, z_2, z_3$  maps to three points that is colinear, then they satisfy a equation

$$f(z) = a\wp(z) + b\wp'(z) + c.$$

If  $b \neq 0$ , then this is a meromorphic function with three poles, thus three zeros, which is exactly  $z_1, z_2, z_3$ , so by (12.5.6.2),  $z_1 + z_2 + z_3 \equiv 0 \pmod{\Lambda}$ . If  $b = 0$ , then  $z_3 = 0$ , which corresponds to the point  $(1, 0, 0) \in \mathbb{P}^2$ , then the same argument shows  $z_1 + z_2 + 0 \equiv 0 \pmod{\Lambda}$ .  $\square$

**Lemma (20.9.8.10) [Isogenies and Homogeneities].** If  $E_1, E_2$  are elliptic curves in  $\mathbb{P}^2$  corresponding to  $\Lambda_1, \Lambda_2$  via (20.9.8.9) resp., then the natural map

$$\{\text{isogenies } E_1 \rightarrow E_2\} \rightarrow \{\text{non-constant holomorphic maps } \varphi : \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2, \varphi(0) = 0\}$$

is a bijection.  $\lrcorner$

*Proof:* By (16.10.5.2), it suffices to show for any lattices  $\Lambda_1, \Lambda_2$  and  $\alpha \in \mathbb{C}^*$  that  $\alpha\Lambda_1 \subset \Lambda_2$ , the map

$$[\wp(z, \Lambda_1), \wp'(z, \Lambda_1), 1] \rightarrow [\wp(\alpha z, \Lambda_2), \wp'(\alpha z, \Lambda_2), 1]$$

is a morphism. For this, notice  $\wp(\alpha z, \Lambda_2)$  is an elliptic function for  $\Lambda_1$ , thus by (12.5.6.7), it is a rational function of  $\wp(z, \Lambda_1)$  and  $\wp'(z, \Lambda_1)$ .  $\square$

**Lemma (20.9.8.11).** Let  $E \in \text{EllCur}/\mathbb{C}$  be contained in  $\mathbb{P}^2$ , defined by a Weierstrass equation, then there exists a lattice  $\Lambda \subset \mathbb{C}$ , unique up to homothety, that the embedding given in (20.9.8.9) induces an isomorphism  $\mathbb{C}/\Lambda \cong E(\mathbb{C})$ .  $\lrcorner$

*Proof:* This is a consequence of (12.5.6.8) and (21.3.1.15).  $\square$

**Thm. (20.9.8.12) [Elliptic Curves and Complex Tori, Weierstrass].** The following categories are equivalent:

- $\text{IsogAbVar}^1/\mathbb{C}$ .
- Category of complex tori with morphisms homomorphisms preserving 0.
- Category of lattices in  $\mathbb{C}$  with homotheties as morphisms.

$\lrcorner$

*Proof:* This is a consequence of (20.9.8.9)(20.9.8.10) and (16.10.5.2).  $\square$

**Cor. (20.9.8.13) [ $j$ -Functions Coincide].** In this correspondence, If  $E \cong E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$  for some  $\tau \in \mathcal{H}$ , then  $E$  is given by the Weierstrass equation

$$W : y^2 = x^4 - \frac{g_2(\Gamma)}{4}x - \frac{g_3(\Gamma)}{4}.$$

Then

$$\Delta(W) = \Delta(\tau), \quad j(E) = j(\tau).$$

$\lrcorner$

*Proof:* Comparing the definitions and use the definitions of  $E_4, E_6, G_4, G_6, g_2, g_3$  (26.3.4.5)(12.5.6.7), we get

$$g_2 = 60G_4 = 120\zeta(4)E_4 = \frac{4}{3}\pi^4 E_4 \text{ (31.3.5.3),}$$

$$g_3 = 140G_6 = 280\zeta(6)E_6 = \frac{8}{27}\pi^6 E_6 \text{ (31.3.5.3).}$$

So by (21.3.1.6),

$$j(W) = j(\tau) = 1728 \frac{E_4(\tau)^3}{E_4^3(\tau) - E_6^2(\tau)}.$$

And for  $\Delta$ , because  $\Delta(\tau) = (2\pi)^{12} E_4^3(\tau)/j(\tau)$  and  $\Delta(W) = c_4^3/j(\tau)$  (21.3.1.5), it suffices to show that  $c_4 = (2\pi)^4 E_4$ . But

$$c_4 = 48g_2 = (2\pi)^4 E_4.$$

$\square$

**Prop. (20.9.8.14).** Let  $E \in \text{EllCur}/\mathbb{C}$  and  $\Lambda$  a lattice in  $\mathbb{C}$  that  $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$ , then

- there is a natural isomorphism  $H_1(E(\mathbb{C}), \mathbb{Z}) \cong \Lambda : \gamma \mapsto \int_\gamma dz$ .
- There is a natural isomorphism  $H_1(E(\mathbb{C}), \mathbb{Z}/(m)) \cong E[m]$ .

$\lrcorner$

*Proof:* Cf. [Sil16]P176.  $\square$

Over  $\mathbb{R}$ 

**Prop. (20.9.8.15).** Let  $E \in \text{EllCur}/\mathbb{R}$ , then there exists a unique  $\tau$  in the set

$$\mathcal{C} = \{it | t \geq 1\} \cup \{e^{i\theta} | \pi/3 \leq \pi/2\} \cup \{1/2 + it | t \geq \sqrt{3}/2\}.$$

s.t.  $j(\tau) = j(E)$ . And by (21.3.6.3), each  $\tau \in \mathcal{C}$  corresponds to exactly two isomorphism classes of elliptic curves over  $\mathbb{R}$ .  $\lrcorner$

*Proof:* Firstly these  $\tau$  satisfies  $j(\tau) \in \mathbb{R}$  because by the power expansions of  $j(\tau)$  (26.3.4.22), for  $\tau = it$  or  $1/2 + it$ ,  $q \in \mathbb{R}$ , thus  $j(\tau) \in \mathbb{R}$ . For  $\tau = e^{i\theta}$ ,  $\overline{j(e^{i\theta})} = j(-e^{-i\theta}) = j(e^{i\theta})$  as  $j \in M_0(\Gamma(1))$ .

Next it can be proven  $j(i\infty) = +\infty$  and  $j(1/2 + i\infty) = -\infty$ , and  $j$  is injective on  $\mathcal{C}$  by (26.3.4.22). Thus the assertion follows by continuity.  $\square$

**Remark (20.9.8.16).** Notice by action of  $\alpha = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \in \text{SL}(2, \mathbb{Z})$  on  $\{e^{i\theta} | \pi/3 \leq \pi/2\}$ , we can also replace  $\mathcal{C}$  by

$$\mathcal{C}' = \{it | t \geq 1\} \cup \{1/2 + it | t > 1/2\}.$$

 $\lrcorner$ **9 char  $k > 0$  Case**

**Def. (20.9.9.1) [Newton-Polygons of Abelian Varieties].** For  $p \in \text{Prime}$ ,  $k \in \text{Field}^p$  and  $A \in \text{AbVar}^g/k$ , define the **Newton-polygon of Abelian variety** of  $A$  to be  $\text{NP}(A) = \text{NP}(A[p^\infty])$  (27.10.1.38). Notice the slopes appear in pairs because  $A$  is isogenous to  $A^\vee$ , and using (27.10.1.37) (27.10.1.3).

$A$  is called an **ordinary Abelian variety** if  $\text{NP}(A) = 0^g 1^g$ . And  $A$  is called a **supersingular Abelian variety** if  $\text{NP}(A) = (\frac{1}{2})^{2g}$ .  $\lrcorner$

**Prop. (20.9.9.2) [Base Change of Isotypic Abelian Varieties].** If  $k'/k \in \text{Field}^{\text{fin}}$  and  $A \in \text{AbVar}/k$  is isotypic (20.9.6.8), then  $A_{k'}$  is also isotypic. (This doesn't holds true for  $k'/k = \mathbb{C}/\mathbb{R}$ ).  $\lrcorner$

*Proof:* We may assume that  $A$  is simple. By (20.9.6.14), it suffices to show that  $\text{End}(A_{k'})_{\mathbb{Q}}$  is simple. By (20.9.6.9), we may assume that  $k'/k$  is algebraic. And because  $\text{End}(A_{k'})$  is f.g. (20.9.6.12), we may assume that  $k'/k$  is finite.

If  $B'$  is a simple isogeny factor of  $A_{k'}$ , then  $\text{Res}_{k'/k}(B')$  is an isogeny factor of  $\text{Res}_{k'/k}(A_{k'}) \sim A^{[k:k]}$ , so  $\text{Res}_{k'/k}(B')$  is isogenous to a power of  $A$ . Then  $(\text{Res}_{k'/k}(B'))_{k'} = \prod_{\sigma \in \text{Gal}(k'/k)} \sigma^*(B') \sim (B')^{[k':k]}$  is isogenous to a power of  $A_{k'}$ . So  $B'$  is isogenous to a non-trivial power of  $A$ , which implies that  $A_{k'}$  is isotypic.  $\square$

**Prop. (20.9.9.3) [Finiteness over Finite Fields].** If  $k \in \text{Field}^{\text{fin}}$ , then for any  $g \in \mathbb{Z}_+$ ,  $\#\text{AbVar}^g/k < \infty$ .  $\lrcorner$

*Proof:* Firstly we show for  $A$  principally polarized by an ample line bundle  $\mathcal{L}$ . Then  $\chi(\mathcal{L}) = 1$  by (20.9.5.2), and  $\mathcal{L}^{\otimes 3}$  is very ample by (20.9.5.8), so it embeds  $A$  as a variety of degree  $3^g \cdot g!$  in  $\mathbb{P}^{3^g - 1}$  by (20.9.5.2). Such a variety is determined by its Chow form, which is a homogenous polynomial of degree  $3^g \cdot g! \cdot d$  in each of  $g + 1$  sets of  $3^g$  variables **?**. And the number of such polynomials are finite.

The general case can be reduced to the principal polarized case by Zarhin's trick (20.9.7.2).  $\square$

**Prop. (20.9.9.4).** Let  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}}$ , and  $(A, \lambda) \in \mathcal{AbVar}^{\text{polar}}/\mathbb{F}_q$  with Rosati involution  $\dagger$ , then

$$\text{Frob}_{q,A}^{\dagger} \circ \text{Frob}_{q,A} = [q].$$

┘

*Proof:* Cf. [Mil08]P76. ?

□

### Honda-Tate Theory

References are [Tat66, Hon68].

**Prop. (20.9.9.5) [Weil-Conjecture for Abelian Varieties].** For  $p \in \text{Prime}$ ,  $r \in \mathbb{Z}_+$ ,  $q = p^r$ ,  $\ell \in \text{Prime} \setminus \ell$ , and  $A \in \mathcal{AbVar}^g/\mathbb{F}_q$ , let  $\varphi_A = \varphi^r$  be the Frobenius of  $A$ , and let  $P_{\varphi}(X)$  be the characteristic polynomial of the action of  $\varphi_A$  on  $T_{\ell}(A)$ , then

- $P_{A,\ell}(X) \in \mathbb{Z}[X]$ , and is invariant of  $\ell$  chosen. Thus we omit  $\ell$  from now on.
- $P_A(X) = \prod_{i=1}^{2g} (X - \alpha_i)$ , where  $\alpha_i \in \mathbb{C}$  and  $|\alpha_i| = q^{1/2}$ .
- $\#A(\mathbb{F}_{q^m}) = \prod_{i=1}^{2g} (1 - \alpha_i^m)$ .

In particular,

$$|\#A(\mathbb{F}_{q^m}) - q^{mg}| \leq 2gq^{m-\frac{1}{2}} + (4g - 2g - 1)q^{m(g-1)}.$$

┘

*Proof:* These follow from (20.9.6.21) and Weil-conjecture (9.6.2.13)(31.1.3.5). □

**Prop. (20.9.9.6) [Tate].** For  $k \in \text{Field}^{\text{fin}}$  and  $A \in \mathcal{AbVar}/k$ , we have

$$Z(\text{End}^0(A)) = \mathbb{Q}(\varphi_{A/k}).$$

┘

*Proof:* Cf. [Mumford, Abelian Varieties]App.I, Thm. 3(a). ?

□

**Thm. (20.9.9.7) [Honda-Tate].** For  $k \in \text{Field}^{\text{fin}}$  and  $A \in \mathcal{AbVar}^{\text{sim},g}/k$ , by (20.9.9.5), for any embedding  $\mathbb{Q}(\varphi_A) \hookrightarrow \mathbb{C}$ ,  $\iota(\alpha)$  is a Weil  $q$ -integer (19.3.1.16), and we get a map of sets

$$A \mapsto [\varphi_A] : \text{IsogAbVar}^{\text{sim}}/k \rightarrow \text{Weil}(\#k^{1/2})/(\sim \text{conjugations}).$$

Then this is a bijection.

┘

*Proof:* The injectivity follows from (21.1.4.5) and the irreducibility of the action of  $\varphi_A$  on  $V_{\ell}(A)$  ?, and the surjectivity is proven by Honda ?? ? (or [Oort, Abelian varieties over finite fields], [Chai-Oort]). The proof of surjectivity uses abelian varieties in characteristic 0 (in fact, it uses descents to number fields of CM abelian varieties over  $\mathbb{C}$ ). □

**Cor. (20.9.9.8).** For  $k \in \text{Field}^{\text{fin}}$ ,  $\ell \in \text{Prime} \setminus \{\text{char } k\}$ , two Abelian varieties  $A, B \in \mathcal{AbVar}^g/k$  are isogenous iff  $\det(\varphi_A|H^1(A_{\bar{k}}, \mathbb{Q}_{\ell})) = \det(\varphi_B|H^1(B_{\bar{k}}, \mathbb{Q}_{\ell}))$ . ┘

*Proof:* ?

□

**Cor. (20.9.9.9)**[Description of  $\text{End}(A)_{\mathbb{Q}}$  in terms of  $\varphi_A$ , Tate[Tat66]]. For  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}_+}$  and  $A \in \text{AbVar}^{\text{sim},g}/\mathbb{F}_q$ , let  $D = \text{End}(A)_{\mathbb{Q}}$ ,  $\varpi = \varphi_A \in D$ , then  $[D : \mathbb{Q}]_{\text{red}} = 2g$ (3.6.4.22), and exactly one of the following happens:

- $\varpi = \sqrt{q} \in \mathbb{Z}_+$ , and  $D$  is the unique central quaternion division algebra over  $\mathbb{Q}$  ramified exactly at  $\{p, \infty\}$ . In which case  $A$  is a supersingular elliptic curve, and  $\text{End}^0(E_{\overline{\mathbb{F}_p}}) = \text{End}^0(E)$ .
- $\varpi = \sqrt{q} \notin \mathbb{Z}_+$ , and  $D$  is the unique central quaternion division algebra over  $\mathbb{Q}(\sqrt{p})$  ramified exactly at the two infinite places. In which case  $A = \text{Res}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(E)$  where  $E \in \text{EllCur}/\mathbb{F}_{q^2}$  is supersingular and  $\text{End}^0(E_{\overline{\mathbb{F}_p}}) = \text{End}^0(E)$ .
- $\mathbb{Q}(\varpi)$  is a CM-field, and  $D \in \text{Az}_{\mathbb{Q}(\varpi)}$  s.t.  $D$  is split at all places away from  $p$ , and for  $v \in S_{\mathbb{Q}(\varpi)}(p)$ ,  $\text{inv}_v(D) = \frac{\text{ord}_v(\varpi)}{\text{ord}_v(q)}[\mathbb{Q}(\varpi)_v : \mathbb{Q}_p] \in \mathbb{Q}/\mathbb{Z}$ .

In particular,  $A$  is a strictly CM Abelian variety(21.2.2.1). ┘

*Proof:* ? Cf.[Tat66] or [Waterhouse and Milne, 1971] or [CCO14]P71. □

## 10 Jacobian Varieties and Abanese Varieties

**Notation (20.9.10.1).**

- Let  $k \in \text{Field}$ .
- ┘

**Prop. (20.9.10.2)**[Picard Schemes of Curves]. Let  $C \in \text{Cur}/k$  be a smooth complete precurve of genus  $g$  with a rational point, then

- $\underline{\text{Pic}}_{C/k}$  is representable by a disjoint union of smooth complete varieties  $\underline{\text{Pic}}_{C/k}^d$ ,  $d \in \mathbb{Z}$  where  $\underline{\text{Pic}}_{C/k}^d$  corresponds to relative line bundles of degree  $d$ .
  - $\underline{\text{Pic}}_{C/k}^0$  is an Abelian variety over  $k$ .
  - for  $d \geq 0$ , the Abel map  $\text{Div}_{C/k}^d \rightarrow \underline{\text{Pic}}_{C/k}^d$  is surjective for  $d \geq g$  and smooth for  $d \geq 2g - 1$ .
  - The morphism  $\text{Div}_{C/k}^g \rightarrow \underline{\text{Pic}}_{C/k}^g$  is birational.
- ┘

*Proof:* Cf.[Sta]0BA0 or [Neron Models] or(20.9.10.3)? . item1 follows from(9.1.11.5) □

**Prop. (20.9.10.3)**[Picard Scheme of Relative Curves]. If  $X/S$  is locally projective, flat with fibers all curves,  $m \in \mathbb{Z}$ , let  $\underline{\text{Pic}}_{X/S}^m$  denote the subfunctor of  $\underline{\text{Pic}}_{X/S}$  representing invertible sheaves  $\mathcal{L}$  with  $\deg(\mathcal{L}) = m$ , then

- $\underline{\text{Pic}}_{X/S}^m$  are clopen subschemes of  $\underline{\text{Pic}}_{X/S}$  of f.t. and form a disjoint cover of it, and forming it commutes with base change.
  - $\underline{\text{Pic}}_{X/S}^{(0)} = \underline{\text{Pic}}_{X/S}^0 = \underline{\text{Pic}}_{X/S}^{\tau}$ , and each  $\underline{\text{Pic}}_{X/S}^m$  is a fppf-torsor under  $\underline{\text{Pic}}_{X/S}^0$ .
  - If  $X/S$  is projective and  $S$  is Noetherian, then each  $\underline{\text{Pic}}_{X/S}^m$  is quasi-projective over  $S$ .
- ┘

*Proof:* Cf.[Kle05]P60. ? □

**Def. (20.9.10.4)**[Jacobians of Curves]. For a smooth complete precurve over  $k$ , its **Jacobian variety**  $\text{Jac}(C)$  is just its Picard variety  $\underline{\text{Pic}}_{C/k}^0$ (20.9.10.2). ┘



**Cor. (20.9.10.5).** For a smooth complete precurve  $C/k$  with a rational point,  $x \in \text{Pic}_{C/k}$  is in  $\text{Jac}(C)$  iff the corresponding line bundle  $x$  on  $C_{k(x)}$  has degree 0, by (20.9.10.2).  $\lrcorner$

**Prop. (20.9.10.6).** If  $C$  is a smooth complete curve of dimension  $g$ , then  $\text{Jac}(C)$  has dimension  $g$ , by (10.9.3.25) and (10.9.3.23).  $\lrcorner$

**Def. (20.9.10.7)[Symmetric Product].** Let  $C$  be a smooth curve, then the symmetric product  $C^{(r)} = C^r/S_r$  is a smooth variety of dimension  $r$  by considering the symmetric functions.  $\lrcorner$

**Prop. (20.9.10.8)[Generic Riemann-Roch].**  $\lrcorner$

**Prop. (20.9.10.9)[Polarization of Jacobians].**  $\lrcorner$

**Prop. (20.9.10.10)[Push and Pull].** For a non-constant morphism  $\varphi : C_1 \rightarrow C_2$  of pointed smooth curves over  $k$ , the push and pull induce maps

$$\varphi_* : \text{Jac}(C_1) \rightarrow \text{Jac}(C_2), \quad \varphi^* : \text{Jac}(C_2) \rightarrow \text{Jac}(C_1).$$

$$\text{s.t. } \varphi_* \circ \varphi^* = [\deg(\varphi)].$$

*Proof:*

$\square$

**Prop. (20.9.10.11)[Abel-Jacobi].** Let  $X/\mathbb{C}$  be a smooth curve with a rational point  $x_0$ , then  $\text{Jac}(C)$  is isomorphic to  $\Omega_{\text{hol}}^1(X)^\vee/H_1(X, \mathbb{Z})$  as complex manifolds via

$$\mathcal{L}(\sum_x n_x x) \mapsto \sum_x n_x \int_{x_0}^x.$$

$\lrcorner$

*Proof:*

$\square$

**Prop. (20.9.10.12).** Let  $C/k$  be a smooth complete curve and  $P \in C(k)$ , then for any  $\ell \in \mathbf{P}$ , the map  $f_P : C \rightarrow \text{Jac}(C)$  induce isomorphisms

$$H^1(\text{Jac}(C), \mathcal{O}_{\text{Jac}(C)}) \cong H^1(C, \mathcal{O}_C), \quad H^1(\text{Jac}(C), \mathbb{Z}_\ell) \cong H^1(C, \mathbb{Z}_\ell).$$

$\lrcorner$

*Proof:* Cf. [Mil08]P114.  $\square$

**Prop. (20.9.10.13)[Surjection by Jacobians].** For any  $A \in \text{AbVar}/k$ , there exists a smooth complete curve  $C$  over  $k$  and a surjection  $\text{Jac}(C) \rightarrow A$ .  $\lrcorner$

*Proof:* We only prove for  $\#k = \infty$ . For  $k$  finite, Cf. [Gabber, On space filling curves and Albanese varieties], [Poonen, Bjorn, Bertini theorems over finite fields. Ann. of Math]. ?

Since elliptic curves are their own Jacobians, we can assume that  $\dim A > 1$ . Choose a projective embedding  $A \rightarrow \mathbb{P}_k^n$ , then by Bertini and the fact  $\#k = \infty$ , there is a hyperplane cut  $A \cap H$  that is also a smooth complete  $k$ -variety. By repeating  $\dim A - 1$  times, we get a  $k$ -curve  $C$  on  $A$ . This curve gives a map  $\text{Jac}(C) \rightarrow A$  by (20.9.10.9) and double duality (20.9.4.3). The image is an Abelian subvariety of  $A$ . If it is not the whole of  $A$ , then by Poincaré's reducibility theorem (20.9.6.10),

there exists another Abelian subvariety  $A_2 \subset A$  s.t.  $\text{Jac}(C) \times A_2 \rightarrow A$  is an isogeny. In particular,  $\text{Jac}(C) \cap A_2$  is finite.

Using the embedding  $C \subset \text{Jac}(C)$ ,  $C$  is regarded as a subscheme of  $\text{Jac}(C)$ . Take  $n \in \mathbb{Z} \cap k^*$  large, the composition  $\text{Jac}(C) \times A_2 \xrightarrow{(1,n)} \text{Jac}(C) \times A_2 \rightarrow A$  is finite, and the inverse image of  $C$  projects to  $[n]^{-1}(A_1 \cap A_2) \subset A_2$ , which is finite and disconnected. So the inverse image of  $C$  is not connected, which contradicts the fact it is an ample divisor by (7.5.4.12) and (7.9.2.24).  $\square$

**Conj. (20.9.10.14) [Resolution Conjecture].** Let  $k \in \mathbf{Field}$ ,  $k = \bar{k}$ , and  $A \in \mathcal{AbVar}/k$ , we can find a surjective homomorphism  $J_1 \rightarrow A$  where  $J_1$  is a Jacobian of a curve by (20.9.10.13). Then the identity component of the kernel is also an Abelian variety. Then we can do the same process again. is it possible to choose the Jacobians s.t. the process terminate after finite steps?  $\perp$

*Proof:*  $\square$

**Thm. (20.9.10.15) [Tonelli].** Let  $k \in \mathbf{Field}$  and  $k = \bar{k}$ , and  $(C, P), (C', P')$  be pointed complete smooth curves over  $k$ . Let  $f : C \rightarrow \text{Jac}(C)$  and  $f' : C' \rightarrow \text{Jac}(C')$  be the maps corresponding to  $P, P'$ . Then if  $\beta : (\text{Jac}(C), \lambda) \cong (\text{Jac}(C'), \lambda')$  is an isomorphism of polarized Jacobians, then

- There exists an isomorphism  $\alpha : C \rightarrow C'$  s.t.  $f' \circ \alpha = \pm \beta \circ f + c$  for some  $c \in \text{Jac}(C')(k)$ .
- If  $g(C) \geq 2$  and  $C$  is non-hyperelliptic, then  $\alpha, \pm 1, c$  are determined by  $\beta, P, P'$ . And if  $g(C) \geq 2$ , then  $\pm 1$  can be arbitrary, and  $\alpha, c$  is determined by  $\beta, \pm 1, P, P'$ .  $\perp$

*Proof:* Cf. [Mil08]P120.  $\square$

**Cor. (20.9.10.16).**

- If  $k \in \mathbf{Field}$ ,  $C, C'$  are complete smooth curves over  $k$  with rational points s.t. their polarized Jacobians are isomorphic, then  $C, C'$  are isomorphic over  $k$ .
- If  $k \in \mathbf{Field}$  is perfect,  $C, C'$  are complete smooth curves of genus  $\geq 2$  over  $k$ , then if their polarized Jacobians are isomorphic, then  $C, C'$  are also isomorphic over  $k$ .  $\perp$

*Proof:* Let  $\beta : (\text{Jac}(C), \lambda) \cong (\text{Jac}(C'), \lambda')$  is an isomorphism, for any  $P \in C(\bar{k}), P' \in C'(\bar{k})$ , there is a unique isomorphism  $\alpha : C_{\bar{k}} \cong C'_{\bar{k}}$  s.t.  $f^{P'} \circ \alpha = \pm \beta \circ f^P + c$ , and if  $C$  is hyperelliptic, take  $\pm = +$ . Then if  $Q, Q'$  are different points,  $f^P = f^Q + d, f^{Q'} = f^{P'} + e$  for some  $d \in \text{Jac}(C)(\bar{k}), e \in \text{Jac}(C')(\bar{k})$ . So  $\alpha$  is invariant of the points  $P, P'$  chosen. In particular,

$$f^{\sigma(P')} \circ \alpha = \sigma(f^{P'}) \circ \alpha = \pm \beta \circ \sigma(f^P) + \sigma(c) = \pm \beta f^{\sigma(P)} + \sigma(c).$$

for any  $\sigma \in \text{Gal}_k$ . So  $\alpha$  is invariant under  $\text{Gal}_k$ , which means  $C, C'$  are isomorphic over  $k$ .  $\square$

### Albanese Varieties

**Prop. (20.9.10.17) [Albanese Varieties].** For  $X \in \mathbf{SmProjVar}/k$  with a rational point  $p$ , the **Albanese variety** of  $(X, p)$  is an Abelian variety  $J_X \in \mathcal{AbVar}/k$  with a morphism  $\nu : X \times X \rightarrow J_X$  s.t.

- $\nu(x, y) + \nu(y, z) = \nu(x, z)$ , and
- Any map  $\varphi : X \times \rightarrow A \in \mathcal{Sch}/k$  for an Abelian variety  $A \in \mathcal{AbVar}/k$  satisfying  $\varphi(x, y) + \varphi(y, z) = \varphi(x, z)$  factors through  $\nu$  by a homomorphism  $J_X \rightarrow A \in \mathcal{AbVar}/k$ .

If  $X$  has a rational point  $p$ , then the map  $X \rightarrow X \times \{p\} \xrightarrow{\nu} J_X$  satisfies the universal property to  $\mathcal{A}b\mathcal{V}ar/k$ . Equivalently, there is an adjunction

$$J_{(-)} : \mathcal{S}m\mathcal{P}roj\mathcal{V}ar^{\text{pointed}}(k) \rightleftarrows \mathcal{A}b\mathcal{V}ar/k : \text{Forgetful Functor}.$$

┘

*Proof:*    **?**

□

### Arithmetics

**Prop. (20.9.10.18).** If  $F \in \mathbf{GField}$  and  $C$  is a complete smooth curve over  $F$  with good reduction at a place  $P$  of  $F$ , then  $\text{Jac}(C)$  has good reduction at  $P$ .    ┘

*Proof:*    The hypothesis implies  $C$  extends to a smooth proper curve  $\mathcal{C}$  over  $\text{Spec } R_P$ . Then the Picard scheme  $\mathcal{J}$  of  $\mathcal{C}/\text{Spec } R_P$  (20.9.10.3) has generic fiber  $\text{Jac}(C)$ , which implies  $\text{Jac}(C)$  has good reduction at  $P$ .    □



## 21 | Rational Points

### 21.1 Arithmetic of Abelian Varieties

References are [Mordell Seminar notes, Bhatt], <http://virtualmath1.stanford.edu/~conrad/mordellsem/>, [Fal83, Mil08].

**Notation (21.1.0.1).**

- Use notations on reductions of schemes defined in [Reductions of Schemes](#).

┘

#### 1 Néron-Models for Abelian Varieties

##### Néron Models for Abelian Varieties

References are [Serre-Tate, Good Reduction of Abelian Varieties], [BLR90] and <http://virtualmath1.stanford.edu/~conrad/mordellsem/Notes/L11.pdf>.

**Prop. (21.1.1.1) [Néron-Models for Abelian Varieties].** Let  $K \in \mathbf{Field}$  and  $A$  be any ring of integers in the field  $K$ , and  $X \in \mathbf{AbVar}/K$ . Then there exists an open subset  $Y = \mathrm{Spec} A_S \subset \mathrm{Spec} A$ , and a scheme  $\tilde{X}$  projective over  $Y$ , and morphisms  $\tilde{m} : \tilde{X} \times \tilde{X} \rightarrow \tilde{X}$  over  $Y$  and  $\tilde{e} : Y \rightarrow \tilde{X}$  that:

- The fiber of  $\tilde{X}$  over a generic point of  $Y$  with the morphisms  $\tilde{m}, \tilde{e}$  are Abelian varieties isomorphic to  $X$ .
- $\tilde{X}$  is a group scheme over  $Y$ , and its fiber over any closed point of  $Y$  with the morphisms  $\tilde{m}, \tilde{e}$  are Abelian varieties.
- The mapping in item1 induces an isomorphism of groups  $\tilde{X}(Y) \cong X(K)$ .

┘

*Proof:* 1: Consider for any qc scheme over a field  $K$ , it is glued together from f.m. affine schemes, and these glueing involves f.m. polynomials and rational transition functions, and the coefficients of them are contained in a subring  $A_S \subset K$  of f.t. over  $A$ . Thus this variety can be seen as a variety over  $A_S$  with the same equations, satisfying item1. The situation is similar for morphisms between qc schemes, in particular  $\tilde{m}$  and  $\tilde{e}$ , thus constructing  $\tilde{X}$ .

2: Cf. [Mumford, P265]. ?

3: Cf. [Mumford, P265]. ?

□

**Cor. (21.1.1.2).** Let  $x \in \tilde{X}(Y)$ , consider  $x$  as a closed subscheme of  $\tilde{X}$  and denote  $n^{-1}(x)$  the closed subscheme in  $\tilde{X}$  the inverse image of  $x$  under the morphism  $[n]_Y$ . Then the natural projection  $n^{-1}(x) \rightarrow Y$  is étale over all points  $y \in Y$  that  $\mathrm{char} k(y) \nmid n$ .

┘

*Proof:* [Mumford, P265].? □

**Cor. (21.1.1.3) [Néron Differentials].** Let  $(S, K)$  be a Dedekind scheme and  $E \in \text{AbVar}/K$  with Néron model  $\mathcal{E}/S$  (20.3.4.1), then  $\mathcal{K}_{\mathcal{E}/S}$  is free of rank 1. A generator of it is called a **Néron differential** of  $E$ . It is defined up to units on  $S$ . ┘

**Thm. (21.1.1.4) [Good Reduction of Abelian Varieties].** Let  $(S, K)$  be a Dedekind scheme,  $A \in \text{AbVar}/K$  and  $\mathcal{A} \in \text{Sch}/S$  is proper smooth with generic fiber  $A$ , then the group structure extends to  $\mathcal{A}$ , making it a Néron model of  $A$ . ┘

*Proof:* Cf. [BLR90]P19 or [L11, Neron Model]P8.? □  
By (20.3.4.8), we may assume that  $S$  is local.

**Thm. (21.1.1.5) [Néron Model of Abelian Varieties].** Let  $(S, K)$  be a Dedekind scheme and  $A \in \text{AbVar}/K$ , then  $A$  has a Néron model  $\mathcal{A}/S$ . Moreover, let  $S' \subset S$  be the set of good points for  $A$ , then  $\mathcal{A}_{S'} \in \text{AbVar}/S'$ . ┘

*Proof:* We show that  $A$  extends to an Abelian scheme over a nbhd of each good point of  $S$  for  $A$ . This will suffice combining with (21.1.1.4)(21.1.1.8)(20.3.4.8) and (20.3.4.9).

By hypothesis,  $A$  extends to a smooth proper scheme  $A_{(s)}$  over  $\mathcal{O}_{S,s}$ , and then we can extend it to a scheme of f.t. over a nbhd of  $s$ . By shrinking if necessary, we can assume it is smooth and proper, by (7.8.4.6). □

**Prop. (21.1.1.6) [Strong Néron Extension Property].** Let  $(S, K)$  be a Dedekind domain, and  $A \in \text{AbVar}/K$  with a Néron model  $\mathcal{A}/S$ , then for any  $\mathcal{Y} \in \text{Sch}^{\text{sm}}/S$  and a rational map  $\varphi : \mathcal{Y}_K \rightarrow A$ , there exists uniquely an  $S$ -morphism  $\mathcal{Y} \rightarrow \mathcal{A}$  extending  $\varphi$ . ┘

*Proof:* This follows from (20.9.1.7). □

### Proof of the Existence of Néron Models for Abelian Varieties

Cf. [BLR]Chap3-6.

**Remark (21.1.1.7).** In this subsection, let  $(\mathcal{O}_K, K, k)$  be a DVR. ┘

**Prop. (21.1.1.8) [Local Néron Models for Abelian Varieties].** (21.1.1.5) is true when  $S$  is a DVR. ┘

**Def. (21.1.1.9) [Weak Néron Models].** For  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ , a **weak Néron model** for  $X$  is a family  $(X_i) \in \text{Sch}^{\text{sm,sep,ft}}/R$  s.t. each  $K^{\text{sh}}$ -point of  $X$  extends to a  $R^{\text{sh}}$ -point of some  $X_i$ . ┘

**Thm. (21.1.1.10).** For  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$ , if  $X(K^{\text{sh}})$  is bounded in  $X$ , then there exists a weak Néron model for  $X$ . ┘

*Proof:* Cf. [BLR90]P74. □

**Prop. (21.1.1.11) [Weak Néron Property].** Let  $X \in \text{Sch}^{\text{sm,sep,ft}}/K$  and  $(X_i)$  is a weak Néron model for  $X$ . If  $Z \in \text{Sch}^{\text{sm}}/\mathcal{O}_K$  s.t.  $Z_k$  is irreducible, then any rational map  $Z_k \rightarrow X_k$  extends to a rational map  $Z \rightarrow X_i$  over  $R$  for some  $i$ . ┘

*Proof:* Cf. [BLR90]P74. □

### Properties of Néron Models

**Prop. (21.1.1.12).** If  $(\mathcal{O}_K, K)$  is a DVR, then for  $\mathcal{G} \in \mathcal{G}\text{rp}^{\text{sm}, \text{ft}}/\mathcal{O}_K$ , the following are equivalent:

- $\mathcal{G}$  is a Néron model of its generic fiber.
- $\mathcal{G}(\mathcal{O}_K^{\text{sh}}) \rightarrow \mathcal{G}(K^{\text{sh}})$  is an isomorphism.
- $\mathcal{G}$  is separated and  $\mathcal{G}(\mathcal{O}_K^{\text{sh}}) \rightarrow \mathcal{G}(K^{\text{sh}})$  is surjective.

┘

*Proof:*  $1 \rightarrow 2$  is trivial.  $2 \rightarrow 3$  is valuation criterion.  $2 \rightarrow 1$  follows from [Weil's Extension Theorem](#) and [Weak Néron Property](#).

For  $3 \rightarrow 2$ : Cf. [BLR]7.1.1. ?

□

**Prop. (21.1.1.13).** If  $R$  has mixed characteristic  $(0, p)$  and absolute ramification  $e < p - 1$ ,  $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0 \in \mathcal{A}\text{bVar}/K$  is an exact sequence, and  $A$  has good reduction, then the sequence of Néron models

$$0 \rightarrow \mathcal{A}' \rightarrow \mathcal{A} \rightarrow \mathcal{A}'' \rightarrow 0$$

is exact.

┘

*Proof:* Cf. [BLR]Prop7.5.4.

□

## 2 Reductions

Cf. [L13, Semistable Reductions for Abelian Varieties].

**Prop. (21.1.2.1).** If  $S \in \text{Sch}$  is an integral schemes with generic fiber  $\eta$ , and  $A, B \in \mathcal{A}\text{bVar}/S$ , then the map  $\text{Hom}(A, B) \rightarrow \text{Hom}(A_\eta, B_\eta)$  is injective with torsion-free cokernels.

┘

*Proof:* Cf. [CCO14]P86.

□

**Prop. (21.1.2.2) [Lifting Endomorphisms to Normal Domains].** If  $R$  is a normal domain with fraction field  $K$ , then the functor  $\mathcal{A}\text{bVar}/R \rightarrow \mathcal{A}\text{bVar}/K$  is fully faithful.

┘

*Proof:* Cf. [Fal83]Section2.1. ?

□

### over DVRs

**Notation (21.1.2.3).**

- Let  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  with  $\kappa = \kappa^{\text{perf}}$ .

┘

**Prop. (21.1.2.4) [Reduction of Faithful].** For  $(R, \kappa) \in \text{locCRing}$ , the functor  $\mathcal{A}\text{bVar}/R \rightarrow \mathcal{A}\text{bVar}/\kappa$  is faithful.

┘

*Proof:* Cf. [CCO14]P44. ?

□

**Prop. (21.1.2.5) [Reduction Types of Abelian Varieties].** Let  $A \in \mathcal{A}\text{bVar}/K$  with Néron model  $\mathcal{A}/\mathcal{O}_K$  (21.1.1.5), then by (20.9.1.20)(10.2.7.2) and (10.3.4.13), there exists a filtration of subgroups

$$\mathcal{A}_k \supset \mathcal{A}_k^0 \supset \mathcal{A}_k^1$$

where  $\mathcal{A}_k^0$  is the identity component of  $\mathcal{A}_k$ , and  $\mathcal{A}_k^1$  is affine commutative s.t.  $\mathcal{A}_k^0/\mathcal{A}_k^1 \in \mathcal{A}\text{bVar}/k$ . Then  $A$  is said to have:

- **good reduction** if  $\mathcal{A}_k \in \text{AbVar}/k$ . This is compatible with the definition as in (20.3.2.2) by (20.3.2.1).
- **semistable reduction** if  $\mathcal{A}_k^1$  is a torus, in which case  $\mathcal{A}_k^0$  is a semi-Abelian variety (20.9.1.23).
- **bad reduction** or **additive reduction** if  $\mathcal{A}^1$  contains a copy of  $\mathbb{G}_a$ .

There are not other possibilities.  $\square$

**Def. (21.1.2.6) [Tamagawa-Numbers].** For  $A \in \text{AbVar}/K$  with Néron model  $\mathcal{A}/\mathcal{O}_K$  (21.1.1.5), define the **component group** of  $A$  to be  $\pi_0(\mathcal{A}_k)$ , denoted by  $\Phi(A)$ . And  $c(A) = \#\Phi(A) < \infty$  is called the **Tamagawa number** of  $A$ .

Then  $c(A)$  equals the number of geo.connected components of  $\mathcal{A}_k$ . And if  $k$  is finite, it also equals the connected components of  $\mathcal{A}_k$  with a rational point.  $\square$

*Proof:* Cf. [Liu Qing, 10.2.21(a)].  $\square$

**Thm. (21.1.2.7) [Semistable Reduction Theorem].** Let  $K \subset_{\text{fin}} K' \in \text{Field}$ , and  $\mathcal{O}_{K'}$  the alg.closure of  $\mathcal{O}_K$  in  $K'$ . For  $A \in \text{AbVar}/K$ ,

- If  $A$  has stable or semistable reduction, then the reduction type of  $A_{K'}$  is the same as that of  $A$ .
- If  $K'/K$  is unramified, then the reduction type of  $A_{K'}$  is the same as that of  $E$ .
- There exists such a finite extension  $K'/K$  s.t.  $A_{K'}$  is of stable or split semistable reduction type.

$\square$

*Proof:* We only prove for elliptic curves. ?

1: For these two cases,  $c_4$  and  $\Delta$  cannot be reduced anymore to another minimal Weierstrass equation by (21.3.4.5).

2: It suffices to consider the additive reduction case: ? The short Weierstrass equation case is clear, for the general case, use Tate's algorithm.

3: Use the Legendre or Deuring form (21.3.1.18)(21.3.1.19) to analyze  $\Delta$  and  $c_4$ .  $\square$

**Thm. (21.1.2.8) [Néron-Ogg-Shafarevich Criterion].** Let  $A \in \text{AbVar}/K$  and  $\ell \in \text{Prime} \setminus \text{char } \kappa$ , then the following are equivalent:

- $A$  has good reduction.
- The module  $A[m](K^{\text{sep}})$  is unramified as a  $\text{Gal}_K$ -module, for all  $m \in \mathbb{Z} \setminus \text{char } \kappa$ .
- The Tate module  $T_\ell(A)$  is unramified for all (some) primes  $\ell \in \text{Prime} \setminus \text{char } \kappa$ .
- The module  $A[m]$  is unramified for infinitely many  $m \in \mathbb{Z} \setminus \text{char } \kappa$ .

$\square$

*Proof:*  $2 \rightarrow 3 \rightarrow 4$  are obvious.

For the rest, Cf. <http://virtualmath1.stanford.edu/~conrad/mordellsem/Notes/L11.pdf> P20 or [Mil08] P141. ?

$1 \rightarrow 2$ : Let  $K'$  be the finite extension of  $K$  generated by  $E[m]$ , then (21.1.2.7) implies  $E_{K'}$  has the same reduction type as  $E$ . Then by (21.3.4.20),  $E[m] \rightarrow \bar{E}_{K'}(k')$  is an injection. So clearly  $I_v$  acts trivially on  $E[m]$ .

$4 \rightarrow 1$ : Let  $m \in \mathbb{Z} \setminus \text{char } k$  s.t.  $E[m]$  is unramified and  $m > \#E(K^{\text{ur}})/E_0(K^{\text{ur}})$ , where  $\#E(K^{\text{ur}})/E_0(K^{\text{ur}}) < \infty$  by (21.3.4.15). Then because  $E[m] \cong (\mathbb{Z}/(m))^2 \subset E(K^{\text{ur}})$ ,  $E_0(K^{\text{ur}})$  contains



some subgroup isomorphic to  $(\mathbb{Z}/(\ell))^2$  for some  $\ell \in \text{Prime} \setminus \text{char } k$ , thus by (21.3.4.20),  $\tilde{E}_{\text{sm}}(\bar{k})$  contains some subgroup isomorphic to  $(\mathbb{Z}/(\ell))^2$ . But by (21.3.1.17), this is possible only when  $\tilde{E}_{\text{sm}} = \tilde{E}$ . Thus  $E_{K^{\text{ur}}}$  has good reduction, so  $E$  also has good reduction by (21.1.2.7).  $\square$

**Cor. (21.1.2.9).** If  $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$  are exact sequences of Abelian varieties over  $K$  and  $A$  has good reduction, then  $A', A''$  also have good reductions.  $\lrcorner$

**Cor. (21.1.2.10) [Isogeny and Good Reductions].** If  $A, A' \in \text{AbVar}/K$  are isogenous, then  $A$  has good reduction iff  $A'$  does.  $\lrcorner$

*Proof:* Use (21.1.2.8) and the fact if  $\ell \in \text{Prime} \setminus \text{char } k$  is prime to  $\deg \varphi$ , then  $\varphi_\ell : T_\ell(A) \rightarrow T_\ell(A')$  is an isomorphism of  $\text{Gal}_K$ -modules.  $\square$

**Prop. (21.1.2.11).** If  $K \in p\text{-LField}$ ,  $\ell \in \text{Prime} \setminus \{p\}$  and  $A \in \text{AbVar}/K$  satisfies  $\rho_{\ell,A}(\text{Gal}_K)$  has finite image, then  $A$  has potential good reduction.  $\lrcorner$

*Proof:* It follows from class field theory that the image of  $I_K$  is a quotient of  $\mathcal{O}_K^*$ , which contains a pro- $p$ -group of finite index, and  $\text{End}(T_\ell(A))$  contains a pro- $\ell$ -group of finite index, so the image is finite. Then  $A$  has potential good reduction by (21.1.2.8).  $\square$

**Thm. (21.1.2.12) [Néron-Ogg-Shafarevich Criterion for Elliptic Curves].** For  $\ell \in \text{Prime} \setminus \{\text{char } k\}$  and  $A \in \text{AbVar}/K$ ,

- $A$  has good reduction iff  $T_\ell(A)$  is an unramified  $\text{Gal}_K$ -module.
- $A$  has good or semistable reduction iff  $I_K$  acts unipotently on  $T_\ell(A)$ .
- $A$  has potential good reduction iff  $T_\ell(E)$  is potentially unramified, i.e.  $\rho_{\ell,A}(I_K)$  is finite.

$\lrcorner$

*Proof:* 1 follows from (21.1.2.8).

2: We only prove for  $\dim A = 1$  case ?: As  $I_K$  acts trivially on  $\det(T_\ell(E))$  by (25.1.6.3), it acts unipotently on  $T_\ell(E)$  iff it has a fixed vector. If it has a fixed vector, then  $\tilde{E}_{\text{sm}}(\bar{k})$  contains a subgroup isomorphic to  $\mathbb{Z}/(\ell)$ , which is possible only if  $E$  has semistable or good reduction, by (21.3.1.17) and (21.1.2.7). The converse is proved the same way as that of (21.1.2.8).

3 follows from (3.2.6.9).  $\square$

**Cor. (21.1.2.13).** If  $\text{char } k = p$  and  $\varphi : E \rightarrow E' \in \text{EllCur}/K$  is an isogeny of degree  $p$ , then  $E$  has good reduction of ordinary type/supersingular type iff  $E'$  does.  $\lrcorner$

*Proof:* By (21.1.2.10) we can assume both  $E, E'$  have good reductions, then their minimal Weierstrass equations are their Néron models, thus by definition  $\varphi$  extends to a map  $\tilde{\varphi} : \tilde{E} \rightarrow \tilde{E}'$ . Let  $\psi = \tilde{\varphi}$ , then we also have  $\tilde{\psi} : \tilde{E}' \rightarrow \tilde{E}$  s.t.  $\tilde{\psi} \circ \tilde{\varphi} = [p]_{\tilde{E}}$ , and  $\tilde{\varphi} \circ \tilde{\psi} = [p]_{\tilde{E}'}$ . Thus  $[p]_{\tilde{E}'} \circ \tilde{\varphi} = \tilde{\varphi} \circ [p]_{\tilde{E}}$ , which implies  $\deg_s([p]_{\tilde{E}}) = \deg_s([p]_{\tilde{E}'})$ , thus we are done by (21.3.3.5).  $\square$

**Prop. (21.1.2.14) [Isogeny and Semistability].** If  $A, B \in \text{AbVar}/k$  are isogenous, then  $A$  is semistable iff  $B$  is semistable.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (21.1.2.15) [Multiplication Maps].** If  $\mathcal{A} \in \text{AbVar}/K$  has semistable reduction  $\mathcal{A}/\mathcal{O}_K$ , then for any  $m \in \mathbb{Z}_+$ ,  $\mathcal{A}[m]$  is a quasi-finite flat separated group scheme over  $\mathcal{O}_K$  (7.4.4.44). Thus by (10.1.2.36), there is a filtration

$$0 \subset \mathcal{A}[m]_t \subset \mathcal{A}[m]^0 \subset F(\mathcal{A}[m]) \subset \mathcal{A}[m],$$

where  $F(\mathcal{A}[m])$  is finite flat over  $\mathcal{O}_K$ , and  $\mathcal{A}[m]_t \subset \mathcal{A}^0$  is the largest torus.  $\lrcorner$

### Good Reductions

**Thm. (21.1.2.16) [Extending Abelian Variety Structures].** Let  $S \in \mathbf{NSch}_{\text{cpct, cntd}}$  and  $X \in \mathbf{SmPrpr}/S$  with a section  $e : S \rightarrow X$ . If for some geometric point  $s \in S$ ,  $X_s \in \mathbf{AbVar}/\kappa(s)$  with unit point  $e(s)$ , then  $X \in \mathbf{AbVar}/S$  with unit section  $e$ .  $\lrcorner$

*Proof:* Cf. [Ngo, Shimura Varieties]P16.  $\square$

**Prop. (21.1.2.17).** Let  $(\mathcal{O}_K, K, k)$  be a DVR,  $A \in \mathbf{AbVar}/K$  with Néron model  $\mathcal{A}/\mathcal{O}_K$ , then the following are equivalent:

- $\mathcal{A}$  is an Abelian scheme.
- $A$  has good reduction over  $\mathcal{O}_K$ .
- $\mathcal{A}_k^0$  is proper hence in  $\mathbf{AbVar}/k$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2, 2 \rightarrow 3$  is clear.

$3 \rightarrow 1$ : By (7.12.5.4)  $\mathcal{A}_0$  (21.1.1.5) is an Abelian scheme, so  $\mathcal{A}_0 = \mathcal{A}$  is the Néron model by (21.1.1.4).  $\square$

**Prop. (21.1.2.18) [Exactness].** Let  $(R, K)$  be a DVR with mixed characteristic  $(0, p)$  and ramification index  $e < p - 1$ . If  $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$  is an exact sequence in  $\mathbf{AbVar}/K$ , and  $A$  has good reduction, then so does  $A'$  and  $A''$  by (21.1.2.8). Then the Néron models

$$0 \rightarrow \mathcal{A}' \rightarrow \mathcal{A} \rightarrow \mathcal{A}'' \rightarrow 0$$

for an exact sequence.  $\lrcorner$

*Proof:* Cf. [BLR90]Prop7.5.4..  $\square$

**Prop. (21.1.2.19) [Extending Polarizations].** Let  $(R, K)$  be a DVR and  $A \in \mathbf{AbVar}/K$  has good reduction over  $R$ , then any polarization  $\lambda : \mathcal{A} \rightarrow \mathcal{A}^\vee$  extends to a polarization on  $\mathcal{A}$ .  $\lrcorner$

*Proof:* See [Artin 1986, 4.4 in Arithmetic Geometry], and [Chai and Faltings 1990]. ?  $\square$

### over Dedekind Domains

**Def. (21.1.2.20) [Local Conductors].** The conductor of  $E$  is defined in [Sil99]P380.  $\lrcorner$

*Proof:*  $\square$

**Def. (21.1.2.21) [Conductor].** For  $F \in \mathbf{GField}$  and  $E \in \mathbf{EllCur}/F$ , the **conductor(elliptic curve)** of  $E$  is defined to be the integral ideal of  $F$  given by

$$N(E/F) = \prod_{v \in \Sigma_F^0} \mathfrak{p}_v^{f_v}$$

where  $f_v$  is the local conductors (21.1.2.20).

If  $F = \mathbb{Q}$ , then the conductor  $N_E = \prod_{p \in \mathbf{P}} p^{f_p}$ , where

$$f_p = \begin{cases} 0, & E \text{ has good reduction at } p \\ 1, & E \text{ has multiplicative reduction at } p \\ 2, & E \text{ has additive reduction at } p \text{ \& } p \neq 2, 3 \\ 2 + \delta_p, & E \text{ has additive reduction at } p \text{ \& } p = 2, 3 \end{cases}$$

where  $0 \leq \delta_2 \leq 6, 0 \leq \delta_3 \leq 3$ .  $\lrcorner$

**Conj. (21.1.2.22)**[Szpiro]. For any  $F \in \mathbf{NField}$  and  $\varepsilon > 0$ , there exists  $C_\varepsilon > 0$  s.t.

$$|\mathrm{Nm}_{F/\mathbb{Q}}(\Delta_E)| \leq C_\varepsilon |\mathrm{Nm}_{F/\mathbb{Q}}(N_E)|^{6+\varepsilon}$$

for any  $E \in \mathbf{EllCur}/F$ , where  $N_E$  is the conductor of  $E$  (21.1.2.21).  $\lrcorner$

*Proof:*  $\square$

**Thm. (21.1.2.23)** [No Everywhere Good Reduction over  $\mathbb{Q}$ , Fontaine(85)/ Brumer-Kramer(01)/ Schoof[Sch05]]. For  $p \in \mathbf{Prime}$ , there exists an Abelian variety  $A \in \mathbf{AbVar}/\mathbb{Q}$  of positive dimension with good reductions outside  $p$  and semi-stable reduction at  $p$  iff  $p \notin \{2, 3, 5, 7, 13\}$ .

In particular, the minimal conductor of an Abelian variety over  $\mathbb{Q}$  is 11, which is given by the elliptic curve  $E: y^2 + y = x^3 - x^2$ .

There are 24 elliptic curves over  $\mathbb{Q}$  with good reductions away from 2.

There are 784 elliptic curves over  $\mathbb{Q}$  with good reduction away from 6.  $\lrcorner$

*Proof:* Firstly, if  $N \in \mathbf{Prime}_{>13}$ , then the modular Jacobian variety  $\mathrm{Jac}(X_0(N))$  has good reduction outside  $N$  and has semistable reduction at  $N$ .

We only prove that there are no elliptic curves with everywhere good reductions. ?

Let  $E: y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$  be an elliptic curve over  $\mathbb{Q}$  with  $a_i \in \mathbb{Z}$  and  $\Delta = \pm 1$ , then  $a_1$  must be odd, otherwise  $\Delta \equiv 5b_6^2 \equiv \pm 1 \pmod{8}$  is impossible. Thus  $c_4 = b_2^2 - 24b_4 \equiv 1 \pmod{8}$ . And  $c_4^3 - c_6^2 = (\pm 12)^3$  shows that  $c_4 \pm 12$  is a square or 3 times a square, which are both impossible by modulo 8.

For good reduction away from 2, 3 ?  $\square$

**Prop. (21.1.2.24)**[Schoof]. Every semi-stable Abelian variety over  $\mathbb{Q}$  that has good reduction outside 11 is isogenous to a power of  $\mathrm{Jac}(X_0(11))$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (21.1.2.25)**[Schoof]. For  $p \in \{2, 3, 5\}$ , there are no Abelian varieties over  $\mathbb{Q}$  of positive dimension that has good reduction outside  $p$  and acquire semi-stable reduction at  $p$  over an extension of  $\mathbb{Q}$  that is at most tamely ramified at  $p$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (21.1.2.26)**[Conductors for Modular Abelian Varieties]. If  $g \in \mathbb{Z}_+$  and  $A \in \mathbf{AbVar}^g/\mathbb{Q}$  s.t.  $L(A; s)$  is an entire analytic  $L$ -function, then we have

$$N(A) \geq e^{1.24g} > 3.$$

*Proof:* Apply the formula (35.4.8.2) at  $s = \sigma \in \mathbb{R}_{>1}$  and take the real part to get

$$\frac{1}{2} \log q + g \mathrm{dlog} \Gamma(\sigma + \frac{1}{2}) - g \log(2\pi) + \mathrm{dlog} L(A; \sigma) \geq 0.$$

Notice that  $\mathrm{Re}(b) = \mathrm{Re}(\rho^{-1})$  is cancelled by (35.4.8.3).  $\lrcorner$

Also by Ramanujan-Peterson bound ( $|\alpha_i(p)| \leq 1$ ), we get  $|\mathrm{dlog} L(A; \sigma)| \leq g \mathrm{dlog} \zeta(\sigma)$ .  
So we get

$$\frac{1}{2g} \log N(A) \geq \log(2\pi) - \mathrm{dlog} \Gamma(\sigma + \frac{1}{2}) - \mathrm{dlog} \zeta(\sigma).$$

If we take  $\sigma = 2.3$ , then the RHS is greater than 0.62.  $\square$

**Prop. (21.1.2.27).** Let  $F \in \mathbf{NField}$  and  $A, B \in \mathbf{AbVar}/F$ , and  $S \subset \Sigma_F$  is a finite set of places of  $F$  containing  $\Sigma_F^\infty$  and all places s.t.  $A$  or  $B$  has bad reduction, then for any  $\ell \in \mathbf{Prime}$  s.t.  $S(\ell) \cap S = \emptyset$ , there exists a finite set  $T = T(S, \ell, g) \in \Sigma_\infty^0$  s.t.  $T \cap (S \cup S(\ell)) = \emptyset$ , and

$$P_\ell(\tilde{A}_v, t) = P_\ell(\tilde{B}_v, t) \Rightarrow A \sim B$$

┘

*Proof:* Cf. [Mil08]P142.  $\square$

### Semi-stable Reductions

**Prop. (21.1.2.28) [Fundamental Semi-stable Reduction Theorem].** For any  $F \in \mathbf{NField}$ ,  $A \in \mathbf{AbVar}/F$ , there exists a finite extension  $L/F$  s.t.  $A_L$  is semistable, by (21.1.2.7).  $\square$

### 3 Néron-Tate Heights

**Prop. (21.1.3.1) [Néron-Tate Heights].** Let  $F \in \mathbf{GField}$ ,  $X \in \mathbf{AbVar}/F$ ,  $\mathcal{L} \in \mathbf{Pic}(X)$ , then there are uniquely defined bilinear form  $b_{\mathcal{L}}$ , additive homomorphism  $l_{\mathcal{L}}$  on  $X(F^{\mathrm{sep}})$  s.t.

$$\hat{h}_{\mathcal{L}}(x) = \frac{1}{2} b_{\mathcal{L}}(x, x) + l_{\mathcal{L}}(x) \sim h_{\mathcal{L}}(x).$$

Such a  $\hat{h}_{\mathcal{L}}(x)$  is called the **Néron-Tate height** of  $\mathcal{L}$ .  $\square$

*Proof:* We want to use (3.2.4.37) for the Weil height (20.2.1.25)  $h_{\mathcal{L}}$  on  $X(F^{\mathrm{sep}})$ : Apply the theorem of the cube (20.9.1.26) to the projections  $\pi_i : X \times X \times X \rightarrow X$  and pullback to  $X$  via the diagonal map, then taking the Weil heights, we will get the desired relation

$$h_{\mathcal{L}}(\sum_{i=1}^3 x_i) - \sum_{1 \leq i < j \leq 3} h_{\mathcal{L}}(x_i + x_j) + \sum_{i=1}^3 h_{\mathcal{L}}(x_i) \sim 0.$$

Thus the condition of (3.2.4.37) is satisfied and we get the desired height  $\hat{h}_{\mathcal{L}}(x)$ .  $\square$

**Remark (21.1.3.2).** Another way to get this is through Tate's limiting argument (21.8.3.2).  $\square$

**Cor. (21.1.3.3).** The Néron-Tate height is a refinement of Weil heights (20.2.1.25) made for Abelian varieties:

- The map  $\hat{h} : \mathbf{Pic}(X) \mapsto \mathbb{R}^{X(\bar{K})} : \mathcal{L} \mapsto \hat{h}_{\mathcal{L}}$  is an additive homomorphism.
- (Symmetry) If  $\varphi : A \rightarrow B$  is a homomorphism of Abelian varieties, then  $\hat{h}_{\varphi^*(\mathcal{L})} = \hat{h}_{\mathcal{L}} \circ \varphi$ .
- (Positivity) Let  $\mathcal{L} \in \mathbf{Pic}(X)$  be even. If  $\mathcal{L}$  is base-free or ample, then  $\hat{h}_{\mathcal{L}} \geq 0$ .
- (Boundedness) Let  $\mathcal{L} \in \mathbf{Pic}(X)$  be even and ample (such an  $\mathcal{L}$  exists by (20.9.1.32)), then  $\hat{h}_{\mathcal{L}}$  induces a symmetric bilinear form on  $X(K)$  satisfying  $\{x \in X(\bar{K}) \mid \deg(x) \leq d, (x, x) < C\}$  is finite for all  $C > 0$

┘

*Proof:* 1, 2 follow from the corresponding property of Weil heights(20.2.1.25) and the uniqueness in(3.2.4.37).

3: Notice that the fact  $\mathcal{L}$  is even implies  $\hat{h}_{\mathcal{L}} = \frac{1}{2}b(x, x)$ . The ample case reduces to the base-free case, because there is a multiple  $m\mathcal{L}$  that is very ample, and then  $\hat{h}_{m\mathcal{L}} = m\hat{h}_{\mathcal{L}}$ . For the base-free case,  $c$  is the pullback of  $\mathcal{O}(1)$  for some morphism  $X \rightarrow \mathbb{P}^n$ , thus  $h_{\mathcal{L}}$  is non-negative by(20.2.1.3), and  $h_{\mathcal{L}} \sim \hat{h}_{\mathcal{L}}$ , thus  $\hat{h}_{\mathcal{L}}$  must also be non-negative.

4 follows from Northcott's theorem(20.2.1.26). □

**Cor. (21.1.3.4) [Positive Definiteness].** For any ample line bundle  $\mathcal{L} \in \text{Pic}(A)$ ,  $b_{\mathcal{L}}$  induces a positive definite symmetric bilinear form on  $A(\bar{F}) \otimes_{\mathbb{Z}} \mathbb{R}$  ┘

*Proof:* We want to use(3.2.4.38). Notice  $b_{\mathcal{L}}$  is just  $2\hat{h}_{\mathcal{L} \otimes [-1]^*\mathcal{L}}$ , thus satisfies the conditions by(21.1.3.3). □

**Prop. (21.1.3.5) [Zero of Heights and Torsion].** If  $F \in \mathbf{GField}$ ,  $A \in \mathbf{AbVar}/F$  and  $\mathcal{L}$  is ample, then  $\hat{h}_{\varphi, \mathcal{L}}(x) = 0$  iff  $x$  is preperiodic, i.e. the sequence  $\{x, \varphi(x), \varphi^2(x), \dots\}$  is finite.

In particular,  $\hat{h}_{\mathcal{L}}(x) = 0$  iff  $x$  is torsion. ┘

*Proof:* Assume  $l \neq 0$ . If  $x$  is preperiodic, then clearly  $\hat{h}_{\varphi, \mathcal{L}}(x) = \lim_{r \rightarrow \infty} \lambda^{-r} h_{\mathcal{L}}(\varphi^r(x)) = 0$ . Conversely, if  $\hat{h}_{\varphi, \mathcal{L}}(x) = 0$ , then  $\hat{h}_{\varphi, \mathcal{L}}(\varphi^r(x)) = 0$  for any  $r$ , then  $|h_{\mathcal{L}}(\varphi^r(x))| \leq |\hat{h}_{\varphi, \mathcal{L}}(\varphi^r(x))| + C(\varphi, \mathcal{L}) = C(\varphi, \mathcal{L})$  is bounded and also  $\varphi^r(x) \in X(k(x))$ , thus has bounded degree. So by Northcott's theorem(20.2.1.26), there are only f.m. such points. □

**Cor. (21.1.3.6) [Kronecker].** For  $\zeta \in \bar{\mathbb{Q}}$ ,  $h^+(\zeta) = 0$ (20.2.1.5). ┘

*Proof:* This is a special case of(21.1.3.5), where  $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^1 : [x, y] \mapsto [x^n, y^n]$  and  $\mathcal{L} = \mathcal{O}(1)$ , thus the preperiodic points are just  $0, \infty$  and all the roots of unity. □

**Prop. (21.1.3.7) [Néron-Tate Pairings].** Let  $F \in \mathbf{NField}$  and  $A \in \mathbf{AbVar}/F$ , then  $A \times \hat{A}$  is also an Abelian variety with the even Poincaré class  $p$ (20.9.4.10), thus generating a function:

$$\hat{h}_{A \times A^\vee, p} : A(\bar{F}) \times \hat{A}(\bar{F}) \rightarrow \mathbb{R}.$$

Then in fact this pairing is bilinear, called the **Néron-Tate pairing**  $\langle \cdot, \cdot \rangle_{A, \text{N-T}}$ .

The Weil-Tate pairing satisfies the functoriality property: If  $f : A \rightarrow B$  is a homomorphism of Abelian varieties, then  $\langle f(a), b \rangle_{B, \text{N-T}} = \langle a, \hat{f}(b) \rangle_{A, \text{N-T}}$ . ┘

*Proof:* Use functoriality of heights for  $A \rightarrow A \times \hat{A}$ , then  $\hat{h}_{A \times \hat{A}, p}((a, 0)) = \hat{h}_{A, 0}(a) = 0$ . Similarly,  $\langle 0, a \rangle_{A, K} = 0$ . Thus  $\hat{h}_{A \times \hat{A}, p}$  is bilinear by(3.2.4.32).

The last assertion follows from(10.9.3.37). □

**Prop. (21.1.3.8).** Let  $\mathcal{L} \in \text{Pic}(A)$  and  $\varphi_{\mathcal{L}} : A \rightarrow \hat{A}$  the associated homomorphism(20.9.4.6), then for any  $a, a' \in A(\bar{K})$ ,

- $\hat{h}_{\tau_a^*, \mathcal{L}}(a) = \hat{h}_{\mathcal{L}}(a) + b(a, a')$ .
- $\hat{h}_{\mathcal{L}}(a) = \langle a, \mathcal{L} \rangle_{A, \text{N-T}}$ .
- $b_{\mathcal{L}}(a, a') = \langle a, \varphi_{\mathcal{L}}(a') \rangle_{A, \text{N-T}} = \hat{h}_{\varphi_{\mathcal{L}}(a')}(a)$ .

┘

*Proof:* 1:  $b_{\mathcal{L}}(a, a') = \widehat{h}_{\mathcal{L}}(a + a') - \widehat{h}_{\mathcal{L}}(a) - \widehat{h}_{\mathcal{L}}(a') = \widehat{h}_{\mathcal{L}}(\tau_{a'}a) - \widehat{h}_{\mathcal{L}}(a) - \widehat{h}_{\mathcal{L}}(a')$ . Then  $b(\cdot, a') + \widehat{h}_{\mathcal{L}}$  is a representative in the class  $h_{\tau_{a'}^*(\mathcal{L})}$  by functoriality. And it is a quadratic function, thus by uniqueness, we are done.

2: Because the pullback of  $p$  to  $A \times \{\mathcal{L}\}$  is  $\mathcal{L}$  by (10.9.3.35), and thus this follows from functoriality and (21.1.3.7).

3: By (20.9.4.6) and item1,  $b(a, a') = \widehat{h}_{\varphi_{\mathcal{L}}(a')}(a)$ . Thus we are done by item2.  $\square$

**Cor. (21.1.3.9)[Regulator].** The Néron-Tate pairing induces a non-degenerate pairing  $A(K)/A(K)_{\text{tor}} \times \widehat{A}(K)/\widehat{A}(K)_{\text{tor}} \rightarrow \mathbb{R}$ . The discriminant of this pairing is called the **regulator** of  $A$ , denoted by  $R(A/K)$ .  $\square$

*Proof:* Let  $\mathcal{L} \in \text{Pic}(A)$  be ample, then  $\varphi_{\mathcal{L}} : A \rightarrow \widehat{A}$  is an isogeny, thus  $\varphi_{\mathcal{L}} : A(K)/A(K)_{\text{tor}} \rightarrow \widehat{A}(K)/\widehat{A}(K)_{\text{tor}}$  is injective, and by (21.7.1.7), the image is a subgroup of  $\widehat{A}(K)/\widehat{A}(K)_{\text{tor}}$  of the same rank, thus we are done by (21.1.3.8) item3 and (21.1.3.4).  $\square$

**Cor. (21.1.3.10).** Let  $A$  be an Abelian variety,  $c \in \text{Pic}(A)$  an even ample class,  $c' \in \text{Pic}(A)$ , then  $\widehat{h}_{c'} = O(\widehat{h}_c^{1/2})$ .  $\square$

*Proof:* By (20.9.4.18),  $\varphi_c$  is isogeny thus surjective, thus we can assume  $c' = \varphi_c(a')$  for some  $a' \in A(K)$ . Then by (21.1.3.8),  $\widehat{h}_{c'}(a) = b_c(a, a')$ . Because  $b_c$  is positive semi-definite by (21.1.3.3), by Cauchy-Schwartz (3.2.4.31),  $|\widehat{h}_{c'}(a)|^2 \leq b(a, a)b(a', a') = 4\widehat{h}_c(a)\widehat{h}_c(a')$ .  $\square$

### Hilbert's Irreducibility Theorem

**Prop. (21.1.3.11)[Runge's Theorem].**  $\square$

**Prop. (21.1.3.12) [Hilbert's Irreducibility Theorem].** Let  $C$  be a smooth irreducible projective curve over a number field  $K$  and let  $f : C \rightarrow \mathbb{P}^1$  be a surjective rational function on  $C$  over  $K$ , then for all  $n \in \mathbb{N}$  except for a set of natural density 0, the divisor  $f^*[n]$  is a prime divisor over  $K$ .  $\square$

*Proof:* Cf.[Diophantine Geometry, P319].  $\square$

### Local Height Pairings

Cf.[Sil99]Chap6 or [Local Heights on Curves, Gross, in Arithmetic Geometry] and [B-G06] or [Introduction to Diophantine Geometry, Lang].

### Canonical Heights on Semi-Abelian Varieties

Cf.[Points of small height on semiabelian varieties]Section3 or [Poonen, Bogomolov plus Mordell-Lang].

**Def. (21.1.3.13)[Canonical Height on Semi-Abelian Varieties].**  $\square$

## 4 Isogenies

**Thm. (21.1.4.1) [Masser-Wüstholz/Pellarin].** Let  $F \in \mathbf{NField}$ ,  $d = [F : \mathbb{Q}]$ . If  $E, E' \in \mathbf{EllCur}/F$  are isogenous, then there exists an isogeny  $\psi : E \rightarrow E'$  over  $F$  with  $\text{degree} \leq \kappa(d)(1 + h^+(j(E)))^2$ , where  $\kappa(d)$  depends only on  $d$  and is effective.  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (21.1.4.2) [Bounded on Rational Isogenies].** Let  $F \in \mathbf{NField}$ ,  $d = [F : \mathbb{Q}]$  and  $E \in \mathbf{EllCur}/F$  without CM, and there is a cyclic isogeny  $\varphi : E \rightarrow E'$  over  $F$  with degree  $\delta$ , then  $\delta \leq \kappa(d)(1 + h^+(j(E)))^2$ .  $\lrcorner$

*Proof:* Let  $\psi : E \rightarrow E'$  be an isogeny of  $\text{degree} \leq \kappa(d)(1 + h^+(j(E)))^2$ . In fact we can assume that  $\psi$  is cyclic. Then  $\varphi^\vee \circ \psi \in \text{End}(E) = \mathbb{Z}$ . Thus  $\varphi = \pm\psi$  has  $\text{degree} \leq \kappa(d)(1 + h^+(j(E)))^2$ .  $\square$

### Tate's Hom Conjecture

References are [Tat66, Fal83].

**Lemma (21.1.4.3) [Faithfulness of  $p$ -Divisible Groups].** Let  $k \in \mathbf{Field}$ ,  $A_1, A_2 \in \mathbf{AbVar}/k$ , and  $\ell \in \mathbf{Prime}$ , the natural map

$$\text{Hom}(A_1, A_2) \otimes \mathbb{Z}_\ell \rightarrow \text{Hom}_{\mathbb{Z}_\ell[\text{Gal}_k]}(A_1[\ell^\infty], A_2[\ell^\infty])$$

is injective with torsion-free cokernel.  $\lrcorner$

*Proof:* Choose a  $\mathbb{Z}$ -basis  $\{f_1, \dots, f_n\}$  of  $\text{Hom}(A_1, A_2)$  (20.9.6.12).

To prove the cokernel is torsion-free, suppose  $c_i \in \mathbb{Z}_\ell$  and  $\varphi \in \text{Hom}_{\mathbb{Z}_\ell[\text{Gal}_k]}(A_1[\ell^\infty], A_2[\ell^\infty])$  s.t.  $\ell^m \varphi = \sum c_i f_i[\ell^\infty]$ , then we may choose  $n_i \in \mathbb{Z}$  s.t.  $c_i = n_i + \ell^m d_i$  with  $d_i \in \mathbb{Z}_\ell$ . Then  $f = \sum n_i f_i \in \text{Hom}(A_1, A_2)$  satisfies

$$f[\ell^\infty] = \ell^m(\varphi - \sum d_i f_i[p^\infty]),$$

which means  $f$  vanishes on  $A_1[\ell^m]$ , so  $f = g \circ [\ell^m]$  for some  $g \in \text{Hom}(A_1, A_2)$ , and then  $\varphi = g[\ell^\infty] + \sum d_i f_i[p^\infty]$  is in the image of the natural map, because  $A[\ell^\infty] \in p\text{Div}/k$ .

To prove injectivity, it suffices to show that if  $c_i \in \mathbb{Z}_\ell$  and  $\sum c_i f_i$  kills  $A[\ell]$ , then  $\ell | c_i$  for each  $i$ . Because if this is proved, then whenever  $\sum c'_i f_i$  kills  $A[\ell^\infty]$ , we get  $\ell | c_i$ , so  $(\sum c'_i f_i) \circ \ell$  kills  $A[p^n]$  for each  $n \in \mathbb{Z}_+$ . But  $[\ell] : A[p^{n+1}] \rightarrow A[p^n]$  is faithfully flat (20.9.6.18), so  $\sum c'_i f_i$  kills  $A[p^\infty]$  too. In this way, we prove that  $c_i = 0$  for each  $i$ .

To prove the desired assertion, we may assume that  $c_i \in \mathbb{Z}$ , and then  $\sum c_i f_i : A_1 \rightarrow A_2$  factors through  $A \xrightarrow{[\ell]} A$  as  $\sum c_i f_i = \ell(\sum b_i f_i)$ , so  $\ell | c_i$  for each  $c_i$ .  $\square$

**Thm. (21.1.4.4) [Tate's Hom Conjecture, Mumford/Tate [Tat66]/Faltings [Fal83]].** If  $k \in \mathbf{Field}$  is f.g. over its prime field and  $A_1, A_2 \in \mathbf{AbVar}/k$ , then

$$\text{Hom}(A_1, A_2) \otimes \mathbb{Z}_\ell \rightarrow \text{Hom}_{\mathbb{Z}_\ell[\text{Gal}_k]}(A_1[\ell^\infty], A_2[\ell^\infty])$$

is an isomorphism.  $\lrcorner$

*Proof:* Cf.[Tat66] and[CCO14]Appendix A.1.?. Passing to  $A \times B$ , we may assume that  $A = B$ .  
First notice that

$$\mathrm{Hom}(A_1, A_2) \otimes \mathbb{Q}_\ell \rightarrow \mathrm{Hom}_{\mathbb{Z}_\ell[\mathrm{Gal}_K]}(A_1[\ell^\infty], A_2[\ell^\infty]) \otimes \mathbb{Q}_\ell$$

is injective(This follows from(21.1.4.3) and the fact  $\mathbb{Q}_\ell$  is flat over  $\mathbb{Z}_\ell$ .), so it suffices to show this is surjective.

?

□

**Cor.(21.1.4.5)[Faltings Isogeny Theorem].** Let  $k \in \mathbf{Field}^{\mathrm{fin}}$  or  $k \in \mathbf{GField}$ ,  $\ell \in \mathbf{Prime}$ ,  $A_1, A_2 \in \mathbf{AbVar}/k$ , then  $A_1, A_2$  are isogenous iff  $T_\ell(A_1) \cong T_\ell(A_2) \in \mathrm{Rep}_{\mathbb{Z}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$ . ┘

## 5 Heights of Abelian Varieties

Cf.[Milne, Abelian Varieties] and [Faltings Height].

**Def.(21.1.5.1)[Faltings Heights].** Let  $F \in \mathbf{NField}$  and  $A \in \mathbf{AbVar}/F$ , take the Néron model  $\mathcal{A}/\mathcal{O}_F$ , then  $\mathcal{K}_{A/\mathcal{O}_F}$  is an invertible sheaf, and thus  $\omega_{A/\mathcal{O}_F}$  is an invertible  $\mathcal{O}_F$ -sheaf, represented by a fractional ideal  $M \subset F$ , and  $M \otimes_R F = \Gamma(A, \mathcal{K}_{A/F})$  by(10.1.1.12).

We define a norm on  $M \otimes_F F_v = \Gamma(A_v, \mathcal{K}_{A_v/F_v})$  for each  $v \in \Sigma_F^\infty$ :

$$\|\omega\|_v = \left( \left( \frac{i}{2} \right)^g \int_{A(\overline{F}_v)} \omega \wedge \overline{\omega} \right)^{1/2}.$$

This make  $\widetilde{M}$  a metrized line bundle on  $\mathcal{O}_F$ (23.2.1.1).

Then we define the **Faltings height** of  $A$ :

$$H_{\mathrm{Fal}}(A) = H((\widetilde{M}, |\cdot|)) \quad (23.2.1.2), \quad h_{\mathrm{Fal}}(A) = \frac{1}{[F : \mathbb{Q}]} \log H_{\mathrm{Fal}}(A).$$

More explicitly, take a non-zero holomorphic  $g$ -form  $\omega$  on  $A$ , and for  $v \in \Sigma_F^0$ , let  $\omega_v$  be a Néron differential on  $A_v$ , then

$$h_{\mathrm{Fal}}(A) = \frac{-1}{[F : \mathbb{Q}]} \left( \sum_{v \in \Sigma_F^0} \log \left| \frac{\omega}{\omega_v} \right|_v + \sum_{v \in \Sigma_F^\infty} \frac{[F_v : \mathbb{Q}_v]}{2} \log \left| \left( \frac{i}{2} \right)^g \int_{A(\overline{F}_v)} \omega \wedge \overline{\omega} \right| \right).$$

┘

**Def.(21.1.5.2)[Stable Faltings Heights].** Let  $F \in \mathbf{NField}$  and  $A \in \mathbf{AbVar}/F$ , then by?? there exists a finite extension  $L/F$  s.t.  $A_L$  is semistable. Then the Néron model of  $A_L$  are stable under base change?. Then we can define the **stable Faltings height**  $h_{\mathrm{st}}(A) = h(A_L)$ , which is invariant of  $L$  chosen. ┘

**Prop.(21.1.5.3).** Let  $F \in \mathbf{NField}$  and  $A \in \mathbf{AbVar}/F$ , then  $h_{\mathrm{Fal}}(A) = h_{\mathrm{Fal}}(A^\vee)$ . ┘

*Proof:* Cf.[Szpiro, La Conjecture de Mordell, Séminaire Bourbaki, 1983/84]. □

**Thm.(21.1.5.4)[Faltings Heights and Isogenies].** Let  $F \in \mathbf{NField}$ ,  $\mathcal{A}_1, \mathcal{A}_2$  be semi-Abelian schemes over  $\mathcal{O}_F$  of relative dimension  $g$  with proper generic fiber  $A_1, A_2 \in \mathbf{AbVar}/F$ . Let  $\varphi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$  be an isogeny with kernel  $\mathcal{G}/\mathcal{O}_F$ , then

$$h_{\mathrm{st}}(A_2) = h_{\mathrm{st}}(A_1) + \frac{1}{2} \log \deg(\varphi) - [F : \mathbb{Q}] \log e^* \Omega_{\mathcal{G}/\mathcal{O}_F}^1.$$

┘



*Proof:* Cf.[Faltings, Lemma5] or [Faltings height].? □

**Prop. (21.1.5.5) [Lower Bounds of Faltings Heights].** There exists  $c \in \mathbb{R}_+$  s.t.  $h_{\text{st}}(A) \leq cg$  for any  $A \in \text{AbVar}^g/\overline{\mathbb{Q}}$ . ┘

*Proof:* Cf.[Bost, Arakelov geometry of abelian varieties, 1996]. □

**Thm. (21.1.5.6) [Finiteness of Heights in an Isogeny Class].** Let  $F \in \text{NField}$  and  $A \in \text{AbVar}/F$  be semistable, then the set of Faltings heights of  $B \in \text{AbVar}/F$  isogenous to  $A$  is finite. ┘

*Proof:* This is the hardest part of the proof.? Cf.[Conrad Seminar, L20]. □

### Modular Heights

**Def. (21.1.5.7) [Modular Heights].** Consider the Siegel modular varieties  $\mathcal{M}_{g,d}/\mathbb{Q}$  parametrizing Abelian schemes(28.1.3.4), then for any  $F \in \text{NField}$ ,  $(A, \lambda) \in \text{AbVar}^{\dim=g, \text{polar}=d}/F$  defines a  $F$ -point  $j((A, \lambda))$  of  $\mathcal{M}^{g,d}$ . Then we can define the **modular height**  $h_M(A, \lambda)$  to be  $h(j((A, \lambda)))$ , where  $h$  is the Weil height associated to the canonical ample divisor on  $\mathcal{M}^{g,d}$ (28.1.3.5), which is defined up to a bounded function on  $\mathcal{M}^{g,d}$ . ┘

**Prop. (21.1.5.8) [Comparison of Heights].** For  $F \in \text{NField}$ , there are constants  $c_1, c_2, c_3$  s.t. for any semistable  $(A, \lambda) \in \text{AbVar}^{\text{polar}}/F$ ,

$$|h_{\text{Fal}}(A) - c_1 h_M(A, \lambda)| < c_2 \log h_M(A, \lambda) + c_3.$$

┘

*Proof:* ? Hard. Cf. [Chai and Faltings, 1990]. □

**Prop. (21.1.5.9) [Height I].** Let  $F \in \text{NField}$  and  $g, d, C \in \mathbb{Z}_+$ , then up to isomorphism, there are only f.m. semistable  $(A, \lambda) \in \text{AbVar}^{g, \text{polar}=d}/k$  s.t.  $h_M(A, \lambda) < C$ . ┘

*Proof:* By the definition of Siegel modular varieties, two objects in  $(A, \lambda) \in \text{AbVar}^{g, \text{polar}=d}/F$  corresponds to the same point of  $\mathcal{M}^{g,d}$ (21.1.5.7) iff they are isomorphic over  $\overline{F}$ . And by Northcott's theorem(20.2.1.26) the image of polarized Abelian varieties with bounded modular heights is finite. Thus the assertion follows from(21.1.7.2). □

**Cor. (21.1.5.10) [Height II].** Let  $F \in \text{NField}$  and  $g, C \in \mathbb{Z}_+$ , then up to isomorphism, there are only f.m. semistable  $A \in \text{AbVar}^g/F$  s.t.  $h_{\text{Fal}}(A) < C$ , by(21.1.5.8).

(Is this true for non-semistable varieties?). ┘

*Proof:* By the proposition(21.1.5.9) and(21.1.5.8), the set of isomorphism classes of semistable Abelian varieties  $(B, \lambda) \in \text{AbVar}^{g, \text{polar}=d}/F$  is finite. Now notice for any  $A \in \text{AbVar}/F$ ,  $B = (A \times A^\vee)^4$  is principally polarized by(20.9.7.2), and  $h(B) = 8h(A)$  by(21.1.5.3), so we are done. □

### Elliptic Curve case

**Prop. (21.1.5.11) [Heights of Elliptic Curves].** For  $F \in \text{NField}$  and  $E \in \text{EllCur}/F$ . For each  $v \in \Sigma_F^\infty$ , let  $\tau_v \in \mathcal{H}$  in the fundamental domain s.t.  $E_{\overline{K}_v} \cong E_{\tau_v}$ . Then

$$12[F : \mathbb{Q}]h_{\text{Fal}}(E/F) = \log \left| \text{Nm}_{F/\mathbb{Q}} N_{E/F} \right| - \sum_{v \in \Sigma_F^\infty} [F_v : \mathbb{Q}_v] \log \left| \Delta(\tau_v)(\text{Im } \tau_v)^6 \right|.$$

┘

*Proof:* Choose any Weierstrass model  $W : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ , and let  $\Delta$  be its discriminant, and take

$$\omega = \Delta \cdot \left( \frac{dx}{2y + a_1x + a_3} \right)^{\otimes 12}.$$

Then notice this  $\omega$  is invariant under change of variables, thus  $-\sum_{v \in \Sigma_F^0} \log |\frac{\omega}{\omega_v}|_v = |\mathrm{Nm}_{F/\mathbb{Q}} N_{E/F}| \text{ ?}$ .  
 $\square$

**Prop. (21.1.5.12).** For  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$ ,

$$O(1) \leq h^+(j(E)) + \frac{1}{[F:\mathbb{Q}]} \log |\mathrm{Nm}_{F/\mathbb{Q}} \Upsilon_{E/F}| - 12h_{\mathrm{Fal}}(E/F) \leq 6 \log(1 + h^+(j(E))) + O(1).$$

In particular, if  $E/F$  is semistable, then

$$|h^+(j(E)) - 12h_{\mathrm{Fal}}(E/F)| \leq 6 \log(1 + h^+(j(E))) + O(1),$$

which is a special case of (21.1.5.8).  $\lrcorner$

*Proof:* Cf. [Heights and Elliptic Curves, P258] ?  $\square$

**Cor. (21.1.5.13).** For any  $\varepsilon \in \mathbb{R}_+$ , we have:  $E \in \mathbf{EllCur}/\mathbb{Q}$ ,

$$h_{\mathrm{Fal}}(E/\mathbb{Q}) + O(1) \leq \frac{1}{12} \max(|j(E)|, |\Delta_E^{\min} j(E)|) \leq (1 + \varepsilon) h_{\mathrm{Fal}}(E/\mathbb{Q}) + O_\varepsilon(1),$$

$$h_{\mathrm{Fal}}(E/\mathbb{Q}) + O(1) \leq \frac{1}{12} \max(|c_4(E)|^3, |c_6(E)|^2) \leq (1 + \varepsilon) h_{\mathrm{Fal}}(E/\mathbb{Q}) + O_\varepsilon(1).$$

$\lrcorner$

*Proof:* By definition,  $h^+(j(E)) + \log |\Upsilon_E| = \log \max(|j\Delta_E^{\min}|, |\Delta_E^{\min}|)$ . So the first inequality follows from (21.1.5.12) and the fact  $\log(1 + h) < \varepsilon h + O_\varepsilon(1)$  for  $h \in \mathbb{R}_+$ .

The second inequality follows from the first by the definition of  $\Delta$  and  $j$  (21.3.1.5).  $\square$

**Cor. (21.1.5.14) [Northcott Property for Elliptic Curves].** For  $d, C \in \mathbb{R}_+$ , there exists only f.m. pairs  $(F, E)$  where  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$  s.t.  $F = \mathbb{Q}(j(E))$ ,  $\deg(F) \leq d$  and  $h_{\mathrm{Fal}}(E/F) \leq C$ .  
 $\lrcorner$

*Proof:* It follows from (21.1.5.12) that for such pairs  $(F, E)$ ,  $h^+(j(E))$  is bounded from above, so by Northcott, there are only f.m. possibilities for  $j(E)$ . Then it suffices to show that for any  $E \in \mathbf{EllCur}/\overline{\mathbb{Q}}$ , there are only f.m.  $E' \in \mathbf{EllCur}/\mathbb{Q}(j(E))$  s.t.  $j(E') = j(E)$  and  $h(E'/\mathbb{Q}(j(E))) \leq C$ .

Denote  $F = \mathbb{Q}(j(E))$ , it follows from (21.1.5.12) that for any such  $E'$ ,  $|\mathrm{Nm}_{F/\mathbb{Q}} \Upsilon_{E'/F}|$  is bounded.

Choose  $S \in \Sigma_F$  s.t.  $S(6) \cup \Sigma_F^\infty \cup \mathrm{Ram}_{E/F} \subset S$  and  $\mathrm{cl}(\mathcal{O}_{F,S}) = 1$ , ?  $\square$

## 6 Small Points

**Notation (21.1.6.1).**

- Let  $F \in \mathbf{NField}$  and  $G \in \mathbf{semiAbVar}/F$  with maximal subtorus  $T$  of dimension  $t$  and quotient  $A = G/T$  of dimension  $g$ .
- Let  $\mathcal{L}$  be the canonical line bundle and  $h$  be a canonical height on  $A$  (21.1.3.13).
- For each  $x \in G(\overline{\mathbb{Q}})$ , use  $\mathcal{O}(x)$  to denote the Galois orbit of  $x$ .

- A **torsion subvariety** is a subvariety  $Y \subset G$  of the form  $Y = x + B$  where  $x$  is a torsion point and  $B \subset G$  is an semi-Abelian subvariety.

┘

**Def. (21.1.6.2) [Small Sequences].** A sequence  $(x_n)_{n \in \mathbb{N}} \subset G(\overline{F})$  is called

- **strict** if no infinite subsequence is contained in a proper torsion subvariety of  $A$ .
- **generic** if no infinite subsequence is contained in a proper subvariety of  $A$ .
- **small** if  $\lim_{n \rightarrow \infty} h(x_n) = 0$ .

┘

**Prop. (21.1.6.3).** If  $X \in \text{Var}/\overline{\mathbb{Q}}$  and  $(x_n) \in X^{\mathbb{N}}$  is a Zariski dense sequence in  $X$ , then  $(x_n)$  has a generic sequence.

┘

*Proof:* This follows from the fact that there are only countably many proper subvarieties of  $X$ .  $\square$

**Thm. (21.1.6.4) [Strong Equidistribution Conjecture, Bilu/Szpiro-Ullmo-Zhang/Kuhne].** For any strict small sequence (21.1.6.2)  $(x_i) \in A^{\mathbb{N}}$ , there is a weak convergence of measures on  $G_{\mathbb{C}_v}^{\text{an}}$ :

$$\frac{1}{\#\mathcal{O}_v(x_i)} \sum_{y \in \mathcal{O}_v(x_i)} \delta_y \rightharpoonup \frac{1}{\deg_{\mathcal{L}}(\overline{G})} c_1(\overline{\mathcal{L}_v})^{\wedge(g+t)}.$$

Notice for  $v \in \inf \Sigma_F$ ,  $c_1(\overline{\mathcal{L}_v})^{\wedge(g+t)}$  is a Haar measure on  $G_{\mathbb{C}_v}^{\text{an}}$ .

┘

*Proof:* Cf. [Points of small height on semiabelian varieties]P2078<sup>?</sup>. The proof is Arakelov, and is similar to that of the Abelian variety case.  $\square$

**Cor. (21.1.6.5) [Mordell-Lang plus Bogomolov, Poonen/Remond].** Let  $\Gamma \subset G(\overline{F})$  be a f.g. subgroup, with divisible hull  $\Gamma'$ . For  $\varepsilon \in \mathbb{R}_+$ , denote  $\mathbb{D}(\Gamma', \varepsilon) \subset G(\overline{F})$  the metric nbhd in the metric defined by the canonical height<sup>?</sup>. Then for any closed subvariety  $X \subset A$ , there exists some  $\varepsilon \in \mathbb{R}_+$  s.t.  $X(\overline{F}) \cap \Gamma'_{\varepsilon}$  is contained in a finite union  $\cup_j Z_j$  where each  $Z_j \subset X_{\overline{F}}$  is a torsion subvariety (21.1.6.1).

┘

*Proof:* <sup>?</sup>. Cf. [Poonen].  $\square$

**Cor. (21.1.6.6) [Bogomolov Conjecture, Ullmo-Zhang].** If  $X \subset A$  is subvariety but not a torsion subvariety, then there exists  $\varepsilon \in \mathbb{R}_+$  s.t.  $\{P \in X(\overline{F}) | h(P) \leq \varepsilon\}$  is not Zariski-dense in  $X$ .

┘

*Proof:* <sup>?</sup>  $\square$

**Cor. (21.1.6.7) [Manin-Mumford-Lang Conjecture, Raynaud].** By Bogomolov conjecture (21.1.6.6) and (21.1.3.5), if  $X \subset G$  is subvariety but not a torsion subvariety (21.1.6.1), then  $X(\overline{\mathbb{Q}}) \cap G(\overline{\mathbb{Q}})_{\text{tor}}$  is not Zariski-dense in  $X$ .

This is a special case of the Mordell-Lang conjecture (21.1.7.5).

┘

**Thm. (21.1.6.8) [Equidistribution of Small Points, Zhang].** If  $(x_n)_{n \in \mathbb{N}}$  is a small and strict sequence in  $A$ , then for any continuous function on  $A^{\text{an}}$ ,

$$\lim_{n \rightarrow \infty} \frac{1}{\#\mathcal{O}(x_n)} \sum_{y \in \mathcal{O}(x_n)} f(y) = \int_{A^{\text{an}}} f dx.$$

┘

*Proof:* Firstly  $(x_n)$  is generic: If there is an infinite subsequence with Zariski closure a proper subvariety  $X \subset A$ , then it follows from Bogomolov's conjecture (21.1.6.6) that  $X$  is a torsion subvariety. Then  $(x_n)$  is not strict, contradiction.  $\square$

**Cor. (21.1.6.9).** Let  $S \subset A^{\text{an}}$  be a closed subset, then  $S$  contains only f.m. non-Galois-conjugate maximal torsion subvarieties. And if  $S'$  is the complement in  $S$  of all the Galois orbits of maximal torsion subvarieties contained in  $S$ , then there exists  $\varepsilon \in \mathbb{R}_+$  s.t. for any  $x \in A(\overline{\mathbb{Q}})$  s.t.  $O(x) \subset S'$ ,  $h(x) > \varepsilon$ .  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (21.1.6.10) [Theorem of Successive Minima, Zhang].** Let  $F \in \mathbf{NField}$ ,  $X \in \mathcal{V}\text{ar}^d/F$ , and  $\mathcal{L}$  an ample line bundle on  $X$  with a semipositive adelic metric  $\|\cdot\|$ . For  $1 \leq i \leq d$ , define

$$\lambda_i = \sup_{Y \subset X, \dim Y = i} \left( \inf_{P \in X(\overline{\mathbb{Q}}) \setminus Y(\overline{\mathbb{Q}})} h_{\mathcal{L}}(P) \right),$$

then

$$\lambda_d \geq h_{\mathcal{L}}(X) \geq \frac{1}{d}(\lambda_1 + \dots + \lambda_d).$$

$\lrcorner$

*Proof:* Cf. [Small Points and Adelic metric, Shouwu Zhang]. ?  $\square$

### Small Points on $\mathbb{G}_m^n$

References are [Sch96] and [B-G06].

### Small Points on Elliptic Curves

**Conj. (21.1.6.11) [Lang's Height Conjecture].** For any  $F \in \mathbf{NField}$ , there exists  $c_1(F), c_2(F) \in \mathbb{R}_+$  s.t. for any  $E \in \mathbf{EllCur}/F$  and  $P \in E(F) \setminus E(F)_{\text{tor}}$ ,

$$h_{\text{N-T}}(P) \geq c_1(F) \log |\text{Nm}_{F/\mathbb{Q}} \Delta_{E/F}^{\min}| + c_2(F). \quad (21.3.4.24)$$

$\lrcorner$

**Thm& Conj. Cor. (21.1.6.12) [Silverman].** This conjecture is true when we restrict to  $E \in \mathbf{EllCur}/F$  with  $j(E) \in \mathbb{Z}$ .  $\lrcorner$

*Proof:* Cf. [Sil86] P263.  $\square$

## 7 Finiteness Theorems

**Prop. (21.1.7.1).** Let  $L/F \in \mathbf{NField}$  be Galois, then for any  $A \in \mathbf{AbVar}/F$ , there exists f.m. isomorphism classes of  $B \in \mathbf{AbVar}/F$  s.t.  $A_L \cong B_L$ .  $\lrcorner$

*Proof:* Cf. [Conrad note, L14] ?  $\square$

**Prop. (21.1.7.2) [Weak Finiteness, Faltings].** Let  $F \in \mathbf{NField}$  and  $(A, \lambda) \in \mathbf{AbVar}^{g, \text{polar}=d}/F$ , then there exists only f.m. isomorphism classes of  $(B, \lambda') \in \mathbf{AbVar}^{g, \text{polar}=d}/F$  s.t.  $(A_{\overline{F}}, \lambda) \cong (B_{\overline{F}}, \mu)$ .  $\lrcorner$

*Proof:* Firstly, by (21.1.2.14), any such  $B$  is also semistable, so the set of good reductions are stable under base change, thus the set of good reductions of  $A$  and  $B$  in  $\Sigma_F$  are the same.

Then by (21.7.1.1) and (20.8.1.8), for  $\ell \in \mathbf{P} \setminus \{2\}$ , there is a finite extension  $L/F$  s.t. every  $\ell$ -torsion points of such  $B$  are in  $B(L)$ .

Now any such  $B$  are isomorphic to  $A$  over  $L$  as above: Given any isomorphism  $\alpha : (A_{\overline{F}}, \lambda) \cong (B_{\overline{F}}, \mu)$ , for  $\alpha \in \text{Gal}_L$ ,  $\alpha$  and  $\sigma(\alpha)$  has the same action on  $\ell$ -torsion points, as all these points are  $L$ -rational, so  $\sigma(\alpha)^{-1} \circ \alpha$  acts trivially on  $A[\ell]$ , which implies by (20.9.7.6) that  $\sigma = \alpha(\sigma)$ , so  $\sigma$  is defined over  $L$  and we are done.

Finally we finish using (21.1.7.1) and (20.9.7.3).  $\square$

**Thm. (21.1.7.3) [Finitely Many Isogenous Varieties, Mazur [Maz78]/Faltings [Fal83]].** Let  $F \in \mathbf{NField}$  and  $A \in \mathcal{AbVar}/F$ , then there are only f.m. isomorphism classes of Abelian varieties  $B \in \mathcal{AbVar}/F$  isogenous to  $A$ .  $\lrcorner$

*Proof:* If  $A$  is semistable, then by (21.1.2.14), any  $B \in \mathcal{AbVar}/F$  isogenous to  $A$  is also semistable. Then we are done by (21.1.5.10) and (21.1.5.6).

For a general  $A$ , there exists a finite extension  $L/F$  s.t.  $A_L$  is semistable by??. Then the general case reduced to the semistable case by using (21.1.7.1).  $\square$

**Thm. (21.1.7.4) [Good Reductions are Rare, Shafarevich/Faltings].** For  $F \in \mathbf{NField}$  and  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$ , there are only f.m. isomorphism classes of  $A \in \mathcal{AbVar}^g/F$  having good reduction at all finite places outside  $S$ .  $\lrcorner$

*Proof:* By (21.1.7.3), it suffices to show that there are only f.m. isogeny classes of  $A \in \mathcal{AbVar}^g/F$  having good reduction at all finite places outside  $S$ . But then we can use (21.1.2.27) and the fact there are only f.m. such polynomials  $P_\ell(A_v, t)$  because of Weil conjecture.  $\square$

### Mordell-Lang Conjecture

**Thm. (21.1.7.5) [Mordell-Lang Conjecture, Laurent [Lau84]/Falting-Vojta/Hindry/Raynaud/McQuillan].**

Let  $F \in \mathbf{NField}$  and  $G/F$  be a semi-Abelian variety. Suppose  $X \subset A$  is a closed subvariety that is not a translate of a semi-Abelian subvariety of  $A$ , and  $\Gamma \subset A(\overline{F})$  is a subgroup s.t.  $\text{rank } \Gamma_{\mathbb{Q}} < \infty$ , with divisible hull  $\Gamma'$ , then  $X(\overline{F}) \cap \Gamma'$  is not Zariski dense in  $X$ .  $\lrcorner$

*Proof:* ?  $\square$

**Remark (21.1.7.6).** This can also be proved by using Zhang's equidistribution theorem (21.1.6.8).  $\lrcorner$

**Conj. Cor. (21.1.7.7) [Multiplicative Manin-Mumford, Conjecture of Lang].** If  $V \subset (\mathbb{C}^\times)^n$  is an algebraic subvariety, then there are f.m. torsion cosets  $X_i \subset V$  that accounts for all the torsion points of  $(\mathbb{C}^\times)^n$  that is contained in  $V$ .  $\lrcorner$

*Proof:* There is a direct proof in [Pila, AWS notes]Thm1.8.

It suffices to prove that if the torsion points are dense in  $V$ , then  $V$  is a torsion coset. In this case,  $V$  is defined over  $\overline{\mathbb{Q}}$ , because the torsion points are roots of unity.  $\square$

## 8 Galois-Cohomologies of Abelian Varieties

**Prop. (21.1.8.1) [Fundamental Exact Sequence].** Let  $k \in \mathbf{Field}$  and  $\varphi : X \rightarrow X' \in \text{IsogAbVar}_k$  be an étale isogeny, then there is an exact sequence

$$0 \rightarrow \ker(\varphi)(k^s) \rightarrow X(k^s) \rightarrow X'(k^s) \rightarrow 0$$

by (10.1.5.6), thus taking Galois cohomology, there is an exact sequence

$$0 \rightarrow X'(k)/\varphi(X(k)) \xrightarrow{\delta} H^1(\text{Gal}_k, X[\varphi]) \rightarrow H^1(\text{Gal}_k, X)[\varphi] \rightarrow 0.$$

┘

**Cor. (21.1.8.2).** By (20.9.6.18), if  $n \in \mathbb{Z} \cap K^\times$ , then there is an exact sequence

$$0 \rightarrow X(k^s)/(n) \xrightarrow{\delta} H^1(\text{Gal}_k, X[n]) \rightarrow H^1(\text{Gal}_k, X)[n] \rightarrow 0.$$

And the first map is described as follows: for  $P \in X(k^s)$ , let  $[n]Q = P$  where  $Q \in X(k^s)$ , then  $P$  is mapped to the 1-cocycle  $f(\sigma) = \sigma(Q) - Q$ . ┘

**Def. (21.1.8.3) [(Classical) Selmer Groups].** Let  $F$  be a global field or the function field of a non-singular smooth curve over an alg.closed field  $k$ ,  $\varphi : A \rightarrow A' \in \text{AbVar}/F$  an étale isogeny, then by restriction (9.8.1.11), there is an exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & A'(F)/\varphi(X(F)) & \longrightarrow & H^1(F, A[\varphi]) & \longrightarrow & H^1(F, A)[\varphi] \longrightarrow 0 \\ & & \downarrow & & \downarrow \text{res} & & \downarrow \text{res} \\ 0 & \longrightarrow & \prod_{v \in \Sigma_F} X'(F_v)/\varphi(A(F_v)) & \longrightarrow & \prod_{v \in \Sigma_F} H^1(F_v, A[\varphi]) & \longrightarrow & \prod_{v \in \Sigma_F} H^1(F_v, A)[\varphi] \longrightarrow 0 \end{array}$$

Where  $v$  are extended to  $F^s$  arbitrarily. Then the  $\varphi$ -**Selmer group** of  $A/F$  is a subgroup of  $H^1(\text{Gal}_F, A[\varphi])$  defined by

$$\text{Sel}_\varphi(A/F) = \ker \left( H^1(F, A[\varphi]) \xrightarrow{\text{res}} \prod_{v \in \Sigma_F} H^1(F_v, A)[\varphi] \right).$$

Notice the kernel is independent of the extension of  $v$  to  $F^{\text{sep}}$ , because any two such extensions induce conjugate groups  $\text{Gal}_{F_v}$  in  $\text{Gal}_F$  and use (9.8.1.13). ┘

**Prop. (21.1.8.4) [Selmer Groups and Šafarevič -Tate Groups].** Let  $C$  be a proper Dedekind scheme or a non-singular curve over a field  $k$  with fraction field  $F$ ,  $\varphi : A \rightarrow A'$  an étale isogeny of Abelian varieties over  $F$ , then there is an exact sequence

$$0 \rightarrow A'(K)/\varphi(A(K)) \rightarrow \text{Sel}_\varphi(A/K) \rightarrow \text{III}(A/K)[\varphi] \rightarrow 0 \quad (9.8.3.28).$$

┘

*Proof:* This exact sequence is a consequence of (21.1.8.3) and (21.1.8.3)(9.8.3.28). ┘

**Remark (21.1.8.5) [Computability].** It is difficult to characterize  $A'(K)/\varphi(A(K)) \subset \text{Sel}_\varphi(A/K)$ . ┘

**Prop. (21.1.8.6) [Sel $_\varphi(A)$  is Finite].** Let  $F \in \text{GField}$  or the function field of a non-singular smooth curve over an alg.closed field  $k$ ,  $\varphi : A \rightarrow A'$  an étale isogeny of Abelian varieties over  $F$ , then if  $S$  is a finite set of places of  $F$  containing all the places dividing  $\deg(\varphi)$ , and the places that  $A$  has bad reductions, then any element in  $\text{Sel}_\varphi(A/F)$  is unramified at all finite places outside  $S$ . In particular,  $\# \text{Sel}_\varphi(A/F) < \infty$  by (9.8.3.11). ┘

*Proof:* By definition of  $\text{Sel}_\varphi$  and (21.1.8.3), if  $\xi \in \text{Sel}_\varphi(A/F)$ , then for any  $v \in \Sigma_F^{\text{fin}}$ , there is some  $P \in A(F_v^s)$  s.t.  $\xi_\sigma = \sigma(P) - P \in A[\varphi]$  for any  $\sigma \in G_v$ . But if  $A$  has good reduction at  $v$  and  $\sigma \in I_v$ , then  $\sigma(P) - P \mapsto \tilde{O} \in \tilde{A}_v$ . But notice  $A[\varphi] \subset A[m]$  which maps injectively into  $\tilde{A}_v(k_v)$  by (21.3.4.20) if  $v$  is not dividing  $\deg(\varphi)$ . Thus in this case,  $\xi_\sigma = 0$  for any  $\sigma \in I_v$ .  $\square$

**Cor. (21.1.8.7) [ $\ell^\infty$ -Selmer Groups].** If  $\ell \in \text{Prime} \setminus \text{char } F$ , then the  $\ell$ -primary part  $\text{III}(A/F)[\ell^\infty] = \varinjlim_{n \geq 1} \text{III}(A/F)[\ell^n]$  is isomorphic to  $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\delta_\ell} \times T_\ell$  for some  $\delta_\ell \in \mathbb{N}$  and  $T_\ell \in \text{Ab}^{\text{fin}}$ . And define

$$\text{Sel}_{\ell^\infty}(A/F) = \varinjlim_{n \geq 1} \text{Sel}_{\ell^n}(A/F),$$

called the  $\ell^\infty$ -Selmer group. Then by taking limits of the exact sequence in (21.1.8.4), there is an exact sequence

$$0 \rightarrow A(F) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \rightarrow \text{Sel}_{\ell^\infty}(A/F) \rightarrow \text{III}(A/F)[\ell^\infty] \rightarrow 0.$$

If we define the  $\ell$ -Selmer rank  $\text{rank}_\ell(A/F) = \text{rank}(A/F) + \delta_\ell$ , then

$$\text{Sel}_{\ell^\infty}(A/F) = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\text{rank}_\ell(A/F)} \cdot \widetilde{\text{III}}(A/F)[\ell^\infty],$$

for some finite group  $\widetilde{\text{III}}(A/F)[\ell^\infty]$ .  $\lrcorner$

**Prop. (21.1.8.8) [ $\text{Sel}_p(A/F)$  and  $\text{Sel}_{p^\infty}(A/F)$ ].** If  $p \nmid \text{char } F$  and  $A(F)[p] = 0$ , then for any  $m \leq n \in \mathbb{Z}_+$ ,

- The inclusion  $A[p^m] \subset A[p^n]$  induces an isomorphism  $H^1(F, A[p^m]) = H^1(F, A[p^n])[p^m]$ .
- $\text{Sel}_{p^m}(A/F) = \text{Sel}_{p^n}(A/F)[p^m] = \text{Sel}_{p^\infty}(A/F)[p^m]$ .

$\lrcorner$

*Proof:* 1: Taking cohomology of the exact sequence  $0 \rightarrow A[p^m](F^s) \rightarrow A[p^n](F^s) \xrightarrow{p^m} A[p^{n-m}](F^s) \rightarrow 0$  implies that

$$H^1(F, A[p^m]) = \ker \left( H^1(F, A[p^n]) \xrightarrow{p^m} H^1(F, A[p^{n-m}]) \right).$$

And replacing  $m$  with  $n - m$  implies that  $A[p^{n-m}] \subset A[p^n]$  induces an inclusion  $H^1(F, A[p^{n-m}]) \subset H^1(F, A[p^n])$ . Then the composition  $A[p^n] \xrightarrow{p^m} A[p^{n-m}] \subset A[p^n]$  induces a composition  $H^1(F, A[p^n]) \xrightarrow{p^m} H^1(F, A[p^{n-m}]) \hookrightarrow H^1(F, A[p^n])$  that equals multiplication by  $m$ . Thus

$$H^1(F, A[p^m]) = \ker \left( H^1(F, A[p^n]) \xrightarrow{p^m} H^1(F, A[p^{n-m}]) \right) = H^1(F, A[p^n])[p^m].$$

2: This follows from 1 and the definition of  $\text{Sel}_{p^m}(A/F)$ .  $\square$

**Prop. (21.1.8.9) [Poonen-Rains[?P-R15]/Bhargava-Kane-Lenstra-Poonen-Rains[BKLPR]].** For  $E \in \text{EllCur}/\mathbb{Q}$  and  $n \in \mathbb{Z}_+$ , there is an exact sequence

$$\begin{array}{ccc} E(\mathbb{Q})/(n) & \longrightarrow & H^1(\mathbb{Q}, E[n]) \\ \downarrow & & \downarrow \beta \\ E(\mathbf{A})/(n) & \xrightarrow{\alpha} & H^1(\mathbf{A}, E[n]) \end{array},$$

and

- Each of  $\text{Im}(\alpha), \text{Im}(\beta)$  is a maximal isotropic subspace of  $H^1(\mathbf{A}, E[n])$  w.r.t.  $Q$ ?
- If  $n \in \text{Prime}$  or the map  $\text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2; \mathbb{Z}/(n))$  is a surjection, then  $\beta$  is injective.

In particular, for  $p \in \text{Prime}$ ,  $\text{Sel}_p(E) = \beta^{-1} \text{Im}(\alpha)$  is the intersection of two maximal isotropic subspaces of  $H^1(\mathbf{A}, E[n])$ .  $\lrcorner$

*Proof:* Cf. [?P-R15] and [BKLPR].?  $\square$

### Tate-Shafarevich Groups

**Conj. (21.1.8.10)** [Tate-Šafarevič]. For any  $A \in \mathcal{AbVar}/F$ , we have  $\#\text{III}(A/F) < \infty$  (9.8.3.28). ┘

*Proof:* □

**Remark (21.1.8.11).** There are still no  $E \in \text{EllCur}/\mathbb{Q}$  with  $\text{rank}_{\text{an}}(E/\mathbb{Q}) \geq 2$  s.t.  $\text{III}(E/\mathbb{Q})$  can be proven to be finite. ┘

**Thm. (21.1.8.12)** [Cassels-Tate Pairing, Tate[Tat63]/Cassels[?Cas65]]. Let  $A \in \mathcal{AbVar}/F$  and  $A^\vee$  its dual, then there is a bilinear **Cassels-Tate pairing**  $\text{III}(A/K) \times \text{III}(A^\vee/K) \rightarrow \mathbb{Q}/\mathbb{Z}$  (9.8.3.28) and the radical on both sides are the set of divisible elements. ┘

*Proof:* Cf.[Birch and Swinnerton-Dyer Conjecture, P10]. ? □

**Cor. (21.1.8.13)** [Principally Polarized Šafarevič Groups, Cassels/Poonen]. If  $K \in \mathcal{AbVar}/F$  has a principal polarization  $\lambda$  that comes from a rational divisor  $D$ , then the pullback pairing  $\text{III}(A/K) \times \text{III}(A/K) \rightarrow \mathbb{Q}/\mathbb{Z}$  is alternating. In particular, if  $\#\text{III}(A/F) < \infty$ , then it is a perfect square, by (3.2.4.40).

In particular, if  $E \in \text{EllCur}/F$  and  $\#\text{III}(E/F) < \infty$ , then it is a perfect square. ┘

*Proof:* ? □

**Remark (21.1.8.14).** For a general Abelian variety  $A$ ,  $\#\text{III}(A/\mathbb{Q})$  may not be a square. ┘

The discussion of Selmer groups and ranks are continued in B-S.D Conjecture 31.4.



## 21.2 Complex Multiplication Theory

*“The theory of complex multiplication...is not only the most beautiful part of mathematics but also of all science.”*

–D. Hilbert

References are [S-T61, CCO14, Mil20b], [Lang, Complex Multiplication], [Abelian varieties with complex multiplication and modular functions, Shimura(1998)] and [The Fundamental Theorem of Complex Multiplication, Milne, 2007].

**Notation (21.2.0.1).**

- Use notations defined in [Abelian Varieties](#).

⌋

Abelian varieties with complex multiplication correspond to special points on the moduli variety of abelian varieties, and their arithmetic is intimately related to that of the values of modular functions and modular forms at those points.

### 1 CM-Algebras and CM-Types

**Def. (21.2.1.1) [CM-Fields].** A **CM-field** is a number field that is a totally imaginary quadratic extension of a totally real field.

⌋

**Prop. (21.2.1.2) [Characterizing CM-Fields].** For  $E \in \mathbf{NField}$ , then  $E$  is a CM-field or totally real iff there exists exactly one  $c_E \in \text{Aut}(E)$  s.t. for any  $\rho \in \text{Hom}(E, \mathbb{C})$ ,  $\rho \circ c_E = \bar{\rho}$ . And it is CM iff  $c_E \neq \text{id}_E$ .

⌋

*Proof:* If  $\rho \circ c_E = \bar{\rho}$  for any  $\rho \in \text{Hom}(E, \mathbb{C})$ , then  $\rho \circ c_E^2 = \rho$ , so  $c_E^2 = \text{id}$ . if  $F$  is the fixed field of  $c_E$ , then  $[E : F] = 2$ , and clearly  $F$  is totally real. Conversely, if  $E$  is a CM-field or totally real, then clearly there exists a unique such  $c_E$ , which is the identity if  $E$  is totally real and the unique involution fixing the totally real subfield  $F$ .

In this case, if  $c_E \neq \text{id}_E$ , then  $\rho(E) \not\subseteq \mathbb{R}$  for any  $\rho \in \text{Hom}(E, \mathbb{C})$ , thus  $E$  is totally imaginary. And if  $c_E = \text{id}_E$ , then  $E$  is also totally real.  $\square$

**Cor. (21.2.1.3).** A finite composite of CM-fields is a CM-field. In particular, the Galois closure of a CM-field is a CM-field.

And the composite of all CM-fields in  $\mathbb{C}$  are the field  $\mathbb{Q}^{\text{cm}} \subset \overline{\mathbb{Q}}$  corresponding to the subgroup of  $\text{Gal}_{\mathbb{Q}}$  generated by all the elements  $\{[\sigma, c] | \sigma \in \text{Gal}_{\mathbb{Q}}\}$ , which is a normal subgroup.  $\square$

*Proof:* The complex involutions on these CM-fields is compatible on their intersections, thus defines a complex involution on their composite.  $\square$

**Cor. (21.2.1.4).** Any CM-field  $E$  is of the form  $E = F(\alpha)$ , where  $F$  is totally real,  $\alpha^2 \in F$ , and  $\rho(\alpha^2) \in \mathbb{R}_-$  for any homomorphism  $\rho : F \rightarrow \mathbb{C}$ .

⌋

*Proof:* If  $F$  is the totally real field contained in  $E$  and  $\alpha \in E$  generates  $E$  over  $F$ , then by completing the square, we can assume  $\alpha^2 \in F$ . Then  $\rho(\alpha^2) \in \mathbb{R}_-$  for any  $\rho : F \rightarrow \mathbb{C}$ , otherwise  $E$  is not totally imaginary.  $\square$

**Def. (21.2.1.5) [CM-Algebras].** A **CM-algebra** is a finite product of CM-fields.

⌋

**Def. (21.2.1.6) [CM-Types].** A **CM-type** for a CM-algebra (21.2.1.5)  $E$  is a subset  $\Phi \subset \text{Hom}(E, \mathbb{C})$  s.t.  $\text{Hom}(E, \mathbb{C}) = \Phi \amalg \Phi \circ c_E$  (21.2.1.2).

A pair  $(E, \Phi)$  where  $E$  is a CM-algebra and  $\Phi$  is a CM-type on  $E$  is called a **CM-pair**.

More generally, if  $E$  is an étale algebra over  $\mathbb{Q}$ , then a **CM-type** on  $E$  is a subset  $\Phi \subset \text{Hom}(E, \mathbb{C})$  s.t.  $\text{Hom}(E, \mathbb{C}) = \sigma\Phi \amalg c \circ \sigma\Phi$  for any  $\sigma \in \text{Gal}_{\mathbb{Q}}$ .  $\perp$

**Prop. (21.2.1.7).** Let  $E$  be an étale algebra over  $\mathbb{Q}$  and  $F$  the product of the largest totally real subfields of the factors of  $E$ , then choosing a CM-type is equivalent to choosing an extension  $\rho'$  to  $E$  of any homomorphism  $\rho : F \rightarrow \mathbb{R}$ . Thus a CM-pair  $(E, \Phi)$  defines an isomorphism

$$E \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow[\cong]{\Phi} \prod_{\rho: F \rightarrow \mathbb{R}} \mathbb{C} \cong \mathbb{C}^{\Phi} : a \otimes r \mapsto (\rho'(a)r)_{\rho}, \quad \Phi = \{\rho' | \rho : F \rightarrow \mathbb{R}\}$$

$\perp$

**Def. (21.2.1.8) [Primitive CM-Fields].** If  $E_0$  is a CM-field and  $E/E_0$  is a finite extension, then a CM-type  $\Phi_0$  on  $E_0$  extends to a CM-type on  $E$  (21.2.1.6) by defining

$$\Phi = \{\varphi \in \text{Hom}(E, \mathbb{C}) | \varphi|_{E_0} \in \Phi_0\}.$$

A **primitive CM-field** is a CM-pair (21.2.1.6)  $(E, \Phi)$  where  $E \in \text{Field}$  and there doesn't exist a proper subfield  $E_0 \subset E$  s.t.  $\Phi$  is the extension of a CM-type on  $E_0$ .  $\perp$

**Prop. (21.2.1.9).** Every CM-pair  $(E, \Phi)$  where  $E$  is a field is an extension of a unique primitive CM-pair  $(E_0, \Phi_0)$  s.t.  $E_0 \subset E$ . Moreover, for any Galois CM-field  $E_1$  containing  $E$ ,  $E$  is the fixed field of  $E$  defined by the subgroup

$$H = \{\sigma \in \text{Gal}(E_1/\mathbb{Q}) : \Phi_1^{\sigma} = \Phi_1\},$$

where  $\Phi_1$  is the extension of  $\Phi$  to  $E_1$ .  $\perp$

*Proof:* If  $E$  is Galois over  $\mathbb{Q}$ , and  $E_0$  is defined as above, then  $\Phi c_E = c\Phi \neq \Phi$ , so  $c_E \notin H$ , thus  $c_E$  acts non-trivially on  $E_0$ . And  $c_E$  preserves  $E_0$ : We need to show that  $\sigma c_E a = c_E a$  for any  $a \in E_0$ . For which it suffices to show that  $c_E \sigma c_E \in H$ . For this, notice

$$\Phi c_E \sigma c_E = c\Phi \sigma c_E = c\Phi c_E = \Phi.$$

Thus  $E_0$  is CM-subfield of  $E$ . And  $\Phi_0 = \Phi|_{E_0}$  is a CM-type on  $E_0$ , because if  $\varphi'|_{E_0} = c\varphi|_{E_0}$  for different  $\varphi, \varphi' \in \Phi$ , then  $c\varphi \in \varphi'H \subset \Phi$ , contradiction.

And if  $E_0$  is extended from another CM-subfield  $E'$ , then any  $\sigma \in \text{Gal}(E/\mathbb{Q})$  fixing  $E'$  will fix  $\Phi$  and lies in  $H$ . This shows  $E_0 \subset E'$ . So  $(E_0, \Phi_0)$  is primitive.

For the general case, it suffices to notice that this  $E_0$  is contained in  $E$  because any  $\sigma \in \text{Gal}(E_1/\mathbb{Q})$  fixing  $E$  will fix  $\Phi_1$  thus fixes  $E_0$  by definition. And  $\Phi$  clearly extends  $\Phi_0$ .  $\square$

**Cor. (21.2.1.10).** A CM-pair  $(E, \Phi)$  where  $E$  is a field is primitive iff there exists a Galois CM-field  $E_1$  containing  $E$  s.t.  $E$  is the fixed field of  $E$  defined by the subgroup

$$H = \{\sigma \in \text{Gal}(E_1/\mathbb{Q}) : \Phi_1^{\sigma} = \Phi_1\},$$

where  $\Phi_1$  is the extension of  $\Phi$  to  $E_1$ .  $\perp$

**Prop. (21.2.1.11) [Shimura-Taniyama].** Let  $E$  be a CM-field,  $F = E^{\mathbf{c}_E}$  (21.2.1.2) and  $E = F(\alpha)$  where  $\alpha^2 \in F$  is totally negative. Then

$$\Phi = \{\varphi \in \text{Hom}(E, \mathbb{C}) : \text{Im}(\varphi(\alpha)) > 0\}$$

is a CM-type on  $E$ . And  $(E, \Phi)$  is a primitive CM-pair iff

- $E = \mathbb{Q}(\alpha)$ .
- $\sigma(\alpha)/\alpha$  is not totally positive for any  $\sigma \in \text{Gal}(E/\mathbb{Q})$ .

┘

*Proof:*  $\Phi$  is a CM-type on  $E$  because for any  $\varphi \in \Phi$ ,  $\varphi \mathbf{c}_E \notin \Phi$  because  $\text{Im}(\varphi \mathbf{c}_E(\alpha)) = -\text{Im}(\varphi(\alpha)) < 0$ .

Let  $F_1$  be the Galois closure of  $F$  and  $E_1 = F_1(\alpha)$ , and  $\Phi_1$  is the extension of  $\Phi$  to  $E_1$ , then

$$\Phi_1 = \{\varphi \in \text{Hom}(E_1, \mathbb{C}) : \text{Im}(\varphi(\alpha)) > 0\}.$$

and if

$$H = \{\sigma \in \text{Gal}(E_1/\mathbb{Q}) : \Phi_1^\sigma = \Phi_1\},$$

then  $H$  is exactly the group of  $\sigma \in \text{Gal}(E_1/\mathbb{Q})$  s.t.  $\sigma(\alpha)/\alpha \in \mathbb{R}_+$ . Then the assertion follows from (21.2.1.10).  $\square$

### Reflex Fields and Reflex Norms

**Def. (21.2.1.12) [Reflex Fields of CM Pairs].** Let  $(E, \Phi)$  be a CM-pair, then the **reflex field**  $E^*$  of  $(E, \Phi)$  is the subfield of  $\overline{\mathbb{Q}}$  generated by the elements  $\left\{ \sum_{\varphi \in \Phi} \varphi(a) \mid a \in E \right\}$ . Then:

- $E^*$  is the fixed field of the group  $\{\sigma \in \text{Gal}_{\mathbb{Q}} \mid \sigma(\Phi) = \Phi\}$ .
- $E^*$  is a CM-field.
- If  $(E_1, \Phi_1)$  is an extension of  $(E, \Phi)$  (21.2.1.8), then the reflex field of  $(E_1, \Phi_1)$  is just  $E^*$ .

┘

*Proof:* 1: The group  $\{\sigma \in \text{Gal}_{\mathbb{Q}} \mid \sigma(\Phi) = \Phi\}$  fixes every element in  $E$ , so it is clearly contained in the fixed group of  $E^*$ , and if  $\sigma$  is in the fixed group of  $E^*$ , then

$$\sum_{\varphi \in \Phi} \varphi(a) = \sum_{\varphi \in \Phi} \sigma \varphi(a)$$

for each  $a \in E$ , so  $\sigma(\Phi) = \Phi$  by the linear independence of characters.

2: For  $\sigma \in \text{Gal}_{\mathbb{Q}}$  and  $a \in E$ ,

$$\mathbf{c}_\sigma \left( \sum_{\varphi \in \Phi} \varphi(a) \right) = \mathbf{c}_\sigma \left( \sum_{\varphi \in \Phi} \sigma \varphi(a) \right) = \sum_{\varphi \in \Phi} \sigma \varphi(\mathbf{c}_\sigma(a)) = \sigma \left( \sum_{\varphi \in \Phi} \varphi(\mathbf{c}_\sigma(a)) \right) = \sigma \mathbf{c}_\sigma \left( \sum_{\varphi \in \Phi} \varphi(a) \right).$$

Thus  $E$  is a CM-field or totally real by (21.2.1.2). The latter case is not possible by item 1 and the fact  $\Phi \circ \mathbf{c}_E \neq \Phi$ .

3 is trivial.  $\square$

**Def. (21.2.1.13) [Reflex CM-Pairs].** Let  $(E, \Phi)$  be a CM-pair that  $E$  is contained in  $\mathbb{Q}$ , let  $E_1$  be the Galois closure of  $E$ , and  $\Phi_1$  the extension of  $\Phi$  to  $E_1$ . Then  $(E_1, \Phi_1^{-1})$  is also a CM-pair. Then the primitive subfield  $(E^*, \Phi^*)$  satisfies  $E^*$  is the reflex field of  $(E, \Phi)$ .

$(E^*, \Phi^*)$  is called the **reflex CM-pair** of  $(E, \Phi)$ .  $\square$

**Prop. (21.2.1.14)[Reflex Modules].** For a CM-pair  $(E, \Phi)$ , the reflex field  $E^*$  (21.2.1.12) is the smallest subfield  $K \subset \overline{\mathbb{Q}}$  s.t. there exists an  $E \otimes K$ -module  $V$  with

$$V \otimes_K \overline{\mathbb{Q}} \cong \prod_{\varphi \in \Phi} \overline{\mathbb{Q}}_{\varphi}$$

as  $E \otimes \overline{\mathbb{Q}}$ -modules. Such a  $E \otimes E^*$ -module  $V_{\Phi}$  is uniquely determined, called the **reflex-module** of  $\Phi$ .  $\lrcorner$

*Proof:* Such a module is equivalent to a  $k$ -vector space with  $E$  acting by  $k$ -linear endomorphism s.t.  $\text{tr}_k(a|V) = \sum_{\varphi \in \Phi} \varphi(a)$  for each  $a \in E$ . Then clearly  $k$  contains each  $\sum_{\varphi \in \Phi} \varphi(a)$  and thus contains  $E^*$ .

Conversely, the  $E \otimes \overline{\mathbb{Q}}$ -vector space  $\prod_{\varphi \in \Phi} \overline{\mathbb{Q}}_{\varphi}$  with  $E$  acting on each component  $\overline{\mathbb{Q}}_{\varphi}$  by  $\varphi$ . This module is  $\text{Gal}(\overline{\mathbb{Q}}/E)$ -equivariant where  $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/E)$  acts on the second component of  $E \otimes \overline{\mathbb{Q}}$  and acts on  $\prod_{\varphi \in \Phi} \overline{\mathbb{Q}}_{\varphi}$  via

$$\sigma(x_{\varphi}) = (\sigma x_{\sigma^{-1} \circ \varphi}).$$

So it descends to a  $E \otimes E^*$ -module by Galois-descent.  $\square$

**Def. (21.2.1.15)[Reflex Norms].** Situation as in (21.2.1.14), the **reflex-norm**

$$\text{Nm}_{\Phi} : \text{Res}_{E/\mathbb{Q}}(\mathbb{G}_m) \rightarrow \text{Res}_{L/\mathbb{Q}}(\mathbb{G}_m)$$

is the determinant of the  $L$ -linear actions by  $E$ .  $\lrcorner$

### Classification of Primitive CM-Pairs

**Prop. (21.2.1.16).** Milne, Prop1.30. ?  $\lrcorner$

**Cor. (21.2.1.17).** Milne, Prop1.31. ?  $\lrcorner$

### Positive Involutions

**Def. (21.2.1.18)[Positive Involutions].** Let  $k \subset \mathbb{R} \in \mathbf{Field}$  and  $B \in \mathbf{Ring}^{\text{fin}}/k$ , then a **positive involution** w.r.t.  $\text{tr}$  is an involution  $\iota$  on  $B$  s.t.

$$\text{tr}(x) = \text{tr}(\iota(x)), \quad \text{tr}(\iota(x) \cdot x) > 0$$

for any nonzero  $x \in B$ .  $\lrcorner$

**Prop. (21.2.1.19)[Positive Involutions are Semisimple].** Let  $k \subset \mathbb{R} \in \mathbf{Field}$  and  $B \in \mathbf{CRing}$  is of f.d. over  $k$  with a positive involution  $\iota$ , then  $B$  is semisimple. And if  $Z$  is the center of  $B$ , then  $Z = \prod_i K_i$  where  $K_i \in \mathbf{NField}$ , and each  $K_i$  are stable under  $\iota$ .  $\lrcorner$

*Proof:* By (3.6.3.8), it suffices to show that  $B$  is Jacobson semisimple: If  $\mathfrak{a}$  is a nonzero nilpotent two-sided ideal of  $B$ ,  $a \neq 0 \in \mathfrak{a}$  is nilpotent, then  $b = a\iota(a) \neq 0$  because  $\text{tr}(a\iota(a)) > 0$ . Thus  $b \neq 0 \in \mathfrak{a}$ , and  $\iota(b) = b$ . So by the same reason  $b^2 \neq 0$ , and so on, so  $b$  is not nilpotent, contradicting (3.6.3.6).

The center is clearly invariant under the isomorphism given, so it is also semisimple, which is then a product of fields. To show each factor is stable under  $\iota$ , notice that if  $1 = \sum_i e_i$  is a decomposition w.r.t. this product, then  $1 = \sum_i \iota(e_i)$ , so  $\{\iota(e_i)\} = \{e_i\}$ . Moreover,  $\iota(e_i) = e_i$  for each  $i$ , because otherwise  $e_i \iota(e_i) = 0$ , and  $\text{tr}(e_i \iota(e_i)) = 0$ .  $\square$

**Prop. (21.2.1.20).** Let  $k \subset \mathbb{R}$  be a field and  $B$  a f.d.  $k$ -algebra with an involution  $\iota$ , then the following are equivalent:

- There is a faithful f.d.  $B$ -module  $V$  with a positive definite symmetric  $k$ -bilinear form  $(-, -) : V \times V \rightarrow k$  s.t.

$$(bu, v) = (u, \iota(b)v), \quad b \in B, u, v \in V.$$

- For any f.d.  $B$ -module  $V$ , there exists a positive definite symmetric  $k$ -bilinear form  $(-, -) : V \times V \rightarrow k$  s.t.

$$(bu, v) = (u, \iota(b)v), \quad b \in B, u, v \in V.$$

- There exists a  $k$ -linear functional  $\text{tr}$  on  $B$  s.t.  $\iota$  is positive w.r.t.  $\text{tr}$ .

⌋

*Proof:* 1  $\rightarrow$  2: As  $B$  is semisimple (21.2.1.19), any f.d.  $B$ -module is a direct summand of a direct sum of the faithful module  $V$ .

2  $\rightarrow$  3: Apply item2 to the  $B$ -module  $B$  itself, then we find a positive definite symmetric  $k$ -bilinear form  $(-, -)$  on  $B$ , and an orthonormal basis  $(e_1, \dots, e_n)$  of  $B$ . Then we define

$$\text{tr} : B \rightarrow \mathbb{Q} : \text{tr}(b) = \sum_i (bb_i, b_i)$$

then  $\text{tr}(b) = \text{tr}(\iota(b))$ , and

$$\text{tr}(b\iota(b)) = \sum_i (b\iota(b)b_i, b_i) = \sum_i (\iota(b)b_i, \iota(b)b_i) > 0.$$

3  $\rightarrow$  1:  $B$  is semisimple by (21.2.1.19), and then we can take  $V = B$ , and the positive definite symmetric  $k$ -bilinear form

$$(x, y) \mapsto \text{tr}_{B/k}(\iota(y)x).$$

then  $(bx, y) = \text{tr}(\iota(y)bx) = \text{tr}(\iota(\iota(b)y)x) = (x, \iota(b)y)$ . □

**Def. (21.2.1.21) [Albert-Algebras].** An **Albert-algebra** is a division ring  $D$  of f.d. over  $\mathbb{Q}$  equipped with a positive involution  $*$  (21.2.1.18). ⌋

**Thm. (21.2.1.22) [Classifying Albert-Algebras].** For each Albert-algebra  $(D, *)$  (21.2.1.21), exactly one of the following happens:

- $D = \mathbb{Z}$  is a totally real field and  $*$  =  $\text{id}_D$ .
- $D$  is a central quaternion algebra over a totally real field  $Z$ , and  $D$  splits at each real place of  $Z$ .
- $D$  is a central quaternion algebra over a totally real field  $Z$ , and  $D$  is non-split at each real place of  $Z$ .
- $D$  is a central division algebra over a CM-field  $Z$ , and for each  $v \in \Sigma_Z^{\text{fin}}$ ,  $\text{inv}_v(D) + \text{inv}_{v*}(D) = 0$ . Moreover,  $D$  splits at  $v \in \Sigma_Z^{\text{fin}}$  if  $v = v^*$ .

⌋

*Proof:* Cf. [Mumford, Abelian varieties] 21.2. ? □

**Prop. (21.2.1.23) [Commutative Albert-Algebras].** Every Albert-algebra  $(A, *)$  (21.2.1.21) is a product of the following two:

- $F$  is a totally real field and  $\iota = \text{id}_F$ .

- $E$  is a CM-field and  $\iota = c_E$ .

┘

*Proof:* Firstly, any product of these two are positive involutions. Conversely, if  $(B, \text{tr})$  is a f.d.  $\mathbb{Q}$ -algebra with a positive involution, then by (21.2.1.19),  $B$  is semisimple, thus a finite product of number fields. Then by (21.2.1.19), each factor is fixed by the involution, so it suffices to consider the case  $B \in \mathbf{NField}$ . Let  $F$  be the fixed field of  $\iota$ . Because the trace form  $\text{tr}_{B/\mathbb{Q}}$  is non-degenerate, we can assume  $\text{tr}(x) = \text{tr}(\alpha x)$  for some  $\alpha \in B^\times$ .

If  $F = B$ , then  $F$  is totally real: if there exists a non-real embedding  $\sigma : F \rightarrow \mathbb{C}$ , then we can use weak approximation to find  $x \in F^\times$  s.t.  $\sigma(\alpha x^2)$  is near  $-1$  and  $\varphi(\alpha x^2)$  is near  $0$  for any other embedding  $\varphi : F \rightarrow \mathbb{C}$ . Then clearly  $\text{tr}(\alpha x^2) < 0$ , contradiction.

If  $F \neq B$ , then  $[B : F] = 2$ , and  $\text{tr}_{B/\mathbb{Q}}(\alpha x \iota(x)) > 0$  for any  $x \in B^\times$ . Take  $x = 1$ , we see  $\text{tr}_{B/\mathbb{Q}}(\alpha) > 0$ . In particular,  $\alpha + \iota(\alpha) \neq 0 \in F$ . Then in particular,  $\text{tr}_{B/\mathbb{Q}}(\alpha x \iota(x)) = \text{tr}_{F/\mathbb{Q}}((\alpha + \iota(\alpha))x \iota(x)) > 0$ . So by the argument above,  $F$  is totally real. And  $B$  is totally imaginary: If there exists a real embedding  $\sigma : B \rightarrow \mathbb{C}$ , then  $\sigma$  and  $\sigma \circ \iota$  corresponds to different places, and by weak approximation, we can find  $x \in B$  s.t.  $\sigma((\alpha + \iota(\alpha))x)$  is near  $-1$  and  $\sigma(\iota(x))$  is near  $1$ , and  $\varphi(x)$  is near  $0$  for any other  $\varphi : B \rightarrow \mathbb{C}$ . Then  $\text{tr}_{F/\mathbb{Q}}((\alpha + \iota(\alpha))x \iota(x)) < 0$ , contradiction. So in this case  $B$  is a CM-field, and  $\iota = c_B$ . □

**Cor. (21.2.1.24).** There exists uniquely a positive involution on any CM-algebra, which is  $c_E$ . ┘

### Weil $q$ -Integers

**Prop. (21.2.1.25) [Weil  $q$ -Integers and CM Fields].** If  $p \in \mathbf{Prime}$ ,  $q \in p^{\mathbb{Z}}$  and  $\varpi \in \text{Weil}(q^{1/2})$ , then  $\mathbb{Q}(\varpi)$  is either isomorphic to  $\mathbb{Q}$  or  $\mathbb{Q}(\sqrt{q})$  or a CM-field. ┘

*Proof:* For any embedding  $\rho : \mathbb{Q}(\pi) \rightarrow \mathbb{C}$ ,

$$\rho(\pi) \overline{\rho(\pi)} = q = \rho(\pi) \rho(q/\pi),$$

so  $\overline{\rho(\pi)} = \rho(q/\pi)$ , and  $E = \mathbb{Q}(\pi)$  is a CM-field with the endomorphism  $c_E : \pi \mapsto q/\pi$ . □

**Prop. (21.2.1.26).** Let  $p \in \mathbf{P}$ ,  $q \in p^{\mathbb{Z}}$ ,  $F \in \mathbf{NField}$  and  $\pi, \pi' \in F \cap \text{Weil}(q^{1/2})$ . If  $\text{ord}_v(\pi) = \text{ord}_v(\pi')$  for any  $v \in \Sigma_F^p$ , then  $\pi/\pi'$  is a roots of unity in  $E$ . ┘

*Proof:* This is because there is an endomorphism of  $F$  (the conjugation) taking  $\pi$  to  $q/\pi$ , so for any  $v \notin \Sigma_F^p$ ,  $\text{ord}_v(\pi) = 0$ . Since the same is true for  $\pi'$ , and  $\text{ord}_v(\pi) = \text{ord}_v(\pi')$  for  $v \in \Sigma_F^\infty$  too, by the definition of Weil  $q$ -integers. Thus  $|\pi/\pi'|_v = 1$  for any  $v \in \Sigma_F$ , and the assertion follows from (19.4.1.20). □

## 2 CM-Abelian Varieties

**Def. (21.2.2.1) [Complex Multiplications].** For  $k \in \mathbf{Field}$  and  $A \in \mathbf{AbVar}/k$ ,

$$[\text{End}(A)_{\mathbb{Q}} : \mathbb{Q}]_{\text{red}} \leq 2 \dim A.$$

And  $A$  is called an Abelian variety with **strict complex multiplication** or a **strictly CM Abelian variety** if the equality holds.

And  $A$  is called an Abelian variety with **complex multiplication** or a **CM-Abelian variety** if  $A_{k^{\text{sep}}}$  is a strictly CM Abelian variety. ┘

*Proof:* Take  $\ell \in \text{Prime} \setminus \{\text{char } k\}$ , then  $\text{End}(A)_{\mathbb{Q}}$  acts faithfully on  $T_{\ell}(A, \mathbb{Q}_{\ell})$  by (21.1.4.3), which has dimension  $2 \dim A$  (20.9.6.19), so we finish by (20.9.6.14) and (3.6.4.25).  $\square$

**Prop. (21.2.2.2) [Complex Multiplication and CM-Algebras].** For  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ , the following are equivalent:

- $A$  is strictly CM.
- $\text{End}(A)_{\mathbb{Q}}$  contains an étale subalgebra of rank  $2 \dim A$  over  $\mathbb{Q}$ .
- For any Weil cohomology  $\mathcal{H}^*$  with coefficients in  $\Omega \in \text{Field}^0$ , the centralizer of  $\text{End}(A)_{\mathbb{Q}}$  in  $\text{End}_{\Omega}(H^1(A))$  is commutative (and equals  $Z(\text{End}(A)) \otimes \Omega$ ).

And if  $P \leq \text{End}(A)_{\mathbb{Q}}$  is an étale subalgebra of rank  $2 \dim A$  over  $\mathbb{Q}$ , then  $P = Z_{\text{End}(A)_{\mathbb{Q}}}(P)$ . If moreover  $P$  is a field, then  $A$  is isotypic, and  $P$  is a maximal commutative subfield of  $\text{End}(A)_{\mathbb{Q}}$ .  $\lrcorner$

*Proof:* 1  $\iff$  2 follows from the fact the maximal étale subalgebra of  $\text{End}(A)_{\mathbb{Q}}$  has degree  $[\text{End}(A)_{\mathbb{Q}} : \mathbb{Q}]_{\text{red}}$  (3.6.4.24).

2  $\iff$  3 follows by noticing  $\dim_{\Omega}(\mathcal{H}^1(A)) = 2 \dim A$ , and  $\text{End}_{\Omega}(A)$  acts faithfully on it  $?$ . Then use (3.6.4.25).

For the last assertion, Cf. [CCO14]P1.3.1.1.  $\square$

**Lemma (21.2.2.3).** Let  $A \in \text{AbVar}/\mathbb{C}$  and  $F \subset \text{End}(A)_{\mathbb{Q}}$ . If  $F$  has a real place, then  $[F : \mathbb{Q}] \mid \dim A$ .  $\lrcorner$

*Proof:*  $?$   $\square$

**Prop. (21.2.2.4) [CM-Abelian Varieties and CM-Algebras, Tate].** For  $k \in \text{Field}$  and  $A \in \text{AbVar}/k$ ,

- if  $A$  is simple, then  $A$  is CM iff  $\text{End}(A)_{\mathbb{Q}}$  is a CM-field of degree  $2 \dim A$  over  $\mathbb{Q}$ .
- If  $A$  is isotypic, then  $A$  is CM iff  $\text{End}(A)_{\mathbb{Q}}$  contains a CM-field of degree  $2 \dim A$  over  $\mathbb{Q}$ , which is invariant under some Rosati involution.
- $A$  is CM iff  $\text{End}(A)_{\mathbb{Q}}$  contains a CM-algebra of degree  $2 \dim A$  over  $\mathbb{Q}$ , which is invariant under some Rosati-involution. And in this case, for  $\ell \in \text{Prime} \setminus \{\text{char } k\}$ ,  $T_{\ell}(A)$  is free of rank 1 over this algebra.

$\lrcorner$

*Proof:* We use (21.2.2.2).

1: It suffices to show that if  $\text{End}(A)_{\mathbb{Q}}$  is a field of degree  $2 \dim A$ , then it is CM.  $?$

2:  $?$  Cf. [CCO14]Thm1.3.4.

3 follows from 2.  $\square$

**Thm. (21.2.2.5) [Grothendieck].** For  $k \in \text{Field}$ , any CM abelian variety over  $k$  (21.2.2.1) is defined over a finite extension of the prime field of  $k$ .  $\lrcorner$

*Proof:* Cf. [C.-F. 2004. The isomorphism classes of abelian varieties of CM-type].  $\square$

### Complex CM-Abelian Varieties

**Prop. (21.2.2.6) [Isogeny and CM-Types].** If  $K \in \text{Field}^0$  s.t.  $K = \overline{K}$  and  $L$  is a CM field and  $A \in \text{AbVar}/K$  has CM with an embedding  $j : L \hookrightarrow \text{End}(A)_{\mathbb{Q}}$ , then the  $L$ -linear isogeny class of  $A$  is uniquely determined by the  $K$ -valued CM-type  $\Phi$  on  $L$  associated to  $(A, j)$ . Moreover, every CM-type on  $L$  arises in this way for some  $(A, j)$  over  $K$ .  $\lrcorner$

*Proof:* Cf. [CCO14]P67. ?

□

**Prop. (21.2.2.7) [CM-Types of Complex Abelian Varieties].** Let  $E$  be a CM-algebra, and  $(A, i : E \rightarrow \text{End}(A)_{\mathbb{Q}}) \in \text{AbVar}^g/\mathbb{C}$  is a CM-Abelian variety, then the action of  $E$  on  $\text{Lie}(A)$  is faithful, so

$$\text{Lie}(A) \cong \bigoplus_{\varphi \in \Phi} \mathbb{C}_{\varphi}$$

where  $\Phi$  is a set of homomorphisms from  $E$  to  $\mathbb{C}$ . Then it follows from the decomposition (16.10.5.5)

$$H_1(A(\mathbb{C}), \mathbb{C}) \cong \text{Lie}(A) \bigoplus \overline{\text{Lie}(A)}$$

and the fact  $H_1(A(\mathbb{C}), \mathbb{Q})$  is a free  $E$ -module of rank 1 (21.2.2.4), we see  $\Phi \coprod \Phi \circ c_E = \text{Hom}(E, \mathbb{C})$ . So  $\Phi$  is a CM-type on  $E$  (21.2.1.6).

Thus  $A$  is said to have **CM-type**  $(E, \Phi)$  if  $\Phi$  is a CM-type on  $E$  and

$$\text{Lie}(A) \cong \prod_{\Phi} \mathbb{C}$$

as  $E$ -algebras. ┘

**Def. (21.2.2.8) [Dual CM-Abelian Varieties].** Let  $E$  be a CM-algebra, and  $(A, i : E \rightarrow \text{End}(A)_{\mathbb{Q}}) \in \text{AbVar}^g/\mathbb{C}$  is a CM-Abelian variety, then we can define a **dual CM-Abelian structure** on  $A^{\vee}$  as

$$j : E \rightarrow \text{End}(A^{\vee})_{\mathbb{Q}} : x \mapsto i(c_E(x))^t.$$

In this way,  $(A^{\vee}, j : E \rightarrow \text{End}(A^{\vee})_0)$  has CM-type  $\Phi$  as well, by the decomposition (16.10.5.5)

$$H_1(A(\mathbb{C}), \mathbb{R}) \cong \text{Lie}(A) \bigoplus \overline{\text{Lie}(A)}$$

and the argument in (21.2.2.7). ┘

**Def. (21.2.2.9) [Reflex Field of a CM-Abelian Variety].** Let  $A \in \text{AbVar}/\mathbb{C}$  be CM, then if  $E_0$  is the center of  $\text{End}(A)_{\mathbb{Q}}$ , by (21.2.2.4) and (21.2.2.7),  $E_0$  is a CM-algebra, and there exists a CM-type  $\Phi_0$  on  $E_0$  s.t. for any CM-pair  $E$  s.t.  $A$  is of CM-type  $(E, \Phi)$ , there  $\Phi$  is the extension of  $\Phi_0$  to  $E$ .

Then the reflex field of  $(E_0, \Phi)$  equals that of  $(E, \Phi)$  for any  $(E, \Phi)$ , called the **reflex field** of  $A$ .

? ┘

**Prop. (21.2.2.10) [Abelian Varieties attached to CM-Pairs].** Let  $(E, \Phi)$  be a CM-pair (21.2.1.6) and  $\Lambda$  is a  $\mathbb{Z}$ -lattice in  $E$ , then

$$\Lambda \otimes_{\mathbb{Z}} \mathbb{R} \cong E \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow[\cong]{\Phi} \prod_{\Phi} \mathbb{C} \text{ (21.2.1.7).}$$

So  $\Lambda \otimes_{\mathbb{Z}} \mathbb{R}$  has a complex structure. Then  $\mathcal{O} = \{x \in E \mid a\Lambda \subset \Lambda\} \subset \text{End}(A_{\Phi})$  is an order in  $E$  (19.3.1.38), and then  $E \subset \text{End}_{\mathbb{Q}}(A_{\Phi})$ .

Then there is a Riemann-form on  $(\Lambda, J_{\Phi})$  whose associated positive Rosati-involution (20.9.8.7) stabilizes  $E$  (and thus equals  $c_E$  on  $E$  by (21.2.1.24)). In particular,  $A_{\Phi}$  is a complex Abelian variety (20.9.8.1). ┘

*Proof:* To give such a Riemann-form, by (16.10.5.9), it suffices to give a non-degenerate bilinear form  $\psi : E \times E \rightarrow \mathbb{Q}$  that satisfies



1.  $\psi(ax, y) = \psi(x, \bar{a}y)$  for  $a, x, y \in E$ .
2.  $\psi(x, y) = -\psi(y, x)$  for  $x, y \in E$ .
3.  $\psi(J_\Phi x, J_\Phi y) = \psi(x, y)$  for  $x, y \in E \otimes_{\mathbb{Q}} \mathbb{R}$ .
4.  $\psi(x, J_\Phi x) > 0$  for any non-zero  $x \in E \otimes_{\mathbb{Q}} \mathbb{R}$ .

Now because the trace form  $\text{tr} : E \times E \rightarrow \mathbb{Q}$  is non-degenerate, non-degenerate  $\mathbb{Q}$ -bilinear forms of  $E$  are exactly of the form

$$\psi(x, y) = \text{tr}_{E/\mathbb{Q}}(\bar{y}\alpha x), \alpha \in E^\times.$$

And the conditions (1) is automatic, and (3) is automatic because in the isomorphism

$$E \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow[\cong]{\Phi} \prod_{\rho: F \rightarrow \mathbb{R}} \mathbb{C} \cong \mathbb{C}^\Phi : a \otimes r \mapsto (\rho'(a)r)_\rho, \quad \Phi = \{\rho' | \rho : F \rightarrow \mathbb{R}\}, \quad (21.2.1.7)$$

$\psi_{\mathbb{R}}$  corresponds to

$$\psi'((x_\varphi), (y_\varphi)) = \sum_{\varphi \in \Phi} \text{tr}_{\mathbb{C}/\mathbb{R}}(\varphi(a)\bar{y}_\varphi x_\varphi)$$

and  $J_\Phi$  corresponds to the isomorphism

$$(x_\varphi)_{\varphi \in \Phi} \mapsto (ix_\varphi)_{\varphi \in \Phi}.$$

Thus it suffices to choose  $\alpha$  s.t.  $\varphi(\alpha) \in i\mathbb{R}_+$  for any  $\varphi \in \Phi$ .

By (21.2.1.4), there exists  $\alpha \in E^\times$  s.t.  $E = F(\alpha)$  s.t.  $\varphi(\alpha) \in i\mathbb{R}$  for any  $\varphi \in \Phi$ , and we can then use weak approximation on  $F$  to modify  $\alpha$  s.t.  $\varphi(\alpha) \in i\mathbb{R}_+$  for any  $\varphi \in \Phi$ . So this  $\alpha$  clearly exists. Moreover, any other element  $\alpha'$  is  $\alpha$  times some element  $a \in F$  that is totally positive.  $\square$

**Thm. (21.2.2.11) [Classifying Complex CM Abelian Varieties up to Isogeny].** For each CM-algebra  $E$ , the map  $(A, i) \rightarrow (E, \Phi)$  (21.2.2.7) defines a bijection between the set of  $E$ -linear isogeny classes of CM-Abelian varieties  $A/\mathbb{C}$  with an embedding  $i : E \rightarrow \text{End}(A)_{\mathbb{Q}}$  and the set of CM-types over  $E$  (21.2.1.6) with inverse  $(E, \Phi) \mapsto (A_\Phi, i_\Phi)$  which is the Abelian variety corresponding to the lattice  $\Lambda = \mathcal{O}_E$  defined in (21.2.2.10), called the **principally-CM Abelian variety** of type  $(E, \Phi)$ .

Moreover, each principally-CM Abelian variety is defined over  $\bar{\mathbb{Q}}$ .  $\lrcorner$

*Proof:* Notice in (21.2.2.10), any two choice of the lattice  $\mathfrak{a}$  give isogenous Abelian varieties, and because  $H_1(A, \mathbb{Q})$  is free of rank 1 over  $E$ , let  $e \in H_1(A, \mathbb{Q})$  be a basis vector, then  $\mathfrak{a}e = H_1(A, \mathbb{Z})$  for some lattice  $\mathfrak{a} \subset E$ . Then this  $e$  defines an isomorphism

$$Ee \cong H_1(A, \mathbb{Z}) \otimes \mathbb{Q} = H_1(A, \mathbb{Q})$$

thus the isomorphism

$$A \cong H_1(A, \mathbb{C})/H_1(A, \mathbb{Z}) \xrightarrow[\cong]{e^{-1}} E \otimes_{\mathbb{Q}} \mathbb{C}/\mathfrak{a} \xrightarrow[\cong]{\Phi} \mathbb{C}^\Phi/\Phi(\mathfrak{a}) \sim \mathbb{C}^\Phi/\Phi(\mathcal{O}_E).$$

This is clearly a bijection.

For the last assertion, Cf. [CCO14]P67?  $\square$

**Prop. (21.2.2.12) [Classifying Simple Complex CM-Abelian Varieties].** Let  $A$  be a simple CM-Abelian variety over  $\mathbb{C}$  and  $E = \text{End}(A)_{\mathbb{Q}}$ , then  $E$  is a CM-field, and the map  $A \mapsto (E, \Phi_A)$  defines a bijection between the set of isomorphism classes of simple complex Abelian varieties with CM and the set of isomorphism classes of primitive CM-paris.  $\lrcorner$

*Proof:*  $E \in \mathbf{Field}$  by (21.2.2.4), and if it is not primitive, then  $(E, \Phi)$  is extended from a primitive CM-field  $(E_0, \Phi_0)$ , and a choice of  $E_0$ -basis of  $E$  defines an embedding  $E \subset \mathrm{Mat}([E : E_0], E_0) \subset \mathrm{End}_0(A_{\Phi_0}^{[E:E_0]})$ , and

$$E \otimes_{\mathbb{Q}} \mathbb{R} = E \otimes_{E_0} (E_0 \otimes_{\mathbb{Q}} \mathbb{R}) \cong E \otimes_{E_0} (\mathbb{C}^{\Phi_0}) \cong \mathbb{C}^{\Phi}$$

shows  $A_{\Phi_0}^{[E:E_0]}$  is also of type  $(E, \Phi)$ . Thus  $A \sim A_{\Phi_0}^{[E:E_0]}$  by (21.2.2.11), contradicting the fact  $A$  is simple.  $\square$

**Cor. (21.2.2.13).** The simple Abelian varieties with complex multiplications are classified up to conjugacy by the  $\mathrm{Gal}_{\mathbb{Q}}$ -orbits of CM-types on  $\mathbb{Q}^{\mathrm{cm}}$ , by (21.2.1.17).  $\lrcorner$

### Good Reductions

**Prop. (21.2.2.14) [Potential Good Reduction of CM-Abelian Varieties].** Let  $E \in \mathbf{NField}$ ,  $A \in \mathbf{AbVar}/\mathbb{C}$  has complex multiplication by  $E$ , then  $A$  is defined over some number field  $F$ . And  $F$  can be chosen s.t.  $A_F$  has good reduction everywhere.  $\lrcorner$

*Proof:* Cf. [Milne, Abelian Varieties, P55] or [Serre-Tate, Good reduction of abelian varieties]Thm6.  $\square$

### 3 Fundamental Theorem of Complex Multiplication

**Thm. (21.2.3.1) [Fundamental Theorem of Complex Multiplication, Shimura-Taniyama].** Let  $A \in \mathbf{AbVar}/\mathbb{F}_q$  be the reduction from an complex Abelian variety of CM-type  $(E, \Phi)$ , then the Weil- $q$ -integers of  $A$  can be constructed as follows:  $\square$

*Proof:* Cf. [Milne, Abelian Varieties, P83].  $\square$

**Thm. (21.2.3.2) [CM-Abelian Varieties are defined over Number Fields, Shimura-Taniyama].** If  $K \in \mathbf{Field}^0$  s.t.  $K = \overline{K}$  and  $A \in \mathbf{AbVar}/K$  has CM, then  $A$  is defined over a number field contained in  $K$ .  $\lrcorner$

*Proof:* Cf. [CCO14]P77.  $\square$

**Prop. (21.2.3.3) [Shimura].** If  $A \in \mathbf{AbVar}/\mathbb{Q}$  is of  $\mathrm{GL}(2)$ -type and  $A_{\overline{\mathbb{Q}}}$  has CM Abelian subvariety, then  $A$  is isogenous to a power of CM elliptic curves.  $\lrcorner$

*Proof:* Cf. [Rib04]Prop5.1.  $\square$

### 4 Main Theorem of Complex Multiplication

References are [CCO14].

**Thm. (21.2.4.1) [Main Theorem of Complex Multiplications, Shimura-Taniyama[S-T61]].** Let  $(A, \alpha : L \rightarrow \mathrm{End}(A)_{\mathbb{Q}})$  be a CM-Abelian variety over  $F \in \mathbf{NField}$  with CM-type  $(L, \Phi)$  s.t.  $L$  is a CM-field, then

- The reflex field  $E = E(L, \Phi)$  is contained in  $F$ .
- There exists a unique algebraic Hecke-character  $\epsilon_A : \mathbf{A}_E^{\times} \rightarrow L^{\times}$  s.t. for each  $\ell \in \mathbf{Prime}$ , the continuous homomorphism  $\psi_{\epsilon_A} : \mathrm{Gal}_F \rightarrow L_{\ell}^{\times}$  attached to  $\epsilon_A$  is equal to the  $\ell$ -adic Galois-representation attached to  $A[\ell^{\infty}]$ .

- The algebraic part of  $\epsilon$  is the map  $\text{Nm}_\Phi \circ \text{Nm}_{F/E}$  (21.2.1.15).
- For each  $v \in \Sigma_F^{\text{fin}}$ ,  $A$  has good reduction at  $v$  iff  $\epsilon_A$  is unramified at  $v$ . In which case, we have  $\epsilon_A(\varpi_v) = F_{\tilde{A}, \kappa_v}^\sim$ , where  $\tilde{A} \in \text{AbVar}/\kappa$  is the reduction of  $A$  at  $v$ .

┘

*Proof:* Cf. [CCO14]P127?.

□

**Thm. (21.2.4.2) [Converse to the Main Theorem of Complex Multiplications, Shimura].** Let  $(L, \Phi)$  be a CM-pair with  $L \in \text{NField}$  of degree  $2g$  with reflex field  $E$ . Let  $F/E \in \text{NField}$  and  $\epsilon : \mathbf{I}_K \rightarrow L^\times$  an algebraic Hecke-character with algebraic part being  $\text{Nm}_\Phi \circ \text{Nm}_{F/E}$  (21.2.1.15), then there exists CM-Abelian variety  $(A, i : E \rightarrow \text{End}(A)_\mathbb{Q}) \in \text{AbVar}^g/K$  s.t.  $\epsilon$  is the algebraic Hecke-character associated to  $(A, i)$  as in (21.2.4.1). Moreover, such  $(A, i)$  is unique up to  $L$ -linear isogenies.

┘

*Proof:* Cf. [CCO14]A.3?.

□

## 5 over Finite Fields

**Thm. (21.2.5.1) [Honda-Tate].** For any  $\kappa \in \text{Field}^{\text{fin}}$  and  $A \in \text{AbVar}/\kappa$ , there exists a finite extension  $\kappa'/\kappa \in \text{Field}$  and some  $(K, \mathcal{O}_K, \kappa) \in p\text{-Field}$  and some  $\mathcal{A} \in \text{AbVar}/\mathcal{O}_K$  s.t.  $\mathcal{A}_{\kappa'} \cong A_{\kappa'}$ .

┘

*Proof:* Cf. [CCO14]P75?.

□

## 6 CM-Liftings

### CM-Liftings of $p$ -Divisible Groups

### CM-Liftings of Abelian Varieties

## 7 Elliptic Curves

References are [Deu58, Ser67].

**Prop. (21.2.7.1) [Complex Elliptic Curves with CM].** If  $E \cong E_\tau \in \text{EllCur}/\mathbb{C}$  has CM, then  $\mathcal{K} = \mathbb{Q}(\tau)$  is an imaginary quadratic extension of  $\mathbb{Q}$  and  $\text{End}(E)$  is isomorphic to an order of  $\mathcal{K}$ .

By (21.3.1.29), if  $E \in \text{EllCur}/\mathbb{C}$  has CM (21.2.2.1), then  $\text{End}(E)$  is an  $\mathbb{Z}$ -order in  $\mathcal{K}$ , which by (19.3.1.53) must be of the form  $\mathcal{O}_{\mathcal{K},f} = \mathbb{Z} + f\mathcal{O}_K$ , where  $f = [\mathcal{O}_K : \text{End}(E)]$ .

┘

*Proof:* Cf. [Sil16]P176?.

□

**Thm. (21.2.7.2) [Classifying CM Elliptic Curves].** Given any imaginary quadratic field  $\mathcal{K}$  and an  $\mathbb{Z}$ -order  $\mathcal{O}_{\mathcal{K},f} \subset \mathcal{K}$ , the isomorphism classes of elliptic curves over  $\mathbb{C}$  with maximal CM by  $\mathcal{O}_{\mathcal{K},f}$  is in bijection with  $\text{Cl}(\mathcal{O}_{\mathcal{K},f})$ , denoted by  $\text{EllCur}_\mathbb{C}(\mathcal{O}_{\mathcal{K},f})$ .

In particular,  $\#\text{EllCur}_\mathbb{C}(\mathcal{O}_{\mathcal{K},f}) < \infty$ . The  $j$ -values of elliptic curves with maximal CM by  $\mathcal{O}_{\mathcal{K},f}$  are called singular moduli **associated to**  $\mathcal{O}_{\mathcal{K},f}$ .

┘

*Proof:* If  $E \cong \mathbb{C}/\Lambda \cong E_\tau$ , then  $\mathbb{Z} + \mathbb{Z}\tau$  is an  $\mathcal{O}_{\mathcal{K},f}$ -module of rank 1, and by definition it is a proper fractional ideal of  $\mathcal{O}_{\mathcal{K},f}$  thus invertible (19.3.1.54). Conversely, any such an invertible fractional ideal  $\mathfrak{a}$  is a lattice in  $\mathcal{K}$  thus defines an elliptic curve  $E_{\mathfrak{a}}$  with  $\text{End}(E_{\mathfrak{a}}) = \mathcal{O}$  by (19.3.1.54). Finally it follows from (16.10.5.2) that two such  $E_{\mathfrak{a}}$  and  $E_{\mathfrak{a}'}$  are isomorphic iff  $\mathfrak{a}, \mathfrak{a}'$  differ by a principle ideal.

□

**Prop. (21.2.7.3).** Let  $\mathcal{K}$  be an imaginary quadratic field,  $E \in \text{EllCur}_{\mathbb{C}}(\mathcal{O}_{\mathcal{K}})$ , and  $\mathfrak{a} \in \text{Ideal}(\mathcal{O}_{\mathcal{K}})$ , then

- $E[\alpha]$  is the kernel of the isogeny  $E \rightarrow \bar{\mathfrak{a}} * E$ .
- $E[\alpha]$  is a free  $\mathcal{O}_{\mathcal{K}}/\mathfrak{a}$ -module of rank 1.
- The degree of isogeny  $E \rightarrow \bar{\mathfrak{a}} * E$  is  $\text{Nm}_{\mathcal{K}/\mathbb{Q}}(\mathfrak{a})$ .
- For  $\alpha \in \mathcal{O}_{\mathcal{K}}^{\times}$ ,  $\deg([\alpha]_E) = \text{Nm}_{\mathcal{K}/\mathbb{Q}}(\alpha)$ .

┘

*Proof:* Cf.[Sil99]P102.

□

**Thm. (21.2.7.4) [Singular Moduli are Integral, Weber-Fueter].** Let  $E \in \text{EllCur}_{\mathbb{C}}(\mathcal{O}_{\mathcal{K},f})$ , then

- $j(E) \in \mathcal{O}_{\overline{\mathbb{Q}}}$ , and  $\mathcal{K}(j(E))$  is the ray class field  $\mathcal{K}_f$  of  $\mathcal{K}$ .
- $\text{Gal}(\mathcal{K}_f/\mathcal{K})$  acts transitively on the  $j$ -values associated to  $\mathcal{O}_{\mathcal{K},f}$  (21.2.7.2).

In particular,  $\deg(j(E)) = \text{cl}(\mathcal{O}_{\mathcal{K},f})$ .

┘

*Proof:* Cf.[Serre] or [Sutherland]L21. or [Silverman]P447 or [Cox, Chap11]. Notice this should be a special case of Shimura-Taniyama.

□

**Cor. (21.2.7.5).** There are 13 elliptic curves  $E \in \text{EllCur}/\mathbb{Q}$  with CM, namely they are associated to  $\mathcal{O}_{\mathcal{K},f}$  with

- $f = 1, \mathcal{K} = \mathbb{Q}(\sqrt{-d})$  with  $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ , with corresponding  $j$ -invariances  
 $j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad 0, \quad -3^3 \cdot 5^3, \quad -2^{15}, \quad -2^{15} \cdot 3^3, \quad -2^{18} \cdot 3^3 \cdot 5^3, \quad -2^{15} \cdot 3^3 \cdot 5^3 \cdot 11^3, \quad -2^{18} \cdot 3^3 \cdot 5^3 \cdot 23^3 \cdot 29^3.$
- $f = 2, \mathcal{K} = \mathbb{Q}(\sqrt{-d})$  with  $d \in \{1, 3, 7\}$ , with corresponding  $j$ -invariances  
 $j = 2^3 \cdot 3^3 \cdot 11^3, \quad 2^4 \cdot 3^3 \cdot 5^3, \quad 3^3 \cdot 5^3 \cdot 17^3.$
- $f = 3, \mathcal{K} = \mathbb{Q}(\sqrt{-3})$  with corresponding  $j$ -invariances  $j = 2^{15} \cdot 3 \cdot 5^3$ .

┘

*Proof:*

□

**Thm. (21.2.7.6) [Hasse].** For  $[\Lambda] \in \text{Pic}(\mathcal{O}_{\mathcal{K},f})$ , denoted  $j(\Lambda) = j(\mathbb{C}/\Lambda)$ , then if  $\mathfrak{p}$  is a prime ideal of  $\mathcal{O}_{\mathcal{K}}$  prime to the conductor and  $\mathfrak{p}_f = \mathfrak{p} \cap \mathcal{O}_{\mathcal{K},f}$ , then

$$(\text{Art}(\text{Frob}_{\mathfrak{p}}))(j(\Lambda)) = j(\Lambda \cdot \mathfrak{p}_f^{-1}).$$

┘

*Proof:* ?

□

**Prop. (21.2.7.7) [CM  $j$ -Invariants are Integral].** For  $K \in p\text{-LField}$  and  $E \in \text{EllCur}/K$  with CM,  $j(E) \in \mathcal{O}_{\overline{K}}$ .

┘

*Proof:*

□

**Prop. (21.2.7.8).** If  $k \in \text{Field}^p$ , and  $E \in \text{EllCur}/k$  has CM, then  $j(E) \in \overline{\mathbb{F}}_q \cap k$ , by (21.3.3.5).

┘

**Thm. (21.2.7.9) [Kronecker's Jugendtraum].** Let  $\mathcal{K}$  be an imaginary quadratic field and  $E \in \text{EllCur}(\mathcal{O}_{\mathcal{K}})$ , then

$$\mathcal{K}^{\text{ab}} = \mathcal{K}(j(E), \{x_{\text{tor}}(E)\}),$$

where  $\{x_{\text{tor}}(E)\}$  is the set of  $x$ -coordinates of torsion points of  $E$ .

┘

*Proof:* ?

□

**CM-Lifting**

**Thm. (21.2.7.10)**[Honda/Chai-Conrad-Oort2014]. For any  $A \in \text{AbVar}/\mathbb{F}_q$ , there exists an isogeny  $A \sim B_0$  s.t.  $B_0$  admits a CM lift to characteristic 0.  $\lrcorner$

*Proof:*  $\square$

**Thm. (21.2.7.11)** [Deuring CM-Lifting Lemma]. Let  $p \in \mathbf{P}$  and  $E \in \text{EllCur}/\overline{\mathbb{F}}_p$ , and  $\alpha_0 \in \text{End}(E_0) \setminus \mathbb{Z}$ , then there exists  $F \in \text{NField}$ ,  $\mathcal{E} \in \text{EllCur}/\mathcal{O}_F$  and  $\alpha \in \text{End}(\mathcal{E})$ ,  $\mathfrak{p} \in \Sigma_F^p$  s.t.

$$\mathcal{E}_{\overline{\kappa(\mathfrak{p})}} \cong E_0$$

s.t.  $\alpha_{\overline{\kappa(\mathfrak{p})}}$  corresponds to  $\alpha_0$ .  $\lrcorner$

*Proof:* Cf.[Suw19]P5?  $\square$

**Hilbert Class Polynomials**

**Def. (21.2.7.12)** [Hilbert Class Polynomials]. Let  $\mathcal{O}$  be a order in an imaginary quadratic field  $\mathcal{K}$ , the **Hilbert class polynomial** associated to  $\mathcal{O}$  is defined to be

$$H_{\mathcal{O}}(X) = \prod_{E \in \text{EllCur}_{\mathbb{C}}(\mathcal{O})} (X - j(E)).$$

And for  $D \in \mathbb{Z}$ ,  $D \equiv 0, 1 \pmod{4}$ , the **Hilbert class polynomial** of discriminant  $D$  is defined to be the class polynomial for the order  $\mathcal{O}_D$ .  $\lrcorner$

**Lemma (21.2.7.13)**. For  $\ell \equiv -1 \pmod{4} \in \mathbf{P} \setminus \{3\}$  and  $D = -\ell$  or  $-4\ell$ ,  $H_D(1728) \equiv 0 \pmod{\ell}$ .  $\lrcorner$

*Proof:* Consider the curve  $E : y^2 = x^3 - x$ , it is supersingular by (21.3.3.12) or (21.2.7.20), and has  $j$ -invariant 1728. Thus  $\varphi_E^2 = -\ell$  by (21.3.3.9)(21.3.3.3). Notice  $E[2] = \{\infty, (0, 0), (\pm 1, 0)\}$ , which are all defined over  $\mathbb{F}_p$ , so  $2 \mid (1 + F)$ , and  $\mathbb{Z}[\frac{1+\varphi_E}{2}] \cong \mathcal{O}_{\ell}$ . Thus by Deuring CM-lifting lemma (21.2.7.11), there is an elliptic curve over some  $\mathcal{O}_F$  with maximal CM by  $\mathcal{O}_{-\ell}$  whose  $j$ -invariant is mapped to  $1728 \pmod{\ell}$ , so  $H_{-\ell}(1728) \equiv 0 \pmod{\ell}$ .

Similarly,  $E$  has endomorphism  $I : (x, y) \mapsto (-x, iy)$ ,  $I^2 = -1$ ,  $IF = -FI$ , so we can lift  $IF$  to get an elliptic curve over some  $\mathcal{O}_F$  with CM by  $\mathcal{O}_{-4\ell}$ . And it is maximal CM, because  $\frac{1+IF}{2} \notin \text{End}(E)$ , as  $(1 + IF)(P = (1, 0)) = (1, 0) + (-1, 0) \neq O$ .  $\square$

**Lemma (21.2.7.14)**. For  $\ell \equiv -1 \pmod{4} \in \mathbf{P} \setminus \{3\}$  and  $D = -\ell$  or  $-4\ell$ , let  $\mathcal{K}_D$  be the ray class field of  $\mathcal{O}_D$ , then for any root  $x_0$  of  $H_D(X)$  s.t.  $x_0 \equiv 1728 \pmod{\ell}$ , there exists a unique  $\mathfrak{l} \in \Sigma_{\mathcal{K}_D}^{\ell}$  s.t.  $x_0 - 1728 \in \mathfrak{l}$ . The  $j$ -invariant of this elliptic curve is mapped to  $1728 \pmod{\ell}$ , so  $H_{-\ell}(1728) \equiv 0 \pmod{\ell}$ .  $\lrcorner$

*Proof:* The existence follows from (21.2.7.13), and if there are two primes  $\mathfrak{l}, \mathfrak{l}'$  that divides  $x - 1728$ , then there exists  $\sigma \neq \text{id}$  s.t.  $x_1 = \sigma(x_1) \equiv 1728 \pmod{\ell}$ . Then there are two elliptic curves  $E_1, E_2/\overline{\mathbb{Q}}$  with  $j$ -invariants  $x_0, x_1$  that reduces to the elliptic  $E/\mathbb{F}_{\ell} : y^2 = x^3 - x$ . Then we get a degree-preserving injection  $\text{Hom}(E_1, E_2) \hookrightarrow \text{End}(E_{\overline{\mathbb{F}}_{\ell}}) = A$ .

For the rest, see [Suw19]P8?  $\square$

**Prop. (21.2.7.15)**. For  $\ell \equiv -1 \pmod{4} \in \mathbf{P} \setminus \{3\}$ , there exists  $R, S \in \mathbb{Z}[X]$  s.t.

$$H_{-\ell}(X) = (X - 1728)R(X)^2, \quad H_{-4\ell}(X) = (X - 1728)S(X)^2.$$

$\lrcorner$

*Proof:* Notation as in (21.2.7.14), there exists an involution  $\tau$  of  $\mathcal{K}_D$  s.t.  $\sigma(x_0) = x_0$ . Thus by the lemma,  $\tau(\mathfrak{l}) = \mathfrak{l}$ , and  $f(\mathfrak{l}/\ell)$  is odd because  $\text{Gal}(\mathcal{K}_D/\mathcal{K})$  does (19.3.3.8). So  $\sigma$  acts trivially on  $\kappa(\mathfrak{l})$ . But  $\tau$  doesn't fix any roots  $x$  of  $H_D(X)$  other than  $x_0$  (because  $x_i$  corresponds to ideal classes of  $\mathcal{O}_D$  where  $x_0$  corresponds to the trivial class, and  $\tau$  acts by  $[I] \mapsto [I]^{-1}$ ), so the other roots come in pairs, and the assertion follows.  $\square$

**Prop. (21.2.7.16).** For  $\ell \equiv -1 \pmod{4} \in \mathbf{P} \setminus \{3\}$ , the only real roots of  $H_\ell(X)$  and  $H_{4\ell}(X)$  are  $j(\frac{1}{2}(1 + \sqrt{-\ell}))$  and  $j(\sqrt{-\ell})$  resp..  $\lrcorner$

*Proof:* For  $D = \ell$  or  $-4\ell$ , since the complex conjugation is compatible with the correspondence in (21.2.7.2), and because  $\overline{I\overline{I}} = \text{Nm}(I)\mathcal{O}_D$ , so the fixed points of  $\mathfrak{c}$  are just 2-torsions in  $\text{Pic}(\mathcal{O}_D)$ . But  $\text{Pic}(\mathcal{O}_D)$  is odd by (19.3.3.8), thus the only real  $j$ -value corresponds to the trivial class.  $\square$

**Cor. (21.2.7.17).** For any  $j \in \mathbb{R}$ , for  $\ell \equiv -1 \pmod{4} \in \mathbf{P} \setminus \{3\}$  sufficiently large,  $H_{-\ell}(j) > 0$  and  $H_{-4\ell}(j) < 0$ .  $\lrcorner$

*Proof:* This is because  $j(\frac{1}{2}(1 + \sqrt{-\ell})) \rightarrow -\infty$  and  $j(\sqrt{-\ell}) \rightarrow \infty$  as  $\ell \rightarrow \infty$ , by the power expansion (26.3.4.22).  $\square$

### Applications

**Prop. (21.2.7.18).** For  $p \in \mathbf{P}$ ,

•

$$\sum_{x \in \mathbb{F}_p} \left( \frac{x^3 + x}{p} \right) = \begin{cases} 0 & , p \equiv 3 \pmod{4} \\ -2a & , p = a^2 + 4b^2, a \equiv 1 \pmod{4} \end{cases}.$$

•

$$\sum_{x \in \mathbb{F}_p} \left( \frac{x^3 + 1}{p} \right) = \begin{cases} 0 & , p \equiv 2 \pmod{3} \\ -2a & , p = a^2 + 3b^2, a \equiv 1 \pmod{3} \end{cases}.$$

•

$$\sum_{x \in \mathbb{F}_p} \left( \frac{x^3 + x}{p} \right) = \begin{cases} 0 & , \left( \frac{p}{7} \right) = -1 \\ -2a & , p = a^2 + 7b^2, \left( \frac{a}{7} \right) = 1 \end{cases}.$$

$\lrcorner$

*Proof:* Because these corresponds to elliptic curves with CM by an imaginary quadratic field with class number 1, thus there exists , so the trace of  $\varphi_p$  can be determined. ( ? How, can it be lifted into  $\mathcal{O}_{\mathcal{K}}?$ ).  $\square$

### Supersingular Primes

References are [Elk87] and [Suw19].

**Lemma (21.2.7.19).** Let  $k \in \text{Field}^p$  and  $E \in \text{EllCur}/k$ ,  $E$  is supersingular iff there exists an order  $\mathcal{O}$  of an imaginary quadratic field  $\mathcal{K}$  s.t.  $\mathcal{O} \subset \text{End}(E)$  and  $p$  doesn't split in  $\mathcal{K}$ .  $\lrcorner$

*Proof:* By (21.3.3.6), if it is ordinary, then  $\text{End}(\mathcal{O})$  is a  $\mathbb{Z}$ -order in an imaginary quadratic field  $\mathcal{K}$ . Then consider the  $p$ -adic representation

$$\mathcal{K} \otimes \mathbb{Q}_p = \text{End}(E) \otimes \mathbb{Q}_p \rightarrow \text{End}_{\mathbb{Q}_p}(V_p(E)) \cong \mathbb{Q}_p.$$

Then this map has a kernel, so  $\mathcal{K} \otimes \mathbb{Q}_p$  is not a field, and  $p$  splits in  $\mathcal{K}$ .  $\square$

**Thm. (21.2.7.20) [Supersingular Criterion for CM Elliptic Curves, Deuring].** Let  $F \in \mathbf{NField}$  and  $E \in \mathbf{EllCur}/F$  with CM by an imaginary quadratic field  $\mathcal{K}$ . If  $\mathfrak{p} \in \Sigma_F^p$  is a good reduction for  $E$ , then  $\tilde{E}/\kappa(\mathfrak{p})$  is ordinary/supersingular iff  $p$  is split/non-split in  $\mathcal{K}$ .  $\lrcorner$

*Proof:* Cf. [Lang, Elliptic Functions, Thm13.12]. ?  $\square$

**Cor. (21.2.7.21) [Supersingular Reductions for CM Elliptic Curves].** For  $E \in \mathbf{EllCur}/\mathbb{Q}$  with CM by  $\mathcal{K}$ , then

$$\#\{p \in \mathbf{P}, p \leq X \mid \tilde{E}/\mathbb{F}_p \text{ is supersingular elliptic}\} = \frac{X}{2 \log X} + O\left(\frac{X}{\log X}\right).$$

$\lrcorner$

*Proof:* This follows from the effective Chebotarev density theorem (19.7.5.2).  $\square$

**Prop. (21.2.7.22) [Elkies].** For any  $E \in \mathbf{EllCur}/\mathbb{Q}$ , there are infinitely many  $p \in \mathbf{P}$  s.t.  $\tilde{E}/\mathbb{F}_p$  is supersingular elliptic. (This is also true if  $\mathbb{Q}$  is replaced by a real field, Cf. [Elkies, 1989]).  $\lrcorner$

*Proof:* By Deuring's CM-lifting lemma (21.2.7.11), for  $D = -\ell$  or  $-4\ell$ ,  $\tilde{E}/\mathbb{F}_p$  has CM by  $\mathcal{O}_D$  iff  $H_D(j(E)) = 0 \in \mathbb{F}_p$ . So by (21.2.7.19),  $p$  is a supersingular prime for  $E$  if

- $H_{-\ell}(j(E)) \in p\mathbb{Z}_p$  or  $H_{-4\ell}(j(E)) \in p\mathbb{Z}_p$ , and
- $\text{ord}_p(\ell)$  is odd or  $-\ell$  is a quadratic non-residue modulo  $p$ .

Suppose that there is a finite set  $S$  containing all the supersingular primes of  $E$ , assume  $2 \in S$ . By Dirichlet's theorem, there exists  $\ell \in \mathbf{P}$  sufficiently large s.t.

$$\ell \equiv 3 \pmod{4}, \quad \left(\frac{p}{\ell}\right) = 1, \quad p \in S.$$

Then by quadratic reciprocity, if  $p \in \mathbf{P}$  and  $H_{-\ell}(j(E)) \in p\mathbb{Z}_p$  or  $H_{-4\ell}(j(E)) \in p\mathbb{Z}_p$ , then  $p \neq \ell$  and  $\left(\frac{p}{\ell}\right) = 1$ , otherwise there would be a new supersingular prime for  $E$  that is not in  $S$ .

As  $H_{-\ell}, H_{-4\ell} \in \mathbb{Z}[X]$ , for any  $p \in \mathbf{P}$ ,

$$H_{-\ell}(j(E))^{-1} \in p\mathbb{Z}_p \iff j(E)^{-1} \in p\mathbb{Z}_p \iff H_{-4\ell}(j(E))^{-1} \in p\mathbb{Z}_p,$$

and if  $j(E)^{-1} \in p$ , then

$$\text{ord}_p(H_{-\ell}(j(E))^{-1}) + \text{ord}_p(H_{-4\ell}(j(E))^{-1}) = \text{ord}_p(j(E)^{-1}) \cdot (\deg(H_{-\ell}) + \deg(H_{-4\ell}))$$

which is even by (21.2.7.15). And  $H_{-\ell}(j(E))H_{-4\ell}(j(E)) < 0$  by (21.2.7.17). So  $H_{-\ell}(j(E))H_{-4\ell}(j(E))$  would be a quadratic non-residue modulo  $\ell$  by our hypothesis. But this contradicts (21.2.7.15).  $\square$

### Without CM case

**Thm. (21.2.7.23) [Serre].** For  $F \in \mathbf{NField}$  and  $E \in \mathbf{EllCur}/F$  without CM,  $\rho_E(\text{Gal}_F) \subset \text{GL}(2; \mathbf{A}^f)$  has finite index.  $\lrcorner$

*Proof:* Cf. [Abelian l-adic representations and elliptic curves, Serre]. or [Bounds for Serre's open image theorem for elliptic curves over number fields].  $\square$

**Conj. (21.2.7.24) [Lang-Trotter].** For  $E \in \text{EllCur}/\mathbb{Q}$  without CM,

$$\#\{p \in \mathbf{P}, p \leq X \mid \tilde{E}/\mathbb{F}_p \text{ is supersingular elliptic}\} \sim \frac{\sqrt{X}}{\log X}.$$

┘

*Proof:*

□

**Prop. (21.2.7.25) [Serre-Elkies].** For  $E \in \text{EllCur}/\mathbb{Q}$  without CM,

$$\#\{p \in \mathbf{P}, p \leq X \mid \tilde{E}/\mathbb{F}_p \text{ is supersingular elliptic}\} \ll X^{3/4}.$$

In particular, supersingular primes have density 0.

┘

*Proof:* [N. D. Elkies. Distribution of supersingular primes. *Asterisque*, (198-200):127–132 (1992), 1991. *Journées Arithmétiques*, 1989].

□

## 8 Singular Moduli

References are [Zag08], [Traces of Singular Moduli, Zagier], and [On Singular Moduli, Gross-Zagier].

**Def. (21.2.8.1) [Singular Moduli].** A **singular modulus** is a number  $j(\tau) \in \mathbb{C}$  where  $\tau \in \mathcal{K} \setminus \mathbb{Q}$  and  $\mathcal{K}$  is an imaginary quadratic field.

┘

**Prop. (21.2.8.2).** Any singular modulus (21.2.8.1) belongs to  $\overline{\mathbb{Q}}$ .

┘

*Proof:*

□

**Remark (21.2.8.3).** The singular moduli generate class fields of imaginary quadratic fields, and can in fact be applied to solve Hilbert’s 12th problem over imaginary quadratic fields.

Zagier’s proof of Borchers’ theorem is the beautiful observation that the generating functions for traces of singular moduli are essentially weight  $3/2$  weakly holomorphic modular forms.

┘

**Def. (21.2.8.4).** For  $d \in \mathbb{Z}_{<0}$ , let  $\mathcal{Q}_d$  be the set of integral binary forms of discriminant  $d$ . And for each  $Q \in \mathcal{Q}_d$ , let  $\tau_Q$  be the root of  $Q$  in  $\mathcal{H}$ .

┘

## 9 CM Modular Forms

**Def. (21.2.9.1) [CM Modular Forms].** A **CM modular form** is a modular form constructed from a binary quadratic form  $Q(a, b)$  and a spherical polynomial  $P(a, b)$  of order  $d$ .

For example,

$$f(z) = \sum_{a \equiv 1 \pmod{4}, 2 \mid b} a q^{a^2 + b^2}$$

is a modular form, which corresponds exactly to  $L(E, s)$ , where  $E \in \text{EllCur}/\mathbb{Q} : y^2 = x^3 + x$ .

┘

*Proof:* Cf. [1-2-3 of Modular Forms, P93].

?

□

**Prop. (21.2.9.2).** For  $n \in 2\mathbb{Z}_+$ ,  $\eta(z)^n$  is a CM modular form iff  $n \in \{2, 4, 6, 8, 10, 14, 26\}$ . For example,

$$\eta(z) =$$

┘

*Proof:* Cf. [Serre1985, Sur la lacunarité des puissances de ].

□

**Prop. (21.2.9.3) [Lacunary of CM Modular Forms].**

┘

*Proof:* Cf. [Serre1981, Quelques applications du théorème de densité de Chebotarev].

□



## 10 Real Multiplications

**Def. (21.2.10.1) [Real Multiplications, Hida].**  $A \in \mathcal{AbVar}/\overline{\mathbb{Q}}$  is said to have **real multiplication** if there exists a totally real field  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = \dim A$  s.t.

$$F \subset \mathrm{End}(A)_{\mathbb{Q}}, \quad F = \mathrm{End}^F(A)_{\mathbb{Q}}.$$

┘

**Def. (21.2.10.2) [Q-Real-Multiplications, Hida].**  $A \in \mathcal{AbVar}/\overline{\mathbb{Q}}$  is said to be a **Q-real-multiplication Abelian variety** if it has real multiplication by a totally real field  $F$  (21.2.10.1) and there is a  $F$ -linear isogeny  $A^{\sigma} \rightarrow A$  for any  $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ . ┘

## 21.3 Arithmetic of Elliptic Curves

References are [Sil99, Sil11, ?SI116]. Should add Sutherland's notes <https://math.mit.edu/classes/18.783/2017/lectures.html> L2-11, 13-14, 17-24. and Snowden's notes <http://www-personal.umich.edu/~asnowden/teaching/2013/679/index.html> L5,6,7,9,11, 14, 15, 17-23.

**Notation (21.3.0.1).**

- Use notations defined in [Abelian Varieties](#).
- Use notations defined in [Global Fields](#).

┘

### 1 Basics

**Def. (21.3.1.1) [Elliptic Schemes].** For  $k \in \mathbf{Field}$ , an **elliptic curve**  $E$  over  $k$  is a complete non-singular curve of genus 1 over a field  $k$ , together with a specified rational pt  $O$ . In (21.3.1.10), we will see an elliptic curve is smooth.

More generally, for  $S \in \mathbf{Sch}$ , an **Elliptic scheme**  $f : E \rightarrow S$  is an Abelian scheme over  $S$  of relative dimension 1 (20.9.1.1). The category of elliptic schemes over  $S$  is denoted by  $\mathbf{EllCur}/S$ . ┘

**Remark (21.3.1.2).** Elliptic curves are Abelian varieties by (21.3.1.10). Thus the theory of Abelian varieties apply to elliptic curves. ┘

**Prop. (21.3.1.3) [Invertible Sheaf].** Let  $S \in \mathbf{Sch}$ ,  $E \in \mathbf{EllCur}/S$ , and  $e \in E(S)$  be the unit, then we can define

$$\omega_{E/S} = e^* \Omega_{E/S} = f_* \Omega_{E/S} \cong R^1 f_* \mathcal{O}_E \in \mathbf{Mod}(\mathcal{O}_S),$$

┘

*Proof:*

□

### Weierstrass Theory

**Def. (21.3.1.4) [Weierstrass Equations].** A **Weierstrass equations** is an equation of the form

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6.$$

┘

**Def. (21.3.1.5) [Important Invariants].** Let  $k \in \mathbf{Field}$ ,  $E \in \mathbf{EllCur}/k$ , given any Weierstrass equation as in (21.3.1.4), we can define

- $b_2 = a_1^2 + 4a_2$ ,  $b_4 = a_1 a_3 + 2a_4$ ,  $b_6 = a_3^2 + 4a_6$ ,  $b_8 = a_1^2 a_6 - a_1 a_3 a_4 + a_2 a_3^2 + 4a_2 a_6 - a_4^2$ .
- $\Delta = -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6$  the **discriminant of  $E$** , to determine whether  $E$  is singular.
- the quantity  $c_4 = b_2^2 - 24b_4$ , used in the singular case to determine iff it has a node or cusp.
- the quantity  $c_6 = -b_2^3 + 36b_2 b_4 - 216b_6$ , used to determine twisted elliptic curves (21.3.6.3).
- the  **$j$ -invariant**  $j = c_4^3/\Delta$ , which is used to in the non-singular case to characterize  $E$ .

They satisfy the equations:

$$4b_8 = b_2 b_6 - b_4^2, \quad 1728\Delta = c_4^3 - c_6^2.$$

┘

**Prop. (21.3.1.6)[Reduced Weierstrass Equations].** Let  $k \in \mathbf{Field}$ ,  $E \in \mathbf{EllCur}/k$ , given any Weierstrass equation as in (21.3.1.4),

- If  $\text{char } k \neq 2$ , we can replace  $(x, y)$  by  $(x, y - (a_1x + a_3)/2)$  to eliminate  $a_1, a_3$  to transform the original Weierstrass equation to

$$y^2 = x^3 + \frac{b_2x^2 + 2b_4x + b_6}{4}.$$

- If  $\text{char } k \neq 2, 3$ , then we can further replace  $(x, y)$  to  $((x - 3b_2)/36, y/216)$  to eliminate  $b_2$  to transform the Weierstrass equation to

$$y^2 = x^3 - 27c_4x - 54c_6.$$

In particular,  $E$  has a Weierstrass equation of the form

$$y^2 = x^3 + a_4x + a_6, \quad \Delta = -16(4a_4^3 + 27a_6^2), \quad j = 1728 \frac{4a_4^3}{4a_4^3 + 27a_6^2}$$

- If  $\text{char } k = 3$ , then  $E$  has a Weierstrass equation of the form

$$y^2 = x^3 + a_2x^2 + a_6, \quad \Delta = -a_2^3a_6, \quad j = -a_2^3/a_6, \text{ or } y^2 = x^3 + a_4x + a_6, \quad \Delta = -a_4^3, \quad j = 0$$

- If  $\text{char } k = 2$ , then  $E$  has a Weierstrass equation of the form

$$y^2 + xy = x^3 + a_2x^2 + a_6, \quad \Delta = a_6, \quad j = 1/a_6, \text{ or } y^2 + a_3y = x^3 + a_4x + a_6, \quad \Delta = a_3^4, \quad j = 0.$$

┘

**Prop. (21.3.1.7)[Explicit Embedding of Elliptic Curves].** Any  $E \in \mathbf{EllCur}/k$  is isomorphic to a plane curve in  $\mathbb{P}_k^2$  defined by an affine Weierstrass equation in  $k[x, y]$ .

And any isomorphism between elliptic curves defined by affine Weierstrass equations over  $k$  and fixes  $[0, 1, 0]$  are linear maps of the form

$$(X, Y) \mapsto (u^2X + r, u^3Y + su^2X + t), \quad u \in k^*, s, r \in k.$$

┘

*Proof:* If  $E$  is an elliptic curve over  $k$ , consider a rational point  $O \in E(k)$ , Riemann-Roch (7.13.2.9) tells  $l(nP) = \deg(nP) = n$  for  $n \geq 1$ . Now  $L(kP) = k$  by Riemann-Roch (7.13.2.9). So we choose a basis  $1, x$  for  $\mathcal{O}_X(2O)$ , and extend it to a basis  $1, x, y$  of  $L(3P)$ . Since  $L(6O) = 6$ , there is a linear relation between the seven elements  $1, x, x^2, xy, y^2, x^3$ . And  $y^2, x^3$  must occur by observing the pole order at  $P$ . Thus by rescaling, we can write the relation as

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

So  $x, y$  defines a rational map of  $E$  to  $\mathbb{P}^2 : a \mapsto (x(a), y(a), 1)$ . This map extends to an embedding of  $E$  into  $\mathbb{P}^2$ , by (7.13.1.16). Moreover, by the above proof, if  $E'$  is another Weierstrass form, then  $1, x'$  are basis for  $\mathcal{O}_X(2O)$  and  $1, x', y'$  are basis for  $\mathcal{O}_X(3O)$ . Thus we have the linear relation between  $(X, Y)$  and  $(X', Y')$ .

To define an Abelian structure on  $E$ , first notice that the map

$$E(k) \rightarrow \text{Cl}^0(E) : Q \mapsto [Q] - [P]$$

is a bijection. Using Riemann-Roch, it is injective because  $l(Q) = 1$  for  $Q \in E(k)$ , and for any divisor  $A$  of degree 0,  $L(A + P) > 0$ , so there exists an effective divisor that is equivalent to  $A + P$ , but this must be a rational point  $Q \in E(k)$ , thus this map is surjective. Thus we can endow  $E(k)$  with a group structure inherited from  $\text{Cl}^0(E)$ . By (21.3.1.9), there is a group variety structure prolonging this construction, so  $E$  is a group scheme.  $\square$

**Remark (21.3.1.8).** There is a more advanced way to prove that this is a group structure, in [Qing Liu]P492. ?

And this group structure coincides with the group structure defined using intersection of a line, because “ $P, Q, R$  is on a line just means that  $[P] + [Q] + [R] - 3[O] = 0 \in \text{Cl}^0(E)$ .”  $\lrcorner$

**Lemma (21.3.1.9) [Explicit Group Structure].** Let  $E$  be an elliptic curve given by a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

then the group addition on  $E \setminus O$  is given by

- $-(x_0, y_0) = (x_0, -y_0 - a_1x - a_3)$ .
- $(x_1, y_1) + (x_2, y_2) = O$  iff  $x_1 = x_2$  and  $y_1 + y_2 + a_1x_2 + a_3 = 0$ . Otherwise

$$(x_1, y_1) + (x_2, y_2) = (\lambda^2 + a_1\lambda - a_2 - x_1 - x_2, -(\lambda + a_1)x_3 - y_3 - a_3)$$

where

$$\lambda = \begin{cases} \frac{y_1 - y_2}{x_1 - x_2} & x_1 \neq x_2 \\ \frac{3x^2 + 2a_2x + a_4 - a_1y_1}{2y_1 + a_1x_1 + a_3} & x_1 = x_2 \end{cases}, \quad \nu = \begin{cases} \frac{y_2x_1 - y_1x_2}{x_1 - x_2} & x_1 \neq x_2 \\ \frac{-x_1^3 + a_4x_1 + 2a_6 - a_3y_1}{2y_1 + a_1x_1 + a_3} & x_1 = x_2 \end{cases}$$

- (Doubling Formula)  $x([2]P) = \frac{x^4 - b_4x^2 - 2b_6x - b_8}{4x^3 + b_2x^2 + 2b_4x + b_6}$ .

Thus the group actions defined in (21.3.1.7) are all morphisms. In particular,  $E$  is a group variety.  $\lrcorner$

*Proof:* Cf. [Sil16]P53.  $\square$

**Cor. (21.3.1.10) [Elliptic Curves and Abelian Varieties].** If  $X$  is an Abelian variety of dimension 1, then  $X$  is an elliptic curve. The converse is also true, by (21.3.1.7). In particular, the theory of Abelian varieties 20.9 applies to elliptic curves. And we will use notations for Abelian varieties.  $\lrcorner$

*Proof:* By (10.1.4.38)(10.1.4.37), the tangent space  $\mathcal{T}_{X/k}$  is trivial of rank 1, thus it is a curve of genus 1 by (7.13.1.25). it is also regular, by (7.6.4.15).  $\square$

**Cor. (21.3.1.11) [Homomorphisms between Elliptic Curves].** A map of schemes between two elliptic curves over  $k$  is a homomorphism iff it preserves the origin.  $\lrcorner$

*Proof:* If  $E, E' \in \text{EllCur}/k$  and  $f : E \rightarrow E'$  is a map of schemes preserving the origin, there is a commutative diagram

$$\begin{array}{ccccc} E(k) & \xrightarrow{\cong} & \text{Cl}^0(E) & \xrightarrow{\cong} & \text{Pic}^0(E) \\ \downarrow f & & & & \uparrow f^* \\ E'(k) & \xrightarrow{\cong} & \text{Cl}^0(E') & \xrightarrow{\cong} & \text{Pic}^0(E') \end{array}.$$

$\square$

**Prop. (21.3.1.12) [Normalized Invariant Differential Form].** Let  $k \in \mathbf{Field}$  and  $E$  a plane cubic over  $k$  defined by the equation  $W : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ . Then the differential form given by

$$\omega = \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y}$$

is a non-zero section of  $\mathcal{K}_{E_{\text{sm}}}$ . It is normalized s.t. at the origin, with the uniformizer  $z = -x/y$ , it has value 1.

Moreover, it is invariant under translation. It is called the **normalized invariant differential form** on  $E$  w.r.t  $W$ .  $\lrcorner$

*Proof:* Firstly we show  $\text{div}(\omega) = 0$ : consider a finite smooth point  $P_0 = (x_0, y_0)$ , then it cannot have pole, otherwise  $P_0$  is singular. Because  $x - x_0 \in L(2O)$ ,  $\text{ord}_{P_0}(x - x_0) \leq 2$ , and if  $\text{ord}_{P_0}(x - x_0) = 2$ ,  $\text{ord}_{P_0}(y - y_0) \geq 2$  also by inspecting the Weierstrass equation. Thus  $\text{ord}_P(\omega) = 0$ . Then consider the situation at  $O$ :  $\text{ord}_O(x) = -2, \text{ord}_O(y) = -3$ , thus  $\text{ord}_O(\frac{dx}{2y+a_1x+a_3}) = 0$  except possibly  $\text{char } k = 2$ . But the same calculation shows  $\text{ord}_O(\frac{dy}{3x^2+2a_2x+a_4-a_1y}) = 0$  except possibly when  $\text{char } k = 3$ . Thus we are done.

Now any translation action induces a map  $\tau_x^*\omega = a(x)\omega$ , where  $a \in K(C)^\times$ . But  $a(x)$  has no pole and zeros, thus  $a(x) \in k^\times$ . But then  $a(x)$  is rational function with no zero and poles, thus  $a(x) = a(O) = 1$ .  $\square$

**Cor. (21.3.1.13).**  $[m]_E^*\omega = \text{pr}_1^*\omega + \text{pr}_2^*\omega$ .  $\lrcorner$

*Proof:* It follows from this and the see-saw principal that  $m_E^*\mathcal{K}_E \cong \text{pr}_1^*\mathcal{K}_E \otimes \text{pr}_2^*\mathcal{K}_E$ .  $\square$

**Prop. (21.3.1.14) [Change of Variables].** A change of variables of the form  $(x, y)$  replaced by  $(u^2x + a, u^3y + bx + c)$  changes  $c_4$  to  $u^{-4}c_4$ ,  $c_6$  to  $u^{-6}c_6$ ,  $\Delta$  to  $u^{-12}\Delta$ , and preserves  $j$ .  $\lrcorner$

*Proof:* This is just a equation of identities, so it suffices to prove for  $k$  of characteristic 0, then in this case, by (21.3.1.6),  $c_4, c_6$  is just the coefficients of the Weierstrass equation transformed into the reduced form (times a constant). So it suffices to prove for transformation between reduced Weierstrass equations. In this case,  $a = b = c = 0$ , and the assertion is clear.  $\square$

**Prop. (21.3.1.15) [Characterizing Singularities].** Let  $k \in \mathbf{Field}$ , then

- The plane cubic  $E$  over  $k$  defined by a Weierstrass formula  $f$  is never singular at  $O = [0, 1, 0]$ , and it is a curve.
- $E$  is smooth iff its determinant  $\Delta \neq 0$ . And in this case it is an elliptic curve by genus formula.
  - $E$  has a node iff  $\Delta = 0$  and  $c_4 \neq 0$ .
  - $E$  has a cusp iff  $\Delta = c_4 = 0$ .
- For  $E, E' \in \mathbf{EllCur}/k$ ,  $E_{\bar{k}} \cong E'_{\bar{k}}$  iff  $j(E) = j(E')$ . In fact, they are isomorphic over a separable field extension  $k'/k$  of degree 24, and if  $j \neq 0, 1728$ , then they are isomorphic over a separable field extension  $k'/k$  of degree 2.

$\lrcorner$

*Proof:* 1: Firstly,  $E$  is never singular at  $[0, 1, 0]$ : On  $U(z)$ , the curve is given by  $z + a_1xz + a_3z^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$ , which is not singular at  $(0, 0)$ . Secondly,  $E$  has genus 1. Finally it is a curve by (3.2.3.14) and checking  $f$  doesn't has a root  $y \in \bar{k}(x)$ : This is true by degree reasons using reduced Weierstrass equations (21.3.1.6).

2: (1): By(21.3.1.10), non-singular is equivalent to smoothness. So we may base change to  $\bar{k}$ . If  $P$  is non-singular, linearly transform it to  $(0,0)$ , then by Jacobian criterion,  $\frac{\partial}{\partial x}f = \frac{\partial}{\partial y}f = 0$ , so  $a_3 = a_4 = a_6 = 0$ , and it can be verified that  $\Delta = 0$ . Conversely, if  $\Delta = 0$ , then use the reduced Weierstrass equations and argue case by case, we can find a singular point.

(2), (3): In this case, assume  $a_3 = a_4 = a_6 = 0$ , then  $c_4 = (a_1^2 + 4a_2)^2$ , and  $E : y^2 + a_1xy - a_2x^2 - x^3 = 0$ , so the assertion is clear.

2: We use the reduced Weierstrass equations, derive an equation between their coefficients, and take a suitable change of variables. Details are omitted.  $\square$

**Prop. (21.3.1.16) [Fields of Moduli].** For any element  $j_0 \in \bar{k}$ , there exists an elliptic curve over  $k(j_0)$  with  $j$ -invariant  $j_0$ . In particular, any elliptic curve over  $k$  is defined over  $k(j(E))$ , by(21.3.1.15).  $\lrcorner$

*Proof:* If  $j_0 \neq 0, 1728$ , then we can take the curve

$$E : y^2 + xy = x^3 - \frac{36}{j_0 - 1728}x - \frac{1}{j_0 - 1728}$$

with  $\Delta(E) = \frac{j_0^3}{(j_0 - 1728)^3}$ ,  $j(E) = j_0$ .

If  $j = 0$  or  $1728$ , we can take one of the curves

$$E : y^2 + y = x^3, \quad \Delta = -27, \quad j = 0$$

$$E : y^2 = x^3 + x, \quad \Delta = -64, \quad j = 1728.$$

Notice for  $\text{char } k = 2$  or  $3$ ,  $0 = 1728$ , and at least one of these are elliptic curves.  $\square$

**Prop. (21.3.1.17) [Singular Weierstrass].** If  $k \in \text{Field}$  is perfect or  $\text{char } k \neq 2, 3$ ,  $E \in \mathbb{P}_k^2$  is a plane cubic given by a Weierstrass equation  $W$  is singular, then

- $E$  has a unique singular point, and it is a rational point.
- $E$  is birational to  $\mathbb{P}_k^1$ , i.e. its non-singular projective model is  $\mathbb{P}^1$ (7.13.1.19).
- $E_{\text{sm}}$  is a commutative  $k$ -group under the group action given in(21.3.1.9) with origin  $O = [0, 1, 0]$ , and  $E_{\text{sm}} \cong \mathbb{G}_m$  if  $E$  has a split node, and  $E_{\text{sm}} \cong \mathbb{G}_a$  if  $E$  has a cusp.  $\lrcorner$

*Proof:* 1: By(21.3.1.15), the singular points must be a finite point, and  $\Delta = 0$ . Over  $\bar{k}$ , the singular points of  $E$  is characterized by the equations  $W = \frac{\partial}{\partial x}W = \frac{\partial}{\partial y}W = 0$ . Then we can use the reduced Weierstrass equations(21.3.1.6): If  $\text{char } k \neq 2, 3$ , then clearly  $(x_0, y_0) \in k^2$ . The perfect case is similar.

2: Because the curve is never singular at  $O$  by(21.3.1.15), by a linear change, we can assume that one of the singular point is  $(0,0)$ . Then by Jacobian criterion, it must be of the form

$$y^2 + a_1xy = x^3 + a_2x^2.$$

Then the projection along  $(0,0)$  is an isomorphism? this is wrong! it may not be rational.

$$E \setminus \{0, \infty\} \rightarrow \mathbb{P}_k^1 \setminus \{[0 : 1]\} : (x, y) \mapsto [x : y]$$

with inverse

$$[1, t] \mapsto (t^2 + a_1t - a_2, t^3 + a_1t^2 - a_2t).$$

$E_{\text{sm}}$  is a group scheme because we can verify on closed points, and the fact any line through  $(0,0)$  has multiplicity more than 1. To determine the group structure, Cf.[Sil16]Exercise3.5.?.  $\square$

**Prop. (21.3.1.18) [Legendre Forms].** Suppose  $\text{char } K \neq 2$ , then

- Every elliptic curve over  $\overline{K}$  is isomorphic to an elliptic curve in **Legendre form**

$$E_\lambda : y^2 = x(x-1)(x-\lambda), \quad \lambda \neq 0, 1 \in \overline{K}.$$

- $\Delta(E_\lambda) = 16\lambda^2(\lambda-1)^2, j(E_\lambda) = 2^8 \frac{(\lambda^2-\lambda+1)^3}{\lambda^2(\lambda-1)^2}, c_4 = 16(1-\lambda(1-\lambda)).$
- If  $K$  is a valued field, we may take  $v(\lambda) \geq 0$  as  $j(E_\lambda) = j(E_{\lambda^{-1}})$ .
- The map  $\overline{K} \rightarrow \overline{K} : \lambda \mapsto j(E_\lambda)$  is 6 to 1 except above the points 0 and 1728, where for  $\text{char } K \neq 2, 3, \#j^{-1}(0) = 2, \#j^{-1}(1728) = 3$ , and for  $\text{char } K = 3, j^{-1}(0 = 1728) = 1$ .

┘

*Proof:* 1: put the Weierstrass equation of  $E$  in the reduced form (21.3.1.6), then it is of the form

$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$

in  $\overline{K}[x, y]$ , and because  $\Delta = 16(e_1 - e_2)^2(e_2 - e_3)^2(e_3 - e_1)^2 \neq 0$ , so  $e_1, e_2, e_3$  are pairwise distinct. Thus the linear transform

$$x \mapsto (e_2 - e_1)x' + e_1$$

makes the Weierstrass equation in the Legendre form with  $\lambda = \frac{e_3 - e_1}{e_2 - e_1}$ .

2: Direct calculation.

3: Directly calculating the differential of  $j$ . ?

□

**Prop. (21.3.1.19) [Duering Normal Form].** Every elliptic curve over  $\overline{K}$ , except possibly  $\text{char } K = 3$  and  $j(E) = 0$ , is isomorphic to an elliptic curve in **Duering normal form**

$$E_\alpha : y^2 + \alpha xy + y = x^3, \quad \alpha \in \overline{K}, \quad \Delta(E_\alpha) = \alpha^3 - 27 \neq 0, \quad j(E_\alpha) = \frac{\alpha^3(\alpha^3 - 24)^3}{\alpha^3 - 27}, \quad c_4 = \alpha(\alpha^3 - 24).$$

And if  $K$  is a valued field, we can take  $v(\alpha) > 0$ .

┘

*Proof:* Use (21.3.1.15), it suffices to show that  $j(\alpha)$  can take any value  $j \in \overline{K}$ .

The last assertion is because if  $v(\alpha) < 0$ , then the product of roots of  $\frac{x(x-24)^3}{x-27} = \frac{\alpha^3(\alpha^3-24)^3}{\alpha^3-27}$  other than  $\alpha$  is  $-72^3/(\alpha-27)$  has positive valuations.

□

**Prop. (21.3.1.20) [Automorphism Group].** Let  $k \in \text{Field}, E \in \text{EllCur}/k$ , then the the automorphic group of  $E$  is

$$\text{Aut}(E) \cong \begin{cases} \mu_2(k) & j(E) \neq 0, 1728 \\ \mu_4(k) & j(E) = 1728, \text{char } k \neq 2, 3 \\ \mu_6(k) & j(E) = 0, \text{char } k \neq 2, 3 \\ D_{12} & j(E) = 0 = 1728, \text{char } k = 3, k = \overline{k} \\ C_3 \rtimes Q_8 & j(E) = 0 = 1728, \text{char } k = 2, k = \overline{k} \end{cases}$$

┘

*Proof:* Any automorphism of  $E$  is defined by a linear maps of the form

$$(X, Y) \mapsto (u^2X + r, u^3Y + su^2X + t), \quad u \in k^\times, s, r \in K.$$

If  $\text{char } k \neq 2, 3$ , use the reduced Weierstrass form  $E : y^2 = x^3 + Ax + B$ , so it is of the form  $(X, Y) \mapsto (u^2X, u^3Y)$ , which is possible iff  $u^{-4}A = A, u^{-6}B = B$ . Thus by arguing case by case, we are done.

The  $\text{char } k = 2, 3$  cases are similar, use reduced Weierstrass forms (21.3.1.6) and argue case by case.

□

**Prop. (21.3.1.21) [Dual Elliptic Curves].** An elliptic curve  $E$  is canonically isomorphic to its dual  $\widehat{E}$ , by the mapping  $P \mapsto \mathcal{L}(P - O)$ .  $\lrcorner$

*Proof:* This morphism is induced by the ample line bundle  $\mathcal{L}(O)$  (20.9.4.6), and it is an isomorphism because it has degree 1, by Riemann-Roch (20.9.5.2).  $\square$

**Cor. (21.3.1.22).** For any  $\mathcal{L} \in \text{Pic}(E)$  of degree  $d$ ,  $\varphi_{\mathcal{L}} = [d] : E \rightarrow E$  via the canonical duality, because  $\text{Pic}^0(E)$  induces trivial maps by (20.9.4.12) and (9.1.11.5).  $\lrcorner$

**Cor. (21.3.1.23) [Characterizing Principal Divisors].** Let  $E \in \text{EllCur}/k$  and  $D = \sum n_i [P_i] \in \text{Div}(E)$ , where  $P_i$  are rational points, then  $D$  is a principal divisor iff  $\sum n_i = 0$  and  $\sum [n_i] P_i = O$ .  $\lrcorner$

*Proof:* Of course  $\sum n_i = 0$ . In this case,  $\sum [n_i] P_i$  is mapped to the line bundle corresponding to  $[D] \in \widehat{E}$  via  $E \rightarrow \widehat{E}$ , so  $[D] = 0$  iff  $\sum [n_i] P_i = O$ .  $\square$

### Isogenies

**Prop. (21.3.1.24).** Let  $\varphi, \varphi' : E' \rightarrow E$  be isogenies of elliptic curves over  $k$ , and let  $\omega$  be an invariant differential on  $E$ , then by (21.3.1.13),

$$(\varphi + \varphi')^* \omega = \varphi^* \omega + (\varphi')^* \omega.$$

**Cor. (21.3.1.25).**  $[m]^* \omega = m\omega$ .  $\lrcorner$

**Cor. (21.3.1.26) [End( $E$ ) is Commutative in Characteristic 0].** Let  $E \in \text{EllCur}/k$  with the normalized invariant differential form  $\omega$ , then there is a ring homomorphism

$$\text{End}(E_{\overline{k}}) \rightarrow \overline{k} : \varphi \mapsto a_{\varphi}, \text{ where } \varphi^* \omega = a_{\varphi} \omega.$$

And the kernel is the set of inseparable endomorphisms of  $E_{\overline{k}}$ . In particular, if  $\text{char } k = 0$ , then  $\text{End}(E_{\overline{k}})$  is commutative.  $\lrcorner$

*Proof:* Cf. [Sil16] P79. ?  $\square$

**Prop. (21.3.1.27) [Dual Isogenies].** Let  $k \in \text{Field}$ ,  $E, E' \in \text{EllCur}/k$  and  $\varphi \in \text{Hom}(E, E')$ , then under the canonical isomorphism (21.3.1.21), the dual map  $\widehat{\varphi}$  can be regarded as a map  $\widehat{\varphi} : E' \rightarrow E$ . Then

$$\widehat{\varphi} \circ \varphi = [\deg(\varphi)]_E, \varphi \circ \widehat{\varphi} = [\deg(\varphi)]_{E'}.$$

$\text{End}(E) \rightarrow \text{End}(E) : \varphi \mapsto \widehat{\varphi}$  is additive, and defines an involution on  $\text{End}(E)$ .  $\lrcorner$

*Proof:* Let  $d = \deg(\varphi)$ . By unwinding definition,  $\widehat{\varphi} \circ \varphi : E \rightarrow E' \rightarrow \widehat{E}' \rightarrow \widehat{E}$  is just  $\varphi_{\varphi^* \mathcal{L}(O)}$ . But for elliptic curves,  $\varphi_{\mathcal{L}}$  only depends on  $\deg(\mathcal{L})$ , thus  $\varphi_{\varphi^* \mathcal{L}(O)} = \varphi_{\mathcal{L}(dO)} = [d] \varphi_{\mathcal{L}(O)}$ . The second assertion follows by noticing  $\varphi = \widehat{\widehat{\varphi}}$  and  $\deg(\varphi) = \deg(\widehat{\varphi})$  (20.9.6.4).  $\square$

**Cor. (21.3.1.28) [End( $E$ )].** Let  $k$  be a field and  $E \in \text{EllCur}/k$ , then  $\text{End}(E) \otimes \mathbb{Q}$  is a division ring.  $\lrcorner$

**Prop. (21.3.1.29) [Classification of End( $E$ )].** Let  $k \in \text{Field}$ ,  $E \in \text{EllCur}/k$ , then  $\text{End}(E)$  has the following three possibilities:

- $\mathbb{Z}$ .
- an  $\mathbb{Z}$ -order in an imaginary quadratic extension over  $\mathbb{Q}$ .
- an  $\mathbb{Z}$ -order in a definite quaternion algebra over  $\mathbb{Q}$ .

And the third case won't happen in characteristic 0, by (21.3.1.26).  $\lrcorner$

*Proof:* This follows from (3.6.10.1) and (21.1.4.3).  $\square$



## 2 Formal Groups of Elliptic Curves

In this subsection, the formal group structure of an elliptic is studied.

In this subsection,  $E$  is an elliptic curve over a field  $K$  given by Weierstrass equation  $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ .

**Prop. (21.3.2.1) [Calculation of Formal Group Law].** Make a change of variable  $z = -x/y, w = -1/y$  to the Weierstrass equation of  $E$ , the equation becomes

$$w = z^3 + (a_1z + a_2z^2)w + (a_3 + a_4z)w^2 + a_6w^3 = f(z, w).$$

Then

- $\hat{\mathcal{O}}_{E,O} \cong K[[z]]$ .
- There is a unique power series  $w(z) \in \mathbb{Z}[a_1, \dots, a_6][[z]]$  satisfying  $w(0) = 0$  and  $w(z) = f(z, w(z))$ .
- $w(z) = z^3(1 + A_1z + A_2z^2 + \dots)$ , where  $A_n$  is a homogenous polynomial in  $a_1, \dots, a_6$ , where  $a_i$  has weight  $i$ .

•

$$\begin{aligned} x(z) &= z/w(z) = z^{-2} - a_1z^{-1} - a_2 - a_3z - (a_4 + a_1a_3)z^2 + \dots \in \mathbb{Z}[a_1, \dots, a_6][[z]] \\ y(z) &= -1/w(z) = -z^{-3} + a_1z^{-2} + a_2z^{-1} + a_3 + (a_4 + a_1a_3)z + \dots \in \mathbb{Z}[a_1, \dots, a_6][[z]] \end{aligned}$$

- The formal group law of  $E$  (20.9.2.1) is given by

$$\begin{aligned} F(z_1, z_2) &= z_1 + z_2 - a_1z_1z_2 - a_2(z_1^2z_2 + z_1z_2^2) \\ &\quad + (-2a_3z_1^3z_2 + (a_1a_2 - 3a_3)z_1^2z_2^2 - 2a_3z_1z_2^3) + \dots \in \mathbb{Z}[a_1, \dots, a_6][[z_1, z_2]]. \end{aligned}$$

- The normalized invariant differential form (21.3.1.12)

$$\begin{aligned} \omega(z) &= \frac{dx(z)}{2y(z) + a_1x(z) + a_3} = [1 + a_1z + (a_1^2 + a_2)z^2 + (a_1^3 + 2a_1a_2 + 2a_3)z^3 \\ &\quad + (a_1^4 + 3a_1^2a_2 + 6a_1a_3 + a_2^2 + 2a_4)z^4 + \dots] dz \in \mathbb{Z}[a_1, \dots, a_6][[z]]dz \end{aligned}$$

•

$$i(z) = \frac{x(z)}{y(z) + a_1x(z) + a_3} = \frac{z^{-2} - a_1z^{-1} + \dots}{-z^{-3} + 2a_1z^{-2} + \dots} \in \mathbb{Z}[a_1, \dots, a_6][[z]].$$

In particular, this formal group law is defined over  $\mathbb{Z}[a_1, \dots, a_6]$ . ┘

*Proof:*  $\omega(z) \in \mathbb{Z}[a_1, \dots, a_6][[z]]dz$  by (10.7.3.18).

For  $F(z_1, z_2)$ : ? □

**Prop. (21.3.2.2) [Inseparable Degree and Heights].** Let  $K$  be a field of characteristic  $p > 0$  and  $E_1/K, E_2/K$  be two elliptic curves, and  $\varphi : E_1 \rightarrow E_2$  a non-zero isogeny of elliptic curves over  $K$ , and let  $\hat{\varphi} : \hat{E}_1 \rightarrow \hat{E}_2$  be the homomorphism of formal group schemes, then

$$\deg_i(\varphi) = p^{\text{ht}(\hat{\varphi})}.$$

┘

*Proof:* Cf. [Sil16] P134. ? □

**Cor. (21.3.2.3).** Let  $K$  be a field of characteristic  $p > 0$ , then  $\text{ht}(\hat{E}) = 1$  or  $2$ , because  $\deg([p]_E) = p^2$  (20.9.6.18). ┘

**3 char  $k > 0$  case****Notation (21.3.3.1).**

- Let  $k \in \mathbf{Field}^p$ ,
- Let  $E \in \mathbf{EllCur}/k$  be given by a Weierstrass equation  $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ .

┘

**Prop. (21.3.3.2).** Let  $k = \mathbb{F}_q$ ,  $\#E(k) \equiv 0 \pmod{4}$ .

┘

*Proof:* There are two involutions on  $E$ :  $(x, y, z) \mapsto (x, -y, z)$ ,  $(x, y, z) \mapsto (z, y/x, x)$ . The fixed points of these elements contain

$$(0, 1, 0), \quad (0, 0, 1), \quad (\varepsilon_1, \varepsilon_2\sqrt{2\varepsilon_1}, 0), \quad (\varepsilon_3\sqrt{-1}, 0, 1).$$

Considering either cases s.t.  $-1 \in \mathbb{F}_q^2$  or  $2 \in \mathbb{F}_q^2$ , the number of these points are divisible by 4, thus we are done.  $\square$

**Prop. (21.3.3.3) [Hasse-Weil].** Let  $p \in \mathbf{P}$ ,  $q = p^r$ ,  $k = \mathbb{F}_q$  and  $E \in \mathbf{EllCur}/k$ , then

- If  $a = 1 + q - \#E(\mathbb{F}_q)$ ,  $\alpha, \beta$  be the two roots of  $T^2 - aT + q = 0$ , then  $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \alpha^n - \beta^n$ .
- $|\#E(\mathbb{F}_q) - 1 - q| \leq 2\sqrt{q}$ .
- The Frobenius map  $\varphi_E : E \rightarrow E$  satisfies  $\varphi^2 - a\varphi + [q] = 0$ .
- $a = 1 + q - \deg(1 - \varphi)$ , and  $[a] = \varphi + \hat{\varphi}$ .

┘

*Proof:* 1, 2 are special cases of (20.9.9.5). 3, 4 follow from (20.9.7.13) and (20.9.6.21).  $\square$

**Cor. (21.3.3.4).** Let  $k = \mathbb{F}_q$ ,  $p \neq 2$  and  $E$  is given by  $y^2 = x^3 + a_2x^2 + a_4x + a_6$ , then

$$\left| \sum_{x \in \mathbb{F}_q} \left( \frac{x^3 + a_2x^2 + a_4x + a_6}{\mathbb{F}_q} \right) \right| \leq 2\sqrt{q}.$$

┘

*Proof:* This is because  $\#E(\mathbb{F}_q) = \sum_{x \in \mathbb{F}_q} \left( 1 + \left( \frac{x^3 + a_2x^2 + a_4x + a_6}{\mathbb{F}_q} \right) \right)$ .  $\square$

**Supersingular Elliptic Curves**

Main references are [Igu58].

**Def. (21.3.3.5) [Supersingular Elliptic Curve].** Let  $p \in \mathbf{Prime}$ ,  $k \in \mathbf{Field}^p$ ,  $F_{E/k,r} : E \rightarrow E^{(p^r)}$  the relative Frobenius (7.2.10.1), and  $\hat{F}_{E/k}$  be its dual. Then the following are equivalent:

- $E[p^r] = 1$  for all  $r \geq 1$ .
- $E[p^r] \not\cong \mathbb{Z}/p^r\mathbb{Z}$  for some  $r \geq 1$ .
- $\hat{F}_{E/k,r}$  is purely inseparable for one (thus all)  $r$ .
- $[p] : E \rightarrow E$  is purely inseparable, and  $j(E) \in \mathbb{F}_{p^2}$ .
- $\text{End}(E)$  is an order in a quaternion algebra.
- The formal group  $\hat{E}/K$  associated to  $E$  has height  $2 (\neq 1)$ .

Such curves are called **supersingular elliptic curves**, otherwise  $E$  is called an **ordinary elliptic curve**.

Being supersingular is stable and reflective under base change.  $\lrcorner$

*Proof:* Cf.[Silverman P144].  $\textcolor{red}{?}$   $\square$

**Prop. (21.3.3.6) [Ordinary Elliptic Curves].** Let  $k \in \mathbf{Field}$ ,  $\text{char } k = p > 0$ ,  $E \in \mathbf{EllCur}/k$  be an ordinary elliptic curve, then

- If  $j(E) \in \overline{\mathbb{F}}_p$ , then  $\text{End}(E)$  is an order in an imaginary quadratic field. In particular, this is the case when  $k$  is a finite field.
- If  $j(E) \notin \overline{\mathbb{F}}_p$ , then  $\text{End}(E) \cong \mathbb{Z}$ .

$\lrcorner$

*Proof:* 1: If  $j(E) \in \overline{\mathbb{F}}_p$ , then  $E$  is defined over some  $\mathbb{F}_q$ . Let  $\varphi = \varphi_q$ , then if  $\varphi^r \in \mathbb{Z}$ ,  $\varphi^r = [\pm q^r]$ , so  $\#E[\varphi^r] = \deg_s(\varphi^r) = 1$  by (20.9.6.2), contradicting the fact  $E$  is ordinary. Thus the assertion follows from (21.3.1.29).

2:  $\textcolor{red}{?}$   $\square$

**Cor. (21.3.3.7).** For  $E \in \mathbf{EllCur}/\overline{\mathbb{F}}_p$ ,  $E$  is ordinary iff  $\text{End}(E)$  is an order in an imaginary quadratic field, and  $E$  is supersingular iff  $\text{End}(E)$  is an order in a quaternion algebra.  $\lrcorner$

**Def. (21.3.3.8) [Hasse Invariant].** The **Hasse invariant** for  $E$  is a number which is 0 if  $E$  is supersingular and 1 if  $E$  is ordinary. The definition of it is in [Katz, p-adic Modular Forms].  $\textcolor{red}{?}$   $\lrcorner$

**Prop. (21.3.3.9) [Supersingular and Trace].** Let  $E \in \mathbf{EllCur}/\mathbb{F}_q$  and  $\varphi_E$  be the Frobenius of  $E/\mathbb{F}_q$ , then

- $E$  is supersingular iff  $\#E(\mathbb{F}_q) \equiv 1 \pmod{p}$ , iff  $a_p = \alpha + \beta \equiv 0 \pmod{p}$  in (21.3.3.3).
- If  $q = p \geq 5$  is a prime, then  $E$  is supersingular iff  $\#E(\mathbb{F}_p) = p + 1$ . This is false for  $p = 2$  or  $p = 3$ .

$\lrcorner$

*Proof:* 1: By (21.3.3.3), let  $a = 1 + q - \#E(\mathbb{F}_q)$ , then  $\widehat{\varphi}_E = [a] - \varphi$  is separable iff  $a = 0 \in \mathbb{F}_q$  by??. Thus the assertion follows from (21.3.3.5).

2: Because  $|\alpha + \beta| \leq 2\sqrt{p} < p$  in this case.  $\square$

**Cor. (21.3.3.10).** Let  $p \geq 5$  and  $E \in \mathbf{EllCur}/\mathbb{F}_p$  be supersingular, then

$$\#E(\mathbb{F}_{p^n}) = \begin{cases} p^n + 1 & , n = 2k + 1 \\ (p^{n/2} - (-1)^{n/2})^2 & , n = 2k \end{cases}$$

$\lrcorner$

*Proof:* This is because  $\alpha = -\beta = \sqrt{p}i$ .  $\square$

**Prop. (21.3.3.11) [Supersingular in Characteristic 2].** Let  $k$  be a field of characteristic 2, then  $E/k$  is supersingular iff  $j(E) = 0$ . In particular,  $y^2 + y = x^3$  is the only supersingular elliptic curve over  $\overline{k}$ .  $\lrcorner$

*Proof:* We use the condition  $\#E(\mathbb{F}_2) = 2$ . We use the reduced Weierstrass equations in (21.3.1.5). By doubling formula (21.3.1.9), this is equivalent to the non-existence of a point  $(x, y) \in E(\overline{k}) \setminus O$  s.t.  $a_1x + a_3 = 0$ . And it can be checked this is true iff  $j(E) = 0$ .  $\square$

**Prop. (21.3.3.12)**[Supersingular Elliptic Curves over Finite Fields, Igusa]. Let  $p \in \text{Prime}$  and  $E \in \text{EllCur} / \overline{\mathbb{F}}_p$ .

1. If  $p = 2$ , the only supersingular elliptic curve over  $\overline{\mathbb{F}}_2$  is isomorphic to  $y^2 + y = x^3$ .
2. If  $p \geq 3$  and  $E$  is given by a Weierstrass form  $y^2 = f(x)$ , then  $E$  is supersingular iff  $\text{Coef}\left(\frac{f(x)^{(p-1)/2}}{x^{p-1}}\right) = 0 \in \overline{\mathbb{F}}_p$ .
3. If  $p \geq 3$  and  $E$  is given by a Legendre form  $y^2 = x(x-1)(x-\lambda)$ , then  $E$  is supersingular iff

$$H_p(\lambda) = \sum_{i=1}^{(p-1)/2} \binom{\frac{(p-1)}{2}}{i}^2 \lambda^i = 0.$$

4. If  $p \geq 3$ , the polynomial  $H_p$  has distinct roots in  $\overline{\mathbb{F}}_p$ , and the number of isomorphism classes of supersingular elliptic curves over  $\overline{\mathbb{F}}_p$  is

$$\begin{cases} 1 & , p = 3 \\ \lfloor \frac{p}{12} \rfloor & , p \equiv 1 \pmod{12} \\ \lfloor \frac{p}{12} \rfloor + 1 & , p \equiv 5, 7 \pmod{12} \\ \lfloor \frac{p}{12} \rfloor + 2 & , p \equiv 11 \pmod{12} \end{cases}$$

5. (Mass Formula)

$$\sum_{E/\overline{\mathbb{F}}_p \text{ supersingular}} \frac{1}{\text{Aut}(E)} = \frac{p-1}{24}.$$

┘

**Remark (21.3.3.13).** This has relations to the class number of normal quaternion algebras over  $\mathbb{Q}$  ramified at  $p$ , Cf. [Igu58].

┘

*Proof:* 1: This follows from (21.3.3.11).

2: Let  $E \in \text{EllCur} / \mathbb{F}_q$ , denote  $A_q = \text{Coef}\left(\frac{f(x)^{(q-1)/2}}{x^{q-1}}\right)$ , then

$$\#E(\mathbb{F}_q) \equiv \sum_{x \in \mathbb{F}_q} \left(1 + \left(\frac{x^3 + a_2x^2 + a_4x + a_6}{\mathbb{F}_q}\right)\right) \equiv 1 - A_q \pmod{p}.$$

Then  $A_q \equiv a_E \pmod{p}$ , and the assertion follows from (21.3.3.9).

Finally, it can be seen from the equation  $f(x)^{(p^{r+1}-1)/2} = f(x)^{(p^r-1)/2} [f(x)^{(p-1)/2}]^{p^r}$  that  $A_{p^{r+1}} = A_{p^r} A_p^{p^r}$ . So by an induction argument,  $A_q = 0$  iff  $A_p = 0$ .

3:

$$\text{Coef}\left(\frac{(x(x-1)(x-\lambda))^{(p-1)/2}}{x^{p-1}}\right) = \text{Coef}\left(\frac{(x-1)^{(p-1)/2}(x-\lambda)^{(p-1)/2}}{x^{(p-1)/2}}\right) = \sum_{i=1}^{(p-1)/2} \binom{\frac{(p-1)}{2}}{i}^2 \lambda^i = H_p(\lambda).$$

- 4: Consider the **Picard-Fuchs differential operator**

$$D = 4t(t-1) \frac{\partial^2}{\partial t^2} + 4(1-2t) \frac{\partial}{\partial t} - 1.$$

Direct calculation shows

$$DH_p(t) = p \sum_{i=1}^{(p-1)/2} (p-2-4i) \binom{\frac{(p-1)}{2}}{i} t^i = 0.$$

Thus possible multiple roots of  $H$  can only be 1 or 0. But  $H_p(0) = 1$ , and  $H_p(1) = \left(\frac{p-1}{2}\right) = (-1)^{\frac{p-1}{2}}$ , thus  $H_p(t)$  has no multiple roots.

Let  $\varepsilon_p(j) = 1$  if the elliptic curve  $E$  with the indicated  $j$ -invariant is supersingular, and  $\varepsilon_p(j) = 0$  otherwise. Notice for  $p \geq 5$ ,  $j : \lambda \rightarrow j(E_\lambda)$  is 6 to 1 except  $\#j^{-1}(0) = 2$  and  $\#j^{-1}(1728) = 3$ , thus by counting the number of zeros of  $H_p(\lambda)$ , the number of supersingular elliptic curves is

$$\frac{1}{6} \left( \frac{p-1}{2} - 2\varepsilon_p(0) - 3\varepsilon_p(1728) \right) + \varepsilon_p(0) + \varepsilon_p(1728) = \frac{p-1}{12} + \frac{2}{3}\varepsilon_p(0) + \frac{1}{2}\varepsilon_p(1728).$$

Thus the assertion follows from determining elliptic curves of  $j$ -invariants 0 and 1728 are supersingular or not, which is done by (21.3.3.14).

5: It follows from the proof of item4 that this number is  $1/2$  of number of roots of  $H_p(\lambda) = \frac{p-1}{24}$ .

□

**Lemma (21.3.3.14) [Examples of Supersingular Elliptic Curves].**

- For  $p \geq 5$ , the elliptic curve  $E : y^2 = x^3 + 1$  with  $j$ -invariant 0 is supersingular iff  $p \equiv 2 \pmod{3}$ .
- For  $p \geq 3$ , the elliptic curve  $E : y^2 = x^3 + x$  with  $j$ -invariant 1728 is supersingular iff  $p \equiv 3 \pmod{4}$ .

┘

*Proof:* These follows from calculating the  $x^{p-1}$ -term of  $(x^3 + 1)^{(p-1)/2}$  or  $(x^3 + x)^{(p-1)/2}$  and (21.3.3.12) item2. □

## 4 Reduction of Elliptic Curves

References are [Liu02].

In this subsection, let  $(S, K)$  be a Dedekind scheme with generic point  $\eta$ .

**Prop. (21.3.4.1) [Minimal Models of Elliptic Curves].** Let  $E \in \text{EllCur}/K$  and  $\mathcal{E}/S$  be a minimal model for  $E$ , then

- $\mathcal{E}_{\text{sm}}(S) = \mathcal{E}(S) = E(K)$ .
- The automorphism  $(m, \text{pr}_2) : E \times_K E \rightarrow E \times_K E$  extends to an automorphism  $t : \mathcal{E} \times_S \mathcal{E}_{\text{sm}} \rightarrow \mathcal{E} \times_S \mathcal{E}_{\text{sm}}$ .
- $t$  induces an automorphism  $\mathcal{E}_{\text{sm}} \times_S \mathcal{E}_{\text{sm}} \rightarrow \mathcal{E}_{\text{sm}} \times_S \mathcal{E}_{\text{sm}}$ , and  $\text{pr}_1 \circ E$  makes  $\mathcal{E}_{\text{sm}}$  a smooth group scheme over  $S$  extending that of  $E$ .

┘

*Proof:* Cf. [Liu02] P492. □

**Thm. (21.3.4.2) [Minimal Models and Néron Models].** Let  $(K, \mathcal{O}_K) \in \text{DVR}$ ,  $E \in \text{EllCur}/K$  with minimal model  $\mathcal{E}/\mathcal{O}_K$  (20.3.7.4), then  $\mathcal{E}_{\text{sm}}$  is a Néron model of  $E$ . ┘

*Proof:* Cf. [Liu, Prop10.2.12] or [BLR, 1.5/1].

To show universal property, let  $X/S$  be smooth, and  $f : X_\eta \rightarrow E$  be a morphism considered as a rational morphism  $X \rightarrow \mathcal{E}_{\text{sm}}$ . Let  $\xi \in X$  be a point of codimension 1, and  $T = \text{Spec } \mathcal{O}_{X,\xi}$ , then by (21.4.6.7) and the proof of (21.3.4.1),  $\mathcal{E} \times_S T \rightarrow T$  is an arithmetic surface with smooth locus  $\mathcal{N} \times_S T$ ?, so by (21.4.2.8),  $\mathcal{N}(T) = \mathcal{N}_T(T) = E(K)$ , which means  $f$  can be extended to  $\xi$ . Then in fact  $f$  is a morphism by (10.1.1.18).  $\square$

### over DVRs

In this subsubsection, let  $(K, \mathcal{O}_K, \mathfrak{m}, \kappa)$  be a DVR,  $S = \text{Spec}(\mathcal{O}_K)$ .

**Prop. (21.3.4.3) [Weierstrass Models are Normal].** Let  $E \in \text{EllCur}/K$ , then any Weierstrass equation for  $E$  over  $R$  is normal.  $\lrcorner$

*Proof:* Cf. [Conrad, Minimal model for Elliptic Curves].  $\square$

**Def. (21.3.4.4) [Minimal Weierstrass Equation].** Let  $E \in \text{EllCur}/K$ , then a **minimal Weierstrass equation** for  $E$  is an Weierstrass equation for  $E$  in  $R[X, Y]$  with determinant  $\Delta$  with minimal valuation.  $\lrcorner$

**Prop. (21.3.4.5) [Uniqueness of Minimal Weierstrass Equations].** Let  $E \in \text{EllCur}/K$ , then  $E$  has a minimal Weierstrass equation, and it is unique up to change of variables of the form

$$(X, Y) \mapsto (u^2X + r, u^3Y + su^2X + t), \text{ where } u \in \mathcal{O}_K^*, r, s, t \in \mathcal{O}_K.$$

Moreover, given any Weierstrass equation, then any change of coordinates that is used to produce a minimal Weierstrass equation is of the form

$$(X, Y) \mapsto (u^2X + r, u^3Y + su^2X + t), \text{ where } u, r, s, t \in R.$$

$\lrcorner$

*Proof:* Cf. [Sil16]P186.  $\square$

**Cor. (21.3.4.6) [Weierstrass Models].** Let  $E \in \text{EllCur}/K$ , a minimal Weierstrass of  $E$  defines a model  $\mathcal{W}/R$  of  $E$ , and it is invariant of the Weierstrass equation chosen, called the **Weierstrass model** of  $E$ .  $\lrcorner$

**Prop. (21.3.4.7) [Weierstrass Model of Elliptic Curves].** If  $E \in \text{EllCur}/K$ , then a Weierstrass equation with  $a_i \in R$  defines a scheme  $\mathcal{W} \in \mathbb{P}_R^2$ . Then

- Both  $\mathcal{W}_{\text{sm}}$  and  $\mathcal{W}$  has generic fiber  $E$ . If  $E$  has good reduction and  $W$  is a minimal Weierstrass equation, then  $\mathcal{W} = \mathcal{W}_{\text{sm}}$  is smooth.
- The natural map  $\mathcal{W}(\mathcal{O}_K) \rightarrow E(K)$  is a bijection. If  $\mathcal{W}$  is regular, then  $\mathcal{W}_{\text{sm}}(R) \rightarrow \mathcal{W}(R)$  is also a bijection.
- The group structure on  $E/K$  extends to a group structure on  $\mathcal{W}_{\text{sm}}/R$ , and the addition further extends to a group action of  $\mathcal{W}_{\text{sm}}/R$  on  $\mathcal{W}/R$ .

$\lrcorner$

*Proof:* Cf. [Mil199]P321.  $\square$

**Prop. (21.3.4.8) [Weierstrass Model and Minimal Models].** If  $E \in \text{EllCur}/K$  with minimal model  $\mathcal{E}$ , then any minimal Weierstrass model  $\mathcal{W}$  can be obtained by blowing down the finitely many components of  $\mathcal{E}_k$  which are disjoint from the closure in  $\mathcal{E}$  of  $\mathbf{0}_E$ . In particular, by (21.3.4.2),  $\mathcal{W}_{\text{sm}}$  is isomorphic to the relative identity component (20.3.2.3)  $\mathcal{N}_0$  of the Néron model of  $E$ .  $\square$

*Proof:* Cf. [Liu Qing, Thm. 9.4.35]  $\square$

**Def. (21.3.4.9) [Reduction Types].** For  $E \in \text{EllCur}/K$ ,

- $E$  has good reduction iff  $(\mathcal{N}_0)_\kappa \in \text{EllCur}/\kappa$  (21.3.4.8).
- $E$  has semistable reduction (21.1.2.5) iff  $(\mathcal{N}_0)_{\kappa^{\text{sep}}} \cong \mathbb{G}_m$ . And it is called a **split multiplicative reduction** if  $(\mathcal{N}_0)_\kappa \cong \mathbb{G}_m$ , otherwise it is called a **non-split multiplicative reduction**.
- $E$  has bad reduction or **additive reduction** iff  $(\mathcal{N}_0)_\kappa \cong \mathbb{G}_a$ .

$\square$

*Proof:* These follow from (21.1.2.5) (21.3.1.17) and the fact  $\mathcal{N}_0 \cong \mathcal{W}_{\text{sm}}$  for any minimal Weierstrass model for  $E$  (21.3.4.8).  $\square$

**Def. (21.3.4.10) [Reduction of Elliptic Curves].** Let  $E \in \text{EllCur}/K$  with Néron model  $\mathcal{N}/\mathcal{O}_K$  (20.3.4.1), then  $\mathcal{N}(\mathcal{O}_K) = E(K)$ , and by infinitesimal lifting property of smoothness?, the reduction map  $r : \mathcal{N}(\mathcal{O}_K) \rightarrow \mathcal{N}_0(\kappa)$  is surjective. Define

$$E_0(K) = r^{-1}(\mathcal{N}_0(\kappa)) \text{ (21.3.4.8)}, \quad E_1(K) = \ker(r).$$

$E_1(K)$  is called the **kernel of the reduction**. In particular,

$$\mathcal{N}_0(\kappa) \cong E_0(K)/E_1(K).$$

$\square$

**Prop. (21.3.4.11) [Formal Group is the Kernel].** Let  $E \in \text{EllCur}/K$ , then with notations as in (21.3.2.1),

$$\widehat{E}(\mathfrak{m}) \rightarrow E_1(K) : z \mapsto [z, -1, w(z)]$$

is an isomorphism of groups.  $\square$

*Proof:* By (21.3.2.1),  $\omega(z) \in z^3 R[[z]]$ , and by the definition of  $w(z)$ ,  $[z, -1, w(z)] \in E_1(K)$ , and it is a group homomorphism by the definition of formal group law associated to  $E$ . It remains to prove that this map is surjective:

Clearly  $\mathbf{0}$  is in the image, suppose  $(x_0, y_0) \in E_1(K)$ , then  $v(x_0) < 0, v(y_0) < 0$ , thus by inspecting the Weierstrass equation,  $v(x_0) = -2r, v(y_0) = -3r$  for some  $r > 0 \in \mathbb{Z}$ . Then  $z_0 = x_0/y_0 \in \mathfrak{m}$ , and  $(x_0, y_0)$  is the image of  $z_0$ .  $\square$

**Prop. (21.3.4.12) [Characterizing Reductions].** Let  $E \in \text{EllCur}/K$  be given by a minimal Weierstrass equation (21.3.4.4)  $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$  over  $\mathcal{O}_K$ , then by (21.3.1.15),

- $\widetilde{E}$  is a good reduction iff  $\Delta \in \mathcal{O}_K^*$ .
- $\widetilde{E}$  is a semistable reduction iff  $\Delta \in \mathfrak{m}$  and  $c_4 \in \mathcal{O}_K^*$ .
- $\widetilde{E}$  is an additive reduction iff  $\Delta, c_4 \in \mathfrak{m}$ .

$\square$

**Prop. (21.3.4.13)[Characterizing Potentially Good Reduction].**  $E$  has potentially good reduction iff  $j(E) \in \mathcal{O}_K$ .

In particular, an elliptic curve with complex multiplication over  $K$  has potentially good reduction.

┘

*Proof:* If  $E_{\bar{k}}$  has good reduction, then  $j(E) = (c'_4)^3/\Delta' \in R'$ . Conversely, if  $j(E) \in R$ , use Legendre form or Deuring form (21.3.1.18)(21.3.1.19), the equation

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2} \text{ or } j = \frac{\alpha^3(\alpha^3 - 24)^3}{\alpha^3 - 27}$$

implies  $\lambda \in \mathcal{O}_K^*$ ,  $\lambda(\lambda - 1) \in \mathcal{O}_K^*$  (or  $\alpha \in \mathcal{O}_K^*$ ,  $\alpha^3 - 27 \in \mathcal{O}_K^*$ ), so  $E_{\bar{k}}$  has good reduction.  $\square$

**Prop. (21.3.4.14) [Tamagawa Numbers].** Let  $E \in \text{EllCur}/K$  with a minimal model  $\mathcal{E}/\mathcal{O}_K$ , then  $c(E)$  equals the number of geo.integral components occuring with multiplicity 1 in  $\mathcal{E}_k$ .  $\square$

*Proof:* Cf.[Liu Qing, 10.2.24].  $\square$

**Prop. (21.3.4.15)[Tamagawa Numbers].** Let  $E \in \text{EllCur}/K$ . If  $E$  has split multiplicative reduction over  $K$ , then  $E(K)/E_0(K)$  is a cyclic group of order  $v(\Delta) = -v(j)$ . And in other cases, the group is finite and has order at most 4.  $\square$

*Proof:*  $\square$

**Cor. (21.3.4.16).** If  $K \in p\text{-Field}$  and  $E \in \text{EllCur}/K$ , then  $E(K)$  contains a subgroup of finite index that is isomorphic to  $\mathcal{O}_K$ .  $\square$

*Proof:* Because  $E(K)/E_0(K)$  and  $E_0(K)/E_1(K) \cong \tilde{E}_{ns}(k)$  are all finite by (21.3.4.15), it suffices to prove that  $E_1(K) \cong \hat{E}(\mathfrak{m})$  has a subgroup of finite index that is isomorphic to  $\mathcal{O}_K$ . But this follows from the fact  $\hat{E}(\mathfrak{m}^r) \cong \mathfrak{m}^r$  for  $r$  large (10.7.4.21).  $\square$

### Tate Algorithm

References are [Sil99].

**Notation (21.3.4.17).**

- Suppose  $\kappa$  is perfect.

┘

**Thm. (21.3.4.18)[Kodaira-Néron].** Suppose  $\kappa = \bar{k}$  and  $E \in \text{EllCur}/K$  with a minimal proper regular model  $\mathcal{E}/\mathcal{O}_K$  (20.3.7.4), then  $\mathcal{C}_\kappa$  is isomorphic to one of the following:

- (Type  $I_0$ ) An elliptic curve.
- (Type  $I_1$ ) A rational curve with a node.
- (Type  $I_n, n \geq 2$ )  $n$  smooth rational curves intersecting transversally at a single point one-by-one in the shape of a  $n$ -gon.
- (Type II) A rational curve with a cusp.
- (Type III) Two non-singular rational curves which intersects tangentially at a single point.
- (Type  $I_0^*$ ) A non-singular rational curve of multiplicity 2 with 4 non-singular rational curves of multiplicity 1 attached.



- (Type  $I_n^*$ ) A chain of  $n + 1$  non-singular rational curves of multiplicity 2, with two non-singular curves of multiplicity 1 attached at each end.
- (Type  $IV^*$ ) ?
- (Type  $III^*$ ) ?
- (Type  $II^*$ ) ?

┘

*Proof:* Cf. [Sil99] P354.

□

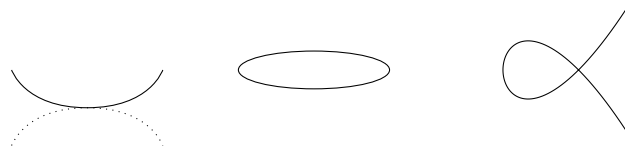


Figure (21.3.4.1): Kodaira Types

**Prop. (21.3.4.19) [Tate Algorithm].** There is an algorithm that determines if a Weierstrass equation for  $E$  is minimal. ?

┘

### Torsion Points

**Prop. (21.3.4.20) [Controlling Torsion Points].** Let  $m \in \mathbb{Z} \cap k^\times$ , then

- $E_1(K)[m] = O$
- The reduction map  $E_0(K)[m] \rightarrow \tilde{E}_{\text{sm}}(k)[m]$  is injective.
- If  $E$  has good reduction and  $K = K^{\text{sep}}$ , then it is also an isomorphism.

┘

*Proof:* 1: By (21.3.4.11),  $E_1(K)[m] \cong \hat{E}(\mathfrak{m})[m]$ , which has only one element because  $[m]_{\hat{E}}$  is an isomorphism (10.7.3.5).

2 follows from 1 and the exact sequence in (21.3.4.10).

3 follows as they have the same cardinality.

□

**Prop. (21.3.4.21) [Controlling Torsion Points].** Let  $\text{char } K = 0$  and  $\text{char } k = p > 0$ , let  $E$  be given by a Weierstrass equation

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \in R[X, Y].$$

Let  $P = (x_0, y_0) \in E(K)$  be a torsion point of exact order  $m$ , then

- If  $m$  is not a  $p$ -power, then  $x_0, y_0 \in R$ .
- If  $m = p^n$ , then  $v(x) \geq -2 \lfloor \frac{v(p)}{p^n - p^{n-1}} \rfloor$ ,  $v(y) \geq -3 \lfloor \frac{v(p)}{p^n - p^{n-1}} \rfloor$ .

┘

*Proof:* Change the Weierstrass equation to a minimal Weierstrass equation, then by (21.3.4.4), the coordinates in the new equation satisfies  $v(x') \leq v(x_0), v(y') \leq v(y_0)$ , thus it suffices to prove for minimal Weierstrass equations.

If  $v(x_0) \geq 0$ , then  $v(y_0) \geq 0$ . Suppose  $v(x_0) < 0$ , then  $v(y_0) < 0$ , and  $v(x_0) = -2r, v(y_0) = -3r$  for some  $r > 0$ . thus  $P \in E_1(K)$  and thus corresponds to  $z \in \hat{E}(\mathfrak{m})$  (21.3.4.11). Then the theorem follows from (10.7.4.20). □

### Over Dedekind Domains

**Def. (21.3.4.22) [Canonical Heights].** For  $F \in \mathbf{NField}$  and  $E \in \mathbf{EllCur}/F$ , by (20.9.10.5) and (20.9.4.12),  $\hat{h}_{\mathcal{L}}$  only depends on  $\deg(\mathcal{L})$ . And by (7.13.2.20), any degree 1 line bundle is ample, and the Néron-Tate bilinear form  $\langle \cdot, \cdot \rangle_{\mathbf{N-T}}$  given by this line bundle is called the **canonical height pairing** of  $E$ . It is a positive-definite quadratic form on  $E(\overline{K}) \otimes \mathbb{R}$  by (21.1.3.4). And for  $P \in E(\overline{K})$ ,  $\langle P, P \rangle_{\mathbf{N-T}}$  is called the **canonical height** of  $P$ .

In particular we can take the line bundle  $\mathcal{O}(O)$ , then  $\langle O, O \rangle_{\mathbf{N-T}} = 0$ , and for any  $P \neq 0$  in  $\mathcal{L}(O)$ ,

$$\langle P, P \rangle_{\mathbf{N-T}} = \lim_{n \rightarrow \infty} h(x([2^n]P))/4^N.$$

┘

*Proof:* For the final assertion, apply (20.2.1.25) to the morphism  $\varphi : E \rightarrow \mathbb{P}^1$  associated to the rational function  $x$ :

$$\langle P, P \rangle_{\mathbf{N-T}} = \lim_{n \rightarrow \infty} \hat{h}_{\mathcal{O}(2O)}([2^n]P)/4^N = \lim_{n \rightarrow \infty} h_{\mathcal{O}_{\mathbb{P}^1}(1)}(\varphi([2^n]P))/4^N = \lim_{n \rightarrow \infty} h(x([2^n]P))/4^N.$$

□

**Prop. (21.3.4.23) [Elliptic Regulators].** Let  $F$  be a number field and  $E \in \mathbf{EllCur}/F$ , then the **elliptic regulator** is the volume of a fundamental domain of  $E(K)/E(K)_{\text{tor}}$  computed using the canonical height. In particular, it is the discriminant of  $(\langle X_i, X_j \rangle_{E,F})_{i,j}$  where  $\{X_i\}$  is a basis for  $E(K)/E(K)_{\text{tor}}$ .

┘

**Def. (21.3.4.24) [Minimal Discriminants].** For  $F \in \mathbf{GField}$  and  $E \in \mathbf{EllCur}/F$ , define the **minimal discriminant**  $\Delta_{E/F}^{\min}$  to be  $\Delta_{E/F}^{\min} = \prod_{v \in \Sigma_F^{\text{fin}}} \mathfrak{p}_v^{\text{ord}(\Delta_v^{\min})}$ , where  $\Delta_v^{\min}$  is the discriminant of a minimal Weierstrass equation for  $E_v$ .

Then for any Weierstrass equation  $W$  for  $E$ , by (21.3.1.14),  $\mathfrak{D}_{E/F} = \Delta_{E/F}^{\min} (\mathfrak{a}_W)^{12}$  for some ideal  $\mathfrak{a}$  of  $F$ , and the ideal class of  $\mathfrak{a}_{\Delta}$  is stable under change of Weierstrass equations, called the **Weierstrass class** of  $E$ , denoted by  $\overline{\mathfrak{a}}_{E/F}$ .

┘

**Def. (21.3.4.25) [Unstable Minimal Discriminants].** For  $F \in \mathbf{GField}$  and  $E \in \mathbf{EllCur}/F$ , suppose  $(j_E) = \mathfrak{a}\mathfrak{b}^{-1}$  where  $\mathfrak{a}, \mathfrak{b} \in \text{Ideal}^*(\mathcal{O}_F)$  are relatively prime, then the **unstable minimal discriminant** of  $E$  is defined to be

$$\Upsilon_{E/F} = \Delta_{E/F}^{\min} \cdot \mathfrak{b}^{-1}.$$

Then

- $\Upsilon_{E/F} \in \text{Ideal}^*(\mathcal{O}_F)$ .
- For  $v \in \Sigma_F^{\text{fin}}$ ,  $v(\Upsilon_{E/F}) = 0$  iff  $E$  has good or semistable reduction at  $v$ .
- $v \in \Sigma_F^{\text{fin}}$ ,  $v(\Upsilon_{E/F}) < 12 + 12v(2) + 6v(3)$ .

┘

*Proof:* 1, 2: These follow from the fact  $j(E) = c_4^3/\Delta$  (21.3.1.5) and (21.3.4.12).

3: For any minimal Weierstrass equation, maybe it follows from the change of variable formula (21.3.1.14)?

□

**Prop. (21.3.4.26) [A Curve Related to Fermat's Last Theorem].** Let  $a, b, c \in \mathbb{Z}^{\times}$  satisfy  $(a, b, c) = 1$ , let  $E$  be the elliptic curve defined by the Weierstrass equation

$$E : y^2 = x(x + a)(x - b),$$

then

- the minimal discriminant (21.3.4.24)  $\Delta_{E/K}^{\min}$  of  $E$  is either  $2^4|abc|^2$  or  $2^{-8}|abc|^2$ .
- $E$  has semistable reduction for any odd prime.

┘

*Proof:* By calculation

$$\begin{aligned} \mathfrak{D}_W &= 2^4(abc)^2, \quad c_4 = 2^4(a^2 + ab + b^2), \quad c_6 = -2^5(2a^3 + 3a^2b + 3ab^2 + 2b^3), \\ (22a^2 - 8ab - 8b^2)c_4 + (a + 2b)c_6 &= 288a^2, \quad -(8a^2 + 8ab - 22b^2)c_4 - (2a + b)c_6 = 288b^2. \end{aligned}$$

1: Any change of variable of  $E$  to the minimal Weierstrass equation satisfies  $u^4|c_4, u^6|c_6$ , so

$$u^4|(288a^2, 288b^2)$$

so  $u = 1$  or  $2$ , and  $\Delta_E^{\min} = u^{-12}\mathfrak{D}_W = 2^4|abc|^2$  or  $2^{-8}|abc|^2$ .

2: If  $p \in S(\Delta_E^{\min}) \setminus \{2\}$ , then  $p$  divides  $a$  or  $b$  or  $c$ . In each case,  $p \nmid c$ , so  $E$  has multiplicative reduction at these  $p$  (21.3.4.12).  $\square$

**Prop. (21.3.4.27) [Global Minimal Weierstrass Equation].** For  $F \in \mathbf{NField}$  and  $E \in \mathbf{EllCur}/F$ ,  $E$  has a global minimal Weierstrass equation if its Weierstrass class is trivial.

In particular, if  $\text{cl}(F) = 1$ , e.g.  $F = \mathbb{Q}$ , then any  $E \in \mathbf{EllCur}/F$  has a global minimal Weierstrass equation.  $\square$

*Proof:* Choose any Weierstrass equation for  $E$  over  $K$ , let

$$x = u_v^2x + r_v, \quad y = u_v^3y + s_vu_v^2x + t_v, \quad \text{where } u_v, r_v, s_v, t_v \in \mathcal{O}_{K_v}$$

be the local change of variable to get the minimal Weierstrass equation at place  $v$ , then by hypothesis,  $(u_v)_v$  is principal, which means there is some  $u \in K$  s.t.  $v(u) = v(u_v)$  for any  $v$ . Then choose by Chinese remainder theorem  $r, s, t \in \mathcal{O}_K$  that is closed to  $r_v, s_v, t_v$  for each relevant  $v$ , then the change of variables

$$x = u^2x + r, \quad y = u^3y + su^2x + t$$

changes the coordinates to a minimal Weierstrass equation.  $\square$

**Cor. (21.3.4.28).** Let  $F \in \mathbf{GField}$  with  $6 \nmid \text{cl}(F)$  and  $E \in \mathbf{EllCur}/F$  with everywhere good reduction, then  $E$  has a minimal model.  $\square$

*Proof:* Because in this case the Weierstrass class  $\mathfrak{a}_\Delta$  is a 12-torsion in  $\text{Cl}(F)$ , which implies it is trivial.  $\square$

**Cor. (21.3.4.29).** If  $F \in \mathbf{NField}$  and  $S$  is a finite set of places of  $K$  s.t.  $\text{Cl}(\mathcal{O}_{F,S}) = 1$ , then any elliptic curve over  $F$  has a minimal Weierstrass equation of the form  $W : y^2 = x^3 + Ax + B$  with  $A, B \in \mathcal{O}_{K,S}$  satisfying  $\mathfrak{D}_W\mathcal{O}_{F,S} = \Delta_{E/F}^{\min}\mathcal{O}_{F,S}$ .  $\square$

*Proof:* The proof is easier than that (21.3.4.27).  $\square$

**Prop. (21.3.4.30).** If  $F \in \mathbf{NField}$  with  $\text{Cl}(F) \neq 1$ , then there exists  $E \in \mathbf{EllCur}/F$  with no global minimal Weierstrass equations.  $\square$

*Proof:* Cf. [Weierstrass equations and the minimal discriminant of an elliptic curve", Mathematika 31 (1984), no. 2, 245–251].  $\square$

**Prop. (21.3.4.31) [Density of Global Minimal Models].**  $\square$

*Proof:* Cf. [The density of elliptic curves having a global minimal Weierstrass equation", J. Number Theory 109 (2004), no. 1, 41–58.].  $\square$

## 5 Tate Curve

**Def. (21.3.5.1).** Define formal power series

$$\begin{aligned} s_k(q) &= \sum_{n \geq 1} \sigma_k(n) q^n = \sum_{n \geq 1} \frac{n^k q^n}{1 - q^n} \in \mathbb{Z}[[q]], \\ a_4(q) &= -5s_3(q), \quad a_6(q) = -\frac{5s_3(q) + 7s_5(q)}{12} \in \mathbb{Z}[[q]], \\ X(u, q) &= \sum_{n \in \mathbb{Z}} \frac{q^n u}{(1 - q^n u)^2} - 2s_1(q), \quad Y(u, q) = \sum_{n \in \mathbb{Z}} \frac{(q^n u)^2}{(1 - q^n u)^3} + s_1(q) \in \mathbb{Z}[u][[q]]. \end{aligned}$$

┘

**Prop. (21.3.5.2).**

$$\begin{aligned} s_1(q) &= \sum_{n \geq 1} \frac{q^n}{(1 - q^n)^2} \in \mathbb{Z}[[q]], \\ X(u, q) &= \frac{u}{(1 - u)^2} + \sum_{n \geq 1} \left[ \frac{q^n u}{(1 - q^n u)^2} + \frac{q^n u^{-1}}{(1 - q^n u^{-1})^2} - \frac{2q^n}{(1 - q^n)^2} \right] \in \mathbb{Z}[u][[q]], \\ Y(u, q) &= \frac{u^2}{(1 - u)^3} + \sum_{n \geq 1} \left[ \frac{(q^n u)^2}{(1 - q^n u)^3} + \frac{q^n u^{-1}}{(1 - q^n u^{-1})^3} + \frac{q^n}{(1 - q^n)^2} \right] \in \mathbb{Z}[u][[q]], \end{aligned}$$

Thus

$$X(qu, q) = X(u, q) = X(u^{-1}, q), \quad Y(qu, q) = Y(u, q) \in \mathbb{Z}[u, u^{-1}][[q]].$$

┘

*Proof:* The first one follows from (10.7.2.2). □

**Prop. (21.3.5.3) [Relation to Weierstrass  $\wp$ -Functions].** Let  $\wp(z; \tau)$  be the Weierstrass  $\wp$ -function associated to the lattice  $\Lambda_z = \text{span}\{1, \tau\}$  (12.5.6.5), then  $X(e^{2\pi iz}, q(\tau))$ ,  $Y(e^{2\pi iz}, q(\tau))$  converges and uniformly and absolutely on  $\mathbb{C} \setminus \Lambda_z$  in  $\mathbb{C}$  to holomorphic functions, and

$$\frac{1}{(2\pi i)^2} \wp(z; \tau) = X(e^{2\pi iz}, q(\tau)) + \frac{1}{12}, \quad \frac{1}{(2\pi i)^3} \wp'(z; \tau) = X(e^{2\pi iz}, q(\tau)) + 2Y(e^{2\pi iz}, q(\tau)).$$

┘

*Proof:* The convergence of  $X, Y$  is clear. By (21.3.5.1),  $X(e^{2\pi iz}, q(\tau))$  is  $\{1, \tau\}$ -doubly periodic, and clearly it has a double pole at the origin. Also the Laurent part of  $X(e^{2\pi iz}, q(\tau))$  at  $z = 0$  is the Laurent part of  $\frac{e^{2\pi iz}}{(1 - e^{2\pi iz})^2}$ , which is  $\frac{1}{(2\pi i)^2 z^2} - \frac{1}{12}$ . Then by comparison with that of  $\wp(z; \tau)$  (12.5.6.6) and use Liouville's theorem.

The second identity follows from applying  $\frac{\partial}{\partial z}$ . □

**Prop. (21.3.5.4) [Formal Identities].** There are identities

1.

$$-a_6(q) + a_4^2(q) + 72a_4a_6 - 64a_4^3 - 432a_6^3 = \Delta(q) \in \mathbb{Z}[[q]].$$

2.

$$Y(u^{-1}, q) = -Y(u, q) - X(u, q) \in \mathbb{Z}[u, u^{-1}][[q]].$$

3.

$$Y(u, q)^2 + X(u, q)Y(u, q) = X(u, q)^3 + a_4(q)X(u, q) + a_6(q) \in \mathbb{Z}[u][[q]].$$

4.

$$(X(u_2, q) - X(u_1, q))^2 X(u_1 u_2, q) = (Y(u_2, q) - Y(u_1, q))^2 + (Y(u_2, q) - Y(u_1, q))(X(u_2, q) - X(u_1, q)) \\ - (X(u_2, q) - X(u_1, q))^2 (X(u_2, q) + X(u_1, q))$$

$$(X(u_2, q) - X(u_1, q))Y(u_1 u_2, q) = -[(Y(u_2, q) - Y(u_1, q)) + (X(u_2, q) - X(u_1, q))]X(u_1 u_2, q) \\ - (Y(u_1, q)X(u_2, q) - Y(u_2, q)X(u_1, q))$$

┘

*Proof:* Verify these in the complex analytic case. **?**

3: By (12.5.6.7) and (21.3.5.3)(26.3.4.5),

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z)^2 - g_3\wp(z)$$

implies the desired equation.

□

**Cor. (21.3.5.5) [Tate Curve].** Notation as in (21.3.5.1),  $\text{Tate}(q) : y^2 + xy = x^3 + a_4(q)x + a_6(q)$  is an elliptic scheme over  $\mathbb{Z}[[q]]$ , called the **Tate curve**. The discriminant and  $j$ -invariant of  $\text{Tate}(q)$  is given by

$$\Delta(\text{Tate}(q)) = \Delta(q) : q \prod_{n \geq 1} (1 - q^n)^{24} \text{ (26.3.4.20)}, \quad j(\text{Tate}(q)) = j(q) \text{ (26.3.4.22)}.$$

And it has the normalized differential form  $\omega = dx/(2y + x)$ . Also denote  $\omega_{\text{can}} = 2\pi i \omega$ .

┘

**Prop. (21.3.5.6) [N-Torsion Points].** It can be shown formally from (21.3.5.10) that

$$\text{Tate}(q^N)[N] = \{(X(\zeta_N^i q^j, q^N), Y(\zeta_N^i q^j, q^N))\} \text{ **?**},$$

which are all defined over  $\mathbb{Z}[[q]][\frac{1}{N}, \zeta_N]$ . In particular, all the level  $N$  structures of  $\text{Tate}(q^N)$  are defined over  $\mathbb{Z}[[q]][\frac{1}{N}, \zeta_N]$ .

┘

*Proof:*

□

**Prop. (21.3.5.7) [Tate Curve over  $\mathbb{C}$ ].** Notation as in (21.3.5.1), then for  $q \in \mathbb{C}, 0 < |q| < 1$ ,

- $E_q : y^2 + xy = x^3 + a_4(q)x + a_6(q)$  is an elliptic curve over  $\mathbb{C}$ , and

$$\varphi : \mathbb{C}^\times / q^{\mathbb{Z}} \rightarrow E_q^{\text{an}} : u \mapsto [X(u, q), Y(u, q), 1]$$

is a complex analytic isomorphism.

- The discriminant of  $E_q$  is given by

$$\Delta(E_q) = \Delta(q) : q \prod_{n \geq 1} (1 - q^n)^{24} \text{ (26.3.4.20)}, \quad j(E_q) = j(q) \text{ (26.3.4.22)}$$

- For any elliptic curve  $E/\mathbb{C}$ , there exists a  $q \in \mathbb{C}^*$ ,  $|q| < 1$  s.t.  $E$  is isomorphic to  $E_q$ .

┘

*Proof:* Cf. [Sil99]P410.?

□

**Prop. (21.3.5.8) [Tate Curve over  $\mathbb{R}$ ].** For  $E \in \text{EllCur}/\mathbb{R}$ , there exists a unique  $q \in \mathbb{R}$ ,  $0 < |q| < 1$  s.t.

- $E$  is isomorphic to  $E_q : y^2 + xy = x^3 + a_4(q)x + a_6(q)$  over  $\mathbb{R}$ .
- the isomorphism composed with

$$\varphi : \mathbb{C}^*/q^{\mathbb{Z}} \cong E_q^{\text{an}} : u \mapsto (X(u, q), Y(u, q))$$

as in (21.3.5.10) induces an isomorphism of complex Lie groups  $\psi : \mathbb{C}^*/q^{\mathbb{Z}} \rightarrow E_{\mathbb{C}}^{\text{an}}$  that commutes with complex conjugation. In particular,  $\psi$  induces an isomorphism of real Lie groups

$$\psi : \mathbb{R}^*/q^{\mathbb{Z}} \cong E_{\mathbb{R}}^{\text{an}}.$$

┘

*Proof:* 1: By (20.9.8.15) and (20.9.8.16), there are exactly on  $\tau \in \mathcal{C}' = \{it|t \geq 1\} \cup \{1/2 + it|t > 1/2\}$  s.t.  $j(\tau) = j(E) \in \mathbb{R}$ , and

$$\mathcal{J} = \{it|t > 0\} \cup \{1/2 + it|t > 0\}.$$

corresponds to  $q \in \mathbb{R}$ ,  $0 < |q| < 1$  via  $\tau \mapsto q = e^{2\pi i \tau}$ . Also there are identifications

$$\begin{bmatrix} 1 & -1 \\ 1 & \end{bmatrix} \{it|t > 1\} = \{it|t < 1\}, \quad \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \{1/2 + it|t > 1/2\} = \{1/2 + it|t < 1/2\}, \quad \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} (i) = \frac{1}{2} + \frac{i}{2},$$

thus it suffices to show that for any two  $q, q' \in \mathcal{J}$  with  $q \neq q'$ ,  $E_q \not\cong E_{q'}$ . It suffices to show for the twists, where we can use (21.3.6.3) to show that their  $c_4, c_6$  changed sign, so they are not isomorphic over  $\mathbb{R}$ .

2: This follows from (21.3.5.10).

□

**Cor. (21.3.5.9) [Connected Components].** Let  $E \in \text{EllCur}/\mathbb{R}$ , then

$$E(\mathbb{R}) \cong \begin{cases} \mathbb{R}/\mathbb{Z} & \Delta(E) < 0 \\ \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/(2) & \Delta(E) > 0 \end{cases}$$

as real Lie groups.

┘

*Proof:* Notice  $\text{sgn}(\Delta) = \text{sgn}(q)$  by (21.3.5.10). The  $q > 0$  case is clear. For  $q < 0$ , notice there is an isogeny  $E_{q^2} \rightarrow E_q$  with kernel  $q$ , so  $E_{\mathbb{R}}^{\text{an}} \cong [\mathbb{R}/\mathbb{Z} \times \mathbb{Z}/(2)]/(1/2, -1) \cong \mathbb{R}/\mathbb{Z}$ .

□

**Prop. (21.3.5.10) [over Complete Valued Fields].** Notation as in (21.3.5.1), let  $K$  be a complete non-Archimedean valued field, then for  $q \in K$ ,  $0 < |q| < 1$ ,

- $E_q : y^2 + xy = x^3 + a_4(q)x + a_6(q) \in \text{EllCur}/K$ . The discriminant of  $E_q$  is given by

$$\Delta(E_q) = \Delta(q) : q \prod_{n \geq 1} (1 - q^n)^{24} \quad (26.3.4.20), \quad j(E_q) = j(q) \quad (26.3.4.22)$$

•

$$\varphi : \overline{K}^\times \rightarrow E_q(\overline{K}) : u \mapsto [X(u, q), Y(u, q), 1]$$

is a continuous surjective homomorphism with kernel  $q^\mathbb{Z}$ , and it is  $\text{Gal}_K$ -invariant, thus for any  $K \subset L \subset \overline{K}$  inducing a continuous surjective homomorphism

$$\varphi : L^\times \rightarrow E_q(L)$$

with kernel  $q^\mathbb{Z}$ .

┘

*Proof:* 1: This follows from (21.3.5.5) by base change.

2: Clearly  $X(u, q), Y(u, q)$  converges for any  $u \in K^\times / q^\mathbb{Z}$ , and by (21.3.5.1) it suffices to consider the case  $|q| < |u| < |q|^{-1}$ . In this case, the formal power series for  $X(u, q), Y(u, q)$  are convergent in the complete field  $K(u)$ , and it follows from (21.3.5.4) that  $\varphi(\overline{K}^*) \in E$ . It is clearly continuous for  $u \neq 1$ , and if  $u$  converges to 1,  $X(u, q)/Y(u, q) \rightarrow 0$  by the expansion in (21.3.5.4), so  $\varphi(1) = O$ .

To show  $\varphi$  is a homomorphism, it suffices to verify for  $1 < |u_1| < |q|^{-1}, |q| < |u_2| < 1$  that  $\varphi(u_1) + \varphi(u_2) = \varphi(u_1 u_2)$ . Notice by the argument above,  $\varphi(u) = O$  iff  $u = 1$ . If  $u_1 = 1$  or  $u_2 = 1$ , then  $\varphi(u_1) + \varphi(u_2) = \varphi(u_1 u_2)$  is clear, and if  $u_1 u_2 = 1$ , then  $\varphi(u_1) + \varphi(u_2) = O$  by the identities in (21.3.5.4) and (21.3.1.9). When  $u_1 \neq 1, u_2 \neq 1, u_1 u_2 \neq 1$ ,  $\varphi(u_1) \neq \varphi(u_2)$ ,  $\varphi(u_1) + \varphi(u_2)$  are determined by polynomial equations with coordinates of  $\varphi(u_1), \varphi(u_2)$ , and we can show this equation holds for  $\varphi(u_1 u_2)$  because it holds in the complex case (21.3.5.10). For the case  $\varphi(u_1) = \varphi(u_2)$ , notice that  $\text{Im}(\varphi)$  is infinite, as for  $|t| < 1$ ,  $|X(1+t)| = |t|^{-2}$  by the formula in (21.3.5.4). Now this case follows from (3.1.2.14).

Next, we show  $\varphi : L^\times \rightarrow E_q(L)$  is surjective. As  $K$  is arbitrary, we may assume  $L = K$ . For this, Cf. [Tate, a review of non-Archimedean Elliptic Functions] ?

□

**Cor. (21.3.5.11).**  $\text{Tate}(q)(\mathbb{Q}((q))) \cong \mathbb{Q}((q))^\times / q^\mathbb{Z}$ .

┘

**Lemma (21.3.5.12).** Let  $\alpha \in K$  with  $|\alpha| > 1$ , then there exists a unique  $q \in K$  s.t.  $|q| < 1$  s.t.  $j(q) = \alpha$ . Moreover,  $q \in \mathbb{Z}[[\frac{1}{\alpha}]]$ .

┘

*Proof:* Let  $f(q) = j(q)^{-1} \in q + q^2 \mathbb{Z}[[q]]$ , thus by (10.7.1.3) there exists a  $g \in q\mathbb{Z}[[q]]$  s.t.  $f(g(q)) = q = g(f(q))$ . So if  $f(q) = \frac{1}{\alpha}$ , then  $q = f(\frac{1}{\alpha})$ . And this  $q$  do satisfies  $j(q) = \alpha$ .

□

**Thm. (21.3.5.13) [p-Adic Uniformization, Tate].** Let  $K$  be a complete valued field,  $E \in \text{EllCur}/K$  with  $|j(E)| > 1$ , then

- There exists a unique  $q \in K$  s.t.  $E_{\overline{K}} \cong (E_q)_{\overline{K}}$ .
- For this  $q$ , the following are equivalent:
  - $E \cong E_q$ .
  - $\gamma(E) = 1$  (21.3.6.3).
  - $E$  has split multiplicative reduction.

┘

*Proof:* 1 follows from (21.3.5.12) and (21.3.1.15).

2: Firstly  $\gamma(E_q) = 1$ , because

$$c_4(E_q) = 1 - 48a_4(q) = 1 + 240s_3(q), \quad -c_6(E_q) = 1 - 72a_4(q) - 864a_6(q) = 1 - 504s_5(q)$$

are all of the form  $1 + 4\alpha$ ,  $|\alpha| < 1$ , thus they are both squares in  $K^\times$ , as  $1 + 4\alpha = [(1 + 4\alpha)^{1/2}]^2$ ??

Next,  $|a_4(q)|, |a_6(q)| \leq 1$ , so  $\tilde{E}_q : y^2 + xy = x^3$  has split multiplicative reduction. Then  $1 \iff 2$  by (21.3.6.3), and to show  $1 \iff 3$ , it suffices to show that if  $E$  has split multiplicative reduction, then  $\gamma(E) = 1$ : Let the valuation ring of  $K$  be  $R$  with maximal ideal  $\mathfrak{m}$  and residue field  $k$ . Let  $y^2 + a_1xy + a_3 = x^3 + a_2x^2 + a_4x + a_6$  be a minimal polynomial. We may assume the singular point is  $(0, 0) \in \tilde{E}(k)$ , then  $a_3 \equiv a_4 \equiv a_6 \equiv 0 \pmod{\mathfrak{m}}$ , thus  $b_4 = a_1a_3 + 2a_4 \equiv 0 \pmod{\mathfrak{m}}$ , and  $c_4 = b_2^2 - 2b_4 \equiv b_2^2 \pmod{\mathfrak{m}}$ . Thus  $b_2 \in R^*$ , and

$$\gamma(E) = -\frac{c_4}{c_6} = \frac{1}{b_2} \frac{1 - 24b_4/b_2^{-2}}{1 - 36b_4/b_2^2 + 216b_6/b_2^3} \pmod{(K^\times)^2}.$$

So by gain,  $\gamma(E) \equiv b_2 \pmod{(K^\times)^2}$ .

Because  $E$  has multiplicative split reduction,  $y^2 + \bar{a}_1xy - \bar{a}_2x^2 = (y - \bar{\alpha}x)(y - \bar{\beta}x) \in k$ ,  $\bar{\alpha} \neq \bar{\beta}$ . Thus by Henselian lifting,  $y^2 + a_1xy - a_2x^2 = (y - \alpha x)(y - \beta x) \in K$  for some  $\alpha, \beta$ , and  $b_2 = a_1^2 + 4a_2 = (\alpha - \beta)^2 \in (K^\times)^2$ .  $\square$

**Cor. (21.3.5.14).** If  $\gamma(E) \neq 1$ , let  $L = K(\sqrt{\gamma(E)})$ , then  $E_L \cong (E_q)_L$ , and

$$E(K) \cong \{u \in L^\times / q^\mathbb{Z} \mid \text{Nm}_{L/K}(u) \in \mathbb{Z}\}.$$

┘

*Proof:* Cf. [Sil99]P444.  $\square$

## 6 Twists of Elliptic Curves

Cf. [Sil16]Chap10.6.

**Prop. (21.3.6.1) [Twists of Elliptic Curves].** Because the category of smooth curves is an algebraic stack, for  $k \in \mathbf{Field}$  and  $C \in \mathbf{Cur}^{\text{sm}}/k$ , the set  $\text{Twist}(C/k)$  of twisted smooth curves of  $C$  is in bijection with  $H^1(k, \text{Isom}(C_{k^s}))$  by (7.1.5.2).

Also the category of elliptic curves (with the origin fixed) is an algebraic stack, thus the set  $\text{Twist}((E, O)/k)$  of twisted elliptic curves of an elliptic curve over  $k$  is in bijection with  $H^1(k, \underline{\text{Aut}}(E))$ .

┘

**Remark (21.3.6.2).** [Sil16]P318,342 has a direct proof, and can find the twist corresponding to a cocycle explicitly. ?  $\square$

**Prop. (21.3.6.3) [Twist((E, O)/k)].** Let  $k \in \mathbf{Field}$ ,  $\text{char } k \neq 2, 3$  and  $E \in \mathbf{EllCur}/k$ . Denote  $n = \# \text{Aut}(E_{\bar{k}})$ , then

- $\text{Twist}((E, O)/k)$  is canonically isomorphic to  $k^\times / (k^\times)^n$ .
- For  $E/k$  given by Weierstrass equation  $y^2 = x^3 + Ax + B$ , the twisted elliptic curve of  $E$  corresponding to  $D \pmod{(K^\times)^n}$  has the Weierstrass equation

$$E^{(D)} : \begin{cases} y^2 = x^3 + D^2Ax + D^3B, & j(E) \neq 0, 1728 \\ y^2 = x^3 + DAx, & j(E) = 1728 \\ y^2 = x^3 + DB, & j(E) = 0 \end{cases}.$$



- In case  $j(E) \neq 0, 1728$ ,  $n = 2$ , define  $\gamma(E) = -c_4/c_6 \in \mathbb{K}^\times/(\mathbb{K}^\times)^2$  (which equals  $A/B$  in the short form), then for  $E, E' \in \mathbf{EllCur}/k$  with  $j(E), j(E') \neq 0, 1728$ ,

$$E \cong E' \iff j(E) = j(E') \text{ \& } \gamma(E) = \gamma(E').$$

┘

*Proof:* This is because the Kummer sequence and (21.3.1.20) implies  $H^1(k, \underline{\mathrm{Aut}}(E)) \cong H^1(k, \mu_n) = k^\times/(k^\times)^n$ . For the corresponding elliptic curve, Cf. [Sil16]P343. ? □

**Def. (21.3.6.4) [Weil-Châtelet Groups].** For  $k \in \mathbf{Field}$ ,  $E \in \mathbf{EllCur}/k$ , define the **Weil-Châtelet Group**  $\mathrm{WC}(E/k)$  to be the isomorphism classes of  $E$ -torsors on  $\mathrm{Sch}_{\mathrm{\acute{e}t}}/k$ .

There is a bijection of pointed sets  $\mathrm{WC}(E/k) \cong H^1(k, E)$ , by (7.3.2.18). So  $\mathrm{WC}(E/k) \subset \mathrm{Twist}(E/k)$  (21.3.6.1). ┘

**Prop. (21.3.6.5).** If  $k \in \mathbf{Field}$ ,  $E \in \mathbf{EllCur}/k$  and  $C \in \mathrm{WC}(E/k)$ , then  $\mathrm{Jac}(C) \cong E$ . ┘

*Proof:* Because  $C$  is a smooth curve, let  $p \in C$  be a geometric point, and take the line bundle  $\mu^* \mathcal{L}(p) - \mathrm{pr}_2^* \mathcal{L}(p) - \mathrm{pr}_1^* \mathcal{L}(O)$ , where  $\mu : E \times C \rightarrow C$  is the action. Then after a base change to  $k(p)$ , this map corresponds to the canonical isomorphism  $E \cong E^\vee$  (21.3.1.21), thus it is also an isomorphism because  $\mathrm{Sch}_X$  is a prestack over  $\mathrm{Sch}_{\mathrm{fpqc}}$ . □

**Prop. (21.3.6.6).** Let  $E \in \mathbf{EllCur}/\mathbb{R}$ , then

$$\mathrm{WC}(E/\mathbb{R}) \cong \begin{cases} 0 & \Delta(E) > 0 \\ \mathbb{Z}/(2) & \Delta(E) < 0 \end{cases}.$$

┘

*Proof:* By (21.3.6.4) and (21.3.5.8), the Galois cohomology of the exact sequence  $1 \rightarrow q^\mathbb{Z} \rightarrow \mathbb{C}^\times \rightarrow E(\mathbb{C}) \rightarrow 1$  says

$$0 \rightarrow \mathrm{WC}(E/\mathbb{R}) \rightarrow H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), q^\mathbb{Z}) \rightarrow H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times),$$

and  $H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), q^\mathbb{Z}) \cong q^\mathbb{Z}/q^{2\mathbb{Z}}$ ,  $H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times) \cong \mathbb{R}^\times/(\mathbb{R}^\times)^2$  by (9.8.3.5)(9.8.1.21). So we are done by observing  $\mathrm{sgn}(\Delta) = \mathrm{sgn}(q)$  by (21.3.5.10). □

**Prop. (21.3.6.7) [Tate Duality on Weil-Châtelet Groups].** Let  $K \in p\text{-}\mathbf{LField}$  and  $E \in \mathbf{EllCur}(K)$ , then there is a perfect pairing of LCA groups

$$E(K) \times \mathrm{WC}(E/K) \rightarrow \mathbb{Q}/\mathbb{Z}.$$

┘

*Proof:* Cf. [Tate. WC-groups over p-adic fields] or [Sil99]Ex.10.24. ? □

**Prop. (21.3.6.8) [Rational Point of Genus 1 Curves].** Let  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $C \in \mathrm{WC}(E/\mathbb{Q})$ . Suppose the following holds:

- $C(\mathbb{Q}_\ell) \neq \emptyset$  for any  $\ell \in \mathbf{Prime}$ .
- Either  $E/\mathbb{Q}$  is semistable or  $\mathrm{rank}_{\mathrm{an}}(E/\mathbb{Q}) \leq 1$ .

Then  $C$  has a rational point over a solvable extension of  $\mathbb{Q}$ . ┘

*s Proof:* [Ciperiani-Wiles, Solvable points on genus one curves]. □

## 7 Integral Points

**Thm.(21.3.7.1) [Finiteness of Integral Points on Elliptic Curves, Siegel].** Let  $F \in \mathbf{NField}, \Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$ , and  $C \in \mathbf{Cur}^{\text{sm}^{\text{Jprpr}}}/F$  with genus 1. Suppose  $f \in R(C) \setminus F$ , then

$$\#\{P \in C(F) : f(P) \in \mathcal{O}_{F,S}\} < \infty.$$

┘

*Proof:* WLOG, we can enlarge  $F$  and  $S$ . So we may assume that  $C(F)$  contains a pole  $Q$  of  $f$  and  $(C, Q)$  is thus an elliptic curve with a Weierstrass form  $y^2 = x^3 + Ax + B$ . So  $f$  can be written as

$$f(x, y) = \frac{\varphi(x) + \psi(x)y}{\eta(x)},$$

where  $\varphi(X), \psi(X), \eta(X) \in F[X]$ . So

$$(f\eta(x) - \varphi(x))^2 = (\psi(x)y)^2 = \psi(x)^2(x^3 + Ax + B).$$

Since  $\text{ord}_Q(x) = -2, \text{ord}_Q(y) = -3, \text{ord}_Q(f) < 0$ , it follows that  $2 \deg(\eta) < \max(2 \deg \varphi, 2 \deg \psi + 3)$ . So regarding the above equation as a equation of  $X$  with  $f$  being constant, then the highest term of  $x$  comes from either  $\psi(x)^2 x^3$  or  $\varphi(x)^2$ , thus it is non-vanishing.

So by enlarging  $S$ , we see that  $x$  is integral over  $\mathcal{O}_{F,S}[f]$ . So if  $P \in C(F)$  s.t.  $f(P) \in \mathcal{O}_{F,S}$ , then  $x$  is integral over  $\mathcal{O}_{F,S}$ , which means  $x \in \mathcal{O}_{F,S}$ .

So we are reduced to proving for  $f = x$ . And this follows from [\(21.6.4.15\)](#). □

## 21.4 Surfaces and Arithmetic Surfaces

References are [Sil99], [Liu Qing].

### Notation (21.4.0.1).

- In this section, let  $S \in \text{Sch}$  be a Dedekind scheme with generic point  $\eta$  or a spectrum of a field, and  $\eta = S$ .

┘

### 1 over Fields

References are [Liu Qing, Algebraic Geometry and Arithmetic Curves].

**Prop. (21.4.1.1) [27-Lines].** Any smooth cubic surface in  $\mathbb{P}_k^3$  contains exactly 27 lines.

┘

*Proof:*

□

**Prop. (21.4.1.2).** Any birational transformation of non-singular surfaces will be factorized into f.m blowing-ups and blowing-downs of points.

┘

*Proof:*

□

**Prop. (21.4.1.3).** Any smooth surface over a field  $k$  is projective.

┘

*Proof:* Cf.[Badescu, Algebraic Surfaces]Thm1.28.

□

**Prop. (21.4.1.4) [Non-Projective Smooth Proper Threefold].** Cf.[Vak17]P671.

┘

*Proof:*

□

### Resolution of Surfaces

Cf.[Sta]Chap51.

### Regular Surfaces

### 2 Fibered Presurfaces

**Def. (21.4.2.1) [fibered presurfaces].** Let  $S \in \text{Sch}$  be Dedekind, a **fibered presurface** over  $S$  is proper **?** flat  $S$ -scheme integral of dimension 2.

┘

**Prop. (21.4.2.2).** Let  $S \in \text{Sch}$  be Dedekind,  $X$  a fibered presurface over  $S$ , then  $X_s$  has dimension 1 for any  $s \in S$ , and  $X_\eta$  is a precurve.

┘

*Proof:* Cf.[Qing Liu]P348.

□

**Example (21.4.2.3).** Let  $q \in \mathbb{Z}_+$  be a square-free integer, then  $\mathcal{C} = \mathbf{Proj}(\mathbb{Z}[X, Y, Z]/(X^q + Y^q + Z^q))$  is a normal fibered presurface over  $\text{Spec } \mathbb{Z}$ .

┘

*Proof:* Cf.[Qing Liu]P455.

□

**Prop. (21.4.2.4) [Horizontal and Vertical Divisors].** Let  $S \in \text{Sch}$  be Dedekind,  $X$  a fibered presurface over  $S$ , then

- If  $x$  is a closed point of  $X_\eta$ , then  $\overline{\{x\}}$  is an integral closed subscheme of  $X$  that is finite surjective over  $S$ , called a **horizontal divisor**.
- If  $D \subset X$  is a prime Weil divisor, then  $D$  is either a horizontal divisor, or an integral component of a special fiber, called a **vertical divisor**.
- If  $x_0 \in X$  is closed, then  $\dim \mathcal{O}_{X, x_0} = 2$ .

Moreover, call a Weil divisor  $D$  on  $X$  horizontal or vertical if it consists of horizontal or vertical prime divisors.  $\lrcorner$

*Proof:* Cf.[Qing Liu]P349.  $\square$

**Prop. (21.4.2.5) [Generically Smoothness].** Let  $S \in \text{Sch}$  be Dedekind and  $\pi : X \rightarrow S$  be a fibered presurface s.t.  $X_\eta$  is smooth, then there is a dense open subscheme  $V \subset S$  s.t.  $\pi^{-1}(V) \rightarrow V$  is smooth.  $\lrcorner$

*Proof:* Cf.[Qing Liu]P352.  $\square$

### Regular Fibered Presurfaces

**Prop. (21.4.2.6) [Regular Fibered Presurfaces are Projective].** Let  $S$  be affine and  $X/S$  be a flat morphism with fibers of dimension 1, then  $X/S$  is projective. In particular, a regular fibered presurface is projective.  $\lrcorner$

*Proof:* Cf.[Qing Liu]P353.  $\square$

**Prop. (21.4.2.7) [Sections are Regular].** Let  $\pi : \mathcal{C} \rightarrow S$  be a regular fibered presurface and  $s \in S$ , then

- If  $s \in S$  is closed and  $x \in \mathcal{C}_s$  is closed, then  $\mathcal{C}_s$  is regular at  $x$  iff  $\pi^\#(\mathfrak{p}_s) \not\subset \mathfrak{m}_x^2$ .
- If  $P : U \rightarrow \mathcal{C}$  is a rational section, then  $\mathcal{C}_P$  is smooth at  $P(s)$  for  $s \in U$ .  $\lrcorner$

*Proof:* 1: This is because  $\mathcal{O}_{\mathcal{C}_s, x} = \mathcal{O}_{\mathcal{C}, x} / \pi^\#(\mathfrak{p}_s)$ ,  $\mathfrak{m}_{\mathcal{C}_s, x} = \mathfrak{m}_{\mathcal{C}, x} / \pi^\#(\mathfrak{p}_s)$  and use the definition of regularity.

2: We show that  $\pi^\#(\mathfrak{p}_s) \not\subset \mathfrak{m}_x^2$ : Suppose otherwise, then

$$\mathfrak{p}_s = (\pi \circ P)^\#(\mathfrak{p}_s) \subset \pi^\#(\mathfrak{m}_x^2) = \mathfrak{p}^2,$$

which is impossible.  $\square$

**Cor. (21.4.2.8).** Let  $\mathcal{C}/S$  be a regular fibered presurface, then  $\mathcal{C}(S) = \mathcal{C}_{\text{sm}}(S)$ . If moreover  $\mathcal{C}/S$  is proper, then  $\mathcal{C}(S) = \mathcal{C}_{\text{sm}}(S) = \mathcal{C}_\eta(K)$ .  $\lrcorner$

**Example (21.4.2.9).**

- $\mathcal{C} \subset \mathbb{P}_{\mathbb{Z}}^2$  defined by the equation  $y^2 = x^3 + 2x^2 + 6$  is a regular fibered presurface over  $\mathbb{Z}$  with 3 singular points.
- $\mathcal{C} \subset \mathbb{P}_{\mathbb{Z}}^2$  defined by the equation  $y^2 = x^3 + 2x^2 + 6$  is not regular at  $(x, y, 2)$ .  $\lrcorner$

*Proof:* 1: The determinant  $\Delta = -2^6 \cdot 3 \cdot 97$ , so it has three singular fibers  $\mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_{97}$ :

$$\mathcal{C}_2 : y^2 = x^3, \quad \mathcal{C}_3 : y^2 = x^2(x+2), \quad \mathcal{C}_{97} : y^2 = (x+66)^2(x+64).$$

By (21.3.1.17), they are singular at a single point, and we check these points are regular in  $\mathcal{C}$ :

For  $\mathcal{C}_2$ , the singular point is defined by  $(x, y, 2)$ , and has residue field  $\mathbb{F}_2$ . To show it is regular, it suffices to show that  $(x, y, 2)/(x, y, 2)^2$  is generated by 2 elements  $x, y$ :  $2 = 3^{-1}(y^2 - x^3 - 3x^2)$ . The  $\mathcal{C}_3, \mathcal{C}_{97}$  cases are similar.  $\square$

### 3 Arithmetic Surfaces

**Def. (21.4.3.1) [Arithmetic Surfaces].** An **arithmetic surface** is a flat of f.t. over  $S$  that is integral normal and excellent, such that its generic fiber is a non-singular projective curve, and its special fibers are unions of curves.  $\lrcorner$

**Prop. (21.4.3.2).** Any arithmetic surface is regular in codimension 1.  $\lrcorner$

### 4 Elliptic Surfaces

In this subsection the elliptic curves over a function field of residue characteristic 0 is studied.

**Notation (21.4.4.1).**

- Let  $k \in \text{Field}^0$ .

**Def. (21.4.4.2) [Elliptic Surfaces].** For  $C \in \text{Cur}^{\text{SmPrpr}}/k$ , an **elliptic surface** over  $C$  is a triple  $(\mathcal{E}, \text{pr}, \sigma_0)$  where

- $\mathcal{E}/k$  is a projective surface.
- $\text{pr} : \mathcal{E} \rightarrow C$  is a morphism s.t. for a.e.  $t \in C(\bar{k})$ , the fiber  $\mathcal{E}_t = \text{pr}^{-1}(t)$  is a non-singular curve of genus 1.
- $\sigma_0 : C \rightarrow \mathcal{E}$  is a section of  $\text{pr}$ .

It is called a **non-singular elliptic surface** if  $\mathcal{E}$  is non-singular.  $\lrcorner$

**Prop. (21.4.4.3).** Let  $F \in \text{NField}$ ,  $C \in \text{Cur}^{\text{SmPrpr}}/F$ ,  $K = R(C)$ , and  $\mathcal{E} \rightarrow C$  a non-split elliptic surface, and  $\delta \in \text{Div}(C)$  s.t.  $\deg(\delta) > 0$ . Then the set of points  $t \in C(\bar{F})$  s.t.

$$\sigma_t : E(K) \rightarrow \mathcal{E}_t(\bar{F})$$

is non-injective is a set of bounded height.  $\lrcorner$

*Proof:*  $\square$

### Split Elliptic Surfaces

**Thm. (21.4.4.4) [Nothcott for non-Split Elliptic Surfaces].** Let  $k = \bar{k}, C \in \text{Cur}^{\text{SmPrpr}}/k$  with  $K = R(C)$ , and  $\mathcal{E} \rightarrow C$  an elliptic surface over  $k$ . Suppose  $\kappa \in \mathbb{R}_+$  s.t.

$$\#\{P \in E(K) : h(P) \leq d\} = \infty,$$

then  $\mathcal{E}$  splits over  $k$ .  $\lrcorner$

*Proof:* By (21.4.4.5), it suffices to show that there exists  $\Gamma \in \mathcal{C}ur^{\text{SmPrpr}}/k$  and  $\varphi : \Gamma \times C \rightarrow \mathcal{E}$  a dominant rational map over  $C$ .

For the rest, Cf. [Sil99]P222. ?

□

**Lemma (21.4.4.5).** Let  $\mathcal{E} \rightarrow C$  be an elliptic surface over  $k$ . Suppose  $\Gamma \in \mathcal{C}ur^{\text{SmPrpr}}/k$  and  $\varphi : \Gamma \times C \rightarrow \mathcal{E}$  is a dominant rational map over  $C$ , then  $\mathcal{E}$  splits over  $k$ . ┘

*Proof:* [Sil99]P228.

□

### Integral Points

**Prop. (21.4.4.6) [Manin].** Let  $C \in \mathcal{C}ur^{\text{SmPrpr}}/k$  with  $K = R(C)$ . Let  $E : y^2 = x^3 + Ax + B$  be an elliptic curve over  $K$  that doesn't split over  $\bar{k}$ . Then for any  $t \in C(k)$ ,  $v_t$  is bounded on  $E(K)$ . ┘

*Proof:* Cf. [Sil99]P275. ?

□

**Thm. (21.4.4.7) [Finiteness of Integral Points on Elliptic Curves].** Let  $C \in \mathcal{C}ur^{\text{SmPrpr}}/k$  of genus  $g$ ,  $K = R(C)$ , and  $S \subset C(\bar{k})$  is a non-empty finite set. Let  $E/K$  be an elliptic curve that doesn't split over  $\bar{k}$ , then for any  $f \in R(E) \setminus K$ ,

$$\#\{P \in E(K) : f(P) \in \mathcal{O}_{K,S}\} < \infty.$$

┘

*Proof:* The same reduction argument as in the proof of (21.3.7.1) shows that we can assume  $f = x$  for some Weierstrass model of  $E$ . Then the result follows from (21.4.4.8). □

**Prop. (21.4.4.8) [Finiteness of Integral Points on Hyperelliptic Curves].** Let  $C \in \mathcal{C}ur^{\text{SmPrpr}}/k$  of genus  $g$ ,  $K = R(C)$ , and  $\varphi \subset_{\text{fin}} S \subset C(\bar{k})$ . Let  $f \in \mathcal{O}_{K,S}[X]$  be a monic polynomial with  $\Delta(f) \in \mathcal{O}_{K,S}^*$ . Suppose  $x, y \in \mathcal{O}_{K,S}$  s.t.  $y^2 = f(x)$ , then

$$\deg(y^4/\Delta) \leq 4n(n-1) \max(2g-2 + \#S, 0).$$

┘

*Proof:* Cf. [The canonical height and integral points on elliptic curves]Prop8.2 ?

The same reduction argument as in the proof of Siegel's theorem (21.3.7.1) shows that the assertion reduces to bounding the degree of  $u \in \mathcal{O}_{F,S}^*$  in the equation

$$u + v = 1.$$

Then this follows from (21.6.4.16). □

## 5 Intersection Theory

**Def. (21.4.5.1) [Canonical Divisors].**  $K_{X/S}$ , Cf. [Qing Liu]P389. ┘

**Def. (21.4.5.2) [Vertically Nef].** Let  $X/S$  be a regular fibered presurface, then  $D \in \text{Cl}(X)$  is called **vertically nef** if  $D \cdot C \geq 0$  for any vertical divisor  $C$ . ┘

**Prop. (21.4.5.3) [Factorization Theorem].** Any birational morphism of regular fibered presurfaces over  $S$  is a finite composition of blowing-up along a single closed point. ┘

*Proof:* Cf. [Qing Liu]P395. □

### Arithmetic Surfaces

**Prop. (21.4.5.4) [Intersection between Vertical Divisors].** Let  $X/S$  be an arithmetic surface and  $s \in S$  a closed point, then for any  $\Gamma \in \text{Div}_s(X)$ ,  $\Gamma \cdot X_s = 0$ . In particular, if  $\Gamma_1, \dots, \Gamma_r$  are irreducible components of  $X_s$  of multiplicities  $d_1, \dots, d_r$ , then

$$\Gamma_i^2 = -\frac{d_j}{d_i} \sum_{j \neq i} \Gamma_i \cdot \Gamma_j \leq 0.$$

┘

## 6 Minimal Arithmetic Surfaces

**Def. (21.4.6.1) [Exceptional Divisors].** Let  $X/S$  be a regular fibered presurface, then an **exceptional divisor** or **(-1)-curve**  $E \subset X$  is a prime divisor s.t. there is a morphism of regular fibered presurfaces  $f : X \rightarrow Y$  s.t.  $f(E)$  is a closed point. Notice an exceptional divisor must be a vertical divisor. ┘

**Prop. (21.4.6.2) [Castelnuovo's criterion, Lichtenbaum/Shafarevich].** Let  $X/S$  be a regular fibered presurface,  $E \subset X_s$  a vertical prime divisor. Let  $k' = H^0(E, \mathcal{O}_E)$ , then  $E$  is an exceptional divisor iff  $E \cong \mathbb{P}_{k'}^1$ . And this is the case if  $E^2 = -[k' : k(s)]$ . ┘

*Proof:* Cf.[Qing Liu]P416. ┘

**Prop. (21.4.6.3) [Characterizing Exceptional Divisors].** Let  $X/S$  be a regular fibered presurface, then for a vertical prime divisor  $E$  on  $X$ ,

- $E$  is exceptional iff  $K_{X/S} \cdot E < 0$  (21.4.5.1) and  $E^2 < 0$ . And in this case,  $K_{X/S} \cdot E = E^2$ .
- If  $\dim S = 0$  and  $H^0(X, [\omega_{X/S}]^q) \neq 0$  for some  $q \geq 1$  or  $\dim S = 1$  and  $p_a(X_\eta) > 0$ , then  $E$  is exceptional iff  $K_{X/S} \cdot E < 0$ . ┘

*Proof:* Cf.[Qing Liu]P417. ┘

**Def. (21.4.6.4) [Minimal Surfaces].** A **relatively minimal surface** over  $S$  is a regular fibered presurface  $\mathcal{C}/S$  s.t. for any other such regular fibered presurface  $\mathcal{C}'$  and any birational morphism  $\mathcal{C} \rightarrow \mathcal{C}'$  is an isomorphism. Equivalently,  $\mathcal{C}$  is a regular proper model that doesn't contain an exceptional curve of the first kind, i.e. cannot be blown down by (21.4.5.3). Cf.[Sta]0C21.

A **minimal surface** over  $S$  is a regular fibered presurface  $\mathcal{C}/S$  s.t. for any other such regular fibered presurface  $\mathcal{C}'$  and any birational map  $\mathcal{C}' \rightarrow \mathcal{C}$  is a birational morphism. In particular, a minimal surface  $\mathcal{C}/S$  is relatively minimal, and  $\text{Aut}_S(\mathcal{C}) \cong \text{Aut}_K(\mathcal{C}_\eta)$  is an isomorphism. ┘

**Cor. (21.4.6.5).** If a relatively minimal arithmetic surface over  $S$  is birational to a minimal surface over  $S$ , then they are isomorphic over  $S$ . ┘

**Prop. (21.4.6.6) [Étale Descent].** Let  $X/S$  be an arithmetic surface with  $p_a(X_\eta) > 0$ , and  $S' \rightarrow S$  is an étale covering or  $S = \text{Spec } R$  where  $R$  is a DVR and  $S' = \text{Spec } \hat{R}$ , then  $X/S$  is minimal iff  $X \times_S S'/S'$  is minimal. ┘

*Proof:* Cf.[Qing Liu]P423. ┘

**Cor. (21.4.6.7).** Let  $X/S$  be an arithmetic surface with  $p_a(X_\eta) > 0$  and  $T \rightarrow S$  is a smooth morphism, then for any  $\xi \in T$  of codimension 1,  $X \times_S \mathcal{O}_{T,\xi}$  is a minimal regular surface over  $\mathcal{O}_{T,\xi}$ . ┘

*Proof:* Cf.[Qing Liu]P424. ┘

### Minimal Models

**Prop. (21.4.6.8) [Minimal Models].** Let  $\mathcal{C}/S$  be a normal fibered presurface, then a **regular model** for  $\mathcal{C}/S$  is a regular fibered presurface  $\mathcal{C}'/S$  with a birational map  $\mathcal{C}' \rightarrow \mathcal{C}$  over  $S$ . Notice  $\mathcal{C}'_\eta \rightarrow \mathcal{C}_\eta$  is a birational map of regular projective precurve, thus an isomorphism.

A (relatively) **minimal model** for  $\mathcal{C}/S$  is a regular model for  $\mathcal{C}/S$  that is (relatively) minimal. Notice if a minimal model exists, it is unique up to a unique isomorphism.  $\lrcorner$

**Lemma (21.4.6.9).** If  $X/S$  is an arithmetic surface, then there exists only f.m. fibers  $X_s$  containing an exceptional divisor.  $\lrcorner$

*Proof:* Cf.[Qing Liu]P420.  $\square$

**Prop. (21.4.6.10) [Blowing Down to Relatively Minimal Surfaces].** If  $X/S$  is an arithmetic surface, then  $X$  is a finite blowing-up of relatively minimal arithmetic surface over  $S$ .  $\lrcorner$

*Proof:* The number of exceptional divisors decreases along contractions **?**  $\square$

**Prop. (21.4.6.11) [Lichtenbaum/Shafarevich].** Let  $X/S$  be an arithmetic surface with  $p_a(X_\eta) \geq 1$ , then  $X$  admits a unique minimal model over  $S$ .  $\lrcorner$

*Proof:* Cf.[Qing Liu]P422.  $\square$

**Remark (21.4.6.12).** This is not true for  $p_a(X_\eta) = 0$ , Cf.[Qing Liu]P422.  $\lrcorner$

**Cor. (21.4.6.13) [Minimal and Relatively Minimal Surfaces].** If  $X/S$  is an arithmetic surface s.t.  $p_a(X_\eta) > 1$ , then  $X$  is minimal iff it is relatively minimal iff  $K_{X/S}$  is vertically nef, by (21.4.6.5) and (21.4.6.4)(21.4.6.3). **?** In the field case, why the hypothesis is satisfied?  $\lrcorner$

**Cor. (21.4.6.14).** If  $X/S$  is a smooth arithmetic surface, then  $X/S$  is relatively minimal by (21.4.5.4) as in this case  $X_s$  is a disjoint union of irreducible curves. And if  $p(X_\eta) > 0$ , then  $X$  is minimal.  $\lrcorner$



## 21.5 Arithmetic of K3 Surfaces

References are [Arithmetic and Geometry of K3 surfaces and Calabi-Yau Threefolds].

### 1 over Finite Fields

**Thm. (21.5.1.1) [Liedtke].** If  $p \in \text{Prime}_{\geq 5}$ , then any supersingular(27.5.7.1) K3 surface over  $\mathbb{F}_p$  is unirational(i.e. has a rational dominant map  $\mathbb{P}^2 \rightarrow X$ ).  $\lrcorner$

*Proof:* Cf.[Supersingular K3 surfaces are unirational].  $\square$

### 2 over Local Fields

### 3 over Global Fields

**Conj. (21.5.3.1) [Shafarevich].** For  $d, g \in \mathbb{Z}_+$ , consider all K3 surfaces defined over a number field of degree  $d$ , then there are only f.m. isomorphism classes among  $\text{NS}(X_{\overline{\mathbb{Q}}})$ .

By [Rational quadratic forms, Cassels]Chap9, Thm1.1, there are only finitely many lattices of bounded rank and discriminant. Thus this conjecture is equivalent to the boundedness of discriminant of  $\text{NS}(X_{\overline{\mathbb{Q}}})$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (21.5.3.2) [Coleman's Conjecture  $\implies$  Shafarevich Conjecture].** The Coleman's conjecture(21.7.11.1) implies Shafarevich's conjecture(21.5.3.1).  $\lrcorner$

**Conj. (21.5.3.3) [Br(K3)].** Let  $d \in \mathbb{Z}_+$ , then there exists  $C(d) \in \mathbb{R}_+$  s.t. if  $F \in \mathbf{NField}$ ,  $[F : \mathbb{Q}] \leq d$  and  $X \in \text{K3}/F$ , then  $\# \text{Br}(X_{\overline{k}})^{\text{Gal}_F} < \infty$ .

This would imply that  $\# \text{Br}(X)/\text{Br}_0(X)$  is uniformly bounded.Cf.[Uniformity Conjectures for Abelian Varieties and K3 Surfaces]P3.  $\lrcorner$

*Proof:*  $\square$

## 21.6 Diophantine Geometry

*“Mathematik ist die Königin von Wissenschaften und Arithmetik die Königin von Mathematik. ...in allen Relationen sie wird zum ersten Rank erlaubt.”*

–C. F. Gauss

References are [Galois Cohomology, Serre]Chap2.4.5, [Wei50, B-G06], [Fundamentals of Diophantine Geometry, Lang].

Arithmetic of varieties(i.e. varieties over absolutely f.g. fields) that are not necessarily Abelian varieties are studied in this section. The Abelian variety case is studied in [Abelian Varieties](#).

### 1 Hilbert’s 10-th Problem

#### Diophantine Sets

**Def. (21.6.1.1) [Diophantine Sets].** For  $m \in \mathbb{Z}_+$ , a subset  $E \subset (\mathbb{Z}_+)^m$  is called a **Diophantine set** if there exists  $n \in \mathbb{N}$  and a polynomial

$$P(T_1, \dots, T_m, X_1, \dots, X_n) \in \mathbb{Z}[T_1, \dots, T_m, X_1, \dots, X_n]$$

s.t.

$$(t_1, \dots, t_m) \in E \iff \exists (x_1, \dots, x_n) \in \mathbb{Z}^n, P(t_1, \dots, t_m, x_1, \dots, x_n) = 0.$$

┘

**Prop. (21.6.1.2).** Any Diophantine set is recursively denumerable([2.7.2.7](#)). ┘

**Prop. (21.6.1.3).** The class of Diophantine sets contains the level sets of integral polynomials, and is closed under finite direct sums, finite intersections and projections. ┘

**Prop. (21.6.1.4) [Diophantine Sets in  $\mathbb{Z}_+$ ].**  $S \subset \mathbb{Z}_+$  is a Diophantine subset iff there exists  $n \in \mathbb{Z}_+$  and  $P \in \mathbb{Z}[X_1, \dots, X_n]$  s.t.

$$S = \{m \in \mathbb{Z}_+ | \exists (X_1, X_2, \dots, X_n) \in \mathbb{Z}^n | P(X_1, \dots, X_n) = m\}$$

┘

*Proof:* One direction is clear. For the other, notice if there exists  $n \in \mathbb{N}$  and  $Q \in \mathbb{Z}[X_1, \dots, X_n]$  s.t.

$$S = \{m \in \mathbb{Z}_+ | \exists (X_1, X_2, \dots, X_n) \in \mathbb{Z}^n | P(m, X_1, \dots, X_n) = 0\},$$

then take  $P(X_0, X_1, \dots, X_n) = X_0(1 - P^2(X_0, X_1, \dots, X_n))$ . ┘

**Prop. (21.6.1.5) [Examples of Diophantine Sets].**

- For  $a, n \in \mathbb{Z}_+$ , suppose

$$(a + \sqrt{a^2 - 1})^n = x_n(a) + y_n(a)\sqrt{a^2 - 1},$$

where  $x_n(a), y_n(a) \in \mathbb{Z}_+$ , then the set  $E = \{(y, n, a) \in (\mathbb{Z}_+)^3 | y = y_n(a)\}$  is a Diophantine set.

- The set  $E_0 = \{(y, n, a) \in (\mathbb{Z}_+)^3 | y = a^n\}$  is a Diophantine set.
- The set  $E = \{(r, k, n) \in (\mathbb{Z}_+)^3 | r = \binom{n}{k}, n \geq k\}$  is a Diophantine set.

- The set  $E = \{(m, k) | m = k!\}$  is a Diophantine set.
- The set  $E = \{(x, y, p, q, k) | \frac{x}{y} = \binom{p/q}{k}, p > qk\}$  is a Diophantine set.

┘

*Proof:* Cf.[Manin, P99].

1:

□

**Cor. (21.6.1.6)[Prime is Diophantine].**  $\text{Prime} \subset \mathbb{Z}_+$  is a Diophantine set.

┘

*Proof:* This is because by Wilson's theorem(2.6.1.35), for  $p \in \mathbb{Z}_+$ ,  $p$  is a prime iff it is the first coordinate of the following Diophantine equation

$$\{(p, f, a) \in \mathbb{Z}_+ | p = f + 1, \quad q = f!, \quad q - ap = 1\}.$$

□

**Thm. (21.6.1.7)[Matiyasevich].** Any recursively denumerable subset of  $(\mathbb{Z}_+)^n$  is a Diophantine subset.

┘

*Proof:* Cf.[M. Davis, J. Matijasevic, and J. Robinson, Hilbert's 10th Problem. Diophantine equations: Positive aspects of a negative solution, in Mathematical Developments from Hilbert's Problems, F. Browder, ed., American Mathematical Societ, Providence, RI, 1976.]

□

**Cor. (21.6.1.8)[Hilbert's 10-th Problem, Matiyasevich].** Hilbert's 10-th problems is undecidable/-computable/recursive(2.7.2.6).

┘

*Proof:*

□

**Def. (21.6.1.9)[Simple Diophantine Sets].** A **simple Diophantine set** is a Diophantine set  $G \subset \mathbb{Z}_+$  s.t.  $\mathbb{Z}_+ \setminus G$  is not Diophantine, and for any infinite Diophantine subset  $D \subset \mathbb{Z}_+$ ,  $D \cap G \neq \emptyset$ .

┘

**Prop. (21.6.1.10).** There exists a simple Diophantine set(21.6.1.9).

┘

### Diophantine Coding

**Prop. (21.6.1.11)[Cantor Numbering of Tuples].** We can encode the pair of natural numbers in the following way:

$$(0, 0), (0, 1), (1, 0), (0, 2), (1, 1), (2, 0), (0, 3), \dots$$

Equivalently, the **Cantor numbering** is defined to be

$$\text{Cant}(a, b) = \frac{(a + b)^2 + 3a + b}{2}.$$

And

$$\text{Cant}(a, b) = c \iff \exists y[(a + y)^2 + 3a + y = 2c], \quad \text{Cant}(a, b) = c \iff \exists x[(x + b)^2 + 3x + b = 2c].$$

Similar, for  $n \in \mathbb{Z}_+$ , we can define numbering for  $n$ -tuples inductively as

$$\text{Cant}_1(a_1) = a_1, \quad \text{Cant}_2(a_1, a_2) = \text{Cant}(a_1, a_2), \quad \text{Cant}_{k+1}(a_1, \dots, a_{k+1}) = \text{Cant}_k(a_1, \dots, a_{k-1}, \text{Cant}_2(a_k, a_{k+1})).$$

┘

### Unsolvability

#### Diophantine Forms of Problems

**Prop. (21.6.1.12) [Riemann Hypothesis].** For a Diophantine form of the negation of Riemann hypothesis, Cf.[The Riemann Hypothesis and Diophantine Equations].  $\lrcorner$

## 2 Conjectures

**Def. (21.6.2.1) [Arithmetic Varieties].** An **arithmetic variety** is a regular scheme that is projective and flat over  $\mathbb{Z}$ .  $\lrcorner$

### Weak Approximation

**Prop. (21.6.2.2).** If  $X$  is a smooth complete intersection of two quadrics in  $\mathbb{P}_F^N$  for  $N \geq 5$ , and  $X(F) \neq \emptyset$ , then  $X$  satisfies weak approximation.  $\lrcorner$

*Proof:* Cf.[Colliot-Thelene-Skorobogatov1987, R-equivalence on cubic bundles of degree 4].  $\square$

### Mazur's Conjecture

**Conj. (21.6.2.3) [Mazur].** Let  $X/\mathbb{Q}$  be a smooth algebraic scheme over a field, then the topological closure of  $X(\mathbb{Q})$  in  $X(\mathbb{R})$  consist of a finite union of connected components.  $\lrcorner$

*Proof:*  $\square$

**Remark (21.6.2.4).** As the kernel of the Brauer-pairing(11.2.7.1) vanishes on all the infinite places, when the Brauer-Manin obstruction is sufficient, Mazur's conjecture holds.  $\lrcorner$

### Artin's Conjecture

**Def. (21.6.2.5) [ $C_d$ -Fields].** A field  $K$  is called  $C_k$  or for any homogenous polynomial  $F(X_1, \dots, X_n)$  of degree  $d$  with coefficient in  $K$  s.t.  $n > d^k$  has a non-zero solution in  $K^n$ .

$C_0$ -fields are just alg.closed fields,  $C_1$ -fields are also called **quasi-algebraically-closed field**.  $\lrcorner$

**Prop. (21.6.2.6).** Algebraic extensions of a  $C_1$  field are  $C_1$ .  $\lrcorner$

*Proof:* For a homogenous polynomial  $F(X_1, \dots, X_n)$ , its coefficient lies in a finite extension of  $K$  contained in  $L$ , so we may assume  $L/K$  is finite. Then choose a basis  $\{e_1, \dots, e_m\}$  of  $L$  over  $K$ , then consider the function

$$f(x_{11}, \dots, x_{1m}, \dots, x_{n1}, \dots, x_{nm}) = \text{Nm}_{L/K}(F(x_{11}e_1 + \dots + x_{1m}e_m, \dots, x_{n1}e_1 + \dots + x_{nm}e_m)),$$

which is a homogenous polynomial of degree  $nm$  with coefficient in  $K$ , because it has values all in  $K$ . So it has a nonzero solution in  $K^{nm}$  by(7.12.2.3), Krull's height theorem and  $k$  is alg.closed.  $\square$

**Thm. (21.6.2.7) [Finite Fields are  $C_1$ , Chevalley-Waring].** Any finite field  $\mathbb{F}_q$  is quasi-algebraically closed. In fact, for any system of homogenous polynomials  $f_i$  of  $n$  variables, if  $\sum_{i=1}^r \deg f_i < d$ , then the number of solutions to this equation on  $\mathbb{F}_q$  is divisible by  $p$ , where  $q$  is a  $p$ -power.  $\lrcorner$

*Proof:* The number of solutions to this system is equivalent to

$$\sum_{x \in \mathbb{F}_q^n} \prod (1 - f_i^{q-1}(x))$$

modulo  $p$ .

But notice that if  $i < q-1$ , then  $\sum_{x \in \mathbb{F}_q} x^i = 0$  in  $\mathbb{F}_q$  by (2.6.1.29), but as the degree of the highest term of  $\sum_{x \in \mathbb{F}_q^n} \prod (1 - f_i^{q-1}(x))$  modulo  $p$  is smaller than  $n(q-1)$ , some  $x_i$  has power smaller than  $q-1$ , thus when summed over  $\mathbb{F}_q$ , it vanishes.  $\square$

**Remark (21.6.2.8).** This may follow from the fact any smooth projective Fano variety over  $\mathbb{F}_q$  has a rational point (21.7.7.3).  $\lrcorner$

**Thm. (21.6.2.9) [Tsen].** If  $K = \overline{K}$ , then algebraic function fields of dimension 1 over  $K$  is quasi- $\text{alg.closed}$  (21.6.2.5).  $\lrcorner$

*Proof:* By (21.6.2.6), it suffice to consider the case  $K = k(t)$  purely transcendental. for a polynomial  $F$  with coefficient in  $k(t)$ , we can assume it has coefficient in  $k[t]$ , then let  $\delta$  be their maximal degree. If substituted with  $X_i = \sum_{j=0}^N a_{ij} t^j$ , the function becomes a system of  $\delta + dN + 1$  homogenous equation with  $n(N+1)$  unknowns  $a_{ij}$ , since  $d < n$ ,  $\delta + dN + 1 < n(N+1)$  for  $N$  large. In this case,  $\square$

**Thm. (21.6.2.10) [Almost all  $\mathbb{Q}_p$  are  $C_1$ , Ax-Kochen].** For any  $d \in \mathbb{Z}_+$ , there is a  $N_d$  that if  $p > N_d$ , then any homogenous polynomial  $f(X_1, \dots, X_n)$  of degree  $d$  with coefficient in  $\mathbb{Q}_p$  that  $d^k < n$  has a non-zero solution in  $\mathbb{Q}_p^n$ .  $\lrcorner$

*Proof:* The proof uses Model theory. ?  $\square$

**Thm. (21.6.2.11) [Lang].** If  $(\mathcal{O}_K, K, \kappa) \in \text{DVR}$  is Henselian with  $\kappa = \overline{\kappa}$ , and  $\widehat{K}$  is separable over  $K$ , then  $K$  is  $C_1$ .

In particular, the maximal unramified extension of a  $p$ -adic local field is  $C_1$ .  $\lrcorner$

*Proof:* [Lang, On quasi algebraic closure].  $\square$

### 3 Rational Points

#### Congruent Number Problem

**Def. (21.6.3.1) [Congruent Numbers].** For  $n \in \mathbb{Z}_+$ ,  $n$  is called a **congruent number** if there exists  $(x, y, z) \in \mathbb{Q}^2$  s.t.  $x^2 + y^2 = z^2$  and  $\frac{xy}{2} = n$ .

By (21.6.5.2), this is equivalent to the existence of  $(r, s, t) \in \mathbb{Z}_+$  s.t.

$$rs(r^2 - s^2) = nt^2.$$

$\lrcorner$

**Prop. (21.6.3.2).** For any  $n \in \mathbb{Q}_+$ , there is a bijection of sets:

$$\{x < y < z \in \mathbb{Q}_+ | x^2 + y^2 = z^2, \frac{xy}{2} = n\} \longleftrightarrow \{a \in \mathbb{Q} | a \in \mathbb{Q}^2, a + n \in \mathbb{Q}^2, a - n \in \mathbb{Q}^2\}$$

given by

$$(x, y, z) \mapsto a = \left(\frac{z}{2}\right)^2, \quad a \mapsto (\sqrt{a+n} - \sqrt{a-n}, \sqrt{a+n} + \sqrt{a-n}, 2\sqrt{a}).$$

In particular,  $n \in \mathbb{Z}_+$  is a congruent number iff there exists  $x \in \mathbb{Q}_+$  s.t.  $x, x+n, x-n$  are all rational squares.  $\lrcorner$

**Prop. (21.6.3.3) [Congruent Numbers and Elliptic Curves].** For any  $n \in \mathbb{Q}_+$ , there is a bijection of sets:

$$\{a, b, c \in \mathbb{Q} \mid a^2 + b^2 = c^2, \frac{ab}{2} = n\} \longleftrightarrow \{x, y \in \mathbb{Q} \mid ny^2 = x^3 - x, y \neq 0\}$$

given by

$$(a, b, c) \mapsto a = \left(-\frac{b}{a+c}, \frac{2}{a+c}\right), \quad (x, y) \mapsto \left(\frac{1-x^2}{y}, \frac{-2x}{y}, \frac{1+x^2}{y}\right).$$

And it can be shown that the elliptic curve  $E^{(n)} : ny^2 = x^3 - x$  satisfies

$$E^{(n)}(\mathbb{Q})_{\text{tor}} = \{O, (0, 0), (1, 0), (-1, 0)\} \text{ ?}.$$

So  $n$  is a congruence number iff  $\text{rank}(E^{(n)}/\mathbb{Q}) \geq 1$ . ┘

**Thm. (21.6.3.4) [Tunnell].** Let  $n \in \mathbb{Z}_+$  be square-free, and  $a = \begin{cases} 1 & , 2 \nmid n \\ 2 & , 2 \mid n \end{cases}$ . Then if  $n$  is a congruent number, we have

$$\#\{x, y, z \in \mathbb{Z} \mid \frac{n}{a} = 2ax^2 + y^2 + 8z^2, 2 \mid z\} = \#\{x, y, z \in \mathbb{Z} \mid \frac{n}{a} = 2ax^2 + y^2 + 8z^2, 2 \nmid z\}.$$

And the converse is also true if we assume the rank part of B-S.D. conjecture for the elliptic curve  $E^{(n)}/\mathbb{Q} : y^n = x^3 - n^2x$ . ┘

*Proof:* □

**Cor. (21.6.3.5) [Fibonacci1220/Fermat1640].**

- 1, 2, 3 are not a congruent number.
  - 5, 6, 7 are congruent numbers.
- ┘

*Proof:* 1: We give a direct proof: If 1 is a congruent number, then by (21.6.3.1), there exists  $r, s \in \mathbb{Z}_+$  s.t.  $\gcd(r, s) = 1$ ,  $r \not\equiv s \pmod{2}$ , and  $rs(r-s)(r+s)$  is a square.

Then  $r, s, r+s, r-s$  are pairwise coprime, so there exists  $x, y, u, v \in \mathbb{Z}_+$  s.t.

$$r = x^2, \quad s = y^2, \quad r+s = u^2, \quad r-s = v^2,$$

and ? □

**Conj. (21.6.3.6) [Congruence Number Problem].** Any  $n \in \mathbb{Z}_+$  s.t.  $n \equiv 5, 6, 7 \pmod{8}$  is a congruence number. ┘

*Proof:* □

**Remark (21.6.3.7).** For  $n \in \mathbb{Z}_+$  square-free, the  $L$ -function of  $E^{(n)}$  has root number

$$\epsilon(n) = \begin{cases} 1 & , n \equiv 1, 2, 3 \pmod{8} \\ -1 & , n \equiv 5, 6, 7 \pmod{8} \end{cases}, \quad N(n) = \begin{cases} 32n^2 & , 2 \nmid n \\ 16n^2 & , 2 \mid n \end{cases}.$$

In particular, the parity conjecture of B-S.D (31.4.1.8) implies the congruence number problem (21.6.3.6). ┘

*Proof:* ? □

**Conj. (21.6.3.8).**

$$\lim_{X \rightarrow \infty} \frac{\#\{n \in \mathbb{Z}_+ | n \equiv 1, 2, 3 \pmod{8} \text{ \& } n \text{ is congruent}\} \cap [1, X]}{\#\{n \in \mathbb{Z}_+ | n \equiv 1, 2, 3 \pmod{8}\} \cap [1, X]} = 0.$$

┘

*Proof:* □

### Cube Sum Problems

**Conj. (21.6.3.9)[Sum of Three Rational Cubes, Heath-Brown].** Any  $n \in \mathbb{Z}_+$  s.t.  $n \not\equiv 4, 5 \pmod{9}$  has infinitely many representations as sums of three integral cubes. ┘

*Proof:* ? □

*Proof:* Cf.[density of zeros of forms for which weak approximation fails, Heath-Brown]. ? □

**Conj.Cor. (21.6.3.10).** The question that whether  $n \in \mathbb{Z}_+$  is a sum of three cubes is decidable. ┘

*Proof:* □

**Prop. (21.6.3.11)[Sum of Two Rational Cubes].** For  $n \in \mathbb{Z}_{\geq 3}$ ,  $n$  is a sum of two rational cubes iff for the elliptic curve over  $\mathbb{Q}$ :

$$E_n : y^2 = x^3 - 432n^2,$$

we have  $\text{rank}(E_n/\mathbb{Q}) \geq 1$ .

And if the B-S.D conjecture holds for  $E_n$ , then any primes  $p \in \text{Prime}$  s.t.  $p \equiv 4, 7, 8 \pmod{9}$  is a sum of three rational cubes. ┘

*Proof:* ? □

**Thm. (21.6.3.12) [Alpöge-Bhargava-Shnidman2022].** The set of natural integers expressible as a sum of two rational cubes has natural density  $\geq 2/21$ , and the set of natural integers not expressible as a sum of two rational cubes has natural density  $\geq 1/6$ . ┘

*Proof:* Cf.[Integers expressible as the sum of two rational cubes]. □

## 4 Integral Points

### Simple Cases

**Thm. (21.6.4.1)[Pell's Equations].** A **Pell's equation** is a Diophantine equation of the form

$$\{x, y \in \mathbb{Z}_+ | x^2 - dy^2 = N\}$$

where  $d \in \mathbb{Z}_+ \setminus (\mathbb{Z}_+)^2$  and  $N \in \mathbb{Z}^\times$ .

- For  $N = 1$ , there exists a solution  $(x_0, y_0)$  s.t. every solution  $(x, y)$  satisfies  $(x + \sqrt{d}y) = (x_0 + \sqrt{d}y_0)^k$  for some  $k \in \mathbb{Z}_+$ . Such a  $(x_0, y_0)$  is called a **fundamental solution**.

- In general, if  $(x_0, y_0)$  is any solution to the equation for  $x^2 - dy^2 = 1$ , then every solution  $(x, y)$  for  $x^2 - dy^2 = N$  satisfies  $(x + \sqrt{d}y) = (x' + \sqrt{d}y')(x_0 + \sqrt{d}y_0)^k$  for some  $k \in \mathbb{N}$ , where  $x', y' \in \mathbb{Z}$ , and

$$(x')^2 - d(y')^2 = N, \quad |x'| \leq \frac{\sqrt{|N|}(\sqrt{u_0} + \sqrt{u_0}^{-1})}{2}, \quad |y'| \leq \frac{\sqrt{|N|}(\sqrt{u_0} + \sqrt{u_0}^{-1})}{2\sqrt{|d|}}, \quad u_0 = x_0 + \sqrt{d}y_0.$$

Moreover, if  $N > 0$ , we even can restrict to  $|y'| \leq \frac{\sqrt{|N|}(\sqrt{u_0} - \sqrt{u_0}^{-1})}{2\sqrt{|d|}}$ . And if  $N < 0$ , then  $y' > 0$ .  $\perp$

*Proof:* 1: If  $N = 1$ , then the solutions of this equations are a subgroup of group of units in  $\mathcal{O}_{\mathbb{Q}(\sqrt{d})}$ . It follows from unit theorem that the group of units in  $\mathcal{O}_{\mathbb{Q}(\sqrt{d})}$  is isomorphic to  $\{\pm 1\} \times \mathbb{Z}$ . So because we are restricted to the case that  $x, y \in \mathbb{Z}_+$ , it follows that the group of solutions is isomorphic to  $\mathbb{Z}_+$ , and it is in fact generated by a single minimal solution  $(x_0, y_0)$ .

2: We use the regulator map

$$\text{Reg} : \mathbb{Q}(\sqrt{d})^\times \rightarrow \mathbb{R}^2 : u \mapsto (\log u, \log \sigma(u)),$$

where  $\sigma$  is the non-trivial element of  $\text{Gal}(\mathbb{Q}(\sqrt{d})/\mathbb{Q})$ . Then  $\text{Reg}(u_0) = \log(u_0)(1, -1)$ , which is linearly independent to  $(1, 1)$ . Thus for any solution  $(x, y)$ , there exists  $c_1, c_2 \in \mathbb{R}$  s.t.

$$L(x + \sqrt{d}y) = c_1(1, 1) + c_2 L(u_0).$$

In particular,  $c_1 = \frac{\log |x + \sqrt{d}y| + \log |x - \sqrt{d}y|}{2} = \frac{\log |N|}{2}$ . And we can find  $k \in \mathbb{Z}$  s.t.  $c_2 = k + \delta$  with  $|\delta| \leq 1/2$ . Therefore if  $x' + \sqrt{d}y' = (x + \sqrt{d}y)(x_0 - \sqrt{d}y_0)^{-k}$ , then

$$\max(\log |x' + \sqrt{d}y'|, \log |x' - \sqrt{d}y'|) \leq \log \sqrt{|N|u_0}.$$

From which we can get the desired bound on  $x', y'$ . Finally, notice the hypothesis  $x, y \in \mathbb{Z}_+$  implies that we can assume  $k \geq 0$ .  $\square$

### Diophantine Sequences

**Def. (21.6.4.2) [Diophantine Sequences].** For  $m \in \mathbb{Z}_+$ , a sequence of positive integers  $a_1 \leq a_2 \leq \dots \leq a_m \in \mathbb{Z}_+$  is called a **Diophantine sequence** of length  $m$  if for any  $1 \leq i < j \leq m$ ,

$$a_i a_j + 1 \in (\mathbb{Z}_+)^2.$$

$\perp$

**Conj. (21.6.4.3).** Every Diophantine sequence of length 3 can be extended to a Diophantine quadruple:  
If  $(a, b, c)$  is a Diophantine triple,

$$ab + 1 = r^2, \quad ac + 1 = s^2, \quad bc + 1 = t^2, \quad a, b, c, r, s, t \in \mathbb{Z}_+,$$

then  $(a, b, c, d)$  is a Diophantine quadruple where  $d = a + b + c + 2abc \pm 2rst$ .

Then it is conjectured that these are the only two ways of extension.  $\perp$



*Proof:* For example,

$$(at \pm rs)^2 - 1 = a^2(bc + 1) + (ac + 1)(ab + 1) \pm 2arst - 1 = a(a + b + c + 2abc \pm 2rst).$$

For the conjecture, ?

□

**Thm. (21.6.4.4) [Baker-Davenport/Kedlaya/Dujella-Pethö].**

- If  $\{1, 3, c\}$  is a Diophantine triple, then there exists  $k \in \mathbb{Z}_+$  s.t.

$$c = c_k = \frac{1}{6}[(2 + \sqrt{3})(7 + 4\sqrt{3})^k + (2 - \sqrt{3})(7 - 4\sqrt{3})^k - 4].$$

Notice  $c_0 = 0, c_1 = 8, c_2 = 120$ .

- $(1, 3, c, d)$  is a Diophantine quadruple iff  $\{c, d\} = \{c_k, c_{k+1}\}$  for some  $k \in \mathbb{Z}_+$ .

┘

*Proof:* 1:

2: Given item1, it suffices to show that if  $k \in \mathbb{Z}_+$ , the set of equations

$$\{d, x, y, z \in \mathbb{Z}_+ | d + 1 = x^2, \quad 3d + 1 = y^2, \quad c_k d + 1 = z^2\}$$

has only solutions  $d \in \{c_{k-1}, c_{k+1}\}$ .

It's equivalent to solve the Pell'e equation set:

$$\{z^2 - c_k x^2 = 1 - c_k, \quad 3z^2 - c_k y^2 = 3 - c_k\}.$$

We first prove for  $k = 1$ :

**Lemma (21.6.4.5) [The  $k = 1$  case, Baker-Davenport].**  $(1, 3, 8, d)$  is a Diophantine sequence iff  $d = 120$ . ┘

*Proof:* If  $(1, 3, 8, n)$  is a Diophantine sequence, then  $n = x^2 - 1$  for some  $x \in \mathbb{Z}_+$ . Then there exists  $y, z \in \mathbb{Z}_+$  s.t.

$$\{3x^2 - 2 = y^2, \quad 8x^2 - 7 = z^2\}.$$

Thus it suffices to show that such a set of equations has solutions in  $\mathbb{Z}_+$  iff  $x \in \{1, 11\}$ .

First, by (21.6.4.1), we can show that positive integral solutions to  $y^2 - 3x^2 = -2$  are all of the form

$$y + \sqrt{3}x = (1 + \sqrt{3})(2 + \sqrt{3})^m, \quad m \in \mathbb{N},$$

thus

$$x = \frac{1 + \sqrt{3}}{2\sqrt{3}}(2 + \sqrt{3})^m - \frac{1 - \sqrt{3}}{2\sqrt{3}}(2 - \sqrt{3})^m.$$

If  $m = 0, x = 1$ , and if  $m = 2, x = 11$ .

Similarly, positive integral solutions to  $z^2 - 8x^2 = -7$  are all of the form

$$z + \sqrt{8}x = (1 + \sqrt{8})(3 + \sqrt{8})^n, \quad n \in \mathbb{N}$$

or

$$z + \sqrt{8}x = (5 + 2\sqrt{8})(3 + \sqrt{8})^n = (-1 + \sqrt{8})(3 + \sqrt{8})^{n+1}, \quad n \in \mathbb{N},$$

thus

$$x = \frac{1 + \sqrt{8}}{2\sqrt{8}}(3 + \sqrt{8})^n - \frac{1 - \sqrt{8}}{2\sqrt{8}}(3 - \sqrt{8})^n$$

or

$$x = \frac{-1 + \sqrt{8}}{2\sqrt{8}}(3 + \sqrt{8})^n - \frac{-1 - \sqrt{8}}{2\sqrt{8}}(3 - \sqrt{8})^n.$$

Let  $P = \frac{1+\sqrt{3}}{2\sqrt{3}}(2 + \sqrt{3})^m$ ,  $m \geq 3$  and  $Q = \frac{1+\sqrt{8}}{2\sqrt{8}}(3 + \sqrt{8})^n$  or  $Q = \frac{-1+\sqrt{8}}{2\sqrt{8}}(3 + \sqrt{8})^n$ , then we want a solution to

$$P + \frac{2}{3P} = Q + \frac{7}{8Q}.$$

For any such solution,  $P > 80$ , and

$$P - Q > \frac{2}{3P} - \frac{2}{3Q} = \frac{2(P - Q)}{PQ},$$

so  $P > Q$ . And also  $Q > P - \frac{7}{8Q} > P - \frac{7}{8}$ , so

$$P - Q = \frac{7}{8Q} - \frac{2}{3P} < \frac{7}{8(P - \frac{7}{8})} - \frac{2}{3P} < \frac{1}{4P}.$$

Thus

$$0 < \log \frac{P}{Q} < -\log\left(1 - \frac{1}{4P^2}\right) < \frac{1}{4P^2} + \frac{1}{16P^4} < 0.26P^{-2},$$

which means that

$$0 < m \log(2 + \sqrt{3}) - n \log(3 + \sqrt{8}) + \log \frac{(1 + \sqrt{3})\sqrt{8}}{(\pm 1 + \sqrt{8})\sqrt{3}} < 0.26P^{-2} < \frac{0.11}{(2 + \sqrt{3})^{2m}}.$$

Then it follows from Baker's theory(19.9.1.11) that  $m < 10^{40}$ .

Finally, using(20.2.4.8) twice, it can be shown that no such  $m \geq 3$  exists(Cf. [B-D69]).  $\square$

After this, assume  $k$  is the smallest natural number s.t. the assertion is wrong. Then  $k \geq 2$  because  $k = 1$  is true by the above lemma(21.6.4.5). Fix  $c = c_k \geq 120$ . The hypothesis implies that there exists  $s, t \in \mathbb{Z}_+$  s.t.  $c + 1 = s^2$ ,  $3c + 1 = t^2$ . So Neither  $c$  nor  $3c$  is a square. Then it follows from(21.6.4.1) that there exists  $z', x' \in \mathbb{Z}$ ,  $m \in \mathbb{N}$ ,  $z'', y'' \in \mathbb{Z}$ ,  $n \in \mathbb{N}$  s.t.

$$z + x\sqrt{c} = (z' + x'\sqrt{c})(s + \sqrt{c})^m, \quad z\sqrt{3} + y\sqrt{c} = (z''\sqrt{3} + y''\sqrt{c})(t + \sqrt{3c})^n,$$

and

$$|z'| \leq \sqrt{c-1} \frac{\sqrt{s+\sqrt{c}} + \sqrt{s+\sqrt{c}}^{-1}}{2} < \sqrt{c} \frac{\sqrt{2s+3}}{2} < \frac{c}{4},$$

$$|z''| \leq \sqrt{3(c-3)} \frac{\sqrt{t+\sqrt{3c}} + \sqrt{t+\sqrt{3c}}^{-1}}{6} < \sqrt{t^2-10} \frac{\sqrt{2t+3}}{6} < \frac{c}{6}.$$

If we define sequences

$$\{v_p\} : v_0 = z', v_1 = sz' + cx', v_{p+2} = 2sv_{p+1} - v_p\},$$

$$\{w_q\} : w_0 = z'', w_1 = tz'' + cx'', w_{q+2} = 2tw_{q+1} - w_q\},$$

then  $z = v_m = w_n$ . Moreover, we can use induction to show that

$$v_{2p} \equiv z' + 2c(p^2 z' + psx') \pmod{c^2}, \quad v_{2p+1} \equiv sz' + c[2p(p+1)sz' + (2p+1)x'] \pmod{c^2}, \quad p \in \mathbb{N}.$$

$$w_{2q} \equiv z'' + 2c(3q^2 z'' + qty'') \pmod{c^2}, \quad w_{2q+1} \equiv tz'' + c[6q(q+1)tz'' + (2q+1)y''] \pmod{c^2}, \quad q \in \mathbb{N}.$$

Then we claim:

**Lemma (21.6.4.6).** Let  $k \in \mathbb{Z}_{\geq 2}$  be the minimal number s.t. there exists  $t \in \mathbb{Z}_+$  s.t.  $(1, 3, c_k, c_t)$  is a Diophantine sequence, then with  $c = c_k$ ,

- If  $v_{2m} = w_{2n}$  has a solution, then  $z' = z'' = \pm 1$ .
- If  $v_{2m+1} = w_{2n}$  has a solution, then  $z'' = sz' = \pm s$ .
- If  $v_{2m} = w_{2n+1}$  has a solution, then  $z' = tz'' = \pm t$ .
- If  $v_{2m+1} = w_{2n+1}$  has a solution, then  $z'/t = z''/s = \pm 1$ .

┘

*Proof:* Cf. [D-P98]P294. ? Use minimality of  $k$ . □

After this, it is easy to show that  $v_{2m} = w_{2n+1}$  or  $v_{2m+1} = w_{2n}$  has no solutions by parity reason (notice  $c$  is even). For the next two cases:

**Lemma (21.6.4.7).**

- If  $v_{2m} = w_{2n}$ , then

$$0 < 2m \log(s + \sqrt{c}) - 2n \log(t + \sqrt{3c}) + \log \frac{\sqrt{3}(\sqrt{c} \pm 1)}{\sqrt{c} \pm \sqrt{3}} < \frac{3}{2}(s + \sqrt{c})^{-4m}.$$

- If  $v_{2m+1} = w_{2n+1}$ , then

$$0 < (2m+1) \log(s + \sqrt{c}) - (2n+1) \log(t + \sqrt{3c}) + \log \frac{\sqrt{3}(2\sqrt{c} \pm t)}{2\sqrt{c} \pm s\sqrt{3}} < 22(s + \sqrt{c})^{-4m-2}.$$

In particular, in both cases,  $m \geq n$ . ┘

*Proof:* Cf. [D-P98]P298. ? The derivation is similar to the  $k = 1$  case. □

**Cor. (21.6.4.8).**

- If  $v_{2m} = w_{2n}$ , and  $m, n \neq 0$ , then  $n > 0.105\sqrt{c}$ .
- If  $v_{2m+1} = w_{2n+1}$ , and  $m, n \neq 0$ , then  $n > 0.156\sqrt[4]{c}$ .

┘

*Proof:* Cf. [D-P98]P300. ? □

Then it follows from (21.6.4.7) and Baker's theory (19.9.1.11) that if  $v_{2m} = w_{2n}$ , then

$$\frac{3}{2}m \log c < 3.822 \cdot 10^{15} \cdot 0.33 \log c \cdot 0.38 \log c \cdot 0.64 \log c \cdot \log(2m),$$

thus

$$\frac{m}{\log(2m)} < 2.045 \cdot 10^{14} (\log c)^2.$$

Combining(21.6.4.8), we get

$$m < 2.045 \cdot 10^{14} \log(2m) [\log(91m^2)]^2,$$

which implies  $m < 9 \cdot 10^{19}$  and so  $c < 8 \cdot 10^{41}$  and  $k \leq 36$ .

Similarly, if  $v_{2m+1} = w_{2n+1}$ , then

$$2m \log c < 3.822 \cdot 10^{15} \cdot 0.33 \log c \cdot 0.38 \log c \cdot 0.73 \log c \cdot \log(2m+1),$$

thus

$$\frac{m}{\log(2m+1)} < 1.75 \cdot 10^{14} (\log c)^2.$$

Combining(21.6.4.8), we get

$$m < 1.75 \cdot 10^{14} \log(2m+1) [\log(1689m^4)]^2,$$

which implies  $m < 4 \cdot 10^{20}$  and so  $c < 5 \cdot 10^{85}$  and  $k \leq 75$ .

So to complete the proof, it suffices to show that:

**Lemma (21.6.4.9).**

- For  $k \leq 36$ , the equation  $v_{2m} = w_{2n}$  with  $0 \leq n \leq m < 9 \cdot 10^{19}$  has only one solution  $m = n = 0$ .
- For  $k \leq 75$ , the equation  $v_{2m+1} = w_{2n+1}$  with  $0 \leq n \leq m < 4 \cdot 10^{20}$  has only one solution  $m = n = 0$  (in which case  $z = 2c \pm st$ , corresponding to  $d = c_{k \pm 1}$ , in view of(21.6.4.3)).

┘

*Proof:* Use(20.2.4.8) twice, with  $M = 9 \cdot 10^{19}$  and  $M = 9$ . Cf.[D-P98]P303.

□

□

**Thm. (21.6.4.10) [Bo-Alain-Volker].** There are no Diophantine sequences of length 5.

┘

*Proof:* Cf.[There is no Diophantine quintuple. (English summary) Trans. Amer. Math. Soc. 371 (2019), no. 9, 6665–6709.]

□

### Siegel's Theorem

References are [Sie35, C-Z02].

**Thm. (21.6.4.11) [Siegel].** Let  $F \in \mathbf{NField}$ ,  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and  $\overline{C} \in \mathcal{C}ur^{\text{smProj}}/F$ . Let  $C \subset \overline{C}$  be an affine non-empty subset. Suppose  $g(\overline{C}) > 0$  or  $\#\overline{C} \setminus C \geq 3$ , then  $\#C(\mathcal{O}_{F,S}) < \infty$ .

┘

*Proof:* Firstly, it suffices to prove for  $\overline{C}$  smooth: As the normalization is finite and birational(7.4.2.8), by omitting f.m. points, we may assume that all rational points of  $C$  lift to rational points of  $C$ . Moreover, if  $\Gamma(C_{\text{sm}}) = \Gamma(C)[f_1, \dots, f_r]$ , where  $f_i$  are integral over  $C_{\text{sm}}$ . We can enlarge  $S$  s.t.  $C$  has an integral model over  $\mathcal{O}_{F,S}$ , and each  $f_i$  is integral over  $\Gamma(C/\mathcal{O}_{F,S})$ . Then each  $\mathcal{O}_{F,S}$ -points of  $C$  lifts to  $C_{\text{sm}}$ , and the assertion reduces to  $C_{\text{sm}}$ .

For  $g(\overline{C}) \geq 2$ , this is a special case of Faltings' Theorem(21.7.10.2).

For  $g = 1$ , the assertion follows from(21.3.7.1).

For  $g = 0$ , then  $\#\tilde{C} \setminus C_{\text{sm}} \geq 3$  (in fact, any other cases can be reduced to the case  $\#\overline{C} \setminus C_{\text{sm}} \geq 3$ , Cf.[B-G06]P184.) Then ?

□

**Remark (21.6.4.12).** By using Baker's theorem, we can make this theorem effective. ?

┘

### $S$ -Unit Equations

References are [Eve86] and [B-G06]

**Thm. (21.6.4.13) [S-Unit Equation, Evertse[Eve86]].** Let  $F \in \mathbf{NField}$ ,  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and  $a, b \in F^\times$ , then the equation

$$aX + bY = 1$$

has at most  $3 \cdot 7^{[F:\mathbb{Q}] + 2\#S}$  many solutions with  $X, Y \in \mathcal{O}_{F,S}^*$ . ┘

*Proof:*

□

**Thm. (21.6.4.14) [Effective  $S$ -Unit Equation].** The  $S$ -unit equation theorem can be made effective, Cf. [Sil99]P289 or [B-G06].? ┘

*Proof:*

□

**Thm. (21.6.4.15) [Finiteness of Integral Points on Hyperelliptic Curves, Siegel].** Let  $F \in \mathbf{NField}$ ,  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and  $f(X) \in \mathbb{F}[X]$  be a separable polynomial of degree  $d \geq 3$ , then the equation

$$Y^2 = f(X)$$

has only f.m. solutions with  $X, Y \in \mathcal{O}_{F,S}^*$ . ┘

*Proof:* Enlarging  $F$ , we may assume that  $f$  splits as  $f(x) = a(x - \alpha_1) \dots (x - \alpha_d)$ . By enlarging  $S$ , we may assume that

- $a \in \mathcal{O}_{F,S}^*$ ,
- $\alpha_i - \alpha_j \in \mathcal{O}_{F,S}^*$  for any  $i \neq j$ ,
- $\text{cl}(\mathcal{O}_{F,S}) = 1$ .

Suppose  $x, y \in \mathcal{O}_{F,S}$  s.t.  $y^2 = f(x)$ , then for any  $\mathfrak{p} \in \text{Spec}(\mathcal{O}_{F,S})$ ,  $\mathfrak{p}$  cannot divide two  $x - \alpha_i$  and  $x - \alpha_j$ . And also  $v_{\mathfrak{p}}(x - \alpha_i)$  is odd. So for each  $i$ , there are  $z_i \in \mathcal{O}_{F,S}$  and  $b_i \in \mathcal{O}_{F,S}^*$  s.t.  $x - \alpha_i = b_i z_i^2$ . By passing to the extension  $E/F$  adjoining all square roots of  $\mathcal{O}_{F,S}$  (which is a finite extension by Dirichlet's  $S$ -unit theorem), we may assume that  $b_i = 1$ . Then

$$\alpha_j - \alpha_i = (z_i - z_j)(z_i + z_j),$$

thus each  $z_i \pm z_j \in \mathcal{O}_{F,S}^*$ . Then the finiteness of  $S$ -unit equation (21.6.4.14)

$$\frac{z_1 \pm z_2}{z_1 - z_3} \mp \frac{z_2 \pm z_3}{z_1 - z_3} = 1$$

implies that there are only f.m. possibilities for  $\frac{z_1 \pm z_2}{z_1 - z_3}$ . Multiplying them we find that there are only f.m. possibilities for  $(\alpha_2 - \alpha_1)/(z_1 - z_3)^2$ . Thus there are only f.m. possibilities for  $z_1 - z_3$ , and the same for  $z_1 = \frac{1}{2}(z_1 - z_3 + \frac{\alpha_3 - \alpha_1}{z_1 - z_3})$ . Thus there are only f.m. possibilities of  $x$ , and also  $y$ . □

**Thm. (21.6.4.16) [S-Unit Equation over Function Fields, Mason/Silverman[Sil84]].** Let  $k \in \mathbf{Field}$ ,  $k = \bar{k}$ ,  $C \in \text{Cur}^{\text{SmPrpr}}/k$  with genus  $g$ ,  $K = R(C)$ , and  $\emptyset \subset_{\text{fin}} S \subset C(k)$ . Suppose  $u, v \in \mathcal{O}_{F,S}^*$  s.t.  $u + v = 1$ , then either  $u \in k^*$  or

$$\deg_s(u) \leq 2g - 2 + \#S.$$

Notice this inequality is sharp by taking  $C = \mathbb{P}^1$ ,  $S = \{0, \infty\} \cup \mu_n$  and  $u = T^n$ . ┘

*Proof:* Assume  $u \notin k^*$ . Choose  $n \in \mathbb{Z}$  s.t.  $(n, \text{char } k) = 1$  if  $\text{char } k > 0$ . Consider  $x = u^{1/n}, y = v^{1/n}$ , and consider  $K' = K(x, y)$ , corresponding to some non-singular complete curve  $C'/k$  of genus  $g'$  with a map  $\varphi : C' \rightarrow C$  whose degree divides  $n^2$ , thus separable.

As  $u, v$  are  $S$ -units,  $\varphi$  are unramified outside  $S$ . Thus the Riemann-Hurewitz(7.13.1.33) implies that

$$2g' - 2 = \deg(\varphi)(2g - 2) + \sum_{p \in \varphi^{-1}(S)} (e_p(\varphi) - 1) \leq \deg(\varphi)(2g - 2 + \#S).$$

There is also a natural map  $\psi : C' \rightarrow F_n$  where  $F_n : x^n + y^n = 1$  is the Fermat curve. Then Riemann-Hurewitz implies that

$$2g' - 2 \geq \deg_s(\psi)(2g(F_n) - 2) = (n^2 - 3n) \deg_s(\psi).$$

Finally, if

$$\lambda : F_n \rightarrow \mathbb{P}^1 : (X, Y) \mapsto X^n$$

is the separable map of degree  $n^2$ , then there is a commutative diagram

$$\begin{array}{ccc} C' & \xrightarrow{\varphi} & C \\ \downarrow \psi & & \downarrow u \\ F_n & \xrightarrow{\lambda} & \mathbb{P}^1 \end{array}$$

which implies that  $n^2 \deg_s(\psi) = \deg_s(u) \deg(\varphi)$ . Combining the above, we get that

$$(1 - 3/n) \deg_s(u) \leq 2g - 2 + \#S.$$

The desired assertion then follows by letting  $n \rightarrow \infty$ . □

### abc-Conjecture

**Conj. (21.6.4.17) [Strong *abc*-Conjecture, Masser-Oesterlé1985].** Given  $F \in \mathbf{NField}$ , for any  $\varepsilon > 0$ , there is a constant  $C_\varepsilon$  that for any  $a, b, c \in \mathcal{O}_F$  that  $a + b = c$ ,

$$H_F([a, b, c]) \leq C_\varepsilon |\text{Nm}_{K/\mathbb{Q}}(\prod_{\mathfrak{p}|abc} \mathfrak{p})|^{1+\varepsilon}.$$

┘

*Proof:* □

## 5 Ternary Diophantine Equations

**Def. (21.6.5.1) [Ternary Diophantine Equations].** A **ternary diophantine equation** is a Diophantine equation of the form

$$\{x, y, z \in \mathbb{Z} \mid Ax^p + By^q = Cz^r\}$$

where  $A, B, C \in \mathbb{Z}^\times$ . A **non-trivial solution** is a solution that  $xyz \neq 0$ . A **primitive solution** is a solution s.t.  $\gcd(x, y, z) = 1$ . ┘

**Prop. (21.6.5.2) [Pythagorean Triples].** A triple  $(x, y, z) \in \mathbb{Z}^3$  s.t.  $\gcd(x, y, z) = 1$  and  $x^2 + y^2 = z^2$  is called a **Pythagorean triple**. Then for any  $a > b \in \mathbb{Z}_+$  s.t.  $\gcd(a, b) = 1$  and  $a \not\equiv b \pmod{2}$ ,

$$(x, y, z) = (a^2 - b^2, 2ab, a^2 + b^2)$$

is a Pythagorean triple. Conversely, any Pythagorean triple is of this form (after possibly switching  $x$  and  $y$ ).  $\square$

*Proof:* It is easy to verify that  $(a^2 - b^2, 2ab, a^2 + b^2)$  is a Pythagorean triple. Conversely, for a Pythagorean triple  $(x, y, z)$ , we may assume  $y \neq 2$ , and  $y^2 = (z - x)(z + x)$ . Notice  $\gcd(z - x, z + x) | 2$ , but  $2 \nmid y$ , so  $\gcd(z - x, z + x) = 1$ . Thus there exists  $r, s \in \mathbb{Z}_+$  s.t.

$$z + x = r^2, z - x = s^2.$$

Then  $r \equiv s \pmod{2}$ . Denote  $a = \frac{r+s}{2}, b = \frac{r-s}{2} \in \mathbb{Z}$ , then  $z = \frac{r^2+s^2}{2} = a^2 + b^2, x = \frac{r^2-s^2}{2} = 2ab$ , and  $y = rs = a^2 - b^2$ .  $\square$

**Def. (21.6.5.3) [Hellegouarch-Frey Curve].** Let  $a, b, c \in \mathbb{Z}$  satisfy  $a^p + b^p = c^p, p \geq 3 \in \text{Prime}, abc \neq 0$ , the **Frey curve** for  $a, b, c$  is the elliptic curve  $E_{a^p, b^p, c^p} \in \text{EllCur}/\mathbb{Q}$  defined by the Weierstrass equation

$$W_{a^p, b^p, c^p} : y^2 = x(x + a^p)(x - b^p).$$

Then it satisfies:

- $\mathfrak{D}_{W_{a^p, b^p, c^p}} = 16a^{2p}b^{2p}(a^p + b^p)^2 = 16(abc)^{2p}$ .
- $\Delta_{E_{a^p, b^p, c^p}/\mathbb{Q}}^{\min}$  satisfies  $\Delta_{a^p, b^p, c^p}^{\min} \geq \frac{|abc|^{2p}}{2^8}$  by (21.3.4.26).

$\square$

*Proof:*  $\square$

**Prop. (21.6.5.4) [Galois Representation of the Hellegouarch-Frey Curve].** Let  $\rho_{a^p, b^p, c^p; p} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2; \mathbb{F}_p)$  be the mod  $p$  Galois representation of  $\text{Gal}_{\mathbb{Q}}$  corresponding to the Hellegouarch-Frey curve  $E_{a^p, b^p, c^p}$  (21.6.5.3), and let  $\bar{\rho}_{a^p, b^p, c^p; p}$  be its reduction modulo  $p$ . Suppose that  $a \equiv -1 \pmod{4}$  and  $2|b$ , then

- $\bar{\rho}_{a^p, b^p, c^p; p}$  is absolutely irreducible.
- $\bar{\rho}_{a^p, b^p, c^p; p}$  is odd.
- $\bar{\rho}_{a^p, b^p, c^p; p}$  is unramified outside  $2p$ , finite flat at  $p$  (30.3.2.3), and semistable at 2?.

$\square$

*Proof:* ? Cf. [Serre, Galois representations arising from Modular Forms] Section 4.1..  $\square$

**Remark (21.6.5.5).** One suspects that such a Galois representation doesn't even exist.  $\square$

**Thm. (21.6.5.6) [Fermat's Last Theorem, Ribet [Rib90]/Wiles [Wil95]/Wiles-Taylor [T-W95]].** If  $p \in \text{Prime}_{\geq 5}$ , then there are no integral solution to the equation  $a^p + b^p = c^p$ .  $\square$

*Proof:* If there exists an integral solution s.t.  $abc \neq 0$ , consider the Hellegouarch-Frey curve  $E_{a^p, b^p, c^p}$  (21.6.5.3), then by (21.6.5.4),  $\bar{\rho}_{a^p, b^p, c^p; p}$  is absolutely irreducible, odd and unramified outside  $2p$ , and flat at  $p$ .

Then it follows from modularity theorem (30.4.3.7) that  $\bar{\rho}_{a^p, b^p, c^p}$  is attached to some newform  $f \in \text{SF}_2^{\text{new}}(\Gamma_0(2p))$ . Then it follows from Ribet's level-lowering theorem (30.3.2.6) that  $\bar{\rho}_{a^p, b^p, c^p} = \bar{\rho}_{g, p}$  for some newform  $g \in \text{SF}_2^{\text{new}}(\Gamma_0(2))$ . But there is no cuspidal form of level  $\Gamma_0(2)$  and weight 2 because  $X_0(2) = \mathbb{P}^1$ , and  $H^0(\mathbb{P}^1, \Omega_{\mathbb{P}^1}^1) = 0$ , contradiction.  $\square$

**Remark (21.6.5.7) [Szpiro's Conjecture and Fermat's Last Theorem].** The Szpiro's conjecture (21.1.2.22) implies Fermat's last theorem for sufficiently large  $n$  (depending on the effectiveness of Szpiro's conjecture).  $\lrcorner$

*Proof:* Let  $a, b, c \in \mathbb{Z}$  satisfy  $a^n + b^n = c^n, n \geq 2, abc \neq 0$ , the conductor  $N_{a,b,c}$  of the Frey's curve (21.6.5.3)  $E_{a,b,c}$  satisfies  $N_{a,b,c} \leq 2^8(abc)$  by (21.3.4.26). Szpiro's conjecture implies

$$\frac{|abc|^{2n}}{2^8} \leq |\Delta_{a,b,c}^{\min}| \leq \kappa_1 N_E^7 \leq 2^{56} \kappa_1 |abc|^7,$$

so  $|abc|^{2n-7} \leq 2^{64} \kappa$ . As  $|abc| \geq 2$ , this gives an upper bound for  $n$ .  $\square$

**Prop. (21.6.5.8) [Fermat's Curve over  $\mathbb{F}_p$ ].** For any  $d \in \mathbb{Z}_+$ , the affine **Fermat's curve**  $X^d + Y^d = 1$  has a solution modulo  $p$  s.t.  $XY \neq 0$ , for any  $p \in \mathbf{Prime}$  sufficiently large.  $\lrcorner$

*Proof:* This follows from Weil's conjecture, because the number of solutions of the Fermat's curve is  $N_p = p + O(\sqrt{p})$ .  $\square$

**Prop. (21.6.5.9) [Fermat's Equation in Function Case].** If  $n \geq 2$ , then any non-trivial solutions to the equation in  $\mathbb{C}[T]$  of

$$X^n + Y^n = Z^n$$

are of the form  $n = 2, X = (a^2 - b^2)/2, Y = ab, Z = (a^2 + b^2)/2$ , where  $a, b, c \in \mathbb{C}[T]$ .  $\lrcorner$

*Proof:* The  $n = 2$  case is easy. If  $n > 2$ , we show there are no non-trivial solutions: Differentiate it to get:

$$X^{n-1}X' + Y^{n-1}Y' = Z^{n-1}Z',$$

And cancelling  $X^{n-1}$ , we get

$$Y^{n-1}(X'Y - Y'X) = Z^{n-1}(X'Z - Z'X).$$

Now  $X'Z - Z'X \neq 0$  because  $X, Z$  are not linearly equivalent, and  $Y, Z$  is coprime, so  $Y^{n-1} | X'Z - Z'X$ . But then if we assume  $\dim Y \geq \dim X$ , then  $(n-1)\dim Y \leq 2\dim Y - 1$ , which implies  $n \leq 2$ , contradiction.  $\square$

**Prop. (21.6.5.10) [Darmon].**

- For  $n \in \mathbb{Z}_{\geq 3}$ , the equation  $\{x, y, z \in \mathbb{Z} | x^n + y^n = 2z^n\}$  has no non-trivial primitive solution.
- For  $n \in \mathbb{Z}_{\geq 4}$ , the equation  $\{x, y, z \in \mathbb{Z} | x^n + y^n = z^2\}$  has no non-trivial primitive solution.
- For  $n \in \mathbb{Z}_{\geq 3}$ , the equation  $\{x, y, z \in \mathbb{Z} | x^n + y^n = z^3\}$  has no non-trivial primitive solution.

$\lrcorner$

*Proof:*

2: The case  $n = 4$  is proved by Fermat. The case  $n = 5, 6, 9$  is proved by [Poonen, Some diophantine equations of the form]. Thus it suffices to prove the case  $n = p \in \mathbf{P}_{\geq 7}$ . If  $2|ab$ , assume  $2|a$ , and  $c \equiv 1 \pmod{4}$ . And if  $2 \nmid ab$ , by replacing  $c$  with  $-c$  if necessary, we can assume  $a \equiv -1 \pmod{4}$ . Then consider the Frey curve  $E \in \mathbf{EllCur}/\mathbb{Q}$  given by the Weierstrass equation

The case  $n = 4$  is proved by Lucas [Theory of Numbers, Dickson]P630. The case  $n = 5$  is proved by [Poonen, Some diophantine equation of the form].  $\square$



**Prop. (21.6.5.11) [Darmon].** For  $p \in \text{Prime}_{\geq 11}$ , the ternary Diophantine equation

$$\{x, y, z \in \mathbb{Z} \mid x^4 - y^4 = z^p\}$$

has no non-trivial primitive solution s.t.  $z \in 2\mathbb{Z}$ . And it has no non-trivial primitive solution if  $p \equiv 1 \pmod{4}$ .  $\perp$

*Proof:* Let  $(a, b, c)$  be a solution of this equation. If  $c$  is odd, we can assume  $a$  is odd and  $b$  is even (otherwise change  $c$  to  $-c$ ). Then  $a - b, a + b, a^2 + b^2$  are all  $p$ -th powers up to powers of 2. Consider the Frey curve  $E \in \text{EllCur}/\mathbb{Q}$  given by the Weierstrass equation

$$W : y^2 = x(x + (a + b)^2)(x - (a - b)^2) = x^3 + 4abx^2 - (a^2 - b^2)^2x.$$

Then

$$j(E) = 2^6 \frac{(a^2 + 3b^2)^3(3a^2 + b^2)^3}{c^{2p}(a^2 - b^2)^3}, \mathfrak{D} = 2^6 c^{2p}(a^2 - b^2)^2.$$

Then the conductor  $N_E$  of  $E$  can be calculated by Tate's algorithm ?.

$$\text{ord}_p(N_E) = \begin{cases} 1 & , p \mid c \\ 0 & , p \neq 2, p \nmid c \\ 5 & , p = 2, 2 \nmid c \end{cases}$$

By a result of Mazur ?,  $\bar{\rho}_{E,2}$  is irreducible, and by [Serre, Sur les repr'esentations modulaires de degre 2 de  $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ ],  $\rho$  is unramified outside  $2p$ , and is finite at  $p$ .  $\square$

### Catalan's Equation

References are [Bil04], [Mih04] and [Catalan's Conjecture, Schoof].

**Notation (21.6.5.12).**

- Use notations on the Cyclotomic fields  $\mathbb{Q}(\zeta_p)$  as in q-Primary Units.

$\perp$

**Thm. (21.6.5.13) [Catalan's Conjecture, Lebesgue/Cassels/Ko/Tijdeman/Mihăilescu].**

$(X, Y, U, V) = (3, 2, 2, 3)$  is the only solution of the equation

$$X^U - Y^V = 1$$

with  $X, Y, U, V \in \mathbb{Z}_+, U, V \geq 2$ .  $\perp$

*Proof:* It suffices to prove for  $p = U, q = V \in \text{Prime}$ ,

The case  $q = 2$  is excluded by (21.6.5.14). The case  $p = 2$  is treated by (21.6.5.15). Thus it suffices to consider  $p, q \geq 3$ , and show that this equation has no non-zero solution in  $\mathbb{Z}$ .

Suppose  $(x, y, p, q)$  is a solution. There is clearly no solution with  $p = q$ , so by changing  $(x, y, p, q)$  to  $(-y, -x, q, p)$  if necessary, we may assume that  $p > q$ . Then by (21.6.5.21), we have  $q > 10^5$ , and by (21.6.5.22), we have  $p \not\equiv 1 \pmod{q}$ . Thus we can use the notation as in q-Primary Units.

Then (21.6.5.24) and (21.6.5.25) implies that  $N^\perp(1 + \mathfrak{c})I_1I_3 = 0$ . As  $R$  is a product of fields, this implies  $I_1I_3 \subset ((1 + \mathfrak{c})N^\perp)^\perp = (1 - \mathfrak{c}) + (N) = I_1I_2I_3$  (33.1.4.6). But because  $I_2$  are prime to  $I_1I_3$ ,

$$1 \subset I_2 + I_1I_3 = I_2 + I_1I_2I_3 = I_2.$$

So  $I_2 = 0$ , which means that  $C = C_p$ . So all the  $p$ -cyclotomic units are  $q$ -primary (33.1.4.4). But then this implies that  $p < q$  by (33.1.4.5), contradiction.  $\square$

**Lemma (21.6.5.14) [Lebesgue1850].** For  $p \in \text{Prime}$ , the equation  $x^p - y^2 = 1$  has no solution in  $\mathbb{Z}_+$ .  
 $\lrcorner$

*Proof:* We may assume that  $p \neq 2$ . It is clear that  $y$  is even and  $x$  is odd. In particular,  $1 + iy$  and  $1 - iy$  are coprime in  $\mathbb{Z}[i]$ . As  $\mathbb{Z}[i]$  is a PID (19.3.1.48), there exists  $c \in \mathbb{Z}[i]$  s.t.  $1 + iy = c^p$ . Thus  $c + \bar{c}|c^p + \bar{c}^p = \pm 2$ . Because  $c + \bar{c}$  is even, this means that  $c + \bar{c} = \pm 2$ . It follows that  $c = \pm(1 + bi)$ .

Because  $y$  is even,  $c$  is not divisible by  $1 + i$ , thus also  $b$  is even. The equation

$$(1 + bi)^p + (1 - bi)^p = \pm 2$$

shows that in fact  $(1 + bi)^p + (1 - bi)^p = 2$ . So

$$\binom{p}{2}(bi)^2 + \binom{p}{4}(bi)^4 + \dots + \binom{p}{p-1}(bi)^{p-1} = 0.$$

But the first term has 2-adic order strictly smaller than the rest: For any  $k \geq 4$  even,

$$\binom{p}{k} \binom{p}{2}^{-1} (bi)^{k-2} = \binom{p-2}{k-2} \frac{2}{k(k-1)} (bi)^{k-2}$$

has positive 2-adic order. Thus the first term vanishes, contradiction.  $\square$

**Lemma (21.6.5.15) [Ko].** The equation  $x^2 - y^n = 1$  has only one solution:  $(x, y, n) = (3, 2, 3)$ .  
 $\lrcorner$

*Proof:* [New Proof of Ko Chao's Theorem]. ?  $\square$

**Lemma (21.6.5.16) [Cassell's Relations].** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , there is no solution with  $(p, \frac{X^p-1}{X-1}) = 1$  or  $(q, \frac{Y^q-1}{Y-1}) = 1$ . In particular, there exists  $a, b \in \mathbb{Z}, u, v \in \mathbb{Z}_+$  s.t.  $(pa, u) = (qb, v) = 1$ , and

$$x - 1 = p^{q-1}a^q, \quad \frac{x^p - 1}{x - 1} = pv^q, \quad y = pav.$$

$$y + 1 = q^{p-1}b^p, \quad \frac{y^q + 1}{y + 1} = qu^p, \quad x = qbu.$$

$\lrcorner$

*Proof:* [Cassels, On the equation  $a^x - b^y = 1$ , 953, 1960] ?.  $\square$

**Lemma (21.6.5.17).** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ ,  $\lambda = \frac{x - \zeta_p}{1 - \zeta_p} \in \mathbb{Z}[\zeta_p]$ , and  $(\lambda)$  is a  $p$ -th power of an ideal of  $\mathbb{Z}[\zeta_p]$ .  
 $\lrcorner$

*Proof:* Since  $p|x - 1$  by Cassel's relation, we have  $\mathfrak{p} = (1 - \zeta_p)|x - \zeta_p$ , but  $\mathfrak{p}^2 \nmid x - \zeta_p$ . Thus  $\lambda$  is an algebraic integer not divisible by  $\mathfrak{p}$ . Thus the same is true for its conjugate  $\lambda^\sigma$ .

Now the identities  $(1 - \lambda^\sigma)\lambda^\sigma - (1 - \lambda^\tau)\lambda^\tau = \zeta_p^\tau - \zeta_p^\sigma$  implies that for  $\sigma \neq \tau \in \text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ , the greatest common divisor of  $\lambda^\sigma$  and  $\lambda^\tau$  divides  $\mathfrak{p}$ . Thus the numbers  $\lambda^\sigma$  are pairwise coprime.

Cassel's relation also implies that  $\prod_\sigma \lambda^\sigma = v^q$ . As the  $\lambda^\sigma$  are pairwise coprime, each of them is a  $q$ -th power of an ideal.  $\square$

**Lemma (21.6.5.18)**[double Wieferich Relations, Mihăilescu]. For any solution  $(x, y, p, q)$  to Catalan's equation(21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we have  $q^2|x$ , and also

$$p^{q-1} \equiv 1 \pmod{q^2}.$$

┘

*Proof:* Firstly we show that for any  $\Theta \in (1 - \mathfrak{c})\mathcal{I}$  where  $\mathcal{I}$  is the Stickelberger ideal(33.3.2.4), the element  $(x - \zeta_p)^\Theta$  is a  $q$ -th power in  $\mathbb{Q}(\zeta_p)$ : Let  $\Theta = (1 - \mathfrak{c})\Theta'$  and  $\lambda = (x - \zeta_p)/(1 - \zeta_p)$ . Then by(21.6.5.17),  $(\lambda) = \mathfrak{a}^q$  for some  $\mathfrak{a} \in \text{Ideal}(\mathbb{Z}[\zeta_p])$ . By Stickelberger's theorem(33.3.2.6),  $\mathfrak{a}^{\Theta'} = (\alpha)$  is principal. So  $(\lambda^{\Theta'}) = (\alpha)^p$ , or equivalently, there exists  $\eta \in \mathbb{Z}[\zeta_p]^*$  s.t.  $\lambda^{\Theta'} = \eta\alpha^p$ . Then we have:

$$(x - \zeta)^\Theta = \left(\frac{1 - \zeta_p}{1 - \zeta_p}\right)^{\Theta'} \frac{\eta}{\bar{\eta}} \left(\frac{\alpha}{\bar{\alpha}}\right)^p.$$

Both  $\frac{1 - \zeta_p}{1 - \zeta_p}$  and  $\frac{\eta}{\bar{\eta}}$  are roots of unity because they are algebraic integers and all conjugates has absolute value 1(19.3.1.12). In particular, they are  $q$ -th powers. So the assertion follows.

Then notice that  $(1 - \zeta_p x)^\Theta$  is also a  $q$ -th power. On the other hand,  $q|x$ (by Cassel's relation(21.6.5.16)) implies that  $(1 - \zeta_p x)^\Theta \equiv 1 \pmod{q}$ . But then because it is a  $q$ -th power and  $q$  is unramified in  $\mathbb{Q}(\zeta_p)$ , we have  $(1 - \zeta_p x)^\Theta \equiv 1 \pmod{q^2}$ .

Notice then that if  $\Theta = \sum_{\sigma} n_{\sigma}\sigma$ ,

$$(1 - \zeta_p x)^\Theta \equiv 1 - x \sum_{\sigma} n_{\sigma} \zeta_p^{\sigma} \pmod{q^2}.$$

If  $q^2 \nmid x$ , then  $q|n_{\sigma}$  for any  $\sigma$ .

But for  $\Theta = (1 - \mathfrak{c})\Theta_2$ (33.3.2.5),  $\Theta = -\sigma_1^{-1} - \dots - \sigma_{(p-1)/2}^{-1} + \sigma_{(p+1)/2}^{-1} + \dots + \sigma_{p-1}^{-1}$ , contradiction.

For the last assertion, notice that by Cassel's relation(21.6.5.16),  $p^{q-1}a^q \equiv -1 \pmod{q^2}$ . Thus  $a^q \equiv -1 \pmod{q}$ . But then  $a^q \equiv -1 \pmod{q^2}$ , which in turn implies that  $p^{q-1} \equiv 1 \pmod{q^2}$ .  $\square$

**Lemma (21.6.5.19)**[Tijdeman's Inequality, [Tij76]]. For any solution  $(x, y, p, q)$  to Catalan's equation(21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we have

$$p \leq 24.34q \left( \max \left( \log \frac{p+1}{\log q} + 0.14, 21 \right) \right)^2 \cdot \log q.$$

┘

*Proof:* Cf.[Catalan's Conjecture, Bilu]P9. ?

$\square$

**Lemma (21.6.5.20).** For any solution  $(x, y, p, q)$  to Catalan's equation(21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , either  $q < 28000$  or  $p \leq 4q^2$ .  $\square$

*Proof:* Use Tijdeman's Inequality, [Tij76]. If  $\log \frac{p+1}{\log q} + 0.14 \leq 21$ , then  $p < 10734q \log q$ . Suppose  $p > 4q^2$ , then  $q < 683.5 \log q$ , which implies that  $q < 28000$ .

If  $\log \frac{p+1}{\log q} + 0.14 \geq 21$ , then

$$p \leq 24.34q \left( \log \frac{p+1}{\log q} + 0.14 \right)^2 \cdot \log q.$$

Suppose  $q \geq 28000$  and  $p > 4q^2$ , then  $0.14 - \log \log q \leq -2.18$ , and

$$\frac{p}{(\log(p+1) - 2.18)^2} \leq 24.34q \log q.$$

But for  $p \geq 67$ , the LHS is an increasing function of  $p$ , so substituting  $p$  by  $4q^2$ , we get

$$q \leq 6.085(\log(4q^2 + 1) - 2.18)^2 \log q.$$

$q \geq 28000$  Implies that  $\log(4q^2 + 1) - 2.27 \leq \log q^2$ , so we get  $q \leq 24.34(\log q)^3$ , which is however not true for  $q \geq 28000$ .  $\square$

**Lemma (21.6.5.21) [Lower Bounds for  $p, q$ , Mignotte-Roy [M-R95]].** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we have

$$p, q \geq 10^5.$$

┘

*Proof:* Cf. [Catalan's equation has no new solutions with either exponent less than 10651, Mignotte-Roy] ?.  $\square$

**Lemma (21.6.5.22).** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we have  $p \not\equiv 1 \pmod{q}$ .  $\square$

*Proof:* Suppose  $p \equiv 1 \pmod{q}$ , then Weiferich's relation (21.6.5.18) will imply that  $p \equiv 1 \pmod{q^2}$ . So  $p \geq 1 + 4q^2$  because  $1 + q^2, 1 + 2q^2, 1 + 3q^2$  are not primes. But (21.6.5.21) together with (21.6.5.20) imply that  $p \leq 4q^2$ , contradiction.

Notice there are purely algebraic proof of this fact, Cf. [Bilu –Catalan without logarithmic forms] or [Mihailescu, On the class group of cyclotomic extensions in the presence of a solution to Catalan's equation]. ?  $\square$

**Lemma (21.6.5.23) [Lower Bounds on  $x$ , Mignotte-Bugeaud].** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we have

$$|x| \geq q^{p-1}.$$

┘

*Proof:* Cassel's relation (21.6.5.16) implies that

$$((-y)^{q-1} - 1) + ((-y)^{q-2} - 1) + \dots + (-y - 1) = q(u^q - 1),$$

thus  $q^{p-1} | (y+1)q(u^q - 1)$ . Then  $u^q \equiv 1 \pmod{q^{p-2}}$ . As the multiplicative group  $(\mathbb{Z}/(q^{p-2}))^*$  has order prime to  $q$  (since  $p > q$ ), this implies that  $u \equiv 1 \pmod{q^{p-2}}$ . In particular,  $u \geq q^{p-2} + 1$ . And then  $|x| = |qbu| \geq q^{p-1}$ .  $\square$

**Lemma (21.6.5.24) [Catalen's Equation and Cyclotomic Units].** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we know that  $p \not\equiv 1 \pmod{q}$ , so we can use notations as in q-Primary Units.

Then for any  $\Theta \in N^\perp(1 + \mathfrak{c})I_1 I_3$ , we have  $(x - \zeta_p)^\Theta \in (\mathbb{Z}[\zeta_p])^q$ .  $\square$

*Proof:* Cf. [Catalan's Conjecture, Bilu] P17. ?  $\square$

**Lemma (21.6.5.25) [Catalen's Equation and Cyclotomic Units].** For any solution  $(x, y, p, q)$  to Catalan's equation (21.6.5.13) with  $p > q \in \text{Prime}_{\geq 3}$ , we know that  $p \not\equiv 1 \pmod{q}$ , so we can use notations as in q-Primary Units.

Assume  $q \geq 7$ , and for some  $\Theta \in N^\perp(1 + \mathfrak{c})$ , we have  $(x - \zeta_p)^\Theta \in (\mathbb{Z}[\zeta_p])^q$ , then  $\Theta = 0$ .  $\square$

*Proof:* Cf.[Catalan's Conjecture, Bilu]P17.?

□

**Conj. (21.6.5.26).** For  $k \in \mathbb{Z}_+$ , there exists  $C(k), N(k) \in \mathbb{Z}_+$  s.t. for any  $k$ -itinerary integral sequence  $U_n, V_m$  and  $C \in \mathbb{Z}$ , there are at most  $N(k)$  solutions  $(n, m)$  to the equation  $U_n - V_m = c$ . ┘

*Proof:* Tentative approach: See [Chim-Pink-Ziegler] [DIOPHANTINE EQUATION  $U^n b^m = c$ ] and [An upper bound for the number of solutions of ternary purely exponential diophantine equations, 1, 2]. Prove for when the determining equation has coefficient too large(which implies that some root is large), there are at most  $N(k)$  solutions.

Even generalize to several series.

□

## 6 Unlikely Intersections

**Def. (21.6.6.1) [Unlikely Intersections].** Let  $S \in \text{Var}^{\text{sm}}/k$ , and  $X, Y$  be irreducible subvarieties of  $S$ . Then an **atypical component** of  $X \cap Y$  is a connected component  $T \subset X \cap Y$  s.t.  $\dim T > \dim X + \dim Y - \dim S$ . ┘

**Conj. (21.6.6.2) [Zilber Conjecture on Intersecting with Tori].** Let  $g \in \mathbb{Z}_+, S = \mathbb{G}_{m, \mathbb{C}}^g$ . For any irreducible subvariety  $X \subset S$ , denote  $X^{\text{atyp}} \subset X$  to be the union of atypical components of  $X \cap T$  where  $T \subset S$  is an algebraic group. Then  $X^{\text{atyp}}$  is Zariski-closed. Equivalently, if  $X$  is not contained in a proper subvariety of  $S$ , then  $X^{\text{atyp}} \subset X$  is not Zariski dense in  $X$ . ┘

*Proof:*

□

**Remark (21.6.6.3).** Zilber's conjecture over  $\overline{\mathbb{Q}}$  is equivalent to Zilber's conjecture over  $\mathbb{C}$ .?

┘

### Algebraic Reparameterization Lemma

**Thm. (21.6.6.4) [Gromov Reparameterization Lemma, Pila-Wilkie].** For any  $r \in \mathbb{N}$  and any strongly bounded definable set  $X \subset M^n$ , there exists an  $r$ -parameterization of  $X$ . ┘

*Proof:*

□

### Unlikely Intersections of O-minimal Definable Sets

References are [P-W06].

**Remark (21.6.6.5).** The principle is that transcendental sets should contain few rational points. ┘

**Notation (21.6.6.6).**

- Use notations on o-minimal sets as in 2.5.

┘

**Def. (21.6.6.7) [Algebraic and Transcendental Parts].** For  $n \in \mathbb{Z}_+, X \subset \mathbb{R}^n$ , the **algebraic part** of  $X$  is defined to be the union of all connected semi-algebraic subset of  $\mathbb{R}^n$  non-isomorphic to a point contained in  $X$ , denoted by  $X^{\text{alg}}$ . And the **transcendental part** of  $X$  is defined to be  $X^{\text{trans}} = X \setminus X^{\text{alg}}$ . ┘

**Lemma (21.6.6.8) [Rational Points Lies on Hypersurfaces].** Let  $Z \subset (0, 1)^n \times M^m$  be a definable family of fiber dimension  $k < n$ , then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $d(\varepsilon, k, n) \in \mathbb{N}$  and  $K(Z, \varepsilon) \in \mathbb{R}_+$  s.t. for any fiber  $X$ , the set

$$\#\{x \in X \cap \mathbb{Q}^2 \mid H_+(x) \leq T\}$$

is contained in at most  $K(Z, \varepsilon)T^\varepsilon$  hypersurfaces of degree at most  $d(\varepsilon, k, n)$ . ┘

*Proof:* Cf.[P-W06]P610? □

**Thm. (21.6.6.9) [Pila-Wilkie].** If  $n, m \in \mathbb{Z}_+$  and  $Z \subset \mathbb{R}^n \times \mathbb{R}^m$  be a definable family (parametrized by the second coordinate), then for any  $\varepsilon \in \mathbb{R}_+$ , there is a definable family  $W(Z, \varepsilon) \subset Z$  and  $c(Z, \varepsilon) \in \mathbb{R}_+$  s.t. for any  $y \in \mathbb{R}^m$ ,  $X_{W,y} \subset (X_{Z,y})^{\text{alg}}$ , and

$$\#\{(X_{Z,y} \setminus X_{W,y}) \cap \mathbb{Q}^2 | H(x) \leq T\} \leq C(Z, \varepsilon) T^\varepsilon.$$

In particular, if  $n \in \mathbb{Z}_+$  and  $X \subset \mathbb{R}^n$  be o-minimal definable, then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C(X, \varepsilon) \in \mathbb{R}_+$  s.t.

$$\#\{x \in (X \setminus X^{\text{alg}}) \cap \mathbb{Q}^2 | H_+(x) \leq T\} \leq C(X, \varepsilon) T^\varepsilon.$$

┘

*Proof:* Cf.[P-W06]?.

Because the height of a point is invariant under the maps  $(x_i) \mapsto (\pm x_i^{\pm 1})$ , we may assume that  $Z \subset [0, 1]^n \times \mathbb{R}^m$ . And by an induction on  $n$ , we may assume that  $Z \subset (0, 1)^n \times \mathbb{R}^m$ . □

**Cor. (21.6.6.10) [Bombieri-Pila].** Let  $f \in C([0, 1])$  be a non-algebraic real-analytic function, then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C(f, \varepsilon) \in \mathbb{R}_+$  s.t.

$$\#\{x \in \Gamma(f) \cap \mathbb{Q}^2 | H(x) \leq T\} \leq C(f, \varepsilon) T^\varepsilon.$$

┘

*Proof:* This follows from the theorem and the fact that globally subanalytic sets form an o-minimal structure(2.5.2.1). □

**Conj. (21.6.6.11) [Pila-Wilkie].** Let  $n \in \mathbb{Z}_+$  and  $X \subset \mathbb{R}^n$  be definable in  $\mathbb{R}_{\text{exp}}$ , then there exists  $c_1(X), c_2(X) \in \mathbb{R}_+$  s.t. for  $T$  large,

$$\#\{x \in (X \setminus X^{\text{alg}}) \cap \mathbb{Q}^2 | H_+(x) \leq H\} \leq c_1(X) (\log T)^{c_2(X)}.$$

┘

*Proof:* □

### Point-Counting approach

Cf. [Pila, AWS notes]Section1?.

## 21.7 (Bounding)Rational Points

References are [Sta], [Abelian Varieties notes Conrad], [BLR90, Sil99, B-G06, Mil08, Sil11, Sil16], [Abelian Variety van der Geer], <http://www-personal.umich.edu/~asnowden/teaching/2013/679/index.html>.

### 1 Mordell-Weil Theorem

**Prop. (21.7.1.1) [Local Chevalley-Weil Theorem for Abelian Varieties].** Let  $S$  be a Dedekind scheme with function field  $F$ , Let  $A$  be an Abelian variety over  $F$  and  $m \in \mathbb{Z} \cap F^*$ . Let  $s \in S$  be a closed point s.t.  $A$  has good reduction over  $\mathcal{O}_{S,s}$  and  $m \neq 0 \in \kappa(s)$ , then for any  $P \in A(\overline{F})$ , the extension  $\kappa(P)/\kappa([m]P)$  is unramified at all places over  $v$ .  $\lrcorner$

*Proof:* By base change, we may assume  $Q = [m]P$  is a rational point. Let  $w$  be a place of  $\kappa(P)$  that  $w|v_s$  with valuation ring  $R_w$ . As  $A$  has good reduction in  $v$ ,  $A$  extends to an Abelian scheme  $\overline{A}$  over  $\mathcal{O}_{S,s}$ , then the valuation criterion of properness shows  $P$  extends to a  $R_w$ -valued point of  $\overline{A}$ . Now the theorem follows from (20.9.6.18) and (7.6.5.16).  $\square$

**Cor. (21.7.1.2).** Let  $F \in \mathbf{GField}$ ,  $X \in \mathbf{AbVar}/F$  and  $n \in \mathbb{Z} \cap F^*$ ,  $L = K(X[n](\overline{F}))$ , i.e. the composite of all fields in  $\overline{F}$  obtained by adjoining  $[n]^{-1}x, x \in X(F)$ , then  $L$  is a finite field extension of  $F$ .  $\lrcorner$

*Proof:* This follows from the proposition and (20.8.1.8).  $\square$

**Lemma (21.7.1.3) [Weak Mordell-Weil Theorem].** Let  $F \in \mathbf{GField}$  and  $X \in \mathbf{AbVar}/F$ , then for  $n \in \mathbb{Z} \cap F^*$ ,  $X(F)/(n)$  is finite.  $\lrcorner$

*Proof:* This follows from the finiteness of the Selmer group (21.1.8.6) and (21.1.8.4).  $\square$

**Thm. (21.7.1.4) [Mordell[Mor22]/Weil[Wei29]].** Let  $F \in \mathbf{GField}$ , then the group  $X(F)$  of rational points of an Abelian variety  $X$  is f.g.  $\lrcorner$

*Proof:* Because of (21.7.1.3) and (21.1.3.3), this follows from (3.2.4.39) applied to  $M = X(F)$  and the symmetric bilinear form on  $X(F)$  given by (21.1.3.3).  $\square$

**Remark (21.7.1.5) [Computability].** The difficulty of computing  $X(F)$  lies entirely at computing  $X(F)/nX(F)$  for some  $n$ , for which, see (21.1.8.5).  $\lrcorner$

**Cor. (21.7.1.6) [Rank].**  $X(F) \cong \mathbb{Z}^r \oplus X(F)_{\text{tor}}$ , where  $X(F)_{\text{tor}}$  is a finite group, and  $r$  is called the **rank** of  $X$ .  $\lrcorner$

**Cor. (21.7.1.7) [Isogenous Varieties have the same Rank].** Let  $F \in \mathbf{GField}$  and  $X, X' \in \mathbf{AbVar}/F$  are isogenous, then  $\text{rank}(X) = \text{rank}(X')$ . In particular,  $\text{rank}(X) = \text{rank}(\widehat{X})$ .  $\lrcorner$

*Proof:* The isogeny implies  $X(F)/X(F)_{\text{tor}} \rightarrow \widehat{X}(F)/\widehat{X}(F)_{\text{tor}}$  is injective. Thus  $\text{rank}(X) \leq \text{rank}(\widehat{X})$ . The converse follows from (20.9.6.3).  $\square$

### 2 Lang-Néron Theorem

References are [Con06].

**Def. (21.7.2.1) [Regular Extensions].** A field extension  $K/k$  is called **primary** if  $k$  is separably closed in  $K$ . It is called **regular** if it is separable and primary.  $\lrcorner$

**Prop. (21.7.2.2).** Let  $K/k$  be a primary extension, then

- If  $A \in \mathcal{A}b\mathcal{V}ar/k$ , then any Abelian subvariety of  $A_K$  is defined over  $k$ .
- If  $A, B \in \mathcal{A}b\mathcal{V}ar/k$ , then any homomorphism  $A_K \rightarrow B_K$  is defined over  $k$ .

┘

*Proof:*

□

**Prop. (21.7.2.3) [ $K/k$ -Images and  $K/k$ -Traces].** For a field extension  $K/k$ , the  $K/k$ -**image** is a functor  $\mathrm{Im}_{K/k} : \mathcal{A}b\mathcal{V}ar/K \rightarrow \mathcal{A}b\mathcal{V}ar/k$  left adjoint to the base change functor, and  $K/k$ -**trace** is a functor  $\mathrm{tr}_{K/k} : \mathcal{A}b\mathcal{V}ar/K \rightarrow \mathcal{A}b\mathcal{V}ar/k$  right adjoint to the base change functor.

Then for  $K/k$  primary,  $\mathrm{Im}_{K/k}$  and  $\mathrm{tr}_{K/k}$  exist, and are resp. left and right inverses to the base change functor. Also there is a natural isomorphism

$$(\mathrm{Im}_{K/k}(A))^\vee \cong \mathrm{tr}_{K/k}(A^\vee).$$

The adjunction maps are denoted by

$$\tau_{A,K/k} : (\mathrm{tr}_{K/k} A)_K \rightarrow A, \quad \lambda_{A,K/k} : A \rightarrow (\mathrm{Im}_{K/k} A)_K.$$

┘

*Proof:* It suffices to prove for  $\mathrm{Im}_{K/k}$ , and  $\mathrm{tr}_{K/k}$  follows by double duality theorem(20.9.4.3). For this, Cf.[Con06]P16?.

□

**Prop. (21.7.2.4).** Let  $K/k$  be a primary extension and  $A \in \mathcal{A}b\mathcal{V}ar/K$ , the unique map  $\mathrm{tr}_{K/k} A \rightarrow \mathrm{Im}_{K/k} A$  descending  $\lambda_{A,K/k} \circ \tau_{A,K/k}$  is an isogeny.

┘

*Proof:* Cf.[Con06]P22?.

□

**Prop. (21.7.2.5).** Let  $K/k$  be a regular extension(21.7.2.1), then for any  $A \in \mathcal{A}b\mathcal{V}ar/K$ , the finite group  $\ker(\tau_{A,K/k})$  is connected and coconnected. In particular,  $\tau_{A,K/k}(K) : \mathrm{tr}_{K/k}(A)(k) \rightarrow A(K)$  is injective.

┘

*Proof:* ?

□

**Thm. (21.7.2.6) [Lang-Néron].** Let  $K/k$  be a f.g. regular extension and  $A \in \mathcal{A}b\mathcal{V}ar/K$ , then  $A(K)/\mathrm{tr}_{K/k}(A)(k)$  is a f.g. Abelian group.

┘

*Proof:* Cf.[Con06]P23?.

□

**Cor. (21.7.2.7).** Let  $K/k$  be a f.g. regular extension and  $A \in \mathcal{A}b\mathcal{V}ar/k$ , then  $A(K)/A(k)$  is a f.g. Abelian group.

┘

**Thm. (21.7.2.8) [Mordell-Weil over Function Fields].** Let  $k \in \mathbf{Field}$ ,  $C/k$  a complete non-singular curve with  $K = R(C)$ , and  $\mathcal{E} \rightarrow C$  an elliptic surface, and let  $E/K$  be the generic fiber. Suppose  $\mathcal{E}/C$  doesn't split or  $k$  is f.g. over its prime field, then  $E(K)$  is f.g.

┘

*Proof:* Cf.[Advanced Arithmetic Elliptic Curves, Silverman]P230. ? Is this a consequence of Lang-Néron?.

□

**Prop. (21.7.2.9) [Grothendieck].** Let  $K \in \mathbf{Field}$ ,  $K = K^s$ , with prime field  $k$ , then any Abelian variety of CM-type over  $K$  is isogenous to an Abelian variety defined over a finite extension of  $k$ .

┘

*Proof:* Cf.[Oort, The isogeny class of a CM-type abelian variety is defined over a finite extension of the prime field].

□



### 3 Elliptic Curves

#### Torsion Points

References are [\[Maz78\]](#).

**Prop. (21.7.3.1) [Controlling Torsion Points].** For  $F \in \mathbf{NField}$  and  $E \in \mathbf{EllCur}/F$  given by a Weierstrass equation

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \in \mathcal{O}_F[X, Y].$$

Let  $P = (x_0, y_0) \in E(F)$  be a torsion point of exact order  $m$ , then

- If  $m$  is not a  $p$ -power, then  $x_0, y_0 \in \mathcal{O}_F$ .
- If  $m = p^n$ , then for any place  $v$  of  $F$ ,  $v(x) \geq -2\lfloor \frac{v(p)}{p^n - p^{n-1}} \rfloor$ ,  $v(y) \geq -3\lfloor \frac{v(p)}{p^n - p^{n-1}} \rfloor$ .

┘

*Proof:* This follows by base change  $E$  to each  $\mathcal{O}_{K,v}$  and use [\(21.3.4.21\)](#). □

**Cor. (21.7.3.2) [Integrality of Torsion Points in  $\mathbb{Q}$ ].** Let  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $P = (x_0, y_0) \in E(\mathbb{Q})$  a torsion point of exact order  $m$ . then if  $m$  is not a  $p$ -power,  $x_0, y_0 \in \mathbb{Z}$ . If  $m = p^n$ , then  $\lfloor \frac{v(p)}{p^n - p^{n-1}} \rfloor = 0$  unless  $p = 2$  and  $n = 1$ . Thus  $x_0 \in \frac{1}{4}\mathbb{Z}$ ,  $y_0 \in \frac{1}{8}\mathbb{Z}$ , and if  $m \geq 3$ ,  $x_0, y_0 \in \mathbb{Z}$ .

The former case can occur, for example

$$E : y^2 + xy = x^3 + 4x + 1, \quad \left(-\frac{1}{4}, \frac{1}{8}\right) \in E(\mathbb{Q})[2].$$

┘

**Prop. (21.7.3.3) [Nagell–Lutz].** Let  $E \in \mathbf{EllCur}/\mathbb{Q}$  be given by a Weierstrass equation

$$E : y^2 = x^3 + Ax + B \in \mathbb{Z}[x, y],$$

Let  $P = (x_0, y_0) \in E(\mathbb{Q})_{\text{tor}}$ , then

- $x_0, y_0 \in \mathbb{Z}$ .
- Either  $y_0 = 0$ , i.e.  $[2](P) = O$  or  $y_0^2$  divides  $4A^3 + 27B^2$ .

┘

*Proof:* 1: Let  $P$  has exact order  $m$ . If  $m = 2$ , then  $y_0 = 0$ , and then  $x_0 \in \mathbb{Z}$  as it is integral over  $\mathbb{Z}$ . If  $m \geq 3$ , the results follows from [\(21.7.3.1\)](#).

2: Suppose  $[2](P) = (x_1, y_1) \neq O$ , then  $y_0 \neq 0$ , and  $x_0, y_0, x_1 \in \mathbb{Z}$ . Now [\(21.3.1.9\)](#) shows  $x_1 = \varphi(x_0)/4\psi(x_0)$  where

$$\varphi(X) = X^4 - 4AX^2 - 8BX + A^2, \quad \psi(X) = X^3 + AX + B.$$

And they satisfy a polynomial equation

$$f(X)\varphi(X) - g(X)\psi(X) = 4A^2 + 27B^2$$

where  $f(X) = 3X^2 + 4A$ ,  $g(X) = 3X^3 - 5AX - 27B$ . Then because  $y_0^2 = x_0^3 + Ax_0 + B$ , we get

$$y_0^2(4f(x_0)x_1 - g(x_0)) = 4A^2 + 27B^2.$$

□

### Ranks

**Conj. (21.7.3.4) [Unboundedness Conjecture].** For  $F \in \mathbf{GField}$ , the rank of elliptic curves over  $F$  can take arbitrary large values.  $\lrcorner$

*Proof:*  $\square$

**Remark (21.7.3.5).** For  $F \in \mathbf{FField}$ , this is proven by Tate-Šafarevič[?T-\vS67] and Ulmer[Ulm02].

For  $F = \mathbb{Q}$ , this is a consequence of the conjecture: for any  $k \in \mathbb{Z}_+$ , there exists a cube-free integer  $D$  s.t.  $D$  can be written as a sum of two positive cubes in at least  $k$  ways. Cf. [Silverman, Integral Points on Curves of Genus 1].  $\lrcorner$

**Prop. (21.7.3.6) [Park-Poonen-Voight-Wood(2016)].** For any  $n \in \mathbb{Z}_+$ , there exists  $F \in \mathbf{NField}$  and  $E \in \mathbf{EllCur} / F$  s.t.  $\text{rank}(E/F) \geq [F : \mathbb{Q}] \geq n$ .  $\lrcorner$

*Proof:* Cf.[A heuristic for boundedness of ranks of elliptic curves]Theorem 12.4.2.  $\square$

**Conj. (21.7.3.7) [Lang].** For any  $\varepsilon > 0$ , there exists a constant  $C_\varepsilon > 0$  s.t. for any  $E \in \mathbf{EllCur} / \mathbb{Q}$ , there exists a basis  $\{P_1, \dots, P_r\}$  of the free part of  $E(\mathbb{Q})_{\text{lf}}$  satisfying

$$\max_{1 \leq i \leq r} \hat{h}(P_i) \leq C_\varepsilon^r |D_{E/\mathbb{Q}}|^{\frac{1}{12} + \varepsilon}.$$

where  $D_{E/\mathbb{Q}}$  is the minimal discriminant of  $E/\mathbb{Q}$  and  $\hat{h}$  is the canonical height.  $\lrcorner$

*Proof:*  $\square$

## 4 Modular Curves

References are [M-T73, Maz76, Maz77, Maz78].

**Thm. (21.7.4.1) [Mazur].** Let  $p \in \mathbf{Prime}_{\geq 5}$ , and  $n$  be the nominator of  $\frac{N-1}{12}$ , then  $\text{Jac}_0(p)(\mathbb{Q})_{\text{tor}}$  is a cyclic group of order  $n$ , generated by the class  $[(0) - (\infty)]$ .  $\lrcorner$

*Proof:* Cf.[Maz78]Chap3.1.2..  $\square$

**Thm. (21.7.4.2) [Mazur].** Let  $N \in \mathbb{Z}_+$  s.t.  $g(X_1(N)) > 0$  (i.e.  $N = 11$  or  $N \geq 13$ ), then  $Y_1(N)(\mathbb{Q}) = \emptyset$ .  $\lrcorner$

*Proof:* Cf.[Maz78]Chap3.5.3.  $\square$

**Thm. (21.7.4.3) [Mazur].** If  $p \in \{11\} \cup \mathbf{Prime}_{\geq 17}$ , then  $\#X_{\text{split}}(p)(\mathbb{Q}) < \infty$ .  $\lrcorner$

*Proof:* Cf.[Maz78]Chap3.6.  $\square$

**Thm. (21.7.4.4) [Mazur/Momose/Merel].** If  $p \in \{11\} \cup \mathbf{Prime}_{\geq 17}$ , then for any  $P \in Y_{\text{split}}(p)(\mathbb{Q})$ ,  $j(P) \in \mathbb{Z}$ .  $\lrcorner$

*Proof:* Cf.[Bilu-Parent].  $\square$

**Def. (21.7.4.5) [Modular Units on  $X_{\text{split}}(p)$ ].** As the  $\lrcorner$

**Prop. (21.7.4.6).** Situation as in ?,  $\tau \in \mathcal{H}$  s.t.  $|q(\tau)| \leq 1/p$ , then

- If  $p|c$ , then

$$\left| \log |U_c(\tau)| - (p-1)^2 \log |q(\tau)| \right| \leq 4\pi^2 \frac{p^2}{\log |q(\tau)^{-1}|} + O(p \log p).$$

- If  $p \nmid c$ , then

$$\left| \log |U_c(\tau)| + 2(p-1) \log |q(\tau)| \right| \leq 8\pi^2 \frac{p^2}{\log |q(\tau)^{-1}|} + O(p).$$

┘

*Proof:*

□

**Prop. (21.7.4.7).** For any  $p \in \text{Prime}_{\geq 3}$  and  $P \in Y_{\text{split}}(\mathbb{C})$ ,

$$\log |j(P)| \leq 2\pi\sqrt{p} + \max \left( 6 \log p, \frac{1}{2(p-1)} \left| \log |U(P)| \right| - 6 \log p \right) + O(1).$$

┘

*Proof:*

□

**Cor. (21.7.4.8).** For any  $p \in \text{Prime}$  and  $P \in Y_{\text{split}}(p)(\mathbb{Z})$ ,  $0 \leq \log |U(P)| \leq 24p \log p$ .

In particular, by (21.7.4.7), there exists constant  $C > 0$  s.t. for any  $p \in \mathbf{P}$  and  $P \in Y_{\text{split}}(p)(\mathbb{Z})$ ,  $\log |j(P)| \leq 2\pi p^{1/2} + 6 \log p + C$ .

┘

*Proof:*

□

**Prop. (21.7.4.9).** There exists an effective constant  $\kappa > 0$  s.t. for any  $p \in \mathbf{P}$  and an elliptic curve  $E \in \text{EllCur}/\mathbb{Q}$  without CM and endowed with a structure of normalizer of split Cartan subgroup in level  $p$ , then  $h^+(j(E)) \geq \kappa p$ .

┘

*Proof:* For such an elliptic curve  $E$  with two  $p$ -isogenies  $\varphi_1 : E \rightarrow E_1, \varphi_2 : E \rightarrow E_2$ . Then  $\varphi = \varphi_1^\vee \circ \varphi_2 : E_1 \rightarrow E_2$  is a cyclic  $p^2$ -isogeny. Then it follows from (21.1.4.2) that  $h^+(j(E)) \geq \kappa_1 p$  for some constant  $\kappa_1$  independent of  $p$  and  $E$ .

(21.1.5.4) shows that  $h_{\text{st}}(E_1) \leq h_{\text{st}}(E) + \frac{1}{2} \log p$ , and (21.1.5.12) shows that for any  $X \in \text{EllCur}/F$ ,

$$|h^+(j(X)) - 12h_{\text{st}}(X/F)| \leq 6 \log \left( 1 + h^+(j(X)) \right) + O(1).$$

Thus the assertion follows.

□

**Thm. (21.7.4.10)** [Bilu-Parent/Bilu - Parent - Rebolledo/Balakrishnan-Dogra-Müller-Tuitman-Vonk]. For  $p \in \text{Prime}$ , every point in  $X_{\text{split}}(p)(\mathbb{Q})$  is either a CM point or a cusp iff  $p > 7$ .

┘

*Proof:* Firstly it follows from (21.7.4.4) (21.7.4.8) and (21.7.4.9) that this holds for  $p$  large. Then ? □

**Lemma (21.7.4.11).**  $X_{\text{split}}(13)(\mathbb{Q})$  consists of 6 CM-points and one cusp. Moreover,  $X_{\text{nonsplit}}(13)(\mathbb{Q})$  consists of 7 CM-points.

┘

*Proof:*

□

### Quotients of Modular Curves

**Thm. (21.7.4.12) [Rational Points of Quotients of Modular Curves].** Situation as in (28.3.5.1), if  $g(X_H(p)) > 0$ , and  $X_H(p)$  is defined over  $\mathbb{Q}$ , then

- $X_H(p) = X_0(p)$  has only f.m.  $\mathbb{Q}$ -points, and they can be calculated.
- $X_H(p) = X_{\text{split}}(p)$  has only f.m.  $\mathbb{Q}$ -points for  $p \neq 13$ .
- For  $X_H(p) = X_{\text{nonsplit}}(p)$ , nothing is known. ?
- $X_H(p) = X_{\text{Sym}_4}(p)$  has no rational points for  $p > 13$ .

┘

*Proof:*

□

**Thm. (21.7.4.13) [Formal Immersions and  $q$ -Expansions, Kamienny].** Let  $A$  be a quotient of  $\text{Jac}_0(p)_{\mathbb{Q}}$  by an ideal  $I \in \text{Ideal}^{\times}(\mathbb{T})$ . Then the following are equivalent:

- $\varphi_{A, \mathbb{F}_{\ell}}^{(d)}$  is formal immersion at  $\infty_{\mathbb{F}_{\ell}}^{(d)}$ .
- There exists  $d$  weight 2 cusp forms  $f_1, \dots, f_d$  annihilated by  $I$  s.t. the vectors  $(a_1(f_i), \dots, a_d(f_i))$  are linearly independent mod  $\ell$ .
- The image of  $T_1, \dots, T_d$  in  $\mathbb{T}/(I + \ell T)$  are linearly independent.

┘

*Proof:* [Merel's theorem on the boundedness of the torsion of elliptic curves] P6. ?

□

**Thm. (21.7.4.14) [Mazur-Kamienny].**

┘

### Isogenies of Elliptic Curves

**Prop. (21.7.4.15) [Potential Good Reductions].** Let  $F \in \mathbb{N}\text{Field}$ ,  $N \in \mathbb{Z}_+$  be square-free and  $v \in \Sigma_F^{\text{fin}}$  with  $e_v(F/\mathbb{Q}) \leq \text{char } v - 1$ . Then for any  $x = (E, C_N) \in X_0(N)(F)$ , if there exists an optimal quotient  $f : \text{Jac}_0(N)^{\text{new}} \rightarrow A \in \text{AbVar}$  s.t.  $f(x) \in A(F)$  is of finite order, then  $E$  has potentially good reduction at  $v$ .

┘

*Proof:* •

□

**Cor. (21.7.4.16) [Potential Good Reductions, Mazur [Maz78]].** For  $p \in \text{Prime}_{\geq 11} \setminus \{13\}$  and  $E \in \text{EllCur}/\mathbb{Q}$ , if  $E$  has a  $\mathbb{Q}$ -rational isogeny of order  $p$ , then  $E$  has potentially good reduction at any  $\ell \in \text{Prime}_{\geq 3}$ .

┘

*Proof:* This is because the Eisenstein quotient  $\text{Jac}_0(N) \rightarrow A_{\text{Eis}}$  satisfies  $\text{rank } A_{\text{Eis}}(\mathbb{Q}) = 0$  (28.3.6.2), so the hypothesis in (21.7.4.15) is automatically satisfied.

□

**Cor. (21.7.4.17) [Potential Good Reductions, Mazur].** If  $\mathcal{K}$  is an imaginary quadratic field, then there exists a finite set  $\emptyset \subset_{\text{fin}} S_{\mathcal{K}} \subset \text{Prime}$  s.t. if  $E \in \text{EllCur}/\mathcal{K}$  has a rational isogeny of order  $p$  for some  $p \in \text{Prime} \setminus S_{\mathcal{K}}$  that is inert in  $\mathcal{K}$ , then  $E$  has potentially good reduction at all  $v \in \Sigma_{\mathcal{K}}^{\text{fin}}$  with  $\text{char } v \geq 5$  (and also at all  $v \in \Sigma_{\mathcal{K}}^{\text{fin}}$  with  $\text{char } v \geq 3$  if 3 is unramified in  $\mathcal{K}$ ).

┘

*Proof:* This follows from (21.7.4.15) and (28.3.6.3).

□

**Thm. (21.7.4.18) [Mazur].** For  $p \in \text{Prime}$  s.t.  $g(X_0(N)) > 0$  (i.e.  $p = 11$  or  $p \geq 17$ ), then there are no  $E \in \text{EllCur}/\mathbb{Q}$  possessing a  $\mathbb{Q}$ -rational isogeny of order  $p$ , unless  $p \in \{11, 17, 19, 37, 43, 67, 163\}$ .

A table of examples of rational isogenies is given in [Maz78].

┘

*Proof:* ? □

**Cor. (21.7.4.19) [Mazur-Serre].** For any  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $p \in \mathbf{Prime}$ , one has the following three possibilities:

- $\bar{\rho}_{E,p} : \mathrm{Gal}_{\mathbb{Q}} \rightarrow \mathrm{GL}(2; \mathbb{F}_p)$  is surjective.
- $\mathrm{Im}(\rho_{E,p})$  is contained in the normalizer of a Cartan subgroup of  $\mathrm{GL}(2; \mathbb{Z}/(p))$ .
- $p \leq 19$  or  $p \in \{7, 43, 67, 163\}$ .

┘

*Proof:* If  $\bar{\rho}_{E,p}$  is not surjective, then by (28.3.5.1), the image is contained in either:

- Normalizer of a Cartan subgroup of  $\mathrm{GL}(2; \mathbb{Z}/(p))$ .
- A Borel subgroup, in which case  $\mathrm{Gal}_{\mathbb{Q}}$  fixes a subgroup of  $E[p]$  of order  $p$  thus  $E$  has an isogeny of order  $p$ , thus  $p \leq 19$  or  $p \in \{7, 43, 67, 163\}$  by (21.7.4.18).
- An exceptional subgroup  $H \leq \mathrm{GL}(2; \mathbb{F}_p)$ , in which case  $p < 17$  by (28.3.5.3).

□

**Cor. (21.7.4.20).** There exists  $C \in \mathbb{Z}_+$  s.t. every  $E \in \mathbf{EllCur}/\mathbb{Q}$  is isogenous to at most  $C$ -many mutually non-isomorphic elliptic curves over  $\mathbb{Q}$ . ┘

*Proof:* ? □

**Cor. (21.7.4.21) [Mazur-Serre].** If  $E \in \mathbf{EllCur}/\mathbb{Q}$  is semistable and  $p \in \mathbf{Prime}_{\geq 11}$ , then  $\bar{\rho}_{E,p}$  is surjective. ┘

*Proof:* ? This is a direct consequence of [Ser72]Prop21 and Mazur's theorem (21.7.5.8). □

**Conj. (21.7.4.22) [Mazur].** Is it true that for any  $F \in \mathbf{NField}$ , the  $j$ -invariants of an elliptic curve  $E \in \mathbf{EllCur}/F$  with an  $F$ -rational cyclic group of order  $N$  s.t.  $g(X_0(N)) \geq 2$  (equivalently,  $N > 21$  and  $N \neq 24, 25, 27, 32, 36, 49$ ) ? ┘

*Proof:* □

**Prop. (21.7.4.23) [Mazur].** For an imaginary quadratic field  $\mathcal{K}$ , there exists a finite set  $\emptyset \subset_{\mathrm{fin}} S_{\mathcal{K}} \subset \mathbf{Prime}$  s.t. for any  $p \in \mathbf{Prime} \setminus S_{\mathcal{K}}$  that is inert in  $\mathcal{K}$ , there are no elliptic curve  $E \in \mathbf{EllCur}/\mathcal{K}$  with a rational isogeny of order  $p$ . ┘

*Proof:* Cf. [Maz78]Prop8.1. □

## 5 Uniformity Conjecture for Torsion Points

**Conj. Cor. (21.7.5.1) [Uniform-Boundedness for Torsion Points of Abelian Varieties].** The Morton-Silverman boundedness conjecture (21.8.4.1) implies that for any  $D, g \in \mathbb{Z}_+$ , there exists  $\eta(D, g) \in \mathbb{Z}_+$  s.t. for any  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = D$  and  $A \in \mathbf{AbVar}/F$ ,  $\#A(F)_{\mathrm{tor}} \leq \eta(D, g)$ . ┘

*Proof:* Cf. [Fak03]P3. ? □

### Mazur's Approach: Eisenstein Quotients

**Prop. (21.7.5.2).** If  $p \in \text{Prime}$  and  $E \in \text{EllCur}/\mathbb{Q}$  has a rational torsion point of order  $p$ , then  $E$  is isogenous to  $E'/\mathbb{Q}$  with a rational point of order  $p$  and  $\mathbb{Q}(E'[p])$  is a ramified extension of  $\mathbb{Q}(\mu_p)$ .  $\lrcorner$

*Proof:* Cf.[Elliptic Curves, Group Schemes and Mazur's Theorem, P26] ?.  $\square$

**Prop. (21.7.5.3).** If  $p \in \text{Prime}_{>11}$  and  $E \in \text{EllCur}/\mathbb{Q}$  has a rational torsion point of order  $p$ , then  $\mathbb{Q}(E[p])$  is an unramified extension of  $\mathbb{Q}(\mu_p)$ .  $\lrcorner$

*Proof:* Cf.[Elliptic Curves, Group Schemes and Mazur's Theorem, P29] ?.  $\square$

**Prop. (21.7.5.4) [Specialization Lemma].**  $\lrcorner$

**Lemma (21.7.5.5) [Mazur].** If  $E \in \text{EllCur}/\mathbb{Q}$  and  $p \in \text{Prime}_{\geq 11} \setminus \{13\}$ , then  $E[p] = 0$ .  $\lrcorner$

*Proof:* If  $0 \neq x \in E[p]$ , then by (21.7.4.16),  $E$  has potentially good reduction at  $p = 3$ . The specialization of  $x$  to  $\tilde{E}_{\mathbb{F}_3}$  is non-zero by the specialization lemma (21.7.5.4).

Then  $\tilde{E}_{\mathbb{F}_3}$  cannot be of additive reduction, because if it is of additive reduction, by looking at the Kodaira-Néron classification of the fiber (21.3.4.18), we see that  $\#\pi_0(\tilde{E}_{\mathbb{F}_3}) \leq 4$ , so the specialization of  $x$  is contained in  $(\tilde{E}_{\mathbb{F}_3})^0$  which is isomorphic to  $\mathbb{G}_a$ , a contradiction because  $p \geq 7$ .

So  $\tilde{E}$  is a good reduction, and thus  $\#\tilde{E}_{\mathbb{F}_3} \leq 7$ , contradiction again.  $\square$

**Lemma (21.7.5.6) [Mazur-Tate [M-T73]].** If  $E \in \text{EllCur}/\mathbb{Q}$ , then  $E[13] = 0$ .  $\lrcorner$

*Proof:* Cf. <http://www-personal.umich.edu/~asnowden/teaching/2013/679/index.html>.  $\square$

**Lemma (21.7.5.7) [Kubert].** For  $E \in \text{EllCur}/\mathbb{Q}$ ,

- $E$  doesn't contain a rational torsion point of exact order  $N$  where  $N \in \{14, 15, 16, 18, 20, 21, 24, 25, 27, 35, 49\}$ .
- $E(\mathbb{Q})_{\text{tor}}$  doesn't contain a subgroup isomorphic to  $\mathbb{Z}/(2) \times \mathbb{Z}/(10)$  or  $\mathbb{Z}/(2) \times \mathbb{Z}/(12)$ .  $\lrcorner$

*Proof:* Cf.[Kubert. Universal bounds on the torsion of elliptic curves.] ?

Firstly, notice for  $N \in \{11, 14, 15, 17, 19, 20, 21, 24, 27, 32, 36, 49\}$ ,  $X_0(N)$  has genus 1 (26.3.3.18). And by taking the cusp as the origin,  $X_0(N)$  is an elliptic curve. Then Kubert finds that  $\text{rank}(X_0(N)) = 0$ , and can find all the rational points. Only for  $N = 21$  or  $N = 27$  there are non-cusp rational points. So it suffices to show that for  $X_0(21)$  and  $X_0(27)$ , the quadratic twists of the elliptic curves corresponding to the rational non-cusp points of  $X_0(N)$  don't have rational 21 torsion-points.

For  $N = 21$ , the non-cusp rational points of  $X_0(N)$  are given by quadratic twists of

$$y^2 = x^3 + 45x - 18$$

$$y^2 = x^3 - 75x - 262$$

$$y^2 = x^3 - 1515x - 46106$$

$$y^2 = x^3 - 17235x - 870894.$$

For  $N = 27$ , the non-cuspidal rational points of  $X_0(N)$  are given by quadratic twists of

$$y^2 + y = x^3 - 30x - 5.$$

For the second case, let  $P$  be a generator of  $\mathbb{Z}/(2)$  and  $Q$  a generator of  $\mathbb{Z}/(10)$  or  $\mathbb{Z}/(12)$ , then the dual map  $E/P \rightarrow E \rightarrow E/Q$  is a cyclic isogeny, thus corresponds to a rational point on  $Y_0(20)$  or  $Y_0(24)$ , which in fact has no rational point.

For  $N = 16$ , ?

For  $N = 18$ , ?

For  $N = 25$ , ?

For  $N = 35$ , let  $E = X_0(35)/w_5$ , where  $w_5$  is the Atkin-Lehner involution. Then  $E$  is an elliptic curve, and  $E(\mathbb{Q}) \cong \mathbb{Z}/(3)$ . Then Kubert computed that the preimages of these points are cusps.  $\square$

**Thm. (21.7.5.8) [Ogg's Conjecture, Mazur[Maz77]].** For  $E \in \text{EllCur}/\mathbb{Q}$ ,  $E(\mathbb{Q})_{\text{tor}}$  can only be isomorphic to one of the following 15 groups:

- $\mathbb{Z}/(n)$  for  $n \in [10]_+$  or  $n = 12$ .
- $\mathbb{Z}/(2) \oplus \mathbb{Z}/(2n)$  for  $n \in [4]_+$ .

Moreover, each of these groups occur, by (21.7.5.9).  $\perp$

*Proof:* This follows from (21.7.5.5)(21.7.5.6) and (21.7.5.7). Notice the existence of elliptic curves with these torsion groups can also be shown using modular curves ?  $\square$

**Prop. (21.7.5.9) [Bounding Torsion Points].** Proposition (21.3.4.20) is useful in controlling torsion points of  $E$ . For example: If  $E/\mathbb{Q}$  is an elliptic curve with Weierstrass equation

- $E : y^2 + y = x^3 - x + 1$ , then  $\Delta = -13 \cdot 47$ , so  $E$  has good reduction modulo 2. But it can be calculated that  $\tilde{E}(\mathbb{F}_2) = \tilde{O}$ ,  $E(\mathbb{Q})[2] = O$ , thus  $E(\mathbb{Q})_{\text{tor}} = O$ .
- $E : y^2 = x^3 + 3$ , then  $\Delta = -2^4 \cdot 3^5$ , so  $E$  has good reduction modulo  $p$  for  $p \geq 5$ . But it can be calculated that  $\#\tilde{E}(\mathbb{F}_5) = 6$ ,  $\#\tilde{E}(\mathbb{F}_7) = 13$ , so  $E(\mathbb{Q})[p] = O$  for any prime  $p$ , and  $E(\mathbb{Q})_{\text{tor}} = O$ . In particular,  $(1, 2) \in E(\mathbb{Q})$  has infinite order.
- $E : y^2 = x^3 + x$ , then  $\Delta = -2^6$ , so  $E$  has good reduction modulo  $p$  for  $p \geq 3$ . But it can be calculated that  $\tilde{E}(\mathbb{F}_3) \cong \mathbb{Z}/(4)$ ,  $\tilde{E}(\mathbb{F}_5) \cong \mathbb{Z}/(2) \times \mathbb{Z}/(2)$ , thus  $E(\mathbb{Q})_{\text{tor}} \cong 0$  or  $\mathbb{Z}/(2)$ . The latter case is right, as  $(0, 0) \in E(\mathbb{Q})$ .
- $E : y^2 = x^3 + 2$ .
- $E : y^2 = x^3 + 8$ .
- $E : y^2 = x^3 + 4$ .
- $E : y^2 = x^3 + 4x$ .
- $E : y^2 - y = x^3 - x^2$ .
- $E : y^2 = x^3 + 1$ .
- $E : y^2 = x^3 - 43x + 166$ . then  $\Delta = -2^{19} \cdot 13$ , thus  $E$  has good reduction modulo 3, 5. But it can be calculated that  $\#\tilde{E}(\mathbb{F}_3) = 7 = \#\tilde{E}(\mathbb{F}_5)$ , thus  $E(\mathbb{Q})_{\text{tor}} \cong 0$  or  $\mathbb{Z}/(7)$ . By (21.7.3.3), for any torsion point  $(x, y)$ ,  $y|2^7$ , thus we can find rational points

$$\{(3, \pm 8), \quad (-5, \pm 16), \quad (11, \pm 32)\}.$$

And using doubling formula (21.3.1.9) for  $P = (3, 8)$ ,

$$x([2](P)) = -5, \quad x([4](P)) = 11, \quad x([8](P)) = 3.$$

Thus  $P$  must have order 7, and  $E(\mathbb{Q})_{\text{tor}} \cong \mathbb{Z}/(7)$ .

- $E : y^2 + 7xy = x^3 + 16x.$
- $E : y^2 + xy + y = x^3 - x^2 - 14x + 29.$
- $E : y^2 + xy = x^3 - 45x + 81.$
- $E : y^2 + 43xy - 210y = x^3 - 210x^2.$
- $E : y^2 = x^3 - 4x.$
- $E : y^2 = x^3 + 2x^2 - 3x.$
- $E : y^2 + 5xy - 6y = x^3 - 3x^2.$
- $E : y^2 + 17xy - 120y = x^3 - 60x^2.$

┘

### Merel's Approach: Winding Quotients

**Thm. (21.7.5.10)** [Uniform Boundedness of Torsion Groups, Merel[Mer96]/Oesterlé]. For any  $d \in \mathbb{Z}_+$  there exists  $N(d) \in \mathbb{Z}_+$  s.t. for any  $F \in \mathbf{NField}$  of degree  $d$  and any  $E \in \mathbf{EllCur}/F$ , we have

$$\#E(F)_{\text{tor}} \leq N(d).$$

┘

*Proof:* By [Mazur and Kamienny95]?, this follows from the lemma(21.7.5.11) below. □

**Lemma (21.7.5.11)** [Merel[Mer96]]. Let  $\mathbb{Q} \neq F \in \mathbf{NField}$  and  $p \in \mathbf{Prime}$ , then for any  $E \in \mathbf{EllCur}/F$ , if  $E(K)[p] \neq 0$  implies  $p < (3^{d/2} + 1)^2$ . ┘

*Proof:* □

### Abelian Variety Case

## 6 Rational Points on Curves(Chabauty)

### Finiteness Results for Curves(Faltings)

**Prop. (21.7.6.1)** [Genus 0 Curves]. For  $F \in \mathbf{GField}$  of characteristic  $\neq 2$  and any smooth complete curve  $C/F$  genus 0 is a conic in  $\mathbb{P}^2$ . Thus by(7.13.8.2), it is isomorphic to  $\mathbb{P}^1$  iff it has a rational point. But by Hasse-Weil principle, this is equivalent to it having a rational point over each local field of  $F$ , and this can be understood via Hilbert symbol. ┘

**Thm. (21.7.6.2)** [Shafarevich's Conjecture, Faltings]. Let  $F \in \mathbf{GField}$  and  $S \subset \Sigma_F$  a finite set of places,  $g \in \mathbb{Z}_+$ , then there are only f.m. isomorphism classes of smooth curves over  $k$  of genus  $g$  having good reduction at all finite places outside  $S$ . ┘

*Proof:* This follows from(20.9.10.18)(20.9.10.16) and(20.9.7.3)(21.1.7.4). □

**Prop. (21.7.6.3)** [Kodaira-Parshin]. Let  $F \in \mathbf{NField}$  and  $S$  is a finite set of places of  $F$  containing all places over 2, then for any complete non-singular curve  $C/F$  of genus  $\geq 1$  having good reduction at non-Archimedean places outside  $S$ , there exists a finite extension  $L/F$  and a constant  $N$ , s.t.: For any  $P \in C(F)$ , there exists a complete smooth curve  $C^P$  over  $L$  and a finite map  $\varphi^P : C^P \rightarrow C_L$ , that satisfies



- $C^P$  has good reduction at places not above  $S$ .
- $g(C^P) \leq N$ .
- $\varphi_P$  is ramified exactly at  $P$ .

┘

*Proof:* Cf. [Mil08]P115, 145. ?

□

**Thm. (21.7.6.4) [Uniform Mordell Conjecture 1922, Faltings 1983].** For  $g \in \mathbb{Z}_{\geq 2}$  and  $d \in \mathbb{Z}$ , there exists  $C(g, d) \in \mathbb{Z}_+$  s.t. for any  $F \in \mathbf{NField}$  with  $\deg(F) = d$  and  $C/F$  be a complete non-singular curve of genus  $g$ , we have

$$\#C(F) \leq C(g, d).$$

┘

*Proof:* We only prove that  $\#C(F)$  is finite ?.

Using the construction of Kodaira-Parshin (21.7.6.3), there is a finite extension  $L/F$  s.t. any  $P \in C(F)$  corresponds to a curve  $C^P$  over  $L$  and a map  $\varphi_P : C^P \rightarrow C_L$ . By Shafarevich's conjecture applied to  $L$  (21.7.6.2), there are only f.m. isomorphism classes of  $C^P$ . But for different  $P, Q$  s.t.  $C^P \cong C^Q$ , the maps  $\varphi_P, \varphi_Q$  are non-isomorphic, as they are ramified at different places. So we are done by de Franchis' theorem (7.13.1.35). □

**Prop. (21.7.6.5) [Manin-Mumford].** Let  $F \in \mathbf{NField}$  and  $C/F$  be a complete non-singular curve of genus  $g \geq 2$ , and  $J$  the Jacobian of  $C$ . Fix an embedding  $C \hookrightarrow J$  over  $F$ , then  $\#C(\overline{F}) \cap J(\overline{F})_{\text{tor}} < \infty$ .  
┘

*Proof:* This follows from the Manin-Mumford-Lang conjecture (21.1.6.7), as  $C$  is not a torsion subvariety. □

### Chabauty-Coleman Methods

The idea is: By the embedding  $X \rightarrow J(X)$ , to calculate  $X(\mathbb{Q})$ , calculate  $J(X)(\mathbb{Q})$  first, then determine which points are in  $X(\mathbb{Q})$ .

**Prop. (21.7.6.6).** Let  $F \in \mathbf{NField}$ ,  $v \in \Sigma_F^{\text{fin}}$  s.t.

- $\text{char } v > 2g$ ,
- $v \in \text{Unr}_{F/\mathbb{Q}}(F)$ .

Let  $g \in \mathbb{Z}_+$ ,  $C \in \mathcal{C}ur^{\text{SmPrpr}}/F$  of genus  $g$  s.t.

- $C$  has good reduction at  $v$ .
- $\text{rank}(J(C)(F)) \leq g - 1$ .

Then  $\#C(F) \leq \|v\| + 2g(\|v\|^{1/2} + 1) - 1$ .  
┘

*Proof:*

□

### Coleman Integration

References are [p-adic abelian integrals and torsion points on curves, Coleman].

**Prop. (21.7.6.7) [p-adic Primitives].** Let  $K \in p\text{-Field}$ ,  $K \subset \mathbb{C}_p$  and  $A \in \mathcal{AbVar}/K$ , then for any  $\omega \in H^0(A, \Omega_{A/K}^1)$ , there exists a unique locally analytic homomorphism  $\lambda_\omega : A(\mathbb{C}_p) \rightarrow \mathbb{C}_p$  s.t.  $d\omega = \omega$ .  
┘

*Proof:* Cf.[p-adic abelian integrals and torsion points on curves, Coleman]Chap2.  $\square$

**Prop. (21.7.6.8)[Coleman Integration].** For  $J \in \mathcal{AbVar}/\mathbb{Q}_p$ , there is a **Coleman integration map**:

$$J(\mathbb{Q}_p) \times H^0(J/\mathbb{Q}_p, \Omega^1) \rightarrow \mathbb{Q}_p : (Q, \omega) \mapsto \int_0^Q \omega.$$

s.t. on an open subset  $U \subset J$ , the integration can be calculated locally by writing  $\omega$  in power series.

⌋

*Proof:*  $\square$

**Prop. (21.7.6.9).** Let  $K \in p\text{-Field}$ ,  $K \subset \mathbb{C}_p$  and  $j : (C, c_0) \rightarrow (A, 0)$  is a morphism over  $K$  of pointed curves into an Abelian variety, both with good reduction. If  $G \subset A(K)$  is a subgroup, denoted  $V_G = \{\omega \in H^0(A, \Omega_{A/K}^1) : \lambda_\omega(G) = 0\}$ . Then any  $x \in j(C(K)) \cap G$  satisfies:  $\int_{c_0}^x \omega = 0$  for any  $\omega \in j^*V_G$ .  $\square$

*Proof:* Cf.[p-adic abelian integrals and torsion points on curves, Coleman]Thm2.7.  $\square$

**Prop. (21.7.6.10)[Upper Bounds for Solutions].** Let  $(K, \mathcal{O}_K, \kappa) \in p\text{-NField}$  and  $\mathcal{C} \in \text{Cur}^{\text{SmPrpr}}/\mathcal{O}_K$  with  $\omega \in H^0(\mathcal{C}, \Omega_{\mathcal{C}/\mathcal{O}_K}^1)$  s.t. the reduction  $\tilde{\omega} \neq 0$ . Let  $\lambda_\omega$  be the  $p$ -adic primitive of  $\omega$  (21.7.6.7), then the number of zeros of  $\lambda_\omega$  in  $C(K)$  is at most

$$\sum_{x \in C(\kappa)} N(\text{ord}_x(\tilde{\omega}), |p|^{1/e_K}).$$

In particular, if  $e_K = 1$ , and  $\text{ord}_x(\tilde{\omega}) \leq p - 3$  for any  $x \in C(\kappa)$ , then the number of zeros of  $\lambda_\omega$  is at most

$$\#\kappa - 1 + 2g(\sqrt{\#\kappa} + 1).$$

⌋

*Proof:*  $\square$

### Chabauty-Kim Method

**Thm. (21.7.6.11) [Chabauty-Coleman].** Let  $X \in \text{Cur}^g/\mathbb{Q}$ , and  $p \in \text{Prime}$  a good reduction of  $X$ . Suppose  $r' < g$  (e.g.  $r < g$ )  $\color{red}?$ , then

•

- If  $p > 2g$ , then  $\#X(\mathbb{Q}) \leq \#X(\mathbb{F}_p) + 2r' \leq \#X(\mathbb{F}_p) + (2g - 2)$ .  $\square$

⌋

## 7 Finite Fields

**Prop. (21.7.7.1) [Esnault].** Let  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}}$ ,  $X \in \text{SmProjVar}/\mathbb{F}_q$  s.t.  $\text{CH}_0(X_{\overline{R(X)}}) = \mathbb{Z}$ , then  $\#X(\mathbb{F}_q) \equiv 1 \pmod{q}$ . In particular,  $X$  has a rational point.  $\square$

*Proof:* Cf.[Weil 1 Proof].  $\square$

**Cor. (21.7.7.2).** Any rationally chain connected varieties over a finite field  $k = \mathbb{F}_q$  s.t.  $\text{CH}_0(X_{\overline{R(X)}}) = \mathbb{Z}$ , then  $\#X(K) \equiv 1 \pmod{q}$ . In particular,  $X$  has a rational point.  $\square$

**Cor. (21.7.7.3) [Manin-Lang].** If  $X/\mathbb{F}_q$  is a smooth projective Fano variety, then  $\#X(\mathbb{F}_q) \equiv 1 \pmod{q}$ .  $\square$

⌋

*Proof:* This is because by (7.12.4.4), any Fano variety is rationally chain connected.  $\square$

## 8 Hilbert's Theorem

Cf.[Mordell-Lang theorem, Serre].

## 9 Vojta's Conjecture

## 10 Bombieri-Lang Conjecture

**Conj. (21.7.10.1) [Bombieri-Lang].** Let  $F \in \mathbf{NField}$  and  $X \in \mathbf{Var}/F$  is of general type (i.e. with maximal Kodaira dimension (7.12.3.6)), then there are f.m. subvarieties  $X_i \subset X$  of lower dimensions s.t.  $X(F) = \cup_i X_i(F)$ .  $\lrcorner$

*Proof:*

□

**Thm. (21.7.10.2) [Mordell Conjecture, Faltings].** By (21.7.6.4), for  $F \in \mathbf{NField}$  and  $C \in \mathbf{SmProjCur}/F$  with genus  $g \geq 2$ , we have  $\#C(F) < \infty$ .  $\lrcorner$

## 11 Coleman's Conjecture

**Conj. (21.7.11.1) [Coleman].** For  $d, g \in \mathbb{Z}_+$ , consider all Abelian varieties  $A$  of dimension  $g$  and defined over a number field of degree  $d$ , then there are only f.m. isomorphism classes among  $\text{End}(A_{\overline{\mathbb{Q}}})$ .  $\lrcorner$

*Proof:*

□

**Prop. (21.7.11.2) [Equivalent Statements of Coleman's Conjecture].** The following are equivalent:

- Coleman's conjecture for  $\text{End}(A_{\overline{\mathbb{Q}}})$  (21.7.11.1).
- Coleman's conjecture for  $\text{End}(A)$ .
- $\text{disc}(A)$  is uniformly bounded for all Abelian varieties  $A$  of dimension  $g$  and defined over a number field of degree  $d$ .
- $\delta(A)$  is uniformly bounded for all Abelian varieties  $A$  of dimension  $g$  and defined over a number field of degree  $d$ .
- $\Delta_A$  is uniformly bounded for all Abelian varieties  $A$  of dimension  $g$  and defined over a number field of degree  $d$ .

□

*Proof:* Cf.[Uniformity Conjectures for Abelian Varieties and K3 Surfaces]Prop3.2, 3.4. □

**Prop. (21.7.11.3).** Coleman's conjecture implies the uniform boundedness conjecture for  $A(F)_{\text{tor}}$ .  $\lrcorner$

*Proof:* Cf.[Conjectures uniformes sur les varietes abeliennes]Thm1.1. ?

□

**Conj. (21.7.11.4) [Br(AbVar)].** Let  $d \in \mathbb{Z}_+$ , then there exists  $C(d) \in \mathbb{R}_+$  s.t. if  $F \in \mathbf{NField}$ ,  $[F : \mathbb{Q}] \leq d$  and  $A \in \mathbf{AbVar}/F$ , then  $\# \text{Br}(A_{\overline{k}})^{\text{Gal}_F} < \infty$ .  $\lrcorner$

*Proof:*

□

## 12 Hilbert's 10-th Problem

**Conj. (21.7.12.1) [Hilbert's 10-th Problem].** Let  $F \in \mathbf{NField}$ ,  $X$  be a projective scheme over  $\mathcal{O}_K$ , it there an algorithm to determine in a finite number of operations whether  $X(K) = \emptyset$ .  $\lrcorner$

*Proof:*

□

**Remark (21.7.12.2) [Mazur-Rubin].** If the BSD conjecture holds for all elliptic curves over any number fields, then Hilbert's 10-th problem has a negative answer for any number field.  $\lrcorner$

*Proof:*

□

## 21.8 Arithmetic Dynamical Systems

References are [Sil07] and [Fak03].

**Def. (21.8.0.1) [Rational Maps].** A map  $\varphi : \mathbb{P}^N \rightarrow \mathbb{P}^M$  is called a **rational map of degree  $d$**  if  $\lrcorner$

**Prop. (21.8.0.2) [Fakhruddin's Trick].** Let  $k \in \mathbf{Field}$ ,  $\#k = \infty$ ,  $X$  a projective variety over  $k$ ,  $\varphi : X \rightarrow X$  a morphism,  $\mathcal{L} \in \mathbf{Pic}(X)$  a very ample line bundle and  $d \in \mathbb{Z}_+$  s.t.  $\varphi^*(\mathcal{L}) \cong \mathcal{L}^{\otimes d}$ , then there exists  $N \in \mathbb{Z}_+$  and an embedding  $\iota : X \rightarrow \mathbb{P}_k^N$  with a morphism  $\psi : \mathbb{P}_k^N \rightarrow \mathbb{P}_k^N$  s.t.  $\psi \circ \iota = \iota \circ \varphi$ .  $\lrcorner$

*Proof:* Cf. [Fak03]P2.  $\square$

**Thm. (21.8.0.3).** For  $k \in \mathbf{Field}$ ,  $k = \bar{k}$ ,  $A \in \mathbf{AbVar}/k$  with a group  $G$  acting freely, then  $X = A/G$  has a line bundle  $\mathcal{L}$  and a morphism  $\varphi : X \rightarrow X$  s.t.  $\varphi^*(\mathcal{L}) \cong \mathcal{L}^{\otimes d}$  for some  $d \in \mathbb{Z}_{\geq 2}$ . And if  $\text{char } k = 0$ , then for any  $X \in \mathbf{SmProj}/k$  with  $\text{Kod}(X) \geq 0$  and a line bundle  $\mathcal{L}$  and a morphism  $\varphi : X \rightarrow X$  s.t.  $\varphi^*(\mathcal{L}) \cong \mathcal{L}^{\otimes d}$  for some  $d \in \mathbb{Z}_{\geq 2}$  is of the form  $X = A/G$  for  $A \in \mathbf{AbVar}/k$  with a group  $G$  acting freely.  $\lrcorner$

*Proof:* Cf. [Fak03]P8.  $\square$

### 1 over Finite Fields

### 2 over Local Fields

### 3 over Global Fields

#### Heights

#### Canonical Heights

**Def. (21.8.3.1) [Quasi-Homogenous Functions].** Let  $N \subset \mathbb{R}$  be a multiplicatively-closed subset of  $\mathbb{R}$  acting on a set  $S \in \mathbf{Set}$ . Then a function  $h : S \rightarrow \mathbb{R}$  is called a **quasi-homogenous function** of degree  $d \in \mathbb{R}_+$  iff for any  $n \in N$ , there exists  $C(n) \in \mathbb{R}_+$  s.t.

$$|h(nx) - n^d h(x)| \leq C(n), \forall x \in S.$$

$\lrcorner$

**Prop. (21.8.3.2) [Tate's Limiting Argument, [Man64]].** Situation as in (21.8.3.1), there exists a unique homogenous function  $\hat{h} : S \rightarrow \mathbb{R}$  of degree  $d$  s.t.  $\hat{h} - h$  is bounded, which is given by

$$\hat{h}(x) = \lim_{|n| \rightarrow \infty} \frac{h(nx)}{n^d}.$$

$\lrcorner$

*Proof:* Cf. [B-G06]P285.  $\square$

**Prop. (21.8.3.3).** For  $F \in \mathbf{NField}$  and  $f : \mathbb{P}_F^1 \rightarrow \mathbb{P}_F^1$  of degree  $> 1$ , then the canonical height

$$\hat{h}_f : \alpha \mapsto \lim_{n \rightarrow \infty} \frac{h(f^{(n)}(\alpha))}{d^n}$$

is the unique function s.t.  $\hat{h}_f \circ f = d\hat{h}_f$ , and  $\hat{h}_f \sim h$ .  $\lrcorner$

**Prop. (21.8.3.4).** For  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$  and  $f = \varphi_{E,2}$ ,  $\hat{h}_f(\text{pr}(P)) = 2\hat{h}_E(P)$  for any  $P \in E(F)$ .  $\lrcorner$

## 4 Uniform-Boundedness

**Conj. (21.8.4.1)**[Uniform-Boundedness Conjecture, Morton-Silverman]. For  $D, N, d \in \mathbb{Z}_+$  and  $d \geq 3$ , there exists  $\kappa(D, N, d) \in \mathbb{Z}_+$  s.t. for any  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = D$  and each morphism  $\psi : \mathbb{P}_F^N \rightarrow \mathbb{P}_F^N$  defined by homogenous polynomials of degree  $d$ ,  $\# \text{PrePer}(\psi; F) \leq \kappa(D, N, d)$ .  $\lrcorner$

*Proof:*  $\square$

**Remark (21.8.4.2).** By[Fak03]P4, by restriction of scalars, it suffice to consider the case  $F = \mathbb{Q}$ .  $\lrcorner$

**Conj. (21.8.4.3).** For  $F \in \mathbf{NField}$  and  $N \in \mathbb{Z}_+, d \in \mathbb{Z}_{\geq 2}$ , there exists a finite set  $S(d, N, F) \subset \mathbb{P}^N(F)$  s.t. for any morphism  $f : \mathbb{P}_F^N \rightarrow \mathbb{P}_F^N$  defined by homogenous polynomials of degree  $d$ ,  $S(d, N, F) \not\subseteq f(\mathbb{P}^N(F))$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (21.8.4.4).** This conjecture is a consequence of the Bombieri-Lang conjecture(21.7.10.1), by[Fak03]P4.  $\lrcorner$

*Proof:*  $\square$

## 5 Arithmetic Dynamical Mordell-Lang Conjecture

References are [Poo14].

**Prop. (21.8.5.1)**[ $p$ -adic Interpolation of Iterates, Poonen]. Let  $K$  be a valued field s.t.  $|p| = 1/p$ , and  $f \in \mathcal{O}_K \langle X_1, \dots, X_d \rangle^d$  (12.2.4.2) satisfies  $f(\underline{X}) \equiv \underline{X} \pmod{p^c}$  for some  $c > 1/(p-1)$ . Then there exists  $g \in \mathcal{O}_K \langle X_1, \dots, X_d, T \rangle$  s.t.  $g(\underline{X}, n) = f^{\circ n}(\underline{X}) \in \mathcal{O}_K \langle \underline{X} \rangle^d$  for any  $n \in \mathbb{N}$ . Moreover,  $g(\underline{X}, T+1) = f(g(\underline{X}, T))$  holds.  $\lrcorner$

*Proof:* The hypothesis implies that  $h(f(\underline{X})) \equiv f(\underline{X}) \pmod{p^c}$  for any  $h(\underline{X}) \in \mathcal{O}_K[\underline{X}]^d$ , and by taking limits this is also true for any  $h \in \mathcal{O}_K \langle \underline{X} \rangle^d$ . Thus if we define the linear operator

$$\Delta(h)(\underline{X}) = h(f(\underline{X})) - f(\underline{X}),$$

then for any  $m \in \mathbb{N}$ ,  $\Delta^m$  maps  $\mathcal{O}_K \langle \underline{X} \rangle^d$  into  $p^{mc} \mathcal{O}_K \langle \underline{X} \rangle^d$ . Then because  $|m!| > p^{-m/(p-1)}$  (2.6.1.33), so the Mahler series

$$g(\underline{X}, T) = \sum_{m \in \mathbb{N}} \binom{T}{m} \Delta^m(\underline{X})$$

converges in  $\mathcal{O}_K \langle \underline{X} \rangle^d$ . And

$$g(\underline{X}, n) = \sum_{m=0}^n \binom{n}{m} \Delta^m(\underline{X}) = (\Delta + \text{id})^n(\underline{X}) = f^{\circ n}(\underline{X}).$$

For the last assertion, the equation clearly holds for  $T \in \mathbb{N}$ , so by comparing coefficients, it holds for any  $T$ .  $\square$

## 6 Dynamics Associated to Algebraic Groups

### Power Maps

### Chebyshev Maps

### Lattès Maps

References are [On Lattès Maps, Milnor].

**Def. (21.8.6.1) [Lattès Maps].** A rational function  $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$  is called a **Latte map** if there exists  $E \in \text{EllCur}/\mathbb{C}$  s.t. it is a quotient of an endomorphism  $\varphi \in \text{End}(E)$  through a finite map  $\text{pr} : E \rightarrow \mathbb{P}^1$ . ┘

**Prop. (21.8.6.2).** For any Lattès map  $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$  corresponding to an endomorphism  $\varphi \in \text{End}(E)$  of degree  $\geq 2$  ?,  $\text{Jul}(f) = \mathbb{P}^1(\mathbb{C})$ . ┘

*Proof:* This is because the Julia set is the closure of repelling periodic points, and the set of periods of  $E \in \text{EllCur}/\mathbb{C}$  are dense and it is expanding everywhere. □

## 7 Equidistributions

## 21.9 Arithmetic Topology

Main references are [Knots and Primes, an Introduction to Arithmetic Topology, Masanori-Morishita], [Knots and Primes, Chao Li], [Kapranov, M.: Analogies between number fields and 3-manifolds. Unpublished Note (1996)], [Kapranov, M.: Analogies between the Langlands correspondence and topological quantum field theory. In: Progress in Math., vol. 131, pp. 119–151. Birkhäuser, Basel (1995)], [Manin, Y., Marcolli, M.: Holography principle and arithmetic of algebraic curves. Adv. Theor. Math. Phys. 5(3), 617–650 (2001)], [Reznikov, A.: Three-manifolds class field theory (Homology of coverings for a nonvirtually b1-positive manifold). Sel. Math. New Ser. 3, 361–399 (1997)], [Reznikov, A.: Embedded incompressible surfaces and homology of ramified coverings of three-manifolds. Sel. Math. New Ser. 6, 1–39 (2000)].

### 1 Knots and Primes



## 22 | Motives

### 22.1 Motives

References are [Sta]Chap45, [Kle94, Sch94, Ful98, Mil12], and [Lectures on Pure Motives, Murre]. [Jannsen, Uwe Motivic sheaves and filtrations on Chow groups1994], [Jannsen, Uwe Equivalence relations on algebraic cycles2000], [Jannsen, Uwe Mixed motives and algebraic  $K$ -theory1990].

**Notation (22.1.0.1).**

- Let  $k \in \text{Field}$ .

┘

#### 1 Correspondences

**Def. (22.1.1.1).** The full subcategory of  $\text{Sch}/k$  consisting of smooth projective schemes over  $k$  is denoted by  $\text{SmProj}/k$ .

Any  $X \in \text{SmProj}/k$  has a decomposition  $X = \coprod_n X_n$  into clopen subschemes that  $X_n$  is equidimensional of dimension  $n$  by (9.1.1.28). Thus we can talk about Rings  $\text{CH}^*(X)$  and  $\text{CH}_{\text{num}}^*(X)$  (9.1.1.29).

┘

**Def. (22.1.1.2) [Adequate Equivalent Relations of Cycles].** An adequate equivalent relation on cycles is an equivalence relation on  $Z^*(X)$  for each  $X \in \text{SmProj}/k$  s.t.

- it is compatible with gradings and additions.
- it is compatible with products:  $Z \sim 0 \implies Z \times Y \sim 0$ .
- it is compatible with intersections:  $Z_1 \sim 0 \implies Z \cdot W = 0$ .
- it is compatible with projections: If  $Z \sim 0 \in Z^*(X \times Y) \implies (\text{pr}_X)_*(Z) \sim 0$ .
- it satisfies moving lemma: Given  $Z, W_1, \dots, W_\ell \in Z^*(X)$ , there exists  $Z' \sim Z$  s.t.  $Z'$  and  $W_i$  intersect properly for any  $i$ .

┘

**Def. (22.1.1.3) [(Rational)Correspondences].** Let  $X, Y \in \text{SmProj}/k$  the group of (rational) correspondences from  $X$  to  $Y$  of degree(codimension)  $r \in \mathbb{Z}$  is defined to be

$$\text{Corr}_{\text{rat}}^r(X, Y) = \bigoplus_d \text{CH}^{d+r}(X_d \times_k Y)_{\mathbb{Q}} \subset \text{CH}^*(X \times Y)_{\mathbb{Q}}.$$

Similarly we can define the groups of **Grothendieck correspondences**  $\text{Corr}_{\text{num}}^r(X, Y) \subset \text{CH}_{\text{num}}^*(X \times Y)_{\mathbb{Q}}$ .

┘

**Def. (22.1.1.4) [Compositions of Correspondences].** Let  $X, Y, Z \in \text{SmProj}/k$ , there is a **composition of correspondences** map

$$\text{Corr}^s(Y, Z) \times \text{Corr}^r(X, Y) \rightarrow \text{Corr}^{r+s}(X, Z) : (c', c) \mapsto c' \circ c = \text{pr}_{13*}(\text{pr}_{23}^* c' \cdot \text{pr}_{12}^* c).$$

Then composition of correspondences are  $\mathbb{Q}$ -linear and associative.  $\lrcorner$

*Proof:* Cf. [Sta]0FG0.  $\square$

**Prop. (22.1.1.5) [Tensor Product of Correspondences].** Let  $X, Y, X', Y' \in \text{SmProj}/k$ , there is a **tensor product of correspondences** map

$$\otimes : \text{Corr}^r(X, Y) \times \text{Corr}^s(X', Y') \rightarrow \text{Corr}^{r+s}(X \times_k X', Y \times_k Y') : (c, c') \mapsto \text{pr}_{13}^*(c) \cdot \text{pr}_{24}^*(c').$$

which is  $\mathbb{Q}$ -linear and associative, and commutes with compositions in both coordinates.  $\lrcorner$

**Prop. (22.1.1.6).**  $\text{Corr}^0(\mathbb{P}_k^1) = \text{CH}^1(\mathbb{P}_k^1 \times \mathbb{P}_k^1)_{\mathbb{Q}} \cong \mathbb{Q} \oplus \mathbb{Q}$ , where a basis is given by  $c_0 = [\{0\} \times \mathbb{P}^1]$  and  $c_2 = [\mathbb{P}^1 \times \{0\}]$ . And the diagonal  $\Delta = c_0 + c_2$ .

Also  $c_0 \circ c_0 = c_0$ ,  $c_2 \circ c_2 = c_2$ ,  $c_0 \circ c_2 = 0 = c_2 \circ c_0$ , so  $\text{Corr}^0(\mathbb{P}_k^1) \cong \mathbb{Q} \oplus \mathbb{Q}$  as a  $\mathbb{Q}$ -algebra.  $\lrcorner$

**Def. (22.1.1.7) [Push and Pull via Correspondences].** Let  $c \in \text{Corr}_{\text{rat}}^r(X, Y)$ , we can define the

- (Pullback):  $\text{CH}_k(Y) \rightarrow \text{CH}_{k-r}(X) : \beta \mapsto c^*(\beta) = \text{pr}_{1,*}(c \cdot \text{pr}_2^* \beta)$ .
- (Pushforward):  $\text{CH}^k(X) \rightarrow \text{CH}^{k+r}(Y) : \alpha \mapsto c_*(\alpha) = \text{pr}_{2,*}(c \cdot \text{pr}_1^* \alpha)$ .

by using stratification and (9.1.1.12)(9.1.1.9).  $\lrcorner$

**Prop. (22.1.1.8) [Correspondences and Chow Groups].**

- There are canonical isomorphisms

$$\text{CH}_{-r}(X)_{\mathbb{Q}} \cong \text{Corr}_{\text{rat}}^r(X, \text{Spec } k)$$

s.t. pullbacks by correspondences correspond to compositions.

- There are canonical isomorphisms

$$\text{CH}^r(X)_{\mathbb{Q}} \cong \text{Corr}_{\text{rat}}^r(\text{Spec } k, X)$$

s.t. pushforwards by correspondences correspond to compositions.  $\lrcorner$

*Proof:* Cf. [Sta]0FG0.  $\square$

**Cor. (22.1.1.9).** Pushforwards and pullbacks by correspondences commute with compositions, by (22.1.1.8) and (22.1.1.4).  $\lrcorner$

**Prop. (22.1.1.10) [Graphs].** For  $f : X \rightarrow Y \in \text{SmProj}/k$ , the transposed graph  $\Gamma_f^t$  defines a correspondence  $[\Gamma_f^t] \in \text{Corr}^0(Y, X)$ .  $[\Gamma_{\text{id}_X}] \in \text{Corr}^0(X, X)$  is denoted by  $[\Delta_X]$ .  $\lrcorner$

**Def. (22.1.1.11) [Transpose].** Let  $X, Y \in \text{SmProj}/k$  be equidimensional, then the isomorphism  $X \times_k Y \cong Y \times_k X$  induces a **transpose isomorphism**

$$(-)^t : \text{Corr}^r(X, Y) \rightarrow \text{Corr}^{\dim X - \dim Y + r}(Y, X).$$

In particular, when  $f : X \rightarrow Y \in \text{SmProj}/k$  and  $X, Y$  are equidimensional, then  $[\Gamma_f^t]^t = [\Gamma_f]$ .  $\lrcorner$

**Prop. (22.1.1.12).** Let  $\alpha \in \text{Corr}^*(X, Y), \beta \in \text{Corr}^*(Y, Z)$ , then

1.  $(\beta \circ \alpha)^t = \alpha^t \circ \beta^t$ .
2. If  $\beta = [\Gamma_g^t]$ , then  $\beta \circ \alpha = (\text{id}_X \times g)^! \alpha$ .
3. If  $\alpha = [\Gamma_f^t]$ , then  $\beta \circ \alpha = (f \times \text{id}_Z)_* \beta$ .
4. If  $\alpha = [\Gamma_f^t], \beta = [\Gamma_g^t]$ , then  $\beta \circ \alpha = [\Gamma_{fg}^t]$ .
5.  $\Delta_Y \circ \alpha = \alpha, \quad \beta \circ \Delta_Y = \beta$ .

┘

*Proof:* 1: This follows from commutativity and naturality of intersection products.

2: By (9.1.3.6) and (9.1.7.9),

$$[\Gamma_g^t] \circ \alpha = \text{pr}_{13*}(\text{pr}_{23}^*[\Gamma_g^t] \cdot \text{pr}_{12}^* \alpha) = \text{pr}_{13*}(1, g, 1)_*((1, g, 1)^! \text{pr}_{12}^* \alpha) = (1, g)^! \alpha.$$

3 is similar to item 2.

4, 5 follow from item 2, 3. □

**Prop. (22.1.1.13) [Push and Pull by Graphs].** Let  $f : X \rightarrow Y \in \text{SmProj}/k$ , then

- Pushforward by  $[\Gamma_f]^t$  agrees with the Gysin map  $f^! : \text{CH}^*(Y) \rightarrow \text{CH}^*(X)$ .
- Pullback by  $[\Gamma_f]^t$  agrees with the pushforward  $f_* : \text{CH}_*(X) \rightarrow \text{CH}_*(Y)$ .

┘

*Proof:* These follows from (22.1.1.8) and (22.1.1.12). □

**Prop. (22.1.1.14).** Let  $f : Y \rightarrow X \in \text{SmProj}/k$  where  $X, Y$  are equidimensional, then there are commutative diagram of correspondences

$$\begin{array}{ccccc} X \times Y & \xrightarrow{[\Gamma_f^t] \otimes \Delta_Y} & Y \times Y & \xrightarrow{[\Gamma_{\Delta_Y}^t]} & Y \\ \downarrow \Delta_X \otimes [\Gamma_f] & & & & \downarrow [Y] \\ X \times X & \xrightarrow{[\Gamma_{\Delta_X}^t]} & X & \xrightarrow{[X]} & \text{Spec } k \end{array}$$

┘

*Proof:* By (22.1.1.12), it suffices to show that  $(\text{id}_X \times f)^!(\Delta_X)_*[X] = (f \times \text{id}_Y)_*(\Delta_Y)_*[Y]$ . But both sides equal  $[\Gamma_f^t]$ .  $((\text{id}_X \times f)^![\Delta_X] = [\Gamma_f^t])$  follows from [Sta]0FF7? □

**Prop. (22.1.1.15) [Virtual Number of Coincidences].** Let  $X, Y \in \text{SmProj}/k, \alpha, \beta \in \text{Corr}^*(X, Y)$ , then

$$\text{pr}_{1*}(\alpha \cdot \beta) = (\beta^t \circ \alpha) \cdot \Delta_X = (\alpha^t \circ \beta) \cdot \Delta_X.$$

In particular, if  $X, Y$  are equidimensional of dimension  $n$ , and  $\alpha, \beta \in \text{Corr}^0(X, Y)$ , then the **virtual number of coincidence** of  $\alpha$  and  $\beta$ , defined as  $\int_{X \times Y} \alpha \cdot \beta$ , equals the virtual number of fixed points of  $\beta^t \circ \alpha$ . ┘

*Proof:* Cf. [Ful98]P309. □

**Def. (22.1.1.16) [Degenerate Correspondences].** Let  $X, Y \in \text{SmProj}/k$ , then the subgroup of **degenerate correspondences** from  $X$  to  $Y$  is the subgroup of  $\text{Corr}^*(X, Y)$  generated by exterior products  $\text{CH}^*(X) \times \text{CH}^*(Y)$  (9.1.7.3). The subgroup of degenerate correspondences from  $X$  to  $Y$  is denoted by  $I(X, Y)$ .

Then  $I(X, X)$  is a two-sided ideal in  $\text{Corr}^*(X, X)$ , and is stable under transpose if  $X$  are equidimensional.  $\lrcorner$

**Def. (22.1.1.17) [Valence].** Let  $X \in \text{SmProj}^n/k$ , then  $\alpha \in \text{Corr}^0(X, X)$  is said to be of **valence**  $\nu$  iff  $\alpha + \nu[\Delta_X] \in I(X, X)$ .  $\lrcorner$

**Def. (22.1.1.18) [Degree of Correspondences].** Let  $X, Y, Z \in \text{SmProj}^n/k$  be irreducible, then for  $\alpha \in \text{Corr}^n(X, Y)$ , the **degrees of**  $\alpha$  is defined to

$$\alpha_*[X] = \text{pr}_{2*}(\alpha) = d_2(\alpha)[Y], \quad \alpha^*[Y] = \text{pr}_{1*}(\alpha) = d_1(\alpha)[X].$$

Then

- Let  $\beta \in \text{Corr}^n(Y, Z)$ , then  $d_i(\beta \circ \alpha) = d_i(\alpha)d_i(\beta)$ .
- If  $P \subset X$  is a rational point, then  $d_1(\alpha) = \int_{X \times Y} \alpha \cdot [P \times Y]$ .

$\lrcorner$

*Proof:* 1 is clear. 2 follows by composing with the correspondence  $[\Gamma_P] \in \text{Corr}^{-n}(P, X)$ .  $\square$

## 2 Weil Cohomology Theories

### Pre-Weil Cohomology Theories

**Prop. (22.1.2.1) [Non-Existence of a  $\mathbb{Q}$ -Cohomology Theory].** For any  $p \in \text{Prime} \cup \{0\}$ , there exists  $k \in \text{Field}^p$  s.t.  $k = \bar{k}$  and there doesn't exist a cohomology theory on the category of smooth projective varieties over  $k$  with coefficients in  $\mathbb{Q}$  that specializes to the étale cohomology with  $\mathbb{Q}_\ell$  coefficients, where  $\ell \in \text{Prime} \setminus \{p\}$ .  $\lrcorner$

*Proof:* For  $p = 0$ , take  $k = \mathbb{C}$ . For a smooth projective variety  $X$  over  $k$ , it is defined over a subfield  $k_0$  finite over  $\mathbb{Q}$ . Let  $\Gamma = \text{Gal}(k/k_0)$ , then  $\Gamma$  acts on  $H^*(X, \mathbb{Q}_\ell)$ . If  $H^*(X, \mathbb{Q}_\ell)$  is specialized from a  $\mathbb{Q}$ -cohomology  $\tilde{H}^*(-, \mathbb{Q})$ , then the continuous  $\Gamma$ -action on  $H_{\text{ét}}^*(X, \mathbb{Q}_\ell)$  stabilizes  $\tilde{H}^*(X, \mathbb{Q})$ . But it is known that

$$H_{\text{ét}}^*(X, \mathbb{Q}_\ell) \cong H_{\text{Betti}}^*(X, \mathbb{Q}) \oplus \mathbb{Q}_\ell,$$

so  $\dim_{\mathbb{Q}} \tilde{H}^*(X, \mathbb{Q}) = \dim H_{\text{Betti}}^*(X, \mathbb{Q}) < \infty$ . Then because infinite Galois group is uncountable,  $\Gamma$  acts  $\tilde{H}^*(X, \mathbb{Q})$  through a finite quotient. But this is not true in general. ? In fact, This contradicts Tate's conjecture.

If  $p > 0$ , let  $k = \overline{F}_p$  and  $E$  be an supersingular elliptic curve over  $\bar{k}$ , which exists by (21.3.3.12), then  $\text{End}(E) \otimes \mathbb{Q}$  is a definite quaternion algebra  $Q$  over  $\mathbb{Q}$ , and  $\dim_{\mathbb{Q}_\ell} H_{\text{ét}}^1(E, \mathbb{Q}_\ell) = 2$  (20.9.6.19), so if  $H^*(X, \mathbb{Q}_\ell)$  is specialized from a  $\mathbb{Q}$ -cohomology  $\tilde{H}^*(-, \mathbb{Q})$ ,  $\dim_{\mathbb{Q}} \tilde{H}^1(E, \mathbb{Q}) = 2$ , and  $\text{End}(E) \otimes \mathbb{Q}$  acts on it. But there is no ring homomorphism  $Q \rightarrow \text{Mat}(2; \mathbb{Q})$ .  $\square$

**Def. (22.1.2.2) [Weil Cohomology Theories].** For  $F \in \text{Field}^0$ , a **pre-Weil cohomology theory** over  $k$  with coefficients in  $F$  is given by a tuple  $(F(1), H^*, \gamma, \text{tr})$ , where

- $F(1)$  is a 1-dimensional vector space over  $F$ . And denote  $F(n) = F(1)^{\otimes n}$ ,  $F(-n) = F(n)^*$ , and for any  $V \in \text{Vect}_F$ , define  $V(n) = V \otimes F(n)$ .

- $H^*$  is a functor  $H^* : \text{SmProj}/k \rightarrow \mathcal{C}\text{Ring}^{\text{gr}}/F$ .  $H^*(X)$  is called the **cohomology ring** of  $X$ , and its multiplication is denoted by  $\cup$ .
- For every  $X \in \text{SmProj}/k$  and  $i \in \mathbb{N}$ ,  $\gamma_X$  is a family of homomorphisms  $\gamma_X : \text{CH}^i(X) \rightarrow H^{2i}(X)(i)$  for each  $i \in \mathbb{N}$ , called the **cycle class maps** of  $X$ .
- For any  $X \in \text{SmProj}^d/k$ ,  $\text{tr}_X$  is a map  $\text{tr}_X : H^{2d}(X)(d) \rightarrow F$ , called the **trace map** of  $X$ . The trace map is sometimes also denoted by  $\int_X$ .

that satisfy the following properties:

**Poincaré duality:** For any  $X \in \text{SmProj}^d/k$ ,

1.  $\dim_F H^i(X) < \infty$  for any  $i$  and  $H^i(X) = 0$  unless  $0 \leq i \leq 2n$ .
2.  $H^i(X) \otimes_F H^{2d-i}(X)(d) \xrightarrow{\cup} H^{2d}(X)(d) \xrightarrow{\text{tr}_X} F$  is a perfect pairing, called the **Poincaré pairing**.

**Künneth formula:** For any  $X, Y \in \text{SmProj}/k$ ,

$$H^*(X) \otimes_F H^*(Y) \rightarrow H^*(X \times_k Y) : (a, b) \mapsto \text{pr}_1^* a \cup \text{pr}_2^* b$$

is an isomorphism in  $\mathcal{C}\text{Ring}^{\text{gr}}/F$ .

**Cycle class map is natural:**

1. For any  $f : X \rightarrow Y \in \text{SmProj}/k$ ,  $\gamma_X(f^! \beta) = f^* \gamma_Y(\beta)$  for  $\beta \in H^*(Y)$  and  $\gamma_Y(f_* \alpha) = f_* \gamma_X(\alpha)$  for  $\alpha \in H^*(X)$ , where  $f_*$  on  $H^*$  is defined in (22.1.2.3).
2.  $\gamma_{X \times Y}(a \cdot b) = \gamma_X(a) \cup \gamma_Y(b)$ .
3.  $\int_{\text{Spec } k} \gamma(\text{Spec } k) = 1$ .

**Weak Lefschetz theorem** For any  $X \in \text{SmProj}^d/k$  and a smooth hyperplane section  $h : W \subset X$ , the induced map  $h^* : H^i(X) \rightarrow H^i(W)$  is an isomorphism for  $i \leq d-2$  and an injection for  $i = d-1$ .

**Strong Lefschetz theorem** For any  $X \in \text{SmProj}^d/k$  and a smooth hyperplane section  $h : W \subset X$ , define the Lefschetz operator

$$L : H^i(X) \rightarrow H^{i+2}(X) : x \mapsto x \cdot \gamma_X(W),$$

then for  $i \in [d]$ ,  $L^{d-i} : H^i(X) \rightarrow H^{2d-i}(X)$  is an isomorphism.

$H^*$  is called a **Weil cohomology theory** if moreover it satisfies:

- For  $X \in \text{SmProj}/k$  and  $\Gamma(X, \mathcal{O}_X) = k'$ , then the natural map  $H^0(\text{Spec } k') \rightarrow H^0(X)$  is an isomorphism.

┘

**Prop. (22.1.2.3)[Pushforward].** Let  $k \in \text{Field}$ ,  $F \in \text{Field}^0$  and  $H^*$  be a pre-Weil cohomology theory over  $k$  with coefficients in  $F$  (22.1.2.2). For  $f : X \rightarrow Y \in \text{SmProj}/k$  where  $\dim X = d$ ,  $\dim Y = e$ , we can define a **pushforward map of cohomology**  $H^{2d-*}(X)(d) \rightarrow H^{2e-*}(Y)(e)$  via Poincaré duality, i.e.

$$\int_X f^* b \cup a = \int_Y b \cup f_* a$$

for any  $a \in H^{2d-i}(X)(d)$ ,  $b \in H^i(Y)$ . Then

- $f_*(f^* b \cup a) = b \cup f_* a$ .
- $g_* f_* = (gf)_*$ .

┘

*Proof:* Use duality and the corresponding properties of  $f^*$ . □

**Prop. (22.1.2.4) [ $H^*(\text{Spec } k)$ ].** Let  $k \in \mathbf{Field}$ ,  $F \in \mathbf{Field}^0$  and  $H^*$  be a pre-Weil cohomology theory over  $k$  with coefficients in  $F$  (22.1.2.2), then  $H^i(\text{Spec } k) = 0$  unless  $i = 0$ , and there is a unique  $F$ -algebra isomorphism  $H^0(\text{Spec } k) \cong F$ , sending  $\gamma([\text{Spec } k])$  to 1 and  $\text{tr}_{\text{Spec } k}$  is identity under this identification.  $\lrcorner$

*Proof:* Cf. [Sta]0FHE.  $\square$

**Prop. (22.1.2.5) [Coproducts].** Let  $k \in \mathbf{Field}$ ,  $F \in \mathbf{Field}^0$  and  $H^*$  be a pre-Weil cohomology theory over  $k$  with coefficients in  $F$  (22.1.2.2), then for  $X, Y \in \mathbf{SmProj}/k$ ,  $H^*(X \amalg Y) \rightarrow H^*(X) \times H^*(Y)$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Sta]0FHJ.  $\square$

**Def. (22.1.2.6) [Lefschetz Operators].**  $\lrcorner$

### Weil Cohomology Theories

**Prop. (22.1.2.7) [Characterizing Weil Cohomology Theories].** Let  $k, F \in \mathbf{Field}$  and  $H^*$  be a pre-Weil cohomology theory, for  $X \in \mathbf{SmProj}/k$  and  $\Gamma(X, \mathcal{O}_X) = k'$ , the following are equivalent:

- There exists f.m. geometric points  $x_1, \dots, x_r \in X$  s.t.  $H^0(X) \rightarrow H^0(x_1) \oplus \dots \oplus H^0(x_r)$  is injective.
- The map  $H^0(\text{Spec } k') \rightarrow H^0(X)$  is an isomorphism.

If these hold, then  $H^0(X)$  is finite étale over  $F$ . Moreover, if  $X$  is equidimensional of dimension  $d$ , then these are further equivalent to

- The classes of closed points of  $X$  generate  $H^{2d}(X)(d)$  as a module over  $H^0(X)$ .

$\lrcorner$

*Proof:*  $\square$

**Prop. (22.1.2.8) [Non-negativeness].** Let  $k, F \in \mathbf{Field}$  and  $H^*$  be a Weil cohomology theory, then for any  $Y \in \mathbf{SmProj}/k$ ,  $H^i(Y) = 0$  for  $i < 0$ .  $\lrcorner$

*Proof:* If  $H^i(Y) \neq 0$  for some  $Y \in \mathbf{SmProj}/k$  and  $i < 0$ , we may assume  $Y$  is irreducible by (22.1.2.5), and also we may assume  $i = -2j$  is even by changing  $Y$  to  $Y \times Y$  using Künneth formula. Then take  $X = Y \times (\mathbb{P}_k^1)^j$ , then Künneth formula shows

$$H^0(Y) \oplus H^i(Y) \otimes H^2(\mathbb{P}_k^1)^{\otimes j} \subset H^0(X),$$

so  $H^0(X)$  cannot be isomorphic to  $H^0(\text{Spec } \Gamma(X, \mathcal{O}_X)) = H^0(\text{Spec } \Gamma(Y, \mathcal{O}_Y)) \cong H^0(Y)$ .  $\square$

**Prop. (22.1.2.9). ?** Let  $k, F \in \mathbf{Field}$  and  $H^*$  be a Weil cohomology theory, then

- If  $X \in \mathbf{SmProj}^d/k$ , then  $\text{tr}_X \circ \gamma = \text{deg} : \text{CH}^d(X) \rightarrow F$  (9.1.1.27).
- If  $X, Y \in \mathbf{SmProj}/k$ , then  $\text{tr}_{X \times_k Y} = \text{tr}_X \otimes \text{tr}_Y$  via Künneth formula (22.1.2.2). In particular, for  $a \in H^{2\dim X}(X)(\dim X)$ ,  $b \in H^*(\dim Y)$ ,  $\text{pr}_{2,*}(a \otimes b) = (\int_X a) b \in H^*(Y)$ .

$\lrcorner$

*Proof:* 1: This holds for  $X = \text{Spec } k$  by hypothesis, and also for any other  $x \in X(k)$  by pushforward.

2: For any  $x \in X(k), y \in Y(k)$ , by item1,  $\gamma([x \times y]) \in H^{\text{top}}(X \times Y)$  is mapped to  $1 \in F$  via  $\text{tr}_{X \times Y}$ , and it equals  $\gamma([x]) \otimes \gamma([y])$  by (22.1.2.2) as  $[a \times b] = \text{pr}_1^!(a) \cdot \text{pr}_2^!(b)$ . The latter is also mapped to  $1 \in F$  by  $\text{tr}_X \otimes \text{tr}_Y$ .

For the last assertion, notice  $\int_Y \text{pr}_{2,*}(a \otimes b) \cup c = \int_{X \times Y} (a \otimes b) \cup (1 \otimes c) = (\int_X a) \int_Y (b \cup c)$   $\square$

**Prop. (22.1.2.10).** If  $Z \in \mathrm{CH}^*(X)$  satisfies  $mZ$  is algebraically trivial for some  $m \in \mathbb{Z}^\times$ , then  $\gamma_X(Z) = 0$ . ┘

*Proof:* Cf. [Algebraic Cycles and the Weil Conjecture, Kleiman, Prop1.2.1]. □

**Prop. (22.1.2.11) [Lefschetz Trace Formula].** Let  $k, F \in \mathbf{Field}$  and  $H^*$  be a Weil cohomology theory, then for  $X \in \mathrm{SmProj}/k$  and  $\alpha, \beta \in H^{2*}(X \times X)(*)$ , then

$$(\alpha \cdot \beta) = \sum_i (-1)^i \mathrm{tr}(\beta \circ \alpha | H^i(X)).$$

┘

*Proof:* Cf. [Sta]0FH0. ? □

**Cor. (22.1.2.12).** Let  $k, F \in \mathbf{Field}$  and  $H^*$  be a Weil cohomology theory, then for  $X \in \mathrm{SmProj}/k$ ,

$$\sum_{i=0}^{2 \dim X} (-1)^i \dim_F H^i(X) = \deg([\Delta_X] \cdot [\Delta_X]) = \deg(c_d(\mathcal{T}_X) \cap [X]).$$

┘

**Conj. (22.1.2.13) [Betti Numbers].** Let  $k, F \in \mathbf{Field}$ ,  $k = \bar{k}$  and  $H^*$  be a Weil cohomology theory, is it true that for a smooth projective variety over  $k$ , the numbers  $\beta_i = \dim_F H^i(X)$  are independent of  $F$  and the cohomology theory? ┘

*Proof:* ? □

### 3 Pure Motives

**Conj. (22.1.3.1) [Category of Motives as a Universal Cohomology Theory].** There should be a category of motives  $\mathrm{Mot}(k)$  s.t.

- $\mathrm{Mot}(k)$  is a neutral Tannakian category (4.2.1.14).
- There is a functor  $h : \mathrm{SmProj}/k \rightarrow \mathrm{Mot}(k)$ .
- Every correspondence  $X \rightarrow Y$  of degree 0 defines a map  $hX \rightarrow hY$ .
- Every Weil cohomology theory (22.1.2.2) on the  $\mathrm{SmProj}/k$  factors uniquely through  $h$ .

┘

*Proof:* ? □

**Def. (22.1.3.2) [Chow Motives].** The category  $\mathrm{Mot}_{\mathrm{rat}}(k)$  of **Chow motives** over  $k$  consists of triples  $(X, e, m)$  where  $X \in \mathrm{SmProj}/k$ ,  $e$  is an idempotent in  $\mathrm{Corr}_{\mathrm{rat}}^0(X, X)_{\mathbb{Q}}$  and  $m \in \mathbb{Z}$ . And morphisms in  $\mathrm{Mot}_{\mathrm{rat}}(k)$  are defined to be

$$\mathrm{Hom}((X, e, m), (Y, f, n)) = f \circ \mathrm{Corr}_{\mathrm{rat}}^{n-m}(X, Y) \circ e.$$

The category  $\mathrm{Mot}_{\mathrm{rat}}^{\mathrm{eff}}(k)$  of **effective Chow motives** over  $k$  is defined to be the full subcategory of  $\mathrm{Mot}_{\mathrm{rat}}(k)$  consisting of triples  $(X, e, 0)$ .

There is a contravariant functor  $\mathrm{SmProj}/k \rightarrow \mathrm{Mot}_{\mathrm{rat}}^{\mathrm{eff}}(k) : X \mapsto (X, [\Delta_X], 0)$  and  $f \mapsto [\Gamma_f^t]$ . ┘

**Def. (22.1.3.3)[Grothendieck Motives].** The category  $\text{Mot}_{\text{num}}(k)$  of **Grothendieck motives** over  $k$  consist of triples  $(X, e, m)$  where  $X \in \text{SmProj}/k$ ,  $e$  is an idempotent in  $\text{Corr}_{\text{num}}^0(X, X)_{\mathbb{Q}}$  and  $m \in \mathbb{Z}$ . And morphisms in  $\text{Mot}_{\text{num}}(k)$  are defined to be

$$\text{Hom}((X, e, m), (Y, f, n)) = f \circ \text{Corr}_{\text{num}}^{n-m}(X, Y) \circ e.$$

The category  $\text{Mot}_{\text{num}}^{\text{eff}}(k)$  of **effective Grothendieck motives** over  $k$  is defined to be the full subcategory of  $\text{Mot}_{\text{num}}(k)$  consisting of triples  $(X, e, 0)$ .  $\lrcorner$

**Prop. (22.1.3.4).** There is a natural functor  $\text{Mot}_{\text{rat}}^{\text{eff}}(k) \rightarrow \text{Mot}_{\text{num}}^{\text{eff}}(k)$  and  $\text{Mot}_{\text{rat}}(k) \rightarrow \text{Mot}_{\text{num}}(k)$ , by definition(22.1.1.3).  $\lrcorner$

**Def. (22.1.3.5)[Tate Twists and the Lefschetz Motive].** In  $\text{Mot}_{\text{rat}}(k)$  or  $\text{Mot}_{\text{num}}(k)$ , define  $\mathbf{1}(n) = (\text{Spec } k, \text{id}, n)$ , and for any object  $M$ , define  $M(n) = M \otimes \mathbf{1}(n)$ .  $\lrcorner$

$\mathbf{1}(-1)$  is also denoted by  $\mathbb{L}$ , called the **Lefschetz motive**.  $\lrcorner$

**Prop. (22.1.3.6).**  $\text{Mot}_{\text{rat}}(k)$  and  $\text{Mot}_{\text{num}}(k)$  are Karoubian categories(4.8.1.16) and symmetric monoidal categories with identity  $\mathbf{1}(0)$ (22.1.3.5).  $\lrcorner$

*Proof:* The symmetric monoidal structure is given by(22.1.1.5). It is clearly an additive category. To show it is Karoubian, for  $M = (X, e, m) \in \text{Mot}_{\text{rat}}(k)$ ,  $p \in \text{End}(M)$  be an idempotent, then  $p = epe$ ,  $N = (X, p, m) \in \text{Mot}_{\text{rat}}(k)$ , and  $p : N \rightarrow M$  is a morphism.  $\square$

**Prop. (22.1.3.7).**  $h(\mathbb{P}_k^1) = \mathbf{1} \oplus \mathbb{L} \in \text{Mot}_{\text{rat}}(k)$ .  $\lrcorner$

*Proof:* Cf.[Sta]0FGG.  $\square$

**Prop. (22.1.3.8) [Mot<sub>rat</sub>(k) as Inverting  $c_2$ ].** For any  $\mathbb{Q}$ -Karoubian category  $\mathcal{C}$  and a symmetric monoidal functor  $f : \text{Mot}_{\text{rat}}^{\text{eff}}(k) \rightarrow \mathcal{C}$  s.t.  $f(c_2) \in f(\mathbb{P}_k^1)$  is an invertible object,  $f$  factors through uniquely through a symmetric monoidal functor  $\text{Mot}_{\text{rat}}(k) \rightarrow \mathcal{C}$ .  $\lrcorner$

*Proof:* Cf.[Sta]0FGH.  $\square$

**Prop. (22.1.3.9)[Left Duals].** Let  $X \in \text{SmProj}^d/k$ , then  $h(X)(d)$  is a left dual of  $h(X)$ . In particular, every element in  $\text{Mot}_{\text{rat}}(k)$  has a left dual.  $\lrcorner$

*Proof:* Cf.[Sta]0FGI, 0FGJ.  $\square$

**Def. (22.1.3.10)[Chow Groups of Motives].** Let  $M = (X, e, m) \in \text{Mot}_{\text{rat}}(k)$ , the  $i$ -th **Chow group of motives** of  $M$  is defined to be

$$\text{CH}^i(M) = e \circ \text{CH}^{i+m}(X)_{\mathbb{Q}} = \text{Hom}(\mathbb{L}, M).$$

Then each  $\text{CH}^i$  defines a functor  $\text{Mot}_{\text{rat}}(k) \rightarrow \text{Vect}_{\mathbb{Q}}$  via pushforwards  $\square$ .

**Prop. (22.1.3.11) [Manin].** If  $M \in \text{Mot}_{\text{rat}}(k)$  and  $c : M \rightarrow N \in \text{Mot}_{\text{rat}}(k)$  satisfies that: For  $X \in \text{SmProj}/k$ , the map  $c \otimes \text{id} : M \otimes h(X) \rightarrow N \otimes h(X)$  induces isomorphisms on Chow groups, then  $c$  is an isomorphism.  $\lrcorner$

*Proof:* Cf.[Sta]0FGN.  $\square$

**Prop. (22.1.3.12)[Weil Cohomologies and Chow Motives].** Suppose  $k = \bar{k}$  and let  $F \in \text{Field}^0$ , then a classical Weil cohomology over  $k$  with coefficients in  $F$  is equivalent to a  $\mathbb{Q}$ -linear monoidal functor  $G : \text{Mat}_{\text{rat}}(k) \rightarrow \mathcal{C}\text{Ring}^{\text{gr}}/F$  together with an isomorphism  $F[2] \rightarrow G(\mathbf{1}(1))$  s.t.

- $G(h(X)) \subset \mathcal{C}\text{Ring}^{\text{gr} \geq 0}/F$ .
- $\dim_F G^0(h(X)) = 1$ .

$\lrcorner$

*Proof:* Cf.[Sta]0FH3.  $\square$



### Theorems

**Thm. (22.1.3.13)** [Voisin-Voevodsky]. For  $X \in \text{SmProj}/k$ ,  $Z_{\text{alg}}^i(X)_{\mathbb{Q}} \subset Z_{\times}^i(X)_{\mathbb{Q}}$ . ┘

*Proof:* Cf. [Lectures on Pure Motives, Murre]P19. □

**Prop. (22.1.3.14)** [Mot<sub>rat</sub>( $k$ ) is not Abelian]. If  $\text{char } k = 0$  or  $\text{char } k$  is not algebraic over its base field, then Mot<sub>rat</sub>( $k$ ) is not an Abelian category. ┘

*Proof:* ? Cf. [Sch94]P11. □

**Thm. (22.1.3.15)** [Mot<sub>num</sub>( $k$ ) is Abelian Semisimple, Jannsen]. Suppose  $k = \bar{k}$ , let  $\sim$  be an adequate equivalence relation (22.1.1.2), and  $F \in \text{Field}^0$ , then the following are equivalent:

- Mot <sub>$\sim$</sub> ( $k$ ) is an Abelian semisimple category.
- $\sim$  is the numerical equivalence relation.
- For all  $X \in \text{SmProj}/k$ ,  $\text{Corr}_{\sim}^0(X, X)_F$  is a f.d. semisimple  $F$ -algebras.

┘

*Proof:* Cf. [Lectures on Pure Motives, Murre]P39 or [Jan92]. □

**Prop. (22.1.3.16)** [Mot<sub>num</sub>( $k$ ) is a Tannakian Category]. The usual symmetric monoidal structure on Mot<sub>num</sub>( $k$ ) is not a Tannakian category, because the trace of  $\Delta_X$  on  $h(X)$  may be negative. But if the Künneth standard conjectures (22.1.5.6) holds ad we change the monoidal structure s.t. the canonical isomorphism is given by

$$h^i(X) \otimes h^i(X) \cong (-1)^{ij} h^j(X) \otimes h^i(X),$$

then Mot<sub>num</sub>( $k$ ) becomes a Tannakian category (4.2.1.13), and it is neutral (4.2.1.14) if  $\text{char } k = 0$ . ┘

*Proof:* □

### Realizations

**Cor. (22.1.3.17)** [Realization Functors]. Suppose Mot( $k$ ) exists (22.1.3.1), then there should be **realization functors**

- $\text{Real}_p : \text{Mot}(k) \rightarrow \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_k)$  s.t.  $H^i(-, \mathbb{Q}_p) = \text{Real}_p \circ h^i$ ,
- $\text{Real}_{\iota, \text{Betti}} : \text{Mot}(k) \rightarrow \text{Hdg}_{\text{int}}$  s.t.  $H_{\iota, \text{Betti}}^i(-, \mathbb{Z})_{\text{lf}} = \text{Real}_{\iota} \circ h^i$ , for each embedding  $\iota : k \rightarrow \mathbb{C}$ .
- $\text{Real}_{\text{dR}} : \text{Mot}(k) \rightarrow \text{Fil}_K$  s.t.  $H_{\text{dR}}^i(-) = \text{Real}_{\text{dR}} \circ h^i$ .

And comparison isomorphisms

$$I_{\ell} : \text{Real}_{\iota} \otimes \mathbb{Q}_{\ell} \cong \text{Real}_{\ell} : \text{Mot}(k) \rightarrow \text{Vect} / \mathbb{Q}_{\ell}$$

$$I_{\text{dR}} : \text{Real}_{\iota, \text{Betti}} \otimes \mathbb{C} \cong \text{Real}_{\text{dR}} \otimes_{k, \iota} \mathbb{C} : \text{Mot}(k) \rightarrow \text{Fil}_{\mathbb{C}}^{\bullet}$$

and the Tate twist (22.1.3.5) acts on the relations via

- $\text{Real}_{\iota, \text{Betti}}(M(r)) = (\Lambda, \Lambda_{\mathbb{C}}^{p, q})$  is given as follows:

$$\Lambda = (2\pi i)^r \text{Real}_{\iota, \text{Betti}}(M), \quad \Lambda_{\mathbb{C}}^{p, q} = (2\pi i)^r \text{Real}_{\iota, \text{Betti}}(M)^{p-r, q-r}.$$

- $\text{Real}_p(M(r)) = \text{Real}_p(M)(r) = \text{Real}_p(M) \otimes \mu_p^{\otimes r}$ .
- $\text{Real}_{\text{dR}}(M(n)) = \text{Real}_{\text{dR}}(M)$ .

┘

*Proof:* It suffices to show that these are all Wei cohomology theories, ? □

**Prop. (22.1.3.18).** By (16.11.6.7), if  $\iota : k \rightarrow \mathbb{C}$  factors through  $\mathbb{R}$ , then

$$I_{\text{dR}} : \text{Real}_{\iota, \text{Betti}} \otimes \mathbb{C} \cong \text{Real}_{\text{dR}} \otimes_{k, \iota} \mathbb{C}$$

identifies  $\mathfrak{c}^* \otimes \mathfrak{c}$  on the LHS with  $\mathfrak{c}$  on the RHS. ┘

**Prop. (22.1.3.19)[Tate Conjecture and Motives].** The Tate conjecture (22.3.2.2) and Grothendieck-Serre conjecture (9.4.6.39) hold iff the functor  $\text{Mot}(k) \rightarrow \mathcal{R}\text{ep}_{\mathbb{Q}_p}(\text{Gal}_K) : M \mapsto H_{\text{ét}}^*(M)$  is fully faithful.

┘

*Proof:* ? □

**Prop. (22.1.3.20).** The functor  $\text{Mot}(k) \rightarrow \text{Hdg}(M)$  is fully faithful iff the Hodge conjecture holds for  $k$ . ┘

*Proof:* Cf. [Hodge-de Rham Structure and Periods of Automorphic Forms, Michael Harries, in Motives 2] Prop 1.2. ? □

### Motives of Curves and Jacobians

**Prop. (22.1.3.21).** Let  $C, C' \in \text{SmProjCur}/k$  with Jacobians  $J, J' \in \text{AbVar}/k$ , then

$$\text{Hom}_{\sim}(h^1(C), h^1(C')) = \text{Hom}(J, J') \otimes \mathbb{Q}.$$

And

$$\text{Hom}_{\sim}(\mathbb{L}, h^1(X)) = \begin{cases} 0 & , \sim = \text{num}, \text{hom} \\ J(k) \otimes \mathbb{Q} & , \sim = \text{rat} \end{cases}.$$

┘

*Proof:* ? □

**Cor. (22.1.3.22)[Abelian Motives].** The full subcategory of  $\text{Mot}_{\text{rat}}(k)$  whose objects are direct summands of the form  $h^1(X)$  for some  $X \in \text{SmProjVar}^1/k$  is equivalent to  $\text{IsogAbVar}_k$ . ┘

### Dimension 1 and Codimension 1 Motives

#### 4 André's Pure Motives

André's category of pure motives is the category of motives modulo homological equivalences with the Lefschetz involutions added. It is known that this is a semisimple Tannakian category.

## 5 Standard Conjectures

References are [Kle94] and [Standard conjectures on algebraic cycles, Grothendieck].

**Remark (22.1.5.1).** The standard conjectures over  $k$  are necessary conditions to make the  $\text{Mot}_{\text{num}}(k)$  a semistable universal cohomology theory(22.1.3.1).  $\lrcorner$

**Conj. (22.1.5.2) [Standard Conjecture D,  $\sim_{\text{hom}} \iff \sim_{\text{num}}$ ].** Suppose  $k = \bar{k}$  and let  $F \in \text{Field}$  and  $H^*$  be a Weil cohomology theory with coefficient in  $F$ (22.1.2.2), then for  $X \in \text{SmProj}/k$ , a cycle  $Z \in \text{CH}^*(X)$  is numerically trivial iff  $\gamma_X(Z) = 0$ . Equivalently, every Weil cohomology factors through  $\text{Mot}_{\text{num}}(k)$ .

This is an existence theorem for algebraic cycles.  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj.Cor. (22.1.5.3) [Lieberman].** If  $\text{char } k = 0$ , this conjecture is true for varieties of dimension  $\leq 4$  and for Abelian varieties(which is implied by Hodge conjecture).  $\lrcorner$

*Proof:* ?  $\square$

### Lefschetz Standard Conjectures

**Conj. (22.1.5.4) [Lefschetz Standard Conjecture].** Suppose  $k = \bar{k}$  and let  $F \in \text{Field}$  and  $H^*$  be a Weil cohomology theory with coefficient in  $F$ (22.1.2.2),  $X \subset \text{SmProj}^d/k$ , then the Lefschetz operator  $L : \text{CH}^i(X) \rightarrow \text{CH}^{i+2}(X)$  satisfies for any  $i \leq d$ ,

$$L^{d-i} : H^i(X) \rightarrow H^{2d-i}(X)$$

is an isomorphism by(22.1.2.2). and for  $i \leq d$  we can define operators

$$\Lambda = (L^{d-i+2})^{-1} \circ L \circ (L^{d-i}) : H^i(X) \rightarrow H^{i-2}(X), \quad \Lambda = L^{n-i} \circ L \circ (L^{d-i+2})^{-1} : H^{2d-i+2}(X) \rightarrow H^{2n-i}(X),$$

then we have the following conjectures:

$A(X, L)$  : The restriction of  $L^{d-i} : \gamma_X(\text{CH}^i(X)) \rightarrow \gamma_X(\text{CH}^{2d-i}(X))$  is an isomorphism.

$B(X)$  :  $\Lambda$  is induced from a correspondence from  $X$  to  $X$  of degree  $-1$ .

$B^c(X)$  : The operator  $\Lambda^c$ (22.1.2.6) is algebraic.  $\lrcorner$

**Conj.Cor. (22.1.5.5) [Relations between Lefschetz Standard Conjectures].** Situation as in(22.1.5.4), we have the following relations between the Lefschetz standard conjecture:

- 
- The following are equivalent:

$$A(X \times X, L \otimes 1 + 1 \otimes L), \quad B(X), \quad B(X \times X).$$

$\lrcorner$

**Conj.Cor. (22.1.5.6) [Künneth Standard Conjecture].** Assume the conjecture  $D$ (22.1.5.2) holds, then it is conjectured that for any  $X \in \text{SmProj}^d/k$ , there exists a decomposition of  $\Delta_X \in \text{Corr}_{\text{num}}^d(X \times X)$  into orthogonal idempotents

$$\Delta_X = h^0(X) + \dots + h^{2d}(X)$$

s.t. it induces the decomposition

$$H^*(X) = H^0(X) \oplus H^1(X) \oplus \dots \oplus H^{2d}(X)$$

for any Weil cohomology theory  $H^*$ .

By Deligne's purity theorem, any homomorphism  $H_{\text{ét}}^i(X) \rightarrow H_{\text{ét}}^j(Y)$  induced by an algebraic cycle is zero unless  $i = j$  ?, thus such a decomposition, if exists, would be unique. ┘

*Proof:* □

**Remark (22.1.5.7) [Murre].** A **Chow-Künneth decomposition** of  $\Delta_X$  is defined to be a decomposition

$$\Delta_X = \sum_{i=0}^{2d} \pi_i \in \text{CH}^{\dim X}(X \times X, \mathbb{Q})$$

that induces the decomposition  $\Delta_X = h^0(X) + \dots + h^{2\dim X}(X)$  in  $\text{CH}_{\text{num}}^{\dim X}(X \times X)$ .

Murre conjectured that such a decomposition exists with certain properties, which is equivalent to the conjectural existence of Beilinson-Bloch filtration on  $\text{CH}^*(X)$ . ? ┘

**Thm& Conj.Cor. (22.1.5.8).** the Lefschetz standard conjecture is proved for

- Curves.
- Abelian varieties and Hilbert schemes of smooth surfaces (where the multiplication map can be used to decompose motives),
- A generalized flag variety  $X = G/P$ .

The Künneth standard conjecture is proved for

- Varieties over finite fields (where the Frobenius map can be used to decompose motives).

┘

*Proof:* ? □

### Hodge Standard Conjectures

**Conj. (22.1.5.9) [Hodge Standard Conjecture].** Suppose  $k = \bar{k}$  and let  $F \in \text{Field}^0$  and  $H^*$  be a Weil cohomology theory over  $k$  with coefficients in  $F$  (22.1.2.2). If  $X \in \text{SmProj}^d/k$ ,  $P^k(X) = \ker(L|H^k(X))$  be the set of primitive cohomology classes,  $A^i(X) = \gamma_X(\text{CH}^i(X))$ , then for any  $i < d/2$ , the  $\mathbb{Q}$ -valued pairing on  $A^i(X) \cap P^{2i}(X)$ :

$$(x, y) \mapsto (-1)^i \langle L^{r-2i}x, y \rangle$$

is positive definite.

Equivalently, the pairing on  $A^i(X) : (x, y) \mapsto \langle x, *y \rangle$  (22.1.2.6) is positive-definite. ┘

*Proof:* □

**Thm& Conj.Cor. (22.1.5.10).** This conjecture is true in characteristic 0 by Hodge theory. This conjecture is shown for surfaces by Grothendieck(1958). This conjecture is shown for Abelian varieties of dimension 4 by Ancona(2020). ┘

*Proof:* ? □

**Prop. (22.1.5.11) [Conjecture  $D$  and Lefschetz Conjectures].** The conjecture  $D(22.1.5.2)$  implies the Lefschetz standard conjecture  $B(X)(22.1.5.9)$ . And the Hodge standard conjecture plus the Lefschetz standard conjecture  $B(X)$  imply conjecture  $D$ .  $\lrcorner$

*Proof:* For  $X \in \text{SmProj}^d/k$ , If  $B(X)$  holds, then by (22.1.5.5),  $A(X, L)$  holds, and  $*$  clearly maps  $\gamma_X(\text{CH}^i(X))$  into  $\gamma_X(\text{CH}^{2d-i}(X))$ .  $\square$

**Prop. (22.1.5.12) [Hodge Conjecture and Lefschetz Standard Conjecture].** If  $\text{char } k = 0$ , the Hodge conjecture (16.11.3.1) implies the Lefschetz and Künneth standard conjectures.  $\lrcorner$

*Proof:*  $\square$

**Prop. (22.1.5.13).** If the Lefschetz standard conjecture and the Hodge standard conjecture hold, then for any  $X \in \text{SmProj}/k$ ,

- $\text{Corr}_{\text{num}}^*(X, X)$  is a semisimple  $\mathbb{Q}$ -algebra.
- (Generalized Riemann Hypothesis) If  $k$  is a finite field with Frobenius  $\varphi$ , then for any Weil cohomology  $H^*$ , the Frobenius action  $\Phi$  on  $H^i(X)$  is semisimple with characteristic polynomial in  $\mathbb{Z}[T]$ , and eigenvalues of absolute value  $q^{i/2}$ .

$\lrcorner$

*Proof:* Cf. [Kle94] P19.  $\square$

**Prop. (22.1.5.14) [Tate Conjecture and Standard Conjectures].** If  $k$  is f.g., the Tate conjecture implies Lefschetz, Künneth standard conjectures (22.1.5.6) and conjecture  $D(22.1.5.2)$  for  $\ell$ -adic étale cohomology.  $\lrcorner$

*Proof:*  $\square$

## 6 Mixed Motives

**Conj. (22.1.6.1) [Mixed Motives, Lichtenbaum(84)/Beilinson[Bei87]/Lichtenbaum(90)].** The motives are conjectured to be the subcategory of semisimple objects in a larger category of **mixed motives**.

There is at present no definition of a category of mixed motives, but several mathematicians have constructed triangulated categories that are candidates to be its derived category; it remains to define a  $T$ -structure on one of these categories whose heart is the category of mixed motives. ?  $\lrcorner$

Beilinson conjectured that his theory would have deep links to Quillen's algebraic  $K$ -theory.

### Motivic t-Structures

References are [Remarks on Grothendieck's standard Conjecture, Beilinson].

### Beilinson-Bloch-Murre Filtration Conjecture

**Conj. (22.1.6.2) [Beilinson].** For  $X \in \text{Var}/k$ , there exists a descending filtration  $\text{Fil}^\bullet \text{CH}^\bullet(X)_{\mathbb{Q}}$  s.t.

- $\text{Fil}^0 \text{CH}^j(X)_{\mathbb{Q}} = \text{CH}^j(X)_{\mathbb{Q}}$ ,  $\text{Fil}^1 \text{CH}^j(X)_{\mathbb{Q}} = \text{CH}^j(X)_{\mathbb{Q}} = \text{CH}^j(X)_{\text{hom } 0, \mathbb{Q}}$ .
- this filtration is multiplicative.
- $\text{gr}^v \text{CH}^j(X)_{\mathbb{Q}}$  only depends on the motive modulo homological equivalence,  $h^{2j-v}(X)$ .

- $\mathrm{Fil}^{j+1} \mathrm{CH}^j(X)_{\mathbb{Q}} = 0$ .

And moreover, if  $k = \mathbb{Q}$ , then  $\mathrm{Fil}^2 = 0$ . ┘

*Proof:* □

## 7 Mixed Tate Motives

References are [Brown].

### Multiple Zeta Values

**Def. (22.1.7.1) [Multiple Zeta Values].** A **multiple zeta value** is a number in  $\mathbb{R}_+$  of the form

$$\zeta(n_1, \dots, n_r) = \sum_{0 < k_1 < \dots < k_r} \frac{1}{k_1^{n_1} \dots k_r^{n_r}} \in \mathbb{R}_+$$

where  $r \in \mathbb{Z}_+$  and  $n_1, \dots, n_r \in \mathbb{Z}_+, n_r \geq 2$ . The **weight** of the tuple  $(n_1, \dots, n_r)$  is defined to be  $w = \sum_i n_i$ , and the **depth** is defined to be  $r$ . ┘

**Prop. (22.1.7.2) [Space of Multiple Zeta Values].** Let  $\mathcal{Z} \subset \mathbb{R}$  be the  $\mathbb{Q}$ -subspace generated by multiple zeta values, then  $\mathcal{Z}$  is closed under multiplication. ┘

*Proof:* ? □

**Prop. (22.1.7.3).** ┘

### Periods and Multiple Zeta Values

**Thm. (22.1.7.4).** The periods of mixed Tate motives over  $\mathbb{Z}$  lies in  $\mathcal{Z}[\frac{1}{2\pi i}]$  (22.1.7.2). ┘

*Proof:* □

**Thm. (22.1.7.5).** Every multiple zeta value of weight  $N \in \mathbb{Z}_+$  is a  $\mathbb{Q}$ -linear combination of

$$\{\zeta(a_1, \dots, a_r) : a_i \in \{2, 3\}, a_1 + \dots + a_r = N\}.$$

┘

*Proof:* □

## 8 Motivic Galois Theory

**Thm. (22.1.8.1) [André].** The motivic Galois group attached to the subcategory  $\mathrm{Mot}^{\mathrm{abCM}}$  of category of André's motives consisting of CM Abelian varieties satisfies

$$G^{\mathrm{abCM}} \cong \mathbb{S}_{\mathrm{Serre}}$$

where  $\mathbb{S}_{\mathrm{Serre}}$  is Serre's pro-torus characterizing CM-types. ┘

*Proof:* □

### Mixed Motivic Galois Groups

**Def.(22.1.8.2)** [**Mixed Motivic Galois Groups**]. If the conjectural category of mixed motives  $\text{MMot}(K)$  exist [\(22.1.6.1\)](#), then the **mixed motivic Galois group**  $G_{\text{Mot},K}$  is defined to be the Tannakian Galois group of the fiber functor  $H_{\text{Betti}} : \text{MMot}(K) \rightarrow \text{Vect}_{\mathbb{C}}$ .  $\lrcorner$

## 9 Periods

**Prop.(22.1.9.1).**  $\lrcorner$

## 22.2 $\mathbb{P}^1 \setminus \{0, 1, \infty\}$

### 1 Dessins d'Enfant

**Def. (22.2.1.1) [Dessins d'Enfant].** A **dessin d'enfant** is a tuple  $(G, \{s_v\}_{v \in V(G)})$  where  $G \in \mathbf{Graph}^{\text{fin}}$  is a connected bipartite simple graph and for each  $v \in V(G)$ ,  $s_v$  is a cyclic ordering of the edges meeting  $v$ .  $\lrcorner$

**Prop. (22.2.1.2) [Dessins d'Enfant and Riemann Surfaces].** If  $X$  be a Riemann surface and  $f : X \rightarrow \mathbb{P}^1(\mathbb{C})$  is a map finite étale map that is unramified outside  $\{0, 1, \infty\}$ . Then we get a dessin d'enfant  $G_X$  by taking  $V(G) = f^{-1}(\{0, 1\})$ , and  $E(G) = f^{-1}([0, 1])$ . Moreover, the cyclic ordering comes from the natural orientation of  $\mathbb{P}^1(\mathbb{C})$ . We called vertices in  $f^{-1}(0)$  the **black vertices** and vertices in  $f^{-1}(1)$  the **white vertices**.

To show that  $G$  is connected, notice there is an action of  $\pi(\frac{1}{2}, \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\})$  on the edges of  $X$ . Where the loop around 0 rotates the edges around each black vertex, and the loop around 1 rotates the edges around each white vertex. Then for any two  $u, v \in V(G)$ , take a path  $\gamma$  in  $X$  connecting  $u$  to  $v$  that avoids  $V(G)$ , then the image of  $\gamma$  is a path in  $\mathbb{P}^1(G)$ . If we “shrink” it in  $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ , we will get a path that loops around 1 or 0 several times, and this indicates a path from  $u$  to  $v$ .

Conversely, for any dessin d'enfant  $G$ , we can use the rotations at black and white vertices to get a action of the free group  $F_2$  on the edges. The action of transitive because  $G$  is connected. Then the stabilizer of either edge is a finite index subgroup, which are all conjugate. By the Galois theory of covering maps, we get a covering  $X \setminus V(G) \rightarrow \mathbb{P}^1 \setminus \{0, 1, \infty\}$ . And it extends to a map  $X \rightarrow \mathbb{P}^1$ .  $\lrcorner$

**Thm. (22.2.1.3) [Belyĭ [Bel79]].** Let  $\overline{\mathbb{Q}} \subset k \in \mathbf{Field}$  and  $X \in \mathbf{SmPrprVar}^1/k$ , then  $X$  is defined over  $\overline{\mathbb{Q}}$  iff it is branched over  $\mathbb{P}^1$  with three branching points. Such a map is called a **Belyĭ map**.

In particular, every algebraic curve over  $\overline{\mathbb{Q}}$  contains an embedded dessin d'enfant by (22.2.1.2).  $\lrcorner$

*Proof:* If  $X$  is defined over  $\overline{\mathbb{Q}}$ , then by (7.13.1.4)  $X$  is projective thus there is a rational map  $f : X \rightarrow \mathbb{P}_{\overline{\mathbb{Q}}}^1$ .

Let  $S \subset \mathbb{P}^1(\overline{\mathbb{Q}})$  be the critical values of  $f$ , then we can find a map  $\varphi : \mathbb{P}_{\overline{\mathbb{Q}}}^1 \rightarrow \mathbb{P}_{\overline{\mathbb{Q}}}^1$  s.t. the critical values of  $\varphi \circ f$  are contained in  $\mathbb{P}^1(\overline{\mathbb{Q}})$ : Enlarge  $S$  s.t.  $S$  is  $\text{Gal}_{\overline{\mathbb{Q}}}$ -invariant.

We use induction on  $\#S$  to show that there exists a map  $\varphi : \mathbb{P}_{\overline{\mathbb{Q}}}^1 \rightarrow \mathbb{P}_{\overline{\mathbb{Q}}}^1$  s.t. the critical values of  $\varphi \circ f$  are in  $\mathbb{P}^1(\mathbb{Q})$ . If  $\#S \leq 1$ , then  $S \subset \mathbb{Q}$  and we are done. If  $\#S > 1$ , then we can define  $P(X) = \prod_{s \in S \setminus \{\infty\}} (X - s) \in \mathbb{Q}[X]$ . Let  $S_1 = P(\{r \in \overline{\mathbb{Q}} : P(r) = 0\})$  be the set of critical values of  $P$ , which is also  $\text{Gal}_{\overline{\mathbb{Q}}}$ -invariant and  $\#S_1 < n - 1$ . Then we can use induction hypothesis to find a  $\varphi'$  that maps  $S$  into  $\mathbb{P}^1(\mathbb{Q})$  and also has critical values in  $\mathbb{Q}$ . Then  $\varphi' \circ \varphi$  satisfies the requirement. Then the induction is complete.

We then find a map  $g : X \rightarrow \mathbb{P}_{\overline{\mathbb{Q}}}^1$  with the set of critical values  $T \subset \mathbb{P}^1(\mathbb{Q})$ . Then we use induction on  $\#T$  to show that there exists a function  $\psi : \mathbb{P}_{\overline{\mathbb{Q}}}^1 \rightarrow \mathbb{P}_{\overline{\mathbb{Q}}}^1$  s.t. the critical values of  $\psi \circ g$  are in  $\{0, 1, \infty\}$ . If  $\#T \leq 3$ , then we can choose  $\psi$  to be a fractional transformation. And if  $\#T > 3$ , we can use a fractional transformation to make  $\{0, 1, \infty\} \subset T$  and  $T$  also contains a point  $\frac{m}{m+n}$  where  $m, n \in \mathbb{Z}_+$ . Consider the polynomial

$$p_{m,n}(z) = \frac{(m+n)^{m+n}}{m^m n^n} z^m (1-z)^n \in \mathbb{Q}[X],$$



then it maps  $\{0, 1, \frac{n}{m+n}, \infty\}$  into  $\{0, 1, \infty\}$  and it has critical values  $\{0, 1, \infty\}$ . Then the set  $T'$  of critical values of  $p_{m,n} \circ g$  has cardinality smaller than  $T$ , which means we can use induction hypothesis to find  $\psi$  s.t.  $\psi \circ p_{m,n} \circ g$  has critical values in  $\{0, 1, \infty\}$ .

Conversely, if  $X$  is branched over  $\mathbb{P}^1$  with three branching points, composing with a fractional transformation, we may assume that  $\varphi : X \rightarrow \mathbb{P}^1$  is finite and branched over  $\{0, 1, \infty\}$ . Then for any  $\sigma \in \text{Aut}(\mathbb{C})$ ,  $\varphi^\sigma$  is also finite and branched over  $\{0, 1, \infty\}$ . But there are only f.m. isomorphism of branched covers with the same branching points by (7.13.1.41). Thus there is a finite subgroup  $H \leq \text{Aut}(\mathbb{C})$  s.t.  $X^\sigma \cong X$  for any  $\sigma \in H$ . By (3.2.8.13),  $\mathbb{C}^H/\mathbb{Q}$  is a finite extension, thus the moduli field of  $X$  is contained in  $\overline{\mathbb{Q}}$ . So, by (10.9.1.2),  $X$  is defined over  $\overline{\mathbb{Q}}$ .  $\square$

**Cor. (22.2.1.4).** A compact Riemann surface  $X$  is defined over  $\overline{\mathbb{Q}}$  iff  $X = \overline{\Gamma \backslash \mathcal{H}}$  for some subgroup  $\Gamma \leq \text{PSL}(2; \mathbb{Z})$  of finite index.  $\lrcorner$

*Proof:* This follows from the fact that  $\mathcal{H}$  is a universal covering space of  $\mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$  with fundamental group  $\text{PSL}(2; \mathbb{Z})$ .  $\square$

### Galois Theory of Dessins

How to effectively whether two dessins belong to the same  $\text{Gal}_{\mathbb{Q}}$ -orbit?

## 2 Mixed Motives over $\mathbb{Z}$

## 3 Multiple Zeta Functions

## 22.3 Tate Conjecture & Hodge Conjecture

References are [Tat94], [Tat65b], [Recent Progress on the Tate Conjecture, Totaro] and [Tate Conjecture over Finite Fields, Milne].

**Notation (22.3.0.1).**

- Let  $k \in \mathbf{Field}$  that is f.g. over its prime field.
- Let  $k^s$  be a separable closure of  $k$ .
- Let  $\ell \in \mathbf{Prime} \setminus \{\text{char } k\}$ .

┘

### 1 Hodge-Tate Classes and Monodromy Groups

**Def. (22.3.1.1) [Hodge-Tate Structure].** A **Hodge-Tate structure** (over  $\mathbb{Q}$ ) is a (pure) Hodge structure  $V \in \mathbf{Hdg}_{\mathbb{Q}}$  together with a continuous action of  $\text{Gal}_{\mathbb{Q}}$ -action on  $V \otimes \mathbb{Q}_{\ell}$ . ┘

**Def. (22.3.1.2) [ $\ell$ -adic Monodromy Groups].** For  $(V, \rho) \in \mathbf{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_k)$ , let  $G_{\ell}(V) \subset \text{GL}(V)_{\mathbb{Q}_{\ell}}$  be the Zariski closure of the image of  $\rho$ , called the  **$\ell$ -adic Monodromy group** of  $V$ . And also define  $H_{\ell}(V) = (G_{\ell}(V) \cap \text{SL}(V))^0$ .

For  $X \in \mathbf{SmProjVar}/k$  and  $r \in \mathbb{N}$ , define the corresponding  $\ell$ -adic monodromy group of  $X$  to be  $G_{\ell}^r(X) = G_{\ell}(H_{\text{ét}}^r(X, \mathbb{Q}_{\ell}))$ . ┘

**Thm. (22.3.1.3) [Serre].** If  $k = F \in \mathbf{NField}$ ,  $X \in \mathbf{AbVar}/F$  and  $V = V_{\ell}(X) \in \mathbf{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_F)$ , then

$$\ker(\text{Gal}_F \rightarrow G_{\ell}(V)/G_{\ell}(V)^0)$$

is independent of  $\ell$ . And there exists a finite extension  $F'/F$  s.t.  $\text{Gal}_{F',\ell} = \text{Gal}_{F,\ell}^0$  for any  $\ell$ . ┘

*Proof:* Cf. [Lettres a Ken Ribet, in Oeuvres, Serre]. □

**Def. (22.3.1.4) [Tate Classes].** For  $X \in \mathbf{SmProjVar}/k$  and  $r \in \mathbb{N}$ , the space  $\text{Tate}^r(X, \ell)$  of **Tate classes** of degree  $m$  is defined to be the subspace of  $H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_{\ell})$  consisting by the elements whose stablizer in  $\text{Gal}_k$  is open. ┘

### 2 Tate Conjectures

*“This stuff is too beautiful not to be true.”*

–J. Tate

**Prop. (22.3.2.1) [ $\ell$ -adic Cycle Classes].** Let  $X \in \mathbf{SmProjVar}/k$ , then there exists a cycle map

$$c^r : \text{CH}^r(X_{k^s}) \rightarrow H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_{\ell}(r)),$$

as étale cohomology is a Weil cohomology theory ?. Let  $\text{Ch}_{\text{hom},\ell}^r(X_{k^s})$  be the image, and  $\text{Ch}_{\text{hom},\ell}^r(X)$  be the image of  $\text{CH}^r(X) \subset \text{CH}^r(X_{k^s})$  under this map. Then there are maps

$$\text{Ch}_{\text{hom},\ell}^r(X_{k^s}) \otimes \mathbb{Q}_{\ell} \subset H_{\text{ét}}^{2r}(\overline{X}, \mathbb{Q}_{\ell}(r)), \quad \text{Ch}_{\text{hom},\ell}^r(X) \otimes \mathbb{Q}_{\ell} \subset H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_{\ell}(r))^{\text{Gal}_k}.$$

┘

**Conj. (22.3.2.2) [Tate].** Situation as in (22.3.2.1), then

$T^r(X/k, \ell)$ :  $\text{Ch}_{\text{hom}, \ell}^r(X) \otimes \mathbb{Q}_\ell = H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_\ell(r))^{\text{Gal}_k}$ .

$E^r(X/k, \ell)$ :  $\text{Ch}_{\text{hom}, \ell}^r(X) \cong \text{Ch}_{\text{num}}^r(X_{k^s})$ . In particular,  $\text{Ch}_{\text{hom}, \ell}^r(X)$  doesn't depend on  $\ell$ .

$SS^r(X/k, \ell)$ :  $\text{Gal}_F$  acts semisimply on  $H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_\ell(r))$ . (If  $\text{char } k = 0$ , this means that  $G_\ell(X)^0$  is reductive).

$S^r(X/k, \ell)$ : The map  $H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_\ell(r))^{\text{Gal}_k} \rightarrow H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_\ell(r))_{\text{Gal}_k}$  is bijective. (If  $\#k < \infty$ , this means that 1 is a semisimple eigenvalue for  $\text{Fr}_X$ ).

┘

*Proof:*

□

**Thm& Conj. Cor. (22.3.2.3).**

1.  $T^r(X/k, \ell)$  holds trivially for  $r \in \{0, \dim X\}$ . In particular,  $T^*(X/k, \ell)$  holds for  $X \in \mathcal{C}_{\text{ur}}^{\text{SmPrpr}}/k$ .
2.  $T^1(A/k, \ell)$  holds for any  $A \in \text{AbVar}/k$ .
3. If  $\text{char } k = 0$ , then  $T^*(A/k, \ell)$  holds for any  $\ell \in \text{Prime}$  and  $A \in \text{AbVar}/F$  whose simple factors are all of dimension  $\leq 2$ .
4. If  $k = \mathbb{Q}$ , then  $T^*(X/k, \ell)$  holds for any  $\ell \in \text{Prime}$  and  $X$  a K3 surface over  $\mathbb{Q}$ .
5. If  $\text{char } k = F \in \text{NField}$ , then  $T^*(A/F)$  holds for any  $A \in \text{AbVar}^g/F$  when  $g \in \text{Prime} \cup \{1\}$ , and largely true for  $g = 4$  by [Hodge classes and Tate classes on simple abelian Fourfolds, Moonen-Zarhin1995].
6.  $T^1(X/k, \ell)$  holds for Hilbert modular surfaces.
7.  $T^1(X/k, \ell)$  holds for many quaternionic Shimura surface.
8.  $T^1(X/k, \ell)$  holds for Picard modular surface.
9.  $T^1(X/k, \ell)$  holds for Siegel modular threefold.

┘

*Proof:*

1. trivial.
2. Cf.[Families of Motives and the Mumford–Tate Conjecture, Moonen]P269 or using the Rosati involution in [Abelian Varieties, Milne]Prop14.2?
3. Cf.[Li-Zhang].? This is deduced from recent progress on Hodge conjectures and Mumford-Tate conjectures.
4. This follows from (22.3.5.1).
- 5.
6. Cf.[Period relations and the Tate conjecture for Hilbert modular surfaces].
7. ?
8. Cf.[Tate classes and arithmetic quotients of the two-balls].
9. Cf.[Tat94]P80.

□

**Conj. (22.3.2.4) [Integral Tate Conjecture].** Situation as in (22.3.2.1), we may ask if

$$\mathrm{Ch}_{\mathrm{hom}, \ell}^r(X) \otimes \mathbb{Z}_\ell = H_{\mathrm{et}}^{2r}(X_{k^s}, \mathbb{Z}_\ell(r))^{\mathrm{Gal}_k}.$$

Cf. [Tate Conjecture, Milne, P3]. ┘

*Proof:* □

**Conj. (22.3.2.5) [Integrated Tate Conjectures].** If  $k'/k$  is a finite Galois extension, then  $T^r(X/k', \ell)$  implies  $T^r(X/k, \ell)$  by Galois descent, so we can state the integrated version of Tate's conjecture: For any  $\Omega \in \mathbf{Field}$  s.t.  $\Omega^{\mathrm{sep}} = \Omega$  and  $X \in \mathbf{SmProjVar}/\Omega$ , the cycle class map

$$\mathrm{cl} : \mathrm{CH}^*(X) \otimes \mathbb{Q}_\ell \rightarrow \mathrm{Tate}^*(X, \ell).$$

is surjective. This is equivalent to  $T^r(X/k, \ell)$  for all f.g. subalgebra of  $\Omega$  (Cf. [Families of Motives and the Mumford–Tate Conjecture, Moonen]P286). ┘

*Proof:* □

**Prop. (22.3.2.6) [Moonen].** If the integrated Tate conjecture (22.3.2.5) holds whenever  $\mathrm{char} k = 0$  and  $X \in \mathbf{SmProjVar}/k$ , then  $SS^*(X/k, \ell)$  holds whenever  $\mathrm{char} k = 0$  and  $X \in \mathbf{SmProjVar}/k$ . ┘

*Proof:* Cf. [A remark on the Tate conjecture, Moonen] ?. □

**Thm. (22.3.2.7).**  $(SS^*(A/k), \ell)$  is true for any Abelian variety  $A \in \mathbf{AbVar}/k$ . ┘

*Proof:* □

**Prop. (22.3.2.8) [( $E^1(X/k)$ ) Holds].** The Kummer exact sequence of étale sheaves on  $X_{k^s} : 0 \rightarrow \mu_{\ell^n} \rightarrow \mathbb{G}_m \xrightarrow{\ell^n} \mathbb{G}_m \rightarrow 0$  induces an injection  $\mathrm{Pic}(\overline{X})/(\ell^n) \hookrightarrow H^2(\overline{X}, \mu_{\ell^n})$ . Taking limits gives an injection  $\mathrm{Pic}(\overline{X}) \otimes \mathbb{Q}_\ell \hookrightarrow H^2(\overline{X}, \mathbb{Q}_\ell(1))$ . And  $c^1$  is just

$$c^1 : Z^1(\overline{X}) \rightarrow \mathrm{Pic}(\overline{X}) \rightarrow \mathrm{Pic}(\overline{X}) \otimes \mathbb{Q}_\ell \hookrightarrow H_{\mathrm{et}}^2(\overline{X}, \mathbb{Q}_\ell(1)).$$

Thus the kernel of  $c^1$  is just the  $\mathrm{Pic}(X)$  modulo  $\ell^\infty$ -divisible elements and torsion elements, but  $\mathrm{Pic}^0(\overline{X})$  is  $\ell^\infty$ -divisible as it is the  $k^s$ -points of an Abelian variety over  $k^s$ . And also  $\mathrm{Pic}(\overline{X})/\mathrm{Pic}_{\mathrm{num}=0}(\overline{X}) \rightarrow \mathrm{Pic}(X_{\overline{k}})/\mathrm{Pic}^\tau(X_{\overline{k}}) = N^1(X_{\overline{k}})$  is injective by [Sta]0CC5 ?, and  $N^1(X_{\overline{k}})$  is finite free, so  $\mathrm{Pic}_{\mathrm{num}=0}(\overline{X}) = \mathrm{Pic}(\overline{X})_{\mathrm{tor}}$ , and  $A_{\mathrm{et}, \ell}^1(\overline{X}) = GH^1(X)$ . ┘

**Prop. (22.3.2.9).** Situation as in (22.3.2.2), let  $X, Y \in \mathbf{SmProjVar}/k$ , then  $T^1(X \times Y, \ell) \iff T^1(X/k, \ell) \wedge T^1(Y/k, \ell)$ . ┘

*Proof:* Cf. [Chao Li] ?. □

**Prop. (22.3.2.10) [Birational Invariance].** Situation as in (22.3.2.2), let  $f : X \rightarrow Y \in \mathbf{SmProjVar}/k$  be a dominant rational map, then  $(T^1(X/k)) \Rightarrow (T^1(Y/k))$ . In particular,  $(T^1)$  is birationally invariant. ┘

*Proof:* Cf. [Tat94]P79. □

**Cor. (22.3.2.11).** A Fermat curve  $S_n : a_0 z_0^n + a_1 z_1^n + a_2 z_2^n + a_3 z_3^n = 0$  is dominated by the product of two curves

$$C_1 : a_0 x_0^n + a_1 x_1^n = y^n, \quad C_2 : a_2 x_2^n + a_3 x_3^n = -z^n$$

by

$$C_1 \times C_2 \rightarrow S : ([x_0, x_1, y], [x_2, x_3, z]) \mapsto [x_0 z, x_1 z, x_2 y, x_3 y],$$

so by (22.3.2.10) and (22.3.2.3),  $(T^1(S_n))$  holds.

More generally,  $T^1(X/k, \ell)$  holds for any  $X$  that is dominated by a product of curves and Abelian varieties.  $\lrcorner$

**Thm. (22.3.2.12)[Tate Implies Künneth Decomposition, Deligne].** If  $d = \dim X$  and  $T^d(X \times X)$  holds, then the Künneth components of the diagonal  $\Delta \in H_{\text{ét}}^{2d}(X \times X, \overline{\mathbb{Q}}_\ell(d))$  are algebraic.  $\lrcorner$

*Proof:* Cf.[Tate1994]P75. ?

$\square$

### 3 over Finite Fields

References are [Milne, Tate Conjecture over Finite Fields], and [Endomorphisms of Abelian varieties over finite fields, Tate, 1966].

**Prop. (22.3.3.1)[ $(T^1)$  and Brauer Groups].** Let  $\#k < \infty$  and  $X \in \text{SmProjVar}/k$ , Tate proved that  $H_{\text{ét}}^{2r}(X_{k^s}, \mathbb{Q}_\ell(r))^{\text{Gal}_k} \cong H_{\text{ét}}^{2r}(X, \mathbb{Q}_\ell(r))$ , and the Kummer sequence gives an exact sequence

$$0 \rightarrow \text{Pic}(X) \otimes \mathbb{Z}_\ell \rightarrow H^2(X, \mathbb{Z}_\ell(1)) \rightarrow T_\ell \text{Br}(X) \rightarrow 0,$$

so the integral  $(T^1(X/k))$  is equivalent to  $T_\ell \text{Br}(X) = 0$ , i.e.  $\# \text{Br}(X)[\ell^\infty] < \infty$ .  $\lrcorner$

*Proof:* Cf.[Chao Li].

$\square$

**Prop. (22.3.3.2)[ $(T^1)$  and BSD].** Let  $\#k < \infty$  and  $C/k$  a non-singular complete curve with function field  $K$ ,  $\mathcal{E}/C$  is a regular elliptic surface with generic fiber  $E \in \text{EllCur}/K$ , then  $(T^1(\mathcal{E}/k))$  is equivalent to the BSD conjecture for  $E/K$ .  $\lrcorner$

*Proof:* Cf.[Chao Li].

$\square$

**Prop. (22.3.3.3)[Grothendieck].** Let  $\#k < \infty$  and  $C$  a smooth complete curve over  $k$  with function field  $K$ ,  $\mathcal{E}/C$  is a regular elliptic surface with generic fiber  $E \in \text{EllCur}/K$ , then

$$\text{Br}(\mathcal{E}/k) \cong \text{III}(E/K).$$

$\lrcorner$

*Proof:* Cf.[Grothendieck, Alexander, Le groupe de Brauer. III. 1968].

$\square$

**Prop. (22.3.3.4)[Tate].** Let  $\#k < \infty$ ,  $X \in \text{SmProjVar}/k$ , then  $(T^r(X/k))$  and  $(E^r(X/k))$  hold iff

$$\text{ord}_{s=r}(Z(X, s)) = -\text{rank CH}_{\text{hom}}^r(X).$$

$\lrcorner$

*Proof:* Cf.[Tat94].

$\square$

**Prop. (22.3.3.5) [Tate Conjecture for Jacobians].** Let  $\#k < \infty$ ,  $C \in \mathcal{C}ur^{\text{SmProj}}/k$ ,  $X$  a smooth projective surface over  $k$ , and  $f : X \rightarrow C$  a morphism with finite fibers. Assume  $X_{k(C)}$  is smooth, and  $J = \text{Jac}(X_{k(C)})$ . Then the following are equivalent:

- $T^1(X/k, \ell)$ .
- $\#\text{Br}(X) < \infty$ .
- $\#\text{III}(J) < \infty$ .
- BSD conjecture holds for  $J/k(C)$ .

⌋

*Proof:* Cf. [Curves and Jacobians over function fields, Ulmer] Prop5.1.2 and Prop6.3.1. □

#### 4 Connection with B-S.D

**Conj. (22.3.4.1) [( $P^r(X/F)$ )].** If  $k = F \in \mathbf{NField}$  and  $X \in \mathbf{SmProjVar}/F$ , then for any  $r \in \mathbb{N}$ ,

$$\text{ord}_{s=1} L(H^{2r}(X)(r); s) = -\text{rank } \text{CH}_{\text{hom}}^r(X).$$

⌋

*Proof:* ? □

**Thm& Conj. Cor. (22.3.4.2).** Let  $F \in \mathbf{NField}$ ,  $E_1, E_2, E_3, E_4 \in \mathbf{EllCur}/F$  and  $A \in \mathbf{AbVar}^2/F$ , then  $(P^r(X/F))$  (22.3.4.1) holds in the following cases:

- $F$  is totally real or imaginary CM, and  $X = E_1^{n_1} \times E_2^{n_2}$  with  $n_1 \in \mathbb{Z}_+, n_2 \in \mathbb{N}$ ,
- $F$  is totally real or imaginary CM, and  $X = E_1^{n_1} \times E_2^{n_2} \times E_3$  with  $n_1 \in \mathbb{Z}_+, n_2 \in \{1, 2\}$ .
- $F$  is totally real or imaginary CM, and  $X = E_1^{n_1} \times E_2^{n_2} \times E_3 \times E_4$  with  $n_1, n_2 \in \{1, 2\}$ .
- $F$  is totally real and  $X \in \{A, A^2\}$ .

⌋

*Proof:* By (22.3.2.3), it suffices to prove that

$$\text{ord}_{s=1} L(H_{\text{ét}}^{2r}(X)(r); s) = -\text{rank } H_{\text{ét}}^{2r}(\overline{X}, \overline{\mathbb{Q}}_\ell(r))^{\text{Gal}_k}$$

for some  $\ell \in \mathbf{Prime}$ .

1: By Künneth formula and the decomposition of  $\bigotimes^{k_i} H_{\text{ét}}^1(E_{i,\overline{F}}, \overline{\mathbb{Q}}_\ell)$  into symmetric powers (because  $\dim H_{\text{ét}}^1(E_{i,\overline{F}}, \overline{\mathbb{Q}}_\ell) = 2$ ), we have an isomorphism of  $\text{Gal}_F$ -representations

$$H_{\text{ét}}^{2r}(X_{\overline{F}}, \overline{\mathbb{Q}}_\ell(r)) \cong \bigoplus_{0 \leq k_i \leq n_i} m_{k_1, k_2} \left( \text{Sym}^{k_1} H_{\text{ét}}^1(E_{1,\overline{F}}, \overline{\mathbb{Q}}_\ell) \otimes \text{Sym}^{k_2} H_{\text{ét}}^1(E_{2,\overline{F}}, \overline{\mathbb{Q}}_\ell) \right) \left( \frac{k_1 + k_2}{2} \right),$$

where  $m_{k_1, k_2} \in \mathbb{N}$  is non-zero only when  $k_1 + k_2 \leq 2r$  a even. So the assertion follows from (30.1.4.5) and (30.1.4.3).

2, 3, 4 are similar, by the same references as above. □

## 5 Hyper-Kähler Varieties

References are [The Tate Conjecture For K3 Surfaces In Odd Characteristic, Pera], [The Tate Conjecture For K3 Surfaces—A Survey Of Some Recent Progress], [2-adic integral canonical models and the Tate conjecture in characteristic 2].

**Thm. (22.3.5.1)[Tate Conjecture for K3 Surfaces].** Tate and Weil’s conjecture holds for  $X \in \text{K3}/\mathbb{Q}$ .  
 $\lrcorner$

*Proof:* Use the Kuga-Satake construction, we get a Kuga-Satake Abelian variety  $A \in \text{AbVar}/F$  for some  $F \in \text{NField}$ . By Tate’s isogeny theorem, any  $\xi \in \text{Tate}^1(X, \ell) \subset (\text{End}(H_{\text{ét}}^1(A_{\mathbb{C}}, \mathbb{Q}_{\ell})))^{\text{Gal}_F}$  is induced by an isogeny of  $A$ , thus is contained in

$$(\text{End}(H_{\text{Betti}}^1(A, \mathbb{Q})))_{\mathbb{Q}_{\ell}}^{(0,0)} \cap H_{\text{ét}}^2(X, \mathbb{Q}_{\ell}(1)) = (H_{\text{Betti}}^2(X, \mathbb{Q}(1)))_{\mathbb{Q}_{\ell}}^{(0,0)}.$$

Then the Lefschetz  $(1, 1)$ -form theorem implies that  $\xi$  is a divisor  $D$  on  $X_{\mathbb{C}}$ . And the  $\text{Aut}(\mathbb{C}/\mathbb{Q})$ -invariance of the  $\ell$ -adic realization of  $D$  implies that  $D$  is defined over  $\mathbb{Q}$ . (In fact, by (22.3.2.5), it suffices to prove over  $\mathbb{C}$ ).  $\square$

**Thm. (22.3.5.2)[Artin-Swinnerton-Dyer/Nygaard/Nygaard-Ogus/Maulik/Pera/Charles/Kim-Pera].**  $(T^1(X/k))$  (22.3.2.2) holds if  $X/k$  is a K3 surface.  
 $\lrcorner$

*Proof:* Cf.[The Tate Conjecture For K3 Surfaces In Odd Characteristic, Pera] and [2-adic integral canonical models and the Tate conjecture in characteristic 2]. ?  $\square$

**Cor. (22.3.5.3)[Lieblich-Maulik-Snowden].** There are only finitely many isomorphism classes of K3 surfaces over a finite field of characteristic  $\geq 5$ .  
 $\lrcorner$

*Proof:* [Finiteness of K3 surfaces and the Tate conjecture, Lieblich-Maulik-Snowden].  $\square$

**Thm. (22.3.5.4) [Moonen].** Let  $\text{char } k = 0$  and  $X \in \text{SmProj}/k$  s.t.  $h^{2,0}(X) = 1$ . If  $\iota : K \rightarrow \mathbb{C}$  is an embedding and  $X_{\mathbb{C}}$  has a smooth deformation s.t. the Hodge structure  $H^2(X_{\mathbb{C}}, \mathbb{Q})$  varies non-trivially, then  $T^1(X/k, \ell)$  holds for any  $\ell$ .  
 $\lrcorner$

*Proof:* Cf.[Tate and Mumford-Tate conjectures in codimension 1 for varieties with  $h^{2,0} = 1$ , Moonen]. ?  $\square$

### Kuga-Satake Construction

**Notation (22.3.5.5).**

- Use notations on Clifford algebras defined in [Clifford Algebras](#).  
 $\lrcorner$

**Def. (22.3.5.6)  $[L_d]$ .** Let  $g \in \mathbb{Z}_{\geq 2}$ ,  $d = 2(g - 1)$  and let  $L_d$  be the complement of a vector of length  $d$  in  $\Lambda_{\mathbb{Q}}^{K3}$ , which has signature  $(2, 19)$ .  
 $\lrcorner$

**Prop. (22.3.5.7).** Any map  $h_{\infty} : \$ \rightarrow SO((\Lambda_d^{K3})_{\mathbb{R}})$  lifts to a map  $h_{\infty} : \$ \rightarrow \text{GSpin}((\Lambda_d^{K3})_{\mathbb{R}})$ . In particular, we have a map of Shimura datum  $\pi : (\text{GSpin}(L_d), \mathbb{D}) \rightarrow \text{SO}(L_d, \mathbb{D})$ .  
 $\lrcorner$

*Proof:* For any  $h \in \mathbb{D} \subset \text{Hom}(\mathbb{C}^\times, \text{SO}(L_d))$  corresponding to a Hodge structure, let  $v = x + yi \in V^{2,0}$  s.t.  $q(x) = 1$ . Because  $q(v) = 0$ , we get  $q(y) = 1$  and  $x \perp y$ , i.e.  $\{x, y\}$  is an orthonormal basis for  $N = V^{2,0} \oplus V^{0,2} \cap V_{\mathbb{R}}$ . So if we let  $J = ab \in \text{Cl}^+(L_d)$ , which is independent of the choice of  $v$ , then  $J^2 = -\text{id}$ ,  $J' = -J$ , and

$$h' : \mathbb{C}^\times \rightarrow \text{Cl}^+(L_d) : a + bi \mapsto a + bJ$$

has image in  $\text{GSpin}(L_d)$ . The conjugation by the image of  $h'$  gives  $N_{\mathbb{C}}$  the Hodge structure of  $(1, -1) \oplus (-1, 1)$  while fixes the orthogonal complement of  $N$ :

$$(\cos \theta + \sin \theta ab)(a + bi)(\cos \theta - \sin \theta ab) = e^{2i\theta}(a + bi).$$

$$(\cos \theta + \sin \theta ab)(a - bi)(\cos \theta - \sin \theta ab) = e^{-2i\theta}(a - bi).$$

So it is just the Hodge structure of  $h$ . Notice both period domain for  $\text{GSpin}$  and  $\text{SO}$  are  $\mathbb{D} = \{\text{negative 2-planes in } L_d\}$ .  $\square$

**Prop. (22.3.5.8) [Kuga-Satake Map].** The form on  $\text{Cl}^+(L_d)$ :

$$\langle x, y \rangle_J = \text{tr}(xJy)$$

is symplectic, unimodular and  $\text{Spin}(L_d)$ -invariant, so left multiplication by  $\text{GSpin}(L_d)$  induces a map  $\text{GSpin}(L_d) \rightarrow \text{GSp}(\text{Cl}^+(L_d), \langle -, - \rangle_J)$ . Moreover,  $(x, y) = \langle x, Jy \rangle_J = -\text{tr}(xJy'J)$  is symmetric and definite, thus  $J$  belongs to  $\mathcal{H}_{220}^\pm$ , because the period space for  $\text{GSp}$  is the space of complex structures  $J$  s.t.  $\langle Ju, Jv \rangle = \langle u, v \rangle$ , and  $\langle -, J- \rangle$  is positive definite. Thus we defined a morphism of Shimura datum

$$(\text{GSpin}(L_d), \mathbb{D}) \rightarrow (\text{GSp}(\text{Cl}^+(L_d), \mathcal{H}_{220}^\pm).$$

┘

*Proof:*

$\square$

**Prop. (22.3.5.9).** By (22.3.5.7) and (22.3.5.8), there are maps between Shimura data:

$$\begin{array}{ccc} \{\text{Sh}_K(\text{GSpin}(\Lambda_d^{\text{K3}}, \mathbb{D}))\} & \xrightarrow{\iota_{\text{KS}}} & \{\text{Sh}_K(\text{GSp}_{221}, \mathbb{H}_{220}^\pm)\} \\ \downarrow \tilde{\pi} & & \\ \{\text{Sh}_K(\text{SO}(\Lambda_d^{\text{K3}}, \mathbb{D}))\} & & \end{array}$$

that induces maps between the corresponding Shimura varieties

$$\begin{array}{ccc} \{\text{Sh}_K(\text{GSpin}(\Lambda_d^{\text{K3}}, \mathbb{D}))\} & \xrightarrow{\iota_{\text{KS}}} & \{\text{Sh}_K(\text{GSp}_{221}, \mathcal{H}_{220}^\pm)\} \\ \downarrow \tilde{\pi} & & \\ \{\text{Sh}_K(\text{SO}(\Lambda_d^{\text{K3}}, \mathbb{D}))\} & & \end{array}$$

Then if we equip the Shimura varieties with models over  $\mathbb{Q}$  where

- $\{\text{Sh}_K(\text{SO}(\Lambda_d^{\text{K3}}, \mathbb{D}))\}$  is associated the moduli space  $\{M_K^{\text{K3}}\}_K$  of K3 surfaces, which is a canonical model.
- $\{\text{Sh}_K(\text{GSp}_{221}, \mathcal{H}_{220}^\pm)\}$  is associated the moduli space  $\mathcal{A}^{g,d}$  of Abelian varieties.
- $\{\text{Sh}_K(\text{GSpin}(\Lambda_d^{\text{K3}}, \mathbb{D}))\}$  is associated the induced model by the model  $\mathcal{A}^{g,d}$ , which is also defined over  $\mathbb{Q}$  and is a canonical model.



┘

*Proof:* The fact that  $M_K^{K3}$  is a canonical model follows from the theorem of complex multiplications on K3 surface, the proof is the same as proving that  $\mathcal{A}^{g,K}$  is the canonical from the CM theory of Abelian varieties. ? □

**Cor. (22.3.5.10) [Kuga-Satake Construction, Deligne[K-S67]].** If  $X \in \text{K3}/\mathbb{Q}$ , then there exists  $F \in \mathbf{NField}$ ,  $A \in \mathbf{AbVar}/F$  and an embedding of Hodge-Tate structures

$$H_{\text{Betti}}^2(X, \mathbb{Q}(1)) \rightarrow \text{End}(H_{\text{Betti}}^1(A, \mathbb{Q})).$$

┘

*Proof:* This follows from (28.1.2.12). To show there is an embedding, notice that the representation  $\text{Std}_{\text{SO}} \circ \pi$  of  $\text{GSpin}(\Lambda_d^{K3})$  induces the pullback of the Hodge-Tate structure  $H_{\text{Betti}}^2(X, \mathbb{Q}(1))$ , and the representation  $\text{End}(\text{Std}_{\text{GSp}} \circ \iota_{\text{KS}})$  of  $\text{GSpin}(\Lambda_d^{K3})$  induces the pullback of the Hodge-Tate structure  $\text{End}(H_{\text{Betti}}^1(A, \mathbb{Q}))$ , so it suffices to show that there is an embedding of representations

$$\text{Std}_{\text{SO}} \circ \pi \hookrightarrow \text{End}(\text{Std}_{\text{GSp}} \circ \iota_{\text{KS}}).$$

For this, let  $\Lambda^\sharp = \Lambda_d^{K3} \oplus \mathbb{Z}$ , then  $\Lambda_d^{K3} \subset (\text{Cl}^+(\Lambda^\sharp))^{\text{Ad}} \subset \text{End}(\text{Cl}^+(\Lambda^\sharp))$ . But  $\text{Cl}^+(\Lambda^\sharp) \cong \text{Cl}^+(\Lambda)^2$  as  $\text{GSpin}$ -representations, so  $\Lambda_d^{K3} \subset \text{End}(\text{Cl}^+(\Lambda_d^{K3}))^4$ . As  $\Lambda_d^{K3}$  is a simple representation, there must be a embedding  $\Lambda_d^{K3} \subset \text{End}(\text{Cl}^+(\Lambda_d^{K3}))$ .

To sum up, For any K3 surface  $X/\mathbb{Q}$  polarized of degree  $d$ , let  $x \in \text{Sh}_K(\text{GSpin}(L_d, \mathbb{D}))(\mathbb{Q})$  be a point that maps under  $\tilde{\pi}$  (22.3.5.9) to the point parametrizing  $X/\mathbb{Q}$ . Then  $\tilde{\iota}(x) \in \mathcal{A}^{g,d}$  is a point parametrizing some Abelian variety  $A/F$  for some  $F \in \mathbf{NField}(K$  needs to be small enough to make  $\mathcal{A}_K^{g,d}$  representable). Then this  $A/F$  is the Kuga-Satake variety of  $X/\mathbb{Q}$ . □

## 6 Mumford-Tate Conjectures

**Conj. (22.3.6.1) [Mumford-Tate Conjectures].** For  $X \in \text{SmProj}/\mathbb{C}$  and  $r \in \mathbb{N}$ , we have the Mumford-Tate group  $\text{MT}^r(X)$  (16.11.2.1) and the  $\ell$ -adic monodromy group  $G_\ell^r(X)$  (22.3.1.2). Then

$$\text{MT}^r(X) \otimes \mathbb{Q}_\ell = G_\ell^r(X)^0.$$

┘

*Proof:* □

**Prop. (22.3.6.2) [Hodge  $\wedge$  Tate  $\implies$  Mumford-Tate Conjecture].** For  $X \in \text{SmProj}/\mathbb{C}$ , if the Hodge conjecture (16.11.3.1) holds for  $X^m$  for any  $m \in \mathbb{Z}_+$  and the Tate conjecture  $T^*(X^m, \ell)$  (22.3.2.2) holds for  $X^m$  for any  $m \in \mathbb{Z}_+$ , then the Mumford-Tate conjecture (22.3.6.1) also holds for  $X^m$  for any  $m \in \mathbb{Z}_+$ . ┘

*Proof:* This is because  $\text{MT}(X)$  and  $G_\ell^0(X)$  are determined by their invariant classes (because they are reductive groups ?), and the Hodge (resp. Tate) conjecture together with the comparison isomorphism implies that these classes are all combinations of cycle classes. □

**Prop. (22.3.6.3) [Mumford-Tate Conjecture  $\implies$  (Hodge  $\iff$  Tate)].** For  $X \in \text{SmProjVar}/\mathbb{C}$  and  $\ell \in \mathbf{Prime}$ , if the Mumford-Tate conjecture holds for  $X$  in codimension  $r$ , then the Hodge conjecture for  $X$  in codimension  $r$  is equivalent to  $(T^r(X, \ell))$ . ┘

*Proof:* If the Mumford-Tate conjecture is true for  $X$ , then the comparison isomorphism  $H_{\text{Betti}}^{2r}(X, \mathbb{Q}(r)) \otimes \mathbb{Q}_\ell \cong H_{\text{ét}}^{2r}(X, \mathbb{Q}_\ell(r))$  implies that

$$\text{Hdg}^r(X) \otimes \mathbb{Q}_\ell \cong \text{Tate}^r(X).$$

So the two conjectures are equivalent.  $\square$

**Prop. (22.3.6.4) [Tate Numbers Bounded by Hodge Numbers].** Let  $X \in \text{SmProj}/\mathbb{Q}$ , then

$$\text{Tate}^r(X) \subset \left( H_{\text{ét}}^{2r}(X, \mathbb{Q}_\ell(r)) \otimes_{\mathbb{Q}_\ell} \mathbb{C}_\ell \right)^{\text{Gal}_{\mathbb{Q}_\ell}} \cong H^r(X_{\mathbb{Q}_\ell}, \Omega_{X_{\mathbb{Q}_\ell}/\mathbb{Q}_\ell}^r)$$

In particular,  $\dim_{\mathbb{Q}_\ell} \text{Tate}^r(X, \ell) \leq h^{r,r}(X_{\mathbb{C}})$ .  $\lrcorner$

*Proof:* This follows from the deRham comparison (27.7.3.3) and invariance of deRham cohomology.  $\square$

## 7 for Drinfeld Modules

**Conj. (22.3.7.1) [Tate Conjecture for  $\varphi$ -Modules].** Let  $F \in \text{FField}$  with prime field  $\mathbb{F}_q$  and  $v \in \Sigma_F$ , then for any f.g. field  $k/\mathbb{F}_q$  and any  $\varphi$ -module  $M$  over  $F \otimes k$ , the natural map

$$F_v \otimes_K M^\varphi \rightarrow ((F \otimes k)_v \otimes_{F_v} M)^\varphi$$

is an isomorphism.  $\lrcorner$

*Proof:*  $\square$

**Remark (22.3.7.2).** It suffices to show that: For any  $A \in \text{Mat}(r; K \otimes k)$ , every solution of the equation

$$AX^\sigma = X$$

for  $X \in (K_v \otimes k)^{\oplus r}$  is a  $K_v$ -linear combination of solutions for  $X \in (K \otimes k)^{\oplus r}$ . Then notice that if  $K'/K$  is a finite extension field with the same base field  $\mathbb{F}_q$ , then if the Tate conjecture holds for  $(K, \pi)$ , then it holds for  $(K', \pi')$  for any  $\pi'|\pi$ , because  $K_\pi \otimes_K K' \cong \prod_{\pi'|\pi} K'_{\pi'}$ .

Thus it suffices to assume that  $K = \mathbb{F}_q(t)$ . and shrinking more, we may assume that  $\mathfrak{m}_v = (t)$ .  $\lrcorner$

**Thm& Conj. Cor. (22.3.7.3) [Taguchi].** The Tate conjecture for  $\varphi$ -modules are true if  $k/\mathbb{F}_q$  is a function field and  $M$  comes from a  $t$ -motive of characteristic different from  $\text{char } v$ , i.e. in case  $K = \mathbb{F}_q(t)$ ,  $M$  is represented by a matrix

$$A \in \text{GL}(r; k[t, \frac{1}{t-\theta}])$$

for some  $\theta \in k^\times$ . (Notice this is slightly more general, because  $A$  may not be in  $\text{Mat}(r; K \times k)$ ).  $\lrcorner$

*Proof:* It suffices to show that for any solution of  $A\hat{X}^\sigma = \hat{X}$  with  $\hat{X} \in (K_v \otimes k)^{\oplus r}$ , we can get a solution  $X \in (K \otimes k)^{\oplus r}$  that is sufficiently close to  $\hat{X}$ . In this way, any  $n$  independent solutions  $\{\hat{X}_i\}$  will give  $n$  independent solutions  $\{X_i\}$ , thus proving the assertion.

For this, Cf. [Tag95].  $\square$

## 22.4 Motivic Cohomology(Voevodsky-Suslin)

References are [Motivic Cohomology, Voevodsky], [ $\mathbb{A}^1$ -Homology of Schemes, Voevodsky], [Bloch-Kato conjecture and motivic cohomology with finite coefficients, Suslin-Voevodsky] and [Voe02].

**Def. (22.4.0.1) [Bloch's Motivic Cohomology Groups].** For  $k \in \mathbf{Field}$  and  $X \in \mathbf{SmProj}/k$ , define the complex

$$C_{\mathrm{Blo},\bullet}^p(X) = C^p(X \times \mathbb{A}^\bullet),$$

and it is a complex via the pullback along the embedding from edges of  $\Delta^n \subset \mathbb{A}^n$ . And in fact the cycles must satisfy that the intersections are all proper. (This is what makes it impossible to work with.)

Then define

$$H_{\mathrm{Mot}}^{2p-q}(X, \mathbb{Z}(p)) = \mathrm{CH}^p(X, q).$$

┘

**Def. (22.4.0.2) [Beilinson's Motivic Cohomology Groups].** For  $X \in \mathbf{SmProj}/\mathbb{Q}$ ,

$$H_{\mathrm{Mot}}^p(X, \mathbb{Q}(q)) = (K_{2q-p}(X)_{\mathbb{Q}})^{(q)},$$

where the RHS is the Adams operator  $q$ -part.

Then this definition is compatible with Bloch's definition (22.4.0.1).<sup>?</sup> In particular,  $H_{\mathrm{Mot}}^{2q}(X, \mathbb{Q}(q)) = \mathrm{CH}^q(X)_{\mathbb{Q}}$ , and

$$\bigoplus \mathrm{CH}^p(X, q) \otimes \mathbb{Q} \cong K_q(X).$$

┘

*Proof:*

□

**Prop. (22.4.0.3) [Comparison Maps].** There are  $\ell$ -adic comparison maps

$$H_{\mathrm{Mot}}^p(X, \mathbb{Q}(q)) \rightarrow H_{\mathrm{\acute{e}t}}^p(X, \mathbb{Q}_{\ell}(q)), \ell \in \mathbf{P},$$

and Deligne comparison maps

$$H_{\mathrm{Mot}}^p(X, \mathbb{Q}(q)) \rightarrow H_{\mathrm{Del}}^p(X, \mathbb{Q}(q)).$$

┘

*Proof:*

□

**Conj. (22.4.0.4) [Integral Motivic Cohomology Groups].** For  $X \in \mathbf{SmProj}/\mathbb{Q}$ , define

$$H_{\mathrm{Mot}}^p(X_{\mathbb{Z}}, \mathbb{Q}(q)) = H_{\mathrm{Mot}}^p(\mathcal{X}, \mathbb{Q}(q)),$$

Where  $\mathcal{X}/\mathbb{Z}$  is a proper flat model of  $X$ , which always exists.<sup>?</sup>

┘

*Proof:* Cf. [Bei85] 2.4.2.<sup>?</sup>

□

**Conj. (22.4.0.5) [Beilinson].** For  $k \in \mathbf{Field}$  and any  $n \in \mathbb{N}$ , there exists sheaf  $\mathbb{Z}(n) \in K(\mathrm{Sh}(\mathrm{Var}_{\mathrm{Zar}}^{\mathrm{sm}, \mathrm{quasi-proj}}/k))$  s.t.

1.  $\mathbb{Z}(0) = \mathbb{Z}[0]$ ,  $\mathbb{Z}(1) = \mathcal{O}^*[-1]$ ,
2.  $H^n(\mathrm{Spec} F, \mathbb{Z}(n)) = K_{\mathrm{Mil}}^n(F)$  for any  $F \in \mathbf{Field}$ .
3.  $H^{2n}(X, \mathbb{Z}(n)) = \mathrm{CH}^n(X)$ .
4.  $H^p(X, \mathbb{Z}(n)) = 0$  for  $p < 0$ .
5. There are spectral sequences

$$H^{p-q}(X, \mathbb{Z}(-q)) \implies K_{-p-q}(X).$$

6. For  $\ell \in \mathfrak{P} \setminus \{\mathrm{char} k\}$ ,

$$\mathbb{Z}(n) \otimes^{\mathbb{L}} \mathbb{Z}/(\ell) \cong \tau_{\leq n}(R\pi_* \mu_\ell^{\otimes n}).$$

7.  $H^i(X, \mathbb{Z}(n)) \otimes \mathbb{Q} \cong H_{\mathrm{Mot}}^i(X, \mathbb{Q}(n))$  (22.4.0.2).

┘

*Proof:* ? Voevodsky constructed candidates  $\mathbb{Z}(n)$  satisfying these conditions except for item 4.  $\square$

**Def. (22.4.0.6) [Absolute Chern Class Maps].** There are Chern class maps:

$$c_{j,m} : K_j(X) \rightarrow H_{\mathrm{Del}}^{2m-j}(X/\mathbb{R}, \mathbb{R}(m)),$$

$$\mathrm{ch}_j = \sum_{m \geq 0} \frac{(-1)^{m-1}}{(m-1)!} c_{j,m} : K_j(X) \otimes \mathbb{Q} \rightarrow \bigoplus_{m \geq 0} H_{\mathrm{Del}}^{2m-j}(X/\mathbb{R}, \mathbb{R}(m)).$$

such that  $\mathrm{ch}_j$  maps  $(K_j(X)_{\mathbb{Q}})^m$  to  $H_{\mathrm{Del}}^{2m-j}(X/\mathbb{R}, \mathbb{R}(m))$ .

┘

*Proof:* Cf. [Notes on Beilinson's Conjecture, Yihang] or [Nekovar].  $\square$

### Curve case

**Prop. (22.4.0.7).** Let  $k \in \mathbf{Field}$ ,  $X$  a non-singular complete curve over  $k$ , and  $C \subset X$  a finite subscheme,  $U = X \setminus C$ . Suppose  $\mathrm{Pic}^0(C)$  is of finite order in  $\mathrm{Jac}(X)$ , then

$$H_{\mathrm{Mot}}^2(U, \mathbb{Q}(2)) = H_{\mathrm{Mot}}^2(X, \mathbb{Q}(2)) + \{\mathcal{O}(U)^\times, k^\times\}.$$

┘

## 22.5 Schemes over $\mathbb{F}_1$

References are [Connes, Schemes over  $\mathbb{F}_1$  and Zeta-Functions], [Sou04].

### 1 Geometry of Monoids

Cf.[Toric Singularities, Kato].

**Notation(22.5.1.1).**

- In this section, monoids are assumed to be commutative and have a zero element and neutral element.

┘

**Def.(22.5.1.2) [Ideals in Monoids].** For  $M \in \mathcal{CMon}^{0,1}$ , an **ideal(of monoids)** of  $M$  is a subset  $I \subset M$  s.t.

- $0 \in I$ .
- $x \in I, y \in M \implies xy \in I$ .

And a **prime ideal(of monoids)** of  $M$  is an ideal  $\mathfrak{p} \subset M$  s.t.  $M \setminus \mathfrak{p}$  is a multiplicative subset of  $M$ .

┘

**Def.(22.5.1.3) [Spectrum of Monoids].** For  $M \in \mathcal{CMon}^{0,1}$  and  $I \in \text{Ideal}(M)$ , let  $D(I)$  be the set of primes ideals of  $M$  not containing  $I$ . Then if  $\text{Spec}(M)$  is the set of prime ideals of  $M$ ,  $D(I)$  defines a the Zariski topology on  $\text{Spec}(M)$ .

┘

**Prop.(22.5.1.4)[Abelian Groups and Monoids].** For  $H \in \mathcal{Ab}$ , denote  $\mathbb{F}_1[H] = H \cup \{0\} \in \mathcal{CMon}^{0,1}$ . Then this defines  $\mathcal{Ab}$  as a full subcategory of  $\mathcal{CMon}^{0,1}$ , and the inclusion is left adjoint to taking invertible elements: For any  $H \in \mathcal{Ab}, M \in \mathcal{CMon}^{0,1}$ ,

$$\text{Hom}_{\mathcal{CMon}^{0,1}}(\mathbb{F}_1[H], M) = \text{Hom}_{\mathcal{Ab}}(H, M^\times).$$

┘

### Mon-Functors

#### $\mathbb{F}_1$ -Schemes

**Def.(22.5.1.5) [ $\mathbb{F}_1$ -Schemes].** An  **$\mathbb{F}_1$ -scheme** is a topological space with a sheaf of monoids that is locally isomorphic to the spectrum of a commutative monoids. The category of  $\mathbb{F}_1$ -schemes is denoted by  $\text{Sch}/\mathbb{F}_1$ .

┘

**Def.(22.5.1.6).** An **finitely generated  $\mathbb{F}_1$ -scheme** is a scheme that has a finite cover by affine schemes  $U_i = \text{Spec } A_i$  s.t. each  $A_i$  are f.g. monoids.

┘

**Def.(22.5.1.7)[Ascend Functor].** The functor  $\mathcal{CMon}^{0,1} \rightarrow \mathcal{CRing} : A \mapsto \mathbb{Z}[A]$  defines a functor

$$- \otimes \mathbb{Z} : \text{Sch}/\mathbb{F}_1 \rightarrow \text{Sch}.$$

┘

**Prop.(22.5.1.8).** For  $X \in \text{Sch}/\mathbb{F}_1$ ,  $X$  is f.g. iff  $X_{\mathbb{Z}}$  is finitely generated.

┘

*Proof:* ?

□

**Def. (22.5.1.9) [Ring Monoids].** For  $R \in \mathcal{C}\text{Ring}$ , define  $\text{Spec } R \in \text{Sch}/\mathbb{F}_1$  to be the affine scheme corresponding to the multiplicative monoid  $(R, \times)$ .

Then we have  $\text{Hom}_{\mathbb{F}_1}(\text{Spec } R, X) = \text{Hom}_{\text{Sch}}(\text{Spec } R, X_{\mathbb{Z}})$  for any  $X \in \text{Sch}/\mathbb{F}_1$ .  $\lrcorner$

**Prop. (22.5.1.10) [Zeta Polynomials].** For any  $X \in \text{Sch}^{\text{fg}}/\mathbb{F}_1$ , there exists  $e \in \mathbb{Z}_+$  and  $N(T) \in \mathbb{Z}[T]$  s.t. for any  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}_+}$ , if  $(q-1, e) = 1$ , then

$$\#X(\mathbb{F}_q) = N(q).$$

Such  $N$  is called the **zeta polynomial** of  $X$ .  $\lrcorner$

*Proof:* This is quite easy, Cf.[Remarks on zeta functions and K-theory over  $\mathbb{F}_1$ , Dietmar].  $\square$

**Def. (22.5.1.11) [Zeta Functions of  $\mathbb{F}_1$ -Schemes].** For  $X \in \text{Sch}^{\text{fg}}/\mathbb{F}_1$ , if  $N_X(T) = a_0 + a_1T + \dots + a_nT^n$  is the zeta polynomial of  $X$  (22.5.1.10), then the **zeta function**( $\mathbb{F}_1$ ) of  $X$  is defined to be

$$\zeta_X(s) = \lim_{q \rightarrow 1} Z_q(X, q^{-s})(q-1)^{N(1)} = s^{a_0}(s-1)^{a_1} \dots (s-n)^{-a_n}.$$

And the **Euler characteristic**( $\mathbb{F}_1$ ) of  $X$  is defined to be

$$\chi(X) = N_X(1) = \sum a_i.$$

**Prop. (22.5.1.12).** This zeta function so defined (22.5.1.11) fulfills the requirement of the motivic zeta function as in [Manin, Zeta functions].  $\textcolor{red}{?}$   $\lrcorner$

**Prop. (22.5.1.13).** Let  $\underline{X}$  be a Mon-scheme and  $X$  its geometric realization, then the restriction of  $\underline{X}$  to  $\mathcal{A}\text{b}$  via  $\mathbb{F}_1$  is:

$$\underline{X}(\mathbb{F}_1[H]) = \prod_{x \in X} \text{Hom}_{\mathcal{A}\text{b}}(\mathcal{O}_{X,x}^\times, H), \quad H \in \mathcal{A}\text{b}.$$

## 2 Zeta Functions

**Conj. (22.5.2.1).** For any  $n \in \mathbb{Z}_+$ , there should be a field  $\mathbb{F}_{1^n}$ , s.t.

- $\text{Spec } \mathbb{Z}$  is over  $\mathbb{F}_{1^n}$ , and
- $\mathbb{F}_{1^n} \otimes_{\mathbb{F}_1} \mathbb{Z} = \mathbb{Z}[T]/(T^n - 1)$ .

*Proof:*  $\square$

**Conj. (22.5.2.2).** There exists a “curve”  $C = \overline{\text{Spec } \mathbb{Z}}$  over  $\mathbb{F}_1$  whose zeta-function is  $Z_C(s) = \Lambda(s)$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (22.5.2.3).** One expects  $\overline{\text{Spec } \mathbb{Z}}$  to be of infinite genus.  $\lrcorner$

*Proof:* [Manin, Zeta Functions and Motives].  $\square$

### 3 Absolute Motives via Zeta Functions

**Remark (22.5.3.1) [Absence of Absolute Motives].** Because of the absence of direct products of  $\mathbb{Z}$ -schemes, we cannot use correspondence to define morphisms between absolute motives. As a result, we can only observe them through the zeta function, and observe the imaginary morphisms between by their induced morphisms on zeta functions.  $\lrcorner$

### 4 K-Theory of $\mathbb{F}_1$ -Schemes

**Def. (22.5.4.1) [ $K_n^+(A)$ ].** For  $A \in \mathcal{CMon}$ , define  $GL(n; A)$  to be the set of group of  $n \times n$  matrices s.t. each row and column has exactly one non-zero element which lies in  $A^\times$ . Then  $GL(n; A) \cong A^\times \rtimes \text{Sym}_n$ , and there are natural embeddings  $GL(n; A) \hookrightarrow GL(n+1; A)$ .

Then we can define  $GL(A) = \varinjlim_n GL(n; A)$ , and define the **K-group of monoids** as  $K_j^+(A) = \pi_j(\mathbf{BGL}(A)^+)$ , where  $+$  is the Quillen's plus construction.  $\lrcorner$

**Cor. (22.5.4.2).**  $K_j(\mathbb{F}_1) = \pi_j^{\text{st}}(\mathbb{S})$ .  $\lrcorner$

*Proof:* ?  $\square$

**Def. (22.5.4.3) [ $K_n^Q(A)$ ].** For  $A \in \mathcal{CMon}$ , let  $\mathcal{P}$  be the category of f.g. pointed projective  $A$ -modules, and  $\mathcal{E}$  be the class of split exact sequences in  $\mathcal{P}$ , then  $(\mathcal{P}, \mathcal{E})$  is a quasi-exact category, and we can define the **K-group of monoids** as  $K_i^Q(A) = K_i(\mathcal{P}, \mathcal{E})$  (11.1.2.5).  $\lrcorner$

**Prop. (22.5.4.4).** For  $A \in \mathcal{Ab}$ ,  $K_n^+(A) = K_n^Q(A)$  (22.5.4.1)(22.5.4.3) for any  $n \in \mathbb{N}$ .

In particular,

$$K_i^+(A) = K_i(A^\times) = \begin{cases} Z \times A^\times & , i = 0 \\ \pi_i^{\text{st}}(\mathbb{S}) & , i > 0 \end{cases}.$$

And there are maps  $K_i^+(A) = K_i(A^\times) \rightarrow K_i^Q(A)$ .  $\lrcorner$

*Proof:* Any projective pointed  $A$ -module is free ?, so the usual proof for  $A \in \mathcal{CRing}$  pass through ? Cf. Higher K-theory, 2).

And then  $\mathcal{P}$  is just the product of the category of pointed sets and  $\{A\}$ . Thus  $\mathbf{BQ}(\mathcal{P}) = \mathbf{BQ}(\text{Set}_{\text{pt}}) \times \mathbf{BA}$ , and the assertion follows.

The map  $K_i(A^\times) \rightarrow K_i^Q(A)$  comes from the fact any projective  $A^\times$ -modules are free.  $\square$





## 23 | Arakelov Geometry

### 23.1 Arithmetic Intersection Theory

#### 1 Néron's Symbol

Cf.[Lang, Arithmetic Geometry] and [Hriljac].

**Prop. (23.1.1.1) [Néron's Symbol].** For  $(K, v)$  a valued field and  $C \in \text{Cur}^{\text{sm}, \text{proper}}/K$  s.t.  $C(K)$  is Zariski-dense in  $C$ , then there is a unique pairing  $\langle D, E \rangle_v \in \mathbb{R}$  for any pair of degree 0 rational divisors  $D, E$  on  $C$  with disjoint divisors s.t.

- This pairing is bilinear and symmetric.
- If  $D = \text{div}(f)$  where  $f \in R(C)^\times$ , then  $\langle D, E \rangle_v = v(f(E))$ .
- For any  $P_0 \in C(K) \setminus \text{Supp}(D)$ , the map

$$C(K) \setminus \text{Supp}(D) \rightarrow \mathbb{R} : P \mapsto \langle D, P - P_0 \rangle_v$$

is continuous and locally bounded.

This pairing is called the **Néron symbol**. ┘

*Proof:* Cf.[Lang, Diophantine Geometry, P281]. □

**Remark (23.1.1.2).** The Néron symbols is compatible with base change of valued fields. So if we consider extending the field  $K$  to its algebraic closure of the completion, we can drop the assumption that  $C(K)$  is Zariski-dense in  $C$ . ┘

**Cor. (23.1.1.3).** Situation as in (23.1.1.1), suppose for any pair of divisors  $D, E \in Z_0(C)_K$  with disjoint supports, there is a real number  $[D, E] \in \mathbb{R}$  satisfying the following:

- The pairing is bilinear and symmetric.
- If  $D = \text{div}(f)$  for  $f \in R(C)^\times$ , then  $[D, E] = v(f(E))$ .
- If  $\tau(D, E) = [D, E] - \langle D, E \rangle_v$ , then for any fixed  $D$ , the number  $[D, E]$  is bounded by  $\deg^+(E)$  (where  $E_{\bar{K}} = E_+ - E_-$  and  $\deg^+(E) = \deg(E_+)$ ). ┘

*Proof:* Cf.[Lang, Diophantine Geometry, P282]. □

**Prop. (23.1.1.4) [Global Néron Symbols].** The sum of the global Néron symbols is just the Néron-Tate pairing on the ┘

### Neron Functions

**Def. (23.1.1.5) [Néron Functions].** For  $(K, v) = (\mathbb{R}, |\cdot|)$  or  $(K, v) = (\mathbb{C}, |\cdot|^2)$ , and  $X \in \mathcal{C}ur^{sm, proper}/K$ , a **Néron function** is an association, which to each divisor  $D \in \text{Div}(X)$  associates a function

$$\lambda_D : X(\mathbb{C}) \setminus \text{Supp}(D) \rightarrow \mathbb{R}$$

satisfying the following:

- $\lambda_D$  depends additively on  $D$ .
- If  $D = \text{div}(f)$  for  $f \in R(X)^\times$ , then  $\lambda_D = -\log |f|_v + \text{constant}$ .
- For  $P, Q \in X(\mathbb{C})$ ,  $\lambda_P(Q) = \lambda_Q(P)$ .
- If  $D = \text{div}(f)$  on an open subset  $U \subset X$ , then  $\lambda_D - \log |f|_v$  is continuous on  $U$ .

┘

## 2 Arithmetic Intersections

References are [Points of small height on semiabelian varieties, Kuhne],

### Arithmetic Intersection on Curves

References are [Ara74], [Intersection Theory of Divisors on an Arithmetic Surface, Arakelov], [Heights and Arakelov's Intersection Theory, Hriljac], [Calculus on Arithmetic Surfaces, Faltings].

**Notation (23.1.2.1).**

- Let  $F \in \mathbf{NField}$ .
- Let  $X \in \mathcal{C}ur^{sm, proper, g}/F$  with a smooth proper model  $\mathcal{X}/\mathcal{O}_F$  with no degenerate fibers.

┘

**Def. (23.1.2.2) [Arakelov-Divisors].** Situation as in (23.1.2.1), choose for each  $v \in \Sigma_F^\infty$  a Hermitian metric  $ds_\alpha^2$  on  $X_v$  s.t.  $X_v$  has volume 1 w.r.t. the corresponding volume element  $d\mu_v$ .

Then a **Arakelov-divisor** on  $\mathcal{X}$  is a formal sum of the form

$$D \in \text{CH}^1(\mathcal{X}) \oplus \bigoplus_{v \in \Sigma_F^\infty} \mathbb{R}X_v.$$

The group of Arakelov-divisors is denoted by  $\text{Div}_{\text{Ar}}(\mathcal{X})$ .

┘

**Def. (23.1.2.3) [Degree of Arakelov-Divisors].** The **degree of Arakelov-divisor**  $D = \sum_{\mathfrak{p} \in \Sigma_F^{\text{fin}}} n_{\mathfrak{p}} \cdot [\mathfrak{p}] + \sum_{v \in \Sigma_F^\infty} \lambda_v[v]$  is defined to be

$$\deg(D) = \sum_{\mathfrak{p} \in \Sigma_F^{\text{fin}}} n_{\mathfrak{p}} \cdot \log \|\mathfrak{p}\| + \sum_{v \in \Sigma_F^\infty} \lambda_v.$$

┘

**Def. (23.1.2.4) [Principle Arakelov Divisors].** Situation as in (23.1.2.2), for any  $\varphi \in R(X)^\times$ , define the **Arakelov divisor** of  $\varphi$  as

$$\text{div}_{\text{Ar}}(\varphi) = \text{div}(\varphi) \oplus \bigoplus_{v \in \Sigma_F^\infty} \left( - \int_{X_v} \log |\varphi| d\mu_v \right) \cdot X_v.$$

The group of c-divisors modulo principle divisors is denoted by  $\text{Pic}_c(\mathcal{X})$ .

┘

**Remark (23.1.2.5) [Alternative Principle Divisors].** But notice that Hriljac defined the Archimedean part differently: For a choice of Néron functions on  $X_v$  for each  $v \in \Sigma_F^\infty$  (23.1.1.5), define

$$\operatorname{div}_{\operatorname{Ar}}(\varphi) = \operatorname{div}(\varphi) \oplus \bigoplus_{v \in \Sigma_F^\infty} \gamma_v(\varphi) \cdot X_v,$$

where  $\gamma_v(\varphi)$  is constant  $-\log |\varphi|_v - \lambda_{\operatorname{div}(\varphi)}$  (23.1.1.5). Why is this compatible with the form above?  $\lrcorner$

**Def. (23.1.2.6) [Canonical Metrics].** Situation as in (23.1.2.2), there is a canonical metric  $ds^2$  on  $X_v$  induced by the invariant metric  $ds_{J,v}^2$  on  $J(X_v)$  defined by the  $\Theta$ -polarization? via the pullback  $X_v \rightarrow J(X_v)$ .  $\lrcorner$

**Def. (23.1.2.7) [Arithmetic Line Bundles].** For  $D = D_f + D_\infty \in \operatorname{Div}_{\operatorname{Ar}}(\mathcal{X})$ , the **Arakelov line bundle**  $\mathcal{O}(E)$  is defined to be the line bundle  $\mathcal{O}(E_f)$  together with the Hermitian metrics on the induced line bundle on  $X_v$  for  $v \in \Sigma_F^\infty$ , where if  $E_\infty = \sum_{v \in \Sigma_F^\infty} r_v \cdot v$ , we multiply the metric on  $\mathcal{O}(E_f)$ ? Cf. [Calculus on Arithmetic Surfaces, Faltings]P391) by  $\exp(-r_v)$ .  $\lrcorner$

**Def. (23.1.2.8) [Arithmetic Intersections].** Let  $D, E$  be disjoint horizontal divisors on  $X$ , then the **Arithmetic intersection** of  $D, E$  is defined to be the sum of finite intersections and infinite intersections, where

- the finite intersections of two divisors in  $X_v$  is defined to be  $\log ||v||$  times the usual intersections?.
- the infinite intersections depends on a choice of Néron functions on  $X_v$  for  $v \in \Sigma_F^\infty$  (23.1.1.5), and is defined by decomposing  $D, E$  into  $\mathbb{C}$ -points  $P_i, Q_j$  and sum up  $\lambda_v(P_i, Q_j) = \lambda_{P_i}(Q_j) = \lambda_{Q_j}(P_i)$ .

And for the Archimedean divisors  $X_v$ , they intersect null with vertical divisors and among themselves, and if  $D$  is a horizontal prime divisor and  $v \in \Sigma_F^\infty$ , then  $(X_v \cdot D) = \deg_K(D)$ .  $\lrcorner$

**Prop. (23.1.2.9).** The intersection number is independent of the metric chosen.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (23.1.2.10) [Pullbacks].** Under a inverse image homomorphism of a finite map of degree  $d$ , the intersection number is multiplied by  $d$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (23.1.2.11) [Rational Invariance of Intersection Numbers].** If  $f \in R(X)^\times$ , then

$$(\operatorname{div}_{\operatorname{Ar}}(f) \cdot E) = 0.$$

$\lrcorner$

*Proof:* Cf. [Hriljac]P33.  $\square$

**Def. (23.1.2.12) [Canonical Divisors].**  $\lrcorner$

**Prop. (23.1.2.13) [Avoiding Special Fibers].** For  $K \in p\text{-LField}$  and  $s \in \text{Spec } \mathcal{O}_K$  be the special point, denote  $\text{Div}_s(X)$  the divisor group generated by components of  $X_s$ . Then the radical of the intersection pairing on  $\text{Div}_s(X) \otimes \mathbb{Q}$  is the space generated by  $X_s$ .

In particular, there exists a unique linear map

$$\Phi_s : \text{Div}_0(X)_K \rightarrow \text{Div}_s(X) \otimes \mathbb{Q}/\mathbb{Q}X_s$$

s.t. for any  $D \in \text{Div}_0(X_K)$ , the divisor  $\overline{D} + \Phi_s(D)$  is orthogonal to  $\text{Div}_s(X)$ .  $\lrcorner$

*Proof:* Cf.[Lang, Arithmetic Geometry, Chap12, Thm4.5].  $\color{red}?$   $\square$

**Cor. (23.1.2.14).** Situation as in(23.1.2.13), if  $D \in \text{Div}_0(X)_K$  is a principle fiber, then  $\overline{D} + \Phi_s(D)$  is also a principle fiber.  $\lrcorner$

**Thm. (23.1.2.15).** Local situation as in(23.1.1.1), for  $D, E \in \text{Div}_0(X)_K$  with disjoint supports, we have

$$\langle D, E \rangle_v = I(\overline{D} + \Phi_s(D), \overline{E}).$$

$\lrcorner$

*Proof:*  $\square$

### Arithmetic Riemann-Roch Formula

**Def. (23.1.2.16) [Euler Character of Arithmetic Modules].** Let  $F \in \text{NField}$  and  $M \in \text{Mod}_{\mathcal{O}_F}^{\text{fg}}$ , together with a Haar measure on  $M \otimes \mathbb{R}$ , then define

$$\chi(M) = -\log \text{Vol}(M \otimes \mathbb{R}/M) + \log(\#M_{\text{tor}}) + \text{rank}_{\mathcal{O}_F} M \cdot \log(F \otimes \mathbb{R}/\mathcal{O}_F),$$

which is independent of the Haar measure chosen.  $\lrcorner$

**Def. (23.1.2.17) [Euler Character of Admissible Line Bundles].** For  $X \in \text{Cur}^{\text{sm,proper}}/F$  with a smooth proper model  $\mathcal{X}/\mathcal{O}_F$ , if  $\mathcal{L}$  is an admissible line bundle on  $\mathcal{X}$ , then we can define

$$\chi(\mathcal{L}) = \chi(H^0(\mathcal{X}, \mathcal{L})) - \chi(H^1(\mathcal{X}, \mathcal{L})).$$

$\lrcorner$

**Thm. (23.1.2.18) [Arithmetic Riemann-Roch Formula, Faltings].** Situation as in(23.1.2.17), then

$$\chi(\mathcal{L}) = \frac{1}{2} \langle \mathcal{L}, \mathcal{L} - \omega_{\mathcal{X}} \rangle + \chi(\mathcal{O}_{\mathcal{X}}).$$

$\lrcorner$

*Proof:* Cf.[Calculus on Arithmetic Surfaces, Faltings]P405.  $\color{red}?$   $\square$

**Cor. (23.1.2.19) [Positivity of Admissible Line Bundles].** Let  $\mathcal{L}$  be an admissible Hermitian line bundle on  $\mathcal{X}$  with  $\langle \mathcal{L}, \mathcal{L} \rangle > 0, \deg(\mathcal{L}) > 0$ , then when  $n$  large,  $\mathcal{L}^{\otimes n}$  is isomorphic to  $\mathcal{O}(D)$  for a positive Arakelov divisor  $D$ .  $\lrcorner$

*Proof:* Cf.[Calculus on Arithmetic Surfaces, Faltings]P406.  $\color{red}?$   $\square$

Arithmetic Hodge Index TheoremArithmetic Noether's FormulaPositivity of the Arithmetic Relative Dualizing Sheaf  $\omega_{\mathcal{X}}$ **3 Green Functions****Notation (23.1.3.1).**

- Let  $(X, \omega)$  be a Kähler manifold.
- 

┘

**Def. (23.1.3.2) [Admissible Green Functions].** For a divisor  $Z \subset X$ , an **admissible Green function** for  $Z$  is a function  $g_Z \in C^\infty(X)$  s.t.

$$\mathrm{dd}^c g_Z + \delta_Z = \omega_Z$$

where  $\omega_Z$  is a harmonic  $(1, 1)$ -form on  $X$ .

Then admissible Green functions exists for any divisor  $Z \subset X$  and is unique up to a locally constant function.

┘

*Proof:*

□

**4  $S$ -Punctured Arithmetic Chow Groups****Notation (23.1.4.1).**

- Let  $F \in \mathbf{NField}$ ,
- Let  $S \in \Sigma_F$  be a finite set of places.
- Let  $\mathcal{X}/\mathrm{Spec} \mathcal{O}_F[S^{-1}]$  be a regular flat proper scheme with smooth generic fiber  $X$ .
- 

┘

**Def. (23.1.4.2) [Arithmetic Chow Groups].** The **arithmetic Chow group**  $\widehat{\mathrm{Ch}}^1(\mathcal{X})_{\mathbb{Q}}$  is defined to be the  $\mathbb{Q}$ -linear combinations of tuples  $(Z, g_Z)$  where  $Z$  is a divisor on  $\mathcal{X}$  and  $g_Z = (g_{Z,v})_{v \in \Sigma_F^\infty}$  is a tuple of Green functions for the divisor  $Z_v(\mathbb{C}) \subset X_v(\mathbb{C})$ , modulo principle divisors  $\mathrm{div}_{\mathrm{Ar}}(f)$  for  $f \in R(X)^\times$ .

┘

**Def. (23.1.4.3) [Admissible Chow Groups].** The **admissible Chow group**  $\widehat{\mathrm{Ch}}^{1, \mathrm{adm}}(\mathcal{X})_{\mathbb{Q}}$  is the subgroup of  $\widehat{\mathrm{Ch}}^1(\mathcal{X})_{\mathbb{Q}}$  generated by all the tuples  $(Z, g_Z)$  where  $g_Z$  is a tuple of admissible Green functions for  $Z$  (23.1.3.2).

And let  $\widehat{\mathrm{Ch}}_\infty^1(\mathcal{X})$  be the subgroup of  $\widehat{\mathrm{Ch}}^{1, \mathrm{adm}}(\mathcal{X})_{\mathbb{Q}}$  generated by the tuples  $(Z, g_Z)$  s.t.  $Z = 0$ . Then there is a natural map

$$\widehat{\mathrm{Ch}}^{1, \mathrm{adm}}(\mathcal{X})_{\mathbb{Q}} / \widehat{\mathrm{Ch}}_\infty^1(\mathcal{X}) \rightarrow \mathrm{Ch}^1(X).$$

┘

**Prop. (23.1.4.4) [Admissible Extension of Divisors].** If  $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_F[S^{-1}]$  is smooth, then the natural map

$$\widehat{\operatorname{Ch}}^{1, \operatorname{adm}}(\mathcal{X})_{\mathbb{Q}} / \widehat{\operatorname{Ch}}_{\infty}^1(\mathcal{X}) \rightarrow \operatorname{Ch}^1(X) \quad (23.1.4.3)$$

is an isomorphism.  $\square$

*Proof:* Cf. [Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang]P18.  $\square$

**Def. (23.1.4.5) [Cycles and Vertical Cycles].** Denote  $\tilde{\mathcal{Z}}_1(\mathcal{X})$  the group of 1-cycles on  $\mathcal{X}$ , and denote  $\mathcal{Z}_1(\mathcal{X})$  the quotient group of  $\tilde{\mathcal{Z}}_1(\mathcal{X})$  by the subgroup generated by vertical principle divisors, i.e.  $\operatorname{div}(f)$  where  $f \in R(Y)^{\times}$  and  $Y \subset X_v$  is a 2-dimensional integral closed subscheme of a special fiber.  $\square$

### S-Punctured Intersection Pairings

References are [Bost-Gillet-Soule, Heights of projective varieties and positive Green forms].

**Def. (23.1.4.6) [S-Punctured Intersection Pairing].** Denote  $\mathbb{R}_S = \mathbb{R} / \{\mathbb{Q}||v||\}_{v \in S}$ , then if  $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_F[S^{-1}]$  is smooth, there is a **punctured arithmetic intersection pairing**

$$\widehat{\operatorname{Ch}}^1(\mathcal{X}) \times \tilde{\mathcal{Z}}_1(\mathcal{X}) \rightarrow \mathbb{R}_S$$

s.t. if  $(Z, g_Z) \in \operatorname{Ch}^1(\mathcal{X})$  and  $C \subset \mathcal{X}$  is a closed curve s.t.  $Z \cap C$  is Artinian, then

$$((Z, g_Z), C) = \log |\mathcal{O}_{Z \cap C}| + \frac{1}{2} \sum_{v \in \Sigma_F} g_v(C_v(\mathbb{C})).$$

$\square$

**Prop. (23.1.4.7) [S-Punctured Admissible Intersection Pairing].** Denote  $\mathcal{Z}_1(\mathcal{X})_{\deg=0}$  the subgroup of  $\mathcal{Z}_1(\mathcal{X})$  generated by 1-cycles with vanishing degree on each connected component of  $X$ , then the intersection pairing (23.1.4.6), when restricted to  $\mathcal{Z}_1(\mathcal{X})_{\deg=0}$ , induces a pairing

$$(-, -) : \widehat{\operatorname{Ch}}^1(\mathcal{X}) / \widehat{\operatorname{Ch}}_{\infty}^1 \times \tilde{\mathcal{Z}}_1(\mathcal{X}) \rightarrow \mathbb{R}_S.$$

So by (23.1.4.4) induces a pairing

$$(-, -)^{\operatorname{adm}} : \operatorname{Ch}^1(X) / \widehat{\operatorname{Ch}}_{\infty}^1 \times \tilde{\mathcal{Z}}_1(\mathcal{X}) \rightarrow \mathbb{R}_S.$$

$\square$

*Proof:* ?  $\square$

## 5 Arithmetic Standard Conjectures

References are [Arithmetic Analogs of the Standard Conjectures, Gillet-Soule].

## 23.2 Arakelov Geometry

### 1 Metrized Line Bundles

**Def. (23.2.1.1) [Metrized Line Bundles].** For  $F \in \mathbf{NField}$ , a **metrized line bundle**  $(\mathcal{L}, |\cdot|)$  is a line bundle on  $\mathcal{O}_F$  together with norms  $\|\cdot\|_v$  on the free  $F_v$ -module  $M \otimes_R F_v$  of rank 1 for  $v \in \Sigma_F^\infty$ .  $\lrcorner$

**Def. (23.2.1.2) [Heights and Degree of Metrized Line Bundles].** Let  $F$  be a number field and  $(\mathcal{L}, |\cdot|)$  be a metrized line bundle on  $\mathcal{O}_F$  (23.2.1.1), for each  $v \in \Sigma_F^0$ ,  $\mathcal{L}_v = \mathcal{O}_{F,v} m_v$  for some  $m_v \in \mathcal{L}_v$ , so we define  $\|m\|_v = |m/m_v|_v$  for  $m \in \mathcal{L}_v$ , and define the **height of  $\mathcal{L}$**  to be

$$H((\mathcal{L}, |\cdot|)) = \prod_{v \in \Sigma_F} \|m\|_v^{-1}, \quad m \neq 0 \in M$$

which is independent of  $m$  by product formula. Also we define

$$h(M) = \frac{1}{[F : \mathbb{Q}]} \log H(M),$$

called the **degree of  $\mathcal{L}$** .  $h((\mathcal{L}, |\cdot|))$  is invariant under change of number fields.  $\lrcorner$





## 24 | Representation Theory

### 24.1 Smooth Representations

*“A tremendous effort has been made by mathematicians for more than a century to clear up the chaos in group theory. Still, we cannot answer some of the simplest questions.”*

–R. Brauer

In this section the representation theory over field  $\mathbb{C}$  are studied (at least over a discrete field of characteristic 0). Modular representations are studied in [Positive Characteristic Representations](#).

**Notation (24.1.0.1).**

- Let  $G \in \text{TopGrp}$ .
- Use notations on non-commutative algebras defined in [More on \(Non-Commutative\) Algebras](#).

┘

#### 1 Topological Representations

**Def. (24.1.1.1) [Representations].** Let  $R$  be a topological ring with a continuous  $G$ -action, then an  **$R$ -representation** of  $G$  is a free topological  $R$ -module  $V$  with an  $R$ -covariant action of  $G$  on  $M$  s.t.  $G \times M \rightarrow M$  is continuous, i.e.  $g(r.m) = g(r).g(m)$  for any  $g \in G, r \in R, m \in M$ . The category of such representations is denoted by  $\text{Rep}_R(G)$ . Also denote  $\text{Rep}(G) \triangleq \text{Rep}_{\mathbb{C}}(G)$  to the category of representations on any Fréchet space over  $\mathbb{C}$ .

An **irreducible representation** is a representation that has no non-trivial invariant closed  $R$ -submodules. A non-irreducible representation is called a **reducible representation**. The category of irreducible  $R$ -representations of  $G$  is denoted by  $\text{Irr}_R(G)$ .

A representation  $(\rho, V) \in \text{Rep}_R(G)$  is called a **finite dimensional representation** if  $\dim_R V < \infty$ . The subcategory of f.d.  $R$ -representations is denoted by  $\text{Rep}_R^{\text{fd}}(G)$ .

An **indecomposable representation** is a representation that is not a direct sum of two sub-representations.

┘

**Def. (24.1.1.2) [Free Representations].** Situation as in (24.1.1.1),  $V \in \text{Rep}_R(G)$  is called a **free  $R$ -representation** if  $V$  is free over  $B$ . And it is trivial iff  $V \cong B^n \in \text{Rep}_B(G)$ .

┘

**Def. (24.1.1.3) [Constructing Representations].** For  $(\rho, M), (\rho', M') \in \text{Rep}_R(G)$ ,

- The **dual representation**  $(\rho^*, M^*) \in \text{Rep}_R(G)$  is defined as follows:  $M^* = \text{Hom}_R(M, R)$ , and  $\rho^*(g)(f)(m) = g(f(g^{-1}(m)))$ . It is  $R$ -semi-linear because

$$\rho^*(g)(f)(r.m) = g(f(g^{-1}(r.m))) = g(g^{-1}(r)f(g^{-1}m)) = g(r).\rho^*(g)(f)(m).$$

- The **tensor product representation**  $(\rho \otimes \rho', M \otimes M')$  is defined as follows:  $g(m \otimes m') = g(m) \otimes g(m')$ .
- The **symmetric tensor representation** and **exterior tensor representation** is defined as tensor product, and symmetric and exterior tensors.  $\lrcorner$

**Def. (24.1.1.4) [Projective Representations].** A **projective representation** of a topological group  $G$  on a TVS  $V$  is a continuous map  $G \times \mathbb{P}(V) \rightarrow \mathbb{P}(V)$ , where the topology on  $\mathbb{P}(V)$  is induced from  $V \setminus \{0\} \subset V$ .  $\lrcorner$

**Prop. (24.1.1.5) [F.D. Schur's Lemma].** Let  $(\pi_1, V_1), (\pi_2, V_2) \in \text{Irr}^{\text{fd}}(G)$ , then  $\text{Hom}_G(V_1, V_2) = \mathbb{C}$  if  $\pi_1 \cong \pi_2$ , and 0 otherwise.  $\lrcorner$

*Proof:* This is because  $\pi_i$  are irreducible as representations with discrete topology of  $G$ : Any subspace of  $V_i$  is closed. So we reduce to (24.1.1.10).  $\square$

**Def. (24.1.1.6) [Types of Representations].** For  $V \in \text{Rep}^{\text{fd}}(G)$ , there are three possibilities:

- $V \not\cong V^*$ , called a **representation of complex type**.
- There is an invariant symmetric form on  $V$  inducing an isomorphism  $V \cong V^*$ , called a **representation of real type**.
- There is an invariant alternating form on  $V$  inducing an isomorphism  $V \cong V^*$ , called a **representation of quaternion type**.

The category of representation of  $\mathbb{K}$ -type is denoted by  $\text{Rep}(G)_{\mathbb{K}}$ .  $\lrcorner$

*Proof:* By Schur's lemma (24.1.1.5), either  $V \not\cong V^*$  or there is a unique isomorphism  $V \cong V^*$  up to scalar, which is equivalent to an invariant form  $B$  on  $V$ . The two coordinate of this invariant form induces two isomorphisms  $V \cong V^*$ , which must be proportional, so  $B(x, x) = cB(x, x)$ , so  $c = 0$  or  $B(x, x) = 0$  for all  $x$ , which means  $B$  is symmetric or alternating.  $\square$

**Def. (24.1.1.7) [Unitary Representations].** Usually we consider unitary representations on a Hilbert space. A **unitary representation** of  $G$  on a Hilbert space  $\mathcal{H} \in \mathcal{H}\text{ilbSpa}$  is defined to be a homomorphism from  $G$  to the group  $U(\mathcal{H})$  of unitary operators of  $\mathcal{H}$  continuous in the strong operator topology (12.7.4.6). Notice by (12.7.4.7), this is equivalent to the unitary and continuous in the weak operator topology.

The category of unitary representations of  $G$  is denoted by  $\text{Rep}^{\text{uni}}(G)$ .  $\lrcorner$

**Def. (24.1.1.8) [Projective Unitary Representation].** A **projective unitary representation** of  $G$  on a Hilbert space  $\mathcal{H} \in \mathcal{H}\text{ilbSpa}$  is defined to be a continuous homomorphism from  $G$  to the group  $PU(\mathcal{H})$  of unitary operators of  $\mathcal{H}$ , where the topology of  $PU(\mathcal{H})$  is induced from the strong operator topology of  $U(\mathcal{H})$  (12.7.4.6). Notice by (12.7.4.7), this is equivalent to the unitary and continuous in the weak operator topology.  $\lrcorner$

**Def. (24.1.1.9) [Character].** Let  $(\rho, V) \in \text{Rep}^{\text{fd}}(G)$ . Then the **character** of  $\chi_\rho$  is defined to be  $\chi_\rho(g) = \text{tr}(\rho(g))$ .

Suppose  $g, h \in G$  are conjugate, then  $\chi_\rho(g) = \chi_\rho(h)$ .  $\lrcorner$

**Prop. (24.1.1.10) [Schur's lemma].** If  $(\pi, V)$  is an at most countable dimensional irreducible  $\mathbb{C}$ -representation of a topological  $\mathbb{C}$ -algebra, then  $\text{End}_A(V) \cong \mathbb{C}$ . In particular this holds for  $\dim A$  countable.  $\lrcorner$

*Proof:* First the dimension of  $\dim_{\mathbb{C}} \text{End}(V)$  is at most countable, because  $V$  is acyclic by irreducibly, so  $\dim_{\mathbb{C}} \text{End}(V) \leq \dim_{\mathbb{C}} V$ . And  $\text{End}(V)$  is a skew field, by irreducibility. So the result follows from (3.6.1.2).  $\square$

**Cor. (24.1.1.11).** Any irreducible representation of a commutative group has dimension 1.  $\lrcorner$

**Cor. (24.1.1.12) [Index 2 Subgroup].** Let  $H$  be an open (normal) subgroup of index 2,  $G = H \rtimes \{\sigma\}$ .

Let  $(\pi, V)$  be an irreducible representation of  $H$  of at most countable dimensional, then

- $\text{res}_H^G \text{ind}_H^G \pi \cong \pi \oplus \pi^\sigma$ .
- $\pi \cong \pi^\sigma$  iff  $\pi$  is the restriction of an irreducible representation of  $G$ , called **type I**. And there are exactly two such extensions to  $G$ .
- $\pi \not\cong \pi^\sigma$  iff  $\text{ind}_H^G \pi$  is an irreducible representation of  $G$ , called **type II**.

All these are true for unitarizable representations.  $\lrcorner$

*Proof:* 1: This is clear.

2: If  $\pi$  is a restriction of an irreducible smooth representation of  $G$ , then  $\sigma$  intertwines  $\pi$  and  $\pi^\sigma$ . Conversely, if  $\pi \cong \pi^\sigma$ , then there is an operator  $A$  on  $V$  that intertwines  $\pi$  and  $\pi^\sigma$ , and then  $A^2 = \text{id}$  by Schur's lemma (24.1.1.10). Let  $\sigma$  acts by  $A$ , then this representation extends to  $G$ .

3: If  $\pi$  extends to  $G$ , then clearly  $\pi$  appear in  $\text{ind}_H^G \pi$  but not surjective, so  $\text{ind}_H^G \pi$  is not irreducible. Conversely, if  $\text{ind}_H^G \pi$  is not irreducible, notice  $\text{res}_H^G \text{ind}_H^G \pi \cong \pi \oplus \pi^\sigma$ , thus  $\pi \cong \pi^\sigma$ , otherwise any  $G$ -invariant subspace can only be  $\pi$  or  $\pi^\sigma$ , so  $\pi$  or  $\pi^\sigma$  extends to representations of  $G$ , contradicting 2.  $\square$

**Prop. (24.1.1.13) [Kolchin].** Let  $k \in \mathbf{Field}$  and  $(\rho, V) \in \text{Rep}_k^{\text{fd}}(G)$  s.t. each  $\rho(g)$  is unipotent, then there exists a basis of  $V$  s.t.  $\rho(G) \in \text{Unip}(n; k)$ .  $\lrcorner$

*Proof:* It suffices to show that there is a non-zero vector of  $V$  fixed by  $G$ . We can pass to  $\bar{k}$ , and also assume that  $\rho$  is simple. Let  $A \subset \text{End}_{\bar{k}}(V)$  be the algebra generated by  $G$ , then  $V$  is simple over  $A$ , and  $\text{End}_A(V) = k$ . Thus by (3.6.2.9),  $A = \text{End}_k(V)$ .

The  $k$ -subspace  $J$  of  $A$  spanned by the elements  $\{g - 1\}_{g \in G}$  is a two-sided ideal of  $A$ , and it consists of elements of trace 0, so it cannot be  $A$ . But  $A$  is simple (3.6.2.21), so  $J = 0$ , which means each  $g$  acts as a scalar, so  $\dim V = 1$ .  $\square$

**Prop. (24.1.1.14) [B-Trivial Representations].** For  $B \in \text{TopCRing}$  with a  $G$ -action,  $n \in \mathbb{N}$ , there is a bijection between the set of equivalence classes of  $B$ -trivial representations  $V$  of  $G$  of rank  $n$  and the category of  $H^1(G, \text{GL}(n; B))$ . Moreover,  $V$  is trivial iff it is mapped to the distinguished point of  $H^1(G, \text{GL}(n; B))$ .  $\lrcorner$

*Proof:* This follows by taking the matrix of  $g \in G$  w.r.t. a  $B$ -basis of  $V$ .  $\square$

**Cor. (24.1.1.15).** Let  $L/K \in \mathbf{GalField}$ , then any  $\rho \in \text{Rep}_L^{\text{fd}}(\text{Gal}(L/K))$  is trivial.  $\lrcorner$

*Proof:* This follows from Hilbert's theorem90 (10.2.5.12).  $\square$

### Admissible Representations

**Def. (24.1.1.16) [Admissible Representations].** Let  $E$  a topological field that  $G$  acts trivially and  $B$  a topological  $E$ -algebra s.t.  $B \in \text{Rep}_E(G)$ . Then  $V \in \text{Rep}_E^{\text{fd}}(G)$  is called a  **$B$ -admissible** representation if  $B \otimes_E V \in \text{Rep}_B(G)$  is trivial, i.e. isomorphic to  $B^{\oplus \dim V}$ .

The category  $\text{Rep}_E^{B\text{-adm}}(G)$  is the full subcategory of  $\text{Rep}_E(G)$  consisting of f.d.  $B$ -admissible  $E$ -representations of  $G$ .  $\lrcorner$

**Prop. (24.1.1.17) [Inclusions and Admissibility].** Let  $E$  be a topological field that  $G$  acts trivially and  $B_1, B_2, B$  a topological  $E$ -algebra s.t.  $B_1, B_2, B \in \text{Rep}_E(G)$ , and  $B_1 \subset B, B_2 \subset B, B_1 \cap B_2 = B_0$ , and  $B^G \subset B_0$ , then

$$\text{Rep}_E^{B_1\text{-adm}}(G) \cap \text{Rep}_E^{B_2\text{-adm}}(G) = \text{Rep}_E^{B_0\text{-adm}}(G).$$

┘

*Proof:* The RHS is contained in LHS trivially. For the converse inclusion, if  $V \in \text{Rep}_E^{B_1\text{-adm}}(G) \cap \text{Rep}_E^{B_2\text{-adm}}(G)$ , there exists elements  $\{u_i\} \subset (B_1 \otimes_E V)^G$  and  $\{v_i\} \subset (B_2 \otimes_E V)^G$  s.t.

$$B \otimes_E V = Bu_1 \oplus \dots Bu_n = Bv_1 \oplus \dots Bv_n.$$

Then the transformation matrix from  $\{u_i\}$  to  $\{v_i\}$  is an element in  $\text{GL}(n; B)$  that is invariant under  $G$ , so contained in  $\text{GL}(n; B^G)$ . Thus it is clear that

$$\{v_i\} \subset (B_2 \otimes_E V)^G \cap (B_1 \otimes_E V)^G = (B_0 \otimes_E V)^G,$$

and then  $B_0 \otimes_E V = B_0v_1 \oplus \dots B_0v_n$ , and  $V$  is  $B_0$ -admissible. □

### Gal<sub>K</sub>-Regularity

**Def. (24.1.1.18) [ $G$ -Regularity].** Situation as in (24.1.1.16), we want to establish a numerical criterion for recognizing  $B$ -admissible representations.  $B$  is called  **$G$ -regular** if it satisfies the following three conditions:

H1 :  $B$  is a domain.

H2 :  $(\text{Frac}(B))^G = B^G$ , in particular,  $B^G$  is a field.

H3 : if  $b \neq 0 \in B$  and  $Eb$  is stable under  $G$ -action, then  $b \in B^*$ .

Notice a field is clearly  $G$ -regular. ┘

**Cor. (24.1.1.19).** Notice that (H3) implies  $B^G \in \text{Field}$ , because for  $b \in B^{\text{Gal}_K}$ ,  $Eb$  is clearly stable under  $G$ -action, thus  $b$  is invertible.

Also the morphism

$$\alpha_B(W) : B \otimes_{B^G} W^G \rightarrow W$$

is injective for all finite free  $W \in \text{Rep}_B(G)$ . In particular, this is true for  $W = B \otimes_E V, V \in \text{Rep}_E^{\text{fd}}(G)$ , and we get a functor

$$D_B : \text{Vect}_E \rightarrow \text{Vect}_{B^G}$$

such that

$$\dim_{B^G} D_B(V) \leq \dim_E V.$$

┘

*Proof:* To show  $\alpha_W$  is injective, it suffices to show a linear basis  $\{e_i\}$  of  $W^G$  over  $B^G$  is linearly independent over  $B$ : Suppose  $\sum a_i e_i = 0$ , where  $a_i \in B$ , with the number of nonzero coefficients minimal, and  $a_1 \neq 0$ , then dividing  $a_1 \in \text{Frac}(B)$ , we assume  $a_1 = 1$ , and then acting by  $g - \text{id}$ , we get

$$\sum (g(a_i) - a_i) e_i = 0$$

and this has smaller non-zero elements, unless  $a_i$  is fixed by  $g$  for any  $g \in G$ , so  $a_i \in \text{Frac}(B)^G = B^G$  by (H2), contradiction. □

**Prop. (24.1.1.20) [B-Admissible Representations].** If  $B$  is  $G$ -regular (24.1.1.18),  $V \in \mathcal{R}\text{ep}_E^{\text{fd}}(G)$  and  $W = B \otimes_E V$ , then the following are equivalent:

- $W$  is trivial, i.e.  $V$  is  $B$ -admissible.
- $\alpha_B(W)$  (24.1.1.19) is an isomorphism.
- $\dim_{B^G} D_B(V) = \dim_E V$ .

┘

*Proof:* 1, 2 are equivalent by (24.1.1.19), as  $B^{\text{Gal}K}$  is a field. Also  $2 \rightarrow 3$  is clear.

$3 \rightarrow 2$ :  $\alpha_W : B \otimes_{B^G} W^G \rightarrow B \otimes_E V$  is a  $B$ -linear morphism of two finite free  $B$ -modules, then it suffices to show the determinant map is an isomorphism. Let  $v_1, \dots, v_d$  be a  $E$ -basis of  $V$  and  $w_1, \dots, w_d$  a  $B^G$ -basis of  $W^G$ . Let  $b$  be the unique element of  $B$  that

$$\alpha_W(v_1) \wedge \dots \wedge \alpha_W(v_d) = bw_1 \wedge \dots \wedge w_d$$

then  $gb = \eta b$  for  $g \in G$  where  $\eta$  is determined by the identity  $\alpha_W(gv_1) \wedge \dots \wedge \alpha_W(gv_d) = \eta \alpha_W(v_1) \wedge \dots \wedge \alpha_W(v_d)$ . Now the  $E$ -space of  $v_1, \dots, v_d$  is  $V$ , which is stable under  $G$  action, thus  $\eta \in E$ , and then by (H3)  $b \in B^*$ , so we are done.  $\square$

**Cor. (24.1.1.21) [ $\mathcal{R}\text{ep}_E^{B\text{-adm}}(G)$ ].** If  $B$  is  $G$ -regular, then

- $\mathcal{R}\text{ep}_E^{B\text{-adm}}(G) \subset \mathcal{R}\text{ep}_E(G)$  is stable under subobjects and quotients.
- $D_B : \mathcal{R}\text{ep}_E^B(G) \rightarrow \text{Vect}_{B^G}$  is exact and faithful.
- $\mathcal{R}\text{ep}_E^{B\text{-adm}}(G) \subset \mathcal{R}\text{ep}_E(G)$  is stable under taking dual and tensor products. And if  $V, V_1, V_2 \in \mathcal{R}\text{ep}_E^{B\text{-adm}}(G)$ , then there is a natural isomorphism

$$D_B(V_1) \otimes D_B(V_2) \cong D_B(V_1 \otimes V_2)$$

and

$$D_B(V) \otimes D_B(V^\vee) \cong D_B(V \otimes V^\vee) \rightarrow D_B(E) = B^G$$

is a perfect pairing between  $D_B(V)$  and  $D_B(V^\vee)$ .

┘

*Proof:* 1: Given an exact sequence  $0 \rightarrow V_1 \rightarrow V \rightarrow V_2 \rightarrow 0 \in \mathcal{R}\text{ep}_E(G)$ , tensoring  $B$  and taking  $G$ -fixed points, we get an exact sequence

$$0 \rightarrow D_B(V_1) \rightarrow D_B(V) \rightarrow D_B(V_2)$$

from which we derive the inequality  $\dim_{B^G} D_B(V) \leq \dim_{B^G} D_B(V_1) + \dim_{B^G} D_B(V_2)$ . Now we have  $\dim_{B^G} D_B(V_i) \leq \dim_E V_i$  by (24.1.1.19), so

$$\dim_{B^G} D_B(V) \leq \dim_{B^G} D_B(V_1) + \dim_{B^G} D_B(V_2) \leq \dim_E V_1 + \dim_E V_2 = \dim_E V.$$

But this is an equality because  $V$  is  $B$ -admissible, thus  $V_1, V_2$  are all  $B$ -admissible, and the exact sequence is in fact an isomorphism by dimension reason.

2:  $D_B$  is faithful because  $B \otimes_{B^G} D_B(V) \cong B \otimes_E V$ .

3: There is a natural map

$$D_B(V_1) \otimes_{B^G} D_B(V_2) = (B \otimes_E V_1)^G \otimes (B \otimes_E V_2)^G \rightarrow (B \otimes_E (V_1 \otimes_E V_2))^G = D_B(V_1 \otimes_E V_2),$$

and  $\dim_{B^G} D_B(V_1 \otimes_E V_2) \leq \dim_E(V_1) \cdot \dim_E(V_2)$ , so it suffices to show that this map is injective. For this, notice that  $D_B(V_1 \otimes_E V_2) \subset B \otimes_E (V_1 \otimes_E V_2)$ , and after tensoring  $B$ ,

$$D_B(V_1) \otimes_{B^G} D_B(V_2) \subset B \otimes_{B^G} (D_B(V_1) \otimes_{B^G} D_B(V_2)) \cong (B \otimes_E V_1) \otimes_B (B \otimes_E V_2) \rightarrow B \otimes_E (V_1 \otimes_E V_2)$$

is an isomorphism.

To show for the dual preserves  $B$ -admissibility, notice that  $\text{Rep}_E^{B\text{-adm}}(G)$  is also stable under exterior products, as exterior products are quotient of tensor products. Notice there is an isomorphism

$$\wedge(V^\vee) \otimes \wedge^{\dim V - 1}(V) \cong V^\vee,$$

so it suffices to show for  $\dim V = 1$ . Let  $v_0$  be an  $E$ -basis of  $V$ ,  $g(v_0) = \eta(g)v_0$ , then  $D_B(V) = B^G(b \otimes v_0)$  for some  $b \neq 0 \in B$ . Thus  $b/g(b) = \eta(g)$ . And it is easy to show that  $D_B(V^\vee) = B^G(b^{-1} \otimes v_0)$ , and the natural pairing is perfect. In general, the pairing is also perfect because perfectness of a pairing can be checked after passing to the determinant space.  $\square$

## Unitary Representations of Locally Compact Groups

For unitary representations of locally compact groups, see [Harmonic Analysis on Locally Compact Groups](#).

## 2 Smooth Representations

**Notation (24.1.2.1).**

- Let  $G \in \text{LCG}$ .

┘

**Prop. (24.1.2.2) [Smooth Representations].** A **smooth representation** of  $G$  on a complex vector space is a continuous representation w.r.t the discrete topology.

The category  $\text{Rep}^{\text{alg}}(G)$  of smooth representations is a full Abelian subcategory of the category of continuous representations on Fréchet spaces, and there is a right adjoint to the forgetful functor:

$$\text{Rep}(G) \rightarrow \text{Rep}^{\text{alg}}(G) : V \mapsto V^{\text{alg}} = \bigcup_{K \subset G \text{ pre-compact open}} V^K.$$

So it preserves injectives and  $\mathcal{M}(G)$  has enough injectives.  $\square$

**Prop. (24.1.2.3) [ $V^{\text{alg}}(G)$  is Dense].** For  $(\pi, V) \in \text{Rep}(G)$ ,  $V^{\text{alg}} \subset V$  (24.1.2.2) is dense.  $\square$

*Proof:* For any pre-compact open subset  $U \subset G$ ,  $\mu(U)^{-1} \mathbf{1}_U \cdot v \in V^{\text{alg}}$  (12.8.3.21), and  $\mu(U)^{-1} \mathbf{1}_U \cdot v$  converges to  $v$  as a net by continuity.  $\square$

**Def. (24.1.2.4) [Equivariant Sheaves].** Let  $a : G \times X \rightarrow X$  be an action of  $G$  on a space  $X$  and  $p : G \otimes X \rightarrow X$  be the projection, then a **equivariant sheaf** on  $X$  is a pair  $(\mathcal{F}, \rho)$ , where  $\mathcal{F}$  is a locally constant complex sheaf on  $X$  and  $\rho$  is an isomorphism of sheaves  $p^*(\mathcal{F}) \cong a^*(\mathcal{F})$  that:

- $\rho$  is identity on  $e \otimes X$ .
- $p_{23}^* \rho \circ (\text{id}_G \times a)^* \rho = (m \times \text{id}_X)^* \rho$  on  $G \times G \times X$ .

┘

**Prop. (24.1.2.5).** If  $X \cong \text{pt}$ , then a equivariant sheaf on  $X$  is equivalent to a representation of  $G$ .  $\square$

*Proof:* For any equivariant sheaf on  $X$ , the pullback are just locally constant functions of  $G$  with value in  $V$ . Then  $\rho$  on each stalk  $g$  defines an action of  $g$  on  $V$ . Compatibility with the  $G$ -action shows that this is a group action. And consider the stalk at  $e$ , because  $\rho$  is id at  $e$ , for each  $v$ , there is an open nbhd  $U$  that  $\rho(u)v = v$  on for  $u \in U$ , thus it is smooth. The converse is obvious.  $\square$

**Def. (24.1.2.6) [Jacquet Functors].** The **Jacquet functor**  $J_G : \mathcal{R}\text{ep}(G) \rightarrow \mathcal{V}\text{ect}_{\mathbb{C}}$  is the functor mapping a representation  $V$  of  $G$  to  $V/V(G)$ , where  $V(G)$  is spanned by  $\pi(g)v - v$ . It is equivalent to the functor  $V \mapsto V \otimes_G \mathbb{C}$ .

Let  $\psi$  be a character of  $G$ , then we can more generally define  $J_{G,\psi} : \mathcal{R}\text{ep}(G) \rightarrow \mathcal{V}\text{ect}_{\mathbb{C}}$  is the functor mapping a representation  $V$  of  $G$  to  $V/V_{G,\psi}$ , where  $V_{G,\psi}$  is spanned by  $\pi(g)v - \psi v$ . It is equivalent to the functor  $V \mapsto V \otimes_G \mathbb{C}_{\psi}$ , or equivalently  $J_{G,\psi} = J_G \circ (\psi^{-1} \otimes -)$ .  $\lrcorner$

**Prop. (24.1.2.7).** If  $G$  is compact, then  $e_G V = V^G = V/V(G)$ .

And if  $G$  is a union of increasing family of compact groups, then  $J_G$  is exact, so is  $J_{G,\psi}$ .  $\lrcorner$

*Proof:*  $J_G$  is clearly right exact. And if  $G$  is compact,  $e_G V = V^G = V/V(G)$ : if  $\pi(G)v = v$ , then  $v = e_G v$ , and if  $e_G v = 0$ , then because  $v$  is smooth,  $\sum_{h_i \in G/K} h_i v = 0$ , thus  $v = \frac{1}{[G:K]}(v - h_i v)$  is in  $V(G)$ .

If  $G$  is a union of compact groups  $K_i$ , then  $V/V(G) = \text{colim } V/V(K_i)$  is exact.  $\square$

**Prop. (24.1.2.8).** If  $G$  is a union of increasing family of compact open subgroups  $\{K_{\alpha}\}$ , then  $v \in V(G, \psi)$  ( $\psi$  can be trivial) iff for some  $K_{\alpha}$ ,  $\int_{K_{\alpha}} \psi^{-1}(h)\pi(h)v dh = 0$ .  $\lrcorner$

*Proof:* We can assume  $\psi$  is trivial. If  $v = \pi(h)w - w$  and  $h \in K_{\alpha}$ , then  $\int_{K_{\alpha}} \psi^{-1}(h)\pi(h)v dh = 0$ . Conversely, by the proof of (24.1.2.7),  $V(G) = \cup V(K_{\alpha}) = \cup \ker(e_{K_{\alpha}})$ , which is equivalent to  $\int_{K_{\alpha}} \pi(h)v dh = 0$ .  $\square$

**Def. (24.1.2.9) [Contragradient Representations].** For  $V \in \mathcal{R}\text{ep}^{\text{alg}}(G)$ , the **contragradient smooth representation**  $V^{\vee} = (V^*)^{\text{alg}}$  is the smooth part of  $V^*$ .  $\lrcorner$

**Prop. (24.1.2.10).** Let  $E$  be a smooth or f.d. representation of  $G$ , then

- $E$  has an irreducible subquotient.
- If  $E$  is f.g., then it has an irreducible quotient.

$\lrcorner$

*Proof:* 2: Use Zorn's lemma for the set of proper  $G$ -subspaces of  $U$ , the union of an ascending chain of proper  $G$ -subspaces is proper, because  $E$  is f.g.. So it has a maximal proper  $G$ -space, and then the quotient is irreducible.

1 follows from item2 by choosing a f.g. submodule.  $\square$

**Def. (24.1.2.11) [Matrix Coefficients of Smooth Representations].** For  $(\pi, V) \in \mathcal{R}\text{ep}^{\text{alg}}(G)$ , a **matrix coefficient** for  $V$  is a function on  $G$  of the form  $g \mapsto (\pi(g)u, v)$ , where  $u \in V$  and  $v \in V^{\vee}$  (24.1.2.9).  $\lrcorner$

**Def. (24.1.2.12) [Induced Representations of Discrete Groups].**  $\lrcorner$

**Prop. (24.1.2.13) [Induced Representation].** Situation as in (24.1.6.40), there are algebraic isomorphisms

$$\text{Ind}_H^G(V) = \text{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[G], V), \quad \text{ind}_H^G(V) = \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} V.$$

I.e., if  $H$  acts on  $V$  by  $\rho$ , then  $\text{ind}_H^G(\rho)$  is the space  $\oplus_{\gamma \in G/H} V_{\gamma}$  where  $V_{\gamma} \cong V \in \text{Mod}_H$ , and that for  $v_{\gamma} \in V_{\gamma}$ ,  $(\text{Ind}(\rho)g)v_{\gamma} = \rho(h)v_{\gamma'} \in V_{\gamma'}$  where  $g\gamma = \gamma'h$  that  $\gamma' \in G/H, h \in H$ .  $\lrcorner$



*Proof:* Choose a set of left coset representatives  $\Gamma$  of  $G/H$ , then  $\Gamma^{-1}$  is a set of right coset representatives for  $H \backslash G$ . Now  $\mathbb{Z}[G] = \bigoplus_{\gamma \in \Gamma} \mathbb{Z}[H]\gamma^{-1}$ , and we define the space  $V_\gamma \cong V$  of maps from  $\mathbb{Z}[G]$  to  $V$  determined by:  $v_\gamma(g) = \rho(g\gamma)v \in V$  if  $g \in H\gamma^{-1}$  and 0 otherwise.

Then  $\text{Ind}(\rho)(V) = \bigoplus_{\gamma \in \Gamma} V_\gamma$  that for  $v_\gamma \in V_\gamma$ ,  $(\text{Ind}(\rho)g)v_\gamma$  is the map  $\mathbb{Z}[G]$  to  $V$  determined by:  $((\text{Ind}(\rho)g)v_\gamma)(g') = \rho(g'\gamma'h)v \in V$  if  $g' \in H(\gamma')^{-1}$ , and 0 otherwise, where  $g\gamma = \gamma'h$ . Then  $(\text{Ind}(\rho)g)v_\gamma = \rho(h)v_{\gamma'}$ , which is the same as the formula for  $\text{Ind}(\rho)$  as in (24.1.2.13). So the map  $v_\gamma \rightarrow v_\gamma : \text{Ind}(\rho) \rightarrow \text{Ind}(\rho)$  is an isomorphism in  $\text{Mod}_{\mathbb{Z}[G]}$ .  $\square$

**Prop. (24.1.2.14) [Projection Formula].** Let  $H \leq G$ , then for any  $\rho \in \text{Rep}(G), \sigma \in \text{Rep}(H)$ , we have

$$\text{ind}_H^G(\rho|_H \otimes \sigma) \cong \rho \otimes \text{ind}_H^G(\sigma).$$

┘

*Proof:* For any  $\tau \in \text{Rep}(G)$ ,

$$\begin{aligned} \text{Hom}_G(\text{ind}_H^G(\rho \otimes \sigma), \tau) &= \text{Hom}_H(\rho \otimes \sigma, \tau) \\ &= \text{Hom}_H(\sigma, \text{Hom}(\rho, \tau)) \\ &= \text{Hom}_G(\text{ind}_H^G(\sigma), \text{Hom}(\rho, \tau)) \\ &= \text{Hom}_G(\rho \otimes \text{ind}_H^G(\sigma), \tau) \end{aligned}$$

Thus the assertion follows from Yoneda lemma.  $\square$

**Prop. (24.1.2.15) [Restrictions, Clifford's Theorem].** If  $k \in \text{Field}, H \trianglelefteq G$  and  $(\rho, V) \in \text{Rep}_k^{\text{fd}}(G)$  is completely reducible, then  $\rho|_H$  is also completely reducible.  $\square$

*Proof:* Use definition (3.6.2.4), we may assume  $V$  is simple.  $G$  maps a simple  $H$ -subrepresentation of  $V$  to another simple  $H$ -subrepresentation of  $V$  because  $H$  is normal in  $G$ . So if  $W$  is the sum of all simple  $H$ -subrepresentations of  $V$ , then  $W \neq 0$  and  $G$  preserves  $W$ , which implies  $W = V$ . And then  $V$  is  $H$ -semisimple by (3.6.2.4).  $\square$

**Prop. (24.1.2.16) [Restrictions].** Let  $k \in \text{Field}$  and  $H \leq G$  is an open normal subgroup of  $G$  of finite index  $n$  s.t.  $n$  is invertible in  $k$ . Let  $(\pi, V) \in \text{Rep}_k^{\text{alg}}(G) \cup \text{Rep}_k^{\text{fd}}(G)$ , then

- If  $\pi|_H$  is completely reducible, then  $\pi$  is also completely reducible.
- If  $(\pi, V)$  is irreducible, then  $\pi|_H = \pi_1 \oplus \dots \oplus \pi_k$  where  $\pi$  are irreducible representations of  $H$  and  $k \leq [G : H]$ .
- If  $(\pi, V)$  is irreducible, the isotropy parts of  $H$  are all of the same dimension, and if there are more than 1 isotropy classes, then  $\pi$  is an induced representation.

Moreover, item 2, 3 are also true for  $H$  open normal in  $G$  s.t.  $HZ(G)$  is of finite index  $n$ .  $\square$

*Proof:* Let  $g_1, \dots, g_n$  be a coset representation of  $G/H$ .

1: If  $V_1$  is a  $G$ -submodule of  $V$ , then  $V_1$  has a complementary  $H$ -submodule  $V_1^\perp$ . Let  $P_0$  be the projection of  $V$  onto  $V_1$  along  $V_1^\perp$ , then  $P = \frac{1}{p} \sum \pi(g_i)P_0\pi(g_i)^{-1}$  is a projection of  $V$  onto  $V_1$  that commutes with  $G$ -action. Therefore the kernel of  $P$  is a  $G$ -submodule of  $V$  that is complementary to  $V_1$ .

2: Let  $v \in V$ , then  $\{\pi(g_i)v\}$  generates  $V$  as a  $H$ -representation, thus by (24.1.2.10) there is an  $H$ -submodule  $V' \subset V$  that  $V/V'$  is irreducible. Let  $V_i = \pi(g_i)V'$ , then  $V/V_i$  are also irreducible representations of  $H$ . Consider the  $H$ -invariant map  $V \mapsto \bigoplus V/V_i$ , its kernel is  $G$ -invariant, thus trivial, thus  $V$  is a submodule of  $\bigoplus V/V_i$ , and the assertion follows.

3: The  $G$ -action permutes with isotropy classes. And if there are more than 1 classes, choose the stabilizer  $G_\rho$  of one  $V[\rho]$ , then  $V = \text{Ind}_{G_\rho}^G(V[\rho])$  by (24.1.2.13).  $\square$



### 3 Finite Groups

References are [Ser77, A-B95]. [群表示论 notes 薛航].

**Notation (24.1.3.1).**

- Let  $G \in \mathcal{G}rp^{\text{fin}}$ .
- Let  $k \in \mathcal{F}ield^0$ , e.g. a subfield of  $\mathbb{C}$ .
- The representations in this subsection is assumed to be of f.d. over  $\bar{k}$ . In particular, there is no need to consider topologies. And we denote  $\mathcal{R}ep(G) = \mathcal{R}ep_{\bar{k}}(G)$ .
- Because a finite group is compact, the representation theory of compact groups [Compact Group](#) apply.
- Because a finite group is locally profinite, the representation theory of locally profinite groups [Smooth Representations](#) apply.

┘

**Prop. (24.1.3.2) [Orthogonality of Characters].** The characters  $\chi_i$  of irreducible representations of  $G$  form a basis of  $ZL^2(G)$  by (24.7.3.30). Also, for  $s \in G$ , let  $c(s)$  be the number of elements in the conjugacy class of  $s$ , then:

- $\int_i |\chi_i(s)|^2 = g/c(s)$ .
- If  $t$  is not conjugate to  $s$ , then  $\sum_i \chi_i(s)^* \chi_i(t) = 0$ .

┘

*Proof:* The last assertion follows from the first one, if you consider a matrix with conjugacy classes as column and characters as rows, and place  $\sqrt{c(s)/g} \chi_i(s)$  in the entries, then it is an orthogonal matrix.  $\square$

**Cor. (24.1.3.3) [Representations Determined by Characters].** Any  $\rho \in \mathcal{R}ep^{\text{fd}}(G)$  is determined by its character, by (3.6.2.13).  $\square$

**Cor. (24.1.3.4) [Number of Representations].**  $\#\hat{G}$  equals the number of conjugacy classes of  $G$ , and  $\sum_{\rho \in \hat{G}} \dim(\rho)^2 = \#G$ .  $\square$

*Proof:* Both  $\{\chi_\pi\}$  and the characteristic functions of the conjugate classes of  $G$  are basis for  $L^2(G)$ . And the second assertion follows from the Peter-Weyl theorem (24.7.3.16) as  $\sum_{\pi \in \hat{G}} d_\pi^2$  is the dimension of  $L^2(G)$ .  $\square$

**Cor. (24.1.3.5).**  $G \in \mathcal{A}b^{\text{fin}}$  iff  $\hat{G} = G^\vee$ .  $\square$

*Proof:* This follows immediately from the equation  $\sum_{\pi \in \hat{G}} d_\pi^2 = |G|$  (24.1.3.4), as  $G$  is Abelian iff it has  $|G|$  conjugacy classes iff  $\hat{G} = G^\vee$  iff  $d_\pi = 1$  for any  $\pi \in \hat{G}$ .  $\square$

**Prop. (24.1.3.6) [Decomposition of Regular Representations].** By (24.7.3.7), the regular representation of  $G$  on  $L^2(G)$  admits a decomposition

$$L^2(G) = \bigoplus_{\rho \in \text{Irr}(G)} \dim(\rho) \rho.$$

┘

**Prop. (24.1.3.7).** If  $G$  is a finite  $p$ -group and  $A$  is a nonzero  $p$ -torsion  $G$ -module, then  $A^G \neq 0$ .  $\square$

*Proof:* We may consider  $A$  generated by a single element. Because  $A$  is  $p$ -torsion,  $|A| = p^n$  for some  $n$ . Now consider the orbit, then if the orbit is not a single element, then its order is divisible by  $p$ , so  $|A^G|$  is divisible by  $p$ . But 0 is fixed, so  $A^G \neq 0$ .  $\square$

### Group Algebra $k[G]$

**Prop. (24.1.3.8) [Total Decomposition of Representations, Maschke].** If  $k \in \text{Field}^p$  and  $(\#G, p) = 1$ , then any  $(\rho, V) \in \text{Rep}_F^{\text{fd}}(G)$  is a direct sum of irreducible representations.  $\lrcorner$

*Proof:* For any subrepresentation  $U \subset V$ , choose an arbitrary projection  $\pi$  of  $U$  to  $V$ , and let  $\rho(v) = 1/\#G \sum g^{-1}\pi(g(v))$ , then it can be checked  $\rho$  commutes with  $G$ -actions, thus its kernel is also a  $G$ -modules, and it is identity on  $V$ , so  $U = V \oplus \ker \rho$ .  $\square$

**Cor. (24.1.3.9) [Semisimple Ring].** If  $k \in \text{Field}$  and  $\#G$  is invertible in  $k$ , then  $k[G]$  is a semisimple ring (3.6.2.17).  $\lrcorner$

**Prop. (24.1.3.10) [Brauer-Nesbitt].** If two finite dimensional semisimple representations over a field has the same characteristic polynomial for any element  $g \in G$ , then they are isomorphic.  $\lrcorner$

*Proof:* Just use the irreducible representations are orthogonal and that they have the same and for char  $p$ , we can use divide by  $p$  and the char poly becomes  $p$ -th power and we can do this forever, contradiction.  $\square$

**Prop. (24.1.3.11) [Integrality of Characters].** For any  $\rho \in \text{Rep}^{\text{fd}}(G)$ , the values of  $\chi_\rho$  are sums of roots of unity.  $\lrcorner$

*Proof:* This is because for any  $g \in G$ ,  $g^m = 1$  for some  $m \in \mathbb{Z}_+$ . Thus  $\rho(g)^k = 1$ , and the eigenvalues of  $\rho(g)$  are all in  $\mu_k$ .  $\square$

**Prop. (24.1.3.12) [Schur Indices].**  $\lrcorner$

### Induced Representations and Mackey Theory

**Remark (24.1.3.13).** The induced representation  $\text{Ind}_H^G$  (24.1.2.13) is the same as the (compact) induction defined in (24.1.6.40). In particular, all the results there holds in the finite group case.  $\lrcorner$

**Prop. (24.1.3.14) [Character of Induced Representations].** Character of induced representations, Cf. [Serre, P30].  $\lrcorner$

**Prop. (24.1.3.15) [Normal Subgroup Dichotomy].** Let  $A \trianglelefteq G$  be a normal subgroup, and  $\rho \in \text{Irr}(G)$ , then

- either there exists  $A \subset H \subsetneq G$  s.t.  $\rho$  is induced from a representation of  $H$ ,
- or  $\rho|_A$  is not isotypic.

$\lrcorner$

*Proof:* Let  $\rho|_A = \otimes \rho_i$  be the decomposition into  $A$ -isotypic representations, then  $G$  permutes these  $V_i$ , and acts transitively because  $\rho$  is irreducible. If  $V = V_1$ , then we are in case 2. Otherwise, let  $H$  be the stabilizer of  $V_i$  under this transition map, then  $\rho = \text{Ind}_H^G(V_1)$ , so we are in case 1.  $\square$

**Cor. (24.1.3.16) [Dimensions Bounds of Representations, Tate].** If  $A \trianglelefteq G$  is an Abelian normal subgroup, then dimension of any irreducible representations  $(\rho, V) \in \text{Rep}^{\text{fd}}(G)$  divides  $[G : A]$ . In particular, they divide  $[G : Z(G)]$ .  $\lrcorner$

*Proof:* Use induction on  $G$  and consider (24.1.3.15). If we are in case 1, then we can use induction hypothesis on  $H$ . If we are in case 2, then there are surjective maps  $G/A \rightarrow \rho(G)/\rho(A)$ , where  $\rho(A) \subset \text{GL}(V)$  are scalar multiplications, thus  $\rho(A) \subset Z(\rho(G))$ .

Then we are reduced to the case  $A = Z(G)$ . Cf. [Ser77]P53. ? □

**Cor. (24.1.3.17).** If  $G$  is a  $p$ -group, then the dimension of any irreducible representations of  $G$  is a  $p$ -power. ┘

**Prop. (24.1.3.18) [Representations of Supersolvable Groups].** If  $G$  is supersolvable, then any irreducible representation of  $G$  is induced from a character of a subgroup of  $G$ . ┘

*Proof:* We may use induction on  $\#G$ . Thus it suffices to consider a faithful representation  $\rho$ . If  $G$  is Abelian this is trivial, and if  $G$  is non-Abelian, then by (3.1.8.10),  $G$  contains a normal Abelian subgroup  $A \leq G$  that is not contained in the center. Because  $\rho$  is faithful, some  $a \in A$  satisfies  $\rho(a)$  is not a scalar multiplication. Thus the restriction of  $\rho$  to  $A$  is not isotypic. □

### Rationality Problems

**Def. (24.1.3.19) [ $R_k(G)$  and  $\overline{R}_k(G)$ ].** Let  $R_k(G) \subset ZC(G)$  be the  $\mathbb{Z}$ -module generated by the characters of  $\rho \in \text{Rep}_k(G)$ , which is a subring of  $R(G)$

Also define the  $\mathbb{Z}$ -module  $\overline{R}_k(G) \subset R(G)$  to be the submodule generated by characters of  $\text{Rep}(G)$  that is valued in  $k$ . Clearly  $R_k(G) \subset \overline{R}_k(G)$ . ┘

**Prop. (24.1.3.20).**  $[\overline{R}_k(G) : R_k(G)] < \infty$ . In fact, we can take a finite extension  $k'/k$  s.t.  $\text{Rep}(K) = \text{Rep}_{k'}(G)$ , and  $d\overline{R}_k(G) \subset R_k(G)$ . ┘

*Proof:*  $k'$  exists because there are only f.m. irreducible representations of  $G$ . Now for any  $\rho \in \text{Rep}_{k'}^{\text{fd}}(G)$ , if  $\rho$  is regarded as a  $k$ -representation, then the new representation has character  $\text{tr}_{k'/k}\chi_\rho$  ? . Thus the last assertion follows. □

**Prop. (24.1.3.21).** Let  $\{\rho_i\}$  be the isomorphism classes of all irreducible representations of  $G$  over  $k$ , then their characters  $\{\chi_i\}$  form an orthogonal basis of  $R_k(G)$ . ┘

*Proof:* □

**Cor. (24.1.3.22).**  $\rho \in \text{Rep}(G)$  is realizable over  $k$  iff its character belongs to  $R_k(G)$ . ┘

*Proof:* One direction is trivial, for the other, if  $\chi \in R_k(G)$ , then  $\chi = \sum n_i \chi_i$ , and  $(\chi, \chi_i) = n_i(\chi_i, \chi_i)$ . As  $(\chi, \chi_i) \geq 0$  as they are representations of  $G$ , we have  $n_i \geq 0$ , thus  $\rho = \sum n_i \rho_i$  is realizable over  $K$ , by (24.1.3.3). □

**Prop. (24.1.3.23).** Let  $m_i$  be the Schur indices of  $G$ , then the characters  $\psi = \chi_i/m_i$  form a basis of  $\overline{R}_K(G)$ . In particular,  $R_k(G) = \overline{R}_k(G)$  iff  $k[G]$  is quasi-split. ┘

*Proof:* Cf. [Ser77]P93. ? □

**Def. (24.1.3.24) [Galois Action on  $G$ ].** Let  $k' = k(\mu_m)$ , where  $m$  divides the orders of all elements in  $G$ , then  $\text{Gal}(k'/k) \leq (\mathbb{Z}/(m))^*$ . Then this group can act on  $G$  by  $\sigma_t(x) = x^n$  as a set, and we call two elements  $s, s' \in G$   $\text{Gal}(k'/k)$ -conjugate iff they are in the same  $\text{Gal}(k'/k)$  orbits of  $G$ . ┘

**Thm. (24.1.3.25) [Galois Descent of Characters].** Situation as in (24.1.3.24), then for  $f \in ZC(G)$  valued in  $k'$ ,

$$\sigma_t(f(g)) = f(g^t)$$

- $f \in R(G) \otimes k$  iff  $\sigma_t(f)(g) = f(g^t)$  for any  $\sigma_t \in \text{Gal}(k'/k)$  and  $g \in G$ .
- $f \in R_k(G) \otimes k$  iff  $f(g) = f(g^t)$  for any  $\sigma_t \in \text{Gal}(k'/k)$  (i.e.  $f(x) = f(y)$  if  $\langle x \rangle = \langle y \rangle \leq G$ ).

In particular,  $R(G) = \overline{R}_k(G)$  iff  $f(x) = f(y)$  for any  $x, y \leq G$  generating the same subgroup of  $G$ .  $\lrcorner$

*Proof:* 1: If  $f = \chi_\rho \in R(G)$ , then  $\sigma_t(f)(g) = \sigma_t \chi_\rho(g) = \sigma_t(\sum \omega_i) = \sum \omega_i^t = \chi_\rho(g^t)$ . Conversely, if  $\sigma_t(f)(g) = f(g^t)$  for any  $\sigma_t \in \text{Gal}(k'/k)$  and  $g \in G$ , then

$$f = \sum_{\rho \in \text{Irr}(G)} c_\rho \chi_\rho, \quad c_\rho = \langle f, \chi_\rho \rangle.$$

And

$$c_\rho = \langle f, \chi_\rho \rangle = \langle \Psi^t f, \Psi^t \chi_\rho \rangle = \langle \sigma_t(f), \sigma_t(\chi_\rho) \rangle = \sigma_t \langle f, \chi \rangle = \sigma_t c_\rho.$$

Thus  $c_\rho \in k$  and  $f \in R(G) \otimes k$ .

2: This is because the fixed points of  $\text{Gal}(k'/k)$  acting on  $R(G) \otimes k$  is just  $\overline{R}_k(G) \otimes k = R_k(G) \otimes k$  (24.1.3.20).  $\square$

**Cor. (24.1.3.26) [Number of  $k$ -Representations].**  $\# \text{Irr}_k(G)$  equals the number of  $(\mathbb{Z}/(m))^*$ -orbits of conjugacy classes of  $G$ , and  $R_k(G)$  forms a basis for the functions on  $G$  that is constant on  $(\mathbb{Z}/(m))^*$ -orbits of conjugacy classes of  $G$ .  $\lrcorner$

**Cor. (24.1.3.27) [Brauer].** If  $m$  is the least common multiples of the orders of the elements of  $G$  and  $\mu_m \subset k$ , then  $R_k(G) = R(G)$ .  $\lrcorner$

*Proof:* By (24.1.3.36), any  $\chi \in R(G)$  is of the form  $\chi = \sum n_i \text{Ind}_{H_i}^G(\varphi_i)$  where  $H_i \leq G$  and  $\varphi_i \in H_i^\vee$ .  $\varphi_i$  are valued in  $\mathbb{Q}(\mu_m)$ , thus  $\varphi_i \in R_k(H_i)$ . Then by (24.1.3.32),  $\chi \in R_k(G)$ .  $\square$

**Cor. (24.1.3.28) [Bounds on Schur Indices].** If  $m$  is the least common multiple of the orders of the elements of  $G$ , then all the Schur indices of  $G$  divides the Euler function  $\varphi(m)$ .  $\lrcorner$

*Proof:* This follows from (24.1.3.27) and (3.6.4.19) by considering the field extension  $k(\mu_m)$ .  $\square$

**Cor. (24.1.3.29) [Rational Characters].** Let  $m \in \mathbb{Z}_+$  be a multiple of all orders of elements in  $G$  and  $f \in ZC(G)$  valued in  $\mathbb{Q}(\mu_m)$ , then by (24.1.3.25),

- $f \in R(G) \otimes \mathbb{Q}$  iff  $\sigma_t(f)(x) = f(x^t)$  for any  $t \in \mathbb{Z} \setminus (m)$ .
- $f \in R_{\mathbb{Q}}(G) \otimes \mathbb{Q}$  iff  $f(x) = f(x^t)$  for any  $t \in \mathbb{Z} \setminus (m)$  (i.e.  $f(x) = f(y)$  if  $\langle x \rangle = \langle y \rangle \leq G$ ).

In particular,  $R(G) = \overline{R}_{\mathbb{Q}}(G)$  iff any two element generating the same subgroup of  $G$  is conjugate.

And  $\# \hat{G}$  equals the number of conjugacy classes of cyclic subgroups of  $G$ .  $\lrcorner$

**Prop. (24.1.3.30) [Reduction to Cyclic Subgroups].**  $R_{\mathbb{Q}}(G) \otimes \mathbb{Q}$  is generated by  $\{\text{Ind}_C^G(\mathbf{1})\}_{C \leq G \text{ cyclic}}$ . In fact, if  $\{C_1, \dots, C_d\}$  is a system of representative of the conjugacy classes of cyclic subgroups of  $G$ , then  $\{\text{Ind}_{C_i}^G(\mathbf{1})\}$  is an orthogonal basis for  $R_{\mathbb{Q}}(G)$ .  $\lrcorner$

*Proof:* It suffices to show that if  $\theta \in R_{\mathbb{Q}}(G)$  and  $\langle \theta, \text{Ind}_C^G(\mathbf{1}) \rangle = 0$ , then  $\chi = 0$ . By Frobenius reciprocity, this follows from (24.1.3.31).  $\square$

**Lemma (24.1.3.31).** If  $\theta \in R_{\mathbb{Q}}(G)$  s.t.  $\sum_{s \in C} \theta(s) = 0$  for any cyclic subgroup  $C \leq G$ , then  $\theta = 0$ .

In particular, for  $V, V' \in \text{Rep}_{\mathbb{Q}}^{\text{fd}}(G)$ ,  $V \cong V'$  iff we have  $\dim V^C = \dim(V')^C$  for any cyclic subgroup  $C \leq G$ .  $\lrcorner$

*Proof:* We use induction on  $\#G$ . For any  $s \in G$ , if  $x \in \langle s \rangle$  generates  $\langle s \rangle$ , then  $\theta(s) = \theta(x)$  because  $x, s$  are  $(\mathbb{Z}/(m))^*$ -conjugate. And if  $\langle x \rangle \subsetneq \langle s \rangle$ , then by induction hypothesis,  $\theta(x) = 0$ . Thus

$$0 = \sum_{x \in \langle s \rangle} \theta(x) = \varphi(\# \langle s \rangle) \varphi(s),$$

and  $\varphi(s) = 0$ . □

### Reduction to Subgroups

**Prop. (24.1.3.32) [Induction and Restriction of Characters].** Let  $H \leq G$ , then the induction induces an additive homomorphism  $\text{Ind}_H^G : R(H) \rightarrow R(G)$ , and restriction induces a ring homomorphism  $R(G) \rightarrow R(H)$ . The formula  $\text{Ind}(\varphi \cdot \text{res}(\psi)) = \text{Ind}(\varphi) \cdot \psi$  shows the image of  $\text{Ind}$  is an ideal of  $R(G)$ . Also by Frobenius reciprocity (24.1.6.43),  $\text{Ind}$  and  $\text{Res}$  are dual to each other:

$$(\varphi, \text{res } \psi)_H = (\text{Ind } \varphi, \psi)_G.$$

┘

**Thm. (24.1.3.33) [Generalized Artin Theorem].** Let  $X$  be a family of subgroups of a finite subgroup  $G$ . Let  $\text{Ind} : \bigoplus_{H \in X} R_k(H) \rightarrow R_k(G)$  be the ring homomorphism induced by induction, then the following properties are equivalent:

- $G$  is the union of conjugates of the subgroups in  $X$ .
- the cokernel of  $\text{Ind}$  is finite.

┘

*Proof:*  $2 \rightarrow 1$ : By the character of induced representations (24.1.3.14), any function in the image of  $\text{Ind}$  vanishes outside the union of conjugates of the subgroups in  $X$ , so if this is not  $G$ , then the cokernel cannot be finite.

$1 \rightarrow 2$ : Notice the duality of  $\text{Ind}$  and  $\text{Res}$  (24.1.3.32), it suffices to show that  $\text{Res}$  is injective, but this is clear. □

**Cor. (24.1.3.34) [Artin Theorem].** Every  $f \in R_k(G)$  is a rational combination of characters induced from  $k$ -characters of cyclic subgroups of  $G$ . ┘

*Proof:* In (24.1.3.33), choose  $X$  to be the family of cyclic subgroups of  $G$ . □

**Def. (24.1.3.35) [( $k, p$ )-elementary Subgroups].** For  $p \in \text{Prime}$ , a subgroup  $H \leq G$  is called a  $(k, p)$ -**elementary subgroup** of  $G$  if  $H = C \rtimes P$  where  $P$  is a  $p$ -subgroup and  $C$  is a cyclic subgroup s.t. for any  $y \in P$ , there exists  $\sigma_t \in \text{Gal}(k'/k)$  (24.1.3.24) s.t.  $xyx^{-1} = x^t$ .

A subgroup  $H \leq G$  is called  $k$ -**elementary** if it is  $(K, p)$ -elementary for some  $p \in \text{Prime}$ .

By (3.1.8.9), any  $k$ -elementary subgroup of  $G$  is nilpotent thus supersolvable (3.1.8.9). ┘

**Thm. (24.1.3.36) [Generalized Brauer's Theorem, [B-T55]].**

- If  $p \in \text{Prime}$  and  $\#G = p^g l$  where  $(l, p) = 1$ , then  $l \cdot \text{id}$  is contained in the image of

$$\text{Ind} : \bigoplus_{H \leq G \text{ } (k,p)\text{-elementary}} R_k(H) \rightarrow R_k(G).$$

In particular, the cokernel is finite and prime to  $p$ .

•

$$\text{Ind} : \bigoplus_{H \leq G \text{ } k\text{-elementary}} R_k(H) \rightarrow R_k(G)$$

is surjective. ┘

*Proof:* Cf.[Serre]P99. ? □

**Cor.(24.1.3.37)[Reduction to Characters, Brauer].**  $R(G)$  is generated by  $\text{Ind}_H^G(\chi)$  where  $H \leq G$  and  $\chi \in H^\vee$ . ┘

**Cor.(24.1.3.38).** Given  $\varphi \in ZC(G)$ ,  $\varphi \in R_k(G)$  iff  $\varphi|_H \in R_k(H)$  for any  $(k, p)$ -elementary subgroup  $H$  of  $G$ . ┘

*Proof:* Let  $\mathbf{1} = \sum_{H \leq G \text{ } k\text{-elementary}} \text{Ind}_H^G f_H$  where  $f_H \in R_k(H)$ , then

$$\varphi = \sum_{H \leq G \text{ } k\text{-elementary}} \text{Ind}_H^G (f_H \cdot \text{Res}_H^G \varphi).$$

Thus the assertion follows. □

**Cor.(24.1.3.39).**

- $\text{Ind} : \bigoplus_{H \leq G \text{ } k\text{-elementary}} R_k(H) \rightarrow R_k(G)$  is surjective.
- If  $k[H]$  is quasi-split for any  $k$ -elementary subgroup of  $G$ , then  $k[G]$  is quasi-split. ┘

*Proof:* Use the same method of proof as [\(24.1.3.38\)](#). □

**Prop.(24.1.3.40)[2-Dimensional Representations of Cyclic or Dihedral Type].** If  $G \in \mathfrak{Grp}^{\text{fin}}$  and  $(\rho, \mathbb{C}^{\oplus 2}) \in \text{Rep}^{\text{fd}}(G)$ , then  $\text{Im}(\rho)$  is isomorphic to a cyclic group or a dihedral group iff  $\rho = \text{Ind}_H^G(\chi)$  where  $H$  is a subgroup of  $G$  of index 2 and  $\chi \in H^\vee$ . Moreover, in this case,  $\chi \not\cong \chi^g$  for any  $g \in G \setminus H$ . ┘

*Proof:* ? □

### Important representations

**Prop.(24.1.3.41)**  $[Q_8]$ . There is a 2-dimensional representation of the quadratic group  $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ :

$$i \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad j \mapsto \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad k \mapsto \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

┘

**Prop.(24.1.3.42).** For  $n \in \mathbb{Z}_+$ , there is a representation of  $\text{Sym}_n$  on the  $n-1$ -dimensional hypersurface  $\sum x_i = 0$ . ┘

## 4 Symmetric Groups

References are [The Symmetric Group, Sagan].

**Prop. (24.1.4.1).** For  $n \in \mathbb{Z}_+$ ,  $\text{Rep}^{\text{fd}}(\text{Sym}_n) = \text{Rep}_{\mathbb{Q}}^{\text{fd}}(\text{Sym}_n)$ . In particular, all characters of representations of  $\text{Sym}_n$  are valued in  $\mathbb{Z}$ .  $\lrcorner$

*Proof:* This follows from (24.1.3.29).  $\square$

**Def. (24.1.4.2) [Standard Representation of  $\text{Sym}_n$ ].** For  $n \in \mathbb{Z}_+$ , there is a standard  $n$ -dimensional representation of  $\text{Sym}_n$  on  $\mathbb{C}[X_1, \dots, X_n]$  s.t.  $\pi(X_i) = X_{\pi(i)}$  for any  $i \in [n]_+$  and  $\pi \in \text{Sym}_n$ .  $\lrcorner$

**Def. (24.1.4.3) [Permutation Modules].** Suppose  $n \in \mathbb{Z}_+$  and  $\lambda \vdash n$ , the **permutation module** of  $\lambda$  is defined to be

$$M^\lambda = \mathbb{C}\{\{t_1\}, \dots, \{t_k\}\}$$

where  $\{t_1\}, \dots, \{t_k\}$  is a complete list of  $\lambda$ -tabloids (37.5.1.2).

As  $\text{Sym}_n$  acts on the set of tableaux of shape  $\lambda$ , and descends to an action on  $\lambda$ -tabloids, we have an action of  $\text{Sym}_n$  on  $M^\lambda$ .

Then  $M^\lambda$  is a cyclic representation, and  $\dim M^\lambda = n!/\lambda!$ .  $\lrcorner$

**Def. (24.1.4.4) [Polytabloids].**  $\lrcorner$

**Prop. (24.1.4.5).** Let  $t$  be a tableau of size  $n$  and  $\pi \in \text{Sym}_n$ , then

- $R_{\pi t} = \pi R_t \pi^{-1}$ ,
- $C_{\pi t} = \pi C_t \pi^{-1}$ ,
- $\kappa_{\pi t} = \pi \kappa_t \pi^{-1}$ ,
- $e_{\pi t} = \pi e_t$ .

In particular,  $S^\lambda$  is a cyclic  $\text{Sym}_n$ -representation.  $\lrcorner$

*Proof:*  $\square$

**Def. (24.1.4.6) [Specht modules  $S^\lambda$ ].** Let  $n \in \mathbb{Z}_+$ ,  $\lambda \vdash n$ , then define the **Specht module**  $S^\lambda \subset M^\lambda$  to be the submodule generated by the polytabloids  $e_t$  where  $t$  are tableaux of shape  $\lambda$ . Then  $S^\lambda$  is a cyclic  $\text{Sym}_n$ -representation by (24.1.4.5).  $\lrcorner$

### Generators of Specht modules

### Branching Rules

### Decomposition of $M^\lambda$

## 5 Character Tables Lists

## 6 Locally Profinite Groups

References are [B-Z77, Ber84a] and [Bernstein, Representation of  $p$ -adic Groups].

**Notation (24.1.6.1).**

- Use notations on locally profinite spaces in [Profinite Spaces](#).
- For structure theory of locally profinite groups, see [Locally Profinite Groups](#).

- Let  $G \in \text{locProfGrp}$ . ┘

**Prop. (24.1.6.2) [Small Group Argument].**  $\text{Rep}^{\text{fd}}(G) = \text{Rep}^{\text{fin,alg}}(G)$ . ┘

*Proof:* This is because a small nbhd of  $\mathbf{1} \in \text{GL}(n; \mathbb{C})$  doesn't contain a subgroup. □

### Invariant Distributions

**Def. (24.1.6.3) [Invariant Distributions].** If  $X \in \text{locProf}$  and  $G \times X \rightarrow X$  is a left action, then  $G$  acts on the left on  $C_c^\infty(X)$  by  $g(f)(x) = f(g^{-1}x)$ , and acts on  $\text{Dist}(G)$  by  $g(T)(f) = T(g^{-1}(f))$ .

For  $\chi \in G^\vee$ , denote  $\text{Dist}(G)^{G,\chi} \triangleq \{T \in \text{Dist}(G) | gT = \chi(g)T\}$ . And denote  $\text{Dist}(G)^G \triangleq \text{Dist}(G)^{G,\mathbf{1}}$ . ┘

**Prop. (24.1.6.4).** Any  $\varphi \in C_c^\infty(G)$  is  $K$ -bi-invariant under some compact open subgroup  $K \leq G$ . ┘

*Proof:*  $\varphi$  must be of the form  $\sum a_i \chi_{U_i}$ , where each  $U_i$  is open compact. Then for each element  $x \in U_i$ , there is an open compact group  $U_x$  that  $U_x x \cup x U_x \subset U_i$ , by (12.10.4.3). Now because  $\text{Supp } f$  is compact, f.m. of the  $U_x x \cap x U_x$  covers  $\text{Supp } \varphi$ , thus we consider their intersection  $\cap U_{x_i}$ , which is a compact open subgroup  $K_0$  that  $\varphi$  is  $K_0$ -bi-invariant. □

**Thm. (24.1.6.5) [Localization Principle].** For  $\text{pr} : Z \rightarrow T \in \text{locProf}$ ,  $C_c^\infty(Z)$  and  $\text{Dist}(Z)$  are  $C_c^\infty(T)$ -modules. Then for any weak\*-closed subspace  $M \leq \text{Dist}(Z)$ , we have

$$M = \overline{\oplus_{t \in T} M \cap \text{Dist}(\text{pr}^{-1}(t))}.$$

┘

*Proof:* Cf. [Ber84a]P59. ? □

**Cor. (24.1.6.6).** Let  $\text{pr} : Z \rightarrow T \in \text{locProf}$  and  $G$  acts on  $Z$  preserving fibers of  $\text{pr}$ . Suppose  $\chi \in G^\vee$  s.t.  $\text{Dist}(\text{pr}^{-1}(t))^{G,\chi} = 0$  for each  $t \in T$ , then we have  $\text{Dist}(Z)^{G,\chi} = 0$ . ┘

**Prop. (24.1.6.7) [Localization Principle, Bernstein-Zelevinsky [B-Z77]].** Let  $p : X \rightarrow Y \in \text{locProf}$  and  $\mathcal{F}$  be a cosmooth  $C^\infty$ -sheaf on  $X$ ?. Let  $G$  act on  $X$  and the sheaf  $\mathcal{F}$  that  $p(gx) = p(x)$ , and  $\chi \in G^\vee$ , then:

- Let  $\Gamma_c(X, \mathcal{F})(\chi)$  be the  $C_c^\infty(X)$ -submodule of  $\Gamma_c(X, \mathcal{F})$  generated by  $gf - \chi(g)^{-1}f, g \in G, f \in \Gamma_c(X, \mathcal{F})$ . Then  $\Gamma_c(X, \mathcal{F})/\Gamma_c(X, \mathcal{F})(\chi)$  is a non-degenerate  $C_c^\infty(Y)$ -module by composing  $p$ , so we can define  $\mathcal{G}$  the sheaf on  $Y$  corresponding to this submodule by (4.4.4.15). Then if  $y \in Y$  and  $Z = p^{-1}(y)$ , the stalk

$$\mathcal{G}_y \cong \Gamma_c(Z, \mathcal{F})/\Gamma_c(Z, \mathcal{F})(\chi).$$

- Assume there are no non-zero distributions  $D \in \text{Dist}(p^{-1}(y), \mathcal{F}|_{p^{-1}(y)})$  that satisfies  $gD = \chi(g)D$  for any  $y \in Y$ , then no such  $D$  exists in  $\mathcal{D}(X, \mathcal{F})$ . ┘

*Proof:* 1: Firstly  $\Gamma_c(X, \mathcal{F})/\Gamma_c(X, \mathcal{F})(\chi)$  is a  $C_c^\infty(Y)$ -module because  $\varphi \circ p$  fixes  $\Gamma_c(X, \mathcal{F})(\chi)$ :

$$(\varphi \circ p)(gf - \chi(g)^{-1}f) = g((\varphi \circ p)f) - \chi(g)^{-1}((\varphi \circ p)f),$$

which uses the condition  $p(gx) = p(x)$ . The non-degeneracy is also clear, by (4.4.4.15).



Secondly  $\Gamma_c(Z, \mathcal{F}) \cong \Gamma_c(X, \mathcal{F})/\Gamma_c(U, \mathcal{F})$  by (4.4.4.12), so  $\Gamma_c(Z, \mathcal{F})/\Gamma_c(Z, \mathcal{F})(\chi)$  is isomorphic the quotient of  $\Gamma_c(X, \mathcal{F})$  by  $\Gamma_c(U, \mathcal{F})$  and  $\Gamma_c(X, \mathcal{F})(\chi)$ , because  $\Gamma_c(U, \mathcal{F})$  is stable under action of  $f \mapsto gf - \chi(g)^{-1}f$  (this uses the condition  $p(gx) = p(x)$ ).

We claim the space  $\Gamma_c(U, \mathcal{F})$  is the space  $L$  generated by elements of the form  $(\varphi \circ p)f$ , where  $\varphi \in C_c^\infty(Y)$ ,  $\varphi(y) = 0$ ,  $f \in \Gamma_c(X, \mathcal{F})$ :  $L$  is clearly contained in  $\Gamma_c(U, \mathcal{F})$ , and if  $f \in \Gamma_c(U, \mathcal{F})$ , then  $\text{Supp } f$  is compact and disjoint from  $Z$ , so there is an open compact subset  $U \subset Y$  containing  $p(\text{Supp } f)$  but not  $y$ . Let  $\varphi = \chi_U$ , then  $f = (\chi_U \circ p)f \in L$ .

Then by (4.4.4.15), the stalk  $\mathcal{G}_y$  is isomorphic to  $M/M(y)$ , where  $M = \Gamma_c(X, \mathcal{F})/\Gamma_c(X, \mathcal{F})(\chi)$ . So  $M(y)$  is just the image of the space  $L$  in  $M$  by (4.4.4.16), hence  $\mathcal{G}_y$  is exactly  $\Gamma_c(Z, \mathcal{F})/\Gamma_c(Z, \mathcal{F})(\chi)$ .

2: this follows from 1, as  $gD = \chi(g)D$  is just saying  $D(g^{-1}f - \chi(g)f) = 0$ , or that  $D$  annihilates  $\Gamma_c(Z, \mathcal{F})/\Gamma_c(Z, \mathcal{F})(\chi) = \mathcal{G}_y$  by item 1. So this is equivalent to  $\mathcal{G}_y = 0$  for any  $y$ , which is equivalent to  $\mathcal{G} = 0$ , as  $\mathcal{G}$  is a sheaf.  $\square$

**Cor. (24.1.6.8) [Invariant Distributions on Orbits].** Let  $\gamma$  be an action of  $G$  on  $X \in \text{locProf}$  and a  $C^\infty(X)$ -sheaf. Assume the action is constructible (4.12.1.21) and there are no  $G$ -invariant distribution on any  $G$ -orbit in  $X$ , then there are no non-zero  $G$ -invariant distribution on  $X$ .  $\lrcorner$

*Proof:* Firstly, by (4.12.1.22) and (4.4.4.6) any orbit is locally profinite. If there is a  $G$ -invariant distribution on  $X$ , we may change  $X$  to  $\text{Supp } T$ , which is  $G$ -invariant, thus by (4.12.1.22) there is an open subset  $U \subset X$  that  $G$  acts regularly, thus we reduce to the regular action case.

Then we can consider  $X \rightarrow X/G$ ,  $X/G$  is locally profinite by (4.12.1.10), so Bernstein-Zelevinsky (24.1.6.7) can be used.  $\square$

**Thm. (24.1.6.9) [Frobenius Descent of Distributions, Bernstein-Zelevinsky [B-Z76]].** Let  $G$  is unimodular and acts transitively on  $Z \in \text{locProf}$ . Let  $\text{pr} : X \rightarrow Z \in \text{locProf}$  be  $G$ -equivariant and  $z \in Z, \chi \in G^\vee$ , then there is a canonical isomorphism

$$\Psi : \text{Dist}(\text{pr}^{-1}(z))^{\text{Stab}_G(z), \chi\nu^{-1}} \cong \text{Dist}(X)^{G, \chi} : \Psi(T)(f) = \int_Z \chi(g_z) T(g_z(f)) d\mu(z),$$

where  $\mu$  is a left  $(G, \nu)$ -equivariant Radon measure on  $Z$  for some  $\nu \in G^\vee$  and  $g_z$  is any element in  $\text{Stab}_G(z)$ .

In particular, for any  $T \in \text{Dist}(\text{pr}^{-1}(z))^{\text{Stab}_G(z), \chi\nu^{-1}}$ , we have  $\text{Supp}(\Psi(T)) = G \cdot \text{Supp}(T)$ .  $\lrcorner$

*Proof:* This is a consequence of Frobenius reciprocity, Cf. [B-Z76] 2.21-2.36.  $\square$

**Cor. (24.1.6.10) [Frobenius Descent and Fourier Transform].** Let  $K \in p\text{-NField}$  and  $V \in \text{Vect}^{\text{fin}}/K$  with a  $G$ -invariant non-degenerate bilinear form  $B$ . Suppose  $G$  is unimodular and acts transitively on  $Z \in \text{locProf}$  with unimodular stabilizers, then for the  $G$ -equivariant map  $\text{pr} : Z \times V$ , we have

$$\mathcal{F}_B \circ \Psi^{-1} = \Psi^{-1} \circ \mathcal{F}_B : \text{Dist}(X)^{G, \chi} \cong \text{Dist}(\text{pr}^{-1}(z))^{\text{Stab}_G(z), \chi}.$$

$\lrcorner$

*Proof:* This follows from the explicit formula of  $\Psi$  in (24.1.6.9).  $\square$

**Lemma (24.1.6.11).** Let  $H \leq G$  a closed subgroup, then  $G/H \in \text{locProf}$  by (12.10.4.4), and the projection  $P : C_c(G) \rightarrow C_c(G/H)$  defined in (12.10.3.5) restricts to a projection  $P : C_c^\infty(G) \rightarrow C_c^\infty(G/H)$ , and it is surjective.  $\lrcorner$

*Proof:* Firstly  $P$  maps  $C_c^\infty(G)$  into  $C_c^\infty(G/H)$  because if  $\varphi$  is left invariant under  $K$ , then  $P\varphi$  is also left invariant under  $K$ . For the surjectivity, let  $\varphi \in C_c^\infty(G/H)$ , then  $V = \text{Supp } \varphi$  is open compact, then by (12.10.4.5) there is an open compact subspace  $V$  that  $p(V) = U$ , Then we can define

$$\psi(x) = \chi_V(x)\varphi(p(x))/P(\chi_V)(p(x)),$$

then it is supported on  $V$ , and  $P(\psi) = \varphi$ . Also, it is locally constant, as is easily verified.  $\square$

**Prop. (24.1.6.12) [Left Invariant Distribution].** Any  $T \in \text{Dist}(G)^G$  (24.1.6.3) is the restriction of a unique left Haar measure  $\mu$  on  $G$ .  $\lrcorner$

*Proof:* Any element  $f \in C_c^\infty(G)$  is of the form  $\sum a_i \lambda(h_i) e_K$  for some  $K$ , by (24.1.6.4), where  $e_K = \mu(K)^{-1} \chi_K$ . Because  $T$  is left-invariant,  $T(f) = \sum a_i T(e_K)$ , and  $\int f d\mu = \sum a_i$ . Thus to show  $T$  is a multiple of  $d\mu$ , it suffices to show  $T(e_K)$  is independent of  $K$ .

If  $K_1 \leq K_2$ , then  $K_2 = \coprod_{i=1}^n a_i K_1$ , where  $n = [K_2 : K_1]$ , so  $e_{K_2} = \mu(K_1)/\mu(K_2) \sum_{i=1}^n \lambda(a_i) e_{K_1}$ . Also  $\mu(K_2) = n\mu(K_1)$  by left invariance. Now it is clear  $T(e_{K_1}) = T(e_{K_2})$  by left invariance of  $T$ . Then for any two  $K, K'$ , we can find an open compact group  $K''$  in their intersection, thus  $T(e_K) = T(e_{K'})$ .  $\square$

**Cor. (24.1.6.13).** If  $\chi \in G^\vee$  and  $T \in \text{Dist}(G)^{G,\chi}$  (24.1.6.3), then there is a unique left Haar measure  $\mu$  on  $G$  that  $T(f) = \int_G \chi(g) f(g) d\mu(g)$ .  $\lrcorner$

*Proof:* Consider the distribution  $T'(f) = T(\chi^{-1} \cdot f)$ , then it is left invariant.  $\square$

**Cor. (24.1.6.14) [Invariant Quotient Distribution].** Let  $H \leq G$  be a closed subgroup. then left  $H$ -invariant measures on  $G$  are identified with measures on  $G/H$ .

And if there exists a left  $G$ -invariant distribution  $D$  on  $G/H$ , then  $G/H$  admits a  $G$ -invariant measure (12.10.3.7), whose restriction to  $C_c^\infty(G/H)$  is  $D$ .  $\lrcorner$

*Proof:* Consider the distribution  $D \circ P^\infty$ , which is a left invariant linear functional on  $C_c^\infty(G)$ , (24.1.6.12) shows that it's the restriction of a Haar measure on  $G$ .

Then we can use the same method as in (12.10.3.7) to show  $\Delta_G|_H = \Delta_H$ , and the surjectivity of  $P^\infty$  (24.1.6.11) shows  $D$  is a quotient measure on  $G/H$ .  $\square$

**Prop. (24.1.6.15) [Gelfand Pairs].** Let  $H \leq G$  be a closed subgroup s.t.  $\Delta(G)|_H = \Delta_H$ . Suppose  $\iota$  is an involution on  $G$  that leaves  $H$  invariant and acts trivially on those distributions on  $G$  with are  $H$ -bi-invariant, then for any  $V \in \text{Irr}^{\text{alg}}(G)$ ,

$$\dim(V^*)^H \cdot \dim((V^\vee)^*)^H \leq 1.$$

$\lrcorner$

*Proof:* Given two  $H$ -invariant maps  $l : C \rightarrow \mathbb{C}, m : V^\vee \rightarrow \mathbb{C}$ , by Frobenius reciprocity, we get two maps  $l' : C_c^\infty(H \backslash G) \rightarrow V, m' : C_c^\infty(H \backslash G) \rightarrow V^\vee$  that are surjective, and give rise to  $B : C_c^\infty(\mathcal{H} \times \mathcal{H} \backslash G \times G) \rightarrow V \times V^\vee \rightarrow \mathbb{C}$ .

Then  $B$  satisfies  $B(f, g) = B(i(g), i(f))$ , where  $i(f)(x) = f(\bar{x}^{-1})$ : There is a commutative diagram

$$\begin{array}{ccc} G \times G & \xrightarrow{(x,y) \mapsto xy^{-1}} & G \\ \downarrow (x,y) \mapsto (\iota(y)^{-1}, \iota(x)^{-1}) & & \downarrow \iota \\ G \times G & \xrightarrow{(x,y) \mapsto xy^{-1}} & G \end{array}$$

and the horizontal arrows identify left  $H$ -invariant right  $G$ -invariant distributions on  $G$  with left  $H \times H$ -invariant distributions on  $G \times G$ .

By the surjectivity of  $l', m'$ ,  $\ker(l'), \ker(m')$  are left and right radicals of  $B$  resp., and the above formula shows they determine each other, and  $l', m'$  are determined by their kernels, thus we are done.  $\square$

**Prop. (24.1.6.16) [Gelfand-Kazhdan].** If  $\gamma$  is an action of  $G$  on  $X \in \text{locProf}$ ,  $\sigma$  is a homeomorphism  $X \cong X$ ,  $\mathcal{F}$  is a  $C^\infty$ -sheaf on  $X$ , and we assume:

- $\gamma$  is constructible,
- for each  $g \in G$ , there is a  $g^\sigma \in G$  that  $\gamma(g)\sigma = \sigma\gamma(g^\sigma)$ .
- For some  $n \geq 0$  and  $g_0 \in G$ ,  $\gamma^n = \gamma(g_0)$ .
- If there is a non-zero  $G$ -invariant  $\mathcal{F}$ -distribution  $T$  on a  $G$ -orbit  $S$ , then  $\sigma(S) = S$  and  $\sigma(T) = T$ .

Then any  $G$ -invariant distribution on  $X$  is invariant under  $\sigma$ .  $\lrcorner$

*Proof:* Let  $T$  be a  $G$ -invariant  $\mathcal{F}$ -distribution that  $\sigma T \neq T$ , then  $n > 1$ , and for any  $n$ -th root of unity  $\xi$ , consider  $T_\xi = \sum \xi^{-i} \sigma^i(T)$ . Then

$$\sigma T_\xi = \xi T_\xi, \quad \sum_{\xi} T_\xi = nT, \quad \sum_{\xi} \xi T_\xi = n\sigma(T).$$

so  $\sum_{\xi} (\xi - 1)T_\xi = n(\sigma(T) - T) \neq 0$ , which shows there is a root  $\xi \neq 1$  that  $T_\xi \neq 0$ . Notice  $T_\xi$  is  $G$ -invariant by condition2. Consider the action  $\sigma_x i = \xi \cdot \sigma$ , then  $T_\xi$  is invariant under  $\sigma_\xi$ .

Let  $G'$  be the semi-direct product of  $G$  with  $\sigma_\xi$ , under the action of  $\sigma_\xi^{-1} g \sigma_\xi = g^\sigma$ , then  $G'$  is locally profinite and acts on  $X, \mathcal{F}$ . Clearly this action is also constructible.

Now for any  $G'$ -orbit  $S'$ , we prove there are no  $G'$ -invariant distribution on  $S'$ , because it is in priori  $G$ -invariant, so some distribution exists on some  $G$ -orbit  $S \subset S'$ , but then condition4 shows  $\sigma$  fixes  $S$ , then  $S = S'$  and  $\sigma(T) = T$ . But  $\sigma_\xi(T) = T$ , contradiction. Finally, (24.1.6.8) shows there are no  $G'$ -invariant distribution on  $X$ , contradicting  $T_\xi$ .  $\square$

**Cor. (24.1.6.17) [Gelfand-Kazhdan].** If  $G$  is  $\sigma$ -compact, and  $\gamma$  is an action of  $G$  on  $X \in \text{locProf}$ ,  $\sigma : X \cong X$  is a homeomorphism, and we assume:

- $\gamma$  is constructible,
- for each  $g \in G$ , there is a  $g^\sigma \in G$  that  $\gamma(g)\sigma = \sigma\gamma(g^\sigma)$ .
- For some  $n \geq 0$  and  $g_0 \in G$ ,  $\sigma^n = \gamma(g_0)$ .
- For any  $x$ ,  $x$  and  $\sigma(x)$  are in the same  $G$ -orbit.

Then any  $G$ -invariant distribution on  $X$  is invariant under  $\sigma$ .  $\lrcorner$

*Proof:* We use (24.1.6.17) and take  $\mathcal{F}$  to be just  $C^\infty(X)$ . Then we need to check condition4: for any  $G$ -orbit  $S$  of  $X$  let  $s \in S$  and  $\text{Stab}(s) = H$ ,  $S \cong G/H$  by (4.12.1.22) and (12.10.4.6). Then (24.1.6.14) shows  $T$  is just the  $G$ -invariant measure on  $G/H : T\varphi = \int_{G/H} \varphi(\gamma(g)s)$ . Now condition4 and2 shows  $\sigma T$  is also  $G$ -invariant, thus  $\sigma T = cT$ . Clearly  $c \geq 0$ , and condition3 shows  $c^n = 1$ , thus  $c = 1$ .  $\square$

## Hecke Algebras

For Hecke algebras of Lie groups, see [Hecke Algebras](#).

**Def. (24.1.6.18) [Hecke Algebras of Locally Profinite Groups].** The set  $\mathcal{H}(G)$  of test functions (4.4.4.11) on  $G$  under convolution is an algebra, called the **Hecke algebra (locally profinite groups)** of  $G$ . And for a compact open subgroup  $K$  of  $G$ ,  $\mathcal{H}_K$  is the subspace of  $K$ -bi-invariant functions in  $\mathcal{H}(G)$ .

Notice  $\mathcal{H} = \cup_K \mathcal{H}_K$  by (24.1.6.4). Also  $\mathcal{H}_K$  has a unit  $e_K = \mu(K)^{-1} \chi_K$ . This is easily verified.

Then  $\mathcal{H}_G$  is an idempotented algebra (3.6.5.2).  $\lrcorner$

*Proof:* Define the set  $\mathcal{E}$  of idempotents in  $\mathcal{H}(G)$  as  $e_K$ , where  $K$  is compact open in  $G$ . The fact that  $\mathcal{H}$  is an idempotented algebra follows from (24.1.6.19) and (24.1.6.21).  $\square$

**Prop. (24.1.6.19) [Point Measure].** Consider the point measure  $\delta_g$  for  $g \in G$ , it is not an element in  $\mathcal{H}(G)$ , but it can convolute on  $\mathcal{H}(G)$ :  $(\delta_g * \varphi)(x) = \varphi(g^{-1}x)$ ,  $(\varphi * \delta_g)(x) = \varphi(xg)$ . Then

- If  $g \in K$ , then  $\delta_g * e_K = e_K * \delta_g = e_K$ .
- $\delta_g * e_K * \delta_{g^{-1}} = e_{gKg^{-1}}$ .
- If  $K = K_1 K_2$  is an open subgroup, then  $e_{K_1} * e_{K_2} = e_K$ . In particular,  $e_K * e_K = e_K$ .

$\lrcorner$

**Remark (24.1.6.20).** In fact we should define the Hecke algebra as locally constant distributions on  $G$ , then there equations are more natural. This algebra is equivalent to Hecke algebra by  $f \mapsto f\mu_G$ .  $\lrcorner$

**Prop. (24.1.6.21).**  $\mathcal{H}_K = e_K * \mathcal{H} * e_K = \mathcal{H}[e_K]$ .  $\lrcorner$

*Proof:* Notice by (24.1.6.19), functions in  $e_K * \mathcal{H} * e_K$  is clearly  $K$ -bi-invariant. For the other direction, notice if  $\varphi$  is left and right  $K$ -invariant, then  $\varphi = e_K * \varphi = \varphi * e_K = e_K * \varphi * e_K$ .  $\square$

**Prop. (24.1.6.22) [Smooth Representations and  $\mathcal{H}$ -Modules].** For  $(\pi, V) \in \text{Rep}^{\text{alg}}(G)$  and any  $v \in V$ ,  $g \mapsto \pi(g)v$  can be regarded as a locally constant function with value in  $V$ , thus for any  $\varphi \in \mathcal{H}_G$ ,  $g \mapsto \varphi(g)\pi(g)v$  is locally constant with compact support, thus we can define a representation of the Hecke algebra  $\mathcal{H}_G$  by

$$\pi(\varphi)v = \int_G \varphi(g)\pi(g)v dg$$

which just has nothing to do with integration, and this is compatible with convolution by formal reason. Then this is a smooth  $\mathcal{H}_G$ -module, and this gives an equivalence between the category of smooth (resp. admissible) non-degenerate representations of  $\mathcal{H}(G)$  (3.6.5.4) and the category of smooth (resp. admissible) representations of  $G$  (24.1.6.29).  $\lrcorner$

*Proof:* For any  $v \in V$ , there is an open compact subgroup  $K$  that  $v \in V[e_K]$ , thus  $e_K v = v$ , so  $V$  is smooth. For the equivalence of categories, for any  $\mathcal{H}(G)$ -module  $V$ , we can define a  $G$ -action by linearly extending the action  $\pi(g)e_K v = (\delta_g * e_K)v$ . Notice by associativity of representation and smoothness, this is well-defined on all of  $V$ , and is a representation of  $G$ . Also it is continuous because  $v = e_K v$  for some  $K$  thus  $\pi(g)v = (\delta_g * e_K)v = e_K v = v$  for any  $g \in K$ . Finally these two functors are inverse to each other is also easily checked.  $\square$

**Cor. (24.1.6.23).**  $(-)^K, (-)^{\text{alg}}$  (24.1.2.2) are both exact, by (3.6.5.3).  $\lrcorner$

**Cor. (24.1.6.24).** For  $(\pi, V) \neq 0 \in \text{Rep}^{\text{alg}}(G)$ , the following are equivalent by (3.6.5.6) and (24.1.6.22):

- $\pi$  is irreducible.
- $V$  is simple as  $\mathcal{H}$ -module.
- $V^K$  is either zero or simple as a  $\mathcal{H}_K$ -module for all open compact subgroups  $K \leq G$ .

┘

**Prop. (24.1.6.25)[Fourier Transform].** If  $G$  is Abelian, the Fourier transform induces an isomorphism of algebras

$$\mathcal{H}(G^\vee) \cong C_c^\infty(G) : \varphi \mapsto \varphi^\vee(\xi) = \int_G \varphi(y) \overline{\langle \xi, y \rangle} d\mu(y).$$

┘

*Proof:* By (12.11.1.9) this is an algebra homomorphism. It remains to show that the image are all locally compact, and this is clear from the fact  $\widehat{G}$  is given the compact-open topology (12.11.1.7).  $\square$

### Admissible Smooth Representations

**Prop. (24.1.6.26).**

- For any compact open subset  $K \leq G$ ,  $V^K = ((V^\vee)^K)^*$ .
- $\text{Hom}_G(V, W^\vee) = \text{Hom}(W, V^\vee)$ .
- $V \hookrightarrow (V^\vee)^\vee$ .

┘

*Proof:* 1: Using (24.1.6.22), because  $(\widetilde{V})^K = V^{*K}$ . There is a homomorphism  $V^{*K} \rightarrow V^{K*}$ , it is injective, because if  $f(v) = 0$  for each  $v \in V^K$ , then  $f(w) = f(e_K v) = 0$ . It is also injective, because for each  $f \in V^{K*}$ , the inverse image is  $g(w) = g(e_K w)$ .

2:  $\text{Hom}(V, \widetilde{W}) = \text{Hom}(V, W^*) = \text{Hom}(V \otimes W, \mathbb{C})$ .

3: by the proof of item1,

$$V^{\vee\vee} = \cup_K \left( (\cup_K V^{*K})^{*K} \right) = \cup_K \left( (\cup_K V^{*K})^{K*} \right) = \cup_K (V^{*K*}) = \cup_K (V^{K**}).$$

So the filtered colimits of the injections  $V^K \rightarrow (V^K)^{**}$  gives an injection  $\cup_K (V^{K**})$ .  $\square$

**Cor. (24.1.6.27)[Contragradient Functor is Exact].** The contragradient functor  $V \mapsto V^\vee$  is exact. Because  $(-)^*$ ,  $(-)^{\text{alg}}$  are all exact (24.1.6.23).  $\square$

**Cor. (24.1.6.28).** If  $P$  is projective in  $\mathcal{M}(G)$ , then  $\widetilde{P}$  is injective in  $M(G)$ .  $\square$

*Proof:*  $\text{Hom}(X, P^\vee) = \text{Hom}(P, X^\vee)$ , and notice that the contragradient functor is exact (24.1.6.27).  $\square$

**Def. (24.1.6.29)[Admissible Representations].** An **admissible smooth representation** of  $G$  is a smooth representation  $V \in \text{Rep}^{\text{alg}}(G)$  s.t.  $\dim V^K < \infty$  for any compact open subgroup  $K \leq G$ . The category of admissible smooth representation of  $G$  is denoted by  $\text{Rep}^{\text{adm}}(G)$ .

Then a smooth representation is admissible iff  $V \cong (V^\vee)^\vee$ .  $\square$

*Proof:* If  $V^K$  is of f.d. for each  $K$ , then by the proof of item3 of (24.1.6.26),  $V \cong \widetilde{\widetilde{V}}$ . Conversely, if  $V \cong \widetilde{\widetilde{V}}$ , then  $V^K \cong V^{K**}$ , thus  $V$  must be finite, by (3.5.3.9).  $\square$

**Cor. (24.1.6.30).** For an irreducible admissible representation, the contragradient is also irreducible admissible.

Extensions of admissible representations are admissible, by (24.1.6.23).  $\square$

**Prop. (24.1.6.31) [Decomposition of Admissible Representations].** Let  $K \leq G$  be a compact open subgroup, then  $(\pi, V) \in \text{Rep}(G)$  is smooth iff it decomposes as

$$V = \bigoplus_{\rho \in \text{Rep}(K)} V^\rho,$$

and  $V$  is admissible iff each  $V^\rho$  are all of f.d. In particular, this shows the two notations of admissibility (locally compact group and real reductive groups) are compatible for smooth representations (24.1.6.29)(24.6.4.2).  $\lrcorner$

**Remark (24.1.6.32).** WARNING: The decomposition theorem of compact groups cannot be directly used, as representation may not be of f.d. so may not be unitarizable.  $\lrcorner$

*Proof:* Firstly  $V \subset \sum_{\rho \in \widehat{K}} V(\rho)$ , because any  $v \in V$  is fixed by some compact open subgroup  $K_0$  of  $K$ , and we can choose  $K_0$  to be normal in  $K$  by (4.12.1.6), so for  $\Gamma = K/K_0$ ,

$$v \in V^{K_0} = \bigoplus_{\rho \in \widehat{\Gamma}} V(\rho) \subset \sum_{\rho \in \widehat{K}} V(\rho).$$

Also this sum is direct, because otherwise  $\sum_{\rho \in S} c_\rho v_\rho = 0$ , but let  $K_0$  be the intersection of kernels of  $\rho$ , then this is an equation of elements in representations of  $\Gamma = K/K_0$  finite, so contradicting (24.1.3.8).

Conversely, if this decomposition holds, then each  $v \in V^\rho$  is smooth, because it is fixed by a small open compact nbhd of  $K$ .

If  $\pi$  is admissible, then  $V(\rho) \subset V^{\ker \rho}$  is of f.d.. Conversely, if  $V$  is not admissible, then  $V^{K_0}$  is of infinite dimensional for some  $K_0$  compact compact normal, so  $V^{K_0}$  decomposes as direct sums of  $V(\rho)$  for  $\rho \in \widehat{K/K_0}$ , thus one of these space must be of infinite dimensional.  $\square$

**Def. (24.1.6.33) [Characters of Admissible Representations].** Let  $(\pi, V) \in \text{Rep}^{\text{adm}}(G)$ , then for any  $\varphi \in \mathcal{H}$ , we have  $\varphi \in \mathcal{H}_K$  for some compact open subset of  $G$ , by (24.1.6.18), so  $\text{Im}(\pi(\varphi)) \subset V^K$ , which is of f.d., so we can define the trace of  $\varphi$  as  $\text{tr}(\pi(\varphi)|V^K)$ . Notice this is independent of  $K$  chosen by linear algebra reasons. And this defines a distribution  $\chi_\pi$  on  $\mathcal{H} : \varphi \mapsto \text{tr}(\pi(\varphi)|V^K)$ , called the **character (of admissible representation)** of  $V$ .  $\lrcorner$

**Cor. (24.1.6.34).**  $\text{Irr}^{\text{alg}}(G)$  has enough injectives.  $\lrcorner$

*Proof:* As  $\text{Irr}^{\text{alg}}(G)$  has enough projectives (3.6.5.9)(24.1.6.22), there is a surjection  $P \rightarrow \tilde{X}$ , thus an injection  $\tilde{X} \hookrightarrow \tilde{P}$  (24.1.6.27). Now  $X \hookrightarrow \tilde{X}$  by (24.1.6.26).  $\square$

### Irreducible Admissible Representations

**Prop. (24.1.6.35) [Separation Lemma].** If  $G$  is  $\sigma$ -compact, then for any  $0 \neq h \in \mathcal{H}(G)$ , there exists  $\rho \in \text{Irr}^{\text{adm}}(G)$  s.t.  $\rho(h) \neq 0$ .  $\lrcorner$

*Proof:* Cf. [Bernstein, P20], [Bernstein-Zelevinsky, P19]. ?  $\square$

**Prop. (24.1.6.36) [Schur's Lemma].**

- If  $G$  is a  $\sigma$ -compact, then any  $V \in \text{Irr}^{\text{alg}}(G)$  is of at most countable dimension. In particular,  $\text{End}_G(V) = \mathbb{C}$  by (3.6.1.2).
- For any  $(\pi, V) \in \text{Irr}^{\text{adm}}(G)$ , we have  $\text{End}_G(V) = \mathbb{C}$ .

$\lrcorner$

*Proof:* 1: If it is of at most countable dimension because if  $\xi$  generate  $V$ , then notice its stablizer is compact open, and  $G$  is  $\sigma$ -compact, so  $V$  is at most countable.

2: Let  $K_0$  be a small open compact subgroup that  $V^{K_0} \neq 0$ , then  $V^{K_0}$  is of f.d. and preserved under  $T \in \text{End}_G(V)$ , thus  $T$  has an eigenvalue  $c$ , thus  $T = cI$  on  $V$ .  $\square$

**Prop. (24.1.6.37).** Let  $(\pi_1, V_1), (\pi_2, V_2) \in \text{Irr}^{\text{alg}}(G)$ . If  $V_1^K \cong V_2^K \neq 0$  as  $\mathcal{H}_K$ -module for some compact open subgroup  $K$  of  $G$ , then  $\pi_1 \cong \pi_2$ .  $\square$

*Proof:* This follows immediately from (3.6.5.7).  $\square$

**Prop. (24.1.6.38) [Characters Determine Irreducible Admissible Representations].** Let  $\pi_1, \dots, \pi_n \in \text{Irr}^{\text{adm}}(G)$  be pairwise inequivalent, then their characters  $\text{tr}(\pi_i)$  are linearly independent. In particular, an irreducible admissible representation is determined by its character.  $\square$

*Proof:* Choose compact open subgroup  $K$  that  $V_i^K \neq 0$  for any  $i$ , the hypothesis together with (24.1.6.37) shows  $V_i^K \cong V_j^K$ . Then we finish by (24.7.3.23).  $\square$

**Prop. (24.1.6.39) [Representations of Product Group].** If  $G_1, G_2 \in \text{locProfGrp}$  and  $(\pi_i, M_i) \in \text{Irr}^{\text{adm}}(G_i)$ , then  $M_1 \otimes M_2 \in \text{Irr}^{\text{adm}}(G_1 \times G_2)$ , and  $\text{Irr}^{\text{adm}}(G_1 \times G_2) = \text{Irr}^{\text{adm}}(G_1) \times \text{Irr}^{\text{adm}}(G_2)$ .  $\square$

*Proof:* By (24.1.6.22) and (3.6.5.15), this follows if we have  $\mathcal{H}_{G_1 \otimes G_2} \cong \mathcal{H}_{G_1} \otimes \mathcal{H}_{G_2}$ . And this fact is easily deduced from (4.4.4.13).  $\square$

### Induced Representations and Mackey Theory

**Def. (24.1.6.40) [Smooth Induced Representations].** Let  $H \leq G$  be a closed subgroup and  $(\pi, V) \in \text{Irr}^{\text{alg}}(H)$ , then we can define the (normalized) **smooth induced representation**  $\text{n-Ind}_H^G \pi$  as the space of locally constant  $f$  on  $G$  with values in  $V$  s.t.

$$f(hg) = \sqrt{\frac{\Delta_G(h)}{\Delta_H(h)}} \pi(h) f(g),$$

with the natural right  $G$ -representation. Notice this is compatible with that of unitary representations of locally compact groups, in (24.7.4.3).

The induced representation  $\text{n-Ind}_H^G \pi$  has a subrepresentation  $\text{n-ind}_H^G \pi$  consisting of functions that is compactly supported in  $H \backslash G$ , called the (normalized) **smooth compactly-induced representation** of  $\pi$ .

We can also define the unnormalized versions  $\text{Ind}_H^G$  and  $\text{ind}_H^G$ .  $\square$

**Remark (24.1.6.41).** The normalized factor here is used to make the induction of a unitarizable representation unitarizable (24.1.6.51).  $\square$

**Prop. (24.1.6.42) [Fixed Points of the Induced Representations].** Situation as in (24.1.6.40). Let  $K \subset G$  be an open compact subgroup, and  $\Omega \subset G$  be a system of coset representations of  $H \backslash G / K$ . For each  $g \in G$ , denote  $N_g = H \cap gKg^{-1}$  the compact open subgroup of  $H$ , then the restriction of functions from  $G$  to  $\Omega$  induces an isomorphism of  $(\text{Ind}_H^G(V))^K$  with the  $\prod_{g \in \Omega} V^{N_g}$ . And in this way,  $(\text{ind}_H^G(V))^K$  is mapped to  $\bigoplus_{g \in \Omega} V^{N_g}$ .  $\square$

**Prop. (24.1.6.43).** Let  $B \leq H \leq G$  be closed subgroups, then

- $\text{Ind}_H^G$  and  $\text{ind}_H^G$  are both exact functors  $\text{Rep}^{\text{alg}}(H) \rightarrow \text{Rep}^{\text{alg}}(G)$ .



- $\text{Ind}_H^G \circ \text{Ind}_B^H = \text{Ind}_B^G$  and  $\text{ind}_H^G \circ \text{ind}_B^H = \text{ind}_B^G$ .
- $\text{Ind}_H^G(\sigma)^\vee = \text{ind}_H^G(\sigma^\vee)$ .
- (Smooth Frobenius Reciprocity) If  $(\pi, W) \in \text{Rep}^{\text{alg}}(G)$ ,  $(\rho, V) \in \text{Rep}^{\text{alg}}(H)$ , then there are functorial isomorphisms

$$\text{Hom}_G(\pi, \text{Ind}_H^G(\rho)) = \text{Hom}_H(\pi|_H, \rho \otimes \sqrt{\frac{\Delta_G}{\Delta_H}}), \quad \text{Hom}_G(\text{ind}_H^G(\sigma), \pi) = \text{Hom}_H(\rho \otimes \sqrt{\frac{\Delta_H}{\Delta_G}}, (\pi^\vee|_H)^\vee)$$

⌋

*Proof:* 1: This follows from the description in (3.6.5.3) and (3.6.5.3) by taking colimits.

2: For simplicity we prove for  $G, H, B$  unimodular. An element in  $\text{Ind}_H^G \circ \text{Ind}_B^H(V)$  is an element  $\varphi : G \mapsto \text{Hom}(H, V)$  that satisfies  $\varphi(hg)(h') = (\pi(h)\varphi(g))(h') = \varphi(g)(hh')$ , thus  $\varphi(g)(h) = \varphi(hg)(1)$ . Let  $\Phi(g) = \varphi(g)(1)$ , then  $\Phi$  satisfies  $\Phi(bg) = \pi(b)\Phi(g)$ . So  $\varphi \mapsto \Phi$  is an isomorphism. ind case follows from item 3.

3: For  $f : G \rightarrow V \in \text{Ind}_H^G(\sigma)$ ,  $f' \in G \rightarrow V^\vee \in \text{ind}_H^G(\sigma^\vee)$ ,  $\langle f, f' \rangle$  is compactly supported in  $H \backslash G$ , and  $\langle \pi(h)f, \pi(h)f' \rangle = \frac{\Delta_G(h)}{\Delta_H(h)} \langle f, f' \rangle$ , thus we can define a  $G$ -invariant pairing

$$\langle f, f' \rangle = \int_{H \backslash G} \langle f(g), f'(g) \rangle d\mu_{H \backslash G}(g) \quad (12.10.3.15).$$

This map is a perfect pairing on the  $K$ -fixed part for any compact open  $K$  using description in (24.1.6.42), thus we are done.

4: Given a  $\Phi : W \rightarrow \text{Ind}_H^G V$ , we have a map  $\varphi : W \rightarrow V : \varphi(w) = \Phi(w)(1)$ , then it is verified that  $\varphi \in C(\pi|_H, \rho \otimes \sqrt{\frac{\Delta_G}{\Delta_H}})$ . Conversely, if  $(\varphi : W \rightarrow V) \in C(\pi|_H, \rho \otimes \sqrt{\frac{\Delta_G}{\Delta_H}})$  is given, we can define  $\Phi : W \rightarrow \text{Ind}_H^G V : \Phi(w)(g) = \varphi(\pi(g)w)$ , then  $\Phi(w) \in \text{Ind}_H^G V$  and this is linear in  $w$  (remember to check smoothness). Finally, it is easily verified these maps are inverse to each other. The second assertion follows from item 3 and (24.1.6.26).  $\square$

**Cor. (24.1.6.44).** If  $H \backslash G$  is compact, then  $\text{Ind}_H^G = \text{ind}_H^G$ , and they map admissible representations to admissible representations, by the description in (3.6.5.3), as  $H \backslash G/K$  is both compact and discrete thus finite.  $\square$

**Prop. (24.1.6.45).** If  $H \trianglelefteq G$  and for any  $\rho \in \text{Rep}(H)$ , let  $\rho^g$  be  $\rho$  twisted by conjugation of  $g \in G$ , then  $\text{Ind}_H^G(\rho) \cong \text{Ind}_H^G(\rho^g)$ , and  $\text{ind}_H^G(\rho) \cong \text{ind}_H^G(\rho^g)$ .  $\square$

*Proof:*  $\square$

**Prop. (24.1.6.46).** Let  $H \leq G, K \trianglelefteq G$  be closed subgroups and  $\rho \in \text{Rep}(H)$ , then

$$(\text{Ind}_H^G(\rho))^K \cong \text{Ind}_{H/H \cap K}^{G/K}(\rho^{H \cap K}) \in \text{Rep}^{\text{alg}}(G/K)$$

⌋

*Proof:*  $\square$

**Prop. (24.1.6.47).** Let  $H \leq G$  be a closed subgroup and  $(\pi, V) \in \text{Irr}^{\text{alg}}(H)$ , then there is a  $G$ -invariant map  $P_\pi : C_c^\infty(G) \otimes V \rightarrow \text{ind}_H^G(\pi) :$

$$P_\pi(\varphi, v)(g) = \int_H \varphi(b^{-1}g) \sqrt{\frac{\Delta_G(b)}{\Delta_H(b)}} \pi(b)v db.$$



And if  $\dim \pi < \infty$  and there is an open compact subgroup  $K$  of  $G$  that  $G = HK$ , then this map is surjective.

Moreover,

$$P(\lambda(b)^{-1}\varphi, v) = \sqrt{\Delta_G(b)\Delta_H(b)}P(\varphi, \pi(b)v), \quad b \in H$$

┘

*Proof:* It can be verified that  $P_\pi(\varphi, v)(bg) = \sqrt{\frac{\Delta_G(b)}{\Delta_H(b)}}\pi(b)P_\pi(\varphi, v)$ .

If  $H \backslash G$  is compact, by (12.10.4.5), there is a compact open subset  $K$  of  $G$  that  $G = HK$ , then for any  $f \in \text{Ind}_H^G(\pi) = \sum v_i \otimes f_i$ , consider  $\varphi_i = \chi_{K^{-1}}f_i$ , then  $\sum P(\varphi_i, v_i) = \sum V(H \cap K)f_i v_i = V(H \cap K)f$ .  $\square$

**Prop. (24.1.6.48) [Mackey's Decomposition].** Let  $H, K \leq G$  be closed subgroups and  $\rho \in \text{Irr}^{\text{alg}}(H)$ . If  $s \in G$ , then we can define a new representation of  $H_s = K \cap sHs^{-1}$  as  $\rho^s(g) \mapsto \rho(s^{-1}gs)$ . Then if we use unnormalized induction,

- If either  $H$  or  $K$  is open in  $G$ , then

$$\text{res}_K^G \text{ind}_H^G \rho \cong \bigoplus_{s \in H \backslash G/K} \text{ind}_{H_s}^K \rho^s$$

- If  $K$  is open in  $G$ , then

$$\text{res}_K^G \text{Ind}_H^G \tau \cong \left( \prod_{s \in H \backslash G/K} \text{Ind}_{H_s}^K \tau^s \right)^\infty$$

┘

*Proof:* Cf. [Yam22].  $\square$

**Cor. (24.1.6.49) [Mackey's Intertwining Theorem].** Let  $H \leq G$  be closed and,  $K \leq G$  be open,  $\sigma$  (resp.  $\tau$ ) be a smooth representation of  $K$  (resp.  $H$ ), define  $\tau^s$  as in (24.1.6.48), then if we use unnormalized induction,

- 

$$\text{Hom}_G(\text{ind}_K^G \sigma, \text{Ind}_H^G \tau) \cong \prod_{s \in H \backslash G/K} \text{Hom}_{H_s}(\sigma, \tau^s).$$

- If moreover  $H \backslash HgK \cong H_s \backslash K$  is compact for any  $g \in G$  (e.g.  $K$  is compact) and  $\sigma$  is f.g. over  $K$ , then

$$\text{Hom}_G(\text{ind}_K^G \sigma, \text{ind}_H^G \tau) \cong \bigoplus_{s \in H \backslash G/K} \text{Hom}_{H_s}(\sigma, \tau^s).$$

┘

*Proof:* 1: This is a direct consequence of Mackey's decomposition (24.1.6.48) and Frobenius reciprocity (24.1.6.43):

$$\begin{aligned} \text{Hom}_G(\text{ind}_K^G \sigma, \text{Ind}_H^G \tau) &\cong \text{Hom}_K(\sigma, \text{res}_K^G \text{Ind}_H^G \tau) \cong \text{Hom}_K(\sigma, \left( \prod_{s \in H \backslash G/K} \text{Ind}_{H_s}^K \tau^s \right)^\infty) \\ &\cong \text{Hom}_K(\sigma, \prod_{s \in H \backslash G/K} \text{Ind}_{H_s}^K \tau^s) \cong \prod_{s \in H \backslash G/K} \text{Hom}_K(\sigma, \text{Ind}_{H_s}^K \tau^s) \\ &\cong \prod_{s \in H \backslash G/K} \text{Hom}_{H_s}(\sigma, \tau^s). \end{aligned}$$

2 is similar.  $\square$

**Cor. (24.1.6.50) [Compact Induction Admissible].** If  $H \leq G$  be a closed subgroup s.t.  $H \backslash G$  is compact, then  $\text{Ind}_H^G$  takes admissible representations to admissible representations.  $\lrcorner$

*Proof:* For any compact open subgroup  $K \leq G$ ,  $H \backslash G/K$  is finite, as  $H \backslash G$  is compact and  $K$  is open, then by Mackey's decomposition (24.1.6.48),  $(\text{Ind}_H^G \rho)^K = \oplus_{s \in H \backslash G/K} \rho^{s^{-1}Ks \cap H}$  is of f.d. if  $\rho$  is admissible.  $\square$

### Unitarizable Admissible Representations

**Prop. (24.1.6.51) [Induction of Unitarizable Admissible Representations].** If  $H \leq G$  is a closed subgroup and  $(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}}(H)$  with  $\mathcal{H}^{\text{alg}} = V \in \text{Rep}^{\text{alg}}(H)$ , then

$$\text{ind}_H^G(V) = \text{ind}_H^G(\mathcal{H})^{\text{alg}} \text{ (24.7.4.3)}.$$

In particular, the induction of a unitarizable representation is unitarizable, by (24.7.4.3) and that fact  $\text{Ind}_H^G(\sigma)^\vee = \text{ind}_H^G(\sigma^\vee)$  (24.1.6.43).  $\lrcorner$

**Prop. (24.1.6.52) [Converse of Schur's Lemma].** If  $(\pi, V) \in \text{Irr}^{\text{adm, uni}}(G)$  and  $\text{Hom}_G(V, V) = \mathbb{C}$ , then  $\pi$  is irreducible.  $\lrcorner$

*Proof:* Cf. [Bump, P523]. ?  $\square$

### Compact Representations

**Def. (24.1.6.53) [Compact Representations].** A **compact representation** of  $G$  is a smooth representation  $(\pi, V) \in \text{Rep}^{\text{alg}}(G)$  s.t. for every  $\xi \in V$  and every compact open subgroup  $K \subset G$ , the function  $D_{\xi, K} : G \rightarrow V : g \mapsto \pi(e_K)\pi(g^{-1})\xi$  has compact support.  $\lrcorner$

**Prop. (24.1.6.54).**  $V \in \text{Rep}^{\text{alg}}(G)$  is compact iff every matrix coefficient of  $V$  is compactly supported. And every f.g. compact representation is admissible.  $\lrcorner$

*Proof:* If  $V$  is compact, for any  $\xi \in V, \xi^\wedge \in V^\wedge, \xi^\wedge \in (V^\wedge)^K$ , then  $\text{Supp } \varphi_{\xi, \xi^\wedge} \in \text{Supp } D_{\xi, K}$  is compact.

For the converse, Cf. [Bernstein Zelevinsky, P26].  $\square$

## 24.2 Positive Characteristic Representations

References are [Ser77, Bon11].

**Notation (24.2.0.1).**

- Use notations on representations defined in [Smooth Representations](#).

┘

### 1 Basics

**Prop. (24.2.1.1) [Dickson].** For any  $p \in \text{Prime}$ , if  $G \subset \text{GL}(2; \overline{\mathbb{F}}_p)$  is a finite subgroup that acts irreducibly on  $\overline{\mathbb{F}}_p^2$ , then it has projective image isomorphic to either a dihedral group,  $\text{Alt}_4$ ,  $\text{Sym}_4$ ,  $\text{Alt}_5$ ,  $\text{PSL}(2; \mathbb{F})$  or  $\text{PGL}(2; \mathbb{F})$ , where  $\mathbb{F} \subset \overline{\mathbb{F}}_p$  is a finite subfield.

If  $p = 2$ , then the projective image must be dihedral.

┘

*Proof:* Cf. [Huppert, B.: Endliche Gruppen I. Springer, Berlin (1967)]2.8.27.

□

### 2 Block Theory

See [Bon11].

### 3 Mixed Characteristic

**Notation (24.2.3.1).**

- Let  $(K, \mathcal{O}_K, \varpi, \kappa) \in p\text{-Field}$ .

┘

**Prop. (24.2.3.2) [Compact Groups Stabilize a Lattice].** Let  $\Gamma \in \text{TopGrp}_{\text{cpct}}$  and  $\rho : \Gamma \rightarrow \text{GL}(n; \overline{\mathbb{Q}}_p)$  be a continuous homomorphism, then there exists a finite extension  $L/\mathbb{Q}_p$  that  $\rho(\Gamma) \subset \text{GL}(n; L)$ , and up to conjugation,  $\rho(\Gamma) \subset \text{GL}(n; \mathcal{O}_L)$ , or equivalently,  $\Gamma$  fixes a  $\mathcal{O}_L$ -lattice.

┘

*Proof:* Notice  $\rho(\Gamma)$  is compact and Hausdorff, so by Baire category theorem, now that  $\text{GL}(n; L)$  is closed in  $\text{GL}(n; \overline{\mathbb{Q}}_p)$  for all  $L/\mathbb{Q}_p$  finite, and all this extensions are countable by primitive element theorem, so there is an  $L$  that  $\rho(\Gamma) \cap \text{GL}(n; L)$  contains an open subset of  $\rho(\Gamma)$ , so it is an open subgroup, thus of finite index, hence by adding all the coset representations into  $L$ , we get a finite extension  $L'$ .

For the second assertion, notice  $\rho(\Gamma)$  is compact in  $\text{GL}(n; L)$ , thus by (20.7.3.10), it is conjugate to  $\text{GL}(n; \mathcal{O}_L)$ .

□

**Def. (24.2.3.3) [Residual Representations].** For any  $\Gamma \in \text{TopGrp}_{\text{cpct}}$  and  $(\rho, V) \in \text{Rep}_K^{\text{fd}}(\Gamma)$ , by (24.2.3.2) we may take a  $\Gamma$ -stable  $\mathcal{O}_K$ -lattice  $\Lambda \subset V$ , and define the **residual representation**  $\bar{\rho}$  to be  $\Gamma \rightarrow \text{GL}(\Lambda/(\varpi_K))$ . Then the semisimplification of  $\bar{\rho}$  is independent of  $\Gamma$  chosen. ?

┘

**Prop. (24.2.3.4).** Situation as in (24.2.3.3),  $\bar{\rho}$  is irreducible iff the only  $\Gamma$ -stable  $\mathcal{O}_L$ -lattices in  $V$  are of the form  $\varpi_K^n \Lambda$  for some  $n \in \mathbb{Z}$ .

┘

**Def. (24.2.3.5) [Residually Disjoint Representations].** Let  $C \in \mathcal{CRing}$  and  $(B, \mathfrak{m}_B, \kappa_B) \subset C$  be a local Henselian ring. For  $G \in \text{Grp}$ , suppose there exist two representations  $(V_1, \rho_1), (V_2, \rho_2) \in \text{Rep}_C^{\text{fd}}(G)$  (24.1.1.1) s.t.

$$\det(1 - gT|V_i) \in B[T]$$

for any  $g \in G$  and  $i \in \{1, 2\}$ . Then they are called **residually disjoint representations** if there exists  $x \in B[G]$  s.t.  $\det(1 - xT|V_1)$  and  $\det(1 - xT|V_2)$  are relatively prime in  $\kappa_B$ .  $\lrcorner$

**Prop. (24.2.3.6).** Situation as in (24.2.3.5), if  $(V_1, \rho_1), (V_2, \rho_2) \in \text{Rep}_C^{\text{fd}}(G)$  are residually disjoint, then there exists  $x_1, x_2 \in B[G]$  s.t.

$$x_i|_{V_j} = \begin{cases} \text{id}_{V_j} & i = j \\ 0 & , i \neq j \end{cases}.$$

 $\lrcorner$ 

*Proof:* This follows from (19.1.2.25).  $\square$

### cde-Triangle

**Prop. (24.2.3.7) [ $p'$ -Groups].** If  $(p, \#G) = 1$ , then

- Each  $\kappa[G]$ -module is projective, and each  $\mathcal{O}_K$ -free  $\mathcal{O}_K[G]$ -module is projective.
- Reduction modulo  $\mathfrak{m}_K$  defines a bijection  $S_K \cong S_\kappa$ .
- After identifying  $S_K$  with  $S_\kappa$ , the matrices  $C, D, E$  are identity matrices.

 $\lrcorner$ 

*Proof:* Cf. [Ser77]P128.  $\color{red}?$   $\square$

**Prop. (24.2.3.8).** Let  $K \subset_{\text{fin}} K' \in \text{Field}$  and  $\theta \in R_{K'}(G)$  s.t.

- $\theta \in \overline{R}_K(G)$ .
- There exists  $n \in \mathbb{Z}_+$  s.t.  $n\theta$  is realized on a projective  $\mathcal{O}_{K'}[G]$ -module.

Then  $x$  can be realized on a unique projective  $\mathcal{O}_K[G]$ -module.  $\lrcorner$

*Proof:* Cf. [Ser77]P134.  $\color{red}?$   $\square$

### Artin and Swan Characters

**Def. (24.2.3.9) [Artin Characters].** and  $L/K$  be an unramified representation with Galois group  $G$ , define a function on  $G$ :

$$a_G(s) = \begin{cases} v(s(\varpi) - \varpi) & , s \neq 1 \\ \sum_{t \neq 1} v(t(\varpi) - \varpi) & , s = 1 \end{cases}.$$

Then  $a_G(s)$  is the character of a representation of  $G$ , and  $a_G$  is called the **Artin character**.

Also define  $\text{Sw}_G = a_G - r_G + 1$ , where  $r_G$  is the regular character, then  $\text{Sw}_G$  is also a character, called the **Swan character**.  $\lrcorner$

*Proof:* Cf. [Ser77]P159.  $\color{red}?$   $\square$

**Prop. (24.2.3.10).** Situation as in (24.2.3.9), then

$$\begin{aligned} \#G \cdot a_G &= \sum_{i=0}^{\infty} \#G_i \cdot \text{Ind}_{G_i}^G(r_{G_i} - 1), \\ \#G \cdot \text{Sw}_G &= \sum_{i=1}^{\infty} \#G_i \cdot \text{Ind}_{G_i}^G(r_{G_i} - 1). \end{aligned}$$

In particular,  $\text{Sw}_G = 0$  iff  $G_1 = 1$ , i.e. iff  $F/E$  is tamely unramified.  $\lrcorner$

*Proof:*

□

**Prop. (24.2.3.11) [Realization of Artin and Swan Characters].** Let  $\ell \in \text{Prime} \setminus \{p\}$ , then

- the Artin character and Swan character can be realized on  $\mathbb{Q}_\ell$ .
- the Swan character can be realized on a projective module over  $\mathbb{Z}_\ell[G]$ , denoted by  $\text{Sw}_G$ .

┘

*Proof:* Because  $a_G = \text{sw}_G + (r_G - 1)$ , it suffices to prove item 2. For this, we use (24.2.3.8). It suffices to show that

$$\#G \cdot \text{Sw}_G = \sum_{i=1}^{\infty} \#G_i \cdot \text{Ind}_{G_i}^G (r_{G_i} - 1)$$

is realizable on a projective  $\mathcal{O}_{K'}[G]$ -module. Because  $(\#G_i, \ell) = 1$ , every  $\mathcal{O}_{K'}[G]$ -module is projective by (24.2.3.7), and thus the induction and also  $\#G \cdot \text{sw}_G$  is projective. □

**Def. (24.2.3.12) [Wild Ramification Index  $b(M)$ ].** Let  $\ell \in \text{Prime} \setminus \{p\}$  and  $M \in \text{Mod}_{\mathbb{F}_\ell}[G]$ , define the **Wild ramification index**

$$b(M) = \left\langle \overline{\text{Sw}_G}, M \right\rangle_{\mathbb{F}_\ell} = \dim \text{Hom}_G(\overline{\text{Sw}_G}, M) = \dim \text{Hom}_{\mathbb{Z}_\ell[G]}(\overline{\text{Sw}_G}, M) \quad (24.2.3.11).$$

Then

- $b$  is additive on  $\text{Mod}_{\mathbb{F}_\ell}[G]$ , because  $\text{Sw}_G$  is projective.
- By (24.2.3.10),

$$b(M) = \sum_{i=1}^{\infty} \frac{\#G_i}{\#G} \dim_{\mathbb{F}_\ell} M/M^{G_i}.$$

- $b(M)$  iff  $G_1$  acts trivially on  $M$ .

┘

## 4 Equal Characteristic

**Notation (24.2.4.1).** Let  $p \in \text{Prime}$  and  $k \in \text{Field}^p$ .

┘

**Prop. (24.2.4.2).** If  $G$  is a  $p$ -group, then  $\text{Irr}_k(G) = \{1\}$ .

┘

*Proof:* For any  $v \in V$ , consider the additive subgroup generated by  $g(s)v$ , then it is a finite group of prime power order. Then (3.1.6.4) shows it has a element other than 0 fixed by all  $G$ , thus it is not irreducible unless trivial representation. □

## 24.3 Representations of Finite Groups of Lie Type

References are [Bon11], [Introduction to Deligne-Lustig Theory, David Schwein], [Representations of finite groups of Lie type, Dign/Michel, 1991].

**Notation (24.3.0.1).**

- Let  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}^+}$ ,  $k \in \text{Field}$ ,  $\#k = q$ ,  $G \in \text{AlgGrp}/k$ .
- Let  $(K, \mathcal{O}_K, \kappa) \in \ell\text{-LField}$ .
- Let  $\ell \in \text{Prime} \setminus p$ .
- Use notations on étale cohomologies defined in [Étale Cohomology Theory](#).

┘

### 1 Finite Groups of Lie Type

**Def. (24.3.1.1) [Finite Groups of Lie Type].** For a reductive group  $G/\bar{k}$  and a Steinberg endomorphism  $F$  of  $G$ , the fixed points  $G(\bar{k})^F$  is called a **finite group of Lie type**.

If  $G$  is defined over some finite subfield  $k$ , i.e.  $G = G_0 \times_k \bar{k}$ , and  $F = \text{Fr}_{G_0}$ , then  $G^F = G_0(k)$  is called a **Chevalley group**. If moreover  $G_0 \in \text{AlgGrp}/k$  is reductive and simply-connected, then  $G_0(k)$  is called a **universal finite group** (of Lie type).

If  $G$  is a finite group of Lie type that is not a Chevalley group, then it is called a **twist group** (of Lie type).

┘

**Prop. (24.3.1.2).** If  $p \in \text{Prime}$  and  $q \in p^{\mathbb{Z}^+}$ ,

- $\# \text{GL}(2; q) = q(q-1)^2(q+1)$ .
- $\# \text{SL}(2; q) = \# \text{PGL}(2; q) = q(q^2-1)$ .
- $\# \text{PSL}(2; q) = \begin{cases} q(q^2-1) & , 2|q \\ \frac{1}{2}q(q^2-1) & , 2 \nmid q \end{cases}$ .

┘

**Thm. (24.3.1.3) [Steinberg].** If  $G \in \text{RedGrp}/\bar{k}$  is simple, and  $F \in \text{End}(G)$  s.t. the fixed point of  $F(\bar{k})$  is finite, then there are exactly the following possibilities:

- (Chevalley Groups)  $F$  is a standard Frobenius:  $A_n, B_n, C_n, D_n, G_2, F_4, E_6, E_7, E_8$ .
- (Steinberg Groups)  ${}^2A_n, {}^2D_n, {}^3D_4, {}^2_6$ .
- (Suzuki Groups)  ${}^2B_2, p = 2$ .
- (Ree Groups)  ${}^2G_2, p = 3$ .
- (Ree Groups)  ${}^2F_4, p = 2$ .

┘

*Proof:*

□

### Examples

**Prop. (24.3.1.4) [SL(n)].**

- $\text{PSL}(2, \mathbb{F}_2) \cong \text{Sym}(3)$ ,
- $\text{PSL}(2, \mathbb{F}_3) \cong \text{Alt}(4)$ ,

- (Jordan1861) For  $k \in \mathbf{Field}$ ,  $\mathrm{PSL}(n, k)$  is a simple group except for  $n = 2$  and  $k \cong \mathbb{F}_2$  or  $\mathbb{F}_3$ .
- $\mathrm{PSL}(2, \mathbb{F}_4) \cong \mathrm{PSL}(2, \mathbb{F}_5) \cong \mathrm{Alt}(5)$ .
- $\mathrm{SU}(n)_{\mathbb{F}_q}$  is a twist of  $\mathrm{SL}(n)_{\mathbb{F}_q}$ , denoted by  ${}^2A_{n-1}(\mathbb{F}_{q^2})$ .

┘

*Proof:* Cf.[DSV03]P74.

□

**Prop.(24.3.1.5).**

┘

## 2 Conjugacy Classes

## 3 Deligne-Lustig Varieties

**Def.(24.3.3.1)**[Deligne-Lustig Varieties].

┘

### Affineness of Deligne-Lustig Varieties

## 4 Deligne-Lustig Theory

Main references are [Finite Groups of Lie Type, Carter], [P. Deligne and G. Lusztig, Representations of reductive groups over finite fields, Ann. of Math. (2) 103 (1976), no.1, 103–161.], [D-L76], [https://www.dpmms.cam.ac.uk/~dbms2/deligne\\_lusztig/](https://www.dpmms.cam.ac.uk/~dbms2/deligne_lusztig/).

Deligne and Lusztig used the  $\ell$ -adic étale cohomology theory to construct all cuspidal representations of  $G(\mathbb{F}_q)$ .

**Remark(24.3.4.1).** Because  $G(\mathbb{F}_q)$  is finite and  $\mathbb{C} \cong \overline{\mathbb{Q}_\ell}$ , we can consider representations  $\mathrm{Rep}(G(\mathbb{F}_q)) = \{\theta : G(\mathbb{F}_q) \rightarrow \mathrm{GL}(\overline{\mathbb{Q}_\ell})\}$ .

┘

**Def.(24.3.4.2)**[Induction Pairs]. A **induction pair** for  $G(\mathbb{F}_q)$  is a pair  $(T, \theta)$ , where  $T \subset G$  is a maximal torus and  $\theta$  is an  $\ell$ -adic character  $T(\mathbb{F}_q) \rightarrow \overline{\mathbb{Q}_\ell}^\times$ .

┘

### Deligne-Lustig Inductions

**Def.(24.3.4.3)**[Deligne-Lustig Inductions]. Let  $(T, \theta)$  be an induction pair(24.3.4.2), the action of  $G(\mathbb{F}_q)$  and  $T(\mathbb{F}_q)$  on  $\tilde{X}_S$  commutes, so for any  $\theta \in \mathrm{Irr}(T(\mathbb{F}_q))$ , the  $\theta$ -isotypic part of  $H_{\mathrm{ét},c}^i(\tilde{X}_T, \overline{\mathbb{Q}_\ell})$  is  $G(\mathbb{F}_q)$ -invariant, thus we can define a **Deligne-Lustig Induction map** that on  $\mathrm{Irr}(G(\mathbb{F}_q))$  is defined by

$$R_T^G : \mathrm{Irr}(T(\mathbb{F}_q)) \rightarrow K_0(\mathrm{Rep}(G(\mathbb{F}_q))) : \theta \mapsto \sum_i (-1)^i H_{\mathrm{ét},c}^i(X_T, \overline{\mathbb{Q}_\ell})^\theta.$$

┘

**Prop.(24.3.4.4).** Let  $(T, \theta)$  be an induction pair, if  $\theta^w \neq \theta$  for any  $w \in W$ , then  $R_T^G(\theta) \in \mathrm{Irr}(G(\mathbb{F}_q))$ .

┘

*Proof:*

□

**Prop.(24.3.4.5).** If  $(T, \theta), (T', \theta')$  are two induction pairs that are not  $G(\mathbb{F}_q)$ -conjugate, then  $(R_T^G(\theta), R_{T'}^G(\theta')) = 0$ .

┘

*Proof:*

□

**Def. (24.3.4.6) [Geometrically Conjugate Pairs].** If  $(T, \theta), (T', \theta')$  are two induction pairs, they are called **geometrically conjugate** if  $(T_{k'}, \theta \circ \text{Nm}_{k'/k}), (T'_{k'}, \theta' \circ \text{Nm}_{k'/k})$  are  $G(k')$ -conjugate for some finite extension  $k'/\mathbb{F}_q$ .  $\lrcorner$

**Prop. (24.3.4.7).** If  $(T, \theta), (T', \theta')$  are two induction pairs that are not geometrically conjugate, then  $R_T^G(\theta), R_{T'}^G(\theta')$  are disjoint.  $\lrcorner$

*Proof:* Cf. [D-L76] Cor6.3.  $\square$

**Def. (24.3.4.8) [Dual Groups].**  $\lrcorner$

**Prop. (24.3.4.9).** The geometric conjugacy classes of induction pairs of  $G$  corresponds to the geometric conjugacy classes of semisimple elements of  $G^*$  (24.3.4.8).  $\lrcorner$

*Proof:*  $\square$

**Def. (24.3.4.10) [Lustig Series].** Let  $S$  be a semisimple conjugacy class of  $G^*$ , the **Lustig series**  $\mathcal{E}(G, S)$  is the set of irreducible representations of  $G$  occurring in  $R_T^G(\theta)$  for any  $(T, \theta)$  in the geometric conjugacy class corresponding to  $S$  via (24.3.4.9). By (24.3.4.7),

$$\text{Irr}(G(\mathbb{F}_q)) = \coprod_{S \in G_{\text{s.st-conj}}^*} \mathcal{E}(G, S).$$

$\mathcal{E}(G, (1))$  is called the set of **unipotent representations**, which are all the irreducible representations appearing in  $R_T^G(1)$  for various  $T$ .  $\lrcorner$

## 5 Howlett-Lehler Theory

This theory decomposes Deligne-Lustig inductions into irreducible components.

## 6 Character Sheaves(Lustig)

References are [Introduction to character sheaves, Lustig], [Algebraic And Geometric Methods In Representation Theory, Lustig, 2014].

**Remark (24.3.6.1).** Let  $(S, \theta)$  be an induction pair (24.3.4.2), there is in fact an isomorphism

$$H_{\text{ét},c}^i(\tilde{X}_S, \overline{\mathbb{Q}}_\ell) \cong H_{\text{ét},c}^i(X_S, \mathcal{F}_\theta) \in \text{Rep}(G(\mathbb{F}_q))$$

where  $\mathcal{F}_\theta$  is the  $\ell$ -adic local system on  $X_S$  corresponding to the  $S(\mathbb{F}_q)$ -torsor and  $\theta$ . Thus much of the Deligne-Lustig theory can be interpreted in the language of sheaves, called **character sheaves**.  $\lrcorner$

## 7 Gelfand-Graev representations

## 8 Alvis-Curtis Duality

Cf. [Duality for representations of a reductive group over a finite field, I, II, Deligne-Lustig].

## 9 Deligne-Lustig Varieties

**Prop. (24.3.9.1).** Let  $\lrcorner$



**10**  $GL(n)$ 

**Notation (24.3.10.1).** The notation is the same as in (24.11.0.1).  $\lrcorner$

**Principal Series Representations**

**Prop. (24.3.10.2).** Let  $(\pi, V) \in \text{Rep}^{\text{fd}}(GL(2, k))$ , then

- if the representation  $(\pi_1, V)$  is defined by  $\pi_1(g) = \pi(g^{-t})$ , then  $\pi_1 \cong \widehat{\pi}$ .
- if  $n = 2$  and  $(\pi, V)$  is irreducible, let  $\omega$  be the central character of  $\pi$ . If  $(\pi_2, V)$  is defined by  $\pi_2(g) = \omega(\deg g)^{-1} \pi(g)$ , then  $\pi_2 \cong \widehat{\pi}$ .

 $\lrcorner$ 

*Proof:* The proof of (24.11.1.10) applies to this case, noticing a finite group is profinite hence locally profinite.  $\square$

**Lemma (24.3.10.3).** Let  $\chi_1, \chi_2, \mu_1, \mu_2$  be characters of  $F^*$ , consider the principal representations  $\mathcal{B}(\chi_1, \chi_2), \mathcal{B}(\mu_1, \mu_2)$  of  $GL(2, k)$  defined in (24.11.5.1), then

$$\dim \text{Hom}_{GL(2; k)}(\mathcal{B}(\chi_1, \chi_2), \mathcal{B}(\mu_1, \mu_2)) = \delta_{\chi_1, \mu_1} \delta_{\chi_2, \mu_2} + \delta_{\chi_1, \mu_2} \delta_{\chi_2, \mu_1}.$$

 $\lrcorner$ 

*Proof:* Let  $\chi, \mu$  be characters of  $B(k)$  defined as in (24.11.5.1), then by ??, the dimension is just the dimension of space of functions  $\Delta : GL(2, k) \rightarrow \mathbb{C}$  that

$$\Delta(b_2 g b_1) = \mu(b_1) \Delta(g) \chi(b_1), \quad b_1, b_2 \in B(k).$$

Then by the Bruhat decomposition (10.4.4.28),  $\Delta$  is determined by its values on 1 and  $w_0$ . Notice that if  $\chi_1 \neq \mu_1$  or  $\chi_2 \neq \mu_2$ , we can use this condition to show  $\Delta(1) = 0$ , and if  $\chi_1 \neq \mu_2$  or  $\chi_2 \neq \mu_1$ , we can use this condition to show  $\Delta(w_0) = 0$ , and also the construction of  $\Delta$  is clear in other cases.  $\square$

**Prop. (24.3.10.4) [Principal-Series].** Let  $\chi_1, \chi_2, \mu_1, \mu_2$  be characters of  $F^*$ , then  $\mathcal{B}(\chi_1, \chi_2)$  is an irreducible representation of degree  $q = |F| + 1$  unless  $\chi_1 = \chi_2$ , in which case it is the direct sum of two irreducible representations having degree 1 and  $q$ . And  $\mathcal{B}(\chi_1, \chi_2) \cong \mathcal{B}(\mu_1, \mu_2)$  iff  $\{\chi_1, \chi_2\} = \{\mu_1, \mu_2\}$ .  $\lrcorner$

*Proof:* Use (24.3.10.3), then

$$\dim \text{End}_{GL(2, k)}(\mathcal{B}(\chi_1, \chi_2)) = 1 + \delta_{\chi_1, \chi_2}.$$

Now by Peter-Weyl, if a representation  $V$  is isomorphic to  $\sum d_i \pi_i$ , where  $\pi_i$  are irreducible, then  $\dim_G(V) = \sum d_i^2$  (24.7.3.6). Then we now  $\mathcal{B}(\chi_1, \chi_2)$  decomposes into two representations if  $\chi_1 = \chi_2$  and is irreducible if  $\chi_1 \neq \chi_2$ .

In case  $\chi_1 = \chi_2$ , there is an invariant subspace of dimension 1, generated by the function  $f(g) = \chi(\deg g)$ . so the rest representation is of dimension  $q$ , because  $\# GL(2, k)/B(k) = q + 1$ .  $\square$

**Steinberg-Representations**

References are [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport]Chap7.

**Def. (24.3.10.5) [Steinberg-Representation].** We call the  $q$ -dimensional subrepresentation of  $\mathcal{B}(1, 1)$  the **Steinberg-representation(of finite groups)**. It is verified that the  $q$  dimensional subrepresentation of  $\mathcal{B}(\chi, \chi)$  is the twist of the Steinberg representation.  $\lrcorner$

### Weil-Representations

**Prop. (24.3.10.6) [Weil-Representation for  $SL(2; k)$ ].** In situation (26.8.4.8), let  $W = L^2(E)$ , define the modified Fourier transform w.r.t.  $\psi$  as in (12.11.1.34):

$$\widehat{\Phi}(x) = q^{-1} \sum_{y \in E} \Phi(y) \psi(\text{tr}(\bar{x}y)).$$

Then there is a **Weil-representation**  $\omega : SL(2, k) \rightarrow \text{End}(W)$  :

$$(\omega(t(a))\Phi)(x) = \Phi(ax), \quad \omega(n(z))\Phi(x) = \psi(zN(x))\Phi(x), \quad (\omega(w_1)\Phi)(x) = \varepsilon \widehat{\Phi}(x).$$

where we are using the presentation as in 5, and  $\varepsilon = 1$  if  $E/F$  is split, and  $-1$  if it is anisotropic.  $\lrcorner$

**Remark (24.3.10.7).** When  $F$  has characteristic  $\neq 2$ , this is just the Weil representation in (26.8.4.7).  $\lrcorner$

*Proof:* This follows from direct verification, Cf. [Bump, P407].  $\square$

**Prop. (24.3.10.8) [Dihedral Weil Representations for  $GL(2, k)$ ].** Situation as in (26.8.4.8), Define

$$W(\chi) = \{\Phi \in W \mid \Phi(yx) = \chi(y)^{-1} \Phi(x), y \in E_1^*\},$$

then by considering the order of  $E_1^*$ ,  $\dim W(\chi) = q + \varepsilon$ . Also it is verified that  $W(\chi)$  is stable under the Weil representation of  $SL(2, k)$  (24.3.10.6).

Now we want to extend this representation to  $GL(2, k)$  by defining

$$(\omega\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right)\Phi)(x) = \chi(b)\Phi(bx)$$

where  $b$  is arbitrary that  $N(b) = a$ .  $\lrcorner$

**Remark (24.3.10.9).** This is related to (26.8.4.10) and Howe-duality.  $\lrcorner$

*Proof:* It must be shown that this is truly a representation of  $GL(2, k)$ , it suffices to show that

$$\omega\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right)\omega(g)\omega\left(\begin{bmatrix} a^{-1} & \\ & 1 \end{bmatrix}\right) = \omega\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right)g\left(\begin{bmatrix} a^{-1} & \\ & 1 \end{bmatrix}\right),$$

where  $g$  is a generator of  $SL(2; k)$ . This is clear for  $g = t(a)$ , and also clear for  $g = n(z)$ . For  $g = w_1$ , it suffices to check

$$\omega\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right) \circ \wedge \circ \omega\left(\begin{bmatrix} a^{-1} & \\ & 1 \end{bmatrix}\right)\Phi(x) = \widehat{\Phi}(ax)$$

which is subtle but also clear.  $\square$

**Prop. (24.3.10.10) [Split Weil-Representations].** In the split case, the character  $\chi$  of  $E^*$  is of the form  $\chi((x, y)) = \chi_1(x)\chi_2(y)$ , and then condition in (26.8.4.8) just says  $\chi_1 \neq \chi_2$ . Then:

$$(\pi(\chi), W(\chi)) \cong \mathcal{B}(\chi_2, \chi_1)$$

$\lrcorner$

*Proof:* The intertwining operator is given by

$$L : W(\chi) \rightarrow \mathcal{B}(\chi_2, \chi_1) : (L\Phi)(g) = (\omega(g)\Phi)((1, 0)).$$

Firstly  $L\Phi \in \mathcal{B}(\chi_2, \chi_1)$  by direct verification for  $t(a), T_1(k)$  and  $n(z)$ . Then it is an intertwining operator is clear.

Then this is an isomorphism because both are of dimension  $q + 1$  (24.3.10.8) and  $\mathcal{B}(\chi_2, \chi_1)$  is irreducible (24.3.10.4).  $\square$

**Def. (24.3.10.11) [Cuspidal Representation].**  $(\pi, V) \in \text{Rep}^{\text{fd}}(GL(2, k))$  is called a **cuspidal representation** if the Jacquet module  $J(V) = 0$ . Notice in this finite case, this is equivalent to  $V$  has no  $N(k)$ -fixed vector, because the contragradient is well-known as  $\text{Rep}(G)$  is semisimple, and the trivial isotropic parts correspond, unlike the  $p$ -adic case.  $\lrcorner$

**Lemma (24.3.10.12).** If  $(\pi, V)$  is a cuspidal representation of  $GL(2, k)$ , then the dimension of  $V$  is a multiple of  $q - 1$ .  $\lrcorner$

*Proof:* Because  $N(k) \cong F$ , any character of  $N(k)$  is of the form  $\psi_a(n(x)) = \psi(ax)$ . Now decompose the contragradient representation  $V^*$  of  $G$  into isotypic parts of  $N(k):V^* = \bigoplus_{a \in F} V^*(a)$ , then the hypothesis implies  $V^*(0) = 0$ .

Notice that the group  $T_1(k)$  acts transitively on the spaces  $V^*(a), a \neq 0$  by  $\widehat{\pi}\left(\begin{bmatrix} t & \\ & 1 \end{bmatrix}\right)l$ , because

$$\begin{bmatrix} t & \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & x/t \\ & 1 \end{bmatrix} = \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \begin{bmatrix} t & \\ & 1 \end{bmatrix}.$$

So the dimension of  $V$  is a multiple of  $q - 1$ .  $\square$

**Prop. (24.3.10.13) [Anisotropic Representation].** In the anisotropic case, the Weil representation  $(\pi(\chi), W(\chi))$  is cuspidal and irreducible.  $\lrcorner$

*Proof:* Suppose it is not cuspidal, then it contains a non-zero  $N(k)$ -fixed vector  $\Phi_0$  (24.3.10.11), which means  $\Phi_0(x) = w(n(z)\Phi_0)(x) = \psi(zN(x))\Phi_0(x)$ . Now  $\Phi_0(0) = 0$  because  $\chi$  is nontrivial on  $E_1^*$ , and if  $x \neq 0$ , then there is a  $z$  that  $\psi(zN(x)) \neq 1$  because  $\psi$  is non-trivial, so  $\Phi_0(x) = 0$  also, so  $\Phi_0 = 0$ , contradiction.

Finally, subrepresentation of cuspidal representation is cuspidal by (24.3.10.11), then  $(\pi(\chi), W(\chi))$  is irreducible, by the fact it is of dimension  $q - 1$  (24.3.10.8) and lemma (24.3.10.12).  $\square$

**Prop. (24.3.10.14) [Classification of Representations of  $GL(2, k)$ ].** There is a list of all irreducible representations of  $GL(2, k)$ :

- $q - 1$  1-dimensional representations  $\chi(\deg g)$ , where  $\chi$  is a character of  $k^\times$ .
- $\frac{(q-1)(q-2)}{2}$  principal series representations of dimension  $q + 1$ .
- $q - 1$  Steinberg representations (with twists) of dimension  $q$ .
- $\frac{q(q-1)}{2}$  cuspidal Weil representations of dimension  $q - 1$ .

$\lrcorner$

*Proof:* All these are irreducible representations by (24.3.10.4) (24.3.10.5) and (24.3.10.13). It suffices to show they are not isomorphic. Notice different kind of representation have different dimensions, thus it suffices to compare the same representations.

For principal series representations  $\mathcal{B}(\chi_1, \chi_2)$ ,  $\chi_1 \neq \chi_2$ , their isomorphisms are known by (24.3.10.4). For Steinberg representations, twisting are clearly different. For cuspidal Weil representations, there are  $q^q - q$  way of choosing  $\chi$  (26.8.4.8) and ? Cf. [Local Langlands For  $GL(2)$ ].

Finally, they are all the irreducible representations by the fact  $\sum d_\sigma^2 = |G| = (q-1)^2 q(q+1)$  (24.1.3.4).  $\square$

### Whittaker Models

**Def. (24.3.10.15) [Whittaker Functionals & Whittaker Models].** Let  $(\pi, V)$  be an irreducible representation of  $GL(2, k)$ , then the notion of **Whittaker functional** and **Whittaker model** is defined as in (24.9.5.2).  $\lrcorner$

**Prop. (24.3.10.16) [Existence and Uniqueness of Whittaker Models].** Let  $\mathcal{G}$  be the representation of  $GL(2, k)$  induced from the character  $\psi_N(n(z)) = \psi(z)$  on  $N(k)$ , then it is multiplicity-free, and every irreducible representation of dimension  $> 1$  occurs in it. Notice that this is just the existence and uniqueness of Whittaker models.  $\lrcorner$

*Proof:* To show multiplicity free, it suffices to show  $\text{End}_{GL(2, k)}(\mathcal{G})$  is commutative, by Shur's lemma (24.7.1.8). Then we use ??, which says this ring is isomorphic to the ring of functions  $\Delta$  on  $G$  that

$$\Delta(n_2 g n_1) = \psi_N(n_2) \Delta(g) \psi_N(n_1).$$

where the multiplication is convolution ??.

Notice it follows from the Bruhat Decomposition (10.4.4.28) that the double coset  $N(k) \backslash GL(2, k) / N(k)$  is uniquely represented by matrices with exactly two non-zero entries. Then it is clear a diagonal coset can support a function  $\Delta$  iff it is a scalar multiple of  $I$ . In other words, the representatives are

$$\begin{bmatrix} a & \\ & a \end{bmatrix}, \quad \begin{bmatrix} & b \\ c & \end{bmatrix}.$$

Consider the involution of  $G$  given by

$$\iota(g) = w_1 g^t w_1^{-1},$$

then it is an anti-involution of  $G$  and it induces isomorphism on  $N(k)$ , so it induces an anti-involution of order two on the ring of functions  $\Delta$  by  $\iota(\Delta)(g) = \Delta(\iota(g))$ . Notice this is an anti-involution because of (12.10.2.5) and the fact a finite group is unimodular. But this anti-involution fixes the representatives as above, so it is in fact identity on these  $\Delta$ , which proves the convolution is commutative.

For the last assertion, just notice the dimension of  $\mathcal{G}$  is  $(q-1)(q^2-1)$ , and the sum dimensions of irreducible representations of dimension  $> 1$  is just  $(q-1)(q^2-1)$  by (24.3.10.14).  $\square$

**Cor. (24.3.10.17).** Frobenius reciprocity (24.7.4.5) implies that the space of Whittaker functionals is of dimension 1 for any irreducible representation of dimension  $> 1$ .  $\lrcorner$

## 11 $SL(n)$

**Def. (24.3.11.1) [Drinfeld Curves].** The **Drinfeld curve** is defined to be the affine plane curve  $\text{Dri} / \mathbb{A}_q^2$

$$\text{Dri} : x^q y - x y^q = 1$$

with completion  $\overline{\text{Dri}} = \text{Dri} \cup \{\infty\} \subset \mathbb{P}_q^2$ .  $\lrcorner$

**Prop. (24.3.11.2).** There is a natural  $\mu_{q+1} \times SL(2, \mathbb{F}_q)$ -action on  $\text{Dri}$ , which induces a map on  $H_{\text{ét},c}^*(\text{Dri}, \mathbb{Q}_\ell)$ . Then we get the Deligne-Lustig induction

$$R_{T'}^G : \mu_{q+1}^* \rightarrow K_0(\text{Rep}^{\text{fd}}(SL(2; \mathbb{F}_q))).$$

┘

*Proof:*

□

| Reps \ Classes               | $\begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$ | $\begin{bmatrix} -1 & \\ & -1 \end{bmatrix}$ | $\begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$ | $\begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$ | $\begin{bmatrix} 1 & -1 \\ & 1 \end{bmatrix}$ | $\begin{bmatrix} -1 & 1 \\ & -1 \end{bmatrix}$ | $\begin{bmatrix} -1 & -1 \\ & -1 \end{bmatrix}$ |
|------------------------------|--------------------------------------------|----------------------------------------------|---------------------------------------------|----------------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| size                         | 1                                          | 1                                            | 6                                           | 4                                            | 4                                             | 4                                              | 4                                               |
| 1                            | 1                                          | 1                                            | 1                                           | 1                                            | 1                                             | 1                                              | 1                                               |
| $\chi$                       | 1                                          | 1                                            | 1                                           | $\omega$                                     | $\omega^2$                                    | $\omega^2$                                     | $\omega$                                        |
| $\chi^2$                     | 1                                          | 1                                            | 1                                           | $\omega^2$                                   | $\omega$                                      | $\omega$                                       | $\omega^2$                                      |
| $\rho$ rational quaternionic | 2                                          | -2                                           | 0                                           | -1                                           | -1                                            | 1                                              | 1                                               |
| $\chi\rho$                   | 2                                          | -2                                           | 0                                           | $-\omega$                                    | $-\omega^2$                                   | $\omega^2$                                     | $\omega$                                        |
| $\chi^2\rho$                 | 2                                          | -2                                           | 0                                           | $-\omega^2$                                  | $-\omega$                                     | $\omega$                                       | $\omega^2$                                      |
| $\pi$                        | 3                                          | 3                                            | -1                                          | 0                                            | 0                                             | 0                                              | 0                                               |

Figure (24.3.11.1): Character Table of  $SL(2, \mathbb{F}_3)$

## 12 $GL(n)$

References are [Representations of the Finite Classical Groups, Zelevinsky, 1981].

**Thm. (24.3.12.1) [Irreducible Characters of  $GL(n; \mathbb{F}_q)$ , Green].**

┘

## 13 $Sp(2n)$

**Thm. (24.3.13.1) [Irreducible Characters of  $Sp(2n, \mathbb{F}_q)$ , Green].**

┘

## 24.4 Representations of Semisimple Lie Algebras

References are [?Eti21] and [Car05].

In this section representations of split semisimple Lie algebras are studied.

**Notation (24.4.0.1).**

- Let  $k \in \text{Field}$ .
- Let  $(\mathfrak{g}, \mathfrak{h}, \Phi, \Lambda, \mathfrak{n}^+, \mathfrak{n}^-)$  be a split semisimple Lie algebra over  $k$ .
- Use notations for split semisimple Lie algebras defined in [Notations for Split Semisimple Lie Algebras](#).

⌋

### 1 Semisimple Representations

**Lemma (24.4.1.1).** Every 1-dimensional representation of  $\mathfrak{g}$  is trivial. ⌋

*Proof:* Such a representation vanishes at  $[\mathfrak{g}, \mathfrak{g}]$ , which equals  $\mathfrak{g}$  by (3.7.2.4). □

**Prop. (24.4.1.2) [Semisimple Lie Algebras and Representations, Weyl].** For an arbitrary  $L \in \text{LieAlg}_k$ ,

- If the adjoint representation  $L \rightarrow \mathfrak{gl}_{\mathfrak{g}}$  is semisimple, then  $L$  is semisimple.
- If  $L$  is semisimple and  $k$  has characteristic 0, then  $\text{Rep}(L)$  is semisimple.

⌋

*Proof:* If the adjoint representation of  $L$  is semisimple, then every ideal of  $L$  has a complement, thus if  $\mathfrak{g}$  is not semisimple, it has a dimension 1 quotient. But notice the Lie algebra  $k$  of dimension

1 has non-semisimple representations, for example  $c \mapsto \begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix}$ , contradiction.

Semisimplicity of a Lie algebra is invariant under base change, also does simplicity of the category of representations (24.4.1.4), so we can assume that  $k$  is alg.closed. Now we need to show that any proper submodule  $W$  of a  $L$ -module  $V$  has a complement.

Assume first that  $\dim V/W = 1$  and  $W$  is a simple  $L$ -module. This implies  $L$  acts trivially on  $V/W$  by (24.4.1.1). Let  $c_V$  be the Casimir element of  $V$  (3.7.9.11), then  $c_V$  is also trivial on  $V/W$ . And  $c_V$  acts as a nonzero scalar on  $W$  as  $W$  is simple, by (3.7.9.11). Then the kernel of  $c_V$  is of dimension 1, and is a  $\mathfrak{g}$ -complement of  $W$  in  $V$ .

Next if  $\dim V/W = 1$  but  $W$  is not simple  $\mathfrak{g}$ -module, then there is a submodule  $W' \subset W$ . By induction, the  $L$ -submodule  $W/W'$  has a complement  $V'/W'$  in  $V/W'$ . Then  $V'/W'$  has dimension 1, thus by induction,  $V' = W' \oplus L$  for some 1-dimensional  $\mathfrak{g}$ -module. Then  $L$  is complementary to  $W$  in  $V$ .

Finally for the general case, let  $L$  acts on  $\text{Hom}_k(V, W)$ , consider the subspaces  $V_1, W_1$  of  $\text{Hom}_k(V, W)$ , where  $V_1$  is the subspace of maps that restriction to  $W$  is a constant multiple of identity, and  $W_1$  is the subspaces of  $W$  consisting of maps vanishing on  $W$ . They are both  $L$ -modules and  $\dim V_1/W_1 = 1$ . Then the above case shows  $V_1 = W_1 \oplus L$  for some 1-dimensional  $L$ -module  $L$ . Because  $L$  acts trivially on  $L$  (24.4.1.1), this means  $L = \mathbb{F}f$  consists of  $L$ -homomorphisms. But  $f|_W$  is non-zero constant, so the kernel of  $f$  is a complement of  $W$  in  $V$ . □

**Cor. (24.4.1.3).** Let  $(V, \rho) \in \text{Rep}(\mathfrak{g})$  and  $f : \mathfrak{g} \rightarrow V$  a linear map that

$$f([x, y]) = \rho(x)f(y) - \rho(y)f(x),$$

then there exists a  $v_0 \in V$  that  $f(x) = \rho(x)v_0$ .  $\lrcorner$

*Proof:* The condition on  $f$  is equivalent to  $(f, \rho) : \mathfrak{g} \rightarrow \mathfrak{af}(V)$  (3.7.1.11) is a homomorphism of Lie algebras. And this induces a representation  $\rho'$  of  $\mathfrak{g}$  on  $V' = V \oplus k$  that  $\rho'(x)(V') \subset V$  for all  $x \in \mathfrak{g}$ . Because  $\mathfrak{g}$  is semisimple, there is a line  $L \subset V'$  that  $V' = V \oplus L$  and  $\mathfrak{g}$  acts trivially on  $L$  (24.4.1.1).

In other words, there is a vector  $(-v_0, 1)$  that  $\rho'(x)(-v_0, 1) = 0$  for all  $x \in \mathfrak{g}$ . So  $f(x) = \rho(x)(v_0)$  for all  $x$ . So we are done.  $\square$

**Prop. (24.4.1.4) [Semisimplicity and Extension].** For an arbitrary  $L \in \text{LieAlg}_k$ , if  $\text{Rep}(L_K)$  is semisimple for some extension field  $K/k$ , then also is  $\text{Rep}(L)$ .  $\lrcorner$

*Proof:* This is because for any  $(V, \rho) \in \text{Rep}_k(L)$ ,  $K \otimes_k \text{End}(\rho) \cong \text{End}(\rho_K)$ , because this is true for  $\text{End}(V)$ , and being  $L$ -equivariant is a linear condition. Then we can use (3.6.2.25) and (3.6.2.26).  $\square$

**Prop. (24.4.1.5) [Semisimple Representations].** For any arbitrary  $L \in \text{Lie}_k$ , then the following conditions on a representation  $L \rightarrow \mathfrak{gl}_V$  are equivalent:

- $\rho$  is semisimple.
- $\rho(L)$  is reductive and its center consists of semisimple endomorphisms.
- $\rho(\mathfrak{r})$  consists of semisimple endomorphisms.
- The restriction of  $\rho$  to  $\mathfrak{r}$  is semisimple.

$\lrcorner$

*Proof:* 1  $\rightarrow$  2: If  $\rho$  is semisimple, then  $\rho(L)$  is reductive by (3.7.4.4). Its center consists of semisimple endomorphisms by [Mil13]P60<sup>?</sup>.

2  $\rightarrow$  3: This is because if  $\rho(L)$  is reductive, then its center equals its radical and contains  $\rho(\mathfrak{r})$ , so  $\rho(\mathfrak{r})$  consists of semisimple endomorphisms.

3  $\rightarrow$  4: [Mil13]P60<sup>?</sup>.

4  $\rightarrow$  1: [Mil13]P60<sup>?</sup>.  $\square$

**Cor. (24.4.1.6).** Let  $\mathfrak{g}$  be any Lie algebra, and  $\rho, \rho' \in \text{Rep}(\mathfrak{g})$ . If  $\rho$  and  $\rho'$  are semisimple, then so are  $\rho \otimes \rho'$  and  $\rho^\vee$ .

In particular, the category  $\text{Rep}(\mathfrak{g})$  of semisimple representations of a Lie algebra  $\mathfrak{g}$  form a Tannakian category, thus there is an algebraic group  $G$  that  $\text{Rep}^{ss}(\mathfrak{g}) = \text{Rep}(G)$ .  $\lrcorner$

*Proof:* Use the third criterion, for any  $x \in \mathfrak{r}$ , as  $\rho(x), \rho'(x)$  are semisimple, so is  $\rho(x) \otimes \rho'(x)$ , so  $\rho \otimes \rho'$  is also semisimple by (24.4.1.5).  $\square$

### Representations of $\mathfrak{sl}_2(\mathbb{C})$

**Def. (24.4.1.7) [Primitive Elements].** For  $V \in \text{Rep}(\mathfrak{sl}_2)$ , an element  $v \in V$  is called **primitive of weight  $\lambda$**  if it is non-zero and  $Xv = 0, Hv = \lambda v$ .  $\lrcorner$

**Prop. (24.4.1.8).** Every non-zero f.d.  $\mathfrak{sl}_2$ -module contains a primitive element.  $\lrcorner$

*Proof:* an element  $e$  is primitive iff the line generated by  $e$  is stable under the action of  $\{X, H\}$ : if  $Xe = \lambda e$  and  $He = \mu e$ , then using the  $[H, X] = 2X$ , we see that  $2\lambda = 0$ , thus  $\lambda = 0$ , and  $e$  is primitive. So each f.d.  $\mathfrak{sl}_2$ -module contains a primitive element, by Lie's theorem (3.7.1.24).  $\square$

**Prop. (24.4.1.9) [Submodule Generated by Primitive Element].** Let  $V \in \text{Rep}(\mathfrak{sl}_2)$  and  $e \in V$  a primitive element of weight  $\lambda$ . Let  $e_n = Y^n e / n!$ , and  $e_{-1} = 0$ , then we have

$$He_n = (\lambda - 2n)e_n, \quad Ye_n = (n+1)e_{n+1}, \quad Xe_n = (\lambda - n + 1)e_{n-1}.$$

┘

*Proof:* By induction on  $n$ ,

$$HY^n e = ([H, Y] + YH)Y^{n-1}e = (\lambda - 2(n-1) - 2)Y^n e = (\lambda - 2n)e.$$

$Ye_n = (n+1)e_{n+1}$  is obvious.

And

$$\begin{aligned} nXe_n &= XYe_{n-1} = [X, Y]e_{n-1} + YXe_{n-1} \\ &= He_{n-1} + (\lambda - n + 2)Ye_{n-2} \\ &= (\lambda - 2n + 2 + (\lambda - n + 2)(n-1))e_{n-1} \\ &= n(\lambda - n + 1)e_{n-1} \end{aligned}$$

□

**Cor. (24.4.1.10).** Situation as in (24.4.1.9). Only two cases arise: either

- The elements  $\{e_n\}$  are linearly independent.
- The elements  $e_0, e_1, \dots, e_m$  are linearly independent, and  $e_{m+1} = e_{m+2} = \dots = 0$ , and weight  $\lambda$  of  $e$  equals  $m$ .

And if  $V$  is f.d., then case 1 cannot happen, and the subspace  $W$  generated by  $e_0, \dots, e_m$  is a  $\mathfrak{g}$ -module and it is irreducible. ┘

*Proof:* Because each  $e_i$  has different eigenvalue under action of  $H$ , thus if they are all nonzero, then they are linearly independent. If  $e_0, e_1, \dots, e_m$  are linearly independent, and  $e_{m+1} = e_{m+2} = \dots = 0$ , then by the proposition,

$$Xe_{m+1} = (\lambda - m)e_m$$

and  $e_{m+1} = 0$  with  $e_m \neq 0$ , thus  $\lambda = m$ . Now the formulas in (24.4.1.9) shows that  $W$  is a  $\mathfrak{g}$ -module. And if  $W' \subset W$  is a subspace invariant under  $\mathfrak{g}$ , then it contains some eigenvalues  $e_k$  of  $H$ , and then the formulas in (24.4.1.9) shows it contains all  $e_0, e_1, \dots, e_m$ , thus  $W' = W$ . Thus  $W$  is irreducible. □

**Prop. (24.4.1.11) [Irreducible Representations of  $\mathfrak{sl}_2(\mathbb{C})$ ].** Let  $W_m = \{e_0, \dots, e_m\}$  be a  $m+1$ -dimensional vector space and  $\mathfrak{sl}_2$  acts on  $W_m$  by

$$He_n = (m - 2n)e_n, \quad Ye_n = (n+1)e_{n+1}, \quad Xe_n = (m - n + 1)e_{n-1}.$$

Then  $W_m$  is a f.d. irreducible representation of  $\mathfrak{sl}_2$ , and any f.d representation of  $\mathfrak{sl}_2$  of dimension  $m+1$  is isomorphic to one of  $W_m$ . ┘

*Proof:* The first assertion follows from (24.4.1.10) and the fact  $e_0$  is a primitive element. For the second assertion, notice any f.d. representation  $W$  of  $\mathfrak{g}$  contains a primitive element (24.4.1.8) thus by (24.4.1.10) generates an irreducible  $\mathfrak{sl}_2$ -submodule  $W_n$ , thus this submodule equals  $W$ , and  $n+1 = m+1$ , thus  $n = m$ . □



**Remark (24.4.1.12).** Notice this is special case of (24.4.2.16).  $\lrcorner$

**Cor. (24.4.1.13).**  $W_0$  is the just trivial action of  $\mathfrak{sl}_2$ ,  $W_1$  is isomorphic to the natural action of  $\mathfrak{sl}_2$  on  $\mathbb{C}^2$ , and  $W_2$  is isomorphic to the adjoint action of  $\mathfrak{sl}_2$  on itself.  $\lrcorner$

*Proof:* In fact,  $W_1$  can be identified with the vector space  $\mathbb{C}\{x, y\}$  where

$$Hx = x, Hy = -y, Yx = y, Yy = 0, Xy = x, Xx = 0.$$

Then the  $m$ -th symmetric tensor of  $W_1$  is isomorphic to the vector space of polynomials in  $x, y$  of degree  $m$ , and by (3.7.9.2),

$$\begin{aligned} H(C_m^k y^k x^{m-k}) &= (m - 2k)(C_m^k y^k x^{m-k}), \\ Y(C_m^k y^k x^{m-k}) &= (k + 1)(C_m^{k+1} y^{k+1} x^{m-k-1}), \\ X(C_m^k y^k x^{m-k}) &= (m - k + 1)(C_m^{k-1} y^{k-1} x^{m-k+1}). \end{aligned}$$

So it is isomorphic to  $W_{m+1}$ .  $\square$

**Cor. (24.4.1.14) [Representations of  $\mathfrak{sl}_2(\mathbb{K})$ ].** Let  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$  and  $V \in \text{Rep}^{\text{fd}}(\mathfrak{sl}_2(\mathbb{K}))$ , then

- $V$  is isomorphic to a direct sum of  $W_m$ s,
- The endomorphism on  $V$  induced by  $H$  is diagonalizable and with integral eigenvalues. if  $\pm n$  are eigenvalues of  $H$ , then so are  $n - 2, n - 4, \dots, 4 - n, 2 - n$ .
- For any  $n \geq 0$ , the linear maps

$$Y^n : V^n \rightarrow V^{-n}, X^n : V^{-n} \rightarrow V^n$$

are isomorphisms. In particular,  $V^{-n}$  and  $V^n$  have the same dimensions.  $\lrcorner$

*Proof:* by Weyl's theorem (24.4.1.2), any representation of  $\mathfrak{sl}_2(\mathbb{K})$  is isomorphic to a direct sum of irreducible representations, and the irreducible representations of  $\mathfrak{sl}_2(\mathbb{C})$  clearly comes from real representations of  $\mathfrak{sl}_2(\mathbb{R})$ , thus irreducible representations of  $\mathfrak{sl}_2(\mathbb{R})$  must be of the same form (otherwise tensor with  $\mathbb{C}$  and decompose, and use conjugation).

Because we can assume  $V$  is one of  $W_m$ , so the other assertions are clear.  $\square$

## 2 Verma Modules

**Def. (24.4.2.1) [Weights].** Let  $V \in \text{Rep}(\mathfrak{g})$ . If  $\lambda \in \mathfrak{h}^\vee$ , a vector  $v \in V$  is said to have weight  $\lambda$  iff  $hv = \lambda(h)v$  for any  $h \in \mathfrak{h}$ . The space of vectors in  $V$  of weight  $\lambda$  is denoted by  $V[\lambda]$ . An **integral weight** is a weight  $\lambda$  of the form  $\lambda = \sum n_i \alpha_i^\vee$ , where  $\alpha_i \in \Phi$ , equivalently its values on  $h_i$  are all integers.

$V$  is said to have a **weight decomposition** if  $V = \bigoplus_{\lambda \in \mathfrak{h}^*} V[\lambda]$ .  $\lrcorner$

**Prop. (24.4.2.2).** Any f.d.  $\mathfrak{g}$ -module  $V$  has a weight decomposition, and the weights are all integral.  $\lrcorner$

*Proof:* For any  $i$ ,  $V$  is a  $\mathfrak{s}_i = \{e_i, h_i, f_i\}$ -module (3.7.3.20), so by the representation theory of  $\mathfrak{sl}_2$ ,  $h_i$  acts semisimply on  $V$ , and the eigenvalues are integers. Thus  $\mathfrak{h}$  acts semisimply on  $V$ , thus there is a weight decomposition, and the weights are all integral.  $\square$

**Def. (24.4.2.3) [Highest Weight Representations].** A vector  $v \in V$  is called a **highest weight vector** if it is a weight vector and  $\mathfrak{n}_+ v = 0$ , or equivalently  $e_i v = 0$  for any  $e_i$ . A **highest weight representation** with highest weight  $\lambda$  is a representation that is generated by a highest weight vector of weight  $\lambda$ .  $\lrcorner$

**Prop. (24.4.2.4).** Any  $V \in \mathcal{R}\text{ep}^{\text{fd}}(\mathfrak{g})$  contains a highest weight vector, thus any irreducible f.d.  $\mathfrak{g}$ -representation is a highest weight representation.  $\lrcorner$

*Proof:* Choose a weight  $\lambda$  of  $V$  s.t.  $\langle \lambda, \rho^\vee \rangle$  is maximal, then  $e_i v$  has weight  $\lambda + \alpha_i$ , thus  $e_i v = 0$  for any  $i$ .  $\square$

### Verma Modules

**Def. (24.4.2.5) [Highest Weight Representations].**  $V \in \mathcal{R}\text{ep}(\mathfrak{g})$  is called a **highest weight representation** with highest weight  $\lambda \in \mathfrak{h}^*$  if it is generated by such a nonzero vector.  $\lrcorner$

**Def. (24.4.2.6) [Verma Modules].** Let  $\lambda \in \mathfrak{h}^*$ ,  $I_\lambda \subset U(\mathfrak{g})$  be the left ideal generated by the elements  $\{h - \lambda(h)\}_{h \in \mathfrak{h}}$  and  $e_i$ , then the **Verma module**  $M_\lambda = U(\mathfrak{g})/I_\lambda$  has a weight decomposition, and it is a highest weight representation with highest vector  $v_\lambda = \bar{1}$  with weight  $\lambda$ .  $\lrcorner$

**Prop. (24.4.2.7) [ $M_\lambda$  is Noetherian].**  $M_\lambda$  satisfies ascending chain condition on f.d. submodules.  $\lrcorner$

*Proof:* This is because  $U(\mathfrak{g})$  is left Noetherian (3.7.8.17) and  $M_\lambda = U(\mathfrak{g})/I_\lambda$ .  $\square$

**Prop. (24.4.2.8).** The map  $\varphi : U(\mathfrak{n}_-) \rightarrow M_\lambda : \varphi(x) = x v_\lambda$  is an isomorphism of left  $U(\mathfrak{n}_-)$ -modules.  $\lrcorner$

*Proof:* This follows from PBW theorem.  $\square$

**Cor. (24.4.2.9).**  $M_\lambda$  has a weight decomposition with  $P(M_\lambda) = \lambda - Q^+(\Phi)$ , and weight spaces of  $M_\lambda$  are all of f.d..  $\lrcorner$

**Prop. (24.4.2.10).** If  $V$  is a  $\mathfrak{g}$ -module and  $v \in V$  is a highest weight vector, then there is a unique homomorphism  $M_\lambda \rightarrow V$  of  $\mathfrak{g}$ -modules that maps  $v_\lambda$  to  $v$ .  $\lrcorner$

**Cor. (24.4.2.11).** Every highest weight representation has a decomposition into f.d. weight spaces. And they have a unique highest weight vector up to a scalar.  $\lrcorner$

**Prop. (24.4.2.12) [Quotient of Verma Modules].**  $M_\lambda$  has a unique irreducible quotient  $L_\lambda$ . In particular,  $L_\lambda$  is a quotient of every highest weight representation of  $\mathfrak{g}$  with weight  $\lambda$ , by (24.4.2.10).  $\lrcorner$

*Proof:* It is easy to show that any submodule of  $M_\lambda$  is a direct sum of its weight spaces. It cannot have weight  $\lambda$ , otherwise it generates  $M_\lambda$  by (24.4.2.11). Then the sum  $J_\lambda$  of all proper submodules of  $M_\lambda$  also cannot have weight  $\lambda$ , thus also proper. Then  $L_\lambda = M_\lambda/J_\lambda$  is the unique irreducible quotient of  $M_\lambda$ .  $\square$

**Cor. (24.4.2.13) [Classification of Irreducible Highest Weight Representations].** Any irreducible highest weight representation of  $\mathfrak{g}$  is of the form  $L_\lambda$  where  $\lambda \in \mathfrak{h}^*$ .  $\lrcorner$

**Prop. (24.4.2.14) [Composition Factors of Verma Modules].** The Verma module  $M_\lambda$  has a finite composition series  $M_\lambda = N_0 \supset N_1 \supset \dots \supset N_r = 0$  that each  $N_i/N_{i+1}$  is irreducible and isomorphic to  $L_{\omega(\lambda+\rho)-\rho}$  for some  $\omega \in W$ .  $\lrcorner$

*Proof:* We can find a (not necessarily finite) series  $N_i$  of submodules of  $M_\lambda$  that the subquotients are irreducible as  $M_\lambda$  satisfies ascending chain condition on submodules (24.4.2.7). As each submodule of  $M_\lambda$  has a weight decomposition, so does the subquotients  $N_i/N_{i+1}$ , and we can choose one of its maximal weight  $\mu$ , which satisfies  $\mu < \lambda$ . Then by the property of  $M_\mu$ ,  $N_i/N_{i+1}$  is isomorphic to the unique irreducible quotient  $L_\mu$  (24.4.2.12).

Now if  $L_\mu$  is a subquotient of  $M_\lambda$ , by considering the central character, we get  $\chi_\lambda = \chi_\mu$ . By (24.4.2.22), this implies that  $\mu = \omega(\lambda + \rho) - \rho$  for some  $\omega \in W$ . Finally, the composition series is finite because  $W$  is finite and each weight space of  $W$  is f.d..  $\square$

**Lemma (24.4.2.15).** Let  $V \in \text{Rep}(\mathfrak{g})$  with weight decomposition into f.d. weight spaces. If  $V$  is a sum of f.d.  $\mathfrak{s}_i$ -modules for each  $i$ , then for each  $\lambda \in P$  and  $w \in W$ ,  $\dim V[\lambda] = \dim V[w\lambda]$ .  $\square$

*Proof:* As  $W$  is generated by  $s_i$ , it suffices to prove for  $w = s_i$ ,  $\dim V[\lambda] = \dim V[s_i\lambda]$ .

If  $(\lambda, \alpha_i^\vee) = m \geq 0$  for some  $i$ , consider the operator  $f_i^m : V[\lambda] \rightarrow V[s_i\lambda]$  is an isomorphism by hypothesis and the representation theory of  $\mathfrak{sl}_2$ . Similarly, if  $(\lambda, \alpha_i^\vee) = -m \leq 0$ , then  $e_i^m : V[\lambda] \rightarrow V[s_i\lambda]$  is an isomorphism.  $\square$

**Prop. (24.4.2.16) [Classification of F.D. Irreducible Representations].** F.d. irreducible representations of  $\mathfrak{g}$  are classified by their highest dominant integral weights  $\lambda \in P^+(\Phi)$  via the bijection  $\lambda \mapsto L_\lambda$ . Moreover, for any  $\mu \in P(\Phi)$  and  $w \in W$ ,  $\dim L_\lambda[\mu] = \dim L_\lambda[w\mu]$ .  $\square$

*Proof:* Firstly if  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ , then  $\lambda \in P^+(\Phi)$  by (24.4.2.2). Now if  $\lambda \in P^+$ , then in  $L_\lambda$ ,  $f_i^{\lambda(h_i)+1} v_\lambda = 0$  for any  $i$ , because by (24.4.1.9),  $e_i f_i^{\lambda(h_i)+1} v_\lambda = 0$ , and  $e_j f_i^{\lambda(h_i)+1} v_\lambda = 0$  for  $j \neq i$  as  $v_\lambda$  is a highest weight vector. So  $f_i^{\lambda(h_i)+1} v_\lambda$  generates a proper submodule of  $L_\lambda$ , which must be 0 as it cannot contain  $v_\lambda$ , and  $L_\lambda$  is irreducible.

Thus  $v_\lambda$  generate the  $\mathfrak{s}_i$ -module of highest weight  $\lambda(h_i)$ , and any element of  $\mathfrak{g}$  generate a f.d.  $\mathfrak{s}_i$ -module, so  $V$  is a sum of f.d.  $\mathfrak{s}_i$ -modules for each  $i$ , thus by lemma (24.4.2.15),  $\dim L_\lambda[\mu] = \dim L_\lambda[w\mu]$ . To show  $L_\lambda$  is of f.d., first notice  $P(L_\lambda) \cap P^+$  is finite, and  $WP^+(\Phi) = P(\Phi)$  (3.9.5.16), and the fact  $P(L_\lambda)$  is  $W$ -invariant.  $\square$

**Prop. (24.4.2.17).**  $L_\lambda^* = L_{-w_0(\lambda)}$ .  $\square$

*Proof:* It suffices to show the lowest weight of  $L_\lambda$  is  $w_0\lambda$ :  $w_0\lambda$  is a weight by (24.4.2.16), and if  $\lambda' < w_0\lambda$ , then  $w_0\lambda' > \lambda$ .  $\square$

**Lemma (24.4.2.18).** The representation  $L_n$  of  $\mathfrak{sl}_2$  is of real type if  $n$  is even and of quaternion type if  $n$  is odd.  $\square$

*Proof:*  $L_n = \text{Sym}^n L_1$ , thus we can take the bilinear form  $\text{Sym}^n B$ , where  $B$  is the alternating form on  $L_1 \cong \mathbb{C}^2$ , which is symmetric if  $n$  is even and alternating if  $n$  is odd.  $\square$

**Prop. (24.4.2.19) [Types of Irreducible Representations].** For  $\lambda \in P^+(\Phi)$  s.t.  $\lambda = -w_0\lambda$  (so  $L_\lambda$  is self-dual by (24.4.2.17)), it is of real type if  $n = (2\rho^\vee, \lambda)$  is even and of quaternion type if  $(2\rho^\vee, \lambda)$  is odd.  $\square$

*Proof:* The number  $n$  is the eigenvalue of  $h$  on  $v_\lambda$ , where  $\{h, e, f\}$  is the principal  $\mathfrak{sl}_2$ -subalgebra, and the other eigenvalues are all smaller. Then  $L_\lambda \cong L_n \oplus \bigoplus_{-i < n} L_{m_i}$ , so the invariant form on  $L_\lambda$  restricts to a non-zero invariant form on  $L_n$ , so the assertion follows from (24.4.2.18).  $\square$

**Prop. (24.4.2.20) [Residual Finiteness].** The homomorphism  $U(\mathfrak{g}) \rightarrow \prod_{\lambda \in P^+} \text{End}(L_\lambda)$  is injective.  $\square$

*Proof:* ?

If  $x \in U(\mathfrak{g})$  is in the kernel, then  $x$  acts trivially on  $\Gamma(G) = \bigoplus_{\lambda \in P_+(\Phi)} L_\lambda \otimes L_\lambda^*$ , where  $x$  acts on  $\Gamma(G)$  by differential operators and acts on  $L_\lambda$  only. Then  $x$  acts trivially on  $\Gamma(G)$ , which means  $x = 0$ .  $\square$

### Central Characters of Verma Modules

**Prop. (24.4.2.21) [Central Character  $\chi_\lambda$ ].** As  $M_{\lambda+\rho}$  is generated by a single highest weight vector,  $Z(U(\mathfrak{g}))$  acts on it by a character  $\chi_\lambda \in \text{Spec } Z(U(\mathfrak{g}))$ . How to determine  $\chi_\lambda$ ?  $\lrcorner$

*Proof:*  $\square$

**Prop. (24.4.2.22).** For  $\lambda, \mu \in \mathfrak{h}^*$ ,  $\text{ch}(M_\lambda) = \text{ch}(M_\mu)$  iff  $\mu = w_\bullet \lambda$  for some  $w \in W$ .  $\lrcorner$

*Proof:* Cf. [Carter]P235. ?  $\square$

**Prop. (24.4.2.23) [Action of Casimir Elements].** Let  $c \in Z(U(\mathfrak{g}))$  be the Casimir element, then  $c$  acts on  $M_{\lambda-\rho}$  as scalar multiplication by

$$\chi_\lambda(c) = \langle \lambda, \lambda \rangle - \langle \rho, \rho \rangle.$$

$\lrcorner$

*Proof:* Cf. [Carter]P239.  $\square$

### Fundamental Representations

**Prop. (24.4.2.24) [Fundamental Representations].** The following are equivalent for  $\omega \in P^+$ :

- $\omega$  is minuscule (3.9.5.35).
- All weights of the representation  $L_\omega$  belongs to the orbit  $W\omega$ .
- The restriction of  $L_\omega$  to  $\mathfrak{s}_i$  is a direct sum of 1-dimensional and 2-dimensional subrepresentations.

Such a representation  $L_\omega$  is called a **fundamental representation** of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* 1  $\rightarrow$  2: By (24.4.2.16), for any weight  $\mu$  of  $L_\omega$ , there is a  $w \in W$  that  $w\mu$  is dominant and  $\omega - w\mu \in Q^+$ . Thus  $w\mu = \omega$  by (3.9.5.37), and  $\mu \in W\omega$ .

2  $\rightarrow$  1: If  $\omega$  is not minuscule, then there is some positive  $\alpha$  that  $(\omega, \alpha^\vee) > 1$ , so  $2(\omega, \alpha) < (\alpha, \alpha)$ , and  $\omega - \alpha$  is a weight of  $L_\omega$  (the weight of  $f_\alpha v_\omega$ , and it is not conjugate to  $\omega$ , as  $(\omega - \alpha, \omega - \alpha) = (\omega, \omega) - 2(\alpha, \omega) + (\alpha, \alpha) < (\omega, \omega)$ ).

2  $\iff$  3: If  $\omega$  is minuscule and  $v \in L_\omega$  be a highest weight vector for  $\mathfrak{s}_i$ , then  $h_{\alpha_i} v = \omega(h_{\alpha_i}) = (\omega, \alpha_i^\vee) = 0$  or  $v$  (3.7.3.21). Thus we get 4 by the representation theory of  $\mathfrak{sl}_2$ . The reverse argument is also true.  $\square$

**Cor. (24.4.2.25).** The character for a fundamental representation  $L_\omega$  is

$$\text{ch}(L_\omega) = \sum_{w \in W} e^{w(\omega)}.$$

$\lrcorner$

**Prop. (24.4.2.26).** Let  $\omega$  be a minuscule weight, then for any dominant integral weight  $\lambda$ ,

$$L_\omega \otimes L_\lambda \cong \bigoplus_{w \in W, \lambda + w\omega \in P^+} L_{\lambda + w\omega}.$$

┘

*Proof:* By Weyl's character formula and (24.4.2.25),

$$\text{ch}(L_\omega \otimes L_\lambda) = \frac{\sum_{\mu \in W\omega} \sum_{w \in W} \varepsilon(w) e^{w(\lambda+\rho)+\mu}}{\Delta} = \frac{\sum_{\gamma \in W\omega} \sum_{w \in W} \varepsilon(w) e^{w(\lambda+\rho+\gamma)}}{\Delta}$$

If  $\lambda + \gamma \notin P^+$ , then  $(\lambda + \gamma, \alpha_i^\vee) < 0$  for some  $i$ . But  $(\gamma, \alpha_i^\vee) \geq -1$  as  $\omega$  is minuscule, so  $(\lambda + \gamma, \alpha_i^\vee) = -1$ , and  $(\lambda + \gamma + \rho, \alpha_i^\vee) = 0$ . Then for such  $\gamma$ , in the summand the  $w$  term cancels with  $ws_i$  term. Thus the summand equals the character for  $\sum_{w \in W} e^{w\omega}$ . Then we finish by (24.4.3.3).  $\square$

**Prop. (24.4.2.27)[Fundamental Representations for Simple Lie Algebras].** By (3.9.6.12),

- for  $B_n$ , the fundamental representation corresponding to the minuscule weights  $\omega_n$  is the **Spin representation** of dimension  $2^n$ , denoted by  $S$ .
- for  $C_n$ , the fundamental representation corresponding to the minuscule weights  $\omega_1$  is the standard representation of  $\mathfrak{sp}_{2n}$ .
- for  $D_n$ , the fundamental representation corresponding to the minuscule weights  $\omega_1, \omega_{n-1}, \omega_n$  are the standard representation and two Spin representations  $S^+, S^-$ .
- for  $E_7$ , the unique fundamental representation has dimension 56<sup>?</sup>.
- for  $E_6$ , the two fundamental representations are dual, and have dimensions 27<sup>?</sup>.

┘

*Proof:* <sup>?</sup>

$\square$

**Prop. (24.4.2.28)[Bott Periodicity for Spin Representations].** Let  $\mathfrak{g} = \mathfrak{so}_m$ , then the behavior of the spin representation of  $\mathfrak{g}$  is

- $S$  is of real type if  $m \equiv 1, 7 \pmod{8}$ .
- $S$  is of quaternionic type if  $m \equiv 3, 5 \pmod{8}$ .
- $S_+, S_-$  are of real type if  $m \equiv 0 \pmod{8}$ .
- $S_+^* \cong S_-$  are of complex type if  $m \equiv 2, 6 \pmod{8}$ .
- $S_+, S_-$  are of quaternionic type if  $m \equiv 4 \pmod{8}$ .

┘

*Proof:* We use (24.4.2.19). If  $\mathfrak{g} = \mathfrak{so}_{2n}$ , then  $\rho^\vee = \rho = \sum \omega_i = (n-1, n-2, \dots, 1, 0)$ . So  $(2\rho^\vee, \omega_{n-1}) = (2\rho^\vee, \omega_n) = \frac{n(n-1)}{2}$ .

If  $\mathfrak{g} = \mathfrak{so}_{2n+1}$ , then it can be verified that  $\rho^\vee = (n, n-1, \dots, 0)$ , so  $(2\rho^\vee, \omega_n) = \frac{n(n+1)}{2}$ .

Also we need to consider  $w_0$ , so  $\mathfrak{so}_{4n+2}$  are of complex type.  $\square$

### 3 Character Formulae

**Def. (24.4.3.1) [Central Characters of a Representation].** For  $V = \oplus_{\mu \in P} V[\mu] \in \mathcal{R}\text{ep}^{\text{fd}}(\mathfrak{g})$ , the **central character** is an analytic function on  $\mathfrak{h}$  that

$$\text{ch}_V(h) = \sum_{\mu \in P} \dim V[\mu] e^{\mu(h)}.$$

In fact  $\chi$  is an element in  $\mathbb{Z}[e^{\alpha_1}, \dots, e^{\alpha_r}, e^{-\alpha_1}, \dots, e^{-\alpha_r}]$ .  $\lrcorner$

#### Weyl's Character Formula

**Thm. (24.4.3.2) [Weyl's Character Formula, Weyl1925].** Let  $\lambda \in P^+$ , then the character of  $L_\lambda \in \mathcal{R}\text{ep}^{\text{fd}}(\mathfrak{g})$  (24.4.2.16) is given by

$$\text{ch}(L_\lambda) = \frac{\sum_{w \in W} \text{sgn}(w) e^{w(\lambda + \rho)}}{e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})}.$$

The denominator  $\Delta_{\mathfrak{g}} = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})$ , called **Weyl's denominator** of  $\mathfrak{g}$ .  $\lrcorner$

*Proof:* We already know that  $\Delta \chi_\lambda \sum_{\mu \in P} c_\mu e^\mu$  is  $W$ -anti-invariant,  $c_{\lambda + \rho} = 1$ , and  $c_\mu = 0$  unless  $\mu \in \lambda + \rho - Q^+$ , so it suffices to show  $c_\mu = 0$  if  $\mu \in P^+ \cap (\lambda + \rho - Q^+)$  and  $\mu \neq \lambda + \rho$ . ?  $\square$

**Cor. (24.4.3.3) [Characters Determine Representations].** Irreducible f.d. representations of a semisimple Lie algebra is determined by its characters, as the maximal exponent of  $\chi_\lambda$  are different for different  $\lambda$ .  $\lrcorner$

**Cor. (24.4.3.4).** For  $\mathfrak{sl}_2(\mathbb{C})$ ,  $\text{ch}_{W_n}(z) = e^{nz} + e^{(n-2)z} + \dots + e^{-nz} = \frac{e^{(n+1)z} - e^{-(n+1)z}}{e^z - e^{-z}}$ . Notice these functions are linearly independent, so by (24.4.1.14), representations of  $\mathfrak{sl}_2(\mathbb{C})$  are determined by their characters.  $\lrcorner$

**Cor. (24.4.3.5) [Weyl Denominator Formula].**

$$\Delta_{\mathfrak{g}} = \sum_{w \in W} \text{sgn}(w) e^{w(\rho)} \quad (24.4.3.2).$$

*Proof:* This follows from the Weyl character formula (24.4.3.2) by setting  $\lambda = 0$ .  $\square$

**Thm. (24.4.3.6) [Weyl Dimension Formula].** For  $\lambda \in P^+(\Phi^\vee)$ ,

$$\dim L_\lambda = \frac{\prod_{\alpha \in \Phi^+} \langle \alpha, \lambda + \rho^\vee \rangle}{\prod_{\alpha \in \Phi^+} \langle \alpha, \rho^\vee \rangle}$$

*Proof:* Choose an element  $h_\rho \in \mathfrak{h}$  that corresponds to  $\rho \in \mathfrak{h}^*$  via the Killing form, then

$$\chi_\lambda(2th_\rho) = \frac{\sum_{w \in W} \text{sgn}(w) e^{2t \langle w(\lambda + \rho), \rho^\vee \rangle}}{\prod_{\alpha \in \Phi^+} (e^{t \langle \alpha, \rho \rangle} - e^{-t \langle \alpha, \rho \rangle})}.$$

The crucial fact is that we can use Weyl's denominator formula (24.4.3.5) on the nominator, to get

$$\chi_\lambda(2th_\rho) = \frac{\prod_{\alpha \in \Phi^+} (e^{t \langle \alpha, \lambda + \rho \rangle} - e^{-t \langle \alpha, \lambda + \rho \rangle})}{\prod_{\alpha \in \Phi^+} (e^{t \langle \alpha, \rho \rangle} - e^{-t \langle \alpha, \rho \rangle})}.$$

Now taking limit  $t \rightarrow 0$ , we get the desired formula.  $\square$

**Remark (24.4.3.7).** The Weyl dimension formula can be used to determine a representation  $V$  is irreducible or not: First calculate the maximal weight, then used the Weyl dimension formula to calculate the dimension to see if it is equal to the dimension of  $V$ .  $\lrcorner$

**Thm. (24.4.3.8) [Kostant's Multiplicity Theorem].** For  $\lambda \in P^+(\Phi)$ ,

$$\dim L_\lambda[\gamma] = \sum_{w \in W} \operatorname{sgn}(w) \mathfrak{P}(w_\bullet \lambda - \gamma).$$

where  $\mathfrak{P}$  is the Kostant's partition function (3.9.5.21).  $\lrcorner$

*Proof:* This follows from plugging in the formula

$$\Delta_{\mathfrak{g}}^{-1} = \frac{1}{e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})} = \sum_{\alpha \in Q(\Phi)} \mathfrak{P}(\alpha) e^{-(\rho + \alpha)}.$$

to Weyl's character formula (24.4.3.2).  $\square$

**Prop. (24.4.3.9) [Steinberg's Multiplicity Formula].** Let  $\lambda, \mu \in P^+$ , then

$$L_\lambda \otimes L_\mu \cong \sum_{\nu \in P^+} c_{\lambda\mu\nu} L_\nu,$$

where

$$c_{\lambda\mu\nu} = \sum_{w, w' \in W} \varepsilon(w) \varepsilon(w') \mathfrak{P}(w_\bullet \lambda + w'_\bullet \mu - \nu).$$

where  $\mathfrak{P}$  is the Kostant's partition function (3.9.5.21).  $\lrcorner$

*Proof:* Cf. [Cartan, P265].  $\square$

**Cor. (24.4.3.10) [Clebsch-Gordan Rule].** The tensor products of representations of  $\mathfrak{sl}_2(\mathbb{C})$  satisfy

$$W_m \otimes W_n \cong \bigoplus_{i=0}^{\min(m,n)} W_{|m-n|+2i}.$$

*Proof:* This follows from (24.4.3.9) or from (24.4.3.4).  $\square$

#### 4 Category $\mathcal{O}$ (Bernstein-Gelfand-Gelfand)

**Def. (24.4.4.1) [ $\mathcal{O}_{\mathfrak{g}}$ ].** The category  $\mathcal{O}$  is the full subcategory of  $\operatorname{Rep}(\mathfrak{g})$  consisting of f.g.  $\mathfrak{g}$ -modules  $M$  with a weight decomposition and  $P(M) \subset \cup_{i=1}^m (\lambda_i - Q^+)$  for f.m. weights  $\lambda_1, \dots, \lambda_m \in \mathfrak{h}^*$ .  $\lrcorner$

**Prop. (24.4.4.2).** For any  $M \in \mathcal{O}_{\mathfrak{g}}$ , the weight spaces of  $M$  are f.d..  $\lrcorner$

*Proof:* By hypothesis we can assume  $v_1, \dots, v_m$  are generators of  $M$  that are also eigenvectors of  $\mathfrak{h}$ . Let  $E = \sum_{i=1}^m U(\mathfrak{h} \oplus \mathfrak{n}_+) v_i = \sum_{i=1}^m U(\mathfrak{n}_+) v_i$  which is of f.d. by hypothesis on weights of  $M$ . On the other hand, the natural map  $U(\mathfrak{n}^-) \otimes E \rightarrow M$  is surjective, so the assertion follows from the fact that the weight spaces of  $U(\mathfrak{n}^-) \otimes E \rightarrow M$  are f.d..  $\square$

**Prop. (24.4.4.3).** For any  $M \in \mathcal{O}_{\mathfrak{g}}$ , the action of  $Z(U(\mathfrak{g}))$  on  $M$  factors through a f.d. quotient.  $\lrcorner$

*Proof:* By (3.7.8.23),  $Z(U(\mathfrak{g}))$  is f.g., so it suffices to show that each  $z \in Z(U(\mathfrak{g}))$  satisfies a polynomial equation on  $M$ . Let  $\mu_1, \dots, \mu_k \in \mathfrak{h}^*$  s.t.  $M$  is generated by  $E = \oplus M[\mu_i]$ . As  $\dim E < \infty$  by (24.4.4.2) and  $Z(U(\mathfrak{g}))$  preserves  $E$ ,  $z$  satisfies a polynomial equation on  $E$ . Hence  $z$  satisfies the same equation on  $M$ , as  $z$  is in the center and  $E$  generates  $M$ .  $\square$

**Prop. (24.4.4.4).** It is clear that every object  $V \in \mathcal{O}_{\mathfrak{g}}$  has a highest weight, thus the simple objects of  $\mathcal{O}$  are just  $L_{\lambda}$  where  $\lambda \in \mathfrak{h}^*$ .  $\lrcorner$

**Prop. (24.4.4.5) [Finite Length].** Every  $M \in \mathcal{O}$  has finite length.  $\lrcorner$

*Proof:*  $\square$

### Category $\mathcal{O}_{\text{int}}$

**Def. (24.4.4.6) [ $\mathcal{O}_{\text{int}}$ ].** The category  $\mathcal{O}_{\text{int}}$  is the subcategory of representations  $V$  of  $\mathfrak{g}$  that with weight decomposition into f.d. weight spaces that the weights  $P(M) \subset \cup_{i=1}^m (\lambda_i - Q^+)$  for f.m. weights  $\lambda_1, \dots, \lambda_m \in P^{\vee}(R)$ .  $\lrcorner$

**Def. (24.4.4.7) [Characters].** The character for a f.d. representation can be extended to category  $\mathcal{O}_{\text{int}}$  as a formal Laurent series

$$\text{ch}(V) \in \mathbb{Z}[[e^{-\alpha_1^{\vee}}, \dots, e^{-\alpha_r^{\vee}}]][e^{\alpha_1^{\vee}}, \dots, e^{\alpha_r^{\vee}}] : \text{ch}(V)(h) = \sum \dim V[\mu] e^{\mu(h)}.$$

**Prop. (24.4.4.8).** The character of  $M_{\lambda}$  is given by

$$\text{ch}(M_{\lambda}) = \frac{e^{\lambda}}{\prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})} = \frac{e^{\lambda+\rho}}{\Delta_{\mathfrak{g}}} \text{ (24.4.3.2)}.$$

**Prop. (24.4.4.9) [Duality Functor].** Let  $\tau : \mathfrak{g} \rightarrow \mathfrak{g}$  be the Cartan involution, then for any  $V \in \mathcal{O}_{\mathfrak{g}}$ , we can define  $V^{\vee} = (V^{\tau})_{\text{fin}}^*$ . Then

- $V^{\vee}$  has the same character and composition factors as  $V$ .
- $(M_{\lambda})^{\vee} = M_{\lambda^{\vee}}, L_{\lambda}^{\vee} = L_{\lambda}$ .
- The assignment  $V \mapsto V^{\vee}$  is an involutive equivalence of categories  $\mathcal{O} \cong \mathcal{O}^{\text{op}}$  which preserves the decomposition into  $\mathcal{O}_{\chi}(S)$ .
- 

*Proof:*  $\square$

**Prop. (24.4.4.10) [Projective Verma Modules].** If  $\lambda \in \mathfrak{h}^*$  is dominant, then  $M_{\lambda-\rho}$  is a dominant object in  $\mathcal{O}$ , and for any  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ ,  $V \otimes M$  is projective.  $\lrcorner$

*Proof:* ? Cf. [Etingof]P53. ?

The second assertion follows from the fact that tensoring  $V$  is left adjoint to tensoring  $V^*$ .  $\square$

**Prop. (24.4.4.11) [Enough Projectives].**  $\lrcorner$



*Proof:*

□

**Cor. (24.4.4.12).** For any  $\lambda \in \mathfrak{h}^*$ ,  $L_\lambda$  has a projective cover  $P_\lambda$ .

For any  $\lambda, \mu \in \mathfrak{h}^*$ , define  $d_{\lambda, \mu} \in \mathbb{N}$  as

$$\delta_{\lambda, \mu} = \dim \operatorname{Hom}(P_\lambda, L_\mu).$$

┘

### Harish-Chandra Bimodules

**Def. (24.4.4.13) [Harish-Chandra Bimodules].** A **Harish-Chandra bimodule** over  $\mathfrak{g}$  is a  $\mathfrak{g}$ -bimodule  $V$  of finite length (as bimodules) s.t. the adjoint action of  $\mathfrak{g}$  (i.e.  $g(v) = gv - vg$ ) is locally finite (i.e. every  $v \in V$  is contained in a f.d.  $\operatorname{ad}(\mathfrak{g})$ -module). The category of Harish-Chandra bimodules over  $\mathfrak{g}$  is denoted by  $\operatorname{H-C}(\mathfrak{g})$ .

┘

**Def. (24.4.4.14) [Maximal Quotients of  $U(\mathfrak{g})$ ].** For any  $\chi \in \operatorname{Hom}(Z(U(\mathfrak{g})), \mathbb{C})$ , define

$$U_\chi(\mathfrak{g}) = U(\mathfrak{g}) / (z - \chi(z) : z \in Z(U(\mathfrak{g}))),$$

called a **maximal quotient of  $U(\mathfrak{g})$** .

┘

**Prop. (24.4.4.15) [Maximal Quotients].** For any maximal quotient  $U_\chi(\mathfrak{g})$  of  $U(\mathfrak{g})$  and any  $V \in \operatorname{Irr}^{\operatorname{fd}}(\mathfrak{g})$ ,  $\dim \operatorname{Hom}_{\mathfrak{g}}(V, U_\chi(\mathfrak{g})) = \dim V[0]$ .

In particular,  $U_\chi(\mathfrak{g})$  is a Harish-Chandra bimodule (24.4.4.13), and  $V \otimes U_\chi(\mathfrak{g})$  is a Harish-Chandra bimodule for any  $V \in \operatorname{Rep}^{\operatorname{fd}}(\mathfrak{g})$ .

┘

*Proof:* This follows from (3.7.8.21) (Notice  $\operatorname{Hom}_{\mathfrak{g}}(V, U_\chi(\mathfrak{g})) \cong Z(U(\mathfrak{g}))^{\oplus \dim V[0]}$ , so after quotient by the central ideal, it is isomorphic to  $\mathbb{C}^{\dim V[0]}$ ). Notice also that  $V \otimes U_\chi$  is f.g. thus of finite length.

□

**Prop. (24.4.4.16).** Every irreducible  $\mathfrak{g}$ -bimodule  $M$  with right infinitesimal character  $\chi$  that is locally finite under the adjoint action of  $\mathfrak{g}$  (e.g. Harish-Chandra bimodules) is a quotient of some  $V \otimes U_\chi(\mathfrak{g})$  for some  $V \in \operatorname{Rep}^{\operatorname{fd}}(\mathfrak{g})$ . In particular, they are Harish-Chandra bimodule by (24.4.4.15).

┘

*Proof:* Let  $V \subset M$  be a non-zero irreducible  $\mathfrak{g}$ -module under the adjoint action, and consider the tensor bimodule  $V^* \otimes M$ . The canonical sub-bimodule  $V^* \otimes V \subset V^* \otimes M$  is  $\operatorname{ad}(\mathfrak{g})$ -invariant (i.e. commutes with  $\mathfrak{g}$ ), so we have a bimodule homomorphism  $\psi : U(\mathfrak{g}) \rightarrow V^* \otimes M$  which descends to a homomorphism  $U_\chi(\mathfrak{g}) \rightarrow V^* \otimes M$ , which gives rise to a map of bimodules  $V \otimes U_\chi(\mathfrak{g}) \rightarrow M$ , which is surjective because  $M$  is irreducible.

□

**Def. (24.4.4.17) [Maps of Finite Type].**

┘

**Prop. (24.4.4.18).** For any  $M, N \in \operatorname{Rep}(\mathfrak{g})$ ,  $V \in \operatorname{Rep}^{\operatorname{fd}}(\mathfrak{g})$ ,

$$\operatorname{Hom}_{\mathfrak{g}}(V, {}_{\operatorname{ad}} \operatorname{Hom}_{\operatorname{fin}}(M, N)) = \operatorname{Hom}_{\mathfrak{g}}(V \otimes M, N).$$

┘

**Prop. (24.4.4.19).** For  $M, V \in \mathcal{O}_{\mathfrak{g}}$  and  $N \in \operatorname{Rep}^{\operatorname{fd}}(\mathfrak{g})$ , then

$$\operatorname{Hom}_{\operatorname{fin}}(M, V \otimes N) = V \otimes \operatorname{Hom}_{\operatorname{fin}}(M, N).$$

┘

*Proof:* □

**Thm. (24.4.4.20) [Duflo-Joseph].** The action homomorphism  $M_{\chi_{\lambda+\rho}} \rightarrow \text{Hom}_{\text{fin}}(M_\lambda, M_\lambda)$  is an isomorphism. And for any  $V \in \text{Rep}^{\text{fin}}(\mathfrak{g})$ , the natural map  $V \otimes U_{\chi_{\lambda+\rho}} \rightarrow \text{Hom}_{\text{fin}}(M_\lambda, V \otimes M_\lambda)$  is an isomorphism. ┘

*Proof:* Cf.[Rep of Lie Groups, Etingof]P59. ? The second assertion follows from the first and (24.4.4.19). □

**Def. (24.4.4.21) [ $H_\lambda$ ].** For  $\lambda \in \mathfrak{h}^*$  and  $\theta \in \mathfrak{h}^*/W_\bullet$ , there is a functor

$$H_\lambda : \mathcal{O} \rightarrow \text{H-C} : X \mapsto \text{Hom}_{\text{fin}}(M_{\lambda-\rho}, X).$$

┘

**Prop. (24.4.4.22).** If  $\lambda$  is dominant, then  $H_\lambda$  is exact. ┘

*Proof:* For any  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ , by (24.4.4.18),

$$\text{Hom}_{\mathfrak{g}}(V, {}_{\text{ad}}H_\lambda(M)) = \text{Hom}_{\mathfrak{g}}(V \otimes M_{\lambda-\rho}, M).$$

So the assertion follows from the fact that  $V \otimes M_{\lambda-\rho}$  is projective by (24.4.4.10). □

**Prop. (24.4.4.23) [Central Character of Harish-Chandra Bi-modules].** Let  $V \in \text{H-C}^{\text{fd}}(\mathfrak{g})$  and  $\lambda \in \mathfrak{h}^*$ , then

- The left central characters occurring in  $V \otimes U_{\chi_\lambda}$  are  $\chi_{\lambda+\nu}$  where  $\nu$  runs through weights of  $V$ .
- Let  $M$  be a left  $\mathfrak{g}$ -module with central character  $\chi_\lambda$ , then the central characters of  $V \otimes M$  are among  $\chi_{\lambda+\nu}$  where  $\nu$  runs through weights of  $V$ .
- If  $M \neq 0 \in \text{H-C}(\mathfrak{g})$  with left and right central characters  $(\chi_\lambda, \chi_\mu)$ , then there exists  $w \in W$  s.t.  $w_\bullet \lambda - \mu \in P$ .

┘

*Proof:* Cf.[Rep of Lie Groups, Etingof]P60. ?

2: This follows from 1 and the isomorphism  $V \otimes M = (V \otimes U_{\chi_\lambda}) \otimes_{U_{\chi_\lambda}} M$ .

3: ? By (24.4.4.16), any irreducible  $\mathfrak{g}$ -bimodule is a quotient of  $V \otimes U_{\chi_\lambda}$  for some  $V \in \text{Rep}(\mathfrak{g})$  and  $\lambda \in \mathfrak{h}^*$ , then □

**Prop. (24.4.4.24).** There is a decomposition of categories

$$\text{H-C}(\mathfrak{g}) = \bigoplus_{\gamma, \lambda} \text{H-C}_{\chi_{\lambda+\gamma}, \chi_\gamma}(\mathfrak{g})$$

where  $\gamma \in P_+$  and  $\lambda \in \mathfrak{h}^*/\text{Stab}_W(\gamma)_\bullet$ . ┘

*Proof:* Cf.[Rep of Lie Groups, Etingof]P60. ? □

### Principle Series

**Def. (24.4.4.25) [Principle Series].** For  $\lambda, \mu \in \mathfrak{h}^*$  with  $\lambda - \mu \in P$ , define the **principle series representation**  $M(\lambda, \mu) = \text{Hom}_{\text{fin}}(M_{\lambda-\rho}, M_{\mu-\rho}^\vee) \in \text{H-C}_{\chi_\mu, \chi_\lambda}(\mathfrak{g})$ . ? ┘

### Projective Functors

**Def. (24.4.4.26)**  $[\text{Rep}(\mathfrak{g})_f]$ . Denote by  $\text{Rep}(\mathfrak{g})_f$  the category of representations of  $\mathfrak{g}$  with locally finite action of  $Z(\mathfrak{g})$ . Then there is a decomposition of categories:

$$\text{Rep}(\mathfrak{g})_f = \bigoplus_{\chi \in \mathfrak{h}^*/W_\bullet} \text{Rep}(\mathfrak{g})_\chi,$$

where  $\text{Rep}(\mathfrak{g})_\chi$  is the subcategory of modules with generalized central character  $\chi$ .  $\lrcorner$

**Def. (24.4.4.27)**  $[F_V]$ . For  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ , there is an endofunctor  $F_V$  of  $\text{Rep}(\mathfrak{g})_f$  given by  $M \mapsto (V \otimes M)$ . When  $M \in \text{Rep}^{\text{fd}}(\mathfrak{g})_{\chi_\lambda}$ ,  $F(M) = (V \otimes U_{\chi_\lambda}) \otimes_{U_{\chi_\lambda}} M$ , so by (24.4.4.23),

$$F(\text{Rep}(\mathfrak{g})_{\chi_\lambda}) \subset \bigoplus_{\beta \in P(V)} \text{Rep}(\mathfrak{g})_{\chi_\lambda + \beta}.$$

$\lrcorner$

**Prop. (24.4.4.28)**. For  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ ,  $F_{V^*}$  is both left and right adjoint to  $F_V$ .  $\lrcorner$

**Def. (24.4.4.29)** **[Projective Functors]**. A **projective functor of representations** is an endofunctor of  $\text{Rep}(\mathfrak{g})_f$  that is isomorphic to a direct sum of some  $F_V$  where  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ .  $\lrcorner$

**Prop. (24.4.4.30)** **[Projective Functors]**.

- For  $\chi \in \mathfrak{h}^*/W$ , the projection functor to the generalized  $\chi$ -part:  $\Pi_\chi : \text{Rep}(\mathfrak{g})_f \rightarrow \text{Rep}(\mathfrak{g})_\chi$  is a projective functor.
- The category of projective functors form a category which is closed under taking compositions, direct summands and finite direct sums.
- Every projective functor has a left and right adjoint, and the left and right adjoints are equal. ?
- Every projective functor  $F$  has a decomposition  $F = \bigoplus_{\theta, \chi \in \mathfrak{h}^*/W} \Pi_\theta \circ F \circ \Pi_\chi$ .  $\lrcorner$

**Def. (24.4.4.31)** **[Projective  $\chi$ -Functors]**. For  $\chi \in \mathfrak{h}^*/W_\bullet$  and a projective functor  $F$  (24.4.4.29), denote  $F(\chi)$  the restriction of  $F$  to  $\text{Rep}(\mathfrak{g})_\chi^1$ , where  $\text{Rep}(\mathfrak{g})_\chi^1$  is the subcategory of  $\text{Rep}(\mathfrak{g})_\chi$  of modules with central character  $\chi$ .

Then a **projective  $\chi$ -functor** is defined to be an functor from  $\text{Rep}(\mathfrak{g})_\chi^1$  to  $\text{Rep}(\mathfrak{g})_f$  that is isomorphic to a direct sum of some  $F_V(\chi)$  where  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ .  $\lrcorner$

**Prop. (24.4.4.32)**. Let  $\lambda \in \mathfrak{h}^*$  be dominant,  $\chi = \chi_\lambda$  and  $F_1, F_2$  be projective  $\chi$ -functors, then the map

$$\text{Hom}(F_1, F_2) \rightarrow \text{Hom}(F_1(M_{\lambda-\rho}), F_2(M_{\lambda-\rho}))$$

is an isomorphism.  $\lrcorner$

*Proof:* It suffices to prove for  $F_i = F_{V_i}(\theta)$ . Denote  $V = V_1^* \otimes V_2$ , then it suffices to show that

$$\text{Hom}(\text{Id}(\theta), F_V(\theta)) \rightarrow \text{Hom}_{\mathfrak{g}}(M_{\lambda-\rho}, V \otimes M_{\lambda-\rho})$$

is an isomorphism. By (3.6.6.2), the LHS is isomorphic to

$$\text{Hom}_{(U(\mathfrak{g}), U_\chi(\mathfrak{g}))}(U_\chi(\mathfrak{g}), V \otimes U_\chi(\mathfrak{g})) = (V \otimes U_\chi(\mathfrak{g}))^{\mathfrak{g}_{\text{ad}}}.$$

But then the desired isomorphism follows from the Duflo-Joseph theorem (24.4.4.20).  $\square$

**Prop. (24.4.4.33) [Lifting Morphisms of Projective Functors].** Let  $F_1, F_2$  be projective functors, then

- Every map  $\varphi : F_1(\chi) \rightarrow F_2(\chi)$  lifts to a map  $\widehat{\varphi} : F_1|_{\text{Reg}(\mathfrak{g})_\chi} \rightarrow F_2|_{\text{Reg}(\mathfrak{g})_\chi}$ .
- If  $F_1 = F_2$  and  $\varphi^2 = \varphi$ , then we can chose  $\widehat{\varphi}$  s.t.  $\widehat{\varphi}^2 = \widehat{\varphi}$ .
- If  $\varphi$  is an isomorphism, then so is  $\widehat{\varphi}$ .

┘

*Proof:*

□

**Cor. (24.4.4.34).** If  $F_1, F_2$  are projective functors, then any isomorphism  $F_1(M_{\lambda-\rho}) \cong F_2(M_{\lambda-\rho})$  lifts to an isomorphism  $F_1|_{\mathcal{R}\text{ep}(\mathfrak{g})_{\chi_\lambda}} \cong F_2|_{\mathcal{R}\text{ep}(\mathfrak{g})_{\chi_\lambda}}$ , and any decomposition  $F_1(M_{\lambda-\rho}) = \oplus M_i$  lifts to a decomposition  $F = \oplus F_i$  where  $F_i$  are projective functors and  $F_i(M_{\lambda-\rho}) = M_i$ .

In particular, any projective  $\chi$ -functor (24.4.4.31) is of the form  $F(\chi)$  for some projective functor  $F$ .

┘

*Proof:* In priori  $F_i$  are just projective  $\chi$ -functors, but they are in fact restrictions of a projective functor  $F_i \circ \Pi_i$  because we have a decomposition for each  $\chi \in \mathfrak{h}^*/W_\bullet$ , so

$$F = \bigoplus_{\chi} \bigoplus_i F_{\chi,i} \circ \Pi_{\chi}.$$

□

**Cor. (24.4.4.35) [Decompositions of Projective Functors].** For any projective functor  $F$  and  $\chi \in \mathfrak{h}^*/W_\bullet$ ,  $F \circ \Pi_{\chi}$  is a finite direct sum of indecomposable projective functors. In particular,  $F = \bigoplus_{\chi \in \mathfrak{h}^*/W_\bullet} F \circ \Pi_{\chi}$  is a direct sum of indecomposable projective functors.

┘

*Proof:* This follows from the fact  $F \circ \Pi_{\theta}(M_{\lambda-\rho}) \in \mathcal{O}$  is a finite sum of indecomposables (finite by (24.4.4.5)). And then we use (24.4.4.34). □

**Prop. (24.4.4.36) [Indecomposable Projective Functors].** If  $F$  is an indecomposable functor, then  $F = F \circ \Pi_{\chi}$ , and

- $F(M_{\lambda-\rho}) = 0$  if  $\chi \neq \chi_{\lambda}$ .
- $F(M_{\lambda-\rho}) = P_{\mu-\rho}$  for some  $\mu \in \mathfrak{h}^*$  if  $\lambda$  is dominant and  $\chi_{\lambda} = \chi$ .

┘

*Proof:* This is because  $F(M_{\lambda-\rho})$  is both indecomposable and projective. ?

□

$K_0(\mathcal{O}_{\mathfrak{g}})$

**Prop. (24.4.4.37) [BGG Reciprocity].** For  $\lambda, \mu \in \mathfrak{h}^*$ , if we define  $d_{\mu,\lambda}^* \in \mathbb{N}$  as

$$\text{ch}(P_{\lambda}) = \sum_{\mu} d_{\lambda,\mu}^* \text{ch}(M_{\lambda}) = \sum_{\mu} d_{\lambda,\mu}^* \frac{e^{\lambda}}{\prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})} \quad (24.4.4.8),$$

then  $d_{\lambda,\mu}^* = d_{\lambda,\mu}$  (24.4.4.12).

┘

*Proof:* Cf. [Etingof] P66. ?

□

**Thm. (24.4.4.38) [Kazhdan-Lusztig Conjecture, Beilinson-Bernstein/Brylinski-Kashiwara].**

?

┘

*Proof:*

□

**Prop. (24.4.4.39)[Grothendieck Group of  $\mathcal{O}_{\mathfrak{g}}$ ].** The Grothendieck group  $K_0(\mathcal{O}_{\mathfrak{g}})$  is freely generated by  $\delta_{\lambda} = [M_{\lambda-\rho}]$ , and we can construct an inner product on  $K_0(\mathcal{O}_{\mathfrak{g}})$  by declaring this basis to be orthonormal.  $\lrcorner$

**Prop. (24.4.4.40).** Since projective functors are left/right adjoint thus exact, they define endomorphisms of  $K(\mathcal{O}_{\mathfrak{g}})$ . For example, if  $V \in \text{Rep}^{\text{fd}}(\mathfrak{g})$ , then

$$[F_V]\delta_{\lambda} = \sum_{\beta} m_V(\beta)\delta_{\lambda+\beta}$$

by (24.4.4.23).

And If  $P \in \mathcal{O}_{\mathfrak{g}}$  is projective, then for any  $M \in \mathcal{O}$ ,

$$([P], [M]) = \dim \text{Hom}(P, M).$$

⌋

*Proof:*

□

**Prop. (24.4.4.41).**

- If  $F_1, F_2$  are projective functors on  $\mathcal{O}$  with  $[F_1] = [F_2]$ , then  $F_1 = F_2$ .
- If  $(F, F')$  is an adjoint pair of projective functors on  $\mathcal{O}$ , then  $([F], [F'])$  is an adjoint pair on  $K(\mathcal{O})$ . In particular, the left and right adjoints of  $[F]$  are equal by (24.4.4.30).  $\lrcorner$

*Proof:* 1: By (24.4.4.32), it suffices to show that  $F_1(M_{\lambda-\rho}) = F_2(M_{\lambda-\rho})$  for any  $\lambda \in \mathfrak{h}^*$  projective. But these two elements are projective and have the same character, so they are equal.

2: It suffices to show that  $([F]x, y) = (x, [F']y)$ . By (24.4.2.22) and (24.4.4.10), it suffices to check for  $x = [P]$  where  $P \in \mathcal{O}$  is projective. But then the assertion follows from (24.4.4.40).  $\square$

## Classifying Harish-Chandra Bimodules

### 5 Lie Algebra Cohomology

References are [Jan16], [Wei94] and [?Eti21].

**Prop. (24.4.5.1)[Chevalley-Eilenberg resolution].**  $\lrcorner$

## 24.5 Representations of $C^*$ -Algebras

References are [ $C^*$ -Algebras, Dixmier].

## 24.6 Smooth Representations of Lie Groups

**Notation(24.6.0.1).**

- Use notations on representation theory defined in [Smooth Representations](#).
- Use notations on Lie groups defined in [Lie Groups](#).
- Let  $G \in \mathcal{L}ieGrp_{\text{cntd}}$  with Lie algebra  $\mathfrak{g}$ .
- Let  $K \leq G$  be a maximal compact subgroup.
- Denote  $\mathcal{Z} = Z(U(\mathfrak{g}))$ .
- Let  $(\pi, V) \in \text{Rep}(G)$ , where  $V$  is a separable Fréchet spaces over  $\mathbb{C}$  (Cf. [Casselman, Canonical extensions of Harish-Chandra modules to representations of  $G$ ] and [Wallach, Real reductive groups, 2]).

┘

References are [\[Kna94\]](#).

### 1 Smooth Vectors

**Prop.(24.6.1.1)[Left Invariant Differential Operators].** Consider its left and right regular action  $\lambda, \rho$  on  $C^\infty(G)$  [\(12.10.1.1\)](#). We may write  $dX$  for  $X \in \mathfrak{g}$  as the representation of  $\mathfrak{g}$  via  $\rho$ , then it commutes with  $\lambda$ . Then it induces a map of  $U(\mathfrak{g})$  to the ring of left  $G$ -invariant differential operators on  $G$ . [\(3.7.8.2\)](#).

┘

**Prop.(24.6.1.2)[Center element Bi-invariant].** For  $D \in Z(U(\mathfrak{g}))$ , the differential operator  $D$  defined in [\(24.6.1.1\)](#) is invariant under both left and right regular representations of  $G$ .

┘

*Proof:* The left invariance is general from [\(24.6.1.1\)](#), for the right invariance, Because  $G$  is connected, it suffices to prove invariance for a nbhd of identity of  $G$ , thus suffices to prove

$$\rho(g_t)D = D\rho(g_t), \quad g_t = \exp(tX).$$

For this, let  $\varphi(g, t) = (\rho(g_t)D - D\rho(g_t))(g)$  and take derivative w.r.t  $t$ , then it reads:

$$\frac{\partial}{\partial t}\varphi(g, t) = (DdX\rho(g_t)f - dX\rho(g_t)Df)(g) = dX\varphi(g, t)$$

because  $dX$  commutes with  $D$ . And also  $\varphi(g, 0) = 0$ , so by lemma [\(24.6.1.3\)](#),  $\varphi(g, t) = 0$  for any  $t, g$ .

□

**Lemma(24.6.1.3).** If  $\varphi \in C^\infty(G \times \mathbb{R})$  satisfies

$$\frac{\partial}{\partial t}\varphi(g, t) = dX.\varphi(g, t)$$

for some  $X \in \mathfrak{g}$  and  $\varphi(g, 0) = 0$ , then  $\varphi = 0$ .

┘

*Proof:* Let  $\varphi_g(u, v) = \varphi(g \exp(uX), v)$ , then the condition says

$$\frac{\partial}{\partial u}\varphi_g(u, v) = \frac{\partial}{\partial v}\varphi_g(u, v),$$

which means  $\varphi_g(u, v) = F(u + v)$ , and the fact  $\varphi_g(u, 0) = 0$  shows  $F = 0$ .

□

**Cor.(24.6.1.4).** If  $G = \text{GL}(2; \mathbb{R})^+$ , then  $\mathfrak{g} = \mathfrak{gl}(2; \mathbb{R})$ , and the Casimir element [\(3.7.8.23\)](#)  $\Delta = -1/2(1/2h^2 + ef + fe)$  corresponds via [\(24.6.1.1\)](#) to a bi-invariant differential operator on  $C^\infty(G)$ , and it is called the **Laplace-Beltrami operator**.

┘

### Smooth Vectors

**Def. (24.6.1.5) [Smooth Vectors].** Define the subspace  $V^\infty \leq V$  of **smooth vectors** as  $V^\infty = \bigcap_n V^n$  where

$$V^0 = \{v \in V, \frac{\partial}{\partial t} \exp(t\xi)v \text{ exists, } \forall \xi \in \mathfrak{g}\}, \quad V^n = \{v \in V^{n-1}, T_\xi(v) \in V^{n-1}, \forall \xi \in \mathfrak{g}\}.$$

And  $V^\infty$  is given the inverse limit topology.

Then there is a Lie algebra action of  $\mathfrak{g}$  on  $V^\infty$ . ┘

**Prop. (24.6.1.6) [Action of Distribution on Smooth Vectors].** There is a continuous action of  $\text{Meas}_c(G)$  on  $V^\infty$ :  $T \mapsto \pi(T)$  compatible with the convolution structure on  $\text{Meas}_c(G)$ . ┘

*Proof:* For  $v \in V^\infty$ , define  $F^v(g) = g(v)$ . For  $T \in \text{Meas}_c(G)$ , define  $\pi(T)v = \langle T, F^v \rangle$ . For example,

- $\pi(\delta_x) = \pi(x)$ .
- $\pi(u) = u$  for  $u \in U(\mathfrak{g})$ .
- $\pi(f\mu_{\text{Haar}})v = \int_G f(g)\pi(g)v dg$  for  $f \in C_c(G)$ .
- $\pi(f * g) = \pi(f)\pi(g)$ .

□

**Cor. (24.6.1.7).** For any  $f \in C_c^\infty(G)$  and  $v \in V$ ,  $f\mu_{\text{Haar}} \cdot (v) \in V^\infty$ . ┘

*Proof:* Notice  $X\pi(f)v = \pi(X * f)v$ , and we calculate  $X * f$ :

$$(X * f, v) = \int \frac{d}{dt} \pi(e^{tX}y)vf(y)dy = \frac{d}{dt} \int f(y)\pi(e^{tX}y)v dy = \frac{d}{dt} \int f(e^{-tX}y)\pi(y)v dy = \pi(f_X)v$$

where  $f_X(g) = \frac{d}{dt}(e^{-tX}g)|_{t=0}$  is smooth. Iterating, we can show  $\pi(f)v \in V^\infty$ . □

**Cor. (24.6.1.8) [Smooth Vectors Dense].**  $V^\infty$  is dense in  $V$ . ┘

*Proof:* Choose a smooth approximate identity (12.10.2.9)  $\{f_n\}$ , then  $f_n \cdot v \in V^\infty$  converges to  $v$  by continuity, and these vectors are smooth by (24.6.1.7). □

**Cor. (24.6.1.9).** For  $V \in \text{Rep}^{\text{fd}}(G)$ ,  $V^\infty = V$ . In particular, if  $G = K$  is compact, then  $V^{\text{fin}} \subset V^\infty$ . ┘

**Prop. (24.6.1.10) [Fréchet Topology on  $V^\infty$ ].** If  $V \in \mathcal{H}\text{ilbSpa}$ , we can define a Fréchet topology on  $V^\infty$  by the seminorms

$$\{\nu_{X_1, \dots, X_n} : v \mapsto \|X_1 \dots X_n \cdot v\|\}_{X_1, \dots, X_n \in \mathfrak{g}}.$$

Then  $V^\infty$  is complete in this topology. ┘

*Proof:* ? □

**Prop. (24.6.1.11) [Smooth Vectors in  $\mathbb{S}^1$ ].** Let  $K = \mathbb{S}^1$  and  $\rho$  be the regular representation of  $\mathbb{S}^1$  on the Hilbert space  $L^2(\mathbb{S}^1) = L^2[0, 2\pi]$ , then the smooth vectors in  $\rho$  are just precisely the elements of  $C^\infty(\mathbb{S}^1)$ . ┘

*Proof:* Take a Fourier expansion  $f(x) = \sum a_n e^{2\pi i n x}$ . Suppose  $f$  is a  $C^1$  vector, then there is a  $g(x) = \sum b_n e^{2\pi i n x}$  that  $\lim \frac{1}{t}(f_t - f) = g$ , so

$$\lim_{n \rightarrow \infty} \sum_n \left| \frac{1}{t}(e^{2\pi i n t} - 1)a_n - b_n \right|^2 = 0.$$

which means  $b_n = 2\pi i n a_n$ .

So if  $f$  is a  $C^\infty$  vector, then  $|a_n|$  decay rapidly, and  $f$  and all its derivatives converge absolutely so  $f$  is smooth. Conversely, integration shows the Fourier coefficients of any smooth function  $f$  decay rapidly. □



### Examples

**Prop. (24.6.1.12)**[Stone-Newmann]. The Schrödinger representation (12.11.1.21)  $\pi$  of the Heisenberg group  $H(\mathbb{R})$  acting on  $L^2(\mathbb{R})$  is unitary and irreducible, where  $\pi(p_a) = e^{iax}$ , and  $\pi(r_b) = T_b$ .  $\lrcorner$

**Prop. (24.6.1.13)**  $[\widetilde{\mathrm{SL}(2; \mathbb{R})}]$ . Let  $\widetilde{\mathrm{SL}(2; \mathbb{R})}$  be the universal covering of  $\mathrm{SL}(2; \mathbb{R})$ . Then  $\mathrm{Rep}^{\mathrm{fd}}(\widetilde{\mathrm{SL}(2; \mathbb{R})}) = \mathcal{R}\mathrm{ep}^{\mathrm{fd}}(\mathrm{SL}(2; \mathbb{R}))$ . In particular,  $\widetilde{SL_2(\mathbb{R})}$  admits no f.d. faithful representations, and the only quotient groups of it that admit f.d. faithful representations are  $SL_2(\mathbb{R})$  and  $\mathrm{PSL}(2; \mathbb{R})$ .  $\lrcorner$

*Proof:* Let  $\rho : \widetilde{\mathrm{SL}(2; \mathbb{R})} \rightarrow \mathrm{GL}(n; \mathbb{R})$  be a representation inducing a real Lie algebra representation  $\rho_*$ . Consider its complexification  $\rho \otimes \mathbb{C} : \widetilde{\mathrm{SL}(2; \mathbb{R})} \rightarrow \mathrm{GL}(n; \mathbb{C})$ . Because  $SL(2; \mathbb{C})$  is simply connected (16.6.3.1),  $\rho_* \otimes \mathbb{C}$  corresponds to a complex representation  $\rho' : \mathrm{SL}(2; \mathbb{C}) \rightarrow \mathrm{GL}(n; \mathbb{C})$  by (16.5.3.13). Now because  $\widetilde{SL_2(\mathbb{R})}$  is simply connected, so there is a real Lie group homomorphism  $\gamma : \widetilde{\mathrm{SL}(2; \mathbb{R})} \rightarrow \mathrm{SL}(2; \mathbb{C})$  that  $\rho' \circ \gamma = \rho$ . But because  $\rho, \rho'$  both commutes with conjugation, so does  $\gamma$ . Thus the image of  $\gamma$  is in  $\mathrm{GL}(2; \mathbb{R})$ , and  $\rho$  factors through  $\mathrm{SL}(2; \mathbb{R})$ .

There is a quotient map  $\pi : \widetilde{\mathrm{SL}(2; \mathbb{R})} \rightarrow \mathrm{PSL}(2; \mathbb{R})$ , and  $\mathrm{PSL}(2; \mathbb{R})$  has trivial center, so the center of  $\widetilde{\mathrm{SL}(2; \mathbb{R})}$  is contained in  $\ker(\pi)$ , which is isomorphic to  $\pi_1(\mathrm{PSL}(2; \mathbb{R})) = \mathbb{Z}$ , and is central in  $\widetilde{SL_2(\mathbb{R})}$  by (16.5.1.6). so the center of  $\widetilde{SL_2(\mathbb{R})}$  is just  $\mathbb{Z}$ . By what has been proved, for any representation of other covering space of  $\mathrm{PSL}_2(\mathbb{R})$ , the induced representation on  $\widetilde{SL_2(\mathbb{R})}$  factors through  $SL_2(\mathbb{R})$ , thus trivial on the subgroup  $2\mathbb{Z} \subset \mathbb{Z}$ , and the original representation factors through  $SL_2(\mathbb{R})$  or  $\mathrm{PSL}_2(\mathbb{R})$ . So the only possibility of faithful representation is  $SL_2(\mathbb{R})$  or  $\mathrm{PSL}_2(\mathbb{R})$ . For  $\mathrm{PSL}_2(\mathbb{R})$ , take the  $\square$

**Prop. (24.6.1.14)**[F.D. Representations of  $\mathrm{SL}(2; \mathbb{K})$ ]. Let  $\mathbb{K} = \mathbb{C}$  or  $\mathbb{R}$ , then the representations of  $\mathrm{SL}(2; \mathbb{K})$  are isomorphic to direct representations  $(\rho_n, V_n)$ , where  $V_n$  =homogeneous polynomials of degree  $n$  in indeterminants  $x, y$ , and

$$\rho_n \left( \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) (x^k y^{n-k}) = (ax + by)^k (cx + dy)^{n-k}.$$

$\lrcorner$

*Proof:* We can check these representations truly induce irreducible representations of their Lie algebras  $\mathfrak{sl}(2; \mathbb{K})$ . Notice  $\mathrm{SL}(2; \mathbb{C})$  is simply connected (16.6.3.1), so (24.6.1.13) and (16.5.3.13) gives the result.  $\square$

## 2 Finite-Dimensional Representations

**Prop. (24.6.2.1)**. Any 1 dimensional representations of  $\mathbb{C}^\times$  is of the form

$$z \mapsto z^\mu \bar{z}^\nu$$

for some  $\mu, \nu \in \mathbb{C}$  s.t.  $\mu - \nu \in \mathbb{Z}$ .  $\lrcorner$

**Prop. (24.6.2.2)**. Finite dimensional representations of a semisimple Lie group is completely reducible.  $\lrcorner$

*Proof:* This is because its Lie algebra representation is completely reducible by (24.4.1.2), and each Lie algebra subrepresentation corresponds to a Lie group subrepresentation.  $\square$

### Archimedean Weil-Groups

References are [Kna94].

**Def. (24.6.2.3) [Archimedean Weil Groups].** Define

$$W_{\mathbb{C}} = \mathbb{C}^{\times}, \quad W_{\mathbb{R}} = \mathbb{C}^{\times} \rtimes \text{Gal}(\mathbb{C}/\mathbb{R}).$$

┘

**Prop. (24.6.2.4) [Characters of  $W_{\mathbb{R}}$ ].** Characters of  $W_{\mathbb{R}}$  are parameterized by a pair  $(t, \epsilon) \in \mathbb{C} \times \{\pm 1\}$  as follows:  $(t, \epsilon)$  corresponds to the character  $\phi_{(\epsilon, t)}$  s.t.

$$\phi_{t, \epsilon}(x) = |x|^t, \quad \phi_{t, \epsilon}(\mathbf{c}) = \epsilon.$$

┘

*Proof:* ?

□

**Prop. (24.6.2.5) [Semisimple Representations of  $W_{\mathbb{R}}$ ].** If  $\rho \in \text{Rep}^{\text{fd}}(W_{\mathbb{R}})$  s.t.  $\rho(w)$  is semisimple for any  $w \in W_{\mathbb{R}}$ , then  $\rho$  is semisimple. And if  $\rho$  is irreducible, then  $\dim(\rho) \leq 2$ .

And equivalence classes of irreducible 2-dimensional semisimple representations of  $W_{\mathbb{R}}$  are parameterized by a pair  $(t, m) \in \mathbb{C} \times \mathbb{Z}_+$  as follows:  $(t, m)$  corresponds to the representation

$$\text{Ind}_{W_{\mathbb{C}}}^{W_{\mathbb{R}}} \chi_{t, m}, \quad \text{where } \chi_{t, m}(z) = |z|^t e^{m \arg(z)}, \forall z \in \mathbb{C}^{\times}.$$

┘

*Proof:* Easy because the restriction of  $\rho$  to  $\mathbb{C}^{\times}$  is diagonalizable.

□

### 3 $(\mathfrak{g}, K)$ -Modules

References are [Representations of Reductive Groups, Gaitsgory].

#### $(\mathfrak{g}, K)$ -Modules

**Def. (24.6.3.1)  $(\mathfrak{g}, K)$ -Modules.** If  $K \leq G$  be a maximal compact Lie subgroup (10.5.5.11), then  $K$  acts on  $\mathfrak{g}$  by Ad. Then a  $(\mathfrak{g}, K)$ -**module** is a  $\mathbb{C}$ -vector space with a  $K$ -finite action and a  $\mathfrak{g}$  action that satisfies:

- for  $k \in K, \eta \in \mathfrak{g}$ ,  $T_k T_{\eta} T_k^{-1} = T_{\text{Ad}_k(\eta)}$ .
- The action of  $\mathfrak{k}$  on  $V$  induced by  $K$  agrees with the restriction of the action of  $\mathfrak{g}$ .

An **admissible  $(\mathfrak{g}, K)$ -module** is a  $(\mathfrak{g}, K)$ -module  $V$  s.t.  $\dim V^{\rho} < \infty$  for any  $\rho \in \text{Irr}(K)$ . ┘

**Def. (24.6.3.2) [Contragradient  $(\mathfrak{g}, K)$ -Module].** If  $M$  is an admissible  $(\mathfrak{g}, K)$ -module, then we can define its **contragradient  $(\mathfrak{g}, K)$ -module** as

$$M^{\vee} = (M^*)^{\text{alg}} = \bigoplus_{\rho \in \text{Irr}(K)} (M^{\rho})^*,$$

which is the  $K$ -finite part of the usual dual of  $M$ , and  $\mathfrak{g}$  clearly acts on it. ┘

( $\mathfrak{g}, K$ )-Modules and  $\mathfrak{g}$ -Modules

**Def. (24.6.3.3).** Let  $K_0$  be the unital component of  $K$ , then there are forgetful functors

$$\mathrm{Mod}_{(\mathfrak{g}, K)} \rightarrow \mathrm{Mod}_{(\mathfrak{g}, K_0)} \rightarrow \mathrm{Mod}_{\mathfrak{g}}.$$

┘

**Prop. (24.6.3.4).** The functor  $\mathrm{Mod}_{(\mathfrak{g}, K_0)} \rightarrow \mathrm{Mod}_{\mathfrak{g}}$  is fully faithful, and its essential image is stable under taking submodules.

┘

*Proof:* Cf.[Gaitsgory P39].

□

**Cor. (24.6.3.5).** If  $M \in \mathrm{Mod}_{(\mathfrak{g}, K)}$  is irreducible as a  $\mathfrak{g}$ -module, then it is irreducible.

┘

**Prop. (24.6.3.6).** The functor  $\mathrm{Mod}_{(\mathfrak{g}, K)} \rightarrow \mathrm{Mod}_{\mathfrak{g}}$  sends f.g. objects to f.g. objects.

┘

*Proof:* By(24.6.3.4), it suffices to consider the functor  $(\mathfrak{g}, K) - \mathrm{mod} \rightarrow (\mathfrak{g}, K_0) - \mathrm{mod}$ . Let  $M$  be f.g.  $(\mathfrak{g}, K)$ -module, and  $\cup M_i = M$  be a chain of  $(\mathfrak{g}, K_0)$ -submodules. Pick  $k \in K$  for each element of  $\pi_0(K)(\mathrm{f.m.})$ , then each  $M'_i = \sum_k k(M_i)$  is a  $(\mathfrak{g}, K)$ -submodule, thus  $M'_i = M$  for some  $i$ . Now we can choose  $j$  large that  $k(M_i) \in M_j$  for any  $k$ , then  $M_j = M$  (because we may choose  $M_i$  be f.g.  $(\mathfrak{g}, K_0)$ -modules) ?.

□

**Cor. (24.6.3.7).** The category  $\mathrm{Mod}_{(\mathfrak{g}, K)}$  is Noetherian.

┘

*Proof:* If  $M \in \mathrm{Mod}_{(\mathfrak{g}, K)}$  is f.g. and  $M_1 \subset M$ , then  $M$  is f.g. as  $\mathfrak{g}$ -module, then  $M_1$  is f.g. as  $\mathfrak{g}$ -module by(3.7.8.17). So clearly it is also f.g. as a  $(\mathfrak{g}, K)$ -module.

□

**Prop. (24.6.3.8).** For an irreducible  $(\mathfrak{g}, K)$ -module  $M$ , the underlying  $\mathfrak{g}$ -module is a direct sum of f.m. irreducible components.

┘

*Proof:* By(24.6.3.4), it suffices to prove  $M$  is a direct sum of f.m. irreducible  $(\mathfrak{g}, K_0)$ -modules.  $M$  is f.g. as a  $(\mathfrak{g}, K_0)$ -modules by the proof of(24.6.3.6), so it has a maximal submodule  $M'$  that  $N = M/M'$  is irreducible. Pick  $k \in K$  for each component of  $K$ , consider

$$M'' = \bigcap_k k(M')$$

which is a proper  $(\mathfrak{g}, K)$ -submodule of  $M$ , so it is 0, Hence the map

$$M \rightarrow \oplus (N)^k$$

is injective, where  $N^k$  is  $N$  twisted by conjugate action of  $k$ , so it is a submodule of a semisimple-module, thus semisimple.

□

Properties of  $(\mathfrak{g}, K)$ -Modules

**Cor. (24.6.3.9) [Schur's Lemma].** Schur's lemma holds for irreducible  $(\mathfrak{g}, K)$ -modules.

┘

*Proof:* It suffices to show any endomorphism  $S$  of an irreducible  $(\mathfrak{g}, K)$ -module  $M$  has an eigenvalue. But  $S$  preserves  $M^\rho$  for any  $\rho$ , and  $M^\rho$  is of f.d, thus it has an eigenvalue over  $\mathbb{C}$ .

□

**Prop. (24.6.3.10).** Any irreducible  $(\mathfrak{g}, K)$ -module has a Banach space structure.

┘

*Proof:*

□

Action of  $Z(U(\mathfrak{g}))$ 

**Prop. (24.6.3.11).** Any  $M \in \text{Mod}_{(\mathfrak{g}, K)}$  decomposes as

$$M \cong \bigoplus_{\chi \in \text{Spec}(Z(U(\mathfrak{g})))} M_\chi$$

s.t.  $Z(U(\mathfrak{g}))$  acts on each  $M_\chi$  with a generalized character  $\chi$ . Let  $\text{Mod}_{(\mathfrak{g}, K)}^\chi$  be the full subcategory of  $(\mathfrak{g}, K)$ -modules on which  $Z(U(\mathfrak{g}))$  acts with a generalized character  $\chi$ . **?** Cf.[Gaitsgory P42].  $\lrcorner$

*Proof:*  $Z(U(\mathfrak{g}))$  commutes with  $G$  thus  $K$  action, so it preserves each  $M^\rho$ , which are of f.d..  $\square$

**Prop. (24.6.3.12).** The category  $\text{Mod}_{(\mathfrak{g}, K)}^\chi$  has only f.m. isomorphism classes of irreducible objects.  $\lrcorner$

*Proof:* Cf.[Gaitsgory].  $\square$

**Prop. (24.6.3.13).** If  $M$  is a f.g.  $(\mathfrak{g}, K)$ -module, then for any  $\rho$  of  $K$ ,  $M^\rho$  is f.g. over  $Z(U(\mathfrak{g}))$ .  $\lrcorner$

*Proof:* Cf.[Gaitsgory P43]. **?**  $\square$

**Prop. (24.6.3.14).** For  $M \in \text{Mod}_{(\mathfrak{g}, K)}^\chi$ , the following are equivalent:

- $M$  is f.g..
- $M$  is of finite length.
- $M$  is admissible.

$\lrcorner$

*Proof:*  $2 \rightarrow 1$  is trivial,  $1 \rightarrow 3$  is by (24.6.3.13) (Notice that  $Z(U(\mathfrak{g}))$  acts via  $\chi$ ).

For  $3 \rightarrow 2$ : Use (24.6.3.12), there are only f.m. irreducible classes  $\rho_\alpha$ , let  $\rho = \bigoplus_\alpha \rho_\alpha$ , then if there is a chain of length  $n$ , then there are at least  $n$  linearly independent morphisms in  $\text{Hom}_K(\rho, M)$ . Thus  $n$  is bounded, because  $\dim_K \text{Hom}_K(\rho, M)$  is finite because  $M$  is admissible.  $\square$

**Cor. (24.6.3.15).** The category  $\text{Mod}_{(\mathfrak{g}, K)}^\chi$  is Artinian (4.8.3.34).  $\lrcorner$

**Cor. (24.6.3.16).** Every irreducible  $(\mathfrak{g}, K)$ -module is admissible.  $\lrcorner$

*Proof:* Firstly irreducible module are in  $\text{Mod}_{(\mathfrak{g}, K)}^\chi$  for some  $\chi$ , and then use the proposition and (24.6.3.8).  $\square$

**Cor. (24.6.3.17) [Harish-Chandra Modules].** For a  $M \in \text{Mod}_{(\mathfrak{g}, K)}$ , the following conditions are equivalent:

- $M$  is f.g. and admissible.
- $M$  is f.g. and its support over  $\text{Spec}(Z(U(\mathfrak{g})))$  is finite.
- $M$  is admissible and its support over  $\text{Spec}(Z(U(\mathfrak{g})))$  is finite.
- $M$  is of finite length.

Then such modules are called a **Harish-Chandra module**.  $\lrcorner$

*Proof:* **?**  $\square$

## 4 Reductive Groups

References are [Vogan, Representations of real reductive Lie groups] and [Gaitsgory].

**Notation (24.6.4.1).**

- Let  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ .
- Let  $G \in \mathcal{R}\text{edGrp}/\mathbb{K}$  (which is relevant, thus the complex representations of  $G$  and  $G(\mathbb{K})$  are the same, by (10.5.5.8))
- Let  $\mathcal{K} \leq G(\mathbb{K})$  be a maximal compact subgroup (10.5.5.21).

┘

**Def. (24.6.4.2) [Admissible Representation].** Define  $V^{\text{alg}}, V^\rho, V^{K\text{-fin}}$  as in (24.7.3.8)(24.6.1.5).

$V$  is called an **admissible representation** of  $G(\mathbb{R})$  if  $\dim V[\rho] < \infty$  for any  $\rho \in \text{Irr}(K)$ . Notice this notion doesn't depend on the choice of  $K$ , by (10.5.5.8). ┘

**Prop. (24.6.4.3).** For any compact open subset  $K \leq G$  and  $\rho \in \text{Irr}(K)$ ,  $V^\infty \cap V[\rho]$  is dense in  $V[\rho]$ . ┘

*Proof:* ?

□

**Cor. (24.6.4.4).** If  $V$  is admissible, then for any compact open subset  $K \leq G$ ,  $V^{K\text{-fin}} \subset V^{\text{alg}}$  by (24.6.1.9). ┘

### Infinitesimal-Equivalence Classes

**Prop. (24.6.4.5) [Infinitesimal-Equivalence].** If  $V \in \text{Rep}^{\text{adm}}(G)$ , then  $V^{\mathcal{K}\text{-fin}} \in \text{Mod}_{(\mathfrak{g}, \mathcal{K})}^{\text{adm}}$  (24.6.3.1), and the map  $V \mapsto V^{\mathcal{K}\text{-fin}}$  induces a functor

$$\text{Rep}^{\text{adm}}(G) \rightarrow \text{Mod}_{(\mathfrak{g}, \mathcal{K})}^{\text{adm}}.$$

More generally, for any  $V \in \text{Rep}(G)$ ,  $V^\infty \cap V^{K\text{-fin}} \in \text{Mod}_{(\mathfrak{g}, \mathcal{K})}$ , but may not be admissible.

Two  $V_1, V_2 \in \text{Rep}^{\text{adm}}(G)$  are called **infinitesimal-equivalent** iff their images under this functor are isomorphic. ┘

*Proof:* By (24.6.4.4),  $\mathfrak{g}$  can act on  $V^{\mathcal{K}\text{-fin}}$ , and we check  $T_k T_\eta T_{k^{-1}} = T_{\text{Ad}_k \eta}$ , which is by definition. □

**Lemma (24.6.4.6) [Real Analytic Vectors, Harish-Chandra].** Let  $V \in \text{Rep}^{\text{adm}}(G)$  and  $v \in V^{K\text{-fin}}$ , then for any  $\eta \in V^*$ , the function  $g \mapsto \eta(g(v))$  is real analytic on  $G$ , i.e.  $v$  is **weakly-analytic**. ┘

*Proof:* Cf. [Gaitsgory P37].

□

**Prop. (24.6.4.7) [ $\text{Rep}^{\text{adm}}(G)$  and  $\text{Mod}_{(\mathfrak{g}, K)}$ ].**

1. For  $V_1, V_2 \in \text{Rep}^{\text{adm}}(G)$ , if  $S : V_1 \rightarrow V_2 \in \text{TVS}$  s.t.  $S(V_1^{K\text{-fin}}) \subset V_2^{K\text{-fin}}$  and induces a map of  $(\mathfrak{g}, K)$ -modules, then  $S$  is a map of  $G$ -representations.
2. If  $V \in \text{Rep}^{\text{adm}}(G)$ ,  $M = V^{K\text{-fin}}$ , then the functors

$$\{V_1 \subset V\} \mapsto \{V_1^{K\text{-fin}} \subset M\} : (M_1 \subset M) \mapsto \overline{M_1} \subset V$$

induces mutually inverse bijections between  $G$ -subrepresentations of  $V$  and  $(\mathfrak{g}, K)$ -submodules of  $M$ .

┘

*Proof:* 1: It suffices to show for  $v_1 \in V^{K\text{-fin}}$  that  $T_g S(v_1) = S(T_g(v_1))$ . By Hahn-Banach it suffices to show for any  $\eta \in V_2^*$ ,

$$\eta(T_g S(v_1)) = \eta(S(T_g(v_1))).$$

Both sides are analytic in  $g$  by (24.6.4.6), so it suffices to show all their derivatives at 1 are equal, and use the fact  $\pi_0(K) \rightarrow \pi_0(G)$  is surjective (10.5.5.11). And the derivatives equal because  $S$  commutes with  $\mathfrak{g}$ -action.

2: Firstly  $\overline{M}_1$  is a  $G$ -representation: because  $\overline{M}_1 = ((M_1)^\perp)^\perp$  by Hahn-Banach. So it suffices to show for  $v_1 \in M_1$ ,  $\eta(g(v_1)) = 0$  for any  $\eta \in (M_1)^\perp$ . Then this uses analyticity (24.6.4.6) as above and the fact  $M_1$  is a  $(\mathfrak{g}, K)$ -subrepresentation.

For the bijection, notice  $V_1^{K\text{-fin}}$  is dense in  $V_1$  by (24.7.3.18). Conversely, for a  $(\mathfrak{g}, K)$ -submodule  $M_1$ , it suffices to show the image of  $T_{\xi_\rho \mu_{\text{Haar}}}(\overline{M}_1) \subset M_1^\rho$ , by (24.7.3.14). However  $T_{\xi_\rho \mu_{\text{Haar}}}(M_1) \in M_1^\rho$  by (24.7.3.14), so this is true by continuity.  $\square$

**Cor. (24.6.4.8) [Irreducibility of  $(\mathfrak{g}, K)$ -Modules].** For  $V \in \text{Rep}^{\text{adm}}(G)$ ,  $V$  is irreducible iff  $V^{\mathcal{K}\text{-fin}}$  is irreducible as  $(\mathfrak{g}, \mathcal{K})$ -modules.  $\text{┘}$

**Cor. (24.6.4.9) [Irreducible Unitary Representation Determined by Finite Part].** If  $V_1, V_2 \in \text{Irr}^{\text{uni}}(G)$  are infinitesimally-equivalent (24.6.4.5), then they are isomorphic.  $\text{┘}$

*Proof:* Firstly they are admissible by (24.7.5.7), so we can talk about their corresponding  $(\mathfrak{g}, K)$ -modules  $M_i$ , then  $V_i$  are the Hilbert space completion of  $M_i$  by (24.6.4.7).

Now each  $V_i$  has Hermitian forms, so  $V_i^\infty \cong (V_i^*)^\infty$ , and if  $S : V_1^\infty \cong V_2^\infty$ , then  $S^* S$  is an automorphism of  $V_1^\infty$ , thus by (24.6.4.8)(24.6.3.9) it is a scalar map, so after a scalar change, we may assume  $S$  preserves Hermitian structure thus induces an isomorphism of vector spaces  $V_1 \cong V_2$  by continuity. So by (24.6.4.7) it is an isomorphism of  $G$ -representations.  $\square$

**Def. (24.6.4.10) [Irreducible Unitarizable  $(\mathfrak{g}, \mathcal{K})$ -Modules].** An irreducible **unitarizable  $(\mathfrak{g}, \mathcal{K})$ -module** is an irreducible  $(\mathfrak{g}, \mathcal{K})$ -module  $M \in \text{Mod}_{(\mathfrak{g}, \mathcal{K})}^{\text{adm}}$  (24.6.3.1) s.t. it is the  $\mathcal{K}$ -finite part of some unitary representation  $V \in \text{Irr}^{\text{uni}}(G(\mathbb{K}))$  (24.6.4.5). By (24.6.4.8)(24.7.5.7) and (24.6.4.8), such a unitary representation  $V$  is uniquely determined if  $M$  is unitarizable.  $\text{┘}$

### Matrix Coefficients

**Def. (24.6.4.11) [Tempered Representations].**  $\pi \in \text{Irr}^{\text{adm}}(G(\mathbb{K}))$  is called an

- **square-integrable representation(Archimedean)** or **discrete series(Archimedean)** if its matrix coefficients are all contained in  $L^2(G(\mathbb{K}))$  for any  $\varepsilon \in \mathbb{R}_+$ .
- **essentially square-integrable representation(Archimedean)** if the restriction of its matrix coefficients of  $\pi$  to  $G(K)^1$  (20.5.1.2) are all contained in  $L^2(G(\mathbb{K})^1)$  for any  $\varepsilon \in \mathbb{R}_+$ .
- **tempered representation(Archimedean)** if its matrix coefficients are all contained in  $L^{2+\varepsilon}(G(\mathbb{K}))$  for any  $\varepsilon \in \mathbb{R}_+$ .
- **essentially tempered representation(Archimedean)** if the restriction of its matrix coefficients of  $\pi$  are all contained in  $L^{2+\varepsilon}(G(\mathbb{K})^1)$  (20.5.1.2) for any  $\varepsilon \in \mathbb{R}_+$ .

┘

**Prop. (24.6.4.12) [Discrete Series are Unitarizable].** Discrete-series of  $G(\mathbb{K})$  (24.6.4.11) can be realized as subrepresentations of  $L^2(G(\mathbb{K}))$ . In particular, they are always unitarizable.  $\text{┘}$

*Proof:* Cf. [Dixmier,  $C^*$ -algebras] Chap14. ?

 $\square$

**Irreducible Admissible  $(\mathfrak{g}, K)$ -Modules of  $\mathrm{GL}(n; \mathbb{R})$** **Thm. (24.6.4.13)** [Classification of  $\mathrm{Irr}_{(\mathfrak{g}, K)}^{\mathrm{adm}}(\mathrm{GL}(n; \mathbb{R}))$ ]. ┘

*Proof:* ? Cf. [Knapp and G. Zuckerman, Classification theorems for representations of semisimple Lie groups]. □

**Prop. (24.6.4.14)** [Lie Theory]. For  $V \in \mathrm{Mod}_{(\mathfrak{g}, K)}^{\mathrm{adm}}(\mathrm{GL}(2; \mathbb{R})^+)$ ,

- $V^k$  is the space of all vectors  $x \in V$  that  $Hx = kx$ .
- If  $x \in V^k$ , then  $Rx \in V^{k+2}, Lx \in V^{k-2}$ .
- If  $0 \neq x \in V^k$ , then  $\mathbb{C}x = V^k, \mathbb{C}R^n x = V^{k+2n}, \mathbb{C}L^n x = V^{k-2n}$  and

$$V = \mathbb{C}x \oplus \bigoplus_{n>0} \mathbb{C}R^n x \oplus \bigoplus_{n>0} \mathbb{C}L^n x.$$

- Suppose  $\Delta = \lambda$  on  $V$ , then if  $x \in V^k$ , then

$$LRx = (-\lambda - \frac{k}{2}(1 + \frac{k}{2}))x, \quad RLx = (-\lambda + \frac{k}{2}(1 - \frac{k}{2}))x.$$

┘

*Proof:* 1: Let  $W = iH$ . If  $x \in V^k$ , then

$$Wx = \frac{\partial}{\partial t} \pi(e^{tW})x = \frac{d}{dt} \pi(k_t)x = \frac{d}{dt} e^{ikt}x = ikx$$

thus  $Hx = kx$ . And we know  $V$  decomposes as direct sums of representations of  $K$  (24.7.3.5), thus the result follows.

2: Clear from (3.7.2.11).

3: Because the RHS is a  $\mathfrak{g}$ -submodule, by representation of  $\mathfrak{sl}_2(\mathbb{C})$  (24.4.1.11). And it is also a  $K$ -subrepresentation, by item1.

4: By (24.4.1.11). □

**Cor. (24.6.4.15)** [Non-Discrete Case]. For any  $\lambda, \mu \in \mathbb{C}$  that  $\lambda \neq \frac{k}{2}(1 + \frac{k}{2})$  for  $k$  even/odd, there exists at most one isomorphism class of irreducible admissible even/odd  $(\mathfrak{g}, K)$ -module  $V$  on which  $\Delta, I$  acts by  $\lambda, \mu$  respectively, and the  $K$ -type is one vector  $f_k$  for each  $k \in \mathbb{Z}$ . ┘

*Proof:* This follows from the classification of representation of  $\mathfrak{sl}(2; \mathbb{C})$  (24.4.1.11). Notice the action of  $K$  is controlled by (24.6.4.14) item1. □

**Cor. (24.6.4.16)** [Discrete Case]. Let  $k \in \mathbb{Z}^+$  and  $\lambda = \frac{k}{2}(1 + \frac{k}{2})$ . Let  $V$  be an irreducible admissible  $(\mathfrak{g}, K)$ -module with parity equals  $k$ , Let  $\Sigma$  be the  $K$ -types of  $V$ , then  $\Sigma$  is one of the following sets:

- $\Sigma^+(k) = \{l \in \mathbb{Z} | l \equiv k \pmod{2}, l \geq k\}$
- $\Sigma^-(k) = \{l \in \mathbb{Z} | l \equiv k \pmod{2}, l \leq -k\}$
- $\Sigma^0(k) = \{l \in \mathbb{Z} | l \equiv k \pmod{2}, -k < l < k\}$

And there are at most one isomorphism class with each  $\Sigma$ . ┘

*Proof:* This follows from the classification of representation of  $\mathfrak{sl}(2; \mathbb{C})$  (24.4.1.11). Notice the action of  $K$  is controlled by (24.6.4.14) item1. □

**Def. (24.6.4.17)**  $[H(s_1, s_2, \varepsilon)]$ . If  $\lambda \geq 1/4$ , let  $s - 1/2$  be the square root of  $1/4 - \lambda$  which is imaginary, and let  $s_1, s_2$  be determined that  $\mu = s_1 + s_2, s = \frac{1}{2}(s_1 - s_2 + 1)$ .

Consider the 1-dimensional representation  $\sigma$  of  $B(\mathbb{R})^+$  that

$$\sigma\left(\begin{bmatrix} y_1 & x \\ & y_2 \end{bmatrix}\right) = \operatorname{sgn}(y_1)^\varepsilon |y_1|^{s_1} |y_2|^{s_2},$$

If  $s_1, s_2$  are purely imaginary (i.e.  $\mu$  is purely imaginary), then this representation is unitary, and we can consider the induced representation on  $\operatorname{GL}(2; \mathbb{R})^+$  (24.7.4.3), then it is a unitary representation of  $\operatorname{GL}(2; \mathbb{R})^+$ . For  $f \in \operatorname{ind}_{B(\mathbb{R})^+}^G \sigma$ ,

$$f\left(\begin{bmatrix} y_1 & x \\ & y_2 \end{bmatrix} g\right) = \operatorname{sgn}(y_1) |y_1|^{s_1+1/2} |y_2|^{s_2-1/2} f(g)$$

and

$$(f_1, f_2) = \frac{1}{2\pi} \int_0^{2\pi} f_1(k_\theta) \overline{f_2(k_\theta)} d\theta.$$

so  $H(s_1, s_2, \varepsilon)$  is identical to  $L^2[-\pi/2, \pi/2]$  by Iwasawa decomposition (16.6.1.3). Then its  $K$ -finite vectors can be determined, which are sums of

$$f_l(g) = u^{s_1+s_2} y^s e^{il\theta}, l \equiv \varepsilon \pmod{2}.$$

so  $H(s_1, s_2, \varepsilon)$  is admissible.

Even if  $\mu$  is not purely imaginary, we can still define  $H(s_1, s_2, \varepsilon)$  as above, and we consider its smooth vectors  $H^\infty(s_1, s_2, \varepsilon)$ , which are just the smooth functions in  $H(s_1, s_2, \varepsilon)$ , by (24.6.1.11).  $\square$

**Prop. (24.6.4.18).** We can define the regular action of  $G$  on  $H(s_1, s_2, \varepsilon)$  (but may not be unitary), and the subspace  $H^\infty(s_1, s_2, \varepsilon)$  is the space of smooth vectors for this representation.  $\square$

*Proof:* It suffices to define the regular action of  $G$  on  $H^\infty(s_1, s_2, \varepsilon)$  and show it is a bounded operator, so that it can be extended to  $H(s_1, s_2, \varepsilon)$  by continuity.

By Cartan decomposition (16.5.5.1),  $G$  is generated by  $K$  and diagonal matrices of positive entries.  $\rho(K)$  clearly preserves the inner product, so it suffices to consider these matrices.

By (26.1.2.2) and the calculation  $d\theta' = y_1 y_2 D(\theta)^{-2} d\theta$ , we have

$$\int_0^{2\pi} |\pi(\operatorname{diag}(y_1, y_2)) f(k_\theta)|^2 d\theta = (y_1 y_2)^{s_1-1/2} \int_0^{2\pi} D(\theta)^{-s_1+s_2+1} |f(k_{\theta'})|^2 d\theta'.$$

where

$$\theta' = \arctan\left(\frac{y_1}{y_2} \tan \theta\right), \quad D(\theta) = \sqrt{y_1^2 \sin^2 \theta + y_2^2 \cos^2 \theta}.$$

$D(\theta)$  is bounded above and below, thus  $\pi(\operatorname{diag}(y_1, y_2))$  is a bounded operator. Also to show this representation is continuous, it suffices to show for  $|f|_2$  small and  $y_1, y_2$  small,  $|\pi(\operatorname{diag}(y_1, y_2)) f|_2$  is small. And this is also a consequence of the above formula.  $\square$

**Lemma (24.6.4.19).** For  $f_l \in H(s_1, s_2, \varepsilon)$  as in (24.6.4.17), we have

$$H f_l = l f_l, \quad R f_l = \left(s + \frac{l}{2}\right) f_{l+2}, \quad L f_l = \left(s - \frac{l}{2}\right) f_{l-2}, \quad \Delta f_l = \lambda f_l, \quad I f_l = \mu f_l$$

$\square$



*Proof:* Clear from (26.1.2.5) and definition of  $f_l$  (24.6.4.17).  $\square$

**Prop. (24.6.4.20) [Existence of  $(\mathfrak{g}, K)$ -Modules].** Let  $s = \frac{1}{2}(s_1 - s_2 + 1)$ ,  $\lambda = s(1 - s)$ ,  $\mu = (s_1 + s_2)$ , then (subquotients) of the  $(\mathfrak{g}, K)$ -module  $\mathfrak{H}$  of  $H(s_1, s_2, \varepsilon)$  afford classes in (24.6.4.15) and (24.6.4.16). More precisely,  $\Delta$  and  $I$  acts by scalars  $\lambda, \mu$  respectively, and

- If  $s$  is not of the form  $k/2$ ,  $k \equiv \varepsilon \pmod{2}$ , then  $\mathfrak{H}$  is irreducible.
- If  $s \geq 1/2$  and  $s = \frac{k}{2}$  where  $k \geq 1$  is an integer that  $k \equiv \varepsilon \pmod{2}$ , then  $\mathfrak{H}$  has two irreducible invariant subspaces  $\mathfrak{H}_+, \mathfrak{H}_-$  with  $K$ -types  $\Sigma_+, \Sigma_-$  respectively, and the quotient  $\mathfrak{H}/\mathfrak{H}_+ \oplus \mathfrak{H}_-$  is irreducible and has  $K$ -type  $\Sigma^0(k)$ .
- If  $s \leq 1/2$  and  $s = 1 - \frac{k}{2}$  where  $k \geq 1$  is an integer that  $k \equiv \varepsilon \pmod{2}$ . Then  $\mathfrak{H}$  has an invariant subspace  $\mathfrak{H}^0$  with  $K$ -types  $\Sigma^0(k)$  and the quotient  $\mathfrak{H}/\mathfrak{H}^0$  decomposes into two irreducible invariant subspaces  $\mathfrak{H}^+, \mathfrak{H}^-$  with  $K$ -types  $\Sigma_+, \Sigma_-$  respectively.

┘

*Proof:* The action of  $H, R, L, \Delta, I$  is all clear from (24.6.4.19), and the decomposition and irreducibility is all clear from the representation theory of  $\mathfrak{sl}_2$  and (24.6.4.15)(24.6.4.16).  $\square$

**Prop. (24.6.4.21) [List of Irreducible Admissible  $(\mathfrak{g}, K)$ -Modules for  $\mathrm{GL}(2; \mathbb{R})^+$ ].** Every irreducible admissible  $(\mathfrak{g}, K)$ -module may be realized as the space of  $K$ -finite vectors in an admissible representation of  $G$  on a Hilbert space. Let  $\lambda, \mu \in \mathbb{C}$ , and  $\varepsilon = 0, 1$ .

- If  $\lambda$  is not of the form  $\frac{k}{2}(1 - \frac{k}{2})$ , where  $k \equiv \varepsilon \pmod{2}$ , then there exists a unique irreducible admissible  $(\mathfrak{g}, K)$ -module of parity  $\varepsilon$  on which  $\Delta, I$  acts by scalars  $\lambda, \mu$ , denoted by  $P_\mu(\lambda, \varepsilon)$ . These are called the **principal series**.
- If  $\lambda = \frac{k}{2}(1 - \frac{k}{2})$  for some  $1 \leq k \equiv \varepsilon \pmod{2}$ , then there exists three (two for  $k = 1$ ) irreducible admissible  $(\mathfrak{g}, K)$ -module of parity  $\varepsilon$  on which  $\Delta, I$  acts by scalars  $\lambda, \mu$ . Their  $K$ -types are  $\Sigma^\pm, \Sigma^0$  respectively. The irreducible admissible  $(\mathfrak{g}, K)$ -modules of  $K$ -types  $\Sigma^\pm$  are denoted by  $D_\mu^\pm(k)$ . If  $k > 1$ ,  $D_\mu^\pm(k)$  are called **discrete-series** and for  $k = 1$  they are called **limits of discrete-series**.

┘

*Proof:* This is a consequence of (24.6.4.15)(24.6.4.16) and (24.6.4.20).  $\square$

**Prop. (24.6.4.22) [List of Irreducible Admissible  $(\mathfrak{g}, K)$ -Modules for  $\mathrm{GL}(2; \mathbb{R})$ ].** Let  $\mu \in \mathbb{C}$ , and  $\varepsilon = 0, 1$ .

- The f.d. representations are obtained by tensoring the symmetric powers of the standard representation of  $G$  with the 1-dimensional representation of the form  $\chi \circ \det$ .
- If  $\chi_1, \chi_2$  are quasi-characters of  $\mathbb{R}^*$  that  $\chi_1 \chi_2^{-1}$  is not of the form  $y \mapsto \mathrm{sgn}(y)^\varepsilon |y|^{k-1}$ , where  $k \equiv \varepsilon \pmod{2}$ , then there is a irreducible  $(\mathfrak{g}, \mathrm{O}(2; \mathbb{R}))$ -module  $\pi(\chi_1, \chi_2)$ .
- If  $\mu \in \mathbb{R}$  and  $k \geq 1$  is an integer, then there are representations  $D_\mu(k) = (D_\mu^+(k) \oplus D_\mu^-(k))$ , called **discrete series** if  $k \geq 2$  and **limits of discrete series** if  $k = 1$ .

┘

*Proof:* Use (24.1.1.12) on  $\mathrm{SO}(2; \mathbb{R}) \subset \mathrm{O}(2, \mathbb{R})$ . It turns out  $P_\mu(\lambda, \varepsilon)$  can be extended in two ways to a representation of  $\mathrm{O}(2, \mathbb{R})$ :  $H(s_1, \varepsilon_1, s_2, \varepsilon_2)$ , corresponding to item2. It is unitarizable as  $H(s_1, s_2, \varepsilon)$  is unitarizable.

The (limits of) discrete series are conjugate in pairs and combined to give a representation of  $\mathrm{O}(2, \mathbb{R})$ , which is item3. It is unitarizable as it is just  $H(s_1, s_2, \varepsilon)/\mathfrak{H}^0$ .

The case  $D_\mu^0(k)$  are also of type  $I$  and can be extended, and this irreducible representation are exactly the smooth parts of the symmetric representation of  $GL(2, \mathbb{R})$  twisted by  $\chi \circ \det$ .  $\square$

**Cor. (24.6.4.23) [Contragradient].** Let  $G = GL(2; \mathbb{R})$ ,  $K = O(2; \mathbb{R})$ ,  $(\pi, V)$  be an irreducible admissible  $(\mathfrak{g}, K)$ -module, then the contragradient  $\hat{\pi}$  is isomorphic to the  $(\mathfrak{g}, K)$ -module  $T'_k = T_{k-t}$  and  $T'_X = T_{X-t}$ .

And it is also isomorphic to  $\pi \otimes (\omega \circ \det)$ .  $\lrcorner$

*Proof:* This is a consequence of the classification of irreducible  $(\mathfrak{g}, K)$ -modules (24.6.4.21), where the  $K$ -type are unchanged, eigenvalues of  $\Delta$  are unchanged, and eigenvalue of  $I$  is changed to  $-\mu$ , so they are isomorphic.  $\square$

$\text{Irr}^{\text{adm}}(GL(n; \mathbb{K}))$

**Prop. (24.6.4.24).** Any  $\pi \in \text{Irr}^{\text{adm}}(GL(n; \mathbb{K}))$  is infinitesimally-equivalent to an isobaric sum  $\boxtimes_i \pi_i$  where each  $\pi_i \in \text{Irr}^{\text{adm}}(GL(n_i; \mathbb{K}))$  is essentially square-integrable.  $\lrcorner$

*Proof:* Cf. [Getz]P284.  $?$   $\square$

**Thm. (24.6.4.25) [Classifying  $\text{Irr}^{\text{adm}}(GL(n; \mathbb{R}))$  up to Infinitesimal-Equivalence].** Let  $n_1, \dots, n_r \in \{1, 2\}$  and  $n = \sum_i n_i$ , and  $\sigma_i \in \text{Irr}^{n_i}(W_{\mathbb{R}})$  with parameter  $t_i$   $?$ , then  $\lrcorner$

*Proof:*  $\square$

**Thm. (24.6.4.26) [Classifying  $\text{Irr}^{\text{adm}}(GL(n; \mathbb{C}))$  up to Infinitesimal Equivalence, Želobenko--Naimark].** Let  $\sigma_1, \dots, \sigma_n$  be 1-dimensional representations of  $\mathbb{C}^\times$  of the form  $\sigma_i(z) = \arg(z)^{l_i} |z|_{\mathbb{C}}^{t_i}$  where  $l_i \in \mathbb{Z}$ ,  $t_i \in \mathbb{C}$  (24.6.2.1) and  $\text{Re}(t_1) \geq \dots \geq \text{Re}(t_n)$ , then:

- $\sigma_1 \boxtimes \dots \boxtimes \sigma_n \in \text{Irr}^{\text{adm}}(GL(n; \mathbb{C}))$  has a unique quotient  $\sigma_1 \boxplus \dots \boxplus \sigma_n$ .
- Each  $\pi \in \text{Irr}^{\text{adm}}(GL(n; \mathbb{C}))$  is infinitesimally equivalent (24.6.4.5) to a representation of the form  $\sigma_1 \boxplus \dots \boxplus \sigma_n$ .
- Two such representations  $\sigma_1 \boxplus \dots \boxplus \sigma_n, \sigma'_1 \boxplus \dots \boxplus \sigma'_n$  are infinitesimal equivalent iff  $(\sigma_1, \dots, \sigma_n)$  is a permutation of  $(\sigma'_1, \dots, \sigma'_n)$ .  $\lrcorner$

*Proof:*  $?$   $\square$

### Langlands Classifications(Archimedean Local Langlands Correspondence)

References are [Lan89, Kal16a], [Langlands-Shelstad. On the definition of transfer factors].

**Prop. (24.6.4.27) [Langlands Correspondence for  $GL(n; \mathbb{C})$ ].** For any  $\phi \in \text{Rep}_{\text{ss}}^{\text{fd}}(W_{\mathbb{C}})$  of the form  $\phi \cong \text{diag}(\sigma_1, \dots, \sigma_n)$ , we define  $\pi(\rho) = \sigma_1 \boxplus \dots \boxplus \sigma_n \in \text{Irr}^{\text{adm}}(GL(n; \mathbb{C}))$  (24.6.4.26). Then this induces a bijection between equivalence classes in  $\text{Rep}^{n, \text{ss}}(W_{\mathbb{C}})$  and equivalence classes in  $\text{Irr}^{\text{adm}}(GL(n; \mathbb{C}))$ , by (24.6.4.26).

And we have

$$L(\rho; s) = L(\pi(\rho); s), \quad \epsilon(\rho, \psi_v; s) = \epsilon(\pi(\rho), \psi_v; s) \text{ (31.1.1.5)(29.5.4.3)}.$$

$\lrcorner$

*Proof:* For the compatibility of  $L$  and  $\epsilon$ -factors, Cf.[Jacquet, Principal  $L$ -Functions].  $\square$

**Remark (24.6.4.28).** Langlands[Lan89] reduced the classification of irreducible admissible representations of  $G(\mathbb{R})$  to classification of the subset of irreducible tempered representations.

The set of irreducible tempered representations of semi-simple groups are classified by [Knapp-Zuckerman1982].  $\lrcorner$

**Thm. (24.6.4.29) [Archimedean Langlands Correspondence].** For  $K \in \{\mathbb{R}, \mathbb{C}\}$  and  $G \in \text{RedGrp}/K$ , the local Langlands correspondence(29.3.2.2) is true for  $G$ . In particular,  $\lrcorner$

*Proof:* ?  $\square$

**Remark (24.6.4.30)[ $L$ -Indistinguishable Representations, Shelstad].** Two phenomena contribute to this  $L$ -indistinguishability(29.3.2.2): discrete series with the same infinitesimal character, and reducibility of standard induced tempered representations. [Shelstad,  $L$ -indistinguishability for real group] has quantified the statement that these are the only contributing factors, and she has explored the consequences of  $L$ -indistinguishability.  $\lrcorner$

## 24.7 Unitary Representations of Locally Compact Groups

References are [Kna01, Fol16] and [Representation of Lie Groups notes, Etingof].

**Remark (24.7.0.1) [Non-compact Lie Groups].** In this section (continuous) representations of (possibly non-compact) LCG are considered. Notice the representations of compact groups is already considered in [Compact Group](#).

The feature of representations of non-compact groups is that most interesting representations are infinite dimensional and involves topologies.  $\lrcorner$

**Notation (24.7.0.2).**

- Use notations on locally compact groups defined in [Harmonic Analysis on Locally Compact Groups](#).
- Use notations on measure theory defined in [Measure Theory and Functions on  \$\mathbb{R}^n\$](#) .
- Let  $G \in \text{LCG}$ .
- Let  $\mu$  be a left Haar measure on  $G$  and  $\rho$  a right Haar measure on  $G$ .

$\lrcorner$

### 1 Basics

References are [Fol16]Chap3.

**Def. (24.7.1.1) [Unitarizable Representations].** A **unitary representations** of  $G$  is a representation  $\pi$  of  $G$  on a Hilbert space  $\mathcal{H}$  s.t.  $\text{Im}(\pi) \subset U(\mathcal{H})$  (12.7.4.7). The category of unitarizable representations of  $G$  is denoted by  $\text{Rep}^{\text{uni}}(G)$ .

$(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}}(G)$  is called an **irreducible unitary representation** if it has no non-trivial invariant closed subspaces. the category of Irreducible unitarizable representations of  $G$  is denoted by  $\text{Irr}^{\text{uni}}(G)$  or  $\hat{G}$ .  $\lrcorner$

**Prop. (24.7.1.2).** For a pre-Hilbert space  $\mathcal{H}$  and an action  $\pi : G \times \mathcal{H} \rightarrow \mathcal{H}$ ,  $\pi$  is continuous iff  $\pi : G \mapsto U(\mathcal{H})$  is continuous in the strong operator topology or the weak operator topology.  $\lrcorner$

*Proof:* By (12.7.4.7), the strong and weak operator topologies on  $U(\mathcal{H})$  are the same.

It is clear that if  $\pi : G \times \mathcal{H} \rightarrow \mathcal{H}$  is continuous then  $\pi : G \rightarrow U(\mathcal{H})$  is continuous in the strong operator topology. Conversely, use the inequality

$$\|\pi(g_1)\varphi_1 - \pi(g_2)\varphi_2\| = \|\varphi_1 - \pi(g_1^{-1}g_2)\varphi_2\| \leq \|\varphi_1 - \varphi_2\| + \|\varphi_2 - \pi(g_1^{-1}g_2)\varphi_2\|.$$

$\square$

**Example (24.7.1.3) [Left and Right Unitary Regular Representations].** The **left regular action** of  $G$  on  $L^2(G)$  is defined to be  $L_y f(x) = f(y^{-1}x)$ . The **right regular action** of  $G$  on  $L^2(G)$  is defined to be  $R_y f(x) = f(xy)$ .

These are continuous unitary representations of  $G$ , and they extends the representations on  $C_c(G)$  (12.10.1.1).

More generally, if  $G$  acts on the right on  $X \in \text{LCH}$  and  $\rho$  a right- $G$ -invariant Radon measure on  $X$  (12.3.1.22), then  $G$  has a right acts on  $L^2(X, \mu)$  by  $R_g(f)(x) = f(xg)$  for any  $f \in L^2(X, \rho)$  and  $g \in G$ .  $\lrcorner$

*Proof:* These follows from (12.10.1.2) and (24.7.1.13), using the fact  $C_c(X)$  is dense in  $L^2(X, \rho)$  (12.3.8.7).  $\square$

**Def. (24.7.1.4) [Intertwining Operators].** if  $\pi_1, \pi_2 \in \text{Rep}^{\text{uni}}(G)$ , then the space  $C(\pi_1, \pi_2)$  of **intertwining operators** of  $\pi_1, \pi_2$  as:

$$C(\pi_1, \pi_2) = \{T : \mathcal{H}_{\pi_1} \rightarrow \mathcal{H}_{\pi_2} : T\pi_1(x) = \pi_2(x)T, \quad \forall x \in G\}.$$

And denote  $C(\pi_1, \pi_1)$  by  $C(\pi_1)$ .  $\lrcorner$

**Lemma (24.7.1.5).** The adjoint operator  $S \mapsto S^*$  induces a bijection between the spaces  $C(\pi_1, \pi_2) \cong C(\pi_2, \pi_1)$ .  $\lrcorner$

**Prop. (24.7.1.6) [Cyclic Representations].** Every  $(\pi, \mathcal{H}_\pi) \in \text{Rep}^{\text{uni}}(G)$  is a direct sum of cyclic representations.  $\lrcorner$

*Proof:* Use Zorn's lemma, because for any proper-contained subspace  $V \subset \mathcal{H}_\pi$ , we can choose a non-zero element  $u \in V^\perp$ .  $\square$

**Lemma (24.7.1.7).** Suppose  $(\pi, \mathcal{H}_\pi) \in \text{Rep}^{\text{uni}}(G)$  and  $M \subset \mathcal{H}_\pi$  is a closed subspace. Let  $P$  be the orthogonal projection onto  $M$ , then  $M$  is invariant under  $\pi$  iff  $P \in C(\pi)$ .  $\lrcorner$

*Proof:* If  $P \in C(\pi)$  and  $v \in M$ , then  $\pi(x)v = \pi(x)Pv = P\pi(x)v \in M$ , so  $M$  is  $\pi$ -invariant. Conversely if  $M$  is  $\pi$ -invariant, then so does  $M^\perp$ , so  $\pi(x)Pv = \pi(x)v = P\pi(x)v$ , and also for  $v \in M^\perp$ , so  $\pi(x)P = P\pi(x)$ , for any  $x$ .  $\square$

**Prop. (24.7.1.8) [Schur's Lemma].**

- $\pi \in \text{Rep}^{\text{uni}}(G)$  is irreducible iff  $C(\pi)$  consists only of scalar multiples of identity.
- If  $\pi_1, \pi_2 \in \text{Irr}^{\text{uni}}(G)$  are non-equivalent, then  $C(\pi_1, \pi_2) = 0$ .  $\lrcorner$

*Proof:* 1: if  $\pi$  is reducible, then it contains a non-trivial projection by lemma (24.7.1.7). Conversely, if  $T \neq cI \in C(\pi)$ , then we consider  $A = \frac{1}{2}(T + T^*)$ ,  $B = \frac{1}{2i}(T - T^*)$ , then at least one of them are not  $cI$ . But they are normal, thus by symbolic calculus (12.9.3.3) any  $\chi_E(A)$  for some  $E \subset \mathbb{R}$  Borel is non-trivial (because the spectrum of  $A$  is not a single point) and commutes with  $\pi$ , so  $\mathcal{H}_\pi$  is reducible by (24.7.1.7) again.

2: By (24.7.1.5), for  $T \in C(\pi_1, \pi_2)$ ,  $T^* \in C(\pi_2, \pi_1)$ ,  $TT^* = cI$ ,  $T^*T = cI$ . so  $T = 0$  or  $c^{-1/2}T$  is unitary, and it is an isomorphism between  $\pi_1, \pi_2$ .  $\square$

**Cor. (24.7.1.9).** if  $G$  is Abelian, then  $\dim \pi = 1$  for any  $\pi \in \text{Irr}^{\text{uni}}(G)$ .  $\lrcorner$

*Proof:* If  $\pi$  is a representation of  $G$ , then any  $\pi(x)$  commutes with  $\pi$ , thus  $\pi(x) = c_x \text{id}$  for some  $c_x$ , so every subspace of  $\mathcal{H}_\pi$  is irreducible, thus  $\dim \mathcal{H} = 1$ .  $\square$

**Prop. (24.7.1.10) [Unitary Representations and  $L^1(G)$ -Representations].** Any  $(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}} G$  determines a non-degenerate \*-representation of  $L^1(G)$  (14.1.1.12) by

$$f \mapsto \pi(f) = \int f(x)\pi(g)d\mu(g).$$

Conversely, any non-degenerate \*-representation of  $L^1(G)$  (14.1.1.12) arises from a unitary representation of  $G$ .  $\lrcorner$

*Proof:* For  $\pi \in \text{Rep}^{\text{uni}}(G)$  and  $f \in L^1(G)$ ,  $u \in \mathcal{H}$ , we can define  $\pi(f)u$  as the vector-valued integrations of  $g \mapsto \pi(g)u$  (12.8.3.20), so

$$(\pi(f)u, v) = \int f(x)(\pi(x)u, v) d\mu(x)$$

for any  $v \in \mathcal{H}$ . Then it is clear that  $u \mapsto \pi(f)u$  is linear, and  $\|\pi(f)u\| \leq \|u\| \|f\|$  by (12.8.3.20). Thus  $\|\pi(f)\| \leq \|f\|_1$ .

For the  $*$ -algebra structure (12.10.2.4), it suffices to prove that

$$\pi(f * f') = \pi(f)\pi(f'), \quad \pi(f^*) = \pi(f)^*,$$

which are true for formal reason:

$$\pi(f * g) = \int \int f(y)g(y^{-1}x)\pi(x)d\mu(y)d\mu(x) = \int \int f(y)g(x)\pi(yx)d\mu(x)d\mu(y) = \pi(f)\pi(g),$$

$$\pi(f^*) = \int \Delta(x^{-1})\overline{f(x^{-1})}\pi(x)dx = \int \overline{f(x)}\pi(x^{-1})dx = \int (f(x)\pi(x))^*dx = \pi(f)^*,$$

and verified by supplying  $u, v$ . For the non-degeneracy, for any  $u \in \mathcal{H}^\times$ , choose a precompact nbhd  $V$  of identity that  $\|\pi(g)u - u\| < \|u\|$  for  $g \in V$ , and let  $f = |V|^{-1}\chi_V$ , then it can be verified that  $\|\pi(f)u\| \neq 0$ .

For the converse, Cf.[Folland P79-81] ? □

**Remark (24.7.1.11) [Left Regular Representations].** For the left regular representation  $\pi_L$  of  $G$  on  $L^2(G)$  (24.7.1.3),  $\pi_L(f)g = f * g$  for any  $f \in L^1(G)$ ,  $g \in L^2(G)$  (12.10.2.7), by (12.10.2.4). ┘

**Prop. (24.7.1.12).** For  $\pi \in \text{Rep}^{\text{uni}}(G)$ ,

- The bicommutant (14.1.2.2) of  $\pi(G)$  and  $\pi(L^1(G))$  are identical.
  - $T \in \mathcal{L}(\mathcal{H})$  intertwines  $\pi$  iff it commutes with every  $\pi(f) \in \pi(L^1(G))$ .
  - A closed subspace  $M$  of  $\mathcal{H}$  is invariant under  $\pi$  iff  $\pi(f)M \subset M$  for any  $f \in L^1(G)$ .
- ┘

*Proof:* 1: Cf.[Folland, P82]. ?

2 follows from 1 noticing the fact that  $T$  commutes with an algebra iff it commutes with its bicommutant.

3 follows from 2 and (24.7.1.7). □

**Prop. (24.7.1.13) [Completion of Unitary Representation].** If  $\mathcal{H}_0$  is a pre-Hilbert space that  $G$  is a topological group acting continuously on  $\mathcal{H}_0$  that preserves the inner product, then if  $\mathcal{H}$  is the Hilbert completion of  $\mathcal{H}_0$ , then the action of  $G$  extends to a continuous unitary action on  $\mathcal{H}$ . ┘

*Proof:* The extension is clear as  $\|\pi(g)f\| = \|f\|$ . For the continuity, for any  $v \in \mathcal{H}$ ,  $g \in G$ ,  $\varepsilon \in \mathbb{R}_+$ , choose  $v_0 \in \mathcal{H}_0$  s.t.  $\|v - v_0\| < \varepsilon/6$ , then there is a nbhd  $W$  of  $g$  that for any  $g_1 \in W$ , we have  $|\pi(g_1)v_0 - \pi(g)v_0| < \varepsilon/3$ .

Then if  $\|v_1 - v\| < \varepsilon/6$  and  $g_1 \in W$ , then

$$\begin{aligned} \|\pi(g_1)v_1 - \pi(g)v\| &= \|\pi(g_1)v_1 - \pi(g_1)v_0 + \pi(g_1)v_0 - \pi(g)v_0 + \pi(g)v_0 - \pi(g)v\| \\ &\leq \|\pi(g_1)v_1 - \pi(g_1)v_0\| + \|\pi(g_1)v_0 - \pi(g)v_0\| + \|\pi(g)v_0 - \pi(g)v\| \\ &= \|v_1 - v_0\| + \|\pi(g_1)v_0 - \pi(g)v_0\| + \|\pi(g)v_0 - \pi(g)v\| \\ &\leq \varepsilon \end{aligned}$$

which shows the action is continuous. □

**Cor. (24.7.1.14).** Given a discrete subgroup  $\Gamma \leq G$ , the right regular action of  $G$  extends to a continuous unitary representation of  $G$  on  $L^2(\Gamma \backslash G, \omega)$ .  $\lrcorner$

*Proof:* Because we can approximate  $f \in L^2(\Gamma \backslash G)$  by compactly supported continuous functions (12.3.8.7), then the  $G$  action is uniformly continuous.  $\square$

**Lemma (24.7.1.15) [Auxiliary Compact Supported Function Approximation].** Let  $G$  be a locally compact Lie group and  $K$  a compact subgroup. If  $(\pi, \mathcal{H})$  is a unitary representation of  $G$  on a Hilbert space, and let  $f \neq 0 \in \mathcal{H}$ , then for any  $\varepsilon > 0$ , there is a  $\varphi \in C_c^\infty(G)$  s.t.  $\pi(\varphi)$  is self-adjoint and  $|\varphi(\rho)f - f| < \varepsilon$ .

Moreover, if  $f \in \mathcal{H}^\xi$  which is the decomposition part for  $K$ , we can assume  $\varphi(kg) = \varphi(gk) = \xi(k)^{-1}\varphi(g)$ . In particular if  $\mathcal{H}^\xi$  is f.d., we find a  $\varphi$  that  $\pi(\varphi)f = f$ .  $\lrcorner$

*Proof:* By continuity, there is a nbhd  $H$  of 1 that  $|\pi(g)f - f| < \varepsilon$ , then we can choose a  $\varphi$  positive real valued with support in  $U$  with integral 1, then  $|\pi(\varphi)f - f| < \varepsilon$  by (12.8.3.22). We can also choose  $\varphi(g) = \varphi(g^{-1})$ , then  $\pi(\varphi)$  is self-adjoint.

For the second case, notice first there is a nbhd  $V$  of 1 that  $kVk^{-1} \in U$  for any  $k \in K$  (4.12.1.6), so let  $\varphi_1$  be a positive real valued function supported in  $V$ , and let

$$\varphi_0(g) = \int_K \varphi_1(kgk^{-1})dk,$$

then  $\varphi_0$  is supported in  $U$  and  $\varphi(kgk^{-1}) = \varphi_0(g)$  for any  $k \in K$ . Assume now that  $\pi(k_\theta) = e^{ik\theta}f$ , then we can use (12.10.3.8) for  $P = G$  to see that

$$\pi(\varphi_0)f = \int_G \varphi_0(h)\pi(h)f dh = \int_G \int_K \varphi_0(hk)\pi(hk)f dk dh = \int_G \int_K \xi(k)\varphi_0(hk)dk \pi(h)f dh = \pi(\varphi)f,$$

where

$$\varphi(g) = \int_K \xi(k)\varphi_0(gk)dk = \int_K \xi(k)\varphi_0(kg)dk.$$

so  $\varphi(k) = \varphi(gk) = \xi^{-1}(k)\varphi(g)$  as required.  $\square$

### Functions of Positive Type

**Def. (24.7.1.16) [Positive-Type Function].** A function of **positive type** on  $G$  is a function  $\varphi \in L^\infty(G)$  that defines a positive linear functional on the  $B^*$ -algebra  $L^1(G)$ , i.e.

$$\int f(x)\overline{f(y)}\varphi(y^{-1}x)d\nu(y)d\mu(x) \geq 0, \text{ where } f \in L^1(G).$$

We denote by  $\text{Pos}(G)$  the set of continuous functions of positive type on  $G$ .  $\lrcorner$

**Prop. (24.7.1.17).** If  $\varphi$  is of positive type, then so does  $\overline{\varphi}$ . (Easy calculation).  $\lrcorner$

**Prop. (24.7.1.18).** If  $\pi \in \text{Rep}^{\text{uni}}(G)$  and  $u \in \mathcal{H}_\pi$ , then  $\varphi(x) = (\pi(x)u, u) \in \text{Pos}(G)$  (24.7.1.16).  $\lrcorner$

*Proof:*  $\varphi$  is continuous by definition, so if  $f \in L^1$ , then

$$\int \int f(x)\overline{f(y)}\varphi(y^{-1}x)d\mu(x)d\mu(y) = \int \int (f(x)\pi(x)u, f(y)\pi(y)u)d\mu(x)d\mu(y) = \|\pi(f)u\|^2 \geq 0.$$

$\square$

**Prop. (24.7.1.19).** If  $f \in L^2(G)$ , then  $f * \tilde{f} \in \text{Pos}(G)$  (12.10.1.3), where  $\tilde{f}(x) = \overline{f(x^{-1})}$ . ┘

*Proof:* Cf.[Folland, P84]. □

**Prop. (24.7.1.20) [Cyclic Representations and Functions of Positive Type].** Any  $\varphi \in \text{Pos}(G)$  arises from a cyclic unitary representation  $\pi_\varphi \in \text{Rep}^{\text{uni}}(G)$  and a cyclic vector  $\varepsilon$  as in (24.7.1.18).

Moreover, if  $\rho, \pi \in \text{Rep}^{\text{uni}}(G)$  are cyclic with cyclic vectors  $u, v$  s.t.  $(\pi(g)u, u) = (\rho(g)v, v)$  for any  $g \in G$ , then there exists  $T \in C(\pi, \rho)$  (24.7.1.4) s.t.  $T(u) = v$ . ┘

*Proof:* Cf.[Folland P84-85]. ?

For the last assertion, by the hypothesis,

$$(\pi(g)u, \pi(h)u) = (\pi(h^{-1}g)u, u) = (\rho(h^{-1}g)v, v) = (\rho(g)v, \rho(h)v),$$

so if we define  $T(\pi(g)u) = \rho(g)v$ , then  $T$  extends by linearity to an isometry between the linear span of  $\{\pi(g)u\}$  to the linear span of  $\{\rho(g)v\}$ . Thus it extends by continuity to a map from  $\mathcal{H}_\pi$  to  $\mathcal{H}_\rho$ . It is easy to verify that  $T \in C(\pi, \rho)$ . □

**Cor. (24.7.1.21).** If  $\varphi \in \text{Pos}(G)$ , then  $\varphi$  can be chosen to be continuous. ┘

**Cor. (24.7.1.22).** If  $\varphi \in \text{Pos}(G)$ , then  $\|\varphi\|_\infty = \varphi(1)$ , and  $\varphi(x^{-1}) = \overline{\varphi(x)}$ . ┘

*Proof:*  $\varphi(x) = (\pi(x)u, u)$  for some representation  $\pi$  and  $u \in \mathcal{H}_\pi$  by (24.7.1.20), so  $|\varphi(x)| \leq \|u\|^2 = \varphi(1)$  and  $\varphi(x^{-1}) = (\pi(x^{-1})u, u) = (u, \pi(x)u) = \overline{\varphi(x)}$ . □

**Def. (24.7.1.23) [Normalized Positive-Type Functions].** We set:

- $\text{Pos}_0(G) = \{\varphi \in \text{Pos}(G) : \|\varphi\|_\infty \leq 1\} = \{\varphi \in \text{Pos}(G) : 0 \leq \varphi(1) \leq 1\}$  (24.7.1.22).
- $\text{Pos}_1(G) = \{\varphi \in \text{Pos}(G) : \|\varphi\|_\infty = 1\} = \{\varphi(1) = 1\}$ .

$\text{Pos}_0(G), \text{Pos}_1(G)$  are weak\*-compact subsets of  $L^\infty(G)$  by Banach-Alaoglu (12.8.4.1). ┘

**Prop. (24.7.1.24) [Cyclic Representations and  $\text{Pos}_1(G)$ ].** It follows from (24.7.1.20) that there is a bijection between  $\text{Pos}_1(G)$  and the set of equivalence classes of cyclic unitary representations of  $G$  with a fixed cyclic generator of unit length. ┘

**Thm. (24.7.1.25) [Extreme Points of  $\text{Pos}_1(G)$ ].** A  $\varphi \in \text{Pos}_1(G)$  is an extreme point iff  $\pi_\varphi$  is irreducible (24.7.1.20). And  $\text{Extrm}(\text{Pos}_0(G)) = \text{Extrm}(\text{Pos}_1(G)) \cup \{0\}$ . ┘

*Proof:* Cf.[Folland P86]. ?

For the last assertion, if  $\varphi_1, \varphi_2 \in \text{Pos}_0(G)$  and  $c_1, c_2 \in \mathbb{R}_+$  s.t.  $c_1 + c_2 = 1$  and  $c_1\varphi_1 + c_2\varphi_2 = 0$ , then  $\varphi_1(1) = \varphi_2(1) = 0$ , thus  $\varphi_1 = \varphi_2 = 0$  by (24.7.1.22). Thus  $0 \in \text{Pos}_0(G)$  is an extreme point. Similarly, if  $c_1\varphi_1 + c_2\varphi_2 \in \text{Pos}_1(G)$ , then  $\varphi_1, \varphi_2 \in \text{Pos}_1(G)$ , thus any  $\varphi \in \text{Pos}_1(G)$  is extreme iff it is extreme in  $\text{Pos}_0(G)$ . Clearly no elements in  $\text{Pos}_0(G) \setminus (\text{Pos}_1(G) \cup \{0\})$  is extreme. □

**Cor. (24.7.1.26).** The convex hull of  $\text{Pos}_1(G)$  is weak\*-dense in  $\text{Pos}_1(G)$ . ┘

*Proof:* Cf.[Fol16]P88 ? □

**Prop. (24.7.1.27) [Two Topologies Coincide].** On  $\text{Pos}_1(G)$ , the compact-open topology coincides with that of the weak\*-topology. ┘

*Proof:* Cf.[Folland Abstract Harmonic Analysis P80]. □



**Prop. (24.7.1.28).** The linear span  $B(G)$  of  $C_c(G) \cap \text{Pos}(G)$  includes all functions of the form  $f * g$  where  $f, g \in C_c(G)$ . And it is dense in  $C_c(G)$  and  $L^p(G)$  for  $p \in (0, \infty)$ .

Denote  $B^p(G) = B(G) \cap L^p(G)$ . ┘

*Proof:* By (24.7.1.19),  $\text{Pos}(G) \cap C_c(G)$  includes all functions of the form  $f * \tilde{f}$  with  $f \in C_c(G)$ , thus its linear span includes all  $f * g$  for  $f, g \in C_c(G)$  by polarization. Thus it is dense in  $C_c(G)$  and  $L^p(G)$  because we can use approximate identity (12.10.2.9). □

**Prop. (24.7.1.29) [Gelfand-Raikov].**  $\hat{G}$  separate points of  $G$ . ┘

*Proof:* Cf. [Folland Abstract Analysis P91]. □

## 2 Dual Spectra

References are [Fol16], [Dixmier, C\*-Algebras], [Fell and Dorian, Representations of \*-algebras, locally compact groups, and Banach \*-algebraic bundles, Vol I].

**Notation (24.7.2.1).**

- Let  $G, G', G_1, G_2 \in \text{LCG}$  be second-countable. ┘

**Prop. (24.7.2.2) [Only Separable Representations].** As  $G$  is second-countable,  $L^p(G)$  is separable for any  $p < \infty$  by taking a approximate identity. In particular, every cyclic representation (e.g. irreducible representations) is separable. In particular, irreducible representations are separable. By (24.7.1.6), it suffices to consider separable representations. ┘

### Group C\*-Algebras

**Def. (24.7.2.3) [Group C\*-Algebras].** For  $f \in L^1(G)$ , define  $\|f\|_* = \sup_{\pi \in \hat{G}} \|\pi(f)\|$  (24.7.1.1)(24.7.1.10). Then this is a seminorm on  $L^1(G)$  and  $\|f\|_* \leq \|f\|_1$  by (24.7.1.10). It will be shown to be a norm in (24.7.2.5). ┘

**Prop. (24.7.2.4).** For any  $f \in L^1(G)$ ,

$$\|f\|_*^2 = \sup_{\pi \in \text{Rep}^{\text{uni}}(G)} \|\pi(f)\|^2 = \sup_{\varphi \in \text{Pos}_1(G)} \int f^* * f \varphi d\mu(g) = \sup_{\varphi \in \text{Extrm}(\text{Pos}_1(G))} \int f^* * f \varphi d\mu(g) \quad (24.7.1.23).$$

┘

*Proof:*  $1 \leq 2$  is trivial, and  $2 \leq 3$  follows from the fact that  $\varphi(g) = (\pi(g)u, u)$  is of positive-type (24.7.1.18) and the fact  $\|\pi(f)u\|^2 = \int f^* * f \varphi d\mu(g)$ .

$3 \leq 4$  follows from (24.7.1.26) and Krein-Milman.  $4 \leq 1$  follows from (24.7.1.20) and (24.7.1.25). □

**Cor. (24.7.2.5) [C\*(G)].**  $\|\cdot\|_*$  (24.7.2.3) is a norm on  $L^1(G)$ , and

$$\|fg\|_* \leq \|f\|_* \|g\|_*, \quad \|f^* * f\|_* = \|f\|_*^2,$$

Thus it can be completed to a C\*-algebra (14.1.1.1), denoted by  $C^*(G)$ . ┘

*Proof:* It suffices to show that if  $f \in L^1(G)$  and  $\|f\|_* = 0$ , then  $f = 0$ . But this is because we consider the left regular unitary action of  $f$  on  $L^1(G)$  (24.7.1.3), then  $\pi(f)g = f * g$  (24.7.1.11). If we take  $g$  to be an approximate identity (12.10.2.9), then we see  $f = 0$  by (12.10.2.10). □

The rest are easy.

**Example (24.7.2.6).** If  $G$  is Abelian, then for any  $f \in L^1(G)$ ,  $\|f\|_* = |f^\vee|_\infty$ , so  $C^*(G) \cong C_0(G^\vee)$  as  $C^*$ -algebras, where  $C_0(G^\vee)$  has the point-wise convergence topology.  $\lrcorner$

**Def. (24.7.2.7) [Primitive Ideals].** By (24.7.1.10) and (24.7.1.12), There is an equivalence of categories

$$\mathcal{R}\text{ep}^{\text{uni}}(G) \cong \mathcal{R}\text{ep}^{\text{non-degenerate}}(C^*(G)). \text{?}$$

identifying irreducible representations.

For  $\pi \in \mathcal{R}\text{ep}^{\text{uni}}(G)$ , define

$$\ker(\pi) = \{f \in C^*(G) | \pi(f) = 0\},$$

which is a closed ideal of  $C^*(G)$ . Ideals of the form  $\ker(\pi)$  for  $\pi \in \widehat{G}$  are called **primitive ideals** of  $C^*(G)$ . The space of all primitive ideals of  $C^*(G)$  is denoted by  $\text{Prim}(G)$ .  $\lrcorner$

**Prop. (24.7.2.8) [Jacobson Topology].** For any non-empty set  $U \subset \text{Prim}(G)$  (24.7.2.7), define

$$\overline{U} = \{J \in \text{Prim}(G) | \cap_{I \in U} I \subset J\},$$

and also set  $\overline{\emptyset} = \emptyset$ . Then for any  $U, V \subset \text{Prim}(G)$ ,

$$U \subset \overline{U}, \quad \overline{\overline{U}} = U, \quad \overline{U \cup V} = \overline{U} \cup \overline{V}.$$

Then by Kuratowski theorem (4.4.1.2), there is a unique topology on  $\text{Prim}(G)$  s.t. in this topology  $\overline{U}$  defined above is the closure of  $U$  for any  $U \subset \text{Prim}(G)$ . This topology is called the **hull-kernel topology** or the **Jacobson-topology** on  $\text{Prim}(G)$ . This topology is has separation property  $T_0$ .  $\lrcorner$

*Proof:* It suffices to prove that  $\overline{U \cup V} = \overline{U} \cup \overline{V}$ .  $U \cup V \subset \overline{U \cup V}$  is clear, and for the converse inclusion, if  $\pi \in \widehat{G}$  satisfies  $\ker(\pi) \notin \overline{U \cup V}$ , then there exists  $f \in \cap_{J \in U} J$  and  $g \in \cap_{J \in V} J$  s.t.  $\pi(f) \neq 0$  and  $\pi(g) \neq 0$ . Take  $u \in \mathcal{H}_\pi$  s.t.  $\pi(f)u \neq 0$ , then since  $\pi$  is irreducible, by (24.7.2.7), there exists  $h \in C^*(G)$  s.t.

$$\pi(g)\pi(h)\pi(f)u \neq 0.$$

But  $g * h * f \in \cap_{J \in U \cup V} J$ , so  $\cap_{J \in U \cup V} J \not\subset \ker(\pi)$ , and hence  $\ker(\pi) \notin \overline{U \cup V}$ .  $\square$

### Fell Topology

**Def. (24.7.2.9) [Fell Topology, Fell [Fel60]].** The map  $\widehat{G} \rightarrow \text{Prim}(G) : \pi \mapsto \ker(\pi)$  (24.7.2.7) is clearly a surjection and pulls back the Jacobson topology (24.7.2.8) to a topology on  $\widehat{G}$ , called the **Fell topology**.  $\lrcorner$

**Prop. (24.7.2.10) [Dixmier].** The Fell topology is the quotient topology of the surjection

$$\text{Extrm}(\text{Pos}_1(G)) \rightarrow \widehat{G} : \varphi \mapsto \pi_\varphi \text{ (24.7.1.24) (24.7.1.25)}$$

$\lrcorner$

*Proof:* Cf. [Dixmier,  $C^*$ -Algebras] Section 3.4.  $\square$

**Prop. (24.7.2.11).** If  $G$  is Abelian, then the Fell topology on  $\widehat{G}$  is just the usual topology defined in (12.11.1.7).

If  $G$  is compact, then the Fell topology is discrete.  $\lrcorner$

*Proof:* For Abelian case, Cf.[Fol16]P226.?

For  $G$  compact, if  $\pi \in \widehat{G}$ , then  $\chi_\pi$  is a scalar product on  $\pi$ , and any  $\pi'(\chi_\pi) = 0$  for any  $\pi' \in \widehat{G} \setminus \{\pi\}$  by Schur orthogonality. Thus  $[\pi]$  is not in the closure of  $\widehat{G} \setminus \{\pi\}$ , which means  $\{\pi\} \subset \widehat{G}$  is open.  $\square$

**Def. (24.7.2.12) [Mackey-Borel Structures, Mackey[Mac57, Mac76]].** For  $n \in \mathbb{Z}_+$ , let  $\mathcal{H}^n = \mathbb{C}^n$  with the natural Hilbert space structure and  $\mathcal{H}^\infty = \ell^2(\mathbb{Z})$ , and for each  $n \in \mathbb{Z}_+ \cup \{\infty\}$ , denote  $R_n(G)$  to be the set of unitary representations of  $G$  on  $\mathcal{H}^n$  (without identifying equivalent representations), and denote  $R(G) = \bigcup_{n \in \mathbb{Z}_+ \cup \{\infty\}} R_n(G)$ .

For each  $n \in \mathbb{Z}_+ \cup \{\infty\}$ , we can define a  $\sigma$ -algebra  $\mathcal{B}_n$  on  $R_n(G)$  to be the smallest algebra s.t. the functions  $\pi \mapsto (\pi(g)u, v)$  are measurable for any  $u, v \in \mathcal{H}^n$ . And let  $\mathcal{B}$  be the colimit  $\sigma$ -algebra on  $R(G)$  (12.3.1.3).

Because  $R(G) \rightarrow \widehat{G}$  is surjective (24.7.2.2), we can define the quotient  $\sigma$ -algebra on  $\widehat{G}$  (12.3.1.3), called the **Mackey-Borel structure** on  $\widehat{G}$ .  $\lrcorner$

**Prop. (24.7.2.13).** The Mackey-Borel structure on  $\widehat{G}$  includes the Borel sets for the Fell topology, but in generally it is strictly larger. For example, the Mackey-Borel structure is separated?? Cf.[Diximier, C\*-Algebras]Section3.8), but every point in  $\widehat{G}$  is a Borel set in the Fell-topology only when  $\widehat{G} \rightarrow \text{Prim}(G)$  is a bijection (24.7.2.9).  $\lrcorner$

*Proof:*

$\square$

### Type 1 Groups

**Def. (24.7.2.14) [Primary Representations].** A **primary representation** or a **factor representation** of  $G$  is a unitary representation  $\pi \in \text{Rep}^{\text{uni}}(G)$  s.t. the von.Newman algebra  $A(\pi)$  generated by  $\pi(G)$  (14.1.2.1) is a factor (14.1.3.2).

In particular, any  $\pi \in \widehat{G}$  is primary, by Schur's lemma (24.7.1.4).  $\lrcorner$

**Def. (24.7.2.15) [Type I Groups].**  $G$  is called a **type I group** if every primary representation of  $G$  (24.7.2.14) is a direct sum of copies of some irreducible representation.  $\lrcorner$

**Example (24.7.2.16) [Type I Groups].**

- Any compact group is of type I, by (24.7.2.29) and (24.7.3.5).
- Any LCA is of type I.
- The Heisenberg groups  $H_n$  are of type I ( Cf.[Fol16]P229).  $\lrcorner$

**Def. (24.7.2.17) [CCR Algebras and GCR Algebras, Kaplansky[Kap51]].** A  $C^*$ -algebra  $A$  is called a **CCR algebra** if for any  $\pi \in \widehat{A}$ ,  $\pi(f)$  is a compact operator for each  $f \in A$ .

A  $C^*$ -algebra  $A$  is called a **GCR algebra** if every non-zero quotient  $C^*$ -algebra of  $A$  possesses a non-zero two-sided ideal of  $A$  that is CCR.  $\lrcorner$

**Def. (24.7.2.18) [CCR Groups and GCR Groups].**  $G$  is called a **CCR group** (reps. **GCR group**) if  $C^*(G)$  is a CCR algebra (resp. **GCR algebra**) (24.7.2.17).  $\lrcorner$

**Thm. (24.7.2.19) [Characterizing Type I Groups, Glimm[Gli61]].** The following are equivalent:

1.  $G$  is of type I.
2. The Fell topology on  $\widehat{G}$  (24.7.2.9) is  $T_0$ .

3. The map  $\widehat{G} \rightarrow \text{Prim}(G)$  is a bijection.
4. The Mackey-Borel structure on  $\widehat{G}$ (24.7.2.12) is countably-separated(12.3.1.5).
5. The Mackey-Borel structure on  $\widehat{G}$  is standard(12.3.1.5).
6. The Mackey-Borel structure on  $\widehat{G}$  coincides with the  $\sigma$ -algebra of Borel sets for the Fell topology.
7. For any  $(\pi, \mathcal{H}_\pi) \in \widehat{G}$ ,  $\pi(C^*(G))$ (24.7.2.3) contains all compact operators on  $\mathcal{H}_\pi$ .
8.  $G$  is GCR.

┘

*Proof:* ? Cf.[Dixmier, C\*-Algebras].

2  $\iff$  3 is trivial by(24.7.2.10) because  $\text{Prim}(G)$  is  $T_0$ (24.7.2.8).

□

**Thm.(24.7.2.20)[CCR Groups are of Type I, Glimm[Gli61]].**  $G$  is a CCR iff the Fell topology on  $\widehat{G}$  is  $T_1$ . In particular, every CCR group is of type I, by(24.7.2.19).

┘

*Proof:* ? Cf.[Dixmier, C\*-Algebras].

□

**Example(24.7.2.21)[CCR Groups].**

┘

**Example(24.7.2.22)[CCR Groups and Type I Groups].**

- If  $G$  is compact or Abelian, then  $G$  is CCR, because any  $\pi \in \widehat{G}$  is of f.d..
- (Harish-Chandra)Every connected semisimple Lie group is CCR.
- (Dixmier-Kirillov)Every connected nilpotent Lie group is CCR.
- A discrete group is of type I iff it admits an Abelian normal subgroup of finite index.

┘

*Proof:* 2: Cf.[HC61].

3: Cf.[Representations of nilpotent Lie groups and their applications, Part 1].

4: Cf.[Tho68].

□

**Prop.(24.7.2.23)[Reductive Groups are of Type 1].** If  $K \in \text{LField}^0$  and  $G \in \text{linAlgGrp}/K$ , then  $G(K)$  is second-countable of type I(24.7.2.15).

┘

*Proof:* Cf.[Sliman. Theorie de Mackey pour les groupes adeliqes]Theorem1.2.3.

□

**Prop.(24.7.2.24).** If  $F \in \text{NField}$  and  $G \in \text{RedGrp}/F$ , then  $G(\mathbf{A}_F)$  is of type I.

┘

*Proof:* Cf.[Spectral theory of automorphic forms, Clozel]Appendix.

□

### Tensor Products

**Lemma(24.7.2.25).** Suppose  $\mathcal{H}_1, \mathcal{H}_2 \in \text{HilbSpa}$  and  $S \subset B(\mathcal{H}_1)$  s.t.  $\Gamma(S) = \mathbb{C}$ . Then

$$\Gamma(S \otimes \text{id}_{\mathcal{H}_2}) = \text{id}_{\mathcal{H}_1} \otimes B(\mathcal{H}_2) \subset B(\mathcal{H}_1 \otimes \mathcal{H}_2)(12.9.4.12).$$

┘

*Proof:* Cf.[?Fi116]P234?.

□

**Prop. (24.7.2.26) [Tensor Products].** Given  $\pi_1 \in \text{Rep}^{\text{uni}}(G_1), \pi_2 \in \text{Rep}^{\text{uni}}(G_2)$ , we have:  $\pi_1 \otimes \pi_2 \in \text{Rep}^{\text{uni}}(G_1 \times G_2)$  is irreducible iff  $\pi_1, \pi_2$  are both irreducible.  $\lrcorner$

*Proof:* One direction is clear. The other direction follows from (24.7.2.25) and Schur's lemma (24.7.1.8).  $\square$

**Prop. (24.7.2.27).** Let  $\mathcal{H}_1, \mathcal{H}_2 \in \mathcal{H}\text{ilbSpa}$  and  $(\pi, \mathcal{H}_1), (\pi', \mathcal{H}_1) \in \text{Irr}^{\text{uni}}(G)$ , then:

- If  $\mathcal{M} \subset \mathcal{H}_1 \otimes \mathcal{H}_2$  is an irreducible subspace for the representation  $\pi \otimes \text{id}_{\mathcal{H}_2}$ , then  $(\pi \otimes \text{id}_{\mathcal{H}_2}, \mathcal{M})$  is equivalent to  $\pi$ .
- If  $\pi \otimes \text{id}_{\mathcal{H}_2} \cong \pi' \otimes \text{id}_{\mathcal{H}_2}$ , then  $\pi \cong \pi'$ .  $\lrcorner$

*Proof:* 1: Cf. [Fol16]P235. ?

2: Take  $v_0 \in \mathcal{H}_2 \setminus \{0\}$  and let  $\mathcal{M} = \mathcal{H}_1 \otimes v_0$ , then  $\mathcal{M}_0$  is invariant and irreducible under  $\pi \otimes I$ , thus  $(\mathcal{M}_0, \pi \otimes 1)$  is isomorphic to  $\pi$ . If  $U$  is an equivalence from  $\pi \otimes \text{id}_{\mathcal{H}_2}$  to  $\pi' \otimes \text{id}_{\mathcal{H}_2}$ , then  $U|_{\mathcal{M}_0}$  is an equivalence from  $(\mathcal{M}, \pi \otimes \text{id}_{\mathcal{H}_2})$  to  $(U(\mathcal{M}), \pi' \otimes \text{id}_{\mathcal{H}_2})$ . The former is equivalent to  $\pi$  and the latter is equivalent to  $\pi'$ , by item 1.  $\square$

**Cor. (24.7.2.28).** For  $\pi_1, \pi'_1 \in \text{Irr}^{\text{uni}}(G_1), \pi_2, \pi'_2 \in \text{Irr}^{\text{uni}}(G_2)$ ,  $\pi_1 \otimes \pi_2 \cong \pi'_1 \otimes \pi'_2$  iff  $\pi_1 \cong \pi'_1, \pi_2 \cong \pi'_2$ .  $\lrcorner$

**Thm. (24.7.2.29) [Primary Representations].** For  $\{\pi_\alpha\}_{\alpha \in I} \subset \text{Irr}^{\text{uni}}(G)$ ,  $\oplus_{\alpha \in I} \pi_\alpha$  is primary (24.7.2.14) iff all the  $\pi_\alpha$  are equivalent.  $\lrcorner$

*Proof:*  $\pi^{\oplus I} \cong \pi \otimes \ell^2(I)$ , so  $C(\pi \otimes \text{id}_{\ell^2(I)}) \cong B(\ell^2(I))$  by Schur's lemma (24.7.1.8) and (24.7.2.25). Conversely, if these  $\pi_\alpha$  are not all equivalent, let  $P$  be a projection onto an isotypic component, then  $P \in C(\oplus_\alpha \pi_\alpha)$ , and for any  $T \in C(\oplus_\alpha \pi_\alpha)$ ,  $T$  maps each isotypic component to itself, by (24.7.2.27) and Schur's lemma (24.7.1.8). Thus  $PT = TP$ , and this implies that  $P$  is in the center of  $C(\oplus_\alpha \pi_\alpha)$ , contradiction.  $\square$

**Thm. (24.7.2.30) [Tensor Products].** If  $G_1, G_2 \in \text{LCG}$  and  $G_1$  is of type I (24.7.2.15), then irreducible unitary representations of  $G_1 \times G_2$  are exactly tensor representations  $\pi_1 \otimes \pi_2$ , where  $\pi_1 \in \text{Irr}^{\text{uni}}(G_1)$  and  $\pi_2 \in \text{Irr}^{\text{uni}}(G_2)$ .  $\lrcorner$

*Proof:*  $\pi_1 \otimes \pi_2$  are irreducible by (24.7.2.26), conversely, for any  $\pi \in (G_1 \times G_2)^\wedge$ , let  $\pi^1(x) = \pi(x, 1)$  and  $\pi^2(y) = \pi(1, y)$ , then  $\pi^1, \pi^2$  are commutative. So if  $T$  is in the center of  $C(\pi^1)$ , then  $T \in C(\pi)$ , which implies  $T \in \mathbb{C}$  by Schur's lemma (24.7.1.8). So  $\pi^1$  is primary. So by hypothesis (24.7.2.15), we can assume  $\mathcal{H}_\pi = \mathcal{H}_1 \otimes \mathcal{H}_2$  and  $\pi^1 \cong \pi_1 \otimes \text{id}_{\mathcal{H}_2}$ . then by Schur's lemma (24.7.1.8) and (24.7.2.25), we can write  $\pi^2 = \text{id}_{\mathcal{H}_1} \otimes \pi_2$  for some representation  $\pi_2$  on  $\mathcal{H}_2$ . Then  $\pi \cong \pi_1 \otimes \pi_2$ .  $\square$

## Direct Integrals

**Def. (24.7.2.31) [Fields of Hilbert Spaces].** For a measurable space  $(A, \mathcal{M})$ , a **measurable field of Hilbert spaces** is a functor  $A \rightarrow \mathcal{H}\text{ilbSpa} : \alpha \mapsto \mathcal{H}_\alpha$  and a countable set of set of vectors  $\{e_i(\alpha)\}_{i \in \mathbb{Z}_+}$  for each  $\alpha \in A$ , s.t.

- The function  $\alpha \mapsto (e_i(\alpha), e_j(\alpha))$  is measurable for any  $i, j \in \mathbb{Z}_+$ .
- $\overline{\text{span}\{e_i(\alpha)\}_{i \in \mathbb{Z}_+}} = \mathcal{H}_\alpha$  for each  $\alpha \in A$ .

For a measurable field of Hilbert spaces  $\{\mathcal{H}_\alpha\}$  on  $A$ , a **measurable Hilbert vector field** is an element  $(f(\alpha))_{\alpha \in A} \in \prod_{\alpha \in A} \mathcal{H}_\alpha$  s.t.  $\alpha \mapsto (f(\alpha), e_i(\alpha))$  is a measurable function for each  $i \in \mathbb{Z}_+$ .

For a measurable field of Hilbert spaces  $\{\mathcal{H}_\alpha\}$  on  $A$ , a **measurable operator field** is an element  $\prod_{\alpha \in A} T_\alpha \in \prod_{\alpha \in A} B(\mathcal{H}_\alpha)$  s.t.  $\alpha \mapsto T(\alpha)f(\alpha)$  is a measurable Hilbert vector field whenever  $f$  is a measurable Hilbert vector field.  $\lrcorner$

**Prop. (24.7.2.32) [Measurable Orthonormal Bases].** Situation as in (24.7.2.31), define  $d(\alpha) = \dim \mathcal{H}_\alpha$ , then for any  $m \in \mathbb{Z}_+ \cup \{\infty\}$ ,  $A_m = \{\alpha \in A \mid d(\alpha) = m\}$  is a measurable subset of  $A$ , and there exists a sequence  $\{u_k\}_{k \in \mathbb{Z}_+}$  of measurable Hilbert vector fields (24.7.2.31) s.t.

- For each  $\alpha \in A$ ,  $\{u_k(\alpha)\}_{k=1}^{d(\alpha)}$  is an orthonormal basis of  $\mathcal{H}_\alpha$ , and  $u_k(\alpha) = 0$  for  $k > d(\alpha)$ .
- For each  $k \in \mathbb{Z}_+$  there exists a measurable partition  $A = \coprod_{l=1}^{\infty} A_l^k$  of  $A$  s.t. on each  $A_l^k$ ,  $u_k(\alpha)$  is a finite linear combination of elements in  $\{e_j(\alpha)\}$  with coefficients depending measurably on  $\alpha$ .

┘

*Proof:* Cf. [Fol16] P239. ?

□

**Cor. (24.7.2.33).** Situation as in (24.7.2.32), a Hilbert vector field  $f$  is a measurable Hilbert vector field (24.7.2.31) iff  $\alpha \mapsto (f(\alpha), u_k(\alpha))$  is a measurable function for each  $k \in \mathbb{Z}_+$ . If  $f, g$  are measurable Hilbert vector fields, then  $\alpha \mapsto (f(\alpha), g(\alpha))$  is a measurable function.

┘

**Cor. (24.7.2.34).** Situation as in (24.7.2.31), a field of operators on  $A$  is measurable iff  $\alpha \mapsto (T(\alpha)e_i(\alpha), e_j(\alpha))$  is a measurable function for any  $\alpha \in A$ .

┘

*Proof:* Use  $(T(\alpha)f(\alpha), e_j(\alpha)) = (f(\alpha), T(\alpha)^*e_j(\alpha))$ , and use (24.7.2.33).

□

**Def. (24.7.2.35) [Direct Integrals of Hilbert Spaces].** Situation as in (24.7.2.31), if  $\mu$  is a measure on  $(A, \mathcal{M})$ , then the **direct integral** (w.r.t.  $\mu$ ) of the spaces  $\mathcal{H}_\alpha$  is the set of measurable Hilbert vector fields  $f$  on  $A$  s.t.

$$\|f\|^2 = \int \|f(\alpha)\|_\alpha^2 d\mu(\alpha) < \infty,$$

denoted by  $\int^\oplus \mathcal{H}_\alpha d\mu(\alpha)$ . Then it is a Hilbert space with inner product

$$(f, g) = \int (f(\alpha), g(\alpha))_\alpha d\mu(\alpha).$$

If  $\mathcal{H}_\alpha = \mathcal{H}$  for each  $\alpha \in A$ , then denote  $L^2(A, \mu; \mathcal{H}) \triangleq \int^\oplus \mathcal{H}_\alpha d\mu(\alpha)$ .

┘

*Proof:* The proof that this is complete is the same as the proof of  $L^2(\mu)$  is complete. ?

□

**Prop. (24.7.2.36).** Situation as in (24.7.2.32), we have an isomorphism of Hilbert spaces

$$\int^\oplus \mathcal{H}_\alpha d\mu(\alpha) \cong L^2(A_\infty, \mu; \ell^2) \oplus \bigoplus_{m \in \mathbb{Z}_+} L^2(A_m, \mu; \mathbb{C}^m).$$

┘

**Def. (24.7.2.37) [Direct Integral of Operators].** Situation as in (24.7.2.35), A measurable field of operators  $\{T(\alpha)\}$  s.t.

$$\|T\|_\infty \triangleq \|\|T(\alpha)\|\|_{L^\infty} < \infty$$

is called an **essentially finite measurable field of operators**, and it defines an operator  $\int^\oplus T(\alpha) d\mu(\alpha)$  on  $\int^\oplus \mathcal{H}_\alpha d\mu(\alpha)$  by mapping  $\{f(\alpha)\}$  to  $\{T(\alpha)f(\alpha)\}$ .

If  $T, S$  are essentially finite measurable field of operators, then

$$\int^\oplus T(\alpha) d\mu(\alpha) + \int^\oplus S(\alpha) d\mu(\alpha) = \int^\oplus (T + S)(\alpha) d\mu(\alpha),$$

$$\int^{\oplus} T(\alpha) d\mu(\alpha) \circ \int^{\oplus} S(\alpha) d\mu(\alpha) = \int^{\oplus} (T \circ S)(\alpha) d\mu(\alpha)$$

$$\left( \int^{\oplus} T(\alpha) d\mu(\alpha) \right)^* = \int^{\oplus} T(\alpha)^* d\mu(\alpha).$$

A special kind of fields of operators is the case when  $T(\alpha)$  are all scalar multiples of  $\text{id}_{\mathcal{H}_\alpha}$ . The set of these operators are called **diagonal operators**.  $\perp$

**Prop. (24.7.2.38).** Situation as in (24.7.2.35), if  $T$  is an essentially finite measurable field of operators, then

$$\left\| \int^{\oplus} T(\alpha) d\mu(\alpha) \right\| = \|T\|_{\infty} \quad (24.7.2.37).$$

In particular,  $T \mapsto \int^{\oplus} T(\alpha) d\mu(\alpha)$  is injective.  $\perp$

*Proof:* Clearly  $\left\| \int^{\oplus} T(\alpha) d\mu(\alpha) \right\| \leq \|T\|_{\infty}$ . Conversely, for any  $f \in \int^{\oplus} \mathcal{H}_\alpha d\mu(\alpha)$  and  $\varphi \in L^\infty(\mu)$ , Cf. [Fol16]P244?  $\square$

### Direct Integral Representations

**Def. (24.7.2.39) [Measurable Fields of Representations].** If  $(A, \mathcal{M}, \mu)$  is a measure space, then a **measurable field of representations** of  $G$  is an element  $\prod_{\alpha \in A} (\pi_\alpha, \mathcal{H}_\alpha) \in \prod_{\alpha \in A} \text{Rep}^{\text{uni}}(G)$  s.t.  $\{\mathcal{H}_\alpha\}$  is a measurable field of Hilbert spaces on  $A$  (24.7.2.31) and  $\alpha \mapsto \pi_\alpha(g)$  is measurable field of operators (24.7.2.31) for any  $g \in G$ .  $\perp$

**Def. (24.7.2.40) [Direct Integral Representations].** Situation as in (24.7.2.39), for any  $g \in G$ , since  $\|\pi_\alpha(g)\| = 1$ , we can form the direct integral operator

$$\pi(g) = \int_A^{\oplus} \pi_\alpha(g) d\mu(\alpha) \quad (24.7.2.37).$$

Then this is a unitary representation of  $G$  on  $\int^{\oplus} \mathcal{H}_\alpha d\mu(\alpha)$  by (24.7.2.37), and it is continuous w.r.t. the strong operator topology by dominated convergence. This  $(\int^{\oplus} \pi_\alpha(g) d\mu(\alpha), \int^{\oplus} \mathcal{H}_\alpha d\mu(\alpha))$  is called the **direct integral representation** of  $\{(\pi_\alpha, \mathcal{H}_\alpha)\}_\alpha$ , denoted by  $\int^{\oplus} (\pi_\alpha, \mathcal{H}_\alpha)$ .  $\perp$

**Prop. (24.7.2.41) [Unitary Representations of Abelian and Compact Groups].** If  $G$  is Abelian or compact, then any  $\pi \in \text{Rep}^{\text{uni}}(G)$  is equivalent to a direct integral of irreducible representations.  $\perp$

*Proof:* Cf. [Fol16]P245.  $\square$

**Thm. (24.7.2.42).** If  $G$  is second-countable and  $(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}}(G)$  where  $\mathcal{H}$  is separable, and  $B$  is a commutative von. Neumann subalgebra of  $C(\pi)$  (14.1.2.1), then there exists a standard measure space  $(A, \mathcal{M}, \mu)$  (12.3.1.5) and a measurable field of representations  $\{(\pi_\alpha, \mathcal{H}_\alpha)\}$  of  $G$  on  $A$  together with an isomorphism  $U : (\pi, \mathcal{H}) \cong \int^{\oplus} (\pi_\alpha, \mathcal{H}_\alpha) d\mu(\alpha)$  s.t.  $UBU^{-1}$  is the algebra of diagonal operators (24.7.2.37) on  $\int^{\oplus} (\pi_\alpha, \mathcal{H}_\alpha) d\mu(\alpha)$ .  $\perp$

*Proof:* Cf. [Fol16]P246. ?  $\square$

**Thm. (24.7.2.43).** Situation as in (24.7.2.42),

- If  $B$  is the center of  $C(\pi)$ , then  $\pi_\alpha$  is primary for  $\mu$ -a.e.  $\alpha \in A$ .



- $\pi_\alpha$  is irreducible for  $\mu$ -a.e.  $\alpha \in A$  iff  $B$  is a maximal commutative von. Neumann subalgebra algebra of  $C(\pi)$ . ┘

*Proof:* Cf. [Fol16]P247. ? □

**Prop. (24.7.2.44) [Standard Measures on  $\hat{G}$ ].** A **standard measure on  $\hat{G}$**  is a Radon measure  $\hat{\mu}$  on  $\hat{G}$  w.r.t. the Mackey-Borel structure (24.7.2.12) s.t. there exists  $A \subset G$  with  $\hat{\mu}(G \setminus A) = 0$  and  $A$  is standard (12.3.1.5) with the Mackey-Borel structure.

By (24.7.2.19), if  $G$  is of type I, then every Radon measure  $\hat{\mu}$  is standard. ┘

**Prop. (24.7.2.45) [Standard Representations, Mackey [Mac58]].** Notation as in (24.7.2.12), let  $\hat{G}_n$  be the image of  $R_n(G)$ , then there is a canonical measurable field of Hilbert spaces on  $\hat{G}$  s.t.  $\mathcal{H}_\alpha = \mathcal{H}_n$  for any  $\alpha \in R_n(G)$ , called the **canonical field of Hilbert spaces**.

If  $A \subset \hat{G}$  be a subset s.t. the Mackey-Borel structure is standard on  $A$  (12.3.1.5), then there is a measurable field of representations  $\{\pi_\alpha\}$  of  $G$  on this canonical field of Hilbert spaces s.t.  $[\pi_\alpha] = \alpha$  for any  $\alpha \in \hat{G}$ . ┘

*Proof:* Cf. [Mac58]Thm10.2. □

### Quasi-Dual Spectra

**Def. (24.7.2.46) [Quasi-Equivalent Representations].**  $\pi, \rho \in \text{Rep}^{\text{uni}}(G)$  are called **quasi-equivalent** if there are no non-zero subrepresentation  $\pi' \subset \pi$  s.t.  $C(\pi', \rho) = 0$  and there are no non-zero subrepresentation  $\rho' \subset \rho$  s.t.  $C(\rho', \pi) = 0$ .

Let  $\check{G}$  be the set of quasi-equivalence class of primary representations of  $G$  (24.7.2.14), called the **quasi-dual spectra** of  $G$ .

There is a natural injection map  $\hat{G} \rightarrow \check{G}$ , which is a bijection iff  $G$  is of type I (24.7.2.15). ┘

**Def. (24.7.2.47) [Mackey-Borel Structures and Standard Measures].**  $\check{G}$  is endowed with a Mackey-Borel structure similar to (24.7.2.12) and we can define standard measures on  $\check{G}$  as in (24.7.2.44). ┘

**Thm. (24.7.2.48) [Central Decompositions, Ernest [Ern62]].** For any  $(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}}(G)$  s.t.  $\mathcal{H}$  is separable, there is a standard measure (24.7.2.47)  $\check{\mu}$  on  $\check{G}$  with a standard set  $A \subset \check{G}$  and a measurable field of representations  $\{(\pi_\alpha, \mathcal{H}_\alpha)\}_\alpha$  of  $G$  on  $A$  (24.7.2.39) together with an isomorphism  $U : (\pi, \mathcal{H}) \cong \int_A^\oplus (\pi_\alpha, \mathcal{H}_\alpha) d\check{\mu}(\alpha)$  (24.7.2.40) s.t.

- The center of  $UC(\pi)U^{-1}$  is the algebra of diagonal operators on  $\int^\oplus (\pi_\alpha, \mathcal{H}_\alpha) d\mu(\alpha)$  (24.7.2.37).
- $[\pi_\alpha] = \alpha$  for  $\check{\mu}$ -a.e.  $\alpha \in \check{G}$ .

And such a measure  $\check{\mu}$  is unique up to equivalence, and  $\{\pi_\alpha\}$  is  $\check{\mu}$ -a.e. uniquely defined. This is called the **central decomposition** of  $\pi$ . ┘

*Proof:* Cf. [Dixmier, C\*-Algebras]Section8.4. or [Pedersen, C\*-Algebras and their Automorphism Groups]Section4.12. □

**Cor. (24.7.2.49) [Decomposition of Unitary Representations].** If  $G$  is of type I (24.7.2.15) and  $(\rho, V) \in \text{Rep}^{\text{uni}}(G)$ , then there exists a positive Radon measure  $\mu$  on  $\hat{G}$  and a measurable multiplicity function  $m : \hat{G} \rightarrow \mathbb{Z}_+ \cup \{\infty\}$  s.t.

$$(\rho, V) = \int_{\hat{G}}^\oplus \left( \hat{\oplus}^{m(\pi)} V_\pi \right) d\hat{\mu}(\pi).$$

This  $\mu$  is unique up to equivalence, and  $m(\pi)$  are  $\mu$ -a.e. uniquely defined. ┘



*Proof:* Cf.[Spectral theory of automorphic forms, Clozel]Thm3.3. ? or [Dixmier, C\*-Algebras]Section8.6.  $\square$

**Remark (24.7.2.50)**[Dixmier[Dix64]]. If  $G$  is not of type I, then there exist mutually singular standard measures  $\mu, \nu$  on  $\hat{G}$  s.t.  $\int^\oplus \pi_\alpha d\mu(\alpha) d\mu(\alpha) = \int^\oplus \pi_\alpha d\nu(\alpha) d\mu(\alpha)$  are equivalent primary representations.  $\lrcorner$

*Proof:* Cf.[Dix64].  $\square$

### Plancherel Theorem

**Notation (24.7.2.51).**

- Let  $G, G', G_1, G_2 \in \text{LCG}$  be second-countable and unimodular.

$\lrcorner$

**Def. (24.7.2.52) [Fourier Transform].** For  $f \in L^1(G)$ , the **Fourier transform** of  $f$  is defined to be the measurable field of operators on  $\hat{G}$  (with the Mackey-Borel structure) given by

$$\mathcal{F}f(\pi) = \int_G f(g)\pi(g^{-1})d\mu(g).$$

? Why is it a measurable field of operators.

Notice that  $\mathcal{F}|_{C_c^\infty(X)}$  is just the dual of the matrix coefficient map

$$m_\pi : \pi \otimes \pi^\vee \rightarrow C^\infty(X).$$

$\lrcorner$

**Def. (24.7.2.53).** Define

$$I^1(G) = L^1(G) \cap L^2(G), \quad I^2(G) = I^1(G) * I^1(G),$$

then  $I^2(G) \subset C_0(G)$  because  $L^2(G) * L^2(G) \subset C_0(G)$  ?.

$\lrcorner$

**Thm. (24.7.2.54) [Plancherel Theorem].** Suppose  $G$  is of type I with a Haar measure  $\mu$ , then there is a unique Haar measure  $\hat{\mu}$  on  $\hat{G}$  s.t. the Fourier transform  $f \mapsto \mathcal{F}f$  (24.7.2.52) maps  $I^1(G)$  (24.7.2.53) into  $\int^\oplus \mathcal{H}_\pi \otimes \mathcal{H}_{\bar{\pi}} d\mu(\pi)$ , and extends to a unitary map

$$\mathcal{F} : L^2(G) \rightarrow \int_{\hat{G}}^\oplus \mathcal{H}_\pi \otimes \mathcal{H}_{\bar{\pi}} d\hat{\mu}(\pi)$$

intertwining the two-sided regular representation  $\tau$  with  $\int^\oplus \pi \otimes \bar{\pi} d\mu(\pi)$ . This  $\hat{\mu}$  is called the **Plancherel measure** on  $\hat{G}$ .

In particular,  $\mathcal{F}f(\pi)$  is Hilbert-Schmidt for  $\hat{\mu}$ -a.e.  $\pi \in \hat{G}$ , and for  $f, g \in I^1(G)$ , we have the **Parseval-Plancherel formula**

$$\int_G f(x)\overline{g(x)}d\mu(x) = \int_{\hat{G}} \langle \mathcal{F}f(\pi), \mathcal{F}g(\pi) \rangle_{\text{H-S}} d\hat{\mu}(\pi).$$

(where for Hilbert-Schmidt pairing notice that  $\bar{\pi} \cong \pi^\vee$ ). And for  $h \in I^2$  (24.7.2.53), we have the **Fourier inversion formula**

$$h(g) = \int_{\hat{G}} \langle \mathcal{F}h(\pi), \pi(g) \rangle_{\text{H-S}} d\hat{\mu}(\pi).$$

$\lrcorner$

*Proof:* ? □

**Prop. (24.7.2.55).** Let  $\rho, \lambda$  be the right and left regular representations of  $G$ , resp., and let  $R, L$  be the von. Neumann algebras generated by  $\rho(G)$  and  $\lambda(G)$  resp. Then  $R = C(\lambda), L = C(\rho)$ . ┘

*Proof:* Cf. [Fol16]P254. ? □

### 3 Compact Group

Cf. [群表示论 notes, 薛航] and [Fo15]Chap5.

**Notation (24.7.3.1).**

- Let  $K \in \text{LCG}$  be compact.

┘

#### Unitary Representations

**Prop. (24.7.3.2) [F.d. Representation is Unitary].** If  $(\pi, V) \in \text{Rep}^{\text{fd}}(K)$ , then there is an inner product on  $V$  s.t. the action of  $K$  is unitary. ┘

*Proof:* Choose an arbitrary inner product  $(\cdot, \cdot)_0$  on  $V$ , then consider

$$(u, v) = \int_G (\pi(x)u, \pi(x)v)_0 dx.$$

where  $dx$  is a Haar measure on  $G$ . Then

$$(\pi(y)u, \pi(y)v) = \int_G (\pi(xy)u, \pi(xy)v)_0 dx = \Delta(y) \int_G (\pi(x)u, \pi(x)v)_0 dx = \int_G (\pi(x)u, \pi(x)v)_0 dx$$

because  $G$  is compact hence unimodular (12.10.1.18). Thus this is an inner product on  $V$  that is invariant under  $G$ . □

**Cor. (24.7.3.3) [F.D. Representation of Compact Groups Totally Decomposable].** Any  $\pi \in \text{Rep}^{\text{fd}}(K)$  is completely reducible. ┘

*Proof:* This is because we can assume this representation is unitary by (24.7.3.2), and then for any subrepresentation we can take the orthogonal complement. □

**Lemma (24.7.3.4).** Let  $(\pi, \mathcal{H}) \in \text{Rep}^{\text{uni}}(K)$  and  $u \neq 0 \in \mathcal{H}$  be a unit vector. If the operator  $T$  on  $\mathcal{H}$  is defined by

$$Tv = \int_G (v, \pi(x)u) \pi(x)u dx,$$

then  $T$  is a positive, non-zero compact operator in  $C(\pi)$ . ┘

*Proof:*

$$(Tv, v) = \int_K (v, \pi(x)u) (\pi(x)u, v) dx = \int_K |(\pi(x)u, v)|^2 dx \geq 0,$$

so it is positive. Moreover, if  $v = u$ , then  $x \mapsto |(\pi(x)u, v)|$  is a positive on a nbhd of 1, so  $T \neq 0$ .

Finally, because  $K$  is compact,  $x \mapsto \pi(x)u$  is uniformly continuous, so for any  $\varepsilon > 0$ , there is a disjoint partition  $E_i$  of  $G$  and  $x_i \in E_i$  that if  $x \in E_i$ , then  $\|\pi(x)u - \pi(x_i)u\| \leq \varepsilon/2$ . Then

$$\|(\pi(x)u) \pi(x)u - (\pi(x_i)u) \pi(x_i)u\| \leq \|(v, [\pi(x) - \pi(x_i)]u) \pi(x)u\| + \|(v, \pi(x_i)u) [\pi(x) - \pi(x_i)]u\| < \varepsilon \|v\|.$$

So consider

$$T_\varepsilon(v) = \sum |E_j|(v, \pi(x_j)u) \pi(x_j)u = \sum \int_{E_i} (v, \pi(x_j)u) \pi(x_j)u dx$$

then  $\|T - T_\varepsilon\| < \varepsilon$ , and  $T_\varepsilon$  has f.d image, thus  $T$  is compact, by (12.8.5.3).

Also  $T \in C(\pi)$  because

$$\pi(y)Tv = \int_G (v, \pi(x)u) \pi(yx)u dx = \int_G (v, \pi(y^{-1}x)u) \pi(x)u dx = \int_G (\pi(y)v, \pi(x)u) \pi(x)u dx = T\pi(y)v.$$

□

**Thm. (24.7.3.5) [Unitary Representations of Compact Groups].** Any  $(\rho, V) \in \text{Rep}^{\text{uni}}(K)$  is an orthogonal sum of irreducible unitary subrepresentations. Moreover, the isotopic part  $V^\pi$  for any irreducible representation  $\pi$  of  $G$  is uniquely determined. And any  $\pi \in \text{Irr}(G)$  is of f.d. (thus unitary by (24.7.3.2)).

*Proof:* By taking orthogonal complements and Zorn's lemma, it suffices to show any unitary representation  $\pi$  has an irreducible subrepresentation. Choose  $T$  as in (24.7.3.4), then  $T$  is compact nonzero self-adjoint, so by Riesz-Fredholm (12.8.5.9) it has a finite-dimensional eigenspace, which is  $\pi$ -invariant, and it clearly has an irreducible subrepresentation by taking orthogonal complements.

For the orthogonality, Let  $V^\pi$  be the linear span of invariant subspaces isomorphic to  $\pi$ , for  $L_1, L_2$  of type  $\pi_1 \neq \pi_2$ , then consider the orthogonal projection  $P$  onto  $L_2$ , then  $P|_{L_1} \in C(\pi_1, \pi_2)$ , which vanishes by Schur (24.7.1.8), so they are orthogonal.

The final assertion follows from (24.7.3.20).

□

**Cor. (24.7.3.6).** The cardinality of irreducible constituents of  $V$  that is isomorphic to  $\pi$  is independent of the decomposition, and it is equal to  $\dim \text{Hom}_G(\pi, \rho)$ , and is denoted by  $\text{mult}(\pi, \rho)$ .

*Proof:* Cf. [Folland, P137].

□

**Cor. (24.7.3.7) [Decomposition of Spectrum].** Let  $H \leq K$  be a closed subgroup, then  $K$  acts unitarily on  $L^2(K/H)$ , and it decomposes as

$$L^2(K/H) \cong \widehat{\bigoplus_{\pi \in \widehat{K}} m_H(\pi) \pi}$$

where  $m_H(\pi) = \dim(\pi^*)^H = \dim \pi^H$ .

□

*Proof:* By (24.7.3.5), it suffices to determine the multiplicity of  $\pi$  in  $L^2(K/H)$ , which is just dimension of  $\text{Hom}_K(\pi, L^2(K/H))$ , which can be viewed as  $G$ -invariant  $L^2$  functions on  $K/H$  with values on  $\pi^*$ . For any such function  $f$ ,  $f(1)$  is  $H$ -invariant, and for any  $H$ -invariant vector  $v$ ,  $g \mapsto gv$  is continuous, thus is  $L^2$ . So the dimension of this space is just  $\dim \pi^H$ .

□

### Matrix Coefficients and Peter-Weyl Theorem

**Def. (24.7.3.8) [Finite Vectors].** For  $\rho \in \widehat{K}$ , denote  $V^\rho = \rho \otimes \text{Hom}_K(\rho, V)$  the  $\rho$ -isotypic component in  $V$ . And let  $V^{\text{fin}} = \bigoplus_\rho V^\rho$  the space of **finite vectors** in  $V$  (24.7.3.5).

□

**Def. (24.7.3.9) [Matrix Coefficients].**  $C(K)$  is a representation of  $K \times K$  by  $((g_1, g_2)f)(g) = f(g_1^{-1}gg_2)$ .

For a f.d. representation  $(\pi, V)$  of a topological group  $G$ , we can view  $\text{End}(V)$  as a representation of  $K \times K$  via

$$(g_1, g_2)S = \pi(g_1)S\pi(g_2^{-1})$$

There is a **matrix coefficient map**:

$$\text{MC}_V : \text{End}(V) \rightarrow C(K), \quad \text{MC}_V(S)(g) = \text{MC}_{S,V}(g) = \text{tr}(\pi(g)S|V)$$

which is a map of  $K \times K$ -representations. And denote  $\mathcal{E}_\pi$  the image of  $\text{MC}_V$ , and denote  $L^2(G)^{\text{alg}} = \bigoplus_{\pi \in \widehat{G}} \mathcal{E}_\pi$ .  $\lrcorner$

**Prop. (24.7.3.10)[Schur Orthogonality Conditions].** Let  $(\pi_1, V_1), (\pi_2, V_2) \in \text{Rep}^{\text{fd}}(K)$ , then for any  $S_i \in \text{End}(V_i)$ ,

$$\int_K \text{MC}_{S_1, V_1}(k) \text{MC}_{S_2, V_2}(k) = 0$$

unless  $V_2 \cong V_1^*$ , in which case

$$\int_K \text{MC}_{S_1, V}(k) \text{MC}_{S_2, V^*}(k) = \frac{1}{\dim V} \text{tr}(S_1 \circ S_2^*|V).$$

In particular,  $\mathcal{E}_\pi$  is orthogonal to  $\mathcal{E}_{\pi'}$  for  $[\pi] \neq [\pi']$ , and if  $\{e_i\}$  is any orthonormal basis of  $V$ ,  $\sqrt{d_\pi} \pi_{ij}$  is an orthonormal basis of  $\mathcal{E}_\pi$ , and  $\mathcal{E}_\pi$  is isomorphic to  $\text{End}(V)$  as  $K \times K$ -representations.  $\lrcorner$

*Proof:* Cf.[Gaitsgory P3].  $\square$

**Cor. (24.7.3.11).**  $\text{MC}_{S,V}(k^{-1}) = \text{MC}_{S^*, V^*}(k)$ .  $\lrcorner$

**Prop. (24.7.3.12).**  $\mathcal{E}_\pi$  is invariant under left and right translations of  $G$ , and it is a two-sided ideal in  $L^1(G)$ .  $\lrcorner$

*Proof:* It can be shown that  $f \star \varphi_{u,v} = \varphi_{u, \pi(\bar{f})v}$ , and  $\varphi_{u,v} \star f = \varphi_{\pi(\widehat{f})u, v}$ , where  $\widehat{f}(x) = f(x^{-1})$ .  $\square$

**Prop. (24.7.3.13).** If  $\rho \in \text{Irr}(K)$  and  $V \in \text{Rep}(K)$ , then for any  $S \in \text{End}(\rho^*)$ , the image of

$$\text{MC}_{S, \rho} \cdot \mu_{\text{Haar}} \in \text{Meas}(K)$$

acting on  $V$  (12.8.3.24) belongs to  $V^\rho$ .  $\lrcorner$

*Proof:* Cf.[Gaitsgory P7].  $\square$

**Cor. (24.7.3.14).** For any  $\rho \in \text{Irr}(K)$ , the element  $\xi_\rho \mu_{\text{Haar}} \in \text{Meas}(K)$  acts in any continuous representation  $V$  as a projection with image equal to  $V^\rho$ .  $\lrcorner$

*Proof:* Directly from the proposition and (24.7.3.27).  $\square$

**Prop. (24.7.3.15).**  $\mathcal{E}$  is an algebra, Cf.[Folland, P141]. It is dense in  $C(K)$ , and dense in  $L^p(K)$  for  $p < \infty$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (24.7.3.16)[Peter-Weyl].**

- $\bigotimes_{\pi \in \widehat{G}} \mathcal{E}_\pi = L^2(K)$ , where  $\mathcal{E}_\pi \cong \text{End}(V_\pi)$ .

- $L^2(K)^{\text{alg}} = \otimes_{\pi \in \widehat{G}} \mathcal{E}_\pi$  identifies with the  $K$ -finite vectors in  $L^2(K)$  w.r.t. the left translation (thus the notation).  $\lrcorner$

*Proof:* 1: By (24.7.3.5), the multiplicity of  $\pi$  in  $L^2(K)$  is just  $\dim \pi$ , and this is just the multiplicity of  $\pi$  in  $\mathcal{E}_\pi$  by (24.7.3.10), so this is a surjection, thus an isomorphism.

2: Clearly every vector in  $\otimes_{\pi \in \widehat{G}} \mathcal{E}_\pi$  is  $K$ -finite. Conversely, if some vector  $v$  is  $K$ -finite, then it generates a finite vector space  $V$  under the left translation action of  $G$ . Now the linear function  $f \mapsto f(1)$  restricts to a linear function  $l$  on  $V$ , and then  $l(L_g^* v) = v(g)$ , so  $v$  is a matrix coefficient of  $V$ , so  $v \in L_{\text{alg}}^2(K)$ .  $\square$

**Cor. (24.7.3.17) [Representations of Product Groups].** Any irreducible representation of a the product group  $G \times H$  for  $G, H$  is of the form  $\rho \boxtimes \psi$  where  $\rho$  and  $\psi$  are irreducible representations of  $G, H$  resp.  $\lrcorner$

*Proof:* By orthogonality of characters (24.7.3.23), there representations are irreducible and different. They are all the representations because they already form a basis in  $L^2(G \times H)$ .  $\square$

**Prop. (24.7.3.18).** For any  $V \in \text{Rep}^{\text{uni}}(K)$ , the subset  $V^{K\text{-fin}}$  is dense in  $V$ .  $\lrcorner$

*Proof:* For any  $v \in V$ , choose a Dirac sequence  $f_n$ , then  $\pi(f_n)v \rightarrow v$ . Then by Peter-Weyl (24.7.3.16), we can choose  $K$ -finite functions  $g_n$  that  $\|g_n - f_n\|_{L^2} < \frac{1}{n}$ . Then  $g_n$  also converges to  $\delta_1$  in the weak topology. Thus

$$\pi(g_n)v \rightarrow v$$

and (24.7.3.13) shows  $\pi(g_n)v \in V^{K\text{-fin}}$ .  $\square$

**Cor. (24.7.3.19).** Matrix coefficients of f.d. representations are dense in  $C(K)$ . (Immediate from the proposition and Peter-Weyl theorem (24.7.3.16).  $\lrcorner$

**Cor. (24.7.3.20).**  $\dim \pi < \infty$  for any  $\pi \in \text{Irr}(K)$ .  $\lrcorner$

*Proof:* This is because  $V^{K\text{-fin}}$  is a sub-representation of  $V$  and it is dense in  $V$ , thus  $V = V^{K\text{-fin}}$  is of f.d. because it is irreducible.  $\square$

### Fourier Analysis on Compact Groups

**Def. (24.7.3.21) [Characters].** If  $V \in \widehat{K}$ , define  $\xi_V = \dim V \cdot \chi_V \in L^2(G)^{\text{alg}}$ .  $\lrcorner$

**Prop. (24.7.3.22).** If we take an invariant inner product on  $V$ , and  $e_i$  an orthonormal basis, then

$$\chi_V(g) = \sum_i (ge_i, e_i),$$

and

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}, \quad \chi_{V \oplus W} = \chi_V + \chi_W, \quad \chi_{V \otimes W} = \chi_V \cdot \chi_W$$

**Cor. (24.7.3.23) [Orthogonality of Characters].** For  $V, W \in \text{Irr}(K)$

$$\int_K \chi_V(k) \chi_W(k^{-1}) = \int_K \chi_V(k) \chi_{W^*}(k) = \int_K \chi_V(k) \overline{\chi_W(k)}$$

so by (24.7.3.10) this equals 1 if  $V \cong W$  and 0 otherwise.  $\lrcorner$

**Prop. (24.7.3.24).** Let  $(\rho, V) \in \text{Rep}^{\text{fd}}(G)$  with character  $\chi$ , then

- $\chi(1) = \dim V$ .
- $\chi(s^{-1}) = \chi(s)^*$ .
- $\chi(sts^{-1}) = \chi(s)$ .

┘

*Proof:* Notice the eigenvalues of  $\rho(g)$  all have absolute value 1, because this representation is unitarizable (24.7.3.2) thus  $\rho^*(g) = \rho(g)^*$ .  $\square$

**Prop. (24.7.3.25) [Fourier Transform on Compact Groups].** Let  $f \in L^2(K)$ , then by Peter-Weyl (24.7.3.16) and (24.7.3.10),

$$f = \sum_{\pi \in \widehat{G}} \sum_{i,j=1}^{d_\pi} c_{ij}^\pi \pi_{ij}, \quad c_{ij}^\pi = d_\pi \int_G f(x) \overline{\pi_{ij}(x)} dx.$$

So

$$f(x) = \sum_{\pi \in \widehat{G}} \sum_{i,j=1}^{d_\pi} d_\pi \int_G f(y) \overline{\pi_{ij}(y)} \pi_{ij}(x) dy = \sum_{\pi \in \widehat{G}} d_\pi f(y) \text{tr}(\pi(y^{-1}x)) = \sum_{\pi \in \widehat{G}} d_\pi (f * \chi_\pi)(y).$$

Thus

$$f = \sum_{\pi \in \widehat{G}} d_\pi f * \chi_\pi,$$

and  $d_\pi f * \chi_\pi$  are the projection of  $f$  onto  $\mathcal{E}_\pi$  (24.7.3.10).  $\square$

**Cor. (24.7.3.26).**  $\xi_V * \xi_W = 0$  unless  $W \cong V$  and  $\xi_V * \xi_V = \xi_V$  (24.7.3.21).  $\square$

**Cor. (24.7.3.27).** For continuous irreducible representations  $V, W$  of  $K$ ,  $\pi(\xi_V) \in \text{End } W$  equals  $\text{Id}_W$  if  $W \cong V$  and zero otherwise.  $\square$

**Prop. (24.7.3.28).** For  $1 \leq p < \infty$ , the spaces  $L^p(K)$  and  $C(K)$  are Banach algebras under convolution (12.10.2.4), and  $ZL^p(K)$  and  $ZC(K)$  are their centers.  $\square$

*Proof:* By (12.10.2.7),  $\|f * g\|_p \leq \|f\|_1 \|g\|_p \leq \|f\|_p \|g\|_p$ , for  $1 \leq p \leq \infty$ , thus  $L^p(G)$  and  $C(G)$  are Banach algebras.

For  $f \in L^p(K)$ ,  $f * g = g * f$  iff

$$\int_K f(xy) g(y^{-1}) dy = \int_K g(y) f(y^{-1}x) dy = \int_K f(yx) g(y^{-1}) dy, \quad a.e. x$$

for any  $g \in L^p(K)$ , which is equivalent to  $f(xy) = f(yx)$  a.e.  $x, y$ . Similarly for  $f \in C(K)$ .  $\square$

**Lemma (24.7.3.29).** If  $f \in ZL^1(K)$  and  $\pi \in \widehat{K}$ , then  $d_\pi f * \chi_\pi = (\int_K f \overline{\chi_\pi}) \chi_\pi$ .  $\square$

*Proof:* For  $x \in G$ ,

$$\pi(f)\pi(x) = \int_K f(y)\pi(yx)dy = \int_K f(yx^{-1})\pi(y)dy = \int_K f(x^{-1}y)\pi(y)dy = \int_K f(y)\pi(xy)dy = \pi(x)\pi(f)$$

so  $\pi(f)$  is a scalar by Schur's lemma (24.7.1.8). Notice  $f'(g) = f(g^{-1}) \in ZL^1(K)$  also, so

$$d_\pi[f * \chi_\pi](x) = d_\pi \int_K f(y^{-1})(\text{tr} \pi)(yx) dy = d_\pi \text{tr} \left( \int_K f(y^{-1}) \pi(yx) dy \right) = \text{tr} \pi(f') \text{tr}(\pi(x)),$$

and  $\text{tr} \pi(f') = \int_K f(y^{-1}) \text{tr} \pi(y) dy = \int_K f \overline{\chi_\pi}$ .  $\square$

**Prop. (24.7.3.30)[Characters form an Orthonormal Basis].**  $\{\chi_\pi | \pi \in \widehat{K}\}$  is an orthonormal basis for  $ZL^2(K)$ .  $\lrcorner$

*Proof:*  $\chi_\pi \in ZC(K) \subset ZL^2(K)$  by (24.7.3.24), and they are orthonormal by (24.7.3.23). They form a basis by (24.7.3.29) and (24.7.3.25).  $\square$

**Cor. (24.7.3.31).** If  $K$  is compact Abelian and  $\mu(G) = 1$ , then  $\widehat{G}$  is an orthonormal basis of  $L^2(G)$ .  $\lrcorner$

**Prop. (24.7.3.32).** The linear spans of  $\{\chi_\pi | \pi \in \widehat{K}\}$  is dense in  $ZC(K)$  and  $ZL^p(K)$  for  $1 \leq p < \infty$ .  $\lrcorner$

*Proof:* Cf. [Fol15] P148.  $\square$

**Prop. (24.7.3.33)[Real Type].** Let  $V \in \text{Rep}(G)$ , the following are equivalent:

- $V = U \otimes_{\mathbb{R}} \mathbb{C}$  is a complexification of a real representation  $U$  of  $G$ .
- $V$  admits a  $G$ -invariant anti-linear endomorphism  $S$  that  $S^2 = 1$ .
- There is a  $G$ -invariant symmetric form  $B$  on  $V$  inducing an isomorphism  $V \cong V^*$ .

$\lrcorner$

*Proof:*  $1 \iff 2$  is clear.

$2 \rightarrow 3$ : By averaging there is a  $G$ -invariant Hermitian form  $h$  on  $V$ , then we can define  $B(u, v) = h(u, Sv)$ . As  $h'(u, v) = h(Sv, Su)$  is also a  $G$ -invariant Hermitian form,  $h(Sv, Su) = \pm h(u, v)$ , so  $B(v, u) = B(u, v)$ .

$3 \rightarrow 1$ :  $h$  induces a  $G$ -isomorphism  $\bar{\varphi}_h : \bar{V} \cong V^*$ . Then  $\sigma = \bar{\varphi}_h \circ \varphi_B$  is a  $G$ -isomorphism  $V \rightarrow \bar{V}$ . Then  $\bar{\sigma} \circ \sigma : V \rightarrow V$  is a  $G$ -isomorphism, thus by Schur's lemma  $\bar{\sigma} \circ \sigma = \lambda$  for some  $\lambda \in \mathbb{C}$ . More explicitly,  $B(v, u) = H(v, \sigma(u))$  for any  $u, v \in V$ . Because  $B$  is symmetric or symplectic,  $B(u, v) = \pm B(v, u)$ , thus  $H(v, \sigma(u)) = \pm H(u, \sigma(v))$ , and

$$\bar{\lambda}H(v, u) = H(v, \sigma^2(u)) = \pm H(\sigma(u), \sigma(v)) = \pm \overline{H(\sigma(v), \sigma(u))} = \overline{H(u, \sigma^2(v))} = \lambda H(v, u).$$

Thus  $\lambda$  is real. And then we can normalize  $\sigma$  that  $\sigma = \pm 1$ . If  $\sigma^2 = 1$ , then consider  $V_0 = \ker(\sigma - 1)$ , then  $iV_0 = \ker(\sigma + 1)$ , and  $V_0 \oplus iV_0 = V$ . If  $\sigma^2 = -1$ , then consider the action of  $\mathbb{H}$  on  $V$  given by  $(\alpha + \beta j)(v) = \alpha v + \beta \sigma(v)$ . It is an action because  $\sigma$  is anti-linear and  $\sigma^2 = -1$ .  $\square$

**Prop. (24.7.3.34)[Quaternion Type].** Let  $G$  be a topological group and  $V$  is a complex representation, then the following are equivalent:

- $V = W_{\mathbb{C}}$  is a restriction of a quaternionic representation  $W$  of  $G$ .
- $V$  admits a  $G$ -invariant anti-linear endomorphism  $S$  that  $S^2 = -1$ .
- There is a  $G$ -invariant alternating form  $B$  on  $V$  inducing an isomorphism  $V \cong V^*$ .

$\lrcorner$

*Proof:*  $1 \iff 2$  is clear. The proof of  $2 \rightarrow 3 \rightarrow 1$  is the same as the proof of (24.7.3.33).  $\square$

**Prop. (24.7.3.35).** An irreducible complex representation  $V$  of  $G$ , then  $V$  is of real/complex/quaternionic type iff  $\text{End}_{\mathbb{R}[G]}(V) \cong M_2(\mathbb{R})/\mathbb{C}/\mathbb{H}$   $\lrcorner$

*Proof:* By (24.7.3.33), if  $V$  is of real type, then  $V \cong U \otimes_{\mathbb{R}} \mathbb{C} \cong V \oplus V$ , then  $\text{End}_{\mathbb{R}[G]}(V) \cong M_2(\mathbb{R})$ .

By (24.7.3.34), if  $V$  is of quaternionic type, then it is a restriction of a quaternionic representation, so  $\mathbb{H} \subset \text{End}_{\mathbb{R}[G]}(V)$ . And for any  $\mathbb{R}$ -linear endomorphism  $f$  of  $V$ ,  $f = \frac{f+ifoi}{2} + \frac{f-ifoi}{2}$  is a sum of  $\mathbb{C}$ -linear and anti-linear endomorphisms of  $V$ . For an anti-linear endomorphism  $g$ ,  $gj$  is  $\mathbb{C}$ -linear, thus  $f \in \mathbb{H}$ .

If  $V$  is of complex type, then there are no invariant anti-linear endomorphisms, thus  $f = \frac{f+ifoi}{2} + \frac{f-ifoi}{2}$  is  $\mathbb{C}$ -linear, so  $\text{End}_{\mathbb{R}[G]}(V) \cong \mathbb{C}$ .  $\square$

**Prop. (24.7.3.36) [Types of Representations].** Let  $V \in \mathcal{R}\text{ep}^{\text{fd}}(K)$  and  $dk$  be the Haar measure on  $K$  with  $\int_K dk = 1$ , then:

$$\int_K \chi(k^2) dk = \begin{cases} 1 & \iff V \text{ is of real type} \\ 0 & \iff V \text{ is of complex type} \\ -1 & \iff V \text{ is of quaternionic type} \end{cases}$$

┘

*Proof:* Notice that  $\chi(k^2) = \chi_{\text{Sym}^2(V)}(k) - \chi_{\wedge^2(V)}(k)$ , so  $\int_G \chi(k^2) dk = \dim \text{Sym}^2(V)^G - \dim \wedge^2(V)^G$ . Then it is clear by (24.1.1.6).  $\square$

#### 4 Induced Representations

**Lemma (24.7.4.1).** If  $H \leq G$  is a closed subgroup and  $q : G \rightarrow G/H$  is the projection, for any  $(\sigma, \mathcal{H}) \in \mathcal{R}\text{ep}^{\text{uni}}(H)$ , let  $\mathcal{F}_0$  be the space of continuous functions  $f : G \rightarrow \mathcal{H}$  that  $q(\text{Supp}(f))$  is compact and

$$f(x\xi) = \sqrt{\frac{\Delta_H(\xi)}{\Delta_G(\xi)}} \sigma(\xi^{-1}) f(x).$$

Then for any  $\alpha : G \rightarrow \mathcal{H}$  that is continuous with compact support, we have

$$f_\alpha(x) = \int_H \sqrt{\frac{\Delta_G(\eta)}{\Delta_H(\eta)}} \sigma(\eta) \alpha(x\eta) d\eta \in \mathcal{F}_0,$$

and is left uniformly continuous w.r.t  $G$ . Moreover, every element in  $\mathcal{F}_0$  arises in this way.  $\square$

*Proof:* Clearly  $q(\text{Supp } f_\alpha) \subset q(\text{Supp } \alpha)$ , and

$$f_\alpha(x\xi) = \int_H \sqrt{\frac{\Delta_G(\eta)}{\Delta_H(\eta)}} \sigma(\eta) \alpha(x\xi\eta) d\eta = \int_H \sqrt{\frac{\Delta_G(\xi^{-1}\eta)}{\Delta_H(\xi^{-1}\eta)}} \sigma(\xi^{-1}\eta) \alpha(x\eta) d\eta = \sqrt{\frac{\Delta_H(\xi)}{\Delta_G(\xi)}} \sigma(\xi^{-1}) f_\alpha(x).$$

For left uniform continuity, Cf. [Folland, P164].

For the surjectivity, if  $f \in \mathcal{F}_0$ , by (12.10.3.6), there exists  $\psi \in C_c(G)$  that  $\int_H \psi(x\eta) d\eta = 1$  for  $x \in \text{Supp } f$ . So we can let  $\alpha = \psi \cdot f$ , then

$$f_\alpha(x) = \int_H \sqrt{\frac{\Delta_G(\eta)}{\Delta_H(\eta)}} \psi(x\eta) \sigma(\eta) f(x\eta) d\eta = \int_H \psi(x\eta) f(x) d\eta = f(x)$$

$\square$

**Remark (24.7.4.2) [Left and Right Compatibility].** If we consider the right action, then it suffices to consider all the functions  $g(x) = f(x^{-1})$ . Then  $g$  satisfied

$$g(\xi x) = \sqrt{\frac{\Delta_G(\xi)}{\Delta_H(\xi)}} \sigma(\xi) g(x).$$

┘



**Def. (24.7.4.3) [Induced Unitary Representations].** For  $(\sigma, V) \in \mathcal{R}\text{ep}^{\text{uni}}(H)$  and  $f, f' \in \text{ind}_H^G \rho$ , consider the function  $g \mapsto (f(g), f'(g))$ , then it is a function on  $\mathcal{S}(G, \mathcal{H})$  (12.10.3.11), thus

$$(f, f') \mapsto \int_{H \backslash G} (f, f') d\nu_{H \backslash G}(g)$$

is a right  $G$ -invariant Hermitian inner product on  $\text{ind}_H^G \rho$ , thus  $\text{ind}_H^G \rho$  is unitarizable, called the **induced unitary representation** of  $\rho$ .  $\perp$

**Prop. (24.7.4.4) [Induction and Restriction].** If  $(\sigma, \mathcal{H}) \in \mathcal{R}\text{ep}^{\text{uni}}(H)$  and  $(\pi, \mathcal{H}') \in \mathcal{R}\text{ep}^{\text{uni}}(G)$ , then  $\text{ind}_H^G(\sigma \otimes \text{res}(\pi)) \cong \text{ind}_H^G(\sigma) \otimes \pi$ .  $\perp$

**Prop. (24.7.4.5) [Frobenius Reciprocity].** If  $G$  is compact group,  $H \leq G$  is a closed subgroup and  $\pi \in \text{Irr}^{\text{uni}}(G)$ ,  $\rho \in \text{Irr}^{\text{uni}}(H)$ , then

$$C(\pi, \text{ind}_H^G(\rho)) = C(\pi|_H, \rho), \quad \text{mult}(\pi, \text{ind}_H^G(\rho)) = \text{mult}(\rho, \pi|_H).$$

$\perp$

*Proof:* It suffices to prove the first one, the second follows from (24.7.1.5) and (24.7.3.6).

$G/H$  admits a  $G$ -invariant measure as  $\Delta_G = \Delta_H = 1$ . For the rest, it is similar to that of (24.1.6.43), Cf. [Folland, P172].  $\square$

## 5 Reductive Groups

**Notation (24.7.5.1).**

- Let  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ .
- Let  $G \in \text{RedGrp}/\mathbb{K}$ .
- Let  $\mathfrak{g} = \text{Lie}(G(\mathbb{K}))$ .
- Use notations on smooth representations of Lie groups defined in [Smooth Representations of Lie Groups](#).

$\perp$

By (24.6.4.9)(24.6.4.5) and the fact that admissible  $(\mathfrak{g}, \mathcal{K})$ -modules can be classified ?, the problem of classifying unitary representations of  $G(\mathbb{K})$  is reduced to the question of which admissible representations are unitary.

**Lemma (24.7.5.2) [Auxiliary Compact Supported Function Approximation].** Let  $G$  be a locally compact Lie group and  $K$  a compact subgroup. If  $\mathcal{H} \in \mathcal{R}\text{ep}^{\text{uni}}(G)$  and  $f \in \mathcal{H} \setminus \{0\}$ , then for any  $\varepsilon \in \mathbb{R}_+$ , there is a  $\varphi \in C_c^\infty(G)$  s.t.  $\pi(\varphi)$  is self-adjoint and  $|\varphi(\rho)f - f| < \varepsilon$ .

Moreover, if  $f \in \mathcal{H}^\xi$  which is the decomposition part for  $K$ , we can assume  $\varphi(kg) = \varphi(gk) = \xi(k)^{-1}\varphi(g)$ . In particular if  $\mathcal{H}^\xi$  is f.d., we find a  $\varphi$  that  $\pi(\varphi)f = f$ .  $\perp$

*Proof:* By continuity, there is a nbhd  $H$  of 1 that  $|\pi(g)f - f| < \varepsilon$ , then we can choose a  $\varphi$  positive real valued with support in  $U$  with integral 1, then  $|\pi(\varphi)f - f| < \varepsilon$  by (12.8.3.22). We can also choose  $\varphi(g) = \varphi(g^{-1})$ , then  $\pi(\varphi)$  is self-adjoint.

For the second case, notice first there is a nbhd  $V$  of 1 that  $kVk^{-1} \in U$  for any  $k \in K$  (4.12.1.6), so let  $\varphi_1$  be a positive real valued function supported in  $V$ , and let

$$\varphi_0(g) = \int_K \varphi_1(kgk^{-1}) dk$$

then  $\varphi_0$  is supported in  $U$  and  $\varphi(kgk^{-1}) = \varphi_0(g)$  for any  $k \in K$ . Assume now that  $\pi(k_\theta) = e^{ik_\theta}f$ , then we can use (12.10.3.8) for  $P = G$  to see that

$$\pi(\varphi_0)f = \int_G \varphi_0(h)\pi(h)f dh = \int_G \int_K \varphi_0(hk)\pi(hk)f dk dh = \int_G \int_K \xi(k)\varphi_0(hk)dk\pi(h)f dh = \pi(\varphi)f$$

where

$$\varphi(g) = \int_K \xi(k)\varphi_0(gk)dk = \int_K \xi(k)\varphi_0(kg)dk$$

so  $\varphi(k) = \varphi(gk) = \xi^{-1}(k)\varphi(g)$  as required.  $\square$

**Prop. (24.7.5.3).** If  $V \in \text{Irr}^{\text{uni}}(G)$ , then the image of the induced action of  $\text{Meas}_c(G)$  is dense in  $\text{End}(V)$  in the strong topology (12.7.3.4).  $\lrcorner$

*Proof:* This follows immediately from the von Neumann theorem (14.1.2.3) and Schur's lemma (24.7.1.8): if we denote the algebra generated by  $\text{Meas}_c(G)$  by  $A$ , then

$$\overline{A} = (A^c)^c = (\mathbb{C})^c = \text{End}(V).$$

$\square$

**Prop. (24.7.5.4).** If  $V \in \text{Rep}(G)$  s.t. the image of the induced action of  $\text{Meas}_c(G)$  is dense in  $\text{End}(V)$  in the strong topology, then for any  $\rho \in \text{Irr}(K)$ , we have

$$\dim(V^\rho) \leq \dim(\rho)^2.$$

$\lrcorner$

*Proof:* Follows directly from the following two lemmas (24.7.5.5)(24.7.5.6).  $\square$

**Lemma (24.7.5.5).** For any  $\rho \in \text{Irr}(K)$ , let  $A_\rho = \xi_\rho \cdot \text{Meas}_c(G) \cdot \xi_\rho$ , this is an algebra that acts on  $V^\rho$  by (24.7.3.14). Then there exists a family of f.d. representations  $\pi$  of  $A_\rho$  that:

- Each  $\pi$  is of dimension  $\leq n = \dim(\rho)^2$ .
- For every element  $a \in A_\rho$ , there exists a  $\pi$  that  $\pi(a)$  is non-trivial.

$\lrcorner$

*Proof:* Consider the set of all irreducible f.d. representations of  $G$ , and  $\pi^\rho$  their  $\rho$ -isotypic parts. Then these are representations of  $A_\rho$ , and for any  $\varphi \in \text{Meas}_c(G)$ , there is a  $\pi$  that  $T_\varphi \neq \text{Id}$ ?, and each  $\pi^\rho$  has dimension  $\leq \dim(\rho)^2$ ?. Cf. [Gaitsgory P46].  $\square$

**Lemma (24.7.5.6).** If  $A$  is an associative algebra equipped with a family of f.d. modules satisfying conditions in (24.7.5.5), then if  $V$  is a representation of  $A$  that the image of  $A$  is dense in  $\text{End}(V)$  in the strong topology, then  $\dim V \leq n$ .  $\lrcorner$

*Proof:* For an associated algebra  $A$ , consider the minimal integer  $r$  that the property  $P(r)$ :

$$\sum_{\sigma \in \Sigma_r} \text{sgn}(\sigma) a_{\sigma(1)} \cdots a_{\sigma(r)} = 0$$

for any  $a_1, \dots, a_r$ , then Amitsur-Levitski showed that for  $A = GL(n, \mathbb{C})$ ,  $r = 2n$  (3.7.10.7).

Now the condition of  $A$  in (24.7.5.5) shows that  $P_{2n}$  is true for  $A$ . If  $\dim V \geq n+1$ , then the image of  $A$  satisfies  $P(2n)$ , so also  $\text{End}(V)$  satisfies  $P(2n)$  because  $A$  is dense in  $\text{End}(V)$ . But  $V$  contains a subgroup  $GL(n+1, \mathbb{C})$ , so it cannot satisfy  $P(2n)$  by (3.7.10.7), contradiction.  $\square$

**Cor. (24.7.5.7)[Irreducible Unitary Representations are Admissible, Harish-Chandra].** If  $V \in \text{Irr}^{\text{uni}}(G)$ , then  $V^{K\text{-fin}} \in \text{Mod}_{(\mathfrak{g}, K)}^{\text{adm}}$ , and  $\dim(V^\rho) \leq \dim(\rho)^2$  for any  $\rho \in \text{Irr}(K)$ . In particular, every unitary irreducible representation of  $G$  is admissible (24.6.4.2).  $\lrcorner$

*Proof:* Follows directly from (24.6.4.8), (24.7.5.3) and (24.7.5.4).  $\square$

**Remark (24.7.5.8).** This only holds for  $G$  real reductive. For example, if  $G = \widetilde{\text{SL}(2; \mathbb{R})}$ , then this Harish-Chandra theorem doesn't hold. One reason is that it doesn't have a non-trivial compact subgroup.  $\lrcorner$

**Prop. (24.7.5.9).** If  $M \in \text{Irr}_{(\mathfrak{g}, K)}^{\text{adm}}$  equipped with an invariant inner product  $((km_1, km_2) = (m_1, m_2), (\xi m_1, m_2) + (m_1, \xi m_2) = 0)$ , then the Hilbert space completion of  $M$  carries a unique unitary  $G$ -representation s.t.  $V^{K\text{-fin}} = M$  as  $(\mathfrak{g}, K)$ -modules.  $\lrcorner$

*Proof:* By (24.7.5.10), the Hermitian form can be extended continuously to the Banach space completion of  $M$ . It suffices to prove the extended Hermitian form is continuous, because then we can choose its completion w.r.t.  $(-, -)$ .

For the invariance, consider  $f(g) = (gm_1, m_2) - (m_1, g^{-1}m_2)$ , then notice  $(a, -)$  are continuous functional on  $V$ , thus by (24.6.4.6) and similar analytic method as in (24.6.4.7) using the invariance of inner product.  $\square$

**Lemma (24.7.5.10).** Situation as in (24.7.5.9),  $M$  has a Banach norm s.t.  $(m, m) \leq \|m\|^2$ .  $\lrcorner$

*Proof:* (24.6.3.10) shows  $M$  does have a Banach norm. Then let  $M \cong V^{K\text{-fin}}$  and  $M^{*\text{alg}} \cong (V^*)^{K\text{-fin}}$ . However the Hermitian form induces  $M \cong M^{*\text{alg}}$ , thus we can form

$$M \xrightarrow{\Delta} M \oplus M \rightarrow V \oplus V^*.$$

let  $V'$  be the closure of the image of  $M$ , then it is a  $G$ -representation by (24.6.4.7), and then

$$(m, m) \leq \|i_1(m)\| \cdot \|i_2(m)\| \leq (\|i_1(m)\| + \|i_2(m)\|)^2.$$

$\square$

**Cor. (24.7.5.11).** The above proposition (24.7.5.9) is true for  $M$  admissible.  $\lrcorner$

**Def. (24.7.5.12) [Admissible Characters].** If  $(\pi, V) \in \text{Irr}^{\text{uni}}(G(K))$  is unitarizable, then for any Hilbert orthonormal basis  $\mathcal{B}_\pi$  of  $V$  and  $f \in C_c^\infty(G(F))$ , the sum

$$\Theta_\pi(f) = \sum_{\varphi \in \mathcal{B}_\pi} (\pi(f)\varphi, \varphi)$$

converges absolutely, and is independent of the basis chosen. The linear functional

$$\Theta_\pi : C_c^\infty \rightarrow \mathbb{C} : f \mapsto \Theta_\pi(f)$$

is called the **admissible character** of  $\pi$ .  $\lrcorner$

*Proof:* Cf. [Knapp, Representation theory of semisimple groups]Thm10.2.  $\square$

### Tempered Representations

**Prop. (24.7.5.13) [Tempered Representations].** Irreducible admissible tempered representations of  $G(\mathbb{K})$  (24.6.4.11) are infinitesimally-equivalent to unitarizable representations.  $\lrcorner$

*Proof:* Cf.[Knapp, Representation theory of semisimple groups]Thm8.53.  $\color{red}?$   $\square$

**Prop. (24.7.5.14) [Tempered Representations and Tempered Distributions].** For  $(\pi, V) \in \text{Irr}^{\text{uni}}(G(K))$ ,  $\pi$  is tempered iff the admissible character  $\Theta_\pi$  of  $\pi$  (24.7.5.12) defines a tempered distribution on a certain Schwartz space on  $G(K)$   $\color{red}?$   $\lrcorner$

*Proof:* Cf.[Knapp, Representation theory of semisimple groups]Thm12.23.  $\color{red}?$   $\square$

**Thm. (24.7.5.15) [Tempered Plancherel Theorem].** There is a Plancherel theorem relating spaces of functions on  $G(K)$  and tempered representations of  $G(K)$ .  $\lrcorner$

*Proof:* Cf.[Wallach. Real reductive groups. 2]13.14.1.  $\color{red}?$   $\square$

### Hecke Algebras

**Prop. (24.7.5.16) [Hecke Algebras of Lie Groups].**

- If  $K$  is a compact Lie group, then the **Hecke algebra**  $\mathcal{H}_K$  is defined to be the ring of smooth functions on  $K$  that is  $K$ -finite under both left and right translations, where the algebra is given by convolution. By Peter-Weyl theorem, these functions are dense in  $C(K)$  and  $L^2(K)$ , and it is an idempotented algebra over  $\mathbb{C}$   $\color{red}?$  Cf.[Bump, P309].
- The **Hecke algebra**  $\mathcal{H}_G$  is defined to be  $\mathcal{H}_G = \mathcal{H}_{\mathcal{K}} \otimes_{U(\mathfrak{k}_{\mathbb{C}})} U(\mathfrak{g}_{\mathbb{C}})$ , where the right action of  $U(\mathfrak{k}_{\mathbb{C}})$  on  $\mathcal{H}_K$  is given by

$$f * D = \rho(D)f,$$

$\color{red}?$  Cf.[Bump, P312]. Why is it independent of  $\mathcal{K}$   $\color{red}?$   $\lrcorner$

*Proof:*  $\square$

**Prop. (24.7.5.17) [Equivalence of Representations Lie Group Case].**

- For a compact Lie group  $K$ , the category  $\mathcal{M}(K)$  of smooth representations of  $\mathcal{H}_K$  is equivalent to the category of unitary representations of  $K$ .
- For a real reductive group  $G$  and a maximal compact subgroup  $K$ , the category of (admissible)( $\mathfrak{g}, K$ )-modules is equivalent to the category of smooth(admissible) modules of  $\mathcal{H}_G$ .  $\lrcorner$

*Proof:* Cf.[Cohomological Induction and Unitary Representations, P75].  $\square$

$\text{Irr}^{\text{uni}}(\text{GL}(n; \mathbb{R})^+)$

Cf.[Vogan, unitary dual of  $\text{GL}(n)$  over an Archimedean field].

**Lemma (24.7.5.18) [Finite Dimensional Case].** The only irreducible f.d. unitary representations of  $\text{GL}(n; \mathbb{R})^+$  are the 1-dimensional characters  $g \mapsto \det(g)^r$  where  $r$  is purely imaginary.  $\lrcorner$

*Proof:* Such a representation defines a continuous map of  $GL(n; \mathbb{R})^+$  into the compact unitary group  $U(n)$ . Now it induces a Lie algebra map  $\mathfrak{sl}_n(\mathbb{R}) \rightarrow \mathfrak{u}_m$ . This map must be trivial because otherwise this is an embedding because  $\mathfrak{sl}_n(\mathbb{R})$  is simple. But this is impossible because the adjoint action of  $\mathfrak{sl}_n(\mathbb{R})$  has real eigenvalues but the adjoint action of  $\mathfrak{u}_m$  are all purely imaginary by (3.7.5.4). So the action is trivial on  $SL(n, \mathbb{R})^+$ , so induces an irreducible representation of  $\det(g)$ , which is clearly 1-dimensional.  $\square$

**Lemma (24.7.5.19).** For  $\mathcal{H} \in \text{Rep}^{\text{uni}}(G)$ , if  $X \in \mathfrak{g}$ , we have

$$(Xu, v) = -(u, Xv),$$

so  $(Xv, w) = -(v, \overline{X}w)$  when complexified. So

$$(Rv, w) = -(v, Lw)$$

for any  $v, w \in \mathcal{H}$ , by (26.1.0.1).  $\lrcorner$

**Lemma (24.7.5.20) [Principal Series].** For the principal series  $P_\mu(\lambda, \varepsilon)$  of  $GL(2; \mathbb{R})^+$ , there exists an irreducible unitary representation in this class if  $\mu$  is purely imaginary and  $\lambda \geq 1/4$  real.  $\lrcorner$

*Proof:* Consider the unitary representation  $H(s_1, s_2, \varepsilon)$  defined in (24.6.4.17), then it is irreducible and its class is  $P_\mu(\lambda, \varepsilon)$  by (24.6.4.20) and (24.6.4.8).  $\square$

**Lemma (24.7.5.21) [Possibilities of Unitary Representations].** Let  $\mathcal{H}$  be a unitary representation of  $GL(2, \mathbb{R})^+$ . Assume  $\Delta, I$  acts by scalars  $\lambda, \mu$  respectively, then

- $\mu$  is purely imaginary and  $\lambda$  is real.
- If the  $(\mathfrak{g}, K)$ -module type of  $\mathcal{H}$  is a principal series  $P_\mu(\lambda, \varepsilon)$ , then  $\lambda > 0$ , and if  $\varepsilon = 1$ ,  $\lambda > 1/4$ .  $\lrcorner$

*Proof:* 1: This follows from (24.7.5.19), as action of  $I$  is skew-symmetric and action of  $\Delta$  is symmetric.

2: By (24.6.4.15),  $V^k \neq 0$  for  $k \equiv \varepsilon \pmod{2}$ , let  $f_k \in V^k$ . Because  $-4\Delta - H^2 + 2H = 4RL$  (26.1.0.1), take  $k = \varepsilon$ , then

$$(-4\lambda - \varepsilon^2 + 2\varepsilon)f_\varepsilon = 4RLf_\varepsilon.$$

But by (24.7.5.19),

$$(4RLf_\varepsilon, f_\varepsilon) = -4(Lf_\varepsilon, Lf_\varepsilon) < 0$$

thus  $4\lambda > 2\varepsilon - \varepsilon^2$ .  $\square$

**Cor. (24.7.5.22) [Reduction of  $\mu$ ].** The infinitesimal equivalence class of representations  $P_\mu(\lambda, \varepsilon)$  or  $D_\mu^\pm(k)$  contains an irreducible unitary representation iff  $\mu$  is purely imaginary and the corresponding class  $P(\lambda, \varepsilon)$  or  $D^\pm(k)$  contains an irreducible unitary representation.  $\lrcorner$

*Proof:*  $\mu$  must be purely imaginary by the proposition. And we may tensoring a unitary representation by a  $\deg(g)^r$ , it is also unitary iff  $r$  is purely imaginary, and this increases the value of action of  $I$  by  $2r$  and doesn't affect  $\mu$  and  $\varepsilon$  because  $\Delta$  has nothing to do with  $u$  (26.1.2.5).  $\square$

**Prop. (24.7.5.23) [Intertwining Integral].** Let  $s = \frac{1}{2}(s_1 - s_2 + 1)$ , define for  $f \in V$ ,

$$M(s) : H^\infty(s_1, s_2, \varepsilon) \rightarrow H^\infty(s_2, s_1, \varepsilon) : (M(s)f)(g) = \int_{N(F)} f(w_0ug)du.$$

Then if  $\text{Re}(s_1 - s_2) > 0$ , the integral is absolutely convergent, and commutes with the action of  $G$ .  $\lrcorner$

*Proof:* Replacing  $f$  with  $\rho(h)f$ , we see that the convergence of  $M(s)(f)(h)$  is equivalent to the convergence of  $M(s)(\rho(h)f)$ , so we assume  $h = 1$ .

We use the identity

$$\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} = \begin{bmatrix} \Delta_x^{-1} & -x\Delta_x^{-1} \\ & \Delta_x \end{bmatrix} k_{\theta_x}$$

similar to (26.1.2.2) where

$$\Delta_x = \sqrt{1+x^2}, \quad \theta(x) = \arctan(-1/x).$$

Then

$$(M(s)f)(1) = \int_{-\infty}^{\infty} (1+x^2)^{-s} f(k_{\theta(x)}) dx$$

which converges for  $s > 1/2$ , that is  $\operatorname{Re}(s_1 - s_2) > 0$ .

To show  $M(s)f \in H(s_2, s_1, \varepsilon)$ , we check

$$(M(s)f)\left(\begin{bmatrix} 1 & \xi \\ & 1 \end{bmatrix} g\right) = (M(s)f)(g), \quad (M(s)f)\left(\begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} g\right) = \operatorname{sgn}(y_1)^\varepsilon |y_1|^{s_2+\frac{1}{2}} |y_2|^{s_1-\frac{1}{2}} (M(s)f)(g).$$

The first one is an easy consequence of change of variable, for the second,

$$\begin{aligned} (M(s)f)\left(\begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} g\right) &= \int f\left(\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} g\right) dx \\ &= \int f\left(\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} \begin{bmatrix} 1 & y_1^{-1}y_2x \\ & 1 \end{bmatrix} g\right) dx \\ &= \frac{y_1}{y_2} \int f\left(\begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} \begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) dx \\ &= \frac{y_1}{y_2} \operatorname{sgn}(y_1)^\varepsilon |y_1|^{s_2-\frac{1}{2}} |y_2|^{s_1+\frac{1}{2}} \int f\left(\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) dx \\ &= \operatorname{sgn}(y_1)^\varepsilon |y_1|^{s_2+\frac{1}{2}} |y_2|^{s_1-\frac{1}{2}} (M(s)f)(g) \end{aligned}$$

To check  $M(s)f$  is smooth, notice that the restriction of  $M(s)f$  to  $K$  equals

$$(M(s)f)(k_t) = \int_{-\infty}^{\infty} (1+x^2)^{-s} f(k_{\theta(x)+t}) dx.$$

The convergence is uniform in  $t$ , thus is smooth in  $t$ .

The commutativity of  $M(s)$  with  $G$ -action is immediate, because left and right action commutes.

□

**Prop. (24.7.5.24).** Let  $f_{k,s}$  be the function  $f_k$  in  $H(s_1, s_2, \varepsilon)$  (24.6.4.17) where  $\operatorname{Re}(s_1 - s_2) > 0$  and  $s = \frac{1}{2}(s_1 - s_2 + 1)$ , then

$$M(s)f_{k,s} = (-i)^k \sqrt{\pi} \frac{\Gamma(s)\Gamma(s-\frac{1}{2})}{\Gamma(s+\frac{k}{2})\Gamma(s-\frac{k}{2})} f_{k,1-s}$$

□

*Proof:* Because  $M(s)$  commutes with  $G$ -action,  $M(s)f_{k,s}$  is a multiple of  $f_{k,1-s}$ , thus it suffices to calculate  $(M(s)f_{k,s})(1)$ , which is

$$(M(s)f)(1) = \int_{-\infty}^{\infty} (1+x^2)^{-s} e^{ik\theta(x)} dx, \quad \theta(x) = \arctan(-1/x)$$

by (24.7.5.23).

This integral then is calculated to be the expression above, by (12.3.9.1).  $\square$

**Lemma (24.7.5.25) [Complementary Series].** For  $\mu$  purely imaginary and  $0 < \lambda < 1/4$  and  $\varepsilon = 0$ , there exists an irreducible unitary representation in this class of the  $(\mathfrak{g}, K)$ -module  $P_\mu(\lambda, 0)$ .  $\lrcorner$

*Proof:* Let  $s_1, s_2$  be complex numbers, we construct first a Hermitian pairing

$$H(s_1, s_2, \varepsilon)_{fin} \times H(-\overline{s_1}, -\overline{s_2}, \varepsilon)_{fin} \rightarrow \mathbb{C} : (f, g') \mapsto \int_K f(k) \overline{g'(k)} dk$$

which is invariant under action of  $G$  by (12.10.3.16). Now let  $s_2 = -\overline{s_1}$ , then  $s = \frac{1}{2}(s_1 - s_2 + 1)$  is real and  $\mu = s_1 + s_2$  is purely imaginary. Then composing this pairing with  $i^\varepsilon M(s) : H(s_1, s_2, \varepsilon)_{fin} \rightarrow H(s_2, s_1, \varepsilon)_{fin} = H(-\overline{s_1}, -\overline{s_2}, \varepsilon)_{fin}$ , then

$$(f, \overline{f'}) = \int_K f(k) \overline{i^\varepsilon M(s) f'(k)} dk.$$

is  $G$ -invariant. We will show that it is positive definite if  $\varepsilon = 0$  and  $1/2 < s < 1$ .

It can be seen from (24.7.5.24) that an orthogonal basis for  $H(s_1, s_2, 0)_{fin}$  under this pairing is  $f_{k,s}$  for  $k$  even. And by (24.7.5.24),

$$(f_{k,s}, f_{k,s}) = (-1)^{k/2} \sqrt{\pi} \frac{\Gamma(s) \Gamma(s - \frac{1}{2})}{\Gamma(s + \frac{k}{2}) \gamma(s - \frac{k}{2})}$$

which is positive for  $1/2 < s < 1$ . Now we obtain a unitary representation of  $G$  on the Hilbert completion of this space (24.7.1.13).

Now we have constructed a unitary representation in the infinitesimal equivalence class  $P_\mu(\lambda, 0)$  with  $\lambda = s(1-s)$ ,  $s = \frac{1}{2}(s_1 - \overline{s_1} + 1)$ , so any  $0 < \lambda < 1/4$  is possible.  $\square$

**Lemma (24.7.5.26).** For any integer  $k$ , there is a bijection between holomorphic functions  $\varphi$  on  $\mathcal{H}$  and smooth functions  $\Phi$  on  $\mathrm{GL}(2; \mathbb{R})^+$  that is invariant under  $Z(R)^+$  and

$$\Phi(gk_\theta) = e^{ik\theta} \Phi(g), \quad L\Phi = 0.$$

$\lrcorner$

*Proof:* The bijection is given by

$$\varphi(z) = y^{-k/2} \Phi \left( \begin{bmatrix} y & x \\ 1 & 1 \end{bmatrix} \right), \quad \Phi(g) = ((y^{k/2} \varphi)[g]_k) i.$$

and the proof is a combination of formal calculation in (26.1.3.10) and (26.1.2.15) forgetting  $\Gamma$ :

$$L_k(y^{k/2} f(z)) = -(z - \overline{z}) \frac{\partial}{\partial \overline{z}} - \frac{k}{2} (y^{k/2} f(z)) = -2iy^{(k+2)/2} \frac{\partial}{\partial \overline{z}} f(z).$$

$\square$

**Lemma (24.7.5.27) [Discrete Series].** For  $k \in \mathbb{Z}_{\geq 2}$ , there exists a unitary representation in the infinitesimal equivalence class  $D^\pm(k)$ , more precisely,

- Let  $L^2(\mathcal{H}, \mu_k)$  be the  $L^2$ -space of holomorphic functions  $f$  on the upper plane  $\mathcal{H}$  w.r.t the measure  $\mu_k = y^k \frac{dx dy}{y^2}$  (12.5.5.3). Then the left action

$$\pi_k(g)f = f[g^{-1}]_k.$$

of  $G$  is unitary and this representation  $\pi_k$  is in the infinitesimal equivalence class  $D^-(k)$ .

- Consider the automorphic of  $\mathrm{GL}(2; \mathbb{R})^+$ :

$$\iota\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a & -b \\ -c & d \end{bmatrix}$$

then the representation  $\pi_k \circ \iota$  is in the infinitesimal equivalence class  $D^+(k)$

┘

*Proof:* The second one follows from the first one, as  $\iota$  interchanges the action of  $K$  thus the  $K$ -types.

For the first one, firstly it is a unitary representation: for  $z' = g(z) = x' + iy'$ , we have

$$y' = \frac{ad - bc}{|cz + d|^2} y$$

and  $\mu_z = \mu_{z'}$ , thus

$$\|\pi(g^{-1})f\|^2 = \int_{\mathcal{H}} |f(z')|^2 \frac{(ad - bc)^k}{|cz + d|^{2k}} y^k \mu_z = \int_{\mathcal{H}} |f(z')|^2 (y')^k \mu_{z'} = \|f\|^2.$$

For the infinitesimal equivalence class, we consider the orthogonal basis  $\varphi_n = \left(\frac{z-i}{z+i}\right)^n \frac{(2i)^k}{(z+i)^k}$  and prove

$$\pi(k_\theta^{-1})\varphi_n = e^{2\pi i(k+2n)\theta} \varphi_n.$$

Then this will determine the  $K$ -type of  $\pi_k$ . This can be proven by direct calculation, Cf.[Ngo, P39].

□

**Cor. (24.7.5.28).** By (24.7.5.26) and the measure  $\mu_k$  we choose, it is clear that there is an isometry between  $L^2(\mathcal{H}, \mu_k)$  and a subspace of  $L^2(G/Z)$  that is compatible with the left  $G$  action on  $L^2(G)$ , but the left and right action on  $L^2(G/Z)$  is isomorphic, as  $f(t) \mapsto f(t^{-1})$  intertwine them, because  $G/Z$  is unimodular. So this representation is **square integrable**, i.e. it can be embedded in  $L^2(G/Z)$ . ┘

**Lemma (24.7.5.29) [Limits of Discrete Series].** There exists a unitary representation in the infinitesimal equivalence class  $D^\pm(1)$ . ┘

*Proof:* These two classes already appear in the unitary representation  $H(0, 0, 1)$ , by (24.6.4.20). □

**Prop. (24.7.5.30) [List of Irreducible Unitary Representations of  $\mathrm{GL}(2, \mathbb{R})^+$ ].** Let  $\mu$  be purely imaginary,

- The 1-dimensional representation  $g \mapsto |\deg(g)|^\mu$ .
- The unitary principal series  $P_\mu(\lambda, \varepsilon)$ , where  $\varepsilon = 0, 1$  and  $\lambda \geq 1/4$ .



- The complementary series representations  $P_\mu(\lambda, 0)$  where  $0 < \lambda < 1/4$ .
- The holomorphic discrete series  $D_\mu^0(k) (k \geq 2)$  and limits of discrete series ( $k = 1$ )  $D_\mu^\pm(k)$ .

Notice each of these infinitesimal equivalence classes of irreducible representations has a unique representative that is a unitary representation by (24.6.4.9).  $\lrcorner$

*Proof:* By (24.6.4.8) and (24.6.4.9), the conclusion follows from the classification of  $(\mathfrak{g}, K)$ -modules (24.6.4.21) and determining which infinitesimal class has a unitary representative, which follows from (24.7.5.18)(24.7.5.21), (24.7.5.20)(24.7.5.22), (24.7.5.25)(24.7.5.27), (24.7.5.29).  $\square$

**Cor. (24.7.5.31).** Similar as in (24.6.4.22), by (24.6.4.8), we can classify all irreducible unitary representations of  $GL(2, \mathbb{R})$ .  $\lrcorner$

## 6 Semisimple Lie Groups

References are [Knapp, Representations of Semisimple Lie Groups].

### Nilpotent Lie Groups

**Prop. (24.7.6.1).** If  $G \in \text{LieGrp}$  is simply-connected and nilpotent, then for any sub-Lie-algebra  $\mathfrak{h} \subset \mathfrak{g}$ , there is a unique connected subgroup  $H \leq G$  with  $\text{Lie}(H) = \mathfrak{h}$ ?. And any  $\lambda \in \mathfrak{g}^*$  s.t.  $\lambda([\mathfrak{h}, \mathfrak{h}]) = 0$  has a unique lifting to a character  $\sigma_\lambda$  of  $H$ ?.  $\lrcorner$

*Proof:*

$\square$

**Thm. (24.7.6.2) [Kirillov].** Let  $G \in \text{LieGrp}$  be simply-connected. For any  $\lambda \in \mathfrak{g}^*$ , let  $\mathfrak{h} \subset \mathfrak{g}$  be a maximal Lie subalgebra s.t.  $\lambda([\mathfrak{h}, \mathfrak{h}]) = 0$ , then there exists  $H = \exp(\mathfrak{h})$  and  $\sigma_\lambda \in H^\vee$  as in (24.7.6.1). Then  $\text{ind}_H^G(\sigma_\lambda)$  is irreducible, and its equivalent class only depends on the orbit of  $\lambda$  under the coadjoint action  $\text{Ad}^*$ . Then the map

$$\mathcal{O}_{\text{Ad}^*} \rightarrow \widehat{G} : \lambda \mapsto \text{ind}_H^G(\sigma_\lambda)$$

is a bijection which is a homeomorphism w.r.t. the natural quotient topology on  $\mathcal{O}_{\text{Ad}^*}$  and the Fell topology on  $\widehat{G}$  (24.7.2.9).  $\lrcorner$

*Proof:* Cf. [Representations of Nilpotent Lie Groups and Their Applications].  $\square$

### Solvable Lie Groups

**Thm. (24.7.6.3)** [Takenouchi[Tak57]/Bernat[Ber65]/Auslander-Kostant[A-K71]]. A simply-connected Lie group is of type I iff the space of coadjoint orbits is countably-separated, and every  $\lambda \in \mathfrak{g}^*$  has lift to the group  $G_\lambda = \{g \in G | \text{Ad}^*(g)\lambda = \lambda\}$ .

This is the case for any exponential solvable Lie groups (i.e. when  $\exp_G$  is an isomorphism).  $\lrcorner$

*Proof:* •

$\square$

## 7 Unitary Groups

**Prop. (24.7.7.1) [Parameters for Representations of  $U(m; \mathbb{R})$ ].** The Langlands parameters of irreducible representations of the definite unitary group  $U(m; \mathbb{R})$  is determined by their restriction to  $W_{\mathbb{C}} \cong \mathbb{C}^{\times}$ , which are up to conjugacy all the morphisms of the form

$$\phi(z) = \text{diag}((z/\bar{z})^{a_1}, \dots, (z/\bar{z})^{a_m}).$$

On the other hand, if we regard  $U(m; \mathbb{R})$  as a subgroup of  $GL(m; \mathbb{C})$ , the restrictions of the rational irreducible algebraic representations  $W_{\underline{k}}$  of  $GL(m; \mathbb{C})$  for any  $k_1 \geq \dots \geq k_m \in \mathbb{Z}$  (whose highest weight w.r.t. the standard Borel subgroup is the character  $\delta_{\underline{k}} : (z_1, \dots, z_m) \mapsto \prod_i z_i^{k_i}$ ) to  $U(m; \mathbb{R})$  form a system of irreducible representations of  $U(m; \mathbb{R})$ .  $W_{\underline{k}}$  is said to have **regular weights** if  $k_1 > k_2 > \dots > k_m$ .

The relation of the two descriptions of  $\text{Irr}^{\text{uni}}(U(m; \mathbb{R}))$  is given by

$$a_i = k_i + \frac{m-1}{2} - i.$$

┘

*Proof:* ? Cf. [Bergeron-Clozel, Spectre automorphe des variétés hyperboliques et applications topologiques] Prop 4.3.2 and □

## 24.8 Lie Algebras II

### 1 Springer Theory

**Notation(24.8.1.1).**

- Let  $G$  be a simply-connected complex Lie group with Lie algebra  $\mathfrak{g}$  and a Borel subgroup  $B$  generated by a maximal torus  $T$  and 1-dimensional subgroups  $\exp(te_i)$  for  $e_i$ .
- Let  $\mathcal{F}$  be the flat variety of  $G$  consisting of Borel subalgebras  $\mathfrak{b} \subset \mathfrak{g}$ .

┘

**Def.(24.8.1.2).**

┘

## 24.9 Admissible Representations of $p$ -adic Reductive Groups

References are [B-Z76, B-Z77, Sil78, Tad86, Kud94, Bum97, P-R08, G-H11], [admissible dual of  $\mathrm{GL}(N)$  via compact open subgroups, Bushnell-Kutzko], [Getz, automorphic Representations], [Bernstein. Representations of  $p$ -adic groups], [Casselman, Introduction to the theory of admissible representations of  $p$ -adic reductive groups].

### Notation(24.9.0.1).

- Use notations on representations of locally profinite groups defined in [Locally Profinite Groups](#).
- Use group-theoretic notations as in [\(3.1.2.15\)](#).
- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \mathfrak{p}_K, \varpi, \kappa) \in p\text{-LField}$ . Let  $q = \#\kappa$ .
- Fix a non-trivial character  $\psi : K \rightarrow \mathbb{C}^\times$ .
- Choose the Haar measure  $dx$  on  $K$  that is self-dual w.r.t.  $\psi$  as in [\(12.11.1.34\)](#).
- Let  $G \in \text{RedGrp}/K$ .
- If  $G$  is quasi-split, choose a Borel subgroup  $B \leq G$  and let  $U = R_u(B), T = B/U$ .
- Let  $\mathfrak{g} = \mathrm{Lie}(G(K))$ .
- Use notations on root data defined in [Reflection Groups and Coxeter Groups](#).
- Let  $Z \in \text{AlgGrp}/K$  be the maximal split torus contained in the center of  $G$ .
- Let  $n \in \mathbb{Z}_+$ .
- For any  $\chi \in (\mathcal{O}_K^*)^\vee$ , by continuity, there is a minimum  $v \in \mathbb{N}$  that  $\chi(1 + \mathfrak{p}^v) = 1$ , and  $\mathfrak{p}^v$  is called the **conductor of character** of  $\chi$ .

┘

### Notation(24.9.0.2). If moreover $G = \mathrm{GL}(n)$ ,

- Use the notations on  $\mathrm{GL}(n)$  defined in [General Linear Groups](#).
- For  $\alpha \in \mathbb{A}^{n-1}(K)$ , let  $\psi_{N,\alpha}$  be a character of  $\mathrm{Unip}(K)$  given by  $\psi_{\mathrm{Unip}}(g) = \sum_i \psi(\alpha_i g_{i,i+1})$ .
- Denote  $\mathcal{K} = \mathrm{GL}(n; \mathcal{O}_K)$ , which is the maximal compact subgroup of  $\mathrm{GL}(n; K)$  up to conjugacy [\(20.7.3.10\)](#).
- For  $k \in \mathbb{Z}_+$ , define subgroups  $\mathrm{Unip}_k(n; K)$  of  $\mathrm{Unip}(n; K)$  as follows:  $\mathrm{Unip}_k(n; K)$  is the group of unipotent matrices that  $v(e_{ij}) \geq k(i-j)$ . Then  $\cup_{k>0} \mathrm{Unip}_k(n, K) = \mathrm{Unip}(n, K)$ . In particular,  $\mathrm{Unip}(n, K)$  is exhausted by compact open subgroups.

┘

## 1 Basics

References are [Representations of  $p$ -adic Groups Bernstein]. [[?Bum98](#)]Chap4.

### Geometry

**Def.(24.9.1.1) [Unramified Quasi-Characters].** A **unramified quasi-character** on  $K$  is a quasi-character  $\chi$  that  $\chi(\mathcal{O}_K^*) = 1$ .

Define  $\alpha(\chi) = \chi(\varpi)$  if  $\chi$  is unramified and 0 otherwise, called the **Satake parameter** of  $\chi$ . ┘

## Hecke Algebras

### Smooth Representations

**Prop. (24.9.1.2) [Irreducible Smooth Representations are Admissible].**  $\text{Irr}^{\text{alg}}(G(K)) = \text{Irr}^{\text{adm}}(G(K))$ .  $\lrcorner$

*Proof:* ? Cf. [Getz]Thm8.3.5. We only prove for  $G = \text{GL}(n)$ .

By (24.9.3.7)(24.9.3.8) and (24.9.3.5), it suffices to prove for  $\pi$  quasi-cuspidal. But then this follows from (24.9.3.9)(24.11.1.13) and (24.1.6.54).  $\square$

## 2 Bernstein Centers

References are [Ber84].

**Def. (24.9.2.1) [Bernstein Centers].** The **Bernstein center**  $G$  is defined to be  $\mathcal{Z}(G(K)) \triangleq \text{End}(\mathbf{1}_{\text{Rep}^{\text{adm}}(G(K))})$ .

For any  $\pi \in \text{Irr}^{\text{adm}}(G(K))$ , elements of  $\mathcal{Z}(G(K))$  acts on  $\pi$  as scalars. Thus there is a map

$$\mathcal{Z}(G(K)) \rightarrow \text{Map}(\text{Irr}^{\text{adm}}(G(K)), \mathbb{C}),$$

called the **infinitesimal character of Bernstein center** of  $G$ .

And we can describe explicitly which functions on  $\text{Irr}^{\text{adm}}(G(K))$  come from elements of the Bernstein center ?.  $\lrcorner$

**Prop. (24.9.2.2).** Bernstein center can also be described as conjugation-invariant distributions acting on  $\mathcal{T}(G(K))$ . ?  $\lrcorner$

## 3 Supercuspidal Representations

### Jacquet-Functors

**Def. (24.9.3.1) [Jacquet-Functors].** Let  $P = LU$  be a parabolic subgroup of  $G$  and  $(\pi, V) \in \text{Rep}^{\text{alg}}(G(F))$ , then

$$V_U = V / (\text{span}\{\rho(u)v - v | u \in U\}) = V/V(U)$$

is naturally a  $L \cong P/U$  representation, so we can define the **Jacquet functor**:

$$J_P^G : \text{Rep}^{\text{alg}}(G(F)) \rightarrow \text{Rep}^{\text{alg}}(L(F)) : V \mapsto V_U.$$

Given a character  $\psi$  of  $U(F)$ , similarly  $V_{U,\psi} = V / (\text{span}\{\rho(u)v - \psi(u)v | u \in U\}) = V/V(U, \psi)$  is a  $Z(K)$ -module, we can also define similarly the **twisted Jacquet functor**

$$J_{P,\psi}^G : \text{Rep}^{\text{alg}}(G(F)) \rightarrow \text{Rep}^{\text{alg}}(Z(K)) : V \mapsto V_{U,\psi}.$$

For convenience, define the **normalized Jacquet functor**:  $r_P^G = \sqrt{\frac{\Delta_P}{\Delta_G}} \otimes J_P^G$ .

$J_P^G, J_{P,\psi}^G, r_P^G$  are exact functors, by (24.1.2.6).  $\lrcorner$

**Prop. (24.9.3.2) [Finite Length Property].**  $J_P^G, J_{P,\psi}^G, r_P^G$  map smooth representations of finite length to smooth representations of finite length.  $\lrcorner$

*Proof:* If  $V$  is a f.g.  $G$ -module, then it is a f.g.  $P$ -module, because  $G = PK$ , and  $Kv_i$  is of f.d. for any  $v \in V$ . Thus the quotient  $V_U$  of  $V$  is also f.g..

For finite length ?, Cf. [Bernstein-Zelevinsky2, P8].  $\square$

### Parabolic Inductions

**Def. (24.9.3.3) [Parabolic Inductions].** Let  $P = LU$  be a parabolic subgroup of  $G$  and denote  $\text{pr} : P \rightarrow P/U \cong L$ . The (normalized) **parabolic induction** functor  $\text{n-Ind}_P^G$  is the functor

$$\text{n-Ind}_P^G : \text{Rep}^{\text{alg}}(L) \rightarrow \text{Rep}^{\text{alg}}(G) : \rho \mapsto \text{n-Ind}_P^G(\rho \circ \text{pr}) \quad (24.1.6.40).$$

There are also other functors

$$\text{n-ind}_P^G : \text{Rep}^{\text{alg}}(L) \rightarrow \text{Rep}^{\text{alg}}(G) : \rho \mapsto \text{n-ind}_P^G(\rho \circ \text{pr}) \quad (24.1.6.40).$$

$$\text{Ind}_P^G : \text{Rep}^{\text{alg}}(L) \rightarrow \text{Rep}^{\text{alg}}(G) : \rho \mapsto \text{Ind}_P^G(\rho \circ \text{pr}) \quad (24.1.6.40).$$

$$\text{ind}_P^G : \text{Rep}^{\text{alg}}(L) \rightarrow \text{Rep}^{\text{alg}}(G) : \rho \mapsto \text{ind}_P^G(\rho \circ \text{pr}) \quad (24.1.6.40).$$

┘

**Prop. (24.9.3.4) [Parabolic Induction and Jacquet Functor].** For any parabolic subgroup  $P$ ,  $\text{n-Ind}_P^G$  (24.9.3.3) is right adjoint to the normalized Jacquet-functor  $r_P^G$  (24.9.3.1). ┘

*Proof:* For  $\sigma \in \text{Rep}^{\text{alg}}(L)$ ,  $\rho \in \text{Rep}^{\text{alg}}(G)$ ,

$$\text{Hom}_{P/U}(\sqrt{\Delta_P} \otimes (\text{res}_P^G(\sigma))_U, \rho) \cong \text{Hom}_P(\text{res}_P^G(\sigma), \sqrt{\delta_P} \otimes (\rho \circ \text{pr})) = \text{Hom}_G(\sigma, \text{n-Ind}_P^G(\rho \circ \text{pr})).$$

□

**Prop. (24.9.3.5) [Admissibility and Finite Length Property].**  $\text{n-Ind}_P^G$  is an exact functor, and it maps admissible representations to admissible representations and maps finite length representations to finite length representations. ┘

*Proof:* The exactness follows from [B-Z77]1.9. ?

The admissibility follows from (24.1.6.50) and the Iwasawa-decomposition.

For finite length. Cf. [Prasad, Representation theory of  $\text{GL}(n)$  over non-Archimedean local fields]Thm4.1 ? or [BZ1]2.26. □

**Prop. (24.9.3.6) [Transitivity of Parabolic Inductions].** ┘

### Supercuspidal Representation

References are [B-Z77, Kud94].

**Def. (24.9.3.7) [Supercuspidal Representations].**  $(\pi, V) \in \text{Rep}^{\text{alg}}(G(K))$  is called a **quasi-cuspidal representation** if for any properly-contained parabolic subgroup  $P = LU$  of  $G$ , we have  $r_P^G(V) = 0$ . A **cuspidal representation** is a representation that is both quasi-cuspidal and admissible. The category of cuspidal representations is denoted by  $\text{Rep}^{\text{cusp}}(G(F))$ . ┘

**Prop. (24.9.3.8) [Supercuspidal Dichotomy].** If  $\pi \in \text{Irr}^{\text{alg}}(G(F))$ , then either  $\pi$  is (quasi-)cuspidal, or  $\pi$  is a subrepresentation of  $I_P^G(\rho)$ , where  $P = LU$  is a proper parabolic subgroup of  $G$ , and  $\rho \in \text{Irr}^{\text{alg}}(L(F))$  is (quasi-)cuspidal. ┘

*Proof:* If  $\pi$  is not (quasi-)cuspidal, choose a minimal parabolic  $P$  that  $r_P^G(\pi) \neq 0$ . As  $r_P^G(\pi)$  is f.g. (24.9.3.2), by (24.1.2.10), it has an irreducible subquotient  $\sigma$ . Then

$$0 \neq \text{Hom}_M(r_P^G \pi, \sigma) = \text{Hom}_G(\pi, I_P^G \sigma).$$

And by minimality,  $\sigma$  is (quasi-)cuspidal by (24.11.2.4). □

**Thm. (24.9.3.9) [Characterizing Supercuspidal Representations, Harish-Chandra].** then for  $(\pi, V) \in \text{Rep}^{\text{alg}}(\text{GL}(n; K))$ , the following are equivalent:

1.  $\pi$  is quasi-cuspidal (24.9.3.7).
2. For any principal congruence subgroup  $\mathcal{K}_m$ , the function  $D_{\xi, K} : G \rightarrow V : g \mapsto \pi(e_K)\pi(g^{-1})\xi$  has compact support modulo  $Z(K)$ .
3. Every matrix coefficient of  $\pi$  is compactly supported modulo  $Z(K)$ .
4. The restriction of  $\pi$  to  $G_n^0$  is compact.

┘

*Proof:*  $2 \rightarrow 3$  is verbatim as that of (24.1.6.54),  $3 \rightarrow 4$  is easy.  $4 \rightarrow 2$  follows from the definition of compact representations and the fact  $\text{GL}(n; K)/G_n^0 Z(K)$  is finite.

$1 \iff 2$ : Cf. [Bernstein-Zelevinsky, P34]. ?

□

**Cor. (24.9.3.10) [Contragradients of Cuspidal Representations].** For  $\pi \in \text{Rep}^{\text{cusp}}(\text{GL}(n; K))$ ,  $\pi^\vee \in \text{Rep}^{\text{cusp}}(\text{GL}(n; K))$  too.

┘

**Prop. (24.9.3.11) [Supercuspidal Representations Exist].** If  $\text{char } K = 0$ , then supercuspidal representations of  $G(K)$  exist.

┘

*Proof:* Cf. [Beuzart-Plessis, A short proof of the existence of supercuspidal representations for all reductive  $p$ -adic groups].

□

### Generalized Steinberg-Representations

**Def. (24.9.3.12) [Generalized Steinberg-Representations].** For a parabolic subgroup  $P \leq G$ , define the generalized **Steinberg-representation**

$$\text{St}_P^G \triangleq \text{Ind}_P^G(\mathbf{1}) / \sum_{Q \supset P} \text{Ind}_Q^G(\mathbf{1}) \in \text{Rep}^{\text{alg}}(G(K)) \quad (24.9.3.3),$$

where  $Q$  runs through all parabolic subgroups containing  $P$ .

For example, if  $G = \text{GL}(2)$ ,  $\text{St}_2$  has length 2, and there are exact sequences:

$$0 \rightarrow 1 \rightarrow \text{val}^{-1/2} \times \text{val}^{1/2} \rightarrow \text{St}_2 \rightarrow 0, \quad 0 \rightarrow \text{St}_2 \rightarrow \text{val}^{1/2} \times \text{val}^{-1/2} \rightarrow 1 \rightarrow 0$$

┘

**Prop. (24.9.3.13) [Generalized Steinberg-Representations].** Let  $K_0^{\text{f.l.}}(G(K))$  be the Grothendieck-group of finite length smooth representations of  $G(K)$ , then

- For any parabolic subgroup  $P \leq G$ ,  $\text{St}_P^G$  (24.9.3.12) is the unique irreducible quotient representation of  $\text{Ind}_P^G(\mathbf{1})$ .
- For any parabolic subgroup  $P \leq G$ , the map  $Q \mapsto \text{St}_Q^G$  is a bijection from the set of parabolic subgroups of  $G$  containing  $P$  to the set of Jordan-Hölder constituents of  $\text{Ind}_P^G(\mathbf{1})$ .
- For parabolic subgroups  $P, P' \leq G$ ,  $P \sim P' \iff \text{St}_P^G \cong \text{St}_{P'}^G$ .
- Let  $\Delta_0$  be a set of simple roots of the relative root system of  $G$ , and for each subset  $I \leq \Delta_0$ , let  $P_I$  be the parabolic subgroup corresponding to  $I$ . Then for each subset  $I \leq \Delta_0$ , there is an exact sequence of smooth representations

$$0 \rightarrow \text{Ind}_G^G(\mathbf{1}) \rightarrow \bigoplus_{\substack{I \subset K \subset \Delta_0 \\ \#\Delta_0 \setminus K=1}} \text{Ind}_{P_K}^G \rightarrow \bigoplus_{\substack{I \subset K \subset \Delta_0 \\ \#\Delta_0 \setminus K=2}} \text{Ind}_{P_K}^G \rightarrow \dots \rightarrow \bigoplus_{\substack{I \subset K \subset \Delta_0 \\ \#K \setminus I=1}} \text{Ind}_{P_K}^G \rightarrow \text{Ind}_{P_I}^G \rightarrow \text{St}_{P_I}^G \rightarrow 0.$$

where the differential is defined by  $\text{?}$ .

┘

*Proof:* Cf.[Period domains over finite and p-adic fields, Dat-Orlik-Rapoport]P304.  $\text{?}$

□

#### 4 Admissible Invariant Distributions(Harish-Chandra)

References are [HC99].

**Notation(24.9.4.1).**

- Let  $\text{char } K = 0$ .

┘

**Def.(24.9.4.2) [Regularity Polynomial].** Let  $l = \text{rank } \mathfrak{g}$  and for  $X \in \mathfrak{g}$ , let  $\eta_{\mathfrak{g}}(X)$  be the coefficient of  $T^l$  in the polynomial  $\det(T - \text{ad}X)$ , then  $\eta_{\mathfrak{g}}$  is a polynomial function on  $\mathfrak{g}$ , whose zero set is the set of singular elements in  $\mathfrak{g}$ .

┘

**Prop.(24.9.4.3) [Finitely Many Nilpotent Orbits].** The set of nilpotent elements in  $\mathfrak{g}$  a union of f.m.  $G(K)$ -orbits, which are called the **nilpotent orbits**.

Since  $\text{char } K = 0$ , the set  $\mathcal{O}(0)$  of nilpotent orbits in  $\mathfrak{g}$  is finite.

And a  $G$ -orbit  $\mathcal{O}$  is nilpotent iff  $0 \in \overline{\mathcal{O}}$ .

┘

*Proof:* Cf.[HC99].  $\text{?}$

□

#### Fourier Transform on Lie Algebras

**Def.(24.9.4.4) [Fourier Transformations].** Let  $\mathfrak{g} = \text{Lie}(G)$  and  $B$  be a  $K$ -valued non-degenerate symmetric  $G$ -invariant bilinear form on  $\mathfrak{g}$   $\text{?}$ . And let  $dX$  be a Haar-measure on  $\mathfrak{g}$ . Then we can define Fourier transform on  $\mathfrak{g}$ : For  $f \in C_c^\infty(\mathfrak{g})$ , denote

$$\mathcal{F}f(Y) = \int_{\mathfrak{g}} f(X)\psi(B(X, Y))dX,$$

then  $\mathcal{F}$  is a linear bijection of  $C_c^\infty(\mathfrak{g})$  with itself, and we can also define the Fourier transform of a distribution on  $\mathfrak{g}$ .

┘

**Prop.(24.9.4.5).** If  $\mathcal{O}$  is a  $G$ -orbit in  $\mathfrak{g}$ , then  $\mathcal{O}$  carries a  $G$ -invariant measure  $\mu_{\mathcal{O}}$ .

┘

*Proof:* Cf.[Rao, Orbital integrals in reductive groups].  $\text{?}$

□

**Prop.(24.9.4.6).** For a  $G$ -orbit  $\mathcal{O}$  in  $\mathfrak{g}$ ,  $\mathcal{F}\mu_{\mathcal{O}}$  is a locally  $L^1$ -function on  $\mathfrak{g}$  that is locally constant on the set of regular elements in  $\mathfrak{g}$ .

┘

*Proof:*  $\text{?}$

□

#### 5 Generic representations

**Notation(24.9.5.1).**

- Let  $G$  be quasi-split.

┘

**Def.(24.9.5.2) [ $\psi$ -Generic Represnetations].** For  $(\pi, V) \in \text{Irr}^{\text{adm}}(G(K))$  and  $\psi \in U(K)^\vee$ ,



- a  **$\psi$ -Whittaker functional** on  $V$  is a  $U(K)$ -equivariant map  $\lambda : V \rightarrow \mathbb{C}_\psi$  (Compare with (26.1.4.9).)
- a  **$\psi$ -Whittaker model** is a subrepresentation of  $\text{Ind}_{U(K)}^{G(K)}(\mathbb{C}_\psi)$  that is isomorphic to  $\pi$  and consists of functions  $\varphi$  of moderate growth.

$\pi$  is said to be a  **$\psi$ -generic representation** if it admits a non-zero  $\psi$ -Whittaker functional. This only depends on the  $(B/U)(K)$ -orbit of  $\psi$ ?

And it is said to be a **generic representation** if it is  $\psi$ -generic for some generic unipotent character  $\psi \in U(K)^\vee$  (10.4.4.23).  $\lrcorner$

**Cor. (24.9.5.3) [Whittaker Functionals and Whittaker Models].** Situation as in (24.9.5.2), if  $\lambda$  is a Whittaker functional of  $\pi$ , then the space of functions

$$W_v(g) = \lambda_v(\pi(g)v)$$

for  $v \in V$  is a Whittaker model for  $\pi$ .  $\lrcorner$

*Proof:* A Whittaker model is equivalent to a  $\text{GL}(n; K)$ -homomorphism  $V \rightarrow \text{Ind}_{\text{Unip}(n; K)}^{\text{GL}(n; K)}(\mathbf{1}(\psi_{\text{Unip}}))$ , which by smooth Frobenius reciprocity (24.1.6.43) is equivalent to a  $\text{Unip}(n; K)$ -homomorphism  $V \rightarrow \mathbf{1}(\psi_{\text{Unip}})$ , which is just a Whittaker functional. So this proposition is equivalent to (24.9.5.5).  $\square$

**Remark (24.9.5.4).** Generic representations are important (at least for  $\text{GL}(n)$ ) because for any  $F \in \mathbf{NField}$  and  $\pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$ , all the local components  $\pi_v$  are generic, by (26.6.4.7).

Whittaker models are important because we can use it to attach local Euler factors for such representations.  $\lrcorner$

**Thm. (24.9.5.5) [Local Multiplicity One, Shalika [Sha74]/Gelfand-Kazhdan [G-K75]].** For  $(\pi, V) \in \text{Irr}^{\text{adm}}(G)$  and a generic unipotent character  $\psi \in U(K)^\vee$  (10.4.4.23), the space of  $\psi$ -Whittaker functionals for  $\pi$  has dimension  $\leq 1$ . Equivalently, there exists at most one  $\psi$ -Whittaker model for  $\pi$ , by (24.9.5.3).  $\lrcorner$

**Remark (24.9.5.6).** The condition that  $\psi$  is generic cannot be removed, by Exercise 11.2 of [Getz].  $\lrcorner$

*Proof:* We only prove for  $G = \text{GL}(n)$ ?

Define another representation  $\pi'(g) = \pi(\iota(g)^{-1})$  (24.11.1.17), then this representation is isomorphic to  $\pi_1$  defined in (24.11.1.10) ( $\pi(w^0)$  induces an isomorphism), which is isomorphic to the contragredient of  $\pi$ , so there is a pairing on  $V$  that

$$(\pi(g)\xi, \eta) = (\xi, \pi(\iota(g))\eta).$$

Now for any smooth functional  $\Lambda$ , there is an element  $[\Lambda]$  that  $(\xi, [\Lambda]) = \Lambda(\xi)$ .

Now for any linear functional  $\Lambda$  on  $V$  and  $\varphi \in \mathcal{H}_G$ , we can define another smooth linear function  $(\Lambda * \varphi)(\xi) = \Lambda(\pi(\varphi)\xi)$ . Then clearly  $\varphi * (\varphi_1 * \varphi_2) = (\Lambda * \varphi_1) * \varphi_2$ . We need the following lemma:

**Lemma (24.9.5.7).**

- $\pi(g)[\Lambda * \varphi] = [\Lambda * \rho(\iota(g)^{-1})\varphi]$ .
- If  $L$  is a smooth functional,  $[L * \varphi] = \pi(\iota(\varphi))[L]$ .
- If  $L$  is a Whittaker functional,  $[\Lambda * \lambda(u)\varphi] = \psi_N(u)[\Lambda * \varphi]$ .

$\lrcorner$

*Proof:* 1:

$$\begin{aligned} (\xi, \pi(g)[\Lambda * \varphi]) &= (\pi(\iota(g))\xi, [\Lambda * \varphi]) = (\Lambda * \varphi)(\pi(\iota(g))\xi) \\ &= \int_G \Lambda(\pi(h)\pi(\iota(g))\xi) \varphi(h) dh = \int_G \Lambda(\pi(h)\xi) \varphi(h\iota(g)^{-1}) dh = (\xi, [\Lambda * \rho(\iota(g)^{-1})\varphi]). \end{aligned}$$

$$2: (\xi, [L * \varphi]) = (L * \varphi)(\xi) = L(\pi(\varphi)\xi) = (\pi(\varphi)\xi, [L]) = (\xi, \pi(\iota(\varphi))[L]).$$

3:

$$\begin{aligned} (\xi, [\Lambda * \lambda(u)\varphi]) &= (\Lambda * \lambda(u)\varphi)(\xi) = \int_G \Lambda(\pi(g)\xi) \varphi(u^{-1}g) dg \\ &= \int_G \Lambda(\pi(u)\pi(g)\xi) \varphi(g) dg \\ &= \psi_N(u)(\Lambda * g)(\xi) = \psi_N(u)(\xi, [\Lambda * \varphi]) \end{aligned}$$

□

Now if  $\Lambda_1, \Lambda_2$  are two Whittaker functionals, we will show they are propositional: we define a distribution  $\Delta$  on  $G$  that  $\Delta(\varphi) = \Lambda_1([\Lambda_1 * \varphi])$ , then by the lemma above, (24.11.1.17) can be applied to  $\bar{\Delta}$  so we have  $\Delta = \iota(\Delta)$ .

Next we show for any linear functional  $\Lambda$ ,  $V = \{[\Lambda * \varphi] | \varphi \in \mathcal{H}\}$ : Notice the RHS is  $G$ -invariant by (24.9.5.7), and it is not empty by smoothness technique. To go further, need another lemma:

**Lemma (24.9.5.8).** If  $\varphi \in \mathcal{H}$  satisfies  $\Lambda_1 * \varphi = 0$ , then  $\Lambda_2 * \varphi = 0$ . ┘

*Proof:* Firstly, all  $\Lambda_1 * \pi(g)\varphi = 0$ , which follows from (24.9.5.7) item1. Hence,

$$\Lambda_2([\Lambda_1 * \lambda(g)\iota(\varphi)]) = \Delta(\iota(\rho(\iota(g)^{-1})\varphi)\varphi) = \Delta(\rho(\iota(g)^{-1})\varphi) = \Lambda_2([\Lambda_1 * \rho(\iota(g)^{-1})\varphi]) = 0.$$

hence by linearity, for any  $\sigma \in \mathcal{H}$ ,  $\Lambda_2([\Lambda_1 * \sigma * \iota(\varphi)]) = 0$ , which by (24.9.5.7) item2 is equivalent to  $\Lambda_2(\pi(\varphi)[\Lambda_1 * \varphi]) = 0 = (\Lambda_2 * \varphi)[\Lambda_1 * \varphi]$ , because  $\Lambda_1 * \sigma$  is smooth. But we know  $[\Lambda * \varphi]$  can be any  $v \in V$ , thus  $\Lambda_2 * \varphi = 0$ . □

By the lemma, we can define a map  $T : V \rightarrow V : T([\Lambda_1 * \varphi]) = [\Lambda_2 * \varphi]$ , which is a  $G$ -homomorphism by (24.9.5.7), and it is defined on all of  $V$ , so  $T\xi = c\xi$  for some  $c$ , then we see  $\Lambda_2 = c\Lambda_1$ , by a smoothness technique. □

**Prop. (24.9.5.9).** A Whittaker functional for  $(\pi, V)$  is the same as a linear functional on  $J_{\text{Unip}(n;K), \psi_N}(V)$ .

In particular, for  $(\pi, V) \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$ ,  $\dim J_{\text{Unip}(n;K), \psi_N}(V) \leq 1$ , by (24.9.5.5). ┘

## Existence of Whittaker Models

### 6 Unramified Representations

**Notation (24.9.6.1).**

- Let  $(G, B, T, W)$  be unramified reductive (20.5.2.25).
- Let  $S \leq T$  be the maximal split torus.
- Let  $U = R_u(B)$ .
- Let  $\mathcal{K} = \mathcal{G}(\mathcal{O}_K) \leq G(K)$  be a hyperspecial subgroup (20.7.2.1).
- Let  $\sigma_K \in \text{Gal}_K$  be a geometric Frobenius.

┘

**Def. (24.9.6.2) [Unramified Hecke Algebras].** Define

$$C^\infty(\mathcal{K} \backslash G(K) / \mathcal{K}) \stackrel{\Delta}{=} \mathcal{H}_{\mathcal{K}}(G(K)) \quad (24.1.6.18),$$

called the **unramified Hecke algebra** or **spherical Hecke algebra** of  $G$ . By (20.7.2.12), for different choice of  $\mathcal{K}$ , the spherical Hecke algebras are conjugate by an elements of  $(G/Z(G))(k)$ .

This unramified Hecke algebra has a spherical idempotent (3.6.5.10)  $e_{\mathcal{K}}$ , where the anti-involution is given by  $?$ . In particular, it is commutative. ┘

**Def. (24.9.6.3) [Unramified Representations].**  $(\pi, V) \in \text{Irr}^{\text{adm}}(G(K))$  is called a  $\mathcal{K}$ -**unramified representation** if it contains a  $\mathcal{K}$ -fixed vector, or equivalently, its corresponding  $\mathcal{H}_{\mathcal{K}}(G(K))$ -module via the correspondence (24.1.6.22) is spherical (3.6.5.11).

And if  $\pi$  is unramified, the dimension of spherical vectors is exactly 1 by (3.6.5.11). The action of  $\mathcal{H}_{\mathcal{K}}(G(K))$  on this subspace is an additive character, called the **spherical character** of  $\pi$  (3.6.5.11). ┘

**Prop. (24.9.6.4) [Unramified Representations and Spherical Characters].** Two unramified irreducible representation  $\pi, \pi' \in \text{Irr}^{\text{adm}}(G)$  are equivalent iff their spherical characters are equal, by (3.6.5.12).

Conversely, Every additive character  $\psi : \mathcal{H}_{\mathcal{K}}(G(K)) \rightarrow \mathbb{C}$  arises as a spherical character for some unramified representation  $\pi \in \text{Irr}^{\text{adm}}(G(K))$  (24.9.6.3). ┘

*Proof:* Cf. [Laumon, Cohomology of Drinfeld modular varieties] Prop D.1.8.  $?$  ┘

**Prop. (24.9.6.5).** By Cartan decomposition (20.7.2.14), the unramified Hecke algebra  $\mathcal{H}_{\mathcal{K}}(G)$  has a basis consisting of elements of the form  $c_{\lambda} = \mathbf{1}_{\mathcal{K}\lambda(\varpi)\mathcal{K}}$  for  $\lambda \in P^+(\Phi(G, \hat{S}))$  ┘

**Prop. (24.9.6.6) [Degrees of Double Cosets].** If  $G$  is split reductive, then for any  $\lambda \in P^+(\Phi)$ , we have

$$\deg(c_{\lambda}) = \sum_{y \in W\lambda W} q^{\ell(y)} / \sum_{w \in W} q^{\ell(w)} = \frac{\#(G/P_{\lambda})(\kappa)}{q^{\dim(G/P_{\lambda})}} \cdot q^{\langle \lambda, 2\rho \rangle}.$$

where  $\ell : X^{\bullet}(\hat{T}) \rtimes W \rightarrow \mathbb{Z}$  is the length function ( $?$  Cf. [Lus83]). ┘

*Proof:* This follows from Bruhat-Tits decomposition  $?$ . ┘

### Satake-Isomorphisms

References are [Gro98], [Getz], [Cartier79].

**Thm. (24.9.6.7) [Chevalley's Restriction Theorem].** For  $(G, T, W) \in \text{RedGrp}/K$  split reductive, there is an isomorphism

$$\text{Spec} \left( \mathcal{O}(T)^{W(G, T)} \right) = T/W(G, T) \cong G/\text{conj} = \text{Spec} \left( \mathcal{O}(G)^G \right).$$

And  $(G/\text{conj})(\mathbb{C})$  is exactly the set of closed conjugacy classes of  $G(\mathbb{C})$ . ┘

*Proof:* Cf. [Steinberg, Regular elements of semisimple algebraic groups] Section 6  $?$  ┘

**Def. (24.9.6.8) [Satake-Transform].** Suppose  $G$  is split reductive. For  $f \in \mathcal{H}_{\mathcal{K}}(G(K); \mathbb{Z})$ , the **Satake transform** of  $f$  is defined to be

$$\mathrm{CT}_p(f)(t) = \delta(t)^{1/2} \int_{U(K)} f(tn) dn, \text{ where } t \in T(K).$$

Then  $\mathrm{CT}_p(f)$  is a function on  $T(K)/T(\mathcal{O}_K) = X_{\bullet}(T)$  with values in  $\mathbb{Z}[q^{1/2}, q^{-1/2}]$ , and is invariant under the action of  $W$ .  $\lrcorner$

*Proof:* ?  $\square$

**Lemma (24.9.6.9) [Satake-Isomorphisms, Satake [Sat63]].** If  $G$  is split reductive, then the Satake transform (24.9.6.8) induces an isomorphism

$$\mathrm{CT}_p : \mathcal{H}_{\mathcal{K}}(G(K); \mathbb{Z}[q^{1/2}, q^{-1/2}]) \cong \mathbb{Z}[X_{\bullet}(T)]^W [q^{1/2}, q^{-1/2}].$$

And if  $\rho \in X^{\bullet}(T)$ , then it induces an isomorphism

$$\mathrm{CT}_p : \mathcal{H}_{\mathcal{K}}(G(K); \mathbb{Z}) \otimes \mathbb{Z}[q] \cong \mathbb{Z}[q][X_{\bullet}(T)]^W.$$

More explicitly, there exists  $b_{\lambda}(\mu) \in \mathbb{Z}$  for any  $\mu, \lambda \in P^+(\Phi^{\vee})$  s.t.

$$\mathrm{CT}_p(c_{\lambda}) = q^{\langle \lambda, \rho \rangle} \mathrm{ch}(V_{\lambda}) + \sum_{\mu < \lambda \in P^+(\Phi^{\vee})} b_{\lambda}(\mu) q^{\langle \lambda, \rho \rangle} \mathrm{ch}(V_{\mu}).$$

$\lrcorner$

*Proof:* ? Cf. [Gro98].

Then the isomorphism follows easily from Cartan decomposition (20.7.2.14).  $\square$

**Thm. (24.9.6.10) [Satake-Isomorphism, Satake-Langlands].** If  $G/K$  is unramified, then there is an isomorphism of algebras

$$\mathrm{CT}_p : \mathcal{H}_{\mathcal{K}}(G(K)) \cong \mathcal{O}(\hat{T} \rtimes \sigma_K)^{\kappa^N}.$$

$\lrcorner$

*Proof:* ? Cf. [Getz].  $\square$

**Cor. (24.9.6.11) [Unramified Langlands-Correspondence].** If  $G$  is split reductive, then the Satake isomorphism (24.9.6.10) induces a bijection between the set of semisimple conjugacy classes of  $\hat{G}(\mathbb{C})$  and  $\mathrm{Irr}^{\mathrm{unr}}(G(K))$  s.t. for any  $\pi \in \mathrm{Irr}^{\mathrm{unr}}(G)$ , the evaluation of the spherical character of  $\pi$  on any  $f \in \mathcal{H}_{\mathcal{K}}(G(K))$  is given by the evaluation of  $\mathrm{CT}_p(f)$  on the Satake parameter  $s(\pi) \in \hat{T}$ .

The **Satake parameter** of  $\pi \in \mathrm{Irr}^{\mathrm{unr}}(G(K))$  is denoted by  $[s(\pi)]$  for some  $s(\pi) \in \hat{G}(\mathbb{C})$ , which is usually chosen in  $\hat{T}(\mathbb{C})$ .  $\lrcorner$

*Proof:* This follows from Chevalley restriction theorem (24.9.6.7), (10.2.1.19) and (24.9.6.4).  $\square$

**Cor. (24.9.6.12).** If  $G_1, G_2 \in \mathrm{RedGrp}/K$  are unramified, and  $\mathcal{K}_1 \leq G_1(K), \mathcal{K}_2 \leq G_2(K)$  are hyperspecial subgroups, then any morphism  ${}^{\mathrm{L}}G_1 \rightarrow {}^{\mathrm{L}}G_2$  induces a morphism of Hecke algebras

$$\mathcal{H}_{\mathcal{K}_2}(G_2(K)) \rightarrow \mathcal{H}_{\mathcal{K}_1}(G_1(K)).$$

$\lrcorner$

*Proof:* ? □

**Prop. (24.9.6.13).** The Satake parameter (24.9.6.11) of the trivial representation of  $G$  is given by  $s(\pi) = 2\rho(q^{1/2})$ . ┘

*Proof:* ? □

**Prop. (24.9.6.14).** Suppose  $G$  is split reductive. For  $s \in \widehat{G}(\mathbb{C})$ , denote  $\pi(s) \in \text{Irr}^{\text{unr}}(G(K))$  corresponding to  $s$  via (24.9.6.11), then  $\pi(s)$  is tempered iff  $\overline{\langle s \rangle} \subset \widehat{G}(\mathbb{C})$  is compact. And Macdonald [Mac71] determined the Plancherel measure on the unramified unitary dual of  $G$ . ┘

*Proof:* ? □

### Kazhdan-Lusztig Polynomials

**Notation (24.9.6.15).**

- In this subsubsection, we assume that  $G$  is split reductive.

┘

**Prop. (24.9.6.16)** [ $d_\lambda \mu$ ]. By (24.9.6.9) there exists  $d_\lambda(\mu) \in \mathbb{Z}$  s.t. for any  $\lambda \in P^+(\Phi^\vee)$ ,

$$q^{\langle \lambda, \rho \rangle} \text{ch}(V_\lambda) = \text{CT}(c_\lambda) + \sum_{\mu < \lambda \in P^+(\Phi^\vee)} d_\lambda(\mu) \text{CT}_p(c_\mu) \quad (24.9.6.5).$$

To understand  $\text{CT}_p$ , it suffices to understand these  $d_\mu(\lambda)$ . ┘

**Prop. (24.9.6.17)** [ $q$ -Analogue of Kostant's Partition Function]. For  $\mu \in Q(\Phi^\vee)$ , denote

$$\widehat{P}_X(\mu) = \sum_{\mu = \sum_{\alpha \in \Phi^+} n(\alpha^\vee) \alpha^\vee} X^{-\sum_{\alpha \in \Phi^+} n(\alpha^\vee)} \in \mathbb{Z}[X, X^{-1}].$$

where  $\mu = \sum_{\alpha \in \Phi^+} n(\alpha^\vee) \alpha^\vee$  is an expression of  $\mu$  as a non-negative sum of positive coroots. Then

$$X^{\langle \mu, \rho \rangle} \widehat{P}_X(\mu)$$

is a polynomial in  $X$  with integer coefficients, and has constant coefficient 1 if  $\mu \geq 0$ .

Moreover,  $\widehat{P}_1(\mu) = \mathfrak{P}(\mu)$  (3.9.5.21). ┘

**Thm. (24.9.6.18)** [Lusztig-Kato Polynomials, Lusztig [Lus83]/Kato [Kat82]]. For any  $\mu < \lambda \in P^+(\Phi^\vee)$ ,

$$P_{\mu, \lambda}(q) \triangleq d_\lambda(\mu) = q^{\langle \lambda - \mu, \rho \rangle} \sum_{w \in W} \text{sgn}(w) \widehat{P}(w(\lambda + \rho^\vee) - (\mu + \rho^\vee))(q) \quad (24.9.6.16).$$

is a Kazhdan-Lusztig polynomial for the affine Weyl group of  $\widehat{G}$ . In particular, by theory of Kazhdan-Lusztig polynomials,  $P_{\mu, \lambda}$  is a polynomial with non-negative integral coefficients of degree  $< \langle \lambda - \mu, \rho \rangle$ . ┘

*Proof:* ? □

The constant coefficients is 1 because  $\langle \lambda - w_\bullet \lambda, \rho \rangle > 0$  unless  $w = \text{id}$ .

**Cor. (24.9.6.19) [A Simple Situation].** Situation as in (24.9.6.18), if  $P_{\mu,\lambda}$  has constant coefficient 1, and  $P_{\mu,\lambda}(1) = \dim V_\lambda[\mu]$  by Kostant's multiplicity theorem (24.4.3.8). In particular, for any  $\mu, \lambda \in P^+(\Phi^\vee)$ ,

$$\dim V_\lambda[\mu] \leq 1 \implies d_{\mu,\lambda} = P_{\mu,\lambda}(q) = \dim V_\lambda[\mu].$$

┘

**Example (24.9.6.20).** If  $G$  is simple adjoint and  $\lambda$  is the highest coroot corresponding to  $\text{Ad} \in \text{Rep}^{\text{fd}}(\widehat{G})$ , then

$$P_{0,\lambda}(q) = \sum_{i=1}^l q^{m_i-1},$$

where  $m_1, \dots, m_l$  are the exponents of  $\mathfrak{g}$  ?.

┘

*Proof:* ? Cf. [Lus83]P226. □

**Example (24.9.6.21) [GL(n)].** If  $G = \text{GL}(n)$ ,  $B = \text{B}(n)$ ,  $T = T(n)$ ,  $\mathcal{K} = \text{GL}(n; \mathcal{O}_K)$ , then  $\mathcal{H}_{\mathcal{K}}(G; \mathbb{Z})$  has a basis given by representatives:

$$\{c_{\lambda_i} : \lambda_i = \text{diag}(\underbrace{1, \dots, 1}_{r \text{ many 1s}}, 0, \dots)\}_{i \in [n-1]_+} \cup \{c_{\lambda_{-1}} : \lambda_{-1} = \text{diag}(-1, \dots, -1)\} \quad (24.9.6.16).$$

Then by (24.9.6.16),

$$\text{CT}_p(c_{\lambda_i}) = q^{i(n-i)/2} \text{ch}(\wedge^r \text{Std}), \quad \text{CT}_p(c_{\lambda_{-1}}) = e^{\lambda_{-1}}.$$

┘

*Proof:*  $\lambda_i$  are minuscule and  $V_{\lambda_i} = \wedge^r \text{Std}$  and  $V_{\lambda_{-1}} = \det^{-1}$ . So we can use (24.9.6.19). □

**Example (24.9.6.22).** For  $G = \text{PGL}(2)$ ,  $\widehat{G} = \widehat{G} = \text{SL}(2)$  and  $X^\bullet(\widehat{T}) = \mathbb{Z}\chi$ , where  $\chi$  is the standard representation of  $\widehat{G}$ . Then in this case  $\dim V_\lambda[\mu] \leq 1$  for any  $\lambda, \mu \in P^+(\Phi^\vee)$ , so by (24.9.6.19), we have:

$$q^{\frac{n}{2}} \text{ch}(\text{Sym}^n \text{Std}) = \text{CT}_p \left( \sum_{m \in [n], m \equiv n \pmod{2}} T_{\varpi^m} \right).$$

┘

**Example (24.9.6.23).** For  $G = \text{GSp}(2n)$  and  $\widehat{G} = \text{GSpin}(2n+1)$  and minuscule coweight  $\omega_n \in X^\bullet(\widehat{T})^+$  (24.4.2.27), by (24.9.6.16),

$$\text{CT}_p(c_{\omega_n}) = q^{\frac{n(n+1)}{4}} \text{ch}(\text{Spin}).$$

┘

**Example (24.9.6.24).** For  $G = \text{PSp}_4 \cong \text{SO}(5)$ ,  $\widehat{G} = \text{Sp}(4) \cong \text{O}(5)$  and the fundamental weight  $\omega_1 = \varepsilon_1 \in X_\bullet(T)$ , we have  $V_{\omega_1} = \text{Std}$ . Thus by (24.9.6.19),

$$q^2 \text{ch}(V_{\omega_1}) = \text{CT}_p(c_{\omega_1}) + 1.$$

┘

**Example (24.9.6.25).** For  $G = G_2$ ,  $\widehat{G} = G_2$  and the fundamental weights  $0 \leq \omega_1 \leq \omega_2$ ,  $V_{\omega_1}$  is the 7-dimensional fundamental representation and  $V_{\omega_2}$  is the 14-dimensional adjoint representation. Thus

$$P_{0,\omega_1}(q) = 1 = P_{\omega_1,\omega_2}(q).$$

And by (24.9.6.20),

$$P_{0,\omega_2}(q) = 1 + q^4,$$

because  $\mathfrak{g}_2$  has exponent  $m_1 = 1, m_2 = 5$ ?. Thus

$$q^3 \operatorname{ch}(V_{\omega_1}) = \operatorname{CT}_p(c_{\omega_1}) + 1, \quad q^5 \operatorname{ch}(V_{\omega_2}) = \operatorname{CT}_p(c_{\omega_2}) + \operatorname{CT}_p(c_{\omega_1}) + (q^4 + 1).$$

┘

### Unramified principal Series

#### Spherical Whittaker Function and Spherical Functions

See [Casselman, W., The unramified principal series of  $p$ -adic groups I: the spherical function, Compositio Math. 40 (1980), 387-406.] and [Casselman, W. and J. Shalika, The unramified principal series of  $p$ -adic groups II: the Whittaker function, Compositio Math. 40 (1980), 207-231.] for more general calculations.

**Def. (24.9.6.26) [Spherical Whittaker Function].** The **spherical Whittaker function** of a generic unramified representation is the spherical vector  $W^0$  in the Whittaker model normalized that  $W_0(1) = 1$ ?.

For  $n = 2$ , we can define another normalization  $W_0(g) = \Lambda(\pi(g)\varphi_K)$ , where  $\Lambda$  is the Whittaker functional defined in (24.11.5.8) and  $\varphi_K$  is the normalized spherical vector defined in (24.11.6.1). ┘

## 7 Iwahori-Spherical Representations

References are [P-R08, Bor76].

**Thm. (24.9.7.1) [Borel-Matsumoto Theorem].** Any Iwahori-spherical representation is a subrepresentation of an unramified principal series. ┘

*Proof:* ? □

## 8 Unitarizable, Tempered & $L^2$ -Representations

References are [Tad86] and [Vogan, The unitary dual of  $GL(\mathbb{R})$  over an Archimedean field].

### Unitarizable Representations

**Prop. (24.9.8.1).** If  $V \in \operatorname{Irr}^{\operatorname{adm}}(G(F))$  is unitarizable, then  $V \cong V^\vee$ . ┘

**Thm. (24.9.8.2) [Harish-Chandra[HC70]/Bernstein(74)].** For any  $\pi \in \operatorname{Irr}^{\operatorname{uni}}(G)$ , we have  $V^{\operatorname{alg}} \in \operatorname{Rep}^{\operatorname{adm}}(G)$ . ┘

*Proof:* ? □

**Thm. (24.9.8.3) [Unitary Representations and Smooth Representations].** For  $V \in \mathcal{R}\text{ep}^{\text{uni}}(G)$ ,  $V$  is irreducible iff  $V^{\text{alg}} \in \mathcal{R}\text{ep}^{\text{alg}}(G(F))$  is irreducible.  $\lrcorner$

**Remark (24.9.8.4).** This is an analogue of the Archimedean case (24.6.4.7)(24.7.5.7).  $\lrcorner$

*Proof:* If  $V$  is reducible, then  $V^{\text{alg}}$  is irreducible by (24.1.2.3). If  $V$  is irreducible and  $\emptyset \neq M_1 \subset M$ ,  $\overline{M_1}$  is  $G$ -stable, thus equals  $V$ . Thus it suffices to show that  $\overline{M_1}^{\text{alg}} = M_1$ . Then it suffices to show for each compact open subset  $K \leq G(F)$  and  $\rho \in \text{Irr}(K)$ , the image of  $T_{\xi_\rho \mu_{\text{Haar}}}(\overline{M_1}) \subset M_1^\rho$ , by (24.7.3.14). However  $T_{\xi_\rho \mu_{\text{Haar}}}(M_1) \in M_1^\rho$  by (24.7.3.14). Because  $\dim M_1^\rho < \infty$  as  $V$  is admissible (24.9.8.2), this holds true by continuity.  $\square$

**Cor. (24.9.8.5) [Irreducibility of Algebraic Representations].** For  $V \in \mathcal{R}\text{ep}^{\text{uni}}(G)$ ,  $V$  is irreducible iff  $V^{\text{alg}}$  is irreducible.  $\lrcorner$

**Prop. (24.9.8.6) [Infinitesimally Equivalent Representations].** The map  $(-)^{\text{alg}} : \mathcal{R}\text{ep}^{\text{uni}}(G(K)) \rightarrow \text{Irr}^{\text{adm}}(G(K))$  (24.9.8.3) is injective.  $\lrcorner$

**Remark (24.9.8.7).** This is an analogue of the Archimedean case (24.6.4.9).  $\lrcorner$

*Proof:* If  $S : V_1^{\text{alg}} \cong V_2^{\text{alg}}$ , because each  $V_i$  has a Hermitian form, we have  $V_i^{\text{alg}} \cong (V_i^*)^{\text{alg}}$ , and  $S^*S$  is an automorphism of  $V_i^{\text{alg}}$ , which is a scalar multiplication by (24.1.6.36). Thus by a possibly scalar change, we may assume that  $S$  preserves the Hermitian structure, and thus defines an isomorphism  $V_1 \cong V_2$  by continuity.  $\square$

**Prop. (24.9.8.8) [Twists and Unitary Central Characters].** For any  $\pi \in \mathcal{R}\text{ep}^{\text{adm}}(G(F))$  that has a central character, there exists a unique positive-real-valued character  $\chi$  of  $G$  s.t.  $\pi \otimes \chi$  has a unitary central character.  $\lrcorner$

*Proof:* Cf. [Admissible representations of  $p$ -adic groups, Casselman] Prop 5.2.5.  $\square$

**Def. (24.9.8.9) [Complementary Series].** A **complementary series representation** of  $G(K)$  is a unitarizable representation of  $G(K)$  that is not tempered.  $\lrcorner$

### Tempered Representations

**Def. (24.9.8.10) [(Essentially) Discrete-Series].** Let  $(\pi, V) \in \text{Irr}^{\text{adm}}(G(K))$  and  $r \in \mathbb{R}_+$ , then  $\pi$  is called a **essentially  $L^r$ -integrable representation** the matrix coefficients for  $V$  are all  $L^r$ -integrable on  $G(K)^1$  (20.5.1.2). It is called a  **$L^r$ -integrable representation** if the matrix coefficients for  $V$  are all  $L^r$ -integrable on  $G(K)$ .

Essentially  $L^2$ -representations are also called **essentially square-integrable representations (non-Archimedean)**, and  $L^2$ -representations are called **square-integrable representations (non-Archimedean)** or **discrete-series (non-Archimedean)**.  $\lrcorner$

**Def. (24.9.8.11) [Tempered Representations].**  $\pi \in \text{Irr}^{\text{adm}}(G(K))$  is called an **essentially tempered representation** if it is a  $L^{2+\varepsilon}$ -integrable representation for any  $\varepsilon > 0$ .

An essentially tempered representation is called a **tempered representation** if moreover the central character  $\omega_\pi$  is unitary.  $\lrcorner$

**Prop. (24.9.8.12) [Tempered Representations are Unitarizable].** Any irreducible tempered representation of  $G(K)$  (24.9.8.11) is unitarizable.  $\lrcorner$



*Proof:* Cf.[Silberger. Introduction to harmonic analysis on reductive  $p$ -adic groups]Cor4.5.13.  $\square$

**Prop. (24.9.8.13) [Characterizing Tempered Representations].** For  $\pi \in \text{Irr}^{\text{adm}}(G(K))$  with unitary central character, the following are equivalent:

- $\pi$  is tempered.
- Each matrix coefficient of  $\pi$  defines a tempered distribution on  $G(K)$ .
- The distribution character of  $\pi$  is tempered.

┘

*Proof:* For explanation and proof, Cf.[Sil78]4.5 and [Sil82]2.6. Maybe can also be found in[HC99].  $\square$

**Prop. (24.9.8.14) [Silberger].** If  $\pi \in \text{Irr}^{\text{adm}}(G(K))$  is essentially  $L^r$ -integrable for some  $r \in \mathbb{R}_+$ , then it is essentially  $L^{r'}$ -integrable for any  $r' \in \mathbb{R}_{\geq r}$ .  $\square$

*Proof:* Cf.[Sil82]2.5.  $\square$

**Thm. (24.9.8.15) [Tempered Plancherel Theorem].** There is a tempered Plancherel theorem relating spaces of functions on  $G(K)$  and tempered representations of  $G(K)$ .  $\square$

*Proof:* Cf.[Waldspurger. La formule de Plancherel pour les groupes  $p$ -adiques].?  $\square$

### Irreducible Generic Unramified Unitarizable Representations

## 9 Langlands Classifications

Generalization of the Langlands classification of  $\text{GL}(n)$  to other reductive groups lacks a complete determination of discrete series. Notice that the classification of supercuspidal representations is already finished in [C.J. Bushnell and P.C. Kutzko, The admissible dual of  $\text{GL}(N)$  via compact open subgroups, Princeton university press, Princeton, (1993).]

### Shalika Germs

### Howe's Theorem

**Thm. (24.9.9.1) [Howe].** For a compact subset  $\omega \subset \mathfrak{g}$ , denote  $J(\omega)$  the space of  $G$ -invariant distributions on  $\mathfrak{g}$  with support in  $\text{Ad}(G)\omega$ .

For a lattice  $L$  in  $\mathfrak{g}$  and a distribution  $T$  on  $\mathfrak{g}$ , denote  $j_L(T)$  the restriction of  $T$  to  $C_c(\mathfrak{g}/L)$ .

Then for any compact subset  $\omega \subset \mathfrak{g}$  and a lattice  $T$  in  $\mathfrak{g}$ , we have  $\dim j_L J(\omega) < \infty$ .  $\square$

*Proof:* Cf.[HC99]Thm12.1?  $\square$

**Remark (24.9.9.2).** Waldspurger[Une formule des traces locale pour les algèbres de Lie  $p$ -adiques] generalized this to the case of weighted orbital integrals.  $\square$

**Thm. (24.9.9.3).** Let  $\omega \subset \mathfrak{g}$  be a compact subset and  $T \in J(\omega)$ (24.9.9.1), then there exists a locally  $L^1$ -function  $F$  on  $\mathfrak{g}$  s.t.

- $\hat{T}(f) = \int_{\mathfrak{g}} F(X) f(X) dX$  for any  $f \in C_c^\infty(\mathfrak{g})$ .
- $F$  is locally constant on  $\text{reg}(\mathfrak{g})$ .
- $|\eta_{\mathfrak{g}}|^{1/2} \cdot F$ (24.9.4.2) is locally bounded on  $\mathfrak{g}$ .

┘

*Proof:* Cf.[HC99]Thm4.4.  $\square$

### Admissible Distributions

**Def. (24.9.9.4) [Admissible Distributions].** For a unimodular  $G \in \text{locProfGrp}$ ,  $K_0 \leq G$  a compact open subgroup and  $\Theta$  a distribution on an open  $G$ -invariant subset  $U$  of  $G$  and  $\gamma \in U$ ,  $\Theta$  is called a  $(G, K_0)$ -**admissible distribution** at  $\gamma$  if

- $\gamma K_0 \subset U$ ,
- For any open subgroup  $K \leq K_0$  and  $\sigma \in \text{Irr}(K)$ , we have  $\Theta * \chi_\sigma = 0$  on  $\gamma K_0$  unless  $G$  intertwines  $\mathbf{1}_{K_0}$  and  $\sigma$ .

Suppose  $\Theta$  is  $(G, K_0)$ -admissible at  $\gamma$  and  $K_1 \leq K_0$  is an open subgroup, then  $\Theta$  is also  $(G, K_1)$ -admissible at  $\gamma$ . And  $\Theta$  is called a  $G$ -**admissible distribution** if it is  $(G, K_0)$ -admissible for some  $K_0$ .  $\lrcorner$

*Proof:*  $\text{?}$  Cf. [HC99]P74.  $\square$

**Prop. (24.9.9.5) [Admissible Characters are Admissible].** For a unimodular  $G \in \text{locProfGrp}$  and  $\pi \in \text{Irr}^{\text{adm}}(G)$ ,  $\Theta_\pi \triangleq \chi_\pi$  is a  $G$ -admissible distribution on  $G$  (24.9.9.4).  $\lrcorner$

*Proof:*  $\text{?}$  Cf. [Clozel, Characters of non-connected, reductive p-adic groups].  $\square$

**Thm. (24.9.9.6) [ $G$ -admissible Distributions around Semisimple Elements, Harish-Chandra].**

Let  $\Theta$  be a  $G$ -invariant distribution on a  $G$ -invariant open subset  $U \subset G$ , and  $\gamma \in U$  is a semisimple element s.t.  $\Theta$  is  $G$ -admissible at  $\gamma$  (24.9.9.4), then  $\Theta$  coincides with a locally  $L^1$ -function around  $\gamma$ .  $\lrcorner$

*Proof:* Cf. [HC99]Thm16.1  $\text{?}$   $\square$

**Thm. (24.9.9.7) [Local Character Expansions, Harish-Chandra].** For  $\pi \in \text{Irr}^{\text{adm}}(G(K))$ , there exists  $c_{\mathcal{O}}(\pi)$  for each  $\mathcal{O} \in \mathcal{O}(0)$  (24.9.4.3) s.t.

$$\chi_\pi(\exp(Y)) = \sum_{\mathcal{O} \in \mathcal{O}(0)} c_{\mathcal{O}}(\pi) \mathcal{F}\mu_{\mathcal{O}}(Y)$$

for any regular  $Y$  in a nbhd of  $0 \in \mathfrak{g}$ .  $\lrcorner$

*Proof:* Cf. [HC99]Thm16.2.  $\square$

**Thm. (24.9.9.8) [Howe].** For  $G = \text{GL}(n)$ , Howe observed that the functions  $\mathcal{F}\mu_{\mathcal{O}}$  in (24.9.9.7) have nice forms, thus showed that  $c_{\mathcal{O}}(\pi) \in \mathbb{Z}$  for any  $\pi \in \text{Irr}^{\text{cusp}}(G(K))$  and  $\mathcal{O} \in \mathcal{O}(0)$ .  $\lrcorner$

*Proof:* Cf. [The Fourier transform and germs of characters, Howe].  $\square$

**Thm. (24.9.9.9) [Harish-Chandra [HC99]].** If  $(\pi, V) \in \text{Irr}^{\text{adm}}(G(K))$ , then  $\chi_\pi$  (24.1.6.33) is represented by a locally integrable function on  $G(K)$  that is locally constant on the regular elements of  $G(K)$ , i.e. for any  $\varphi \in \mathcal{H}(G(K))$ , we have

$$\text{tr}(\pi(\varphi)) = \int_G \chi_\pi(g) \varphi(g) dg.$$

And  $\chi_\pi$  is stable under conjugation by  $G(K)$ . Moreover,  $|D_G|^{1/2} \chi_\pi \text{?}$  is locally bounded on  $G$ .  $\lrcorner$

*Proof:* Cf. [HC99]Thm16.3.  $\text{?}$   $\square$

### Formal Degrees

**Thm. (24.9.9.10) [Harish-Chandra].** There exists  $c \in \mathbb{R}_+$  s.t.  $c_0(\pi) = c \cdot d(\pi)$  for any  $\pi \in \text{Irr}^{\text{cusp}, \text{uni}}(G(K))$ , where  $d(\pi)$  is the **formal degree** of  $\pi$ .  $\lrcorner$

*Proof:* ? Cf. [HC99]Thm22.3  $\square$

**Prop. (24.9.9.11) [Formal Degrees, Harish-Chandra-Rogawski].** Harish-Chandra derived a formula for the leading term  $c_0(\pi)$  for  $\pi \in \text{Irr}^{\text{cusp}}(G(K))$ , and Rogawski proved it for any  $\pi \in \text{Irr}^{\text{disc}}(G(K))$ .  $\lrcorner$

*Proof:* Cf. [HC99]Thm22.3 and [Rogawski, An application of the wilding to orbital integrals]. ?  $\square$

**Prop. (24.9.9.12) [Howe/Harish-Chandra[HC99]].** It is possible to normalize the Haar measure on  $G(K)/Z$  s.t.  $d(\pi) \in \mathbb{Z}$  for any  $\pi \in \text{Irr}^{\text{cusp}, \text{uni}}(G(K))$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (24.9.9.13) [Huntsinger].** For  $\pi \in \text{Irr}^{\text{temp.}}(G(K))$ ,  $c_0(\pi) = 0$  iff  $\pi \notin \text{Irr}^{\text{disc}}(G(K))$ .  $\lrcorner$

*Proof:* [Vanishing of the leading term in Harish-Chandra's local character expansion].  $\square$

**Thm. (24.9.9.14).** Rodier showed that  $\pi \in \text{Irr}^{\text{adm}}(G(K))$  is generic iff  $c_{\mathcal{O}}(\pi) \neq 0$  for some regular nilpotent orbit  $\mathcal{O}$ . And Mœglin-Waldspurger showed that if  $\mathcal{O}$  is maximal among the nilpotent orbits s.t.  $c_{\mathcal{O}}(\pi) \neq 0$ , then  $c_{\mathcal{O}}(\pi)$  is related to the dimension of a degenerate or generalized Whittaker models.  $\lrcorner$

*Proof:* Cf. [Rodier, Modèle de Whittaker et caractères de représentations] and [Mœglin-Waldspurger, Modèles de Whittaker dégénérés pour des groupes  $p$ -adiques].  $\square$

## 24.10 Supercuspidal Representations of $p$ -adic Reductive Groups

### 1 $GL(n)$

References are [Local Langlands Correspondence for  $GL(2)$ ], [Bushnell and P. Kutzko. The admissible dual of  $GL(N)$  via compact open subgroups].

**Prop. (24.10.1.1)[Bushnell-Kutzko].** All cuspidal representations of  $GL(n, F)$  can be constructed by induction from open subgroups.  $\square$

*Proof:* Cf.[Bushnell, C. and P. Kutzko, The admissible dual of  $GL(N)$  via compact open subgroups, Princeton University Press, Princeton (1993).]  $\square$

### 2 $SL(n)$

[Bushnell and P. Kutzko. The admissible dual of  $SL(N)$ ]

## 24.11 Admissible Representations of $\mathrm{GL}(n)$ over $p$ -adic Fields

References are [B-Z76, B-Z77, Tad86, Kud94, Bum97, P-R08, G-H11], [admissible dual of  $\mathrm{GL}(N)$  via compact open subgroups, Bushnell-Kutzko]

**Notation(24.11.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \mathfrak{p}_K, \varpi) \in p\text{-LField}$ .
- Fix a non-trivial character  $\psi : K \rightarrow \mathbb{C}^\times$ .
- Choose the Haar measure  $dx$  on  $K$  that is self-dual w.r.t.  $\psi$  as in(12.11.1.34).
- Use group-theoretic notations as in(3.1.2.15).
- Use notations on admissible representations of  $p$ -adic reductive groups defined in [Admissible Representations of  \$p\$ -adic Reductive Groups](#).
- Let  $n \in \mathbb{Z}_+$ .
- Use the notations on  $\mathrm{GL}(n)$  defined in [General Linear Groups](#).
- For  $\alpha \in \mathbb{A}^{n-1}(K)$ , let  $\psi_{\mathrm{Unip}, \alpha}$  be a character of  $\mathrm{Unip}(n; K)$  given by  $\psi_{\mathrm{Unip}}(g) = \sum_i \psi(\alpha_i g_{i, i+1})$ . If  $\alpha = (1, \dots, 1)$ , then  $\psi_{\mathrm{Unip}, \alpha}$  is just denoted by  $\psi_{\mathrm{Unip}}$ .
- Denote  $\mathcal{K} = \mathrm{GL}(n; \mathcal{O}_K)$ , which is the maximal compact subgroup of  $\mathrm{GL}(n; K)$  up to conjugacy(20.7.3.10).
- Denote  $G_n^0 = \det^{-1}(\mathcal{O}_K^*) \subset \mathrm{GL}(n; K)$ .
- Define subgroups  $\mathrm{Unip}^k(n, K)$  of  $\mathrm{Unip}(n, K)$  as  $\mathrm{Unip}^k(n, K)$  is the group of unipotent matrices that  $v(e_{ij}) \geq k(i - j)$ . Then  $\cup_{k>0} \mathrm{Unip}^k(n, K) = \mathrm{Unip}(n, K)$ . In particular,  $\mathrm{Unip}(n, K)$  is exhausted by compact open subgroups.
- For any  $\chi \in (\mathcal{O}_K^*)^\vee$ , by continuity, there is a minimum  $v \in \mathbb{N}$  that  $\chi(1 + \mathfrak{p}^v) = 1$ , and  $\mathfrak{p}^v$  is called the **conductor of character** of  $\chi$ .

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### 1 Basics

References are [Representations of  $p$ -adic Groups Bernstein]. [Bum97]Chap4.

#### Geometry

**Prop.(24.11.1.1).**  $\mathrm{GL}(n; K)$  is unimodular, and the modular function of  $B(K)$  is  $\Delta_B\left(\begin{bmatrix} x & y \\ & z \end{bmatrix}\right) = \frac{z}{x}$  by(12.10.1.20). Also denote  $\delta = \Delta_B^{-1}$ .

┘

**Prop.(24.11.1.2)[Diagonal Quasi-Characters].** Let  $\chi_1, \dots, \chi_n$  be quasi-characters of  $K^\times$ , then define two quasi-characters  $\chi, \chi'$  on  $T(K)$ :

$$\chi(\mathrm{diag}(y_1, \dots, y_n)) = \chi_1(y_1) \cdots \chi_n(y_n), \quad \chi'(\mathrm{diag}(y_1, \dots, y_n)) = \chi_n(y_1) \cdots \chi_1(y_n).$$

When  $n = 2$ , the quasi-character  $\chi$  on  $T(K)$  is called **regular** if  $\chi_1 \neq \chi_2$ , and **dominant** if  $|\chi_1(\varpi)| < |\chi_2(\varpi)|$ .

┘

**Def. (24.11.1.3) [Principal Congruence Subgroups].** There is a map  $\pi : \mathcal{K} \rightarrow \mathrm{GL}(n; \mathcal{O}_K/(\varpi^m))$ , the kernel of which is called the **principal congruence subgroup** of level  $m$ , denoted by  $\mathcal{K}_m$ .  $\lrcorner$

**Def. (24.11.1.4) [Partitions and Level Subgroups].** Let  $\alpha = (n_1, \dots, n_k)$  be a partition of  $n$ , define

$$\mathrm{GL}(\alpha) = \mathrm{GL}(n_1) \times \dots \times \mathrm{GL}(n_k), \quad \mathcal{K}_\alpha = \mathcal{K} \cap \mathrm{GL}_\alpha(F), \quad B_\alpha = B \cap \mathrm{GL}_\alpha, \quad D_\alpha = D \cap \mathrm{GL}_\alpha.$$

And for a principal congruence subgroup  $\mathcal{K}_m$  of  $\mathrm{GL}(n; K)$  (24.11.1.3),  $\mathcal{K}_m \cap \mathrm{GL}_\alpha(K)$  is called a principal congruence subgroup of  $\mathrm{GL}_\alpha(F)$ .

Denote  $U_\alpha$  the subgroup of  $U_n$  That  $a_{ij} = 0$  for  $i, j$  in the same segment of  $\alpha$  and  $i \neq j$ . Let  $P_\alpha = \mathrm{GL}_\alpha \ltimes U_\alpha$  be the parabolic subgroup corresponding to .

For  $\beta \prec \alpha$ , denote

$$U_\beta(\alpha) = U_\beta \cap \mathrm{GL}_\alpha, \quad P_\beta(\alpha) = P_\beta \cap \mathrm{GL}_\alpha = \mathrm{GL}_\beta \ltimes U_\beta(\alpha).$$

Let  $N = \mathcal{K}_m \cap G_\alpha$  be a principal congruence subgroup, denote

$$N_\beta^+(\alpha) = N \cap U_\beta(\alpha), \quad N_\beta^-(\alpha) = N \cap \bar{U}_\beta(\alpha), \quad N_\beta^0(\alpha) = N \cap \mathrm{GL}_\beta(F).$$

Moreover, define  $\mathcal{K}_0(\mathfrak{a}) = \pi^{-1}(B(\mathcal{O}_F/\mathfrak{a}))$ ,  $\mathcal{K}_1(\mathfrak{a}) = \pi^{-1}(N(\mathcal{O}_F/\mathfrak{a}))$ . In particular,  $\mathcal{K}_0(\mathfrak{p})$  are called the **Iwahori subgroups**, a vector is called **Iwahori fixed** iff it is  $\mathcal{K}_0(\mathfrak{p})$ -fixed.  $\lrcorner$

**Def. (24.11.1.5) [Mirabolic Levels].** For  $m \in \mathbb{N}$ , the **mirabolic subgroups**  $\mathcal{K}_{0,1}(m)$  of  $\mathrm{GL}(n; \mathcal{O}_K)$  is defined to be the inverse image of the mirabolic subgroup  $M_n(\mathcal{O}_K/\mathfrak{p}_K^m)$  (3.1.2.15) under the map  $\mathrm{GL}(n; \mathcal{O}_K) \rightarrow \mathrm{GL}(n; \mathcal{O}_K/\mathfrak{p}_K^m)$ .  $\lrcorner$

**Prop. (24.11.1.6) [Iwahori Factorizations].** For  $1 \neq \mathfrak{a} \in \mathrm{Ideal}^\times \mathcal{O}_K$ , there are **Iwahori factorizations**

$$K_0(\mathfrak{a}) = N_-(\mathfrak{a})T(\mathcal{O})N(\mathcal{O}), \quad K_1(\mathfrak{a}) = N_-(\mathfrak{a})T(\mathfrak{a})N(\mathcal{O}).$$

$$N = N_\beta^-(\alpha)N_\beta^0(\alpha)N_\beta^+(\alpha) = N_\beta^+(\alpha)N_\beta^0(\alpha)N_\beta^-(\alpha).$$

$\lrcorner$

*Proof:* This is by column and row reduction. Cf.[Bernstein-Zelevinsky 1, P32]. ?  $\square$

**Cor. (24.11.1.7).** Denote  $K_0 = K_0(\mathfrak{a})$  or  $K_1(\mathfrak{a})$ , and  $T_0 = T(\mathcal{O})$  or  $T(\mathfrak{a})$  respectively, then we can decompose the Haar measure on  $K_0$  as

$$\int_{K_0} \varphi(k) dk = \int_{N_-(\mathfrak{a})} \int_{T_0} \int_{N(\mathcal{O})} \varphi(n_- t_0 n) dn dt_0 dn_- = \int_{N_-(\mathfrak{a})} \int_{T_0} \int_{N(\mathcal{O})} \varphi(n t_0 n_-) dn dt_0 dn_-.$$

$\lrcorner$

*Proof:* Use (12.10.3.9), noticing all groups here are compact.  $\square$

**Cor. (24.11.1.8) [Iwahori-Bruhat Decomposition].**

$$\mathrm{GL}(2; K) = B(K)\mathcal{K}_0(\mathfrak{p}) \coprod B(K)w_0\mathcal{K}_0(\mathfrak{p}) \text{ (24.11.1.4).}$$

? How about the general case?  $\lrcorner$

*Proof:* By pulling back the Bruhat decomposition of  $\mathrm{GL}(2; \kappa)$  to  $\mathcal{K} = \mathrm{GL}(2; \mathcal{O}_K)$ , we have

$$\mathcal{K} = K_0(\mathfrak{p}) \cup K_0(\mathfrak{p})w_0K_0(\mathfrak{p}).$$

The Iwahori factorization shows  $K_0(\mathfrak{p}) = (K_0(\mathfrak{p}) \cap B(K))N_-(\mathfrak{p})$ . Then this implies

$$\mathcal{K} = K_0(\mathfrak{p}) \cup (K_0(\mathfrak{p}) \cap B(K))w_0K_0(\mathfrak{p}),$$

because  $w_0^{-1}N_-(\mathfrak{p})w_0 \in K_0(\mathfrak{p})$ . Then Iwasawa decomposition  $G = B\mathcal{K}$  gives the desired result. Notice this decomposition is clearly disjoint.  $\square$

**Prop. (24.11.1.9).**

- If an open subgroup  $N \leq \mathrm{GL}(n; K)$  contains  $\mathrm{Unip}(n; K)$ , then it must contain  $\mathrm{SL}(n; K)$ .
- Any open normal subgroup  $N \trianglelefteq \mathrm{GL}(n; K)$  must contain  $\mathrm{SL}(n; K)$ .

┘

*Proof:* 1: By Bruhat decomposition (10.4.4.28), it suffices to show that  $N$  contains  $B(n; K) \cap \mathrm{SL}(n; K)$ . And this is easy to see.

2: this normal subgroup contains any unipotent matrix close enough to  $\mathbf{1}$ . But by the theory of Jordan forms (3.5.4.2), any unipotent subgroup is conjugate to a matrix close enough to  $\mathbf{1}$ . Thus  $\mathrm{Unip}(n; K) \subset N$ . and then by item1 it contains  $\mathrm{SL}(n; K)$ .  $\square$

### Hecke Algebras

**Prop. (24.11.1.10) [Contragradient Representations, Gelfand-Kazhdan[G-K75]].** If  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ , then:

- If  $\pi_1$  is defined by  $\pi_1(g) = \pi(g^{-t})$ , then  $\pi^\vee \cong \pi_1$ .
- If  $n = 2$  and  $\omega$  is the central character of  $\pi$ , then  $\pi^\vee \cong \pi(\omega^{-1})$ .

┘

*Proof:* 1: It is clear that the character (24.1.6.33) of a representation is conjugation invariant, thus by (24.11.1.18) it is transpose invariant.

Now the character of  $\pi_1$  is  $\chi_1(\varphi) = \chi(\varphi'')$ , where  $\varphi''(g) = \varphi(g^{-t})$ , and this equals  $\chi(\varphi')$  where  $\varphi'(g) = \varphi(g^{-1})$ , because the character is transpose invariant. It is also clear that  $\widehat{\varphi}' = \pi(\varphi)^t$  on a finite space  $V^K$ , then  $\widehat{\chi}(\varphi) = \chi(\varphi') = \chi_1(\varphi)$ , so by (24.1.6.38)  $\pi \cong \pi_1$ .

- 2: If  $n = 2$ , we use further the property that if  $g = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ , then

$$g^{-1} = (\deg g)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = (\deg g)^{-1} \begin{bmatrix} & 1 \\ -1 & \end{bmatrix} g^t \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}^{-1}$$

then the assertion is clear using item1.  $\square$

**Cor. (24.11.1.11).** Let  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ , then  $\pi$  is irreducible iff  $\pi^\vee$  is irreducible.  $\lrcorner$

### Smooth Representations

**Def. (24.11.1.12)[Twists].** Let  $(\pi, V) \in \text{Irr}^{\text{alg}}(\text{GL}(n; K))$ , for any quasi-character  $\chi$  of  $K^\times$ , denote  $\pi(\chi)$  the representation  $\pi \otimes (\chi \circ \det)$ . And for  $s \in \mathbb{C}$ , denote  $\pi(s)$  the representation of  $\pi$  tensored by the 1-dimensional representation  $g \mapsto |\det(g)|^s$ .  $\lrcorner$

**Lemma (24.11.1.13).** Let  $(\pi, V) \in \text{Rep}^{\text{alg}}(\text{GL}(n; K))$ , then  $\pi|_{G_n^0}$  splits into a finite direct sum of irreducible representations of  $G_n^0$ .  $\lrcorner$

*Proof:* This follows from (24.1.2.16).  $\square$

**Lemma (24.11.1.14).** If  $(\pi, V) \in \text{Rep}^{\text{alg}}(\text{GL}(n; K))$  satisfies  $V$  is generated by  $V^{K_m}$ , then for any submodule  $V'$ ,  $V'$  is generated by  $V^N \cap V'$ .  $\lrcorner$

*Proof:* Cf. [B-Z76]P38.  $\square$

**Prop. (24.11.1.15)[Finite Length Representations, Howe].** Let  $(\pi, V) \in \text{Rep}^{\text{alg}}(\text{GL}(n; K))$ , then  $\pi$  is of finite length iff it is admissible and f.g..  $\lrcorner$

*Proof:* A finite length representation is admissible by (24.9.1.2) and (24.1.6.30). For the converse, as  $V$  is f.g., take  $N$  compact open s.t.  $V$  is generated by  $V^N$ , and  $\dim V^N < \infty$  as  $V$  is admissible. Then by (24.11.1.13):  $V' \rightarrow V' \cap V^N$  is an injection from the set of subrepresentations of  $V$  to the set of subspaces of  $V^N$ , thus  $V$  is of finite length.  $\square$

### Invariant Distributions

**Thm. (24.11.1.16)** [Mirabolic-Invariant Distributions, Bernstein[Ber84a]].  
 $\text{Dist}(\text{GL}(n; K))^{\text{Ad}_{M_n(K)}} = \text{Dist}(\text{GL}(n; K))^{\text{GL}(n; K)}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (24.11.1.17)[Transpose Invariant Distribution].** If  $\Delta \in \mathcal{D}(\text{GL}(n; K))$  is a distribution that satisfies

$$\lambda(u)\Delta = \psi_N(u)^{-1}\Delta, \quad \rho(u)\Delta = \psi_N(u)\Delta,$$

where  $\psi$  is defined in (26.1.4.9), then  $\Delta$  is stable under involution  $\iota : \text{GL}(n; K) \rightarrow \text{GL}(n; K) : \iota(g) = w^0 g^t w^0$  (24.11.0.1).  $\lrcorner$

*Proof:* Firstly notice that  $\iota$  fixes  $\text{Unip}(n; K)$ , and  $\psi_N(\iota(g)) = \psi_N(g)$ , so  $\iota(\Delta)$  also satisfies these equations, so we can replace  $\Delta$  by  $\Delta - \iota(\Delta)$ , then assume  $\iota(\Delta) = -\Delta$  and prove  $\Delta = 0$ .

Consider the group  $G$  that is a semi-direct product

$$1 \rightarrow \text{Unip}(n; K) \times \text{Unip}(n; K) \rightarrow G \rightarrow \mathbb{F}_2$$

and  $\iota \in \mathbb{F}_2$  acts on  $\text{Unip}(n; K) \times \text{Unip}(n; K)$  by  $(u_1, u_2) \mapsto (\iota(u_2)^{-1}, \iota(u_1)^{-1})$ .

Define a character  $\chi$  on  $G$  by  $\chi((u_1, u_2)) = \psi_N(u_1)^{-1}\psi_N(u_2)$ ,  $\chi(\iota) = -1$ , then  $G$  acts on  $\text{GL}(n, K)$  by

$$\sigma((u_1, u_2)) = \lambda(u_1)\rho(u_2), \quad \sigma(\iota) = \iota$$

then the conditions are summarized into a single condition:

$$\sigma(g)\Delta = \chi(g)\Delta.$$



We only prove for  $n = 2$ : ?

Consider the action of  $\text{Unip}(n; K) \times \text{Unip}(n; K)$  on  $GL(2; K)$  by left-right action, then we can use (24.1.6.16) and (24.1.6.17), because the action is constructive, by (20.5.2.10), and  $\iota \text{Unip}(n; K) \iota = \text{Unip}(n; K)$ , and  $\iota$  preserves orbits except for  $\text{diag}\{a, d\}, a \neq d$ .

But there are no desired distribution on this orbit: this orbit is homeomorphic to  $N(K)$  via  $u \mapsto u \text{diag}(a, d)$ , and the distribution is transferred to a left invariant distribution, thus by (24.1.6.12) it is just the Haar measure

$$\Delta(f) = c_1 \int_{\text{Unip}(n; K)} f(u \begin{bmatrix} a & b \\ & d \end{bmatrix}) \psi_{\text{Unip}}(u) du.$$

We notice using a right-invariant version of (24.1.6.12) that

$$\Delta(f) = c_2 \int_{N(K)} f\left(\begin{bmatrix} a & b \\ & d \end{bmatrix} u\right) \psi_N(u) du = c_2 \int_{N(K)} f(u \begin{bmatrix} a & b \\ & d \end{bmatrix}) \psi_N\left(\begin{bmatrix} a & b \\ & d \end{bmatrix}^{-1} u \begin{bmatrix} a & b \\ & d \end{bmatrix}\right) du$$

as  $N(K)$  is unimodular. Notice now  $c_1 \psi_N(u) = c_2 \psi_N\left(\begin{bmatrix} a & b \\ & d \end{bmatrix}^{-1} u \begin{bmatrix} a & b \\ & d \end{bmatrix}\right)$  cannot happen for all  $u$ , as this implies  $c_1 = c_2$  by choosing  $u = I$ , and then  $\psi(x) = \psi(ax/d)$ , which is impossible by (12.11.1.36). So if some  $u$  this is not equal, then we find a function supported at a nbhd of  $u$ , then this two distributions cannot be equal.  $\square$

**Prop. (24.11.1.18) [Transpose Invariant Distributions].** If  $D \in \text{Dist}(GL(n; K))^{\text{Ad}_G}$  (24.1.6.3), then it is also invariant under transpose.  $\lrcorner$

*Proof:* This follows from (24.1.6.17), as we look at the conjugate action of  $GL(n; K)$  on itself, with  $\sigma$  being the transposition. Conjugate action is constructive, by (20.5.2.10), and  $g^\sigma = g^{-t}$ , and a matrix is conjugate to its transpose (3.5.4.19).  $\square$

### Finite dimensional Representations

**Prop. (24.11.1.19).**  $\text{Irr}^{\text{fin, alg}}(GL(n; K)) = GL(n; K)^\vee$ .  $\lrcorner$

*Proof:* By the no-small-subgroup argument, the kernel of this representation contains an open normal subgroup. But any open normal subgroup of  $GL(n; V)$  contains  $SL(n; V)$  by (24.11.1.9), thus this is a representation of  $K^\times$ , which must be of 1-dimensional by (24.1.6.36).  $\square$

**Lemma (24.11.1.20) [2-Dimensional Smooth Representation of  $K^\times$ ].** Any  $\rho \in \text{Rep}^{\text{alg, dim}=2}(K^\times)$  is one of the following:

- $\rho(t) = \text{diag}(\xi(t), \xi'(t))$ , where  $\xi, \xi'$  are two quasi-characters of  $K^\times$ .
- $\rho(t) = \xi(t) \begin{bmatrix} 1 & v(t) \\ & 1 \end{bmatrix}$ , where  $\xi$  is a quasi-character of  $K^\times$ .

$\lrcorner$

*Proof:* There exists a 1-dimensional invariant subspace spanned by  $x$  by (24.1.6.36), on which  $K^\times$  acts by a quasi-character  $\xi$ , and consider the quotient space, on which  $K^\times$  acts by a quasi-character  $\xi'$ . choose  $y$  that is linearly-independent of  $x$ , then  $\rho(t)y = \xi'(t)y + \lambda(t)x$ , and

$$\lambda(tu) = \xi'(u)\lambda(t) + \lambda(u)\xi(t)$$

which is symmetric in  $t, u$ .

If  $\xi \neq \xi'$ ,

$$\lambda(t)(\xi(u) - \xi'(u)) = \lambda(u)(\xi(t) - \xi'(t))$$

therefore  $\lambda(t) = C(\xi(t) - \xi'(t))$ , so  $z = y - Cx$  is fixed by  $\rho(K^\times)$ .

If  $\xi = \xi'$ , then  $\lambda/\xi$  is an additive character of  $K^\times$ , thus it is trivial on  $\mathcal{O}^*$ , so  $\lambda(t) = cv(t)$ .  $\square$

### Zelevinsky-Segments

**Def. (24.11.1.21) [Zelevinsky-Segments].** For  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$ , a **Zelevinsky-segment**  $\Delta(\pi, m)$  is an ordered  $m$ -tuple  $(\pi, \pi(1), \dots, \pi(m-1))$  of Tate twists of  $\pi$ .

For two segments  $\Delta(\pi_1, m)$  and  $\Delta(\pi_2, n)$ , they are called **linked segments** if  $\Delta(\pi_1, m) \subsetneq \Delta(\pi_2, n)$ ,  $\Delta(\pi_2, n) \subsetneq \Delta(\pi_1, m)$ , and  $\Delta(\pi_1, m) \cup \Delta(\pi_2, n)$  is also a segment. And  $\Delta(\pi_1, m)$  is said to **precedes**  $\Delta(\pi_2, n)$  if  $\pi_2 = \pi_1(m)$ .  $\lrcorner$

**Def. (24.11.1.22) [Dual Segments].** For a segment  $\Delta(\pi, m) = (\pi, \pi(1), \dots, \pi(m-1))$ , its **dual segment**  $\Delta^\vee$  is the segment  $(\pi^\vee(1-m), \pi^\vee(2-m), \dots, \pi) = \Delta(\pi^\vee(1-m), m)$ .  $\lrcorner$

**Def. (24.11.1.23) [ $\pi(\Delta)$ ].** Let  $\Delta = \Delta(\pi, m)$  be a Zelevinsky-segment of length  $m$  and degree  $mn$ , we can define  $\pi(\Delta) \in \text{Irr}^{\text{adm}}(\text{GL}(mn; K))$  by

$$\pi(\Delta) \stackrel{\triangle}{=} \pi \times \pi(1) \times \dots \times \pi(m-1) \stackrel{\triangle}{=} \text{n-Ind}_{P(n, n, \dots, n)}^{\text{GL}(mn; K)} (\otimes \pi_i)$$

$\lrcorner$

**Def. (24.11.1.24) [Zelevinsky-Conditions].** A tuple of Zelevinsky-segments  $(\Delta_1, \dots, \Delta_r)$  (24.11.1.21) is said to satisfy the **Zelevinsky condition** if for each  $i < j$ ,  $\Delta_i$  doesn't precedes  $\Delta_j$  (24.11.1.21).  $\lrcorner$

## 2 Supercuspidal Representations

### Jacquet-Functors

**Notation (24.11.2.1).**

- Use notations on Jacquet functors defined in [Jacquet-Functors](#).

$\lrcorner$

**Prop. (24.11.2.2).** For any  $a \in \mathbb{G}_m^{n-1}(F)$ ,

$$J_{\text{Unip}(n; K), \psi_a, \text{Unip}(n; K)}(V) \cong J_{\text{Unip}(n; K), \psi_{\text{Unip}(n; K)}}(V) \text{ (24.11.0.1)}$$

$\lrcorner$

*Proof:* This is because  $\pi(a)$  maps  $V(\text{Unip}(n; K), \psi_a, \text{Unip})$  isomorphically to  $V(\text{Unip}(n; K), \psi_{\text{Unip}})$ , thus induces an isomorphism  $J_{\text{Unip}(n; K), \psi_a, \text{Unip}}(V) \cong J_{\text{Unip}(n; K), \psi_{\text{Unip}}}(V)$ .  $\square$

**Lemma (24.11.2.3) [Jacquet].** If  $(\pi, V) \in \text{Rep}^{\text{alg}}(\text{GL}(n; K))$ , then using the notations as in (24.11.1.7), where  $\mathfrak{a}$  is a proper ideal,  $V^{K_0}$  and  $V^{N_-(\mathfrak{a})T_0}$  have the same image in the Jacquet module  $J(V)$ .  $\lrcorner$

*Proof:* One inclusion is trivial, for the other, if  $x \in V^{N_-(\mathfrak{a})T_0}$  then  $x_1 = \int_{K_0} \pi(k) x dk$  lies in  $V^{K_0}$ , so it suffices to show  $x$  and  $x_1$  have the same image in  $J(V)$ . But by (24.11.1.7)

$$x_1 = \int_{N_-(\mathfrak{a})} \int_{T_0} \int_{N(\mathcal{O})} \pi(nt_0 n_-) x dn dt_0 dn_- = \int_{N(\mathcal{O})} \pi(n) x dn$$

and notice  $p(\pi(n)x) = p(x)$ .  $\square$

**Prop. (24.11.2.4)[Jacquet Modules are Admissible].** If  $(\pi, V) \in \mathrm{Rep}^{\mathrm{alg}}(\mathrm{GL}(2; K))$ , then

- Using the notation as in (24.11.1.7), if  $\mathfrak{a}$  is a proper ideal, then the projection map  $p : V \mapsto J(V)$  induces a surjection of  $V^{K_0} \rightarrow J(V)^{T_0}$ .
- If  $(\pi, V)$  is admissible, then  $J(V)$  is also admissible representation of  $T(F)$ .

? For  $\mathrm{GL}(n; K)$  case, Cf.[Bernstein-Zelevinsky1, P33].? ┘

*Proof:* 2 follows from 1 because we can choose  $T_0 = T(\mathfrak{a})$  to be arbitrarily small, then it is of f.d., thus admissible.

For 1, notice first that any  $x \in J(V)^{T_0}$  is an image of a  $x_1 \in V^{T_0}$ , because  $p(\pi(t)x) = \pi(t)p(x) = p(x)$ , thus we can choose  $x_1 = \frac{1}{V(T_0)} \int_{T_0} \pi(t)x dt$ .

Thus for any f.d. subspace  $\overline{U}$  of  $J(V)^{T_0}$ , we can find a f.d.  $U \subset V^{T_0}$  that is mapped isomorphically onto  $\overline{U}$ . Now  $U$  is fixed by some  $N_-(\mathfrak{p}^n)$  for  $n$  large, so  $U$  is fixed by  $N_-(\mathfrak{p}^n)T_0$ . Notice  $\mathfrak{a} = \mathfrak{p}^m$  for some  $m$ , and

$$\pi(d)N_-(\mathfrak{p}^n)T_0\pi(d)^{-1} = N_-(\mathfrak{a})T_0, \quad d = \begin{bmatrix} \varpi^{n-m} & \\ & 1 \end{bmatrix},$$

so  $\pi(d)U$  is stabilized by  $N_-(\mathfrak{a})T_0$ . Hence by lemma (24.11.2.3),  $\pi(d)p(U) = p(\pi(d)U) \subset p(V^{K_0})$ , so the dimension of  $\overline{U}$  is bounded by dimension of  $V^{K_0}$ , so we can choose  $\overline{U}$  just to be  $J(V)^{T_0}$ . Now  $\pi(d)$  commutes with  $T(F)$ , so we have  $\pi(d)J(V)^{T_0} = J(V)^{T_0} \subset p(V^{K_0})$ . The reverse containment is clear. □

### Supercuspidal Representation

References are [B-Z77, Kud94].

**Prop. (24.11.2.5)[Representation Lifted from Finite Case].** Let  $(\pi_0, V_0) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2; \mathbb{F}_q))$ , then it is representation of  $\mathcal{K}$  by the projection  $\mathcal{K} \rightarrow \mathrm{GL}(2; \mathbb{F}_q)$ , with central character  $\omega_0$ . Extend  $\omega_0$  to a character  $\omega$  of  $K^\times$ , then extend  $\pi_0$  to a representation of  $\mathcal{K}Z(K)$  with central character  $\omega$ . Finally let  $(\pi, V) = \mathrm{ind}_{\mathcal{K}Z(K)}^{\mathrm{GL}(2; K)} \pi_0$ . Then  $\pi$  is a unitarizable cuspidal irreducible admissible representation of  $\mathrm{GL}(2; K)$ . ┘

*Proof:* To show it is admissible, we use Mackey's intertwining formula (24.1.6.49). By (24.1.6.31), it suffices to show for any  $\rho \in \mathrm{Irr}^{\mathrm{adm}}(\mathcal{K})$ ,  $\dim \mathrm{Hom}_{\mathcal{K}}(\pi_0, \rho) < \infty$ . By  $p$ -adic Cartan decomposition (24.1.6.49), a representative set for  $\mathcal{K} \backslash G / \mathcal{K}Z(K)$  is  $\left\{ \begin{bmatrix} 1 & \\ & \varpi^n \end{bmatrix}, 0 \leq n \in \mathbb{Z} \right\}$ . Notice  $\pi_0$  is of f.d., thus it suffices to show that only f.m.  $\mathrm{Hom}_{K_0(\mathfrak{p}^n)}(\pi_0, \rho^{\varpi^n}) \neq 0$ . For this, notice that by continuity, for  $n$  large,  $\rho^{\varpi^n} \left( \begin{bmatrix} 1 & b \\ & 1 \end{bmatrix} \right) = \mathrm{id}$  for any  $b \in \mathcal{O}$ , thus any  $\varphi \in \mathrm{Hom}_{K_0(\mathfrak{p}^n)}(\pi_0, \rho^{\varpi^n}) \neq 0$  factors through the Jacquet module of  $\pi$ , which is 0 as  $\pi_0$  is cuspidal.

By (24.1.6.51),  $\pi$  is unitarizable. And then by (24.1.6.52), it suffices to show that  $\dim \mathrm{End}(\pi) = 1$ . By Mackey theory again,

$$\mathrm{Hom}_G(\pi, \pi) \subset \mathrm{Hom}_G(\pi, \mathrm{Ind}_{\mathcal{K}Z(K)}^{\mathrm{GL}(2, K)} \pi_0) = \prod_{n \geq 0} \mathrm{Hom}_{K_0(\mathfrak{p}^n)}(\pi_0, \rho^{\varpi^n})$$

by the same reason as above, for  $n > 0$ , this is 0, and for  $n = 0$ , this has dimension 1 because  $\pi_0$  is an irreducible representation of  $K$ .

It remains to show  $\pi$  is cuspidal: Use Mackey theory again. By  $p$ -adic Cartan decomposition (20.7.2.15), a representative set for  $\mathcal{K} \backslash G / \mathcal{K} Z(K)$  is  $\left\{ \begin{bmatrix} 1 & \\ & \varpi^n \end{bmatrix}, 0 \leq n \in \mathbb{Z} \right\}$ . Now it suffices to show  $\text{Hom}_{\text{Unip}(2,K)}(\pi_0, 1) = 0$ . But this is because  $\pi_0$  is cuspidal.  $\square$

**Prop. (24.11.2.6) [f.m.  $K$ -Spherical Supercuspidal Representations].** Let  $K$  be a compact open subgroup of  $\text{GL}(n; K)$ , then there are only finitely many isomorphism classes of irreducible supercuspidal representations of  $G \circ$  with a  $K$  fixed vector.  $\perp$

*Proof:* ? Cf. [Bernstein. Harvard lecture notes on representations of  $p$ -adic reductive groups] Proposition 21 in Section 3.  $\square$

### 3 Generic representations

**Prop. (24.11.3.1).**

- The definition of generic representations of  $\text{GL}(n; K)$  is independent of  $\psi$  chosen, by (24.11.2.2).
- For a quasi-character  $\chi$  of  $K^\times$ ,  $\pi$  is generic iff  $\pi(\chi)$  is generic, as  $J_{\text{Unip}(n,K), \psi_{\text{Unip}}}(\pi) = J_{\text{Unip}(n,K), \psi_{\text{Unip}}}(\pi(\chi))$ .  $\perp$

**Prop. (24.11.3.2) [Contragradient of Generic Representations].** The contragradient of a generic representation  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  is also generic. In fact, if  $\mathcal{W}(\pi; \psi)$  is a Whittaker model for  $\pi$  w.r.t.  $\psi$ , then  $\widetilde{W} = \{W(w^0 g^{-t}) | W \in \mathcal{W}(\pi; \psi)\}$  (24.11.0.1) is a Whittaker model for  $\widetilde{\pi}$  w.r.t.  $\overline{\psi}$ .  $\perp$

*Proof:* This follows from (24.11.1.10).  $\square$

### Existence of Whittaker Models

**Thm. (24.11.3.3) [Generic Representations of  $\text{GL}(n)$ ].**  $Q(\Delta_1, \dots, \Delta_k) \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  is generic iff no two of  $\Delta_i$  are linked, in which case

$$\pi = Q(\Delta_1) \times \dots \times Q(\Delta_k).$$

In particular,  $\pi$  is generic iff its Gelfand-Kirillov dimension is maximal among those irreducible admissible representations with the same cuspidal supports.  $\perp$

*Proof:* Cf. [Induced Representations of Reductive  $p$ -adic Groups, Zelevinsky(1980)]. ?  $\square$

**Prop. (24.11.3.4) [Gauges for Whittaker Models].** If  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  is unitary and generic, then there exists a gauge function, i.e. a function  $\xi > 0$  on  $\text{GL}(n; K)$  that is invariant under the action of center, left action of  $\text{Unip}(n; K)$  and right action of  $\mathcal{K}$ , and given on  $A$  by

$$\xi(a) = \varphi(\alpha_1(a), \dots, \alpha_n(a)) |\alpha_1 \dots \alpha_n(a)|^{-t}$$

for some  $t \in \mathbb{R}_{\geq 0}$  and  $\varphi \geq 0 \in \mathcal{S}(K^{r-1})$ , s.t. any  $W \in \mathcal{W}(\pi; \psi)$  is bounded by  $\xi$ .  $\perp$

*Proof:* Cf. [Automorphic forms on  $\text{GL}(3)$ , 1] Prop 2.3.6 ?  $\square$

**Def. (24.11.3.5) [Sheaf on  $F$  associated to  $V$ ].** As  $\mathrm{Unip}(2) \cong \mathbb{A}^1$ , any smooth representation  $V$  of  $\mathrm{Unip}(2, K)$  corresponds to a smooth  $\mathcal{H}(F)$ -module by (24.1.6.22), thus we can view it as a  $C_c^\infty(F)$ -module by

$$\varphi(v) = \int_{\mathrm{Unip}(2, K)} \widehat{\varphi}(x) \pi(u) v dx.$$

Then this module is smooth thus non-degenerate by (24.1.6.22), then we can define  $\mathcal{S}(V)$  the sheaf associated to  $V$ , as in (4.4.4.15).  $\lrcorner$

**Cor. (24.11.3.6).** Let  $V$  be a smooth  $B(K)$ -sheaf and let  $a \in K$ , then the stalk

$$\mathcal{S}(V)_a \cong \begin{cases} J(V) & a = 0 \\ J_{\psi_a}(V) \cong J_\psi(V) & a \neq 0 \end{cases}.$$

$\lrcorner$

*Proof:* By definition (4.4.4.15), the stalk is  $V$  modulo the subgroup consisting of elements  $v$  that  $\chi_U \cdot v = 0$ , where  $U = a + \mathfrak{p}^k$  for some large  $k$ . Back the definition of  $\mathcal{S}(V)$ , consider the Fourier transform

$$\widehat{\chi_{a+\mathfrak{p}^k}}(x) = \overline{\psi(ax)} V(\mathfrak{p}^k) \chi_{\mathfrak{p}^{n-k}}(x)$$

where  $\mathfrak{p}^n$  is the conductor of  $\psi$ . Thus

$$\chi_{a+\mathfrak{p}^k} v = C \int_{\mathfrak{p}^{n-k}} \overline{\psi(ax)} \pi \left( \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \right) v dx = 0$$

for large  $k$ , which is equivalent to  $v \in V_{N, \psi_a}$  by (24.1.2.8). Finally,  $J_{\psi_a}(V) \cong J_\psi(V)$  by (24.11.2.2).  $\square$

**Cor. (24.11.3.7) [Existence of Whittaker Functional].**  $(\pi, V) \in \mathrm{Rep}^{\mathrm{adm}}(\mathrm{GL}(2; K))$  has a Whittaker functional unless it factors through the determinant map. In particular, if  $\pi$  is irreducible, then  $\dim J_\psi(V) = 1$  iff  $\dim V > 1$ .  $\lrcorner$

*Proof:* The existence of a Whittaker functional is equivalent to the fact  $J_\psi(V) \neq 0$ , which is the stalk of the sheaf  $\mathcal{S}(V)$ . If it vanishes, then  $\mathcal{S}(V)$  is a skyscraper sheaf by (24.11.3.6), and by the correspondence of  $\mathcal{S}(V)$  and  $V$  (4.4.4.15),  $V$  equals  $\Gamma(F, \mathcal{S}(V)) = J(V)$ , thus  $\mathrm{Unip}(2, K)$  acts trivially on  $V$ .

So also all the conjugates of  $\mathrm{Unip}(2, K)$  acts trivially, so  $\mathrm{SL}(2; K)$  acts trivially, by 5, thus the representation factors through the quotient  $K^\times$ , the rest is clear.

Finally, if it is irreducible, then it factors through the determinant map iff  $V \cong \mathbb{C}$  and  $\pi(g) = \chi(\det(g))$ , by (24.1.6.36). So  $J_\psi(V) = 0$  as  $\psi$  is non-trivial.  $\square$

### Kirillov Model

References are [Ber84a]

**Def. (24.11.3.8) [Kirillov Models].** For  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ , a **Kirillov model**  $\mathcal{K}(\pi)$  is a subrepresentation of  $\mathrm{Ind}_{\mathrm{Unip}(n, K)}^{M_n(K)}(\psi)$  that is isomorphic to  $\pi|_{M_n(F)}$ .

When  $n = 2$ , it is equivalently a subspace of  $C^\infty(K^\times)$  with an action of  $M_2(F)$  that satisfies

$$\left[ \pi \left( \begin{bmatrix} a & b \\ & 1 \end{bmatrix} \right) \varphi \right](x) = \psi(bx) \varphi(ax)$$

and isomorphic to  $(\pi, V)$  via the isomorphism  $v \mapsto \varphi_v$ .  $\lrcorner$

*Proof:* The last assertion is because  $M_2(F) = \text{Unip}(2, K)T_1(F)$ , thus the value of a function in  $\text{Ind}_{N(K)}^{B_1(F)}(\psi)$  is determined by its restriction on  $T_1(F) \cong K^\times$ .  $\square$

**Prop. (24.11.3.9).** The action of  $M_n(K)$  on  $\text{ind}_{\text{Unip}(n, K)}^{M_n(K)}(\psi)$  as defined in (24.11.3.8) is irreducible.  $\lrcorner$

*Proof:* ?

Let  $U$  be a non-zero invariant subspace of  $\text{ind}_{\text{Unip}(n, K)}^{M_n(K)}(\psi)$ , we will show that for any  $a \in K^\times$ ,  $U$  contains  $\chi_U$  for any sufficiently small nbhd  $U$  of  $a$ . Let  $\varphi \in U$  and  $\varphi(b) \neq 0$ , then by action of  $\begin{bmatrix} b/a & \\ & 1 \end{bmatrix}$ , we may assume  $\varphi(a) \neq 0$ .

If  $f \in C_c^\infty(F)$ ,  $f$  acts on  $\varphi$  via

$$\pi(f)\varphi = \int_F f(x)\pi\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}\right)\varphi dx,$$

which in fact a finite sum of elements in  $N(K)v$ . Then  $\pi(f)\varphi(y) = \int_F f(x)\psi(xy)\varphi(y)dx = \widehat{f}(y)\varphi(y)$ . By the isomorphism  $\mathcal{H}_F \cong C_c^\infty(F)$  (24.1.6.25), we can choose  $f$  that  $\widehat{f} = \varphi(a)^{-1}\chi_U$  for a small nbhd  $U$  of  $a$  that  $\varphi$  is constant on  $U$ , then  $\pi(f)\varphi = \chi_U$ .  $\square$

**Lemma (24.11.3.10).** For  $(\pi, V) \in \text{Irr}^{\text{adm}}(\text{GL}(2; K))$ , if  $\dim V = \infty$ , then  $V^{\text{Unip}(2, F)} = 0$ .  $\lrcorner$

*Proof:* Cf. [Bump, P464] ?

The stabilizer of  $v$  is open, thus by (24.11.1.9) it is fixed by all  $\text{SL}(2; K)$ . Also it is fixed by the center  $Z(K)$ . Now  $\text{GL}(2; K)/\text{SL}(2; K)Z(K) \cong K^\times/(K^\times)^2$  which is finite by (19.1.5.13), thus the invariant subspace generated by  $v$  is of f.d., contradiction.  $\square$

**Lemma (24.11.3.11).** If  $(\pi, V) \in \text{Irr}^{\text{adm}}(\text{GL}(2; K))$ , and  $\Lambda$  is a Whittaker functional on  $V$ , then for  $v \neq 0 \in V$ , there exists some  $a \in K^\times$  that  $\Lambda(\pi\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right)v) \neq 0$ .  $\lrcorner$

*Proof:* For any  $a \in K^\times$ , it is clear that the kernel of the map  $v \mapsto \Lambda(\pi\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right)v)$  contains  $V_{N, \psi_a}$  ?? but by local multiplicity one (24.9.5.5),  $\dim J_{\psi_a}(V) \leq 1$ , and this map is non-trivial because  $\Lambda$  is non-trivial, thus the kernel is exactly  $V_{N, \psi_a}$ .

Thus if  $\Lambda(\pi\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right)v) = 0$ , then  $v$  is the section of  $\mathcal{S}(V)$  that vanishes at all  $a \neq 0$ , thus for any  $x \in F$ ,  $v' = v - \pi\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}\right)v$  vanishes at every  $a \in F$ , thus  $v' = 0$ . Then  $v = 0$  by (24.11.3.10).  $\square$

**Prop. (24.11.3.12)[Kirillov Model and Whittaker Model].** For  $(\pi, V) \in \text{Irr}^{\text{adm}}(\text{GL}(2; K))$ , if it has a Whittaker functional  $\Lambda$ , then it has a Whittaker model  $\mathcal{W}$  consisting of functions  $W_v(g) = \Lambda(\pi(g)v)$ , and we can define functions on  $K^\times$  by

$$\varphi_v(a) = W_v\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right).$$

Then  $G$  acts on  $\mathcal{K}(\pi)$  by acting on the subscript. Then it is a Kirillov model for  $V$ , consisting of functions that is compactly supported on  $K$ .

Conversely, if  $\{\varphi_v | v \in V\}$  is a Kirillov model for  $(\pi, V)$ , then we can construct functions on  $\text{GL}(2; K)$  by  $W_v(g) = \varphi_{\pi(g)v}(1)$ . Then this is a Whittaker model for  $V$ .  $\lrcorner$

*Proof:* The equations can be checked by hand. To given an action  $\pi(\varphi_v) = \varphi_{\pi(g)v}$  on  $\{\varphi_v | v \in V\}$ , we need to show that  $W_v \mapsto \varphi_v$  is injective, which is true by (24.11.3.11). Then  $v \mapsto \pi_v$  is an isomorphism.

It remains to show that  $\varphi_v \in C_c^\infty(K^\times)$ . As  $\pi$  is a smooth representation, for any  $v$ ,  $W_v$  is stable under  $N(\mathfrak{p}^k)$  for some  $k$ , thus by equation above,  $\varphi_v(a) = \psi_a(n)\varphi_v(a)$  for all  $a$  and  $n \in N(\mathfrak{p}^k)$ . Then if  $|a|$  is sufficiently large,  $\psi_a(n) \neq 0$  for some  $n \in N(\mathfrak{p}^k)$ , implying  $\varphi_v(y) = 0$ .  $\square$

**Prop. (24.11.3.13) [Kirillov Model and Jacquet Functor, Gelfand-Kirillov [G-K75]].** Let  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}, \mathrm{generic}}(\mathrm{GL}(2; K))$ , having a Kirillov model by (24.11.3.12), then there is an exact sequence of vector spaces

$$0 \rightarrow \mathrm{ind}_{\mathrm{Unip}(n, K)}^{M_n(K)}(\psi) \rightarrow \mathcal{K}(\pi) \rightarrow J_{\mathrm{Unip}(n; K)}(\pi) \rightarrow 0.$$

┘

*Proof:*  $V_N$  is generated by elements of the form  $v' = \pi\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}\right)v - v$ . Notice  $\varphi_{v'}(y) = (\psi(xy) - 1)\varphi_v(y)$ , as  $\psi$  is continuous, when  $|y|$  is small,  $\varphi_{v'}(y) = 0$ .

To show that  $C_c^\infty(K^\times) \subset V_N$ , notice  $V_N$  is non-zero, because  $\dim V = \infty$  but  $\dim J(V)$  is finite (24.11.5.14). Notice  $V_N$  is stable under  $B(K)$  action, thus it must be  $C_c^\infty(K^\times)$  by (24.11.3.9).  $\square$

**Prop. (24.11.3.14).** If  $(\pi, V)$  is an irreducible generic admissible representation of  $\mathrm{GL}(2; K)$  thus having a Kirillov model. If  $\chi$  is a quasi-character of  $T(F)$  that  $\pi(t)\bar{\varphi}_v = (\delta^{1/2}\chi)(t)\bar{\varphi}_v \in J(V)$ , then for  $|t|$  small,  $\varphi_v(t)$  is a constant multiple of  $|t|^{1/2}\chi_1(t)$ .  $\square$

*Proof:* Let  $t_0 \in \mathfrak{p} \setminus \mathfrak{p}^2$ , then

$$\pi\left(\begin{bmatrix} t_0 & \\ & 1 \end{bmatrix}\right)\varphi - |t_0|^{1/2}\chi_1(t_0)\varphi \in V_N,$$

thus by (24.11.3.13), there exists some  $\varepsilon(t_0) > 0$  that

$$\varphi(tu) - |t|^{1/2}\chi_1(t)\varphi(u) = 0$$

for  $t = t_0$  and  $|u| \leq \varepsilon(t_0)$ . Because both sides are locally constant on  $t$ , by the compactness of  $\mathfrak{p} \setminus \mathfrak{p}^2$ , there exists some  $\varepsilon > 0$  that for the equation is true for all  $t \in \mathfrak{p} \setminus \mathfrak{p}^2$  and  $|u| \leq \varepsilon$ .

Now any element  $t \in \mathfrak{p}$  can be factored into product of elements in  $\mathfrak{p} \setminus \mathfrak{p}^2$ , so by induction this is true for any  $t \in \mathfrak{p}$ . Thus we are done.  $\square$

**Lemma (24.11.3.15) [Gelfand Uniqueness Principle].** Let  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(2; K))$ , and let  $\chi$  be a quasi-character of  $K^\times$ , then there are at most two essentially different values of  $s$  that there are at least two linear functionals  $L : V \rightarrow \mathbb{C}$  that satisfies

$$L(\pi\left(\begin{bmatrix} y & \\ & 1 \end{bmatrix}\right)v) = \chi(y)|y|^s L(v).$$

┘

**Remark (24.11.3.16).** In fact, for any  $s$ , there exists at most one such linear functional?  $\square$

*Proof:* If  $\dim V = 1$ , this is trivial. Otherwise  $\dim V = \infty$  and  $(\pi, V)$  has a Kirillov model. Identify  $V$  with its Kirillov model. Suppose  $L_1, L_2$  are two linear functionals that satisfies the equation, consider their restriction to  $V_N = C_c^\infty(K^\times)$ , on which  $B_1(F)$  acts, thus  $L_1, L_2$  are linearly dependent when restricted to  $V_N$  by (24.1.6.13). Thus there exists constants  $c_1, c_2$  that  $c_1 L_1 + c_2 L_2$  factors through  $J(V)$ .  $\dim J(V) \leq 2$  by (24.11.5.14). So by (24.11.1.20) for all but two possible choices of  $s$ ,  $c_1 L_1 + c_2 L_2 = 0$ , thus there are two linear functionals only for possibly two choices of  $s$ .  $\square$



### Newvectors

References are [Schmidt, Some remarks on local newforms for  $\mathrm{GL}(2)$ ].

**Prop. (24.11.3.17)** [Newvectors, Gelfand-Kazhdan[G-K71]]. For  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ , there exists  $m \in \mathbb{N}$  s.t.  $V^{\mathcal{K}_{0,1}(m)} \neq 0$  (24.11.1.5). And if  $m_0$  is the smallest such  $m$ ,  $V^{\mathcal{K}_{0,1}(m_0)}$  is denoted by the space of **newvectors** for  $\pi$ .  $\lrcorner$

*Proof:* ? Cf.[G-H11]P44.  $\square$

**Thm. (24.11.3.18)** [New Vectors are Unique, Casselman[Cas73]/Jacquet-Shapiro-Shalika[JSS81]]. For  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ , the space of new vectors is 1-dimensional (24.11.3.17). And the smallest  $m$  s.t.  $V^{\mathcal{K}_{0,1}(m_0)} \neq 0$  is just the conductor  $c(\pi)$  (29.3.1.31).  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (24.11.3.19).** For any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ , there exists an irreducible representation of  $\mathrm{GL}(n; \mathcal{O}_K)$  appearing in  $\pi|_{\mathrm{GL}(n; \mathcal{O}_K)}$  with multiplicity 1.  $\lrcorner$

*Proof:* This follows from (24.1.6.31) and (24.11.3.18).  $\square$

**Prop. (24.11.3.20).** The conductor (29.3.1.31) of an irreducible principle series  $\pi(\chi_1, \chi_2)$  is equal to  $c(\chi_1) + c(\chi_2)$ .

The conductor of the Steinberg representation is 1. The conductor of a supercuspidal representation is at least 2.

The conductor of the central character of a supercuspidal representation with conductor  $r$  is at most  $\lfloor \frac{r}{2} \rfloor$ .  $\lrcorner$

*Proof:* Cf.[Schmidt, Some remarks on local newforms for  $\mathrm{GL}(2)$ ] and [Atkin-Li, Twists of newforms and pseudo-eigenvalues of W-operators]. ?  $\square$

## 4 Bernstein-Zelevinsky Classification

References are [B-Z76, B-Z77, Zel80].

**Prop. (24.11.4.1)** [Composition Factors and Permutations]. Let  $n \in \mathbb{Z}_+$  and  $\kappa = (\kappa_1, \dots, \kappa_r), \lambda = (\lambda_1, \dots, \lambda_s)$  be two ordered partition of  $n$ . Let  $(\pi_i, V_i) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(\kappa_i; K))$  (24.1.6.40) and  $(\pi'_j, V'_j) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(\lambda_j; K))$  (24.1.6.40), then the following are equivalent:

- $r = s$  and  $(\kappa, \{\pi_i\})$  is a permutation of  $(\lambda_j, \{\pi'_j\})$ .
- There is a non-zero intertwining operator

$$\pi_1 \times \dots \times \pi_r \rightarrow \pi'_1 \times \dots \times \pi'_s.$$

- Some subquotient of  $\pi_1 \times \dots \times \pi_r$  is isomorphic to some subquotient of  $\pi'_1 \times \dots \times \pi'_s$ .
- The composition factors of  $\pi_1 \times \dots \times \pi_r$  and  $\pi'_1 \times \dots \times \pi'_s$  are the same.  $\lrcorner$

*Proof:* Cf. Theorem 2.9(a) in [B-Z77]. ?

Notice  $\pi_1 \times \dots \times \pi_r$  is admissible of finite length by (24.9.3.5).  $\square$

**Cor. (24.11.4.2).**  $\mathrm{Hom}(\chi_1 \times \dots \times \chi_n, \mu_1 \times \dots \times \mu_n) \neq 0$  only if  $\{\chi_1, \dots, \chi_n\} = \{\mu_1, \dots, \mu_n\}$ .  $\lrcorner$



**Thm. (24.11.4.3).** For any Zelevinsky segment  $\Delta$  of length  $m$  (24.11.1.21), the representation  $\pi(\Delta)$  has length  $2^{m-1}$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (24.11.4.4) [Zelevinsky-Quotient Representations].** Let  $\Delta$  be a Zelevinsky-segment, then  $\pi(\Delta)$  has a unique irreducible subrepresentation  $Z(\Delta)$  and a unique irreducible quotient representation  $Q(\Delta)$ , called the **Zelevinsky-quotient representation** associated to  $\Delta$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (24.11.4.5) [Langlands Quotients].** Let  $(\Delta_1, \dots, \Delta_r)$  be a tuple of Zelevinsky segments satisfying the Zelevinsky condition (24.11.1.24), then  $Q(\Delta_1) \times \dots \times Q(\Delta_r) \in \text{Rep}^{\text{adm}}(GL(\sum n_i m_i; K))$  admits a unique irreducible quotient  $Q(\Delta_1, \dots, \Delta_r)$ , called the **Langlands quotient**. Moreover,  $Q(\Delta_1, \dots, \Delta_r)$  is independent of the order of  $\Delta_1, \dots, \Delta_r$  (as long as they satisfy the Zelevinsky condition (24.11.1.24)).  $\lrcorner$

*Proof:*  $\square$

**Thm. (24.11.4.6) [Bernstein-Zelevinsky [B-Z77]].** Any  $(\pi, V) \in \text{Irr}^{\text{adm}}(GL(n; K))$  is isomorphic to one of the form  $\pi \cong Q(\Delta_1, \dots, \Delta_r)$  where  $\Delta_i = \Delta(\pi_i, m_i)$  and  $\sum n_i m_i = n$  for a unique tuple of segments  $(\Delta_1, \dots, \Delta_r)$  satisfying Zelevinsky condition up to permutation.  $\lrcorner$

*Proof:*  $\square$

**Thm. (24.11.4.7).** Let  $(\Delta_1, \dots, \Delta_r)$  be a tuple of segments satisfying the Zelevinsky condition, then  $Q(\Delta_1, \dots, \Delta_r)$  is irreducible iff no two elements are linked.  $\lrcorner$

*Proof:*  $\square$

**Prop. (24.11.4.8) [Zelevinsky].**  $Q(\Delta_1, \dots, \Delta_r)^\vee = Q(\Delta_1^\vee, \dots, \Delta_r^\vee)$ .  $\lrcorner$

*Proof:* Cf. [Ze] 3.3., 9.4. ?  $\square$

**Prop. (24.11.4.9).** Dual statements of (24.11.4.5)(24.11.4.6)(24.11.4.7)(24.11.4.8) hold, i.e.  $Z(\Delta_1) \times \dots \times Z(\Delta_r)$ , which admits a unique irreducible subrepresentation  $Z(\Delta_1, \dots, \Delta_r)$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (24.11.4.10) [Supercuspidal Supports].** For a  $\pi = Q(\Delta_1, \dots, \Delta_r) \in \text{Irr}^{\text{adm}}(GL(n; K))$ , where  $\Delta_i = (\pi_i, m_i)$ , define the **supercuspidal supports** of  $\pi$  as the set

$$\text{Supp}(\pi) = \{\pi_i(j)\}_{1 \leq i \leq r, 1 \leq j \leq m_i - 1}.$$

$\lrcorner$

**Prop. (24.11.4.11) [Constituents of  $Q(\Delta_1) \times \dots \times Q(\Delta_r)$ ].** The constituents of  $Q(\Delta_1) \times \dots \times Q(\Delta_r) \in \text{Rep}^{\text{adm}}(GL(n; K))$  are exactly the representations  $Q(\Delta'_1, \dots, \Delta'_r)$ , where the multiset  $\{\Delta'_1, \dots, \Delta'_r\}$  are obtained from the multiset  $\{\Delta_1, \dots, \Delta_r\}$  by successively replacing two linked segments  $\{\Delta^1, \Delta^2\}$  by  $\{\Delta^1 \cap \Delta^2, \Delta^1 \cup \Delta^2\}$ .  $\lrcorner$

*Proof:* Cf. [Zel80] Thm 7.1 ?  $\square$

### Zelevinsky Involutions

**Prop. (24.11.4.12).** For  $n \in \mathbb{Z}_+$ , denote  $\mathcal{R}_n(K) = \text{Rep}^{\text{adm, finite length}}(\text{GL}(n; K))$  and  $\mathcal{R}(K) = \bigoplus_{n \in \mathbb{N}} \mathcal{R}_n(K)$ . Then

$$([\pi_1], [\pi_2]) \mapsto [\pi_1 \times \pi_2]$$

defines a graded ring structure on  $\mathcal{R}(K)$ . And then  $\mathcal{R}(K)$  is isomorphic to the ring of polynomials on the set of Zelevinsky segments.  $\lrcorner$

*Proof:* Cf. [Ze]1.9.  $\square$

**Prop. (24.11.4.13).**

- The map  $t : \mathcal{R}(K) \rightarrow \mathcal{R}(K) : Q(\Delta) \mapsto Z(\Delta)$  defines a graded involution of  $\mathcal{R}(K)$ .
- $t$  perserves irreducible representations.
- $t(Q(\Delta_1, \dots, \Delta_r)) = Z(\Delta_1, \dots, \Delta_r)$ ,  $t(Z(\Delta_1, \dots, \Delta_r)) = Q(\Delta_1, \dots, \Delta_r)$ . In particular,  $t$  commutes with contragredients, by (24.11.4.8)(24.11.4.9).  $\lrcorner$

*Proof:*  $\square$

### Examples

**Prop. (24.11.4.14)[Principal Series].** For  $n = 2$ , the principle series (24.11.5.1)  $\chi_1 \boxtimes \chi_2$  are irreducible except the following two cases:

- If  $\chi_2 = \chi_1(1)$ , then  $\chi_1 \times \chi_2$  has a 1-dimensional invariant subspace and the quotient representation is irreducible.
- If  $\chi_2 = \chi_1(-1)$ , then  $\chi_1 \times \chi_2$  has an irreducible invariant subspace of codimension 1.

In these cases, the infinity-dimensional irreducible representations are called the **Steinberg representations**  $\text{St}_2(\chi_1, \chi_2)$ .  $\lrcorner$

*Proof:* Firstly we prove if  $\mathcal{B}(\chi_1, \chi_2)$  has a non-trivial subspace, then it has a non-trivial subspace of dimension 1 or codimension 1: Let  $V'$  be the invariant subspace and  $V''$  the quotient space, by the exactness of Jacquet functor (24.9.3.1) and (24.9.5.9), at least one of  $J_\psi(V')$ ,  $J_\psi(V'')$  vanishes. If  $J_\psi(V') = 0$ , then by (24.11.3.7) it factors through the determinant map, thus it has a 1-dimensional invariant space. If  $J_\psi(V'') = 0$ , then we can use (24.11.5.4) and (24.1.6.27) to dualize.

Next, if  $\mathcal{B}(\chi_1, \chi_2)$  has a 1-dimensional subspace  $V = \{f\}$ , then  $\pi(g)f = \rho(\det(g))f$  for some quasi-character  $\rho$  of  $K^\times$ . Now consider the fact  $f \in \mathcal{B}(\chi_1, \chi_2)$ , take  $b = \text{diag}(y, y^{-1})$ , then  $(\delta^{1/2}\chi)(b) = 1$ , showing  $\chi_1\chi_2^{-1} = |\cdot|^{-1}$ . The codimension1 case is dual by (24.1.6.27) and (24.11.5.4), so  $\mathcal{B}(\chi_1, \chi_2)$  is irreducible except when  $\chi_1\chi_2^{-1} = |\cdot|^{\pm 1}$ .

Finally, if  $\chi_1\chi_2^{-1} = |\cdot|^{-1}$ , let  $\chi_1 = \chi|\cdot|^{-1/2}$ , then  $f(g) = \chi(\det(g))$  is an invariant 1-dimensional subspace, and this is the only 1-dimensional invariant subspace that factors through the determinant map, because if  $f(g) = \chi'(\det(g)) \in \mathcal{B}(\chi_1, \chi_2)$ , then  $\chi' = \chi_1|\cdot|^{-1/2} = \chi_2|\cdot|^{-1/2} = \chi$ .

Also the quotient representation is irreducible because the same argument by Jacquet module shows if it is non-irreducible, then it has an invariant subspace of dimension 1 that  $G$  action factors through the determinant map or of codimension 1, which means there are two invariant 1-dimensional subspace that the  $G$ -action factors through the determinant map, or a invariant subspace of codimension1 of  $\mathcal{B}(\chi_1, \chi_2)$ , the latter case contradicting the argument above. For the former case, we get a 2-dimensional subrepresentation of  $\mathcal{B}(\chi_1, \chi_2)$  that factors through the determinant map.

The argument above shows it is generated by  $\chi(\det())$  and some function  $f$ , but  $f$  must satisfy  $f(g) = \chi(\det(g))f(1) + a$ , which is possible only if  $a = 0$ , so  $f = \chi \circ \det$ , contradiction.

The codimension 1 case is dual by (24.1.6.27) and (24.11.5.4).  $\square$

**Cor. (24.11.4.15) [Classification of  $\text{Irr}^{\text{adm}}(GL(2; K))$ ].** The  $n = 2$  case is particularly clear: any  $\pi \in \text{Irr}^{\text{adm}}(GL(2; K))$  has the following possibilities:

1.  $\pi$  is supercuspidal.
2.  $\pi = Q(\chi \otimes \text{val}^{-1/2}, \chi \otimes \text{val}^{1/2})$ , which equals  $\text{St}_2(\chi)$ .
3.  $\pi = \chi_1 \times \chi_2$ , where  $\chi_1, \chi_2$  are not linked, which is the irreducible principal series.
4.  $\pi = Q(\chi \otimes \text{val}^{1/2}, \chi \otimes \text{val}^{-1/2})$ , which equals  $\mathbf{1}(\chi)$ .

$\lrcorner$

**Def. (24.11.4.16) [Generalized Speh-Representations].** A **generalized Speh-representation** of  $GL(n; K)$  is a unitary representation of the form

$$\text{Speh}(\pi_0; t) \triangleq Q(\pi_0(\frac{t-1}{2}), t) \text{ (24.11.4.4)}.$$

where  $td = n$  and  $\pi_0 \in \text{Irr}^{\text{cusp, uni}}(GL(d; K))$ . Notice that when  $\pi_0 = \mathbf{1}$ , then  $\text{Speh}(\pi_0; t) = \text{St}(\mathbf{1}; t)$  are generalized steinberg-representations (24.9.3.12).  $\lrcorner$

## 5 Principal Series Representations

**Def. (24.11.5.1) [Principal Series Representations].** Given a diagonal quasi-character  $\chi = (\chi_1, \dots, \chi_n)$  of  $T(K)$ ,  $\chi_1 \times \dots \times \chi_n \in \text{Rep}^{\text{alg}}(GL(n; K))$  is called a **principal series representation** of  $G$ . It is admissible of finite length by (24.9.3.5).  $\lrcorner$

**Lemma (24.11.5.2).** By (24.1.6.47), we have a map  $P : C_c^\infty(G(K)) \rightarrow \chi_1 \times \dots \times \chi_n$ :

$$(P\varphi)(g) = \int_{B(K)} \varphi(b^{-1}g)(\delta^{1/2}\chi)(b)db$$

Then this map is intertwining and surjective. Moreover, we have

$$P(\lambda(b)^{-1}\varphi) = (\delta^{-1/2}\chi)(b)P(\varphi), \quad b \in B(K)$$

$\lrcorner$

**Prop. (24.11.5.3).** The representation  $\chi_1 \times \dots \times \chi_n$  admits at most one Whittaker functional. In other words,  $\dim J_\psi(\chi_1 \times \dots \times \chi_n) \leq 1$ . in fact, it is exactly one, as will be shown by (24.11.4.14) and (24.11.3.7).  $\lrcorner$

*Proof:* Let  $\Lambda : V \rightarrow \mathbb{C}$  be a Whittaker functional, then we define a distribution  $\Delta$  on  $GL(2; K)$  as  $\Delta(\varphi) = \Lambda(P\varphi)$ . Then

$$\lambda(b)\Delta = (\delta^{-1/2}\chi)(b)\Delta, b \in B(K), \quad \rho(n)\Delta = \psi_N(n)^{-1}\Delta, n \in N(K)$$

by (24.11.5.2). Because  $P$  is surjective (24.11.5.2), it suffices to show that these  $\Delta$  are unique up to scalar.

Consider the left-right action of  $B(K) \times N(K)$  on  $G(F)$ , then there are  $n!$  orbits  $B(K)WN(K)$  by Bruhat decomposition(10.4.4.28). For  $w \neq 1$ , the orbit  $B(K)wN(K)$  is isomorphic to  $B(K) \times N(K)$  via  $(b, n) \mapsto bwn^{-1}$ , thus the restriction of  $\Delta$  on  $B(K)wN(K)$  must be of the form

$$\Delta_1(\varphi) = C \int_{B(K)} \int_{N(K)} \varphi(bwn^{-1}) \psi_N(n) (\delta^{1/2} \chi^{-1})(b) db dn$$

by(24.1.6.12). This is the unique map that satisfies the condition.

As for the distribution on  $B(K)$ , the same reasoning shows the restriction of  $\Delta$  on  $B(K)$  must be of the form

$$\Delta_2(\varphi) = C \int_{B(K)} \varphi(b) (\delta^{1/2} \chi^{-1})(b) db,$$

but it satisfies  $\rho(n)\Delta_2 = \Delta_2$ , so there are no distribution on  $B(K)$ . Finally, (4.4.4.12) gives us the result.  $\square$

**Cor.(24.11.5.4).**  $(\chi_1 \times \dots \times \chi_n)^\vee \cong \chi_1^{-1} \times \dots \times \chi_n^{-1}$ .  $\lrcorner$

*Proof:* If  $f \in \mathcal{B}(\chi_1, \chi_2)$ ,  $f' \in \mathcal{B}(\chi_1^{-1}, \chi_2^{-1})$ , then the pairing  $(f, f') = \int_K f(k) f'(k) dk$  is  $G$ -invariant by(12.10.3.16), so this defines a smooth functional  $l_{f'}$ , and it is non-degenerate, the mapping  $f' \mapsto l_{f'}$  is injective(by letting  $f = 1/f'$  on a  $B(K)$ -orbit that  $f'$  is non-zero) from  $\mathcal{B}(\chi_1^{-1}, \chi_2^{-1})$  to  $\mathcal{B}(\chi_1, \chi_2)^\vee$ . Now by symmetry the other side is also injective, and we are done because  $(V^\vee)^\vee \cong V$ (24.1.6.29).  $\square$

### Intertwining Integrals and Whittaker Functionals

**Prop.(24.11.5.5)[Intertwining Integral].** Let  $\xi_i$  be two characters of  $K^\times$ , and  $\chi_i = |\cdot|^{s_i} \xi_i$ ,  $(\pi, V) = \chi_1 \times \chi_2$ ,  $(\pi', V') = \chi_2 \times \chi_1$ . Define for  $f \in V$ ,

$$Mf : \mathrm{GL}(2; K) \rightarrow \mathbb{C} : Mf(g) = \int_{\mathrm{Unip}(2; K)} f(w_0 u g) du$$

then if  $\mathrm{Re}(s_1 - s_2) > 0$ , the integral is absolutely convergent,  $Mf \in V'$ , and  $M$  is a nonzero intertwining, so  $V \cong V'$  if they are both irreducible.  $\lrcorner$

*Proof:* Should compare this proof with(24.7.5.23).

For the convergence, it suffices to check for  $\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}$ ,  $|x|$  large, but then

$$f\left(\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) = f\left(\begin{bmatrix} x^{-1} & -1 \\ & x \end{bmatrix} \begin{bmatrix} 1 & \\ x^{-1} & 1 \end{bmatrix} g\right) = |x|^{-1} (\chi_1^{-1} \chi_2)(x) f\left(\begin{bmatrix} 1 & \\ x^{-1} & 1 \end{bmatrix} g\right)$$

and because  $f$  is locally constant, when  $|x|$  is large,  $f\left(\begin{bmatrix} 1 & \\ x^{-1} & 1 \end{bmatrix} g\right) = f(g)$ , thus the convergence on the unbounded region is dominated by

$$\int_{|x| > q^N} |x|^{-1} |(\chi_1^{-1} \chi_2)(x)| dx = \int_{|x| > q^N} |x|^{-s_1 + s_2 - 1} dx,$$

which converges for  $\mathrm{Re}(s_1 - s_2) > 0$ .

To show  $Mf \in V'$ , we need to check

$$Mf\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) = Mf(g), \quad Mf\left(\begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} g\right) = |y_1/y_2|^{1/2} \chi_2(y_1) \chi_1(y_2) Mf(g).$$

The first is trivial and for the second:

$$Mf\left(\begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} g\right) = \int_F f\left(\begin{bmatrix} y_2 & \\ & y_1 \end{bmatrix} \begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & y_2 y_1^{-1} x \\ & 1 \end{bmatrix} g\right) dx$$

so it is clear.

Finally  $M$  is clearly intertwining, and it is not trivial by looking at the function  $f$ :

$$f = \left| \frac{y_1}{y_2} \right|^{\frac{1}{2}} \chi_1(y_1) \chi_2(y_2) \chi_{\mathcal{O}_F}(x), \text{ where } g = \begin{bmatrix} y_1 & z \\ & y_2 \end{bmatrix} \begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}$$

and vanish if  $g \in B(K)$ . Notice the representation of  $g$  is unique by Bruhat decomposition. Then  $Mf(1) = 1$ .  $\square$

**Prop. (24.11.5.6) [Analytic Continuation of Intertwining Integral].** Let  $\xi_i$  be two characters of  $K^\times$ ,  $\chi_i = |\cdot|^{s_i} \xi_i$ ,  $(\pi_{s_1, s_2}, V_{s_1, s_2}) = \chi_1 \times \chi_2$ ,  $(\pi'_{s_2, s_1}, V'_{s_2, s_1}) = \chi_2 \times \chi_1$ .

Notice that by Iwasawa decomposition (20.7.2.16), an arbitrary  $f \in \mathcal{B}(\chi_1, \chi_2)$  is determined by its restriction on  $K$ , and if a function  $f_0$  on  $K$  satisfies

$$f_0\left(\begin{bmatrix} y_1 & x \\ & y_2 \end{bmatrix} k\right) = \xi_1(y_1) \xi_2(y_2) f_0(k), \quad y_1, y_2 \in \mathcal{O}^*,$$

then  $f_0$  can extend uniquely to an element  $f_{s_1, s_2} \in V_{s_1, s_2}$  for any  $s_1, s_2$ , called the **flat sections** of  $f_0$ .

Then the intertwining integral  $Mf_{s_1, s_2}$  defined in (24.11.5.5) is analytic for the dominant  $\chi$  in the flat family, and has an analytic continuation to all  $s_1, s_2$  that  $\chi_1 \neq \chi_2$ , and defines a nonzero operator  $V_{s_1, s_2} \rightarrow V'_{s_2, s_1}$ .  $\square$

*Proof:* The proof is parallel to that of (24.11.5.5).

$$Mf_{s_1, s_2}(g) = \int_{|x| \leq q^N} f_{s_1, s_2}\left(\begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) dx + \int_{x \geq q^{N+1}} |x|^{-s_1 + s_2 - 1} (\xi_1^{-1} \xi_2)(x) dx f_{s_1, s_2}(g)$$

the first term can easily be extended, and the second term vanishes if  $\xi_1^{-1} \xi_2$  ramifies, and equals something like a multiple of  $\frac{(\chi_1^{-1} \chi_2(\varpi))^{N+1}}{1 - \chi_1^{-1} \chi_2(\varpi)}$ , which extends unless  $\chi_1 = \chi_2$ .

For the intertwining property, it is because equalities maintain along analytic continuation, in particular, it is non-trivial as  $Mf_{s_1, s_2}(1) = 1$ .  $\square$

**Cor. (24.11.5.7).**  $Z(\chi_1, \chi_2) \cong Z(\chi_2, \chi_1)$ ,  $\sigma(\chi_1, \chi_2) \cong \sigma(\chi_2, \chi_1)$ . If  $\mathcal{B}(\chi_1, \chi_2)$  is irreducible, then  $\pi(\chi_1, \chi_2) \cong \pi(\chi_2, \chi_1)$ . Moreover, there are no other isomorphisms between these representations.  $\square$

*Proof:* It suffices to show there are no other isomorphisms, and this is by (24.11.4.2).  $\square$

**Prop. (24.11.5.8) [Whittaker functional of  $\mathcal{B}(\chi_1, \chi_2)$ ].** There is a Whittaker functional on  $\mathcal{B}(\chi_1, \chi_2)$  defined by

$$\Lambda(f) = \int_F f(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx.$$

This integral is absolutely convergent if  $\chi$  is dominant and is analytic for the flat family of  $\chi$ , by method the same as in the proof of (24.11.5.5). And this can also be extended to all  $\chi$  (as flat section (24.11.5.6)) by defining

$$\Lambda(f) = \lim_{k \rightarrow \infty} \int_{\mathfrak{p}^{-k}} f(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx.$$

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**Remark (24.11.5.9).** Compare with (26.6.4.3).

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*Proof:* This makes sense because it stabilize as  $k \rightarrow \infty$ : when  $k$  is large,

$$\int_{\mathfrak{p}^{-k-1} \setminus \mathfrak{p}^{-k}} f(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx = q^{-k-1} f(1) \int_{\mathfrak{p}^{-k-1} \setminus \mathfrak{p}^{-k}} \psi(-x) (\chi_1^{-1} \chi_2)(x) dx$$

By continuity (See the proof of (24.11.5.5)). If  $\psi(t) \neq 0$ , choose  $k$  large that  $\chi_1^{-1} \chi_2(x) = \chi_1^{-1} \chi_2(x+t)$  for any  $x \in \mathfrak{p}^{-k-1} \setminus \mathfrak{p}^{-k}$ , thus this integral vanishes.  $\square$

**Remark (24.11.5.10).** For general  $n$ , this method won't work, and another method is used in Bernstein, J., Letter to Piatetski-Shapiro (1985). To appear in Cogdell and Piatetski-Shapiro (book in preparation.) to extend analytically.

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**Prop. (24.11.5.11).** If  $\chi_1 \times \chi_2$  is irreducible, with the notations in (24.11.5.5), the intertwining integral  $M : \chi_1 \times \chi_2 \rightarrow \chi_2 \times \chi_1$  satisfies

$$\Lambda' \circ M = \xi_1 \xi_2^{-1} (-1) \gamma(1 - s_1 + s_2, \xi_1^{-1} \xi_2, \psi) \Lambda,$$

where  $\Lambda$  is the Whittaker functional defined in (24.11.5.8), and  $\gamma(s, \chi, \psi)$  is the local constant defined in (29.5.2.8).  $\square$

*Proof:* Cf. [Bump, P485]. ?

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**Cor. (24.11.5.12) [Composition of Intertwining Integrals].** If  $\mathcal{B}(\chi_1, \chi_2)$  is irreducible,  $M : \mathcal{B}(\chi_1, \chi_2) \rightarrow \mathcal{B}(\chi_2, \chi_1)$ ,  $M' : \mathcal{B}(\chi_2, \chi_1) \rightarrow \mathcal{B}(\chi_1, \chi_2)$  are intertwining integrals, then

$$M' \circ M = \gamma(1 - s_1 + s_2, \xi_1^{-1} \xi_2, \psi) \gamma(1 + s_1 - s_2, \xi_1 \xi_2^{-1}, \psi) \cdot \text{Id}.$$

If  $\mathcal{B}(\chi_1, \chi_2)$  is reducible,  $M' \circ M = 0$ .

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*Proof:* The irreducible case follows from the proposition. For the reducible case, let  $\chi_1 \chi_2^{-1} = |\cdot|$ , the intertwining integral  $\mathcal{B}(\chi_2, \chi_1) \rightarrow \mathcal{B}(\chi_1, \chi_2)$  is non-zero, so the image is either  $\mathcal{B}(\chi_1, \chi_2)$  or  $\pi(\chi_1, \chi_2)$ . In the first case,  $\mathcal{B}(\chi_2, \chi_1)$  has a infinite dimensional proper quotient, contradiction. The intertwining integral  $\mathcal{B}(\chi_1, \chi_2) \rightarrow \mathcal{B}(\chi_2, \chi_1)$  is non-zero, so the image is either  $\mathcal{B}(\chi_2, \chi_1)$  or  $\sigma(\chi_1, \chi_2)$ . In the first case,  $\mathcal{B}(\chi_1, \chi_2)$  has a 1-dimensional quotient, contradiction.  $\square$

### Jacquet Module

**Prop. (24.11.5.13) [Jacquet Module of  $\chi_1 \times \dots \times \chi_n$ ].** Let  $\chi_i$  be quasi-characters of  $K^\times$ , then the Jacquet module of  $\chi_1 \times \dots \times \chi_n$  has dimension  $n!$ .

And if  $n = 2$ , the representation of  $T(F)$  on the Jacquet module is isomorphic to the representation

$$t \mapsto \begin{cases} \begin{bmatrix} (\delta^{1/2}\chi)(t) & \\ & (\delta^{1/2}\chi')(t) \end{bmatrix}, & \chi_1 \neq \chi_2 \\ (\delta^{1/2}\chi)(t) \begin{bmatrix} 1 & v(t_1/t_2) \\ & 1 \end{bmatrix}, & \chi_1 = \chi_2 = \chi \end{cases}.$$

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*Proof:* Firstly we show  $\dim J(\mathcal{B}(\chi_1, \chi_2)) = 2$ : Let  $\Lambda : V \rightarrow \mathbb{C}$  be a functional that  $\Lambda(\rho(n)v) = \Lambda(v)$  for  $n \in N(K)$ , then we define a distribution  $\Delta$  on  $GL(2, K)$  as  $\Delta(\varphi) = \Lambda(P\varphi)$ . Then

$$\lambda(b)\Delta = (\delta^{-1/2}\chi)(b)\Delta, b \in B(K), \quad \rho(n)\Delta = \Delta, n \in N(K)$$

by (24.11.5.2). Because  $P$  is surjective (24.11.5.2), it suffices to show that there are exactly two linearly independent such  $\Delta$ .

Consider the left-right action of  $B(K) \times N(K)$  on  $GL(2; K)$ , then there are  $n!$  orbits  $B(K)WN(K)$  by Bruhat decomposition (10.4.4.28). For  $w \neq 1$ , the orbit  $B(K)w \text{Unip}(K)$  is isomorphic to  $B(K) \times \text{Unip}(K)$  via  $(b, n) \mapsto bwn^{-1}$ , thus the restriction of  $\Delta$  on it must be of the form

$$\Delta_1(\varphi) = C \int_{B(K)} \int_{\text{Unip}(K)} \varphi(bwn^{-1})(\delta^{1/2}\chi^{-1})(b) db dn$$

by (24.1.6.12).

As for the distribution on  $B(K)$ , the same reasoning shows the restriction of  $\Delta$  on  $B(K)$  must be of the form

$$\Delta_2(\varphi) = C \int_{B(K)} \varphi(b)(\delta^{1/2}\chi^{-1})(b) db,$$

and it truly satisfies  $\rho(n)\Delta_2 = \Delta_2$ . Finally, (4.4.4.12) gives us the result.

Next we consider  $J(V)$  as a 2-dimensional  $T(F)$ -representation must be of the two forms in (24.11.1.20), it suffices to distinguish these two cases.

Denote  $V = \chi_1 \boxtimes \chi_2$ . Consider for any quasi-character  $\mu$  of  $T(F)$ ,

$$\text{Hom}_{T(F)}(J(V), \delta^{1/2}\mu) \cong \text{Hom}_{B(K)}(V, \delta^{1/2}\mu) \cong \text{Hom}_{GL(2, K)}(V, \mathcal{B}(\mu_1, \mu_2)).$$

So (24.11.4.2)(24.11.5.7) can be used, so if  $\chi_1 \neq \chi_2$ , there are two  $\mu$  that can make the Hom group non-vanish, so it is the first case. If  $\chi_1 = \chi_2$ , then  $V$  is irreducible, and there is only one  $\mu$  that makes this Hom group non-vanish, and the Hom group is of dimension 1 by Schur's lemma, so it is the second case.  $\square$

**Prop. (24.11.5.14) [Jacquet Modules are Finite Dimensional].** For  $(\pi, V) \in \text{Irr}^{\text{adm}}(GL(n; K))$ , we have  $\dim J_{\text{Unip}(n, K)}(V) \leq n!$ , and if it is nonzero, then  $\pi$  is a subrepresentation of some  $\chi_1 \times \dots \times \chi_n$ .

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*Proof:* If  $J_{\text{Unip}(n, K)}(V) \neq 0$ , then  $\pi$  is isomorphic to a subrepresentation of some  $\chi_1 \times \dots \times \chi_n$  by the adjunction (24.9.3.4). Then by exactness of Jacquet functor (24.9.3.1),  $J(V)$  is a subspace of  $J(\chi_1 \times \dots \times \chi_n)$ , which has dimension  $n!$  (24.11.5.13).  $\square$

**Prop. (24.11.5.15) [Jacquet Functor of  $\sigma(\chi_1, \chi_2)$ ].** Suppose  $\chi_1, \chi_2$  are quasi-characters of  $K^\times$  that  $\chi_1\chi_2^{-1} = | - |^{-1}$ , then  $\mathcal{B}(\chi_1, \chi_2)$  is reducible, then the Jacquet modules of  $\pi(\chi_1, \chi_2)$  and  $\sigma(\chi_1, \chi_2)$  are 1-dimensional, and the characters of  $T(F)$  they afford are  $\delta^{1/2}\chi$  and  $\delta^{1/2}\chi'$ .  $\square$

*Proof:* Clearly the Jacquet module of a representation  $\mathbb{1}(\chi)$  is of dimension 1, so by the exactness of Jacquet functor (24.9.3.1) and (24.11.5.13), the Jacquet functor for  $\sigma(\chi_1, \chi_2)$  is also of dimension 1.

For the determination of the Jacquet module, notice for any representation  $(\pi, V)$  of  $GL(2, K)$ ,

$$\mathrm{Hom}_{T(F)}(J(V), \delta^{1/2}\mu) \cong \mathrm{Hom}_{B(K)}(V, \delta^{1/2}\mu) \cong \mathrm{Hom}_{GL(2, K)}(V, \mathcal{B}(\mu_1, \mu_2)).$$

So for  $\pi = \pi(\chi_1, \chi_2)$ ,  $\mathrm{Hom}_{T(F)}(J(V), \delta^{1/2}\mu) = \mathbb{C}$  for  $\mu = \chi$ , so  $J(\pi(\chi_1, \chi_2))$  is  $\delta^{1/2}\chi$ . For  $\pi = \sigma(\chi_1, \chi_2)$ , then  $\mathrm{Hom}_{T(F)}(J(V), \delta^{1/2}\mu) = \mathbb{C}$  for  $\mu = \chi'$  by (24.11.5.7), so  $J(\sigma(\chi_1, \chi_2))$  is  $\delta^{1/2}\chi'$ .  $\square$

### Kirillov Modules of Principal Series

**Prop. (24.11.5.16) [Kirillov Model of Principal Series].** Let  $\pi(\chi_1, \chi_2)$  be an irreducible principal series,

- If  $\chi_1 \neq \chi_2$ , then the Kirillov model of  $\pi(\chi_1, \chi_2)$  consists of smooth functions  $\varphi$  on  $K^\times$  that is compactly supported on  $F$  and  $\varphi(t)$  is a linear combination of the function  $|t|^{1/2}\chi_1(t)$  and  $|t|^{1/2}\chi_2(t)$  when  $|t|$  is small.
- If  $\chi_1 = \chi_2$ , then the Kirillov model of  $\pi(\chi_1, \chi_1)$  consists of smooth functions  $\varphi$  on  $K^\times$  that is compactly supported on  $F$  and  $\varphi(t)$  is a linear combination of the function  $|t|^{1/2}\chi_1(t)$  and  $v(t)|t|^{1/2}\chi_1(t)$  when  $|t|$  is small.

$\square$

*Proof:* First assume that  $\chi_1 = \chi_2$ , then by (24.11.5.13), there are two functions  $\varphi_1, \varphi_2$  that satisfies

$$\pi(t)\bar{\varphi}_1 = (\delta^{1/2}\chi)(t)\bar{\varphi}_1, \quad \pi(t)\bar{\varphi}_2 = (\delta^{1/2}\chi)(t)\bar{\varphi}_2 + v(t_1/t_2)(\delta^{1/2}\chi)(t)\bar{\varphi}_1.$$

Then by (24.11.3.14),  $\varphi_1(u) = C|u|^{1/2}\chi_1(u)$  for  $|u|$  small, and because  $\bar{\varphi}_1 \neq 0 \in J(V)$ ,  $C \neq 1$  by (24.11.3.13), and we can assume that  $C = 1$ . Then for any  $t_0 \in \mathfrak{p} \setminus \mathfrak{p}^2$ ,

$$\pi\left(\begin{bmatrix} t_0 & \\ & 1 \end{bmatrix}\right)\varphi_2 - |t_0|^{1/2}\chi_1(t_0)[\varphi_1 + \varphi_2] \in V_N.$$

Thus by (24.11.3.13), there exists some constant  $\varepsilon(t_0)$  that

$$\varphi_2(tu) = |t|^{1/2}\chi_1(t)\varphi_2(u) + |tu|^{1/2}\chi_1(tu).$$

is true for  $t = t_0$  and  $|u| \leq \varepsilon(t_0)$ . Notice both sides are locally constant on  $t$ , so this is true for  $t$  near  $t_0$ . Because  $\mathfrak{p} \setminus \mathfrak{p}^2$  is compact, there is a  $\varepsilon$  that this is true for any  $t \in \mathfrak{p} \setminus \mathfrak{p}^2$  and  $|u| \leq \varepsilon$ .

Now any element  $t \in \mathfrak{p}$  can be factored into product of elements in  $\mathfrak{p} \setminus \mathfrak{p}^2$ , so by induction

$$\varphi_2(tu) = |t|^{1/2}\chi_1(t)\varphi_2(u) + v(t)|tu|^{1/2}\chi_1(tu).$$

for any  $t \in \mathfrak{p}$  and  $|u| \leq \varepsilon$ . So the theorem is true for  $\varphi_1$  and  $\varphi_2$ , thus true for any other function because they differs from a linear combination of these two by an element in  $V_N$ , which vanishes for  $|t|$  small, by (24.11.3.13).

Case1 is similar and easier.  $\square$



**Prop. (24.11.5.17) [Kirillov Model of Steinberg Representations].** Let  $\sigma(\chi_1, \chi_2)$  be a Steinberg representation of  $\mathrm{GL}(2; K)$ , where  $\chi_1\chi_2^{-1} = |\cdot|^{-1}$ , then the Kirillov model of  $\sigma(\chi_1, \chi_2)$  consists of smooth functions  $\varphi$  on  $K^\times$  that is compactly supported on  $F$  and  $\varphi(t)$  is a constant multiple of  $|t|^{1/2}\chi_2(t)$  when  $|t|$  is small.  $\square$

*Proof:* The proof is the same as that of (24.11.5.16), with (24.11.5.15) used instead of (24.11.5.13).  $\square$

## 6 Unramified Representations

**Def. (24.11.6.1) [Normalized Spherical Vector].** Let  $\chi_1, \dots, \chi_n$  be nonramified quasi-characters of  $K^\times$ , then  $\mathcal{B}(\chi_1, \dots, \chi_n)$  contains a  $K$ -fixed vector  $\varphi_K$  that is defined to be

$$\varphi_K(bk) = (\delta^{1/2}\chi)(b), b \in B(K), k \in \mathcal{K}$$

Notice this is well-defined, because for  $u \in B(K) \cap \mathcal{K}$ ,  $(\delta^{1/2}\chi)(u) = 1$ .  $\varphi_K$  is  $\mathcal{K}$ -spherical, and we refer to it the **normalized spherical vector** in  $V$ . Such a  $\mathcal{B}(\chi_1, \dots, \chi_n)$  is called a **unramified principal series** if it is irreducible.

Notice when  $n = 2$  and  $\chi_2 = \chi_1 \otimes \mathrm{val}$ , this spherical vector just spans the 1-dimensional invariant subspace in (24.11.4.14).  $\square$

**Cor. (24.11.6.2).** When  $n = 2$ , under the Satake isomorphism (29.3.2.9), the Hecke operators  $T(\mathfrak{p}) = \begin{bmatrix} \varpi & \\ & 1 \end{bmatrix}$  and  $R_p = \varpi I$  are mapped to

$$q^{1/2}(t_1 + t_2), \quad t_1 t_2,$$

resp.  $\square$

*Proof:* For any unramified spherical representation with Satake parameters  $\alpha_1, \alpha_2$ , using the representative of  $\mathcal{K} \begin{bmatrix} \varpi & \\ & 1 \end{bmatrix} \mathcal{K}$  over  $\mathcal{K}$  as in (26.3.2.8),

$$\begin{aligned} (T(\mathfrak{p})\varphi_K)(1) &= \int_{\mathcal{K}} \begin{bmatrix} \varpi & \\ & 1 \end{bmatrix}_{\mathcal{K}} \varphi_K(g) dg = \sum_{\gamma \in K} \begin{bmatrix} \varpi & \\ & 1 \end{bmatrix}_{K/K} \varphi_K(\gamma) \\ &= (\Delta^{1/2}\chi) \left( \begin{bmatrix} 1 & \\ & \varpi \end{bmatrix} \right) + q(\Delta^{1/2}\chi) \left( \begin{bmatrix} \varpi & \\ & 1 \end{bmatrix} \right) = q^{1/2}(\alpha_1 + \alpha_2) \end{aligned}$$

and similarly for  $R(\mathfrak{p})$ .  $\square$

**Cor. (24.11.6.3) [Unramified Representation of  $\mathrm{GL}(n; K)$ , Zelevinsky].** Every irreducible admissible unramified representation (24.9.6.3)  $(\pi, V) \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$  is of the form  $Q(\chi_1, \chi_2, \dots, \chi_n)$  as in Bernstein-Zelevinsky Classification, where  $\chi_i$  are unramified quasi-characters, representing Zelevinsky segments of length 1.  $\square$

*Proof:* Use the Satake isomorphism to find Satake parameters of unramified quasi-characters  $\chi_i$  that makes the representation of  $\mathcal{H}_K \cong \mathbb{C}[t_1, t_1^{-1}, \dots, t_n, t_n^{-1}]^W$  isomorphic to that of  $\pi$ , then by (3.6.5.3),  $\chi_1 \times \chi_2 \times \dots \times \chi_n$  has a unique irreducible subquotient that has the same spherical character as  $\pi$ , then by (3.6.5.12), this subquotient is isomorphic to  $\pi$ . Now use the fact we can change the order of  $\chi_i$  s.t. it is just  $Q(\Delta_1, \Delta_2, \dots, \Delta_n)$ .  $\square$

**Prop. (24.11.6.4)[Intertwining Operator on Spherical Vector].** For  $\chi_1, \chi_2$  unramified with Satake parameter  $\alpha_i$ , consider  $M : \mathcal{B}(\chi_1, \chi_2) \rightarrow \mathcal{B}(\chi_2, \chi_1)$ , then

$$M\varphi_{K,\chi} = \frac{1 - q^{-1}\alpha_1\alpha_2^{-1}}{1 - \alpha_1\alpha_2^{-1}}\varphi_{K,\chi'}.$$

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*Proof:* Clearly this equation is true up to scalar because  $M$  is intertwining, so it suffices to calculate  $(M\varphi_K)(1)$ . For this, we assume  $\chi$  is dominant (24.3.2.10) because we can use analytic continuation. Then

$$(M\varphi_K)(1) = \int_F \varphi_K(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) dx.$$

The integral is 1 on  $\mathcal{O}$ , and for  $m > 0$ , on  $\mathfrak{p}^{-m} \setminus \mathfrak{p}^{-m+1}$ , by (24.11.5.5), it equals  $q^{-m}\alpha_1^m\alpha_2^{-m}q^m(1-q^{-1})$ , so the total sum is  $1 + (1 - q^{-1})\alpha_1\alpha_2^{-1}/(1 - \alpha_1\alpha_2^{-1})$ .  $\square$

**Prop. (24.11.6.5).** If  $\chi_1, \chi_2$  are nonramified,  $V = \mathcal{B}(\chi_1, \chi_2)$ ,  $K_0(\mathfrak{p})$  is the Iwahori subgroup (24.11.1.4), then the composition

$$V^{K_0(\mathfrak{p})} \hookrightarrow V \rightarrow J(V)$$

is an isomorphism. In particular, it is of dimension 2.  $\square$

*Proof:* Firstly notice  $V^{K_0(\mathfrak{p})}$  has dimension  $\leq 2$ , because of the decomposition (24.11.1.8) and definition, and  $J(V)$  has dimension 2 by (24.11.5.13), so it suffices to show the map is surjective. The image is just  $J(V)^{T(\mathcal{O})}$  by (24.11.2.4). But by (24.11.5.13) and the fact  $\chi_i$  are nonramified, all of  $J(V)$  are  $T(\mathcal{O})$ -fixed, thus  $J(V)^{T(\mathcal{O})} = J(V)$ .  $\square$

**Cor. (24.11.6.6)[Casselman Basis].** When  $\chi_1 \neq \chi_2$ , we can easily find a basis of  $V^{K_0(\mathfrak{p})}$ , that is

$$L_1(\varphi) = \varphi(1), \quad L_0(\varphi) = (M\varphi)(1)$$

where  $M$  is intertwining integral defined in (24.11.5.6). These are  $N(K)$ -invariant, thus define functionals on  $J(V) \cong V^{K_0(\mathfrak{p})}$ . They are linearly independent checked on  $T(F)$ .

Then the dual basis  $\varphi_0, \varphi_1 \in V^{K_0(\mathfrak{p})}$  are called the **Casselman basis**.

$\varphi_0$  has a simple form: using Iwahori-Bruhat decomposition (24.11.1.8),

$$\varphi_0(g) = \begin{cases} (\Delta^{1/2}\chi)(b) & g = bw_0k \in B(K)wK_0(\mathfrak{p}) \\ 0 & \text{otherwise} \end{cases}.$$

┘

*Proof:* It is easily verified that this formula is well-defined and defines a Iwahori-fixed vector, so it suffices to evaluate it by  $L_i$ . Clearly  $L_1(\varphi_0) = 0$ , and for  $L_0(\varphi_0)$ , notice that

$$w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \in B(K)w_0K_0(\mathfrak{p})$$

iff  $x \in \mathcal{O}$ , so  $L_0(\varphi_0) = 1$ .  $\square$

**Lemma (24.11.6.7).** When  $\chi_1 \neq \chi_2$  nonramified and  $m \geq 0$ , the function

$$F_m(g) = \int_{\mathcal{O}} \varphi_K(g \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} a_m) dx, \quad a_m = \begin{bmatrix} \varpi^m & \\ & 1 \end{bmatrix}$$

is an Iwahori-fixed vector, and

$$F_m = q^{-m/2} \alpha_2^m M(\varphi_K)(1) \varphi_0 + q^{-m/2} \alpha_1^m \varphi_1$$

┘

*Proof:*

$$\int_{K_0(\mathfrak{p})} \pi(ka_m) \varphi_K dk = \int_{N_-(\mathfrak{p})} \int_{T(\mathcal{O})} \int_{N(\mathcal{O})} \varphi_K(gn_- t_0 n a_m) dn dt_0 dn_- \quad (24.11.1.6)$$

noticing that  $a_m^{-1} t_0 n_- a_m \in K$  as  $m \geq 0$ , so the integrand is independent of  $t_0, n_-$  and equals  $F_m$ , and it is clearly Iwahori-fixed. And

$$c_1 = L_1(F_m) = F_m(1) = \varphi_K(a_m) = (\Delta^{1/2} \chi)(a_m) = q^{-m/2} \alpha_1^m.$$

Similarly, for  $\chi$  dominant,

$$c_0 = L_0(F_m) = \int_{\text{Unip}(K)} F_m(w_0 n) dn = \int_{\text{Unip}(K)} \int_{\text{Unip}(\mathcal{O})} \varphi_K(w_0 n n' a_m) dn' dn.$$

By Fubini, the  $n'$  can be omitted, thus it equals  $(M\varphi_K)(a_m) = (\Delta^{1/2} \chi')(a_m)$ . □

### Spherical Whittaker Function and Spherical Functions

See [Casselman, W., The unramified principal series of  $p$ -adic groups I: the spherical function, Compositio Math. 40 (1980), 387-406.] and [Casselman, W. and J. Shalika, The unramified principal series of  $p$ -adic groups II: the Whittaker function, Compositio Math. 40 (1980), 207-231.] for more general calculations.

**Def. (24.11.6.8) [Spherical Whittaker Function].** The **spherical Whittaker function** of a generic unramified representation is the spherical vector  $W^0$  in the Whittaker model normalized that  $W_0(\mathbf{1}) = 1$ .

For  $n = 2$ , we can define another normalization  $W_0(g) = \Lambda(\pi(g)\varphi_K)$ , where  $\Lambda$  is the Whittaker functional defined in (24.11.5.8) and  $\varphi_K$  is the normalized spherical vector defined in (24.11.6.1). ┘

*Proof:* To prove that  $W^0(\mathbf{1}) \neq 0$ , Cf. [C-S80]. ? □

**Prop. (24.11.6.9).** We may assume that the conductor of  $\psi$  is  $\mathcal{O}$ , because any other character is of the form  $x \mapsto \psi(ax)$ , thus

$$W_0(g, \psi_a) = \int_F \varphi_K(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g) \psi(-ax) dx = |a|^{-1/2} \chi_2(a) W_0\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix} g, \psi\right),$$

(and also by analytic continuity). In particular,  $\mu(\mathcal{O}) = 1$  (19.4.2.6).

Notice that

$$W_0\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \begin{bmatrix} z & \\ & z \end{bmatrix} gk\right) = \psi(x) \omega(z) W_0(g), \quad z \in K^\times, k \in K,$$

where  $\omega = \chi_1 \chi_2$ , so to calculate  $W_0$ , it suffices to compute  $W_0\left(\begin{bmatrix} \varpi^m & \\ & 1 \end{bmatrix} = a_m\right)$ . ┘

**Remark (24.11.6.10).** We want to calculate the spherical Whittaker function  $W_0$  explicitly because it is used to evaluate the local part of the global L-function (29.5.5.5).  $\lrcorner$

**Lemma (24.11.6.11) [Calculating  $W_0$ ].** For  $\mathcal{B}(\chi_1, \chi_2)$  with Satake parameters  $\alpha_1, \alpha_2$ , the spherical Whittaker function satisfies

$$W_0(1) = 1 - q^{-1}\alpha_1\alpha_2^{-1}, \quad W_0(a_m) = 0$$

for  $m < 0$ .  $\lrcorner$

*Proof:* As in the proof of (24.11.5.5), because  $\varphi_{\mathcal{K}}$  is  $\mathcal{K}$ -invariant, we have

$$\begin{aligned} \int_{\mathcal{O}} \varphi_{\mathcal{K}}(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx &= 1, \\ \int_{\mathfrak{p}^{-1} \setminus \mathcal{O}} \varphi_{\mathcal{K}}(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx &= q^{-1}\alpha_1\alpha_2^{-1} \int_{\mathfrak{p}^{-1} \setminus \mathcal{O}} \psi(-x) dx = -q^{-1}\alpha_1\alpha_2^{-1}, \\ \int_{\mathfrak{p}^{-k-1} \setminus \mathfrak{p}^{-k}} \varphi_{\mathcal{K}}(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx &= q^{-k-1}\alpha_1^{k+1}\alpha_2^{-k-1} \int_{\mathfrak{p}^{-k-1} \setminus \mathfrak{p}^{-k}} \psi(-x) dx = 0, \quad k \geq 1. \end{aligned}$$

For  $W_0(a_m)$ , choose  $x \in \mathcal{O}$  that  $\psi(\varpi^m x) \neq 1$ , then

$$W_0(a_m) = W_0(a_m \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) = W_0(\begin{bmatrix} 1 & \varpi^m \\ & 1 \end{bmatrix} a_m) = \psi(\varpi^m x) W_0(a_m)$$

so  $W_0(a_m) = 0$ .  $\square$

**Lemma (24.11.6.12) [Functional Equation].** For fixed  $g$ ,  $W_0(g)$  is an analytic function of  $\alpha_1, \alpha_2$ , and  $(1 - q^{-1}\alpha_1\alpha_2^{-1})^{-1}W_0(g)$  is symmetric in  $\alpha_1, \alpha_2$ .  $\lrcorner$

*Proof:* The Whittaker functional is defined by calculated by continuation (24.11.5.8), and for  $\chi$  dominant,  $W_0$  is analytic (24.11.5.8), thus by calculation in (24.11.6.11), its value at 1 is identically 1, thus so does its analytic continuation. Then to check after switching  $\alpha_1, \alpha_2$  it is the same meromorphic function, it suffices to show for all irreducible principal series  $\mathcal{B}(\chi_1, \chi_2)$ . But in this case,  $\mathcal{B}(\chi_1, \chi_2) \cong \mathcal{B}(\chi_2, \chi_1)$ , so their Whittaker model thus spherical Whittaker function differ only by a scalar, thus the same.  $\square$

**Prop. (24.11.6.13) [Calculating  $W_0$ ].**

$$(1 - q^{-1}\alpha_1\alpha_2^{-1})^{-1}W_0(a_m) = \begin{cases} q^{-m/2} \frac{\alpha_1^{m+1} - \alpha_2^{m+1}}{\alpha_1 - \alpha_2} & m \geq 0 \\ 0 & m < 0 \end{cases}$$

$\lrcorner$

*Proof:* By?? we can assume  $\chi$  is dominant and use analytic continuation.

$$W_0(a_m) = \int_F F_m(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx,$$

by (24.11.5.8) (24.11.6.7) and a change of variable. Then use (24.11.6.7), we see that

$$W_0(a_m) = C_1 q^{-m/2} \alpha_1^m + C_0 q^{-m/2} \alpha_2^m$$

where

$$C_0 = (M\varphi_K)(1) \int_F \varphi_0(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx$$

And  $\int_F \varphi_0(w_0 \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}) \psi(-x) dx = 1$ , by the same consideration as in the proof of (24.11.6.6).

Finally,  $(M\varphi_K)(1)$  can be calculated by (24.11.6.4), and the requirement of (24.11.6.12) will determine  $C_1$ .  $\square$

**Prop. (24.11.6.14) [Spherical Function].** For a spherical irreducible admissible representation  $(\pi, V)$  of  $GL(2; K)$ , its contragradient  $\widehat{V}$  is also spherical (3.6.5.11). Then we define the **spherical function**

$$\sigma(g) = (\pi(g)v, \widehat{v})$$

where  $v, \widehat{v}$  are spherical and normalized that  $\langle v, \widehat{v} \rangle = 1$ . This is bi-invariant under  $K$ -action By (24.11.6.3).

By (24.11.6.3),  $(\pi, V)$  is of the form  $\chi(\det(g))$  or  $\pi(\chi_1, \chi_2)$  for  $\pi_i$  nonramified. In the former case the spherical function is just  $\chi \circ \det$ . Now we only consider the latter interesting case, for which there is a spherical functional  $\widehat{v} : \varphi \mapsto \int_K \varphi(k) dk$ , and it is 1 on the normalized spherical vector  $v$  (24.11.6.1), and

$$\sigma(g) = \int_K (\pi(g)\varphi_K)(k) dk = \int_K \varphi_K(kg) dk$$

┘

**Prop. (24.11.6.15) [Macdonald Formula].** The spherical function for unramified principal series behave well under  $Z(K)$ -action and is  $K$ -binvariant, so in order to compute it, it suffices to compute its value on  $a_m$ . We have:

$$\sigma(a_m) = \frac{q^{-m/2}}{1+q^{-1}} \left[ \alpha_1^m \frac{1-q^{-1}\alpha_2\alpha_1^{-1}}{1-\alpha_2\alpha_1^{-1}} + \alpha_2^m \frac{1-q^{-1}\alpha_1\alpha_2^{-1}}{1-\alpha_1\alpha_2^{-1}} \right].$$

┘

*Proof:* First notice that

$$\int_K F_m(k) dk = \int_K \int_{\mathcal{O}} \varphi_K(k \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} a_m) dx dk = \int_K \varphi_K(ka_m) dk$$

by a change of variable. Then by (24.11.6.7), this equals

$$\left[ \int_K \varphi_0(k) dk \right] q^{-m/2} \alpha_2^m \frac{1-q^{-1}\alpha_1\alpha_2^{-1}}{1-\alpha_1\alpha_2^{-1}} + \left[ \int_K \varphi_1(k) dk \right] q^{-m/2} \alpha_1^m.$$

Next we calculate  $\int_K \varphi_0(k) dk$  directly: By (24.11.6.6), this equals the volume of  $K \cap (K_0(\mathfrak{p})w_0K_0(\mathfrak{p})) = K_0(\mathfrak{p})w_0K_0(\mathfrak{p})$ .  $K/K_0(\mathfrak{p}) \cong GL(2; \mathbb{F}_q)/B(\mathbb{F}_q)$  has cardinality  $q+1$ , and by pulling back the Bruhat decomposition of  $GL(2, \mathbb{F}_q)$ ,  $K_0(\mathfrak{p})w_0K_0(\mathfrak{p})$  consists of  $q$  left cosets of  $K_0(\mathfrak{p})$ . So

$$\int_K \varphi_0(k) dk = \frac{q}{1+q}.$$

Finally the expression is symmetric in  $\alpha_1, \alpha_2$ , because the spherical vectors in  $V, \widehat{V}$  are unique (3.6.5.11), and  $\widehat{\mathcal{B}(\chi_1, \chi_2)} = \mathcal{B}(\chi_2, \chi_1)$ . Also, the expression is a combination of  $\alpha_i^m$  and the coefficient is independent of  $m$ , so it can be determined as above.  $\square$

## 7 Iwahori-Spherical Representations

References are [P-R08, Bor76].

**Thm. (24.11.7.1) [Iwahori-Spherical Representations, Casselman].**  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  is Iwahori-spherical iff the supercuspidal support of  $\pi$  consists of unramified quasi-characters of  $K^\times$ .

This can be generalized to other groups. ?

*Proof:*

□

## 8 Unitarizable, Tempered & $L^2$ -Representations

References are [Tad86] and [Vogan, The unitary dual of  $\text{GL}(n; \mathbb{R})$  over an Archimedean field].

**Remark (24.11.8.1) [Casselman].** For any  $\pi \in \text{Rep}^{\text{alg}}(\text{GL}(n; K))$ , there exists a unique real-valued quasi-character  $\chi$  of  $K$  that the central character of  $\pi \otimes (\chi \circ \det)$  is unitary.

*Proof:*

□

**Thm. (24.11.8.2) [Tempered Representations of  $\text{GL}(n; K)$ , Jacquet/Zelevinsky [Zel80]].** Let  $n \in \mathbb{Z}_+$ , then  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  is

- essentially square-integrable iff it is of the form  $Q(\Delta)$  for a single Zelevinsky segment  $\Delta$ .
- discrete series iff it is of the form  $Q(\Delta)$  for a single Zelevinsky segment, and the central character of  $\sigma(\frac{m-1}{2})$  is unitary, where  $\Delta = \Delta(\sigma, m)$ .
- tempered iff  $\pi = Q(\Delta_1, \dots, \Delta_r) = Q(\Delta_1) \times \dots \times Q(\Delta_r)$ , where each  $Q(\Delta_i)$  is a discrete series. In particular, tempered representations are generic by (24.11.3.3).

□

*Proof:* Cf. [Generic Representations, Jacquet(1977)] and [B-Z77].

1: [Zel80]9.3.

2: This follows from item1 and (24.9.8.8)(24.11.1.19).

□

**Cor. (24.11.8.3) [Classification of Irreducible tempered representations of  $\text{GL}(2; K)$ ].** Irreducible tempered representations of  $\text{GL}(2; K)$  are one of the following form:

- Cuspidal representation with unitary central characters.
- Steinberg representations  $\text{St}_2(\chi)$  with unitary central characters  $\chi$ .
- Irreducible principal representations  $\chi_1 \times \chi_2$  with  $\chi_1, \chi_2$  unitary.

□

*Proof:* This follows from (24.11.8.2) and (24.11.4.15).

□

## Unitarizable Representations

References are [Ber84a, Tad86].

**Prop. (24.11.8.4) [Matrix Coefficients of Unitarizable Representations are Bounded].** Every matrix coefficient function of a unitarizable representation on  $\text{GL}(n; K)$  is a bounded function on  $\text{GL}(n; K)$ .

□

*Proof:*  $|(\pi(g)u, v)| \leq \|u\| \cdot \|v\|$ .  $\square$

**Prop. (24.11.8.5) [Tempered Representations are Unitarizable].** Any irreducible tempered representation of  $GL(n; K)$  is unitarizable.  $\lrcorner$

*Proof:* Firstly, any discrete series are unitarizable: Cf. [G-H11]1, P364.  $\color{red}?$

And the parabolic induction of a unitarizable representation is also unitarizable by (24.11.4.2). Thus the assertion follows from (24.11.8.2).  $\square$

**Thm. (24.11.8.6) [Bernstein [Ber84a]].** Let  $(\pi, V) \in \text{Irr}^{\text{adm}}(GL(n; K))$ , then each  $M_n$ -invariant pairing  $B : V \times V^\vee \rightarrow \mathbb{C}$  is proportional to the standard pairing.  $\lrcorner$

*Proof:* Cf. [Ber84a]Thm5.1  $\color{red}?$   $\square$

**Cor. (24.11.8.7) [Bernstein [Ber84a]].** Any irreducible unitarizable representation  $\pi$  of  $GL(n; K)$  is irreducible when restricted to  $M_n$ .  $\lrcorner$

*Proof:* Cf. [Ber84a]Thm5.4  $\color{red}?$   $\square$

**Cor. (24.11.8.8).** If  $P \leq G$  is a levi subgroup and  $(\pi, V) \in \text{Irr}^{\text{uni}}(P(K))$ , then  $I_{P(K)}^{G(K)}(\pi)$  is irreducible.  $\lrcorner$

*Proof:* Cf. [Ber84a]Section8.2.  $\square$

**Prop. (24.11.8.9) [Bounds on Parameters of Principal Series of  $GL(n; K)$ ].** Let  $\chi_1 \times \dots \times \chi_n$  be a unitarizable principle series of  $GL(n; K)$  and  $|\chi_i| = \|\sigma_i$  for  $\sigma_i \in \mathbb{R}$ . Denote  $\rho_i = (n - 2i + 1)/2$ , then for any  $k \in [n]_+$ ,

$$|\sigma_1 + \dots + \sigma_k| \leq \rho_1 + \dots + \rho_k.$$

Moreover, if  $\pi$  is generic, unramified and unitarizable, then  $|\sigma_i| < 1/2$  for any  $i$ .  $\lrcorner$

*Proof:* Cf. [G-H11]2, P79. and [J-S81]P515.  $\color{red}?$   $\square$

**Lemma (24.11.8.10) [Possibility of Unitarizable principal Series for  $GL(2)$ ].** If  $\mathcal{B}(\chi_1, \chi_2)$  is unitarizable, then either  $\chi_1, \chi_2$  are all characters or  $\chi_1 = \overline{\chi_2}^{-1}$ .  $\lrcorner$

*Proof:* There is an anti-linear  $GL(2; K)$ -map  $\mathcal{B}(\chi_1, \chi_2) \rightarrow \mathcal{B}(\overline{\chi_1}, \overline{\chi_2})$  which is conjugation, so if there is a  $GL(2; K)$ -invariant Hermitian pairing  $(-, -)$  on  $\chi_1 \times \chi_2$ , then  $(f_1, f_2) \mapsto (f_1, \overline{f_2})$  will be a non-degenerate  $GL(2; K)$ -invariant bilinear pairing

$$(\chi_1 \times \chi_2) \times (\overline{\chi_1} \times \overline{\chi_2}) \rightarrow \mathbb{C}.$$

So  $\mathcal{B}(\chi_1, \chi_2) \cong \mathcal{B}(\overline{\chi_1}^{-1}, \overline{\chi_2}^{-1})$ , so  $\chi_i$  are characters or  $\chi_1 = \overline{\chi_2}^{-1}$  by (24.11.4.2).  $\square$

**Lemma (24.11.8.11).** Let  $\chi \in K^\vee$  and  $s \in \mathbb{R} \setminus \{0, 1/2\}$  (so that  $\chi(s) \times \chi(-s)$  is irreducible), then  $\chi(s) \times \chi(-s)$  is unitarizable iff  $0 < |s| < 1/2$ .  $\lrcorner$

*Proof:* For  $|s| < 1/2$ , we show  $\mathcal{B}(\chi_s, \chi_{-s})$  is unitary: Modify the intertwining operator  $M_s^* = (1 - q^{-2s})M_s$  (24.11.5.6), then it is definable for  $s = 0$ , by calculation in (24.11.5.6), so the modified Hermitian product

$$(-, -)^* = (1 - q^{-2s})(-, -)$$

is defined at  $s = 0$ , and is positive/negative definite, because in this case it is irreducible and unitarizable (24.11.8.5). The eigenvalue of this Hermitian form deforms continuously, and it is never zero because it is non-degenerate as said above (only when  $s \neq 1/2$  because we need  $M_s$  to be isomorphism), so it is always definite and  $\mathcal{B}(\chi_s, \chi_{-s})$  is unitarizable.  $\square$

**Prop. (24.11.8.12) [Complementary Series for  $\mathrm{GL}(2)$ ].** An irreducible principal series  $\mathcal{B}(\chi_1, \chi_2)$  is unitary iff either  $\chi_1, \chi_2$  are all characters, or there is a unitary character  $\chi_0$  and  $s \in (0, 1/2)$  that  $\chi_1 = \chi_0(s), \chi_2 = \chi_0(-s)$ , which are exactly the complementary series (24.9.8.9).  $\lrcorner$

*Proof:* This follows immediately from the lemmas (24.11.8.5)(24.11.8.10)(24.11.8.11) above.  $\square$

**Lemma (24.11.8.13) [Basic Type Unitarizable Representations of  $\mathrm{GL}(n; K)$ ].**

- Let  $m, d \in \mathbb{Z}_+$ , and  $\pi$  a discrete series of  $\mathrm{GL}(m; K)$ . Then

$$u(\pi; d) = Q(\pi(\frac{d-1}{2}), \dots, \pi(-\frac{d-1}{2}))$$

is an irreducible unitarizable representation of  $\mathrm{GL}(md; K)$ .

- $I_{P_{md, md}(K)}^{\mathrm{GL}(2md; K)}(u(\pi; d)(s), u(\pi(s); d)(-s)) = u(\pi(s); d) \times u(\pi(-s); d)$ , where  $\pi$  is a discrete series of  $\mathrm{GL}(n; K)$  and  $s \in (0, 1/2)$ .  $\lrcorner$

*Proof:* Why are they unitarizable? Cf. [Tad86] ThmA.8 and Prop4.2. ?  $\square$

**Thm. (24.11.8.14) [Classification of Unitarizable Representations of  $\mathrm{GL}(n; K)$ ].** Let  $n \in \mathbb{Z}_+$  and  $\kappa = (\kappa_1, \dots, \kappa_r)$  be an ordered partition of  $n$ . If  $\pi_i$  are irreducible unitarizable representations of  $\mathrm{GL}(\kappa_i, K)$  as described in (24.11.8.13), then  $I_{P_\kappa(K)}^{\mathrm{GL}(n; K)}(\pi_1, \dots, \pi_r) = \pi_1 \times \dots \times \pi_r$  is an irreducible unitarizable representation of  $\mathrm{GL}(n; K)$ .

And conversely, any irreducible unitarizable representation of  $\mathrm{GL}(n; K)$  is one of this form, up to permutation of  $\pi_i$ .  $\lrcorner$

*Proof:* Cf. [Tad86] Thm. D. ?  $\square$

**Lemma (24.11.8.15) [Basic Type Generic Unitarizable Representations of  $\mathrm{GL}(n; K)$ ].** The following representations of  $\mathrm{GL}(n; K)$  are irreducible, generic and unitarizable:

- Discrete series, by (24.11.8.2) and (24.11.8.5).
- $I_{P_{n, n}(K)}^{\mathrm{GL}(2n; K)}(\pi(s), \pi(-s))$ , where  $\pi$  is a discrete series of  $\mathrm{GL}(n; K)$  and  $s \in (0, 1/2)$ .  $\lrcorner$

*Proof:* 2: Notice if  $\pi = Q((\pi_1, m))$ , then this is just  $Q((\pi_1 \otimes |\det|^s, m)) \times Q((\pi_1 \otimes |\det|^{-s}, m))$ , thus generic by (24.11.3.3). It is unitarizable by (24.11.8.13).  $\square$

**Thm. (24.11.8.16) [Classification of Generic Unitarizable Representations of  $\mathrm{GL}(n; K)$ ].** Let  $n \in \mathbb{Z}_+$  and  $\kappa = (\kappa_1, \dots, \kappa_r)$  be an ordered partition of  $n$ . If  $\pi_i$  are irreducible generic unitarizable representations of  $\mathrm{GL}(\kappa_i, K)$  as described in (24.11.8.15), then  $I_{P_\kappa(K)}^{\mathrm{GL}(n; K)}(\pi_1, \dots, \pi_r) = \pi_1 \times \dots \times \pi_r$  is an irreducible generic unitarizable representation of  $\mathrm{GL}(n; K)$ .

And conversely, any irreducible generic unitarizable representation of  $\mathrm{GL}(n; K)$  is one of this form, up to permutation of  $\pi_i$ .  $\lrcorner$

*Proof:* This follows from (24.11.8.14) and (24.11.3.3).  $\square$

### Irreducible Generic Unramified Unitarizable Representations



## 24.12 Local Langlands for $\mathrm{GL}(n)$

References are [B-K93, Hen93, Kud94, Rog00, H-T01, Wed08, Sch13, Bus19].

**Notation (24.12.0.1).**

- Let  $p \in \text{Prime}$ .
- Fix  $\ell \in \text{Prime} \setminus \{p\}$  and an isomorphism  $\iota_\ell : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-NField}$ .
- Fix a square root  $|\cdot|_K^{1/2} : K^\times \rightarrow \overline{\mathbb{Q}_\ell}^\times$  s.t.  $\iota \circ |\cdot|_K^{1/2}$  is valued in  $\mathbb{R}_{>0}$ .
- Let  $\mathbf{1} \neq \psi \in K^\vee$ .
- Let  $(K_r, \mathcal{O}_r, \varpi_K, \kappa_r)/K$  be an unramified extension of degree  $r$ .

┘

### 1 Local Langlands for $p$ -adic $\mathrm{GL}(n)$

References are [Kud94, Yao17, Wed08].

**Prop. (24.12.1.1) [LLC for  $\mathrm{GL}(1)$ ].** Local class field theory told us that there is an isomorphism  $\mathrm{Art}_K : K^\times \cong W_K^{\mathrm{ab}} \cong \mathrm{WD}_K^{\mathrm{ab}}$ ,

Notice by Schur's lemma, any irreducible smooth representation of  $K^\times$  is 1-dimensional and factors through some  $U_K$ . And a Weil-Deligne representation is now a continuous map  $W_K^{\mathrm{ab}} \rightarrow \mathbb{C}^\times$ . but it must factor through some  $U_K$ , so these two sets are bijective by  $\mathrm{Art}_K$ . ┘

**Thm. (24.12.1.2) [LLC for  $\mathrm{GL}(n)$ , Hasse(30)/Tunnell(78)/Kutzko(80)/Harris-Taylor[H-T01]/Henniart(84, 86, 88, 93, 00)[Hen00].** There exists a unique collection of bijections  $\{\mathrm{rec}_n\}_{n \in \mathbb{Z}_+}$  between sets:

$$\mathrm{rec}_n : \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K)) \xrightarrow{\cong} \mathfrak{wd}_{\varphi\text{-s.st}}^n(W_K)$$

satisfying the following properties:

1. For a quasi-character  $\chi$  of  $K^\times$ ,  $\mathrm{rec}_1(\chi) = \chi \circ \mathrm{Art}_K^{-1}$  (19.7.2.11).
2. For a quasi-character  $\chi$  of  $K^\times$  and  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ ,

$$\mathrm{rec}_n(\pi(\chi)) = \mathrm{rec}_n(\pi) \otimes \mathrm{rec}_1(\chi).$$

3. For any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$  with central character  $\omega$ ,

$$\det(\mathrm{rec}_n(\pi)) = \mathrm{rec}_1(\omega).$$

4. For any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ ,  $\mathrm{rec}_n(\pi^\vee) = \mathrm{rec}_n(\pi)^*$ .
5. For any two  $\pi_1 \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n_1; K))$ ,  $\pi_2 \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n_2; K))$ ,

$$L(\pi_1 \times \pi_2; s) = L(\mathrm{rec}_{n_1}(\pi_1) \otimes \mathrm{rec}_{n_2}(\pi_2); s), \quad \epsilon(\pi_1 \times \pi_2, \psi; s) = \epsilon(\mathrm{rec}_{n_1}(\pi_1) \otimes \mathrm{rec}_{n_2}(\pi_2); s) \text{ (29.3.1.29) (31.1.1.3)}.$$

Moreover, these bijections also preserve two more invariants:

1. For any  $L/K \in \text{GalField}^{\mathrm{fin}}$ ,  $\{\mathrm{rec}_{L,n}\}$  and  $\{\mathrm{rec}_{K,n}\}$  preserves  $\mathrm{Gal}(L/K)$ -actions?.
2. Conductor:  $c(\pi) = c(\mathrm{rec}_n(\pi))$  (29.3.1.31) (31.1.1.3).

3. Depth:  $d(\pi) = d(\text{rec}_n(\pi))$ ? Cf.[Moy-Prasad]

Should be written more clearly like [Kaletha-Shin]Theorem1.2.4.?

┘

*Proof:* The uniqueness follows from(24.12.1.5)?.

The existence of  $\{\text{rec}_n\}$  satisfying the first 5 properties are constructed in Harris-Taylor's Proof.

The conductors are preserved because of the definition(29.3.1.31) and(31.1.1.4) and the fact  $\text{rec}_n$  preserves  $\epsilon$ -factors.

For depth:?

For Galois invariance:?

□

**Prop.(24.12.1.3) [Reduction to Supercuspidal Representations, Bernstein-Zelevinsky[B-Z77]].** By Bernstein-Zelevinsky classification4, it suffices to construct  $\text{rec}_n$  for irreducible cuspidal representations, then for any  $\pi = Q(\Delta_1, \dots, \Delta_r) \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$ , where  $\Delta_i = \Delta_i(\pi_i, m_i)$ , we can define

$$\text{rec}_n(\pi) = \bigoplus_{i=1}^r \text{rec}_{n_i}(\pi_i) \otimes \text{Sp}(m_i) \in \mathfrak{w}\mathfrak{d}_{\varphi\text{-s.st}}^n(W_K)(25.1.2.23).$$

┘

*Proof:* Check this satisfies the properties 1-5.?

□

**Cor.(24.12.1.4).** If  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  and  $\text{rec}_n(\pi) = (\rho_0, V, N)$ , then  $\rho_0$  only depends on the supercuspidal support of  $\pi$ (24.11.4.10).

┘

**Thm.(24.12.1.5)[Uniqueness of the Correspondence, Henniart[Hen93]].** For  $n \in \mathbb{Z}_{\geq 2}$ , suppose  $\pi, \pi' \in \text{Irr}^{\text{cusp}}(\text{GL}(n; K))$  s.t. for any  $m \in [n-1]_+$  and  $\tau \in \text{Irr}^{\text{cusp}}(\text{GL}(m; K))$ , we have

$$\epsilon(\pi \times \tau, \psi; s) = \epsilon(\pi' \times \tau, \psi; s)$$

then  $\pi \cong \pi'$ .

┘

*Proof:* Cf.[Hen93].

□

**Remark(24.12.1.6).** If one only requires that

$$L(\pi_1; s) = L(\text{rec}_{n_1}(\pi_1); s), \quad \epsilon(\pi_1; s) = \epsilon(\text{rec}_{n_1}(\pi_1); s),$$

then  $\sigma_n$  cannot be determined uniquely.

┘

**Prop.(24.12.1.7) [Gamma Factor Determines Representations].** For  $(\pi_1, V_1), (\pi_2, V_2) \in \text{Irr}^{\text{adm, generic}}(\text{GL}(2; K))$ , if  $\gamma(\pi_1, \xi, \psi; s) = \gamma(\pi_2, \xi, \psi; s)$  for any quasi-character  $\xi$  of  $K^\times$ , then  $\pi_1 \cong \pi_2$ .

┘

*Proof:* Identify  $V_i$  with their Kirillov models, and let  $V_0 = V_1 \cap V_2$ . Then it suffices to show that  $\pi_1(w_1), \pi_2(w_1)$  acts the same on  $V_0$ : This is because  $V_0$  contains  $C_c^\infty(K^*)$ , and  $B(K)$  action on  $V_1, V_2$  are the same by(24.11.3.8), and  $GL(2, F)$  is generated by  $B(F)$  and  $w_1$ . Then  $V_0 = V_1 = V_2$  as they are irreducible representations.

For  $\varphi \in V$ , let  $\varphi_i = \pi_i(w_1)\varphi$ , then it suffices to show  $\varphi_1(1) = \varphi_2(1)$ : For other  $a$ ,

$$\varphi_i(a) = (\pi_i \left( \begin{bmatrix} a & \\ & 1 \end{bmatrix} \begin{bmatrix} & 1 \\ -1 & \end{bmatrix} \right) \varphi)(1) = (\pi_i \left( \begin{bmatrix} & 1 \\ -1 & \end{bmatrix} \begin{bmatrix} 1 & \\ & a \end{bmatrix} \right) \varphi)(1)$$

and  $\pi_1\left(\begin{bmatrix} 1 & \\ & a \end{bmatrix}\right)\varphi = \pi_2\left(\begin{bmatrix} 1 & \\ & a \end{bmatrix}\right)\varphi$ .  
For  $n \in \mathbb{Z}$ , let

$$F_\xi(n) = \int_{\mathfrak{q}^{-n} \setminus \mathfrak{q}^{-n+1}} (\varphi_1(y) - \varphi_2(y)) \xi(y) d^\times y,$$

then  $F_\xi(0)$  only depends on the restriction of  $\xi$  to  $\mathcal{O}^*$ , and  $F_\xi(0) = 0$  for all but f.m. characters  $\xi$  of  $\mathcal{O}^*$ :  $\varphi_1 - \varphi_2$  is a locally constant function that is fixed by  $U \subset \mathcal{O}^*$ , so in order  $F_\xi(0) \neq 0$ , at least  $\xi$  should be trivial on  $U$ , and there are f.m. such characters. Thus the hypothesis of Fourier transform on  $\mathcal{O}^*$  is satisfied (12.11.1.24) and

$$\varphi_1(1) - \varphi_2(1) = \sum_{\xi \in (\mathcal{O}^*)^\wedge} F_\xi(0).$$

The hypothesis and the functional equation (29.5.5.9) shows  $Z(s, \varphi_1, \xi) = Z(s, \varphi_2, \xi)$  for any quasi-character  $\xi$  of  $F^*$ , so

$$\sum_n F_\xi(n) \|v\|^{-sn} = Z(s, \varphi_1, \xi) - Z(s, \varphi_2, \xi) = 0$$

and  $F_\xi(n) = 0$  for  $n$  sufficiently small, thus  $F_\xi(n) = 0$  for all  $n$ . In particular,  $F_\xi(0) = 0$ , for any  $\xi$ .  $\square$

**Thm. (24.12.1.8) [Injectivity of the Correspondence].** Any set of maps  $\text{rec}_n : \text{Irr}^{\text{cusp}}(GL(n; K)) \rightarrow \mathfrak{wd}_{\varphi\text{-s.st}}^n(W_K)$  satisfying the properties listed in (24.12.1.2) must be injective.  $\lrcorner$

*Proof:* By the definition of Rankin-Selberg  $L$ -factors (29.3.1.32) and (31.1.1.3),

$$\text{rec}(\pi) = \text{rec}(\pi') \iff \text{ord}_{s=0} L(\text{rec}(\pi)^\vee \otimes \text{rec}(\pi'); s) < 0 \iff L(\pi^\vee \times \pi'; s) < 0 \iff \pi = \pi'.$$

$\square$

**Thm. (24.12.1.9) [Surjectivity, Numerical Local Langlands Correspondence, Henriart [Hen88]].** For any  $n \in \mathbb{Z}_+$ ,  $d \in \mathbb{N}$  and quasi-character  $\chi$  of  $K^\times$ , the subset of  $\text{Irr}^{\text{adm, cusp}}(GL(n; K))$  of representations with conductor  $d$  and central character  $\omega$  is finite, and the subset of  $\text{Irr}_{\varphi\text{-s.st}}^n(\text{WD}_K)$  of representations with conductor  $d$  and determinant  $\omega$  is finite. Moreover, this two sets have the same cardinality.

In particular, any map  $\text{rec}_n : \text{Irr}^{\text{cusp}}(GL(n; K)) \rightarrow \text{Irr}_{\varphi\text{-s.st}}^n(\text{WD}_K)$  satisfying properties listed in (24.12.1.2) must be surjective.  $\lrcorner$

*Proof:* ?  $\square$

### Properties of the Correspondence

**Prop. (24.12.1.10) [Properties of the Correspondence].**  $\pi \in \text{Irr}^{\text{adm}}(GL(n; K))$  is

- supercuspidal iff  $\text{rec}_n(\pi)$  is irreducible, by (25.1.2.19) and Bernstein-Zelevinsky Classification.
- essentially square-integrable iff  $\text{rec}_n(\pi)$  is indecomposable, by (25.1.2.23) and Bernstein-Zelevinsky Classification.
- (Gross-Prasad-Rallis) generic iff  $L(\text{Ad}(\text{rec}_n(\pi)); s)$  has no pole at  $s = 1$ , by (29.3.1.34).
- unramified iff  $\text{rec}_n^\ell(\pi)$  is unramified (25.1.2.22). Moreover,  $\text{rec}_n^\ell(Q(\chi))(\sigma_K) \sim \text{diag}(\chi_1, \dots, \chi_n)$ , by (24.12.1.3).

- Iwahori-spherical iff  $\text{rec}_n(\pi)$  is  $I_K$ -spherical (25.1.2.22), by (24.11.7.1).
- tempered iff the representation  $\eta \in \text{Rep}^{\text{fd}}(L_F)$  corresponding to  $\text{rec}_n(\pi)$  (29.3.1.15) satisfies  $\eta(W_F)$  is bounded iff  $\eta(\sigma_K)$  has only eigenvalues on the unit circle.
- depth-0 iff  $\text{rec}_n^\ell(\pi)$  is tamely ramified.

⌋

*Proof:* For temperedness, Cf. [Wed08]?: Because

$$\text{rec}(\Delta_1 \boxtimes \dots \boxtimes \Delta_n) = \text{rec}(Q(\Delta_1)) \oplus \dots \oplus \text{rec}(Q(\Delta_n)),$$

we are reduced to the case that  $\pi = Q(\Delta)$  and to prove that  $\pi$  is a discrete series iff  $\eta(\sigma_K)$  has only eigenvalues on the unit circle. Now  $\pi$  is square-integrable iff its central character  $\omega_\pi$  is unitary. But  $\omega_\pi \circ \text{Art}_K^{-1} = \det(\text{rec}_n(\pi))$  by (24.12.1.2), so the assertion follows by noticing that all the absolutely values of eigenvalues of  $\eta(w)$  are equal (Making explicit the equivalence (29.3.1.15)).

For depth-0: ?

□

**Prop. (24.12.1.11) [LLC for  $\text{GL}(2)$ ].** Let  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$ ,

1. If  $\pi$  is a supercuspidal, then  $\text{rec}_2(\pi) = (\rho, 0)$ ,  $\rho$  irreducible, by (24.12.1.10).
2. If  $\pi$  is a principal series  $\chi_1 \boxtimes \chi_2$ , then  $\text{rec}_2(\pi) = (\chi_1 \oplus \chi_2, 0)$ .
3. If  $\pi = \chi$ , then  $\text{rec}_2(\pi) = (\chi \oplus \chi(1), 0)$ .
4. If  $\pi = \text{St}_2(\chi)$ , then  $\text{rec}_2(\chi) = \text{Sp}(2)(\chi) = (\chi \oplus \chi(1), N)$  (25.1.2.23), where  $N$  sends  $\chi$  to  $\chi(1)$  and vanish on  $\chi(1)$ .

⌋

## 2 Harris-Taylor's Proof

The sketch is as follows:

- Choose a CM field  $F$  and  $w \in \Sigma_F^{\text{fin}}$  s.t.  $F_w \cong K$ .
- Look at the cohomology of projective  $(n-1)$ -dimensional PEL-type Shimura pro-variety  $X = \varprojlim_{m \in \mathbb{Z}_+} X_m$  associated to the reductive group  $G/\mathbb{Q}$  s.t.

$$G \otimes \mathbb{Q}_p \cong \mathbb{Q}_p^\times \times \text{GL}(n; K) \times \mathbf{G}_{D_p}$$

- For each  $\xi \in \text{Irr}_{\mathbb{Q}}^{\text{fd}}(G)$ , we can associate a lisse  $\overline{\mathbb{Q}_\ell}$ -sheaf  $L_\xi$  on  $X$ . The cohomologies

$$H^i(X, L_\xi) = \varinjlim_{U \in G(\mathbf{A}_{F,f})} H_{\text{ét}}^i((X_U)_{\overline{F}}, L_\xi)$$

are infinite-dimensional  $\overline{\mathbb{Q}_\ell}$ -representations of  $G(\mathbf{A}_{F,f}) \times \text{Gal}_F$ . We can choose  $\xi$  s.t.  $H^i(X, L_\xi) = 0$  for  $i \neq n-1$ .

- There is a map

$$R_\xi : \text{Irr}^{\text{adm}}(G(\mathbf{A}_f)) \rightarrow \text{Rep}^{\text{fd}}(W_K) : \Pi \mapsto \text{Hom}_{G(\mathbf{A}_{F,f})}(\Pi, H^{n-1}(X, L_\xi)).$$

- The decomposition  $G(\mathbf{A}_f) = \mathbb{Q}_p^\times \times \text{GL}(n; K) \times (\text{other terms})$  gives a decomposition  $\Pi = \Pi_0 \otimes \Pi_w \otimes \Pi^w$ . For any  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n; K))$ , we can find such a  $\Pi$  s.t.  $\Pi_w$  is an unramified twist of  $\pi$ ,  $\Pi_0$  is unramified and  $R_\xi(\Pi) \neq 0$ .

- Choose an integral model  $\tilde{X}_m$  of  $X_m \otimes F_v$  and consider the completions  $R_{K,n,m}$  of the local rings of a certain stratum of the special fibre of  $\tilde{X}_m$ , which is the deformation space of the unique 1-dimensional special formal  $\mathcal{O}_K$ -module  $\Sigma_{K,n,m}$  of  $\mathcal{O}_K$ -height  $n$  with Drinfeld level- $m$  structure over  $\bar{\kappa}$ . These spaces have canonical vanishing cycle sheaves  $\psi_{n,m}^i$  and their limit  $\psi_n^i$  are endowed with an admissible action of the subgroup  $A_{K,n} \subset \mathrm{GL}(n; K) \times D_{K,1/n}^\times \times W_K$  consisting of elements  $(\gamma, \delta, \sigma)$  s.t.

$$|\det \gamma|^{-1} \cdot |\mathrm{Nmrd}(\delta)| \cdot |\mathrm{Art}_K^{-1} \sigma| = 1$$

- (Harris-Taylor Construction Theorem B) Local Jacquet-Langlands theory associates to each  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n; K))$  a  $\rho = \mathrm{J-L}(\pi^\vee) \in \mathrm{Irr}^{\mathrm{adm}}(D_{K,1/n}^*)$ . Then there exists  $r_\ell(\pi) \in \mathrm{Rep}_{\mathbb{Q}_\ell}^n(W_K)$  s.t.

$$[\pi \otimes r_\ell(\pi)]^{\mathrm{s.st}} = \sum_{i=0}^{n-1} (-1)^{n-1-i} [\mathrm{Hom}_{D_{K,1/n}^*}(\psi_n^i, \mathrm{J-L}(\pi^\vee))]^{\mathrm{s.st}}.$$

and

$$n[R_\xi(\Pi) \otimes \chi(\Pi_0 \circ \mathrm{Nm}_{K/\mathbb{Q}_p})]^{\mathrm{s.st}} \in \mathbb{Z}[r(\pi)]^{\mathrm{s.st}}.$$

- To show the last assertion, one gives a description of  $H^{n-1}(X, L_\xi)^{\mathbb{Z}_p^\times}$  in which  $[\mathrm{Hom}_{D_{K,1/n}^*}(\psi_n^i, \mathrm{J-L}(\pi^\vee))]$  occurs (Theorem D).
- Define  $\mathrm{rec}_n(\pi) = r_\ell(\pi^\vee(\frac{1-n}{2}))$ .

### 3 Scholze's Proof

**Remark (24.12.3.1).** The local Langlands correspondence for essentially- $L^2$  and generalized Speh representations (24.11.4.16) of  $\mathrm{GL}(n; K)$  are realized in the cohomology of the moduli space of 1-dimensional  $p$ -divisible groups of height  $n$ .

To construct Galois representations, Harris and Taylor consider the cohomology of unitary Shimura varieties that split at  $p$ , uses it to construct  $\ell$ -adic Galois representations associated to certain cuspidal automorphic representations of  $\mathrm{GL}(n)$ , and proves a local-global compatibility result at places of bad reduction. This is achieved by counting points on Igusa varieties.

This method and its generalizations allow one to compute the trace of arbitrary Hecke correspondences at  $p$  on a Shimura variety. But Scholze restrict attention to Hecke operators at  $p$  coming from the maximal compact subgroup and reduce all counting problems to counting problems for the maximal compact level structure, for which one can just appeal Kottwitz's point-counting methods.

┘

**Thm. (24.12.3.2)** [Test Function Characterization of Local Langlands, Scholze [Sch13]].

- (a) For each  $n \in \mathbb{Z}_+$ , there is a unique map

$$\sigma_n : \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K)) \rightarrow \mathfrak{w}\mathfrak{d}_{\psi\text{-s.st}}^n(W_K)$$

s.t. for any  $\tau \in W_K$  and any  $h \in C_c^\infty(\mathrm{GL}(n; \mathcal{O}_K))$ , we have

$$\mathrm{tr}(f_{\tau,h}|\pi) = \mathrm{tr}(\tau|\sigma_n(\pi))\mathrm{tr}(h|\pi) \quad (24.12.5.4),$$

**?** Explain why must choose  $h$ . Write  $\mathrm{rec}'(\pi) = \sigma_n(\pi)(\frac{1-n}{2})$ .

- (b) If  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$  is a constituent of  $\pi_1 \times \dots \times \pi_r$ , then  $\mathrm{rec}'(\pi) = \mathrm{rec}'(\pi_1) \oplus \dots \oplus \mathrm{rec}'(\pi_r)$ .

- (c)  $\text{rec}'$  induces a bijection between  $\text{Irr}^{\text{cusp}}(\text{GL}(n; K))$  and  $\text{Irr}_{\varphi\text{-s.st}}^n(W_K)$ .
- (d) The restriction of  $\text{rec}'$  to supercuspidal representations is compatible with twists, central characters, duals, and  $L$ - and  $\epsilon$ -factors of pairs, hence  $\text{rec}' = \text{rec}$  as in (24.12.1.2), by (24.12.1.5).  $\lrcorner$

*Proof:* (a) and (b) follow from (24.12.3.3) and (24.12.3.5).

For (c), use computation of  $I_K$ -invariant nearby cycles for simple Shimura varieties. This computation leads to a direct proof of the bijective correspondence for supercuspidal representations, without using the numerical local Langlands correspondence. ?

Finally, for the proof of (d): By Brauer induction and linearity, it suffices to assume that  $\pi$  is induced from characters. It suffices to show that: For any  $\pi_1 \in \text{Irr}^{\text{adm}}(\text{GL}(n_1; K))$ ,  $\pi_2 \in \text{Irr}^{\text{adm}}(\text{GL}(n_2; K))$ , there exists  $F \in \mathbf{NField}$  with  $w \in \Sigma_F^{\text{fin}}$  s.t.  $K \cong F_w$ , and two potentially Abelian  $\pi_i \in \text{Irr}^{\text{auto}}(\text{GL}(n_i)/F)$ ,  $\pi_2 \in \text{Irr}^{\text{auto}}(\text{GL}(n_2)/F)$  s.t.  $(\Pi_i)_w$  is an unramified twist of  $\pi_i$ . Cf. proof of [H-T01] VII.2.10. ?

Then the compatibility follows from Henniart's method of twisting with highly ramified characters, cf. Corollary 2.4 of [Hen00]. ?  $\square$

**Prop. (24.12.3.3) [Dévissage for Constructing Galois Representations].** For  $n \in \mathbb{Z}_+$ , suppose (a) and (b) of (24.12.3.2) hold for all  $n' < n$  and the following hold:

1. If  $\pi = \pi_1 \times \dots \times \pi_r \in \text{Rep}^{\text{adm}}(\text{GL}(n; K))$  where  $\pi_i \in \text{Irr}^{\text{adm}}(\text{GL}(n_i; K))$ , then

$$\text{tr}(f_{\tau, h}|\pi) = \text{tr} \left( \tau \Big| \bigoplus_{1 \leq i \leq r} \sigma(\pi_i) \left( \frac{n - n_i}{2} \right) \right) \text{tr}(h|\pi).$$

2. For  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  that is either essentially- $L^2$  or a generalized Speh representation (24.11.4.16), then there exists a virtual finite dimensional representation  $\sigma(\pi)$  of  $W_K$  with  $\mathbb{Q}_+$  coefficients of dimension  $n$  s.t.

$$\text{tr}(f_{\tau, h}|\pi) = \text{tr}(\tau|\sigma(\pi)) \text{tr}(h|\pi).$$

3. If  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n; K))$ , then  $\sigma(\pi)$  is a genuine representation of  $W_K$ .

Then (a) and (b) of (24.12.3.2) hold true for  $n$ , by defining  $\sigma(\pi)$  as follows: If  $\pi$  has supercuspidal support  $\{\pi_1, \dots, \pi_r\}$  (with multiplicity), then we define

$$\sigma_n(\pi) = \bigoplus \sigma(\pi_i) \left( \frac{1 - n_i}{2} \right).$$

$\lrcorner$

*Proof:* Firstly, the uniqueness of  $\sigma_n$  follows from the fact the elements mapping to a positive power of  $\sigma_K$  determines a Weil-Deligne representation up to semisimplification. ?

What is Bernstein center ? Let  $f_{\tau}$  be the function on the Bernstein center that acts through the scalar  $\text{tr}(\tau|\sigma(\pi) \left( \frac{n-1}{2} \right))$  on any  $\pi \in \text{Irr}^{\text{adm}}(\pi)$ . ?

Then to prove (a) and (b) of (24.12.3.2), it suffices to show that

$$\text{tr}(f_{\tau, h}|\pi) = \text{tr}(f_{\tau} * h|\pi)$$

for any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ . If  $\pi$  is induced or supercuspidal, then this is true by the hypothesis listed. By (24.12.3.4), it suffices to check for any generalized Speh representation  $\pi = \mathrm{St}(\pi_0; t)$  (24.11.4.16). Choose  $h_0 \in C_c^\infty(\mathrm{GL}(n; K))$  as in item1 of (24.12.3.4), then

$$\mathrm{tr}(f_{\tau, h_0} | \pi') = \mathrm{tr}(f_\tau * h_0 | \pi')$$

holds for any tempered  $\pi' \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ : This is true ?  $\square$

**Lemma (24.12.3.4) [Testing on Tempered Representations and Generalized Speh Representations].**

- Let  $d \in \mathbb{Z}_+, t \in \mathbb{Z}_{\geq 2}, n = dt, \pi_0 \in \mathrm{Irr}^{\mathrm{cusp}, \mathrm{uni}}(\mathrm{GL}(d; K))$  and  $\pi = \mathrm{St}(\pi_0, t)$ . Then there exists  $h \in C_c^\infty(\mathrm{GL}(n; \mathcal{O}_K))$  s.t.  $\mathrm{tr}(h | \pi) = 0$  for any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; F))$  that is tempered but not of the form  $\pi = \pi_0(iy_1) \times \dots \times \pi_0(iy_t)$  where  $y_i \in \mathbb{R}$ . And for these  $\pi$ ,  $\mathrm{tr}(h | \pi) \neq 0$ .
- If  $h \in C_c^\infty(\mathrm{GL}(n; K))$  s.t. for all  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(d; K))$  that is tempered but non-square-integrable or  $\pi = \mathrm{St}(\pi_0, t)$  for some  $d \in \mathbb{Z}_+, t \in \mathbb{Z}_{\geq 2}, n = dt, \pi_0 \in \mathrm{Irr}^{\mathrm{cusp}, \mathrm{uni}}(\mathrm{GL}(d; K))$ ; then  $\mathrm{tr}(h | \pi) = 0$  for any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$ .

┘

*Proof:* Kazhdan's density theorem ? and  $\square$

**Prop. (24.12.3.5) [Constructing Galois Representations].** The hypotheses of (24.12.3.3) are true for any  $n \in \mathbb{Z}_+$ . In particular, (a) and (b) of (24.12.3.2) hold true for any  $n \in \mathbb{Z}_+$ .  $\square$

*Proof:* 1 is proved in Theorem 6.4. of [Sch13], by relating the deformation spaces of one-dimensional  $p$ -divisible groups to the deformation spaces of their infinitesimal parts. ?

2. Firstly by (24.12.6.3),  $K$  can be realized as  $K = F_w$  with notation as in (24.12.6.1) and (24.12.6.1). Then we can apply (24.12.7.8) to  $\pi^\vee$  to get  $\pi_f \in \mathrm{Irr}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  s.t.

- $H_\xi^*(\pi_f) \neq 0$ .
- $\pi_{p,0}$  is unramified.
- $\pi_w = \pi \otimes \chi$  for some unramified quasi-character  $\chi$  of  $K^\times$ .

Then we apply (24.12.7.7) to get

$$\mathrm{tr}(f_{\tau, h}^\vee | \pi^\vee \otimes \chi) = \frac{1}{a(\pi_f)} \mathrm{tr}(\tau([H_\xi[\pi_f]] \otimes \chi_{\pi_{p,0}}) \mathrm{tr}(h^\vee | \pi^\vee \otimes \chi)).$$

Then we can take  $\sigma(\pi) = \frac{1}{a(\pi_f)} [H_\xi[\pi_f]]^\vee \otimes \chi_{\pi_{p,0}}^{-1} \otimes \chi^{-1}$  ?, which has dimension  $n$  by (24.12.6.7).

3: It suffices to show that  $\sigma(\pi) \in K_0(\mathrm{Rep}^{\mathrm{fd}}(W_K))$ . For this, Cf. [Sch13]P705 ?  $\square$

### Bijjective Correspondence for Supercuspidal Representations

**Thm. (24.12.3.6) [Supercuspidal Representations are Realized on Lubin-Tate Spaces].** Let  $[R\psi_n]$  be the alternating sum of the global section of the vanishing cycles for the Lubin-Tate tower, then it's endowed with an admissible action of the subgroup  $A_{K,n} \subset \mathrm{GL}(n; K) \times D_{K,1/n}^\times \times W_K$  consisting of elements  $(\gamma, \delta, \sigma)$  s.t.

$$|\det \gamma|^{-1} \cdot |\mathrm{Nmrd}(\delta)| \cdot |\mathrm{Art}_K^{-1} \sigma| = 1$$

.

Let  $\rho \in \text{Irr}^{\text{adm}}(D_{K,1/n}^*)$  s.t.  $\pi = \text{J-L}(\rho) \in \text{Irr}^{\text{cusp}}(\text{GL}(n; K))$ . Then

$$[R\psi](\rho) = (-1)^{n-1} \pi^\vee|_{\text{GL}(n; \mathcal{O}_K)} \otimes \sigma(\pi) \in K_0 \left( \text{Rep}^{(\text{adm}, \text{cont})}(\text{GL}(n; \mathcal{O}_K) \times W_K) \right) \quad (24.12.3.3) \text{?}.$$

┘

*Proof:* Cf.[Sch13]P704.?

This follows from (24.12.5.7).? □

**Def. (24.12.3.7)[Functorial Extension of Class Field Theory].** A **functorial extension of class field theory** is a family of maps

$$\text{rec}'_n : \text{Irr}^{\text{adm}}(\text{GL}(n; K)) \rightarrow \text{Rep}^n(W_K)$$

for each  $K \in p\text{-NField}$  and  $n \in \mathbb{Z}_+$ , s.t. the following holds:

- For a quasi-character  $\chi$  of  $K^\times$ ,  $\text{rec}'_1(\chi) = \chi \circ \text{Art}_K^{-1}$  (19.7.2.11).
- If  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  is a constituent of  $\pi_1 \boxtimes \dots \boxtimes \pi_t$ , then  $\sigma(\pi) = \sigma(\pi_1) \oplus \dots \oplus \sigma(\pi_t)$ .
- For a quasi-character  $\chi$  of  $K^\times$  and  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$ ,

$$\text{rec}'_n(\pi(\chi)) = \text{rec}'_n(\pi) \otimes \text{rec}'_1(\chi).$$

- If  $K'/K$  is a cyclic Galois extension of prime degree and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n; K))$  with  $\Pi = \text{BC}_K^{K'}(\pi) \in \text{Irr}^{\text{adm}}(\text{GL}(n; K'))$ , then  $\text{rec}'(\Pi) = \text{rec}'(\pi)|_{W_{K'}}$ .
- If  $\pi \in \text{Irr}^{\text{adm}}(\text{GL}(n; K))$  and  $\text{rec}'(\pi)$  is unramified, then  $\pi$  is Iwahori-spherical.

┘

**Thm. (24.12.3.8)[Bijective Correspondence for Supercuspidal Representations].** Suppose  $\text{rec}'_n$  is a functorial extension of local class field theory (24.12.3.7), then

$$\text{rec}'_n : \text{Irr}^{\text{cusp}}(\text{GL}(n; K)) \cong \text{Rep}^n(W_K)$$

is a bijection for any  $K \in p\text{-NField}$  and  $n \in \mathbb{Z}_+$ . ┘

*Proof:* ? Cf.[Sch13]P707. □

#### 4 $p$ -divisible Groups and Vanishing Cycles

**Def. (24.12.4.1)[ $p$ -divisible Groups].** Let  $S \in \text{Sch}/\mathcal{O}_K$  s.t.  $p$  is locally nilpotent?, then a  $\varpi$ -divisible  $\mathcal{O}_K$ -module  $H$  over  $S$  is a  $p$ -divisible group  $G \in p\text{Div}(S)$  together with  $\iota : \mathcal{O}_K \rightarrow \text{End}_S(H)$  s.t. the two induced actions of  $\mathcal{O}_K$  on  $\text{Lie}_S(G)$  are equal.

For any  $\varpi$ -divisible  $\mathcal{O}_K$ -module  $G/S$ , we have  $[K : \mathbb{Q}_p] \mid \text{ht}(G)$ ?. So we can define the  $\mathcal{O}_K$ -**height**  $\text{ht}_{\mathcal{O}_K}(G) = \text{ht}(G)/[K : \mathbb{Q}_p]$ . ┘

**Def. (24.12.4.2)[Drinfeld-level- $m$ -Structures].** A **Drinfeld-level- $m$ -structure** on a 1-dimensional  $\varpi$ -divisible  $\mathcal{O}_K$ -module  $H/S$  is a tuple of sections  $(X_1, \dots, X_n) \in H[\varpi^m]$  s.t.

$$\sum_{i_1, \dots, i_n \in \mathcal{O}_K/(\varpi^m)} [i_1 X_1 + \dots + i_n X_n] = [H[\varpi^m]]$$

as relative Cartier divisors over  $S$ . ┘



### D-Groups

**Def. (24.12.4.3) [Local Simple Unitary Groups].** For  $D'/\mathbb{Q}_p$  is a semisimple algebra with a maximal order  $\mathcal{O}_{D'}$ , define  $D = \mathrm{Mat}(n; K) \times D'$  with maximal order  $\mathcal{O}_D = \mathrm{Mat}(n; \mathcal{O}_K) \times \mathcal{O}_{D'}$ .

Define  $\mathbf{G}_{D'} = \mathrm{Res}_{\mathcal{O}_K/\mathbb{Z}_p}(\mathcal{O}_D^*)$  and  $\mathbf{G}_D = \mathrm{Res}_{\mathcal{O}_K/\mathbb{Z}_p} \mathrm{GL}(n) \times \mathbf{G}_{D'}$ .  $\lrcorner$

**Def. (24.12.4.4) [D-Groups].** Let  $S \in \mathrm{Sch}$  s.t.  $p$  is locally nilpotently, then a  $D'$ -group over  $S$  is an étale  $p$ -divisible group  $H$  over  $S$  together with an action

$$\iota : \mathcal{O}_{D'}^{\mathrm{op}} \rightarrow \mathrm{End}(H)$$

s.t.  $H[p]$  is free of rank 1 over  $\mathcal{O}_{D'}^{\mathrm{op}}/(p)$ .  $\lrcorner$

**Def. (24.12.4.5)  $((\mathcal{O}_K, D')$ -Groups).** For  $S \in \mathrm{Sch}/\mathcal{O}_K$ , an  $(\mathcal{O}_K, D')$ -group over  $S$  is a  $p$ -divisible group  $\underline{H}/S$  with an action

$$\iota : \mathcal{O}_D^{\mathrm{op}} \rightarrow \mathrm{End}(\underline{H})$$

s.t.  $\underline{H}$  decomposes as a product  $\underline{H} = H^{\oplus n} \times H'$  under the action of  $\mathcal{O}_D^{\mathrm{op}} \cong \mathrm{Mat}(n; \mathcal{O}_K)^{\mathrm{op}} \times \mathcal{O}_{D'}^{\mathrm{op}}$  where  $H$  is a 1-dimensional  $\varpi$ -divisible  $\mathcal{O}_K$ -module of height  $n$  and  $H'$  is a  $D'$ -group (24.12.4.4).

And a **level- $m$ -structure** on a  $(\mathcal{O}_K, D')$ -group  $\underline{H}$  is a Drinfeld-level- $m$ -structure on  $H$  and an isomorphism of  $\mathcal{O}_{D'}^{\mathrm{op}}$ -modules  $\mathcal{O}_{D'}^{\mathrm{op}}/(p^m) \cong H'[p^m]$  ?.  $\lrcorner$

**Prop. (24.12.4.6) [Dieudonné Parameters].** An  $(\mathcal{O}_K, D')$ -group  $\underline{H}$  over  $\kappa_r$  is equivalent to a Dieudonné module  $\underline{M}$  over  $W(\kappa_r)$  s.t. there is an isomorphism of  $\mathcal{O}_D \otimes W(\kappa_r)$ -module

$$\underline{M} \cong \mathcal{O}_D \otimes W(\kappa_r).$$

Thus  $\underline{M}$  is represented by an element  $\delta_0 \in \mathbf{G}_D(W(\kappa_r)[\frac{1}{p}])$  up to  $\sigma_K$ -conjugacy. This  $\delta_0$  is called the **Dieudonné paramter** of  $\underline{H}$ .  $\lrcorner$

### Deformations of $p$ -adic Spaces

**Def. (24.12.4.7)  $[\overline{H}_\beta]$ .** Take  $r \in \mathbb{Z}_+$  and a 1-dimensional  $\varpi$ -divisible  $\mathcal{O}_K$ -module  $\overline{H}_\beta$  of height  $n$  over  $\kappa_r$ , whose Dieudonné module is given by an element

$$\beta \in \mathrm{GL}(n; \mathcal{O}_{K_r}) \mathrm{diag}(\varpi_K, 1, \dots, 1) \mathrm{GL}(n; \mathcal{O}_{K_r})$$

up to  $\sigma_K$ -conjugacy.  $\lrcorner$

**Def. (24.12.4.8) [Deformation Spaces].** Let  $\mathrm{Spf} R_\beta$  be the formal deformation space of  $\overline{H}_\beta$  as  $\varpi$ -divisible  $\mathcal{O}_K$ -modules over  $\mathcal{O}_{K_r}$ , with universal deformation  $H_\beta$ . ?

For  $m \in \mathbb{Z}_+$ , let  $R_{\beta, m}$  be the covering of  $R_\beta$  parametrizing Drinfeld-level- $m$ -structures on  $H_\beta$ .  $\lrcorner$

**Prop. (24.12.4.9).**

- $R_\beta \cong \mathcal{O}_{K_r}[[T_1, \dots, T_{n-1}]]$ .
  - $R_{\beta, m}/R_\beta$  is a finite Galois covering with Galois group  $\mathrm{GL}(\mathcal{O}_K/(\varpi^m))$ , étale in the generic fiber.
  - $R_{\beta, m}$  is regular.
- $\lrcorner$

*Proof:* Cf. [Sch13]P672.  $\square$

**Def. (24.12.4.10) [Deformation Spaces].** For a  $(\mathcal{O}_K, D')$ -group  $\overline{H}$  over  $\kappa_r$ , there is a deformation space  $\mathrm{Spf} R_{\overline{H}}$  of deformations of  $\overline{H}$  over  $\mathcal{O}_{K_r}$  with universal deformation  $\underline{H}$ , and a covering map  $R_{\overline{H},m}/R_{\overline{H}}$  parametrizing level- $m$ -structures on the universal deformation.  $\lrcorner$

**Prop. (24.12.4.11) [Comparison of Deformation Spaces].** If  $\overline{H} \cong \overline{H}_\beta^{\oplus n} \otimes \overline{H}'$ , then

- The canonical map  $\mathrm{Spf} R_{\overline{H}} \rightarrow \mathrm{Spf} R_\beta$  is an isomorphism.
- The canonical map  $\mathrm{Spf} R_{\overline{H},m} \rightarrow \mathrm{Spf} R_{\beta,m} \times_{\mathrm{Spf} R_\beta} (H'[p^m])^*$  is an isomorphism, where  $(H'[p^m])^*$  is the set of  $\mathcal{O}_D^{\mathrm{op}}$ -generators of  $H'[p^m]$ . In particular,  $\mathrm{Spf} R_{\overline{H},m}/\mathrm{Spf} R_{\overline{H}}$  is a finite Galois cover with Galois group  $\mathrm{GL}(n; \mathcal{O}_K/(\varpi^m)) \times (\mathcal{O}_{D'}/(p^m))^*$ .  $\lrcorner$

*Proof:* This follows from the fact there are no deformations of a  $D'$ -group, by rigidity of étale covers.  $\square$

**Def. (24.12.4.12) [Lubin-Tate Tower].** For  $\beta$  basic, the tower

$$(\mathrm{Spf} R_{n,m})_{m \in \mathbb{Z}_+} = \left( \mathrm{Spf} R_{\beta,m} \otimes_{\mathcal{O}_r} \check{\mathcal{O}}_K \right)_{m \in \mathbb{Z}_+}$$

is called the **Lubin-Tate tower** of height  $n$ . This doesn't depend on  $\beta$ , because there exists a unique 1-dimensional formal  $\mathcal{O}_K$ -module of height  $n$  over  $\overline{\kappa}$  and a unique Drinfeld-level- $m$ -structure for any  $m$ .  $\lrcorner$

**Prop. (24.12.4.13).** There is a compatible action of  $A_{K,n} \subset \mathrm{GL}(n; K) \times D_{K,1/n}^\times \times W_K$  consisting of elements  $(\gamma, \delta, \sigma)$  s.t.

$$|\det \gamma|^{-1} \cdot |\mathrm{Nmrd}(\delta)| \cdot |\mathrm{Art}_K^{-1} \sigma| = 1$$

on the Lubin-Tate tower, where

- $\mathrm{GL}(n; K)$  acts on the Drinfeld-level-structures,
- $D_{K,1/n}^\times$  acts on the trivialization of the special fiber.
- $W_K$  acts as Frobenius.  $\lrcorner$

## Vanishing Cycles

**Def. (24.12.4.14) [Vanishing Cycles].** For  $\mathcal{X} \in \mathrm{Sch}^{\mathrm{ft}}/\mathcal{O}_{\check{K}}$  with generic fiber  $X/\check{K}$ , the formalism of vanishing cycles is used to relate the cohomology of  $\overline{X}$  and the cohomology of  $X_{K^s}$  together the action of  $I_K$ .

Consider the diagram  $\overline{X} \xrightarrow{i} X \xleftarrow{\bar{j}} X_{K^s}$  and  $\Lambda \in \mathcal{A}b^{\mathrm{fin}}$ , then the **sheaf of vanishing cycle** of  $\mathcal{X}/\mathcal{O}_{\check{K}}$  is defined to be

$$\Psi_{\mathcal{X}}^\bullet(\Lambda) = i^* R^\bullet \bar{j}_* \Lambda \in \mathrm{Sh}(\overline{X}_{\mathrm{\acute{e}t}}),$$

which has an action of  $I_K$ .  $\lrcorner$

**Prop. (24.12.4.15) [Proper case].** If  $\mathcal{X}/\mathcal{O}_{\check{K}}$  is proper, then proper base change implies  $i^* : H_{\mathrm{\acute{e}t}}^p(\mathcal{X}, R^q \bar{j}_* \Lambda) \cong H_{\mathrm{\acute{e}t}}^p(\overline{X}, i^* R^q \bar{j}_* \Lambda)$ . And the spectral sequence for  $\bar{j}$  implies

$$H_{\mathrm{\acute{e}t}}^p(\overline{X}, \Psi^n(\Lambda)) \implies H_{\mathrm{\acute{e}t}}^{p+q}(X_{K^s}, \Lambda).$$

which is  $I_K$ -invariant. If moreover,  $\mathcal{X}/\mathcal{O}_K$  is smooth, then by (9.4.6.36), this spectral sequence degenerate at page 2 and  $\Psi^n(\Lambda) = \begin{cases} \Lambda & , n = 0 \\ 0 & , n \geq 2 \end{cases}$ , inducing an isomorphism

$$H_{\text{ét}}^n(\overline{X}, \Lambda) \implies H_{\text{ét}}^n(X_{K^s}, \Lambda).$$

where  $I_K$  acts trivially. ┘

**Def. (24.12.4.16) [Formal Vanishing Cycles].** The vanishing cycle is defined verbatim in the Berkovich formal schemes setting, and the resulting sheaves are called **formal vanishing cycles**. But Scholze didn't pass to the maximal unramified extension (the Lubin-Tate tower) as Harris-Taylor did, and we can get more informations. ┘

**Thm. (24.12.4.17) [Formal Vanishing Cycles on the Deformation Tower].** Consider the formal vanishing cycles on the tower  $\{\text{Spf } R_{\beta, m}\}_{m \in \mathbb{Z}_+}$ :

$$\Psi_{\beta}^i = \varinjlim_{m \in \mathbb{Z}_+} H^0(\overline{\text{Spf } R_{\beta, m}}, \Psi_{\text{Spf } R_{\beta, m}}^i \overline{\mathbb{Q}_{\ell}}).$$

Then

$$H^0(\overline{\text{Spf } R_{\beta, m}}, R^i \psi_{\text{Spf } R_{\beta, m}} \overline{\mathbb{Q}_{\ell}}) \in \text{Rep}^{\text{fd}}(W_{K_r} \times \text{GL}(n; \mathcal{O}_K/(\varpi^m))),$$

and it vanishes unless  $i \in [0, n-1]$ .

In particular  $\Psi_{\beta}^i, [\Psi_{\beta}] \in \text{Rep}^{(\text{cont}, \text{adm})}(W_{K_r} \times \text{GL}(n; \mathcal{O}_K))$ . ┘

*Proof:* Cf. [Sch13] P673. □

**Prop. (24.12.4.18) [Isomorphism of Formal Vanishing Cycles].** Define the formal vanishing cycles as

$$\Psi_{\underline{H}}^i = \varinjlim_{m \in \mathbb{Z}_+} H^0(R_{\underline{H}}, \Psi_{\text{Spf } R_{\underline{H}, m}}^i \overline{\mathbb{Q}_{\ell}}),$$

then they has an action of  $W_{K_r} \times \mathbf{G}_D(\mathbb{Z}_p)$ . And by (24.12.4.11), there is a  $W_{K_r} \times \mathbf{G}_D(\mathbb{Z}_p)$ -equivariant isomorphism

$$\Psi_{\underline{H}}^i \cong \Psi_{\beta}^i \otimes C_c^{\infty}(\mathcal{O}_{D'}^*; \overline{\mathbb{Q}_{\ell}}).$$

┘

*Proof:* □

**Def. (24.12.4.19) [Vanishing Cycles on the Lubin-Tate Tower].** Define the vanishing cycles as

$$\Psi_n = \varinjlim_{m \in \mathbb{Z}_+} H^0(\overline{\text{Spf } R_{n, m}}, \Psi_{\text{Spf } R_{n, m}}^i \overline{\mathbb{Q}_{\ell}}),$$

with an action of  $A_{K, n}$  induced by (24.12.4.13). Then  $\Psi_n \in \text{Rep}^{(\text{adm}, \text{alg}, \text{cont})}(\text{GL}(n; \mathcal{O}_K) \times \mathcal{O}_{D_{K, 1/n}}^* \times I_K)$ , and it vanishes unless  $i \in [0, n-1]$ . ┘

*Proof:* The smoothness of  $\mathcal{O}_{D_{K, 1/n}}^*$ -action follows from comparison theorem Corollary 4.5 of [Vanishing cycles for formal schemes. 2, Berkovich], Cf. proof of [H-T01], Lemma II.2.8. The rest follows from (24.12.4.17). □

## 5 Local Cyclic Base Change

**Prop. (24.12.5.1).**  $K_r/K$  be the unramified extension of degree  $r$ , then  $\mathfrak{N}_{\sigma^{-1}} : \mathrm{GL}(n; K_r) \rightarrow \mathrm{GL}(n; K)$  defines an injection from the  $\sigma$ -conjugacy classes of  $\mathrm{GL}(n; K_r)$  to the conjugacy classes of  $\mathrm{GL}(n; K)$  (10.5.1.14). In fact, if  $\gamma = \mathfrak{N}_{\sigma^{-1}}(x)$ , then  $G_{x,\delta}$  is an inner form of  $G_\gamma$ .  $\lrcorner$

*Proof:* Cf. [Arthur-Clozel]P3.  $\square$

**Prop. (24.12.5.2) [Cyclic Base Change].** If  $\gamma = \mathrm{Nm}(x)$ , we can define orbital integrals and twisted orbital integrals:

$$\mathrm{TOrb}_\delta(\varphi) = \int_{G_{\delta,\sigma} \backslash \mathrm{GL}(n; K_r)} \varphi(g^{-1} \delta g^\sigma) \frac{dg_r}{dt},$$

$$\mathrm{Orb}_\gamma(f) = \int_{G_\gamma \backslash \mathrm{GL}(n; K)} \varphi(g^{-1} \delta g) \frac{dg}{dt}.$$

Then for any  $\varphi \in C_c^\infty(\mathrm{GL}(n; K_r))$ , there exists  $f \in C_c^\infty(\mathrm{GL}(n; K))$  s.t. for any regular  $\gamma \in \mathrm{GL}(n; K)$ ,

$$\mathrm{Orb}_\gamma(f) = \begin{cases} \mathrm{TOrb}_{\gamma,\sigma}(\varphi) & , \gamma = \mathrm{Nm} \delta, \delta \in \mathrm{GL}(n; K_r) \\ 0 & , \text{otherwise} \end{cases}.$$

$\lrcorner$

*Proof:* Cf. [Arthur-Clozel]P20.  $\square$

### Test Functions

**Def. (24.12.5.3) [Test Functions].** For any  $\tau \in \sigma_k^r I_K$  and  $h \in \mathcal{T}(\mathrm{GL}(n; \mathcal{O}_K); \mathbb{Q})$ , define

$$\psi_{\tau,h}(\beta) = \begin{cases} \mathrm{tr}(\tau \times h^\vee | [\Psi_\beta]) & , \beta \in \mathrm{GL}(n; \mathcal{O}_{K_r}) \mathrm{diag}(\varpi_K, 1, \dots, 1) \mathrm{GL}(n; \mathcal{O}_{K_r}) \\ 0 & , \text{otherwise} \end{cases} \quad (24.12.4.17).$$

Then  $\varphi_{\tau,h} \in C_c^\infty(\mathrm{GL}(n; K_r); \mathbb{Q})$ , and is independent of  $\ell$ .  $\lrcorner$

*Proof:* Cf. [Sch13]P674 ?. This uses algebrization theorem and a result of Mieda on independence of  $\ell$ .  $\square$

**Def. (24.12.5.4) [Base Change Test Function,  $f_{\tau,h}$ ].** Define  $f_{\tau,h} \in \mathcal{T}(\mathrm{GL}(n; K); \mathbb{Q})$  s.t. it has matching twisted orbital integral with  $\varphi_{\tau,h} \in C_c^\infty(\mathrm{GL}(n; K_r); \mathbb{Q})$  (24.12.5.2) w.r.t. the Haar measures neutralized by  $\mathrm{GL}(n; \mathcal{O}_K)$ .

Notice  $f$  is not unique but its orbital integrals are, thus its trace on any representation  $\rho \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n; K))$  is well-defined by Weyl's integration formula ?.  $\lrcorner$

**Def. (24.12.5.5).** For any  $\tau \in W_{K_r}$  and  $h \in C_c^\infty(\mathrm{GL}(n; \mathcal{O}_K); \mathbb{Q})$ ,  $h' \in C_c^\infty(\mathcal{O}_{D'}^*; \mathbb{Q})$ , define the function

$$\varphi_{\tau,h,h'}(\delta_0) = \mathrm{tr} \left( \tau \times h^\vee \times h' \mid [\Psi_{\overline{H}}] \right) \quad (24.12.4.18).$$

if  $\overline{H}$  has Dieudonné paramter  $\delta_0$  (24.12.4.6), and 0 if there is not such  $\overline{H}$ . Then by (24.12.5.3),  $\varphi_{\tau,h,h'} \in C_c^\infty(\mathbf{G}_D(W(\kappa_r)[\frac{1}{p}]))$  and is independent of  $\ell$ .  $\lrcorner$

**Prop. (24.12.5.6) [Base Change Test Functions].**  $\varphi_{\tau,h,h'} \in C_c^\infty(\mathbf{G}_D(W(\kappa_r)[\frac{1}{p}]); \mathbb{Q})$  corresponds to  $f_{\tau,h}^\vee \times h' \in C_c^\infty(\mathbf{G}_D(\mathbb{Q}_p); \mathbb{Q})$ .  $\lrcorner$

*Proof:* Cf.[Sch13]P688.?

□

**Prop.(24.12.5.7)[Relation to Lubin-Tate space].** Since  $\beta$  is basic, we can associate  $d = \mathrm{Nm}(\beta) \in D_{K,1/n}^\times$  with the same characteristic polynomial. Then for any  $\tau \in W_{K_\tau}$  and  $h \in C_c^\infty(\mathrm{GL}(n; \mathcal{O}_K))$ , we have

$$\varphi_{\tau,h}(\beta) = \mathrm{tr} \left( \tau \times d^{-1} \times h^\vee |[R\psi] \right).$$

┘

*Proof:* Cf.[Sch13]P704.

□

## 6 Simple Shimura Varieties

**Notation(24.12.6.1).**

- Let  $F_0 \in \mathbf{NField}$  be totally real with  $2|[F_0 : \mathbb{Q}]$ .
- Let  $\tau \in \Sigma_{F_0}^\infty$  and  $x_0 \in \Sigma_{F_0}^{\mathrm{fin}}$ .
- Let  $\mathcal{K} \subset \mathbb{C}$  be an imaginary quadratic field s.t. the rational prime below  $x_0$  splits in  $\mathcal{K}$ .
- Let  $F = F_0\mathcal{K}$  and  $S_F(x_0) = \{x, x^c\}$ .
- Let  $w \in \Sigma_F^{\mathrm{fin}}$  with  $u = w \cap \mathcal{K}$  and  $p = w \cap \mathbb{Q}$  s.t.  $p$  is split in  $\mathcal{K}$  and  $w \notin \{x, x^c\}$  (in particular,  $\mathbb{D}$  is split at  $w$ ).
- Let  $K = F_w$  with residue field  $\kappa$ . Let  $n \in \mathbb{Z}_+$ .

┘

**Prop.(24.12.6.2)[Division Rings and Simple Unitary Groups].** There exists  $\mathbb{D} \in \mathbf{Az}_F$  of dimension  $n^2$  with an involution  $*$  of second kind, and a homomorphism  $h_0 : \mathbb{C} \rightarrow \mathbb{D}_\tau$  s.t.

- $x \mapsto h_0(\mathbf{i})^{-1}x^*h_0(\mathbf{i})$  is a positive involution on  $\mathbb{D}_\mathbb{R}$ .
- $\mathbb{D}$  splits at all places  $v \in \Sigma_F \setminus \{x, x^c\}$ .
- Define  $\mathbf{G}_0 \subset \mathrm{Res}_{F/F_0}(\mathbb{D}^*) \in \mathbf{AlgGrp}/F_0$  representing the functor

$$\mathbf{G}_0 : R \mapsto \{g \in (\mathbb{D} \otimes_{F_0} R)^* | gg^* = 1\},$$

then  $\mathbf{G}_0$  is quasi-split at all non-split places of  $F_0$ , unitary of signature  $(1, n-1)$  at  $\tau$  and unitary of signature  $(0, n)$  at all other infinite places of  $F_0$ .

Define

$$\mathbf{G} \in \mathbf{RedGrp}/\mathbb{Q} : R \mapsto \{g \in (\mathbb{D} \otimes R)^* | gg^* \in R^*\},$$

together with the amplitude map  $\nu : \mathbf{G} \rightarrow \mathbf{G}_m$  with kernel  $\mathbf{G}_1$ .

┘

*Proof:* Take  $\mathbb{D} = B^{\mathrm{op}}$  as in [H-T01]Lemma 1.7.1?.

□

**Prop.(24.12.6.3)[Globalization].** Any  $K \in p\text{-}\mathbf{NField}$  can be realized as  $K = F_w$  for some  $F$  and  $w$  with the setting as in(24.12.6.1) and(24.12.6.1).

┘

*Proof:*

□

**Prop. (24.12.6.4)**[Shimura Varieties]. Situation as in (24.12.6.2), we can regard  $h_0$  as a map  $\mathbb{S} \rightarrow \mathbf{G}_{\mathbb{R}}$ . Then the datum  $(\mathbf{G}, h^{-1})$  defines a tower of Shimura varieties

$$\varprojlim_{\mathcal{K} \subset \mathbf{G}(\mathbf{A}_f)} \mathrm{Sh}_{\mathcal{K}}(\mathbf{G}, h^{-1})$$

with reflex field  $E = \tau(F)$ . ┘

*Proof:* By [Kot92] and [Sch13]P691. □

### Automorphic Vector Bundles

**Thm. (24.12.6.5)**[Matsushima's formula].

$$\mathrm{inj} \lim_U H^i(X_U(\mathbb{C}), \mathcal{L}_{\iota_{\ell}(\xi)}^{\mathrm{top}}) \cong \bigoplus_{\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathbf{G}/\mathbb{Q})} \pi_f \otimes H^i(\mathrm{Lie} \mathbf{G}(\mathbb{R}), U, \pi_{\infty} \otimes \iota_{\ell}(\xi))^{\# \ker^1(\mathbb{Q}, \mathbf{G})}.$$

┘

*Proof:* □

**Def. (24.12.6.6)**[Cohomology of Automorphic Vector Bundles]. For any  $\xi \in \mathrm{Irr}_{\overline{\mathbb{Q}_{\ell}}}(\mathbf{G})$ , we can get a lisse sheaf  $\mathcal{L}_{\xi, \mathcal{K}} \subset \mathrm{Loc}_{\overline{\mathbb{Q}_{\ell}}}^{\mathrm{\acute{e}t}}(\mathrm{Sh}_{\mathcal{K}}(\mathbf{G}))$  for any open compact  $\mathcal{K} \subset \mathbf{G}(\mathbf{A}_f)$  that is small enough. The action of  $\mathbf{G}(\mathbf{A}_f)$  on  $\mathrm{Sh}_{\mathcal{K}}(\mathbf{G})$  extends to  $\mathcal{L}_{\xi, \mathcal{K}}$ , and we can consider the cohomologies

$$H_{\xi}^* = \varinjlim_{\mathcal{K}} H_{\mathrm{\acute{e}t}}^*(\mathrm{Sh}_{\mathcal{K}}(\mathbf{G})_{\overline{F}}, \mathcal{L}_{\xi, \mathcal{K}}) \in \mathrm{Rep}^{(\mathrm{cont}, \mathrm{adm})}(\mathrm{Gal}_F \times \mathbf{G}(\mathbf{A}_f)).$$

Then:

- There is a decomposition  $H_{\xi}^i = \bigoplus_{\pi_f} \pi_f \otimes H_{\xi}^i[\pi_f]$ , where  $\pi_f$  runs over  $\mathrm{Irr}_{\overline{\mathbb{Q}_{\ell}}}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  and  $H_{\xi}^i[\pi_f] \in \mathrm{Rep}_{\overline{\mathbb{Q}_{\ell}}}^{\mathrm{fd}}(\mathrm{Gal}_F)$ .

•

$$\dim H_{\xi}^i[\pi_f] = \# \ker^1(\mathbb{Q}, \mathbf{G}) \sum_{\pi_{\infty}} m_{\tau}(\iota_{\ell}(\pi) \otimes \pi_{\infty}) \dim H^i(\mathrm{Lie} \mathbf{G}(\mathbb{R}), U_{\tau}, \pi_{\infty} \otimes \iota_{\ell}(\xi)),$$

where  $\pi_{\infty}$  runs through  $\mathrm{Irr}(\mathbf{G}(\mathbb{R}))$  and  $m(\pi)$  is the multiplicity of  $\pi$  in  $\mathcal{A}(\mathbf{G}/\mathbb{Q})$ .

- $H_{\xi}^i[\pi_f] \in \mathrm{Rep}_{\overline{\mathbb{Q}_{\ell}}}^{\mathrm{fd}}(\mathrm{Gal}_F)$  is pure of weight  $i + w(\xi)$  and deRham.

•

┘

*Proof:* Cf. [H-T01]P104. ? □

**Prop. (24.12.6.7)**[Kottwitz]. For any  $\pi_f \in \mathrm{Irr}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  and  $\xi \in \mathrm{Irr}_{\overline{\mathbb{Q}_{\ell}}}^{\mathrm{adm}}(\mathbf{G})$ , consider

$$H_{\xi}^*[\pi_f] \in K_0(\mathrm{Rep}(\mathrm{Gal}_F)) \text{ (24.12.6.6)}.$$

Then either  $\pi_f$  only appears in odd dimensions or only appears in even dimensions. Thus  $\pm H_{\xi}^*[\pi_f]$  is a genuine representation.

And there exists  $a(\pi_f) \in \mathbb{N}$  s.t.  $\dim(\pm H_{\xi}^*[\pi_f]) = a(\pi_f)n$  for any  $\xi$ . ┘

*Proof:* Cf. [Kot92] ?. □

### Integral Models

**Prop. (24.12.6.8) [Integral Models for  $\mathbf{G}$ ].** Notation as in (24.12.6.1), if we denote  $D' = \prod_{w'|u, w' \neq w} \mathbb{D}_{w'}$  and  $D = \prod_{w'|u} \mathbb{D}_{w'}$ , and choose a maximal order  $\mathcal{O}_{\mathbb{D}_p}$  for any  $\mathfrak{p} \in S_F(u)$ , then  $\mathcal{O}_D = \mathrm{Mat}(n; \mathcal{O}_K) \times \mathcal{O}_{D'}$ , putting us in the situation of (24.12.4.3). And we have:

$$\mathbf{G}_{\mathbb{Q}_p} = (\mathbf{G}_D)_{\mathbb{Q}_p} \times \mathbf{G}_m = \mathrm{Res}_{K/\mathbb{Q}_p} \mathrm{GL}(n) \times (\mathbf{G}_{D'})_{\mathbb{Q}_p} \times \mathbf{G}_m \text{ (24.12.4.3)}.$$

In particular, this gives an integral model of  $\mathbf{G}_{\mathbb{Q}_p}$  over  $\mathbb{Z}_{(p)}$ .

Define the maximal subgroup

$$\mathcal{K}_p^0 = \mathrm{Mat}(n; \mathcal{O}_K) \times \mathcal{O}_{D'}^* \times \mathbb{Z}_p^* \in \mathbf{G}(\mathbb{Z}_p),$$

and for any  $m \in \mathbb{Z}_+$ , define the congruence subgroups

$$\mathcal{K}_p^m = (1 + \varpi^m \mathrm{Mat}(n; \mathcal{O}_K)) \times (1 + p^m \mathcal{O}_{D'}) \times \mathbb{Z}_p^* \subset \mathbf{G}(\mathbb{Z}_p).$$

┘

**Prop. (24.12.6.9) [PEL-Type Integral Models].** Consider the functor on  $\mathrm{NSch}/\mathcal{O}_K$  mapping  $S \in \mathrm{NSch}/\mathcal{O}_K$  to isomorphism classes of tuples  $(A, \lambda, \iota, \bar{\eta}^p, \bar{\eta}_p)$  where

- $A \in \mathrm{AbVar}/S$  is a projective Abelian scheme of relative dimension  $n^2[F_0 : \mathbb{Q}]$  up to prime-to- $p$  isogeny.
- $\lambda : A \rightarrow A^\vee$  is a polarization of degree prime to  $p$ .
- $\iota$  is a  $*_{D^{\mathrm{op}}}$  homomorphism  $\mathcal{O}_{D^{\mathrm{op}}} \rightarrow \mathrm{End}_S(A)$  satisfying the determinant condition.
- $\bar{\eta}^p$  is a  $\mathcal{K}^p$ -level structure away from  $p$ .
- $\eta_p$  a level- $m$ -structure on  $\underline{H}(A)$  where  $\underline{H}(A) \oplus \underline{H}(A)^\vee \cong T_p(A)$  corresponding to  $\mathcal{O}_{D^{\mathrm{op}}} \otimes \mathbb{Z}_p \cong \mathcal{O}_D^{\mathrm{op}} \times \mathcal{O}_D$ .

and when  $\mathcal{K}^p$  is small, this functor is represented by a projective scheme  $\mathrm{Sh}_{\mathcal{K}^p, m}$ , and the generic fiber of  $\mathrm{Sh}_{\mathcal{K}^p, m}$  is a disjoint union of  $\#\mathrm{III}^1(\mathbb{Q}, \mathbf{G})$  copies of the canonical model  $\mathrm{Sh}_{\mathcal{K}_p^m, \mathcal{K}^p}(\mathbf{G}, \{h^{-1}\})$  localized at  $w$ .

┘

*Proof:* Cf. [H-T01] Section 3.1 and 3.4. □

**Prop. (24.12.6.10) [Group Actions].** For any  $\xi \in \mathrm{Irr}_{\overline{\mathbb{Q}_\ell}}^{\mathrm{adm}}(\mathbf{G})$ , we can associate lisse  $\overline{\mathbb{Q}_\ell}$ -sheaves on  $\mathrm{Sh}_{\mathcal{K}^p, m}$  compatible with the sheaf  $\mathcal{F}_{\xi, \mathcal{K}_p^m, \mathcal{K}^p}$  on the generic fiber.

And there is an obvious action of  $\mathcal{O}_D^* \times \mathbf{G}(\mathbf{A}_f^p)$  on these integral models compatible with these lisse sheaves. ┘

*Proof:* □

## 7 Global methods

**Notation (24.12.7.1).**

- Use notations defined in [Simple Shimura Varieties](#).

┘

### Langlands-Kottwitz Method

**Def. (24.12.7.2).** Define

$$f = h^\vee \times h' \times \mathbf{1}_{\mathbb{Z}_p^*} \times f^p \in C_c^\infty(\mathbf{G}(\mathbf{A}_f))$$

where

$$h \in C_c^\infty(\mathrm{GL}(n; \mathcal{O}_K); \mathbb{Q}), \quad h' \in C_c^\infty(\mathcal{O}_{D'}^*; \mathbb{Q}), \quad f^p = \frac{1}{\mathrm{Vol}(K^p)} \mathbf{1}_{K^p g^p K^p}$$

where  $g^p \in \mathbf{G}(\mathbf{A}_f^p)$  and  $K^p \subset \mathbf{G}(\mathbf{A}_f^p)$  is a sufficiently small compact open subgroup.

Fix  $m \in \mathbb{Z}_+$  s.t.  $h^\vee \times h' \times \mathbf{1}_{\mathbb{Z}_p^*}$  is bi- $\mathcal{K}_p^m$ -invariant. ┘

For  $\tau \in W_K^+$  with  $\deg(\tau) = r$ , we want to evaluate  $\mathrm{tr}(\tau \times f|[H_\xi])$  via Lefschetz trace formula. ?

**Prop. (24.12.7.3).**

$$\pi_* R\psi \mathcal{L}_{\xi, \mathcal{K}^p, m} = (\pi_* R\psi \overline{\mathbb{Q}_\ell}) \otimes \mathcal{L}_{\xi, \mathcal{K}^p, 0}.$$

┘

**Lemma (24.12.7.4).** There is a canonical isomorphism

$$\hat{\mathcal{O}}_{\mathrm{Sh}_{K^p, 0, x}} \cong R_{\underline{H}_y} \otimes \check{\mathcal{O}}_K.$$

┘

*Proof:* Cf. [Sch13]P697. ?

□

**Thm. (24.12.7.5).**

$$\mathrm{tr}(\tau \times h^\vee \times h' \times \mathbf{1}_{\mathbb{Z}_p^*} \times f^p|[H_\xi]) = \sum_{(\gamma_0: \gamma, \delta)} c(\gamma_0: \gamma, \delta) \mathrm{Orb}_\gamma(f^p) \mathrm{TOrb}_{\delta\sigma_0}(\varphi_{\tau, h, h'} \times \mathbf{1}_{p^{-1}W(\kappa_r)^*}) \mathrm{tr}(\gamma_0|\xi).$$

where

- $\gamma_0 \in \mathbf{G}(\mathbb{Q})$  is a semisimple element that becomes elliptic in  $\mathbf{G}(\mathbb{R})$ ,
- $\gamma \in \mathbf{G}(\mathbf{A}_f^p)$  s.t.  $\gamma \sim \gamma_0 \in \mathbf{G}(\overline{\mathbb{Q}_\ell})$  for any  $\ell \in \mathrm{Prime} \setminus \{p\}$ ,
- $\delta \in \mathbf{G}(F)$  with conditions.

┘

*Proof:* Cf. [Sch13]P698.

□

**Cor. (24.12.7.6) [Test Function Trace Formula].**

$$n\mathrm{tr}(\tau \times h^\vee \times h' \times \mathbf{1}_{\mathbb{Z}_p^*} \times f^p|[H_\xi]) = \mathrm{tr}(\mathbf{1} \times f_{\tau, h}^\vee \times h' \times \mathbf{1}_{p^{-r'}\mathbb{Z}_p^*} \times f^p|[H_\xi]).$$

┘

*Proof:* Cf. [Sch13]P698.

□



### Cohomology of Shimura Varieties

**Prop. (24.12.7.7).** Notation as in [Simple Shimura Varieties](#), assume that  $\pi_f \in \mathrm{Irr}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  s.t.  $H_\xi^*(\pi_f) \neq 0$  [\(24.12.6.6\)](#), and in the decomposition  $\pi_p = \pi_w \otimes \pi_p^w \otimes \pi_{p,0}$  corresponding to

$$\mathbf{G}(\mathbb{Q}_p) = \mathrm{GL}(n; F_w) \times (D')^\times \times \mathbb{Q}_p^\times \text{ [\(24.12.6.8\)](#),$$

assume  $\pi_{p,0}$  is unramified and let  $\chi_{\pi_{p,0}} = (\pi_{p,0} \circ \mathrm{Art}_{\mathbb{Q}_p}^{-1})|_{W_K}$  be a quasi-character of  $W_K$ . Then for any  $\tau \in W_K^+$  and  $h \in C_c^\infty(\mathrm{GL}(n; \mathcal{O}_K))$ , we have

$$\mathrm{tr}(f_{\tau,h}^\vee | \pi_w) = \frac{1}{a(\pi_f)} \mathrm{tr}(\tau | [H_\xi[\pi_f]] \otimes \chi_{\pi_{p,0}}) \mathrm{tr}(h^\vee | \pi_w).$$

┘

*Proof:* Take  $h' \in C_c^\infty(\mathcal{O}_{D'}^*; \mathbb{Q})$  s.t.  $\mathrm{tr}(h' | \pi_p^w) = 1$ , and take  $m \in \mathbb{Z}_+$  s.t. both  $h^\vee \times h' \times \mathbf{1}_{\mathbb{Z}_p^*}$  and  $f_{\tau,h}^\vee \times h' \times \mathbf{1}_{p^{-r'}\mathbb{Z}_p}^*$  are both  $\mathrm{bi}\text{-}\mathcal{K}_p^m$ -invariant [\(24.12.6.8\)](#), and  $\pi_p^{\mathcal{K}_p^m} \neq 0$ . And we take  $\mathcal{K}^p \subset \mathbf{G}(\mathbf{A}_f^p)$  s.t.  $(\pi_f^p)^{\mathcal{K}^p} \neq 0$ , and let  $\mathcal{K} = \mathcal{K}^p \mathcal{K}_p^m$ .

Because  $\dim H^*(\mathrm{Sh}_{\mathcal{K}}, \mathcal{L}_{\xi, \mathcal{K}}) < \infty$ , there are only f.m.  $\pi_f \in \mathrm{Irr}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  that is invariant under  $\mathcal{K}$  and  $H_\xi^*[\pi_f] \neq 0$  [?](#). So we can choose  $f^p \in C_c^\infty(\mathbf{G}(\mathbf{A}_f^p))$  bi-invariant under  $\mathcal{K}^p$  s.t.  $\mathrm{tr}(f^p | \pi_f^p) = 1$ , and satisfying the following: whenever  $\pi'_f \in \mathrm{Irr}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  is  $\mathcal{K}$ -spherical and  $\mathrm{tr}(f^p | \pi'^p_f) \neq 0$ , we have  $\pi'^p_f \cong \pi_f^p$ , thus also  $\pi_f \cong \pi'_f$  by [\(24.12.7.12\)](#).

Applying [\(24.12.7.6\)](#), we get

$$n \mathrm{tr}(\tau | [H_\xi^*[\pi_f]]) \mathrm{tr}(h^\vee | \pi_w) = na(\pi_f) \mathrm{tr}(f_{\tau,h}^\vee | \pi_w) \pi_{p,0}(p^{-r'}),$$

and the assertion follows. □

**Thm. (24.12.7.8) [Globalization, Harris-Taylor].** If  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n)/K)$  that is either essentially square-integrable or generalized Speh [\(24.11.4.16\)](#), then there exists  $\pi_f \in \mathrm{Irr}^{\mathrm{adm}}(\mathbf{G}(\mathbf{A}_f))$  and  $\xi \in \mathrm{Irr}_{\mathbb{Q}_\ell}(\mathbf{G})$  s.t. (notation as in [\(24.12.7.7\)](#)):

- $H_\xi^*(\pi_f) \neq 0$  [\(24.12.6.6\)](#).
- $\pi_{p,0}$  is unramified.
- $\pi_w$  is an unramified twist of  $\pi$ .

┘

*Proof:* Cf. [\[H-T01\] 6.2.5 and 6.2.11](#) [?](#). □

**Cor. (24.12.7.9) [A variant Globalization].** If  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(\mathrm{GL}(n)/K)$  is essentially square-integrable, then there exists  $\Pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$  s.t.

- $\Pi^\vee \cong \Pi^c$ .
- $\Pi$  is regular  $C$ -algebraic [\(30.1.3.2\)](#).
- $\Pi_x$  is supercuspidal.
- $\Pi_w$  is an unramified twist of  $\pi$ .

┘

*Proof:* Cf. [\[H-T01\] Cor. 6.2.6](#) [?](#). □

**Prop. (24.12.7.10) [Clozel's Base Change].** Suppose  $\pi \in \text{Irr}^{\text{cusp}}(\mathbf{G}/\mathbb{Q})$  is cohomological for  $\xi' \in \text{Irr}_{\mathbb{C}}(\mathbf{G})$ , then there is a unique  $\text{BC}(\pi) = (\psi, \Pi) \in \text{Irr}^{\text{auto}}(\text{Res}_{\mathcal{K}/\mathbb{Q}} \mathbf{G}_m \times \text{Res}_{F/\mathbb{Q}} D^{\times})$  s.t.

- $\psi = \psi_{\pi}|_{\mathbf{I}_{\mathcal{K}}}^{\mathbf{c}}$ .
- If  $p \in \text{Prime}$  is split in  $\mathcal{K}$ , then  $\text{BC}(\pi)_p = \text{BC}(\pi_p)$ .
- For a.e.  $p \in \text{Prime}$  that is inert in  $\mathcal{K}$ , we have  $\text{BC}(\pi_p) = \text{BC}(\pi)_p$ .
- $\Pi$  is cohomological for  $\xi'_{\mathcal{K}}?$ .
- $\psi|_{\mathcal{K}_{\infty}^{\times}}^{\mathbf{c}} = \xi'|_{\mathcal{K}_{\infty}^{\times}}^{-1}$ .
- $\psi_{\Pi}|_{\mathbf{I}_E} = \psi^{\mathbf{c}}/\psi$ .

┘

*Proof:* Cf. [H-T01]6.2.1.?

□

**Cor. (24.12.7.11).** It follows from strong multiplicity one (29.3.6.7) and the theorem that: if  $\pi, \pi' \in \text{Irr}^{\text{auto}}(\mathbf{G}/\mathbb{Q})$  are cohomological for  $\xi' \in \text{Irr}_{\mathbb{C}}(\mathbf{G})$  and  $\pi_p \cong \pi'_p$  for a.e.  $p \in \text{Prime}$ , then  $\pi_p \cong \pi'_p$  for any  $p \in \text{Prime}$  that is split in  $\mathcal{K}$ .

┘

**Cor. (24.12.7.12) [Harris-Taylor].** If  $\pi, \pi' \in \text{Irr}^{\text{adm}}(\mathbf{G}(\mathbf{A}_f))$  and  $\xi \in \text{Irr}_{\overline{\mathbb{Q}_{\ell}}}(\mathbf{G})$  s.t.

- $\pi^p \cong (\pi')^p$ .
- $[H_{\xi}^*[\pi_f]] \neq 0$ .
- $[H_{\xi}^*[\pi'_f]] \neq 0$ .

Then  $\pi \cong \pi'$ .

┘

*Proof:* It follows from (24.12.6.6) that both  $\pi, \pi'$  are automorphic and cohomological for  $\iota_{\ell}(\xi)$ . Then the assertion follows from (24.12.7.11), noticing that  $p$  is split in  $\mathcal{K}$  (24.12.6.1). □

**Prop. (24.12.7.13) [Clozel's Descent].** Suppose  $\Pi \in \text{Irr}^{\text{auto}}(D^{\times}/F)$  and  $\psi$  is a Hecke character of  $\mathcal{K}$  s.t.

- $\Pi \cong \Pi^*$ .
- $\psi_{\Pi}|_{\mathbf{I}_{\mathcal{K}}} = \psi^{\mathbf{c}}/\psi$ .
- $\Pi$  is cohomological for  $\xi'_{\mathcal{K}}$ .
- $\xi'|_{\mathcal{K}_{\infty}^{\times}}^{-1} = \psi|_{\mathcal{K}_{\infty}^{\times}}^{\mathbf{c}}$ .

Then there is a  $\pi \in \text{Irr}^{\text{auto}}(\mathbf{G}/\mathbb{Q})$  s.t.

- $\text{BC}(\pi) = (\psi, \Pi)$ .
- $\pi$  is cohomological for  $\xi'$ .
- $\dim H_{\iota_{\ell}^{-1}(\xi')}^{-1}[\iota_{\ell}^{-1}\pi_f] \neq 0$ .

┘

*Proof:* Cf. [H-T01]6.2.9.?

□

**Prop. (24.12.7.14) [Descending from  $D^{\times}$  to  $\mathbf{G}$ ].** Let  $\Pi \in \text{Irr}^{\text{auto}}(D^{\times})$  s.t.  $\Pi \cong \Pi^*$  and  $\Pi$  is regular  $C$ -algebraic, then there exists a Hecke character  $\psi$  of  $\mathcal{K}$  and  $\xi' \in \text{Irr}(\mathbf{G}/\mathbb{C})$  s.t.

- $\psi_{\Pi}|_{\mathbf{I}_{\mathcal{K}}} = \psi^{\mathbf{c}}/\psi$ .
- $\Pi$  is cohomological for  $\xi'_{\mathcal{K}}?$ .

- $\xi'|_{\mathcal{K}_\infty^\times}^{-1} = \psi|_{\mathcal{K}_\infty^\times}^c$ .
- $\psi(\mathcal{O}_{\mathcal{K},x_0}^\times) = 1$ .

┘

*Proof:* Cf. [H-T01] Lemma 6.2.10. ?

□

**Thm. (24.12.7.15)** [Jacquet-Langlands plus Base Change, Harris-Taylor]. Let  $\Pi \in \text{Irr}^{\text{cusp}}(GL(n)/F)$  s.t.

- $\Pi^\vee \cong \Pi^c$ .
- $\Pi$  is regular  $C$ -algebraic (30.1.3.2).
- $\Pi_x$  is square-integrable.

Then there exists

- Some  $\pi_f \in \text{Irr}^{\text{adm}}(\mathbf{G}(\mathbf{A}_f))$ ,
- Some  $\xi \in \text{Rep}^{\text{fd}}(\mathbf{G})$  s.t.  $H_\xi^*[\pi_f] \neq 0$  (24.12.6.6).
- Some algebraic Hecke character  $\psi$  of  $\mathcal{K}$  (19.4.1.27) satisfying the following: For any  $w \in \Sigma_F^{\text{fin}}$  with  $u = w \cap \mathcal{K}$  and  $p' = w \cap \mathbf{Q}$  s.t.  $p'$  is split in  $\mathcal{K}$  and  $w \notin \{x, x^c\}$ , we have

$$\pi_w \cong \Pi_w, \quad \pi_{p',0} \cong \psi_u \text{ (24.12.7.7)}.$$

Moreover, for any  $p' \in \text{Prime}$  that is split in  $\mathcal{K}$ , we can arrange  $(\pi_f, \xi, \psi)$  s.t.  $\pi_{p',0}$  is unramified.

┘

*Proof:* Use global Jacquet-Langlands correspondence (29.3.6.7) to obtain  $\rho \in \text{Irr}^{\text{auto}}(\mathbb{D}^\times)$  s.t.  $J\text{-L}(\rho) \cong \Pi$  (24.12.6.2), then  $\rho \cong \rho^*$  and  $\rho_\infty$  is regular  $C$ -algebraic. Then (24.12.7.13) (24.12.7.14) and the properties of the base-change map (24.12.7.10) finish the proof. □

**Cor. (24.12.7.16)** [Theorem 10.6 of Scholze, Deligne-Brylinski (86)/Carayol (86)/Harris-Taylor [H-T01]]. Let  $\Pi \in \text{Irr}^{\text{cusp}}(GL(n)/F)$  s.t.

- $\Pi^\vee \cong \Pi^c$ .
- $\Pi$  is regular  $C$ -algebraic (30.1.3.2).
- $\Pi_x$  is square-integrable.

Then there exists  $a \in \mathbb{Z}_+$  and  $R(\Pi) \in \text{Rep}_{\mathbb{Q}_\ell}^{\text{an}}(\text{Gal}_K)$  s.t. for any  $v \in \Sigma_F^{\text{fin}} \setminus S(\ell)$ , we have

$$R(\Pi)|_{W_{F_v}} = a\sigma_{n,v}(\Pi_v) \text{ (24.12.3.2)}.$$

Moreover, for each  $v \in \Sigma_F^{\text{fin}}$ , the representation  $\Pi_v$  is tempered.

┘

*Proof:* This is Theorem C of Harris-Taylor. ?

Take  $\pi_f, \xi, \psi$  as in (24.12.7.15),  $a = a(\pi_f)$  (24.12.7.7). Take  $\chi_\psi$  be a character of  $\text{Gal}_F$  corresponding to  $\psi$  under the global class field theory. And let

$$R(\Pi) = [H_\xi^*(\pi_f)]^\vee \otimes \chi_\psi^{-1}.$$

Then the assertion is trivial if  $v \in \Sigma_F^{\text{fin}} \setminus (S_F(\ell) \cup \{x, x^c\})$  is split over  $F_0$  and  $\pi_{p',0}$  is unramified, where  $p' = v \cap \mathbf{Q}$ . ? Then by Chebotarev density theorem, this determines  $\frac{1}{a}R(\Pi)$  uniquely. ? But for any such  $p' \in \text{Prime}$ , we can arrange that  $\pi_{p',0}$  is unramified, so this condition can be dropped.

Now for an arbitrary  $v$ , we can take a real quadratic field  $R$  that is linearly disjoint with  $F$  s.t. the rational prime under  $x$  splits in  $R$ , and the completion of  $R$  and  $\mathcal{K}$  at the rational prime under  $v$  are isomorphic. Let  $\tilde{F}_0 = F_0 R$  and  $\tilde{F} = F R$ , and choose  $\tilde{v}$  and  $\tilde{x}$  above  $v$  and  $x$  in  $\tilde{F}$  s.t.  $\tilde{v} \notin \{\tilde{x}, \tilde{x}^c\}$ .

Then we can take  $\tilde{\Pi} \in \text{Irr}^{\text{auto}}(\text{GL}(n)/\tilde{F})$  to be the cyclic base change of  $\Pi$ , which continues to have the properties listed. Hence we get an  $R(\tilde{\Pi})$  that satisfies the assertion for any  $\tilde{v} \in \Sigma_F^{\text{fin}} \setminus (S_{\tilde{F}}(\ell) \cup \{\tilde{x}, \tilde{x}^c\})$  split over  $\tilde{F}_0$ . By Chebotarev density theorem,  $\frac{1}{a}R(\tilde{\Pi}) = \frac{1}{a}R(\Pi)|_{\text{Gal}_{\tilde{F}}}$ . Then applying the above split case to  $\tilde{v}$  finishes the proof.

Using  $p$ -adic Hodge theory, it would be no problem to show that one can choose  $a = 1$ , cf. proof of Proposition VII.1.8 in [H-T01]. Also, this holds for any CM field  $F$ , cf. proof of Theorem VII.1.9 of [H-T01]. ?  $\square$

**Cor. (24.12.7.17).** Let  $\Pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  s.t.

- $\Pi^\vee \cong \Pi^c$ .
- $\Pi(\frac{n-1}{2})$  is regular  $C$ -algebraic(30.1.3.1).
- $\Pi_x$  is square-integrable for some  $x \in \Sigma_F^{\text{fin}}$  that is split over  $F_0$ .

Then there exists  $a \in \mathbb{Z}_+$  and  $R(\Pi) \in \text{Rep}_{\mathbb{Q}_\ell}^{\text{an}}(\text{Gal}_K)$  s.t. for any  $v \in \Sigma_F^{\text{fin}} \setminus S(\ell)$ , we have

$$R(\Pi)|_{W_{F_v}} = a \text{rec}'_v(\Pi_v)$$

┘

*Proof:* It suffices to find a Hecke character  $\chi$  of  $F$  s.t.  $\chi^{-1} = \chi^c$  and  $\chi_\infty(\frac{n-1}{2})$  is  $C$ -algebraic, which can be done as in the proof of Corollary 7.2.8 of [H-T01]. ?  $\square$

## 8 Lubin-Tate Space

**Prop. (24.12.8.1)[Relation to Lubin-Tate Towers].** Let  $\rho \in \text{Irr}^{\text{adm}}(D_{K,1/n}^\times)$  s.t.  $\text{J-L}(\rho)$  is supercuspidal, then

$$[R\psi][\rho] \cong (-1)^{n-1} \pi^\vee|_{\text{GL}(n; \mathcal{O}_K)} \times \sigma_n(\pi)$$

as a virtual representation of  $\text{GL}(n; \mathcal{O}_K) \times W_K$ . ┘

*Proof:* ?  $\square$

**Cor. (24.12.8.2).** For any  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n; K))$ ,  $\sigma_n(\pi)$  is a genuine representation of  $W_K$ . ┘

*Proof:* By(24.11.3.19), there exists an irreducible representation of  $\text{GL}(n; \mathcal{O}_K)$  that occurs in  $\pi^\vee|_{\text{GL}(n; \mathcal{O}_K)}$  with multiplicity 1. Then by(24.12.8.1),  $\square$

**Prop. (24.12.8.3) [Bijective Correspondence for Supercuspidal Representations].** The correspondences  $\text{rec}'_n$  defined in(24.12.3.2) is a functorial extension of class field theory(24.12.3.7). In particular, it induces a bijection

$$\text{rec}'_n : \text{Irr}^{\text{cusp}}(\text{GL}(n; K)) \cong \text{Rep}^n(W_K)$$

for any  $n \in \mathbb{Z}_+$ . ┘

*Proof:* ? Cf.[Sch13]P705.  $\square$

## 9 Inertial Local Langlands Correspondence

**Prop. (24.12.9.1) [Inertial Correspondence].** Under the local langlands correspondence (24.12.1.2), if  $\pi, \pi' \in \text{Irr}^{\text{adm}, \text{generic}}(GL(n))$ ,

- $\pi|_{\mathcal{K}} \cong \pi'|_{\mathcal{K}}$  iff  $\text{rec}_n(\pi)|_{I_K} \cong \text{rec}_n(\pi')|_{I_K}$ .
- (Paskunas) Let  $\tau$  be a Weil-Deligne inertia type of  $I_K$  that extends to an  $n$ -dimensional irreducible F-semisimple representation of  $W_K$ , then there exists a unique  $\sigma_\tau \in \text{Irr}^{\text{alg}}(K)$  s.t. for any infinite-dimensional  $\pi \in \text{Irr}^{\text{adm}}(GL(n; K))$ ,  $\pi|_{\mathcal{K}}$  contains  $\sigma_\tau$  iff  $\text{rec}_n(\pi)|_{I_K} \cong \sigma$ .

┘

*Proof:*

□

### Types

References are [Paskunas, Unicity of types for supercuspidal representations of  $GL_N$ ] and [Bushnell, The Admissible Dual of  $GL(N)$  via Compact Open Subgroups].

### Cuspidal Representations

### Simple Characters

### Tame Lifting

## 10 Local Langlands for Laurent Series

References are [Laumon-Rapoport-Stuhler,  $\mathcal{D}$ -elliptic sheaves and the Langlands correspondence].



## 25 | Galois Representations

### 25.1 Galois Representations(Basics)

References are [Theories of  $p$ -adic Galois Representations, Fontaine/Yi Ouyang] and [Tay04, Con06, Yao17], and [Abelian  $\ell$ -adic Representations and Elliptic Curves, Serre].

**Notation(25.1.0.1).**

- Use notations on representations defined in [Smooth Representations](#).
- Use notations on positive characteristic representations defined in [Positive Characteristic Representations](#).
- Use notations on Galois cohomology defined in [Cohomology of Arithmetic Fields](#).

┘

**Remark(25.1.0.2).** For  $K \in \mathbf{Field}$ ,  $E \in \mathbf{Field}^{\text{top}}$ , the theory of Galois representations study the category  $\mathcal{R}\text{ep}_E^{\text{fd}}(\text{Gal}_K)$ .

┘

#### 1 Basics

**Def.(25.1.1.1)[Galois Representations].** For any  $K \in \mathbf{Field}$ ,  $p \in \mathbf{Prime}$ ,

- an **Artin representation** of  $K$  is a f.d. smooth  $\mathbb{C}$ -representation of  $\text{Gal}_K$ .
- a  **$p$ -adic Galois representation** of  $K$  is a f.d. continuous  $\mathbb{Z}_p$ -representation of  $\text{Gal}_K$ . If  $K \in p\text{-LField}$ , this is called a  **$p$ -adic local Galois representation**(of  $K$ ).
- a  **$p$ -adic integral Galois representation** of  $K$  is a f.d. continuous  $\mathbb{Q}_p$ -representation of  $\text{Gal}_K$ . If  $K \in p\text{-LField}$ , this is called a  **$p$ -adic local integral Galois representation**(of  $K$ ).
- if  $K \in p\text{-LField}$ ,  $\ell \in \mathbf{Prime} \setminus \{p\}$ , then an  **$\ell$ -adic Galois representation** of  $K$  is a f.d. continuous  $\mathbb{Q}_\ell$ -representation of  $\text{Gal}_K$ .

┘

**Cor.(25.1.1.2)[Artin Representations as  $\ell$ -adic(or  $p$ -adic) Representations].** A choice of isomorphism(19.1.2.18)  $\overline{\mathbb{Q}}_p \cong \overline{\mathbb{Q}}_\ell \cong \mathbb{C}$  induces a bijection between Artin representations and  $\ell$ -adic(or  $p$ -adic) representations with open kernels.

┘

**Prop.(25.1.1.3)[Brauer-Nesbitt].** If two  $n$ -dimensional representations have the same char polynomial and  $\text{char } k = 0$ , or  $\text{char } k > n$  and they have the same character, then their semisimplification are the same.

┘

*Proof:* The proof is not hard, use the Artin-Wedderburn theorem, and the fact the representation may not be semisimple.  $\square$

**Def. (25.1.1.4) [Restricted Local Galois Representations].** By (19.3.1.24), if  $F \in \mathbf{GField}$ , for any  $v \in \Sigma_F$ , a Galois representation  $\rho$  of  $F$  can be restricted to a Galois representations  $\rho_v$  of  $\mathrm{Gal}_{F_v}$ , called the **restricted local Galois representation**.  $\lrcorner$

**Def. (25.1.1.5) [Cyclotomic Characters and Tate Twists].** For  $K \in \mathbf{Field}$  and  $p \in \mathbf{Prime}$ , the  $p$ -adic **cyclotomic character**  $\chi_p$  of  $\mathrm{Gal}_K$  is the character corresponding to the 1-dimensional  $p$ -adic Tate module of  $\mathbb{G}_{a,K}$ .

For any  $(\rho, V) \in \mathrm{Rep}_{\mathbb{Z}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$  or  $(\rho, V) \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$ , we can define the **Tate twist**  $\rho(n)$  for any  $n \in \mathbb{Z}$  as the representation twisted by  $n$ -th power of the cyclotomic character  $\chi_p^n$ .  $\lrcorner$

**Def. (25.1.1.6) [Tate Duals].** For any  $(\rho, V) \in \mathrm{Rep}_{\mathbb{Z}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$  or  $(\rho, V) \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$ , the **Tate dual**  $V^D$  of  $V$  is defined to be  $V^D = V^\vee(1)$ .  $\lrcorner$

## 2 $\ell$ -adic Local Galois Representations

**Def. (25.1.2.1) [Notations].**

- Let  $p \in \mathbf{Prime}$ ,
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-LField}$  with  $\#\kappa = q$ .
- Let  $\ell \in \mathbf{Prime} \setminus \{p\}$ .
- $t_\ell : I_K/R_K \rightarrow \mathbb{Z}_\ell$  be the additive  $\ell$ -adic tame character (19.1.5.25).
- Let  $\sigma_K \in \mathrm{Gal}_K$  be a fixed geometric Frobenius element.
- Let  $\chi = \chi_\ell$  be the unramified cyclotomic character (25.1.1.5)  $\widehat{\mathbb{Z}} \cong \mathrm{Gal}_K/I_K \rightarrow \mathbb{Z}_\ell : \sigma_K \mapsto q$ .  $\lrcorner$

**Def. (25.1.2.2) [Properties of Galois Representations].** For  $A \in \mathbf{TopCRing}$  and  $\rho \in \mathrm{Rep}_A^{\mathrm{fd}}(\mathrm{Gal}_K)$ ,

- $\rho$  is called an **odd representation** if  $\det \rho(c) = -1$ , and an **even representation** otherwise.
- $\rho$  is said to be an **unramified Galois representation** if  $\rho(I_K) = 1_A$ .
- $\rho$  is said to be a **flat Galois representation** if for any Artinian quotient  $A/I$  of  $A$ , the quotient representation  $\bar{\rho}|_{\mathrm{Gal}_K}$  is isomorphic to the representation of  $\mathrm{Gal}_K$  on the geometric points of a finite flat group scheme over  $\mathcal{O}_K$ .
- If  $\rho \in \mathrm{Rep}_{\mathbb{Q}_\ell}^2(\mathrm{Gal}_K)$ , then it is said to be a **Steinberg-type Galois representation** if its corresponding Weil-Deligne representation (25.1.2.26) is isomorphic to  $(\chi_\ell \otimes \mathrm{id}, \mathbb{Q}_\ell^2, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix})$ .  $\lrcorner$

**Prop. (25.1.2.3).** If  $(\rho, V) \in \mathrm{Rep}^n(\Gamma)$ , then it has a filtration  $0 = V_0 \subset V_1 \subset \dots \subset V_n = V$  where  $V_{i+1}/V_i$  is irreducible, then there is a semisimplification of  $\rho$ , which is  $\rho^{\mathrm{ss}} = \oplus V_{i+1}/V_i$ .  $\lrcorner$

**Prop. (25.1.2.4) [Induced Galois Representations].** Let  $n \in \mathbb{Z}_+$  s.t.  $p \nmid n$ , then any  $(\rho, V) \in \mathrm{Irr}^n(\mathrm{Gal}_K)$  is induced from a character  $\chi$  of  $L^\times$  for a field extension  $L/K$  of degree  $n$ .  $\lrcorner$

*Proof:* Suppose  $\chi$  factors through a normal subgroup  $\mathrm{Gal}(L/K)$ . It suffices to show  $\rho$  is an induced representation from a subgroup, then use induction. Suppose it is not, then consider the filtration of Galois groups  $R_{L/K} \subset I_{L/K} \subset \mathrm{Gal}(L/K)$ .  $R_v$  is a pro- $p$ -group, thus all irreducible f.d. representations are  $p$ -powers, by (24.1.3.17), thus it is 1-dimensional. Now  $R_v$  is a normal subgroup, by (24.1.2.15),  $R_v$  acts by scalars on  $V$ . Now the exact sequence  $1 \rightarrow R_v \rightarrow I_v \rightarrow I_v/R_v \rightarrow 0$  is a



split exact sequence(19.1.4.18), and  $I_v/R_v$  is cyclic, thus  $\rho|_{I_v}$  also contains a character, and the same argument as above shows  $I_v$  acts by characters. Then a twist of  $\rho$  factors through  $G/I_v$ , which is a cyclic group, contradiction.  $\square$

### Artin and Swan Conductors

**Def.(25.1.2.5) [Local Artin and Swan Conductors, Artin[Art31]].** For  $(\rho, V) \in \text{Rep}^{\text{fd}}(\text{Gal}_K)$ , define the **Artin conductor(local)**

$$a(\rho) = a(\chi_\rho) = \sum_{i \in \mathbb{N}} \frac{1}{[G_0 : G_i]} \dim(V/V^{G_i})$$

and the **Swan conductor(local)**

$$b(\rho) = b(\chi_\rho) = \sum_{i \in \mathbb{Z}_+} \frac{1}{[G_0 : G_i]} \dim(V/V^{G_i}).$$

where  $G_i$  are the higher ramification groups of  $G$ (19.1.4.20).  $\lrcorner$

**Prop.(25.1.2.6) [Functoriality of Artin  $L$ -Factors, Artin[Art31]].** The Artin and Swan conductors(25.1.2.5) satisfy the following functorial properties:

- $L_v(F, \mathbf{1}; s) = L_v(\mathbf{1}; s)$ (31.1.1.1).
- For  $\rho, \rho' \in \text{Rep}^{\text{fd}}(\text{Gal}_K)$ , we have

$$a(\rho \oplus \rho') = a(\rho) + a(\rho'), \quad b(\rho \oplus \rho') = b(\rho) + b(\rho')$$

- If  $K \subset_{\text{fin}} M$ ,  $\rho \in \text{Rep}^{\text{fd}}(\text{Gal}_M)$ , then

$$a(\text{Ind}_K^M \rho) = v(\mathfrak{d}_{M/K}) \dim \rho + a(\rho) \cdot f_{M/K}.$$

$\lrcorner$

*Proof:* ?  $\square$

**Prop.(25.1.2.7) [Conductor-Discriminant Formula].** For  $L/F \in \text{GalField}^{\text{fin}}$ ,

$$\text{disc}(L/F) = \prod_{\rho \in \text{Irr}(\text{Gal}(L/F))} \mathfrak{f}(F, \rho)^{\dim \rho}.$$

$\lrcorner$

*Proof:* Cf.[Neu99]P534.  $\square$

### Weil Representations

References are [Wed08, Yao17, G-R10].

**Prop.(25.1.2.8) [Primitive Weil Representation].**  $\pi \in \text{Rep}^{\text{fd}}(W_K)$  is called **primitive** if it is not induced from a representation of a proper subgroup. Then for any primitive representation of  $W_F$ , it factors through  $C_{E/F}$  for some finite extension  $E/F$  by continuity, and by Clifford's theorem(24.1.2.15), the normal subgroup  $C_E$  acts by scalars, so the image of  $W_{E/F}$  in  $\text{GL}(V)$  is a finite group.  $\lrcorner$

**Def. (25.1.2.9) [Twisting Character of  $W_K$ ].** For  $s \in \mathbb{C}$ , define the quasi-character  $\omega_s : W_F \rightarrow \mathbb{C}^\times$  :  
 $x \mapsto \|x\|^s$ .  $\lrcorner$

**Prop. (25.1.2.10) [Twisting of Weil Representation, Deligne].**  $\rho \in \text{Irr}(W_K)$  factors through  $\text{Gal}_K$  iff  $\#\rho(W_K) < \infty$ .

For any  $\rho \in \text{Irr}(W_K)$ , there exists some  $s \in \mathbb{C}$  that  $\omega_s \otimes \rho$  factors through  $W_K \rightarrow \text{Gal}_K$ .  $\lrcorner$

*Proof:*  $\ker(\rho|_{I_K})$  is an open subgroup of  $I_K$ . Conjugation by  $\sigma$  induces a permutation on the finite subgroup  $\rho(I_K)$ , so there is some  $n$  that  $\rho^n = \text{id}$  on  $\rho(I_K)$ , which means  $\rho(\sigma^n)$  commutes with each elements of  $W_K$ , thus it is a scalar by Schur's lemma. Thus we can chose  $s$  that  $(\omega_s \otimes \rho)(\sigma^n) = 0$ , then  $\omega_s \otimes \rho$  is trivial on  $U = \langle \sigma^n, \ker(\rho|_{I_K}) \rangle$ , which is a subgroup of  $W_F$  of finite index. Thus the image is finite.

Also the closed subgroup  $\overline{U} = \overline{\langle \sigma^n, \ker(\rho|_{I_K}) \rangle} \subset \text{Gal}_K$  if of finite index, and  $\overline{U} \cap W_K = U$ , thus  $W_K/U \cong \text{Gal}_K/\overline{U}$ , and this representation extends to  $\text{Gal}_K$ .  $\square$

**Cor. (25.1.2.11) [Galois-Type Representations].** As  $W_K$  is dense in  $\text{Gal}_K$ ,  $\text{Rep}^{\text{fd}}(\text{Gal}_K) \subset \text{Rep}^{\text{fd}}(W_K)$  as a full subcategory. The representations in this subcategory are called **representations of Galois-type**.

(25.1.2.10) shows  $\text{Irr}(\text{Gal}_K)$  is almost the same as  $\text{Irr}(W_K)$ .  $\rho \in \text{Irr}(W_K)$  is of Galois-type iff it has a finite image.  $\lrcorner$

**Prop. (25.1.2.12) [Semisimple Weil Representations].** Let  $L \in \text{Field}$ ,  $\overline{L} = L$  and  $\rho \in \text{Rep}_L^{\text{fd}}(W_K)$  that  $\#\rho(I_K) < \infty$ , the following are equivalent:

- $\rho(\sigma)$  is semisimple for any  $\sigma \in W_K$ .
- $\rho(\sigma_K)$  is semisimple.
- $\rho$  is a semisimple representation.

$\lrcorner$

*Proof:* Because  $\langle \rho(\sigma_K) \rangle$  is a subgroup of  $\rho(W_K)$  of finite index, 1 and 3 are equivalent by (24.1.2.16).

Notice that for any  $x \in W_K$ , there exists  $n \in \mathbb{Z}_+$  s.t.  $\rho(x^n)$  lies in the center of  $\rho(W_K)$ : ?

Then there exists ? Cf. [Yao17]P4.  $\square$

**Def. (25.1.2.13) [ $\ell$ -Integral Representations].** For  $\rho \in \text{Rep}_{\mathbb{Q}_\ell}^{\text{fd}}(W_K)$ , by (25.1.2.10), the eigenvalues of  $r$  are  $\ell$ -adic units iff the eigenvalues of  $\rho(\sigma) \in \mathcal{O}_{\mathbb{Q}_\ell}^*$ , iff the characteristic polynomial of  $\rho(\sigma)$  are in  $\mathcal{O}_{\mathbb{Q}_\ell}[T]$  and  $|\det(\rho(\sigma))| = 1$ . Such a representation is called an  **$\ell$ -integral representation**.  $\lrcorner$

**Def. (25.1.2.14) [Types of 2-Dimensional Representations].** By (25.1.2.8), any  $\rho \in \text{Rep}^2(W_F)$  has finite image in  $\text{PGL}(2; \mathbb{C}) \cong \text{SO}(3; \mathbb{R})$  (16.6.1.13). Thus by conjugacy into  $\text{SO}(3; \mathbb{R})$ , the image is isomorphic to a cyclic, dihedral, tetrahedral, octahedral or icosahedral group by (15.4.1.8), called the **type of Artin representation**.  $\lrcorner$

**Def. (25.1.2.15) [Dihedral Weil Representation].** A **dihedral Weil representation** is a representation  $W_F \rightarrow \text{GL}(2; \mathbb{C})$  that is induced from a quasi-character of  $W_E$  for some quadratic extension  $E/F$ . (What's the relation with (25.1.2.14)?)  $\lrcorner$

**Prop. (25.1.2.16) [Induced Weil Representations].** If  $(p, n) = 1$  and  $K \in p\text{-NField}$ , then any  $\rho \in \text{Irr}^n(W_K)$  is induced from a quasi-character of  $W_L$  (equivalently  $E^\times$ ) where  $L/K \in p\text{-LField}$ ,  $[L : K] = n$ .  $\lrcorner$

*Proof:* By (24.7.4.4), the class of such induced extensions are stable under twisting by  $\omega_s$ , thus using (25.1.2.10), we can assume that the representation factors through  $\text{Gal}_F$ , thus through  $\text{Gal}(K/F)$  for some finite extension  $K/F$ , thus the assertion follows from (25.1.2.4).  $\square$

### Weil-Deligne Representations

References are [G-R10].

**Def. (25.1.2.17)[Weil-Deligne Groups].** The **Weil-Deligne group**  $\mathrm{WD}_K$  is the group scheme  $W_K \rtimes \mathbb{G}_a \in \mathrm{GrpSch}/\mathbb{Q}$  given by the action

$$wxw^{-1} = |w|x.$$

┘

**Def. (25.1.2.18)[Weil-Deligne Representations, Weil[Wei51]/Deligne].** For  $L \in \mathrm{Field}^0$ , the category  $\mathfrak{wd}_L(W_K)$  of **Weil-Deligne representations** of  $W_K$  over  $L$  consists of triples  $(\rho, V, N)$ , where  $(\rho, V) \in \mathrm{Rep}_L^{\mathrm{alg}, \mathrm{fin}}(W_K)$  and  $N \in \mathrm{End}(V)$  s.t.

$$\mathrm{Ad}(\rho(x))N = |x|.N, \text{ where } x \in W_K.$$

Equivalently, a Weil-Deligne representation is a  $(\rho, V) \in \mathrm{Rep}_L^{\mathrm{alg}, \mathrm{fin}}(W_K)$  together with a map  $N : V \mapsto V(1) \in \mathrm{Rep}_L^{\mathrm{alg}, \mathrm{fin}}(W_K)$  (25.1.1.5).

Equivalently, a Weil-Deligne representation is a representation of the group scheme  $\mathrm{WD}_K$  (25.1.2.17) over  $L$ , by ?.

$\mathfrak{wd}_{\mathbb{C}}(W_K)$  is also denoted by  $\mathfrak{wd}(W_K)$ .

┘

**Prop. (25.1.2.19).** For  $L \in \mathrm{Field}^0$  and  $(\rho, V, N) \in \mathfrak{wd}_L(W_K)$ ,  $N$  is nilpotent.

And  $(\rho, V, N)$  is irreducible iff  $N = 0$  and  $\rho$  is irreducible.

┘

*Proof:* Because  $q^{-1} \cdot N = \sigma N \sigma^{-1} \cong N$ , if  $\lambda$  is an eigenvalue of  $N$ , then  $\lambda/p$  is also an eigenvalue of  $N$ . So  $N$  has no non-zero eigenvalues, thus nilpotent. The last assertion follows from the fact  $\ker N$  is stable under  $W_K$ .  $\square$

**Remark (25.1.2.20).**  $\ell$ -adic Weil-Deligne representations are exactly the continuous  $\ell$ -adic representations of  $W_K$ , as will be illustrated in (25.1.2.26).  $\square$

**Def. (25.1.2.21).** Define tensor product and inner Homs:  $(\rho_1, V_1, N_1) \otimes (\rho_2, V_2, N_2) = (\rho_3, V_3, N_3)$  where

$$V_3 = V_1 \otimes V_2, \quad \rho_3(x)(v_1 \otimes v_2) = \rho_1(x)v_1 \otimes \rho_2(x)v_2, \quad N_3(v_1 \otimes v_2) = N_1(v_1) \otimes v_2 + v_1 \otimes N_2(v_2).$$

and also  $\mathrm{Hom}((\rho_1, V_1, N_1), (\rho_2, V_2, N_2)) = (\rho_3, V_3, N_3)$  where

$$V_3 = \mathrm{Hom}(V_1, V_2), \quad (\rho_3(x)\varphi)(v_1) = \rho_2(x)(\varphi(\rho_1(x)^{-1}(v_1))), \quad (N_3\varphi)(v_1) = N_2(\varphi(v_1)) - \varphi(N_1(v_1)).$$

And also the dual  $\rho^\vee = \mathrm{Hom}(\rho, \mathbf{1})$ .

┘

**Def. (25.1.2.22) [Properties of Weil-Deligne Representations].**  $\rho = (\rho_0, V, N) \in \mathfrak{wd}_L(W_K)$  is called

- **Frobenius-semisimple** if  $\rho_0(\sigma_K)$  is semisimple.
- **unramified** if  $\rho_0(I_K) = \mathbf{1}$  and  $N = \mathbf{0}$ .
- **$I_K$ -spherical** if  $\rho_0(I_K) = \mathbf{1}$ .

┘

**Def. (25.1.2.23)**  $[\mathrm{Sp}(m)]$ . For  $m \in \mathbb{Z}_+$ , define the representation  $\mathrm{Sp}_m(\rho) \in \mathfrak{wd}(W_K)$  given by

$$\mathrm{Sp}(m) = \mathbb{C}\{e_0, \dots, e_{m-1}\}$$

with

$$\mathrm{Sp}_m(x)(e_i) = |x|^i e_i, \quad N e_i = e_{i+1} (e_m = 0)$$

Then  $(\rho, V, N) \in \mathfrak{wd}(W_K)$  is indecomposable iff it is of the form  $\rho_0 \otimes \mathrm{Sp}_m$  where  $\rho_0 \in \mathrm{Irr}(W_K)$  and  $m \in \mathbb{Z}_+$ .  $\lrcorner$

**Thm. (25.1.2.24)** **[Grothendieck's  $\ell$ -adic Monodromy Theorem]**. For  $\mathbb{Q}_\ell \subset_{\mathrm{fin}} L$  and  $(\rho, V) \in \mathrm{Rep}_L^{\mathrm{fd}}(W_K)$ ,

- There exists an open subgroup  $I'_K \leq I_K$  and a uniquely determined nilpotent operator  $N \subset \mathrm{End}(V)$  s.t. for all  $\sigma \in I'_K$ ,  $\rho(\sigma) = \exp(t_\ell(\sigma)N)$ .
- For any  $x \in W_K$ ,

$$\rho(x)N\rho(x^{-1}) = |x|.N.$$

$\lrcorner$

*Proof:* The uniqueness of  $N$  is clear. Conversely, by continuity  $\mathrm{Gal}_K$  stabilizes an  $\mathcal{O}_L$ -lattice of  $V$ , and reduction modulo  $\ell$  gives a representation  $\bar{\rho}$ .  $\ker(\bar{\rho}) \subset \mathrm{Gal}_K$  corresponds to a finite extension  $K''/K$ . Since the kernel of  $\mathrm{GL}(\Lambda) \rightarrow \mathrm{GL}(\Lambda/(\ell))$  is pro- $\ell$ , we see that action of  $\rho|_{I_{K''}}$  must factor through the tame character  $t_\ell : I_{K''} \rightarrow \mathbb{Z}_\ell(1)$ . Because for any  $x \in \mathrm{Gal}_{K''}$ ,  $t_\ell(x\sigma x^{-1}) = \chi_{\mathrm{cycl}}(x)t_\ell(\sigma)$  (19.1.5.25), so if  $a$  is an eigenvalue of  $\sigma$ , then  $a^{\chi(x)}$  is also an eigenvalue of  $\sigma$  for any  $x \in \mathrm{Gal}_{K''}$ . This implies  $a$  is a root of unity, and implies that there exists a finite extension  $K'/K''$  s.t.  $\rho(I_{K'})$  are all unipotent. Let  $\sigma' \in I_{K'}$  s.t.  $\rho(\sigma')$  is a topological generator of  $\rho(I_{K'})$ , and define  $N = \log(\rho(\sigma'))$ , then this  $N$  satisfies the property.  $\square$

**Cor. (25.1.2.25)** **[Potentially Unramified]**. If  $\mathbb{Q}_\ell \subset_{\mathrm{fin}} L$  and  $\rho \in \mathrm{Rep}_L^{\mathrm{alg}, \mathrm{fin}}(W_K)$  is semisimple, then  $\#\rho(I_K) < \infty$ .  $\lrcorner$

*Proof:* Choose a finite extension  $K'/K$  s.t.  $I_{K'} \subset I'_K$  as in (25.1.2.24), then  $\rho|_{I_{K'}}$  is both unipotent and semisimple (24.1.2.15).  $\square$

**Thm. (25.1.2.26)** **[ $\ell$ -adic Weil-Deligne Representations, Deligne]**. For  $\mathbb{Q}_\ell \subset_{\mathrm{fin}} L$ , there is an equivalence of categories

$$\mathrm{WD}_\ell : \mathrm{Rep}_L^{\mathrm{fd}}(W_K) \cong \mathfrak{wd}_L(W_K) : (\rho, V) \mapsto (\rho_\sigma, V, N)$$

where  $N$  is given in (25.1.2.24), and for any  $x \in I_K$ ,

$$\rho_\sigma(\sigma_K^n x) = \rho(\sigma_K^n x) \exp(-t_\ell(x)N).$$

Moreover, by (25.1.2.11) and (25.1.2.13), this map induces an equivalence:

$$\mathrm{WD}_\ell : \mathrm{Rep}_L^{\mathrm{fd}}(\mathrm{Gal}_K) \cong \mathfrak{wd}_L^{\ell\text{-int}}(W_K).$$

And  $\rho$  is unramified iff  $\mathrm{WD}_\ell(\rho)$  is unramified.  $\lrcorner$

*Proof:* The image is a Weil-Deligne representation by (25.1.2.24).  $\mathrm{WD}_\ell$  is a functor and is an equivalence by the uniqueness of  $N$  (25.1.2.24).  $\square$

**Cor. (25.1.2.27)**. For  $\mathbb{Q}_\ell \subset_{\mathrm{fin}} L$  and  $\rho \in \mathrm{Rep}_L^{\mathrm{fd}}(\mathrm{Gal}_K)$  with  $\mathrm{WD}_\ell(\rho) = (\rho_0, N)$  (25.1.2.26), the following are equivalent:

- $\rho$  is potentially semisimple.
- $\rho$  is  $\sigma_K$ -semisimple and  $\#\rho(I_K) < \infty$ .
- $\rho_0$  is  $\sigma_K$ -semisimple and  $N = 0$ .

┘

*Proof:* 1 and 3 are equivalent by (25.1.2.26). 3 implies 2 is obvious. To show 2 implies 1, notice  $\rho \otimes \bar{L}$  is a semisimple representation of  $\langle \rho(\sigma_K) \rangle$ , which is a subgroup of finite index of  $\rho(\text{Gal}_K)$ , so by (24.1.2.16),  $\rho \otimes \bar{L}$  is a semisimple representation of  $\text{Gal}_K$ .  $\square$

**Def. (25.1.2.28) [Induction of Weil-Deligne Representations].** For  $L/K \in p\text{-LField}$  and let  $(\rho_0, V, N) \in \mathfrak{w}\mathfrak{d}(W_L)$ , define the **induction**

$$\text{Ind}_K^L(\rho_0, N) = (\text{Ind}_L^K \rho_0, \tilde{N}), \quad \text{where } (\tilde{N}f)(g) = |g|.N(f(g)) \quad \forall f : W_K \rightarrow V \in \text{Ind}_K^L(V).$$

And restrictions are defined routinely.  $\square$

### 3 Mod $\ell$ Local Galois Representations

**Notation (25.1.3.1).** Use notation as in (25.1.2.1).  $\square$

**Def. (25.1.3.2) [Decomposed Generically].** An unramified representation

$$\bar{\rho} : \text{Gal}_K \rightarrow \text{GL}(n, \overline{\mathbb{F}}_\ell)$$

is called **decomposed generically** if the eigenvalues  $\{\lambda_1, \dots, \lambda_n\}$  of  $\bar{\rho}(\varphi)$  satisfies  $\lambda_i/\lambda_j \notin \{1, q\}$  for any  $1 \leq i \neq j \leq n$ .  $\square$

### 4 Global Galois Representations

**Def. (25.1.4.1) [Global Artin and Swan Conductors].** Let  $L/F \in \text{GField}$  and  $(\rho, V) \in \text{Rep}^{\text{fd}}(\text{Gal}(L/F))$ . For  $v \in \Sigma_F^{\text{fin}}$ , let  $G_v = G_{v,0} \supset G_{v,1} \supset \dots \supset G_{v,m} = 0$  be higher ramification groups at  $v$ , then the **Artin conductor(global)** of  $\rho$  is defined to be the ideal in  $\mathcal{O}_F$ :

$$\mathfrak{f}(\rho) = \prod_{v \in \Sigma_F^{\text{fin}}} \mathfrak{p}_v^{f_v(\rho)}, \quad f_v(\rho) = f(\rho_v) \quad (25.1.2.5).$$

And the **Swan conductor(global)**

$$\mathfrak{b}(\rho) = \prod_{v \in \Sigma_F^{\text{fin}}} \mathfrak{p}_v^{b_v(\rho)}, \quad b_v(\rho) = b(\rho_v) \quad (25.1.2.5).$$

┘

**Prop. (25.1.4.2) [Conductor-Discriminant-Formula].** For any Galois extension of global fields  $L/F$ ,

$$\mathfrak{d}_{L/F} = \prod_{\rho \in \text{Irr}(\text{Gal}(L/F))} \mathfrak{f}(F, \rho)^{\chi_\rho(1)}$$

┘

*Proof:* Cf. [Neu99]P534.  $\square$

## 5 Deforming Galois Representations(Mazur)

**Def.(25.1.5.1) [ $p$ -Finiteness Condition,  $\Phi_p$ ].**  $G \in \text{ProfGrp}$  is said to satisfy the  $p$ -finiteness condition  $\Phi_p$  if the following equivalent conditions hold:

- The pro- $p$ -completion of  $G$  is topologically f.g..
- The Abelianization  $G_0$  of the pro- $p$ -completion of  $G$  is finite over  $\mathbb{Z}_p$ .
- $\# \text{Hom}(G_0, \mathbb{F}_p) < \infty$ .

┘

**Prop.(25.1.5.2).** Algebraic fundamental groups of smooth varieties over  $\mathbb{Z}$  satisfy the  $p$ -finiteness condition.

In particular, if  $K \in p\text{-LField}$ ,  $\text{Gal}_K$  satisfies the  $p$ -finiteness condition. And if  $F \in \text{NField}$  and  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F$ , then  $\text{Gal}_{F,S}$  satisfies the  $p$ -finiteness condition.

┘

*Proof:* ?

□

**Prop.(25.1.5.3).** Let  $(A, k)$  be a Noetherian complete local ring and  $\Pi \in \text{ProfGrp}$ , then for any continuous representation  $\rho : \Pi \rightarrow \text{GL}(n; A)$ , the residue representation  $\bar{\rho}$  is absolutely irreducible iff

$$r : A[[\Pi]] \rightarrow \text{Mat}(n; A)$$

is subjective.

┘

*Proof:*

□

**Def.(25.1.5.4) [Residual Representation].** Let  $F \in \text{NField}$  and  $W \in \text{Irr}_{\mathbb{Q}_\ell}(\text{Gal}_F)$ . Choose any invariant  $\overline{\mathcal{O}_\ell}$ -lattice  $L \subset W$ , we can consider the residual representation  $\text{Gal}_F \rightarrow \text{GL}(L) \rightarrow \text{GL}(L/(p))$ . Then this residual representation is uniquely determined up to semisimplification, called the **residual representation** of  $W$ .

┘

*Proof:* ?

□

**Def.(25.1.5.5) [Deformations of Representations].** Let  $G \in \text{Grp}$ ,  $\text{pr} : A_1 \rightarrow A$  be a map of Noetherian complete local rings that induces an isomorphism on residue fields, and  $\rho \in \text{Rep}_{A_0}^{\text{fd}}(G)$ , then a **deformation representation** of  $\rho$  to  $A_1$  is an equivalence class of  $\rho_1 \in \text{Rep}_{A_1}^{\text{fd}}(G)$  s.t.  $\text{GL}(\text{pr}) \circ \rho_1 = \rho$ , where two such  $\rho_1$  are equivalent iff they are conjugate by an element in  $\ker(\text{GL}(\text{pr}))$ .

┘

**Def.(25.1.5.6) [Galois Deformation Theorems].** Fix  $\ell \in \text{Prime}$ ,  $F \in \text{NField}$ , and  $\bar{\rho} : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{F}_\ell})$ , we want to lift  $\bar{\rho}$  to a compatible Galois representation of  $\text{Gal}_F$  over some number field  $E$ . Consider the following hypothesizes:

**Residual Conditions** Conditions on the representation  $\bar{\rho}$ , e.g. irreducible/surjective.

**Semi-Global Conditions** Conditions on the lifting  $\rho : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}_\ell})$ , e.g. fixed determinant/self-dual/skew-self-dual.

**Local Conditions at  $w$**  Conditions on the restriction of the lifting  $\rho : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}_\ell})$  to  $\text{Gal}_{F_w}$  for some  $w \in \Sigma_F$ .

Then a **deformation problem theorem** is a theorem of the following form:

If  $\bar{\rho}$  satisfies the **Residual Conditions**, and if some lifting  $\rho : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}_\ell})$  satisfies the **Semi-Global Conditions** and **Local Conditions at  $w$**  for any  $w \in \Sigma_F$ , and is strongly potentially automorphic(30.1.4.2) over a number field  $E$  with some conditions, then any lifting satisfying the **Semi-Global Conditions** and **Local Conditions at  $w$**  for any  $w \in \Sigma_F$  is strongly potentially automorphic over the same number field  $E$ .

┘

### Deformation Spaces

**Thm. (25.1.5.7) [Representability of Deformation Problems].** Let  $N \in \mathbb{Z}_+$  and  $\bar{\rho} : \text{Gal}_{F,S} \rightarrow \text{GL}(N; k)$  be absolutely irreducible, then the deformation problem:

$$\{ \text{Noetherian complete local rings with residue field } k \} \rightarrow \text{Set} : A \mapsto \{ \text{Set of liftings of } \bar{\rho} \text{ to } A \}$$

is representable by a universal object

$$\rho : \text{Gal}_{F,S} \rightarrow \text{GL}(N; R(\bar{\rho})).$$

┘

*Proof:* ?

□

**Prop. (25.1.5.8) [Schlessinger's Criteria].**

┘

**Remark (25.1.5.9).** Determination of the Krull dimension of  $R(\bar{\rho})$  is equivalent to the determination of the truth or falsity of the Leopoldt Conjecture for  $p$  and the number field  $K$ . ?

┘

### Zariski Tangent Spaces

## 6 Representations from Geometry

**Prop. (25.1.6.1).** Let  $k \in \text{Field}$  and  $E \in \text{EllCur}/k$ , then for any  $N \in \mathbb{Z} \cap k^\times$ , there is an action of  $\text{Gal}_k$  on  $E[N] \cong (\mathbb{Z}/(N))^2$ , giving a representation

$$\rho_{E,N} : \text{Gal}_k \mapsto \text{GL}(2, \mathbb{Z}/(N)).$$

Let  $\bar{\rho}_{E,N}$  denote the representation  $\text{Gal}_k \mapsto \text{GL}(2, \mathbb{Z}/(N))/\{\pm 1\}$ . Then  $\rho_{E,N}$  is surjective iff  $\bar{\rho}_{E,N}$  is surjective.

┘

*Proof:* Notice if  $\bar{\rho}_{E,N}$  is surjective, then either  $\begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$  or  $\begin{bmatrix} & 1 \\ -1 & \end{bmatrix}$  is in the image, thus  $-1$  is in the image.

□

**Def. (25.1.6.2) [Tate modules representations].** For  $\ell \in \mathbf{P}$ , the action of  $\text{Gal}_k$  on the Tate module  $T_\ell(E)$  (20.9.6.19) is denoted by  $\rho_{E,\ell^\infty}$ . Equivalently,

$$\rho_{E,\ell^\infty} = \varprojlim_{n \in \mathbb{Z}_+} \rho_{E,\ell^n}.$$

If  $\text{char } k = 0$ , we can assemble all the  $T_\ell(E), \ell \in \mathbf{P}$  together to get a representation

$$\rho_E : \text{Gal}_k \rightarrow \text{End}(E(k^s)_{\text{tor}}) \cong \text{GL}(2, \hat{\mathbb{Z}}).$$

┘

**Prop. (25.1.6.3) [Determinant of Tate Modules].** If  $(K, k)$  be a CDVR and  $E \in \text{EllCur}/K, \ell \in \mathbf{P} \setminus \{\text{char } k\}$ , then  $\det(\rho_{E,\ell^\infty}) \cong \chi_\ell$  (25.1.1.5), by Weil pairing (20.9.7.10).

┘

**Prop. (25.1.6.4) [Serre].** If  $p \in \text{Prime}$  and  $\mathbb{Q}_p \subset_{\text{fin}} K$  is a local field with ramification index  $e$ . Let  $E \in \text{EllCur}/K$  with semistable reduction, then if  $2e < p - 1$ , then the image of  $I_K$  under the projective representation associated to  $\rho_{E,p}$  contains an element of order  $\geq (p - 1)/e$ .

┘

*Proof:* ? □

**Thm. (25.1.6.5) [Serre].** Assume  $E \in \text{EllCur}/\mathbb{Q}$  is non-CM, then  $\text{Im}(\rho_E) \subset \text{GL}(2, \widehat{\mathbb{Z}})$  has finite index.  
In particular, there exists  $N \in \mathbb{Z}_+$  s.t.  $\rho_{E,p}$  is surjective for any  $p \geq N$ . ┘

*Proof:* Cf. [Abelian l-adic representations and elliptic curves, Serre]. ? □

**Conj. (25.1.6.6) [Serre's Uniformity Problem].** There exists a number  $N \in \mathbb{Z}_+$  s.t. for any  $E \in \text{EllCur}/\mathbb{Q}$  non-CM,  $\rho_{E,p}$  is surjective for any  $p \in \text{Prime}_{\geq N}$ . ┘

*Proof:* □



## 25.2 $p$ -adic Local Galois Representations

References are [Berger, Galois representations and  $(\varphi, \Gamma)$ -modules], [Ser98, Car19]. and [notes on  $p$ -adic Hodge, Conrad], [notes on  $p$ -adic Hodge, Serin Hong].

### Notation (25.2.0.1).

- This is a continuation of the section [Galois Representations\(Basics\)](#).
- Use notations defined in [Smooth Representations](#).
- Use notations defined in [Untils and Fargues-Fontaine Curves](#).
- Let  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa) \in p\text{-Field}$ ,
- $K_0 = W(\kappa)[\frac{1}{p}]$  its maximal unramified subextension of  $K$ .
- The Frobenius action on  $K_0$  is denoted by  $\varphi_0$ .

┘

### 1 $\mathbb{C}_K$ -Admissibility

#### $\mathbb{C}_K$ -Admissibility

**Prop. (25.2.1.1) [Variant of Hilbert's Theorem90].** Any  $V \in \text{Rep}_{\widehat{K^{\text{ur}}}}^{\text{fd}}(\text{Gal}(K^{\text{ur}}/K))$  is trivial. In particular, any unramified f.d. representation of  $\text{Gal}_K$  is  $\widehat{K^{\text{ur}}}$ -admissible thus  $\mathbb{C}_p$ -admissible, which is a special case of (25.2.1.2). ┘

*Proof:* Denote by  $\mathcal{O}$  the ring of integers of  $\widehat{K^{\text{ur}}}$  and  $\mathfrak{m}$  the maximal ideal, Let  $W$  be a f.d.  $\widehat{K^{\text{ur}}}$ -semi-linear representation,  $(v_{1,0}, \dots, v_{d,0})$  a basis of  $W$  over  $\widehat{K^{\text{ur}}}$  and  $\mathcal{O}_W$  the  $\mathcal{O}$ -span of  $(v_{1,0}, \dots, v_{d,0})$ , then we are going to construct a sequence of tuples  $(v_{1,n}, \dots, v_{d,n})$  that  $v_{i,n+1} \equiv v_{i,n} \pmod{\mathfrak{m}^n}$  and  $\text{Frob}_q(v_{i,n}) \equiv v_{i,n} \pmod{\mathfrak{m}^n}$  for all  $i$  and  $n$ .

Use induction on  $n$ : the case  $n = 1$  follows from the fact  $\mathcal{O}_W/\mathfrak{m}\mathcal{O}_W$  is trivial as a  $\bar{k}$ -semi-linear representation of  $\text{Gal}_k$ . To prove this, notice there is a finite extension  $l$  of  $k$  and an  $l$ -semi-linear representation  $W_L$  of  $\text{Gal}(l/k)$  that  $\bar{k} \otimes_l W_L \cong \mathcal{O}_W/\mathfrak{m}\mathcal{O}_W$ , then the assertion follows from Hilbert's theorem90(10.2.5.12).

For general  $n$ , we are looking for vectors  $w_1, \dots, w_d \in \mathcal{O}_W$  that  $\text{Frob}_q(v_{i,n} + \pi^n w_i) \equiv v_{i,n} + \pi^n w_i \pmod{\mathfrak{m}^{n+1}}$ , which is equivalent to  $\text{Frob}_q \bar{w}_i - \bar{w}_i = \frac{\text{Frob}_q v_{i,n} - v_{i,n}}{\pi^n}$  in  $\mathcal{O}_W/\mathfrak{m}\mathcal{O}_W$ . To prove this, notice  $\text{Frob}_q - \text{id}$  is surjective on  $\mathcal{O}_W/\mathfrak{m}\mathcal{O}_W$ , which follows from the fact  $\mathcal{O}_W/\mathfrak{m}\mathcal{O}_W$  is trivial as proved above and  $\text{Frob}_q - \text{id}$  is surjective on  $\bar{k}$ .

Now  $v_{i,n}$  are Cauchy sequences and they converges to a tuple  $v_i$  that  $G_{K^{\text{ur}}/K}$  acts trivially and it is an  $\mathcal{O}$ -basis of  $\mathcal{O}_W$ , as its reduction modulo  $\mathfrak{m}$  is a basis of  $\mathcal{O}_W/\mathfrak{m}\mathcal{O}_W$ , so it is a  $\widehat{K^{\text{ur}}}$ -basis of  $W$ .  $\square$

**Prop. (25.2.1.2) [ $\mathbb{C}_p$ -Admissibility].** For  $(\rho, V) \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$ , the following are equivalent:

- $V$  is  $\mathbb{C}_p$ -admissible.
- $\#\rho(I_K) < \infty$ .
- $V$  is  $L\widehat{K^{\text{ur}}}$ -admissible for some finite extension  $L/K$ .

┘

*Proof:* Cf. [p-adic Period Rings Intro, P18]. ?

2  $\rightarrow$  3: This follows from (25.2.1.1).  $\square$

**Cor. (25.2.1.3)** [ $H_{\text{cont}}^0(\text{Gal}_K, \mathbb{C}_p(\psi))$ , **Sen-Tate**].  $H_{\text{cont}}^0(\text{Gal}_K, \mathbb{C}_p(\psi)) = \begin{cases} K & , \# \psi(I_K) < \infty \\ 0 & , \# \psi(I_K) = \infty \end{cases}$ . In particular,

$$H_{\text{cont}}^0(\text{Gal}_K, \mathbb{C}_p(m)) = \begin{cases} K & , m = 0 \\ 0 & , m \neq 0 \end{cases}.$$

┘

*Proof:* This follows from (24.1.1.20) and (25.2.1.2). For the last assertion, the cyclotomic extension of  $K$  thus also the cyclotomic character of  $G_K$  is infinitely unramified, thus  $\chi_{\text{cycl}}^s$  factors through a finite quotient iff  $s = 0$ .  $\square$

**Cor. (25.2.1.4)** [**Potentially Unramified**]. If  $\eta : \text{Gal}_K \rightarrow \mathbb{Z}_p^*$  is a character and there is  $y \in \mathbb{C}_K^\times$  that  $\eta(g) = g(y)/y$ , then there exists a finite Abelian extension  $L$  of  $K$  that  $\eta|_{G_L}$  is unramified, i.e.  $\eta$  is **potentially unramified**.  $\square$

**Cor. (25.2.1.5)**. For any  $n, m \in \mathbb{Z}$ ,

$$\text{Hom}_{\text{Gal}_K}(\mathbb{C}_K(n), \mathbb{C}_K(m))$$

is of one-dimensional over  $K$  if  $n = m$ , and vanishes otherwise.  $\square$

*Proof:* Let  $W = \text{Hom}_{\mathbb{C}_p}(\mathbb{C}_K(n), \mathbb{C}_K(m)) = \mathbb{C}_K(m - n)$ , then the desired space is  $W^{\text{Gal}_K}$ , and the assertion follows from (25.2.1.3).  $\square$

### $H_{\text{cont}}^1$ of $\text{Gal}_K$ -Actions on $\mathbb{C}_p(\psi)$

**Def. (25.2.1.6)** [**Notations**]. Let  $K_\infty/K$  an Abelian extension s.t.  $\Gamma(K_\infty/K) \cong \mathbb{Z}_p$ , and  $H_K = \text{Gal}_{K_\infty}$ . Such  $K_\infty$  always exists, for example we can take a subextension of  $K_\infty = K(p^{\frac{1}{p^\infty}})$ .

Let  $\Gamma_m = \Gamma_0^m$  and  $K_m$  the fixed field of  $\Gamma_m$ .

Decompose  $\Gamma = \Sigma \times \Gamma_0$  and let  $\gamma$  be a topological generator of  $\Gamma_0$ , then every element of  $\Gamma_0$  can be written as  $\gamma^t$  for some  $t \in \mathbb{Z}_p$ . Denote  $\gamma_s = \gamma^{p^s}$ .  $\square$

**Lemma (25.2.1.7)**. Giving an  $\sigma \in \text{Gal}(K/\mathbb{Q}_p)$ , if  $x, y \in \mathfrak{m}_{\mathbb{C}_p}$  that  $x \equiv y \pmod{\pi_K^n}$ , then  $[\pi_K]^\sigma(x) \equiv [\pi_K]^\sigma(y) \pmod{\pi_K^{n+1}}$ , where  $f^\sigma$  is given by action of  $\sigma$  on the coefficients.  $\square$

*Proof:* This is because the coefficients of  $[\pi_K]^\sigma$  are divisible by  $\pi_K$  except for degree  $q$ , where it is  $x^q - y^q = (x - y)(x^{q-1} + x^{q-2}y + \dots + y^{q-1})$  which is divisible by  $\pi_K^{n+1}$  because the residue field of  $K$  is of order  $q$ .  $\square$

**Prop. (25.2.1.8)**. If we let the action of  $\sigma \in \text{Gal}(K/\mathbb{Q}_p)$  on the residue field giving by  $\bar{\sigma} : \kappa \rightarrow \bar{\mathbb{F}}_p : x \mapsto x^{q_\sigma}$ , where  $q_\sigma = p^{n_\sigma}$  is a  $p$ -power, given an element  $\eta = (\eta_0, \eta_1, \dots) \in T(G)$ , we have  $\eta^{q_\sigma} \equiv [\pi_K]^\sigma(\eta_{n+1}^{q_\sigma}) \pmod{\pi_K}$ , hence the above lemma (25.2.1.7) shows that  $[\pi_K^n]^\sigma \eta_n^{q_\sigma} \equiv [\pi_K^{n+1}]^\sigma(\eta_{n+1}^{q_\sigma}) \pmod{\pi_K^{n+1}}$ , so  $[\pi_K^n]^\sigma(\eta_n^{q_\sigma})$  is a Cauchy sequence, converging to an element  $\mu_\sigma$  (don't care about  $\eta$ ).

If  $g \in \text{Gal}_K$ , then  $g(\eta_n) = [\chi_K(g)](\eta_n)$ , hence take  $q_\sigma$ -th power,  $g(\eta_n^{q_\sigma}) \equiv [\chi_K(g)]^\sigma(\eta_n^{q_\sigma}) \pmod{\pi_K}$ , then

$$[\chi_K(g)]^\sigma [\pi_K^n]^\sigma(\eta_n^{q_\sigma}) \equiv [\pi_K^n]^\sigma g(\eta_n^{q_\sigma}) = g([\pi_K^n]^\sigma \eta_n^{q_\sigma}) \pmod{\pi_K}.$$

hence by limiting,  $g(\mu_\sigma) = [\chi_K(g)]^\sigma(\mu_\sigma)$ .  $\square$

**Lemma (25.2.1.9).**

$$v_p(\mu_\sigma) = \begin{cases} \frac{q_\sigma}{e_K(q-1)} + \frac{1}{e_K} & n(\sigma) \neq 0 \\ \frac{1}{e_K(q-1)} + v_p(\sigma(\pi_K) - \pi_K) & n(\sigma) = 0 \end{cases}$$

┘

*Proof:* By (10.7.3.30), we know the Newton polygon of  $[\pi_K^n]^\sigma$ . When  $n(\sigma) \neq 0$ ,  $v(\eta_1^{q_\sigma}) = \frac{q_\sigma}{e_K(q-1)} > \frac{1}{e_K(q-1)}$ , so the valuation of  $[\pi_K]^\sigma(\eta_1^{q_\sigma})$  equals the valuation of its degree 1 term, which is  $v(\pi_K \eta_1^{q_\sigma}) = \frac{q_\sigma}{e_K(q-1)} + \frac{1}{e_K}$ . Now we have by (25.2.1.8), we have  $[\pi_K]^\sigma \eta^{q_\sigma} \equiv [\pi_K^2]^\sigma(\eta_2^{q_\sigma}) \pmod{\pi_K^2}$ , and  $\frac{q_\sigma}{e_K(q-1)} + \frac{1}{e_K} < 2/e_K$ , so valuation already stable at degree 1, and  $v(\mu_\sigma) = v([\pi_K]^\sigma(\eta_1^{q_\sigma}))$ .

If  $q_\sigma = 1$ , it's more delicate, because degree 1 and degree  $q$  term has the same minimal valuation, so they may jump to higher valuations. Notice  $[\pi_K^n](\eta_n) = 0$ , so  $[\pi_K^n]^\sigma(\eta_n) = ([\pi_K^n]^\sigma - [\pi_K^n])(\eta_n)$ . And we have by (19.1.4.22), for  $x \in \mathcal{O}_K$ ,  $v(\sigma(x) - x) \geq v(x) + v(\frac{\sigma(\pi_K)}{\pi_K} - 1) + \delta_{v(x),0}v(\pi_K)$ , with equality when  $v_p(x) = q/e_K$ . So by the Newton polygon, the minimum valuation of the coefficient of  $[\pi_K^n]^\sigma - [\pi_K^n]$  appear at degree  $p^{n-1}$  and possibly  $p^n$ . The valuation of  $\eta_n$  is too small ( $\frac{1}{e_K p^{n-1}(p-1)}$ ) that we don't need to consider other degrees but can assure that degree  $p^{n-1}$  is of minimum valuation, which is  $v(\eta_n^{p^{n-1}}) + v(\sigma(\pi_K) - \pi_K) = \frac{1}{e_K(q-1)} + v_p(\sigma(\pi_K) - \pi_K)$ .  $\square$

**Prop. (25.2.1.10).** For any  $\sigma \in \text{Gal}(K/\mathbb{Q}_p) \setminus \{\text{id}\}$ , there is an element  $\alpha_\sigma \in \mathbb{C}_p^*$  that  $\sigma \circ \chi_K(g) = g(\alpha_\sigma)/\alpha_\sigma$  for all  $g \in \text{Gal}_K$ , where  $\chi_K$  is the Lubin-Tate character.  $\square$

*Proof:* We let  $\alpha_\sigma = \log_{\mathcal{F}_\pi}^\sigma(\mu_\sigma)$ , by (25.2.1.9),  $1/e_K < \mu_\sigma < \infty$ , so by the Newton polygon analysis of  $\log_{\mathcal{F}_\pi}$  (10.7.3.31),  $\alpha_\sigma$  has the same valuation of  $\mu_\sigma$ , in particular,  $\alpha_\sigma \neq 0$ . Then

$$g(\alpha_\sigma) = \log_{\mathcal{F}_\pi}^\sigma(g(\mu_\sigma)) = (\log_{\mathcal{F}} \circ [\chi_K(g)])^\sigma(\mu_\sigma) = (\chi_K(g) \cdot \log_{\mathcal{F}_\pi})^\sigma(\mu_\sigma) = \sigma(\chi_K(g)) \cdot \alpha_\sigma.$$

$\square$

**Cor. (25.2.1.11).**  $\log_p(\sigma(\chi_K(g))) = g(\log(\alpha_\sigma)) - \log_p(\alpha_\sigma)$ .  $\square$

**Prop. (25.2.1.12)** [ $H_{\text{cont}}^1(\text{Gal}_K, \mathbb{C}_p(\psi))$ , Tate [Tat67]]. Let  $\psi : \text{Gal}_K \rightarrow \mathbb{Z}_p^*$  be a character, then we can form a representation  $\mathbb{C}_p(\psi)$  of  $\text{Gal}_K$  on  $\mathbb{C}_p$  that  $\rho(\sigma)(x) = \psi(\sigma)\sigma(x)$ . This is an action because  $\text{Gal}_K$  acts trivial on  $\mathbb{Z}_p^*$ . Then we have

$$H_{\text{cont}}^1(\text{Gal}_K, \mathbb{C}_p(\psi)) = \begin{cases} 0 & , \# \psi(I_K) = \infty \\ \text{a } K\text{-vector space of dimension 1} & , \# \psi(I_K) < \infty \end{cases}.$$

┘

*Proof:* If  $\# \psi(I_K) < \infty$ , then by (25.2.1.2),  $\mathbb{C}_p(\psi) \cong \mathbb{C}_p$  as  $\text{Gal}_K$ -representations. So we may assume  $\psi = \mathbf{1}$ . And we take  $K_\infty$  to be any  $\mathbb{Z}_p$ -extension of  $K$ .

If  $\# \psi(I_K) = \infty$ , then we can take a finite extension  $K'/K$  that is contained in the fixed field  $K_\psi$  of  $\ker(\psi)$  such that  $\text{Gal}(K_\psi/K') \cong \mathbb{Z}_p$  ?. Notice that there is an inf-res exact sequence

$$0 \rightarrow H_{\text{cont}}^1(\Gamma(K'/K), H^0(\text{Gal}_{K'}, \mathbb{C}_\infty(\psi))) \rightarrow H_{\text{cont}}^1(\text{Gal}_K, \mathbb{C}_p(\psi)) \rightarrow H_{\text{cont}}^1(\text{Gal}_{K'}, \mathbb{C}_p(\psi)),$$

So by (25.2.1.3), it suffices to show that  $H_{\text{cont}}^1(\text{Gal}_{K'}, \mathbb{C}_p(\psi)) = 0$ . For this reason, we may change notation to assume that  $K_\infty = K_\psi$  and  $K_\psi = K$ .

Now there is an inf-res exact sequence

$$0 \rightarrow H_{\text{cont}}^1(\text{Gal}(K_\infty/K), \widehat{K_\infty}(\psi)) \rightarrow H_{\text{cont}}^1(\text{Gal}_K, \mathbb{C}_p(\psi)) \rightarrow H_{\text{cont}}^1(H_K, \mathbb{C}_p(\psi)),$$

$\psi$  is trivial on  $H_K$ , so  $\mathbb{C}_p(\psi) \cong \mathbb{C}_p$  as  $H_K$ -representations, thus we may assume  $\psi = \text{id}$ . Let  $f \in Z_{\text{cont}}^1(H_K, \mathbb{C}_p(\psi))$  be a cocycle, as  $H_K$  is compact,  $f(H_K) \in p^{-k}\mathcal{O}_{\mathbb{C}_p}$  for some integer  $k$ . So the lemma below (25.2.1.13) shows that we can move  $f$  cohomologously to higher valuation, i.e.  $f(g) = \sum x_i - g(\sum x_i)$ , so  $f$  is a coboundary. This means that  $H_{\text{cont}}^1(H_K, \mathbb{C}_p(\psi)) = 0$ .

Thus we are reduced to the study of the first term. Let  $\gamma$  be a topological generator of  $\Gamma(K_\infty/K)$ , by (9.8.5.4) we know  $H_{\text{cont}}^1(\Gamma_K, \widehat{K}_\infty(\psi)) = \widehat{K}_\infty(\psi)/1 - \gamma$ .

For  $n$  large, we have a decomposition  $\widehat{K}_\infty(\psi) = K_n(\psi) \oplus X_n(\psi)$  by (19.1.5.38), and  $1 - \gamma^n$  is invertible on  $X_n(\psi)$ . Now  $1 - \gamma^n = (1 - \gamma)(1 + \gamma + \dots + \gamma^{n-1})$ , so  $1 - \gamma$  is also invertible in  $X_n(\psi)$ . And on  $K_n(\psi)$ , if  $\psi$  is of infinite order, then  $1 - \gamma$  is injective, because if  $x$  is fixed, then  $x = \psi(\gamma)^n \gamma^n(x) = \psi(\gamma)^N x$ , which implies  $\psi(\gamma)$  is of finite order. But it is also surjective because it is a  $K$ -linear mapping of  $K_n$ . So  $\widehat{K}_\infty(\psi)/1 - \gamma = 0$ . On the other hand, if  $\psi = \mathbf{1}$  then  $K_n/1 - \gamma = K_n/\ker(\text{tr}_{K_n/K}) = K$ .  $\square$

**Lemma (25.2.1.13).** If  $f : H_K \rightarrow p^n\mathcal{O}_{\mathbb{C}_p}$  is a continuous cocycle, then there exists a  $x \in p^{n-1}\mathcal{O}_{\mathbb{C}_p}$  that the cohomologous cocycle  $g \mapsto f(g) - (x - g(x))$  has values in  $p^{n+1}\mathcal{O}_{\mathbb{C}_p}$ .  $\lrcorner$

*Proof:*  $p^{n+2}\mathcal{O}_{\mathbb{C}_p}$  is open in  $p^n\mathcal{O}_{\mathbb{C}_p}$ , so there is a finite extension  $L/K$  that  $f(H_L) \in p^{n+2}\mathcal{O}_{\mathbb{C}_p}$ . By (19.1.5.32), there is a  $z \in \mathcal{O}_{L_\infty}$  that  $\text{tr}_{L_\infty/K_\infty}(z) = p$ , so there is a  $y \in p^{-1}\mathcal{O}_{L_\infty}$  that  $\text{tr}_{L_\infty/K_\infty}(y) = 1$ .

Now for a set of representatives  $Q$  of  $H_K/H_L$ , denote  $x_Q = \sum_{h \in Q} h(y)f(h)$ , then for  $g \in H_K$ ,  $g(Q)$  is also a set of representative, and

$$g(x_Q) = \sum_{h \in Q} gh(y)gf(h) = \sum_{h \in Q} gh(y)(f(gh) - f(g)) = x_{g(Q)} - f(g),$$

because  $\text{tr}(y) = 1$ . So  $f(g) - (x_Q - g(x_Q)) = x_{g(Q)} - x_Q$ . The RHS is in  $p^{n+1}\mathcal{O}_{\mathbb{C}_p}$ , because: if we let  $gh_i = h_{g(i)}a_i$ , where  $a_i \in H_L$ , then  $x_{g(Q)} - x_Q = \sum h_{g(i)}(y)f(h_{g(i)}a_i) - \sum h_{g(i)}(y)f(h_{g(i)}) = \sum h_{g(i)}(y)h_{g(i)}(f(a_i))$ , which is in  $p^{n+1}$  because  $h_{g(i)}(y) \in p^{-1}\mathcal{O}_{\mathbb{C}_p}$  and  $f(a_i) \in p^{n+2}\mathcal{O}_{\mathbb{C}_p}$  by the choice of  $L$ .  $\square$

**Prop. (25.2.1.14).** The 1-dimensional  $K$ -vector space  $H_{\text{cont}}^1(\text{Gal}_K, \mathbb{C}_p)$  is generated by the cocycle  $[g \mapsto \log_p \chi(g)]$ .  $\lrcorner$

*Proof:* By the proof of (25.2.1.12), we know that  $H^1(\Gamma_K, K_n) \xrightarrow{f} H^1(\text{Gal}_K, \mathbb{C}_p)$  is an isomorphism. for  $\alpha \in K$ , if  $\chi(g) = \gamma^k$ , then  $f(\alpha)(g) = (1 + \gamma + \dots + \gamma^{k-1})(\alpha) = k\alpha = \alpha \cdot \log_p(\chi(g))/\log_p(\gamma)$ . So by continuity,  $f$  is a multiple of  $[g \mapsto \log_p(\chi(g))]$ .  $\square$

**Lemma (25.2.1.15).** And  $f \in \text{Hom}(I_K^{\text{ab}}, \mathbb{Q}_p)$  is of the form  $f(g) = \text{tr}_{K/\mathbb{Q}_p}(\beta_f \log_p \chi_K(g))$  for some  $\beta_f \in K$ .  $\lrcorner$

*Proof:* By (19.7.2.28),  $\chi_K$  is a canonical isomorphism  $I_K^{\text{ab}} \cong \mathcal{O}_K^*$ . Any  $f \in \text{Hom}(\mathcal{O}_K^*, \mathbb{Q}_p)$  is of the form  $f(y) = \text{tr}_{K/\mathbb{Q}_p}(\beta_f \log_p(y))$  for some  $\beta_f \in K$ , because: by (19.1.5.16), when  $n$  is large,  $\log_p$  is a bijection between  $U_K^n$  and  $\pi_K^n \mathcal{O}_K$ .

$\pi_K^n \mathcal{O}_K \rightarrow \mathbb{Q}_p$  can be extended to a map  $K \rightarrow \mathbb{Q}_p$  as  $\mathbb{Q}_p$  is divisible. Now trace is a invertible bilinear form on  $K$ , so the assertion is true on  $U_K^n$  for some  $n$ , and because  $U_K^n$  is of finite index in  $\mathcal{O}_K^*$  and  $\mathbb{Q}_p$  is of char 0, this is true for all  $\mathcal{O}_K^*$ .  $\square$

**Prop. (25.2.1.16).** The map  $H^1(\text{Gal}_K, \mathbb{Q}_p) \rightarrow H^1(\text{Gal}_K, \mathbb{C}_p)$  is given as follows: as  $f \in H^1(\text{Gal}_K, \mathbb{Q}_p)$  must factor through  $\text{Gal}_K^{\text{ab}}$ , if the restriction of  $f$  to  $I_K^{\text{ab}}$  corresponds to  $\beta_f$ , then  $f$  maps to  $\beta_f : (g \mapsto \log_p \chi(g))$ .  $\lrcorner$

*Proof:*  $f(g) = \text{tr}_{K/\mathbb{Q}_p}(\beta_f \log_p \chi_K(g))$  on  $I_K$ , but this map extends to map on  $G_K$ . So  $f(g) = \text{tr}_{K/\mathbb{Q}_p}(\beta_f \log_p \chi_K(g)) + c(g)$  for a unramified map  $c$  on  $G_K$ .

Now by (9.8.5.3),  $H^1(G, \widehat{\mathbb{Q}}_p^{ur}/\mathbb{Q}_p)$  vanish because  $H^1(G, \overline{\mathbb{F}}_p)$  vanish (9.8.3.1), so there is a  $z \in \widehat{\mathbb{Q}}_p^{ur}$  that  $c(g) = g(z) - z$ . And

$$\text{tr}_{K/\mathbb{Q}_p}(\beta_f \log_p \chi_K(g)) = \sum_{\sigma} \sigma(\beta_f \log_p \chi_K(g)) = \beta_f \text{tr}_{K/\mathbb{Q}_p}(\log_p \chi_K(g)) + \sum_{\sigma} (\sigma(\beta_f) - \beta_f) \sigma(\log_p \chi_K(g)).$$

Notice (25.2.1.10) gives a  $\beta_{\sigma}$  that  $\sigma(\log_p \chi_K(g)) = g(\beta_{\sigma}) - \beta_{\sigma}$ , and  $\text{tr}_{K/\mathbb{Q}_p}(\log_p \chi_K(g)) = \log_p \chi(g)$  because  $(N_{K/\mathbb{Q}_p} \chi_K(g))^{-1} = (\chi(g))^{-1}$ , as they both correspond via local CFT to the element in  $G_K^{ab}$  which acts by  $g$  on  $L_{\pi}$  and id on  $K^{ur}$ . Thus the result.  $\square$

## 2 Hodge-Tate Representations

References are [Sen80] and [Car19].

### Hodge-Tate Representations

**Def. (25.2.2.1) [ $B_{\text{H-T}}$ ].** Let  $B_{\text{H-T}} = \mathbb{C}_K[t, t^{-1}]$ ,  $B'_{\text{H-T}} = \mathbb{C}_K((t))$ , and let  $G_K$  acts on it by  $g(at^i) = g(a)\chi_{\text{cycl}}(g)t^i$ . In addition, there is a filtration on  $B'_{\text{H-T}}$  given by  $\text{Fil}^m B'_{\text{H-T}} = t^m \mathbb{C}_p[[t]]$ , then the graded ring of  $B'_{\text{H-T}}$  is isomorphic to  $B_{\text{H-T}}$ . (25.2.1.3) shows that  $B_{\text{H-T}}^{\text{Gal}_K} = (B'_{\text{H-T}})^{\text{Gal}_K} = K$ .

$B_{\text{H-T}}$  and  $B'_{\text{H-T}}$  are  $\text{Gal}_K$ -regular (24.1.1.18).  $\lrcorner$

*Proof:*  $B'_{\text{H-T}}$  is  $\text{Gal}_K$ -regular because it is a field. For  $B_{\text{H-T}}$ ,  $B_{\text{H-T}} \subset \text{Frac}(B_{\text{H-T}}) \subset B'_{\text{H-T}}$ , taking  $\text{Gal}_K$ -fixed points shows (H2). For (H3), if  $\mathbb{Q}_p x$  is stable under  $\text{Gal}_K$  and  $x$  is not of the form  $at^i$ , then we can get a non-trivial  $\text{Gal}_K$ -fixed point of  $\mathbb{C}_K(j-i)$ , which is impossible by (25.2.1.3).  $\square$

**Cor. (25.2.2.2) [Hodge-Tate Representations].** Let  $W \in \text{Rep}_{\mathbb{C}_K}(\text{Gal}_K)$ . For  $k \in \mathbb{Z}$ , let

$$W\{k\} = \{x \in W \mid g(x) = \chi_{\text{cycl}}^k(g)x\} \subset W(-k)$$

then

$$\bigoplus_{k \in \mathbb{Z}} (\mathbb{C}_K(k) \otimes_K W\{k\}) \rightarrow W$$

is injective.  $W$  is called a **Hodge-Tate representation** if this is an isomorphism.  $\lrcorner$

*Proof:* Notice  $B_{\text{H-T}} \cong \bigoplus_{m \in \mathbb{Z}} \mathbb{C}_K(m) \in \text{Rep}_{\mathbb{C}_K}(\text{Gal}_K)$ , so

$$B_{\text{H-T}} \otimes_K (B_{\text{H-T}} \otimes_{\mathbb{C}_K} W)^{\text{Gal}_K} \cong B_{\text{H-T}} \otimes_K \bigoplus_{m \in \mathbb{Z}} (\mathbb{C}_K(m) \otimes_K W\{m\}) \hookrightarrow B_{\text{H-T}} \otimes_{\mathbb{C}_K} W.$$

is injective by (24.1.1.19).  $\square$

**Cor. (25.2.2.3) [Hodge-Tate Representations].**  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$  is called a **Hodge-Tate representation** if it is  $B_{\text{H-T}}$ -admissible. The category of Hodge-Tate representations are denoted by  $\text{Rep}_{\mathbb{Q}_p}^{\text{H-T}}(\text{Gal}_K)$ .

Then  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$  is Hodge-Tate iff it is  $B'_{\text{H-T}}$ -admissible iff  $\mathbb{C}_K \otimes_{\mathbb{Q}_p} V$  is Hodge-Tate thus decomposes as

$$\mathbb{C}_K \otimes_{\mathbb{Q}_p} V \cong \mathbb{C}_K(n_1) \oplus \dots \oplus \mathbb{C}_K(n_d) \in \text{Rep}_{\mathbb{C}_K}(\text{Gal}_K).$$

$\lrcorner$

*Proof:* If  $\mathbb{C}_K \otimes_{\mathbb{Q}_p} V$  is Hodge-Tate, then clearly  $\dim_K (B_{\text{H-T}} \otimes_{\mathbb{Q}_p} V)^{\text{Gal}_K} = \dim_{\mathbb{Q}_p} V$ , thus  $V$  is  $B_{\text{H-T}}$ -admissible (24.1.1.20). Conversely,  $\dim_K (B_{\text{H-T}} \otimes_{\mathbb{Q}_p} V)^{\text{Gal}_K} = \dim_{\mathbb{Q}_p} V = d$  implies  $V$  is Hodge-Tate by (25.2.2.2). The equivalence with  $B'_{\text{H-T}}$ -admissibility is similar.  $\square$

**Def. (25.2.2.4) [Hodge-Tate Weights].** For  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{H-T}}(\text{Gal}_K)$ ,  $V$  is said to have **Hodge-Tate weight**  $i$  with multiplicity  $d_i$  if  $\dim_K W\{i\} = d_i$ .

$V$  is called **Hodge-Tate regular** if each Hodge-Tate weight is multiplicity-free.

In particular,  $\mathbb{Q}_p(n)$  has a single Hodge-Tate weight  $n$ .  $\lrcorner$

**Prop. (25.2.2.5).** If  $K' \subset \overline{K} \in \text{Field}$ , then for  $W \in \text{Rep}_{\mathbb{C}_K}^{\text{fd}}(\text{Gal}_K)$ , the natural maps

$$K' \otimes_K D_K(W) \rightarrow D_{K'}(W), \quad \widehat{K^{\text{ur}}} \otimes_K D_K(W) \rightarrow D_{\widehat{K^{\text{ur}}}}(W)$$

are isomorphisms. In particular,

$$\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K) \cap \text{Rep}_{\mathbb{Q}_p}^{\text{H-T}}(\text{Gal}_{K'}) = \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K) \cap \text{Rep}_{\mathbb{Q}_p}^{\text{H-T}}(I_K) = \text{Rep}_{\mathbb{Q}_p}^{\text{H-T}}(\text{Gal}_K).$$

$\lrcorner$

*Proof:* For  $K' \subset \overline{K}$ ,  $D_K(W) = D_{K'}(W)^{\text{Gal}(K'/K)}$ , thus the isomorphism follows from Galois descent (24.1.1.15). For  $\widehat{K^{\text{ur}}}$ , Cf. [Conrad, P20] ?  $\square$

### 3 Sen-Tate Theory

#### Colmez-Sen-Tate Conditions

#### Sen's Theory

**Remark (25.2.3.1).** Sen's theory goes further than  $\text{Rep}_{\mathbb{Q}_p}^{\text{H-T}}(\text{Gal}_K)$  and to study  $\text{Rep}_{\mathbb{C}_K}^{\text{fd}}(\text{Gal}_K)$ .  $\lrcorner$

**Prop. (25.2.3.2) [Hilbert's Theorem90 for  $\text{Gal}_{K_\infty}$ ].** Any  $W \in \text{Rep}_{\mathbb{C}_p}^{\text{fd}}(\text{Gal}_K)$  is trivial as a  $\mathbb{C}_p$ -semi-linear representation of  $\text{Gal}_{K_\infty}$ . In particular, there is an isomorphism

$$\mathbb{C}_K \otimes_{\widehat{K_\infty}} W^{\text{Gal}_{K_\infty}} \cong W \in \text{Rep}_{\mathbb{C}_K}(\text{Gal}_{K_\infty}).$$

$\lrcorner$

*Proof:* The proof is similar to that of (25.2.1.1).

Let  $\mathcal{O}_W$  be any  $\mathcal{O}_{\mathbb{C}_p}$ -lattice in  $W$ . Firstly we construct a  $\mathbb{C}_p$ -basis  $w_1, \dots, w_d$  of  $W$  that  $w_i \in \mathcal{O}_W$  and  $gw_i \equiv w_i \pmod{p^2 \mathcal{O}_W}$  for all  $g \in \text{Gal}_{K_\infty}$  and  $p\mathcal{O}_W \subset \mathcal{O}_{\mathbb{C}_p} w_1 \oplus \dots \oplus \mathcal{O}_{\mathbb{C}_p} w_d$ .

By continuity there is a finite Galois extension  $L/K$  that ? Cf. [p-adic Hodge Intro, P23].  $\square$

**Prop. (25.2.3.3).** By (25.2.3.2), the next step of Sen's theory is to study  $W \in \text{Rep}_{\widehat{K_\infty}}(\Gamma_K)$ . To this attempt, Sen considers the subspace of  $\Gamma_K$ -finite vectors in  $W$ , they form a vector space  $W_0 \subset W$  over  $K_\infty$ . Then there exists an integer  $r$  and a basis  $(v_1, \dots, v_d)$  of  $W$  that the  $K_r$ -span of  $v_i$  are stable under  $\Gamma_K$ -actions.

Obviously, these  $v_i$  are  $G_K$ -finite, thus in particular

$$\widehat{K_\infty} \otimes_{K_\infty} W_0 \cong W.$$

$\lrcorner$

*Proof:* Cf.[p-adic Hodge Intro, P25].? □

**Cor. (25.2.3.4) [Sen's Operator].** Combining the previous two propositions, let  $W \in \text{Rep}_{\mathbb{C}_p}^{\text{fd}}(\text{Gal}_K)$ , denote  $\widehat{W}_\infty = W^{G_{\overline{K}/K_\infty}}$  and  $W_\infty$  the set of  $\Gamma_K$ -finite vectors of  $\widehat{W}_\infty$ , then

$$\mathbb{C}_p \otimes_{K_\infty} W_\infty \cong W.$$

Let  $v_1, \dots, v_d$  be given by (25.2.3.3),  $W_r = \bigoplus K_r v_i$ . For  $g \in G_K$ , let  $\rho_W(g)$  be the endomorphism of  $W_r$  given by action of  $g$ , then  $\rho_W(\gamma_s)$  is linear when  $s \geq r$ , and because  $\gamma_s$  converges to id,  $\log \rho_W(\gamma_s)$  is defined for  $s$  large. Then **Sen's operator**  $\Phi_W$  is defined to be  $\Phi_W = \frac{\log \rho(\gamma_s)}{p^s}$ , or equivalently  $\Phi_W(v) = \lim_{t \rightarrow 0} \frac{\gamma^t(v) - v}{t}$ .

Sen's operator is defined over  $K$ , as it commutes with  $\Gamma_K$  seen from the limit form, and its kernel is the  $\mathbb{C}_p$ -subspace of  $W$  generated by elements invariant under  $\Gamma_K$ . ┘

*Proof:* It is evident that fixed points of  $\Gamma_K$  are killed by  $\Phi_W$ . Conversely, the kernel of  $\Phi_W$  on  $W_\infty$  is stable under action of  $G_K$ , thus is a sub-representation of  $\Gamma_K$  in  $W_\infty$ , and because  $W_\infty$  consists of finite vectors, the  $G_K$ -action is continuous w.r.t the discrete topology, and Hilbert's theorem90(10.2.5.12) shows this subspace is generated by elements invariant under  $G_K$ . □

**Prop. (25.2.3.5) [Sen's Category].** Let  $\text{Sen}(K, K_\infty)$  be the category of f.d.  $K_\infty$ -vector spaces equipped with an endomorphism defined over  $K$ , then the construction sending  $W$  to  $(W_\infty, \Phi_W)$  induces a functor

$$\text{Sen} : \text{Rep}_{\mathbb{C}_p}^{\text{fd}}(\text{Gal}_K) \rightarrow \text{Sen}(K, K_\infty).$$

This functor commutes with direct sums, and also under tensor products, with the Sen's operator given by

$$\Phi_{W \otimes W'} = \Phi_W \otimes \text{id}_{W'} + \text{id}_W \otimes \Phi_{W'}.$$

This functor is faithful, but in general not full. However, it reflects isomorphisms. ┘

*Proof:* This functor is faithful because  $W_\infty$  generates  $W$  as a  $\mathbb{C}_p$ -vector space. To show it reflects isomorphism, Let  $f : W_\infty \rightarrow W'_\infty$  be an isomorphism commuting with Sen's operator, then it extends by linearity to an isomorphism of  $\mathbb{C}_p$ -vector spaces  $W \rightarrow W'$ , and  $f$  is  $\Gamma_s$ -equivariant for some  $s$  by the definition of Sen's operator. Then considering the space of  $\Gamma_s$ -equivariant  $\mathbb{C}_p$ -linear morphisms from  $W$  to  $W'$ , Hilbert's theorem90 shows there is a basis  $f_i$  consisting of  $G_K$ -equivariant morphisms. Then it remains to show there exists a linear  $K$ -combination of  $f_i$  that is invertible. This is possible because it is true for  $K_s$ , as  $f$  is invertible, and  $K$  is an infinite field. □

**Cor. (25.2.3.6).**  $W \in \text{Rep}_{\mathbb{C}_p}^f(\text{Gal}_K)$  is trivial iff  $\Phi_W = 0$ . ┘

*Proof:* As  $\mathcal{S}$  reflects isomorphisms, compare with the trivial representation  $\mathbb{C}_p^d$ . □

**Prop. (25.2.3.7) [Hodge-Tate Representations and Sen's Operators].** A representation  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G_K)$  is Hodge-Tate iff the Sen's operator  $\Phi_{\mathbb{C}_p \otimes_{\mathbb{Q}_p} V}$  is semisimple with eigenvalues in  $\mathbb{Z}$ . For a general  $V$ , the eigenvalues of  $\Phi_{\mathbb{C}_p \otimes_{\mathbb{Q}_p} V}$  is called the **generalized Hodge-Tate weights** of  $V$ . ┘

*Proof:* If  $V$  is a Hodge-Tate representation, then clearly  $\Phi_W(v) = \lim_{t \rightarrow 0} \frac{\gamma^t(v) - v}{t}$  acts by  $k$  on  $\mathbb{C}_p(k)$ , thus it is semisimple with eigenvalues in  $\mathbb{Z}$ . Conversely, on the  $i$ -eigenspace of  $\Phi$ , tensoring  $\chi_{\text{cycl}}^{-i}$ ,  $\Phi$  acts trivially, then because the kernel of  $\Phi$  are the fixed points of  $\Gamma_K$ (25.2.3.4), thus this eigenspace is isomorphic to  $\mathbb{C}_p(i)^d$ , so  $V$  is Hodge-Tate. □



#### 4 Fontaine's Rings

**Notation (25.2.4.1).** In this subsection, we denote  $B_{\text{dR}}, B_{\text{crys}}$  etc. to denote Fontaine's ring w.r.t. the perfectoid field  $\mathbb{C}_K$  defined in [Fontaine's Period Rings](#).  $\lrcorner$

**Lemma (25.2.4.2).** Let  $\varepsilon = (\dots, \varepsilon_1, \varepsilon_0) \in \mathcal{O}_{\mathbb{C}_K}^\flat$  s.t.  $\varepsilon_0 = 1$  and  $\varepsilon_1 \neq 1$ , then  $|\varepsilon - 1| = \frac{p}{p-1}$ .  $\lrcorner$

*Proof:*

$$|\varepsilon - 1| = |(\varepsilon - 1)^\sharp|_{\mathbb{C}_K} = |\lim_{n \rightarrow \infty} (\varepsilon_n - 1)^{p^n}|_{\mathbb{C}_K} = \lim_{n \rightarrow \infty} p^n |\varepsilon_n - 1| = \lim_{n \rightarrow \infty} \frac{p^n}{p^{n-1}(p-1)} = \frac{p}{p-1}$$

□

**Prop. (25.2.4.3) [ $\mathbb{Q}_p$ -Line in  $B_{\text{dR}}$ ].**  $\theta([\varepsilon] - 1) = \varepsilon_0 - 1 = 0$ , so  $[\varepsilon] - 1 \in \ker \theta$ , and we can define

$$t_\varepsilon = \log([\varepsilon]) = \sum_{n \geq 1} (-1)^{n-1} \frac{([\varepsilon] - 1)^n}{n} \in B_{\text{dR}}^+.$$

Then this is a uniformizer in the CDVR  $B_{\text{dR}}^+$ . Moreover, any other choice of  $\varepsilon$  is of the form  $\varepsilon' = \varepsilon^a$  for  $a \in \mathbb{Z}_p$ , then  $t_{\varepsilon'} = at_\varepsilon$ , and  $\gamma(t_\varepsilon) = \chi_{\text{cycl}}(\gamma)t$  for any  $\gamma \in \text{Gal}_K$ .  $\lrcorner$

*Proof:*  $t_\varepsilon$  is a uniformizer because  $[\varepsilon] - 1$  is:  $[\varepsilon^{1/p}] - 1$  is a unit in  $B_{\text{dR}}$ , and

$$\eta = \frac{[\varepsilon] - 1}{[\varepsilon^{1/p}] - 1} = 1 + [\varepsilon^{1/p}] + \dots + [\varepsilon^{(p-1)/p}]$$

is distinguished, because if  $\eta = \sum [c_n]p^n$ , consider reducing to the residue field:  $W(\mathcal{O}_{\mathbb{C}_K}^\flat) \rightarrow W(\mathcal{O}_{\mathbb{C}_K}^\flat/t)$ , then  $\bar{\varepsilon} = 1$  by (25.2.4.2), and  $\bar{\eta} = p$ , thus  $|c_0| < 1, |c_1 - 1| < 1$ , so it is distinguished (5.5.4.20), thus a uniformizer by (27.7.1.2).

For the last assertion, by the formal property of log, it suffices to show that if  $a_i \rightarrow a \in \mathbb{Z}_p$ , then  $[\varepsilon^{a_i}] \rightarrow [\varepsilon^a] \in B_{\text{dR}}$ . Then it suffices to show that for  $a \in \mathbb{Z}_p, |a|$  small,

$$|[\varepsilon^a] - 1| \rightarrow 0.$$

And this can be done with the topology given in (27.7.1.20)?  $\lrcorner$

**Cor. (25.2.4.4).**  $\text{gr}(B_{\text{dR}}) \cong B_{\text{H-T}}$ .  $\lrcorner$

**Cor. (25.2.4.5).**  $B_{\text{dR}}$  is  $\text{Gal}_K$ -regular, but  $B_{\text{dR}}^+$  is not  $\text{Gal}_K$ -regular.  $\lrcorner$

*Proof:*  $B_{\text{dR}}$  is  $\text{Gal}_K$ -regular because it is a field.  $B_{\text{dR}}^+$  is not  $\text{Gal}_K$ -regular because  $\mathbb{Q}_p t_\varepsilon$  is stable under  $\text{Gal}_K$ -action but  $t_\varepsilon$  is not invertible in  $B_{\text{dR}}^+$ .  $\lrcorner$

**Prop. (25.2.4.6) [Galois Actions].**  $\text{Gal}_K$  acts on  $\mathcal{O}_{\mathbb{C}_K}/(p)$  thus acts on  $\mathcal{O}_{\mathbb{C}_K}^\flat$  and on  $A_{\text{inf}}$ . Then Fontaine's functor  $\theta : A_{\text{inf}} \rightarrow \mathcal{O}_{\mathbb{C}_K}$  is  $\text{Gal}_K$ -equivariant, thus  $\ker \theta$  is  $\text{Gal}_K$ -stable, so  $\text{Gal}_K$ -acts on  $B_{\text{dR}}^+$  and  $B_{\text{dR}}$ , and  $B_{\text{dR}} \rightarrow \mathbb{C}_K$  is  $\text{Gal}_K$ -equivariant.  $\lrcorner$

**Prop. (25.2.4.7).** There is a canonical lifting of  $\bar{K} \rightarrow \mathbb{C}_K$  along  $B_{\text{dR}}^+ \rightarrow \mathbb{C}_K$ , and it is  $\text{Gal}_K$ -equivariant. However, this embedding is not continuous, thus there is no embedding  $\mathbb{C}_K \subset B_{\text{dR}}^+$ .  $\lrcorner$



*Proof:*  $K_0 = W(k)[\frac{1}{p}] \subset W(\mathcal{O}_{\mathbb{C}_K}^\flat)[\frac{1}{p}] = B_{\text{inf}} \subset B_{\text{dR}}$ , and it follows from Hensel's lemma that any element in  $\overline{K}$  lifts uniquely to an element of  $B_{\text{dR}}^+$ , so  $\overline{K} \subset B_{\text{dR}}^+$ , and is  $\text{Gal}_K$ -invariant, by uniqueness and the fact  $B_{\text{dR}} \rightarrow \mathbb{C}_K$  is  $\text{Gal}_K$ -equivariant (25.2.4.6).

For the last assertion, if the embedding is continuous, the  $B_{\text{dR}}^+ \rightarrow \mathbb{C}_K$  has a section, and the filtration splits so  $B_{\text{dR}} \cong B_{\text{H-T}}$ .  $\square$

**Prop. (25.2.4.8).**

- $K = (B_{\text{dR}}^+)^{\text{Gal}_K} = B_{\text{dR}}^{\text{Gal}_K}$
- $K_0 = B_{\text{crys}}^{\text{Gal}_K}$ , and the canonical morphism  $K \otimes_{K_0} B_{\text{crys}} \rightarrow B_{\text{dR}}$  is injective.
- $\mathbb{Q}_p = B_e^{\text{Gal}_K}$ .
- $\mathbb{Q}_p = (\text{Fil}^0 B_\mu)^{\text{Gal}_K}$  for  $\mu = \text{crys}$  or  $\mu \geq 1$ .

┘

*Proof:* 1: Firstly  $K \subset B_{\text{dR}}$  and is invariant under  $\text{Gal}_K$  by (25.2.4.7). On the other hand, the exact sequence

$$0 \rightarrow \text{Fil}^{m+1} B_{\text{dR}} \rightarrow \text{Fil}^m B_{\text{dR}} \rightarrow \mathbb{C}_p(m) \rightarrow 0 \quad (25.2.4.13).$$

induces an injection

$$B_{\text{dR}}^{\text{Gal}_K} \cap \text{Fil}^m B_{\text{dR}} / B_{\text{dR}}^{\text{Gal}_K} \cap \text{Fil}^{m+1} B_{\text{dR}} \hookrightarrow \mathbb{C}_p(m)^{\text{Gal}_K}.$$

Thus  $B_{\text{dR}}^{\text{Gal}_K} = B_{\text{dR}}^{\text{Gal}_K} = K$ .

2: The injectivity of  $K \otimes_{K_0} B_{\text{crys}} \rightarrow B_{\text{dR}}$  Cf. [Laurent Fargues and Jean-Marc Fontaine Prop10.2.8].

3: From 2 and notice  $B_e = B_{\text{crys}}^{\varphi=\text{id}}$ .

4: Cf. [Period Rings, P45]. ?

┘

**Lemma (25.2.4.9).** Let  $W \in \text{Vect}_K^{\text{fd}}$ , then the map:

$$\{\text{Filtrations on } W\} \rightarrow \{\text{Gal}_K\text{-stable } B_{\text{dR}}^+\text{-lattice in } W \otimes_K B_{\text{dR}}\} : \text{Fil} \mapsto \text{Fil}^0(W \otimes_K B_{\text{dR}})$$

is bijective and the inverse is given by  $\Gamma \mapsto \{(t^n \Gamma)^{\text{Gal}_K} \subset (B_{\text{dR}} \otimes_{B_{\text{dR}}^+} \Gamma)^{\text{Gal}_K} = W\}_{n \in \mathbb{Z}}$ .  $\square$

*Proof:* Cf. [Laurent Fargues and Jean-Marc Fontaine Prop10.4.3]. ?  $\square$

### deRham and Crystalline Representations

**Def. (25.2.4.10) [deRham, Crystalline and Semistable Representations].** Situation as in (25.2.4.8), for  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$ ,

- $V$  is called a **deRham representation** iff  $V$  is  $B_{\text{dR}}$ -admissible??, or equivalently

$$\dim_K(B_{\text{dR}} \otimes_{\mathbb{Q}_p} V)^{\text{Gal}_K} = \dim_{\mathbb{Q}_p} V.$$

The category of deRham representations of  $\text{Gal}_K$  are denoted by  $\text{Rep}_{\mathbb{Q}_p}^{\text{dR}}(\text{Gal}_K)$ .

- $V$  is called a **crystalline representations** iff  $V$  is  $B_{\text{crys}}$ -admissible, or equivalently

$$\dim_{K_0}(B_{\text{crys}} \otimes_{\mathbb{Q}_p} V)^{\text{Gal}_K} = \dim_{\mathbb{Q}_p} V.$$

The category of crystalline representations of  $\text{Gal}_K$  are denoted by  $\text{Rep}_{\mathbb{Q}_p}^{\text{crys}}(\text{Gal}_K)$ .  $\square$

**Def. (25.2.4.11) [ $\text{Rep}'_{B_e}(\text{Gal}_K)$ ].** Denote by  $\text{Rep}'_{B_e}(\text{Gal}_K)$  the category of finite locally free  $B_e$ -modules  $M$  with a semi-linear  $\text{Gal}_K$ -action that there exists a  $\text{Gal}_K$ -invariant  $B_{\text{dR}}^+$ -lattice  $\Gamma \in M \otimes_{B_e} B_{\text{dR}}$  that  $G_K$  acts continuously.  $\square$

### Summary

**Prop. (25.2.4.12) [Diagram of Inclusions].**

$$\begin{array}{ccccccccc}
 B_{\text{inf}}^+ & \longrightarrow & B_{\mu}^+ & \hookrightarrow & B_{\text{crys}}^+ & \hookrightarrow & B_{\mu}^+ & \hookrightarrow & B_{\text{max}}^+ & \hookrightarrow & B_{\text{dR}}^+ \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & B_{\mu} & \hookrightarrow & B_{\text{crys}} & \hookrightarrow & B_{\mu} & \hookrightarrow & B_{\text{max}} & \hookrightarrow & B_{\text{dR}} \\
 & & (\mu > p-1) & & & & (1 \leq \mu \leq p-1) & & = B_1^+ & & 
 \end{array}$$

┘

**Prop. (25.2.4.13) [Properties of  $B_{\text{dR}}$ ].**

- $B_{\text{dR}}$  is a discretely valued field with residue field  $\mathbb{C}_K$  and valuation ring  $B_{\text{dR}}^+$ .
- $B_{\text{dR}}$  is an algebra over  $\widehat{\overline{K^{\text{ur}}}} = \overline{K} \widehat{K^{\text{ur}}}$  but not over  $\mathbb{C}_K$ .
- $B_{\text{dR}}$  has a special uniformizer  $t_{\varepsilon}$  s.t.  $\mathbb{Q}_p t_{\varepsilon} \cong \mathbb{Q}_p(1) \in \mathcal{R}\text{ep}_{\mathbb{Q}_p}(\text{Gal}_K)$ .
- $B_{\text{dR}}$  has a filtration  $\{\text{Fil}^m B_{\text{dR}} = t_{\varepsilon}^m B_{\text{dR}}^+\}$ , and  $B_{\text{dR}}^{\text{gr}} \cong B_{\text{H-T}}$  as graded rings.
- $B_{\text{dR}}^{\text{Gal}_K} = K$ .

┘

*Proof:* 1 follows from (27.7.1.18), 2 follows from (25.2.4.7). 3 follows from (25.2.4.3). 4 follows from (25.2.4.4). 5 follows from (25.2.4.8).  $\square$

**Prop. (25.2.4.14) [Properties of  $B_{\text{crys}}$ ].**

- $B_{\text{crys}}$  is an algebra over  $\widehat{K^{\text{ur}}}$ .
- $B_{\text{crys}}$  has a Frobenius endomorphism  $\varphi$ .
- There is a canonical embedding  $B_{\text{crys}} \otimes_{K_0} K \hookrightarrow B_{\text{dR}}$ , and  $t_{\varepsilon} \in B_{\text{crys}}$ .
- $B_{\text{crys}}^{\text{Gal}_K} = K_0$ .
- $(B_{\text{crys}} \cap B_{\text{dR}}^+)^{\varphi=1} = \mathbb{Q}_p$ .

┘

*Proof:* ?  $\square$

## 5 deRham Representations

**Lemma (25.2.5.1).** Let  $V$  be a finite  $\mathbb{Q}_p$ -vector space with an action of  $\text{Gal}_K$ , then the action is continuous iff the induced action on  $V \otimes_{\mathbb{Q}_p} B_{\text{dR}}^+$  is continuous.  $\square$

*Proof:* This is because the action of  $\text{Gal}_K$  on  $B_{\text{dR}}^+$  is continuous, and  $V$  has the induced topology in  $V \otimes_{\mathbb{Q}_p} B_{\text{dR}}^+$ .  $\square$

**Prop. (25.2.5.2) [ $\mathbb{C}_K$ -admissible Representations are deRham].**  $\mathcal{R}\text{ep}_{\mathbb{Q}_p}^{\mathbb{C}_K\text{-adm}}(\text{Gal}_K) \subset \mathcal{R}\text{ep}_{\mathbb{Q}_p}^{\text{dR}}(\text{Gal}_K)$ .  $\square$

*Proof:* For  $V \in \mathcal{R}\text{ep}_{\mathbb{Q}_p}^{\mathbb{C}_K}$ , by (25.2.1.2), there exists a finite extension  $L/K$  s.t.  $V$  is  $L\widehat{K^{\text{ur}}}$ -admissible. Thus  $V$  is deRham as  $L\widehat{K^{\text{ur}}} \subset B_{\text{dR}}$  (25.2.4.13).  $\square$

**Prop. (25.2.5.3) [Potentially deRham are deRham].** Let  $K' \subset \mathbb{C}_K$  be another  $p$ -adic field, then

$$\mathrm{Rep}_{\mathbb{Q}_p}(\mathrm{Gal}_K) \cap \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_{K'}) = \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_K).$$

In particular, being deRham is not sensible to ramifications, which is a bad feature compared to being crystalline or semistable.  $\lrcorner$

*Proof:* Because  $\widehat{K^{\mathrm{ur}}} \subset \widehat{(K')^{\mathrm{ur}}}$  is of finite degree, it suffices to prove for two cases:  $K'/K$  is finite or  $K' = \widehat{K^{\mathrm{ur}}}$ . But the finite case follows from Galois descent the same as??. The second case follows from [Conrad, P80] ?  $\square$

**Prop. (25.2.5.4) [Filtered  $D_{\mathrm{dR}}$ ].** For  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$ , there is a finite filtration  $\mathrm{Fil}$  on  $D_{\mathrm{dR}}(V)$  s.t.

$$\mathrm{Fil}^m D_{\mathrm{dR}}(V) = (t^m B_{\mathrm{dR}} \otimes_E V)^{\mathrm{Gal}_K} \subset D_{\mathrm{dR}}(V).$$

$\lrcorner$

**Prop. (25.2.5.5) [deRham Representations are Hodge-Tate].** For  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$ ,

- there is an injection of graded vector spaces

$$\mathrm{gr}(D_{\mathrm{dR}}(V)) \hookrightarrow D_{\mathrm{H-T}}(V),$$

- If  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_K)$ , the map in item1 is an isomorphism, and  $V$  is Hodge-Tate.
- If  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_K)$ ,

$$B_{\mathrm{dR}} \otimes_K D_{\mathrm{dR}}(V) \cong B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V$$

identifies filtrations.  $\lrcorner$

*Proof:* 1: Consider the exact sequences

$$0 \rightarrow \mathrm{Fil}^{m+1} B_{\mathrm{dR}} \rightarrow \mathrm{Fil}^m B_{\mathrm{dR}} \rightarrow \mathbb{C}_p(m) \rightarrow 0 \text{ (25.2.4.13)}.$$

Tensoring  $V$  and taking  $\mathrm{Gal}_K$ -invariants give injections

$$h : \mathrm{gr}^m(D_{\mathrm{dR}}(V)) \hookrightarrow V(m)^{\mathrm{Gal}_K},$$

giving the injection  $\mathrm{gr}(D_{\mathrm{dR}}(V)) \hookrightarrow D_{\mathrm{H-T}}(V)$ .

2: If  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_K)$ , this is an isomorphism by dimension reason, and  $V$  is Hodge-Tate by dimension reason.

3: Firstly notice  $\mathrm{Fil}^m(B_{\mathrm{dR}} \otimes_K D_{\mathrm{dR}}(V)) \subset \mathrm{Fil}^m(B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V)$  is trivial, thus it suffices to show that the induced map

$$f : \mathrm{gr}(B_{\mathrm{dR}} \otimes_K D_{\mathrm{dR}}(V)) \rightarrow \mathrm{gr}(B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V) = B_{\mathrm{H-T}} \otimes_{\mathbb{Q}_p} V$$

is an isomorphism. But notice

$$B_{\mathrm{H-T}} \otimes \mathrm{gr}(D_{\mathrm{dR}}(V)) \xrightarrow{g} \mathrm{gr}(B_{\mathrm{dR}} \otimes_K D_{\mathrm{dR}}(V)) \xrightarrow{f} B_{\mathrm{H-T}} \otimes_{\mathbb{Q}_p} V$$

equals  $B_{\mathrm{H-T}} \otimes h$ , so  $g$  is an isomorphism because it is surjective, and thus  $f$  is also an isomorphism.  $\square$

**Cor. (25.2.5.6).** 1-dimensional Hodge-Tate representations are deRham.  $\lrcorner$

*Proof:* This is because if  $V \cong \mathbb{Q}_p(\psi)$  where  $\psi$  is a character of  $\text{Gal}_K$ , and  $\mathbb{C}_p \otimes_E V \cong \mathbb{C}_p(m)$ , then by Sen-Tate (25.2.1.3),  $\psi(-m)$  is potentially unramified, thus  $\mathbb{C}_p$ -admissible by (25.2.1.2), and thus deRham (25.2.5.2).  $\square$

**Remark (25.2.5.7) [ $D_{\text{dR}}$  Insensitive to Ramifications].**  $D_{\text{dR}}$  is far from fully faithful. In fact, any unramified representation  $V$  is deRham by (25.2.5.3), and  $D_{\text{dR}}(V)$  is a simple filtration with graded ring  $K^d[0]$ , but  $V$  can be different from trivial representation.  $\lrcorner$

**Prop. (25.2.5.8).** The functor  $D_{\text{dR}} : \text{Rep}_{\mathbb{Q}_p}^{\text{dR}}(\text{Gal}_K) \rightarrow \text{Fil Vect}_K$  is faithful and exact, and commutes with taking tensor products and duals. Moreover, the perfect pairing in (24.1.1.21) is a perfect pairing of filtered vector spaces.  $\lrcorner$

*Proof:* Let  $0 \rightarrow V_1 \rightarrow V \rightarrow V_2 \rightarrow 0 \in \text{Rep}_{\mathbb{Q}_p}^{\text{dR}}(\text{Gal}_K)$  be an exact sequence, then  $V_i$  are also Hodge-Tate by (25.2.5.5), so there is an exact sequence

$$0 \rightarrow \text{gr}(D_{\text{dR}}(V_1)) \rightarrow \text{gr}(D_{\text{dR}}(V)) \rightarrow \text{gr}(D_{\text{dR}}(V_2)) \rightarrow 0,$$

showing that

$$0 \rightarrow D_{\text{dR}}(V_1) \rightarrow D_{\text{dR}}(V) \rightarrow D_{\text{dR}}(V_2) \rightarrow 0$$

is an exact sequence of filtered vector spaces.

For tensor product, it suffices to show that

$$\text{gr}(D_{\text{dR}}(V_1) \otimes D_{\text{dR}}(V_2)) \rightarrow \text{gr}(D_{\text{dR}}(V_1 \otimes V_2))$$

is an isomorphism, which reduces to

$$D_{\text{H-T}}(V_1) \otimes D_{\text{H-T}}(V_2) \cong D_{\text{H-T}}(V_1 \otimes V_2) \text{ (24.1.1.21)}.$$

For perfect pairing, it suffices to show that the map

$$D_{\text{dR}}(V^\vee) \rightarrow D_{\text{dR}}(V)^\vee$$

is an isomorphism of filtered vector spaces. But then it reduces to the isomorphism

$$D_{\text{H-T}}(V^\vee) \cong D_{\text{H-T}}(V)^\vee \text{ (24.1.1.21)}.$$

$\square$

**Prop. (25.2.5.9) [Extensions of deRham Representations].** If  $0 \rightarrow V_1 \rightarrow V \rightarrow V_2 \rightarrow 0$  is an exact sequence in  $\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$  s.t.  $V_1, V_2$  are deRham, and the Hodge-Tate weights of  $V_1$  are strictly larger than that of  $V_2$ , then  $V$  is deRham.

In particular, any upper-triangular representation with diagonal  $(\mathbb{Q}_p(a_1), \mathbb{Q}_p(a_2), \dots, \mathbb{Q}_p(a_n))$  with  $a_1 > a_2 > \dots > a_n$  is deRham.  $\lrcorner$

*Proof:* By twisting, we may assume that all Hodge-Tate weights of  $V_1$  are positive and Hodge-Tate weights of  $V_2$  are non-positive. There is an exact sequence

$$0 \rightarrow D_{\text{dR}}(V_1) \rightarrow D_{\text{dR}}(V) \rightarrow D_{\text{dR}}(V_2) = \text{Fil}^0 D_{\text{dR}}(V_2)$$

so it suffices to show that  $\mathrm{Fil}^0 D_{\mathrm{dR}}(V) \rightarrow \mathrm{Fil}^0 D_{\mathrm{dR}}(V_2)$  is surjective. But it follows from (9.8.4.4) that there is an exact sequence

$$\mathrm{Fil}^0 D_{\mathrm{dR}}(V) \rightarrow \mathrm{Fil}^0 D_{\mathrm{dR}}(V_2) \rightarrow H_{\mathrm{cont}}^1(\mathrm{Gal}_K, B_{\mathrm{dR}}^+ \otimes_{\mathbb{Q}_p} V_1).$$

So it suffices to show that  $H^1(\mathrm{Gal}_K, B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1) = 0$ . The exact sequence

$$0 \rightarrow t_\varepsilon^{m+1} B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1 \rightarrow t_\varepsilon^m B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1 \rightarrow \mathbb{C}_p(m) \otimes_{\mathbb{Q}_p} V_1 \rightarrow 0$$

induces a surjection  $H_{\mathrm{cont}}^1(\mathrm{Gal}_K, t_\varepsilon^{m+1} B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1) \twoheadrightarrow H_{\mathrm{cont}}^1(\mathrm{Gal}_K, t_\varepsilon^m B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1)$  by hypothesis. Notice  $B_{\mathrm{dR}}^+$  is  $t_\varepsilon$ -complete, so we can use approximation technique similar to (9.8.5.3) to show that  $H_{\mathrm{cont}}^1(\mathrm{Gal}_K, t_\varepsilon^m B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1) = 0$ .  $\square$

**Remark (25.2.5.10).** For an example of  $V \in \mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{H-T}}(\mathrm{Gal}_K)$  that is not deRham, Cf.[Conrad, P78].  $\lrcorner$

### $B_{\mathrm{dR}}$ -Representations

Fontaine in [Arithmétique des représentations galoisiennes  $p$ -adiques, 2004] studied  $\mathcal{R}\mathrm{ep}_{B_{\mathrm{dR}}}(\mathrm{Gal}_K)$  in similar spirit of Sen's theory. He firstly showed that any  $B_{\mathrm{dR}}$ -representations descends to a  $K_\infty((t))$ -representation  $W$ , and similar to Sen's operator, Fontaine defined a  $K_\infty$ -linear derivative  $\nabla_W : W \rightarrow W$ .

## 6 Crystalline Representations

References are [Crystalline Representations and F-crystals, Kisin].

**Prop. (25.2.6.1).** Let  $L/K$  be a finite field extension, then

$$\mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}(\mathrm{Gal}_K) \cap \mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{crys}}(\mathrm{Gal}_L) = \mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{crys}}(\mathrm{Gal}_K).$$

$\lrcorner$

*Proof:* Cf.[Period Rings, P52]. ?

$\square$

**Prop. (25.2.6.2) [Unramified Representations are crystalline].**

$$\mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{fd}, \mathrm{ur}}(\mathrm{Gal}_K) \subset \mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{fd}, \mathbb{C}_p\text{-adm}, \mathrm{crys}}(\mathrm{Gal}_K).$$

$\lrcorner$

*Proof:* By (25.2.1.1), any unramified  $V \in \mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$  is  $\widehat{K}^{\mathrm{ur}}$ -admissible, thus  $B_{\mathrm{crys}}$ -admissible as  $\widehat{K}^{\mathrm{ur}} \subset B_{\mathrm{crys}}$  (25.2.4.14). And  $V$  is also  $\mathbb{C}_p$ -admissible by (25.2.1.1).

To show the converse inclusion, use (25.2.1.2)(24.1.1.17) and the fact  $L\widehat{K}^{\mathrm{ur}} \cap B_{\mathrm{crys}} = \widehat{K}_0^{\mathrm{ur}}$  ?.  $\square$

**Remark (25.2.6.3).** For examples of  $V \in \mathcal{R}\mathrm{ep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_K)$  that is not crystalline, Cf.[Period Rings, P52]. ?

$\lrcorner$

**Prop. (25.2.6.4) [Crystalline Representation].** The functor

$$\mathcal{D} : \mathcal{R}\mathrm{ep}_{B_e}(G_K)' \rightarrow \varphi\text{-Mod}_{K_0} : W \mapsto (W \otimes_{B_e} B_{\mathrm{crys}})^{\mathrm{Gal}_K}$$

are left adjoint to the functor

$$\mathcal{V} : \varphi\text{-Mod}_{K_0} \rightarrow \mathcal{R}\mathrm{ep}_{B_e} G_K' : (D, \varphi_D) \mapsto (D \otimes_{K_0} B_{\mathrm{crys}})^{\varphi_D \otimes \varphi = \mathrm{id}}$$

Moreover,  $\mathcal{V}$  is fully faithful,  $\mathrm{id} \cong \mathcal{D} \circ \mathcal{V}$ ,  $\mathcal{V} \circ \mathcal{D} \hookrightarrow \mathrm{id}$ , and  $M \in \mathcal{R}\mathrm{ep}_{B_e}(\mathrm{Gal}_K)$  is in the image of  $\mathcal{V}$  iff  $\mathcal{V}(\mathcal{D}(M)) \cong M$ .  $\lrcorner$

*Proof:* Cf.[Laurent Fargues and Jean-Marc Fontaine Prop10.2.12].  $\square$

**Cor.(25.2.6.5).** In particular, a  $B_e$ -representation is crystalline iff it is in the image of  $\mathcal{V}$ . Now we define a Vector bundle  $\mathcal{E}$  on  $X$  to be crystalline iff the  $H^0(X \setminus \{\infty\}, \mathcal{E})$  is crystalline.  $\lrcorner$

## 7 Semi-stable Representations

**Thm.(25.2.7.1) [ $p$ -adic Monodromy Theorem, Colmez/André-Kedlaya-Mebkhout].**  $V \in \mathcal{R}\text{ep}_{\mathbb{Q}_p}(\text{Gal}_K)$  is deRham iff it is potentially semistable.  $\lrcorner$

*Proof:* ?  $\square$

**Cor.(25.2.7.2).** For  $V \in \mathcal{R}\text{ep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$ , we have the following implications:

$$\begin{array}{ccccc}
 \text{unramified} = \mathbb{C}_p\text{-adm} + \text{crystalline} & \xRightarrow{\quad} & \text{crystalline} & \xRightarrow{\quad} & \text{semi-stable} \\
 \Downarrow & & \Downarrow & & \Downarrow \\
 \mathbb{C}_p\text{-adm} = \text{pot.unramified} = \text{Hodge-Tate with weights } 0 & \implies & \text{pot.crys} & \implies & \text{de Rham} = \text{pot.semi-stable} \\
 & & & & \Downarrow \\
 & & & & \text{Hodge-Tate}
 \end{array}$$

$\lrcorner$

**Prop.(25.2.7.3) [deRham Representation gives Weil-Deligne Representations].** For  $K \in p\text{-LField}$ , there is a functor

$$\text{WD} : \mathcal{R}\text{ep}_{\mathbb{Q}_\ell}^{\text{dR}}(\text{Gal}_K) \rightarrow \mathfrak{wd}(W_K).$$

$\lrcorner$

*Proof:* ? Cf.[Fontaine-Mazur, Geometric Galois Representations]  $\square$

## Ordinary Representations

**Def.(25.2.7.4) [Ordinary Galois-Representations].** Let  $K \in p\text{-LField}$  and  $n \in \mathbb{Z}_+$ , then a continuous representation  $\rho : \text{Gal}_K \rightarrow \text{GL}(n; \overline{\mathbb{Q}_p})$  is called an **ordinary Galois-representation** if the following conditions hold:

- $\rho$  can be upper-triangularized with diagonal entries given by  $\text{diag}(\chi_1, \chi_2, \dots, \chi_n)$ .
- There exists  $b \in (\mathbb{Z}^n)^{\text{Hom}(K, \overline{\mathbb{Q}_p})}$  s.t.  $b_{v,1} < b_{v,2} < \dots < b_{v,n}$  for all  $v \in \text{Hom}(K, \overline{\mathbb{Q}_p})$ , and

$$(\chi_i \circ \text{Art}_K)(\alpha) = \prod_{v \in \text{Hom}(K, \overline{\mathbb{Q}_p})} v(\alpha)^{b_{\tau, i}}$$

for  $\alpha$  in an open subgroup  $U \leq K^\times$ .

And  $\rho$  is called an **ss-ordinary Galois-representation** if we can choose  $U = \mathcal{O}_K$ . And  $\rho$  is called an **cr-ordinary Galois-representation** if it is ordinary and crystalline(in which case it is also ss-ordinary?).  $\lrcorner$

**Prop. (25.2.7.5)[Ordinary Representations are DeRham].** Situation as in (25.2.7.4), if  $\rho : \text{Gal}_K \rightarrow \text{GL}(n; \overline{\mathbb{Q}_p})$  is ordinary (resp. ss-ordinary), then it is deRham (resp. semi-stable), with

$$\text{H-T}_v(\rho) = \{-b_{v,1}, -b_{v,2}, \dots, -b_{v,n}\}$$

for each  $v \in \text{Hom}(K, \overline{\mathbb{Q}_p})$ .

If  $\rho$  is ss-ordinary and for each  $i \in [n-1]_+$  there exists some  $v \in \text{Hom}(K, \overline{\mathbb{Q}_p})$  s.t.  $b_{v,i} + 1 < b_{v,i+1}$ , then  $\rho$  is cr-ordinary.  $\lrcorner$

*Proof:* Cf. [Nek,  $p$ -adic height pairings] Prop. 1.24, 1.26 and 1.28, and [Gee-Geraghty, Companion forms for unitary and symplectic groups] Lemma 3.1.4 ?.  $\square$

## 8 Barsotti-Tate Representations

### 9 $(\varphi, \Gamma)$ -Modules

References are [Fontaine90: Représentations  $p$ -adiques des corps locaux], [Fontaine94a: Le corps des périodes  $p$ -adiques] and [Fontaine94b: Représentations  $p$ -adiques semi-stables]. I'm mostly following [Berger, Galois representations and  $(\varphi, \Gamma)$ -modules].

**Def. (25.2.9.1) [ $\varphi$ -Modules].** Let  $\varphi_0 : A \rightarrow A \in \mathcal{C}\text{Ring}$  and  $M \in \text{Mod}_A$ , then an additive map  $\varphi : M \rightarrow M$  is called  $\varphi_0$ -**semi-linear** iff  $\varphi(am) = \varphi_0(a) \cdot \varphi(m)$  for  $a \in A$ . A  $\varphi$ -**module** over  $(A, \varphi_0)$  is an  $A$ -module  $M$  with a  $\varphi_0$ -semi-linear  $\varphi$ . The category of  $\varphi$ -modules over  $A$  is denoted by  $\varphi\text{-Mod}_A$ .

Giving a  $A$ -module  $M$  and a  $\varphi : M \rightarrow M$ , there is a map  $\Phi : A \otimes_{\varphi_0, A} M = (\varphi_0)_* M \rightarrow M : \lambda \otimes m \rightarrow \lambda \varphi(m)$ , which is a  $A$ -module map iff  $\varphi$  is  $\varphi_0$ -semi-linear.

If we define a ring  $A[\varphi]$  as the free group  $A[X]$  modulo the relation  $Xa = \varphi(a)X$  and ring relations in  $A$ , then it is a ring, and a  $\varphi$ -module over  $(A, \varphi)$  is equivalent to a left  $A[\varphi]$ -module.

Thus  $\varphi\text{-Mod}_A$  is a Grothendieck Abelian category with tensor products, and moreover, the kernel as  $A[\varphi]$ -module is the same as the kernel as a  $A$ -module.  $\lrcorner$

**Def. (25.2.9.2).** If there is a map  $\alpha : (A_1, (\varphi_0)_1) \rightarrow (A_2, (\varphi_0)_2)$  that commutes with  $(\varphi_0)_i$ , then we have a **pullback** from  $\Phi\mathcal{M}_1$  to  $\Phi\mathcal{M}_2$ :  $\alpha^*(M) = A_2[\varphi] \otimes_{A_1[\varphi]} M$  (25.2.9.1).  $\lrcorner$

**Def. (25.2.9.3) [Étale  $\varphi$ -Modules].** If  $A$  is Noetherian, then a  $\varphi$ -module  $M$  is called **étale** iff it is f.g. and the corresponding  $\Phi : (\varphi_0)_* M \rightarrow M$  in (25.2.9.1) is an isomorphism. The subcategory of étale  $\varphi$ -modules is denoted by  $\varphi\text{-Mod}^{\text{ét}}(A)$ .

In case when  $\varphi_0$  is a bijection,  $\Phi$  is a bijection iff  $\varphi$  is a bijection.  $\lrcorner$

*Proof:* If  $\varphi_0$  is a bijection, then  $M \rightarrow (\varphi_0)_* M$  is a bijection by  $\lambda \otimes m \rightarrow \varphi_0^{-1}(\lambda)m$ .  $\square$

**Prop. (25.2.9.4).** If  $A$  is Noetherian and  $A_{\varphi_0}/A$  is flat, then  $\varphi\text{-Mod}^{\text{ét}}(A)$  is a Tannakian category.  $\lrcorner$

*Proof:* 0 is the zero object, the canonical sum&product are clearly étale. And we need to check the kernel and cokernel are étale. But we have an exact sequence  $0 \rightarrow \ker \rightarrow M \rightarrow N \rightarrow \text{Coker} \rightarrow 0$  so we tensor with  $A_{\varphi_0}$  to get a morphism of sequences that  $\varphi_{0*} M \rightarrow M$ ,  $\varphi_{0*} N \rightarrow N$  are both bijective, so by 5-lemma, it is bijection at kernel and cokernel, so they are étale.  $\square$

**Def. (25.2.9.5) [Dual Étale  $\varphi$ -Modules].** For  $E \in \text{Field}$ ,  $M \in \varphi\text{-Mod}^{\text{ét}}(E)$ , the isomorphism  $\Phi : (\varphi_0)_* M \cong M$  induces an isomorphism

$$\Phi^t : M^\vee \cong ((\varphi_0)_* M)^\vee = (\varphi_0)_* M^\vee.$$

Thus the dual  $\Phi^{-t} : (\varphi_0)_* M^\vee \cong M^\vee$  shows  $M^\vee$  is also an étale  $\varphi$ -module.  $\lrcorner$

**Prop. (25.2.9.6)** [ $\mathbb{F}_p$ -Representations and Étale  $\varphi$ -Modules]. Let  $E \in \mathbf{Field}^p$ , then

- For any  $V \in \mathcal{R}\mathrm{ep}_{\mathbb{F}_p}^{\mathrm{fd}}(\mathrm{Gal}_E)$ ,  $V$  is  $E^{\mathrm{sep}}$ -admissible, and

$$D_{E^{\mathrm{sep}}}(V) = (E^{\mathrm{sep}} \otimes V)^{\mathrm{Gal}_E}$$

has a  $\varphi$ -action, and it is an étale  $\varphi$ -module.

- For any  $M \in \varphi\text{-Mod}(E)$ ,

$$\mathbb{V}(M) = (E^{\mathrm{sep}} \otimes_E M)^{\varphi=\mathrm{id}}$$

is a  $\mathbb{F}_p$ -representation of  $\mathrm{Gal}_E$ , and there is an injection

$$\alpha_M : E^{\mathrm{sep}} \otimes_{\mathbb{F}_p} \mathbb{V}(M) \hookrightarrow E^{\mathrm{sep}} \otimes_E M.$$

- These two functors define an equivalence of Tannakian categories (4.2.1.13)

$$D_{E^{\mathrm{sep}}} : \mathcal{R}\mathrm{ep}_{\mathbb{F}_p}(\mathrm{Gal}_E) \cong \varphi\text{-Mod}^{\mathrm{\acute{e}t}}(E) : \mathbb{V}.$$

┘

*Proof:* 1:  $V$  is  $E^{\mathrm{sep}}$ -admissible by (24.1.1.15). To show it is étale, it suffices to show that  $\varphi : D_{E^{\mathrm{sep}}}(V) \rightarrow D_{E^{\mathrm{sep}}}(V)$  is bijective. Let  $e_1, \dots, e_n$  be a basis of  $D_{E^{\mathrm{sep}}}(V)$ , and  $v_1, \dots, v_n$  be a basis of  $V$ , then  $\underline{e} = \underline{v}B$  for some matrix  $B \in \mathrm{GL}(n; E^{\mathrm{sep}})$ . Then if  $[\varphi]\underline{e} = A\underline{e}$  for  $A \in \mathrm{Mat}(n; E)$ , then  $A = B^{-1}\varphi(B)$ , and  $\det(A) = \det(B)^{p-1} \neq 0$ , so  $\varphi$  is bijective.

2: It suffices to show that if  $v_1, \dots, v_h \in \mathbb{V}(M)$  are linearly dependent over  $E^{\mathrm{sep}}$ , then they are dependent over  $\mathbb{F}_p$ . For this, we use induction on  $h$ : Suppose  $\sum \lambda_i v_i = 0$ , we may assume that  $v_h = -1$ , so  $v_h = \sum_{i=1}^{h-1} \lambda_i v_i$ , and by an action of  $\varphi$ , then  $\sum_{i=1}^{h-1} (\lambda_i^p - \lambda_i) v_i = 0$ , so by induction hypothesis,  $\lambda_i \in \mathbb{F}_p$ .

3: For  $V \in \mathcal{R}\mathrm{ep}_{\mathbb{F}_p}(\mathrm{Gal}_E)$ , because  $E$  is  $E^{\mathrm{sep}}$ -admissible,

$$\mathbb{V}(D_{E^{\mathrm{sep}}}(V)) = (E^{\mathrm{sep}} \otimes_E D_{E^{\mathrm{sep}}}(V))^{\mathrm{Gal}_E} \cong (E^{\mathrm{sep}} \otimes_{\mathbb{F}_p} V)^{\mathrm{Gal}_E} = V.$$

Conversely, there is a

To show these are isomorphisms of Tannakian categories, one can easily show that both  $D_{E^{\mathrm{sep}}}$  and  $\mathbb{V}$  preserve tensor products and they identify identity elements  $\mathbb{F}_p$  and  $E$ .  $\square$

**Cor. (25.2.9.7).** For  $E \in \mathbf{Field}^p$  and  $d \in \mathbb{Z}_+$ ,  $\mathcal{R}\mathrm{ep}_{\mathbb{F}_p}^d(\mathrm{Gal}_E)$  is in bijection with the conjugacy classes of matrices in  $\mathrm{GL}(d; E)$  where

$$A \sim B \iff \exists P \in \mathrm{GL}(d; E), B = P^{-1}AP.$$

┘

### Galois-Representations and Étale $\varphi$ -Modules

**Notation (25.2.9.8).**

- Let  $\kappa \in \mathbf{Field}^p$ ,  $\mathcal{O}_K \triangleq \mathrm{Coh}(\kappa)$  (5.5.2.30), and  $K = \mathcal{O}_K[\frac{1}{p}]$ . There is a natural Frobenius  $\varphi : K \rightarrow K$ .
- Obsolete: We used  $E = \kappa$ ,  $\mathcal{O}_{\mathcal{E}} = \mathcal{O}_K$ .

┘



**Prop. (25.2.9.9).** By the functoriality of Cohen rings ?, if  $\mathcal{O}_{\check{K}} = \text{Coh}(\bar{\kappa})\mathcal{E}^{\text{ur}} = \mathcal{O}_{\mathcal{E}^{\text{ur}}}[\frac{1}{p}]$ , then there is a bijection  $\text{Gal}(\mathcal{E}^{\text{ur}}/\mathcal{E}) \cong \text{Gal}_E$ . Thus there are  $\text{Gal}_K$ -action and  $\varphi$ -action on  $\mathcal{O}_{E^{\text{ur}}}, \mathcal{E}^{\text{ur}}$  and by continuity extends to actions on  $\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}}, \widehat{\mathcal{E}^{\text{ur}}}$ , and

$$(\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}})^{\text{Gal}_E} = \mathcal{E}, \quad (\mathcal{E}^{\text{ur}})^{\text{Gal}_E} = \mathcal{O}_E, (\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}})^{\varphi=\text{id}} = \mathbb{Q}_p, \quad (\mathcal{E}^{\text{ur}})^{\varphi=\text{id}} = \mathbb{Z}_p.$$

┘

*Proof:* ?

□

**Prop. (25.2.9.10).** For  $M \in \varphi\text{-Mod}^{\text{ft}}(\mathcal{O}_{\mathcal{E}})$ ,  $M$  is étale over  $\mathcal{O}_E$  iff  $M/(p)$  is étale over  $E$ .

┘

*Proof:* This is because étale is equivalent to the matrix of  $\varphi$  is a bijection, which is equivalent to its reduction modulo  $p$  is a bijection.

□

**Def. (25.2.9.11) [Effective  $\varphi$ -Modules].** An **effective  $\varphi$ -module** over  $\mathcal{E}$  is a  $\varphi$ -module  $(D, \varphi) \in \varphi\text{-Mod}_{\mathcal{E}}$  s.t. there is a complete  $\mathcal{O}_E$ -lattice  $M$  of  $D$  that  $\varphi(M) \subset M$ .

┘

**Def. (25.2.9.12) [Stably-Étale  $\varphi$ -Modules].** A **stably-étale  $\varphi$ -module** over  $\mathcal{E}$  is a  $\varphi$ -module over  $\mathcal{E}$  s.t. there exists a  $\varphi$ -stable  $\mathcal{O}_{\mathcal{E}}$ -lattice in  $\mathcal{E}$  that is an étale  $\varphi$ -module over  $\mathcal{O}_E$ . Then the category of stably-étale  $\varphi$ -modules is a Tannakian category, denoted by  $\varphi\text{-Mod}^{\text{st.ét}}(\mathcal{O}_{\mathcal{E}})$ .

┘

*Proof:* For  $\mathcal{O}_{\mathcal{E}}$  this follows from (25.2.9.4), and for  $\mathcal{E}$ , notice if  $D = M[\frac{1}{p}]$ ,  $D' = M'[\frac{1}{p}]$ , then

$$\text{Hom}_{\varphi\text{-Mod}^{\text{ét}}(\mathcal{E})}(D, D') = \text{Hom}_{\varphi\text{-Mod}^{\text{ét}}(\mathcal{O}_{\mathcal{E}})}(L, L')[\frac{1}{p}]$$

and  $p : D' \rightarrow D'$  is an isomorphism, so it follows that  $\varphi\text{-Mod}^{\text{ét}}(\mathcal{O}_{\mathcal{E}})$  is also Abelian.

□

**Prop. (25.2.9.13).** Any  $V \in \text{Rep}_{\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}}}(\text{Gal}_E)$  is trivial.

┘

*Proof:* Cf.[Fontaine-Ouyang]P34. ?

□

**Thm. (25.2.9.14) [Classification of  $\text{Rep}_{\mathbb{Z}_p}(\text{Gal}_{\kappa})$ ].** For  $V \in \text{Rep}_{\mathbb{Z}_p}(\text{Gal}_{\kappa})$ ,

$$\mathbb{M}(V) = (\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}} \otimes_{\mathbb{Z}_p} V)^{\text{Gal}_E}$$

is an étale  $\varphi$ -module over  $\mathcal{O}_{\mathcal{E}}$ , and for any  $M \in \varphi\text{-Mod}^{\text{ét}}(\mathcal{O}_{\mathcal{E}})$ ,

$$\mathbb{V}(M) = (\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}} \otimes_{\mathcal{O}_{\mathcal{E}}} M)^{\varphi=\text{id}}$$

is a  $\mathbb{Z}_p$ -representation of  $\text{Gal}_E$ . And these two functors define an equivalence of categories:

$$\mathbb{M} : \text{Rep}_{\mathbb{Z}_p}(\text{Gal}_E) \cong \varphi\text{-Mod}^{\text{ét}}(\mathcal{O}_{\mathcal{E}}) : \mathbb{V}.$$

┘

*Proof:* To show  $\mathbb{M}(V)$  is étale, Cf.[Fontaine]P35.

By (25.2.9.15), we have an isomorphism

$$\mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}} \otimes_{\mathbb{Z}_p} \mathbb{V}(M) \cong \mathcal{O}_{\widehat{\mathcal{E}^{\text{ur}}}} \otimes_{\mathcal{O}_{\mathcal{E}}} M,$$

and by (25.2.9.13),

$$\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathcal{O}_{\mathcal{E}}} \mathbb{M}(V) \cong \mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathbb{Z}_p} V.$$

Thus

$$\begin{aligned} \mathbb{V}(\mathbb{M}(V)) &= (\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathcal{O}_{\mathcal{E}}} \mathbb{M}(V))^{\varphi=\text{id}} \cong (\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathbb{Z}_p} V)^{\varphi=\text{id}} = V(25.2.9.9), \\ \mathbb{M}(\mathbb{V}(M)) &= (\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathbb{Z}_p} \mathbb{V}(M))^{\text{Gal}_E} \cong (\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathcal{O}_{\mathcal{E}}} M)^{\text{Gal}_E} = M(25.2.9.9). \end{aligned}$$

□

**Lemma (25.2.9.15).** Situation as in (25.2.9.14),

$$\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathbb{Z}_p} V(M) \cong \mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathcal{O}_{\mathcal{E}}} M.$$

┘

*Proof:* Cf. [Fontaine-Ouyang]P36. ?

□

**Thm. (25.2.9.16) [Classification of  $\text{Rep}_{\mathbb{Q}_p}(\text{Gal}_E)$ ].**

- For  $V \in \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_E)$ ,

$$\mathbb{D}(V) = (\widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathbb{Q}_p} V)^{\text{Gal}_E}$$

is an stably-étale  $\varphi$ -module over  $\mathcal{E}$ , and there is a natural isomorphism

$$\widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathcal{E}} \mathbb{D}(V) \cong \widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathbb{Q}_p} V.$$

- for any  $D \in \varphi\text{-Mod}^{\text{st.ét}}(\mathcal{E})$ ,

$$\mathbb{V}(D) = (\widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathcal{E}} D)^{\varphi=\text{id}}$$

is a  $\mathbb{Q}_p$ -representation of  $\text{Gal}_E$ , and there is a natural isomorphism

$$\widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathbb{Q}_p} \mathbb{V}(D) \cong \widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathcal{E}} D,$$

- These two functors define an equivalence of Abelian Tannakian categories (4.2.1.13):

$$\mathbb{M} : \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_E) \cong \varphi\text{-Mod}^{\text{sst.ét}}(\mathcal{E}) : \mathbb{D}.$$

┘

*Proof:* For any  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_E)$ , by (24.2.3.2), there exists a stable  $\mathbb{Z}_p$ -lattice, thus

$$\widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathbb{Q}_p} V = (\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathbb{Z}_p} T) \left[ \frac{1}{p} \right], \quad \mathbb{D}(V) = \mathbb{D}(T) \left[ \frac{1}{p} \right] = \mathcal{E} \otimes_{\mathcal{O}_{\mathcal{E}}} \mathbb{D}(T).$$

and for any  $D \in \varphi\text{-Mod}^{\text{st.ét}}(\mathcal{E})$ , there exists an  $\mathcal{O}_{\mathcal{E}}$ -lattice stable under  $\varphi$ , which is an étale  $\varphi$ -module over  $\mathcal{O}_E$ . Thus

$$\widehat{\mathcal{E}}^{\text{ur}} \otimes_{\mathcal{E}} D = (\mathcal{O}_{\widehat{\mathcal{E}}^{\text{ur}}} \otimes_{\mathcal{O}_{\mathcal{E}}} M) \left[ \frac{1}{p} \right], \quad \mathbb{V}(D) = V(M) \left[ \frac{1}{p} \right] = \mathbb{Q}_p \otimes_{\mathbb{Z}_p} \mathbb{V}(M).$$

Thus it is clear that the assertion follows from that of (25.2.9.14). □

**Cor. (25.2.9.17).** Étale modules over  $E$  of rank  $d$  are of the form  $M_A = \oplus_{i=1}^d E e_i$  with

$$\varphi : M \rightarrow M : \varphi(\lambda e_j) = \lambda^p \sum_{i=1}^d a_{ij} e_i.$$

Isomorphism classes of  $d$ -dimensional  $\mathbb{Q}_p$ -representations of  $\text{Gal}_E$  are in bijection with the isomorphism classes of matrixes in  $\text{GL}(d; \mathcal{O}_{\mathcal{E}})$  where

$$A \sim B \iff \exists P \in \text{GL}(d; \mathcal{E}), B = P^{-1} A \varphi(P).$$

┘

$(\varphi, \Gamma)$ -Modules

**Def. (25.2.9.18) [  $(\varphi, \Gamma)$ -modules ].** If  $A \in \mathcal{TopCRing}$  with a Frobenius ring map  $\varphi_0 : A \rightarrow A \in \mathcal{TopCRing}$ , and  $A$  has an action of  $\Gamma \in \mathcal{TopGrp}$  that commutes with  $\varphi_0$ , then a  $(\varphi, \Gamma)$ -**module**  $M$  is a  $\varphi$ -module  $M$  over  $A$  with a semi-linear action of  $\Gamma$  that commutes with  $\varphi$ .

If  $A$  is complete and  $\varphi$  is flat, then an **étale**  $(\varphi, \Gamma)$ -**module**  $M$  is a  $(\varphi, \Gamma)$ -module that the  $\varphi$ -module structure is étale (25.2.9.3).

Similar to  $\varphi$ -modules,  $(\varphi, \Gamma)$ -modules form a Tannakian category. ? ⌋

**Thm. (25.2.9.19) [Classification of  $\mathcal{Rep}_{\mathbb{Q}_p}(\text{Gal}_K)$ ].** Cf. [Fontaine, Ouyang]. ⌋

Overconvergent  $(\varphi, \Gamma)$ -ModulesFiltered  $(\varphi, N)$ -Modules

**Def. (25.2.9.20) [  $(\varphi, N, \text{Gal}(L/K))$ -Modules ].** Let  $(L, \kappa_L)/(K, \kappa) \in \mathcal{GalField}$  and  $L_0 = L \cap K_0$ , then the category  $(\varphi, N)\text{-Mod}_{L/K}$  of  $(\varphi, N, \text{Gal}(L/K))$ -modules consists of f.d.  $L_0$ -spaces  $V_0$  with

- a  $\varphi_0$ -semi-linear endomorphism,
- a  $L_0$ -linear endomorphism  $N$ ,
- a semi-linear continuous action of  $\text{Gal}(L/K)$  (w.r.t the discrete topology).

That satisfies:

- $N\varphi = p\varphi N$ ,
- $N, \varphi$  commutes with  $\text{Gal}(L/K)$ -actions.

Notice that the condition implies  $N$  maps the subspace of  $\varphi$ -slope  $l$  to the subspace of  $\varphi$ -slope  $l - 1$ , in particular,  $N$  is nilpotent. ⌋

**Prop. (25.2.9.21).** Situation as in (25.2.9.20),  $(\varphi, N)\text{-Mod}_{L/K}$  is an Abelian Tannakian category, where the actions on the tensor is defined to as follows:

$$\varphi(v \otimes w) = \varphi(v) \otimes \varphi(w), \quad N(v \otimes w) = N(v) \otimes w + v \otimes N(w), \quad g(v \otimes w) = g(v) \otimes g(w).$$

⌋

**Def. (25.2.9.22) [Filtered  $(\varphi, N, \text{Gal}(L/K))$ -Modules, Fontaine[Fon94a]].** Let  $L/K$  be a Galois extension with residue field  $\kappa_L$  and  $L_0 = L \cap K_0$ . Then the category  $(\varphi, N)\text{-Fil Mod}_{L/K}$  of **filtered**  $(\varphi, N, \text{Gal}(L/K))$ -**module** consists of tuples

$$(V, \varphi, N, \text{Gal}(L/K), \text{Fil}^\bullet)$$

where

- $(V, \varphi, N, \text{Gal}(L/K)) \in (\varphi, N)\text{-Mod}_{L/K}$  (25.2.9.20),
- $(V \otimes_{L_0} L, \text{Fil}^\bullet) \in \text{Fil}_L$ .

The category  $\varphi\text{-Fil Mod}_{L/K}$  of **filtered  $\varphi$ -modules(isocrystals)** over  $K$  is the Abelian subcategory of  $(\varphi, N, \text{Gal}(L/K))\text{-Fil Mod}_{L/K}$  with  $N = 0$ .  $\varphi\text{-Fil Mod}_{L/K}$  is also denoted by  $\varphi\text{-Fil Mod}_K$ . ⌋

**Prop. (25.2.9.23) [HN-Formalism for Filtered  $\varphi$ -Modules].** The category  $\varphi\text{-Fil Mod}_{L/K}$  (25.2.9.22) is a HN-formalism where  $\mathcal{A}$  is the Abelian category  $\varphi\text{-Mod}(L_0)$  (25.2.9.20), the rank is defined as usual and the degree is defined to be

$$\deg((V, \varphi, \text{Fil}^\bullet)) = t_{\text{H-T}}(V_L, \text{Fil}^\bullet) - t_N(V, \varphi)$$

where  $t_{\text{H-T}}$  is the Hodge-Tate degree (4.3.5.27) and  $t_N = v_p(\det(\varphi|V))$ . This is a HN-formalism. ⌋

*Proof:* The proof is clear, the same as that of (4.3.5.27).  $\square$

**Def. (25.2.9.24) [Weakly Admissible  $\varphi$ -Modules].** The category  $\varphi\text{-Fil Mod}_{L/K}^{\text{weak. adm}}$  of **weakly admissible  $\varphi$ -modules** is the subcategory of  $\varphi\text{-Fil Mod}_{L/K}$  consisting of objects that are semistable of slope 0 w.r.t the HN-formalism (25.2.9.23). It is an Abelian category by (4.3.5.33).

More explicitly, it is a filtered  $\varphi$ -module  $(V, \varphi, \text{Fil}^\bullet)$  s.t. for any non-empty subobject  $(V', \varphi', \text{Fil}'^\bullet)$ , we have

$$\sum_{i \in \mathbb{Z}} i \dim_L \text{gr}_{\text{Fil}'^\bullet}^i(V' \otimes_{L_0} L) \leq v_p(\det(\varphi|V)),$$

and the equality holds for  $(V, \varphi, \text{Fil}^\bullet)$ .  $\lrcorner$

**Prop. (25.2.9.25)**  $[\varphi\text{-Fil Mod}_{L/K}^{\text{weak. adm}}$  **is Tannakian, Faltings[Fal95]].** The subcategory  $(\varphi, N)\text{-Fil Mod}_{L/K}^{\text{weak. adm}} \subset (\varphi, N)\text{-Fil Mod}_{L/K}$  is stable under tensor products and duals. In particular,  $(\varphi, N)\text{-Fil Mod}_{L/K}^{\text{weak. adm}}$  is an Tannakian category.  $\lrcorner$

*Proof:* [Tensor Products in  $p$ -adic Hodge, Totaro, P9,12].  $\square$

**Cor. (25.2.9.26).** The subcategory  $\varphi\text{-Fil Mod}_K^{\text{weak. adm}} \subset \varphi\text{-Fil Mod}_K$  is stable under tensor products and duals.  $\lrcorner$

**Cor. (25.2.9.27).** Tensor product of semistable filtered isocrystals in  $(\varphi, N)\text{-Fil Mod}_{L/K}$  is also semistable.  $\lrcorner$

*Proof:* This is because shifting the filtration of a semistable filtered isocrystal of slope  $c$  down by  $c$  gives a weakly admissible filtered isocrystal.  $\square$

**Def. (25.2.9.28) [Admissible Filtered  $\varphi$ -Modules].**

$$D_{\text{crys}} : \text{Rep}_{\mathbb{Q}_p}^{\text{crys}}(\text{Gal}_K) \rightarrow \varphi\text{-Fil Mod}_K^{\text{weak. adm}}$$

is a fully faithful tensor functor. Objects in the image are called **admissible filtered  $\varphi$ -modules**.  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (25.2.9.29) [Weakly Admissible Implies Admissible, Colmez-Fontaine[C-F00]].** The natural map  $D_{\text{crys}}$  (25.2.9.28) induces an isomorphism of Tannakian categories

$$\text{Rep}_{\mathbb{Q}_p}^{\text{crys}}(\text{Gal}_K) \cong \varphi\text{-Fil Mod}_K^{\text{weak. adm}}.$$

*Proof:*  $\square$

## 10 $(\varphi, \Gamma)$ -Modules over the Robba Ring

**Notation (25.2.10.1).**

- Let  $p \in \text{Prime}$ ,  $K \in p\text{-NField}$ ,  $\Gamma = \mathbb{Z}_p^*$ .
- Use notations on Robba rings define in Robba Ring.
- Let  $H_p = \ker(\text{cycl}_p) \subset \text{Gal}_{\mathbb{Q}_p}$ .

$\lrcorner$

**Prop. (25.2.10.2)**[Fontaines Ring  $B^{\dagger, \text{rig}}$ ]. There exists a  $(\text{Gal}_{\mathbb{Q}_p}, K)$ -regular ring  $B^{\dagger, \text{rig}}$  with an action of  $\text{Gal}_{\mathbb{Q}_p}$  and Frobenius  $\varphi$  s.t.  $(B^{\dagger, \text{rig}})^{H_p} = \mathcal{R}_K$ , and we can define

$$D_{\dagger, \text{rig}} : \text{Rep}_K^{\text{fd}}(\text{Gal}_K) \rightarrow (\varphi, \Gamma)\text{-Mod}^{\text{ét}}(\mathcal{R}_K) : V \mapsto (V \otimes_{\mathbb{Q}_p} B^{\dagger, \text{rig}})^{H_p}.$$

┘

*Proof:* Cf.[Ber02]Section3.4.?

□

**Thm. (25.2.10.3)**[Colmez]. The functor  $D_{\dagger, \text{rig}}$  induces a tensor equivalence

$$\text{Rep}_K^{\text{fd}}(\text{Gal}_{\mathbb{Q}_p}) \cong (\varphi, \Gamma)\text{-Mod}^{\text{ét, slope}=0}(\mathcal{R}_K).$$

┘

*Proof:* Cf.[P. Colmez, Série principale unitaire pour  $\text{GL}_2(\mathbb{Q}_p)$  et représentations triangulines de dimension 2]Prop2.7.?

□

### Triangularizable $(\varphi, \Gamma)$ -Modules

**Def. (25.2.10.4)** [Classification of 1-Dimensional  $(\varphi, \Gamma)$ -Modules]. Let  $\delta : \mathbb{Q}_p^* \rightarrow K^*$  be a continuous character, then there is a 1-dimensional  $(\varphi, \Gamma)$ -module over  $\mathcal{R}_K$  with basis  $x$  s.t.  $\varphi(a) = \delta(p)a$ ,  $\gamma(a) = \delta(\gamma)a$ . Then

- Any  $(\varphi, \Gamma)$ -module of rank 1 over  $\mathcal{R}_K$  is isomorphic to  $R_K(\delta)$  for a unique  $\delta$ . And it is isoclinic of slope  $v(\delta(p))$ ?
- $\text{Ext}_{(\varphi, \Gamma)}(\mathcal{R}_K(\delta_2), \mathcal{R}_K(\delta_1))$  has dimension 1 except when  $\delta_1 \delta_2^{-1} = x^{-i}$  or  $\chi_{\text{cycl}} x^i$  for  $i \in \mathbb{N}$  (where  $x : \mathbb{Q}_p^* \rightarrow K^*$  is the natural inclusion), in which case it has dimension 2.

┘

*Proof:* Cf.[P. Colmez, S'erie principale unitaire pour  $\text{GL}_2(\mathbb{Q}_p)$  et repr'esentations triangulines de dimension 2]Section0.1.?

□

**Def. (25.2.10.5)** [Triangularizable Representations]. A  $(\varphi, \Gamma)$ -module  $D$  over  $\mathcal{R}_K$  is called a **triangularizable  $(\varphi, \Gamma)$ -module** if there is a complete flag of  $(\varphi, \Gamma)$ -submodules which are direct summands of  $D$ .

A Galois representation  $V \in \text{Rep}_K^{\text{fd}}(\text{Gal}_{\mathbb{Q}_p})$  is called triangularizable if  $D_{\dagger, \text{rig}}(V)$  is triangularizable.

For any triangularizable  $(\varphi, \Gamma)$ -module  $D$  over  $\mathcal{R}_K$ , the graded pieces  $\text{gr}_i(D)$  are isomorphic to  $\mathcal{R}_K(\delta_i)$  where  $\delta_i : \mathbb{Q}_p^\times \rightarrow L^\times$  by (25.2.10.4). These characters  $\delta_i$  are called the **parameters of  $D$** . ┘

### $p$ -adic Hodge Theory of $(\varphi, \Gamma)$ -Modules

**Thm. (25.2.10.6)** [Comparison of Fontaine's Functors]. For  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_{\mathbb{Q}_p})$  and  $* \in \{\text{crys}, \text{dR}, \text{Sen}\}$ ,

$$D_*(D_{\dagger, \text{rig}}(V)) = D_*(V).$$

┘

*Proof:* Cf.[Ber02]Thm0.2 and Section5.3, [L. Berger, E'quations diff'erentielles  $p$ -adiques et  $(\varphi, N)$ -modules filtr'es] and [P. Colmez, Les conjectures de monodromie  $p$ -adiques]Prop5.6.?

□

### 25.3 $p$ -adic Differential Equations

References are [Kedlaya].

## 25.4 $p$ -adic Families of Galois Representations

References are [L-Z17].

**Notation (25.4.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $k \in p\text{-Field}$ .
- Let  $K \in \text{PerfdField}$  s.t.  $k(\mu_{p^\infty}) \subset K \subset \mathbb{C}_K$ .
- Use notations on adic spaces defined in [Adic Spaces](#).
- Use notations on perfectoid spaces defined in [Perfectoid Spaces \(Scholze\)](#).
- Let  $\epsilon = \epsilon_p = (1, \zeta_p, \zeta_{p^2}, \dots) \in \mathcal{O}_{K^\flat}$ .
- Let  $X \in \text{AdicSpa}^{\text{loc.ft}}/k$ .
- For any local system  $\mathbb{L} \in \text{LocSys}(X_{\text{ét}})$ , let  $\widehat{\mathbb{L}}$  be the pullback to  $X_{\text{pro-ét}}$ .

┘

**Thm. (25.4.0.2) [Rigidity for  $p$ -adic Local Systems, Liu-Zhu [L-Z17]].** Let  $X \in \text{AdicVar}/k$  and  $\mathbb{L} \in \text{LocSys}_{\mathbb{Q}_p}^{\text{ét}}(X)$ . If  $x \in |X|_0$  and  $\mathbb{L}_{\bar{x}} \in \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_{\kappa(x)})$  is deRham, then  $\mathbb{L}_{\bar{y}} \in \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_{\kappa(x)})$  is deRham for any  $y \in |X|_0$ . Moreover,  $\mathbb{L}_{\bar{y}}$  has the same Hodge-Tate weights as  $\mathbb{L}_{\bar{x}}$ .

┘

*Proof:* By resolution of singularity, Cf. [Canonical desingularization in characteristic zero by blowing up the maximum strata of a local invariant]?, it suffices to prove for  $X$  smooth.

Then this follows from (25.4.2.2)?.

□

**Remark (25.4.0.3).** The semi-stable version of this result is false. But a pot. semi-stable version could be true.

┘

**Thm. (25.4.0.4) [Local Constancy of  $\ell$ -adic Local Systems, Kisin].** For  $X \in \text{Sch}^{\text{ft}}/k$ , any  $\mathbb{L} \in \text{LocSys}_{\mathbb{Z}_\ell}(X)$  is locally constant in the  $p$ -adic topology.

┘

*Proof:* [Kisin, Local constancy in  $p$ -adic families of Galois representations (1999)].

□

**Thm. (25.4.0.5) [Rigidity for  $p$ -adic Local Systems, Liu-Zhu [L-Z17]].** Let  $F \in \text{NField}$  and  $X \in \text{Var}/F$  with generic point  $\eta$  and  $\mathbb{L} \in \text{LocSys}_{\mathbb{Q}_p}(X_{\text{ét}})$ . If  $x \in |X|$  and  $\mathbb{L}_{\bar{x}} \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_{\kappa(x)})$  is geometric (30.1.2.1), then  $\mathbb{L}_{\bar{y}} \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_{\kappa(x)})$  is geometric for any  $y \in |X|$ .

┘

*Proof:* Cf. [L-Z17] P335.?

□

**Thm. (25.4.0.6) [Geometric Local System on Shimura Varieties, Liu-Zhu [L-Z17]].** let  $(G, X)$  be a Shimura datum and  $\text{Sh}_{\mathcal{K}}(G, X)$  be its associated Shimura variety over the reflex field  $E$ .

Let  $V \in \text{Rep}_{\mathbb{Q}}^{\text{fd}}(G)$  be trivial on the largest anisotropic subtorus  $Z_G^s$  in the center of  $G$  that is  $\mathbb{R}$ -split, which induces a Betti local system  $\mathbb{L}_V \in \text{LocSys}_{\mathbb{Q}}(\text{Sh}_{\mathcal{K}}(G, X))$ , and an étale local system  $\mathbb{L}_{V,p} \in \text{LocSys}_{\mathbb{Q}_p}(\text{Sh}_{\mathcal{K}}(G, X))$ .

Then for any  $x \in |\text{Sh}_{\mathcal{K}}(G, X)|_0$ ,  $(\mathbb{L}_{V,p})_{\bar{x}} \in \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_{E(x)})$  is geometric (30.1.2.1).

┘

*Proof:* ?

□

## 1 $p$ -adic Simpson Correspondence

References are [Faltings, A  $p$ -adic Simpson correspondence] and [L-Z17].

## 2 $p$ -adic Riemann-Hilbert Correspondence

**Thm. (25.4.2.1) [ $p$ -adic Riemann-Hilbert Correspondence, geometric version, Liu-Zhu [L-Z17]].** Let  $X \in \text{AdicSmVar}/k$  and  $\mathbb{L} \in \text{LocSys}_{\mathbb{Q}_p}(X_{\text{ét}})$ , then

- $R^i \nu'_*(\widehat{\mathbb{L}} \otimes \mathcal{O}_{\mathbb{B}_{\text{dR}}}^{[a,b]}) = 0$  for  $i \in \mathbb{Z}_+$ , and the functor

$$\mathcal{RH} : \text{LocSys}_{\mathbb{Q}_p}(X_{\text{ét}}) \rightarrow \text{Gal}(K/k)\text{-FilVect}^{\nabla, \text{Griffith transverse}} / \mathcal{X} : \mathbb{L} \mapsto (\mathcal{RH}(\mathbb{L}), \nabla_{\mathbb{L}})$$

is a tensor functor, where  $\mathcal{RH}(\mathbb{L}) \stackrel{\Delta}{=} \nu'_*(\widehat{\mathbb{L}} \otimes \mathcal{O}_{\mathbb{B}_{\text{dR}}})$  and

$$\nabla_{\mathbb{L}} : \mathcal{RH}(\mathbb{L}) \rightarrow \mathcal{RH}(\mathbb{L}) \otimes_{\mathcal{O}_{\mathcal{X}}} \Omega_{\mathcal{X}/\mathbb{B}_{\text{dR}}}$$

is an integrable connection satisfying the Griffith transversality condition?.

- There is a canonical isomorphism

$$(\text{gr}^0 \mathcal{RH}(\mathbb{L}), \text{gr}^0(\mathbb{L})) \cong (\mathcal{H}(\mathbb{L}), \vartheta_{\mathbb{L}})$$

compatible with Higgs fields on both sides, where  $\mathcal{H}(\mathbb{L}) = R\nu'_*(\widehat{\mathbb{L}} \otimes \mathcal{O}_{\mathbb{C}_K})$ .

- There is a natural isomorphism

$$\nu!^* \mathcal{RH}(\mathbb{L}) \otimes_{\nu^* \mathcal{O}_{\mathcal{X}}} \mathcal{O}_{\mathbb{B}_{\text{dR}}} |_{(X_K)_{\text{pro-ét}}} \cong (\widehat{\mathbb{L}} \otimes \mathcal{O}_{\mathbb{B}_{\text{dR}}}) |_{(X_K)_{\text{pro-ét}}}.$$

- If  $f : X \rightarrow Y \in \text{AdicSmVar}/k$ , then there is a natural isomorphism

$$f^*(\mathcal{RH}(\mathbb{L}), \nabla_{\mathbb{L}}) \cong (\mathcal{RH}(f^*\mathbb{L}), \nabla_{f^*\mathbb{L}})$$

- Assume  $f : X \rightarrow Y \in \text{AdicSmVar}/k$  is smooth and proper and  $\mathbb{L} \in \text{LocSys}_{\mathbb{Z}_p}(X_{\text{ét}})$  s.t.  $R^i f_* \mathbb{L} \in \text{LocSys}_{\mathbb{Z}_p}(Y_{\text{ét}})$  for each  $i$ , then there is a natural isomorphism

$$R^i f_{\text{dR}*}(\mathcal{RH}(\mathbb{L}), \nabla_{\mathbb{L}}) \cong (\mathcal{RH}(R^i f_* \mathbb{L}), \nabla_{R^i f_* \mathbb{L}}).$$

⌋

*Proof:* Cf. [L-Z17]P325. □

**Thm. (25.4.2.2) [ $p$ -adic Riemann-Hilbert Correspondence, arithmetic version, Liu-Zhu [L-Z17]].** For  $X \in \text{AdicSmVar}/k$  and  $\mathbb{L} \in \text{LocSys}_{\mathbb{Q}_p}(X_{\text{ét}})$ ,

- $(D_{\text{dR}}^i(\mathbb{L}), \nabla_{\mathbb{L}})$  is a vector bundle with an integrable connection on  $X$ . And it vanishes if  $i \geq 2$ .
- If  $f : X \rightarrow Y \in \text{AdicSmVar}/k$ , then there is a canonical isomorphism

$$f^*(D_{\text{dR}}^i(\mathbb{L}), \nabla_{\mathbb{L}}) \cong (D_{\text{dR}}^i(f^*\mathbb{L}), \nabla_{f^*\mathbb{L}}).$$

- If  $x \in |X|_0$  s.t.  $\mathbb{L}_{\bar{x}}$  is deRham, then the spectral sequence computing  $D_{\text{dR}}^i(\mathbb{L})$  associated to the natural filtration on  $\mathcal{O}_{\mathbb{B}_{\text{dR}}}$  degenerates at  $E_1$ -page, and induces a filtration  $\text{Fil}^\bullet$  on  $D_{\text{dR}}^i(\mathbb{L})$  by subbundles.

Moreover,  $\nabla_{\mathbb{L}}$  satisfies Griffith transversality with this filtration, and the isomorphism in item2 preserves this filtration.



- $\mathbb{L}$  is deRham in the sense of [Scholze]Def8.3, and  $(D_{\mathrm{dR}}^0(\mathbb{L}), \nabla, \mathrm{Fil}^\bullet)$  is the associated filtered  $\mathcal{O}_X$ -module with an integrable connection defined in [Scholze]Def7.4. Moreover,  $D_{\mathrm{dR}}^1(\mathbb{L}) \cong D_{\mathrm{dR}}^0(\mathbb{L})$ .
- $D_{\mathrm{dR}}^0$  is a tensor functor from  $\mathrm{LocSys}_{\mathbb{Q}_p}^{\mathrm{dR}}(X_{\mathrm{\acute{e}t}})$  to the category of filtered  $\mathcal{O}_X$ -modules with an integrable connection satisfying the Griffith transversality.

┘

*Proof:* Cf. [L-Z17]P328.

□

### 3 Logarithmic Riemann-Hilbert correspondences

#### 4 Relative Fontaine-Mazur Conjecture

**Conj.(25.4.4.1) [Relative Fontaine-Mazur Conjecture, Liu-Zhu[L-Z17]].** Let  $F \in \mathbf{NField}$  and  $X \in \mathcal{V}\mathrm{ar}/F$  with generic point  $\eta$  and  $\mathbb{L} \in \mathrm{LocSys}_{\mathbb{Q}_p}^{\mathrm{\acute{e}t}}(X)$ . If  $x \in |X|$  and  $\mathbb{L}_{\bar{x}} \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_{\kappa(x)})$  is geometric(30.1.2.1), then there exists  $Y \in \mathrm{Sch}^{\mathrm{ft}}/\kappa(\eta)$  s.t.  $\mathbb{L}_{\bar{\eta}}$  appears in as a subquotient of  $H_{\mathrm{\acute{e}t}}^*(Y_{\bar{\eta}})$  up to Tate twists.

┘

*Proof:*

□



## 26 | Automorphic Representations

### 26.1 Archimedean Automorphic Representations

References are [HC68, P-R94, Bor97, Bum97], [Automorphic Forms and  $L$ -Functions for the Group  $\mathrm{GL}(n; \mathbb{R})$ , Goldfeld], [Classification of Irreducible Representations of Real Algebraic Groups, Langlands] and [course notes given by Liang Xiao], [Classical Automorphic Forms, Iwaniec].

**Notation (26.1.0.1).**

- Let  $K \in \{\mathbb{R}, \mathbb{C}\}$ .
- Let  $G$  be a real reductive group.
- Define a height on  $G$  is induced from an inclusion

$$(\rho, \rho^{-1}) : G \rightarrow \mathrm{Mat}(n; \mathbb{R}) \times \mathrm{Mat}(n; \mathbb{R}) : g \mapsto (g, g^{-1}).$$

where  $\rho$  is a faithful representation of  $G$ . This choice of  $\rho$  is not important here.

- Use notations on arithmetic of algebraic groups defined in [Arithmetic of Algebraic Groups](#).
- Use notations on arithmetic subgroups defined in [Arithmetic Subgroups\(Fundamental Domains\)](#).
- Let  $Z(G)$  be a greatest  $\mathbb{Q}$ -split torus in the center of  $G$ .
- Let  $\mathcal{K} \leq G(\mathbb{R})$  be a maximal compact subgroup.
- If  $G = \mathrm{GL}(n; K)^+$ , denote  $G_1 = \mathrm{SL}(n; K)$ ,  $\mathcal{K} = \mathrm{SO}(n; K)$ ,  $B$  the upper triangular matrices,  $\mathrm{Unip}(n; K)$  the group of unipotent upper triangular matrices in  $G$ ,  $T$  the group of diagonal matrices.
- Define right regular action  $\rho$  of  $G$  on  $C^\infty(G)$ , and also the left regular action  $\lambda$ . We will write  $dX$  for  $X \in \mathfrak{g}$  as the representation of Lie algebra of  $G$  via  $\rho$ , then it commutes with  $\lambda$ . So it induces a map of  $U(\mathfrak{g})$  to the ring of left  $G$ -invariant differential operators on  $G$  [\(24.6.1.1\)](#).
- Denote  $\mathcal{Z} = Z(U(\mathfrak{g}))$ .
- Fix an additive character  $\psi : K \rightarrow \mathbb{C}^\times$ .
- Define a character  $\psi_{\mathrm{Unip}}$  on  $\mathrm{Unip}(n; K)$  by

$$\varphi_{\mathrm{Unip}}(u) = \psi\left(\sum_{i=1}^{n-1} u_{i,i+1}\right).$$

- Let  $\chi \in \Gamma^\vee$ .

- Let  $\omega$  be a character of the center  $Z(\mathbb{R}) \subset G$ . Assume that  $\omega(-1) = \chi(-1)$ .

┘

**Def. (26.1.0.2) [Archimedean Norm].** Take an embedding  $G \hookrightarrow \mathrm{GL}(n)$  and define the norm on  $G$  by

$$\|g\| = \max(|g_{ij}|, |g_{ij}^{-1}|).$$

┘

**Prop. (26.1.0.3).** Choose an Archimedean norm (26.6.0.4), then there exists  $C \in \mathbb{R}_+$  s.t.

$$\|xy\| \leq C \|x\| \|y\|.$$

┘

## 1 Basics

**Def. (26.1.1.1) [Form Spaces].** Let  $C^\infty(\Gamma \backslash G, \chi, \omega)$  be the space of smooth functions  $f \in C^\infty(G)$  s.t.

$$f(\gamma g) = \chi(\gamma) f(g), \text{ where } \gamma \in \Gamma, g \in G,$$

$$f(zg) = \omega(z) f(g), \text{ where } z \in Z(\mathbb{R}), g \in G.$$

Let the subspace  $C_c^\infty(\Gamma \backslash G, \chi, \omega)$  be those functions  $f$  that  $|f|$  is compactly supported in  $G_1$ . Similarly we define  $C(\Gamma \backslash G, \chi, \omega)$  and  $C_c(\Gamma \backslash G, \chi, \omega)$ .

**Def. (26.1.1.2) [Archimedean Automorphic Forms].** Let the space of **automorphic forms**  $\mathcal{A}(\Gamma \backslash G, \chi, \omega)$  be the subspace of  $C^\infty(\Gamma \backslash G, \chi, \omega)$  (26.1.1.1) consisting of  $\mathcal{K}$ -finite and  $\mathcal{Z}$ -finite functions satisfies the **condition of moderate growth**:  $|F(g)| < C \|g\|^N$  (26.1.0.2) for some  $C \in \mathbb{R}_+, N \in \mathbb{Z}$ .

And  $f$  is said to be **rapidly decreasing** if for any  $N \in \mathbb{Z}$ , there exists  $C_N \in \mathbb{R}_+$  s.t.  $|F(g)| < C_N \|g\|^{-N}$  (26.1.0.2).

When  $\omega$  is trivial on  $Z^+(\mathbb{R})$ , let  $\mathcal{A}(\Gamma \backslash G, \chi, k)$  be the subspace of  $\mathcal{A}(\Gamma \backslash G, \chi)$  consisting of functions  $f$  that  $\rho(k_\theta)(f) = e^{ik\theta} f$ .

Automorphic forms are real analytic, by (26.1.1.15).

┘

**Def. (26.1.1.3) [Cuspidal Forms for  $\mathrm{GL}(2)$ ].** If  $\infty$  is a cusp of  $\Gamma$ , then  $\{\pm 1\}\Gamma_\infty = \{\pm 1\} \left\langle \begin{bmatrix} 1 & r \\ & 1 \end{bmatrix} \right\rangle$ , so a  $F \in \mathcal{A}(\Gamma \backslash G, \chi, \omega)$  is called **cuspidal** at  $\infty$  iff  $\chi(\tau_r) \neq 1$  or

$$\int_0^r F\left(\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g\right) dx = 0.$$

for any  $g \in G$ . Notice this is independent of  $r$  chosen.

More generally, if  $a$  is a cusp of  $\Gamma$ , then choose  $\xi \in \mathrm{SL}(2; \mathbb{R})$  that  $\xi(\infty) = a$ , then for  $F \in \mathcal{A}(\Gamma \backslash G, \chi, \omega)$ ,  $F'(g) = F(\xi g) \in \mathcal{A}(\Gamma' \backslash G, \chi', \omega)$ , where  $\Gamma' = \xi^{-1}\Gamma\xi$ ,  $\chi'(\gamma') = \chi(\xi\gamma'\xi^{-1})$  (Because left and right actions commute). Then  $F$  is called **cuspidal** at  $a$  iff  $F'$  is cuspidal at  $\infty$ .

The subspace of cuspidal forms is denoted by  $\mathcal{A}_0(\Gamma \backslash G, \chi, \omega) \subset \mathcal{A}(\Gamma \backslash G, \chi, \omega)$ .

┘

### $L^p$ -Spaces

**Def. (26.1.1.4) [ $L^p$ -Spaces].** For  $p \in \mathbb{R}_{\geq 1}$ ,  $L^p(\Gamma \backslash G, \chi, \omega)$  is defined to be the space of functions on  $G$  that satisfies conditions in (26.1.1.1) and is  $L^p$ -integrable on  $\Gamma \backslash Z(\mathbb{R}) \backslash G(\mathbb{R})$  (because it can descend) with the quotient Haar measure (16.6.1.3). ┘

$L_0^p(\Gamma \backslash G, \chi, \omega)$  is the subspace of  $L^p(\Gamma \backslash G, \chi, \omega)$  consisting of cuspidal elements, where cuspidality is defined the same way as in (26.1.1.3) but in the sense that holds for a.e.  $g$ .

When  $G = \mathrm{GL}(2)$ ,  $L^p(\Gamma \backslash \mathrm{GL}(2), \chi, \omega, k)$  be the subspace of  $L^2(\Gamma \backslash \mathrm{GL}(2), \chi, \omega)$  consisting of functions  $F$  that  $\rho(k_\theta)F = e^{ik\theta}F$ . ┘

**Prop. (26.1.1.5).** For  $p \in \mathbb{R}_{\geq 1}$ , the space  $C_c(\Gamma \backslash G, \chi, \omega)$  is dense in  $L^p(\Gamma \backslash G, \chi, \omega)$ . ┘

*Proof:* Firstly we show  $C_c(\Gamma \backslash G, \chi)$  is dense in  $L^2(\Gamma \backslash G, \chi)$ . Let  $\mathcal{F}$  be a Poincaré fundamental domain for  $\Gamma$  in  $\mathrm{SL}(2; \mathbb{R})$  (26.1.1.10), then elements in  $C_c(\mathcal{F})$  can be extended to elements of  $C_c(\Gamma \backslash G, \chi)$ , and in this way,  $L^2(\Gamma \backslash G, \chi)$  is identified with  $L^2(\mathcal{F})$ , so the claim follows from (12.3.8.7). □

**Cor. (26.1.1.6).** The right regular action of  $G$  extends to a continuous unitary representation of  $G$  on  $L^2(\Gamma \backslash G, \chi, \omega)$ , and  $L_0^2(\Gamma \backslash G, \chi, \omega)$  is invariant under this representation. ┘

*Proof:* We must verify continuity, and this is clear using the proposition because we can choose a compact supported function  $f$  to approximate, then the right action is uniformly continuous. The unitarity is clear. The invariance of  $L_0^2(\Gamma \backslash G, \chi)$  is clear. □

**Remark (26.1.1.7).** Can this be extended to arbitrary locally compact group  $G$ ? ?, compare with (24.7.1.14). ┘

**Prop. (26.1.1.8) [ $L^p$ -Functions are of Moderate Growth].** Let  $p \in \mathbb{R}_{\geq 1}$  and  $f \in L^p(\Gamma \backslash G, \chi, \omega)$ . If  $f$  is  $\mathcal{K}$ -finite and  $\mathcal{Z}$ -finite, then  $f$  is automatically of moderate growth (26.1.1.2), thus an automorphic form. ┘

*Proof:* Because  $Z(\mathbb{R})\Gamma \backslash G(\mathbb{R})$  has finite volume ?, it suffices to prove for  $p = 1$ . For this, Cf. [BJ79]. ? □

### Siegel Sets

**Def. (26.1.1.9) [Siegel Sets].** ┘

**Prop. (26.1.1.10) [Poincaré Fundamental Domain].** Fundamental domain for  $\Gamma \subset \mathrm{GL}(2; \mathbb{R})$  acting on  $\mathcal{H}$  is defined in (12.10.3.4). There is a well-shaped open subset  $F$  with a set  $F \subset F' \subset \bar{F}$  that  $F'$  is a fundamental domain for  $\Gamma$  acting on  $\mathcal{H}$ .

Notice that if  $\mathcal{F}$  is a fundamental domain for  $\Gamma$  in  $\mathcal{H} \cong \mathrm{SL}(2; \mathbb{R})/\mathrm{SO}(2)$ , then the inverse image of  $\mathcal{F}$  is a fundamental domain for  $\Gamma$  in  $\mathrm{SL}(2; \mathbb{R})$ , because a.e.  $z \in \mathcal{H}$  is not fixed by any  $\gamma \in \Gamma$ . ┘

*Proof:* Choose a  $z \in \mathcal{H}$  which is not fixed by any  $\gamma \in \Gamma$ , then for any  $\gamma \in \Gamma$ , draw the circle of points that have the same distance to  $z$  and  $\gamma(z)$ , then the intersection of all the part containing  $z$  is a fundamental domain. ? □

**Prop. (26.1.1.11) [Siegel].** If a Poincaré fundamental domain  $\Omega$  has finite area, then  $\partial\Omega$  is a union of f.m. geodesics, and  $\partial\bar{\Omega} \cap X$  is finite, and  $\Gamma(\partial\bar{\Omega} \cap X)$  is the set of cusps for  $\Gamma$ . ┘

*Proof:* The volume of  $\Omega$  has a relation with the angles of geodesics of  $\Omega$ , so it can only have f.m. vertices with interior angle  $< 0.9\pi$ . And the  $\Gamma$ -orbits of these angles intersect  $\bar{\Omega}$  at f.m. vertices, which is because they consists of  $\Gamma$ -conjugates with the same distance with  $z_0$ , and a compact set meets f.m. geodesics of  $\Omega$ , because  $\Gamma z_0$  is discrete in  $\mathcal{H}$ . In particular,  $\Gamma$  has f.m. vertices in the boundary.

Now if  $\Gamma$  has infinitely many vertices, there is a vertex  $b$  that all its  $\Gamma$ -conjugates have angles  $> 0.9\pi$ , then consider the  $\Gamma$ -conjugates of  $\Omega$  with a vertex  $b$ , then all their angle at  $b > 0.9\pi$ , but this cannot be possible geometrically.

For any cusp  $x$  of  $\Gamma$ , suppose the Poincare fundamental domain is defined using  $z$ , notice there is a  $\gamma \in \Gamma$  that  $d(\gamma x, z)$  attains minimum, then  $\gamma x$  must be in the boundary of  $\Gamma$ , as  $d(\gamma_0 z, \gamma x) = d(z, \gamma x)$ , and no other  $z$  are closer to  $\gamma x$ .  $\square$

**Prop. (26.1.1.12)[Siegel Set and Fundamental Set].** Siegel set is a nicely shaped substitutes for a fundamental domain:

- Let  $a_1, \dots, a_n \in \mathbb{R} \cup \{\infty\}$  be a representation of the  $\Gamma$ -orbits of cusps of  $\Gamma$ (26.1.1.11), let  $\xi \in SL(2, \mathbb{R})$  be chosen that  $\xi_i(a_i) = \infty$ . If  $c > 0, d > 0$  be chosen suitably, then the set  $\cup \xi_i^{-1} \mathcal{F}_{c,d}$  contains a fundamental domain of  $\Gamma$ .
- Suppose  $\infty$  is a cusp of  $\Gamma$ , then if  $d$  is large enough, then  $\mathcal{F}_d^\infty$  contains a fundamental domain for  $\Gamma$ .

┘

*Proof:* 1:  $\xi_i \Gamma \xi_i^{-1}$  contains a unipotent subgroup generated by  $\begin{bmatrix} 1 & \delta_i \\ & 1 \end{bmatrix}$ , so if  $d \geq \delta_i$ , then  $\xi_i^{-1} \mathcal{F}_{c,d}$  contains a nbhd of the cusp  $a_i$  in the fundamental domain  $F$  of  $\Gamma$ . so  $F - \cup \xi_i^{-1} \mathcal{F}_{c,d}$  is precompact in  $\mathcal{H}$ , by (26.1.1.11). Now if  $c = 0, d = \infty$ , then  $F - \cup \xi_i^{-1} \mathcal{F}_{c,d} = \emptyset$ , then this is true for some  $c, d$ .

For 2: because  $\infty$  is a cusp, we may assume  $F \in \mathcal{H} \cap \{x > 0\}$ . If  $d$  is large, then  $d$  will contain each of the pieces  $\xi_i^{-1} \mathcal{F}_{c,d} \cap F$  in item 1.  $\square$

**Prop. (26.1.1.13)[Compatibility of Growth Conditions].** Let  $G/\mathbb{R}$  be semisimple,  $f$  be a function on  $\Gamma \backslash G(\mathbb{R})$ , then the following are equivalent:

- $f$  is of moderate growth/rapidly decreasing (26.1.1.2).
- $f$  is of moderate growth/rapidly decreasing on each cusp of  $\Gamma$ ?
- $f$  is of moderate growth/rapidly decreasing on each Siegel set  $\mathfrak{S}_{i,c,d}$ .

┘

*Proof:* Cf. [Bor97] P5.11. ? We only prove for  $SL(2; \mathbb{R})$ .  $\square$

### Harish-Chandra Theorem

**Thm. (26.1.1.14)[Harish-Chandra].** If  $f \in C^\infty(G)$  is both  $K$ -finite and  $\mathcal{Z}$ -finite, then

- $f$  is real analytic.
- $U(\mathfrak{g})f$  is an admissible  $(\mathfrak{g}, K)$ -module.
- There exists  $\alpha \in C_c^\infty(G)$  that

$$\alpha(kgk^{-1}) = \alpha(g), \forall k \in \mathcal{K}, \quad f * \alpha = f.$$

- If  $|f(g)| < C \|g\|^N$ , then all  $U(\mathfrak{g})f$  satisfies similar inequalities with the same  $N$ , i.e.  $f$  is uniformly of moderate growth.

⌋

*Proof:* We prove only for  $G = \mathrm{SL}(2)$ . See [Harish-Chandra, Discrete series for semisimple Lie groups, II. Explicit determination of the characters. Acta. Math. 116 (1966)] or [Gan's notes, P24] for the general case. ?

Because  $W$  lies in the Lie algebra of  $K$  and  $W = iH$ , the hypothesis implies  $\mathcal{R}f$  is f.d., where  $\mathcal{R} = \mathbb{C}[\Delta, H]$ . Let  $V$  be the smallest closed  $G$ -invariant subspace of  $C^\infty(G)$  containing  $f$  and let  $V_0 = U(\mathfrak{g})f$ .

We first prove that

$$V_0 = \bigoplus_{n=-\infty}^{\infty} (V_0 \cap V(n)).$$

Notice there is a continuous projection of  $V$  onto  $V(n)$ :

$$E_n \varphi(g) = \frac{1}{2\pi} \int_0^{2\pi} e^{-in\theta} \varphi(gk_\theta) d\theta,$$

because  $f$  is  $K$ -finite, there is an  $N$  s.t.  $f = \sum_{n=-N}^N E_n f$ . Then notice any  $Df$  is also  $K$ -finite because  $D$  is combination of polynomials in  $R, L, H$  and  $R, L, H$  shift the weights, so the LHS is contained in the RHS. It's left to show now  $E_n Df \in V_0$  for any  $n$ :  $E_n Df$  can be extracted using Lagrange polynomial in  $H$ , so it is clearly in  $V_0$ .

Next we show  $V_0 \cap V(n)$  is of f.d.: Let  $f_1, \dots, f_k$  be a basis of  $\mathcal{R}f$ , because each  $f_k$  is  $K$ -finite. so if we use the decomposition of  $U(\mathfrak{sl}_2(\mathbb{C}))$  (3.7.8.26), only the  $R, L$  shifting of the  $E_n f$  will be considered, and clearly each  $V_0 \cap V(n)$  is of f.d.

Now we show  $f$  is analytic: because  $f$  is  $\mathcal{Z}$ -finite, there is an equation  $P(\Delta)f = 0$  where  $P$  is a monic function, and because  $\Delta$  commutes with  $E_k$  and  $f$  is  $K$ -finite,  $P(\Delta)E_k f = 0$ . Now the  $P(\Delta)E_k f = P(\Delta_k)E_k f$  by (26.1.2.5) and (26.1.3.23), and  $P(\Delta_k)$  is an elliptic operator, so  $E_k f$  is analytic by (13.1.8.5).

Now we show  $V$  is the closure of  $V_0$ : Suppose not, then by Hahn-Banach there is a non-zero continuous linear functional  $\Lambda$  on  $V$  that  $\Lambda(V_0) = 0$ . If  $F \in C^\infty(G)$ , let  $\varphi_F(g) = \Lambda(\rho(g)F)$ , and let  $\varphi = \varphi_f$ . Clearly  $dX\varphi_F = \varphi_{dXF}$ , so  $D\varphi_F = \varphi_{DF}$ , which implies  $\varphi$  is  $\mathcal{Z}$ -finite and smooth, and also  $K$ -finite because  $f$  does. So by what we have proved,  $\varphi$  is analytic. But now  $\varphi$  is analytic and  $D\varphi(1) = \varphi_{DF}(1) = 0$  because  $Df \in V_0$ , so  $\varphi = 0$  by Taylor expansion. So  $\Lambda(\rho(g)f) = 0$  for any  $g$ , contradiction because  $\rho(g)f$  is dense in  $V$ .

So actually  $V(n) \subset V_0$ , because  $E_n V_0 = V_0 \cap V(n) \subset V(n)$  is dense, and  $V_0 \cap V(n)$  is of f.d., so  $V(n) \subset V_0 = V_n$  (12.8.1.8) and  $V_0 = \bigoplus V(n)$  is an admissible  $(\mathfrak{g}, K)$ -module.

Let  $J$  be the convolution algebra (because  $G$  is unimodular) of functions  $\alpha$  that  $\alpha(kgk^{-1}) = \alpha(g)$ , then it can be checked that convolution  $- * \alpha$  commutes with action of  $K$ , so  $f * J$  is in the same  $K$ -type space as  $f$ , thus  $K$ -finite and in a f.d. space.

Now we can approximate  $f$  by  $f * J$ : choose a Dirac sequence  $\{\alpha_n\} \in C_c^\infty(G_1)$ , we may replace  $\{\alpha_n\}$  by the function  $\beta_n(g) = \int_K \alpha_n(k^{-1}gk) dk$  to obtain a Dirac sequence in  $J$ . Then  $f * \alpha_n \rightarrow f$  uniformly on compact sets. But  $f * J$  is f.d., so there are some  $f * \alpha = f$ .

Finally for the growth estimate, it suffices to check for  $D \in \mathfrak{g}$ . Then  $dX(f) = dX(f * \alpha) = f * (dX\alpha)$ , from which the estimate is clear.  $\square$

**Cor.(26.1.1.15) [Automorphic Forms Generate Admissible  $(\mathfrak{g}, K)$ -Modules].** The space  $\mathcal{A}(\Gamma \backslash G, \chi, \omega)$  and  $\mathcal{A}_0(\Gamma \backslash G, \chi, \omega)$  are stable under the action of  $U(\mathfrak{g})$ , and for  $f \in \mathcal{A}(\Gamma \backslash G, \chi, \omega)$ ,  $f$  is analytic and  $U(\mathfrak{g})f$  is an admissible  $(\mathfrak{g}, K)$ -module.

Moreover, if  $f$  satisfies condition of moderate growth (26.1.1.2) and  $D \in U(\mathfrak{g})$ , then  $Df$  satisfies similar conditions with the same constant  $N$ .  $\lrcorner$

*Proof:* It suffices to prove for  $\mathcal{A}(\Gamma \backslash G, \chi, \omega)$  because the cuspidality condition is clearly preserved by right action.

We want to use Harish-Chandra theorem (26.1.1.14) on  $G(K)/Z(K)$ . Now  $|f|$  is constant on each fiber of  $G(K) \rightarrow G(K)/Z(K)$ . it suffices to compare the norm on  $G(K)$  and  $G(K)/Z(K)$ .

We do this only for  $\mathrm{GL}(2; \mathbb{R})$ ?  $\color{red}$

The condition of moderate growth is compatible because the minimal  $\|g\|$  in a  $Z(\mathbb{R})$ -orbit is achieved when  $\deg g = 1$  (26.1.1.2). So all the assertion follows from that of (26.1.1.14) and its proof.  $\square$

## 2 $\mathrm{GL}(n)$

**Notation (26.1.2.1).** Let  $G = \mathrm{GL}(n)$ . And if  $n = 2$ ,

- $G$  acts on  $\mathcal{H}$  through linear fractional transformation (16.6.1.10) and fixes the measure  $y^{-2} dx dy$ . We will denote  $a, b, c, d$  the linear functionals on  $\mathrm{Mat}(2; \mathbb{R})$  s.t. for  $\gamma \in \mathrm{Mat}(2, \mathbb{R})$ ,  $\gamma = \begin{bmatrix} a(\gamma) & b(\gamma) \\ c(\gamma) & d(\gamma) \end{bmatrix}$ .
- Use the Lie algebra notations (3.7.2.11):

$$H = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad R = \frac{1}{2} \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}, \quad L = \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & -1 \end{bmatrix} \quad W = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = iH.$$

and the Casimir element in  $\mathcal{Z}$ :

$$\Delta = -\frac{1}{4}(H^2 + 2RL + 2LR).$$

- Let  $\Gamma \leq G$  be a discrete subgroup s.t.  $\mathrm{Vol}(\Gamma \backslash \mathcal{H}) < \infty$ . we may also assume that  $-1 \in \Gamma \subset \mathrm{SL}(2; \mathbb{R})$ .  $\lrcorner$

**Prop. (26.1.2.2).**

$$k_\theta \begin{bmatrix} y_1 & \\ & y_2 \end{bmatrix} = \begin{bmatrix} y_1 y_2 D(\theta)^{-1} & \xi D(\theta)^{-1} \\ & D(\theta) \end{bmatrix} k_{\theta'}.$$

where

$$\theta' = \arctan\left(\frac{y_1}{y_2} \tan \theta\right), \quad D(\theta) = \sqrt{y_1^2 \sin^2 \theta + y_2^2 \cos^2 \theta}, \quad \xi = (y_2^2 - y_1^2) \sin \theta \cos \theta.$$

$\lrcorner$

*Proof:* Hint: find  $\theta'$  first.  $\square$

**Prop. (26.1.2.3) [Gelfand].** Let  $G = \mathrm{GL}(n; \mathbb{R})^+$ ,  $\mathcal{K} = \mathrm{SO}(n)$ , denote  $C_c^\infty(\mathcal{K} \backslash G / \mathcal{K})$  to be the smooth functions  $\varphi \in C^\infty(G)$  that  $\varphi(k_1 g k_2) = \varphi(g)$ , then  $C_c^\infty(\mathcal{K} \backslash G / \mathcal{K})$  is commutative.  $\lrcorner$

*Proof:* Consider the map  $\varphi \mapsto \widehat{\varphi} : \widehat{\varphi}(g) = \varphi(g^t)$ , then it is an anti-involution of  $C_c^\infty(K \backslash G / K)$ :

$$(\widehat{\varphi_1 * \varphi_2})(g) = \int_G \varphi_1(g^t h) \varphi_2(h^{-1}) dh = \int_G \widehat{\varphi_2}(h^{-t}) \widehat{\varphi_1}(h^t g) dh = \int_G \widehat{\varphi_2}(h) \widehat{\varphi_1}(h^{-1} g) dh = (\widehat{\varphi_2} * \widehat{\varphi_1})(g)$$

But we find  $\widehat{\varphi} = \varphi$ , because we can use (16.5.5.1),  $\varphi(g) = \varphi(d) = \widehat{\varphi}(d) = \widehat{\varphi}(g)$ .  $\square$



**Prop. (26.1.2.4).** Let  $G = \mathrm{GL}(2; \mathbb{R})^+$ ,  $\mathcal{K} = \mathrm{SO}(2)$ , let  $\sigma \in K^\vee$ , then  $C_c^\infty(\mathcal{K} \backslash G / \mathcal{K}, \sigma)$  is commutative.  
 $\perp$

*Proof:* The proof is the same as that of (26.1.2.3), but modified as

$$\widehat{\varphi}(g) = \varphi\left(\begin{bmatrix} -1 & \\ & 1 \end{bmatrix} g^t \begin{bmatrix} -1 & \\ & 1 \end{bmatrix}\right).$$

□

**Prop. (26.1.2.5).** In the coordinate (16.6.1.3), we have the following equation:

$$\frac{\partial}{\partial R} = e^{2i\theta} \left( iy \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + \frac{1}{2i} \frac{\partial}{\partial \theta} \right), \quad \frac{\partial}{\partial L} = e^{-2i\theta} \left( -iy \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} - \frac{1}{2i} \frac{\partial}{\partial \theta} \right), \quad \frac{\partial}{\partial H} = -i \frac{\partial}{\partial \theta}$$

So in particular

$$\Delta = -y^2 \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + y \frac{\partial^2}{\partial x \partial \theta}.$$

□

*Proof:*  $H = -iW$ , and  $\exp(tW) = k_t = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$ , so  $dW = \partial/\partial\theta$  is clear.

For  $dR$ , first notice that

$$k_\theta \exp(tR) = \exp(e^{2i\theta} R) k_\theta,$$

as

$$\begin{aligned} k_\theta \exp(tR) k_{-\theta} &= C^{-1} \begin{bmatrix} e^{i\theta} & \\ & e^{-i\theta} \end{bmatrix} \exp(te) \begin{bmatrix} e^{-i\theta} & \\ & e^{i\theta} \end{bmatrix} C = C^{-1} \begin{bmatrix} e^{i\theta} & \\ & e^{-i\theta} \end{bmatrix} \begin{bmatrix} 1 & t \\ & 1 \end{bmatrix} \begin{bmatrix} e^{-i\theta} & \\ & e^{i\theta} \end{bmatrix} C \\ &= C^{-1} \begin{bmatrix} 1 & e^{2i\theta} t \\ & 1 \end{bmatrix} C \\ &= \exp(e^{2i\theta} t R) \end{aligned}$$

where  $C$  is the Cayley transformation, notation as in (3.7.2.11),

Now

$$(dRf)(g) = \frac{d}{dt} f(bk_\theta \exp(tR)) = \frac{d}{dt} f(b \exp(e^{2i\theta} t R) k_\theta) = e^{2i\theta} \frac{d}{dt} f(b \exp(tR) k_\theta)$$

Then notice

$$R = 1/2H + \begin{bmatrix} 0 & i \\ 0 & 0 \end{bmatrix} + 1/2 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \exp(tR) \sim k_{t/2} + \begin{bmatrix} 1 & it \\ & 1 \end{bmatrix} + \begin{bmatrix} e^{t/2} & \\ & e^{-t/2} \end{bmatrix}$$

so we get the desired result.

For  $dL$  the calculation is similar to that of  $dR$ . □

**Def. (26.1.2.6) [Cusps].** A **cusp** of  $\Gamma$  is a point in  $\mathbb{P}^1(\mathbb{R})$  whose stablizer in  $\Gamma$  contains a non-trivial parabolic element (16.6.1.2).

Let  $\infty$  be a cusp, then  $\{\pm 1\}\Gamma_\infty = \{\pm 1\}\langle \begin{bmatrix} 1 & r \\ & 1 \end{bmatrix} \rangle$  for some  $r > 1$ . Then if  $\Gamma_\infty = \langle -\begin{bmatrix} 1 & r \\ & 1 \end{bmatrix} \rangle$ , then it is called an **irreducible cusp**, otherwise it is called a **regular cusp**. Similarly, for any other cusp  $a = \xi(\infty)$ , where  $\xi \in \mathrm{SL}(2; \mathbb{R})$ , we called  $a$  is regular/irregular cusp iff  $\infty$  is regular/irregular cusp w.r.t.  $\Gamma' = \xi^{-1}\Gamma\xi$ .  
 $\perp$

Define  $C(\Gamma \backslash G, \chi, k)$  or  $C_c(\Gamma \backslash G, \chi, k)$  to be the subspace of  $C(\Gamma \backslash G, \chi)$  or  $C_c(\Gamma \backslash G, \chi)$  consisting of functions  $F$  s.t.  $\rho(k_\theta)F = e^{ik\theta}F$ .

When  $\omega$  is the character of  $Z(\mathbb{R})$  that  $\omega(Z(\mathbb{R}^+)) = 1$ , then we denote  $C^\infty(\Gamma \backslash G, \chi) = C^\infty(\Gamma \backslash G, \chi, \omega)$ .  $\lrcorner$

**Prop. (26.1.2.7) [Hecke Operators].** Similar as in (26.3.2.4), we can define the action of  $\mathcal{R}_N$  on  $C^\infty(\Gamma_0(N) \backslash \mathcal{G}, \chi_d)$ :

$$T_\alpha(f)(g) = \sum_i \chi_d(\alpha_i)^{-1} f(\alpha_i g), \quad \text{if } \Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i.$$

and preserves automorphic or cuspidal forms.  $\lrcorner$

*Proof:* This is an action because if  $\Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i$ ,  $\Gamma_0(N) \beta \Gamma_0(N) = \coprod_j \Gamma_0(N) \beta_j$ , then  $[\alpha][\beta] = \coprod_{i,j} \Gamma_0(N) \alpha_i \beta_j$ .

Also  $f(\gamma x) = \chi(\gamma) f(x)$  for  $\gamma \in \Gamma_0(N)$  because  $\Gamma_0(N) \backslash \Gamma_0(N) \alpha \Gamma_0(N)$  is right  $\Gamma_0(N)$ -invariant.  $\square$

$\Gamma \backslash \mathcal{H}$

**Def. (26.1.2.8) [Holomorphic Right Weight Action].** Besides the right weight action (26.1.3.19), there is a right action of  $\mathrm{GL}(2; \mathbb{R})^+$  on  $C^\infty(\mathcal{H})$ :

$$f[\gamma]_k(z) = \deg(\gamma)^{k-1} (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right).$$

*Proof:* This is an action because?  $\square$

**Remark (26.1.2.9).** This action is related to that of (26.1.3.19) via (26.1.3.22).  $\lrcorner$

**Lemma (26.1.2.10).** If  $f$  is a holomorphic function on  $\mathcal{H}$  and  $\gamma \in \mathrm{SL}(2; \mathbb{R})$  satisfies  $\gamma^n \neq 1$  for some  $n \in \mathbb{Z}^\times$  and  $f[\gamma]_k = f$ , then  $f = 0$ .  $\lrcorner$

*Proof:* Use a Cayley transformation  $\mathcal{H} \rightarrow \mathbb{D}$  to map the fixed point to origin, then  $\gamma$  corresponds to  $\mathrm{diag}(\alpha, \alpha^{-1})$ , where  $\alpha$  is not a root of unity. Then if  $f(z) = \sum a_n z^n$ , the formula  $f[\gamma]_k = f$  says  $a_n \alpha^{2n+k} = a_n$ . Because  $\alpha^{2n+k} \neq 1$  for any  $n$ ,  $a_n = 0$  for any  $n$ , thus  $f = 0$ .  $\square$

**Def. (26.1.2.11) [Form Spaces].** if  $\Gamma$  is a discrete subgroup of  $G_1$ , let  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  be the space of smooth functions on  $\mathcal{H}$  s.t.

$$f[\gamma]_k = \chi(\gamma) f, \quad \forall \gamma \in \Gamma \text{ (26.1.2.8)}.$$

And elements in  $C^\infty(\Gamma \backslash \mathcal{H}, 1, 0)$  are called **automorphic forms**.

Let the subspace  $C_c^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  be those functions  $f$  that  $|f|$  is compactly supported in  $\Gamma \backslash \mathcal{H}$ .  $\lrcorner$

**Def. (26.1.2.12) [Behavior at Cusps].** If  $\infty$  is a cusp of  $\Gamma$ , then  $\Gamma$  contains some  $\tau_r = \begin{bmatrix} 1 & r \\ & 1 \end{bmatrix}$  for  $r > 0$ ,

then a continuous function  $f(x + iy)$  on  $\mathcal{H}$  is called

- **of at most linear exponential growth** at  $\infty$  iff  $|f(x + iy)| = O(e^{Ny})$  for  $y \rightarrow \infty$  for some  $N \in \mathbb{Z}_+$ .
- **of moderate growth** at  $\infty$  iff  $|f(x + iy)| = O(y^N)$  for  $y \rightarrow \infty$  for some  $N \in \mathbb{Z}_+$ .
- **decay rapidly** at  $\infty$  iff  $|f(x + iy)| \leq y^{-N}$  for any  $N \in \mathbb{Z}_+$ .

- **cuspidal** at  $\infty$  iff either  $\chi(\tau_r) \neq 1$  or  $\int_0^r f(z+u)du = 0$  for any  $z \in \mathcal{H}$ .

If  $f$  is meromorphic on  $\mathcal{H}$  then we have a Fourier expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n z / r} = \sum_{n=-\infty}^{\infty} a_n q^n = T(q).$$

Then  $f$  is called **meromorphic/holomorphic/vanishes** at the cusp  $\infty$  iff  $T(q)$  does at  $q = 0$ .

The general cusp case is reduced to the  $\infty$  case the same way as in (26.1.1.3). Notice this is independent of possible  $r$  chosen.  $\perp$

**Prop. (26.1.2.13) [Hecke Operators].** Similar as in (26.3.2.4), we can define the action of  $\mathcal{R}_N$  on  $C^\infty(\Gamma_0(N) \backslash \mathcal{H}, \chi_d, k)$ :

$$T_\alpha(f) = \sum_i \chi_d(\alpha_i)^{-1} f[\alpha_i]_k \quad (26.1.2.8), \text{ where } \Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i.$$

and preserves automorphic or cuspidal automorphic forms.  $\perp$

*Proof:* This is an action because if  $\Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i$ ,  $\Gamma_0(N) \beta \Gamma_0(N) = \coprod_j \Gamma_0(N) \beta_j$ , then  $[\alpha][\beta] = \coprod_{i,j} \Gamma_0(N) \alpha_i \beta_j$ .

Also  $T_\alpha(f)[\gamma]_k = T_\alpha(f)$  for  $\gamma \in \Gamma_0(N)$  because  $\Gamma_0(N) \backslash \Gamma_0(N) \alpha \Gamma_0(N)$  is right  $\Gamma_0(N)$ -invariant.  $\square$

### $L^2$ -Spaces

**Def. (26.1.2.14) [Peterson Inner Product].** Notice that if  $f, g \in C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  (26.1.2.11), then  $f \bar{g} y^k$  is invariant under action of  $\Gamma$ , so we define the Hilbert space  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$  as the square-integrable functions in the inner product (26.1.0.1):

$$(f, g)_{\text{Pet}} = \int_{\Gamma \backslash \mathcal{H}} f(z) \bar{g}(z) y^k \frac{dx dy}{y^2}.$$

$\perp$

**Prop. (26.1.2.15) [Two Form Spaces Equal].** There is an isomorphism of Hilbert spaces

$$\sigma_k : C^\infty(\Gamma \backslash \mathcal{H}, \chi, k) \cong C^\infty(\Gamma \backslash G, \chi, k) : (\sigma_k f)(g) = f[g]_k(i).$$

that preserves the inner products thus induces an isomorphism

$$\sigma_k : L^2(\Gamma \backslash \mathcal{H}, \chi, k) \cong L^2(\Gamma \backslash G, \chi, k)$$

And we have (26.1.3.23):

$$\sigma_{k+2} R_k = dR \sigma_k, \quad \sigma_{k-2} L_k = dL \sigma_k, \quad \sigma_k \Delta_k = \Delta \sigma_k.$$

Also cuspidality and the behaviors at the cusps are compatible.

Also if  $\Gamma = \Gamma_0(N)$ , this isomorphism is compatible with the Hecke operator actions (26.1.2.13) and (26.1.2.7).  $\perp$

*Proof:* It can be verified that the inverse of  $\sigma_k$  is given by

$$f(z) = y^{-k/2} F\left(\begin{bmatrix} y & x \\ & 1 \end{bmatrix}\right)$$

More precisely, if coordinates (16.6.1.3),

$$F(u, x, y, \theta) = y^{k/2} f(x + iy) e^{ik\theta}, \quad f(z) = y^{-k/2} F(1, x, y, 0).$$

Check the left action of  $\Gamma$  and  $Z^+(\mathbb{R})$ : For  $\Gamma$  actions (26.1.2.8) and (26.1.1.1), this is because  $\sigma_k(f)(\gamma g) = f[\gamma g]_k(i) = \sigma_k(f[\gamma]_k)$ . Finally check that  $\sigma_k$  preserves inner product, which is immediate from (26.1.2.14) (26.1.1.4) and (16.6.1.3).

The equations between  $R_k, L_k, \Delta_k$  and  $R, L, \Delta$  are easily verified from (26.1.2.5) and (26.1.3.23).

The action of Hecke operators are compatible by the form  $(\sigma_k f)(g) = (f|_k g)(i)$  as above.

Compatibility of behaviors at cusps: For cuspidality, compare (26.1.2.12) and (26.1.1.3). For growth conditions, this follows from (26.1.1.13).  $\square$

### Cuspidal Representations

### Unramified Representations

### Unitarizable, Tempered and $L^2$ -Representations

**Thm. (26.1.2.16) [Classification of  $\text{Irr}^{\text{auto}}(\text{GL}(2)/\mathbb{Q})$ ].** Let  $\chi(-1) = (-1)^\varepsilon$ ,  $\varepsilon = 0, 1$ , (26.1.5.2) shows the representation  $\mathcal{H} = L^2_0(\Gamma \backslash G, \chi)$  decomposes into Hilbert space direct sums of irreducible representations, and  $\Delta$  acts as real scalars  $\lambda$  on each irreducible subspace (24.6.4.21), and  $\mu$  acts by 0. So comparing the classification of representations of  $\text{GL}(2; \mathbb{R})$  (24.7.5.30), we can list what representation can appear in it by looking at eigenvalues  $\lambda$  of  $\Delta$ :

- $\mathbf{1}$  is only f.d. irreducible unitary subrepresentation of  $G$  occurring in  $\mathcal{H}$ .
- If  $\lambda \neq \frac{k}{2}(1 - \frac{k}{2})$  for any  $k \in \varepsilon + 2\mathbb{N}$ , then  $\lambda \geq \frac{\varepsilon}{2}(1 - \frac{\varepsilon}{2})$ , and this subrepresentation is isomorphic to  $P(\lambda, \varepsilon)$ . And let  $k' \equiv \varepsilon \pmod{2}$  be any integer, then the multiplicity of  $P(\lambda, \varepsilon)$  is equal to the multiplicity of the eigenvalue  $\lambda$  of  $\Delta_{k'}$  on  $L^2(\Gamma \backslash \mathcal{H}, \chi, k')$  because  $L^2(\Gamma \backslash \mathcal{H}, \chi, k') \cong L^2(\Gamma \backslash G, \chi, k')$  by (26.1.2.15).
- If  $\lambda = \frac{k}{2}(1 - \frac{k}{2})$  for some  $k \in \varepsilon + 2\mathbb{N}$ , then this subrepresentation is isomorphic to either  $D^+(k)$  or  $D^-(k)$ , and  $D^\pm(k)$  have the same multiplicity in  $\mathcal{H}$ , equal to the dimension  $\dim(M_k(\Gamma, \chi))$  (26.3.1.3) of holomorphic modular forms of weight  $k$  for  $\Gamma$ .  
Also if  $k' \equiv \varepsilon \pmod{2}$  is any integer  $\geq k$  (resp.  $\leq -k$ ), then the multiplicity of  $D^+(k)$  (resp.  $D^-(k)$ ) is equal to the multiplicity of the eigenvalue  $\lambda$  of  $\Delta_{k'}$  on  $L^2(\Gamma \backslash \mathcal{H}, \chi, k')$  because  $L^2(\Gamma \backslash \mathcal{H}, \chi, k') \cong L^2(\Gamma \backslash G, \chi, k')$  by (26.1.2.15).

┘

*Proof:* These follow from classification of irreducible representations of  $\text{GL}(2; \mathbb{R})^+$  (24.7.5.31).

Relation with modular forms: by (24.6.4.16), the multiplicity of  $D^+(k)$  equals the dimension of the  $\frac{k}{2}(1 - \frac{k}{2})$ -eigenspace of  $\Delta_k$  on  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$ , and (26.1.3.10) shows the dimension of space of functions annihilated by  $\Delta_k$  equals the dimension of modular forms of weight  $k$ . Finally notice that complex conjugation interchanges  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$  and  $L^2(\Gamma \backslash \mathcal{H}, \chi, -k)$  and  $\overline{\Delta_k} = \Delta_{-k}$  (26.1.3.23), so the multiplicity of  $D^+(k)$  and  $D^-(k)$  equal.  $\square$

### 3 Maass Forms

#### Maass Forms

**Def. (26.1.3.1) [Maass Operator].** The **Maass differential operators** on  $C^\infty(\mathcal{H})$  is defined to be

$$R_k = 2i \frac{\partial}{\partial z} + ky^{-1} = i \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) + ky^{-1}, \quad L_k = -2iy^2 \frac{\partial}{\partial \bar{z}} = -iy^2 \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

and

$$\Delta_k = -y^2 \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + iky \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) + \frac{k}{2} \left( 1 - \frac{k}{2} \right) = -L_{k+2} R_k - \frac{k}{2} \left( 1 + \frac{k}{2} \right) = -R_{k-2} L_k + \frac{k}{2} \left( 1 - \frac{k}{2} \right).$$

In particular,

$$\Delta_{k+2} \circ R_k = R_k \circ \Delta_k, \quad \Delta_{k-2} \circ L_k = L_k \circ \Delta_k.$$

And  $\Delta_k$  is a symmetric (unbounded) operator on  $L^2(\mathcal{H})$  (26.1.2.14) with domain  $C_c^\infty(\mathcal{H})$ . ┘

*Proof:* For the formula: ?

For the symmetry: composed with the measure, the order 2 part is just the ordinary Laplacian, and the order 1 part becomes  $iky \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$ , then notice

$$\int_{\mathcal{H}} iky \left[ \left( \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) f \bar{g} + f \left( \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \bar{g} \right] y^k \frac{dx dy}{y^2} = i \int_{\mathcal{H}} d \left( y^{k-1} f \bar{g} dy + y^{k-1} f \bar{g} dx \right) = 0$$

as  $f, g$  are compactly supported. □

**Prop. (26.1.3.2) [Maass Operator and Weight Action].** For  $f \in C^\infty(\mathcal{H})$ ,  $g \in G$ ,

$$(R_k f)[g]_{k+2} = R_k(f[g]_k), \quad (L_k f)[g]_{k-2} = L_k(f[g]_k), \quad (\Delta_k f)[g]_k = \Delta_k(f[g]_k) \quad (26.1.2.8)$$

┘

*Proof:* For  $R_k$ , because scalar doesn't matter, we may assume  $g \in \text{SL}(2; \mathbb{R})$ , and let  $w = \frac{az+b}{cz+d}$ , then

$$\frac{\partial}{\partial z} = \frac{\partial w}{\partial z} \frac{\partial}{\partial w} = (cz+d)^{-2} \frac{\partial}{\partial w}, \quad (w - \bar{w}) = \frac{z - \bar{z}}{|cz+d|^2}.$$

So

$$(w - \bar{w}) \frac{\partial}{\partial w} = \left( \frac{(cz+d)^2}{|cz+d|^2} \right) (z - \bar{z}) \frac{\partial}{\partial z}.$$

And for any  $\varphi \in C^\infty(\mathcal{H})$ ,

$$(z - \bar{z}) \frac{\partial}{\partial z} \left( \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \varphi \right) = (z - \bar{z}) \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \frac{\partial \varphi}{\partial z} + \frac{k}{2} \left[ \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2} - \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \right] \varphi.$$

This is because

$$\frac{\partial}{\partial z} |cz+d| = \frac{c}{2} \frac{|cz+d|}{cz+d}, \quad \frac{\partial}{\partial z} \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k = -\frac{k}{2} c(z-\bar{z}) \frac{(c\bar{z}+d)^k}{|cz+d|^k (cz+d)} = \frac{k}{2} \left[ \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2} - \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \right].$$

Thus,

$$\begin{aligned} R_k(f|_k g) &= [(z - \bar{z}) \frac{\partial}{\partial z} + \frac{k}{2}] \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k f(w) = [(z - \bar{z}) \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \frac{\partial}{\partial z} + \frac{k}{2} \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2}] f(w) \\ &= \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2} [(w - \bar{w}) \frac{\partial}{\partial w} + \frac{k}{2}] f(w) = ((R_k f)|_{k+2} g)(z). \end{aligned}$$

The case of  $L_k$  is similar, and the formula for  $\Delta_k$  follows. □

**Cor. (26.1.3.3).** The operator  $R_k, L_k, \Delta_k$  maps functions between  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  and arises and decreases weights respectively.  $\lrcorner$

**Prop. (26.1.3.4).** For  $f \in C^\infty(\Gamma \backslash \mathcal{H}, \chi, k), g \in C^\infty(\Gamma \backslash \mathcal{H}, \chi, k+2)$ , if one of them decays rapidly at all cusps, then

$$(R_k f, g) = (f, -L_{k+2} g) \quad (26.1.2.14).$$

In particular,  $\Delta_k = -L_{k+2} R_k - k$  is symmetric on the subspace of  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$  generated by rapidly decaying functions ( $\Delta_k$  is unbounded and defined only on  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  now, but will be extended to  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$  in (26.1.5.5) when  $\Gamma \backslash \mathcal{H}$  is compact).  $\lrcorner$

*Proof:* Let  $\omega = y^k f(z) \overline{g(z)} d\bar{z}$ . It can be shown by definition and using (12.4.1.9) that  $\omega(\gamma z) = \omega(z)$ , so  $\omega$  descends to a differential form on  $\Gamma \backslash \mathcal{H}$ , and because one of  $f, g$  decays rapidly at all cusps,

$$\begin{aligned} 0 &= \int_{\Gamma \backslash \mathcal{H}} d(y^k f(z) \overline{g(z)} d\bar{z}) = - \int_{\Gamma \backslash \mathcal{H}} \left[ \frac{\partial}{\partial y} (y^k f \bar{g}) + i \frac{\partial}{\partial x} (y^k f \bar{g}) \right] dx \wedge dy \\ &= - \int_{\Gamma \backslash \mathcal{H}} \left[ \left( i y^{k+2} \frac{\partial}{\partial x} f + y^{k+2} \frac{\partial f}{\partial y} + k y^{k+1} f \right) \bar{g} - \overline{\left( i y^{k+2} \frac{\partial g}{\partial x} - y^{k+2} \frac{\partial g}{\partial y} \right) f} \right] \frac{dx dy}{y^2} \\ &= \int_{\Gamma \backslash \mathcal{H}} [y^{k+2} (R_k f) \bar{g} + y^k f \overline{(L_{k+2} g)}] \frac{dx dy}{y^2}. \end{aligned}$$

so the conclusion follows.  $\square$

**Cor. (26.1.3.5).** If  $\lambda$  is an eigenvector of  $\Delta_k$  on  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$ , then either  $\lambda = \frac{l}{2}(1 - \frac{l}{2})$ , where  $1 \leq l \leq k$  and  $l \equiv k \pmod{2}$ , or  $\lambda \geq \frac{\varepsilon}{2}(1 - \frac{\varepsilon}{2})$ , where  $\varepsilon = (\text{rem } ()k \pmod{2})$ . In particular, eigenvalues of  $\Delta_0$  are  $\geq 0$  and eigenvalues of  $\Delta_1$  are  $\geq 1/4$ .  $\lrcorner$

*Proof:* This can be done by repeatedly using  $\Delta_{k-2} L_k = L_k \Delta_k$  to reduce to the case  $k = \varepsilon$ : If  $L^{ol} f = 0$  for some  $l$ , then  $\lambda = \frac{l}{2}(1 - \frac{l}{2})$ , and if  $L^{ol} f \neq 0$ , then reduce to  $0 \leq k \leq 1$  case, and we can use (24.7.5.21). Alternatively, this can be deduced from (26.1.2.16).  $\square$

**Prop. (26.1.3.6) [Bol's Identity].** Let  $D = \frac{1}{2\pi i} \frac{\partial}{\partial z} = q \frac{\partial}{\partial q}$ , then for any  $k \in \mathbb{Z}_+$ ,

$$D^{\circ(k-1)} = \frac{1}{(-4\pi)^{k-1}} R_{2-k}^{\circ(k-1)}.$$

$\lrcorner$

*Proof:* Cf. [Period functions for Maass wave forms] (4.15).  $\square$

**Def. (26.1.3.7) [Maass Forms].** A twisted **Maass form** of weight  $k$  and parameter  $s$  for  $\Gamma$  is an element in  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  (26.1.2.11) that is an eigenform for  $\Delta_k$  of eigenvalue  $\lambda = \frac{1}{4} - s^2$  and is of moderate growth at cusps of  $\Gamma$  (26.1.2.12). The space of Maass forms are denoted by  $\mathcal{A}_{\text{Maass}}(\Gamma \backslash \mathcal{H}, \chi, k)$ . The space of cuspidal Maass forms are denoted by  $\mathcal{A}_{\text{cusp Maass}}(\Gamma \backslash \mathcal{H}, \chi, k)$ .

A Maass form of weight 0 and  $\chi = 1$  is sometimes just called a Maass form. The space of Maass forms is denoted by  $\mathcal{A}_{\text{Maass}}(\Gamma \backslash \mathcal{H})$ .  $\lrcorner$

**Def. (26.1.3.8) [Weak Maass Forms].** A **weak Maass form** of weight  $k$  and parameter  $s$  for  $\Gamma$  is an element in  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  (26.1.2.11) that is an eigenform for  $\Delta_k$  and is of at most linear exponential growth at cusps of  $\Gamma$  (26.1.2.12).  $\lrcorner$

**Prop. (26.1.3.9) [Hecke Operators on Maass Forms].** If  $\Gamma = \Gamma_0(N)$ , the space of Maass forms is stable under the action of the Hecke algebra on  $C^\infty(\Gamma_0(N) \backslash \mathcal{H}, \chi, k)$  (26.1.3.21), by (26.1.3.2).  $\lrcorner$

**Prop. (26.1.3.10) [Holomorphic Modular Forms as Maass Forms].**

$$\mathrm{MF}_k(\Gamma, \chi) = \mathcal{A}_{\mathrm{Maass}}(\Gamma \backslash \mathcal{H}, \chi, k)^{L_k=0}, \quad \mathrm{SF}_k(\Gamma, \chi) \cong \mathcal{A}_{\mathrm{cusp Maass}}(\Gamma \backslash \mathcal{H}, \chi, k)^{L_k=0}.$$

$\lrcorner$

*Proof:* This is because  $L_k = 2iy^2 \frac{\partial}{\partial \bar{z}}$  (26.1.3.23). Also, if  $f$  is a holomorphic modular form, then each  $y^{k/2} f[\gamma]_k$  is bounded by a polynomial of  $y$  at  $\infty$ , and conversely, if  $f(q)$  is bounded by a polynomial of  $\log(1/|q|)$ , then ?? shows  $f$  is holomorphic at each cusp.  $\square$

**Prop. (26.1.3.11) [Non-holomorphic Eisenstein series as Maass Forms].** The non-holomorphic Eisenstein series  $E(\nu + 1/2; s)$  is a Maass form of parameter  $\nu$  for  $\Gamma(1)$ .  $\lrcorner$

*Proof:*  $E(s, \nu + 1/2)$  is automorphic and of moderate growth by (26.9.2.3) and (26.9.2.1).

To show  $E(s, \nu + 1/2)$  is an eigenfunction of  $\Delta$ , if  $\mathrm{Re}(\nu) > 1/2$ , notice  $\Delta(y^{\nu+1/2}) = (\frac{1}{4} - \nu^2)y^{\nu+1/2}$ , and  $\Delta$  is invariant under action of  $\mathrm{SL}(2; \mathbb{Z})$ , thus each  $\mathrm{Im}(\gamma(y^{\nu+1/2}))$  is an eigenfunction for  $\Delta$  of eigenvalue  $\frac{1}{4} - \nu^2$ , so the same is true for  $E(s, \nu + 1/2)$ .

For general  $\nu$ , this can be seen from the Fourier coefficients of  $E(s, \nu + 1/2)$  (26.9.2.2): ?  $\square$

**Prop. (26.1.3.12) [Maass Forms as Automorphic Forms].** If  $f \in \mathcal{A}_{\mathrm{Maass}}(\Gamma \backslash \mathcal{H}, \chi, k)$ , then  $\sigma_k(f) \in C^\infty(\Gamma \backslash G, \chi, k)$  (26.1.2.15), and is an eigenform of  $\Delta$ . In particular it is  $K$ -finite,  $\mathcal{Z}$ -finite, of moderate growth by (26.1.2.15), and it is cuspidal iff  $\sigma_k(f)$  is cuspidal, so  $\sigma_k(f) \in \mathcal{A}(\Gamma \backslash G, \chi, k)$ . And  $f$  is cuspidal iff  $\sigma_k(f)$  is cuspidal.

In fact, there are a bijections

$$\mathcal{A}_{\mathrm{Maass}}(\Gamma \backslash \mathcal{H}, \chi, k, \lambda) \cong \mathcal{A}(\Gamma \backslash G, \chi, k, \lambda), \quad \mathcal{A}_{\mathrm{Maass}}(\Gamma \backslash \mathcal{H}, \chi, k) \cong \mathcal{A}(\Gamma \backslash G, \chi, k)^{\Delta\text{-s.st}}$$

$$\mathcal{A}_{\mathrm{cusp Maass}}(\Gamma \backslash \mathcal{H}, \chi, k) \cong \mathcal{A}_0(\Gamma \backslash G, \chi, k), \quad \mathcal{A}_{\mathrm{cusp Maass}}(\Gamma \backslash \mathcal{H}, \chi) \cong \mathcal{A}_0(\Gamma \backslash G, \chi) = L_0^2(\Gamma \backslash G, \chi)^{K\text{-fin}}.$$

this is because when  $f$  is cuspidal,  $\Delta$  is semisimple on the finite dimensional subspace  $\mathcal{Z}f$  because of the spectral decomposition of  $L_0^2(\Gamma \backslash G, \chi, k)$  (26.1.5.8).  $\lrcorner$

**Cor. (26.1.3.13) [Modular Forms as Automorphic Forms].** Combine this and (26.1.3.10), we get a bijection

$$\mathrm{MF}_k(\Gamma, \chi) \cong \mathcal{A}(\Gamma \backslash G, \chi, k)^{L=0}$$

that induces a bijection

$$\mathrm{SF}_k(\Gamma, \chi) \cong \mathcal{A}_0(\Gamma \backslash G, \chi, k)^{L=0}.$$

$\lrcorner$

**Cor. (26.1.3.14) [Finitely Many Maass Forms of Given Type].**  $\dim \mathcal{A}_{\mathrm{Maass}}(\Gamma_0(N) \backslash \mathcal{H}, \chi, k, \lambda) < \infty$ .  $\lrcorner$

*Proof:* This follows from (26.1.1.15).  $\square$

**Conj. (26.1.3.15) [Selberg Conjecture].** Let  $\Gamma$  be a congruence subgroup, for any  $f \in \mathcal{A}_{\mathrm{cusp Maass}}(\Gamma \backslash \mathcal{H}, \chi, k)$ , the eigenvalue  $\lambda$  of  $\Delta_0$  satisfies  $\lambda \geq 1/4$ , or equivalently, the parameter  $s$  is purely imaginary. In this case, the cuspidal representation generated by  $f$  is tempered.

This conjecture is true for  $\Gamma = \Gamma(1)$  but wrong for some other congruence subgroups.  $\lrcorner$

*Proof:*  $\square$

### Harmonic Maass Forms

References are [Harmonic Maass Forms, Mock Modular Forms, And Quantum Modular Forms, Ono], [Harmonic Maass Forms and Mock Modular Forms, theory and applications, Ono].

**Def. (26.1.3.16) [Harmonic Maass Forms].** For  $k \in \frac{1}{2}\mathbb{Z}$ , the space  $H_k^!(\Gamma, \chi)$  of **harmonic Maass forms of moderate growth** of weight  $k$  consists of weak Maass forms (26.1.3.8) of eigenvalue  $\frac{k}{2}(1 - \frac{k}{2})$ .

The space  $H_k(\Gamma, \chi)$  of **harmonic Maass forms** consists of harmonic Maass forms of moderate growth s.t. for each cusp  $a$ , there exists a Laurent polynomial  $P_{f,a}(\tau) \in \mathbb{C}[q^{-1}]$  s.t.

$$f|_k(\tau) - P_{f,a}(\tau) = O(e^{-\varepsilon y}) \quad (26.1.2.8)$$

as  $y \rightarrow \infty$ , where  $\varepsilon \in \mathbb{R}_+$ . Such  $P_{f,a}$  is called the **principle part** of  $f$  at  $a$ .

$\text{MF}_k^!(\Gamma, \chi) \subset H_k(\Gamma, \chi)$ , by (26.1.3.10). ┘

**Prop. (26.1.3.17).** ┘

**Def. (26.1.3.18) [Mock Modular Forms].** A **Mock modular form** is the holomorphic part  $f^+$  of a harmonic Maass form  $f$  s.t.  $f^- \neq 0$ . ┘

### Hecke Theory of Maass Forms

#### Obsolete Original Form Spaces

**Def. (26.1.3.19) [Right Weight Action].** There are right actions of  $\text{GL}(2, \mathbb{R})^+$  on  $C^\infty(\mathcal{H})$  defined to by

$$(f|_k \gamma)(z) = \left( \frac{c\bar{z} + d}{|cz + d|} \right)^k f\left( \frac{az + b}{cz + d} \right)$$

┘

*Proof:* It is an action because ? □

**Def. (26.1.3.20) [(Original)Form Spaces].** if  $\Gamma$  is a discrete subgroup of  $G_1$ , let  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  be the space of smooth functions on  $\mathcal{H}$  that

$$f|_k \gamma = \chi(\gamma) f, \quad \forall \gamma \in \Gamma.$$

And elements in  $C^\infty(\Gamma \backslash \mathcal{H}, 1, 0)$  are called **automorphic functions**.

Let the subspace  $C_c^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  be those functions  $f$  that  $|f|$  is compactly supported in  $\Gamma \backslash \mathcal{H}$ . ┘

**Prop. (26.1.3.21) [Hecke Operators].** Similar as in (26.3.2.4), we can define the action of  $\mathcal{R}_N$  on  $C^\infty(\Gamma_0(N) \backslash \mathcal{H}, \chi_d, k)$ :

$$T_\alpha(f) = \sum_i \chi_d(\alpha_i)^{-1} f|_k \alpha_i \quad (26.1.3.19), \quad \text{if } \Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i.$$

and preserves automorphic or cuspidal automorphic forms. ┘

*Proof:* This is an action because if  $\Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i$ ,  $\Gamma_0(N) \beta \Gamma_0(N) = \coprod_j \Gamma_0(N) \beta_j$ , then  $[\alpha][\beta] = \coprod_{i,j} \Gamma_0(N) \alpha_i \beta_j$ .

Also  $T_\alpha(f)|_k \gamma = T_\alpha(f)$  for  $\gamma \in \Gamma_0(N)$  because  $\Gamma_0(N) \backslash \Gamma_0(N) \alpha \Gamma_0(N)$  is right  $\Gamma_0(N)$ -invariant. □



**Prop. (26.1.3.22)[Relations between Original and Classical Form Spaces].** The relation between the two definitions of form space  $C^\infty(\Gamma \backslash H, \chi, k)$  in (26.1.2.11) and (26.1.3.20) is that:

If  $f(z)$  is a classical automorphic function, then  $f'(z) = y^{k/2}f(z)$  is an original automorphic function. And this is a bijection.  $\square$

*Proof:*

$$(y^{k/2}f)|_k\gamma(z) = \left(\frac{c\bar{z}+d}{|cz+d|}\right)^k y^{k/2}f\left(\frac{az+b}{cz+d}\right) \cdot \frac{\det(\gamma)^{k/2}}{|cz+d|^k} = y^{k/2} \frac{\det(\gamma)^{k/2}}{(cz+d)^k} f\left(\frac{az+b}{cz+d}\right) = y^{k/2}f[\gamma]_k(z)$$

$\square$

**Def. (26.1.3.23)[Original Maass Operator].** The Maass differential operators on  $C^\infty(\mathcal{H})$  is defined to be

$$R_k = (z - \bar{z})\frac{\partial}{\partial z} + \frac{k}{2} = iy\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} + \frac{k}{2}, \quad L_k = -(z - \bar{z})\frac{\partial}{\partial \bar{z}} - \frac{k}{2} = -iy\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} - \frac{k}{2}$$

and

$$\Delta_k = -y^2\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + ik y \frac{\partial}{\partial x} = -L_{k+2}R_k - \frac{k}{2}\left(1 + \frac{k}{2}\right) = -R_{k-2}L_k + \frac{k}{2}\left(1 - \frac{k}{2}\right)$$

$\Delta_k$  is a symmetric(unbounded) operator on  $L^2(\mathcal{H})$  with domain  $C_c^\infty(\mathcal{H})$ .  $\square$

*Proof:* For the formula: **?**

For the symmetry: composed with the measure, the order2 part is just the ordinary Laplacian, and the order 1 part becomes  $iy^{-1}\frac{\partial}{\partial x}$ , then notice

$$\int_{\mathcal{H}} iy^{-1}\left(\frac{\partial f}{\partial x}\bar{g} + f\frac{\partial \bar{g}}{\partial x}\right)dx dy = i \int_{\mathcal{H}} d(y^{-1}f\bar{g}dy) = 0$$

as  $f, g$  are compactly supported.  $\square$

**Prop. (26.1.3.24)[Maass Operator and Weight Action].** For  $f \in C^\infty(\mathcal{H}), g \in G$ ,

$$(R_k f)|_{k+2}g = R_k(f|_k g), \quad (L_k f)|_{k-2}g = L_k(f|_k g), \quad (\Delta_k f)|_k g = \Delta_k(f|_k g)$$

$\square$

*Proof:* For  $R_k$ , because scalar doesn't matter, we may assume  $g \in SL_2(\mathbb{R})$ , and let  $w = \frac{az+b}{cz+d}$ , then

$$\frac{\partial}{\partial z} = \frac{\partial w}{\partial z} \frac{\partial}{\partial w} = (cz+d)^{-2} \frac{\partial}{\partial w}, \quad (w - \bar{w}) = \frac{z - \bar{z}}{|cz+d|^2}.$$

So

$$(w - \bar{w})\frac{\partial}{\partial w} = \left(\frac{cz+d}{|cz+d|^2}\right)(z - \bar{z})\frac{\partial}{\partial z}.$$

And for any smooth function  $\varphi \in C^\infty(\mathcal{H})$ ,

$$(z - \bar{z})\frac{\partial}{\partial z}\left(\left(\frac{c\bar{z}+d}{|cz+d|}\right)^k \varphi\right) = (z - \bar{z})\left(\frac{c\bar{z}+d}{|cz+d|}\right)^k \frac{\partial \varphi}{\partial z} + \frac{k}{2}\left[\left(\frac{c\bar{z}+d}{|cz+d|}\right)^{k+2} - \left(\frac{c\bar{z}+d}{|cz+d|}\right)^k\right]\varphi.$$

This is because

$$\frac{\partial}{\partial z}|cz+d| = \frac{c}{2} \frac{|cz+d|}{cz+d}, \quad \frac{\partial}{\partial z} \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k = -\frac{k}{2} c(z-\bar{z}) \frac{(c\bar{z}+d)^k}{|cz+d|^k(cz+d)} = \frac{k}{2} \left[ \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2} - \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \right].$$

Thus,

$$\begin{aligned} R_k(f|_k g) &= [(z-\bar{z}) \frac{\partial}{\partial z} + \frac{k}{2}] \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k f(w) = [(z-\bar{z}) \left( \frac{c\bar{z}+d}{|cz+d|} \right)^k \frac{\partial}{\partial z} + \frac{k}{2} \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2}] f(w) \\ &= \left( \frac{c\bar{z}+d}{|cz+d|} \right)^{k+2} [(w-\bar{w}) \frac{\partial}{\partial w} + \frac{k}{2}] f(w) = ((R_k f)|_{k+2} g)(z). \end{aligned}$$

The case of  $L_k$  is similar, and  $\Delta_k$  follows.  $\square$

**Cor. (26.1.3.25).** The operator  $R_k, L_k, \Delta_k$  maps functions between  $C^\infty(\Gamma \backslash \mathcal{H}, \chi, k)$  and arises and decreases weights respectively.  $\lrcorner$

## 4 Whittaker Models

**Def. (26.1.4.1) [Whittaker Function Space].** For  $\psi \in \widehat{\mathbb{R}}$ , denote  $W_\psi =$  the space of smooth functions  $f$  on  $\mathrm{GL}(n; \mathbb{R})^+$  that satisfies

$$f(ug) = \psi_{\mathrm{Unip}(n)}(u)f(g), \quad u \in \mathrm{Unip}(n)$$

A function  $f \in W$  is called:

- of **moderate growth** if for any compact set  $\Omega \subset G$ ,  $|f(\begin{bmatrix} y & \\ & 1 \end{bmatrix} g)| < C|y|^k$  for some  $C, k > 0$  for any  $g \in \Omega$  when  $|y| \rightarrow \infty$ .
- **rapidly decreasing** if for any compact set  $\Omega \subset G$  and  $k \in \mathbb{R}$ ,  $|f(\begin{bmatrix} y & \\ & 1 \end{bmatrix} g)| < C|y|^k$  for some  $C$  for any  $g \in \Omega$  when  $|y| \rightarrow \infty$ .  $\lrcorner$

**Lemma (26.1.4.2) [Whittaker Function].** Let  $\mu, \lambda \in \mathbb{C}$  and  $k \in \mathbb{Z}$ . Let  $W(\lambda, \mu, k)$  be the space of functions  $f \in W(26.1.4.1)$  on  $G$  s.t.  $\Delta f = \lambda f$ ,  $If = \mu f$ ,  $f \in (C^\infty(G))^k$  and  $f$  is of moderate growth, then  $W(\lambda, \mu, k)$  is 1-dimensional, and functions in this space are actually rapidly decreasing and analytic.

Moreover, the operators  $R, L$  map  $W(\lambda, \mu, k)$  into  $W(\lambda, \mu, k+2)$ ,  $W(\lambda, \mu, k-2)$  respectively.

Also, for  $|y| \rightarrow 0$  and  $\mu$  imaginary,  $W(\lambda, \mu, k)$  is bounded by a  $|y|^{-1/2}$  for some  $\varepsilon > 0$ .  $\lrcorner$

*Proof:* For  $f \in W(\lambda, \mu, k)$ , in coordinate (16.6.1.3), we have

$$f(g) = u^\mu \psi(x) e^{ik\theta} w(y), \quad w(y) = f\left(\begin{bmatrix} y^{1/2} & \\ & y^{-1/2} \end{bmatrix}\right)$$

Thus it suffices to study the behavior of  $w(y)$ . By the expression of  $\Delta$  (26.1.2.5), if  $\psi(x) = e^{iax}$ , then

$$w'' + (-a^2 + \frac{k}{2y} + \frac{\lambda}{y^2})w = 0$$

And this is the Whittaker's equation, and the only moderate growth function is rapidly decreasing and analytic. This is by direct calculation in [A course of Modern Analysis, Whittaker/Watson(1927)]<sup>?</sup>.

For the action of  $R, L$ , they preserve  $W$  because they are right actions. And  $Rf, Lf$  have the same eigenvalue of  $\Delta, I$  because  $\Delta, I$  are in the center of  $U(\mathfrak{g})$ . They shift the weight by (26.1.2.15) and (26.1.3.24).  $\square$

**Prop. (26.1.4.3) [Uniqueness of Whittaker Models for  $GL(2; \mathbb{R})$ ].** For  $(\pi, V) \in \text{Irr}^{\text{adm}}((\mathfrak{g}, K))$ , there exists at most one space  $W(\pi, \psi) \in W$  consisting of  $K$ -finite functions  $f \in W$  that is of moderate growth, and is invariant under the action of  $U(\mathfrak{g})$  and  $K$ , and is infinitesimal equivalent to  $(\pi, V)$ .

Moreover, functions in  $W(\pi, \psi)$  are actually rapidly decreasing and analytic. The space  $W(\pi, \psi)$  is called the **Whittaker model** of  $\pi$ , if it exists.  $\lrcorner$

*Proof:* By (24.6.3.9),  $\Delta, I$  acts by scalars  $\lambda, \mu$  on  $V$ . By (24.6.4.14) or (24.6.4.22) if  $V^k \neq 0$ , then  $\dim V^k = 1$ . If  $V^k \neq 0$ , then the image of  $V^k$  under the isomorphism with  $W(\pi, \psi)$  is in the space  $W(\lambda, \mu, k)$ . Thus  $W(\pi, \psi)$  is the direct sum of the  $W(\lambda, \mu, k)$  for all  $k$  that  $V^k \neq 0$ , so uniquely determined by (26.1.4.2). And the rapid decreasing and analytic properties are also consequences of (26.1.4.2).  $\square$

**Prop. (26.1.4.4).** If  $(\pi, V) \in \text{Irr}^{\text{adm}}((\mathfrak{g}, K))$  has a Whittaker model, then there exists some  $\xi \in V$  that  $W_\xi(1) \neq 0$ .  $\lrcorner$

*Proof:* Because  $K$  intersect each connected component of  $G$ , we can assume some  $W_\xi$  is non-zero on  $K^0$ , thus by analyticity (26.1.4.3), its derivatives at 1 are not identically 0, thus there exists some  $D \in U(\mathfrak{g})$  that  $DW_\xi(1) = W_{D\xi}(1) \neq 0$ .  $\square$

**Prop. (26.1.4.5) [Whittaker Models for  $GL(n; \mathbb{C})$ ].** This result is true if  $\mathbb{R}$  is replaced by  $\mathbb{C}$ .  $\lrcorner$

*Proof:* Cf. [Automorphic Forms on  $GL(2)$ , Jacquet/Langlands (1970) Thm5.3. P232]<sup>?</sup>.  $\square$

**Prop. (26.1.4.6) [Fourier-Maass Expansion].**  $\lrcorner$

*Proof:*  $\square$

**Cor. (26.1.4.7) [Rapidly Decreasing].** Cuspidal automorphic forms are rapidly decreasing at any Siegel set.  $\lrcorner$

*Proof:* Cf. [Gan's Notes, P26]<sup>?</sup>.  $\square$

**Prop. (26.1.4.8) [Existence of Whittaker Models].** The Whittaker model exists for the  $\lrcorner$

**Prop. (26.1.4.9) [Local Multiplicity One Theorem, Shalika [Sha74]].** Given a unitary irreducible representation  $V$  of  $GL(n; F)$ , a **Whittaker functional** on  $V^\infty$  is a  $\text{Unip}(n; F)$ -map  $\lambda : V^\infty \rightarrow \psi_N$ .

Then the space of Whittaker functionals on  $V^\infty$  is at most 1-dimensional.  $\lrcorner$

*Proof:* <sup>?</sup>  $\square$

## 5 Spectral Problems

### The Spectral Problems for $\Gamma \backslash \mathcal{H}$ Compact

**Def. (26.1.5.1).** In this subsection we assume  $\Gamma \backslash G_1$  is compact, or equivalently  $\Gamma \backslash \mathcal{H}$  is compact, because  $K$  is compact. This condition makes  $C^\infty(\Gamma \backslash \mathcal{H}, \chi) = L^2(\Gamma \backslash \mathcal{H}, \chi)$ , and will make the decomposition having only discrete parts(26.1.5.2).  $\lrcorner$

**Prop. (26.1.5.2) [ $L^2(\Gamma \backslash G, \chi)$  Totally Decomposable].** The space  $L^2(\Gamma \backslash G, \chi)$  decomposes into a Hilbert space direct sum of subspaces that are invariant and irreducible under the right regular action  $\rho$ .  $\lrcorner$

*Proof:* This follows from(29.1.4.2).  $\square$

**Lemma (26.1.5.3).** Let  $\sigma$  be the character on  $K$  that  $\sigma(k_\theta) = e^{-ik\theta}$ ,  $C_c^\infty(K \backslash G/K, \sigma)$  is commutative by(26.1.2.4), let  $\xi$  be a character of it. Let  $H(\xi)$  be the subspace of  $f \in L^2(\Gamma \backslash G, \chi, k)$  that  $\pi(\varphi)f = \xi(\varphi)f$  for  $\varphi \in C_c^\infty(K \backslash G/K, \sigma)$ .

Then  $H(\xi)$  are of f.d. and different  $H(\xi), H(\eta)$  are orthogonal that  $\oplus_\xi H(\xi) = L^2(\Gamma \backslash G, \chi, k)$ .  $\lrcorner$

*Proof:* Suppose  $0 \neq f \in H(\xi)$ , then by(24.7.5.2), we can find a  $\varphi \in C_c^\infty(K \backslash G/K, \sigma)$  s.t.  $\rho(\varphi)f \neq 0$ . And by hypothesis  $\rho(\varphi)f = \xi(\varphi)f$ , thus  $\xi(\varphi) \neq 0$ , and  $f$  is an eigenvalue of  $\rho(\varphi)$  which is compact and self-adjoint, so the  $\xi(\varphi)$ -eigenspace of  $\rho(\varphi)$  is f.d. and  $H(\varphi)$  is contained in this space, thus f.d.

To show the orthogonality, choose  $\varphi \in C_c^\infty(K \backslash G/K, \sigma)$  that  $\xi(\varphi) \neq \eta(\varphi)$ . Considering  $\varphi = \varphi_1 + i\varphi_2$ , where  $\rho(\varphi_1), \rho(\varphi_2)$  are both self-adjoint, then we may assume  $\varphi$  is self-adjoint. Then  $H(\xi), H(\eta)$  are contained in different eigenspaces of  $\rho(\varphi)$ , so they are orthogonal.

Finally for the direct sum, it suffices to show that if  $f$  is orthogonal to all  $H(\xi)$ , then  $f = 0$ . Given  $f$ , let  $\varphi_0 \in C_c^\infty(K \backslash G/K, \sigma)$  be chosen that  $\rho(\varphi_0)f$  is near  $f$  that  $\rho(\varphi_0)f, f$  are not orthogonal(24.7.5.2).

Consider the eigenspace decomposition of  $\rho(\varphi_0)$  on  $L^2(\Gamma \backslash G, \chi, k)$ , then  $f = f_0 + f_1 + f_2 + \dots$ , then  $\rho(\varphi_0)f = \lambda_1 f + \lambda_2 f + \dots$ . Because  $f$  is not orthogonal to  $\rho(\varphi_0)f$ , thus  $f_i$  is not orthogonal to  $f$  for some  $i \geq 1$ . Let  $V$  be the  $\lambda_i$ -eigenspace of  $\rho(\varphi_0)$ , then  $V$  is f.d. and invariant under  $\rho(\varphi)$  for all  $\varphi \in C_c^\infty(K \backslash G/K, \sigma)$  because  $C_c^\infty(K \backslash G/K, \sigma)$  is commutative(26.1.2.4). So  $V$  is a direct sum of elements of the spaces  $H(\xi)$ , so  $V$  is orthogonal to  $f$ , contradiction.  $\square$

**Prop. (26.1.5.4).** The space  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$  decomposes into a Hilbert space direct sum of eigenspaces for  $\Delta_k$ .  $\lrcorner$

*Proof:* By(26.1.2.15), it suffices to prove for  $L^2(\Gamma \backslash G, \chi, k)$  and  $\Delta$ . Because  $\Delta$  are in the center of  $U(\mathfrak{g})$ ,  $H(\xi)$  are all  $\Delta$ -invariant. So we finish by the lemma(26.1.5.3), as each  $H(\xi)$  is f.d. so  $\Delta$  is a self-adjoint operator on  $H(\xi)$ , because  $C_c^\infty(\Gamma \backslash G, \chi)$  is dense(26.1.1.5). So it is a direct sum of  $\Delta$ -eigenspaces.  $\square$

**Prop. (26.1.5.5).**

- The eigenvalues  $\lambda_i$  of  $\Delta_k$  on  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$  tends to  $\infty$ , and satisfies  $\sum \lambda_i^{-2} < \infty$ .
- The laplacian  $\Delta_k$  has an extension to a self-adjoint operator on the Hilbert space  $L^2(\Gamma \backslash \mathcal{H}, \chi, k)$ .  $\lrcorner$

*Proof:* Cf.[Bump P185].?  $\square$

### General Spectral Problems

**Prop. (26.1.5.6)** [Gelfand, Graev and Piatetski-Shapiro]. Let  $\varphi \in C_c^\infty(G)$ , then

- There exists a constant  $C(\varphi)$  that for all  $f \in L_0^2(\Gamma \backslash G, \chi, \omega)$ , we have  $\|\rho(\varphi)f\|_{C(G)} \leq C(\varphi)\|f\|_2$ .
- $\rho(\varphi)$  is a compact operator on  $L_0^2(\Gamma \backslash G, \chi, \omega)$ .

Notice this generalizes (29.1.4.8).  $\square$

*Proof:* We may assume  $\Gamma$  has cusps, otherwise this is proved in (29.1.4.8). Conjugating  $\Gamma$  by an element of  $\mathrm{SL}(2; \mathbb{R})$ , we may assume that  $\infty$  is a cusp for  $\Gamma$ , and  $\Gamma_\infty$  is generated by  $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ . Then it suffices to prove that

$$\sup_{g \in \mathcal{G}_{c,d}} |\rho(\varphi)f(g)| \leq C_0 \|f\|_2,$$

because we can do the same for other cusps of  $\Gamma$ , and use (26.1.1.12) to show that  $\sup_{g \in F} |\rho(\varphi)f(g)| \leq C_0 \|f\|_2$ , hence for all  $g \in \mathcal{H}$ .

Now let  $\varphi_\omega(g) = \int_{R^*} \varphi(zg)\omega(z)dz$ , then

$$\begin{aligned} \rho(\varphi)f(g) &= \int_{Z(\mathbb{R}) \backslash G} f(gh)\varphi_\omega(h)dh \\ &= \int_{Z(\mathbb{R}) \backslash G} f(h)\varphi_\omega(g^{-1}h)dh \\ &= \int_{\Gamma_\infty Z(\mathbb{R}) \backslash G} \sum_{\gamma \in \Gamma_\infty} f(\gamma h)\varphi_\omega(g^{-1}\gamma h)dh \\ &= \int_{\Gamma_\infty Z(\mathbb{R}) \backslash G} K(g, h)f(h)dh, \end{aligned}$$

where  $K(g, h) = \sum_{\gamma \in \Gamma_\infty} \chi(\gamma)\varphi_\omega(g^{-1}\gamma h)$ . Then we can estimate the kernel  $K(g, h)$  and show it decays rapidly for  $h$  when  $g$  is fixed, and this will give the desired result, Cf. [Bump, P286] ?.

2: Let  $X(\Gamma)$  be the space obtained by compactifying  $\Gamma \backslash G_1$  by adjoining cusps, and let  $\Sigma$  be the image of the unit ball in  $L_0^2(\Gamma \backslash G, \chi, \omega)$  under  $\rho(\varphi)$ , then we can extend each  $|\rho(\varphi)f|$  to  $X(\Gamma)$  that it vanish at the cusps, by the corollary to item1 below (26.1.5.7), then  $\Sigma$  is bounded in the  $L^\infty$ -norm by item1, and it is also equicontinuous, because its derivatives can be bounded uniformly for  $f$ :

$$(X\rho(\varphi)f)(g) = \rho(\varphi_X)f(g), \quad \varphi_X(g) = \frac{\partial}{\partial t}\varphi(\exp(-tX)g).$$

so the conclusion of item1 applied to  $\varphi_X$ s shows the uniformly boundedness. Now Arzela-Ascoli shows that  $\Sigma$  is precompact in  $C(X(\Gamma))$ , thus also in  $L^2(X(\Gamma))$ .  $\square$

**Cor. (26.1.5.7).** If  $\varphi \in C_c^\infty(G)$  and  $f \in L_0^2(\Gamma \backslash G, \chi, \omega)$  (26.1.1.4), then  $\rho(\varphi)f$  is smooth and rapidly decreasing at cusps.  $\square$

*Proof:* This is contained in the proof of (26.1.5.6) above, Cf. [Bump, P286] ?.  $\square$

**Prop. (26.1.5.8)** [ $L_0^2(\Gamma \backslash G, \chi, \omega)$  **Totally Decomposable**]. The space  $L_0^2(\Gamma \backslash G, \chi, \omega)$  decomposes into a Hilbert space direct sum of irreducible subrepresentations of  $G$ , and if  $H$  is such a subrepresentation, then the  $H^{K-\text{fin}} \subset H$  is dense, and  $H^{K-\text{fin}} \in \mathrm{Irr}^{\mathrm{adm}}((\mathfrak{g}, K))$  is contained in  $\mathcal{A}_0(\Gamma \backslash G, \chi, \omega)$ .

Moreover,  $\mathcal{A}_0(\Gamma \backslash G, \chi, \omega)$  is just the smooth part of  $L_0^2(\Gamma \backslash G, \chi, \omega)$ .  $\square$

*Proof:* The proof is exactly the same as that of (26.1.5.2), but where we use (26.1.5.6) in place of (29.1.4.8). Lemma (24.7.5.2) is indispensable.

$H^{K-\text{fin}} \in \text{Irr}^{\text{adm}}(\mathfrak{g}, K)$  by (24.7.5.7). To show it is contained in  $\mathcal{A}_0(\Gamma \backslash G, \chi, \omega)$ , it suffices show that  $H^k \subset \mathcal{A}_0(\Gamma \backslash G, \chi, \omega)$  for any  $k$ . For this, if  $0 \neq f \in H_k$ , choose by (24.7.5.2) a function  $\varphi \in C_c^\infty(G)$  that  $\rho(\varphi)f \neq 0$ , thus we can assume  $\rho(\varphi)(f) = f$  by (24.7.5.2) and (29.1.4.8), as  $\dim H^k < \infty$  by (24.7.5.4). Then by (26.1.5.7)  $f$  is smooth and decay rapidly, and it is clearly  $K$ -finite and  $\mathcal{Z}$ -finite, thus  $f \in \mathcal{A}_0(\Gamma \backslash G, \chi, \omega)$ .

The last assertion follows as any cuspidal form decays rapidly thus is contained in  $L_0^2(\Gamma \backslash G, \chi, \omega)$  (26.1.4.7).  $\square$

**Cor. (26.1.5.9) [Finite Multiplicity, Harish-Chandra].**

- $L_0^2(\Gamma \backslash G, \chi, \omega) = \bigoplus_{\pi} \pi^{m(\pi)}$ , where  $m(\pi) \in \mathbb{Z}_+$ .
- Let  $\mathcal{A}(\Gamma \backslash G, \chi, \omega, \lambda)$  be the  $\lambda$ -eigenspace of  $\Delta$  on  $\mathcal{A}(\Gamma \backslash G, \chi, \omega)$ , and  $\mathcal{A}(\Gamma \backslash G, \chi, \omega, \lambda, k)$  be the  $k$ -part in  $K$ -decomposition of  $\mathcal{A}(\Gamma \backslash G, \chi, \omega, \lambda)$ , and  $\mathcal{A}_0(\Gamma \backslash G, \chi, \omega, \lambda, k)$  the cuspidal part, then  $\dim \mathcal{A}_0(\Gamma \backslash G, \chi, \omega, \lambda, k) < \infty$ .

┘

*Proof:* 1: Let  $(\pi, V)$  be an irreducible unitary representation of  $G$ , let  $k$  be chosen that  $V^k \neq 0$ , let  $0 \neq \xi \in V^k$ , then by (24.7.5.2) there is a  $\varphi \in C_c^\infty(G)$  that  $\pi(\varphi)\xi = \xi$  and  $\pi(\varphi)$  is self-adjoint by (24.7.5.2) and (29.1.4.8), as  $\dim V^k < \infty$  by (24.7.5.4). Now consider for any continuous linear map  $T : V \rightarrow L_0^2(\Gamma \backslash G, \chi, \omega)$ ,  $T\xi$  lies in the 1-eigenspace of the compact self-adjoint operator  $\rho(\varphi)$ , which is of f.d. as  $\rho(\varphi)$  is compact by (26.1.5.6). Of course,  $T$  is determined by  $T\xi$  because  $V$  is irreducible, so these  $T$  form a f.d. vector space, so the multiplicity is finite.

2: This follows from the first, because by (26.1.5.8) the  $K$ -finite parts of an irreducible subrepresentation of  $L_0^2(\Gamma \backslash G, \chi, \omega)$  are just the space of cuspidal forms contained in it, and  $\lambda, k$  determined the action of  $\Delta, I$ , so there are at most onw representation of  $G$  that satisfies these, by classification in (24.6.4.22) and (24.6.4.9), and they appear for finite multiplicity by item1, and for each of them, the  $k$ -part is of f.d., because they are admissible (24.7.5.7). So the conclusion follows.  $\square$

**Thm. (26.1.5.10) [Fundamental Theorem of Harish-Chandra].** In fact, Harish-Chandra showed that for any finite codimensional ideal  $J \subset \mathcal{Z}$ , The subspace  $\mathcal{A}(\Gamma \backslash G, \chi, \omega, J)$  of automorphic forms annihilated by  $J$  is an admissible  $(\mathfrak{g}, K)$ -module. In particular, any  $\pi \in \text{Irr}^{\text{adm}}(\mathfrak{g}, K)$  appears in  $\mathcal{A}(\Gamma \backslash G, \chi, \omega, J)$  with finite multiplicity. ?  $\square$

*Proof:* ?  $\square$

### Eisenstein Series

**Prop. (26.1.5.11).** Let  $\mathcal{A}^2(\Gamma \backslash G, \chi, \omega) = \mathcal{A}(\Gamma \backslash G, \chi, \omega) \cap L^2(\Gamma \backslash G, \chi, \omega)$ , then this is exactly the space of smooth  $K$ -finite  $\mathcal{Z}$ -finite vectors in the discrete spectrum  $L_{\text{disc}}^2(\Gamma \backslash G, \chi, \omega)$ .  $\square$

*Proof:*  $\square$

## 26.2 Spectral Theory of Automorphic Forms

References are [Spectral Methods in Automorphic Forms, Iwaniec].

### 1 Quantum Unique Ergodicity

**Thm. (26.2.1.1) [Holowinsky].** Let  $f$  be a Hecke cusp form of weight  $k$  for  $\mathrm{SL}(2; \mathbb{Z})$ , and let  $F_k(z) = y^{k/2} f(z)$ , then

- If  $\varphi$  is a Maass cusp form which is also an eigenfunction of all the Hecke operators, then

$$|\langle \varphi F_k, F_k \rangle| \ll_{\varphi, \varepsilon} (\log k)^{-\frac{1}{30} + \varepsilon}.$$

- If  $\psi \in C_c^\infty(0, \infty)$ , then

$$\left| \langle E(\cdot|\psi) F_k, F_k \rangle - \frac{3}{\pi} \langle E(\cdot|\psi), 1 \rangle \right| \ll_{\psi, \varepsilon} (\log k)^{-\frac{2}{15} + \varepsilon}.$$

┘

*Proof:*

□

**Cor. (26.2.1.2).** When  $k \geq \infty$ , the zeros of the Hecke cusp form of weight  $k$  become equidistributed w.r.t. the measure  $\frac{3}{\pi} \frac{dx dy}{y^2}$ . ┘

*Proof:* Cf. [RUDNICK, On the asymptotic distribution of zeros of modular forms].

□

**Prop. (26.2.1.3) [Rudnick].** If the quantum unique ergodicity is true for  $\mathrm{GL}(2)$ , then the zeros of these eigenforms are equidistributed w.r.t.  $d\tilde{v}$ . ┘

## 26.3 Modular Forms I

References are [Kat73, Miy06, R-S11, ?D-S16, BGHZ, Mil17c].

**Notation (26.3.0.1).**

- Use notations on automorphic forms defined in [Archimedean Automorphic Representations](#).

┘

### 1 Modular Forms

#### Siegel Modular Forms

**Def. (26.3.1.1) [Siegel Modular Forms].** Let  $\Gamma \subset \mathrm{Mp}(2; \mathbb{R})$  be a subgroup commensurable with  $\mathrm{Mp}(2; \mathbb{Z})$ , and let  $(\rho, V_\rho) \in \mathrm{Rep}^{\mathrm{fd}}(\Gamma)$  that factors through a finite quotient of  $\Gamma$ , and let  $k \in \frac{1}{2}\mathbb{Z}$ , then a **Siegel modular form** of weight  $k$  and type  $\rho$  for  $\Gamma$  is a holomorphic function  $f \in \mathcal{O}(\mathcal{H}; V_\rho)$

s.t. for  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \pm\sqrt{c\tau + d} \in \Gamma$ ,

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (\pm\sqrt{c\tau + d})^{2k} \rho_M(\gamma) f(\tau).$$

┘

**Def. (26.3.1.2) [Holomorphic Siegel Modular Forms]. ?**

The space of holomorphic Siegel modular forms is denoted by  $\mathrm{MF}_k(\Gamma, \rho)$ .

┘

**Def. (26.3.1.3) [Siegel Modular Forms(of half Integral Weights)].** Let  $\Gamma \subset \mathrm{Mp}(2; \mathbb{R})$  be a discrete subgroup commensurable with  $\mathrm{Mp}(2; \mathbb{Z})$ , and let  $(\rho, V_\rho) \in \mathrm{Rep}^{\mathrm{fd}}(\Gamma)$  that factors through a finite quotient of  $\Gamma$ , and let  $k \in \frac{1}{2}\mathbb{Z}$ , then the space  $\mathrm{MF}_k^!(\Gamma, \rho)$  of (weakly holomorphic) **Siegel modular form** of weight  $k$  and type  $\rho$  for  $\Gamma$  consists of functions  $f \in C^\infty(\mathcal{H}; V_\rho)$  s.t.

- $f$  is holomorphic.
- (weak modularity) For  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \pm\sqrt{c\tau + d} \in \Gamma$ ,

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (\pm\sqrt{c\tau + d})^{2k} \rho_M(\gamma) f(\tau).$$

- $f$  is meromorphic at the cusps(26.1.2.12), i.e. for any  $g = f[\gamma]_k$ , the coefficients of the expansion

$$g(\tau) = \sum_{n \in \mathbb{Q}} \sum_{\gamma} c_{n,\gamma} q^n e_\gamma$$

vanishes for  $n$  small. Equivalently,  $f[\gamma]_k$  are all meromorphic at  $\infty$  for  $\gamma \in \mathrm{Mp}(\mathbb{Q})$ .

And a meromorphic modular form  $f$  is called a **holomorphic modular form** iff it is holomorphic and holomorphic at cusps(26.1.2.12). The subspace of holomorphic modular forms in  $\mathrm{MF}^!(\Gamma, \rho)$  is denoted by  $\mathrm{MF}(\Gamma, \rho)$ .

A meromorphic modular form of weight 0 is called a **modular function**.

Denote  $\mathrm{MF}_k^!(\Gamma, \mathbf{1}) = \mathrm{MF}_k^!(\Gamma)$ ,  $\mathrm{MF}_k(\Gamma, \mathbf{1}) = \mathrm{MF}_k(\Gamma)$ , and moreover we denote by  $\mathrm{SF}_k(\Gamma, \rho)$  the set of **holomorphic cusp forms** consisting of all  $f \in \mathrm{MF}_k(\Gamma, \rho)$  that vanishes at the cusps(26.1.2.12), and denote  $\mathrm{SF}_k(\Gamma, \mathbf{1}) = \mathrm{SF}_k(\Gamma)$ .

Denote by  $\mathrm{MF}^!(\Gamma) = \bigoplus_{k \geq 0} \mathrm{MF}_k^!(\Gamma)$  the graded ring of meromorphic modular forms,  $\mathrm{MF}(\Gamma) = \bigoplus_{k \geq 0} \mathrm{MF}_k(\Gamma)$  the graded algebra of holomorphic modular forms for  $\Gamma$ ,  $\bigoplus_{k \geq 0} \mathrm{SF}_k(\Gamma)$  is a graded ideal of  $\mathrm{MF}(\Gamma)$ .

┘



**Prop. (26.3.1.4) [Fourier Expansions of Modular Forms].** Let  $N \in \mathbb{Z}_+, k \in \mathbb{N}$  and  $\Gamma \subset \Gamma(1)$  be a modular form of level  $N$ . Then  $f \in \Omega(\mathcal{H}, \mathbb{C})$  is in  $\text{MF}_k(\Gamma)$  iff it is weakly modular, and there exists some  $r \in \mathbb{Z}_+$  s.t.  $f$  has a Fourier expansion

$$f(\tau) = \sum_{n \in \mathbb{N}} a_n e^{2\pi i n / N},$$

where  $a_n = O(n^r)$ . ┘

*Proof:* Cf. [Diamond, modular forms] P17. □

**Thm. (26.3.1.5) [Valence Formula].** If  $f \in \text{MF}_{2k}^!(\Gamma)$ , then

$$\frac{1}{2} \sum_{Q \text{ elliptic points with period 2}} \text{ord}_Q(f) + \frac{1}{3} \sum_{Q \text{ elliptic points with period 3}} \text{ord}_Q(f) + \sum_{Q \text{ others}} \text{ord}_Q(f) = kd/6,$$

where the sum is over a set of representatives for points in  $\Gamma \backslash \mathcal{H}^*$ . ┘

*Proof:* Cf. [Milne, P53]. ? □

**Def. (26.3.1.6) [Hauptmodul].** Let  $\Gamma$  be a discrete subgroup of  $\text{SL}(2; \mathbb{R})$  with exactly one cusp,  $k > 0$ , a **Hauptmodul** for  $\Gamma$  is an element in  $M^!(\Gamma)$  s.t. it is holomorphic on  $\mathcal{H}$  and has a simple pole at the cusp with residue 1, i.e. its Fourier series at the cusp is of the form

$$f(\tau) = q^{-1} + a_0 + a_1 q + \dots$$

┘

**Prop. (26.3.1.7) [Petersson Inner Product].** For a congruence subgroup  $\Gamma \subset \Gamma(1)$  (20.4.5.9) and  $k \in \mathbb{Z}_+, f \in \text{MF}_k(\Gamma, \chi)$  and  $g \in \text{SF}_k(\Gamma, \chi)$ ,  $f \bar{g} y^k$  is invariant under the action of  $\Gamma$ , thus we can define

$$(f, g)_{\text{Pet}} = (f, g)_{\Gamma, k} \triangleq \frac{1}{[\text{SL}(2, \mathbb{Z}) : \{\pm 1\}\Gamma]} \int_{\Gamma \backslash \mathcal{H}} f \bar{g} y^k \frac{dx dy}{y^2},$$

which is finite, and restricts to an inner product on  $S_k(\Gamma, \chi)$ . Moreover, this inner form is invariant of the  $\Gamma$  chosen. ┘

*Proof:* To show it is finite, notice that  $\Gamma \backslash \mathcal{H}$  is a finite translations of the fundamental domain of  $\Gamma(1)$ , and for  $\alpha \in \text{SL}(2; \mathbb{Z})$ ,

$$\int_{\alpha(D)} f \bar{g} y^k \frac{dx dy}{y^2} = \int_D f \circ \alpha \cdot \overline{g \circ \alpha} y(\alpha(z))^k \frac{dx dy}{y^2} = \int_D f[\alpha]_k \cdot \overline{g[\alpha]_k} y^k \frac{dx dy}{y^2},$$

and  $f[\alpha]_k, g[\alpha]_k$  are modular forms for  $\Gamma' = \alpha^{-1} \Gamma \alpha, \chi'(\gamma') = \chi(\alpha \gamma \alpha^{-1})$ , thus it suffices to prove the integral is finite on the fundamental domain  $F$  for  $\Gamma(1)$  (20.4.5.17). For this, notice  $g$  is of exponential decay for  $y$ , thus the integral is bounded by  $\int_{y_1}^{\infty} e^{-cy} y^{k-2} dy < \infty$ . □

**Prop. (26.3.1.8).** If  $\alpha \in \text{GL}(2, \mathbb{R})^+, f \in \text{MF}_k(\Gamma)$  and  $g \in \text{SF}_k(\alpha^{-1} \Gamma \alpha)$ , then

$$(f[\alpha]_k, g)_{\alpha^{-1} \Gamma \alpha} = (f, g[\alpha^{-1}]_k)_{\Gamma}.$$

┘

*Proof:* This is because for there is an isomorphism  $\alpha : \alpha^{-1}\Gamma\alpha\backslash\mathcal{H} \cong \Gamma\backslash H$ , and for any left  $\Gamma$ -invariant function  $\varphi$ ,

$$\int_{\alpha^{-1}\Gamma\alpha\backslash H} \varphi(\alpha(\tau))d\mu(\tau) = \int_{\Gamma\backslash H} \varphi(\tau)d\mu(\tau)$$

and  $(f\overline{g[\alpha^{-1}]y^k})(\alpha\tau) = (f[\alpha]_k\overline{gy^k})(\tau)$ .  $\square$

**Prop.(26.3.1.9) [Lower Bound on Petersson Inner Products].** For any  $N \in \mathbb{Z}_+$  and  $f = \sum_{n \in \mathbb{Z}_+} a_n q^n \in \text{SF}(\Gamma(N))$ ,

$$[\Gamma(1) : \Gamma(N)] \cdot \|f\|_{\text{Pet}} \geq \sum_{n \in \mathbb{Z}_+} \frac{|a_n|^2}{4\pi n e^{4\pi n}}.$$

┘

*Proof:* Integrate on the region  $\{\tau = x + iy : |x| \leq 1/2, y \geq 1\}$ .  $\square$

**Prop.(26.3.1.10) [Twisted Modular Forms].** For  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}(N)$ , we can define  $\text{MF}_k(N, \chi) = \text{MF}_k(\Gamma_0(N), \chi_d)$  (26.3.1.3)(20.4.5.10) and  $\text{SF}_k(N, \chi) = \text{SF}_k(\Gamma_1(N)) \cap \text{MF}_k(N, \chi)$ . Then there are direct sum decompositions:

$$\text{MF}_k(\Gamma_1(N)) = \bigoplus_{\chi \in \text{Diri}(N), \chi(-1) = (-1)^k} \text{MF}_k(N, \chi), \quad \text{SF}_k(\Gamma_1(N)) = \bigoplus_{\chi \in \text{Diri}(N), \chi(-1) = (-1)^k} \text{SF}_k(N, \chi).$$

Moreover, the summands are orthogonal w.r.t. the Petersson inner product (26.3.1.7).  $\square$

*Proof:* Cf.[Diamond, P418].  $\square$

**Cor.(26.3.1.11) [Reduction to  $\Gamma_0(N)$ ].** For any congruence subgroup  $\Gamma(N)$ , there exists  $\alpha \in \text{GL}(2, \mathbb{Q})^+$  s.t.  $\text{diag}(1, N)\Gamma_1(N)\text{diag}(1, 1/N) \subset \Gamma(N)$ . Thus if  $f \in \text{MF}_k(\Gamma(N))$ ,  $f[\text{diag}(1, N)]_k \subset \text{MF}_k(\Gamma_1(N))$ , which is a combination of modular forms in  $\text{MF}_k(\Gamma_0(N), \chi)$ . Thus the study of congruence modular forms reduces to the study of  $\text{MF}_k(N, \chi)$ .  $\square$

### Cusp Forms

**Def.(26.3.1.12) [Poincaré Series].** For any Fuchsian subgroup  $\Gamma \subset \Gamma(1)$ ,  $T = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$ . Let  $h$  be the minimal positive integer that  $T^h \subset \Gamma$ , and define  $\Gamma_0$  the subgroup of  $\Gamma$  generated by  $T^h$ . Then we define the **Poincaré series of weight  $2k$  and character  $n$  for  $\Gamma$**  to be the series

$$\varphi_{k,n}(z) = \sum_{\Gamma_\infty \backslash \Gamma'} \frac{e^{\frac{2\pi i n \gamma(z)}{h}}}{(cz + d)^{2k}},$$

where  $\Gamma'$  is the image of  $\Gamma$  in  $\Gamma(1)/\{\pm 1\}$ .  $\square$

**Prop.(26.3.1.13).** For  $k \in \mathbb{Z}_+, n \in \mathbb{N}$ , The Poincaré series (26.3.1.12)  $\varphi_{2k,n}$  converge absolutely on compact subsets of  $\mathcal{H}$ , and  $\varphi_{2k,n} \in \text{MF}_{2k}(\Gamma)$ . Moreover,

- $\varphi_{2k,0}(\tau)$  vanishes at all finite cusps, and  $\varphi_{2k,0}(i\infty) = 1$ .
- For  $n \geq 1$ ,  $\varphi_{2k,n}(\tau) \in \text{SF}_k(\Gamma)$ .

┘

*Proof:* Cf.[Gunning, Lectures on Modular Forms]3.9. ?  $\square$

**Prop. (26.3.1.14) [Cusp Forms are Spanned by Poincaré Series].** For any Fuchsian subgroup  $\Gamma \subset \Gamma(1)$  and  $k \in \mathbb{Z}_+, n \in \mathbb{N}$ , if  $f = \sum_{n \in \mathbb{Z}_+} a_n e^{2\pi i n z/h} \in \text{SF}_k(\Gamma)$ , then

$$(f, \varphi_{2k,n})_{\text{Pet}} = \frac{h^{2k}(2k-2)!}{(4\pi)^{2k-1}} n^{1-2k} a_n.$$

Thus the Poincaré series  $\{\varphi_{2k,n}(z)\}_{n \in \mathbb{Z}_+}$  span  $S_{2k}(\Gamma)$ . ┘

*Proof:* Cf. [Gunning, Lectures on Modular Forms] 3.11.Thm 5. ? □

### Modular Forms of Half Integral Weights

References are [Modular Forms of Half Integral Weight, Shimura].

**Prop. (26.3.1.15) [Theta Functions].** It follows from (26.8.2.5) that

$$\theta\left(\frac{\tau}{4\tau+1}\right) = \theta\left(-\frac{1}{4(-1/(4\tau)-1)}\right) = \sqrt{2i\left(\frac{1}{4\tau}+1\right)} = \theta\left(-\frac{1}{4\tau}-1\right) = \sqrt{4\tau+1}\theta(\tau).$$

Thus by (20.4.5.12),  $\theta^2 \in \text{MF}_1(\Gamma_0(4), \chi_{-4})$ . ┘

**Prop. (26.3.1.16) [Dedekind  $\eta$ -Function].** The Dedekind  $\eta$ -function (10.7.2.6)

$$\eta(\tau) = e^{2\pi i \tau/24} \prod_{n \in \mathbb{Z}_+} (1 - e^{2\pi i n \tau})$$

belongs to  $\text{MF}_{1/2}(\Gamma(1), \chi)$ . ┘

*Proof:* To show this, it suffices to show

$$\eta(\gamma(z)) = \varepsilon(\gamma)(cz+d)^{1/2}\eta(z)$$

for every  $\gamma \in \Gamma(1)$ , where  $\varepsilon(\gamma)$  is a 24-th root of unity. Because  $[\gamma]_k$  is an action (26.3.1.3), by 5, it suffices to show for  $S = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$  and  $T = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$ . The case for  $T$  is clear from the last expression of  $\eta(z)$ . For  $S$ , it suffices to show that

$$\eta(-1/\tau) = \sqrt{\frac{\tau}{i}} \cdot \eta(\tau).$$

This is true when  $\tau = i$ . So it suffices to show its logarithm derivative:

$$\frac{\eta'(-1/z)}{\eta(-1/z)} z^{-2} = \frac{1}{2z} + \frac{\eta'(z)}{\eta(z)}.$$

And notice

$$\frac{\eta'(z)}{\eta(z)} = \frac{2\pi i}{24} (1 - 24 \sum_{n \in \mathbb{Z}_+} \frac{ne^{2\pi i n z}}{1 - e^{2\pi i n z}}) = \frac{2\pi i}{24} (1 - 24 \sum_{n \in \mathbb{Z}_+} \sigma(n) e^{2\pi i n z}) = \frac{2\pi i}{24} E_2(z).$$

So the assertion follows from (26.3.4.14). □

**Thm. (26.3.1.17) [Serre-Stark].** Every modular forms of weight  $1/2$  (for any level) is a linear combination of unary theta functions. ┘

## 2 Hecke Algebras

**Def. (26.3.2.1) [Hecke Algebra of Level  $N$ ].** For  $N \in \mathbb{Z}_+$ , the **Hecke algebra of level  $N$**  is defined to be

$$\mathcal{H}_N = \prod_{p \in \text{Prime} \setminus S(N)} \mathcal{H}_{\mathcal{K}_p}(\text{GL}(2; \mathbb{Q}))$$

It is commutative ┘

### Hecke Operators

**Prop. (26.3.2.2).** If  $\Gamma_1, \Gamma_2$  are congruence subgroups of  $X$ , then for any  $\alpha \in \text{GL}^+(2; \mathbb{Q})$ , the double coset

$$\Gamma_1 \alpha \Gamma_2 = \coprod_i \Gamma_1 \beta_i$$

for some  $\beta_i \in \text{GL}^+(2, \mathbb{Q})$ . Then we can define a map

$$\text{Div}(X(\Gamma_1)) \rightarrow \text{Div}(X(\Gamma_2)) : [\tau] \mapsto [\beta_i \tau],$$

which induces a morphism  $\text{Jac}(\Gamma_1) \rightarrow \text{Jac}(\Gamma_2)$  ?. ┘

**Prop. (26.3.2.3) [Action on Modular Forms].** Let  $\Gamma_1, \Gamma_2$  be congruence subgroups of  $\text{GL}^+(2, \mathbb{R})$ , then ┘

**Prop. (26.3.2.4) [Hecke Operators].** The Hecke algebra  $\Gamma \backslash \text{GL}^+(2; \mathbb{Q}) / \Gamma$  acts on  $\text{MF}_k(\Gamma)$  via

$$T_\alpha(f) = \sum_i f[\alpha_i]_k \text{ (26.1.2.8)}, \text{ where } \Gamma \alpha \Gamma = \coprod_i \Gamma \alpha_i.$$

And it preserves  $\text{SF}_k(\Gamma)$ . ┘

*Proof:* This is an action because if  $\Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i$ ,  $\Gamma_0(N) \beta \Gamma_0(N) = \coprod_j \Gamma_0(N) \beta_j$ , then  $[\alpha][\beta] = \coprod_{i,j} \Gamma_0(N) \alpha_i \beta_j$ .

Also  $T_\alpha(f)[\gamma]_k = T_\alpha(f)$  for  $\gamma \in \Gamma_0(N)$  because  $\Gamma_0(N) \backslash \Gamma_0(N) \alpha \Gamma_0(N)$  is right  $\Gamma_0(N)$ -invariant. And they are holomorphic at cusps because  $f$  is holomorphic at cusps thus all  $f[\alpha_i]_k$  are holomorphic at cusps. Moreover, if  $f$  vanishes at cusps, then all  $f[\alpha_i]_k$  vanishes at cusps. □

### $\Gamma_1(N)$ cases

References are [D-S16].

**Def. (26.3.2.5) [Twisted Hecke Actions on  $\text{MF}_k(N, \chi)$ ].** For any  $N \in \mathbb{Z}_+$  and  $f \in \text{MF}_k(N, \chi)$ ,  $\mathcal{H}_N$  acts on  $f$  by

$$T_\alpha(f) = \sum_i \chi_d(\alpha_i)^{-1} f[\alpha_i]_k, \text{ where } \Gamma_0(N) \alpha \Gamma_0(N) = \coprod_i \Gamma_0(N) \alpha_i.$$

This twists makes  $T_\alpha(f) \in \text{MF}_k(N, \chi)$ . It also preserves  $\text{SF}_k(N, \chi)$ . ┘

**Prop. (26.3.2.6) [Adjoint of Hecke Operators].** With respect to the Petersson inner product on  $\text{SF}_k(\Gamma_1(N))$ , for any  $p \in \text{Prime} \setminus S(N)$ ,

$$\langle p \rangle^* = \langle p \rangle^{-1}, \quad T(p)^* = \langle p \rangle^{-1} T(p).$$

┘

*Proof:* Cf.[?D-S16]P187. □

**Cor. (26.3.2.7) [Hecke Eigenforms].** There is a basis consisting of eigenfunctions for  $\mathcal{H}_N$ , called the **Hecke eigenforms**. ┘

*Proof:* This is because  $\mathcal{H}_N$  is commutative(26.3.2.1) and they act by self-adjoint operators(26.3.2.6). □

**Def. (26.3.2.8)  $[T(p)]$ .** For  $p \in \text{Prime}$ ,

$$\Gamma_1(N) \begin{bmatrix} 1 & \\ & p \end{bmatrix} \Gamma_1(N) = \begin{cases} \coprod_{j \in [p-1]} \Gamma_1(N) \begin{bmatrix} 1 & j \\ & p \end{bmatrix} & , p|N \\ \coprod_{j \in [p-1]} \Gamma_1(N) \begin{bmatrix} 1 & j \\ & p \end{bmatrix} \coprod \Gamma_1(N) \begin{bmatrix} m & n \\ N & p \end{bmatrix} \begin{bmatrix} p & \\ & 1 \end{bmatrix} & , p \nmid N \end{cases}.$$

Thus for any  $f \in \text{MF}_k(N, \chi)$ ,

$$T(p)f = \begin{cases} \sum_{j \in [p-1]} f \left[ \begin{bmatrix} 1 & j \\ & p \end{bmatrix} \right]_p & , p|N \\ \sum_{j \in [p-1]} f \left[ \begin{bmatrix} 1 & j \\ & p \end{bmatrix} \right]_p + f \left[ \begin{bmatrix} m & n \\ N & p \end{bmatrix} \begin{bmatrix} p & \\ & 1 \end{bmatrix} \right]_k & , p \nmid N. \end{cases}$$

┘

*Proof:* Cf.[?D-S16]P171 ?. □

**Cor. (26.3.2.9).**

1.  $T(1) = \text{id}$ .
2. If  $(m, n) = 1$ , then  $T(m)T(n) = T(mn)$ .
3. If  $p$  is a prime that  $(p, N) = 1$  and  $n \geq 1$ , then  $T(p^n) \cdot T(p) = T(p^{n+1}) + pR(p) \cdot T(p^{n-1})$ , where  $R(p)$  is the coset  $\Gamma_1(N) \begin{bmatrix} p & \\ & p \end{bmatrix} \Gamma_1(N)$ .

4.  $\mathcal{H}_N$  is generated by  $T(p), R(p), R(p)^{-1}$  for  $p \in \text{Prime} \setminus S(N)$ .

So we can define  $\mathcal{H}'_N = \mathbb{Z}\{T(p), R(p), R(p)^{-1}\}_{p \in \text{Prime}}$ , called the **extended Hecke algebra of level  $N$** . It is also commutative. ┘

*Proof:* 1: Trivial.

2: This follows from Chinese remainder theorem.

3: Omitted.

4: This follows from  $p$ -adic Cartan decomposition. □

**Prop. (26.3.2.10) [Explicit Formula for Hecke Operators].** Let  $f = \sum_{m \in \mathbb{Z}} a_m(f) q^m \in \text{MF}_k(N; \chi)$ , then for any  $m, n \in \mathbb{Z}_+$ ,

$$a_m(T_n(f)) = \sum_{d|(m,n)} \chi(d) d^{k-1} a_{mn/d^2}(f).$$

┘

*Proof:* Because  $T(n)$  is multiplicative(26.3.2.9), it suffices to prove for  $n = p^k$  where  $p \in \text{Prime}, k \in \mathbb{Z}_+$ . □

**Cor. (26.3.2.11).**  $a_1(T_n(f)) = a_n(f)$  for any  $f \in \text{SF}_k(\Gamma_1(N))$  and  $n \in \mathbb{Z}_+$ . In particular, there is a perfect pairing

$$\text{SF}_k(N, \chi) \times (\mathcal{H}_N)_{\mathbb{C}} \rightarrow \mathbb{C} : (f, t) \mapsto a_1(t(f)).$$

┘

### New Forms

**Def. (26.3.2.12) [Old and New Forms].** For  $d|N$ , define be the subspaces of  $\text{SF}_k(N, \chi)$ :

$$\text{SF}_k(N, \chi)[d] = \{f + g[\text{diag}(d, 1)]_k \text{ where } f, g \in S_k(N/d, \chi_0), \chi = \chi_0 \mathbf{1}_d\},$$

and let

$$\text{SF}_k(N, \chi)^{\text{old}} = \sum_{p \in S(N)} \text{SF}_k(N, \chi)[p],$$

called the space of **old forms of level  $N$** . Also the space

$$\text{SF}_k(N, \chi)^{\text{new}} = (\text{SF}_k(N, \chi)^{\text{old}})^{\perp},$$

called the space of **new forms of level  $N$** .

┘

**Remark (26.3.2.13).** The dimension of newforms and oldforms are calculated by [Dimensions of the Spaces of Cusp Forms and Newforms on  $\Gamma_0(N)$  and  $\Gamma_1(N)$ ].

┘

**Thm. (26.3.2.14) [Atkin-Lehler [A-L70]].** If  $f = \sum_{n \in \mathbb{Z}_+} a_n(f)q^n \in \text{SF}_k(\Gamma_1(N))$  satisfies  $a_n(f) = 0$  for any  $n \in \mathbb{Z}_+$  s.t.  $(n, N) = 1$ , then  $f \in \text{SF}_k(\Gamma_1(N))^{\text{old}}$  (26.3.2.12).

┘

*Proof:* Cf. [D-S16]P191.

□

**Prop. (26.3.2.15) [Eigenform Basis].**  $\text{SF}_k(N, \chi)^{\text{old}}$  and  $\text{SF}_k(N, \chi)^{\text{new}}$  are both stable under  $\mathcal{H}'_N$  (26.3.2.1). In particular, they have a orthogonal basis of Hecke eigenforms for the Hecke operators, by (26.3.2.6) (26.3.2.7).

$f \in \text{SF}_k(N, \chi)$  is called a **normalized newform** if it is a Hecke eigenform and  $a_1(f) \neq 0$ . Equivalently, it satisfies

1.  $a_1(f) = 1$ .
2.  $a_{p^r}(f) = a_p(f)a_{p^{r-1}}(f) - \chi(p)p^{k-1}a_{p^{r-2}}(f)$  for all  $p \in \text{Prime}$  and  $r \in \mathbb{Z}_{\geq 2}$ .
3.  $a_{mn}(f) = a_m(f)a_n(f)$  for all  $m, n \in \mathbb{Z}_+$  s.t.  $(m, n) = 1$ .

┘

*Proof:* Cf. [D-S16]P189?

For the equations, Cf. [D-S16]P199.

□

**Prop. (26.3.2.16) [Normalized Newforms].** If  $0 \neq f = \sum a_n(f)q^n \in \text{SF}_k(N, \chi)^{\text{new}}$  (26.3.2.12) is an eigenform for all  $\{T(n)\}_{n \in \mathbb{Z}_+ \setminus (N)}$  with eigenvalues  $\lambda(n)$ , then  $a_1(f) \neq 0$ . And if  $a_1(f) = 1$ , then  $T(m)f = a_m(f)f$  for all  $m \in \mathbb{Z}_+$ . Such  $f$  is called a **Normalized newform**.

┘

*Proof:* By (26.3.2.10),  $a_1(T(n)f) = a_n(f)$ . So if  $c_1(f) = 0$ , then  $c_n(f) = 0$  for any  $n \in \mathbb{Z}_+ \setminus (N)$ , which implies that  $f$  is an oldform by (26.3.2.14), contradiction. Thus  $a_1(f) \neq 0$ .

Assume  $a_1(f) = 1$ . For any  $m \in \mathbb{Z}_+$ , denote  $g_m = T(m)f - a_m(f)f$ , then  $g_m \in \text{SF}_k(N, \chi)^{\text{new}}$ , and it is an eigenform for all  $\{T(n)\}_{n \in \mathbb{Z}_+ \setminus (N)}$ , but  $a_1(g_m) = 0$ . So  $g_m = 0$  by the argument as above. Thus  $T(m)f = a_m(f)f$  for any  $m \in \mathbb{Z}_+$ .

□

**Prop. (26.3.2.17)**[Standard Basis for  $\mathrm{SF}_k(N, \chi)$ ]. The set

$$\{f(n\tau) : n, M \in \mathbb{Z}_+, nM|N, f \in \mathrm{SF}_k(M, \chi_0)^{\mathrm{new}} \text{ is a normalized newform, } \chi = \chi_0 \mathbf{1}_{N/M}\}$$

is a basis of  $\mathrm{SF}_k(N, \chi)$ .  $\lrcorner$

*Proof:* It is clear from the definition of oldforms and newforms (26.3.2.12) that these elements span  $\mathrm{SF}_k(N, \chi)$ . To show that they are linearly independent, suppose there is a non-trivial linear relation

$$\sum_{i,j} c_{i,j} f_i(n_{ij}\tau)$$

with minimal possible term. Then all  $f_i$  have the same eigenvalues away from  $N$ , because otherwise we can use the operator  $T_p - a_p(f_1)$  to get a shorter linear relation. But then all  $f_i$  are equal by strong multiplicity one  $?$ .  $\square$

**Cor. (26.3.2.18).** For any normalized eigenform  $g \in \mathrm{SF}_k(N, \chi)$ , there exists unique  $M \in \mathbb{Z}_+$  s.t.  $M|N$  and  $\chi_0 \in \mathrm{Diri}(M)$  s.t.  $\chi_0$  lifts to  $\chi$ , and a unique normalized newform  $f \in \mathrm{SF}_k(M, \chi_0)^{\mathrm{new}}$  s.t.  $a_p(f) = a_p(g)$  for all  $p \in \mathrm{Prime} \setminus S(N/M)$ .  $\lrcorner$

*Proof:* The uniqueness follows from strong multiplicity one  $?$ .  $\square$

### 3 Dimension Formulae

**Def. (26.3.3.1)**[Notations]. In this subsection,

- $\Gamma$  is a **congruence subgroup** of  $\mathrm{SL}(2, \mathbb{Z})$ .
- $g$  is the genus of  $X(\Gamma)$ .
- $d$  is the degree of the map  $X(\Gamma) \rightarrow X(1)$  which is equal to  $[\mathrm{SL}(2, \mathbb{Z}) : \{\pm 1\}\Gamma]$ .
- $\varepsilon_2$  is the number of elliptic points with period 2.
- $\varepsilon_3$  is the number of elliptic points with period 3.
- $\varepsilon_\infty$  is the number of cusps.
- $\varepsilon_\infty^{\mathrm{reg}}$  is the number of regular cusps (26.1.2.6).
- $\varepsilon_\infty^{\mathrm{irr}}$  is the number of irregular cusps.

$\lrcorner$

**Prop. (26.3.3.2)**[Genus Formula].

$$g = 1 + \frac{d}{12} - \frac{\varepsilon_2}{4} - \frac{\varepsilon_3}{3} - \frac{\varepsilon_\infty}{2}.$$

$\lrcorner$

*Proof:* Cf. [Diamond, P68].  $?$   $\square$

**Thm. (26.3.3.3)**[Dimension Formulae for  $k$  Even]. For  $k \in 2\mathbb{Z}$ ,

$$\dim(\mathrm{MF}_k(\Gamma)) = \begin{cases} \frac{(k-1)d}{12} + (\frac{1}{4} - \{\frac{k}{4}\})\varepsilon_2 + (\frac{1}{3} - \{\frac{k}{3}\})\varepsilon_3 + \frac{1}{2}\varepsilon_\infty & k \geq 2 \\ 1 & k = 0 \\ 0 & k < 0 \end{cases}$$

and

$$\dim(\mathrm{SF}_k(\Gamma)) = \begin{cases} \frac{(k-1)d}{12} + (\frac{1}{4} - \{\frac{k}{4}\})\varepsilon_2 + (\frac{1}{3} - \{\frac{k}{3}\})\varepsilon_3 - \frac{1}{2}\varepsilon_\infty & k \geq 4 \\ g & k = 2 \\ 0 & k \leq 0 \end{cases}$$

┘

*Proof:* Cf.[Diamond P87].

□

**Cor.(26.3.3.4).** For any Fuchsian group  $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$  and  $k \in \mathbb{Z}$ ,

$$\dim \mathrm{MF}_k(\Gamma) \leq \frac{\mathrm{Vol}(\mathrm{SL}(2; \mathbb{R})/\Gamma)}{4\pi} \cdot k + 1.$$

In particular,  $\mathrm{MF}_0(\Gamma) = 0$ .

┘

*Proof:* Cf.[Modular Forms, 1-2-3]P12.

□

**Cor.(26.3.3.5).** For any Fuchsian subgroup  $\Gamma \subset \mathrm{SL}(2; \mathbb{R})$ , any three  $f, g, h \in \mathrm{MF}_*(\Gamma)$  are not alg.independent.

┘

*Proof:* Otherwise for  $n$  large,  $\dim \mathrm{MF}_n(\Gamma) = \Omega(n^2)$ .

□

**Prop.(26.3.3.6)[Dimension Formulae for  $k$  Odd].** For  $k$  odd,

- if  $k < 0$  or  $-1 \in \Gamma$ , then  $\mathrm{MF}_k(\Gamma) = \mathrm{SF}_k(\Gamma) = 0$ .
- If  $k \geq 3$ ,  $-1 \notin \Gamma$ , then

$$\dim(\mathrm{MF}_k(\Gamma)) = (k-1)(g-1) + \lfloor \frac{k}{3} \rfloor \varepsilon_3 + \frac{k}{2} \varepsilon_\infty^{\mathrm{reg}} + \frac{k-1}{2} \varepsilon_\infty^{\mathrm{irr}}$$

$$\dim(\mathrm{SF}_k(\Gamma)) = (k-1)(g-1) + \lfloor \frac{k}{3} \rfloor \varepsilon_3 + (\frac{k}{2} - 1) \varepsilon_\infty^{\mathrm{reg}} + \frac{k-1}{2} \varepsilon_\infty^{\mathrm{irr}}$$

- If  $k = 1$ ,  $-1 \notin \Gamma$ , then

$$\dim(\mathrm{MF}_1(\Gamma)) \begin{cases} = \varepsilon_\infty^{\mathrm{reg}}/2 & \varepsilon_\infty^{\mathrm{reg}} > 2g-2 \\ \geq \varepsilon_\infty^{\mathrm{reg}}/2 & \varepsilon_\infty^{\mathrm{reg}} \leq 2g-2 \end{cases}, \quad \dim(\mathrm{SF}_1(\Gamma)) = \dim(\mathrm{MF}_1(\Gamma)) - \varepsilon_\infty^{\mathrm{reg}}/2.$$

┘

*Proof:* Cf.[Diamond P91].

□

### Explicite Dimension Formulae for $\Gamma(N), \Gamma_1(N), \Gamma_0(N)$

**Lemma(26.3.3.7).** For  $N \in \mathbb{Z}_{\geq 2}$ ,

$$\mathrm{SL}(2, \mathbb{Z}) \rightarrow \mathrm{SL}(2, \mathbb{Z}/(N))$$

is surjective.

┘

*Proof:* Cf.[?D-S16]P101.

□

**Prop.(26.3.3.8)[Degrees for  $\Gamma(N), \Gamma_1(N), \Gamma_0(N)$ ].**



- The degree of the mapping  $X(N) \rightarrow X(1)$  is

$$d = d_N = [\mathrm{SL}(2, \mathbb{Z}) : \{\pm 1\}\Gamma(N)] = \begin{cases} 1/2N^3 \prod_{p|N} (1 - 1/p^2) & N > 2 \\ 6 & N = 2 \end{cases}$$

- There is a coset decomposition  $\Gamma_1(N) = \coprod_{j=1}^N \Gamma(N) \begin{bmatrix} 1 & j \\ 0 & 1 \end{bmatrix}$ , in particular,  $d = [\mathrm{SL}(2, \mathbb{Z}) : \{\pm 1\}\Gamma_1(N)] = d_N/N, N \geq 2$ .
- There is a coset decomposition  $\Gamma_0(N) = \coprod_{y \bmod N, (y, N)=1} \Gamma_1(N) \begin{bmatrix} x & k \\ N & y \end{bmatrix}$ , where  $x, k$  are chosen that  $xy - kN = 1$ . In particular,  $d = [\mathrm{SL}(2, \mathbb{Z}) : \Gamma_0(N)] = 2d_N/N\varphi(N), N \geq 2$ .

┘

*Proof:* 1: By (26.3.3.7), the map  $\mathrm{SL}(2, \mathbb{Z}) \rightarrow \mathrm{SL}(2, \mathbb{Z}/N\mathbb{Z})$  is surjective, so  $[\Gamma(N) : \Gamma(1)] = \#\mathrm{SL}(2, \mathbb{Z}/N\mathbb{Z})$ . Let  $N = \prod p_i^{n_i}$ , then  $\mathrm{SL}(2, \mathbb{Z}/N\mathbb{Z}) \cong \prod_i \mathrm{SL}(2, \mathbb{Z}/p_i^{n_i}\mathbb{Z})$ , and calculating depending on  $e_{11}$  is invertible or  $e_{12}$  is invertible,  $\#\mathrm{SL}(2, \mathbb{Z}/p_i^{n_i}\mathbb{Z}) = 2(p^{n_i} - p^{n_i-1})p^{2n_i} - (p^{n_i} - p^{n_i-1})^2 p^{n_i} = p_i^{3n_i} - p_i^{3n_i-2}$ . Thus  $\#\mathrm{SL}(2, \mathbb{Z}/(N)) = N^3 \prod_{p|N} (1 - p^{-2})$ , and we get the desired formula.  $\square$

**Prop. (26.3.3.9) [Elliptic Points for  $\Gamma(N), \Gamma_1(N), \Gamma_0(N)$ ].** For  $N \in \mathbb{Z}_+$ ,

- There are on elliptic points for  $\Gamma(N)$  for  $N \geq 2$ .
- There are on elliptic points for  $\Gamma_1(N)$  for  $N \geq 4$ .
- The elliptic points of period 2 for  $\Gamma_0(N)$  corresponds to the ideals  $J \subset \mathrm{Ideal}(\mathbb{Z}[i])$  s.t.  $\mathbb{Z}[i]/J \cong \mathbb{Z}/(N)$ , and the elliptic points of period 3 for  $\Gamma_0(N)$  corresponds to the ideals  $J \subset \mathrm{Ideal}(\mathbb{Z}[\omega])$  s.t.  $\mathbb{Z}[\omega]/J \cong \mathbb{Z}/(N)$ .

┘

*Proof:* Cf. [Diamond]P56, 95. ?

$\square$

**Lemma (26.3.3.10).** Let  $s = a/c, s' = a'/c' \in \mathbb{P}^1(\mathbb{Q})$  with  $(a, c) = (a', c') = 1$ , then for any  $\gamma \in \mathrm{SL}(2, \mathbb{Z})$ ,

$$s' = \gamma(s) \iff \begin{bmatrix} a' \\ c' \end{bmatrix} = \pm \gamma \begin{bmatrix} a \\ c \end{bmatrix}.$$

┘

*Proof:*

$\square$

**Lemma (26.3.3.11).** Let  $s = a/c, s' = a'/c' \in \mathbb{P}^1(\mathbb{Q})$  with  $(a, c) = (a', c') = 1$ , then

- $\Gamma(N)s' = \Gamma(N)s \iff \begin{bmatrix} a' \\ c' \end{bmatrix} \equiv \pm \begin{bmatrix} a \\ c \end{bmatrix} \pmod{N}$
- $\Gamma_1(N)s' = \Gamma_1(N)s \iff \begin{bmatrix} a' \\ c' \end{bmatrix} \equiv \pm \begin{bmatrix} a + jc \\ c \end{bmatrix} \pmod{N}$ , for some  $j \in \mathbb{Z}$ .
- $\Gamma_0(N)s' = \Gamma_0(N)s \iff \begin{bmatrix} ya' \\ c' \end{bmatrix} \equiv \begin{bmatrix} a + jc \\ cy \end{bmatrix} \pmod{N}$ , for some  $y$  relatively prime to  $N$  and some  $j \in \mathbb{Z}$ .

┘

*Proof:* Use(26.3.3.10).

1: One direction is clear, for the other, If  $\begin{bmatrix} a' \\ c' \end{bmatrix} \equiv \pm \begin{bmatrix} a \\ c \end{bmatrix} \pmod{N}$ , take  $b, d$  that  $ac - bd = 1$ , then  $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  maps  $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$  to  $\begin{bmatrix} a \\ c \end{bmatrix}$ . As  $\Gamma(N)$  is normal in  $\Gamma(1)$ , it suffices to prove for  $\begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ . Then  $a' \equiv 1 \pmod{N}$ , take integers  $\beta, \delta$  that  $a'\delta - c'\beta = (1 - \alpha')/N$ , let  $\gamma = \begin{bmatrix} a' & \beta N \\ c' & \delta N \end{bmatrix}$ , then  $\gamma \in \Gamma(N)$  and  $\begin{bmatrix} a' \\ c' \end{bmatrix} = \gamma \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ .

2 follows from 1 and the coset decomposition  $\Gamma_1(N) = \coprod_{j=1}^N \Gamma(N) \begin{bmatrix} 1 & j \\ & 1 \end{bmatrix}$ .

3: There is a coset decomposition  $\Gamma_0(N) = \coprod_{y \in (\mathbb{Z}/(N))^*} \Gamma_1(N) \begin{bmatrix} x & k \\ N & y \end{bmatrix}$ , where  $x, k$  are chosen that  $xy - kN = 1$ . Then

$$\begin{aligned} \Gamma_0(N)s' = \Gamma_0(N)s &\iff \begin{bmatrix} a' \\ c' \end{bmatrix} \equiv \pm \begin{bmatrix} xa + kc + jcy \\ cy \end{bmatrix} \pmod{N}, \exists j, (y, N) = 1 \\ &\iff \begin{bmatrix} ya' \\ c' \end{bmatrix} \equiv \begin{bmatrix} a + jc \\ cy \end{bmatrix} \pmod{N}, \exists j, (y, N) = 1 \end{aligned}$$

□

**Prop. (26.3.3.12)[Cusps for  $\Gamma(N), \Gamma_1(N), \Gamma_0(N)$ ].** For  $N \in \mathbb{Z}_+$ ,

- $\varepsilon_\infty(\Gamma_0(N)) = \sum_{d|N} \varphi((d, N/d))$ .
- 

┘

*Proof:* 1: By(26.3.3.11), if  $\Gamma_0(N)s' = \Gamma_0(N)s$ , then  $\begin{bmatrix} ya' \\ c' \end{bmatrix} \equiv \begin{bmatrix} a + jc \\ c \end{bmatrix} \pmod{N}$ , which means first  $(c, N) = (c', N)$ . So we consider the equivalence classes with  $(c, N) = d$ . In this case any equivalence class has a representative that  $c = d$ . Consider the equivalence relations between them, then  $\begin{bmatrix} a' \\ d \end{bmatrix}$  represents the same cusp as  $\begin{bmatrix} a \\ d \end{bmatrix}$  iff  $(y_0 + iN/d)a' \equiv a + jd \pmod{N}$  for some  $i, j$ , which is equivalent to  $a' \equiv y_0 a \pmod{(c, N, a'N/d) = (d, N/d)}$ . So there are  $\sum_{d|N} \varphi((d, N/d))$  many equivalence classes(cusps). □

**Prop. (26.3.3.13)[Regular Cusps].** All the cusps of  $\Gamma_0(N)$  and  $\Gamma(N)$  are regular. The only irregular cusp of  $\Gamma_1(N)$  are  $s = 1/2$  for  $N = 4$ . ┘

*Proof:* Cf.[?D-S16]P103. □

**Thm. (26.3.3.14)[Lists of Statistics for  $\Gamma_0(N), \Gamma_1(N), \Gamma(N)$ ].** For  $N \in \mathbb{Z}_+$ ,

| $\Gamma$                     | $d$                        | $\varepsilon_2$                                                                    | $\varepsilon_3$                                                                    | $\varepsilon_\infty$                            |
|------------------------------|----------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------|
| $\Gamma_0(N), N > 2$         | $\frac{2d_N}{N\varphi(N)}$ | $\begin{cases} \prod_{p N}(1 + (\frac{-1}{p})) & 4 \nmid N \\ 0 & 4 N \end{cases}$ | $\begin{cases} \prod_{p N}(1 + (\frac{-3}{p})) & 9 \nmid N \\ 0 & 9 N \end{cases}$ | $\sum_{d N} \varphi((d, N/d))$                  |
| $\Gamma_1(2)(= \Gamma_0(2))$ | 3                          | 1                                                                                  | 0                                                                                  | 2                                               |
| $\Gamma_1(3)$                | 4                          | 0                                                                                  | 1                                                                                  | 2                                               |
| $\Gamma_1(4)$                | 6                          | 0                                                                                  | 0                                                                                  | 3                                               |
| $\Gamma_1(N), N > 4$         | $d_N/N$                    | 0                                                                                  | 0                                                                                  | $\frac{1}{2} \sum_{d N} \varphi(N)\varphi(N/d)$ |
| $\Gamma(1)(= \Gamma_1(1))$   | 1                          | 1                                                                                  | 1                                                                                  | 1                                               |
| $\Gamma(N), N > 1$           | $d_N$                      | 0                                                                                  | 0                                                                                  | $d_N/N$                                         |

where

$$d_N = \begin{cases} 1/2 \prod_{p|N} N^3(1 - 1/p^2) & N > 2 \\ 6 & N = 2 \end{cases}.$$

┘

*Proof:* Cf.[Diamond P107].

The numbers  $\varepsilon_2(\Gamma_0(N))$  and  $\varepsilon_3(\Gamma_0(N))$  follow from (26.3.3.9) and the informations on  $\mathbb{Z}[i]$  and  $\mathbb{Z}[\omega]$  (19.3.1.49).  $\square$

**Example (26.3.3.15) [Statistics for  $X_0(p)$ ].** For  $p \in \text{Prime}$ ,

- $\Gamma(1) = \coprod_j \Gamma_0(p)\alpha_j$ , where  $\alpha_j = \begin{bmatrix} 1 & \\ j & 1 \end{bmatrix}$  when  $j \in \{0, \dots, p-1\}$  and  $\alpha_\infty = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$ .
- $\varepsilon_\infty = 2$ .
- $\varepsilon_2(\Gamma_0(p)) = 1 + \left(\frac{-1}{p}\right)$ .
- $\varepsilon_3(\Gamma_0(p)) = 1 + \left(\frac{-3}{p}\right)$ .
- $d = p + 1$ .
- $g = \begin{cases} \lfloor \frac{p+1}{12} \rfloor - 1 & , p \equiv 1 \pmod{12} \\ \lfloor \frac{p+1}{12} \rfloor & , \text{otherwise} \end{cases}$ .

┘

*Proof:* Only the first one need a proof: ?

$\square$

**Cor. (26.3.3.16) [List of Dimension Formulae for  $\Gamma_0(N), \Gamma_1(N), \Gamma(N)$ ].** For  $N \in \mathbb{Z}_+$ ,

| $\Gamma$                   | $g(X(\Gamma))$                                                        | $\dim(M_k(\Gamma)) \& \dim(S_k(\Gamma)),$<br>$2 k, k \geq 2$             | $\dim(M_k(\Gamma)) \& \dim(S_k(\Gamma)),$<br>$2 k+1, k \geq 3$           |
|----------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| $\Gamma_0(N), N > 2$       | long expression, cf. (26.3.3.2)                                       | expression too long                                                      | expression too long                                                      |
| $\Gamma_1(2)(\Gamma_0(2))$ | 0                                                                     | $\lfloor \frac{k}{4} \rfloor \pm 1$                                      | 0                                                                        |
| $\Gamma_1(3)$              | 0                                                                     | $\lfloor \frac{k}{3} \rfloor \pm 1$                                      | $\lfloor \frac{k}{3} \rfloor \pm 1$                                      |
| $\Gamma_1(4)$              | 0                                                                     | $\frac{k-1 \pm 3}{2}$                                                    | $\frac{k-1 \pm 2}{2}$                                                    |
| $\Gamma_1(N), N > 4$       | $1 + \frac{d_N}{12N} - \frac{1}{4} \sum_{d N} \varphi(d)\varphi(N/d)$ | $\frac{(k-1)d_N}{12N} \pm \frac{1}{4} \sum_{d N} \varphi(d)\varphi(N/d)$ | $\frac{(k-1)d_N}{12N} \pm \frac{1}{4} \sum_{d N} \varphi(d)\varphi(N/d)$ |
| $\Gamma(1)(\Gamma_1(1))$   | 0                                                                     | 1                                                                        | 1                                                                        |
| $\Gamma(2)$                | 0                                                                     | $\frac{k-1 \pm 3}{2}$                                                    | 0                                                                        |
| $\Gamma(N), N > 2$         | $1 + \frac{d_N(N-6)}{12N}$                                            | 0                                                                        | $\frac{(k-1)d_N}{12N} \pm \frac{d_N}{2N}$                                |

┘

*Proof:* These follow from (26.3.3.2)(26.3.3.3)(26.3.3.6)(26.3.3.14) and (26.3.3.13). □

**Remark (26.3.3.17) [Dimension for  $\text{MF}_1$ ].** The only case that is not calculated is the dimensions of  $\dim(M_1(\Gamma))$ . But we may use (26.3.3.4) to get that

$$\dim \text{MF}_1(\Gamma_1(N)) \leq \frac{d_N}{12N} + 1.$$

┘

**Cor. (26.3.3.18) [Low Genus Cases].** For  $N \in \mathbb{Z}_+$ ,

- $g(X(N)) = 0$  iff  $N \in \{1, 2, 3, 4, 5\}$ .
- $g(X(N)) = 1$  iff  $N = 6$ .
- $g(X_1(N)) = 0$  iff  $N \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$ .
- $g(X_1(N)) = 1$  iff  $N \in \{11, 14, 15\}$ .
- $g(X_0(N)) = 0$  iff  $N \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 16, 18, 25\}$ .
- $g(X_0(N)) = 1$  iff  $N \in \{11, 14, 15, 17, 19, 20, 21, 24, 27, 32, 36, 49\}$ .

┘

*Proof:* If  $N = \prod_p p^{e_p}$ ,

1, 2:

3, 4:

$$g(X_1(N)) = 1 + \frac{1}{24} \prod_{p|N} p^{2e_p} \left(1 - \frac{1}{p^2}\right) - \frac{1}{4} \prod_p p^{e_p-2} (p-1)[(p+1) - e_p(p-1)],$$

so if  $g(X_1(N)) = 0$ ,

$$\prod_p \frac{p^{e_p}(p+1)}{[(p+1) + e_p(p-1)]} < 6.$$

Then it can be verified that  $e_2 \leq 3, e_3 \leq 2, e_5 \leq 1, e_7 \leq 1, e_{11} \leq 1, e_p = 0$  for  $p > 11 \in \mathbf{P}$ . Then the assertion follows. 4 is similar.

5, 6: ?

□

### Low Dimensional Tabulatetes

**Prop. (26.3.3.19).** Let  $N, k \in \mathbb{Z}_+$  s.t.  $2k(N+1) = 24$ , then

$$\text{SF}_{2k}(\Gamma_0(N)) = \text{SF}_{2k}(\Gamma_1(N)) = \mathbb{C}.g(\tau).$$

where  $g(\tau) = (\eta(\tau)\eta(N\tau))^k$ . ┘

*Proof:* Cf. [Koblitz]P130 or [Diamond]P75.

By ?,  $g(\tau)^{N+1} = \Delta(\tau)\Delta(N\tau) \in \text{SF}_{24}(\Gamma_0(N))$ , and it is non-vanishing on  $\mathcal{H}$  as  $\Delta$  does (26.3.4.23). So  $(f(\tau)/g(\tau))^{N+1} \in \text{MF}_0^!(\Gamma_0(N))$ .

But  $\text{ord}_{\tau=\infty} g(\tau)^{N+1} = N+1$ , and

$$g^{N+1}[S]_{24}(\tau) = N^{-12} q_N^{N+1} \prod [(1 - q_N^n)^{24} (1 - q_N^n)^{24}],$$

so  $\text{ord}_{\tau=0} g(\tau)^{N+1} = N+1$ . So  $(f(\tau)/g(\tau))^{N+1} \in \text{MF}_0(\Gamma_0(N)) = \mathbb{C}$  (26.3.3.4). □

#### 4 Elliptic Eisenstein Series

**Def. (26.3.4.1) [Eisenstein Series].** Let  $\Gamma$  be a congruence subgroup of  $\Gamma(1)$ , define the space of elliptic **Eisenstein series**  $\text{Eis}_k(\Gamma, \chi)$  to be the orthogonal complement of  $\text{SF}_k(\Gamma, \chi)$  in  $\text{MF}_k(\Gamma, \chi)$ . Also denote  $\text{Eis}_k(\Gamma) = \text{Eis}_k(\Gamma, 1)$ .  $\lrcorner$

**Prop. (26.3.4.2) [Dimensions of Eisenstein Series].** Notation as in (26.3.3.1), by (26.3.3.3) and (26.3.3.6), the dimensions of the space of Eisenstein series satisfy:

$$\dim(E_k(\Gamma)) = \begin{cases} \varepsilon_\infty & k \geq 4, 2|k \\ \varepsilon_\infty - 1 & k = 2 \\ 1 & k = 0 \\ \varepsilon_\infty^{\text{reg}} & k \geq 3, 2|k+1, -1 \notin \Gamma \\ \varepsilon_\infty^{\text{reg}}/2 & k = 1, -1 \notin \Gamma \\ 0 & k < 0 \text{ or } 2|k+1, -1 \in \Gamma \end{cases}$$

And also the dimension of  $E_k(\Gamma_0(N)), E_k(\Gamma_1(N)), E_k(\Gamma(N))$  can be read off from (26.3.3.16).  $\lrcorner$

#### Eisenstein Series for $\Gamma(1)$

**Def. (26.3.4.3).** In this subsection, denote  $q = e^{2\pi i \tau}$ .  $\lrcorner$

**Prop. (26.3.4.4) [Weakly Modular Forms and Lattices].** Let  $\mathcal{L}$  be the set of lattices in  $\mathbb{C}$ . If  $F : \mathcal{L} \rightarrow \mathbb{C}$  is a function of weight  $2k$ , i.e.  $F(\lambda\Gamma) = \lambda^{-2k}F(\Gamma)$  for  $\lambda \in \mathbb{C}^*$ , then  $f(z) = F(\Gamma(z, 1))$  is a weakly modular form on  $\mathcal{H}$  for  $\Gamma(1)$  of weight  $2k$ , and this is a bijection between functions of weight  $2k$  on  $\mathcal{L}$  and weakly modular functions on  $\mathcal{H}$  for  $\Gamma(1)$  of weight  $2k$ .  $\lrcorner$

*Proof:* By the hypothesis, there is a function  $f$  on  $\mathcal{H}$  that for any  $w_1, w_2$  with  $w_1/w_2 \in \mathcal{H}$ ,

$$F(\Gamma(w_1, w_2)) = w_2^{-2k} f(w_1/w_2).$$

Then the invariance of  $F$  under  $\text{SL}(2, \mathbb{Z})$  action implies  $f$  is weakly modular of weight  $2k$ .  $\square$

**Prop. (26.3.4.5) [Eisenstein Series for  $\text{SL}(2, \mathbb{Z})$ ].** Let  $k \in 2\mathbb{Z}_{\geq 2}$  and  $\Gamma$  a lattice of  $\mathbb{C}$ , define the **Eisenstein series of weight  $k$**  to be

$$G_k(\Gamma) = \sum_{\omega \in \Lambda, \omega \neq 0} \frac{1}{\omega^k},$$

and also for a complex number  $z$ , let  $\Lambda_z$  be the lattice generated by 1 and  $z$ , and let  $G_k(z) = G_k(\Gamma_z)$ . Then  $G_k(z) \in \text{MF}_k(\Gamma(1))$ , and

$$G_k(z) = 2\zeta(k) + 2 \frac{(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n$$

And denote

$$E_k(z) = G_k(z)/2\zeta(k) = \sum_{(m,n) \in \mathbb{Z}^2 \setminus (0,0), \gcd(m,n)=1} \frac{1}{(m\tau + n)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n$$

the **normalized Eisenstein series**.  $\lrcorner$

*Proof:*  $G_k(z)$  is weakly modular of weight  $k$  by (26.3.4.4). For the expansion, notice by (12.4.3.11),

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n=1}^{\infty} \left( \frac{1}{z+n} + \frac{1}{z-n} \right),$$

and by definition

$$z \cot(\pi z) = \pi i - \frac{\pi i}{1-q} = \pi i - 2\pi i \sum_{n=1}^{\infty} q^n$$

where  $z \in \mathcal{H}$ .

Taking  $(k-1)$ -th derivative of this, we get

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} n^{k-1} q^n,$$

thus

$$\begin{aligned} G_k(z) &= \sum_{(m,n) \neq (0,0)} \frac{1}{(n\tau + m)^k} \\ &= 2\zeta(k) + 2 \sum_{n=1}^{\infty} \sum_{m \in \mathbb{Z}} \frac{1}{(n\tau + m)^k} \\ &= 2\zeta(k) + \frac{2(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sum_{a=1}^{\infty} a^{k-1} q^{an} \\ &= 2\zeta(k) + 2 \frac{(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \end{aligned}$$

Finally the assertion about  $E_k(z)$  follows from (31.3.5.3).

To show the Eisenstein series is orthogonal to any cusp form, notice for any cusp form  $f \in \mathcal{S}_k(\Gamma(1))$ ,

$$f(\gamma\tau)(\operatorname{Im}(\gamma\tau))^k = f(z)\overline{\tau(\gamma, z)}^{-k} \operatorname{Im}(z)^k$$

and by (26.3.4.5)

$$G_k(\tau) = \sum_{(m,n) \neq 0 \in \mathbb{Z}^2} \frac{1}{(m\tau + n)^k} = \sum_{\Gamma(1)_{\infty} \setminus \Gamma(1)} \frac{1}{\tau(\gamma, \tau)^k},$$

so

$$\begin{aligned} \int_{\Gamma(1) \setminus \mathcal{H}} f(\tau) \overline{G_k(z)} y^k \frac{dx dy}{y^2} &= \int_{\Gamma(1) \setminus \mathcal{H}} \sum_{\Gamma(1)_{\infty} \setminus \Gamma(1)} f(z) \overline{\tau(\gamma, z)}^{-k} y^k \frac{dx dy}{y^2} \\ &= \int_{\Gamma(1) \setminus \mathcal{H}} \sum_{\Gamma(1)_{\infty} \setminus \Gamma(1)} f(\gamma z) (\operatorname{Im}(\gamma z))^k \frac{dx dy}{y^2} \\ &= \int_{\Gamma(1)_{\infty} \setminus \mathcal{H}} f(z) (\operatorname{Im} z)^k \frac{dx dy}{y^2} \\ &= \int_0^{\infty} \left[ \int_0^1 f(x + iy) dx \right] y^{k-2} dy = 0 \end{aligned}$$

as  $f$  is a cusp form. □

**Cor. (26.3.4.6).** By (26.3.4.5) and (10.7.1.12):

$$\begin{aligned} E_4(z) &= 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n)q^n, & E_6(z) &= 1 - 504 \sum_{n=1}^{\infty} \sigma_5(n)q^n \\ E_8(z) &= 1 + 480 \sum_{n=1}^{\infty} \sigma_7(n)q^n, & E_{10}(z) &= 1 - 264 \sum_{n=1}^{\infty} \sigma_9(n)q^n \\ E_{12}(z) &= 1 + \frac{65520}{691} \sum_{n=1}^{\infty} \sigma_{11}(n)q^n, & E_{14}(z) &= 1 - 24 \sum_{n=1}^{\infty} \sigma_{13}(n)q^n. \end{aligned}$$

┘

**Cor. (26.3.4.7) [Ramanujan Identities].** By (26.3.4.8),

$$\dim \text{MF}_6(\Gamma(1)) = \dim \text{MF}_8(\Gamma(1)) = \dim \text{MF}_{10}(\Gamma(1)) = \dim \text{MF}_{14}(\Gamma(1)) = 1,$$

so there are identities:

$$E_4^2 = E_8, \quad E_4 E_6 = E_{10}, \quad E_6 E_8 = E_4 E_{10} = E_{14}$$

which give equations like:

$$\begin{aligned} \sigma_7(n) &= \sigma_3(n) + 120 \sum_{m=1}^{n-1} \sigma_3(m) \sigma_3(n-m) \\ 11\sigma_9(n) &= 21\sigma_5(n) - 10\sigma_3(n) + 1054 \sum_{m=1}^{n-1} \sigma_3(m) \sigma_5(n-m). \end{aligned}$$

┘

**Cor. (26.3.4.8) [Modular Forms for  $\Gamma(1)$ ].**

$$\text{MF}_*(\Gamma(1)) = \mathbb{C}[E_4, E_6], \quad \text{SF}_*(\Gamma(1)) = \Delta \cdot \text{MF}_*(\Gamma(1)).$$

Thus for  $k \geq 4$  even,

$$\dim(S_k(\Gamma(1))) = \begin{cases} \lfloor \frac{k}{12} \rfloor - 1 & k \equiv 2 \pmod{12} \\ \lfloor \frac{k}{12} \rfloor & \text{otherwise} \end{cases}.$$

$$\text{MF}_k(\Gamma(1)) = \text{MF}_k(\Gamma(1)) \oplus \mathbb{C}E_k.$$

┘

*Proof:* Cf. [Diamond P88] or [Koblitz, Modular Forms]P118. □

**Cor. (26.3.4.9).**  $\text{MF}(\Gamma(1)) = \text{MF}(\Gamma(1), \mathbb{Z})_{\mathbb{C}}$ ,  $\text{SF}(\Gamma(1)) = \text{SF}(\Gamma(1), \mathbb{Z})_{\mathbb{C}}$  (26.3.5.4). ┘

*Proof:* By (26.3.4.20),  $E_4, E_6$  have Fourier coefficients in  $\mathbb{Z}$ . □

### Eisenstein Series for $\Gamma_0(N)$

Cf. [Diamond] or [Koblitz]P131 or [Hida, L-Functions and Eisenstein Series]P127.

**Prop. (26.3.4.10).** For  $k, N \in \mathbb{Z}_+, k \geq 3$  and  $\chi \in \text{Diri}(N)$  s.t.  $\chi(-1) = (-1)^k$ , there is an Eisenstein series

$$G_{k,\chi}(\tau) = \frac{1}{2}L(1-k, \chi) + \sum_{n \in \mathbb{Z}_+} \left( \sum_{d|n} \chi(d) d^{k-1} \right) e^{2\pi i \tau} \in \text{MF}_k(\Gamma_0(N), \chi).$$

┘

*Proof:*

□

**Prop. (26.3.4.11) [Connection with Dedekind eta-Functions].** For  $p \in \text{Prime}_{\geq 3}$ , let

$$h(\tau) = \prod_{a_2=1}^{p-1} G_3^{(0,a_2)((\bmod p))}(\tau), \quad f(\tau) = \prod_{a_2=1}^{\frac{p-1}{2}} G_3^{(0,a_2)((\bmod p))}(\tau),$$

then

- $h(\tau) = (-1)^{(p-1)/2} f(\tau)$ ,
- $f(\tau) \in \text{MF}_{3(p-1)/2}(\Gamma_0(p), \left(\frac{-}{p}\right))$ ,  $h(\tau) \in \text{MF}_{3(p-1)}(\Gamma_0(p))$ .
- $f(\tau)$  is a constant multiple of  $(\eta^p(\tau)/\eta(p\tau))^3$ .

┘

*Proof:* Cf. [Koblitz]P135, 136. ?

□

### General Eisenstein Series

#### Quasimodular Forms

**Def. (26.3.4.12) [Almost Modular Forms].** For a discrete subgroup  $\Gamma \subset \text{SL}(2; \mathbb{R})$ , let  $\widehat{M}_k^{\leq p}(\Gamma)$  be the space of functions in  $C^\infty(\mathcal{H})$  of the form  $F(\tau) = \sum_{r=0}^p f_r(\tau)(-4\pi y)^{-r}$ , where  $f_r$  are holomorphic on  $\mathcal{H}$  and holomorphic at the cusps, and  $F[\gamma]_k = F$  for any  $\gamma \in \Gamma$ .

Denote  $\widehat{\text{MF}}_k(\Gamma) = \cup_{p=0}^\infty \widehat{\text{MF}}_k^{\leq p}(\Gamma)$  and the graded ring  $\widehat{\text{MF}}_*(\Gamma) = \oplus_{k \geq 0} \widehat{\text{MF}}_k(\Gamma)$ , called the space of **almost modular forms**.

┘

**Def. (26.3.4.13) [Quasi-Modular Forms].** The graded ring  $\widetilde{\text{MF}}_*(\Gamma)$  of **quasi-modular forms** is defined to be the constant terms( $f_0$ ) of functions in  $\widehat{\text{MF}}_*(\Gamma)$ . Any almost modular function is determined by its constant term, so  $\widetilde{\text{MF}}_*(\Gamma)$  is canonically isomorphic to  $\widehat{\text{MF}}_*(\Gamma)$ . ?

┘

*Proof:*

□

### Eisenstein Series of Weight 2

**Def. (26.3.4.14) [ $E_2(\tau)$ ].** Denote

$$G_2(\tau) = \frac{1}{2} \sum_{n \neq 0} \frac{1}{n^2} + \frac{1}{2} \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(m\tau + n)^2}.$$



This summation is convergent but not absolutely convergent, and

$$E_2(\tau) = \frac{6}{\pi^2} G_2(\tau) = 1 - 24q - 72q^2 + \dots \in \widetilde{M}_2^{\leq 1}(\Gamma(1))$$

with

$$G_2^*(\tau) = G_2(\tau) - \frac{\pi}{2\operatorname{Im}(\tau)}, \quad E_2^*(\tau) = \frac{6}{\pi^2} G_2^*(\tau) = E_2(\tau) - \frac{3}{\pi\operatorname{Im}(\tau)} \in \widehat{M}_2^{\leq 1}(\Gamma(1)).$$

In particular,  $E_2(\tau)$  satisfies

$$E_2[\gamma]_2(\tau) = E_2(\tau) + \frac{12}{2\pi i} \frac{c}{c\tau + d}, \quad \gamma \in \Gamma(1)$$

┘

*Proof:* For  $\varepsilon \in \mathbb{R}_+$ , define

$$G_{2,\varepsilon}(\tau) = \sum_{(m,n) \in \mathbb{Z}^2 \setminus (0,0)} \frac{1}{(m\tau + n)^2 |m\tau + n|^\varepsilon},$$

then it is absolutely convergent and

$$G_{2,\varepsilon}(\gamma(\tau)) = (cz + d)^2 |cz + d|^{2\varepsilon} G_{2,\varepsilon}(\tau).$$

Then it suffices to show that  $\lim_{\varepsilon \rightarrow 0} G_{2,\varepsilon}(\tau) = G_2^*(\tau)$ .

Define

$$I_\varepsilon(\tau) = \int_{-\infty}^{\infty} \frac{dt}{(z + t)^2 |z + t|^{2\varepsilon}},$$

then

$$G_{2,\varepsilon}(\tau) - \sum_{m \in \mathbb{Z}_+} I_\varepsilon(m\tau) = \sum_{n \in \mathbb{Z}_+} \frac{1}{n^{2+2\varepsilon}} + \sum_{m \in \mathbb{Z}_+} \left( \sum_{n \in \mathbb{Z}} \frac{1}{(m\tau + n)^2 |m\tau + n|^{2\varepsilon}} - \int_n^{n+1} \frac{dt}{(m\tau + t)^2 |m\tau + t|^{2\varepsilon}} \right).$$

This sum is absolutely convergent for  $\varepsilon > -1/2$ , thus when  $\varepsilon \rightarrow 0$ , it just equals  $G_2(\tau)$ .

On the other hand, for  $\varepsilon > -1/2$ ,

$$I_\varepsilon(x + iy) = \int_{-\infty}^{\infty} \frac{dt}{(x + t + iy)^2 ((x + t)^2 + y^2)^\varepsilon} = \frac{I(\varepsilon)}{y^{1+2\varepsilon}},$$

where  $I(\varepsilon) = \int_{-\infty}^{\infty} (t + i)^{-2} (t^2 + 1)^{-\varepsilon} dt$ . We have  $I(0) = 0$ , and

$$I'(0) = \int_{-\infty}^{\infty} \frac{\log(t^2 + 1)}{(t + i)^2} dt = \left( \frac{1 + \log(t^2 + 1)}{t + i} - \arctan t \right) \Big|_{-\infty}^{\infty} = -\pi.$$

For  $\varepsilon \in \mathbb{R}_+$ ,  $\sum_{m \in \mathbb{Z}_+} I_\varepsilon(m\tau) = I(\varepsilon) \zeta(1 + 2\varepsilon) / y^{1+2\varepsilon}$ , so  $\lim_{\varepsilon \rightarrow 0} \sum_{m \in \mathbb{Z}_+} I_\varepsilon(m\tau) = -\pi/2y$ . □

### Eisenstein Series of Weight 1

**Prop. (26.3.4.15).** For  $N \in \mathbb{Z}_+$ ,  $k = 1$ , we can modify to give a definition of

$$G_{1,\chi}(\tau) \in \text{MF}_1(\Gamma_0(N), \chi).$$

┘

**Prop. (26.3.4.16) [Eisenstein Series of Weight 1].** For  $N \in \mathbb{Z}_+$ , the set

$$\{E^{\psi,\varphi,1}\}$$

is a basis for  $\text{Eis}_1(\Gamma_1(N))$ . In fact, for any  $\chi \in \text{Diri}(N)$ , the set

$$\{E^{\psi,\varphi,1}, \varphi\psi = \chi\}$$

is a basis for  $\text{Eis}_1(\Gamma(N), \chi)$  (26.3.4.1).

┘

*Proof:* Cf. [Diamond]P141. □

### Derivatives

**Prop. (26.3.4.17) [Derivations of Modular Forms].** Let  $\Gamma$  be a discrete subgroup of  $\text{SL}(2; \mathbb{R})$  and  $f \in \text{MF}_k(\Gamma)$ , then

$$Df \triangleq \frac{1}{2\pi i} \frac{\partial}{\partial \tau} f = q \frac{\partial}{\partial q} f = \sum_{n=1}^{\infty} n a_n q^n$$

satisfies

$$(Df)[\gamma]_{k+2}(\tau) = (Df)(\tau) + \frac{k}{2\pi i} \frac{c}{cz+d} f(z), \quad \gamma \in \Gamma.$$

In particular, by (26.3.4.14),  $\theta_k(f) = D(f) - \frac{k}{12} E_2 f \in \text{MF}_{k+2}(\Gamma)$ , called the **Serre derivation** of  $f$ .

┘

*Proof:* This follows from differentiating the equation  $f(\frac{az+b}{cz+d}) = (cz+d)^k f(z)$ . □

**Cor. (26.3.4.18).**

$$D(E_2) = \frac{E_2^2 - E_4}{12}, \quad D(E_4) = \frac{E_2 E_4 - E_6}{3}, \quad D(E_6) = \frac{E_2 E_6 - E_4}{2}$$

┘

*Proof:*  $D(E_4)$  and  $D(E_6)$  follows from (26.3.4.17) by comparing coefficients, for  $D(E_2)$ , differentiating the equation  $E_2(\frac{a\tau+b}{c\tau+d}) = (c\tau+d)^2 E_2(\tau) + \frac{12}{2\pi i} c(c\tau+d)$ , we get

$$D(E_2)[\gamma]_4(z) = D(E_2)(z) + \frac{2c}{2\pi i} \frac{E_2(\tau)}{c\tau+d} + \frac{12c^2}{(2\pi i)^2} \frac{1}{(c\tau+d)^2}.$$

Thus by (26.3.4.14), we see  $D(E_2) - \frac{1}{12} E_2^2 \in \text{MF}_4(\Gamma(1))$ , then we can compare coefficients. □

**Prop. (26.3.4.19) [Structure of  $\widetilde{M}_*(\Gamma)$ ].** Let  $\Gamma$  be a discrete subgroup of  $\text{SL}(2; \mathbb{R})$ , then

- $D(\widetilde{M}_k^{\leq p}(\Gamma)) \subset \widetilde{M}_{k+2}^{\leq p+1}(\Gamma)$ . In particular,  $D(\widetilde{M}_*(\Gamma)) \subset \widetilde{M}_*(\Gamma)$ .

- If  $\varphi \in \widetilde{M}_2^{\leq 1}(\Gamma)$ , then for any  $k, p \in \mathbb{N}$ ,  $\widetilde{M}_k^{\leq p}(\Gamma) = \bigoplus_{r=0}^p M_{k-2r}(\Gamma)\varphi^r$ . In particular,  $\widetilde{M}_*(\Gamma(1)) = \mathbb{C}[E_2, E_4, E_6]$ .
- If  $\varphi \in \widetilde{M}_2^{\leq 1}(\Gamma)$ , then for  $p, k \geq 0$ ,

$$\widetilde{M}_k^{\leq p}(\Gamma) = \begin{cases} \bigoplus_{r=0}^p D^r(M_{k-2r}(\Gamma)), & p < k/2 \\ \bigoplus_{r=0}^{k/2-1} D^r(M_{k-2r}(\Gamma)) \oplus \mathbb{C} \cdot D^{k/2-1}\varphi, & p \geq k/2 \end{cases}$$

┘

*Proof:* Cf. [BGHZ]P59.

□

### $\Delta$ -Function and $j$ -Function

**Def. (26.3.4.20) [Discriminant Function].** The modular **discriminant function**

$$\Delta(q(\tau)) = \frac{(2\pi)^{12}}{1728} (E_4^3 - E_6^2) = \sum_{n \in \mathbb{Z}_+} \tau(n) q^n = (2\pi)^{12} (q - 24q^2 + 252q^3 - 1472q^4 + 4830q^5 - 6048q^6 - 16744q^7 + \dots)$$

by (26.3.4.6). So  $\Delta(z) \in \text{SF}_{12}(\Gamma(1))$ , and the coefficients  $\tau(n)$  are called the **Ramanujan  $\tau$ -function**.

┘

**Cor. (26.3.4.21).**

$$E_6^2 - E_{12} = -(1008 + \frac{24 \cdot 2730}{691}) \frac{\Delta(z)}{(2\pi)^{12}}.$$

In particular,  $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ .

┘

*Proof:* Consider  $E_6^2 - E_{12} \in \text{SF}_{12}(\Gamma(1))$ , and  $\text{SF}_{12}(\Gamma(1))$  is 1-dimensional and generated by  $\Delta$  (26.3.4.8), thus  $E_6^2 - E_{12} = c\Delta$  for some constant  $c \in \mathbb{C}$ . By comparing the degree 1 term of both sides:

$$\begin{aligned} \frac{\Delta(z)}{(2\pi)^{12}} (q(\tau)) &= \frac{1}{1728} (E_4^3(q) - E_6^2(q)) = q - 24q^2 + \dots \\ E_{12}(\tau) &= 1 + \frac{24 \cdot 2730}{691} \sum_{n=1}^{\infty} \sigma_{11}(n) q^n, \quad E_6(z) = 1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n \end{aligned}$$

by (26.3.4.6) and (26.3.4.20), so  $c = -1008 - \frac{24 \cdot 2730}{691}$ . Then the formula

$$691(E_6^2 - E_{12}) = 691c\Delta,$$

modulo 691 gives  $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ .

□

**Cor. (26.3.4.22) [ $j$ -Function].** By (26.3.1.5), the discriminant function  $\Delta(z)$  has exactly one simple zero at  $\infty$ , thus we can define the  **$j$ -function** on  $\mathcal{H}$  as

$$j : \mathcal{H} \rightarrow \mathbb{C}, j(q(\tau)) = 1728 \frac{E_4(\tau)^3}{E_4^3(\tau) - E_6^2(\tau)} = \frac{1}{q} + 744 + 196884q + 21493760q^2 + 864299970q^3 + \dots \in \frac{1}{q} + \mathbb{Z}[[q]]$$

which is an automorphic function for  $\Gamma(1)$ , and it induces an isomorphism

$$j : X(1) \cong \mathbb{P}_{\mathbb{C}}^1.$$

In particular,  $\text{MF}_0^!(\Gamma(1)) = \mathbb{C}(j(z))$ .

┘

*Proof:* To show it surjects onto  $\mathbb{C}$ , notice it induces a holomorphic map  $X(1) \cong \mathbb{P}^1 \rightarrow \mathbb{P}^1$ , and it maps  $\infty$  to  $\infty$  with no ramification, thus it has degree 1. So it is surjective.  $\square$

**Prop. (26.3.4.23).**

$$\Delta(q(\tau)) = (2\pi)^{12} \eta(q(\tau))^{24} = q \prod_{n=1}^{\infty} (1 - q^n)^{24}.$$

In particular,  $\Delta(\tau)$  is non-vanishing on  $\tau \in \mathcal{H}$ .  $\lrcorner$

*Proof:* By (26.3.1.16),  $\eta^{24}(\tau) \in \text{SF}_{12}(\Gamma(1))$ , which has dimension 1 and is spanned by  $\Delta(\tau)$  (26.3.4.8). Thus the assertion follows by comparing coefficients.  $\square$

### Zeros of Eisenstein Series

**Thm. (26.3.4.24) [Rankin-Swinnerton-Dyer].** If  $n \in \mathbb{N}$ ,  $s \in \{2, 3, 4, 5, 0, 7\}$  and  $k = 6n + s \geq 2$ , then the Eisenstein series  $E_{2k} \in M_{2k}(\Gamma(1))$  has at least weighted zero exactly  $s/12$  at  $i$  and  $\omega$  (which suffices to determine its zeros), and all the other  $n$  zeros are non-elliptic and located on arc between  $i$  and  $\omega$ .  $\lrcorner$

*Proof:* Firstly, by the valence formula (26.3.1.5) and by rationality reason,  $E_{2k}$  has at least weighted zero  $s/6$  at  $i$  and  $\omega$ . So by valence formula, it suffices to show that  $E_{2k}$  has at least  $n$  zeros on the arc between  $i$  and  $\omega$ .

Let

$$\begin{aligned} F_{2k}(\theta) &= \frac{1}{2} e^{ik\theta} E_{2k}(e^{i\theta}) \\ &= \frac{1}{2} \sum_{(c,d) \in \mathbb{Z}^2 \setminus (0,0), \gcd(c,d)=1} \frac{1}{(ce^{i\theta/2} + de^{-i\theta/2})^{2k}} \\ &= 2 \cos(k\theta) + \frac{1}{2} \sum_{(c,d) \in \mathbb{Z}^2 \setminus (0,0), c^2+d^2>1, (c,d)=1} \frac{1}{(ce^{i\theta/2} + de^{-i\theta/2})^{2k}}. \end{aligned}$$

Then it suffices to show that  $F_{2k}(\theta)$  has at least  $n$  zeros when  $\theta \in (\pi/2, 2\pi/3)$ . For this, we may assume  $n \geq 1$ . Notice for these theta,

$$|ce^{i\theta/2} + de^{-i\theta/2}|^2 = c^2 + 2cd \cos \theta + d^2 \geq \frac{1}{2}(c^2 + d^2),$$

and for  $N \in \mathbb{Z}_{\geq 5}$ , the number of integral solutions to  $c^2 + d^2 = N$  is at most  $2(2\sqrt{N} + 1) \leq 5\sqrt{N}$ .

$$\begin{aligned} \left| \frac{1}{2} \sum_{c^2+d^2>1, (c,d)=1} \frac{1}{(ce^{i\theta/2} + de^{-i\theta/2})^{2k}} \right| &\leq 1 + 3^{-k} + 4\left(\frac{5}{2}\right)^{-k} + \sum_{N=10}^{\infty} 5\sqrt{N} \left(\frac{N}{2}\right)^{-k} \\ &\leq 1 + 3^{-k} + 4\left(\frac{2}{5}\right)^k + \frac{5 \cdot 9^{\frac{3}{2}}}{k - \frac{3}{2}} \left(\frac{2}{9}\right)^k \\ &< 1.0214 \end{aligned}$$

Thus, when  $m \in \mathbb{Z}$  and  $k/2 \leq m \leq 2/3k$ ,  $F_{2k}(m\pi/k)$  is strictly positive or negative according as  $m$  is even or odd. So by intermediate property (3.2.10.9), there are at least  $\lfloor \frac{2k}{3} \rfloor - \lceil \frac{k}{2} \rceil = n$  zeros when  $\theta \in (\pi/2, 2\pi/3)$ .  $\square$

## 5 Arithmetics

**Prop. (26.3.5.1) [Number Field of  $f$ ].** If  $f \in \text{SF}_k(\Gamma_1(N))$  is a normalized eigenform, then the coefficients  $a_n(f)$  are algebraic integers, and generate a number field  $K_f$ , called the **number field of  $f$** .  
 $\perp$

*Proof:* Cf. [Diamond, P238].  $\square$

**Prop. (26.3.5.2).** if  $f \in \text{SF}_2(N, \chi)$  is a normalized eigenform, then for any  $\sigma \in \text{Gal}_{\mathbb{Q}}$ ,  $f^\sigma$  is also a normalized eigenform in  $\text{SF}_2(N, \chi^\sigma)$ , where  $\chi^\sigma(n) = \chi(n)^\sigma$ . And if  $f$  is a newform, then so is  $f^\sigma$ .  $\perp$

*Proof:* Cf. [Diamond, P239].  $\color{red}?$   $\square$

**Cor. (26.3.5.3) [Integrality of Modular Forms].**  $S_2(\Gamma_1(N))$  has a basis of modular forms with integral coefficients.  $\perp$

*Proof:* Let  $f \in S_2(\Gamma_1(N))$  be a newform of level  $M|N$ , and  $K_f$  the number field of  $f$ . Let  $\alpha_1, \dots, \alpha_d$  be a  $\mathbb{Z}$ -basis of  $\mathcal{O}_K$ , and  $\Sigma^\infty = \{\sigma_1, \dots, \sigma_d\}$  the set of embeddings of  $K$  into  $\mathbb{C}$ . Let

$$g_i = \sum_{j=1}^d \sigma_j(\alpha_i) f^{\sigma_j},$$

then by linear independence of characters, the matrix  $(\sigma_i(\alpha_j))$  is non-degenerate, so  $\text{span}\{g_1, \dots, g_d\} = \text{span}\{f, \dots, f^{\sigma_d}\}$ , and  $g_i \in S_2(\Gamma_1(N))$  by (26.3.5.2). Then by (26.3.2.12), applying this for all old and new forms of  $S_2(\Gamma_1(N))$ , the assertion follows.  $\square$

**Prop. (26.3.5.4) [Integral Modular Forms].** If  $A \subset \mathbb{C}$  be a subring, denote  $\text{MF}_k(\Gamma; A) = \text{MF}_k(\Gamma) \cap A[[q]]$ .  $\perp$

## 6 Modular Forms Mod $p$

**Notation (26.3.6.1).**

- Let  $p \in \text{Prime}$ .
- Let  $F \in \text{NField}$  and  $v \in S_F(p)$ .
- Let  $\Gamma \in \text{SL}(2; \mathbb{Z})$  be a congruence subgroup.

$\perp$

**Def. (26.3.6.2) [Modular Forms Mod  $p$ ].** Denote  $\text{MF}_k(\Gamma; \kappa_v) = M_k(\Gamma; \mathcal{O}_F) \otimes \kappa_v$ , called the space of **modular forms mod  $p$  of weight  $k$** .  $\perp$

**Prop. (26.3.6.3) [Surm[Stu87]].** If  $k \in \mathbb{Z}_+$ ,  $f, g \in \text{MF}_k(\Gamma, \mathcal{O}_F)$  satisfies their first  $\lfloor k[\Gamma(1) : \Gamma]/12 + 1 \rfloor$  terms of Fourier coefficients are equal modulo  $\mathfrak{m}_v$ , then  $f \equiv g \pmod{\mathfrak{m}_v}$ .  $\perp$

*Proof:*

$\square$

**Prop. (26.3.6.4) [Serre's Equality].** There is an isomorphism  $\text{MF}_{p+1}(\Gamma(1); \mathbb{F}_p) \cong \text{MF}_2(\Gamma_0(p); \mathbb{F}_p)$ .  $\perp$

*Proof:* Because  $E_k(z)1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n)q^n$ , and by Kummer's congruence (33.1.1.16)  $\text{ord}_p(B_{p-1}) = -1$ , thus  $E_{p-1} \pmod{1} \pmod{p}$ . Then multiplying by  $E_{p-1} : M_2(\Gamma_0(p), \mathbb{F}_p) \rightarrow M_{p+1}(\Gamma_0(p), \mathbb{F}_p)$  raises the level by  $p-1$ . Then we compose with the natural averaging map  $M_{p+1}(\Gamma_0(p), \mathbb{F}_p) \rightarrow M_{p+1}(\text{SL}(2, \mathbb{Z}), \mathbb{F}_p)$ , which is dual to the natural inclusion  $M_{p+1}(\text{SL}(2, \mathbb{Z}), \mathbb{F}_p) \rightarrow M_{p+1}(\Gamma_0(p), \mathbb{F}_p)$ .

Why isomorphism.  $\color{red}?$   $\square$

## 7 Non-Congruent Modular Forms

Cf.[Non-Congruent Modular Forms, Ling Long].

**Thm. (26.3.7.1) [Unbounded Denominators Conjecture].** Let  $N$  be any positive integer and let  $f(\tau) \in \mathbb{Z}[[q^{1/N}]]$  for  $q = e^{\pi i \tau}$  be a holomorphic function on the upper half plane. Suppose there exists an integer  $k$  and a finite index  $\Gamma \subset SL(2, \mathbb{Z})$  that  $f$  is  $\Gamma$ -invariant, and  $f$  is meromorphic at the cusps of  $\Gamma$ , then  $f(\tau)$  is a modular form for some congruent subgroup of  $SL(2, \mathbb{Z})$ .  $\lrcorner$

*Proof:* Cf.[the Unbounded Denominator Conjecture].  $\square$

## 26.4 Modular Forms II

References are [\[Shi78\]](#).

**Notation(26.4.0.1).**

- Use notations as in [Archimedean Automorphic Representations](#).

┘

### 1 Hilbert Modular Forms

**Notation(26.4.1.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(F, \mathcal{O}_F) \in \text{NField}$  be totally real.
- Let  $d = [F : \mathbb{Q}]$ .
- Let  $\Sigma_F^\infty = \{\tau_1, \dots, \tau_d\}$ .
- Let  $\mathfrak{c} \in \text{Gal}_F$  be a complex conjugation.
- Let  $h = \text{cl}^+(F)$ .
- Let  $\text{GL}(2; F)^+$  be the set of matrices in  $\text{GL}(2; F)$  with totally real determinant.
- Let  $\mathfrak{n} \in \text{Ideal}^\times(\mathcal{O}_F)$ .
- Let  $\mathfrak{n}^p \in \text{Ideal}^\times(\mathcal{O}_F)$  be the smallest ideal containing  $\mathfrak{n}$  and prime to  $(p)$ .
- Let  $\{\mathfrak{t}_\kappa\}_{\kappa \in \text{Cl}^+(F)} \subset \text{Ideal}^\times(F)$  be a representative for  $\text{Cl}^+(F)$ . s.t.  $(\|\mathfrak{t}_\kappa\|, \|\mathfrak{n}\|p) = 1$  for each  $\kappa$ .
- Let  $\psi_0$  be a character of  $(\mathcal{O}_F/\mathfrak{n})^*$ .
- Let  $\psi$  be a character of  $\text{Cl}^{\text{n}\infty}(F)$  s.t.  $\psi$  restricts to  $\psi_0$ .

┘

### Archimedean Hilbert Modular Forms

**Def.(26.4.1.2)[Congruence Subgroups].** For  $\mathfrak{c} \in \text{Ideal}^\times(\mathcal{O}_F)$ ,  $\mathfrak{b} \in I(F)$  and  $\psi_0 \in ((\mathcal{O}_F/\mathfrak{c})^*)^\vee$ , denote

$$\Gamma(\mathfrak{b}, \mathfrak{c}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{GL}(2; F)^+, a, d \in \mathcal{O}_F, b \in \mathfrak{b}^{-1}, c \in \mathfrak{c}\mathfrak{b}^{-1}, ad - bc \in \mathcal{O}_F^\times \right\},$$

┘

**Def.(26.4.1.3) [Archimedean Hilbert Modular Forms].** For  $\mathbf{k} = (k_1, \dots, k_d) \in \mathbb{Z}^d$ ,  $\mathfrak{c} \in \text{Ideal}^\times(\mathcal{O}_F)$ ,  $\mathfrak{b} \in I(F)$  and  $\psi_0 \in ((\mathcal{O}_F/\mathfrak{c})^\times)^\vee$ , an **Archimedean Hilbert modular form** of weight  $\mathbf{k}$  w.r.t.  $\Gamma(\mathfrak{b}, \mathfrak{c})$  with nebentypus  $\psi_0$  is a holomorphic function  $f \in \mathcal{O}(\mathcal{H}^d)$  s.t.

- $f[\alpha]_{\mathbf{k}} = \psi_0(a)f$  for each  $\alpha \in \Gamma(\mathfrak{b}, \mathfrak{c})$  [\(26.4.1.2\)](#),
- Because  $f$  is invariant under the translation by elements in  $\mathfrak{b}^{-1}$ , it has a Fourier expansion

$$f(z) = \sum_{\mu \in \mathfrak{b}\mathfrak{d}_F^{-1}} c(\mu, f) e^{2\pi i \sum_{i=1}^n \tau_i(\mu) z_i}.$$

Then we require that  $f$  is holomorphic at cusps, i.e.  $c(\mu, f) = 0$  unless  $\mu$  is totally non-negative.

The space of such forms is denoted by  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0)$ .

And  $f$  is called an **Archimedean Hilbert cusp form** if the constant term of  $f[\alpha]_k$  vanishes for each  $\alpha \in \mathrm{GL}(2; F)$ . The space of Archimedean Hilbert cusp forms is denoted by  $\mathrm{SF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0)$ .  $\lrcorner$

**Prop. (26.4.1.4).** Situation as in (26.4.1.3), we have:

- $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0) = \mathrm{SF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0)$  unless  $k_1 = \dots = k_d$ .
- $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0) = 0$  unless  $k_1 = \dots = k_d \geq 0$  or  $k_i \in \mathbb{Z}_+$  for each  $i$ .

$\lrcorner$

*Proof:* Cf. [Shi78]P640. ?

$\square$

**Def. (26.4.1.5) [Fields of Definition of  $f$ , Klingen(1960)].** Situation as in (26.4.1.3), for  $f \in \mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0)$ , let  $\mathbb{Q}(f) = \mathbb{Q}\{c(\mu, f) | \mu \in \mathfrak{b}\mathfrak{d}_F^{-1}, \mu \gg 0\}$ , then  $\mathbb{Q}(f)$  is f.g. over  $\mathbb{Q}$  and contains  $c(0, f)$ .  $\lrcorner$

*Proof:* Cf. [Shi78]P642. ?

To show  $c(0, f) \in \mathbb{Q}(f)$ , because there are no constant modular forms, the linear functionals  $c(\mu; -)$  for  $\mu \in \mathfrak{b}\mathfrak{d}_F^{-1}, \mu \gg 0$  generates  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}); \psi_0)^\vee$ . Let  $B^\vee$  be a basis of  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}); \psi_0)^\vee$  consisting of linear functionals  $c(\mu; -)$ , then the  $\mathbb{Q}$ -span of the dual basis  $B$  of  $B^\vee$  in  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}); \psi_0)$  contains  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0; \mathbb{Q})$ . And  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0; \mathbb{Q})$  generates  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0)$  by (26.4.1.6), so the  $\mathbb{Q}$ -space of  $B$  is just  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0; \mathbb{Q})$ . But then  $c(0; -)$  also take  $\mathbb{Q}$ -values on  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0; \mathbb{Q})$ , thus it is in the linear span of linear functions of the form  $c(\mu; -)$  for  $\mu \gg 0$ .  $\square$

**Prop. (26.4.1.6) [Rational Hilbert Modular Forms, Shimura[Shi78]].** Situation as in (26.4.1.3), for  $A \subset \mathbb{C} \in \mathcal{C}\mathrm{Ring}$ , denote  $\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0; A)$  the space of Hilbert modular forms with Fourier coefficients in  $A$ , then

$$\mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0; \mathbb{Z}) \otimes \mathbb{C} = \mathrm{MF}_k(\Gamma(\mathfrak{b}, \mathfrak{c}), \psi_0).$$

$\lrcorner$

*Proof:* ? Cf. [Shi78]Prop1.7.

$\square$

### Hilbert Modular Forms

**Def. (26.4.1.7) [Hilbert Modular Forms].** For  $\mathfrak{n} \in \mathrm{Ideal}^\times(\mathcal{O}_F)$ , a **Hilbert modular form** of level  $\mathfrak{n}$  and nebentypus  $\psi_0$  is a tuple  $\mathbf{f} = (f_\kappa)_{\kappa \in \mathrm{Cl}^+(F)}$  where  $f_\kappa \in \mathrm{MF}_k(\Gamma(t_\kappa \mathfrak{d}, \mathfrak{n}), \psi_0)$  (26.4.1.3).

The space of such functions is denoted by  $\mathrm{MF}_k(\mathfrak{n}, \psi_0)$ , i.e.

$$\mathrm{MF}_k(\mathfrak{n}, \psi_0) = \prod_{\kappa \in \mathrm{Cl}^+(F)} \mathrm{MF}_k(\Gamma(t_\kappa \mathfrak{d}, \mathfrak{n}), \psi_0).$$

Also define the space of **Hilbert cusp forms**

$$\mathrm{SF}_k(\mathfrak{n}, \psi_0) = \prod_{\kappa \in \mathrm{Cl}^+(F)} \mathrm{SF}_k(\Gamma(t_\kappa \mathfrak{d}, \mathfrak{n}), \psi_0),$$

and we denote

$$\mathrm{MF}_k(\mathfrak{n}) = \bigoplus_{\psi \in ((\mathcal{O}_F/\mathfrak{n})^*)^\vee} \mathrm{MF}_k(\mathfrak{n}, \psi_0), \quad \mathrm{SF}_k(\mathfrak{n}) = \bigoplus_{\psi \in ((\mathcal{O}_F/\mathfrak{n})^*)^\vee} \mathrm{SF}_k(\mathfrak{n}, \psi_0).$$

$\lrcorner$



**Def. (26.4.1.8) [Dirichlet Series of Hilbert Modular Forms].** Let  $\mathbf{f} = (f_\kappa)_{\kappa \in \text{Cl}^+(F)} \in \text{MF}_k(\mathbf{n})$ . Any  $\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)$  is of the form  $(\mu)\mathfrak{t}_i^{-1}$  for some  $i$  and some totally positive  $\mu$  by (26.4.1.1). For  $\mathfrak{a} \in I(F)$ , define

$$c(\mathfrak{a}, \mathbf{f}) = \begin{cases} c(\mu, f_\kappa) \|\mathfrak{t}_i\|^{-k/2} & , \mathfrak{a} = (\mu)\mathfrak{t}_\kappa^{-1} \in \text{Ideal}^\times(\mathcal{O}_F) \\ 0 & , \text{otherwise} \end{cases}, \quad c_\kappa(0, \mathbf{f}) = c(0, f_\kappa) \|\mathfrak{t}_\kappa\|^{-k/2}.$$

Then the **Dirichlet series** of  $\mathbf{f}$  is defined by

$$L(\mathbf{f}; s) = \sum_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)} \frac{c(\mathfrak{a}, \mathbf{f})}{\|\mathfrak{a}\|^s}.$$

And if  $\mathbf{g}$  is another modular form, let the **Dirichlet series** of  $\mathbf{f} \otimes \mathbf{g}$  is given by

$$L(\mathbf{f} \otimes \mathbf{g}; s) = \sum_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)} \frac{c(\mathfrak{a}, \mathbf{f})c(\mathfrak{a}, \mathbf{g})}{\|\mathfrak{a}\|^s}.$$

┘

### Hecke Operators

**Def. (26.4.1.9) [Hecke Operators].** There are commutative operators  $\{T_n(\mathfrak{a}), S_n(\mathfrak{a})\}_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)}$  on  $\text{MF}_k(\mathbf{n})$  with relations

$$\begin{aligned} T_n(\mathfrak{a})T_n(\mathfrak{b}) &= \sum_{\mathfrak{a}+\mathfrak{b} \subset \mathfrak{c}} \|\mathfrak{c}\|^{k-1} S_n(\mathfrak{c})T_n(\mathfrak{c}^{-2}\mathfrak{a}\mathfrak{b}). \\ \sum_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)} T_n(\mathfrak{a}) \|\mathfrak{a}\|^{-s} &= \prod_{v \in \Sigma_F^{\text{fin}}} \left(1 - T_n(\mathfrak{p}_v) \|v\|^{-s} + S_n(\mathfrak{p}_v) \|v\|^{k+1-2s}\right)^{-1}. \\ c(\mathfrak{a}, \mathbf{f}|T_n(\mathfrak{b})) &= \sum_{\mathfrak{a}+\mathfrak{b} \subset \mathfrak{c}} \|\mathfrak{c}\|^{k-1} c(\mathfrak{c}^{-2}\mathfrak{a}\mathfrak{b}, \mathbf{f}|S_n(\mathfrak{c})). \\ c_\kappa(0, T(\mathfrak{a})\mathcal{F}) &= \sum_{\mathfrak{b}|\mathfrak{a}, [\mathfrak{b}]=[\mathfrak{t}_\kappa], [\mathfrak{a}]=[\mathfrak{t}_\kappa]} \frac{\|\mathfrak{t}_\kappa\|^{k/2}}{\|\mathfrak{t}_\kappa \mathfrak{b}^2 \mathfrak{a}^{-1}\|^{k/2}} \lambda_{\mathfrak{b}} \|\mathfrak{b}\| c_{\kappa \mathfrak{b}^2 \mathfrak{a}^{-1}}(0, \mathcal{F}). \end{aligned}$$

┘

*Proof:* Cf. [Wil86].

□

**Prop. (26.4.1.10) [Central Characters and Eigenvalues].** For  $\mathbf{n} \in \text{Ideal}^\times(\mathcal{O}_F)$ , if  $\mathbf{f} \in \text{MF}_k(\mathbf{n}, \psi_0)$  is a common eigenvalue of  $S(\mathfrak{p})$  for each  $\mathfrak{p} \in \Sigma_F^{\text{fin}} \setminus S_F(\mathbf{n})$ , then there exists an Artin Hecke character  $\psi$  of  $F$  s.t.  $S(\mathfrak{a})\mathbf{f}(z) = \psi(\mathfrak{a})f(z)$  for each  $\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)$ .

Define

$$\text{MF}_k(\mathbf{n}, \psi) = \{\mathbf{f} \in \text{MF}_k(\mathbf{n}, \psi_0) : S(\mathfrak{a})\mathbf{f} = \psi(\mathfrak{a})\mathbf{f}, \forall \mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)\} \quad (26.4.1.9)$$

and

$$\text{SF}_k(\mathbf{n}, \psi) = \{\mathbf{f} \in \text{SF}_k(\mathbf{n}, \psi_0) : S(\mathfrak{a})\mathbf{f} = \psi(\mathfrak{a})\mathbf{f}, \forall \mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)\} \quad (26.4.1.9)$$

And  $\mathbf{f}$  is said to have central character  $\psi$  if  $\mathbf{f} \in \text{MF}_k(\mathbf{n}, \psi)$ . And  $f \in \text{MF}_k(\mathbf{n}; \psi)$  is called a normalized **Hilbert eigenform** if it is a common eigenvalue for all these operators  $\{T_n(\mathfrak{a}), S_n(\mathfrak{a})\}_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)}$  and  $c(\mathcal{O}_F, \mathbf{f}) = 1$ . In this case,  $S(\mathfrak{a})\mathbf{f} = \psi(\mathfrak{a})\mathbf{f}$  and  $T(\mathfrak{a})\mathbf{f} = c(\mathfrak{a}, \mathbf{f})\mathbf{f}$  for each  $\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)$ .

┘

*Proof:* Cf.[Shi78]Prop2.1? □

**Prop. (26.4.1.11) [Algebraicity of Eigenvalues].** For any  $\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)$ , the eigenvalues of  $T(\mathfrak{a})$ (26.4.1.9) are algebraic integers. Moreover, if  $k_1 = \dots = k_d$ , then the eigenvalues of  $T'(\mathfrak{a})$  are algebraic integers. ┘

*Proof:* Cf.[Shi78]P650. ? □

**Prop. (26.4.1.12) [Galois Conjugates].** The Galois conjugate of a Hilbert modular form is also a Hilbert modular form. ┘

*Proof:* Cf.[Shi78]? □

**Prop. (26.4.1.13) [Rational Hilbert Modular Forms, Shimura].** For  $A \subset \mathbb{C} \in \mathcal{CRing}$ , denote  $\text{MF}_k(\mathfrak{n}, \psi; A)$  the space of Hilbert modular forms with Fourier coefficients in  $A$ , then

$$\text{MF}_k(\mathfrak{n}; \mathbb{Z}) \otimes \mathbb{C} = \text{MF}_k(\mathfrak{n}).$$

┘

*Proof:* ? Cf.[Shimura, some arithmetic properties of modular forms of one and several variables]Thm7. □

**Def. (26.4.1.14) [Peterson Inner Product].** For  $\Gamma \subset \text{GL}(2; F)^\times$  be a congruence subgroup(26.4.1.2) and  $f, g \in \text{MF}_k(\Gamma)$  s.t. either  $f$  or  $g$  is a cusp form, define

$$\langle f, g \rangle_{\text{Pet}} = \frac{1}{\text{Vol}(\Gamma \backslash \mathcal{H}^d)} \int_{\Gamma \backslash \mathcal{H}^d} \overline{f(z)} g(z) y^k d\mu(z).$$

And for  $\mathfrak{c} \in \text{Ideal}^\times(\mathcal{O}_F)$  and  $\mathbf{f} = (f_\kappa), \mathbf{g} = (g_\kappa) \in \text{MF}_k(\mathfrak{c})$  s.t. either  $\mathbf{f}$  or  $\mathbf{g}$  is a cusp form, define

$$\langle \mathbf{f}, \mathbf{g} \rangle_{\text{Pet}} = \sum_{\kappa \in \text{Cl}^+(F)} \langle f_\kappa, g_\kappa \rangle_{\text{Pet}}.$$

┘

**Prop. (26.4.1.15) [Adjoint of Hecke Operators].** If  $y \in Y$  and  $(\det y, \mathfrak{c}) = 1$ , then

$$\psi^*(\det y) \langle \mathbf{f} | WyW, \mathbf{g} \rangle_{\text{Pet}} = \langle \mathbf{f}, \mathbf{g} | WyW \rangle_{\text{Pet}}$$

for any  $\mathbf{f}, \mathbf{g} \in \text{MF}_k(\mathfrak{c}, \psi)$ . ┘

*Proof:* Cf.[?Hil78]P652. □

**Cor. (26.4.1.16).** If  $(\mathfrak{m}, \mathfrak{c}) = 1$  and  $\mathbf{f} \in \text{MF}_k(\mathfrak{c}, \psi)$  satisfies  $T(\mathfrak{m})\mathbf{f} = \lambda(\mathfrak{m})\mathbf{f}$ , then  $\lambda(\mathfrak{m}) = \psi^*(\mathfrak{m})\overline{\lambda(\mathfrak{m})}$ . ┘

*Proof:* ? □

**Prop. (26.4.1.17) [Hilbert Newforms].** Given  $\mathbf{f} \in \text{MF}_k(\mathfrak{n}, \psi)$  and  $\mathfrak{q} \in \text{Ideal}^\times(\mathcal{O}_F)$ , there exists a unique  $\mathbf{f}|_{\mathfrak{q}} \in \text{MF}_k(\mathfrak{n}\mathfrak{q}, \psi)$  s.t.

$$c(\mathfrak{a}, \mathbf{f}|_{\mathfrak{q}}) = c(\mathfrak{a}\mathfrak{q}^{-1}, \mathbf{f}).$$

for any  $\mathfrak{a} \in \text{Ideal}(\mathcal{O}_F)$ .

For  $\mathfrak{n} \in \text{Ideal}^\times(\mathcal{O}_F)$ , denote

$$\text{SF}_k(\mathfrak{n}, \psi)^{\text{old}} = \{\mathbf{g}|_{\mathfrak{q}_v} : v \in \Sigma_F^{\text{fin}}, v|\mathfrak{n}, \mathbf{g} \in \text{SF}_k(\mathfrak{n}/\mathfrak{q}_v, \psi)\}.$$

and denote  $\text{SF}_k(\mathfrak{n}, \psi)^{\text{new}}$  the orthogonal complement of  $\text{SF}_k(\mathfrak{n}, \psi)^{\text{old}}$  in  $\text{SF}_k(\mathfrak{n}, \psi)$ .

By (26.4.1.15),  $\text{SF}_k(\mathfrak{n}, \psi)^{\text{new}}$  is stable under  $T(\mathfrak{p})$  for a.e.  $\mathfrak{p} \in \Sigma_F^{\text{fin}}$ , so  $\text{SF}_k(\mathfrak{n}, \psi)^{\text{new}}$  has basis consisting of forms that are eigenforms for a.e.  $T(\mathfrak{p})$ . Then by [Miyake, On automorphic forms on  $\text{GL}(2)$  and Hecke operators]?, this basis consists of eigenforms. An eigenform in  $\text{SF}_k(\mathfrak{n}, \psi)^{\text{new}}$  is called a **normalized Hilbert newform** if  $c(\mathcal{O}_F, \mathbf{f}) = 1$ .

If  $\mathfrak{f}$  is a newform in  $\text{SF}_k(\mathfrak{n}, \psi)^{\text{new}}$ ,  $\mathbf{f}$  is said to have **conductor**  $\mathfrak{n}$ . ┘

*Proof:* Cf. [Shi78]P651. □

**Def. (26.4.1.18) [Fields of Definition of  $\mathbf{f}$ ].** Suppose  $k_1 \equiv \dots \equiv k_d$  and  $\mathbf{f} \in \text{SF}_k(\mathfrak{c}, \psi)^{\text{new}}$  is a newform, denote  $\mathbb{Q}(\mathbf{f}) = \mathbb{Q}\{c(\mathfrak{a}, \mathbf{f}) | \mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)\}$  (26.4.1.8) with ring of integer  $\mathcal{O}_{\mathbf{f}}$ . Then

- $\mathbb{Q}(\mathbf{f})$  contains the values  $\psi^*(\mathfrak{m})$  for each  $\mathfrak{m} \in \text{Ideal}^\times(\mathcal{O}_F)$ .
- $\mathbb{Q}(\mathbf{f})$  is generated by  $C(\mathfrak{p}, \mathbf{f})$  for a.e.  $\mathfrak{p} \in \Sigma_F^{\text{fin}}$ .
- $\mathbb{Q}(\mathbf{f}) \in \text{NField}$  and is either CM or totally real. ┘

*Proof:* ? Cf. [Shi78]P654. □

### Adelic Hilbert Modular Forms

Cf. [Shi78](2.15).

**Prop. (26.4.1.19) [Local Component of Adic Hilbert Modular Forms].** Let  $v \in \Sigma_F^{\text{fin}} \setminus S_F(\mathfrak{n})$ ,  $\psi : \text{Cl}^{\text{fin}}(F) \rightarrow \mathbb{C}^\times$ , and  $\mathbf{f} \in \text{SF}_k^0(\mathfrak{n}\mathfrak{p}_v, \psi)$  be a normalized newform, then  $\pi_{\mathbf{f}, v}$  is an unramified twist of the Steinberg representation and

$$c(\mathfrak{p}_v, \mathbf{f})^2 = \psi(\mathfrak{p}_v) \|v\|^{k-2}.$$

Conversely, if  $\mathbf{f} \in \text{SF}_k^0(\mathfrak{n}\mathfrak{p}_v, \psi)$  is an eigenform and the above equation holds, then  $\mathfrak{p}_v$  divides the conductor of  $\mathbf{f}$ . ┘

*Proof:* ? Cf. [VIEWING MODULAR FORMS AS AUTOMORPHIC REPRESENTATIONS]. □

## 2 $\Lambda$ -adic Hilbert Modular Forms

**Notation (26.4.2.1).**

- Let  $p \in \text{Prime}$ .
- Let  $q = p^{1+\delta_{p,2}}$ .
- Let  $\omega$  be the Teichmüller character  $(\mathbb{Z}/(q))^\times \rightarrow \mathbb{Z}_p^*$ .
- Let  $(K, \mathcal{O}, \mathfrak{m}, \kappa) \in p\text{-NField}$ .
- Let  $\mathbb{Q}_\infty$  be the cyclotomic  $\mathbb{Z}_p$ -extension of  $\mathbb{Q}$ .
- Let  $\gamma \in \text{Gal}(F_\infty/F)$  be a topological generator and  $u = \chi_{\text{cycl}}(\gamma)$ .
- Let  $\Lambda_{\mathcal{O}} = \mathcal{O}[[T]] \cong \mathbb{Z}_p[\text{Gal}(\mathbb{Q}_\infty/\mathbb{Q})]$ ,  $F_\Lambda = \text{Frac}(\Lambda_{\mathcal{O}})$ . ┘

**Def. (26.4.2.2) [Specialization Maps].** Define

$$\mathfrak{X} = \{(k, \zeta) \in \mathbb{Z}_{\geq 2} \times \mu_{p^\infty} : \zeta^{p^r} = 1\},$$

and for each  $(k, \zeta) \in \mathfrak{X}$ , define a  $\mathcal{O}$ -homomorphism

$$v_{k, \zeta} : \Lambda_{\mathcal{O}} \rightarrow \mathcal{O}[\zeta] : T \mapsto \zeta u^{k-2} - 1 \quad (26.4.2.1),$$

and denote  $P_{k, \zeta}$  the kernel  $(T - \zeta u^{k-2} - 1)$ , which is a prime divisor  $1q2$  of  $\Lambda_{\mathcal{O}}$ .  $\lrcorner$

**Def. (26.4.2.3) [ $\Lambda$ -adic Hilbert Modular Forms].** A  $\Lambda$ -adic Hilbert modular form  $\mathcal{F}$  of level  $\mathfrak{n}$  is given by a set of elements

$$\begin{cases} c(\mathfrak{a}, \mathcal{F}) \in \Lambda_{\mathcal{O}} & , \mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F) \\ c_\kappa(0, \mathcal{F}) \in \Lambda_{\mathcal{O}} & , \kappa \in \text{Cl}^+(F) \end{cases}.$$

s.t. for any  $(k, \zeta) \in \mathfrak{X}$  (26.4.2.1) with the exact order of  $\zeta$  being  $p^r$ , there exists a  $\mathbf{f}_{k, \zeta} \in \text{MF}_k(\mathfrak{n}p^r; \mathcal{O}[\zeta])$ , with

$$c(\mathfrak{a}, \mathbf{f}_{k, \zeta}) = v_{k, \zeta}(c(\mathfrak{a}, \mathcal{F})), \quad \mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F), \quad c_\kappa(0, \mathbf{f}_{k, \zeta}) = v_{k, \zeta}(c_\kappa(0, \mathcal{F})) \quad (26.4.2.2) (26.4.1.8).$$

The space of  $\Lambda$ -adic Hilbert modular forms with level  $\mathfrak{n}$  is denoted by  $\text{MF}(\mathfrak{n}; \Lambda_{\mathcal{O}})$ , and define the subspace of  $\Lambda$ -adic Hilbert cusp forms  $\text{SF}(\mathfrak{n}; \Lambda_{\mathcal{O}})$  to be the subspace of  $\text{MF}(\mathfrak{n}; \Lambda_{\mathcal{O}})$  consisting of forms  $\mathcal{F}$  s.t.  $c_\kappa(0, \mathcal{F}) = 0$  for each  $\kappa \in \text{Cl}^+(F)$ .  $\lrcorner$

**Def. (26.4.2.4) [Hecke Action on  $\Lambda$ -adic Hilbert Modular Forms].** For  $\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)$ , if

$$\|\mathfrak{a}\|^{k-2} S(\mathfrak{a}) \mathbf{f}_{k, \zeta} = v_{k, \zeta}(\lambda_{\mathfrak{a}} \mathbf{f}_{k, \zeta})$$

holds for each  $(k, \zeta) \in \mathfrak{X}$ , then we define the **Hecke operator**

$$S(\mathfrak{a}) \mathcal{F} = \lambda_{\mathfrak{a}} \mathcal{F}.$$

If  $S(\mathfrak{a})$  exists for each  $\mathfrak{a} \in \text{Ideal}^*(\mathcal{O}_F)$ , then there exists a  $\Lambda$ -adic Hilbert modular form  $\mathcal{G}$  s.t.

$$\mathbf{g}_{k, \zeta} = T(\mathfrak{a}) \mathbf{f}_{k, \zeta}$$

for each  $\mathfrak{a} \in \text{Ideal}^*(\mathcal{O}_F)$  and  $(k, \zeta) \in \mathfrak{X}$ . So we define the **Hecke operator**

$$T(\mathfrak{a}) \mathcal{F} = \mathcal{G},$$

which is given explicitly as

$$c(\mathfrak{m}, T(\mathfrak{n}) \mathcal{F}) = \sum_{\mathfrak{m} + \mathfrak{n} \subset \mathfrak{a}} \lambda_{\mathfrak{a}} \|\mathfrak{a}\| c(\mathfrak{a}^{-2} \mathfrak{m} \mathfrak{n}, \mathcal{F}), \quad \mathfrak{m}, \mathfrak{n} \in \text{Ideal}^\times(\mathcal{O}_F),$$

$$c_\kappa(0, T(\mathfrak{a}) \mathcal{F}) = \sum_{\mathfrak{b} | \mathfrak{a}, [\mathfrak{b}] = [t_b], [\mathfrak{a}] = [t_a]} \lambda_{\mathfrak{b}} \|\mathfrak{b}\| c_{\kappa b^2 a^{-1}}(0, \mathcal{F}).$$

by (26.4.1.9).  $\lrcorner$

**Def. (26.4.2.5) [Central Characters and Cusp Forms].** For any limit character

$$\chi : \varprojlim_r \mathrm{Cl}^{\mathrm{np}^r \infty}(F) \rightarrow \Lambda_{\mathcal{O}},$$

a  $\Lambda$ -adic Hilbert modular form  $\mathcal{F}$  is said to have central character  $\psi$  if  $S(\mathfrak{a})^* \mathcal{F}$  (26.4.2.4) exists for each  $\mathfrak{a} \in \mathrm{Ideal}^\times(\mathcal{O}_F)$  and  $\lambda_{\mathfrak{p}_v} = \chi(\mathfrak{p}_v)$  for any  $v \in \Sigma_F^{\mathrm{fin}} \setminus S_F(\mathrm{np})$ .

The space of  $\Lambda$ -adic Hilbert modular forms with level  $\mathfrak{n}$  and central character  $\chi$  is denoted by  $\mathrm{MF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}})$ , and let  $\mathrm{SF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}})$  be the subspace of  $\mathrm{MF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}})$  consisting of  $\Lambda$ -adic Hilbert cusp forms.

Then  $\mathrm{MF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}})$  is a torsion-free  $\Lambda$ -module.  $\lrcorner$

*Proof:*  $\mathrm{MF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}})$  is a torsion-free because if  $g \in \Lambda_{\mathcal{O}}^\times$  s.t.  $g(T)\mathcal{F} = 0$ , then  $g(\zeta - 1)v_{2,\zeta}(\mathcal{F}) = 0$  for a.e.  $\zeta \in \mu_{p^\infty}$  (26.4.2.2). Thus  $\mathcal{F} = 0$  by Weierstrass preparation theorem (19.1.7.7).  $\square$

**Def. (26.4.2.6) [ $\Lambda$ -adic Hilbert Modular Forms over Extension Fields].** For any finite extension  $L/F_{\Lambda_{\mathcal{O}}}$  and any limit character  $\chi : \varprojlim_r \mathrm{Cl}^{\mathrm{np}^r \infty}(F) \rightarrow \Lambda_{\mathcal{O}}$ , define

$$\mathrm{MF}(\mathfrak{n}, \chi; \mathcal{O}_L) = \mathrm{MF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}}) \otimes \mathcal{O}_L, \quad \mathrm{SF}(\mathfrak{n}, \chi; \mathcal{O}_L) = \mathrm{SF}(\mathfrak{n}, \chi; \Lambda_{\mathcal{O}}) \otimes \mathcal{O}_L.$$

$$\mathrm{MF}(\mathfrak{n}, \chi; L) = \mathrm{MF}(\mathfrak{n}, \chi; \mathcal{O}_L) \otimes L, \quad \mathrm{SF}(\mathfrak{n}, \chi; L) = \mathrm{SF}(\mathfrak{n}, \chi; \mathcal{O}_L) \otimes L.$$

By Galois theory and (26.4.1.12), this definition only depends on  $L$  but not on  $\mathcal{O}_K$ .  $\lrcorner$

**Def. (26.4.2.7) [Fields of Definition of  $\mathcal{F}$ ].** Situation as in (26.4.2.5), for a normalized eigenform  $\mathcal{F} \in \mathrm{MF}(\mathfrak{m}; \chi; \Lambda_{\mathcal{O}})$ , let  $M_{\mathcal{F}}$  be the field generated by Hecke eigenvalues of  $\mathcal{F}$  over  $\Lambda_{\mathcal{O}}$ , with integral ring  $\Lambda_{\mathcal{F}}$ . It is a finite extension of  $\Lambda_{\mathcal{O}}$ . ?  $\lrcorner$

### Ordinary Hilbert Modular Forms

**Def. (26.4.2.8) [Ordinary Hilbert Modular Forms].**  $\mathbf{f} \in \mathrm{MF}_k(\mathfrak{n}, \psi)$  is called a **ordinary modular form** at  $\lambda \in \Sigma_F^{\mathrm{fin}}$  if for any  $v \in S_F(\|\lambda\|)$ , the equation

$$T^2 - c(\mathfrak{p}_v, \mathbf{f})T + \psi(\mathfrak{p}_v) \|v\|^{k-1}$$

has at least one root  $\alpha_v(f)$  that is a unit mod  $\lambda$ .

Notice that for  $k \geq 2$ , this is equivalent to  $v_\lambda(c(\mathfrak{p}_v, \mathbf{f})) = 0$  for each  $v \in S_F(\|\lambda\|)$ .  $\lrcorner$

**Def. (26.4.2.9) [Hida's Stabilization Operator  $e$ ].** Denote  $U_p = T_{\mathrm{np}}(p)$  and define the operator

$$e = \varprojlim_r U_p^{r!}$$

which acts on  $\mathrm{MF}_k(\mathrm{np}^r, \mathcal{O})$  with the uniform topology for any  $r, k \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* For  $m \in \mathbb{Z}_+$  large, the action of  $U_p^{p^m}$  on  $A = \mathrm{MF}_k(\mathrm{np}^r, \mathcal{O})/\mathfrak{m}_K \mathrm{MF}_k(\mathrm{np}^r, \mathcal{O})$  is semisimple, so a suitable power  $U_p^M$  is idempotent on  $A$ . But then  $U_p^{p^n M}$  is an idempotent on  $\mathrm{MF}_k(\mathrm{np}^r, \mathcal{O})/\mathfrak{m}_K^n \mathrm{MF}_k(\mathrm{np}^r, \mathcal{O})$ . Then  $e$  acts on  $\mathrm{MF}_k(\mathrm{np}^r, \mathcal{O})$  by taking limits over  $n$ .  $\square$

**Def. (26.4.2.10) [ $p$ -Stabilized Newforms].** A  $p$ -stabilized newform is a Hilbert modular eigenform that is a newform away from  $p$ , and for  $\mathfrak{p} \in S_F(p)$ , they are eigenforms for the operator  $U_p$  (26.4.2.9) with unit eigenvalues.

A  $\lambda$ -stabilized newform of weight  $k \geq 2$  is an eigenform of the form  $e\mathbf{f}$  (26.4.2.9) where  $\mathbf{f}$  is a normalized Hilbert newform (26.4.1.17).

The Dirichlet series of

If  $k \geq 2$  and  $\mathbf{f} \in \text{MF}(\mathbf{n}, K)$  is ordinary, then  $e\mathbf{f}$  is an eigenform of level  $\mathbf{m}\mathfrak{P}$  where  $\mathfrak{P}$  is the product of primes above  $p$  that doesn't divide  $\mathbf{m}$ .

Denote

$$\text{MF}_k^0(\mathbf{n}, \psi) \triangleq e \text{MF}_k(\mathbf{n}, \psi), \quad \text{SF}_k^0(\mathbf{n}, \psi) \triangleq e \text{SF}_k(\mathbf{n}, \psi).$$

┘

**Def. (26.4.2.11) [Cyclotomic Characters].** A limit character  $\chi : \varprojlim_r \text{Cl}^{np^r\infty}(F) \rightarrow \Lambda_{\mathcal{O}}$  is called **cyclotomic** if for each  $(k, \zeta) \in \mathfrak{X}$ , we have

- $\chi_{k, \zeta} = (v_{k, \zeta} \circ \chi) \omega^{2-k}$  factor through  $I^{np^r\infty} \rightarrow \text{Gal}(F^{n\infty}(\mu_{p^r})/F)$ .
- There exists  $c \in \mathbb{R}_+$  s.t.  $\# \text{Im}(\chi_{k, \zeta}) \geq cp^r$ .

To give an example of cyclotomic character, for a ray class character  $\psi : \text{Cl}^{n\infty} \rightarrow \mathbb{C}^\times$ , let  $\mathcal{O}_\psi$  be the extension of  $\mathbb{Z}_p$  generated by  $\text{Im}(\psi)$ , then there is a limit character

$$\psi : \varprojlim_r I^{np^r} \rightarrow \Lambda_{\mathcal{O}_\psi} : \mathfrak{a} \mapsto \psi(\mathfrak{a})(1+T)^{\log_u(\|\mathfrak{a}\|)}.$$

┘

### $p$ -Stabilized $\Lambda$ -adic Newforms

**Def. (26.4.2.12) [ $p$ -Stabilized(Ordinary)  $\Lambda$ -adic Modular Forms].** Hida's stabilization operator  $e$  (26.4.2.9) extends to an idempotent operator on  $\text{MF}(\mathbf{n}, \psi; \Lambda_{\mathcal{O}})$  compatible with specializations.

Denote  $\text{MF}^0(\mathbf{n}, \psi; \Lambda_{\mathcal{O}}) = e \text{MF}(\mathbf{n}, \psi; \Lambda_{\mathcal{O}})$ , called the space of **ordinary  $\Lambda$ -adic modular forms**.

┘

*Proof:* For  $(k, \zeta) \in \mathfrak{X}$  with  $\zeta$  of exact order  $p^r$ , let  $\pi_{k, \zeta} \in \Lambda \in \Lambda_{\mathcal{O}}$  be a minimal polynomial of  $\zeta u^{k-2} - 1$  over  $K$ , then the kernel of

$$v_{k, \zeta} : \text{MF}(\mathbf{n}, \psi; \Lambda_{\mathcal{O}}) \rightarrow \text{MF}_k(\mathbf{n}p^r, \mathcal{O}[\zeta])$$

are those with coefficients in  $\pi_{k, \zeta} \Lambda_{\mathcal{O}}$ . So we can define  $e\mathcal{F}$  modulo  $(\pi_{k, \zeta})$ , and define  $e\mathcal{F} \in \text{MF}(\mathbf{n}, \psi; \Lambda_{\mathcal{O}})$  as their limit, because  $\cap \pi_{k, \zeta} = 0$ . □

**Thm. (26.4.2.13) [Hida].** If  $\psi$  is cyclotomic ?, then  $\text{MF}^0(\mathbf{n}, \psi; \Lambda_{\mathcal{O}})$  is a f.g.  $\Lambda_{\mathcal{O}}$ -module. ┘

*Proof:* Let  $\mathcal{F}_1, \dots, \mathcal{F}_t \in \text{MF}^0(\mathbf{n}, \chi; \Lambda_{\mathcal{O}})$  be independent over  $\Lambda$ , and let  $N = \Lambda\{F_1, \dots, F_t\}$  and  $W = (N \otimes F_{\Lambda}) \cap \text{MF}^0(\mathbf{n}, \chi; \Lambda_{\mathcal{O}})$ , then  $W/N$  is a torsion  $\Lambda$ -module, and if  $\mathfrak{a}_1, \dots, \mathfrak{a}_t \in \text{Ideal}^\times(\mathcal{O}_F)$  s.t.  $D = \det(c(\mathfrak{a}_i, \mathcal{F}_j)) \in \Lambda^\times$ , then  $W/N$  is annihilated by  $D$ . Because  $\Lambda$  is Noetherian and  $\text{MF}^0(\mathbf{n}, \psi; \Lambda_{\mathcal{O}})$  is torsion-free (26.4.2.5), it suffices to show that  $t$  is bounded.

Consider the maps

$$v_{k, 1} : N/P_{k, 1}N \rightarrow \text{MF}_k^0(\mathbf{n}p, \mathcal{O}),$$

When  $k$  is sufficiently large s.t.  $v_{k, 1}(D) \neq 0$ , this map is injective, but the dimension of the RHS is bounded independent of  $k$  by (26.4.2.14), which shows that  $t$  is bounded. □

**Lemma (26.4.2.14).** For fixed  $\mathbf{n}$ ,  $\dim \text{MF}_k^0(\mathbf{n}, K)$  is bounded independent of  $k$ . ┘

*Proof:* ? Cf.[Wil88]P539. It is proven by group cohomology method.

Firstly the dimension from Eisenstein series are bounded by ? (or use the description of this space in [W1]Prop1.5.)  $\square$

**Cor. (26.4.2.15).** Denote

$$\mathrm{SF}^0(\mathfrak{np}^\infty, \psi; \Lambda) = \bigcup_{r \in \mathbb{N}} \mathrm{SF}^0(\mathfrak{np}^r, \psi; \Lambda),$$

then  $\mathrm{SF}^0(\mathfrak{np}^\infty, \psi)$  if f.g. over  $\Lambda_{\mathcal{O}}$ . In fact, if  $\mathfrak{c}(\psi)|\mathfrak{n}$ , then  $\mathrm{SF}^0(\mathfrak{np}^\infty, \psi; \Lambda) = \mathrm{SF}^0(\mathfrak{np}, \psi; \Lambda)$  is f.g. over  $\Lambda_{\mathcal{O}}$  by (26.4.2.13).  $\lrcorner$

*Proof:* If  $\mathfrak{c}(\psi)|\mathfrak{n}$  and  $\mathcal{F} \in \mathrm{SF}^0(\mathfrak{np}^\infty, \psi; \Lambda)$ , it suffices to show that for any  $(k, \zeta) \in \mathfrak{X}$ , we have  $\mathbf{f}_{k, \zeta} \in \mathrm{SF}_k(\mathfrak{np}^{r+1})$ .  $\square$

**Prop. (26.4.2.16).** If  $\mathcal{F}$  is given by a set of elements

$$\begin{cases} c(\mathfrak{a}, \mathcal{F}) \in \Lambda_{\mathcal{O}} & , \mathfrak{a} \in \mathrm{Ideal}^\times(\mathcal{O}_F) \\ c_\kappa(0, \mathcal{F}) \in \Lambda_{\mathcal{O}} & , \kappa \in \mathrm{Cl}^+(F) \end{cases}.$$

s.t. there exists  $M \in \mathbb{Z}_+$  s.t. for any  $(k, \zeta) \in \mathfrak{X}$  (26.4.2.1) with the exact order of  $\zeta$  being  $p^r$  and  $k \geq M$ , there exists a  $\mathbf{f}_{k, \zeta} \in \mathrm{MF}_k^0(\mathfrak{np}^r, \psi_{k, \zeta}; \mathcal{O}[\zeta])$  (26.4.2.11), with

$$c(\mathfrak{a}, \mathbf{f}_{k, \zeta}) = v_{k, \zeta}(c(\mathfrak{a}, \mathcal{F})), \quad \mathfrak{a} \in \mathrm{Ideal}^\times(\mathcal{O}_F), \quad c_\kappa(0, \mathbf{f}_{k, \zeta}) = v_{k, \zeta}(c_\kappa(0, \mathcal{F})) \text{ (26.4.2.2) (26.4.1.8)},$$

then  $\mathcal{F} \in \mathrm{MF}^0(\mathfrak{n}; \chi; \Lambda_{\mathcal{O}})$ .  $\lrcorner$

*Proof:* Cf.[Wil90]P505.  $\square$

### $\Lambda$ -adic Newforms

**Def. (26.4.2.17) [ $\Lambda$ -adic Hilbert Newforms].** A normalized  $\Lambda$ -adic eigenform  $\mathcal{F} \in \mathrm{SF}^0(\mathfrak{np}^\infty, \chi; \Lambda_{\mathcal{O}_L})$  is called a  **$\Lambda$ -adic Hilbert newform** of level  $\mathfrak{n}$  if for a.e.  $(k, \zeta) \in \mathfrak{X}$  and any lifts  $\tilde{v}_{k, \zeta}$  of  $v_{k, \zeta}$  to  $L$ , we have  $\tilde{v}_{k, \zeta}(\mathcal{F})$  is a  $p$ -stabilized newform of level divisible by  $\mathfrak{n}^p$ .

The space generated by normalized  $\Lambda$ -adic newforms of level  $\mathfrak{mp}^\infty$  and level  $\chi$  is denoted by  $\mathrm{SF}^0(\mathfrak{np}^\infty, \chi; \Lambda_{\mathcal{O}_L})^{\mathrm{new}}$ .  $\lrcorner$

**Prop. (26.4.2.18).** A normalized  $\Lambda$ -adic eigenform  $\mathcal{F} \in \mathrm{MF}^0(\mathfrak{np}^\infty, \chi; \Lambda_{\mathcal{O}_L})$  is a  $\Lambda$ -adic newform (26.4.2.17) if  $\tilde{v}_{k, \zeta}(\mathcal{F})$  is a  $p$ -stabilized newform of level divisible by  $\mathfrak{n}^p$  for any infinite set of primes  $\tilde{v}_{k, \zeta}$ .  $\lrcorner$

*Proof:* Cf.[Wil88]P552. ?  $\square$

**Prop. (26.4.2.19) [A Newform Basis for  $\mathrm{MF}^0(\mathfrak{np}^\infty, \chi; L)$ ].** Let

$$U = \{\mathcal{F}(\mathfrak{a}z) : \mathfrak{a}\mathfrak{m}p^\infty | \mathfrak{np}^\infty, \mathcal{F} \in \mathrm{MF}^0(\mathfrak{mp}^\infty, \chi)^{\mathrm{new}} \text{ is a normalized newform}\},$$

and  $L/F_\Lambda$  is a finite extension s.t. all normalized newforms in  $U$  are defined (which exists by (26.4.2.15)). Then  $U$  spans  $\mathrm{MF}^0(\mathfrak{np}^\infty, \chi; L)$ .  $\lrcorner$

*Proof:* Cf.[Wil88]P552. ? We only prove for cusp forms.  $\square$

**Def. (26.4.2.20) [Λ-adic ℓ-newforms].** Suppose  $\mathcal{O}$  contains the image of  $\chi$  and  $L/F_\Lambda$  is a finite extension.  $\ell \in \Sigma_F^{\text{fin}} \setminus S(\mathfrak{np})$ , there are two maps

$$\varphi_1, \varphi_2 : \text{SF}^0(\mathfrak{np}^\infty, \chi; L) \rightarrow \text{SF}^0(\mathfrak{nl}p^\infty, \chi; L) : \varphi_1(\mathcal{F}) = \mathcal{F}, \quad \varphi_2(\mathcal{F}) = \mathcal{F}|_\ell.$$

Denote  $\text{SF}^0(\mathfrak{nl}p^\infty, \chi; L)^{\ell\text{-old}} = \text{Im}(\varphi_1) + \text{Im}(\varphi_2)$ .

If  $L$  contains all the eigenvalues for the eigenforms in  $\text{SF}^0(\mathfrak{nl}p^\infty, \chi; L)$ , then we can define

$$\text{SF}^0(\mathfrak{nl}p^\infty, \chi; L)^{\ell\text{-new}} = L \cdot \{ \mathcal{F}(\mathfrak{a}z) : \mathfrak{a}mp^\infty | \mathfrak{np}^\infty, \ell | \mathfrak{m}, \mathcal{F} \in \text{SF}^0(\mathfrak{mp}^\infty, \chi)^{\text{new}} \}.$$

and by (26.4.2.19), it satisfies

$$\text{SF}^0(\mathfrak{nl}p^\infty, \chi; L) = \text{SF}^0(\mathfrak{nl}p^\infty, \chi; L)^{\ell\text{-old}} \oplus \text{SF}^0(\overline{\mathfrak{nl}}, \chi; L)^{\ell\text{-new}}$$

For general  $L$ , we can define

$$\text{SF}^0(\mathfrak{nl}p^\infty, \chi; L)^{\ell\text{-new}} = \ker(U_\ell^2 - \chi(\ell)) \in \text{SF}^0(\mathfrak{nl}p^\infty, \chi; L),$$

by (26.4.1.19). ┘

**Def. (26.4.2.21) [Congruence Modules, Ribet].** Situation as in (26.4.2.20), for  $\mathcal{F} \in \text{SF}^0(\mathfrak{np}^\infty, \chi)^{\ell\text{-new}}$ , define  $\Lambda_{\mathcal{F}}$ -modules

$$\begin{aligned} H_{\mathcal{F}}(\ell) &= \{ \mathcal{H} \in \text{SF}^0(\mathfrak{nl}p^\infty, \chi, M_{\mathcal{F}})^{\ell\text{-new}} : \mathcal{H} = \mathcal{G} - u\mathcal{F} - v\mathcal{F}|_\ell, \\ &\quad \mathcal{G} \in \text{SF}^0(\mathfrak{nl}p^\infty, \chi; \Lambda_{\mathcal{F}}), \mathcal{F} \in \text{SF}^0(\mathfrak{np}^\infty, \chi; \Lambda_{\mathcal{F}}), u, v \in M_{\mathcal{F}} \}. \end{aligned}$$

$$C_{\mathcal{F}}(\ell) = H_{\mathcal{F}}(\ell) / \left( \text{SF}^0(\mathfrak{nl}p^\infty, \chi; M_{\mathcal{F}})^{\ell\text{-new}} \cap \text{SF}^0(\mathfrak{nl}p^\infty, \chi; \Lambda_{\mathcal{F}}) \right) \quad ?$$

where  $C_{\mathcal{F}}(\ell)$  is called the **congruence module** for  $\mathcal{F}$ , which measures how far the direct sum decomposition above fails to be a direct decomposition over  $\mathcal{O}_L$ .

Moreover, for any finite extension  $L/M_{\mathcal{F}}$ , we can define analogously  $H_L(\ell)$  and  $C_L(\ell)$ . ┘

**Prop. (26.4.2.22).** There is a  $\Lambda_{\mathcal{F}}$ -module map

$$\gamma : C_{\mathcal{F}}(\ell) \rightarrow M_{\mathcal{F}}/\Lambda_{\mathcal{F}} : \mathcal{H} = \mathcal{G} - u\mathcal{F} - v\mathcal{F}|_\ell \mapsto u.$$

Then this map is injective and the image is annihilated by  $w_\ell = (\alpha(\mathfrak{p}_\ell, \mathcal{F})^2 - \chi(\ell))(\beta(\mathfrak{p}_\ell, \mathcal{F})^2 - \chi(\ell)) \in \Lambda_{\mathcal{F}}$ . In particular,  $\gamma$  defines an isomorphism of  $C_{\mathcal{F}}(\ell)$  onto a fractional ideal  $\mathfrak{a}_{\mathcal{F}, \ell}$  of  $\Lambda_{\mathcal{F}}$  contained in  $w_\ell^{-1}\Lambda_{\mathcal{F}}$ . ┘

*Proof:* This map is well-defined because if  $\mathcal{H} = \mathcal{G} - u\mathcal{F} - v\mathcal{F}|_\ell = \mathcal{G}' - u'\mathcal{F} - v'\mathcal{F}|_\ell$ , then  $u - u' \in \Lambda_{\mathcal{F}}$  and  $v - v' \in \Lambda_{\mathcal{F}}$  by (26.4.2.19).

If  $\mathcal{H} = \mathcal{G} - u\mathcal{F} - v\mathcal{F}|_\ell$  and  $u \in \Lambda_{\mathcal{F}}$ , we show that  $v \in \Lambda_{\mathcal{F}}$ : In this case

$$v(U_\ell^2 - \chi(\ell))\mathcal{F}|_\ell = vc(\ell, \mathcal{F})\mathcal{F} - v\psi(\ell)(\|\ell\| + 1)\mathcal{F}|_\ell$$

has coefficients in  $\Lambda_{\mathcal{F}}$ . Thus  $v \in \Lambda_{\mathcal{F}}$ .

Apply  $U = (U_\ell^2 - \alpha_\ell(\mathcal{F})^2)(U_\ell^2 - \beta_\ell(\mathcal{F})^2)$  to the equation, where  $\alpha_\ell(\mathcal{F}), \beta_\ell(\mathcal{F})$  are the two roots of  $T^2 - c(\ell, \mathcal{F})T + \chi(\ell)\|\ell\| = 0$ , then  $U\mathcal{F} = U\mathcal{F}|_\ell = 0$ , and  $U\mathcal{H} = w_\ell\mathcal{H}$  by definition. So  $w_\ell\mathcal{H}$  has coefficients in  $\Lambda_{\mathcal{F}}$ . But then by (26.4.2.19),  $w_\ell u, w_\ell v \in \Lambda_{\mathcal{F}}$ . ┘

**Conj. (26.4.2.23).** Situation as in (26.4.2.22),  $w_\ell \mathfrak{a}_{\mathcal{F}, \ell} \not\subseteq P_i$  for  $P_i$  not above  $p$ . ┘



*Proof:* ?

□

**Thm& Conj.Cor.(26.4.2.24) [Congruence Theorem, Wiles].** This is true for  $F = \mathbb{Q}$  by [Ribet, Congruence relations between modular forms] and [Diamond].

In general, there exists  $v \in \Lambda_{\mathcal{F}}$  and  $c \in \mathbb{Z}$  depending on  $\mathcal{F}$  s.t.

$$\operatorname{div}(w_{\ell} \mathbf{a}_{\mathcal{F}, \ell}) \leq c \operatorname{div}(1 + \|\ell\|) + \operatorname{div}(v).$$

┘

*Proof:* ?

□

### 3 Eisenstein Series

**Def.(26.4.3.1) [Hilbert Eisenstein Series].** Let  $\varepsilon : \operatorname{Cl}^{\infty}(F) \rightarrow \mathbb{C}^{\times}$  with parity  $(-1)^k$  for each  $v \in \Sigma_F^{\infty}$  (and  $\mathfrak{n} \neq \mathbf{1}$  when  $F = \mathbb{Q}$ ), then there exists a **Hilbert modular Eisenstein series**  $E_{k, \varepsilon} \in \operatorname{MF}_k(\mathfrak{n}, \varepsilon; \overline{\mathbb{Q}})$  (26.4.1.10) s.t.

$$L(E_{k, \varepsilon}; s) = \zeta_F(s) L_{\mathfrak{n}}(\varepsilon; s + 1 - k).$$

with constant terms are given by  $c_{\kappa}(E_{k, \varepsilon}) = 2^{-d} L(\varepsilon; 1 - k)$ .

┘

*Proof:* ? Cf.[Shi78]Prop3.4.

□

### $\Lambda$ -adic Eisenstein Series

**Prop.(26.4.3.2) [ $\Lambda$ -adic Eisenstein Series].** Suppose  $\mathfrak{n} \neq \mathbf{1}$  if  $F = \mathbb{Q}$ , for each  $\chi : \operatorname{Cl}^{\mathfrak{n}}(F) \rightarrow \mathbb{C}^{\times}$ , there exists a  $\Lambda$ -adic Hilbert Eisenstein series  $\mathcal{E}_{\chi} \in \operatorname{MF}^0(\mathfrak{n}p, \chi; F_{\Lambda_{\chi}})$  s.t. for each  $(k, \zeta) \in \mathfrak{X}$ ,

$$v_{k, \zeta}(\mathcal{E}_{\chi}) = E_{k, \chi \rho \omega^{2-k}},$$

where  $\omega$  is the Teichmuller Hecke character,  $\rho$  is the Hecke character s.t.  $\rho(\mathfrak{a}) = \zeta^{\log_p(\|\mathfrak{a}\|)}$ .

┘

*Proof:* Cf.[Wil88]Prop1.3.1 and 1.3.2. ?

Here we define  $\mathcal{E}_{\chi}$  by its associated formal Dirichlet series:

$$L(\mathcal{E}_{\chi}; s) = \zeta_F(s) L(\chi; s - 1).$$

and constant terms

$$C_{\kappa}(0, \mathcal{E}_{\chi}) = 2^{-d} G_{\chi}(T) / H_{\chi}(T),$$

where

$$G_{\chi}(T) = G_{\chi \omega^2}(u^2(1 + T) - 1), \quad H_{\chi}(T) = H_{\chi \omega^2}(u^2(1 + T) - 1),$$

where for each Hecke character  $\psi$ ,  $G_{\psi} \in \mathbb{Z}_p[\psi][[T]]$  and  $H_{\psi} \in \mathbb{Z}_p[\psi][T]$  is defined in (33.2.1.5) s.t.

$$L_p(\psi; 1 - s) = G_{\psi}(u^s - 1) / H_{\psi}(u^s - 1).$$

□

**Def. (26.4.3.3) [Eisenstein Ideals].** Let  $I_{\text{Eis}} \subset \mathbf{T}(26.4.5.2)$  be the ideal generated by

$$\{(T(\mathfrak{m}) - c(\mathfrak{m}, \mathcal{E}_\chi)), \hat{G}_\chi^0(T)\},$$

where

$$\hat{G}_\chi^0(T) = \varpi^{-\mu} \prod_{\zeta \in \mu_{p^\infty}} (T + 1 - \zeta u^{-1})^{-s_\zeta} G_{\chi\omega^2}(u^2(T + 1) - 1),$$

where  $\varpi$  is a chosen uniformizer of  $\mathcal{O}_\chi$ , and the integers  $\mu, \zeta$  are chosen s.t.  $\hat{G}_\chi^0(T)$  is an integral power series with common factor 1 and  $\hat{G}_\chi^0(\zeta u^{-1} - 1) \neq 0$  for all  $\zeta \in \mu_{p^\infty}$ . Notice that this is a finite product.  $\lrcorner$

**Thm. (26.4.3.4).** If  $\chi \neq \omega_{\text{Tei}}^{-2}$  or  $L_p(\mathbf{1}_p; 1) = \infty$ , then

$$\mathbf{T}/I_{\text{Eis}} \cong \Lambda_\chi / (\hat{G}_\chi^0(T)) (26.4.3.3).$$

$\lrcorner$

*Proof:* Cf. [Wil90]P508 ?.

$\square$

## 4 Embedding into $\Lambda$ -adic Modular Forms

**Thm. (26.4.4.1) [Embedding into  $\Lambda$ -adic Modular Forms, Hida [Hid86]/Wiles [Wil88]].** For a  $p$ -stabilized normalized newform (26.4.2.10)  $\mathbf{f} \in \text{SF}_k^0(\mathfrak{n}, \psi; \mathcal{O})^{\text{new}}$ , there exists a finite extension  $L/F_{\Lambda_{\mathcal{O}}}$  and an eigenform  $\mathcal{F} \in \text{SF}^0(\mathfrak{n}p^\infty, \psi\omega^{k-2}; \mathcal{O}_L)$  s.t.  $\mathcal{F} \equiv \mathbf{f} \pmod{\tilde{P}_k}$  for some prime  $\tilde{P}_{k,1} \in \text{Spec}(\mathcal{O}_L)$  over  $P_k \in \text{Spec}(\Lambda)$ .  $\lrcorner$

*Proof:* ? Cf. [Wil88]P546.

Let me briefly mention how to deduce lifting for all  $k$ : Wiles defined a Eisenstein series  $\mathbf{E}$  of level  $p$ -power and all constant terms 1, Then he find eigenforms with level dividing  $\mathfrak{n}p^\infty$  that is  $p$ -near to  $\mathbf{e}(\mathbf{f}\mathbf{E}^{p^m})$ , which is  $p$ -near to  $\mathbf{f}$ . These forms have large weights, so they have liftings. But there are only f.m.  $\Lambda$ -adic eigenforms of level  $\mathfrak{n}p^\infty$  (26.4.2.13), so there are infinitely liftings to a single  $\Lambda$ -adic eigenforms  $\mathcal{F}$ . Then take limit  $m$ , we see that  $\mathcal{F}$  is a lifting of  $\mathbf{f}$ .

extensions Because there are only f.m.  $\Lambda$ -adic eigenforms with  $\mathfrak{n}p^\infty$ -level (26.4.2.13),  $\square$

**Lemma (26.4.4.2).** Suppose  $\mathcal{O}$  is large enough s.t. it contains all the eigenvalues of the eigenforms in  $\text{SF}_k(\mathfrak{n}, \psi)$ , then  $\lrcorner$

*Proof:*

$\square$

**Lemma (26.4.4.3).** Let  $v \in \Sigma_F^{\text{fin}} \setminus (p)$  and  $\mathbf{g} \in \text{SF}_k(\mathfrak{n}, \psi; \mathcal{O})$  is a normalized eigenform, then each root of  $T^2 - c(\mathfrak{p}_v, \mathbf{p})T + \psi(\mathfrak{p}_v) \|v\|^{k-1}$  is either 0 or a unit modulo  $\mathfrak{m}$ .  $\lrcorner$

*Proof:* ? Cf. [Wil88]P550.

$\square$

## 5 Hecke Algebras and Eisenstein Ideals

**Def. (26.4.5.1) [Hecke Algebra  $\mathbf{T}$ ].** Let  $\mathbf{T}$  be the subring of  $\text{End}(\text{SF}^0(\mathfrak{np}^\infty, \chi; F_{\Lambda_\chi}))$  generated over  $\Lambda_\chi$  by all the Hecke operators  $T_{\mathfrak{np}(\mathfrak{m})}$  where  $\mathfrak{m}$  runs through all the integral ideals of  $\mathcal{O}_F$ .  $\lrcorner$

**Prop. (26.4.5.2) [Hecke Algebra  $\mathbf{T}$ ].** Let  $\chi$  be a primitive even ray class character of  $\text{Cl}^n(F)$  and  $\{\mathcal{F}_i\}$  be a set of normalized newforms in  $\text{SF}^0(\mathfrak{mp}^\infty, \chi; F_{\Lambda_\chi})^{\text{new}}$  where  $\mathfrak{mp}^\infty | \mathfrak{np}^\infty$ , and let  $\{\mathcal{F}_i\}_{i \in I}$  represents the equivalence classes of conjugation over  $F_{\Lambda_\chi}$ .

Let  $A = \prod_i L_{\mathcal{F}_i}$  be the  $F_{\Lambda_\chi}$ -algebra, then the Hecke algebra  $\mathbf{T}$  (26.4.5.1) acts on  $A$  via

$$\mathbf{T} \mapsto \text{diag}(c(\mathcal{O}_F, T\mathcal{F}_i)).$$

(Notice that to be naturally defined, need to check that  $T\mathcal{F}_i = 0$  implies  $T\mathcal{F}_i^\sigma = 0$  ?). Then this map induces an isomorphism

$$\mathbf{T} \otimes_{\Lambda_\chi} L \cong A \otimes_{\Lambda_\chi} L \cong \prod_{\mathcal{F}_i} L,$$

where  $L/F_{\Lambda_\chi}$  is a finite extension s.t. all the  $\mathcal{F}_i$  are defined (which exists by (26.4.2.15)), by (26.4.2.19) and strong multiplicity one. Then by comparing dimension, we get an isomorphism

$$\mathbf{T} \otimes_{\Lambda_\chi} F_{\Lambda_\chi} \cong A.$$

$\lrcorner$

**Prop. (26.4.5.3).** Situation as in (26.4.5.2), denote  $W = A \otimes A$  on which  $\text{Gal}_F$  acts via  $\oplus_i \rho_{\mathcal{F}_i}$ . This action commutes with the action of  $\mathbf{T}$ .

If  $\theta$  is an irreducible divisor of  $\hat{G}_\chi^0(T)$  (26.4.3.3), which determines a prime ideal  $\mathfrak{P} \in \text{Spec}(\mathbf{T})$  via (26.4.3.4). Then there is an inclusion of local rings  $\Lambda_{\chi, \theta} \subset \mathbf{T}_{\mathfrak{P}}$  s.t.

$$\Lambda_{\chi, \theta} / (\hat{G}_\chi^0(T)) \cong \mathbf{T}_{\mathfrak{P}} / I_{\text{Eis}}.$$

$\lrcorner$

*Proof:* ?

$\square$

**Prop. (26.4.5.4) [Two Types of  $\mathbf{T}_{\mathfrak{P}}$ -Modules].** Situation as in (26.4.5.3), if  $\mathcal{L} \subset W_{\mathfrak{P}}$  is a stable lattice (i.e. a f.g.  $\text{Gal}_F$ -stable  $\mathbf{T}_{\mathfrak{P}}$ -module s.t.  $\mathcal{L} \otimes F_{\Lambda_\chi} = W_{\mathfrak{P}}$ ), then any irreducible subquotient of  $\mathcal{L}/\mathfrak{P}\mathcal{L}$  is one of the following type:

- $V \cong \mathbf{T}_{\mathfrak{P}}/\mathfrak{P}$  where  $\text{Gal}_F$  acts trivially,
- $V \cong \mathbf{T}_{\mathfrak{P}}/\mathfrak{P}$  where  $\text{Gal}_F$  acts via character  $\eta$ , where  $\eta(-1)$  corresponds to  $\chi$  (26.4.2.11) via class field theory.

These two types are called **type 1** and **type  $\eta$**  modules resp. They are disjoint types because  $\eta$  is odd and  $2 \notin \mathfrak{P}$  ?. And a  $\mathbf{T}_{\mathfrak{P}}[\text{Gal}_F]$ -module is called **type 1** and **type  $\eta$**  modules resp. if all their irreducible components are **type 1** and **type  $\eta$**  resp..  $\lrcorner$

*Proof:* ? Cf. [Wil90]P511.  $\square$

**Lemma (26.4.5.5) [Type 1 Deprived Lattice].** Situation as in (26.4.5.4), given any stable lattice  $\mathcal{L} \subset W_{\mathfrak{P}}$ , there exists a sublattice  $\mathcal{L}' \subset \mathcal{L}$  s.t.  $\mathcal{L}/\mathcal{L}'$  is of type 1 and any such sublattice of  $\mathcal{L}$  with this property is contained in  $\mathcal{L}'$ . In particular,  $\mathcal{L}'$  has no type 1 quotients, and it is called the **type 1 deprived lattice** of  $\mathcal{L}$ .  $\lrcorner$

*Proof:* ? Cf.[Wil90]P511. □

**Prop. (26.4.5.6).** Situation as in (26.4.5.4), there exists a stable lattice  $\mathcal{L} \subset W_{\mathfrak{P}}$  s.t.  $X = \mathcal{L}/I_{\text{Eis}}\mathcal{L}$  (26.4.3.3) has no type 1 quotient and there is an exact sequence of  $\mathbf{T}_{\mathfrak{P}}[\text{Gal}_F]$ -modules:

$$0 \rightarrow C \rightarrow X \rightarrow M \rightarrow 0$$

s.t.

- $C \cong \mathcal{L}^+/I_{\text{Eis}}\mathcal{L}^+$  is of type 1 (26.4.5.4).
- $M$  if of type  $\eta$  (26.4.5.4) and  $M \cong \mathbf{T}_{\mathfrak{P}}/I_{\text{Eis}}$  as  $\mathbf{T}_{\mathfrak{P}}$ -modules. ┘

*Proof:* ? Cf.[Wil90]P514. □

**Prop. (26.4.5.7).** Situation as in (26.4.5.6), if we denote  $H_{\infty} = F_{\chi}(\mu_{p^{\infty}})$  and let  $N_{\infty}$  be the splitting field of the exact sequence  $0 \rightarrow C \rightarrow X \rightarrow M \rightarrow 0$  over  $H_{\infty}$ , then there is a pairing

$$G \times M \rightarrow C : (\sigma, m) \mapsto \sigma \tilde{m} - \tilde{m}, \quad \tilde{m} \text{ any lifting of } m \text{ in } X.$$

Then this pairing induces a  $\text{Gal}_{H_{\infty}/F}$ -equivariant map

$$\varphi : G \rightarrow \text{Hom}_{\mathbf{T}_{\mathfrak{P}}/I_{\text{Eis}}}(M, C),$$

which induces an isomorphism of  $\mathbf{T}_{\mathfrak{P}}$ -modules  $\tilde{G} = \varphi(G)_{\theta} = \text{Hom}_{\mathbf{T}_{\mathfrak{P}}/I_{\text{Eis}}}(M, C) \cong C$ . ┘

*Proof:* Cf.[Wil90]P514. ? □

**Cor. (26.4.5.8).** Situation as in (26.4.5.7), we have

$$\text{Fitt}_{\Lambda_{\theta}}(\tilde{G}) \subset (\hat{G}_{\chi}^0(T)).$$

In particular, the characteristic polynomial of  $\gamma - 1$  on  $\tilde{G}$  is divisible in  $\Lambda_{\chi, \theta}$  by

$$\hat{G}_{\chi}^0(u^{-1}(1+T)^{-1} - 1).$$

┘

## 6 Ramanujan-Peterson Conjecture for Hilbert Modular Forms

**Thm. (26.4.6.1) [Brylinki-Labesse].** For  $\mathbf{f} \in \text{SF}_k^0(\mathfrak{m}; \chi)$ , the roots of  $T^2 - c(\ell, \mathbf{f})T + \chi(\ell) \|\ell\|^{k-1}$  are  $\|\ell\|$ -Weil numbers for  $\ell \in \Sigma_F^{\text{fin}}$  outside a finite set depending on  $\mathbf{m}$ . ┘

*Proof:* Cf.[Wil88]P555. □

## 7 Pseudo-Characters

**Lemma (26.4.7.1) [Families of Galois Representations, Wiles[Wil88]].** Let  $\{\mathbf{a}_i\}_{i \in \mathbb{N}}$  be an infinite set of prime divisors of  $\mathcal{O}_L$ , and for each  $i \in \mathbb{N}$ , we are given a continuous Galois representation

$$\rho_i : \text{Gal}_F \rightarrow \text{GL}(2; \widetilde{\mathcal{O}_L/\mathfrak{a}_i})$$

s.t.

- Each  $\rho_i$  is unramified outside a set  $S_i \subset \Sigma_F^{\text{fin}}$  of analytic density 0,
- For any  $v \in \Sigma_F^{\text{fin}}$ , there exists  $a_v, \varepsilon_v \in \mathcal{O}_F$  s.t.

$$\text{tr}\rho_i(\sigma_v) = a_v, \quad \det \rho_i(\sigma_v) = \varepsilon_v$$

whenever  $v \notin \Sigma_i$ .

- There exists a complex conjugation  $\mathbf{c}$  that is odd for each  $\rho_i$ .

Then there exists a continuous semisimple representation  $\rho : \text{Gal}_F \rightarrow \text{GL}(2; L)$  s.t.

$$\text{tr}\rho(\sigma_v) = a_v, \quad \det \rho(\sigma_v) = \varepsilon_v$$

for  $v \in \Sigma_F^{\text{fin}} \setminus \cup_i S_i$ .

Moreover, if  $S = \cup_i S_i$  has analytic density 0, then  $\rho$  is uniquely determined, and  $\rho$  is irreducible iff some  $\rho_i$  is.  $\square$

*Proof:* We may exclude primes above  $p$  from  $\{\mathfrak{a}_i\}$ . Since  $\mathcal{O}_L/\mathfrak{a}_i$  is a DVR, we can fix a basis of  $(\widetilde{\mathcal{O}_L/\mathfrak{a}_i})^2$  s.t.  $\rho_i(\mathbf{c}) = \text{diag}(-1, 1)$ . Firstly we assume  $p \neq 2$ .

Denote  $\rho_i(\sigma) = \begin{bmatrix} a_\sigma^i & b_\sigma^i \\ c_\sigma^i & d_\sigma^i \end{bmatrix}$ , and let  $x_{\sigma, \tau}^i = b_\sigma^i c_\tau^i$ .

For any  $R \in \mathbb{C}\text{Ring}$ , consider the following data:  $(a_\sigma, d_\sigma, x_{\sigma, \tau})$  where

- $a_\tau, d_\tau$  are continuous function from  $\text{Gal}_F$  to  $R$ ,
- $x_{\sigma, \tau}$  is a continuous function from  $\text{Gal}_F \times \text{Gal}_F$  to  $R$ .

s.t.

- $a_{\sigma\tau} = a_\sigma a_\tau + x_{\sigma, \tau}$ .
- $d_{\sigma\tau} = d_\sigma d_\tau + x_{\tau, \sigma}$ .
- $x_{\sigma\tau, \rho\gamma} = a_\sigma a_\gamma x_{\tau, \rho} + a_\sigma d_\rho x_{\tau, \gamma} + d_\tau d_\rho x_{\sigma, \gamma}$ .
- $a_1 = d_1 = 1, x_{\sigma, 1} = x_{1, \sigma} = 0$ .
- $x_{\sigma, \tau} x_{\rho, \eta} = x_{\sigma, \eta} x_{\rho, \tau}$ .

called a **pseudo-character** of  $\text{Gal}_F$  valued in  $R$ . Then each  $\rho_i$  gives a pseudo-character of  $\text{Gal}_F$  valued in  $\mathcal{O}_L/\mathfrak{a}_i$ . Moreover, they are in fact valued in  $\mathcal{O}_L/\mathfrak{a}_i$ :

$$a_\sigma = \frac{1}{2}(\text{tr}\rho(\sigma) - \text{tr}\rho(\sigma\mathbf{c})), \quad d_\sigma = \frac{1}{2}(\text{tr}\rho(\sigma) + \text{tr}\rho(\sigma\mathbf{c})), \quad x_{\sigma, \tau} = a_{\sigma\tau} - a_\sigma a_\tau.$$

We use induction on  $r$  to construct pseudo-representations  $\alpha_r$  valued in  $\mathcal{O}_L/(\mathfrak{a}_1 \cap \dots \cap \mathfrak{a}_r)$  s.t.  $\alpha_r \equiv \tilde{\rho}_i \pmod{\mathfrak{a}_i}$  for each  $i \in [r]_+$ . Suppose  $\alpha_r$  is constructed, we first notice that  $\alpha_r \equiv \tilde{\rho}_{r+1} \pmod{(\mathfrak{a}_1 \cap \dots \cap \mathfrak{a}_r, \mathfrak{a}_{r+1})}$ , because their trace of  $\sigma_v$  on them are equal for a.e.  $v \in \Sigma_F^{\text{fin}}$ , and the data is determined by the traces as above. Then using the exact sequence

$$0 \rightarrow \mathcal{O}_L/\mathfrak{m}_1 \cap \mathfrak{m}_2 \rightarrow \mathcal{O}_L/\mathfrak{m}_1 \oplus \mathcal{O}_L/\mathfrak{m}_2 \rightarrow \mathcal{O}_L/(\mathfrak{m}_1, \mathfrak{m}_2) \rightarrow 0,$$

we can find a common lift  $\alpha_{r+1}$  valued in  $\mathcal{O}_L/(\mathfrak{a}_1 \cap \dots \cap \mathfrak{a}_{r+1})$  of  $\alpha_r$  and  $\tilde{\rho}_{r+1}$ .

Taking limits  $r \rightarrow \infty$ , we get a pseudo-representation  $\alpha$  valued in  $\mathcal{O}_L$  because  $L$  is a local ring. Then we construct a representation  $\rho : \text{Gal}_F \rightarrow \text{GL}(2; \mathcal{O}_L)$  with  $\tilde{\rho} = \alpha$ : If  $x_{\sigma, \tau} = 0$  for each  $\sigma, \tau \in \text{Gal}_F$ , then we take  $\rho(\sigma) = \text{diag}(a_\sigma, d_\sigma)$ , and if  $x_{\sigma_0, \tau_0} \neq 0$ , then we denote

$$c_\tau = x_{\sigma_0, \tau}, \quad b_\sigma = x_{\sigma, \tau_0}/x_{\sigma_0, \tau_0},$$

then we can check  $\rho(\sigma) = \begin{bmatrix} a_\sigma & b_\sigma \\ c_\sigma & d_\sigma \end{bmatrix}$  is a representation with  $\tilde{\rho} = \alpha$  ?. Then we consider the semisimplification of  $\rho$ .

Finally for  $p = 2$ , we only have  $\rho_i(\sigma)$  take values in  $\frac{1}{2}\mathcal{O}_L/\mathfrak{a}_i$ . So we can modify the definition of pseudo-representation to be the data  $(2a_\sigma, 2d_\sigma, 4x_{\sigma,\tau}, \text{tr}\sigma)$  and modify the relations in the obvious way and include the relation  $2\text{tr}\sigma = (2a_\sigma) + (2d_\sigma)$ . The rest goes through as before. ?

The uniqueness of  $\rho$  follows from Chebotarev density theorem, and the irreducibility is easy.  $\square$

**Remark (26.4.7.2).** If  $p \neq 2$  and  $\mathcal{O}_L$  has the property that each prime divisor is principal (e.g. it is a UFD), then we may take  $\rho$  to be valued in  $\mathcal{O}_L$ : Let  $\mathfrak{c}$  be the ideal generated by all the elements  $\{x_{\sigma_0,\tau}\}_{\tau \in \text{Gal}_F}$ , and let  $\text{div } \mathfrak{c} = \text{div } \theta$  for some  $\theta \in \mathcal{O}_L^\times$ , then we can modify the definition of  $\rho$  to be

$$c_\sigma = x_{\sigma_0,\tau}/\theta, \quad b_\sigma = x_{\sigma,\tau_0}\theta/x_{\sigma_0,\tau_0} \in \mathcal{O}_L.$$

┘

## 8 Galois Representations attached to Hilbert Modular Forms

**Thm. (26.4.8.1)** [Galois Representations attached to Cuspidal Newforms, Eischler-Shimura/Deligne/Deligne-Serre/Shimura/Rogawski-Tunnell [R-T83]/Carayol [Car86]/Mazur-Wiles [M-W86]/Wiles [Wil88]]. For any normalized newform  $f \in \text{SF}(\mathfrak{n}, \psi)^{\text{new}}$  and  $\lambda \in \Sigma_{K_f}^{\text{fin}}$  which is ordinary for  $\mathbf{f}$  (26.4.2.8), there exists a unique continuous irreducible Galois representation

$$\rho_{\mathbf{f},\lambda} : \text{Gal}_F \rightarrow \text{GL}(2; \mathcal{O}_{\mathbf{f},\lambda})$$

s.t. for any  $v \in \Sigma_F^{\text{fin}} \setminus S_F(\mathfrak{n} \parallel \lambda)$ ,  $\rho_{\mathbf{f},\lambda}$  is unramified at  $v$ , and

$$\text{tr}(\sigma_v | \rho_{\mathbf{f},\lambda}) = c(\mathfrak{p}_v, \mathbf{f}), \quad \det(\sigma_v | \rho_{\mathbf{f},\lambda}) = \psi(v) \|v\|^{k-1}.$$

Moreover,

1. If  $k = 1$ , then

- $\#\text{Im}(\rho_{\mathbf{f},\lambda}) < \infty$ , thus lifts to an Artin representation  $\rho_{\mathbf{f}}$ ,
- $\rho_{\mathbf{f}}$  is unramified outside  $\mathfrak{c}$ .
- $\text{tr}(\sigma_v | \rho_{\mathbf{f}}) = c(v, \mathbf{f})$  and  $\det(\sigma_v | \rho_{\mathbf{f}}) = \psi(v)$  for each  $v \in \Sigma_F^{\text{fin}} \setminus S_F(\mathfrak{c})$ .

2. For  $v \in S_F(\parallel \lambda \parallel)$ ,  $\rho_{\mathbf{f},\lambda}|_{\text{Gal}_v}$  is equivalent to the form  $\begin{bmatrix} \varepsilon_1 & * \\ & \varepsilon_2 \end{bmatrix}$ , where  $\varepsilon_2$  is unramified and  $\varepsilon_2(\sigma_v) = \alpha_v(\mathbf{f})$  (26.4.2.8).

3. (Local-global compatibility) For any  $v \in \Sigma_F^{\text{fin}} \setminus S_F(\parallel \lambda \parallel)$ ,  $\text{WD}_v(\rho_v) \cong \text{rec}(\pi_{\mathbf{f},v})(-\frac{1}{2})$  up to semisimplification.

┘

*Proof:* Firstly, it suffices to prove that a representation  $\rho_\lambda : \text{Gal}_F \rightarrow \text{GL}(2; F')$  for some extension  $F'/\mathbb{Q}(\mathbf{f})$  exists, because if such  $\rho$  exists, it is irreducible by (26.4.8.5), and it must be odd ?, so we can choose a basis s.t.  $\rho(\mathfrak{c}) = \text{diag}(-1, 1)$ . Then by the calculation in the proof of (26.4.7.1) and Chebotarev density theorem,  $\rho$  is valued in  $\text{GL}(2; \mathbb{Q}(\mathbf{f})_\lambda)$  (26.4.1.18). Then by (24.2.3.2),

By (26.4.4.1), we can lift  $\mathbf{f}$  to an eigenform  $\mathcal{F} \in \mathrm{SF}^0(\mathfrak{n}p^\infty, \psi\omega^{k-2}, \mathcal{O}_L)$  for some finite extension  $L/F_\Lambda$  s.t.  $\mathbf{f} = \mathcal{F}(\bmod \tilde{P}_k)$  for some  $\tilde{P}_k$  above  $P_k = (T - (u^{k-2} - 1)) \in \mathrm{Spec}(\mathcal{O}_\Lambda)$ . The  $\Lambda$ -adic newform associated to  $\mathcal{F}$  gives a representation  $\rho_{\mathcal{F}}$  by (26.4.8.6). The reduction of  $\rho_{\mathcal{F}}$  gives the desired representation  $\rho_{\mathbf{f},\lambda}$ . (notice that the semisimplification doesn't depend on the stable lattice  $\mathcal{L}$ ) Cf. [Wil88]P562.

For the “moreover” part:

1. See (26.4.8.4).
2. This follows from (26.4.8.6).
3. If  $\mathfrak{p}_v \nmid \mathfrak{n}$ , this is immediate from the characteristic polynomial relations above, and if  $\pi_{\mathbf{f},v}$  is not a principle representation, this follows from Carayol's result. And by twisting  $\mathbf{f}$  by a quasicharacter, we can assume that  $c(\mathfrak{p}_v, \mathbf{f}) \neq 0$ .

Lift  $\mathbf{ef}$  to a  $\Lambda$ -adic eigenform  $\mathcal{F}$  using (26.4.4.1), and let  $\mathcal{F}^*$  be the newform part of  $\mathcal{F}$ . Then  $U(\mathfrak{p}_v)\mathcal{F}^* = c(\mathfrak{p}_v, \mathcal{F})$  if  $\mathfrak{p}_v$  divides the level of  $\mathcal{F}^*$ , and  $T(\mathfrak{p}_v)\mathcal{F}^* = (c(\mathfrak{p}_v, \mathcal{F}) + \beta(\mathfrak{p}_v, \mathcal{F}))\mathcal{F}^*$ , where  $\beta(\mathfrak{p}_v, \mathcal{F}) = c(\mathfrak{p}_v, \mathcal{F})^{-1}\chi(\mathfrak{p}_v)\|v\|$ .

Recall that the Galois representation attached to  $\mathcal{F}^*$  is constructed by congruences  $\{\mathcal{F}^* \equiv \mathcal{G}_i(\bmod P_i)\}_{i \in \mathbb{Z}_+}$  for some set of eigenforms  $\mathcal{G}_i$ , and we may assume that  $c(\mathfrak{p}_v, \mathcal{G}_i) \neq 0$  for each  $i$ . Then Carayol's method applies to each specialization  $\mathbf{g}$  of these  $\mathcal{G}_i$  because  $\mathbf{g}$  will be Steinberg at some finite prime (Cf. Lemma 1.4.3), so  $\mathrm{WD}_v(\rho_{\mathbf{g}})^{\varphi\text{-s.st}} \cong \mu_1 \oplus \mu_2$  where  $\mu_2$  is unramified,  $\mu_2(\sigma_v) = \alpha_v(\mathbf{g})$  and  $\mu_1\mu_2 = \chi(1)$ . Then the same proof as in (26.4.8.7) shows that the same holds for  $\mathcal{G}_i$ . Then this holds for  $\mathcal{F}^*$  modulo  $P_i$ , and by specialization, it holds for  $\mathbf{f}$ .

□

**Lemma (26.4.8.2) [Weight 2 base case, Carayol [Car86]].** Situation as in (26.4.8.1), if either  $[F : \mathbb{Q}]$  is odd or  $\pi_{\mathbf{f},w}$  is supercuspidal or special for some  $w \in \Sigma_F^{\mathrm{fin}}$ , then there exists a continuous Galois representation

$$\rho_{\mathbf{f},\lambda} : \mathrm{Gal}_F \rightarrow \mathrm{GL}(2; \mathcal{O}_{\mathbf{f},\lambda})$$

s.t. for any  $v \in \Sigma_F^{\mathrm{fin}} \setminus S_F(\mathfrak{n} \|\lambda\|)$ ,  $\rho_{\mathbf{f},\lambda}$  is unramified at  $v$ , and

$$\mathrm{tr}(\sigma_v | \rho_{\mathbf{f},\lambda}) = c(\mathfrak{p}_v, \mathbf{f}), \quad \det(\sigma_v | \rho_{\mathbf{f},\lambda}) = \psi(v) \|v\|^{k-1}.$$

If moreover we have  $k = 2$ , then for any  $v \in S_F(\|\lambda\|)$  s.t.  $\pi_{\mathbf{f},v}$  is a principle representation,  $\rho_{\mathbf{f},\lambda}|_{\mathrm{Gal}_v}$  is equivalent to the form  $\begin{bmatrix} \varepsilon_1 & * \\ & \varepsilon_2 \end{bmatrix}$ , where  $\varepsilon_2$  is unramified and  $\varepsilon_2(\sigma_v) = \alpha_v(\mathbf{f})$  (26.4.2.8).  $\square$

*Proof:* Cf. [Wil88]Lemma 2.1.5. and [W1]Thm 2.2.

For the moreover part:

If  $[F : \mathbb{Q}]$  is odd, choose a quaternion algebra  $B \in \mathrm{Az}_F$  which is split at an infinite place  $\tau : F \rightarrow \mathbb{C}$ , ramified at all the remaining infinite places and split at all finite places. Then [Hida, abelian varieties with complex multiplication as factors of the Jacobians of Shimura curve] proved that

**Lemma (26.4.8.3).** There exists  $A_{\mathbf{f}} \in \mathrm{AbVar}^{[\mathbb{Q}(\mathbf{f}) : \mathbb{Q}]}$  together with an embedding  $\theta : \mathbb{Q}(\mathbf{f}) \hookrightarrow \mathrm{End}_F(A_{\mathbf{f}})$  s.t. for each  $\lambda \in \Sigma_F^{\mathrm{fin}}$ , the associated  $\lambda$ -adic representation is just  $\rho_{\mathbf{f},\lambda}$  as in (26.4.8.1).  $\square$

If  $v \nmid \mathfrak{n}$ , consider the endomorphism ring  $E$  generated by  $\mathcal{O}_{\mathbf{f}}$  and  $\sigma_v$  acting on  $(A_{\mathbf{f}})_{\kappa(v)}$ , then the  $p$ -adic representation attached to the Dieudonné modular  $D(T_p A_{\mathbf{f}})$  and the  $\ell$ -adic representation of  $A_{\mathbf{f}}$  has the same trace on  $E$ .

□

**Lemma (26.4.8.4) [Weight 1 case].** If  $k = 1$ , the “moreover” part of the theorem holds.  $\lrcorner$

*Proof:* This follows from Deligne-Serre method ?. The main part is to construct a mod  $p$  Galois representation attached to  $\mathbf{f}$ . But in this case  $\rho_{\mathbf{f}}$  is ordinary outside  $\mathfrak{c}$  (26.4.2.8) because  $\psi(\mathfrak{p}_v)$  is a unit, so we can construct it by the main part of the theorem.  $\square$

**Lemma (26.4.8.5).** Situation as in (26.4.8.1), when  $\rho_{\mathbf{f}, \lambda}$  exists, it is absolutely irreducible.  $\lrcorner$

*Proof:* Cf. [Wil86] prop 2.1 ?

If it is not irreducible, then its semisimplification is two characters. But it is locally algebraic, i.e. they can be extended to Hecke characters. Thus the  $L$ -function of  $\mathbf{f}$  is the product of two Hecke  $L$ -functions with some local factors deleted. But then some twist of  $f$  has a pole, contradicting Jacquet-Langlands.  $\square$

### Galois Representations attached to $\Lambda$ -adic Modular Forms

**Thm. (26.4.8.6) [Galois Representations attached to  $\Lambda$ -adic Modular Forms, Hida [Hid86]/Wiles [Wil88]].** Suppose  $L/F_{\Lambda}$  is a finite extension and  $\mathcal{F} \in \mathrm{SF}^0(\mathfrak{n}p^{\infty}, \chi; L)$  is a  $\Lambda$ -adic newform (26.4.2.17), then there exists a unique continuous irreducible Galois representation

$$\rho_{\mathcal{F}} : \mathrm{Gal}_F \rightarrow \mathrm{GL}(2; L)$$

s.t. for any  $v \in \Sigma_{K_{\mathbf{f}}}^{\mathrm{fin}} \setminus S_{K_{\mathbf{f}}}(\mathfrak{n}p)$ ,  $\rho_{\mathcal{F}}$  is unramified at  $v$ , and

$$\mathrm{tr}(\sigma_v | \rho_{\mathcal{F}}) = c(\mathfrak{p}_v, \mathcal{F}), \quad \det(\sigma_v | \rho_{\mathcal{F}}) = \chi(\mathfrak{p}_v) \|\sigma_v\|.$$

Moreover, for  $v \in S_F(p)$ ,  $\rho_{\mathcal{F}}|_{\mathrm{Gal}_v}$  is equivalent to the form  $\begin{bmatrix} \varepsilon_1 & * \\ & \varepsilon_2 \end{bmatrix}$ , where  $\varepsilon_2$  is unramified and  $\varepsilon_2(\sigma_v) = \alpha_v(\mathcal{F})$  (26.4.2.8).  $\lrcorner$

*Proof:* If  $\rho_{\mathcal{F}}$  exists, then it is irreducible by (26.4.8.5). Then  $\rho_{\mathcal{F}}$  is unique by Chebotarev density and (3.6.2.13). Next we prove existence:

Choose  $\ell \in \Sigma_F^{\mathrm{fin}} \setminus S_F(\mathfrak{m}np)$ , and let  $L/M_{\mathcal{F}}$  be a finite extension containing all the Hecke eigenvalues of the newforms in  $\mathrm{SF}^0(\mathfrak{n}lp^{\infty}, \chi)^{\ell\text{-new}}$ . Let  $\mathbf{T} \subset \mathrm{End}(\mathrm{SF}^0(\mathfrak{n}lp^{\infty}, \chi; L)^{\ell\text{-new}})$  denote the endomorphism ring generated over  $\mathcal{O}_L$  by the Hecke operators  $\{T(\mathfrak{m}) : \ell \nmid \mathfrak{m}\}$ .

Let  $I = \mathrm{Ann}_{\mathbf{T}}(C_L(\ell)) \subset \mathbf{T}$  (26.4.2.21). Since  $T(\mathfrak{m}) - c(\mathfrak{m}, \mathcal{F}) \in I$  for each  $\mathfrak{m}$  prime to  $\ell$  (by definition (26.4.2.21)), we see that  $\mathbf{T}/I \cong \mathcal{O}_L/\mathfrak{b}_{\ell}$  for some  $\mathfrak{b}_{\ell} \in \mathrm{Ideal}^{\times}(\mathcal{O}_F)$ . Since  $C_{\mathcal{F}}(\ell) \otimes \mathcal{O}_L \subset C_L(\ell)$ , we have

$$\mathrm{div}_L(\mathfrak{b}_{\ell}) \geq -\mathrm{div}_L(\mathfrak{a}_{\mathcal{F}, \ell}) \geq \mathrm{div}_L(w_{\ell}) - \mathrm{div}_L(v) - c \mathrm{div}_L(1 + \|\ell\|) \quad (\star)$$

for some fixed  $v \in \Lambda_{\mathcal{F}}$  and  $c \in \mathbb{Z}$  by (26.4.2.22) and (26.4.2.24), where  $w_{\ell} = (\alpha_{\ell}(\mathcal{F})^2 - \chi(\mathfrak{p}_{\ell}))(\beta_{\ell}(\mathcal{F})^2 - \chi(\mathfrak{p}_{\ell}))$ .

Let  $\{\mathcal{F}_i(\mathfrak{a}_{ij}z)\}_{i,j}$  be a newform basis for  $\mathrm{SF}^0(\mathfrak{n}lp^{\infty}, \chi; L)^{\ell\text{-new}}$  where  $\mathcal{F}_i$  are newforms of level  $\mathfrak{m}$  s.t.  $\ell \nmid \mathfrak{m} \nmid lp^{\infty}$  (26.4.2.19), inducing an isomorphism  $\mathrm{SF}^0(\mathfrak{n}lp^{\infty}, \chi; L)^{\ell\text{-new}} \cong A = \oplus_{i,j} L$ . Then  $\mathbf{T}$  acts on  $A$ , then  $A \cong T \otimes L$  as a  $T$ -module, because the eigenvalues are different.

Now let  $\mathrm{Gal}_F$  acts on  $W = A \oplus A$  through  $\bigoplus_{i,j} \rho_{\mathcal{F}_i} \otimes L$ , where  $\rho_{\mathcal{F}_i}$  exists because a.e. specialization of  $\mathcal{F}_i$  is  $\ell$ -new ?, thus Steinberg at  $\ell$  by (26.4.1.19), so we can use lemma (26.4.8.7) below. This action commutes with the action of  $\mathbf{T}$  ?.



Suppose  $Q$  be a prime of  $\mathcal{O}_L$  containing  $\mathfrak{b}_\ell$ , with  $Q \cap \mathbf{T} = \mathfrak{P}$ . Pick a basis of  $W_{\mathfrak{P}}$  as  $T_{\mathfrak{P}} \otimes L$ -module consisting of  $\mathfrak{c}$ , then we get a representation

$$\rho : \text{Gal}_F \rightarrow \text{GL}(2; \mathbf{T}_{\mathfrak{P}} \otimes L)$$

and a pseudo-representation  $\tilde{\rho}$  valued in  $\mathbf{T}_{\mathfrak{P}}$  as in the proof of (26.4.7.1). By reducing modulo  $I$ , we get a pseudo-representation  $\tilde{\rho}$  valued in  $\mathbf{T}_{\mathfrak{P}}/I \cong \mathcal{O}_{L,Q}/\mathfrak{b}_\ell$ , s.t. for each prime  $\mathfrak{q} \nmid n\ell p$ ,  $\tilde{\rho}$  has the right trace and determinant modulo  $I$ . And by the proof of (26.4.7.1),  $\tilde{\rho}(\text{mod } I)$  takes value in  $\Lambda_{\mathcal{F},P}/\mathfrak{b}_\ell \cap \Lambda_{\mathcal{F},P}$ , thus reducing modulo  $P$ , we get a representation  $\rho_P : \text{Gal}_F \rightarrow \text{GL}(2; \widetilde{\Lambda_{\mathcal{F}}/P})$  with the right trace and character away from  $n\ell p$ .

Now by (26.4.8.9), we can pick an infinite set  $A = \{(P_i, \ell_i)\}$  of pairs where  $\ell_i \in \Sigma_F^{\text{fin}} \setminus S_F(n\ell p)$ ,  $P_i$  are prime divisors of  $\Lambda_{\mathcal{F}}$ , and  $v_{P_i}(w_{\ell_i}) > 0$ , s.t.  $S = \{\ell_i : \ell_i \in A\}$  has density zero, and we can assume  $v_{P_i}(\mathfrak{b}_{\ell_i}) > 0$  by  $(\star)$ . Then by the argument above, we obtain representations

$$\rho_i : \text{Gal}_F \rightarrow \text{GL}(2; \Lambda_{\mathcal{F}}/P_i).$$

unramified outside  $n\ell_i p$  and having desired characteristic polynomial of  $\sigma_v$  when  $v \in \Sigma_F^{\text{fin}} \setminus S_F(n\ell_i p)$ . Then by (26.4.7.1), we get a semisimple continuous Galois representation

$$\rho_A : \text{Gal}_F \rightarrow \text{GL}(2; M_{\mathcal{F}})$$

unramified outside  $S \cup S_F(n\ell p)$  and for  $v \in \Sigma_F^{\text{fin}} \setminus S \cup S_F(n\ell p)$ , the characteristic polynomial of  $\sigma_v$  is  $T^2 - c(\mathfrak{p}_v, \mathcal{F})T + \chi(\mathfrak{p}_v) \|v\|$ .

But since  $S$  has density zero, for any  $\ell \in S$ ,  $\rho_S \cong \rho_{S \setminus \{\ell\}}$ . Thus  $\rho$  is unramified outside  $S_F(n\ell p)$ , with the right characteristic polynomial of  $\sigma_v$ .

To show that  $\rho_{\mathcal{F}}|_{\text{Gal}_v}$  is of the required form, notice when modulo  $P_i$  for each  $i$ , it is of the required form, and we use the same method as in the proof of (26.4.8.7). ?  $\square$

**Lemma (26.4.8.7).** Situation as in (26.4.8.6), the theorem holds if either of the following holds:

- $[F : \mathbb{Q}]$  is odd,
- For infinitely many  $(k, \zeta) \in \mathfrak{X}$ , there exists a lifting  $\tilde{v}_{k,\zeta}(\mathcal{F})$  with the property that the associated  $\pi = \otimes'_v \pi_v \in \text{Irr}^{\text{auto}}(\text{GL}(2)/F)$  has some finite component that is Steinberg or supercuspidal.

┘

*Proof:* Cf. [W1188]P565. ?

Each of the two hypotheses implies representations  $\text{Gal}_F \rightarrow \text{GL}(2; \mathcal{O}_L/\tilde{P}_{k,\zeta})$  associated to  $\tilde{v}_{k,\zeta}(\mathcal{F})$  for infinitely many  $(k, \zeta) \in \mathfrak{X}$  by work of Carayol (26.4.8.2), and these representation are all odd, thus  $\rho_{\mathcal{F}}$  exists by (26.4.7.1).

For  $v \in S_F(p)$ , see the restriction of  $\rho_{\mathcal{F}}$  to  $\text{Gal}_v$ , let  $\varepsilon_2$  be the unramified representation of  $\text{Gal}_v$  s.t.  $\alpha(\sigma_v) = \alpha_v(\mathcal{F})$ , consider the two representations  $\rho$  and  $\varepsilon_2^{-1} \det \rho \oplus \varepsilon_2$ , then they have the same trace on  $\text{Gal}_v$  modulo  $\tilde{P}_{k,\zeta}$  for infinitely many  $(k, \zeta) \in \mathfrak{X}$ , so their traces are equal, and these two representations are isomorphic up to semisimplification by (3.6.2.13). It suffices to show that  $\rho|_{\text{Gal}_v}$  has a subrepresentation isomorphic to  $\varepsilon_2^{-1} \det \rho$ . If not,  $\varepsilon_2^{-1} \det \rho \neq \varepsilon_2$ , and  $\rho|_{\text{Gal}_v}$  is isomorphic to a representation of the form  $\sigma \mapsto \begin{bmatrix} \varepsilon_2(\sigma) & b_\sigma \\ & d_\sigma \end{bmatrix}$ . Then for infinitely many  $(k, \zeta) \in \mathfrak{X}$ ,  $\rho(\text{mod } \tilde{P}_{k,\zeta})$  is given by Carayol's method, and  $\varepsilon_2^{-1} \det \rho \neq \varepsilon_2(\text{mod } \tilde{P}_{k,\zeta})$ . In this case, (26.4.8.2) shows that  $\rho|_{\text{Gal}_v}$  has a subrepresentation isomorphic to  $\varepsilon_2^{-1} \det \rho$  ?, thus  $b_\sigma = 0$  for all  $\sigma$ , and  $\rho|_{\text{Gal}_v}$  is semisimple, thus also has a subrepresentation isomorphic to  $\varepsilon_2^{-1} \det \rho$ .  $\square$

**Lemma (26.4.8.8) [An Infinite Set of Primes].** ┘

*Proof:*

□

**Lemma (26.4.8.9) [An infinite Set of Divisors].** As  $\ell$  varies, there are infinitely many prime divisor  $P$  of  $\Lambda_{\mathcal{F}}$  dividing  $w_{\ell}$ . ┘

*Proof:* Suppose that when  $\ell$  varies, there exists a finite set of prime divisors  $P_i$  of  $\Lambda_{\mathcal{F}}$  and  $N_1 \in \mathbb{Z}_+$  s.t.  $\text{div } w_{\ell} \leq \sum N_1 \sum_i P_i$ , then the same holds true for  $\text{Nm}_{\Lambda_{\mathcal{F}}/\Lambda_{\mathcal{X}}} w_{\ell}$  in  $\Lambda_{\mathcal{X}}$ . Then there exists  $k \in \mathbb{Z}_+$  s.t.  $v_p(\text{Nm}(w_{\ell})(\zeta - 1)) < k$  where  $\zeta$  is as in (26.4.8.8). But for  $\ell \in S_k$  as in lemma (26.4.8.8), we have  $v_p(w_{\ell}(\zeta - 1)) \geq k$ , contradiction.

Then if the assertion is not true, then there exists a prime divisor  $P$  of  $\Lambda_{\mathcal{F}}$  s.t. for any  $k, n \in \mathbb{Z}_+$ ,  $v_P(w_{\ell}) \geq k$  for  $\ell$  in a positive portion of primes in  $S_n$  (26.4.8.8).

But this cannot happen: ?

□

### Hecke Action on Newforms

**Def. (26.4.8.10).** ┘

## 26.5 Siegel Modular Forms

### 1 Metaplectic Groups

**Def. (26.5.1.1) [Metaplectic Groups].** The group  $\mathrm{SL}(2, \mathbb{R})$  has a double cover  $\widetilde{\mathrm{Sp}}(2; \mathbb{R})$  called the **metaplectic group**:

$$\widetilde{\mathrm{Sp}}(2; \mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \pm\sqrt{c\tau + d} \right\} \in \mathrm{SL}(2; \mathbb{R}) \rtimes \mathcal{O}(\mathcal{H}),$$

where  $\mathrm{SL}(2; \mathbb{R})$  acts on  $\mathcal{O}(\mathcal{H})$  as usual.

And  $\mathrm{Mp}(2; \mathbb{Z}) \subset \mathrm{Mp}(2; \mathbb{R})$  is the subgroup consisting of elements of the form

$$\mathrm{Mp}(2, \mathbb{Z}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \pm\sqrt{c\tau + d} \right\} \in \mathrm{SL}(2; \mathbb{Z}) \rtimes \mathcal{O}(\mathcal{H}).$$

And for any subgroup  $H \subset \mathrm{SL}(2; \mathbb{R})$ , denote  $\widetilde{H}$  the inverse image of  $H$  in  $\mathrm{Mp}(2, \mathbb{R})$ . Also denote  $\mathrm{Mp}(2; \mathbb{Z}) = \widetilde{\mathrm{SL}(2; \mathbb{Z})}$ , called the **metaplectic modular group**.  $\lrcorner$

### 2 Examples

### 3 Serre Duality

**Thm. (26.5.3.1) [Borcherds].** Situation as in (26.3.1.3), and suppose  $\rho = \sigma_k$  on  $\Gamma \cap \widetilde{\mathrm{SO}}(2)$ , then  $\mathrm{MF}_{2-k}^!(\Gamma; \rho)$  is dual to  $\mathrm{MF}_k(\Gamma; \rho^*)$  via the sum of the perfect pairings at cusps

$$\bigoplus_{k \in \text{cusp}} \mathrm{PowSer}_k(\Gamma, \rho^*) \times \bigoplus_{k \in \text{cusp}} \mathrm{SingLaur}_k(\Gamma, \rho) \rightarrow \mathbb{C} : \langle (f_k), (\varphi_k) \rangle = \sum_k \mathrm{res}(f_k \varphi_k \, \mathrm{dlog} \, q_k).$$

$\lrcorner$

*Proof:*

$\square$

## 26.6 Adelic Automorphic Representations

References are [B-J79, Bum97, Gan07, G-H11, G-H24], [Automorphic Forms on  $GL(2)$ , Jacquet and Langlands(1970)]. [Introduction to Langlands Program, Cogdell].

**Notation (26.6.0.1).**

- Fix  $F \in \mathbf{NField}$ .
- Use notations on ideles and adeles defined in [Adeles and Ideles](#).
- Fix  $\psi = \prod_v \psi_v$  an additive character of  $\mathbf{A}_F/F$ .
- Fix a connected reductive group  $G \in \mathbf{RedGrp}/F$ .
- Fix a maximal compact subgroup  $\mathcal{K}_\infty \leq G(F \otimes \mathbb{R})$  (the choice of  $\mathcal{K}_\infty$  is not so important because any two maximal compact subgroups of  $G(F \otimes \mathbb{R})$  are conjugate (10.5.5.21)), a compact open subgroup  $\mathcal{K}_f \leq G(\mathbf{A}_{F,f})$ , and denote  $\mathcal{K} = \mathcal{K}_f \mathcal{K}_\infty \leq G(\mathbf{A}_F)$ .
- Use notations on arithmetics of algebraic groups defined in [Arithmetic of Algebraic Groups](#).
- Let  $Z$  be the largest  $F$ -split torus contained in the center of  $G$ .
- Use notations on smooth representations of  $p$ -adic groups defined in [Admissible Representations of  \$p\$ -adic Reductive Groups](#).
- Use notations on Archimedean automorphic forms defined in [Archimedean Automorphic Representations](#).
- If  $G = GL(n)$ , use notations on subgroups of  $GL(n)$  defined in [Group-Theoretic Notations for  \$GL\(n\)\$](#) , and use notations on smooth representations of  $GL(n)$  defined in [Admissible Representations of  \$GL\(n\)\$  over  \$p\$ -adic Fields](#).
- Fix an admissible character  $\omega \in ([Z]_F)^\vee$ . Notice if  $G = GL(n)$ ,  $\omega$  is just a Hecke character.

┘

**Def. (26.6.0.2) [Lie Algebra].** Let  $\mathfrak{g}_\infty = \mathrm{Lie}(G(\mathbf{A}_\infty))$ ,  $\mathcal{Z}(\mathfrak{g}) = Z(U(\mathfrak{g}_\infty))$ .

┘

**Def. (26.6.0.3).** For  $\alpha \in F^{n-1}$ , let  $\psi_{N,\alpha}$  be a character of  $N(F)$  given by  $\psi_N(g) = \sum_i \psi(\alpha_i g_{i,i+1})$ . For  $\alpha = (1, \dots, 1)$ , denote  $\psi_{N,\alpha}$  by  $\psi_N$ .

┘

**Def. (26.6.0.4) [Adelic Norm].** Take an embedding  $G \hookrightarrow GL(n)$  over  $F$ . Define the norm on  $G(\mathbf{A}_F)$  by

$$\|g\| = \prod_v \max(|g_{ij}|_v, |g_{ij}^{-1}|_v).$$

Notice that if  $v \in \Sigma_F^{\mathrm{fin}}$  and  $g \in GL(m; \mathcal{O}_v)$ , then  $\|g\|_v = 1$ , so it is definable. And  $\|g\|$  is bounded below by a positive constant  $c$ .

┘

**Prop. (26.6.0.5).** Choose an Adelic norm (26.6.0.4), then there exists  $C \in \mathbb{R}_+$  s.t.

$$\|xy\| \leq C \|x\| \|y\|.$$

┘

## 1 Automorphic Representations

### Admissible Representations and Tensor Product Theorem

**Def. (26.6.1.1) [Admissible Representations].** A **smooth representation** of  $G(\mathbf{A}_F)$  is defined to be a TVS  $V \in \text{TVS}/\mathbb{C}$  with a commuting smooth  $G(\mathbf{A}_{F,f})$ -action and a  $(\mathfrak{g}_\infty, K_\infty)$ -module structure (10.5.5.22). The category of smooth representations of  $G(\mathbf{A}_F)$  is denoted by  $\text{Rep}^{\text{alg}}(G/F)$ .

Any  $(\pi, V) \in \text{Rep}^{\text{alg}}(G(\mathbf{A}_F))$  induces a representation of  $\mathcal{K} = \mathcal{K}_f \times \mathcal{K}_\infty$ . Then this representation is called an **admissible representation** iff every vector is  $\mathcal{K}$ -finite, and  $\dim V^\rho < \infty$  for any  $\rho \in \text{Irr}(\mathcal{K})$ . By (24.1.6.31), it can be checked that this is equivalent to: for any irreducible representation  $\rho_\infty$  of  $K_\infty$  and an open compact subgroup  $U \subset \mathcal{K}_f$ ,  $\dim V^{\rho_\infty} \cap V^U < \infty$ . The category of admissible representations of  $G(\mathbf{A}_F)$  is denoted by  $\text{Rep}^{\text{adm}}(G/F)$ .  $\lrcorner$

**Def. (26.6.1.2) [Restricted Tensor Representation].** Given a set of locally compact groups  $G_v$  and a.e. their compact subgroups  $K_v$ ,  $(\rho_v, V_v) \in \text{Rep}^{\text{alg}}(G_v)$ ,  $\xi_v^0 \in V_v^{K_v}$  are given for a.e.  $v$ , then we can define the **restricted tensor representation**

$$(\rho, V) = \bigotimes_v' (\rho_v, V_v, \xi_v^0)$$

of  $G(\mathbf{A}_F) = \prod' (G_v, K_v)$  on  $\bigotimes_v' V_v$  by

$$\rho(\bigotimes_v g_v)(\bigotimes_v \xi_v) = \bigotimes_v \rho_v(g_v) \xi_v.$$

$\lrcorner$

**Def. (26.6.1.3) [Global Hecke Algebra].** For  $v \in \Sigma_F$ , let  $\mathcal{H}_{G(F_v)}$  be the Hecke algebras constructed in (24.1.6.18) and (24.7.5.16). As  $\mathcal{H}_{G(F_v)}$  has a spherical idempotent  $e_v^0 = e_{K_v}$  (26.6.1.1), we can define the **global Hecke algebra**

$$\mathcal{H}_{G(\mathbf{A}_F)} = \bigotimes_v' (\mathcal{H}_{G(F_v)}, e_v^0).$$

Then by definition of representations in (26.6.1.17) and (24.7.5.17)(24.1.6.22),

$$\text{Rep}^{\text{alg}}(G(\mathbf{A}_F)) = \text{Rep}^{\text{alg}}(\mathcal{H}_{G(\mathbf{A}_F)}).$$

$\lrcorner$

**Thm. (26.6.1.4) [Tensor Product Theorem, Flath [Fla79]].** For  $(\rho, V) \in \text{Irr}^{\text{adm}}(G/F)$ , there exists uniquely for each  $v \in \Sigma_F^\infty$  a  $(\rho_v, V_v) \in \text{Irr}^{\text{adm}}((\mathfrak{g}_\infty, K_v))$ , and for each  $v \in \Sigma_F^{\text{fin}}$  a  $(\pi_v, V_v) \in \text{Irr}^{\text{alg}}(G(F_v))$  s.t. for a.e.  $v$ ,  $V_v$  contains a non-zero  $\mathcal{K}_v$ -fixed vector  $\xi_v^0$ , and

$$(\rho, V) = \bigotimes_v' (\rho_v, V_v, \xi_v^0).$$

Conversely, any such a restricted tensor product is in  $\text{Irr}^{\text{adm}}(G/F)$ .  $\lrcorner$

*Proof:* By considering the global Hecke algebra, this follows immediately from (26.6.1.3) and (24.7.5.17)(24.1.6.22).  $\square$

**Cor. (26.6.1.5) [Contragradient Representations].** For any  $\rho = \bigotimes_v \rho_v \in \text{Irr}^{\text{adm}}(G/F)$ , we can define the **contragradient representation** of  $\rho$  as  $\rho^\vee = \bigotimes_v \rho_v^\vee$  (24.1.2.9)  $\in \text{Irr}^{\text{adm}}(G/F)$ .  $\lrcorner$

**Cor. (26.6.1.6)** [Irreducible Representations of  $K$ ].

$$\mathrm{Irr}(K) = \bigotimes_v^{\prime} \mathrm{Irr}(K_v).$$

┘

*Proof:* This is similar to the proof of (26.6.1.4). □

### Automorphic Representations

**Def. (26.6.1.7)** [ $L^2$ -Space]. Define  $L^2(G/F, \omega)$  the space of all measurable functions  $\varphi$  on  $G(\mathbf{A}_F)$  s.t.

$$\varphi(zg) = \omega(z)\varphi(g), \forall z \in Z(\mathbf{A}), \quad \varphi(\gamma g) = \varphi(g), \forall \gamma \in G(F)$$

and  $\varphi$  is square integrable module the center:

$$\int_{[G]_F} |\varphi(g)|^2 dg < \infty.$$

w.r.t. the Tamagawa measure on  $G(\mathbf{A}_F)$  (20.5.4.5).

Then the right action of  $G(\mathbf{A}_F)$  on  $L^2(G/F, \omega)$  is continuous, by (24.7.1.14).

We also denote  $L^2([G]) \triangleq L^2(G/F, \mathbf{1})$ . ┘

**Remark (26.6.1.8)** [Motivation]. To motivate this definition, notice in the isomorphism

$$L^2(\mathrm{SL}(2; \mathbb{Z}) \backslash \mathcal{H}) \cong L^2(\mathrm{SL}(2; \mathbb{Q}) \backslash \mathrm{SL}(2; \mathbf{A}) / \mathrm{SO}(2; \mathbb{R}) \mathcal{K}),$$

removing  $K$  gives cusp modular functions of all level, and removing  $\mathrm{SO}(2; \mathbb{R})$  gives the Hilbert space of all cusp forms ?. Thus  $L^2(G(F) \backslash G(\mathbf{A}_F))$  gives all the informations needed. ┘

**Def. (26.6.1.9)** [Cuspidality]. A measurable function  $\varphi : [G]_F \rightarrow \mathbb{C}$  is called **cuspidal** iff for any proper parabolic subgroup  $P = MU$ , where  $U$  is the unipotent radical, we have

$$\int_{[U]_F} \varphi(ng) dn = 0 \quad (20.5.4.3)$$

for a.e.  $g \in A_G \backslash G(\mathbf{A}_F)$ . Notice this integral is convergent when  $\varphi$  is continuous because  $[U]$  is compact by (20.5.4.8). (Why is it definable for measurable functions ?)

In fact, it suffices to check for a set of representative of conjugacy classes of maximal parabolic subgroups  $P$ , by [HC68] ?.

The closed space of all cuspidal elements in  $L^2([G])$  is denoted by  $L_0^2([G], \omega)$  or  $L_{\mathrm{cusp}}^2([G], \omega)$ .  $L_0^2([G], \omega)$  is stable under the right action of  $A_G \backslash G(\mathbf{A}_F)$ . ┘

**Example (26.6.1.10)**. A function  $\varphi \in L^2(\mathrm{GL}(n)/F, \omega)$  is cuspidal (26.6.1.9) iff

$$\int_{\mathrm{Mat}_{r \times s}(F) \backslash \mathrm{Mat}_{r \times s}(\mathbf{A}_F)} \varphi \left( \begin{bmatrix} \mathbf{1}_r & X \\ & \mathbf{1}_s \end{bmatrix} g \right) dX = 0$$

a.e.  $g$  for any  $r + s = n, 1 < r < n$ . ┘

**Def. (26.6.1.11)** [Adelic Smooth Functions].

- The space  $C^\infty(G(\mathbf{A}_F))$  of **smooth functions on  $G(\mathbf{A}_F)$**  is defined to be the restricted tensor product  $\bigotimes' C^\infty(G(F_v))$  w.r.t.  $f_{v,0} = \mathbf{1}_{\mathcal{K}_v}$ .
- The space  $\mathcal{T}(G(\mathbf{A}_F))$  of **adelic compactly supported smooth functions** or **adelic test functions** on  $G(\mathbf{A}_F)$  is defined to be the restricted tensor product  $\bigotimes' \mathcal{T}(G_v)$  w.r.t.  $f_{v,0} = \mathbf{1}_{\mathcal{K}_v}$ .  $\mathcal{T}(G(\mathbf{A}_F))$  acts on  $L^2(G(F) \backslash G(\mathbf{A}_F), \omega)$  on the right.
- $G(\mathbf{A}_F)$  acts on  $C^\infty(G(\mathbf{A}_F))$  by right translation, and induces an action of  $\mathfrak{g}_\infty$  thus an action of  $U(\mathfrak{g}_\infty)$  on it. A function  $f$  is called  **$\mathcal{Z}(\mathfrak{g})$ -finite** if it is contained in a f.d. space that is invariant under the action of  $\mathcal{Z}(\mathfrak{g})$  (26.6.0.2).
- A function  $f : G(\mathbf{A}_F) \rightarrow \mathbb{C}$  is said to **have moderate growth** iff  $|f(g)| \leq C \|g\|^N$  (26.6.0.4) for some constant  $C > 0$  and any  $g \in G(\mathbf{A}_F)$ .
- A function  $f : G(\mathbf{A}_F) \rightarrow \mathbb{C}$  is said to be **rapidly decreasing** if for any  $k \in \mathbb{Z}_+$ ,  $|f(g)| \leq C_k \|g\|^{-k}$  for any  $g \in G(\mathbf{A}_F)$  and some constant  $C_k$ .

┘

**Prop. (26.6.1.12) [Moderate Growth and Siegel-Sets].** A function  $\varphi : [G] \rightarrow \mathbb{C}$  is of moderate growth (on  $G(\mathbf{A}_F)$ ) (26.6.1.11) if for any Siegel set  $\mathcal{S}$ , there exists  $(c, r) \in \mathbb{R}_+ \times \mathbb{R}$  s.t.

$$|\varphi(nmsk)| \leq c\alpha(s)^r$$

for all  $(n, m, s, k) \in \Omega_N \times \Omega_M \times A_{T_0}(t) \times \mathcal{K}$  and  $\alpha \in \Delta$ .

┘

*Proof:* Cf. [G-H24]P162.  $\mathcal{S}$

□

**Def. (26.6.1.13) [Adelic Automorphic Forms].** We denote by  $\mathcal{A}(G/F, \omega)$  the space of **adelic automorphic forms** consisting of smooth functions  $\varphi$  on  $G(\mathbf{A}_F)$  (26.6.1.11) s.t.

1.  $\varphi(z\mathbf{g}) = \omega(z)\varphi(\mathbf{g}), \forall z \in Z(\mathbf{A}_F)$ .
2.  $\varphi(\gamma\mathbf{g}) = \varphi(\mathbf{g})$ , where  $\mathbf{g} \in G(\mathbf{A}_F), \gamma \in G(F)$ .
3.  $\varphi$  is  $\mathcal{K}$ -finite and  $\mathcal{Z}(\mathfrak{g})$ -finite (26.6.1.11).
4.  $\varphi$  of moderate growth (26.6.1.11).

And denote by  $\mathcal{A}_0(G/F, \omega)$  the automorphic forms that is cuspidal in sense of (26.6.1.9), called the space of **cuspidal forms**. If we want to emphasize  $\mathcal{K}$  then we write  $\mathcal{A}(G/F, \mathcal{K}, \omega) \triangleq \mathcal{A}(G/F, \omega)$ .

$\mathcal{A}(G/F, \omega)$  is a  $(\mathfrak{g}_\infty, \mathcal{K}_\infty)$ -module (24.6.3.1), and  $G(\mathbf{A}_{F,f})$  acts smoothly on it making it an admissible  $G(\mathbf{A}_{F,f})$ -representation (24.1.6.29):

$$(\mathfrak{g}, \mathcal{K}_\infty) \times G(\mathbf{A}_{F,f}) \times \mathcal{A}(G/F, \omega) \rightarrow \mathcal{A}(G/F, \omega) : (X, k, \mathbf{g}, \Phi) \mapsto (x \mapsto X\Phi(xk\mathbf{g})).$$

These two actions are commutative, and the subspace  $\mathcal{A}_0(G/F, \omega)$  is stable under both actions. Thus  $\mathcal{A}(G/F, \omega), \mathcal{A}_0(G/F, \omega) \in \text{Rep}^{\text{adm}}(G/F)$  (26.6.1.1). ┘

**Prop. (26.6.1.14) [Adelic Automorphic Forms and Archimedean Automorphic Forms].** Suppose  $f$  is killed by an ideal  $J_\infty$  of finite codimension in  $\mathcal{Z}(\mathfrak{g})$ , and is invariant under the action of a compact open subset  $\mathcal{K}_f \leq G(\mathbf{A}_{F,f})$ . Let

$$G(F) \backslash G(\mathbf{A}_F) / \mathcal{K}_f = \coprod_i \Gamma_i \backslash G(F \otimes \mathbb{R})$$

as in (20.5.3.24), then there is an isomorphism of  $(\mathfrak{g}_\infty, K_\infty)$ -modules

$$\mathcal{A}(G(F)\backslash G(A), \omega, J_\infty)^{\mathcal{K}_f} \cong \bigoplus_i \mathcal{A}(\Gamma_i \backslash G(A_\infty), \mathbf{1}, \omega_\infty, J_\infty) \quad (26.1.5.10),$$

┘

**Cor. (26.6.1.15).** If a smooth function  $f$  on  $G(\mathbf{A}_F)$  satisfies item 1-3 in (26.6.1.13) and  $f \in L^2(G/F, \omega)$  (26.6.1.7), then  $f$  is automatically of moderate growth (26.6.1.11) and thus an automorphic form. ┘

*Proof:* This follows from (26.6.1.14) and (26.1.1.8). ┐

**Prop. (26.6.1.16) [Analytic Properties].** Any automorphic form  $f$  is real analytic when restricted to  $G(\mathbf{A}_\infty)$ , and is of moderate growth.

Any cusp form  $f$  is rapidly decreasing. In particular,  $f \in L^2_{\text{cusp}}(G/F)$ . ┘

*Proof:* It is real analytic by (26.1.1.15). The growth condition reduces immediately to the Archimedean case (26.1.4.7). ┐

**Def. (26.6.1.17) [Automorphic Representations].**  $\mathcal{A}(G/F, \omega), \mathcal{A}_0(G/F, \omega) \in \text{Rep}^{\text{adm}}(G/F)$  by definition (26.6.1.13). So  $\pi \in \text{Irr}^{\text{adm}}(G/F)$  is called

- an **automorphic representation** of  $G$  over  $F$  if  $\pi$  can be realized as a subquotient of  $\mathcal{A}(G/F, \omega)$  for some  $\omega$ ,
- an **automorphic cuspidal representation** of  $G$  over  $F$  if  $\pi$  can be realized as a subrepresentation of  $\mathcal{A}_0(G(F)\backslash G(\mathbf{A}_F), \omega)$  for some  $\omega$ .
- an **tempered automorphic representation** of  $G$  over  $F$  if  $\pi$  is an automorphic representation s.t. for each  $v \in \Sigma_F$ ;  $\pi_v \in \text{Irr}^{\text{adm}}(G(F_v))$  (or  $\text{Mod}^{\text{adm}}_{(\mathfrak{g}_v, \mathcal{K}_v)}$ ) is tempered (24.9.8.11) (24.6.4.11).

The category of automorphic representations of  $G(\mathbf{A}_F)$  is denoted by  $\text{Irr}^{\text{auto}}(G/F, \omega)$ . And the category of cuspidal representations of  $G(\mathbf{A}_F)$  is denoted by  $\text{Irr}^{\text{cusp}}(G/F, \omega)$ . ┘

**Remark (26.6.1.18).** There is a general method of constructing subrepresentations of  $\mathcal{A}(G(F)\backslash G(\mathbf{A}_F), \omega)$  using Eisenstein series ?, but no known methods for constructing subrepresentations of  $\mathcal{A}_0(G(F)\backslash G(\mathbf{A}_F), \omega)$ . ┘

**Thm. (26.6.1.19) [Automorphic Representations are Admissible, Harish-Chandra].**

$$\text{Irr}^{\text{cusp}}(G/F, \omega) \subset \text{Irr}^{\text{auto}}(G/F, \omega) \subset \text{Irr}^{\text{adm}}(G/F).$$

┘

*Proof:* Let  $V_1 \subset V_2 \subset \mathcal{A}(G(F)\backslash G(A), \omega)$  be submodules s.t  $V_2/V_1 \cong \pi \in \text{Irr}^{\text{auto}}(G/F, \omega)$ , then we can assume  $V_2$  is generated by an element  $f \in V_2 \setminus V_1$ , because otherwise change  $V_2$  by the submodule  $V'_2$  generated by  $f$  and  $V_1$  by  $V_1 \cap V'_2$ .

Suppose  $f$  is killed by an ideal  $J_\infty$  of finite codimension in  $\mathcal{Z}(\mathfrak{g})$ , and is invariant under the action of a compact open subset  $\mathcal{K}_f \leq G(\mathbf{A}_{F,f})$ . Let

$$G(F)\backslash G(\mathbf{A}_F)/\mathcal{K}_f = \coprod_{i=1}^h \Gamma_i \backslash G(F \otimes \mathbb{R})$$



as in (20.5.3.24), then there is an isomorphism of  $(\mathfrak{g}_\infty, \mathcal{K}_\infty)$ -modules

$$\mathcal{A}(G(F) \backslash G(\mathbf{A}_F), \omega, J_\infty)^{\mathcal{K}_f} \cong \bigoplus_{i=1}^h \mathcal{A}(\Gamma_i \backslash G(F \otimes \mathbb{R}), \mathbf{1}, \omega_\infty, J_\infty) \quad (26.1.5.10),$$

which is admissible by the fundamental theorem of Harish-Chandra (26.1.5.10). Thus  $V_2 \subset \mathcal{A}(G(F) \backslash G(\mathbf{A}_F), \omega, J_\infty)^{\mathcal{K}_f}$  is also admissible, and so is  $\pi$ .  $\square$

**Def. (26.6.1.20) [Galois Conjugate of Automorphic Representations].** For  $\pi \in \text{Irr}^{\text{auto}}(G/F)$  and any  $\sigma \in \text{Aut}(F)$ , defined  $\sigma(\pi)$  to be the set of functions  $\sigma(\pi)(g)(x) = g(\sigma^{-1}(x))$ . Then  $\sigma(\pi)$  is also automorphic.  $\lrcorner$

*Proof:* If  $\pi$  is realized in the subspace of  $\mathcal{A}(G/F, \omega)$  of the form  $\{f(g)\}_{f \in V}$ , then  $\sigma(\pi)$  is realized in the subspace of  $\mathcal{A}(G/F, \omega)$  of the form  $\{f(\sigma(g))\}_{f \in V}$ .  $\square$

### Discrete Automorphic Representations

**Def. (26.6.1.21) [Discrete (Unitary) Automorphic Representations].** A **discrete automorphic representations** of  $G(\mathbf{A}_F)$  is an irreducible unitary representation  $\pi$  of  $G(\mathbf{A}_F)$  that is equivalent to a subrepresentation of  $L^2(G/F, \omega)$ .

And it is called a **discrete cuspidal automorphic representation** if it is equivalent to a subrepresentation of  $L^2_{\text{cusp}}(G/F, \omega)$ . We will see in (26.6.2.5) that a discrete cuspidal automorphic representation is naturally in bijection with cuspidal automorphic representation.  $\lrcorner$

**Prop. (26.6.1.22) [Discrete-Series Representations].** If  $(\pi, \mathcal{H})$  is a discrete automorphic representation of  $G(\mathbf{A}_F)$  (26.6.1.21), then  $V^{\mathcal{K}\text{-fin}}$  is an  $A_G$ -trivial automorphic representation of  $G$  over  $F$  (26.6.1.17). And two discrete automorphic representations  $(\pi_1, \mathcal{H}_1), (\pi_2, \mathcal{H}_2)$  of  $G(\mathbf{A}_F)$  are equivalent iff  $V_1^{\mathcal{K}\text{-fin}} \cong V_2^{\mathcal{K}\text{-fin}} \in \text{Irr}^{\text{auto}}(G/F)$ .

So we can define a **discrete-series representation** of  $G$  over  $F$  to be an automorphic representation  $(\pi, V) \in \text{Irr}^{\text{auto}}(G/F)$  that is equivalent to the  $\mathcal{K}$ -finite part of some discrete automorphic representation of  $G$  (26.6.1.21).  $\lrcorner$

*Proof:* Cf. [Getz]P172. ?  $\square$

**Def. (26.6.1.23) [Weakly-Embedded Representations].** A discrete-series  $\pi \in \text{Irr}^{\text{uni}}(A_G \backslash G(\mathbf{A}_F))$  is called a **weakly-embedded representation** if there exists a **tempered embedding**  $\pi^{\mathcal{K}\text{-fin}} \hookrightarrow C^\infty([G]_F)$  s.t. the image is  $L^{2+\varepsilon}$  for any  $\varepsilon \in \mathbb{R}_+$ .  $\lrcorner$

### Parabolic Inductions

References are [Lan79].

**Notation (26.6.1.24).**

- Let  $G \in \text{RedGrp}/F$ .

$\lrcorner$

**Prop. (26.6.1.25) [Parabolic Inductions, Langlands].** Let  $P \leq G$  be a parabolic subgroup with a Levi subgroup  $M$ , and  $\sigma = \otimes'_v \sigma_v \in \text{Irr}^{\text{cusp}}(M/F)$ , then

$$I_{P(\mathbf{A}_F)}^{G(\mathbf{A}_F)}(\sigma) = \bigotimes_v^I I_{P(F_v)}^{G(F_v)}(\sigma_v) \in \text{Rep}^{\text{adm}}(G(\mathbf{A}_F)) \quad (26.6.1.1).$$

Then for a.e.  $v \in \Sigma_F$ ,  $\sigma_v$  has a unique unramified constituent  $\pi_v^0$ , and the constituents of  $I_{P(\mathbf{A}_F)}^{G(\mathbf{A}_F)}(\sigma)$  are exactly those representations  $\pi = \otimes'_v \pi_v$  where  $\pi_v$  is a constituent of  $I_{P(F_v)}^{G(F_v)}(\sigma_v)$  and  $\pi_v = \pi_v^0$  for a.e.  $v$ .  $\lrcorner$

*Proof:* Cf.[Automorphic Representations, Langlands]  $\square$

**Remark (26.6.1.26) [Cuspidal Support is not Unique].** The pair  $(M, \sigma)$  may not be unique up to conjugacy. It is unique for  $\mathrm{GL}(n)$ , by [J-S81a]Thm4.4, but in general it is false. For example, Waldspurger showed for  $G = \mathrm{PGSp}(4)$  that there are cuspidal representations  $\pi$  that are abstractly isomorphic to a subquotient of some  $I_P(\sigma)$  with  $\sigma$  cuspidal on  $M = \mathrm{GL}(2)$ . These are called **CAP representations**.  $\lrcorner$

**Thm. (26.6.1.27) [Any Automorphic Representation comes from a Parabolic Induction].**  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(G(\mathbf{A}_F))$  is automorphic (26.6.1.17) iff it is a constituent of  $I_P^G(\sigma)$  for some parabolic subgroup  $P \leq G$  with a Levi subgroup  $M$  and  $\sigma \in \mathrm{Irr}^{\mathrm{cusp}}(M/F)$ .  $\lrcorner$

*Proof:* Cf.[Automorphic Representations, Langlands]  $\square$

**Def. (26.6.1.28) [Isobaric Automorphic Representations].** By (26.6.1.25) and (26.6.1.27), any  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$  is a subquotient of  $I_{P(\mathbf{A}_F)}^{G(\mathbf{A}_F)}(\sigma; \lambda)$  where  $P = LU \leq G$  is a parabolic,  $\sigma \in \mathrm{Irr}^{\mathrm{cusp}}(L/F)$  and  $\lambda \in (\mathfrak{a}_L)_{\mathbb{C}}^*$  (Cf. [G-H24]Chap10). Then  $\pi$  is called an **isobaric automorphic representation** if  $\pi$  is the canonical subquotient (the Langlands quotient (26.6.1.25)), i.e.  $\pi_v = \pi_{1,v} \boxplus \dots \boxplus \pi_{r,v}$  for any  $v \in \Sigma_F$ . An isobaric automorphic representation is denoted by

$$\pi = \pi_1 \boxplus \dots \boxplus \pi_r.$$

If  $G = \mathrm{GL}(n)$ , then by LLC,  $\pi$  being isobaric is equivalent to the  $L$ -parameter of  $\pi_v$  being the direct sum of  $L$ -parameters of  $\pi_v$  for each  $v \in \Sigma_F$ .  $\lrcorner$

### Central Extensions

**Prop. (26.6.1.29).** If  $1 \rightarrow \mathbb{G}_m \rightarrow G' \rightarrow G \rightarrow 1 \in \mathrm{RedGrp}/F$  is a central extension, then automorphic forms on  $G$  are identified with automorphic forms on  $G'$  that are trivial on  $\mathbb{G}_m$ .  $\lrcorner$

*Proof:* ?  $\square$

## 2 Cuspidal Spectra

**Thm. (26.6.2.1) [Gelfand-Piatetski-Shapiro].** The subspace  $L_{\mathrm{cusp}}^2([G]) \subset L^2([G])$  is closed, and  $R_{\mathrm{cusp}}(f)$  is a trace class for each  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$ .  $\lrcorner$

*Proof:* Cf.[Getz]P241.  $\square$

**Cor. (26.6.2.2) [Gelfand-Piatetski-Shapiro].** There is a Hilbert space direct sum decomposition

$$L_{\mathrm{cusp}}^2([G]) = \widehat{\bigoplus_{\pi \in \mathrm{Irr}^{\mathrm{uni}}(A_G \backslash G(\mathbf{A}_F))} m(\pi) \pi},$$

where  $m(\pi) < \infty$  for any  $\pi \in \mathrm{Irr}^{\mathrm{uni}}(A_G \backslash G(\mathbf{A}_F))$ .  $\lrcorner$

*Proof:* Cf.[Getz]P241.  $\square$

**Lemma (26.6.2.3) [Gelfand-Graev-Piatetski-Shapiro].** Let  $\varphi \in C_c^\infty(G(\mathbf{A}_F))$ , then

- There exists a constant  $C(\varphi)$  that for all  $f \in L_{\text{cusp}}^2(G/F, \omega)$ , we have  $\|\rho(\varphi)f\|_\infty \leq C(\varphi) \|f\|_2$ .
- $\rho(\varphi)$  is a compact operator on  $L_{\text{cusp}}^2(G/F, \omega)$ .

┘

*Proof:* The proof is the same as that of (26.1.5.6), but use global Siegel sets (20.5.4.16), Cf. [Bump, P297] ? .

□

**Thm. (26.6.2.4) [Cuspidal Spaces Totally Decomposable, Gelfand-Graev-Piatetski-Shapiro].** The space  $L_{\text{cusp}}^2(G(F) \backslash G(\mathbf{A}_F), \omega)$  (26.6.1.7) decomposes into a Hilbert space direct sum of irreducible invariant subspaces over  $G(\mathbf{A}_F)$ .

┘

*Proof:* The proof is exactly the same as (26.1.5.2), but where we use (26.6.2.3) in place of (29.1.4.8) and lemma ?? in place of lemma (24.7.5.2).

□

**Thm. (26.6.2.5) [Irreducible Cuspidal Representations are Admissible, Gelfand-Graev-Piatetski-Shapiro [GGS69]].** If  $(\pi, V) \in \text{Irr}^{\text{uni}}(G(\mathbf{A}_F))$  is contained in the decomposition of  $\mathcal{H} = L_{\text{cusp}}^2(G/F, \omega)$  in (26.6.2.4), then  $V^{\mathcal{K}\text{-fin}} \subset V$  is dense,  $V^{\mathcal{K}\text{-fin}} \subset \mathcal{A}_0(G/F, \omega)$  (26.6.1.13) and  $(\pi, V^{\mathcal{K}\text{-fin}}) \in \text{Irr}^{\text{adm}}(G/F, \omega)$ .

And there is a decomposition

$$L_{\text{cusp}}^2(G(F) \backslash G(\mathbf{A}_F), \omega)^{\mathcal{K}\text{-fin}} = \mathcal{A}_0(G(F) \backslash G(\mathbf{A}_F), \omega) \cong \bigoplus_{\pi \in \text{Irr}^{\text{cusp}}(G/F, \omega)} m(\pi) \pi,$$

with  $m(\pi) < \infty$ .

In particular, discrete cuspidal automorphic forms for  $G$  are naturally in bijection with cuspidal automorphic forms for  $G$  (26.6.1.21), compatible with the decomposition (26.6.2.2).

┘

*Proof:* Cf. [G-H24] P265.

This is general by (24.7.5.7), and for the containment  $V^{\mathcal{K}\text{-fin}} \subset \mathcal{A}_0(G(F) \backslash G(\mathbf{A}_F), \omega)$ , use a similar argument as in (26.1.5.8) using a similar lemma as lemma (26.1.5.7).

$V^{\mathcal{K}\text{-fin}}$  is admissible by (26.6.1.19).  $m(\pi) < \infty$  by the similar proof as in (26.1.5.9). ?

□

### 3 $\text{Irr}^{\text{auto}}(\text{GL}(n)/F)$

References are [Bum97] Chap3, [G-H11, G-H11b].

#### Basics

**Prop. (26.6.3.1).**  $\text{GL}(n; \mathbf{A}_F)$  is unimodular.

┘

*Proof:* This is because  $\text{GL}(n; F_v)$  is unimodular for any  $v \in \Sigma_F$  and because we can calculate the restricted product measure by (12.10.3.24).

□

**Def. (26.6.3.2) [Level Subgroups].** For  $\mathfrak{m} \in \text{Ideal}^\times(\mathcal{O}_F)$ , define compact open subgroups

$$\mathcal{K}_0(\mathfrak{m}) = \text{pr}^{-1}(B(\mathcal{O}_F/\mathfrak{m})), \quad \mathcal{K}_{0,1}(\mathfrak{m}) = \text{pr}^{-1}(M(\mathcal{O}_F/\mathfrak{m})) \subset \text{GL}(2; \mathbf{A}_{F,f}) \text{ (24.11.1.5)}.$$

┘

**Prop. (26.6.3.3).** If  $F = \mathbb{Q}$ , then the inclusion induces homeomorphisms

$$\Gamma_0(N) \backslash \mathrm{GL}(2; \mathbb{R})^+ \cong \mathrm{GL}(2; \mathbb{Q}) \backslash \mathrm{GL}(2; \mathbf{A}) / \mathcal{K}_0(N).$$

$$\Gamma_0(N) \backslash \mathrm{SL}(2; \mathbb{R}) \cong \mathrm{GL}(2; \mathbb{Q}) Z(\mathbf{A}) \backslash \mathrm{GL}(2; \mathbf{A}) / \mathcal{K}_0(N)$$

Thus the definition of congruence subgroups(20.4.5.9) are compatible with that of(20.5.3.24)  $\lrcorner$

*Proof:* Because  $\mathrm{cl}(\mathbb{Q}) = 1$  and  $\det(\mathcal{K}_0(N)) = \prod_{v \in \Sigma_F^{\mathrm{fin}}} \mathcal{O}_v^*$ , item3 of(20.5.3.23) shows

$$\mathrm{GL}(2; \mathbf{A}) = \mathrm{GL}(2; \mathbb{Q}) \mathrm{GL}(2; \mathbb{R}) \mathcal{K}_0(N) = \mathrm{GL}(2; \mathbb{Q}) \mathrm{GL}(2; \mathbb{R})^+ \mathcal{K}_0(N).$$

So

$$\mathrm{GL}(2; \mathbb{Q}) \backslash \mathrm{GL}(2; \mathbf{A}) \cong \mathrm{GL}(2; \mathbb{Q}) \backslash \left( \mathrm{GL}(2; \mathbb{Q}) \mathrm{GL}(2; \mathbb{R})^+ \mathcal{K}_0(N) \right) \cong \Gamma_0(N) \backslash \left( \mathrm{GL}(2; \mathbb{R})^+ \mathcal{K}_0(N) \right).$$

Now taking quotient on the right by  $\mathcal{K}_0(N)$  gives the desired assertion.

2 follows from 1 by modulo the center.  $\square$

**Cor. (26.6.3.4).** The quotient space  $\mathrm{GL}(n; F) Z(\mathbf{A}_F) \backslash \mathrm{GL}(n; \mathbf{A}_F)$  has finite measure.  $\lrcorner$

*Proof:* For the general case, Cf.[Humphreys, Arithmetic Groups, (1980)?].

Because  $\mathcal{K}_0(N)$  is compact, it suffices to prove that  $\mathrm{GL}(n; F) Z(\mathbf{A}_F) \backslash \mathrm{GL}(n; \mathbf{A}_F) / \mathcal{K}_0(N)$  has finite measure(because  $\mathrm{GL}(n; F)$  and  $\mathrm{GL}(n; A)$  are both unimodular, the measure is compatible). But this space is homeomorphic to  $\Gamma_0(N) \backslash \mathrm{SL}(2; \mathbb{R})$ , which has finite measure because  $\Gamma(1) \backslash \mathrm{SL}(2, \mathbb{R})$  does and  $\Gamma_0(N)$  is of finite index in  $\Gamma(1)$ .  $\square$

**Prop. (26.6.3.5) [Contragradient Representations].** For  $(\pi, V) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F, \omega)$ ,

- its contragradient  $(\pi^\vee, V^\vee) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F, \omega^{-1})$ , and  $V^\vee$  can be chosen to be the space of all functions  $g \mapsto \varphi(g^{-t})$  where  $\varphi \in V$ .
- if  $n = 2$ , then  $\pi^\vee \cong \pi \otimes (\omega^{-1} \circ \det)$ .

$\lrcorner$

*Proof:* It suffices to analyze each place. Notice for cuspidal representations, it suffices to show for finite places, because we can use strong multiplicity one(26.6.5.1).

For admissible case we only prove for  $n = 2$  and  $F$  totally real(the problem is we haven't studied  $\mathbb{C}$ ).?

Let  $\hat{V}$  be the space of functions of the form  $\hat{\varphi}(g) = \varphi(g^{-t})$ , then the right action of  $G$  on  $\hat{V}$  corresponds back to the restriction of the right action composed with the automorphism  $g \mapsto g^{-t}$ . Thus the result follows from(24.11.1.10) and its Archimedean analogy for  $(\mathfrak{g}_\infty, K_\infty)$ -modules(24.6.4.23).  $\square$

**Def. (26.6.3.6) [Global New Vectors].** For  $(\pi, V) \cong \otimes'_v (\pi_v, V_v, \xi_v^0) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$ , a **new vector(global)**  $\xi$  for  $\pi$  is defined to be a restricted tensor product of local new vectors of  $\pi_v$ (24.11.3.17) s.t.  $\xi_v$  is the distinguished vector  $\xi_v^0$  for a.e.  $v \in \Sigma_F$ , and for  $v \in \Sigma_F^\infty$ ,  $\xi_v$  generates an irreducible representation of  $\mathcal{K}_v$ .  $\lrcorner$

Adelization

**Prop. (26.6.3.7) [Global Hecke Algebra].** There are isomorphisms

$$\Gamma_0(N) \backslash G_0(N) / \Gamma_0(N) \text{ (20.4.5.13)} \cong \prod'_{v \in \text{Prime} \setminus S(N)} \mathcal{K}_v \backslash \text{GL}(2; \mathbb{Q}_v) / \mathcal{K}_v.$$

$$\Gamma_0(N) \backslash G_0(N) \cong \prod'_{v \in \text{Prime} \setminus S(N)} K_v \backslash \text{GL}(2; \mathbb{Q}_v)$$

which induces an isomorphism  $\mathcal{R}_N \cong \prod_{v \in \text{Prime} \setminus S(N)} \mathcal{H}_{\mathcal{K}_v}$  (26.3.2.1)(24.9.6.3).  $\lrcorner$

**Prop. (26.6.3.8) [Adelization of Maass Forms].** Let  $\chi \in \text{Diri}(N)$  and  $\omega$  be the adelized Hecke character of  $\chi$  (19.4.1.24). Define a character  $\chi_d$  on  $\Gamma_0(N)$  by

$$\chi_d \left( \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = \chi(d),$$

and define a character  $\lambda_d$  of  $\mathcal{K}_0(N)$  (26.6.3.2) by

$$\lambda_d \left( \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = \prod_{v \in S(N)} \omega_v(d_v).$$

By (26.6.3.3), there is a homeomorphism

$$\Gamma_0(N) \backslash \text{GL}(2; \mathbb{R})^+ \cong \text{GL}(2; \mathbb{Q}) \backslash \text{GL}(2; \mathbf{A}) / \mathcal{K}_0(N) \cong \text{GL}(2; \mathbb{Q}) \backslash \left( \text{GL}(2; \mathbb{Q}) \text{GL}(2; \mathbb{R})^+ \mathcal{K}_0(N) \right) / \mathcal{K}_0(N),$$

which induces an isomorphism

$$f \mapsto \Phi_f : C^\infty(\Gamma_0(N) \backslash \text{GL}(2; \mathbb{R})^+, \chi_d) \cong C^\infty(\text{GL}(2)/\mathbb{Q}, \omega)[\lambda_d] \text{ (26.1.1.1)(26.6.1.13)(26.6.3.2)}$$

$$\Phi_f(\gamma g_\infty k_0) = f(g_\infty) \lambda_d(k_0), \text{ where } \gamma \in \text{GL}(2; \mathbb{Q}), g_\infty \in \text{GL}(2; \mathbb{R})^+, k_0 \in \mathcal{K}_0(N).$$

And this map clearly identifies  $\mathcal{K}_\infty$ -finiteness properties because the images are  $\mathcal{K}_{0,1}(N)$ -fixed, and also identifies  $\mathcal{Z}$ -finiteness properties and moderate growth conditions, thus induces an isomorphism

$$\mathcal{A}(\Gamma_0(N) \backslash \text{GL}(2; \mathbb{R})^+, \chi_d) \cong \mathcal{A}(\text{GL}(2)/\mathbb{Q}, \omega)[\lambda_d] \text{ (26.1.1.2)(26.6.1.13)}$$

compatible with  $(\mathfrak{g}_\infty, \mathcal{K}_\infty)$ -action.

This map also identifies cuspidality, thus it induces an isomorphism

$$\mathcal{A}_0(\Gamma_0(N) \backslash \text{GL}(2; \mathbb{R})^+, \chi_d) \cong \mathcal{A}_0(\text{GL}(2; \mathbb{Q}) \backslash \text{GL}(2; \mathbf{A}), \omega)[\lambda_d].$$

And by (26.1.3.10) also induces isomorphisms

$$\text{MF}_k(\Gamma_0(N), \chi) \cong \mathcal{A}(\text{GL}(2; \mathbb{Q}) \backslash \text{GL}(2; \mathbf{A}), \omega)^{L=0}[\lambda_d](k),$$

$$\text{SF}_k(\Gamma_0(N), \chi) \cong \mathcal{A}_0(\text{GL}(2; \mathbb{Q}) \backslash \text{GL}(2; \mathbf{A}), \omega)^{L=0}[\lambda_d](k),$$

where  $(k)$  means  $\mathcal{K}_\infty$  acts by  $k_\theta \mapsto e^{k\theta}$ .  $\lrcorner$

*Proof:* Firstly  $\Phi_f$  is well-defined because if  $g_\infty, g'_\infty \in \mathrm{GL}(2; \mathbb{R})^+$  satisfies  $g'_\infty = \gamma g_\infty k_0$  for some  $\gamma \in \mathrm{GL}(2; \mathbb{R})$  and  $k_0 \in \mathcal{K}_0(N)$ , then by (26.6.3.3),  $\gamma \in \Gamma_0(N)$  and  $g'_\infty = \gamma g_\infty$ . So  $f(g'_\infty) = f(g_\infty) \lambda_d(k_0)$  by (19.4.1.24).

To show that  $\Phi_f(zg) = \omega(z)\Phi_f(g)$  for  $z \in Z(\mathbf{A})$ , notice by (19.4.1.20),  $\mathbf{I} = \mathbb{Q}^\times \times \mathbb{R}_+ \times \widehat{\mathbb{Z}}^\times$ , and for these elements  $\Phi_f(zg) = \omega(z)\Phi_f(g)$ .

It identifies moderate growth because of (26.6.0.5).

To show cuspidality, notice  $x \mapsto \gamma \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g$  is left-invariant under  $a$  if  $g^{-1} \begin{bmatrix} 1 & a \\ & 1 \end{bmatrix} g \in \mathcal{K}_0(N)$ . but such  $a$  is open in  $\mathbf{A}$ , thus contains some open compact subgroup  $U_f \subset \mathbf{A}_f$ . Notice by strong approximation,  $\mathbf{A} = \mathbb{Q} \times \mathbb{R} \times \widehat{\mathbb{Z}}$ , thus there is an isomorphism  $\mathbb{R}/M \cong \mathbb{Q} \backslash \mathbf{A} / \widehat{\mathbb{Z}}$ . Now let  $g = \gamma g_\infty k_0$ , then

$$\int_{\mathbf{A}/\mathbb{Q}} \Phi_f \left( \begin{bmatrix} 1 & a \\ & 1 \end{bmatrix} g \right) = C \int_{\mathbb{R}/M} \Phi_f \left( \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g \right) dx = C \lambda(k_0) \int_0^M f(\gamma^{-1} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \gamma g_\infty) dx,$$

which vanishes for each  $g$  iff  $f$  is cuspidal at every cusp. For similar reason,  $\Phi_f$  is of moderate growth iff  $f$  is of moderate growth at every cusp.  $\square$

**Prop. (26.6.3.9) [Strong Multiplicity One for Cusp Forms].**

- Let  $f$  be a cuspidal eigenfunction in  $\mathcal{A}_0(\Gamma_0(N) \backslash \mathrm{GL}(2; \mathbb{R})^+, \chi)$  for Hecke operators  $\{T_p : p \in \text{Prime} \setminus S(N)\}$ , then  $\Phi_f$  is contained in a unique irreducible subspace  $(\pi_f, V_f) \subset L_0^2(\mathrm{GL}(2; \mathbb{Q}) \backslash \mathrm{GL}(2; \mathbf{A}), \omega)$ .
- If  $f, g$  are cuspidal eigenfunctions for a.e. Hecke operators in  $\mathcal{A}_0(\Gamma_0(N) \backslash \mathrm{GL}(2; \mathbb{R})^+, \chi)$  with the same eigenvalues, then  $f = g$ .

In particular, any Hecke eigenform  $\pi_f$  generates a cuspidal representation  $\pi_f$ .  $\lrcorner$

*Proof:* 2: By (26.6.2.4), it suffices to consider the projection of  $\Phi_f$  to any  $(\pi, V) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/\mathbb{Q})$ . By (26.6.3.7),  $\Phi_f$  is an eigenform of the Hecke operator  $T_p$  for  $p \in \text{Prime} \setminus S(N)$ . But it is also eigenvalues of the Hecke operator  $R_p$ , and  $\mathcal{H}_{\mathbb{Q}_p}$  is generated by  $T_p, R_p, R_p^{-1}$  by (26.3.2.9), so  $\pi_p$  is unramified with determined eigenvalues for a.e.  $p \in \text{Prime}$ . But then  $\pi_p$  is determined by (3.6.5.12), and  $(\pi, V)$  is determined by the strong multiplicity one theorem (26.6.5.1).

1 follows from 2 because if  $\pi$  is a component of  $L_0^2(\mathrm{GL}(2; \mathbb{Q}) \backslash \mathrm{GL}(2; \mathbf{A}), \omega)$  that the projection of  $\Phi_f$  is non-zero, then it has the same Hecke eigenvalue as  $\Phi_f$ , so it must be just  $\Phi_f$  by item 2.

Cf. [Bump, P344].  $\square$

**Prop. (26.6.3.10) [Compatibility of Hecke Operators].** Cf. [Viewing modular forms as automorphic representations] Chap 5. ?  $\lrcorner$

**Prop. (26.6.3.11) [Local Properties of Adelize Cusp Modular Forms].** If  $f(z) = \sum_{n \in \mathbb{Z}_+} a_n q^n \in \mathrm{SF}_k(N, \chi_0)$  is a normalized eigenform with associated automorphic representation  $\pi_f = \otimes'_p \pi_{f,p}$ , then

1.  $\pi_{f,\infty}$  is a discrete series of weight  $k$ .
2. For  $p \in \text{Prime} \setminus S(N)$ ,  $\pi_{f,p}$  is an unramified principal series with  $T_p$  and  $S_p$  acting via multiplication by  $a_p p^{1-k/2}$  and  $\psi(p)$ . Thus the Satake parameters are the two roots of  $T^2 - a_p T + \chi(p) p^{k-1} = 0$ .
3. If  $p|N$ ,  $N = p^r N'$  with  $(p, N') = 1$ ,  $f$  is  $p$ -new and  $f = f'(\chi)$  with  $f'$   $p$ -primitive (i.e. has the least conductor among its twists) and  $c(\psi_p \chi_p^{-1}) = r$ , then  $\pi_{f,p} \cong \pi(\chi_1 \chi, \chi_2 \chi)$  is a principal series where  $\chi_2$  is unramified,  $\chi_2(p) = a_p p^{(1-k)/2}$  and  $\chi_1 = \psi_p \chi_2^{-1}$ .

4. If  $N = pN'$  with  $(p, N') = 1$  and  $\psi_p$  is unramified, then  $\pi_{f,p}$  is the twist of the Steinberg representation by an unramified character  $\chi$  with  $\chi(p) = a_p p^{(2-k)/2}$ .
5. In all other cases,  $\pi_{f,p}$  is supercuspidal, and can be determined by algorithm in [Loeffler-Weinstein, computation of local components of a newform].

┘

*Proof:* 1: This follows from the fact  $L\Phi_f = 0$ ???: The hypothesis implies that  $\pi_{f,p}$  is invariant under the action of  $\mathrm{GL}(2; \mathbb{Z}_p)$ , thus must be an unramified representation. And we can identify the Satake parameters via (26.6.3.7).

3: In this case  $\pi_{f,p}$  cannot be a supercuspidal representation because of (24.11.3.20), and it cannot be a twist of Steinberg representation because  $f$  is  $p$ -primitive.

4: In this case  $\pi_{f,p}$  cannot be a supercuspidal representation because the conductor of a supercuspidal representation is at least 2 (24.11.3.20), and it cannot be a principal series because  $\psi_p$  is unramified. □

**Prop. (26.6.3.12) [Local Properties of Adelize Maass Forms].** Notice if  $f$  comes from a Maass form, then [G-H11] documented the properties of the adelized Maass form  $\Phi_f$  in detail. ┘

**Prop. (26.6.3.13) [Newforms and New Vectors].** Let  $\Phi : \mathrm{GL}(2; \mathbf{A}) \rightarrow \mathbb{C}$  be an adelic cusp form with character  $\omega$ , then the following are equivalent:

- $\Phi$  is a newform in some  $(\pi, V) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/\mathbb{Q})$ .
- $\Phi$  is the adelization of some newform  $f$  (26.6.3.8).

┘

*Proof:* Cf. [G-H11]P45? □

### Classification of $\mathrm{Irr}_{\mathrm{disc}}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$

**Thm. (26.6.3.14) [Mœglin-Waldspurger].** For  $N \in \mathbb{Z}_+$ ,  $\mathrm{Irr}_{\mathrm{disc}}^{\mathrm{auto}}(\mathrm{GL}(N)/F)$  consists exactly of isobaric automorphic representations (26.6.1.28) of the form

$$\pi\left(\frac{n-1}{2}\right) \boxplus \pi\left(\frac{n-3}{2}\right) \boxplus \pi\left(\frac{1-n}{2}\right)$$

where  $m|N$  and  $n = N/m$ , and  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(m)/F)$ . And these representations are pairwise different. ┘

*Proof:* Cf. [Mœglin-Waldspurger89]? □

**Def. (26.6.3.15) [ $\mathcal{A}(\mathrm{GL}(n)/F)$ ].** For  $N \in \mathbb{Z}_+$ , we define  $\mathcal{A}(\mathrm{GL}(n)/F)$  to be the set of isobaric automorphic representations of the form

$$\pi\left(\frac{n-1}{2}\right) \boxplus \pi\left(\frac{n-3}{2}\right) \boxplus \pi\left(\frac{1-n}{2}\right)$$

where  $m|N$  and  $n = N/m$ , and  $\pi \in \mathrm{Irr}_{\mathrm{disc}}^{\mathrm{auto}}(\mathrm{GL}(m)/F)$ . In particular, we have  $\mathrm{Irr}_{\mathrm{disc}}^{\mathrm{auto}}(\mathrm{GL}(N)/F) \subset \mathcal{A}(\mathrm{GL}(n)/F)$  by (26.6.3.14). ┘



#### 4 Whittaker-Models

**Def. (26.6.4.1) [Whittaker-Models].** The notion of **Whittaker-model** and **Whittaker functional** w.r.t. a generic character  $\psi$ , genericness is defined the same as in the local case (24.9.5.2).  $\lrcorner$

**Prop. (26.6.4.2) [Uniqueness of Global Whittaker Models].** For  $(\pi, V) \in \text{Irr}^{\text{adm}}(\text{GL}(2)/F)$ ,  $(\pi, V)$  has a Whittaker model w.r.t  $\psi$  iff each  $(\pi_v, V_v)$  (26.6.1.4) has a Whittaker model  $\mathcal{W}(\pi_v, \psi_v)$ . If this is the case, then  $\mathcal{W}(\pi, \psi)$  is unique and

$$\mathcal{W}(\pi, \psi) = \bigotimes'_v (W_v, W_v^0),$$

where  $W_v^0$  are the unique spherical function (24.11.6.13) of  $\mathcal{W}_v$  normalized s.t.  $W_v^0(\mathcal{K}_v) = 1$ .  $\lrcorner$

*Proof:* Let  $(\pi, V) = \otimes'_v (\pi_v, V_v)$  w.r.t a.e. spherical vectors  $\xi_v^0$  by tensor product theorem (26.6.1.4), and  $(\pi_v, V_v)$  are spherical a.e.  $v$ .

Firstly if every  $(\pi_v, V_v)$  has a Whittaker model, the  $W$  in the proposition is truly a Whittaker model: functions in  $\mathcal{W}(\pi, \psi)$  are clearly smooth and  $K$ -finite, and they have moderate growth because each local part is, and for a.e.  $v$ ,  $\mathcal{W}_v(\pi, \psi)$  is compactly supported on  $|x|_v \leq 1$ . And there is a canonical isomorphism of  $V$  onto  $\mathcal{W}(\pi, \psi)$  by letting  $(W_\xi)_v = W_v^0$  if  $\xi_v = \xi_v^0$  and

$$W_\xi(g) = \prod_v W_{v, \xi_v}(g_v).$$

This is definable because  $g_v \in K_v$  for a.e.  $v$  thus  $W_{v, \xi_v}(g_v) = W_v^0(g_v) = 1$  for a.e.  $v$ .

Because the local Whittaker models are all rapidly decreasing, and the spherical Whittaker model vanishes for  $|\xi_v|_v > 1$ , thus  $W$  is also rapidly decreasing, and is a Whittaker model for  $\pi$ .

Secondly if  $\mathcal{W}$  is a Whittaker model for  $(\pi, V)$ , denote  $\xi \mapsto W_\xi$  the isomorphism of  $V$  onto  $\mathcal{W}$ . Notice there exists some  $\xi \in V$  that  $W_\xi(1) \neq 0$ : if  $W_\xi(g_\infty g_f) \neq 0$ , then  $W_{\pi(g_f)\xi}(g_\infty) \neq 0$ , and argue the same way as in (26.1.4.4). We may assume  $\xi^0 = \otimes_v \xi_v^0$  that  $W_{\xi^0}(1) = 1$ .

Consider the "pullback" of  $W$  to  $\text{GL}(n; F_v)$  via  $\xi$ , then they are clearly Whittaker models for  $(\pi_v, V_v)$  thus unique (20.7.3.10) (26.1.4.3), and also  $W_{v, \xi_v^0}(1) = 1$ . So  $W_v^0$  exists a.e. uniquely. Now we prove  $W$  is of the form we said above, this will prove uniqueness.

Then we need to prove

$$W_\xi(g) = \prod_v W_{v, \xi_v}(g_v)$$

We only need to prove for one  $\xi = \xi^0$ , because  $\mathcal{W}, \mathcal{W}(\pi_v, \psi_v)$  are both irreducible. And also we can assume  $g_v = 1, a.e.$ , because  $(W_\xi)_v(g_v) = W_{v, \xi_v}(g_v) = 1, a.e.$ . Then we are in the finite case and we can multiply by scalars at f.m.  $v$  s.t. this equation is true and nonzero for  $g$ .  $\square$

**Thm. (26.6.4.3) [Shalika-Expansions and Existence of Whittaker-Models, Shalika [Sha74]].**

For  $(\pi, V) \in L_{\text{cusp}}^2(\text{GL}(n)/F)$  and  $\Phi \in V$ , define

$$W_{\psi, \Phi}(g) = \int_{\text{Unip}(n, F) \setminus \text{Unip}(n; \mathbf{A}_F)} \Phi(ug) \psi_{\text{Unip}}(-u) du,$$

then  $\text{GL}(n; \mathbf{A}_F)$  acts on these functions, and they form a Whittaker model  $\mathcal{W}(\pi, \psi)$ .  $W_{\pi, \Phi}(g)$  is called the **Fourier-Whittaker coefficients** for  $\Phi$ .

And we have a Fourier-expansion formula:

$$\Phi(g) = \sum_{\gamma \in \text{Unip}(n-1; F) \setminus \text{GL}(n-1; F)} W_{v, \psi_v}(\Phi) \left( \begin{bmatrix} \gamma & \\ & 1 \end{bmatrix} g \right),$$

which converges absolutely and uniformly on compact subsets of  $\text{GL}(n; \mathbf{A}_F)$ .  $\lrcorner$



**Remark (26.6.4.4).** This is false for some other groups, for the reason that the Fourier-inversion may not be true, thus it may happen that all the Fourier coefficients vanish. For example, holomorphic Siegel modular forms on  $\mathrm{Sp}(n)$  don't have Whittaker models.  $\lrcorner$

*Proof:* We only prove for  $n = 2$ . For general case, see [Cogdell.  $L$ -functions and converse theorems for  $\mathrm{GL}(n)$ ]Section1.1.?

For any  $g \in \mathrm{GL}(n; \mathbf{A}_F)$ , consider  $F(n) = \varphi(ng)$  on  $\mathrm{Unip}(\mathbf{A}_F)$ , then it is a continuous function on  $\mathrm{Unip}(F) \backslash \mathrm{Unip}(\mathbf{A}_F)$ , and  $\mathrm{Unip}(F) \backslash \mathrm{Unip}(\mathbf{A}_F)$  is compact(19.4.1.3), so by Fourier inversion formula(12.11.1.18),

$$F(x) = \sum_{\alpha \in F} C(\alpha) \psi(\alpha x), \quad C(\alpha) = \int_{\mathrm{Unip}(n, F) \backslash \mathrm{Unip}(n, A)} \varphi \left( \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g \right) \psi(-\alpha x) dx.$$

Now  $C(0) = 0$  because  $\varphi$  is cuspidal, and if  $\alpha \in F^\times$ , as  $\varphi$  is automorphic,

$$\begin{aligned} C(\alpha) &= \int_{\mathrm{Unip}(n; F) \backslash \mathrm{Unip}(n; \mathbf{A}_F)} \varphi \left( \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} g \right) \psi(-\alpha x) dx \\ &= \int_{A/F} \varphi \left( \begin{bmatrix} 1 & \alpha x \\ & 1 \end{bmatrix} \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} g \right) \psi(-\alpha x) dx = W_\varphi \left( \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} g \right). \end{aligned}$$

Substitute  $x = 0$ , we get  $\Phi(g) = \sum_{\alpha \in F^\times} W_\Phi \left( \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} g \right)$ .

Now we show  $\{W_\Phi\}$  is Whittaker model: they all satisfies  $W_\Phi(ng) = \psi(n)W_\Phi(g)$  by construction, and because  $W_{X\varphi} = XW_\varphi$ ,  $\rho(g)W_\varphi = W_{\rho(g)\varphi}$ , it is clear that this space is invariant under action of  $\mathrm{GL}(2, A_f)$  and  $(\mathfrak{g}_\infty, K_\infty)$ , and also it is of moderate growth in  $y$  because  $\varphi$  does(26.6.1.13) and  $N(F) \backslash N(A)$  is compact, and it consists of  $K$ -finite vectors because  $V$  is admissible(26.6.2.5). Finally, the Fourier inversion shows that  $\varphi \mapsto W_\varphi$  is non-zero, thus injective.  $\square$

**Cor. (26.6.4.5) [Cuspidal Forms Decay Rapidly].** Any cuspidal form on  $\mathrm{GL}(2; \mathbf{A}_F)$  is rapidly decreasing for  $|y| \rightarrow \infty$ .

Moreover, it also decays rapidly when  $|y| \rightarrow 0$ .  $\lrcorner$

**Remark (26.6.4.6).** This is proved already in(26.6.1.16).  $\lrcorner$

*Proof:* To show it is rapidly decreasing, use the Fourier expansion formula, and the fact the Whittaker model is product of local Whittaker models. Also for  $v \in \Sigma_F^{\mathrm{fin}}$ , the local Kirillov model is compactly supported(24.11.3.12), and for  $v \in \Sigma_F^\infty$ , the Whittaker model is rapidly decreasing(26.1.4.3), notice for any  $g \in \mathrm{GL}(2, A)$ , for any  $C$ , the number of  $a \in K^*$  that  $\varphi \left( \begin{bmatrix} a & \\ & 1 \end{bmatrix} g \right) \neq 0$  and  $|a|_\infty < C$  is bounded by a polynomial of  $C$ , thus it is rapidly decreasing.

For  $|y| \rightarrow 0$ , as  $\varphi$  is automorphic, notice

$$\varphi \left( \begin{bmatrix} y & \\ & 1 \end{bmatrix} \right) = \varphi(w_0 \begin{bmatrix} 1 & \\ & y \end{bmatrix} w_0) = \omega(y)(\pi(w_0)\varphi) \left( \begin{bmatrix} y^{-1} & \\ & 1 \end{bmatrix} \right).$$

$\square$

**Cor. (26.6.4.7).** Every local component of a cuspidal representation of  $\mathrm{GL}(n; \mathbf{A}_F)$  is generic, by(26.6.4.2).  $\lrcorner$

## 5 Multiplicity One Theorems

**Thm. (26.6.5.1) [Strong Multiplicity One for  $\mathrm{GL}(n)$ , Shalika/Jacquet-Shalika].** If  $(\pi, V), (\pi', V') \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F, \omega)$  (26.6.1.22) satisfy  $\pi_v \cong \pi'_v$  a.e.  $v$ , then  $\pi \cong \pi'$ , and  $V = V' \subset L_{\mathrm{disc}}^2(\mathrm{GL}(n; F) \backslash \mathrm{GL}(n; \mathbf{A}_F))$ .

The same holds true if  $\pi, \pi'$  are isobaric automorphic representations (26.6.1.28).  $\square$

*Proof:* We only prove for  $n = 2$  and the case  $\pi_1, \pi_2$  are cuspidal and isomorphic on Archimedean places ?. For general  $n$ , Cf. [Euler products and the classification of automorphic representations 1, Jacquet-Shalika]P553 ?, [Representations of the Group  $\mathrm{GL}(n; K)$  where  $K$  is a local field, Gelfand-Kazhdan], [Euler Subgroups, in Lie Groups and their Representations, Piatetski-Shapiro(1975)]. For the Archimedean places, Cf. [Base Change for  $\mathrm{GL}(2)$ , Langlands(1980)Lemma3.1]

Firstly if  $\pi_v \cong \pi'_v$  for every place  $v$ , then their corresponding Whittaker model is the same (multiplied by a scalar) by (26.6.4.2). Then by (26.6.4.3) we have a Fourier expansion formula

$$\varphi(g) = \sum_{\alpha \in F^\times} W_\varphi \left( \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} g \right).$$

Thus  $V = V'$ .

In case that  $\pi_v \cong \pi'_v$  only outside a finite set  $S$ , we choose functions  $W_v, W'_v$  in the local Whittaker model for  $\pi_v, \pi'_v$  s.t. if  $v \notin S$ ,  $W_v = W'_v$  and  $W_v$  is the unique spherical function normalized that  $W_v(K_v) = 1$  a.e.  $v$ , and if  $v \in S$ , they are chosen that

$$F(y) = W_v \left( \begin{bmatrix} y & \\ & 1 \end{bmatrix} \right) = W'_v \left( \begin{bmatrix} y & \\ & 1 \end{bmatrix} \right) \in C_c^\infty(F_v^\times),$$

which is possible by (24.11.3.13). And we define

$$\varphi(g) = \sum_{\alpha \in F^*} W \left( \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} g \right), \quad \varphi'(g) = \sum_{\alpha \in F^*} W' \left( \begin{bmatrix} \alpha & \\ & 1 \end{bmatrix} g \right),$$

$$W(g) = \prod_v W_v(g_v), \quad W'(g) = \prod_v W'_v(g_v).$$

as in (26.6.4.3).

Then we claim  $\varphi = \varphi'$  on  $\mathrm{GL}(2; \mathbf{A}_F)$ :  $\varphi \in V, \varphi' \in V'$  are automorphic, thus  $\varphi = \varphi'$  on  $\mathrm{GL}(2; F) \backslash \mathrm{GL}(2; \mathbf{A}_F^S)$  by continuity, which is just  $\mathrm{GL}(2; \mathbf{A}_F)$  by weak approximation, so we have a  $\varphi \in V \cap V'$ , meaning  $V = V'$ .  $\square$

**Thm. (26.6.5.2) [Multiplicity One for  $\mathrm{SL}(2)$ , Ramakrishnan].** For  $F \in \mathbf{NField}$ , in the spectral decomposition

$$L_{\mathrm{cusp}}^2(\mathrm{SL}(2; F) \backslash \mathrm{SL}(2; \mathbf{A}_F)) = \bigoplus_{m \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{SL}(2)/F)} m_\pi \pi,$$

we have  $m_\pi \leq 1$  for any  $\pi$ .  $\square$

*Proof:* ?  $\square$

## 6 Ramanujan-Conjecture

**Conj. (26.6.6.1) [Generalized Ramanujan-Conjecture].** If  $G \in \text{RedGrp}/F$ , and  $\pi \in \text{Irr}^{\text{cusp, loc. generic}}(G/F, \omega)$ , then  $\pi$  is tempered (24.9.8.11) (24.6.4.11).  $\lrcorner$

*Proof:* ?  $\square$

**Conj. Cor. (26.6.6.2) [Ramanujan].** If  $\pi = \otimes' \pi_v \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F, \omega)$ , then for any  $v \in \Sigma_F$ ;  $\pi_v$  is tempered. This should follow from the generalized Ramanujan-conjecture (26.6.6.1), as any  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F, \omega)$  is generic by (26.6.4.3).  $\lrcorner$

*Proof:*  $\square$

**Prop. (26.6.6.3).** The Ramanujan-conjecture (26.6.6.1) for  $\text{GL}(n)$  means that: for  $v \in \Sigma_F^{\text{fin}}$ , each eigenvalue of  $c_v(\pi)$  has absolute value 1, and for  $v \in \Sigma_F^\infty$ , it means that  $\text{Re}(\mu_{j, \pi_v}) = 0$ .  $\lrcorner$

*Proof:* ?  $\square$

**Conj. Cor. (26.6.6.4) [Selberg Eigenvalue Conjecture].** For  $F = \mathbb{Q}$  and  $G = \text{GL}(2)$ , the Ramanujan conjecture for the infinite place (26.6.6.2) is equivalent to the following: For any  $N \in \mathbb{Z}_+$  and any eigenvalue  $\lambda$  of the Laplacian on  $L_0^2(\Gamma_0(N) \backslash \mathcal{H})$ , we have  $\lambda \geq 1/4$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (26.6.6.5).** The Ramanujan conjecture (26.6.6.1) implies the Ramanujan-Petersson conjecture (29.5.4.9).  $\lrcorner$

*Proof:* The Ramanujan-Petersson conjecture says the eigenvalue  $\lambda_p$  of  $T_p$  on  $f$  satisfies  $|\lambda_p| \leq 2p^{1/2}$ . Consider the irreducible cuspidal representation  $\pi_f$  generated by  $\Phi_f$  (26.6.3.9), then  $T_p$  is the same as the eigenvalue of the local Hecke algebra  $T_p$  on  $\Phi_f$ . But  $\pi_{f,p}$  is unramified, thus by (24.11.8.12) it is unitary principal or complementary. But notice if the Satake parameters of  $\pi_{f,p}$  are  $\alpha_i$ , then  $\lambda_p = p^{1/2}(\alpha_1 + \alpha_2)$ , Then it suffices to show that  $|\alpha_1 + \alpha_2| \leq 2$ , which is equivalent to  $\pi_p$  being tempered (24.11.8.12).  $\square$

## 7 Eisenstein Series

References are [Bump, Chap3].

Langlands' theory of Eisenstein series reduces the study of all the automorphic representations to the discrete ones (26.6.1.17).

## 26.7 $(\mathfrak{g}, \mathcal{K}_\infty)$ -Cohomologies

References are [B-W00].

**Notation (26.7.0.1).**

- Let  $G, G_1, G_2 \in \text{RedGrp}/\mathbb{Q}$ .
- Let  $\mathcal{K}_\infty \leq G(\mathbb{R}), \mathcal{K}_{i\infty} \leq G_i(\mathbb{R})$  be maximal compact subgroups.
- Let  $\mathfrak{g} = \text{Lie}(G_\mathbb{R}), \mathfrak{k} = \text{Lie}(\mathcal{K}_\infty)$ .
- Let  $\theta : \mathfrak{g} \rightarrow \mathfrak{g}$  be a Cartan involution s.t.  $\mathfrak{g}^\theta = \mathfrak{k}$ .<sup>?</sup>
- Let  $\mathfrak{p} = \mathfrak{g}^{\theta=-1}$ .
- Let  $A_G \leq G(\mathbb{R})$  be the identity component in the real topology of the maximal  $\mathbb{Q}$ -split torus in  $Z(G)$  with Lie algebra  $\mathfrak{a}_G$ .
- Let  $\mathcal{K}_f \subset G(\mathbf{A}_f)$  be a compact open subgroup, and  $\mathcal{K} = \mathcal{K}_f \mathcal{K}_\infty$ .
- Let  $X = A_G \backslash G(\mathbb{R}) / \mathcal{K}_\infty$  and let  $\text{Sh}_\mathcal{K}(G) = G(\mathbb{Q}) \backslash (X \times G(\mathbf{A}_f) / \mathcal{K}_f)$  be the Shimura manifold.

┘

### 1 $(\mathfrak{g}, \mathcal{K}_\infty)$ -Cohomologies

**Def. (26.7.1.1)  $[(\mathfrak{g}, \mathfrak{k})\text{-Cohomologies}]$ .** For  $\mathcal{A} \in \text{Mod}_\mathfrak{g}$  and  $q \in \mathbb{N}$ , let

$$C^q(\mathfrak{g}, \mathfrak{k}; \mathcal{A}) = \text{Hom}_\mathfrak{k}(\wedge^q(\mathfrak{g}/\mathfrak{k}), \mathcal{A}),$$

where the action of  $\mathfrak{k}$  on  $\mathfrak{g}/\mathfrak{k}$  is the adjoint action  $\text{ad}$ . Define

$$d : C^q(\mathfrak{g}, \mathfrak{k}; \mathcal{A}) \rightarrow C^{q+1}(\mathfrak{g}, \mathfrak{k}; \mathcal{A}) :$$

$$(df)(x_0, \dots, x_q) = \sum_{i=1}^q (-1)^i x_i \cdot f(x_0, \dots, \widehat{x}_i, \dots, x_q) + \sum_{1 \leq i < j \leq q} (-1)^{i+j} f([x_i, x_j], x_0, \dots, \widehat{x}_i, \dots, \widehat{x}_j, \dots, x_q).$$

Then this  $d$  is well-defined and  $d \circ d = 0$ , thus making  $C^\bullet(\mathfrak{g}, \mathfrak{k}; \mathcal{A})$  into a complex. Denote  $H^q(\mathfrak{g}, \mathfrak{k}; \mathcal{A})$  its  $q$ -th cohomology group.

┘

**Def. (26.7.1.2)  $[(\mathfrak{g}, \mathcal{K}_\infty)\text{-Cohomologies}]$ .** For  $\mathcal{A} \in \text{Mod}_{(\mathfrak{g}, \mathcal{K}_\infty)}$ ,  $\mathcal{K}_\infty$  acts on  $C^q(\mathfrak{g}, \mathfrak{k}; \mathcal{A})$  (26.7.1.1) by adjoint action on  $\mathfrak{k}$  and action on  $\mathcal{A}$ .<sup>?</sup>, thus we can define

$$C^\bullet(\mathfrak{g}, \mathcal{K}_\infty; \mathcal{A}) = C^q(\mathfrak{g}, \mathfrak{k}; \mathcal{A})^{\mathcal{K}_\infty}.$$

Then  $d$  defines a differential on  $C^\bullet(\mathfrak{g}, \mathcal{K}_\infty; \mathcal{A})$ , making it a complex. Its  $q$ -th cohomology group is denoted by  $H^q(\mathfrak{g}, \mathcal{K}_\infty; \mathcal{A})$ , called the  $(\mathfrak{g}, \mathcal{K}_\infty)$ -cohomology of  $\mathcal{A}$ .

┘

**Thm. (26.7.1.3) [Künneth Formula].** If  $\mathcal{A}_i$  are admissible  $(\mathfrak{g}_i, \mathcal{K}_i)$ -modules, then for any  $n \in \mathbb{N}$ , we have natural isomorphisms

$$H^n(\mathfrak{g}_1 \oplus \mathfrak{g}_2, \mathcal{K}_{1\infty} \times \mathcal{K}_{2\infty}; \mathcal{A}_1 \otimes \mathcal{A}_2) = \bigoplus_{p+q=n} H^p(\mathfrak{g}_1, \mathcal{K}_1; \mathcal{A}_1) \otimes H^q(\mathfrak{g}_2, \mathcal{K}_2; \mathcal{A}_2).$$

┘

*Proof:* Cf.[B-W00]Section1.1.3. ?

□

**Thm. (26.7.1.4) [Poincaré Duality].** If  $\mathcal{A}$  is an admissible irreducible  $(\mathfrak{a}_G \backslash \mathfrak{g}, \mathcal{K}_\infty)$ -module, then for any  $q \in \mathbb{N}$  we have

$$H^q(\mathfrak{a}_G \backslash \mathfrak{g}, \mathcal{K}_\infty; \mathcal{A}) \cong H^{\dim X - q}(\mathfrak{a}_G \backslash \mathfrak{g}, \mathcal{K}_\infty; \mathcal{A}^\vee)^\vee.$$

┘

*Proof:* Cf.[B-W00]Prop1.7.6.

□

**Prop. (26.7.1.5).** Let  $V \in \text{Irr}(G)$ , then

$$H^\bullet(\mathfrak{g}, \mathcal{K}_\infty; V \otimes \mathbb{C}) = \begin{cases} 0 & , V \neq \mathbf{1} \\ (\wedge^\bullet \mathfrak{p})^{\mathcal{K}_\infty} & , V = \mathbf{1} \end{cases} \quad (26.7.0.1).$$

┘

*Proof:* Cf.[B-W00]Cor2.3.2.

□

**Thm. (26.7.1.6) [Concentration of  $(\mathfrak{g}, \mathcal{K}_\infty)$ -Cohomologies].** For  $V \in \text{Irr}(G)$ , let

$$\mathcal{P}(V) = \{\theta\text{-stable parabolic subalgebra } \mathfrak{q} \leq \mathfrak{g}_\mathbb{C} : V^{\text{rad}(\mathfrak{q})} = 1\}.$$

and let

$$c(V) = \min\{\dim(\text{rad}(\mathfrak{q}) \cap \mathfrak{p}) : \mathfrak{q} \in \mathcal{P}(V)\}.$$

Then if  $\mathfrak{g}$  is semisimple,  $\pi_\infty \in \text{Irr}_{(\mathfrak{g}, \mathcal{K}_\infty)}$ ,  $V \in \text{Irr}(G)$  s.t. the largest subalgebra of  $\mathfrak{g}$  acting trivially on  $\pi_\infty$  is contained in  $\mathfrak{k}$ , then

$$H^i(\mathfrak{g}, \mathcal{K}_\infty; \pi_\infty \otimes V)$$

vanishes for any  $i < c(V^*)$ .

┘

*Proof:* Cf.[B-W00]Thm2.10.1.

□

## 2 Cohomology of Shimura Manifolds

References are [?B-K00]Section7.2.

## 26.8 Theta Correspondence

References are [Pra93, Pra98], [Correspondances de Howe sur un corps  $p$ -adique, Moeglin-Vineras-Waldspurger] and [Local Theta Correspondence notes, Gan-Kudla-Takeda].

### 1 Weil Representations

**Notation (26.8.1.1).**

- Use notations on Schrödinger representations defined in [Schrödinger Representations](#).
- Let  $K \in \mathbf{Field}$  be a finite field or a local field, and  $\text{char } K \neq 2$ .
- If  $K \in p\text{-LField}$ , denote it by  $(K, \mathcal{O}_K, \mathfrak{m}_K, \varpi_K)$ .
- Let  $\psi \in (K^\times)^\vee$ .
- Let  $W$  be a f.d. symplectic space over  $K$ .

⌋

**Def. (26.8.1.2) [Heisenberg Groups].** The **Heisenberg group**  $H(W)$  of  $W$  is defined to be the set  $W \times K$  endowed with the group law:

$$(w_1, t_2) \cdot (w_2, t_2) = (w_1 + w_2, t_1 + t_2 + \frac{1}{2}(w_1, w_2)).$$

In particular,  $H(W)$  is a central extension of  $W$  by  $K$ :

$$0 \rightarrow K \rightarrow H(W) \rightarrow W \rightarrow 0.$$

Especially, we can take any  $V \in \mathcal{V}\text{ect}^{\text{fd}}/K$  and  $W = V \times V^*$ , with the symplectic pairing given by

$$\left( (v_1, f_1), (v_2, f_2) \right) = f_2(v_1) - f_1(v_2).$$

In this case, the corresponding Heisenberg group is denoted by  $H(V)$ .

⌋

**Prop. (26.8.1.3) [Schrödinger representations].** Let  $n \in \mathbb{Z}_+$ ,  $V \in \mathcal{V}\text{ect}_K^n$ . Then there is a surjective homomorphism

$$\tau : H(V) \rightarrow A(V) : (v^*, v, x) \mapsto (v^*, v, \psi(x)\psi(-\langle v, v^* \rangle)) \quad (12.11.1.20).$$

$A(V)$  acts on  $L^2(V)$  by the Schrödinger representation (12.11.1.21):  $(\rho(v^*, v, t)\Phi)(u) = t\psi(-2\langle u, v^* \rangle)\Phi(u + v)$ , and this induces an action  $\pi$  of  $H$  on  $L^2(G)$ :  $(\rho(v^*, v, x)\Phi)(u) = t\psi(-2\langle u, v^* \rangle)\Phi(u + v)$ . ⌋

**Prop. (26.8.1.4) [Heisenberg Group for Quadratic Spaces].** If  $V$  is a quadratic space, then we can identify  $V$  with  $V^*$ . Thus  $\text{SL}(2; K)$  acts on  $H(V)$  by

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} (v_1, v_2, x) = (av_1 + bv_2, cv_1 + dv_2, x),$$

and acts on  $A(V)$  by

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} (v_1, v_2, t) = (av_1 + bv_2, cv_1 + dv_2, t\psi(-acB(v_1, v_1) - bdB(v_2, v_2))).$$

These two actions are compatible with  $\tau$ .

$O(V)$  acts on  $H(V)$  by  $g(v_1, v_2, x) = (gv_1, gv_2, x)$ , and similarly acts on  $A(V)$ , compatible with  $\tau$ .

Recall the Fourier transform on  $V$  is w.r.t. the pairing  $(v, v^*) \mapsto \psi(-2B(v, v^*))$ .  $\lrcorner$

**Prop. (26.8.1.5) [Projective Weil Representation].** By (3.5.11.26), for any symplectic space  $W$ , we may choose isotropic subspaces  $X, Y$  of  $W$  s.t.  $W = X \oplus Y$ . Then the pairing induces an isomorphism  $Y \cong X^*$ . Then  $H(W) \cong H(X)$ , with a Schrödinger representation  $\pi$  on  $L^2(X)$  (26.8.1.3).

Then there exists a unitary projective representation  $\omega_1$  of  $\mathrm{Sp}(W)$  on  $L^2(X)$  realized on the Bruhat-Schwartz space  $\mathcal{S}(X)$  s.t.

- For  $g \in \mathrm{Sp}(W), h \in H(X)$ ,  $\omega_1(g)\pi(h)\omega_1(g)^{-1} = \pi(g(h))$ .
- $(\omega_1(\begin{bmatrix} \mathbf{1} & B \\ & \mathbf{1} \end{bmatrix})\Phi)(v) = \psi(\frac{X^t B X}{2})\Phi(v)$ .
- $(\omega_1(\begin{bmatrix} A & \\ & A^{-t} \end{bmatrix})\Phi)(v) = |\det A|^{1/2} \chi(a) \Phi(av)$ , where  $\chi(a) \in \mathbb{T}^1$ .
- $\omega_1(\begin{bmatrix} & \mathbf{1} \\ -\mathbf{1} & \end{bmatrix})\Phi = \gamma(B)\Phi^\vee$ , where  $\gamma(B) \in \mu_8$ .

$\lrcorner$

*Proof:* Because the action of  $\mathrm{Sp}(W)$  on  $A(X)$  preserves the center, thus by (12.11.1.21), for any  $g \in \mathrm{Sp}(W)$ , there exists a unitary automorphism  $\omega_1(g)$  of  $L^2(X)$  that

$$\omega_1(g)\pi(h)\omega_1(g)^{-1} = \pi(g(h)).$$

To check the properties of  $\omega_1$ , it suffices to check this equation for  $h = (v, 0, 1)$  or  $(0, v, 1)$  (because these elements generate  $A(X)$ ), using the fact  $\rho$  action on  $L^2(X)$  is irreducible (12.11.1.21). ?  $\square$

**Thm. (26.8.1.6) [Weil Representation].** The projective representation  $\omega_1$  of  $\mathrm{Sp}(W)$  lifts uniquely to a true representation  $\omega$  of  $\mathrm{Mp}(W)$ , called the (local) **Weil representation**. Sometimes we also call the smooth part of this representation the Weil representation.

Notice that this representation depends on  $\psi$ .  $\lrcorner$

*Proof:* Cf. [LINEAR SYMMETRIES OF FREE BOSON FIELDS] Thm 5.1.  $\square$

**Prop. (26.8.1.7) [Weil Representations and Restrictions].** Let  $K/k$  be a separable extension of fields, then  $\psi_K : x \mapsto \psi(\mathrm{tr}(x))$  is a non-trivial character of  $K$ . And if  $(W, (-, -))$  is a f.d. symplectic space over  $K$ , then  $(\mathrm{res}_{K/k} W, \mathrm{tr}_{K/k}(-, -))$  is a symplectic space over  $k$ , and we have an inclusion  $\mathrm{Sp}(W) \subset \mathrm{Sp}(\mathrm{res}_{K/k}(W))$ .

Then the restriction of the Weil representation for  $\mathrm{Sp}(\mathrm{res}_{K/k}(W))$  on  $\mathrm{Sp}(W)$  is just the Weil representation (w.r.t. the character  $\psi_K$ ).  $\lrcorner$

*Proof:* There is a trace map  $\mathrm{tr} : H(W) \rightarrow H(\mathrm{res}_{K/k}(W))$ , and for the Schrödinger representation  $\pi : H(\mathrm{res}_{K/k} W) \rightarrow U(L^2(V))$  is the Schrödinger representation of  $H(W)$  with character  $\psi_K$ . Then the rest is easy.  $\square$

**Def. (26.8.1.8) [Lattice Model of Weil Representations].** Let  $\mathcal{L}$  be an integral lattice of  $W$ , and denote  $\mathcal{S}_{\mathcal{L}, \psi}(W)$  the space of locally constant compactly supported functions on  $W$  s.t.

$$f(x + a) = \psi((x, a))f(x)$$

for any  $x \in W$  and  $a \in \mathcal{L}$ . And let  $\mathcal{S}_m(W)$  be the subspace of  $\mathcal{S}_{\mathcal{L},\psi}(W)$  consisting of functions that are supported on  $\varpi^{-m}\mathcal{L}$ . Also let  $\mathcal{S}_m(W)^+$  and  $\mathcal{S}_m(W)^-$  be the subspace of odd and even functions resp..

Then it is easy to see that  $\dim \mathcal{S}_m(W) = \#(\varpi^{-m}\mathcal{L}/\mathcal{L}) = q^{2mn}$ .

Then there is a model of the Weil representation of  $\mathrm{Sp}(W)$  on  $\mathcal{S}_{\mathcal{L},\psi}(W)$ , s.t. for  $g \in \mathrm{Sp}(\mathcal{L})$ ,  $(gf)(x) = f(gx)$ .  $\lrcorner$

*Proof:* ? Cf.[MVW].  $\square$

**Prop. (26.8.1.9) [Defining Fields of the Weil Representation].** The Weil representation of  $\mathrm{SL}(2)$  can be defined over a number field because it is sum of its even and odd pieces, both of which are defined over number fields: the even piece because it occurs in an explicit principal series, and the odd piece because it is induced from compact open subgroup.  $\lrcorner$

*Proof:* ?  $\square$

### $\mathcal{K}$ -Types of Weil Representations

**Prop. (26.8.1.10) [Maximal Subgroups].** Suppose  $p \in \mathrm{Prime}_{\geq 3}$  and  $k \in p\text{-LField}$ , then the metaplectic covering of  $\mathrm{Sp}(W)$  splits over  $\mathrm{Sp}(\mathcal{L})$ ?, where  $\mathrm{Sp}(\mathcal{L})$  is the maximal compact subgroup of  $\mathrm{Sp}(W)$  stabilizing a lattice  $\mathcal{L}$  of  $W$ ?.  $\lrcorner$

Let  $\{\Gamma(i)\}$  be the standard filtration on  $\mathrm{Sp}(\mathcal{L})$  s.t.  $\Gamma(0) = \mathrm{Sp}(\mathcal{L})$ , and  $\Gamma(i) = \ker(\mathrm{Sp}(\mathcal{L}) \rightarrow \mathrm{Sp}(\mathcal{L}/\varpi^i))$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (26.8.1.11) [ $K$ -Types of the Weil Representation].** Suppose  $\psi$  has trivial conductor, the restriction of the Weil representation  $\omega$  of  $\mathrm{Sp}(W)$  to the maximal compact subgroup  $\mathrm{Sp}(\mathcal{L})$  (by (26.8.1.10)) decomposes as

$$\omega = \omega_0 \oplus \bigoplus_{m \in \mathbb{Z}_+} (\omega_{2m}^+ \oplus \omega_{2m}^-),$$

where  $\omega_0$  is the trivial representation, and for  $m \in \mathbb{Z}_+$ ,  $\omega_{2m}^\pm$  are irreducible representations of  $\mathrm{Sp}(\mathcal{L}/(\varpi^{2m}))$  of dimension  $\frac{1}{2}[q^{mn} - q^{(m-1)n}]$ .  $\lrcorner$

*Proof:* Denote  $\omega_{2m}^\pm = \mathcal{S}_m^\pm / \mathcal{S}_{m-1}^\pm$  (26.8.1.8). Clearly  $\mathrm{Sp}(\mathcal{L})$  leaves the spaces  $\mathcal{S}_m^\pm$  invariant, and also  $\Gamma(2m)$  acts trivially on  $\mathcal{S}_m$ .

For  $m \in \mathbb{Z}_+$ , notice  $\Gamma(m)/\Gamma(2m)$  is Abelian. To show that  $\omega_{2m}^\pm$  are irreducible representations, look at its restriction to  $\Gamma(m)/\Gamma(2m)$ . For any  $g = 1 + \varpi^m\gamma \in \Gamma(m)$ ,  $f \in \mathcal{S}_m$ ,  $x \in \varpi^{-m}\mathcal{L}$ ,

$$(gf)(x) = f(gx) = f(x + \varpi^m\gamma x) = \psi((x, \varpi^m\gamma x))f(x) = \psi((x, gx))f(x).$$

So the characters appearing in the decomposition of  $\omega_{2m}^\pm$  belongs to  $\{\psi_x\}_{x \in \varpi^{-m}\mathcal{L}}$ , where  $\psi_x(g) = \psi((x, gx))$ .

Notice that  $\psi_x = \psi_y$  iff  $x \equiv \pm y \pmod{\mathcal{L}}$ ?. And if  $f$  is a vector s.t.  $\Gamma(m)/\Gamma(2m)$  acts by  $\psi_x$ , then for any  $A \in \mathrm{Sp}(\mathcal{L})$ ,  $Af$  is a vector s.t.  $\Gamma(m)/\Gamma(2m)$  acts by  $\psi_{Ax}$ . Also notice  $\mathrm{Sp}(\mathcal{L})$  acts transitively on  $\varpi^{-m}\mathcal{L} \setminus \varpi^{-m+1}\mathcal{L}$ ?, so any irreducible representation of  $\mathrm{Sp}(\mathcal{L})/\Gamma(2m)$  restricted to  $\Gamma(m)/\Gamma(2m)$  must has dimension at least  $\frac{1}{2}[q^{mn} - q^{(m-1)n}]$ , which is exactly the dimension of  $\omega_{2m}^\pm$ . So  $\omega_{2m}^\pm$  are irreducible.  $\square$



### Character Formula for Weil Representations of $\mathrm{SL}(2)$

**Prop. (26.8.1.12) [Weil Representation associated to Quadratic Fields].** Let  $L/K$  be an extension of local fields with quadratic character  $\omega_{L/K} : K^\times \rightarrow \{\pm 1\}$ . The norm 1 elements  $E^1$  can be regarded as elements in  $\mathrm{SL}(2; F)$ , and the metaplectic cover of  $\mathrm{SL}(2; F)$  splits over  $L^1$ ?. In fact, for any character  $\gamma$  of  $L^\times$  whose restriction to  $K^\times$  is  $\omega_{L/K}$ , one can construct such a splitting.

Let  $\psi \in K^\vee$  and  $\delta \in L^\times$  s.t.  $\mathrm{tr}_{L/K}(\delta) = 0$ . Then  $\delta$  gives a symplectic structure on  $E$  by  $(x, y) = \mathrm{tr}(\delta x \bar{y})$ . Let  $\omega(\psi, \delta)$  be the Weil representation of  $\mathrm{Sp}(E) \cong \mathrm{SL}(2; F)$ , and let  $\omega(\psi, \delta, \gamma)$  be the restriction of  $\omega(\psi, \delta)$  to  $E^1$  via the splitting induced by  $\gamma$ .  $\lrcorner$

**Prop. (26.8.1.13) [Rogawski].** Let  $\chi \in (E^1)^\vee$ , then  $\chi$  appears in  $\omega(\psi, \delta, \gamma)$  iff

$$\varepsilon(\gamma \chi_E^{-1}, \psi_E) = \chi(-1) \gamma(2\delta),$$

where  $\chi_E(x) = \chi(x/\bar{x})$ ,  $\psi_E(x) = \psi(\mathrm{tr}(x))$ .  $\lrcorner$

*Proof:* Cf. [The multiplicity formula for A-packets] Prop 3.4.  $\square$

### General Weil Representations

References are [Splitting metaplectic covers of dual reductive pairs, Kudla].

**Notation (26.8.1.14).**

- Let  $\mathbb{C} \neq K \in \mathbf{LField}$ .
  - Let  $D/K$  be a division algebra with an involution  $\tau$  and center  $L$  s.t.  $L^\tau = K$  (there are three cases).
  - Let  $\epsilon \in \{\pm 1\}$ .
- $\lrcorner$

**Def. (26.8.1.15) [Tensor Symplectic Vector Spaces].** Let  $W = D^{2n} = X \oplus Y$  be the left  $D$ -vector space with a  $\epsilon$ -skew-Hermitian form

$$\langle (x_1, y_1), (x_2, y_2) \rangle = y_2^{\tau t} x_1 - \epsilon x_2^{\tau t} y_1.$$

and  $G = \mathrm{U}(W)$ .

Let  $(V, (-, -))$  be a right  $D$ -vector space of dimension  $m$  with a  $\epsilon$ -Hermitian form.

Then define

$$\mathbb{W} = V \otimes_D W, \langle -, - \rangle = \kappa \mathrm{trrd}_{D/F} ((-, -) \otimes \langle -, - \rangle^\tau).$$

where  $\kappa = 1$  if  $D = F$  and  $\kappa = 1/2$  otherwise, is a symplectic vector space of dimension  $2nmd$ .  $\lrcorner$

## 2 Theta Functions

### Lattices

**Def. (26.8.2.1) [Quadratic Lattices].** Let  $V$  be a real inner product space with an Haar measure  $\mu$  normalized that for an orthonormal basis  $e_i$  of  $V$ ,  $V/\mathbb{Z}\{e_i\}$  has volume 1, then we can identify  $V$  with  $V^*$  by this inner product. Let  $\Gamma$  be a full lattice, then its  $\mathbb{Z}$ -dual  $\Gamma^D$  is defined to be the lattice  $V^D = \{y \in V \mid (x, y) \in \mathbb{Z}, \forall x \in \Gamma\}$ . the **level** of  $\Gamma$  is defined to be the minimal  $N \in \mathbb{Z}_+$  s.t.  $N.V^D \subset V$ . It is called a **unimodular lattice** if  $\mathrm{Vol}(V/\Gamma) = 1$ .

$\Lambda$  is called a **even lattice** iff  $(x, x) \in 2\mathbb{Z}$  for any  $x \in \Gamma$ .  $\lrcorner$

**Def. (26.8.2.2) [Theta Functions].** Situation as in (26.8.2.1), for any  $x_0 \in \Gamma$ , we can define **theta functions**

$$\theta_{\Gamma, x_0} : \mathcal{H} \times \mathbb{C} \rightarrow \mathbb{C} : \theta_{\Gamma, x_0}(\tau, z) = \sum_{x \in \Gamma} e^{-\pi i[(x, x)\tau + (x, x_0)z]}.$$

and

$$\theta_{\Gamma}(\tau) = \theta_{\Gamma, 0}(\tau, 0), \quad \Theta_{\Gamma}(t) = \theta_{\Gamma}(it) = \sum_{x \in \Gamma} e^{-\pi t(x, x)}.$$

┘

**Cor. (26.8.2.3).** With the notation as in (26.8.2.2), the theta function satisfies

$$\Theta_{\Gamma}(t) = \frac{t^{-n/2}}{\mu(V/\Gamma)} \Theta_{\Gamma^D}(t^{-1})$$

┘

*Proof:* Notice that  $\Theta_{s\Gamma}(t) = \Theta_{\Gamma}(s^2 t)$ , so this formula follows from (12.11.3.16) applied to  $t^{-1/2}\Gamma$  and  $f(x) = e^{-\pi(x, x)}$ .  $\square$

**Thm. (26.8.2.4) [Hecke-Schoenberg].** Given  $k \in \mathbb{Z}_+$  and a lattice  $\Gamma$  of rank  $k$ , level  $N$  and discriminant  $\Delta$ , then

$$\theta_{\Gamma}(\tau) \in \text{MF}_k(\Gamma_0(N), \chi_D) \text{ (26.8.2.2)}.$$

┘

*Proof:* Cf. [Classical Automorphic Forms, Iwaniec] Chap 10.  $\square$

**Cor. (26.8.2.5) [Simple Theta Function].** The simple theta function  $\theta$  defined by

$$\theta(\tau) = \sum_{n \in \mathbb{Z}} e^{2\pi i n^2 \tau}$$

satisfies

$$\theta(\tau + 1) = \theta(\tau), \quad \theta\left(-\frac{1}{4\tau}\right) = \sqrt{-2i\tau} \theta(\tau).$$

┘

### Self-Dual Lattices

**Def. (26.8.2.6) [Self-Dual Lattices].** Situation as in (26.8.2.2), a **self-dual lattice** is a lattice  $\Gamma$  in  $V$  that  $V^D = V$ . Equivalently, if  $\{f_i\}$  is a  $\mathbb{Z}$ -basis of  $\Gamma$ , then the matrix  $A = ((e_i, e_j))_{i,j}$  is a matrix with integer coefficients and determinant 1. The last equivalence is because  $\Gamma' \subset \Gamma$  equals  $\Gamma$  if  $\mu(V/\Gamma) = \mu(V/\Gamma')$ , but this is equivalent to  $\mu(V/\Gamma) = 1$ , because  $\mu(V/\Gamma) \cdot \mu(V/\Gamma') = 1$ .  $\square$

**Example (26.8.2.7) [ $E_{8k}$ ].** Let  $V = \mathbb{R}^{8k}$  with the canonical inner product, denote  $E_{8k}$  the set of vectors  $\sum_i x_i e_i$  in  $V$  that

$$2x_i \in \mathbb{Z}, \quad x_i - x_j \in \mathbb{Z}, \quad \sum_{i=1}^{8k} x_i \in 2\mathbb{Z}.$$

Notice  $E_8$  is compatible with the root system  $E_8$  (3.9.6.3), and the root system just consists of all vectors in  $E_8$  of length  $\sqrt{2}$ .

Then  $E_{8k}$  is a self-dual and even.  $\square$

**Cor. (26.8.2.8).** Let  $k \geq 2$ , then all the vectors in  $E_{8k}$  of length  $\sqrt{2}$  are  $\{\pm e_i \pm e_j | i \neq j\}$ .  $\lrcorner$

**Remark (26.8.2.9).** For more examples of self-dual lattices, see [Ser73] Chap5.  $\lrcorner$

**Prop. (26.8.2.10) [Theta Function for Self-Dual Even Lattices].** Let  $\Gamma \subset V$  be a self-dual even lattice (26.8.2.6), then the dimension  $n$  of  $V$  is divisible by 8, and the theta function  $\theta_\Gamma(z) \in M_{n/2}(\Gamma(1))$ .  $\lrcorner$

*Proof:* We first show that

$$\theta_\Gamma(-1/z) = (-iz)^{n/2} \theta_\Gamma(z)$$

and because both sides are holomorphic functions on  $\mathcal{H}$ , it suffices to show this for  $z = it, t > 0$ . Thus it suffices to show

$$\Theta_\Gamma(t^{-1}) = t^{-n/2} \Theta_\Gamma(t).$$

And this is just (26.8.2.3), because  $\Gamma$  is self-dual (26.8.2.6).

Then  $\theta_\Gamma[ST]_{n/2} = (-i)^{n/2} \theta_\Gamma(20.4.5.17)$ , but  $(ST)^3 = 1$ , so  $(-i)^{3n/2} = 1$ , so  $8|n$ , and  $\theta_\Gamma \in M_{n/2}(\Gamma(1))$ .  $\square$

**Thm. (26.8.2.11) [Siegel's Mass Formula].** Cf. [Chao Li].  $\lrcorner$

**Prop. (26.8.2.12) [Theta Function for Non-Even Self-Dual Lattices].** If we consider theta function for non-even self-dual lattices in  $\mathbb{R}^n$ , then we get a modular form of weight  $n/2$  w.r.t. the subgroup of  $\mathrm{SL}(2, \mathbb{Z})$  generated by the elements  $S$  and  $T^2$ . This image of this subgroup has index 3 in  $\mathrm{PSL}(2, \mathbb{Z})$ , and it has two cusps, thus two Eisenstein series.

In particular, we can apply this to the lattice  $\{e_i\}$ , and use this information to obtain formula giving the number of ways to represent an integer into a sum of  $n$  squares.  $\lrcorner$

*Proof:*  $\square$

## Twisted Theta Functions

### 3 Jacobi Forms

Cf. [Jacobi Forms, Zagier].

**Def. (26.8.3.1) [Jacobi Forms].** A **Jacobi form** for  $\mathrm{SL}(2, \mathbb{Z})$  is a holomorphic function  $\varphi : \mathcal{H} \times \mathbb{C} \rightarrow \mathbb{C}$  s.t.

- For any  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ ,

$$\varphi\left(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d}\right) = (c\tau + d)^k e^{2\pi i \frac{mcz}{c\tau + d}} \varphi(\tau, z).$$

- For any  $\lrcorner$

**Prop. (26.8.3.2).** The theta functions  $\theta_{\Gamma, x_0}(\tau, z)$  defined in (26.8.2.2) is a Jacobi form of weight  $\mathrm{rank} \Gamma / 2$  and index  $(x_0, x_0)$ .  $\lrcorner$

#### 4 Local Theta Correspondence

**Notation (26.8.4.1).**

- Use notations as in 1.
- Let  $(K, \mathcal{O}_K, \mathfrak{p}, \varpi) \in p\text{-LField}$ .
- Let  $\psi \in (K^\times)^\vee$ .
- Let  $W$  be a symplectic space over  $K$ .

┘

**Def. (26.8.4.2) [Howe Dual Pairs].** A pair of reductive subgroups  $(G_1, G_2)$  of  $\mathrm{Sp}(W)$  is called a **Howe dual pair** if

$$Z_{\mathrm{Sp}(W)}(G_1) = G_2, \quad Z_{\mathrm{Sp}(W)}(G_2) = G_1.$$

┘

**Example (26.8.4.3) [Howe Dual Pairs].**

- Let  $V$  be an orthogonal space and  $W$  a symplectic space, then  $V \otimes W$  is naturally a symplectic space, and there is a map  $O(V) \times \mathrm{Sp}(W) \rightarrow \mathrm{Sp}(V \otimes W)$ , which makes  $(O(V), \mathrm{Sp}(W))$  a Howe dual pair. ?
- Let  $K/k$  be a quadratic extension,  $V$  a Hermitian space over  $K$ ,  $W$  a skew-Hermitian space over  $K$ , then  $V \otimes W$  is naturally a symplectic space given by

$$(v_1 \otimes w_1, v_2 \otimes w_2) = \mathrm{tr}_{K/k}((v_1, v_2) \cdot (w_1, w_2)).$$

And there is a map  $O(V) \times \mathrm{Sp}(W) \rightarrow \mathrm{Sp}(V \otimes W)$ , which makes  $(O(V), \mathrm{Sp}(W))$  a Howe dual pair. ?

┘

**Def. (26.8.4.4) [Seesaw Pairs, Kudla].** Two Howe dual pairs  $(G, G'), (H, H')$  in  $\mathrm{Sp}(W)$  is called a (Kudla) **seesaw pair** if  $H \subset G$  and  $G' \subset H'$ .

┘

**Example (26.8.4.5) [Seesaw Pairs].**

- $$\begin{array}{ccc} \mathrm{Sp}(W_1 \oplus W_2) & & O(V) \times O(V) \\ & \searrow & \swarrow \\ \mathrm{Sp}(W_1) \times \mathrm{Sp}(W_2) & & O(V) \end{array} .$$
- $$\begin{array}{ccc} O(V_1 \oplus V_2) & & \mathrm{Sp}(W) \times \mathrm{Sp}(W) \\ & \searrow & \swarrow \\ O(V_1) \times O(V_2) & & \mathrm{Sp}(W) \end{array} .$$
- $$\begin{array}{ccc} O(V_1 \oplus V_2) & & \mathrm{Sp}(W) \times \mathrm{Sp}(W) \\ & \searrow & \swarrow \\ O(V_1) \times O(V_2) & & \mathrm{Sp}(W) \end{array} .$$

┘

**Prop. (26.8.4.6) [Local Theta Lifts].** Let  $k \in p\text{-LField}$ . For any Howe dual pair  $(G_1, G_2)$  of  $\text{Sp}(W)$ , let  $\tilde{G}_i$  be the inver image of  $G_i$ , then  $\tilde{G}_1, \tilde{G}_2$  commute. For any  $\pi \in \text{Irr}^{\text{alg}}(\tilde{G}_1)$ , consider the restriction  $\omega$  of the smooth Weil representation on  $\tilde{G}_1 \times \tilde{G}_2$ , and define

$$A(\pi) = \omega / \cap \{ \ker \varphi : \varphi \in \text{Hom}_{\tilde{G}_1}(\omega, \pi) \}.$$

Then  $A(\pi) \in \text{Irr}^{\text{alg}}(\tilde{G}_1 \times \tilde{G}_2)$ , which is of the form  $\pi \otimes \theta_0(\pi)$ , where  $\theta_0(\pi) \in \text{Irr}^{\text{alg}}(\tilde{G}_2)$ .

Then  $\theta_0(\pi)$  has finite length. And if  $p \neq 2$ , then it has a unique irreducible quotient, denoted by  $\theta(\pi)$ , called the **local theta lift** of  $\pi$ .  $\lrcorner$

*Proof:* Cf.[Waldspurger, Demonstration d'une conjecture de duality de Howe dans le cas  $p$ -adique,  $p \neq 2$ ].  $\square$

### Examples

**Prop. (26.8.4.7) [Even Dimensional case].** If  $\dim V = 2n$  and denote  $\Delta = (-1)^n \det(B) \in F^\times / (F^\times)^2$ ,  $\chi : F^\times \rightarrow \{\pm 1\}$  the quadratic character  $a \mapsto (\Delta, a)_F$  (19.5.5.5), and  $\gamma(B)$  as defined in (19.5.5.12), then the projection representation of  $\text{SL}(2; F)$  in (26.8.1.5) is a true representation.

It suffices to check the relations (3.1.5.9). Use (19.5.5.17) to show that  $\gamma(B)^2 = \chi(-1)$ . And it suffices to show that

$$w_1 \begin{bmatrix} a^{-1} & \\ & a \end{bmatrix} \begin{bmatrix} 1 & -a \\ & 1 \end{bmatrix} w_1 \begin{bmatrix} 1 & -a^{-1} \\ & 1 \end{bmatrix} \Phi = \begin{bmatrix} 1 & a \\ & 1 \end{bmatrix} w_1 \Phi$$

The LHS equals  $\gamma(B)^2 |a|^{-n} \mathcal{F}(\Phi * F_{-a^{-1}B})$ , which by (19.5.5.12) and (19.5.5.16) equals  $\gamma(B) \mathcal{F}(\Phi) F_{aB}$ , which is just the RHS.  $\lrcorner$

**Def. (26.8.4.8) [Setup for Dihedral Weil Representation].** Let  $E$  be a 2-dimensional commutative semisimple algebra over  $F$ , then  $E = F \oplus F$  and  $F$  embeds diagonally, called the **split case**; or  $E$  is the unique quadratic extension of  $F$ , called the **anisotropic case**. Let the automorphism of  $E$  given by

$$x \mapsto \bar{x} : \begin{cases} (\xi, \eta) = (\eta, \xi), & E = F \oplus F \\ \text{The non-trivial Galois automorphism,} & E \text{ is a field} \end{cases}.$$

Let  $\text{tr}(x) = x + \bar{x}$ ,  $N(x) = x\bar{x}$ . Let  $E_1^* = \{x \in E^* | N(x) = 1\}$ .

There is an embedding  $\iota$  of  $E$  into  $GO(E) \subset \text{GL}(2; F)$ : for  $E = F \oplus F$ , it embeds diagonally, and for  $E$  a field,  $E$  embeds by left action on itself. Notice  $\iota(\overline{x})(\overline{y}) = \iota(\bar{x})(y)$ , thus  $GO(E) \cong E^* \rtimes \{c\}$  ?.

Let  $\psi$  be a character of  $E^*$  that is non-trivial on  $E_1^*$ .  $\lrcorner$

**Prop. (26.8.4.9) [Theta Correspondence of Dihedral Representations].** Let  $\psi$  be a character of  $E^*$ , then by (24.1.1.12),  $\psi$  extends to a representation of  $GO(E)$  iff  $\psi(x) = \psi(\bar{x})$ , which is equivalent to  $\psi(x) = 1$  on  $E_1^*$ , by Hilbert's theorem90 or direct inspection. Then by (24.1.1.12), in the other case,  $\text{ind}_{E^*}^{GO(E)} \psi$  is an irreducible representation of  $GO(E)$ , and Howe duality predicts an irreducible smooth representation of  $GL(2, F)$ . In case  $E$  is a field or  $F$  is finite, we can construct this representation directly, in (26.8.4.10) and (24.3.10.8).  $\lrcorner$

**Prop. (26.8.4.10) [Dihedral Representations].** Let  $E/F$  be a quadratic extension of non-Archimedean local fields, then  $E$  is a quadratic space over  $F$  by the norm form, and let  $\xi$  be a

quasi-character of  $E^*$  that doesn't factor through the norm map  $N : E^* \rightarrow F^*$ . Let  $U_{\xi, \psi}$  be the space of functions  $\Phi \in C_c^\infty(E)$  that satisfy

$$\Phi(yv) = \xi(y)^{-1} \Phi(v), \quad \forall y \in E^*, N(y) = 1,$$

and let  $\chi : F^* \rightarrow \{\pm 1\}$  be the quadratic character attached to the extension  $E/F$  (19.7.2.14), and let  $GL(2, F)_+$  be the subgroup of  $GL(2, F)$  that the determinants are norms from  $E$ , which is an open normal subgroup of index 2, then there exists an irreducible admissible representation  $\omega_{\xi, \psi}$  of  $GL(2, F)_+$  on  $U_{\xi, \psi}$  s.t.

$$(\omega_{\xi, \psi} \left( \begin{bmatrix} a & \\ & 1 \end{bmatrix} \right) \Phi)(v) = |a|^{1/2} \xi(b) \Phi(bv), \quad \forall b \in E^*, N(b) = a \in F^*$$

$$(\omega_{\xi, \psi} \left( \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \right) \Phi)(v) = \psi(xN(v)) \Phi(v)$$

$$(\omega_{\xi, \psi} \left( \begin{bmatrix} a & \\ & a^{-1} \end{bmatrix} \right) \Phi)(v) = |a| \chi(a) \Phi(av)$$

$$(\omega_{\xi, \psi}(w_1) \Phi) = \gamma(N) \hat{\Phi}.$$

Then the representation  $\omega_\xi$  of  $GL(2, F)$  induced from this representation of  $GL(2, F)_+$  is irreducible and cuspidal.  $\perp$

**Remark (26.8.4.11).** This dihedral representation is the only cuspidal representation of  $GL(2; F)$ , by [Tunnell 1978 or 1979].  $\textcolor{red}{?}$   $\perp$

*Proof:* Notice in this case, the character  $\chi$  defined in (26.8.4.7) is the same as the quadratic character corresponding to  $E/F$ : let  $E = F(\sqrt{D})$ , then  $N_{E/F} = \langle 1, -D \rangle$ , thus  $(a, D)_F = 1$  iff  $\langle a, D, -1 \rangle$  is universal iff  $a$  is represented by  $\langle 1, -D \rangle$ , which is equivalent to  $a \in N_{E/F}(E^*)$ . Then by (26.8.4.7), we have an action of  $SL(2, F)$  on  $L^2(E)$ , and we can extend it to  $GL(2, F)_+$  satisfying all these equations (  $\textcolor{red}{?}$  by direct verification). And it can be verified  $U_{\xi, \psi}$  is stable under this action.

We show this action is smooth: Every  $\varphi \in U_{\xi, \psi}$  is stable under some  $K(\mathfrak{p}^n)$  for  $n$  large: By (24.11.1.6), it suffices to prove it is stable under  $N_-(\mathfrak{p}^n)$ ,  $T(\mathfrak{p}^n)$  and  $N(\mathfrak{a})$ . By action of  $w_1$ , it suffices to show it is stable under the latter two. But this is clear from the equations above.

Similarly, to show it is admissible, it suffices to show for any ideal  $\mathfrak{a}$ , the space of functions fixed by  $N(\mathfrak{a})$  and  $N_-(\mathfrak{a})$  is of f.d.: It is clear from equation 2 that if  $\varphi$  is fixed by  $N(\mathfrak{a})$ , then there is a fractional ideal  $\mathfrak{a}'$  that  $\text{Supp}(\varphi) \subset N^{-1}(\mathfrak{a}')$ . Similarly for  $\mathcal{F}(\varphi)$ . These two conditions means  $\varphi$  has bounded support and fixed by some open subgroup, these functions are f.d..

To show  $\omega_{\xi, \psi}$  and  $\omega_\xi$  are irreducible: Let  $B_1(F)_+$  be the subgroup of  $B_1(F)$  consisting of matrices  $\left\{ \begin{bmatrix} a & b \\ & 1 \end{bmatrix} \mid a \in N_{E/F}(E^*) \right\}$ , let  $\mathcal{K}_+$  the restriction of  $\omega_{\xi, \psi}$  to  $B_1(F)_+$ . Define a map

$$\Lambda : C_c^\infty(N(E^*)) \rightarrow \mathcal{K}_+ : (\Lambda \varphi)(v) = \xi(v)^{-1} |N(v)|^{-1/2} \varphi(N(v)),$$

then this is an isomorphism, and is  $B_1(F)$ -invariant, where  $B_1(F)$  acts on  $C_c^\infty(N(E^*))$  as in (24.11.3.8). Thus by Mackey decomposition (24.1.6.48),

$$\text{res}_{B_1(F)} \text{ind}_{GL(2; F)_+}^{GL(2; F)} \omega_{\xi, \psi} \cong \text{ind}_{B_1(F)_+}^{B_1(F)} \omega_{\xi, \psi} \cong \text{ind}_{N(F)}^{B_1(F)} \psi,$$

which is irreducible by (24.11.3.9). thus  $\omega_\xi$  is irreducible, thus also does  $\omega_{\xi, \psi}$ .

A byproduct of the above argument is that  $\omega_\xi$  is cuspidal.  $\square$

### Theta Correspondence and Langlands Functoriality

One would like to understand the theta correspondence in terms of Langlands' functoriality

## 5 Global Theta Correspondence

**Notation (26.8.5.1).**

- Let  $F \in \mathbf{NField}$ ,
- Let  $W = X \oplus Y$  be a non-degenerate symplectic space over  $F$  where  $X, Y$  are isotropic subspaces.

┘

### Global Weil Representations

**Def. (26.8.5.2) [Global Weil Representations].** The restricted tensor product of the local projective Weil representations of  $\mathrm{Sp}(W_v)$  on  $\mathcal{S}(X(F_v))$  is a projective representation of  $\mathrm{Sp}(W)(\mathbf{A}_F)$  on  $\mathcal{S}(X(\mathbf{A}_F))$ .

Then this projective representation lifts to a true representation of  $\mathrm{Mp}(W)(\mathbf{A}_F)$  on  $\mathcal{S}(X(\mathbf{A}_F))$ , called the (global) **Weil representation**.

┘

*Proof:*

□

**Prop. (26.8.5.3) [Weil].** The covering  $\mathrm{Mp}(W)(\mathbf{A}_F) \rightarrow \mathrm{Sp}(W)(\mathbf{A}_F)$  splits over  $\mathrm{Sp}(W)(F)$ .

┘

*Proof:*

□

### Theta Lifts

**Def. (26.8.5.4) [Theta Kernels].** The **theta kernel** on  $\mathcal{S}(X(\mathbf{A}_F))$  is defined by

$$\Theta(g, \Phi)(x) = \sum_{y \in X(F)} \omega(g)\Phi(x+y) \text{ where } \Phi \in \mathcal{S}(X(\mathbf{A}_F)), g \in \mathrm{Mp}(W)(\mathbf{A}_F), x \in X(\mathbf{A}_F).$$

Then this is well-defined, and ?

┘

**Def. (26.8.5.5) [Theta Functions].** For any  $\Phi \in \mathcal{S}(X(\mathbf{A}_F))$ , there is a function

$$\theta_\Phi : \mathrm{Sp}(W)(F) \backslash \mathrm{Mp}(W)(\mathbf{A}_F) \rightarrow \mathbb{C} : g \mapsto \theta_\Phi(g) = \Theta(g, \Phi)(0) \text{ (26.8.5.4)}.$$

Then this is an automorphic function. ? It has the property  $\theta_\Phi(hg) = \theta_{\omega(h)\Phi}(g)$  for  $h \in \mathrm{Mp}(W)(\mathbf{A}_F)$ .

┘

*Proof:* To show the sum is convergent: ?

□

**Def. (26.8.5.6) [Theta Lifts].** Let  $(G_1, G_2)$  be a Howe dual pair in  $\mathrm{Sp}(W)$  (26.8.4.2), and  $f \in \mathcal{A}_0(G_1(F) \backslash G_1(\mathbf{A}_F))$ ,  $\Phi \in \mathcal{S}(X(\mathbf{A}_F))$ , define

$$\theta_\Phi(f) : G_2(F) \backslash G_2(\mathbf{A}_F) \rightarrow \mathbb{C} : g_2 \mapsto \theta_\Phi(f)(g_2) = \int_{G_1(F) \backslash G_1(\mathbf{A}_F)} \theta_\Phi(g_1 g_2) f(g_1) dg_1 \text{ (26.8.5.5)}.$$

Then this is an automorphic function ? . Moreover, the space  $\{\theta_\varphi(f) : \varphi \in \mathcal{S}(X)(\mathbf{A}_F)\}$  is  $\tilde{G}_2(\mathbf{A}_F)$  invariant by (26.8.5.5).

Denote

$$\Theta(\mathcal{A}, \psi) = \{\theta_\Phi(f) : \Phi \in \mathcal{S}(X)(\mathbf{A}_F), f \in \mathcal{A}_0(G_1(F) \backslash G_1(\mathbf{A}_F))\} \in \mathcal{A}(G_2(F) \backslash G_2(\mathbf{A}_F)),$$

called the **theta lifts of cuspidal representations** of  $G_1$ . And any irreducible subquotient of it is called a representation obtained via **theta correspondence**.  $\lrcorner$

**Remark (26.8.5.7) [Questions].** 1: When is  $\Theta(\mathcal{A}, \psi)$  non-zero?

2: What is the relation of the space of theta lifts to local theta correspondences.  $\lrcorner$

**Thm. (26.8.5.8) [Rallis].** If all the theta lifts (26.8.5.6) are cusp forms, then the space generated by them is irreducible and identifies with  $\otimes_v (\pi_v^\vee)$ .  $\lrcorner$

*Proof:* Cf. [Rallis, On Howe Duality Conjecture].  $\square$

**Thm. (26.8.5.9) [Theta Correspondence for  $(\mathrm{PGL}(2), \widetilde{\mathrm{SL}(2)})$ , Waldspurger].** For any  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{PGL}(2)/F)$ ,  $\lrcorner$

*Proof:* ?  $\square$

### Orthogonal Theta Lifts

**Notation (26.8.5.10).**

- Let  $F \in \mathbf{NField}$ .
- Let  $\psi_0 : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$  be the standard additive character.
- Let  $V \in \mathrm{QuadSpa}^m/F$  be non-degenerate with  $m$  even.
- Let  $r$  be the Witt index of  $V$  (19.5.2.5).
- Then  $(O(V), \mathrm{Sp}(n))$  is a Howe dual pair. Denote  $H = O(V), G = \mathrm{Sp}(n)$ .
- Let  $\mathcal{K} \in \mathrm{Sp}(n; \mathbf{A}_F)$  be the maximal compact subgroup ?.
- Denote  $\varrho_n = \frac{n+1}{2}, s_0 = \frac{m}{2} - \varrho_n$ .

$\lrcorner$

**Def. (26.8.5.11) [Standard Maximal Compact Groups].**  $\mathrm{Sp}(n)$  is defined over  $\mathbb{Z}$ , and for any  $v \in \Sigma_F$ , define the standard maximal compact subgroups  $\mathcal{K}_v$  of  $\mathrm{Sp}(n; F_v)$  to be

$$\mathcal{K}_v = \begin{cases} \mathrm{Sp}(n; \mathcal{O}_{F_v}) & , v \in \Sigma_F^{\mathrm{fin}} \\ \mathrm{Sp}(n; \mathcal{O}_{F_v}) \cap \mathrm{Sp}(2n) & , v \in \Sigma_F^\infty \end{cases}.$$

And denote  $\mathcal{K} = \prod_{v \in \Sigma_F} \mathcal{K}_v$ .

Then any  $g \in G(\mathbf{A}_F)$  can be written as  $g = pk$ , where  $p = \begin{bmatrix} A & * \\ 0 & A^{-t} \end{bmatrix} \in P_{\mathrm{Sie}}(\mathbf{A}_F)$  and  $k \in \mathcal{K}$ .

Then we denote

$$|A(g)|_{\mathbf{A}} = |\det(A)|_{\mathbf{A}},$$

which is well-defined.  $\lrcorner$



**Def. (26.8.5.12) [Siegel Theta Functions].** For  $\Phi \in S(V^n(\mathbf{A}_F))$  and  $g \in G(\mathbf{A}_F), h \in H(\mathbf{A}_F)$ , the **Siegel Theta function**(26.8.5.5) is

$$\theta_\Phi(g, h) = \sum_{x \in V^n} \omega(g, h) \Phi(x),$$

which is smooth of moderate growth?, and the **Siegel Theta lifts** is

$$\text{Int}(g, \Phi) = \int_{H(F) \backslash H(\mathbf{A}_F)} \theta(g, h; \Phi) dh \text{ (26.8.5.12)}.$$

where  $dh$  is normalized s.t.  $\text{Vol}(H(F) \backslash H(\mathbf{A}_F)) = 1$ . Then this integral is absolutely convergent if  $2\varrho_n < m - r$  or  $r = 0$ (26.8.5.10).  $\lrcorner$

*Proof:* Cf.[Wei65]Prop8.?.  $\square$

**Def. (26.8.5.13) [Degenerate principal Series].** Denote

$$I_v(\chi; s) = I_P^G(\chi(s)),$$

which consists of  $\mathcal{K}$ -finite functions  $f^{(s)}$  on  $\text{Sp}(n; \mathbf{A}_F)$  s.t.

$$f^{(s)}(pg) = \chi_{\text{disc}(V)}(\det(A)) |\det(A)|^m f(g), \text{ where } p = \begin{bmatrix} A & * \\ & A^{-t} \end{bmatrix} \in P(\mathbf{A}_F).$$

Where  $\lrcorner$

### Unitary Theta Lifts

**Notation (26.8.5.14).**

- Let  $E/F \in \mathbf{NField}$  be a CM field.
- Let  $\mathbf{1} \neq \psi : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$ .
- Let  $\chi_{E/F}$  be the quadratic Hecke character associated to  $E/F$ .
- Let  $V \in \text{HermSpa}^m(E/F)$  be non-degenerate.
- Let  $r$  be the Witt index of  $V$ (19.5.2.5).
- Let  $\chi$  be a Hecke character of  $E$  s.t.  $\chi|_{\mathbf{I}_F} = \chi_{E/F}^m$ .
- Then  $(\text{U}(V), \text{U}(n, n)_F)$ (10.5.3.19) is a Howe dual pair. Denote  $H = \text{U}(V), G = \text{U}(n, n)$ .  $G$  has a model over  $\mathcal{O}_F$ .
- Let  $P_{\text{Sie}} \subset G$  be the Siegel parabolic subgroup.
- Let  $S(V^n(\mathbf{A}_F))$  be the subspace of  $\mathcal{S}(V^n(\mathbf{A}_F))$  consisting of functions that corresponds to polynomials in the Fock model?.
- $(G, H)$  is a Howe dual pair, and let  $\omega_{\psi, \chi}$  be the Weil representation of  $G(\mathbf{A}_F) \times H(\mathbf{A}_F)$  acting on  $\mathcal{S}(V^n(\mathbf{A}_F))$ .
- Denote  $\varrho_n = \frac{n+1}{2}, s_0 = \frac{m}{2} - \varrho_n$ .

$\lrcorner$

**Def. (26.8.5.15) [Standard Maximal Compact Groups].**  $U(n, n)_F$  is defined over  $\mathcal{O}_F$ , and for any  $v \in \Sigma_F$ , define the standard maximal compact subgroups  $\mathcal{K}_v$  of  $U(n, n)(F_v)$  to be

$$\mathcal{K}_v = \begin{cases} U(n, n)(\mathcal{O}_{F_v}) & , v \in \Sigma_F^{\text{fin}} \\ U(n, n)(\mathcal{O}_{F_v}) \cap U(2n) & , v \in \Sigma_F^\infty \end{cases}.$$

And denote  $\mathcal{K} = \prod_{v \in \Sigma_F} \mathcal{K}_v$ .

Then any  $g \in G(\mathbf{A}_F)$  can be written as  $g = pk$ , where  $p = \begin{bmatrix} A & * \\ 0 & \mathbf{c}(A)^{-t} \end{bmatrix} \in P_{\text{Sie}}(\mathbf{A}_F)$  and  $k \in \mathcal{K}$ .

Then we denote

$$|A(g)|_{\mathbf{A}} = |\det(A)|_{\mathbf{A}},$$

which is well-defined. ┘

**Def. (26.8.5.16) [Local Degenerate Principal Series].** Let  $v \in \Sigma_F$  be inert for  $E/F$ , thus  $E_v/F_v$  is a quadratic extension. Denote

$$I_v(\chi; s) = I_{P(F_v)}^{G(F_v)}(\chi(s)),$$

which consists of  $\mathcal{K}_v$ -finite functions  $f^{(s)}$  on  $G(F_v)$  s.t.

$$f^{(s)}(pg) = \chi_v(\det(A)) |\det(A)|_v^{s + \frac{n}{2}} f^{(s)}(g), \text{ where } p = \begin{bmatrix} A & * \\ & \mathbf{c}(A)^{-t} \end{bmatrix} \in P(F_v).$$

A **standard holomorphic section**  $f^{(s)}$  is a holomorphic section s.t. its restriction to  $\mathcal{K}_v$  is independent of  $s$ .

Let  $R_v(V, \chi)$  denote the image of the intertwining operator

$$S(V^n) \rightarrow I_v(\chi) : \Phi_v \mapsto \theta_{\Phi_v}(g, 1) \text{ (26.8.5.5),}$$

┘

## 6 Borchers Products

References are [Borchers] and [Borchers Products on  $0(2, 1)$  and Chern Classes of Heegner Divisors].

**Def. (26.8.6.1) [Modular Forms with Heegner Points].** A meromorphic modular function  $f$  for  $SL(2; \mathbb{Z})$  is said to have a Heegner divisor if its zeros and poles are supported at the cusp  $i\infty$  and CM points. ┘

## 26.9 Eisenstein Series

*“...I had not recognized in 1966, when I discovered after many months of unsuccessful search a promising definition of automorphic  $L$ -function, what a fortunate, although, and this needs to be stressed, unforeseen by me, or for that matter anyone else, blessing it was that it lay in the theory of Eisenstein series.”*

– R. P. Langlands.

**Remark (26.9.0.1).** The motivation to study Eisenstein series is that  $L$ -functions appear as constant terms of Eisenstein series.  $\lrcorner$

### 1 Eisenstein Series for $P_{1,n-1}$

References are [J-S76].

**Def. (26.9.1.1)** [Eisenstein Series for  $P_{1,n-1}$ ].  $\lrcorner$

### 2 Non-Holomorphic Eisenstein Series for $GL(2)$

References are [Bump, Chap3] and [Diamond]Chap4.10.

**Def. (26.9.2.1)** [Non-Holomorphic Eisenstein Series]. The non-holomorphic Eisenstein series of weight  $s$  is defined to be

$$E^*(z, s) = \frac{1}{2} L_{\mathbb{R}}(2s) \sum_{(m,n) \neq 0 \in \mathbb{Z}^2} \frac{(\operatorname{Im}(z))^s}{|mz + n|^{2s}} = L_{\mathbb{R}}(2s) \sum_{\gamma \in \overline{\Gamma}_{\infty} \backslash \overline{\Gamma}(1)} (\operatorname{Im}(\gamma(z)))^s$$

It is absolutely convergent for  $\operatorname{Re}(s) > 1$ , and is automorphic for  $\Gamma(1)$  (26.1.3.20).  $\lrcorner$

**Prop. (26.9.2.2)** [Fourier Expansion of  $E(z, s)$ ]. The Fourier coefficients of  $E(z, s)$  is of the form

$$E^*(z, s) = \sum_{n=-\infty}^{\infty} a_n(y, s) e^{2\pi i r x},$$

where

$$a_0(y, s) = \Lambda(2s)y^s + \Lambda(2-2s)y^{1-s}$$

and for  $n \neq 0$ ,

$$a_n(y, s) = 2|n|^{-s+1/2} \sigma_{2s-1}(|n|) \sqrt{y} K_{s-1/2}(2\pi|n|y),$$

where  $K_{s-1/2}(z)$  is the K-Bessel function (12.6.3.1). In particular,

$$E^*(z, s) = \Lambda(2s)y^s + \Lambda(2-2s)y^{1-s} + 4\sqrt{y} \sum_{n \in \mathbb{Z}_+} n^{\frac{1}{2}-s} \sigma_{2s-1}(n) K_{s-\frac{1}{2}}(2\pi ny) \cos(2\pi nx)$$

$\lrcorner$

*Proof:*

$$a_r(y, s) = \int_0^1 E^*(x + iy, s) e^{-2\pi i r x} dx = \int_0^1 \frac{1}{2} \pi^{-s} \Gamma(s) \sum_{(m,n) \neq 0 \in \mathbb{Z}^2} \frac{y^s}{|mz + n|^{2s}} e^{-2\pi i r x} dx.$$

The term with  $m = 0$  only contributes to  $a_0$ , and equals  $\pi^{-s}\Gamma(s)\zeta(2s)y^s$ . For the rest we may assume  $m > 0$  by symmetry, then they contribute to

$$\begin{aligned} & \pi^{-s}\Gamma(s)y^s \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} \int_0^1 [(mx+n)^2 + m^2y^2]^{-s} e^{-2\pi irx} dx \\ &= \pi^{-s}\Gamma(s)y^s \sum_{m=1}^{\infty} \sum_{n \bmod m} \int_{-\infty}^{\infty} [(mx+n)^2 + m^2y^2]^{-s} e^{-2\pi irx} dx \\ &= \pi^{-s}\Gamma(s)y^s \sum_{m=1}^{\infty} m^{-2s} \sum_{n \bmod m} e^{2\pi irn/m} \int_{-\infty}^{\infty} (x^2 + y^2)^{-s} e^{-2\pi irx} dx \end{aligned}$$

Notice  $\sum_{n \bmod m} e^{2\pi irn/m} = \begin{cases} m & m|r \\ 0 & \text{otherwise} \end{cases}$ , thus the calculation reduces to (12.6.3.5).  $\square$

**Prop. (26.9.2.3) [Functional Equation].**  $E^*(z, s)$  has meromorphic continuation to all  $s \in \mathbb{C}$ , and it is analytic except at  $s = 1$  or  $s = 0$ , where the residue at  $s = 1$  is  $1/2$  for any  $z$ , and it satisfies the functional equation

$$E^*(z, s) = E^*(z, 1 - s).$$

and  $E(x + iy, s) = O(y^\sigma)$  for  $y \rightarrow \infty$ , where  $\sigma = \max(\operatorname{Re}(s), 1 - \operatorname{Re}(s))$ .  $\lrcorner$

*Proof:* This follows from the Fourier expansion of  $E^*(z, s)$  (26.9.2.2). Each term  $a_r(y, s)$  has analytic extension to all  $s$ , except that  $a_0$  has simple poles at  $s = 0$  and  $s = 1$  by (35.5.2.5) (the pole at  $s = 1/2$  was neutralized). And the functional equation and the convergence is clear from the properties of K-Bessel functions (12.6.3.2). For the residue, it suffices to show the residue of  $\pi^{-s/2}\Gamma(s/2)\zeta(s)$  at 0 is  $-1$  (35.5.2.5).  $\square$

**Prop. (26.9.2.4) [Kronecker's First Limit Formula].**

$$2E^*(z, s) = \frac{1}{s-1} + \gamma_0 - \log(4\pi y |\eta(z)|^4) + O(s-1).$$

And a similar formula is true for  $s = 0$ , by the functional equation (26.9.2.3).  $\lrcorner$

*Proof:* Cf. [Kronecker's First Limit Formula, Revisited].  $\square$

**Prop. (26.9.2.5) [Eisenstein Series of Mixed Type].** The Eisenstein series of mixed type is defined to be

$$E_{k,s}(z) = \sum_{(m,n) \in \mathbb{Z}^2 \setminus (0,0)} \frac{y^s}{(mz+n)^k |mz+n|^{2s}}, k \in \mathbb{Z}, s \in \mathbb{C}.$$

Then

$$R_k(y^{k/2} E_{k,s}(z)) = (k+s)y^{(k+2)/2} E_{k+2,s}(z).$$

and  $y^{k/2} E_{k,s}(z)$  is a Maass form in  $(z)$  of weight  $k$ . Its better to consult the paper <https://arxiv.org/abs/1803.08210> for the Fourier expansion of  $E_{k,s}$ .  $\lrcorner$

*Proof:*

$$((z - \bar{z}) \frac{\partial}{\partial z} + k/2) \left[ \frac{\left( \frac{z - \bar{z}}{2i} \right)^{s+k/2}}{(mz+n)^{s+k} (m\bar{z}+n)^s} \right]$$

$$\begin{aligned}
&= (z - \bar{z}) \left( \frac{\frac{s+k/2}{2i} \left( \frac{z-\bar{z}}{2i} \right)^{s-1+k/2} (mz+n)^{s+k} - \left( \frac{z-\bar{z}}{2i} \right)^{s+k/2} m(s+k)(mz+n)^{s+k-1}}{(m\bar{z}+n)^s (mz+n)^{2s+2k}} \right) + \frac{k}{2} \frac{\left( \frac{z-\bar{z}}{2i} \right)^{s+k/2}}{(mz+n)^{s+k} (m\bar{z}+n)^s} \\
&= \frac{\left( \frac{z-\bar{z}}{2i} \right)^{s+k/2}}{(mz+n)^{s+k+1} (m\bar{z}+n)^s} [(s+k/2)(mz+n) - (z-\bar{z})m(s+k) + \frac{k}{2}(mz+n)] \\
&= (k+s) \frac{\left( \frac{z-\bar{z}}{2i} \right)^{s-1+(k+2)/2}}{(mz+n)^{s+k+1} (m\bar{z}+n)^{s-1}} \\
&= (k+s) y^{(k+2)/2} E_{k+2, s-1}(z)
\end{aligned}$$

□

**Prop. (26.9.2.6) [Rankin-Selberg].** Let  $\varphi \in C^\infty(\Gamma(1) \backslash \mathcal{H})$  be decreasing rapidly along  $y \rightarrow \infty$  and  $\varphi(z) = \sum_{n \in \mathbb{Z}} \varphi_n(y) e^{2\pi i n x}$  be its Fourier expansion. Call  $\varphi_0$  the constant term of  $\varphi$ . Consider the Mellin transform

$$M(s, \varphi_0) = \int_0^\infty \varphi_0(y) y^s \frac{dy}{y} \quad (12.5.4.2), \quad \Lambda(s, \varphi_0) = \Lambda(2s) M(s-1, \varphi_0).$$

As  $\varphi(s, \varphi_0)$  is bounded as a function on  $y$  and decay rapidly,  $M(s, \varphi)$  is absolutely convergent for  $\operatorname{Re}(s) > 0$ .

Then

$$\Lambda(s, \varphi_0) = \int_{\Gamma(1) \backslash \mathcal{H}} E^*(z, s) \varphi(z) \frac{dx dy}{y^2}$$

and thus has a meromorphic continuation to all  $s$  and satisfies a functional equation, with at most simple poles at  $s = 0$  or  $1$ ,

$$\operatorname{Res}_{s=1} \Lambda(s, \varphi_0) = \frac{1}{2} \int_{\Gamma(1) \backslash \mathcal{H}} \varphi(z) \frac{dx dy}{y^2}.$$

┘

*Proof:* Using (26.9.2.1),

$$\begin{aligned}
\int_{\Gamma(1) \backslash \mathcal{H}} \varphi(z) \frac{dx dy}{y^2} &= \Lambda(2s) \sum_{\gamma \in \Gamma_\infty \backslash \Gamma(1)} \int_{\Gamma(1) \backslash \mathcal{H}} (\operatorname{Im}(\gamma(z)))^s \varphi(\gamma(z)) \frac{dx dy}{y^2} \\
&= \Lambda(2s) \int_{\Gamma_\infty \backslash \mathcal{H}} \operatorname{Im}(z)^s \varphi(z) \frac{dx dy}{y^2} \\
&= \Lambda(2s) \int_0^\infty y^{s-1} \int_0^1 \varphi(x + iy) dx \frac{dy}{y} \\
&= \Lambda(2s) \int_0^\infty \varphi_0(y) y^{s-1} \frac{dy}{y} \\
&= \Lambda(s, \varphi_0)
\end{aligned}$$

The assertion about the meromorphic continuation and the residue follows from the equation above and (26.9.2.3). □

## 26.10 Automorphic Forms on Unitary Groups

**Prop. (26.10.0.1)**[Ramakrishnan, 2000]. Multiplicity one theorem is true for cuspidal representations on  $\mathrm{SL}(2)$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (26.10.0.2)** [Blasius, 1994]. Multiplicity one theorem is false for cuspidal representations on  $\mathrm{SL}(n)$ ,  $n \geq 3$ .  $\lrcorner$

*Proof:*  $\square$

### 1 Definite Unitary Groups

References are [Automorphic Forms on Unitary Groups, Eischen].

Unitary groups have associated Shimura varieties, which provide convenient structure for studying algebraic aspects of automorphic forms. We have substantial results about Galois representations associated to automorphic forms on unitary groups (e.g. [Ski12, Che04, Che09, CH13, Har10]). In addition, we have convenient representations of the  $L$ -functions associated automorphic forms on unitary groups, which are useful both for proving analytic properties and for extracting algebraic information (and even  $p$ -adic properties, as seen in [EHLS20]). Working with unitary groups has enabled major developments, including a proof of the main conjecture of Iwasawa Theory for  $\mathrm{GL}_2$  [SU14] and the rationality of special values of certain automorphic  $L$ -functions (including [Shi00, Har97, Har08, Har84, Bou15]), as well as progress toward cases of the Bloch – Kato conjecture (including [SU06, Klo09, Klo15, Wan19]), and the Gan – Gross – Prasad conjecture (many recent developments, including [Xue14, Xue19, Zha14, Liu14, Yun11, JZ20, He17, BP20, BPLZZ21]).

**Def. (26.10.1.1)** [Definite Unitary Groups]. Let  $m \in \mathbb{Z}_+$ ,  $E/\mathbb{Q}$  be a CM field and  $G \in \mathrm{AlgGrp}/\mathbb{Q}$  be a unitary group attached to  $E/\mathbb{Q}$  and the central simple algebra  $\mathbb{D} \cong \mathrm{Mat}(m; E)$  with an involution (10.5.3.21).

Then for any  $p \in \mathrm{Prime}$ ,

- If  $p = \mathfrak{p}_v \mathfrak{p}_{v'}$  is a splitting in  $E$ , then  $v : E \rightarrow \mathbb{Q}_p$  induces an isomorphism  $G(\mathbb{Q}_p) \cong \mathrm{GL}(m; \mathbb{Q}_p)$ .
- If  $p$  is inert or ramified in  $E$ , then  $G(\mathbb{Q}_p)$  is a unitary group attached to  $E_p/\mathbb{Q}_p$ .
- Each embedding  $E \rightarrow \mathbb{C}$  gives an isomorphism between  $G(\mathbb{R})$  and a Archimedean unitary group  $\mathrm{U}(p, q)$ , where  $p + q = m$  is the signature.

$G$  is called a **definite unitary group** if  $G(\mathbb{R})$  is of signature  $(n, 0)$  or  $(0, n)$ .  $\lrcorner$

**Def. (26.10.1.2)** [Quasi-Split Definite Unitary Groups]. Let  $E/\mathbb{Q}$  be a CM field,  $m \in \mathbb{Z}_+$ , and let  $\mathrm{U}(m) \in \mathrm{AlgGrp}/\mathbb{Q}$  be the unitary group attached to the positive-definite  $\mathfrak{c}$ -Hermitian form

$$q((z_1, \dots, z_m)) = \sum_{i=1}^m \mathrm{Nm}(z_i)$$

as in (26.10.1.1), then

- $\mathrm{U}(m)$  is a definite unitary group.
- If  $p \in \mathrm{Prime}$  doesn't split in  $E$  and  $m \not\equiv 2 \pmod{4}$ , then  $\mathrm{U}(m)(\mathbb{Q}_p)$  is the quasi-split unitary group attached to  $E_p/\mathbb{Q}_p$ .

- If  $m \not\equiv 2 \pmod{4}$ , then  $U(m)$  is the unique  $m$ -variable unitary group attached to  $E/\mathbb{Q}$  that is quasi-split at every finite place and compact at infinity. And if  $m \equiv 2 \pmod{4}$ , then there is no group with these properties.

┘

*Proof:* This follows easily from (19.5.5.11). For  $m \equiv 2 \pmod{4}$ , this follows from Hasse's principle ?.

□

**Prop. (26.10.1.3).** For any definite unitary group  $G$  and  $\pi \in \text{Irr}^{\text{auto}}(G/\mathbb{Q})$ ,  $\pi_\infty$  is algebraic and  $\pi_f$  is defined over  $\overline{\mathbb{Q}}$ .

┘

### Galois Representations attached to Automorphic Representations

**Notation (26.10.1.4).**

- Situation as in (26.10.1.1), let  $G \in \text{AlgGrp}/\mathbb{Q}$  be a definite unitary group.
- Fix a prime  $p \in \text{Prime}$  that is split in  $E$ .
- Fix an embedding  $\iota_p : \overline{\mathbb{Q}} \rightarrow \overline{\mathbb{Q}_p}$ , and  $v \in S_E(p)$  the place of  $E$  induced by  $\iota_p$ .
- Fix a prime  $\ell \in \text{Prime}$  that is non-split in  $E$ .
- 

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**Def. (26.10.1.5) [Unramified Representations of  $G(\mathbb{Q}_\ell)$ ].** If  $\ell$  is unramified in  $E$ , then  $G(\mathbb{Q}_\ell)$  is an unramified unitary group ?, so we can talk about unramified representations of  $G(\mathbb{Q}_\ell)$ .

If  $\ell$  is inert in  $E$ , then we can also talk about unramified representations of  $G(\mathbb{Q}_\ell)$  meaning a representation of  $G(\mathbb{Q}_\ell)$  that has a non-zero vector invariant under a **very special maximal compact subgroup** ?, Cf. [B-C09].

┘

**Conj. (26.10.1.6) [Galois Representations attached to Automorphic Representations of  $U(m)$ ].** Let  $\pi \in \text{Irr}^{\text{auto}}(U(m)/\mathbb{Q})$  s.t.  $\pi_\infty$  has regular weight, then there is a continuous semisimple Galois representation

$$\rho_\pi : \text{Gal}_E \rightarrow \text{GL}(m; \overline{\mathbb{Q}_p})$$

s.t.

- If  $\ell = \mathfrak{p}_x \mathfrak{p}_{x'}$  is split in  $E$  and  $\pi_\ell$  is unramified, then  $\rho_\pi$  is unramified above  $\ell$  and  $\det(1 - \sigma_x | \rho_\pi)$  is given by the characteristic polynomial of the Langlands conjugacy class of  $\pi_x(\frac{1-m}{2})$ .
- If  $\pi_\ell$  is unramified, then  $\rho_\pi$  is unramified at each prime over  $\ell$ .
- If  $\ell = \mathfrak{p}_x \mathfrak{p}_{x'}$  splits in  $E$ , then the nilpotent monodromy operator of the Weil-Deligne representation attached to  $\rho_\pi|_{W_{E_x}}$  is in the closure of the conjugacy class of  $N(\pi_x(\frac{1-m}{2}))$  in  $\text{Mat}(m; \overline{\mathbb{Q}_p})$ .
- If  $x$  is a prime over  $\ell$  and  $\pi_\ell$  is a non-monodromic principle series (Cf. [B-C09] P166 ?), then the Weil-Deligne representation attached to  $\rho_\pi|_{W_{E_x}}$  has trivial monodromy.
- The  $p$ -adic representation  $\rho_\pi|_{\text{Gal}_{E_v}}$  is deRham, and its Hodge-Tate weights are

$$\{-a_1, \dots, -a_m\} + \frac{m-1}{2}$$

where the restriction to  $\mathbb{C}^\times$  of the  $L$ -parameter of  $\pi_\infty$  (24.7.7.1) is given by  $z \mapsto \text{diag}((z/\bar{z})^{a_1}, \dots, (z/\bar{z})^{a_m})$ .

┘

## 2 Quaternionic Modular Forms

References are [Modular Forms on Exceptional Groups, Pollack].



## 27 | $p$ -adic Geometry

### 27.1 $\mathbb{F}_p$ -Algebraic Stacks

References are [B-S17, Zhu17].

**Notation (27.1.0.1).**

- Let  $p \in \text{Prime}$ .
- let  $k \in \text{Field}^p$  s.t.  $k = k^{\text{perf}}$ .

┘

**Prop. (27.1.0.2).** Let  $R \rightarrow S \in \text{Perf}^p$  be injective and integral, and let  $M \in \text{Mod}_R$  s.t.  $M \otimes_R S$  is flat over  $S$ , then  $M$  is flat over  $R$ .

┘

*Proof:* Cf. [Zhu17]P477.

□

### 1 Perfect Algebraic Stacks

#### Perfect Algebraic Spaces

**Def. (27.1.1.1) [Perfect Schemes].**  $X \in \text{Sch}/\mathbb{F}_p$  is called a **perfect algebraic space** if the Frobenius is an isomorphism on it.

Let  $\text{PerfSch}/k$  be the site of perfect qcqs  $\mathbb{F}_p$ -schemes endowed with the v-topology (7.1.4.44).

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**Def. (27.1.1.2) [Perfection].** There is a **perfection of scheme** functor

$$(-)^{\text{perf}} : \text{Sch}/k \rightarrow \text{PerfSch}^p/k : X \mapsto X^{\text{perf}} \triangleq \varprojlim (\cdots \xrightarrow{\varphi_X} X \xrightarrow{\varphi_X} X).$$

$X^{\text{perf}}$  can also be defined by glueing of the perfection functors  $R \rightarrow R_{\text{perf}}$  on affine open subschemes (5.5.1.3) as it commutes with colimits.

In particular,  $(-)^{\text{perf}}$  commutes with inverse limits.

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**Prop. (27.1.1.3) [Perfections are Universal Homeomorphism].** For  $X \in \text{Sch}/k$ ,  $X^{\text{perf}} \rightarrow X$  is a universal homeomorphism. In particular, by (7.1.4.23) and (7.2.10.3), there is an equivalence of categories:

$$(-)^{\text{perf}} : X_{\text{ét}} \cong (X^{\text{perf}})_{\text{ét}}.$$

┘

*Proof:* Cf. [Sta]0BRD.?

□

**Prop. (27.1.1.4) [Perfect Algebraic Spaces as Functors].** The composition of functors between categories

$$\mathcal{Y} : \mathcal{A}lgSp^{\text{perf}}/k \rightarrow \mathcal{A}lgSp_k \rightarrow \mathcal{S}h(\mathcal{A}ff_{\text{fpqc}}/k) \rightarrow \mathcal{S}h(\mathcal{P}erf\mathcal{A}ff_{\text{fpqc}}/k)$$

is fully faithful. ⌋

*Proof:* Cf. [Zhu17]469. □

**Prop. (27.1.1.5) [Perfection and Properties of Morphisms, Zhu [Zhu17]/Bhatt-Scholze [B-S17]].**

For  $f : X \rightarrow Y \in \mathcal{A}lgSp/k$ , the following properties holds true for  $f$  iff it holds true for  $f^{\text{perf}}$ :

1. Quasi-compact.
2. Quasi-separated.
3. Affine.
4. Quasi-finiteness.
5. Separated.
6. Integral.
7. Universally closed.
8. Universal homeomorphism.

The following properties holds for  $f^{\text{perf}}$  if it holds for  $f$ :

1. (Closed/Open)immersion.
2. Étale .
3. (Faithfully)Flat.

⌋

*Proof:* Cf. [B-S17]3.4 or [Zhu17]LemmaA.7. ?

The quasi-finiteness follows from the fact that  $X^{\text{perf}} \rightarrow X$  is a homeomorphism (27.1.1.3). □

**Prop. (27.1.1.6) [Perfection and Line Bundles].** If  $X \in \mathcal{S}ch_{\text{qcqs}}/\mathbb{F}_p$ , the perfection  $X^{\text{perf}} \rightarrow X$  induces an isomorphism  $\text{Pic}(X)_{[1/p]} \cong \text{Pic}(X^{\text{perf}})$ . ⌋

**Prop. (27.1.1.7) [Perfection and Ampleness].** If  $X \in \mathcal{S}ch_{\text{qcqs}}/k$  and  $\mathcal{L} \in \text{Pic}(X)$ , then  $\mathcal{L}$  is ample iff the pullback to  $X^{\text{perf}}$  is ample. ⌋

*Proof:* Using the identification

$$\Gamma(X^{\text{perf}}, \mathcal{L}^{\otimes n}) \cong \varinjlim_{f \mapsto f^p} \Gamma(X, \mathcal{L}^{\otimes np^m}) \quad \text{?}$$

this follows from (27.1.1.3) and affineness part of (27.1.1.5). □

**Prop. (27.1.1.8) [Universal Homeomorphisms and Isomorphisms].** If  $f : X \rightarrow Y \in \mathcal{P}erf\mathcal{S}ch_{\text{qcqs}}/k$  is a universal homeomorphism, then  $f$  is an isomorphism. In particular, by (27.1.1.5), for  $f_0 : X_0 \rightarrow Y_0 \in \mathcal{S}ch/\mathbb{F}_p$ ,  $f_0$  is a universal homeomorphism iff  $f_0^{\text{perf}}$  is an isomorphism.

The same statement holds for  $f : X \rightarrow Y \in \mathcal{A}lgSp^{\text{perf, pfp}}/k$  that is separated. ⌋

*Proof:* ? Cf. [B-S17]P345 or [Zhu17]P472. □

**Prop. (27.1.1.9) [Limits of Perfect Schemes].** If  $(X_i)$  is a cofiltered inverse system in  $\mathcal{P}\text{erfSch}_{\text{qcqs}}$  with affine transition maps, then the limit  $X \triangleq \varprojlim_i X_i$  exists in  $\mathcal{P}\text{erfSch}_{\text{qcqs}}$ , and agrees with the limit taken in  $\mathcal{S}\text{ch}_{\text{qcqs}}$ .

And the pullback functor defines an equivalence of categories:

$$2\text{-}\varinjlim \mathcal{P}\text{erfSch}^{\text{pfp}}/X_i \cong \mathcal{P}\text{erfSch}^{\text{pfp}}/X.$$

┘

*Proof:* Use the arguments in [EGA4]Section8.8. ?

□

**Prop. (27.1.1.10) [Perfect Base Change].** If

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array} \in \mathcal{P}\text{erfSch}_{\text{qcqs}}/k$$

is Cartesian, then for any  $K \in D_{\text{Qcoh}}(X)$ , the base change map (7.3.3.18)

$$\text{L}g^* \text{R}f_* K \rightarrow \text{R}f'_* \text{L}g' K$$

is an isomorphism.

┘

*Proof:* It suffices to prove for  $X$  affine. Let  $X = \text{Spec } B, Y = \text{Spec } A, Y' = \text{Spec } A',$  and  $X' = \text{Spec } B \otimes_A A',$  then it suffices to prove that

$$K \otimes_A^{\text{L}} A' \cong K \otimes_B^{\text{L}} B'.$$

This follows from the fact  $B \otimes_A A' = B \otimes_A^{\text{L}} A',$  by (5.5.1.7).

□

**Prop. (27.1.1.11) [Cartier-Isomorphisms].** If

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*Proof:*

□

**Cor. (27.1.1.12) [Affine Line case].** Let  $R$  be an  $\mathbb{F}_p$ -algebra and

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### Perfect Algebraic Stacks

**Def. (27.1.1.13) [Perfect Algebraic Stacks].** A **perfect algebraic stack** over  $k$  is a stack  $\mathcal{X} \in \mathcal{A}\text{lgSta}/k$  s.t.

- The diagonal  $\Delta_{\mathcal{X}}$  is represented by a perfect algebraic space over  $k$ .
- There exists  $U \in \mathcal{A}\text{lgSp}^{\text{perf}}/k$  and a perfectly smooth surjective map (it is automatically representable). Any such  $U$  is called a **smooth atlas** of  $\mathcal{X}$ .

(27.1.1.1).

The category of perfect algebraic stacks over  $k$  is denoted by  $\mathcal{A}\text{lgSta}^{\text{perf}}/k$ .

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**Remark (27.1.1.14).** WARNING: Unlike the case of algebraic spaces, perfect algebraic stacks may not be an algebraic stack, and the perfection of an algebraic stack may not exist. ?

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**Def. (27.1.1.15) [Properties of Algebraic Stacks].**  $X \in \mathcal{A}\text{lgSta}^{\text{perf}}/k$  is called

- A **quasi-compact perfect algebraic stack** if there exists a quasi-compact smooth atlas  $U \rightarrow X$  (27.1.1.13).
- A **perfect algebraic stack (locally) of f.t.** if there exists a smooth atlas  $U \rightarrow X$  s.t.  $U$  is (locally)perfectly of f.t. over  $k$  (27.1.2.2).  
And a morphism  $f : X \rightarrow Y \in \text{Sta}(\text{Perf}_{\text{fpqc}}/k)$  is called
- A **separated morphism of perfect algebraic stacks** if the diagonal  $\Delta_f$  is perfectly proper.
- A **quasi-separated morphism of perfect algebraic stacks** if the diagonal  $\Delta_f$  is quasi-compact and quasi-separated.
- A **Deligne-Mumford morphism of perfect algebraic stacks** if the diagonal is étale .

┘

## 2 Perfect Algebraic Spaces Perfectly of Finite Type

**Prop. (27.1.2.1)[Perfectly Finitely Presented Morphisms].**  $f : X \rightarrow Y \in \text{PerfSch}_{\text{qcqs}}/k$  (27.1.1.1), the following are equivalent:

- Any open affine subscheme  $\text{Spec } B \subset X$  mapping to an open affine subscheme  $\text{Spec } A \subset Y$ ,  $A \rightarrow B$  is perfectly f.p. (5.5.1.5).
- There is an open affine covering  $\text{Spec } A_i \rightarrow X$  mapping to an open affine covering  $\text{Spec } B_i \rightarrow Y$  that  $B_i \rightarrow A_i$  are all perfectly f.p.
- For any cofiltered system  $\{Z_i\} \in \text{PerfSch}/Y$  with affine transition maps, there is a bijection  $\varinjlim \text{Hom}_Y(Z_i, X) \cong \text{Hom}_Y(\varprojlim Z_i, X)$ .

And in this case,  $f$  is called a **perfectly finitely presented morphism**.

In particular, being perfectly finitely presented is local on the base and target.

┘

*Proof:* Cf. [B-S17]P346?.

□

**Def. (27.1.2.2)[Perfect Algebraic Spaces locally perfectly of Finite Type].** A **perfect algebraic space locally perfectly of finite type** over  $k$  is a perfect algebraic space  $X \in \text{AlgSp}^{\text{perf}}/k$  s.t. there exist an étale affine cover  $\{U_i\}$  of  $X$  s.t. each  $U_i$  is the perfection of an affine scheme of finite type over  $k$ . And it is moreover called a **perfect algebraic space perfectly of finite type** if it is moreover quasi-compact.

A **perfect algebraic space perfectly of finite presentation** over  $k$  is a perfect algebraic space  $X \in \text{AlgSp}^{\text{perf}}/k$  that is quasi-compact and perfectly of f.t..

┘

**Prop. (27.1.2.3)[Finite Presented Models].** Any  $f : X \rightarrow Y \in \text{AlgSta}^{\text{perf}, \text{pfp}}/k$  (27.1.2.2) is a perfection of some  $f' \in \text{AlgSp}^{\text{pf}}/k$ .

┘

*Proof:* Cf. [Zhu17]P472.

□

**Prop. (27.1.2.4).** If  $f : X \rightarrow Y \in \text{PerfSch}_{\text{qcqs}}/k$  is perfectly of finite presentation, perfectly proper and surjective with geometrically connected fibers, then we have  $Rf_* \mathcal{O}_X = \mathcal{O}_Y$ .

┘

*Proof:* Cf. [B-S17]P364?.

□

**Cor. (27.1.2.5)[Isomorphism can be Checked Pointwise].** If  $f : X \rightarrow Y \in \text{PerfSch}^{\text{pfp}, \text{proper}}/k$  s.t.  $f_{\bar{y}}$  is an isomorphism for each geometric point  $\bar{y}$  of  $Y$ , then  $f$  is an isomorphism.

┘

*Proof:* By (27.1.2.3) we can choose a finite type model  $f_0$  of  $f$ , then  $f_0$  is proper and quasi-finite by (27.1.1.5), thus finite, in particular affine. But it follows from (27.1.2.4) that  $Rf_* \mathcal{O}_X = \mathcal{O}_Y$ .

□

**Def. (27.1.2.6) [Perfectly Proper Morphisms].** A morphism  $f : X \rightarrow Y \in \mathcal{AlgSta}^{\text{perf}, \text{pfp}}/k$  is called a **perfectly proper morphism** if it is separated and universally closed.  $\lrcorner$

**Prop. (27.1.2.7).** If  $f : X \rightarrow Y \in \mathcal{AlgSp}^{\text{perf}, \text{pfp}}/k$  is a perfectly proper morphism (27.1.2.6) with geometrically connected fibers, then the natural map  $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Zhu17]P473.  $\square$

**Prop. (27.1.2.8).** If  $f : X \rightarrow Y \in \text{PerfSch}_{\text{qcqs}}/k$  is perfectly of finite presentation, perfectly proper and surjective with geometrically connected fibers, then  $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  is an isomorphism, and the pullback functor  $\text{Coh}_{\text{lf}}(Y) \rightarrow \text{Coh}_{\text{lf}}(X)$  is fully faithful. The same assertion holds for  $\text{Coh}_{\text{lf}}(W_n(Y)) \rightarrow \text{Coh}_{\text{lf}}(W_n(X))$  or  $\text{Coh}_{\text{lf}}(W(Y)) \rightarrow \text{Coh}_{\text{lf}}(W(X))$ .  $\lrcorner$

*Proof:* Using Hom-bundle, it suffices to prove that  $\mathcal{E} \rightarrow f_*f^*\mathcal{E}$  is an isomorphism for each  $\mathcal{E} \in \text{Coh}_{\text{lf}}(Y)$ . For this it suffices to show for  $X, Y$  affine. Choosing a finite type model  $f_0 : X_0 \rightarrow Y_0$  of  $f$  (27.1.2.3), it suffices to show that  $\mathcal{O}_{Y_0} \rightarrow (f_0)_*\mathcal{O}_{X_0}$  is an isomorphism after perfection. But  $\text{Spec}((f_0)_*\mathcal{O}_{X_0}) \rightarrow Y_0$  is finite because of properness, and surjective, and is universally injective because  $f$  has geometrically connected fibers, so it is a universal homeomorphism (7.4.5.25), and its perfection is an isomorphism by (27.1.1.8).  $\square$

**Thm. (27.1.2.9) [Fibral Criterion for Descending Vector Bundles].** If  $f : X \rightarrow Y \in \mathcal{AlgSp}^{\text{perf}, \text{pfp}, \text{proper}}/k$  is surjective s.t. all geometric fibers are connected, then for each  $\mathcal{E} \in \text{Coh}_{\text{lf}}(X)$ ,  $\mathcal{E}$  descends to  $Y$  iff  $\mathcal{E}_{\bar{y}}$  is trivial for each geometric point  $\bar{y}$  of  $Y$ .  $\lrcorner$

*Proof:* Cf. [B-S17]P366.  $\square$

**Def. (27.1.2.10) [Perfectly Smooth Morphisms].** A morphism  $f : X \rightarrow Y \in \mathcal{AlgSp}^{\text{perf}, \text{pfp}}/k$  is called a **perfectly smooth morphism** if for each  $x \in X$ , there exists an étale nbhd  $(U, u) \in \text{PerfSch}$  of  $(X, x)$  and an étale nbhd  $(V, v) \in \text{PerfSch}$  of  $(Y, f(x))$  s.t.  $(U, u) \rightarrow (Y, f(x))$  factors through  $(U, u) \xrightarrow{h} (V, v) \rightarrow (Y, f(x))$ , and  $h$  factors as  $h = \text{pr} \circ h'$ , where  $h' : U \rightarrow V \times (\mathbb{A}^n)^{\text{perf}}$  is étale and  $\text{pr} = \text{pr}_1 : V \times (\mathbb{A}^n)^{\text{perf}} \rightarrow V$ .

A morphism  $f : X \rightarrow Y \in \mathcal{AlgSta}^{\text{perf}, \text{pfp}}/k$  is called a **perfectly smooth morphism** if there exists a smooth atlas  $U \rightarrow X$  s.t.  $U \rightarrow X \rightarrow Y$  is perfectly smooth (which is automatically representable?).

Notice if  $f$  is the perfection of a smooth morphism, then it is perfectly smooth, by (7.6.6.7).  $\lrcorner$

**Prop. (27.1.2.11).** If  $G \in \mathcal{AlgGrp}/k$ , then the stack  $\mathbf{BG}^{\text{perf}}$  is a perfectly smooth perfect algebraic stack over  $k$  (27.1.2.10).  $\lrcorner$

*Proof:* ?  $\square$

## Vector Bundles

**Prop. (27.1.2.12) [Models for Vector Bundles].** Let  $X \in \mathcal{AlgSp}^{\text{perf}, \text{pfp}}/k$  and  $\mathcal{E} \in \text{Coh}_{\text{lf}}(X)$ , then there exists  $X_0 \in \mathcal{AlgSp}^{\text{pf}}/k$  and  $\mathcal{E}_0 \in \text{Coh}_{\text{lf}}(X_0)$  s.t.  $X = X_0^{\text{perf}}$  and  $\mathcal{E} = (X \rightarrow X_0)^*\mathcal{E}_0$ .  $\lrcorner$

*Proof:* Cf. [Zhu17]P473?.  $\square$

**Cor. (27.1.2.13) [Relative Grassmannian].** If  $X \in \mathcal{AlgSp}^{\text{perf}, \text{pfp}}/k$  and  $\mathcal{E} \in \text{Coh}_{\text{lf}}^n(X)$ ,  $i \in \mathbb{N}$ , then the functor that assigns to each  $f : Y \rightarrow X \in \text{PerfSch}_{\text{qcqs}}/X$  the set of rank  $i$  quotients  $\mathcal{Q}$  of  $f^*\mathcal{E}$  is represented by a perfect algebraic space  $\text{Gr}^{\text{perf}}(\mathcal{E}, i)$  which is the perfection of a proper scheme over  $X$ . And if  $X \in \text{PerfSch}^{\text{pfp}}/k$ , then  $\text{Gr}^{\text{perf}}(\mathcal{E}, i)$  is the perfection of a smooth projective scheme over  $X$ .  $\lrcorner$

*Proof:* If  $(X_0, \mathcal{E}_0)$  is a model of  $(X, \mathcal{E})$  (27.1.2.12), then the functor is represented by  $\mathrm{Gr}^{\mathrm{perf}}(i, \mathcal{E}) \stackrel{\Delta}{=} \mathrm{Gr}(\mathcal{E}_0, i)^{\mathrm{perf}}$  (10.9.2.14) (the proof of this fact should be similar to the proof of (27.1.2.12)).  $\square$

### 3 Perfect Group Schemes

**Prop. (27.1.3.1).** For any  $H \in \mathrm{PerfSch}^{\mathrm{pfp}}/k$  (27.1.2.2), there exists  $H_0 \in \mathrm{AlgGrp}^{\mathrm{sm}}/k$  s.t.  $H = H_0^{\mathrm{perf}}$ . In particular,  $H$  is perfectly smooth (27.1.2.10).  $\lrcorner$

*Proof:* Cf. [Zhu17]P475.  $\square$

**Thm. (27.1.3.2) [Quotients by Perfect Groups].** Suppose  $H \in \mathrm{GrpSch}^{\mathrm{perf}, \mathrm{pfp}}/k$  acts freely on  $X \in \mathrm{PerfAff}^{\mathrm{pfp}}/k$  (i.e.  $(\lambda, \mathrm{id}) : H \times X \rightarrow X \times X$  is a monomorphism). then  $X/H \stackrel{\Delta}{=} \mathcal{Y}_X / \mathcal{Y}_H \in \mathrm{Sh}(\mathrm{PerfAff}_{\mathrm{fpqc}}/k)$  (27.1.1.4) is representable by a perfect scheme perfectly of finite presentation over  $k$ . If in addition  $(\lambda, \mathrm{id})$  is a closed embedding, then  $X/H \in \mathrm{AlgSp}^{\mathrm{perf}, \mathrm{sep}}/k$ .  $\lrcorner$

*Proof:* Cf. [Zhu17]P476.  $\square$

**Prop. (27.1.3.3) [Group Actions].** Suppose  $H \in \mathrm{GrpSch}_{\mathrm{cnd}}^{\mathrm{perf}, \mathrm{pfp}}/k$  acts on  $X \in \mathrm{AlgSp}^{\mathrm{perf}, \mathrm{sep}, \mathrm{pfp}}/k$ . If  $x \in |X|_0$  and  $H_x$  is the stablizer of  $x$  in  $H$ , then  $H/H_x \rightarrow X$  (27.1.3.2) is a locally closed embedding.  $\lrcorner$

*Proof:* Cf. [Zhu17]P477.  $\square$

**Prop. (27.1.3.4) [Perfect Torsors].** For  $H \in \mathrm{GrpSch}^{\mathrm{aff}}/k$  and  $X \in \mathrm{AlgSp}/k$ , If  $E/X \in \mathrm{Sh}(\mathrm{Aff}_{\mathrm{fpqc}}/k)$  is an  $H$ -torsor (7.1.1.16), then  $E$  is represented by an algebraic space affine over  $X$ . And if  $H, X$  are both perfect, then  $E$  is also perfect.  $\lrcorner$

*Proof:* This is because  $E^{\mathrm{perf}} \rightarrow E$  is a morphism of  $H$ -torsors, thus an isomorphism.  $\square$

**Prop. (27.1.3.5) [De-perfection of Perfect Torsors].** For  $H \in \mathrm{GrpSch}^{\mathrm{aff}, \mathrm{sm}}/k$  and  $X \in \mathrm{AlgSp}^{\mathrm{perf}}/k$ , then the functor  $(-)^{\mathrm{perf}} : \mathrm{Tors}_H(X) \rightarrow \mathrm{Tors}_{H^{\mathrm{perf}}}(X)$  is an equivalence of groupoids.  $\lrcorner$

*Proof:* Cf. [Zhu17]P468.  $\square$

### 4 $\ell$ -Adic Sheaves

**Notation (27.1.4.1).**

- Let  $\ell \in \mathrm{Prime} \setminus \{p\}$ .

**Prop. (27.1.4.2) [Constructible  $\ell$ -adic Sheaves].** For  $X \in \mathrm{AlgSp}^{\mathrm{perf}, \mathrm{pfp}}/k$ , we can define  $D_{\mathrm{const}}^b(X, \overline{\mathbb{Q}}_\ell)$ . And if  $\varepsilon : X \rightarrow X'$  is a model for  $X$ , then there is a pair of equivalence of categories:

$$\varepsilon^* : D_{\mathrm{const}}^b(X', \overline{\mathbb{Q}}_\ell) \cong D_{\mathrm{const}}^b(X, \overline{\mathbb{Q}}_\ell) : \varepsilon_*$$

One can define six operations between these categories, and usual proper base change or smooth base change holds for perfectly proper or perfectly smooth maps.  $\lrcorner$

*Proof:* Cf. [Zhu17]P478.  $\square$

**Prop. (27.1.4.3) [Perverse Sheaves].** For  $X \in \mathcal{A}lgSp^{\text{perf}, \text{pfp}}/k$ , proper base change and smooth base change hold. In particular, we can define the category  $\text{Perv}_\ell(X)$  of perverse sheaves on  $X$ .

And we have the notion of the Goresky-Macpherson intermediate extension, and for every  $X$ , the intersection cohomology sheaf  $\text{IC}_{X, \ell}$ . The restriction of  $\text{IC}_{X, \ell}$  to each perfectly smooth open subset  $U$  is canonically isomorphic to  $\overline{\mathbb{Q}}_\ell[2 \dim X](\dim X)$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (27.1.4.4).**  $\lrcorner$

**Prop. (27.1.4.5) [Chern Classes for Homogenous Spaces].** For  $G \in \mathcal{A}lgGrp_{\text{cntd}}/k$  and  $X \in \mathcal{A}lgSp^{\text{perf}, \text{sep}, \text{pfp}}/k$ , we can define the characteristic class

$$c(E) : R_{G, \ell} \rightarrow H_{\text{ét}}^*(X_{\overline{k}}, \overline{\mathbb{Q}}_\ell).$$

$\lrcorner$

*Proof:* Cf. [Zhu17]P479.  $\square$

**Prop. (27.1.4.6) [Lefschetz Trace Formulae].** Lefschetz trace formulae (9.6.2.13) holds for morphism of separated pfp perfect algebraic spaces over  $\mathbb{F}_q$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (27.1.4.7) [Equivariant Perverse Sheaves].** For  $J \in \mathcal{G}rpSch^{\text{perf}, \text{aff}}/k$  and  $X \in \mathcal{A}lgSp^{\text{perf}, \text{sep}, \text{pfp}}/k$  with an action  $\lambda$  of  $J$ , the category  $\text{Perv}_{J, \ell}(X)$  of  **$J$ -equivariant perverse sheaves** on  $X$  consists of pairs  $(A, \varphi)$  where  $A \in \text{Perv}_\ell(X)$  and  $\varphi : \text{pr}_2^* A \cong \lambda^* A \in \text{Perv}_\ell(F)$  is an isomorphism.

For a closed normal subgroup  $J_1 \trianglelefteq J$ ,

- If  $J_1$  acts freely on  $X$ , then  $\text{pr} : X \rightarrow X/J_1$  induces an equivalence of categories

$$\text{pr}^*[\dim J_1] : \text{Perv}_{J/J_1, \ell}(X/J_1) \cong \text{Perv}_{J, \ell}(X).$$

- If  $J_1$  is cntd and acts trivially on  $X$ , then there is a natural equivalence of categories:

$$\text{Perv}_{J/J_1, \ell}(X) \cong \text{Perv}_{J, \ell}(X).$$

$\lrcorner$

*Proof:* Cf. [Zhu, An introduction to affine Grassmannians and the geometric Satake equivalence]A.1.4.  $\square$

**Prop. (27.1.4.8) [Equivariant Cohomologies].** Situation as in (27.1.4.7), for  $A \in \text{Perv}_J(X)$ , we can define the  **$J$ -equivariant cohomology**  $H_J^*(X_{\overline{k}}, A)$ . ? Cf. [Zhu17]P481.

Then for a closed normal subgroup  $J_1 \trianglelefteq J$ ,

- If  $J_1$  acts freely on  $X$ , then for each  $A \in \text{Perv}_{J/J_1, \ell}(X/J_1)$ , there are natural isomorphisms

$$H_{J, \ell}^*(X_{\overline{k}}, \text{pr}^* A) \cong H_{J/J_1, \ell}^*((X/J_1)_{\overline{k}}, A).$$

- If  $J_1$  is the perfection of a unipotent group and acts trivially on  $X$ , then for each  $A \in \text{Perv}_{J, \ell}(X)$ , there are natural isomorphisms

$$H_{J, \ell}^*(X_{\overline{k}}, A) \cong H_{J/J_1, \ell}^*(X_{\overline{k}}, A).$$

$\lrcorner$

*Proof:* Cf. [Zhu17]P482.  $\square$

## 27.2 Formal and Rigid Geometry

References are [BGR84, Bos15].

**Notation (27.2.0.1).**

- Let  $(K, |\cdot|_K, \mathcal{O}_K, \kappa)$  be a complete non-Archimedean valued field.
- Use notations on affinoid  $K$ -algebras defined in [Affinoid Algebras](#).

┘

### 1 Affinoid $K$ -Spaces

**Def. (27.2.1.1) [Affinoid  $K$ -Space].** An affinoid  $K$ -algebra  $A$  can be viewed as the function ring on the space  $\mathrm{Sp} A$  of maximal ideals of  $A$  with the usual Zariski topology called the **affinoid  $K$ -space associated to  $A$** . A morphism of affinoid algebras induce a map on their affinoid spectra. This is because residue fields of maximal ideals are finite over  $K$ . So we *define* the category of affinoid  $K$ -spaces as the opposite category of affinoid  $K$ -algebras.

┘

**Cor. (27.2.1.2).** The category of affinoid spaces admits fiber products, because of (12.2.4.31).

┘

**Prop. (27.2.1.3).** By the properties of a Jacobson space (4.12.2.25)(4.12.2.22), the affinoid  $K$ -space has good properties w.r.t. closed, open hence irreducible compared to  $\mathrm{Spec} A$  in Zariski topology. In particular, it is a Noetherian space.

┘

**Def. (27.2.1.4) [Canonical Topology].** The affinoid  $K$ -space has another topology, called the **canonical topology**, generated by  $X(f, \varepsilon) = \{x | f(x) \leq \varepsilon\}$  as a subbasis. And this topology is in fact generated by  $X(f) = X(f, 1)$  as a subbasis.

┘

*Proof:* For the last assertion, notice  $f(x)$  assume value in  $|\overline{K}|$ , which is dense in  $\mathbb{R}_+$ , so we can assume  $\varepsilon \in |\overline{K}|$  (by approximation from below), hence  $\varepsilon^n = |c|$ , where  $c \in K$ , so  $X(f, \varepsilon) = X(f^n, c) = X(c^{-1}f^n)$ .  $\square$

**Prop. (27.2.1.5).**  $\{x | f(x) = \varepsilon\}$  is open in  $\mathrm{Sp} A$ .

┘

*Proof:* We let  $f(x) = \varepsilon$  and  $k = A/\mathfrak{m}_x$ , let the minipoly of  $f$  in  $A/\mathfrak{m}_x$  be  $P$  of degree  $n$ , and let  $g = P(f)$ , then  $g(x) = 0$ , and if  $|g(y)| < \varepsilon^n$ , then  $|f(y)| = \varepsilon$ , otherwise  $|f(y) - \alpha_i| \geq |\alpha_i| = \varepsilon$  for every root  $\alpha_i$  of  $P$ , hence  $|P(f(y))| \geq \varepsilon^n$ , contradiction.  $\square$

**Cor. (27.2.1.6).** By the proof, we have,  $X(f_1, \dots, f_r)$ ,  $f_i \in \mathfrak{m}_x$  forms a basis of  $x$  in  $\mathrm{Sp} A$ . (Replace every  $X(f_i)$  by  $\{y | |f_i(y)| = \varepsilon\}$ , then by some  $X(g_i)$  for  $g_i \in \mathfrak{m}_x$ ).

┘

**Def. (27.2.1.7) [Affinoid Subdomain].** For an affinoid  $K$ -space  $X$ , a subset  $U$  is called a **affinoid subdomain** of  $X$  if there is an closest affinoid space map  $X' \rightarrow X$  with image in  $U$ , i.e. any other these maps factor through it. The definition is weird but the situation is clarified by the following proposition.

┘

**Prop. (27.2.1.8).** For an affinoid subdomain  $i : X' \rightarrow X$ ,

- $i$  is injective and  $\mathrm{Im} i = U$ .
- $i^*$  induce an isomorphism  $A/\mathfrak{m}_{i(x)}^k \cong A'/\mathfrak{m}_x^k$ .
- $\mathfrak{m}_x = \mathfrak{m}_{i(x)} A'$ .

┘



*Proof:* Consider a point  $y \in U$ , there is a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{i^*} & A' \\ \downarrow & \swarrow \alpha & \downarrow \\ A/\mathfrak{m}_y^n & \xrightarrow{\sigma} & A'/\mathfrak{m}_y^n A' \end{array}.$$

Then there is a map  $\alpha : A' \rightarrow A/\mathfrak{m}_y^n$  that makes the upper diagram commutative by universal property of subdomain, and the lower triangle is commutative by universal properties again. Then we see  $\sigma$  is surjective and notice the kernel of the projection is  $\mathfrak{m}_y A'$  is in the kernel of  $\alpha$ , thus  $\sigma$  is injective.

Now the case  $n = 1$  shows  $\mathfrak{m}_y A'$  is maximal, hence  $i$  is surjective and the inverse image is just one point.  $\square$

**Prop. (27.2.1.9) [Special Subdomains].** There are three special affinoid subdomain of  $X$ : **Weierstrass domain**  $X(f_1, \dots, f_r)$ , **Laurent domain**  $X(f_1, \dots, f_r, g_1^{-1}, \dots, g_s^{-1})$ , **rational domain**  $X(\frac{f_1}{f_0}, \dots, \frac{f_r}{f_0}) = \{x \mid |f_i(x)| \leq |f_0(x)|\}$  for  $(f_0, \dots, f_r) = (1)$ . They are all open by (27.2.1.5).  $\lrcorner$

*Proof:* The Weierstrass domain corresponds to  $A \rightarrow A\langle X_1, \dots, X_r \rangle / (X_i - f_i)$ .

The Laurent domain corresponds to  $A \rightarrow A\langle X_1, \dots, X_{r+s} \rangle / (X_i - f_i, 1 - X_{r+j} g_j)$ .

The rational domain corresponds to  $A \rightarrow A\langle X_1, \dots, X_r \rangle / (f_i - f_0 X_i)$ .

They are affinoid subdomains is in fact, easily checked.  $\square$

**Lemma (27.2.1.10).** Weierstrass domain are Laurent, and Laurent domain are rational, this is because intersection of rational domains are rational.

Any rational domain is a Weierstrass domain of a Laurent domain.  $\lrcorner$

*Proof:* Notice a Laurent subdomain is a finite intersections of  $X(\frac{f}{1})$  and  $X(\frac{1}{g})$ , so it is rational.

For a rational domain  $U$ ,  $f_0$  is a unit in  $\mathcal{O}(U)$ , hence its inverse has a bounded value, then  $|cf_0| > 1$  for some  $c \in K^*$ . Hence  $U$  is Weierstrass in  $X((cf_0)^{-1})$ .  $\square$

**Cor. (27.2.1.11) [Pullback & Composition of Affinoid Subdomain].** The pullback (hence intersections) of affinoid subdomains is affinoid subdomain and it is just the set-theoretic inverse image, and specialness are preserved.

The affinoid subdomain of an affinoid subdomain is affinoid subdomain, and Weierstrassness and rationalness are preserved (while Laurentness not).  $\lrcorner$

*Proof:* Pullback: fiber product exist in the category of affinoid  $K$ -spaces, then the universal property is checked. The set-theoretic property follows from (27.2.1.8).

Speciality: Clear.

Transitivity: Clear by universal property.

For the speciality, if  $V = X(f_i)$ ,  $U = V(g_j)$  is Weierstrass, then because by (12.2.4.27)  $A$  is dense in  $A\langle f_i \rangle$ , we can replace  $g_j$  by elements from  $A$ , by adding elements of small sup-norm, because valuation is non-Archimedean. Then  $U = X(f_i, g_j)$ . For the rational subdomain  $V = X(\frac{f_1}{f_0}, \dots, \frac{f_r}{f_0})$ , use (27.2.1.10), it suffices to prove for  $U = V(g)$  or  $U = V(g^{-1})$ . For this, notice the image of  $A[f_0^{-1}]$  is dense in  $A\langle \frac{f_1}{f_0}, \dots, \frac{f_r}{f_0} \rangle$ , by (12.2.4.27), so as before, we change  $g$  that it  $g_0 f_0^n g \in A$  for some  $n$ . Now

$$V(g) = V \cap \{x \in X \mid |g_0(x)| \leq |f_0^n(x)|\}, \quad V(g^{-1}) = V \cap \{x \in X \mid |g_0(x)| \geq |f_0^n(x)|\}.$$

But now  $f_0^n$  is a unit in  $A\langle \frac{f_1}{f_0}, \dots, \frac{f_r}{f_0} \rangle$ , so  $|f(x)|_{\sup} \geq |c|$  for some  $c \in K^*$ , so

$$V(g) = V \cap X\left(\frac{g_0}{f_0^n}, \frac{c}{f_0^n}\right), \quad V(g^{-1}) = V \cap X\left(\frac{f_0^n}{g_0}, \frac{c}{g_0}\right).$$

is rational in  $X$ .  $\square$

**Cor. (27.2.1.12).** For a special subdomain  $U$  of  $X$ , the canonical topology induces the canonical topology of  $U$ , by the transitivity property of affinoid subdomains and (27.2.1.10). In fact, by (27.2.1.14), any affinoid subdomain is open and the topology coincides.  $\lrcorner$

**Prop. (27.2.1.13).** Let  $\varphi : Y = \mathrm{Sp} B \rightarrow X = \mathrm{Sp} A$  be a morphism, if  $x$  is a point of  $X$  that  $A/\mathfrak{m}_x \rightarrow B/\mathfrak{m}_x B$  is a surjection, then there is an affinoid nbhd  $U$  of  $x$  that  $\varphi$  restricts to a closed immersion on  $\varphi^{-1}(U)$ . If  $A/\mathfrak{m}_x^n \cong B/\mathfrak{m}_x^n$  for all  $n$ , then there is an affinoid nbhd  $U$  of  $x$  that  $\varphi$  restricts to an isomorphism  $\varphi^{-1}(U) \cong U$ .  $\lrcorner$

*Proof:* Cf.[Rigid and Formal Geometry P57].  $\square$

**Cor. (27.2.1.14).** Every affinoid subdomain of  $X$  is open and has the restriction topology of  $X$  (canonical topology), because it satisfies the second condition of (27.2.1.13), by (27.2.1.8).  $\lrcorner$

**Lemma (27.2.1.15).** If  $f \in A\langle X_1, \dots, X_n \rangle$  is  $X_n$ -distinguished of order  $\leq s$  for each element of  $\mathrm{Sp} A$ , then the set of elements that  $f$  is  $X_n$ -distinguished of exact order  $s$  is a rational subdomain of  $A$ .  $\lrcorner$

*Proof:* Let  $f = \sum f_v X_n^v$ , let the constant coefficient of  $f_v$  be  $a_v$ , then the set is in fact  $U = \{x \in \mathrm{Sp} A \mid |a_v(x)| \leq |a_s(x)|\}$ . This is because, if  $f$  is distinguished of order  $s_x$  at  $x$ , then  $a_{s_x} \neq 0$  because  $f_{s_x}$  is a unit, and  $|a_v|_x \leq |f_v|_x \leq |f_{s_x}|_x = |a_{s_x}|_x$  for  $v \leq s_x$  and strict inequality holds for  $v > s_x$ . In particular,  $a_0, \dots, a_s$  cannot have a common zero, so it is truly a rational subdomain.  $\square$

**Prop. (27.2.1.16).** If  $f \in A\langle X_1, \dots, X_n \rangle$  is  $X_n$ -distinguished of order  $s$  for each element of  $\mathrm{Sp} A$ , then the map

$$A\langle X_1, \dots, X_{n-1} \rangle \rightarrow A\langle X_1, \dots, X_n \rangle / (f)$$

is finite.  $\lrcorner$

*Proof:* Cf.[Rigid And Formal Geometry P79].  $\square$

### Presheaf of Affinoid Functions

**Def. (27.2.1.17)[G-Topology on Affinoid Spaces].** The **weak Grothendieck category** (affine topology) on an affinoid space  $X$  has coverings defined by the finite cover between affinoid subdomains, called **affinoid coverings**.

The **strong Grothendieck category** (fpqc topology) on an affinoid space  $X$  is defined by: objects are unions of affinoid subdomains  $U = \cup U_i$  that for any morphism from an affinoid space  $\varphi : Z \rightarrow U \subset X$ , the pullback covering  $\cup \varphi^{-1}(U_i)$  has a finite subcover by affinoid subdomains. A covering is defined by the same finiteness property.

The strong Grothendieck topology satisfies completeness conditions  $G_0, G_1, G_2$  defined in (7.1.1.12), as easily verified.

The weak Grothendieck topology is a temporary notion, it will be obsolete after Tate's acyclicity theorem is proved. Admissible opens and admissible coverings are notions w.r.t. the strong Grothendieck topology.  $\lrcorner$

*Proof:* The weak Grothendieck category is a Grothendieck category by (27.2.1.11). The strong Grothendieck category is a Grothendieck category because: the finiteness condition lifts along base change, and also for base change, because we can first choose a finite subcover, then choose a finite subcover of the base change covering of that finite covering.  $\square$

**Def. (27.2.1.18) [Rational Coverings and Laurent Coverings].** For  $n$  functions  $f_1, \dots, f_n$  without common zeros, the rational subdomains  $U_i = X(\frac{f_1}{f_i}, \dots, \frac{f_n}{f_i})$  is an affinoid covering, called a **rational covering**. For  $n$  functions  $f_1, \dots, f_n$ , there is a **Laurent covering**  $X(\prod f_i^{\varepsilon_i}), \varepsilon_i = \pm 1$ .  $\lrcorner$

**Prop. (27.2.1.19).** Morphisms of affinoid spaces are continuous in weak Grothendieck topology by (27.2.1.11). It is also continuous in the strong Grothendieck topology, as one can check the finiteness conditions.  $\lrcorner$

**Prop. (27.2.1.20).** Let  $X$  be an affinoid  $K$ -space, for any  $f \in \mathcal{O}_X(X)$ , consider the following sets:

$$U_1 = \{x : |f(x)| < 1\}, \quad U_2 = \{x : |f(x)| > 1\}, \quad U_3 = \{x : |f(x)| > 0\}.$$

Then any finite union of sets of the form is admissible, and any finite cover by finite union of sets of the form is an admissible covering.  $\lrcorner$

*Proof:* We first show that  $U_1$  is admissible open, the others are similar. Let  $\varepsilon_n$  be an ascending sequence of elements in  $\sqrt{|K^*|}$  converging to 1, then  $U_1 = \cup_n X(\varepsilon_n^{-1}f)$  is a union of open subsets because  $\varepsilon_n \in \sqrt{|K^*|}$ . Now for any affinoid space  $Z$  mapping into  $U_1$ ,  $|\varphi^*(f)(z)|_{\sup} < 1$  for all  $z \in Z$ , thus by maximum principle (12.2.4.22),  $|f|_{\sup} < 1$ , thus the cover  $U_1 = \cup_n X(\varepsilon_n^{-1}f)$  can be refined by a finite cover. This proves that it is admissible open.

For the admissibility of covering, the proof is similar, but use the following lemma (27.2.1.21).  $\square$

**Lemma (27.2.1.21).** For any affinoid  $K$ -algebra  $A$ , if  $f_i, g_j, h_k$  are system of functions on  $A$  that: for every  $x \in A$ , either  $|f_i(x)| < 1$ ,  $|g_j| > 1$  or  $h_k(x) > 0$ , then we can replace  $>, <$  by  $\geq, \leq$  and elements in  $\sqrt{|K^*|}$  that the same condition is true.  $\lrcorner$

*Proof:* Cf. [Rigid and Formal Geometry P97].  $\square$

**Cor. (27.2.1.22).** The strong Grothendieck category is finer than the Zariski category, because any standard affine open set is of the form  $U_3$  and also Zariski covering is open covering because  $\mathrm{Sp}(A)$  is Noetherian (12.2.4.17).  $\lrcorner$

**Def. (27.2.1.23) [Presheaf of Affinoid Functions].** There is a **presheaf of affinoid functions** defined on the weak Grothendieck topology because of the universal property of the affinoid subdomains.

Then the stalk  $\mathcal{O}_{X,x}$  are local ring with maximal ideal  $\mathfrak{m}_x \mathcal{O}_{X,x}$ . Hence let  $X = \mathrm{Sp} A$ , the stalk map factor thorough  $A \rightarrow A_{\mathfrak{m}_x} \hookrightarrow \mathcal{O}_{x,X}$ , and

$$A/\mathfrak{m}_x^n \cong A_{\mathfrak{m}_x}/\mathfrak{m}_x^n A_{\mathfrak{m}_x} \cong \mathcal{O}_{X,x}/\mathfrak{m}_x^n \mathcal{O}_{X,x}$$

so it induces isomorphisms between their  $\mathfrak{m}_x$ -adic completions.  $\lrcorner$

*Proof:* By (27.2.1.8), there is an isomorphism  $K' = \mathcal{O}_X(X)/\mathfrak{m}_x \cong \mathcal{O}_X(U)/\mathfrak{m}_x \mathcal{O}(U)$ . Take the converse and pass to direct colimit (it is exact),  $\mathcal{O}_{X,x}/\mathfrak{m}_x \mathcal{O}_{X,x} \cong K'$ . This map will be regarded as evaluation at  $x$ . The kernel  $\mathfrak{m}_x \mathcal{O}_{X,x}$  is a maximal ideal. There are no other maximal ideals in  $\mathcal{O}_{X,x}$  because if  $f$  is not in the kernel, then  $f(x) \neq 0$ , and multiply by an element in  $K^*$ , it can be made  $|f(x)| \geq 1$ , and then  $U(f^{-1})$  is an affinoid subdomain containing  $x$  that  $f$  is invertible in it.

For the second assertion, for an affinoid subdomain  $\mathrm{Sp} A'$ , there are maps

$$A/\mathfrak{m}^n \rightarrow A'/\mathfrak{m}^n \rightarrow \mathcal{O}_{X,x}/\mathfrak{m}^n \mathcal{O}_{X,x}.$$

We first show these are isomorphisms: the first map is an isomorphism by (27.2.1.8), then take direct colimit, the composition map is also isomorphism.

$A/\mathfrak{m}_x^n \cong A_{\mathfrak{m}_x}/\mathfrak{m}_x^n A_{\mathfrak{m}_x}$  is classical.

$A_{\mathfrak{m}_x} \hookrightarrow \mathcal{O}_{x,X}$  is injective because by Krull's intersection theorem (5.2.2.15),  $A_{\mathfrak{m}_x} \hookrightarrow \mathcal{O}_{x,X} \rightarrow \widehat{\mathcal{O}_{X,x}} \cong \widehat{A_{\mathfrak{m}_x}}$  is injective.  $\square$

**Cor. (27.2.1.24).**  $f \in A = \mathcal{O}_X(X)$  vanish iff it vanish at every stalk, this is because  $A \rightarrow \prod_{\mathfrak{m}} A_{\mathfrak{m}} \rightarrow \prod \mathcal{O}_{X,x}$  is injective.  $\lrcorner$

**Cor. (27.2.1.25).** Giving a covering of affinoid subdomain of an affinoid space  $X_i \rightarrow X$ , then  $\mathcal{O}_X(X) \rightarrow \prod \mathcal{O}_{X_i}(X_i)$  is an injection. (This is because the kernel vanishes at each stalk.)  $\lrcorner$

**Cor. (27.2.1.26).** For a subdomain of an affinoid space  $X$ , the corresponding ring map is flat.  $\lrcorner$

*Proof:* Cf.[Formal and Rigid Geometry P68].  $\square$

**Prop. (27.2.1.27).** The stalk  $\mathcal{O}_{X,x}$  is Noetherian, in particular, it is  $\mathfrak{m}$ -adically separated by Krull's intersection theorem(5.2.2.15).  $\lrcorner$

*Proof:* First it is  $\mathfrak{m}$ -adically separated, because by(27.2.1.23), for a  $f \in \cap \mathfrak{m}^n \mathcal{O}_{X,x}$ , we can choose an affinoid subdomain  $\text{Sp } A$  that  $f \in A$ (27.2.1.8), then  $f \in \mathfrak{m}^n A$ , so by Krull's intersection theorem(5.2.2.15), we have  $f = 0$  in  $A_{\mathfrak{m}}$ .

In the same way, any f.g. ideal  $\mathfrak{a}$  of  $\mathcal{O}_{X,x}$  is  $\mathfrak{m}$ -adically closed, this is because it is generated by an ideal in the affinoid algebra of a nbhd, and then  $\mathcal{O}_{x,X}/\mathfrak{a}$  is separated as the stalk of an affinoid algebra  $A'/\mathfrak{a}'$ .

Now pass a chain of f.g. ideals to their completion, then that chain is stationary because  $\hat{\mathcal{O}}_{X,x} = \hat{A}_{\mathfrak{m}_x}$  is Noetherian(5.1.1.45). And now this chain is also stationary because ideals are closed in  $\mathfrak{m}$ -adic topology.  $\square$

### Locally Closed Immersions

**Def. (27.2.1.28) [Immersion].** A morphism of affinoid spaces is called a **closed immersion** iff the corresponding ring map is surjective. It is called a **locally closed immersion** iff it is injective and the stalk map are all surjective. It is called an **open immersion** iff it is injective and the corresponding stalk maps are isomorphism. All these notions are stable under compositions.

An affinoid subdomain is an open immersion by(27.2.1.8)(27.2.1.23) and(27.2.1.27).  $\lrcorner$

**Lemma (27.2.1.29).** Base change by affinoid subdomain of closed/locally closed/open immersions are of the same type.  $\lrcorner$

*Proof:* This is obvious for locally close and open, because affinoid subdomains are open(27.2.1.14), for the closed immersion, use(12.2.4.33).  $\square$

**Prop. (27.2.1.30).** A closed immersion of affinoid spaces is equivalent to a locally closed immersion that the corresponding ring map is finite.  $\lrcorner$

*Proof:* Cf.[Rigid and Formal Geometry P70]. A closed immersion  $X' \rightarrow X$  is a locally closed immersion because the canonical topology of  $\text{Sp } A$  restricts to the canonical topology on  $\text{Sp } A/\mathfrak{a}$ (27.2.1.4), then use(12.2.4.33), and the fact direct limit is exact.  $\square$

**Prop. (27.2.1.31)[Clopen Immersion].** The image of an open and closed immersion is Zariski closed and open. In particular, it is a Weierstrass subdomain.  $\lrcorner$

*Proof:* Cf.[Rigid and Formal Geometry P71].  $\square$

**Def. (27.2.1.32).** A **Runge immersion** is a closed immersion followed by an open immersion of Weierstrass subdomain. Runge immersion is stable under base change of affinoid subdomains by(27.2.1.29)  $\lrcorner$

**Prop. (27.2.1.33) [Equivalent Definition of Runge Immersions].** For a morphism  $\sigma : A \rightarrow A'$ ,  $\mathrm{Sp} A' \rightarrow \mathrm{Sp} A$  is a Runge immersion iff  $\sigma(A)$  is dense in  $A'$  iff  $\sigma(A)$  contains a set of affinoid generator of  $A'$  over  $A$ .  $\lrcorner$

*Proof:* For a Runge immersion,  $\sigma(A)$  is dense in  $A'$ , because this is true for Weierstrass subdomain and closed immersion.

If  $\sigma(A)$  is dense in  $A'$ , then by (12.2.4.29), we can modify a set of affinoid generators by a set of affinoid generators in  $\sigma(A)$ .

If  $h_i$  is a set of affinoid generators in  $\sigma(A)$ , then  $A \rightarrow A\langle h_i \rangle \rightarrow A'$  is a Runge immersion.  $\square$

**Cor. (27.2.1.34).** Runge immersion is stable under composition.  $\lrcorner$

**Prop. (27.2.1.35).** An open and Runge immersion is an immersion of Weierstrass subdomain.  $\lrcorner$

*Proof:* By localizing on this Weierstrass subdomain, and notice Weierstrass subdomain is stable under composition (27.2.1.11), we reduce to clopen immersion case, and result follows by (27.2.1.31).  $\square$

**Lemma (27.2.1.36) [Extension of Runge Immersion].** For a morphism of affinoid spaces  $X' \rightarrow X = \mathrm{Sp} A$ , if  $f_1, \dots, f_n, g$  generate  $A$ , for  $\varepsilon \in \sqrt{\{|K^*|\}}$ , denote  $X_\varepsilon = \{x \mid |f_i(x)| \leq \varepsilon |g|\}$ , this is a rational subdomain. The inverse image of  $X_\varepsilon$  is  $X'_\varepsilon$ , then if  $X'_{\varepsilon_0} \rightarrow X_{\varepsilon_0}$  is a Runge immersion for some  $\varepsilon_0$ , then there is a  $\varepsilon > \varepsilon_0$  that  $X'_\varepsilon \rightarrow X_\varepsilon$  is also a Runge immersion.  $\lrcorner$

*Proof:* Cf. [Rigid and Formal Geometry P73].  $\square$

**Prop. (27.2.1.37) [Gerritzen-Grauert].** For a locally closed immersion  $\varphi : X' \rightarrow X$ , there is a finite cover of  $X$  of rational subdomains  $X_i$  that  $\varphi^{-1}(X_i) \rightarrow X_i$  are Runge immersions.  $\lrcorner$

*Proof:* Cf. [Formal and Rigid Geometry P79].  $\square$

**Cor. (27.2.1.38) [Gerritzen-Grauert].** Any affinoid subdomain is equivalent to a finite union of rational subdomains.  $\lrcorner$

*Proof:* An affinoid subdomain is an open immersion by (27.2.1.28), so  $\varphi^{-1}(X_i) \rightarrow X_i$  is open and Runge, so it is Weierstrass by (27.2.1.35). In particular,  $X \cap X_i$  is rational in  $X$  by transitivity, thus the result.  $\square$

### Tate's Acyclicity

**Lemma (27.2.1.39) [Reduction of Weak Grothendieck Topology].**

- Every affinoid covering has a refinement of rational covering.
  - For every rational covering, there is a Laurent covering  $\{V_i\}$  that restriction on each  $V_i$  is rational covering generated by units.
  - Every rational covering generated by units has a refinement of Laurent covering.
- $\lrcorner$

*Proof:* 1: By (27.2.1.38), we can assume the covering consists of rational subdomains  $U_i = X(\frac{f_{i1}, \dots, f_{ii_k}}{f_{i0}})$ , then consider the elements  $f_{v_1 \dots v_n} = \prod_{i=1}^n f_{iv_i}$ , where at least some  $v_i = 0$ . Denote the set of these elements by  $I$ .

Firstly, these elements has no common zero on  $X$ , thus generating a rational covering of  $X$ : for any  $x \in U_i$ ,  $f_{i0}$  doesn't vanish at  $x$ , thus the product  $\prod_{j \neq i} f_{jv_j}$  vanishes for all choices of  $v_j$ , but this is impossible because for each  $j$ ,  $(\{f_{ik}\}_k) = (1)$ .

Secondly, this is a refinement of  $U_i$ : We show  $X_{f_{v_1 \dots v_n}} \subset U_k$  where  $v_k = 0$ . For this, consider  $x \in X_{f_{v_1 \dots v_n}}$ , then  $x \in U_j$  for some  $j$ . If  $j = k$ , we are done, otherwise,

$$|f_{v_1 \dots \mu_k \dots v_n}(x)| \leq |f_{v_1 \dots 0 \dots \mu_k \dots v_n}(x)| \leq |f_{v_1 \dots v_n}(x)|.$$

Where the last inequality is because  $(v_1, \dots, 0, \dots, \mu_k, \dots, v_n)$  has a 0, thus  $f_{v_1 \dots 0 \dots \mu_k \dots v_n} \in I$ .

2: For a rational covering,  $f_i$  is invertible in the ring of  $U = X(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i})$ , thus it has a inverse that attains maximum value on  $U$  (12.2.4.22). Hence there is a  $c \in K^*$  that  $|c|^{-1} < \inf(\max\{|f_i(x)|\})$ .

I claim the Laurent covering w.r.t. the elements  $cf_0, \dots, cf_n$  satisfies the requirement. Because for example, on  $V = X((cf_0) \cdots (cf_s)(cf_{s+1})^{-1} \cdots (cf_n)^{-1})$ ,  $|f_i(x)| < |f_j(x)|$  for  $i \leq s < j$ , so the covering restricted to  $V$  is just the rational covering generated by  $f_{s+1}, \dots, f_n$ , and they are all invertible in  $\mathcal{O}(V)$ .

3: In fact the Laurent covering generated by the element  $f_i f_j^{-1}$  for  $i < j$  is a refinement of the rational covering generated by  $f_1, \dots, f_n$ , because in any one of this Laurent subdomains  $V$ , for any two  $i, j$ , either  $|f_i(x)| < |f_j(x)|$  or  $|f_j(x)| < |f_i(x)|$  for all  $x \in V$ , so there is a maximal one  $f_s$ , then  $V \subset X(\frac{f_0}{f_s}, \dots, \frac{f_n}{f_s})$ .  $\square$

**Prop. (27.2.1.40) [Tate's Acyclicity Theorem].** The presheaf of affinoid functions on an affinoid space  $X = \text{Sp } A$  is a sheaf w.r.t the weak Grothendieck category. In fact, for any  $A$ -module  $M$ , the presheaf  $\widetilde{M} = M \otimes_A \mathcal{O}_X$  is a sheaf w.r.t. the weak Grothendieck topology, called the **quasi-coherent** sheaf on  $X$ .

Moreover, for any finite cover of affinoid subdomains, the Čech cohomology group  $\check{H}^q(\text{Sp } A, \widetilde{M})$  vanish for  $q \neq 0$ .  $\perp$

*Proof:* It suffices to prove the last assertion. First reduce to the case of Laurent covering by (27.2.1.39) and (7.3.2.10)(7.3.2.11). Noticing the base change invariance of the specialities of affinoid subdomains (27.2.1.11). Even more, by (7.3.2.12) and an induction process, it suffices to prove for the simple Laurent covering  $X(f), X(f^{-1})$ .

It suffices to prove for the sheaf of affinoid functions  $\mathcal{O}_X$ , because for any Qco sheaf  $\widetilde{M}$ , choose a free resolution of  $M$ , then use dimension shifting, notice the covering is finite (the flatness of the algebra map (27.2.1.26) is used to deduce the long exact sequence).

For the sheaf  $\mathcal{O}_X$ , the main tool is the following commutative diagram:

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 & & (X-f)A\langle X \rangle \times (1-fY)A\langle Y \rangle & \xrightarrow{\delta''} & (X-f)A\langle X, X^{-1} \rangle & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \\
 0 \longrightarrow & A & \xrightarrow{\epsilon'} & A\langle X \rangle \times A\langle Y \rangle & \xrightarrow{\delta'} & A\langle X, X^{-1} \rangle & \longrightarrow 0 \\
 & \parallel & & \downarrow & & \downarrow & \\
 0 \longrightarrow & A & \xrightarrow{\epsilon} & A\langle f \rangle \times A\langle f^{-1} \rangle & \xrightarrow{\delta} & A\langle f, f^{-1} \rangle & \longrightarrow 0
 \end{array}$$

where  $\delta'$  is given by  $(h_1(X), h_2(Y)) \mapsto h_1(X) - h_2(X^{-1})$ , and  $\delta''$  is induced by  $\delta'$ . The columns are all exact, and the first row and the second row are exact.  $\epsilon$  is injective by (27.2.1.25). Then the last row is also exact, by spectral sequence.  $\square$

**Prop. (27.2.1.41) [Strong/Weak Topos The same].** If  $X$  is an affinoid  $K$ -space, the category of sheaves w.r.t the strong Grothendieck topology is equivalent to the category of sheaves w.r.t. the weak Grothendieck topology by pushforward and pullback of sheaves by (7.1.2.25) because the strong and weak Grothendieck category satisfies the conditions.

In particular, this applies to the case  $\mathcal{O}_X$  by (27.2.1.40), the resulting sheaf is called the **sheaf of rigid analytic functions** on  $X$ , also denoted by  $\mathcal{O}_X$ .  $\square$

## 2 Rigid Analytic Spaces

**Def. (27.2.2.1) [G-Ringed K-Spaces].** A **G-ringed  $K$ -space** is a pair  $(X, \mathcal{O}_X)$  where  $X$  is a  $G$ -topological space (7.1.1.11) and  $\mathcal{O}_X$  is a sheaf of  $K$ -algebras. It is called **local G-ringed  $K$ -space** if the stalks are all local rings. Their morphisms are defined routinely.  $\square$

**Prop. (27.2.2.2) [Morphisms Between Affinoid Spaces].** An affinoid  $K$ -space with the sheaf of analytic functions  $(X, \mathcal{O}_X)$  (27.2.1.41) is an example of local  $G$ -ringed  $K$ -space (27.2.1.23).

A continuous homomorphism of rings induces a local  $G$ -ringed morphism. And all morphisms come from these.

Moreover, an affinoid  $K$ -space is a complete  $G$ -ringed  $K$ -space (i.e. rigid) (7.1.1.12).  $\square$

*Proof:* It is a  $G$ -space by (27.2.1.40)(27.2.1.41), morphisms by (27.2.1.19), notice the  $\mathfrak{m}_x$  generate the maximal ideal of  $\mathcal{O}_{X,x}$  (27.2.1.23), so the morphism is local.

To show all morphisms are like these, we need to show a morphism  $\sigma : A \rightarrow B$  gives at most one  $\mathrm{Sp} B \rightarrow \mathrm{Sp} A$ : the morphism is local, so it maps  $\mathfrak{m}_{\varphi(x)}$  to  $\mathfrak{m}_x$ , and from the commutative diagram

$$\begin{array}{ccc} A/\mathfrak{m}_{\varphi(x)} & \longrightarrow & B/\mathfrak{m}_x \\ \downarrow \cong & & \downarrow \cong \\ \mathcal{O}_{\varphi(x), \mathrm{Sp} A}/\mathfrak{m}_{\varphi(x)} \mathcal{O}_{\varphi(x), \mathrm{Sp} A} & \longrightarrow & \mathcal{O}_{x, \mathrm{Sp} B}/\mathfrak{m}_x \mathcal{O}_{x, \mathrm{Sp} B} \end{array}$$

(27.2.1.23) shows  $\mathfrak{m}_{\varphi(x)}$  is mapped into  $\mathfrak{m}_x$ , so  $\mathfrak{m}_{\varphi(x)} = (\sigma^*)^{-1}\mathfrak{m}_x$ , which shows  $\varphi$  is unique set-theoretically, and on the level of structure sheaf, the uniqueness of  $\mathcal{O}_{\mathrm{Sp} A}(V) \rightarrow \mathcal{O}_{\mathrm{Sp} B}(\varphi^{-1}(V))$  is unique by the definition of affinoid subdomain (27.2.1.7).  $\square$

**Def. (27.2.2.3) [Rigid Spaces].** The category of **rigid (analytic) space** is a full subcategory of local  $G$ -ringed  $K$ -spaces (27.2.2.1) s.t. it is  $G_0, G_1, G_2$ -complete (7.1.1.12), and it has an admissible covering  $\{X_i \rightarrow X\}_i$  s.t.  $(X_i, \mathcal{O}_X|_{X_i})$  are affinoid  $K$ -spaces (27.2.1.1) with the strong  $G$ -topology (27.2.1.17).

It follows easily that an admissible open subset of a rigid space is again rigid.  $\square$

**Prop. (27.2.2.4) [Glueing Rigid Spaces].** Glueing rigid analytic spaces is legitimate, so does glueing morphisms on the source.  $\square$

*Proof:* First glue the set, then use (7.1.1.14) to glue  $G$ -topology, finally the glue of structure sheaf is similar to (7.1.5.3).  $\square$

**Cor. (27.2.2.5) [Spectrum Adjointness].** If  $X$  is rigid and  $Y$  is affinoid, then  $\mathrm{Hom}(X, Y) \cong \mathrm{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ .  $\square$

*Proof:* This follows from (27.2.2.2) and glueing (27.2.2.4).  $\square$



**Prop. (27.2.2.6) [Fiber Products].** Fiber products exist in the category of rigid analytic space. This is fiber product of affinoid spaces are affinoid so we can glue them by universal property, the same as (7.2.7.15).  $\square$

**Prop. (27.2.2.7).** An affinoid space is connected in the weak Grothendieck topology iff it is connected in the strong Grothendieck topology iff it is connected in the Zariski topology.  $\square$

*Proof:* Firstly the weak and strong are equal because any strong covering of  $X$  has a refinement of weak covering, and a weak covering is a strong covering. So it suffices to prove the equivalence of the last two.

One direction is trivial, for the other direction, use Tate's acyclicity, if  $X_1, X_2 \rightarrow \mathrm{Sp} A$ ,  $X_1 \cap X_2 = \emptyset$ , then  $A = \mathcal{O}_X(X_1) \times \mathcal{O}_X(X_2)$ , so  $\mathrm{Spec} A$  is not connected, neither do  $\mathrm{Sp} A$ .  $\square$

**Prop. (27.2.2.8).** We can define the connected components of  $X$  as the equivalence classes of elements that can be reached using connected admissible open subsets of  $X$ . Then the connected components are admissible and forms an admissible cover of  $X$ .  $\square$

*Proof:* Notice that there exists a finite covering consisting of connected Zariski subsets, by (27.2.2.7) and the fact  $\mathrm{Sp} A$  has f.m. connected components because  $\mathrm{Spec} A$  does as  $A$  is Noetherian (12.2.4.13) and (27.2.1.3).

Thus we are done, because by (27.2.1.22), a Zariski covering is admissible, and clearly the connected components of  $X$  are just this Zariski covering.  $\square$

### Rigid GAGA

**Lemma (27.2.2.9).** Let  $Z \in \mathrm{Aff}^{\mathrm{ft}}/K$ , and  $Y$  a rigid  $K$ -space, then the set of morphisms of local  $G$ -ringed spaces  $(Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$  corresponds to  $K$ -algebra morphisms from  $\mathcal{O}_Z(Z)$  to  $\mathcal{O}_Y(Y)$ .  $\square$

*Proof:* Cf. [Formal and Rigid Geometry P111].  $\square$

**Def. (27.2.2.10) [Rigid Analytification].** There is a partial functor  $X^{\mathrm{rig}}$  from the category of schemes  $X$  locally algebraic over a valued field  $K$  to the category of rigid analytic  $K$ -spaces that are right adjoint to the forgetful functor from the category of rigid  $K$ -spaces to local ringed  $K$ -space, called the **rigid GAGA functor**.

The existence of this functor is proven in (27.2.2.14).  $\square$

**Def. (27.2.2.11) [Analytification of Affine Schemes].** Let  $T_n(r)$  be the elements  $\sum a_v \zeta^v$  in  $T_n$  that  $\lim a_v r^{|v|} = 0$ . Then choose a  $c \in K$ ,  $|c| > 1$ , define  $T_n^{(i)} = T_n(|c|^i)$ . Then  $T_n^{(i)} = K\langle c^{-i}X_1, \dots, c^{-i}X_n \rangle$ , so clearly  $\mathrm{Sp}(T_n^{(i)})$  is an affinoid subdomain of  $\mathrm{Sp}(T_n^{(i+1)})$  by (27.2.1.9). Thus there is a chain of inclusions of affinoid subdomains:

$$B^n = \mathrm{Sp}(T_n^{(0)}) \hookrightarrow \mathrm{Sp}(T_n^{(1)}) \hookrightarrow \mathrm{Sp}(T_n^{(2)}) \hookrightarrow \dots$$

Then we can use (27.2.2.4) to glue them together as  $\mathbb{A}_K^{n, \mathrm{rig}}$ .  $\square$

**Prop. (27.2.2.12).** The maximal spectrum  $\mathrm{Spm}(K[X_i]) = \cup_n \mathrm{Sp}(T_n^{(i)})$  as sets.  $\square$

*Proof:* It suffices to show the following two.

- For any maximal ideal  $\mathfrak{m} \subset T_n$ ,  $\mathfrak{m}' = \mathfrak{m} \cap K[X_i]$  is maximal.
- For any maximal ideal  $\mathfrak{m}' \subset K[X_i]$ , there is some  $N$  that  $\mathfrak{m}' T_n^{(i)}$  is maximal in  $T_n^{(i)}$  for all  $i > N$ .



For 1: Consider the  $K \subset K[X_i]/\mathfrak{m}' \subset T_n/\mathfrak{m}$ ,  $T_n/\mathfrak{m}$  is a finite extension of  $K$  by (12.2.4.11), then so does  $K[X_i]/\mathfrak{m}'$ , by (5.2.1.3). To prove  $\mathfrak{m}' = \mathfrak{m} \cap K[X_i]$ , consider the following diagram:

$$\begin{array}{ccc} K[X_i]/\mathfrak{m}' & \longrightarrow & T_n/\mathfrak{m}'T_n \\ \parallel & & \downarrow \\ K[X_i]/\mathfrak{m}' & \longrightarrow & T_n/\mathfrak{m} \end{array}$$

As  $K[X_i]/\mathfrak{m}'$  is finite over  $K$ , it is complete, but  $K[X_i]$  is dense in  $T_n$ , thus the horizontal maps are surjective. But then the lower horizontal is isomorphism, then the upper horizontal is also isomorphism, and then the vertical map is isomorphism, thus we are done.

For 2,  $K[X_i]/\mathfrak{m}'$  is a finite extension of  $K$ , thus has a unique valuation, let  $N$  be large that  $|\overline{X}_i| \leq |c|^N$ , then for  $i > N$ , the quotient map factors uniquely as  $K[X_i] \rightarrow T_n^{(i)} \rightarrow K/\mathfrak{m}'$ . Then the kernel  $\mathfrak{m}$  of  $T_n^{(i)}$  is a maximal ideal (same reason as before) that satisfies  $\mathfrak{m} \cap K[X_i] = \mathfrak{m}'$ . Then we finish by item 1.  $\square$

**Cor. (27.2.2.13) [Analytification for Affine Schemes].** Similarly, for an affine scheme  $Z = \text{Spec } K[X_i]/\mathfrak{a}$  of f.t over  $K$ , we construct its analytification  $Z^{\text{rig}}$  as the glue of the inclusions:

$$\text{Sp}(T_n^{(0)}/\mathfrak{a}) \hookrightarrow \text{Sp}(T_n^{(1)}/\mathfrak{a}) \hookrightarrow \dots$$

Then  $Z^{\text{rig}}$  is the analytification of  $K[X_i]/\mathfrak{a}$ .

And we see from the proof of (27.2.2.12) the maximal spectrum  $\text{Max}(K[X_i]/\mathfrak{a}) = \cup_n \text{Spa}(T_n^{(i)}/\mathfrak{a})$  as sets.  $\lrcorner$

*Proof:* The canonical map  $K[X_i]/\mathfrak{a} \rightarrow T_n^{(i)}/\mathfrak{a}$  glue together to be a morphism  $\mathcal{O}_Z(Z) \rightarrow \mathcal{O}_{Z^{\text{rig}}}(Z^{\text{rig}})$ , which by (27.2.2.9) corresponds to a map  $Z^{\text{rig}} \rightarrow Z$  of local ringed spaces.

Now any other morphism  $Y \rightarrow Z$  from a rigid  $K$ -space  $Y$  to  $Z$ , choose an affinoid  $K$ -space covering  $Y_i$  of  $Y$ , then the map  $Y_i \rightarrow Z$  corresponds by (27.2.2.9) to a morphism  $\sigma : K[X_i]/\mathfrak{a} \rightarrow \mathcal{O}_{Y_i}(Y_i)$ , thus if we choose  $i$  large enough that  $|\sigma(X_i)| \leq |c|^i$ , then  $\sigma$  can be extended uniquely to

$$K[X_i]/\mathfrak{a} \rightarrow T_n^{(i)}/\mathfrak{a} \xrightarrow{\sigma} \mathcal{O}_{Y_i}(Y_i),$$

By the universality of affinoid subdomains. This  $\sigma$  corresponds to a morphism  $Y_i \rightarrow \text{Sp}(T_n^{(i)}) \rightarrow Z^{\text{rig}}$ , and these clearly glue together to give a morphism  $Y \rightarrow Z^{\text{rig}}$ , thus proving the universal property.  $\square$

**Prop. (27.2.2.14) [General Analytification].** For  $X \in \text{Sch}^{\text{loc.ft}}/K$ , choose an affine covering  $\{Z_i\}$  of  $X$  and consider the analytifications of  $Z_i$  (27.2.2.13), then  $Z_i \cap Z_j$  obviously has the inverse image as the rigid analytification by universal property, thus unique, so the analytifications of  $Z_i$  can be glued to an analytification of  $X$ .

Moreover,  $|X^{\text{rig}}| = |X|_0$ , because this is the case for  $Z_i$  (27.2.2.11).  $\lrcorner$

**Prop. (27.2.2.15).** Rigid analytification preserves fiber products.  $\lrcorner$

*Proof:* This follows from the construction of fibered product of schemes (7.2.7.15), so we only need to prove the affine case. For this, Cf. [U. Köpf, Über eigentliche Familien algebraischer Varietäten über affinoiden Räumen. Schriftenr, Satz 1.8].  $\square$

**Prop. (27.2.2.16) [Stalks].** For a point  $z \in Z^{\text{rig}}$ , the completions of  $\mathcal{O}_{Z^{\text{rig}},z}$  and  $\mathcal{O}_{Z,z}$  are the same.  $\lrcorner$

*Proof:* Cf. [U. Köpf, Über eigentliche Familien algebraischer Varietäten über affinoiden Räumen. Schriftenr, Satz 2.1].  $\square$

### A way to construct Rigid Analytic Spaces

**Prop. (27.2.2.17).** Let  $n \in \mathbb{Z}_+$ ,  $K \in p\text{-NField}$  and  $K' \subset \mathbb{C}_K$  be a complete field containing  $K$ . Suppose  $X \subset \mathbb{P}^n$ ,  $T \subset \mathbb{P}^m$  are projective schemes, and  $H \subset (X \times T)_{K'}$ . For  $t \in T(K)$ , denote  $H_t \subset X_K$  be the fiber, then there is an admissible open subset of  $X_{K'}^{\text{rig}}$  with underlying set

$$X(\mathbb{C}_p) \setminus \bigcup_{t \in T(K)} H_t.$$

┘

*Proof:* For each  $z \in \mathbb{P}^n(\mathbb{C}_p)$  or  $z \in \mathbb{P}^m(\mathbb{C})$ , choose an affine unimodular representative  $\tilde{z} \in \mathbb{A}^{n+1}(\mathbb{C}_p)$  or  $\tilde{z} \in \mathbb{A}^{m+1}(\mathbb{C}_p)$  with  $\max(|\tilde{z}_i|) = 1$ .

Let  $H$  be defined by a family of homogeneous polynomials

$$f_\alpha(X_0, \dots, X_n, T_0, \dots, T_m)$$

with  $\mathcal{O}_{K'}$ -coefficients, then the functions  $|f_\alpha(\tilde{x}, \tilde{t})|$  are independent of the representatives chosen, so denoted by  $|f_\alpha(x, t)|$ . For  $\varepsilon \in \mathbb{R}_+$  and  $t \in T(K)$ , consider the tubular nbhd

$$H_t(\varepsilon) = \{x \in X(\mathbb{C}_p) : |f_\alpha(x, t)| \leq \varepsilon, \forall \alpha\}.$$

Then it is easy to see that when  $t' \in T$  with  $\|\tilde{t}' - \tilde{t}\|$  small enough, we will have  $H_{t'}(\varepsilon) \subset H_t(\varepsilon)$ . Because the set of unimodular representatives in  $K^{m+1} \setminus \{0\}$  is compact, there exists a finite set  $S_\varepsilon \subset T(K)$  s.t.

$$\bigcup_{t \in S_\varepsilon} H_t(\varepsilon) = \bigcup_{t \in T(K)} H_t(\varepsilon).$$

Then  $X(\mathbb{C}_p) \setminus \bigcup_{t \in T(K)} H_t(\varepsilon)$  is an admissible open subset of  $X^{\text{rig}}$ , which is because for each  $\alpha$ ,

$$\{x \in \mathbb{P}^n(\mathbb{C}_p) : |f_\alpha(x)| \geq \varepsilon\}$$

is admissible open, as it is on the  $n$  admissible opens of  $\mathbb{P}^n$ .

Now define  $X_\varepsilon = X(\mathbb{C}_p) \setminus \bigcup_{t \in T(K)} H_t(\varepsilon)$ , then by the definition of strong Grothendieck topology, we need to show that any map

$$f : Y \rightarrow \bigcup_{\varepsilon \in \mathbb{R}_+} X_\varepsilon = X(\mathbb{C}_p) \setminus \bigcup_{t \in T(K)} H_t$$

from some affinoid space  $Y$  factors through some  $X_\varepsilon$ . Now for each  $t \in T(K)$ , there exists  $\varepsilon_t \in \mathbb{R}_+$  s.t.  $f(Y) \subset X(\mathbb{C}_p) \setminus H_t(\varepsilon_t)$  by maximum principal(12.2.4.22). But by the continuity argument of  $H_t(\varepsilon)$  as above, we can choose  $\varepsilon_{t'} = \varepsilon_t$  for  $t'$  sufficiently near to  $t$ . Thus we can choose a common  $\varepsilon$  for each  $t \in T(K)$ , and the assertion is proved.  $\square$

### 3 Coherent Sheaves on Rigid Spaces

**Prop. (27.2.3.1).** For an affinoid  $K$ -space  $X$ , there is a Qco module construction  $M \rightarrow M \otimes_A \mathcal{O}_X$  as in(27.2.1.40) in the weak Grothendieck topology, and it extends uniquely to a sheaf w.r.t. the strong Grothendieck topology by(27.2.1.41), also denoted by  $M \otimes_A \mathcal{O}_X$ . This is a faithfully exact, fully faithful functor between Abelian categories from  $\mathcal{A}b$  to  $\mathcal{O}_X$ -modules, and it preserves tensor product and direct sums.  $\square$

*Proof:* Because  $\Gamma(X, M \otimes_A \mathcal{O}_X) = M$  and obviously fully faithful, this map is fully faithful, and it is exact because the restriction map of an affinoid subdomain is flat (27.2.1.26), and shification is exact.  $\square$

**Def. (27.2.3.2) [Coherent Sheaves].** For an  $\mathcal{O}_X$ -module  $\mathcal{F}$  on a rigid space  $X$ , **finite type, of finite presentation, coherence** are defined w.r.t the strong topology as  $X$  is a ringed site. All these notions are stable under passing to an admissible open subspaces.  $\lrcorner$

*Proof:* For the passing of coherence to admissible open subspaces, use the fact that restriction maps are flat (27.2.1.26).  $\square$

**Cor. (27.2.3.3).** Notice  $\mathcal{O}_X^n = A^n \otimes_A \mathcal{O}_X$ , by (27.2.3.1) and the fact  $A$  is Noetherian, passing to a refinement covering,  $\mathcal{F}$  is coherent iff there is an admissible affinoid covering  $\mathfrak{U}: X_i \rightarrow X$  that  $F|_{X_i}$  is associated to a finite  $\mathcal{O}_{X_i}$ -module. In this case,  $\mathcal{F}$  is said to be  **$\mathfrak{U}$ -coherent**. Thus the coherent sheaves form a weak Serre subcategory of  $\mathcal{O}_X$ -modules.

In particular,  $\mathcal{O}_X$  is coherent.  $\lrcorner$

**Prop. (27.2.3.4).** If  $\mathcal{F}, \mathcal{G}$  are all  $\mathfrak{U}$ -coherent modules, then:

- $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$  and  $\mathcal{F} \oplus \mathcal{G}$  are  $\mathfrak{U}$ -coherent,
- if  $\mathcal{F} \rightarrow \mathcal{G}$  is a  $\mathcal{O}_X$ -module morphism, then the kernel and image are all  $\mathfrak{U}$ -coherent.
- If  $\mathcal{I}$  is a  $\mathfrak{U}$ -coherent sheaf of ideal of  $\mathcal{O}_X$ , then  $\mathcal{I}\mathcal{F}$  is  $\mathfrak{U}$ -coherent.

$\lrcorner$

*Proof:* The first and the second are consequences of (27.2.3.1), noticing  $A_i$  is Noetherian. The third is an image of  $\mathcal{I} \otimes_{\mathcal{O}_X} \mathcal{F}$ .  $\square$

**Lemma (27.2.3.5).** If  $\mathcal{F}$  is  $\mathfrak{U}$  coherent for a simple Laurent covering  $\mathfrak{U}$ , then  $H^1(\mathfrak{U}, \mathcal{F}) = 0$ .  $\lrcorner$

*Proof:* The goal is to show any element in  $\mathcal{F}(U_1 \cap U_2)$  can be represented by  $u_1 + u_2$ , where  $u_i \in \mathcal{F}(U_i)$ . Let  $U_1 = \text{Sp } A\langle f \rangle, U_2 = \text{Sp } A\langle f^{-1} \rangle, U_1 \cap U_2 = \text{Sp } A\langle f, f^{-1} \rangle$ . Now  $A\langle f \rangle = A\langle X \rangle / (X - f), A\langle f^{-1} \rangle = A\langle Y \rangle / (Yf - 1), A\langle f, f^{-1} \rangle = A\langle X, Y \rangle / (X - f, Yf - 1)$ , and we endow them with the residue norm.

Now we want to give norms to  $M_1 = \mathcal{F}(U_1), M_2 = \mathcal{F}(U_2), M_{12} = \mathcal{F}(U_1 \cap U_2)$ .  $M_i$  are finite  $\mathcal{O}_X(U_i)$ -modules, so there are elements  $v_i, w_j, i \leq m, j \leq n$  that generate  $M_1, M_2$  respectively. So there are attached morphisms

$$(A\langle f \rangle)^m \rightarrow M_1, \quad (A\langle f^{-1} \rangle)^n \rightarrow M_2, \quad (A\langle f, f^{-1} \rangle)^m \rightarrow M_{12}$$

And endow them with the residue norm, which is complete..

Notice that to prove the assertion, it suffice to show for each  $\varepsilon > 0$ , there is an  $\alpha$  that for each  $u \in M_{12}$ , there are  $u_1$  and  $u_2$  in  $M_i$  respectively that  $|u_i| < \alpha|u|$  and  $|u - u_1 - u_2| < \varepsilon|u|$ , because then we can use iteration and completeness to get the result.

Giving  $\beta > 1$ , any  $g \in A\langle f, f^{-1} \rangle$  can be lifted to an element  $\sum c_{ij} X^i Y^j$  that  $|c_{ij}| \leq \beta|g|$ . Then by regrouping terms that  $i \geq j$  or  $i < j$ , there are two element  $g^+ \in A\langle f \rangle$  and  $g^- \in A\langle f^{-1} \rangle$  that  $g^+ + g^-$  restricts to  $g$  on  $U_1 \cap U_2$ , and  $|g^+|, |g^-| \leq \beta|g|$ .

Now that  $\mathcal{F}$  is coherent, so  $v_i$  and  $w_j$  both generate  $M_{12}$  separately. Then there are equations  $v_i = \sum c_{ij} w_j$  and  $w_i = \sum d_{ij} v_j$ , where  $c_{ij}, d_{ij} \in A\langle f, f^{-1} \rangle$ . The image of  $A\langle f \rangle$  is dense in  $A\langle f, f^{-1} \rangle$  (12.2.4.27), so there are elements  $c'_{ij} \in A\langle f^{-1} \rangle$  s.t.  $\max_{ijl} |c_{ij} - c'_{ij}| |d_{jl}| < \beta^{-2} \varepsilon$ .

Now I claim the above approximation process is true for  $\alpha = \beta^2 \max(|c'_{ij}| + 1)$ . For this, notice for any  $u = \sum a_i v_i$  with  $a_i \in A\langle f, f^{-1} \rangle$ , which we may assume  $|a_i| \leq \beta|u|$  by the definition of the norm on  $M_{12}$ , then  $a_i = a_i^+ + a_i^-$ , that  $|a_i^*| \leq \beta|a_i|$ . Consider the following element

$$u^+ = \sum a_i^+ v_i \in M_1, \quad u^- = \sum a_i^- \sum c'_{ij} w_j \in M_2$$

Then it is easily verified that  $|u^*| < \alpha|u|$ , and

$$u - u^- - u^+ = \sum \sum a_i^- (c_{ij} - c'_{ij}) w_j = \sum \sum \sum a_i^- (c_{ij} - c'_{ij}) d_{jl} v_l.$$

which has norm smaller than  $\max |a_i^- (c_{ij} - c'_{ij}) d_{jl}| \leq \beta^2 |u| \cdot \beta^{-2} \varepsilon = \varepsilon |u|$ , finishing the proof.  $\square$

**Prop. (27.2.3.6) [Kiehl].** An  $\mathcal{O}_X$ -module  $\mathcal{F}$  on an affinoid  $K$ -space  $\mathrm{Sp} A$  is coherent iff it is associated to a finite  $A$ -module.  $\lrcorner$

*Proof:* The converse is obvious, for the other direction, by (27.2.1.39), it suffices to prove for  $\mathfrak{U}$  a Laurent covering, and further, it suffices to prove for the simplest Laurent covering  $X(f), X(f^{-1}) \rightarrow X$  because:  $U(f, g) \cup U(f, g^{-1}) \cup U(f^{-1}, g) \cup U(f^{-1}, g^{-1}) = (U(f, g) \cup U(f, g^{-1})) \cup (U(f^{-1}, g) \cup U(f^{-1}, g^{-1}))$ .

Thus the above lemma shows that  $H^1(\mathfrak{U}, \mathcal{F}) = 0$ . Now I prove that for any finite affinoid covering  $\mathfrak{U} = \cup \mathrm{Sp} A_i$ , if  $H^1(\mathfrak{U}, \mathcal{F}) = 0$  for any coherent sheaf  $\mathcal{F}$ , then any  $\mathfrak{U}$ -coherent sheaf  $\mathcal{F}$  is associated to a finite  $A$ -module, this will finish the proof.

Consider any maximal ideal  $\mathfrak{m}_x$  of  $A$ ,  $\mathfrak{m}_x \otimes_A \mathcal{O}_X$  is a coherent sheaf as  $\mathfrak{m}_x$  is finite because  $A$  is Noetherian, so there is a short exact sequence

$$0 \rightarrow \mathfrak{m}_x \mathcal{F} \rightarrow \mathcal{F} \rightarrow \mathcal{F}/\mathfrak{m}_x \mathcal{F} \rightarrow 0$$

of  $\mathfrak{U}$ -coherent sheaves, because  $A/\mathfrak{m}_x$  is a field, thus flat.

Now for any affinoid space  $U'$  in  $U_i$  for some  $i$ , the section of this exact sequence is exact, because the ring morphism associated to an affinoid subdomain is flat (27.2.1.26). In particular, this can be applied to any intersections of  $U_i$ , in particular the Čech complex of these sheaves. Then the long exact sequence and the fact  $H^1(\mathfrak{U}, \mathfrak{m}_x \mathcal{F}) = 0$  shows

$$0 \rightarrow \mathfrak{m}_x \mathcal{F}(X) \rightarrow \mathcal{F}(X) \rightarrow \mathcal{F}/\mathfrak{m}_x \mathcal{F}(X) \rightarrow 0$$

Next we want to show  $\mathcal{F}/\mathfrak{m}_x \mathcal{F}(X) \rightarrow \mathcal{F}/\mathfrak{m}_x \mathcal{F}(U_i)$  is isomorphism for any  $x \in U_i$ . To prove this, first for any affinoid subspace  $U' = \mathrm{Sp} B$  contained in some  $U_j$ , let  $U' \cap U_i = \mathrm{Sp} B_i$ ,  $\mathcal{F}|_{U'} = M' \otimes_A \mathcal{O}_{U'}$ , we show  $\mathcal{F}/\mathfrak{m}_x \mathcal{F}(U') \cong \mathcal{F}/\mathfrak{m}_x \mathcal{F}(U' \cap U_i)$ , this is equivalent to

$$M'/\mathfrak{m}_x M' \rightarrow M'/\mathfrak{m}_x M' \otimes_B B_i = M'/\mathfrak{m}_x M' \otimes_{B/\mathfrak{m}_x B} B_j/\mathfrak{m}_x B_j$$

is an isomorphism. But  $B/\mathfrak{m}_x B \cong B_j/\mathfrak{m}_x B_j$ : This is true when  $x \in U'$  by (27.2.1.8), and they are both trivial ring if  $x \notin U'$ . Then look at the morphism of Čech complex induced by  $\mathcal{F}/\mathfrak{m}_x \mathcal{F} \rightarrow \mathcal{F}/\mathfrak{m}_x \mathcal{F}|_{U_i}$ , then it is an isomorphism, by what we just proved, so its  $H^0$  is also isomorphism, which is  $\mathcal{F}/\mathfrak{m}_x \mathcal{F}(X) \cong \mathcal{F}/\mathfrak{m}_x \mathcal{F}(U_i)$ .

Finally, by the commutative diagram 
$$\begin{array}{ccc} \mathcal{F}(X) & \longrightarrow & \mathcal{F}/\mathfrak{m}_x \mathcal{F}(X) \\ \downarrow & & \downarrow \\ \mathcal{F}(U_i) & \longrightarrow & \mathcal{F}/\mathfrak{m}_x \mathcal{F}(U_i) \end{array}$$
, the right vertical arrow is isomorphism, so if denote  $\mathcal{F}(U_i)$  by  $M_i$ , then  $\mathcal{F}(X)$  generate  $M_i/\mathfrak{m}_x M_i$  for every  $x$ , then consider

$L = M_i/\mathcal{F}(X)$ , then  $\mathfrak{m}_x L = L$  for every  $x$ , then by Nakayama, for each maximal ideal  $\mathfrak{m}$ , there is a  $m \in \mathfrak{m}$  that  $(1 + m)L = 0$ , so  $\text{Ann}(L) = (1)$ , so  $L = 0$ , i.e.  $\mathcal{F}(X)$  generate  $M_i$  for each  $i$ .

Now choose  $f_i$  in  $\mathcal{F}(X)$  that generate  $M_i$  simultaneously, then the map  $\mathcal{O}_X^n \rightarrow \mathcal{F}$  is a surjection of  $\mathfrak{U}$ -coherent sheaves, its kernel  $\mathcal{G}$  is also coherent by (27.2.3.4), now all the above argument works for  $\mathcal{G}$ , so there is a surjection  $\mathcal{O}_X^m \rightarrow \mathcal{G}$ , so  $\mathcal{O}_X^m \rightarrow \mathcal{O}_X^n \rightarrow \mathcal{F} \rightarrow 0$ , so  $\mathcal{F}$  is associated to the cokernel of the map  $A^m \rightarrow A^n$ .  $\square$

**Cor. (27.2.3.7).** Coherence for a  $\mathcal{O}_X$ -module on a rigid  $K$ -space is affinoid local on the target.  $\lrcorner$

### Cohomology on Rigid Analytic Spaces

**Lemma (27.2.3.8).** The category of  $\mathcal{O}_X$ -modules on a rigid  $K$ -space is a Grothendieck category by (4.8.3.29).  $\lrcorner$

**Def. (27.2.3.9) [Derived Cohomologies].** Consider the right derived functor for  $\Gamma$  and more general  $f_*$ , these are left exact by (7.1.2.9). Then  $R^p f_* \mathcal{F} = (f_* \mathcal{H}^p(\mathcal{F}))^\sharp$  by Grothendieck spectral sequence.

The Čech-to-Derived spectral sequence (7.3.2.13) is applied:  $H^p(\{U_i \rightarrow U\}, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(U, F)$ ,  $\check{H}^p(U, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(U, F)$  and  $\check{H}^1(U, F) \cong H^1(U, F)$ .

In particular, if  $H^q(U_{i_0 i_1 \dots i_r}, \mathcal{F}) = 0, q > 0$ , then  $H^p(\{U_i \rightarrow U\}, \mathcal{F}) = H^p(U, \mathcal{F})$  (7.3.2.15). And it is enough to have  $\check{H}^q(U_{i_0 i_1 \dots i_r}, \mathcal{F}) = 0, q > 0$  by (7.3.2.16).  $\lrcorner$

**Cor. (27.2.3.10).** A Qco sheaf on an affinoid space has vanishing higher sheaf cohomology by Tate's acyclicity (27.2.1.40) and (7.3.2.16).  $\lrcorner$

### Properties of Rigid $K$ -Spaces

This subsection is strongly suggested to read after reading the parallel part of schemes.

**Def. (27.2.3.11).** A morphism is called a **closed immersion** if there is an admissible affinoid covering that it restricts to a closed immersion of affinoid spaces (It is compatible with definition (27.2.1.28) before by (27.2.3.15)). It is called an **open immersion** iff it is injective and the corresponding stalk maps are isomorphisms. The **(quasi-)separatedness, quasi-compactness, finiteness** are defined similarly as for schemes.  $\lrcorner$

**Lemma (27.2.3.12) [Nike's Trick].** In a rigid analytic  $K$ -space  $X$  and  $\text{Sp } A, \text{Sp } B$  be affinoid subspaces, then there is an admissible affinoid covering of  $\text{Sp } A \cap \text{Sp } B$ .  $\lrcorner$

*Proof:* This is analogous to the scheme case (7.4.1.1), but the proof is different:  $X$  has an admissible covering, this restricts to an admissible covering of  $\text{Sp } A \cap \text{Sp } B$ , and any admissible covering can be refined by an affinoid admissible covering.  $\square$

**Prop. (27.2.3.13) [Affinoid Communication Theorem].** A property  $P$  of affinoid open subsets of  $X$  is called **affinoid local** if:  $\text{Sp } A$  has  $P \Rightarrow$  all affinoid subdomains of  $\text{Sp } A$  has  $P$ , and any admissible affinoid cover of  $\text{Sp } A$  has  $P \Rightarrow \text{Sp } A$  has  $P$ . Notice a stalk-wise property is obviously affine-local.

Now if we call  $X$  has  $\tilde{P}$  if there is an admissible affinoid covering  $A_i \rightarrow X$  that  $A_i$  has  $P$ . Then the following are equivalent:

- all open affinoid subsets of  $X$  has  $P$ .
- all open subspace of  $X$  has  $\tilde{P}$ .
- $X$  has a cover of open subspaces that has  $\tilde{P}$ .

- $X$  has  $\tilde{P}$ .

┘

*Proof:* The proof is the same as the scheme case (7.4.1.2). □

**Prop. (27.2.3.14).** Separated morphism is quasi-separated because closed immersion is affinoid hence quasi-compact (27.2.1.3). ┘

**Prop. (27.2.3.15) [Finite Morphism].** For a morphism  $\varphi : X \rightarrow Y$  of rigid  $K$ -spaces

- It is finite iff the inverse image of any affinoid space is affinoid, and  $\varphi_* \mathcal{O}_Y$  is a coherent  $\mathcal{O}_X$ -module. In particular, finiteness is local on the target because coherence do.
- It is closed immersion iff it is finite and  $\mathcal{O}_Y \rightarrow \varphi_* \mathcal{O}_X$  is surjective, this shows the definition of closed immersion is compatible with before.

┘

*Proof:* Coherence is affinoid local on the target by Kiehl's theorem, so it suffices to prove the inverse image of any affinoid space is affinoid for a finite morphism: Consider any affinoid subdomain  $U \subset X$  with inverse image  $\varphi^{-1}(U)$ , by Kiehl's theorem,  $B = \mathcal{O}_X(f^{-1}(U))$  is finite over  $A = \mathcal{O}_Y(U)$ , thus can be given an affinoid  $K$ -algebra structure (12.2.4.34). Now

$$\varphi^{-1}(U) \xrightarrow{\chi} \mathrm{Sp} B \xrightarrow{\rho'} \mathrm{Sp} A$$

$\chi$  is locally an isomorphism, as  $\rho$  is finite, so  $\chi$  is an isomorphism.

The second assertion is because locally  $\mathcal{O}_Y, \varphi_* \mathcal{O}_X$  are both Qco so surjectivity is equivalent to the global section is surjective (27.2.3.1). □

**Prop. (27.2.3.16).** Closed/Open immersion, quasi-compactness, (quasi-)separatedness are all local on the target, and stable under base change. ┘

*Proof:* Closed immersion is local on the target because finiteness do and surjectiveness of  $\mathcal{O}_Y \rightarrow \varphi_* \mathcal{O}_X$  is checked locally. Open immersion is local on the target because stalk and injectivity are all checked locally.

Then Closed/Open immersion are stable under base change because the affinoid case is true (27.2.1.29).

Quasi-compact is easily seen local on the target and stable under base change.

(Quasi-)Separatedness is local on the target because closed immersion and quasi-compact do.

(Quasi-)Separatedness is stable under base change because closed immersion and quasi-compact do, because diagonal commutes with base change (4.1.1.49). □

**Prop. (27.2.3.17).** Morphisms between affinoid  $K$ -spaces are separated. Moreover, because of localness, any finite morphism is separated. ┘

*Proof:* The diagonal is  $\mathrm{Sp} A \rightarrow \mathrm{Sp} A \hat{\otimes}_B A$ , whose ring map is surjective. □

**Prop. (27.2.3.18).** By (4.1.1.51), for  $X \rightarrow S$  and  $Y \rightarrow S$ , the map  $X = X \times_Y Y \rightarrow X \times_S Y$  is an immersion. It is closed immersion if  $Y \rightarrow S$  is separated, and it is qc if  $Y \rightarrow S$  is quasi-separated. ┘

**Cor. (27.2.3.19).** If  $s : S \rightarrow X$  is a section of  $f : X \rightarrow S$ , the above proposition applies to this case, because  $S = S \times_X X \rightarrow S \times_S X = X$ . ┘

**Prop. (27.2.3.20).** A morphism is quasi-separated iff there is an admissible affinoid covering  $W_i$  that, for any two affinoid open  $U, V$  that are mapped to an affinoid open, their intersection is quasi-compact. This is because quasi-compact is local on the target.

A morphism is separated iff there is an admissible affinoid covering  $W_i$  that, for any two affinoid open  $U, V$  that are mapped to an affinoid open, their intersection is affinoid open and  $\mathcal{O}(U) \hat{\otimes}_{\mathcal{O}(W_i)} \mathcal{O}(V) \rightarrow \mathcal{O}(U \cap V)$  is surjective. This is because closed immersion is local on the target (27.2.3.16).  $\lrcorner$

**Cor. (27.2.3.21).** If  $g \circ f$  is (quasi-)separated, then so is  $f$ .  $\lrcorner$

**Cor. (27.2.3.22).** If  $X$  is (quasi-)separated, then  $X \rightarrow Y$  is (quasi-)separated.  $\lrcorner$

### Étale Topology on Rigid Analytic Spaces

**Def. (27.2.3.23) [Étale Sites].** The étale sites of rigid analytic spaces has coverings given by  $Y \rightarrow X$  that is étale and there exists an admissible affinoid covering  $\{Y_i \rightarrow Y\}$  s.t.  $\{Y_i \rightarrow X\}$  is an admissible covering of  $X$ .  $\lrcorner$

**Prop. (27.2.3.24).** Any étale surjective map  $Y \rightarrow X$  factors uniquely as  $Y \rightarrow X' \rightarrow X$  where  $Y \rightarrow X'$  is an étale covering and  $X' \rightarrow X$  is étale and bijective on points.  $\lrcorner$

*Proof:* ?  $\square$

**Def. (27.2.3.25) [Lisse  $\mathbb{Q}_p$ -Sheaves].** Locally constant sheaves of  $\mathbb{Z}/(p^n)$ -modules on a rigid analytic space is defined as in étale sites of schemes, and so is the definition of sheaves of  $\mathbb{Z}_p$ -modules.

And lisse  $\mathbb{Q}_p$ -sheaves are lisse  $\mathbb{Z}_p$ -sheaves on an étale covering of  $X$  (27.2.3.23), with the Hom set tensored with  $\mathbb{Q}_p$ .

$\text{LocSys}_{\mathbb{Q}_p}^{\text{ét}}(X)$  is a tensor category, and each  $x \in |X|$  defines a fiber functor valued in  $\text{Vect}^{\text{fin}}/\mathbb{Q}_p$ .

$\lrcorner$

## 4 Proper Mapping Theorem

**Def. (27.2.4.1) [Relatively Compact Subsets].** For a rigid space  $X$  over affinoid space  $Y$ , if  $U \subset U' \subset X$  be affinoid subspaces,  $U$  is called **relatively compact** in  $U'$  iff there is a set of affinoid generators  $f_i$  of  $\mathcal{O}_X(U')$  over  $\mathcal{O}_Y(Y)$  that  $|f_i(x)| < 1$  on  $U$ . This is denoted by  $U \Subset_Y U'$ .  $\lrcorner$

**Prop. (27.2.4.2).** If  $X_1, X_2$  are affinoid spaces over an affinoid space  $Y$ , and  $U_i$  are affinoid space of  $X_i$ , then

- if  $U_1 \Subset_Y X_1$ , then  $U_1 \times_Y X_2 \Subset_{X_2} X_1 \times_Y X_2$ .
- if  $U_i \Subset_Y X_i$ , then  $U_1 \times_Y U_2 \Subset_Y X_1 \times_Y X_2$ .
- If  $U_i \Subset_Y X_i$ , and  $X_i$  are affinoid subspaces of a rigid space separable over  $Y$ , then  $U_1 \cap U_2 \Subset_Y X_1 \cap X_2$ .
- If  $U_1 \Subset_Y X_1$ , and  $i : T \rightarrow X_1$  is a closed immersion, then  $i^{-1}(U_1) \Subset_Y i^{-1}(X_1)$ .

The proof is easy. For the last one, should notice  $|f(x)| = |f(i(x))|$ , because it is closed immersion, so the residue field is the same.  $\lrcorner$

**Def. (27.2.4.3) [Proper Morphism].** A **proper** morphism  $\varphi : X \rightarrow Y$  of rigid  $K$ -spaces is a separated morphism that there is an admissible affinoid covering  $Y_i$  of  $Y$  that there are two admissible affinoid coverings  $X_{ij}, X'_{ij}$  of  $\varphi^{-1}(Y_i)$  that  $X_{ij} \Subset_{Y_i} X'_{ij}$  for any  $i, j$ .  $\lrcorner$



**Prop. (27.2.4.4).** Properness is stable under base change and composition ┘

*Proof:* The base change follows directly from (27.2.4.2).

For the composition, Cf.[Formal and Rigid Geometry P131](difficult). □

**Prop. (27.2.4.5).** Properness is local on the target. ┘

*Proof:* This is because separatedness is local on the target (27.2.3.16) and the second condition of properness is itself local. □

**Prop. (27.2.4.6).** If  $g \circ f : X \rightarrow Y \rightarrow Z$  is proper and  $g$  is separated, then  $f$  is proper. ┘

*Proof:* By ??,  $\tau : X \rightarrow X \times_Z Y$  is closed immersion, and  $f$  is separated by (27.2.3.21). Now proper is local, so we may assume  $Z$  is affinoid, so there are two admissible covering  $X_i, X'_i$  of  $X$  that  $X_i \subseteq_Z X'_i$ , and choose an admissible affinoid covering  $Y_i \rightarrow Y$ , then  $X_j \times_Z Y_i, X'_j \times_Z Y_i$  are admissible coverings of  $Y_i$  that is  $X_j \times_Z Y_i \subseteq_{Y_i} X'_j \times_Z Y_i$ . And it can be pulled back to an affinoid admissible coverings of  $f^{-1}(Y_i)$  that  $\tau^{-1}(X_j \times_Z Y_i) \subseteq_{Y_i} \tau^{-1}(X'_j \times_Z Y_i)$ , because  $\tau$  is closed immersion. So  $X \rightarrow Y$  is proper. □

**Prop. (27.2.4.7).** Finite morphism is proper, in particular, closed immersion is proper. ┘

*Proof:* Finite morphism is separated by (27.2.3.17), and locally, assume both space are affinoid,  $X = \text{Sp } B \rightarrow \text{Sp } A = Y$ , then  $B$  is a finite  $A$ -module, in priori a f.g.  $A$ -algebra, so there is a set of generators  $f_i$  of  $B$  over  $A$  that (by multiplying a constant in  $K^*$ )  $|f_i|_{\text{sup}} < 1$  (12.2.4.22), so  $X \subseteq_Y X$ , hence it is proper. □

**Prop. (27.2.4.8)[Proper and Analytification].** For a morphism between schemes locally of f.t. over  $K$ , it is proper iff its rigid analytification is proper. ┘

*Proof:* Cf.[U. Köpf, Über eigentliche Familien algebraischer Varietäten über affinoiden Räumen. Schriftenr. Math. Inst. Univ. Münster, 2. Serie, Heft 7 (1974) Satz 2.16]. □

For the following: Cf.[Formal and Rigid Geometry P132].

**Prop. (27.2.4.9)[Proper Mapping theorem, Kiehl].** The higher direct images of a proper map of rigid analytic spaces takes coherent sheaves to coherent sheaves. ┘

*Proof:* □

**Prop. (27.2.4.10).** For a scheme  $X$  locally of f.t. over  $K$ , an  $\mathcal{O}_X$ -module  $\mathcal{F}$  on  $X$  gives rise to an  $\mathcal{O}_{X^{\text{rig}}}$ -module on  $X^{\text{rig}}$ , and it is coherent iff  $\mathcal{F}$  is coherent. ┘

*Proof:* □

**Prop. (27.2.4.11).** For a proper scheme over  $K$ ,  $H^q(X, \mathcal{F}) \cong H^q(X^{\text{rig}}, \mathcal{F}^{\text{rig}})$  for  $\mathcal{F}$  coherent. ┘

*Proof:* □

**Prop. (27.2.4.12).** When  $X$  is proper, coherent sheaves on  $X^{\text{rig}}$  corresponds to coherent sheaves on  $X$ . This gives an analog of Chow's theorem when applied to  $X = \mathbb{P}_K^n$  and  $\mathcal{F}'$  is a sheaf of ideals in  $\mathcal{O}_{X^{\text{rig}}}$ . ┘

*Proof:* □



## 5 Formal Schemes

References for this subsection is [Har77, Bos15] and [Topics in Algebraic Geometry, Illusie].

**Def. (27.2.5.1) [Formal Spectrum  $\mathrm{Spf} A$ ].** For  $(A, \mathfrak{a}) \in \mathrm{AdicCRing}(12.2.1.10)$  s.t.  $A$  is  $\mathfrak{a}$ -adically complete and separated, define  $\mathrm{Spf} A$  be the ringed space with underlying topological space  $\mathrm{Spec}(A/\mathfrak{a})$  (open primes of  $A$ ), and the structure sheaf  $\mathcal{O}$  that  $\mathcal{O}(D(f)) = A\langle f^{-1} \rangle$  (12.2.1.13), called the **formal spectrum of  $A$** .  $\lrcorner$

*Proof:* To construct this sheaf, we first check that  $\mathcal{O}(D(f)) = A\langle f^{-1} \rangle = \varprojlim_n (A/\mathfrak{a}^n[f^{-1}])$  defines a sheaf on the site of subspaces of  $\mathrm{Spf} A$  of the form  $D(f)$ : For any open covering  $\{D(f_i)\}$  of  $D(f)$ , there are exact sequences:

$$0 \rightarrow (A/\mathfrak{a}^n)_f \rightarrow \prod_i (A/\mathfrak{a}^n)_{f_i} \rightarrow \prod_{i,j} (A/\mathfrak{a}^n)_{f_i f_j}$$

by (5.4.2.3). Then we take inverse limit, which is exact by Mittag-Leffler (5.9.3.2), to get an exact sequence, which is just the sheaf condition of  $\mathcal{O}$ . Then, we can use (7.1.2.25) to extend this sheaf to a sheaf on  $\mathrm{Spf} A$ .

? Or we notice that  $\mathcal{O} = \varprojlim_n \mathcal{O}_{\mathrm{Spec}(A/\mathfrak{a}^n)}$ .  $\square$

**Prop. (27.2.5.2) [Stalks of  $\mathrm{Spf} A$ ].** Let  $x \in \mathrm{Spf} A$  correspond to a prime  $\mathfrak{p}_x$  in  $A$ . Then the stalk of  $\mathrm{Spf} A$  at  $x$  is just  $\mathcal{O}_x = \varinjlim_{x \in D(f)} A\langle f^{-1} \rangle$ , which is a local ring with maximal ideal  $\mathfrak{m}_x$  containing  $\mathfrak{p}_x \mathcal{O}_x$ . Moreover,  $\mathfrak{m}_x = \mathfrak{p}_x \mathcal{O}_x$  iff  $\mathfrak{a}$  is f.g.. So  $\mathrm{Spf} A$  is a local ringed space, called the **affine formal scheme** of  $A$ .  $\lrcorner$

*Proof:* Cf. [Bos15] P159. ?  $\square$

**Remark (27.2.5.3).** In many case, for example in Scholze's treatment of the  $p$ -adic geometry, the ideal of definition  $\mathfrak{a}$  is assumed to be f.g., because we need this to show that the  $\mathfrak{a}$ -adic completion of  $A[f^{-1}]$  is  $(\mathfrak{a})$ -adically complete (5.2.3.6), so that we can interpret  $D(f) \subset \mathrm{Spf} A$  as an affine formal scheme  $\mathrm{Spf}(A\langle f^{-1} \rangle)$  (12.2.1.14).

Another way to get around this finiteness condition is to consider formal spectrum for a larger class of rings, called the **admissible rings**, which is a complete and separated topological ring with a basis consisting of open ideals, and with an ideal of definition  $\mathfrak{a}$  that  $\mathfrak{a}^n \rightarrow 0$  for  $n \rightarrow 0$ .  $\lrcorner$

**Prop. (27.2.5.4) [Affine Formal Adjointness].** Morphisms between local-topological-ringed spaces  $\mathrm{Spf} B \rightarrow \mathrm{Spf} A$  corresponds to continuous homomorphisms  $A \rightarrow B$ .  $\lrcorner$

**Def. (27.2.5.5) [Formal Schemes].** The category of **formal schemes** is the full subcategory of the category of local-topological-ringed topological spaces  $(X, \mathcal{O}_X)$  consisting of objects that is locally isomorphic to an affine formal scheme  $\mathrm{Spf} A$  (27.2.5.1).

The category of formal schemes contains the category of schemes, by mapping  $\mathrm{Spec} A$  to  $\mathrm{Spf} A$ , where  $A$  is endowed with the discrete topology.  $\lrcorner$

**Prop. (27.2.5.6) [Glueing and Fiber Products].** Formal Schemes can easily be glued, and also spectrum adjointness holds as in (27.2.2.5). Finally there are fibered products, constructed as in (7.2.7.15), where the affine case corresponds to completed tensor product (12.2.1.15).  $\lrcorner$

**Def. (27.2.5.7) [Formal Completion of Schemes Along a Closed Subscheme].** For  $X \in \mathrm{Sch}$  and  $Y \subset X$  a closed subscheme, consider the sheaf  $\mathcal{O}_Y^\wedge$  defined by restricting the projective limit  $\varprojlim_n \mathcal{O}_X/\mathcal{I}(Y)^n$  to  $Y$ , then  $(Y, \mathcal{O}_Y^\wedge)$  is a formal scheme (27.2.5.5), called the **formal completion** of  $X$  along  $Y$ .  $\lrcorner$

**Def. (27.2.5.8) [Properties of Morphisms between Formal Schemes].** A morphism  $f : \mathfrak{X} \rightarrow \mathfrak{Y}$  between formal schemes is called **finite type/étale/smooth/etc.** if for each  $Z \in \text{Sch}$ , the fiber product  $\mathfrak{X} \times_{\mathfrak{Y}} Z \in \text{Sch}$ , and  $\mathfrak{X} \times_{\mathfrak{Y}} Z \rightarrow Z$  satisfies the desired property.  $\lrcorner$

**Def. (27.2.5.9) [Underlying Reduced Schemes].** For each formal scheme  $\mathfrak{X}$ , there exists a unique reduced scheme  $\mathfrak{X}_{\text{red}} \in \text{Sch}_{\text{red}}$  with a morphism of local-topological-ringed spaces  $\mathfrak{X}_{\text{red}} \rightarrow \mathfrak{X}$  s.t. for each  $Z \in \text{Sch}_{\text{red}}$ , the natural map  $\text{Hom}(Z, \mathfrak{X}_{\text{red}}) \rightarrow \text{Hom}(Z, \mathfrak{X})$  is a bijection.  $\lrcorner$

*Proof:* ?  $\square$

**Def. (27.2.5.10) [Formally Locally Finite Type Morphisms].** A morphism  $f : \mathfrak{X} \rightarrow \mathfrak{Y}$  between formal schemes is called **formally of finite type**(resp. **formally locally of finite type**) if the morphism  $f_{\text{red}} : \mathfrak{X}_{\text{red}} \rightarrow \mathfrak{Y}_{\text{red}} \in \text{Sch}$  is of finite type(resp. locally of finite type).  $\lrcorner$

### Noetherian Adic Formal Schemes

Cf.[Hartshorne Chap2.9] and [Sta]30.23. and Illusie.

**Def. (27.2.5.11) [Noetherian Formal Adic Schemes].**  $\lrcorner$

**Def. (27.2.5.12) [Coherent Sheaves on Noetherian Formal Adic Schemes].**  $\lrcorner$

**Prop. (27.2.5.13).** Let  $A$  be a Noetherian ring and  $I$  an ideal, let  $X$  be a proper scheme over  $A$  and  $\mathcal{F}$  a coherent sheaf on  $X$ , then for any  $p \geq 0$ , the inverse systems  $(H^p(X, I^n \mathcal{F}))$  and  $(I^n H^p(X, \mathcal{F}))$  are isomorphic pro- $A$ -modules.  $\lrcorner$

*Proof:* Cf.[Sta]02OA.  $\square$

**Thm. (27.2.5.14) [Theorem of Formal Functions].** Let  $A$  be a Noetherian ring with an ideal  $I$ ,  $X$  be a proper scheme over  $A$  and  $\mathcal{F}$  a coherent scheme on  $X$ . Then for any  $p \geq 0$ , the system of maps

$$H^p(X, \mathcal{F})/I^n H^p(X, \mathcal{F}) \rightarrow H^p(X, \mathcal{F}/I^n \mathcal{F})$$

induce an isomorphism of limits

$$H^p(X, \mathcal{F})^\wedge \cong \varprojlim_n H^p(X, \mathcal{F}/I^n \mathcal{F}).$$

*Proof:* Cf.[Sta]02OC.  $\square$

**Prop. (27.2.5.15) [Grothendieck's Existence Theorem].** Cf.[Sta]0883.  $\lrcorner$

**Prop. (27.2.5.16) [Grothendieck's Algebrization Theorem].** Cf.[Sta]089A.  $\lrcorner$

### Admissible Formal Schemes

**Remark (27.2.5.17) [Setup].** For a good theory of admissible formal schemes, let the base ring  $R$  be an adic ring with an ideal of definition  $I$  that satisfies either one of the following situation:

- $R$  is an adic valuation ring with a principal ideal of definition.
- $R$  is Noetherian and has no  $I$ -torsion.

$\lrcorner$

**Def. (27.2.5.18)** [Underlying Reduced Schemes]. ┘

**Def. (27.2.5.19)** [Admissible Formal Schemes]. Let  $R$  be an adic ring, then a formal  $R$ -scheme  $X$  is called **locally formally of finite type/finite presentation/admissible** if there is an open affine covering  $\text{Spf } A_i$  of  $X$  that  $A_i$  satisfies those properties (12.2.1.16).

$X$  is called **formally of finite type** if it is locally of topologically finite type and quasi-compact. It is called **formally of finite presentation** if it is locally of topologically finite presentation, quasi-compact and quasi-separated. ┘

**Prop. (27.2.5.20)**. A map  $f : X \rightarrow Y$  of  $R$ -formal schemes is locally formally of finite type (27.2.5.19) iff  $f_{\text{red}} : X_{\text{red}} \rightarrow Y_{\text{red}}$  (27.2.5.18) is locally of finite type. ┘

*Proof:* ? □

**Prop. (27.2.5.21)** [Induced Admissible Formal Subscheme]. Let  $X$  be a formal  $R$ -scheme that is locally of topologically finite type, and let  $\mathcal{O}_X$  be its structure sheaf. Then we can look at the ideal sheaf  $\mathcal{I} \subset \mathcal{O}_X$  consisting of all elements locally killed by a power of  $I^n$ . This is a qco sheaf, as  $\mathcal{I}(U) = \{f \in \mathcal{O}(U) \mid I^n f = 0 \text{ for some } n\}$ , because the quotient by the RHS is locally topologically of finite type and has no  $I$ -torsion, thus admissible, by (12.2.1.18), and admissibility is local (12.2.1.19).

In particular we can take the closed subscheme  $X_{\text{adm}} \subset X$  corresponding to  $\mathcal{I}$ , then it is an admissible formal scheme, called the **induced admissible formal subscheme** of  $X$ . ┘

**Prop. (27.2.5.22)**. ┘

**Def. (27.2.5.23)** [Admissible Formal Blowing-up]. ┘

### Formal Models of Rigid Analytic Spaces

**Notation (27.2.5.24)**.

- Let  $(\mathcal{O}_K, K)$  be a complete valuation field of height 1. ┘

**Prop. (27.2.5.25)** [Raynaud's Generic Fiber Functor]. The functor  $A \mapsto A \otimes_{\mathcal{O}_K} K$  from the category of  $\mathcal{O}_K$ -algebras topologically of finite type to the category of affinoid  $K$ -algebras ? extends to a functor

$$(-)^{\text{rig}} : \mathfrak{X} \rightarrow \mathfrak{X}^{\text{rig}}$$

from the category of admissible formal schemes over  $\mathcal{O}_K$  to  $\text{Rig Spa}/K$ .  $\mathfrak{X}^{\text{rig}}$  is called the **Raynaud generic fiber** of  $\mathfrak{X}$ . ┘

*Proof:* Cf. [Bos15]P174. ? □

**Def. (27.2.5.26)** [Formal Models]. In view of (27.2.5.25), one would like to describe all formal  $\mathcal{O}_K$ -schemes s.t. the generic fiber  $\mathfrak{X}^{\text{rig}}$  is isomorphic to a given rigid  $K$ -space  $\mathfrak{X}_K$ . But first notice  $A \mapsto A \otimes_{\mathcal{O}_K} K$  kills all  $\mathcal{O}_K$ -torsion, in particular the generic fiber functor only depends on the induced admissible formal scheme  $\mathfrak{X}_{\text{adm}}$  (27.2.5.21).

So given  $\mathfrak{X} \in \text{Rig Spa}/K$ , any admissible formal  $\mathcal{O}_K$ -scheme  $\mathfrak{X}$  satisfying  $\mathfrak{X}^{\text{rig}} \cong \mathfrak{X}_K$  is called a **formal  $R$ -model** of  $\mathfrak{X}_K$ . ┘

**Prop. (27.2.5.27)** [Raynaud(1972)]. The Raynaud's generic fiber functor maps admissible formal blowing-ups of admissible formal schemes (27.2.5.23) to isomorphisms of rigid analytic spaces, and it induces an equivalence of categories between:

- The category of quasi-compact admissible formal  $\mathcal{O}_K$ -schemes (27.2.5.19), localized by admissible formal blowing-ups.
- $\text{Rig Spa}^{\text{qcqs}}/K$ .

⌋

*Proof:* ?

□

### Functorial Viewpoint

**Prop. (27.2.5.28) [Functors].** Let  $(K, \mathcal{O}_K, \varpi_K) \in \text{CDVR}$ , then a formal  $\mathcal{O}_K$ -scheme is determined by the functor it determines on  $\text{NilSch}/\mathcal{O}_K$  (the category of  $\mathcal{O}_K$ -schemes s.t.  $\varpi_K$  is locally nilpotent). ⌋

*Proof:* ?

□

## 6 Berkovich's Rigid Spaces

References are [Berkovich, Spectral theory and analytic geometry over non-Archimedean fields].

**Def. (27.2.6.1) [Berkovich's Spectra].** For a Banach ring  $A$ , the **Berkovich spectrum**  $M(A)$  is defined to be the set of all bounded multiplicative semi-norms on  $A$  (19.1.1.3) (including Archimedean norms), endowed with the weakest topology s.t. the functions  $|\cdot| \mapsto |f|$  is continuous for any  $f \in A$ . ⌋

**Def. (27.2.6.2) [Comparing Affinoid Algebras and Berkovich's Spectra].** If  $K \in p\text{-Field}$  and  $A$  is an affinoid Tate  $K$ -algebra, then  $\text{Spm}(A)$  with the canonical topology (27.2.1.4) is naturally a subspace of  $M(A)$  (27.2.6.1), by identifying  $\mathfrak{m} \in \text{Spm}(A)$  with the unique bounded multiplicative semi-norm on  $A$  with kernel  $\mathfrak{m}$ , because  $\dim_K A/\mathfrak{m} < \infty$  (12.2.4.11). In this way,  $\text{Spm}(A)$  is identified with set of bounded multiplicative semi-norms  $|\cdot|$  on  $A$  s.t.  $\dim_K A/\ker(|\cdot|) < \infty$ . ⌋

**Def. (27.2.6.3) [Berkovich's Spaces].** The Berkovich spectrum  $M(A)$  of a Banach ring (27.2.6.1) is endowed with affinoid subdomains as in rigid affinoid cases, and a **Berkovich's analytic space** is a locally ringed  $G$ -spaces that are locally isomorphic to Berkovich spectra of Banach rings. ⌋

**Def. (27.2.6.4) [Strict Berkovich Spaces].** A **strict Berkovich space** is a Berkovich space (27.2.6.3). ⌋

**Example (27.2.6.5) [Berkovich's Affine Spaces].** For  $K \in p\text{-Field}$ , the **Berkovich's analytic affine space** is defined to be

$$\mathbb{A}_K^n = \{\text{multiplicative seminorms on } K[T_1, \dots, T_n] \text{ whose restriction to } K \text{ is bounded}\}$$

together with the weakest topology defined by  $f \in K[T_1, \dots, T_n]$ .

Then it is a strict Berkovich space (27.2.6.4) as an increasing union of open balls, i.e Berkovich's version of affinoid Tate spaces. ⌋

**Prop. (27.2.6.6).** For  $K \in p\text{-Field}$ ,

- Any Berkovich spaces over  $K$  are arcwise-connected.
- If  $X = M(A)$  where  $A$  is an affinoid  $K$ -algebra, then  $\dim |X| = \dim A$ .
- Smooth Berkovich spaces are locally contractible.

⌋

*Proof:*    ? □

**Prop. (27.2.6.7)[Strict Berkovich Analytic Spaces as Rigid Analytic Spaces].** For  $K \in p\text{-Field}$ , there is a functor from the category of separated strict Berkovich spaces over  $K$  to the category of separated rigid analytic spaces that

$$X \mapsto X_0 = \{x \in X \mid [\kappa(x) : K] < \infty\}.$$

And this functor is fully faithful, and preserves tensor products. ┘

*Proof:*    ? □

## 27.3 Adic Spaces

References are [Hub93], [Hub96], [Mor19], [Bha17], [Wed14], [S-W20], [Sch12].

**Notation (27.3.0.1).**

- Use notations as in [Topological Commutative Algebra\(Adic Rings\)](#).

┘

### 1 (Continuous)Valuation Spectrums

References are [Mor19].

**Def. (27.3.1.1) [Riemann-Zariski Space].** Let  $K \in \mathbf{Field}$  and  $A \subset K \in \mathbf{CRing}$ , the Riemann-Zariski space  $RZ(K, A)$  is defined to be the space of all valuation subrings of  $K$  containing  $A$  with the topology generated by

$$U(x_1, \dots, x_n) = \{P \in RZ(K, A) \mid x_1, \dots, x_n \in P\}.$$

$RZ(K, 0)$  is also denoted by  $RZ(K)$ .  $RZ(K, A)$  is just isomorphic to  $\mathrm{Spa}(K, A^{itc})$ , so it is spectral by (27.3.2.4). ┘

**Cor. (27.3.1.2).** Clearly the specialization relations of  $RZ(K, A)$  is identical to inclusion relations. ┘

**Def. (27.3.1.3) [Valuation Spectrum].** For  $A \in \mathbf{CRing}$ , the **valuation spectrum**  $\mathrm{Spv}(A) \in \mathbf{Top}$  is the set of equivalent classes of valuations on  $A$ , topologized by the open subsets

$$\mathrm{Spv}(A)(\frac{f}{g}) = \{x \in \mathrm{Spa}(A) \mid |f(x)| \leq |g(x)| \neq 0\}.$$

$\mathrm{Spv}(A)$  is spectral, with sub-basis generated by  $\mathrm{Spv}(\frac{f}{g})$

There is a kernel map  $\ker : \mathrm{Spv}(A) \rightarrow \mathrm{Spec} A$  sending a valuation to its kernel (support). Then this map is continuous, and the fiber of this map over  $\mathfrak{p}$  is just isomorphic to the Riemann-Zariski space  $RZ(k(\mathfrak{p}))$ .

Moreover, the map  $\ker : \mathrm{Spv}(A) \rightarrow \mathrm{Spec} A$  is spectral, as the kernel of  $D(f)$  is  $U(\frac{f}{f})$ . ┘

#### Specialization Relations in $\mathrm{Spv}(A)$

**Def. (27.3.1.4) [Vertical Specializations].** Let  $x, y \in \mathrm{Spv}(A)$ . We say that  $x$  is a vertical specialization of  $y$  if  $x$  is a specialization of  $y$  and  $\mathfrak{p}_x = \mathfrak{p}_y$ . ┘

#### Valuations with Support Conditions

**Def. (27.3.1.5) [ $\mathrm{Spv}(A, J)$ ].** We define a space  $\mathrm{Spv}(A, J) \subset \mathrm{Spv}(A)$  by

$$\mathrm{Spv}(A, J) = \{x \in \mathrm{Spv}(A) \mid r(x) = x\} = \{x \in \mathrm{Spv}(A) \mid c\Gamma_x(J) = \Gamma\}.$$

with the subset topology from  $\mathrm{Spv}(A)$ . ┘

**Prop. (27.3.1.6).**

- $\mathrm{Spv}(A, J)$  is a spectral space.

- A basis of quasi-compact open subsets for the topology is given by the sets  $U(\frac{f_1, \dots, f_n}{g})$  where  $J \subset \sqrt{(f_1, \dots, f_n)}$ .
- The retraction  $r : \mathrm{Spv}(A) \rightarrow \mathrm{Spv}(A, J)$  is a spectral map.

┘

*Proof:* Cf.[Mor19]P52.

□

### Continuous Valuation Spectrum

**Def. (27.3.1.7) [Continuous Valuations].** For  $A \in \mathrm{HubCRing}$ , define  $\mathrm{Cont}(A) \in \mathcal{T}\mathrm{op}$  as the subspace of  $\mathrm{Spv}(A)$ (27.3.1.3) consisting of continuous valuations on  $A$ . Then

$$\mathrm{Cont}(A) = \{x \in \mathrm{Spv}(A, A^{00}A) \mid x(A^{00}) < 1\}$$

┘

*Proof:* Cf.[Mor19]P64.

□

**Cor. (27.3.1.8) [Cont(A) is Spectral].** For any Huber ring  $A$ ,  $\mathrm{Cont}(A)$  is a spectral space, with a basis of quasi-compact open subsets given by  $U(\frac{f_1, \dots, f_n}{g})$ , where  $A^{00} \subset \sqrt{(f_1, \dots, f_n)}$ , or equivalently  $(f_1, \dots, f_n)$  is open??.

┘

*Proof:* This is because

$$\mathrm{Cont}(A) = \mathrm{Spv}(A, A^{00}A) - \bigcup_{g \in A^{00}} U(\frac{1}{g}),$$

which is an open subset of  $\mathrm{Spv}(A, A^{00}A)$ , thus the assertion follows from(27.3.1.6) and(4.12.3.9). □

## 2 Adic Spectra

**Def. (27.3.2.1) [Adic Spectra].** For any  $(A, A^+) \in \mathrm{HubPair}$ (12.2.5.13), the **adic spectrum** of  $(A, A^+)$  is defined to be

$$\mathrm{Spa}(A, A^+) = \{x \in \mathrm{Cont}(A) \mid x(A^+) \leq 1\}.$$

And for  $A \in \mathrm{HubCRing}$ , denote  $\mathrm{Spa} A = \mathrm{Spa}(A, A^0)$ , where  $A^0$  is the ring of power-bounded elements(12.2.5.8).

The shape of these open sets is dictated by the desired properties that both  $\{x \mid f(x) \neq 0\}$  and  $\{x \mid f(x) \leq 1\}$  be open. These desiderata combine features of classical algebraic geometry and rigid geometry, respectively.

This adic spectrum construction(27.3.2.1) defines a functor

$$\mathrm{Spa} : \mathrm{HubPair}^{\mathrm{op}} \rightarrow \mathcal{T}\mathrm{op}.$$

And for any ring of integers  $A^+$  of  $A$ ,  $\mathrm{Spa} A = \mathrm{Spa}(A, A^0) \hookrightarrow \mathrm{Spa}(A, A^+)$  is an immersion of spaces, by(12.2.5.13). ┘

**Def. (27.3.2.2) [Kernel map].** Taking kernels of valuations gives a map  $\ker : \mathrm{Spa}(A, A^+) \rightarrow \mathrm{Spec} A$ . This map is continuous, as the inverse image of  $D(f)$  is  $\mathrm{Spa}(A, A^+)(\frac{f}{f})$ . We call a subset a **Zariski open subset** of  $\mathrm{Spa}(A, A^+)$  iff it is open in the initial topology along  $\ker$ . ┘

**Def. (27.3.2.3) [Rational Subsets].** A rational subset of  $\mathrm{Spa}(A, A^+)$  is defined to be

$$\mathrm{Spa}(A, A^+)(\frac{f_1, \dots, f_n}{g}) = \{x \in \mathrm{Spa}(A, A^+) | x(f_i) \leq x(g)\},$$

where  $(f_i)$  is an open ideal.  $\lrcorner$

**Prop. (27.3.2.4) [Adic Spectra are Spectral].** The adic spectrum  $\mathrm{Spa}(A, A^+)$  is spectral, and the rational subsets (27.3.2.3) form a basis consisting of quasi-compact open subsets.  $\lrcorner$

*Proof:* Firstly  $\mathrm{Spa}(A, A^+)$  is closed in the constructible topology of  $\mathrm{Cont}(A)$ : for any  $a \in A$ ,

$$\{x \in \mathrm{Cont}(A) | |a|_x \leq 1\} = U(\frac{1, a}{1})$$

is a quasi-compact open subset of  $\mathrm{Cont}(A)$ , so constructible, thus  $\mathrm{Spa}(A, A^+) = \cap_{a \in A^+} \{x \in \mathrm{Cont}(A) | |a|_x \leq 1\}$  is closed in the constructible topology.

So the assertions follow from (4.12.3.9) and (27.3.1.8).  $\square$

**Remark (27.3.2.5).** In spite of the proposition that adic spectra are spectral, it is very different from classical algebraic geometry. For example, the generalizations of a point  $y$  is totally ordered (localization of the valuation ring), but this nearly never happen for an affine variety.  $\lrcorner$

**Remark (27.3.2.6).** The rational subsets are stable under intersection, and form a basis for the topology of  $\mathrm{Spa}(A, A^+)$ . But in general  $\mathrm{Spa}(\frac{f}{g})$  is not quasi-compact, in particular,  $\ker : \mathrm{Spa}(A, A^+) \rightarrow \mathrm{Spec} A$  is not quasi-compact.  $\lrcorner$

**Cor. (27.3.2.7) [Detecting Nilpotence Locally].** If  $(A, A^+)$  is an affinoid Tate ring and  $f \in A$ , then  $f \in A^{00}$  iff  $|f(x)|^n \rightarrow 0$  for all  $x \in \mathrm{Spa}(A, A^+)$ .  $\lrcorner$

*Proof:* If  $f$  is topological nilpotent, then  $f^N \in tA^+$  for some  $n$ , so  $|f(x)|^{nN} \leq |t(x)|^n \rightarrow 0$  because  $x$  is continuous. Conversely, if  $|f(x)|^n \rightarrow 0$  for all  $x$ , then  $X = \cup_n X(\frac{f^n}{t})$ . But  $X$  is quasi-compact, so  $|f(x)|^n \leq |t(x)|$  for all  $x$ , for some  $n$ . So by (27.3.2.17)  $f^n \in tA^+$ . Now  $A^+$  is a filtered colimits of rings of definitions (12.2.5.14), so  $f^n \in tA_0$  for some  $tA_0$ , which shows that  $f \in A^{00}$ .  $\square$

**Def. (27.3.2.8) [Specialization Map].** The specialization map

$$\mathrm{Sp} : \mathrm{Spa}(A, A^+) \rightarrow \mathrm{Spec}(A^+/A^{00})$$

that maps a point  $x$  to the inverse image of the maximal ideal of  $R_x$  along the valuation map  $A^+ \rightarrow R_x$  it corresponds. It clearly lies in  $\mathrm{Spec}(A^+/A^{00})$  as any pseudo-uniformizer is mapped to a pseudo-uniformizer in  $R_x$  thus in the maximal ideal.

This map is continuous and spectral, unlike that the kernel map (27.3.2.6): the inverse image of a  $D(f)$  for  $f \in A^+$  is the set of points  $x \in \mathrm{Spa}(A, A^+)$  that  $x(f)$  is a unit, i.e.  $|x(f)| = 1$ . As  $|x(f)| \leq 1$  for all  $f \in A^+$ , this set is just  $\mathrm{Spa}(A, A^+)(\frac{1}{f})$ , so specialization map  $\mathrm{sp}$  is both continuous and spectral.  $\lrcorner$

### Completed Residue Fields

**Def. (27.3.2.9) [Completed Residue Fields].** Let  $x \in \mathrm{Spa}(A, A^+)$ , then we denote by  $k(x)$  the fraction field of  $A/\mathfrak{p}_x$ , with a valuation ring  $k^+(x)$ . If  $x$  is not analytic, we set  $\widehat{\kappa}(x) = \kappa(x)$ ,  $\widehat{\kappa}^+(x) = \widehat{k}^+(x)$ . If  $x$  is analytic, we set  $\widehat{\kappa}(x) = \kappa(x)^\wedge$ ,  $\widehat{\kappa}^+(x) = \kappa^+(x)^\wedge$ .  $(\widehat{\kappa}(x), \widehat{\kappa}^+(x))$  is called the **completed residue field** of  $x$ .  $\lrcorner$



**Prop. (27.3.2.10)[Vertical Generalizations].** The morphism

$$\mathrm{Spa}(\kappa(x), \kappa(x)^+) \rightarrow \mathrm{Spa}(A, A^+)$$

induces a homeomorphism onto the set of vertical generalizations of  $x$ .

$x$  is analytic iff  $\kappa(x)$  is microbial.  $\lrcorner$

*Proof:* For the first assertion, if  $x$  is not analytic, then  $\mathrm{Spa}(k(x), k(x)^+) \cong RZ(k(x), k(x)^+)$  is homeomorphic to the vertical generalizations of  $x$ .

If  $x$  is analytic, then by (27.3.2.17),  $\mathrm{Spa}(\kappa(x), \kappa(x)^+) \rightarrow \mathrm{Spa}(k(x), k(x)^+)$  is a homeomorphism. And now we need to check more that if  $R \in RZ(k(x), k(x)^+)$  corresponds to a vertical generalization  $y$ , then  $|\cdot|_R$  is continuous iff  $|\cdot|_y$  is continuous. For this, see [Mor19]107.

The second assertion follows from (12.2.6.7).  $\square$

**Prop. (27.3.2.11)[Residue of Rational Subsets].** Let  $f : (A, A^+) \rightarrow (A \langle \frac{T}{s} \rangle, A \langle \frac{T}{s} \rangle^+)$  be a rational localization (27.3.2.3), and  $y \in \mathrm{Spa}(A \langle \frac{T}{s} \rangle, A \langle \frac{T}{s} \rangle^+)$  with  $x = \mathrm{Spa}(f)(y)$ , then the canonical map  $(k(x), k(x)^+) \rightarrow (k(y), k(y)^+)$  induces an isomorphism of Huber pairs  $(\kappa(x), \kappa(x)^+) \cong (\kappa(y), \kappa(y)^+)$  after completion.  $\lrcorner$

*Proof:* Cf. Morel P108.  $\color{red}{?}$   $\square$

**Def. (27.3.2.12)[Adic Points].** An **adic point** is an adic space  $\mathrm{Spa}(K, K^+)$  where

- $(K, K^+) \in \mathrm{HubPair}$ ,
- $K \in \mathbf{Field}$ , and  $K$  is either complete non-Archimedean or discrete,
- $K^+$  is an open and bounded valuation subring (hence integrally closed).

An **analytic adic point** is an adic point s.t.  $K$  is complete non-Archimedean.  $\lrcorner$

**Remark (27.3.2.13).** The adic point is not a point in general. In fact, if  $K$  is non-Archimedean,  $\mathrm{Spa}(K, K^+)$  is totally ordered by inclusion  $\color{red}{?}$ , with a unique closed point corresponding to  $K^+$  and a unique generic point corresponding to  $\mathcal{O}_K$ .  $\lrcorner$

**Prop. (27.3.2.14)[Valuation Ring Characterization of Spa].** For  $(A, A^+) \in \mathrm{HubPair}$ , there is a natural bijection between  $\mathrm{Spa}(A, A^+)$  and the set of maps  $\varphi : (A, A^+) \rightarrow (K, K^+)$  that  $\mathrm{Spa}(K, K^+)$  is an adic point where  $K_x = \kappa(x)$  and  $x$  corresponds to the image of the closed point of  $\mathrm{Spa}(K, K^+)$  under the map  $\mathrm{Spa}(\varphi)$ . And  $x$  is analytic iff the corresponding adic point  $\mathrm{Spa}(K, K^+)$  is analytic.  $\lrcorner$

*Proof:* Let  $\varphi : (A, A^+) \rightarrow (K, K^+)$  be a map of Huber pairs, then correspond the maps gives a continuous valuation on  $A$  which is in  $\mathrm{Spa}(A, A^+)$  and  $\mathfrak{p}_x = \ker(\varphi)$ . Notice  $x$  is the image of the closed point of  $\mathrm{Spa}(K, K^+)$ . And we get a map  $k(x) \rightarrow K$  such that  $k(x)^+ = k(x) \cap K^+$ , and that has dense image by assumption, so after completion (when non-Archimedean) induces an isomorphism  $(\kappa(x), \kappa(x)^+) \cong (K, K^+)$ . Thus this is a bijection of sets.  $\square$

**Prop. (27.3.2.15)[Uniformity].** If  $A$  is uniform, then the map

$$A \rightarrow \prod_{x \in |\mathrm{Spa}(A, A^+)|} \kappa(x)$$

is a homeomorphism of  $A$  onto its image, where  $\kappa(x)$  is the completed residue field (27.3.2.9).  $\lrcorner$

*Proof:* [Berkovich, Étale cohomology for non-Archimedean analytic spaces].  $\color{red}{?}$   $\square$

**Cor. (27.3.2.16).** Let  $\tilde{\mathcal{O}}_X$  be the sheafification of  $\mathcal{O}_X$ , then if  $A$  is uniform,  $A \rightarrow H^0(X, \tilde{\mathcal{O}}_X)$  is injective.  $\lrcorner$

*Proof:* In fact,  $A \rightarrow H^0(X, \tilde{\mathcal{O}}_X) \rightarrow \prod_x \kappa(x)$  is injective.  $\square$

### Properties of Adic Spectrums

**Prop. (27.3.2.17) [Properties of Adic Spectrums].**

- The completion map  $(A, A^+) \rightarrow (\hat{A}, \hat{A}^+)$  induces an homeomorphism on the adic spectrums that preserves rational subsets.
- $\text{Spa}(A, A^+)$  vanishes iff its completion  $\hat{A}$  vanishes.
- (Adic Nullstellensatz)  $A^+ = \{f \in A \mid |x(f)| \leq 1, \forall x \in \text{Spa}(A, A^+)\}$ .
- If  $A$  is complete, then  $f \in A$  is a unit iff  $|f|_x \neq 0$  for all  $x \in \text{Spa}(A, A^+)$ .
- If  $A$  is Tate, then  $f$  is topologically nilpotent iff  $|f|_x^n \rightarrow 0$  for any  $x \in \text{Spa}(A, A^+)$ .

┘

*Proof:* 1: Use the valuation ring characterization (27.3.2.14), a point of  $x$  determined a continuous map  $(A, A^+) \rightarrow (K, K^+)$ . Now this extends to a map under completion, thus determines a point of  $\text{Spa}(\hat{A}, \hat{A}^+)$ , so the Spa map is surjective. And injectivity follows from the fact  $A$  is dense in  $\hat{A}$ .

For the homeomorphism, just notice that if  $f_i - f'_i, g - g' \in t^{N+1}\hat{A}$ , then

$$\text{Spa}\left(\frac{f_1, \dots, f_n}{g}\right) = \text{Spa}\left(\frac{f'_1, \dots, f'_n}{g'}\right).$$

where  $f_n = t^N$ . So now  $A$  is dense in  $\hat{A}$ , if we choose  $f_i, g \in A$ , then this rational subset is clearly induced from  $A$ .

2: Cf. [Mor19]P104.

3: Cf. [Mor19]P91.

4: Cf. [Mor19]P106.

5: Cf. [Mor19]P106.

□

**Cor. (27.3.2.18) [Generalizations in Adic Spectrum].** The above proposition shows that the generalization relations of Spa are easily determined, for an element  $y$ , all generalizations of  $y$  are in bijection with  $\text{Spec}(R_y/(t))$  as a poset, thus totally ordered, and each  $y$  has a unique generic point as generalization, because it is microbial.

Moreover,  $\text{Spa}(A, A^0)$  is closed under generalization in  $\text{Spa}(A, A^+)$ , and they have the same set of generic points.

┘

*Proof:* The last assertion is because the generalizations of a point  $y$  is just valuation rings containing  $R_y$ , and  $R_y$  contains  $A^0/\mathfrak{p}_y$ , so does its generalizations. And for any generic point  $x \in \text{Spa}(A, A^+)$ ,  $A^0$  is mapped to the valuation ring  $R_x$ , because it is a rank 1 valuation, so if  $t^k f^{\mathbb{N}} \subset R_x$ , then  $f \in R_x$  because otherwise we have  $|t| < |f^{-n}|$  for  $n$  large.

□

**Def. (27.3.2.19) [Specialization Map].** The specialization map

$$\text{Sp} : \text{Spa}(A, A^+) \rightarrow \text{Spec}(A^+/A^{00})$$

that maps a point  $x$  to the inverse image of the maximal ideal of  $R_x$  along the valuation map  $A^+ \rightarrow R_x$  it corresponds. It clearly lies in  $\text{Spec}(A^+/A^{00})$  as any pseudo-uniformizer is mapped to a pseudo-uniformizer in  $R_x$  thus in the maximal ideal.

This map is continuous and spectral, unlike that the kernel map (27.3.2.6): the inverse image of a  $D(f)$  for  $f \in A^+$  is the set of points  $x \in \text{Spa}(A, A^+)$  that  $x(f)$  is a unit, i.e.  $|x(f)| = 1$ . As  $|x(f)| \leq 1$  for all  $f \in A^+$ , this set is just  $\text{Spa}(A, A^+)(\frac{1}{f})$ , so specialization map  $\text{sp}$  is both continuous and spectral.

┘

**Prop. (27.3.2.20) [Maximal Hausdorff Quoteint].** Let  $X = \mathrm{Spa}(A, A^+)$  be an affinoid Tate space, the if  $\overline{X}$  is the quotient of  $X$  by the equivalence relation generated by specialization, then  $\overline{X}$  is the Hausdorffization of  $X$ , i.e.  $\overline{X}$  is Hausdorff.  $\lrcorner$

*Proof:* To show  $\overline{X}$  is Hausdorff, if  $x, y \in X$  is not mapped to the same point in  $\overline{X}$ , then by (27.3.2.18), we may assume  $x, y$  is generic in  $X$ , and  $\overline{\{x\}} \cap \overline{\{y\}} = \emptyset$ . Now we must find two disjoint open subsets of  $x, y$  that is stable under specialization. Cf. [Bhatt Perfectoid Spaces P75].  $\square$

### Constructions of Adic Spectrums

**Prop. (27.3.2.21) [Direct Limits of Uniform Affinoids].** The direct limits exists in the category of uniform affinoid Tate rings. and  $A^+ = \mathrm{colim} A_i^+$ .

Moreover,

$$|\mathrm{Spa}(A, A^+)| \cong \varinjlim_i |\mathrm{Spa}(A_i, A_i^+)|$$

as a homeomorphism of spectral spaces, and each rational subset of  $\mathrm{Spa}(A, A^+)$  is pulled back from some rational subset of  $\mathrm{Spa}(A_i, A_i^+)$ .

The same conclusion also hold in the category of complete uniform affinoid Tate rings (For the homeomorphism, (27.3.2.17) is used).  $\lrcorner$

*Proof:* Suppose the colimit index has a minimal element  $i_0$ , let  $t$  be a pseudo-uniformizer, then each  $A_i^+$  is a ring of definition with pseudo-uniformizer  $t$ . Now we set  $A = \mathrm{colim} A_i$  with ring of definitions  $A^+ = \mathrm{colim} A_i^+$ , then  $A^+$  is integrally closed in  $A$ , thus  $(A, A^+)$  is truly a uniform affinoid Tate ring. Now we check it is the colimit: For any compatible map  $(A_i, A_i^+) \rightarrow (B, B^+)$ , there is a map  $f : (A, A^+) \rightarrow (B, B^+)$  as abstract rings. We check it is continuous: we may assume  $B^+$  is the ring of definition, then  $t^n A^+ \subset f^{-1}(t^n B^+)$ , thus it is continuous.

For the adic spectrum, now a point  $x \in \mathrm{Spa}(A, A^+)$  is determined by the map of uniform affinoid Tate rings  $(A, A^+) \rightarrow (k(\mathfrak{p}), R_x)$ , and by the universal property, it is defined by a compatible set of maps  $(A_i, A_i^+) \rightarrow (k(\mathfrak{p}), R_x)$ . Now it is easy to see the desired bijection of topological spaces, as the elements defining rational subsets are pullbacks from some  $A_i$ .  $\square$

**Prop. (27.3.2.22) [Perfection of Adic Spectrum].** Let  $(A, A^+)$  be an affinoid Tate ring of char  $p$ , then

- The Frobenius map induces a homeomorphism on the adic spectrum of  $(A, A^+)$ .
- If  $(A, A^+)$  is uniform, then there is a perfection functor, which is left adjoint to the forgetful functor from the category of perfect uniform affinoid Tate rings to the category of affinoid Tate rings. And it is just  $(A_{\mathrm{perf}}, A_{\mathrm{perf}}^+)$ .
- The natural map  $(A, A^+) \rightarrow (A_{\mathrm{perf}}, A_{\mathrm{perf}}^+)$  induces a homeomorphism on the adic spectrum.  $\lrcorner$

*Proof:* 1: The Frobenius pulls a valuation a multiple of itself, thus equals itself.

2: Clearly  $A_{\mathrm{perf}}^+$  is integrally closed in  $A^+$  and is in  $A_{\mathrm{perf}}^0$ . It suffices to show  $(A_{\mathrm{perf}}, A_{\mathrm{perf}}^+)$  is uniform, but this is because  $A_{\mathrm{perf}}^0 = (A_{\mathrm{perf}})^0 \subset (t^{-n} A_0)_{\mathrm{perf}} = t^{-n} (A_0)_{\mathrm{perf}}$ .

3: The Spa map is checked to be continuous and injective, for the converse, using the microbial valuation ring characterization, a point  $x$  of  $\mathrm{Spa}(A, A^+)$  corresponds to a map  $A^+$  to a complete field  $k^+(x)$ , thus a map  $A_{\mathrm{perf}}^+ \rightarrow k^+(x)_{\mathrm{perf}}$ , now  $k^+(x) \rightarrow k^+(x)_{\mathrm{perf}}$  is faithfully flat that preserves pseudo-uniformizers, thus it is a point  $y$  that maps to  $x$ .  $\square$

### 3 Affinoid Adic Spaces

**Lemma (27.3.3.1) [Functions on Rational Subsets].** If  $X = \mathrm{Spa}(A, A^+)$  is an affinoid adic space, and  $U$  is a rational subset, then there is a unique complete affinoid Tate ring  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  over  $(A, A^+)$  s.t. the map of spectra

$$\mathrm{Spa}(\mathcal{O}_X(U), \mathcal{O}_X^+(U)) \rightarrow \mathrm{Spa}(A, A^+)$$

is universal w.r.t. all the complete affinoid Tate space with a map to  $X$  that has image in  $U$ .

And in this case, this Spa map is a homeomorphism identifying the rational subsets contained in  $U$  to rational subsets of  $\mathrm{Spa}(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$ . In particular,  $U$  is quasi-compact.  $\lrcorner$

*Proof:* ? See Hub94 P1.3 for the proof in the Huber ring case.

Choose a ring of definition  $(A_0, t)$ , and  $U = \mathrm{Spa}(A, A^+)(\frac{f_1, \dots, f_n}{g})$  for  $f_i, g \in A_0$ , and  $f_n = t^{??}$ , and let  $B = A[g^{-1}]$  and  $B_0 = A_0[\frac{f_i}{g}]$ . Then  $B = B_0[t^{-1}]$  (notice that  $A_0[t^{-1}] = A$ ). So  $B$  is a Tate  $A$ -algebra with ring of definition  $(B_0, t)$ . Now if  $B^+$  is the integral closure of the subring of  $B$  generated by  $A^+[\frac{f_i}{g}]$ , then  $(B, B^+)$  is an affinoid Tate ring. Set  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  to be its completion.

By construction  $\mathrm{Spa}(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  maps into  $U$ , because  $x(g) \neq 0$  because  $g$  is a unit, and  $|x(f_i)| \leq |x(g)|$  as  $f_i/g \in B^+$ .

Now check universal property: if  $\mathrm{Spa}(C, C^+)$  maps into  $U$ , then  $g$  is a unit in  $C$  by (27.3.2.17), and then  $f_i/g \in C^+$  by (27.3.2.17) again. Now  $C^0$  is the filtered colimit of all rings of definition, so there is a ring of definition  $C_0$  that contains  $A_0$  and all  $f_i/g$  ((12.2.5.17) is used). Then this gives a map of affinoid Tate rings that maps  $B_0$  into  $C_0$ , and when passed to completion, induces a map  $\mathcal{O}_X(U) \rightarrow C$  of Tate algebras. Now also  $B^+$  is mapped into  $C^+$  because  $C^+$  is integrally closed, so we are done.

For the last assertion, by (27.3.2.17), we only have to prove  $\mathrm{Spa}(B, B^+) \rightarrow U$  is a homeomorphism preserving rational subsets, for this, the injectivity is clear as  $B$  is a localization of  $A$ . And the surjectivity follows immediately from the valuation ring characterization and universal property. Continuity is also clear.

For the openness, for any rational subset  $X(\frac{f_1, \dots, f}{g})$  of  $X = \mathrm{Spa}(B, B^+)$ , because  $B = A[g^{-1}]$ ,  $g$  is unit in  $B$ , we can assume that  $f_i, g \in A_0$ . Now we show  $U \cap \mathrm{Spa}(A, A^+)(\frac{f_1, \dots, f}{g})$  is rational, for this, it suffices to add a  $t^N$  to  $f_i$ , and this is possible, as  $X(\frac{f_1, \dots, f}{g})$  is quasi-compact by (27.3.2.4). (This is in fact similar to the proof that continuous bijection from compact to Hausdorff is homeomorphism).  $\square$

**Remark (27.3.3.2).** The proof goes through with complete replaced by Zariski or Henselian, because we only use item5 of (27.3.2.17), which is true for all Zariski pairs.

And by looking at the construction, if a rational subset  $U$  has a representation  $X(\frac{f_1, \dots, f_n}{g})$ , then  $f_i \in \mathcal{O}_X^+(U)$ , and  $g$  is invertible in  $\mathcal{O}_X(U)$ .  $\lrcorner$

#### Stalks

**Def. (27.3.3.3) [Stalks].** The stalks of an affinoid adic space is defined as in the case of schemes, i.e. the colimit of the function ring of rational subsets containing  $x$ , without topology, and similarly for the **integral stalk**. notice that the function rings are defined by universal property w.r.t to complete Huber pairs, so the stalks only depend on the completion of  $(A, A^+)$ .  $\lrcorner$

**Lemma (27.3.3.4).** Let  $U$  be an open subset of  $X = \mathrm{Spa}(A, A^+)$  and  $f, g \in \mathcal{O}_X(U)$ , then  $V = \{x \in U \mid |f|_x \leq |g|_x \neq 0\}$  is an open subset of  $X$ .  $\lrcorner$

*Proof:* Cf. Morel P116. ?

□

**Prop. (27.3.3.5) [Valuations on the Stalks].** Let  $X = \mathrm{Spa}(A, A^+)$  be an affinoid adic space, and  $x \in X$ , then:

- There is a valuation  $x$  on  $\mathcal{O}_{X,x}$  extending that on  $A$ , and  $\mathcal{O}_{X,x}^+ = \{f \in \mathcal{O}_{X,x} \mid |f(x)| \leq 1\}$ .
- $\mathcal{O}_{X,x}$  is local with maximal ideal  $\mathfrak{m}_x = \ker x$ , and  $\mathcal{O}_{X,x}^+$  is local with maximal ideal  $\{f \in \mathcal{O}_{X,x} \mid |f(x)| < 1\}$ .
- If  $\kappa(x)$  is the residue field of  $\mathcal{O}_{X,x}$  and  $\kappa(x)^+$  be the image of  $\mathcal{O}_{X,x}^+$  in  $\kappa(x)$ , then  $k^+(x)$  is naturally a valuation ring, and  $(\kappa(x), \kappa(x)^+)$  is an affinoid field over  $(A, A^+)$ . In particular, there is an isomorphism between the residue fields of  $\mathcal{O}_{X,x}^+$  and  $k^+(x)$ .
- If  $\varphi : (A, A^+) \rightarrow (B, B^+)$  is a morphism of Huber pairs and  $y \in \mathrm{Spa}(B, B^+)$  is a point that  $\mathrm{Spa}(\varphi)(y) = x$ , then the morphism of rings  $\mathrm{Spa}(\varphi)_x^b : \mathcal{O}_{X,x} \rightarrow \mathcal{O}_{Y,y}$  induced by  $\mathrm{Spa}(\varphi)$  is such that  $|\cdot|_x \circ \mathrm{Spa}(\varphi)_x^b = |\cdot|_y$ . In particular,  $\mathrm{Spa}(\varphi)_x^b$  is a morphism of local rings. Also it sends  $\mathcal{O}_{X,x}^+$  to  $\mathcal{O}_{Y,y}^+$  and this is a morphism of local rings.

If moreover  $A$  is Tate, then we have:

- The ring  $\mathcal{O}_{X,x}^+$  is  $t$ -adically Henselian, and  $\mathcal{O}_{X,x}^+ \rightarrow k^+(x)$  induces an isomorphism after  $t$ -adic completion.
- The pairs  $(\mathcal{O}_{X,x}^+, \mathfrak{m}_x)$  and  $(\mathcal{O}_{X,x}, \mathfrak{m}_x)$  are Henselian.

┘

*Proof:* ? Morel P115.

1: Consider the  $t$ -adic completion of the valuation ring  $R_x$  corresponding to  $x$ , then  $(\widehat{k}(\mathfrak{p}_x), \widehat{R}_x)$  is an affinoid Tate ring over  $(A, A^+)$  that is mapped to  $x$  (and its generalizations), thus by universal property, there are unique maps from every rational subsets containing  $x$  to  $(\widehat{k}_x, \widehat{R}_x)$ , thus inducing a map  $(\mathcal{O}_{X,x}, \mathcal{O}_{X,x}^+) \rightarrow (\widehat{k}_x, \widehat{R}_x)$ , which induces the desired valuation. And also we have  $\mathcal{O}_{X,x}^+ \subset \{f \in \mathcal{O}_{X,x} \mid |f(x)| \leq 1\}$ , for the converse, if  $|f(x)| \leq 1$ , then  $U(\frac{f-1}{1})$  is rational subsets in  $U$  containing  $x$ , so by (27.3.3.2),  $f \in \mathcal{O}_X(V)^+$ , thus  $f \in \mathcal{O}_{X,x}^+$ .

2: for  $g$  not in  $\mathfrak{m}_x$ ,  $|g(x)| > |t(x)|^n$  for some  $n$ , so  $g$  is invertible in  $U(\frac{t^n}{g})$  by (27.3.3.2), hence invertible in  $\mathcal{O}_{X,x}$ . Similarly for  $\mathcal{O}_{X,x}^+$ , as  $g$  is invertible in  $U(\frac{1}{g})$ .

3: This is clear from the construction of the valuation on  $\mathcal{O}_{X,x}$  in item 1.

4:

5: As filtered colimits of Henselian pair is Henselian (5.3.10.3) and the function ring is complete, the stalk is Henselian. As for the completion, notice  $\mathfrak{m}_x \subset \mathcal{O}_{X,x}^+$  and is  $t$ -divisible, thus  $\mathcal{O}_{X,x}^+$  has the same  $t$ -adic completion as  $k^+(x)$ .

6: We first prove  $(\mathcal{O}_{X,x}^+, t)$  is Henselian, for this, it suffices to prove  $(\mathcal{O}_X^+(U), t)$  is Henselian, by (5.3.10.3). And  $\mathcal{O}_X^+(U)$  is a filtered colimits of rings of definitions (12.2.5.14) and they are  $t$ -adically complete hence Henselian, so we are done by (5.3.10.3) again. Then so does  $(\mathcal{O}_{X,x}, \mathfrak{m}_x)$  because the property of being Henselian only depends on  $I$  (5.3.10.10). □

**Cor. (27.3.3.6).** By the construction of the valuation on the stalk, we have an inclusion of rings  $k(\mathfrak{p}_x) \subset k(x) \subset \widehat{k}(x)$  that has the same completions, where the first is induced by the compatible map  $A \rightarrow \mathcal{O}_X(X) \rightarrow \mathcal{O}_X(U)$ . ┘

**Def. (27.3.3.7) [Structure Presheaves].** Now by the universal property of function rings (27.3.3.1), we have a map between them induced by inclusion of rational subsets, so we can define the **structure**

**presheaf** to be

$$\mathcal{O}_X(W) = \lim_{U \subset W \text{ rational}} \mathcal{O}_X(U),$$

and similarly for the **integral structure sheaf**  $\mathcal{O}_X^+$ .

Then there is a valuation of a point on  $\mathcal{O}_X(W)$  by passing to the stalk, and

$$\mathcal{O}_X^+(W) = \{f \in \mathcal{O}_X(W) \mid |f(x)| \leq 1, \forall x \in W\}.$$

because this is true for all rational subsets by adic nullstellensatz(27.3.2.17).

A Huber ring  $(A, A^+)$  is called **sheafy** iff the structure sheaf  $\mathcal{O}_X$  on  $X = \text{Spa}(A, A^+)$  is a sheaf. In this case,  $\mathcal{O}_X^+$  is also a sheaf by the above formula, so sheafyness is a property that only depends on  $A$ .  $\lrcorner$

### Criteria for Sheafyness

**Def. (27.3.3.8) [Stably Uniform Huber Pair].** For  $(A, A^+) \in \text{HubPair}$  s.t.  $A$  is analytic and  $X = \text{Spa}(A, A^+)$ , it is called **stably uniform** if  $\mathcal{O}_X(U)$  is uniform for all rational subsets  $U \subset X = \text{Spa}(A, A^+)$ .  $\lrcorner$

**Prop. (27.3.3.9).** Let  $A \rightarrow B$  be a continuous map of Huber rings which splits in the category of topological  $A$ -modules, then  $A$  is stably uniform.  $\lrcorner$

*Proof:* The splitting means the map is strict(12.2.1.6), so  $A$  is also uniform. Also the splitting is preserved under completed tensor product with rational localization, so  $A$  is stably uniform.  $\square$

**Prop. (27.3.3.10) [Examples of Sheafy Huber Rings].** Let  $(A, A^+)$  be a complete Huber pair,

1. (Schemes) If  $A$  is discrete, then  $A$  is sheafy.
2. (Formal Schemes) If  $A$  has a Noetherian ring of definition, then  $A$  is sheafy.
3. (Rigid Spaces, Fargues-Fontaine Curves) If  $A$  is Tate and strongly Noetherian(12.2.4.26), then  $A$  is sheafy.
4. (Perfectoid Spaces) If  $A$  is analytic and  $(A, A^+)$  is stably uniform(27.3.3.8), then  $A$  is sheafy and acyclic.

$\lrcorner$

*Proof:* Cf. Hub94 T2.2, [Ked19]1.7.

We use the arguments following(27.3.4.33).

If  $A$  is analytic and  $(A, A^+)$  is stably uniform. Firstly the ideals  $(f - gT), (g - fT)$  are closed in their rings, Cf.[Ked19]L1.5.26.

Then we have a diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & 0 & \longrightarrow & B\langle T \rangle \oplus B\langle T^{-1} \rangle & \xrightarrow{(x,y) \mapsto x+T^{-1}y} & B\langle T, T^{-1} \rangle \longrightarrow 0 \\
 & & \downarrow & & \downarrow (f-gT, g-fT^{-1}) & & \downarrow \times(f-gT) \\
 0 & \longrightarrow & B & \longrightarrow & B\langle T \rangle \oplus B\langle T^{-1} \rangle & \xrightarrow{(x,y) \mapsto x-y} & B\langle T, T^{-1} \rangle \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & B & \longrightarrow & B\langle \frac{f}{g} \rangle \oplus B\langle \frac{g}{f} \rangle & \longrightarrow & B\langle \frac{f}{g}, \frac{g}{f} \rangle \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

where the columns and the first two rows are exact, thus the third row are exact in the middle and right by spectral sequence. Also it is exact at the left by (27.3.2.16).  $\square$

**Remark (27.3.3.11).** Notice that these contains nearly everything of interest, so Scholze comments that we can somehow pretend that non-sheafy Huber rings don't appear in nature.  $\lrcorner$

**Remark (27.3.3.12).** Stably uniform is hard to check in practice, so a recent paper of Hansen-Kedlaya [Sheafyness Criterion for Huber Rings] studied another class of sousperfectoid rings which can be splitly embedded into a perfectoid ring, and another class of diamantine rings, which involves a condition on the cohomology of pro-étale site of  $A$ , closely related to properties of diamonds.  $\lrcorner$

**Prop. (27.3.3.13) [Non-examples of Sheafy Huber Rings].** Cf. Hub94 end of section1. ?  $\lrcorner$

## 4 Adic Spaces

**Def. (27.3.4.1) [Huber category].** The category  $\mathcal{V}^{pre}$  is a category that the objects are triples  $(X, \mathcal{O}_X, \{v_x\}_{x \in X})$  where  $X \in \mathbf{Top}$ ,  $\mathcal{O}_X$  is a sheaf of topological rings on  $X$ , and  $v_x$  are continuous valuations on the stalk  $\mathcal{O}_{X,x}$  with support  $\mathfrak{m}_x$  for each  $x \in X$ .

And a morphism in  $V^{pre}$  is a pair  $(f, f^\flat)$  where  $f : X \rightarrow Y$  is a map of topological spaces and  $f^\flat : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$  is a morphism of presheaves of topological rings, and the induced morphism of  $f^\flat$  on the stalks are compatible with the valuations.

The **Huber category**  $V$  is the full subcategory of  $V^{pre}$  whose objects are triples  $(X, \mathcal{O}_X, v_x)$  in  $V^{pre}$  that  $\mathcal{O}_X$  is a sheaf.  $\lrcorner$

**Prop. (27.3.4.2) [Sheaf of Integral Elements].** For any  $(X, \mathcal{O}_X, \{v_x\}_{x \in X}) \in \mathcal{V}$ , we can define a **sheaf of integral elements**  $\mathcal{O}_X^+$  s.t. for any open subset  $U \subset X$ ,

$$\mathcal{O}_X^+(U) = \{f \in \mathcal{O}_X(U) : |f(x)| \leq 1, \forall x \in U\}.$$

It is clearly a sheaf.  $\lrcorner$

**Def. (27.3.4.3) [Open Immersions].** An **open immersion** in  $V^{pre}$  is a homeomorphism onto an open subset that induces an isomorphism of presheaves.  $\lrcorner$



**Def. (27.3.4.4)[Adic Spaces].** The category of **affinoid adic spaces** is the full subcategory of the Huber category (27.3.4.1) consisting of objects isomorphic to an adic spectrum of a Huber pair  $(A, A^+)$ , and the category of **adic spaces** is the full subcategory of the Huber category (27.3.4.1) whose objects are locally isomorphic to an affinoid adic space. The category of adic spaces is denoted by  $\text{AdicSpa}$ .  
 $\lrcorner$

**Prop. (27.3.4.5)[Adic Spaces are Locally Spectral].** Adic spaces are locally spectral, and morphisms of adic spaces are spectral.  $\lrcorner$

*Proof:*  $\square$

**Prop. (27.3.4.6)[Adic Spectrum Adjointness].** For any  $(R, R^+) \in \text{HubPair}$  and  $Y \in \text{AdicSpa}$ , there is a natural isomorphism

$$\text{Hom}(Y, \text{Spa}(R, R^+)) \cong \text{Hom}((R, R^+), (\mathcal{O}_Y(Y), \mathcal{O}_Y^+(Y))).$$

$\lrcorner$

*Proof:* It suffices to show for  $Y = (S, S^+)$  affine, because an morphism from  $Y \rightarrow \text{Spa}(R, R^+)$  is glued from local morphisms, and  $\mathcal{O}_Y$  is a sheaf.

For this, Cf. [Huber94]Prop2.1(2).  $\color{red}{?}$   $\square$

**Def. (27.3.4.7)[Noetherian Adic Spaces].** A **locally Noetherian adic space** is an adic space that is locally of the form  $\text{Spa}(A, A^+)$ , where  $A$  is a strongly Noetherian Tate  $K$ -algebra. A **Noetherian adic space** is a locally Noetherian adic space that is qcqs. In particular, noetherian adic spaces are spectral.  $\lrcorner$

**Def. (27.3.4.8)[Uniform Adic Spaces].**  $X \in \text{AdicSpa}$  is called a **uniform adic space** if for any open affinoid  $U = \text{Spa}(R, R^+) \subset X$ , the Huber ring  $R$  is uniform.  $\lrcorner$

### Cartier Divisors and Closed Immersions

**Def. (27.3.4.9)[Cartier Divisors].** Let  $X$  be a uniform analytic adic space, then a (effective) **Cartier divisor** on  $X$  is an ideal sheaf  $\mathcal{I} \subset \mathcal{O}_X$  that is locally free of rank 1. The support of a Cartier divisor is  $\text{Supp}(\mathcal{O}_X/\mathcal{I})$ .

The support  $Z$  of a Cartier divisor is a nowhere dense closed subset of  $X$ , and the map  $I \mapsto \mathcal{I} = i\mathcal{O}_X$  induces a bijection between invertible ideals  $I \subset R$  that  $V(I)$  is nowhere dense in  $X$  and Cartier divisors on  $X$ .  $\lrcorner$

*Proof:* By (27.3.4.24), any Cartier divisor is of the form  $I \otimes_R \mathcal{O}_X$  for some invertible ideal  $I \subset R$ . We need to show that  $\varphi : I \otimes_R \mathcal{O}_X \rightarrow \mathcal{O}_X$  is injective iff  $V(I) \subset X$  is nowhere dense.

By restriction, we can assume  $I = (f)$  is principle, and if  $V(f)$  contains an open subset, then on an open rational subset,  $f = 0$  by uniformity (27.3.2.15), so  $\varphi$  is not injective. Conversely, if  $V(f)$  is nowhere dense, we show  $f$  is a nonzero-divisor: if  $fg = 0$  and  $g \neq 0$ , then  $U = U(\frac{g}{g})$  is contained in  $V(f)$  and is open, so  $U = 0$ , which implies  $g = 0$  by uniformity (27.3.2.15), contradiction.  $\square$

**Prop. (27.3.4.10).** Let  $X \in \text{AdicSpa}$  be uniform (27.3.4.8) and  $\mathcal{I} \subset \mathcal{O}_X$  a Cartier divisor with support  $Z$  and  $j : U = X \setminus Z \rightarrow X$ . There are injective maps of sheaves

$$\mathcal{O}_X \hookrightarrow \varinjlim_n \mathcal{I}^{-n} \hookrightarrow j_*\mathcal{O}_U.$$

$\lrcorner$



*Proof:* This is local, it suffices to check this is for any affine subscheme of  $X$ , so we can assume  $X = \mathrm{Spa}(R, R^+)$  and  $\mathcal{I} = f\mathcal{O}_X$  for some nonzero-divisor  $f \in R$  whose vanishing locus  $Z$  is nowhere dense, and check the global sections:

$$R \rightarrow R[\frac{1}{f}] \rightarrow \Gamma(U, \mathcal{O}_U).$$

Then it suffices to show  $R \rightarrow \Gamma(U, \mathcal{O}_U)$  is injective. But if  $g$  vanishes on  $U$ , then the vanishing locus of  $g$  is a closed subset containing  $U$ , which implies that it is all of  $X$  as  $Z$  is nowhere dense, so  $g = 0$  by uniformity(27.3.2.15).  $\square$

**Def. (27.3.4.11) [Meromorphic Along the Cartier Divisor].** A function  $f \in H^0(U, \mathcal{O}_U)$  is **meromorphic along** the Cartier divisor  $\mathcal{I} \subset \mathcal{O}_X$  if it lifts to  $H^0(X, \varinjlim_n \mathcal{I}^{-n})$ .  $\lrcorner$

**Def. (27.3.4.12) [Closed Cartier Divisors].** On a uniform analytic adic space  $X$ , a **closed Cartier divisor** is a Cartier divisor  $\mathcal{I} \subset \mathcal{O}_X$  with support  $Z$  that  $(Z, \mathcal{O}_X/\mathcal{I}, |\cdot|_x, x \in Z)$  is an adic space.  $\lrcorner$

**Prop. (27.3.4.13) [Closed Cartier Divisor and Closed Immersion].** On a uniform analytic adic space  $X$ , a Cartier divisor  $\mathcal{I} \subset \mathcal{O}_X$  is closed iff the map  $\mathcal{I}(U) \hookrightarrow \mathcal{O}_X(U)$  has closed image for all open affinoid  $U \subset X$ . In this case, for all open affinoid  $U = \mathrm{Spa}(R, R^+) \subset X$ , the intersection  $U \cap Z = \mathrm{Spa}(S, S^+)$  is an affinoid adic space, where  $S = R/I$  and  $S^+$  is the integral closure of the image of  $R^+$  in  $S$ .  $\lrcorner$

*Proof:* The property of having closed image can be checked locally as affinoid subspace is quasi-compact. The fact  $\mathcal{O}_X/\mathcal{I}$  is a  $(\mathcal{O}_X/\mathcal{I})(U_i)$  is separated for an open covering  $\{U_i\}$  of  $X$ , so at  $\mathcal{I}(U_i)$  is closed in  $\mathcal{O}_X(U_i)$ .

Conversely, if  $\mathcal{I}(U) \hookrightarrow \mathcal{O}_X(U)$  has closed image for all open affinoid  $U \subset X$ , we want to show  $(Z, \mathcal{O}_X/\mathcal{I}, |\cdot|_x, x \in Z)$  is an adic space: We can construct this locally, so if  $X = \mathrm{Spa}(R, R^+)$ ,  $S = R/I$  is separated and complete, thus a complete Huber ring, and consider the quotient Huber ring(12.2.5.19)  $(S, S^+)$ , then  $\mathrm{Spa}(S, S^+) \rightarrow \mathrm{Spa}(R, R^+)$  is a closed immersion of topological spaces with image  $V(I)$ .

What's left to show is that the map of presheaves  $\mathcal{O}_X/\mathcal{I} \rightarrow \mathcal{O}_Z$  is an isomorphism, and also  $\mathcal{O}_X/\mathcal{I}(U) = \mathcal{O}_X(U)/\mathcal{I}(U)$ . To show this, we first show the presheaf  $\mathcal{O}_X/\mathcal{I}$  is a sheaf: this is by  $3 \times 3$ -lemma:  $\mathcal{I}$  is locally  $(f)$ , and for a rational covering  $U_i \rightarrow X$ ,

$$\mathcal{O}_X(X)/f \rightarrow \prod_i \mathcal{O}_X(U_i)/f \Rightarrow \prod_{ij} \mathcal{O}_X(U_{ij})/f$$

is exact.

And to show  $\mathcal{O}_X/\mathcal{I} \rightarrow \mathcal{O}_Z$  is an isomorphism, it suffices to notice taking localization and taking quotient commutes, because they are both defined by universal properties.(12.2.5.19)(12.2.5.32).  $\square$

**Remark (27.3.4.14).** Even if  $A$  is Tate and stably uniform, and  $f \in A$  is a nonzero-divisor that  $fA \subset A$  is closed, it may not be true that  $\mathcal{O} = f\mathcal{O}_X \hookrightarrow \mathcal{O}_X$  is a closed Cartier divisor on  $X = \mathrm{Spa}(A, A^+)$ . This is because there may be rational localization  $(A, A^+) \rightarrow (B, B^+)$  that  $fB$  is not closed in  $B$ . Cf. [Ked19]P16.  $\lrcorner$

### Examples of Adic Spaces

**Prop. (27.3.4.15) [Examples of Adic Spaces].**

- (Adic Closed Unit Disk) The space  $\mathrm{Spa}(\mathbb{Z}[T]) = \mathrm{Spa}(\mathbb{Z}[T], \mathbb{Z}[T])$  represents the functor  $X \mapsto \mathcal{O}_X^+(X)$ .
- (Adic Affine Line) The functor  $X \mapsto \mathcal{O}_X(X)$  is also representable, by  $\mathrm{Spa}(\mathbb{Z}[T], \mathbb{Z})$ . Notice for any non-Archimedean field  $K$ ,

$$\mathrm{Spa}(\mathbb{Z}[T], \mathbb{Z}) \times \mathrm{Spa} K = \cup_{n \geq 1} \mathrm{Spa} K \langle \varpi^n T \rangle = \varinjlim_{n, T \mapsto \varpi T} \mathrm{Spa} K \langle T \rangle.$$

This is because For any Huber pair  $(R, R^+)$  over  $(K, \mathcal{O}_K)$ ,  $R = \cup_n \varpi^{-n} R^+$  because  $\varpi$  is topologically nilpotent.

- (The Open Unit Disk) Let  $D = \mathrm{Spa} \mathbb{Z}[[T]]$ , then for any non-Archimedean field  $K$ ,  $D_K = D \times \mathrm{Spa} K$  represents the functor that maps a  $K$ -algebra  $R$  to all its elements of norm  $\leq 1$ . Then this the open disk over  $K$ . And  $D_K$  is also represented by

$$\cup_{n \geq 1} \mathrm{Spa} K \left\langle T, \frac{T^n}{\varpi} \right\rangle.$$

- (The Punctured Open Unit Disk) Let  $\mathbb{D}^* = \mathrm{Spa} \mathbb{Z}((T))$ , then

$$D_K^* = D^* \times \mathrm{Spa} K = \mathbb{D}_K \left\langle \frac{T}{T} \right\rangle = \mathbb{D}_K \setminus \{0\}$$

┘

**Prop. (27.3.4.16) [The Open Unit Disk over  $\mathbb{Z}_p$ ].** Consider  $X = \mathbb{Z}_p[[T]]$  with the  $(p, T)$ -adic topology, there is exactly one non-analytic point  $x = x_{\mathbb{F}_p}$ . Let  $X = \mathrm{Spa}(\mathbb{Z}_p[[T]])$  and  $\mathcal{Y} = X \setminus \{x_{\mathbb{F}_p}\}$ , then for a point  $x \in \mathcal{Y}$ ,  $T(x)$  and  $p(x)$  cannot both be 0 by (12.2.6.2).

Then there exists a unique continuous map

$$\kappa : |\mathcal{Y}| \rightarrow [0, \infty]$$

characterized by the following property: For any rational number  $m/n < r$ ,  $|T(x)|^n > |p(x)|^m$ , and for any rational number  $m/n > r$ ,  $|T(x)|^n < |p(x)|^m$ . This map  $\kappa$  is surjective. ┘

*Proof:* Let  $\tilde{x}$  be a maximal vertical generalization of  $x$ , then it has rank 1 and we can assume  $\tilde{x}$  is real valued by (12.2.3.9). Now we define

$$\kappa(x) = \frac{\log |T(\tilde{x})|}{\log |p(\tilde{x})|},$$

This is definable as  $T(\tilde{x})$  and  $p(\tilde{x})$  cannot both be 0.

The uniqueness of  $\kappa(x)$  follows from the fact  $|T(x)|^n > |p(x)|^m$  implies to  $|T(\tilde{x})|^n \geq |p(\tilde{x})|^m$  because  $\tilde{x}$  is a generalization of  $x$ .

To show  $\kappa$  satisfies the condition, if  $m/n < r$ , then  $|T(\tilde{x})|^n > |p(\tilde{x})|^m$ , so  $|T(x)|^n > |p(x)|^m$  because  $x, \tilde{x}$  define the same topology. And it is continuous because

$$\kappa^{-1}((-\infty, r)) = \cup_{m/n < r} U\left(\frac{T^n}{p^m}\right).$$

To show the surjectivity, if  $\kappa = [x_0 : x_1]$ , define the valuation as

$$v\left(\sum a_{ij} p^i T^j\right) = \sup_{a_{ij} \neq 0} e^{-x_0 i - x_1 j}$$

where  $(a_{ij})^p = 1$ . ┘

### Construction of Adic Spaces

**Def. (27.3.4.17)** [Fiber Products of Adic Spaces and Schemes]. Cf. [Wed14]P91. ┘

**Def. (27.3.4.18)** [Adic Spaces attached to Schemes]. ┘

**Def. (27.3.4.19)** [Adic Spaces attached to Formal Schemes]. Cf. [Wed14]. ┘

**Def. (27.3.4.20)** [Adic Spaces attached to Rigid Analytic Spaces]. Cf. [Wed14]. ┘

### Sheaves and Vector Bundles

**Def. (27.3.4.21).** Let  $(A, A^+)$  be a Huber pair and  $X = \mathrm{Spa}(A, A^+)$ , let  $\widetilde{M} = M \otimes_A \mathcal{O}_X$  be the presheaf on  $X$ . ┘

**Prop. (27.3.4.22).** If  $A$  is sheafy, then for any finite projective  $A$ -module  $M$ , the presheaf  $\widetilde{M}$  is a sheaf on  $X = \mathrm{Spa}(A, A^+)$ , and  $H^i(U, \mathcal{F}) = 0$  for any rational subset of  $X$  and  $i > 0$ . ┘

*Proof:* Because  $M$  is a direct sum of a finite free  $A$ -module, then we reduce to the case  $\mathcal{O}_X$  is sheafy. □

**Remark (27.3.4.23).** This is a partial analogy with Tate's acyclicity theorem in rigid analytic geometry (27.2.1.40)(27.2.3.10), but it only holds for f.p. modules, not even f.g. modules. One impediment is that the rational localization map are generally not flat. To get around this, Kedlaya defined a category of pseudo-coherent modules, with the property that even when flatness fails, tensoring is also exact in this category. ┘

**Prop. (27.3.4.24)** [Vector Bundles]. Let  $(A, A^+)$  be a sheafy analytic Huber pair and  $X = \mathrm{Spa}(A, A^+)$ , then the functor  $M \rightarrow \widetilde{M}$  from the category of finite projective  $A$ -modules to the category  $\mathrm{Vect}_X$  of locally finite free  $\mathcal{O}_X$ -modules is an equivalence of categories. In particular,  $\mathrm{Vect}_X$  only depends on  $A$ . ┘

*Proof:* Cf. [Ked19].P40 ? □

### Pre-Adic Spaces

References are [S-W20]L3 and [Ked19].

**Def. (27.3.4.25)** [Pre-Adic Spaces]. ┘

**Remark (27.3.4.26).** Pre-adic spaces is an approach to work around the failure of sheafyness of general Huber pair, with techniques from algebraic stacks. ┘

### Analytic Points

**Prop. (27.3.4.27)** [Analytic and Tate Rings]. Let  $(A, A^+)$  be a complete Huber pair, then

- The Huber ring  $A$  is analytic iff all points of  $\mathrm{Spa}(A, A^+)$  are analytic.
- A point  $x \in \mathrm{Spa}(A, A^+)$  is analytic iff there is a rational nbhd  $U \subset \mathrm{Spa}(A, A^+)$  that  $\mathcal{O}_X(U)$  is Tate.

┘

*Proof:* 1:  $x$  is non-analytic iff  $A^{00} \subset \mathfrak{p}_x$  by (12.2.6.5), so all points are analytic iff  $A^{00}A = A$ , which means  $A$  is analytic.

2: Let  $x$  be an analytic point, then there exists  $f \in I$  where  $I$  is an ideal of definition that  $|f|_x \neq 0$ . Now  $\{g \in A \mid |g(x)| \leq |f(x)|\}$  is open (because  $|\cdot|_x$  is continuous), so contains some  $I^n$ . Now let  $I^n = (g_1, \dots, g_k)$ , then  $U(\frac{g_1, \dots, g_k}{f})$  is a rational subset. Then in  $\mathcal{O}_X(U)$ ,  $f$  is a unit, but also it is topologically nilpotent, because it is contained in  $I$ .

Conversely, if  $x \in X$  has a rational nbhd  $U = U\left\langle \frac{T}{s} \right\rangle$  such that  $\mathcal{O}_X(U)$  is Tate, and  $x$  is not analytic, then  $\mathfrak{p}_x$  contains an ideal of definition  $I \subset A_0$ . Now let  $f \in \mathcal{O}_X(U)$  be a topologically nilpotent unit, then there exists  $m \geq 1$  that  $f^m$  lies in the closure of  $IA_0[t/s \mid t \in T]$  in  $\mathcal{O}_X(U)$ . Since  $x \in U$ , the valuation  $|\cdot|_x$  extends to  $\mathcal{O}_X(U)$ , and since  $|\cdot|_x$  is continuous,  $|f^m(x)| = 0$ , contradiction, as  $f$  is a unit in  $\mathcal{O}_X(U)$ .  $\square$

**Def. (27.3.4.28)[Analytic Points].** Let  $X$  be a pre-adic space, then a point  $x \in X$  is called an **analytic point** if there is some open affinoid nbhd  $U = \text{Spa}(A, A^+) \subset X$  of  $x$  that  $A$  is Tate. And  $X$  is called **analytic** if all of its points are analytic.  $\lrcorner$

**Prop. (27.3.4.29).** Let  $f : X \rightarrow Y$  be a map of analytic pre-adic spaces, then  $|f| : |X| \rightarrow |Y|$  is generalizing. If  $f$  is quasi-compact and surjective, then  $|f|$  is a quotient map.  $\lrcorner$

*Proof:* Cf. [Hub96] 1.1.10. and [Étale Cohomology of Diamonds, L2.5].  $\square$

**Def. (27.3.4.30).** A map  $f : Y \rightarrow X$  of pre-adic spaces is called **analytic** if it carries analytic points to analytic points.  $\lrcorner$

### Proof of Acyclicity and Sheafyness by Čech Reduction

**Def. (27.3.4.31) [Standard Rational Coverings].** Let  $X = \text{Spa}(A, A^+)$  and  $f_1, \dots, f_n \in A$  which generates the unit ideal, then the sets  $X(\frac{f_1, \dots, f_n}{f_i})$  covers  $X$ , called the **standard rational covering** of  $X$ . And if  $n = 2$ , it is called a **standard binary rational covering** of  $X$ .

There are special types of standard binary rational coverings: if  $f_1 = f, f_2 = 1$ , then it is called a **simple Laurent covering**. If  $f_1 = f, f_2 = 1 - f$ , then it is called a **simple balanced covering**.  $\lrcorner$

**Lemma (27.3.4.32) [Reduction of Coverings].** Let  $A$  be an analytic Huber ring,

- (Huber) Every open covering of  $X$  can be refined by some standard rational covering.
- (Gabber-Ramero) Every open covering of a rational subspace of  $X$  can be refined by some compositions of simple Laurent coverings and simple balanced coverings.

$\lrcorner$

*Proof:* 1: Cf. [Ked19] P28.

2: Cf. [Ked19] P29.  $\square$

**Prop. (27.3.4.33) [Čech Reduction].** By a Čech cohomological argument the same as Tate's acyclicity theorem in rigid geometry (27.2.1.40), it suffices to prove any presheaf is a sheaf or any sheaf is acyclic on simple Laurent coverings and simple balanced coverings.

That is, for every rational localization  $(B, B^+)$  over  $(A, A^+)$ , every pair  $f, g \in B$  that  $g = f$  or  $1 - f$ , if the sequence

$$0 \rightarrow B \rightarrow B\left\langle \frac{f}{g} \right\rangle \oplus B\left\langle \frac{g}{f} \right\rangle \rightarrow B\left\langle \frac{f}{g}, \frac{g}{f} \right\rangle \rightarrow 0,$$

- is exact at exact at left and middle, then  $\mathcal{O}_X$  is sheafy.

- is exact, then  $\mathcal{O}_X$  is acyclic.

Also remember the following equations:

$$B \left\langle \frac{f}{g} \right\rangle = B \langle T \rangle / \overline{(f - gT)}, \quad B \left\langle \frac{g}{f} \right\rangle = B \langle T \rangle / \overline{(g - fT)}, \quad B \left\langle \frac{f}{g}, \frac{g}{f} \right\rangle = B \langle T, T^{-1} \rangle / \overline{(f - gT)}$$

┘

**Prop. (27.3.4.34) [Properties of (Finite)Étale Maps].**

- (Finite)Étale maps are invariant under compositions and pullbacks.
- If  $g$  and  $gf$  are (finite)étale, then so is  $f$ .
- $f$  is (finite)étale iff  $f^\flat$  is (finite)étale.

┘

*Proof:* Cf. [S-W20]P65. □

## 5 Étale Sites

**Def. (27.3.5.1) [Finite Étale Maps of Adic Spaces].** A **finite étale map of Huber pairs** is a map  $(A, A^+) \rightarrow (B, B^+) \in \text{HubPair}$  s.t.  $A \rightarrow B$  is finite étale, and  $B^+$  is the integral closure of  $A^+$  in  $B$ .

A **standard étale map of Huber pairs** is a map  $(A, A^+) \rightarrow (B, B^+) \in \text{HubPair}$  that is a finite étale map followed by a rational localization (27.3.2.3).

A map  $f : X \rightarrow Y \in \text{AdicSpa}$  is called a **finite étale morphism of adic spaces** if there is a cover of  $Y$  by affinoid subspaces  $\{V_i \subset Y\}$  s.t.  $U_i = f^{-1}(V_i)$  are all affinoids, and the map  $(\mathcal{O}_Y(V_i), \mathcal{O}_Y^+(V_i)) \rightarrow (\mathcal{O}_X(U_i), \mathcal{O}_X^+(U_i))$  is finite étale. Write  $Y_{\text{fét}}$  for the category of all such maps.

A map  $f : X \rightarrow Y \in \text{AffndSpa}$  is called a **standard étale morphism of affinoid spaces** if the map  $(\mathcal{O}_Y(V_i), \mathcal{O}_Y^+(V_i)) \rightarrow (\mathcal{O}_X(U_i), \mathcal{O}_X^+(U_i))$  is standard étale. ┘

**Prop. (27.3.5.2).** Being a finite étale map of adic spaces is local on the target. ┘

*Proof:* ? □

**Def. (27.3.5.3) [Étale Maps of Adic Spaces].** An **étale map of adic spaces**  $X \rightarrow Y \in \text{AdicSpa}$  is a map s.t. for any  $x \in X$ , there exists an open subspace  $x \in U$  and an open subspace  $f(U) \subset V$  together with a  $W \in \text{AdicSpa}$  that  $f : U \rightarrow V$  factors through an open embedding  $U \rightarrow W$  and a finite étale map  $W \rightarrow V$  (27.3.5.1). ┘

## 6 Pro-Étale Sites

**Def. (27.3.6.1) [Underlying Spaces of Pro-Étale Objects].** For  $X \in \text{AdicSpa}$  and  $U = \varprojlim U_i \in \text{pro}(X_{\text{ét}})$ , the **underlying space** of  $U$  is defined to be  $|U| = \varprojlim |U_i|$ . ┘

*Proof:* This is well-defined because the underlying spaces of adic spaces are compatible with fiber products. □

**Def. (27.3.6.2) [(Pro-)Étale Morphisms of Pro-Étale Objects].** For  $X \in \text{AdicSpa}$ , a morphism  $U \rightarrow V \in \text{pro}(X_{\text{ét}})$  is called a **étale/finite étale morphism** if it is the base change of an étale/finite étale morphism  $U_0 \rightarrow V_0 \in X_{\text{ét}}$  (27.3.5.1)(27.3.5.3). And it is called a **pro-étale morphism** if  $U = \varprojlim U_i$  where each  $U_i \in \text{pro}(X_{\text{ét}})$  is étale over  $V$  and the transition maps  $U_i \rightarrow U_j$  are finite étale and surjective. ┘

**Prop. (27.3.6.3) [Properties of Pro-Étale Morphisms].** For  $X \in \text{AdicSpa}$ ,

- Finite étale/étale/pro-étale morphisms in  $\text{pro}(X_{\text{ét}})$  are stable under base change and compositions, and if  $U \rightarrow V$  is pro-étale, then  $|U \times_V W| \rightarrow |U| \times_{|V|} |W|$  is surjective.
- (Compact Open Subsets) If  $U \rightarrow X$  is pro-étale and  $W \subset |U|$  is a qc open subset, then there is some  $V \in \text{pro}(X_{\text{ét}})$  with  $V \rightarrow X$  pro-étale with an étale map  $V \rightarrow U$  which is a homeomorphism onto  $W$ , and satisfies that: If  $f : V' \rightarrow U \in \text{pro}(X_{\text{ét}})$  and  $|f|$  factors through  $W$  with  $V' \rightarrow X$  pro-étale, then  $f$  factors through  $V$ .
- Pro-étale maps in  $\text{pro}(X_{\text{ét}})$  are open.
- Any surjective étale/finite étale morphism  $U \rightarrow V \in \text{pro}(X_{\text{ét}})$  is a pullback from a surjective étale/finite étale morphism in  $X_{\text{ét}}$ .

┘

*Proof:* Cf. [Scholze13]P15. ?

□

**Def. (27.3.6.4) [Pro-Étale Sites].** For  $X \in \text{AdicSpa}$ , the **pro-étale site**  $X_{\text{pro-ét}}$  is defined to be the full subcategory of  $\text{pro}(X_{\text{ét}})$  consisting of objects that are pro-étale over  $X$  (27.3.6.2), together with coverings given by a family of pro-étale morphisms (27.3.6.2)  $\{f_i : U_i \rightarrow U\}$  s.t.  $|U| = \cup f_i(|U_i|)$ . This is truly a site by (27.3.6.3).

The **pro-finite-étale site**  $X_{\text{pro-fét}}$  is defined to be the category  $\text{pro}(X_{\text{ét}})$  with coverings given by a family of open morphisms  $\{f_i : U_i \rightarrow U\}$  s.t.  $|U| = \cup f_i(|U_i|)$  (27.3.6.1). This is truly a site by (27.3.6.3).

We use the notation  $U = \varprojlim U_i \in X_{\text{pro-ét}}$  (resp.  $X_{\text{pro-fét}}$ ) to indicate that  $U_i \in X_{\text{ét}}$  and  $U = \varprojlim U_i \in \text{pro}(X_{\text{ét}})$ .

┘

**Prop. (27.3.6.5).** Let  $X \in \text{NAdicSpa}_{\text{cntd}} \text{Spa} \cup \text{NSch}_{\text{cntd}}$ , then there is a canonical equivalence of sites

$$X_{\text{pro-fét}} \cong \pi_1(X, \bar{x})\text{-Prof}.$$

┘

*Proof:*

□

**Prop. (27.3.6.6).** For  $X \in \text{AdicSpa}$ , a morphism  $f : U \rightarrow V \in X_{\text{pro-fét}}$  is open iff it is a pro-étale morphism in  $X_{\text{pro-ét}}$ . In particular, coverings in  $X_{\text{pro-fét}}$  and  $X_{\text{pro-ét}}$  are compatible, and there is a map of sites  $X_{\text{pro-ét}} \rightarrow X_{\text{pro-fét}}$ .

┘

*Proof:* By (27.3.6.3), pro-étale maps are open. For the converse, use (27.3.6.5) and [Scholze13]Lemma3.6. ?

□

**Prop. (27.3.6.7).**  $X_{\text{ét}}$  has arbitrary finite limits.

┘

*Proof:*

□

**Prop. (27.3.6.8) [Quasi-Compactness and Quasi-Separatedness].** For  $X \in \text{NAdicSpa}_{\text{cntd}} \text{Spa} \cup \text{NSch}_{\text{cntd}}$ ,

- If  $U = \lim U_i \rightarrow X$  is a pro-étale presentation of  $U \in X_{\text{pro-ét}}$  s.t. each  $U_i$  is affinoid, then  $U$  is a qcqs object in  $X_{\text{pro-ét}}$ . These objects are generating, and stable under fiber products.
- $\text{Sh}(X_{\text{pro-ét}})$  is algebraic (Cf. SGA 4 VI, Definition 2.3 ?).

- $U \in X_{\text{pro-ét}}$  is qc/qs iff  $|U|$  is. And a morphism  $f : U \rightarrow V \in X_{\text{pro-ét}}$  is qc/qs iff  $|f|$  is.
- Let  $U = \varprojlim U_i \rightarrow V$  be an inverse limit of surjective finite étale maps, then  $U$  is qc/qs iff  $V$  is.
- the site  $X_{\text{pro-ét}}$  is qs/qcqs? iff  $|X|$  is.

┘

*Proof:* Cf.[Scholze13]Prop3.12.?

□

**Prop. (27.3.6.9)[Enough Points for  $X_{\text{pro-ét}}$ ].** For  $X \in \text{AdicSpa}$ ,  $X_{\text{pro-ét}}$  has enough points, given by profinite covers of geometric points.

┘

*Proof:*

□

**Prop. (27.3.6.10)[Cohomology on Pro-Étale Objects].** Suppose  $X$  be a locally Noetherian scheme or a locally Noetherian adic space,  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , and  $U = \varprojlim U_j \in X_{\text{pro-ét}}$  is qcqs, then for any  $i \in \mathbb{N}$ ,

$$H^i(U, \nu^* \mathcal{F}) = \varinjlim H^i(U_j, \mathcal{F}).$$

┘

*Proof:*

□

**Cor. (27.3.6.11).** Suppose  $X$  be a locally Noetherian scheme or a locally Noetherian adic space, and  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , then

- the adjunction maps  $\mathcal{F} \rightarrow \nu_* \nu^* \mathcal{F} \rightarrow R\nu_* \nu^* \mathcal{F}$  are isomorphisms. In particular, there is an embedding of topoi  $\nu_* : \text{Sh}(X_{\text{ét}}) \rightarrow \text{Sh}(X_{\text{pro-ét}})$ .
- If  $f : X \rightarrow Y$  is a qcqs morphism, then for any  $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ , the base change morphism

$$\nu_Y^* Rf_{\text{ét},*} \mathcal{F} \rightarrow Rf_{\text{pro-ét},*} \nu_X^* \mathcal{F}$$

is an isomorphism.

┘

*Proof:* 1: By (7.3.1.7),  $R^i \nu_* \nu^* \mathcal{F}$  is the sheaf on  $X_{\text{ét}}$  associated to the sheaf  $U \mapsto H^i(U, \nu^* \mathcal{F})$ . But when  $U$  is affinoid,  $H^i(U, \nu^* \mathcal{F}) = H^i(U, \mathcal{F})$  by (27.3.6.10), which implies the assertion, as for  $i \in \mathbb{Z}_+$ , any section is locally vanishing (7.3.2.14).

2: By the same argument, we can show that for any  $i \in \mathbb{Z}_+$ , the  $i$ -th cohomology of both sides are equal to the shiffication of the presheaf taking any qcqs  $U = \varprojlim U_i \in X_{\text{pro-ét}}$  to  $\varinjlim H^i(U_i \times_Y X, \mathcal{F})$ .

□

**Prop. (27.3.6.12).** Should be put somewhere else?? Let  $T \in \text{Site}$  and  $\mathcal{F}_i \in \text{Sh}(T)$  be an inverse system. Assume that there is a basis  $B$  of  $T$  s.t. for any  $U \in B$ ,  $R^1 \varprojlim \mathcal{F}_i(U) = 0$ , and  $H^j(U, \mathcal{F}_i) = 0$  for  $j \in \mathbb{Z}_+$ , then  $R^j \varprojlim \mathcal{F}_i = 0$  for any  $j \in \mathbb{Z}_+$ , and  $(\varprojlim \mathcal{F}_i)(U) = \varprojlim \mathcal{F}_i(U)$  for any  $U \in T$ . Moreover,  $H^j(U, \varprojlim \mathcal{F}_i) = 0$  for  $U \in B$  and  $j \in \mathbb{Z}_+$ .

┘

*Proof:* Cf.[Scholze13]Lemma3.18.?

□

### Structure Sheaves

**Def. (27.3.6.13) [Structure Sheaves].** Let  $X \in \mathrm{NAdicSpa} / \mathrm{Spa}(\mathbb{Q}_p, \mathbb{Z}_p)$ , then

- The sheaf  $\mathcal{O}_X = \nu^* \mathcal{O}_{X_{\text{ét}}}$  is called the **structure sheaf** on  $X_{\text{pro-ét}}$ , and it has a **subring of integral elements**  $\mathcal{O}_X^+ = \nu^* \mathcal{O}_{X_{\text{ét}}}^+$ .
- $\widehat{\mathcal{O}}_X^+ = \varprojlim_n \mathcal{O}_X^+ / (p^n)$  is called the **integral completed structure sheaf** on  $X_{\text{pro-ét}}$ , and  $\widehat{\mathcal{O}}_X = \widehat{\mathcal{O}}_X^+[\frac{1}{p}]$  is called the **completed structure sheaf** on  $X_{\text{pro-ét}}$ .

┘

**Prop. (27.3.6.14).** Let  $X \in \mathrm{NAdicSpa} / \mathrm{Spa}(\mathbb{Q}_p, \mathbb{Z}_p)$  and  $U \in X_{\text{pro-ét}}$ .

- For any  $x \in |U|$ , there is a natural continuous valuation  $|\cdot|_x : f \mapsto |f(x)|$  on  $\mathcal{O}_X(U)$ .
- $\mathcal{O}_X^+(U) = \{f \in \mathcal{O}_X(U) : \forall x \in |U|, |f(x)| \leq 1\}$ .
- For any  $n \in \mathbb{Z}_+$ , the map of sheaves  $\mathcal{O}_X^+ / (p) \rightarrow \widehat{\mathcal{O}}_X^+ / (p)$  is an isomorphism, and  $\widehat{\mathcal{O}}_X^+(U)$  is flat and  $p$ -adically complete.
- For any  $x \in |U|$ , the valuation  $|\cdot|_x$  extends to a continuous valuation on  $\widehat{\mathcal{O}}_X(U)$ .
- $\widehat{\mathcal{O}}_X^+(U) = \{f \in \widehat{\mathcal{O}}_X(U) : \forall x \in |U|, |f(x)| \leq 1\}$ . In particular,  $\widehat{\mathcal{O}}_X^+(U) \subset \widehat{\mathcal{O}}_X(U)$  is integrally closed.

┘

*Proof:* Cf. [Scholze]P22. ? Firstly these assertions are local w.r.t.  $U$ , so we may assume that  $U$  is qcqs (27.3.6.8), and choose a pro-étale presentation  $U = \varprojlim U_i \rightarrow X$ .

1: A point  $x \in |U|$  is given by a compatible system of points  $x_i \in |U_i|$ , giving a compatible system of continuous valuations on  $\mathcal{O}_{X_{\text{ét}}}(U_i)$ . But  $\mathcal{O}_X(U) = (\nu^* \mathcal{O}_{X_{\text{ét}}})(U) = \varinjlim \mathcal{O}_{X_{\text{ét}}}(U)$ , so these valuations give a valuation on  $\mathcal{O}_X(U)$ . □

**Remark (27.3.6.15).** It is not clear whether for any  $U \in X_{\text{pro-ét}}$ , the triple  $(|U|, \widehat{\mathcal{O}}_X|_{|U|}, \{|\cdot|_x\}_{x \in |U|})$  is an adic space. But for any  $U \in X_{\text{pro-ét}}$ , there is a covering by pro-étale perfectoid spaces over  $X$  s.t. on this covering, the assertion is true (27.4.4.5). ┘



## 27.4 Perfectoid Spaces(Scholze)

References are [Sch12].

**Remark(27.4.0.1).** The theory of perfectoid spaces presents a general framework to reduce problems about algebraic varieties in characteristic 0 to algebraic varieties in characteristic  $p$ .  $\lrcorner$

**Notation(27.4.0.2).**

- Use notations on perfectoid fields defined in [Perfectoid Fields](#).
- Use notations on adic spaces defined in [Adic Spaces](#).

$\lrcorner$

### 1 Basics

#### Affinoid Perfectoid Spaces

**Def.(27.4.1.1) [Perfectoid Spaces].** For any  $(R, R^+) \in \mathcal{P}\text{erfd}$ , we associated to it an affinoid adic space  $\text{Spa}(R, R^+)$ , called an **affinoid perfectoid space**.

Tate's acyclicity(27.4.2.19) shows that the adic spectra of perfectoid affinoid algebras are sheafy, so we can defined the category of **perfectoid spaces**  $\mathcal{P}\text{erfdSpa}$  to be the full subcategory of  $\text{AdicSpa}$  consisting of spaces locally isomorphic to affinoid perfectoid spaces.  $\lrcorner$

**Remark(27.4.1.2).** Notice that it is not true that if  $(A, A^+) \in \text{HubPair}$  and  $\text{Spa}(A, A^+) \in \mathcal{P}\text{erfdSpa}$ , then  $A \in \mathcal{P}\text{erfd}$ . Thus there is ambiguity to the term affinoid perfectoid spaces. But we always use this to mean the affinoid adic space associated to a perfectoid Huber pair.  $\lrcorner$

*Proof:* Cf.[Ked19]P62.  $\square$

**Prop.(27.4.1.3).**  $\mathcal{P}\text{erfdSpa}/\mathbb{F}_p$  is stable under absolute products.  $\lrcorner$

*Proof:* Cf.[Sch17]P71.  $\square$

**Prop.(27.4.1.4) [Fiber Products of Perfectoid Spaces].**  $\mathcal{P}\text{erfdSpa}_K$  admits fiber products, which is compatible with that in  $\text{AdicSpa}/K$ .  $\lrcorner$

*Proof:* Perfectoid spaces are constructed by glueing, thus it suffices to show that the category of perfectoid  $K^\flat$ -algebras has fiber pushouts, by tilting equivalence. For this, if  $X = (A, A^+), Y = (B, B^+), Z = (C, C^+)$ , define  $X \otimes_Y Z = (D, D^+)$ , where  $D$  is the completion of  $A \otimes_B C$ , and  $D^+$  is the completion of the integral closure of  $A^+ \otimes_{B^+} C^+$  in  $D$ . Then  $D$  is a perfect  $K^\flat$ -algebra, and it is truly the filtered colimits, by(12.2.5.23).

Notice that we don't even need a base field  $K$ , Cf.[Ked19]P57.  $\square$

#### Tiltings

**Prop.(27.4.1.5) [Tilting Rational Subsets].** For  $(K, K^+) \in \mathcal{P}\text{erfdField}$  and  $(R, R^+) \in \mathcal{P}\text{erfd}_K$ ,

- The  $\sharp$  map induces an isomorphism  $X = \text{Spa}(R, R^+) \cong X^\flat = \text{Spa}(R^\flat, R^{b+})$  that identifies rational open subsets.
- For a rational subset  $U$  with tilting  $U^\flat$ , the complete affinoid Tate algebra  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  is perfectoid over  $R$  with tilt  $(\mathcal{O}_{X^\flat}(U^\flat), \mathcal{O}_{X^\flat}^+(U^\flat))$ .

$\lrcorner$

*Proof:* This follows from (27.4.1.9).  $\square$

**Lemma (27.4.1.6) [Huber's Presheaf in Char  $p$ ].** Assume  $\text{char} K = p$  and  $U = X(\frac{f_1, \dots, f_n}{g})$  is a rational subset that  $f_i, g \in R^+$  and  $f_n = \varpi^N$ , then:

- Consider the subring  $R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}]$ , its  $\varpi$ -adic completion  $(R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}])^a$  is a perfectoid  $K^{0a}$ -algebra.
- The map  $R^+[X_i^{\frac{1}{p^\infty}}] \rightarrow R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}]$  has kernel containing and almost equal to  $I = (g^{\frac{1}{p^m}} X_i^{\frac{1}{p^m}} - f_i^{\frac{1}{p^m}})$ .
- $\mathcal{O}_X(U)$  is a perfectoid  $K$ -algebra and  $\mathcal{O}_X(U)^{0a} \cong (R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}])^a$ .  $\perp$

*Proof:* 1: The ring  $R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}]$  is perfect and  $\varpi$ -torsion-free, and  $R^+$  is semi-perfect, thus its completion is clearly a perfectoid  $K^{0a}$ -algebra by (12.2.9.1).

2: Clearly  $I \subset \ker$  and notice  $R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}][\varpi^{-1}] = R[g^{-1}]$  as  $f_n = \varpi^N$ , so  $I[\varpi^{-1}] = \ker[\varpi^{-1}]$ . Now consider the mapping

$$P_0 = R^+[X_i^{\frac{1}{p^\infty}}]/I \rightarrow R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}]$$

Now this map is an isomorphism after inverting  $\varpi$ , so the kernel is  $\varpi^\infty$ -torsion. But we have  $I = I^{[p]}$  because  $R^+$  is semi-perfect, so  $P_0$  is perfect, so the kernel must be almost zero.

3: Consider the inclusion  $R^+[\frac{f_i}{g}] \hookrightarrow R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}]$ , we show the cokernel is killed by  $\varpi^{nN}$ : as  $f_n = \varpi^N$ ,

$$\varpi^{nN} \prod_{i=1}^n (\frac{f_i}{g})^{\frac{1}{p^{a_i}}} = \prod_{i=1}^n (f_i^{\frac{1}{p^{a_i}}} g^{1-\frac{1}{p^{a_i}}}) \frac{f_n}{g} \in R^+[\frac{f_i}{g}].$$

So these two ring has the same  $\varpi$ -adic completion, the first one is just  $\mathcal{O}_X(U)$  by the construction (27.3.3.1), so  $\mathcal{O}_X(U)$  is perfectoid  $K$ -algebra, and the isomorphism is by tilting equivalence  $\text{perf}_K \cong \text{perf}_{K^{0a}}$  (12.2.9.5).  $\square$

**Lemma (27.4.1.7) [Huber's Presheaf in Char 0].** Let  $U = X(\frac{f_1, \dots, f_n}{g})$  is a rational subset that  $f_i, g$  are perfect elements in  $R^+$ ,  $f_i = a_i^\sharp, g = b^\sharp$ , and  $f_n = \varpi^N$ , so  $f_i, g$  have compatible  $p^n$ -th roots, then let  $U^b = X^b(\frac{f_1, \dots, f_n}{g})$  be the tilting of  $U$ ,  $U$  is the inverse image of  $U^b$  along the map  $X \rightarrow X^b$ . Then the conclusion of (27.4.1.6) is also true, and moreover,  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  tilts to  $(\mathcal{O}_{X^b}(U^b), \mathcal{O}_{X^b}^+(U^b))$ .  $\perp$

*Proof:* 1, 2 of (27.4.1.6): Notation as before, there is a map  $P_0 = R^+[X_i^{\frac{1}{p^\infty}}]/I \rightarrow R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}]$ , and an inclusion  $R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}] \rightarrow \mathcal{O}_X^+(U)$ . Now write  $(S, S^+)$  for the untilt of the perfectoid  $(R^b, R^{b+})$ -algebra  $(\mathcal{O}_{X^b}(U^b), \mathcal{O}_{X^b}^+(U^b))$ , then by the tilting process (12.2.9.13),  $\text{Spa}(S, S^+)$  maps into  $U$ , so by the universal property, there is a map

$$\mu : (\mathcal{O}_X(U), \mathcal{O}_X^+(U)) \rightarrow (S, S^+).$$

Consider the composition

$$P_0 \xrightarrow{a_0} R^+[(\frac{f_i}{g})^{\frac{1}{p^\infty}}] \xrightarrow{d_0} S^+,$$

we prove their completion gives the same  $K^{0a}$ -algebras(notice  $S^+$  is already complete):  $a_0$  is surjective, thus so does its completion, the map  $d_0 \circ a_0$  is almost isomorphism modulo  $\varpi$  by(27.4.1.6) item2 and tilting equivalence, so does its completion. Now(5.7.3.2) tells us the completion of  $d_0 \circ a_0$  is almost isomorphism, so does  $a$  and  $d$  as  $a$  is surjective.

By the way, we know that  $R^+\langle(\frac{f_i}{g})^{\frac{1}{p^\infty}}\rangle[\varpi^{-1}]$  is the untilt of  $\mathcal{O}_{X^b}^+(U^b)$ .

3 of(27.4.1.6) is proved as before.

For the tilting, by the above, we already know  $R^+\langle(\frac{f_i}{g})^{\frac{1}{p^\infty}}\rangle$  tilts to the perfectoid  $K^{b0a}$ -algebra  $\mathcal{O}_{X^b}(U^b)^{0a}$ , and by item3  $\mathcal{O}_X(U)$  tilts to the perfectoid  $K^b$ -algebra  $\mathcal{O}_{X^b}(U^b)$ . Now the question is the tilt of  $\mathcal{O}_X^+(U)$ , notice as in the proof of item1, there is a natural map

$$(R^+\langle(\frac{f_i}{g})^{\frac{1}{p^\infty}}\rangle[\varpi^{-1}], R^+\langle(\frac{f_i}{g})^{\frac{1}{p^\infty}}\rangle) \rightarrow (\mathcal{O}_X(U), \mathcal{O}_X^+(U)),$$

whose tilting gives by universality of Huber's presheaf a map

$$\xi : (\mathcal{O}_{X^b}(U^b), \mathcal{O}_{X^b}^+(U^b)) \rightarrow (\mathcal{O}_X(U), \mathcal{O}_X^+(U))^b.$$

These two map  $\mu, \xi$  are inverse to each other, showing that the tilting of  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  is  $(\mathcal{O}_{X^b}(U^b), \mathcal{O}_{X^b}^+(U^b))$ .  $\square$

**Lemma(27.4.1.8)[Approximation Lemma].** Assume  $R = K\langle T_0^{\frac{1}{p^\infty}}, \dots, T_n^{\frac{1}{p^\infty}} \rangle$ ,  $f \in R^0$  is homogenous of degree  $d \in \mathbb{N}[p^{-1}]$ , then for any  $c > 0, \varepsilon > 0$ , there exists some  $g_{c,\varepsilon} \in R^{b0}$  homogeneous of degree  $d$  that

$$|(f - g^\sharp)(x)| \leq |\varpi|^{1-\varepsilon} \max\{|f(x)|, |\varpi|^c\}.$$

In particular, if  $\varepsilon < 1$ , then

$$\max\{|f(x)|, |\varpi|^c\} = \max\{|g_{c,\varepsilon}^\sharp(x)|, |\varpi|^c\}.$$

┘

*Proof:* Cf.[Scholze Perfectoid Spaces, Lemma6.5] ?.  $\square$

**Prop.(27.4.1.9).** For any  $R \in \text{Perfd}_K$ ,

- The same conclusion of(27.4.1.8) holds.
- For  $f, g \in R$ , there exist  $a, b \in R^b$  that  $X(\frac{f, \varpi^c}{g}) = X(\frac{a^\sharp, \varpi^c}{b^\sharp})$ . In particular, any rational subsets  $U$  of  $X$  comes from  $X^b$ , thus(27.4.1.7) applies for  $U$ .
- For any  $x \in X$ ,  $\kappa(x)^\wedge \in \text{PerfdField}$ .
- $X \rightarrow X^b$  is a homeomorphism preserving rational subsets.

┘

*Proof:* 1: Using the tilting equivalence, we can write  $f = g_0^\sharp + \varpi g_1^\sharp + \dots + \varpi^c g_c^\sharp + f_{c+1} \varpi^{c+1}$ , let  $f_0 = g_0^\sharp + \varpi g_1^\sharp + \dots + \varpi^c g_c^\sharp$ . Then notice that the solution  $g_{c,\varepsilon}$  for  $f_0$  is suitable for  $f$  as well. So now if we consider the mapping

$$\mu : K\langle T_0^{\frac{1}{p^\infty}}, \dots, T_n^{\frac{1}{p^\infty}} \rangle \rightarrow R : T_i \rightarrow g_i^\sharp,$$

together with its tilting

$$\mu^b : K^b \langle T_0^{\frac{1}{p^\infty}}, \dots, T_n^{\frac{1}{p^\infty}} \rangle \rightarrow R^b.$$

Then for the approximation  $g_{c,\varepsilon}^\#$  for  $f' = \sum \varpi^i T_i$ ,  $\mu^b(g)$  is what we are searching for.

2: Using 1, we find  $a, b \in R^b$  that  $|(g - b^\#)(x)| < \max\{|g(x)|, \varpi^c\}$  and  $\max\{|f(x)|, |\varpi|^c\} = \max\{|a^\#(x)|, |\varpi|^c\}$  for all  $x \in \text{Spa}(R, R^+)$ . Then it is routine to check that  $X(\frac{f, \varpi^c}{g}) = X(\frac{a^\#, \varpi^c}{b^\#})$ .

3: By (27.3.3.5),  $\widehat{k^+(x)}$  equals the completion of the colimit  $\text{colim } \mathcal{O}_X^+(U)$  over rational subsets  $U$  containing  $x$ . As these are all perfectoid  $K^{0a}$ -algebras (12.2.9.14), and completion of the filtered limits of perfectoid  $K^{0a}$ -algebras is perfectoid (12.2.9.10), we know that  $\widehat{k^+(x)}$  is perfectoid over  $K^{0a}$ , thus inverting  $\varpi$  shows  $k(x)$  is also perfectoid over  $K$ .

4: this is injective because  $X$  is  $T_0$  and a rational subset is the untilt of a rational subset of  $X^b$  by item 2. For surjectivity, a point of  $X^b$  determines a continuous map  $(R^b, R^{b+}) \rightarrow (\widehat{k(x)}, \widehat{k^+(x)})$ , thus by untilting (12.2.9.13) corresponds to a map  $(R, R^+) \rightarrow (L, L^+)$ , then  $(L, L^+)$  is a perfectoid field, by (12.2.9.15), so it corresponds to a point  $y \in X$ . Then it is clear that  $y$  maps to  $x$ , because the

$$\begin{array}{ccc} R^b & \xrightarrow{\#} & R \\ \downarrow & & \downarrow \\ \kappa(x)^\wedge & \xrightarrow{\#} & L \end{array} \quad \text{is commutative.}$$

□

**Prop. (27.4.1.10) [Tilting Equivalence for Perfectoid Spaces].** Given  $K \in \text{PerfdField}$ , then for any  $X \in \text{PerfdSpa}/K$ , there is a unique  $X^b \in \text{PerfdSpa}/K^b$  that satisfies:  $X(R, R^+) \cong X^b(R^b, R^{b+})$  functorially, called the **tilt** of  $X$ . Moreover,  $|X| \cong |X^b|$ .

When  $X$  is affinoid, this tilting coincides with that of (27.4.1.5).

And this tilting induces an equivalence between the category of perfectoid spaces over  $K$  and perfectoid spaces over  $K^b$ . ┘

*Proof:* Firstly, the universal property truly determines the tilt  $X^b$  uniquely: if there are two tilts  $X_1, X_2$ , as they are locally affinoid perfectoid like  $\text{Spa}(R^b, R^{b+})$  by (27.4.1.5), the immersion map  $\text{Spa}(R^b, R^{b+}) \rightarrow X_1$  determines via the functorial isomorphism a morphism  $\text{Spa}(R^b, R^{b+}) \rightarrow X_2$ . Now  $X_1$  has a sheaf structure, so these morphisms glue to give a morphism  $X_1 \rightarrow X_2$ . The same argument shows conversely there is a morphism  $X_2 \rightarrow X_1$ , and they are clearly converse to each other, so  $X_1 \cong X_2$ .

The construction of  $X$  is just the glueing of the tilting of the affinoid perfectoid spaces, as the tilting defined in (27.4.1.5) is a functor. The universal property is verified by just checking on the affinoid perfectoid spaces, as we can glue using the sheaf property. For the affinoid case, we should use (27.3.4.6). The last assertion is by (27.4.1.5). □

## 2 Properties of Perfectoid Spaces

### Totally Disconnected Spaces

**Def. (27.4.2.1) [Totally Disconnected Spaces].** A perfectoid space  $X$  is called **totally disconnected** if it is qcqs and any open covering  $\{U_i \rightarrow X\}$  splits, i.e.  $\coprod U_i \rightarrow X$  splits, or equivalently, there is a refinement covering  $\{V_i \rightarrow X\}$  that  $X \cong \coprod V_i$ .

A perfectoid space  $X$  is called **strictly totally disconnected** if it is qcqs and every étale cover splits. ┘

**Prop. (27.4.2.2).** If  $X \in \mathcal{P}\text{erfdSpa}$  is qcqs, then  $X$  is totally disconnected iff all its connected components are of the form  $\text{Spa}(K, K^+)$  where  $(K, K^+) \in \mathcal{P}\text{erfdField}$ . And it is strictly totally disconnected if moreover  $K = \overline{K}$ .  $\lrcorner$

*Proof:* Cf.[Sch17].P29, P35.  $\square$

**Prop. (27.4.2.3).** If  $X \in \mathcal{P}\text{erfdSpa}$  is a totally disconnected, then  $X$  is affinoid.  $\lrcorner$

*Proof:* Cf.[Sch17].P30.  $\textcolor{red}{?}$   $\square$

### Injections

**Def. (27.4.2.4) [Injections].** A map  $f : X \rightarrow Y \in \mathcal{P}\text{erfdSpa}$  is called an **injection** if it is a monomorphism.  $\lrcorner$

**Prop. (27.4.2.5) [Residue Field Map is Injection].** Let  $X$  be a perfectoid space and  $x \in X$ , giving rise to a map of residue fields

$$i_x : \text{Spa}(\kappa(x), \kappa(x)^+) \rightarrow X,$$

then  $i_x$  is an injection of perfectoid spaces.  $\lrcorner$

*Proof:* To show this, firstly we can replace  $X$  with an affinoid nbhd of  $X$ . Then notice that  $\text{Spa}(\kappa(x), \kappa(x)^+)$  is the filtered limit over all rational nbhds  $U$  of  $x$  in  $X$ , and for each  $U$ ,  $U \rightarrow X$  is an injection by definition(27.3.3.1), so  $i_x$  is also an injection.  $\square$

**Prop. (27.4.2.6) [Characterizations of Injections].** Let  $f : Y \rightarrow X$  be a map of perfectoid spaces, then the following conditions are equivalent:

- $f$  is an injection.
- For any  $(K, K^+) \in \mathcal{P}\text{erfdField}$ , the map of sets  $f_* : Y(K, K^+) \rightarrow X(K, K^+)$  is injective.
- The map  $|f| : |Y| \rightarrow |X|$  is injective, and for all rank 1 point  $y \in Y$  with image  $f(y) = x \in X$ , the map of completed residue fields  $\kappa(x) \rightarrow \kappa(y)$  is an isomorphism.
- The map  $|f| : |Y| \rightarrow |X|$  is injective, and  $f$  is final in the category of maps  $Z \rightarrow X$  that  $|Z| \rightarrow |X|$  factors through the map  $|Y| \rightarrow |X|$ .

In particular, by item4, an injection of perfectoid spaces is determined by its topological map.  $\lrcorner$

*Proof:*  $4 \rightarrow 1 \rightarrow 2$  is trivial. For the rest, Cf.[Sch17]P21.  $\square$

**Prop. (27.4.2.7) [Injection and Base Change].**

- Let  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$  be an injection, and  $X' \rightarrow X$  any map of perfectoid spaces, then the pullback  $f' : Y' = Y \times_X X' \rightarrow X'$  is also an injection, and the induced map

$$|Y'| \rightarrow |Y| \times_{|X|} |X'|$$

is a homeomorphism.

- A map of perfectoids spaces is an injection iff it is universally injective.  $\lrcorner$

*Proof:* Cf.[Sch17]P24.  $\square$

### Immersion

**Def. (27.4.2.8) [Immersion].**  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$  is called an **immersion of perfectoid spaces** if  $f$  is an injection and  $|f| : |Y| \rightarrow |X|$  is a locally closed immersion. If  $|f|$  is moreover closed or open, then it is called closed/open immersion.  $\lrcorner$

**Def. (27.4.2.9) [Zariski-Closed Immersion].** For  $f : Z \rightarrow X \in \mathcal{P}\text{erfdSpa}$  where  $X = \text{Spa}(R, R^+)$  is affinoid,

- $f$  is called a **Zariski-closed immersion of perfectoid spaces** if  $f$  is a closed immersion and  $|Z| = V(I) \subset |X|$ , where  $I \subset R$  is an ideal.
- $f$  is called a **strongly Zariski-closed immersion of perfectoid spaces** if  $Z = \text{Spa}(S, S^+)$  is affinoid,  $R \rightarrow S$  is surjective, and  $S^+$  is the closure of  $R^+$  in  $S$ .

$\lrcorner$

**Prop. (27.4.2.10).**

- If  $f$  is strongly Zariski-closed, then  $f$  is Zariski-closed, in particular a closed immersion.
- If  $f$  is Zariski-closed, then  $Z$  is affinoid.
- If  $X$  is of characteristic  $p$ , and  $f$  is Zariski-closed, then  $f$  is strongly Zariski-closed.

$\lrcorner$

*Proof:* Cf. [Sch12] Section 2.5.  $\square$

**Prop. (27.4.2.11).** For  $Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$ , the diagonal map  $\Delta_f : Y \rightarrow Y \times_X Y$  is an immersion.  $\lrcorner$

*Proof:* Clearly  $\Delta_f$  is an injection, thus it suffices to show that  $|\Delta_f|$  identifies  $Y$  with a locally closed subset of  $|Y \times_X Y|$ . This can be checked locally on the target, so we can assume  $X = \text{Spa}(R, R^+)$  and  $Y = \text{Spa}(S, S^+)$ , then the diagonal map is strongly Zariski closed, as  $S \hat{\otimes}_R S \rightarrow S$  is surjective and maps the integral closure of  $S^+ \hat{\otimes}_{R^+} S^+ \rightarrow S^+$  onto  $S^+$ . Thus by (27.8.1.11),  $\Delta_f$  is a closed immersion in this case.  $\square$

**Def. (27.4.2.12) [Separated Map].** A map  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$  is called separated if  $\Delta_f$  is a closed immersion.  $\lrcorner$

**Prop. (27.4.2.13) [Valuation Criterion].** For  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$ , the following are equivalent:

- $f$  is separated.
- $|\Delta_f| : |Y| \rightarrow |Y \times_X Y|$  is a closed immersion.
- $|f|$  is quasi-separated, and for any perfectoid adic field  $(K, K^+)$  and any diagram

$$\begin{array}{ccc} \text{Spa}(K, \mathcal{O}_K) & \longrightarrow & Y \\ \downarrow & \nearrow & \downarrow f \\ \text{Spa}(K, K^+) & \longrightarrow & X \end{array}$$

$\lrcorner$

### Almost Acyclicity

**Def.(27.4.2.14) [ $p$ -Finite Tate Ring].** Denote  $L = \mathbb{F}_p[[t]]_{\text{perf}}^\wedge[t^{-1}]$ , an  $\mathbb{F}_p[t]$  algebra  $A^+$  is called **algebraically admissible** if it is f.p., reduced,  $t$ -torsion-free, and integrally closed in  $A^+[t^{-1}]$ . A perfectoid affinoid  $L$ -algebra  $(R, R^+)$  is called  **$p$ -finite** if it is the completion of the perfection(27.3.2.22) of a uniform Tate ring of the form  $(A^+[t^{-1}], A^+)$ , where  $A^+$  is algebraically admissible.  $\lrcorner$

**Lemma(27.4.2.15) [Tate's Acyclicity for Classical Affinoid Algebra].** If  $A^+$  is an algebraically admissible  $\mathbb{F}_p[t]$ -algebra, then  $(A^+[t^{-1}], A^+)$  is a uniform affinoid Tate algebra(because it is finite), and:

- For any rational subset  $U \subset X$ , the structure presheaf  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  is also uniform, and it is a perfection of an algebraically admissible  $\mathbb{F}_p[t]$ -algebra, so  $\mathcal{O}_X^+(U) = \mathcal{O}_X(U)^0$ .
- For any covering  $\mathfrak{U} : U_i \rightarrow X$  of rational subsets, the Čech cohomology groups  $H^i(\mathfrak{U}, \mathcal{O}_X^+)$  are all killed by  $t^N$  for  $N$  large.
- $(A, A^+)$  is sheafy, with  $H^i(X, \mathcal{O}_X^+)$  being  $t^\infty$ -torsion for all  $i$ .

$\lrcorner$

*Proof:* ?

$\square$

**Lemma(27.4.2.16) [Tate's Acyclicity for  $p$ -Finite Perfectoid Algebras].** Let  $(R, R^+)$  be a  $p$ -finite perfectoid  $L$ -algebra that comes from the completion of perfection of  $(A, A^+)$ , then:

- The map  $X = \text{Spa}(R, R^+) \rightarrow Y = \text{Spa}(R, R^+)$  is a homeomorphism.
- For rational subset  $V \subset Y$  with preimage  $U \subset X$ ,  $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$  is the completion of the perfection of  $(\mathcal{O}_Y(V), \mathcal{O}_Y^+(V))$ .
- For any covering  $\mathfrak{U} : U_i \rightarrow X$  of rational subsets, the Čech cohomology groups  $H^i(\mathfrak{U}, \mathcal{O}_X^0)$  are all almost zero.
- $(R, R^+)$  is sheafy, with  $H^i(X, \mathcal{O}_X^+)$  almost zero for all  $i > 0$ .

$\lrcorner$

*Proof:* 1: This is because the adic spectrum is insensitive for perfection(27.3.2.22) and completion(27.3.2.17).

2: This is by the universal property, as these two are both the universal elements for the complete and affinoid adic spaces mapping to  $X$  that factors through  $U$ .

3: The complex calculating  $H^i(\mathfrak{U}, \mathcal{O}_X^0)$  is just the complex calculating  $H^i(\mathfrak{U}, \mathcal{O}_Y^+)$  under completion of perfection((27.3.2.21) used). So(27.4.2.15) and(27.4.2.17)(applied to every element) shows that the perfection makes the complex almost acyclic, and this is preserved under completion as  $(-)^a$  is exact.

4: The complex calculating  $H^i(\mathfrak{U}, \mathcal{O}_X)$  is just the complex calculating  $H^i(\mathfrak{U}, \mathcal{O}_X^+)$  inverting  $t$ , thus they are all 0 as localization is exact. For the second, it is because of item3 and the fact  $\mathcal{O}_X^+$  is almost isomorphic to  $\mathcal{O}_X^0$ (12.2.9.14).  $\square$

**Lemma(27.4.2.17).** Let  $A$  be a ring with an element  $t$  that admits compatible  $p^n$ -th roots, then for an  $A$ -module  $M$  that  $t^N M = 0$ , consider the Frobenius pushforward  $M \rightarrow F_* M$ , then the colimit  $\text{colim}_{F_*} F_*^n M$  is naturally a module over  $A_{\text{perf}}$ , and it is annihilated by  $t^{\frac{1}{p^n}}$  for all  $n$ .  $\lrcorner$

*Proof:* The  $A_{\text{perf}}$  structure is natural, and notice  $F_*^k M$  is annihilated by  $t^{\frac{N}{p^k}}$ , thus naturally the colimit is annihilated by  $t^{\frac{1}{p^n}}$  for all  $n$ .  $\square$

**Prop. (27.4.2.18) [Noetherian Approximation in Char  $p$ ].** If  $K \in \mathcal{P}\text{erfdField}^p$  with pseudo-uniformizer  $t$ , then  $K$  is an extension of  $L = \mathbb{F}_p[[t]]_{\text{perf}}[t^{-1}]$ , and If  $A$  is an  $K^0$ -perfectoid algebra that is integrally closed in  $A[t^{-1}]$ , then:

- $A$  is a completion of a filtered colimit  $(\text{colim}_i B_i)^\wedge$  that  $B_i$  are  $p$ -finite, that induces an homeomorphism

$$\text{Spa}(A[t^{-1}], A) \cong \lim_i \text{Spa}(B_i[t^{-1}], B_i)$$

that each rational subset of  $\text{Spa}(A[t^{-1}], A)$  comes from a rational subset of some  $\text{Spa}(B_i[t^{-1}], B_i)$ .

- If  $U_i \subset \text{Spa}(B_i[t^{-1}], B_i)$  is a compatible system of rational subsets that corresponds to  $U \subset \text{Spa}(A, A^+) = X$ , then

$$(\mathcal{O}_X(U), \mathcal{O}_X^+(U)) \cong \varinjlim_j (\mathcal{O}_j(U_j), \mathcal{O}_j^+(U_j))^\wedge.$$

⌋

*Proof:* 1:  $A = \text{colim}_i A_i$ , where  $A_i$  are all the f.p.  $\mathbb{F}_p[t]$ -algebras in  $A$ . Then each  $A_i$  is reduced (as  $A$  is complete and integrally closed in  $A[t^{-1}]$ ) and  $t$ -torsion-free, and we can assume they are integrally closed in  $A_i[t^{-1}]$  because  $A$  does, by passing to their integral closure.

Then applying the  $(-)_{\text{perf}}$  functor gives  $\text{colim}_i (A_i)_{\text{perf}} = A$ , as  $A$  is perfect, and applying the completion gives

$$(\text{colim}_i \widehat{A_i})^\wedge = A,$$

as  $A$  is already complete, so we are done.

2: This is immediate from 1 and (27.3.2.21).

3: This is because by universal property for Huber presheaves, there are pushouts diagrams

$$\begin{array}{ccccccc} (B_i[t^{-1}], B_i) & \longrightarrow & (B_j[t^{-1}], B_j) & \longrightarrow & \dots & \longrightarrow & (A[t^{-1}], A) \\ \downarrow & & \downarrow & & & & \downarrow \\ (\mathcal{O}_{X_i}(U_i), \mathcal{O}_{X_i}^+(U_i)) & \longrightarrow & (\mathcal{O}_{X_j}(U_j), \mathcal{O}_{X_j}^+(U_j)) & \longrightarrow & \dots & \longrightarrow & (\mathcal{O}_X(U), \mathcal{O}_X^+(U)) \end{array}$$

So the conclusion follows as colimits commutes with colimits. □

**Prop. (27.4.2.19) [Almost Acyclicity for Affinoid Perfectoids].** Fix  $K \in \mathcal{P}\text{erfdField}$  and  $(R, R^+) \in \mathcal{P}\text{erfd}^{\text{aff}}/K$  with adic spectrum  $X = \text{Spa}(R, R^+)$ , then

- $(R, R^+)$  is sheafy.
- $\mathcal{O}_X^+(X) = R^+$ , and  $H^i(X, \mathcal{O}_X^+)$  is almost zero for any  $i \in \mathbb{Z}_+$ .
- $\mathcal{O}_X(X) = R$ , and  $H^i(X, \mathcal{O}_X) = 0$  for any  $i \in \mathbb{Z}_+$ .

⌋

*Proof:* As in the proof of (27.4.2.16), it suffices to prove  $\mathcal{O}_X^0$  is almost exact w.r.t any covering  $\mathfrak{U}$ . For this, notice each term is  $\varpi$ -adically complete and flat by (27.4.1.5), so it suffices to prove it is almost exact modulo  $\varpi$  (5.7.3.2). Then by the tilting equivalence, it suffices to prove for  $X^\flat$ . So we may assume at first that  $K$  is of char  $p$ . Then we may replace  $K$  by  $L = \mathbb{F}_p[[t]]_{\text{perf}}[t^{-1}]$ .

But then Noetherian approximation (27.4.2.18) shows that the rational subrings are completion of filtered colimits of  $p$ -finite  $K$ -algebras ((27.3.2.21) used), and then we reduced to the  $p$ -finite case, as in the proof of (27.4.2.16). □



### 3 Étale Sites of Perfectoid Spaces

**Def. (27.4.3.1) [Strongly Finite Étale Maps].** For convenience, for perfectoid affinoid  $K$ -algebras, we call a map of Huber pairs  $(A, A^+) \rightarrow (B, B^+)$  **strongly finite étale** if it is finite étale and  $B^{+a}$  is almost finite étale over  $A^{+a}$ .

A map  $f : X \rightarrow Y \in \text{PerfdSpa}$  is called **strongly finite étale** if there is a cover of  $Y$  by affinoids  $V_i \subset Y$  s.t.  $U_i = f^{-1}(V_i)$  are all affinoids, and the map  $(\mathcal{O}_Y(V_i), \mathcal{O}_Y^+(V_i)) \rightarrow (\mathcal{O}_X(U_i), \mathcal{O}_X^+(U_i))$  is strongly finite étale. For  $X \in \text{AdicSpa}$ , denote by  $X_{\text{sfét}}$  the category of strongly finite étale spaces over  $X$ .

A map  $X \rightarrow Y \in \text{PerfdSpa}$  is called **strongly étale** if for any  $x \in X$ , there exists an open subspace  $U \subset X$  containing  $x$  and an open subspace  $V \subset Y$  containing  $f(U)$  together with a  $W \in \text{PerfdSpa}$  that  $f : U \rightarrow V$  factors through an open embedding  $U \rightarrow W$  and a finite étale map  $W \rightarrow V$ .

Eventually we will prove that if  $(A, A^+)$  is perfectoid, then any finite étale map  $(A, A^+) \rightarrow (B, B^+)$  is strongly finite étale.  $\perp$

**Prop. (27.4.3.2) [Strongly Finite Étale is Local].** If  $f : X \rightarrow Y$  is a strongly finite étale map that  $Y = \text{Spa}(A, A^+)$  is an affinoid perfectoid, then  $X$  is also affinoid perfectoid, and the structure map  $(\mathcal{O}_X(X), \mathcal{O}_X^+(X)) \rightarrow (\mathcal{O}_Y(Y), \mathcal{O}_Y^+(Y))$  is strongly finite étale.  $\perp$

*Proof:* By (27.4.3.9), it suffices to prove in char  $p$ . Then we can replace  $K$  by  $L = \widehat{\mathbb{F}_p[[t]]}_{\text{perf}}[t^{-1}]$ . Then by Noetherian approximation (27.4.2.18), we can assume that  $Y$  is a limit of  $p$ -finite affinoids  $\text{Spa}(B_i, B_i^+)$ . As both rational subsets and finite étale algebras pass through filtered colimit, and adic spectrum is quasi-compact (27.3.2.4), we can assume that a finite étale cover of  $Y$  arises through base change of some  $\text{Spa}(B_i, B_i^+)$ . So it suffices to prove the proposition in case of  $Y$   $p$ -finite. Then  $Y$  is a completion of perfection of some algebraically admissible ring over  $\mathbb{F}_p[t]$ . Then by the above argument again, we can assume that  $Y$  is algebraically admissible.

Now a classical theorem (Cf. [Étale cohomology of rigid analytic varieties and adic spaces, Huber 1.6.6(2)]) shows that the finite étale cover of  $Y$  is global finite étale  $\text{Spa}(S, S^+) \rightarrow \text{Spa}(R, R^+)$  in this case. Notice the strongness is not needed because we are working in char  $p$ , where almost purity theorem is already proven.  $\square$

**Cor. (27.4.3.3).** For an affinoid perfectoid space  $Y = \text{Spa}(R, R^+)$ , the functor  $X \mapsto \mathcal{O}_X^+(X)$  defined an equivalence of categories  $Y_{\text{sfét}} \cong R_{\text{afét}}^{+a}$ , and the functor  $X \mapsto \mathcal{O}_X(X)$  gives a fully faithful functor  $Y_{\text{sfét}} \rightarrow R_{\text{fét}}$ .  $\perp$

**Prop. (27.4.3.4) [Properties of Étale Maps].**

- Strongly étale/strongly finite étale maps of perfectoid spaces are stable under base change, and if  $Z \rightarrow Y$  is strongly étale, then  $|X \times_Z Y| \rightarrow |X| \times_{|Z|} |Y|$  is surjective.
- If  $X = \text{Spa}(A, A^+)$ ,  $Y = \text{Spa}(B, B^+)$ ,  $Z = \text{Spa}(C, C^+)$  are all affinoid perfectoids and  $Z \rightarrow Y$  is strongly finite étale, then  $X \times_Z Y = \text{Spa}(D, D^+)$ , where  $D = A \otimes_B C$  and  $D^+$  is the integral closure of  $C^+$  in  $D$ .
- For any diagram  $X \rightarrow Y \leftarrow Z \in \text{AdicSpa}/K$  where  $X \rightarrow Y$  is étale/finite étale and  $Z \in \text{PerfdSpa}$ , the fiber product  $X \times_Y Z$  is also a perfectoid space, and  $X \times_Y Z \rightarrow Z$  is étale/finite étale. Moreover, the map  $|X \times_Y Z| \rightarrow |X| \times_{|Y|} |Z|$  is surjective.
- If  $X = \text{Spa}(A, A^+)$ ,  $Y = \text{Spa}(B, B^+)$ ,  $Z = \text{Spa}(C, C^+) \in \text{AdicSpa}^{\text{aff}}/K$  and  $X \rightarrow Y$  is finite étale, then  $X \times_Z Y = \text{Spa}(D, D^+)$ , where  $D = A \otimes_B C$  and  $D^+$  is the integral closure of  $C^+$  in  $D$ .

*Proof:* Cf.[Scholze Perfectoid Spaces]P40. ┘

**Prop. (27.4.3.5).** Strongly (finite)étale maps of perfectoid spaces are stable under compositions. ┘

*Proof:* Cf.[Scholze Perfectoid Spaces]P42. ? ┘

**Def. (27.4.3.6) [Étale Sites of Perfectoid Spaces].** For  $X \in \text{AdicSpa}$ , the étale site of  $X$  is the category  $X_{\text{ét}}$  of perfectoid spaces that is étale over  $X$  with coverings given by topological coverings. This is truly a site by (27.4.3.4), and it is functorial in  $X \in \text{AdicSpa}$ . ┘

**Prop. (27.4.3.7) [Gabber-Ramero].** If  $A$  is a finite  $K^0$ -algebra that is  $\varpi$ -adically Henselian, then

$$A[\varpi^{-1}]_{\text{fét}} \cong \widehat{A}[\varpi^{-1}]_{\text{fét}}.$$

*Proof:* Cf.[Almost Ring Theory P5.4.53]. ┘

**Cor. (27.4.3.8) [Finite Étale Covers and Direct Limits of Complete Uniform Rings].** Let  $(A_i, A_i^+)$  be a filtered system of complete uniform affinoid  $K$ -algebras, and  $(A, A^+)$  be their colimit in the category of complete uniform affinoid Tate rings, then

$$2\text{-colim}_i A_{i,\text{fét}} \cong A_{\text{fét}}$$

as categories. ┘

*Proof:* By (27.3.2.21),  $A^+$  is the  $\varpi$ -adic completion of the algebraic colimit  $B^+$  of  $A_i^+$ , and  $A = A^+[\varpi^{-1}]$ . Each  $A_i$  is complete and  $\varpi$ -torsion-free, thus the colimit is Henselian and  $\varpi$ -torsion-free (5.3.10.3)(5.3.10.6). Then the proposition (27.4.3.7) shows that  $B^+[\varpi^{-1}]_{\text{fét}} \cong A_{\text{fét}}$ . Now it remains to show that

$$2\text{-colim}_i A_{i,\text{fét}} \cong B^+[\frac{1}{\varpi}]_{\text{fét}},$$

which is because étale sites commutes with taking filtered colimits ┘

### Final Proof of Almost Purity Theorem

**Prop. (27.4.3.9).** There is an equivalence of categories  $X_{\text{sfét}} \cong X_{\text{sfét}}^{\flat}$ . ┘

*Proof:* For this, use (27.4.3.3), (27.4.1.10) and the proven part of (12.2.10.1) and notice that

$$A_{\text{afét}}^{+a} = A_{\text{afét}}^{0a} \cong A_{\text{afét}}^{\flat 0a} = A_{\text{afét}}^{\flat +a},$$

and the integral closure clearly corresponds. (It get around the problem that  $R_{\text{fét}}^{0a} \rightarrow R_{\text{fét}}$  hasn't been proven essentially surjective). ┘

**Prop. (27.4.3.10) [Final Proof of Almost Purity Theorem].** For  $(R, R^+) \in \text{Perfd}/K$ , if  $S \in R_{\text{fét}}$ , then the integral closure of  $S^+$  in  $R^+$  lies in  $R_{\text{afét}}^{+a}$ , and this gives an inverse to the morphism  $d$  in (12.2.10.1), thus finishing the proof of almost purity theorem. ┘

*Proof:* Continuing the proof of(12.2.10.1), it suffices to show that  $d : R_{\text{afét}}^{+a} \rightarrow R_{\text{fét}}$  is essentially surjective, because  $S^+ \rightarrow \bar{S} \subset R^+$  is the only possible inverse, by the almost purity theorem in char  $p$  and tracing the tilting equivalence(12.2.9.6). Given(27.4.3.3), it suffices to prove that for  $X = \text{Spa}(R, R^+)$ , the prestacks  $X_{\text{sfét}} \cong X_{\text{fét}}$ , where  $X_{\text{sfét}}(U) = \mathcal{O}_X^{+a}(U)_{\text{sfét}}$ , and  $X_{\text{fét}}(U) = \mathcal{O}_X(U)_{\text{fét}}$ .

We use(7.1.3.24), firstly  $X_{\text{sfét}}$  is a stack, by(27.4.3.2), and for each  $U$ ,  $X_{\text{sfét}}(U) \rightarrow X_{\text{fét}}(U)$  is fully faithful by almost purity theorem(12.2.10.1).  $X_{\text{fét}}$  is separated by(27.3.4.24), because the structure section of an element  $S \in X_{\text{fét}}$  is determined its value on the stalk.

Its left to prove that their stalks are equal, for this, use the formula

$$\text{colim}_{x \in U} (\mathcal{O}_X(U), \mathcal{O}_X^+(U)) = (\widehat{\kappa(x)}, \widehat{\kappa^+(x)})$$

in the category of complete uniform affinoid  $K$ -algebras(they are all perfectoids(27.4.1.5) thus uniform), by definition. So we get by(27.4.3.8):

$$\text{colim}_{x \in U} \mathcal{O}_X(U)_{\text{fét}} \cong \widehat{k(x)}_{\text{fét}},$$

and by(27.4.3.8) together with the proven part of almost purity theorem(12.2.10.1):

$$\text{colim}_{x \in U} \mathcal{O}_X^+(U)_{\text{afét}} \cong \text{colim}_{x^b \in U^b} \mathcal{O}_X^{b+}(U^b)_{\text{afét}} \cong \kappa^+(x^b)_{\text{afét}} \cong \kappa^+(x)_{\text{afét}}.$$

Now we have already proved the almost purity over fields(12.2.10.1) which says  $\kappa(x^b)_{\text{afét}} \cong \kappa^+(x)_{\text{afét}}$ , so their stalks are the same.  $\square$

**Cor.(27.4.3.11) [Invariance of Étale Sites under Tilting].** There is a natural isomorphism of categories  $X_{\text{ét}} \cong X_{\text{ét}}^b$ , by almost purity theorem(12.2.10.1) and the localness of étale maps.  $\lrcorner$

**Thm.(27.4.3.12) [Tate's Almost Acyclicity, Scholze[Sch12]].** For any  $X \in \text{PerfdSpa}$ , the functor  $U \mapsto \mathcal{O}_X(U)$  is a sheaf on  $X_{\text{ét}}$ , and  $H_{\text{ét}}^i(X, \mathcal{O}_X^+)$  is almost zero if  $X$  is affinoid perfectoid.  $\lrcorner$

*Proof:* Cf.[Scholze Perfectoid Spaces]P43.  $\textcolor{red}{?}$   $\square$

## 4 Perfectoid Pro-Étale Morphisms

References are [Scholze, Rigid Analytic Varieties].

**Def.(27.4.4.1) [Perfectoid Pro-Étale Morphisms].** For  $X \in \text{AdicSpa}$  and  $U \in X_{\text{pro-ét}}$ , then

- $U$  is called **affinoid pro-étale perfectoid** over  $X$  if  $U$  has a pro-étale presentation(27.3.6.2)  $U = \varprojlim U_i \rightarrow X$  by affinoids  $U_i = \text{Spa}(R_i, R_i^+)$  s.t.  $(R, R^+) = (\varinjlim (R_i, R_i^+))^\wedge \in \text{Perfd}_K$ .
- $U$  is called **pro-étale perfectoid** over  $X$  if  $U$  has an open cover(27.3.6.3) by affinoid pro-étale perfectoids  $V \subset U$ .

$\lrcorner$

**Prop.(27.4.4.2) [Associated Perfectoid Spaces].** Let  $X \in \text{AdicSpa}$  and  $U = \varprojlim U_i \in X_{\text{pro-ét}}$  affinoid pro-étale perfectoid over  $X$ , where  $U_i = \text{Spa}(R_i, R_i^+)$  and  $(R, R^+) = (\varinjlim (R_i, R_i^+))^\wedge$ , then we can associate  $U^\wedge = \text{Spa}(R, R^+)$ . In this way, we get a functor from affinoid pro-étale perfectoid spaces to affinoid perfectoid spaces.  $\lrcorner$

**Prop.(27.4.4.3).** Let  $X \in \text{AdicSpa}$  be locally Noetherian,  $U \in X_{\text{pro-ét}}$  be perfectoid over  $X$ , and  $V \rightarrow U$  pro-étale, then  $V$  is also perfectoid over  $X$ .  $\lrcorner$

*Proof:*

□

**Prop. (27.4.4.4) [Colmez].** Let  $X$  be a locally Noetherian adic space over  $\mathrm{Spa}(K, K^+)$ , then the set of affinoid pro-étale perfectoid spaces over  $X$  form a basis for  $X_{\mathrm{pro-ét}}$ . ┘

*Proof:* Cf. [Scholze, rigid analytic varieties] P25. □

**Prop. (27.4.4.5).** Assume  $X \in \mathrm{AdicSpa}$  and  $U \in X_{\mathrm{pro-ét}}$  is affinoid pro-étale perfectoid over  $X$  with  $U^\wedge = \mathrm{Spa}(R, R^+)$ , then

- For any  $b \in K^+$ ,  $\mathcal{O}_X^+(U)/(b) = R^+/(b)$ , and this is almost equal to  $(\mathcal{O}_X^+(b))(U)$ .
- For any nonzero  $b_1, b_2 \in K^+$  s.t.  $|b_1| < |b_2|$ , the image of  $(\mathcal{O}_X^+(b_1))(U)$  in  $(\mathcal{O}_X^+(b_2))(U)$  is equal to  $R^+/(b_2)$ .
- $\widehat{\mathcal{O}}_X^+(U) = R^+$ ,  $\widehat{\mathcal{O}}_X(U) = R$ .
- $\widehat{\mathcal{O}}_X^+(U)$  is the completion of  $\mathcal{O}_X^+(U)$ .
- For any  $i \in \mathbb{Z}_+$ , the cohomology groups  $H^i(U, \widehat{\mathcal{O}}_X^+)$  are almost zero.

In particular, for any affinoid pro-étale perfectoid  $U \in X_{\mathrm{pro-ét}}$ ,  $(|U|, \mathcal{O}_X|_{|U|}, \{|-|_x\}_{x \in U})$  is a perfectoid adic space. ┘

*Proof:*

□

**Prop. (27.4.4.6).** Assume  $X \in \mathrm{AdicSpa}$  and  $U \in X_{\mathrm{pro-ét}}$  is affinoid pro-étale perfectoid over  $X$  with  $U^\wedge = \mathrm{Spa}(R, R^+)$ . Let  $\mathbb{L} \in \mathbb{F}_p\text{-LocSys}$ , then

$$H^i(U, \mathbb{L} \otimes \mathcal{O}_X^+/(p))$$

is almost zero for  $i \in \mathbb{Z}_+$ , and almost finitely generated over  $R^+/(p)$  for  $i = 0$ . ┘

*Proof:* Cf. [Scholze P27]. □

### Odd Definition

Cf. [Sch12].

**Def. (27.4.4.7) [Original Definition of Perfectoid Pro-Étale Morphisms].** For  $X \in \mathrm{PerfdSpa}$  and  $\{X_\alpha\}$  a filtered inverse system of Noetherian adic spaces (27.3.4.7) with a compatible system of maps  $\varphi_\alpha : X \rightarrow X_\alpha$ , we write

$$X \sim \varprojlim X_\alpha$$

if  $|X| \cong \varprojlim |X_\alpha|$ , and for any  $x \in X$ , the map of residue fields  $\varinjlim \kappa(\varphi_i(x)) \rightarrow \kappa(x)$  have dense images. In particular,  $X$  is qcqs, by (4.12.3.15). ┘

**Prop. (27.4.4.8).** Suppose  $X \in \mathrm{PerfdSpa}$  and  $X \sim \varprojlim X_i$ . If  $Y \rightarrow X_i$  is an étale map of Noetherian adic spaces, then  $Y \times_{X_i} X \sim \varprojlim_{j \geq i} Y \times_{X_i} X_j$ . ┘

*Proof:* Cf. [Scholze Perfectoid] P44. □

**Prop. (27.4.4.9).** If  $X \in \mathrm{PerfdSpa}$  and  $X \sim \varprojlim X_i$ , then

$$\mathrm{Sh}(X_{\mathrm{ét}}) \cong \varinjlim \mathrm{Sh}(X_{i,\mathrm{ét}}).$$

┘

*Proof:* Cf.[Scholze Perfectoid]P44. □

**Cor.(27.4.4.10).** If  $X \in \mathcal{P}\text{erfdSpa}$  and  $X \sim \varprojlim X_i$ , then for any sheaf  $\mathcal{F}_i \in \mathcal{S}h(X_{i,\text{ét}})$  with pullbacks  $\mathcal{F}_j \in \mathcal{S}h(X_{j,\text{ét}})$  and  $\mathcal{F} \in \mathcal{S}h(X)$ , then the natural mapping

$$\varinjlim_{j \geq i} H^i(X_{j,\text{ét}}, \mathcal{F}_j) \rightarrow H^i(X, \mathcal{F})$$

is an isomorphism for any  $n \in \mathbb{N}$ . ┘

*Proof:* Cf.[Scholze Perfectoid]P44. □

**Cor.(27.4.4.11).** If  $X \in \mathcal{P}\text{erfdSpa}$  and  $X \sim \varprojlim X_i$  and all the transition maps are homeomorphisms and induce purely inseparable maps on completed residue fields, then  $X_{\text{ét}}$  is isomorphic to  $X_{i,\text{ét}}$  for any  $i$ . ┘

*Proof:* Cf.[Scholze Perfectoid]P44. □

## 5 Applications

### Weight Monodromy Conjecture

**Thm.(27.4.5.1)** [Sch[Sch12]]. Let  $K \in p\text{-NField}$  and  $X \in \mathcal{S}m\mathcal{P}rpr\mathcal{V}ar/K$  is a set-theoretic complete intersection in a smooth projective toric variety, then the weight monodromy conjecture is true for  $X$ .( ? i.e. for any  $i \in \mathbb{N}$ , the monodromy filtration on the étale cohomology group  $H_{\text{ét}}^i(X_{\overline{K}}, \mathbb{Q}_\ell)$  is pure of weight  $i$ ). ┘

*Proof:* ? □

### Commutative Algebra

The proof of Hochster's direct summand conjecture(André and Bhatt).

The proof of Hochster's conjecture on the existence and weak functoriality of big Cohen-Macaulay algebras(André, Heitmann-Ma).

The  $p$ -adic analogue of Kunz's characterization of regular rings through their Frobenius endomorphism(Bhatt-Iyengar-Ma).

The proof of the Auslander–Goldman–Grothendieck purity conjecture on the Brauer group of schemes (Česnavičius).

## 27.5 Crystalline Cohomology

References are [Sta] and [Berthelot-Ogus, Notes on Crystalline Cohomology]

**Notation (27.5.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $\text{NilSch}$  be the subcategory of  $\text{Sch}$  consisting of schemes s.t.  $p$  is locally nilpotent.

┘

### 1 PD-Schemes

**Def. (27.5.1.1) [PD-Schemes].** A **pd-scheme** is a triple  $(S, \mathcal{I}, \gamma)$  where

- $S \in \text{Sch}$ .
- $\mathcal{I}$  is a  $\mathbb{Q}\text{co}$  sheaf of ideals on  $S$ .
- $\gamma$  is a pd-structure on  $\mathcal{I}$  (5.6.1.1).

A morphism of pd-schemes is a morphism that all the structure morphisms are morphisms of pd-structures.

Equivalently a pd-scheme is a triple  $(S_0, S, \gamma)$  where  $S_0$  is a closed subscheme of  $S$  with ideal sheaf  $\mathcal{I}$  s.t.  $(S, \mathcal{I}, \gamma)$  is a pd-scheme.

┘

**Def. (27.5.1.2) [PD-Thickening].** A **pd-thickening** is a triple  $(U, T, \delta)$  where  $T$  is a thickening of  $U$  (7.8.6.1) with sheaf of ideals  $\mathcal{I}$  (i.e.  $U = \mathbf{Spec}(\mathcal{I})$ ) s.t.  $(T, \mathcal{I}, \delta)$  is a pd-scheme (27.5.1.1).

┘

**Prop. (27.5.1.3) [Pullback of pd-Thickenings].** Pullback of pd-thickenings (27.5.1.2) exist in the category of pd-schemes.

┘

*Proof:* Cf. [Sta] 07ME.

□

### 2 Crystalline Sites

**Def. (27.5.2.1) [Coverings].** A family of morphisms  $\{(U_i, T_i, \delta_i) \rightarrow (U, T, \delta)\}$  of pd-thickenings (27.5.1.2) is called a Zariski/smooth/étale/syntomic/fppf... covering iff

- $U_i = U \otimes_T T_i$ ,
- $\{T_i \rightarrow T\}$  is a Zariski/smooth/étale/syntomic/fppf... covering of  $T$ .

┘

**Def. (27.5.2.2) [Crystalline Sites].** For a pd-scheme  $(S, \mathcal{I}, \gamma)$  over  $\mathbb{Z}_{(p)}$  (27.5.1.1),  $S_0 = V(\mathcal{I}) \subset S$  and  $X \in \text{NilSch}/S_0$ , the **big crystalline site**  $(X/S)_{\text{Crys}}$  consists of pd-thickenings (27.5.1.2)  $(U, T, \delta)$  over  $(S, \mathcal{I}, \delta)$  and a morphism  $U \rightarrow X \in \text{Sch}/S$ , and the topology is the Zariski-topology (27.5.2.1). In fact for any  $(U, T, \delta) \in (X/S)_{\text{Crys}}$ ,  $p$  is locally nilpotent in  $T$ , by (5.6.1.8).

$$\begin{array}{ccccc} U & \longrightarrow & X & \longrightarrow & S_0 \\ \downarrow & & & & \downarrow \\ T & \longrightarrow & & & S \end{array}$$

The **crystalline site**  $(X/S)_{\text{crys}}$  is the strictly full subcategory of  $(X/S)_{\text{Crys}}$  consisting of objects  $(U, T, \delta)$  s.t.  $U \in X_{\text{Zar}}$ .

Notice the **structure sheaf**  $\mathcal{O}_{X/S} : (U, T, \delta) \in (X/S)_{\text{Crys}} \mapsto \Gamma(T, \mathcal{O}_T)$  is a sheaf of rings on  $(X/S)_{\text{Crys}}$  or  $(X/S)_{\text{Crys}}$ . There is another sheaf of rings given by  $\mathcal{O}_X : (U, T, \delta) \in (X/S)_{\text{Crys}} \mapsto \Gamma(U, \mathcal{O}_U)$ , and there is a natural map  $\mathcal{O}_{X/S} \rightarrow \mathcal{O}_X$  with kernel  $\mathcal{J}_{X/S}$  equipped with a pd-structure.  $\perp$

**Remark (27.5.2.3) [Infinitesimal Sites].** Situation as in (27.5.2.2), we can also define the (big) **infinitesimal site**  $(X/S)_{\text{inf}}$  by similar definition but without pd-structures.  $\perp$

**Prop. (27.5.2.4) [Finite Limits].** The category  $(S/X)_{\text{Crys}}$  has all finite non-empty limits, and the forgetful functor  $(U, T, \delta) \rightarrow U$  preserves finite non-empty limits.  $\perp$

*Proof:* Cf. [Sta]07I9. ?  $\square$

**Prop. (27.5.2.5) [Comparing with Zariski-Sites].** The functors

$$u_{X/S} : (X/S)_{\text{Crys}} \rightarrow X_{\text{Zar}} : (U, T, \delta) \rightarrow U, \quad U_{X/S} : (X/S)_{\text{Crys}} \rightarrow \text{Sch}_{\text{Zar}}/X : (U, T, \delta) \mapsto U$$

are cocontinuous (easy to verify) and commutes with products and fibered products by (27.5.2.4), thus defines a morphism of topoi

$$u_{X/S} : \text{Sh}((X/S)_{\text{Crys}}) \rightarrow \text{Sh}(X_{\text{Zar}})$$

by (7.1.2.21), which is functorial in  $X/S$ .  $\perp$

**Prop. (27.5.2.6) [Functoriality of Crystalline Sites].** For a morphism of pd-scheme  $(S', \mathcal{I}', \gamma') \rightarrow (S, \mathcal{I}, \gamma)$  over  $\mathbb{Z}_{(p)}$  (27.5.1.1), denote  $S_0 = V(\mathcal{I}) \subset S$ ,  $S'_0 = V(\mathcal{I}') \subset S'$ . Suppose there is a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ S'_0 & \longrightarrow & S_0 \end{array} \in \text{Sch}$$

and assume that  $p$  is locally nilpotent on  $X'$  and  $X$ , then there are morphisms of topoi

$$f_{\text{Crys}} : \text{Sh}((X'/S')_{\text{Crys}}) \rightarrow \text{Sh}((X/S)_{\text{Crys}}), \quad f_{\text{crys}} : \text{Sh}((X'/S')_{\text{crys}}) \rightarrow \text{Sh}((X/S)_{\text{crys}}).$$

$\perp$

*Proof:* Cf. [Sta]07IK. ?  $\square$

**Prop. (27.5.2.7) [Sheaves on Crystalline Sites].** Situation as in (27.5.2.2), denote  $\mathcal{C} = (X/S)_{\text{Crys}}$  or  $(X/S)_{\text{crys}}$ . For each  $(U, T, \delta) \in \mathcal{C}$  and  $\mathcal{F} \in \text{Sh}(\mathcal{C})$ , the restriction

$$\mathcal{F}_T : W \in T_{\text{Zar}} \mapsto \mathcal{F}(U \cap W, W, \delta|_W)$$

is an Abelian sheaf on  $T_{\text{Zar}}$ . And for  $f : (U, T, \delta) \rightarrow (U', T', \delta') \in \mathcal{C}$ , there is a canonical comparison map

$$c_f : f^{-1}\mathcal{F}_{T'} \rightarrow \mathcal{F}_T$$

whose adjoint  $\mathcal{F}_{T'} \rightarrow f_*\mathcal{F}_T$  on each  $W' \in T_{\text{Zar}}$  is induced by

$$f_{f^{-1}(W')} : (U \cap f^{-1}(W'), f^{-1}(W'), \delta|_{f^{-1}(W')}) \rightarrow (U' \cap W', W', \delta|_{W'}).$$

Notice  $c_f$  is an isomorphism if  $f$  is an open immersion. Conversely, if we are given a sheaf  $f_T \in \text{Sh}(T_{\text{Zar}})$  for each  $(U, T, \delta) \in \mathcal{C}$  and comparison maps  $c_f$  for each morphism  $f : (U, T, \delta) \rightarrow (U', T', \delta') \in \mathcal{C}$  s.t.

- $c_f$  is an isomorphism if  $f$  is an open immersion.
- Certain cocycle conditions hold?.

Then these  $f_T$  comes from a sheaf  $f \in \text{Sh}(\mathcal{C})$ . The proof is the same as that of [Sta]0213?.

**Prop. (27.5.2.8)[Crystalline Topoi are Sub-Canonical].** Situation as in (27.5.2.2),  $\text{Sh}((X/S)_{\text{crys}})$  is sub-canonical.

*Proof:* ?

**Def. (27.5.2.9)[Affine Crystalline Site].** Let  $(A, I, \gamma)$  be a pd-structure that  $A$  is a  $\mathbb{Z}_{(p)}$ -algebra, and  $C$  is an  $A/I$ -algebra that  $p$  is nilpotent in  $C$ , then the **affine crystalline site**  $(C/A)_{\text{Crys}}^{\text{aff}}$  is the site whose object are pd-structures  $(B, J, \delta)$  over  $(A, I, \gamma)$  that  $p$  is nilpotent in  $B$  (5.6.1.8), together with a map of rings  $C \rightarrow B/J$  over  $A/I$ , and  $(C/A)_{\text{Crys}}^{\text{aff}}$  the full subcategory of  $(C/A)_{\text{Crys}}^{\text{aff}}$  consisting of objects  $(B, J, \delta)$  s.t.  $C \rightarrow B/J$  is an isomorphism.

Notice for any object  $(B, J, \delta)$  in  $(C/A)_{\text{Crys}}$ ,  $J$  is nilpotent, by (5.6.1.8).

### Sheaf of Differentials

**Def. (27.5.2.10) [PD- $S$ -Derivations].** For  $\mathcal{F} \in \text{Sh}(\mathcal{O}_{X/S, \text{crys}})$ , an  $S$ -derivation  $D : \mathcal{O}_{X/S} \rightarrow \mathcal{F}$  is a map of sheaves that for any object  $(U, T, \delta)$  of  $(X/S)_{\text{crys}}$ , the map  $D : \Gamma(T, \mathcal{O}_T) \rightarrow \Gamma(T, \mathcal{F})$  is a pd-derivation over  $\Gamma(V, \mathcal{O}_V)$  for any open subset  $V \subset S$  that  $T \rightarrow S$  factors through  $V$ .

**Def. (27.5.2.11) [Zariski-Sheaf of Modules of PD-Differentials].** Similar to the construction of sheaf of differentials (7.2.4.2), for each pd-structure  $(U, T, \delta)$  over  $S \in \text{Sch}$ , we can construct the **sheaf of pd-differentials**  $\Omega_{T/S, \delta}$  on  $T_{\text{Zar}}$ , which is a quotient of the sheaf of differentials  $\Omega_{X/S}$  by the submodule generated by local sections  $\{d_{T/S}(\delta_n(t)) - \delta_{n-1}(t)d_{T/S}(t)\}_{n \in \mathbb{Z}_+}$ , so it is affine-locally given by the module of pd-differentials (5.6.2.1).

**Def. (27.5.2.12) [Crystalline Sheaf of Modules of PD-differentials].** For a pd-scheme  $(S, \mathcal{I}, \gamma)$  over  $\mathbb{Z}_{(p)}$  (27.5.1.1),  $S_0 = V(\mathcal{I}) \subset S$  and  $X \in \text{NilSch}/S_0$ , there exists  $\Omega_{X/S, \text{crys}} \in \mathcal{QCoh}(\mathcal{O}_{X/S, \text{crys}})$  s.t. for any  $(U, T, \delta) \in (X/S)_{\text{crys}}$ , we have

$$\Omega_{X/S, \text{crys}}|_{(U, T, \delta)} = \Omega_{T/S, \delta} \quad (27.5.2.11).$$

*Proof:* By (27.5.2.7) and (27.5.3.2), it suffices to check that for each morphism  $f : (U, T, \delta) \rightarrow (U', T', \delta')$  in  $(X/S)_{\text{crys}}$ , the morphism of  $\mathcal{O}_{T'}$ -modules  $c_f : f^* \mathcal{F}_{T'} = f^{-1} \mathcal{F}_{T'} \otimes \mathcal{O}_T \rightarrow \mathcal{F}_T$  is an isomorphism of  $\mathcal{O}_T$ -modules.

**Prop. (27.5.2.13)[Functorial First-Order PD-Thickenings].** Let  $(U, T, \delta) \in (X/S)_{\text{crys}}$ ,  $\mathcal{J}$  the ideal sheaf of  $U$ , we define a first order thickening  $T'$  of  $T$ : let  $\mathcal{O}_{T'} = \mathcal{O}_T \oplus \Omega_{T/S, \delta}$  with the algebraic structure that  $\Omega_{T/S, \delta}^2 = 0$ , and let  $\mathcal{J}' = \mathcal{J} \otimes \Omega_{T/S, \delta}$ , and define the pd-structure as

$$\delta'_n(f, \omega) = (\delta_n(f), \delta_{n-1}(f)\omega).$$

Then  $(U, T, \delta')$  is a pd-thickening and  $(U, T, \delta) \rightarrow (U, T', \delta')$  is a morphism in  $(X/S)_{\text{crys}}$ . Moreover, there are two ring maps

$$p_0, p_1 : \mathcal{O}_T \rightarrow \mathcal{O}_{T'} : p_0(f) = (f, 0), \quad p_1(f) = (f, d_{T/S, \delta}(f))$$



Then we get two contraction of the morphism  $T \rightarrow T'$ , and  $p_0^* - p_1^*$  is the universal derivation  $d_{T/S, \delta}$  included in  $\mathcal{O}_{T'}$ .

This construction is functorial in  $T/S$  by (27.5.2.12) and hence gives a functor of sites  $(X/S)_{\text{crys}} \rightarrow (X/S)_{\text{crys}}$ .  $\perp$

*Proof:* The verification of the pd-axioms in in [Sta]07HH.  $\square$

**Prop. (27.5.2.14) [Functorial Second-Order PD-Thickenings].** There is a further thickening  $T''$  of  $T'$ , which is a second order thickening of  $T$ :

$$\Omega_{T''} = \mathcal{O}_T \oplus \Omega_{T/S, \delta} \oplus \Omega_{T/S, \delta} \oplus \Omega_{T/S, \delta}^2$$

with the algebra structure given by

$$(f, \omega_1, \omega_2, \eta)(f', \omega'_1, \omega'_2, \eta') = (ff', f\omega'_1 + f'\omega_1, f\omega'_2 + f'\omega_2, f\eta' + f'\eta + \omega_1 \wedge \omega'_2 + \omega'_1 \wedge \omega_2)$$

Let  $\mathcal{J}'' = \mathcal{J} \oplus \Omega_{T/S, \delta} \oplus \Omega_{T/S, \delta} \oplus \Omega_{T/S, \delta}^2$ , then there is a PD-structure on  $\mathcal{J}''$  given by

$$\delta_n''(f, \omega_1, \omega_2, \eta) = (\delta_n(f), \delta_{n-1}(f)\omega_1, \delta_{n-1}(f)\omega_2, \delta_{n-1}(f)\eta + \delta_{n-1}(f)\omega_1 \wedge \omega_2).$$

This construction is functorial in  $T/S$  by (27.5.2.12) and hence gives a functor of sites  $(X/S)_{\text{crys}} \rightarrow (X/S)_{\text{crys}}$ .  $\perp$

*Proof:* For the details, Cf. [Sta]07J3.  $\square$

### 3 Crystals

**Def. (27.5.3.1) [Crystals].** In situation (27.5.2.2), denote  $\mathcal{C} = (X/S)_{\text{Crys}}$  or  $(X/S)_{\text{crys}}$ ,  $\mathcal{F} \in \text{Mod}(\mathcal{O}_{X/S})$  is called

- an  $\mathcal{O}_{X/S}$ -**crystal** if for any morphism  $f : (U, T, \delta) \rightarrow (U', T', \delta')$  in  $(X/S)_{\text{crys}}$ , the morphism of  $\mathcal{O}_{T'}$ -modules  $c_f : f^* \mathcal{F}_{T'} = f^{-1} \mathcal{F}_{T'} \otimes \mathcal{O}_T \rightarrow \mathcal{F}_T$  is an isomorphism of  $\mathcal{O}_T$ -modules.
- **locally quasi-coherent** if  $\mathcal{F}_T \in \mathcal{QCoh}(\mathcal{O}_T)$  for any  $(U, T, \delta) \in (X/S)_{\text{crys}}$ .
- **quasi-coherent** if  $f \in \mathcal{QCoh}(\mathcal{C})$  as defined in (7.2.2.25).

In particular,  $\mathcal{O}_{X/S}$  is a crystal in  $\mathcal{O}_{X/S}$ -modules.  $\perp$

#### Crystals in Qco modules

**Def. (27.5.3.2) [Quasi-Coherent Crystals].** Situation as in (27.5.3.1), an  $\mathcal{O}_{X/S}$ -module  $\mathcal{F}$  is called a **quasi-coherent crystal** if it satisfies the following equivalent conditions:

- $\mathcal{F} \in \mathcal{QCoh}(\mathcal{C})$ .
- $\mathcal{F}$  is locally Qco (27.5.3.1) and it is a crystal in  $\mathcal{O}_{X/S}$ -modules.

Moreover,  $\mathcal{F}$  is called a **crystal in finite locally free modules** if moreover  $\mathcal{F}$  is finite locally free.  $\perp$

*Proof:* Cf. [Sta]07IT?.  $\square$

**Def. (27.5.3.3) [Notations in Polynomial case].** If in situation (27.5.2.9),  $S = \text{Spec } A, X = \text{Spec } C$ , we let  $P \rightarrow C$  be a surjection of  $A$ -algebras with  $P = A[X_i]$ , and the kernel is  $J$ . Set  $D = D_{P, \gamma}(J)^\wedge$

be the  $p$ -adically completed pd-envelope, then  $(D, \widehat{J}, \widehat{\gamma})$  admits a natural pd-structure, by (5.6.1.15). Let  $D_e = D/p^e$  and  $J_e$  the image of  $\widehat{J}$  in  $D_e$ . Denote

$$\Omega_D = (\Omega_{D/A, \gamma})^\wedge = (\Omega_{D_{P, \gamma}(J)/A, \bar{\gamma}})^\wedge = \lim_e \Omega_{D(n)_e/A, \bar{\gamma}} \quad (5.6.2.4).$$

By (5.6.2.4),  $\Omega_{D_{P, \gamma}(J)/A, \bar{\gamma}} = \Omega_{P/A} \otimes_P D_{P, \gamma}(J)$  which is free over  $D_{P, \gamma}(J)$  on  $dx_i$ , so  $\Omega_D$  is topologically free over  $D$  on  $dx_i$ , and there is a universal derivation  $d : D \rightarrow \Omega_D$ .

Now let  $J(n) = \ker(P \otimes_A \otimes_A \dots \otimes_A P \rightarrow C)$  where the tensor has  $n+1$  factors, and

$$D(n) = (D_{P \otimes_A \otimes_A \dots \otimes_A P/A, \gamma}(J(n)))^\wedge$$

with divided ideals  $\widehat{J}(n)$ , and also  $D(n)_e = D(n)/p^e$ ,  $T(n)_e = \text{Spec } D(n)_e$ , then  $(X, T(n)_e, \bar{\gamma}(n))$  is a pd-thickening by (5.6.1.8) for  $e$  large (5.6.1.15), by (5.6.1.8) as  $p$  is nilpotent in  $X$ . And

$$\Omega_{D(n)} = (\Omega_{D(n)/A, \bar{\gamma}(n)})^\wedge.$$

Then  $D(0) = D, D(1), \dots$  form a cosimplicial pd-structures.

Let  $\mathcal{F}$  be a sheaf of  $\mathcal{O}_{X/S}$ -modules, denote

$$M(n) = \lim_e (\Gamma((X, T(n)_e, \bar{\gamma}(n)), \mathcal{F})).$$

┘

**Prop. (27.5.3.4).** Notation as in (27.5.3.3), there is an isomorphism

$$D(n) \cong (D\langle \xi_i(j) \rangle)^\wedge$$

where  $\xi_i(j) = X_i \otimes 1 \otimes \dots \otimes 1 - 1 \otimes 1 \otimes \dots \otimes X_i \otimes 1 \otimes \dots \otimes 1$ .

┘

*Proof:* There is an isomorphism  $P \otimes_A \dots \otimes_A P \cong P[\xi_i(j)]$ , and  $J(n)$  is just generated by  $JP \otimes_A \dots \otimes_A P + (\xi_i(j))$ , so this theorem follows from (5.6.1.20).  $\square$

### Crystalline Connections

**Def. (27.5.3.5) [deRham Complexes for  $(X/S)_{\text{crys}}$ ].** Situation as in (27.5.2.2), if we define  $\Omega_{X/S, \delta}^i = \wedge^i \Omega_{X/S, \delta}$  (27.5.2.12), then by (9.2.1.2), the universal  $S$ -derivative  $d_{X/S}$  give rises to the **pd-deRham-complex**

$$\mathcal{O}_{X/S} \rightarrow \Omega_{X/S, \delta}^1 \rightarrow \Omega_{X/S, \delta}^2 \rightarrow \dots$$

as  $\mathcal{O}_{X/S}$ -modules on  $(X/S)_{\text{crys}}$ .

┘

*Proof:* The verification of the condition for the quotient  $\Omega_X \rightarrow \Omega_{X/S, \delta}$  is routine.  $\square$

**Def. (27.5.3.6) [Connections on  $(X/S)_{\text{crys}}$ ].** Situation as in (27.5.2.2), we define the notion of connection on  $\mathcal{F} \in \text{Sh}(\mathcal{O}_{X/S, \text{crys}})$  w.r.t. the sheaf of differentials  $\Omega_{X/S, \delta}$  (27.5.2.12), as in (9.2.1.5), i.e. a map

$$\nabla : \mathcal{F} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_{X/S, \text{crys}}} \Omega_{X/S, \delta} \in \text{Sh}((X/S)_{\text{crys}})$$

s.t. for any local section  $s$  of  $\mathcal{F}$  and local section  $f$  of  $\mathcal{O}_{X/S, \text{crys}}$ , we have

$$\nabla(f \otimes s) = f \otimes \nabla(s) + s \otimes d_{X/S} f \quad (27.5.3.5).$$

┘

**Prop.(27.5.3.7) [Canonical Integrable Connections attached to Crystals].** Any  $\mathcal{F} \in \text{Crys}(\mathcal{O}_{X/S, \text{crys}})$  is equipped with a canonical integrable connection (27.5.3.6).  $\lrcorner$

*Proof:* For any  $(U, T, \delta) \in (X/S)_{\text{crys}}$ , consider the first order thickening  $(U, T', \delta')$  given in (27.5.2.13), then there are two projections  $p_0, p_1 : T' \rightarrow T$  and a inclusion  $i : T \rightarrow T'$ , then by the property of crystals we get isomorphisms

$$p_0^* \mathcal{F}_T \xrightarrow{c_0} \mathcal{F}_{T'} \xleftarrow{c_1} p_1^* \mathcal{F}_T$$

then  $\nabla(s) = p_1^*(s) - c_1^{-1} c_0(p_0^*(s))$  vanishes after pulling back to  $T$  via  $i^*$ , so it is in the kernel of  $i^*$ , which is

$$\mathcal{F}_T \otimes_{\mathcal{O}_T} \Omega_{T/S}$$

by the construction of  $T'$  (27.5.2.13). This  $\nabla$  is functorial in  $T$  as everything is functorial, hence gives a connection on  $\mathcal{F}$ .

For the integrability, Cf. [Sta]07J6.  $\color{red}?$   $\square$

**Cor.(27.5.3.8) [PD-deRham-Complexes].** For  $\mathcal{F} \in \text{Crys}(\mathcal{O}_{X/S, \text{crys}})$ , we can define a pd-deRham-complex

$$\mathcal{F} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_{X/S}} \Omega_{X/S, \delta} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_{X/S, \delta}} \Omega_{X/S, \delta}^2 \rightarrow \dots \in \text{Sh}((X/S)_{\text{crys}}).$$

attached to the canonical integral connection (27.5.3.7).  $\lrcorner$

**Thm.(27.5.3.9) [Integrable Connections and PD-Stratifications].** If  $S \in \text{Sch}/\mathbb{Q}$  and  $X \in \text{Sch}^{\text{sm}}/S$ , then for  $(\mathcal{V}, \nabla) \in \text{Conn}(X/S)$ ,  $\nabla$  is integrable iff it extends to a stratification  $\color{red}?$ .

If  $S \in \text{Sch}^p$  and  $X \in \text{Sch}^{\text{sm}}/S$ , then for  $(\mathcal{V}, \nabla) \in \text{Conn}(X/S)$ ,  $\nabla$  is integrable iff it extends to a pd-stratification  $\color{red}?$ .  $\lrcorner$

*Proof:*  $\color{red}?$   $\square$

### Crystals and Connections

**Thm.(27.5.3.10) [Qco Crystals and Topologically Quasi-Nilpotent Connections, Grothendieck [Gro74]].** Notation as in (27.5.3.3), there is an equivalent between  $\mathcal{QCoh}(\mathcal{O}_{X/S, \text{crys}})$  and the category of pairs  $(M, \nabla)$  s.t.

- $M$  is a  $p$ -adically complete  $D$ -module.
- $\nabla : M \rightarrow M \hat{\otimes}_D \Omega_D$  is an integrable connection.
- $\nabla$  is **topological quasi-nilpotent**: for any  $m \in M$ , there are only f.m. pairs  $(i, k)$  that  $\nabla_{\partial/\partial x_i}^k(m) \notin pM$ .

$\lrcorner$

*Proof:* For a crystal, we associate to it  $M = M(0)$  defined in (27.5.3.3). let  $\Gamma((X, T(0)_e, \bar{\gamma}(0)), \mathcal{F}) = M_e$ , then because  $\mathcal{F}$  is a crystal in qco sheaves,  $\mathcal{F}_{T(0)_e} = \widetilde{M_e}$ , and  $M_e = M_{e+1}/p^e M_{e+1}$ , thus  $M_e = M/p^e M$  and  $M$  is  $p$ -adically complete by (5.2.3.6). By evaluating the natural connection defined in (27.5.3.8) on  $T_e$  and take limit, then we get an integral connection  $\nabla : M \rightarrow M \hat{\otimes}_D \Omega_D$ .

To show this integral is topologically quasi-nilpotent, we can do the same thing for  $M = M(n)$  for any  $n$ , and using the crystal property of  $\mathcal{F}$  and take limits, we get isomorphisms

$$M \hat{\otimes}_{D, p_0} D(1) \rightarrow M(1) \rightarrow M \hat{\otimes}_{D, p_1} D(1)$$

For the rest, Cf. [Sta]07JG.

To prove the functor is an equivalence, Cf. [Sta]07JH.  $\square$

**Def. (27.5.3.11) [Notations in Smooth Case].** Situation as in (27.5.3.3), but this time we choose a smooth  $A$ -algebra  $P'$  and  $A \rightarrow P' \rightarrow C$  with  $\ker(P' \rightarrow C) = J'$  and do the same as (27.5.3.3) again, to get

$$D' = D_{P', \gamma}(J')^\wedge$$

and

$$\Omega_{D'} = (\Omega_{D'/A, \gamma})^\wedge = (\Omega_{D_{P', \gamma}(J)/A, \bar{\gamma}})^\wedge = \lim_e \Omega_{D'(n)_e/A, \bar{\gamma}}$$

┘

**Thm. (27.5.3.12) [Crystals and Connections in Smooth case].** Situation as in (27.5.3.3), we can find a  $P = A[X_i]$  that there are maps  $a : D \rightarrow D', b : D' \rightarrow D$  between the completed pd-envelope of  $P, P'$  that  $a \circ b = \text{id}$  and compatible with the maps  $D \rightarrow C$  and  $D' \rightarrow C$ , such that the base change along  $a, b$  induces an equivalence of categories between the categories of modules with an integrable connection over  $D$  as in (27.5.3.10) and the category of modules over  $D'$  with an integrable connection. ┘

*Proof:* We can find  $P$  that  $P \rightarrow C$  factors through a surjection  $P \rightarrow P'$ , hence we get a surjection  $a : D \rightarrow D'$  the left adjointness of pd-envelope. Let  $e$  large that  $D'_e$  is a pd-thickening of  $C$  over  $A$  by (5.6.1.15)(5.6.1.8), then the kernel of  $D_e \rightarrow D'_e$  is nilpotent by (5.6.1.8), hence by the strongly lifting property (5.4.5.16) of the smooth ring map  $P \rightarrow P'$  (5.4.5.21) w.r.t. the thickening  $D_e \rightarrow D'_e$ , we find a lift  $P' \rightarrow D_e$ .

Notice  $D_{e+i+1} \rightarrow D_{e+i} \times_{D'_{e+i}} D'_{e+i+1}$  is surjective with nilpotent kernels (because  $p^{e+i}D \rightarrow p^{e+i}D'$  is surjective), we can use the smooth of  $A \rightarrow P$  to lift inductively the map  $P' \rightarrow D_e$  to a map  $P' \rightarrow D$ , thus by universal property of completed pd-envelope extends to a map  $b : D' \rightarrow D$ . It is clear that  $a \circ b = \text{id}$ .

For the equivalence of categories, Cf. [Sta]07L5. ?

□

**Remark (27.5.3.13).** In fact this proposition holds with  $P'$  being any ring that  $A \rightarrow P'$  satisfies the strong lifting property (5.4.5.16). In particular, this holds for ind-smooth  $A$ -algebras (9.5.1.5). ┘

## 4 Crystalline Cohomologies

Cf. [Sta]Chap55.24.

**Def. (27.5.4.1) [Crystalline Cohomologies].** For a pd-scheme  $(S, \mathcal{I}, \gamma)$  over  $\mathbb{Z}_{(p)}$  with  $S_0 = V(\mathcal{I}) \subset S$  and  $X \in \text{NilSch}/S_0$ , define

$$H_{\text{crys}}^i(X/S) \triangleq H^i((X/S)_{\text{crys}}, \mathcal{O}_{X/S}) \text{ (27.5.2.2).}$$

called the **crystalline cohomology** of  $X/S$ . ┘

**Def. (27.5.4.2) [Frobenius-Actions].** Situation as in (27.5.4.1), if  $(S, \mathcal{I}, \gamma) = (\text{Spec } A, \tilde{I}, \gamma)$  is affine,  $p \in I$  and  $\varphi : A \rightarrow A \in \mathcal{C}\text{Ring}$  is a Frobenius-map lifting the Frobenius-map on  $A/(p)$ . Then for each  $X \in \text{Sch}/S_0$ , by functoriality (27.5.2.6), the absolute Frobenius-map induces a morphism of topoi

$$(\varphi_X)_{\text{crys}} : (X/S)_{\text{crys}} \rightarrow (X/S)_{\text{crys}},$$

┘

**Def. (27.5.4.3) [ $F$ -Crystals].** Situation as in (27.5.4.2), an  $F$ -crystal on  $X/S$  is a pair  $(\mathcal{E}, \varphi_{\mathcal{E}})$  where  $\mathcal{E}$  is a crystal in finite locally free  $\mathcal{O}_{X/S}$ -modules (27.5.3.2) together with a map

$$\varphi_{\mathcal{E}} : \varphi_{\text{crys}}^* \mathcal{E} \rightarrow \mathcal{E} \text{ (27.5.4.2).}$$

The category of  $F$ -crystals on  $X/S$  is denoted by  $\text{F-Crys}(X/S)$ .

A **non-degenerate  $F$ -crystal** on  $X/S$  is an  $F$ -crystal on  $X/S$  that there exists a map  $V : \mathcal{E} \rightarrow \varphi_{\text{crys}}^* \mathcal{E}$  s.t.  $V \circ F_{\mathcal{E}} = p^i \text{id}$  for some  $i \in \mathbb{N}$ .  $\lrcorner$

**Thm. (27.5.4.4) [Cohomology of  $F$ -Crystals].** Situation as in (27.5.4.2), if  $A$  is  $p$ -adically complete and Noetherian and  $X/S_0$  is proper and smooth, then for each  $(\mathcal{E}, \varphi_{\mathcal{E}}) \in \text{F-Crys}(X/S)$  is non-degenerate (27.5.4.3), the canonical map

$$\varphi_{\mathcal{E}} \circ \varphi_{\text{crys}}^* : \text{R}\Gamma((X/S)_{\text{crys}}, \mathcal{E}) \otimes_{A, \varphi}^{\text{L}} A \rightarrow \text{R}\Gamma((X/S)_{\text{crys}}, \mathcal{E})$$

is an isomorphism after inverting  $p$ .  $\lrcorner$

*Proof:* Cf. [Sta]07N5. ?  $\square$

**Def. (27.5.4.5) [Higher Direct Images].** Let  $(S, \mathcal{I}, \gamma) \rightarrow (S', \mathcal{I}', \gamma')$  be a morphism of PD-schemes over  $\mathbb{Z}_{(p)}$  and  $f : X/S_0 \rightarrow X'/S'_0$  be a morphism of schemes that  $p$  is locally nilpotent on  $X$  and  $X'$ . For the rest, Cf. [Sta]07MJ. ?  $\lrcorner$

### $W(\kappa)$ -Crystalline Cohomologies

**Def. (27.5.4.6) [Crystalline Cohomologies for Schemes over  $\mathbb{F}_p$ ].** For  $\kappa \in \text{Perf}^p$  and  $X \in \text{Sch}/\kappa$ , define

$$H_{\text{crys}}^i(X/W(\kappa)) \triangleq \varprojlim_{n \in \mathbb{Z}_+} H_{\text{crys}}^i(X/W_n(\kappa)) \text{ (27.5.4.1).}$$

If  $X/\kappa$  is smooth and proper, then each  $H_{\text{crys}}^i(X/W(\kappa))_{\text{lf}}$  with the relative Frobenius-map is an  $F$ -crystal over  $\kappa$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (27.5.4.7) [Crystalline Cohomologies and Albanese-Varieties].** For  $\kappa \in \text{Perf}^p$  and  $X \in \text{Sch}^{\text{sm, proper}}/\kappa$ , the map  $X \rightarrow \text{Alb}(X)$  induces an isomorphism

$$H_{\text{crys}}^i(\text{Alb}(X)/W(\kappa)) \cong H_{\text{crys}}^i(X/W(\kappa)),$$

and  $H_{\text{crys}}^i(\text{Alb}(X)/W(\kappa))$  is the Dieudonné-module associated to  $\text{Alb}(X)[p^\infty] \in p\text{Div}/\kappa$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (27.5.4.8).** For  $\kappa \in \text{Perf}^p$  and  $X \in \text{SmPrprVar}/\kappa$ , we have

$$H_{\text{crys}}^*(X/\kappa) = H_{\text{dR}}^*(X/\kappa),$$

and there is a universal coefficient formula

$$0 \rightarrow H_{\text{crys}}^i(X/W(\kappa)) \otimes \kappa \rightarrow H_{\text{crys}}^i(X/\kappa) \rightarrow \text{Tor}_1^{W(\kappa)}(H_{\text{crys}}^{i+1}(X/W(\kappa), \kappa)) \rightarrow 0.$$

$\lrcorner$

*Proof:* ? □

**Thm. (27.5.4.9) [*W*( $\kappa$ )-Cohomology is a Weil-Cohomology Theory, Katz-Messing].** For  $\kappa \in \text{Perf}^p$ ,

- For  $X \in \text{SmPrprVar}/\kappa$ , each  $H_{\text{crys}}^i(X/W(\kappa))$  is of f.t. over  $W(\kappa)$ , and vanishes for  $i > 2 \dim X$ .
- For  $X \in \text{SmPrprVar}/\kappa$ , if  $X$  is projective or has a smooth proper lifting to  $W(\kappa)$ , then

$$\dim_{W(\kappa)[\frac{1}{p}]} H_{\text{crys}}^*(X/W(\kappa))[\frac{1}{p}] = \dim_{\mathbb{Q}_\ell} H_{\text{ét}}^i(X_{\bar{\kappa}}, \mathbb{Q}_\ell).$$

- The functor

$$\text{SmProjVar}/\kappa \rightarrow H_{\text{crys}}^*(X/W(\kappa))[\frac{1}{p}]$$

is a Weil-cohomology theory.

- If  $\kappa \in \mathbf{Field}^{\text{fin}}$  and  $X \in \text{SmProjVar}/\kappa$ , then

$$Z_X(T) = \prod_{i \in [2d]} \det \left( 1 - \varphi_\kappa^* T | H_{\text{crys}}^*(X/W(\kappa))[\frac{1}{p}] \right)^{(-1)^{i+1}}$$

coincides with the zeta-function defined using étale cohomology. In particular,  $Z_X(T) \in \mathbb{Q}(T) \cap Z[[T]]$ . ⌋

*Proof:* ? [Katz-Messing. Some Consequences of the Riemann Hypothesis for Varieties over Finite Fields]. □

### deRham-Witt Complexes

References are [Illusie. De Rham-Witt complexes and p-adic Hodge theory], [crystalline cohomology, mini course, Haoyang Guo] and

**Def. (27.5.4.10) [Witt-Sheaves].** For  $X \in \text{Sch}$  and  $n \in \mathbb{Z}_+$ , if  $\mathcal{F} \in \text{Sh}(X_{\text{Zar}})$  is a sheaf of rings on  $X$ , then the functor

$$W_n(\mathcal{F}) : U \in X_{\text{Zar}} \rightarrow W_n(\mathcal{F}(U))$$

is a sheaf of rings on  $X_{\text{Zar}}$ , called the **truncated Witt-sheaf** of  $\mathcal{F}$ . And we can define

$$W(\mathcal{F}) \stackrel{\Delta}{=} \varprojlim_{n \in \mathbb{Z}_+} W_n(\mathcal{F}),$$

called the **Witt-sheaf** of  $\mathcal{F}$ . And we call  $\mathbf{Spec}(W_n(\mathcal{O}_X))$  the **Witt-scheme** over  $X$ . ⌋

*Proof:* ? □

**Prop. (27.5.4.11) [Canonical PD-Structures on Witt-Schemes].** For  $X \in \text{Sch}^p$  and  $n \in \mathbb{Z}_+$ , then  $(W_n(X), \text{Ver}(W_n(\mathcal{O}_X)))$  has a canonical pd-structure given by

$$\gamma_k(\text{Ver}(x)) = \frac{p^{k-1}}{k!} \text{Ver}(x^m).$$

compatible with that of  $(\mathbb{Z}_{(p)}, (p), \gamma)$ . ⌋

*Proof:* To check that this is a pd-structure, by functoriality and universality, it suffices to use the Witt-coordinate. Then the assertion is clear from the definition of  $\text{Ver}$  (5.5.2.13).  $\square$

**Def. (27.5.4.12)[deRham-Witt Complexes, Langer-Zink].** For  $X \rightarrow S \in \text{Sch}^p$ , there is a projective system of dg-algebras of  $\text{Qco } W(\mathcal{O}_X)$ -sheaves  $\{W_n(\Omega_{X/S}^\bullet)\}$  (27.5.4.10) with operators

$$F : W_{n+1}\Omega_{X/S}^\bullet \rightarrow W_n\Omega_{X/S}^\bullet, \quad \text{Ver} : W_n\Omega_{X/S}^\bullet \rightarrow W_{n+1}\Omega_{X/S}^\bullet$$

s.t. for each local sections  $x, y$  of  $W_\bullet(\Omega_{X/S}^\bullet)$  and local section  $a$  of  $\Omega_{X/S}^\bullet$ , we have

- $d_{X/S} \circ F = pF \circ d_{X/S}$ .
- $F \circ d_{X/S}([a]) = [a^{p-1}]d_{X/S}([x])$ .
- $F \circ d_{X/S} \circ \text{Ver} = d_{X/S}$ .
- $F(x)F(y) = F(xy)$ .
- $\text{Ver} \circ d_{X/S} = pd_{X/S} \circ \text{Ver}$ .
- $\text{Ver}(F(x)y) = x \text{Ver}(y)$ .
- $F \circ \text{Ver} = p$ .

┘

*Proof:* ?

$\square$

**Prop. (27.5.4.13).**

- The construction of  $W_\bullet(\Omega_{X/S}^\bullet)$  (27.5.4.12) is functorial in  $X/S \in \text{Sch}^p$ .
- For each  $f : X \rightarrow Y \in \text{Sch}^{\text{ét}}/S$  and  $n \in \mathbb{Z}_+$ , the induced morphism of Witt-schemes  $W_n(X) \rightarrow W_n(Y)$  is also étale, and there is a canonical isomorphism

$$W_n(\mathcal{O}_X) \otimes_{W_n(\mathcal{O}_Y)} W_n(\Omega_{Y/S}^\bullet) \cong W_n(\Omega_{X/S}^\bullet).$$

┘

*Proof:* ?

$\square$

**Thm. (27.5.4.14) [Crystalline and deRham-Witt Comparison].** Suppose  $\kappa \in \text{Perf}^p$  and  $X \in \text{Sch}^{\text{sm}}/\kappa$ , then there is a canonical isomorphism in the category of projective  $W_n(\kappa)$ -modules:

$$R\mathbf{u}_{X/W_n(\kappa),*} \Omega_{X/W_n(\kappa)} \cong W_n(\Omega_{X/\kappa}^\bullet) \text{ (27.5.2.5)(27.5.4.12)}.$$

┘

*Proof:* [Haoyang Guo notes]P18. ?

$\square$

**Cor. (27.5.4.15).** Suppose  $\kappa \in \text{Perf}^p$  and  $X \in \text{Sch}^{\text{sm,proper}}/\kappa$ , there is a canonical isomorphism

$$R\Gamma_{\text{crys}}(X/W) \cong R\Gamma(X, W\Omega_{X/\kappa}^\bullet),$$

which is represented by a perfect complex, with

$$R\Gamma_{\text{crys}}(X/W) \otimes_{W(\kappa)}^L \kappa = R\Gamma(X, \Omega_{X/\kappa}^\bullet).$$

┘

## 5 $F$ -(Iso)Crystals

References are [Slope Filtrations of  $F$ -Crystals, Katz] and [Period domains over finite and  $p$ -adic fields, Dat-Orlik-Rapoport]Chap8.

**Notation (27.5.5.1).**

- Let  $p \in \text{Prime}$ .
- Let  $R_0 \in \mathcal{CRing}^p$ ,  $S_0 = \text{Spec } R_0$ .

┘

**Def. (27.5.5.2) [( $F$ -)Crystals and ( $F$ -)Isocrystals].**

- A **locally free crystal over  $S_0$**  is a finite projective module over  $\text{Coh}(R_0)$ . (5.5.2.30).
- An **isocrystal on  $S_0$**  is an object in the category of locally free crystals on  $S_0$  up to isogeny, i.e. a module over  $\text{Coh}(R)[\frac{1}{p}]$ .
- An (non-degenerate)  **$F$ -crystal on  $S_0$**  is a pair  $(\mathcal{E}, \varphi_{\mathcal{E}})$  where  $\mathcal{E}$  is a locally free crystal over  $S_0$  with a morphism  $\varphi_{\mathcal{E}} : \mathcal{E}^{(p)} \rightarrow \mathcal{E}$  which admits an inverse up to a power of  $p$ . The category of  $F$ -crystals on  $S_0$  is denoted by  $\text{F-Crys}(S_0)$ .
- An  **$F$ -isocrystal on  $S_0$**  is a pair  $(\mathcal{M}, \varphi_{\mathcal{M}})$  where  $\mathcal{M}$  is an isocrystal over  $S_0$  and  $\varphi : \mathcal{M}^{(p)} \cong \mathcal{M}$  is an isomorphism of isocrystals over  $S_0$ . The category of  $F$ -isocrystals over  $S_0$  is denoted by  $\text{F-Isoc}(S)$ .

┘

**Remark (27.5.5.3).** If we denote  $R = \text{Coh}(R_0)$  the Cohen ring (5.5.2.30) and  $S = \text{Spec } R$ , then  $\text{Coh}(R_0)/(p) \cong R_0$ , and a locally free crystal over  $S_0$  is exactly a finite projective  $\mathcal{O}_{S_0/S}$ -crystal (27.5.3.1).

┘

**Prop. (27.5.5.4).** If  $k \in \text{Perf}^p$  and  $X \in \text{Sch}^{\text{sm,proper}}/k$ , then

$$\left( H_{\text{crys}}^i(X/W(\kappa))_{\text{lf}}, \varphi^* \right)$$

is a  $F$ -crystal over  $\kappa$  (27.5.5.2).

┘

*Proof:* This follows from (27.5.4.4).

□

**Lemma (27.5.5.5) [Fitting's Lemma for  $F$ -Crystals].** For  $\kappa \in \text{Perf}^p$  and  $M \in \text{F-Crys}(\kappa)$  (27.5.5.2), one has  $M = M_0 \oplus M_{>0}$ , where  $\varphi$  is bijection on  $M_0$  and topologically nilpotent on  $M_{>0}$ .  $M_0$  is called the **étale-part of the  $F$ -crystal  $M$** , and  $M_{>0}$  is called the **local-part of the  $F$ -crystal  $M$** .

┘

*Proof:* We consider  $M/p^n M$ , then by (3.2.4.5) under slight modification, we have a decomposition for  $M/p^n M$ . This decompositions for different  $n$  are compatible, so taking an inverse limit gives a decomposition of  $M$  it self.

□

**Prop. (27.5.5.6).**  $\text{F-Isoc}(S_0)$  is a neutral Tannakian-category.

┘

*Proof:* ?

□



### Isocrystals

**Notation (27.5.5.7).**

- Let  $(K, \mathcal{O}_K, \kappa) \in p\text{-Field}$ .
- $K_0 = W(\kappa)[\frac{1}{p}]$  be the maximal unramified subextension of  $K$ .
- The arithmetic Frobenius action on  $\check{K}/K$  is denoted by  $\varphi_0$ .
- We abbreviate F-crystals to crystals and F-isocrystals to isocrystals.

┘

**Def. (27.5.5.8) [Isocrystals and  $\varphi$ -Modules].** An F-isocrystal over  $\kappa$  (27.5.5.2) is the same thing as a  $\varphi$ -module  $(V, \varphi)$  over  $(K_0, \varphi_0)$  (25.2.9.1). The height of  $V$  is defined to be  $\text{ht}(V) = \text{rank } V$ .

┘

**Def. (27.5.5.9) [First Newton-Slopes of Isocrystals].** For  $D \in \varphi\text{-Mod}(K_0)$ , choose a  $W(\kappa)$ -module  $W \subset D$ , and let  $a_n \in \mathbb{Z}$  be the maximum integer that  $\varphi^n(M) \subset p^{a_n}M$ , then we have  $a_{m+n} \geq a_m + a_n$ , thus by (12.3.5.2), we have  $a_n/n$  converges to  $\sup a_n/n = \lambda$ .  $\lambda$  doesn't depend on  $M$  because of the cofinality of lattices, and it is called the **first Newton-slope** of  $V$ .

┘

**Cor. (27.5.5.10).** For  $(V, \varphi) \in \text{F-Isoc}(\kappa)$ , we have  $\lambda(V, p^m \varphi^n) = n\lambda(V, \varphi) + m$ .

┘

**Lemma (27.5.5.11).** If  $D \in \varphi\text{-Mod}(K_0)$ , and  $M$  be a  $W(\kappa)$ -lattice of  $D$  s.t.  $\varphi^{h+1}(M) \subset p^{-1}M$ , where  $h$  is the dimension of  $D$ , then  $D$  is effective (25.2.9.11).

┘

*Proof:* Let  $M_j = M + \varphi(M) + \dots + \varphi^j(M)$ , then  $M_j/M \subset p^{-1}M/M$ , which is a  $k$ -vector space of dimension  $h$ , then  $M_j = M_{j+1}$  for some  $j$ , hence  $M_j$  is stable under  $\varphi$ .  $\square$

**Prop. (27.5.5.12).** For  $D \in \varphi\text{-Mod}(K_0)$ ,  $\lambda = s/r$  for some  $r \in [h]_+$  and  $s \in \mathbb{Z}$ . And  $\lambda(D) \geq 0$  (27.5.5.9) iff  $D$  is effective (25.2.9.11).

┘

*Proof:* If  $D$  is effective, then  $a_n \geq 0$ , conversely, if  $a_n \geq 1$ , then  $M' = M + \varphi(M) + \dots + \varphi^{n-1}(M)$  is stable under  $\varphi$ , so  $D$  is effective.

For the first assertion, we first notice, if  $\lambda > 0$ , then  $\varphi$  is nilpotent on  $M/pM$ , which is a  $k$ -vector space of dimension  $h$ , then  $\varphi^h = 0$  on  $M/pM$ , so  $\lambda \geq 1/h$ .

Now we find  $s, r$  that  $|r\lambda - s| \leq 1/(h+1)$ , and  $\tilde{\varphi} = p^{-s}\varphi^r$  has  $|\tilde{\lambda}| \leq 1/(h+1)$ , so (27.5.5.11) shows that  $\tilde{\varphi}$  is effective, hence  $\tilde{\varphi} \geq 0$ , and by what we have proved,  $\tilde{\varphi} = 0$ , hence it is  $\lambda = s/r$ .  $\square$

**Def. (27.5.5.13) [Isoclinic Isocrystals].** For  $V \in \varphi\text{-Mod}(\check{K}_0)$  of height  $h$  (27.5.5.8), if  $V$  admits a  $W(\kappa)$ -lattice  $M \subset V$  and  $s \in \mathbb{Z}_+, r \in \mathbb{Z}$  s.t.  $\varphi^s(M) = p^rM$ , then  $\lambda(V) = r/s = d/h$  (27.5.5.9) where  $d = \text{ord}(\det(\varphi))$ . Conversely, if  $\lambda(V) = \text{ord}(\det(\varphi))/h$ , then there exists a  $W(\kappa)$ -lattice  $M \subset V$  s.t.  $\varphi^s(M) = p^rM$ .

This is independent of  $M$  chosen because  $\lambda(V)$  is independent of  $V$ , in which case  $V$  is called **isoclinic isocrystal** of slope  $\lambda(V) = d/h \in \mathbb{Q}$ .

┘

*Proof:* If  $\varphi^s(M) = p^rM$ , then it follows directly from the definition that  $\lambda(V) = r/s$  (27.5.5.9). And then  $rh = s \text{ord}(\det(\varphi))$ .

Conversely, if  $\lambda(V) = d/h$ , then it is to see easy from (27.5.5.11) and (27.5.5.10) that  $F^h(M) \subset p^dM$  where  $d = \text{ord}(\det(\varphi))$ . But then by calculating determinant it follows that  $F^h(M) = p^dM$ .  $\square$

**Prop. (27.5.5.14) [Isoclinic Isocrystals].**

- If  $V \in \text{F-Isoc}(\kappa)$  is isoclinic, then any sub- $\varphi$ -module or quotient- $\varphi$ -module of  $V$  is also isoclinic of the same slope.
- If  $V_1, V_2 \in \text{F-Isoc}(\kappa)$  are isoclinic with different slopes, then  $\text{Hom}_{\varphi\text{-Mod}(K_0)}(V_1, V_2) = 0$ .

┘

*Proof:* 1: Let  $M \subset V$  be a  $W(\kappa)$ -lattice with  $\varphi^h(M) = p^d M$ . If  $V' \subset V$  is a sub- $\varphi$ -module, and  $M' = M \cap V'$ , then  $\varphi^h(M') = p^d M'$ , which implies  $V'$  is isoclinic of the same slope. If  $V''$  is a quotient of  $V$ , then the image  $M''$  of  $M$  in  $V''$  is also a lattice and satisfies  $\varphi^h(M'') = p^d M''$ .

2: This follows from item 1. □

**Prop. (27.5.5.15) [Isoclinic Decompositions, Dieudonné-Manin].** Any  $V \in \text{F-Isoc}(\kappa)$  is a finite sum of isoclinic isocrystals (27.5.5.13). This is called the **isoclinic decomposition** of  $V$ .

Using the isoclinic decomposition, we can define the **Newton-slopes**  $\lambda_1 \leq \lambda_2 \leq \dots \leq \text{lb}d_h$  of  $V$  and the **Newton-polygon**  $\text{NP}(V)$  of  $V$ . ┘

*Proof:* We use the  $\tilde{\varphi}$  as in the proof of (27.5.5.12), we see that  $M$  has a decomposition  $M_0 \oplus M_{>0}$  by (27.5.5.5), and  $M_0 \neq 0$  by definition. Then we use induction to get the result. □

**Def. (27.5.5.16)  $[E_\lambda]$ .** For  $\lambda = s/r \in \mathbb{Q}$  with  $s \in \mathbb{Z}_+$  and  $(s, r) = 1$ , we define a  $\varphi$ -module  $E_\lambda$  over  $\mathbb{Q}_p$  s.t.

- $E_\lambda = \bigoplus_{i=0}^{r-1} K e_i$ ,
- $\varphi(e_i) = e_{i+1}$ , and  $\varphi(e_{r+1}) = p^s e_0$ .

Then  $E_\lambda$  is irreducible, by (27.5.5.13) and (27.5.5.14). ┘

**Thm. (27.5.5.17) [Dieudonné-Manin].** The category  $V \in \varphi\text{-Mod}(\check{K}_0)$  is semi-simple with simple objects given by  $\{E_\lambda\}_{\lambda \in \mathbb{Q}}$  (27.5.5.16). And the category  $\varphi\text{-Mod}^{\text{eff}}(\check{K}_0)$  is semi-simple with simple objects given by  $\{E_\lambda\}_{\lambda \in \mathbb{Q}_{\geq 0}}$ . ┘

*Proof:* By (27.5.5.15) we assume  $V$  is isoclinic. Cf. [Abelian Varieties, Geer]P262? □

**Prop. (27.5.5.18) [Isocrystals over Finite Fields].** Let  $m \in \mathbb{Z}_+$  and  $(V, \varphi) \in \text{F-Isoc}(\mathbb{F}_{p^m})$  has height  $h$ , then  $\pi = \varphi^m : V \rightarrow V$  is a  $\mathbb{F}_{p^m}$ -linear isomorphism. Suppose  $\{\alpha_1, \dots, \alpha_h\} \subset \overline{\mathbb{Q}_p}$  are the roots of  $\pi$  counted with multiplicity, then the slopes of  $V$  are just  $\{\text{ord}_p(\alpha_1)/m, \dots, \text{ord}_p(\alpha_h)/m\}$ .

In particular, if  $\text{char}(\pi) = c_h T^h + \dots + c_1 T + c_0$ , then the Newton-polygon  $\text{NP}(V)$  is just given by the lower convex hull of the points  $\{(i, \text{ord}_p(c_{h-i})/m)\}_{i \in [h]}$ . ┘

*Proof:* By (27.5.5.15), we may assume that  $V$  is isoclinic of slope  $\lambda = d/h$ . Then by replacing  $\varphi$  by  $\varphi' = p^{-d}\varphi^h$ ,  $(V, \varphi')$  is isoclinic of slope 0 and  $\pi' = p^{-dm}\pi^h$ , and each  $p^{-dm}\alpha^h$  is a root of  $\pi'$ . Because there exists a  $W(\kappa)$ -lattice s.t.  $\varphi'(M) = M$ , we see that  $\text{ord}_p(p^{-dm}\alpha^h) = 0$ , thus  $\text{ord}_p(\alpha_i)/m = d/h = \lambda$ . □

**Def. (27.5.5.19) [Tate-Twist Isocrystals].** The Tate object  $\mathbf{1}(n)$ ,  $n \in \mathbb{Z}$  is the 1-dimensional isocrystal over  $K_0$  that  $\varphi = p^n \sigma$ , so it is of slope  $n$ . And the **Tate-twist isocrystal** is tensoring by  $\mathbf{1}(n)$ . It preserves rank and shifts Hodge-Tate weight by  $n$ . ┘

**Def. (27.5.5.20) [Hodge-Slopes].** If  $(M, \varphi)$  is a F-crystal of height  $h$  over  $\kappa$ , then there exists ordered bases  $\{e_1, \dots, e_h\}, \{f_1, \dots, f_h\}$  of  $M$  s.t.  $\varphi(e_i) = p^{\mu_i} f_i$  for some  $\mu_i \in \mathbb{Z}$  s.t.  $\mu_1 \leq \mu_2 \leq \dots \leq \mu_h$ . These  $r_i$  are called the **Hodge-slopes** of  $M$ , with  $r_1 = \lambda(V)$  (27.5.5.9). And the polygon starting from  $(0, 0)$  using segments of slopes  $\mu_i$  are called the **Hodge-polygon** of  $(M, \varphi)$ . ┘

**Prop. (27.5.5.21) [Hodge-Polygons and Newton-Polygons].** If  $(M, \varphi)$  is a F-crystal of height  $h$  over  $\kappa$ , then the Newton-polygon  $\text{NP}(M)$  lies above the Hodge-polygon of  $M$  (27.5.5.20), with the same start point  $(0, 0)$  and same end point  $(h, \text{ord}(\det(\varphi)))$ .  $\lrcorner$

*Proof:* Let  $\mu_1 \leq \mu_2 \leq \dots \leq \mu_h$  be the Hodge-slopes of  $M$  (27.5.5.20) and let  $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_h$  be the Newton-slopes (27.5.5.15), then for each  $r \in [h]_+$ , we see that

$$\mu_1 + \dots + \mu_r = \mu_1(\wedge^r(M, \varphi)), \quad \lambda_1 + \dots + \lambda_r = \lambda_1(\wedge^r(M, \varphi)) \text{ (27.5.5.9)}.$$

where for Newton-slopes, we use the fact that if  $n$  is a common denominator of  $\lambda_i$ , then  $\varphi^n$  is given by the diagonal matrix  $(p^{n\lambda_1}, \dots, p^{n\lambda_h})$ . Then it suffices to show that  $\mu_1 \leq \lambda_1$ , which is clear from the definition (27.5.5.9).  $\square$

### Dieudonné-Modules

**Def. (27.5.5.22) [Dieudonné-Rings and  $E(R)$ ].** For  $R \in \mathcal{CRing}$ , define  $E(R)$  to be the ring  $[[V]]W(R)[F]$  subject to the following identities:

- $FV = VF = p$ .
- For each  $\alpha \in W(R)$  we have  $\alpha V = V\alpha^\sigma, F\alpha = \alpha^\sigma F, V\alpha F = \alpha^\tau$ .

And the **Dieudonné-ring**  $D(R)$  is defined to be subring

$$D(k) = W(R)[F, V] \subset E(R) \text{ (27.5.5.22)}.$$

$\lrcorner$

**Def. (27.5.5.23) [Dieudonné-Modules].** A left  $E(R)$ -module  $M$  is called

- A **uniform  $E(R)$ -module** if it is separated and complete in the  $V$ -adic topology.
- A **reduced  $E(R)$ -module** if it is a uniform  $E(R)$ -module s.t.
  - $V\gamma = 0 \implies \gamma = 0$  for any  $\gamma \in M$ .
  - $M/VM$  is a free  $R$ -module.
- A **Dieudonné-module** if it is a reduced and  $M/VM$  is finite free over  $R$ .

$\lrcorner$

**Prop. (27.5.5.24) [Dieudonné-Modules over Perfect Rings].** If  $\kappa \in \mathcal{Perf}^p$ , a **Dieudonné-module** over  $\kappa$  is a triple  $(P, F, V)$  where

- $P$  is a finite free  $W(\kappa)$ -module,
- $F : P \rightarrow P$  is an additive map s.t.  $F(\xi x) = \varphi(\xi)F(x)$  for each  $\xi \in W(\kappa), x \in P$ .
- $V : P \rightarrow P$  is an additive map s.t.  $V(\xi x) = \varphi^{-1}(\xi)V(x)$  for each  $\xi \in W(\kappa), x \in P$ .

s.t.  $VF = FV = p$ .  $\lrcorner$

**Prop. (27.5.5.25) [Representing Dieudonné-Modules].** Let  $M$  be the  $E(R)$ -module defined by the generators  $\{e_i\}_{i \in [n+h]}$  subjecting to the relations

$$Fe_i = \sum_j \alpha_{ij} e_j, \quad i \in [n]_+,$$

$$e_i = V\left(\sum_j \alpha_{ij} e_j\right), \quad n+1 \leq i \leq n+h.$$

for some  $\alpha_{ij} \in W(R)$ . Then  $M$  is a Dieudonné-module over  $E(R)$ . And any Dieudonné-module over an Artinian  $k$ -algebra can be written in this form. Such an Dieudonné-module is associated to a  $p$ -divisible group iff  $\det(a_{ij})$  is invertible in  $W(R)$ , in which case  $M$  is free over  $W(R)$  generated by  $\{e_i\}_{i \in [n+h]_+}$ .

And if we define the **dual Dieudonné-module**  $M^t$  to be the  $E(R)$ -module defined by the generators  $\{f_i\}_{i \in [n+h]}$  subjecting to the relations

$$f_i = V\left(\sum_j \alpha'_{ij} f_j\right), \quad i \in [n]_+,$$

$$Ff_i = \sum_j \alpha'_{ij} e_j, \quad n+1 \leq i \leq n+h.$$

where  $(\alpha'_{ij}) = (\alpha_{ij})^{-1}$ , then  $M^t$  is associated to the dual  $p$ -divisible group of  $G$  (Cf. [Tate]) associating to  $M$ .  $\lrcorner$

*Proof:* Cf. [Norman-Oort]P418. ?  $\square$

## 6 Computations

**Prop. (27.5.6.1) [Affine Thickening is Acyclic].** If  $\mathcal{F} \in \mathcal{Sh}(\mathcal{O}_{X/S, \text{crys}})$  is locally Qco, then  $H^p((U, T, \delta), \mathcal{F}) = 0$  for any  $p > 0$  and  $U$  or  $T$  affine.  $\lrcorner$

*Proof:* Firstly notice  $U$  is affine iff  $T$  is affine, by (7.8.6.2), then we use (7.3.2.16) with  $\mathfrak{G}$  being the affine thickenings and Cov the affine coverings of affine thickenings, then Cov is cofinal, and it suffices to check that  $\check{H}^q(T, \mathcal{F}) = 0$  for an affine thickening  $T$  and  $q > 0$ , and this is just the usual cohomology for Qco sheaves as the affine covering is cofinal, and it follows from (7.7.1.1).  $\square$

**Lemma (27.5.6.2).** Situation as in (27.5.3.3), then the morphism

$$\left(\text{colim}_e h_{(X, T_e, \bar{\delta})}\right)^\sharp \rightarrow * \in \mathcal{Sh}^{\text{Set}}((X/S)_{\text{crys}})$$

is surjective.  $\lrcorner$

*Proof:* We need to show that for any  $(U, B, \delta) \in (X/A)_{\text{Crys}}$ , there is an Zariski covering  $(U_i, B_i, \delta)$  of it that there are maps  $(U_i, B_i, \delta) \rightarrow (X, T_{e_i}, \bar{\delta})$  that are compatible, But this is in fact equivalent to the existence of a morphism  $(U, B, \delta) \rightarrow (X, \text{Spec } D, \delta)$  of pd-structures. For this, notice the morphism  $U \rightarrow X$  can be extended to a morphism  $X \rightarrow \text{Spec } P$  by strong lifting property (5.4.5.16) of smooth morphism (5.4.5.21), and this extends to the desired morphism by the universal property of pd-envelope and the fact  $p$  is locally nilpotent on  $B$  (27.5.2.2) thus  $B$  is locally  $p$ -complete.  $\square$

**Lemma (27.5.6.3).** Let  $K' = (\text{colim}_e h_{(X, T_e, \bar{\delta})})^\sharp$ , then the product sheaf  $(K')^n$  is in fact isomorphic to  $(\text{colim}_e h_{(X, T(n)_e, \bar{\delta})})^\sharp$  on  $(X/S)_{\text{crys}}$ .  $\lrcorner$

*Proof:* This follows from the definition and the universal properties of completion, pd-envelope and  $P \otimes_A \otimes_A \dots \otimes_A P$  is a coproduct. Compare with the proof of (27.5.6.2).  $\square$

**Prop. (27.5.6.4).** Situation as in (27.5.3.3), if  $\mathcal{F}$  is locally Qco and satisfies: for any morphism  $f : (U, T, \delta) \rightarrow (U', T', \delta') \in (X/S)_{\text{crys}}$  that  $f : T \rightarrow T'$  is a closed immersion the map  $f^* \mathcal{F}_{T'} \rightarrow \mathcal{F}_T$  is surjective, then the complex

$$M(0) \rightarrow M(1) \rightarrow \dots$$

computes  $R\Gamma((X/S)_{\text{crys}}, \mathcal{F})$ . Moreover,

$$R\Gamma((X/S)_{\text{crys}}, \mathcal{F} \otimes_{\mathcal{O}_{X/S}} \Omega_{X/S, \delta}^i) = 0$$

for  $i > 0$ . ┘

*Proof:* We use (7.3.1.19) for the presheaf  $K' = \text{colim}_e h_{(X, T_e, \bar{\delta})}$  and  $K = *$ , which satisfies the condition by (27.5.6.2). Then we get a spectral sequence

$$E_1^{p,q} = H^q(K'_p, \mathcal{F}) \Rightarrow H^{p+q}(K, \mathcal{F}).$$

Notice  $K'_n = (\text{colim}_e h_{(X, T(n)_e, \bar{\delta})})^\#$  by (27.5.6.3), so the cohomology

$$R\Gamma(K'_n, \mathcal{F}) = R\lim_e (\Gamma((X, T(n)_e, \bar{\gamma}(n)), \mathcal{F}))$$

Now the surjectivity  $f^* \mathcal{F}_{T'} \rightarrow \mathcal{F}_T$  is equivalent to the surjectivity  $\mathcal{F}_{T'} \rightarrow f_* \mathcal{F}_T$ , so there is an exact sequence of Qco  $\mathcal{T}'$ -sheaves

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{F}_{T'} \rightarrow f_* \mathcal{F}_T \rightarrow 0$$

which implies that  $\mathcal{F}((U', T', \delta')) \rightarrow \mathcal{F}((U, T, \delta))$  is surjective, by (7.7.4.2) and (7.7.1.3).

Then by (5.9.3.3),

$$R\Gamma(K'_n, \mathcal{F}) = R\lim_e (\Gamma((X, T(n)_e, \bar{\gamma}(n)), \mathcal{F})) = M(n)$$

thus we are done. □

**Lemma (27.5.6.5).** Situation as in (27.5.3.3), the complex  $\Omega_{D(\bullet)}$  is homotopic to 0 as a  $D(\bullet)$ -cosimplicial module. ┘

*Proof:* This complex is the  $p$ -adic completion of the base change of the cosimplicial module  $M_\bullet = (\Omega_{P^{\otimes \bullet} A/A})$  under the cosimplicial ring map  $P^{\otimes \bullet} A \rightarrow D(\bullet)$ . Then it suffices to show  $M_\bullet$  is homotopic to 0. For this, the whole thing can be written down clearly, Cf. [Sta]07LA. □

**Lemma (27.5.6.6).** In situation (27.5.3.3), for any cosimplicial module  $M_*$  over the cosimplicial ring  $D(*)$  and  $i > 0$ , the cosimplicial module

$$M_0 \hat{\otimes}_{D(0)} \Omega_{D(0)}^i \rightarrow M_1 \hat{\otimes}_{D(1)} \Omega_{D(1)}^i \rightarrow \dots$$

is homotopic to 0. ┘

*Proof:* □

### Crystal Case

**Prop. (27.5.6.7).** Situation as in (27.5.3.3), if  $\mathcal{F} \in \mathcal{QCoh}(\mathcal{O}_{X/S, \text{crys}})$ , and let  $(M, \nabla)$  be the corresponding module with connection over  $D$  by (27.5.3.10), then the complex

$$M \hat{\otimes}_D \Omega_D^\bullet$$

computes  $R\Gamma((X/S)_{\text{crys}}, \mathcal{F})$ . ┘

*Proof:* Use the spectral sequence associated to the double complex

$$K^{a,b} = M \hat{\otimes}_D \Omega_{D(b)}^a$$

Then the rows  $K^{a,\bullet}$  is acyclic for  $a > 0$  by (27.5.6.6) and (5.8.2.21), and  $K^{0,\bullet}$  is quasi-isomorphic to  $R\Gamma((X/S)_{\text{crys}}, \mathcal{F})$  by (27.5.6.4). Now we look at the other direction, (5.6.2.7) and (27.5.3.4) show that each of the  $b$  maps  $D \rightarrow D(b)$  determines the same quasi-isomorphism

$$M \hat{\otimes}_D \Omega_D^* \cong M \hat{\otimes}_{D(b)} \Omega_{D(b)}^*$$

as their inverse is given by the same  $D(b) \rightarrow D$ . Then it is clear that the  $E_2$  page in this direction is  $H^a(M \hat{\otimes}_D \Omega_D^*)$  in the zero-th row and vanish otherwise, so we get the desired isomorphism by edge morphisms.  $\square$

**Thm. (27.5.6.8) [de Rham Comparison for Crystalline Cohomology].** In situation (27.5.3.11), if  $\mathcal{F} \in \mathcal{QCoh}((X/S)_{\text{crys}})$  with  $(M', \nabla')$  the corresponding module with connection over  $D'$  (27.5.3.12), then the complex

$$M \hat{\otimes}_{D'} \Omega_{D'}^\bullet$$

computes  $R\Gamma((X/S)_{\text{crys}}, \mathcal{F})$ .  $\dashv$

*Proof:* Let  $b : D' \rightarrow D, a : D \rightarrow D'$  be the maps defined in (27.5.3.12), then by (27.5.6.7), it suffices to prove the base change along  $a, b$  induces quasi-isomorphisms

$$M \hat{\otimes}_D \Omega_D^\bullet \cong M \hat{\otimes}_{D'} \Omega_{D'}^\bullet.$$

$a \circ b$  is trivial, thus it suffices to prove that  $b \circ a$  induces an automorphism of  $M \hat{\otimes}_D \Omega_D^\bullet$ . In fact, this is true for any morphism  $\rho : D \rightarrow D$  of pd-algebras over  $A$  compatible with the map  $D \rightarrow C$ :

Write  $\rho(x_i) = x_i + z_i$ , where  $z_i \in J$  because  $\rho$  is compatible with  $D \rightarrow C$ . Then we can factor  $\rho$  as

$$D \xrightarrow{\sigma} D\langle \xi_i \rangle^\wedge \xrightarrow{\tau} C$$

where  $\sigma(x_i) = x_i + \xi_i$  and  $\tau(\xi_i) = z_i$ .

Notice that there exists an automorphism  $\alpha$  of  $D\langle x_i \rangle^\wedge$  that maps  $x_i$  to  $x_i - \xi_i$  and  $\xi_i$  to  $\xi_i$ . (Such a map exists because by universal property, it suffices to give a map of pairs

$$(P, J) \rightarrow D_{P,\gamma}(J)\langle \xi_i \rangle = D_{P[\xi_i],\gamma}(JP[\xi_i] + (\xi_i))$$

by (5.6.1.20), and we surely have.

Now  $\alpha$  is an automorphism, we have a quasi-isomorphism

$$M \hat{\otimes}_D \Omega_D^* \cong M \hat{\otimes}_{D,\sigma} \Omega_{D\langle \xi_i \rangle^\wedge}^*$$

by (5.6.2.7). Also  $\tau$  induces an isomorphism because it has a right inverse, which is an isomorphism by (5.6.2.7) again, so  $\rho$  induces an isomorphism.  $\square$

**Cor. (27.5.6.9) [Crystalline-de Rham Comparison modulo  $p$ ].** In situation (27.5.3.11), if  $R$  is a smooth  $A/p$ -algebra, then there is a natural quasi-isomorphism

$$R\Gamma_{\text{crys}}(R/A) \otimes_A^L A/p \cong \Omega_{R/(A/p)}^\bullet$$

of commutative algebra objects.  $\dashv$

*Proof:* we choose  $P$  to be a smooth lift of  $R$  to  $A$ , then  $J = IP$ , and by (5.6.1.14), the pd-structure extends to  $P$ , thus  $D_{P/A, \gamma} = P$ , and notice  $\Omega_{P/A}^i$  is finite projective hence flat, so  $\Omega_{P/A}^\bullet$  is  $K$ -flat (7.3.3.2), and the left side of is just  $\widehat{\Omega}_{P/A}^\bullet \otimes_A A/p = \widehat{\Omega}_{R/(A/p)}^\bullet$  by (27.5.6.8)(5.6.2.2) and (5.4.3.6).  $\square$

**Cor. (27.5.6.10) [deRham Comparison for Crystalline Cohomology].** Suppose  $\kappa \in \mathcal{P}\text{erf}^p$  and  $X' \in \text{Sch}^{\text{sm}, \text{proper}}/W(\kappa)$  with  $X = X' \otimes_{W(\kappa)} \kappa$ , then we have a canonical isomorphism

$$H_{\text{crys}}^i(X/W(\kappa)) = H_{\text{dR}}^i(X'/W).$$

┘

*Proof:* Cf. [Haoyang Guo notes]P14.  $\square$

## 7 Supersingular Varieties

**Def. (27.5.7.1) [Supersingular Varieties].** For  $k \in \text{Field}^p$ ,  $X \in \text{SmProjVar}/k$  is called a **supersingular variety** iff for any  $i$ , the slopes of the crystalline cohomology  $H^i(X/W(k))$  are all  $i/2$ . When  $\#k < \infty$ , this is equivalent to that the Frobenius eigenvalues of  $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$  are all equal to  $(\#k)^{i/2}$  times a root of unity.  $\square$

*Proof:*  $\square$

## 27.6 Prismatic Cohomology(Bhatt-Scholze)

Main references are [Sta], [Prisms and Prismatic Cohomology, Bhatt-Scholze], [Notes on Prismatic Cohomology, Bhatt], [Prismatic Cohomology notes, Kedlaya].

### 1 Prisms

**Def. (27.6.1.1) [Prisms].** A **prism** is a pair  $(A, I)$  where  $A$  is a  $\delta$ -ring  $A$  and  $I \in \text{Ideal}(A)$ , s.t.

- $V(I) \subset \text{Spec } A$  is a Cartier divisor,
- $A$  is derived  $(p, I)$ -complete, and
- $p \in I + \varphi(I)A$ .

A prism  $(A, I)$  is called **perfect prism** if  $A$  is a perfect  $\delta$ -ring(5.5.4.25). it is called **bounded prism** if  $A/I$  has bounded  $p^\infty$ -torsion. It is called **crystalline prism** if  $I = (p)$ . It is called **orientable** if  $I$  is principal. It is called **oriented** if  $I = (d)$  for some  $d \in A^\times$ .

A map of prisms  $f : (A, I) \rightarrow (B, J)$  is called **(faithfully)flat map of prisms** iff the map  $A \rightarrow B$  is  $(p, I)$ -completely (faithfully)flat(5.9.7.1).  $\lrcorner$

**Lemma(27.6.1.2).** Let  $A$  be a  $\delta$ -ring and  $I$  be a locally principal ideal contained in  $\text{rad}(A)$  that  $(p, I) \subset \text{rad } A$ , then the following are equivalent:

- $p \in I^p + \varphi(I)A$ .
- $p \in I + \varphi(I)A$ .
- There is a f.f. morphism of  $\delta$ -rings  $A \rightarrow A'$  where  $A'$  is a finite product of localizations of  $A$   $\varphi$ -stable multiplicative subsets that  $IA'$  is generated by a distinguished element  $d$  and  $d, p \in \text{rad}(A')$ .

$\lrcorner$

*Proof:*  $1 \rightarrow 2$  is trivial, for  $2 \rightarrow 3$ : Choose  $(g_1, \dots, g_r) = A$  that  $IA_{g_{r_i}}$  is principal. Let  $B = \prod_{i=1}^r A_{g_i}$ , so  $A \rightarrow B$  is f.f. and  $IB = (f)$  is principal. Let  $A'$  be the localization of  $B$  along the ideal  $(p, f)$ (5.1.1.34), then  $p, f \in \text{rad } A'$ . Then  $A \rightarrow A'$  is still f.f., because it is flat hence the image is stable under generalization by(5.4.1.18), so it must be all of  $\text{Spec } A$  because it contains  $(p, I)$  by construction, and  $(p, I) \in \text{rad } A$  by hypothesis.

Now because  $p \in \text{rad } A$ , each localization of  $A$  has a compatible  $\delta$ -structure, and  $\tilde{A}$  is a finite product of localizations of  $A$ , thus it has a  $\delta$ -structure, and  $d$  is distinguished by(5.5.4.23).

$3 \rightarrow 1$ : We need to check that  $p = 0$  in  $A/(I^p + \varphi(I)A)$ , but this can be checked after base change to  $A'$ , which is  $p = 0 \in A'/(d^p, \varphi(d))$ . This is true, because  $d^p = d^p + p\delta(d)$  and  $\delta(d)$  is a unit.  $\square$

**Remark(27.6.1.3) [Examples of Prisms].**

- If  $A$  is a  $p$ -torsionfree and  $p$ -adically complete  $\delta$ -ring  $A$ , the pair  $(A, (p))$  is a bounded crystalline prism.
- $q$ -de Rham cohomology(5.5.4.24) determines a bounded prism. (completeness and boundedness is clear, and  $p \in (d, \varphi(d))$  because?)
- Breuil-Kisin cohomology(5.5.4.24) determines a bounded prism. (boundedness is clear, and?)
- $A_{\text{inf}}$ -cohomology(5.5.4.24) determines a bounded prism. (The same reason as item2).

$\lrcorner$



**Prop. (27.6.1.4) [Universal Oriented Prism].** Let  $A_0 = \mathbb{Z}_{(p)}\{d, \delta(d)^{-1}\}$  be the localization  $\delta$ -ring (5.5.4.15) of the free  $\mathbb{Z}_{(p)}$ - $\delta$ -ring on the variable  $d$ , and let  $A$  be the derived  $(p, d)$ -completion of  $A_0$  (it is discrete by (5.9.7.7)), and let  $I = (d)$ , then  $(A, I)$  is a bounded oriented prism, and it is the initial object in the category of bounded oriented prisms.

Moreover, the sequence  $(p, d)$  is regular and the Frobenius  $\varphi : A/p \rightarrow A/p$  is  $d$ -completely flat.  $\lrcorner$

*Proof:* It is clearly a prism and universal. For the assertions, firstly we show  $A/(p, d) = A_0/(p, d)$ : notice  $A \otimes_{A_0}^L A_0/(p, d) = A_0/(p, d)$  by (5.9.7.4), so we can replace  $\otimes^L$  with  $\otimes$ . Similarly for  $A/(p, d^p)$  because  $A$  is  $(p, d^p)$ -complete. Now this map is .

$(p, d)$  is regular by (5.9.7.7) applied to  $(\mathbb{Z}_{(p)}[d], A)$ ?, for the last assertion, it suffices to show  $A/(p, d) \xrightarrow{\text{Frob}} A/(p, d^p)$  is f.f..  $\square$

**Prop. (27.6.1.5).** Let  $(A, (d))$  be the universal prism (27.6.1.4) and let  $B = A\{\frac{\varphi(d)}{p}\}^\wedge$  (derived  $(p, d)$ -completion), then  $B$  is  $(p, d)$ -complete,  $p$ -torsionfree and it equals the derived  $(p, d)$ -completion of the pd-envelope  $D_{A, \delta}((d))$  of  $(A, (d))$ . In particular,  $(B, (p))$  is a crystalline prism, by (5.5.4.18).  $\lrcorner$

*Proof:*  $B$  is  $p$ -complete by (5.9.6.16), as  $A$  is  $p$ -torsionfree. Also  $d^p = p(\frac{\varphi(d)}{p} - \delta(d))$ , so  $B$  is also  $(p, d)$ -complete. The last assertion follows from (5.6.1.25) and (27.6.1.4).  $\square$

**Cor. (27.6.1.6) [Connection with Crystalline Prism].** In the above situation,  $B$  is also a  $\delta$ -ring by (5.5.4.17), and both  $\varphi(d), p$  are distinguished in  $B$  and  $\varphi(d)$  divides  $p$  in  $B$ , so by (5.5.4.22)  $(\varphi(d)) = (p)$ .

Now the composition of maps of  $\delta$ -rings (5.5.4.5)  $\alpha : A \xrightarrow{\varphi} A \rightarrow B$  promotes to a morphism of prisms (27.6.1.5)  $(A, (d)) \rightarrow (B, (p))$  which decompose as

$$(A, (d)) \xrightarrow{\varphi} (A, \varphi(d)) \rightarrow (B, (\varphi(d))) = (B, (p))$$

$\lrcorner$

**Prop. (27.6.1.7) [Rigidity of Maps].** If  $(A, I) \rightarrow (B, J)$  is a morphism of prisms, then  $I \otimes_A B = J$ , in particular,  $IB = J$ . Conversely, if  $B$  is a derived  $(p, I)$ -complete  $\delta$ - $A$ -algebra, then  $(B, IB)$  is a prism iff  $B[I] = 0$ .  $\lrcorner$

*Proof:* For the first assertion, it suffices to show that  $I \otimes_A B \rightarrow J$  is surjective, because they are both invertible sheaves on  $B$ . Choose f.f. ring morphisms  $A \rightarrow A', B \rightarrow B'$  as in (27.6.1.2) and there is a morphism  $A' \rightarrow B'$  extending  $A \rightarrow B$ , (by taking  $B$  as the localization of  $A' \times_A B$  along  $(p, J)$  and do the construction again). Then  $IB' \subset JB'$  are an inclusion of principal ideals generated by distinguished elements, thus they are equal, by (5.5.4.22), finally we use faithfully flatness.

For the second assertion, notice  $B[I] = 0$  iff  $I \otimes_A B \rightarrow IB$  is an isomorphism. If  $(B, IB)$  is a prism, then clearly  $I \otimes_A B \rightarrow IB$  is an isomorphism, because they are both invertible sheaves. The converse is also trivial.  $\square$

**Prop. (27.6.1.8) [Prism is Nearly Principal].** Let  $(A, I)$  be a prism, then the ideal  $\varphi(I)A$  is a principal ideal, and any generator is a distinguished element. In particular, if it is a perfect prism, then  $I = (f)$  where  $f$  is a distinguished element.  $\lrcorner$

*Proof:* It suffices to prove  $\varphi(I)A$  is generated by a single distinguished element, and then use (5.5.4.21). By (27.6.1.2), we can assume  $p = a + b$  where  $a \in I^p, b \in \varphi(I)A$ . Now we show  $b$  generate  $\varphi(I)A$ : choose a f.f. map as in (27.6.1.2), and then it suffices to check that  $b : A' \rightarrow \varphi(I)A'$

is surjective. Now  $\varphi(I)A' = (d)$ ,  $a = xd^p$ ,  $b = y\varphi(d)$ , so it suffices to show  $y$  is a unit in  $A'$ . Now  $(p, d) \in \text{rad } A'$ , it suffices to show  $A'/(p, f, y) = 0$ . If not, we localize along  $(p, f, y)$ , then we may assume  $(p, f, y) \in \text{rad } A'$ .

The equation  $p = a + b$  implies  $p(1 - y\delta(d)) = d^p(x + y)$ , and the left side is distinguished because  $p$  does and  $1 - y\delta(d)$  is a unit, by (5.5.4.21). Then (5.5.4.22) shows  $d^{p-1}(x + y)$  is a unit, then so does  $d$ , contradicting  $d \in \text{rad } A'$ .  $\square$

**Remark (27.6.1.9).** Notice the proof goes through even  $I$  is only a locally principal ideal of  $A$ .  $\lrcorner$

**Cor. (27.6.1.10).** If  $(A, I)$  is a prism, the invertible  $A$ -modules  $\varphi^*(I) = I \otimes_{A, \varphi} A$  and  $I^p$  are trivial.  $\lrcorner$

*Proof:* Cf. [Prisms, Scholze, P25].  $\square$

**Prop. (27.6.1.11) [Properties of Bounded Prisms].** Let  $(A, I)$  be a bounded prism, then

1. Any derived  $(p, I)$ -complete and  $(p, I)$ -completely flat  $A$ -complex  $M \in D(A)$  is discrete and  $(p, I)$ -complete. For any  $n \geq 0$ , we have  $M[I^n] = 0$  and  $M/I^n M$  has bounded  $p^\infty$ -torsion.
2.  $A$  is  $(p, I)$ -complete,  $A/I$  is  $p$ -adically complete,  $A[I^n] = 0$  and  $A/I^n$  has bounded  $p^\infty$ -torsions.
3. The category of (faithfully)flat prisms over  $(A, I)$  identifies with the category of  $(p, I)$ -completely (faithfully)flat  $\delta$ - $A$ -algebras  $B$  by the bijection  $B \leftrightarrow (B, IB)$ .
4. (Bounded Prisms are fpqc-Locally Orientable) There is a  $(p, I)$ -completely faithfully flat  $\delta$ - $A$ -algebra  $B$  that  $IB = (d)$ , where  $d$  is distinguished and determines a nonzero divisor of  $B$ . Also  $(B, IB)$  is bounded.  $\lrcorner$

*Proof:* 1: By (5.9.7.7).

2 follows from 1. For  $A/I$ , it is derived  $p$ -complete by (5.9.6.8), then it is  $p$ -adically complete by (5.9.6.16).

3: By definition, a (faithfully)flat  $(A, I)$ -prism is  $(p, I)$ -completely (faithfully)flat (5.9.7.1). Conversely, by (27.6.1.7), it suffices to show that  $B[I] = 0$ , and this follows from item 1.

4: We may choose  $B$  to be the derived  $(p, I)$ -completion of the f.f.  $\delta$ -ring defined in (27.6.1.2), then it is also  $(p, I)$ -completely faithfully flat (5.9.7.4), and by item 2 and 3 it determines a bounded prism  $(B, IB)$ .  $\square$

**Prop. (27.6.1.12) [The Site of Bounded Prisms].** The prismatic site is the opposite category of the category of bounded prisms where the covers are determined by f.f. map of prisms.

Then the functors that maps  $(A, I)$  to  $A$  or  $A/I$  are sheaves on this cohomology with vanishing higher cohomologies.  $\lrcorner$

*Proof:* To show this is a site, we need to check the base change of covers. If  $(C, IC) \xleftarrow{c} (A, IA) \xrightarrow{b} (B, IB)$  is a diagram that  $b$  is f.f., then we let  $D$  be the derived  $(p, I)$ -completion of  $B \otimes_A^L C$ , then  $C \rightarrow D$  is also  $(p, I)$ -completely f.f. by (5.9.7.5), so by (27.6.1.11) and (27.6.1.7)  $D$  is discrete and  $(D, ID)$  is a bounded prism over  $(A, IA)$ . It is clear this is a base change in the category of bounded prisms.

The assertion about cohomology follows from [Scholze, Prism, 3.12].  $\square$

### Prismatic Envelopes

**Prop. (27.6.1.13) [Prismatic Envelopes].** Let  $(A, I)$  be a prism, then the forgetful functor from the prisms over  $(A, I)$  to  $\delta$ -pairs over  $(A, I)$  admits a left adjoint, called the **prismatic envelope** which maps  $(B, J)$  to  $B\{\frac{J}{I}\}^\wedge$ .  $\lrcorner$

*Proof:* If we can construct this locally, then we can construct it globally by gluing and the universal property, so we can localize and assume  $I = (d)$  where  $d$  is distinguished. Let  $B'$  be the free  $\delta$ -ring over  $A$  generated by  $\{x/d | x \in J\}$  (5.5.4.10) and  $B_1$  the derived  $(p, d)$ -completion module of  $B$  which is a  $\delta$ -algebra by (5.5.4.18).

If  $d$  is torsion-free in  $B_1$ , then  $(B_1, (d))$  is a prism that satisfies the universal property. Otherwise we choose the maximal  $d$ -torsion-free quotient (5.1.1.16) (5.5.4.14) and taking the derived  $(p, d)$ -completion module, and we can do this to  $\aleph_0$ , where we take the filleted colimit, then it is  $d$ -torsion-free and  $(p, d)$ -complete, by (5.9.6.5) and any prism over  $A$  map factors through this chain.  $\square$

**Cor. (27.6.1.14).** In the above situation, if  $(B, J)$  is  $(p, I)$ -completely flat over  $A$ , and  $J = (I, x_1, \dots, x_n)$  where  $(x_1, \dots, x_n)$  is a  $(p, I)$ -completely regular sequence w.r.t.  $A$ , then the prismatic envelope of  $(B\{\frac{J}{I}\}^\wedge, IB\{\frac{J}{I}\}^\wedge)$  is flat over  $(A, I)$  (27.6.1.1).

Moreover, It is compatible with completed derived base change on  $(A, I)$ , by universal properties and the fact the completed derived base change of it is discrete (5.9.7.5). Also, it is compatible with completed derived base change along a  $(p, I)$ -completely flat map  $(B, J) \rightarrow (B', J')$ .  $\perp$

*Proof:* It suffices to check locally for  $I = (d)$  that  $B_1 = (B\{\frac{x_1}{d}, \dots, \frac{x_n}{d}\})^\wedge$  as a simplicial  $\delta$ -ring is  $(p, I)$ -completely flat over  $A$ , then it is discrete and is torsionfree by (27.6.1.11), thus it is a prism over  $(A, (d))$  by (27.6.1.7). And for the flat localization, notice the image of  $x_1, \dots, x_n$  is also  $(p, I)$ -completely regular w.r.t.  $A$ .

Consider the following diagram of derived  $(p, d)$ -complete simplicial  $\delta$ -rings:

$$\begin{array}{ccccccc}
 \mathbb{Z}_p\{z\}^\wedge & \xrightarrow{z \mapsto d} & A & \longrightarrow & B & \longrightarrow & B\{\frac{x_1}{d}, \dots, \frac{x_n}{d}\}^\wedge = C \\
 \downarrow z \mapsto \varphi(y) & & \downarrow & & \downarrow & & \downarrow \\
 \mathbb{Z}_p\{y\}^\wedge & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & B'\{\frac{x_1}{\varphi(d)}, \dots, \frac{x_n}{\varphi(d)}\}^\wedge = C' \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & D = A'\{\frac{\varphi(y)}{p}\}^\wedge & \longrightarrow & B'' & \longrightarrow & B''\{\frac{x_1}{p}, \dots, \frac{x_n}{p}\}^\wedge = C''
 \end{array}$$

where each square is completed derived tensor product. Notice the last term has denominator  $p$  because  $\varphi(y)$  and  $p$  are both distinguished in  $\pi_0(D)$ , so by (5.5.4.22)  $\frac{\varphi(y)}{p}$  is a unit in  $D$ .

The leftmost arrow is  $(p, z)$ -completely f.f. by (5.5.4.10) (5.9.7.4), so all the vertical arrow in the upper row is  $(p, z)$ -completely flat by (5.9.7.5). Now the map  $D \rightarrow C''$  is  $(p, z)$ -completely flat by (5.6.1.28), noticing that the conditions holds, by (5.9.7.5).

Now by definition the  $(p, d)$ -completely flatness is defined by flatness after base change to  $Kos(A, p, d)$  (5.8.3.5), it suffices to show that there is a map  $D \rightarrow Kos(A', p, d)$ . To show this, it suffices to assume  $A' = \mathbb{Z}_p\{y\} = \mathbb{Z}_p[y, y_1, \dots, y_n, \dots]$  and base change. In this case,  $p, y$  is a regular sequence, so by (5.6.1.25)  $D$  is the derived completion of  $D_{\mathbb{Z}_p\{y\}}((y))$ , and  $Kos(A', p, y) = \mathbb{F}_p[y_1, \dots, y_n]$ , so  $A' \rightarrow Kos(A', p, d)$  factors through  $D$  by universal property.  $\square$

## 2 Perfect Prisms

**Prop. (27.6.2.1) [Properties of Perfect Prisms].** Let  $(A, I)$  be a perfect prism (27.6.1.1), then:

- $I = (d)$  where  $d$  is distinguished and is a nonzero-divisor.
- $A$  is  $p$ -torsionfree and  $p$ -adically complete, hence there is a natural isomorphism  $A \cong W(A/p)$  of  $\delta$ -rings.

- $A/I[p^\infty] = A/I[p]$ , and  $A/p[I^\infty] = A/p[I]$ . In particular,  $(A, I)$  is bounded.
- $A$  is  $(p, I)$ -complete.

⌋

*Proof:* 1:  $I$  is principal by (27.6.1.8).  $d$  is distinguished by (5.5.4.23), it is nonzero-divisor by definition of prisms.

2: This is because  $A$  is  $p$ -torsionfree by (5.5.4.27) and thus  $p$ -adically complete by (5.9.6.16), then  $A \cong W(A/p)$  by the equivalence in (5.5.4.28).

3: Use item2, then  $A/p$  is perfect by (5.5.4.28), thus  $A/p[I^\infty] = A/p[I]$ .  $A/d[p^\infty] = A/d[p]$  follows from (5.5.4.30).

4: This follows from item3 and (27.6.1.11). □

**Prop. (27.6.2.2) [Perfection of Prisms].** There is a **perfection of prisms** functor that maps a prism  $(A, I)$  to a perfect prism  $(A_\infty, IA_\infty)$  left adjoint to the inclusion functor. ⌋

*Proof:* Let  $A'_\infty = A_{\text{perf}}$  be the perfection of  $A$  as a  $\delta$ -ring (5.5.4.26), and  $A_\infty$  be the derived  $(p, I)$ -complete of  $A'_\infty$  as a  $\delta$ -ring (5.5.4.18), then the universal property follows from that of derived completion and perfection once we proved that  $A_\infty$  is perfect and  $IA_\infty = (d)$  where  $d$  is a nonzero-divisor.

$A_\infty$  is perfect because the Frobenius is isomorphism on  $A_{\text{perf}}$  and derived  $(p, I)$ -completion and  $(p, \varphi(I))$ -completion coincide (they have the same radical (5.9.6.15)).

$A_\infty$  is  $p$ -adically complete because it is  $p$ -torsionfree (5.5.4.27), and then use (5.9.6.16).

Now (27.6.1.9) and the fact  $A_\infty$  is perfect shows  $IA_\infty = (d)$  where  $d$  is distinguished, and (5.5.4.30) shows  $d$  is a nonzero-divisor, so we are done. □

**Prop. (27.6.2.3) [Perfect Prisms are Final].** Let  $(A, I)$  be a perfect prism, then for any prism  $(B, J)$ , a map  $A/I \rightarrow B/J$  will induce a map of prisms  $(A, I) \rightarrow (B, J)$ . ⌋

*Proof:* Cf. [B-S19] 4.8.

This map will induce a map  $A \cong W((A/I)^\flat) \rightarrow W((B/J)^\flat)$ . And there is a Fontaine's functor  $W((B/J)^\flat) \rightarrow B$  (12.2.9.7) which is a  $\delta$ -map. Thus we obtain a map  $A \rightarrow B$ . And this map can be seen to be lifting  $A/I \rightarrow B/J$ . □

### 3 Integral Perfectoid Rings

**Def. (27.6.3.1) [Integral Perfectoid Rings].** A commutative ring  $R$  is called a **integral perfectoid ring** if it has the form  $A/I$  for a perfect prism  $(A, I)$ . An equivalent definition of an integral perfectoid ring is given in (27.6.3.6). ⌋

**Def. (27.6.3.2) [Special Fiber].** For an integral perfectoid ring  $R$ , then **special fiber** of  $R$  is defined to be  $\overline{R} = R/\sqrt{pR}$ . It is perfect, by (27.6.3.5). ⌋

**Prop. (27.6.3.3) [Perfect Prisms and Integral Perfectoid Rings].** The mapping  $(A, I) \rightarrow A/I$  defines an equivalence of categories between perfect prisms and integral perfectoid rings, where the converse is given by  $R \mapsto (A_{\text{inf}}(R), \ker(A_{\text{inf}}(R) \rightarrow R))$  (5.5.1.15). ⌋

*Proof:* To show  $A \cong A_{\text{inf}}(A/I)$ , by (5.5.4.28), it suffices to show there is a natural isomorphism  $A/p \cong (A/I)^\flat$ . By (5.5.1.18),  $(A/I)^\flat$  identifies with  $d$ -adic completion of  $A/p$ . Then it suffices to show  $A/p$  is  $I$ -adically complete, which is by (5.9.6.16), as  $A/p$  is derived  $d$ -complete because  $A$  does (5.9.6.8).

Now we have the Fontaine's map  $A = A_{\inf}(A/I) \rightarrow A/I$ , this map is surjective because  $\varphi : A/(p, I) \rightarrow A/(p, I)$  is surjective as  $A/p$  is perfect. Also this is just the quotient map  $A \rightarrow A/I$  because they are equal when modulo  $p$ , and then use(5.5.4.28).  $\square$

**Cor.(27.6.3.4).** Any perfect  $\mathbb{F}_p$ -algebra is an integral perfectoid ring corresponding to a crystalline prism, by(5.5.4.28).

Any integral perfectoid ring is  $p$ -adically complete, by(27.6.1.11).  $\lrcorner$

**Prop.(27.6.3.5).** If  $R$  is a perfectoid ring, then

- $R$  is semiperfect.
- There exists an element  $\varpi \in R$  that admits a compatible system  $\varpi^{1/p^n}$  of  $p$ -power roots s.t.  $\varpi = pu$  for a unit  $u$  and the kernel of the Frobenius  $\varphi : R/p \rightarrow R/p$  is generated by  $\varpi^{1/p}$ .
- $\sqrt{p}R = \cup_n(\varpi^{1/p^n})$ , and it is flat.
- $R[p] = R[\sqrt{p}R]$ .

$\lrcorner$

*Proof:* 1: Let  $R = A/d$  where  $A = A_{\inf}(R)$ , then  $R/p = A/(p, d)$ , so  $\varphi_{R/p}$  is surjective, as  $A/p = R^b$  is perfect.

2: Notice  $d = [a_0] - pu$  for a unit  $u \in A$  by(5.5.4.30), then we can take  $\varpi$  to be the image of  $[a_0]$  in  $R$ , and then  $\varpi^{1/p^n} = [a_0^{1/p^n}]$ .

3: firstly the LHS contains the RHS, and  $R/\cup_n(\varpi^{1/p^n})$  is perfect hence reduced, so the two sides are equal. To check it is flat, we need to check that  $M \otimes_R^L \sqrt{p}R$  is discrete, or equivalently  $M \otimes_R^L \bar{R} \in D^{\geq -1}$  where  $\bar{R} = R/\sqrt{p}R$  is perfect. But there is a distinguished triangle  $M \otimes_{\mathbb{Z}_p}^L \mathbb{Q}_p/\mathbb{Z}_p[-1] \rightarrow M \rightarrow M[\frac{1}{p}](\mathbb{Q}_p \text{ is } \mathbb{Z}_p \text{ flat (5.4.1.6)})$  and the fact  $\bar{R}[\frac{1}{p}] = 0$ , it suffices to prove  $M \otimes_{\mathbb{Z}_p}^L \mathbb{Q}_p/\mathbb{Z}_p \otimes_R^L \bar{R} \in D^{\geq -2}(R)$ . Now  $M \otimes_{\mathbb{Z}_p}^L \mathbb{Q}_p/\mathbb{Z}_p$  has cohomology groups  $p^\infty$ -torsion, so using canonical truncation, it suffices to show  $M \otimes_R^L \bar{R} \in D^{\geq -1}$  for any  $p^\infty$ -torsion  $M$ . Because tensoring commutes with filtered colimits, it suffices to show for  $M$  an  $R/p^n$ -module. Now using exact sequences like

$$0 \rightarrow M[p]/M[p^2] \rightarrow M/M[p^2] \rightarrow M[p]/M[p^2] \rightarrow 0,$$

we can reduce to the case  $M$  is a  $R/p$ -module.

Now there is a commutative diagram

$$\begin{array}{ccc} A_{\inf}(R) & \longrightarrow & R \\ \downarrow & & \downarrow \\ A_{\inf}(\bar{R}) & \longrightarrow & \bar{R} \end{array}$$

and  $d$  is  $p$ -torsionfree in both  $A_{\inf}(R)$  and  $A_{\inf}(\bar{R})$ (5.5.4.30), and  $d = [a_0] - pu = pu \in W(\bar{R})$ (as  $a_0 = 0 \in \bar{R}$ ), so  $\bar{R} = W(\bar{R})/d$  and this is a Tor-independent pushout square. Thus  $M \otimes_R^L \bar{R} \cong M \otimes_{W(R^b)}^L W(\bar{R})$ . As  $p$  is nonzero divisor in both  $A_{\inf}(R)$  and  $A_{\inf}(\bar{R})$ , and  $pM = 0$ , we have a similar diagram quotient by  $p$ , and by the same reason  $M \otimes_{W(R^b)}^L W(\bar{R}) \cong M \otimes_{R^b}^L \bar{R}$ . Now the kernel of  $R^b \rightarrow \bar{R}$  is of the form  $(f^{1/p^\infty})$ , where  $f$  corresponds to  $(\omega^{1/p^n})$ , so the claim follows from(5.5.1.7).

4: notation as in the proof of item2, it suffices to show the  $A$ -module  $R[p]$  is annihilated by  $[a_0^{1/p^n}]$  for  $n \geq 0$ . But  $R[p] = A/d[p] = A/p[d] = R^b[d]$ , and  $d = [a_0]$  on  $R^b$ , which is perfect, so we are done.  $\square$

**Prop. (27.6.3.6) [Equivalent Definition of Integral Perfectoid Rings].** A commutative ring  $R$  is an integral perfectoid ring iff the following are satisfied:

- $R$  is  $p$ -adically complete and  $R/p$  is semiperfect.
- The kernel of  $\theta_R : A_{\text{inf}}(R) \rightarrow R$  (notice  $R$  is  $p$ -adically complete) is principal.
- There exists some  $\varpi \in R$  that  $(\varpi^p) = (p)$ .

And if  $R$  is  $p$ -torsionfree, the condition2 can be replaced by:  $R$  is  $p$ -normal??.

*Proof:* If  $R$  is an integral perfectoid ring, then these are true by (27.6.3.5). Now if these are satisfied, then firstly  $\theta$  is surjective by (5.5.1.16). Next, let  $d \in A_{\text{inf}}(R)$  be the generator of  $\theta$ , we show  $A_{\text{inf}}(R)$  is derived  $(p, d)$ -complete. it is derived  $p$ -complete by (5.9.6.16). Now  $R^b$  is derived  $d$ -complete by (5.5.1.18). By induction and (5.9.6.8),  $A_{\text{inf}}(R)/p^n$  are all derived  $d$ -complete, and also by induction  $A_{\text{inf}}(R)/p^n$  has bounded  $d^\infty$ -torsion, as  $R^b$  is perfect. Then  $A_{\text{inf}}(R)/p^n$  are all  $d$ -adically complete. Thus

$$A_{\text{inf}}(R) = \varprojlim_n A_{\text{inf}}(R)/p^n = \varprojlim_n \varprojlim_m A_{\text{inf}}(R)/(p^n, d^m),$$

which means  $A_{\text{inf}}(R)$  is  $(p, d)$ -adically complete thus derived  $(p, d)$  adically complete.

Then it suffices to show  $d$  is distinguished, by (5.5.4.30). Let  $\varpi^p = up$ , and lift  $\varpi, u$  to  $x, v \in A_{\text{inf}}(R)$ . Since  $A_{\text{inf}}(R)$  is  $d$ -adically complete,  $v$  is unit in  $A_{\text{inf}}(R)$ . Then  $d|x^p - pv$  and  $x^p - pv$  is distinguished (5.5.4.30). Now  $d, p \in \text{rad}(A_{\text{inf}}(R))$  as  $A_{\text{inf}}(R)$  is  $(d, p)$ -adically complete, so by (5.5.4.22),  $d$  is distinguished.

Now if  $R$  is a  $p$ -torsionfree integral perfectoid ring, if  $x \in R[\frac{1}{p}]$  satisfies  $x^p \in R$ , let  $n \geq 0$  minimal that  $y = \varpi^n x \in R$ , then we show  $n = 0$ : if  $n > 0$ , then  $(\varpi^n x)^p = \varpi^{np} x^p \in \varpi^{np} R$ . Then we get  $\varpi^n x \in \varpi^n R$ , and then  $x \in R$  as  $R$  is  $p$ -torsionfree.

Conversely, we use condition1, 2, 4 to prove 3: We first show the kernel of  $\varphi : R/p \rightarrow R/p$  is generated by  $\varpi$  as in condition4: if  $x^p \in pR = \varpi^p R$ , then  $(x/\varpi)^p = y$ , thus  $x \in \varpi R$  by hypothesis. Since  $R/p$  is semiperfect,  $\overline{\varpi}$  admits a compatible  $p^n$ -th roots  $\{\overline{\varpi}^{1/p^n}\}$ . It can be shown by induction that  $\ker(\varphi^n) = (\overline{\varpi}^{1/p^n})$ . This implies that the kernel of  $\overline{\theta}_R : R^b \rightarrow R/p$  is generated by the element  $\varpi^b$  determined by the system  $\{\overline{\varpi}^{1/p^n}\}$ . As  $W(R^b)$  and  $R$  are both  $p$ -torsionfree and  $p$ -adically complete, they kernel of  $\theta_R$  is generated by any element in the kernel that lifts  $\varpi^b$ . In particular, the kernel is principal.  $\square$

**Prop. (27.6.3.7) [Pushout of Integral Perfectoid Rings].** Integral perfectoid rings are closed under pushouts in the category of derived  $p$ -complete rings: i.e. if  $C \leftarrow A \rightarrow B$  are maps of integral perfectoid rings, then  $B \widehat{\otimes}_A^L C$  is also an integral perfectoid ring.  $\square$

*Proof:* Let  $R = C^b \otimes_{A^b}^L B^b = C^b \otimes_{A^b} B^b$  which is a perfect ring, by (5.5.1.3) and (5.5.1.7). Then we have

$$W(R) = W(A^b) \widehat{\otimes}_{W(A^b)}^L W(B^b),$$

because this can be checked via derived Nakayama (5.9.6.10). Now use the fact  $A = W(A^b)/d$  for some distinguished element  $d$ , and then  $B = W(B^b)/d, C = W(C^b)/d$  by rigidity (27.6.1.7), and  $d$  is nonzero-divisor in  $W(B^b), W(C^b), W(R)$  by (5.5.4.28), so taking derived base change along  $W(A^b) \rightarrow A$ , we get

$$D = W(R)/d = B \widehat{\otimes}_A^L C,$$

and  $W(R)$  is a perfect prism, by (27.6.1.7), so  $D = W(R)/d$  is an integral perfectoid ring and equals  $B \widehat{\otimes}_A^L C$ .  $\square$



**Cor. (27.6.3.8).** The category of integral perfectoid rings is closed under arbitrary colimits and products in the category of derived  $p$ -complete rings.

But it is not closed under equalizers: Notice by Ax-Sen-Tate,  $(\mathcal{O}_{\mathbb{C}_p})^{\text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)} = \mathbb{Z}_p$ , but  $\mathbb{Z}_p$  is not an integral perfectoid by (27.6.3.6).  $\lrcorner$

*Proof:* It suffices to show it is closed under products and sums ?.  $\square$

**Prop. (27.6.3.9) [Gluing].** Let  $R$  be an integral perfectoid ring,  $\overline{R} = R/\sqrt{p}R$ ,  $S = R/R[\sqrt{p}R]$ ,  $\overline{S} = S/\sqrt{p}S$ , then  $\overline{R}, S, \overline{S}$  are perfectoids, and the square

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ \overline{R} & \longrightarrow & \overline{S} \end{array}$$

is both a homotopy fiber square (5.9.1.2) and pullback square. Moreover,

- $S$  is  $p$ -torsionfree.
- $\sqrt{p}R$  maps isomorphically onto  $\sqrt{p}S$ .
- $R[\sqrt{p}R]$  maps isomorphically to  $\ker(\overline{R} \rightarrow \overline{S})$ , thus  $x \mapsto x^p$  is bijective on  $R[\sqrt{p}R]$ .

In particular, any integral perfectoid ring is a fiber product of integral rings that is either perfect or  $p$ -torsionfree.  $\lrcorner$

*Proof:* By (27.6.3.5),  $R[\sqrt{p}R] = R[p^\infty]$ . In particular,  $S$  is  $p$ -torsionfree. Now if we know this is a homotopy fiber square, then we get 2, 3 by comparing the kernel. And if we know the kernel, then this is a pushout square by (5.1.1.26). So it suffices to show this is a homotopy fiber square.

Let  $d = [a_0] - pu$  for a distinguished element of  $A = A_{\text{inf}}(R)$  that  $R = A/(d)$ , and the ideal  $I = (a_0^{1/p^\infty}) \subset R^\flat$  and  $J = R^\flat[I]$ . Then the square

$$\begin{array}{ccc} W(R^\flat) & \longrightarrow & W(R^\flat/J) \\ \downarrow & & \downarrow \\ W(R^\flat/I) & \longrightarrow & W(R^\flat/I + J) \end{array}$$

is a homotopy fiber square: all vertices are  $p$ -torsionfree and  $p$ -adically complete, and the square gives a fiber square when modulo  $p^n$  (use induction on  $n$  and (5.5.1.8)), and then take derived  $p$ -completions.

Next we apply ?  $\square$

**Cor. (27.6.3.10).** Integral perfectoid rings are reduced.  $\lrcorner$

*Proof:* By (27.6.3.9), we may assume  $R$  is  $p$ -torsionfree or perfect. Thus it suffices to assume  $R$  is  $p$ -torsionfree. Let  $\varpi \in R$  that  $\varpi^p = pu$  as in (27.6.3.6). If  $x^p = 0$ , we show inductively that  $x \in \varpi^n R$ . If  $x = \varpi^n y$ , then  $y^p = 0$  as  $R$  is  $p$ -torsionfree. Now the kernel of Frobenius  $\varphi : R/p \rightarrow R/p$  is generated by  $\varpi$ , thus we have  $y \in \varpi R$ , so we can use induction.  $\square$

## 4 Prismatic Site

**Remark (27.6.4.1).** In this subsection, fix a bounded prism  $(A, I)$ , all formal schemes over  $A$  are assumed to have the  $(p, I)$ -adic topology, and formal schemes over  $A/I$  are assumed to have the  $p$ -adic topology.  $\lrcorner$

**Def. (27.6.4.2) [Prismatic Site].** Let  $(A, I)$  be a bounded prism and  $X$  be a smooth  $p$ -adic formal scheme over  $A/I$ , let  $(X/A)_\Delta$  be the site whose objects are bounded prisms  $(B, IB)$  over  $(A, I)$  together with a map  $\mathrm{Spf}(B/IB) \rightarrow X$  over  $A/I$ . The morphisms are the natural one, and the coverings in  $(X/A)_\Delta$  are f.f. maps of prisms  $(B, IB) \rightarrow (C, IC)$ . There are structure sheaves  $\mathcal{O}_\Delta((B, IB)) = B$  and  $\overline{\mathcal{O}}((B, IB)) = B/IB$ . They are sheaves by (27.6.1.12).

Thus  $\mathcal{O}_\Delta$  is valued in  $(p, I)$ -complete  $\delta$ - $A$ -algebras and  $\overline{\mathcal{O}}_\Delta$  is valued over  $p$ -complete algebras (5.9.6.16).  $\lrcorner$

**Def. (27.6.4.3) [Perfect Prismatic Site].** The **perfect prismatic site**  $(X/A)_\Delta^{\mathrm{perf}}$  is the full subcategory of  $(X/A)_\Delta$  consisting of perfect prisms. By (27.6.3.3), objects in this site are equivalent to the category of perfectoid rings  $R$  over  $A/I$  with a map  $\mathrm{Spf} R \rightarrow X$ .  $\lrcorner$

**Remark (27.6.4.4).** If we further restrict to the site of perfect prisms  $(S, I)$  that  $S/I$  is integrally closed in  $S/i[\frac{1}{p}]$ , then we will get the notion of diamond of  $(X[\frac{1}{p}], X)$ , in sense of [Sch12].  $\lrcorner$

**Def. (27.6.4.5) [Absolute Prismatic Site].** For a  $p$ -adic formal scheme  $X$  the **absolute prismatic site** consisting of bounded prisms  $(B, J)$  with a map  $\mathrm{Spf} B/J \rightarrow X$ .  $\lrcorner$

**Prop. (27.6.4.6) [Prismatic Site and Étale Site].** Let  $\mathcal{FSch}/X$  be the category of  $p$ -adic formal schemes over  $X$  with the étale topology, then there is a natural functor  $\mu : (X/A)_\Delta \rightarrow \mathcal{FSch}/X$  sending  $(B, IB)$  over  $X$  to  $\mathrm{Spf} B/IB \rightarrow X$ .

This functor is cocontinuous: for any  $p$ -completely étale map  $B/IB \rightarrow C$ , it is a derived  $p$ -completion of some étale map  $B/IB \rightarrow \overline{C}'$  by (5.9.7.9), and this can be lifted to a map  $B \rightarrow S_C^l$  by (27.6.1.11)(5.3.10.6) and (5.3.10.9), and we choose the  $(p, I)$ -completion of  $S_C$ , then it is a prism that lifts  $C$ . Thus by (7.1.2.21) defines a morphism of topoi:

$$\mu : \mathrm{Sh}((X/A)_\Delta) \rightarrow \mathrm{Sh}(\mathcal{FSch}/X).$$

Also there is a natural map of topoi  $\mathrm{Sh}(\mathcal{FSch}/X) \rightarrow \mathrm{Sh}(X_{\mathrm{\acute{e}t}})$  by restriction, by (5.9.7.1). So we get a morphism of topoi

$$\nu : \mathrm{Sh}((X/A)_\Delta) \rightarrow \mathrm{Sh}(X_{\mathrm{\acute{e}t}}).$$

In particular, for any étale formal scheme  $U$  over  $X$ , by definition of  ${}_s\mu$  (7.1.2.11), for any sheaf  $\mathcal{F}$ ,

$$(\nu_*\mathcal{F})(U/X) = H^0((U/A)_\Delta, \mathcal{F}|_{(U/A)_\Delta}).$$

$\lrcorner$

**Cor. (27.6.4.7) [Prismatic Complex and Hodge-Tate Complex].** In the above situation, we define **prismatic complexes**

$$\Delta_{X/A} = R\nu_*\mathcal{O}_\Delta \in D(X_{\mathrm{\acute{e}t}}, A)$$

and the **Hodge-Tate complex**

$$\overline{\Delta}_{X/A} = R\nu_*\overline{\mathcal{O}}_\Delta \in D(X_{\mathrm{\acute{e}t}}, \mathcal{O}_X).$$

The Frobenius action on  $\mathcal{O}_\Delta$  induces a  $\varphi$ -semi-linear map  $\Delta_{X/A} \rightarrow \Delta_{X/A}$ . And there is a relation

$$\overline{\Delta}_{X/A} \cong \Delta_{X/A} \otimes_A^L A/I \in D(X_{\mathrm{\acute{e}t}}, A/I)$$



by the Grothendieck spectral sequences associated to the diagram of functors:

$$\begin{array}{ccc} \mathrm{Mod}((X/A)_\Delta, \mathcal{O}_\Delta) & \xrightarrow{\otimes_{\mathcal{O}_\Delta} \overline{\mathcal{O}}_\Delta} & \mathrm{Mod}((X/A)_\Delta, \overline{\mathcal{O}}_\Delta) \\ \downarrow \Gamma((X/A)_\Delta, -) & & \downarrow \Gamma((X/A)_\Delta, -) \\ \mathrm{Mod}(X_{\mathrm{\acute{e}t}}, A) & \longrightarrow & \mathrm{Mod}(X_{\mathrm{\acute{e}t}}, A/I) \end{array}$$

┘

### Affine Case

**Def. (27.6.4.8) [Situation].** In this subsection, fix a bounded prism  $(A, I)$  and a  $p$ -completely smooth(5.9.7.1)  $A/I$ -algebra  $R$  (or equivalently the  $p$ -adic completion of an étale  $A$ -algebra, by(5.9.7.9) and(5.9.6.16), and they define the same site by the universal property of completion).

┘

**Prop. (27.6.4.9) [Prismatic Site over Affine Formal Scheme].** The **prismatic site** of  $R$  relative to  $A$ , denoted by  $(R/A)_\Delta$  is the site whose objects are bounded prisms over  $(A, I)$  together with an  $A/I$ -algebra map  $R \rightarrow B/IB$ . And it is endowed with the indiscrete topology(7.1.1.2), so sheaves on this site is just presheaves.

There are two natural sheaves on this site,  $\mathcal{O}_\Delta$  maps a prism  $(B, IB)$  to  $B$  which is valued in  $(p, I)$ -complete  $\delta$ - $A$ -algebras, and  $\overline{\mathcal{O}}_\Delta$  which maps a prism  $(B, IB)$  to  $B/IB$  which is valued in  $p$ -complete  $R$ -algebras(5.9.6.16)

┘

**Prop. (27.6.4.10) [Compare with Indiscrete Topology].** If  $(X/A)'_\Delta \subset (X/A)_\Delta$  is a continuous map of sites that the former is endowed with the indiscrete topology, then it is a morphism of sites, by(7.1.2.14), so this induces a morphism of topoi

$$\mathrm{Sh}((X/A)_\Delta) \rightarrow \mathrm{Sh}((X/A)'_\Delta)$$

by(7.1.2.20), then we have the Leray spectral sequence(7.3.1.9)

$$E_2^{p,q} = H^p(\mathcal{C}', R^q f_*(\mathcal{F}^\bullet)) \Rightarrow H^{p+q}(\mathcal{C}, \mathcal{F}^\bullet).$$

and by(7.3.1.7)(7.3.1.6) and(27.6.1.12), we have a natural isomorphism

$$R\Gamma((X/A)'_\Delta, \mathcal{F}) \rightarrow R\Gamma((X/A)_\Delta, \mathcal{F})$$

for  $\mathcal{F} = \mathcal{O}_\Delta$  or  $\overline{\mathcal{O}}_\Delta$ .

┘

### How to Compute the Prismatic Complex in the Affine Case

**Lemma (27.6.4.11) [Weakly Final Object].** Let  $(A, I)$  be a prism and let  $R$  be a  $p$ -completely smooth  $A/I$ -algebra, then the category  $(R/A)_\Delta$  admits a weakly final object. Moreover, we can choose it to be flat over  $(A, I)$ .

┘

*Proof:* Let  $F_0$  be the derived  $(p, I)$ -completion of a free  $\delta$ -ring over  $A$  on the set  $R$ , then there is a surjection of  $A$ -algebras  $F_0 \rightarrow R$ , with kernel  $J$  derived  $(p, I)$ -complete<sup>?</sup>. Then(27.6.1.13) applied to the  $\delta$ -ring  $(F_0, J)$  gives a prism  $(F, IF)$  over  $(A, I)$ , and by construction it is an object of  $(R/A)_\Delta$ . And it is weakly final because of the universal properties of  $F_0$ (5.5.4.10) and  $F$ (27.6.1.13).

For the flatness, we temporarily call a  $\delta$ -pair  $(B, J)$  **good** if

- $B$  is  $(p, I)$ -completely flat over  $A$  and  $J$  is  $(p, I)$ -complete.
- the prismatic envelope is flat over  $(A, I)$  and its formation commutes with completed derived base change on a  $(p, I)$ -completely flat map  $B \rightarrow B'$ .

Then we need to show that  $(F_0, J)$  is good. We have the following observations:

- Good pairs are stable under filtered colimit in the category of  $\delta$ -pairs  $(B, J)$  that  $J$  is derived  $(p, I)$ -complete. (Because filtered colimits of flat modules are flat(5.4.1.6).)
- If  $(B, J)$  is a  $\delta$ -pair over  $A$  with  $B$  completely  $(p, I)$ -flat over  $A$ , and  $B \rightarrow B'$  is a  $(p, I)$ -completely f.f. map that  $(B', (JB')^\wedge)$  is good, then  $(B, J)$  is good. (This follows from(27.6.1.13) and the f.f. descent(27.6.1.13).)

Then we can write  $B$  as a filtered colimit of  $(p, I)$ -complete algebras  $B_s \twoheadrightarrow R$ , and the kernel of each of them is locally generate by a  $(p, I)$ -completely regular sequence, so we can use the observations to pass to f.f. localization and filtered colimit to show that  $(B, J)$  is good. Cf.[Prism, Scholze, 3.14]. ?

□

**Lemma (27.6.4.12) [Products].** The category  $(R/A)_\Delta$  admits products. ┘

*Proof:* For  $\delta$ -rings  $B, C \in (R/A)_\Delta$ , we can take the  $\delta$ -ring colimit  $D_0 = B \otimes_A C$ (5.5.4.13), but it may not be compatible with  $R$ -actions. Instead, let  $J$  be the kernel of the natural map

$$D_0 \rightarrow D_0/ID_0 \rightarrow B/IB \otimes_{A/IA} C/IC \rightarrow B/IB \otimes_R C/IC,$$

then  $(D_0, J)$  is a  $\delta$ -ring over  $(A, I)$ , and then we can use(27.6.1.13) to get a prism  $(D, ID)$  over  $(A, I)$ , then the maps  $R \rightarrow B/IB \rightarrow D_0/ID_0 \rightarrow D/ID$  and  $R \rightarrow C/IC \rightarrow D_0/ID_0 \rightarrow D/ID$  are equal(because they all factor through  $D/J$ ), thus giving a product object in  $(R/A)_\Delta$ . □

**Prop. (27.6.4.13) [Čech-Alexander Construction for Prismatic Cohomology].**

By(7.3.1.20)(27.6.4.10) and the lemmas(27.6.4.11)(27.6.4.12) above, the prismatic complex  $\Delta_{R/A}$  is represented by the complex

$$F^0 \rightarrow F^1 \rightarrow F^2 \rightarrow \dots$$

In particular,  $F^0 = F$  as constructed in(27.6.4.11) and each  $F^n$  are  $(p, I)$ -completely  $A$ -flat,  $I$ -torsion-free and  $(p, I)$ -complete  $\delta$ -rings by(27.6.1.11) and(27.6.1.14).

Moreover, this complex is the prismatic envelope functor applied to the Čech nerve of  $A \rightarrow F_0$ , w.r.t. the Čech nerve  $F^\bullet$  of ideals  $J^\bullet = \ker(F^\bullet \rightarrow F_0 \rightarrow R)$ , because prismatic envelope is a left adjoint hence commutes with tensor product of pairs(27.6.1.13). ┘

**Cor. (27.6.4.14).**  $\Delta_{R/A}$  is derived  $(p, I)$ -complete and  $\bar{\Delta}_{R/A}$  is derived  $p$ -complete, because each term of the complex  $F^\bullet$  is derived  $(p, I)$ -complete, so does its cohomology groups(5.9.6.8), thus so does  $\Delta_{R/A}$  itself. Similarly for  $\bar{\Delta}_{R/A}$ , because each term of  $F^\bullet/I$  is  $p$ -complete(27.6.4.13) hence derived  $p$ -complete, by(5.9.6.16). ┘

## 5 Hodge-Tate Comparison

**Def. (27.6.5.1) [Breuil-Kisin Twist].** Let  $I$  be an invertible ideal of  $A$ , then for any  $A/I$ -module  $M$ , we define the **Breuil-Kisin twist** of  $M$  as  $M\{n\} = M \otimes (I/I^2)^n$ . Notice this definable for  $n \in \mathbb{Z}$  because  $I/I^2$  is an invertible  $\mathcal{O}_{A/I}$ -module, by definition(27.6.1.1). Also it is definable in the level of  $D(A/I)$ , as  $(I/I^2)^n$  is locally free thus flat. ┘

**Def. (27.6.5.2)[Completed de Rham Complex].** The **completed de Rham complex** is the derived  $p$ -completion of the de Rham complex of  $\Omega_{X/(A/I)}$ . We will use the derived  $p$ -completed de Rham complex in the sequel. It has a property that it coincides with the  $p$ -completion of its separate terms by (5.9.7.8) and (5.9.6.16) and the fact  $\Omega_{X/(A/I)}$  is finite projective hence flat (5.4.5.12).

In fact, as  $\Omega_{X/S}$  is finite locally free, the derived  $p$ -completion is just by tensoring  $- \otimes_{R_0} R$ , where  $R_0$  is a smooth  $A$ -algebra that  $R_0^\wedge = R$  by (5.9.7.9). In particular, the completed de Rham complex is compatible with base change and  $p$ -completely étale extension, because the ordinary de Rham complex does (5.4.3.6)(5.4.7.6).  $\lrcorner$

**Prop. (27.6.5.3) [Hodge-Tate Comparison Theorem].** We have a structure map  $\eta^0 : \mathcal{O}_X \rightarrow H^0(\bar{\Delta}_{X/A})$ , and  $H^\bullet(\bar{\Delta}_{X/A})$  is a dga by (5.9.1.3) applied to  $M^\bullet = \Delta_{X/A}$  and (5.9.2.1) noticing  $\bar{\mathcal{O}}_\Delta$  is a sheaf of algebras. then the universal property of de Rham complex (9.2.1.4) and lemma (27.6.5.4) shows  $\eta^0$  extends to a map

$$\eta_R^\bullet : \Omega_{X/(A/I)}^\bullet \rightarrow H^\bullet(\bar{\Delta}_{X/A})\{\bullet\}$$

of sheaves of  $A/I$ -dgas on  $X_{\text{ét}}$ .

Then this is an isomorphism of differential graded  $A/I$ -algebra. In particular  $\bar{\Delta}_{X/A} \in D(X_{\text{ét}}, A/I)$  is a perfect complex with  $H^i(\bar{\Delta}_{X/A}) \cong \Omega_{X/(A/I)}^i\{-i\}$ .  $\lrcorner$

*Proof:* The proof of the isomorphism is given in (27.6.6.10).  $\square$

**Lemma (27.6.5.4).** For any local section  $f \in \mathcal{O}_X(U)$ , the differential  $\beta_I(f) \in H^1(\bar{\Delta}_{X/A})\{1\}(U)$  squares to 0.  $\lrcorner$

*Proof:* This follows from (27.6.6.9) in the affine line case because  $H^2(\bar{\Delta}_{X/A})\{1\}(U) = 0$ , and for the general case, use the étale localization (27.6.6.2) and base change theorem (27.6.6.1), noticing that cup product survives through derived tensor product.  $\square$

**Cor. (27.6.5.5)[Base Change].** The formation of  $\Delta_{X/A} \in D(X_{\text{ét}}, A)$  commutes with base change along a bounded prism  $(A, IA) \rightarrow (B, IB)$ : Let  $g : X_B = X \otimes_{\text{Spf } A/IA} \text{Spf } B/IB \rightarrow X$  be the projection, then

$$(g^* \Delta_{X/A})^\wedge \cong \Delta_{X_B/B}, \quad (g^* \bar{\Delta}_{X/A})^\wedge \cong \bar{\Delta}_{X_B/B}$$

where  $()^\wedge$  is the derived  $(p, I)$ -completion or  $p$ -completion.  $\lrcorner$

*Proof:* Because both side are derived  $(p, I)$ -complete, we take their cylinder object and by derived Nakayama w.r.t.  $B/(p, I)$  (5.9.6.10) it suffices to show the second, which is true because this is true for completed de Rham complexes (27.6.5.2).  $\square$

## 6 Proof of Hodge-Tate Comparison

The strategy is as follows: we study the affine case to construct and prove the Hodge-Tate comparison isomorphism in the affine case, and then this gives the construction of Hodge-Tate map in the general case, then we can also prove the general Hodge-Tate comparison by localizing at affine subschemes, so the affine case is important.

To prove the Hodge-Tate comparison isomorphism in the affine case, we use étale localization to reduce to the polynomial case, and then use flat base change to reduce to the oriented case. Then we use a slick strategy to reduce to the crystalline case, and finally reduce to crystalline comparison.

**Lemma (27.6.6.1) [Base Change].** Let  $R$  be a  $p$ -completely smooth  $A/I$ -algebra and  $(A, I) \rightarrow (A', I')$  be a map of bounded prisms that  $A \rightarrow A'$  has finite  $(p, I)$ -complete Tor amplitude (5.9.7.1). If  $R' = R \hat{\otimes}_A^L A'$  for the the base change, then the natural map induces an isomorphism

$$\Delta_{R/A} \hat{\otimes}_A^L A' \cong \Delta_{R'/A'}, \quad \bar{\Delta}_{R/A} \hat{\otimes}_A^L A' \cong \bar{\Delta}_{R'/A'}$$

┘

*Proof:* We use the Čech nerve of a weakly final object (27.6.4.13) to compute the cohomology, then we notice the  $(p, I)$ -completed base change  $-\hat{\otimes}_A^L A'$  applied termwise to the Čech nerve of  $A \rightarrow F^0$  is the Čech nerve of  $A \rightarrow F^0 \hat{\otimes}_A A'$ , which is weakly final in  $(R'/A')_\Delta$ , by the universal property and the fact  $(p, I)$ -completed base change is a left adjoint.

Finally we use (5.9.7.8) to see that this termwise completed derived base change just represents  $\Delta_{R/A} \hat{\otimes}_A^L A'$  because each term of the prismatic envelope  $F$  is  $(p, I)$ -completely flat over  $A$  and the completed derived base change of  $F^n$  is discrete by (27.6.4.13).  $\square$

**Lemma (27.6.6.2) [Étale Localization].** Let  $R \rightarrow S$  be a  $p$ -completely étalemap of  $p$ -completely smooth algebras, then the natural map

$$\bar{\Delta}_{R/A} \hat{\otimes}_R^L S \rightarrow \bar{\Delta}_{S/A}$$

is an isomorphism.  $\square$

*Proof:* Firstly the forgetful functor  $(R/A)_\Delta \rightarrow (S/A)_\Delta$  has a right adjoint, described as follows: a prims  $(B, IB) \in (B/S)_\Delta$  induces a  $p$ -completely étalemap of (discrete)rings  $B/IB \rightarrow B/IB \hat{\otimes}_R^L S$  by (5.9.7.5), and by Elkik's algebrization (5.9.7.9), this is a derived  $p$ -completion of some étalemap  $B/IB \rightarrow T_0$ , and we can lift it to some étalemap  $B \rightarrow S_0$  by Henselian pair property (5.9.6.12)(5.3.10.9), then we can also take the derived  $(p, I)$ -completion (discrete by (27.6.1.11))  $S_B$  of  $S_0$ , then  $S_B/IS_B \cong B/IB \otimes_R^L S$ ?. So we have the following base change diagram:

$$\begin{array}{ccccc} B & \longrightarrow & S_0 & \longrightarrow & S_B \\ \downarrow & & \downarrow & & \downarrow \\ B/IB & \longrightarrow & T_0 & \longrightarrow & B/IB \hat{\otimes}_R^L S \end{array}$$

For the adjointness, it suffices for every prism  $(T, IT)$  with morphisms  $B \rightarrow T, B/IB \otimes_R^L S \rightarrow T/IT$ , we can lift to a map  $S_B \rightarrow T$ . But if we consider the  $p$ -completely étalemap  $T \rightarrow T \hat{\otimes}_B B_S$  and its base change, it suffices to find a section of this map, and this is by Henselian pair  $(T, IT)$  (5.3.10.6)(27.6.1.11)(5.3.10.9). Moreover,  $S_B$  has a  $\delta$ -structure by (5.5.4.19) so it is clearly a prism, and the right adjoint  $F$  just takes  $(T, IT) \rightarrow (S_B, IS_B)$ .

This right adjoint preserves weakly final objects and products, and it is just the completed derived tensor  $-\hat{\otimes}_R^L S$  when modulo  $I$  by construction, so when combined with (5.9.7.8) we get the conclusion.  $\square$

**Remark (27.6.6.3).** Notice these two lemmas (27.6.6.1)(27.6.6.2) are consequences of Hodge-Tate comparison isomorphism, once we proved it!  $\square$

**Crystalline Comparison in Characteristic  $p$** 

**Prop. (27.6.6.4) [Crystalline Comparison in Characteristic  $p$ ].** Let  $(A, (p))$  be a crystalline prism and let  $I \subset A$  be a pd-ideal with  $p \in I$ , in particular the Frobenius  $A/p \rightarrow A/p$  factors through  $A/I$  by (5.6.1.6), inducing a map  $\psi = \psi_I : A/I \rightarrow A/p$ . Let  $R$  be a smooth  $A/I$ -algebra and let  $R^{(1)} = R \otimes_{A/I, \psi} A/p$ , then there is a canonical

$$\Delta_{R^{(1)}/A} \cong R\Gamma_{\text{crys}}(R/A)$$

of  $E_\infty$ - $A$ -algebras compatible with Frobenius action.  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (27.6.6.5).** If we have a smooth  $R$  over  $A/p$  and let  $\tilde{R} = R \otimes_{A/p} A/I$ , then  $\tilde{R}^{(1)} = \varphi_* R$ , and we can apply this theorem to  $\tilde{R}$  to get a canonical isomorphism

$$\varphi^* \Delta_{R/A} \cong R\Gamma_{\text{crys}}(\tilde{R}/A)$$

of  $E_\infty$ - $A$ -algebras compatible with Frobenius action.  $\lrcorner$

**Lemma (27.6.6.6) [Hodge-Tate Comparison for the Affine Line over  $(\mathbb{Z}_p, (p))$ ].** If  $(A, (p))$  is a  $p$ -torsionfree crystalline prism and  $R = \mathbb{F}_p\langle X \rangle$ , then the Hodge-Tate map constructed in (27.6.5.3) is an isomorphism.  $\lrcorner$

*Proof:* WARNING: This proof will not use the construction of the Hodge-Tate map in degree  $> 1$  and this lemma will be used in the proof of Hodge-Tate map in the general case, so there is no cycle in the reasoning.

The map

$$\Omega_{R^{(1)}/(A/p)}^\bullet \rightarrow H^*(R\Gamma_{\text{crys}}(R/A) \otimes_A^L A/p)\{*\} \cong H^*(\Delta_{R^{(1)}/A} \otimes_A^L A/I)\{*\} = H^*(\bar{\Delta}_{R^{(1)}/A})\{*\}$$

is an isomorphism by Cartier isomorphism?, where the middle is by prismatic-crystalline comparison (27.6.6.4).

It suffices to check this is the Hodge-Tate map for  $R^{(1)}/(A/p)$ . But the Hodge-Tate map induces by the inclusion  $\mathcal{O}_X \rightarrow H^0(\bar{\Delta}_{X/S})$ . It is fairly easy to check the proof of (27.6.6.4) and (27.5.6.9) the composition is also the canonical one. Finally if we choose  $\mathbb{F}_p\langle X \rangle = R = R^{(1)}$ , then we get the desired Hodge-Tate isomorphism.  $\square$

**Lemma (27.6.6.7) [Hodge-Tate Comparison for  $(\mathbb{Z}_p, (p))$ ].** If  $(A, (p))$  is a  $p$ -torsionfree crystalline prism and  $R = \mathbb{F}_p[X_1, \dots, X_n]$ , then the Hodge-Tate map constructed in (27.6.5.3) is an isomorphism.  $\lrcorner$

*Proof:* The proof is the same as that of (27.6.6.6), notice now we already have the Hodge-Tate Comparison map.  $\square$

**Direct proof of the Hodge-Tate comparison for  $(\mathbb{Z}_p, (p))$** 

The proof of (27.6.6.6) and (27.6.6.7) is sloppy because we somehow lose track of whether the composition morphism is the Hodge-Tate comparison map. As the situation is so explicit, we decided to give a direct proof.

### Mix Characteristic Case

**Prop. (27.6.6.8) [Comparing with the Characteristic  $p$ ].** Let  $A$  be a universal oriented prism in any characteristic and  $A \rightarrow B$  as in (27.6.1.5), let  $\alpha : (A, (d)) \xrightarrow{\varphi} (A, (d)) \rightarrow (B, (p))$ , then  $\alpha$  is a map of prisms by (27.6.1.6), and:

1.  $\alpha/p$  factors as  $A/p \rightarrow A/(p, d) \xrightarrow{\varphi} A/(p, d^p) \rightarrow B/p = D_{A/p}((d))$  where the first map has finite Tor amplitude and the last two maps are f.f., thus  $\alpha/p$  has finite Tor amplitude.
2. The functor  $\widehat{\alpha}^* : D_{\text{comp}}(A, (p, d)) \rightarrow D_{\text{comp}}(B, (p))$  reflects isomorphisms.
3. For any  $p$ -completely smooth  $A/I$ -algebra  $R$ , let  $R_B = R \widehat{\otimes}_A B$ , then the map

$$\widehat{\alpha}^* \Delta_{R/A} \rightarrow \Delta_{R_B/B}$$

is an isomorphism. ⌋

*Proof:* 1: It factors because  $d^p = p! \gamma_p(d) \in pB$ , and  $B/p = D_{A/p}((d))$  by (5.6.1.19), noticing  $D_A((d)) = D_A((p, d))$ . The first map is of finite Tor amplitude (5.9.7.1) because  $d$  is a nonzero-divisor in  $A/p$  by (27.6.1.4).  $\varphi$  is f.f. because it is a base change of  $\varphi_{A/p}$  and the latter is f.f. (27.6.1.4). The last one is f.f. because it is a free summand as  $D_{A/p}((d)) = A/p[X_1, X_2, \dots]/(a^p, X_1^p, X_2^p, \dots)$  by (5.6.1.24).

2: Because  $D_{\text{comp}}(A, I)$  is a weak Serre subcategory of  $D(A)$  (5.9.6.8), to show it reflects isomorphisms, by item 1 and derived Nakayama applied to  $-\otimes_B^L B/p$ , it suffices to show if  $X \in D_{\text{comp}}(A, (p, d))$ , if  $X \otimes_A^L A/p \otimes_{A/p}^L A/(p, d) = 0$ , then  $X = 0$ , but  $X \otimes_A^L A/p \otimes_{A/p}^L A/(p, d) = X \otimes_A^L A/(p, d)$ , and this follows from derived Nakayama again.

3:  $\alpha$  has finite  $(p, d)$  Tor amplitude by (5.9.7.3) and (27.6.1.4), so 3 follows from (27.6.6.1). □

### Final Proof

**Prop. (27.6.6.9) [Hodge-Tate Comparison in the Affine Line Case].** In situation (27.6.4.8), If  $R = A/I\langle X \rangle$ ,  $\eta_R^0 : R \rightarrow H^0(\overline{\Delta}_{R/A})$  and the twisted morphisms  $\eta_R^1\{-1\} : \Omega_{R/(A/I)}^1\{-1\} \rightarrow H^1(\overline{\Delta}_{R/A})$  defined by the universal property of  $\Omega_{R/(A/I)}$  are isomorphisms, and  $H^i(\overline{\Delta}_{R/A}) = 0$  for  $i > 1$ .

In particular, by étalelocalization of prismatic cohomology (27.6.6.2), lemma (27.6.5.4) holds for any  $p$ -completely smooth algebra  $R$  over  $A/I$ . But this doesn't say that higher cohomologies vanish for any  $R$ , this is because cup product can survive derived tensor but cohomology groups cannot. ⌋

*Proof:* In this case,  $\Omega_{R/S}$  is topologically free over  $R$ , thus we can choose a map

$$\eta : R \oplus \Omega_{R/A}^1\{-1\}[-1] \rightarrow \overline{\Delta}_{R/A}$$

lifting  $\eta_R^0 \oplus \eta_R^1\{-1\}$ .

Firstly if  $(A, I) = (\mathbb{Z}_p, (p))$ , this case is done by (27.6.6.6).

Next, if  $(A, (d))$  is oriented, then there is a map of prisms from the universal oriented prism (27.6.1.4)  $A_0 \rightarrow A$ , then we have a pushout diagram of simplicial commutative rings

$$\begin{array}{ccc} A_0 & \longrightarrow & A \\ \downarrow \alpha & & \downarrow \beta \\ \mathbb{Z}_p & \longrightarrow & D_0 \longrightarrow E = A \widehat{\otimes}_{A_0}^L D_0 \end{array}$$

and  $(E, (p))$  is a simplicial prism.

Now the warning is what we have done so far can all be extended to the derived algebraic geometry setting, or at least to the "animated commutative algebra" setting!

We denote the composite of the lower row as  $\gamma$ , because  $\widehat{\alpha}^*$  reflects isomorphisms, so does  $\widehat{\beta}^*$ , and we can show  $\widehat{\beta}^* \overline{\Delta}_{R/A} \cong \widehat{\gamma}^* \overline{\Delta}_{\mathbb{F}_p\langle X \rangle/\mathbb{Z}_p}$ , and identifies the Hodge-Tate map: This is because we are in the polynomial case, so we can make the construction of  $\Delta_{R/A}$  clear: we just take the  $F^0$  to be the derived  $(p, I)$ -completion of the free  $\delta$ -rings  $A\{X\}$  in the construction of weakly final object(27.6.4.11), then the resulting Čech-Alexander complex is free and compatible with base change in the complex level.

Also  $\beta$  has finite  $(p, d)$ -Tor amplitude because  $\alpha$  has because  $(p, d)$  is regular in  $A_0$ (27.6.1.4), also  $\gamma$  has finite  $p$ -Tor amplitude because  $E$  is  $p$ -torsionfree by(27.6.1.11), and use(5.9.7.3). So we are reduced to the  $(\mathbb{Z}_p, (p))$  case, which we have done.

Finally, for a general bounded prism  $(A, I)$ , we can reduce to the oriented case by base change along the f.f. extension defined in(27.6.1.11), then we reduce to the orientable case by(27.6.6.1).  $\square$

**Prop.(27.6.6.10)[Hodge-Tate Comparison Theorem in General].** The map

$$\eta_R^\bullet : \Omega_{X/(A/I)}^\bullet \rightarrow H^\bullet(\overline{\Delta}_{X/A})\{\bullet\}$$

constructed in(27.6.5.3) is an isomorphism of sheaves of  $A/I$ -dgas on  $X_{\text{ét}}$ .  $\lrcorner$

*Proof:* Because we already have the Hodge-Tate comparison map, it suffices to prove the theorem for affine subscheme  $\text{Spf } R$ , and because both prismatic cohomology and de Rham complex are étalelocal(27.6.6.2)(27.6.5.2), it suffices to prove for the polynomial case. In this case,  $\widehat{\Omega}_{R/A}^i$  is topological free over  $R$ , then we can lift the Hodge-Tate map(27.6.5.3) to the level of chain complexes:

$$\eta : \otimes_{i=0}^n \Omega_{R/(A/I)}^i \{-i\}[-i] \rightarrow \overline{\Delta}_{R/A}.$$

And then the rest is the same as the proof of(27.6.6.9), where in the  $(\mathbb{Z}_p, (p))$  case, we use(27.6.6.7) instead of(27.6.6.6).  $\square$

### de Rham Comparisons

**Prop.(27.6.6.11)[de Rham Comparison].** In situation(27.6.4.2), if  $W(A/I)$  is  $p$ -torsionfree, then there is a natural isomorphism

$$\Delta_{X/A} \widehat{\otimes}_{A, \varphi}^L A/I \cong \Omega_{X/(A/I)}^\bullet$$

of commutative algebra objects in  $D(A/I)$ .  $\lrcorner$

*Proof:* Cf.[Scholze, Prism, 6.4].

It suffices to construct locally a functorial isomorphism  $\Delta_{R/A} \widehat{\otimes}_{A, \varphi}^L A/I \cong \Omega_{R/(A/I)}^\bullet$  and then glue.

Let  $A \rightarrow W(A/I)$  be the canonical map, and  $\psi : A \xrightarrow{\varphi} A \rightarrow W(A/I)$ , then it takes  $I$  into  $(p)$ ? But the map  $\psi/p : A/I \rightarrow W(A/I)/p = A/I$  factors through  $A/(p, I)$ , thus  $R'$  also equals  $R/p \otimes_{A/(p, I)} W(A/I)/p$ , so there is a base change diagram:

$$\begin{array}{ccccc} R & \longrightarrow & R/p & \longrightarrow & R^{(1)} = R \otimes_{A/I, \psi} W(A/I)/p \\ \uparrow & & \uparrow & & \uparrow \\ A/I & \longrightarrow & A/(p, I) & \xrightarrow{\psi_{(p, [p])}} & W(A/I)/p = A/I \end{array}$$



Base change for prismatic cohomology (27.6.5.5) gives an isomorphism

$$\Delta_{R/A} \widehat{\otimes}_{A,\psi}^L W(A/I) \cong \Delta_{R'/W(A/I)}.$$

Note that  $W(A/I) \rightarrow A/(p, I)$  is a pd-thickening with ideal  $(p, [p])$ , so we can use crystalline comparison w.r.t. the crystalline prism  $(W(A/I), (p))$  to show that

$$\Delta_{R^{(1)}/W(A/I)} \cong R\Gamma_{\text{crys}}((R/p)/W(A/I)).$$

Then finally we use the crystalline de Rham comparison to get the desired result.  $\square$

**Remark (27.6.6.12).** The technical condition  $W(A/I)$  is  $p$ -torsionfree can be removed, by [Scholze, Prism, 15.4].  $\lrcorner$

## 7 Derived de Rham Cohomology

**Def. (27.6.7.1) [Derived de Rham Cohomology].** For an  $\mathbb{F}_p$ -algebra  $k$ , the **derived de Rham cohomology** functor  $dR_{-/k} : \mathcal{CAlg}/k \rightarrow D(k)$  is the left derived functor of the functor  $\mathcal{P}oly_k \rightarrow D(k)$  given by  $R \rightarrow \Omega_{R/k}^\bullet$  via (5.8.4.2).  $\lrcorner$

**Prop. (27.6.7.2) [Derived Cartier Isomorphism].**  $\lrcorner$

### Regular Semiperfect Rings

**Def. (27.6.7.3) [Regular Semiperfect Rings].** Let  $k$  be a perfect ring, a **regular semiperfect ring** over  $k$  is an  $k$ -algebra of the form  $R/I$  where  $R$  is a perfect  $k$ -algebra and  $I$  is an ideal generated by a regular sequence.  $\lrcorner$

**Prop. (27.6.7.4).** Let  $k$  be a perfect field and  $S$  be a regular semiperfect ring, then  $\lrcorner$

## 8 Derived Prismatic Cohomology

**Prop. (27.6.8.1) [Derived Hodge-Tate Comparison].**  $\lrcorner$

## 9 $q$ -de Rham Cohomology

**Prop. (27.6.9.1) [Hodge-Tate isomorphism via  $q$ -de Rham Complex].** Cf. [Bhatt, Prism, 5.3.9.].  $\lrcorner$

## 10 Étale Comparison

**Prop. (27.6.10.1) [Frobenius Fixed Points].** Fix an  $\mathbb{F}_p$ -algebra  $B$  with an element  $t$ , let  $D(B[F])$  be the **derived category of Frobenius  $B$ -modules**: this is a category whose objects are  $(M, \varphi)$  where  $M \in D(B)$  and  $\varphi$  is a morphism  $M \rightarrow M \otimes_{B,\varphi}^L B$  in  $D(B)$ . And let  $D_{\text{comp}}(B[F])$  be the full subcategory spanned by pairs  $(M, \varphi)$  where  $M \in D_{\text{comp}}(B, (t))$  (5.9.6.4).

Given  $(M, \varphi) \in D_{\text{comp}}(B[F])$ , let  $M^{\varphi=1} = R\text{Hom}_{D(B[F])}((B, \varphi), (M, \varphi)) \in D(F_p)$ , called the **Frobenius fixed pts** of  $M$ .  $\lrcorner$

**Prop. (27.6.10.2).** Fix an  $\mathbb{F}_p$ -algebra  $B$  with an element  $t$ , then



- The functor  $D_{\text{comp}}(B[F]) \rightarrow D(\mathcal{F}_p)$  given by  $M \rightarrow M^{\varphi=1}$  and  $M \mapsto (M[t^{-1}])^{\varphi=1}$  commute with colimits.
- For any  $(M, \varphi) \in D_{\text{comp}}(B[F])$  and

⌋

## 11 Almost Purity

## 27.7 $p$ -adic Hodge Theory

Main references are [Berger, Galois representations and  $(\varphi, \Gamma)$ -modules], [Car19]. and [notes on  $p$ -adic Hodge, Conrad]. [notes on  $p$ -adic Hodge, Serin Hong].

**Notation(27.7.0.1).**

- Use notations from [p-adic Local Galois Representations](#).
- Use notations from [Formal and Rigid Geometry](#).

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### 1 Fontaine's Period Rings

#### Fontaine's Ring $\mathbf{A}_{\text{inf}}$

**Def.(27.7.1.1) [Fontaine's Ring  $\mathbf{A}_{\text{inf}}$ ].** Let  $C^\flat$  be a perfectoid field of characteristic  $p$ , for any untilt  $K$  of  $C^\flat$ , the Fontaine's ring  $\mathbf{A}_{\text{inf}} = \mathbf{A}_{\text{inf}}(\mathcal{O}_K) = W(\mathcal{O}_K^\flat) = W(\mathcal{O}_{C^\flat})$  (5.5.1.15). Also denote  $B_{\text{inf}} = \mathbf{A}_{\text{inf}}[\frac{1}{p}]$ .

the set of all the char0 untilts  $K$  of  $C^\flat$  is denoted by  $Y$ . and  $Y_{[a,b]}$  denotes those untilts that  $a \leq |p|_K \leq b$ . ┘

**Prop.(27.7.1.2).** By (12.2.9.7), if  $K \in \text{Perfd}^0 \cap \text{Field}^0$ , with tilt  $C^\flat$ , then there is a diagram

$$\begin{array}{ccc} \mathbf{A}_{\text{inf}} & \xrightarrow{\theta} & \mathcal{O}_K \\ \downarrow & & \downarrow \\ \mathcal{O}_{C^\flat} & \xrightarrow{\bar{\theta}} & \mathcal{O}_K/(p) \end{array},$$

where the Fontaine's map  $\theta$  (5.5.1.15) is surjective, and  $\ker \theta$  is generated by some distinguished element  $\xi = [t] - pu$  where  $u \in \mathbf{A}_{\text{inf}}^*$ . Moreover, any distinguished element in the kernel is a generator. ┘

*Proof:*  $\theta$  is surjective by (5.5.1.16). By (12.2.8.11), there exists  $t \in \mathbf{A}_{\text{inf}}$  s.t.  $t^\sharp = pu'$  for some  $u'$  invertible in  $\mathcal{O}_K$ . Thus  $u' = \theta(u)$  for some invertible  $u \in \mathbf{A}_{\text{inf}}$ , then  $\theta([t] - pu) = 0$ . And  $\xi$  generates the kernel because it generates after modulo  $\varpi$ , and use the fact  $\mathcal{O}_K$  is  $p$ -complete.

For the last assertion, use (27.7.1.4), which shows that if  $\xi'$  is another distinguished element in  $\ker \theta$ ,  $\mathbf{A}_{\text{inf}}/(\xi')$  is an integral domain of dimension 1. So  $(\xi) = (\xi')$  as  $\mathcal{O}_K$  is not a field.  $\square$

**Lemma(27.7.1.3).** If  $R \in \mathcal{CRing}$ ,  $x, y \in R$ , and  $x \in R^\times$  and  $R$  is  $x$ -adically complete Hausdorff, and  $y$  is not a zero-divisor in  $R/x$  and  $R/x$  is  $y$ -adically complete Hausdorff, then the same is true with  $x, y$  interchanged. ┘

*Proof:* ?  $\square$

**Prop.(27.7.1.4) [Untilts and Distinguished Elements].** (27.7.1.2) shows that for any  $K \in \text{Untilt}(C^\flat)$ , the kernel is generated by a distinguished element. Conversely, for any distinguished element  $\xi$ ,  $\mathbf{A}_{\text{inf}}/(\xi)$  can be identified with the valuation ring  $\mathcal{O}_K$  of a perfectoid field  $K$ . and

$$\mathcal{O}_C^\flat = \mathbf{A}_{\text{inf}}/p \rightarrow \mathbf{A}_{\text{inf}}/(\xi, p) \cong \mathcal{O}_K/p$$

exhibits  $K$  as an untilt of  $C^\flat$ . ┘

*Proof:* May assume  $\xi = [t] - up$  and  $t \neq 0$ . Consider the mapping  $\theta : \mathbf{A}_{\text{inf}} \rightarrow \mathbf{A}_{\text{inf}}/(\xi) = \mathcal{O}_K$ , and denote  $\theta([x])$  by  $x^\sharp$ .

Firstly, we can apply lemma(27.7.1.3) to  $\xi$  and  $p$  to conclude that  $\mathbf{A}_{\text{inf}}$  is  $\xi$ -complete and  $\xi$ -torsion-free, and  $\mathcal{O}_K$  is  $p$ -adically complete and  $p$ -torsion-free.

Now for any  $y \in \mathcal{O}_K$  is  $p$ -adically complete, there is a  $x \in \mathcal{O}_C^\flat$  that  $(y) = (x^\sharp)$ : multiplying  $p$ -power, we can assume  $y$  is not divisible by  $p$ , and there is a  $x$  that  $y \equiv x^\sharp \pmod{p}$ , thus  $x$  is not divisible by  $t$ . Now  $t = xx'$  for some  $x' \in \mathfrak{m}_C^\flat$ , thus  $y = x^\sharp + t^\sharp w = x^\sharp(1 + x'w)$ , and  $1 + x'w$  is invertible in  $\mathcal{O}_K$ .

Next we prove  $\mathcal{O}_K$  is an integral domain: It suffices to show any  $y \neq 0 \in \mathcal{O}_K$  is not a zero-divisor. We can assume  $y = x^\sharp$ , by what just proved, and then  $x$  divides  $t^n$  for some  $n$ , so it suffices to consider  $y = t^{n\sharp} = p^n$ , and  $p^n$  is not a zero-divisor by what just proved.

Now we can endow  $\mathcal{O}_K$  with the valuation  $|y| = |x|_{C^\flat}$  for  $y = x^\sharp u$ , and extend it to the quotient field  $K$ . Then this is a Non-Archimedean valuation and the residue field has char  $p$  because  $|p| < 1$ , and  $K$  has char 0, because  $p \neq 0$  in  $K$ . And it is  $p$ -adically complete.

Finally,  $\mathcal{O}_K/(p) \cong \mathbf{A}_{\text{inf}}/(\xi, p) = \mathcal{O}_{C^\flat}/\pi$ , so the Frobenius is surjective, thus  $K = K(\mathcal{O}_K)$  is a perfectoid field.  $\square$

**Cor. (27.7.1.5).** The correspondence  $\xi \mapsto \text{Frac}(\mathbf{A}_{\text{inf}}/(\xi))$  induces a bijection

$$\{\text{Distinguished elements}\}/\text{units} \cong \{\text{Untilts of } C^\flat\}/\text{isomorphisms}.$$

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**Prop. (27.7.1.6) [ $\mathbf{A}_{\text{inf}}$  as Holomorphic Function in  $p$ ].** Any element in  $\mathbf{A}_{\text{inf}}$  can be written uniquely as a unique Teichmuller representation  $[c_0] + [c_1]p + [c_2]p^2 + \dots$ . Now we can regard these elements as holomorphic functions on  $B(0, 1)$ , and any untilts  $K$  of  $\mathcal{O}_{C^\flat}$  can be regarded as points in  $B(0, 1)$ , where  $\mathbf{A}_{\text{inf}}$  take value  $c_0^\sharp + c_1^\sharp p + \dots \in \mathcal{O}_K$  at the point  $K$ .

This map can in fact be extended to  $\mathbf{A}_{\text{inf}}[\frac{1}{[t]}, \frac{1}{p}]$  s.t.

$$\mathbf{A}_{\text{inf}} \hookrightarrow \mathbf{A}_{\text{inf}}[\frac{1}{[t]}] \hookrightarrow \mathbf{A}_{\text{inf}}[\frac{1}{[t]}, \frac{1}{p}] \rightarrow K.$$

called the **evaluation map**.

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### Fontain's Ring $B$

**Def. (27.7.1.7) [Fontaine's Ring  $B$ ].** If compared to the complex case, the elements of  $\mathbf{A}_{\text{inf}}$  are just elements  $\sum a_n z^n$  that  $|a_n| \leq 1$ , this are not all the holomorphic functions on  $B(0, 1)$ , which is

$$\left\{ \sum_{n \in \mathbb{Z}} a_n z^n \mid \limsup_{n \rightarrow \infty} |a_n| \leq 1, \quad \lim_{n \rightarrow \infty} |a_{-n}|^{1/n} = 0 \right\}.$$

This leads to an enlargement of  $\mathbf{A}_{\text{inf}}$ :

For  $0 < a \leq b < 1$  in the value group of  $C^\flat$ ,  $\pi_a, \pi_b \in C^\flat$  with  $|\pi_a| = a, |\pi_b| = b$ , define

$$B_{[a,b]} = \mathbf{A}_{\text{inf}}[\frac{[\pi_a]}{p}, \frac{p}{[\pi_b]}]^\wedge [p^{-1}],$$

this is definable at any untilts  $K$  that  $a \leq |p|_K \leq b$ .

Then  $B_{[a,b]}$  is an algebra over  $\mathbf{A}_{\text{inf}}[\frac{1}{[t]}, \frac{1}{p}]$ , and define  $B = \varprojlim B_{[a,b]}$ .  $\square$

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**Prop. (27.7.1.8) [Gauss Norm].** Any element  $f$  in  $\mathbf{A}_{\text{inf}}[\frac{1}{[t]}, \frac{1}{p}]$  is of the form  $\sum_{n \gg -\infty} [c_n]p^n$ , where  $\{|c_n|\}$  is bounded. So we can define the valuation  $|f|_\rho = \sup\{|c_n|\rho^n\}$ , it is realizable by some term  $|a_n|\rho^n$ . Notice that for an until  $y = (K, \iota)$ , if  $\rho = |p|_K$ , then  $|f(y)| \leq |f|_\rho$ .

Then this is a non-Archimedean valuation on  $\mathbf{A}_{\text{inf}}[\frac{1}{[t]}, \frac{1}{p}]$ .  $\lrcorner$

*Proof:* Firstly  $|f + g|_\rho \leq \max\{|f|_\rho, |g|_\rho\}$  for every  $\rho$  that is generic for  $f + g$  and in the value group of  $C^b$ : In this case,

$$|f + g|_\rho = |(f + g)(y)| \leq \max\{|f(y)|, |g(y)|\} \leq \max\{|f|_\rho, |g|_\rho\}$$

for some point  $y$  by (27.7.1.8), then by continuity and (27.7.1.9), this is true for any  $\rho$ .

The same method shows that  $|f|_\rho |g|_\rho = |fg|_\rho$ .  $\square$

**Lemma (27.7.1.9) [Generic Norms].**  $\rho \in (0, 1)$  is called a **generic value** for  $f$  iff the valuation is realized exactly once. Notice if  $\rho$  is generic for  $f$  and in the value group of  $C^b$ , then  $|f|_\rho = |f(y)|$  for some  $y$  (Choose  $K = \mathbf{A}_{\text{inf}}/([c] - p)$  where  $|c|_{C^b} = \rho$ ).

For any  $f$ , the numbers  $\rho$  that  $\rho$  is not generic for  $f$  is discrete in  $\rho$ .  $\lrcorner$

*Proof:* Consider the Newton polygon of  $f$ , then only the slopes of the Newton polygon are not generic.  $\square$

**Lemma (27.7.1.10).** If  $y = (K, \iota) \in Y$  and  $|p|_K = \rho$ , then  $|f(y)| \leq |f|_\rho$ , and equality holds if either  $\rho$  is generic or  $f$  is invertible.  $\lrcorner$

**Prop. (27.7.1.11) [Valuation Map].** For  $0 < a \leq b < 1$  in the value group of  $C^b$ ,  $|\pi_a| = a, |\pi_b| = b$ ,

$$\mathbf{A}_{\text{inf}}\left[\frac{[\pi_a]}{p}, \frac{p}{[\pi_b]}\right] = \left\{f \in \mathbf{A}_{\text{inf}}\left[\frac{1}{[t]}, \frac{1}{p}\right] : |f|_a \leq 1, |f|_b \leq 1\right\} = V_0.$$

Thus the ring  $B_{[a,b]}$  is identified with the completion of  $\mathbf{A}_{\text{inf}}[\frac{1}{[t]}, \frac{1}{p}]$  w.r.t the valuation  $|\cdot|_a + |\cdot|_b$  (19.1.6.4). In particular, for any point  $y$  that  $a \leq |p|_K \leq b$ , the valuation map (27.7.1.6) can be extended to a map

$$B_{[a,b]} \rightarrow K.$$

$\lrcorner$

*Proof:* Notice  $V_0$  is a subring by (27.7.1.8), so clearly  $\mathbf{A}_{\text{inf}}[\frac{[\pi_a]}{p}, \frac{p}{[\pi_b]}] \subset V_0$ .

For the reverse containment, notice that  $\{|c_n|\}$  is bounded, so there is an  $m$  that  $\pi_b^m c_n \in C^b$  for any  $n$ . Now

$$f = \sum_{n < m} [c_n]p^n + \left( \sum_{n \geq 0} [c_{n+m} \pi_b^m] p^n \right) \left( \frac{p}{[\pi_b]} \right)^m,$$

so it suffices to prove the case  $f$  has finite presentation. Now  $c_n \pi_a^n, c_n \pi_b^n \in \mathcal{O}_C^b$ , thus  $[c_n]p^n = [c_n \pi_a^n] \left( \frac{\pi_n}{p} \right)^{-n} = [c_n \pi_b^n] \left( \frac{p}{\pi_b} \right)^n \in \mathbf{A}_{\text{inf}}[\frac{[\pi_a]}{p}, \frac{p}{[\pi_b]}]$ , where  $n \geq 0$  or  $n \leq 0$ . Thus the inverse containment is true.  $\square$

**Prop. (27.7.1.12) [Topology of  $B$ ].** For  $0 < a \leq c \leq b < 1$ ,  $|f|_c \leq \max\{|f|_a, |f|_b\}$  (trivial), thus the Fontaine's ring  $B$  can be realized as the completion of  $\mathbf{A}_{\text{inf}}$  w.r.t. all these norms, and endowed with the topology of  $p$ -adic Fréchet space.  $\lrcorner$

*Proof:* Cf. [Conrad, P65]. ?  $\square$

**Prop. (27.7.1.13) [Teichmuller Expansion].** An infinite sum  $f = \sum [a_n]p^n$  converges in  $B$  iff it converges in any norm  $|\cdot|_\rho$  for  $0 < \rho < 1$ , which is equivalent to

$$\limsup_{n \geq 0} |c_n|_{C^b}^{1/n} \leq 1, \quad \lim_{n \rightarrow \infty} |c_{-n}|_{C^b}^{1/n} = 0.$$

This is analogous to the complex case (12.4.3.4). However, for now, we don't know iff every element of  $B$  is of this form, and whether the representation is unique?  $\lrcorner$

**Prop. (27.7.1.14) [Frobenius Action].** Notice the Frobenius action of  $C^b$  extends to a Frobenius action on the Witt vector  $\mathbf{A}_{\text{inf}}$ , and it satisfies

$$|\varphi(f)|_{\rho^p} = (|f|_\rho)^p,$$

Thus induces an isomorphism  $B_{[a,b]} \cong B_{[a^p, b^p]}$ . Passing to the limit, we get an automorphism of  $B$ , denoted also by  $\varphi$ .  $\lrcorner$

### the Field $B_{\text{dR}}$

**Prop. (27.7.1.15) [Untilts with Roots of Unity].** Let  $\mathbb{Q}_p^{\text{cycl}} = \mathbb{Q}_p(\mu_{p^\infty})^\wedge$ , and  $\varepsilon = (1, \mu_p, \mu_{p^2}, \dots)$  be a compatible  $p^n$ -th roots of unity that is an element of  $(\mathbb{Q}_p^{\text{cycl}})^b$ . Then  $\varepsilon - 1$  is a pseudo-uniformizer of  $(\mathbb{Q}_p^{\text{cycl}})^b$ . For any untilts  $K$  of  $C^b$  and an embedding of  $\mathbb{Q}_p^{\text{cycl}}$  in  $K$ , the tilting maps  $\varepsilon - 1$  to a pseudo-uniformizer of  $C^b$ . This induces a bijection:

$$\{\text{Untilts } (K, \iota) \text{ of } C^b \text{ with an embedding } \mathbb{Q}_p^{\text{cycl}} \hookrightarrow K\} \cong \{x \in C^b \mid 0 < |x - 1| < 1\}.$$

$\lrcorner$

*Proof:* In fact, the left hand side is equivalent to  $K$  has a compatible  $p^n$ -th roots of unity, and we want to prove that for any  $x$  in the right hand side, there is a unique untilts  $K$  that  $(x^{\frac{1}{p^k}})^\sharp$  is a compatible primitive roots of unity, and this is equivalent to  $(x^{\frac{1}{p}})^\sharp$  satisfies  $1 + x + \dots + x^{p-1} = 0$ , and further equivalent to  $\theta : \mathbf{A}_{\text{inf}} \rightarrow \mathcal{O}_K$  annihilates  $1 + [x^{\frac{1}{p}}] + \dots + [x^{\frac{p-1}{p}}]$ .

It suffices to show  $\xi = 1 + [x^{\frac{1}{p}}] + \dots + [x^{\frac{p-1}{p}}]$  is distinguished (27.7.1.5). Let  $\xi = \sum [c_n]p^n$ , consider reducing to the residue field:  $W(\mathcal{O}_{C^b}) \rightarrow W(\mathcal{O}_{C^b}/\mathfrak{m}_{C^b})$ , then  $\bar{x} = 1$ , and  $\bar{\xi} = p$ , thus  $|c_0| < 1, |c_1 - 1| < 1$ , so it is distinguished (5.5.4.20).  $\square$

**Cor. (27.7.1.16).** Considering different  $p^n$ -th roots of unities, there is a correspondence:

$$\{\text{Char 0 Untilts } (K, \iota) \text{ of } C^b \text{ with a compatible } p^n\text{-th roots of unity}\} \cong \{x \in C^b \mid 0 < |x - 1| < 1\} / \mathbb{Z}_p^*.$$

where  $\mathbb{Z}_p^*$  acts by exponentiation (12.2.8.8).

Furthermore, there is a correspondence:

$$\{\text{Char 0 Untilts } (K, \iota) \text{ of } C^b \text{ with a compatible } p^n\text{-th roots of unity}\} / \varphi_{C^b}^{\mathbb{Z}} \cong \{x \in C^b \mid 0 < |x - 1| < 1\} / \mathbb{Q}_p^*.$$

where the inverse is given by  $x \mapsto$  the vanishing locus of  $\log([x]) \in B$ .  $\lrcorner$

*Proof:* The only thing needed to be proven is the inverse is given by  $N(\log([x]))$ . Notice for any untilts  $K$ ,  $|(x^{p^n})^\sharp - 1| < |p|_K^{1/(p-1)}$  for  $n$  large, then  $\log((x^{p^n})^\sharp) = 0$  iff  $(x^{p^n})^\sharp = 1$  by Newton polygon. Now  $x^\sharp \neq 1$  because  $x \neq 1$  and  $\sharp$  is injective. Hence composing  $\varphi^n$  for some unique  $n$ , we can assume  $x^\sharp = 1, x^{\frac{1}{p}} \neq 1$ , thus it corresponds an untilt as in (27.7.1.15).  $\square$

**Def. (27.7.1.17)**  $[B_{\text{dR}}^+]$ .  $p$  is not a zero-divisor in  $\mathbf{A}_{\text{inf}}/(\xi^n)$ , as in the proof pf(27.7.1.4), so we can define

$$B_{\text{dR}}^+ = \varprojlim_n \mathbf{A}_{\text{inf}}/(\xi^n) \left[ \frac{1}{p} \right]$$

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**Prop. (27.7.1.18)** **[Fontaine's Ring  $B_{\text{dR}}$ ]**.  $B_{\text{dR}}^+$  is a CDVR with  $\xi$  a uniformizer and the residue field  $K$ . Hence we can define  $B_{\text{dR}} = \text{Frac}(B_{\text{dR}}^+)$ . ┘

*Proof:* Firstly  $\xi$  is not a zero divisor in  $B_{\text{dR}}^+$ , because if  $\xi x = 0, x = (x_n)$ , then for any  $n > 0$ , and some  $k$  that  $p^k x_n \in \mathbf{A}_{\text{inf}}/(\xi^n)$ , so  $p^k x_n$  is annihilated by  $\xi$  in  $\mathbf{A}_{\text{inf}}/(\xi^n)$ , thus  $p^k x_n = \xi^{n-1} y_n$  for some  $y_n$ , because  $\xi$  is a non-zero-divisor in  $\mathbf{A}_{\text{inf}}$  (27.7.1.4). So  $p^{n-1} x_{n-1} = 0 \in \mathbf{A}_{\text{inf}}/(\xi^n)$ , thus  $x_{n-1} = 0$ , because  $p$  is non-zero-divisor in  $\mathbf{A}_{\text{inf}}/(\xi^n)$  (27.7.1.4).

Next there is a map  $B_{\text{dR}}^+ / (\xi^m) \rightarrow \mathbf{A}_{\text{inf}} / (\xi^m) [p^{-1}]$ . This is an isomorphism: it is clearly a surjection, and if  $x = (x_n)$  is mapped to 0, then for each  $n \geq m$ , we choose  $p^{k(n)} x_n = 0 \in \mathbf{A}_{\text{inf}}/(\xi^n)$ , then  $p^{k(n)} x_n = \xi^m y_n$  for a unique  $y_n \in \mathbf{A}_{\text{inf}}/(\xi^{n-m})$ . So  $x = \xi^m \cdot (\frac{y_n}{p^{k(n)}}) \in \xi^m B_{\text{dR}}^+$ . (Notice the uniqueness of  $y_n$  shows  $(\frac{y_n}{p^{k(n)}})$  is an element in  $B_{\text{dR}}^+$ ).

Then it follows  $B_{\text{dR}}^+ \cong \varprojlim_m B_{\text{dR}}^+ / (\xi^m)$ , which shows that  $B_{\text{dR}}^+$  is  $\xi$ -adically complete, and  $m = 1$  shows the residue field is equal to  $K$ . ┘

**Remark (27.7.1.19)**.  $\mathbf{A}_{\text{inf}}/(\xi^n) \left[ \frac{1}{p} \right] = \mathbf{A}_{\text{inf}}/(\xi^n) \left[ \frac{1}{[t]} \right]$ , so if  $\text{char } k = p$ , then  $B_{\text{dR}}^+$  is just  $W(C^\flat)$ .

Thus  $B_{\text{dR}}^+$  should be thought as the completed local ring at the point  $y = (K, \iota)$ . ┘

**Prop. (27.7.1.20)** **[Topology on  $B_{\text{dR}}$ ]**. The Gauss norms give  $\mathbf{A}_{\text{inf}}$  a topology, giving  $B_{\text{dR}}$  a topology. Then  $B_{\text{dR}}$  is complete in this topology, and  $B_{\text{dR}} \rightarrow K$  is continuous.

With this topology,  $B_{\text{dR}}$  is abstractly isomorphic to  $\mathbb{C}_p((T))$ , but not topological isomorphic to it. ┘

*Proof:* Cf.[Conrad, P65] or [ $p$ -adic Period Rings]P42. ?

$B_{\text{dR}}$  is abstractly isomorphic to  $\mathbb{C}_p((T))$  by Cohn structure theorem ?, but [Colmez, Une construction de  $B_{\text{dR}}$ ] proved that  $\bar{K}$  is dense in  $B_{\text{dR}}$ , so it cannot be topological isomorphic to  $\mathbb{C}_p((T))$ . ┘

**Prop. (27.7.1.21)** **[The Stalk Map]**. Notice  $\mathbf{A}_{\text{inf}} = \varprojlim_n \mathbf{A}_{\text{inf}}/(\xi^n)$  as  $\xi$  is distinguished, thus there is a natural injection  $\mathbf{A}_{\text{inf}} \rightarrow B_{\text{dR}}^+$ , whose composition with  $B_{\text{dR}}^+ \rightarrow B_{\text{dR}}^+ / \xi \cong K$  maps  $p, [t]$  to  $p, t^\sharp$ , which shows  $p, [t]$  are invertible in  $B_{\text{dR}}^+$ , so there is a map

$$e : \mathbf{A}_{\text{inf}} \left[ \frac{1}{p}, \frac{1}{[t]} \right] \rightarrow B_{\text{dR}}^+.$$

In case  $a \leq |p|_K \leq b$ , this can be further extended to a map  $e : B_{[a,b]} \rightarrow B_{\text{dR}}^+$  (The stalk map). ┘

*Proof:* It suffices to prove for  $a = |p|_K = b$  because the topology is stronger. In this case, choose  $t = p^\flat \in C^\flat$ , then  $|t|_{C^\flat} = |p|_K$ , thus  $\bar{e}_n$  determined a map

$$\mathbf{A}_{\text{inf}} \left[ \frac{[t]}{p}, \frac{p}{[t]} \right] \rightarrow B_{\text{dR}}^+ / (\xi^n) \cong (\mathbf{A}_{\text{inf}} / \xi^n) [p^{-1}],$$

It suffices to prove the image is contained in  $p^{-k}(\mathbf{A}_{\text{inf}} / \xi^n)$  for some  $k = k(n)$ , because then  $\bar{e}_n$  is  $p$ -adically continuous, and extends to map of  $B_{[a]} = (\mathbf{A}_{\text{inf}} \left[ \frac{[t]}{p}, \frac{p}{[t]} \right])_p^\wedge \rightarrow (\mathbf{A}_{\text{inf}} / (\xi^n)) [p^{-1}]$ , which is compatible w.r.t  $n$ , thus gives a map  $B_{[a]} \rightarrow B_{\text{dR}}^+$ .

For this, consider  $f = \bar{e}_n(\frac{[t]}{p})$ ,  $g = \bar{e}_n(\frac{p}{[t]})$ , then their reduction under  $B_{\text{dR}}^+(\xi^n) \rightarrow B_{\text{dR}}^+/\xi \cong K$  is in  $\mathcal{O}_K \cong \mathbf{A}_{\text{inf}}/(\xi)$ , thus

$$f = f_1 + \frac{\xi}{p^c} f_2, \quad g = g_1 + \frac{\xi}{p^c} g_2$$

for  $f_1, f_2, g_1, g_2 \in \mathbf{A}_{\text{inf}}/(\xi^n)$  for some  $c$ . Then any

$$f^m = (f_1 + \frac{\xi}{p^c} f_2)^m = \sum_{i=0}^{m-1} C_m^i f_1^{m-i} (\frac{\xi}{p^c} f_2)^i \in p^{-nc}(\mathbf{A}_{\text{inf}}/(\xi^n)).$$

Thus  $\bar{e}_n(\mathbf{A}_{\text{inf}}[\frac{[t]}{p}, \frac{p}{[t]}]) \in p^{-nc}(\mathbf{A}_{\text{inf}}/(\xi^n))$ .  $\square$

**Cor. (27.7.1.22).** The stalk map  $e : B_{[a,b]} \rightarrow B_{\text{dR}}^+$  composed with the map  $B_{\text{dR}}^+/\xi \cong K$  are in fact equivalent to the valuation map (27.7.1.11).  $\lrcorner$

### $B_{\text{crys}}$

Cf. [Notes on  $p$ -adic Hodge, Serin Hong].

**Def. (27.7.1.23)** [ $B_{\text{crys}}$ ].  $\lrcorner$

### $B_{\text{st}}, B_e$

## 2 $C_{\text{dR}}$ -Theorem

**Thm. (27.7.2.1)** [ $C_{\text{dR}}$ , Faltings/Tsuji]. If  $X \in \text{SmPrpr}/K$ , then for any  $r \in \mathbb{N}$ , there exists a canonical isomorphism

$$\gamma_{\text{dR}}(X) : B_{\text{dR}} \otimes_K H_{\text{dR}}^r(X/K) \cong B_{\text{dR}} \otimes_{\mathbb{Q}_p} H_{\text{ét}}^r(X_{\bar{K}}, \mathbb{Q}_p).$$

which identifies filtrations and  $\text{Gal}_K$ -actions on both sides. Moreover  $\gamma_{\text{dR}}$  is functorial in  $X$ .  $\lrcorner$

*Proof:* Cf. [Faltings,  $p$ -adic Hodge Theory]. or [p-adic Hodge for Rigid Analytic Varieties, Scholze], [BMS18]P104.  $\square$

**Cor. (27.7.2.2)** [deRham Comparison for Étale Cohomologies].  $H_{\text{ét}}^r(X_{\bar{K}}, \mathbb{Q}_p) \in \text{Rep}_{\mathbb{Q}_p}^{\text{dR}}(\text{Gal}_K)$ , and

$$H_{\text{dR}}^r(X) \cong D_{B_{\text{dR}}}(H_{\text{ét}}^r(X_{\bar{K}}, \mathbb{Q}_p)), \quad H^{n-p}(X; \Omega_X^p) \cong \text{gr}^p H_{\text{dR}}^r(X).$$

Also by taking the gradation of (27.7.2.1), by (25.2.5.5), there is a Hodge-like decomposition

$$\mathbb{C}_K \otimes_{\mathbb{Q}_p} H_{\text{ét}}^r(X_{\bar{K}}, \mathbb{Q}_p) \cong \bigoplus_{a+b=r} \mathbb{C}_K(-a) \otimes_K H^b(X, \Omega_X^a).$$

This shows we can recover the de Rham cohomology of  $X$  from the étale cohomology, and the Hodge-Tate weights of  $H_{\text{ét}}^r(X_{\bar{K}}, \mathbb{Q}_p)$  lies in in  $[-r, 0]$ .  $\lrcorner$

**Remark (27.7.2.3).** Is this true for non-proper varieties with logarithmic poles?  $\lrcorner$

**Example (27.7.2.4)** [Elliptic Curve Case, Tensoring  $\mathbb{C}_K$  Lost Informations]. For  $E \in \text{EllCur}/K$  with multiplicative reduction and  $j(E) > 1$ , by (21.3.5.13) and (21.3.5.10),

$$E(\bar{K}) \cong \bar{K}^\times / q^{\mathbb{Z}}$$

as  $\text{Gal}_K$ -representations for some  $q \in K^\times$ . Thus  $T_p(E) \cong q^{\mathbb{Q}_p/\mathbb{Z}_p}$  and there exists an exact sequence

$$0 \rightarrow \mathbb{Z}_p(1) \rightarrow T_p(E) \rightarrow \mathbb{Z}_p \rightarrow 0.$$

Then this sequence doesn't split when tensoring  $\bar{K}$ , but split when tensoring  $\mathbb{C}_K$ , by (27.7.3.3).  $\lrcorner$

*Proof:* Suppose it splits after tensoring  $\overline{K}$ , then it splits after tensoring some finite extension  $K'$ . Then by projection of  $K'$  onto  $\mathbb{Q}$ , we see

$$0 \rightarrow \mathbb{Q}_p(1) \rightarrow V_p(E) \rightarrow \mathbb{Q}_p \rightarrow 0$$

is splitting as a  $\text{Gal}_{K'}$ -representations. But this is not true, as any system of roots of  $p$   $\square$

$C_{\text{dR}}$ -theorem(27.7.2.1) implies any representation of  $\text{Gal}_K$  of the form  $H_{\text{ét}}^r(X_{\overline{K}}, \mathbb{Q}_p)$  is deRham, thus it is natural to consider the converse:

### 3 $C_{\text{crys}}$ -Theorem

**Thm. (27.7.3.1) [Hyodo-Kato Isomorphism].** If  $X \in \text{Sch}^{\text{sm}, \text{proper}}/K$  has good reduction  $\mathcal{X}/\mathcal{O}_K$ , let  $\overline{\mathcal{X}} = \mathcal{X}_k$ , then for any  $r \in \mathbb{N}$ ,  $H_{\text{crys}}^r(\overline{\mathcal{X}}) \in \varphi\text{-Mod}_{W(k)}$ , and there is an isomorphism

$$K \otimes_{W(k)} H_{\text{crys}}(\overline{\mathcal{X}}) \cong H_{\text{dR}}^r(X).$$

Then this isomorphism descends  $H_{\text{dR}}^r(X)$  to  $K_0$ , and this  $K_0$ -structure is independent of the smooth model  $\mathcal{X}/\mathcal{O}_K$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (27.7.3.2) [ $C_{\text{crys}}$ , Faltings].** If  $X \in \text{Sch}^{\text{sm}, \text{proper}}/K$  has good reduction  $\mathcal{X}/\mathcal{O}_K$ , let  $\overline{\mathcal{X}} = \mathcal{X}_k$ , then for any  $r \in \mathbb{N}$ , there exists a canonical isomorphism

$$\gamma_{\text{dR}}(X) : B_{\text{dR}} \otimes_K H_{\text{dR}}^r(X/K) \cong B_{\text{dR}} \otimes_{\mathbb{Q}_p} H_{\text{ét}}^r(X_{\overline{K}}, \mathbb{Q}_p).$$

which respect  $\text{Gal}_K$ -actions and Frobenius-actions on both sides. Moreover  $\gamma_{\text{crys}}$  is functorial in  $X$ , and  $B_{\text{dR}} \otimes \gamma_{\text{crys}} = \gamma_{\text{dR}}$ (27.7.2.1).  $\lrcorner$

*Proof:* Cf.[Faltings, Crystalline Representations and  $p$ -adic Galois Representations].  $?$   $\square$

**Cor. (27.7.3.3) [Crystalline Comparison for Étale Cohomologies].** If  $X \in \text{Sch}^{\text{sm}, \text{proper}}/K$  has good reduction  $\mathcal{X}/\mathcal{O}_K$ , then for any  $r \in \mathbb{N}$ ,  $H_{\text{ét}}^r(X_{\overline{K}}, \mathbb{Q}_p) \in \text{Rep}_{\mathbb{Q}_p}^{\text{dR}}(\text{Gal}_K)$ , and

$$H_{\text{crys}}^r(\mathcal{X}_k) \cong D_{\text{crys}}(H_{\text{ét}}^r(X_{\overline{K}}, \mathbb{Q}_p)).$$

$?$   $\lrcorner$

### 4 Rigid Analytic Varieties(Scholze)

References are [Sch13].

**Notation (27.7.4.1).**

- Let  $(K, \mathcal{O}_K, \kappa) \in p\text{-Field}$ .
  - Use notations on adic spaces defined in [Adic Spaces](#).
- $\lrcorner$

**Thm. (27.7.4.2) [Finiteness of Étale Cohomologies].** If  $K = \overline{K}$ ,  $X \in \text{rigSmPrprVar}/K$ , and  $\mathbb{L} \in \text{LocSys}_{\mathbb{F}_p}^{\text{ét}}(X)$ , then  $\dim_{\mathbb{F}_p} H_{\text{ét}}^i(X, \mathbb{L}) < \infty$  for any  $i \in \mathbb{N}$ , and vanishes for  $i > 2 \dim X$ .  $\lrcorner$

*Proof:*  $\square$



**Thm. (27.7.4.3) [Pro-Étale Sites are Locally Contractible].** If  $K = \overline{K}$  and  $X/K$  a connected affinoid rigid analytic variety, then for  $x \in X(K)$  and any  $p$ -torsion local system  $\mathbb{L}$  on  $X$ , the natural map

$$H_{\text{cont}}^i(\pi_1(X, x), \mathbb{L}_x) \rightarrow H_{\text{ét}}^i(X, \mathbb{L})$$

is an isomorphism.  $\lrcorner$

*Proof:* Cf. [Scholze, Rigid Analytic Varieties]P25. ?

It suffices to show that the natural map  $H^i(X_{\text{fét}}, \mathbb{L}) \rightarrow H^i(X_{\text{ét}}, \mathbb{L})$  is an isomorphism.  $\square$

**Thm. (27.7.4.4).** If  $K = \overline{K}$  and  $f : X \rightarrow Y$  be a proper smooth morphism of rigid analytic varieties over  $K$ , and  $\mathbb{L}$  an  $\mathbb{F}_p$ -local system on  $X_{\text{ét}}$ , then for any  $i \in \mathbb{N}$ , there is an almost isomorphism

$$(R^i f_{\text{ét},*} \mathbb{L}) \otimes \mathcal{O}_Y^+ / (p) \cong R^i f_{\text{ét},*} (\mathbb{L} \otimes \mathcal{O}_X^+ / (p)).$$

In particular, if  $X/K$  is a proper smooth rigid analytic variety, then there is an almost isomorphism

$$H_{\text{ét}}^i(X, \mathbb{L}) \otimes \mathcal{O}_K / (p) \cong H_{\text{ét}}^i(X, \mathbb{L} \otimes \mathcal{O}_X^+ / (p)).$$

$\lrcorner$

*Proof:*

$\square$

**Cor. (27.7.4.5).** If  $K = \overline{K}$  and  $X/K$  is a proper smooth rigid analytic variety, then there is an almost isomorphism

$$H_{\text{ét}}^i(X, \mathbb{L}) \otimes \mathbb{B}_{\text{dR}}^+ \cong H_{\text{pro-ét}}^i(X, \mathbb{L} \otimes \mathbb{B}_{\text{dR}}^+).$$

$\lrcorner$

*Proof:*

$\square$

**Thm. (27.7.4.6) [deRham Lisse Sheaves].** For  $X \in \text{rigSmPrprVar}/K$ , there is a fully faithful functor from the category of filtered  $\mathcal{O}_X$ -modules with an integral connection satisfying the Griffiths transversality ? to  $\text{LocSys}_{\mathbb{B}_{\text{dR}}}/X$ .

A **de Rham lisse sheaf** is a lisse  $\mathbb{Z}_p$ -sheaf  $\mathbb{L}$  s.t. the associated  $\mathbb{B}_{\text{dR}}$ -local system  $\mathbb{L} \otimes \mathbb{B}_{\text{dR}}$  lies in the essential image of this functor.  $\lrcorner$

*Proof:*

$\square$

**Thm. (27.7.4.7) [ $C_{\text{dR}}$  for Rigid Analytic Varieties].** If  $X \in \text{rigSmPrprVar}/K$  and  $\mathbb{L}$  a lisse  $\mathbb{Z}_p$ -sheaf on  $X$ , then there is a  $\text{Gal}_K$ -equivariant isomorphism

$$H_{\text{ét}}^i(X_{\overline{K}}, \mathbb{L}) \otimes B_{\text{dR}}^+ \cong H_{\text{ét}}^i(X_{\overline{K}}, \mathbb{L} \otimes B_{\text{dR}}^+).$$

And if  $\mathbb{L}$  is deRham(27.7.4.6) with associated filtered module with integral connection  $(\mathcal{E}, \nabla, \text{Fil}^\bullet)$ , then

- the Hodge-deRham spectral sequence

$$H_{\text{Hdg}}^{i,j}(X, \mathcal{E}) \implies H_{\text{dR}}^{i+j}(X, \mathcal{E})$$

degenerates,

- $H_{\text{ét}}^i(X_{\overline{K}}, \mathbb{L}) \in \text{Rep}_{\text{dR}}^{\text{fd}}(\text{Gal}_K)$  with associated filtered  $K$ -vector space  $H_{\text{dR}}^i(X, \mathcal{E})$ ,
- there is a  $\text{Gal}_K$ -equivariant isomorphism

$$H_{\text{ét}}^i(X_{\overline{K}}, \mathbb{L}) \otimes \mathbb{C}_K \cong \bigoplus_j H_{\text{Hdg}}^{i-j,j}(X, \mathcal{E}) \otimes \mathbb{C}_K(-j).$$

In particular, this applies to  $\mathbb{L} = \overline{\mathbb{Z}_p}$  and  $\mathcal{E} = \mathcal{O}_X$ . ┘

*Proof:* ?

3 follows from 1 and 2. □

**Thm. (27.7.4.8).** Let  $f : X \rightarrow Y$  be a smooth proper morphism of smooth rigid analytic varieties over  $K$  and  $\mathbb{L}$  a lisse  $\mathbb{Z}_p$ -sheaf on  $X$  which is deRham corresponding to the filtered module with integral connection  $(\mathcal{E}, \nabla, \text{Fil}^\bullet)$ . Assume  $R^i f_{\text{pro-ét},*} \mathbb{L}$  is a lisse  $\mathbb{Z}_p$ -sheaf on  $Y$  (e.g.  $f : X \rightarrow Y \in \text{SmPrpr}/K$ ), then  $R^i f_{\text{pro-ét},*} \mathbb{L}$  is also deRham, with the associated filtered module with integral connection given by  $R^i f_{\text{dR},*}(\mathcal{E}, \nabla, \text{Fil}^\bullet)$ . ┘

*Proof:* □

## 5 Integral $p$ -adic Hodge Theory

**Notation (27.7.5.1).**

- Let  $(C, \mathcal{O}_C, \kappa)$  be the algebraic closure of a  $p$ -adic field. ┘

**Thm. (27.7.5.2) [Properties of  $\mathbf{A}_{\text{inf}}$ -Cohomologies].** For a proper smooth formal scheme  $\mathfrak{X}/\mathcal{O}_C$ , there is a complex  $R\Gamma_{\text{inf}}(\mathfrak{X}) \in \text{Perf}(\mathbf{A}_{\text{inf}})$ , functorial in  $\mathfrak{X}$ , satisfying the following:

- (Étale Specialization)  $R\Gamma_{\mathbf{A}_{\text{inf}}}(\mathfrak{X}) \otimes_{\mathbf{A}_{\text{inf}}}^{\mathbb{L}} W(C^\flat) \cong R\Gamma_{\text{ét}}(\mathfrak{X}_C, \mathbb{Z}_p) \otimes_{\mathbb{Z}_p}^{\mathbb{L}} W(C^\flat)$ .
- (Crystalline Specialization)  $R\Gamma_{\mathbf{A}_{\text{inf}}}(\mathfrak{X}) \otimes_{\mathbf{A}_{\text{inf}}}^{\mathbb{L}} W(k) \cong R\Gamma_{\text{crys}}(\mathfrak{X}_k/W(k))$ .
- (deRham Specialization)  $R\Gamma_{\mathbf{A}_{\text{inf}}}(\mathfrak{X}) \otimes_{\mathbf{A}_{\text{inf}}}^{\mathbb{L}} \mathcal{O} \cong R\Gamma_{\text{dR}}(\mathfrak{X}/\mathcal{O})$ . ┘

*Proof:* □

**Thm. (27.7.5.3) [Torsions in Crystalline and Étale Cohomologies].** Let  $\mathfrak{X}/\mathcal{O}_C$  be a proper smooth formal scheme and  $i \in \mathbb{Z}_{\geq 0}$ . Assume  $H_{\text{crys}}^i(\mathfrak{X}_k/W(k))_{\text{tor}} = 0$ , then  $H_{\text{ét}}^i(\mathfrak{X}_C, \mathbb{Z}_p)_{\text{tor}} = 0$ . ┘

*Proof:* □

## 27.8 Pro-Étale Sites on $\mathcal{P}\text{erfd}$ and Diamonds

References are [Sch17] and [Notes on Diamonds, David Hansen].

### 1 Properties of Perfectoid Spaces

#### Totally Disconnected Spaces

**Def. (27.8.1.1) [Totally Disconnected Spaces].**  $X \in \mathcal{P}\text{erfdSpa}$  is called **totally disconnected** if it is qcqs and any open covering  $\{U_i \rightarrow X\}$  splits, i.e.  $\coprod U_i \rightarrow X$  splits, or equivalently, there is a refinement covering  $\{V_i \rightarrow X\}$  that  $X \cong \coprod V_i$ .

A perfectoid space  $X$  is called **strictly totally disconnected** if it is qcqs and every étale cover splits.  $\lrcorner$

**Prop. (27.8.1.2).** Let  $X$  be a qcqs perfectoid space, then  $X$  is totally disconnected iff all its connected components are of the form  $\text{Spa}(K, K^+)$  where  $(K, K^+)$  are perfectoid affinoid fields. And it is strictly totally disconnected if moreover  $K$  are all alg.closed.  $\lrcorner$

*Proof:* Cf. [Sch17].P29, P35.  $\square$

**Prop. (27.8.1.3).** if  $X$  is a totally disconnected perfectoid space, then  $X$  is affinoid.  $\lrcorner$

*Proof:* Cf. [Sch17].P30.  $\square$

#### Injections

**Def. (27.8.1.4) [Injections].**  $f : X \rightarrow Y \in \mathcal{P}\text{erfdSpa}$  is called an injection if for any perfectoid space  $Z$ ,  $f_* : \text{Hom}(Z, X) \rightarrow \text{Hom}(Z, Y)$  is an injection.  $\lrcorner$

**Prop. (27.8.1.5) [Residue Field Map is Injection].** Let  $X$  be a perfectoid space and  $x \in X$ , giving rise to a map of residue fields

$$i_x : \text{Spa}(\kappa(x), \kappa(x)^+) \rightarrow X,$$

then  $i_x$  is an injection of perfectoid spaces.  $\lrcorner$

*Proof:* To show this, firstly we can replace  $X$  with an affinoid nbhd of  $X$ . Then notice that  $\text{Spa}(\kappa(x), \kappa(x)^+)$  is the filtered limit over all rational nbhds  $U$  of  $x$  in  $X$ , and for each  $U$ ,  $U \rightarrow X$  is an injection by definition (27.3.3.1), so  $i_x$  is also an injection.  $\square$

**Cor. (27.8.1.6).** In particular, if  $X$  is qcqs and has a unique closed point  $x \in X$ , then  $X = \text{Spa}(\kappa(x), \kappa(x)^+)$ , as in this case  $X$  is the only quasi-compact open subset containing  $x$ .  $\lrcorner$

**Prop. (27.8.1.7) [Characterizations of Injections].** For  $f : X \rightarrow Y \in \mathcal{P}\text{erfdSpa}$ , the following are equivalent:

- $f$  is an injection.
- For any perfectoid adic field  $(K, K^+)$ , the map of sets  $f_* : Y(K, K^+) \rightarrow X(K, K^+)$  is an injection.
- The map  $|f| : |Y| \rightarrow |X|$  is injective, and for all rank 1 point  $y \in Y$  with image  $f(y) = x \in X$ , the map of completed residue fields  $\kappa(x) \rightarrow \kappa(y)$  is an isomorphism.
- The map  $|f| : |Y| \rightarrow |X|$  is injective, and  $f$  is final in the category of maps  $Z \rightarrow X$  that  $|Z| \rightarrow |X|$  factors through the map  $|Y| \rightarrow |X|$ .

In particular, by item4, an injection of perfectoid spaces is determined by its topological map.  $\lrcorner$

*Proof:* Cf.[Sch17]P21.  $4 \rightarrow 1 \rightarrow 2$  is trivial.  $\square$

**Prop.(27.8.1.8)[Injection and Base Change].**

- Let  $f : Y \rightarrow X$  be an injection of perfectoid spaces, and  $X' \rightarrow X$  any map of perfectoid spaces, then the pullback  $f' : Y' = Y \times_X X' \rightarrow X'$  is also an injection, and the induced map

$$|Y'| \rightarrow |Y| \times_{|X|} |X'|$$

is a homeomorphism.

- A map of perfectoids spaces is an injection iff it is universally injective.  $\lrcorner$

*Proof:* Cf.[Sch17]P24.  $\square$

### Immersions

**Def.(27.8.1.9)[Immersion**s].  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$  is called an immersion if  $f$  is an injection and  $|f| : |Y| \rightarrow |X|$  is a locally closed immersion. If  $|f|$  is moreover closed or open, then it is called closed/open immersion.  $\lrcorner$

**Def.(27.8.1.10)[Zariski-Closed Immersion**s]. Let  $f : Z \rightarrow X = \text{Spa}(R, R^+) \in \mathcal{P}\text{erfdSpa}$ , then

- $f$  is called a Zariski-closed immersion if  $f$  is a closed immersion and  $|Z| = V(I) \subset |X|$ , where  $I \subset R$  is an ideal.
- $f$  is called a strongly Zariski-closed immersion if  $Z = \text{Spa}(S, S^+)$  is affinoid perfectoid,  $R \rightarrow S$  is surjective, and  $S^+$  is the closure of  $R^+$  in  $S$ .  $\lrcorner$

**Prop.(27.8.1.11).**

- If  $f$  is strongly Zariski closed, then  $f$  is Zariski closed, in particular a closed immersion.
- If  $f$  is Zariski closed, then  $Z$  is affinoid.
- If  $X$  is of characteristic  $p$ , and  $f$  is Zariski closed, then  $f$  is strongly Zariski closed.  $\lrcorner$

*Proof:* Cf.[Sch17] Section2.5.  $\square$

**Prop.(27.8.1.12).** For any map of perfectoid spaces  $Y \rightarrow X$ , the diagonal map  $\Delta_f : Y \rightarrow Y \times_X Y$  is an immersion.  $\lrcorner$

*Proof:* Clearly  $\Delta_f$  is an injection, thus it suffices to show that  $|\Delta_f|$  identifies  $Y$  with a locally closed subset of  $|Y \times_X Y|$ . This can be checked locally on the target, so we can assume  $X = \text{Spa}(R, R^+)$  and  $Y = \text{Spa}(S, S^+)$ , then the diagonal map is strongly Zariski closed, as  $S \hat{\otimes}_R S \rightarrow S$  is surjective and maps the integral closure of  $S^+ \hat{\otimes}_{R^+} S^+ \rightarrow S^+$  onto  $S^+$ . Thus by(27.8.1.11),  $\Delta_f$  is a closed immersion in this case.  $\square$

**Def.(27.8.1.13)[Separated Map].**  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$  is called separated if  $\Delta_f$  is a closed immersion.  $\lrcorner$

**Prop.(27.8.1.14)[Valuation Criterion].** For  $f : Y \rightarrow X \in \mathcal{P}\text{erfdSpa}$ , the following are equivalent:

- $f$  is separated.
- $|\Delta_f| : |Y| \rightarrow |Y \times_X Y|$  is a closed immersion.
- $|f|$  is quasi-separated, and for any perfectoid adic field  $(K, K^+)$  and any diagram

$$\begin{array}{ccc} \mathrm{Spa}(K, \mathcal{O}_K) & \longrightarrow & Y \\ \downarrow & \nearrow \text{dotted} & \downarrow f \\ \mathrm{Spa}(K, K^+) & \longrightarrow & X \end{array}$$

there exists at most one dotted arrow making the diagram commutative.

□

*Proof:* The equivalence of 1 and 2 is by (27.8.1.12).

2  $\rightarrow$  3: If  $|\Delta_f|$  is a closed immersion, then it is in particular quasi-compact, thus  $f$  is quasi-separated. Now if there are two dotted-arrow making this diagram commutative, they define a point  $z \in (Y \times_X Y)(K, K^+)$  s.t.  $z|_{\mathrm{Spa}(K, \mathcal{O}_K)} \in \Delta_f(Y)(K, \mathcal{O}_K)$ . But  $|\Delta_f|(|Y|)$  is closed in  $|Y \times_X Y|$ , so  $z$  maps  $\mathrm{Spa}(K, K^+)$  into  $|\Delta_f|(|Y|)$  iff it maps  $\mathrm{Spa}(K, \mathcal{O}_K)$  into  $|\Delta_f|(|Y|)$ , as  $\mathrm{Spa}(K, \mathcal{O}_K) \subset \mathrm{Spa}(K, K^+)$  is dense??. Now  $\Delta_f$  is an injection, so by (27.8.1.7),  $z$  factors through  $\Delta_f$  thus the two maps are equal.

3  $\rightarrow$  2: The condition implies that  $|\Delta_f| : |Y| \rightarrow |Y \times_X Y|$  is a quasi-compact locally closed immersion of locally spectral spaces which is moreover specializing. But because  $|\Delta_f|$  is quasi-compact, the image of  $|\Delta_f|$  is pro-constructible, and it is also closed under specialization, thus it is closed, Cf. [Sch17]P25. □

## 2 Pro-Étale Site and v-Site

**Prop. (27.8.2.1).** if  $(R, R^+)$  is the completed filtered colimit of perfectoid Huber pairs  $(R_i, R_i^+)$ ,  $X_i = \mathrm{Spa}(R_i, R_i^+)$ ,  $X = \mathrm{Spa}(R, R^+)$ , then base change induce equivalences of categories:

- $2 - \lim_i X_{i, \text{fét}} \cong X_{\text{fét}}$ .
- $2 - \lim_i X_{i, \text{ét}}^{\text{aff}} \cong X_{\text{ét}}^{\text{aff}}$ .
- $2 - \lim_i X_{i, \text{ét}, qcqs} \cong X_{\text{ét}, qcqs}$ .
- $2 - \lim_i X_{i, \text{ét}, qc, sep} \cong X_{\text{ét}, qc, sep}$ .

□

*Proof:* Cf. [Sch17]P27.

1 follows from almost purity theorem. □

**Def. (27.8.2.2) [Pro-Étale Morphism].** A **pro-étale map of perfectoid Huber pairs** is a map  $(A, A^+) \rightarrow (B, B^+)$  s.t.  $(B, B^+)$  is the completed filtered colimit of étale maps  $(A, A^+) \rightarrow (A_i, A_i^+)$  (27.3.5.1).

A morphism of perfectoid spaces is **pro-étale** if there is an affinoid covering  $V_i = \mathrm{Spa}(R_i, R_i^+)$  of  $X$  that  $f^{-1}(V_i)$  have coverings  $U_{ij} = \mathrm{Spa}(R_{ij}, R_{ij}^+)$  that  $(R_i, R_i^+) \rightarrow (R_{ij}, R_{ij}^+)$  are all pro-étale.

In fact, by (27.8.2.7), if this is true for one affinoid covering  $V_i$  of  $X$ , then this is true for any affinoid covering of  $X$ .

? Is this compatible with the definition in (27.3.6.2)? □

**Prop. (27.8.2.3).** If  $S$  is a profinite set and  $X$  is a perfectoid space, then we can define a new perfectoid space  $X \times \underline{S}$  as the inverse limit of  $X \times \underline{S}_i$ , where  $\underline{S} = \varprojlim S_i$ . Then  $X \times \underline{S}$  is pro-étale over  $X$ . □

**Prop. (27.8.2.4) [Immersion are Pro-Étale].** If  $f : Z \hookrightarrow X$  be a Zariski closed immersion with image  $V(I)$ , then  $f$  is affinoid pro-étale. Then  $f(Z)$  can be written as the intersection of rational subsets

$$U_{f_1, \dots, f_n} = \{|f_1|, \dots, |f_n| \leq |\varpi|\}$$

for various  $n$  and  $f_1, \dots, f_n \in I$ . Then  $Z = \varprojlim U_{f_1, \dots, f_n} \rightarrow X$ , as it is a closed immersion thus an injection (27.8.1.11), which shows  $Z \rightarrow X$  is pro-étale.

In particular, an immersion is also pro-étale because pro-étale can be checked analytically locally.

┘

**Cor. (27.8.2.5) [Diagonal Map is Pro-Étale].** If  $f : Y \rightarrow X$  is a map of perfectoid spaces, then  $\Delta_f : Y \rightarrow Y \times_X Y$  is pro-étale, by (27.8.1.12). ┘

**Prop. (27.8.2.6).** If  $X$  is an affinoid perfectoid space, then the functor

$$\mathrm{Pro}(X_{\text{ét}}^{\text{aff}}) \rightarrow X_{\text{pro-ét}}^{\text{aff}} : \varprojlim_i (X_i) \mapsto \varprojlim_i X_i$$

is an equivalence of categories. ┘

*Proof:* This map is essentially surjective by definition, To show it is fully faithful, it suffices to show that for  $Y = \varprojlim_i \mathrm{Spa}(Y_i, Y_i^+)$ ,  $Z = \varprojlim_j \mathrm{Spa}(Z_j, Z_j^+)$ ,

$$\mathrm{Hom}_X(Y, Z) = \varprojlim_j \varinjlim_i \mathrm{Hom}(Y_i, Z_j).$$

We need to show for any  $Z \rightarrow X$ ,

$$\mathrm{Hom}_X(Y, Z) = \varinjlim_i \mathrm{Hom}(Y_i, Z).$$

Because  $2 - \lim_i X_{i, \text{ét}, qcqs} \cong X_{\text{ét}, qcqs}$  (27.8.2.1), we have

$$\mathrm{Hom}_X(Y, Z) = \mathrm{Hom}_Y(Y, Y \times_X Z) = \varinjlim_i \mathrm{Hom}_{Y_i}(Y_i, Y_i \times_X Z) = \varinjlim_i \mathrm{Hom}_X(Y_i, Z).$$

□

**Prop. (27.8.2.7) [Properties of Pro-Étale Morphisms].**

- (Affinoid)Pro-Étale maps are stable under composition and pullbacks.
- Let  $f : Y \rightarrow X$ ,  $f' : Y' \rightarrow X$  be (affinoid)pro-étale, then any map  $g : Y \rightarrow Y'$  over  $X$  is also (affinoid)pro-étale.
- For any affinoid perfectoid space  $X$ , the category  $X_{\text{pro-ét}}^{\text{aff}}$  has all finite limits.

┘

*Proof:* 1: Composition is obvious. pro-étale maps are stable under pullbacks because étale maps do.

2: We can factor  $g$  as a section of the map  $Y \times_X Y' \rightarrow Y$  and the projection map  $Y \times_X Y' \rightarrow Y'$ . Thus it suffices to show a section of a pro-étale map is pro-étale. But if  $Y = \varprojlim_i Y_i \rightarrow X$  is pro-étale, then a section is given by compatible sections  $s_i : X \rightarrow Y_i$ . Then  $X = \varprojlim_i (X \times_{Y_i} Y) \rightarrow X$  is pro-étale.

3: This is because  $X_{\text{ét}}^{\text{aff}}$  has finite limits, because it has a final object and fiber products (12.2.9.17).

□

**Def. (27.8.2.8) [Big Pro-Étale Sites].** Consider the following categories:

- $\mathcal{P}\text{erfdSpa}$ , the category of perfectoid spaces.
- $\mathcal{P}\text{erf}$ , the category of perfectoid spaces of characteristic  $p$ .
- $X_{\text{proét}}$ , the category of perfectoid spaces pro-étale over  $X$ , where  $X$  is a perfectoid space.
- $X_{\text{proét}}^{\text{aff}}$ , the category of affinoid perfectoid spaces pro-étale over  $X$ .

The **big pro-étale site** is the category  $\mathcal{P}\text{erfdSpa}$  endowed with the topology that a family of maps  $\{f_i : Y_i \rightarrow X\}$  is a covering if all  $f_i$  are pro-étale and for any quasi-compact open subset  $U \subset X$ , there is a finite set  $J \subset I$  and quasi-compact opens  $V_j \subset Y_j$  that  $U = \cup_{j \in J} f_j(V_j)$ .

These truly form sites by (27.8.2.7).  $\lrcorner$

**Cor. (27.8.2.9).** The presheaf  $\mathcal{O} : X \mapsto \mathcal{O}_X(X)$ ,  $\mathcal{O}^+ : X \mapsto \mathcal{O}_X^+(X)$  on the big étale site are sheaves. If  $X$  is affinoid perfectoid, then  $H^i(X_{\text{pro-ét}}, \mathcal{O}) = 0$  for  $i > 0$ , and  $H^i(X_{\text{pro-ét}}, \mathcal{O}^+)$  is almost zero for  $i > 0$ . Moreover, the big pro-étale site is subcanonical.  $\lrcorner$

*Proof:* Firstly we can assume  $X$  is affinoid because we already know  $\mathcal{O}, \mathcal{O}^+$  are sheaves w.r.t. the analytic topology. Let  $Y \rightarrow X$  be an affinoid pro-étale covering of  $X$ , where  $Y = \text{Spa}(R_\infty, R_\infty^+)$ ,  $(R_\infty, R_\infty^+) = (\varinjlim_i (R_i, R_i^+))^\wedge$ . Then fixing a pseudo-uniformizer  $\varpi$  of  $R$ , the complexes

$$0 \rightarrow R^+/\varpi \rightarrow R_j^+/\varpi \rightarrow \cdots$$

is almost exact, because  $H^i(X_{\text{ét}}, \mathcal{O}_X^+/\varpi)$  is almost zero for  $i > 0$ . Now take a direct limit over  $i$ , then

$$0 \rightarrow R^+/\varpi \rightarrow R_\infty^+/\varpi \rightarrow \cdots$$

is almost exact. Now by induction on  $n$ , we can prove

$$0 \rightarrow R^+/\varpi^n \rightarrow R_\infty^+/\varpi^n \rightarrow \cdots$$

is almost exact. Then by passing to the direct limit ,

$$0 \rightarrow R^+ \rightarrow R_\infty^\infty \rightarrow \cdots$$

is almost exact. Then by inverting  $\varpi$ ,

$$0 \rightarrow R \rightarrow R_\infty \rightarrow \cdots$$

is exact. These give us the desired results. Notice  $\mathcal{O}^+$  is a sheaf because it is the elements of valuations  $\leq 1$  everywhere by (27.3.3.7).

For the final assertion, if  $\{Y_i \rightarrow Y\}$  is a pro-étale covering of  $Y$ , and  $g_i : Y_i \rightarrow X$  are maps that agree on  $Y_i \times_X Y_j$ , then firstly we can glue these maps together topologically to a map  $|Y| \rightarrow |X|$ . So this problem can be considered locally on  $X$ , so we may assume  $X = \text{Spa}(R, R^+)$  is affinoid, and maps  $(R, R^+) \rightarrow (\mathcal{O}(Y_i), \mathcal{O}^+(Y_i))$  that agree on the overlap (27.3.4.6), then they glue to a map  $(R, R^+) \rightarrow (\mathcal{O}(Y), \mathcal{O}^+(Y))$ , as  $\mathcal{O}, \mathcal{O}^+$  are all sheaves, and this gives a morphism  $Y \rightarrow X$ .  $\square$

**Prop. (27.8.2.10) [Strictly Totally Disconnected Pro-Étale Cover].** Let  $X$  be an affinoid perfectoid space, then there is an affinoid perfectoid space  $\tilde{X}$  with an affinoid pro-étale surjective and universally open map  $\tilde{X} \rightarrow X$  that  $\tilde{X}$  is strictly totally disconnected.  $\lrcorner$

*Proof:* Cf. [Sch17] P35.  $\square$

**Prop. (27.8.2.11).** The presheaf  $\mathcal{O}, \mathcal{O}^+$  are sheaves w.r.t. the  $v$ -topology. Moreover, the  $v$ -site is subcanonical.  $\lrcorner$

*Proof:* Cf.[Sch17]P40.  $\square$

**Prop. (27.8.2.12).** Let  $X$  be an affinoid perfectoid space, then  $H_v^i(X, \mathcal{O}) = 0$  for  $i > 0$ , and  $H_v^i(X, \mathcal{O}^+)$  is almost zero for  $i > 0$ .  $\lrcorner$

*Proof:* Cf.[Sch17]P41.  $\square$

### Descent on Pro-Étale Site

**Prop. (27.8.2.13) [Descent May Fail].** The question is whether the fibered category over  $\text{Perfd}$ :

$$X \mapsto \{\text{Perfectoid Spaces } Y \rightarrow X\}$$

is a stack for the pro-étale topology. This fails in general. An evidence is that the fibered category

$$X \mapsto \{\text{Affinoid Perfectoid Spaces } Y \rightarrow X\}$$

is not a stack on the category of affinoid Perfectoid spaces with the analytic topology, let along the pro-étale topology:

Let  $X = \text{Spa } K\langle x, y \rangle$ , and  $V \subset X$  be  $\{(x, y) \mid |x| = 1 \text{ or } |y| = 1\}$ , then  $V$  is covered by two affinoids, but is not an affinoid itself:  $H^1(V, \mathcal{O}_V) = \bigoplus_{m, n > 0} Kx^{-m}y^{-n} \neq 0$ . But there is a standard rational covering  $X = \cup_i X(\frac{f_1, f_2, f_3}{f_i})$ , where  $f_1 = \varpi, f_2 = x, f_3 = y$ , and

$$U_0 \cap V = \emptyset, \quad U_1 \cap V = \{|x| = 1\} \subset U_1, \quad U_2 \cap V = \{|y| = 1\} \subset U_2$$

are all affinoid. There is a similar example in the perfectoid case, thus

$$X \mapsto \{\text{Affinoid Perfectoid Spaces } Y \rightarrow X\}$$

is not a stack.  $\lrcorner$

**Prop. (27.8.2.14) [Pro-étale is not Pro-étale Local].** There is an example of a non-pro-étale map that is pro-étale locally pro-étale.  $\lrcorner$

*Proof:* Cf.[S-W20]P66.  $\square$

**Prop. (27.8.2.15) [Characterization of Locally pro-étale Maps].** For  $f : X \rightarrow Y \in \text{AffPerfdSpa}$ , the following are equivalent:

- There exists an affinoid pro-étale cover  $Y' \rightarrow Y$  s.t. the base change  $X' = X \times_Y Y' \rightarrow Y'$  is pro-étale.
- For all geometric points  $\text{Spa } C \rightarrow Y$ ,  $X \times_Y \text{Spa } C = \text{Spa } C \times \underline{S}$  for some profinite set  $S$ .

$\lrcorner$

*Proof:* Cf.[S-W20]P66.  $\square$

**Prop. (27.8.2.16) [Descent].** Descent data of the the fibered category

$$X \mapsto \{\text{Perfectoid Spaces } Y \rightarrow X\}$$

of a perfectoid space  $Y' \rightarrow X'$  along a pro-étale cover  $X' \rightarrow X$  is effective in the following cases:



- If  $X, X', Y'$  are affinoids and  $X$  is totally disconnected.
- if  $f$  is separated and pro-étale and  $X$  is strictly totally disconnected.
- If  $f$  is separated and étale. In particular, the fibered category

$$X \mapsto \{\text{separated étale } X \rightarrow Y\}$$

is a stack over the category of perfectoid spaces with the pro-étale topology.

- If  $f$  is finite étale. In particular, the fibered category

$$X \mapsto \{\text{finite étale } X \rightarrow Y\}$$

is a stack over the category of perfectoid spaces with the pro-étale topology.

⌋

*Proof:* Cf. [Sch17] P9.3, 9.6, 9.7. □

### 3 Morphisms of v-Stacks

**Def. (27.8.3.1) [Étale Morphisms of Stacks].** Let  $f : Y' \rightarrow Y$  be a map of pro-étale stacks on  $\mathcal{P}\text{erfd}$ ,

- Assume  $f$  is locally separated (i.e. there is an open cover of  $Y'$  over which  $f$  becomes separated), then  $f$  is called **quasi-pro-étale** if for any strictly totally disconnected perfectoid space  $X$  and a map  $X \rightarrow Y$ , the pullback  $Y' \times_Y X$  is representable and  $Y' \times_Y X \rightarrow X$  is pro-étale.
- Assume  $f$  is locally separated, then  $f$  is called **étale** if for any perfectoid space  $X$  and a map  $X \rightarrow Y$ , the pullback  $Y' \times_Y X$  is representable and  $Y' \times_Y X \rightarrow X$  is étale.
- $f$  is called **finite étale** if for any perfectoid space  $X$  and a map  $X \rightarrow Y$ , the pullback  $Y' \times_Y X$  is representable and  $Y' \times_Y X \rightarrow X$  is finite étale.

⌋

### 4 Diamonds

**Remark (27.8.4.1) [Motivation of Diamonds].** The idea of diamonds is that there should be a functor

$$\diamond : \{\text{analytic adic spaces over } \mathbb{Z}_p\} \rightarrow \{\text{diamonds}\}$$

that forgets the structure morphism to  $\mathbb{Z}_p$ . For a perfectoid space  $X$ ,  $X \mapsto X^{\text{flat}}$  has this property, so this functor should coincide on these objects. Now any analytic adic space  $X$  over  $\mathbb{Z}_p$  is pro-étale locally perfectoid:

$$X = \text{Coeq}(\tilde{X} \times_X \tilde{X} \rightrightarrows X),$$

where  $\tilde{X} \rightarrow X$  is a pro-étale perfectoid cover. The equivalence relations  $R = \tilde{X} \times_X \tilde{X}$  is also perfectoid, so this functor should send  $X$  to  $\text{Coeq}(R^b \rightrightarrows \tilde{X}^b)$ .

For example, if  $X = \text{Spa}(\mathbb{Q}_p)$ , then a pro-étale cover of  $X$  is  $\text{Spa}((\mathbb{Q}_p^{\text{cycl}})^\wedge)$ , and then  $R = \tilde{X} \times_X \tilde{X} = \tilde{X} \times \mathbb{Z}_p^*$  by Galois theory, and then  $\mathbb{Q}_p^\diamond$  should be defined as the coequalizer of  $\text{Spa}((\mathbb{Q}_p^{\text{cycl}})^b)/\mathbb{Z}_p^*$ , whose meaning is explained in (27.8.4.11). ⌋

**Def. (27.8.4.2) [Diamonds].** A **diamond** is a pro-étale sheaf  $\mathcal{D}$  on  $\mathcal{P}\text{erfdSpa}$  that can be written as  $\mathcal{D} = X/R$ , where

- $X \in \mathcal{P}\text{erfd}$ ,

- $R$  is a pro-étale equivalence relation in  $X \times X$  (i.e. an equivalence relation that the maps  $s, t : R \rightarrow X$  are pro-étale),
- $R$  is representable.

┘

**Prop. (27.8.4.3).** Let  $X \in \mathcal{P}\text{erfd}$  and  $R \subset X \times X$  a representable pro-étale equivalence relation, then

- The quotient sheaf  $Y = X/R$  is a diamond.
- The natural map of sheaves  $R \rightarrow X \times_Y X$  is an isomorphism.
- Let  $\tilde{X} \rightarrow X$  be a pro-étale cover by a perfectoid space  $\tilde{X}$ , and  $\tilde{R} = R \times_{X \times X} (\tilde{X} \times \tilde{X})$  the induced equivalence relation, then  $\tilde{R}$  is a representable pro-étale equivalence relation of  $\tilde{X}$ , and the natural map  $\tilde{Y} = \tilde{X}/\tilde{R} \rightarrow Y = X/R$  is an isomorphism.
- The map  $X \rightarrow Y$  is quasi-pro-étale (27.8.3.1).

┘

*Proof:* 1 is by definition.

For 2, firstly  $R \rightarrow X \times_Y X$  is injective as subsheaves of  $X \times X$ . Next, if  $Z \rightarrow X \times_Y X$  is any map from a perfectoid space  $Z$ , then we have two maps  $a, b : Z \rightarrow X$  that their composition with  $X \rightarrow Y$  agree. This means after passing to a pro-étale covering  $\tilde{Z} \rightarrow Z$ , the composition map  $\tilde{Z} \rightarrow Z \rightarrow X \times X$  factors through  $R$ . Now this map  $\tilde{Z} \rightarrow R$  descends to a map  $Z \rightarrow R$ , because pro-étale site is subcanonical and the two projection maps  $\tilde{Z} \times_Z \tilde{Z} \rightarrow R$  coincides because they do after compositing with  $R \hookrightarrow X \times X$ .

3: Firstly  $\tilde{R}$  is representable as fiber products of representable objects (27.4.1.4), and the two projections are pro-étale, as they are compositions of base changes of pro-étale morphisms. Also the map  $\tilde{Y} \rightarrow Y$  of pro-étale sheaves is surjective, as the composition  $\tilde{X} \rightarrow X \rightarrow Y$  is.

To show  $\tilde{Y} \rightarrow Y$  is injective, let  $Z$  be a perfectoid space with two maps  $Z \rightarrow \tilde{Y}$  that coincide after compositing with  $\tilde{Y} \rightarrow Y$ , we need to show  $a = b$ . Now because pro-étale site is subcanonical, it suffices to show this after replacing  $Z$  with a pro-étale cover  $\tilde{Z}$ , such that  $a, b$  factors over  $\tilde{a}, \tilde{b} : \tilde{Z} \rightarrow \tilde{X}$ . The associated map  $\tilde{Z} \rightarrow \tilde{X} \times \tilde{X} \rightarrow X \times X$  factors through  $R$  by item 2, so we get a map  $\tilde{Z} \rightarrow R \times_{X \times X} (\tilde{X} \times \tilde{X}) = \tilde{R}$ , which means  $\tilde{a}, \tilde{b}$  induces the same map  $\tilde{Z} \rightarrow \tilde{Y}$ . So we are done.

4: By 3, we can replace  $X$  by  $\tilde{X} = \coprod_i U_i \rightarrow X$ , where  $U_i$  is an affinoid cover of  $\tilde{X}$ , and  $R$  by the induced equivalence relation in  $\tilde{X} \times \tilde{X}$ , and to pro-étale is analytically local. In this way, we may assume  $R \subset X \times X$  and  $X \times X \rightarrow X$  are separated.

Let  $X'$  be a strictly totally disconnected perfectoid space and  $X' \rightarrow X$  be a map. Because  $X \rightarrow Y$  is surjective as sheaves on  $\mathcal{P}\text{erf}$ , there is a pro-étale cover  $\tilde{X}' \rightarrow X'$  and a map  $\tilde{X}' \rightarrow X$  lying over  $X' \rightarrow Y$ . We can assume  $\tilde{X}'$  is affinoid. Let  $W = X' \times_Y X \rightarrow X'$  be the fiber product, then

$$\tilde{X}' \times_{X'} W = \tilde{X}' \times_X (X \times_Y X) = \tilde{X}' \times_X R$$

is representable and pro-étale over  $\tilde{X}'$ , and also separated. So by (27.8.2.16),  $W$  is also representable, pro-étale and separated over  $X$ . □

**Cor. (27.8.4.4) [Equivalent Characterization of Diamonds].** Let  $Y$  be a pro-étale sheaf on  $\mathcal{P}\text{erf}$ , then  $Y$  is a diamond iff there is a surjective quasi-pro-étale morphism  $X \rightarrow Y$  from a perfectoid space  $X$ . If  $X$  is a disjoint union of strictly totally disconnected spaces, then  $R \subset X \times_Y X \subset X \times X$  is a representable pro-étale equivalence relation with  $Y = X/R$ . ┘

*Proof:* If  $Y$  is a diamond, then  $X \rightarrow Y$  is quasi-pro-étale by (27.8.4.3). Conversely, if there is a quasi-pro-étale morphism  $X \rightarrow Y$ , by (27.8.2.10), we can assume that  $X$  is a disjoint union of strictly

disconnected spaces. In this case, by the definition of auasi-pro-étale,  $R$  is representable and the projections  $R = X \times_Y X \rightarrow X$  is pro-étale, and  $X/R \cong Y$ : it suffices to show this map is injective: if  $a, b : Z \rightarrow X$  are two maps that coincide after composing with  $X \rightarrow Y$ , then after replacing to a pro-étale covering  $\tilde{Z} \rightarrow Z$ , we can lift to maps  $(\tilde{a}, \tilde{b}) : \tilde{Z} \rightarrow R$ . And this map descend to a map  $Z \rightarrow R$ , because the pullback to  $\tilde{Z} \times_Z \tilde{Z} \rightarrow R$  coincides as they do after composing with  $R \hookrightarrow X \times X$ . So  $X/R \rightarrow Y$  is injective.  $\square$

**Cor. (27.8.4.5).**

- Let  $Y$  be a pro-étale sheaf on  $\mathcal{P}\text{erfd}$  and there is a quasi-pro-étale map  $Y' \rightarrow Y$ , where  $Y'$  is a diamond, then  $Y$  is also a diamond.
- Let  $f : Y' \rightarrow Y$  be a quasi-pro-étale map of pro-étale sheaves on  $\mathcal{P}\text{erfd}$  and  $Y$  is a diamond, then  $Y'$  is also a diamond.

┘

*Proof:* 1: By (27.8.4.4), we can choose a quasi-pro-étale map  $X \rightarrow Y'$  where  $X$  is a perfectoid space, then  $X \rightarrow Y' \rightarrow Y$  is also quasi-pro-étale, so  $Y$  is a diamond by (27.8.4.4) again.

2: Choose a surjective quasi-pro-étale map  $X \rightarrow Y$  where  $X$  is a perfectoid space, then  $X' \times_X Y$  is representable and  $X' \rightarrow Y'$  is quasi-pro-étale and surjective, thus  $Y'$  is a diamond by (27.8.4.4).  $\square$

**Lemma (27.8.4.6) [Product of Perfectoid Spaces].** The product of two perfectoid spaces of char  $p$  is also a perfectoid space.  $\square$

*Proof:* Cf. [Sch17]P71.  $\square$

**Prop. (27.8.4.7) [Absolute Product of Diamonds].** Let  $\mathcal{D}, \mathcal{D}'$  be diamonds, then the product sheaf  $\mathcal{D} \times \mathcal{D}'$  is also a diamond.  $\square$

*Proof:* Let  $\mathcal{D} = X/R$  and  $\mathcal{D}' = X'/R'$ , then  $R \times R', X \times X'$  are also representable by (27.8.4.6),  $R \times R' \rightarrow (X \times X') \times (X \times X')$  is also an injection, and the projections are pro-étale. Thus  $\mathcal{D} \times \mathcal{D}' = (X \times X')/(R \times R')$  is a diamond, by (27.8.4.3).  $\square$

**Prop. (27.8.4.8) [Fiber Products].** Fiber products exist in the category of diamonds.  $\square$

*Proof:* Let  $Y_1 \rightarrow Y_2 \leftarrow Y_3$  be a diagram of diamonds. Choose representations  $Y_i = X_i/R_i$ , and after replacing  $X_1, X_2$  with a pro-étale covering using (27.8.4.3), we can assume there are maps  $X_i \rightarrow X_3$  lying over  $Y_i \rightarrow Y_3$ . Moreover, we can replace  $X_i$  by  $X_i \times_{Y_3} X_3$  to assume that  $X_i \rightarrow Y_1 \times_{Y_3} X_3$  is surjective in the pro-étale topology.

In this case, the map  $X_1 \times_{X_3} X_2 \rightarrow Y_1 \times_{Y_3} Y_2$  is surjective in the pro-étale topology, and the equivalence relation  $R_4 = X_4 \times_{Y_4} X_4$  can be calculated to be  $R_4 = R_1 \times_{R_3} R_2$ , which is representable. It remains to see that  $R_4 \rightarrow X_4$  is pro-étale. But  $R_1 \times R_2 \rightarrow X_1 \times X_2$  is pro-étale, so does its base change  $R_1 \times_{X_3} R_2 \rightarrow X_1 \times_{X_3} X_2 = X_4$ , and also  $R_4 = R_1 \times_{R_3} R_2 \rightarrow R_1 \times_{X_3} R_2$  is pro-étale because it is the base change of  $R_3 \rightarrow R_3 \times_{X_3} R_3$ , which is pro-étale by (27.8.2.5).  $\square$

**Prop. (27.8.4.9).** Let  $Y$  be a diamond, then  $Y$  is a sheaf for the v-topology.  $\square$

*Proof:* Cf. [Sch17]P54.  $\square$

**Def. (27.8.4.10) [Underlying Spaces of Diamonds].** Any diamond has an underlying space  $|\mathcal{D}| = |X|/|R|$ .  $\square$

*Proof:* ?  $\square$

$\mathrm{Spd}(\mathbb{Q}_p)$

**Prop. (27.8.4.11).** Let  $\mathrm{Spd}(\mathbb{Q}_p)$  be defined as

$$\mathrm{Spa}((\mathbb{Q}_p^{\mathrm{cycl}})^{\flat})/\mathbb{Z}_p^* = \mathrm{Spa}(\mathbb{F}_p((t^{1/p^\infty}))/\mathbb{Z}_p^*,$$

where  $\mathbb{Z}_p^*$  acts on  $\mathbb{F}_p((t^{1/p^\infty}))$  via  $\gamma(t) = (1+t)^\gamma - 1$  (27.2.8.17), i.e. it is the coequalizer of

$$\mathbb{Z}_p^* \times \mathrm{Spa}(\mathbb{F}_p((t^{1/p^\infty}))) \rightrightarrows \mathrm{Spa}(\mathbb{F}_p((t^{1/p^\infty}))),$$

where one map is projection and the other map is the group action. To show this is diamond, we first need to verify this is an injection thus an equivalence relation, which follows from (27.8.4.12).  $\lrcorner$

**Lemma (27.8.4.12).** Consider the map

$$g : \mathbb{Z}_p^* \times \mathrm{Spa}(\mathbb{F}_p((t^{1/p^\infty}))) \rightrightarrows \mathrm{Spa}(\mathbb{F}_p((t^{1/p^\infty}))) \times \mathrm{Spa}(\mathbb{F}_p((t^{1/p^\infty})))$$

where the first map is group action and the second map is projection, then this is an injection map.  $\lrcorner$

*Proof:* For any adic point  $(K, K^+)$ ,  $\mathbb{Z}_p^*$  acts freely on  $K^{00}$ , thus the map is an injection, by (27.8.1.7).  $\square$

**Prop. (27.8.4.13) [Torsor over Affinoid Perfectoid Space].** Let  $G$  be a profinite group and  $f : \mathcal{F}' \rightarrow \mathcal{F}$  be a  $G$ -torsor, with  $G$  profinite, then for any affinoid  $X = \mathrm{Spa}(B, B^+)$  and any morphism  $X \rightarrow \mathcal{F}$ , the pullback  $\mathcal{F}' \times_{\mathcal{F}} X$  is representable by a perfectoid affinoid  $X' = \mathrm{Spa}(A, A^+)$ , where  $(A, A^+)$  is the completed filtered colimit of  $(A_H, A_H^+)$ , where for each open normal subgroup  $H$  of  $G$ ,  $A_H/B$  is a finite étale  $G/H$ -torsor.  $\lrcorner$

*Proof:* If  $H$  is an open normal subgroup of  $G$ , then  $\mathcal{F}'/\underline{H} \rightarrow \mathcal{F}$  is a  $G/H$ -torsor, and  $\mathcal{F}' = \varprojlim_H \mathcal{F}'/\underline{H}$ , thus we reduce to the case  $G$  is finite. But for this case, we use the fact  $\{\text{perfectoid spaces finite étale over } X\}$  is a stack (27.8.2.16), and the definition of  $G$ -torsor.  $\square$

**Prop. (27.8.4.14) [Description of  $\mathrm{Spd}(\mathbb{Q}_p)$ ].** If  $X = \mathrm{Spa}(R, R^+)$  is an affinoid perfectoid space of characteristic  $p$ , then  $(\mathrm{Spd} \mathbb{Q}_p)(X)$  is the set of isomorphism classes of data of the following shape.

- A  $\mathbb{Z}_p^*$ -torsor  $R \rightarrow \tilde{R}$ , i.e.  $\tilde{R} = (\varinjlim_i R_i)^\wedge$ , where  $R_n/R$  is finite étale with Galois group  $(\mathbb{Z}/p^n\mathbb{Z})^*$ .
- A topological nilpotent unit  $t \in \tilde{R}$  s.t. for all  $\gamma \in \mathbb{Z}_p^*$ ,  $\gamma(t) = (1+t)^\gamma - 1$ .

$\lrcorner$

*Proof:* Notice  $\mathrm{Spa}(\mathbb{Q}_p^{\mathrm{cycl}})^{\flat} \rightarrow \mathrm{Spd}(\mathbb{Q}_p)$  is a  $\mathbb{Z}_p^*$ -torsor by definition, so for any morphism  $X \rightarrow \mathrm{Spd}(\mathbb{Q}_p)$ , the pullback  $\mathrm{Spa}(\mathbb{Q}_p^{\mathrm{cycl}})^{\flat} \times_{\mathrm{Spd}(\mathbb{Q}_p)} X \rightarrow X$  is also a  $\mathbb{Z}_p^*$ -torsor, so it is isomorphic to a torsor  $\mathrm{Spa}(\tilde{R}, \tilde{R}^+) \rightarrow \mathrm{Spa}(R, R^+)$  by (27.8.4.13), and the map  $\mathrm{Spa}(\mathbb{Q}_p^{\mathrm{cycl}})^{\flat} \times_{\mathrm{Spd}(\mathbb{Q}_p)} X \rightarrow X \rightarrow \mathrm{Spa}(\mathbb{Q}_p^{\mathrm{cycl}})^{\flat}$  is an  $\mathbb{Z}_p^*$ -equivariant map, which is equivalent to a topologically nilpotent element  $t \in \tilde{R}$  that is equivariant, i.e.  $\gamma(t) = (1+t)^\gamma - 1$ .

Conversely, for any such a  $\mathbb{Z}_p^*$ -torsor  $R \rightarrow \tilde{R}$ , an equivariant element  $t \in \tilde{R}$  descends to a map  $X \rightarrow \mathrm{Spd}(\mathbb{Q}_p)$ .  $\square$

**Prop. (27.8.4.15).**  $\mathrm{PerfdSpa}/\mathbb{Q}_p$  is equivalent to the category of perfectoid spaces  $X \in \mathrm{Perf}_p$  together with a map  $X \rightarrow \mathrm{Spd}(\mathbb{Q}_p)$ .  $\lrcorner$

*Proof:* Consider the category of triples  $(X^\sharp, X, \iota)$ , where  $X^\sharp$  is a perfectoid space over  $\mathbb{Q}_p$ ,  $X$  is a perfectoid space of characteristic  $p$ ,  $\iota : X^\sharp \cong X$  is an isomorphism. A map of triples is a tuple  $(f^\sharp, f) : (X^\sharp, X, \iota) \rightarrow (Y^\sharp, Y, \iota')$  that  $\iota' \circ f^\sharp = f \circ \iota$ .

This category is equivalent to the category of perfectoid spaces over  $\mathbb{Q}_p$  by the forgetful functor, where the quasi-inverse is given by  $X^\sharp \mapsto (X^\sharp, X^\sharp, \text{id}_{X^\sharp})$ . And this category is also fibered in equivalent relations over  $\mathcal{P}\text{erf}$ . So it is equivalent to a presheaf  $\text{Untilt}_{\mathbb{Q}_p}$  which maps  $X$  to the isomorphism classes of untelts  $(X^\sharp, \iota)$  over  $\mathbb{Q}_p$ , where  $\iota : X^\sharp \cong X$  is an isomorphism, by (4.2.3.29).

Similarly we can show define a functor  $\text{Untilt}$  which maps  $X$  to the isomorphism classes of untelts  $(X^\sharp, \iota)$  (of whatever characteristic), where  $\iota : X^\sharp \cong X$  is an isomorphism.

Let  $X = \text{Spa}(R, R^+)$  be an affinoid perfectoid space of characteristic  $p$ . If  $X^\sharp = \text{Spa}(R^\sharp, R^{\sharp+})$  is an untelt. Let  $\tilde{X}^\sharp = X^\sharp \otimes_{\mathbb{Q}_p} \mathbb{Q}_p^{\text{cycl}}$ , then  $\tilde{X}^\sharp \rightarrow X^\sharp$  is a pro-étale  $\mathbb{Z}_p^*$ -torsor, whose tilt  $\tilde{X} \rightarrow X$  is a pro-étale  $\mathbb{Z}_p^*$ -torsor equipped with a  $\mathbb{Z}_p^*$ -equivariant map  $\tilde{X} \rightarrow \text{Spa}(\mathbb{Q}_p^{\text{cycl}})^\flat$  this is a morphism  $X \rightarrow \text{Spd}(\mathbb{Q}_p)$  by (27.8.4.14).

Conversely, let  $\tilde{X} \rightarrow X$  be a pro-étale  $\mathbb{Z}_p^*$ -torsor and  $\tilde{X} \rightarrow \text{Spa}(\mathbb{Q}_p^{\text{cycl}})^\flat$  a  $\mathbb{Z}_p^*$ -equivariant map, then by tilting equivalence there exists a morphism  $\tilde{X}^\sharp \rightarrow \text{Spa}(\mathbb{Q}_p^{\text{cycl}})$  which is also  $\mathbb{Z}_p^*$ -equivariant. The equivariance means that it is a descent datum along  $\tilde{X} \rightarrow X$ , so it descends to an untelt  $X^\sharp$  of  $X$  over  $\mathbb{Q}_p$ .

Finally, for general affinoid perfectoid space  $X$ , as  $\text{Untilt}_{\mathbb{Q}_p}$  and  $\text{Spd}(\mathbb{Q}_p)$  are all sheaves on  $\mathcal{P}\text{erf}$ , the above construction can be glued to give an isomorphism between them.  $\square$

**Prop. (27.8.4.16) [Untelts is a Sheaf].**  $\text{Untilt}$  is a v-sheaf on  $\mathcal{P}\text{erfd}$ . So does  $\text{Untilt}_{\mathbb{Q}_p}$ , because the invertibility of  $p$  can be verified locally as  $\mathcal{O}$  is sheaf.  $\lrcorner$

*Proof:* Firstly  $\text{Untilt}$  is clearly an analytic sheaf, so it suffices to show that if  $X = \text{Spa}(R, R^+)$  is a perfectoid space of characteristic  $p$  with a v-cover  $Y = \text{Spa}(S, S^+) \rightarrow X$  and  $Y^\sharp = \text{Spa}(S^\sharp, S^{\sharp+})$  is an untelt of  $Y$  that the corresponding two untelts of  $Z = Y \times_X Y$  agree, then there is a unique untelt  $X^\sharp = \text{Spa}(R^\sharp, R^{\sharp+})$  whose pullback to  $Y$  is  $Y^\sharp$ .

Cf. [Sch17]P86.  $\square$

### Spatial Diamonds

**Def. (27.8.4.17) [Spatial Diamonds].** A **spatial diamond** is a qcqs diamond  $D$  s.t.  $|D|$  has a basis of opens  $|U|$  s.t.  $U \subset D$  is a qc open immersion.  $\lrcorner$

*Proof:*  $\square$

### Diamonds associated to Perfectoid Spaces

**Def. (27.8.4.18).** For  $Y \in \mathcal{P}\text{erfdSpa}$ ,  $Y^\diamond$ : is the map  $X \mapsto$  untelts of  $X$  with a map to  $Y$ .

It is a locally spatial diamond, s.t.  $|Y^\diamond| = |Y|$ .  $\lrcorner$

**Example (27.8.4.19).**  $\text{Spa}(R, R^+)^\diamond = \text{Spa}(R^\sharp, R^{\sharp+})$ .  $\lrcorner$

## 27.9 Untilts and Fargues-Fontaine Curves

References are [FF curves, Lurie], [FF Curve, Johannes], [The Fargues-Fontaine Curve and Diamonds, Morrow], [Fargues-Fontaine. Courbes et fibrés vectoriels en théorie de Hodge  $p$ -adique].

### 1 Schematic Fargues-Fontaine Curves

**Def. (27.9.1.1) [Fargues-Fontaine Curve].** The sum  $\bigoplus_n B^{\varphi=p^n}$  is a graded ring. In fact, it is non-negatively graded (27.9.1.33), and we define the **Fargues-Fontaine curve** as the projective scheme

$$\mathbf{Proj} \left( \bigoplus_{n \geq 0} B^{\varphi=p^n} \right).$$

┘

**Def. (27.9.1.2) [Formal Logarithm].** For  $x \in 1 + \mathfrak{m}_{C^b}$ ,  $[x] - 1 = [x - 1] + \sum_{n>0} [c_n] p^n$ , thus  $|[x] - 1|_\rho \geq |x - 1| > 0$ , thus the formal logarithm

$$\log([x]) = \sum_{k>0} \frac{(-1)^{k+1}}{k} ([x] - 1)^k$$

converges for every Gauss norm  $|\cdot|_\rho$ , thus converges to some element in  $B$ . And clearly  $\varphi(\log([x])) = p \log([x])$ , thus  $\log([x]) \in B^{\varphi=p}$ . And  $\log([xy]) = \log([x]) \log([y])$ . ┘

**Prop. (27.9.1.3) [Artin-Hasse Exponential].** There is another way of constructing elements in  $B^{\varphi=p}$ , which is

$$T : a \in \mathfrak{m}_{C^b} \mapsto \sum_n \frac{[a^{p^n}]}{p^n}.$$

We want to relate this one to the formal logarithm:

There is a bijection of sets  $\mathfrak{m}_{C^b} \cong 1 + \mathfrak{m}_{C^b}^b$  that  $\log([E(a)]) = T(a)$ , which is defined by the **Artin-Hasse exponential**

$$E(x) = \prod_{(d,p)=1} \left( \frac{1}{1-x^d} \right)^{\frac{\mu(d)}{d}}.$$

┘

*Proof:* Firstly, it has coefficients in  $\mathbb{Z}_{(p)}$ , because  $(1-x^d)^{\frac{1}{d}} = \sum (-1)^k C_{\frac{1}{d}}^k x^{kd}$  has coefficient in  $\mathbb{Z}_{(p)}$ . And  $[1-x] = \lim_k (1 - [x^{p^{-k}}])^{p^k}$ , so

$$\log \left( \left[ \prod_{(d,p)=1} \left( \frac{1}{1-x^d} \right)^{\frac{\mu(d)}{d}} \right] \right) = \sum_{(d,p)=1} \frac{\mu(d)}{d} \log \left( \frac{1}{[1-d]} \right) = \sum_{(d,p)=1} \mu(d) \sum_{\alpha \in p^{-n}\mathbb{Z}} \frac{[x^{d\alpha}]}{d\alpha}$$

Notice the right hand side stabilizes for any term  $[x^\beta]$ , and if  $\beta \neq \frac{1}{p^k}$ , it will vanish, thus for  $x \in \mathfrak{m}_{C^b}$ , it converges, and the sum equals  $\sum_n \frac{[x^{p^n}]}{p^n}$ . ┘

**Cor. (27.9.1.4).** The set of elements of the form  $\sum_n \frac{[a^{p^n}]}{p^n}$  is closed under addition. ┘

### Valuation Function

**Def. (27.9.1.5) [Exponential Valuation].** For any positive real number  $s$ , define a valuation on  $\mathbf{A}_{\inf}[\frac{1}{p}, \frac{1}{[t]}]$  by the formula  $v_s(f) = -\log |f|_{\exp(-s)}$ , then it is a valuation by (27.7.1.8).

If  $f$  has a Teichmüller expansion  $\sum_{n \gg -\infty} [c_n] p^n$ , then

$$\|f\|_{\rho} = \sup\{|c_n|_{C^b} \rho^n\}, \quad v_s(f) = \inf\{v(c_n) + ns\}.$$

┘

**Prop. (27.9.1.6).** For any  $f \neq 0 \in \mathbf{A}_{\inf}[\frac{1}{p}, \frac{1}{[t]}]$ ,  $s \mapsto v_s(f)$  is a concave function in  $s$  which is piecewise linear with integral slopes. ┘

*Proof:* Consider the Newton Polygon. ┐

**Lemma (27.9.1.7).** If  $s > 0$  and  $f_n$  is a Cauchy sequence in  $\mathbf{A}_{\inf}[\frac{1}{p}, \frac{1}{[t]}]$  for the norm  $|\cdot|_{\exp(-s)}$  and doesn't converge to 0, then the sequences

$$v_s(f_n), \quad \partial_- v_s(f_n), \quad \partial_+ v_s(f_n)$$

stabilize. ┘

*Proof:* Easy, Cf. [FF Curve, Lurie]P44. ┐

**Prop. (27.9.1.8).** If  $0 < a \leq b < 1$ , and  $f_n$  is a Cauchy sequence in  $\mathbf{A}_{\inf}[\frac{1}{p}, \frac{1}{[t]}]$  and doesn't converge to 0 for either the norm  $|\cdot|_a$  or  $|\cdot|_b$ , then the sequence of functions  $s \mapsto v_s(f)$  stabilizes on  $[-\log(b), -\log(a)]$ . ┘

*Proof:* Assume  $f_n$  doesn't converge to 0 for the form  $|\cdot|_b$ , then by (27.9.1.7), the sequences  $v_s(f_n), \partial_+ v_s(f_n)$  converges, thus  $v_s(f_n)$  is bounded uniformly, thus  $v_s(f)$  is bounded.

Then choose  $N$  large that  $|f - f_m|_{\rho}$  very small for any  $m > N$  and  $a \leq \rho \leq b$ , then  $v_s(f) = v_s(f_m)$  for any  $a \leq s \leq b$ , thus it stabilizes. ┐

**Cor. (27.9.1.9).** For  $f \in B^{\times}$ , the construction  $s \mapsto v_s(f)$  is a concave function in  $s$  with piecewise linear function with integral slopes. This is analogous to the Hadamard three circle theorem (12.5.2.13). ┘

*Proof:* This is true for  $f \in B_{[a,b]}$ , because any  $f$  is a limit of a sequence  $f_n$  in both the norm  $|\cdot|_a$  and  $|\cdot|_b$ , so by the proposition, there for  $n$  large,  $v_s(f) = v_s(f_n)$  on  $[-\log(b), -\log(a)]$ , thus the conclusion is true by (27.9.1.6). And for  $f \in B$ , for any interval  $[a, b]$  we can do the same, thus the conclusion is true on each interval, thus it is true. ┐

### Metric Structures on $Y$

**Def. (27.9.1.10) [Metric on  $\overline{Y}$ ].** Let  $\overline{Y} = Y \cup \{0\}$  be the isomorphism classes of untilts of  $C^b$ , where 0 corresponds to  $C^b$  itself.

(27.7.1.5) show  $\overline{Y}$  corresponds to distinguished elements in  $\mathbf{A}_{\inf}$  up to units. So for any  $x, y \in \overline{Y}$ , we let  $d(x, y) = |\xi_x(y)|_{K_y} \leq 1$ . Then this is a metric, and it is non-Archimedean. ┘

*Proof:* Firstly, if  $d(x, y) = 0$ , then  $\xi_x$  divides  $\xi_y$ , which is equivalent to  $(\xi_x) = (\xi_y)$ , by (5.5.4.21).

Secondly, for any  $x, y$ , since  $C^b$  is alg.closed, we can assume  $\xi_x(y) = c^{\sharp}$  for some  $c \in C^b$ . Notice  $\xi(y) = t^{\sharp} + pu(y)$  is in  $\mathfrak{m}_K$ , thus  $c \in \mathfrak{m}_{C^b}$ . So  $\xi_x - [c]$  is also a distinguished element and vanishes at  $y$ , so we may assume that  $\xi_y = \xi_x - [c]$  by (5.5.4.21) again. Then

$$d(y, x) = |\xi_y(x)|_{K_x} = |c^{\sharp}|_{K_x} = |c|_{C^b} = |c^{\sharp}|_{K^y} = d(x, y).$$

Finally it is non-Archimedean because any valuation field  $K$  is non-Archimedean. ┐



**Prop. (27.9.1.11) [ $\bar{Y}$  is Complete].**  $\bar{Y}$  is complete w.r.t this metric.  $\lrcorner$

*Proof:* Given a Cauchy sequence of points  $y_n$  in  $\bar{Y}$ , as in the proof of (27.9.1.10), we can assume that  $\xi_{y_n} = \xi_{y_{n-1}} + [c_n]$  for some  $c_n \in \mathfrak{m}_{C^b}$ , and  $|c_n|_{C^b} = d(y_{n-1}, y_n)$ . Now  $\mathbf{A}_{\text{inf}}$  is  $[t]$ -adically complete for a uniformizer  $t \in C^b$ , thus  $\sum [c_n]$  is definable in  $\mathbf{A}_{\text{inf}}$ , and  $\xi = \xi_0 + \sum [c_n]$  is also distinguished, and corresponds to a point  $y$  which  $y_n$  clearly converges to.  $\square$

### Divisors

**Lemma (27.9.1.12).**  $B_{[a,b]}$  is an integral domain.  $\lrcorner$

*Proof:* By (27.9.1.9), the valuation function  $v_s(f)$  and  $v_s(g)$  are bounded, thus it is clear that  $v_s(fg)$  is also finite, so  $fg \neq 0$ .  $\square$

**Prop. (27.9.1.13) [Divisors].** Assume  $C^b$  is alg.closed, then for any  $f \in B_{[a,b]}$  and  $y = (K, \iota) \in Y_{[a,b]}$ , we define the **order of vanishing**  $\text{ord}_K(f) \in \mathbb{Z} \cup \{\infty\}$  as the valuation of  $e_K(f) \in B_{\text{dR}}^+(K)$ . Then

- if  $f \neq 0 \in B_{[a,b]}$ , then  $\text{ord}_K(f) < \infty$  for each  $K \in Y_{[a,b]}$ , and there are only f.m.  $K$  that  $\text{ord}_K(f) \neq 0$ . In particular,  $B_{[a,b]}$  is an integral domain.
- if  $x, y \neq 0 \in B_{[a,b]}$ , then  $x$  divides  $y$  iff  $\text{ord}_K(x) \geq \text{ord}_K(y)$  for each  $K \in Y_{[a,b]}$ .

Thus for each  $f \in B_{[a,b]}$ , we can define the **divisor** of  $K$  as the formal sum  $\sum_{K \in Y_{[a,b]}} \text{ord}_K(f) K$ , and this is also definable for  $f \in B$ , but it may be an infinite but locally finite sum.  $\lrcorner$

*Proof:* Firstly, by (27.9.1.17) and (27.9.1.18), if  $\text{div}(f) \cap Y_{[a,b]} \neq \emptyset$ , then there is a distinguished element  $\xi$  that  $f = \xi f_1$ . And we can iterate this, and eventually end up with  $f = \xi_1 \dots \xi_n f_n$  that  $\text{div}(f_n) \cap Y_{[a,b]} = \emptyset$ , by (27.9.1.19), so by (27.9.1.20),  $f_n$  is invertible in  $B$ , so  $\text{div}(f)$  is finite. And if  $\text{div}(g) \geq \text{div}(f)$ , then  $g$  also divides  $\xi_1 \dots \xi_n$ , so  $g$  divides  $f$ .  $\square$

**Remark (27.9.1.14).** Notice by (27.7.1.22), for a  $f \in B_{[a,b]}$ ,  $\text{ord}_K(f) > 0$  iff  $f(y) = 0 \in K$ .  $\lrcorner$

**Cor. (27.9.1.15) [ $B$  is Integral Domain].**  $B$  is an integral domain, and if  $C^b$  is alg.closed, then  $x$  is divisible by  $y$  if and only if  $\text{div}(x) \geq \text{div}(y)$ .  $\lrcorner$

**Cor. (27.9.1.16).**  $B$  is integrally closed.  $\lrcorner$

*Proof:*  $B$  is an integral domain by (27.9.1.15), it is integrally closed because if  $f/g$  is integral over  $B$ , then there image in  $B_{\text{dR}}^+(y)$  is integral over  $B_{\text{dR}}^+(y)$  for all  $y \in Y$ , thus in  $B_{\text{dR}}^+(y)$  because it is a valuation ring, and then  $f$  is divisible by  $g$  by (27.9.1.15).  $\square$

**Prop. (27.9.1.17) [Examples of Divisors].**

- For a distinguished element  $\xi$ , if  $\xi = up$ , then  $\xi$  is invertible in  $B$ , thus  $\text{div}(\xi) = 0$ . Otherwise  $\xi$  defines a char0 untilts  $K$  of  $C^b$ , and  $\xi$  is a uniformizer of  $B_{\text{dR}}^+(K)$ , and it doesn't divides other distinguished elements (5.5.4.21), thus  $\text{div}(\xi) = K$ .
- $\text{div}(\log([x])) = \sum_{n \in \mathbb{Z}} \varphi^n(K)$ .

$\lrcorner$

*Proof:* 2:  $\log([x])$  vanishes at a single  $\varphi$ -orbits of  $Y$ , and one of them is given by the distinguished element  $\xi = 1 + [x^{1/p}] + \dots + [x^{p-1/p}] = \frac{[x]-1}{[x^{1/p}]-1}$ . Notice  $[x^{1/p}] - 1$  is mapped to an invertible element in  $K$ , thus it is invertible in  $B_{\text{dR}}^+(K)$ , so  $[x] - 1$  is associated to  $\xi$ , and notice

$$\log([x]) = \sum_{k>0} \frac{(-1)^{k+1}}{k} ([x] - 1)^k \equiv [x] - 1 \pmod{([x] - 1)^2},$$



so  $\text{ord}_K(\log([x])) = 1$ , and because  $\varphi(\log([x])) = p \log([x])$ ,  $\text{ord}_{\varphi^n(K)}(\log([x])) = 1$  for any  $n$ , so we are done.  $\square$

**Lemma (27.9.1.18).** Let  $C^\flat$  be alg.closed. If  $\xi$  is a distinguished element of  $\mathbf{A}_{\text{inf}}$  vanishes at a point  $y \in Y_{[a,b]}$  and  $g \in B_{[a,b]}$  also vanishes at  $y$ , then  $g$  is divisible by  $\xi$  in  $B_{[a,b]}$ .  $\lrcorner$

*Proof:* If  $g \in \mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{[\pi]}]$ , then this is easy by (27.9.1.14) and  $\mathbf{A}_{\text{inf}}/(\xi) = \mathcal{O}_K$  (27.7.1.4).

Now generally  $g$  is a limit of  $g_n \in \mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{[\pi]}]$ , so  $g(y)$  is the limit of  $g_n(y) \in K$ . Now  $g(y) = 0$ , so  $\lim_n g_n(y) = 0$ . Now  $K$  is alg.closed by (12.2.8.16), so we can let  $g_n(y) = c_n^\sharp$ , so  $c_i$  converges to 0 in  $C^\flat$ . So  $\{[c_n]\}$  converges to 0 in norm  $|\cdot|_a$  and  $|\cdot|_b$ , so we can replace  $g_n$  by  $g_n - [c_n]$  and assume  $g_n(y) = 0$ .

Now the first part shows  $g_n = \xi h_i$ , and now  $h_i$  is a Cauchy sequence for both  $|\cdot|_a$  and  $|\cdot|_b$ , so converges to some  $h$ , and then  $g = \xi h$ .  $\square$

**Lemma (27.9.1.19).** Given (27.9.1.18), we have a strategy of proving (27.9.1.13), that is, decomposing  $f, g$  into distinguished elements, but we need to show this decomposition is finite. And this is true:

If  $f \neq 0 \in B_{[a,b]}$ , denote  $\beta = -\log(b), \alpha = -\log(a)$  and let  $N = \partial_- v_\beta(f) - \partial_+ v_\alpha(f) \geq 0$ , then  $f$  cannot be divisible by a product of  $\xi_1, \dots, \xi_{N+1}$  of  $N+1$  distinguished elements.  $\lrcorner$

*Proof:* By (5.5.4.20), if  $\xi$  is distinguished, then  $v_s(\xi) = \max\{s, v(v_0)\}$ . Now  $v(v_0) = v(t^\sharp) = v(|p|_K)$  in  $\mathcal{O}_K = \mathbf{A}_{\text{inf}}/([t] - up)$ , so if  $K$  corresponding to  $\xi$  belongs to  $Y_{[a,b]}$ , then  $v(v_0) \in [\beta, \alpha]$ , so  $\partial_- v_\beta(\xi) = 1, \partial_+ v_\alpha(\xi) = 0$ .

So if  $f = \xi_1, \dots, \xi_{N+1}u$ , then  $N(f) \geq \sum N(\xi_i) \geq N+1$ .  $\square$

**Lemma (27.9.1.20) [Valuation Funtion And Invertibility].** Let  $C^\flat$  be alg.closed and  $f \neq 0 \in B_{[a,b]}$ , then the following are equivalent:

- $f$  is invertible.
- $\partial_- v_\beta(f) = \partial_+ v_\alpha(f)$ .
- $\text{div}(f) \cap Y_{[a,b]} = \emptyset$ .

$\lrcorner$

*Proof:*  $2 \rightarrow 3$ : by (27.9.1.19).

$1 \rightarrow 2$ : Because  $N(f) + N(f^{-1}) = 0$ , and  $N(f) \geq 0, N(f^{-1}) \geq 0$ , so  $N(f) = 0$ .

$2 \rightarrow 1$ : Assume first that  $f = \sum_{n \gg -\infty} [c_n] p^n \in \mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{[\pi]}]$ , then the hypothesis just says that  $s \rightarrow v_s(f)$  is linear in a small nbhd of  $[\beta, \alpha]$ , that is, there is a  $n_0$  that  $v(c_n) + ns > v(c_{n_0}) + n_0 s$  for all  $n \neq n_0$  and  $s \in [\beta, \alpha]$ .

Now we can normalize  $f$  that  $n_0 = 0$  and  $c_0 = 1$ , so  $|f - 1|_\rho < 1$  for all  $\rho \in [\beta, \alpha]$ , so  $f - 1$  is topologically nilpotent in  $B_{[a,b]}$ , and thus  $f$  is invertible.

Generally,  $f$  is a limit of a sequence  $f_n \in \mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{[\pi]}]$ , and by (27.9.1.8) we can assume the hypothesis holds for all  $f_n$ . Then  $f_n$  is invertible, and it is easily shown that  $f_n^{-1}$  is a Cauchy sequence in  $B_{[a,b]}$ , so converges to some  $f^{-1}$ .

$3 \rightarrow 2$ : Firstly, if  $\partial_- v_\beta(f) > \partial_+ v_\alpha(f)$ , then we must have  $\partial_- v_\rho(f) > \partial_+ v_\rho(f)$  for some  $s$ , so wlog, we can assume  $a = b = s$ , and we need to show  $f$  vanishes at some point in  $Y_{\text{exp}(s)}$ . Now combining with (27.9.1.18) and (27.9.1.19), this is equivalent to another statement that any element  $y \in B_{[\rho, \rho]}$  has a decomposition  $y = g\xi_1 \dots \xi_n$  where  $\xi_k$  corresponds to points in  $Y_\rho$  and  $g$  is invertible in  $B_\rho$ . The proof is finished at (27.9.1.30).  $\square$

### Primitive Elements and the Proof of $3 \rightarrow 2$ of The Lemma on Valuation Function and Invertibility

**Def. (27.9.1.21).** Let  $C^b$  be alg.closed, an element in  $B_{[\rho, \rho]}$  is called **good** iff it has a decomposition as in the proof of  $3 \rightarrow 2$  of (27.9.1.20).  $\perp$

**Prop. (27.9.1.22) [Approximating Zero].** If  $f$  is a good element having  $n$ -zeros on  $Y_\rho$ , and  $g \in B_\rho$  that  $|f - g|_\rho < |f|_\rho$ , then for any zero  $y$  of  $g$  on  $Y_\rho$ , there exists a zero  $y'$  of  $f$  on  $Y_\rho$  that  $d(y', y) < \rho \left( \frac{|f - g|_\rho}{|g|_\rho} \right)^{1/n}$ .  $\perp$

*Proof:*  $|f - g|_\rho \geq |(f - g)(y)|_K = |f(y)|_K$ . Now  $f = g\xi_1 \dots \xi_n$ , and  $\xi$  corresponds to  $y_i$ , then

$$|f(y)|_K = |g(y)|_K |\xi_1(y)|_K \dots |\xi_n(y)|_K = \frac{|f|_\rho}{\prod_i |\xi_i|_\rho} \prod_i d(y_i, y) = |f| \prod_i \frac{d(y_i, y)}{\rho}.$$

(Notice  $|g|_\rho = |g(y)|_K$  because  $g$  is invertible (27.7.1.10)) and  $d(y_i, y) \leq \rho$ . So at least one  $\xi$  satisfies the desired inequality.  $\square$

**Cor. (27.9.1.23).** If  $f \in B_\rho$  is given by a Cauchy sequence of good elements, and  $\partial_- v_\rho(f) > \partial_+ v_\rho(f)$ , then  $f$  has a root on  $Y_\rho$ .  $\perp$

*Proof:* By (27.9.1.7), passing to a subsequence, we may assume

$$v_s(f) = v_s(f_n), \quad \partial_- v_s(f) = \partial_- v_s(f_n), \quad \partial_+ v_s(f) = \partial_+ v_s(f_n), \quad |f_{n+1} - f_n|_\rho < |f|_\rho.$$

Let  $n = \partial_- v_s(f) - \partial_+ v_s(f) > 0$ , then each  $f_i$  has exactly  $n$  roots on  $Y_\rho$ , and applying (27.9.1.22), we can find successively roots  $y_n$  of  $f_n$  that  $d(y_{n+1}, y_n) \leq \rho \left( \frac{|f_{n+1} - f_n|_\rho}{|f|_\rho} \right)^{1/n}$ , so the sequence  $\{y_n\}$  is Cauchy and converges to some point  $y \in \bar{Y}$ , so

$$|f_i(y)|_K \leq |f_i|_\rho \frac{d(y_i, y)}{\rho} = |f|_\rho \frac{d(y_i, y)}{\rho} \rightarrow 0.$$

so  $f(y) = 0$ .  $\square$

**Def. (27.9.1.24) [Primitive Elements].** An element  $f = \sum_{n \geq 0} [c_n] p^n \in \mathbf{A}_{\text{inf}}$  is called **primitive of degree  $d$**  if  $c_0 \neq 0, |c_d| = 1$  for some smallest element  $d$ .

Clearly an element is distinguished of degree 1 iff it is distinguished and corresponds to an untilts of  $X^b$  of char 0.  $\perp$

**Prop. (27.9.1.25).**

- Any element  $f \in \mathbf{A}_{\text{inf}}$  of finite Teichmüller expansion can be written uniquely as  $f = p^m [c] g$ , where  $c \in C^b$  and  $g$  is primitive.
- For an element  $f \in \mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{\ell}]$ ,  $f$  can be written as  $p^m [c] g$  iff  $v_s(f)$  consists of f.m. line segments iff  $\sup\{|c_n|\}$  is achieved by some  $n$ .
- If  $f = gh$  in  $\mathbf{A}_{\text{inf}}$  is primitive, then  $g, h$  are also primitive, and  $\deg(f) = \deg(g) + \deg(h)$ .  $\perp$

**Prop. (27.9.1.26).** Let  $f = \sum [c_n] p^n \in \mathbf{A}_{\text{inf}}$  be primitive of degree  $d > 0$ , and let  $\lambda \in (0, 1)$  be the number that  $s = -\log(\lambda)$  is the minimal number that  $v_s(f)$  is non-differentiable at, i.e.  $s$  is  $-1$  times the slope of the line segment on the left of  $v(c_d)$ . Then  $f$  has a zero on  $Y_\lambda$ .  $\perp$

*Proof:* By (27.9.1.27) there is a  $y \in Y_\lambda$  that  $|f(y_1)| \leq \lambda^{d+1}$ , and then (27.9.1.28) shows we can find successively  $y_n$  that

$$d(y_n, y_{n+1}) \leq \lambda^{1+\frac{d}{n}}, \quad |f(y_n)| \leq \lambda^{d+m}.$$

So  $y_n$  is a Cauchy sequence thus converges to some  $y$ , and then  $f(y) = 0$ .  $\square$

**Lemma (27.9.1.27) [Lemma for Approaching a Zero].** If  $C^b$  is alg.closed and  $f \in \mathbf{A}_{\text{inf}}$  is primitive of degree  $d > 0$ , and let  $\lambda$  as in (27.9.1.26), then there is a point  $y \in Y_\lambda$  that  $|f(y)|_{K_y} \leq \lambda^{d+1}$ .  $\lrcorner$

*Proof:* Let  $f = \sum [c_n]p^n$ , we may assume  $c_d = 1$ , and let  $F = x^d + c_{d-1}x^{d-1} + \dots + c_0$ , then the largest valuation of the roots of  $F$  on  $C^b$  is  $\lambda$ , by Newton polygon. Let  $r$  be such a root, then  $c_i$  is divisible by  $r^{d-i}$ , and let  $\xi = p - [r]$  be a distinguished element of  $\mathbf{A}_{\text{inf}}$  and corresponds to an untilt  $K$ , then  $|p|_K = \lambda$ , and

$$p^{-d}f(y) = \sum_{n \geq 0} c_n^\# p^{n-d} \equiv \sum_{i=0}^d \left(\frac{c_i}{r^{d-i}}\right)^\# \mod p = (r^{-d}F(r)) \mod p = 0$$

thus  $f(y)$  is divisible by  $p^{d+1}$ , which is equivalent to  $|f(y)|_K \leq \lambda^{d+1}$ .  $\square$

**Lemma (27.9.1.28) [Lemma for Approaching a Zero].** Situation as in (27.9.1.27), if  $y \in Y_\lambda$  and  $|f(y)| = \lambda^d \cdot \alpha$ , then there is a  $y'$  that  $d(y, y') \leq \lambda \cdot \alpha^{1/d}$  that  $|f(y')| \leq \lambda^{d+1}\alpha$ .  $\lrcorner$

*Proof:* Since  $\mathbf{A}_{\text{inf}}$  is  $\xi$ -complete and every element of  $\mathbf{A}_{\text{inf}}/\xi \cong \mathcal{O}_K$  belongs to the image of  $\sharp : \mathcal{O}_C \rightarrow \mathcal{O}_K$ , thus by induction, we can write  $f = \sum_{n \geq 0} [c_n]\xi^n$ . Because  $f$  is primitive of degree  $d$ , we may assume  $c_d = 1$ , and  $|c_0|_{C^b} = |f(y)|_K = \lambda^d \alpha$ .

Let  $F(x) = c_0 + c_1x + \dots + c_{d-1}x^{d-1} + x^d$ , because  $C^b$  is alg.closed, let  $r$  be a root of minimal absolute value, then  $|r|_{C^b}^m |c_m|_{C^b} \leq \lambda^n \alpha$ , in particular  $|r|_{C^b} \leq \lambda \alpha^{1/n}$ . So let  $\xi' = \xi - [r]$ , then  $\xi$  is also distinguished, and  $d(y', y) = |r|_{C^b} \leq \lambda \alpha^{1/n}$  (27.9.1.10), and  $d(0, y') = \lambda$ , and  $\xi(y') = r^\sharp$ .

Now

$$\frac{(f(y'))^\sharp}{c_0^\sharp} = \sum_{n \geq 0} \frac{c_n^\sharp}{c_0^\sharp} \xi(y')^n = \sum_{n \geq 0} \left(\frac{c_n r^n}{c_0}\right)^\sharp \equiv \sum_{i=0}^n \left(\frac{c_n r^n}{c_0}\right)^\sharp \mod p = \left(\frac{F(r)}{c_0}\right)^\sharp = 0,$$

So  $|f(y')|_{K'} \leq |c_0^\sharp|_{K'} |p|_{K'} = |c_0|_{C^b} \lambda = \lambda^{d+1} \alpha$ .  $\square$

**Cor. (27.9.1.29) [Primitive Elements Decompose as Distinguished Elements].** If  $f \in \mathbf{A}_{\text{inf}}$  is a primitive element of degree  $d > 0$ , then  $f$  admits a factorization as products of distinguished elements  $\xi$  corresponding to points in  $Y$ .  $\lrcorner$

*Proof:* Use induction on  $d$ . If  $d = 1$ , then  $f$  is distinguished by (27.9.1.24), and if  $d > 1$ , then by (27.9.1.26),  $f = \xi g$ , so  $g$  is primitive of degree  $d - 1$  by (27.9.1.25), so induction is finished.  $\square$

**Prop. (27.9.1.30) [Finite Teichmuller Expansion is Good].** Any element of finite Teichmuller expansion is good.

In particular, because any element of  $B_\rho$  can be approximated by elements in  $\mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{\ell}]$ , and such element can be approximated by elements of finite Teichmuller expansion, by (27.9.1.23), we finishes the proof of  $3 \rightarrow 2$  of (27.9.1.20).  $\lrcorner$

*Proof:* If  $f$  has finite Teichmuller expansion, then  $f = p^m [c]g$ , where  $g$  is primitive of degree  $d$ . If  $d = 0$ , then  $g$  is invertible in  $\mathbf{A}_{\text{inf}}$ , thus  $f$  is invertible in  $B_\rho$ . Otherwise, we can use (27.9.1.29) to factorize  $g$  into distinguished elements, and the elements that corresponds to points outside  $Y_\rho$  is invertible in  $B_\rho$  because  $v_s(\xi) = \max\{s, v(v_0)\}$  and  $2 \rightarrow 1$  of (27.9.1.20), so  $f$  is good.  $\square$

### Bounded Meromorphic Functions

**Prop. (27.9.1.31).** If  $f \in B$ , then  $f \in \mathbf{A}_{\text{inf}}$  iff  $|f|_{\rho} \leq 1$  for any  $0 < \rho < 1$ .

Easily we can get characterization of  $f$  being in  $\mathbf{A}_{\text{inf}}[\frac{1}{p}]$ ,  $\mathbf{A}_{\text{inf}}[\frac{1}{\ell}]$  or  $\mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{\ell}]$ .  $\lrcorner$

*Proof:* One direction is trivial, for the other, by (27.9.1.32), we can find successively  $f_n$  that  $f = \sum_{i < n} [c_i]p^i + f_n$ , and  $|f_n|_{\rho} \leq \rho^n$  for all  $0 < \rho < 1$ . So  $f_n$  converges to 0 in any norm  $\rho$ , thus it converges to 0 in  $B$ , and  $f = \sum_{n \geq 0} [c_n]p^n \in \mathbf{A}_{\text{inf}}$ .  $\square$

**Lemma (27.9.1.32).** If  $f \in B$  satisfies  $|f|_{\rho} \leq \rho^m$  for all  $0 < \rho < 1$ , then there is a  $c \in \mathcal{O}_{C^b}$  that  $f = [c]p^m + g$  that  $|g|_{\rho} \leq \rho^{m+1}$  for all  $0 < \rho < 1$ .  $\lrcorner$

*Proof:* Replace  $f$  by  $\frac{f}{p^m}$ , we may assume  $m = 0$ . Choose a sequence  $f_i$  in  $\mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{\ell}]$  converging to  $f$  in  $B$ , where  $f_i = \sum_{n > -\infty} [c_{n,i}]p^n$ .

Firstly we want to truncate  $f_i$  with the positive part  $f_i^+$ . Notice for each  $\rho$  and any  $0 < \varepsilon < 1$ , because  $\lim |f_i - f|_{\varepsilon\rho} = 0$ , thus for  $i$  large,  $|f_i|_{\varepsilon\rho} = |f|_{\varepsilon\rho} \leq 1$ , thus  $|c_{-n,i}|_{C^b} \rho^{-n} \leq \varepsilon^n < \varepsilon \leq \varepsilon$ , so  $|f_i - f_i^+|_{\rho} < \varepsilon$  for  $i$  large, so  $\lim f_i^+ = f$  also in  $B$ .

Secondly,  $|c_{0,i} - c_{0,j}|_{C^b} \leq |f_i - f_j|_{\rho}$  for each  $\rho$ , thus  $c_{0,i}$  is Cauchy in  $C^b$  thus converges to some  $c \in C^b$ , and when  $i$  is large,  $|c_{0,i}|_{C^b} \leq |f_i|_{\rho} = |f|_{\rho} \leq 1$ , so  $c \in \mathcal{O}_{C^b}$ . Now let  $g_i = \sum_{n > 0} [c_{n,i}]p^n$ , then  $g_i$  is also Cauchy in  $B$  for any norm  $|\rho|$  and converges to some  $g$ , and  $f = g + [c]$ .

It's left to check  $|g|_{\rho} \leq \rho$ : each  $v_s(g_i)$  has positive slopes, then so does  $v_s(g)$  because by (27.9.1.7),  $v_s(g_i)$  stabilizes to  $v_s(g)$  uniformly on compact intervals. So if  $v_s(g) < s - \varepsilon$  for some  $s$ , then  $v_{\varepsilon}(g) \leq v_s(g) - (s - \varepsilon) < 0$ , but this cannot happen because  $v_{\varepsilon}(g) \leq \max\{v_{\varepsilon}(f), -\log |c|_{C^b}\} \geq 0$ .  $\square$

### Eigenspaces of Frobenius

**Prop. (27.9.1.33).**

- The vector space  $B^{\varphi=p^n}$  vanish for  $n < 0$ .
- The canonical map  $\mathbb{Q}_p \rightarrow B^{\varphi=\text{id}}$  is an isomorphism.

$\lrcorner$

*Proof:* 1: Consider  $v_{ps}(\varphi(f)) = pv_s(f)$  (27.7.1.14),  $v_s(p^n f) = ns + v_s(f)$ , so if  $\varphi(f) = p^n f$ , then

$$pv_{s/p}(f) = v_s(\varphi(f)) = v_s(p^n f) = ns + v_s(f).$$

Let  $h(s) = \partial_+ v_s(f)$ , then  $h(s/p) = n + h(s)$ , but  $h$  must be non-increasing (27.9.1.6), so  $n \geq 0$ .

2: Firstly we prove  $B^{\varphi=\text{id}}$  is a field: by (27.9.1.15), it suffices to show that  $\text{div}(f) = 0$  for  $f \neq 0 \in B^{\varphi=\text{id}}$ . If  $\text{div}(f) \neq 0$ , because  $f$  is fixed by  $\varphi$ , so  $\text{div}(f) \geq \sum_{n \in \mathbb{Z}} \varphi^n(y)$  for some  $y$ , and  $\sum_{n \in \mathbb{Z}} \varphi^n(y) = \text{div}(\log([\varepsilon]))$  for some  $\varepsilon \in 1 + \mathfrak{m}_{C^b}$  because  $K$  is alg.closed and by (27.7.1.16). So by (27.9.1.15) again  $f = g \log([\varepsilon])$ , and  $g \in B^{\varphi=p^{-1}}$  by (27.9.1.2), then  $g = 0$  by item 1.

3: From (27.9.1.34) and (27.9.1.31),  $f \in \mathbf{A}_{\text{inf}}[\frac{1}{p}]$ , thus  $f = \sum_{n > -\infty} [c_n]p^n$ , so  $\varphi(f) = f$  shows  $c_n^p = c_n$ , which is equivalent to  $c_n \in \mathbb{F}_p$ . So  $f \in W(\mathbb{F}_p)[\frac{1}{p}] = \mathbb{Q}_p$ .  $\square$

**Lemma (27.9.1.34).** If  $f \neq 0 \in B^{\varphi=\text{id}}$ , then there is an integer  $n$  that  $|f|_{\rho} = \rho^n$ .  $\lrcorner$

*Proof:* Notice  $|f|_{\rho}^p = |\varphi(f)|_{\rho^p} = |f|_{\rho^p}$ , so  $v_{ps}(f) = pv_s(f)$ , differentiation shows that  $\partial_- v_{ps}(f) = \partial_- v_s(f)$ . This is for all  $s < 0$ , and  $\partial_- v_s(f)$  is non-decreasing, thus it is constant, and  $v_{ps}(f) = pv_s(f)$  shows  $v_s(f) = ns$  for some integer  $n$ .  $\square$

**Cor. (27.9.1.35).** For  $n \geq 0$ , any element  $f \in B^{\varphi=p^n}$  factors uniquely up to action of  $\mathbb{Q}_p^*$  as  $\lambda \log([\varepsilon_1]) \dots \log([\varepsilon_n])$  where  $\lambda \in B^{\varphi=\text{id}}$ ,  $0 < |\varepsilon_i - 1| < 1$ .  $\lrcorner$

*Proof:* The existence is by (27.9.1.30)

For the uniqueness: it suffices to prove  $\log([\varepsilon])$  is a prime element in  $\oplus_{n \geq 0} B^{\varphi=p^n}$ . For this, notice for any  $f \in B^{\varphi=p^n}$ ,  $\text{div}(f)$  is fixed by  $\varphi$ , and  $\text{div}(\log([\varepsilon]))$  is a single orbit of  $\varphi$ , thus by (27.9.1.15), if  $\log([\varepsilon])$  divides  $fg$ , then  $\log([\varepsilon])$  divides  $f$  or  $g$ .  $\square$

### Applications

**Cor. (27.9.1.36).** If  $C^b$  is alg.closed, then every untilts  $K$  of  $C^b$  belongs to the vanishing locus of  $\log([x])$  for some  $x \in C^b$  that  $0 < |x - 1| < 1$ , and the map

$$\psi : 1 + \mathfrak{m}_{C^b} \rightarrow K : y \mapsto \log(y^\sharp)$$

is surjective with kernel generated by  $x$  (as a  $\mathbb{Q}_p$ -subspace of  $1 + \mathfrak{m}_{C^b}$ ).  $\lrcorner$

*Proof:* By (12.2.8.16), any untilts of  $C^b$  is alg.closed, thus it has a compatible  $p^n$ -th roots of unity. So it belongs to some locus of  $\log([x])$  by (27.7.1.16). Now if  $|z| < |p|_K^{1/(p-1)}$ , then  $z = \log(\exp(z))$ , and  $\exp = y^\sharp$  for some  $y$  because  $K$  is alg.closed. So  $\psi$  contains sufficiently small elements, but it is a map of  $\mathbb{Q}_p$ -vector spaces, thus it is surjective. For the kernel, if  $\log(y^\sharp) = 0$ , then  $\log([y])$  vanish on  $K$ , thus by (27.7.1.16),  $y, x$  is in the same  $\mathbb{Q}_p$ -vector space.  $\square$

**Cor. (27.9.1.37).** If  $C^b$  is alg.closed, then the map

$$1 + \mathfrak{m}_{C^b} \xrightarrow{\log([x])} B^{\varphi=p}$$

is an isomorphism.  $\lrcorner$

*Proof:* Firstly any untilts of  $C^b$  is alg.closed by (12.2.8.16). It is injective because of the correspondence (27.7.1.16), and for the surjectivity, for each  $f \in B^{\varphi=\text{id}}$ , if  $f = 0$ , then  $f = \log([1])$ , and if  $f \neq 0$ , then notice  $\text{div}(f) \neq \emptyset$ , because in this case  $f$  is invertible in  $B$  by (27.9.1.15), thus  $f^{-1} \in B^{\varphi=p^{-1}}$ , so  $f^{-1} = 0$  by (27.9.1.33), contradiction.

Now if  $\text{ord}_K(f) \geq 1$ , then  $\text{ord}_{\varphi^n(K)}(f) \geq 1$  for any  $n \in \mathbb{Z}$  since  $\varphi(f) = pf$ . Consider  $\text{div}(\log([x])) = \sum_{n \in \mathbb{Z}} \varphi^n(K)$  (27.9.1.17), then  $f$  is divisible by  $\log([x])$  by (27.9.1.15),  $f = \log([x])g$ , then  $g \in B^{\varphi=\text{id}}$ , then  $g \in \mathbb{Q}_p^*$  by (27.9.1.33), thus  $f = \log([x^g])$ .  $\square$

**Cor. (27.9.1.38) [Filtration on  $B_{\text{dR}}$ ].** By (27.9.1.17) and (27.9.1.36), we see that for any untilt  $K$  of  $C^b$ , there is a unique up to  $\mathbb{Q}_p$ -constant  $\varepsilon$  that  $t = \log([\varepsilon])$  is the uniformizer of  $B_{\text{dR}}(y)$ . In fact, this  $\varepsilon$  can be to be  $\varepsilon = (1, \xi_p, \dots, x_{p^n}, \dots)$ , where  $\xi_{p^n}$  is a compatible roots of unity in the alg.closed field  $K$ .

Now we prefer to use the filtration  $\text{Fil}^n = t^{-n} B_{\text{dR}}^+$  on  $B_{\text{dR}}$  because it is  $G_{\mathbb{Q}_p}$  invariant, as  $\varepsilon$  does.  $\lrcorner$

**Prop. (27.9.1.39).** Let  $C^b$  be alg.closed, then any point  $x$  of the Fargues-Fontaine curve  $X_{FF}$  that is not the generic point corresponds to the prime  $x_K = (\log([\varepsilon]))$  (27.9.1.35) where  $K \in \text{div}(\log([\varepsilon]))$  (27.9.1.36). And the residue field of  $x_K$  can be identified to  $K$ .  $\lrcorner$

*Proof:* By (27.9.1.35), we can cover  $X_{FF}$  by affine schemes of the form  $\text{Spec}(R_f = B[f^{-1}]^{\varphi=\text{id}})$  for  $f \in B^{\varphi=p}$ , now for any prime  $\mathfrak{p} \subset R_f$ , let  $\frac{g}{f^n} \in \mathfrak{p}$ , then  $g = \lambda \log([\varepsilon_1]) \dots \log([\varepsilon_n])$ , thus some  $\frac{\log([\varepsilon])}{f} \in \mathfrak{p}$ . Let  $K$  be a point that  $\log([\varepsilon])$  vanish (27.7.1.16), then we claim  $(\log([\varepsilon])/f)$  is maximal.

In fact, we may assume  $f$  doesn't vanish on  $K$ , otherwise  $\log([\varepsilon])/f$  is a unit, then there is a map  $\rho : B[f^{-1}]^{\varphi=1} \subset B[f^{-1}] \rightarrow K$ , and this map is surjective with kernel  $(\log([\varepsilon])/f)$ : it is surjective even on  $f^{-1}B^{\varphi=p}$  by (27.9.1.36), and if  $\log([\varepsilon_1])/f$  is mapped to 0, then  $\log([\varepsilon_1])$  differs from  $\log([\varepsilon])$  by some  $Q_p^*$  by (27.7.1.16).  $\square$

**Cor. (27.9.1.40).** If  $C^b$  is alg.closed, there is a bijection of sets:

$$Y/\varphi_{C^b}^{\mathbb{Z}} \cong \{\text{Closed points of } X_{FF}\}.$$

by (27.7.1.16).  $\lrcorner$

**Cor. (27.9.1.41).**  $X_{FF}$  is a Dedekind scheme (7.4.2.15).  $\lrcorner$

*Proof:* Let  $\text{Spec}(R_f = B[f^{-1}]^{\varphi=\text{id}})$ , two elements  $f = \log([\varepsilon]), g = \log([\mu])$  can cover it. The proof of (27.9.1.39) shows that every prime ideals of  $R_f$  is maximal principal, in particular f.g, thus by (5.1.1.50), it is Noetherian. And it has Krull dimension 1 and it is regular because all of its maximal ideals are principal, hence normal (5.3.5.33). So  $X_{FF}$  is a Dedekind scheme.  $\square$

## 2 Line Bundles and Filtrations

**Def. (27.9.2.1).** By (27.9.1.35), the graded algebra  $\oplus_{n \geq 0} B^{\varphi=p^n}$  is generated over  $\mathbb{Q}_p$  by  $B^{\varphi=p}$ , so we can define the Serre twisting sheaf  $\mathcal{O}(1)$  on  $X_{FF}$ , which is a line bundle, and on an open affine scheme  $U = X - \{x\}$ , where  $x$  corresponds to  $\log([\varepsilon])$ ,  $\mathcal{O}(1)(U) = (B[\frac{1}{\log([\varepsilon])}])^{\varphi=p}$ . Similarly we can define  $\mathcal{O}(m)$ , and  $\mathcal{O}(m) = \mathcal{O}(1)^m$ .  $\lrcorner$

**Lemma (27.9.2.2).** There is an isomorphism  $\text{Div}(X) \rightarrow \text{Pic}(X)$  that maps each  $x$  to the inverse of its ideal sheaf (7.5.3.15) (5.3.5.21). And there is also a degree map  $\text{Div}(X) \rightarrow \mathbb{Z}$ . Then:

$$\begin{array}{ccc} \text{Div}(X) & \xrightarrow{\deg} & \mathbb{Z} \\ & \searrow & \downarrow \rho \\ & & \text{Pic}(X) \end{array}$$

commutes.  $\lrcorner$

*Proof:* It suffices to show that any  $\mathcal{O}(x)$  is isomorphic to  $\mathcal{O}(1)$ . As  $\log([\varepsilon])$  is a global section of  $\mathcal{O}(1)$  that vanishes of order 1 at  $x$ , it induces an isomorphism  $\mathcal{O}(1) \cong \mathcal{O}(x)$  by (7.5.3.15).  $\square$

**Lemma (27.9.2.3) [Cohomology of Line Bundles].** For any integer  $m$ ,  $B^{\varphi=p^m} \rightarrow H^0(X, \mathcal{O}(m))$  is an isomorphism and  $H^i(X, \mathcal{O}(m)) = 0$  for  $i > 0, m > 0$ .  $\lrcorner$

*Proof:* This is trivial using Čech cohomology, as  $\oplus_{n \geq 0} B^{\varphi=p^n}$  is PID, so  $X$  is separated.  $\square$

**Prop. (27.9.2.4).** The construction induces an isomorphism  $\rho : \mathbb{Z} \cong \text{Pic}(X) : m \mapsto \mathcal{O}(m)$ .  $\lrcorner$

*Proof:* By lemma (27.9.2.2),  $\rho$  is surjective because  $\text{Div}(X) \rightarrow \text{Pic}(X)$  does, and it is injection because if  $\mathcal{O}(m) \cong \mathcal{O}(n)$ , then tensoring  $\mathcal{O}(-m)$ , we can assume  $\mathcal{O} \cong \mathcal{O}(-k)$ , but they have different global sections by lemma (27.9.2.3) and (27.9.1.33) (27.9.1.35).  $\square$

### Harder-Narasimhan Filtration of Vector Bundles

**Prop. (27.9.2.5) [Harder Narasimhan Formalism for  $\mathcal{V}\text{ect}_X$ ].** For a vector bundle  $L$  on  $X$ , we can define  $\deg(L) = n$  iff  $L \cong \mathcal{O}(n)$  (27.9.2.2), and for a vector bundle  $E$ , define  $\deg(E) = \deg(\wedge(E))$ . And define the generic rank on the category of coherent sheaves on  $X$ . Then this is a Harder-Narasimhan formalism on  $\mathcal{C} = \mathcal{V}\text{ect}_X$  with  $\mathcal{A} = \mathcal{V}\text{ect}_{R(X)}$ .  $\lrcorner$

*Proof:* Only the last axiom needs proof, but if  $\mathcal{E}' \subsetneq \mathcal{E}$ , notice  $\wedge \mathcal{E}' \subsetneq \wedge \mathcal{E}$  (The stalks are PID), so by taking their top exterior power product, we reduce to the case of line bundles.

But  $\mathcal{O}(m)$  cannot map into  $\mathcal{O}(n)$  if  $m > n$  and must be isomorphism if  $m = n$ , by tensoring  $\mathcal{O}(-m)$  and looking at global sections (27.9.2.3), so the assertion is true.  $\square$

**Cor. (27.9.2.6).** Every vector bundle  $\mathcal{E}$  on  $X$  has a unique functorial Harder-Narasimhan filtration, by (4.3.5.22).  $\lrcorner$

### 3 Base Change of Fields

**Prop. (27.9.3.1) [Base Change].** Let  $C^\flat$  be alg.closed. For any finite extension  $E$  of  $\mathbb{Q}_p$  of degree  $n$ ,  $\text{Spec } E \rightarrow \text{Spec } \mathbb{Q}_p$  is finite étale, and finite locally free of degree  $n$ , so does  $X_E = X \otimes_{\mathbb{Q}_p} E \rightarrow X$  (7.6.2.23).

In particular,  $X_E$  is also a Dedekind scheme<sup>?</sup>. For any closed point  $x$  of  $X$  corresponding to an untilt  $K$  of  $C^\flat$ , which is alg.closed, the fiber of  $X_E$  over  $x$  is identical to the spectrum of  $E \otimes_{\mathbb{Q}_p} K \cong K^n$  as  $K$  is alg.closed.

In this situation and use (27.9.1.40), we see that the closed points of  $X_E$  are in bijection with isomorphism classes of  $(K, \iota, u)$  module  $\varphi$ -actions, where  $(K, \iota)$  is an untilt of  $C^\flat$ , and  $u : E \rightarrow K$  is an embedding of  $E$  into  $K$  over  $\mathbb{Q}_p$ , isomorphism classes of these triples are denoted by  $Y_E$ .  $\lrcorner$

**Prop. (27.9.3.2).** By (27.9.3.1) and flat base change (7.7.5.1), we know  $H^0(X_E, \mathcal{O}_{X_E}) = E$ , in particular  $X_E$  is connected.  $\lrcorner$

**Lemma (27.9.3.3).** If  $E$  is unramified of degree  $n$  over  $\mathbb{Q}_p$ , then  $E \cong W(\mathbb{F}_{p^n})[\frac{1}{p}]$ . In particular,

$$\text{Hom}_{\mathbb{Q}_p}(E, K) \cong \text{Hom}_{\mathbb{Z}_p}(W(\mathbb{F}_{p^n}), \mathcal{O}_K) \cong \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_{p^n}, \mathcal{O}_K/p \cong \mathcal{O}_{C^\flat}/[t]) \cong \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_{p^n}, C)$$

where the last isomorphism is by Henselian lemma.

Therefore,  $Y_E \cong Y \otimes \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_{p^n}, C)$ , and

$$\text{Closed points of } Y_E \cong Y_E/\varphi^{\mathbb{Z}} \cong (Y \otimes \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_{p^n}, C))/\varphi^{\mathbb{Z}} \cong Y/\varphi^{n\mathbb{Z}}.$$

$\lrcorner$

**Prop. (27.9.3.4).** If  $E$  is unramified of degree  $n$  over  $\mathbb{Q}_p$  and  $U \neq X$  is an affine open defined by a homogenous element  $t$ , then

$$U_E = \text{Spec} \left( (B[t^{-1}] \otimes_{\mathbb{Q}_p} E)^{\varphi=\text{id}} \right) = \text{Spec}(B[t^{-1}]^{\varphi^n=1}),$$

where  $\varphi$  acts trivially on  $E$ .  $\lrcorner$



*Proof:* Each  $u \in \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_{p^n}, C^b)$  induces a map  $W(\mathbb{F}_p) \rightarrow W(\mathcal{O}_{C^b}) = \mathbf{A}_{\text{inf}} \rightarrow B$ , which extends to a map  $\bar{u} : E \rightarrow B[t^{-1}]$ . and induces a map  $q_u : B[t^{-1}] \otimes_{\mathbb{Q}_p} E \rightarrow B[t^{-1}]$ . Now

$$B[t^{-1}] \otimes_{\mathbb{Q}_p} E \rightarrow \prod_{\text{Hom}_{\mathbb{F}_p}(\mathbb{F}_{p^n}, C)} B[t^{-1}]$$

is an isomorphism, which is just because  $x^{p^n} - x$  splits in  $B[t^{-1}]$ .

And under this isomorphism, the action of  $\varphi$  is

$$\varphi((f_0, \dots, f_{n-1}) = (\varphi(f_{n-1}), \varphi(f_0), \dots, \varphi(f_{n-2})),$$

so proposition is clear.  $\square$

**Cor. (27.9.3.5).** Fix now a finite extension  $E/\mathbb{Q}_p$  with uniformizer  $\pi$  that has ramification degree  $e$  and inertia degree  $d$ , and  $E_0$  is the maximal unramified subextension, then there are maps  $E_0 \rightarrow B$  by (27.9.3.4), fix forever one of them  $p_u$ , this induces a map

$$B[\frac{1}{t}] \otimes_{\mathbb{Q}_p} E \rightarrow B[\frac{1}{t}] \otimes_{E_0} E$$

and this induces an isomorphism

$$(B[\frac{1}{t}] \otimes_{\mathbb{Q}_p} E)^{\varphi=\text{id}} = (B[\frac{1}{t}] \otimes_{\mathbb{Q}_p} E_0)^{\varphi=\text{id}} \otimes_{E_0} E = (B[\frac{1}{t}] \otimes_{E_0} E)^{\varphi^d=\text{id}}$$

$\lrcorner$

**Def. (27.9.3.6)**  $[Y_E^0]$ . Define  $Y_E^0 \subset Y_E = \text{triples } (K, \iota, u)$ , where  $(K, \iota)$  is an untillt of  $C^b$ , and  $u : E \rightarrow K$  is an embedding that  $u|_{E_0}$  is identical to  $e_K \circ p_u : E_0 \rightarrow B \rightarrow K$ . Notice  $Y_E^0$  is not stable under the Frobenius, but it is stable under  $\varphi^d$ , and induces an isomorphism

$$Y_E^0 / \varphi^{d\mathbb{Z}} \cong Y_E / \varphi^{\mathbb{Z}}.$$

$\lrcorner$

**Prop. (27.9.3.7).** Notice for an element  $y$  of  $Y_E^0$ , the map  $u : E \rightarrow K_y = B_{\text{dR}}^+(y)/\xi$  extends uniquely to a map  $E \rightarrow B_{\text{dR}}^+(y)$  that is compatible with  $e_K \circ p_u : E_0 \rightarrow B_{\text{dR}}^+(y)$ , because  $E$  is separable over  $E_0$ . i.e.

$$\begin{array}{ccc} E_0 & \xrightarrow{e_K \circ p_u} & B_{\text{dR}}^+(y) \\ \downarrow & \nearrow \tilde{u} & \downarrow \\ E & \xrightarrow{u} & K \end{array}$$

Then this defines a map  $B \otimes_{E_0} E \rightarrow B_{\text{dR}}^+(y)$ , also called the stalk map.  $\lrcorner$

**Prop. (27.9.3.8).** For any finite extension  $E/\mathbb{Q}_p$ , the degree map  $\deg : \text{Pic}(X_E) \cong \mathbb{Z}$  is an isomorphism.  $\lrcorner$

*Proof:* It suffices to show that  $\mathcal{O}_{X_E}(x) \cong \mathcal{O}_{X_E}(x')$  for each pair of closed points  $x, x'$  of  $X_E$ .

We attempt to construct a line bundle  $\mathcal{O}_{X_E}(1)$  on  $X_E$  that  $\mathcal{O}_{X_E}(1)(U_E) = (B[\frac{1}{t}] \otimes_{E_0} E)^{\varphi^d=\pi}$ , because  $\mathcal{O}_{X_E}(U_E) = (B[\frac{1}{t}] \otimes_{E_0} E)^{\varphi^d=1}$ .

We show simultaneously that  $\mathcal{O}_{X_E}(1)$  is a line bundle and it is isomorphic to  $\mathcal{O}_{X_E}(x)$  for any closed point  $x \in X_E$ : For any  $x \in X_E$  corresponding to a  $\varphi^d$ -orbit of  $Y_E^0$ , let  $f$  be the element



constructed by lemma(27.9.3.9) below, we show that for any affine open  $U = D(t)$ , multiplying by  $f : \mathcal{O}_{X_E}(x)(U_E) \rightarrow \mathcal{O}_{X_E}(1)(U_E)(\star)$  is an isomorphism:

Notice  $B \otimes_{E_0} E$  is free over  $B$ , let  $N(f) \in B$  be its norm, the norm is local, so for each  $y \in Y$ ,  $N(f)_y = \prod_{\bar{y}} f_{\bar{y}}$ , where  $\bar{y} \in Y_E$  are over  $y$ , so it vanishes with order 1 in a  $\varphi^d$ -orbit of  $Y$  (order 1 because  $f$  only vanishes at  $\bar{y}$  in the orbit corresponding to  $x$ ), and then  $N(f)\varphi(N(f)) \dots \varphi^{d-1}(N(f))$  vanishes at a single  $\varphi$ -orbit of  $Y$  with order 1, thus equals  $u \log([\varepsilon])$  for some  $\varepsilon \in \mathfrak{m}_{C^b}$ , by (27.9.1.17)(27.9.1.13). In particular,  $y$  divides  $\log([\varepsilon])$ .

Now if  $x \notin U_E$ , then  $\log([\varepsilon])$  divides  $t$ , so  $f$  divides  $t$ , thus  $f$  is invertible in  $B[\frac{1}{t}] \otimes_{E_0} E$ , thus  $(\star)$  is an isomorphism.

Otherwise if  $x \in U_E$ , then choose some  $x'$  not in  $U_E$ , then the same argument shows that  $f'$  is invertible in  $B[\frac{1}{t}] \otimes_{E_0} E$ , so  $f/f' \in (B[\frac{1}{t}] \otimes_{E_0} E)^{\varphi=\text{id}}$  that vanishes with a single zero at  $x$ , so multiplying by  $f'/f$  defines an isomorphism  $\mathcal{O}_{X_E}(U_E) \cong \mathcal{O}_{X_E}(x)(U_E)$ , so it suffices to show the composition

$$\mathcal{O}_{X_E}(U_E) \xrightarrow{f'/f} \mathcal{O}_{X_E}(x)(U_E) \xrightarrow{f} \mathcal{O}_{X_E}(1)(U_E)$$

is an isomorphism, but this reduces to the first case.  $\square$

**Lemma (27.9.3.9) [Uniformizer Existence].** If  $x$  be a closed point of  $X_E$  corresponding to an orbit of  $\varphi$  in  $Y_E$  thus an orbit  $S$  of  $\varphi^d$  in  $Y_E^0$ , then there is an element  $f \in (B \otimes_{E_0} E)^{\varphi^d=\pi}$  that  $\text{ord}_{\bar{y}}(f) = 1$  if  $\bar{y} \in S$ , and 0 otherwise.  $\lrcorner$

*Proof:* The map defined (27.9.3.18) composed with the Teichmuller section(27.9.3.16) in fact has image in  $(B \otimes_{E_0} E)^{\varphi^n=\pi}$  because  $[\pi] = \pi t + t^{p^n} = \varphi^n$  on  $\mathcal{O}_{C^b}$ , and it is an isomorphism of  $\mathcal{O}_E$  modules. Now there are commutative diagrams:

$$\begin{array}{ccc} G_{LT}(\mathcal{O}_{C^b}) & \xrightarrow{\sigma} & G_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E) \xrightarrow{\log_G} B \otimes_{E_0} E \\ & & \downarrow \qquad \qquad \downarrow \\ & & G_{LT}(\mathcal{O}_K) \xrightarrow{\log_G} K \end{array}$$

The map  $G_{LT}(\mathcal{O}_{C^b}) \rightarrow G_{LT}(\mathcal{O}_K)$  has kernel  $\mathcal{O}_E u$  for some  $u$ , thus we can let  $f = \log_G(\sigma(u))$ , then the image of  $f \in K$  is 0, which means  $f$  has a zero at the point  $y \in Y_E^0$ . And (27.9.3.13) shows the the zeros of  $f$  is just the  $\varphi^d$ -orbit containing  $y$ .  $\square$

### Lubin-Tate Formal Groups and the Proof of the Lemma of Uniformizer

**Prop. (27.9.3.10).** The ring  $B \otimes_{E_0} E$  is an integral domain.  $\lrcorner$

*Proof:* Cf.[Lurie P95]. In fact this is the ramified Witt vector, which is by the same reason as before an integral domain, Cf.[FF Curve Johannes].  $\square$

**Cor. (27.9.3.11).** If  $f \neq 0 \in B \otimes_{E_0} E$ , then  $N_{E/E_0}(f) \neq 0 \in B$ , in particular, the vanishing locus of  $f$  is finite.  $\lrcorner$

**Cor. (27.9.3.12).** If  $f, g \in B \otimes_{E_0} E$ , then  $f$  is divisible by  $g$  iff for each  $\bar{y} \in Y_E^0$ ,  $\text{ord}_{\bar{y}}(f) \geq \text{ord}_{\bar{y}}(g)$ .  $\lrcorner$

*Proof:* If  $\text{ord}_{\bar{y}}(f) \geq \text{ord}_{\bar{y}}(g)$ , suppose  $N_{E/E_0}(g) = gh$ , then multiplying by  $h$ , we can assume  $g \in B$ . Now  $f$  is written uniquely as  $f_0 + f_1\pi + \dots + f_{e-1}\pi^{e-1}$  where  $f_k \in B$ , thus it suffices to show  $f_i$

is divisible by  $g$ , which is equivalent to  $\text{ord}_y(f) \geq \text{ord}_y(g)$  for each  $y \in Y$ , by (27.9.1.15). Now if  $\text{ord}_y(g) = n$ , the hypothesis shows  $f$  vanishes in

$$\prod_{\bar{y} \rightarrow y} B_{\text{dR}}^+(y)/\xi^n = (B_{\text{dR}}^+(y)/\xi^n) \otimes_{E_0} E = B_{\text{dR}}^+(y)/\xi^n + \pi B_{\text{dR}}^+(y)/\xi^n + \dots + \pi^{e-1} B_{\text{dR}}^+(y)/\xi^n$$

thus  $\text{ord}_y(f) \geq n = \text{ord}_y(g)$ .  $\square$

**Cor. (27.9.3.13).** If  $f \in (B \otimes_{E_0} E)^{\varphi^n = \pi}$ , then the vanishing locus of  $f$  is a single  $\varphi^d$ -orbit, and all zeros are simple.  $\lrcorner$

*Proof:* Set  $N_{E/E_0}(f) = f'$  and  $N_{E/E_0}(\pi) = \pi'$ , then  $f$  belongs to  $B^{\varphi^d = \pi'}$ , and its divisor is just the image of divisor of  $f$  in  $Y_E^0$ . So it suffices to show that  $f'$  vanishes on a single  $\varphi^{d\mathbb{Z}}$ -orbit.

Now for  $0 < \rho < 1$ ,

$$\rho^{p^d} |f'|_{\rho^{p^d}} = |\pi' f'|_{\rho^{p^d}} = |f'^{\varphi^d}|_{\rho^{p^d}} = |f'|_{\rho}^{p^d},$$

thus

$$p^d s + v_{p^d s}(f') = p^d v_s(f')$$

for each  $s > 0$ , differentiating, we get

$$1 + \partial_- v_s(f') = \partial_- v_s(f')$$

Now the divisor of  $f'$  is  $\varphi^{d\mathbb{Z}}$ -invariant, and it has exactly one zero on any annulus  $(\rho^n, \rho]$  (27.9.1.19), thus its divisor is a single  $\varphi^d$ -orbit.  $\square$

**Def. (27.9.3.14) [Universal Lubin-Tate Formal Group].** Recall that if  $E$  is a finite extension of  $\mathbb{Q}_p$  with uniformizer  $\pi$ , for a  $\mathcal{O}_E$ -algebra  $A$  complete w.r.t  $\pi$ ,  $G_{LT}(A)$  is the Lubin-Tate formal group, with elements the topological nilpotent elements of  $A^*$ .

Now we define the **universal cover of Lubin-Tate formal group**  $\widetilde{GT}$  as the functor

$$A \mapsto \lim \left( \dots \xrightarrow{[\pi]} G_{LT}(A) \xrightarrow{[\pi]} G_{LT}(A) \right).$$

$\lrcorner$

**Prop. (27.9.3.15).**

- Notice for  $K$  an alg.closed extension of  $E$ ,  $G_{LT}(\mathcal{O}_K)$  is in bijection with  $\mathfrak{m}_K$ , and the kernel of  $[\pi^n]$  on  $G_{LT}(\mathcal{O}_K)$  has order  $\mathcal{O}_E/\pi^n$ , thus the kernel of  $\widetilde{G}_{LT}(\mathcal{O}_K) \rightarrow G_{LT}(\mathcal{O}_K)$  is a 1-dimensional  $\mathcal{O}_E$ -module.
- If  $\pi$  vanishes on  $A$  and  $A$  is perfect, then  $[\pi] = \pi t + t^q = t^q$  on  $A$ , so it is just the Frobenius, and  $\widetilde{G}_{LT}(A) \rightarrow G_{LT}(A)$  is a bijection.
- $\widetilde{G}_{LT}(A) \rightarrow \widetilde{G}_{LT}(A/I)$  is an isomorphism for  $\pi \in I$  and  $A$  is  $I$ -adic.
- $\widetilde{G}_{LT}(A) \rightarrow \widetilde{G}_{LT}(A/I)$  is an isomorphism for any ideal  $I$  that  $I + (\pi) \neq (1)$ , because both of them is isomorphic to  $\widetilde{G}_{LT}(A/(I + (\pi)))$ .

$\lrcorner$

*Proof:* For 3, it suffices to prove that  $\widetilde{G}_{LT}(A/I^{n+1}) \rightarrow \widetilde{G}_{LT}(A/I^n)$  for  $n \geq 1$ . Notice  $F(u, v) \equiv u + v \pmod{I^{2n}}$ , so we have an exact sequence

$$0 \rightarrow I^n/I^{n+1} \rightarrow G_{LT}(A/I^{n+1}) \rightarrow G_{LT}(A/I^n) \rightarrow 0.$$

In particular the kernel is annihilated by  $\pi$ , so there is a commutative diagram

$$\begin{array}{ccccc} \cdots & \xrightarrow{\pi} & G_{LT}(A/I^{n+1}) & \xrightarrow{\pi} & G_{LT}(A/I^{n+1}) \\ & \nearrow & \downarrow & \nearrow & \downarrow \\ \cdots & \xrightarrow{\pi} & G_{LT}(A/I^{n+1}) & \xrightarrow{\pi} & G_{LT}(A/I^{n+1}) \end{array}$$

which show that  $\tilde{G}_{LT}(A/I^{n+1}) \cong \tilde{G}_{LT}(A/I^n)$ .  $\square$

**Cor. (27.9.3.16) [Teichmüller Section].** Consider the  $\mathcal{O}_E$ -algebra  $\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E$ . Because there are isomorphism  $\mathcal{O}_{E_0}/p \cong \mathcal{O}_E/\pi$ , we have an isomorphism

$$C^\flat \cong \mathbf{A}_{\text{inf}}/p \cong (\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E)/(\pi).$$

Now (27.9.3.15) shows the diagram

$$\begin{array}{ccc} \tilde{G}_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E) & \xrightarrow{\cong} & \tilde{G}_{LT}(\mathcal{O}_{C^\flat}) \\ \downarrow & & \downarrow \cong \\ G_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E) & \longrightarrow & G_{LT}(\mathcal{O}_{C^\flat}) \end{array}$$

So the lower horizontal map is surjective, and it even has a canonical section  $\sigma$ , called the **Teichmüller section**.  $\lrcorner$

**Cor. (27.9.3.17).** Given a point of  $Y_E^0$  which corresponds to an untilt of  $C^\flat$  together with a  $E_0$ -map  $E \rightarrow K$ , then this gives a commutative diagram

$$\begin{array}{ccc} \tilde{G}_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E) & \xrightarrow{\cong} & \tilde{G}_{LT}(\mathcal{O}_K) \\ \downarrow & & \downarrow \\ G_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E) & \longrightarrow & G_{LT}(\mathcal{O}_K) \end{array}$$

where the right vertical arrow is surjective with kernel free of rank 1 over  $\mathcal{O}_E$ . So this together with (27.9.3.16) shows there is a surjection  $G_{LT}(\mathcal{O}_{C^\flat}) \rightarrow G_{LT}(\mathcal{O}_K)$  with kernel a rank-1  $\mathcal{O}_E$ -module.  $\lrcorner$

**Prop. (27.9.3.18).** There is a canonical  $\mathcal{O}_E$ -module map

$$G_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E) \xrightarrow{\log_G} B \otimes_{E_0} E.$$

and it is equivariant w.r.t  $\varphi$ .  $\lrcorner$

*Proof:*  $G_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E)$  are in bijection with the maximal ideal of  $\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E$ , and  $\log_G(x)$  is of the form  $x + \frac{c_2}{2}x^2 + \dots + \frac{c_n}{n}x^n + \dots$ , with  $c_n \in \mathcal{O}_E$ .

Now for  $x \in G_{LT}(\mathbf{A}_{\text{inf}} \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_E)$ , we show that  $\log_G(x)$  converges in  $B \otimes_{E_0} E = B + \pi B + \dots + \pi^{e-1}B$ : Let  $c_n x^n = \sum a_{n,i} \pi^i$ , then we need to show  $a_{n,i}/n$  converges to 0 for each of the norm  $|\cdot|_\rho$ . And this is because if  $x = x_0 + \pi y_0$ , then for  $n \geq em$ ,  $|x|_\rho \leq \max\{|x_0|_\rho^{em}, \rho^m\}$ , which decays exponential in  $n$ , and  $|\frac{1}{n}|_\rho$  decays linearly in  $n$ .  $\square$

**Prop. (27.9.3.19).** The map  $\log_G(\sigma(\cdot)) : G_{LT}(\mathcal{O}_{C^\flat}) \rightarrow (B \otimes_{E_0} E)^{\varphi^n = \pi}$  as in (27.9.3.9) is an isomorphism.  $\lrcorner$

*Proof:* For surjectivity, as any  $f \in (B \otimes_{E_0} E)^{\varphi^n = \pi}$  vanishes at a single  $\varphi^{d\mathbb{Z}}$ -orbit, then by (27.9.3.9) we can find a  $\log_G(u)$  that vanishes at the same locus, so  $f = \log(u)\lambda$  where  $\lambda$  is a unit in  $B \otimes_{E_0} E$  (27.9.3.12), so

$$\lambda \in (B \otimes_{E_0} E)^{\varphi^n = \text{id}} = (B \otimes_{\mathbb{Q}_p} E)^{\varphi = \text{id}} \text{ (27.9.3.5)} = B^{\varphi = \text{id}} \otimes_{\mathbb{Q}_p} E = E.$$

For injectivity, we proved in (27.9.3.9) that each  $\log_G(\sigma(u))$  only vanishes at a single  $\varphi^d$ -orbit in  $Y_E^0$ , so it cannot be 0, which vanishes at all points.  $\square$

**Cor. (27.9.3.20).** There are canonical bijections

$$\{\text{Closed Points of } X_E\} \cong \{\varphi^{d\mathbb{Z}}\text{-orbits of } Y_E^0\} \cong ((B \otimes_{E_0} E)^{\varphi^n = \pi} \setminus \{0\})/E^* \cong (G_{LT}(\mathcal{O}_{C^\flat}) - \{0\})/E^*.$$

by (27.9.3.9)(27.9.3.19), (27.9.3.12).  $\lrcorner$

### Vector Bundles and Base Change

**Prop. (27.9.3.21) [Vector Bundles on the Cover].** Let  $\pi : X_E \rightarrow X$  be the covering map, for any vector bundle  $\mathcal{E}$  on  $X_E$ ,  $\pi_*(\mathcal{E})$  is a vector bundle on  $X$ , and this induces an equivalence of categories:

$$\text{Vect}_{X_E} \cong \{X\text{-Bundles with an } E\text{-action}\}.$$

Now define  $\deg(\mathcal{E}) = \deg(\pi_*\mathcal{E})$ , and  $\text{slope}(\mathcal{E}) = \frac{\deg(\mathcal{E})}{\text{rank}(\mathcal{E})} = \frac{1}{n}\text{slope}(\pi_*\mathcal{E})$ .

Then  $\mathcal{E}$  is semistable of slope  $\lambda$  iff  $\pi_*\mathcal{E}$  is semistable of slope  $\lambda/n$ .  $\lrcorner$

*Proof:* One direction is clear, for the other, if  $\mathcal{F} = \pi_*\mathcal{E}$  is not semistable, choose its HN-filtration, then  $\lambda_1 > \lambda/n$ . Now the action of  $E$  on  $\mathcal{F}$  preserves the HN-filtration, thus  $\mathcal{F}_1$  is an  $E$ -vector bundle, thus by the correspondence above,  $\mathcal{F}_1 = \pi_*\mathcal{E}'$  for some subbundle  $\mathcal{E}' \subset \mathcal{E}$ , and clearly this contradicts the semistability of  $\mathcal{E}$ .  $\square$

**Cor. (27.9.3.22).** For any integral number  $d, n$  with  $n > 0$ , there exists a semistable vector bundle on  $X$  with rank  $n$  and degree  $d$ .  $\lrcorner$

*Proof:* Let  $E$  be an extension of  $\mathbb{Q}_p$  of degree  $n$ , then  $\pi_*(\mathcal{O}_{X_E}(d))$  is semistable of rank  $n$  and degree  $d$ , by (27.9.3.1)(7.6.2.22), because  $\mathcal{O}_{X_E}(d)$  is a line bundle (27.9.3.8) so clearly semistable, and it is of degree  $d$  because  $\text{?}$ .  $\square$

### Isocrystals and Classification of Semistable Vector Bundle over $X$

**Prop. (27.9.3.23).** Let  $k = \overline{\mathbb{F}}_p \in C^\flat$ , then there is an inclusion  $W(k) \rightarrow \mathbf{A}_{\text{inf}}$ , which extends to a map  $K \rightarrow B$ . Now given an isocrystal  $V$  over  $k$ , denote  $\mathcal{E}_V$  the coherent sheaf on  $X$  defined by the graded module  $\bigoplus_{n \geq 0} \text{Hom}_K(V, B)^{\varphi = p^n}$ . In other words, on an affine open subscheme  $U = D(t)$ ,  $\mathcal{E}_V(U) = \{\varphi\text{-equivariant } K\text{-linear maps } V \rightarrow B[\frac{1}{t}]\}$ .

And when  $V = E_{m/n}$  is the simple isocrystal, then  $\mathcal{E}_V$  is denoted by  $\mathcal{O}(\frac{m}{n})$ .  $\lrcorner$

**Prop. (27.9.3.24).** In fact we have  $\mathcal{O}(\frac{m}{n})(U) \cong (B[t^{-1}])^{\varphi^n = p^m} = (\rho_*\mathcal{O}(m))(U)$ , where  $\rho : X_E \rightarrow X$ , and  $E$  is an unramified extension of  $\mathbb{Q}_p$ .  $\lrcorner$

**Prop. (27.9.3.25)**[Classification of Semistable Vector Bundles over  $X$ ]. For every vector bundle on  $X$ , the HN-filtration splits non-canonically, and the map  $V \mapsto \mathcal{E}_V$  induces an equivalence of categories between

$$\{\text{Isoclinic Isocrystals of slope } \mu\}^{\text{op}} \rightarrow \{\text{Semistable vector bundles on } X \text{ of slope } \mu\}$$

┘

*Proof:* Cf.[FF Curve, Johannes].?

□

**Cor. (27.9.3.26).** Any two semistable vector bundles of slope  $\lambda$  over  $X$  is isomorphic, and a semistable vector bundle of slope 0 is trivial.

┘

**Prop. (27.9.3.27).** If  $\mathcal{E}, \mathcal{E}'$  be semistable vector bundles on  $X$  of slopes  $\mu, \mu'$ , then  $\mathcal{E} \otimes \mathcal{E}'$  is semistable of slope  $\mu + \mu'$ .

┘

*Proof:* We can assume  $\mathcal{E} = \rho_* \mathcal{O}_{X_E}(d)$  for an unramified extension  $E/\mathbb{Q}_p$  by (27.9.3.26), and then  $\mathcal{E} \otimes \mathcal{E}' = \rho_*(\mathcal{O}_X(d) \otimes \rho^* \mathcal{E}')$ . Since  $\rho_*, \rho^*$  preserves semistability (by (27.9.3.21) and ?). So it suffices to prove  $\mathcal{O}(d) \otimes -$  preserves semistability, but this is clear, as  $\mathcal{O}(d)$  shifts degree. □

### Diamonds Definitions

**Def. (27.9.3.28)** [Diamond]. A **diamond**  $X$  is a sheaf (of sets) on  $\text{PerfdSpa}_{\text{pro-ét}}$  of the form  $X = \textstyle\bigcup_Z / R$ , where  $Z \in \text{PerfdSpa}$  and  $R \in Z \times Z$  is a reasonable representable equivalence relation?.

┘

**Prop. (27.9.3.29)**[Scholze]. Let  $\underline{R} = (R, R^+) \in \text{HubPair}$ , then

$$\text{Spd}(\underline{R}) = \mathcal{Z} \mapsto \{\text{untilts of } \text{Spa}(\underline{R}) \text{ over } \mathcal{Z}\}$$

is a diamond (27.9.3.28).

And this construction can be glued to give diamond  $X^\diamond$  of any adic space  $X$ , which is a sheaf. ┘

**Def. (27.9.3.30)** [Adic Fargues-Fontaine Curve]. Let  $\mathcal{Y}$  be the adic space  $\text{Spa}(\mathbf{A}_{\text{inf}})$  removing the vanishing locus of  $p$  and  $[t]$ , then by what we proved, the Frobenius act totally discontinuous on  $\mathcal{Y}$ , thus the quotient  $\mathcal{X}^{FF}$  is an adic space, the FF-curve. ┘

**Prop. (27.9.3.31).** There is an isomorphism of diamonds:

$$\mathcal{Y}^\diamond \cong \text{Spd}(C^\flat) \times \text{Spd}(\mathbb{Q}_p), \quad \mathcal{X}^{FF, \diamond} \cong \text{Spd}(C^\flat)/\varphi^{\mathbb{Z}} \times \text{Spd}(\mathbb{Q}_p)$$

More generally, over any Huber pair  $\underline{R}$ , there is a relative FF-curve which is defined by

$$\text{Spd}(\underline{R}) \times \text{Spd}(\mathbb{Q}_p)/\varphi^{\mathbb{Z}}$$

┘

*Proof:*  $C^\flat$  is a perfectoid of char  $p$ , so for a perfect Huber pair,  $\underline{S}$  point of  $C^\flat$  is just a morphism  $u : (C^\flat, \mathcal{O}_{C^\flat}) \rightarrow (S, S^+)$ . And a  $\mathbb{Q}_p$  point is just a char0 untilts  $\underline{T}$  of  $\underline{S}$ .

So for each pairs  $(T, u)$ , we need to find a morphism  $\mathbf{A}_{\text{inf}}[\frac{1}{p}, \frac{1}{[t]}] \rightarrow T$ . For this, consider

$$\mathbf{A}_{\text{inf}} = W(\mathcal{O}_{C^\flat}) \rightarrow W(S^+) \cong W(T^\flat) \xrightarrow{\theta_T} T$$

This is a bijection, as we proved in the beginning (27.7.1.2). □

**Prop. (27.9.3.32).** There is a morphism of ringed spaces  $\mathcal{X}^{FF} \rightarrow X^{FF}$  that regard  $\mathcal{X}^{FF}$  as the rigid analytification of  $\mathcal{X}$ , so they have the same category of vector bundles and cohomology, prove by Kedlaya-Liu. ┘

## 4 Applications

**Prop. (27.9.4.1).** The FF curve  $X$  is geometrically simply-connected, i.e. the projection defines an isomorphism of étale groups  $\pi_1(X) \rightarrow \pi_1(\mathrm{Spec} \mathbb{Q}_p) = \mathrm{Gal}_{\mathbb{Q}_p}$ .

Equivalently, the pullback defines an equivalence of étale sites.  $\lrcorner$

*Proof:* Let  $\tilde{X} \rightarrow X$  be an finite étale morphism, we want to prove that  $\tilde{X} = X \otimes_{\mathrm{Spec} \mathbb{Q}_p} \mathrm{Spec}(E)$  for some étale  $\mathbb{Q}_p$ -algebra. Let  $\mathcal{A} = \rho_* \mathcal{O}_{\tilde{X}}$ , and  $E = H^0(X, \mathcal{A}) = H^0(\tilde{X}, \mathcal{O}_{\tilde{X}})$ . Now it suffices to show  $\mathcal{A} = E \otimes_{\mathbb{Q}_p} \mathcal{O}_X$ , which shows  $\tilde{X} = X \otimes_{\mathrm{Spec} \mathbb{Q}_p} \mathrm{Spec}(E)$ , and forces  $E$  be an étale  $\mathbb{Q}_p$ -algebra by fpqc descent ?. Equivalently,  $\mathcal{A}$  is trivial, and this is equivalent to  $\mathcal{A}$  being semistable of slope 0 by (27.9.3.26).

Because  $\rho$  is finite étale, the trace pairing  $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A} \xrightarrow{tr} \mathcal{O}_X$  is non-degenerate (check on stalks), which induces an isomorphism  $\mathcal{A} \cong \mathcal{A}^\vee$ , so  $\deg(\mathcal{A}) = 0$ , and if  $\mathcal{A}$  is not semistable, let  $\mathcal{A}'$  be the first term of the HN-filtration of  $\mathcal{A}$ , then it is of slope  $\lambda > 0$ . So  $\mathcal{A}' \otimes \mathcal{A}'$  is of slope  $2\lambda$  by (27.9.3.27), so the composite  $\mathcal{A}' \otimes \mathcal{A}' \hookrightarrow \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$  must be 0 by (4.3.5.32), which is impossible, because if  $U$  is an affine open that  $\mathcal{A}$  has a section, then this says  $U \otimes_X \tilde{X}$  has a section  $s$  that  $s^2 = 0$ . But  $U \otimes_X \tilde{X}$  is reduced (check on stalks).  $\square$

**Cor. (27.9.4.2).**

- The projection map induces equivalence  $\mathrm{Rep}_{\mathcal{A}\mathrm{b}^{\mathrm{fin}}}(\mathrm{Gal}_{\mathbb{Q}_p}) \cong \mathrm{LocSys}^{\mathrm{ét}}(X)$ .
- For  $M \in \mathrm{Rep}_{\mathcal{A}\mathrm{b}^{\mathrm{fin}}}(\mathrm{Gal}_{\mathbb{Q}_p})$ , the natural map

$$H^*(\mathrm{Gal}_{\mathbb{Q}_p}, M) \rightarrow H_{\mathrm{ét}}^*(X, u^* M)$$

is an isomorphism for  $* = 0, 1$ .  $\lrcorner$

*Proof:* 1 is trivial. For 2, Cf. [Lurie P102].  $\square$

## 5 Weakly Admissible $\Rightarrow$ Admissible

**Def. (27.9.5.1) [Notations].** Let  $K$  be a finite extension of  $\mathbb{Q}_p$ , and  $K_0 = W(k)[\frac{1}{p}]$  be the maximal unramified subextension in  $K$ , let  $C = \mathbb{C}_K$  and  $F = C^\flat$ . Denote by  $\infty \in X$  the closed point determined by  $C$ , which is just the vanishing locus of the Galois stable line  $\mathbb{Q}_p t$ , where  $t = \log([\varepsilon])$  and  $\varepsilon = (1, \xi_p, \xi_{p^2}, \dots) \in C^\flat$  (27.9.1.36).

Notice  $\mathrm{Gal}_K$  acts on  $\mathbb{Q}_p \cdot \log([\varepsilon])$  by the cyclotomic character  $\chi_{\mathrm{cycl}}$ . Recall

$$B_{\mathrm{dR}}^+ = \widehat{\mathcal{O}}_X(\infty), \quad B_{\mathrm{crys}}^+, \quad B_{\mathrm{crys}} = B_{\mathrm{crys}}^+[t^{-1}], \quad B_e = H^0(X \setminus \{\infty\}, \mathcal{O}_X) = (B_{\mathrm{crys}})^{\varphi=\mathrm{id}}.$$

$\lrcorner$

**Def. (27.9.5.2) [Equivariant Action of  $\mathrm{Gal}_K$  on Vector Bundles].** Recall an equivariant action of  $\mathrm{Gal}_K$  on a bundle  $\mathcal{E}$  on  $X$  is a data of isomorphisms  $\sigma^*(\mathcal{E}) \cong \mathcal{E}$  that  $c_{\sigma\tau} = c_\tau \circ \tau^*(c_\sigma)$ . Notice any equivariant action of  $\mathrm{Gal}_K$  on  $\mathcal{E}$  induces a semilinear  $\mathrm{Gal}_K$  action on  $\mathcal{E}_\infty^\wedge = \mathcal{E} \otimes_{\mathcal{O}_X} B_{\mathrm{dR}}^+$ , and here we require this action is continuous. The category of equivariant  $\mathrm{Gal}_K$ -bundles are denoted by  $\mathrm{Vect}_X^{\mathrm{Gal}_K}$ .  $\lrcorner$

**Cor. (27.9.5.3).** By the slope 0 case of the classification of vector bundles on  $X$  (27.9.3.25) and (25.2.5.1), we see that the functor:

$$\mathrm{Rep}_{\mathbb{Q}_p}(\mathrm{Gal}_K) \rightarrow \mathrm{Vect}_X^{\mathrm{Gal}_K} : V \mapsto V \otimes_{\mathbb{Q}_p} \mathcal{O}_X$$

is fully faithful with essential image the category  $\mathrm{Vect}_X^{\mathrm{Gal}_K, \mathrm{sst}, 0}$  of all  $\mathrm{Gal}_K$  vector bundles on  $X$  that the underlying bundle is semistable of slope 0, i.e. trivial (27.9.3.26).  $\square$

**Prop. (27.9.5.4).** There is a pullback diagram of categories:

$$\begin{array}{ccc} \varphi\text{-Fil Mod}_{K/K_0} & \longrightarrow & \varphi\text{-Mod}_{K_0} \\ \downarrow \mathcal{E}(-) & & \downarrow \mathcal{V} \\ \mathrm{Vect}_X^{\mathrm{Gal}_K} & \longrightarrow & \mathrm{Rep}_{B_e}(\mathrm{Gal}_K) \end{array}$$

where  $\mathcal{E}(-)$  maps a  $\varphi$ -filtered module  $(D, \varphi_D, \mathrm{Fil})$  to the bundle that is the bundle  $(\widetilde{D}, \varphi_D)$  modified so that the fiber at  $\infty$  is  $\mathrm{Fil}^0(D_K \otimes_K B_{\mathrm{dR}})$ .  $\square$

*Proof:* 1: By lemma (25.2.4.9),  $\varphi\text{-Fil Mod}_{K/K_0}$  is equivalent to a  $\varphi$ -module  $V$  with a  $\mathrm{Gal}_K$ -stable  $B_{\mathrm{dR}}^+$ -lattice in  $(V \otimes_{K_0} K) \otimes_K B_{\mathrm{dR}}^+ = V \otimes_{K_0} B_{\mathrm{dR}}^+$ .

2: By (25.2.6.4),  $\varphi\text{-Mod}$  is a full subcategory of  $\mathrm{Rep}_{B_e} \mathrm{Gal}_K$ , where the  $\mathrm{Gal}_K$ -stable  $B_{\mathrm{dR}}^+$ -lattice is choose to be  $V \otimes_{K_0} B_{\mathrm{dR}}^+$ .

3: Clearly there is a functor

$$\mathrm{Vect}_X^{\mathrm{Gal}_K} \rightarrow \mathrm{Rep}_{B_e}(\mathrm{Gal}_K) : \mathcal{E} \rightarrow H^0(X \setminus \{\infty\}, \mathcal{E}).$$

and (7.5.1.48) says in this case  $Bun_X^{\mathrm{Gal}_K}$  is equivalent to a  $B_e$ -module with continuous  $\mathrm{Gal}_K$ -actions and and a  $B_{\mathrm{dR}}^+$ -module with continuous  $\mathrm{Gal}_K$ -actions that they corresponds as a  $B_{\mathrm{dR}}$ -module with continuous  $\mathrm{Gal}_K$ -actions.

4: The compatibility in 3 just says that the  $B_{\mathrm{dR}}^+$ -lattice choosen in the definition of (25.2.4.11) just comes from that of 2, so this diagram is clearly a pullback.  $\square$

**Lemma (27.9.5.5).** Let  $(V, \mathrm{Fil}^\bullet) \in \mathrm{FilVect}_K$  and  $W = \mathrm{Fil}(V \otimes_K B_{\mathrm{dR}})$ , if  $V \otimes B_{\mathrm{dR}}^+ = \{e_1, \dots, e_n\}$  and  $\mathrm{Fil}^0(V \otimes_K B_{\mathrm{dR}}) = \{t^{-a_1}e_1, \dots, t^{-a_n}e_n\}$ , then the Hodge polygon of  $\mathrm{Fil}^\bullet$  has slopes  $(a_1, \dots, a_n)$ .  $\square$

*Proof:* Use (25.2.4.9), notice  $(t^a B_{\mathrm{dR}}^+)^{\mathrm{Gal}_K} = 0$  for  $a > 0$ , as in the proof of (25.2.4.8).  $\square$

**Lemma (27.9.5.6).** The functor

$$\mathcal{E}(-) : \varphi\text{-Fil Mod}_{K/K_0} \rightarrow \mathrm{Vect}_X^{\mathrm{Gal}_K}$$

defined in (27.9.5.4) preserves degree and HN-filtration, where the HN-filtration on the RHS is induces by the HN-filtration on  $Bun_X$  by canonicity.  $\square$

*Proof:*  $\deg(\mathcal{E}((D, \varphi_D, \mathrm{Fil}^\bullet))) = \deg(\mathcal{E}(D, \varphi_D)) - \dim_K[D \otimes_{K_0} B_{\mathrm{dR}}^+ : \mathrm{Fil}^0(D_K \otimes_K B_{\mathrm{dR}})]$   
 $= \deg(D_K, \mathrm{Fil}) - \deg(D, \varphi_D)$ .

Now the degree correspond, for the invariance of HN-filtration, it suffices to prove the subobjects are in bijection: Given a subobjects of  $\mathcal{E}(V)$ , we want to show it is a  $\mathcal{E}(V')$ , but this is because on the affine open  $\mathrm{Spec}(B_e)$ , by (25.2.6.4) any subbundle is also crysatalline, i.e. comes from  $\varphi\text{-Mod}_{K_0}$ .  $\square$

**Prop. (27.9.5.7) [Weakly Admissible implies Admissible].** The category of crystalline Galois representations of  $\mathrm{Gal}_K$  is equivalent to the category  $\varphi\text{-FilMod}_{K/K_0}^{\text{weak, adm}}$  of weakly admissible filtered  $\varphi$ -modules for  $K$ .  $\square$

*Proof:* By definition of weakly admissible and (27.9.5.6), there is a pullback diagram

$$\begin{array}{ccc} \varphi\text{-FilMod}_{K/K_0}^{\text{weak, adm}} & \longrightarrow & \varphi\text{-FilMod}_{K/K_0} \\ \downarrow & & \downarrow \mathcal{E}(-) \\ \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}} \mathrm{Gal}_K \cong \mathcal{V}\mathrm{ect}_X^{\mathrm{Gal}_K, \text{semi st}, 0} & \longrightarrow & \mathcal{V}\mathrm{ect}_X^{\mathrm{Gal}_K} \end{array}$$

Adjunction with (27.9.5.4), we get another pullback diagram

$$\begin{array}{ccccc} \varphi\text{-FilMod}_{K/K_0}^{\text{weak, adm}} & \longrightarrow & \varphi\text{-FilMod}_{K/K_0} & \longrightarrow & \varphi\text{-Mod}_{K_0} \\ \downarrow & & \downarrow \mathcal{E}(-) & & \downarrow \mathcal{V} \\ \mathrm{Rep}_{\mathbb{Q}_p}(\mathrm{Gal}_K) \cong \mathcal{V}\mathrm{ect}_X^{\mathrm{Gal}_K, \text{semi st}, 0} & \longrightarrow & \mathcal{V}\mathrm{ect}_X^{\mathrm{Gal}_K} & \longrightarrow & \mathrm{Rep}_{B_e}(\mathrm{Gal}_K) \end{array}$$

But this pullback is just the category of crystalline representations: by (27.9.5.3), for  $V \in \mathrm{Rep}_{\mathbb{Q}_p} \mathrm{Gal}_K$ , the condition  $\mathcal{V}(\mathcal{D})(M) \cong M$  in (25.2.6.4) is just saying that  $V$  is in the image of  $\mathcal{V}$  iff

$$(V \otimes_{\mathbb{Q}_p} B_{\mathrm{crys}})^{\varphi=\mathrm{id}} \otimes_{B_e} B_{\mathrm{crys}} \cong V \otimes_{\mathbb{Q}_p} B_{\mathrm{crys}},$$

which is equivalent to  $V$  being crystalline.  $\square$



## 27.10 Barsotti-Tate Groups

*“Although  $p$ -divisible groups are interesting enough in their own right, our main motivation for studying them has been their applications to abelian varieties.”*

–J. Tate

References are [Zin84].

**Notation (27.10.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $R \in \mathcal{CRing}$ .
- Let  $S \in \mathcal{Sch}$ .

┘

### 1 Barsotti-Tate Groups

References are [Tat67, Mes72, CCO14], [Finite Flat Schemes and  $p$ -divisible Groups, Jakob], [ $p$ -divisible Groups, Morrow], [ $p$ -divisible Groups, Demazure], [P-Divisible Groups and Displays, Zink] and [notes on Tate’s  $p$ -divisible Groups].

**Notation (27.10.1.1).**

- Let  $\kappa \in \text{Perf}^p$ .
- Let  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa) \in p\text{-Field}$ .

┘

### Barsotti-Tate Groups

**Def. (27.10.1.2) [Barsotti-Tate Groups].** A  $p$ -divisible group or a **Barsotti-Tate group** of height  $h$  over  $S$  is an direct system  $(G_n)_{n \in \mathbb{Z}_+} \in \text{finAbSch}/S$  s.t. for each  $n \in \mathbb{Z}_+$ ,

- $G_n = G_{n+1}[p^n]$ .
- $[p] : G_{n+1} \rightarrow G_n$  is faithfully flat.

Equivalently, a  $p$ -divisible group over  $S$  is a fppf-sheaf  $G$  over  $S$  s.t.

- $G$  is  $p$ -divisible:  $[p]_G$  is an epimorphism.
- $G$  is  $p$ -torsion:  $G = \varinjlim_n G[p^n]$
- Each  $G[p^n]$  are representable as sheaves on  $\mathcal{Sch}_{\text{fppf}}/S$ . (Notice that it suffices to show that  $G[p]$  is representable, because we have the exact sequence  $0 \rightarrow G[p] \rightarrow G[p^{n+1}] \rightarrow G[p^n] \rightarrow 0$  and we can use (10.1.1.36)).

For  $G \in p\text{Div}/S$ , there exists a locally constant function  $h : S \rightarrow \mathbb{N}$  s.t. the rank of fiber of  $G[p]$  at  $s \in S$  is  $p^{h(s)}$ . Then it follows from the exact sequence  $0 \rightarrow G[p] \rightarrow G[p^{n+1}] \rightarrow G[p^n] \rightarrow 0$  that  $\text{rank } G[p^n]_s = p^{nh(s)}$  for each  $n \in \mathbb{Z}_+$ .

For each locally constant function  $h : S \rightarrow \mathbb{N}$ , the category of  $p$ -divisible groups over  $S$  of height  $h$  is denoted by  $p\text{Div}^h/S$ . ┘

**Example (27.10.1.3) [Barsotti-Tate Groups].**

- $\mathbb{G}_m[p^\infty] \cong \mu_{p^\infty} \in p\text{Div}^1/\mathbb{Z}$ ?

- If  $S \in \text{Sch}$  and  $A \in \text{AbVar}^g/S$ , then  $X = A[p^\infty] \in p\text{Div}^{2g}/S$ , by (20.9.6.18). And there is a canonical isomorphism  $A[p^\infty]^D \cong A^\vee[p^\infty]$ , by (20.9.6.5).  $\lrcorner$

**Def. (27.10.1.4) [Barsotti-Tate Groups over Formal Schemes].** If  $(\mathfrak{X}, \mathcal{I})$  is a locally Noetherian adic scheme, then  $\mathfrak{X} = \varinjlim_{n \in \mathbb{Z}_+} \mathfrak{X}_n$  where  $\mathfrak{X}_n = \text{Spec}(\mathcal{O}_{\mathfrak{X}}/\mathcal{I}^n)$ . Then a  **$p$ -divisible group** over  $\mathfrak{X}$  is a sequence  $X = (X_n)$  where  $X_n \in p\text{Div}/\mathfrak{X}_n$  together with isomorphism  $X_{n+1} \times_{\mathfrak{X}_{n+1}} \mathfrak{X}_n \cong X_n$ .  $\lrcorner$

**Prop. (27.10.1.5).** If  $\mathfrak{X} = \text{Spf}(A, I)$  is a Noetherian affine adic scheme, then the functor  $X \mapsto (X \bmod I^n)_{n \in \mathbb{Z}_+}$  is an equivalence between  $p\text{Div}/\text{Spec } A$  and  $p\text{Div}/\text{Spf } A$  (27.10.1.4).  $\lrcorner$

*Proof:* Cf. [Messing, The Crystals Associated to Barsotti-Tate Groups].  $\square$

**Def. (27.10.1.6) [Serre-Dual Barsotti-Tate Groups].** For  $G \in p\text{Div}/S$ , the **Serre-dual  $p$ -divisible group** is defined to be the directed system  $G^D \triangleq (G[p^n]^D)$ , where the transition map is the Cartier-dual of  $[p] : G[p^n] \rightarrow G[p^n]$ . Then  $G \in p\text{Div}/S$ ?  $\lrcorner$

**Def. (27.10.1.7) [Quotients of  $p$ -Divisible Groups and Exact Sequences].** If  $H \subset G \in p\text{Div}/S$  and  $\text{ht}(H) = h' \in \mathbb{N}$ ,  $\text{ht}(G) = h \in \mathbb{N}$ , then  $G/H \triangleq (G[p^n]/H[p^n])_{n \in \mathbb{Z}_+}$  is also a  $p$ -divisible group over  $S$ .

An exact sequence of  $p$ -divisible group

$$0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0 \in p\text{Div}/S$$

is exact iff

$$0 \rightarrow G'[p^n] \rightarrow G[p^n] \rightarrow G''[p^n] \rightarrow 0 \in \text{GrpSch}/S$$

is an exact sequence for each  $n \in \mathbb{Z}_+$ . In particular, the Serre-dual of an exact sequence of  $p$ -divisible groups (27.10.1.6) is an exact sequence of  $p$ -divisible groups, by Cartier-duality.  $\lrcorner$

*Proof:* Cf. [Chai-Conrad-Oort14] 1.4.3.5.  $\square$

**Def. (27.10.1.8) [Connected  $p$ -Divisible Groups].** Let  $(\mathcal{O}_K, \kappa)$  be a complete Noetherian local ring of mixed characteristic  $(0, p)$ , then  $G = (G_n)_{n \in \mathbb{Z}_+} \in p\text{Div}/\mathcal{O}_K$  is called **connected  $p$ -divisible group** or a **formal  $p$ -divisible group** if each  $G_n$  is connected, or equivalently each  $G_n$  has infinitesimal special fibers???. The category of connected  $p$ -divisible groups over  $\mathcal{O}_K$  is denoted by  $p\text{Div}^f/\mathcal{O}_K$ .  $\lrcorner$

**Prop. (27.10.1.9) [Formal  $p$ -Divisible Groups and Formal Groups].** Let  $(\mathcal{O}_K, \kappa)$  be a complete Noetherian local ring of mixed characteristic  $(0, p)$ , then  $\Gamma \mapsto \Gamma[p^\infty]$  is an equivalence between the category of commutative formal Lie-groups over  $\mathcal{O}_K$  (27.10.4.7) s.t.  $[p]$  is an isogeny and  $p\text{Div}^f/\mathcal{O}_K$  (27.10.1.8).  $\lrcorner$

*Proof:* Cf. [ $p$ -divisible Groups, Tate] or [notes on Tate's  $p$ -divisible groups].  $\square$

**Prop. (27.10.1.10) [Formal  $p$ -Divisible Groups].** For  $R \in \mathcal{CRing}$  and  $0 \neq G \in p\text{Div}/R$ ,  $G$  is a smooth formal group iff  $G[p]$  is infinitesimal (10.1.2.3), i.e. has connected fibers. This implies that each  $G[p^n]$  is infinitesimal, and  $p$  is nilpotent in  $R$ .  $\lrcorner$

*Proof:* ?  $\square$

**Def. (27.10.1.11) [Étale Barsotti-Tate Groups].** An **étale  $p$ -divisible group** over  $S$  is a  $p$ -divisible group  $X \in p\text{Div}/S$  s.t.  $X[p]$  is étale over  $S$ . By (10.1.2.21), in this case  $X[p^n]$  étale over  $S$  for each  $n \in \mathbb{Z}_+$ .  $\perp$

**Def. (27.10.1.12) [Toroidal Barsotti-Tate Groups].** A **toroidal  $p$ -divisible group** over  $S$  is a  $p$ -divisible group  $X \in p\text{Div}/S$  s.t.  $X[p]$  is of multiplicative-type (10.1.2.20). By (10.1.2.21), in this case  $X[p^n]$  is of multiplicative-type over  $S$  for each  $n \in \mathbb{Z}_+$ .

For  $X \in p\text{Div}/S$ ,  $X$  is toroidal iff  $X^D$  is étale (27.10.1.11), by (10.1.2.20).  $\perp$

**Prop. (27.10.1.13).** If  $S \in \text{Sch}^p$  and  $X \in p\text{Div}/S$  is toroidal (27.10.1.12), then  $X$  is formal (27.10.1.10).  $\perp$

*Proof:* This follows from (10.1.2.14).  $\square$

**Prop. (27.10.1.14) [Connected-Étale Exact Sequences, Messing [Mes72]].** For  $G \in p\text{Div}/S$ , the following statements are equivalent:

- $G$  is an extension of an étale  $p$ -divisible group by a formal  $p$ -divisible group.
- $G[p]$  is an extension by an étale finite group scheme by a infinitesimal finite group scheme (10.1.2.3).
- The function  $s \mapsto \#G[p](\bar{s})$  is a locally constant function on  $S$ .

In particular, this holds if  $S$  is the spectrum of a local ring, then we take  $1 \rightarrow G^0 \rightarrow G \rightarrow G^{\text{ét}} \rightarrow 1$ ?  $\perp$

*Proof:*  $1 \iff 2$  is clear.

$2 \rightarrow 3$ : If  $1 \rightarrow N_1 \rightarrow G[p] \rightarrow N_2 \rightarrow 1$  is an exact sequence with  $N_1$  infinitesimal and  $N_2$  étale, then  $\#G[p](\bar{s}) = \#N_2(\bar{s})$  is locally constant in  $s$  because  $N_2$  is étale.

$3 \rightarrow 2$ : Cf. [Messing. The crystals associated to Barsotti-Tate groups] 2.4.8?  $\square$

**Prop. (27.10.1.15) [Connected-Étale Splittings].** For  $k \in \text{Field}^p$  s.t.  $k = k^{\text{sep}}$  and  $G \in p\text{Div}/k$ , the connected-étale exact sequence (27.10.1.14) uniquely splits, and there is a canonical decomposition

$$X = X_{cc} \times X_{ce} \times X_{ee} \in p\text{Div}/k.$$

$\perp$

*Proof:* Cf. [Oort, Commutative group schemes] Section 1.2.?  $\square$

**Def. (27.10.1.16) [Tate-Modules].** For  $k \in \text{Field}$  and  $G \in p\text{Div}/k$ , the **Tate-module** of  $G$  is defined to be

$$\text{Tate}_p(G) \stackrel{\triangle}{=} \text{Hom}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}, G) = \varprojlim_n G[p^n](k^{\text{sep}}),$$

and the **Tate-comodule** of  $G$  is defined to be

$$\Phi_p(G) = \varinjlim_n G[p^n](k^{\text{sep}}).$$

Then  $\text{Tate}_p(G) \cong \text{Tate}_p(G^{\text{ét}})$  is a free  $\mathbb{Z}_p$ -module of rank  $\text{ht}(G^{\text{ét}})$ ?  $\perp$

**Prop. (27.10.1.17) [Discriminants].** Let  $(R, \kappa)$  be a complete Noetherian local ring with  $\text{char } \kappa = p$  and  $G \in p\text{Div}/R$ . let  $A_{p^n} = \Gamma(G[p^n])$ , then

$$\text{disc}_R(A) = (p^{\dim G \cdot np^{h_n}}) \in \text{Ideal}(R).$$

┘

*Proof:* Cf. [p-divisible Groups, Tate] or [notes on Tate's p-divisible groups].

□

**Def. (27.10.1.18) [Tangent Spaces].** The **tangent space**  $\text{Lie}(G)$  of  $G \in p\text{Div}/\mathcal{O}_K$  is defined to be the Lie algebra of the formal Lie group associated to  $G^0$  (27.10.1.9). ┘

**Thm. (27.10.1.19) [p-Divisible Groups Determined by Generic Fibers].** If  $\text{char } K = 0$ , then the functor  $p\text{Div}/\mathcal{O}_K \rightarrow p\text{Div}/K$  is fully faithful. ┘

*Proof:* ?

□

### Isogenies between $p$ -Divisible Groups

**Def. (27.10.1.20) [Isogenies and Quasi-Isogenies].** For  $S \in \text{Sch}$ , an **isogeny between  $p$ -divisible groups** over  $S$  is a morphism between  $p$ -divisible groups with the same height s.t. the kernel is a finite group scheme.

A **quasi-isogeny between  $p$ -divisible groups**  $X, Y \in p\text{Div}/S$  is a global section  $\varphi$  of the Zariski-sheaf  $\underline{\text{Hom}}_S(X, Y) \otimes \mathbb{Q}$  over  $S$  s.t. locally on  $S$  there exists  $n \in \mathbb{Z}$  s.t.  $p^n \varphi$  is an isogeny. ┘

**Prop. (27.10.1.21) [Quotients by Finite Group Schemes].** Let  $X \in p\text{Div}/S$  and  $N \leq X$  a finite group scheme over  $S$ .

- If  $N \leq X[p^r]$  for some  $r \in \mathbb{Z}_+$ , then for each  $n \leq m \in \mathbb{Z}_+$ , the natural map

$$(X[p^{n+r}]/N)[p^n] \rightarrow (X[p^{m+r}]/N)[p^n]$$

is an isomorphism.

- The fppf-quotient  $X/N \in \text{Sh}(\text{Sch}_{\text{fppf}}/S)$  is represented by an ind-scheme  $Y \in \text{Sch}/S$  which is locally of the form  $Y = ((X[p^{n+r}]/N)[p^n])$  where  $r \in \mathbb{Z}_+$  is chosen s.t.  $[p^r]_N = 0$ . ┘

*Proof:* The question is local on  $S$ , so we may assume that  $N \leq X[p^r]$ . It suffices to show that if we define  $Y_n = (X[p^{n+r}]/N)[p^n] \in \text{Sh}(\text{Sch}_{\text{fppf}}/S)$ , then each  $Y_n$  are finite group scheme over  $S$ , and  $Y_n \rightarrow Y_m[p^n]$  are isomorphisms for each pair  $n \leq m \in \mathbb{Z}_+$ .

To check  $Y_n$  are finite group scheme over  $S$ , notice  $[p^n] : X[p^{n+r}]/N \rightarrow X[p^n]/N$  is a surjection of fppf-sheaves, so it is faithfully flat by (10.1.2.8). So the kernel  $Y_n$  is finite locally free over  $S$ ?.

To check  $Y_n \rightarrow Y_m[p^n]$  are isomorphisms, because they are finite locally free, it suffices to check on geometric fibers, and then we can use either Tate-modules or Dieudonné-theory. □

**Prop. (27.10.1.22) [Inverse Isogenies].** Let  $h : S \rightarrow \mathbb{Z}$  be a locally constant function and  $\varphi : X \rightarrow Y \in p\text{Div}^h/S$  be an isogeny with kernel  $N = \ker(\varphi) \in \text{finAbSch}/S$ , and the induced map  $\varphi : X/N \rightarrow Y$  (27.10.1.21) is an isomorphism.

And if  $N$  is killed by  $[p^m]$  for some  $m \in \mathbb{Z}_+$ , then there exists an isogeny  $\psi : Y \rightarrow X \in p\text{Div}^h/S$  s.t.  $\varphi \circ \psi = [p^n]_Y$ ,  $\psi \circ \varphi = [p^n]_X$ . ┘

*Proof:* To check  $\varphi : X/N \cong Y$ , it suffices to check on  $[p^n]$ -torsions for each  $n \in \mathbb{Z}_+$ . For this, it suffices to check on geometric points ?, and then we can use either Tate-modules or Dieudonné-theory.

Now we can define  $\psi : Y \xrightarrow{\varphi^{-1}} X/N \rightarrow X/X[p^n] \xrightarrow{[p^n]} X$ , so  $\psi \circ \varphi = [p^n]$ , and

$$(\varphi \circ \psi) \circ \varphi = \varphi \circ [p^n] = [p^n] \circ \varphi,$$

showing that  $\varphi \circ \psi = [p^n]$  as well, because  $\varphi$  is faithfully flat. □

**Thm. (27.10.1.23) [Kernel of Dual Isogenies].** Let  $\varphi : X \rightarrow Y \in p\text{Div}/S$  be an isogeny, then  $\varphi^D : Y^D \rightarrow X^D$  is also an isogeny, with a natural isomorphism  $\ker(\varphi^D) \cong \ker(\varphi)^D$ . ┘

*Proof:* Cf. [CCO14]P152. □

**Prop. (27.10.1.24) [Rigidity of Quasi-Isogenies between Barsotti-Tate Groups].** Let  $S \in \text{NSch}$  s.t.  $p$  is locally nilpotent on  $S$ , and  $X, X' \in p\text{Div}/S$ , then every  $\bar{\rho} \in \text{Hom}(X \times_S S_{\text{red}}, X' \times_S S_{\text{red}})$  extends uniquely to a  $\rho \in \text{Hom}(X, X')$ . Moreover,  $\rho$  is a quasi-isogeny if  $\bar{\rho}$  is. ┘

*Proof:* Cf. [Andre, period maps]P73. ? □

**Prop. (27.10.1.25) [Isogenies between Abelian Varieties and  $p$ -Divisible Groups].** Let  $g \in \mathbb{Z}_+$  and  $\varphi : A \rightarrow B \in \text{AbVar}/S$ , then  $\varphi$  is an isogeny iff  $f = \varphi[p^\infty] : A[p^\infty] \rightarrow B[p^\infty] \in p\text{Div}/S$  (27.10.1.3) is an isogeny (27.10.1.20). ┘

*Proof:* Cf. [CCO14]P150 ?. □

**Prop. (27.10.1.26) [Complex Multiplications for  $p$ -Divisible Groups].** Let  $\kappa \in \text{Perf}^p$  and  $h \in \mathbb{Z}_+$ ,  $G \in p\text{Div}^h/\kappa$ , then

- If  $F \leq \text{End}^0(G)$  is a commutative semi-simple  $\mathbb{Q}_p$ -subalgebra, then  $[F : \mathbb{Q}_p] \leq h$ , with equality iff  $M^*(G_\kappa)[\frac{1}{p}]$  is free of rank 1 as a  $W(\kappa) \otimes_{\mathbb{Z}_p} F$ -module.
- If the equality as above holds, then  $Z_{\text{End}^0(G)}(F) = F$ . If moreover the maximal  $\mathbb{Z}_p$ -order  $\mathcal{O}_F \leq F$  lies in  $\text{End}(G)$ , then  $M^*(G_\kappa)$  is free of rank 1 as a  $W(\kappa) \otimes_{\mathbb{Z}_p} \mathcal{O}_F$ -module. ┘

*Proof:* Cf. [CCO14]Prop1.4.3.9. □

**Prop. (27.10.1.27) [Being Isogeny is a Clopen Condition].** For  $S \in \text{NSch}$  and  $f : X \rightarrow Y \in p\text{Div}/S$ , the subset  $U = \{s \in S \mid \varphi_s \text{ is an isogeny}\}$  is clopen in  $S$ , and  $\varphi_U : X_U \rightarrow Y_U \in p\text{Div}/U$  is an isogeny. ┘

*Proof:* Cf. [CCO14]P153 ?. □

### Classical Dieudonné-Theory

**Thm. (27.10.1.28) [Dieudonné-Theory for  $p$ -Divisible Groups].** There is an additive anti-equivalence between Abelian-categories:

$$M^* : p\text{Div}/\kappa \rightarrow \text{Mod}_{D(\kappa)}^{W(\kappa)\text{-lf}} : G \mapsto M^*(G) \stackrel{\Delta}{=} \varprojlim_n M^*(G_n) \in \text{Mod}_{D(\kappa)}$$

called the **contravariant Dieudonné functor**, with inverse given by ?,

Moreover, by counting  $W(\kappa)$ -lengths, we see

- $\text{rank}_{W(\kappa)} M^*(G) = \text{ht}(G)$  (27.10.1.2).
- $M^*(G)/(p^n) \rightarrow M^*(G_n)$  for each  $n \in \mathbb{Z}_+$ .
- If  $\kappa' \in \mathcal{P}\text{erf}^p/\kappa$ , then there are natural isomorphisms

$$M^*(G_{\kappa'}) = M^*(G) \otimes_{W(\kappa)} W(\kappa').$$

- The relative Frobenius-morphism  $F_{G/\kappa} : G \rightarrow G^{(p)}$  corresponds to

$$M^*(G^{(p)}) \cong \varphi_*(M^*(G)) \xrightarrow{F} M^*(G),$$

and the Verschiebung-morphism  $\text{Ver}_{G/\kappa} : G^{(p)} \rightarrow G$  corresponds to

$$M^*(G) \xrightarrow{V} \varphi_*(M^*(G)) = M^*(G^{(p)}).$$

Moreover,  $G$  is connected iff  $F$  is nilpotent on  $M^*(G)$ .

- There are natural isomorphism  $M^*(G^D) \cong M_*(G) \triangleq \text{Hom}_{W(\kappa)}(M^*(G), W(\kappa))$ , and

$$F^D(f) : m \mapsto \sigma(f(V(m))), \quad V^D(f) : m \mapsto \sigma^{-1}(f(F(m))).$$

$M_*(G) \triangleq M^*(G^D)$  is called the **covariant Dieudonné module** of  $G$ . Equivalently (Cf. [C0014]P349), there are natural perfect pairings

$$\langle -, - \rangle_{M^*(G)} : M^*(G) \times M^*(G^D) \rightarrow W(\kappa)$$

s.t.

$$\langle Fx, y \rangle_{M^*(G)} = \langle x, Vy \rangle_{\varphi_* M^*(G)}, \text{ where } x \in \varphi_* M^*(G), y \in M^*(G^D),$$

where  $\langle -, - \rangle_{\varphi_* M^*(G)}$  is the composition

$$\varphi_* M^*(G) \times \varphi_* M^*(G) \xrightarrow{\langle -, - \rangle_{M^*(G)}} \varphi_*(W(\kappa)) \xrightarrow{\varphi} W(\kappa).$$

And this pairing satisfies a double duality.

- There are natural isomorphisms

$$\text{Lie}(G)^\vee \cong M^*(G)/FM^*(G), \quad \text{Lie}(G) \cong \ker(V : M_*(G) \rightarrow \varphi_* M_*(G)).$$

respecting extension of perfect base fields. ┘

*Proof:* This follows directly from (10.1.2.27). □

**Cor. (27.10.1.29).** For  $G \in p\text{Div}/\kappa$ ,  $G$  is connected (27.10.1.8) iff  $F$  is topologically nilpotent on  $M^*(G)$ , by (10.1.2.28). ┘

**Thm. (27.10.1.30)[Dieudonné Theory for Formal Groups].** A **formal group functor** is a functor from the category of finite-length  $B$ -rings to  $\mathcal{G}\text{rp}$ . And a **formal group** is a formal group functor that is representable by a formal scheme.

A formal group is **formally smooth** if for any finite-length  $B$ -ring  $A$  and  $I \in \text{Ideal}(A)$  s.t.  $I^2 = (0)$ , the map  $G(R) \rightarrow G(R/I)$  is surjective.

Then for each  $R \in \mathcal{C}\text{Ring}^p$ , there is an equivalence from the category of smooth commutative formal groups over  $R$  to the category of reduced  $E(R)$ -modules (27.5.5.23). Moreover, If  $M$  corresponds to  $G$ , then  $M/VM \cong T_e(G)$ . In particular,  $\dim G < \infty$  iff  $M$  is a Dieudonné-module. ┘

*Proof:* ?

□

**Prop. (27.10.1.31) [Finite Group Schemes of Order  $p$ ].**

- $M^*(\mathbb{Z}/(p))$  is given by  $\kappa e$  with  $Fe = e, Ve = 0$ .
- $M^*(\mu_p)$  is given by  $\kappa e$  with  $Fe = 0, Ve = e$ .
- $M^*(\alpha_p)$  is given by  $\kappa e$  with  $Fe = 0, Ve = 0$ .

┘

*Proof:* This is because the relative Frobenius kills  $\mu_p$  and  $\alpha_p$  but not  $\mathbb{Z}/(p)$ , and we use the facts  $\mu_p^D = \mathbb{Z}/(p), \alpha_p^D = \alpha_p$  (10.1.2.11)(10.1.2.12). □

**Example (27.10.1.32) [Dieudonné-Modules].**

- $M^*((\mathbb{Q}_p/\mathbb{Z}_p)_\kappa) = W(\kappa)$ .
- $M^*(\mu_{p^\infty, \kappa}) = W(\kappa)(-1)$ , i.e.  $F = p\varphi, V = \text{Ver}$ .
- If  $A \in \mathcal{A}b\mathcal{V}ar/\kappa$  and  $X = A[p^\infty]$ , then there is a canonical isomorphism  $M^*(X) \cong H_{\text{crys}}^1(A/W(\kappa))$ .

┘

*Proof:* 3: ?

□

**Prop. (27.10.1.33) [ $\alpha$ -Group Schemes].** For  $k \in \mathbf{Field}^p$  and  $G \in p\text{-finAbSch}/k$ ,  $G \cong \alpha_p^r$  for some  $r \in \mathbb{Z}_+$  iff

$$F_{G/k} = 0, \text{Ver}_{G/k} = 0.$$

Such a group scheme is called a  **$\alpha$ -group scheme** over  $k$ . ┘

*Proof:* If  $k = k^{\text{perf}}$ , this is easy using Dieudonné-theory (27.10.1.31). In general, we use the fact  $\text{End}_k(\alpha_p) = \mathbb{G}_a$  (10.1.1.8) and the fact  $H^1(\text{Gal}_k, \text{GL}(n)) = 0$ . □

**Def. (27.10.1.34) [ $a$ -Numbers].** For  $k \in \mathbf{Field}^p$  and  $G \in \text{Sh}(\text{Sch}_{\text{fppf}}/k)$  s.t.  $G[p]$  is representable by a f.t.  $K$ -group, then for any  $k''/k'/k \in \mathbf{Field}^p$  s.t.  $k' = (k')^{\text{perf}}, k'' = (k'')^{\text{perf}}$ , there is an isomorphism

$$k'' \otimes_{k'} \text{Hom}((\alpha_p)_{k'}, G_{k'}) \cong \text{Hom}((\alpha_p)_{k''}, G_{k''}),$$

where  $k'$ -vector space structure on  $\text{Hom}((\alpha_p)_{k'}, G_{k'})$  comes from  $\alpha_p$ . So we can define the  **$a$ -number of group scheme** of  $G$  to be

$$a(G) = a(G[p]) = \dim_{\bar{k}} \text{Hom}((\alpha_p)_{k'}, G_{k'}),$$

┘

*Proof:* Using contravariant Dieudonné-theory (10.1.2.27), if  $M = M^*(G[p]_{k'})$ , then by (27.10.1.31) we see that

$$\text{Hom}((\alpha_p)_{k'}, G_{k'}) \cong \text{Hom}(M/(FM + VM), \kappa'),$$

and the assertion clearly follows. □

**Prop. (27.10.1.35) [ $a$ -Numbers of  $p$ -Divisible Groups].** For  $k \in \mathbf{Field}^p$  and  $N \in \text{B-T}_1/k$  (27.10.1.41), we have

$$a(N) = a(N^D).$$

In particular, for  $X \in p\text{Div}/k$ , we have  $a(X) = a(X^D)$  by (27.10.1.42). ┘

*Proof:* We may assume  $\kappa = \kappa^{\text{perf}}$ . Then using contravariant Dieudonné-theory(10.1.2.27), if  $M = M^*(N)$ , then we have

$$a(N) = \dim_{\kappa} M/(FM + VM), \quad a(N^D) = \dim_{\kappa} M[F] \cap M[V],$$

But by definition of  $B\text{-}T_1$ (27.10.1.41), we see that  $M[F] = VM$ ,  $M[V] = FM$ , and there is an exact sequence

$$0 \rightarrow VM/(FM \cap VM) \rightarrow M/FM \rightarrow M/(FM + VM) \rightarrow 0 \in \mathcal{V}\text{ect}/\kappa,$$

so

$$a(N) = \dim M = \dim FM - (\dim VM - a(N^D)) = a(N^D) + \dim M - \dim FM - \dim M[F] = a(N^D).$$

□

**Prop. (27.10.1.36) [ $a$ -Numbers are Upper Semi-Continuous].** For  $S \in \mathcal{S}\text{ch}^p$  and  $G \in \text{finAbSch}/S$  killed by  $[p]$ , the function  $s \mapsto a(G_s)$  is upper semi-continuous??, i.e. for each  $n \in \mathbb{N}$ ,  $\{s \in S | a(G_s) \geq n\}$  is closed in  $S$ . ┘

*Proof:* Cf.[CCO14]P145?.

□

### Isocrystals and Dieudonné-Modules

**Thm. (27.10.1.37) [Dieudonné-Manin].** For each  $s \in \mathbb{Z}_+$  and  $r \in [s]$  s.t.  $(r, s) = 1$ , define the Dieudonné-module

$$M_{r/s} \triangleq D(\mathbb{F}_p)/D(\mathbb{F}_p)(F^{s-r} - V^r)(27.5.5.22),$$

and  $X_{r/s} \in p\text{Div}/\mathbb{F}_p$  s.t.  $D^*(X_{r/s}) = M_{r/s}$ . Then

- $X_{r/s}$  is of dimension  $r$  and height  $s$ .
- $X_{r/s}^D = X_{(s-r)/s}$ .

And for each  $k \in \mathbf{Field}^p$  s.t.  $k = \bar{k}$ ,

- $X_{r/s,k}$  is isogenous to  $X_{r'/s',k}$  iff  $r/s = r'/s'$ ,
- The isogeny category of  $p$ -divisible groups over  $k$  is semi-simple, with simple objects given by these  $X_{r/s,k}$ .

┘

*Proof:* Denote  $e_0 = V^{r-1}$ ,  $e_1 = pV^{r-2}$ ,  $\dots$ ,  $e_r = p^{r-1}$ ,  $e_{r+1} = p^{r-1}F$ ,  $\dots$ ,  $e_{s-1} = p^{r-1}F^{s-r}$ , then  $M_{r/s} = \mathbb{Z}_p\{e_0, p^{-1}e_1, \dots, p^{-r}e_r, p^{-r}e_{r+1}, \dots, p^{-r}e_{s-1}\}$  with

$$Fe_0 = e_1, \dots, Fe_{s-2} = e_{s-1}, Fe_{s-1} = p^r e_0, \quad Ve_0 = p^{-r+1}e_{s-1}, Ve_1 = pe_0, \dots, Ve_{s-1} = pe_{s-2}.$$

So

$$\begin{aligned} M_{r/s}^{\vee} &= \{e_0^{\vee}, pe_1^{\vee}, \dots, p^r e_r^{\vee}, p^r e_{r+1}^{\vee}, \dots, p^r e_{s-1}^{\vee}\} \\ &= \mathbb{Z}_p\{p^r e_r^{\vee}, p^{-1}(p^{r+1}e_{r+1}^{\vee}), \dots, p^{-s+r+1}(p^{s-1}e_{s-1}^{\vee}), p^{-s+r}(p^{s-r}e_0), p^{-s+r}(p^{s-r+1}e_1), \dots, p^{-s+r}(p^{s-1}e_{r-1})\} \end{aligned}$$

with

$$\begin{aligned} F(p^r e_r^{\vee}) &= (p^{r+1}e_{r+1}^{\vee}), \dots, F(p^{s-1}e_{s-1}^{\vee}) = p^{s-r}e_0^{\vee}, F(p^{s-r}e_0^{\vee}) = (p^{s-r+1}e_1^{\vee}), \\ F(p^{s-r+1}e_1^{\vee}) &= (p^{s-r+2}e_2^{\vee}), \dots, F(p^{s-1}e_{r-1}^{\vee}) = p^{s-r}(p^r e_r^{\vee}), \end{aligned}$$

so  $M_{r/s}^{\vee} \cong M_{(s-r)/s}$ .

The rest follows from Dieudonné-theory(27.10.1.28) and(27.5.5.17). □



**Cor. (27.10.1.38)[Slopes and Newton-Polygons].** For any  $k \in \mathbf{Field}^p$  and  $X \in p\mathrm{Div}/k$ , there exists a unique finite subset  $\{\lambda_i\} \subset \mathbb{Q} \cap [0, 1]$  s.t.  $X_{\bar{k}}$  is isogenous to  $\prod_i X_{\lambda_i}^{e_i}$  for some  $e_i \in \mathbb{Z}_+$ . These  $\lambda_i$  are called the **slope of  $p$ -divisible group** of  $X$ . And  $X$  is called an **isoclinic  $p$ -divisible group** if  $\{\lambda_i\}$  is a singleton, equivalently,  $M^*(X_{\bar{k}})$  is an isoclinic (27.5.5.13).

And the lower convex polygon  $\mathrm{NP}(X)$  on the lattice  $\mathbb{Z}^2$  drawn using segments of slope  $\lambda_i = r_i/s_i$  and horizontal length  $e_i s_i$  is called the **Newton-polygon** of  $X$ .  $\lrcorner$

**Cor. (27.10.1.39).** If  $k \in \mathbf{Field}^p$  and  $0 \neq X \in p\mathrm{Div}/k$ , then  $\mathrm{End}(X)$  is a finite free  $\mathbb{Z}_p$ -module.  $\lrcorner$

*Proof:* It suffices to prove for  $k = \bar{k}$  and also assume that  $X = X_\lambda$  for some  $\lambda \in \mathbb{Q} \cap [0, 1]$  (27.10.1.37).

? Cf. [Chai-Oort, Moduli of abelian varieties and  $p$ -divisible groups] or [Demazure, Lectures on  $p$ -divisible Groups].  $\square$

**Thm. (27.10.1.40)[Slope Filtrations, Grothendieck-Zink].** For any  $k \in \mathbf{Field}^p$  and  $X \in p\mathrm{Div}/k$ , there is a filtration

$$0 = Y_0 \leq Y_1 \leq \dots \leq Y_t = X \in p\mathrm{Div}/k$$

s.t.  $Y_i/Y_{i-1}$  is isoclinic of slope  $\lambda_i$  for each  $i \in [t]_+$ , and  $\lambda_1 > \lambda_2 > \dots > \lambda_t$ .  $\lrcorner$

*Proof:* Cf. [Zink, On the slope filtration] Thm7.  $\square$

### Truncated Barsotti-Tate Groups

References are [Displays and formal  $p$ -divisible groups, Lau].

**Def. (27.10.1.41)[Truncated Barsotti-Tate Groups].** For  $n \in \mathbb{Z}_+$ , the category  $\mathrm{B-T}_n/S$  of **truncated Barsotti-Tate group** of level  $n$  over  $S$  is the subcategory of  $p\text{-finAbSch}/S$  consisting of objects  $G$  satisfying necessary (but may not be sufficient) conditions s.t.  $G \cong X[p^n]$  for some  $X \in p\mathrm{Div}/S$ . Cf. [Illusie, Deformation des groupes de Barsotti-Tate] and [Messing, The Crystals associated to Barsotti-Tate Groups] ?.

For example, for  $k \in \mathbf{Field}^p$ ,  $\mathrm{B-T}_1/k$  consists of  $N \in \mathrm{finAbSch}/k$  killed by  $[p]$  s.t. the complex

$$N \xrightarrow{F_{N/k}} N(p) \xrightarrow{\mathrm{Ver}_{N/k}} N$$

is exact. By (10.1.2.27), this is equivalent to

$$M^*(N) \xrightarrow{V} \varphi_* M^*(N) \xrightarrow{F} M^*(N)$$

being exact.  $\lrcorner$

**Prop. (27.10.1.42)[Truncations of  $p$ -Divisible Groups].** For  $k \in \mathbf{Field}^p$  and  $X \in p\mathrm{Div}/k$ , we have  $N = X[p] \in \mathrm{B-T}_1/k$ .  $\lrcorner$

*Proof:* We may base change to  $\bar{k}$ . Then by Dieudonné theory (27.10.1.28),  $V \circ F = p$  on  $M^*(X)$ , which is a finite free  $W(k)$ -module, so  $F, V$  are injective, and there is an exact sequence

$$M^*(X)/(p) \xrightarrow{V} \varphi_* M^*(X)/(p) \xrightarrow{F} M^*(X)/(p),$$

proving  $X[p] \in \mathrm{B-T}_1$  (27.10.1.41).  $\square$

### Hodge-Tate Decomposition

**Prop. (27.10.1.43) [Hodge-Tate Decomposition].** For  $G \in p\text{Div}/\mathcal{O}_K$ , there is an isomorphism of  $\text{Gal}_K$ -representations:

$$\text{Tate}_p(G) \otimes \mathbb{C}_K \cong \text{Lie}(G)(K) \otimes \mathbb{C}_K(1) \bigoplus \text{Lie}(G^*)(K)^* \otimes \mathbb{C}_K(1).$$

┘

*Proof:* Cf. [p-divisible Groups, Morrow].

□

### $p$ -Divisible Groups over $\mathcal{O}_{\mathbb{C}_p}$

.

## 2 Crystalline Dieudonné-Theory

References are [Mes72] and [deJong, crystalline Dieudonné-Theory].

**Notation (27.10.2.1).**

- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-Field}$  with  $e(K) < p$ . In particular,  $(\text{Spec } \kappa \rightarrow \text{Spec } \mathcal{O}_K)$  is a pd-structure (5.6.1.1).
- Let  $S \in \text{NSch}$  s.t.  $p$  is locally nilpotent on  $S$ .

┘

**Def. (27.10.2.2) [Dieudonné-Crystals].** For  $X \in p\text{Div}/S$ , the **Dieudonné-crystal** over  $S$  is the crystalline Ext-sheaf

$$\mathbf{D}(X) \triangleq \mathcal{E}xt^1(X_{\text{Crys}}, \mathcal{O}_{S/\mathcal{O}_K}).$$

Then  $\mathbf{D}(X)$  is a finite locally free crystal over  $S$  (27.5.3.1) of rank  $\text{ht}(X)$ , called the **Dieudonné-crystal** associated to  $X$ , which depends contra-variantly on  $X$ .

And for each  $(T_0, T, \gamma) \in (S/\mathcal{O}_K)_{\text{Crys}}$ , we have  $\mathbf{D}(X)_{(T_0, T)} = \mathbf{D}(X_{T_0})_{(T_0, T)}$ , in particular,  $\mathbf{D}(X)$  only depends on  $X \times_S S_{\text{red}}$ .

┘

*Proof:* ?

□

**Prop. (27.10.2.3) [ $p$ -Divisible Groups and Isocrystals].** If  $S_0 \in \text{Sch}^p$ , then for each  $X \in p\text{Div}/S$ , the relative Frobenius makes  $\mathbf{D}(X)$  a  $F$ -crystal over  $S_0$  (27.5.5.2). And  $X \mapsto \mathbf{D}(X)$  (27.10.2.2) induces an equivalence between  $p\text{Div}^{\text{ét}}/S_0$  and the category of  $F$ -crystals over  $S_0$  s.t.  $\varphi$  is an isomorphism.

┘

*Proof:* Cf. [deJong, crystalline Dieudonné-Theory].

□

**Prop. (27.10.2.4) [Hodge-Filtrations].** For  $X \in p\text{Div}/S$ , there is a canonical locally split exact sequence

$$0 \rightarrow \omega_X \xrightarrow{\triangle} \text{Lie}(X)^\vee \rightarrow \mathbf{D}(X)_S \rightarrow \text{Lie } X^\vee \rightarrow 0 \in \text{Mod}(\mathcal{O}_S),$$

which is compatible with base change, called the **Hodge-filtration** associated to  $X$ .

If  $A \in \text{AbVar}/S$  s.t.  $X = A[p^\infty]$ , then this Hodge-filtration is just the Hodge-filtration

$$0 \rightarrow \omega_A \rightarrow H_{\text{dR}}^1(A/S) \rightarrow \text{Lie}(A^\vee) \rightarrow 0.$$

┘

*Proof:* ?

□

**Def. (27.10.2.5)[Universal Extensions of  $p$ -Divisible Groups].** Suppose  $p$  is locally nilpotent on  $S$  and  $X \in p\text{Div}/S$ , a **universal extension** of  $X$  is an extension of commutative group schemes

$$0 \rightarrow \tilde{V} \rightarrow E \rightarrow X \rightarrow 0 \in \mathcal{A}b\mathcal{S}ch/S$$

where  $V \in \mathcal{Q}Coh(\mathcal{O}_S)$ , with the universal property that: For any vector group  $M \in \mathcal{Q}Coh(\mathcal{O}_S)$ , the natural map

$$\text{Hom}_{\mathcal{O}_S}(V, M) \rightarrow \text{Ext}_S^1(X, \tilde{M})$$

is an isomorphism.

┘

**Thm. (27.10.2.6)[Universal Extensions Exist, Messing, [Mes72]].** Situation as in (27.10.2.5), there exists a universal extension

$$0 \rightarrow \tilde{V}(X) \rightarrow E(X) \rightarrow X \rightarrow 0 \in \mathcal{A}b\mathcal{S}ch/S,$$

with  $V(X) \cong \omega_{G^p}$ . This universal extension is functorial in  $X$ , and commutes with base changes. ┘

*Proof:* Cf. [Mes72] 4.1.10.

□

**Prop. (27.10.2.7)[Crystal in Group Schemes].** Suppose  $p$  is locally nilpotent on  $S$  and  $X \in p\text{Div}/S$ , then Messing defined

- A crystal in group schemes  $\mathbf{E}(X)$ .
- A crystal in formal Lie-groups  $\overline{\mathbf{E}(X)}$ .
- A crystal in finite locally free modules  $\mathbf{D}(X)$ .

which are functorial in  $G$ , using universal extensions of  $p$ -divisible groups (27.10.2.6). ┘

*Proof:* ? Cf. [Mes72] or [Haoran Wang, period morphisms].

□

### 3 Displays and Dieudonné-Theory

References are [Zin01a, Zin02, B-P20].

**Notation (27.10.3.1).**

- Use notations on divided power rings from Divided Power Algebras.
- Let  $\text{Nil}$  be the subcategory of  $\mathcal{C}\text{Ring}$  consisting of rings s.t.  $p$  is nilpotent on it.

┘

**Def. (27.10.3.2)[ $\mathcal{Z}$ -Rings].** Let  $\mathcal{Z}$  be the subcategory of  $\text{Nil}$  consisting of local rings  $(R, \mathfrak{m}, \kappa)$  s.t.  $\kappa$  is perfect and there exists  $N \in \mathbb{Z}_+$  s.t.  $\mathfrak{m}^N = 0$ .

Let  $\tilde{\mathcal{Z}}$  be the subcategory of  $\mathcal{C}\text{Ring}$  consisting of  $p$ -complete separated rings  $A$  s.t.  $A/(p^n) \in \mathcal{Z}$  for any  $n \in \mathbb{Z}_+$ . ┘

### Frames

**Def. (27.10.3.3) [Quasi-Frames].** For  $R \in \text{Nil}$ , a **quasi-frame** over  $R$  is a tuple  $(A, \kappa, \sigma, \sigma_1)$  where

- $A \in \text{TopCRing}$  is a  $p$ -adically complete and separated ring.
  - $\kappa : A \rightarrow W(R)$  is a ring homomorphism s.t.  $\alpha : A \xrightarrow{\kappa} W(R) \rightarrow R$  is surjective with kernel  $J$ .
  - $\varphi : A \rightarrow A$  is a ring homomorphism and  $\varphi_1 : J \rightarrow A$  is a  $\varphi$ -semi-linear map s.t.  $\varphi|_J = p \cdot \varphi_1$ .
- s.t.
- The image of  $J$  in  $A/(p)$  consists of nilpotent elements. In particular,  $J \subset \text{rad}(A)$ .
  - $\kappa(\varphi(a)) = F(\kappa(a))$  for any  $a \in A$ .
  - $\text{Ver}(\kappa(\varphi_1(u))) = \kappa(u)$  for any  $u \in J$ .

┘

**Cor. (27.10.3.4).** If  $R \in \text{Nil}$  and  $A$  is a quasi-frame for  $R$  (27.10.3.3), then any finite flat  $R$ -module can be lifted to a finite flat  $A$ -module.

┘

*Proof:* Cf. [Zin01a]P496. □

**Def. (27.10.3.5) [Frames].** For  $R \in \text{CRing}$  that is  $p$ -adically complete and separated, a **frame for**  $R$  is a triple  $(A, \alpha, \varphi)$  where

- $A \in \text{TopCRing}$  is a  $p$ -complete separated ring that is  $p$ -torsion-free.
- $\alpha : A \rightarrow R$  is a surjection with kernel  $J$  s.t.  $(A, J)$  is a divided power ring (5.6.1.1).
- $\varphi : A \rightarrow A$  is a ring homomorphism inducing the Frobenius on  $A/(p)$ .

┘

**Example (27.10.3.6) [Frames].**

- If  $k \in \text{Field}^p$ ,  $R = k[T_1, \dots, T_d]$ ,  $A = C(k) \langle T_1, \dots, T_d \rangle$  (5.5.2.30)(12.2.1.9), then  $(A, (p), \sigma)$  is a frame for  $R$  where  $\sigma$  is an Frobenius lift of  $C(k)$ .
- (Breuil's Dieudonné-frame) Let  $(R, \varpi, \kappa)$  be a CDVR of characteristic  $(0, p)$  and  $C(\kappa)$  the Cohen ring of  $k$  (5.5.2.30) with a lift of Frobenius  $\sigma$ . Then the minimal polynomial of  $\varpi$  over  $C(\kappa)$  is an Eisenstein polynomial of degree  $e$ . Cf. [Zin01]P496.

┘

**Prop. (27.10.3.7) [Frames and Quasi-Frames].** For  $R \in \text{Nil}$ , there is a functor from the category of frames over  $R$  (27.10.3.5) to the category of quasi-frames over  $R$  (27.10.3.3) mapping  $(A, \alpha, \varphi)$  to  $(A, \kappa, \varphi_1, \varphi)$  defined as follows:

- $A = A$ .
- $\kappa : A \xrightarrow{\delta_A} W(A) \xrightarrow{W(\alpha)} W(R)$ , where  $\delta_A$  is the Cartier-homomorphism (5.5.2.20).
- $\varphi = \varphi$ .
- $\varphi_1 = p^{-1}\varphi|_J$ .

┘

*Proof:* Cf. [Zin01a]Prop1.5. ? □

**Def. (27.10.3.8) [Dieudonné Frames for  $\mathcal{Z}$ -Rings].** For  $R \in \mathcal{Z}$  (27.10.3.2), a **Dieudonné frame** for  $R$  is a frame  $(A, \alpha, \sigma)$  for  $R$  (27.10.3.5) s.t.  $A \in \tilde{\mathcal{Z}}$ , and the Cartier morphism  $A \rightarrow W(A)$  factors through  $\tilde{W}(A)$  ?.

┘

**Example (27.10.3.9) [Dieudonné Frames for  $\mathcal{Z}$ -Rings].** Cf. [Zin01a]P494. ┘

**Displays**

Cf.[The display of a formal  $p$ -divisible group, Zink].

**Def. (27.10.3.10) [Displays, Zink[Zin02]].** For  $R \in \mathcal{CRing}$  that is  $p$ -adically complete and separated, a **display** over  $R$  is a tuple  $\mathcal{P} = (P, Q, F, F_1, V^\sharp)$  where

- $P$  is a finite projective  $W(R)$ -module.
- $Q \subset P$  is a sub- $W(R)$ -module containing  $\text{Ver}(W(R))P$ .
- $F : P \rightarrow P, F_1 : Q \rightarrow P$  are  $\varphi$ -semi-linear maps of  $W(R)$ -modules.

s.t.

- $P/Q$  is a finite projective  $R$ -module.
- $F_1(\text{Ver}(w).x) = wF(x)$  for each  $w \in W(R)$  and  $x \in P$ .
- $P$  is generated by  $F_1(Q)$  as a  $W(R)$ -module.
- $V^\sharp : P \rightarrow W(R) \otimes_{\varphi, W(R)} P$  is a  $W(R)$ -linear homomorphism s.t.

$$V^\sharp(wF(x)) = pw \otimes x, \quad V^\sharp(wF_1(y)) = w \otimes y$$

for any  $w \in W(R), x \in P$  and  $y \in Q$ .

And a display is called a **nilpotent display** if moreover it satisfies the nilpotence condition:

- There exists  $n \in \mathbb{N}$  s.t. the map

$$V^{\sharp(n)} = \varphi^{(n-1)*}(V^\sharp) \circ \varphi^{(n-2)*}(V^\sharp) \circ \dots \circ V^\sharp : P \rightarrow W(R) \otimes_{\varphi^n, W(R)} P$$

is congruent to 0 modulo the ideal  $I(R) + pW(R)$ .

The category of displays over  $R$  is denoted by  $\text{Dsp}/R$ , and the subcategory of nilpotent displays over  $R$  is denoted by  $\text{NilDsp}/R$ .

For  $\mathcal{P} = (P, Q, F, F_1, V^\sharp) \in \text{Dsp}(R)$ ,  $\text{rank}_{W(R)}(P)$  is called the **height of display** of  $\mathcal{P}$ ,  $P/Q$  is called the **Lie-algebra of display** of  $\mathcal{P}$ , and  $\text{rank}_R(P/Q)$  is called the **dimension of display** of  $\mathcal{P}$ .  $\lrcorner$

*Proof:* It is shown in [Zin02]Lemma10 that given  $(P, Q, F, F_1)$ , there exists a unique  $V^\sharp$ .  $\square$

**Remark (27.10.3.11).** For  $\mathcal{P} = (P, Q, F, F_1, V^\sharp) \in \text{Dsp}(R)$  and each  $x \in Q$ , we have

$$F(x) = 1F(x) = F_1(\text{Ver}(1)x) = FV(1)F_1(x) = pF_1(x).$$

$\lrcorner$

**Prop. (27.10.3.12) [Displays and Dieudonné Modules].** If  $R = \kappa \in \mathbf{Field}^p$  s.t.  $\kappa = \kappa^{\text{perf}}$ , a display  $(P, Q, F, F_1) \in \text{Dsp}/\kappa$  (27.10.3.10) is the same as a Dieudonné module  $(P, F, V)$  over  $\kappa$  (27.5.5.23) where  $Q = V(P)$  and  $F_1 = V^{-1}$ . And the display is nilpotent iff  $V$  is topologically nilpotent.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (27.10.3.13) [Nilpotent Displays and Formal  $p$ -Divisible Groups, Zink[Zin02]/Lau[Lau08]].** Situation as in (27.10.3.10), there is a functor

$$\text{B-T} : \text{NilDsp}/R \rightarrow p\text{Div}^{\text{frml}}/R,$$

s.t.

- For  $\mathcal{P} = (P, Q, F, F_1) \in \text{Dsp}/R$ , we have  $\text{Lie}(\text{B-T}(\mathcal{P})) = P/Q$ .  
s.t.  $\text{ht}(\text{B-T}(P)) = \text{rank}(P)$ . The inverse is given by

$$\mathbf{D} : p\text{Div}^{\text{frml}}/R \rightarrow \text{NilDsp}/R : ?$$

In particular, by (27.10.3.12), for  $\kappa \in \mathbf{Field}^p$  s.t.  $\kappa = \kappa^{\text{sep}}$ , there is an equivalence of categories:

$$\text{B-T} : (\text{Nilpotent})\text{Dieudonné modules}/\kappa \cong p\text{Div}^{\text{frml}}/\kappa.$$

⌋

*Proof:* ? Cf. [?zin02]Thm81 where it is proved that B-T is faithful, and fully faithful if  $\text{rad}(R)$  is nilpotent. And [Displays and formal p-divisible groups, Lau] proved the equivalence. ? . □

### Windows

References are [Zink, Windows for displays of p-divisible groups].

**Def. (27.10.3.14) [A-Windows].** If  $R \in \text{Nil}(27.10.3.1)$  and  $(A, J, \kappa)$  is a quasi-frame for  $R(27.10.3.3)$ , a **Dieudonné A-window** over  $R$  is a tuple  $\mathcal{M} = (M, M_1, \Phi, \Phi_1, \Psi)$  where

- $M$  is a finite projective  $A$ -module.
- $M_1 \subset M$  is a  $A$ -submodule containing  $JM$ .
- $\Phi : M \rightarrow M, \Phi_1 : M_1 \rightarrow M$  are two  $\sigma$ -semi-linear maps.

s.t.

- $M/M_1$  is a finite flat  $R$ -module.
- 

$$\Phi(m_1) = p\Phi_1(m_1), \text{ where } m_1 \in M_1.$$

$$\Phi_1(um) = \sigma_1(u)\Phi(m), \text{ where } u \in J, m \in M.$$

- The union of  $\Phi_1(M_1)$  and  $\Phi(M)$  generates  $M$  as an  $A$ -module.

And a Dieudonné  $A$ -window is called a **A-window** if moreover it satisfies the nilpotence condition:

- For the unique  $\sigma$ -semi-linear map  $\Psi : M \rightarrow M$  s.t.

$$\Psi(\Phi_1(m_1)) = m_1, \text{ where } m_1 \in M_1, \quad \Psi(\Phi(m)) = pm, \text{ where } m \in M, ?$$

we have  $\Psi^N(M) \subset JM$  for some  $N \in \mathbb{Z}_+$ .

The category of  $A$ -windows is denoted by  $\text{Win}/A$ . It is an exact category, where the exact sequences are exact sequences of windows  $M$  inducing exact sequence on  $M_1$ . ⌋

**Cor. (27.10.3.15) [A-Windows for Frames].** If  $R \in \mathbb{C}\text{Ring}$  is  $p$ -adically complete and separated, and  $(A, \alpha, \varphi)$  is a frame for  $R(27.10.3.5)$ , then an  $A$ -window over  $R(27.10.3.14)$  is equivalent to a triple  $(M, M_1, \Phi)$  where

- $M \in \text{Mod}_A$  is a finite projective  $A$ -module.
- $M_1 \subset M$  is a sub- $A$ -module containing  $JM$ .
- $\Phi : M \rightarrow M$  is a  $\varphi$ -semi-linear map.

s.t.

- $M/M_1$  is a finite projective  $R$ -module.

- $\Phi(M_1) \subset pM$  and  $M$  is generated by  $\Phi(M)$  and  $p^{-1}\Phi(M_1)$  as an  $A$ -module.
- The last condition implies that  $1 \otimes \Phi : \varphi_* M \rightarrow M$  contains  $pM$ , thus it is injective, and we can define  $\Psi = M \rightarrow pM \xrightarrow{(1 \otimes \Phi)^{-1}} \varphi_* M$ , which is the unique  $\sigma$ -semi-linear map s.t.  $\Psi(\Phi(m)) = pm$  for any  $m \in M$ . Then we require  $\Psi^N(M) \subset (p, J)M$  for some  $N \in \mathbb{Z}_+$ .

┘

*Proof:* This is because  $\Phi_1$  is determined by  $\Phi$  and  $\sigma_1$  is determined by  $\sigma$ , as  $A$  is torsion-free (27.10.3.5). □

**Remark (27.10.3.16).** Situation as in (27.10.3.15), if  $(A, \alpha, \varphi)$  satisfies  $\varphi(J)$  generates  $pA$  as  $A$ -modules, then condition 2 can be simplified to requiring that  $p^{-1}\Phi(M_1)$  generates  $M$ , because  $A\Phi(M) \subset (p^{-1}\varphi(J))\Phi(M)$ , which is contained in  $p^{-1}\Phi(M_1)$ . ┘

**Thm. (27.10.3.17) [A-Windows and Displays].** If  $R \in \mathcal{CRing}$  is  $p$ -adically complete and separated and  $A$  is a quasi-frame for  $R$  (27.10.3.3), then there is a exact functor

$$\text{Dsp} : \text{Win}/A \rightarrow \text{Dsp}/R \text{ (27.10.3.14) (27.10.3.10)}$$

mapping  $(M, M_1, \Phi, \Phi_1, \Psi) \in \text{Win}/A$  to  $(P, Q, F, F_1, V^\sharp) \in \text{Dsp}/R$  where

- $P = W(R) \otimes_{\kappa, A} M$ .
- $Q = \ker \left( P = W(R) \otimes_{\kappa, A} M \xrightarrow{w_0 \otimes 1} R \otimes_{\alpha, A} M \rightarrow M/M_1 \right)$ .
- $F_1 : Q \rightarrow P$  is the unique functor s.t.

$$F_1(x \otimes m_1) = \varphi(x)\Phi_1(m_1), \quad F_1(\text{Ver}(x) \otimes m) = x \otimes \Phi(m)$$

for each  $x \in W(R), m_1 \in M_1, m \in M$ .

- $V^\sharp : W(R) \otimes_{\kappa, A} M \rightarrow W(R) \otimes_{\varphi, W(R)} W(R) \otimes_{\kappa, A} M$  is the unique  $W(R)$ -linear map s.t.

$$V^\sharp(w \otimes \Phi(m)) = pw \otimes 1 \otimes m, \quad V^\sharp(w \otimes \Phi_1(m_1)) = w \otimes 1 \otimes m_1$$

for all  $w \in W(R), m \in M, m_1 \in M_1$ , which exists by [Zin02]Lemma10.

If  $A$  is a frame for  $R$ , then we consider  $\text{Dsp}$  for the quasi-frame associated to  $A$  (27.10.3.7), then  $\text{Dsp}$  is an equivalence of categories, with inverse given by

$$A\text{-Win} : \mathcal{P} \in \text{Dsp}/R \mapsto (M = \mathbf{D}_{\mathcal{P}}(A), M_1, \Phi)$$

where  $\mathbf{D}_{\mathcal{P}}$  is the Dieudonné crystal associated to  $\mathcal{P}$  (Cf. [Zin02]Def2.6). ┘

*Proof:* The existence of  $F_1$  is proved in [Zin01a]P498. ?

The equivalence is proved in [Zin01a]Thm1.6. ? □

**Cor. (27.10.3.18) [Formal  $p$ -Divisible Groups and  $A$ -Windows].** Situation as in (27.10.3.17), there is an equivalence of exact categories

$$G \stackrel{\triangle}{=} \text{B-T} \circ \text{Dsp} : \text{Win}/A \rightarrow p\text{Div}^{\text{frml}}/R \text{ (27.10.3.17) (27.10.3.13)}$$

with inverse given by the covariant Dieudonné functor

$$\mathbf{D} = A\text{-Win} \circ \mathbf{D} : p\text{Div}^{\text{frml}}/R \cong A\text{-Windows}/R : X \mapsto \mathbf{D}(X)(A) = \varprojlim_n \mathbf{D}(X)(A/(p^n)) \text{ (27.10.3.17) (27.10.3.13)}.$$

s.t.

- If  $R' \in \mathcal{CRing}/R$  is  $p$ -adically complete and separated, and  $A'$  is a frame over  $R'$ , and there is a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & R \\ \downarrow & & \downarrow \\ A' & \longrightarrow & R' \end{array}$$

s.t.  $A \rightarrow A'$  is compatible with  $\varphi$ , then for any  $\mathcal{M} \in \text{Win}/A$ , there is a canonical isomorphism

$$G(\mathcal{M}) \otimes_R R' \cong G'(\mathcal{M} \otimes_A A') \in p\text{Div}^{\text{frml}}/R'.$$

┘

*Proof:* ? The equivalence follows from [Zin01a]Cor1.7. [Lau] and [Zin02]Thm94, Thm97.

The exactness follows from [Kisin, Modularity of 2-adic Barsotti-Tate representations]P1.1.4. ? and the natural isomorphism  $\mathbf{D}(G(\mathcal{M})) \cong \mathcal{M}$ .

The base change compatibility follows from the commutative diagram of Frobenius-compatible maps

$$\begin{array}{ccccc} A & \xrightarrow{\delta} & W(A) & \longrightarrow & W(R) \\ \downarrow & & \downarrow & & \downarrow \\ A' & \xrightarrow{\delta} & W(A') & \longrightarrow & W(R') \end{array} \quad (5.5.2.20)$$

and [Zin02]Cor86 ?.

□

**Cor. (27.10.3.19) [Examples of Windows].** Let  $R \in \mathcal{CRing}$  be a  $p$ -adically complete and separated.

- If  $(A, J, \alpha, \varphi)$  is a frame for  $R$  (27.10.3.5), then  $\mathcal{M} = (M, M_1, \Phi) = (A, J, \varphi)$  is a Window over  $A$ , with  $G(\mathcal{M}) = \mathbb{G}_m[p^\infty]$ .
- If  $R \in \mathcal{CRing}$  is  $p$ -torsion-free, then  $(B, J, \varphi)$  is a frame for  $R$  ?, and then the subcategory of  $p\text{Div}/S$  consisting of objects with formal special fibers is equivalent to  $B\text{-Win}/S$ .

┘

*Proof:* 1: By base change compatibility, it suffices to show this for  $A = R = \mathbb{Z}_p$ , and by (27.10.3.18) it suffices to check using classical Dieudonné theory ?.

2: Cf. [Zin01a]Prop1.8 ?.

□

**Thm. (27.10.3.20) [A-Windows and Displays over  $\mathcal{Z}$ -Rings].** If  $(R, \mathfrak{m}) \in \mathcal{Z}$  (27.10.3.2) and  $A \rightarrow R$  is a Dieudonné frame (27.10.3.8), then there is an equivalence DD from the category of Dieudonné  $A$ -windows to the category of Dieudonné displays.

┘

*Proof:* Cf. [Zin01a]Thm3.2. ?

□

**Cor. (27.10.3.21) [ $p$ -Divisible Groups and  $A$ -Windows over  $\mathcal{Z}$ -Rings].** If  $(R, \mathfrak{m}) \in \mathcal{Z}$  (27.10.3.2) with  $\mathfrak{m}$  nilpotent and  $(A, J, \sigma)$  is a Dieudonné frame for  $R$  (27.10.3.8), then the category of Dieudonné  $A$ -windows over  $R$  is equivalent to  $p\text{Div}/R$ .

In particular, if  $\kappa \in \mathbf{Field}^p$  is perfect, then

$$\mathbf{D} : p\text{Div}/\kappa \cong \text{Dieudonné modules}/W(\kappa) : X \mapsto \mathbf{D}(X).$$

(covariant Dieudonné modules).

┘

*Proof:* Cf. [Zin01a]Thm6. ?

□



#### 4 Deformations of $p$ -Divisible Groups

**Prop. (27.10.4.1) [Base Change of  $p$ -Divisible Groups].** Let  $(\mathcal{O}_K, \mathfrak{m}_K, \kappa) \in \text{locCRing}^{(0,p)}$ , the functor  $p\text{Div}/\mathcal{O}_K \rightarrow p\text{Div}/\kappa$  is faithful.  $\lrcorner$

*Proof:* Cf. [CCO14]P45?  $\square$

**Remark (27.10.4.2).** The Noetherian hypothesis is necessary, for example when  $\mathcal{O}_K = \mathcal{O}_{\mathbb{Q}(\mu_{p^\infty})}$ ,  $f : \mathbb{Q}_p/\mathbb{Z}_p \rightarrow \mu_{p^\infty} : p^{-n} \rightarrow \xi_{p^n}$  is an isomorphism on the generic fiber but trivial on the special fiber.  $\lrcorner$

**Thm. (27.10.4.3) [Serre-Tate, Katz [Kat81]].** Suppose  $p$  is locally nilpotent on  $S$ , denote  $\text{Def}_{p\text{Div}}(S)$  the category of triples  $(\bar{A}, X, \epsilon)$  where

- $\bar{A} \in \text{AbVar}/S_{\text{red}}$ .
- $X \in p\text{Div}/S$ .
- $\epsilon : \bar{A}[p^\infty] \cong \Lambda_S S_{\text{red}} \in p\text{Div}/S_{\text{red}}$ .

Then the functor

$$\text{AbVar}/S \rightarrow \text{Def}_{p\text{Div}}(S) : A \mapsto (\bar{A} = A_{\text{red}}, X = A[p^\infty], \epsilon : A_{\text{red}}[p^\infty] \cong A[p^\infty] \times_S S_{\text{red}})$$

is an equivalence of categories.  $\lrcorner$

**Prop. (27.10.4.4) [Barsotti-Tate Groups are Locally Infinitesimally Liftable].** Suppose  $p$  is locally nilpotent on  $S$ , and  $X \in p\text{Div}/S$ , then there exists a open covering  $\{U_i \rightarrow S\}$  of  $S$  s.t. for each  $i$  and each nilpotent thickening  $U_i \rightarrow U$ , there exists a  $\tilde{X} \in p\text{Div}/U$  s.t.  $\tilde{X} \times_U U_i \cong X \times_S U_i$ .  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (27.10.4.5) [Grothendieck-Messing, Messing [Mes72]].** If  $p$  is locally nilpotent on  $S$ ,  $(S_0 \rightarrow S)$  is a nilpotent pd-thickening and  $X_0 \in p\text{Div}/S_0$ , a sub- $\mathcal{O}_S$ -module  $\text{Fil}^1 \subset \mathbf{D}(X_0)_{(S_0 \rightarrow S)}$  is called an **admissible submodule** if it is locally a direct factor, and it reduces to  $\omega_X \subset \mathbf{D}(X)_{S_0}$  (27.10.2.4). And an **admissible pair** is a pair  $(X_0, \text{Fil}^1)$  where  $X_0 \in p\text{Div}/S_0$  and  $\text{Fil}^1$  is an admissible submodule of  $\mathbf{D}(X_0)_{(S_0 \rightarrow S)}$ .

Then the functor

$$X \mapsto (X_0 = X \times_S S_0, \omega_X)$$

is an equivalence between the category  $p\text{Div}/S$  and the category of admissible pairs  $(X_0, \text{Fil}^1)$ .

In particular, by passing to limits, we get similar equivalence for  $(S_0 \rightarrow S)$  where  $S$  is a formal adic scheme.  $\lrcorner$

*Proof:* ?  $\square$

The faithfulness follows from the rigidity property (27.10.1.24).  $\square$

**Thm. (27.10.4.6) [Deformation Space of  $p$ -Divisible Groups].** For a complete local noetherian ring  $(\Lambda, \kappa)$  and  $X_0 \in p\text{Div}/\kappa$  with height  $h$  and dimension  $d$ , let  $\mathcal{C}_\Lambda$  be the category of Artinian local  $\Lambda$ -rings  $R$  with residue field  $\kappa$ , then we have the **deformation functor of  $p$ -divisible groups**

$$\text{Def}_\Lambda(X_0) : \mathcal{C}_\Lambda \rightarrow \text{Set} : (R, \kappa) \mapsto \{\text{deformations of } X_0 \text{ to } R\},$$

then  $\text{Def}_\Lambda(X_0)$  is pro-represented by a power series ring over  $\Lambda$  in  $d(h-d)$ -variables. And the tangent space of this functor is canonically isomorphic to  $\text{Lie}(X_0^D) \otimes_\kappa \text{Lie}(X_0)$  as a  $\kappa$ -vector space.  $\lrcorner$

*Proof:* Cf. [CCO14]P49?  $\square$

### Infinitesimal Deformations

**Def. (27.10.4.7) [Pointed Formal Lie-Varieties and Formal Lie-Groups].** A **pointed formal Lie-variety** over  $S$  is a fppf-sheaf  $X \in \mathbf{Sh}^{\text{Set}}(\mathbf{Sch}_{\text{fppf}}/S)$  together with a section  $\epsilon : \mathbf{pt}_S \rightarrow X \in \mathbf{Sh}(\mathbf{Sch}_{\text{fppf}}/S)$  s.t. Zariski-locally we have

$$X \cong \varinjlim_{n \in \mathbb{Z}_+} \text{Spec } \mathcal{O}_X[[T_1, \dots, T_r]] / (T_1, \dots, T_r)^n$$

for some  $r \in \mathbb{Z}_+$ , and  $\epsilon$  is given by  $(T_i \mapsto 0)$ .

A **formal Lie-group** over  $S$  is a fppf-sheaf  $X \in \mathbf{Sh}(\mathbf{Sch}_{\text{fppf}}/S)$  with the unit section which makes it a pointed formal Lie-variety over  $S$ .  $\lrcorner$

**Prop. (27.10.4.8) [Formal Lie Groups associated to Barsotti-Tate Groups].** Suppose  $p$  is locally nilpotent on  $S$ , and  $X \in p\text{Div}/S$ , then  $\overline{X} \triangleq \varinjlim_k \text{Inf}^k(X) \subset X$  is a formal Lie-group over  $S$  (27.10.4.7), called the **formal Lie-group** associated to  $X$ . Then  $\overline{X}$  is functorial in  $X$  and stable under base changes.  $\lrcorner$

**Def. (27.10.4.9) [ $p$ -Divisible Formal Groups].** A  **$p$ -divisible formal group** over  $S$  is a formal group  $G$  over  $S$  (27.10.4.7) s.t.  $[p] : G \rightarrow G$  is an isogeny (10.7.6.1). Equivalently, if  $[p] : G \rightarrow G$  is Zariski-locally represented by formal power series

$$f_1(\mathbf{X}), \dots, f_n(\mathbf{X}) \in R[[\mathbf{X}]]$$

s.t.  $f_i(\mathbf{0}) = 0$  for each  $i \in [n]_+$ . Then by (10.7.6.1),  $G$  is a  $p$ -divisible formal group if each  $X_i$  is nilpotent in

$$R[[\mathbf{X}]] / (\{f_i(\mathbf{X})\}_{i \in [n]_+}).$$

The base change of a  $p$ -divisible formal group is also a  $p$ -divisible formal group.  $\lrcorner$

**Thm. (27.10.4.10) [Weierstrass Preparation Theorem].** If  $G$  is a  $p$ -divisible formal group over  $R$ , then  $[p]_G^* : R[[\mathbf{X}]] \rightarrow R[[\mathbf{X}]]$  is finite and faithfully flat, and  $\ker([p]_G)$  is finite locally free over  $R$ .  $\lrcorner$

*Proof:* ?. Cf. [Zin84]P88  $\square$

**Prop. (27.10.4.11).** For  $k \in \mathbf{Field}^p$ , if  $G$  is a formal group law of dimension 1 over  $k$  s.t.  $[p]_G \neq 0$ , then  $G$  is a  $p$ -divisible formal group.  $\lrcorner$

*Proof:* Clear by definition (27.10.4.9).  $\square$

**Def. (27.10.4.12) [Heights of  $p$ -Divisible Groups].** If  $R \in \mathcal{CRing}/\mathbb{Z}_{(p)}$  and  $G$  is a  $p$ -divisible formal group over  $R$ , then  $G$  is said to be of height  $h$  if  $\varphi_{K/(p)}$  is of height  $p$  (10.7.6.2).  $\lrcorner$

**Prop. (27.10.4.13).** If  $R \in \mathcal{CRing}/\mathbb{Z}_{(p)}$  and  $G$  is a  $p$ -divisible formal group over  $R$ . If there is an isogeny between  $G_1$  and  $G_2$ , then  $\text{ht}(G_1) = \text{ht}(G_2)$ , by (10.7.6.3).  $\lrcorner$

**Prop. (27.10.4.14) [Rigidity of  $p$ -Divisible Groups].** If  $R \in \mathcal{CRing}/\mathbb{Z}_{(p)}$ ,  $\mathfrak{a} \in \text{Ideal}(R)$  is nilpotent,  $R_1 = R/\mathfrak{a}$ , and  $G_1, G_2$  are formal groups over  $R$  s.t.  $G_1$  is  $p$ -divisible, then  $\text{Hom}(G_1, G_2)$  is torsion-free and

$$\text{Hom}(G_1, G_2) \otimes \mathbb{Q} = \text{Hom}(G_{1,R_1}, G_{2,R_1}) \otimes \mathbb{Q}.$$

$\lrcorner$

*Proof:* Cf. [Zin02]P100. ?  $\square$

## 5 Breuil-Kisin Modules and $p$ -Divisible Groups

References are [CCO14] and [Breuil, C.: Groupes  $p$ -divisibles, groupes finis et modules filtrés], [Kisin, crystalline representations and F-crystal], [Kisin, Modularity of 2-adic Barsotti-Tate representations].

**Notation (27.10.5.1).**

- Let  $\kappa \in \text{Perf}^p$ .
- Let  $K_0 = W(\kappa)[\frac{1}{p}]$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa)$  be a totally ramified finite extension of  $K_0$ .
- Let  $E(u)$  be the minimal polynomial of  $\varpi_K$  over  $W(\kappa)$ , which is a Eisenstein polynomial.
- Let  $e = e(K/K_0) = \deg(E(u))$ .

┘

**Def. (27.10.5.2) [Breuil-Kisin Modules].** Let  $\mathfrak{S} \triangleq W(\kappa)[[u]]$ , called the **Breuil-Kisin module**, and let  $\varphi : \mathfrak{S} \rightarrow \mathfrak{S}$  be the local endomorphism s.t.  $\varphi(u) = u^p$  and  $\varphi|_{W(\kappa)} = \text{id}$ .

Let  $S$  be the  $p$ -adic completion of the subring

$$W(\kappa)[u] \left[ \left\{ \frac{E(u)^n}{n!} \right\}_{n \in \mathbb{Z}_+} \right] \subset K_0[u].$$

Then there is a natural inclusion  $\mathfrak{S} \hookrightarrow S$ , and  $\varphi_{\mathfrak{S}}$  extends uniquely to a local endomorphism of  $S$ .<sup>?</sup>

Let  $\text{Fil}^1(S)$  be the kernel of the surjective  $W(\kappa)$ -linear local homomorphism  $\epsilon : S \rightarrow \mathcal{O}_K$  s.t.  $\epsilon(u) = \varpi_K$ , which is the  $p$ -adic completion of the ideal of  $S$  generated by the set  $\{E(u)^n/n!\}_{n \in \mathbb{Z}_+}$ . Then  $(S, \text{Fil}^1(S))$  is a frame for  $\mathcal{O}_K$  (27.10.3.5), called **Breuil's Dieudonné frame**.

Notice  $\varphi(E(u))/p$  is invertible in  $S$  because

$$\varphi(E(u))/p - \text{Coef}(E(u))/p \in \text{Fil}^1(S) + uW(\kappa)[u] + pW(\kappa)[u].$$

┘

**Def. (27.10.5.3) [B-T $_{\mathfrak{S}}^{\varphi, f}$ ].** B-T $_{\mathfrak{S}}^{\varphi, f}$  is defined to be the additive category of triples  $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \psi_{\mathfrak{M}})$  where

- $\mathfrak{M}$  is a finite free  $\mathfrak{S}$ -module.
- $\varphi_{\mathfrak{M}}$  is a  $\varphi$ -semi-linear endomorphism of  $\mathfrak{M}$  s.t.  $\text{Im} \left( \varphi_* \mathfrak{M} \xrightarrow{1 \otimes \varphi_{\mathfrak{M}}} \mathfrak{M} \right)$  contains  $E(u)\mathfrak{M}$ .
- The hypothesis on  $\varphi_{\mathfrak{M}}$  implies that we have a map  $E(u)\mathfrak{M} \xrightarrow{\varphi_{\mathfrak{M}}^{-1}} \varphi_* \mathfrak{M}$ , and  $\psi_{\mathfrak{M}}$  is defined to be  $\psi_{\mathfrak{M}} : \mathfrak{M} \xrightarrow{E(u) \cdot} E(u)\mathfrak{M} \xrightarrow{\varphi_{\mathfrak{M}}^{-1}} \varphi_* \mathfrak{M}$ .

And  $\mathfrak{M}$  is called a **formal Breuil-Kisin module** if there exists some  $n \in \mathbb{Z}_+$  s.t.

$$\psi_{\mathfrak{M}}^{(n)} = (\varphi^{(n-1)*} \psi_{\mathfrak{M}}) \circ (\varphi^{(n-2)*} \psi_{\mathfrak{M}}) \circ \dots \circ \psi_{\mathfrak{M}} : \mathfrak{M} \rightarrow \mathfrak{S} \otimes_{\varphi^{(n)*}, \mathfrak{S}} \mathfrak{M}$$

is congruent to 0 modulo the ideal  $(p, u)$ . The category of formal Breuil-Kisin modules over  $\mathfrak{S}$  is denoted by B-T $_{\mathfrak{S}}^{\varphi, f}$ .

And  $\mathfrak{M}$  is called an **étale Breuil-Kisin module** (resp. **multiplicative Breuil-Kisin module**) if  $\text{Im} \left( \varphi_* \mathfrak{M} \xrightarrow{1 \otimes \varphi_{\mathfrak{M}}} \mathfrak{M} \right) = E(u)\mathfrak{M}$  (resp.  $\mathfrak{M}$ ).<sup>?</sup>

┘

### Breuil-Kisin Modules and $p$ -Divisible Groups

**Thm. (27.10.5.4)** [Formal Breuil-Kisin Modules and  $S$ -Windows]. There is an functor

$$S\text{-Win} : \mathbf{B}\text{-}\mathbf{T}_{/\mathfrak{S}}^{\varphi, f} \rightarrow \mathbf{Win}/S \text{ (27.10.5.3)(27.10.3.15)}$$

that is exact and preserves exactness, constructed as follows: for  $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \psi_{\mathfrak{M}}) \in \mathbf{B}\text{-}\mathbf{T}_{/\mathfrak{S}}^{\varphi, f}$ , define

- $\mathcal{M} = S \otimes_{\varphi, \mathfrak{S}} \mathfrak{M}$ ,
- and define  $\mathcal{M}_1 \subset \mathcal{M}$  as the preimage of  $\mathrm{Fil}^1(S) \otimes_{\mathfrak{S}} \mathfrak{M}$  under the map  $1 \otimes \varphi_{\mathfrak{M}} : \mathcal{M} = S \otimes_{\varphi, \mathfrak{S}} \mathfrak{M} \rightarrow S \otimes_{\mathfrak{S}} \mathfrak{M}$ ,
- And define  $\Phi : \mathcal{M} \rightarrow \mathcal{M}$  to be the  $\varphi$ -semi-linear map that maps  $s \otimes m$  to  $\varphi(s) \otimes \varphi_{\mathfrak{M}}(m)$ .

Then  $S\text{-Win}(\mathfrak{M}) = (\mathcal{M}, \mathcal{M}_1, \Phi)$ , and we have

$$\mathcal{M}/\mathcal{M}_1 \cong (1 \otimes \varphi_{\mathfrak{M}})(\varphi_* \mathfrak{M})/E(u)\mathfrak{M}.$$

┘

*Proof:* Firstly,  $\mathcal{M}/\mathcal{M}_1 \cong (1 \otimes \varphi_{\mathfrak{M}})(\varphi_* \mathfrak{M})/E(u)\mathfrak{M}$  by [Kisin, Modularity and moduli of finite flat group schemes]1.1.15<sup>?</sup>, so the quotient is a finite free  $\mathcal{O}_K$ -module.

To check  $\Phi(\mathcal{M}_1) \subset p\mathcal{M}$ , Cf. [Kis09c]P1.2.5.<sup>?</sup>

□

**Thm. (27.10.5.5).** There are equivalences of categories:

$$\begin{array}{ccccc} & & p\mathrm{Div}_{\mathrm{Kis}} & & \\ & \nearrow & & \searrow & \\ \mathbf{B}\text{-}\mathbf{T}_{/\mathfrak{S}}^{\varphi, f} & \xrightarrow{S\text{-Win}, \cong} & \mathbf{Win}/S \cong & \xrightarrow{\mathbf{B}\text{-}\mathbf{T} \circ \mathrm{Dsp}, \cong} & p\mathrm{Div}^{\mathrm{frml}}/\mathcal{O}_K \text{ (27.10.3.18)(27.10.5.4).} \end{array}$$

And when  $p \geq 3$ , there are equivalences of categories:

$$\begin{array}{ccccc} & & p\mathrm{Div}_{\mathrm{Kis}} & & \\ & \nearrow & & \searrow & \\ \mathbf{B}\text{-}\mathbf{T}_{/\mathfrak{S}}^{\varphi} & \xrightarrow{S\text{-Win}} & \mathbf{Win}/S \cong & \xrightarrow{\mathbf{B}\text{-}\mathbf{T} \circ \mathrm{Dsp}} & p\mathrm{Div}/\mathcal{O}_K \text{ (27.10.5.4)} \end{array}$$

┘

*Proof:* Cf. [Kisin, Modularity of 2-adic Barsotti-Tate representations]1.2.8.<sup>?</sup> and [Kisin, crystalline representations and F-crystals].

□

**Cor. (27.10.5.6).** There is a commutative diagram of functors

$$\begin{array}{ccccccc}
 & & & & p\mathrm{Div}_{\mathrm{Kis}}, \cong & & \\
 & & & & \curvearrowright & & \\
 \mathrm{B-T}_{/\mathfrak{S}}^{\varphi, f} & \xrightarrow{S\text{-Win}, \cong} & \mathrm{Win}/S & \xrightarrow{\mathrm{Dsp}} & \mathrm{Dsp}/\mathcal{O}_K & \xrightarrow{\mathrm{B-T}, \cong} & p\mathrm{Div}^{\mathrm{frml}}/\mathcal{O}_K \\
 & \searrow \mathrm{CF}_{\mathrm{Dsp}} & & \downarrow \mathrm{BC}_{\mathrm{Dsp}} & & & \downarrow -\otimes \kappa \\
 & & & \mathrm{Dsp}/\kappa & \xrightarrow{\mathrm{B-T}_{\kappa}, \cong} & & p\mathrm{Div}^{\mathrm{frml}}/\kappa \\
 & \searrow \mathrm{CF}_{p\mathrm{Div}} & & & & & \\
 & & & & \curvearrowleft & & 
 \end{array}$$

┘

### Modules over $\mathfrak{S}$ and Commutative Finite $p$ -Groups

**Def. (27.10.5.7)**  $[\mathrm{Mod}_{\mathfrak{S}}^c]$ .  $\mathrm{Mod}_{\mathfrak{S}}^c$  is defined to be the additive category of triple  $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \psi_{\mathfrak{M}})$  where

- $\mathfrak{M} \in \mathrm{Mod}_{\mathfrak{S}}^{\mathrm{fg}}$  killed by a power of  $p$ , and having projective dimension  $\leq 1$  over  $\mathfrak{S}$ .
- $\varphi_{\mathfrak{M}}$  is a  $\varphi$ -semi-linear endomorphism of  $\mathfrak{M}$  s.t.  $\mathrm{Im} \left( \varphi_* \mathfrak{M} \xrightarrow{\varphi_{\mathfrak{M}}} \mathfrak{M} \right)$  contains  $E(u)\mathfrak{M}$ .
- The hypothesis on  $\varphi_{\mathfrak{M}}$  implies that we have a map  $E(u)\mathfrak{M} \xrightarrow{\varphi_{\mathfrak{M}}^{-1}} \varphi_* \mathfrak{M}$ , and  $\psi_{\mathfrak{M}}$  is defined to be  $\psi_{\mathfrak{M}} : \mathfrak{M} \xrightarrow{E(u)\cdot} E(u)\mathfrak{M} \xrightarrow{\varphi_{\mathfrak{M}}^{-1}} \varphi_* \mathfrak{M}$ .
- And  $\mathfrak{M}$  is called a **connected  $\mathfrak{S}$ -module** if there exists some  $n \in \mathbb{Z}_+$  s.t.

$$\psi_{\mathfrak{M}}^{(n)} = (\varphi^{(n-1)*} \psi_{\mathfrak{M}}) \circ (\varphi^{(n-2)*} \psi_{\mathfrak{M}}) \circ \dots \circ \psi_{\mathfrak{M}} : \mathfrak{M} \rightarrow \mathfrak{S} \otimes_{\varphi^{(n)}, \mathfrak{S}} \mathfrak{M}$$

is congruent to 0 modulo the ideal  $(p, u)$ .

┘

**Thm. (27.10.5.8)**  $[\mathrm{Mod}_{\mathfrak{S}}^c \text{ and Connected Commutative Finite } p\text{-Groups}]$ . There is an equivalence of categories

$$\mathcal{G}_{\mathrm{Kis}} : \mathrm{Mod}_{\mathfrak{S}}^c \cong p\text{-GrpSch}_{\mathrm{cntd}}^{\mathrm{fin}}/\mathcal{O}_K \text{ (27.10.5.7),}$$

where  $p\text{-GrpSch}_{\mathrm{cntd}}^{\mathrm{fin}}/\mathcal{O}_K$  is category of connected commutative finite flat group schemes of  $p$ -power order.

And the functor

$$\mathrm{MCF} = \mathrm{M}_* \circ (- \otimes \kappa) \circ \mathcal{G}_{\mathrm{Kis}} : \mathrm{Mod}_{\mathfrak{S}}^c \rightarrow \mathrm{Mod}_{D(\kappa)}^c \text{ (10.1.2.27),}$$

where  $\mathrm{Mod}_{D(\kappa)}^c$  is the Abelian category of  $D(\kappa)$ -modules of finite  $W(\kappa)$ -length, can be explicitly described in [CCO14] Appendix B.4.17. ?

┘

*Proof:* Cf. [Kisin, Modularity of 2-adic Barsotti-Tate representations] P361. □

**Thm. (27.10.5.9)** [**Mod $_{\mathfrak{S}}$  and Commutative Finite  $p$ -Groups**]. When  $p \geq 3$ , there is an equivalence of categories

$$\mathcal{G}_{\text{Kis}} : \text{Mod}_{\mathfrak{S}} \cong p\text{-GrpSch}^{\text{fin}}/\mathcal{O}_K \text{ (27.10.5.7),}$$

where  $p\text{-GrpSch}^{\text{fin}}/\mathcal{O}_K$  is category of commutative finite flat group schemes of  $p$ -power order.  $\lrcorner$

*Proof:* Cf. [Kisin, crystalline representations and F-crystal].  $\square$

**Def. (27.10.5.10)** [**Finite Flat and Connected Galois-Representations**]. A **finite flat and connected Galois-representation** is a finite  $p$ -group  $V \in \mathcal{A}b^{\text{fin}}$  with an action of  $\text{Gal}_K$  s.t. there exists a finite flat connected group scheme  $\mathcal{G} \in \text{GrpSch}/\mathcal{O}_K$  s.t.  $G(\mathcal{O}_{K^{\text{sep}}}) \cong V$ .  $\lrcorner$

**Prop. (27.10.5.11)** [Breuil[Bre02]/Kisin[Kis09c]]. The restriction functor from the category of finite flat and connected  $\text{Gal}_K$ -representations to the category of  $\text{Gal}_{K_{\infty}}$ -Representations is fully faithful.  $\lrcorner$

*Proof:* Cf. [Kis09c]P605.  $\square$

## 28 | Shimura Varieties

### 28.1 Shimura Varieties

References are [Del71b, Del79b, Mil05, Mil11, Lan20], <http://virtualmath1.stanford.edu/~conrad/shimsem/>, [Canonical Models of Mixed Shimura Varieties and Automorphic Vector Bundles, Milne].

**Notation (28.1.0.1).**

- Use notations on reductive groups defined in [Group Theory III: Reductive Groups](#).
- Use notations on symmetric spaces defined in [Locally Symmetric Spaces](#).
- Use notations on Abelian varieties defined in [Abelian Varieties](#).

┘

#### 1 Shimura Varieties

**Def. (28.1.1.1) [Shimura Data].** A **Shimura datum** is a pair  $(G, \mathcal{N})$  where  $G \in \text{RedGrp}/\mathbb{Q}$  and  $X$  is a  $G(\mathbb{R})$ -conjugacy class of homomorphisms  $\mathbb{S} \rightarrow G_{\mathbb{R}}$  (7.1.5.33) s.t. for each  $h \in X$ ,

- Under  $\text{Ad} \circ h : \mathbb{S} \mapsto \text{GL}(\mathfrak{g})$ ,  $\mathfrak{g}$  acquires a Hodge structure of type  $(-1, 1), (0, 0), (1, -1)$  (16.11.1.4).
- $\text{ad}(h(i))$  is a Cartan involution of  $(G_{\text{ad}})_{\mathbb{R}}$ . ?
- $G_{\text{ad}}$  has no  $\mathbb{Q}$ -factor  $H$  s.t.  $H(\mathbb{R})$  is compact.

A **morphism between Shimura-data**  $\varphi : (G_1, X_1) \rightarrow (G_2, X_2)$  is a morphism  $\varphi : G_1 \rightarrow G_2 \in \text{RedGrp}/\mathbb{Q}$  mapping  $X_1$  into  $X_2$ . ┘

**Def. (28.1.1.2) [Weight homomorphisms].** Given a Shimura datum  $(G, \mathcal{N})$  (28.1.1.1), for each  $h \in \mathcal{N}$ , the map  $w_h : \mathbb{G}_m \xrightarrow{w} \mathbb{S} \rightarrow G_{\mathbb{R}}$  (7.1.5.33) factors through the center of  $G_{\mathbb{R}}$  because it acts trivially on  $\mathfrak{g}$  (10.4.2.15), thus is independent of  $h$  chosen, called the **weight homomorphism of  $G$**  (7.1.5.33). ┘

**Def. (28.1.1.3) [Hodge-Cocharacters].** For a Shimura-datum  $(G, X)$  (28.1.1.1) and  $h \in X$ , using the isomorphism  $\mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_{m, \mathbb{C}} \times \mathbb{G}_{m, \mathbb{C}} : z \mapsto (z, \bar{z})$  (7.1.5.33), we get a  $\mathbb{G}(\mathbb{R})$ -conjugacy class of homomorphisms

$$\mu_h : \mathbb{G}_{m, \mathbb{C}} \rightarrow \mathbb{S}_{\mathbb{C}} \rightarrow \mathbb{G}_{\mathbb{C}} : z \mapsto (z, 1) \mapsto \mu(z, 1),$$

called the (conjugacy of) **Hodge-cocharacter** of  $h$ . The conjugacy class of  $\mu_h$  is independent of  $\mu$  chosen. ┘

**Def. (28.1.1.4) [Neat Compact Open Subgroups].** ┘

**Def. (28.1.1.5)** [Shimura Varieties according to Deligne]. For any Shimura datum  $(G, D)$  (28.1.1.1) and compact open subset  $\mathcal{K} \leq G(\mathbf{A}_f)$ , define

$$\mathrm{Sh}_{\mathcal{K}}(G, X)(\mathbb{C}) \triangleq G(\mathbb{Q}) \backslash X \times G(\mathbf{A}_f) / \mathcal{K}.$$

Then  $S_{\mathcal{K}}(G, X)$  is a complex manifold.  $\lrcorner$

*Proof:*

$\square$

**Thm. (28.1.1.6)** [Algebraic Structures, Satake/Baily-Borel[B-B66]/Borel[Bor72]]. Situation as in (28.1.1.5), when  $K$  is neat (28.1.1.4),  $S_{\mathcal{K}}(G)(\mathbb{C})$  is the analytification of a canonical quasi-projective variety  $\mathrm{Sh}_{\mathcal{K}}(G)_{\mathbb{C}} \in \mathrm{Var}/\mathbb{C}$ .

In particular, if  $\mathcal{K}' \subset \mathcal{K}$  are two neat compact open subgroups, then there is a natural finite étale map

$$\mathrm{Sh}_{\mathcal{K}'}(G)_{\mathbb{C}} \rightarrow \mathrm{Sh}_{\mathcal{K}}(G)_{\mathbb{C}} \in \mathrm{Var}/\mathbb{C}.$$

$\lrcorner$

*Proof:* ?

$\square$

### Connected Shimura Varieties

**Def. (28.1.1.7)** [Connected Shimura Data]. A **connected Shimura datum** is a pair  $(G, D)$  where  $G \in \mathrm{RedGrp}^{\mathrm{ss}}/\mathbb{Q}$  and  $D$  a  $G^{\mathrm{ad}}(\mathbb{R})^0$ -conjugacy classes of homomorphisms  $h : \mathbb{S} \rightarrow G_{\mathbb{R}}^{\mathrm{ad}}$  (7.1.5.33) s.t. for each  $h \in D$ ,

**SU1:** Under  $\mathrm{Ad} \circ h : \mathbb{S} \mapsto \mathrm{GL}(\mathfrak{g})$ ,  $\mathfrak{g}$  acquires a Hodge structure of type  $(-1, 1), (0, 0), (1, -1)$  (16.11.1.4).

**SU2:**  $\mathrm{ad}(h(i))$  is a Cartan involution on  $G_{\mathbb{R}}^{\mathrm{ad}}$ .

**SU3:**  $G^{\mathrm{ad}}$  has no  $\mathbb{Q}$ -factor  $H$  s.t.  $H(\mathbb{R})$  is compact.

A **morphism of connected Shimura data**  $(G_1, D_1) \rightarrow (G_2, D_2)$  is a morphism  $\varphi : G_1 \rightarrow G_2$  mapping  $D_1$  into  $D_2$ .

In this case, each  $h \in D$  factors through  $\mathbb{S} \rightarrow \mathrm{U}(1)$  (7.1.5.33) by (28.1.1.2), thus  $D$  is equivalent to a  $G^{\mathrm{ad}}(\mathbb{R})^0$ -conjugacy class of homomorphisms  $h : \mathrm{U}(1) \rightarrow G_{\mathbb{R}}^{\mathrm{ad}}$ .  $\lrcorner$

**Prop. (28.1.1.8).** Let  $H$  be an adjoint real Lie group and  $u : \mathrm{U}(1) \rightarrow H$  a homomorphism satisfying SU1, SU2, then the following are equivalent:

- $u(-1) = 1$ .
- $u$  is trivial.
- $H$  is compact.

$\lrcorner$

*Proof:* 1  $\iff$  2: If  $u(-1) = 1$ , then  $u$  factors through  $\mathrm{U}(1) \xrightarrow{\times 2} \mathrm{U}(1)$ , so the characters  $z, z^{-1}$  cannot occur in  $\mathrm{Ad} \circ u$ . The converse is trivial.

1  $\rightarrow$  3: by (10.5.5.19),  $H$  is compact iff  $\mathrm{ad}(u(-1)) = 1$ , which is equivalent to  $u(-1) = 1$  as  $H$  is adjoint.

3  $\rightarrow$  1: ?

$\square$

**Prop. (28.1.1.9)** [Shimura-Data a la Shimura]. A connected Shimura-datum is equivalent to the following data:



- $G \in \mathcal{AlgGrp}/\mathbb{Q}$  semisimple of non-compact type.
- $D$  a Hermitian symmetric domain.
- an action of  $G(\mathbb{R})^0$  on  $D$  via a surjective homomorphism  $G^{\text{ad}}(\mathbb{R})^0 \rightarrow \text{Hol}(D)^0$  with compact kernel.

┘

*Proof:* Cf. [Mil17b]P45. ?

□

**Prop. (28.1.1.10).** Let  $(G, D)$  be a connected Shimura datum and  $X$  be the  $G^{\text{ad}}(\mathbb{R})$ -conjugacy class of homomorphisms  $\mathbb{S}^1 \rightarrow \mathbb{G}_{\mathbb{R}}$  containing  $D$ , and  $D$  is a connected component of  $X$ , with stabilizer  $G^{\text{ad}}(\mathbb{R})^0 \subset G^{\text{ad}}(\mathbb{R})$ .

┘

*Proof:* Cf. [Mil17b]P45.

□

**Def. (28.1.1.11) [Connected Shimura Varieties].** Let  $(G, D)$  be a connected Shimura datum (28.1.1.7), then the map  $G^{\text{ad}}(\mathbb{R})^0 \rightarrow \text{Hol}(D)^0$  as in (28.1.1.9) has compact kernel, so for any arithmetic subgroup  $\Gamma \subset G^{\text{ad}}(\mathbb{R})^0 \cap G^{\text{ad}}(\mathbb{Q})$  (20.4.0.2), the image  $\bar{\Gamma}$  is an arithmetic subgroup of  $\text{Hol}(D)^0$  (20.4.4.1), thus we can apply Baily-Borel compactification (16.2.7.4) to get an algebraic structure on  $D(\Gamma) = \Gamma \backslash D = \bar{\Gamma} \backslash D$ . Moreover, if  $\Gamma' \subset \Gamma \subset G^{\text{ad}}(\mathbb{R})^0 \cap G^{\text{ad}}(\mathbb{Q})$  and  $\bar{\Gamma}', \bar{\Gamma}$  are both torsion-free, then by Borel (16.2.7.5), there is a morphism  $D(\Gamma') \rightarrow D(\Gamma)$ .

So for any  $\Gamma \in G^{\text{ad}}(\mathbb{R})^0 \cap G^{\text{ad}}(\mathbb{Q})$  s.t.  $\bar{\Gamma}$  is torsion-free,  $D(\Gamma)$  is called a **connected Shimura variety relative to  $(G, D)$** . And the inverse images of these Shimura varieties are called the **connected Shimura variety attached to  $(G, D)$** , denoted by  $\text{Sh}^0(G, D)$ .

┘

**Prop. (28.1.1.12) [Galois Actions on Shimura Varieties].**

┘

### Morphism between Shimura-Varieties

**Prop. (28.1.1.13) [Kisin [Kis10]].** If

- $i : (G_1, X_1) \rightarrow (G_2, X_2)$  is an embedding of Shimura-data,
- $\mathcal{K}_{2,p} \leq G_2(\mathbb{Q}_p)$  is a compact open subgroup,
- $\mathcal{K}_{1,p} = i^{-1}(\mathcal{K}_{2,p}) \leq G_1(\mathbb{Q}_p)$ ,
- $\mathcal{K}_1^p \leq G_1(\mathbf{A}_f^p)$ ,

then there exists  $\mathcal{K}_2^p \leq G_2(\mathbf{A}_f^p)$  containing  $i(\mathcal{K}_1^p)$ , s.t.  $i$  induces an embedding

$$\text{Sh}_{\mathcal{K}_1}(G_1, X_1) \hookrightarrow \text{Sh}_{\mathcal{K}_2}(G_2, X_2).$$

┘

*Proof:* Cf. [Kis10]P

□

## 2 Canonical Models

**Notation (28.1.2.1).**

- Let  $(G, X)$  be a Shimura-datum.
- let  $\mathcal{K} \leq G(\mathbf{A}_f)$  be a compact open subgroup.

┘

**Def. (28.1.2.2) [Reflex Fields].** There exists a **reflex field**  $E = E(G, X) \in \mathbf{NField}$ . ┘

**Cor. (28.1.2.3).** If  $G_{\mathbb{Q}_p}$  is unramified, then  $E(G, X)/\mathbb{Q}$  (28.1.2.2) is unramified at  $p$ . ┘

*Proof:* Cf. [Milne, The points on a Shimura variety modulo a prime of good reduction]4, 4.7. ? □

**Thm. (28.1.2.4) [Canonical Models of Shimura Varieties, Shimura[Shi70]/Shih[Shi78a]/Deligne[Del79b]/Milne[Mil83]/Borovoi[Bor84, Bor87]].** For a Shimura datum  $(G, X)$  (28.1.1.1) and any neat compact open subgroup  $\mathcal{K} \leq G(\mathbf{A}_f)$  (28.1.1.4), there exists a canonical quasi-projective variety

$$\mathrm{Sh}_{\mathcal{K}}(G) \in \mathcal{V}\mathrm{ar}/E(G, X) \text{ (28.1.2.2)}$$

s.t.  $\mathrm{Sh}_{\mathcal{K}}(G)(\mathbb{C})$  (28.1.1.5) is isomorphic to the complex analytification of  $\mathrm{Sh}_{\mathcal{K}}(G)$ .

They satisfy certain additional properties qualifying them as the **canonical model of Shimura varieties**, which is uniquely characterized by the reciprocity law at the special points. ?

In particular, for any two neat compact open subgroups  $\mathcal{K}' \leq \mathcal{K}$ ,  $\mathrm{Sh}_{\mathcal{K}'}(G)(\mathbb{C}) \rightarrow \mathrm{Sh}_{\mathcal{K}}(G)(\mathbb{C})$  is the complex analytification of a unique finite étale map

$$\mathrm{Sh}_{\mathcal{K}'}(G) \rightarrow \mathrm{Sh}_{\mathcal{K}}(G) \in \mathcal{V}\mathrm{ar}/E(G, X).$$

┘

*Proof:* ? □

**Remark (28.1.2.5) [Canonical Models of Connected Shimura Varieties].** The connected Shimura varieties are also defined over a number field, but the number field may depend on the arithmetic subgroup  $\Gamma$ . For the example,  $\Gamma(N) \backslash \mathcal{H}$  only has a model over  $\mathbb{Q}(\mu_N)$ . ┘

**Remark (28.1.2.6).** Delete this ?. The method of constructing the canonical model is the following: When our Shimura variety arises naturally as a moduli space over  $\mathbb{C}$ , then we can use the action of  $\mathrm{Gal}(\mathbb{C}/E)$  on the  $\mathbb{C}$ -points to define a model of the variety over a specific number field.

The theory of complex multiplication will give us an explicit description of the action on certain special points, called the **reciprocity law** at these points, which will determine the model uniquely. ┘

**Remark (28.1.2.7).** Delete this ?. A heuristic reason that a Shimura variety has a canonical model is that if it is defined only on a transcendental field, then it can be spread out to give a flat family of varieties. But there are only countably many arithmetic locally symmetric varieties up to isomorphism, so there cannot be a ┘

**Prop. (28.1.2.8) [Morphisms between Shimura-Varieties, Deligne[Del71b]].** If  $i : (G_1, X_1) \rightarrow (G_2, X_2)$  is morphism between Shimura-data and  $\mathcal{K}_i \leq G_i(\mathbf{A}_f)$  are compact open subgroups s.t.  $i(\mathcal{K}_1) \leq \mathcal{K}_2$ , then  $i$  induces a map

$$\mathrm{Sh}_{\mathcal{K}_1}(G_1, X_1) \rightarrow \mathrm{Sh}_{\mathcal{K}_2}(G_2, X_2) \in \mathcal{V}\mathrm{ar}/E(G_1, X_1).E(G_2, X_2).$$

And for any given  $\mathcal{K}_1 \leq G(\mathbf{A}_f)$ , we can choose  $\mathcal{K}_2$  s.t. this map is a closed embedding. ┘

*Proof:* Cf. [Del71]Section5.4. and 1.15. □

**Prop. (28.1.2.9)[Connected Components of Canonical Models].** If  $\mathcal{K} = \mathcal{K}_p \mathcal{K}^p$  where  $p \in \text{Prime}$  and  $\mathcal{K}_p \leq G(\mathbb{Q}_p)$ ,  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  and  $\mathcal{K}_p$  is hyperspecial, then the geometrically connected components of  $\text{Sh}_{\mathcal{K}}(G, X)$  is defined over a an extension of  $E(G, X)$  (28.1.2.2) unramified at all places over  $p$ .  $\lrcorner$

*Proof:* Cf. [Kis10]P990.  $\square$

**Lemma (28.1.2.10).** Situation as in (28.1.2.9),  $G(\mathbf{A}_f^p)$  acts transitively on

$$\pi_0(\text{Sh}_{\mathcal{K}_p}(G, X)_{\mathbb{C}}) \stackrel{\Delta}{=} \varprojlim_{\mathcal{K}^p} \pi_0(\text{Sh}_{\mathcal{K}_p \mathcal{K}^p}(G, X)_{\mathbb{C}}).$$

$\lrcorner$

*Proof:* Cf. [Kis10]P991.  $\square$

**Lemma (28.1.2.11).** For any  $H \in \text{RedGrp}/\mathbb{Z}_{(p)}$ , we have

$$H(\mathbb{Z}_p) \left( H(\mathbb{Q}) \cap H(\mathbb{R})^0 \right) = H(\mathbb{Q}_p).$$

$\lrcorner$

*Proof:* Cf. [Kis10]P991.  $\square$

### Hodge-Tate Structure on Shimura Varieties

**Prop. (28.1.2.12)[Family of Hodge-Tate Structures].** Let  $G \in \text{AlgGrp}/\mathbb{Q}$  be reductive and  $\rho : G \rightarrow \text{GL}(V)$  a faithful representation, and  $h \in \mathbb{D} \subset \text{Hom}(\mathbb{C}^*, \mathbb{G}_{\mathbb{R}})$  a set of conjugacy classes. Suppose the Shimura variety  $\{G(\mathbb{Q}) \backslash \mathbb{D} \times G(\mathbf{A}_f)/K\}_K$  has a model  $M_K$  over  $\mathbb{Q}$ , then we can define

- A  $\mathbb{Q}$ -local system

$$\mathcal{L}_{\rho} = G(\mathbb{Q}) \backslash V \times \mathbb{D} \times G(\mathbf{A}_f)/K$$

on  $M_K(\mathbb{C})$ ,

- A filtration  $\mathcal{L}_{\rho} \otimes \mathcal{O}_{M_K(\mathbb{C})} \supset \mathcal{F}^p \supset \mathcal{F}^{p+1} \dots$  of holomorphic vector bundles s.t. pointwise it is given by the Hodge filtration associated to  $\rho \circ h$ .
- Let  $V_{\mathbb{Z}} \subset V$  be a lattice s.t.  $V_{\mathbb{Z}}$  is fixed by  $K$ , then for  $N \in \mathbb{Z}_+$ , we can define.  $V_{\mathbb{Z}/(N)}$ -principal cover

$$\widetilde{M}_{\rho, K, N} = (M_K \cap K(N) \times V_{\mathbb{Z}/(N)})/K \rightarrow M_K.$$

So

$$\left( \varprojlim_n \widetilde{M}_{\rho, K, \ell^n}(\mathbb{C}) \right) \left[ \frac{1}{\ell} \right] \cong \mathcal{L}_{\rho} \otimes \mathbb{Q}_{\ell}.$$

Then for any  $x \in M_K(\mathbb{Q})$ , we get a Hodge-Tate structure  $\mathcal{L}_{\rho, x}$  where the Hodge structure on  $\mathcal{L}_{\rho, x} \otimes \mathbb{C}$  comes from the Hodge filtration  $\mathcal{F}_x^p$ , and the Galois structure on  $\mathcal{L}_{\rho, x} \otimes \mathbb{Q}_{\ell}$  comes from the Galois action on the fiber of the system  $\{\widetilde{M}_{\rho, K, \ell^n}\}_n$  over  $x$  (as for usual Shimura varieties).

Moreover, this construction is compatible with morphisms of Shimura varieties induces by a map between groups.  $\lrcorner$

*Proof:*  $\square$

**Example (28.1.2.13).** For the Siegel modular variety  $\mathcal{A}_K^{g, d}$  associated to  $(\text{GSp}_{2g}, \mathcal{H}^{\pm})$  and a point  $x \in \mathcal{A}_K^{g, d}(\mathbb{Q})$  parametrizing a triple  $(A, E, \xi)$ , we have a canonical isomorphism  $H^1(A, \mathbb{Q}) \cong \mathcal{L}_{\text{Std}_{\text{GSp}, x}}$ .  $\lrcorner$

**Example (28.1.2.14).** For the model  $M_K^{\text{K3}}(7.14.6.1)$  of  $\text{Sh}_K(\text{SO}(\Lambda_d^{\text{K3}}), \mathbb{D})$  and a point  $x \in M_K^{\text{K3}}(\mathbb{Q})$  parametrizing a triple  $(X, E, \xi)$ , we have a canonical isomorphism  $H^2(X, \mathbb{Q}(1)) \cong \mathcal{L}_{\text{Std}_{\text{SO}, x}}$ .  $\lrcorner$

### 3 Siegel Modular Varieties

**Def. (28.1.3.1) [Siegel Modular Shimura-Data].** Consider the homomorphism  $h_0 : \mathrm{U}(1) \rightarrow \mathrm{Sp}(n)_{\mathbb{R}} :$

$$x + iy \mapsto \begin{bmatrix} x\mathbf{1}_n & -y\mathbf{1}_n \\ y\mathbf{1}_n & x\mathbf{1}_n \end{bmatrix}, \text{ then}$$

$$Z_{\mathrm{Sp}(n; \mathbb{R})}(h_0) = \left\{ \begin{bmatrix} A & B \\ -B & A \end{bmatrix} \mid A, B \in \mathrm{Mat}(n; \mathbb{R}) \right\} \cong \mathrm{U}(n).$$

In particular,  $\mathrm{Sp}(n; \mathbb{R}).h_0 \cong \mathcal{H}_n$  (16.2.6.6), and

$$(\mathrm{Sp}(n), \mathcal{H}_n)$$

is a connected Shimura datum (28.1.1.7).

Similarly, if  $h_0 : \$ \rightarrow \mathrm{GSp}(n)_{\mathbb{R}} : x + iy \mapsto \begin{bmatrix} x\mathbf{1}_n & -y\mathbf{1}_n \\ y\mathbf{1}_n & x\mathbf{1}_n \end{bmatrix}$ , then

$$Z_{\mathrm{GSp}(n; \mathbb{R})}(h_0) = \left\{ \begin{bmatrix} A & B \\ -B & A \end{bmatrix} \mid A, B \in \mathrm{Mat}(n; \mathbb{R}) \right\} \cong \mathrm{U}(n),$$

$\mathrm{GSp}(n; \mathbb{R}).h_0 \cong \mathcal{H}_n^{\pm}$  (16.2.6.6), and

$$(\mathrm{GSp}(n), \mathcal{H}_n^{\pm})$$

is a Shimura datum (28.1.1.1). ┘

*Proof:* If  $\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \mathbf{1}_n & \mathbf{1}_n \\ \mathbf{1}_n & \mathbf{1}_n \end{bmatrix} = \begin{bmatrix} \mathbf{1}_n & \mathbf{1}_n \\ \mathbf{1}_n & \mathbf{1}_n \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ , then  $A = D, B = -C$ . Conversely,

$$\begin{bmatrix} A & B \\ -B & A \end{bmatrix} \begin{bmatrix} x\mathbf{1}_n & -y\mathbf{1}_n \\ y\mathbf{1}_n & x\mathbf{1}_n \end{bmatrix} = \begin{bmatrix} x\mathbf{1}_n & -y\mathbf{1}_n \\ y\mathbf{1}_n & x\mathbf{1}_n \end{bmatrix} \begin{bmatrix} A & B \\ -B & A \end{bmatrix}$$

for any  $A, B \in \mathrm{Mat}(n; \mathbb{R})$  and  $x, y \in \mathbb{R}$ .

To verify that  $(\mathrm{Sp}(n), \mathcal{H}_n)$  is a connected Shimura datum ? □

### Siegel Modular Varieties

**Prop. (28.1.3.2) [Abelian Varieties with Polarizations and Level Structures].** For  $n, N \in \mathbb{Z}_+$  and  $d_1 | d_2 | \dots | d_n \in \mathbb{Z}_+$  s.t.  $(d_n, N) = 1$ , consider the functor

$$\mathcal{A} : \mathrm{Sch}/\mathbb{Z}[\frac{1}{Nd_n}] \rightarrow \mathrm{Set}$$

s.t. for any  $S \in \mathrm{Sch}/\mathbb{Z}[\frac{1}{Nd_n}]$ ,  $\mathcal{A}(S)$  is the set of isomorphism classes of triples  $(A, \lambda, \eta)$  where

- $A \in \mathrm{AbVar}^n/S$ ,
  - $\lambda : A \rightarrow A^{\vee}$  is a polarization of type  $D = \mathrm{diag}(d_1, \dots, d_n)$ .
  - $\eta : ((\mathbb{Z}/(N))^{2g}, \mathrm{Std}) \cong (A[N], e_N^{\lambda})$  is a symplectic similitude.
- ┘

**Thm. (28.1.3.3) [ $\mathcal{A}$  is Representable].** Situation as in (28.1.3.2), if  $N$  is large enough w.r.t.  $D$  (when  $d_n = 1$ , it suffices to have  $N \geq 3$ ), then  $\mathcal{A}$  is representable by a smooth quasi-projective scheme over  $\mathbb{Z}[\frac{1}{Nd_n}]$ .  $\square$

*Proof:* For  $S \in \text{Sch}$ ,  $(A, \lambda) \in \text{AbVar}^{g,D}/S$ , let  $\mathcal{L}^\Delta(\lambda) = (\text{id}_A, \lambda)^* p_A$ . Then locally on étale nbhd, if  $\lambda = \lambda_{\mathcal{L}}$  for an relative ample line bundle  $\mathcal{L}$ , then

$$\mathcal{L}^\Delta(\lambda) = [-1]^* \mathcal{L} \otimes \mathcal{L} \text{ (20.9.4.9)}$$

is also relatively ample (7.5.4.19), and

$$\lambda_{\mathcal{L}^\Delta(\lambda)} = 2\lambda_{\mathcal{L}} \text{ (20.9.4.16)}.$$

Thus  $\mathcal{L}^\Delta(\lambda)^{\otimes 3}$  is relatively very ample (20.9.5.8), and  $R^i \text{pr}_* \mathcal{L}^{\otimes 3} = 0$  for  $i \geq 1$ . Let  $\mathcal{M} = \text{pr}_* \mathcal{L}^{\otimes 3} \in \text{QCoh}^{\text{free}, 6^n d}/S$ , where  $d = \prod d_i$ .

Define a new functor  $\mathcal{H}$  that is the same as  $\mathcal{A}$  except for that we require a projective trivialization of  $\mathcal{M}$ . Then  $\mathcal{H} \rightarrow \mathcal{A}$  is a  $\text{PGL}(6^n d - 1)$ -torsor. For  $X \in \mathcal{H}(S)$ ,  $\text{pr}_* \mathcal{L}^{\otimes 3}$  provides a projective embedding  $X \subset \mathbb{P}_S^{6^n d - 1}$ , and for any  $r \in \mathbb{Z}_+$ ,  $\text{rank pr}_* \mathcal{L}^{\otimes 3r} = 6^n d r^n$ , so there is a natural transformation

$$\mathcal{H} \rightarrow \underline{\text{Hilb}}_{\mathbb{P}^{6^n d - 1}}^{Q(t), 1} : X \mapsto (X, e), \quad Q(t) = 6^n d t^n.$$

Then this is an open subfunctor whose image are contained in the moduli of pointed smooth subschemes of  $\mathbb{P}^{6^n d - 1}$ : It is injective because any pointed smooth subscheme has at most one Abelian variety structure (20.9.1.5), and its image is a union of components of the smooth locus of  $\underline{\text{Hilb}}_{\mathbb{P}^{6^n d - 1}}^{Q(t), 1}$ : This is because of (21.1.2.16).

Then  $\mathcal{H}$  is a smooth quasi-projective variety. For the rest, it needs to use GIT quotient of  $\mathcal{H}$  by  $\text{PGL}(6^d)$ , using the level structure to show stability properties, which I will not cover here. Cf. [Ngo, Shimura Varieties] P15  $\color{red}{?}$ .  $\square$

**Prop. (28.1.3.4) [Siegel Modular Varieties].** For  $k \in \text{Field}$ , the étale shification  $\mathcal{M}_{\text{ét}}^{g,d}$  of the functor

$$\mathcal{M}^{g,d} : \text{Sch}/k \rightarrow \text{Set} : X \mapsto \text{AbVar}^{g, \text{polar}=d}/X$$

is representable by a variety  $\mathcal{M}^{g,d} \in \text{Var}/k$ .  $\square$

*Proof:*  $\color{red}{?}$   $\square$

**Prop. (28.1.3.5) [Canonical Ample Divisor on  $\mathcal{M}^{g,d}$ ].** There exists a canonical ample divisor on  $\mathcal{M}^{g,d}$  given by the determinant of the sheaf of invariant differentials on the universal Abelian variety  $\mathcal{A}$ .  $\square$

*Proof:*  $\square$

**Prop. (28.1.3.6).** The  $j$ -invariant of elliptic schemes define an isomorphism  $\mathcal{M}_{\mathbb{Q}}^{1,1} \rightarrow \mathbb{A}_{\mathbb{Q}}^1$ .  $\square$

*Proof:*  $\color{red}{?}$   $\square$

**Prop. (28.1.3.7).** Let

$$(V, \psi) = (\mathbb{Q}^{2g}, \langle (a_i), (b_i) \rangle) = \sum_{i=1}^g (a_i b_{g+i} - a_{g+i} b_i),$$

and  $\tilde{G} = \text{GSp}(V)$ , then the Hermitian symmetric domain  $\tilde{X}$  is the Siegel double space, and for any neat compact open subgroup  $\tilde{\mathcal{K}} \subset \tilde{G}(\mathbf{A}_f)$ , the corresponding Shimura variety  $\tilde{S}_{\tilde{\mathcal{K}}}$  is the moduli space of principally polarized  $g$ -dimensional Abelian varieties with level- $\tilde{\mathcal{K}}$ -structure, which has a model over the reflex field  $\mathbb{Q}$ .  $\square$

#### 4 PEL-Type Shimura Varieties

References are [Kot92a] and [Lan, Arithmetic compactification of PEL-type Shimura varieties].

**Def. (28.1.4.1) [PEL-Data].** A **rational PEL-datum** is a tuple  $(B, *, V, \langle -, - \rangle, h_0)$  where

- $B$  is a f.d. semisimple algebra over  $\mathbb{Q}$ .
- $V \in \mathcal{V}\text{ect}^{\text{fin}}/B$ .
- $*$  is a positive involution on  $\mathcal{O}$ , i.e.  $\text{tr}_{B \otimes \mathbb{R}/\mathbb{R}}(x^*x) > 0$  for any  $x \in (B \otimes \mathbb{R})^\times$ .
- $\langle -, - \rangle$  is a non-degenerate  $\mathbb{Q}$ -valued alternating pairing on  $V$  s.t.

$$\langle b^*x, y \rangle = \langle x, by \rangle, \text{ where } x, y \in V, b \in B.$$

- $h_0 : \mathbb{C} \rightarrow \text{End}_{B \otimes \mathbb{R}}(V_{\mathbb{R}})$  is an  $\mathbb{R}$ -algebra homomorphism s.t.
  - For any  $z \in \mathbb{C}$  and  $x, y \in V_{\mathbb{R}}$ ,  $\langle h_0(z)x, y \rangle = \langle x, h_0(\bar{z})y \rangle$ .
  - The symmetric  $\mathbb{R}$ -bilinear pairing  $\langle -, h_0(i)(-) \rangle$  on  $V_{\mathbb{R}}$  is positive definite.

And an **integral PEL-datum** is a rational PEL-datum together with an order  $\mathcal{O}_B \subset B$  invariant under  $*$  and an  $\mathcal{O}_B$ -lattice  $\Lambda \subset V$ .

And a  **$p$ -integral PEL-datum** is a rational PEL-datum together with a  $\mathbb{Z}_{(p)}$ -order  $\mathcal{O}_B \subset B$  whose  $p$ -adic completion is a maximal order in  $B_{\mathbb{Q}_p}$  and an  $\mathcal{O}_B$ -lattice  $\Lambda \subset V_{(p)}$  ? Cf. [Kot92a]Section5 and [Lan13]Section1.4.  $\lrcorner$

**Def. (28.1.4.2) [Group of Similitudes].** For a rational PEL-datum (28.1.4.1), define  $G = \text{GU}_B(V) \in \text{AlgGrp}/\mathbb{Q}$  s.t. for any  $R \in \mathbb{C}\text{Ring}/\mathbb{Q}$  we have

$$\text{GU}_B(V)(R) = \{x \in \text{End}_{B \otimes R}(V \otimes R) : \exists c(x) \in R^*, \langle xu, xv \rangle = c(x) \langle u, v \rangle, \text{ for any } u, v \in V_{\mathbb{R}}\}.$$

Thus there is a natural map  $G \rightarrow \mathbb{G}_m$ , with kernel  $G_1 = \text{U}_B(V)$ .

For an integral PEL-datum, we can extend  $G, G_1$  to an algebraic group over  $\mathbb{Z}$ . And for a  $p$ -integral PEL-datum, we can extend  $G, G_1$  to an algebraic group over  $\mathbb{Z}_{(p)}$ .  $\lrcorner$

**Def. (28.1.4.3) [Reflex Fields of  $h_0$ ].** Cf. [Lan20]P53.  $\lrcorner$

**Prop. (28.1.4.4) [PEL Structures on Abelian Varieties].** Given an integral PEL-datum  $(\mathcal{O}, *, L, \langle -, - \rangle, h_0)$  (28.1.4.1), we get a tuple

$$(A_0, \lambda_0, i_0, (\alpha_{0,n}, \nu_{0,n}))$$

which is an abelian variety with a polarization, an endomorphism structure, and a level structure, such that  $\text{Lie}(A_0)$  satisfies the Lie algebra condition defined by  $h_0$ .  $\lrcorner$

*Proof:* Cf. [Lan20].  $\square$

**Prop. (28.1.4.5) [Conjugacy in  $G$ ].** Situation as in (28.1.4.2), if  $Q \in \text{Field}^0$  is algebraically closed, then  $x, y \in G(Q)$  (28.1.4.2) are conjugate iff they have the same similitude and they are conjugate in  $\text{End}_B(V)(Q)$ .  $\lrcorner$

*Proof:* Cf. [Kot92]P395 ?  $\square$

### PEL-Type Shimura Varieties

**Def. (28.1.4.6) [PEL-Type Shimura-Data].** Given a rational PEL-datum  $(B, *, V, \langle -, - \rangle, h_0)$  (28.1.4.1), define  $G = \mathrm{GU}(V)$  (28.1.4.2), and then  $h_0 : \mathbb{C} \rightarrow G(\mathbb{R})$  defines a manifold  $\mathcal{D} = G(\mathbb{R}).h_0$ .

When  $(G, \mathcal{D})$  is a Shimura-datum, it is called a **PEL-type Shimura datum**. And for each neat open compact subgroup  $\mathcal{K} \leq G(\mathbf{A}_f)$ , the associated Shimura variety  $\mathrm{Sh}_{\mathcal{K}}(G, \mathcal{D})$  is called a **PEL-type Shimura variety**. And the reflex field for  $(G, \mathcal{D})$  coincides with that of  $h_0$  (28.1.4.3)?.

┘

**Example (28.1.4.7).** Siegel modular varieties  $(\mathrm{GSp}(n), \mathcal{H}_n^{\pm})$  are PEL-type Shimura-varieties.  
 $(\mathrm{GU}(a, b), \mathcal{H}_{a,b}^{\pm})$  are PEL-type Shimura-varieties. ┘

## 5 Hodge-Type Shimura Varieties

**Def. (28.1.5.1) [Hodge-Type Shimura Varieties].** A **Hodge-Type Shimura datum** is a Shimura datum  $(G, X)$  (28.1.1.1) with an embedding of Shimura data

$$(G, X) \hookrightarrow (\mathrm{GSp}(2g), \mathcal{H}_{2g}^{\pm}),$$

for some  $g \in \mathbb{Z}_+$ , called a **Siegel-embedding**.

A **Hodge-Type Shimura-variety** is a Shimura-variety associated to a Hodge-type Shimura datum (28.1.1.5). In particular, a Hodge-Type Shimura-variety has a universal Abelian scheme over it. ┘

**Example (28.1.5.2) [Hodge-Type Shimura Data].**

- PEL-type Shimura data are Hodge-type Shimura varieties.
- $\mathrm{GSpin}$ -Shimura data are of Hodge-type but not of PEL types.

┘

**Remark (28.1.5.3).** Hodge-Type Shimura varieties parametrize polarized abelian varieties equipped with some Hodge cycles and level structures. ┘

**Prop. (28.1.5.4).** If  $(G, X)$  is a Shimura datum of Hodge type with an embedding  $(G, X) \hookrightarrow (\mathrm{Sp}_{2g}, \mathbb{D}_{2g})$ , then there exists a neat subgroup  $\widetilde{K} \subset \widetilde{G}(\mathbf{A}^f)$ ,  $K = \widetilde{K} \cap G(\mathbf{A}^f)$  s.t. there is a closed embedding of Shimura varieties

$$S_K \hookrightarrow \widetilde{S}_K \otimes_{\mathbb{Q}} E.$$

┘

*Proof:* [Deligne71, Prop1.15]. ┘

## 6 Abelian-Type Shimura Varieties

**Def. (28.1.6.1) [Abelian-Type Shimura Varieties].** An **Abelian-type Shimura datum** is a Shimura datum  $(G, X)$  (28.1.1.1) s.t. there is a Shimura datum  $(G_1, X_1)$  with a central isogeny  $G_1^{\mathrm{A}} \rightarrow G^{\mathrm{A}}$  inducing an isomorphism

$$(G_1^{\mathrm{ad}}, X_1^{\mathrm{ad}}) \rightarrow (G^{\mathrm{ad}}, X^{\mathrm{ad}}).$$

An **Abelian-type Shimura variety** is a Shimura variety associated to an Abelian-type Shimura datum (28.1.1.5). ┘

**Example (28.1.6.2) [Abelian-Type Shimura Data].**

- Shimura curves are of Abelian-type but not Hodge-type.
- Shimura data of type  $A, B, C, D^{\mathbb{R}}$  are all of Abelian-type.
- Shimura data of type  $E_6, E_7$  are not of Abelian-type.

┘

*Proof:*

□

## 7 Complex Multiplication

## 8 Automorphic Vector Bundles

## 9 Families of Abelian Varieties

References are [Mum66, Mum69].

## 10 André-Oort Conjecture

**Def. (28.1.10.1) [Special Subvarieties].** If  $\varphi : (G_1, X_1) \rightarrow (G_2, X_2)$  is a embedding of Shimura-data (28.1.1.1), and  $\mathcal{K}_i \leq G_i(\mathbf{A}_f)$  are compact open subgroups s.t.

$$\mathrm{Sh}_{\mathcal{K}_1}(G_1)_{\mathbb{C}} \rightarrow \mathrm{Sh}_{\mathcal{K}_2}(G_2)_{\mathbb{C}}$$

is a closed embedding, then  $\mathrm{Sh}_{\mathcal{K}_1}(G_1)_{\mathbb{C}}$  is called a **special subvariety** of  $\mathrm{Sh}_{\mathcal{K}_2}(G_2)_{\mathbb{C}}$ .

If  $G_1$  is a torus, then  $\mathrm{Sh}_{\mathcal{K}_1}(G_1)_{\mathbb{C}}$  is called a **CM-point** of  $\mathrm{Sh}_{\mathcal{K}_2}(G_2)_{\mathbb{C}}$ .

For a given  $\mathcal{K}_1 \leq G_1(\mathbf{A}_f)$ , there always exists  $\mathcal{K}_2 \leq G_2(\mathbf{A}_f)$  s.t.  $\mathrm{Sh}_{\mathcal{K}_1}(G_1)_{\mathbb{C}}$  is a special subvariety of  $\mathrm{Sh}_{\mathcal{K}_2}(G_2)_{\mathbb{C}}$ . ┘

*Proof:* Cf. [Del71] Prop 1.15. □

**Conj. (28.1.10.2) [André-Oort].** Let  $S$  be a Shimura-variety. Let  $V \subset S$  be a subvariety, then there are only f.m. maximal special subvarieties contained in  $V$ . ┘

*Proof:*

□



## 28.2 Compactifications of Shimura Varieties

See [Pink, Arithmetical compactification of mixed Shimura varieties], [Borel-Ji, Compactifications of symmetric and locally symmetric spaces], [Arithmetic compactifications of PEL-type Shimura varieties, Lan].

**Prop. (28.2.0.1).**  $\mathrm{Sh}_{\mathcal{K}}(G)$  is projective iff  $[G, G]$  is anisotropic. ┘

*Proof:* ?

□

**1 Minimal Compactifications(Satake-Baily-Borel)**

**2 (Smooth)Toroidal Compactifications**

**3 Borel-Serre Compactifications**

Reductive Borel-Serre Compactifications

## 28.3 Modular Curves and Shimura Curves

References are [D-R73, Kat73, Maz77, Maz78, K-M85].

**Notation (28.3.0.1).**

- Let  $F \in \mathbf{NField}$ .
- Use notations on quaternion algebras defined in [Arithmetic of Quaternions and Octonions](#).

┘

### 1 Moduli Space of Elliptic Curves

**Def. (28.3.1.1) [ $\Gamma(N)$ -Structure].** Let  $S \in \mathbf{Sch}$  and  $E \in \mathbf{EllCur}/S$ , a

- $\Gamma(N)$ -**structure** on  $E/S$  is a homomorphism  $\varphi : (\mathbb{Z}/(N))^2 \rightarrow \ker([N])(S)$  s.t. there is an equality of effective Cartier divisors

$$E[N] = \sum_{(a,b) \in (\mathbb{Z}/(N))^2} [\varphi((a,b))].$$

- $\Gamma_1(N)$ -**structure** on  $E/S$  is a homomorphism  $\varphi : \mathbb{Z}/(N) \rightarrow \ker([N])(S)$  s.t. the effective Cartier divisor

$$\sum_{a \in \mathbb{Z}/(N)} [\varphi(a)].$$

is a subgroup scheme of  $E$ .

- **balanced**  $\Gamma_1(N)$ -**structure** on  $E/S$  is an exact sequence

$$0 \rightarrow K \rightarrow E[N] \rightarrow K' \rightarrow 0$$

of group schemes over  $S$  s.t.  $K, K'$  both has rank  $N$  over  $S$ .

- $\Gamma_0(N)$ -**structure** on  $E/S$  is a finite subgroup scheme  $K \subset E[N]$  over  $S$  that is cyclic of rank  $n$ .

┘

**Prop. (28.3.1.2) [Functors].**

┘

**Prop. (28.3.1.3) [Weil-Pairing].** Let  $S \in \mathbf{Sch}$  and  $E \in \mathbf{EllCur}/S$  with a  $\Gamma(N)$ -structure,

┘

**Prop. (28.3.1.4) [Moduli Interpretation of Modular Curves].** The functor  $\mathbf{Sch}/\mathbb{Q} \rightarrow \mathbf{Set}$  that maps  $S$  to the isomorphism classes of elliptic curves over  $S$  with a  $\Gamma(N)$ -structure is representable by

┘

**Def. (28.3.1.5) [Atkin-Lehner Involutions].**

┘

### Representability

**Prop. (28.3.1.6) [Level Structure Schemes].** For any  $S \in \mathbf{Sch}$ ,  $E \in \mathbf{EllCur}/S$ , and  $M, N \in \mathbb{Z}_+$  invertible on  $S$ , there exists a moduli scheme  $S(M, N)$  that is finite étale over  $S$ , which represents the moduli problem:

$$S(M, N)(T) = \{\text{monomorphism } \alpha : (\mathbb{Z}/(M) \times \mathbb{Z}/(N))_T \hookrightarrow E_T\}.$$

And denote  $S(N) = S(N, N)$ ,  $S_1(N) = S(N, 1)$ ,  $S'_1(N) = S(1, N)$ .

And also denote the  $S_0(N)$  parametrizing  $\Gamma_0(N)$ -structure. Then it is a quotient of  $S_1(N)$ .

┘

*Proof:* ? □

**Prop. (28.3.1.7)**  $[\mathcal{E}_{0,N}]$ . If  $k \in \mathbf{Field}$  and  $(N, \text{char } k) = 1$ , then the moduli problem  $\mathcal{E}_{0,N}$  of the set of elliptic curves with a cyclic  $N$ -subgroup has a coarse moduli scheme  $M$  over  $k$ . And if  $\text{char } k = 0$ ,  $M$  is canonically isomorphic to  $Y_0(N)_{\mathbb{Q}} \otimes k$ , and the map

$$\mathcal{E}_{0,N}(k) \rightarrow M(k) \cong Y_0(N)_{\mathbb{Q}}(k) \xrightarrow{(j, j_N)} \text{Spec } k[X, Y]/\Phi_m(X, Y).$$

is given by  $(E, S) \mapsto (j(E), j(E/S))$ . ┘

*Proof:* ? □

**Prop. (28.3.1.8)**  $[\mathcal{E}_N]$ . If  $k \in \mathbf{Field}$  and  $\mathbb{Q}(\zeta) \subset k$ , then the moduli problem  $\mathcal{E}_N$  of the set of elliptic curves with a  $\Gamma(N)$ -structure with Weil pairing  $\zeta_N$  has a coarse moduli scheme  $M$  over  $k$ . And if  $k = \mathbb{C}$ ,  $M$  is canonically isomorphic to  $Y(N)$ .

And for  $k = \mathbb{Q}(\zeta_N)$ ,  $M$  has good reduction over places away from  $N$ . ┘

*Proof:* ? □

**Cor. (28.3.1.9)**  $[\mathcal{E}_2]$ . The moduli problem  $\mathcal{E}_2$  in fact has  $\mathbb{A}^1$  as the fine moduli scheme over  $\mathbb{Q}$ : There is an elliptic curve  $E \in \mathbf{EllCur}/\mathbb{A}^1$  given by

$$W_\lambda : y^2 = x(x-1)(x-\lambda), \quad \lambda \in \mathbb{A}^1.$$

And it has a  $\Gamma(2)$ -structure given by the points  $((0,0), (1,0))$ . And it can be checked that this is a fine moduli problem. ? ┘

*Proof:* □

## 2 Modular Curves

**Prop. (28.3.2.1)** **[Shimura Curves]**. For any compact open subgroup  $K \subset \text{GL}(2, \mathbf{A}_f)$ , there exists a modular curve  $M_K$  defined over  $\mathbb{Q}$  s.t.

$$M_K(\mathbb{C}) = \text{GL}(2, \mathbb{Q}) \backslash \mathcal{H}^\pm \times \text{GL}(2, \mathbf{A}_f) / K.$$

And there is a compactification  $M_K \hookrightarrow \overline{M}_K$ , where  $\overline{M}_K$  is a smooth projective curve over  $\mathbb{Q}$ .

In particular, we can define  $X(N)$ ,  $X_1(N)$  and  $X_0(N)$ . ┘

*Proof:* □

**Prop. (28.3.2.2)** **[Integral Modular Curves  $Y(N)$  and  $X(N)$ ]**. For  $N \geq 3$ , the functor “isomorphism classes of elliptic curves with a level  $N$  structure” is representable by a scheme  $Y(N)$  and a universal elliptic curve  $E^{\text{uni}}/Y(N)$  with a level  $N$  structure  $\alpha_N$ .

$Y(N)$  is an affine smooth precurve over  $\mathbb{Z}[\frac{1}{N}]$ , finite flat of degree  $\# \text{GL}(2, \mathbb{Z}/(N))/\{\pm 1\}$  over the affine  $j$ -line  $\mathbb{Z}[\frac{1}{N}, j]$ , and étale away from from 0 and 1728.

The normalization  $X(N)$  of the projective  $j$ -line  $\mathbb{P}_{\mathbb{Z}[\frac{1}{N}]}^1$  in  $Y(N)$  is a proper smooth precurve over  $\mathbb{Z}[\frac{1}{N}]$ .

$\Gamma(X(N)) = \mathbb{Z}[\frac{1}{N}, \zeta_N]$ , and  $X(N) \otimes_{\mathbb{Z}[\frac{1}{N}]} \mathbb{Z}[\frac{1}{N}, \zeta_N]$  (resp.  $Y(N) \otimes_{\mathbb{Z}[\frac{1}{N}]} \mathbb{Z}[\frac{1}{N}, \zeta_N]$ ) is a disjoint union of  $\varphi(n)$  affine (resp. proper) smooth curve over  $\mathbb{Z}[\frac{1}{N}, \zeta_N]$ .

Let  $X(N)^\infty = X(N) \setminus Y(N)$ , then  $(X(N)^\infty)_{\mathbb{Z}[\frac{1}{N}, \zeta_N]}$  is a disjoint union of points, called the **cusps** of  $X(N)$ . And they naturally correspond to the set of isomorphism classes of level  $N$  structures on  $\text{Tate}(q)_{\mathbb{Z}((q)) \otimes \mathbb{Z}[\frac{1}{N}, \zeta_N]}$ .

The completion of  $X(N)$  along any of the cusp is isomorphic to  $\text{Spec } \mathbb{Z}[\frac{1}{N}, \zeta_N][[q]]$ . The completion of the  $j$ -line along  $\infty$  is also isomorphic to  $\text{Spec } \mathbb{Z}[\frac{1}{N}, \zeta_N][[q]]$  via  $j \mapsto q^{-1} + 744 + \dots$ . And the projection of these two completions along  $X(N) \rightarrow \mathbb{P}_{\mathbb{Z}[\frac{1}{N}]}^1$  is just  $q \mapsto q^N$ .

There is a unique invertible sheaf  $\omega_{\overline{E^{\text{uni}}}/X(N)}$  on  $X(N)$  whose restriction on  $Y(N)$  is  $\omega_{E^{\text{uni}}/Y(N)}$ , and at each cusp, it is  $\mathbb{Z}[\frac{1}{N}, \mu_N]$  multiples of the canonical differential of  $\text{Tate}(q^N)$  via the identification as above.  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (28.3.2.3) [Cusps].** Let  $M_K^\infty = \overline{M_K} \setminus M_K$  be the cusps, then

$$M_K^\infty(\mathbb{C}) = (\pm \text{Unip}(\widehat{\mathbb{Z}})) \backslash \text{GL}(2, \widehat{\mathbb{Z}}) / K,$$

and  $\text{Aut}(\mathbb{C}/\mathbb{Q})$  acts on the cusps by

$$[k]^\tau = [\text{diag}(\chi_{\text{cycl}}(\tau), 1)k], \quad k \in \text{GL}(2, \widehat{\mathbb{Z}}).$$

$\lrcorner$

*Proof:* ?  $\square$

**Prop. (28.3.2.4).** If  $M, N \in \mathbb{Z}_+, M + N \geq 5$ , the functor “isomorphism classes of elliptic curves with an injection of  $\mathbb{Z}/(M) \times \mathbb{Z}/(N)$ ” is represented by a scheme  $Y(M, N)$  and a universal elliptic curve  $E^{\text{uni}}/Y(M, N)$  with a level structure  $\alpha : \mathbb{Z}/(M) \times \mathbb{Z}/(N) \hookrightarrow Y(M, N)$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (28.3.2.5).** The restriction map

$$H_{\text{Mot}}^2(\overline{M}, \mathbb{Q}(2)) \rightarrow H_{\text{Mot}}^2(M, \mathbb{Q}(2))$$

is injective.  $\lrcorner$

*Proof:* Cf. [Scholl, Beilinson’s theorem on Modular Curves]P276.  $\square$

**Def. (28.3.2.6) [Ordinary Locus].** Let  $Y(N)^{\text{ord}}$  be the complement of  $Y(N)_{\mathbb{Z}}$  of the finite set of supersingular points in characteristic dividing  $N$ . Then  $Y(N)^{\text{ord}}$  is smooth over  $\mathbb{Z}[\mu_N]$ .  $\lrcorner$

*Proof:* Cf. [Katz-Mazur]Cor10.9.2. ?  $\square$

### Manin-Drinfeld Theorem

**Thm. (28.3.2.7) [Manin-Drinfeld Theorem].** Let  $\mathcal{C} \in \text{Pic}^0(M_K^c \otimes \mathbb{C})$  be the subgroup of classes of divisors supported on the cusps of  $M_K^c(\mathbb{C})$ , then  $\mathcal{C}$  is finite.  $\lrcorner$

*Proof:*  $\square$

### 3 Complex Modular Curves and Jacobians

**Prop. (28.3.3.1)[Local Picture].** Let  $D$  be the unit disk and  $\Delta$  be a finite group acting on  $D$  and fixing 0, then by Schwarz lemma,  $\Delta$  is a finite subgroup of  $\text{Aut}(D, 0) \cong \mathbb{R}/\mathbb{Z}$ , so it is a finite cyclic group. If  $|\Delta| = m$ , then  $z^m$  is invariant under  $\Delta$  and defines a function on  $\Delta \backslash D$ . It is a homeomorphism from  $\Delta \backslash D$  to  $D$ , thus defines a complex structure on  $\Delta \backslash D$ .

Let  $X = \{z \in \mathbb{C} \mid \text{Im}(z) > c\}$  and  $h$  an integer. Let  $\mathbb{Z}$  acts on  $X$  by  $nz = z + nh$ . This action can extend to  $X^* = X \cup \{\infty\}$ , and we can consider the quotient space  $\mathbb{Z} \backslash X^*$ . The function  $q(z) = e^{2\pi iz/h}$  is a homeomorphism from  $\mathbb{Z} \backslash X^*$  onto the open disk of radius  $e^{-2\pi c/h}$  and center 0, which defines a complex structure on  $\mathbb{Z} \backslash X^*$ .  $\perp$

**Lemma (28.3.3.2)[Modular Curves for  $\Gamma(1)$ ].** Let  $\mathcal{H}^* = \mathcal{H} \cup \{\infty\}$ , the Riemann surface  $\Gamma(1) \backslash \mathcal{H}^*$  is compact and of genus 0, so isomorphic to the Riemann surface.  $\perp$

*Proof:* We first define a complex structure on  $\Gamma(1) \backslash \mathcal{H}$ : Let  $p: \mathcal{H} \rightarrow \Gamma(1) \backslash \mathcal{H}$  be the quotient map, and let  $Q$  be a point of  $\mathcal{H}$  mapping to  $P$ . If  $Q$  is not an elliptic point, then we can choose a nbhd of  $Q$  that maps isomorphically to a nbhd of  $P$ , so we can define the complex structure near  $P$ .

If  $Q = i$ , then the map  $z \mapsto \frac{z-i}{z+i}$  maps some open nbhd of  $i$  to an open disk  $D'$  with center 0, and the action of  $S$  is transformed to the action  $z \mapsto -z$ . By the local picture,  $f(z) = (\frac{z-i}{z+i})^2$  is invariant under action of  $S$  and defines a complex structure near  $p(i)$ . Similarly, if  $Q = \rho$ , then  $g(z) = (\frac{z-\rho}{z-\bar{\rho}})^3$  is invariant under the action of  $ST$ , and defines a complex structure near  $p(\rho)$ .

The space  $\Gamma \backslash \mathcal{H}$  we get is not compact, and it can be compactified by adding a point  $\infty$ , the resulting space is compact because it is a quotient space of  $\bar{D} \cup \{\infty\}$ , which is compact. And we give a complex structure on the resulting space: The function  $q = e^{2\pi iz}$  is a function mapping a nbhd of  $\infty$  in the fundamental domain to an open disk with center 0, and thus giving a complex structure near  $\infty$ .

It can be seen directly that  $\Gamma(1) \backslash \mathcal{H}^*$  is homeomorphic to a sphere, and then use the fact any Riemann surface of genus 0 is isomorphic to  $S^1$  (7.13.8.2).  $\square$

**Prop. (28.3.3.3)[Modular Curves for  $\Gamma$ ].** For a congruence group  $\Gamma$ , the quotient space  $\Gamma \backslash \mathcal{H}$  can be compactified to a Riemann surface  $X(\Gamma)$  by adjoining  $n$  points, where  $n$  is the number of cusps of  $\Gamma$ .  $\perp$

*Proof:* The quotient space  $\Gamma \backslash \mathcal{H}$  can be given a complex structure exactly the same way as  $\Gamma(1) \backslash \mathcal{H}$ , and it can be compactified by adding  $\mathbb{P}^1(\mathbb{Q})$ , which has only f.m. orbits under action of  $\Gamma$ , by (20.4.5.18) and the fact  $\Gamma$  has finite index in  $\Gamma(1)$ . For the complex structure: if  $h$  is the smallest positive integer that  $T^h \in \Gamma$ , then  $q = e^{2\pi iz/h}$  is a homeomorphism of a nbhd of  $\infty$  in  $\Gamma \backslash \mathcal{H}$  to some open disk of 0, thus defines a complex structure near  $\infty$ . For any other cusps  $\alpha$ , let  $\gamma \in \Gamma(1)$  satisfies  $\alpha = \gamma(\infty)$ , then  $z \mapsto q(\gamma^{-1}(z))$  defines a complex structure near  $\alpha$ .  $\square$

**Def. (28.3.3.4)[Notations].** Let  $\Gamma$  be a congruence subgroup, then we denote

$$Y(\Gamma) = \Gamma \backslash \mathcal{H}, \quad X(\Gamma) = \Gamma \backslash \mathcal{H}^*.$$

Also abbreviate  $Y(\Gamma(N))$  to  $Y(N)$ ,  $X(\Gamma(N))$  to  $X(N)$ , and  $Y(\Gamma_0(N))$  to  $Y_0(N)$ , and  $X(\Gamma_0(N))$  to  $X_0(N)$ .  $\perp$

**Prop. (28.3.3.5)[Modular Form as Differential Forms].** A holomorphic modular form of degree  $k$  in  $M_k(\Gamma)$  is just a holomorphic differential form on  $X(\Gamma)$  of degree  $k$ , and its dimension can be calculated, see [Dimension Formulae](#).

Similarly, an automorphic function for  $\Gamma$  is the same as a function on  $X(\Gamma)$  (28.3.3.3). In particular, if it is holomorphic and vanishes at cusps, then it is constant.  $\perp$

*Proof:* ? □

**Cor. (28.3.3.6) [Hauptmodul].** Let  $\Gamma$  be a congruence subgroup with only 1 cusp, a **Hauptmodul** is the unique meromorphic modular form that has only a simple pole at the cusp with residue 1. ┘

### Over $\mathbb{C}$

**Prop. (28.3.3.7).** Let  $\mathcal{E}_{0,N}$  be the functor of elliptic curves with a cyclic subgroup of order  $N$ , then there is a bijection

$$Y_0(N) = \Gamma_0(N) \backslash \mathcal{H} \cong \mathcal{E}_{0,N}(\mathbb{C}) : z \mapsto (\mathbb{C}/\Gamma(z, 1), \Gamma(z, 1/N)/\Gamma(z, 1)).$$

┘

*Proof:* Cf. [Diamond or Milne]. □

**Prop. (28.3.3.8).** Let  $\mathcal{E}_N$  be the functor of elliptic curves with a  $\Gamma(N)$  structure of Weil pairing  $\zeta_N$ , then there is a bijection

$$Y(N) = \Gamma(N) \backslash \mathcal{H} \cong \mathcal{E}_{0,N}(\mathbb{C}) : z \mapsto (\mathbb{C}/\Gamma(z, 1); z/N, 1/N).$$

┘

*Proof:* Cf. [Diamond]. □

### Modular Equations

**Prop. (28.3.3.9) [ $X_0(N)$ , Modular Equations].** The field  $C(X_0(N))$  of modular functions for  $\Gamma_0(N)$  is generated by  $j(z)$  and  $j(Nz)$  over  $\mathbb{C}$ , and the minimal polynomial  $F(j, Y)$  of  $j(Nz)$  over  $\mathbb{C}(j)$  has degree  $d = [\mathrm{PSL}(2, \mathbb{Z}) : \bar{\Gamma}_0(N)]$  (26.3.3.1), and  $F(j, Y) \in \mathbb{Z}[j, Y]$ .

When  $N > 1$ ,  $F(X, Y)$  is symmetric in  $X, Y$ , and when  $N = p$  is a prime,

$$F(X, Y) \equiv X^{p+1} + Y^{p+1} - X^p Y^p - XY \pmod{p}.$$

┘

*Proof:* Cf. [Milne, Modular Forms]P88. □

**Cor. (28.3.3.10).** The field of functions on  $X(1)$  is  $\mathbb{C}(j)$ . ┘

**Thm. (28.3.3.11).** For  $E, E' \in \mathrm{EllCur}/\mathbb{C}$ , there exists a cyclic isogeny  $\alpha : E \rightarrow E'$  of degree  $m$  iff  $\Phi_m(j(E), j(E')) = 0$ . ┘

*Proof:* Cf. [Cox, P213]. □

### Action of Hecke Algebras

**Prop. (28.3.3.12) [Double Coset Operators].** There is an action of  $\mathcal{R}$  on  $X_1(\Gamma)$  ┘

### Modular Forms

**Prop. (28.3.3.13) [Modular Forms and Homology].** For  $N \in \mathbb{Z}_+$ , the pairing

$$\langle -, - \rangle : \mathrm{SF}_2(\Gamma_0(N)) \times H_{1, \mathrm{Betti}}(X_0(N), \mathbb{Z}) \rightarrow \mathbb{C} : \langle f, \gamma \rangle = 2\pi i \int_{\gamma} f(z) dz$$

is non-degenerate, and for each  $n \in \mathbb{Z}_+$ , we have

$$\langle T_n(f), \gamma \rangle = \langle f, T_n(\gamma) \rangle.$$

┘

*Proof:* Cf. [Modular forms, a computational approach]P138. ?

This is Poincaré duality, and the second assertion follows from the modular definition of  $T_n$ .  $\square$

### Modular Jacobians

**Def. (28.3.3.14) [Modular Jacobians].** Define  $J_1(N) = \mathrm{Jac}(X_1(N))$ ,  $J_0(N) = \mathrm{Jac}(X_0(N))$ , which is defined over the same field as the defining field of  $X_1(N)$  or  $X_0(N)$ .  $\square$

### Eichler-Shimura Relations

#### 4 Modular Symbols

References are [Man72, MTT86]. [Universal Fourier expansions of modular forms, Merel]. [Modular Forms: A Computational Approach, Steins]Chap8.

**Notation (28.3.4.1).**

- Let  $\Gamma \leq \Gamma(1)$  be a finite-index subgroup.

┘

**Def. (28.3.4.2) [Modular Symbols].** Let  $\mathbf{M}_2$  be the free Abelian group generated by the symbols  $\{\alpha, \beta\}$  where  $\alpha, \beta \in \mathbb{P}^1(\mathbb{Q})$ , subject to the relations

$$\{\alpha, \beta\} + \{\beta, \gamma\} + \{\gamma, \alpha\} = 0, \text{ where } \alpha, \beta, \gamma \in \mathbb{P}^1(\mathbb{Q}),$$

called the set of **modular symbols** (for  $\Gamma(1)$ ). There is a left action of  $\mathrm{GL}(2; \mathbb{Q})$  on  $\mathbf{M}_2$  s.t.

$$g\{\alpha, \beta\} = \{g\alpha, g\beta\}, \text{ where } g \in \mathrm{GL}(2; \mathbb{Q}), \alpha, \beta \in \mathbb{P}^1(\mathbb{Q}),$$

For  $k \in \mathbb{Z}_{\geq 2}$ , define  $\mathbf{M}_k = \mathrm{Sym}^{k-2}(\mathbb{Z}^{\oplus 2})^{\vee} \otimes \mathbf{M}_2$ , called the **modular symbols** of weight  $k$ .

The action of  $\mathrm{GL}(2; \mathbb{Q})$  on  $\mathbf{M}_2$  and the natural action of  $\mathrm{GL}(2; \mathbb{Q})$  on  $\mathbb{Z}^{\oplus 2}$  defines an action of  $\mathrm{GL}(2; \mathbb{Q})$  on  $\mathbf{M}_k$ .

Define  $\mathbf{M}_k(\Gamma) = (\mathbb{Z} \otimes_{\mathbb{Z}[\Gamma]} \mathbf{M}_k)_{\mathrm{lf}}$ .  $\square$

**Def. (28.3.4.3) [Boundary and Cuspidal Modular Symbols].** Situation as in (28.3.4.2), let

$$\mathbf{B}_k = \mathrm{Sym}^{k-2}(\mathbb{Z}^{\oplus 2})^{\vee} \otimes \mathbb{Z}[\mathbb{P}^1(\mathbb{Q})],$$

called the set of **boundary symbols**, then there is a natural action of  $\mathrm{GL}(2; \mathbb{Q})$  on  $\mathbf{B}_k(\Gamma)$ , and denote

$$\mathbf{B}_k(\Gamma) = (\mathbb{Z} \otimes_{\mathbb{Z}[\Gamma]} \mathbf{B}_k)_{\mathrm{lf}}.$$

Then there is a natural map

$$b_k : \mathbf{M}_k(\Gamma) \rightarrow \mathbf{B}_k(\Gamma) : P \otimes \{\alpha, \beta\} \mapsto P \otimes ([\beta] - [\alpha]).$$

And denote  $\mathbf{S}_k(\Gamma) = \ker(b_k)$ .

For any  $R \in \mathcal{CRing}$ , denote

$$\mathbf{M}_k(\Gamma; R) = \mathbf{M}_k(\Gamma) \otimes R, \quad \mathbf{S}_k(\Gamma; R) = \mathbf{S}_k(\Gamma) \otimes R$$

┘

**Prop. (28.3.4.4) [Modular Symbols and Homology].** The natural homomorphism

$$\varphi : \mathbf{M}_2(\Gamma) \rightarrow H_{1, \text{Betti}}(X(\Gamma), \text{cusp}(\Gamma); \mathbb{Z})$$

is an isomorphism, and induces an isomorphism

$$\mathbf{S}_2(\Gamma) \cong H_{1, \text{Betti}}(X(\Gamma); \mathbb{Z}).$$

┘

*Proof:*  $\mathbf{M}_2(\Gamma) \cong \mathbb{Z} \otimes_{\mathbb{Z}[\Gamma]} \mathbf{M}_2$ . Notice  $\mathbf{M}_2 \rightarrow \mathbf{B}_2 \rightarrow \mathbb{Z} \rightarrow 0$  is a free  $\mathbb{Z}[\Gamma]$ -resolution of  $\mathbb{Z}$ , so

$$H_{1, \text{Betti}}(X(\Gamma), \text{cusp}(\Gamma); \mathbb{Z}) = H_1(\mathbb{Z} \otimes_{\mathbb{Z}[\Gamma]} C_\bullet(\mathcal{H}^*, \mathbb{P}^1(\mathbb{Q}))) = H_0(\mathbb{Z} \otimes_{\mathbb{Z}[\Gamma]} \mathbf{M}_2) = \mathbf{M}_2(\Gamma).$$

Similarly,

$$H_{1, \text{Betti}}(X(\Gamma); \mathbb{Z}) = H_1(\mathbb{Z} \otimes_{\mathbb{Z}[\Gamma]} C_\bullet(\mathcal{H}^*)) = H_1((\mathbf{M}_2 \rightarrow \mathbf{B}_2) \otimes_{\mathbb{Z}[\Gamma]} \mathbb{Z}) = \mathbf{S}_2(\Gamma).$$

□

**Prop. (28.3.4.5) [Manin[Man72]].** The homomorphism of left  $\mathbb{Z}[\Gamma(1)]$ -modules:  $\mathbb{Z}[\Gamma(1)] \rightarrow \mathbf{M}_2 : \gamma \mapsto \{\gamma(0), \gamma(\infty)\}$  is surjective with kernel  $\langle 1 + \sigma, 1 + \tau + \tau^2 \rangle$ , where  $\sigma = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}, \tau = \begin{bmatrix} 1 & -1 \\ 1 & \end{bmatrix}$ . ┘

*Proof:* Cf.[modular forms, a computational approach]P43. ?

This map factors through the right action of the left ideal  $\langle 1 + \sigma, 1 + \tau + \tau^2 \rangle$  because for any  $\gamma \in \text{SL}(2; \mathbb{Z})$ ,

$$[\gamma] + [\gamma\sigma] = \{\gamma(0), \gamma(\infty)\} + \{\gamma(\infty), \gamma(0)\} = 0,$$

$$[\gamma] + [\gamma\tau] + [\gamma\tau^2] = \{\gamma(0), \gamma(\infty)\} + \{\gamma(\infty), \gamma(1)\} + \{\gamma(1), \gamma(-)\} = 0.$$

It is surjective because the  $\Gamma(1)$ -action on  $\mathbb{P}^1(\mathbb{Q})$  is two-transitive.

To show the kernel is just  $I = \langle 1 + \sigma, 1 + \tau + \tau^2 \rangle$ , suppose  $g = \sum n_i \gamma_i = 0$ . Notice that  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \sigma = \begin{bmatrix} b & -a \\ d & -c \end{bmatrix}$ , so we may assume that  $c \geq |d|$  for each  $\gamma_i$ . Let  $\gamma$  be with the largest  $c$ , then there must be another  $\gamma'$  with negative coefficients with the same  $(a : c)$ . Thus it must be  $\begin{bmatrix} a & b-a \\ c & d-c \end{bmatrix}$  and  $c > |d| > 0$ . But then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} a & b-a \\ c & d-c \end{bmatrix} = - \begin{bmatrix} a & b-a \\ c & d-c \end{bmatrix} \tau \sigma + \begin{bmatrix} a & b-a \\ c & d-c \end{bmatrix} = - \begin{bmatrix} a & b-a \\ c & d-c \end{bmatrix} \tau^2 = \begin{bmatrix} b-a & -b \\ d-c & -d \end{bmatrix} \pmod{I},$$

Thus we eliminated at least one term  $\gamma$  with the largest  $c$ . Continuing this way, we can only get 0. Thus  $g \in I$ . □



**Cor. (28.3.4.6)** [Manin Symbols, Manin[Man72]]. The map  $\Gamma(1) \rightarrow \mathbf{M}_2(\Gamma) : \gamma \mapsto [\gamma] \triangleq \gamma \cdot \{0, \infty\}$  induces an map

$$[-] : \mathbb{Z}[\Gamma \backslash \Gamma(1)] \rightarrow \mathbf{M}_2(\Gamma),$$

and it induces an isomorphism of modules

$$\mathbb{Z}[\Gamma \backslash \Gamma(1)] / \langle 1 + \sigma, 1 + \tau + \tau^2 \rangle \cong \mathbf{M}_2(\Gamma),$$

where  $\sigma = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}, \tau = \begin{bmatrix} 1 & -1 \\ 1 & \end{bmatrix}$ . These  $[\gamma]$  are called **Manin symbols**.  $\lrcorner$

### overconvergent Modular Symbols

References are [efficient calculation of Stark-Heegner points via overconvergent modular symbols, Darmon-Pollack].

## 5 $X(p)$

[Maz78]Intro contains a lot numerical data for the geometry of  $X_0(p)$  and its minimal quotients.

### Minimal Quotients of $X(p)$

**Prop. (28.3.5.1)** [Minimal Quotients of  $X(p)$ ]. For  $p \in \text{Prime}_{\geq 5}$ , any properly-contained subgroup of  $\text{GL}(2; \mathbb{F}_p)$  is conjugate to a subgroup of the following:

- $H = \text{B}(\mathbb{Z}/(p))$ , and  $X_H(p) = X_0(p)$ , with field of definition  $\mathbb{Q}$ .
- $H$  is the normalizer of a split Cartan subgroup  $\text{diag}(*, *) \amalg \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \text{diag}(*, *)$ , and  $X_H(p) = X_{\text{split}}(p)$ , with field of definition  $\mathbb{Q}$ . Moreover,

$$g_{\text{split}}(p) = \frac{11 + (p-8)p - 4\left(\frac{-3}{p}\right)}{24}.$$

- $H$  is the normalizer of a non-split Cartan subgroup  $\mathbb{F}_{p^2}^\times \subset \text{GL}(2; \mathbb{F}_p)$ , and  $X_H(p) = X_{\text{nonsplit}}(p)$ , with field of definition  $\mathbb{Q}$ . Moreover,

$$g_{\text{nonsplit}}(p) = \frac{23 + (p-10)p + 6\left(\frac{-1}{p}\right) + 4\left(\frac{-3}{p}\right)}{24}.$$

- $H$  is the inverse image of  $\text{Sym}_4 \subset \text{PGL}(2; \mathbb{F}_p)$ , and  $X_H(p) = X_{\text{Sym}_4}(p)$ , with field of definition  $\begin{cases} \mathbb{Q} & , p \equiv \pm 3 \pmod{8} \\ \mathbb{Q}(\chi_p) & , p \equiv \pm 1 \pmod{8} \end{cases}$ .
- $H$  is the inverse image of  $\text{Alt}_5 \subset \text{PGL}(2; \mathbb{F}_p)$  if  $p \equiv \pm 1 \pmod{5}$ , and  $X_H(p) = X_{\text{Alt}_5}(p)$ , with field of definition  $\mathbb{Q}(\chi_p)$ .
- $H$  is the inverse image of  $\text{Alt}_4 \subset \text{PGL}(2; \mathbb{F}_p)$ , and  $X_H(p) = X_{\text{Alt}_4}(p)$ , with field of definition  $\mathbb{Q}(\chi_p)$ .

The last 3 cases are called **exceptional subgroups**.  $\lrcorner$

*Proof:* Cf.[Maz78] ?. □

**Cor. (28.3.5.2).**  $X_{\text{split}}(p)$  corresponds to the problem of classifying elliptic curves with an unordered pair of  $p$ -isogenies. And  $X_{\text{nonsplit}}(p)$  corresponds to the problem of classifying elliptic curves with a chosen subfield  $\mathbb{F}_{p^2}$  in  $\text{GL}(2; E[p])$  (But the inclusion may be up to conjugacy). ┘

**Thm. (28.3.5.3) [Serre].** For  $p \in \text{Prime}_{\geq 17}$  and  $H \subset \text{GL}(2; \mathbb{F}_p)$  exceptional (28.3.5.1),  $X_H(p)$  has no  $p$ -adic rational points. ┘

*Proof:* Cf.[Maz77]Intro. ? □

## 6 Modular Jacobians

### Eisenstein Quotients

**Def. (28.3.6.1) [Eisenstein Quotients].** ┘

**Prop. (28.3.6.2).** The Eisenstein quotient (28.3.6.1)  $\text{Jac}_0(N)^{\text{new}} \rightarrow A_{\text{Eis}}$  satisfies  $\text{rank } A_{\text{Eis}}(\mathbb{Q}) = 0$ . ┘

*Proof:* ? Cf.[?Ma77]3.3.1. □

**Prop. (28.3.6.3).** For any imaginary quadratic field  $\mathcal{K}$ , there exists a finite set  $\emptyset \subset_{\text{fin}} S_{\mathcal{K}} \subset \text{Prime}$  s.t. if  $p \in \text{Prime} \setminus S_{\mathcal{K}}$  is inert in  $\mathcal{K}$ , then there is a non-trivial optimal quotient ?  $A$  of  $\text{Jac}_0(N)$  s.t.  $\text{rank } A_{\mathcal{K}} = 0$ . ┘

*Proof:* Cf.[Maz77]. □

### Winding Quotients

**Def. (28.3.6.4) [Winding Quotients].** For  $N \in \mathbb{Z}_+$ , the **winding quotients**  $M_0(N)$  is the maximal quotient of  $\text{Jac}_0(N)$  with Mordell rank 0. ┘

**Prop. (28.3.6.5) [Kolyvagin-Logachev].** For  $N \in \mathbb{Z}_+$ , asymptotically  $\dim M_0(N) \geq \dim \text{Jac}_0(N)/4$ . ┘

*Proof:* ? □

## 7 Moduli Space of Fake Elliptic Curves

References are [Buz97].

**Notation (28.3.7.1).**

- Let  $D/\mathbb{Q}$  be an indefinite non-split quaternion algebra.
  - Let  $\mathcal{O}_D$  be a maximal order of  $D$ .
  - Let  $R = \prod_{p \in \text{Unr}(D)} \mathbb{Z}_p$ , and fix an isomorphism  $\kappa : \mathcal{O}_D \otimes \mathbb{R} \cong \text{Mat}(2; \mathbb{R})$ .
  - Let  $\ell \in \text{Unr}(D)$ .
- ┘

**Prop. (28.3.7.2).** For any  $M \in \mathbb{Z}$  s.t.  $(M, \text{disc}(D)) = 1$ , the isomorphism  $\kappa : \mathcal{O}_D \otimes \mathbb{R} \cong \text{Mat}(2; \mathbb{R})$  (28.3.7.1) defines a morphism  $u_M : \mathcal{O}_D^\times(\mathbf{A}_f) \rightarrow \text{GL}(2; \mathbb{Z}/(M))$ . ┘

**Def. (28.3.7.3) [Fake Elliptic Curves].** For  $S \in \text{Sch}/\mathbb{Z}[\frac{1}{\text{disc}(D)}]$ , a **fake elliptic curve** over  $S$  is an pair  $(A/S, \iota)$  where  $A \in \text{AbVar}^2/S$ , and  $\iota : \mathcal{O}_D \rightarrow \text{End}_S(A)$  is an injection.  $\lrcorner$

**Def. (28.3.7.4) [Naive Full Level- $M$ -Structures].** For  $S \in \text{Sch}/\mathbb{Z}[\frac{1}{d}]$ ,  $A \in \text{fEllCur}/S$  and  $M \in \mathbb{Z}$  s.t.  $(M, \text{disc}(D)) = 1$ , a naive **full level- $M$ -structure** on  $A$  is defined to be an isomorphism

$$\alpha : (\mathcal{O}_D/(M))_S \cong A[M]$$

preserving the left action of  $\mathcal{O}_D$ . If there exists a full level- $M$ -structure on  $A$ , then  $M$  is invertible on  $S$ .

For  $\delta \in \mathcal{O}_D/(M)$ , right multiplication by  $\delta$  defines an endomorphism  $r_\delta$  of  $(\mathcal{O}_D/(M))_S$ . Thus  $(\mathcal{O}_D/(M))^*$  acts on the left on the set of full level- $M$ -structures on  $A$  by  $\delta \cdot \alpha = \alpha \circ r_\delta$ .  $\lrcorner$

**Def. (28.3.7.5) [Naive Level- $H$ -Structures].**  $\lrcorner$

**Def. (28.3.7.6)  $[\tilde{e}]$ .** Let  $\tilde{e} \in \mathcal{O}_D \otimes \mathbb{Z}_\ell$  be an element that corresponds to  $\text{diag}(1, 0) \in \text{Mat}(2; \mathbb{Z}_\ell)$  via  $\kappa$ . It is an idempotent acting on each  $A \in \text{fEllCur}^D/S$  where  $S \in \text{Sch}/\mathbb{Z}[\frac{1}{\text{disc}(D)}]$ .  $\lrcorner$

**Prop. (28.3.7.7) [Ordinary and Supersingular Fake Elliptic Curves].** For  $p \in \text{Prime}$  and  $k \in \text{Field}^p$ , the  $\ell$ -divisible group of  $A_{\bar{k}}$  is isomorphic to  $E[\ell^\infty]^2$  for some  $E \in \text{EllCur}/\bar{k}$ . Then  $A$  is called an **ordinary fake elliptic curve** (a resp. **supersingular fake elliptic curve**) if  $E$  is ordinary or supersingular.  $\lrcorner$

*Proof:* ?  $\square$

**Def. (28.3.7.8)  $[\Gamma_0(\ell), \Gamma_1(\ell)$ -structures].** For  $S \in \text{Sch}/\mathbb{Z}[\frac{1}{\text{disc}(D)}]$  and  $A \in \text{fEllCur}^D/S$ ,

- a  $\Gamma_0(\ell)$ -**structure** on  $A/S$  is an isogeny of false elliptic curves  $A \rightarrow B \in \text{fEllCur}^D/S$  of degree  $\ell$ .
- $\Gamma_1(\ell)$ -**structure** on  $A/S$  is a pair  $(\varphi, P)$ , where
  - $\varphi : A \rightarrow B \in \text{fEllCur}^D/S$  is an isogeny of false elliptic curves  $A \rightarrow B \in \text{fEllCur}^D/S$  of degree  $\ell$ .
  - $P \in \ker(e| \ker(\varphi))(S)$  is a generator of  $\ker(e| \ker(\varphi))$ .
- A **balanced**  $\Gamma_1(\ell)$ -**structure** on  $A/S$  is a triple  $(\text{pr}, P, P')$  where
  - $\varphi : A \rightarrow B \in \text{fEllCur}^D/S$  is an isogeny of false elliptic curves  $A \rightarrow B \in \text{fEllCur}^D/S$  of degree  $\ell$ .
  - $P \in \ker(e| \ker(\varphi))(S)$  is a generator of  $\ker(e| \ker(\varphi))$ .
  - $P' \in \ker(e| \ker(\varphi^\vee))(S)$  is a generator of  $\ker(e| \ker(\varphi^\vee))$ .

And if  $S \in \text{Sch}/\mathbb{Z}[\frac{1}{\text{disc}(D)}][\mu_\ell]$ ,

- A **canonical balanced**  $\Gamma_1(\ell)$ -**structure** on  $A/S$  is a triple  $(\text{pr}, P, P')$  where
  - $\varphi : A \rightarrow B \in \text{fEllCur}^D/S$  is an isogeny of false elliptic curves  $A \rightarrow B \in \text{fEllCur}^D/S$  of degree  $\ell$ .
  - $P \in \ker(e| \ker(\varphi))(S)$  is a generator of  $\ker(e| \ker(\varphi))$ .
  - $P' \in \ker(e| \ker(\varphi^\vee))(S)$  is a generator of  $\ker(e| \ker(\varphi^\vee))$ .

s.t.  $\langle P, P' \rangle_\varphi = \zeta_\ell$ . ?  $\lrcorner$

**Remark (28.3.7.9) [Comparison with Modular Curves].** If  $A[\ell] \cong E[\ell]^2$  for  $E \in \text{EllCur}/S$ , then the  $\Gamma_0(\ell), \Gamma_1(\ell)$ -structures defined here are identical to the  $\Gamma_0(\ell), \Gamma_1(\ell)$ -structures defined in [Katz-Mazur] (For  $\Gamma_0(N)$ , use [Katz-Mazur] Cor6.8.7..  $\lrcorner$

## 8 Shimura Curves

**Def. (28.3.8.1) [Standard Compact Open Subgroups].** For any  $M \in \mathbb{Z}_+$  s.t.  $(M, \text{disc}(D)) = 1$ , define subgroups of  $\mathcal{O}_D^\times(\mathbf{A}_f)$ :

$$V_0(M) = u_M^{-1}(B(\mathbb{Z}/(M))), \quad V_1(M) = u_M^{-1}(M(\mathbb{Z}/(M))), \quad V_M = \ker(u_M) \text{ (28.3.7.2)}.$$

┘

**Prop. (28.3.8.2) [No Fixed Naive Level- $V_1(N)$ -Structures].** For  $N \in \mathbb{Z}_{\geq 4} \setminus (\text{disc}(D))$  and  $k \in \mathbf{Field}$  s.t.  $k = \bar{k}$  and  $N \text{disc}(D) \neq \text{char}(k)$ , if  $A/k, \iota \in \mathbf{fEllCur}^D/k$  and  $\alpha$  is level- $V_1(N)$ -structure on  $A/k$  (28.3.8.1) (28.3.7.5), then the only automorphism of  $(A/k, \iota)$  fixing  $\alpha$  is the identity. ┘

*Proof:* ? Cf. [Buz97]P595. ┐

**Def. (28.3.8.3) [Compact Open Subgroups].** Let  $U_\infty \subset (D \times \mathbb{R})^\times$  corresponding to  $\text{SO}(2; R) \subset \text{GL}(2; \mathbb{R})$ .

Let  $U \subset \mathcal{O}_D^\times(\mathbf{A}_f)$  be a compact open subgroup s.t.

- $\det(U^\times) = \mathbb{Z}^\times$ .
- $U$  is maximal at primes dividing  $\text{disc}(D)$ , i.e.  $U = \prod_{p \in S(\text{disc}(D))} (\mathcal{O}_D \otimes \mathbb{Z}_p)^\times \times U^d$ , where  $U^d \subset (\mathcal{O}_D \times R)^\times$ .
- $U \subset V_1(N)$  (28.3.8.1) for some  $N \in \mathbb{Z}_{\geq 4}$  s.t.  $(N, \text{disc}(D)) = 1$ .

where the first condition makes the generic fiber connected, the third condition makes the moduli problem representable by schemes. ┘

**Prop. (28.3.8.4).** Fixing the compact open subgroups  $U$  in (28.3.8.3), then there exists  $X^D(U) \subset \mathbf{Cur}/\mathbb{C}$  s.t.

$$X^D(U)(\mathbb{C}) \cong D^\times \backslash D^\times(\mathbf{A}) / U.U_\infty.$$

┘

*Proof:* ? ┐

**Thm. (28.3.8.5) [Integral Model of Shimura Curves, Boutot].** Fixing the compact open subgroups  $U$  in (28.3.8.3), and let  $M_U$  be the smallest number in  $\mathbb{Z}_+ \setminus (\text{disc}(D))$  s.t.  $V_M \subset U$  (28.3.8.1).

Define a fibered category  $\mathcal{X}^D(U) \in \mathbf{FibCat}_{\mathbf{fppf}}^{\text{grpd}} / \mathbb{Z}[\frac{1}{M_U \text{disc}(D)}]$  s.t. for each  $S \in \mathbf{Sch}/\mathbb{Z}[\frac{1}{M_U \text{disc}(D)}]$ ,  $X^D(U)(S)$  is the set of triples  $(A/S, \iota_A, \alpha)$  where  $(A, \iota_A) \in \mathbf{fEllCur}^D/S$  (28.3.7.3) and  $\alpha$  is a level- $U$ -structure on  $A/S$  (28.3.7.5), where the Cartesian morphisms are given by  $(A/S, \iota_A, \alpha) \rightarrow (B/T, \iota_B, \beta)$  where

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ S & \longrightarrow & T \end{array} \text{ is Cartesian diagram.}$$

- The induced map  $a : \text{End}_T(B) \rightarrow \text{End}_S(A)$  satisfies  $\iota_A = a \circ \iota_B$ .
- $\alpha$  is the level structure induced by  $\beta$ .

Then  $\mathcal{X}^D(U)$  is an algebraic stack (10.8.6.4) proper smooth of relative dimension 1 over  $\mathbb{Z}[\frac{1}{M_U \text{disc}(D)}]$ .

Moreover, because  $U$  satisfies the properties in (28.3.8.3),  $\mathcal{X}^D(U)$  is representable by a regular scheme smooth projective variety over  $\mathbb{Z}[\frac{1}{M_U \text{disc}(D)}]$ , with generic fiber  $X^D(U)$  (28.3.8.4). ┘

*Proof:* ? Cf. [Buz97]P597. □

**Cor. (28.3.8.6).** For two compact open subgroups  $U_1 \leq U_2$  satisfying the conditions in (28.3.8.3), the canonical morphism  $\mathcal{X}^D(U_1) \rightarrow \mathcal{X}^D(U_2) \in \mathcal{S}\mathrm{ch}/\mathbb{Z}[\frac{1}{M_{U_1} \mathrm{disc}(D)}]$  is étale. ┘

*Proof:* ? □

### New Structures at $\ell$

**Notation (28.3.8.7).**

- Let  $\ell \in \mathrm{Prime} \setminus S(\mathrm{disc}(D))$ .

┘

## 28.4 André-Oort Conjecture

References are [Canonical Heights on Shimura Varieties and the André-Oort Conjecture].

**Def. (28.4.0.1) [Special Subvarieties].** ? ┘

**Thm. (28.4.0.2) [André-Oort Conjecture].** Let  $V$  be an irreducible closed algebraic subvariety of  $\mathcal{A}_g$ , then  $V$  contains only f.m. maximal special subvarieties ? ┘

*Proof:* ? Cf.[Ullmo and A. Yafaev, Galois orbits and equidistribution of special sub- varieties: towards the André-Oort Conjecture]. □

### 1 André-Oort Conjecture for $\mathcal{A}_g$

**Thm. (28.4.1.1) [Lower Bounds for Fields of Moduli].** For any  $g \in \mathbb{Z}_+$ , there exists  $\delta_g \in \mathbb{R}_+$  s.t. for any primitive CM-type  $(E, \Phi)$  of degree  $2g$  and an Abelian variety  $A$  with CM by  $(E, \Phi)$ , then the fields of moduli  $\mathbb{Q}(A)$  of  $A$  satisfies

$$[\mathbb{Q}(A) : \mathbb{Q}] \gg |\text{disc}(E)|^{\delta_g}.$$

┘

*Proof:* ? □

**Thm. (28.4.1.2) [Galois Points of Special Points are Large].** For any  $g \in \mathbb{Z}_+$ , there exists  $b_g, C_g \in \mathbb{R}_+$  s.t.

$$(\#\{\text{Gal}_{\mathbb{Q}} .x\})^{b_g} \geq C_g |\text{disc}(Z(\text{End}(A_x)))|.$$

┘

*Proof:* This follows from (28.4.1.1) by Cf.[André-Oort Conjecture For  $\mathcal{A}_g$ ]P385 ? □

**Thm. (28.4.1.3) [André-Oort Conjecture for  $\mathcal{A}_g$ ].** The André-Oort Conjecture (28.4.0.2) for  $\mathcal{A}_g$  holds for any  $g \in \mathbb{Z}_+$ . ┘

*Proof:* This follows from (28.4.1.2) by [Ax-Lindemann for  $\mathcal{A}_g$ ]P673 ? □

### 2 Zilber-Pink Conjecture

#### Zilber-Pink Conjecture

**Conj. (28.4.2.1) [Zilber-Pink].** If  $X$  is a mixed Shimura variety and  $V \subset X$  is a subvariety, then the union of atypical components(i.e. is the component of the intersection of  $V$  with a special subvariety with atypical dimension) of  $V$  in  $X$  is a finite union. ┘

*Proof:*

□

**Thm& Conj.Cor. (28.4.2.2) [Special case of Zilber-Pink Conjecture].** If  $V \subset Y(1)^3$  is a curve that is not contained in any proper special subvarieties of  $X$ , then the intersection of  $V$  with the union of special subvarieties of  $X$  of codimension  $\geq 2$  is a finite set. ┘

*Proof:*    ? □

**Thm& Conj.Cor. (28.4.2.3) [Lang's Theorem].** If  $V \subset (\mathbb{C}^\times)^3$  is a curve that is not contained in any proper special subvariety, then there are only f.m. points of  $V$  that satisfy two independent multiplicative conditions. ┘

*Proof:*    Cf.[Bombieri-Masser-Zannier, On unlikely intersections of complex varieties with tori]. □

## 28.5 Integral Models

References are [Kot92a], [Lan, Arithmetic Compactifications of PEL-Type Shimura Varieties] and [Kisin, Integral models for Shimura varieties of Abelian type].

**Notation (28.5.0.1).**

- Use notations on reductive groups defined in [Group Theory III: Reductive Groups](#).
- Use notations on symmetric spaces defined in [Locally Symmetric Spaces](#).
- Use notations on Abelian varieties defined in [Abelian Varieties](#).
- Use notations on Shimura-varieties defined in [Shimura Varieties](#).
- Let  $(G, X)$  be a Shimura-datum.
- Let  $E = E(G, X)$  be the reflex field.
- Let  $p \in \text{Prime}$ .

┘

**Conj. (28.5.0.2) [Smooth Canonical Integral Models, Conjectured by Langlands/Milne].** If  $\mathcal{K}_p \leq G(\mathbb{Q}_p), \mathcal{K}^p \leq G(\mathbb{A}_f^p), \mathcal{K} = \mathcal{K}_p \mathcal{K}^p \leq G(\mathbb{A}_f)$  s.t.  $\mathcal{K}_p$  is hyperspecial, and  $\lambda \in S_E(p)$ , then  $\text{Sh}_{\mathcal{K}}(G, X)$  admits a canonical integral model [?](#) over  $\mathcal{O}_{(\lambda)}$ .

┘

*Proof:*

□

### 1 PEL-Type Shimura Varieties

**Def. (28.5.1.1) [Good Primes].**  $p \in \text{Prime}$  is said to be **good** for an integral PEL-datum  $(\mathcal{O}_B, *, L, \langle -, - \rangle, h_0)$  [\(28.1.4.1\)](#) if

- $p$  is unramified in  $\mathcal{O}_B$ , i.e.  $p \nmid \text{disc}(\mathcal{O}_B)$ .
- $p \nmid [L^\vee : L]$ .
- $p \neq 2$  if  $(\mathcal{O}_B \otimes R, *)$  has a simple factor isomorphic to the matrix algebra over  $\mathbb{H}$  with its canonical positive involution (type D).
- There are no non-trivial level-structures at  $p$ .

┘

### PEL-Type Moduli Problems

**Prop. (28.5.1.2) [PEL-Type Integral Models, Zink[Zin82]/Langlands-Rapoport[L-R87]].** Given an integral PEL-datum  $(\mathcal{O}_B, *, \Lambda, \langle -, - \rangle, h_0)$  [\(28.1.4.1\)](#), let  $\mathcal{K} = \mathcal{K}_p \mathcal{K}^p$  where  $\mathcal{K}_p = G(\mathbb{Z}_p)$ , there is a clopen embedding

$$\text{Sh}_{\mathcal{K}}(G, \mathcal{D})(\mathbb{C}) \subset \mathcal{S}_{\mathcal{K}}$$

as complex manifolds, where  $\mathbf{M}_{\mathcal{K}}$  is a moduli space of polarized abelian varieties with an action of  $B$  respecting the determinant condition, equipped with  $\mathcal{K}$ -level structure, up to isogeny. And  $\mathbf{M}_{\mathcal{K}}$  is a disjoint union of  $\text{III}(G/\mathbb{Q})$ -many such  $\text{Sh}_{\mathcal{K}}(G, \mathcal{D})(\mathbb{C})$ .

When  $\mathcal{K}$  is neat,  $\mathbf{M}_{\mathcal{K}}$  is a finite covering of some Siegel moduli scheme. And when  $(G_i, D_i)$  are all Shimura data, this  $\mathbf{M}_{\mathcal{K}}$  has a canonical integral model, and serves as an integral model of  $\text{Sh}_{\mathcal{K}}(G, \mathcal{D})$ .

┘



*Proof:* ? □

**Prop. (28.5.1.3) [*p*-Integral Models].** Given a *p*-integral PEL-datum  $(\mathcal{O}_B, *, \Lambda \langle -, - \rangle, h_0)$  (28.1.4.1) with reflex field  $E \in \mathbf{NField}$  (28.1.4.3), and any  $\mathfrak{p} \in S_E(p)$ , there is a tower of schemes

$$\varprojlim_{\mathcal{K}^p} \mathcal{S}_{\mathcal{K}_p \mathcal{K}^p} \in \mathbf{Sch}/(\mathcal{O}_E)_{\mathfrak{p}}$$

s.t.

- The base change of  $\mathcal{S}_{\mathcal{K}_p \mathcal{K}^p}$  to  $\mathcal{O}_{E_{\mathfrak{p}}}$  classifies PEL-triples  $(A, \bar{\lambda}, \bar{\eta}^p)$  where  $A$  is an  $\mathcal{O}_B$ -Abelian variety,  $\bar{\lambda}$  is a  $\mathbb{Q}$ -homogenous  $\mathcal{O}_B$ -polarization, and  $\bar{\eta}^p$  is a  $\mathcal{K}^p$ -level structure? .
- The base change of  $\mathcal{S}_{\mathcal{K}_p \mathcal{K}^p}$  to  $\mathbb{C}$  is  $\mathcal{S}_{\mathcal{K}_p \mathcal{K}^p}$  (28.5.1.2).

┘

*Proof:* ? [Kot92]. □

**Remark (28.5.1.4).** Moduli problems defined by Drinfeld level structures are generally not even flat over the base rings. ┘

**Def. (28.5.1.5) [Hecke Operators].** ┘

**Notation (28.5.1.6).** ┘

**Prop. (28.5.1.7) [RSZ Unitary Shimura varieties].** Suppose

- Let  $E$  be an imaginary quadratic field.
- Let  $p \in \mathbf{Prime}_{\geq 3}$  be an inert prime with  $(p) = \mathfrak{p}_v$ .
- Let  $V$  be a  $2r + 1$ -dimensional Hermitian space over  $E$  with signature  $(1, 2r)$  at infinity and is unramified at  $p$ .
- Let  $G = \mathbf{U}(V)$  and
- $\mathcal{K} = \mathcal{K}^p \mathcal{K}_p \leq G(\mathbf{A}_f)$  where  $\mathcal{K}_p$  is the hyperspecial subgroup fixing a self-dual  $\mathbb{Z}_{p^2}$ -lattice  $\Lambda \subset V_p$ .

then the Shimura variety  $\mathrm{Sh}_{\mathcal{K}}(G, \mathcal{N})$  has a PEL-type integral model as follows: Fix a CM elliptic curve  $A_0$  with full  $\mathcal{O}_E$ -action, and a prime-to- $p$  principal polarization  $\lambda_0$  of  $A$  defined over the Hilbert class field  $E^1$  of  $E$ . Then the integral model  $\mathcal{S}_{G, \mathcal{K}, \mathcal{O}_{E^1, (p)}} \in \mathbf{Sch}/\mathcal{O}_{E, (p)}$  represents the functor:  $S \in \mathbf{Sch}/\mathcal{O}_{E, (p)} \mapsto \{(A, \lambda, \eta)\}$ , where

- $A \in \mathbf{AbVar}^{2r+1}/S$  with a faithful action by  $\mathcal{O}_E$  satisfying Kottwitz's determinant condition of signature  $(1, 2r)$ .
- $\lambda : A \rightarrow A^{\vee}$  is a prime-to- $p$  principal polarization s.t. the Rosati-involution of  $\lambda$  induces complex conjugation on  $\mathcal{O}_E$ .
- $\eta$  is a section of the étale sheaf

$$\mathrm{Isom}_{\mathrm{Herm}}(V \otimes \mathbf{A}_f^p, \mathrm{Hom}_{\mathcal{O}_E}(V_{\mathrm{\acute{e}t}}(A_0)^p, V_{\mathrm{\acute{e}t}}(A)^p))/\mathcal{K}^p.$$

parametrizing  $\mathcal{K}^p$ -orbits of Hermitian isomorphisms, where the Hermitian structure on the latter is given by

$$\langle x, y \rangle = \lambda_0^{\vee} \circ y^{\vee} \circ \lambda \circ x, \text{ where } x, y \in \mathrm{Hom}_{\mathcal{O}_E}(V_{\mathrm{\acute{e}t}}(A_0)^p, V_{\mathrm{\acute{e}t}}(A)^p).$$

┘

### Automorphic Sheaves

**Def. (28.5.1.8) [Automorphic Sheaves].** Situation as in (28.5.1.3). Let  $L(\xi) \in \mathbf{NField}$  and  $\xi \in \mathrm{Irr}_{L(\xi)}(G)$ , and  $\lambda \in \Sigma_E^{\mathrm{fin}} \setminus S_E(p)$ . Then we get a  $\lambda$ -adic sheaf  $\mathcal{L}_{\xi, \mathcal{K}^p}$  on  $\mathcal{S}_{\mathcal{K}_p \mathcal{K}^p}$  for each neat  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  s.t.

- If  $\mathcal{K}_1^p, \mathcal{K}_2^p \leq G(\mathbf{A}_f^p)$  are neat and  $g \in G(\mathbf{A}_f^p)$  s.t.  $g^{-1}\mathcal{K}_1 g \leq \mathcal{K}_2$ , there is a canonical identification

$$[g]^* \mathcal{L}_{\xi, \mathcal{K}_1^p} = \mathcal{L}_{\xi, \mathcal{K}_2^p}$$

satisfying certain compatibility conditions. ┘

*Proof:* Cf. [Kot92]Section6, [Pink. On  $\ell$ -adic sheaves on Shimura varieties and their higher direct images in the Baily-Borel compactification]Section5.1. ? Notice that we need to use the fact  $A_{G_{\mathbb{R}}} = (A_G)_{\mathbb{R}}$  (which holds for any Hodge-type Shimura-varieties) to show that for any  $\mathcal{K}_1^p \trianglelefteq \mathcal{K}^p \leq G(\mathbf{A}_f^p)$ ,  $\mathcal{S}_{\mathcal{K}_p \mathcal{K}_1^p} \rightarrow \mathcal{S}_{\mathcal{K}_p \mathcal{K}^p}$  is Galois with Galois group  $\mathcal{K}^p / \mathcal{K}_1^p$ . □

## 2 Integral Models by Taking Closure

**Def. (28.5.2.1) [Locally Healthy Schemes].** ? ┘

**Def. (28.5.2.2) [Healthy Extension Property].** For  $p \in \mathbf{Prime}$ , a pro-scheme  $X/\mathbb{Z}_{(p)}$  is said to have the **healthy extension property** if for any  $S \in \mathrm{Sch}/\mathbb{Z}_{(p)}$  regular locally-healthy (28.5.2.1), any map  $S_{\mathbb{Q}} \rightarrow X$  extends to a map  $S \rightarrow X$ . ┘

**Def. (28.5.2.3) [Canonical Integral Models].** A model  $\mathcal{S}_{\mathcal{K}_p}$  for  $\mathrm{Sh}_{\mathcal{K}_p}$  over  $\mathbb{Z}_{(p)}$  is called an **integral canonical model** if it is regular, locally-healthy (28.5.2.1) and satisfies the healthy extension property (28.5.2.2).

If  $\mathcal{S}_{\mathcal{K}_p}$  is a canonical integral model for  $\mathrm{Sh}_{\mathcal{K}_p}$  over  $\mathbb{Z}_{(p)}$ , then for any compact open subset  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$ ,  $\mathcal{S}_{\mathcal{K}_p \mathcal{K}^p} = \mathcal{S}_{\mathcal{K}_p} / \mathcal{K}^p$  is called a canonical integral model for  $\mathrm{Sh}_{\mathcal{K}_p \mathcal{K}^p}$ . ┘

**Prop. (28.5.2.4) [Canonical Integral Models of Hodge-Type, Kisin[?Kis10]/Kim-Madapusi.Pera].** If  $(G, X)$  is of Hodge-type, choose a Siegel-embedding  $i : (G, X) \rightarrow (\mathrm{GSp}(V, \psi), \mathcal{H}^{\pm})$  s.t.  $\mathcal{G}(\mathbb{Z}_p) = G(\mathbb{Q}_p) \cap \mathrm{GSp}(V_{\mathbb{Z}_p}, \psi)$  for a self-dual  $\mathbb{Z}_p$ -lattice  $V_{\mathbb{Z}_p} \subset V_{\mathbb{Q}_p}$ , and that  $\mathcal{K}^p = \mathcal{K}'^p \cap G(\mathbf{A}_f^p)$ , then the closure of  $\mathrm{Sh}_{\mathcal{K}_p}(G, X)$  in  $\mathcal{S}_{\mathcal{K}'^p}(\mathrm{GSp}(V), \psi)$  is a canonical integral model for  $\mathrm{Sh}_{\mathcal{K}_p}(G, X)$ . ┘

## 3 Abelian-Type Shimura Varieties

References are [?Kis10].

**Thm. (28.5.3.1) [Canonical Integral Models of Abelian-Type, Vasiu07/Kisin[?Kis10]].** If  $(G, \mathcal{N})$  is a Shimura-variety of Abelian-type s.t.  $G$  splits over a tamely-ramified extension of  $\mathbb{Q}_p$ , and  $\mathcal{K}_p$  is parahoric,  $\mathcal{K}^p$  is sufficiently small, then

- For each  $v \in S_{E(G, \mathcal{N})}(p)$ , the  $E_v$ -scheme  $\mathrm{Sh}_{\mathcal{K}_p}(G, X)$  admits a  $G(\mathbf{A}_f^p)$ -equivariant extension to a flat  $\mathcal{O}_{E_v}$ -scheme  $\mathcal{S}_{\mathcal{K}_p}(G, X)$ , satisfying extension property w.r.t. mixed-characteristic DVRs.
- Any sufficiently small  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  acts freely on  $\mathcal{S}_{\mathcal{K}_p}(G, X)$ , and the quotient  $\mathcal{S}_{\mathcal{K}}(G, X) = \mathcal{S}_{\mathcal{K}_p}(G, X) / \mathcal{K}^p$  is a finite  $\mathcal{O}_{E_{\lambda}}$ -scheme extending  $\mathrm{Sh}_{\mathcal{K}}(G, X)$ . ┘

*Proof:* ? □

**Remark (28.5.3.2) [Properness].** [Madapusi Pera, Toroidal compactifications of integral models of Shimura varieties of Hodge type] proved that if  $\mathrm{Sh}_{\mathcal{K}}(G, \mathcal{N})$  is proper, then  $\mathcal{S}_{\mathcal{K}}(G, \mathcal{N})$  is proper over  $\mathcal{O}_{E(G, \mathcal{N})_{(v)}}$ . ┘

## 4 Local Models

**Thm. (28.5.4.1) [Local Models, Conjectured by Rapoport [Rap95], Pappas-Zhu [P-Z13]].** Situation as in (28.5.3.1), Rapoport conjectured that the canonical integral model  $\mathcal{S}_{\mathcal{K}}(G, X, \mathcal{K}_p)/\mathcal{O}_{E_v}$  should be modeled by some group-theoretically defined local model  $\mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}}/\mathcal{O}_{E_v}$  attached to  $(G_{\mathbb{Q}_p}, X, \mathcal{K}_p)$ , s.t.

- $\mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}}$  is projective and flat over  $\mathrm{Spec} \mathcal{O}_{E_v}$ .
- $\mathcal{G}_{\mathcal{K}_p}$  acts on  $\mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}}$ .
- $\mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}} \otimes E \cong G_{\mathbb{Q}_p}/P(\mu_h)$  is the partial flag variety associated to  $(G_{\mathbb{Q}_p}, X)$ .
- $\mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}}(\bar{\kappa}(v))$  can be precisely group-theoretically described.
- The action of  $\mathcal{G}_{\mathcal{K}_p} \otimes \kappa(v)$  on  $\mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}} \otimes \kappa(v)$  has finite many orbits  $\mathcal{O}_w$  indexed by  $w \in \mathcal{K}_p \backslash \mathrm{Adm}(\mu^*)/\mathcal{K}_p$ . And  $\mathcal{O}_w \subset \mathrm{cl}(\mathcal{O}_{w'})$  iff  $w \preceq w' \in W_{\mathcal{K}_p} \backslash \widetilde{W}/W_{\mathcal{K}_p}$  (? Cf. [H-R17]P322).

Cf. [Pappas, On the arithmetic moduli schemes of PEL Shimura varieties].

When  $G$  splits over a tamely-ramified extension of  $\mathbb{Q}_p$ , such a local model is constructed by [Pappas-Zhu], which is an orbit closure in an affine Grassmannian. ┘

*Proof:* ? □

**Prop. (28.5.4.2) [Local Models for Abelian-Type Shimura Varieties, Kisin [Kis10]/Pappas-Zhu].** Situation as in (28.5.3.1), suppose  $p \nmid \#\pi_1([G, G])$  and that either  $(G^{\mathrm{ad}}, X^{\mathrm{ad}})$  has no factor of type  $D^H$  or  $G$  is unramified over  $\mathbb{Q}_p$ , then there is a smooth **Kottwitz-Rapoport map** of stacks

$$\lambda_{\mathcal{K}_p} : \mathcal{S}_{\mathcal{K}}(G, X) \rightarrow \left[ \mathcal{G}_{\mathcal{O}_{E_v}}^{\mathrm{ad}, 0} \backslash \mathbf{M}_{G, X, \mathcal{K}_p}^{\mathrm{loc}} \right] \rightarrow W_{\mathcal{K}_p} \backslash \widetilde{W}/W_{\mathcal{K}_p} \text{ (28.5.4.1)}.$$

In particular, the special fiber of  $\mathcal{S}_{\mathcal{K}}(G, X)$  is reduced, and the strict Henselization of the local rings have normal and Cohen-Macaulay irreducible components. ┘

*Proof:* ? □

## 28.6 Rapoport-Zink Spaces and Mixed-Characteristic Shtukas

References are [Kot85, Rap95, R-R96, R-Z96, Tot96, Kot97, And03], [Period domains over finite and  $p$ -adic fields, Dat-Orlik-Rapoport].

**Notation(28.6.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-NField}$ . (It cannot be relaxed to  $K \in p\text{-Field}$ ).
- Let  $K_0 = W(\bar{\kappa})[\frac{1}{p}]$ .
- The arithmetic Frobenius of  $\text{Gal}(K^{\text{sh}}/K)$  is denoted by  $\varphi_0$ .
- Let  $\text{NilSch}$  be the subcategory of  $\text{Sch}$  consisting of schemes s.t.  $p$  is locally nilpotent.
- We abuse notation by calling  $F$ -isocrystals(27.5.5.2) as isocrystals.
- Use notations on  $p$ -adic Galois representations defined in [p-adic Local Galois Representations](#).
- Let  $G \in \text{RedGrp}/K$ .

┘

### 1 $B(G)$ and Isocrystals with Structures

**Def.(28.6.1.1)** [ $B(G)$ ]. Denote  $B(G) = H^1(\text{Gal}_{\check{K}}, G(\check{K}))$ , which is equal to the set of  $\varphi_0$ -conjugacy classes of  $G(\check{K})$ , i.e.  $x \sim y \in G(K^{\text{sh}})$  iff  $x = gx\sigma(g)^{-1}$  for some  $g \in G(K^{\text{sh}})$ .

More generally, for any  $L \subset \check{K}$  complete and Galois over  $K$ , we can define  $B(G; L)$  to be the  $\varphi_0$ -conjugacy classes of  $G(L)$ .

┘

**Prop.(28.6.1.2).** The exact sequence

$$1 \rightarrow \text{Gal}_{\check{K}} \rightarrow W_K \rightarrow \text{Gal}_{\kappa} \rightarrow 1$$

induces a bijection  $B(G) \xrightarrow{\inf} H^1(W_K, G(K^{\text{sep}}))$ . And the inclusions  $W_K \leq \text{Gal}_K$  and  $G(K^{\text{sep}}) \leq G(K^{\text{sep}})$  induce an injection

$$H^1(K, G) \hookrightarrow B(G).$$

┘

*Proof:* The first identity follows from Serre-Hochschild spectral sequence(9.8.1.14) and the Steinberg's theorem(10.5.2.3)  $H^1(\check{K}, G_{\check{K}}) = \mathbf{1}$ .

To show the injection, it suffices to show that for any finite separable extension  $K'/K \in p\text{-LField}$ ,  $H^1(\text{Gal}(K'/K), G(K')) \hookrightarrow H^1(W_K, G(K^{\text{sep}}))$  is injective. But this is the inflation map(9.8.1.14) for the exact sequence

$$1 \rightarrow W_{K'} \rightarrow W_K \rightarrow \text{Gal}(K'/K) \rightarrow 1,$$

which is injective.

□

**Prop.(28.6.1.3)** [Long Exact Sequences]. If  $1 \rightarrow G_1 \rightarrow G_2 \rightarrow G_3 \rightarrow 1 \in \text{RedGrp}/K$  is an exact sequence, then there is an exact sequence of pointed sets

$$1 \rightarrow G_1(K) \rightarrow G_2(K) \rightarrow G_3(K) \rightarrow B(G_1) \rightarrow B(G_2) \rightarrow B(G_3) \rightarrow 1.$$

┘

*Proof:* It follows from Steinberg's theorem (10.5.2.3) that  $G_1(K^{\text{sh}}) \rightarrow G_2(K^{\text{sh}}) \rightarrow G_3(K^{\text{sh}}) \rightarrow 1$  is exact. Taking  $H^1(\text{Gal}_\kappa, -)$  gives the assertion by (9.8.4.4), where  $B(G_2) \rightarrow B(G_3)$  is clear by (28.6.1.1).  $\square$

**Prop. (28.6.1.4) [Restrictions].** If  $K'/K \in p\text{-Field}$  is a finite separable extension and  $G \in \text{RedGrp}/K'$ , then there is a Shapiro isomorphism

$$B(\text{Res}_{K'/K} G) \cong B(G).$$

┘

*Proof:* ?  $\square$

**Prop. (28.6.1.5) [Change of Alg. Closed Fields].** If  $\kappa'/\kappa \in \text{Field}$  s.t.  $\kappa' = \overline{\kappa'}$ , then the natural map

$$B(G) \rightarrow B(G, W(\kappa')[\frac{1}{p}K]) \quad (28.6.1.1)$$

is a bijection.  $\square$

*Proof:* Cf. [Rapoport-Zink] 1.16. ?

This follows from (28.6.1.21).  $\square$

**Remark (28.6.1.6).** When  $G = \text{GL}(n)_K$ ,  $B(G)$  are in bijection with the isomorphism classes of  $n$ -dimensional  $\varphi_0\text{-}\check{K}$ -spaces, so the Dieudonné-Manin classification of isocrystals can be translated to a classification of  $\sigma$ -conjugacy classes of  $\text{GL}(n; K_0)$ . See (28.6.1.45).  $\square$

**Def. (28.6.1.7) [Kottwitz-Maps].** The Kottwitz-map  $G(\check{K}) \rightarrow \pi_1(G)_{I_K}$  (20.5.2.12) factors through a map  $B(G) \rightarrow \pi_1(G)_{\text{Gal}_K}$ , also called the **Kottwitz-map** of  $G$ . It is surjective if  $\kappa$  is a cd-1 field, by (20.5.2.19).  $\square$

**Prop. (28.6.1.8) [Kottwitz-Maps via Crossed Modules].** Cf. [Leb99] or [R-V14] P7.  $\square$

### Slope Homomorphisms

**Def. (28.6.1.9) [Slope Homomorphisms].** For any  $b \in G(\check{K})$ , there is a morphism  $\nu_b \in X_*(G_{\check{K}})_{\mathbb{Q}}$  called the **slope homomorphism** associated to  $b$ , which is defined as follows:

For any  $(\rho, V) \in \text{Rep}_K^{\text{fd}}(G)$ , there is an associated isocrystal defined in (28.6.1.44), then there is a morphism  $\nu_\rho \in X_*(G_{\check{K}})_{\mathbb{Q}}$  that makes  $\mathbf{D}(\mathbb{Q})$  acts on the slope- $\lambda$  component  $V_\lambda$  of  $V$  by the character  $\lambda \in \mathbb{Q} = X^\bullet(\mathbf{D}(\mathbb{Q}))$  (27.5.5.15).

Then for any  $x \in \mathbf{D}(R)$ , the mapping  $\rho \rightarrow \nu_\rho(x)$  gives an automorphism of the standard fiber functor on  $\text{Rep}_K^{\text{fd}}(G)$ , so by Tannakian duality corresponds to a unique element  $y \in G(R)$  that  $\rho(y) = \nu_\rho(x)$  for any  $\rho$ . The homomorphism  $x \mapsto y$  is functorial in  $R$  and thus defines an element  $\nu \in X_*(G_{\check{K}})_{\mathbb{Q}}$ .  $\square$

**Prop. (28.6.1.10) [Characterizing the Slope Morphism].** The slope morphism associated to  $b \in G(K^{\text{sh}})$  can be characterized intrinsically to be the unique morphism  $\nu \in X_*(G_{\check{K}})_{\mathbb{Q}}$  s.t.: There exists some  $s \in \mathbb{Z}_+, c \in G(K^{\text{sh}})$  that

- $s\nu \in X_*(G_{\check{K}})$ ,
- $c(s\nu)c^{-1}$  is defined over the fixed field of  $\sigma^s$ .

- $c(b\sigma)^s c^{-1} = c(s\nu)(\varpi) c^{-1} \sigma^s \in G(\check{K}) \rtimes \text{Gal}_\kappa$ , i.e.  $cb\sigma(b) \dots \sigma^{s-1}(b) \sigma^s(c)^{-1} = c(s\nu)(\varpi_K) c^{-1}$ .

┘

*Proof:* Cf. [Kot85]P13.

The uniqueness follows by choosing an embedding  $G \leq \text{GL}(n)$  ?.

□

**Cor. (28.6.1.11).** For any  $b \in G(\check{K})$ , we have:

- $\nu_{\sigma(b)} = \sigma(\nu_b)$ .
- $\nu_{gb\sigma(g)^{-1}} = g\nu_b g^{-1}$ .
- $\sigma(\nu_b) = b\nu_b b^{-1}$ .

In particular, the  $G(\check{K})$ -conjugacy classes of  $\nu_b$  depends only on the  $\sigma$ -conjugacy classes of  $b$ , and is invariant under  $\text{Gal}_\kappa$ . ┘

*Proof:* 1: This follows from the fact  $(V \otimes_{\mathbb{Q}_p} K_0, b\sigma) \xrightarrow{\sigma, \cong} (V \otimes_{\mathbb{Q}_p} K_0, \sigma(b)\sigma)$ .

2: This follows from the fact  $(V \otimes_{\mathbb{Q}_p} K_0, b\sigma) \xrightarrow{\rho(g), \cong} (V \otimes_{\mathbb{Q}_p} K_0, gb\sigma(g)^{-1}\sigma)$  by (28.6.1.44).

3: This follows from item2 and the fact  $b = b\sigma(b)\sigma(b)^{-1}$ . ┘

□

**Cor. (28.6.1.12) [Newton-Maps].** For a maximal torus  $T \leq G$  with Weyl group  $W(G, T)$ ,  $b \mapsto \nu_G(b)$  (28.6.1.9) induces a map

$$\nu_G : B(G) \rightarrow \mathcal{N}(G) \triangleq \left( X_*(G_{K^{\text{sh}}})_{\mathbb{Q}} / G(K^{\text{sh}}) \right)^{\text{Gal}_\kappa} \cong \left( X_*(T_{K^{\text{sh}}})_{\mathbb{Q}}^+ \right)^{\text{Gal}_\kappa}$$

called the **Newton-map**, and it is functorial in  $G$ . ┘

**Cor. (28.6.1.13).** For  $b \in B(G)$ ,  $\nu_G(b) = 1$  iff  $b$  is in the image of  $H^1(K, G)$  (28.6.1.2). ┘

*Proof:* ?

□

**Prop. (28.6.1.14) [Newton-Maps and Kottwitz-Maps].** For any  $G$  and a maximal torus  $T \leq G$ , we can define the natural quotient map

$$\mathcal{N}(G) \cong \left( X_*(T_{K^{\text{sh}}})_{\mathbb{Q}}^+ \right)^{\text{Gal}_\kappa} \rightarrow (X_\bullet(T)_{\mathbb{Q}} / \mathbb{Q} \cdot \Phi^\vee(G, T))^{\text{Gal}_K} \quad (28.6.1.12),$$

which induces a functorial map

$$\delta_G : \mathcal{N}(G) \rightarrow \pi_1(G)^{\text{Gal}_K} \otimes \mathbb{Q} \quad (10.4.5.24).$$

Then there is a commutative diagram

$$\begin{array}{ccc} B(G) & \xrightarrow{\nu_G} & \mathcal{N}(G) \\ \downarrow \kappa_G & & \downarrow \delta_G \\ \pi_1(G)_{\text{Gal}_K} & \longrightarrow & \pi_1(G)^{\text{Gal}_K} \otimes \mathbb{Q} \end{array} \quad (28.6.1.12)(28.6.1.7).$$

┘

*Proof:* ? Cf. [Kot85].

□

**Prop. (28.6.1.15) [Newton Map Constancy].** If  $S \in \text{N\acute{S}ch}/\kappa$  is connected and  $M \in \text{F-Isoc}_G(S)$  (28.6.1.44) s.t.  $\nu \circ b_M$  (28.6.1.45) is constant, then  $b_M$  is constant. ┘

*Proof:* Cf.[RR96, P173].?

□

**Cor. (28.6.1.16).** If  $S \in \text{NSch}/\kappa$  and  $M \in \text{F-Isoc}_G(S)$ , and  $[b_0] \in B(G)_b$ , then

$$\{s \in S \mid [b_M(s)] = [b_0]\} \subset S$$

is closed.

┘

*Proof:*

□

**Def. (28.6.1.17) [Kottwitz-Decent Elements].**  $[b] \in B(G)$  (28.6.1.1) is called **Kottwitz-decent** if there is some  $s \in \mathbb{Z}_+$  and some  $b \in [b]$  that  $s\nu_b \in X_*(G_{\check{K}})$  and

$$(b\varphi_0)^s = (s\nu_b)(\varpi_K)\varphi_0^s \in G(K_0) \rtimes \text{Gal}_\kappa.$$

This doesn't depend on the choice of  $b$  by (28.6.1.11).

┘

**Prop. (28.6.1.18) [Kottwitz].** Any  $[b] \in B(G)$  is Kottwitz-decent (28.6.1.17).

┘

*Proof:* Cf.[Kot85].?

□

**Prop. (28.6.1.19).** If  $b \in G(\check{K})$  satisfies the Kottwitz-descent condition for  $s$  in (28.6.1.17), then  $b \in G(K_s)$  and  $\nu_b$  is defined over  $K_s$ , the fixed field of  $\varphi_0^s$ . In particular,  $b$  is Kottwitz-decent for any  $s' \in s\mathbb{Z}$ .

┘

*Proof:* Set  $b_s = b\sigma(b) \dots \sigma^{s-1}(b)$ , then iterating (28.6.1.11),  $b_s \nu^{\sigma^s} b_s^{-1} = \nu$ . And we have  $b_s = \nu^s(p)$ , so  $\nu^{\sigma^s} = \nu$ , so  $\nu$  is defined over  $\mathbb{Q}_{p^s}$ .

To show the first assertion, notice  $(b\sigma)(b\sigma)^s = (b\sigma)^s(b\sigma)$  shows

$$\nu^s(p)\sigma^s b\sigma = b\sigma \nu^s(p)\sigma^s = \nu^s(p)b\sigma^{s+1} \text{ (28.6.1.11)}.$$

and then  $b\sigma^s = \sigma^s b$ .

□

**Cor. (28.6.1.20).** If  $b_1, b_2 \in [b]$  are Kottwitz-decent w.r.t the same  $s \in \mathbb{Z}_+$ , then they are conjugate w.r.t.  $G(K_s)$ , where  $K_s$  is the fixed field of  $\varphi_0^s$ .

In particular, for any descent  $b \in G(\mathbb{Q}_{p^s})$  and any  $V \in \text{Rep}_{\mathbb{Q}_p}(G)$ , the induced isocrystal (28.6.1.44) is defined over the field  $\mathbb{Q}_{p^s}$ , and it only depends on  $[b] \in B(G)$ .

┘

*Proof:* Suppose  $b_2 = gb_1\sigma(g)^{-1}$ , then  $\nu_2 = g\nu_1 g^{-1}$ , and the descent equations are

$$(b_1\sigma)^2 = s\nu_1(p)\sigma^s, \quad g(b_1\sigma)^s g^{-1} = gs\nu_1(p)g^{-1}\sigma^s.$$

Comparing these two,  $g$  commutes with  $\sigma^s$ , so  $g \in G(K_0 \cap \mathbb{Q}_{p^s})$ .

□

**Prop. (28.6.1.21) [Forms  $J_b$  and Conjugacy Classes].** For  $b_1, b_2 \in G(\check{K})$ , the functor

$$J(R) = \{g \in G(R \otimes_K \check{K}) \mid g(b_1\sigma) = (b_2\sigma)g\}$$

is representable by a smooth affine scheme over  $K$ . When  $b_1 = b_2 = b$ , denote the scheme by  $J_b$ .

Assume  $b_1, b_2 \in G(W(\kappa')[\frac{1}{p}])$  where  $\kappa' = \bar{\kappa}' \subset \kappa$ , and  $J'$  the corresponding smooth affine scheme, then  $J' \rightarrow J$  is an isomorphism. In particular,  $B(G) \rightarrow B'(G)$  is injective, and it is surjective if  $G$  is connected and  $\kappa = \bar{\kappa}$ .

Notice if  $b_1, b_2$  are  $\sigma$ -conjugate, then  $J_{b_1} \cong J_{b_2}$ .

┘

*Proof:* Choose an embedding  $G \leq \mathrm{GL}(n)_{\mathbb{Q}_p}$  and let  $G$  be defined by functions  $f_1, \dots, f_k$ , consider the functor:

$$F(R) = \{g \in \mathrm{Mat}(n; R \otimes_{\mathbb{Q}_p} K_0) \mid g = b\sigma(g)b^{-1}\},$$

then it is representable by an affine space by (28.6.1.22) applied to the  $\sigma$ -linear map  $g \mapsto b\sigma(g)b^{-1}$ .

So there is a f.d.  $\mathbb{Q}_p$ -vector space  $W \subset \mathrm{Mat}(n; K_0)$  that  $F(R) = W \otimes_{\mathbb{Q}_p} R$ . Choose a basis  $(A_i)$  of  $W$ , then  $J(R)$  is just the subfunctor of  $r_i \in R$  that

$$f_k\left(\sum r_i A_i\right) = 0, \quad \det\left(\sum r_i A_i\right) \neq 0.$$

Taking a basis of  $K_0$  over  $\mathbb{Q}_p$ , then these are polynomials with coefficients in  $\mathbb{Q}_p$ . It is automatically smooth by Cartier's theorem (10.1.4.3).

The assertion about base change follows from (28.6.1.22).

The surjectivity follows from (28.6.1.18) and (28.6.1.19).  $\square$

**Lemma (28.6.1.22).** Let  $N$  be a f.d. isocrystal over  $K_0$  w.r.t.  $\varphi_0^s$  for some  $s \neq 0$ , then the following functor

$$F : \mathcal{CRing}_{\mathbb{Q}_p} \rightarrow \mathcal{Ab} : R \mapsto (V \otimes_{\mathbb{Q}_p} R)^{\varphi=\mathrm{id}}$$

is representable by a vector space over  $\mathbb{Q}_p$ .  $\lrcorner$

*Proof:*  $F(R) = V^{\varphi=\mathrm{id}} \otimes_{\mathbb{Q}_p} R$ , so it suffices to show  $\dim_{\mathbb{Q}_p} V^{\varphi=\mathrm{id}} < \infty$ . Firstly assume that  $L = \overline{L}$ , then this is a consequence of Dieudonné-Manin classification (27.5.5.17). This functor  $F$  doesn't depend on  $k$  as long as  $k = \overline{k}$ : if  $k'/k$  is a field extension and  $k' = \overline{k'}$ , then the corresponding functor  $F'$  defined by  $N \otimes_{W(k)[\frac{1}{p}]} W(k)[\frac{1}{p}]$  coincide with  $F$ . (This is also by Dieudonné-Manin classification.)  $\square$

**Cor. (28.6.1.23) [ $J_b$  are Inner Forms of  $Z_G(\nu_b)$ ].** Assume  $[b] \in B(G)$  is Kottwitz-decent for  $s \in \mathbb{Z}_+$  (28.6.1.17), then  $J_b$  is a  $K_s/K$ -inner form of the centralizer  $Z_G(\nu_b)$  (28.6.1.19), where  $K_s$  is the fixed field of  $\sigma^s$ .  $\lrcorner$

*Proof:* The descent equation shows  $b\sigma(b) \dots \sigma^{s-1}(b) = s\nu(\varpi_K)$ , so the adjoint  $b_{\mathrm{ad}} : g \mapsto (b\sigma)g(b\sigma)^{-1} = b\sigma(g)b^{-1}$  defines an element in  $H^1(\mathrm{Gal}(K_s/K), \mathrm{Aut}(Z_G(s\nu(p))(K_s)))$ , because

$$\sigma^k b_{\mathrm{ad}} : g \mapsto ((\sigma(b\sigma^{-1}(g))b^{-1})) = \sigma^k(b)\sigma(g)\sigma^k(b)^{-1}.$$

So

$$b_{\mathrm{ad}} \circ \sigma(b_{\mathrm{ad}}) \circ \dots \circ \sigma^{s-1}(b_{\mathrm{ad}}) : g \mapsto b_s g b_s^{-1} = s\nu(p)g(s\nu(p))^{-1} = g.$$

So it defines an inner form, which is just

$$J'(R) = G_{s\nu(p)}(K_s)^{b_{\mathrm{ad}}\sigma} = \{g \in Z_G(s\nu_b)(R \otimes_K K_s) \mid g(b\sigma) = (b\sigma)g\}.$$

Now it suffices to show  $J'(R)$  is just  $J(R)$  defined in (28.6.1.21). For this, notice any  $g \in J(R)$  commutes with  $b\sigma$  thus commutes with  $s\nu(p)$  by (28.6.1.11), and the descent condition  $(b\sigma)^n = s\nu(p)\sigma^n$  shows it commutes with  $\sigma^n$ , so  $g \in J'(R)$ .  $\square$

**Prop. (28.6.1.24) [Fibers of Newton-Maps].** For  $b \in B(G)$ , denote  $\mathcal{F}_b = \nu_G^{-1}(\nu_G(b))$  (28.6.1.12), then there is a natural bijection  $\mathcal{F}_b = H^1(K, J_b)$  (28.6.1.21), where for any  $b' \in \mathcal{F}_b$ , the corresponding  $J_b$ -torsor is  $J_{b',b}$  (28.6.1.21). And the Abelian group structure on  $\mathcal{F}$  induced from  $H^1(K, J_b)$  ? is independent of  $b \in \mathcal{F}$  chosen.  $\lrcorner$



*Proof:* Cf.[Rapoport-Richartz]P162<sup>?</sup>.

Firstly,  $J_{b',b}$  is a  $J_b$ -torsor because it is non-empty: Replacing  $b$  by some  $\sigma$ -conjugate element, we may assume that  $\nu_b = \nu_{b'}$ , and by (28.6.1.18) there exists  $s \in \mathbb{Z}_+$  s.t.

$$b\sigma(b) \dots \sigma^{s-1}(b) = (s\nu_b)(\varpi_K) = b'\sigma(b') \dots \sigma^{s-1}(b').$$

then by (28.6.1.19),  $s\nu_b$  is defined over  $K_s$ , the fixed field of  $\varphi_0^s$ , and  $b, b' \in G(K_s)$ . Then it follows from [Rational conjugacy classes in reductive groups, Kottwitz]5.2.2<sup>?</sup> that there exists some  $G \in K_s \otimes_K K^{\text{sep}}$  s.t.  $b' = gb\sigma(g)^{-1}$ . Thus  $J_{b',b}(K^{\text{sep}}) \neq \emptyset$ .

For any  $c \in H^1(K, J_b)$ ,  $c$  trivializes over some finite extension  $K'/K \subset \check{K}/K$ . But  $H^1(\text{Gal}(K'/K), J_b(K')) \subset H^1(\text{Gal}(K'/K), G(\check{K} \otimes_K K')) = 0$  by Shapiro's theorem (9.8.2.8), thus there exists  $g \in G(\check{K} \otimes_K K')$  s.t.

$$c_\tau = g^{-1}g^\tau, \text{ where } \tau \in \text{Gal}(K'/K).$$

Put  $b' = gb\sigma(g)^{-1}$ , then  $b' \in G(\check{K})$ , and  $[J_{b',b}] = c$ <sup>?</sup>. Thus  $\mathcal{F}_b \rightarrow H^1(K, J_b)$  is surjective.

To prove injectivity, if  $b', b'' \in \mathcal{F}_b$  s.t.  $[J_{b',b}] = [J_{b'',b}]$ , then

$$J_{b',b} \times^{J_b} J_{b,b''} \cong J_{b',b''}$$

is a trivial  $J_{b'}$ -torsor<sup>?</sup>, which means there exists  $g \in G(\check{K})$  s.t.  $b'' = gb'\sigma(g)^{-1}$ .

For the last assertion, Cf.[R-R96]P163. □

**Cor. (28.6.1.25).** For  $n \in \mathbb{Z}_+$ ,  $\nu_{\text{GL}(n)}$  is injective. ⌋

### Kottwitz-Basic Elements

**Prop. (28.6.1.26) [Kottwitz-Basic Elements].** The following are equivalent for  $b \in G(\check{K})$ :

- The slope morphism  $\nu_b$  (28.6.1.9) factors through the center  $Z(G)$  of  $G$ .
- $b$  is  $\sigma$ -conjugate to an element in  $T(\check{K})$  where  $T$  is an elliptic maximal torus of  $G$  (10.4.2.25).
- $J_b$  (28.6.1.21) is a  $\check{K}/K$ -inner form on  $G$ .

In this case,  $b$  and its conjugacy class  $[b]$  are called **Kottwitz-basic elements**. The set of Kottwitz-basic classes in  $B(G)$  is denoted by  $B(G)_{\text{bas}}$ .

And if  $b \in G(\check{K})$  is Kottwitz-basic, then there exists an isomorphism  $u : (J_b)_{\check{K}} \cong G_{\check{K}}$  s.t.  $u(\varphi_0(x)) = b\varphi_0(u(x))b^{-1}$  for all  $x \in J(\check{K})$ . Such a pair  $(u, b)$  is called a  **$B$ -inner twist** of  $G$ . ⌋

*Proof:* 1  $\iff$  3 follows from (28.6.1.23) and (28.6.1.18).

2  $\rightarrow$  1: If  $T \leq G$  is a elliptic maximal torus, then for  $b \in T(\check{K})$ ,  $\nu_b \in V_*(T) \subset V_*(Z(G))$  because  $T/Z(G)$  is anisotropic, thus  $b$  is Kottwitz-basic.

1  $\rightarrow$  2: Cf.[Kot85]P215.<sup>?</sup>

For the last assertion, Cf.[Kot85]P215.<sup>?</sup> □

**Prop. (28.6.1.27).**  $H^1(K, G) \leq B(G)_{\text{bas}}$ , by (28.6.1.2)(28.6.1.14) and (28.6.1.13). ⌋

**Prop. (28.6.1.28) [Kottwitz-Maps and Tate-Nakayama Isomorphisms].** If  $T$  a torus over  $K$ , and  $K'/K \in \text{GalField}^{\text{fin}}$  a splitting field of  $T$ , then  $\kappa_T : B(T) \rightarrow (X_\bullet(T))_{\text{Gal}_K}$  is an isomorphism, and

there is a commutative diagram

$$\begin{array}{ccc} \widehat{H}^{-1}(\mathrm{Gal}(K'/K), X_{\bullet}(T)) & \xrightarrow{\mathrm{TN}} & H^1(\mathrm{Gal}(K'/K), T(K')) \xrightarrow{\cong} H^1(K, T) \\ \downarrow & & \downarrow \\ (X_{\bullet}(T))_{\mathrm{Gal}_K} & \xleftarrow{\kappa_G, \cong} & B(T), \end{array}$$

where TN is the Tate-Nakayama isomorphism. Recall that  $H_{\mathrm{Tate}}^{-1}(\mathrm{Gal}(K'/K), X_{\bullet}(T)) = \{\lambda \in X_{\bullet}(T) \mid \mathrm{tr}_{K'/K} \lambda = 0\} / \{\sigma(\mu) - \mu\}$ . ?

In particular,  $\kappa_T(H^1(K, T)) = ((X_{\bullet}(T))_{\mathrm{Gal}_K})_{\mathrm{tor}}$  because the image of the canonical map

$$H_{\mathrm{Tate}}^{-1}(\mathrm{Gal}(K'/K), X_{\bullet}(T)) \rightarrow (X_{\bullet}(T))_{\mathrm{Gal}_K}$$

is exactly  $((X_{\bullet}(T))_{\mathrm{Gal}_K})_{\mathrm{tor}}$ . ⌋

*Proof:* Firstly we prove that  $\kappa_T : B(T) \rightarrow (X_{\bullet}(T))_{\mathrm{Gal}_K}$  is an isomorphism: If  $T = \mathbb{G}_m$ , then  $\kappa_G$  is surjective by (20.5.2.17), and to prove injectivity, it suffices to show that  $\beta : \mathcal{O}_{\check{K}}^* \rightarrow \mathcal{O}_{\check{K}}^* : y \mapsto y\varphi_0(y)$  is surjective. Because  $\mathcal{O}_{\check{K}}^*$  is complete and  $\beta(U_n) \subset U_n$ , it suffices to prove that  $\beta$  induces isomorphisms on  $U_n/U_{n+1}$ , which is clear because

$$\beta : \bar{\kappa}^{\times} \rightarrow \bar{\kappa}^{\times} : x \mapsto x^{1-\#\kappa},$$

$$\beta : \bar{\kappa} \rightarrow \bar{\kappa} : x \mapsto x - x^{\#\kappa}$$

are both surjective.

Now Kottwitz-maps are compatible with restrictions (20.5.2.16), so the assertion also holds for induced tori (10.2.3.11). Now by (10.2.3.13) we can take an exact sequence  $1 \rightarrow T_0 \rightarrow T_1 \rightarrow T \rightarrow 1$  where  $T_0, T_1 \in \mathcal{A}lg\mathcal{G}rp/k$  are tori and  $T_1$  is an induced tori, then we get a commutative diagram

$$\begin{array}{ccccccc} B(T_0) & \longrightarrow & B(T_1) & \longrightarrow & B(T) & \longrightarrow & 1 \\ \downarrow \kappa_{T_0} & & \downarrow \kappa_{T_1} & & \downarrow \kappa_T & & \\ \pi_1(T_0)_{\mathrm{Gal}_K} & \longrightarrow & \pi_1(T_1)_{\mathrm{Gal}_K} & \longrightarrow & \pi_1(T)_{\mathrm{Gal}_K} & \longrightarrow & 1 \end{array},$$

where the rows are exact by (28.6.1.3) and (10.4.5.26). Then we see  $\kappa_T$  is surjective, and thus  $\kappa_{T_0}$  is also surjective, and then  $\kappa_T$  is injective.

For the commutativity: ? Cf. [Kot85] 2.7. □

**Lemma (28.6.1.29).** Situation as in (28.6.1.26), if  $\pi_1([G, G]) = 1$ , then  $G \rightarrow G/[G, G]$  induces a bijection  $B(G)_{\mathrm{bas}} \cong B(G/[G, G])$ . ⌋

*Proof:* Firstly, if  $G$  is semisimple and simply-connected, then we can choose an anisotropic maximal torus.  $T \leq G$  by (20.7.7.4), and then  $B(T) \rightarrow G(G)_b$  is surjective by (28.6.1.26), and  $H^1(K, T) \cong B(T)$  because  $X_{\bullet}(T)_{\mathrm{Gal}_K}$  is torsion. So it follows from the commutative diagram

$$\begin{array}{ccc} H^1(F, T) & \longrightarrow & B(T) \\ \downarrow & & \downarrow \\ H^1(F, G) & \longrightarrow & B(G) \end{array}$$

that  $B(G)_{\text{bas}} = 1$ .

Take an elliptic maximal torus  $T \leq G$  (20.7.7.4), then  $B(T) \rightarrow B(G)_{\text{bas}}$  is surjective, and  $T_{\text{der}} = T \cap [G, G]$  is connected because of the commutative diagram

$$\begin{array}{ccc} X_{\bullet}(T) & \longrightarrow & X_{\bullet}(T/T_{\text{der}}) \\ \downarrow & & \downarrow \cong \\ \pi_1(G) & \xrightarrow{\cong} & \pi_1(G/[G, G]) \end{array}.$$

So it follows from (28.6.1.3) that  $B(T) \rightarrow B(G)_b \rightarrow B(G/[G, G])$  is surjective.

And if  $b, b' \in B(G)_b$  maps to the same element in  $B(G/[G, G])$ , because  $1 \rightarrow [G, G](\check{K}) \rightarrow G(\check{K}) \rightarrow (G/[G, G])(\check{K}) \rightarrow 1$  is exact by Steinberg's theorem, we can replace  $b'$  by some  $\sigma$ -conjugate element s.t.  $b' = xb$  for some  $x \in [G, G](L)$ . Then it can be shown that for  $u : (J_b)_{\check{K}} \cong G_{\check{K}}$  as in (28.6.1.26), the  $y = u^{-1}(x) \in B((J_b)_{\text{der}})_b$  ?, which is trivial by what we proved above. So  $y = c\varphi_0^{-1}(c)$  for some  $c \in J_{\text{der}}(\check{K})$ . Then  $b' = u(c)b\varphi_0(u(c))^{-1}$ , so  $b = b'$ .  $\square$

**Thm. (28.6.1.30) [Basic Elements and Kottwitz-Maps].** The Kottwitz-map  $\kappa_G$  (20.5.2.12) induces isomorphisms

$$B(G)_{\text{bas}} \cong \pi_1(G)_{\text{Gal}_K}, \quad H^1(K, G) \cong (\pi_1(G)_{\text{Gal}_K})_{\text{tor}} \quad (28.6.1.1).$$

In particular,  $B(G)_b$  is trivial if  $G$  is semisimple and simply-connected.

And for a basic element  $b$ , the cohomology class of  $J_b$  is the image of  $[b]$  under the map

$$B(G)_{\text{bas}} \rightarrow B(G_{\text{ad}}) \cong H^1(K, G_{\text{ad}}).$$

┘

*Proof:* If  $[G, G]$  is simply-connected, then  $\pi_1(G) \cong \pi_1(G/[G, G])$  and  $B(G)_b \cong B(G/[G, G])$  by (28.6.1.29), also if we take an elliptic maximal torus  $T \leq G$  (20.7.7.4), then  $G/[G, G] \cong T/T_{\text{sim.cntd}}$ , so  $H^1(K, G) \rightarrow H^1(K, G/[G, G])$  is surjective because  $H^2(K, T_{\text{sim.cntd}}) = 0$  (10.5.2.8), and it is injective by (10.5.2.7). So the assertion follows from (28.6.1.28).

Finally take a central extension  $1 \rightarrow Z \rightarrow \tilde{G} \rightarrow G \rightarrow 1$  where  $Z$  is an anisotropic torus and  $[\tilde{G}, \tilde{G}]$  is simply-connected (which exists by [STABLE TRACE FORMULA: CUSPIDAL TEMPERED TERMS, Kottwitz] Lemma 6.2 ?). Then  $\pi_1(\tilde{G}) \cong \pi_1(G)$ ,  $B(Z)$  acts on  $B(\tilde{G})_b$  and induces a quotient isomorphism  $B(\tilde{G})_b/B(Z) \cong B(G)_b$ , and  $H^1(K, Z) \rightarrow H^1(K, \tilde{G}) \rightarrow H^1(K, G) \rightarrow 1$  is exact because  $H^2(K, Z) = 0$  (10.5.2.8). Notice the following exact sequences:

$$\pi_1(Z)_{\text{Gal}_K} \rightarrow \pi_1(\tilde{G})_{\text{Gal}_K} \rightarrow \pi_1(G)_{\text{Gal}_K} \rightarrow 1,$$

$$\pi_1(Z)_{\text{Gal}_K} \rightarrow \left( \pi_1(\tilde{G})_{\text{Gal}_K} \right)_{\text{tor}} \rightarrow (\pi_1(G)_{\text{Gal}_K})_{\text{tor}} \rightarrow 1,$$

where the second one follows from the fact that  $\pi_1(Z)_{\text{Gal}_K}$  is torsion because  $Z$  is anisotropic. Then the assertion follows from the assertions for  $\tilde{G}$  and  $Z$ .

For the last assertion, Cf. [Kot85] Chap 5. ? or see (28.6.1.24).  $\square$

**Def. (28.6.1.31) [G-Regular Basic Elements].** If  $G$  is quasi-split and  $M \leq G$  be a Levi-subgroup and  $S \leq M$  the maximal central split torus. For  $b \in B(M)_{\text{bas}}$  with slope homomorphism  $\nu_b : \mathbf{D}(\mathbb{Q}) \rightarrow S$ , the centralizer  $Z_G(\nu_b)$  is a Levi subgroup containing  $M$ . Then  $b$  is called a **G-regular basic element** if  $Z_G(\nu_b) = M$ . ┘

**Prop. (28.6.1.32).** For  $G$  quasi-split,

- Any  $b \in G(\check{K})$  is  $\sigma$ -conjugate to a  $G$ -regular basic element of some Levi-subgroup of  $G$  (28.6.1.31).
- If  $M_1, M_2 \leq G$  are Levi-subgroups and  $b_i \in B(M_i)$  are  $G$ -regular basic elements, then  $b_1, b_2$  has the same image in  $B(G)$  iff there exists  $g \in G(K)$  s.t.  $gM_1g^{-1} = M_2$  and  $\text{Ad}(g)b_1 = b_2$ .  $\lrcorner$

*Proof:* 1: Consider the slope homomorphism  $\nu_b \in X_*(G_{\check{K}})_{\mathbb{Q}}$ , then the conjugacy class of  $\nu_b$  is invariant under action of  $\text{Gal}_{\check{K}}$  by (28.6.1.11). So there exists  $g \in G(\check{K})$  s.t.  $\text{Ad}(g)\nu_b \in X_*(G)_{\mathbb{Q}}$  (for example, the unique dominant element in the conjugacy class of  $\nu_b$  does). Replacing  $b$  by  $gb\sigma(g)^{-1}$ , we may assume that  $\nu_b \in X_*(G)_{\mathbb{Q}}$ . Define  $M = Z_G(\nu_b)$ , then it follows from (28.6.1.11) that  $b \in M$ , and it is clearly a  $G$ -regular basic element of  $M$ , by (28.6.1.26).

2: One direction is clear. For the other direction, as in the proof of item1, for  $i \in [2]_+$ , let  $m_i \in M_i(\check{K})$  s.t.  $[m_i] = b_i \in B(M_i)$  and  $\nu_{m_i}$  are defined over  $K$ . Suppose  $m_2 = hm_1\sigma(h)^{-1}$  for some  $h \in G(\check{K})$ , then  $\nu_{m_1} = \text{Ad}(h)\nu_{m_2}$ , and then  $\nu_{m_2} = \text{Ad}(g)\nu_{m_1}$  for some  $g \in G(K)$  for the same reason as in the proof of item1. Then  $m_2 = (hg^{-1})(gm_1g^{-1})\sigma(hg^{-1})^{-1}$ , thus  $\text{Ad}(g)b_1 = b_2$ .  $\square$

**Prop. (28.6.1.33)[Image of Newton-Maps].** Suppose  $G$  is quasi-split. If  $M \leq G$  is a Levi subgroup, denote  $\varphi_M$  the following map:

$$\pi_1(M)_{\text{Gal}_K} \rightarrow \pi_1(M/[M, M])_{\mathbb{Q}}^{\text{Gal}_K} = \pi_1(Z_M)_{\mathbb{Q}}^{\text{Gal}_K} \rightarrow \mathcal{N}(G) \quad (28.6.1.12),$$

then the image of  $\nu_G : B(G) \rightarrow \mathcal{N}(G)$  (28.6.1.12) is

$$\text{Im}(\nu_G) = \bigcup_{M \leq G \text{ Levi}} \text{Im}(\varphi_M).$$

$\lrcorner$

*Proof:* This follows from (28.6.1.32) and functoriality that  $\text{?}$   $\square$

**Cor. (28.6.1.34)[Newton-Polygons for  $\text{GL}(n)$ ].** For  $n \in \mathbb{Z}_+$ , the image of  $\nu_{\text{GL}(n)}$  consists of sequences of the form  $(\underbrace{\lambda_1, \dots, \lambda_1}_{n_i\text{-many}}, \dots, \underbrace{\lambda_r, \dots, \lambda_r}_{n_r\text{-many}})$  s.t.  $\sum_i n_i = n$  and  $\lambda_i \in \mathbb{Q}$  and  $n_i\lambda_i \in \mathbb{Z}$  for all  $i$ .  $\lrcorner$

*Proof:* This follows from (28.6.1.33) because the Levi subgroups of  $\text{GL}(n)$  are all conjugate to elements of the form  $\prod_i \text{GL}(n_i)$  where  $\sum_i n_i = n$ .  $\square$

### Partial-Ordering on $\mathcal{N}(G)$

**Prop. (28.6.1.35)[Partial-Ordering on  $B(G)$  and  $\mathcal{N}(G)$ ].** For a maximal torus  $T \leq G$  and a system of positive roots on  $\Phi(G, T)$ , define a partial-ordering on  $\mathcal{N}(G)$  (28.6.1.14) as follows: For  $\nu, \nu' \in \mathcal{N}(G)$ ,

$$\nu \preceq \nu' \iff W(G, T) \cdot \nu \preceq W(G, T) \cdot \nu' \in X_{\bullet}(T) \quad (3.9.5.34).$$

And define a partial-ordering on  $B(G)$  as follows: For  $b, b' \in B(G)$ ,

$$b \preceq b' \iff \kappa_G(b) = \kappa_G(b') \ \& \ \nu_b \preceq \nu_{b'} \in \mathcal{N}(G).$$

Then for  $b, b' \in B(G)$ ,

1. If  $\kappa_G(b) = \kappa_G(b')$  and  $b$  is Kottwitz-basic (28.6.1.26), then  $b \preceq b'$ . In particular, Kottwitz-basic elements are minimal w.r.t. this partial ordering.
2.  $X_b = \{b' \in B(G) : b' \preceq b\}$  if finite.
3. If  $G = \mathrm{GL}(n)$ , then  $b \preceq b'$  iff the Newton-polygon of  $b$  lies above the Newton-polygon of  $b'$  and has the same end points.

┘

*Proof:* 3: In this case, if  $\nu = (\lambda_1, \dots, \lambda_n)$ , then  $\delta\nu = \sum_i \lambda_i$ . And by recalling the co-roots of  $\mathrm{GL}(n)$ , we see for  $\lambda_1 \leq \dots \leq \lambda_n, \lambda'_1 \leq \dots \leq \lambda'_n$ ,

$$(\lambda_1, \dots, \lambda_n) \preceq (\lambda'_1, \dots, \lambda'_n) \iff \sum_{i=1}^h \lambda_i = \sum_{i=1}^h \lambda'_i, \text{ \& \> } \sum_{i=1}^r \lambda_i \geq \sum_{i=1}^r \lambda'_i \text{ where } r \in [n]_+.$$

2: By choosing an embedding  $G \leq \mathrm{GL}(n)$ , this follows from?? and the fact all the Newton-polygons has integer break points by (28.6.1.34).

1: This follows from the fact that  $X_\bullet(Z(G))$  are minimal elements in the positive Weyl-Chamber  $C_+(\Phi^+)$ .  $\square$

**Def. (28.6.1.36) [Hodge-Points].** Let  $G/K$  be unramified and  $\Lambda_0 \in \mathcal{B}(G) \subset \mathcal{B}(G_{\check{K}})$  is a hyperspecial point with corresponding parahoric subgroup  $\mathcal{K}_\Lambda$ , and let  $\mathcal{V} = G(K) \cdot \Lambda_0 \subset \mathcal{B}(G_{\check{K}})$ . Then  $G(\check{K}) \rtimes \mathrm{Gal}_K$  acts on  $\mathcal{V}$ . Then by Cartan-decomposition

$$G(L) \backslash (\mathcal{V} \times \mathcal{V}) = \mathcal{K}_\Lambda \backslash G(L) / \mathcal{K}_\Lambda = X_*(S) / W(G, S) \text{ (20.7.2.14),}$$

where  $S \leq G$  is a maximal split torus. Then we get a map  $\mathrm{inv} : \mathcal{V} \times \mathcal{V} \rightarrow X_\bullet(S) / W(G, S)$ .

For  $\Lambda \in \mathcal{V}$  and  $b \in G(\check{K})$ , define  $\mu_\Lambda(b) = \mathrm{inv}(\Lambda, b\varphi_0(\Lambda))$ , called the **Hodge-point** associated to  $\Lambda$  and  $b$ .

Then

$$\mu_{g\Lambda}(gb\varphi_0(g)^{-1}) = \mu_\Lambda(b), \text{ where } g \in G(\check{K}).$$

$$\mu_{\varphi_0(\Lambda)}(\varphi_0(b)) = \varphi_0(\mu_\Lambda(b)).$$

┘

**Thm. (28.6.1.37) [Mazur's Theorem, Mazur [Kat79]/Rapoport-Richartz [R-R96]].** Situation as in (28.6.1.36). Similar as in (28.6.1.14), any  $\mathcal{N} \in X_\bullet(S) / W(G, S)$  defines an element  $\mathcal{N}_\Lambda^\natural \in \pi_1(G)_{\mathrm{Gal}_K}$ , and by averaging the dominant element over  $\mathrm{Gal}_K$  an element  $\overline{\mathcal{N}}_\Lambda \in \mathcal{N}(G)$ . Then

- $\kappa_G(b) = \mathcal{N}_\Lambda(b)^\natural \in \pi_1(G)_{\mathrm{Gal}_K}$ .
- $\nu_G(b) \preceq \overline{\mathcal{N}}_\Lambda(b) \in \mathcal{N}(G)$  (28.6.1.35).

┘

*Proof:* Cf. [R-R96] P179. ?  $\square$

### Acceptable Elements

**Notation (28.6.1.38).**

- Let  $G \in \mathrm{RedGrp}/K$ .

┘

**Def. (28.6.1.39) [Acceptable Elements].** Suppose  $G$  is quasi-split,  $T \leq G$  a maximal torus and a system of positive roots on  $\Phi(G, T)$  (for non-quasi-split group, we need to choose a quasi-split inner form  $G^*$  and use the fact that  ${}^L G \cong {}^L G^*$  (10.4.6.2)).

For  $\mu \in X_\bullet(T)_{\mathbb{Q}}^{+, \text{Gal}_\kappa}$ ,  $[b] \in B(G)$  is called an **acceptable element** for  $\mathcal{N}$  if  $\nu_{[b]} \preceq \bar{\mu}$  (28.6.1.12)(28.6.1.37).

Notice if  $S \leq G$  is a maximal split torus, then  $[b]$  being acceptable for  $\mathcal{N}$  is equivalent to  $\nu_{[b]}$  lying in the convex hull of  $\Phi(G, S) \cdot \bar{\mu} \in V(S)_{\mathbb{R}}$  by (3.9.5.34). The set of acceptable elements for  $\mu$  is denoted by  $A(G, \mu)$ .

And  $[b] \in B(G)$  is called a **neutral acceptable element** for  $\mu$  if it is acceptable for  $\mu$  and  $\kappa_G([b]) = \mu^\natural \in \pi_1(G)_{\text{Gal}_K}$  (28.6.1.7)(28.6.1.37). The set of neutral acceptable elements for  $\mu$  is denoted by  $B(G, \mu)$ .  $\lrcorner$

**Cor. (28.6.1.40).** For  $\mu \in X_\bullet(T)_{\mathbb{Q}}^{+, \text{Gal}_\kappa}$ ,

- $B(G, \mu)$  (28.6.1.39) contains a unique basic element, by (28.6.1.30).
- $0 < \#B(G, \mu) \leq \#A(G, \mu) < \infty$ , by (28.6.1.35).
- $A(G, \mu)$  and  $B(G, \mu)$  only depends on  $\mu^\natural \in \pi_1(G)_{\text{Gal}_K}$ .
- There is a map  $c : A(G, \mu) \rightarrow H^1(K, G) : ([b], \mu) \mapsto \kappa_G([b]) - \mu^\natural$  by (28.6.1.30), and the kernel is exactly  $B(G, \mu)$ .  $\lrcorner$

### F-Isocrystals with $G$ -Structures

**Def. (28.6.1.41) [C-isocrystals].** For a Tannakian category  $\mathcal{C}$  over  $\mathbb{Q}_p$  and  $S \in \text{Sch}/\kappa$ , a **C-isocrystal** over  $S$  is an exact faithful  $\mathbb{Q}_p$ -linear tensor functor (4.2.1.2)

$$M : \mathcal{C} \rightarrow \text{F-Isoc}(S) \text{ (27.5.5.2)}.$$

The category of  $\mathcal{C}$ -isocrystals over  $S$  is denoted by  $\text{F-Isoc}_{\mathcal{C}}(S)$ . It is a groupoid.  $\lrcorner$

*Proof:* It is a groupoid by [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport]P110.  $\square$

**Def. (28.6.1.42) [Isocrystals with  $G$ -structures].** Given  $S \in \text{Sch}/\kappa$ , an **isocrystal with  $G$ -structures** over  $S$  is a  $\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G)$ -isocrystal over  $S$  (28.6.1.41)

$$M : \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G) \rightarrow \text{F-Isoc}(S).$$

which commutes with fiber functors to  $\text{Vect}_S$ . The category of isocrystals with  $G$ -structures over  $S$  is denoted by  $\text{F-Isoc}_G(S)$ .  $\lrcorner$

**Prop. (28.6.1.43).** The natural functor  $\text{F-Isoc}_G(\kappa) \rightarrow \text{F-Isoc}_{\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G)}(\kappa)$  is fully faithful, and it is an equivalence iff  $H^1(K_0, G) = 1$ .  $\lrcorner$

*Proof:* Cf. [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport]P223.  $\square$

**Prop. (28.6.1.44) [Associated Isocrystals with  $G$ -Structure].** For  $b \in G(K_0)$ , there is a functor

$$\mathcal{I}_b : \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G) \rightarrow \text{F-Isoc}(\kappa) : V \mapsto (V \otimes_{\mathbb{Q}_p} K_0, \rho(b) \circ (\text{id} \otimes \sigma)).$$

If  $g \in G(K_0)$  and  $b' = gb\sigma(g)^{-1}$ , then multiplying by  $g$  implies a natural isomorphism between  $\mathcal{I}_b$  and  $\mathcal{I}_{b'}$ .  $\lrcorner$

**Prop. (28.6.1.45) [Isocrystals and  $B(G)$ ].** The map  $b \mapsto \mathcal{I}_b$  (28.6.1.44) induces an isomorphism of groupoids

$$B(G_{K_0}) \cong \text{F-Isoc}_G(\kappa) \text{ (28.6.1.1)}.$$

Thus for any  $S \in \text{Sch}/\kappa$  and  $M \in \text{F-Isoc}_G(S)$ , we get a function

$$b_M : S(\kappa) \rightarrow B(G_{K_0}) : s \mapsto [b]_s \text{ (28.6.1.1)}.$$

┘

*Proof:* ? Cf. [RR96, P171] or [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport]P222. □

**Def. (28.6.1.46) [Augmented Group Schemes over Isocrystals].** An **affine group scheme over  $\text{F-Isoc}(\kappa)$**  is a pair  $(G, \varphi)$  where  $G \in \text{Aff}/K_0$  and  $\varphi : (\varphi_0)_*(G) \rightarrow G$  is an isomorphism of schemes over  $K_0$  s.t.  $\Gamma(G)$  is an increasing union of f.g.  $\varphi$ -stable subspaces.

Let  $\omega_{\text{Isom}}$  be the natural fiber functor  $\text{F-Isoc}(\kappa) \rightarrow \text{Vect}_{K_0}$ , then there is a natural isomorphism  $\varphi_\kappa : (\varphi_0)_*\omega_{\text{Isom}} \rightarrow \omega_{\text{Isom}}$  given by the  $\sigma$ -equivariant identity map  $V \rightarrow \sigma_*(V)$  for any  $V \in \text{F-Isoc}(\kappa)$ .

Let  $D_\kappa = \underline{\text{Aut}}^\otimes(\omega_{\text{Isom}})$  be the **fundamental group scheme** of  $\text{F-Isoc}(\kappa)$ , then it becomes an affine group scheme over  $\text{F-Isoc}(\kappa)$ .

An **augmented group scheme over  $\text{F-Isoc}(\kappa)$**  is a triple  $(G, \varphi, \nu)$  where  $(G, \varphi)$  is an affine group scheme over  $\text{F-Isoc}(\kappa)$  and  $\nu : (D_\kappa, \varphi_\kappa) \rightarrow (G, \varphi)$  is affine group schemes over  $\text{F-Isoc}(\kappa)$ . ┘

### Specializations of Isocrystals

References are [Gro74, R-R96, Kat79, Zin01, O-Z02].

### Weakly-Admissible Pairs

**Prop. (28.6.1.47) [Associated Filtered Isocrystals and Weakly-Admissible Pairs].** Let  $K$  be a totally ramified field extension of  $K_0$ , and  $\mu : \mathbb{G}_{m,K} \rightarrow G_K$  be a cocharacter over  $K$ , then the associated isocrystal over  $K_0$  prolongs to a filtered isocrystal over  $K$  (25.2.9.22),

$$\mathcal{I}_{\mu,b} : \text{Rep}_{\mathbb{Q}_p}(G) \rightarrow \varphi\text{-Mod Fil}_K : V \mapsto (V \otimes_{\mathbb{Q}_p} K_0, \rho(b) \circ (\text{id} \otimes \sigma), \text{Fil}_\mu^\bullet),$$

where the filtration comes from  $\mu$  by weight-filtrations (because  $\mathbb{G}_m$  is diagonalizable (10.2.3.2)).  $(\mu, b)$  is called a **weakly admissible pair** (resp. **admissible pair**) if for any  $V \in \text{Rep}_{\mathbb{Q}_p}(G)$ , the filtered isocrystal  $\mathcal{I}_b(V)$  is weakly admissible (25.2.9.24) (resp. admissible (25.2.9.28)).

It suffices to check this condition for one faithful representation  $V$ . ┘

*Proof:* This is because for a faithful representation  $V$ , any  $\mathbb{Q}_p$ -representation appears as a direct summand of  $V^{\otimes n} \otimes \widehat{V}^{\otimes m}$  (10.3.1.16). Then the assertion follows from the fact direct summands and tensor products of weakly admissible filtered isocrystals are weakly admissible (25.2.9.26). □

**Def. (28.6.1.48) [Equivalent Pairs].** Two pairs  $(\mu, b), (\mu', b')$  as in (28.6.1.47) are called **equivalent** if there exists  $g \in G(K_0)$  s.t.  $b' = gb\sigma(g)^{-1}$ , and the cocharacters  $\mu', g\mu g^{-1}$  define the same filtration on  $\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G)$ . Equivalent,  $\mathcal{I}_{b,\mu} = \mathcal{I}_{b',\mu'}$ . ┘



## 2 Period Domains

References are [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport].

**Notation (28.6.2.1).**

- Denote  $\kappa = \overline{\mathbb{F}_p}$ .
- Let  $K \in p\text{-LField}$ .
- Let  $\mathcal{N} \in X_\bullet(G)_{\mathbb{Q}}/G(\overline{K})$  be a conjugacy class of cocharacters defined over some reflex field  $E \in p\text{-NField}$  (10.3.2.5),
- Let  $\check{E} = EK_0$ .

┘

**Def. (28.6.2.2) [Associated Partial Flag Variety].** There is a faithful tensor functor associated to  $\mathcal{N}$ :

$$\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G) \rightarrow \text{Gr}_E^{\mathbb{Z}} \rightarrow \text{FilVect}_E$$

Consider the functor

$$\mathcal{C}\text{Ring}_E \rightarrow \text{Set} : R \mapsto \{\text{the equivalence classes in the } G(R)\text{-conjugacy class in } \mathcal{N}_R\},$$

then for each  $\mu \in \mathcal{N}$ , there is a closed algebraic subgroup  $P(\mu) \leq G_E$ :

$$P(\mu)(R) = \{g \in G(R) \mid g\mu_R g^{-1} \text{ is equivalent to } \mu_R\}$$

then the functor above is representable by the  $G_E$ -homogenous variety  $\mathcal{F}_{\mathcal{N}} \cong G_E/P(\mu) \in \text{Sch}/E$ . In other words, it is the par-equivalence classes of  $\mu$ . ┘

**Prop. (28.6.2.3).**  $\mathcal{F}_{\mathcal{N}}$  is a smooth projective variety. In particular  $P(\mu)$  is a parabolic subgroup (28.6.2.2). ┘

*Proof:* Take a faithful representation  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G)$  and consider  $\text{Flag}_{\mu}(V)$  the partial flag variety over  $\mathbb{Q}_p$  which associates to any  $\mathbb{Q}_p$ -algebra  $R$  the set of filtrations  $\text{Fil}^\bullet$  of  $V \otimes_{\mathbb{Q}_p} R$  s.t.  $\text{gr}^i \text{Fil}^i(R)$  are direct summands and  $\text{rank Fil}^i = \dim \text{Fil}_{\mu}^i(V)$ . Then  $\text{Flag}_{\mu}(V)$  is a projective variety, by classical results ?, and there is a closed immersion ?

$$\mathcal{F}_{\mathcal{N}} \hookrightarrow \text{Flag}_{\mu}(V)_E,$$

because the filtrations on other representations are determined by the filtration on  $V$  by (10.3.1.16).

Cf. [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport] P121. ┘

**Thm. (28.6.2.4) [ $p$ -adic Period Spaces, Rapoport-Zink [R-Z96]].** For any finite extension  $L/\check{E}$  and  $b \in G(\check{K})$ , a filtration  $\xi \in \mathcal{F}_{\mathcal{N}}(L)$  (28.6.2.2) is called a **weakly admissible point** if the pair  $(\xi, b)$  is weakly admissible (28.6.1.47). The set of such points are denoted by  $\mathcal{F}_{\mu, b}^{\text{weak. adm}}(K)$ . Then

- $\mathcal{F}_{\mu, b}^{\text{weak. adm}}(\mathbb{C}_K)$  is an admissible open subset of  $(\mathcal{F}_{\mu})_{\check{E}}^{\text{rig}}$ . In particular, it has a rigid analytic structure  $(\mathcal{F}_{\mu, b}^{\text{weak. adm}})_{\check{E}}^{\text{rig}}$ , called the  **$p$ -adic period space** or the  **$p$ -adic symmetric space** associated to  $(G, \{\mu\}, b)$ . If  $b' = gb\sigma(g)^{-1}$ , then  $\xi \mapsto g^{-1}\xi$  induces an isomorphism from  $(\mathcal{F}_{\mu, b}^{\text{weak. adm}})_{\check{E}}^{\text{rig}}$  to  $(\mathcal{F}_{\mu, b'}^{\text{weak. adm}})_{\check{E}}^{\text{rig}}$ .



- Moreover, if  $b$  is Kottwitz-decent w.r.t.  $s \in \mathbb{Z}_+$  (28.6.1.17), then  $\mathcal{F}_{\mu,b}^{\text{weak.adm}}(\mathbb{C}_{K_0})$  is defined over  $E\mathbb{Q}_{p^s}$ , and if  $b' = gb\sigma(g)^{-1}$  for  $g \in G(\mathbb{Q}_p^s)$  (28.6.1.20), then  $\xi \mapsto g^{-1}\xi$  induces an isomorphism from  $(\mathcal{F}_b^{\text{weak.adm}})^{\text{rig}}_{E\mathbb{Q}_{p^s}}$  to  $(\mathcal{F}_{b'}^{\text{weak.adm}})^{\text{rig}}_{E\mathbb{Q}_{p^s}}$ .
- $J_b(\mathbb{Q}_p) \leq G(\check{K})$  (28.6.1.21) acts on  $(\mathcal{F}_\mu)^{\text{rig}}_{\check{E}}$ , and preserves  $(\mathcal{F}_{\mu,b}^{\text{weak.adm}})^{\text{rig}}_{\check{E}}$ .

For more properties of  $\mathcal{F}_{b,\mathcal{N}}^{\text{weak.adm}}$ , Cf.[Period domains over finite and p-adic fields, Dat-Orlik-Rapoport]P273.  $\square$

*Proof:* 2: By (28.6.1.18),  $b$  is Kottwitz-decent with some descent equation  $(b\sigma)^s = \nu_b^s(p)\sigma^s$ , in particular,  $b \in G(\mathbb{Q}_{p^s})$  (28.6.1.19). Let  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(G)$  be a faithful representation, then

$$(V \otimes K_0, b(\text{id} \otimes \varphi_0)) \cong (V \otimes \mathbb{Q}_{p^s}, b(\text{id} \otimes \varphi_0)) \otimes K_0.$$

Let  $V \otimes \mathbb{Q}_{p^s} = \oplus_{\lambda \in \mathbb{Q}} V_\lambda$  be the isotypical decomposition of  $V$  (27.5.5.15), then the functor  $T'$ :

$$\mathcal{C}\text{Ring}/\mathbb{Q}_{p^s} \rightarrow \text{Set} : R \mapsto \{V' \subset V \otimes R \mid V' \text{ is a direct summand and } V' = \oplus_\lambda V'_\lambda \cap (V \otimes R)\}$$

is representable by a disjoint union of closed subsets of Grassmannians of  $V \otimes \mathbb{Q}_{p^s}$ . Then  $T'$  descends to a  $\mathbb{Q}_p$ -variety  $T$  s.t.

$$T(\mathbb{Q}_p) = \{b\varphi_0\text{-invariant subspaces of } V \otimes \mathbb{Q}_{p^s}\},$$

because  $b\varphi_0$  defines a descent datum  $\color{red}?$ .

Recall the closed immersion  $\mathcal{F}_\mu \hookrightarrow \text{Flag}_\mu(V)_E$ , and consider the closed subscheme

$$H \subset (\text{Flag}_\mu(V) \times T) \otimes \mathbb{Q}_{p^s}$$

consisting of pairs  $(\text{Fil}^\bullet, V')$  where  $\text{Fil}^\bullet$  is a filtration of  $V \otimes R$  and  $V' = \oplus_\lambda V'_\lambda$  is an  $R$ -direct summand of  $(V \otimes \mathbb{Q}_{p^s}) \otimes R = (\oplus_\lambda V_{p^s,\lambda}) \otimes R$  s.t.

$$\sum_i i \text{rank gr}^i(\text{Fil}^\bullet \cap V') > \sum_{\lambda \in \mathbb{Q}} \lambda \text{rank}(V'_\lambda).$$

$H$  is truly a closed subscheme because the LHS equals  $\sum_{i=-N}^\infty \text{rank}(\text{Fil}^i \cap V') - N \text{rank}(V')$  for  $N'$  sufficiently large, so  $H$  is defined by a Schubert-type condition  $\color{red}?$ .

By property of  $T$  and comparing the above formulas with (25.2.9.24), we see the set of weakly admissible filtrations in  $\text{Flag}_\mu(V)$  is given by

$$\text{Flag}_\mu(V)(\mathbb{C}_p)^{\text{weak.adm}} = \text{Flag}_\mu(V)(\mathbb{C}_p) \setminus \bigcup_{t \in T(\mathbb{Q}_p)} H_t.$$

And this is a rigid analytic subvariety over  $\mathbb{Q}_{p^s}$  by (27.2.2.17). Finally, the assertion follows by intersecting  $\text{Flag}_\mu(V)(\mathbb{C}_p)^{\text{weak.adm}}$  with  $\mathcal{F}_\mu \subset \text{Flag}_\mu(V) \otimes E$ .

1 follows from item2 because we can always find a  $\sigma$ -conjugate  $g_1 b \sigma(g_1)^{-1}$  of  $b$  in  $B(\text{GL}(V))$  that is Kottwitz-descent, by (28.6.1.17), and then we use the  $G$ -equivariant property in item2 to see that

$$\mathcal{F}_{\mu,b}^{\text{weak.adm}} = (\mathcal{F}_\mu)^{\text{rig}}_{\check{E}} \cap g_1 \mathcal{F}_{\mu,b,\text{GL}(V)}^{\text{weak.adm}}$$

is admissible open in  $(\mathcal{F}_\mu)^{\text{rig}}_{\check{E}}$ .

3 follows from item1, 2 and the definition (28.6.1.21).  $\square$

**Prop. (28.6.2.5) [Period Domains and Acceptable Points].** For  $b \in G(\check{F})$  and  $\mathcal{N} \in X_\bullet(G)_{\mathbb{Q}}/G(\overline{K})$ ,  $[b] \in A(G, \mathcal{N})$  (28.6.1.39) iff  $\check{F}(G, b, \mathcal{N})^{\text{weak. adm}} \neq \emptyset$ .  $\lrcorner$

*Proof:* Cf. [Period domains over finite and p-adic fields, Dat-Orlik-Rapoport] 9.5.10. ?  $\square$

**Thm. (28.6.2.6) [Period Domains and GIT Semistable Points, Rapoport-Zink [R-Z96]/Totaro [Tot96]].** There exists a unique ample line bundle on  $(\mathcal{F}_\mu)_{\check{E}}$  that is homogenous under the derived group  $(G^{\text{der}})_{\check{E}}$ .

For any finite extension  $K/\check{E}$ , we have

$$\mathcal{F}_{\mu, b}^{\text{weak. adm}}(K) = (\mathcal{F}_\mu)_{\check{E}}^{\text{s. st}}(\mathcal{L})(K).$$

where  $(\mathcal{F}_\mu)_{\check{E}}^{\text{s. st}}(\mathcal{L})$  is the open subset of points that are semistable under the action of  $T_{\check{E}} \cap (G^{\text{der}})_{\check{E}}$  on  $((\mathcal{F}_\mu)_{\check{E}}, \mathcal{L})$ .  $\lrcorner$

*Proof:* ?  $\square$

**Conj. (28.6.2.7) [Rapoport-Zink].** Situation as in (28.6.2.4), there exists a rigid étale bijective homomorphism

$$\left(\check{\mathcal{F}}_{b, \mathcal{N}}^{\text{weak. adm}}\right)^{\text{rig}} \rightarrow \left(\check{\mathcal{F}}_{b, \mathcal{N}}^{\text{weak. adm}}\right)^{\text{rig}}$$

and a tensor functor

$$\text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K) \rightarrow \text{LocSys}_{\mathbb{Q}_p}^{\text{ét}} \left( \left(\check{\mathcal{F}}_b^{\text{weak. adm}}\right)^{\text{rig}} \right)$$

s.t. at a weakly admissible point  $(\xi, b)$ , the fiber of the image of  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_K)$  is the underlying  $\mathbb{Q}_p$ -vector space of the crystalline Galois representation corresponding to  $\mathcal{I}_{\mu, b}(V)$  (28.6.1.47) via (25.2.9.29).  $\lrcorner$

*Proof:*  $\square$

### Another viewpoint

References are [Toward a Theory of Local Shimura Varieties, Rapoport-Viehmann].

**Def. (28.6.2.8) [PD-Pairs].** A **PD-pairs** is a pair  $(\mathbf{G}, \mathcal{N})$  where  $\mathbf{G} = (G, \varphi, \nu)$  is a reductive augmented group scheme over  $\text{F-Isoc}(\kappa)$  (28.6.1.46) and  $\mathcal{N} \in X_\bullet(G)_{\mathbb{Q}}/G(\overline{K})$  is a conjugacy class of cocharacters.  $\lrcorner$

### Period Domains for $\text{GL}(n)$

**Def. (28.6.2.9) [Drinfeld Half Space].** For  $G = \text{GL}(n)_K, b = 1 \in B(G_{K_0})$  and  $\mathcal{N} = G(\overline{K}).(n - 1, \underbrace{-1, \dots, -1}_{n-1 \text{ many}})$ , we have

$$\Omega_K^n \triangleq \check{F}(G, 1, \mu)^{\text{weak. adm}} = \mathbb{P}_F^{n-1} \setminus \bigcup_{H \subset F^n \text{ hypersurface}} \mathbb{P}(H).$$

*Proof:* ?  $\square$

### 3 Rapoport-Zink Spaces

**Notation (28.6.3.1).**

- Let  $\kappa = \overline{\mathbb{F}_p}$ .

⌋

#### Rapoport-Zink Data

**Def. (28.6.3.2) [Rapoport-Zink Data of EL-Type].** A simple rational Rapoport-Zink datum of EL-type is a tuple  $\mathcal{D} = (K, B, V, \mathcal{N}, [b])$  where

- $K \in p\text{-NField}$ .
- $B \in \text{Az}_K$ .
- $V \in \text{Vect}^{\text{fin}}/B$ .
- $\mathcal{N} \in X_\bullet(G)_{\mathbb{Q}}/G(\overline{K})$  is minuscule<sup>?</sup>, where  $G = \text{GL}_B(V) \in \text{AlgGrp}/\mathbb{Q}_p$ .
- $[b] \in A(G, \mathcal{N})$  (28.6.1.39).

s.t.  $V$  is of weight 0 and 1 under  $\mu$ .

And an **simple integral Rapoport-Zink datum of EL-type** is a simple Rapoport-Zink datum of EL-type  $\mathcal{D} = (K, B, V, \mathcal{N}, [b])$  together with a maximal order  $\mathcal{O}_B$  of  $B$  and an  $\mathcal{O}_B$ -lattice  $\Lambda \subset V$ .

⌋

**Def. (28.6.3.3) [Rapoport-Zink Data of PEL-Type].** When  $p \neq 2$ , a simple rational Rapoport-Zink datum of PEL-type is a tuple  $\mathcal{D} = (K, B, V, (-, -), *, \mathcal{N}, [b])$  where

- $K \in p\text{-NField}$ .
- $B \in \text{Az}_K$ .
- $V \in \text{Vect}^{\text{fin}}_B$ .
- $(-, -)$  is an alternating  $\mathbb{Q}_p$ -valued bilinear form on  $V$ .
- $*$  is an involution on  $B$  s.t.

$$(xu, v) = (u, x^*v), \text{ where } x \in B, u, v \in V.$$

- $\mathcal{N} \in X_\bullet(G(\mathcal{D}))_{\mathbb{Q}}/G(\overline{K})$  is minuscule<sup>?</sup>, where  $G(\mathcal{D}) \in \text{AlgGrp}/\mathbb{Q}_p$  s.t. for any  $R \in \mathbb{C}\text{Ring}/\mathbb{Q}_p$ ,

$$G(\mathcal{D})(R) = \{g \in \text{GL}_{B \otimes_{\mathbb{Q}_p} R}(V \otimes_{\mathbb{Q}_p} R) \mid \exists c(g) \in R^*, (gu, gv) = c(g)(u, v) \text{ for all } u, v \in V \otimes_{\mathbb{Q}_p} R\}.$$

equipped with a map  $c : G \rightarrow \mathbb{G}_m$ .

- $[b] \in A(G, \mathcal{N})$  (28.6.1.39).

s.t.  $V$  is of weight 0 and 1 under  $\mu$ , and  $(\mathbb{G}_m)_{\overline{K}} \xrightarrow{\mu} G_{\overline{K}} \xrightarrow{c} \mathbb{G}_m$  is the identity.

And a **simple integral Rapoport-Zink datum of PEL-type** is a simple Rapoport-Zink datum of PEL-type  $\mathcal{D}$  together with a maximal order  $\mathcal{O}_B$  of  $B$  and an  $\mathcal{O}_B$ -lattice  $\Lambda \subset V$  s.t.  $\varpi\Lambda \subset \Lambda^\vee \subset \Lambda$ , where  $\varpi$  is a uniformizer in  $\mathcal{O}_B$ <sup>?</sup>.

⌋

**Prop. (28.6.3.4) [Associated Isocrystals].** Situation as in (28.6.3.2) or (28.6.3.3), for any  $b \in [b]$ , we can associate to any  $b \in [b]$  an isocrystal

$$(N(V), \Phi) = (V \otimes_{\mathbb{Q}_p} K_0, b(\text{id} \otimes \sigma), V_E^\bullet).$$

This isocrystal is equipped with an action of  $B$ , and in the PEL case an alternating bilinear form

$$\psi : N(V) \otimes N(V) \rightarrow \mathbf{1}(n).$$

where  $n = v_p(c(b))$ . In fact, we can find some unit  $u$  that  $c(b) = p^n u \sigma(u)^{-1}$ , then the pairing is defined as

$$\psi(v, v') = u^{-1}(v, v'),$$

any other choices of  $u$  multiplies  $\psi$  by an element in  $\mathbb{Z}_p^*$ .

We will fix in addition a conjugacy class of cocharacters  $\mu : \mathbb{G}_m \rightarrow G$  defined over a field  $E$ , and the associated homogenous algebraic variety  $\mathcal{F}$  defined over  $E$  of filtrations (28.6.2.2).  $\mathcal{F}$  is equipped with a  $B$ -action, as  $G \in \mathrm{GL}_B(V)$ .

Notice in the PEL case, these filtrations satisfy  $\mathcal{F}^i = (\mathcal{F}^{m-i+1})^\perp$ , where  $m = c \circ \mu \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) \cong \mathbb{Z}$ . This is due to the fact  $(kv, kw) = k^m(v, w)$  and the fact the pairing is non-degenerate.  $\perp$

**Prop. (28.6.3.5) [Local Reflex Fields].** For a conjugacy class of cocharacters  $\{\mu\}$  defined over  $E$  and  $\mu_0 \in \{\mu\}$ , its corresponding filtration  $\mathcal{F}_0^\bullet$ . The reflex field  $E$  in (28.6.3.3) can be described as the field of definition of the isomorphism class of  $\mathcal{F}_0^\bullet$  as a  $B$ -invariant filtration, or equivalently as the finite extension of  $\mathbb{Q}_p$  generated by the traces

$$\mathrm{tr}(d; \mathrm{gr}_{\mathbb{F}_0}^i(V \otimes_{\mathbb{Q}_p} \overline{\mathbb{Q}_p})), d \in B, i \in \mathbb{Z}.$$

And the filtration  $\mathcal{F}$  is described as the functor that for any  $E$ -algebra  $R$ ,  $\mathcal{F}(R)$  is the set of filtrations  $\mathcal{F}^\bullet$  of  $V \otimes_{\mathbb{Q}_p} R$  by  $R$ -modules that are direct summands that

$$\mathrm{tr}(d; \mathrm{gr}_{\mathcal{F}}^i(V \otimes_{\mathbb{Q}_p} R)) = \mathrm{tr}(d; \mathrm{gr}_{\mathbb{F}_0}^i(V \otimes_{\mathbb{Q}_p} \overline{\mathbb{Q}_p})).$$

and moreover in the PEL case satisfies  $\mathcal{F}^i = (\mathcal{F}^{m-i+1})^\perp$ .  $\perp$

*Proof:* 1: The field of definition  $E$  of the conjugacy class  $\{\mu\}$  is determined by Tannakian duality, so it suffices to check over which field these two filtrations are isomorphic as  $G$ -filtrations, but  $G$  is just the group fixing the  $B$ -module structure, so it suffices to show they are equivalent as  $B$ -modules, which is then determined by the traces, by (3.6.2.27).

2: It suffices to show  $\mathcal{F}$  is a homogenous space under  $G$ . We restrict to the PEL case, the EL case is simpler. After base change from  $\mathbb{Q}_p$  to  $\overline{\mathbb{Q}_p}$ , the data decomposes to the following types:

A:  $B = \mathrm{End}(W) \times \mathrm{End}(W^\vee)$  where  $W$  is a f.d.  $\overline{\mathbb{Q}_p}$ -vector space and  $(u, v)^* = (v^t, u^t)$ .

And  $V = W \otimes V' \oplus W^\vee \otimes V'^\vee$  where the pairing is natural and makes the sum orthogonal.

$$G = \{(1 \otimes g, c \cdot (1 \otimes g^{-t}) | g \in \mathrm{GL}(V'), c \in X(G)\}$$

C:  $B = \mathrm{End}(W)$  where  $W$  is a f.d.  $\overline{\mathbb{Q}_p}$ -vector space equipped with a symmetric bilinear form  $(-, -)_W$  and  $*$  is the transposition w.r.t it.

And  $V = W \otimes V'$  where  $V'$  is equipped with an alternating form  $(-, -)_{V'}$  that  $(-, -)_V = (-, -)_W \otimes (-, -)_{V'}$ .

$$G = \{cg | g \in \mathrm{Sp}(V'), c \in X(G)\}$$

BD: As in (C), except that  $(-, -)_W$  is skew-symmetric and  $(-, -)_{V'}$  is symmetric.

$$G = \{cg | g \in \mathrm{SO}(V'), c \in C(G)\}$$

Under this decomposition, the functor  $\mathcal{F}$  in the proposition is represented by products of partial flags of  $V$ :

- $(A) : \mathcal{F}^i = W \otimes (\mathcal{F}')^i \oplus W^\vee \otimes ((\mathcal{F}')^{m+1-i})^\perp$  and the correspondence  $\mathcal{F}^\bullet \mapsto (\mathcal{F}')^\bullet$  identifies  $\mathcal{F}$  with the partial flag variety of  $V'$  with fixed dimensions  $\dim((\mathcal{F}')^i)$ .
- $(B, CD) : \mathcal{F}^\bullet = W \otimes (\mathcal{F}')^\bullet$  and  $\mathcal{F}$  is identified with the partial flag variety of  $V'$  of fixed dimensions  $\dim((\mathcal{F}')^i)$  and  $(\mathcal{F}')^i = ((\mathcal{F}')^{m+1-i})^\perp$ .

The  $(A)$  case  $G$  clearly acts transitively on  $\mathcal{F}$ , and the  $(B, CD)$  case  $(\mathcal{F}')^i$  is isotropic for  $i \geq (m+1)/2$ , and it determines all other components, so  $G$  acts transitively, by Witt's theorem(19.5.2.3).

The reason is(3.6.4.27) and the fact representations of  $B$  is semisimple, then contemplating on the pairing condition.  $\square$

**Prop. (28.6.3.6) [Examples of PEL Type].** Let  $B = D$  be the quaternion algebra over  $\mathbb{Q}_p$  and  $*$  be the involution, i.e.

$$D = \mathbb{Q}_{p^2}[\Pi], \quad \Pi^2 = p, \quad \Pi a = \sigma(a)\Pi$$

and

$$a^* = \sigma(a), a \in \mathbb{Q}_{p^2}, \quad \Pi^* = \Pi.$$

Let  $(V, \iota)$  be a free  $D$ -module of rank  $n$  with a non-degenerate bilinear form satisfying the conditions in??. Then  $G$  is a non-trivial inner form of the group  $GSp_{2n}$  of symmetric similitudes:

Firstly  $\mathbb{Q}_{p^2} \otimes K_0 \cong K_0 \oplus K_0$ , then  $\mathbb{Q}_{p^2}$  acts on  $K_0 \oplus K_0$  by  $a(x, y) = ax, \sigma(a)y$ . As  $V$  is a  $\mathbb{Q}_{p^2}$ -vector space, there is a decomposition

$$V = V_0 \oplus V_1$$

where  $\mathbb{Q}_{p^2}$  acts on  $V_i$  by  $a(v) = v \cdot \sigma^i(a)$ , then  $G_{K_0}$  is just  $GSp(2n)_{K_0}$ , and  $G \neq GSp(2n)$  as the Galois action  $\sigma$  on  $\mathbb{Q}_{p^2} \otimes K_0$  and  $K_0 \cong K_0 \oplus K_0$  are different.

Take  $b \in G(K_0)$  the element with  $c(b) = p$  and the corresponding isocrystal  $(N, \Phi)$  is isotypical of slope  $1/2$ .  $N$  decomposes as  $N_0 \oplus N_1$ . Notice now  $\Pi$  and  $\Phi = b\sigma$  interchanges  $N_i$ , and  $\Pi\Phi = \Phi\Pi$ . Also  $N_i$  is isotropic: For  $v, w \in N_i, a \in \mathbb{Q}_{p^2}$ ,

$$a(v, w) = (av, w) = (\iota(\sigma^i(a))v, w) = (v, \iota(\sigma^{i+1}(a))w) = (v, \sigma(a)w) = \sigma(a)(v, w)$$

so  $(v, w) = 0$ .

We can define a new non-degenerate alternating form

$$\langle -, - \rangle : N_0 \times N_0 \rightarrow K_0 : \langle v, v' \rangle = (v, \Pi v')$$

and also a  $\sigma$ -linear endomorphism of  $N_0$ :  $\Phi_0 = \Pi^{-1} \circ \Phi|_{N_0}$ . From the condition,  $v_p(\det \Phi_0) = 0$ , and  $\Phi$  has all the slopes 0. Also  $\langle \Phi_0 v, \Phi_0 w \rangle = \sigma(\langle v, w \rangle)$ , as

$$\langle \Phi_0 v, \Phi_0 w \rangle = (\Pi^{-1} \Phi v, \Phi w) = (\Pi^{-1} b \sigma v, b \sigma w) = \sigma(v, \Pi w) = \sigma(\langle v, w \rangle).$$

so this alternating form is defined over  $\mathbb{Q}_p$ , denoted by  $(V_0, \langle -, - \rangle)$ , and  $\Phi_0$  corresponds to  $\sigma$ . Then  $J_b = GSp(V_0, \langle -, - \rangle)$ .

Next we consider

$$(0) = \mathcal{F}_0^2 \subset \mathcal{F}_0^1 \subset \mathcal{F}_0^0 = V \otimes \overline{\mathbb{Q}_p}$$

be a filtration where  $\mathcal{F}_0^1$  be a  $D$ -invariant Lagrangian subspace. This corresponds to a cocharacter  $\mu \rightarrow G$ , and  $\mathcal{F}$  is just the  $\mathbb{Q}_{p^2}$  variety of  $D$ -invariant Lagrangian subspaces of  $V_{\mathbb{Q}_{p^2}}$ . By (28.6.3.5), the Shimura field is  $\mathbb{Q}_p$ .

Let  $\mathcal{F} \subset \mathcal{F}(K)$  where  $K/K_0$  is a field extension, then

$$\mathcal{F} = \mathcal{F}_0 \oplus \mathcal{F}_1$$

where  $\mathcal{F}_i \in N_0 \otimes_{K_0} K$ , as  $\mathcal{F}$  is  $\Pi$ -invariant. Now  $\mathcal{F}_0$  is also a Lagrangian subspace of  $(V_0, \langle -, - \rangle)$ .  $\mathcal{F}(K)$  identifies the  $K$ -points of the Grassmannian of Lagrangian subspaces of  $(V_0, \langle -, - \rangle)$ .  $\lrcorner$

**Cor. (28.6.3.7).** Under the above identification, the subset  $\mathcal{F}^{\text{weak.adm}}(K)$  of the Grassmannian of Lagrangian spaces  $\mathcal{F}$  of  $(V_0 \otimes K, \langle -, - \rangle)$  is characterized by  $\mathcal{F}$  satisfying the the following conditions:

For all totally isotropic subspaces  $W_0 \subset V_0$ , we have  $\dim_K \mathcal{F} \cap (W_0 \otimes K) \leq 1/2 \dim W_0$ .  $\lrcorner$

*Proof:* It's clear  $\mu(N, \Phi, \mathcal{F}) = 0$ , so weakly-admissibility is equivalent to semi-stability. The uniqueness of the HN-filtration of  $\mathcal{F}$  implies its  $D$ -invariance, thus semi-stability is equivalent to the fact that for any subspace  $P \subset N$  stable under  $\Phi$  and  $D$ -action, we have

$$\dim_K(\mathcal{F} \cap (P \otimes_{K_0} K)) \leq v_p(\det(\Phi; P)).$$

Now  $\Phi$  is isotypical with slope  $1/2$ ,  $v_p(\det(\Phi; P)) = \frac{1}{2} \dim P$ , and the  $D$ -invariance of  $P$  is equivalent to  $P = P_0 \oplus P_1$  and the  $\Phi$ -invariance of  $P$  is equivalent to the  $\Phi_0$ -invariance of  $P_0$ , i.e.  $P_0$  is a  $\mathbb{Q}_p$ -rational subspace  $W_0 \subset V_0$ .

Finally we show it suffices to check for totally isotropic subspaces: Let  $W'_0$  be the radical of  $W_0$ , then there is a non-singular alternating form on  $W_0/W'_0$ , then the image of  $\mathcal{F}'_0 \cap (P \otimes_{K_0} K)$  in this quotient is a totally isotropic space, thus has dimension  $\leq \frac{1}{2} \dim(W_0/W'_0)$ . then it suffices to check the condition for  $W'_0$ .  $\square$

### Rapoport-Zink Spaces of PEL-Type

**Def. (28.6.3.8)** [Rapoport-Zink Spaces for  $\text{GL}(n)$  and  $\text{GSp}(n)$ , Rapoport-Zink[R-Z96]]. For  $\kappa = \overline{\mathbb{F}}_p$  and  $X_0 \in p\text{Div}/\kappa$ , the **Rapoport-Zink space**  $\text{RZ}(X_0)$  is a formal scheme over  $\text{Spf}(W(\kappa))$  that represents the functor that assigns to each  $S \in \text{NilSch}/W(\kappa)$  the set of isomorphism classes of pairs  $(X, \rho)$  where

- $X \in p\text{Div}/S$ .
- $\rho : X_0 \times_{\kappa} \overline{S} \rightarrow X \times_S \overline{S}$  is a quasi-isogeny (27.10.1.20), where  $\overline{S} = S \otimes_{W(\kappa)} \kappa$ .

And if  $\lambda_0 : X \rightarrow X^\vee$  is a principal polarization, then the **Rapoport-Zink space of symplectic-type**  $\text{RZ}(X_0, \lambda_0)$  is the formal scheme over  $\text{Spf}(W(\kappa))$  that represents the functor that assigns to each  $S \in \text{NilSch}/W(\kappa)$  the set of isomorphism classes of triples  $(X, \lambda, \rho)$  where

- $X \in p\text{Div}/S$ .
- $\lambda : X \rightarrow X^\vee$  is a principal polarization.
- $\rho : X_0 \times_{\kappa} \overline{S} \rightarrow X \times_S \overline{S}$  is a quasi-isogeny that respects polarizations up to a scalar, i.e. there exists  $c(\rho) \in \Gamma(\overline{S}, \mathbb{Q}_p^\times)$  s.t.

$$\lambda \circ \rho = c^{-1}(\rho) \lambda \circ \rho_0.$$

Then  $\text{RZ}(X_0)$  and  $\text{RZ}(X_0, \lambda_0)$  are formally smooth and locally formally of f.t. over  $\text{Spf}(W(\kappa))$ , with a closed embedding  $\text{RZ}(X_0, \lambda_0) \hookrightarrow \text{RZ}(X_0)$ .  $\lrcorner$

*Proof:* ? Cf. [R-Z96] Chap2.  $\square$

**Thm. (28.6.3.9) [Rapoport-Zink Spaces of EL and PEL-Types, Rapoport-Zink[R-Z96]].**

Let  $\mathcal{D} = (K, \mathcal{O}_B, \Lambda, \mathcal{N}, [b])$  be a simple integral Rapoport-Zink datum of EL-type or PEL-type  $\mathcal{D} = (K, \mathcal{O}_B, \Lambda, (-, -), *, \mathcal{N}, [b])$  be a simple rational Rapoport-Zink datum of PEL-type. Let  $E \in p\text{-NField}$  be the local reflex field of  $\mathcal{D}$  (28.6.3.5), and fix  $b \in [b]$ . Suppose there exists  $X_0 \in p\text{Div}/\kappa$  with associated isocrystal isomorphic to  $(N(V), \Phi)$  (28.6.3.4) (and in the PEL case, endow  $X_0$  with the  $\mathcal{O}_B$ -polarization defined by the alternating form on  $V \otimes_{\mathbb{Q}_p} K_0$ ).

Then there is a **Rapoport-Zink space of PEL-type**  $\text{RZ}(\mathcal{D})$  which is a formal scheme locally formally of f.t. over  $\text{Spf}(\mathcal{O}_{\check{E}})$  represents the functor that assigns to each  $S \in \text{NilSch}/\mathcal{O}_{\check{E}}$  the set of isomorphism classes of triples  $(X_{\mathcal{L}}, \rho_{\mathcal{L}})$  where

- $X_{\Lambda} \in p\text{Div}/S$  with an  $\mathcal{O}_B$ -action, for each  $\Lambda \in \mathcal{L}$ .
- $\rho_{\Lambda} : X_0 \times_{\kappa} \bar{S} \rightarrow X_{\Lambda} \times_S \bar{S}$  is an  $\mathcal{O}_B$ -quasi-isogeny for each  $\Lambda \in \mathcal{L}$ , where  $\bar{S} = S \otimes_{\mathcal{O}_{\check{E}}} \kappa$ .

satisfying:

- (Kottwitz Condition[?Kot92b]) For each  $\Lambda \in \mathcal{L}$  and  $d \in B$ ,  $\det_{\mathcal{O}_S}(d | \text{Lie}(X_{\Lambda})) = \det_K(d | V_E^0/V_E^1)$ , where  $V_E^{\bullet}$  is the filtration induced by  $\mu$ .
- If  $M(X_{\Lambda})$  is the Lie algebra of the universal extension of  $X_{\Lambda}$  ?, then locally on  $S$  there is an  $\mathcal{O}_B$ -isomorphism  $M(X_{\Lambda}) \cong \Lambda \otimes \mathcal{O}_S$ .
- If  $\Lambda \subset \Lambda' \in \mathcal{L}$ , the quasi-isogeny  $\rho_{\Lambda'} \circ \rho_{\Lambda}^{-1}$  lifts to an isogeny  $X_{\Lambda'} \rightarrow X_{\Lambda}$  of height  $\log_p(\#\Lambda'/\Lambda)$ .

And  $J_b(\mathbb{Q}_p)$  (28.6.1.21) acts on the left on  $\text{RZ}(\mathcal{D})$  by

$$g.(X_{\Lambda}, \rho_{\Lambda}) = (X_{\Lambda}, \rho_{\Lambda} \circ g^{-1}).$$

⌋

*Proof:* Cf. [R-Z96] Chap3. ?

□

**Prop. (28.6.3.10)**  $[\Delta_{G(\mathcal{D})}]$ . Situation as in (28.6.3.9), denote  $\Delta_{G(\mathcal{D})} \triangleq X^*(G)^{\vee}$  (28.6.3.3), then  $J_b(\mathbb{Q}_p)$  (28.6.1.21) acts on  $\Delta_{G(\mathcal{D})}$  by translation ?. And there is a canonical  $J_b(\mathbb{Q}_p)$ -equivariant map  $\kappa : \text{RZ}(\mathcal{D}) \rightarrow \Delta_{G(\mathcal{D})}$ .

⌋

*Proof:* ? Cf. [R-Z96] Chap3.

□

**Thm. (28.6.3.11).** Situation as in (28.6.3.9),  $\text{RZ}(\mathcal{D})$  is equipped with canonical Weil descent datum from  $\text{Spf } \mathcal{O}_{\check{E}}$  to  $\text{Spf } \mathcal{O}_{\check{E}}$ .

⌋

*Proof:* ? Cf. [R-Z96] Chap3.

□

**Example (28.6.3.12) [Rapoport-Zink Spaces of PEL-Type].**

- (Drinfeld) If  $K \in p\text{-NField}$  and  $B \in \text{Az}_K$  is a division algebra with invariant  $1/d$  with maximal order  $\mathcal{O}_B$ , then the classifying space for quasi-isogenies of special formal  $\mathcal{O}_B$ -modules  $(X, \rho)$  over schemes  $S \in \text{NilSch}/\mathcal{O}_{\check{K}}$  is a EL-type Rapoport-Zink space which is isomorphic to

$$\text{RZ}(\mathcal{D}) = \coprod_{n \in \mathbb{Z}} \hat{\Omega}_K^d \times_K \check{K},$$

where the disjoint sum decomposition is induced by the height of  $\rho$ .

- (Lubin-Tate) If  $K \in p\text{-NField}$  and  $B = K$ , then the EL-type Rapoport-Zink space  $\text{RZ}(\mathcal{D})$  classifies quasi-isogenies of formal  $\mathcal{O}_K$ -modules  $(X, \rho)$  over schemes  $S \in \text{NilSch}/\mathcal{O}_{\check{K}}$  of dimension 1 and height  $d$ . Then Lubin-Tate theory implies there is a non-canonical isomorphism

$$\text{RZ}(\mathcal{D}) \cong \coprod_{n \in \mathbb{Z}} \text{Spf}(\mathcal{O}_{\check{E}})[[T_1, \dots, T_{d-1}]].$$

- If  $K = \mathbb{Q}_p = B$ , then the EL-type Rapoport-Zink space  $\mathrm{RZ}(\mathcal{D})$  classifies quasi-isogenies of ordinary  $p$ -divisible groups over schemes  $S \in \mathrm{NilSch}/\mathcal{O}_{\check{K}}$  of dimension  $n$  and height  $2n$ . And it is isomorphic to

$$\mathrm{RZ}(\mathcal{D}) = \coprod_{(\mathrm{GL}(n, \mathbb{Q}_p)/\mathrm{GL}(n, \mathbb{Z}_p))^2} \mathrm{Spf}(W(\kappa))[[T_{11}, T_{12}, \dots, T_{nn}]].$$

┘

*Proof:* ?

□

### Rapoport-Zink Spaces of Hodge-Type

**Def. (28.6.3.13) [Unramified Local Shimura Data of Hodge-Type].** An unramified **local Shimura datum of Hodge-type** is an unramified local Shimura datum  $(G, [b], \mathcal{N})$  over  $\mathbb{Q}_p$  (28.6.5.1) s.t. there exists a **local Hodge embedding datum**  $(\Lambda, \iota, b, \mu)$  where

- $\Lambda \in \mathrm{Vect}^{\mathrm{fin}}/\mathbb{Z}_p$ .
- $\iota : G \rightarrow \mathrm{GL}(\Lambda)$  is a closed embedding of schemes over  $\mathbb{Z}_p$ .
- $\mu : (\mathbb{G}_m)_{\mathcal{O}_E} \rightarrow G_{\mathcal{O}_E}$  is a representative of  $\mathcal{N}$ , where  $E$  is the local reflex field of  $\mathcal{N}$ .

s.t.

- $\iota \circ \mu : (\mathbb{G}_m)_{\mathcal{O}_E} \rightarrow \mathrm{GL}(n)_{\mathcal{O}_E}$  is diagonalizable to the inverse of the minuscule cocharacter

$$a \mapsto \mathrm{diag}(\underbrace{a, \dots, a}_{r \text{ many}}, \underbrace{1, \dots, 1}_{n-r \text{ many}}).$$

for some  $r \in [n-1]_+$ .

- $b$  is a  $G(W(\kappa))$ - $\sigma$ -conjugacy class of elements  $b \in G(W(\kappa))\mu^\sigma G(W(\kappa))$  of  $[b]$ .

┘

**Prop. (28.6.3.14) [Associated  $p$ -Divisible Groups].** Given an unramified local Shimura-Hodge datum  $(G, b, \mu, C)$  (28.6.3.13), there exists a unique  $X_0 = X_0(G, b, \mu, \Lambda) \in p\mathrm{Div}/\kappa$  whose covariant Dieudonné module is  $\mathbf{D}(X_0)(W(\kappa)) = \Lambda \otimes_{\mathbb{Z}_p} W(\kappa)$  with Frobenius  $F = b \circ \sigma$ .

Moreover, the Hodge-filtration  $VD_\kappa \subset D_\kappa = \mathbf{D}(X_0)(\kappa)$  is induced by a conjugate of  $\mu_\kappa : (\mathbb{G}_m)_\kappa \rightarrow G_\kappa$ .

┘

*Proof:* Cf. [Howard-Pappas17]P13. ?

□

**Prop. (28.6.3.15) [Local Hodge-Tensors].** Situation as in (28.6.3.14), there is a finite list  $(s_\alpha)$  of tensors in the total tensor space  $C^\otimes$  s.t.  $G$  is the closed subscheme of  $\mathrm{GL}(\Lambda)$  fixing these  $s_\alpha$ .

Let  $X_0 = X_0(G, b, \mu, \Lambda) \in p\mathrm{Div}/\kappa$  as in (28.6.3.14), we get Frobenius-invariant tensors

$$t_{\alpha,0} = s_\alpha \otimes 1 \in D^\otimes \otimes_{\mathbb{Z}_p} W(\kappa) = (\mathbf{D}(X_0)(W(\kappa)))^\otimes$$

which uniquely determine morphisms of crystals  $t_{\alpha,0} : \mathbf{1} \rightarrow \mathbf{D}(X_0)^\otimes$  s.t.  $t_{\alpha,0}[\frac{1}{p}]$  are Frobenius-equivariant.

┘

*Proof:* Cf. [Kis10]Prop1.3.2.

□



**Def. (28.6.3.16) [Hodge-Type Rapoport-Zink Spaces].** Situation as in (28.6.3.15), consider the functor

$$\mathrm{RZ}_G^{\mathrm{Nil}} = \mathrm{RZ}_{G,b,\mu,\Lambda,(s_\alpha)}^{\mathrm{Nil}} : \mathrm{NNil}/W(\kappa) \rightarrow \mathrm{Set}$$

that assigns to each  $R \in \mathrm{NNil}/W(\kappa)$  the set of isomorphism classes of triples  $(X, \rho, (t_\alpha))$  where

- $(X, \rho) \in \mathrm{RZ}(X_0)(R)$ .
- $(t_\alpha)$  is a tuple of morphisms of crystals  $t_\alpha : \mathbf{1} \rightarrow \mathbf{D}(X)^\otimes$  over  $\mathrm{Spec}(R)$  s.t.  $t_\alpha[\frac{1}{p}]$  are Frobenius-equivariant.

s.t.

1. The pullback of  $t_\alpha$  to  $\mathrm{Spec}(R/(p))$  is identified with  $t_{\alpha,0}$  under the isomorphism of isocrystals

$$\mathbf{D}(\rho) : \mathbf{D}(X_{R/(p)})^\otimes[\frac{1}{p}] \cong \mathbf{D}(X_0 \times_\kappa R/(p))^\otimes[\frac{1}{p}]$$

2. The sheaf of  $G_{W(\kappa)}$ -sets

$$\mathcal{F} = \mathrm{Isom}_{t_\alpha, s_\alpha \otimes 1}(\mathbf{D}(X), D \otimes_{\mathbb{Z}_p} R)$$

over  $(R/W(\kappa))_{\mathrm{Crys}}$  (27.5.2.2) given by isomorphisms, is a fppf-crystal of  $G_{W(\kappa)}$ -torsors.

3. There exists an étale cover  $U \rightarrow \mathrm{Spec}(R)$  and a section  $f \in \mathcal{F}(U)$  s.t. the Hodge-filtration

$$\mathrm{Fil}^1(X_U) \subset \mathbf{D}(X_U) \xrightarrow{f} D \otimes_{\mathbb{Z}_p} \mathcal{O}_U$$

is induced by a cocharacter that is  $G(W(\kappa))$ -conjugate to  $\mu$ .

┘

**Remark (28.6.3.17).**

- If  $R \in \mathrm{Nil}/W(\kappa)$  s.t.  $R/(p)$  is of f.t. over  $\kappa$ , it suffices to check condition 1 over one closed point of each connected component of  $\mathrm{Spec}(R)$ , by the proof of [Howard-Pappas]Lemma3.2.8?.
- Condition 2+3 implies that the sheaf of isomorphisms between  $\mathbf{D}(X)$  and  $D \otimes_{\mathbb{Z}_p} R$  identifying  $t_\alpha$  with  $s_\alpha \otimes 1$  and identifying the Hodge-filtration is a  $P_\mu \times_{W(\kappa)} R$ -torsor?.

┘

**Def. (28.6.3.18) [Formally Smooth Hodge-Type Rapoport-Zink Spaces].** Situation as in (28.6.3.15), consider the functor

$$\mathrm{RZ}_{G,\mu,b}^{\mathrm{fsm}} : \mathrm{NNil}^{\mathrm{fsm}}/W(\kappa) \rightarrow \mathrm{Set}$$

that assigns to each  $A \in \mathrm{NNil}^{\mathrm{fsm}}/W(\kappa)$  with an ideal of definition  $I$  the set

$$\mathrm{RZ}_{G,\mu,b}^{\mathrm{fsm}}(A) = \varprojlim_n \mathrm{RZ}_G^{\mathrm{Nil}}(A/I^n).$$

┘

**Prop. (28.6.3.19) [Action of  $J_b(\mathbb{Q}_p)$ ].** Situation as in (28.6.3.15), the action of  $J_b(\mathbb{Q}_p)$  on  $X_0$  preserves the tensors  $t_{\alpha,0}$ , thus acts on the functors  $\mathrm{RZ}_{G,\mu,b}^{\mathrm{Nil}}$  (28.6.3.16) and  $\mathrm{RZ}_{G,\mu,b}^{\mathrm{fsm}}$  (28.6.3.18) on the left. ? Cf. [Howard-Pappas]P17.

┘

**Conj. (28.6.3.20) [Representability of Rapoport-Zink Spaces, Bültel-Pappas[B-P20]].** If  $-1$  is not a slope of  $\mathrm{Ad}^G(b)$ ?, then the functor  $\mathrm{RZ}_{G,\mu,b}$  is representable by a formal scheme that is formally smooth and formally locally of f.t. over  $W(\kappa)$ .

┘

*Proof:* ? □

**Thm& Conj.Cor. (28.6.3.21)** [Bütel-Pappas[B-P20]]. Situation as in (28.6.3.15), if  $\iota(b)$  has no slope 0, then the restriction of  $\mathrm{RZ}_{G,\mu,b}$  to  $\mathrm{NilSch}/W(\kappa)$  is representable by a formal scheme that is formally smooth and formally locally of f.t. over  $\mathrm{Spf}(W(\kappa))$ . ┘

*Proof:* ? □

**Thm& Conj.Cor. (28.6.3.22)** [Representability in the Hodge-Type case, Howard-Pappas]. Situation as in (28.6.3.15), if  $(G, b, \mu, \Lambda)$  comes from the fiber of a Hodge-type Shimura variety, then the functor  $\mathrm{RZ}_{G,b,\mu}^{\mathrm{fsm}}$  is representable by a closed formal subscheme  $\mathrm{RZ}_{G,b,\mu}$  of  $\mathrm{RZ}(X_0)$  over  $\mathrm{Spf}(W(\kappa))$  that is formally smooth and formally locally of f.t., with a left action of  $J_b(\mathbb{Q}_p)$ . ┘

*Proof:* ? □

### Special Fibers

**Def. (28.6.3.23)** [Classical Affine Deligne-Lusztig Sets]. Given an unramified local Shimura datum  $(G, b, \mu, \Lambda)$  (28.6.3.13), the (classical) **Affine Deligne-Lusztig set** is defined to be

$$X_{G,b,\mu}(\kappa) = \{g \in G(K_0) | g^{-1}b\sigma(g) \in G(W(\kappa))\mu(p)G(W(\kappa))\}/G(W(\kappa)).$$

┘

**Def. (28.6.3.24)** [Refined Affine Deligne-Lusztig Sets]. Situation as in (28.6.3.23), if  $\kappa'/\kappa$  is a f.g. field extension, consider  $W' = C(\kappa')$  the Cohen ring with a lift of Frobenius  $\sigma$  and  $K' = W'[\frac{1}{p}]$ . Consider the set

$$X' = \{g \in G(K') | g^{-1}b\sigma(g)\mu(p)^{-1} \in G(W')\}$$

and

$$Q_\mu(W') = G(W') \cap \mu^{\sigma^{-1}}(p)^{-1}G(W')\mu^{\sigma^{-1}}(p) \subset G(K').$$

Then the right translation of  $Q_\mu(W')$  on  $G(K')$  preserves the set  $X'$ . And we can define the **refined affine Deligne-Lusztig set** as the quotient

$$X_{G,b,\mu,\sigma}(\kappa') = \{g \in G(K') | g^{-1}b\sigma(g)\mu(p)^{-1} \in G(W')\}/Q_\mu(W').$$

and also the (naive) **refined affine Deligne-Lusztig set**

$$X_{G,b,\mu,\sigma}^{\mathrm{naive}}(\kappa') = \{g \in G(K') | g^{-1}b\sigma(g) \in G(W')\mu(p)G(W')\}/G(W') \subset G(K')/G(W').$$

┘

**Prop. (28.6.3.25)**. Situation as in (28.6.3.24),

- Sending  $gQ_\mu(W')$  to  $gG(W')$  defines an injection

$$\phi(\kappa') : X_{G,b,\mu,\sigma}(\kappa') \hookrightarrow X_{G,b,\mu,\sigma}^{\mathrm{naive}}(\kappa')$$

which is a bijection if  $\kappa'$  is perfect.

- If  $b' = h^{-1}b\sigma(h)$  for some  $h \in G(K_0)$ , then  $g \mapsto hg$  defines a bijection

$$X_{G,b',\mu,\sigma}(\kappa') \cong X_{G,b,\mu,\sigma}(\kappa').$$

- If  $\kappa'$  is perfect, then  $g \mapsto \sigma^{-1}(b^{-1}g)$  defines a bijection

$$X_{G,b,\mu^\sigma}(\kappa') \cong X_{G,b,\mu}(\kappa').$$

- If  $\iota : G \rightarrow \mathrm{GL}(n)_{\mathbb{Z}_p} \in \mathcal{A}\mathrm{lgGrp}/\mathbb{Z}_p$  is a closed embedding, then  $\iota$  defines an injection

$$\iota : X_{G,b,\mu,\sigma}(\kappa') \hookrightarrow X_{\mathrm{GL}(n),\iota(b),\iota\circ\mu}(\kappa').$$

┘

*Proof:* Cf.[Howard-Pappas]P18. □

**Prop. (28.6.3.26) [F.G. Fields as Limits of Smooth Algebras].** If  $\kappa'/\kappa$  is a f.g. field extension with Cohen ring  $C(\kappa')$  and a lift of Frobenius  $\sigma$ , then we can write  $\kappa' = \varinjlim R$  as a filtered colimits of smooth  $\kappa$ -algebras each having a finite  $p$ -basis? and for each  $R$  there exists a  $W(\kappa)$ -flat formally smooth  $p$ -adically complete and separated lift  $\tilde{R}$  of  $R$  with an injection  $\tilde{R} \subset C(\kappa')$  lifting  $R \rightarrow \kappa'$  s.t.  $\sigma(\tilde{R}) \subset \tilde{R}$ . ┘

*Proof:* Cf.[Howard-Pappas]P20?. □

**Thm. (28.6.3.27) [Field-Valued Points of  $\mathrm{RZ}_G^{\mathrm{Nil}}$ ].** Situation as in (28.6.3.15), if  $\kappa'/\kappa$  is a f.g. field extension with Cohen ring  $W(\kappa')$  and a lift of Frobenius  $\sigma$ . Suppose that the Dieudonné module structure on  $D_{W(\kappa)}$  determined by  $\iota(b) \in \mathrm{GL}(C_{W(\kappa)})$  has no zero slopes (i.e.  $X_0$  is formal, Cf.[Howard-Pappas]P20), then there are natural bijections

$$\pi : \varinjlim_R \mathrm{RZ}_G^{\mathrm{Nil}}(R) \cong \mathrm{RZ}_G^{\mathrm{Nil}}(\kappa') \cong X_{G,b,\mu^\sigma,\sigma}(\kappa') \quad (28.6.3.24).$$

In particular, we have a bijection

$$\pi : \mathrm{RZ}_G(\kappa') \cong X_{G,b,\mu^\sigma,\sigma}(\kappa') \quad (28.6.3.22).$$

┘

*Proof:* Cf.[Howard-Pappas]Thm2.4.10.?. □

For the last assertion, notice the reduced scheme of  $\mathrm{RZ}_G$  is locally of f.t. (28.6.3.22)?, so

$$\mathrm{RZ}_G(\kappa') = \varinjlim_R \mathrm{RZ}_G(R) = \varinjlim_R \mathrm{RZ}_G^{\mathrm{Nil}}(R) = X_{G,b,\mu^\sigma,\sigma}(\kappa').$$

□

**Thm. (28.6.3.28) [Unramified Hodge-Type case, Zhu[Zhu17]].** Fix  $(\mathbb{X}_0, \varepsilon)$  as above?, there is a canonical isomorphism  $X_\mu(b) \cong \overline{\mathcal{M}}_\mu(b)^{\mathrm{perf}}$ . In particular,  $\dim \overline{\mathcal{M}}_\mu(b)_{\mathrm{red}} = \dim X_\mu(b) = \langle \rho, \mu - \nu_b \rangle - \frac{1}{2} \mathrm{def}_G(b)$  (28.7.3.4). ┘

*Proof:* Cf.[Zhu17]P463.?. □

### Unitary Rapoport-Zink Spaces

**Notation (28.6.3.29).**

- Let  $p \in \text{Prime}_{\geq 3}$ .
- Let  $(F_0, \kappa, \varpi) \in p\text{-LField}$ ,
- Let  $F/F_0$  be an unramified quadratic extension with non-trivial automorphism  $\sigma$ ,
- Let  $d = [F_0 : \mathbb{Q}_p]$ .
- Let  $\check{F}$  be the maximal unramified extension of  $F$  with residue field  $\mathbb{F}$ ,
- 

┘

**Def. (28.6.3.30) [Unitary Rapoport-Zink Spaces].** Let  $\mathcal{N}_{n,F/F_0}$  be the formal scheme over  $\check{F}$  which classifies quadruples  $(X, \lambda, \iota, \rho)$  over a scheme  $S$  which is  $p$ -nilpotent, where

- $X$  is a  $p$ -divisible group over  $S$  of relative dimension  $n$  and absolute height  $2nd$ .
- $\lambda : X \rightarrow X^\vee$  is a principle  $\mathcal{O}_{F_0}$ -polarization.
- $\iota : \mathcal{O}_F \subset \text{End}_S(X)$  s.t. the induced action of  $\mathcal{O}_{F_0}$  is via the structure morphism  $\mathcal{O}_{F_0} \subset \mathcal{O}_S$ .
- $\rho : X \times_S \bar{S} \rightarrow \mathbb{X}_n \times \bar{S}$  is a quasi-isogeny of height 0, where  $(\mathbb{X}_n, \lambda_0, \iota_0)$  is a supersingular basic **framing object** over  $\mathbb{F}_p$  **?**.

which satisfies:

- The Rosati involution  $\text{Ros}_\lambda$  on  $\mathcal{O}_F$  equals the non-trivial Galois automorphism in  $\text{Gal}(F/F_0)$ .
- The **Kottwitz condition of signature**  $(n-1, 1)$ : For any  $a \in \mathcal{O}_F$ ,

$$\text{char}(\iota(a) | \text{Lie}(X); T) = (T - a)^{n-1}(T - \bar{a}).$$

┘

**Prop. (28.6.3.31).**  $\mathcal{N}_n$  is formally locally of finite type and formally smooth over  $\text{Spf } \mathcal{O}_{\check{F}}$  of relative dimension  $n-1$ .

┘

*Proof:* **?**

□

**Prop. (28.6.3.32).**  $\mathcal{N}_1 \cong \text{Spf } \mathcal{O}_{\check{F}}$ .

┘

*Proof:* **?**

□

**Prop. (28.6.3.33) [Automorphism Group of the Framing].** There is a unique triple  $(\mathbb{E}, \iota_E, \lambda_E)$  over  $\mathbb{F}$  with signature  $(1, 0)$ .

Set

$$\mathbb{V}_n = \text{Hom}_{\mathcal{O}_F}^0(\mathbb{E}, \mathbb{X}_n),$$

which belongs to  $\text{Herm}_n(F/F_0)$ , with the Hermitian form given by

$$(x, y) = \lambda_E^{-1} \circ y^\vee \circ \lambda_{\mathbb{X}_n} \circ x \in \text{End}_{\mathcal{O}_F}^0(\mathbb{E}) \cong F.$$

Then it is the unique  $n$ -dimensional Hermitian space without a self-dual  $\mathcal{O}_F$ -lattice.

And denote

$$\text{Aut}^0(\mathbb{X}_n, \iota_{\mathbb{X}_n}, \lambda_{\mathbb{X}_n}) = \{g \in \text{End}_{\mathcal{O}_F}^0(\mathbb{X}_n, \lambda_{\mathbb{X}_n}), g^\vee \circ \lambda_{\mathbb{X}_n} \circ g = \lambda_{\mathbb{X}_n}\} \cong U(\mathbb{V}_n)(F_0).$$

Thus  $U(\mathbb{V}_n)(F_0)$  acts on  $\mathcal{N}_n$  by changing the framing.

┘

## 4 Period Morphisms

**Def. (28.6.4.1) [Period Morphisms].** The map

$$\mathrm{pr} = \mathrm{pr}_1 \times \mathrm{pr}_2 : \check{M}^{\mathrm{rig}} \rightarrow \check{\mathcal{F}}^{\mathrm{rig}} \times \Delta_{\mathcal{D}} \quad (28.6.3.10)$$

is called the **period morphism** associated to moduli problem  $\check{M}$ . ┘

## 5 Local Shimura Varieties

**Def. (28.6.5.1) [Local Shimura Data].** A **local Shimura datum** over  $K$  is a triple  $(G, [b], \mathcal{N})$  where  $G \in \mathrm{RedGrp}/K$ ,  $[b] \in B(G)$  (28.6.1.5),  $\mathcal{N} \in X_{\bullet}(G)/G(\bar{K})$  with reflex field  $E \in p\text{-NField}$  (10.3.2.5) s.t.

- $\mathcal{N}$  is minuscule ?.
  - $[b] \in B(G, \mathcal{N})$  (28.6.1.39).
- ┘

**Conj. (28.6.5.2) [Local Shimura Varieties, Rapoport [Rap95, ?R-V14]].** There should be a way to associate to a local Shimura datum  $(G, [b], \mathcal{N})$  over  $K$  (28.6.5.1) a tower of rigid analytic spaces over  $\mathrm{Sp}(\check{E})$ :

$$\mathfrak{M}(G, [b], \mathcal{N}) = \left\{ \mathfrak{M}^{\mathcal{K}} \right\}_{\mathcal{K}} \in \mathrm{RigSpa} / \check{E},$$

where  $\mathcal{K}$  ranges over all compact open subsets of  $G(K)$ , s.t.

- Each  $\mathfrak{M}^{\mathcal{K}}$  is equipped with an action of  $J_{[b]}(K)$  (28.6.1.21) .
- $G(K)$  acts on the tower as Hecke correspondences.
- The tower has a Weil-descent datum from  $\check{E}$  to  $E$ .
- There exists an étale and partially proper  $J_{[b]}(K) \times G(K)$ -equivariant **period morphism** of towers of rigid analytic spaces

$$(\check{\pi}^{\mathcal{K}}, \kappa^{\mathcal{K}}) : \mathfrak{M}^{\mathcal{K}} \rightarrow \mathcal{F}(G, [b], \mathcal{N})^{\mathrm{weak.adm}} \times \Delta_G \quad (28.6.2.4)(28.6.3.10),$$

compatible with Weil-descent data, where  $J_b(K)$  acts on  $\mathfrak{M}^{\mathcal{K}} \times \Delta_G$  diagonally and on  $\mathcal{F}(G, [b], \mathcal{N})^{\mathrm{weak.adm}}$  (28.6.2.4) and  $G(K)$  acts on  $\mathfrak{M}^{\mathcal{K}}$  and  $\Delta_G$ . And the image of  $(\check{\pi}^{\mathcal{K}}, \kappa^{\mathcal{K}})$  is of the form  $\mathcal{F}(G, [b], \mathcal{N})^{\mathrm{weak.adm}} \times \Delta'$  where  $\Delta' \subset \Delta_G$  is a union of cosets of a subgroup of finite index in  $\Delta_G$ .

- The fiber of  $\check{\pi}^{\mathcal{K}}$  in  $\mathfrak{M}^{\mathcal{K}}$  is isomorphic to  $G(K)/\mathcal{K}$ , compatible with  $\mathcal{K}$ .
  - There should be an open adic subspace  $\mathcal{F}(G, [b], \mathcal{N})^a \subset \mathcal{F}(G, [b], \mathcal{N})^{\mathrm{weak.adm}}$  stable under the action of  $J_{[b]}(K)$  and a  $J_{[b]}(K)$ -equivariant  $G(K)$ -torsor on it s.t. the tower  $\mathfrak{M}^{\mathcal{K}}$  arises from a system of level- $\mathcal{K}$ -structures.
- ┘

*Proof:*

□

## 6 Cohomology of Local Shimura Varieties

References are [?R-V14].

**Notation (28.6.6.1).**

- Let  $\ell \in \text{Prime} \setminus \{p\}$ .

┘

**Def. (28.6.6.2) [étale Cohomology of Local Shimura Varieties].** Let  $(G, [b], \mathcal{N})$  be a local Shimura datum (28.6.5.1) giving rise to a tower of local Shimura variety  $\mathfrak{M}(G, b, \mathcal{N})$  (28.6.5.2). For each compact open subgroup  $\mathcal{K} \leq G(K)$ , the cohomology groups

$$H_{\text{ét},c}^n \left( \left( \mathfrak{M}^{\mathcal{K}} \right)_{\mathbb{C}_E}, \overline{\mathbb{Q}_\ell} \right), \text{ where } n \in \mathbb{N}$$

which vanishes for  $n > 2 \dim \mathfrak{M}^{\mathcal{K}}$ .

┘

*Proof:* ? Cf. [Fargues, Cohomologie des espaces de modules de groupes p-divisibles et correspondances de Langlands locales].  $\square$

**Remark (28.6.6.3).** The action of  $W_{\mathcal{K}}$  may not be continuous, Cf. [Fargues, Cohomologie des espaces de modules de groupes p-divisibles et correspondances de Langlands locales] 4.4.4.  $\square$

**Prop. (28.6.6.4) [Étale Cohomology of Local Shimura-Varieties].** Situation as in (28.6.6.2), the cohomologies  $H_{\text{ét},c}^n \left( \left( \mathfrak{M}^{\mathcal{K}} \right)_{\mathbb{C}_E}, \overline{bbQ_\ell} \right)$  naturally have actions of  $J_{[b]}(K)$  by functoriality, and actions of  $I_E$ , which extends to an action of  $W_E$  using the Weil-descent datum (28.6.5.2). And

- The actions of  $J_{[b]}(K)$  is smooth, and  $H_{\text{ét},c}^n \left( \left( \mathfrak{M}^{\mathcal{K}} \right)_{\mathbb{C}_E}, \overline{\mathbb{Q}_\ell} \right)$  is a f.g.  $J_{[b]}(K)$ -module.
- The action of  $I_E$  is smooth and continuous.
- For any  $\rho \in \text{Irr}^{\text{adm}}(J_{[b]}(K))$ ,  $\mathcal{K} \leq G(K)$  and  $i, j \in \mathbb{Z}$ ,

$$H^{i,j}(\mathfrak{M}^{\mathcal{K}})[\rho] \triangleq \text{Ext}_{J_{[b]}(K)}^j \left( H_{\text{ét},c}^n \left( \left( \mathfrak{M}^{\mathcal{K}} \right)_{\mathbb{C}_E}, \overline{\mathbb{Q}_\ell} \right), \rho \right)$$

is a finite rank  $\mathbb{Q}_\ell$ -space, and vanishes for  $j$  bigger than the semi-simple rank of  $J_{[b]}$ .

- For any  $\rho \in \text{Irr}^{\text{adm}}(J_{[b]}(K))$  and  $i, j \in \mathbb{Z}$ ,

$$H^{i,j}(G, [b], \mathcal{N})[\rho] \triangleq \varinjlim_{\mathcal{K} \leq G(K)} H^{i,j}(\mathfrak{M}^{\mathcal{K}})[\rho]$$

is an admissible  $G(K)$ -module and a continuous  $W_E$ -module.

┘

*Proof:* ? Cf. [R-V14].  $\square$

**Def. (28.6.6.5).** Situation as in (28.6.6.4), denote the alternating sum

$$[H^\bullet(G, [b], \mathcal{N})[\rho]] \triangleq \sum_{i,j \in \mathbb{N}} (-1)^{i+j} H^{i,j}(G, [b], \mathcal{N})[\rho] (-\dim \mathfrak{M})$$

as a virtual  $G(K) \times W_E$ -module (in the Grothendieck group of smooth finite length representations).

┘

### Kottwitz Conjecture

**Notation (28.6.6.6).**

- Let  $b$  be basic.

┘

**Conj. (28.6.6.7) [conjectured by Kottwitz [Rap95, ?R-V14]].** Suppose

- $G^* \in \text{RedGrp}/\mathbb{Q}_p$ ,  $b^* \in B(G^*)_b$  and  $G = J_{b^*} \in \text{RedGrp}/\mathbb{Q}_p$ .
- $(G, [b], \mathcal{N})$  be a local Shimura-datum (28.6.5.1) giving rise to a tower of local Shimura variety  $\mathfrak{M}(G, b, \mathcal{N})$  (28.6.5.2).
- $\phi \in \Phi^{\text{disc}}(G(K))$  (29.3.1.4).

Recall the f.d.  $\overline{\mathbb{Q}_\ell}$ -dimensional representation  $r_{\mathcal{N}}$  of  $\widehat{G} \times W_E$  (? Cf. [Kottwitz, Shimura varieties and twisted orbital integrals] 2.1.2), we get a representation  $r_{\mathcal{N}} \circ \phi_E$  of  $C_\phi \times W_E$  via

$$r_{\mathcal{N}} \circ \phi_E(s, w) = r_{\mathcal{N}}(s\varphi(w)).$$

Then for each  $\pi\rho \in \Pi_\phi(J)$  (29.3.4.7), we have

$$[H^\bullet(G, [b], \mathcal{N})[\rho]] = \sum_{\pi \in \Pi_\phi(G)} \pi \boxtimes \text{Hom}_{C_\phi^\sharp}(\iota_{\mathfrak{m}}(\pi)^\vee \otimes \iota_{\mathfrak{m}}(\rho), r_{\mathcal{N}} \circ \phi_E) \left(-\frac{\dim \mathfrak{M}(G, b, \mathcal{N})}{2}\right) \quad (28.6.6.5)$$

┘

*Proof:* ? Cf. [?R-V14] for known cases.

□

**Thm& Conj. Cor. (28.6.6.8) [Harris-Taylor [H-T01] Thm 12.1.3].** Situation as in (28.6.6.7), if  $G = G^* = \text{GL}(n)$  and  $\mathcal{N} = G(K).(1, 0, \dots, 0)$  and  $[b] \in B(G, \mathcal{N})$  is the unique basic element (28.6.1.40), then the Kottwitz's conjecture (28.6.6.7) holds.

┘

*Proof:* ?

□

**Thm& Conj. Cor. (28.6.6.9) [Shin].** Situation as in (28.6.6.7), if  $G = G^* = \text{Res}_{K/\mathbb{Q}_p} \text{GL}(n) \in \text{RedGrp}/\mathbb{Q}_p$  where  $F/\mathbb{Q}_p$  is an unramified extension, then the Kottwitz's conjecture holds for any local Shimura-datum  $(G, [b], \mathcal{N})$ .

┘

*Proof:* ? [Shin, cohomology of Rapoport-Zink spaces of EL-type] Cor 1.3. The proof uses point-counting argument on Igusa varieties.

□

### Harris Conjecture

**Conj. (28.6.6.10) [non-Basic Cohomologies, Harris [Har01]].**

┘

## 28.7 Affine Deligne-Lusztig Varieties

References are [Zhu17, B-S17] and [Xuhua He, some results on affine Deligne-Lusztig Varieties].

**Notation (28.7.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $k \in \text{Field}^p$  s.t.  $k = k^{\text{perf}}$ .
- Use notations on perfect schemes defined in [F<sub>p</sub>-Algebraic Stacks](#).

┘

### 1 Families of Torsion $W(k)$ -Modules

**Def. (28.7.1.1) [Types of Torsion  $W(k)$ -Modules].**

- For any non-increasing sequence of non-negative integers  $\lambda = (\lambda_1, \dots, \lambda_n)$ , a f.g.  $p$ -power torsion  $W(k)$ -module  $Q$  is said to have type  $\lambda$  if we have  $Q \cong \oplus_j W(k)/(p^{\lambda_j})$ .
- For  $X \in \text{PerfSch}/k$ , a f.g.  $p$ -power torsion  $W(\mathcal{O}_X)$ -module  $Q$  is said to have type  $\lambda$  (resp. type  $\preceq \lambda$ ) if  $\lambda(Q \otimes_{W(\mathcal{O}_X)} W(\kappa(x))) = \lambda$  (resp.  $\lambda(Q \otimes_{W(\mathcal{O}_X)} W(\kappa(x))) \preceq \lambda$ ) for each  $x \in \text{Spec}(R)$ .

┘

**Prop. (28.7.1.2).** If  $R \in \text{Perf}$  and  $Q$  is a f.g.  $p$ -power torsion  $W(R)$ -module of type  $\lambda$  (28.7.1.1), then  $p^i Q / p^{i+1} Q \in \text{Mod}_R^{\text{f.proj}}$ .

┘

*Proof:* Cf. [B-S17]P369?.

□

**Def. (28.7.1.3) [Isogenies of Finite Projective  $W(R)$ -Modules].** For  $R \in \text{Perf}/k$ , an **isogeny of finite projective  $W(R)$ -modules** is a map  $f : N \rightarrow M \in \text{Mod}_{W(R)}^{\text{f.proj}}$  s.t. it admits an inverse up to multiplication by a power of  $p$ .

┘

**Prop. (28.7.1.4) [Projective Dimension 1].** For  $R \in \text{Perf}/k$ , a f.g.  $p$ -power torsion  $W(R)$ -module  $Q$  is of projective dimension 1 (projective resolutions) iff it is isomorphic to the cokernel of an isogeny of finite projective  $W(R)$ -modules.

┘

*Proof:* If  $Q$  is a cokernel of an isogeny of f.p.  $W(R)$ -modules, then it has projectively dimension  $\leq 1$ . And clearly it cannot have projective dimension 0. Conversely if  $Q$  is of projective dimension 1, pick a surjection  $W(R)^{\oplus n} \rightarrow Q$ , then the kernel  $N$  is projective over  $W(R)$ , and is finite projective by looking at characteristic 0 points of  $\text{Spec}(W(R))$ , because these points are dense.

□

**Prop. (28.7.1.5).** For  $R \in \text{Perf}/k$ ,  $Q \in \text{Mod}_R^{\text{ft}}$  is projective iff it is of projective dimension 1 as  $W(R)$ -modules.

┘

*Proof:* Firstly  $R = W(R)/(p)$  is of projective dimension 1, so any free  $R$ -module is of projective dimension 1, thus also any projective module is of projective dimension 1.

Conversely, if  $Q$  is of projective dimension 1, then? Cf. [B-S17]P372.

□



### Demazure-Schemes

**Def. (28.7.1.6) [Demazure-Schemes].** Let  $X \in \text{PerfSch}/\mathbb{F}_p$  and  $\mathcal{Q} \in \mathcal{QCoh}(W(\mathcal{O}_X))$  be f.g.  $p$ -power torsion and of projective dimension 1, then for each non-increasing sequence of non-negative integers  $\lambda = (\lambda_1, \dots, \lambda_n)$ , there exist a **Demazure-scheme**

$$\text{Dem}_\lambda(\mathcal{Q}) \in \text{PerfSch}^{\text{pfp, proper}}/X$$

parametrizing decreasing filtrations  $\text{Fil}^\bullet \mathcal{Q}$  of  $\mathcal{Q}$  s.t. each  $\text{gr}^i \mathcal{Q} \in \text{Mod}_{\mathcal{O}_X}^{\text{f.proj}}$  is of rank  $n_\lambda(i) \triangleq \max\{j \in \mathbb{Z}_+ \mid i < \lambda_j\}$ . ┘

*Proof:* Cf. [B-S17]P373? □

**Prop. (28.7.1.7) [Properties of Demazure-Schemes, Katz[Kat79]/Bhatt-Scholze[B-S17]].** Situation as in (28.7.1.6),

1. The image of the morphism  $\text{Dem}_\lambda(\mathcal{Q}) \rightarrow X$  is the locus

$$X_{\preceq \lambda} \triangleq \left\{ x \in X \mid \lambda \left( \mathcal{Q} \otimes_{W(\mathcal{O}_X)} W(\kappa(x)) \right) \preceq \lambda \right\}.$$

2. If  $\mathcal{Q}$  is of type  $\lambda$  (28.7.1.1), then  $\text{Dem}_\lambda(\mathcal{Q}) \cong X$ .
3. If  $\text{Dem}_\lambda(\mathcal{Q}) \times_X \bar{x} \neq \emptyset$  for a geometric point  $\bar{x}$  of  $X$ , then

$$\text{R}\Gamma \left( \text{Dem}_\lambda(\mathcal{Q}) \times_X \bar{x}, \mathcal{O}_{\text{Dem}_\lambda(\mathcal{Q}) \times_X \bar{x}} \right) = \kappa(\bar{x}).$$

4.  $X_{\preceq \lambda}$  is closed in  $|X|$ . ┘

*Proof:* Cf. [B-S17]P374? □

- 4: This follows from item1 and the definition (27.1.2.6). □

## 2 Witt-vector Affine Grassmannian

**Notation (28.7.2.1).**

- Fix  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa) \in p\text{-Field}$ . ┘

**Def. (28.7.2.2) [Witt-Vector Affine Grassmannians].** For any non-increasing sequence of non-negative integers  $\lambda = (\lambda_1, \dots, \lambda_n)$ , let  $\text{Gr}_{\preceq \lambda}$  be the functor

$$\text{PerfSch}/\kappa \rightarrow \text{Set} : X \mapsto \left\{ \begin{array}{l} \text{finite projective } W(\mathcal{O}_X)\text{-submodules } \mathcal{E} \subset W(\mathcal{O}_X)^n \text{ s.t.} \\ \beta: \mathcal{E} \rightarrow W(\mathcal{O}_X)^n \text{ is an isogeny (28.7.1.3) and } \mathcal{Q} \triangleq \text{Coker}(\beta) \text{ is finite of type } \leq \lambda \text{ (28.7.1.1)} \end{array} \right\}.$$

Then  $\text{Gr}_{\preceq \lambda}$  is a v-sheaf by [B-S17]4.1. and 4.4.? ┘

**Def. (28.7.2.3) [Demazure-Resolutions].** The **Demazure-resolution** of  $\text{Gr}_{\preceq \lambda}$  is the map

$$\psi_{\text{Gr}_{\preceq \lambda}} : \widetilde{\text{Gr}}_{\preceq \lambda} \triangleq \text{Dem}_\lambda(\mathcal{Q}) \rightarrow \text{Gr}_{\preceq \lambda} \in \text{Sh}(\text{PerfSch}_v/\kappa) \text{ (28.7.2.2) (28.7.1.6),}$$

where  $\mathcal{Q} \in \mathcal{QCoh}(W(\mathcal{O}_{\text{Gr}_{\preceq \lambda}}))$  is the universal cokernel of  $W(\mathcal{O}_{\text{Gr}_{\preceq \lambda}})$  (28.7.2.2), which is  $p$ -power torsion and of projective dimension 1? ┘

$\psi_{\text{Gr}_{\preceq \lambda}}$  is representable by (28.7.1.6), and is surjective, perfectly f.p. and perfectly proper, and is an isomorphism on  $\text{Gr}_\lambda \subset \text{Gr}_{\preceq \lambda}$ , by (28.7.1.7), and  $\text{R}\psi_* \mathcal{O}_{\text{Dem}_\lambda(\mathcal{Q})} = \mathcal{O}_X$  (when base changed to each  $S \in \text{PerfSch}/\text{Gr}_{\preceq \lambda}$ ), by (27.1.2.4). ┘

**Prop. (28.7.2.4)** [ $\widetilde{\text{Gr}}_{\preccurlyeq \lambda}$  is Representable, Zhu[Zhu17]/Bhatt-Scholze[B-S17]].  $\widetilde{\text{Gr}}_{\preccurlyeq \lambda}$  (28.7.2.3) is representable by the perfection of a projective smooth  $\mathbb{F}_p$ -scheme,  $\lrcorner$

*Proof:* By definition (28.7.1.6) and (28.7.1.5),  $\widetilde{\text{Gr}}_{\preccurlyeq \lambda}$  classifies the sequence of  $\lambda_1$ -tuples  $\mathcal{E}_1 \subset \mathcal{E}_2 \subset \dots \subset \mathcal{E}_0 = W(\mathcal{O}_X)^n \in \text{Mod}_{W(\mathcal{O}_X)}^{\text{f.proj}}$  s.t. each  $\mathcal{Q}_i = \mathcal{E}_i/\mathcal{E}_{i+1} \in \text{Mod}_{\mathcal{O}_X}^{\text{f.proj}}$  is of rank  $n_\lambda(i)$  (28.7.1.6).

Then  $\widetilde{\text{Gr}}_{\preccurlyeq \lambda}$  is representable by a successive perfect Grassmannian varieties: Define  $X_0 = \text{Spec}(\mathbb{F}_p)$ ,  $\mathcal{E}_0 = W(\mathcal{O}_{X_0})^{\lambda_1}$ , and define inductively  $\text{pr}_i : X_{i+1} \rightarrow X_i$  and  $\mathcal{E}_i \in \text{Mod}_{W(\mathcal{O}_{X_{i+1}})}^{\text{f.proj}}$  by defining

$$X_{i+1} = \underline{\text{Quot}}(\mathcal{E}_i/(p), n_\lambda(i))$$

which is representable by the perfection of a smooth projective scheme over  $X_i$  (27.1.2.13), with universal quotient  $\mathcal{Q}_i^{\text{univ}} \in \text{Mod}_{\mathcal{O}_X}^{\text{f.proj}}$  and

$$\mathcal{E}_{i+1} = \ker(\text{pr}_i^* \mathcal{E}_i \rightarrow \mathcal{Q}_i^{\text{univ}}),$$

which is also finite projective over  $W(\mathcal{O}_{X_{i+1}})$  by (28.7.1.5).  $\square$

**Thm. (28.7.2.5)** [Representability of  $\text{Gr}_{\text{GL}(n)}$ , Haboush[Hab05]/Kreidl[Kre14]/Zhu[Zhu17]/Bhatt-Scholze[B-S17]]. Situation as in (28.7.2.2), then

- $\text{Gr}_{\leq \lambda}$  is representable by the perfection of a projective  $\mathbb{F}_p$ -scheme,
- There is a natural ample line bundle  $\mathcal{L}_\lambda \in \text{Pic}(\text{Gr}_{\preccurlyeq \lambda})$  s.t.  $\tilde{\mathcal{L}}_\lambda \triangleq \psi_{\text{Gr}_{\preccurlyeq \lambda}}^* \mathcal{L} \in \text{Pic}(\widetilde{\text{Gr}}_{\preccurlyeq \lambda})$  is given by

$$\tilde{\mathcal{L}}_\lambda = \bigotimes_{i \in [\lambda_1 - 1]} \det_{\mathcal{O}_X} \mathcal{Q}_i,$$

- For each  $\mu \preccurlyeq \lambda \in X_\bullet^+(\text{GL}(n))$ , the restriction of  $\mathcal{L}_\lambda$  to  $\text{Gr}_{\preccurlyeq \mu}$  is equal to  $\mathcal{L}_\mu$ .  $\lrcorner$

*Proof:* Cf. [B-S17] Thm 8.8. and Section 8.4. ?  $\square$

**Lemma (28.7.2.6)** [ $p$ -adic Jet Spaces]. For each  $\mathcal{X} \in \text{Sch}^{\text{ft}}/\mathcal{O}_K$ , by work of Greenberg ?, there exists  $L_h^+(\mathcal{X}) \in \text{Aff}^{\text{ft}}/\kappa$  for each  $h \in \mathbb{Z}_+$ , s.t. for each  $R \in \mathbb{C}\text{Ring}/k$ , we have

$$L^h \mathcal{X}(R) = \mathcal{X}(W_{\mathcal{O}_K, h}(R)).$$

And if we denote  $L^+ \mathcal{X} \triangleq \varprojlim_{h \in \mathbb{Z}_+} L^h \mathcal{X} \in \text{Aff}/k$ , then for each  $R \in \mathbb{C}\text{Ring}/k$ , we have

$$L^+ \mathcal{X}(R) = \mathcal{X}(W_{\mathcal{O}_K}(R)).$$

called the  $p$ -adic jet space of  $\mathcal{X}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (28.7.2.7)** [Witt-Vector Loop Groups]. For  $X \in \text{Aff}^{\text{ft}}/K$ , the (Witt-vector)  $p$ -adic loop group of  $X$  is the functor

$$\text{LX} : \text{Perf}/\kappa \rightarrow \mathcal{G}\text{rp} : R \mapsto X \left( W_{\mathcal{O}_K}(R) \left[ \frac{1}{p} \right] \right),$$

Then

- $LX$  is representable by a strict ind-(perfect affine scheme)(i.e. it is an inductive limit of perfect affine schemes along closed immersions).
- If  $X \in \text{Aff}^{\text{ft}}/K$  with an affine integral model  $\mathcal{X} \in \text{Sch}^{\text{ft}}/\mathcal{O}_K$ , then  $L^+\mathcal{X} \subset LX \in \text{PerfSch}/\kappa$ (28.7.2.6) is a closed subscheme.
- If  $X \rightarrow Y \in \text{Aff}^{\text{ft}}/K$  is a closed embedding, then  $LX \rightarrow LY \in \text{PerfSch}/\kappa$  is a closed embedding.

⌋

*Proof:* 1: By Weil-restriction it suffices to prove for  $K$  absolutely-unramified. If  $X = \mathbb{A}^1$ , then  $LX$  is representable by  $\varinjlim_{n \in \mathbb{Z}_+} (\mathbb{A}^n)^{\text{perf}}$ , because by (5.5.3.12), each element  $x \in W_{\mathcal{O}_K}(R)[\frac{1}{p}]$  can be written uniquely as

$$x = \sum_{n \gg -\infty} [x_n]p^n.$$

And then also  $L\mathbb{A}^n$  is representable. Now if  $X = \text{Spec}(K[X_1, \dots, X_n]/(f_1, \dots, f_m))$ , then  $LX$  is a closed subscheme of  $L\mathbb{A}^n$ , because for each  $f \in K[X_1, \dots, X_n]$ , by (5.5.3.10), we have

$$f\left(\sum [x_{1i}]p^i, \dots, \sum [x_{ni}]p^i\right) = \sum_{j \gg -\infty} \left[f^{(j)}(x_{1l}, \dots, x_{nl})^{1/p^j}\right] p^j$$

where each  $f^{(j)}$  is a polynomial of the variables  $\{X_{kl}\}_{k \in [n], l \in \mathbb{Z}}$ .

2, 3: These follow by similar arguments as above.

□

**Prop. (28.7.2.8).** If  $\mathcal{G} \in \text{GrpSch}^{\text{aff,ft}}/\mathcal{O}_K$ ,  $h \in \mathbb{N}$  and  $\mathcal{G}^{(h)}$  is the  $h$ -th congruence group scheme of  $\mathcal{G}$  (defined using dilatations?), then the natural map  $\mathcal{G}^{(h)} \rightarrow \mathcal{G}$  identifies

$$L^+\mathcal{G}^{(h)} = \ker(L^+\mathcal{G} \rightarrow L^h\mathcal{G}).$$

as perfect schemes.

⌋

*Proof:* ?

□

**Def. (28.7.2.9) [Witt-Vector Affine Grassmannian for General Groups].** For  $G \in \text{RedGrp}/K$  with a smooth affine integral model  $\mathcal{G} \in \text{GrpSch}^{\text{sm,aff}}/\mathcal{O}_K$ , define

$$\text{Gr}_{\mathcal{G}} \triangleq LG/L^+\mathcal{G} \in \text{Sh}(\text{PerfAff}_{\text{fpqc}}/k) \text{ (28.7.2.7)(28.7.2.6),}$$

called the (Witt-vector)**affine Grassmannian** for  $\mathcal{G}$ .

If we denote  $\mathcal{E}_0$  the trivial  $\mathcal{G}$ -torsor over  $W_{\mathcal{O}_K}(R)$ , then for each  $R \in \text{Perf}/k$ , we have

$$\text{Gr}_{\mathcal{G}}(R) = \left\{ (\mathcal{E}, \beta) \middle| \mathcal{E} \in \text{Tors}_{\mathcal{G}}(W_{\mathcal{O}_K}(R)), \beta \in \text{Isom} \left( \mathcal{E}|_{\text{Spec}(W_{\mathcal{O}_K}(R)[\frac{1}{p}])}, \mathcal{E}_0|_{\text{Spec}(W_{\mathcal{O}_K}(R)[\frac{1}{p}])} \right) \right\} / \text{Isom}$$

In particular, if  $\mathcal{G} = \text{GL}(n)_{\mathcal{O}_K}$ , then  $\text{Gr}_{\text{GL}(n)}(R)$  classifies finite projective  $W_{\mathcal{O}_K}(R)$ -submodules  $M \subset W_{\mathcal{O}_K}(R)^{\oplus n}[\frac{1}{p}]$  s.t.  $M[\frac{1}{p}] = W_{\mathcal{O}_K}(R)^{\oplus n}[\frac{1}{p}]$ .

⌋

*Proof:* To prove the moduli interpretation, we denote

$$\text{Gr}'_{\mathcal{G}}(R) \triangleq \left\{ (\mathcal{E}, \beta) \middle| \mathcal{E} \in \text{Tors}_{\mathcal{G}}(W_{\mathcal{O}_K}(R)), \beta \in \text{Isom} \left( \mathcal{E}|_{\text{Spec}(W_{\mathcal{O}_K}(R)[\frac{1}{p}])}, \mathcal{E}_0|_{\text{Spec}(W_{\mathcal{O}_K}(R)[\frac{1}{p}])} \right) \right\} / \text{Isom},$$

and also

$$L'G(R) \triangleq \left\{ (\mathcal{E}, \beta, \varepsilon) \middle| (\mathcal{E}, \beta) \in \text{Gr}'_{\mathcal{G}}(R), \varepsilon \in \text{Isom}(\mathcal{E}_0, \mathcal{E}) \right\} / \text{Isom},$$

Then the assertion follows if we prove that

- $L'G$  is an  $L^+\mathcal{G}$ -torsor over  $\mathrm{Gr}'_{\mathcal{G}}$ .
- There exists an  $L^+\mathcal{G}$ -equivariant isomorphism  $LG \cong L'G$ .

For the first assertion, notice that for each  $\mathcal{G}$ -torsor  $\mathcal{E} \in \mathrm{Tors}_{\mathcal{G}}(\mathrm{Spec} W_{\mathcal{O}_K}(R))$ , there exists an étale cover of  $R$  s.t.  $\mathcal{E}$  is trivial, then  $\mathcal{E}$  is trivial over  $\mathrm{Spec} W_{\mathcal{O}_K}(R')$  by formal smoothness. Thus the first assertion holds.

For the second assertion, the isomorphism is given by  $A \mapsto (\mathcal{E}_0, A, \mathrm{id})$  and the inverse is given by  $(\mathcal{E}, \beta, \varepsilon) \mapsto \beta \circ \varepsilon|_{\mathrm{Spec}(W(\mathcal{O}_K)[\frac{1}{p}])}$ .  $\square$

**Lemma (28.7.2.10).** Any smooth affine group scheme  $\mathcal{G}$  over a Dedekind domain  $A$  admits a linear representation  $\rho : \mathcal{G} \rightarrow \mathrm{GL}(n)_A$  s.t.  $\mathrm{GL}(n)_A/\mathcal{G}$  is quasi-affine. And if  $\mathcal{G} \in \mathrm{RedGrp}/A$ , then we can choose  $\rho$  s.t.  $\mathrm{GL}(n)_A/\mathcal{G}$  is affine.  $\lrcorner$

*Proof:* ? Cf. [P-R08]Thm1.b and [Alper, Adequate moduli spaces and geometrically reductive group schemes]Cor9.7.7.  $\square$

**Lemma (28.7.2.11).** If  $\rho : \mathcal{G} \hookrightarrow \mathrm{GL}(n) \in \mathrm{GrpSch}/\mathcal{O}_K$  is an embedding s.t.  $\mathrm{GL}(n)/\mathcal{G}$  is (quasi-)affine, then  $\mathrm{Gr}_{\mathcal{G}} \rightarrow \mathrm{Gr}_{\mathrm{GL}(n)}$  is a (locally)closed embedding.  $\lrcorner$

*Proof:* Cf. [P-R08]Thm1.4?.  $\square$

**Prop. (28.7.2.12) [Representability of  $\mathrm{Gr}_{\mathcal{G}}$ , Zhu[Zhu17]/Bhatt-Scholze[B-S17]].** Situation as in (28.7.2.9),  $\mathrm{Gr}_{\mathcal{G}}$  (28.7.2.9) is an increasing union of perfections of quasi-projective schemes over  $k$ , along closed immersions.

And if  $\mathcal{G}$  is a parahoric group scheme, then  $\mathrm{Gr}_{\mathcal{G}}$  can be written as an increasing union of perfections of projective schemes over  $k$  along closed immersions, and  $\mathrm{Gr}_{\mathcal{G}}$  is called a **affine partial flag variety** in this case.  $\lrcorner$

*Proof:* By Weil-restriction, it suffices to prove for  $K$  absolutely unramified. Then we can embed  $\mathcal{G}$  into  $\mathrm{GL}(n)_{\mathcal{O}_K}$  s.t.  $\mathrm{GL}(n)_{\mathcal{O}_K}/\mathcal{G}$  is quasi-affine by (28.7.2.10), and the  $\mathrm{Gr}_{\mathcal{G}}$  is a locally closed subscheme of  $\mathrm{Gr}_{\mathrm{GL}(n)_{\mathcal{O}_K}}$ , and the assertion follows from (28.7.2.5).

If  $\mathcal{G}$  is parahoric, then [Zhu17]1.4.2. shows that  $\mathrm{Gr}_{\mathcal{G}}$  is ind-proper?. Thus it is ind-projective.?  $\square$

**Remark (28.7.2.13)[Differences with affine Grassmannians in equal characteristic].** ? There is no analogue of the Birkhoff decomposition for  $p$ -adic groups, it is not clear whether one can construct the “big open cell” in the mixed characteristic affine Grassmannian. This is related to the lack of a Beauville-Laszlo style description of the mixed characteristic affine Grassmannian via a “global curve”.

There is no “rotation” torus acting on the mixed characteristic affine Grassmannian.  $\lrcorner$

### Schubert-Varieties

**Notation (28.7.2.14).**

- Let  $K \in p\text{-LField}$ .
- Let  $\mathcal{G} \in \mathrm{RedGrp}/\mathcal{O}_K$  with  $G = \mathcal{G}_K$ .

$\lrcorner$

**Def. (28.7.2.15) [Schubert-Varieties].** For  $\mu \in X_\bullet(T)^+$ , define the (spherical) **Schubert-variety**

$$\mathrm{Gr}_{\preccurlyeq \mu} \triangleq \{(\mathcal{E}, \beta) \in \mathrm{Gr} \mid \mathrm{inv}(\beta) \preccurlyeq \mu\},$$

which is a closed subset of  $|\mathrm{Gr}_{\mathcal{G}}|$  and is given the induced reduced structure as a closed subscheme.

And we also define the **Schubert-cell**

$$\mathrm{Gr}_{\mu} \triangleq \{(\mathcal{E}, \beta) \in \mathrm{Gr} \mid \mathrm{inv}(\beta) = \mu\},$$

which is an open subscheme of  $\mathrm{Gr}_{\preccurlyeq \mu}$ . ┘

**Prop. (28.7.2.16) [Convolution Maps].** Let  $\mathrm{mul}_{\mu_\bullet} : \mathrm{Gr}_{\mu_\bullet} \rightarrow \mathrm{Gr}_{\mathcal{G}}$  be the convolution map. Then  $\mathrm{mul}_{\mu_\bullet}$  is perfectly proper and semi-small, i.e.

$$\dim \mathrm{Gr}_{\mu_\bullet} \times_{\mathrm{Gr}_{\mathcal{G}}} \mathrm{Gr}_{\mu_\bullet} = \dim \mathrm{Gr}_{\mu_\bullet}.$$

┘

*Proof:* Cf. [X-Z17]P25? □

**Def. (28.7.2.17) [Satake-Correspondences].** For two sequences of dominant weights  $\lambda_\bullet, \mu_\bullet$ , define

$$\mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0 \triangleq \mathrm{Gr}_{\lambda_\bullet} \times_{\mathrm{mul}_{\lambda_\bullet}, \mathrm{Gr}_{\mathcal{G}}, \mathrm{mul}_{\mu_\bullet}} \mathrm{Gr}_{\mu_\bullet}.$$

And the correspondence

$$\mathrm{Gr}_{\lambda_\bullet} \xleftarrow{h_{\lambda_\bullet}^-} \mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0 \xrightarrow{h_{\mu_\bullet}^-} \mathrm{Gr}_{\mu_\bullet}$$

is called the **Satake-correspondence** for Affine Grassmannian.

We also define

$$\mathrm{Gr}_{\nu_\bullet; \lambda_\bullet|\mu_\bullet; \xi_\bullet}^0 \triangleq \mathrm{Gr}_{\nu_\bullet} \widetilde{\times} \mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0 \widetilde{\times} \mathrm{Gr}_{\xi_\bullet}.$$

┘

**Prop. (28.7.2.18) [Satake-Correspondences].** Situation as in (28.7.2.17), we have

•

$$\dim \mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0 \leq \frac{1}{2} (\dim \mathrm{Gr}_{\preccurlyeq \lambda_\bullet} + \dim \mathrm{Gr}_{\preccurlyeq \mu_\bullet}) = \langle \rho, |\lambda_\bullet| + |\mu_\bullet| \rangle.$$

• There is a canonical isomorphism

$$\mathrm{Hom}_{\mathrm{Perv}(\mathrm{Gr}_{\mathcal{G}})}((\mathrm{mul}_{\lambda_\bullet})_* \mathrm{IC}_{\lambda_\bullet}, (\mathrm{mul}_{\mu_\bullet})_* \mathrm{IC}_{\mu_\bullet}) = \mathrm{Corr}_{\mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0}((\mathrm{Gr}_{\lambda_\bullet}, \mathrm{IC}_{\lambda_\bullet}), (\mathrm{Gr}_{\mu_\bullet}, \mathrm{IC}_{\mu_\bullet})).$$

• There is a canonical isomorphism

$$\mathrm{Hom}_{\mathrm{Perv}(\mathrm{Gr}_{\mathcal{G}})}((\mathrm{mul}_{\lambda_\bullet})_* \mathrm{IC}_{\lambda_\bullet}, (\mathrm{mul}_{\mu_\bullet})_* \mathrm{IC}_{\mu_\bullet}) = H_{\langle 2\rho, |\lambda_\bullet| + |\mu_\bullet| \rangle}^{\mathrm{B-M}}(\mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0).$$

┘

*Proof:* Cf. [X-Z17]P26? □

**Def. (28.7.2.19) [Satake-Cycles].** Situation as in (28.7.2.17), an irreducible component of  $\mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0$  of dimension  $\langle \rho, |\lambda_\bullet| + |\mu_\bullet| \rangle$  is called a **Satake-cycle**. The set of Satake-cycles of  $\mathrm{Gr}_{\lambda_\bullet|\mu_\bullet}^0$  is denoted by  $\mathcal{S}_{\lambda_\bullet|\mu_\bullet}$ . ┘

### Mirkovic-Vilonen Cycles

**Notation (28.7.2.20).**

- Let  $\mathcal{G} \in \text{RedGrp}/\mathcal{O}_K$  and  $G = \mathcal{G}_K$ . Let  $(\mathcal{B}, \mathcal{T}) \leq G$  be a Borel-pair, and  $\mathcal{U} = R_u(\mathcal{B})$ . Let  $U = \mathcal{U}_K, B = \mathcal{B}_K, T = \mathcal{T}_K$ .

┘

**Def. (28.7.2.21) [Mirkovic-Vilonen Cycles].** For  $\lambda \in X_\bullet(T)$ , define the locally closed subscheme

$$S_\lambda = \varpi^\lambda \cdot LU \cdot L^+ \mathcal{G} / L^+ \mathcal{G} \subset \text{Gr}_{\mathcal{G}},$$

which is the attractor for the  $\mathbb{G}_m$ -action  $\lambda : \mathbb{G}_m \rightarrow L^+ \mathcal{T}$ . Then  $\text{pr}_\lambda : LU \rightarrow S_\lambda : u \mapsto \varpi^\lambda u$  realizes  $LU$  as a  $L^+ \mathcal{U}$ -torsor over  $S_\lambda$ .

Then for any  $\mu \in X_\bullet(T)^+$ ,  $S_\lambda \cap \text{Gr}_{\leq \mu}$  is equidimensional of dimension  $\langle \rho, \lambda + \mu \rangle$ , and the number of irreducible components equals  $\dim V_\mu[\lambda]$ . The set of the irreducible components of  $S_\lambda \cap \text{Gr}_{\leq \mu}$  are denoted by  $\mathbf{M}\text{-}\mathbf{V}_\mu[\lambda]$ , called the set of **Mirkovic-Vilonen cycles**.

┘

*Proof:* ? Cf. [Zhu17] Section 2.2.1, Cor 2.8., [CHKR] Lemma 2.17.4, Proposition 5.4.2, [Mirkovic-Vilonen07] Theorem 3.2(a). □

**Prop. (28.7.2.22).** Situation as in (28.7.2.21),

- Each irreducible component of  $S_\lambda \cap \text{Gr}_\mu$  is geometrically irreducible.
- $\varpi^\lambda$  is contained in each irreducible component of  $S_\lambda \cap \text{Gr}_\mu$ .
- Each irreducible component of  $S_\lambda \cap \text{Gr}_\mu$  is stable under the action of  $L^+ \mathcal{B}$ .
- The set  $\mathbf{M}\text{-}\mathbf{V}_\mu[\lambda]$  is independent of the choice of  $B$  up to canonical isomorphism.

┘

*Proof:* Cf. [X-Z17] P29 ? □

**Prop. (28.7.2.23) [Convoluting Product of Mirkovic-Vilonen Cycles].** Let  $\lambda_\bullet$  be a finite sequence of coweights of  $G$ , and define

$$S_{\lambda_\bullet} \triangleq S_{\lambda_1} \tilde{\times} S_{\lambda_2} \tilde{\times} \dots \tilde{\times} S_{\lambda_n} \subset \text{Gr}_{\mathcal{G}} \tilde{\times} \dots \tilde{\times} \text{Gr}_{\mathcal{G}},$$

then there is an isomorphism

$$S_{\lambda_\bullet} \xrightarrow{\widetilde{\text{mul}}}: (\varpi^{\lambda_1} x_1, \dots, \varpi^{\lambda_n} x_n) \mapsto \left( \varpi^{\lambda_1} x_1, \varpi^{\lambda_1 + \lambda_2} (\varpi^{-\lambda_2} x_1 \varpi^{\lambda_2}) x_2, \dots, \dots \right),$$

where  $x_i \in LU$ .

The set of irreducible components of  $S_{\lambda_\bullet} \cap \text{Gr}_{\mu_\bullet} \cong (S_{\lambda_1} \cap \text{Gr}_{\mu_1}) \tilde{\times} \dots \tilde{\times} (S_{\lambda_n} \cap \text{Gr}_{\mu_n})$  is given by  $\prod_i \mathbf{M}\text{-}\mathbf{V}_{\mu_i}[\lambda_i]$ . ┘

*Proof:* Cf. [X-Z17] P30 ? □

**Prop. (28.7.2.24) [Mirkovic-Vilonen Cycles and Satake-Cycles].** For  $\nu, \mu, \nu + \lambda \in X_\bullet(T)^+$ , there is a unique injective map

$$i_\nu^{\mathbf{M}\text{-}\mathbf{V}} : \mathbb{S}_{(\nu, \mu)|\lambda + \nu} \rightarrow \mathbf{M}\text{-}\mathbf{V}_\mu[\lambda] \text{ (28.7.2.19) (28.7.2.21)}$$

s.t.

$$\mathbf{a} \cap S_{\mu, \lambda} = (S_\lambda \cap G_\lambda) \tilde{\times} i_\nu^{\mathbf{M}\text{-}\mathbf{V}}(\mathbf{a}) \subset \text{Gr}_{\mathcal{G}} \tilde{\times} \text{Gr}_{\mathcal{G}}.$$

And this map is compatible with the canonical bijection  $\mathbf{M}\text{-}\mathbf{V}_\mu^{B_1}[\lambda] \cong \mathbf{M}\text{-}\mathbf{V}_\mu^{B_2}[\lambda]$  (28.7.2.22) for two different Borel-subgroups  $B_1, B_2$ . ┘

*Proof:* Cf.[?X-Z17]P30?.

□

**Prop. (28.7.2.25)[Xiao-Zhu[?X-Z17]].** For each simple root  $\alpha$ , denote  $M_\alpha$  the standard Levi-subgroup with roots  $\{\pm\alpha\}$ , then the map

$$\prod \theta_{M_\alpha} : S_\lambda \rightarrow \prod_i S_{M_\alpha, \lambda}$$

induces a dominant map

$$(S_\lambda \cap \mathrm{Gr}_\mu)^b \rightarrow \prod_\alpha S_{M_\alpha, \lambda} \cap \mathrm{Gr}_{M_\alpha, \lambda + \varepsilon_\alpha(\mathbf{b})\alpha^\vee},$$

with geometrically connected fibers.

┘

*Proof:* Cf.[?X-Z17]P39?.

□

### 3 Affine Deligne-Lusztig Varieties

**Notation (28.7.3.1).**

- Fix  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa) \in p\text{-LField}$ .
- Let  $(G, B, T) \in \mathcal{R}\mathrm{edGrp}/K$  be unramified.

┘

**Prop. (28.7.3.2)[Affine Deligne-Lusztig Varieties].** There is a natural map

$$\mathrm{inv} : \mathrm{Gr}_G \rightarrow X_\bullet(T)^+ : \mathrm{LG}/\mathrm{L}^+\mathcal{G} \rightarrow \mathrm{L}^+\mathcal{G} \backslash \mathrm{LG}/\mathrm{L}^+\mathcal{G} \cong X_\bullet(T)^+ \text{ (28.7.2.9)}.$$

For  $b \in G(\check{K})$  and  $\mu \in X_\bullet(T)^+$ , define the Schubert variety

$$\mathrm{Gr}_{\preccurlyeq \mu} = \mathrm{inv}^{-1}(\preccurlyeq \mu),$$

and define the ADLV

$$X_\mu(b) = \left\{ g \in \mathrm{Gr}_G \mid \mathrm{inv} \left( g^{-1} b \varphi_K(g) \right) \preccurlyeq \mu \right\},$$

which parametrizes the moduli problem that assigns to each  $R \in \mathcal{P}\mathrm{erf}/\kappa$  the isomorphism classes of commutative diagrams of modifications of  $G$ -torsors on  $\mathrm{Spec} W_{\mathcal{O}_K}(R)$ :

$$\begin{array}{ccc} \varphi_K \mathcal{E}_1 & \xrightarrow{\beta_1} & \mathcal{E}_1 \\ \downarrow \varphi_K \beta_0 & & \downarrow \beta_0 \\ \varphi_K \mathcal{E}_0 & \xrightarrow{b \cdot} & \mathcal{E}_0, \end{array}$$

s.t.  $\mathrm{inv}(\beta_1) \preccurlyeq \mu$ .

Then there is a Cartesian diagram of perfect schemes:

$$\begin{array}{ccc} X_\mu(b) & \xrightarrow{g \mapsto (g, g^{-1} b \varphi_K(g))} & \mathrm{Gr} \tilde{\times} \mathrm{Gr}_{\preccurlyeq \mu} \\ \downarrow & & \downarrow \mathrm{pr}_1 \times m \\ \mathrm{Gr} & \xrightarrow{\mathrm{id} \times b \sigma_p} & \mathrm{Gr} \times \mathrm{Gr}. \end{array}$$

Then  $J_b$  acts on  $X_\mu(b)$  by left multiplication.  $\lrcorner$

**Prop. (28.7.3.3) [Non-Emptiness of  $X_\mu(b)$ ].** Situation as in (28.7.3.2),  $X_\mu(b) \neq \emptyset$  iff  $[b] \in B(G, \mu)$ .  $\lrcorner$

*Proof:* Cf. [X-Z17]P47.  $\square$

**Thm. (28.7.3.4) [Dimension of  $X_\mu(b)$ , conjectured by Rapoport [Rap05] and Görtz-Haines-Kottwitz-Reuman [GHKR], Viehmann [Vie06, Vie08, Vie08a]/Hamacher [Ham15, Ham15a]/Zhu [Zhu17]].** Situation as in (28.7.3.2), we have

$$\dim X_\mu(b) = \langle \rho, \mu - \nu_b \rangle - \frac{1}{2} \text{def}_G(b).$$

? Cf. [GHKR] for definition of  $\text{def}_G(b)$ .  $\lrcorner$

*Proof:* Cf. [Zhu17]P456.  $\square$

**Def. (28.7.3.5) [ADLV and Schubert-Varieties].** For  $b \in G(\check{K})$  and  $\mu, \nu \in X_\bullet(T)^+$ , define

$$X_{\mu, \nu}(b) = X_\mu(b) \cap \text{Gr}_{\preceq \nu} \in \text{Gr}_G,$$

which is the fiber product

$$\begin{array}{ccc} X_{\mu, \nu}(b) & \xrightarrow{g \mapsto (g, g^{-1}b\varphi_K(g))} & \text{Gr}_{\preceq \nu} \tilde{\times} \text{Gr}_{\preceq \mu} \\ \downarrow & & \downarrow \text{pr}_1 \times m \\ \text{Gr}_{\preceq \nu} & \xrightarrow{\text{id} \times b\sigma_p} & \text{Gr}_{\preceq \nu} \times \text{Gr}. \end{array}$$

$X_{\mu, \nu}(b)$  is equipped with an action of  $J_b(K) \cap G(\mathcal{O}_{\check{K}})$ , and the isomorphism class of the pair  $(X_{\mu, \nu}(b), J_b(K) \cap G(\mathcal{O}_{\check{K}}))$  depends only on  $b$  up to  $\varphi_K$ -conjugacy by elements of  $G(\mathcal{O}_{\check{K}})$ .  $\lrcorner$

## 4 Irreducible Components of ADLV

### Unramified ADLV

**Notation (28.7.4.1).**

- Fix  $(K, \mathcal{O}_K, \mathfrak{m}_K, \kappa) \in p\text{-LField}$ .
- Let  $(G, B, T) \in \mathcal{R}\text{edGrp}/K$  be unramified.

$\lrcorner$

**Def. (28.7.4.2) [Unramified Elements].** An element  $b \in B(G)$  is called an **unramified element** if it is contained in the image of the natural map  $B(T) \rightarrow B(G)$ , where  $T$  is a maximal torus of  $G_{\check{\mathbb{Q}}_p}$ .  $\lrcorner$

**Prop. (28.7.4.3).** The map  $B(G) \rightarrow X_\bullet(T)_{\check{\mathbb{Q}}}^{+, \text{Gal}_\kappa} \times \pi_1(G)_{\text{Gal}_\kappa}$  identifies  $B(G)_{\text{unr}}$  with  $X_\bullet(T)_{\text{Gal}_\kappa}^+$ .  $\lrcorner$

*Proof:* Cf. [X-Z17]P49.  $\square$



**Prop. (28.7.4.4).** The unramified element  $[\varpi^\tau]$  belongs to  $B(G, \mu)$  iff

$$\bigoplus_{\lambda \in \tau + (\sigma_p - 1)X_\bullet(T)} V_\mu[\lambda] \neq 0.$$

┘

*Proof:* ?

□

**Prop. (28.7.4.5) [Tate-Subspaces].** Define

$$\Lambda = \{\lambda \in X_\bullet(T) \mid \bar{\lambda} \in X_\bullet(Z(G))_{\mathbb{Q}}\},$$

where  $\bar{\lambda}$  is the average of Galois-conjugates of  $\lambda$ . And for each  $V \in \text{Rep}^{\text{fd}}(\widehat{G})$ , denote

$$V^{\text{Tate}} \triangleq \bigoplus_{\lambda \in \Lambda} V[\lambda].$$

┘

**Lemma (28.7.4.6).** For  $\mu \in X_\bullet(T)^+$ , there exists some  $\tau \in \Lambda$  s.t.

$$V_\mu^{\text{Tate}} = \bigoplus_{\lambda \in \tau + (\sigma_p - 1)X_\bullet(T)} V_\mu[\lambda] \text{ (28.7.4.5)}.$$

┘

*Proof:* Cf. [?X-Z17]P50. ?

□

**Prop. (28.7.4.7) [Unramified Basic Elements].** For  $\mu \in X_\bullet(T)^+$  and the basic element  $[b] \in B(G, \mu)$  (28.6.1.40), the following are equivalent:

- $[b]$  is unramified.
- $V_\mu^{\text{Tate}} \neq 0$ .
- $\text{def}_{\mathcal{G}}([b]) \neq 0$ .
- $\dim X_\mu([b]) = \langle \rho, \mu \rangle$ .
- $\mu_{\text{ad}}$  is trivial on  $Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_K}$ .

If these conditions hold, and in addition  $Z(G)$  is connected, then  $[b] = [\varpi^\tau]$  for some  $\tau \in X_\bullet(Z(G))$ .

┘

*Proof:* Cf. [?X-Z17]P50. ?

□

### Irreducible Components of Unramified ADLVs

**Notation (28.7.4.8).**

- Fix  $\mu \in X_\bullet(T)^+$ .
- Fix  $\tau \in X_\bullet(T)$  s.t.  $\tau^\natural \in X_\bullet(T)_{\text{Gal}_K}^+$  and  $[\varpi^\tau] \in B(G, \mu)$ .

┘

**Def. (28.7.4.9) [Mirkovic-Vilonen Cycles].**  $\mathbf{M}\text{-}\mathbf{V}_\mu[\lambda]$ .

And define

$$\mathbf{M}\text{-}\mathbf{V}_\mu^{\tau^\natural} \triangleq \bigcup_{\lambda \in X_\bullet(T), \lambda^\natural = \tau^\natural} \mathbf{M}\text{-}\mathbf{V}_\mu[\lambda].$$

┘

**Prop. (28.7.4.10).** ┘

**Cor. (28.7.4.11) [Unramified ADLVs via Semi-Infinite Orbits].** For  $\mathbf{b} \in \mathbf{M}\text{-}\mathbf{V}_\mu^{\tau \natural}$  (28.7.4.9), denote

$$X_\mu^{\mathbf{b}}(\varpi^\tau) \triangleq \overline{\bigcup_{\nu \in X_\bullet(T)} Y_\nu^{\mathbf{b}}} \subset X_\mu(\varpi^\tau) \text{ (28.7.3.2),}$$

then

- $X_\mu^{\mathbf{b}}(\varpi^\tau)$  is a union of certain irreducible components of  $X_\mu(\varpi^\tau)$ .
- $X_\mu^{\mathbf{b}}(\varpi^\tau) \neq X_\mu^{\mathbf{b}'}(\varpi^\tau)$  if  $\mathbf{b} \neq \mathbf{b}' \in \mathbf{M}\text{-}\mathbf{V}_\mu^{\tau \natural}$ , and

$$X_\mu(\varpi^\tau) = \bigcup_{\mathbf{b} \in \mathbf{M}\text{-}\mathbf{V}_\mu^{\tau \natural}} X_\mu^{\mathbf{b}}(\varpi^\tau).$$

- $B_\tau(K)$  ? acts transitively on the set of irreducible components of  $X_\mu^{\mathbf{b}}(\varpi^\tau)$ .
- $X_\mu^{\mathbf{b}}(\varpi^\tau)$  is equidimensional of dimension  $\langle \rho, \mu - \tau \rangle$ .

┘

*Proof:* Cf. [X-Z17]P55 ?. □

**Thm. (28.7.4.12) [Irreducible Components of Unramified ADLV].** Consider the action of  $J_\tau(K) \triangleq J_{\varpi^\tau}(K)$  on  $X_\mu(\tau)$  (28.7.4.11), then

- The stabilizer of each irreducible component is a hyperspecial subgroup.
- The subset  $X_\mu^{\mathbf{b}}(\varpi^\tau) \subset X_\mu(\varpi^\tau)$  is invariant under the action of  $J_\tau(K)$ , and  $J_\tau(K)$  acts transitively on the set of irreducible components of  $X_\mu^{\mathbf{b}}(\varpi^\tau)$ .
- If  $Z(G)$  is connected, then there is a  $J_\tau(K)$ -equivariant bijection between the set of irreducible components of  $X_\mu^{\mathbf{b}}(\varpi^\tau)$  and

$$\mathbf{M}\text{-}\mathbf{V}_\mu^{\tau \natural} \times \text{HS}_\tau : (\mathbf{b}, x) \mapsto X_\mu^{\mathbf{b}, x}(\varpi^\tau),$$

where  $\text{HS}_\tau$  is the set of hyperspecial subgroups of  $J_\tau(K)$ , and for each  $x \in \text{HS}_\tau$ ,  $X_\mu^{\mathbf{b}, x}(\varpi^\tau)$  is the irreducible component of  $X_\mu^{\mathbf{b}}(\varpi^\tau)$  whose stabilizer is the hyperspecial subgroup of  $J_\tau(K)$  corresponding to  $x$  ?. ┘

*Proof:* ? Cf. [X-Z17]P63. □

## 28.8 $p$ -adic Uniformizations

References are [Schottky Groups and Mumford Curves, 1980], [Mihran Papikian, Non-archimedean uniformization and monodromy pairing].

**Notation (28.8.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-Field}$ .
- Use notations on simplicial complexes defined in [Simplicial Complexes](#).
- Let  $\kappa \in \text{Field}^p$ .
- Let  $K_0 = W(\kappa)[\frac{1}{p}]$ .

┘

### 1 Mumford Curves

**Def. (28.8.1.1)[ $p$ -adic Schottky Groups].** A  $p$ -adic Schottky group is a discrete, f.g. free subgroup  $\Gamma \subset \text{PGL}(2; K)$ . ┘

**Prop. (28.8.1.2).** A subgroup  $\Gamma \leq \text{PGL}(2; K)$  is a  $p$ -adic Schottky group [\(28.8.1.1\)](#) if it is discrete, f.g. and torsion-free. ┘

*Proof:*  $\Gamma$  acts on the Bruhat-Tits building  $\Delta$  of  $\text{PGL}(2; \mathbb{Q}_p)$ , and the stabilizer is  $\Gamma \cap \text{PGL}(2; \mathbb{Z}_p)$ , which is a finite group as  $\Gamma$  is discrete and  $\text{PGL}(2; \mathbb{Z}_p)$  is compact. Thus it is trivial as  $\Gamma$  is torsion-free. Thus  $\Gamma$  acts freely on  $\Delta$ , so  $\Gamma$  must be a free group. ? ┘

**Def. (28.8.1.3)[Limit Points].** For a  $p$ -adic Schottky group  $\Gamma \leq \text{PGL}(2; K)$  [\(28.8.1.1\)](#),  $\Gamma$  acts on  $\mathbb{P}^1(\mathbb{C}_p)$ . Denote  $\Lambda(\Gamma)$  the set of limit points of this action, i.e. the points  $z$  of the form  $z = \lim g_n(z_0)$  for a sequence of distinct elements  $g_n \in \Gamma$ . Denote  $\Omega(\Gamma) = \mathbb{P}^1(K) \setminus \mathcal{L}(\Gamma)$ .

$\Gamma$  is called a **discontinuous  $p$ -adic Schottky group** if  $\Omega(\Gamma) \neq \emptyset$ . It is called a **continuous  $p$ -adic Schottky group** otherwise. ┘

**Thm. (28.8.1.4)[Mumford].** Let  $\Gamma$  is a  $p$ -adic Schottky group of rank  $g$ , then there is a smooth complete curve  $X_\Gamma \in \text{Cur}^{\text{SmProj}, g}/K$  with an analytic isomorphism  $\Omega_\Gamma/\Gamma \cong X_\Gamma(\mathbb{C}_p)$ . Such a curve  $X_\Gamma$  is called a  $p$ -adic **Mumford curve**. ┘

*Proof:*

□

**Thm. (28.8.1.5) [Mumford].** Let  $\Gamma$  is a  $p$ -adic Schottky group of rank  $g \geq 2$  [\(28.8.1.1\)](#), then the corresponding Mumford curve  $X_\Gamma$  has split degenerate stable reduction (several  $\mathbb{P}^1$  glued together with ordinary double points). Conversely, and smooth complete curve over  $K$  with split degenerate stable reduction is a Mumford curve.

Where split degenerate reduction means that the normalization of all components of  $\tilde{X}$  are  $\mathbb{F}_p$ -rational, and all nodes are  $\mathbb{F}_p$ -rational with two  $\mathbb{F}_p$ -rational branches. ┘

*Proof:*

□

**Remark (28.8.1.6).** There are clear duality between  $\Delta/\Gamma$  and the reduction graph of  $C$ , Cf. [M-P05]P419.  $\lrcorner$

**Prop. (28.8.1.7) [Herrlich].** Suppose  $p \in \text{Prime}$  and  $X/\mathbb{Q}_p$  is a Mumford curve with genus  $g \geq 2$ , then

$$\# \text{Aut}(X) \leq \begin{cases} 84(g-1) & , p = 2 \\ 24(g-1) & , p = 3 \\ 30(g-1) & , p = 5 \\ 12(g-1) & , p = 3 \end{cases}$$

 $\lrcorner$ 

*Proof:*

 $\square$ 

## 2 Drinfeld Symmetric Spaces

The Mumford-Kurihara-Mustafin uniformization is a procedure to construct nice analytic varieties by taking discrete Schottky-type quotients of a certain  $p$ -adic analogue of symmetric space called Drinfeld symmetric space  $\Omega$ .

### Mumford-Kurihara-Mustafin Uniformizations

## 3 Rapoport-Zink Uniformization

References are [R-Z96].

**Thm. (28.8.3.1) [PEL-Types].** Situation as in (28.5.1.3), let  $\kappa = \overline{\mathbb{F}_p}$ , and fix a point  $(A_0, \bar{\lambda}_0, \bar{\eta}^p) \in \mathcal{S}_{\mathcal{K}}(\kappa)$  with associated isocrystal  $N_0$  ?. Fix an isomorphism  $N_0 \cong V \otimes_{\mathbb{Q}_p} K_0$  that respects the actions of  $B \otimes K_0$  and the alternating bilinear forms on both sides, then the Frobenius action on  $N_0$  is identified to  $b(\text{id} \otimes \sigma)$  for some  $b \in G(K_0)$ .

Suppose  $b$  is basic. If  $\text{RZ}(\mathcal{D})$  is the PEL-type Rapoport-Zink space associated to the integral PEL-datum  $(\mathcal{O}_B, *, L, \langle -, - \rangle, h_0)$  (28.5.1.2), with an action of  $J(\mathbb{Q}_p)$ , then

- The set of points  $(A, \bar{\lambda}, \bar{\eta}^p) \in \mathcal{S}_{\mathcal{K}}(\kappa)$  s.t.  $(A, \bar{\lambda})$  is isogenous to  $(A_0, \bar{\lambda}_0)$  is a closed subset  $Z \subset (\mathcal{S}_{\mathcal{K}})_{\mathcal{O}_{E_v}}$ .
- If  $\mathcal{S}_{\mathcal{K}}|Z$  is the formal completion of  $(\mathcal{S}_{\mathcal{K}})_{\mathcal{O}_{E_v}}$  along  $Z$ , then there is an isomorphism of formal schemes over  $\text{Spf}(\mathcal{O}_{E_v})$ :

$$I(\mathbb{Q}) \backslash \left[ \text{RZ}(\mathcal{D}) \times G(\mathbf{A}_f^p) / \mathcal{K}^p \right] \cong \mathcal{S}_{\mathcal{K}}|Z.$$

where  $I(\mathbb{Q})$  is group of quasi-isogenies of  $(A_0, \bar{\lambda}_0)$  which is an inner form of  $G$  s.t.  $I_{\mathbb{Q}_p} \cong J_b$  and  $I_{\mathbf{A}_f^p} \cong G_{\mathbf{A}_f^p}$ .  $\lrcorner$

*Proof:* ?

 $\square$ 

**Thm. (28.8.3.2) [Unramified Hodge-Types].**  $\lrcorner$

**Thm. (28.8.3.3) [General Hodge-Types].** [GLEASON-LIM-XU] proved that the Newton strata of the integral models for Shimura varieties at parahoric level constructed by [Pappas-Rapoport] satisfy  $p$ -adic uniformization.  $\lrcorner$

## 28.9 Special Fibers of Shimura Varieties

References are [Rapoport, A guide to the reduction modulo  $p$  of Shimura varieties], [Kisin, Mod  $p$  points on Shimura varieties of abelian type].

**Notation(28.9.0.1).**

- Let  $p \in \text{Prime}$ .

┘

### 1 Local Shtuka

**Notation(28.9.1.1).**

- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-LField}$ .
- Let  $(G, B, T) \in \mathcal{R}\text{edGrp}/K$  be unramified.

┘

### Local Hecke-Stacks

**Def.(28.9.1.2) [Local Hecke-Stacks].** For a sequence of dominant weights  $\mu_\bullet = (\mu_1, \dots, \mu_t)$  of  $G$ , we define the **local Hecke-stack** as the prestack

$$\text{Hk}_{\mu_\bullet}^{\text{loc}} \triangleq [\text{L}^+ \mathcal{G} \backslash \text{Gr}_{\mu_\bullet}],$$

which assigns to each  $R \in \mathcal{P}\text{erf}/\kappa$  the groupoid of chains of modifications of  $\mathcal{G}$ -torsors over  $\text{Spec } W_{\mathcal{O}}(R)$ :

$$\mathcal{E}_t \xrightarrow{\beta_t} \mathcal{E}_{t-1} \xrightarrow{\beta_{t-1}} \dots \xrightarrow{\beta_2} \mathcal{E}_1 \xrightarrow{\beta_1} \mathcal{E}_0$$

of relative positions  $\preccurlyeq \mu_t, \dots, \preccurlyeq \mu_1$  resp.. And there are maps of prestacks

$$t_{\leftarrow} : \text{Hk}_{\mu_\bullet}^{\text{loc}} \rightarrow \mathbf{BL}^+ \mathcal{G} : \left( \mathcal{E}_t \xrightarrow{\beta_t} \mathcal{E}_{t-1} \xrightarrow{\beta_{t-1}} \dots \xrightarrow{\beta_1} \mathcal{E}_0 \right) \mapsto \mathcal{E}_t,$$

$$t_{\rightarrow} : \text{Hk}_{\mu_\bullet}^{\text{loc}} \rightarrow \mathbf{BL}^+ \mathcal{G} : \left( \mathcal{E}_t \xrightarrow{\beta_t} \mathcal{E}_{t-1} \xrightarrow{\beta_{t-1}} \dots \xrightarrow{\beta_1} \mathcal{E}_0 \right) \mapsto \mathcal{E}_0.$$

Similarly, we can define

$$\text{Hk}_{\lambda_\bullet | \mu_\bullet}^{\text{loc}} \triangleq [\text{L}^+ \mathcal{G} \backslash \text{Gr}_{\lambda_\bullet | \mu_\bullet}^0], \quad \text{Hk}_{\nu_\bullet; \lambda_\bullet | \mu_\bullet; \xi_\bullet}^{0, \text{loc}} \triangleq [\text{L}^+ \mathcal{G} \backslash \text{Gr}_{\nu_\bullet; \lambda_\bullet | \mu_\bullet; \xi_\bullet}^0] \quad (28.7.2.17).$$

By taking the  $\text{L}^+ \mathcal{G}$ -quotient of the Satake-correspondence (28.7.2.17), we get a **Satake-correspondence** on local Hecke-stacks

$$\text{Hk}_{\lambda_\bullet}^{\text{loc}} \xleftarrow{h_{\lambda_\bullet}^{\leftarrow}} \text{Hk}_{\lambda_\bullet | \mu_\bullet}^{0, \text{loc}} \xrightarrow{h_{\mu_\bullet}^{\rightarrow}} \text{Hk}_{\mu_\bullet}^{\text{loc}}.$$

┘

**Def.(28.9.1.3) [Hecke-Correspondences and Satake-Correspondences].** There is a **Hecke-correspondence of local Hecke-stacks**

$$\text{Sht}_{\lambda_\bullet}^{\text{loc}} \xleftarrow{\overleftarrow{h}_{\lambda_\bullet}^{\text{loc}}} \text{Sht}_{\lambda_\bullet | \mu_\bullet}^{\text{loc}} \xrightarrow{\overrightarrow{h}_{\mu_\bullet}^{\text{loc}}} \text{Sht}_{\mu_\bullet}^{\text{loc}},$$

where  $\text{Sht}_{\lambda_\bullet|\mu_\bullet}^{\text{loc}}$  is the prestack that assigns to each  $R \in \text{Perf}/\kappa$  the isomorphism classes of commutative diagrams of modifications of  $\mathcal{G}$ -torsors over  $\text{Spec } W_{\mathcal{O}_K}(R)$ :

$$\begin{array}{ccccccccccc} \mathcal{E}'_s & \xrightarrow{\beta'_s} & \mathcal{E}'_{s-1} & \xrightarrow{\beta'_{s-1}} & \dots & \xrightarrow{\beta'_2} & \mathcal{E}'_1 & \xrightarrow{\beta'_1} & \mathcal{E}'_0 & & \\ \downarrow \beta & & & & & & & & \downarrow \varphi_K \beta & & \\ \mathcal{E}_t & \xrightarrow{\beta_t} & \mathcal{E}_{t-1} & \xrightarrow{\beta_{t-1}} & \dots & \xrightarrow{\beta_2} & \mathcal{E}'_1 & \xrightarrow{\beta_1} & \mathcal{E}_0 & & \end{array}$$

And for each  $\nu \in X_\bullet(T)^+$ , we can define  $\text{Sht}_{\lambda_\bullet|\mu_\bullet}^{\nu, \text{loc}}$  as the closed sub-prestack of  $\text{Sht}_{\lambda_\bullet|\mu_\bullet}^{\text{loc}}$  consisting of diagrams of modifications s.t.  $\text{inv}(\beta) \preccurlyeq \nu$ . And for  $\nu = 0$ , we get the **Satake-correspondence of local Hecke-stacks**

$$\text{Sht}_{\lambda_\bullet}^{\text{loc}} \xleftarrow{\overleftarrow{h}_{\lambda_\bullet}^{\text{loc}}} \text{Sht}_{\lambda_\bullet|\mu_\bullet}^{0, \text{loc}} \xrightarrow{\overrightarrow{h}_{\mu_\bullet}^{\text{loc}}} \text{Sht}_{\mu_\bullet}^{\text{loc}},$$

┘

**Cor. (28.9.1.4).**  $\text{Sht}_{\lambda_\bullet|\mu_\bullet}^{\text{loc}}$  is non-empty iff  $B(G, -|\lambda_\bullet|) \cap B(G, -|\mu_\bullet|) \neq \emptyset$ , by (28.7.3.3). ┘

**Def. (28.9.1.5) [Convolutions Maps].** For each  $m \in \mathbb{N} \cup \{\infty\}$  which is  $(\nu_\bullet, \mu_\bullet, \xi_\bullet)$ -large, there is a convolution map

$$\text{mul}_{\nu_\bullet, \mu_\bullet, \xi_\bullet} : \text{Hk}_{\nu_\bullet, \mu_\bullet, \xi_\bullet}^{\text{loc}(m)} \rightarrow \text{Hk}_{\nu_\bullet, |\mu_\bullet|, \xi_\bullet}^{\text{loc}(m)}$$

which is perfectly proper? ┘

### Moduli of Local Shtukas

**Def. (28.9.1.6) [Moduli of Local Shtukas].** For each sequence  $\mu_\bullet$  of dominant coweights, let  $\text{Sht}_{\mu_\bullet}^{\text{loc}}$  be the prestack over  $\text{Perf}/\kappa$  that assigns to each  $R \in \text{Perf}/\kappa$  the isomorphism classes of modifications

$$\mathcal{E}_t \xrightarrow{\beta_t} \mathcal{E}_{t-1} \xrightarrow{\beta_{t-1}} \dots \xrightarrow{\beta_1} \mathcal{E}_1 \xrightarrow{\beta_0} \varphi_K \mathcal{E}_t.$$

In particular, there is a Cartesian diagram of prestacks

$$\begin{array}{ccc} \text{Sht}_{\mu_\bullet}^{\text{loc}} & \xrightarrow{\varphi^{\text{loc}}} & \text{Hk}_{\mu_\bullet}^{\text{loc}} \\ \downarrow & & \downarrow t \leftarrow \times t \rightarrow \\ \text{BL}^+ G & \xrightarrow{1 \times \varphi_K} & \text{BL}^+ G \times \text{BL}^+ G \end{array} \quad (28.9.1.2).$$

┘

**Example (28.9.1.7).** If  $\mathcal{G} = \text{GL}(n)$  and  $K$  is absolutely unramified, then a theorem of Gabber? shows that  $\text{Sht}_{\omega_i}^{\text{loc}}$  is isomorphic to the prestack of  $p$ -divisible groups of dimension  $i$  and height  $n$ .

$$\text{Sht}_0^{\text{loc}} \cong \text{B}(G(\mathcal{O}_K)).?$$

┘

### Moduli of Restricted Local Shtukas

**Def. (28.9.1.8) [Moduli of Restricted Local Shtukas].** For a sequence  $\mu_\bullet$  of dominant coweights and  $m, n \in \mathbb{N}$  s.t.  $m - n$  is  $\mu_\bullet$ -large<sup>?</sup>, the moduli prestack  $\mathrm{Sht}_{\mu_\bullet}^{\mathrm{loc}(m,n)}$  of  $(m, n)$ -restricted local iterated shtukas is defined to be the stack that classifies for each  $R \in \mathcal{P}\mathrm{erf}/k$  the data of

- An  $R$ -point of  $\mathrm{Hk}_{\mu_\bullet}^{\mathrm{loc}(m)}$ .
- An isomorphism

$$\psi : {}^\sigma(\mathcal{E}|_{D_n}) \cong (\mathcal{E}_{\rightarrow}|_{D_m})|_{D_n} \quad ?$$

of  $\mathrm{L}^n G$ -torsors over  $\mathrm{Spec} R$ . In particular, there is a Cartesian diagram of stacks

$$\begin{array}{ccc} \mathrm{Sht}_{\mu_\bullet}^{\mathrm{loc}(m,n)} & \xrightarrow{\varphi^{\mathrm{loc}(m,n)}} & \mathrm{Hk}_{\mu_\bullet}^{\mathrm{loc}(m)} \\ \downarrow & & \downarrow t_{\leftarrow} \times \mathrm{res}_n^m \circ t_{\rightarrow} \\ \mathrm{BL}^n G & \xrightarrow{1 \times \sigma} & \mathrm{BL}^n G \times \mathrm{BL}^n G \end{array}$$

┘

*Proof:* Cf. [X-Z17]P75<sup>?</sup>.

□

## 2 Cohomological Correspondences of Local Shtukas

**Def. (28.9.2.1)  $[[\widehat{G}\sigma_p/\widehat{G}]]$ .**

┘

**Thm. (28.9.2.2) [Geometric Satake-Correspondences, Xiao-Zhu].** There exists a functor  $S : \mathrm{Coh}_{fr}^{\widehat{G}}(\widehat{G}\sigma_p) \rightarrow \mathrm{Perv}^{\mathrm{Corr}}(\mathrm{Sht}_{\overline{\kappa}}^{\mathrm{loc}})$  making the following diagram commutative:

$$\begin{array}{ccc} \mathrm{Rep}^{\mathrm{fd}}(\widehat{G}) & \xrightarrow{\mathrm{Sat}} & \mathrm{Perv}(\mathrm{Hk}_{\overline{\kappa}}^{\mathrm{loc}}) \\ \downarrow & & \downarrow (\Phi^{\mathrm{loc}})^* \\ \mathrm{Coh}_{fr}^{\widehat{G}}(\widehat{G}\sigma_p) & \xrightarrow{S} & \mathrm{Perv}^{\mathrm{Corr}}(\mathrm{Sht}_{\overline{\kappa}}^{\mathrm{loc}}). \end{array}$$

where  $\mathrm{Coh}_{fr}^{\widehat{G}}(\widehat{G}\sigma_p)$  is the full subcategory of the category of coherent sheaves on  $[\widehat{G}\sigma_p/\widehat{G}]$  spanned by the pullback of coherent sheaves along the natural projection  $[\widehat{G}\sigma_p/\widehat{G}] \rightarrow \mathbf{B}\widehat{G}$ .

Moreover,  $S(\mathcal{O}_{[\widehat{G}\sigma_p/\widehat{G}]}) = \delta_1$ , and the map

$$S : \mathbf{J} \triangleq \Gamma([\widehat{G}\sigma_p/\widehat{G}], \mathcal{O}_K) \rightarrow \mathrm{End}_{\mathrm{Perv}^{\mathrm{Corr}}(\mathrm{Sht}_{\overline{\kappa}}^{\mathrm{loc}})}(\delta_1) \cong \mathcal{H}_{\mathcal{K}}(G(K)) \otimes \overline{\mathbb{Q}_\ell}.$$

coincides with the Satake-isomorphism<sup>?</sup>.

┘

*Proof:* Cf. [X-Z17]P125<sup>?</sup>.

□

**Thm. (28.9.2.3) [Xiao-Zhu].** There is a canonical map

$$\begin{aligned} \mathrm{Spc} : \mathrm{Hom}_{\mathrm{Coh}_{fr}^{\widehat{G}}(\widehat{G}\sigma_p)}(\tilde{V}_i, \tilde{V}_j) &\xrightarrow{S} \mathrm{Hom}_{\mathrm{Perv}^{\mathrm{Corr}}(\mathrm{Sht}_{\overline{\kappa}}^{\mathrm{loc}})}(S(\tilde{V}_i), S(\tilde{V}_j)) \quad (28.9.2.2) \\ &\rightarrow \mathrm{Hom}_{\mathcal{H}^p(G)}(\mathrm{H}_{\mathrm{ét},c}(\mathrm{Sh}_{\mu_i, \overline{\kappa(v)}}), \mathrm{H}_{\mathrm{ét},c}(\mathrm{Sh}_{\mu_j, \overline{\kappa(v)}})) \end{aligned}$$

compatible with compositions.

┘

*Proof:* Cf.[?X-Z17]P123?.

□

**Conj.(28.9.2.4)**[ $S = T$  **Theorem, Xiao-Zhu**]. The action of  $\mathbf{J}$  on  $H_{\text{ét},c}(\text{Sh}_{\mu,\kappa(v)})$  coincides with the usual Hecke-algebra action, via the Satake-isomorphism. Cf.[?X-Z17]P125.

┘

*Proof:*

□

**Thm& Conj.Cor.(28.9.2.5)**. If  $\text{Sh}_{\mathcal{K}}(G, \mu)$  is a Shimura-set, then the conjecture(28.9.2.4) holds.

┘

*Proof:* Cf.[?X-Z17]P126?.

□

**Cor.(28.9.2.6)**. If the conjecture(28.9.2.4) holds, then the map  $\text{Spc}(28.9.2.3)$  induces a map

$$\text{J-L}_{\mu_i, \mu_j} : H_{\text{ét},c}(\text{Sh}_{\mu_i, \kappa(v)}) \otimes_{\mathbf{J}} \text{Hom}(\tilde{V}_i, \tilde{V}_j) \rightarrow H_{\text{ét},c}(\text{Sh}_{\mu_j, \kappa(v)}).$$

┘

### 3 Langlands-Rapoport Conjecture

**Notation(28.9.3.1)**.

- Let  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  be a compact open subgroup.
- Let  $\mathcal{K}_p \leq G(\mathbb{Q}_p)$  be a parahoric subgroup.
- Let  $\mathcal{G}_{\mathcal{K}_p}/\mathbb{Z}_p$  be the Bruhat-Tits group scheme corresponding to  $\mathcal{K}_p$ .
- Denote  $\check{\mathcal{K}}_p \triangleq \mathcal{G}_{\mathcal{K}_p}(\mathcal{O}_{\check{\mathbb{Q}}_p})$ .

┘

**Def.(28.9.3.2)** [**Kottwitz-Rapoport Strata**]. Situation as in(28.5.4.2), for each  $w \in \mathcal{K}_p \backslash \text{Adm}(\mu^*)/\mathcal{K}_p$ , define

$$\text{K-R}_{\mathcal{K}_p, w} \triangleq (\lambda_{\mathcal{K}_p} \otimes \kappa(v))^{-1}(\mathcal{O}_w) \subset \mathcal{S}_{\mathcal{K}}(G, X),$$

which is a locally closed subscheme of  $\mathcal{S}_{\mathcal{K}}(G, X)$ .(Cf.[Haines, T.: Introduction to Shimura varieties with bad reduction of parahoric type]?)

┘

**Conj.(28.9.3.3)**[**Basic Non-Emptiness of Kottwitz-Rapoport Strata, He-Rapoport**[H-R17]].

Situation as in(28.9.3.2), if  $\tau = \tau_X \in \widetilde{W}$  is the unique element of length zero mapping to the image of  $X$  in  $\pi_1(G)_{I_{\mathbb{Q}_p}}$  (Cf.[P-R08]Appendix Lemma14?), and  $\mathcal{K}_p = \mathcal{I}_p \leq G(\mathbb{Q}_p)$  is an Iwahori-subgroup, then  $\text{K-R}_{\mathcal{I}_p, \tau}$ (28.9.3.2) intersects with every geometrically connected component of  $\mathcal{S}_{\mathcal{I}_p}(G, X)$ .

┘

*Proof:*

□

**Conj.Cor.(28.9.3.4)** [**Non-Emptiness of Kottwitz-Rapoport Strata, He-Rapoport**[H-R17]].

Situation as in(28.9.3.2), if conjecture(28.9.3.3) holds, then each Kottwitz-Rapoport Strata  $\text{K-R}_{\mathcal{K}_p, w}$  is non-empty and intersects with each geometrically connected component of  $\mathcal{S}_{\mathcal{K}}(G, X)$ .

┘

*Proof:* Cf.[H-R17]P325?.

□



**Conj. (28.9.3.5) [Newton-Strata, He-Rapoport].** Situation as in (28.5.4.2), there should be a **Newton-map**

$$\delta_{\mathcal{K}_p} : \overline{\mathcal{S}}_{G,X,\mathcal{K}} \rightarrow B(G, \mu^*)$$

s.t. for each  $[b] \in B(G)$ ,  $\mathcal{N}_{\mathcal{K}_p,[b]} \triangleq \delta_{\mathcal{K}_p}^{-1}([b])$  is a locally closed subscheme of  $\overline{\mathcal{S}}_{G,X,\mathcal{K}}$ , called the **Newton-stratum** of  $\overline{\mathcal{S}}_{G,X,\mathcal{K}}$ , and for each  $[b] \in B(G_{\mathbb{Q}_p}, X)$ , we have

$$\text{cl}(\mathcal{N}_{\mathcal{K}_p,[b]}) \subset \bigcup_{[b'] \preceq [b]} \mathcal{N}_{\mathcal{K}_p,[b']} \quad (28.6.1.35).$$

┘

*Proof:*

□

**Conj. (28.9.3.6) [Central Leaves, He-Rapoport].** Situation as in (28.5.4.2), let

- $\check{\mathcal{K}}_{p_\varphi} \leq \check{\mathcal{K}}_p \times \check{\mathcal{K}}_p$  be the graph of the Frobenius  $\varphi$ ,
- $G(\check{\mathbb{Q}}_p)/\check{\mathcal{K}}_{p_\varphi}$  the set of  $\check{\mathcal{K}}_p$ - $\sigma$ -conjugacy class of  $G(\check{\mathbb{Q}}_p)$ ,
- $\check{\mathcal{K}}_{p_1}$  be the pro-unipotent radical of  $\check{\mathcal{K}}_p$ .

Then there should be a natural **central leave map**

$$\Upsilon_{\mathcal{K}_p} : \overline{\mathcal{S}}_{G,X,\mathcal{K}_p} \rightarrow G(\check{\mathbb{Q}}_p)/\check{\mathcal{K}}_{p_\varphi}$$

making the following diagram commutative:

$$\begin{array}{ccccc}
 & & G(\check{\mathbb{Q}}_p)/(\check{\mathcal{K}}_p)_\varphi(\check{\mathcal{K}}_{p_1} \times \check{\mathcal{K}}_{p_1}) & \xrightarrow{\text{nat}_{\mathcal{K}_p}} & \check{\mathcal{K}}_p \backslash G(\check{\mathbb{Q}}_p)/\check{\mathcal{K}}_p \cong W_{\mathcal{K}_p} \backslash \widetilde{W}/W_{\mathcal{K}_p} \\
 & \nearrow v_{\mathcal{K}_p} & \uparrow & \nearrow \lambda_{\mathcal{K}_p} (28.5.4.2) & \nearrow \ell_{\mathcal{K}_p} \\
 \overline{\mathcal{S}}_{G,X,\mathcal{K}} & \xrightarrow{\Upsilon_{\mathcal{K}_p}} & G(\check{\mathbb{Q}}_p)/(\check{\mathcal{K}}_p)_\varphi & & \\
 & \searrow \delta_{\mathcal{K}_p} (28.9.3.5) & \downarrow d_{\mathcal{K}_p} & & \\
 & & B(G) & & 
 \end{array}$$

s.t.

- $\text{Im}(\Upsilon_{\mathcal{K}_p}) = \ell_{\mathcal{K}_p}^{-1}(\text{Im}(\lambda_{\mathcal{K}_p}))$ .

┘

*Proof:*

□

**Conj.Cor. (28.9.3.7).** Situation as in (28.9.3.2), if conjecture (28.9.3.3) and the joint stratification exist (28.9.3.6), then

$$\mathrm{Im}(\Upsilon_{\mathcal{K}_p}) = \bigcup_{w \in \mathcal{K}_p \backslash \mathrm{Adm}(\mu^*) / \mathcal{K}_p} \check{\mathcal{K}}_p w \check{\mathcal{K}}_p / \check{\mathcal{K}}_{p,\varphi}.$$

┘

*Proof:* This follows from (28.9.3.4) and (28.9.3.6). □

**Conj.Cor. (28.9.3.8) [Non-Emptiness of Newton-Strata].** Situation as in (28.9.3.5), for each  $[b] \in B(G_{\mathbb{Q}_p}, \mu^*)$ ,  $\mathcal{N}_{\mathcal{K}_p, [b]}$  intersects with each geometrically connected component of  $\overline{\mathcal{S}}_{G, X, \mathcal{K}}$ . ┘

*Proof:* It suffices to prove for  $\mathcal{K}_p = \mathcal{I}_p$  Iwahori, Cf. [H-R17]P27? □

**Conj.Cor. (28.9.3.9) [Closure Relation between Newton-Strata].** Situation as in (28.9.3.5), for  $[b], [b'] \in B(G_{\mathbb{Q}_p}, \mu^*)$ ,

$$\mathcal{N}_{\mathcal{K}_p, [b]} \cap \mathrm{cl}(\mathcal{N}_{\mathcal{K}_p, [b']}) \neq \emptyset \iff [b] \preccurlyeq [b'].$$

┘

*Proof:* Cf. [H-R17]P328? □

### EKOR-Strata

**Thm. (28.9.3.10) [Lusztig/He].** Notation as in (28.9.3.6), let  $\mathcal{I}_p$  be an Iwahori-subgroup contained in  $\mathcal{K}_p$ , then

- For each  $x \in {}^K \widetilde{W}$ , we have  $\check{\mathcal{K}}_{p,\varphi}(\check{\mathcal{K}}_{p1} x \check{\mathcal{K}}_{p1}) = \check{\mathcal{K}}_{p,\varphi}(\check{\mathcal{I}}_p x \check{\mathcal{I}}_p)$ .
- $G(\check{\mathbb{Q}}_p) = \coprod_{x \in {}^{\mathcal{K}_p} \widetilde{W}} \check{\mathcal{K}}_{p,\varphi}(\check{\mathcal{K}}_{p1} x \check{\mathcal{K}}_{p1}) = \coprod_{x \in {}^{\mathcal{K}_p} \widetilde{W}} \check{\mathcal{K}}_{p,\varphi}(\check{\mathcal{I}}_p x \check{\mathcal{I}}_p)$ .
- $\mathrm{nat}_{\mathcal{I}_p}$  (28.9.3.6) is bijective.

┘

*Proof:* Cf. [H-R17]P329? □

**Def. (28.9.3.11) [EKOR-Strata, He-Rapoport [H-R17]].** Situation as in (28.5.4.2), if the joint stratification exists (28.9.3.6), then for each  $x \in \mathcal{K}_p \mathrm{Adm}(\mu^*) \mathcal{K}_p \cap {}^{\mathcal{K}_p} \widetilde{W}$ , we define

$$\mathrm{EKOR}_{\mathcal{K}_p, x} \triangleq v_{\mathcal{K}_p}^{-1}(x) \subset \overline{\mathcal{S}}_{G, X, \mathcal{K}},$$

called the **Ekedahl-Kottwitz-Oort-Rapoport stratum** or **EKOR-stratum** attached to  $x$ . By definition (28.9.3.6), each EKOR-stratum is contained in some Kottwitz-Rapoport-stratum.

Then each EKOR-stratum is locally closed in  $\overline{\mathcal{S}}_{G, X, \mathcal{K}_p}$ , and for each  $x \in \mathcal{K}_p \mathrm{Adm}(\mu^*) \mathcal{K}_p \cap {}^{\mathcal{K}_p} \widetilde{W}$ , we have

$$\mathrm{cl}(\mathrm{EKOR}_{\mathcal{K}_p, x}) = \coprod_{x' \in {}^{\mathcal{K}_p} \widetilde{W}, x' \preccurlyeq_{\mathcal{K}_p, \varphi} x} \mathrm{EKOR}_{\mathcal{K}_p, x'}.$$

┘

*Proof:* Cf. [H-R17]P334? □

**Prop. (28.9.3.12) [EKOR-Strata and Newton-Strata, He-Rapoport [H-R17]].** Situation as in (28.9.3.11), each Newton stratum  $\mathcal{N}_{\mathcal{K}_p, [b]}$  for  $[b] \in B(G_{\mathbb{Q}_p}, \mu^*)$  contains some EKOR-stratum (28.9.3.11). ┘

*Proof:* [H-R17]P335? □

**Conj.Cor.(28.9.3.13) [Finiteness of Fibers, He-Rapoport[H-R17]].** Situation as in (28.9.3.11), if  $\mathcal{I}_p \leq \mathcal{K}_p$  is an Iwahori-subgroup, then for each  $x \in \mathcal{K}_p \text{Adm}(\mu^*) \mathcal{K}_p \cap {}^{\mathcal{K}_p} \widetilde{W}$ ,

$$\pi_{\mathcal{I}_p \mathcal{K}_p, \mathcal{K}} : \text{K-R}_{\mathcal{I}_p, x} \rightarrow \text{EKOR}_{\mathcal{K}_p, x}$$

is a finite morphism. In particular,  $\dim \text{K-R}_{\mathcal{I}_p, x} = \dim \text{EKOR}_{\mathcal{K}_p, x}$ . ┘

*Proof:* Cf.[H-R17]P337. □

## 4 Abelian-Type Shimura Varieties

See [Kisin, Mod  $p$  points on Shimura varieties of abelian type]. Kisin proved a variant of the Langlands-Rapoport conjecture for all Abelian-type Shimura varieties at hyperspecial levels at  $p \in \text{Prime}_{\geq 3}$ .

### Unramified cases

**Def.(28.9.4.1) [ $I_x$ , Kisin].** For each  $x \in |\text{Sh}_{\mu, K}|_0$ , Kisin defined some group schemes:

- $I_{\ell/\kappa(x)} \leq \text{GL}(H^1(A_{\overline{x}}, \mathbb{Q}_{\ell})) \times \text{GL}(\mathbb{Q}_{\text{ét}}(1)_{\ell})$ ?
- The sequence  $\{I_{\ell/\kappa(x)_n}\}$  stabilizes to a group scheme  $I_{\overline{x}, \ell}$ .
- $I_{p/\kappa(x)} \leq \text{GL}(\mathbb{D}(A_x)) \times \text{GL}(\mathcal{O}_{\text{crys}}(1)(W(\kappa(x))))$ ?
- The sequence  $\{I_{p/\kappa(x)_n}\}$  stabilizes to a group scheme  $I_{\overline{x}, p}$ .
- $I_{/\kappa(x)} \leq \text{End}(A_x) * \times \mathbb{G}_m$ ?
- The sequence  $\{I_{/\kappa(x)_n}\}$  stabilizes to a group scheme  $I_{\overline{x}}$ .

Then for each  $v \in \text{Prime}$ , there is a natural isomorphism

$$I_{\overline{x}} \otimes \mathbb{Q}_v \cong I_{\overline{x}, v},$$

and  $I$  is an inner form of a Levi-subgroup  $I_0 \leq G$ , which can be described explicitly in [?X-Z17]P115? ┘

*Proof:* Cf.[Kisin, Mod  $p$  points on Shimura varieties of abelian type]Section2.1.2?, Section2.3 and [?X-Z17]P114. □

### Localization Maps to Shtukas

**Notation(28.9.4.2).**

- Let  $\text{Sh}_{\mu} \triangleq \left( \mathcal{S}_{\mathcal{K}_p \mathcal{K}^p}(G, \{\mu\}) \otimes \kappa(v) \right)^{\text{perf}}$  where  $\mathcal{K}_p$  is hyperspecial.
- ┘

**Prop.(28.9.4.3) [Localization Maps].** If we denote  $\mathcal{E} \triangleq (\varphi_K^{-1})^* \mathcal{E}_{\text{crys}}$ , then there is a local  $G$ -shtuka

$$\beta : \mathcal{E} \rightarrow {}^{\varphi_K} \mathcal{E},$$

over  $\text{Sh}_{\mu, K}$  s.t.  $\text{inv}(\beta) = \mu$ , which corresponds to a morphism of prestacks

$$\text{loc}_v : \text{Sh}_{\mu} \rightarrow \text{Sht}_{\mu}^{\text{loc}}.$$

And for any  $m, n \in \mathbb{N}$  s.t.  $m - n$  is  $\mu$ -large<sup>?</sup>, the composition

$$\mathrm{loc}_v(m, n) : \mathrm{Sh}_\mu \rightarrow \mathrm{Sht}_\mu^{\mathrm{loc}} \xrightarrow{\mathrm{res}_{m,n}} \mathrm{Sht}_\mu^{\mathrm{loc}(m,n)}$$

is perfectly smooth. ┘

*Proof:* Cf.[?X-Z17]P112<sup>?</sup>. □

**Def. (28.9.4.4) [Newton-Maps].** There is a **Newton-map**

$$\nu_\mu : \mathrm{Sht}_\mu^{\mathrm{loc}}(\bar{k}) \rightarrow B(G, \mu^*).$$

Cf.[?X-Z17]P69. ┘

**Prop. (28.9.4.5) [Newton-Strata in the Hyperspecial case].** Consider the map

$$\nu_\mu : \mathrm{Sh}_{\mu, \mathcal{K}}(\overline{\kappa(v)}) \xrightarrow{\mathrm{loc}_v} \mathrm{Sh}_\mu^{\mathrm{loc}}(\overline{\kappa(v)}) \xrightarrow{\nu_\mu} B(G_{\mathbb{Q}_p}, \mu^*) \text{ (28.9.4.4)(28.9.4.3),}$$

then it satisfies the requirement of Newton-map in (28.9.3.5). ┘

*Proof:* Cf.[Wortmann, The  $\mu$ -ordinary locus for Shimura varieties of Hodge type]<sup>?</sup>. □

**Prop. (28.9.4.6) [Non-Emptiness of Newton-Strata].** Each Newton-stratum  $\mathrm{Sh}_{\mathcal{K}_p, [b']}$  (28.9.4.5) is non-empty. ┘

*Proof:* <sup>?</sup> Cf.[Kisin-Pera-Shin]. [?X-Z17]P114 for a proof in the unramified basic locus case.<sup>?</sup> □

### Rapoport-Zink Uniformizations

**Prop. (28.9.4.7) [ $I_x$  for Basic Elements].** For each  $x \in |\mathrm{Sh}_{\mu, \mathcal{K}}|_0$ , we get an element  $b_x \in G(W(\kappa(x)))\mu^*(p)G(W(\kappa(x)))$  which is well-defined up to  $\sigma$ -conjugacy by an element of  $G(W(\kappa(x)))$ .

If  $[b_x] \in B(G_{\mathbb{Q}_p, \mu^*})$  is basic, then  $I_{\bar{x}}$  is an inner form of  $G$ . In particular,  $I_{\bar{x}, p} \cong J_{b_x}$  is an inner form of  $G_{\mathbb{Q}_p}$ . (Notice that in general  $I_{\bar{x}, p} \subsetneq J_{b_x}$ ).

In particular, if  $[b_x] \in B(G_{\mathbb{Q}_p}, \mu^*)$  is basic, then there exists a unique inner twist  $(I_x, \Psi_x) \in H^1(\mathbb{Q}, G_{\mathrm{ad}})$  s.t.

- it is trivial at  $\ell \notin \{p, \infty\}$ ,
  - it corresponds to  $[(b_x)_{\mathrm{ad}}] \in B(G_{\mathrm{ad}, \mathbb{Q}_p})_{\mathrm{bas}} \cong H^1(\mathbb{Q}_p, G_{\mathrm{ad}})$  at  $p$ ,
  - $(I_x)_{\mathbb{R}}$  is compact modulo center.
- ┘

*Proof:* Cf.[?X-Z17]P117 and [Kisin17]Cor2.3.5.<sup>?</sup> □

**Cor. (28.9.4.8).** If  $[b_x] \in B(G_{\mathbb{Q}_p}, \mu^*)$  is basic and unramified, then the inner twisting  $(I_x, \Psi_x)$  is trivial at all finite places and  $(I_x)_{\mathbb{R}}$  is compact modulo center.

Notice that this is a special case of [?X-Z17]2.1.6. and 4.2.10. ┘

*Proof:* Cf.[?X-Z17]P118. □

**Thm. (28.9.4.9) [Unramified Hodge-Type Rapoport-Zink Uniformizations].** The basic locus  $\mathrm{Sh}_{\mu, \mathcal{K}_p, \mathrm{bas}}$  contains a unique isogeny class, and the map

$$I_x(\mathbb{Q}) \backslash X_{\mu^*}(b_x) \times G(\mathbf{A}_f^p) / \mathcal{K}^p \rightarrow (\mathrm{Sh}_{\mu, \mathcal{K}})_{\bar{\kappa}(v)}$$

maps isomorphically onto  $\mathrm{Sh}_{\mu, \mathcal{K}, \mathrm{bas}}^{\mathrm{perf}}$ . ┘

*Proof:* Cf.[?X-Z17]P118<sup>?</sup>. □

References are [EKOR STRATA FOR SHIMURA VARIETIES WITH PARAHORIC LEVEL STRUCTURE, Shen-Yu-Zhang].

**Notation (28.9.5.1).**

- Let  $G = \mathrm{GL}(2)$  and  $(G, X)$  the usual Shimura-datum.
- Let  $\mathcal{K}_p \leq G(\mathbb{Q}_p)$ ,  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  and  $\mathcal{K} = \mathcal{K}_p \mathcal{K}^p \leq G(\mathbf{A}_f)$ .

└

References are [cohomology of certain PEL-type Shimura varieties, Mantovan].

## Inner Forms Computation

**Prop. (28.9.7.1)[Inner Twistings Trivial at Finite Places].** If  $G, G' \in \text{RedGrp}/\mathbb{Q}$  s.t.  $G_{\mathbf{A}_f} \cong G'_{\mathbf{A}_f}$ .

Then  $G$  and  $G'$  has the same pinned Langlands dual groups  $(\widehat{G}, \widehat{B}, \widehat{T}, \{\widehat{X}_\alpha\})$ , and the action of  $\text{Gal}_{\mathbb{Q}_\ell}$  on  $(\widehat{G}, \widehat{B}, \widehat{T}, \{\widehat{X}_\alpha\})$  induced by  $G$  or  $G'$  are the same, so the action of  $\text{Gal}_{\mathbb{Q}}$  are also the same.

So we can choose an inner twisting  $\Psi : G_{\overline{\mathbb{Q}}} \rightarrow G'_{\overline{\mathbb{Q}}}$  s.t. the cohomology class of  $(G', \Psi)$  in  $H^1(\mathbb{Q}, G_{\text{ad}})$  is trivial at all finite places. Thus  $\kappa_{G_{\text{ad}, \mathbb{R}}}((G'_{\mathbb{R}}, \Psi_{\mathbb{R}}))$  is trivial on  $Z(\hat{G}_{\text{sim, cntd}})^{\text{Gal}_{\mathbb{Q}}}$ .

Conversely, if  $(G'_\mathbb{R}, \Psi_\mathbb{R})$  is an inner twisting of  $G_\mathbb{R}$  s.t.  $\kappa_{G_\mathbb{R}}([G'_\mathbb{R}, \Psi_\mathbb{R}])$  is trivial on  $Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_\mathbb{Q}}$ , then there exists a unique inner twisting  $(G, \Psi) \in H^1(\mathbb{Q}, G)$  which restricts to  $(G'_\mathbb{R}, \Psi_\mathbb{R})$  at  $\infty$  and trivial at all finite places.  $\square$

*Proof:* The uniqueness follows from Hasse's principle for adjoint groups(10.5.2.13), and the existence follows from(20.5.5.4).  $\square$

**Prop. (28.9.7.2) [Compact Inner Forms for Shimura-Data].** If  $(G, h)$  is a Shimura-datum, then  $\text{ad}(h(i))$  is an involution on  $G_{\text{ad}}$ , and also defines a cohomology class in  $H^1(\mathbb{R}, G_{\text{ad}, \mathbb{R}})$  as in (10.2.5.8), corresponding to the inner form

$$G_c = \{g \in G \otimes \mathbb{C} | h(\mathbf{i})\iota(g) = gh(\mathbf{i})\}.$$

Then we have

$$\kappa_{G_{\text{ad},\mathbb{R}}}([G_C, \text{ad}(h(i))]) = \mu_{h,\text{ad}}|_{Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_{\mathbb{R}}}} \in X^\bullet\left(Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_{\mathbb{R}}}\right),$$

where the Hodge-cocharacter  $\mu_{h,\text{ad}} \in X_\bullet(G_{\text{ad}})$  (28.1.1.3) is regarded as a character of  $\hat{T}_{\text{sim.cntd.}}$ .  $\square$

*Proof:* Cf. [X-Z17]P16.

☐

**Cor.(28.9.7.3)[Real Canonical Inner Twisting via Compact Inner Forms].** Let  $(G, h), (G', h')$  be two real Shimura-data s.t. the compact modulo center forms  $G_c, G'_c$  (28.9.7.2) are isomorphic, then there is a canonical inner twisting  $\Psi : G_{\mathbb{C}} \rightarrow G'_{\mathbb{C}}$  s.t.

$$\kappa_{G_{\text{ad}}}([G', \Psi]) = \mu_{h, \text{ad}}|_{Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_{\mathbb{R}}}} - \mu_{h', \text{ad}}|_{Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_{\mathbb{R}}}} \in X^{\bullet}\left(Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}_{\mathbb{R}}}\right).$$

┘

*Proof:* Cf. [X-Z17]P17?.

□

**Cor. (28.9.7.4) [Locally Trivial Compact Inner Forms].** Let  $(G, h)$  be a Shimura-datum.

If  $(G'_R, h')$  is a pair where  $G'_R \in \text{RedGrp}/\mathbb{R}$  and  $h \in \text{Hom}(\$, G'_R)$  satisfying (W) and (R) as in (X-Z17)P16) whose compact inner form defined in (28.9.7.2) is isomorphic to the one for  $G_R$ , and

$$\mu_{h,\text{ad}}|_{Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}\mathbb{Q}}} = \mu_{h',\text{ad}}|_{Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}\mathbb{Q}}},$$

then up to isomorphism there is a unique inner form  $(G', \Psi') \in H^1(\mathbb{Q}, G_{\text{ad}})$  that is trivial at all finite places and restricts to the canonical inner twisting  $(G'_R, \Psi_R)$  (28.9.7.3) at  $\infty$ .

In particular, if  $\mu_{h,\text{ad}}|_{Z(\widehat{G}_{\text{sim.cntd}})^{\text{Gal}\mathbb{Q}}} = 0$ , then up to isomorphism there is a unique inner twisting  $(G', \Psi) \in H^1(\mathbb{Q}, G_{\text{ad}})$  trivial at all finite places s.t.  $G'_R$  is compact modulo center. ┘

*Proof:* This follows from (28.9.7.1)(28.9.7.2) and (28.9.7.3). □

### Cohomological Correspondences

**Notation (28.9.7.5).**

- Let  $(G_1, \mathcal{N}_1), (G_2, \mathcal{N}_2)$  be two Shimura-data s.t.  $(G_1)_{\mathbf{A}_f} \cong (G_2)_{\mathbf{A}_f}$ .
- For  $i \in \{1, 2\}$ , let  $E_i$  be the reflex field of  $(G_i, \mathcal{N}_i)$ . Let  $E = E_1 E_2$ .
- By (28.9.7.1) there exists an inner twisting  $\Psi : (G_1)_{\overline{\mathbb{Q}}} \rightarrow (G_2)_{\overline{\mathbb{Q}}}$  s.t. there exists  $h \in (G_2)_{\text{ad}}(\overline{\mathbf{A}_f})$  and  $\theta : (G_1)_{\mathbf{A}_f} \cong (G_2)_{\mathbf{A}_f}$  s.t.

$$\Psi = \text{int}(h) \circ \theta_{\overline{\mathbf{A}_f}}.$$

- Let  $p \in \text{Prime}$  and  $v \in S_E(p)$ .
- Let  $\mathcal{K}_1 = \mathcal{K}_{1,p} \mathcal{K}_1^p \leq G_1(\mathbf{A}_f)$  be a neat compact subgroup and denote  $\mathcal{K}_2 \triangleq \theta(\mathcal{K}_1)$ .
- Denote  $\mathbf{J} \triangleq \mathbf{J}_{\widehat{G}} = \overline{\mathbb{Q}_\ell}[\widehat{G}\sigma_p]^{\widehat{G}}$  (28.9.2.1).

┘

**Situation (28.9.7.6) [Hypothesis].**

- $\text{Sh}_{K_i}(G_i, \mathcal{N}_i)$  has a canonical integral model over  $\mathcal{O}_{E_i(v)}$ . Let  $\text{Sh}_{\mu_i, \mathcal{K}_i}$  be the perfection of the special fiber of the canonical integral model. Then there exists a morphism  $\text{loc}_v : \text{Sh}_{\mu_i, \mathcal{K}_i} \rightarrow \text{Sht}_{\mu_i}^{\text{loc}}$  s.t.

$$\text{loc}_v^{(m,n)} \triangleq \text{res}_{m,n} \circ \text{loc}_v : \text{Sh}_{\mu_i, \mathcal{K}_i} \rightarrow \text{Sht}_{\mu_i}^{\text{loc}(m,n)}$$

is perfectly smooth for any pair  $(m, n) \in \mathbb{Z}$  s.t.  $m - n$  is  $\mu_i$ -large?.

- Assume  $\mu_1|_{Z(\widehat{G})^{\text{Gal}\mathbb{Q}_p}} = \mu_2|_{Z(\widehat{G})^{\text{Gal}\mathbb{Q}_p}}$ , then there exists a perfect ind-scheme  $\text{Sh}_{\mu_1|\mu_2}$  fitting into the following commutative diagram

$$\begin{array}{ccccc} \text{Sh}_{\mu_1, \mathcal{K}_1} & \longleftarrow & \text{Sh}_{\mu_1|\mu_2} & \longrightarrow & \text{Sh}_{\mu_2, \mathcal{K}_2} \\ \downarrow \text{loc}_v & & \downarrow & & \downarrow \text{loc}_v \\ \text{Sh}_{\mu_1}^{\text{loc}} & \longleftarrow & \text{Sh}_{\mu_1|\mu_2}^{\text{loc}} & \longrightarrow & \text{Sh}_{\mu_2}^{\text{loc}} \end{array}$$

s.t. both squares are Cartesian.

⌋

**Thm. (28.9.7.7) [Xiao-Zhu].** Item 1 of Hypothesis(28.9.7.6) holds for Shimura-data of Hodge-type. And item2 of Hypothesis(28.9.7.6) is expected to hold when

$$\mathrm{Hom}_{\mathrm{coh}_{f_r}(\widehat{G}_{\sigma_p})}(\widetilde{V}_{\mu'}, \widetilde{V}_{\mu}) \neq 0.$$

And it is verified when  $(G_1, \mathcal{N}_1) = (G_2, \mathcal{N}_2)$  or  $\mathrm{Sh}_{\mathcal{K}}(G, n)$  is a Shimura set, or  $(G, \mathcal{N})$  is of PEL-type.

⌋

*Proof:* ?

□

## 8 Basic Loci

**Notation (28.9.8.1).**

- Let  $(G, \mathcal{N})$  be a Hodge-type Shimura-datum.  $Z(G)$  is connected,
- Let  $p \in \mathrm{Prime}_{\geq 2}$  s.t.  $V_{\mu^*}^{\mathrm{Tate}} \neq 0$ ?. In particular, the basic element  $b \in B(G_{\mathbb{Q}_p}, \mu^*)$  is unramified by(28.7.4.7). As  $Z(G)$  is connected, we can write  $b = [\varpi^{\tau^*}]$  for some  $\tau^* \in X_{\bullet}(Z(G))$ , by(28.7.4.7).
- Let  $E$  be the reflex field of  $(G, \mathcal{N})$ , and let  $v \in S_E(p)$ .
- Let  $\lambda_b \in X_{\bullet}(T)_{\mathrm{Gal}_{\kappa}}$  be the element associated to  $b$  via the identification(28.7.4.3).
- By(28.7.4.7) and(28.9.7.4), there exists an inner twisting  $\Psi : G_{\overline{\mathbb{Q}}} \rightarrow G'_{\overline{\mathbb{Q}}} \in H^1(\mathbb{Q}, G_{\mathrm{ad}})$  s.t.  $\Psi$  is trivial at all finite places and  $G'_{\mathbb{R}}$  is compact modulo center.
- Let  $\mathcal{K} = \mathcal{K}_p \mathcal{K}^p \leq G(\mathbf{A}_f)$  is a neat open compact subgroup s.t.  $\mathcal{K}_p$  is hyperspecial.
- Let  $\mathrm{Sh}_{\mu}$  be the special fiber of the Shimura variety  $\mathrm{Sh}_{\mathcal{K}}(G, \mathcal{N})$  over  $\mathcal{O}_{E_v}$ . Denote  $d = \langle 2\rho, \mu \rangle = \dim \mathrm{Sh}_{\mu}$ .

⌋

**Prop. (28.9.8.2) [Irreducible Components of Basic Loci, Xiao-Zhu].** Let  $x \in \left| \left( \mathrm{Sh}_{\mu, \mathcal{K}_p} \right)_{\mathrm{bas}} \right|_0$ , and  $b_x \in G(\check{K})$  be a representative for  $\mathrm{loc}_p(x) \in |\mathrm{Sh}_{\mu}^{\mathrm{loc}}|_0$ . Define

$$\mathbf{M}\text{-}\mathbf{V}_{\mu^*}^{\mathrm{Tate}} \triangleq \coprod_{\lambda \in X_{\bullet}(T)^+, \lambda^{\natural} = \lambda_b} \mathbf{M}\text{-}\mathbf{V}_{\mu^*}[\lambda] \quad (28.9.8.1),$$

then by(28.7.4.11)?, there is a canonical decomposition

$$X_{\mu^*}(b_x) = \bigcup_{\mathbf{b} \in \mathbf{M}\text{-}\mathbf{V}_{\mu^*}^{\mathrm{Tate}_p}} X_{\mu^*}^{\mathbf{b}}(b_x).$$

Thus by Rapoport-Zink uniformization(28.9.4.9), we can decompose  $\mathrm{Sh}_{\mu, \mathrm{bas}}$  as the union of

$$\mathrm{Sh}_{\mu, \mathrm{bas}}^{\mathbf{b}} \triangleq I_x(\mathbb{Q}) \backslash X_{\mu^*}^{\mathbf{b}}(b_x) \times G(\mathbf{A}_f^p) / \mathcal{K}^p$$

for  $\mathbf{b} \in \mathbf{M}\text{-}\mathbf{V}_{\mu^*}^{\mathrm{Tate}}$ . This decomposition is independent of the choice of  $x$  and  $b_x$ ?.

Then by(28.7.4.11),  $(\mathrm{Sh}_{\mu, \mathcal{K}, \mathrm{bas}})_{\overline{\kappa_v}}$  is equidimensional of dimension  $d/2$ (28.9.8.1), and there is a bijection between the set of geometric irreducible components of  $\mathrm{Sh}_{\mu, \mathcal{K}, \mathrm{bas}}$  and  $G'(\mathbb{Q}) \backslash G(\mathbf{A}_f) / \mathcal{K} \times \mathbf{M}\text{-}\mathbf{V}_{\mu^*}^{\mathrm{Tate}_p}$ ?.

⌋

**Prop.(28.9.8.3)** [Class Maps and Geometric Jacquet-Langlands Maps]. The cycle class map( ? Cf.[?X-Z17]A.2.18) induces

$$\mathrm{cl}(\mathbf{b}) \triangleq \mathrm{cl}(\mathrm{Sh}_{\mu, \mathrm{bas}}^{\mathbf{b}}) : H_d^{\mathrm{B-M}} \left( \left( \mathrm{Sh}_{\mu, \mathrm{bas}}^{\mathbf{b}} \right)_{\overline{\kappa(v)}} \right) \cong C(G'(\mathbb{Q}) \backslash G'(\mathbf{A}_f) / \mathcal{K}) \rightarrow H_{\mathrm{ét}, c}^d \left( \mathrm{Sh}_{\mu, \kappa(v)}, \overline{\mathbb{Q}_\ell}(d/2) \right),$$

and we denote

$$\mathrm{cl}(\mathrm{Sh}_{\mu, \mathrm{bas}}) : \bigoplus_{\mathbf{b}} H_d^{\mathrm{B-M}} \left( \left( \mathrm{Sh}_{\mu, \mathrm{bas}}^{\mathbf{b}} \right)_{\overline{\kappa(v)}} \right) \rightarrow H_{\mathrm{ét}, c}^d \left( \mathrm{Sh}_{\mu, \kappa(v)}, \overline{\mathbb{Q}_\ell}(d/2) \right).$$

Then there is a commutative diagram

$$\begin{array}{ccc} H_{\mathrm{ét}, c}^0 \left( \mathrm{Sh}_{\tau_{\mathbf{b}}, \kappa(v)} \right) & \xrightarrow{\cong} & H_d^{\mathrm{B-M}} \left( \mathrm{Sh}_{\mu, \mathrm{bas}, \kappa(v)}^{\mathbf{b}} \right) \\ & \searrow \mathrm{J-L}_{\tau_{\mathbf{b}}, \mu}(\mathbf{a}_{in}) & \swarrow \mathrm{cl}(\mathbf{b}) \\ & H_{\mathrm{ét}, c}^d \left( \mathrm{Sh}_{\mu, \kappa(v)}, \overline{\mathbb{Q}_\ell}(d/2) \right) & \end{array}$$

┘

*Proof:* Cf.[?X-Z17]P127 ?.

□

### Hilbert Modular Surfaces

**Notation(28.9.8.4).**

- Let  $F$  be a real quadratic field.
- $p \in \mathrm{Prime}_{\geq 3}$  is inert in  $F$ , with  $(p) = \mathfrak{p}_v$ .
- Let  $\sigma_p \in \mathrm{Gal}_{\mathbb{Q}_p}$  be a geometric Frobenius.
- Let  $G = \mathrm{Res}_{F/\mathbb{Q}} \mathrm{PGL}(2)$ .
- Let  $\mathcal{K}_p \leq G(\mathbb{Q}_p)$  be a hyperspecial subgroup,  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  sufficiently small compact open and  $\mathcal{K} = \mathcal{K}^p \mathcal{K}_p \leq G(\mathbf{A}_f)$ .

┘

The Langlands dual group  $\widehat{G} = \mathrm{SL}(2) \times \mathrm{SL}(2)$ , and  $\sigma_p$  acts via interchanging the two factors. The Hodge cocharacter is given by  $\mu = (1, 1) \in \widehat{T} \cong \mathbb{Z}^2$ ,  $\mu^* = \mu$ , thus  $V_{\mu^*} = \mathrm{Std}^* \times \mathrm{Std}^*$ .

**Prop.(28.9.8.5).** Let  $\mathrm{Sh}_{\mathcal{K}}(G, X)$  be the Hilbert modular surface, which admits an integral model  $\mathcal{S}_{G, \mathcal{K}}$  over  $\mathbb{Z}_{(p)}$ , and denote  $\mathcal{S}_{G, \mathcal{K}} = \mathcal{S}_{G, \mathcal{K}} \otimes \mathbb{F}_{p^2}$  the special fiber.

In this case,

$$\Lambda^{\mathrm{Tate}_p} = \{ \lambda \in X^\bullet(\widehat{T}) \mid \lambda + \sigma_p(\lambda) \in X_\bullet(Z_G) \} = \{ (m, -m) \}$$

and

$$V_{\mu^*}^{\mathrm{Tate}_p} = V_{\mu^*}(\lambda_1) \oplus V_{\mu^*}(\lambda_2), \text{ where } \lambda_1 = (-1, 1), \quad \lambda_2 = (1, -1).$$

┘

Notice that  $\lambda_1(p) = \lambda_2(p) \in B(G, \mu^*)$  is just the unramified basic element because  $\sum_{i=0}^m \sigma_p^i(\lambda)$  is just a multiple of the newton polygon  $\nu_{\lambda(p)}$  of  $\lambda(p)$ .([X-Z]Prop4.2.10).



**Prop. (28.9.8.6) [Rapoport-Zink Uniformization].** Let  $G'$  be the compact inner form of  $G$ , which is  $G' = \text{Res}_{F/\mathbb{Q}} \mathbb{P}B^*$ , where  $B \in \text{QuatAlg}_F$  is ramified exactly at the two Archimedean places.

Let  $\overline{\mathcal{S}}_{G,\mathcal{K}} = \mathcal{S}_{G',\mathcal{K}} \otimes \mathbb{F}_{p^2}$  be the special fiber of the discrete Shimura variety.

There are natural Jacquet-Langlands correspondences (Cf. [X-Z]7.3.4).

$$\begin{array}{ccc} X_{\mathbf{b}_i} \triangleq G'(\mathbb{Q}) \backslash X_{\mu^*}^{\mathbf{b}_i, \min}(1) \times G'(\mathbf{A}_f) / \mathcal{K} & & \\ \swarrow \text{(\mathbb{P}^1)^{perf-bundles}} & & \searrow \\ \overline{\mathcal{S}}_{G',\mathcal{K}} & & \overline{\mathcal{S}}_{G,\mathcal{K}}^{\text{perf}} \end{array}, \quad i \in \{1, 2\}$$

s.t.  $X_{\mathbf{b}_1} \cup X_{\mathbf{b}_2} \rightarrow \overline{\mathcal{S}}_{G,\mathcal{K},b}^{\text{perf}}$  is the surjective Rapoport-Zink uniformization.

Where  $\mathbf{b}_i$  are the MV-cycles. ┘

**Prop. (28.9.8.7) [Mirkovic-Vilonen Cycles].** So each

$$\mathbf{M}\text{-}\mathbf{V}_{\mu^*}[\lambda_i] = \{\mathbf{b}_i\}$$

is a singleton, and by philosophy of Xiao-Zhu4.2.10 we can choose the minimal dominant coweights

$$\nu_{\mathbf{b}_1} = (1, 0), \quad \nu_{\mathbf{b}_2} = (0, 1)$$

s.t.  $\lambda_i + \nu_i^* - \sigma_p(\nu_i^*) = \tau_i^* = 0$  is dominant, and  $\mathbf{b}_i$  is in the image of

$$i_{\nu_i^*}^{\mathbf{M}\text{-}\mathbf{V}} : \mathbb{S}_{(\nu_i^*, \mu^*)|\lambda_i + \nu_i^*} \rightarrow \mathbf{M}\text{-}\mathbf{V}_{\mu^*}[\lambda_i],$$

i.e. there exist  $\mathbf{a}_i \subset \mathbb{S}_{(\nu_i^*, \mu^*)|\lambda_i + \nu_i^*}$  s.t.

$$\mathbf{a}_i \cap (S_{\mu^*} \widetilde{\times} S_{\lambda_i}) = (S_{\nu_i^*} \cap \text{Gr}_{\nu_i^*}) \widetilde{\times} \mathbf{b}_i.$$

$X_{\mu^*}^{\mathbf{b}_i, \min}(\tau_i) = (\mathbb{P}^1)^{\text{perf}}$ , where  $\tau_i(p) = \mathbf{1}$  is just the unramified minimal element in  $B(G, \mu^*)$ . ┘

**Prop. (28.9.8.8) [Irreducible Components of ADLV].** Then we recall that  $X_{\mu^*}^{\mathbf{b}_i, \min}(\tau_i) \subset X_{\mu^*}(\tau_i)$  is the unique irreducible component of maximal dimension of the fiber product

$$\begin{array}{ccc} X_{\mu^*, \nu_i^*}^{\mathbf{a}_i}(\tau_i) & \longrightarrow & \mathbf{a}_i \subset \text{Gr}_{(\nu_i^*, \mu^*)|\lambda_i + \nu_i^*}^0 \\ \downarrow & & \downarrow \text{pr}_1 \times \text{pr}_3 \\ \text{Gr}_{\nu_i^*} & \xrightarrow{\text{id} \times \tau_i(p) \sigma_p} & \text{Gr}_{\nu_i^*} \times \text{Gr}_{\lambda_i + \nu_i^*} \end{array}$$

Because in this case  $\nu_i$  are all minuscule, this fiber product is already irreducible, and equals the perfection of the Deligne-Lusztig variety of  $\text{PGL}(2; \mathbb{F}_{p^2})$ , which is just  $(\mathbb{P}_{\mathbb{F}_{p^2}}^1)^{\text{perf}}$ . ┘

These geometric Jacquet-Langlands correspondences

$$\begin{array}{ccc} & X_{\mathbf{b}_i} & \\ \swarrow \text{(\mathbb{P}^1)^{perf-bundles}} & & \searrow \\ \overline{\mathcal{S}}_{G',\mathcal{K}} & & \overline{\mathcal{S}}_{G,\mathcal{K}}^{\text{perf}} \end{array}$$

induce natural  $\mathcal{H}_{\mathcal{K}}$ -equivariant maps

$$\mathrm{J}\text{-}\mathrm{L}_{\tau_i, \mu}(\mathbf{a}_{i, \mathrm{in}}) : \mathrm{H}_{\mathrm{\acute{e}t}}^0(\overline{\mathcal{S}}_{G', \mathcal{K}, \overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_{\ell}) \cong \mathrm{H}_{\mathrm{\acute{e}t}}^0(X_{\mathbf{b}_i, \overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_{\ell}) \xrightarrow{\mathrm{Gys}_{\mathbf{b}_i}} \mathrm{H}_{\mathrm{\acute{e}t}, c}^2(\overline{\mathcal{S}}_{G, \mathcal{K}, \overline{\mathbb{F}}_p}^{\mathrm{perf}}, \overline{\mathbb{Q}}_{\ell}),$$

$$\mathrm{J}\text{-}\mathrm{L}_{\mu, \tau_j}(\mathbf{a}_{i, \mathrm{out}}) : \mathrm{H}_{\mathrm{\acute{e}t}, c}^2(\overline{\mathcal{S}}_{G, \mathcal{K}, \overline{\mathbb{F}}_p}^{\mathrm{perf}}, \overline{\mathbb{Q}}_{\ell}) \xrightarrow{\mathrm{Res}_{\mathbf{b}_j}} \mathrm{H}_{\mathrm{\acute{e}t}, c}^2(X_{\mathbf{b}_j, \overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_{\ell}) \cong \mathrm{H}_{\mathrm{\acute{e}t}}^0(\overline{\mathcal{S}}_{G', \mathcal{K}, \overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_{\ell}).$$

I'll explain the name in another example.

**Prop. (28.9.8.9) [Intersection Matrix].** The composition of these two maps for  $i, j \in \{1, 2\}$  gives a matrix

$$\begin{bmatrix} -2p & T_p \\ T_p & -2p \end{bmatrix} : C(G'(\mathbb{Q}) \backslash G'(\mathbf{A}_f) / \mathcal{K}, \overline{\mathbb{Q}}_{\ell}) \rightarrow C(G'(\mathbb{Q}) \backslash G'(\mathbf{A}_f) / \mathcal{K}, \overline{\mathbb{Q}}_{\ell}),$$

where  $T_p = [\mathcal{K}_p \begin{bmatrix} p & \\ & 1 \end{bmatrix} \mathcal{K}_p]$ . So on the  $\pi_f$ -components it is given by

$$\begin{bmatrix} -2p & \alpha + \beta \\ \alpha + \beta & -2p \end{bmatrix}.$$

whose determinant is  $-(\alpha - \beta)^2$ . So if  $\alpha \neq \beta$ , the union map

$$\mathrm{cl} = \mathrm{Gys}_{\mathbf{b}_1} \oplus \mathrm{Gys}_{\mathbf{b}_2} : H_1^{\mathrm{B-M}}(\overline{\mathcal{S}}_{G, \mathcal{K}, b, \overline{\mathbb{F}}_p}^{\mathrm{perf}})[\pi_f] \rightarrow \mathrm{H}_{\mathrm{\acute{e}t}}^2(\overline{\mathcal{S}}_{G, \mathcal{K}, \overline{\mathbb{F}}_p}^{\mathrm{perf}}, \overline{\mathbb{Q}}_{\ell})[\pi_f]$$

is injective.  $\lrcorner$

**Remark (28.9.8.10) [The split case].** If  $p \in \mathrm{Prime}$  is split in  $F$ , then  $\Lambda^{\mathrm{Tate}_p} = \{(0, 0)\}$ , and  $V_{\mu^*}^{\mathrm{Tate}_p} = 0$ . As expected, The basic locus in the special fiber at  $p$  is 0-dimensional, by [Hilbert modular varieties of low dimension]Section8.2.  $\lrcorner$

### Geometric Satake for Unitary Groups

**Notation (28.9.8.11).**

- Let  $n \in \mathbb{Z}_+, r = \lfloor \frac{n}{2} \rfloor$ .
- Denote  $i^{\vee} = n + 1 - i$ .
- Let  $V_p$  be the standard Hermitian space over  $\mathbb{Q}_{p^2}$  with basis  $\{e_i\}$  s.t.  $(e_i, e_{j^{\vee}}) = \delta_{ij}$ .
- Let  $G = \mathrm{U}(V_{p^2})$  be the unramified unitary group.
- The stabilizer  $\mathcal{K}_p$  of the lattice  $\Lambda = \oplus \mathbb{Z}_{p^2} e_i$  is a hyperspecial subgroup of  $G$ .
- Let  $T \leq G$  be the diagonal maximal torus. And let  $S = \mathrm{diag}(a_1, a_2, \dots, a_2^{-1}, a_1^{-1}) \subset T$  be the maximal split subtorus.

$\lrcorner$

The Langlands dual group of  $G$  is  $(\widehat{G}, \widehat{B}, \widehat{T}) = (\mathrm{GL}(n), \mathrm{B}(n), \mathrm{T}(n))$ , and  $X^{\bullet}(\widehat{T}) = \mathbb{Z}\varepsilon_1 \oplus \dots \mathbb{Z}\varepsilon_n$ . Then  $\sigma_p$  acts on  $\widehat{G}$  via

$$\sigma_p(A) = JA^{-t}J^{-1}, \text{ where } J = \begin{bmatrix} & & & & 1 \\ & & & -1 & \\ & & \ddots & & \\ & & & & \\ (-1)^n & & & & \end{bmatrix} \text{ ?}$$

In particular,  $\sigma_p(\varepsilon_i) = -\varepsilon_{i^\vee}$ .  
 $\widehat{S} = \widehat{T}/(\sigma_p - 1)$ , and

$$X^\bullet(\widehat{S}) = X^\bullet(\widehat{T})^{\sigma_p} = \oplus_{i=1}^r \mathbb{Z}(\varepsilon_i - \varepsilon_{i^\vee}).$$

And the relative Weyl group  $W_0 = W^{\sigma_p} = \{(i, i^\vee), (i, j)(i^\vee, j^\vee) | i, j \in [n]_+\} \cong \{\pm 1\}^r \rtimes \text{Sym}_r$ .

Then we have

$$\mathbf{J} = J_{\widehat{G}} = \overline{\mathbb{Q}_\ell}[\widehat{G}\sigma_p]^{\widehat{G}} \cong \overline{\mathbb{Q}_\ell}[X^\bullet(\widehat{S})]^{W_0} \cong \overline{\mathbb{Q}_\ell}[\mathfrak{S}_1, \dots, \mathfrak{S}_r],$$

where  $\mathfrak{S}_i$  are elementary symmetric polynomials of  $e^{\varepsilon_1 - \varepsilon_{1^\vee}} + e^{\varepsilon_{1^\vee} - \varepsilon_1}, \dots, e^{\varepsilon_r - \varepsilon_{r^\vee}} + e^{\varepsilon_{r^\vee} - \varepsilon_r}$ .

**Prop. (28.9.8.12) [Relative Root System].** The relative root system  $\Phi_{\text{rel}}^\vee \subset X^\bullet(\widehat{S}) = X^\bullet(\widehat{T})^{\sigma_p}$  consists of the  $W_0$ -orbit of  $\varepsilon_1 - \varepsilon_{1^\vee}$  and the  $W_0$ -orbit of  $\varepsilon_1 - \varepsilon_{1^\vee} - \varepsilon_2 + \varepsilon_{2^\vee}$  (when  $r \geq 2$ ), so we can construct a discriminant functions in  $\mathbf{J}$ :

$$\text{disc}_1 = \prod_{1 \leq i \leq r} (A - 1)(A^{-1} - 1) = \sum_{i=0}^r (-1)^{r-i} 2^i \mathfrak{S}_{r-i}, \text{ where } A = e^{\varepsilon_i - \varepsilon_{i^\vee}}$$

Via routine calculation, we can prove that

$$\text{disc}_1 = \sum_{a=0}^r (-1)^{r-a} (2a+1) \chi_{V_{r-a}}, \text{ where } V_a = \text{End}(\wedge^a \text{Std}).$$

┘

$\lambda_{\underline{m}} = m_1(\varepsilon_1 - \varepsilon_{1^\vee}) + m_2(\varepsilon_2 - \varepsilon_{2^\vee}) + \dots \in X^\bullet(\widehat{S})$  is dominant iff  $m_1 \geq m_2 \geq \dots$ . Denote

$$\lambda_i = (\varepsilon_1 - \varepsilon_{1^\vee}) + \dots + (\varepsilon_i - \varepsilon_{i^\vee}), \text{ where } i \in [r]_+,$$

then the spherical algebra  $\mathcal{H}_{\mathcal{K}_p}(G)$  is generated by  $\{\mathbf{1}_{\mathcal{K}_p \lambda_i(p)}\}$  as a  $\overline{\mathbb{Q}_\ell}$ -algebra by the Cartan decomposition (20.7.2.14).

If we fix  $\sqrt{p} \in \overline{\mathbb{Q}_\ell}$ , then there is a **Satake isomorphism**

$$CT_p : \mathcal{H}_{\mathcal{K}}(G) \cong \overline{\mathbb{Q}_\ell}[X^\bullet(\widehat{S})]^{W_0} = \mathbf{J},$$

moreover, there is a geometric Satake isomorphism with commutative diagram

$$\begin{array}{ccc} V \in K_0(\text{Rep}^{\text{fd}}(\text{L}G)) & \xrightarrow{\chi_V \text{ restricts to } \widehat{G}\sigma_p} & \overline{\mathbb{Q}_\ell}[X^\bullet(\widehat{S})]^{W_0} = \mathbf{J} \ni \text{ch}_V \\ \downarrow \text{g.Sat} & & \uparrow CT_p \\ \text{Sat}(V) \in \text{Perv}_{L^+(G)}^0(\text{Gr}_G) & \xrightarrow{\text{traces of Frobenius}} & \mathcal{H}_{\mathcal{K}}(G) \ni h_V \end{array}$$

Then via geometric Satake, we can calculate  $h_{V_a}$  and thus also  $CT_p^{-1}(\mathfrak{D}) \in \mathcal{H}_{\mathcal{K}}(G)$ .

**U(1, 2r) Shimura Varieties****Notation (28.9.8.13).**

- Let  $E$  be an imaginary quadratic field.
- Let  $p \in \text{Prime}_{\geq 3}$  be an inert prime with  $(p) = \mathfrak{p}_v$ .
- Let  $V$  be a  $2r + 1$ -dimensional Hermitian space over  $E$  with signature  $(1, 2r)$  at infinity and is unramified at  $p$ .
- Let  $G = \text{U}(V)$  and
- $\mathcal{K} = \mathcal{K}^p \mathcal{K}_p \leq G(\mathbf{A}_f)$  where  $\mathcal{K}_p$  is the hyperspecial subgroup fixing a self-dual  $\mathbb{Z}_{p^2}$ -lattice  $\Lambda \subset V_p$ .

┘

**Prop. (28.9.8.14).** The Hodge cocharacter  $\mu = \varepsilon_1 \in X^\bullet(\widehat{T})$ ,  $\mu^* = -\varepsilon_{2r+1}$ .

$V_{\mu^*}^{\text{Tate}_p} = V_{\mu^*}[\lambda]$  is 1-dimensional, where  $\lambda = -\varepsilon_{r+1}$ . So  $\lambda(p) \in B(G, \mu^*)$  is the unramified basic element. And  $MV_{\mu^*}^{\text{Tate}_p} = MV_{\mu^*}(\lambda) = \{\mathbf{b}\}$  is a singleton.

┘

**Prop. (28.9.8.15) [Satake-Cycles].** We can write  $\lambda = \sigma_p(\nu^*) - \nu^* + \tau^*$ , where

$$\nu = \nu_{\mathbf{b}} = \varepsilon_1 + \dots + \varepsilon_r, \quad \tau = \tau_{\mathbf{b}} = \varepsilon_1 + \dots + \varepsilon_{2r+1} \in X_\bullet(Z_G).$$

So there is a Satake-cycle  $\mathbf{a} \in \mathbb{S}_{(\nu^*, \mu^*)|\lambda+\nu^*}$  s.t.

$$\mathbf{a} \cap S_{\nu^*, \lambda} = (S_{\nu^*} \cap \text{Gr}_{\nu^*}) \widetilde{\times} \mathbf{b}. \quad (\mathbf{b} \subset S_\lambda \cap \text{Gr}_{\mu^*}).$$

┘

**Prop. (28.9.8.16).** There is a natural correspondence

$$\begin{array}{ccc} & \text{Gr}_{(\nu^*, \mu^*)|\lambda+\nu^*}^0 & \\ \swarrow & & \searrow \\ \text{Gr}_{(\nu^*, \mu^*)} & & \text{Gr}_{\lambda+\nu^*} \end{array}$$

So any  $\mathbf{a} \in \mathbb{S}_{(\nu^*, \mu^*)|\lambda+\nu^*}$  gives a correspondence between perverse sheaves on  $\text{Gr}_{(\nu^*, \mu^*)}$  and  $\text{Gr}_{\lambda+\nu^*}$ . Because geometric Satake equivalence preserves tensor product, this gives two non-zero homomorphisms (notice  $\tau^* + \sigma_p(\nu^*) = \lambda + \nu^*$ ).

$$\mathbf{a}_{in} : V_{\sigma_p(\nu)} \otimes V_\tau \otimes V_{\nu^*} \rightarrow V_\mu, \quad \mathbf{a}_{out} : V_{\sigma_p(\nu^*)} \otimes V_\mu \otimes V_\nu \rightarrow V_\tau.$$

Thus induces two maps

$$\Xi_\nu(\mathbf{a}_{in}) \in \text{Hom}(\widetilde{V}_\tau, \widetilde{V}_\mu), \quad \Xi_{\nu^*}(\mathbf{a}_{out}) \in \text{Hom}(\widetilde{V}_\mu, \widetilde{V}_\tau),$$

given as follows: at the point  $g\sigma_p \in \widehat{G}\sigma_p$ , it is given by

$$\Xi_\nu(\mathbf{a}_{in})(g\sigma_p) : V_\tau \rightarrow V_\nu \otimes V_\tau \otimes V_{\nu^*} \xrightarrow{g\sigma_p \otimes \text{id} \otimes \text{id}} V_{\sigma_p(\nu)} \otimes V_\tau \otimes V_{\nu^*} \xrightarrow{\mathbf{a}_{in}} V_\mu,$$

$$\Xi_{\nu^*}(\mathbf{a}_{out})(g\sigma_p) : V_\mu \rightarrow V_{\nu^*} \otimes V_\mu \otimes V_\nu \xrightarrow{g\sigma_p \otimes \text{id} \otimes \text{id}} V_{\sigma_p(\nu^*)} \otimes V_\mu \otimes V_\nu \xrightarrow{\mathbf{a}_{out}} V_\tau.$$

Then up to a constant in  $\overline{\mathbb{Q}_\ell}^\times$ , we have

$$\mathfrak{D} = \Xi_{\nu^*}(\mathbf{a}_{out}) \circ \Xi_\nu(\mathbf{a}_{in}) = (-1)^r \text{disc}_1 \in \text{Hom}(\widetilde{V}_\tau, \widetilde{V}_\tau) \cong \mathbf{J}.$$

(in general, we need to take projective determinant).

┘

*Proof:* Cf.[On vector-valued twisted conjugate invariant functions on a group]1.0.2 and 6.4.2.  $\square$

**Prop. (28.9.8.17).**

$$\mathbf{a} \subset \mathrm{Gr}_{(\nu^*, \mu^*)|\tau^* + \sigma_p(\nu^*)}^0 \subset \mathrm{Gr}_{\nu^*} \times \mathrm{Gr}_{\tau^* + \sigma_p(\nu^*)}$$

is  $L^+G$  equivariant isomorphic to

$$H = \{(\ell_1, \ell_2) : \ell_1 \subset \ell_2\} \subset \mathrm{Gr}(r, \Lambda/(p)) \times \mathrm{Gr}(r+1, \Lambda/(p))$$

And the map  $\mathrm{Gr}_{\nu^*} \rightarrow \mathrm{Gr}_{\tau^* + \sigma_p(\nu^*)}$  given by  $g \mapsto \tau^*(p)\sigma_p(g)$  is identified with the map

$$\ell \subset \Lambda/(p) \mapsto \mathrm{Frob}_p(\ell)^\perp.$$

⌋

*Proof:* As  $\nu^* = -\varepsilon_1^\vee - \varepsilon_2^\vee - \dots - \varepsilon_r^\vee$  is minuscule,

$$\mathrm{Gr}_{\nu^*}(R) = \{\Lambda \otimes W(R) \subset L \subset \frac{1}{p}\Lambda \otimes W(R) \mid \mathrm{rank}_R L/\Lambda \otimes W(R) = r\} = \mathrm{Gr}(r, \Lambda/(p)).$$

Similarly  $\tau^* + \sigma(\nu^*) = -\varepsilon_1^\vee - \varepsilon_2^\vee - \dots - \varepsilon_{(r+1)}^\vee$  is minuscule, so  $\mathrm{Gr}_{\tau^* + \sigma(\nu^*)} \cong \mathrm{Gr}(r+1, \Lambda/(p))$ .

The map  $\mathrm{Gr}_{\nu^*} \rightarrow \mathrm{Gr}_{\tau^* + \sigma(\nu^*)}$  given by  $g \mapsto \tau(p)\sigma_p(g)$  send  $L$  to  $\frac{1}{p}\mathrm{Frob}_R^*(L)^\vee$ . ?

The orbits of  $G(\mathbb{F}_{p^2})$ -action on  $\mathrm{Gr}(r, \Lambda/(p)) \times \mathrm{Gr}(r+1, \Lambda/(p))$  is parametrized by  $a = \dim(\ell_1 \cap \ell_2)$ . There is only one closed orbit with dimension  $r^2 + 2r$ , so it must be  $\mathbf{a}$ .  $\square$

**Prop. (28.9.8.18)[Relavent ADLV].** In this case, the ADLV is given by

$$X_{\mu^*}(\tau^*) = \{g \in \mathrm{Gr}_G : g^{-1}\tau^*(p)\sigma_p(g) \in L^+G \backslash \mu^*(p)/L^+(G)\},$$

and an irreducible component  $X_{\mu^*}(\tau^*)^{\mathbf{b}, \min}$  is given by the Cartesian diagram

$$\begin{array}{ccc} X_{\mu^*}(\tau^*)^{\mathbf{b}, \min} & \longrightarrow & \mathbf{a} \subset \mathrm{Gr}_{(\nu^*, \mu^*)|\tau + \sigma_p(\nu)}^0 \\ \downarrow & & \downarrow \mathrm{pr}_1 \times m \\ \mathrm{Gr}_{\nu^*} & \xrightarrow{\mathrm{id} \times \tau^*(p)\sigma_p} & \mathrm{Gr}_{\nu^*} \times \mathrm{Gr}_{\tau^* + \sigma_p(\nu^*)} \end{array}$$

which by the argument above is just

$$\{\ell \subset \Lambda/(p) : \ell \in \mathrm{Frob}(\ell)^\perp\}.$$

which is the Deligne-Lusztig variety for  $\mathrm{GU}(2r+1, \mathbb{F}_p)$ . For example, if  $r = 1$ , this is just the Fermat curve

$$\{(x, y, z) \in \mathbb{P}^2 \mid x^{p+1} + y^{p+1} + z^{p+1} = 0\}.$$

$\mathcal{K}_p$  acts on this Deligne-Lusztig variety through  $\mathrm{GL}(2r+1, \mathbb{F}_p)$ , and then there is a  $G(\mathbb{Q}_p)$ -equivariant map

$$G(\mathbb{Q}_p) \times^{\mathcal{K}_p} X_{\mu^*}^{\mathbf{b}, \min}(\tau^*) \rightarrow X_{\mu^*}(\tau^*)$$

that induces an isomorphism on irreducible components. (By [X-Z]Prop4.4.14).  $\square$

**Prop. (28.9.8.19) [Geometric Jacquet-Langlands Correspondence].** Let  $V'$  be the  $(2r + 1)$ -dimensional Hermitian space over  $E$  that is isomorphic to  $E$  at all finite places and of signature  $(2r + 1, 0)$  at the infinite place. And denote  $G' = \mathrm{U}(V')$ . Then there is a Rapoport-Zink uniformization ([X-Z] 7.2.16, 7.2.17):

$$\begin{array}{ccc} & G'(\mathbb{Q}) \backslash (G'(\mathbb{Q}_p) \times^{\mathcal{K}_p} X_{\mu^*}^{\mathbf{b}, \min}(\tau^*) \times G'(\mathbf{A}_f^p) / \mathcal{K}^p) & \\ \swarrow & & \searrow \\ \overline{\mathcal{S}}_{\tau, \kappa(v)} \triangleq \overline{\mathcal{S}_{\mathcal{K}}(G', \{\tau_{\mathbf{b}}\})}^{\mathrm{perf}} & & \overline{\mathcal{S}}_{\mu, \kappa(v)} \triangleq \overline{\mathcal{S}_{\mathcal{K}}(G, \{\mu\})}^{\mathrm{perf}} \end{array}$$

where the right map is surjective onto the basic Newton strata  $\mathcal{N}_b$  that induces a bijection on the set of irreducible components. (here we may need to base change to the common reflex field of  $G$  and  $G'$ )

Then this correspondence induces a geometric Jacquet-Langlands correspondence

$$\mathrm{J-L}_{\tau, \mu} : \mathrm{H}_{\mathrm{\acute{e}t}}^0(\overline{\mathcal{S}}_{\tau, \bar{\kappa}(v)}) \otimes_{\mathbf{J}} \mathrm{Hom}(\tilde{V}_{\tau}, \tilde{V}_{\mu}) \rightarrow \mathrm{H}_{\mathrm{\acute{e}t}, c}^r(\overline{\mathcal{S}}_{\mu, \bar{\kappa}(v)}, \overline{\mathbb{Q}_{\ell}}(d/2)).$$

Thus evaluating at  $\Xi_{\nu}(a_{in}) \in \mathrm{Hom}(\tilde{V}_{\tau}, \tilde{V}_{\mu})$ , we get a  $\mathbf{J}$ -equivariant map

$$\mathrm{J-L}_{\tau, \mu}(a_{in}) : \mathrm{H}_{\mathrm{\acute{e}t}}^0(\overline{\mathcal{S}}_{\tau, \bar{\kappa}(v)}) \rightarrow \mathrm{H}_{\mathrm{\acute{e}t}, c}^d(\overline{\mathcal{S}}_{\mu, \bar{\kappa}(v)}, \overline{\mathbb{Q}_{\ell}}(d/2))$$

And there is a commutative diagram

$$\begin{array}{ccc} \mathrm{H}_{\mathrm{\acute{e}t}}^0(\overline{\mathcal{S}}_{\tau, \bar{\kappa}(v)}) & \xrightarrow{\cong} & H_r^{\mathbf{B-M}}(\mathcal{N}_b) \\ & \searrow \mathrm{J-L}_{\tau, \mu}(a_{in}) & \swarrow \mathrm{cl}_{\mathbf{b}} \\ & \mathrm{H}_{\mathrm{\acute{e}t}, c}^r(\overline{\mathcal{S}}_{\mu, \bar{\kappa}(v)}, \overline{\mathbb{Q}_{\ell}}(r/2)) & \end{array}$$

Similarly, we have maps  $\mathrm{cl}_{\mathbf{b}}^{\vee}$  and  $\mathrm{J-L}_{\mu, \tau}(a_{out})$ , and

$$\mathrm{J-L}_{\mu, \tau_{\mathbf{b}}}(a_{out}) \circ \mathrm{J-L}_{\tau, \mu}(a_{in}) \in \mathrm{Mat}_{C(G'(\mathbb{Q}) \backslash G'(\mathbf{A}_f) / \mathcal{K}, \overline{\mathbb{Q}_{\ell}})}(\mathcal{H}_{\mathcal{K}_p}(G))$$

encodes the intersection matrix of the basic locus. ┘

**Thm. (28.9.8.20) [The Intersection Matrix].** The composition

$$\mathrm{J-L}_{\mu, \tau_{\mathbf{b}}} \circ \mathrm{J-L}_{\tau_{\mathbf{b}}, \mu}$$

is given by the Hecke operator

$$h = p^{r(r+1)} C T_p^{-1}(\mathfrak{D}) = \sum_{i=0}^r (-1)^r (2i+1) p^{i(i+1)} \sum_{j=0}^{r-i} \binom{2r+1-2j}{r-i-j} T_{p, j} \in \mathcal{H}_{\mathcal{K}_p}(G).$$

Which means for  $G'(\mathbb{Q})g\mathcal{K}, G'(\mathbb{Q})g'\mathcal{K} \in G'(\mathbb{Q}) \backslash G'(\mathbf{A}_f) / \mathcal{K}$ , if we choose  $g, g'$  s.t.  $g^p = (g')^p$ , then the corresponding irreducible components  $X_g$  and  $X_{g'}$  intersection with multiplicity  $h(g^{-1}g')$ . ┘

*Proof:* By geometric Jacquet-Langlands correspondence, the determinant is just  $C T_p^{-1}$  of

$$\mathfrak{D} = \Xi_{\nu^*}(\mathbf{a}_{out}) \circ \Xi_{\nu}(\mathbf{a}_{in}) \in \mathrm{Hom}(\tilde{V}_{\tau}, \tilde{V}_{\tau}) \cong \mathbf{J},$$

□

**Cor. (28.9.8.21).** If  $r = 1$ , then  $h = T_{p,1} + (1+p)(1-2p)$ . So if  $g, g' \in G(\mathbf{A}_f)$  has the same prime-to- $p$  components, then

$$X_g \cdot X_{g'} = \begin{cases} 1 & , g_p \in g'_p \mathcal{K}_p \text{diag}(p, 1) \mathcal{K}_p \\ (1+p)(1-2p) & , g_p \in g'_p \mathcal{K}_p \\ 0 & , \text{otherwise} \end{cases}.$$

┘

## 9 Tate conjecture for Special Fibers of Shimura Varieties

**Notation (28.9.9.1).**

- Let  $(G, \mathcal{N})$  be a Hodge-type Shimura-datum s.t.  $Z(G)$  is connected,
- Let  $p \in \mathbf{Prime}_{\geq 2}$  s.t.  $V_{\mu^*}^{\text{Tate}} \neq 0$  ?.
- Let  $\mathcal{K} = \mathcal{K}_p \mathcal{K}^p \leq G(\mathbf{A}_f)$  is a neat open compact subgroup s.t.  $\mathcal{K}_p$  is hyperspecial.

┘

**Thm. (28.9.9.2) [Xiao-Zhu].** 1. There is a unique inner form  $G'$  of  $G$  that is trivial at all finite places and  $G'(\mathbb{R})$  is compact modulo center.

2.  $\text{Sh}_{\mu, \text{bas}}$  is pure of dimension  $d/2$  ?. And there is a  $\mathcal{H}_{\mathcal{K}}(G)$  equivariant isomorphism

$$H_d^{\text{B-M}} \left( (\text{Sh}_{\mu, \text{bas}})_{\overline{\kappa(v)}} \right) \cong C(G'(\mathbb{Q}) \backslash G'(\mathbf{A}_f) / \mathcal{K}, \overline{\mathbb{Q}_{\ell}}) \otimes_J V_{\mu^*}^{\text{Tate}_p}.$$

3. If  $\pi_f$  is an irreducible model of  $\mathcal{H}_{\mathcal{K}}(G)$  whose Satake-parameter of  $(\pi_f)_p$  is generic w.r.t.  $V_{\mu}$  ?, then the cycle class map ?

$$\text{cl} : H_d^{\text{B-M}} \left( (\text{Sh}_{\mu, \text{bas}})_{\overline{\kappa(v)}} \right) \otimes \overline{\mathbb{Q}_{\ell}} \rightarrow H_{\text{ét}, c}^d \left( (\text{Sh}_{\mu})_{\overline{\kappa(v)}}, \overline{\mathbb{Q}_{\ell}}(d/2) \right)$$

is injective when restricted to  $H_d^{\text{B-M}} \left( (\text{Sh}_{\mu, b})_{\overline{\kappa(v)}} \right) [\pi_f]$ .

4. If  $\text{Sh}_{\mathcal{K}}(G, \mathcal{N})$  is a Kottwitz-arithmetic variety ? (e.g. this is the case when  $[G, G]$  is simply-connected and anisotropic, and there exists  $p' \in \mathbf{Prime} \setminus \{p\}$  s.t.  $(\pi_f)_{p'}$  is an unramified twist of Steinberg-representations), then the  $\pi_f^p$ -isotypical component of the cycle class map  $\text{cl}$  is surjective to

$$\sum_{\pi_p} T^p \left( \pi_p \pi_f^p, \overline{\mathbb{Q}_{\ell}} \right) \otimes \pi_p \pi_f^p,$$

if the Satake-parameters of  $\{\pi_p\}$  are all strongly general w.r.t.  $V_{\mu}$ . In particular, the Tate-conjecture holds for these  $\pi_f^p$ .

┘

*Proof:* ?

□

## 28.10 Perfectoid Shimura Varieties

References are [Sch15] and [Caraiani, Lecture notes on Perfectoid Shimura Varieties].

### 1 Derived Hecke Algebras



## 28.11 Unitary Shimura Varieties

### 1 $U(1, 2r)$ -Shimura Varieties

References are [Vollaard, supersingular locus of the Shimura variety of  $GU(1, s)$ ], [Vollaard-Wedhorn, supersingular locus of the Shimura variety of  $GU(1, n1)$ ].

### Picard Modular Surfaces

Picard modular surfaces are unitary Shimura varieties for  $U(1, 2)$ .

### 2 $U(2, s)$ -Shimura Varieties

### 3 RSZ Shimura Varieties

**Notation(28.11.3.1).**

- Let  $F/F_0$  be a CM number field with a CM-type  $\Phi$ ,
- Let  $\sqrt{\Delta} \in F^\times$  be a totally imaginary element.
- Let  $V \in \text{Herm}_n(F/F_0)$  with signature  $\{(n-1, 1)_{\varphi_0}, (n, 0)_{\varphi \neq \varphi_0}\}$ .
- 

┘

**Prop.(28.11.3.2)[RSZ Shimura Varieties].** Define the Shimura datum

$$(\tilde{G}, \{\tilde{h}\}) = (Z^{\mathbb{Q}} \times \text{res}_{F_0/\mathbb{Q}} U(V), h_{Z^{\mathbb{Q}}} \times h),$$

and let  $\mathbf{F}$  be the reflex field.

For open compact subgroups  $K_{Z^{\mathbb{Q}}} \subset Z^{\mathbb{Q}}(\mathbf{A}_f)$ ,  $K \subset U(V)(\mathbf{A}_{F_0})$  and  $\tilde{K} = K_{Z^{\mathbb{Q}}} \times K$ , there is a Shimura variety over  $\mathbf{F}$ :

$$\text{Sh}_{\tilde{K}}(\tilde{G}, \{\tilde{h}\}) = \text{Sh}_{K_{Z^{\mathbb{Q}}}}(Z^{\mathbb{Q}}, \{h_{Z^{\mathbb{Q}}}\}) \times \text{Sh}_K(\text{res}_{F_0/\mathbb{Q}} U(V), \{h\})$$

┘

*Proof:* Cf.[Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang]P6. **?** □

**Def.(28.11.3.3)[Bad Primes].** For any Shimura variety  $M$  as defined in(28.11.3.2), we can define a set  $\mathfrak{d}$  of bad primes of **Prime**, ┘

**Prop.(28.11.3.4)[Moduli Problem  $\mathcal{M}_0$ ].** With  $\mathfrak{d}$  defined in(28.11.3.3), consider the moduli functor  $\mathcal{M}_0$  on  $\text{NSch}/\text{Spec } \mathcal{O}_{\mathbf{F}}[\mathfrak{d}^{-1}]$  which associates with  $S$  the set of isomorphism classes of tuples  $(A_0, \iota_0, \lambda_0, \overline{\eta}_0)$ , where

- $A_0$  is an Abelian scheme over  $S$  of relative dimension  $[F_0 : \mathbb{Q}]$ ,
- $\iota_0 : \mathcal{O}_F[\mathfrak{d}^{-1}] \rightarrow \text{End}_S(A_0)[\mathfrak{d}^{-1}]$  is an  $\mathcal{O}_F[\mathfrak{d}^{-1}]$ -action s.t. the Kottwitz condition for  $\Phi$  is satisfied: For any  $a \in \mathcal{O}_F[\mathfrak{d}^{-1}]$ ,

$$\text{char}(\iota_0(a)|\text{Lie}(A_0)) = \prod_{\varphi \in \Phi} (T - \varphi(a)).$$

- $\lambda_0$  is an prime-to- $\mathfrak{d}$  principle polarization of  $A_0$  s.t.  $\text{Ros}_{\lambda_0}(\iota(a)) = \iota(\overline{a})$ .

- $\overline{\eta}_0$  is a  $K_{Z^{\mathfrak{q}}, \mathfrak{d}}$ -level structure ?.

Then when  $K_{Z^{\mathfrak{q}}}$  is neat,  $\mathcal{M}_0$  is representable by a finite étale scheme over  $\mathrm{Spec} \mathcal{O}_F[\mathfrak{d}^{-1}]$ . ┘

*Proof:* Cf.[Howard, Complex multiplication cycles and Kudla-Rapoport divisors]Prop3.1.2. □

**Thm. (28.11.3.5)[Integral Models of RSZ Shimura Varieties].** Situation as in (28.11.3.2), consider the functor  $\mathcal{M} = \mathcal{M}_{\widetilde{K}}$  which associates to each  $S \in \mathrm{NSch} / \mathrm{Spec} \mathcal{O}_{\mathbf{F}}[\mathfrak{d}^{-1}]$  the set of isomorphism classes of tuples  $(A_0, \iota_0, \lambda_0, \overline{\eta}_0, A, \iota, \lambda, \overline{\eta})$ , where

- $(A_0, \iota_0, \lambda_0, \overline{\eta}_0) \in \mathcal{M}_0(S)$  (28.11.3.4),
- $\iota : \mathcal{O}_F[\mathfrak{d}^{-1}] \rightarrow \mathrm{End}(A)[\mathfrak{d}^{-1}]$  is an action satisfying the Kottwitz condition of signature

$$((n-1, 1)_{\varphi_0}, (n, 0)_{\varphi \in \Phi \setminus \varphi_0}).$$

- $\lambda$  is an prime-to- $\mathfrak{d}$  principle polarization of  $A$  s.t.  $\mathrm{Ros}_{\lambda}(\iota(a)) = \iota(\overline{a})$ .
- $\overline{\eta}$  is a  $K_{\mathfrak{d}}$ -orbit of isometries of Hermitian modules over  $S$ :

$$\eta : V_{\mathfrak{d}}(A_0, A) \cong V(F_{0, \mathfrak{d}}) \text{.} \text{ ?}$$

- The Eisenstein condition for every place  $v \nmid \mathfrak{d}$  of  $F_0$  ? Cf.[On Shimura varieties for unitary groups]Sec5.2.

Then  $\mathcal{M}$  is representable by a scheme that is separated of f.t. and smooth of relative dimension  $n-1$  over  $\mathrm{Spec} \mathcal{O}_{\mathbf{F}}[\mathfrak{d}^{-1}]$  when  $\widetilde{K}$  is neat. And the generic fiber of  $\widetilde{M}$  is isomorphic to the canonical model Shimura variety  $M$  as in (28.11.3.2). ┘

*Proof:* Cf.[Weil Representations and Arithmetic Fundamental Lemma]Thm6.2. and [Arithmetic diagonal cycles on unitary Shimura varieties]Prop3.5. ? □

## 28.12 p-adic Modular Forms

References are [C-M98, Ma118], [BDP].

### 1 Arithmetic Modular Forms

References are [BDP].

**Def. (28.12.1.1) [Modular Forms].** For  $R_0 \in \mathcal{CRing}$ ,  $N \in \mathbb{Z}_+$ ,  $k \in \mathbb{N}$ , the space  $M(R_0, N, k)$  of **modular forms of level  $N$  and weight  $k$  over  $R_0$**  consists of rules  $f$  that functorially associates a section  $f(E/S, \alpha_N) \in \Gamma(S, \omega_{E/S}^{\otimes k})$  to each datum  $(E/S, \alpha_N)$ , where  $S \in \mathcal{Sch}/R_0$ ,  $E \in \mathbf{EllCur}/S$ , and  $\alpha_n : E/S[n] \cong (\mathbb{Z}/(n))_S^2$  is an isomorphism.

Such a modular form is automatically meromorphic at cusps, simply because  $\text{Tate}(q)$  is an Elliptic curve over  $\mathbb{Z}[[q]]$ .  $\lrcorner$

**Def. (28.12.1.2) [ $q$ -Expansions and Holomorphic Modular Forms].** For  $f \in M(R_0, N, k)$  (28.12.1.1), if  $R_0 \in \mathcal{CRing}/\mathbb{Z}[\frac{1}{N}, \mu_N]$ , the  **$q$ -expansion** of  $f$  is defined to be

$$f((\text{Tate}(q^N), \alpha_N)_{R_0}) \in (\mathbb{Z}((q)) \otimes R_0) \omega_{\text{can}}.$$

where  $\text{Tate}(q)$  is the Tate curve, and  $\alpha_N$  is any level  $N$ -structure on  $\text{Tate}(q^N)$ .

For any  $R_0 \in \mathcal{CRing}$ ,  $f \in M(R_0, N, k)$  is said to be **holomorphic at cusps** if the  $q$ -expansion of  $f$  restricted to  $R_0[\frac{1}{N}, \mu_N]$  is an element is contained in  $(\mathbb{Z}[[q]] \otimes R_0) \omega_{\text{can}}$  for any choice of  $\alpha_N$ . The space of holomorphic modular forms is denoted by  $S(R_0, N, k)$ .  $\lrcorner$

**Def. (28.12.1.3) [Kodaira-Spencer Maps].** If  $T \in \mathcal{Sch}$ ,  $S/T$  is smooth, and  $f : E \rightarrow S \in \mathbf{EllCur}/S$ , then there is an exact sequence

$$0 \rightarrow f^* \Omega_{S/T} \rightarrow \Omega_{E/T} \rightarrow \Omega_{E/S} \rightarrow 0$$

thus an exact sequence

$$0 \rightarrow T_{E/S} \rightarrow T_{E/T} \rightarrow f^* T_{S/T} \rightarrow 0.$$

Thus inducing a homomorphism

$$T_{S/T} \cong f_* f^* T_{S/T} \rightarrow R^1 f_* T_{E/S} = \text{Lie}(E/S)^2 R^1 f_* \Omega_{E/S}^1 \cong \text{Lie}(E/S)^2.$$

Thus inducing a **Kodaira-Spencer map** of  $\mathcal{O}_S$ -modules

$$\text{KS}_{E/S} : \omega_{E/S}^2 \rightarrow \Omega_{S/T}^1.$$

And if  $S/T \subset \overline{S}/T$ ,  $\overline{S} \setminus S = S^\infty$ ,  $E/S \subset \overline{E}/\overline{S}$  is an extension to a curve of genus 1, and the identity  $e \in E(S)$  also extends to a map  $\bar{e} \in \overline{E}(\overline{S})$ , with  $\bar{\omega}_{\overline{E}/\overline{S}} = \bar{e}^* \Omega_{\overline{E}/\overline{S}}^1$  extending  $\omega_{E/S}$ , then the Kodaira-Spencer map extends to a map

$$\text{KS}_{\overline{E}/\overline{S}} : \omega_{\overline{E}/\overline{S}}^2 \rightarrow \Omega_{\overline{S}/T}^1(\log S^\infty).$$

$\lrcorner$

**Prop. (28.12.1.4) [Kodaira-Spencer Map for Modular Curves].**

- If  $T = \operatorname{Spec} \mathbb{Q}$ ,  $S = Y(N)$ ,  $N \geq 3$ , and  $E/S$  is the universal curve, then the Kodaira-Spencer map (28.12.1.3) is an isomorphism. And on  $q$ -expansions of modular forms along any cusp  $t$ , it is of the form

$$f(q^{1/N})(\operatorname{dlog} t)^2 \mapsto f(q^{1/N}) \operatorname{dlog} q.$$

- If  $\bar{S} = X(N)_{\mathbb{Q}}$ , and  $\overline{E^{\operatorname{uni}}}$  is the regular minimal model of the universal elliptic curve, then the extended Kodaira-Spencer map (28.12.1.3) is an isomorphism. In fact, over  $\mathbb{Z}[\frac{1}{N}, \zeta_N]$  and at any cusp, the square of the canonical differential  $\omega_{\operatorname{can}}$  on  $\operatorname{Tate}(q^N)$  is mapped to the  $N \operatorname{dlog} q$ .
- If  $S = Y(N)^{\operatorname{ord}}$ , and  $T = \operatorname{Spec} \mathbb{Z}[\mu_N]$ , then  $S/T$  is smooth by, and the Kodaira-Spencer map on the  $q$ -expansions of modular forms is of the form

$$f(q^{1/N})(\operatorname{dlog} t)^2 \mapsto f(q^{1/N}) \operatorname{dlog} q = N f(q^{1/N}) \operatorname{dlog}(q^{1/N}),$$

so we can define a map  $\frac{1}{N} \operatorname{KS}_{Y(N)/\mathbb{Z}[\mu_N]}$ .

┘

*Proof:* [Katz, p-adic properties of modular forms and modular curves]A1. Prop3.17. and Cor3.18. ?

□

**Cor.(28.12.1.5) [Modular Forms as Sections on Modular Curves].** If  $N \in \mathbb{Z}_{\geq 3}$  and  $R_0 \in \mathbb{C}\operatorname{Ring}/\mathbb{Z}[\frac{1}{N}]$ , there are isomorphisms

$$S(R_0, N, k) \cong \Gamma(X(N), \bar{\omega}^k \otimes \Omega_{X(N)/\mathbb{Q}}^1 \otimes_{\mathbb{Z}[\frac{1}{N}]} R_0) \subset \Gamma(X(N), \bar{\omega}^k \otimes_{\mathbb{Z}[\frac{1}{N}]} R_0) \cong M(R_0, N, k),$$

where the middle inclusion is given by Kodaira-Spencer isomorphism (28.12.1.4).

Thus in general, for any  $\mathbb{Z}[\frac{1}{N}]$ -module  $K$ , we can define a **modular form of level  $N$  and weight  $k$  with coefficients in  $M$  which is holomorphic at cusps** to be a section in

$$\Gamma(X(N), \omega^k \otimes_{\mathbb{Z}[\frac{1}{N}]} M).$$

┘

**Thm.(28.12.1.6) [ $q$ -Expansion Principle].** For  $N \in \mathbb{Z}_{\geq 3}$ ,  $M \in \operatorname{Mod}_{\mathbb{Z}[\frac{1}{N}]}$ , and  $f$  a modular form of level  $N$  and weight  $k$  with coefficients in  $M$  which is holomorphic at cusps. Suppose on each of the  $\varphi(N)$  connected components of  $X(N)_{\mathbb{Z}[\frac{1}{N}]} \mathbb{Z}[\frac{1}{N}, \zeta_N]$ , there exists at least one cusp that the  $q$ -expansion of  $f$  vanishes, then  $f$  vanishes.

┘

*Proof:* Cf. [Kat73]P84. ?

□

**Cor.(28.12.1.7).** Situation as in (28.12.1.6), if  $N \subset M \in \operatorname{Mod}_{\mathbb{Z}[\frac{1}{N}]}$ , and on each of the  $\varphi(N)$  connected components of  $X(N)_{\mathbb{Z}[\frac{1}{N}]} \mathbb{Z}[\frac{1}{N}, \zeta_N]$ , there exists at least one cusp that the  $q$ -expansion of  $f$  is contained in  $N \otimes_{\mathbb{Z}[\frac{1}{N}]} \mathbb{Z}[\frac{1}{N}, \zeta_N] \otimes \mathbb{Z}[[q]]$ , then  $f$  is a modular form with coefficients in  $N$ .

┘

*Proof:* The image of  $f$  in  $\Gamma(X(N), \omega^k \otimes_{\mathbb{Z}[\frac{1}{N}]} M/N)$  is trivial.

□

## 2 Overconvergent Modular Forms

**Notation (28.12.2.1).**

- Let  $p \in \operatorname{Prime}$ .
- Let  $N \in \mathbb{Z}_+ \setminus (p)$ .
- Let  $\Lambda_N = \mathbb{Z}_p[(\mathbb{Z}/N)^* \times \mathbb{Z}_p^*]$ .

┘

### 3 Congruences of Modular Forms

## 28.13 Completed Cohomologies and Eigenvarieties

References are [C-E12].

- 1 Admissible Representations
- 2 Completed Cohomologies
- 3 Eigenvarieties

## 28.14 Drinfeld Modules

References are [Elliptic Modules, Drinfeld] and [Introduction to Drinfeld Modules, Hayes].





## 29 | Global Langlands Conjecture

### 29.1 Trace Formulae

*“It is rare that you can take a proof in the geometric setting and convert it to the genuine number theoretic setting. That is what has transpired through Ngô’s achievement (the proof of the fundamental lemma). Ngô has provided a bridge and now everybody is using this bridge.”*

– P. Sarnak.

References are [Introductory notes on the trace formula, Lapid], [Trace formula, Whitehouse] and [Introduction to Trace Formula, Arthur].

#### 1 Orbital Integrals

##### Local Orbital Integrals

**Notation(29.1.1.1).**

- Let  $K \in \mathbf{LField}$ .
- Let  $G \in \mathbf{AlgGrp}^{\mathrm{sm}}/K$  acts on  $X \in \mathbf{Sch}^{\mathrm{ft,sep}}/K$  on the right.
- Let  $\chi : G(K) \rightarrow \mathbb{C}^\times$  be a quasi-character.

┘

**Def.(29.1.1.2)[Relavent Elements].**  $\gamma \in X(K)$  is called a  **$\chi$ -relevant element** if  $\chi(Z_G(\gamma)(K)) = 1$ .

┘

**Def.(29.1.1.3)[Relatively Unimodular Elements].**  $\gamma \in X(K)$  is called a **relatively unimodular element** if  $\Delta_G|_{Z_G(\gamma)(K)} = \Delta_{Z_G(\gamma)(K)}$ .

┘

**Def.(29.1.1.4) [Local Orbital Integrals].** For a strongly-regular semi-simple element  $\gamma \in G(K)$  (10.5.1.17) and  $h \in \mathcal{T}(G(K))$ , if a Haar measure on  $G(K)$  and a Haar measure on  $Z_{G(K)}(\gamma)$  are chosen, then we can define the **orbital integrals**

$$\mathrm{Orb}_\gamma(h) = \int_{Z_{G(K)}(\gamma) \backslash G(K)} h(x^{-1}\gamma x) d(x^{-1}\gamma x),$$

┘

*Proof:*  $Z_{G(K)}(\gamma) \backslash G(K)$  has the quotient measure because both  $Z_G(\gamma)$  and  $G$  are reductive thus unimodular (12.10.1.22)(12.10.3.7).

And because

□

**Def. (29.1.1.5) [Invariant Distributions].**  $I \in \text{Dist}(G(K))$  is called an **invariant distribution** if  $I(f^g) = I(f)$  for any  $f \in \mathcal{T}(G(K))$  and  $g \in G(K)$ .  $\lrcorner$

**Thm. (29.1.1.6) [Density Theorem, Harish-Chandra].**  $I \in \text{Dist}(G(K))$  is invariant (29.1.1.5) iff  $I(f) = 0$  for any  $f \in \mathcal{T}(G(K))$  s.t.  $\text{Orb}_\gamma(f) = 0$  for any strongly-regular semi-simple element  $\gamma \in G(K)$ .  $\lrcorner$

*Proof:* Cf. [Harish-Chandra, Admissible Invariant Distributions].  $\textcolor{red}{?}$   $\square$

### Relative Orbital Integrals

**Def. (29.1.1.7) [Invariant Theory].** Let  $K \in p\text{-LField}$  and  $H \in \text{RedGrp}/K$ , and  $V$  an affine  $K$ -variety with an action of  $H$ . Then the categorical quotient of  $V$  by  $H$  is the affine variety  $V//H = \text{Spec}(\mathcal{O}(V)^H)$  with a quotient map  $\text{pr} : V \rightarrow V//H$ .

Then for  $x \in V(K)$ ,  $x$  is called a

- **$H$ -semi-simple element** if  $Hx$  is closed in  $V$  (or equivalently  $H(K)x \in V(K)$  is closed in the analytic topology  $\textcolor{red}{?}$ ).
- **$H$ -regular element** if  $\text{Stab}_H(x)$  has minimal dimension.  $\lrcorner$

**Prop. (29.1.1.8) [Relatively Semisimple Elements are Relatively Unimodular].** If  $G^0 \in \text{RedGrp}$  and  $X$  is affine, then every relatively semisimple element (10.2.1.18) is relatively unimodular, by (12.10.1.22) and (10.2.1.20).  $\lrcorner$

**Def. (29.1.1.9) [Local Relative Orbital Integrals].** For  $f \in \mathcal{T}(X(K))$  and a  $\chi$ -relevant relatively semi-simple (10.2.1.18) and relatively unimodular (29.1.1.3) element  $\gamma \in X(K)$ , define the **local relative orbital integral**

$$\text{ROrb}_\gamma(f) = \int_{H_\gamma(K) \backslash G(K)} f(\gamma g) \chi^{-1}(g) \frac{d\rho(g)}{d\rho(g_\gamma)} \quad (12.10.3.7).$$

**Prop. (29.1.1.10).** If  $I \leq H \in \text{AlgGrp}^{\text{sm}}/K$ , then the map  $\text{cl} : (I \backslash H)(K) \rightarrow \mathcal{D}(I, H; K)$  is continuous with discrete topology on  $\mathcal{D}(I, H; K)$ . In particular, the image of  $I(K) \backslash H(K)$  in  $(I \backslash H)(K)$  is closed.  $\lrcorner$

*Proof:*  $\textcolor{red}{?}$  Cf. [Getz]P449.  $\square$

**Prop. (29.1.1.11) [Convergence of  $\text{ROrb}_\gamma(f)$ ].** Situation as in (29.1.1.9), if  $H_\gamma$  is smooth, then  $\text{ROrb}_\gamma(f)$  is absolutely convergent.  $\lrcorner$

*Proof:* Because  $\frac{d\rho(h)}{d\rho(h_\gamma)}$  is a Radon measure (12.3.1.22), it suffices to show that the map  $G_\gamma(K) \backslash G(K) \rightarrow X(K)$  is proper (thus  $f(\gamma g)$  has compact support). It suffices to show that  $G_\gamma(K) \backslash G(K) \rightarrow X(K)$  is a closed embedding. For this, because  $O(\gamma)$  is closed, we may assume  $X = O(\gamma) = G_\gamma \backslash G$  (10.2.1.11). And then the assertion follows from (29.1.1.10).  $\square$

### Adelic Orbital Integrals

**Def. (29.1.1.12) [Adelic Orbital Integrals].**  $\text{Orb}^A(f) \textcolor{red}{?}$ .  $\lrcorner$

## 2 Spectral Expansions

**Def. (29.1.2.1)** [Automorphic Kernel Functions, Selberg[Sel56]]. For  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$ , the kernel function for the right action  $R(f)$  of  $f$  on  $L^2(G/F, \mathbf{1})$  is

$$\mathbf{K}_f(x, y) = \sum_{\gamma \in G(F)} f(x^{-1}\gamma y) \in C^\infty([G] \times [G]),$$

called the **automorphic kernel function** of  $f$ . ┘

*Proof:* For any  $\Phi \in L^2(G/F, \mathbf{1})$ ,

$$\begin{aligned} R(f)\Phi(x) &= \int_{A_G \backslash G(\mathbf{A}_F)} f(y)\varphi(xy)dy \\ &= \int_{A_G \backslash G(\mathbf{A}_F)} f(x^{-1}y)\varphi(y)dy \\ &= \int_{[G]} \left( \sum_{\gamma \in G(F)} f(x^{-1}\gamma y) \right) \varphi(y)dy \end{aligned}$$

Using unfolding, Cf.[G-H24]P244<sup>?</sup>.

To show that  $\mathbf{K}_f$  is smooth, notice that for  $\Omega_1, \Omega_2 \in A_G \backslash G(\mathbf{A}_F)$  and  $(x, y) \in \Omega_1 \times \Omega_2^{-1}$ , the only non-zero summands of  $\mathbf{K}_f(x, y)$  corresponds to the  $\gamma \in G(F) \cap \Omega_1 \text{Supp}(f)\Omega_2$ , which is finite.

To show that  $(y \mapsto \mathbf{K}_f(x, y)) \in L^2([G])$ , notice

$$\begin{aligned} \int_{[G]} \mathbf{K}_f(x, y) \overline{\mathbf{K}_f(x, y)} dy &= \int_{[G]} \left( \sum_{\gamma \in G(F)} f(x^{-1}\gamma y) \right) \left( \sum_{\gamma' \in G(F)} \overline{f(x^{-1}\gamma' y)} \right) dy \\ &= \sum_{\gamma \in G(F)} \int_{A_G \backslash G(\mathbf{A}_F)} f(x^{-1}\gamma y) \overline{f(x^{-1}y)} dy \end{aligned}$$

Using unfolding, Cf.[G-H24]P244<sup>?</sup>. It is clear that this is a finite sum of integrals, thus it is finite. □

**Prop. (29.1.2.2).** For  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$ , the restriction of the right action  $R(f)$  of  $f$  on  $L^2(\pi)$  for each  $\pi \in \text{Irr}^{\text{cusp, disc}}(G/F)$  has a kernel function  $\mathbf{K}_{\pi(f)}$ , and we have

$$\mathbf{K}_{f, \text{cusp}}(x, y) = \sum_{\pi \in \text{Irr}^{\text{cusp, disc}}(G/F)} \mathbf{K}_{\pi(f)}(x, y).$$

┘

**Cor. (29.1.2.3)** [Diagonal Spectral side]. For  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$ , we have

$$\sum_{\pi \in \text{Irr}^{\text{cusp, disc}}(G/F)} \int_{[G]} \mathbf{K}(x, x) dx = \sum_{\pi \in \text{Irr}^{\text{cusp, disc}}(G/F)} m(\pi) \text{tr}(\pi(f)).$$

But notice that this integral is rarely convergent. ┘

*Proof:* Cf.[Brislaw. Traceable integral kernels on countably generated measure spaces]Cor3.2.<sup>?</sup> □

**Def. (29.1.2.4) [Relative Traces].** For  $\pi \in \text{Irr}^{\text{auto}}(G/F, \mathbf{1})$  and  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$ , define the **relative trace** of  $f$  w.r.t.  $H$  and  $\chi$  (29.1.3.2) as

$$\text{Rtr}(\pi(f)) = \int_{A_{G,H}H(F) \backslash H(\mathbf{A}_F)} \mathbf{K}_{\pi(f)}(h_1, h_2) \chi^{-1}(h_1, h_2) d(h_1, h_2).$$

? What is the Haar measure on  $A_{G,H}H(F) \backslash H(\mathbf{A}_F)$ ? ┘

### Full Expansions

Using Langlands' decomposition of  $L^2([G])(\text{Cf. [G-H24]Chap10 ?})$ , we can give a full expansion of automorphic kernel functions:

**Thm. (29.1.2.5).** Fix a minimal parabolic subgroup  $P_0 \leq G$  with Levi decomposition  $P_0 = L_0 U_0$ , and for any standard parabolic subgroup  $P \leq G$ , assume that the Levi decomposition  $P = LU$  is chosen s.t.  $L_0 \leq L$ .

Then for any  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$ , the automorphic kernel function  $\mathbf{K}_f(x, y)$  (29.1.2.1) satisfies: For all  $x \in [G]$ ,

$$\mathbf{K}_f(x, y) = \sum_P n_P^{-1} \sum_{\sigma \in \text{Irr}^{\text{cusp, disc}}(L/F, \mathbf{1})} \int_{i(\mathfrak{a}_L^G)^*} \sum_{\Phi \in \text{Ob}(\sigma)} \text{Eis}(x, I(\sigma, \lambda)(f)\Phi, \lambda) \overline{\text{Eis}(y, \Phi, \lambda)} d\lambda.$$

where

- $P$  runs through all standard parabolic subgroups of  $G$ .
  - $\text{Ob}(\sigma)$  is an orthonormal basis for  $I_P^G(L_{\text{disc}}^2(L/F, \mathbf{1})[\sigma])$  consisting of vectors in the subspace  $I_P^G(L_{\text{disc}}^2(L/F, \mathbf{1})[\sigma])^0$  (Cf. [G-H24]P274 ?).
- ┘

*Proof:* ? Cf. [G-H24]Chap10 for notations, and Cf. [Arthur78. A trace formula for reductive groups. 1]Section4 and [Arthur. An introduction to the trace formula]Thm7.2. □

### Isolation of Cuspidal Spectra

References are [Beuzart-Plessis, Y. Liu, W. Zhang, and X. Zhu. Isolation of cuspidal spectrum, with application to the Gan-Gross-Prasad conjecture].

## 3 Relative Trace Formula

References are [On the Gross-Prasad conjecture for unitary groups, Jacquet-Rallis], [A guide to the relative trace formula, Jacquet].

**Notation (29.1.3.1).**

- Let  $F \in \text{NField}$ .
  - Let  $G \in \text{RedGrp}/F$ .
- ┘

### Simple Trace Formulae

**Situation (29.1.3.2).** Let  $H \leq G \times G$  be a subgroup s.t.  $H^0 = H_u \times H_r$  is a direct product of a reductive group and a unipotent group. In particular,  $H(\mathbf{A}_F)$  is unimodular.

Consider the restriction of the right action of  $G \times G$  on  $G$  to  $H$ .

Denote

$$A_{G,H} = H(\mathbf{A}_F) \cap (A_G \times A_G), \quad A = H(\mathbf{A}_F) \cap \Delta(A_G).$$

Let  $\chi : A_{G,H} H(F) \backslash H(\mathbf{A}_F) \rightarrow \mathbb{C}^\times$  be a quasi-character. ┘

**Situation (29.1.3.3).** Situation as in (29.1.3.2), assume that  $H_r(\mathbf{A}_F) = (H_r(\mathbf{A}_F) \cap A_{G^2}).(H_r(\mathbf{A}_F) \cap G^2(\mathbf{A}_F)^1)$ . ┘

**Cor. (29.1.3.4).** If assumption (29.1.3.3) holds, then

$$H(\mathbf{A}_F) = (H(\mathbf{A}_F) \cap A_{G^2}).(H(\mathbf{A}_F) \cap G^2(\mathbf{A}_F)^1).$$

┘

*Proof:* Cf. [G-H24]P476. □

### Geometric side

**Def. (29.1.3.5) [Elliptic Strongly Regular Semi-Simple Part of the Trace Formulae].** For  $f \in \mathcal{T}(G(\mathbf{A}_F))$  (26.6.1.11), the **elliptic strongly regular semi-simple part** of the geometric side of the trace formula for  $f$  is defined to be

$$\text{Orb}_{\text{s.reg,ell}}^{\mathbf{A}}(f) = \sum_{\gamma \in G(F)_{\text{s.reg,ell}} / \text{Ad}(G(F))} \tau(Z_G(\gamma)) \text{Orb}_{\gamma}^{\mathbf{A}}(f) \quad (29.1.1.12).$$

? Absolutely convergence. ┘

### General Trace Formula

**Thm. (29.1.3.6) [A Simple Trace Formulae].** Let  $f \in \mathcal{T}(A_G \backslash G(\mathbf{A}_F))$  s.t.  $R(f)$  has cuspidal image ? and assume that

- If the  $H(\mathbf{A}_F)$ -orbit of  $\gamma$  intersects  $\text{Supp}(f)$ , then  $\gamma$  is relatively elliptic, relatively unimodular, and the  $H_r$ -orbit of  $\gamma$  is closed.

Then

$$\sum_{\pi} \text{Rtr}(\pi(f)) = \sum_{[\gamma]} \tau(H_{\gamma}) \text{ROrb}_{\gamma}(f) \quad (29.1.2.4),$$

where

- the LHS is summation over  $(H, \chi)$ -distinguished cuspidal representation of  $G$ ,
- and the RHS is summation over relevant classes  $[\gamma]$  that are relatively unimodular, relatively elliptic, and s.t. the  $H_r$ -orbit of  $\gamma$  is closed.
- Both sides are absolutely convergent. ┘

## Petersson-Bruggeman-Kuznetsov trace formula

### Twisted Trace Formulae

#### 4 Selberg Trace Formulae

[Selberg] proved the case where  $G$  is semisimple and  $\Gamma \leq G(\mathbb{R})$  is compact.

Selberg's trace formula is a far-reaching non-commutative generalization of the Poisson summation formula(12.11.1.31). It underlines a duality between geometric and spectral objects.

**Notation(29.1.4.1).**

- Let  $G \in \text{LCG}$ .
- Let  $\Gamma \subset G$  be a discrete topological subgroup.

┘

### Compact Quotient case

**Prop.(29.1.4.2) [ $L^2(\Gamma \backslash G)$  Totally Decomposable].** Suppose  $G$  is unimodular and  $\Gamma \backslash G$  is compact,  $\chi \in \Gamma^\vee$ , then the space  $L^2(\Gamma \backslash G, \chi)$  decomposes as

$$L^2(\Gamma \backslash G, \chi) = \bigoplus_{\pi \in \widehat{G}} m_\pi V_\pi$$

that each  $m_\pi$  is finite.

┘

*Proof:* Let  $\Sigma$  be the set of sums of irreducible invariant subspaces of  $L^2(\Gamma \backslash G, \chi)$  that is mutually orthogonal. then choose by Zorn's lemma a maximal one in  $\Sigma$ , and we prove the orthogonal complement  $H = 0$  otherwise we construct an irreducible subspace of  $H$ .

Let  $f \neq 0 \in H$ , choose by(24.7.5.2) and(29.1.4.8) a  $\varphi \in C_c(G)$  that  $\rho(\varphi)$  is compact self-adjoint and  $\rho(\varphi)f \neq 0$ . So  $\rho(\varphi)$  has a non-zero eigenvalue and the eigenspace  $L$  is of f.d..

Let  $L_0$  be a minimal nonzero subspace of  $L$  that is an intersection of  $L$  with a nonzero closed invariant subspace of  $\mathcal{H}$ , and let  $V$  be the intersection of all closed invariant subspaces  $W$  of  $H$  that  $L_0 = L \cap W$ . We show  $V$  is irreducible, if not, then  $V = V_1 \cap V_2$ , and if  $0 \neq f_0 \in L_0$ , then  $f_0 = f_1 + f_2$  and both  $f_1, f_2$  are eigenfunctions of  $\rho(\varphi)$  of eigenvalue  $\lambda$ . Now if  $f_1 \neq 0$ , then by minimality,  $V_1 \cap L = L_0$ .

The finiteness of  $m_\pi$  follows from the fact that  $\rho(f)$  is Hilbert-Schmidt for every  $f \in C_c(G)$ (29.1.4.3). □

**Prop.(29.1.4.3) [Integral Operators].** If  $\Gamma \backslash G$  is compact, consider the right action of  $G$  on  $L^2(\Gamma \backslash G)$ (24.7.1.14), let  $\varphi \in C_c(G)$ , then  $\varphi$  can act on  $L^2(\Gamma \backslash G, \chi)$  by(12.8.3.24), and:

- $\rho(\varphi)$  is an integration operator, in particular Hilbert-Schmidt and compact.
- If  $\varphi(g^{-1}) = \overline{\varphi(g)}$ , then  $\rho(\varphi)$  is self-adjoint.

┘

*Proof:* 1:

$$(\rho(\varphi)f)(g) = \int_G f(h)\varphi(g^{-1}h)dh = \int_{\Gamma \backslash G} \left( \sum_{\gamma \in \Gamma} f(\gamma h)\varphi(g^{-1}\gamma h) \right) dh = \int_{\Gamma \backslash G} f(h)K_\varphi(g, h)dh$$

where

$$K_\varphi(g, h) = \sum_{\gamma \in \Gamma} \varphi(g^{-1}\gamma h).$$

Because  $\varphi$  is compactly supported, this is a smooth function in  $g$  and  $h$ , in particular square integrable on  $\Gamma \backslash G$  compact. And  $\rho(\varphi)(f)(g)$  is smooth in  $g$  because  $f \in L_1(\Gamma \backslash G, \chi)$  as  $\Gamma \backslash G$  is compact, and  $K(g, h)$  is smooth in  $g$ .

It is Hilbert-Schmidt and compact by ?

2 is easy. □

**Prop. (29.1.4.4) [Trace of  $\rho(\varphi)$ ].** If  $\varphi = \varphi_1 * \varphi_2$  where  $\varphi_i \in C_c(\Gamma \backslash G)$ , then  $\rho(\varphi) = \rho(\varphi_1)\rho(\varphi_2)$  (12.8.3.24) and hence a trace class (12.9.4.6). And its integral kernel is

$$K_\varphi(x, y) = \int_{\Gamma \backslash G} K_{\varphi_1}(x, z) K_{\varphi_2}(z, y) dz.$$

and

$$\text{tr} \rho(\varphi) = \int_{\Gamma \backslash G} K_\varphi(x, x) dx = \int_{\Gamma \backslash G} \int_{\Gamma \backslash G} K_{\varphi_1}(x, y) K_{\varphi_2}(y, x) dx dy.$$

┘

*Proof:* This follows from (12.9.4.9). □

**Cor. (29.1.4.5) [the Geometric Side of Trace Formula].** If  $G$  is unimodular, let  $c(\Gamma)$  be a set of representatives for the conjugacy classes of  $\Gamma$ , then

$$\text{tr}(\rho(\varphi)) = \sum_{\gamma \in c(\Gamma)} \int_{\Gamma \gamma \backslash G} \varphi(g^{-1}\gamma g) dg.$$

And if  $G_\gamma$  is unimodular for every  $\gamma \in \Gamma$ , then

$$\text{tr}(\rho(\varphi)) = \sum_{\gamma \in c(\Gamma)} \text{Vol}(\Gamma_\gamma \backslash G_\gamma) \int_{G_\gamma \backslash G} \varphi(g^{-1}\gamma g) dg.$$

┘

*Proof:*

$$K_f(x, x) = \sum_{\gamma \in \Gamma} f(x^{-1}\gamma x) = \sum_{\gamma \in c(\Gamma)} \sum_{\delta \in \Gamma_\gamma \backslash \Gamma} f(x^{-1}\delta^{-1}\gamma\delta x),$$

so we have

$$\int_{\Gamma \backslash G} K_f(x, x) dx = \int_{\Gamma \backslash G} \sum_{\gamma \in c(\Gamma)} \sum_{\delta \in \Gamma_\gamma \backslash \Gamma} f(x^{-1}\delta^{-1}\gamma\delta x) = \sum_{\gamma \in c(\Gamma)} \int_{\Gamma_\gamma \backslash G} f(x^{-1}\gamma x) dx.$$

□

**Cor. (29.1.4.6) [Selberg Trace Formula for  $\Gamma \backslash G$  Compact].** Suppose  $G$  be a unimodular,  $\Gamma \backslash G$  is compact,  $G_\gamma$  is unimodular for every  $\gamma \in \Gamma$ .

For any  $\varphi \in C_c(G)$  and  $\gamma \in \Gamma$ , denote  $\text{Orb}(\gamma, \varphi) = \int_{G_\gamma \backslash G} \varphi(g^{-1}\gamma g) dg$ , then  $\rho(\varphi)$  is a trace class and

$$\sum_{\pi \in \widehat{G}} m_\pi \text{tr}(\pi(\varphi)) = \text{tr}(\rho(\varphi)) = \sum_{\gamma \in c(\Gamma)} \text{Vol}(\Gamma_\gamma \backslash G_\gamma) \text{Orb}(\gamma, \varphi).$$

┘

*Proof:* Follows from (26.1.5.2) and (29.1.4.5). □

$G = \mathrm{GL}(2; \mathbb{R})$  case

**Lemma (29.1.4.7).** Let  $G = \mathrm{GL}(2; \mathbb{R})$ ,  $K = \mathrm{SL}(2; \mathbb{R})$ ,  $\Gamma \backslash G$  be compact,  $\rho$  be the principal series  $P(\lambda, 0)$  (24.7.5.20) of  $G$ , and  $s = \frac{1}{2}(s_1 - s_2 + 1)$ ,  $\lambda = s(1 - s)$ ,  $\mu = (s_1 + s_2)$ , then for any  $f \in C_c^\infty(K \backslash G/K)$ ,  $\rho(f) \in V_\rho^K$ , which has dimension 1, so  $f$  is a trace class and

$$\mathrm{tr}(\rho(f)) = \int \int f \left( \begin{bmatrix} e^{u/2} & x \\ 0 & e^{-u/2} \end{bmatrix} \right) e^{\frac{us}{2}} du dx.$$

┘

*Proof:* The trace of  $\rho(f)$  is just the scalar by which  $\rho(f)$  acts on a non-zero vector of  $\rho^K$ . Take  $\varphi \in \rho^K \subset H(s_1, s_2, 0)$  normalized that  $\varphi(I) = 1$ , then

$$(\rho(f)\varphi)(I) = \int_G f(g)\varphi(g)dg = \int_K \int_A \int_N f(ank)\varphi(ank)dAdNd\theta = \int_A \int_N f(an)\varphi(an)dAdN.$$

□

**Prop. (29.1.4.8).** Let  $G = \mathrm{GL}(2; \mathbb{R})$  and consider the right action of  $G$  on  $L^2(\Gamma \backslash G, \chi)$  (26.1.1.6), let  $\varphi \in C_c^\infty(G)$ , then  $\varphi$  can act on  $L^2(\Gamma \backslash G, \chi)$  by (12.8.3.24), and:

- $\rho(\varphi)$  is an integration operator, in particular Hilbert-Schmidt and compact. And  $\mathrm{Im}(\rho(\varphi)) \subset C^\infty(\Gamma \backslash G, \chi)$ .
- If  $\varphi(g^{-1}) = \overline{\varphi(g)}$ , then  $\rho(\varphi)$  is self-adjoint.
- If  $\varphi(k_\theta g) = e^{-tk\theta} \varphi(g)$ , then  $\mathrm{Im}(\rho(\varphi)) \subset C^\infty(\Gamma \backslash G, \chi, k)$ .

Compare with (29.1.4.3).

┘

*Proof:* These follows from (29.1.4.3).

□

**Cor. (29.1.4.9).** Let  $G = \mathrm{GL}(2; \mathbb{R})$  and  $\Gamma \subset G$  is a discrete subgroup that  $\Gamma \backslash G$  is compact. If  $\mathcal{H}$  be a nonzero closed  $G$ -subrepresentation of  $L^2(\Gamma \backslash G, \chi)$ , then  $\mathcal{H}$  decomposes as  $\oplus_k \mathcal{H}^k$  w.r.t. action of  $\mathrm{SO}(2; \mathbb{R})$ . And if  $\mathcal{H}^k \neq 0$ , then  $\Delta$  has a nonzero eigenvector in  $H^k \cap C^\infty(\Gamma \backslash G, \chi)$ .

┘

*Proof:* The decomposition is clear from (24.7.3.4). It's left to show  $\Delta$  has an eigenvalue in  $H_k \cap C^\infty(\Gamma \backslash G, \chi)$ . By lemma (24.7.5.2) above, for  $f_0 \in H^k$ , there is a  $\varphi \in C_c^\infty(G)$  s.t.  $\rho(\varphi)f_0 \neq 0$ , and  $\varphi(k_\theta g) = e^{-ik\theta} \varphi(g)$ . So (29.1.4.8) shows  $\rho(\varphi)$  maps  $H$  into  $H^k \cap C^\infty(\Gamma \backslash G, \chi)$  and induces a compact self-adjoint operator on  $H^k$ . So we can choose a f.d. eigenspace of it. Notice  $\Delta$  commutes with the action  $\rho(\varphi)$ , so  $\Delta$  fixes this eigenspace, thus it has an eigenvalue on  $H_k \cap C^\infty(\Gamma \backslash G, \chi)$ .

□

**Applications**

The Selberg trace formula for  $\mathrm{GL}(2; \mathbb{R})$  could be extended to the Hecke operators  $T_p$  on  $L^2(\Gamma \backslash \mathcal{H})$ , thus It yields a finite closed formula for the trace of any Hecke operator on  $\mathrm{SF}_{2k}(\Gamma)$ .

**5 Eichler-Selberg Trace Formulae**

Trace formula for acting on the space of sections of the automorphic line bundle.



## 6 Langlands Trace Formula

Langlands set about constructing continuous spectra for any locally symmetric space  $\backslash X$  of finite volume. The problem is to show that absolutely convergent Eisenstein series have analytic continuation to meromorphic functions, whose values at imaginary arguments exhaust the continuous spectrum.

More importantly, Langlands considered comparison of trace formulae, which leads to important applications.

### Jacquet-Langlands

The Jacquet-Langlands correspondence is a comparison of trace formulae between a non-compact modular curve and a compact modular curve, yielding a reciprocity law.

### Cyclic Base Change and Langlands-Tunnell

The cyclic base change is a comparison of trace formulae between a modular curve and a higher dimensional locally symmetric space  $\Gamma_E \backslash \mathcal{H}_E$  associated to a cyclic extension  $E/\mathbb{Q}$ , yielding a reciprocity law.

## 7 Arthur-Selberg Trace Formulae

A key difficulty in extending Selberg's work to arbitrary reductive groups  $G$  was proving the analytic continuation of Eisenstein series in general and using them to decompose  $L^2([G])$ .

Arthur considered for any reductive group  $G/\mathbb{Q}$  and  $f \in C_c^\infty(A_G \backslash G(\mathbf{A}_F))$  the convolution operator  $R(f)$  on the Hilbert space  $L^2([G])$ , which has an orthogonal decomposition

$$R(f) = R_{\text{disc}}(f) \oplus R_{\text{cont}}(f).$$

The Arthur trace formula is a formula for the trace of the operator  $R_{\text{disc}}(f)$  (Which is of trace class by Muller?). On the geometric side,

- The main terms are indexed by  $\mathbb{Q}$ -elliptic conjugacy classes in  $G(\mathbb{Q})$ .
- The hyperbolic terms are indexed by hyperbolic conjugacy classes in  $G(\mathbb{Q})$ , which are parametrized by elliptic conjugacy classes in Levi subgroups of  $G$ .
- The continuous spectra, which are parametrized by discrete spectra of Levi subgroups of  $G$ .

## 8 Relative Trace Formulae(Jacquet)

**Notation(29.1.8.1).**

- Let  $F \in \mathbf{GField}$ .
- Let  $G \in \mathbf{RedGrp}/F$  be anisotropic.
- Let  $H_1, H_2 \leq G \in \mathbf{AlgGrp}_{\text{cntd}}/F$ .
- Let  $\theta_i : H_i(F) \backslash H_i(\mathbf{A}_F) \rightarrow \mathbb{C}^\times$  be characters.

┘

**Lemma (29.1.8.2).** For  $f \in C_c(G(\mathbf{A}_F))$ , consider the integral

$$\begin{aligned}
 & \int_{H_1(F) \backslash H_1(\mathbf{A}_F)} \int_{H_2(F) \backslash H_2(\mathbf{A}_F)} K_f(h_1, h_2) \theta_1(h_1) \overline{\theta_2(h_2)} dh_2 dh_1 \\
 &= \sum_{\gamma \in H_1(F) \backslash G(F)} \int_{H_1(\mathbf{A}_F)} \int_{H_2(F) \backslash H_2(\mathbf{A}_F)} f(h_1^{-1} \gamma h_2) \theta_1(h_1) \overline{\theta_2(h_2)} dh_2 dh_1 \\
 & \textcolor{red}{?} = \sum_{\gamma \in H_1(F) \backslash G(F) / H_2(F)} \int_{H_{2,\gamma}(F) \backslash H_2(\mathbf{A}_F)} \int_{H_1(\mathbf{A}_F)} f(h_1^{-1} \gamma h_2) \theta_1(h_1) \overline{\theta_2(h_2)} dh_1 dh_2 \\
 &= \sum_{\gamma \in H_1(F) \backslash G(F) / H_2(F)} \text{Vol}(H_{2,\gamma}(F) \backslash H_{2,\gamma}(\mathbf{A}_F)) \int_{H_{2,\gamma}(\mathbf{A}_F) \backslash H_2(\mathbf{A}_F)} \int_{H_1(\mathbf{A}_F)} f(h_1^{-1} \gamma h_2) \theta_1(h_1) \overline{\theta_2(h_2)} dh_1 dh_2,
 \end{aligned}$$

where  $H_{2,\gamma} = \gamma^{-1} H_1 \gamma \cap H_2$ . And in this sum,  $\gamma$  contributes if  $\theta_1(\gamma h_2 \gamma^{-1}) = \theta_2(h_2)$  for any  $h_2 \in H_{2,\gamma}(\mathbf{A}_F)$ , in which case  $\gamma$  is said to be **relevant (in the trace formula)**.

On the other hand,

$$\begin{aligned}
 \rho(f) \varphi_1(g_2) &= \int_{G(F) \backslash G(\mathbf{A}_F)} K_f(g_1, g_2) \varphi_1(g_1) dg_1 \\
 &= \sum_{\pi \in \text{Irr}^{\text{auto}}(G/F)} \sum_{\varphi_2 \in \pi} \sum_{\text{orthogonal}} \int_{G(F) \backslash G(\mathbf{A}_F)} \int_{G(F) \backslash G(\mathbf{A}_F)} K_f(g_1, g_2) \varphi_1(g_1) \varphi_2(g'_2) dg'_2 \overline{\varphi_2(g'_2)} dg_1 \\
 &= \sum_{\pi \in \text{Irr}^{\text{auto}}(G/F)} \sum_{\varphi_2 \in \pi} \sum_{\text{orthogonal}} (K_f, \overline{\varphi_1} \times \overline{\varphi_2}) \overline{\varphi_2(g'_2)}
 \end{aligned}$$

So it follows that

$$\begin{aligned}
 K_f(g_1, g_2) &= \sum_{\pi \in \text{Irr}^{\text{auto}}(G/F)} \sum_{\varphi_1, \varphi_2 \in \pi} \sum_{\text{orthogonal}} (K_f, \overline{\varphi_1} \times \overline{\varphi_2}) \overline{\varphi_1(g_1) \varphi_2(g_2)} \\
 &= \sum_{\pi \in \text{Irr}^{\text{auto}}(G/F)} \sum_{\varphi \in \pi} \sum_{\text{orthogonal}} \overline{\varphi(g_1)} \rho(f) \varphi(g_2).
 \end{aligned}$$

Thus

$$\begin{aligned}
 & \int_{H_1(F) \backslash H_1(\mathbf{A}_F)} \int_{H_2(F) \backslash H_2(\mathbf{A}_F)} K_f(h_1, h_2) \theta_1(h_1) \overline{\theta_2(h_2)} dh_2 dh_1 \\
 &= \int_{H_1(F) \backslash H_1(\mathbf{A}_F)} \int_{H_2(F) \backslash H_2(\mathbf{A}_F)} \left( \sum_{\pi \in \text{Irr}^{\text{auto}}(G/F)} \sum_{\varphi \in \pi} \sum_{\text{orthogonal}} \overline{\varphi(h_1)} \rho(f) \varphi(h_2) \right) \theta_1(h_1) \overline{\theta_2(h_2)} dh_2 dh_1 \\
 &= \sum_{\pi \in \text{Irr}^{\text{auto}}(G/F)} \sum_{\varphi \in \pi} \sum_{\text{orthogonal}} \left( \int_{H_1(F) \backslash H_1(\mathbf{A}_F)} \overline{\varphi(h_1)} \theta_1(h_1) dh_1 \right) \left( \int_{H_2(F) \backslash H_2(\mathbf{A}_F)} \varphi(h_2) \overline{\theta_2(h_2)} dh_2 \right)
 \end{aligned}$$

□

**Prop. (29.1.8.3) [Relative Trace Formula].** If we denote

$$a(\gamma) = \text{Vol}(H_{2,\gamma}(F) \backslash H_{2,\gamma}(\mathbf{A}_F)),$$

$$I_\gamma(\theta_1, \theta_2; f) = \int_{H_{2,\gamma}(\mathbf{A}_F) \backslash H_2(\mathbf{A}_F)} \int_{H_1(\mathbf{A}_F)} f(h_1^{-1} \gamma h_2) \theta_1(h_1) \overline{\theta_2(h_2)} dh_1 dh_2,$$

and

$$P_{H_i, \theta_i}(\varphi) = \int_{H_i(F) \backslash H_i(\mathbf{A}_F)} \varphi(h_i) \overline{\theta_i(h_i)} dh_i,$$

$$I_\pi(\theta_1, \theta_2; f) = \sum_{\varphi \in \pi \text{ orthogonal}} P_{H_1, \theta_1}(\varphi) P_{H_2, \theta_2}(\rho(f)\varphi).$$

Then comparing the equations in (29.1.8.2) gives:

$$\sum_{\gamma \in H_1(F) \backslash G(F) / H_2(F) \text{ relevant}} a(\gamma) I_\gamma(\theta_1, \theta_2; f) = \sum_{\pi \in \widehat{G}} I_\pi(\theta_1, \theta_2; f)$$

┘

**Remark (29.1.8.4).** One can deduce the Selberg trace formula for  $G(F) \backslash G(\mathbf{A}_F)$  (29.1.4.6) from this relative trace formula. Cf. [Ike08].

┘

### Removing Assumptions

References are [On the fine spectral expansion of Jacquet's relative trace formula, Lapid].

## 9 Comparison of Trace Formulae

**Notation (29.1.9.1).**

- Let  $F \in \mathbf{GField}$ .
- Let  $G, G' \in \mathcal{RedGrp}/F$  be anisotropic.
- Let  $H_1, H_2 \leq G \in \mathcal{AlgGrp}_{\text{cntd}}/F$ ,
- Let  $H'_1, H'_2 \leq G' \in \mathcal{AlgGrp}_{\text{cntd}}/F$ ,
- Let  $\theta_i : H_i(F) \backslash H_i(\mathbf{A}_F) \rightarrow \mathbb{C}^\times$  be characters.
- Let  $\theta'_i : H'_i(F) \backslash H'_i(\mathbf{A}_F) \rightarrow \mathbb{C}^\times$  be characters.

┘

**Conj. (29.1.9.2) [Comparison of Trace Formulae].** Suppose there exists a bijection

$$\{ \gamma \in H_1(F) \backslash G(F) / H_2(F) \text{ relevant} \} \cong \{ \gamma' \in H'_1(F) \backslash G(F) / H'_2(F) \text{ relevant} \}$$

with the following properties:

- (matching) For any  $f \in C_c(G(\mathbf{A}_F))$ , there exist  $f' \in C_c(G'(\mathbf{A}_F))$  s.t.  $I_\gamma(\theta_1, \theta_2, f) = I_{\gamma'}(\theta'_1, \theta'_2, f')$ .
- (fundamental lemma) For a.e.  $v \in \Sigma_F$ , there exists an isomorphism of Hecke algebras

$$\mathcal{H}_{\mathcal{K}_v}(G_v) \cong \mathcal{H}_{\mathcal{K}'_v}(G'_v).$$

Then it's expected that there exists a correspondence of the  $L$ -packets of  $G(\mathbf{A}_F)$  and  $G'(\mathbf{A}_F)$  s.t.

$$I_\Pi^\kappa(\theta_1, \theta_2; f) = I_{\Pi'}^{\kappa'}(\theta'_1, \theta'_2; f')$$

where  $\Pi, \Pi'$  are  $L$ -packets of  $G(\mathbf{A}_F), G'(\mathbf{A}_F)$  resp.,  $\kappa, \kappa'$  are certain functions on  $\Pi$  and  $\Pi'$  resp. and

$$I_\Pi^\kappa(\theta_1, \theta_2; f) = \sum_{\pi \in \Pi} \kappa(\pi) I_\pi(\theta_1, \theta_2; f).$$

┘

*Proof:*

□

**Remark (29.1.9.3).** This would imply that there exists certain relation between period integrals for  $G(\mathbf{A}_F)$  and  $G'(\mathbf{A}_F)$ . And it is possible to deduce period formula for  $G(\mathbf{A}_F)$  from period formula for  $G'(\mathbf{A}_F)$ . ┘

## 10 Weyl Laws

Trace formulae are useful for proving asymptotic for the number of such forms as a function of their Laplacian eigenvalue.

**Thm. (29.1.10.1) [Weyl Law for Cusp Forms on Congruence Quotients, Lindenstrauss-Venkatesh].** Let ┘

## 29.2 Stable Trace Formulae

References are [Harris, stable trace formula], [Kottwitz84] and [Langlands-Shelstad87], [Kottwitz-Shelstad99], [Kal16, Kal24] and [Cohomologie, stabilisation et changement de base, Labesse1999]. [Global rigid inner forms and multiplicities of discrete automorphic representations, Kaletha18].

**Notation (29.2.0.1).**

- Use notations on automorphic representations defined in [Adelic Automorphic Representations](#).  $\lrcorner$

### Stably Invariant Distributions

**Def. (29.2.0.2) [Stably Invariant Distributions].** For  $v \in \Sigma_F$ ,  $I \in \text{Dist}(G(F_v))$  is called a **stably invariant distribution** if it satisfies: For any  $f \in \mathcal{T}(G(F_v))$ , if  $\text{SO}_{\text{rb}}(f) = 0$  [\(29.2.4.6\)](#) for any strongly-regular semi-simple element  $\gamma \in G(F_v)$  [\(10.5.1.17\)](#), then we have  $I(f) = 0$ .  $\lrcorner$

**Prop. (29.2.0.3).** For  $v \in \Sigma_F$ , if  $I \in \text{Dist}(G(F_v))$  is represented by a function  $\phi$ , then  $I$  is stably invariant [\(29.2.0.2\)](#) if  $\phi$  is stably invariant.  $\lrcorner$

*Proof:*  $\color{red}{?}$  Use Weyl integration formula.  $\square$

**Prop. (29.2.0.4) [Fourier Transform of Stably Invariant Distributions, Waldspurger].** The Fourier transform of a stably invariant distribution [\(29.2.0.2\)](#) is also stably invariant.  $\lrcorner$

*Proof:* Cf. [Waldspurger. Le lemme fondamental implique le transfert].  $\square$

## 1 Endoscopy

References are [\[Kot84\]](#), [Scholze-Shin], and

**Notation (29.2.1.1).**

- Let  $F \in \mathbf{NField} \cup \mathbf{LField}^0$ ,
- Let  $G \in \mathbf{RedGrp}/F$  s.t.  $\pi_1([G, G]) = 1$ .  $\lrcorner$

**Def. (29.2.1.2) [ $\mathfrak{R}(I_0/F)$ ].** For  $F \in \mathbf{NField}$ ,  $G \in \mathbf{RedGrp}/F$  and  $\gamma_0 \in G(F)$  semi-simple elliptic, denote  $I_0 = Z_G(\gamma_0)^0 \in \mathbf{RedGrp}/F$  [\(10.3.3.5\)](#). Then there is an exact sequence

$$1 \rightarrow Z(\widehat{G}) \rightarrow Z(\widehat{I_0}) \rightarrow Z(\widehat{I_0})/Z(\widehat{G}) \rightarrow 1,$$

inducing a map  $\pi_0 \left( Z(\widehat{I_0})/Z(\widehat{G}) \right)^{\text{Gal}_F} \rightarrow H^1(F, Z(\widehat{G}))$ . Notice  $\left( Z(\widehat{I_0})/Z(\widehat{G}) \right)^{\text{Gal}_F}$  is finite because  $X^\bullet(Z(\widehat{I_0})^{\text{Gal}_F}) = \pi_1(I_0)_{\text{Gal}_F}$  is finite as a quotient of  $X_\bullet(T)_{\text{Gal}_F}$ .

We define  $\mathfrak{R}(\gamma_0)$  to be the set of elements in  $\pi_0 \left( Z(\widehat{I_0})/Z(\widehat{G}) \right)^{\text{Gal}_F}$  whose image under this map belongs to  $\text{III}(Z(\widehat{G})/F)$ . Equivalently, by Chebotarev density theorem,

$$\mathfrak{R}(I_0/F) = \left[ \bigcap_{v \in \Sigma_F} \left( Z(\widehat{I_0})^{\text{Gal}_{F_v}} Z(\widehat{G}) \right) \right] / Z(\widehat{G}).$$

In particular,  $\#\mathfrak{K}(I_0/F) < \infty$ , and there is an exact sequence

$$1 \rightarrow \pi_0(Z(\widehat{G})^{\text{Gal}_F}) \rightarrow \pi_0(Z(\widehat{I}_0)^{\text{Gal}_F}) \rightarrow \mathfrak{K}(I_0/F) \rightarrow \text{III}(Z(\widehat{G})/F) \rightarrow \text{III}(Z(\widehat{I}_0)/F).$$

where the first injection because  $\gamma$  is elliptic implies  $X_\bullet(Z(\widehat{I}_0)/Z(\widehat{G}))^{\text{Gal}_F} = 0$ ?  $\lrcorner$

*Proof:* ? Cf.[Kot84]Cor2.3.  $\square$

**Prop. (29.2.1.3)[Kottwitz].** Situation as in(29.2.1.2). Suppose  $\gamma = (\gamma_v) \in G(\mathbf{A}_F)$  s.t.  $\gamma_v \sim_{\text{st}} \gamma_0$  for all  $v \in \Sigma_F$ , then the  $G(\mathbf{A}_F)$ -conjugacy class of  $\gamma$  contains some element of  $G(F)$  iff

$$\sum_{v \in \Sigma_F} \langle \text{inv}(\gamma_0, \gamma), - \rangle|_{\mathfrak{K}(I_0/F)} = 0 \text{ (10.5.1.6),}$$

where the paring is induced by the Tate-Nakayama duality(? Cf.[Kot84]). And in this case, there are

$$\#\mathfrak{K}(I_0/K)\tau(G)\tau(I_0)^{-1}$$

many  $G(F)$ -conjugacy classes of elements in  $G(F)$  in the  $G(\mathbf{A}_F)$ -conjugacy class of  $\gamma$ .  $\lrcorner$

*Proof:* Cf.[Kot86]Prop6.6, and section9.?  $\square$

### Endoscopic Triples

**Def. (29.2.1.4)[Endoscopic Triples].** An **endoscopic triple** of  $G/F$  is a triple  $(H, s, \eta)$  where

- $H \in \mathcal{R}\text{edGrp}/F$  is a quasi-split reductive group.
- $\eta : \widehat{H}_{\mathbb{C}} \rightarrow \widehat{G}_{\mathbb{C}} \in \mathcal{R}\text{edGrp}/\mathbb{C}$  is an embedding.
- $s \in \left(Z(\widehat{H}_{\mathbb{C}})/Z(\widehat{G}_{\mathbb{C}})\right)^{\text{Gal}_F}$ .

s.t.

- $\eta$  identifies  $\widehat{H}_{\mathbb{C}}$  with  $Z_{\widehat{G}_{\mathbb{C}}}(s)^0$ .
- The conjugacy class of  $\eta$  is defined over  $F$ , i.e. for any  $\sigma \in \text{Gal}_F$ , there exists  $x_\sigma \in \widehat{G}(\mathbb{C})$  s.t.  $\sigma_{\widehat{G}} \circ \eta \circ \sigma_{\widehat{H}}^{-1} = \text{ad}(x_\sigma)(\eta)$ . In particular,  $\eta$  extends to a homomorphism  ${}^L\eta : {}^LH_{\mathbb{C}} \rightarrow {}^LG_{\mathbb{C}}$  commuting with  $s$ (Cf.[Langlands. Stable conjugacy: definitions and lemmas]Prop1?).
- The image of  $s$  in  $H^1(F, Z(\widehat{G}))$  via the connecting homomorphism associated to the exact sequence  $1 \rightarrow Z(\widehat{G}) \rightarrow Z(\widehat{H}) \rightarrow Z(\widehat{G})/Z(\widehat{H})$  is contained in  $\text{III}(Z(\widehat{G})/F)$ .

And  $(H, s, \eta)$  is called an **elliptic endoscopic triple** if  $\left(Z(\widehat{H})^{\text{Gal}_F}\right)^0 = \left(Z(\widehat{G})^{\text{Gal}_F}\right)^0$ .

Let  $\mathfrak{E}(G/F)$ (resp.  $\mathfrak{E}_{\text{ell}}(G/F)$ ) be a fixed set of representatives of the isomorphism classes of (elliptic)endoscopic triples of  $G$  over  $F$ .  $\lrcorner$

**Remark (29.2.1.5)[Extended Endoscopic Triples].** In general if  $\pi_1([G, G]) \neq 1$ , then an extension  $\eta : {}^LH_{\mathbb{C}} \rightarrow {}^LG_{\mathbb{C}}$  may not exist, and endoscopic triples with such an extension is called an **extended endoscopic triples**. Let  ${}^L\mathfrak{E}(G/F)$ (resp.  ${}^L\mathfrak{E}_{\text{ell}}(G/F)$ ) be a fixed set of representatives of the isomorphism classes of extended (elliptic)endoscopic triples of  $G$  over  $F$ .  $\lrcorner$

**Def. (29.2.1.6)[Unramified Endoscopic Triples].** If  $F = K \in p\text{-NField}$  and  $G \in \mathcal{R}\text{edGrp}/K$ , then  $(H, s, \eta) \in \mathfrak{E}(G/K)$  is called an **unramified endoscopic triple** if  $H$  is unramified and  $\sigma_{\widehat{G}} \circ \eta \circ \sigma_{\widehat{H}}^{-1} = \eta$  for any  $\sigma \in I_K$ .  $\lrcorner$

**Prop. (29.2.1.7)[Outer Automorphic Groups of Endoscopic triples].** Situation as in (29.2.1.4), a isomorphism between two endoscopic triples  $(H, s, \eta), (H', s', \eta')$  is a pair  $(\alpha, g)$  where  $\alpha : H' \cong H \in \text{RedGrp}/F$  is an isomorphism s.t.

- $\eta \circ \alpha = \text{Ad}(g) \circ \eta'$ .
- Conjugation by  $g$  as in item1 takes the image of  $s'$  in  $\pi_0 \left( \left( Z(\widehat{H}')/Z(\widehat{G}) \right)^{\text{Gal}_F} \right)$  to the image of  $s$  in  $\pi_0 \left( \left( Z(\widehat{H})/Z(\widehat{G}) \right)^{\text{Gal}_F} \right)$ .

Then  $H_{\text{ad}}(F) \trianglelefteq \text{Aut}((H, s, \eta))$ , and define the **outer automorphism group**

$$\text{OAut}((H, s, \eta)) = \text{Aut}((H, s, \eta))/H_{\text{ad}}(F) \leq \text{OAut}(H).$$

┘

**Def. (29.2.1.8)[ $i(G, H)$ ].** For each  $(H, s, \eta) \in \mathfrak{E}(G/F)$  (29.2.1.4), denote

$$i(G, H) = \tau(G)\tau(H)^{-1} \# \text{OAut}((H, s, \eta)) \text{ (29.2.1.7) (20.5.4.8)}.$$

┘

**Example (29.2.1.9) [Endoscopic Triples for  $\text{SL}(2)$ ].** For  $G = \text{SL}(2)$ , there are three isomorphism classes of endoscopic triples:

- The trivial endoscopic triple  $(G, 1, \text{id})$ .
- The elliptic endoscopic triple  $(T, s, \eta)$  where  $T$  is an anisotropic torus of dimension 1,  $\eta : \mathbb{C}^\times \rightarrow \text{PGL}(2; \mathbb{C})$  is the diagonal torus  $\text{diag}(x, 1)$ , and  $s = -1$ .
- The non-elliptic endoscopic triple  $(\mathbb{G}_m, s, \eta)$  where  $\eta : \mathbb{C}^\times \rightarrow \text{PGL}(2; \mathbb{C})$  is the diagonal torus  $\text{diag}(x, 1)$ , and  $s \in \text{PGL}(2; \mathbb{C}) \setminus \{1\}$ .

┘

*Proof:* ?

□

**Def. (29.2.1.10)[Admissible Isomorphisms].** For  $(H, s, \eta) \in \mathfrak{E}(G/F)$ , take a maximal torus  $T_H \leq H$ , then we may take a maximal torus  $T \leq g$  s.t.  $\eta(\widehat{T}_H) = \widehat{T}$ , and we get an isomorphism  $\varphi : T_H \rightarrow T$ . For any maximal torus  $T'_H \leq H$  and a maximal torus  $T' \leq G$ , an **admissible isomorphism** between  $T'_H$  and  $T'$  is an isomorphism  $T'_H \rightarrow T'$  which is a composition of  $\varphi$  with  $\text{Ad}(h) : T'_H \cong T_H$  and  $\text{Ad}(g) : T \cong T'$  for some  $h \in H(F^{\text{sep}})$  and  $g \in G(F^{\text{sep}})$ .

┘

**Thm. (29.2.1.11).** If  $G$  is quasi-split and  $(H, s, \eta) \in \mathfrak{E}(G/F)$ , then for any maximal torus  $T_H \leq H$ , there exists an admissible isomorphism  $\varphi : T_H \rightarrow T$  (29.2.1.10) for some maximal torus  $T \leq G$ .

┘

*Proof:* Cf. [Kot82] Cor2.2. ?

□

**Def. (29.2.1.12) [Related Semi-Simple Elements].** For  $(H, s, \eta) \in \mathfrak{E}(G/F)$ , two elements  $\gamma_H \in H(F)_{\text{ss}}, \gamma \in G(F)_{\text{ss}}$  are called **associated semi-simple elements** if there exists an admissible isomorphism  $\varphi : T_H \rightarrow T$  between maximal tori of  $H$  and  $G$  (29.2.1.10) mapping  $\gamma_H$  to  $\gamma$ .

Notice when  $\gamma$  is strongly regular, such an admissible isomorphism is unique if it exists (because  $Z_G(\gamma)$  is commutative), denoted by  $\varphi_{\gamma_H, \gamma}$ .

┘

**Def. (29.2.1.13)[ $G$ -Regular Semi-Simple Elements].** For  $(H, s, \eta) \in \mathfrak{E}(G/F)$ , a semi-simple element  $\gamma_H \in H(F)$  is called a  **$G$ -regular element** if it is related to a regular semi-simple element  $\gamma \in G(F)$ . And it is called a **strongly  $G$ -regular element** if it is related to a strongly regular semi-simple element  $\gamma \in G(F)$ .

┘

**Def. (29.2.1.14) [(G, H)-Regular Elements].** Situation as in (29.2.1.12), the  $\varphi(T_H) \leq G_{F^{\text{sep}}}$  is a maximal torus, so we get a map  $j^* : \Phi(G, T) \rightarrow X^\bullet(T_H)$ . We call  $\gamma_H$  an **(G, H)-regular element** if  $\alpha(\gamma_H) \neq 1$  whenever  $\alpha \in \Phi(G, T) \setminus (\varphi^*)^{-1}\Phi(H, T_H)$ . This notion doesn't depend on  $T_H$  and  $\varphi$  chosen?  $\lrcorner$

*Proof:* Cf. [Kot86]P378?  $\square$

**Prop. (29.2.1.15).** Situation as in (29.2.1.10), suppose  $\gamma_H \in H(F^{\text{sep}})$  is semi-simple elliptic and  $(G, H)$ -regular (29.2.1.14), and  $\gamma_0 \in G(F)$  comes from  $\gamma_H$ . Denote  $I_0 \triangleq Z_G(\gamma_0)^0 \in \text{RedGrp}/F$  and  $I_{\gamma_H} \triangleq Z_H(\gamma_H)^0 \in \text{RedGrp}/F^{\text{sep}}$  (10.3.3.5), then  $\varphi$  extends to an isomorphism  $I_{\gamma_H} \rightarrow (I_0)_{F^{\text{sep}}}$ , unique up to conjugacy by  $\varphi(T_H)$ ?  $\lrcorner$

*Proof:* Cf. [Kot86]Section3.  $\square$

### Langlands-Shelstad Transfer for Quasi-Split Groups

References are [L-S87].

**Def. (29.2.1.16) [a-Data, Langlands-Shelstad [L-S87]].**  $\lrcorner$

**Def. (29.2.1.17) [ $\chi$ -Data, Langlands-Shelstad [L-S87]/Kottwitz-Shelstad [K-S99]].**  $\lrcorner$

**Def. (29.2.1.18) [Langlands-Shelstad Transfer Factors for Quasi-Split Groups, Langlands-Shelstad [L-S87]/Kottwitz-Shelstad/Kaletha].** Suppose  $K \in \text{LField}$  and  $G \in \text{RedGrp}/K$  is quasi-split. Given  $(H, s, {}^L\eta) \in {}^L\mathfrak{E}(G/K)$  and a Whittaker datum  $\mathfrak{m}$  (10.4.4.23), there is a function

$$\Delta^{\mathfrak{m}, H, G} = \epsilon \cdot \Delta_I^{-1} \cdot \Delta_{II} \cdot \Delta_{III_2} \cdot \Delta_{IV} : H(K)_{\text{s.reg}} \times G(K)_{\text{s.reg}} \rightarrow \mathbb{C} \text{ (10.5.1.17),}$$

called the **transfer factor** between  $H$  and  $G$  s.t.  $\Delta(\gamma_H, \gamma) \neq 0$  iff  $\gamma_H$  and  $\gamma$  are related (29.2.1.12), and in which case it is defined as follows: Choose a pinning  $(T_0, B_0, \{X_\alpha\}_{\alpha \in \Delta_0})$  and  $1 \neq \psi_K \in (K^\times)^\vee$  s.t. the generic unipotent character  $\psi : (B_0)_u \rightarrow \mathbb{C}^\times$  determined by these choices belongs to  $\mathfrak{m}$ .

- $\epsilon = w(V, \psi_K)$  is the root number, where  $V$  is the degree 0 virtual Galois-representation  $V = X_\bullet(T_0)_{\mathbb{C}} - X_\bullet(T_0^H)_{\mathbb{C}}$ , where  $T_0^H$  is a maximally-split maximal torus of  $H$ .
- $\Delta_I(\gamma_H, \gamma) = \left\langle \lambda, \widehat{\varphi}_{\gamma_H, \gamma}^{-1}(s) \right\rangle$  (29.2.1.12), where  $\lambda \in H^1(F, T_{\gamma, \text{sim.cntd}})$  is the Langlands-Shelstad splitting invariant?,  $\widehat{\varphi}_{\gamma_H, \gamma}^{-1}(s) \in \left( \widehat{T}_\gamma / Z(\widehat{G}) \right)^{\text{Gal}_K}$  and the pairing is given by Tate-Nakayama duality?.
- $\Delta_{II}(\gamma_H, \gamma) = \prod_\alpha \chi_\alpha \left( \frac{\alpha(\gamma)-1}{a_\alpha} \right)$  (29.2.1.16)(29.2.1.17), where  $\alpha$  runs through the  $\text{Gal}_F$ -orbit of  $\Phi(G, T_\gamma) \setminus \left( \varphi_{\gamma_H, \gamma}^* \right)^{-1} \Phi(H, T_{\gamma_H})$ ,
- $\Delta_{III_2}(\gamma_H, \gamma) = \theta(\gamma)$ , where  $\theta \in (T_\gamma(F))^\vee$  is constructed as follows:?
- $\Delta_{IV}(\gamma_H, \gamma) = \prod_{\alpha \in \Phi(G, T_\gamma) \setminus \Phi(H, T_{\gamma_H})} |\alpha(\gamma) - 1|^{1/2}$ .

*Proof:* ? Cf. [Kal24]P21.  $\square$



**Cor. (29.2.1.19)**[ $\kappa$ -Behavior of Transfer Factors]. Situation as in (29.2.1.18), if  $\gamma, \gamma'$  are stably conjugate, then

$$\Delta^{\mathfrak{m}, H, G}(\gamma_H, \gamma') = \Delta^{\mathfrak{m}, H, G}(\gamma_H, \gamma) \left\langle \text{inv}(\gamma, \gamma'), \widehat{\varphi}_{\gamma_H, \gamma}^{-1}(s) \right\rangle \quad (10.5.1.6).$$

In particular,  $\Delta^{\mathfrak{m}}(\gamma_H, \gamma)$  only depends on the stable-conjugacy class of  $\gamma_H$  and the conjugacy class of  $\gamma$ .  $\lrcorner$

*Proof:* Cf. [Kal24]  $\color{red}?$ .  $\square$

**Thm. (29.2.1.20)**[Normalization of Transfer Factors]. When  $F \in \mathbf{NField}$ , for each elliptic extended endoscopic triple  $(H, s, {}^L\eta) \in {}^L\mathfrak{E}_{\text{ell}}(G/F)$  (29.2.1.5), we can normalize the Langlands-Shelstad transfer factor  $\Delta_v$  (29.2.1.18) for each  $v \in \Sigma_F$  s.t. for each pair of related (29.2.1.12) strongly regular semi-simple elliptic elements  $\gamma_H \in H(F)$  and  $\gamma \in G(F)$ , we have

$$\prod_{v \in \Sigma_F} \Delta_v(\gamma_H, \gamma) = \left\langle \text{inv}(\gamma_H, \gamma), \widehat{\varphi}_{\gamma_H, \gamma}^{-1}(s) \right\rangle \quad (29.2.1.12)(10.5.1.6),$$

where the pairing is given by Tate-Nakayama duality  $\color{red}?$ .  $\lrcorner$

*Proof:* Cf. [Kal24] P50  $\color{red}?$ .  $\square$

**Prop. (29.2.1.21)**[Transfer Factors for Unramified Groups, Hales]. Cf. [Hales, A simple definition of transfer factors for unramified groups]  $\color{red}?$ . Hales gave explicit formula for the transfer factor, because in this case  $a$ -data and  $\chi$ -data can be chosen more or less canonically in this case  $\lrcorner$

*Proof:*  $\color{red}?$   $\square$

**Thm. (29.2.1.22)** [Langlands-Shelstad Transfer Conjecture, Shelstad82/Bouaziz94/ Waldspurger97/Ngô [Ngô10]]. For  $K \in \mathbf{LField}$  and  $G \in \mathbf{RedGrp}/K$ , suppose

- $(H, s, {}^L\eta) \in {}^L\mathfrak{E}(G/K)$  (29.2.1.4),
- A normalization of transfer factors  $\Delta_{H, G}^{\mathfrak{m}}$  between  $H$  and  $G$  is fixed as in (29.2.1.18),
- Haar measures on  $G(K)$  and  $H(K)$  are fixed,

then for any  $f \in \mathcal{T}(G(K))$ , there exists  $f^H \in \mathcal{T}(H(K))$  s.t. for any strongly  $G$ -regular semi-simple element  $\gamma_H \in H(K)$  (29.2.1.13), we have

$$\text{SOrb}_{\gamma_H}(f^H) = \sum_{\gamma \in G(K)_{\text{s.reg}}/\text{Ad}(G(K))} \Delta_{H, G}^{\mathfrak{m}}(\gamma_H, \gamma) \text{Orb}_{\gamma}(f) = \Delta_{H, G}^{\mathfrak{m}}(\gamma_H, \gamma_0) \text{SOrb}_{\gamma_0}^{\widehat{\varphi}_{\gamma_H, \gamma_0}^{-1}(s)}(f),$$

which is a finite sum, where

- $\gamma_0 \in G(K)_{\text{s.reg}}/\text{Ad}(G(K))$  is an arbitrary element.
- $\Delta_{H, G}^{\mathfrak{m}}(\gamma_H, \gamma) = 0$  unless  $\gamma_H$  and  $\gamma$  are related (29.2.1.18), in which case we are using Haar measures on  $Z_G(\gamma)$  and  $Z_H(\gamma_H)$  that are compatible via the unique admissible isomorphism  $\varphi_{\gamma_H, \gamma}$  (29.2.1.12) to define the orbital integrals.

•

$$\text{SOrb}_{\gamma_0}^{\kappa}(f) \triangleq \sum_{\substack{\gamma \in G(K)_{\text{s.reg}}/\text{Ad}(G(K)) \\ \gamma \sim_{\text{st}} \gamma_0}} \langle \text{inv}(\gamma_0, \gamma), \kappa \rangle \text{Orb}_{\gamma}(f).$$

Any such  $f^H$  is called a **Langlands-Shelstad transfer** of  $f$  to  $H$ .  $\lrcorner$

*Proof:* In the real case this is proven by [Shelstad,  $L$ -indistinguishability for real groups] and [Bouaziz, Integrales orbitales sur les groupes de Lie reductifs].

In the  $p$ -adic case this follows from (29.2.3.7)(29.2.3.6) and (29.2.3.5).

The second identity follows from (29.2.1.19). This sum is finite by (10.5.1.6) and the fact  $\#H^1(F, T_{\gamma_0}) < \infty$  (10.5.2.9).  $\square$

**Remark (29.2.1.23).** The stable orbital integral of  $f^H$  at regular elements can be expressed in terms of the stable orbital integrals at strongly regular semi-simple elements by continuity, Cf. [LS87]Section 4.3.. And the stable orbital integral of  $f^H$  at singular elements can be related to the stable orbital integrals at regular elements, Cf. [Kot88]Section 3.  $?$   $\lrcorner$

**Thm. (29.2.1.24) [Fundamental Lemma, Ngô [Ngô10]].** Situation as in (29.2.1.22), if

- $K \in p\text{-NField}$ ,
- $G$  is unramified and  $(H, s, {}^L\eta)$  is unramified (29.2.1.6),
- $\mathcal{K} \leq G(K), \mathcal{K}_H \leq H(K)$  are hyperspecial maximal subgroups,
- The Haar measures on  $G(K)$  and  $H(K)$  are neutralized at  $\mathcal{K}, \mathcal{K}_H$  resp.,
- The transfer factor  $\Delta$  between  $H$  and  $G$  is canonically normalized as in (29.2.1.21).

then  $\mathbf{1}_{\mathcal{K}_H}$  is a Langlands-Shelstad transfer of  $\mathbf{1}_{\mathcal{K}}$ .  $\lrcorner$

*Proof:*  $?$  Cf. [ngo10]. This was reduced from the case of groups to the case of Lie algebras via a Harish-Chandra descent argument. The Lie algebra version was proved by Ngô for local fields of Laurent series, and then transferred to characteristic zero.  $\square$

**Cor. (29.2.1.25) [Adelic Langlands-Shelstad Transfer].** An analogue of (29.2.1.22) holds over  $\mathbf{A}_f^p$ , where the adelic  $(G, H)$ -regular elements are defined in [Kot90]P178  $?$ , and the transfer factors between  $H$  and  $G$  are suitably defined  $?$ .  $\lrcorner$

### Endoscopy over $\mathbb{R}$

**Def. (29.2.1.26).** For  $(H, s, \eta) \in \mathfrak{E}_{\infty}(G)$  and  $\lrcorner$

**Def. (29.2.1.27) [Elliptic Tori of  $G_{\mathbb{R}}$  and  $H_{\mathbb{R}}$ ].** We say an elliptic torus of  $G_{\mathbb{R}}$  comes from  $H_{\mathbb{R}}$  if  $?$   $\lrcorner$

### Unstable Cyclic Base Changes

**Thm. (29.2.1.28) [Fundamental Lemma for Unstable Cyclic Base Changes].** For  $(H, s, \eta) \in \mathfrak{E}_p(G)$  (29.2.1.4), suppose

- $(H, s, \eta)$  is unramified (29.2.1.6),
- $s \in Z(\hat{H})^{\text{Gal}_{\mathbb{Q}_p}}$ .

Let  $R = \text{Res}_{\mathbb{Q}_{p^n}/\mathbb{Q}_p}(G)$ , then  $\lrcorner$

## 2 Endoscopy for Non-Quasi-Split Groups

### Pure Inner Forms

**Def. (29.2.2.1) [Pure Refined Extended Endoscopic Triples].**  $?$

Let  ${}^L\mathfrak{E}(G/F)$  (resp.  ${}^L\mathfrak{E}_{\text{ell}}(G/F)$ ) be a fixed set of representatives of the isomorphism classes of (elliptic) pure refined extended endoscopic triples of  $G$  over  $F$ .  $\lrcorner$

**Def. (29.2.2.2) [Transfer Factors for Pure Inner Forms, Kal11].** Suppose  $K \in \text{LField}$ ,  $G^* \in \text{RedGrp}/K$  is quasi-split with a Whittaker datum  $\mathfrak{m}$ . For any pure refined extended endoscopic triple  $(H, \dot{s}, {}^L\eta)$  for  $G^*/K$  (29.2.2.1) and a pure inner twist  $(G, \xi, z)$  of  $G^*$  (10.2.5.11), define the transfer factor

$$\Delta^{\mathfrak{m}, H, z, G} : H(K)_{\text{s.reg}} \times G(K)_{\text{s.reg}} \rightarrow \mathbb{C}^\times$$

as follows: If  $\gamma_H \leq H(K)_{\text{s.reg}}$  and  $\gamma \in G(K)_{\text{s.reg}}$  are not related (29.2.1.12), then  $\Delta^{\mathfrak{m}, H, z, G}(\gamma_H, \gamma) = 0$ . And if  $\gamma_H, \gamma$  are related, then there exists  $\gamma^* \in G^*(K)$  and  $g \in G^*(K^{\text{sep}})$  s.t.  $\gamma = \xi(g\gamma^*g^{-1})$  by Steinberg's theorem (Cf. [Kot82] Section 2). Then  $\gamma_H$  and  $\gamma^*$  are related. And it can be verified that  $(g^{-1}z(\sigma)\sigma(g))_{\sigma \in \text{Gal}_K}$  is a 1-cocycle of  $\text{Gal}_K$  in  $Z_{G^*}(\gamma^*)$  whose cohomology class is denoted by  $\text{inv}[z](\gamma^*, \gamma) \in H^1(K, Z_{G^*}(\gamma^*))$ .

Then we define

$$\Delta^{\mathfrak{m}, H, z, G}(\gamma_H, \gamma) = \Delta^{\mathfrak{m}, H, G}(\gamma_H, \gamma^*) \left\langle \text{inv}[z](\gamma^*, \gamma), \widehat{\varphi}_{\gamma_H, \gamma^*}^{-1}(\dot{s}) \right\rangle \quad (29.2.1.12).$$

where the pairing is via the Tate-Nakayama isomorphism (10.5.2.2).

It can be checked that  $\Delta^{\mathfrak{m}, H, z, G}$  is indeed a transfer factor by [Kaletha, Endoscopic character identities for depth-zero supercuspidal L-packets] Prop 2.2.2.  $\textcolor{red}{?}$   $\lrcorner$

*Proof:*

$\square$

### Isocrystals

**Def. (29.2.2.3) [Transfer Factors via Isocrystals].**  $\Delta^{\mathfrak{m}, H, b}$ . Cf. [Kaletha, Supercuspidal L-packets via isocrystals].  $\textcolor{red}{?}$   $\lrcorner$

### Rigid Inner Forms

**Def. (29.2.2.4).**  $\lrcorner$

**Def. (29.2.2.5) [Transfer Factors via Rigid Inner Forms].**  $\Delta^{\mathfrak{m}, H, z, G}$ .

It can be checked that  $\Delta^{\mathfrak{m}, H, z, G}$  is indeed a transfer factor by [Kal16] Section 5.3.  $\textcolor{red}{?}$  And it is invariant under automorphisms of  $(H, \dot{s}, \eta)$ .  $\lrcorner$

## 3 Endoscopy for Lie Algebras

References are [Waldspurger, Une formule des traces locale pour les algebres de Lie p-adiques] and [Waldspurger, Le lemme fondamental implique le transfert].

### Langlands-Shelstad-Waldspurger Transfer Factors

**Def. (29.2.3.1) [Langlands-Shelstad-Waldspurger Transfer Factors for Lie Algebras, conjectured by Waldspurger].** For  $K \in \text{LField}^0$ ,  $G \in \text{RedGrp}/K$  and  $(H, s, {}^L\eta) \in {}^L\mathfrak{E}(G/K)$ , denote  $\mathfrak{g} = \text{Lie}(G)$ ,  $\mathfrak{h} = \text{Lie}(H)$ , then Waldspurger defined the transfer factor between  $\mathfrak{h}$  and  $\mathfrak{g}$ :

$$\Delta_{\mathfrak{h}, \mathfrak{g}} : \mathfrak{h}(K)_{\text{s.reg}} \times \mathfrak{g}(K)_{\text{s.reg}} \rightarrow \mathbb{C},$$

analogous to Langlands-Shelstad transfer factors (29.2.1.18).  $\lrcorner$

*Proof:* Cf.[Waldspurger, Une formule des traces locale pour les algebres de Lie p-adiques] and [Waldspurger, Le lemme fondamental implique le transfert]. and [Kottwitz, Harmonic analysis on semisimple p-adic Lie algebras]Section3. ?  $\square$

**Thm. (29.2.3.2) [Langlands-Shelstad-Waldspurger Matching Conjectures for Lie Algebras, conjectured by Waldspurger].** For  $K \in \mathbf{LField}^0$  and  $G \in \mathbf{RedGrp}/K$ , denote  $\mathfrak{g} = \mathrm{Lie}(G)$ . Suppose

- $(H, s, {}^L\eta) \in {}^L\mathfrak{E}(G/K)$ (29.2.1.4), and  $\mathfrak{h} = \mathrm{Lie}(H)$ .
- A normalization of transfer factors  $\Delta_{\mathfrak{h}, \mathfrak{g}}$  between  $\mathfrak{h}(K)$  and  $\mathfrak{g}(K)$  is fixed as in(29.2.1.18),
- Haar measures on  $\mathfrak{g}(K)$  and  $\mathfrak{h}(K)$  are fixed,

then for any  $f \in \mathcal{T}(\mathfrak{g}(K))$ , there exists  $f^{\mathfrak{h}} \in \mathcal{T}(\mathfrak{h}(K))$  s.t. for any strongly  $G$ -regular semi-simple element  $X_{\mathfrak{h}} \in \mathfrak{h}(K)$ (29.2.1.13), we have

$$\mathrm{SOrb}_{X_{\mathfrak{h}}}(f^{\mathfrak{h}}) = \sum_{X \in \mathfrak{g}(K)_{\mathrm{s.reg}}/\mathrm{Ad}(G(K))} \Delta_{H,G}^{\mathfrak{m}}(X_{\mathfrak{h}}, X) \mathrm{Orb}_X(f),$$

which is a finite sum, where  $\Delta_{\mathfrak{h}, \mathfrak{g}}^{\mathfrak{m}}(X_{\mathfrak{h}}, X) = 0$  unless  $X_{\mathfrak{h}}$  and  $X$  are related(29.2.1.18), in which case we are using Haar measures on  $Z_G(X)$  and  $Z_H(X_{\mathfrak{h}})$  that are compatible via the unique admissible isomorphism  $\varphi_{X_{\mathfrak{h}}, X}$ (29.2.1.12) to define the orbital integrals.

Any such  $f^{\mathfrak{h}}$  is called **Langlands-Shelstad-Waldspurger transfer** of  $f$  to  $\mathfrak{h}(K)$ .  $\lrcorner$

**Thm. (29.2.3.3) [Weak Transfer Conjectures for Lie Algebras, conjectured by Waldspurger].**

For  $K \in \mathbf{LField}^0$  and  $G \in \mathbf{RedGrp}/K$  with  $\mathfrak{g} \triangleq \mathrm{Lie}(G)$ , there exists  $c \in \mathbb{C}^{\times}$  s.t. for any  $G$ -regular semisimple element  $X_{\mathfrak{h}} \in \mathfrak{h}(K)$  and  $f \in \mathcal{T}(\mathfrak{g}(K))$  with a Langlands-Shelstad-Waldspurger transfer  $f^{\mathfrak{h}} \in \mathcal{T}(\mathfrak{h}(K))$ (29.2.3.2), we have

$$\mathcal{FSOrb}_{X_{\mathfrak{h}}}(f^{\mathfrak{h}}) = c \sum_{X \in \mathfrak{g}(K)_{\mathrm{s.reg}}/\mathrm{Ad}(G(K))} \Delta_{\mathfrak{h}, \mathfrak{g}}^{\mathfrak{m}}(X_{\mathfrak{h}}, X) \mathcal{FOrb}_X(f).$$

Notice that the Fourier transform of a stably invariant distribution is also stably invariant(29.2.0.4).  $\lrcorner$

*Proof:* Cf.[Waldspurger, Une formule des traces locale pour les algebres de Lie p-adiques].  $\square$

**Thm. (29.2.3.4) [Transfer Conjectures for Lie Algebras, conjectured by Waldspurger].** For

$K \in \mathbf{LField}^0$  and  $G \in \mathbf{RedGrp}/K$  with  $\mathfrak{g} \triangleq \mathrm{Lie}(G)$ , there is another version of the transfer conjecture that says there exists  $c \in \mathbb{C}^{\times}$  s.t. for any  $G$ -regular semisimple element  $X_{\mathfrak{h}} \in \mathfrak{h}(K)$  we have

$$i_{\mathfrak{h}}^{\mathfrak{g}}(\mathrm{SO}_{X_{\mathfrak{h}}}) = c \sum_{X \in \mathfrak{g}(K)_{\mathrm{s.reg}}/\mathrm{Ad}(G(K))} \Theta_X,$$

where  $i_{\mathfrak{h}}^{\mathfrak{g}}$  is the endoscopic induction functor.

This statement implies the matching conjecture for  $\mathfrak{g}$ (29.2.3.2) and the weak transfer conjecture(29.2.3.3).  $\lrcorner$

*Proof:* ? Cf.[Kottwitz, Harmonic analysis on semisimple p-adic Lie algebras] and [Waldspurger, Une formule des traces locale pour les algebres de Lie p-adiques]13.7(8).

The proof uses Harish-Chandra's theorem that Shalika germs appear as coefficients in the Lie algebra analog of his local character expansion. Thus the transfer conjecture implies that  $\kappa$ -Shalika germs on  $\mathfrak{g}(K)$  can be expressed in terms of stable Shalika germs on  $\mathfrak{h}(K)$ , and this, by work of [Langlands-Shelstad, Descent for transfer factors]Section2.4, implies matching for both  $G(K)$  and  $\mathfrak{g}(K)$ .  $\square$

**Cor. (29.2.3.5) [Matching Conjectures for  $G$  and  $\mathfrak{g}$ , conjectured by Waldspurger].** For  $K \in \mathbf{LField}^0$  and  $G \in \mathbf{RedGrp}/K$  with  $\mathfrak{g} \triangleq \mathrm{Lie}(G)$ , the Langlands-Shelstad transfer conjecture for  $G$  (29.2.1.22) implies the Langlands-Shelstad-Waldspurger matching conjecture for  $\mathfrak{g}$ .

Conversely, the Langlands-Shelstad-Waldspurger matching conjecture for all reductive Lie algebras over  $K$  implies the transfer conjecture for all reductive groups over  $K$ .  $\lrcorner$

*Proof:* Cf. [Waldspurger, Une formule des traces locale pour les algebres de Lie p-adiques].  $\square$

**Thm. (29.2.3.6) [Fundamental Lemmas for Lie Algebras, Conjectured by Waldspurger, Ngô [Ngô10]].** Situation as in (29.2.3.2), if

- $K \in p\text{-NField}$ ,
- $G$  is unramified and  $(H, s, {}^L\eta)$  is unramified (29.2.1.6),
- $\mathfrak{k} \leq \mathfrak{g}(K)$ ,  $\mathfrak{k}_H \leq \mathfrak{h}(K)$  are hyperspecial parahoric subalgebras?,
- The Haar measures on  $\mathfrak{g}(K)$  and  $\mathfrak{h}(K)$  are neutralized at  $\mathfrak{k}$ ,  $\mathfrak{k}_H$  resp.,
- The transfer factor  $\Delta$  between  $H$  and  $G$  is canonically normalized as in (29.2.1.21)?.

then  $\mathbf{1}_{\mathfrak{k}_H}$  is a Langlands-Shelstad-Waldspurger transfer of  $\mathbf{1}_{\mathfrak{k}}$ .  $\lrcorner$

*Proof:* Cf. [Ngô10].  $\square$

**Thm. (29.2.3.7) [Fundamental Lemma Implies Transfer Conjecture, Waldspurger].** If the fundamental lemmas for Lie algebras (29.2.3.6) are true, then the transfer conjectures for Lie algebras (29.2.3.4) are true.  $\lrcorner$

*Proof:* Cf. [Waldspurger, Le lemme fondamental implique le transfert]. The proof uses global methods.  $\square$

## 4 Stabilization of Trace Formulae

**Def. (29.2.4.1) [Strongly Regular Elliptic Part of the Trace Formulae].**  $\lrcorner$

### Twisted Orbital Integrals

**Def. (29.2.4.2) [Twisted Orbital Integrals].** Situation as in (30.5.3.4), fix Haar measures on  $I_0(\mathbf{A}_f^p)$  and  $I_0(\mathbb{Q}_p)$  and transfer them to  $I(\mathbf{A}_f^p) \cong Z_{G(\mathbf{A}_f^p)}(\gamma)$  and  $I(\mathbb{Q}_p) \cong J_\delta(\mathbb{Q}_p)$ .

Fix the Haar measure on  $G(\mathbf{A}_f^p)$  neutralized by  $\mathcal{K}^p$  and a Haar measure on  $G(\mathbb{Q}_{p^n})$  neutralized by  $\mathcal{G}(\mathbb{Z}_{p^n})$ . Define the orbital integral  $\mathrm{Orb}_\gamma(f^{p,\infty})$  (29.1.1.4), where  $f^{p,\infty} = \mathbf{1}_{\mathcal{K}^p g^{-1} \mathcal{K}^p} \in C_c^\infty(G(\mathbf{A}_f^p))$ .

And define the **twisted orbital integral**

$$\mathrm{TOrb}_\delta(\phi_n) = \int_{J_\delta(\mathbb{Q}_p) \backslash G(\mathbb{Q}_{p^n})} \phi_n(y^{-1} \delta \sigma(y)) dy, \text{ where } \phi_n = \mathbf{1}_{\mathcal{G}(\mathbb{Z}_{p^n}) \mu(p^{-1}) \mathcal{G}(\mathbb{Z}_{p^n})} \in C_c^\infty(G(\mathbb{Q}_{p^n}))$$

where  $\mu$  is a representative of the Hodge cocharacter  $\mu_h$  that is integral(?) [Kottwitz, Shimura varieties and twisted orbital integrals] Lemma 1.1.3). Define

$$\mathrm{Orb}(\gamma_0, \gamma, \delta) = \mathrm{Vol}(I(\mathbb{Q}) \backslash I(\mathbf{A}_f)) \mathrm{Orb}_\gamma(f^{p,\infty}) \mathrm{TOrb}_\delta(\phi_n)$$

which doesn't depend on the Haar measures on  $I_0(\mathbf{A}_f)$ ?.  $\lrcorner$

*Proof:* Cf. [Zhu20]P14? □

**Prop. (29.2.4.3) [Volume Constants  $c(\gamma_0, \gamma, \delta)$ ].** Situation as in (29.2.4.2), define

$$c(\gamma_0, \gamma, \delta) = \text{Vol}(I(\mathbb{Q}) \backslash I(\mathbf{A}_f)) \# \ker(\text{III}(I_0/\mathbb{Q}) \rightarrow \text{III}(G/\mathbb{Q})).$$

which depends only on the Haar measure on  $I_0(\mathbf{A}_f)$ ?. Then we have

$$c(\gamma_0, \gamma, \delta) = \frac{\tau(G) \# \mathfrak{K}(I_0/\mathbb{Q})}{\text{Vol}(A_G \backslash I(\mathbb{R}))} \quad (29.2.1.2)$$

where the Haar measure on  $I_{\mathbb{R}}$  is chosen s.t. its transfer to  $(I_0)_{\mathbb{R}}$  ([Kot88]P631) product with the Haar measure on  $I_0(\mathbf{A}_f)$  is the Tamagawa-measure on  $I_0(\mathbf{A})$  (20.5.4.5). ┘

*Proof:* By definition (20.5.4.8) we have

$$\text{Vol}(I(\mathbb{Q}) \backslash I(\mathbf{A}_f)) = \frac{\tau(I)}{\text{Vol}(A_G \backslash I(\mathbb{R}))} \quad ?.$$

Also we have  $\tau(I) = \tau(I_0)$  by (20.5.4.10), and

$$\tau(G) = \frac{\#(\pi_1(G)_{\text{Gal}_F})_{\text{tor}}}{\# \text{III}(Z(\widehat{G})/\mathbb{Q})}, \quad \tau(I_0) = \frac{\#(\pi_1(I_0)_{\text{Gal}_F})_{\text{tor}}}{\# \text{III}(Z(\widehat{I}_0)/\mathbb{Q})}$$

by (20.5.4.11). So it suffices to show that

$$\frac{\# \ker(\text{III}(I_0/\mathbb{Q}) \rightarrow \text{III}(G/\mathbb{Q})) \# \text{III}(G/\mathbb{Q}) \# (\pi_1(I_0)_{\text{Gal}_F})_{\text{tor}}}{\# \text{III}(Z(\widehat{I}_0)/\mathbb{Q}) \# (\pi_1(G)_{\text{Gal}_F})_{\text{tor}} \# \mathfrak{K}(I_0)} = 1.$$

But this follows from (29.2.1.2) and duality between  $\text{III}(G/\mathbb{Q})$  and  $\text{III}(Z(\widehat{G})/\mathbb{Q})$ ?. □

**Prop. (29.2.4.4) [Only Virtual Abelian Triples Contribute].** Let  $(\gamma_0, \gamma, \delta) \in \text{KT}_p(G, \mu)$  s.t.

- $\kappa_{\mathfrak{p}} \subset \mathbb{F}_{p^n}$ ,
- $\alpha(\gamma_0, \gamma, \delta) = 1$  (30.5.3.3),
- $\text{Orb}(\gamma_0, \gamma, \delta) \neq 0$  (29.2.4.2),

then  $(\gamma_0, \gamma, \delta)$  is geometric (30.5.3.6). ┘

*Proof:* Prove  $(\gamma_0, \gamma, \delta)$  satisfies (30.5.3.7). Cf. [Kot92]P441?. □

### Local Stable Orbital Integrals

**Def. (29.2.4.5) [Kottwitz-Signs].** Let  $K \in p\text{-NField}$ ,  $G \in \text{RedGrp}/K$  and  $\gamma \in G(K)$  a semi-simple element. If  $\gamma' \in G(K)$  is stably conjugate to  $\gamma$ , then  $Z_G(\gamma')^0$  is an inner form of  $Z_G(\gamma)^0$  defined by the cocycle associated to  $\gamma'$  under the natural map  $H^1(K, Z_G(\gamma)^0) \rightarrow H^1(K, Z_G(\gamma)_{\text{ad}})$ ?. ?

There is a **Kottwitz-sign map**?

$$e : H^1(K, Z_G(\gamma)_{\text{ad}}^0) \rightarrow \{\pm 1\}.$$

┘

*Proof:* Cf.[?Kot83]?

□

**Def.(29.2.4.6) [Local Stable Orbital Integrals].** Situation as in(29.2.4.5), for  $h \in \mathcal{T}(G(K))$ , the (local) **stable orbital integral** is defined by

$$\mathrm{SOrb}_\gamma(h) \triangleq \sum_{\gamma'} e(\gamma') \mathrm{Orb}_{\gamma'}(h) \text{ (29.1.1.4),}$$

where  $\gamma'$  ranges through a set of representatives for the conjugacy classes of rational elements in the stable-conjugacy class of  $\gamma$ .

? Cf.[Kot86]Section5.2. for the stable orbital integral for non-strongly-regular semi-simple elements  $\gamma$ . ┘

*Proof:* Cf.[Labesse99]P67. □

## Adelic Stable Orbital Integrals

### 5 Pseudo-Stablizations

References are [Kottwitz. On the  $\lambda$ -adic representations associated to some simple Shimura varieties] and [Scholze-Shin].

## Stabilized Trace Formulae

**Thm.(29.2.5.1)[Stabilized Arthur-Selberg Trace Formulae, Kottwitz].** For each  $f \in \mathcal{T}(G(\mathbf{A}_F))$ , the elliptic part of the stabilized Arthur-Selberg trace formula for  $G$  is

$$\mathrm{Orb}_{\mathrm{s.reg,ell}}^{\mathbf{A}}(f) = \sum_{(H,s,\eta) \in \mathfrak{C}_{\mathrm{ell}}(G)} i(G, H) \mathrm{ST}_{\mathrm{ell}}^{H(\mathbf{A}_F),*}(f^H) \text{ (29.1.3.5)(29.2.1.4)(29.2.1.8),}$$

where

- $f^H \in C_c^\infty(H(\mathbf{A}_F))$  is the product of Langlands-Shelstad transfers of  $f$  (29.2.1.22),
- the distribution

$$\mathrm{ST}_{\mathrm{ell}}^{H(\mathbf{A}_F),*}(h) = \tau(H) \sum_{\substack{\gamma_H \in H(F)_{\mathrm{ell,ss}}/\sim_{\mathrm{st}} \\ \gamma_H \text{ is strongly } (G,H)\text{-regular}}} \mathrm{SOrb}_{\gamma_H}^{\mathbf{A}}(h) \text{ (29.2.4.6), where } h \in \mathcal{T}(H(\mathbf{A}_F)) \text{ (29.2.1.14)}$$

is the  $(G, H)$ -**regular elliptic part of the stable trace formula** for  $H$ .

? What is the Haar measure? ┘

*Proof:* ?, Cf.[Harris. An introduction to the stable trace formula]Thm6.2, P30, and [Kot90]P189. and [Kal24].

By(10.5.1.18), if  $\gamma, \gamma' \in G(F)$  are strongly regular semi-simple and they are stably conjugate, then there is a canonical isomorphism  $Z_G(\gamma) \cong Z_G(\gamma')$ . Thus for any we have

$$\mathrm{Orb}_{\mathrm{s.reg,ell}}^{\mathbf{A}}(f) = \sum_{\gamma_0 \in G(F)_{\mathrm{s.reg,ss}}/\sim_{\mathrm{st}}} \tau(Z_G(\gamma_0)) \sum_{\substack{\gamma \in G(F)/\mathrm{Ad}(G(F)) \\ \gamma \sim_{\mathrm{st}} \gamma_0}} \mathrm{Orb}_\gamma^{\mathbf{A}}(f).$$

And it follows from (10.5.1.6) that the cohomological class of the map from the set of conjugacy class in the stable-conjugacy class of  $\gamma_0$  to the set of  $\mathbf{A}_F$ -conjugacy class in the adelic stable-conjugacy class of  $\gamma_0$  is the map

$$\ker \left( H^1(F, Z_G(\gamma_0)) \rightarrow H^1(F, G) \right) \xrightarrow{\alpha_{\gamma_0}} \ker \left( H^1(F, Z_G(\gamma_0)(\overline{\mathbf{A}}_F)) \rightarrow H^1(F, G(\overline{\mathbf{A}}_F)) \right).$$

This is a map of Abelian groups by (28.6.1.30)(Cf. [Kot86] ?), thus

$$\text{Orb}_{\text{s.reg,ell}}^{\mathbf{A}}(f) = \sum_{\gamma_0 \in G(F)_{\text{s.reg,ss}} / \sim_{\text{st}}} \tau(Z_G(\gamma_0)) \# \ker(\alpha_{\gamma_0}) \sum_{a \in \text{Im}(\alpha_{\gamma_0})} \text{Orb}_{a\gamma_0}^{\mathbf{A}}(f),$$

where  $a\gamma_0$  is any element in  $G(F)$  s.t.  $\text{inv}_{\mathbf{A}}(\gamma_0, a\gamma_0) = a$  (10.5.1.18). □

## 6 Arthur's Multiplicity Formula

References are [Arthur13] and [Kal16a].

**Thm& Conj.Cor. (29.2.6.1) [KMSW14, Endoscopic classification of representations: Inner forms of unitary groups].** proved Arthur's multiplicity formula for pure inner forms of unitary groups.

[Taibi, Arthur's multiplicity formula for certain inner forms of special orthogonal and symplectic groups] proved Arthur's multiplicity formula for some non-quasi-split symplectic and orthogonal groups.

[Hiroshi Ishimoto24] proved Arthur's multiplicity formula for general orthogonal groups of odd dimensions. ┘

*Proof:* ? □



## 29.3 Langlands-Arthur Conjectures

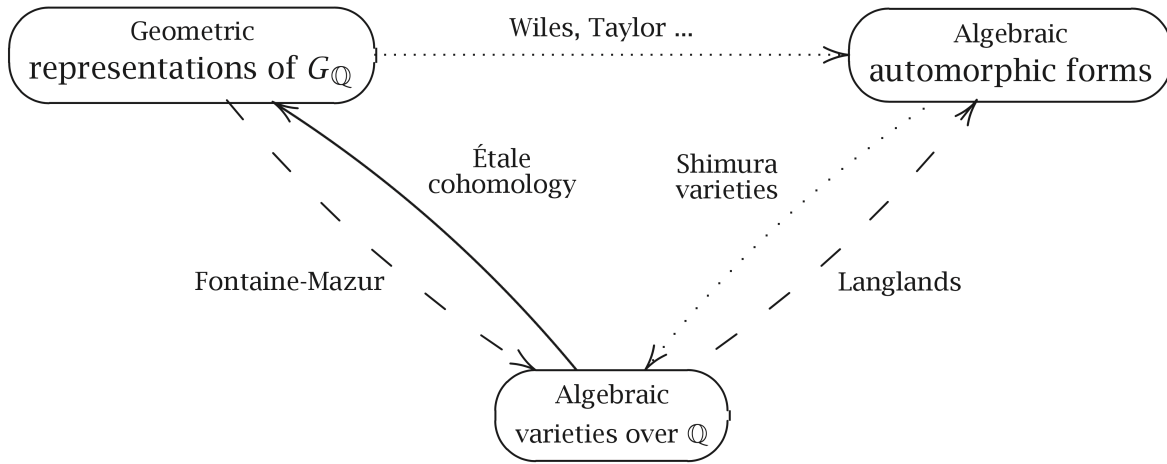


diagram borrowed from [\[Kis07\]](#)

References are [\[Lan70, Art89, ?Kna96, Art03, Cog07, Bor79\]](#).

### Notation (29.3.0.1).

- Let  $F \in \mathbf{GField}$ .
- Let  $G, G' \in \mathbf{RedGrp}/F$ .
- Use notations on automorphic representations defined in [Adelic Automorphic Representations](#).
- The Langlands  $L$ -groups  ${}^L G$  are in their Weil-forms.

┘

## 1 Local $L$ -Parameters and Arthur-Parameters

### Transfers of Levi-Subgroups

References are [\[Endoscopic Classification of Representations, Inner Forms of Unitary Groups, Kaletha-Shin\]Section0.4](#).

This subsection should be moved elsewhere.

**Def. (29.3.1.1) [Transfers of Levi-Subgroups].** Let  $F \in \mathbf{Field}^0$  and  $G^* \in \mathbf{RedGrp}/F$  be quasi-split and  $\Xi : G^* \rightarrow G$  be an equivalence class of inner twist. For a Levi-subgroup  $M^* \leq G^*$ ,  $M^*$  is said to **transfer to  $G$  w.r.t.  $\Xi$**  if there exists a Levi-subgroup  $M \leq G$  and  $\xi \in \Xi$  which restricts to an inner twist  $M^* \rightarrow M$ .

┘

**Prop. (29.3.1.2) [Local-Global Principle of Transfers].** For  $F \in \mathbf{NField}$ , suppose  $G^* \in \mathbf{RedGrp}/F$  be quasi-split and  $\Xi : G^* \rightarrow G$  be an equivalence class of inner twist. Then for a Levi-subgroup  $M^* \leq G^*$ ,  $M^*$  transfers to  $G$  w.r.t.  $\Xi$  [\(29.3.1.1\)](#) iff  $M_v^*$  transfers to  $G_v$  for each  $v \in \Sigma_F$ .

┘

*Proof:* Cf. [\[Endoscopic Classification of Representations, Inner Forms of Unitary Groups, Kaletha-Shin\]P22](#) ?.

□

### Local $L$ -Parameters

**Def. (29.3.1.3) [Local Langlands-Groups].** The local langlands groups  $L_{F_v}$  are defined as the group scheme over  $\mathbb{C}$ :

$$L_{F_v} = \begin{cases} W_{F_v} & , v \in \Sigma_F^\infty \\ W_{F_v} \times \mathrm{SL}(2) & , v \in \Sigma_F^{\mathrm{fin}} \end{cases}.$$

┘

**Def. (29.3.1.4) [Local  $L$ -Parameters].** A local  $L$ -parameter for  $G$  is a homomorphism

$$\phi_v : L_{F_v} \rightarrow {}^L G_{\mathbb{C}} \in \mathrm{GrpSch}/\mathbb{C} \text{ (29.3.1.3) (10.4.6.3)}$$

over  $W_{F_v}$  (29.3.1.3) s.t.

- It is trivial on some open subgroup of  $I_{F_v}$ .
- For any  $w \in W_{F_v}$ ,  $\phi_v(w) \in {}^L G(\mathbb{C})$  is semisimple (10.3.3.1). (Equivalently,  $\phi_v|_{W_{F_v}}$  is Frobenius-semisimple (25.1.2.22), by [Tate, number theoretic background, Section 4.1.3] ?)
- (relevant) For any parabolic subgroup  $P = LU$  of  ${}^L G_{\mathbb{C}}$  (10.4.6.8),  $\mathrm{Im}(\phi_v) \subset L$  implies  $P$  is relevant (10.4.6.9). (This is automatic if  $G$  is quasi-split, by (10.4.6.9)).

Any such  $\phi_v$  defined an 1-cocycle in  $\alpha_{\phi_v} \in H^1(L_{F_v}, \widehat{G})$ . And  $\phi_v$  is called a

- **discrete  $L$ -parameter** or **irreducible  $L$ -parameter** if  $C_{\widehat{G}(\mathbb{C})}(\mathrm{Im}(\phi_v))^0 = Z(\widehat{G}_{\mathbb{C}})^0$ , i.e.  $\mathrm{Im}(\phi_v)$  doesn't lie in any proper parabolic subgroup of  ${}^L G_{\mathbb{C}}$  ?.
- **unramified  $L$ -parameter** if  $v \in \Sigma_F^{\mathrm{fin}}$ ,  $G_{F_v}$  is unramified, and  $\phi_v$  is trivial on  $I_{F_v} \times \mathrm{SL}(2)$ .
- **tempered  $L$ -parameter** or a **bounded  $L$ -parameter** if  $\phi_v(W_{F_v})$  projects to a bounded subset of  $\widehat{G}(\mathbb{C})$ .
- a **supercuspidal  $L$ -parameter** if  $v \in \Sigma_F^{\mathrm{fin}}$  and  $\phi_v$  factors through the projection  $L_{F_v} \rightarrow W_{F_v}$ .

┘

**Remark (29.3.1.5).** In many cases (e.g.  $\mathrm{GL}(n)$ ,  $\mathrm{U}(3)$ ,  $\mathrm{GSp}(4)$ ), discrete  $L$ -parameters are exactly those whose  $L$ -packets consist entirely of supercuspidal representations. ?

┘

**Def. (29.3.1.6) [Equivalent Langlands-Parameters].** Two local  $L$ -parameters  $L_{F_v} \rightarrow {}^L G_{\mathbb{C}}$  (29.3.1.4) are called **equivalent local  $L$ -parameters** if they are conjugate by an element of  $\widehat{G}(\mathbb{C})$ .

The notions of discrete/unramified/tempered  $L$ -parameters are preserved under equivalence relations.

The set of equivalent classes of local  $L$ -parameters for  $G(F_v)$  is denoted by  $\Phi_v(G)$ . The set of equivalence classes of local tempered  $L$ -parameters for  $G(F_v)$  is denoted by  $\Phi_v^{\mathrm{temp}}(G)$ . The set of equivalence classes of discrete  $L$ -parameters for  $G(F_v)$  is denoted by  $\Phi_v^2(G)$ .

┘

**Def. (29.3.1.7) [Infinitesimal Characters].** For  $v \in \Sigma_F^{\mathrm{fin}}$ , two local  $L$ -parameters  $\phi_1, \phi_2$  are said to have the **infinitesimal characters** if their restrictions to  $W_{F_v}$  are equal.

┘

**Def. (29.3.1.8) [Centralizer of Component Groups].** For a local  $L$ -parameter  $\phi_v$  (29.3.1.4), denote

$$C_{\phi_v} \triangleq C_{\widehat{G}_{\mathbb{C}}}(\mathrm{Im}(\phi_v)), \quad \overline{C}_{\phi_v} = C_{\phi_v} / Z(\widehat{G}_{\mathbb{C}})^{\mathrm{Gal}_F}, \quad A_{\phi_v} = \pi_0(\overline{C}_{\phi_v}).$$

It is clear that the isomorphism classes of  $C_{\phi_v}$  and  $A_{\phi_v}$  only depend on the equivalence classes of  $\phi_v$  (29.3.1.6).

┘

**Prop. (29.3.1.9) [Centralizer of Component Groups].** For a local  $L$ -parameter  $\phi_v$  (29.3.1.4),

- $(C_{\phi_v})^0 \in \text{RedGrp}/\mathbb{C}$
- If  $v \in \Sigma_F^\infty$ , then  $A_{\phi_v}$  is always an elementary 2-group.

┘

*Proof:* 1: ? Cf. [Kot84] Section 10.

2: ? This follows from work of Harish-Chandra and Knapp-Zuckerman on the classification of discrete series, and more generally of tempered representations, of real semi-simple groups.  $\square$

**Example (29.3.1.10).** If  $G = \text{GL}(n)$  and  $\phi \in \Phi_v(G)$ , then  $A_{\phi_v} = \mathbf{1}$  (29.3.1.7).  $\square$

*Proof:* The definition of  $L$ -parameters implies that  $\phi_v$  is completely reducible ?, thus  $C_{\phi_v}$  is a product of general linear groups, thus connected.  $\square$

**Prop. (29.3.1.11) [Inner Forms and Restrictions].**

- If  $G^* \in \text{RedGrp}/F_v$  is a quasi-split inner form of  $G_{F_v}$ , then  $\Phi_v(G) \subset \Phi_v(G^*)$ , by (10.4.6.10).
- If  $L/F_v$  is a separable extension and  $G' \in \text{RedGrp}/L$  with  $G = \text{Res}_{L/F_v}(G')$ , then there is a canonical bijection  $\Phi(G) \cong \Phi(G')$ .

┘

*Proof:* 2: Cf. [Bor79] P40. ?  $\square$

**Example (29.3.1.12) [Conjugacy Classes for Galois Representations].** For  $G = e$ , the  $L$ -parameter of a  $n$ -dimensional Galois representation  $\rho$  of  $\text{Gal}(L/F_v)$  gives a family of conjugacy classes  $\{\sigma_v(\rho)\} \in \text{GL}(n; \mathbb{C})$  for  $v \in \text{Unr}_{L/F_v}(F)$ . And such a representation  $\rho$  is determined by this family, by Chebotarev density theorem.  $\square$

**Prop. (29.3.1.13) [Galois Action of Langlands Parameters].** Let  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  and  $v \in \Sigma_F^{\text{fin}}$  s.t.  $\pi_v$  is unramified, then for any  $\eta \in \text{Aut}(E)$ , the conjugate representation  $\eta(\pi)_{\eta(v)}$  (26.6.1.20) is also unramified, and  $c_{\eta(v)}(\eta(\pi)) = c_v(\pi)$ .  $\square$

*Proof:* Let  $B_v$  be the Borel subgroup of  $G_v = \text{GL}(F_v)$ ,  $\chi$  a character of  $B_v$ , and  $I_v = \text{Ind}_{B_v}^{G_v}(\chi) \in \text{Rep}^{\text{adm}}(G_v)$ , then  $\eta(I_v) = I_{\eta(v)} = \text{Ind}_{B_{\eta(v)}}^{G_{\eta(v)}}(\chi \circ \eta^{-1}) \in \text{Rep}^{\text{adm}}(G_{\eta(v)})$ , thus the conjugate of the unique quotient of  $I_v$  is the unique quotient of  $I_{v'}$ . Then the assertion follows from the fact  $\eta(\text{Frob}_v) = \text{Frob}_{\eta(v)}$ .  $\square$

### Local $L$ -Parameters and Complex Weil-Deligne Representations

**Prop. (29.3.1.14) [Weil-Deligne Representations into  $G$ ].** For  $K \in p\text{-LField}$  and  $G \in \text{LieGrp}/\mathbb{C}$  with  $G^0$  reductive, a **Weil-Deligne representation** of  $W_K$  into  $G$  is a triple  $(\rho, N)$  where  $\rho : W_K \rightarrow G$  is a continuous homomorphism s.t.  $\rho(\sigma_K)$  is semisimple in  $G$ , and  $N \in \mathfrak{g}$  s.t.

$$\text{Ad}(\rho(x))N = |x|.N, \text{ where } x \in W_K.$$

Two Weil-Deligne representations of  $W_K$  are called **equivalent** iff they are conjugate by an element of  $G^0$ .  $\square$

**Prop. (29.3.1.15) [Complex Weil-Deligne Representations].** For  $G \in \text{LieGrp}/\mathbb{C}$  with  $G^0$  reductive, there is a bijection between

- The equivalence classes of Weil-Deligne representations of  $W_K$  into  $G$  (29.3.1.14).
- The  $G^0$ -conjugacy classes of homomorphisms  $\phi : W_K \times \mathrm{SL}(2; \mathbb{C}) \rightarrow G$  that is continuous and Frobenius-semisimple on  $W_K$  and algebraic on  $\mathrm{SL}(2; \mathbb{C})$ ,

given by:

$$(\phi, V) \mapsto (\rho_0, V, N) : \rho_0(w) = \phi(w, \mathrm{diag}(|w|^{\frac{1}{2}}, |w|^{-\frac{1}{2}})), \text{ where } N = \mathrm{Lie}(\phi) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

┘

*Proof:* Cf. [G-R10].

□

### Local $L$ -Parameters for Classical Groups

References are [GGP12].

**Def. (29.3.1.16) [Self-Dual  $L$ -Parameters].** For  $V \in \mathcal{V}\mathrm{ect}^{\mathrm{fin}}/\mathbb{C}$ , a local  $L$ -parameter  $\phi_v : L_{F_v} \rightarrow \mathrm{GL}(V)$  (29.3.1.4) is called a

- **orthogonal  $L$ -parameter** if there is an isomorphism of Weil-Deligne representations  $f : V \cong V^\vee \in \mathfrak{wd}(W_{F_v})$  (29.3.1.15) s.t.  $f^\vee = f$ .
- **symplectic  $L$ -parameter** if there is an isomorphism of Weil-Deligne representations  $f : V \cong V^\vee \in \mathfrak{wd}(W_{F_v})$  (29.3.1.15) s.t.  $f^\vee = -f$ .

┘

**Prop. (29.3.1.17).** Situation as in (29.3.1.16), if  $\phi_v$  is orthogonal or symplectic, then up to conjugation by  $\mathrm{GL}(V)$ , there are only one  $f : V \cong V^\vee$  s.t.  $f^\vee = \epsilon(V)f$ .

┘

*Proof:* Because  $\phi_v$  is semisimple (29.3.1.4) and  $V \cong V^\vee$ , we can decompose  $V$  as

$$V = \left( \bigoplus m_i V_i \right) \oplus \left( \bigoplus m_j (P_j \oplus P_j^\vee) \right),$$

where  $\{V_i, P_j\}$  are distinct irreducible representations of  $L_{F_v}$  s.t.  $V_i$  are self-dual and  $P_j \not\cong P_j^\vee$ . It suffices to prove for  $V = m_i V_i$  or  $V = m_j (P_j \oplus P_j^\vee)$ .

In the first case, use the fact that any two non-degenerate bilinear forms on  $\mathbb{C}^{\oplus m_i}$  of the same sign are conjugate under the action of  $\mathrm{GL}(m_i)$ .

In the second case, notice that use the fact that any two perfect pairing  $\mathbb{C}^{\oplus m_j} \times \mathbb{C}^{\oplus m_j} \rightarrow \mathbb{C}$  are conjugate under the action of  $\mathrm{GL}(m_j) \times \mathrm{GL}(m_j)$ . □

**Def. (29.3.1.18) [Conjugate-Dual  $L$ -Parameters].** For  $L/K \in \mathrm{LField}$  with  $\mathrm{Gal}(L/K) = \{\mathrm{id}, \sigma\}$  and  $V \in \mathcal{V}\mathrm{ect}^{\mathrm{fin}}/\mathbb{C}$ , a local  $L$ -parameter  $\phi : L_L \rightarrow \mathrm{GL}(V)$  (29.3.1.4) determines an induced representation

$$\mathrm{Ind}_{L_L}^{L_K}(V) = \{f : L_L \rightarrow V \mid f(\tau s) = \tau(f(s)), \text{ where } \tau \in L_L, s \in L_K\} \in \mathfrak{wd}(W_K).$$

and we define the conjugate  $L$ -parameter  $\phi^\sigma$  on the space

$$V^\sigma = \{f : L_L \setminus L_K \rightarrow V \mid f(\tau s) = \tau(f(s)), \text{ where } \tau \in L_L, s \in L_K\} \in \mathfrak{wd}(W_L).$$

Then there is a canonical isomorphism  $(M^\sigma)^\vee \cong (M^\vee)^\sigma$  induced by the perfect pairing

$$M^\sigma \times (M^\vee)^\sigma \rightarrow \mathbb{C} : (f, f^\vee) \mapsto \langle f(s), f^\vee(s) \rangle, \text{ where } s \in W_L \setminus W_K,$$

which doesn't depends on  $s$  chosen. And there is a canonical isomorphism  $(M^\sigma)^\sigma \cong M : F \mapsto F(s)(s^{-1})$ , for any  $s \in W_L \setminus W_K$ , which doesn't depends on  $s$  chosen.

Then  $\phi$  is called a

- **conjugate-orthogonal  $L$ -parameter** if there is an isomorphism of Weil-Deligne representations  $f : V^\sigma \cong V^\vee \in \mathfrak{wd}(W_L)$  s.t.  $(f^\vee)^\sigma = f$ .
- **conjugate-symplectic  $L$ -parameter** if there is an isomorphism of Weil-Deligne representations  $f : V^\sigma \cong V^\vee \in \mathfrak{wd}(W_L)$  s.t.  $(f^\vee)^\sigma = -f$ .

Fixing a  $s \in W_L \setminus W_K$ , this is equivalent to a non-degenerate bilinear form  $B : V \times V \rightarrow \mathbb{C}$  s.t.

$$B(\tau m, s\tau s^{-1}n) = B(m, n), \quad B(n, m) = \epsilon B(m, s^2n).$$

by [GGP12]P10?.

┘

**Cor. (29.3.1.19) [Conjugate-Dual Quasi-Characters].** Situation as in (29.3.1.18), if  $\phi = \chi$  is a quasi-character of  $L^\times$ , then

- $\chi$  is conjugate-orthogonal iff  $\chi(K^\times) = 1$ .
- $\chi$  is conjugate-symplectic iff  $\chi|_{K^\times} \neq 1$  and  $\chi(\mathcal{N}_{L/K}) = 1$ .

┘

*Proof:* Fixing  $s \in W_L \setminus W_K$ ,  $\chi$  is conjugate-dual with sign  $\epsilon$  if the bilinear form  $B : \mathbb{C}(\chi) \times \mathbb{C}(\chi) \rightarrow \mathbb{C}$  satisfies  $B(\tau m, s\tau s^{-1}n) = B(m, n)$ ,  $B(n, m) = \epsilon B(m, s^2n)$ , which is equivalent to  $\chi(\mathcal{N}_{L/K}) = 1$  and  $\epsilon = \chi(s^2) = \chi(\text{Ver}(s)) = \chi(\text{Art}^{-1}(s))$  by (19.7.2.12), and  $s \in K^\times$  generates  $K^\times/\mathcal{N}_{L/K}$ . Thus the assertion follows.  $\square$

**Prop. (29.3.1.20).** Situation as in (29.3.1.18), if  $\phi$  is conjugate-orthogonal or conjugate-symplectic, then up to conjugation by  $\text{GL}(V)$ , there are only one  $f : V^\sigma \cong V^\vee$  s.t.  $(f^\vee)^\sigma = \epsilon(V)f$ .  $\square$

*Proof:* The proof is totally analogous to that of (29.3.1.17).  $\square$

**Prop. (29.3.1.21).** Situation as in (29.3.1.18),  $\phi$  is conjugate-dual with sign  $\epsilon$  iff  $W = \text{Ind}_{L_L}^{L_K}(V) \in \mathfrak{wd}(W_L)$  is self-dual with sign  $\epsilon$  and has maximal isotropic subspace  $V$  (as the subspace consisting of  $f$  supported on  $L_K$ ).  $\square$

*Proof:* Fixing  $s \in W_K \setminus W_L$ . If  $B$  is a non-degenerate bilinear form on  $V$  s.t.

$$B(\tau m, s\tau s^{-1}n) = B(m, n), \quad B(n, m) = \epsilon B(m, s^2n),$$

then  $W = V \oplus s^{-1}V$  can be given a non-degenerate bilinear form  $B'$  by decreeing  $V, s^{-1}V$  as isotropic subspaces and

$$B'(m, s^{-1}m') = \epsilon B(s^{-1}m', m) = B(m, m').$$

Then it is preserved by  $L_K$ .

Conversely, if  $B'$  is a self-dual non-degenerate bilinear form on  $W$  of sign  $\epsilon$  that is preserved by  $W_L$  and isotropic on  $V$ , then we can define

$$B(m, m') = B'(m, s^{-1}m'),$$

then it is a non-degenerate bilinear form on  $V$  s.t.

$$B(\tau m, s\tau s^{-1}n) = B(m, n), \quad B(n, m) = \epsilon B(m, s^2n),$$

$\square$

**Prop. (29.3.1.22)**[Local  $L$ -Parameters for Symplectic and Orthogonal Groups]. If  $V$  is a symplectic space over  $F_v$  of dimension  $2n$ , then the equivalence classes of Local  $L$ -parameters for  $\mathrm{Sp}(V)$  is in bijection with the isomorphism classes of orthogonal representations of  $L_{F_v}$  of dimension  $2n + 1$  s.t.  $\det(M) = 1$ .  $\lrcorner$

*Proof:*

□

**Prop. (29.3.1.23)**[Local  $L$ -Parameters for Unitary Groups].  $\lrcorner$

### Local Arthur-Parameters

**Def. (29.3.1.24)**[Local Arthur-Parameters]. For  $v \in \Sigma_F$ , a **local Arthur-parameter** for  $G_v$  is a continuous homomorphism

$$\psi_v : L_{F_v} \times \mathrm{SL}(2; \mathbb{C}) \rightarrow {}^L G(\mathbb{C}).$$

s.t.  $\psi_v|_{L_{F_v}}$  is a bounded  $L$ -parameter (29.3.1.4).

Two local Arthur-parameters of  $G_v$  are called **equivalent local Arthur-parameters** if they are conjugate by an element of  $\widehat{G}(\mathbb{C})$ . The set of conjugacy classes of local Arthur-parameters of  $G_v$  is denoted by  $\Psi_v(G)$ .

And the set of continuous homomorphisms

$$\psi_v : L_{F_v} \times \mathrm{SL}(2; \mathbb{C}) \rightarrow {}^L G(\mathbb{C}).$$

up to  $\widehat{G}(\mathbb{C})$ -conjugacy is denoted by  $\Psi_v^+(G)$ . In particular, we have  $\Psi_v(G) \subset \Psi_v^+(G)$ .  $\lrcorner$

**Def. (29.3.1.25)**[Generic Local Arthur-Parameters]. A **generic local Arthur-parameter**  $\psi$  is a local Arthur-parameter (29.3.1.24) that is trivial on  $1 \times \mathrm{SL}(2; \mathbb{C})$ .  $\lrcorner$

**Prop. (29.3.1.26)**[Associated Local  $L$ -Parameters]. Any  $\psi \in \Psi_v^+(G)$  (29.3.1.24) defines an **associated local  $L$ -parameter**  $\phi_{\psi_v} \in \Phi_v(G)$  (29.3.1.4) by the formula

$$\phi_{\psi_v}(w) = \psi_v(w, \mathrm{diag}(|w|^{1/2}, |w|^{-1/2})),$$

where the norm  $|\cdot| : L_F \rightarrow \mathbb{R}^\times$  is given by the composition  $L_F \rightarrow W_{F_v}^{\mathrm{ab}} \xrightarrow{\mathrm{rec}^{-1}} F_v^\times \rightarrow \mathbb{R}_+$ .  $\lrcorner$

**Def. (29.3.1.27)**[Local Arthur-Parameters for  $\mathrm{GL}(n)$ ]. For  $\lrcorner$

### Local $L$ -Factors and $\epsilon$ -Factors

**Notation (29.3.1.28).**

- $1 \neq \psi = \otimes' \psi_v \in \mathrm{Hom}(F \backslash \mathbf{A}_F, \mathbb{C})$ .
- $\pi = \otimes' \pi_v, \pi' = \otimes' \pi'_v \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$ .

□

**Prop. (29.3.1.29)**[Automorphic  $L, \epsilon$ -Factors for  $\mathrm{GL}(n)$ ]. Suppose  $G = \mathrm{GL}(n)$  and  $v \in \Sigma_F^{\mathrm{fin}}$ .

1. If  $n = 1$ , then the  $L$ -factors and  $\epsilon$ -factors are defined as in Tate's thesis.
2. If  $n \geq 2$  and  $\pi_v \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n; F_v))$ , then  $L_v(\pi_v; s) = 1$ , and  $\epsilon(\pi, \psi_v; s)$  is given by generalized Gauss sums. Cf.[Gauss sums and local constants for  $\mathrm{GL}(N)$ , Bushnell] **?**.

3. If  $\Delta = (\sigma; r)$  and  $\pi_v = Q(\Delta)$  (24.11.8.2), then

$$L_v(\pi_v; s) = L_v(\sigma(r-1); s) = L_v(\sigma; s+r-1),$$

and

$$\epsilon_v(\pi_v; \psi_v; s) = \prod_{i=0}^{r-1} \epsilon(\sigma(i); \psi_v; s) \prod_{i=1}^{r-2} \frac{L(\sigma^\vee(-i); -s)}{L(\sigma(i); s)} = \prod_{i=0}^{r-1} \epsilon(\sigma; \psi_v; s+i) \prod_{i=1}^{r-2} \frac{L(\sigma^\vee; -s-i)}{L(\sigma; s+i)}.$$

4. If  $\pi_v = Q(\Delta_1, \dots, \Delta_r)$  in Bernstein-Zelevinsky Classification, then

$$L_v(\pi_v; s) = \prod_{i=1}^r L_v(Q(\Delta_i); s), \quad \epsilon_v(\pi_v; \psi_v; s) = \prod_{i=1}^r \epsilon_v(Q(\Delta_i); \psi_v; s).$$

┘

*Proof:* 2: Cf.[On the local Langlands conjecture for  $\mathrm{GL}(n)$ : The cyclic case, Henniart]P153?  $\square$

**Example (29.3.1.30) [Local  $L$ -Factors for  $\mathrm{GL}(2)$ ].** Given  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(2)/F, \omega)$ , for  $v \in \Sigma_F^{\mathrm{fin}}$ , the local  $L$ -factor is defined to be

$$L(\pi_v; s) = \begin{cases} 1 & , v \in S_{\mathrm{cusp}}(\pi) \\ \frac{1}{(1 - \alpha_1 \|v\|^{-s})(1 - \alpha_2 \|v\|^{-s})} & (29.3.2.10) \quad , \pi_v = \chi_1 \times \chi_2 \\ \frac{1}{1 - \alpha \|v\|^{-s-1}} & , \pi_v = \mathrm{St}_2(\chi) \end{cases}.$$

It follows from (24.11.5.14) and (24.11.4.14) that any irreducible representation of  $\mathrm{GL}(2; F_v)$  is one of the form above.  $\square$

**Prop. (29.3.1.31) [Conductors for  $\mathrm{GL}(n)$ ].** If  $G = \mathrm{GL}(n)$  and  $v \in \Sigma_F^{\mathrm{fin}}$ , then

$$\epsilon_v(\pi_v, \psi_v; s) = \epsilon(\pi_v, \psi_v; 0) \|v\|^{-(m_v + n \cdot c_v(\psi_v))s},$$

where  $m \in \mathbb{N}$ , called the **conductor (admissible representation)** of  $\pi$ , denoted by  $c_v(\pi)$ .

And if for  $t \in \mathbb{N}$ , we let  $\mathcal{K}(t)$  be the inverse image of the mirabolic subgroup in  $\mathrm{GL}(n; \mathcal{O}_{F_v}/\mathfrak{p}_v^t)$ , then  $m$  equals the minimal  $t$  s.t.  $\pi_v$  has a  $\mathcal{K}(t)$ -fixed vector.

In particular,  $m = 0$  iff  $\pi$  is unramified.  $\square$

*Proof:* This follows from (24.11.3.18).  $\square$

**Prop. (29.3.1.32) [Rankin-Selberg  $L, \epsilon$ -Factors for  $\mathrm{GL}(n)$ , Jacquet-Piatetski-Shapiro-Shalika [JSS81]].** For  $v \in \Sigma_F^{\mathrm{fin}}$ ,

1. If  $\sigma, \sigma' \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n; F_v))$ , then

$$L(\sigma \times \sigma'; s) = \prod_{\chi \in F_v^\times \rightarrow \mathbb{C}^\times \text{ unramified s.t. } (\sigma')^\vee(\chi) = \sigma} L(\chi; s),$$

2. If  $r \leq r' \in \mathbb{Z}_+$ ,  $\Delta = (\sigma, r)$ ,  $\Delta' = (\sigma', r')$  and  $\pi_v = Q(\Delta)$ ,  $\pi'_v = Q(\Delta')$  (24.11.8.2), then

$$L(\pi_v \times \pi'_v; s) = \prod_{i=1}^r L(\sigma \times \sigma'; s + r + r' - 1 - i),$$

$$\epsilon(\pi_v \times \pi'_v, \psi_v; s) = \prod_{i=1}^r \left( \prod_{j=0}^{r+r'-2i} \epsilon(\sigma \times \sigma', \psi_v; s + i + j - 1) \right) \cdot \prod_{j=0}^{r+r'-2i-1} \frac{L(\sigma^\vee \times (\sigma')^\vee; -s - i - j + 1)}{L(\sigma \times \sigma'; s + i + j - 1)}$$

3. If  $\pi_v = Q(\Delta_1, \dots, \Delta_r)$ ,  $\pi'_v = Q(\Delta'_1, \dots, \Delta'_{r'})$ , then

$$L(\pi_v \times \pi'_v; s) = \prod_{i=1}^r \prod_{j=1}^{r'} L(Q(\Delta_i) \times Q(\Delta'_j); s), \quad \epsilon(\pi_v \times \pi'_v; s) = \prod_{i=1}^r \prod_{j=1}^{r'} \epsilon(Q(\Delta_i) \times Q(\Delta'_j); s).$$

And in general there are no easy formulas for  $\epsilon(\pi \times \pi'; \psi_v; s)$ ?

Notice that if we set  $\pi' = \mathbf{1}$ , then we get the  $L$ -factors and  $\epsilon$ -factors of  $\pi$ , as defined in (29.3.1.29).  $\lrcorner$

**Cor. (29.3.1.33).** If  $\pi_v, \pi'_v$  are unramified with Langlands parameters  $c(\pi_v)$  and  $c(\pi'_v)$ , then

$$L(\pi_v \times \pi'_v; s) = \det(1 - c(\pi_v) \otimes c(\pi'_v) q^{-s})^{-1}.$$

$\lrcorner$

**Cor. (29.3.1.34).** For  $\pi_v \in \text{Irr}^{\text{adm}}(\text{GL}(n; F_v))$ ,  $L(\pi \times \pi^\vee; s)$  has no pole at  $s = 1$  iff  $\pi$  is generic.  $\lrcorner$

*Proof:* It suffices to check that two Zelevinsky segments  $\Delta, \Delta'$  are linked iff  $L(Q(\Delta) \times Q(\Delta')^\vee; s)$  has a pole at  $s = 1$ . And this fact is easy to check.  $\square$

**Def. (29.3.1.35) [Archimedean  $L$ -Factors and  $\epsilon$ -Factors for  $\text{GL}(n)$ ].** Suppose  $G = \text{GL}(n)$ .

- For  $v \in \Sigma_F^{\mathbb{C}}$ , define the  $L$ -factors and  $\epsilon$ -factors as follows: If  $\pi_v \in \text{Irr}^{\text{adm}}(\text{GL}(n; \mathbb{C}))$  corresponds to  $\phi_v = \text{diag}(\sigma_{1,v}, \dots, \sigma_{n,v}) \in \text{Rep}^{n, \text{s.st}}(W_{\mathbb{C}})$  in the local Langlands correspondence (24.6.4.27), and  $\sigma_{i,v}(z) = (\frac{z}{|z|})^{l_{i,v}} |z|_{\mathbb{C}}^{t_{i,v}}$  where  $l_{i,v} \in \mathbb{Z}, t_{i,v} \in \mathbb{C}$  (24.6.2.1) then define

$$L_v(\pi_v; s) = L_v(\phi_v; s), \quad \epsilon_v(\pi_v, \psi_v; s) = \epsilon_v(\phi_v, \psi_v; s) \quad ?$$

- For  $v \in \Sigma_F^{\mathbb{R}}$ , define the  $L$ -factors and  $\epsilon$ -factors as follows:  $?$

$\lrcorner$

**Def. (29.3.1.36) [Automorphic  $L, \epsilon$ -factors, Langlands].** Assuming the local Langlands conjecture (29.3.2.2), for  $v \in \Sigma_F$ , there exists a unique  $\phi_v \in \Phi_v(G)$  (29.3.1.4) s.t.  $\pi_v \in \Pi_{\phi_v}(G)$ . Then for any  $\rho_v \in \text{Rep}^{\text{fd}}({}^L G)$ , we can define the **local automorphic  $L$ -factors** and **local automorphic  $\epsilon$ -factors**

$$L_v(\pi_v, \rho_v; s) = L_v(\rho_v \circ \phi_v; s), \quad \epsilon_v(\pi_v, \rho_v, \psi_v; s) = \epsilon_v(\rho_v \circ \phi_v, \psi_v; s) \quad (31.1.1.3),$$

Each  $\epsilon(\pi_v, \rho_v, \psi_{F_v}; s)$  is a monomial in  $\|v\|^{-s}$ .  $\lrcorner$

*Proof:*  $?$   $\square$

## 2 Local Langlands Conjectures and Arthur Conjectures for Quasi-Split Groups

References are [Kaletha, local Langlands conjectures for non-quasi-split groups].

**Conj. Cor. (29.3.2.1) [Local Functorial Lifting, Langlands [Lan70]].** Suppose  $v \in \Sigma_F$  s.t.  $G'_{F_v}$  is quasi-split. If  $\varphi: {}^L G' \rightarrow {}^L G \in \text{GrpSch}/\overline{\mathbb{Q}}$  respects the projection to  $\text{Gal}_F$ , then there is a correspondence  $R_\varphi: \text{Irr}^{\text{adm}}(G'(K)) \rightarrow \text{Irr}^{\text{adm}}(G(K))$  s.t. if  $\pi_v = R_\varphi(\pi'_v)$ , then for any  $\rho_v \in \text{Rep}^{\text{fd}}({}^L G_{F_v})$ , we have

$$L_v(\pi_v, \rho_v; s) = L_v(\pi'_v, \rho_v \circ \varphi; s), \quad \epsilon_v(\pi_v, \rho_v, \psi_K; s) = \epsilon_v(\pi'_v, \rho_v \circ \varphi, \psi_K; s) \quad (29.3.1.36).$$

Such a  $\pi$  is called a **functorial lifting** of  $\pi'$ .

Notice that is a consequence of (29.3.2.2) because if  $\pi'_v \in \Pi_{\phi_v}(G')$ , then we can just take any  $\pi \in \Pi_{\varphi \circ \phi_v}(G)$ .  $\lrcorner$



*Proof:*

□

**Conj.Cor.(29.3.2.2)**[Local Langlands Conjecture for Quasi-Split Groups, Langlands[Lan70]].

Suppose  $G_{F_v}$  is quasi-split, then there is a map

$$\mathrm{LL}_v : \mathrm{Irr}_v^{\mathrm{adm}, \mathrm{temp.}}(G) \rightarrow \Phi_v^{\mathrm{temp.}}(G) \quad (29.3.1.4)$$

with finite fibers  $\Pi_{\phi_v} \stackrel{\Delta}{=} \mathrm{LL}_v^{-1}(\phi_v)$  s.t.

1. (Unramified Representations) If  $\phi_v$  is unramified, then each  $\pi \in \Pi_{\phi_v}$  is  $\mathcal{K}_\pi$ -unramified for some hyperspecial maximal subgroup  $\mathcal{K}_\pi \leq G(F_v)$ , and for every hyperspecial maximal subgroup  $\mathcal{K} \leq G(F_v)$  there exists a unique  $\mathcal{K}$ -spherical representation  $\pi_{\mathcal{K}, \phi_v} \leq \Pi_{\phi_v}(G)$ , and the correspondence  $\phi_v \mapsto \pi_{\mathcal{K}, \phi_v}$  is given by the Satake isomorphism(Cf.[Getz]?).
2. (Essentially  $L^2$ -Representations) For  $\phi_v \in \Phi_v(G)$ , the following are equivalent:
  - Some  $\pi_v \in \Pi_{\phi_v}(G)$  is essentially- $L^2$ (20.5.1.2)(24.9.8.10).
  - Any  $\pi_v \in \Pi_{\phi_v}(G)$  is essentially- $L^2$ .
  - $\mathrm{Im}(\phi_v)$  is not contained in any proper Levi subgroup of  ${}^L G_{\mathbb{C}}$ (10.4.6.8).
3. (Temperedness) For  $\phi_v \in \Phi_v(G)$  s.t.  $\phi_v(G_a) = \{1\}$ , the following are equivalent:
  - Some  $\pi \in \Pi_{\phi_v}(G)$  is tempered.
  - Any  $\pi \in \Pi_{\phi_v}(G)$  is tempered.
  - $\phi_v(W_{F_v})$  is bounded.
4. (Parabolic Inductions) If  $\phi_v$  comes from  $\phi_{M,v} \in \Phi_v^{\mathrm{temp.}}(M)$  for some Levi subgroup  $M \leq G$  via the map  ${}^L M \rightarrow {}^L G$ (?, Cf.[G-H24]Chap7.4.), then  $\Pi(\phi_v)$  consists of irreducible subquotients of representations that are parabolically induced from elements of  $\Pi_{\phi_{M,v}}$ .
5. If  $\pi \in \Pi_{\phi_v}(G)$ , then  $\pi$  has central character  $\chi_{\phi_v}$ ?
6. If  $\phi_v, \phi'_v \in \Phi_v(G)$  and  $\phi'_v = \alpha \cdot \phi_v$ ?, then  $\Phi_{\phi'_v} = \Phi_{\phi_v} \otimes \pi_\alpha$ ?
7. (Functoriality) If  $H \in \mathcal{R}\mathrm{edGrp}/F_v$  with a homomorphism  $\eta : H \rightarrow G_{F_v} \in \mathcal{R}\mathrm{edGrp}/F_v$  with commutative kernel and cokernel, and  $\phi_v \in \Phi_v(G)$  with  $\phi'_v = {}^L \eta \circ \phi_v$ , then the pullback of any  $\pi \in \Pi_{\phi_v}(G)$  to  $H(F_v)$  is a direct sum of representations  $\pi' \in \Pi_{\phi'_v}(H)$ .
8. (from Langlands)For any  $\phi_v \in \Phi_v(G)$ , there exists a  $\pi_v(\phi_v) \in \mathrm{Irr}^{\mathrm{adm}}(G(F_v))$  s.t. for any  $\rho_v \in \mathrm{Rep}^{\mathrm{fd}}({}^L G)$ , we have

$$L_v(\pi_v(\phi_v), \rho_v; s) = L_v(\rho_v \circ \phi_v; s), \quad \epsilon_v(\rho_v, \pi_v(\varphi_v), \psi_v; s) = \epsilon_v(\rho_v \circ \phi_v, \psi_v; s) \quad (29.3.1.36)(29.3.1.29).$$

The fibers of  $\mathrm{LL}_v$  are called **local  $L$ -packets** for  $G$ . Two different  $\pi_1, \pi_2$  in the same  $L$ -packet are called  **$L$ -indistinguishable**. ┘

*Proof:* ? Cf.[Bor79]P44. □

**Remark(29.3.2.3).** We only study tempered  $L$ -parameters because the Langlands classification reduces the problem of studying  $\mathrm{Irr}^{\mathrm{adm}}(G(F_v))$  to the study of  $\mathrm{Irr}^{\mathrm{adm}, \mathrm{temp.}}(G(F_v))$ .? ┘

**Conj.(29.3.2.4)**[Refined Local Langlands Conjecture for Quasi-Split Groups]. Suppose  $G_{F_v}$  is quasi-split and  $\mathfrak{m}$  is a Whittaker-datum for  $G_{F_v}$ (10.4.4.23), then the local Langlands correspondence(29.3.2.2) holds and for any  $\phi_v \in \Phi_v^{\mathrm{temp.}}(G)$ ,

- (Langlands-Shelstad[She79]/Lusztig[Lus83a]) There is an injection

$$\iota_{\mathfrak{m}} : \Pi_{\phi_v} \hookrightarrow \text{Irr}(\pi_0(\overline{C}_{\phi_v})) \quad (29.3.1.8).$$

which is moreover bijective if  $v \in \Sigma_F^{\text{fin}}$ .

- (Generic Packet Conjecture, Shahidi[Sha90])  $\Pi_{\phi_v}$  contains a unique  $\mathfrak{m}$ -generic representation (24.9.5.2), and is mapped to the trivial representation by  $\iota_{\mathfrak{m}}$ .

Moreover, for each semi-simple element  $s \in C_{\phi_v}$ , denote

$$\Theta_{\phi_v}^s \stackrel{\Delta}{=} \sum_{\pi \in \Pi_{\phi_v}} \text{tr}(\iota_{\mathfrak{m}}(\pi)(s)) \Theta_{\pi} \quad (24.9.9.5),$$

called the  $s$ -**stable character** associated to  $\phi_v$ , and also denote  $S\Theta \stackrel{\Delta}{=} \Theta_{\phi_v}^1$ , then we have:

- (Endoscopic Character Identities, Kaletha[Kal16a]) If  $(H, s, {}^L\eta) \in {}^L\mathfrak{E}(G/F_v)$  (29.2.1.5) and  $\phi_v = {}^L\eta \circ \phi_v^H$  for some  $\phi_v^H \in \Phi_v^{\text{temp.}}(H)$ , then for any  $f_v \in \mathcal{T}(H(F_v))$  with Langlands-Shelstad transfer  $f_v^H \in \mathcal{T}(H(F_v))$  w.r.t.  $\Delta^{\mathfrak{m}, H, G}$  (29.2.1.22), we have

$$\Theta_{\phi_v}^s(f_v) = S\Theta_{\phi_v^H}(f_v^H).$$

In particular, by definition of Langlands-Shelstad transfer, we see  $S\Theta_{\phi_v}$  is a stable distribution (29.2.0.2). ┘

*Proof:* ? Cf. [Shelstad, Tempered endoscopy for real groups] for a complete treatment for  $v \in \Sigma_F^{\mathbb{R}}$ , and [Art13] for symplectic and orthogonal groups, and [Kal19] for toral supercuspidal representations of fairly general  $p$ -adic groups. [Atobi, uniqueness of generic representations in an L-packet] proved that the  $\mathfrak{m}$ -generic representation must be unique. □

**Conj. (29.3.2.5) [Non-Archimedean LLC for non-Quasi-Split Groups].** ┘

*Proof:* Cf. [Kaletha, The local Langlands conjectures for non-quasi-split groups]. □

**Conj. Cor. (29.3.2.6) [Local Langlands Twists].** Suppose  $G, G' \in \text{RedGrp}/K$ ,  $G$  is quasi-split and  $G'$  is an inner twist of  $G$ , then  ${}^L G = {}^L G'$ . Is there a correspondence  $\text{Irr}(G'_F) \rightarrow \text{Irr}(G_F)$  s.t. if  $\pi = R(\pi')$ , then  $L(s, \rho, \pi) = L(s, \rho, \pi')$  for any representation  $\rho$  of  ${}^L G$ ? ┘

*Proof:* □

### Unramified LLC: Satake Isomorphisms

References are [Gro98].

**Notation (29.3.2.7).**

- Let  $v \in \Sigma_F^{\text{fin}}$  s.t.  $G_{F_v}$  is unramified.
- Let  $\mathcal{K}_v \subset G(F_v)$  be a hyperspecial maximal compact subgroup.
- Let  $(B, T)$  be a Borel-pair of  $G_{F_v}$ .
- Let  $T_d$  be the maximal  $k$ -split sub- $k$ -torus of  $T$ .

- Let  $W$  be the Weyl group of  $(B, T)$  and let  $W_d \leq W$  be the subgroup consisting of elements that preserves  $T_d$ .

┘

**Thm. (29.3.2.8) [Satake Isomorphism].** There is a canonical isomorphism

$$\mathcal{H}_{\mathcal{K}_v}(G(F_v); \mathbb{C}) \cong \mathbb{C}(X_{\bullet}(T_d))^{W_d}$$

┘

*Proof:* Cf. [Cartier, Representations of  $p$ -adic groups: a survey 1979] Sec 4.2. ?

□

**Prop. (29.3.2.9) [Langland's Formation of the Satake Isomorphism].** If  $G$  is split, then  $\mathcal{H}_K$  is isomorphic to ring generated by the characters of irreducible analytic f.d. representations of  ${}^L G$ , and the unramified representations of  $G(K)$  are parametrized by the semisimple conjugacy classes in  ${}^L G^0$ .

If  $G$  is unramified, let  $\Phi$  be the Frobenius element in  $\text{Gal}_F$ , and consider the cosets  $\widehat{G}(\mathbb{C}) \times \Phi \subset {}^L G$ , then  $\mathcal{H}_K$  is isomorphic to the ring of functions on this coset that are generated by the restrictions of characters of irreducible representations of  ${}^L G$ , and the unramified representations of  $G(F)$  are parametrized by the semisimple  $\widehat{G}(\mathbb{C})$ -conjugacy classes in  ${}^L G(\mathbb{C})$  in this coset (i.e. an element of the form  $c = t \times \Phi$  where  $t \in \widehat{G}(\mathbb{C})$  is semi-simple).

┘

*Proof:* ?

□

**Cor. (29.3.2.10) [Satake Isomorphisms for  $\text{GL}(n)$ ].** There is an isomorphism

$$S : \mathcal{H}_{\mathcal{K}} \cong \mathbb{C}[t_1, t_1^{-1}, \dots, t_n, t_n^{-1}]^W$$

s.t. if  $\varphi \in \mathcal{H}_{\mathcal{K}}$ , for any spherical principal series  $\pi(\chi_1, \dots, \chi_n)$  and a spherical vector  $v$ ,  $\pi(\varphi)v = S(\varphi)(\varpi^{\frac{n-1}{2}}\alpha_1, \dots, \varpi^{\frac{1-n}{2}}\alpha_n)$ , where  $\alpha_i = \chi(\chi_i)$  are the Satake parameters of  $\chi_i$ .

┘

*Proof:* Cf. [Spherical representations and the Satake isomorphism] ?

□

**Thm. (29.3.2.11) [Unramified Langlands Correspondence, Gantmacher-Langlands].** For  $\pi_v \in \text{Irr}^{\text{adm}, \text{unr}}(G(F_v))$ , there exists an unramified  $L$ -parameter (29.3.1.4)

$$\rho_{\pi} : W_{F_v} \rightarrow {}^L G(\mathbb{C})$$

s.t.

This induces a bijection between  $\text{Irr}^{\text{adm}, \text{unr}}(G(F_v))$  and conjugacy classes of  ${}^L G(\mathbb{C})$  s.t. the image in  $\widehat{G}(\mathbb{C})$  is semisimple, and the image in  $W_F$  equals  $\sigma_v$ . The class  $c_v(\pi)$  corresponding to  $\pi$  is called the **Satake-parameter** of  $\pi$ .

For example, if  $G = \text{GL}(n)$ , then any  $\pi \in \text{Irr}^{\text{adm}, \text{unr}}(G(F_v))$  is a Zelevinsky quotient  $Q(\chi_1, \chi_2, \dots, \chi_n)$  by (24.11.6.3), and the conjugacy classes corresponding to it is just  $\text{diag}(\chi_1(\varpi), \dots, \chi_n(\varpi)) \in \text{GL}(n; \mathbb{C}) = \widehat{G}$ .

┘

*Proof:* Cf. [Bor79] Sec 6. ?

□

**Def. (29.3.2.12) [Arithmetic Local Representations].** Any unramified  $\pi_v \in \text{Irr}^{\text{adm}}(G(F_v))$  induces a map  $\mathcal{H}_{\mathcal{K}}(G(F_v)) \rightarrow \mathbb{C}$  by acting on a  $\mathcal{K}$ -fixed vector in  $\pi$ . Then for  $E \in \mathbf{NField}$ ,

- $\pi_v$  is said to be a **admissible representation defined over  $E$**  the induced map  $\mathcal{H}_{\mathcal{K}}(G(F_v); \mathbb{Q}) \rightarrow \mathbb{C}$  is valued in  $E$ .
- $\pi_v$  is said to be a **admissible representation with Satake parameter defined over  $E$**  the induced map  $\mathbb{Q}(X_*(T_d))^{W_d} \rightarrow \mathbb{C}$  induced by the Satake isomorphism (29.3.2.8) is valued in  $E$ .

The two definitions coincides if  $\delta \in X^*(T)$ , where  $\delta$  is half the sum of the positive roots of  $G$ .  $\lrcorner$

*Proof:* Cf. [B-G14]P144. ?  $\square$

**Example (29.3.2.13).** Situation as in (29.3.2.12), if  $G = \mathrm{GL}(n)$  and  $(B, T)$  is the canonical Borel-pair, then the Satake parameter of  $\pi_v$  is defined over  $E$  iff the Langlands conjugacy class  $c(\pi_v)$  (29.3.2.11) is defined over  $E$  (i.e. the characteristic polynomial has coefficients in  $E$ ).  $\lrcorner$

**Def. (29.3.2.14) [Local Unramified  $L$ -Packets].**  $\lrcorner$

### Local Arthur-Packets

Local Arthur-packets deal with the representation that are far from tempered. References are [Arthur, Unipotent automorphic representations: conjectures], [Arthur, Unipotent automorphic representations: global motivations].

**Conj. (29.3.2.15) [Local Arthur-Packets].** To every local Arthur-parameter  $\psi_v$  (29.3.1.24) we should naturally associate a finite subset  $\Pi_{\psi_v}(G) \subset \mathrm{Irr}^{\mathrm{uni}}(G(F_v))$  s.t.

- The  $L$ -parameters of the representations in  $\Pi_{\psi_v}(G)$  all have the same infinitesimal character (29.3.1.7).
- $\Pi_{\psi_v}(G)$  contains the  $L$ -packet corresponding to the  $L$ -parameter associated to  $\psi$  (29.3.1.26) (29.3.2.2), i.e.  $\Pi_{\phi_{\psi_v}}(G) \subset \Pi_{\psi_v}(G)$ .
- (Parabolic Inductions) If  $\psi_v$  has image in a Levi subgroup  ${}^L M \subset {}^L G$ , then  $\Pi_{\psi_v}$  consists of the irreducible summands of parabolically normalized-induced representations from the corresponding packet of the Levi subgroup  $M \leq G$ .

•

called the **local Arthur-Packet** associated to  $\psi_v$ .  $\lrcorner$

**Remark (29.3.2.16).** Local Arthur-packets are not, in general, disjoint, and their union will not be  $\mathrm{Irr}^{\mathrm{uni}}(G(F_v))$ .  $\lrcorner$

### Known cases

**Thm. (29.3.2.17) [Non-Archimedean LLC for  $\mathrm{GL}(n)$ , Harris-Taylor [H-T01]/Henniart].** The local Langlands correspondence (29.3.2.2) is true for  $\mathrm{GL}(n)$  over a non-Archimedean local field with the local Langlands groups defined in (29.3.1.3), by (24.12.1.2).  $\lrcorner$

**Thm. (29.3.2.18) [Non-Archimedean LLC for  $\mathrm{SO}(2n+1)$ , Jiang-Soudry].** The local Langlands correspondence for generic supercuspidal representations of  $\mathrm{SO}(2n+1)$  over a non-Archimedean local field is true.  $\lrcorner$

*Proof:* Cf. [The local converse theorem for  $\mathrm{SO}(2n+1)$  and applications]. ?

They deduced a strong multiplicity one result and deduced the LLC for  $\mathrm{SO}(2n+1)$  from that of  $\mathrm{GL}(n)$ .  $\square$

**Thm. (29.3.2.19) [LLC for Classical Groups, Arthur/Mok/Atobe].** Suppose that

- $F \in p\text{-NField}$ .
- Arthur's local claims in [Arthur, The endoscopic classification of representations] are true.
- The stabilization of the twisted trace formula is valid (Cf. [Mœglin-Waldspurger. Stabilisation de la formule des traces tordue, 1&2]).

Then the local Langlands conjecture (29.3.2.2) item 1, 2 are true, and conjecture (29.3.2.4) is true for split groups  $\mathrm{Sp}(2n)$ ,  $\mathrm{SO}(2n+1)$  and quasi-split groups  $\mathrm{U}(n)$ . And a slightly-weaker version of these results hold for quasi-split  $\mathrm{SO}(2n)$ .

And the generic packet conjecture (29.3.2.4) holds for all these cases.  $\lrcorner$

*Proof:* Cf. [Arthur, The endoscopic classification of representations] Thm 1.5.1., [Mok15], [Atobe, On the uniqueness of generic representations in an  $L$ -packet].  $\square$

**Prop. (29.3.2.20) [Explicit LLC].** Under some restriction on  $G$ , [Kaletha, Supercuspidal  $L$ -packets] explicitly constructed certain supercuspidal  $L$ -packet for  $L$ -parameters that are trivial on the  $\mathrm{SL}(2; \mathbb{C})$ -factor. These supercuspidal  $L$ -packet satisfy certain

For any  $G$ , [Fargues-Scholze] less-explicitly constructed  $L$ -parameters for supercuspidal representation of  $G$ .  $\lrcorner$

**Thm. (29.3.2.21) [Archimedean LLC].** For  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ , the local Langlands conjecture (29.3.2.2) (29.3.2.4) and the genericity conjecture (29.3.2.4) are proved for any  $G \in \mathrm{RedGrp}/\mathbb{K}$  and any tempered  $L$ -parameters, see [Langlands Classifications \(Archimedean Local Langlands Correspondence\)](#).  $\lrcorner$

### LLC for $\mathrm{GL}(n)$ over Fields of Laurent-Series

References are [G. Laumon, M. Rapoport, U. Stuhler:  $D$ -elliptic sheaves and the Langlands correspondence]. [V.G. Drinfeld: Elliptic modules, Mat. USSR Sbornik 23, pp. 561–592 (1974)], [V.G. Drinfeld: Elliptic modules II, Mat. USSR Sbornik 31, pp. 159–170 (1977)].

## 3 Global $L$ -Packets and Arthur-Packets

References are [Endoscopic Classification of Representations, Inner Forms of Unitary Groups, Kaletha-Shin].

### Global Formal Parameters

**Conj. (29.3.3.1) [Global Langlands-Groups].** There should exist a locally compact group  $L_F$  with a surjective map  $L_F \rightarrow \mathrm{Gal}_F$ , and for any  $v \in \Sigma_F$ , there are commutative diagrams

$$\begin{array}{ccc} L_{F_v} & \longrightarrow & \mathrm{Gal}_{F_v} \\ \downarrow & & \downarrow \\ L_F & \longrightarrow & \mathrm{Gal}_F \end{array} \quad (29.3.1.3).$$

And the group  $L_F$  should satisfy: For any quasi-split reductive group  $G/F$ , there are compatible surjective maps  $\varphi \twoheadrightarrow c, \pi_v \twoheadrightarrow \varphi_v$ , where  $\varphi : L_F \rightarrow {}^L G$  is a conjugacy class of maps over  $\mathrm{Gal}_F$  and  $\varphi : L_{F_v} \rightarrow {}^L G_v$  is a conjugacy class of maps over  $\mathrm{Gal}_{F_v}$ .  $\lrcorner$

*Proof:*

□

**Conj. (29.3.3.2)[Candidates for global Langlands Groups].** It is conjectured that the global Langlands group  $L_F$  (29.3.3.1) should be the infinite fiber product of non-split extensions of  $W_F$  by compact semisimple simply-connected Lie groups  $K_c$ . ┘

**Def. (29.3.3.3) [Automorphic Conjugacy Classes].** Let  $F \in \mathbf{NField}$ ,  $G \in \mathbf{RedGrp}/F$  and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$ , then there exists a finite set  $\Sigma_F^\infty \subset_{\mathrm{fin}} S \subset \Sigma_F$  s.t.  $G$  and  $\pi$  are unramified outside  $S$ . Then the Satake-isomorphisms for each  $v \in \Sigma_F \setminus S$  gives a semi-stable  $\widehat{G}(\mathbb{C})$ -conjugacy class  $c(\pi_v) \in {}^L G(\mathbb{C})$  (29.3.2.9), and we call

$$c^S(\pi) \triangleq \{c(\pi_v)\}_{v \in \Sigma_F \setminus S}$$

the string of conjugacy data attached to  $\pi$ . The set of images of the near-equivalence classes of these elements  $c^S(\pi)$  is denoted by  $\mathcal{C}_{\mathrm{auto}}(G)$ . ┘

**Def. (29.3.3.4)[Global Formal Parameters for  $\mathrm{GL}(n)$ ].** For each  $n \in \mathbb{Z}_+$  and  $F \in \mathbf{NField}$ , we define the following set of formal parameters for  $\mathrm{GL}(n)$ :

$$\Psi_{\mathrm{cusp}}(n; F) \subset \Psi_{\mathrm{sim}}(n; F) \subset \Psi(n; F)$$

where

- $\Psi_{\mathrm{cusp}}(n; F) \triangleq \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$ .
- $\Psi_{\mathrm{sim}}(n; F)$  is the set of formal symbols  $\pi(r)$  where  $m, r \in \mathbb{Z}_+$  s.t.  $mr = n$  and  $\pi \in \Psi_{\mathrm{cusp}}(m; F)$ . (Notice that  $r$  represents the irreducible representation  $\mathrm{Sp}(n)$  of  $\mathrm{SL}(n)$  of dimension  $r$ ).
- $\Psi(n; F)$  is the set of formal sums  $\psi = l_1\psi_1 \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r$  where  $l_i \in \mathbb{Z}_+$  and  $\psi_k \in \Psi_{\mathrm{sim}}(n_k; F)$  s.t.  $\sum l_k n_k = n$ .

It follows from Mœglin-Waldspurger theorem (26.6.3.14) that the map

$$\Psi_{\mathrm{sim}}(n; F) \rightarrow \mathrm{Irr}_{\mathrm{disc}}^{\mathrm{cusp}}(\mathrm{GL}(n)/F) : \psi \mapsto \pi_\psi \triangleq \pi\left(\frac{n-1}{2}\right) \boxplus \pi\left(\frac{n-3}{2}\right) \boxplus \pi\left(\frac{1-n}{2}\right)$$

is a bijection. And also the map

$$\Psi(n; F) \rightarrow \mathcal{A}(\mathrm{GL}(n)/F) \text{ (26.6.3.15)} : l_1\psi_1 \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r \mapsto \boxplus_{i=1}^r \pi_{\psi_i}^{\boxplus(l_i)} = I \left( \bigotimes_{i=1}^r \pi_{\psi_i}^{\boxplus(l_i)} \right)$$

is a bijection by the definition (26.6.3.15).

A **generic formal parameter** for  $\mathrm{GL}(n)$  is called a formal parameter  $\psi \in \Psi(n; F)$  of the form

$$\psi = l_1\psi_1 \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r$$

where  $\psi_k = \pi_k(r_k)$  and  $r_k = 0$  for each  $k \in [r]_+$ . Denote

$$\Phi_{\mathrm{cusp}}(n; F) \subset \Phi_{\mathrm{sim}}(n; F) \subset \Phi(n; F)$$

the set of cuspidal/simple/ $\emptyset$ -formal parameters. ┘

**Def. (29.3.3.5) [Localization of Formal Parameters].** For  $n \in \mathbb{Z}_+$ ,  $F \in \mathbf{NField}$  and  $v \in \Sigma_F$ , we can associate a localization map

$$\Psi(n; F) \rightarrow \Psi^+(\mathrm{GL}(n; F_v)) \text{ (29.3.1.24)} : l_1\psi_1 \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r \mapsto \bigoplus_{k=1}^r \psi_{k,v}^{\otimes(r_i)},$$

where  $\psi_{k,v} = \mathrm{rec}(\pi_{k,v}) \otimes \mathrm{Sp}(r_k) \in \Psi^+(\mathrm{GL}(n_i r_i; F_v))$  if  $\psi_k = \pi_k(r_k)$  and  $\pi_k \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n_i)/F)$ .  $\lrcorner$

**Def. (29.3.3.6) [Conjugate-Self-Dual Formal Parameters].** Let  $E/F$  be an extension of degree  $\leq 2$  with  $c \in \mathrm{Gal}(E/F)$ . For each  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/E)$ , denote  $\pi^\star = (\pi^\vee)^c$  the conjugate-dual of  $\pi$ . For each

$$\psi = l_1\psi_1 \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r \in \Psi(n; F) \text{ (29.3.3.4),}$$

denote

$$\psi^\star = l_1\psi_1^\star \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r^\star$$

where  $\psi_k^\star = \pi_k^\star(r_k)$  if  $\psi_k = \pi_k(r_k)$ .

The set of conjugate-self-dual formal parameters in  $\Psi(n; E)$  is denoted by  $\tilde{\Psi}(n; E)$ . Moreover we denote  $\tilde{\Psi}_{\mathrm{ell}}(n; E) \subset \tilde{\Psi}(n; E)$  the set of formal parameters

$$\psi = l_1\psi_1 \boxplus l_2\psi_2 \boxplus \dots \boxplus l_r\psi_r \in \Psi(n; F)$$

s.t.  $l_k = 1$  and  $\psi_k^\star = \psi_k$  for each  $k$ .

And we denote  $\tilde{\Psi}_{\mathrm{sim}}(n; E) \triangleq \Psi_{\mathrm{sim}}(n; E) \cap \tilde{\Psi}(n; E)$  and also  $\tilde{\Phi}(n; E) \triangleq \Phi(n; E) \cap \tilde{\Psi}(n; E)$ .  $\lrcorner$

**Def. (29.3.3.7) [Formal Parameters for  $\mathbf{U}_{E/F}(n)$ ].** Let  $E/F$  be an extension of degree  $\leq 2$  with  $c \in \mathrm{Gal}(E/F)$ , and denote  $G^\star = \mathbf{U}_{E/F}(n)$ , we have the following sets of formal parameters for  $\mathbf{U}_{E/F}(n)$ :

$$\Psi_{\mathrm{sim}}(G^\star, \eta_\chi) \subset \Psi_2(G^\star, \eta_\chi) \subset \Psi_{\mathrm{ell}}(G^\star, \eta_\chi) \subset \Psi_{\mathrm{disc}}(G^\star, \eta_\chi) \subset \Psi(G^\star, \eta_\chi),$$

where  $\Psi(G^\star, \eta_\chi)$  is the set of pairs  $(\psi^n, \tilde{\psi})$  where  $\psi^n \in \tilde{\Psi}(n; F)$  and

$$\tilde{\psi} : \mathcal{L}_{\psi^n} \times \mathrm{SL}(2; \mathbb{C}) \rightarrow {}^L G^\star(\mathbb{C}) \text{ ?}$$

is a  $\hat{\mathbf{U}}_{E/F}(n; F)$ -conjugacy class of  $L$ -homomorphisms s.t.  $\psi^n = \eta_\chi \circ \tilde{\psi}$  ?.

? Cf. [Endoscopic Classification of Representations, Inner Forms of Unitary Groups, Kaletha-Shin]P69.  $\lrcorner$

**Thm. (29.3.3.8) [Localization of Formal Parameters for  $\mathbf{U}_{E/F}(n)$ , Mok].** Situation as in (29.3.3.7), for each  $\kappa \in \{\pm 1\}$  and  $\psi = (\psi^n, \tilde{\psi}) \in \Psi(G^\star, \eta_\kappa)$ , there exists  $\psi_v \in \Psi_{\mathrm{uni}}^+(G_v^\star)$  s.t.  $\psi_v^n = \eta_{\chi_\kappa} \circ \psi_v$  (29.3.3.5), and the isomorphism class of  $\psi_v$  is uniquely determined, called the **localization of the formal parameter**  $\psi$ .  $\lrcorner$

*Proof:* [Mok]2.4.10, 2.4.11.  $\square$

**Def. (29.3.3.9) [Relevant Formal Parameters for  $\mathbf{U}_{E/F}(n)$ ].** Situation as in (29.3.3.7), let  $(G, \xi)$  is an inner form of  $G^\star$ . Suppose  $\kappa \in \{\pm 1\}$  and  $\psi = (\psi^N, \tilde{\psi}) \in \Psi(G^\star, \eta_\kappa)$ , then  $\psi$  is said to be a **relevant formal parameter** if the localization  $\psi_v$  (29.3.3.8) is  $(G_v, \xi_v)$ -relevant (29.3.1.4) for each  $v \in \Sigma_F$ .  $\lrcorner$



### Global Arthur-Parameters

**Def. (29.3.3.10) [Global Arthur-Parameters].** Assuming the existence of global Langlands group  $L_F$  (29.3.3.1), a **global Arthur-parameter** is a continuous homomorphism

$$\psi : L_F \times \mathrm{SL}(2; \mathbb{C}) \rightarrow {}^L G_{\mathbb{C}}$$

s.t.

- $\psi|_{\mathrm{SL}(2; \mathbb{C})}$  is holomorphic and the image is in  $\widehat{G}(\mathbb{C})$ .
- For  $w \in L_F$ , the image of  $\psi(w)$  in the quotient  $W_F$  of  ${}^L G$  is the same as the image of  $w$  in the quotient  $L_F \rightarrow W_F$ .
- $\psi(L_F)$  is bounded in  ${}^L G/Z(\widehat{G}(\mathbb{C}))$ .
- $\psi$  is relevant, i.e. any parabolic subgroup of  ${}^L G$  (10.4.6.8) containing  $\mathrm{Im}(\psi)$  is defined over  $F$ .

⌋

**Remark (29.3.3.11).** Notice that the last assertion is automatic if  $G$  is quasi-split because any parabolic subgroup is defined over  $F$ . ?

⌋

**Def. (29.3.3.12) [Equivalence of Arthur-parameters].** Two global Arthur-parameters (29.3.3.10) are called **equivalent Arthur-parameters** if they are conjugate by an element of  $\widehat{G}(\mathbb{C})$  up to a cocycle of  $L_F$  in  $Z(\widehat{G}(\mathbb{C}))$  that is locally trivial at every place of  $F$ .

⌋

**Def. (29.3.3.13) [Generic and Discrete Paramaters].** A global Arthur-parameter  $\psi$  (29.3.3.10) is called **discrete global Arthur-parameter** if  $C_{\widehat{G}(\mathbb{C})}(\mathrm{Im}(\psi))^0 \subset Z(\widehat{G}(\mathbb{C}))$ , or equivalently  $\mathrm{Im}(\psi)$  is not contained in any proper relevant parabolic subgroup of  ${}^L G$  ?.

A global Arthur-parameter  $\psi$  (29.3.3.10) is called **generic global Arthur-parameter** if it is trivial on  $\mathrm{SL}(2; \mathbb{C})$ .

⌋

**Prop. (29.3.3.14) [Associated Global  $L$ -Parameters].** A global Arthur-parameter  $\psi$  defines an **associated global  $L$ -parameter**  $\phi$  (29.3.3.4) by

$$\phi(w) = \psi(w, \mathrm{diag}(|w|^{1/2}, |w|^{-1/2})).$$

where  $|-| : L_F \rightarrow \mathbb{R}^\times$  is given by the composition  $L_F \rightarrow W_F^{\mathrm{ab}} \xrightarrow{\mathrm{rec}^{-1}} \mathbf{I}_F \rightarrow \mathbb{R}_+$ . (Cf. [?assoaictieL0eireinsies]).

⌋

### Global $L$ -Packets and Arthur-Packets

References are [Art89, B-C09].

**Def. (29.3.3.15) [Weak Global  $L$ -Packets].**  $\pi, \pi' \in \mathrm{Irr}^{\mathrm{adm}}(G/F)$  are called **weakly  $L$ -indistinguishable** if there exists  $\Sigma_F^\infty \subset_{\mathrm{fin}} S \subset \Sigma_F$  s.t.  $\pi^S \cong (\pi')^S \in \mathrm{Irr}^{\mathrm{adm}}(G(\mathbf{A}_F^S))$ . An equivalence classes of weakly- $L$ -indistinguishable admissible representations in  $\mathrm{Irr}^{\mathrm{adm}}(G/F)$  is called a **weak global  $L$ -packet**. A weak global  $L$ -packet containing an automorphic representation (26.6.1.17) is called a **weak automorphic  $L$ -packet**.

⌋

**Def. (29.3.3.16) [Global  $L$ -Packets].** It is possible that  $\pi, \pi'$  have the same  $L$ -function, and for any  $r \in \mathrm{Rep}^{\mathrm{fd}}({}^L G)$  and any reductive group  $H$  over  $F$  and  $\sigma \in \mathrm{Irr}^{\mathrm{auto}}(H/F)$  together with  $r_1 \in \mathrm{Rep}^{\mathrm{fd}}({}^L H)$ , we have

$$L_S(\pi \otimes \sigma, r \otimes r_1; s) = L_S(\pi' \otimes \sigma, r \otimes r_1; s).$$

In this case,  $\pi, \pi'$  are called to be in the same global  **$L$ -packet**.

⌋



**Thm. (29.3.3.17) [Isobaric Automorphic Representations, Jacquet-Shalika].** Any weak automorphic  $L$ -packet of  $\mathrm{GL}(n)/F$  contains a unique isobaric element (26.6.1.28).  $\lrcorner$

*Proof:* Cf. [J-S81, J-S81a] ?  $\square$

**Prop. (29.3.3.18) [Which Finite Representation Appear in Automorphic Representations].** If  $G \in \mathrm{RedGrp}_{\sim, \mathrm{cnd}}^{\mathrm{ss}}/F$  and  $\emptyset \subset_{\mathrm{fin}} \Sigma_F$  s.t.  $G(\mathbf{A}_{F,S})$  is non-compact, then any  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(G(\mathbf{A}_{F,S}))$  is a limit (in a suitable sense ?) of local components at  $S$  of automorphic representations of  $G/F$ .  $\lrcorner$

*Proof:* Cf. [Clozel, Spectral theory of automorphic forms] Section 3.3.  $\square$

### Global Arthur-Packets for Unitary Groups

**Remark (29.3.3.19) [Arthur's Program].** Arthur's program asks the question of how to compute the multiplicity  $m_\pi$  of any discrete series  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$  in  $L^2(G(F) \backslash G(\mathbf{A}_F), \omega)$  (26.6.1.17).

Arthur's conjecture predicts the  $m_\pi$  and also gives a general hint about how the discrete spectrum of  $G$  is constructed from the tempered one, and even from the cuspidal ones of subgroups of  $G$ .

The problem with  $L$ -packets that motivated the introduction of Arthur-packets is that it is not always possible to construct a non-trivial linear combination of the characters of its members that is a stable distribution. This problem does not arise for tempered  $L$ -packets.

As long as we are interested in the discrete spectrum and with non-tempered representations, the Arthur-functoriality is better behaved than the  $L$ -packets. ?  $\lrcorner$

**Conj. (29.3.3.20) [Global Arthur Packets].** Let  $G^* \in \mathrm{RedGrp}/F$  be quasi-split and  $(G, \xi) \in H^1(F, G_{\mathrm{ad}}^*)$ . A **global Arthur-packet**  $\Pi_\psi$  associated to a discrete global Arthur-parameter  $\psi$  (29.3.3.10) is a subset of  $\mathrm{Irr}^{\mathrm{uni}}(G(\mathbf{A}_F), \omega)$  defined by

$$\Pi_\psi(G, \xi) = \{\pi = \otimes'_v \pi_v : \pi_v \in \Pi_{\psi_v}(G_v) \text{ (29.3.2.15)}, \pi_v \text{ unramified for a.e. } v \in \Sigma_F\},$$

where we are using  $\xi$  to identify  ${}^L G$  and  ${}^L G^*$ . There should be a direct integral decomposition

$$L^2(G(F) \backslash G(\mathbf{A}_F)) \cong \int \mathcal{A}_{[\psi]} d\hat{\mu}(\psi),$$

where  $\mathcal{A}_{[\psi]}$  is a subspace of  $\mathcal{A}(G/F)$  whose irreducible subquotients are all contained in the Arthur-packet  $\Pi_\psi$ , and  $d(\hat{\psi})$  is a natural measure class on the set of conjugacy classes of Arthur-parameters.

Two Arthur-parameters should be sent to the same Arthur-packet iff they are equivalent, and the correspondence above should send tempered Arthur-parameters (29.3.3.13) to tempered Arthur-packets, in which case it should coincide with the theory of Langlands. Note that this requirement, in the case  $G = \mathrm{GL}(m)$  is the generalized Ramanujan conjecture. ?  $\lrcorner$

*Proof:* ?  $\square$

**Thm & Conj. Cor. (29.3.3.21) [Arthur's Multiplicity Formula for Unitary Groups].** Let  $E/F \in \mathrm{GField}$  be a quadratic extension and  $\kappa \in \{\pm 1\}$ . Fix  $\chi_\kappa \in \mathcal{Z}_E^\kappa$  ? and let  $(G, \xi)$  be a pure inner form of  $G^* = \mathrm{U}_{E/F}(n)$ , choose  $z$  s.t.  $(G, \xi, z)$  is an extended pure inner twist, then there is a  $G(\mathbf{A}_F)$ -module isomorphism

$$L_{\mathrm{disc}}^2(G(F) \backslash G(\mathbf{A}_F)) \cong \bigoplus_{\psi \in \Psi_2(G^*, \eta_{\chi_\kappa})} L_{\mathrm{disc}, \psi}^2(G(F) \backslash G(\mathbf{A}_F)),$$

where  $L_{\text{disc},\psi}^2(G(F)\backslash G(\mathbf{A}_F))$  is a full representative of irreducible discrete representations  $\pi = \otimes'_v \pi_v$  s.t. the  $L$ -parameter of  $\pi_v$  is  $\phi_{\psi_v}$  for a.e.  $v \in \Sigma_F$ .

And when  $\psi$  is generic(29.3.3.4), we have a  $G(\mathbf{A}_F)$ -module isomorphism

$$L_{\text{disc},\psi}^2(G(F)\backslash G(\mathbf{A}_F)) \cong \bigoplus_{\pi \in \Pi_{\psi}(G,\xi,\epsilon_{\psi})} \pi,$$

where

- For each  $v \in \Sigma_F$ , in [Kaletha-Shin]section1.6.4? there is attached a local Arthur-packet  $\Pi_{\psi_v}(G_v, \xi_v)$  equipped with a map

$$\Pi_{\psi_v}(G_v, \xi_v) \rightarrow \text{Irr}(S_{\psi}^{\natural}, \chi_z) : \pi_v \mapsto \langle -, \pi_v \rangle_{\xi_v, z_v},$$

where  $\psi_v$  is the localization of  $\psi$ (29.3.3.8).

- $\Psi_{\psi}(G, \xi) = \{\pi = \otimes'_v \pi_v : \pi_v \in \Pi_{\psi_v}(G_v, \xi_v), \langle -, \pi_v \rangle_{\xi_v, z_v} = 1, a.e.v\}$ .
- $\epsilon_{\psi} : \overline{\mathcal{S}}_{\psi} \rightarrow \{\pm 1\}$  is a character defined by symplectic root numbers, which is trivial if  $\psi$  is generic. (Cf.[Mok]Section2.5. and [Arthur]Section1.5.),
- $\Pi_{\psi}(G, \xi, \epsilon_{\psi}) = \{\pi = \otimes'_v \pi_v \in \Pi_v(G, \xi) \mid \langle -, \pi \rangle_{\xi} = \epsilon_{\psi}\}$ , where  $\langle -, \pi \rangle_{\xi} = \prod_v \langle -, \pi_v \rangle_{\xi_v, z_v}$ .

┘

*Proof:* Cf.[Endoscopic Classification of Representations, Inner Forms of Unitary Groups, Kaletha-Shin]Thm1.7.1?.

□

## 4 Non-Quasi-Split Groups

References are [Vog93, GGP12, Kal16a].

### Local Conjectures

**Notation(29.3.4.1).**

- Let  $K \in \text{LField}^0$ ,
- Let  $G \in \text{RedGrp}/K$  be quasi-split.

┘

**Prop.(29.3.4.2).** Any pure inner form  $G'$  of  $G$  has the same center  $Z(G)$ (10.2.5.10), and the group

$$G'_{\text{ad}}(K)/G'(K) \cong \ker(H^1(K, Z(G)) \rightarrow H^1(K, G'))(10.2.5.3)$$

is independent of  $G'$  and acts on  $\text{Irr}^{\text{adm}}(G'(K))$ .

For any  $L$ -parameter  $\phi$  for  $G$ , the Tate duality

$$H^1(K, Z) \times H^1(K, \pi_1(\widehat{G})) \rightarrow \mathbb{C}^{\times}$$

and the coboundary map  $C_{\phi} \rightarrow H^1(k, \pi_1(\widehat{G}))$  induces a pairing

$$H^1(K, Z) \times A_{\phi} \rightarrow \mathbb{C}^{\times}.$$

Thus any  $g \in G'_{\text{ad}}(K)$  gives a character  $\eta_g$  of  $A_{\phi}$  via  $G'_{\text{ad}}(K) \rightarrow H^1(K, Z(G))$ .

┘

*Proof:* ?

□

**Conj.(29.3.4.3) [Langlands-Vogan Parametrization for Pure Inner Forms, Langlands/Vogan[Vog93]].** Let  $G \in \text{RedGrp}/K$  be quasi-split and  $\mathfrak{m}$  be a Whittaker-datum for  $G$ (10.4.4.23)., then:

1. For any pure inner twisting  $(G', \xi, z)$  of  $G$ (10.2.5.11) and  $\pi \in \text{Irr}^{\text{adm}}(G'(K))$ , we can associate an equivalence class of local  $L$ -parameters  $\phi_\pi \in \Phi(G)$ (29.3.1.6). And conversely, each  $\phi \in \Phi(G)$ (29.3.1.4) determines a finite set

$$\Pi_\phi = \coprod_{(G', \xi, z) \in H^1(K, G)} \Pi_\phi(G', \xi) \subset \coprod_{(G', \xi, z) \in H^1(K, G)} \text{Irr}^{\text{adm}}(G'(K)),$$

called the **Vogan-packet(local)** corresponding to  $\phi$ .

2.  $\Pi_\phi$  contains at most one  $\mathfrak{m}$ -generic representation(24.9.5.2). If it contains a  $\mathfrak{m}$ -generic representation,  $\Pi_\phi$  is called a **generic Vogan-packet**. And it is conjectured that this is true iff  $\text{ord}_{s=1} L(\phi, \text{Ad}; s) \geq 0$ , in particular, independent of  $\mathfrak{m}$  chosen. Notice by generic packet conjecture(29.3.2.4), a tempered  $L$ -packet is generic.
3. There is a bijection

$$\iota_{\mathfrak{m}} : \Pi_\phi \cong \text{Irr}(\pi_0(C_{\phi_v})) \text{ (29.3.1.4)}$$

making the following diagram commutative:

$$\begin{array}{ccc} \Pi_\phi & \xrightarrow{\iota_{\mathfrak{m}}} & \text{Irr}(\pi_0(C_\phi)) \\ \downarrow & & \downarrow \\ H^1(F, G) & \xrightarrow{\kappa_G} & \pi_0 \left( Z(G)^{\text{Gal}_K} \right)^\vee \end{array}$$

where the left vertical map maps  $\Pi_\phi(G', \xi)$  to the cohomology class of  $(G', \xi)$ , and the right vertical map maps  $\rho \in \text{Irr}(\pi_0(C_\phi))$  to its restriction to  $\pi_0 \left( Z(\hat{G})^{\text{Gal}_F} \right)$ , which is a character.

And if  $\Pi_\phi$  is  $\mathfrak{m}$ -generic, then the unique  $\mathfrak{m}$ -generic representation  $\pi \in \Pi_\phi$  corresponds to the trivial representation of  $\pi_0(C_\phi)$ . For  $\rho \in \text{Irr}(\pi_0(C_\phi))$ , denote  $\pi_{\mathfrak{m}}(\phi, \rho) = \iota_{\mathfrak{m}}^{-1}(\rho)$ .

4.  $\Pi_\phi \cap \text{Irr}^{\text{adm}}(G(K))$  is stable under action of  $G^{\text{ad}}(K)$ , and  $G^{\text{ad}}(K)$  acts transitively on the set of different generic representations of  $G(K)$  in  $\Pi_\phi \cap \text{Irr}^{\text{adm}}(G(K))$ . And for any  $g \in G'_{\text{ad}}(K)$ , the action of  $g$  on  $\Pi_\phi \cap \text{Irr}^{\text{adm}}(G'(K))$  corresponds to tensoring the character  $\eta_g$  on  $\text{Irr}(A_\phi)$  via any  $\iota_{\mathfrak{m}}$ .
5. (Central Characters) All representations in  $\pi \in \Pi_\phi$  have the same central character  $\omega_\phi$ (29.3.4.2) determined by  $\phi$  via recipe in [G-R10]Section8?.
6. (Endoscopic Character Identities, Kaletha[Kal16a]) If  $(H, \dot{s}, {}^L\eta) \in {}^L\dot{\mathfrak{E}}(G/K)$ (29.2.2.1) and  $\phi^H \in \Phi^{\text{temp.}}(H)$  s.t.  $\phi = {}^L\eta \circ \phi^H$ , then for any  $(G', \xi, z) \in H^1(K, G)$  and  $\Delta^{\mathfrak{m}, H, z}$ -matching functions  $f_H \in \mathcal{T}(H(K))$ ,  $f' \in \mathcal{T}(G'(K))$ (29.2.2.2), we have

$$S\Theta_{\phi^H}(f^H) = \Theta_{\phi, (G', \xi, z)}^{\dot{s}}(f') \stackrel{\triangle}{=} e(G') \sum_{\pi \in \Pi_\phi(G', \xi)} \text{tr}(\iota_{\mathfrak{m}}(\pi)(\dot{s})) \Theta_\pi(f') \text{ (29.3.2.4)}.$$

where  $\Theta_{\phi, (G', \xi, z)}^{\dot{s}}$  is called the  **$\dot{s}$ -stable character** of  $(G', \xi)$  associated to  $\phi$ , and denote  $S\Theta_{\phi, (G', \xi, z)} \stackrel{\triangle}{=} \Theta_{\phi, (G', \xi, z)}^1$ , called the **stable character**

⌋

*Proof:* ?

If  $K \in \{\mathbb{R}, \mathbb{C}\}$ , the bijection  $\iota_{\mathfrak{m}}$  is constructed in [Vog93]Thm6.3 using Shelstad's work.  $\square$

**Thm& Conj.Cor.(29.3.4.4)** [GL( $n$ ) case]. If  $G = \mathrm{GL}(n)$ , then

- $A_{\phi} = 1$  for any  $L$ -parameter  $\phi$ .
- There are no non-trivial pure inner forms of  $G$ .
- The adjoint action of  $G^{\mathrm{ad}}(K)$  is trivial, because  $H^1(K, Z(G)) = 1$ .
- There is a unique orbit of generic unipotent characters(24.11.2.2), and  $\Pi_{\phi}$  is generic iff  $\mathrm{ord}_{s=1} L(\phi, \mathrm{Ad}; s) \geq 0$ .
- The Langlands-Vogan parametrization conjecture is equivalent to the Langlands conjecture for  $G$ , and is true by(29.3.2.17).

⌋

**Conj.Cor.(29.3.4.5)** [Sp( $V$ ) cases].  $\square$

*Proof:* •  $\square$ 

**Conj.Cor.(29.3.4.6)** [SO( $V$ ) cases].  $\square$

*Proof:* •  $\square$ 

**Conj.(29.3.4.7)** [Langlands Conjecture via Isocrystals, Langlands/Rapoport[Rap95]/Kaletha[Kal14]]. Let  $G^* \in \mathcal{R}\mathrm{edGrp}/K$  be quasi-split and  $\mathfrak{m}$  a Whittaker-datum for  $G^*$ (10.4.4.23), then:

1. For any  $B$ -inner twisting  $(G', \xi, b) \in B(G^*)_b$ (28.6.1.26) and  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(G'(K))$ , we can associate an equivalence class of local  $L$ -parameters  $\phi_{\pi} \in \Phi(G)$ (29.3.1.6). And conversely, each  $\phi \in \Phi(G)$  determines a finite set

$$\Pi_{\phi} = \coprod_{(G', \xi, b) \in B(G^*)_b} \Pi_{\phi}(G', \xi) \subset \coprod_{(G', \xi, b) \in B(G^*)_b} \mathrm{Irr}^{\mathrm{adm}}(G'(K)),$$

called the **extended Vogan-packet(local)** corresponding to  $\phi$ .

2.  $\Pi_{\phi}$  contains at most one  $\mathfrak{m}$ -generic representation(24.9.5.2). If it contains a  $\mathfrak{m}$ -generic representation,  $\Pi_{\phi}$  is called a **generic extended Vogan-packet**. And it is conjectured that this is true iff  $\mathrm{ord}_{s=1} L(\phi, \mathrm{Ad}; s) \geq 0$ , in particular, independent of  $\mathfrak{m}$  chosen. Notice by generic packet conjecture(29.3.2.4), a tempered Vogan-packet is generic.
3. There is a bijection

$$\iota_{\mathfrak{m}} : \Pi_{\phi} \cong \mathrm{Irr}(\pi_0(C_{\phi_v})) \text{ (29.3.1.4)}$$

making the following diagram commutative:

$$\begin{array}{ccc} \Pi_{\phi} & \xrightarrow{\iota_{\mathfrak{m}}} & \mathrm{Irr}(\pi_0(C_{\phi_v}^{\natural})) \\ \downarrow & & \downarrow \\ B(G)_b & \xrightarrow{\kappa_G} & X^{\bullet}(Z(G)^{\mathrm{Gal}_K}) \end{array}$$

where  $C_\phi^\natural = C_\phi / (C_\phi \cap [\widehat{G}_{\mathbb{C}}, \widehat{G}_{\mathbb{C}}])$ , and the left vertical map sends  $\Pi_\phi(G', \xi, b)$  to  $b$ , and the right vertical map sends  $\rho \in \text{Irr}(C_\phi^\natural)$  to its restriction to  $Z(\widehat{G})^{\text{Gal}_F}$ , which is a character.

And if  $\Pi_\phi$  is  $\mathfrak{m}$ -generic, then the unique  $\mathfrak{m}$ -generic representation  $\pi \in \Pi_\phi$  corresponds to the trivial representation of  $C_\phi^\natural$ . For  $\rho \in \text{Irr}(C_\phi^\natural)$ , denote  $\pi_\mathfrak{m}(\phi, \rho) = \iota_\mathfrak{m}^{-1}(\rho)$ .

4.  $\Pi_\phi \cap \text{Irr}^{\text{adm}}(G(K))$  is stable under action of  $G^{\text{ad}}(K)$ , and  $G^{\text{ad}}(K)$  acts transitively on the set of different generic representations of  $G(K)$  in  $\Pi_\phi \cap \text{Irr}^{\text{adm}}(G(K))$ . And for any  $g \in G'_{\text{ad}}(K)$ , the action of  $g$  on  $\Pi_\phi \cap \text{Irr}^{\text{adm}}(G'(K))$  corresponds to tensoring the character  $\eta_g$  on  $\text{Irr}(A_\phi)$  via any  $\iota_\mathfrak{m}$ .
5. (Central Characters) All representations in  $\pi \in \Pi_\phi$  have the same central character  $\omega_\phi$  (29.3.4.2) determined by  $\phi$  via recipe in [G-R10]Section8?.
6. (Endoscopic Character Identities, Kaletha[Kal16a]) If  $(H, s, {}^L\eta) \in {}^L\mathfrak{E}(G^*/K)$  and  $\phi^H \in \Phi^{\text{temp.}}(H)$  s.t.  $\phi = {}^L\eta \circ \phi^H$ , then for any  $(G', \xi, b) \in B(G^*)_b$  and  $\Delta^{\mathfrak{m}, H, b}$ -matching functions  $f_H \in \mathcal{T}(H(K))$ ,  $f' \in \mathcal{T}(G'(K))$  (29.2.2.3), we have

$$\text{S}\Theta_{\phi^H}(f^H) = e(G) \sum_{\pi \in \Pi_\phi(G', \xi, z)} \text{tr}(\pi(s)) \Theta_\pi(f') \quad (29.3.2.4).$$

┘

*Proof:*

□

**Remark (29.3.4.8).** Kottwitz showed that the association  $B(G^*)_b \rightarrow H^1(K, G^*_{\text{ad}}) : b \mapsto J_b$  is surjective if  $Z(G^*)$  is connected. ? But when  $G^*$  is semi-simple,  $B(G^*)_b$  can only give pure inner forms. ┘

**Thm& Conj.Cor. (29.3.4.9) [Kaletha, Supercuspidal L-packets via isocrystals].** and [Kaletha, Epipelagic L-packets and rectifying characters] verified this conjecture for regular depth-zero supercuspidal parameters, and epipelagic parameters. ? ┘

*Proof:*

□

**Conj. (29.3.4.10) [Langlands Conjecture via Rigid Inner Forms, Langlands/Kaletha[Kal16a]].** ┘

*Proof:* ?

□

**Thm& Conj.Cor. (29.3.4.11) [Real Langlands Conjecture, Shelstad/Kaletha[Kal16]].** If  $K = \mathbb{R}$ , then the Langlands conjecture via rigid inner forms holds. ┘

*Proof:* Cf. [Kal16]Section5.6.

□

**Thm. (29.3.4.12) [Rigid Inner Forms vs Isocrystals].** If the Langlands conjecture via isocrystals (29.3.4.7) holds for all quasi-split  $G^* \in \text{RedGrp}/K$  with connected centers, then the Langlands conjecture via rigid inner forms (29.3.4.10) holds for all quasi-split  $G^* \in \text{RedGrp}/K$ . ┘

*Proof:* Cf. [Kaletha, Rigid inner forms vs isocrystals].

□

## Global Vogan-Packets

### 5 Global Langlands Functoriality

References are [Kna97, Rog97].

**Conj. (29.3.5.1) [Global Functorial Lifting, Langlands[Lan70]].** Suppose  $G, G' \in \text{RedGrp}/F$  are quasi-split. Suppose  $\varphi : {}^L G' \rightarrow {}^L G \in \text{AlgGrp}/\overline{\mathbb{Q}}$  preserves the projection to  $\text{Gal}_F$ . If  $w'$  is a prime of  $L'$ ,  $w = w' \cap L, v = w \cap F$ , then  $\varphi$  determines a homomorphism  $\varphi_v : {}^L G'_{L'_{w'}/F_v} \rightarrow {}^L G_{L_w/F_v}$  that makes the diagram

$$\begin{array}{ccc} {}^L G'_{L'_{w'}/F_v} & \longrightarrow & \text{Gal}(L'_{w'}/F_v) \\ \downarrow \varphi_v & & \downarrow \\ {}^L G_{L_w/F_v} & \longrightarrow & \text{Gal}(L_w/F_v) \end{array}$$

commutative.

Then for  $\pi' = \otimes_v \pi'_v \in \text{Irr}^{\text{auto}}(G'/F)$ , is it true that  $\pi = \otimes_v R_{\varphi_v}(\pi'_v) \in \text{Irr}^{\text{cusp}}(G/F)$  (29.3.2.1)? Such an automorphic representation is called a **functorial lifting** or  $\pi'$ .

Notice that, for  $v \in \Sigma_F^{\text{fin}}$  s.t.  $G_{F_v}, G'_{F_v}, \pi_v, \pi'_v$  are all unramified, this is equivalent to  $c_v(\pi)$  is  $\widehat{G}'(\mathbb{C})$ -conjugate to  $\varphi(c_v(\pi'))$  (29.3.2.11).  $\perp$

**Remark (29.3.5.2).** This conjecture reduces the conjecture from a general group to the case of  $\text{GL}(n; F)$ , which can be handled, such as by [Zeta Functions of Simple Algebras, Godement-Jacquet].  $\perp$

**Conj. Cor. (29.3.5.3) [Automorphic Weil Representations].** Suppose  $G/F$  is quasi-split and splits over a Galois extension  $L/F$ . Let  ${}^L U_L$  be a maximal compact subgroup of  ${}^L G_F$ . Let  $\varphi : W_{L/F} \rightarrow {}^L U_F$  be a homomorphism that makes the diagram

$$\begin{array}{ccc} W_{L/F} & \longrightarrow & \text{Gal}(L/F) \\ \downarrow \varphi & & \downarrow \\ {}^L U_F & \longrightarrow & \text{Gal}(L/F) \end{array}$$

commutative. If  $\mathfrak{P}'$  is a prime of  $K'$  and  $\mathfrak{p} = \mathfrak{P}' \cap F$ , then  $\varphi_{\mathfrak{p}} = \varphi \circ \alpha_{\mathfrak{p}}$  takes  $W_{K'_{\mathfrak{P}'}/F_{\mathfrak{p}}}$  into  ${}^L U_{F_{\mathfrak{p}}}$ .

If  $\pi(\varphi) = \otimes_{\mathfrak{p}} \pi(\varphi_{\mathfrak{p}})$ , then  $\pi(\varphi)$  is an automorphic representation, called the **Automorphic lift** of  $\rho$ .  $\perp$

*Proof:*  $\square$

**Conj. Cor. (29.3.5.4) [Automorphic Galois Representations for  $\text{GL}(n)$ , Artin-Langlands[Lan70]].** In (29.3.5.1), take  $G' = e$  and  $G = \text{GL}(n)$ , then  $\pi'$  is trivial, and  $\rho : \text{Gal}(E/F) \rightarrow \text{GL}(n; \mathbb{C})$  is an injective homomorphism for some finite extension  $E/F$ . Then the conjecture asserts that there exists  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  s.t.

$$c_v(\pi) = \varphi(\text{Frob}_v) \quad (29.3.1.12)$$

for any  $v \notin \text{Ram}(F)$ . Such a representation  $\pi$  is called an **automorphic Galois representation**.

Notice that in fact it suffices to show that  $c_v(\varphi) = \varphi(\text{Frob}_v)$  is true for a.e.  $v \in \Sigma_F^{\text{fin}}$ . And it is unique if it exists, by strong multiplicity theorem for  $\text{GL}(n)$  (26.6.5.1).  $\perp$

*Proof:*

□

**Thm& Conj.Cor. (29.3.5.5).**

1. For  $n = 1$ , the Artin-Langlands conjecture(29.3.5.4) is equivalent to the Kronecker-Weber theorem, or more generally the Artin reciprocity law, thus it is true.
2. If the image of  $\rho$  is nilpotent, then this is true by cyclic automorphic induction for  $\mathrm{GL}(n)$  proved by Arthur-Clozel.?
3. If  $n = 2$  and  $\mathrm{Im}(\rho)$  is solvable(i.e. not  $\mathrm{Alt}_5$ (25.1.2.14)), then this is true.
4. if  $n = 2$  and the image of  $\mathrm{Gal}_{\mathbb{Q}}$  in  $\mathrm{PGL}(2; \mathbb{C})$  is isomorphic to  $\mathrm{Alt}_5$ , then under mild conditions, this is proved in [On icosahedral Artin Representations].?

┘

*Proof:* 2:

3: When  $n = 2$ , by(25.1.2.14), it suffices to consider the case that  $\rho$  is of cyclic/dihedral/tetrahedral/octahedral type.

If  $\rho$  is of cyclic or dihedral type, then by(24.1.3.40),  $\rho = \mathrm{Ind}_E^F(\chi)$  for some quadratic extension  $E/F$  and  $\chi \in (\mathrm{Gal}_E)^\vee$  s.t.  $\chi^\tau \not\cong \chi$  where  $\tau \in \mathrm{Gal}(E/F) \setminus \{\mathrm{id}\}$ . Then by global class field theory and cyclic automorphic induction(29.3.5.15),  $\pi(\rho) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/F)$  exists.

If  $\mathrm{Im}(\mathrm{Ad} \circ \rho) \cong \mathrm{Alt}_4$ , then

If  $\mathrm{Im}(\mathrm{Ad} \circ \rho) \cong \mathrm{Sym}_4$ , then

4:

□

**Lemma (29.3.5.6).** Let  $\rho : \mathrm{Gal}_F \rightarrow \mathrm{GL}(2; \mathbb{C})$  be of tetrahedral type and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(2)/F)$  s.t. the following holds:

- $\mathrm{BC}_F^E(\pi) = \pi(\rho_E)$ ,
- $\omega_\pi = \det(\rho)$ ,
- $\mathrm{Ad}(\pi) = \pi(\mathrm{Ad} \circ \rho)$ .

Then  $\pi = \pi(\rho)$ .

┘

*Proof:*

□

**Lemma (29.3.5.7).** Let  $\rho : \mathrm{Gal}_F \rightarrow \mathrm{GL}(2; \mathbb{C})$  and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(2)/F)$  s.t.  $\mathrm{BC}_F^E(\pi) = \pi(\rho|_E)$  and  $\mathrm{BC}_F^K(\pi) = \pi(\rho|_K)$ , then  $\pi = \pi(\rho)$ .

┘

*Proof:*

□

**Remark (29.3.5.8) [Representations of Icosahedral Type].** The case of representations of icosahedral type is particularly hard because Maass forms with eigenvalue  $1/4$  are very hard to access: Given an even (projective)  $A_5$  Galois representation, we don't even know how to prove that there exists a corresponding Maass form with the right Laplacian eigenvalue.

┘

**Thm& Conj.Cor. (29.3.5.9) [Odd dim 2-Representations, Khare-Wintenberger[K-W09b, K-W09c]/ Pilloni-Stroh[P-S16a]].** The Artin-Langlands conjecture(29.3.5.4) holds if  $F$  is totally real and  $\rho$  is odd.

┘

*Proof:*

□

**Prop. (29.3.5.10) [Automorphic for Virtual Representations, Jacquet-Shalika[?J-K81a]].** Let  $L/F \in \mathbf{NField}$  and  $\pi \in \mathrm{Irr}(\mathrm{Gal}(L/F))$  s.t.  $\pi = \sigma - \tau$  for some automorphic representations  $\sigma$  and  $\tau$  (29.3.5.4), then  $\pi$  is also automorphic.  $\lrcorner$

*Proof:* Cf.[?J-K81a]P812.  $\square$

**Def. (29.3.5.11) [Potentially Abelian Representations].** For  $F \in \mathbf{NField}$  and  $n \in \mathbb{Z}_+$ , a  $\Pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$  is called **potentially Abelian** if there exists  $\Sigma \in \mathcal{R}\mathrm{ep}^n(W_F)$  s.t.  $\Sigma|_{W_{F_v}} = \mathrm{rec}_v(\Pi_v)$  for any  $v \in \Sigma_F$ .  $\lrcorner$

### Representation Lifting

**Conj.Cor. (29.3.5.12) [Tensor Product Lifting].** Let  $n, m \in \mathbb{Z}_+$  and  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F), \pi' \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(m)/F)$ . Let  $n_1, \dots, n_r \in \mathbb{Z}_+$  and  $\pi_j \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n_j)/F)$  for any  $j \in [r]_+$ , then denote

$$\pi \boxtimes \pi' = \otimes_{j=1}^r \pi_j$$

if  $\sum_j n_j = mn$ , and

$$c(\pi_v) \otimes c(\pi'_v) \cong \otimes_{j=1}^r c(\pi_{j,v})$$

for a.e.  $v \in \Sigma_F^{\mathrm{fin}}$ .

If such  $n_j, \pi_j$  exists, then  $\pi \boxtimes \pi'$  is well-defined. Then it is conjectured that  $\pi \boxtimes \pi'$  is well-defined for any  $\pi, \pi'$ .  $\lrcorner$

*Proof:*  $\square$

*Proof:*  $\square$

**Conj.Cor. (29.3.5.13) [Global Inner Twists].** If (29.3.2.6) is true, if  $G, G' \in \mathbf{AlgGrp}/F$  and  $G$  is quasi-split and  $G'$  is an inner twist of  $G$ , suppose  $\pi' = \otimes_{\mathfrak{p}} \pi'_{\mathfrak{p}}$  is an automorphic representation of  $G'_F$ , is it true that  $\pi = \otimes_{\mathfrak{p}} R(\pi'_{\mathfrak{p}})$  is an automorphism representation of  $G_F$ ?  $\lrcorner$

*Proof:*  $\square$

### Automorphic Base Change and Inductions

**Conj.Cor. (29.3.5.14) [Automorphic Inductions].** Suppose  $L/F$  be a finite extension with  $[L:F] = d$ . Let  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/L)$ , then an automorphic form  $\Pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(np)/F)$  is said to be the **automorphic induction** of  $\pi$  if

$$L(\Pi_v; s) = \prod_{w|v} L(\pi_w; s)$$

for a.e.  $v \in \Sigma_F^{\mathrm{fin}}$ . Such an automorphic induction is unique if it exists by strong multiplicity one, denoted by  $\mathrm{AI}_F^L(\pi)$ .

Then it is conjectured that such a base change always exists.  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj.Cor. (29.3.5.15) [Cyclic Automorphic Induction, Hecke/Maass/Jacquet-Langlands/Shalika-Tanaka[S-T69]/Jacquet-Piatetski-Shapiro-Shalika/Arthur-Clozel].** If  $E/F$  is a cyclic extension, then automorphic induction exists. Moreover, if  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/E)$ , then  $\mathrm{AI}_E^F(\pi)$  is cuspidal unless there exists  $\tau \neq \mathrm{id} \in \mathrm{Gal}(E/F)$  s.t.  $\pi \cong \sigma(\pi)$ .  $\lrcorner$



*Proof:*

□

**Prop. (29.3.5.16) [Unramified Langlands Class of Automorphic Inductions].** Let  $L/F$  be a finite extension with  $[L : F] = d$ . If  $v \in \Sigma_F^{\text{fin}}$  is unramified in  $L$  and  $\pi_w$  is unramified for any  $w|v$ , then if  $\Pi = \text{AI}_E^F(\pi)$  exists, we can describe the the Langlands class of  $\Pi_v$  as follows: Denote  $d_w = [L_w : F_v]$  and suppose  $c_w(\pi) \sim \text{diag}(z_{w1}, \dots, z_{wn})$ , then  $c_c(\Pi)$  is the semisimple conjugacy class with eigenvalues the union of  $d_w$ -th roots of  $\{z_{wj}\}$ .  $\lrcorner$

*Proof:* This is because

$$L(\pi_w; s) = \prod_{j=1}^n (1 - z_{wj} \|w\|^{-s})^{-1} = \prod_{j=1}^n \prod_{k=1}^d (1 - z_{wj}^{1/d_w} \zeta_{d_w}^k \|v\|^{-s})^{-1}.$$

□

**Conj.Cor. (29.3.5.17) [Automorphic Base Change].** Suppose  $L/F$  be a finite extension with  $[L : F] = d$ . Let  $\pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$ , then an automorphic form  $\Pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/L)$  is called a **base change lift** of  $\pi$  if for a.e.  $v \in \Sigma_F^{\text{fin}}$  and  $w \in S_L(v)$  we have

$$c_w(\Pi) = c_v(\pi)^{[w:v]}.$$

Such an automorphic induction is unique if it exists by strong multiplicity one, denoted by  $\text{BC}_F^L(\pi)$ .

Then it is conjectured that such a base change always exists.  $\lrcorner$

*Proof:*

□

**Thm& Conj.Cor. (29.3.5.18) [Cyclic Base Change Lifts, Langlands[Lan80]/Arthur-Clozel[A-C89]].** Suppose  $L/F$  is a cyclic extension of order  $\ell \in \text{Prime}$ , then

- For any  $\pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$ ,  $\text{BC}_F^L(\pi)$  exists. By iteration, we see that in fact base change lifts exist for any solvable extension.
- For  $\pi, \pi' \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$ ,  $\text{BC}_F^L(\pi) \cong \text{BC}_F^L(\pi')$  iff there exists a character  $\psi$  of  $\mathbf{I}_F/F^\times \text{Nm}_{L/F}(\mathbf{I}_L)$  s.t.  $\pi = \pi' \otimes \psi$ .
- For  $\Pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/L)$ , there exists  $\pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$  s.t.  $\Pi = \text{BC}_F^L(\pi)$  iff  $\sigma(\Pi) \cong \Pi$  for any  $\Pi \in \text{Gal}(E/F)$ .

□

*Proof:* ?

□

**Thm& Conj.Cor. (29.3.5.19) [Cubic Base Change Lifting, Jacquet-Piatetski-Shapiro-Shalika].** For any cubic extension  $L/F$  and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(2)/F)$ , the base change lifting  $\text{BC}_F^L(\pi) \in \text{Irr}^{\text{cusp}}(\text{GL}(2)/L)$  exists.  $\lrcorner$

*Proof:* Cf.[Relevement cubique non normal].

□

**Example (29.3.5.20).** Suppose  $L/F \in \text{GField}$  and  $\chi$  is a Hecke character of  $F$ , then  $\text{BC}_F^L(\chi) = \pi \circ \text{Nm}_{E/F}$ .

And for  $\pi \in \text{Irr}^{\text{auto}}(\text{GL}(n; F))$ , we have  $\text{BC}_F^L(\pi \otimes \chi) = \text{BC}_F^L(\pi) \otimes (\chi \circ N_{L/F})$ .  $\lrcorner$

**Remark (29.3.5.21).** For a general group  $G$ , there are no local-global compatibility between automorphic base change representations, unless for tempered representations. ?  $\lrcorner$

### Examples

**Thm. (29.3.5.22) [Lafforgue].** Lafforgue proved a form of global correspondence as in (29.3.7.2) for  $\mathrm{GL}(n)$  over a function field, but with the Langlands group and motivic Galois group replaced by  $\mathrm{Gal}_F$ , and with complex representations replaced by  $\ell$ -adic representations.  $\lrcorner$

*Proof:* Cf. [Lafforgue, Chtoucas de Drinfeld et correspondance de Langlands].  $\square$

**Thm. (29.3.5.23) [Classical Groups].** Let  $F \in \mathbf{NField}$ ,  $G = \mathrm{GL}(n)$ , and  $G'$  a split form of  $\mathrm{SO}(2n+1)$ ,  $\mathrm{SO}(2n)$  or  $\mathrm{Sp}(2n)$ , then  $\widehat{G}' = \mathrm{Sp}(2n; \mathbb{C})$ ,  $\mathrm{SO}(2n+1; \mathbb{C})$  or  $\mathrm{SO}(2n; \mathbb{C})$ , with the standard representation  $\rho_{\mathrm{Std}}$ . Then in this case, [Cogdell-Kim-Shapiro-Shahidi] established the functoriality for generic  $\pi' \in \mathrm{Irr}^{\mathrm{auto}}(G')$ .  $\lrcorner$

*Proof:* Cf. [Functoriality for Classical Groups, Cogdell-Kim-Shapiro-Shahidi]. ?

The proof sketch is as follows: Firstly we have local partial liftings  $\Pi_v$  by LLC for  $\mathrm{GL}(n)$  (29.3.2.17), then we take the tensor product  $\Pi = \otimes_v \Pi_v$  and use Langlands-Shahidi methods to analyze their  $L$ -functions and show that  $L(\Pi \times \pi'; s)$  has nice analytic properties for any  $\pi' \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(m)/F)$ . Then we use Cogdell-Shapiro converse theorem to conclude that  $\Pi$  is an automorphic representation. Finally, [Ginzburg-Rallis-Soudry01] shows that a weak lift is a strong lift.  $\square$

**Thm. (29.3.5.24) [Adjoint Action Lifting of  $\mathrm{GL}(2)$ , Gelbart-Jacquet].** Consider the adjoint representation  $\mathrm{Ad} : \mathrm{GL}(2) \rightarrow \mathrm{GL}(3)$ , then for any  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(2)/F)$ , there exists  $\mathrm{Ad}(\pi) \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(3)/F)$  s.t.  $\mathrm{Ad}(c(\pi)) = c(\mathrm{Ad}(\pi))$ . Moreover,  $\mathrm{Ad}(\pi)$  is cuspidal iff  $\pi$  is not of the form  $AI_F^E(\theta)$  for some quadratic extension  $E/F$  and some Hecke character  $\theta$  of  $E$ .  $\lrcorner$

*Proof:* Cf. [A relation between automorphic representations of  $\mathrm{GL}(2)$  and  $\mathrm{GL}(3)$ , Gelbart-Jacquet]. The last assertion combines ??  $L$ -indistinguishability for  $SL(2)$ .  $\square$

**Thm. (29.3.5.25) [Symmetric Power Lifting, Gelbart-Jacquet [G-J78]/Kim-Shahidi02/Kim03].**

Let  $F \in \mathbf{NField}$ , then

- Kim established automorphy of  $\mathrm{Sym}^k(\pi)$  for  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/F)$  and  $k \in [4]_+$ .
- If  $F$  is totally real and  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/F)$  s.t.  $\pi_v$  is essentially- $L^2$  for all  $v \in \Sigma_F^\infty$ , then Newton-Thorne established automorphy of  $\mathrm{Sym}^k(\pi)$  for all  $k \in \mathbb{Z}_+$ .

$\lrcorner$

*Proof:* Cf. [Kim. Functoriality for the exterior square of  $\mathrm{GL}(4)$  and the symmetric fourth of  $\mathrm{GL}(2)$ ]. [Newton-Thorne22].  $\square$

**Thm. (29.3.5.26) [Tensor-Product Lifting, Ramakrishnan [Ram00]/Kim-Shahidi02/Kim03].**

Let  $F \in \mathbf{NField}$ , then Kim-Shahidi established functoriality for the following cases:

- $G = \mathrm{GL}(2k)$ ,  $G' = \mathrm{GL}(2) \times \mathrm{GL}(k)$  where  $k \in [3]_+$ , and  $\rho : \widehat{G}' = \mathrm{GL}(2; \mathbb{C}) \times \mathrm{GL}(k; \mathbb{C}) \rightarrow \mathrm{GL}(2k; \mathbb{C}) = \widehat{G}$  is the tensor product representation.
- $G = \mathrm{GL}(4)$ ,  $G' = \mathrm{GL}(2)$ , and  $\rho : \mathrm{GL}(2; \mathbb{C}) \rightarrow \mathrm{GL}(4; \mathbb{C})$  is irreducible.

$\lrcorner$

*Proof:* Cf. [Kim. Functoriality for the exterior square of  $\mathrm{GL}(4)$  and the symmetric fourth of  $\mathrm{GL}(2)$ ].  $\square$

**Thm. (29.3.5.27) [Jacquet-Shalika].** Let  $n \in \mathbb{Z}_+$  and  $\pi, \pi' \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n))$ . If  $c(\pi_v) \otimes c(\pi_v^\vee) \cong c(\pi'_v) \otimes c(\pi'_v^\vee)$ , then  $\pi \cong \pi'$ .  $\lrcorner$

*Proof:* Cf. [On Euler products and the classification of automorphic representations 1, 2, Jacquet-Shalika].  $\square$

## 6 Jacquet-Langlands Correspondences

References are [Trace Formula and the Proof of the Global Jacquet-Langlands Correspondence].

**Notation (29.3.6.1).**

- Use notations on Azumaya algebras over arithmetic fields defined in [Brauer-Grothendieck Groups](#).

┘

### Representations of $D^\times$

**Notation (29.3.6.2).**

- Let  $K \in \text{LField}$ .
- Let  $D \in \text{Az}_K$ .

┘

**Prop. (29.3.6.3).**  $D^\times/K^\times$  is compact.

┘

*Proof:* Use the description of  $D$  in [\(11.2.3.5\)?](#).

□

**Cor. (29.3.6.4).** Any  $\pi \in \text{Irr}^{\text{adm}}(D^\times)$  is essentially square-integrable [\(24.9.8.10\)](#).

┘

### Local Jacquet-Langlands Correspondence

**Thm. (29.3.6.5)** [Local Jacquet-Langlands Correspondence, Jacquet-Langlands [\[J-L70\]](#)/ Rogawski [\[Rog83\]](#)/Deligne-Kazhdan-Vigneras [\[DKV84\]](#)]. If  $D \in \text{Az}_K$  has index  $\text{ind}_K(D) = n$ , then there is a bijection

$$\text{J-L}_D : \text{Irr}^{\text{adm}}(D^\times) \rightarrow \text{Irr}^{\text{ess}, L^2}(\text{GL}(n; K))$$

s.t.

- $\chi_\rho = (-1)^{n-1} \chi_{\text{J-L}_D(\rho)}$ .
- $\omega_\rho = \omega_{\text{J-L}(\rho)}$ .
- $L(\rho; s) = \pm L(\text{J-L}(\rho); s)$  and  $\epsilon(\pi; s) = \pm \epsilon(\text{J-L}(\rho); s)$ , where the definition is given in [\[G-J72\]?](#)
- For any quasi-character  $\chi$  of  $K^\times$ , we have  $\text{J-L}(\rho(\chi)) = \text{J-L}(\rho)(\chi)$ .
- $\text{J-L}(\rho^\vee) = \text{J-L}(\rho)^\vee$ .

┘

*Proof:*

□

### Global Jacquet-Langlands Correspondence

**Notation (29.3.6.6).**

- Let  $F \in \text{NField}$ .
- Let  $D \in \text{Az}_F$  of dimension  $n^2$ .

┘

**Thm. (29.3.6.7)** [Global Jacquet-Langlands Correspondence, Vigneras].

- For  $\rho \in \text{Irr}^{\text{auto}}(D^\times/F)$ , there exists a unique  $\text{J-L}(\rho) \in \text{Irr}^{\text{disc}}(\text{GL}(n)/F)$  s.t.  $\text{J-L}(\rho)_v \cong \rho_v$  for any  $v \in \text{Unr}_F(D)$ .

- If  $v \in \text{Ram}_F(D)$  and  $\text{J-L}(\rho_v) = \text{St}_{s_v}(\sigma_v)$  for some  $s_v|n$  and  $\sigma_v \in \text{Irr}^{\text{cusp}}(\text{GL}(n/s_v; F_v))$ , then  $\text{J-L}(\rho)_v$  is isomorphic to either  $\text{St}_{s_v}(\sigma_v)$  or  $Q((\sigma_v, s_i))$ .
- The image of J-L is the set of  $\Pi \in \text{Irr}^{\text{disc}}(\text{GL}(n)/F)$  s.t. for any  $v \in \text{Ram}_F(D)$ ,  $\Pi_v$  is isomorphic to  $\text{St}_{s_v}(\sigma_v)$  or  $Q((\sigma_v, s_i))$  for some  $s_i|n$  and  $\sigma_v \in \text{Irr}^{\text{cusp}}(\text{GL}(n/s_v; F_v))$ .
- (Strong Multiplicity One) If  $\rho_1, \rho_2 \in \text{Irr}^{\text{auto}}(D^\times/F)$  s.t.  $\rho_{1x} \cong \rho_{2x}$  for a.e.  $x \in \Sigma_F$ , then  $\rho_1 = \rho_2$ .  $\lrcorner$

*Proof:* Cf. [H-T01]P195.  $\color{red}?$   $\square$

**Cor. (29.3.6.8).** Suppose  $\rho \in \text{Irr}^{\text{auto}}(D^\times/F)$ , the following are equivalent:

- $\text{J-L}(\pi) \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$ .
- $\rho_x$  is generic for some  $x \in \Sigma_F \setminus \text{Ram}_F(D)$ .
- $\rho_x$  is generic for any  $x \in \Sigma_F \setminus \text{Ram}_F(D)$ .  $\lrcorner$

*Proof:* Combine the theorem with the main theorem of [Moeglin-Waldspurger] on residual spectrum of  $\text{GL}(n)$ .  $\color{red}?$   $\square$

**Def. (29.3.6.9) [Adjunct Representations].** If  $F/F_0$  is a CM field with  $\text{Gal}(F/F_0) = \{1, \mathbf{c}\}$ , then for  $\rho \in \text{Irr}^{\text{auto}}(D^\times/F)$ , we can define the **adjunction automorphic representation**  $\rho^* \in \text{Irr}^{\text{auto}}(D^\times/F)$  to be

$$\rho^*(g) = \rho(g^{-*}).$$

Then via the Jacquet-Langlands correspondence,  $\text{J-L}(\Pi^*) \cong \text{J-L}(\rho)^\vee \circ \mathbf{c}$ , by strong multiplicity theorem.  $\lrcorner$

**Thm. (29.3.6.10) [generalized Global Jacquet-Langlands Correspondence for Discrete Series, Badulescu(08)/Badulescu-Renard(10)].** For  $k \in \mathbb{Z}_+$ , there exists an injection

$$\text{J-L}_D : \text{Irr}^{\text{disc}}(\text{GL}(k; D)) \rightarrow \text{Irr}^{\text{disc}}(\text{GL}(kn; F))$$

s.t.  $\text{J-L}_D(\rho)_v = \text{J-L}_{D_v}(\rho_v)$  (29.3.6.5) for any  $v \in \Sigma_F$ .  $\lrcorner$

*Proof:*  $\square$

The case  $n = 1, d = 2$

## 7 Motivic Galois Groups

**Def. (29.3.7.1) [Motivic Galois Groups].** The motivic group  $\mathcal{G}_F$  is a proalgebraic group whose representations parameterizes motives.  $\lrcorner$

**Conj. (29.3.7.2) [Motives and Automorphic Forms, Langlands].** There is a commutative diagram of continuous homomorphisms

$$\begin{array}{ccc} L_F & \xrightarrow{\Phi} & \mathcal{G}_F \\ & \searrow \quad \swarrow & \\ & \text{Gal}_F & \end{array}$$

together with

$$\begin{array}{ccc} L_{F_v} & \xrightarrow{\Phi} & \mathcal{G}_{F_v} \\ & \searrow \quad \swarrow & \\ & \text{Gal}_{F_v} & \end{array} .$$

In particular, for any motive  $m$ , there exists an automorphic representation  $\pi$  of  $\text{GL}(n)$  s.t. the family of conjugacy classes  $c(\pi) = \{c_v(\pi)\}$  in  $\text{GL}(n; \mathbb{C})$  associated to  $\pi$  equals the family of conjugacy classes  $c(m) = \{c_v(m)\}$  in  $\text{GL}(n; \mathbb{C})$  obtained from  $m$  which is the image of Frobenius in the family of  $p$ -adic representations attached to  $m$  (for unramified  $v$ ).  $\lrcorner$

*Proof:*

□

## 8 Using Shimura Varieties

The idea is as follows:

## 9 Geometric Langlands Correspondence

References are [Informal Introduction to Geometric Langlands, Gaiitsgory].

## 29.4 Categorical Local Langlands Conjecture

References are [Zhu17].

### 1 Geometric Satake-Isomorphism

**Notation (29.4.1.1).**

- Let  $k \in \mathbf{Field}^p$  s.t.  $k = \overline{k}$ .
- Let  $K_0 = W(k)[\frac{1}{p}]$ .
- Let  $K_0 \subset_{\text{fin}} K \in p\text{-}\mathbf{Field}$ .
- Let  $\mathcal{G} \in \mathbf{RedGrp}/\mathcal{O}_K$ .

┘

**Def. (29.4.1.2) [Equivariant Perverse Sheaves on  $\mathbf{Gr}_{\mathcal{G}}$ ].** Recall that  $\mathbf{Gr}_{\mathcal{G}}$  is an inductive limit of  $L^+\mathcal{G}$ -invariant closed subschemes  $\mathbf{Gr}_{\preccurlyeq \mu}$ , and the action of  $L^+\mathcal{G}$  on each  $\mathbf{Gr}_{\preccurlyeq \mu}$  factors through some  $L^h\mathcal{G}$  which is a perfectly of f.t. (28.7.2.6) ?, thus we can define the category  $\text{Perv}_{L^+\mathcal{G}}(\mathbf{Gr}_{\preccurlyeq \mu})$  of  $L^+\mathcal{G}$ -equivariant perverse sheaves on  $\mathbf{Gr}_{\preccurlyeq \mu}$ ,

Denote

$$\text{Perv}_{L^+\mathcal{G}}(\mathbf{Gr}_{\mathcal{G}}) \triangleq \varinjlim_{\mu} \text{Perv}_{L^+\mathcal{G}}(\mathbf{Gr}_{\preccurlyeq \mu}).$$

the category of  $L^+\mathcal{G}$ -equivariant perverse sheaves on  $\mathbf{Gr}_{\mathcal{G}}$ .

┘

*Proof:*

□

**Prop. (29.4.1.3) [Intersection Cohomologies].** Denote  $\text{IC}_{\mu}$  the intersection cohomology sheaf on  $\mathbf{Gr}_{\preccurlyeq \mu}$ , then  $\text{IC}_{\mu}|_{\mathbf{Gr}_{\mu}} = \overline{\mathbb{Q}_{\ell}}[(2\rho, \mu)]$ , and its restriction to each stratum ? is constant.

Then  $\text{Perv}_{L^+\mathcal{G}}(\mathbf{Gr}_{\mathcal{G}})$  is a semi-simple category, with irreducible objects exactly those  $\text{IC}_{\mu}$ .

┘

*Proof:* Cf. [Zhu17]P430 ?.

□

## 29.5 Automorphic $L$ -Functions

References are [Bor79, Bum97, Cog00] and [Gelbart-Shahidi, Analytic Properties of Automorphic  $L$ -Functions], [On the Ramanujan conjecture and finiteness of poles for certain  $L$ -functions, Shahidi].

### Notation (29.5.0.1).

- Use notations defined in [Adelic Automorphic Representations](#).
- Use notations defined in [Archimedean  \$L\$ -Factors](#).
- Fix  $F \in \mathbf{GField}$ .
- Fix an additive character  $0 \neq \psi = \otimes'_{v \in \Sigma_F} \psi_v : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$ .
- Fix a linear algebraic group  $G \in \mathcal{AlgGrp}/F$  with center  $Z$ , and let  $\mathcal{K} \subset G(\mathbf{A}_F)$  be a hyperspecial compact subgroup (20.5.3.4).
- Fix a central character  $\omega : Z(F) \backslash Z(\mathbf{A}_F) \rightarrow \mathbb{C}^\times$ .
- Let  $\pi \in \mathrm{Irr}^{\mathrm{adm}}(G(\mathbf{A}_F))$ .

⌋

### 1 Analytic $L$ -Functions

**Def. (29.5.1.1)** [Analytic  $L$ -Functions, Iwaniec-Kowalski [I-K04]]. An **analytic  $L$ -function** of degree  $d \in \mathbb{Z}_+$  is a collection of a tuple

$$(L(f; s), L_\infty(f; s), N(f), \epsilon(f))$$

- $L(f; s)$  is a Dirichlet series

$$L(f; s) = \sum_{n \in \mathbb{Z}_+} a_n(f) n^{-s} = \prod_{p \in \mathbf{Prime}} L_p(f; s) = \prod_{p \in \mathbf{Prime}} (1 - \alpha_1(p) p^{-s})^{-1} \dots (1 - \alpha_d(p) p^{-s})^{-1}$$

which converges absolutely for  $\mathrm{Re}(s) > 1$ , where  $\lambda_f(n) \in \mathbb{C}$ ,  $\lambda_f(1) = 1$ ,  $\alpha_i(p) \in \mathbb{C}$  s.t.  $|\alpha_i(p)| < p$ . These  $\alpha_i(p)$  are called the **local parameters at  $p$**  of  $L(f; s)$ .

- $L_\infty(f; s)$  is a **gamma-factor** s.t.

$$L_\infty(f; s) = \pi^{-\frac{ds}{2}} \prod_{i=1}^d \Gamma\left(\frac{s + \kappa_i}{2}\right),$$

where  $\kappa_i \in \mathbb{C}$  s.t.  $\mathrm{Re}(\kappa_i) \geq -1$  and  $\overline{\{\kappa_i\}_i} = \{\kappa_i\}_i$ , called the **local parameters at  $\infty$**  of  $L(f; s)$ .

- $N(f) \in \mathbb{Z}_+$  is a **conductor (of  $L$ -function)** of  $L(f; s)$  s.t.  $\alpha_i(p) \neq 0$  for  $p \nmid N(f)$  and any  $i \in [d]_+$ . Any  $p \in \mathbf{Prime} \setminus S(N(f))$  is said to be **unramified for the  $L(f; s)$** .
- A **root number**  $w(f) \in \mathbb{C}^\times$ , s.t.  $|w(f)| = 1$ .

s.t. the **completed  $L$ -function**

$$\Lambda(f; s) = N(f)^{\frac{s}{2}} L_\infty(f; s) L(f; s)$$

extends to a meromorphic function of order  $\leq 1$  (12.4.3.18) for  $s \in \mathbb{C}$ , with at most poles at  $s = 0$  and  $s = 1$ , and satisfies a functional equation

$$\Lambda(f; s) = w(f) \Lambda(\bar{f}; 1 - s),$$

where

$$L(\bar{f}; s) = \sum_{n \in \mathbb{Z}_+} \bar{\lambda}_f(n) n^{-s} \prod_{p \in \text{Prime}} (1 - \bar{\alpha}_1(p) p^{-s})^{-1} \dots (1 - \bar{\alpha}_d(p) p^{-s})^{-1}, \quad \Lambda(\bar{f}; s) = L_\infty(f; s) L(\bar{f}; s)$$

$L(f; s)$  is said to be an **entire analytic  $L$ -function** if  $\Lambda(f; s)$  is entire.

More generally, for  $a \in \mathbb{R}$ ,  $(L(f; s), L_\infty(f; s), N(f), \epsilon(f))$  is called an  **$a$ -shifted analytic  $L$ -function** if

$$(L(f; s + a), L_\infty(f; s + a), N(f), \epsilon(f))$$

is an analytic  $L$ -function.  $\lrcorner$

**Def. (29.5.1.2) [Self-Dual Analytic  $L$ -functions].** For an analytic  $L$ -function  $L(f; s)$ , if  $L(\bar{f}; s) = L(f; s)$ , then  $L(f; s)$  is said to be a **self-dual analytic  $L$ -function**. For a self-dual analytic  $L$ -function, we have  $w(f) \in \{\pm 1\}$ , called the **sign of functional equation**.  $\lrcorner$

**Prop. (29.5.1.3) [Constructing New Analytic  $L$ -Functions].** If

$$(L(f; s), L_\infty(f; s), N(f), w(f)), \quad (L(g; s), L_\infty(g; s), N(g), w(g))$$

are two analytic  $L$ -functions of degree  $n, m$  resp., then

- $(L(\bar{f}; s), L_\infty(f; s), N(f), \bar{w}(f))$  is another analytic  $L$ -function of degree  $n$ , called the **conjugate analytic  $L$ -function**.
- $(L(f; s)L(g; s), L_\infty(f; s)L_\infty(g; s), N(f)N(g), w(f)w(g))$  is another analytic  $L$ -function of degree  $n + m$ .
- If  $L(f; s)$  is entire, then for any  $t \in \mathbb{R}$ ,  $(L(f; s + it)L(\bar{f}; s - it), L_\infty(f; s + it)L_\infty(f; s - it), N(f)^2, 1)$  is another analytic  $L$ -function of degree  $2n$ .  $\lrcorner$

**Def. (29.5.1.4) [Trivial Zeros].** For an analytic  $L$ -function  $L(f; s)$ , because  $\Lambda(f; s)$  is holomorphic away from  $\{0, 1\}$ , the poles of  $L_\infty(f; s)$  are zeros of  $L(f; s)$ , except for  $s = 0$ . These are exactly the points

$$\{\{\kappa_i\} - 2\mathbb{N}\} \setminus \{0\},$$

called the set of **trivial zeros**. The other points are called **non-trivial zeros** of  $L(f; s)$ , which must lie in the critical strip  $0 \leq \text{Re}(s) \leq 1$ .  $\lrcorner$

**Def. (29.5.1.5) [Rankin-Selberg Convolutions].** If  $f, g$  are two analytic  $L$ -functions of degree  $n, m$  resp. with local parameters  $\{\kappa_i\}, \{\nu_j\}$  at  $\infty$  resp. and local parameters  $\{\alpha_i(p)\}, \{\beta_j(p)\}$  at  $p \in \text{Prime}$  resp., then  $f, g$  are said to have **Rankin-Selberg convolution** if there exists some analytic  $L$ -function  $(L(f \otimes g; s), L_\infty(f \otimes g; s), N(f \otimes g), w(f \otimes g))$  of degree  $de$  s.t.

- For any  $p \in \text{Prime} \setminus S(N(f)N(g))$ ,

$$L_p(f \otimes g; s) = \prod_{i=1}^d \prod_{j=1}^e (1 - \alpha_i(p)\beta_j(p)p^{-s})^{-1}.$$

- $L_\infty(f \otimes g; s) = \pi^{-\frac{des}{2}} \prod_{i=1}^d \prod_{j=1}^e \Gamma\left(\frac{s + \mu_{ij}}{2}\right)$ , s.t.

$$\text{Re}(\mu_{ij}) \leq \text{Re}(\kappa_i) + \text{Re}(\nu_j), \quad |\mu_{ij}| \leq |\kappa_i| + |\nu_j|.$$



- $N(f \otimes g) | N(f)^e N(g)^d$ .
- $L(f \otimes g; s)$  has a pole at  $s = 1$  if  $g = \bar{f}$ .

Similarly, we can define tensor product of  $n$  analytic  $L$ -functions.  $\lrcorner$

**Prop. (29.5.1.6).** Situation as in (29.5.1.5), if  $L(f \otimes \bar{f}; s)$  or  $L(f \otimes f; s)$  exist, then for any  $i \in [d]_+$ , we have

$$|\alpha_i(p)| < \sqrt{p}, \quad \forall p \in \mathbf{Prime}, \text{ and } \operatorname{Re}(\kappa_i) > -\frac{1}{2}.$$

$\lrcorner$

**Def. (29.5.1.7) [Symmetric Power  $L$ -Functions].** For an analytic  $L$ -functions  $f$  of degree  $n$  with local parameters  $\{\kappa_i\}$  at  $\infty$  and local parameters  $\{\alpha_i(p)\}$  at  $p \in \mathbf{Prime}$ , then  $f$  are said to have  $n$ -th **symmetric power  $L$ -function** if there exists some analytic  $L$ -function

$$(L(\operatorname{Sym}^n f; s), L_\infty(\operatorname{Sym}^n f; s), N(\operatorname{Sym}^n f), w(\operatorname{Sym}^n f))$$

of degree  $\binom{d+n-1}{n}$  s.t.

- For any  $p \in \mathbf{Prime} \setminus S(N(f))$ ,

$$L_p(f \otimes g; s) = \prod_{a_i \in \mathbb{N}, a_1 + \dots + a_d = n} (1 - \alpha_1(p)^{a_1} \dots \alpha_d(p)^{a_d} p^{-s})^{-1}.$$

- $L_\infty(f \otimes g; s)$ . ?
- $N(f \otimes g)$ . ?

Similarly we can define **exterior power  $L$ -functions**.  $\lrcorner$

### Automorphic $L$ -Functions

**Def. (29.5.1.8) [Automorphic  $L$ -Functions].** Assuming the local Langlands conjecture (29.3.2.2), for  $\rho \in \operatorname{Rep}^{\text{fd}}({}^L G)$ , we can formally define

$$L(\pi, \rho; s) = \prod_{v \in \Sigma_F^{\text{fin}}} L_v(\pi_v, \rho_v; s), \quad \Lambda(\pi, \rho; s) = \prod_{v \in \Sigma_F} L_v(\pi_v, \rho_v; s), \quad \epsilon(\pi, \rho; s) = \prod_{v \in \Sigma_F^{\text{fin}}} \epsilon(\pi_v, \rho_v, \psi_v; s) \quad (29.3.1.36),$$

called the **Langlands  $L$ -functions** and the  **$\epsilon$ -factors(global)** attached to  $\pi$  and  $\rho$ .

The product for  $\epsilon(\pi, \rho; s)$  is a finite product and it is non-vanishing, because  $\epsilon(\pi_v, \rho_v, \psi_v; s) = 1$  if  $\mathfrak{c}(\psi_v) = \mathcal{O}_{F_v}$  and  $\pi_v$  is unramified. Notice  $\epsilon(\pi, \rho; s)$  doesn't depends on  $\psi$ .  $\lrcorner$

**Prop. (29.5.1.9) [Convergence for  $L$ -Functions].** Situation as in (29.5.1.8), if  $\pi$  is unitarizable, then  $L(\pi, \rho; s)$  is absolutely convergent to an analytic function for  $\operatorname{Re}(s)$  sufficiently large.

More generally, if  $P = LM \leq G$  is a parabolic subgroup,  $\sigma \in \operatorname{Irr}^{\text{adm}}(M(\mathbf{A}_F))$  and  $\pi$  is a constituent of  $I_{P(\mathbf{A}_F)}^G(\sigma)$ , then  $L(\pi, \rho; s)$  is absolutely convergent to an analytic function for  $\operatorname{Re}(s)$  sufficiently large.  $\lrcorner$

*Proof:* Cf. [Bor79]P50, 52. or [Euler Products, 1971] or [Shahidi, Eisenstein series and automorphic  $L$ -functions]Section2.5.. Prove first for unitarizable representations and then use parabolic induction.  $\square$

**Conj. (29.5.1.10)** [Meromorphic Continuations, Langlands[Lan70]]. Situation as in (29.5.1.8), if  $\pi \in \text{Irr}^{\text{auto}}(G/F)$ , then  $L(\pi, \rho; s)$  is absolutely convergent to an analytic function for  $\text{Re}(s)$  sufficiently large by (29.5.1.9) and (26.6.1.27).

Then it is conjectured to extend to a meromorphic function for all  $s \in \mathbb{C}$  with only a finite number of poles and bounded on vertical strips, and satisfies the functional equation

$$\Lambda(\pi, \rho; s) = \epsilon(\pi, \rho; s) \Lambda(\pi^\vee, \rho^\vee; 1 - s).$$

Such a meromorphic function  $L(\pi, \rho; s)$  is called an **automorphic  $L$ -function**. ┘

*Proof:* □

**Remark (29.5.1.11).** For some cases of this conjectures, Cf. [Bor79]P53. ┘

**Remark (29.5.1.12).** This meromorphic continuation conjecture is in fact related to global Langlands conjecture, Cf. [Cogdell-Kim-Murty, Lectures on automorphic L-functions], [Functoriality for the classical groups], [Kaletha. Genericity and contragredience in the local Langlands correspondence], [Adams-Vogan. Contragredient representations and characterizing the local Langlands correspondence], [Cai-Freidberg-Ginzburg-Kaplan, Doubling constructions and tensor product L-functions: the linear case].? ┘

## 2 Tate's Thesis (Godement-Jacquet for $\text{GL}(1)$ )

References are [Tat67b, R-V99, Poo15].

### Global Zeta Functions

**Lemma (29.5.2.1)** [Dedekind Zeta Function, Dedekind[Ded64]]. The **Dedekind zeta function** for  $F$  is defined by

$$\zeta(F; s) = \prod_{\mathfrak{p} \in \Sigma_F^{\text{fin}}} \frac{1}{1 - \|\mathfrak{p}\|^{-s}}.$$

It converges absolutely for  $\text{Re}(s) > 1$ . Notice if  $F \in \mathbf{FField}$ , then

$$\zeta(F; s) = \sum_{\mathfrak{a} \in \text{Ideal}(\mathcal{O}_F)} \frac{1}{\|\mathfrak{a}\|^s}$$
┘

*Proof:* It suffices to show  $\prod_{\mathfrak{p} \in \Sigma_F^{\text{fin}}} \frac{1}{1 - \|\mathfrak{p}\|^{-\sigma}} < +\infty$  for  $\sigma > 1$ .

If  $F \in \mathbf{NField}$ , let  $d = [F : \mathbb{Q}]$ , this is bounded by

$$\prod_p (1 - p^{-\sigma})^{-d} = \left( \sum_{n \geq 1} n^{-\sigma} \right)^d < \infty.$$

If  $F \in \mathbf{FField}$ , let the number of irreducible polynomial modulo  $p$  be  $N_n$ , then by the same argument, it suffices to prove the convergence of

$$\prod_{v \in \Sigma_F^{\text{fin}}} \frac{1}{1 - \frac{1}{\|v\|^\sigma}} = \prod_{n \geq 1} \left( \frac{1}{1 - q^{-n\sigma}} \right)^{N_n}.$$

This is convergent iff  $\sum_{n \geq 1} N_n q^{-n\sigma}$  is convergent (12.4.3.8), but the latter is bounded by

$$\sum_{n \geq 1} q^n q^{-n\sigma} = \sum_{n \geq 1} q^{-n(\sigma-1)} < \infty.$$

□

**Def. (29.5.2.2) [Zeta Functions].** If  $\Phi \in \mathcal{S}(F)$  and  $\chi$  is a Hecke character of  $F$ ,  $s \in \mathbb{C}$ , the **global zeta function** is defined to be

$$\zeta(\chi, \Phi; s) = \int_{\mathbf{I}_F} \Phi(\mathbf{a}) \chi(\mathbf{a}) \|\mathbf{a}\|^s d^\times \mathbf{a}.$$

When  $\Phi = \otimes_v \Phi_v$  is a pure tensor,

$$\zeta(\chi, \Phi; s) = \prod_{v \in \Sigma_F} \zeta_v(s, \chi_v, \Phi_v) = \prod_{v \in \Sigma_F} \int_{F_v^\times} \Phi_v(x) \chi_v(x) |x|_v^s d^\times x,$$

where the **local zeta functions**  $\zeta_v(\chi_v, \Phi_v; s)$  converge to a holomorphic function for  $\operatorname{Re}(s) > 0$  and the integral is absolutely convergent for  $\operatorname{Re}(s) > 1$ . ┘

*Proof:* For the local zeta function:  $\Phi_v$  is rapidly decreasing, thus it suffices to integrate the part where  $|x|_v < 1$ . Then  $\Phi_v$  is bounded on this compact region, thus it suffices to evaluate  $\int_{|x|_v \leq 1} |x|_v^s d^\times x < \infty$  for  $\operatorname{Re}(s) > 0$ , which can be done case by case.

To show the global integral converges, notice the local integrals is the same as the local  $L$ -factors for a.e.  $v \in \Sigma_F$  by (29.5.2.3), so we can use Fubini and notice that  $\prod_v L(\chi_v; s)$  converges by comparison with the Dedekind zeta function (29.5.2.1). □

**Lemma (29.5.2.3) [Unramified Factors].** For any place  $v$  that unramified in the sense of (19.4.2.19) and  $\Phi_v = \mathbf{1}_{\mathcal{O}_v}$ ,

$$\zeta_v(\chi_v, \Phi_v; s) = L(\chi_v; s) = (1 - \chi_v(\varpi) \|\mathfrak{p}\|^{-s})^{-1} \quad (31.1.1.1).$$

┘

*Proof:* Notice the a.e.  $v \in \Sigma_F$  satisfies  $v$  that unramified in the sense of (19.4.2.19) and  $\Phi_v = \mathbf{1}_{\mathcal{O}_v}$ , and for such a  $v$ ,

$$\zeta_v(\chi_v, \Phi_v; s) = \sum_{k \in \mathbb{N}} \int_{v(x)=k} \chi_v(x) |x|_v^s d^\times x = \sum_{k \in \mathbb{N}} (\chi_v(\varpi_v) \|v\|^{-s})^k = (1 - \chi_v(\varpi) \|v\|^{-s})^{-1}.$$

□

**Thm. (29.5.2.4) [Global Functional Equation].** The global zeta function (29.5.2.2) can be extended to a meromorphic function for all  $s$ , and it has poles iff  $\chi(x) = |x|^\lambda$  for some  $\lambda \in i\mathbb{R}$ . In which case, it only has poles at

$$s = \begin{cases} -\lambda, 1 - \lambda & , F \in \mathbf{NField} \\ -\lambda + \frac{2\pi ni}{\log(\#F_0)}, 1 - \lambda + \frac{2\pi ni}{\log(\#F_0)} & , F \in \mathbf{FField} \end{cases}$$

with residue  $-k\Phi(0)$  and  $k\Phi^\vee(0)$  resp., where  $k = \operatorname{Vol}(\mathbf{I}_F^1/F^\times)$  (19.4.2.20), and essentially bounded on the vertical strips away from the poles. And we have functional equations

$$\zeta(\chi, \Phi; s) = \zeta(\chi^{-1}, \Phi^\vee; 1 - s).$$

┘

*Proof:* Consider the exact sequence

$$1 \rightarrow \mathbf{I}_F^1 \rightarrow \mathbf{I}_F \rightarrow |\mathbf{I}_F| \rightarrow 1.$$

Let  $\mathbf{I}_F^t$  be the inverse image of  $t \in |\mathbf{I}_F|$ , where  $|\mathbf{I}_F| = \mathbb{R}_+$  or  $q^{\mathbb{Z}}$ , with Haar measure  $dt/t$  or  $\log q$  times the counting measure (also denoted  $dt/t$ ), and choose Haar measure  $d^\times x$  on each  $\mathbf{I}_F^t$  compatible with  $d^\times \mathfrak{a}$  and  $dt/t$ . In particular, if  $|a_t| = t$ ,

$$\int_{|\mathbf{I}_F|} \int_{\mathbf{I}_F^1} \Phi(a_t x) \chi(a_t x) t^s d^\times x \frac{dt}{t} = \int_{\mathbf{I}_F} \Phi(\mathfrak{a}) \chi(\mathfrak{a}) |\mathfrak{a}|^s d^\times \mathfrak{a}.$$

Then the measure  $d^\times x$  and the counting measure on  $F^\times$  induces a measure  $d^\times x$  on  $F^\times \backslash \mathbf{I}_F^1$ , in particular, for any  $f \in C(\mathbf{C}_F)$ , we have

$$\int_{\mathbf{I}_F^t / F^\times} f(x) d^\times x = \int_{\mathbf{I}_F^1 / F^\times} f(a_t x) d^\times x = \int_{\mathbf{I}_F^1 / F^\times} f\left(\frac{a_t}{x}\right) d^\times x = \int_{\mathbf{I}_F^{1/t} / F^\times} f\left(\frac{1}{x}\right) d^\times x. \quad (\star)$$

Denote  $\zeta_t(\chi, \Phi; s) = \int_{\mathbf{I}_F^t} \Phi(x) \chi(x) |x|^s d^\times x$ , then

$$\zeta(\chi, \Phi; s) = \int_0^1 \zeta_t(\chi, \Phi; s) \frac{dt}{t} + \int_1^\infty \zeta_t(\chi, \Phi; s) \frac{dt}{t} = J + I$$

where if  $|\mathbf{I}_F| = q^{\mathbb{Z}}$  the value at 1 is counted half-half at this two part. Now for the  $I$ -part, if  $\operatorname{Re}(s)$  is smaller, it is smaller, thus  $I$  extends to a holomorphic function to all  $s \in \mathbb{C}$ . For the  $J$ -part, by lemmas (29.5.2.6) (29.5.2.5),

$$\begin{aligned} J &= \int_0^1 \zeta_{1/t}(\widehat{\Phi}, \chi^{-1}; 1-s) \frac{dt}{t} + \left[ \int_0^1 (k\widehat{\Phi}(0) \left(\frac{1}{t}\right)^{1-s} - k\Phi(0)t^s) \frac{dt}{t} \right] \delta_{c, |\cdot|} \\ &= \int_1^\infty \zeta_t(\widehat{\Phi}, \chi^{-1}; 1-s) \frac{dt}{t} + \left[ \int_0^1 (k\widehat{\Phi}(0)t^{s-1} - k\Phi(0)t^s) \frac{dt}{t} \right] \delta_{c, |\cdot|} \\ &= I(\chi, \Phi; s) + I(\chi^{-1}, \widehat{\Phi}; 1-s) + k\delta_{c, |\cdot|} \left[ \int_0^1 (\widehat{\Phi}(0)t^{s-1} - \Phi(0)t^s) \frac{dt}{t} \right] \end{aligned}$$

So it can be extended, and the final part is

$$\frac{k\widehat{\Phi}(0)}{s-1} - \frac{k\Phi(0)}{s}$$

when  $F \in \mathbf{NField}$ . And when  $F \in \mathbf{FField}$ , it equals

$$k \log q \left[ \widehat{\Phi}(0) \left( -\frac{1}{2} + \sum_{n=0}^\infty (q^{-n})^{s-1} \right) - \Phi(0) \left( -\frac{1}{2} + \sum_{n=0}^\infty (q^{-n})^s \right) \right] = \frac{k \log q}{2} \left( \widehat{\Phi}(0) \frac{1+q^{1-s}}{1-q^{1-s}} + \Phi(0) \frac{1+q^s}{1-q^s} \right).$$

Now clearly  $\zeta(\chi, \Phi; s) = \zeta(\chi^{-1}, \Phi^\vee; 1-s)$ , and it has the desired residues at 1 and  $|\cdot|$ . □

**Lemma (29.5.2.5).**  $\int_{\mathbf{I}_F^t / F^\times} \chi(x) |x|^s d^\times x = kt^s$  if  $\chi = \mathbf{1}$  and 0 otherwise. ┘

*Proof:*  $\int_{\mathbf{I}_F^t / F^\times} \chi(x) |x|^s d^\times x = \chi(a_t) t^s \int_{\mathbf{I}_F^1 / F^\times} \chi(x) d^\times x$ ,  $\mathbf{I}_F^t / F^\times$  is compact (19.4.1.10) and  $\chi$  is trivial on  $\mathbf{I}_F^1 / F^\times$  iff  $\chi = \mathbf{1}$ , thus we can use (12.10.1.14). □

**Lemma (29.5.2.6).**  $\zeta_t(\chi, \Phi; s) + \Phi(0) \int_{\mathbf{I}_F^t/F^\times} \chi(x)|x|^s d^\times x = \zeta_{1/t}(\widehat{\Phi}, \chi^{-1}; 1-s) + \widehat{\Phi}(0) \int_{\mathbf{I}_F^{1/t}/F^\times} \chi(x)|x|^s d^\times x.$   $\lrcorner$

*Proof:*

$$\begin{aligned}
 \zeta_t(\chi, \Phi; s) + \Phi(0) \int_{\mathbf{I}_F^t/F^\times} \chi(x)|x|^s d^\times x &= \int_{\mathbf{I}_F^t/F^\times} \left( \sum_{a \in F^\times} \Phi(ax) \right) \chi(ax) |ax|^s d^\times x + \int_{\mathbf{I}_F^t/F^\times} \Phi(0) \chi(x) |x|^s d^\times x \\
 &= \int_{\mathbf{I}_F^t/F^\times} \left( \sum_{a \in F} \Phi(ax) \right) \chi(x) |x|^s d^\times x \\
 (19.4.2.22) \quad &= \int_{\mathbf{I}_F^t/F^\times} \frac{1}{|x|} \left( \sum_{a \in K} \widehat{\Phi}(a/x) \right) \chi(x) |x|^s d^\times x \\
 &\stackrel{(\star)}{=} \int_{\mathbf{I}_F^{1/t}/F^\times} \left( \sum_{a \in F} \widehat{\Phi}(ay) \right) |y| \chi^{-1}(y) |y|^{-s} d^\times y \\
 &= \zeta_{1/t}(\widehat{\Phi}, \chi^{-1}; 1-s) + \widehat{f}(0) \int_{\mathbf{I}_F^{1/t}/F^\times} \chi(x) |x|^s d^\times x
 \end{aligned}$$

$\square$

### Local Functional Equations

**Lemma (29.5.2.7).** For any  $\Phi_v, \Psi_v \in \mathcal{S}(F_v)$ ,  $\chi_v \in (F_v^\times)^\vee$ ,  $s \in \mathbb{C}$ ,

$$\frac{\zeta_v(\chi_v, \Phi_v; s)}{\zeta_v(\chi_v^{-1}, \widehat{\Phi}_v; 1-s)} = \frac{\zeta_v(\chi_v, \Psi_v; s)}{\zeta_v(\chi_v^{-1}, \widehat{\Psi}_v; 1-s)}.$$

$\lrcorner$

*Proof:* ?

$$\begin{aligned}
 \zeta(f, c) \zeta(\widehat{g}, \bar{c}) &= \int_{K^\times} f(\alpha) c(\alpha) \left( \int_{K^\times} \widehat{g}(\beta) c^{-1}(\beta) |\beta| d\beta \right) d\alpha \\
 &= \int \int f(\alpha) \widehat{g}(\beta) c(\alpha \beta^{-1}) |\beta| d\alpha d\beta \\
 (Fubini) \quad &= \int \int f(\alpha) \widehat{g}(\alpha \beta) c(\beta^{-1}) |\alpha \beta| d\alpha d\beta \\
 &= \int \left( \int f(\alpha) \widehat{g}(\alpha \beta) |\alpha| d\alpha \right) |\beta| c(\beta^{-1}) d\beta
 \end{aligned}$$

And notice

$$\int_{K^\times} f(\alpha) \widehat{g}(\alpha \beta) |\alpha| d\alpha = C \cdot \int_{K^+ \setminus \{0\}} f(\xi) \widehat{g}(\xi \beta) d\xi = C \cdot \int_{K^+} \int_{K^+} f(\xi) g(\eta) e^{-2\pi i \Lambda(\xi \beta \eta)} d\eta d\xi,$$

which is clearly symmetric in  $f$  and  $g$ . So the conclusion follows.  $\square$

**Thm. (29.5.2.8) [Local Functional Equations].** For any  $\Phi_v \in \mathcal{S}(F_v)$ ,  $\chi_v \in (F_v^\times)^\vee$ , the local zeta function  $\zeta_v(\chi_v, \Phi_v; s)$  (29.5.2.2) can be extended to a meromorphic function to all  $s \in \mathbb{C}$  that is analytic for  $\operatorname{Re}(s) > 0$ , and there is a function  $\gamma_v(\chi_v, \psi_v; s)$ , meromorphic for  $s$  and independent of  $f$ , such that

$$\zeta_v(\chi_v^{-1}, \widehat{\Phi}_v; 1-s) = \gamma_v(\chi_v, \psi_v; s) \zeta_v(\chi_v, \Phi_v; s).$$

$\lrcorner$

*Proof:* By (29.5.2.2), both  $\zeta_v(\chi_v^{-1}, \widehat{\Phi}_v; 1-s)$  and  $\zeta_v(\chi_v, \Phi_v; s)$  are holomorphic in  $0 < \operatorname{Re}(s) < 1$ , so we define  $\gamma(\chi_v, \psi_v; s)$  in this region, and try to extend it. Now (29.5.2.7) shows this is invariant of  $\Phi_v$ , so we can take any  $\Phi_v$  we want. Notice if  $\Phi_v = 0$  on a nbhd of 0, then  $\zeta_v(\chi, \Phi_v; s)$  is entire. Then because Fourier transform preserves  $\mathcal{S}(F_v)$  (19.4.2.11), we can take  $\Phi_v$  to be 0 around 0 or  $\widehat{\Phi}_v$  to be 0 around 0, then  $\gamma_v(\chi_v, \psi_v; s)$  is analytic on both  $\operatorname{Re}(s) > 0$  and  $\operatorname{Re}(s) < 1$ .  $\square$

**Prop. (29.5.2.9).** Let  $v \in \Sigma_F^{\text{fin}}$  and  $\operatorname{Re}(s) < 1$ , then for  $N \in \mathbb{Z}_+$  sufficiently large,

$$\gamma_v(s, \chi_v, \psi_v) = \int_{\mathfrak{p}^{-N}} |x|^{-s} \chi_v^{-1}(x) \psi_v(x) dx.$$

┘

*Proof:* In (29.5.2.8), take  $\Phi_v = \mathbf{1}_{1+\mathfrak{p}^N}$ , then  $\widehat{\Phi}_v = \mathbf{1}_{\mathfrak{p}^{-N}} \cdot \psi_v$ .  $\square$

**Prop. (29.5.2.10).**

- $\gamma(1-s, \chi^{-1}, \psi) = \chi(-1)/\gamma(s, \chi, \psi)$ .
- $\rho(\bar{c}) = c(-1)\overline{\rho(c)}$

┘

*Proof:* 1:

$$\zeta(f, c) = \rho(c)\zeta(\widehat{f}, \widehat{c}) = \rho(c)\rho(\widehat{c})\zeta(\widehat{\widehat{f}}, c) = \rho(c)\rho(\widehat{c})c(-1)\zeta(f, c)$$

2:

$$\zeta(f, c) = \rho(c)\zeta(\widehat{f}, \widehat{c}), \quad \overline{\zeta(f, c)} = \zeta(\bar{f}, \bar{c}) = \rho(\bar{c})\zeta(\widehat{\bar{f}}, \widehat{\bar{c}})$$

And

$$\widehat{\bar{f}}(\xi) = \int \bar{f}(\eta) e^{-2\pi i \Lambda(\xi \eta)} d\eta = \overline{\int f(\eta) e^{2\pi i \Lambda(\xi \eta)} d\eta} = \widehat{f}(-\xi)$$

so

$$\rho(\bar{c})\zeta(\widehat{\bar{f}}, \widehat{\bar{c}}) = \rho(\bar{c})c(-1)\zeta(\widehat{\bar{f}}, \widehat{\bar{c}}) = \rho(\bar{c})c(-1)\overline{\zeta(\widehat{f}, \widehat{c})}$$

Thus  $\rho(\bar{c}) = c(-1)\overline{\rho(c)}$ .  $\square$

**Cor. (29.5.2.11).** Because when  $\sigma(c) = \frac{1}{2}$ ,  $\widehat{c} = \bar{c}??$ , we have  $|\rho(c)| = 1$  in this case.  $\square$

### 3 Hecke $L$ -Functions(GL(1) Case)

#### Local Factors

**Prop. (29.5.3.1) [Local  $L$ -Factors are Common Divisors].** Situation as in (29.5.2.8),  $\frac{\zeta_v(\chi_v, \Phi_v; s)}{L(\chi_v; s)}$  is entire for any  $\Phi_v \in \mathcal{S}(F_v)$ , and if  $v \in \Sigma_F^{\text{fin}}$ , it is a rational function of  $\|v\|^{-s}$ .

Moreover, if  $\psi$  is the standard additive character and we take

$$\Phi_v(x) = \begin{cases} \mathbf{1}_{\mathfrak{c}(\psi_v)\mathfrak{c}(\chi_v)^{-1}} \cdot \psi_v(x) & , v \in \Sigma_F^{\text{fin}} \\ x^\epsilon e^{-\pi x^2} & , v \in \Sigma_F^{\mathbb{R}}, \chi_v(x) = |x|^\nu \operatorname{sgn}(x)^\epsilon, \nu \in i\mathbb{R}, \epsilon \in \{0, 1\} \\ \pi^{-1} \bar{x}^k e^{-2\pi |x|_v} & , v \in \Sigma_F^{\mathbb{C}}, \chi_v(x) = |x|^\nu e^{ik \arg(x)}, \nu \in i\mathbb{R}, k \in \mathbb{N} \\ \pi^{-1} x^{-k} e^{-2\pi |x|_v} & , v \in \Sigma_F^{\mathbb{C}}, \chi_v(x) = |x|^\nu e^{ik \arg(x)}, \nu \in i\mathbb{R}, k \in \mathbb{Z}_-, \end{cases}$$

then

$$\frac{\zeta_v(\chi_v, \Phi_v; s)}{L(\chi_v; s)} \stackrel{(31.1.1.1)}{=} \begin{cases} \|\mathfrak{d}_v\|^{s-1/2} & , v \in \Sigma_F^{\text{fin}}, \chi_v \text{ unramified} \\ \|\mathfrak{d}_v\|^s \|\mathfrak{c}(\chi_v)\|^s g(\chi, \psi_{\pi^{c(\psi_v)-c(\chi_v)}}) & , v \in \Sigma_F^{\text{fin}}, \chi_v \text{ ramified} \\ 1 & , v \in \Sigma_F^\infty \end{cases} \in \mathbb{C}^\times.$$

In particular, they can all be non-zero constants.  $\lrcorner$

*Proof:* For  $v \in \Sigma_F^{\text{fin}}$ ,

$$\zeta_v(\chi_v, \Phi_v; s) = \sum_{k \in \mathbb{Z}} \|v\|^{ks} \int_{v(x)=-k} \Phi_v(x) \chi_v(x) d^\times x.$$

As  $\Phi_v \in C_c^\infty(F_v)$ , the summand is 0 for  $k$  large. And for  $k$  small,  $\Phi_v(x)$  is constant. If  $\chi_v$  is ramified, then these terms are 0, thus it is a polynomial in  $\ll v \gg^{-s}$ , and for  $\chi_v$  unramified, these terms are a geometric series, which is easily seen to be a rational function of  $\|v\|^{-s}$ , and has the same pole as  $L(\chi_v; s)$  (31.1.1.1).

For  $v \in \Sigma_F^{\text{R}}$ , the poles of  $\zeta_v(\chi_v, \Phi_v; s)$  are the same as poles of  $\int_{|x| \leq 1} \Phi_v(x) \chi_v(x) |x|^s dx$ , because  $\int_{|x| \geq 1} \Phi_v(x) \chi_v(x) |x|^s dx$  is convergent for any  $s \in \mathbb{C}$ . For the former, we can write  $\Phi_v$  as a sum of an odd function and an even function. Only the part with the same parity of  $\chi_v$  will be non-zero, and for that part, by (12.4.3.5), its poles are all simple and are poles of  $L(\chi_v; s)$  (31.1.1.1).

For  $v \in \Sigma_F^{\text{C}}$ , if  $\chi_v(x) = |x|^\nu e^{ik \arg(x)}$ ,  $\nu \in i\mathbb{R}$ ,  $k \in \mathbb{Z}$ ,

$$\zeta_v(\chi_v, \Phi_v; s) = \frac{1}{2\pi} \int_0^\infty \int_0^{2\pi} r^{2v+2s} e^{ik\theta} \Phi_v(re^{i\theta}) d\theta \frac{dr}{r} = \int_0^\infty r^{2v+2s} \varphi(r) \frac{dr}{r},$$

where

$$\varphi(r) = \frac{1}{2\pi} \int_0^{2\pi} e^{ik\theta} \Phi_v(re^{i\theta}) d\theta.$$

Suppose  $\Phi_v(x) = \sum_{n,m \in \mathbb{N}} a(n,m) x^n \bar{x}^m$ , then  $\varphi(r) = \sum_{m-n=k} a(n,m) r^{m+n}$ . Thus by (12.4.3.5), its poles are  $\{s | 2s + 2v - (|k| + 2l)\}$ , which are all simple and are poles of  $L(\chi_v; s)$  (31.1.1.1).

For the last assertion, we need direct calculations, Cf. [R-V99]P249 (notice that the complex place differ by a multiple of 2).  $\square$

**Prop. (29.5.3.2) [Compatibility of Local  $\epsilon$ -Factors].** Situation as in (29.5.2.8), there exists a non-vanishing holomorphic function  $\epsilon_v(\chi_v, \psi_v)$  that

$$\frac{\zeta_v(\chi_v^{-1}, \widehat{\Phi}_v; 1-s)}{L(\chi_v^{-1}; 1-s)} = \epsilon_v(\chi_v, \psi_v; s) \frac{\zeta_v(\chi_v, \Phi_v; s)}{L(\chi_v; s)}.$$

and  $\frac{\zeta_v(\chi_v, \Phi_v)}{L_v(\chi_v; s)}$  is holomorphic. Moreover,  $\epsilon_v(\chi_v, \psi_v; s)$  is of the form  $ab^s$  for  $a \in \mathbb{C}^\times$ ,  $b \in \mathbb{R}_+$ . And  $\epsilon(\chi_v, \psi_v) = 1$  if  $v$  is unramified in the sense of (19.4.2.19).  $\lrcorner$

*Proof:* Such an  $\epsilon_v(\chi_v, \psi_v; s)$  exists by (29.5.2.8). It is holomorphic and non-vanishing by (29.5.3.1) as both  $\frac{\zeta_v(\chi_v^{-1}, \widehat{\Phi}_v; 1-s)}{L(\chi_v^{-1}; 1-s)}$  and  $\frac{\zeta_v(\chi_v, \Phi_v; s)}{L(\chi_v; s)}$  are holomorphic and for any  $s_0 \in \mathbb{C}$ ,  $\Phi_v$  can be chosen to make either one of them non-vanishing at  $s_0$ .

To show it is of the form  $ab^s$ : If  $v \in \Sigma_F^{\text{fin}}$ , it is a rational function in  $\|v\|^{-s}$  with no zeros or poles, so it must be of the form  $ab^s$ . And if  $v$  is unramified, then we can take  $\Phi_v = \mathbf{1}_{\mathcal{O}_{F,v}}$ ,  $\Phi_v^\vee = \mathbf{1}_{\mathcal{O}_{F,v}^\vee}$  by (19.4.2.5), and both sides are 1 by (29.5.2.3).

If  $v \in \Sigma_F^\infty$  and  $\psi$  is the standard additive character, then we can take  $\Phi_v$  defined in (29.5.3.1) to do the calculation to show it directly? Cf. [R-V99]P249 (notice that the complex place differ by a multiple of 2):

$$\Phi_v^\vee(x) = \begin{cases} \mu_v(\mathfrak{c}(\psi_v)\mathfrak{c}(\chi_v)^{-1}) \cdot \mathbf{1}_{1+\mathfrak{c}(\chi_v)} & , v \in \Sigma_F^{\text{fin}} \\ i^\epsilon \cdot x^\epsilon e^{-\pi x^2} & , v \in \Sigma_F^{\mathbb{R}}, \chi_v(x) = \text{sgn}(x)^\epsilon, \epsilon \in \{0, 1\} \\ i^k \cdot \pi^{-1} x^k e^{-2\pi|x|_v} & , v \in \Sigma_F^{\mathbb{C}}, \chi_v(x) = |x|^\nu e^{ik \arg(x)}, \nu \in i\mathbb{R}, k \in \mathbb{N} \\ i^{-k} \cdot \pi^{-1} \bar{x}^{-k} e^{-2\pi|x|_v} & , v \in \Sigma_F^{\mathbb{C}}, \chi_v(x) = |x|^\nu e^{ik \arg(x)}, \nu \in i\mathbb{R}, k \in \mathbb{Z}_-. \end{cases}$$

So it follows from (29.5.3.1) that  $\epsilon_v$  is the same as that defined in (31.1.1.1).  $\square$

### Hecke $L$ -Functions

**Prop. (29.5.3.3)** [Hecke  $L$ -Functions, Hecke [Hec18, Hec20]]. For a Hecke character  $\chi = \prod_v \chi_v$  on  $F$  and  $s \in \mathbb{C}$ , we define the **Hecke  $L$ -function** and **completed Hecke  $L$ -function** as

$$L(\chi; s) = \prod_{v \in \Sigma_F^{\text{fin}}} L(\chi_v; s), \quad \Lambda(\chi; s) = \prod_{v \in \Sigma_F} L(\chi_v; s) \quad (31.1.1.1),$$

which converges for  $\text{Re } s > 1$  and has a meromorphic continuation to all  $s \in \mathbb{C}$ .

Also, we can define the **global  $\epsilon$ -factor** for  $\chi$  as

$$\epsilon(\chi; s) = \prod_{v \in \Sigma_F} \epsilon_v(\chi_v, \psi_v; s).$$

All but f.m. of the product equals 1 by (29.5.3.2), and they are of the form  $ab^s$ , so  $\epsilon(s, \chi)$  is also of the form  $ab^s$ , where  $a \in \mathbb{C}^*, b \in \mathbb{R}_+$ . In particular, it is holomorphic and non-vanishing. The fact  $\epsilon(s, \chi)$  is independent of  $\psi$  can be seen from (29.5.3.4).  $\square$

**Prop. (29.5.3.4)** [Meromorphic Extension of Hecke  $L$ -Functions, Hecke [Hec18, Hec20]]. For a Hecke character  $\chi$  of  $F$ ,

1.  $L(\chi; s)$  has meromorphic continuation to all  $s \in \mathbb{C}$ , and there is a functional equation

$$\Lambda(\chi; s) = \epsilon(\chi; s) \Lambda(\chi^{-1}; 1 - s),$$

where  $\epsilon(\chi; s)$  is of the form  $ab^s$ ,  $a \in \mathbb{C}^\times, b \in \mathbb{R}_+$ .

2.  $\Lambda(\chi; s)$  is essentially bounded on the vertical strips away from the poles.
3.  $\Lambda(\chi; s)$  has poles iff  $\chi(x) = |x|^\lambda$  for some  $\lambda \in i\mathbb{R}$ . In this case,
  - If  $F \in \mathbf{NField}$ , then it only has simple poles at  $s \in \{-\lambda, 1-\lambda\}$ , with residue  $-k \|\mathfrak{d}\|^{1/2} \pi^{-r_2}$  and  $k\pi^{-r_2}$  resp., where  $k = \text{Vol}(F^\times \backslash \mathbf{I}_F^1)$  (19.4.2.20).
  - If  $F \in \mathbf{FField}$ , it has poles at  $s \in \{-\lambda + \frac{2\pi ni}{\log(\#F_0)}, 1 - \lambda + \frac{2\pi ni}{\log(\#F_0)}\}_{n \in \mathbb{Z}}$ , with residue  $-k \|\mathfrak{d}\|^{1/2}$  and  $k$  resp., where  $k = \text{Vol}(F^\times \backslash \mathbf{I}_F^1)$  (19.4.2.20) resp..
4. If  $F \in \mathbf{NField}$ ,  $\zeta_F(s)$  only have a simple pole at  $s = 1$ , with residue  $k$ , and

$$\lim_{s \rightarrow 0} s^{-(r_1+r_2-1)} \zeta_F(s) = -\frac{\text{cl}(F) \text{Reg}(F)}{w(F)}.$$

$\square$



*Proof:* 1: The functional equation follows from the definition of local  $\epsilon$ -factors(29.5.3.2) and(29.5.2.4). To show it is essentially bounded on vertical strips, use

$$\frac{\Lambda(\chi; s)}{\zeta(\chi, \Phi; s)} = \prod_{v \in \Sigma_F} \frac{L(\chi_v; s)}{\zeta_v(\chi_v, \Phi_v; s)}$$

where we can take  $\Phi_v$  s.t. each fraction is of the form  $ab^s$ , and a.e. term is 1, by(29.5.3.2).

2:  $L(\chi; s)$  is essentially bounded on vertical strips because  $\zeta(\chi, \Phi; s)$  does(29.5.2.4).

3: To find the poles and residues, choose  $\Phi_v$  as in(29.5.3.1), then  $\zeta(\chi, \Phi; s)$  is a constant multiple of  $\Lambda(\chi; s)$ , and the poles and residues of can be read from that of  $\zeta(\chi, \Phi; s)$ (29.5.2.4). Then we use(29.5.2.4), the assertion about poles follows.

For the residue, it suffices to calculate for  $\chi = \mathbf{1}$ : we take  $\psi$  to be the standard additive character on  $\mathbf{A}_F$ (19.4.2.16), then by calculation the chosen  $\Phi$  satisfies:  $\Phi(0) = \pi^{-r_2}$  and  $\Phi^\vee(0) = \|\mathfrak{d}\|^{1/2} \pi^{-r_2}$ , using(29.5.3.2) and(19.4.2.5)(19.4.2.4). Thus

$$\Lambda_F(s) = \zeta(\mathbf{1}, \Phi; s) \|\mathfrak{d}\|^{1/2-s} \text{ (29.5.3.1).}$$

Thus the residues of  $\Lambda_F(s)$  at  $s = 0$  and  $s = 1$  resp. are

$$-k \|\mathfrak{d}\|^{1/2} \pi^{-r_2}, \quad k \pi^{-r_2}.$$

4: Use item3 and(31.1.1.1) and properties of the  $\Gamma$ -factors(12.6.1.19)(12.6.1.3). □

## 4 Godement-Jacquet Theory(Principal $L$ -Functions)

References are [G-J72, Jac79] and [G-H11]Chap11,15.

### Global Zeta Integrals

**Def.(29.5.4.1) [Godement-Jacquet Zeta Integrals].** Let  $n \in \mathbb{Z}_{\geq 2}$  and let  $(\pi, V) \in \text{Irr}^{\text{cusp}}(\text{GL}(n; \mathbf{A}_F))$ . For any matrix coefficient  $\beta$  of  $\pi$  and  $\Phi \in \mathcal{S}(\text{GL}(n; \mathbf{A}_F))$ , consider the global **Godement-Jacquet Zeta Integral**

$$Z(\beta, \Phi; s) = \int_{\text{GL}(n; \mathbf{A}_F)} \Phi(g) \beta(g) |\det(g)|^{s + \frac{n-1}{2}} d^\times g,$$

which is absolutely convergent for  $\text{Re}(s)$  sufficiently large?.

If  $\beta = \prod_{v \in \Sigma_F} \beta_v$ ,  $\Phi = \prod_{v \in \Sigma_F} \Phi_v$  are pure, then there is a factorization

$$Z(\beta, \Phi; s) = \prod_{v \in \Sigma_F} Z_v(\beta_v, \Phi_v; s), \quad Z_v(\beta_v, \Phi_v; s) = \int_{\text{GL}(n; \mathbf{A}_F)} \Phi_v(g) \beta_v(g) |\det(g)|_v^{s + \frac{n-1}{2}} d^\times g.$$

┘

*Proof:*

□

**Thm.(29.5.4.2)[Global Functional Equations].**

┘

*Proof:*

□

### Local Zeta Functions

**Thm. (29.5.4.3) [Local Functional Equations].** Situation as in (29.5.4.1), for  $v \in \Sigma_F$ ,

- The local zeta integral  $Z_v(\beta_v; \Phi_v; s)$  converges absolutely for  $\text{Re}(s)$  sufficiently large, and extends to an entire function. And it is a rational function of  $\|v\|^{-s}$  if  $v \in \Sigma_F^{\text{fin}}$ .
- There exists  $\Phi_v$  and  $\beta_v$  s.t.  $Z_v(\beta_v; \Phi_v; s) = 1$ .
- These local zeta integrals has a common divisor  $L_v(\pi; s)$  in the sense that

$$\frac{Z_v(\beta_v; \Phi_v; s)}{L_v(\pi; s)}$$

is always entire, and equals to 1 for some choice of  $\Phi_v, \beta_v$ . Moreover, if  $v \in \Sigma_F^{\text{fin}}$ ,  $L_v(\pi; s) = Q(\|v\|^{-s})^{-1}$  where  $Q \in \mathbb{C}[X]$  is a monomial polynomial.

- There exists a function  $\gamma_p(\pi; s)$  s.t. there is a functional equation

$$Z_v(\beta_v^\vee, \widehat{\Phi}_v; 1-s) = \gamma_p(\pi; s) Z_p(\beta_v, \Phi_v; s), \quad \widehat{\Phi}(x) = \int_{\text{GL}(n; F_v)} \Phi_v(y) \text{tr}(yx^t) dy.$$

where  $dy$  is the self-dual measure on  $\text{Mat}(n \times n; F_v)$ .

- There exists a local (non-vanishing) root number  $\epsilon_v(\pi, \psi_v; s)$  s.t. there is a functional equation

$$\frac{Z_v(\beta_v^\vee, \widehat{\Phi}_v; 1-s)}{L_v(\pi^\vee; 1-s)} = \epsilon_v(\pi, \psi_v; s) \frac{Z_v(\beta_v; \Phi_v; s)}{L_v(\pi; s)}.$$

And  $\epsilon_v(\pi; s)$  is a monomial in  $\|v\|^{-s}$  if  $v \in \Sigma_F^{\text{fin}}$ , while it is a constant if  $v \in \Sigma_F^\infty$ .

┘

*Proof:* ?.

Cf. [Jacquet, principal  $L$ -Functions].

Firstly the above statement make sense for reducible representations too. And notice that if  $(\pi', V')$  is a subquotient of  $(\pi, V)$ , then the statements for  $\pi$  imply statements for  $\pi'$ . This is because matrix coefficients of  $\pi'$  are all matrix coefficients of  $\pi$  ?.

Then we reduce to the case that  $\pi_v$  is cuspidal:

For  $\pi$  cuspidal, Cf. [G-J11] Prop11.13.2. Prop11.8.3. Prop11.13.5. ?

To show that  $\epsilon_v(\pi, \psi_v; s)$  is a constant, prove that it is a polynomial in  $s$ , and then change  $\Phi$  to  $\Phi^\vee$ . □

**Remark (29.5.4.4).** The proof doesn't apply to  $n = 1$  case. ┘

**Prop. (29.5.4.5) [Local Zeta Function and Parabolic Inductions].** ┘

### Principle $L$ -Functions

**Thm. (29.5.4.6) [Principle  $L$ -Functions are Analytic, Jacquet-Langlands [J-L70]/Godement-Jacquet [G-J72]].** Let  $n \in \mathbb{Z}_{\geq 2}$  and let  $(\pi, V) \in \text{Irr}^{\text{auto}}(\text{GL}(n; \mathbf{A}_F))$  is non-trivial, then  $L(\pi; s)$  has an analytic continuation to an entire function on  $\mathbb{C}$ .

In fact,  $\Lambda(\pi; s)$  extends to a meromorphic function for  $s \in \mathbb{C}$  and satisfies the function equation

$$\Lambda(\pi; s) = \epsilon(\pi; s) \Lambda(\pi^\vee; 1-s).$$

Moreover, if  $F \in \mathbf{NField}$ , then  $\Lambda(\pi; s)$  has f.m. poles and is bounded away from poles on vertical strips. And if  $F \in \mathbf{GField}$  with constant field  $\Omega$ , then  $\Lambda(\pi; s)$  is a rational function of  $(\#\Omega)^{-s}$ . ┘

*Proof:* Cf.[G-J72].<sup>?</sup> The non-cuspidal case reduces to the cuspidal case by parabolic induction, see[Jac79].

The boundedness on vertical strips follows from the Phragmen-Lindelöf principle.<sup>?</sup>  $\square$

**Prop. (29.5.4.7) [Root Numbers and Conductors].** The  $\epsilon$ -factor in (29.5.4.6) is of the form

$$\epsilon(\pi; s) = w(\pi)N(\pi)^{s-\frac{1}{2}}$$

where  $w(\chi) = \epsilon(\chi; \frac{1}{2})$  satisfies  $|w(\pi)| = 1$  and  $N(\pi)$  is the **conductor of automorphic representation** of  $\pi$ , which is the product of local conductors (29.3.1.31).  $\lrcorner$

*Proof:* <sup>?</sup>  $\square$

How to calculate the  $\epsilon$ -factors and root numbers.<sup>?</sup> Cf.[Character theory and Artin  $L$ -functions(1977)].

### Principle $L$ -Functions for Modular Forms

**Prop. (29.5.4.8) [Trivial Bound].** If  $f \in \text{SF}_k(\Gamma(N))$ , suppose  $f(z) = \sum_{n \geq 0} a_n e^{2\pi i n z / N}$ , then  $|a_n| = O(n^{k/2})$ .  $\lrcorner$

*Proof:*  $a_n = \int_0^N f(x + iy) e^{-2\pi i n(x+iy)/N} dx$  for any  $y$ , so let  $y = N/n$ , then  $a_n = \int_0^N f(x + iN/n) e^{-2\pi i n x / N} dx$ . As  $f$  is a cusp modular form,  $(\text{Im}(z))^{k/2} |f(z)|$  is cuspidal automorphic function, thus bounded on  $\mathcal{H}$ , thus  $|a_n| \leq C(N/n)^{\frac{k}{2}}$ .  $\square$

**Thm. (29.5.4.9) [Ramanujan-Petersson Conjecture].** Let  $f \in \text{SF}_k(\Gamma(1))$  is a Hecke eigenform, then for any  $p \in \text{Prime}$ , we have  $|a_p(f)| \leq 2p^{(k-1)/2}$ .  $\lrcorner$

*Proof:* By (26.3.2.16) this means the eigenvalue  $\lambda_p$  of  $f$  w.r.t.  $T_p$  satisfies  $|\lambda_p| \leq 2p^{1/2}$ . This is a consequence of Weil conjecture and modularity.<sup>?</sup>  $\square$

**Def. (29.5.4.10) [ $L$ -functions associated to Modular Forms, Hecke [Hec37, Hec37a]].** Each  $f \in \text{MF}_k(\Gamma_1(N))$  has an associated  $L$ -functions: if  $f = \sum_{n=0}^{\infty} a_n q^n$ , define

$$L(f; s) = \sum_{n=1}^{\infty} a_n n^{-s}.$$

$\lrcorner$

**Prop. (29.5.4.11) [Functional Equations of Modular Forms, Hecke [Hec37, Hec37a]].** Let  $(a_n)_{n \in \mathbb{N}} \subset \mathbb{C}$  s.t.  $a_n = O(n^M)$  for some integer  $M$ . Given  $\lambda > 0, k > 0, C = \pm 1$ , write

$$\varphi(s) = \sum_{n=1}^{\infty} a_n n^{-s}, \quad \Phi(s) = \left(\frac{2\pi}{\lambda}\right)^{-s} \Gamma(s) \varphi(s), \quad f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z / \lambda}.$$

Then the following conditions are equivalent:

- The function  $\Lambda(s) = \Phi(s) + \frac{a_0}{s} + \frac{C a_0}{k-s}$  can be analytically continued to a holomorphic function of the whole plane which is bounded on vertical strips, and it satisfies the functional equation

$$\Phi(s) = C \Phi(k-s).$$

- In the upper half plane,  $f$  satisfies the functional equation

$$f(-1/z) = C\left(\frac{z}{i}\right)^k f(z).$$

⌋

*Proof:* Notice first

$$\int_0^\infty e^{-2\pi nt/\lambda t^s} \frac{dt}{t} = \left(\frac{2\pi}{n\lambda}\right)^{-s} \int_0^\infty e^{-t} t^s \frac{dt}{t} = \left(\frac{2\pi}{n\lambda}\right)^{-s} \Gamma(s).$$

Thus for  $\operatorname{Re}(s)$  large,

$$\Phi(s) = \int_0^\infty (f(it) - a_0) t^s \frac{dt}{t} = \int_1^\infty (f(it) - a_0) t^s \frac{dt}{t} + \int_1^\infty \left(f\left(\frac{i}{t}\right) - a_0\right) t^{-s} \frac{dt}{t}.$$

If 2 holds, then  $f(\frac{i}{t}) = Ct^k f(it)$ , and

$$\begin{aligned} \Phi(s) &= \int_1^\infty (f(it) - a_0) t^s \frac{dt}{t} + \int_1^\infty (Ct^k f(it) - a_0) t^{-s} \frac{dt}{t} \\ &= \int_1^\infty (f(it) - a_0) t^s \frac{dt}{t} + C \int_1^\infty (f(it) - a_0) t^{k-s} \frac{dt}{t} + \int_1^\infty (Ct^{k-s} a_0 - t^{-s} a_0) \frac{dt}{t}. \end{aligned}$$

The first two integral is absolutely convergent for any  $s$  and then can be extended to an analytic function of the whole plane, and the final term equals

$$-\left(\frac{a_0}{s} + \frac{Ca_0}{k-s}\right)$$

when  $\operatorname{Re}(s)$  is large, so it can be extended to a meromorphic function on the whole plane. And  $\Phi(s) = \Phi(k-s)$  follows easily from the equation above. Also it is bounded on vertical strips because it attain maximum at the real axis.

Conversely, first notice it suffices to prove for real  $y > 0$ ,

$$f\left(\frac{i}{y}\right) = Cy^k f(iy)$$

because this implies these two holomorphic functions on  $\mathcal{H}$  coincide on the imaginary axis, so they must be equal. Notice

$$\int_0^\infty (f(iy) - a_0) y^s \frac{dy}{y} = \Phi(s)$$

converges absolutely for  $\operatorname{Re}(s)$  sufficiently large, so for  $\sigma > 0$  sufficiently large, Mellin inversion formula(12.5.4.2) shows

$$\begin{aligned} f(iy) - a_0 &= \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \Phi(s) y^{-s} ds = \frac{C}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \Phi(k-s) y^{-s} ds \\ &= \frac{C}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \left(\Lambda(k-s) - \frac{Ca_0}{s} - \frac{a_0}{k-s}\right) y^{-s} ds \end{aligned}$$

Notice  $\Phi(\sigma + it)$  decays exponentially for fixed  $\sigma = \alpha$  sufficiently large, because of the Stirling formula(12.6.1.16), and for  $\sigma = \beta$  sufficiently small,  $\Phi(s) = C\Phi(k-s)$  also shows  $\Phi(\sigma + it)$  decays exponentially. And also  $\frac{Ca_0}{\sigma+it} + \frac{a_0}{k-\sigma-it} = O(t^{-1})$  for  $\sigma = \alpha$  or  $\beta$ , so  $\Lambda(\sigma + it) = O(t^{-1})$  for  $\sigma$  large

or small, and also the hypothesis shows  $\Lambda$  is bounded on the strip  $\alpha < \operatorname{Re}(z) < \beta$ , thus by (12.4.5.7),  $\Lambda(\sigma + it) \rightarrow 0$  for  $t \rightarrow 0$  and  $\sigma$  in any compact set. So we can move the integration of  $\Lambda(s)$  to the left or to the right. Then

$$\begin{aligned}
 f(iy) - a_0 &= \frac{C}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} (\Lambda(k-s) - \frac{Ca_0}{s} - \frac{a_0}{k-s}) y^{-s} ds \\
 &= \frac{C}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} y^{-k} \Lambda(s) y^s ds - \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} (\frac{a_0}{s} + \frac{Ca_0}{k-s}) y^{-s} ds \\
 &= \frac{Cy^{-k}}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \Phi(s) y^s ds + \frac{Cy^{-k}}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} (\frac{a_0}{s} + \frac{Ca_0}{k-s}) y^s ds + \frac{1}{2\pi i} \int_{k-\sigma+i\infty}^{k-\sigma-i\infty} (\frac{a_0}{k-s} + \frac{Ca_0}{s}) y^{s-k} ds \\
 &= Cy^{-k} (f(\frac{i}{y}) - a_0) + \frac{y^{-k}}{2\pi i} \int_{\gamma} (\frac{Ca_0}{s} + \frac{a_0}{k-s}) y^s ds \\
 &= Cy^{-k} f(\frac{i}{y}) - Ca_0 y^{-k} + y^{-k} (Ca_0 - a_0 y^k) = Cy^{-k} f(\frac{i}{y}) - a_0
 \end{aligned}$$

So we are done.  $\square$

**Def. (29.5.4.12) [L-Functions associated to  $\operatorname{SF}_k(1)$ ].** Let  $a_1, a_2, \dots$  be a sequence of complex numbers such that  $a_n = O(n^M)$  for some  $M > 0$ . Let

$$L(s) = \sum a_n n^{-s}, \quad \Lambda(s) = L_{\mathbb{C}}(s) L(s), \quad f(z) = \sum a_n e^{-2\pi i n z}.$$

More generally, let  $m > 0$  and  $\chi$  a primitive character mod  $m$ , then we define **twisted  $L$ -function** as

$$L(f, \chi; s) = \sum a_n \chi(n) n^{-s}, \quad \Lambda(f, \chi; s) = L_{\mathbb{C}}(s) L(f, \chi; s)$$

Notice if  $f(x) \in S_k(\Gamma(1))$ , and  $\pi_f \in \operatorname{Irr}^{\operatorname{auto}}(\operatorname{GL}(2)/\mathbb{Q})$  corresponding to  $f$  via (26.6.3.8), then

$$L(f, \chi; s) = L(\pi_f, \chi; s).$$

┘

**Prop. (29.5.4.13) [Functional Equation associated to  $\operatorname{SF}_k(N; \psi)$ ].** Notation as in (29.5.4.12). If  $f(z) \in \operatorname{SF}_k(N, \psi)$  (26.3.1.3). Let  $w_N$  be the Fricke involution (20.4.5.11), and notice  $\psi(a_\gamma) = \overline{\psi(d_\gamma)}$ , so

$$f[w_N]_k[\gamma]_k = f[w_N \gamma w_N^{-1}]_k[w_N]_k = \overline{\psi(d)} f[w_N].$$

Thus  $g = f[w_N]_k \in S_k(N, \overline{\psi})$ .

Now if

$$f(z) = \sum a_n e^{2\pi i n z/N}, \quad g(z) = \sum b_n e^{2\pi i n z/N},$$

and  $\chi \in \operatorname{Diri}'(D)$ , define  $L(s, f, \chi), L(s, g, \overline{\chi}), \Lambda(s, f, \chi), \Lambda(s, g, \overline{\chi})$  as in (29.5.4.12), then  $\Lambda(s, f, \chi)$  extends to an analytic function for all  $s \in \mathbb{C}$ , and there are functional equations

$$\Lambda(f, \chi; s) = i^k \chi(N) \psi(D) \frac{\tau(\chi)^2}{D} (D^2 N)^{-s+k/2} \Lambda(g, \overline{\chi}; k-s), \quad (29.1)$$

where  $\tau(\chi)$  is the Gauss sum of  $\chi$  (35.3.6.8).

In particular, if  $D = 1$  (and thus  $\chi = \mathbf{1}$ ), then

$$\Lambda(f; s) = i^k N^{-s+k/2} \Lambda(g; k-s). \quad (29.2)$$

┘

*Proof:* Let

$$f_\chi(z) = \sum \chi(n) a_n q^n, \quad g_{\bar{\chi}}(z) = \sum \overline{\chi(n)} b_n q^n.$$

Use (35.3.6.13) on  $f_\chi(z)$ , we get

$$f_\chi = \frac{\chi(-1)\tau(\chi)}{D} \sum_{m \in (\mathbb{Z}/D\mathbb{Z})^*} \overline{\chi(m)} f\left[\begin{smallmatrix} D & m \\ & D \end{smallmatrix}\right]_k$$

Now

$$\begin{aligned} f_\chi\left[\begin{smallmatrix} & -1 \\ D^2N & \end{smallmatrix}\right]_k &= f_\chi\left[\begin{smallmatrix} & -1/DN \\ D & \end{smallmatrix}\right]_k = \frac{\chi(-1)\tau(\chi)}{D} \sum_{m \in (\mathbb{Z}/D\mathbb{Z})^*} \overline{\chi(m)} g[w_N^{-1}]_k \left[\begin{smallmatrix} D & m \\ & D \end{smallmatrix}\right]_k \left[\begin{smallmatrix} & -1/DN \\ D & \end{smallmatrix}\right]_k \\ &= \frac{\chi(-1)\tau(\chi)}{D} \sum_{m \in (\mathbb{Z}/D\mathbb{Z})^*} \overline{\chi(m)} g\left[\begin{smallmatrix} D & -r \\ -Nm & s \end{smallmatrix}\right] \left[\begin{smallmatrix} D & r \\ & D \end{smallmatrix}\right]_k \end{aligned}$$

where  $(r, s)$  are integers chosen that  $Ds - rNm = 1$ . Thus  $\overline{\chi(m)} = \chi(-N)\chi(r)$ , and because  $g \in M_k(N, \bar{\psi})$ ,

$$f_\chi\left[\begin{smallmatrix} & -1 \\ D^2N & \end{smallmatrix}\right]_k = \frac{\chi(N)\tau(\chi)}{D} \sum_{r \in (\mathbb{Z}/D\mathbb{Z})^*} \chi(r)\psi(D)g\left[\begin{smallmatrix} D & r \\ & D \end{smallmatrix}\right]_k.$$

Compare this with the formula

$$g_{\bar{\chi}} = \frac{\chi(-1)\tau(\bar{\chi})}{D} \sum_{m \in (\mathbb{Z}/D\mathbb{Z})^*} \chi(m)g\left[\begin{smallmatrix} D & m \\ & D \end{smallmatrix}\right]_k,$$

we get

$$f_\chi\left[\begin{smallmatrix} & -1 \\ D^2N & \end{smallmatrix}\right]_k = \chi(-N)\psi(D)\frac{\tau(\chi)}{\tau(\bar{\chi})}g_{\bar{\chi}} = \chi(N)\psi(D)\frac{\tau(\chi)^2}{D}g_{\bar{\chi}} \text{ (35.3.6.10)(35.3.6.12)}. \quad (29.3)$$

Now similar to the proof of (29.5.4.11),

$$\Lambda(s, f, \chi) = \int_0^\infty f_\chi(iy)y^s \frac{dy}{y}.$$

So when  $\text{Re}(s)$  is large,

$$\begin{aligned} &\chi(N)\psi(D)\frac{\tau(\chi)^2}{D}\Lambda(s, g, \bar{\chi}) \\ &= \chi(N)\psi(D)\frac{\tau(\chi)^2}{D} \int_0^\infty g_{\bar{\chi}}(iy)y^s \frac{dy}{y} \\ &= \int_0^{D\sqrt{N}} (D^2N)^{-k/2}(iy)^{-k} f_\chi\left(\frac{1}{-D^2Niy}\right)y^s \frac{dy}{y} + \chi(N)\psi(D)\frac{\tau(\chi)^2}{D} \int_{D\sqrt{N}}^\infty g_{\bar{\chi}}(iy)y^s \frac{dy}{y} \\ &= \int_{D\sqrt{N}}^\infty (D^2N)^{k/2}t^k i^{-k} f_\chi(it)(D^2Nt)^{-s} \frac{dt}{t} + \chi(N)\psi(D)\frac{\tau(\chi)^2}{D} \int_{D\sqrt{N}}^\infty g_{\bar{\chi}}(iy)y^s \frac{dy}{y} \\ &= i^{-k}(D^2N)^{k/2-s} \int_{D\sqrt{N}}^\infty f_\chi(it)t^{k-s} \frac{dt}{t} + \chi(N)\psi(D)\frac{\tau(\chi)^2}{D} \int_{D\sqrt{N}}^\infty g_{\bar{\chi}}(iy)y^s \frac{dy}{y} \end{aligned}$$

Both integral are absolutely convergent for any  $s$ . And similarly when  $\operatorname{Re}(s)$  is small,

$$\begin{aligned} i^{-k}(D^2N)^{k/2-s}\Lambda(k-s, f, \chi) &= i^{-k}(D^2N)^{k/2-s} \int_0^\infty f_\chi(it) t^{k-s} \frac{dt}{t} \\ &= i^{-k}(D^2N)^{k/2-s} \int_{D\sqrt{N}}^\infty f_\chi(it) t^{k-s} \frac{dt}{t} + \chi(N)\psi(D) \frac{\tau(\chi)^2}{D} \int_{D\sqrt{N}}^\infty g_{\overline{\chi}}(iy) y^s \frac{dy}{y} \end{aligned}$$

Thus we get the desired result.  $\square$

**Cor. (29.5.4.14) [Central Values].** As  $f[w_N^2]_k = (-1)^k f$ , if  $f \in \operatorname{SF}_k(\Gamma_1(N))$  satisfies  $f[w_N]_k = cf$ ,  $c = \epsilon i^k$ ,  $\epsilon \in \{\pm 1\}$ , then

$$N^{s/2}\Lambda(f; s) = \epsilon(-1)^k N^{(k-s)/2}\Lambda(f; k-s).$$

So  $\operatorname{ord}_{s=1/2} L(f; s)$  is even if  $\epsilon = (-1)^k$  and odd if  $\epsilon = (-1)^{k+1}$ .

In particular, if  $N = 1$  and  $k \equiv 2 \pmod{4}$ ,  $L(1/2, f) = 0$ .

It is conjectured that  $L(f; 1/2) \neq 0$  if  $4|k$ , Cf.[Modular Forms, Ribet] P148. ?  $\lrcorner$

### Principle $L$ -Functions for Maass Forms

See Wei's notes.

### Principle $L$ -Functions for Quaternion Algebras

**Notation (29.5.4.15).**

- Let  $Q \in \operatorname{QuatAlg}/F$ .
- Use notations on quaternion algebras defined in [Arithmetic of Quaternions and Octonions](#).  $\lrcorner$

**Prop. (29.5.4.16) [Meromorphic Extension of  $L$ -Functions].**  $\lrcorner$

*Proof:* Cf.[Vigneras]P51 ?  $\square$

**Prop. (29.5.4.17).**

$$\Lambda(Q; s) = \Lambda_F(2s)\Lambda_F(2s-1)J(Q; 2s),$$

where

$$J(Q; 2s) = \prod_{v \in \operatorname{Ram}_F(Q)} J_v(Q; s), \text{ where } J_v(Q; s) = \begin{cases} 1 - \|v\|^{1-s} & , v \in \Sigma_F^{\text{fin}} \\ s-1 & , v \in \Sigma_F^\infty \end{cases}.$$

In particular, by [\(29.5.3.4\)](#).

$$\operatorname{ord}_{s=1/2} \Lambda(Q; s) = -2 + \# \operatorname{Ram}_F(Q).$$

*Proof:* ?  $\square$

## 5 Doubling Methods( $\mathrm{GL}(n) \times \mathrm{GL}(m)$ )

References are [Cog00], [Bump, Rankin-Selberg method, an introduction] and [Analytic Theory of  $L$ -functions for  $\mathrm{GL}(n)$ ].

The tensor product  $L$ -function on  $\mathrm{GL}(n) \times \mathrm{GL}(m)$ : Take a cusp form  $f$  on  $\mathrm{GL}(n)$  and  $f'$  on  $\mathrm{GL}(m)$  with  $n > m$ , integrate

$$\int_{\mathrm{GL}(m;F) \backslash \mathrm{GL}(m;\mathbf{A}_F)} P f(g) f'(g) |\det(g)|^{s-1/2} dg.$$

where  $P$  is the projection operator brilliantly designed s.t. this integral resolves to the product of Whittaker models for  $f$  and  $f'$ .

**Conj. (29.5.5.1) [Rankin-Selberg  $L$ -function].** For a global field  $F$ , assume the Langlands functoriality, for any  $\pi_1 \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  and  $\pi_2 \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(m)/F)$ , their product  $\pi_1 \times \pi_2 \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n) \times \mathrm{GL}(m)/F)$ , and there is a tensor product map  $\otimes : \mathrm{GL}(n) \times \mathrm{GL}(m) \rightarrow \mathrm{GL}(mn)$ , which by functorial lifting (29.3.5.1) gives us an automorphic representation  $\pi_1 \boxtimes \pi_2 \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(mn)/F)$ , whose  $L$ -function is denoted by  $L(s, \pi \times \pi')$  And for a finite set  $S$  of places of  $F$  large enough,

$$L_S(s, \pi_1 \times \pi_2) = \prod_{v \notin S} \prod_{i=1}^n \prod_{j=1}^m \frac{1}{1 - \alpha_i(\pi_{1,v}) \beta(\pi_{2,v}) q_v^{-s}}$$

called the (partial) **Rankin-Selberg  $L$ -function** of  $\pi_1, \pi_2$ . ┘

Examples of  $L$ -functions constructed using Eulerian Integrals include:

- (Doubling Method) Standard  $L$ -function on classical groups (Shapiro-Rallis). This construction works for non-generic representations as well.
- (Generalized doubling method) Tensor product  $L$ -function on  $\mathrm{GL}(n) \times G$  with  $G$  a classical group (Cai-Friedberg-Ginzburg-Kaplan).
- Symmetric square on  $\mathrm{GL}(n)$  (Shimura/Gelbart-Jacquet/Piatetski-Shapiro/Patterson/Bump-Ginzburg(1992)).
- Exterior square on  $\mathrm{GL}(n)$  (Jacquet-Shalika/Bump-Ginzburg(1992)).
- Symmetric cube on  $\mathrm{GL}(2)$  (Bump-Ginzburg-Hoffstein(1990)).
- Symmetric fourth power on  $\mathrm{GL}(2)$  (Ginzburg-Rallis).
- Standard  $L$ -function of  $E_6$  and  $E_7$  (Ginzburg).
- Standard  $L$ -function of  $G_2$  (Ginzburg-Hundley/Gurevich-Segal).

### Eulerian Integrals on $\mathrm{GL}(n) \times \mathrm{GL}(m)$ (Jacquet-Shapiro-Shalika)

References are [JSS83, Bum97, Cog00].

#### **Notation (29.5.5.2).**

- Let  $n' \leq n \in \mathbb{Z}_+$ .
- For each  $v \in \Sigma_F$ , let  $\psi_v$  be a non-trivial additive character of  $F_v$ .
- Let  $(\pi, V) \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$ ,  $(\pi', V') \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n')/F)$ .
- Let  $\omega_\pi$  be the central character of  $\pi$ , and  $\omega_{\pi'}$  the central character of  $\pi'$ .



- If  $n' = 1$ , write  $\xi \triangleq \pi'$  as a Hecke character.

┘

**Def. (29.5.5.3)[Unramified Places].** For  $(\pi, V) \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$ ,  $(\pi', V') \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$ , pure cuspidal forms  $\Phi = \otimes_v \Phi_v \in V$ ,  $\Phi' = \otimes_v \Phi'_v \in V'$ , we call a place  $v \in \Sigma_F^{\text{fin}}$  **unramified for**  $(\pi, \Phi, \xi)$  if

- $\mathfrak{c}(\psi_v) = \mathcal{O}_v$ .
- $\pi_v, \pi'_v$  are unramified principal series.
- $\Phi_v$  (resp.  $\Phi'_v$ ) is the spherical function in  $\pi_v$  (resp.  $\pi'_v$ ) normalized s.t.  $W_{\psi_v, \Phi_v}$  (resp.  $W_{\psi_v, \Phi'_v}$ ) is the normalized spherical Whittaker function.

Then  $v$  is unramified a.e. by (26.6.4.2) and (26.6.4.3).

┘

**Def. (29.5.5.4)[Zeta Functions for  $\text{GL}(2) \times \text{GL}(1)$ ].** For any  $\Phi \in V$ , consider the zeta integral

$$\zeta(\Phi, \xi; s) = \int_{\mathbf{C}_F} \Phi\left(\begin{bmatrix} y & \\ & 1 \end{bmatrix}\right) \|y\|^{s-1/2} \xi(y) d^\times y$$

It is absolutely convergent by (26.6.4.5) and (19.4.1.10).

If  $\Phi = \otimes \Phi_v$  is a pure tensor, by (26.6.4.3), this zeta integral equals

$$\int_{\mathbf{I}_F} W_{\psi_v, \Phi_v}\left(\begin{bmatrix} y & \\ & 1 \end{bmatrix}\right) \|y\|^{s-1/2} \xi(y) d^\times y = \prod_{v \in \Sigma_F} \int_{F_v^\times} W_{\psi_v, \Phi_v}\left(\begin{bmatrix} y_v & \\ & 1 \end{bmatrix}\right) |y_v|^{s-1/2} \xi_v(y_v) d^\times y_v = \prod_{v \in \Sigma_F} \zeta_v(\Phi_v, \xi_v; s),$$

where for each  $v \in \Sigma_F$ , the **local zeta functions**

$$\zeta_v(\Phi_v, \xi_v; s) = \int_{F_v^\times} W_{\psi_v, \Phi_v}\left(\begin{bmatrix} y_v & \\ & 1 \end{bmatrix}\right) |y_v|^{s-1/2} \xi_v(y_v) d^\times y_v$$

converges absolutely to a holomorphic function for  $\text{Re}(s) > 1/2$ , and the integral is absolutely convergent when  $\text{Re}(s) > 3/2$ , using (12.10.3.7).

┘

*Proof:* We analyze the integrand for the local zeta integrals:

For  $v \in \Sigma_F^{\text{fin}}$ , the integrand is compactly supported on (24.11.3.12), and because  $\pi_v$  is unitary, use (24.11.5.17) and (24.11.8.12) to analyze the Kirillov model of  $\pi_v$ , we see the integral converges absolutely for  $\text{Re}(s) > 1/2$ .

For  $v \in \Sigma_F^\infty$ ,  $W_{\psi_v, \Phi_v}$  decays rapidly for  $|y| \rightarrow \infty$ , and for  $|y| \rightarrow 0$ , by (26.1.4.2), it is bounded by  $|y|^{-1/2}$ , thus it also converges absolutely for  $\text{Re}(s) > 1/2$ .

By (29.5.5.5), the local factor is the same as the local  $L$ -factor a.e.  $v$ , and  $|\alpha_i| < \|v\|^{1/2}$  by (24.11.8.12), thus it converges for  $\text{Re}(s) > 3/2$  by comparison with the Dedekind Zeta function (29.5.2.1).  $\square$

**Prop. (29.5.5.5)[Unramified Factors].** If  $v$  is unramified in the sense of (29.5.5.3), then for  $\text{Re}(s) > 1/2$  (thus for all  $s \in \mathbb{C}$ , as they are both meromorphic),

$$\zeta_v(\Phi_v, \pi'_v; s) = L(\pi_v, \pi'_v; s) \text{ (29.3.1.32).}$$

┘

*Proof:* With the unramified hypothesis, there is an explicit formula for  $W_v$  in terms of the Satake parameters  $\alpha_1, \alpha_2$  (24.11.6.13): if  $\text{ord}_v(y) = m$ , then

$$W_v\left(\begin{bmatrix} y & \\ & 1 \end{bmatrix}\right) = \begin{cases} \|v\|^{-m/2} \frac{\alpha_1^{m+1} - \alpha_2^{m+1}}{\alpha_1 - \alpha_2}, & m \geq 0 \\ 0 & , \text{otherwise} \end{cases}.$$

$$\begin{aligned} \zeta_v(W_v, \xi_v; s) &= \sum_{m=0}^{\infty} \|v\|^{-m/2} \frac{\alpha_1^{m+1} - \alpha_2^{m+1}}{\alpha_1 - \alpha_2} \|v\|^{m/2 - ms} \xi(\varpi_v)^m \\ &= \frac{1}{(1 - \alpha_1 \xi(\varpi_v) \|v\|^{-s})(1 - \alpha_2 \xi(\varpi_v) \|v\|^{-s})} = L(\pi_v, \chi_v; s) \end{aligned}$$

□

**Prop. (29.5.5.6)[Global Functional Equation].** In spite of the convergence problem of the product of local zeta integrals, the global zeta integral is absolutely convergent for any  $s$  as  $W_\Phi$  decay rapidly ?. Moreover, because  $\Phi$  is automorphic, we have

$$\begin{aligned} \zeta(\Phi, \xi; s) &= \int_{\mathbf{C}_F} \Phi(w_1 \begin{bmatrix} y & \\ & 1 \end{bmatrix}) |y|^{s-1/2} \xi(y) d^\times y \\ &= \int_{\mathbf{C}_F} \Phi\left(\begin{bmatrix} 1 & \\ & y \end{bmatrix} w_1\right) |y|^{s-1/2} \xi(y) d^\times y \\ &= \int_{\mathbf{C}_F} (\pi(w_1)\Phi)\left(\begin{bmatrix} y & \\ & 1 \end{bmatrix}\right) |y|^{-s+1/2} (\xi\omega)^{-1}(y) d^\times y \\ &= \zeta(\pi(w_1)\Phi, \xi^{-1}\omega^{-1}; 1-s) \end{aligned}$$

┘

### Local Functional Equation

**Prop. (29.5.5.7).** Let  $v \in \Sigma_F$ . If  $n = n'$ , for any  $\Phi_v \in V_v, \Phi'_v \in V'_v$  and  $\varphi_v \in \mathcal{S}(F_v^n)$ , define the local zeta integral

$$\zeta_v(\Phi_v, \Phi'_v, \varphi_v; s) = \int_{\text{Unip}(n; F_v) \backslash \text{GL}(n; F_v)} W_{\psi_v, \Phi_v}(g) W_{\psi_v^{-1}, \Phi'_v}(g) \varphi_v((0, \dots, 0, 1)g) |\det(g)|^s dg,$$

where  $dg$  is the  $\text{GL}(n; F_v)$ -invariant measure induced from that of  $\text{GL}(n; F_v)$  and  $\text{Unip}(n; F_v)$  normalized s.t.  $\mathcal{K}$  and  $\mathcal{K} \cap \text{Unip}(n'; F_v)$  has volume 1 by ?.

If  $n > n'$ , then we can define for  $j = 0, \dots, n - n' - 1$  that

$$\zeta_{v,j}(\Phi_v, \Phi'_v; s) = \int_{\text{Unip}(n'; F_v) \backslash \text{GL}(n'; F_v)} \int_{\text{Mat}(j \times n'; F_v)} W_{\psi_v, \Phi_v}\left(\begin{bmatrix} g & 0 & 0 \\ x & \mathbf{1}_j & 0 \\ 0 & 0 & \mathbf{1}_{n-n'-j} \end{bmatrix}\right) W_{\psi_v^{-1}, \Phi'_v}(g) |\det(g)|^{s - \frac{n-n'}{2}} dx dg,$$

where  $dg$  is the  $\text{GL}(n'; F_v)$ -invariant measure induced from that of  $\text{GL}(n'; F_v)$  and  $\text{Unip}(n'; F_v)$  normalized s.t.  $\mathcal{K}$  and  $\mathcal{K} \cap \text{Unip}(n'; F_v)$  has volume 1 by ?. How does the local zeta integral connects to the global zeta integral ?

┘

**Prop. (29.5.5.8) [Local Zeta Function as Common Divisors, Jacquet-Shalika [J-S81, J-S81a]].**

Situation as in (29.5.5.4),

- The local zeta integral  $\zeta_v(\Phi_v, \Phi'_v; s)$  is absolutely convergent for  $\operatorname{Re}(s)$  sufficiently large, and defines a holomorphic function there, with a meromorphic continuation to all  $s \in \mathbb{C}$ .
- If  $\pi_v, \pi'_v$  are unitary, then the integral converges for  $\operatorname{Re}(s) \geq 1$ , normally on compact subsets.
- We can take specific  $\Phi_v \in V_v$  and  $\Phi'_v \in V'_v$  s.t.  $\frac{\zeta_v(\Phi_v, \Phi'_v; s)}{L_v(\pi_v, \pi'_v; s)}$  (29.3.1.32) is of the form  $ab^s$  for some  $a \in \mathbb{C}^\times, b \in \mathbb{R}_+$ .
- If  $v \in \Sigma_F^{\text{fin}}$ , we have

$$\frac{\zeta_v(\Phi_v, \Phi'_v; s)}{L_v(\pi_v, \pi'_v; s)} \in \mathbb{C}[\|v\|^s, \|v\|^{-s}].$$

and  $\Phi_v, \Phi'_v$  can be chosen s.t.  $\zeta_v(\Phi_v, \Phi'_v; s) = L(\pi_v, \pi'_v; s)$ .

The same holds true for  $n = n'$ .

┘

*Proof:* Cf. [J-S81a] P782. We only prove for  $\operatorname{GL}(2)$ , Cf. [Jacquet-Langlands,  $\operatorname{GL}(2)$ ].

This is by direct calculation: If  $V$  is cuspidal, then  $V = C_c^\infty(K^\times)$  by (24.11.3.13). If  $V$  is not cuspidal, then by (24.11.5.14) it is  $\pi(\chi_1, \chi_2)$  or  $\sigma(\chi_1, \chi_2)$ , whose elements are known (24.11.5.16) (24.11.5.17). For  $\operatorname{Re}(s)$  sufficiently large, the  $|t| > q^{-k}$  part can contribute to any  $p(q^{-s})$  for  $p \in \mathbb{C}[q^s, q^{-s}]$  with  $\deg \leq k-1$ , and the  $|t| \leq q^{-k}$  part is of the form

$$\int_{\mathfrak{p}^k \setminus \{0\}} (\chi_i \xi)(y) |y|^s d^\times y = \sum_{n \geq k} ((\chi \xi)(\varpi) q^{-s})^n \int_{\mathcal{O}^*} (\chi_i \xi)(y) d^\times y = C \frac{\alpha_i^k q^{-ks}}{1 - \alpha_i q^{-s}},$$

where  $C \neq 0$  iff  $\chi_i \xi$  is unramified, or of the form

$$\int_{\mathfrak{p}^k \setminus \{0\}} v(y) (\chi_i \xi)(y) |y|^s d^\times y = \sum_{n \geq k} n ((\chi \xi)(\varpi) q^{-s})^n \int_{\mathcal{O}^*} (\chi_i \xi)(y) d^\times y = C \frac{\alpha_i^k q^{-ks}}{(1 - \alpha_i q^{-s})^2},$$

where  $C \neq 0$  iff  $\chi_i \xi$  is unramified.

For the last assertion, choose the compactly supported part of  $\Phi$  suitably.

□

**Prop. (29.5.5.9) [Local Functional Equations].** For  $v \in \Sigma_F$ , the local zeta integral  $\zeta_v(\Phi_v, \xi_v; s)$  defined in (29.5.5.4) has a meromorphic continuation to all  $s$ , and there exists a meromorphic function  $\gamma_v(x, \pi_v, \xi_v, \psi_v)$  s.t.

$$\zeta_v(\pi(w_1)\Phi_v, \xi_v^{-1}\omega_v^{-1}; 1-s) = \gamma_v(\pi_v, \xi_v, \psi_v; s) \zeta_v(\Phi_v, \xi_v; s).$$

┘

*Proof:* This follows by similar method as the proof of (29.5.2.8) and by evaluating  $\gamma_v(\pi_v, \xi_v, \psi_v; s)$  explicitly, using methods of Weil representation parallel to the proof of (29.5.5.10), Cf. [Jacquet-Langlands (1970), P37].

There are easier way to prove this when  $v \in \Sigma_F^{\text{fin}}$ : We show that both sides are linear functionals on  $\Phi \in V$  that satisfies

$$L\left(\pi\left(\begin{bmatrix} y & \\ & 1 \end{bmatrix}\right)\Phi\right) = \xi^{-1}(y) |y|^{-s+1/2} L(\Phi).$$

This is true for RHS by a change of variable and analytic continuation (29.5.5.8), and then also true for LHS.

Thus for general  $s$ , (24.11.3.15) shows two sides differ by a scalar  $\gamma(s, \pi, \xi, \psi)$ . By (29.5.5.8), for general  $s$ , both sides can be non-zero, thus  $\gamma(s, \pi, \xi, \psi)$  is non-zero, and thus a meromorphic function of  $s$ .  $\square$

**Prop. (29.5.5.10) [Gamma Factor Commutes with Parabolic Induction].** ? See. [JSS83, J-S90].

For example, if  $v \in \Sigma_F^{\text{fin}}$  and  $\pi = \chi_1 \times \chi_2 \in \text{Irr}^{\text{adm}}(\text{GL}(2; F_v))$ , then

$$\gamma(\pi, \xi, \psi; s) = \gamma(\xi\chi_1, \psi; s)\gamma(\xi\chi_2, \psi; s).$$

┘

*Proof:* Cf. [Bump, P548]. ? This uses the concrete realization of the Whittaker model of  $\mathcal{B}(\chi_1, \chi_2)$  as a quotient of the Weil representation in the split case.  $\square$

### Rankin-Selberg $L$ -Functions( $\text{GL}(n) \times \text{GL}(m)$ case)

**Prop. (29.5.5.11) [Local  $\epsilon$ -Factors].** Situation as in (29.5.5.9),

If  $n = n'$ , the  $\epsilon$ -factor defined in (29.3.1.32) satisfies

$$\frac{\zeta_v(\pi(w_n)\Phi_v, \pi(w_n)\Phi'_v, \varphi_v^\wedge; 1-s)}{L_v(\pi_v^\vee \times \pi_v'^\vee; s)} = \omega_{\pi'}(-1)^n \epsilon_v(\pi_v \times \pi_v', \psi_v; s) \frac{\zeta_v(\Phi_v, \Phi'_v, \varphi_v; s)}{L_v(\pi_v \times \pi_v'; s)}, \quad \varphi^\wedge(x) = \int_{F_v^n} \varphi(y) \psi(y^t x) dy,$$

and  $\frac{\zeta(\Phi_v, \xi_v; s)}{L_v(\pi_v, \xi_v; s)}$  is holomorphic. Moreover,  $\epsilon_v(\pi_v, \pi_v', \psi_v; s)$  is of the form  $ab^s$  for  $a \in \mathbb{C}^*$ ,  $b \in \mathbb{R}_+$ , where  $b \in \|v\|^{\mathbb{Z}_+}$  if  $v \in \Sigma_F^{\text{fin}}$ . And  $\epsilon_v(\pi_v \times \pi_v', \psi_v; s) = 1$  if  $v$  is unramified in the sense of (29.5.5.3).

If  $n > n'$ , the  $\epsilon$ -factor defined in (29.3.1.32) satisfies

$$\frac{\zeta_{v, n-n'-1-j}(\pi(\begin{bmatrix} 1_{n'} & 0 \\ 0 & w_{n-n'} \end{bmatrix} w_n) \Phi_v, \pi(w_1) \Phi'_v; 1-s)}{L_v(\pi_v^\vee \times \pi_v'^\vee; s)} = \omega_{\pi'}(-1)^{n-1} \epsilon_v(\pi_v \times \pi_v', \psi_v; s) \frac{\zeta_{v,j}(\Phi_v, \Phi'_v; s)}{L_v(\pi_v \times \pi_v'; s)}.$$

┘

*Proof:* ? We only prove for  $n = 2$ .

Such a meromorphic  $\epsilon_v(\pi_v, \xi_v, \psi_v; s)$  exists by (29.5.5.9). It is holomorphic and non-vanishing by the same reason as in (29.5.3.2) as both  $\frac{\zeta(\Phi_v, \xi_v; s)}{L_v(\pi_v, \xi_v; s)}$  and  $\frac{\zeta(\pi(w_1)\Phi_v, \xi_v^{-1}; 1-s)}{L_v(\pi_v, \xi_v^{-1}; s)}$  are holomorphic by (29.5.5.8) and for any  $s_0 \in \mathbb{C}$ ,  $\Phi_v$  can be chosen to make either one of them non-vanishing at  $s_0$ . To show it is of the form  $ab^s$ : If  $v \in \Sigma_F^{\text{fin}}$ , it is a rational function in  $\|v\|^{-s}$  with no zeros or poles, so it must be of the form  $ab^s$ . And if  $v$  is unramified, then we can let  $\Phi_v$  be the spherical function in  $\pi_v$  normalized s.t.  $W_{v, \varphi_v}(1) = 1$ , then  $\pi(w_1)\Phi_v$  is the normalized spherical function of  $\hat{\pi}_v$ . Then by (19.4.2.5), both sides are 1 by (29.5.5.5).

If  $v \in \Sigma_F^\infty$ , then we can take  $\Phi_v$  as in (29.5.5.8) to do the calculation to show it directly, Cf. [Jacquet-Langlands, GL(2)]. ?  $\square$

**Prop. (29.5.5.12) [Rankin-Selberg  $L$ -Functions].** for any  $\pi = \otimes' \pi_v \in \text{Irr}^{\text{adm}}(\text{GL}(n; \mathbf{A}_F))$  and  $\pi' = \otimes' \pi'_v \in \text{Irr}^{\text{adm}}(\text{GL}(n'; \mathbf{A}_F))$ , define

$$L(\pi \times \pi'; s) = \prod_{v \in \Sigma_F^{\text{fin}}} L_v(\pi_v \times \pi'_v; s), \quad \Lambda(\pi \times \pi'; s) = \prod_{v \in \Sigma_F} L_v(\pi_v \times \pi'_v; s), \quad \epsilon(\pi \times \pi'; s) = \prod_{v \in \Sigma_F} \epsilon_v(\pi_v \times \pi'_v, \psi_v; s).$$

Notice that  $\epsilon(\pi \times \pi'; s)$  is definable because a.e. term is 1, and doesn't depend on  $\psi$ . ?

If  $\pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$  and  $\pi' \in \text{Irr}^{\text{auto}}(\text{GL}(n')/F)$ , then  $L(\pi \times \pi'; s)$  is absolutely convergent for  $\text{Re}(s) > 1$ , and defines a holomorphic function there.  $\square$

*Proof:* Cf. [J-S81]Thm5.3 and (5.3.2)(5.3.5). ?

For definition of completed  $L$ -functions for  $\mathrm{GL}(n)$ , Cf. [Gelbart-Jacquet, A relation between automorphic representations of  $\mathrm{GL}(2)$  and  $\mathrm{GL}(3)$ ] and [Jacquet, Principal  $L$ -functions of the linear group]. ? □

**Thm. (29.5.5.13)[Global Functional Equations].** Let  $\Lambda(\pi, \pi'; s)$  be the completed  $L$ -function defined in (29.5.5.12), then the global  $\epsilon$ -factor  $\epsilon(\pi, \pi'; s)$  (29.3.1.32) is independent of  $\psi$ , and  $L(\pi, \pi'; s)$  satisfies a functional equation

$$\Lambda(\pi, \pi'; s) = \epsilon(\pi, \pi'; s) \Lambda(\pi^\vee, \pi'^\vee; s).$$

┘

*Proof:* This follows from (29.5.5.11) and (29.5.5.6). □

**Thm. (29.5.5.14)[Poles on  $\mathrm{Re}(s) = 1$ , Jacquet-Shalika [J-S81a]].** Suppose  $\pi, \pi'$  are unitary, then

- If  $n > n'$ ,  $L(\pi \times \pi'; s)$  has no pole on the line  $\mathrm{Re}(s) = 1$ .
- If  $n = n'$ , the poles of  $L(\pi \times \pi'; s)$  on the line  $\mathrm{Re}(s) = 1$  are all simple poles at  $s$  where  $\pi(s-1) \cong \pi'^\vee$ . In particular,  $L(\pi \times \pi'; s)$  has a pole at  $s = 1$  iff  $\pi' \cong \pi^\vee$ .

┘

*Proof:* Cf. [J-S81a]P795, P802. ? □

### Rankin-Selberg for Modular Forms

**Notation (29.5.5.15).**

- Use notations on Eisenstein series as in Eisenstein Series.

┘

**Prop. (29.5.5.16)[Convolution of Cusp Forms].** If  $f(z) = \sum A(n)q^n \in S_k(\Gamma(1))$ ,  $g(z) = \sum B(n)q^n \in M_k(\Gamma(1))$ , define

$$L(f \times g; s) = \zeta(2s - 2k + 2) \sum A(n)B(n)n^{-s}, \quad \Lambda(f \times g; s) = (2\pi)^{-2s} \Gamma(s) \Gamma(s - k + 1) L(f \times g; s),$$

then  $L(s, f \times g)$  is absolutely convergent for  $s$  large,  $\Lambda(s, f \times g)$  has a meromorphic continuation to all  $s$ , analytic except for at most simple poles at  $s = k$  or  $s = k - 1$ , and it satisfies a functional equation

$$\Lambda(s, f \times g) = \Lambda(2k - 1 - s, f \times g).$$

and the residue of  $\Lambda$  at  $s = k$  is  $\frac{1}{2}\pi^{1-k}(f, g)$  (26.3.1.7).

Moreover, if  $f, g$  are Hecke eigenforms, let

$$1 - A(p)X + p^{k-1}X^2 = (1 - \alpha_1(p)X)(1 - \alpha_2(p)X),$$

$$1 - B(p)X + p^{k-1}X^2 = (1 - \beta_1(p)X)(1 - \beta_2(p)X),$$

then  $L(s, f \times g)$  has an Euler product formula

$$L(s, f \times g) = \prod_p \prod_{i=1}^2 \prod_{j=1}^2 \frac{1}{1 - \alpha_i(p)\beta_j(p)p^{-s}}.$$

┘

*Proof:* We may assume  $f, g$  are Hecke eigenforms (26.3.2.16), so  $A(n), B(n)$  are real. Let  $\varphi(z) = f(z)\overline{g(z)}y^k$ , then  $\varphi$  satisfies the condition of the Rankin-Selberg method (26.9.2.6), thus we calculate

$$\begin{aligned}\varphi_0(y) &= \int_0^1 f(x+iy)\overline{g(x+iy)}y^k dx = \sum_{n=0}^{\infty} \int_{m=0}^{\infty} \int_0^1 A(n)B(m)e^{2\pi i(n-m)x}e^{-2\pi(n+m)y}y^k dx \\ &= \sum_{n=0}^{\infty} A(n)B(n)e^{-4\pi ny}y^k\end{aligned}$$

and

$$M(s, \varphi_0) = \sum_{n=0}^{\infty} A(n)B(n) \int_0^{\infty} e^{-4\pi ny}y^{s+k} \frac{dy}{y} = (4\pi)^{-s-k} \Gamma(s+k) \sum_{n=0}^{\infty} A(n)B(n)n^{-s-k}$$

and

$$\Lambda(s, \varphi_0) = 4^{-s-k+1} \pi^{-2s-k+1} \Gamma(s) \Gamma(s+k-1) \zeta(2s) \sum_{n=1}^{\infty} A(n)B(n)n^{-s-k+1}.$$

Thus

$$\Lambda(s-k+1, \varphi_0) = \pi^{k-1} \Lambda(s, f \times g),$$

and the assertion follows from (26.9.2.6). The residue is also clear.

The Euler product formula follows from (10.7.2.3) applied to  $z = p^{-s}$  for all  $p \in \mathbf{P}$ .  $\square$

**Remark (29.5.5.17) [Averaged Ramanujan Conjecture].** By the Ramanujan conjecture (29.5.4.9), for any cuspidal form  $f = \sum a(n)q^n$ ,  $L(s, f) = \sum a_n n^{-s}$  is convergent for  $\operatorname{Re}(s) > \frac{k+1}{2}$ . But we can prove this directly: as  $\sum a_n^2 n^{-s}$  is convergent for  $\operatorname{Re}(s) > k$ , and  $|a(n)| \leq \max(n^{\frac{k-1}{2}}, n^{-\frac{k-1}{2}} |a(n)|^2)$ .  $\perp$

## 6 Langlands-Shahidi Method

Construct  $L$ -functions by analytic continuation of constant terms of Eisenstein Series. See [Shahidi, Automorphic  $L$ -functions, a survey], and [Shahidi, Eisenstein series and automorphic  $L$ -functions], [Cogdell-Kim-Murty, Lectures on automorphic  $L$ -functions].

## 7 Exterior Square $L$ -Functions

There are two Rankin-Selberg constructions of exterior square  $L$ -functions, which can be found in [Jacquet-Shalika1990, Exterior square  $L$ -functions] and [Bump-Friedberg1990, The “exterior square” automorphic  $L$ -functions on  $\operatorname{GL}(n)$ ].

Here are some examples of  $L$ -functions:

- $\pi_1 = \pi, \pi_2 = \pi^\vee$ , then for  $\operatorname{Ram}(\pi) \subset_{\text{fin}} S \subset \Sigma_F$ ,

$$L_S(s, \pi \times \pi^\vee) = \prod_{v \notin S} \prod_{1 \leq i, j \leq n} \frac{1}{1 - \alpha_i(\pi_v) \alpha_j^{-1}(\pi_v) q_v^{-s}}.$$

In this case, this is the functorial lifting of

$$\operatorname{GL}(n) \rightarrow \operatorname{GL}(n) \times \operatorname{GL}(n) \rightarrow \operatorname{GL}(n^2) : g \mapsto (g, g^{-t}) \mapsto g \otimes g^{-t}$$

by (26.6.3.5), which is isomorphic to the conjugation action of  $\mathrm{GL}(n)$  on  $\mathrm{Mat}(n)$ . This action decomposes as a trivial action and a  $n^2 - 1$ -dimensional representation  $\mathrm{Ad}^0$ . Thus there should be a decomposition

$$L_S(s, \pi \times \pi^\vee) = \zeta_{F,S}(s) L_S(s, \pi, \mathrm{Ad}^0)$$

where  $\zeta_{F,S}(s)$  is the partial zeta function on  $F$ .

- $\pi_1 = \pi = \pi_2$ , then for  $\mathrm{Ram}(\pi) \subset_{\mathrm{fin}} S \subset \Sigma_F$ ,

$$L_S(s, \pi \times \pi) = \prod_{v \notin S} \prod_{1 \leq i, j \leq n} \frac{1}{1 - \alpha_i(\pi_v) \alpha_j(\pi_v) q_v^{-s}}.$$

In this case, this is the functorial lifting of

$$\mathrm{GL}(n) \xrightarrow{\Delta} \mathrm{GL}(n) \times \mathrm{GL}(n) \rightarrow \mathrm{GL}(n^2) : g \mapsto (g, g^{-1}) \mapsto g \otimes g^{-1}$$

by (26.6.3.5), which decomposes as a  $\frac{1}{2}n(n+1)$ -dimensional symmetric square representation  $\mathrm{Sym}^2$ , and a  $\frac{1}{2}n(n-1)$ -dimensional exterior square representation  $\wedge^2$ . Thus there should be a decomposition

$$L_S(s, \pi \times \pi^\vee) = L_S(s, \mathrm{Sym}^2 \pi) L_S(s, \wedge^2 \pi),$$

where  $L_S(\mathrm{Sym}^2 \pi; s)$  and  $L_S(\wedge^2 \pi; s)$  are called the **symmetric square  $L$ -functions** and **exterior square  $L$ -functions** of  $\pi$ .

## 8 Symmetric Power $L$ -Functions

The construction of symmetric square  $L$ -functions can be found in [Symmetric square  $L$ -functions on  $\mathrm{GL}(r)$ , Bump-Ginzburg(1992)] and [Symmetric power  $L$ -functions for  $\mathrm{GL}(2)$ , Shahidi]. For symmetric powers, Cf.[Kim-Shahidi]. Their method uses the unitary dual of  $\mathbf{G}_2$ .

## 9 Triple Products(Garrett)

Cf.[Bump], [Garrett] and [S-R87] and Harris/Ikeda/Ramakrishnan/Orloff.

The construction of the  $L$ -function involves integrating against Eisenstein series on  $\mathrm{GSp}(4)$ .

Other triple convolutions, of type  $\mathrm{GL}(2) \times \mathrm{GL}(3) \times \mathrm{GL}(m)$ , where  $m=3, 4$ , or  $5$  may be obtained from Shahidi's method by means of the exceptional groups E6, E7, and E8. These  $L$ -functions have not yet been achieved as Rankin-Selberg convolutions.

**Thm. (29.5.9.1)** [Integral Representations of Triple Product  $L$ -Functions]. ┘

*Proof:* Cf.[Garrett, Decomposition of Eisenstein series: Rankin triple products] and [S-R87]. ? □

## 29.6 Converse Theorems

### 1 Converse Theorem for $GL(2)$

**Prop. (29.6.1.1) [Converse Theorem (Correspondence for  $\Gamma(1)$ ), Hecke].** If  $f \neq 0 \in \text{SF}_k(\Gamma(1))$  (so  $2|k$ ), then  $\Lambda(f; s)$  (29.5.4.12) has analytic continuation to all  $s \in \mathbb{C}$ , is bounded on vertical strips, and satisfies a functional equation

$$\Lambda(s, f) = (-1)^{k/2} \Lambda(k - s, f).$$

Conversely, let  $a_0, a_1, \dots$  be a sequence of complex numbers that  $a_n = O(n^M)$  for some  $M > 0$ . Let  $f(s), L(s), \Lambda(s)$  be defined in (29.5.4.12) and  $\Lambda(f; s)$  has analytic continuation to all  $s$  and satisfies the above functional equation, then  $f(z) \in \text{SF}_k(\Gamma(1))$ .

Moreover, in this case,  $f$  is a normalized Hecke eigenform (26.3.2.16) iff  $L(s, f)$  has as Euler product formula

$$L(s) = \prod_p (1 - a_p p^{-s} + p^{k-1-2s})^{-1}$$

for  $s$  sufficiently large, where  $1 - a_s X + p^{k-1} X^2 = (1 - aX)(1 - \bar{a}X)$ , where  $|a| = p^{\frac{k-1}{2}}$ .  $\lrcorner$

*Proof:* The first part is a direct consequence of (29.5.4.13). Notice  $a_n$  is bounded by polynomials in  $n$  by (29.5.4.8).

When  $s$  is large, the Euler product is absolutely convergent, thus it suffices to compare the coefficients. By (26.3.2.16) and (26.3.2.9),  $f$  being a normalized Hecke eigenform is equivalent to

$$\begin{cases} c_m c_n = c_{mn}, & (m, n) = 1 \\ c_p c_{p^n} = c_{p^{n+1}} + p^{2k-1} c_{p^{k-1}} \end{cases}$$

which is equivalent to

$$(1 - a_p X + p^{k-1} X^2) \left( \sum_{r=0}^{\infty} a_{p^r} X^r \right) = 1.$$

$\square$

**Thm. (29.6.1.2) [Converse Theorem for Modular Forms, Hecke/Weil [Wei67]].** Let  $N > 0$  and  $\psi \in \text{Diri}(N)$ . Suppose  $a_n, b_n$  are two sequence of complex numbers that  $|a_n|, |b_n| = O(n^M)$  for some positive integer  $M$ . If  $(D, N) = 1$  and  $\chi$  is a primitive character mod  $D$ , let

$$L_1(\chi; s) = \sum \chi(n) a_n n^{-s}, \quad L_2(s, \bar{\chi}) = \sum \overline{\chi(n)} b_n n^{-s},$$

and  $\Lambda_i(s, \chi_i) = L_{\mathbb{C}}(s) L_i(s, \chi_i)$ .

Now if for  $D$  equals to a.e.  $p$  and any primitive character mod  $D$ ,  $\Lambda_i(s, \chi)$  has analytic continuation to all  $s$ , are bounded on vertical strips, and satisfy the functional equation

$$\Lambda_1(s, \chi) = i^k \chi(N) \psi(D) \frac{\tau(\chi)^2}{D} (D^2 N)^{-s+k/2} \Lambda_2(k - s, \bar{\chi}),$$

where  $\tau(\chi)$  is the Gauss sum of  $\chi$  (35.3.6.8), then  $f(z) = \sum a_n e^{2\pi i n z} \in \text{MF}_k(N, \psi)$ .  $\lrcorner$

*Proof:* Let

$$f_{\chi}(z) = \sum \chi(n) a_n q^n, \quad g_{\bar{\chi}}(z) = \sum \overline{\chi(n)} b_n q^n.$$



We first show equation 29.3 holds. As in the proof of (29.5.4.11), it suffices to show the functional equation it is true on the positive imaginary axis. If  $\sigma = \operatorname{Re}(s)$  is sufficiently large, then

$$\int_0^\infty f_\chi(iy) y^s \frac{dy}{y} = \Lambda_1(s, \chi).$$

so by Mellin inversion formula

$$f_\chi(iy) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \Lambda_1(s, \chi) y^{-s} ds = i^k \chi(N) \psi(-D) \frac{\tau(\chi)^2}{D} \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} (D^2 N)^{-s+k/2} \Lambda_2(k-s, \bar{\chi}) ds$$

Same argument of Phragmén-Lindelöf principle as in proof of (29.5.4.11) shows  $\Lambda_2(\sigma + it, \bar{\chi})$  converges to 0 for  $t \rightarrow \infty$ , uniformly on any compact subset, so we can move the integral horizontally and make a change of variable  $s \mapsto k-s$  to get

$$\begin{aligned} f_\chi(iy) &= i^k \chi(N) \psi(-D) \frac{\tau(\chi)^2}{D} (D^2 N)^{-k/2} y^{-k} \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \Lambda_2(k-s, \bar{\chi}) (D^2 N y)^s ds \\ &= i^k \chi(N) \psi(-D) \frac{\tau(\chi)^2}{D} (D^2 N)^{-k/2} y^{-k} g_{\bar{\chi}}\left(\frac{i}{D^2 N y}\right) \end{aligned}$$

which is equivalent to 29.3.

The rest is to manipulate  $2 \times 2$  matrices and use (26.1.2.10) to show that  $g \in \operatorname{MF}_k(N, \bar{\psi})$ , Cf. [Bump, P62] ?  $\square$

**Remark (29.6.1.3).** For  $\Gamma(1)$ , the function only requires one functional equation, and we used the fact  $\Gamma(1)$  is generated by  $S$  and  $T$ . But  $\Gamma_0(N)$  is not generated by two elements, so we must assume functional equations for the twists  $L(f, \chi; s)$  too.  $\lrcorner$

## 2 Converse Theorems for $\operatorname{GL}(n)$

References are [C-S94, C-S99, C-S02].

**Def. (29.6.2.1) [Setup].** Let  $n \in \mathbb{Z}_{\geq 2}$  and  $\Pi = \otimes' \Pi_v \in \operatorname{Irr}^{\operatorname{adm}}(\operatorname{GL}(n; \mathbf{A}_F))$ , there are associated local  $L$  and  $\epsilon$ -functions

$$\Lambda(\Pi; s) = \prod_{v \in \Sigma_F} L(\Pi_v; s), \quad \epsilon(\Pi; \psi; s) = \prod_{v \in \Sigma_F} \epsilon(\Pi_v, \psi_v; s) \quad (29.5.5.12).$$

Assume that:

- $\Lambda(\Pi; s)$  converges absolutely for  $s \in \mathbb{C}$  with  $\operatorname{Re}(s)$  sufficiently large.
- The central character  $\omega_\Pi$  of  $\Pi$  is invariant under  $F^\times$ .

Then  $\epsilon(\Pi, \psi; s)$  doesn't depend on  $\psi$ .

And for  $m \in [n-1]_+$  and  $\pi' = \otimes'_v \pi_v \in \operatorname{Irr}^{\operatorname{cusp}}(\operatorname{GL}(m)/F)$ , we can also formally define

$$\Lambda(\Pi \times \pi'; s) = \prod_{v \in \Sigma_F} L_v(\Pi_v \times \pi'_v; s), \quad \epsilon(\Pi \times \pi', \psi; s) = \prod_{v \in \Sigma_F} \epsilon(\Pi_v \times \pi'_v; s) \quad (29.5.5.12),$$

and similarly for the contragradient

$$\Lambda(\Pi^\vee \times (\pi')^\vee; s), \quad \epsilon(\Pi^\vee \times (\pi')^\vee, \psi; s).$$

Then:

- $\Lambda(\Pi \times \pi'; s)$  and  $\Lambda(\Pi^\vee \times (\pi')^\vee; s)$  converges absolutely for  $\operatorname{Re}(s) \gg 0$ ,
- $\epsilon(\Pi \times \pi', \psi; s)$  and  $\epsilon(\Pi^\vee \times (\pi')^\vee, \psi; s)$  are independent of  $\psi$ .

┘

*Proof:* Cf.[Cogdell-Shapiro, Converse Theorems for  $\mathrm{GL}(n)$ ]

□

**Def. (29.6.2.2) [Nice Properties of  $L$ -Functions].** Situation as in (29.6.2.1), we say that  $L(\Pi \times \pi'; s)$  is **nice  $L$ -function** if

- $\Lambda(\Pi \times \pi'; s)$  and  $\Lambda(\Pi^\vee \times (\pi')^\vee; s)$  extends to an analytic function for  $s \in \mathbb{C}$ .
- $\Lambda(\Pi \times \pi'; s)$  and  $\Lambda(\Pi^\vee \times (\pi')^\vee; s)$  are bounded on vertical strips of finite width.
- There is a functional equation

$$\Lambda(\Pi \times \pi'; s) = \epsilon(\Pi \times \pi'; s) \Lambda(\Pi^\vee \times (\pi')^\vee; s).$$

┘

**Thm. (29.6.2.3) [Converse Theorem for  $\mathrm{GL}(n)$  for Ramified Twists, Jacquet-Shapiro-Shalika [JSS79, JSS79a]/Cogdell-Shapiro [C-S99]].** Situation as in (29.6.2.1). Assume  $n \geq 3$ . Let  $\emptyset \subsetneq_{\mathrm{fin}} S \subset \Sigma_F^{\mathrm{fin}}$  and suppose that  $L(\Pi \times \pi'; s)$  is nice (29.6.2.2) for any  $\pi' \in \operatorname{Irr}^{\mathrm{cusp}}(\mathrm{GL}(m)/F)$  which is unramified at  $v \in S$  and  $m \leq n-2$ , then there exists  $\Pi' \in \operatorname{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  s.t.  $\Pi'_v \cong \Pi_v$  for any  $v \in \Sigma_F \setminus S$ .

┘

*Proof:*

□

**Thm. (29.6.2.4) [Converse Theorem for  $\mathrm{GL}(n)$  for Unramified Twists, Hecke/Weil/-Jacquet-Langlands [J-L70]/Cogdell-Shapiro [C-S94]].** Situation as in (29.6.2.1), Let  $\Sigma_F^\infty \subsetneq_{\mathrm{fin}} S \subset \Sigma_F$  and suppose that

- $\mathrm{Cl}(\mathcal{O}_{F,S}) = 1$ .
- $L(\Pi \times \pi'; s)$  is nice (29.6.2.2) for any  $\pi' \in \operatorname{Irr}^{\mathrm{cusp}}(\mathrm{GL}(m)/F)$  which is unramified outside  $S$  and  $m \leq n-1$ .

then there exists  $\Pi' \in \operatorname{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  s.t.  $\Pi'_v \cong \Pi_v$  for any  $v \in S \cup (\operatorname{Unr}(\Pi) \cap \operatorname{Unr}(\Pi'))$ .

┘

*Proof:*

□

### Conjectures

**Conj. (29.6.2.5) [Converse Theorem for  $\mathrm{GL}(n)$ , Jacquet].** Situation as in (29.6.2.1). Suppose  $L(\Pi \times \pi'; s)$  is nice (29.6.2.2) for any  $m \in [\lfloor \frac{n}{2} \rfloor]_+$  and  $\pi' \in \operatorname{Irr}^{\mathrm{cusp}}(\mathrm{GL}(m)/F)$ , then  $\Pi \in \operatorname{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$ . ┘

*Proof:*

□

**Conj. (29.6.2.6) [Converse Theorem for  $\mathrm{GL}(n)$ , Cogdell-Shapiro [C-S94]].** Situation as in (29.6.2.1). Suppose  $L(\Pi \otimes \omega; s)$  is nice (29.6.2.2) for any Hecke character  $\omega$ , then there exists  $\Pi' \in \operatorname{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  s.t.  $\Pi'_v \cong \Pi_v$  for any  $v \in \operatorname{Unr}(\Pi) \cap \operatorname{Unr}(\Pi')$ . Moreover,  $L(\Pi \otimes \omega; s) = L(\Pi' \otimes \omega; s)$  and  $\epsilon(\Pi \otimes \omega; s) = \epsilon(\Pi' \otimes \omega; s)$ . ┘

*Proof:*

□

**Thm& Conj. Cor. (29.6.2.7).** This conjecture is true for  $n \leq 3$  and in this case we can take  $\Pi' = \Pi$ , by (29.6.2.4) and (29.6.2.5). For  $n \geq 4$ , taking  $\Pi = \Pi'$  might not be possible. ┘

[Jacquet-Langlands] proved converse theorems for  $\mathrm{GL}(2)$ .

## 29.7 Automorphic Representations of $\mathrm{SO}(n)$

References are [Ginzburg-Rallis-Soudry, Generic automorphic forms on  $\mathrm{SO}(2n+1)$ : functorial lift to  $\mathrm{GL}(2n)$ ], [Cogdell-Kim-Piatetski-Shapiro-Shahidi, On lifting from classical groups to  $\mathrm{GL}(N)$ ],

the well-known Piatetski-Shapiro program to prove the Langlands functorial lift from irreducible generic cuspidal automorphic forms on the classical groups to the general linear group via the converse theorem method succeeded in the case of  $\mathrm{SO}(2n+1)$  by using the analytic properties of automorphic  $L$ -functions developed by the Langlands-Shahidi method.

### 1 Endoscopy for $\mathrm{SO}(2n+1)$

**Thm. (29.7.1.1) [Endoscopy for  $\mathrm{SO}(2n+1)$ ].** There is a bijective functorial correspondence between generic cuspidal representations  $\tau \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{SO}(2m+1)/F)$  and families of distinct cuspidal symplectic representations  $\pi_i \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2n_i)/F)$  with  $n_1 + \dots + n_k = m$  s.t.

$$L(\pi \times \tau; s) = \prod_{i=1}^k L(\pi \times \pi_i; s).$$

for any  $r \in \mathbb{Z}_+$  and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(r)/F)$ . ┘

*Proof:* ? □

## 29.8 Endoscopic Classifications

References are [Jacquet-Langlands70], [Trace Formula, an introduction, Arthur], [Problems beyond endoscopy, Arthur] and [Getz].

The goal is to classify automorphic representations of classical groups in terms of automorphic representations of  $\mathrm{GL}(n)$ .

### 1 Arthur's Multiplicity Conjectures

**Conj.(29.8.1.1)**[Arthur's Multiplicity Conjecture]. ┘

**Thm& Conj.Cor.(29.8.1.2).** Arthur's multiplicity conjectures are verified for quasi-split classical groups by [Arthur, endoscopic classification of representations] and [Endoscopic classification of representations of quasi-split unitary groups, Mok]. Some cases of non-quasi-split groups are known by [Endoscopic Classification of Representations: Inner Forms of Unitary Groups, Kaletha-Minguez-Shin-White] and [Taibi. Arthur's multiplicity formula for certain inner forms of special orthogonal and symplectic groups]. ┘

*Proof:* ? □

### 2 Comparison of Trace Formulae

The approach is comparison of trace formulae. It's complicated because one needs to use stable trace formula and also the twisted trace formula.

The theory of endoscopy represents a precise measure, in both geometric and spectral terms, of the failure of geometric conjugacy to imply conjugacy. ?

### 3 Jacquet-Langlands Transfer

### 4 Twisted Endoscopy

### 5 Applications

1. A complete classification of automorphic representations of  $G$  will lead to a sharp Weyl's law for the Shimura variety  $X_\Gamma = \gamma \backslash G(\mathbb{R})/K$ . Notice [Lindenstrauss-Venkatesh] has proved a crude Weyl's law for general non-compact

2. Endoscopic classification will imply some cases of Langlands functoriality.

### 6 Beyond Endoscopy

## 29.9 Relative Langlands Duality

References are [S-V17] and [BZSV].

**Notation (29.9.0.1).**

- Let  $k \in \mathbf{Field}^0$ .
- Let  $(G, \Phi, B, T, \Phi, \Delta, W) \in \mathbf{RedGrp}/F$  be split. Let  $U$  be the unipotent of  $B$  and  $L = B/U$ .
- Use notations on spherical varieties defined in [Spherical Varieties](#).
- Let  $X = H \backslash G$  be a spherical variety for  $G$  (29.9.1.14) with an open orbit  $Bx_0 \subset X$ .
- Use notations on Langlands-Arthur conjectures defined in [Langlands-Arthur Conjectures](#).

┘

### 1 Spherical Varieties

References are [Luna-Vust theory of spherical embeddings, Knop] and [L-V83].

**Notation (29.9.1.1).**

- Let  $k \in \mathbf{Field}^0$ . Notice that there is a corresponding theory over characteristic  $p$ .
- Let  $(G, \Phi, B, T, \Phi, \Delta, W) \in \mathbf{RedGrp}/F$  be split. Let  $U$  be the unipotent of  $B$  and  $L = B/U$ .
- Let  ${}^L G$  be the Langlands dual group of  $G$  with a canonical maximal torus  ${}^L T$ .  ${}^L G$  is canonical up to a conjugation by elements of  ${}^L T$ . ?

┘

**Def. (29.9.1.2)[Spherical Varieties].** A (homogenous)spherical variety for  $G$  is a quasi-affine normal variety  $X \in \mathbf{Var}/k$  with a right  $G$ -action s.t. it is  $G$ -homogenous space  $X = H \backslash G$  and  $B$  has a dense orbit  $X^\circ \subset X$ . Such an  $H$  is called a **spherical subgroup** of  $G$ .

More generally, if  $G$  is not necessarily split and  $B \leq G_{\bar{k}}$  is a Borel subgroup, then a spherical variety for  $G$  is a separated normal scheme of f.t.

┘

**Prop. (29.9.1.3)[Semi-Invariant Functions].** We have  $R(X) = R(G)^H$ . ?

The multiplicative group of non-zero rational  $B$ -eigenfunctions is denoted by  $R(X)^{(B)}$ , and the group of  $B$ -eigencharacters is denoted by  $\chi(X) \subset B(k)^\vee$ .

Because  $X$  has a dense  $B$ -orbit, we have an exact sequence

$$1 \rightarrow k^\times \rightarrow R(X)^{(B)} \rightarrow \chi(X) \rightarrow 1.$$

Denote  $\Lambda_X = \chi(X)^\vee$  and  $\mathfrak{a}_X = \Lambda_X \otimes \mathbb{Q}$ .

Let  $\mathcal{V}(X)$  or  $\mathfrak{a}_X^+$  be the subspace of  $\mathfrak{a}_X$  generated by restrictions of  $G$ -invariant  $\mathbb{Q}$ -valued valuations on  $R(X)$  that is trivial on  $k^\times$ , and  $\mathfrak{a}_X^+ \cap \Lambda_X$  the monoid of  $\mathbb{Z}$ -valued valuations. Thus  $\mathfrak{a}_X^+$  is a finite free Abelian group ? , whose rank is called the **spherical rank of  $X$** .

┘

*Proof:* The kernel of  $R(X)^{(B)} \rightarrow \chi(X)$  are  $B$ -invariant rational functions on  $X$ . Since  $B$  has an open orbit on  $X$ , they are all constant.  $\square$

**Def. (29.9.1.4)[ $B$ -Invariant Divisors and Colors].** Denote  $\mathcal{D}(X)$  the set of all  $B$ -stable prime divisors on  $X$ . Because  $X$  has an open  $B$ -orbit, we know  $\#\mathcal{D}(X) < \infty$ .

A **spherical color** of  $X$  is a  $B$ -stable divisor of  $X$  that is not  $G$ -stable.

Any  $D \in \mathcal{D}(X)$  induces a discrete valuation  $v_D$  on  $R(X)$  that is trivial on  $k^\times$  by (5.3.5.33), thus induces an element  $\rho_{v_D} \in \Lambda_X$ .

┘

**Thm. (29.9.1.5) [Sumihiro].** If  $G$  is a torus acting on a normal variety  $X$ , then every point  $x \in X$  is contained in a  $G$ -stable affine open subset.  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (29.9.1.6).** If  $G$  acts on a normal variety  $X$  and  $Y \subset X$  is an orbit, then there exists a  $B$ -stable affine open subset  $X_0 \subset X$  s.t.

- $X_0 \cap Y \neq \emptyset$ .
- Any  $f \in k[X_0 \cap Y]^{(B)}$  extends to  $f' \in k[X_0]^{(B)}$ .

$\lrcorner$

*Proof:* Cf.[Luna-Vust theory of Spherical Embeddings, Knop]P3.  $\square$

**Cor. (29.9.1.7) [Existence of Semi-Invariant Functions].** If  $f \in \mathcal{O}(B.x_0)$  and  $v_0$  is a  $G$ -invariant  $\mathbb{Q}$ -valued valuation on  $R(X)$ , then there exists  $f' \in R(X)^{(B)}$  s.t.

- $v_0(f') = v_0(f)$ .
- $v(f') \geq v(f)$  for any  $G$ -invariant  $\mathbb{Q}$ -valued valuation on  $R(X)$ .
- $\bar{v}_D(f') \geq \bar{v}_D(f)$  for any  $D \in \mathcal{D}(X)$  (29.9.1.4).

$\lrcorner$

*Proof:* Cf.[Luna-Vust theory of Spherical Embeddings, Knop]P5.  $\square$

**Cor. (29.9.1.8) [Luna-Vust [L-V83] Prop7.4].** The restriction from the set of  $G$ -invariant  $\mathbb{Q}$ -valued valuations on  $R(X)$  that is trivial on  $k^\times$  to  $\mathfrak{a}_X^+$  (29.9.1.3) are injective.  $\lrcorner$

*Proof:* If  $v_0, v_1$  are  $G$ -invariant  $\mathbb{Q}$ -valued valuations on  $R(X)$  that is trivial on  $k^\times$ , we can take  $f \in k[B.x_0]$  s.t.  $v_0(f) < v_1(f)$  (because  $R(X) = \text{Frac}(k[B.x_0])$ ). Then by (29.9.1.7), there exists  $f' \in R(X)^{(B)}$  s.t.  $v_0(f') = v_0(f) < v_1(f) \leq v_1(f')$ . Thus the restrictions of  $v_0$  and  $v_1$  to  $R(X)^{(B)}$  are different.  $\square$

**Def. (29.9.1.9) [Associated Parahoric Subgroups].** The **associated parahoric subgroup** of  $X$  is the subgroup  $P(X) = \{g \in G \mid X^\circ g = X^\circ\} \leq G$ .

Fix  $x_0 \in X^\circ(k)$  with stabilizer  $H \leq G$ , then  $X = H \backslash G$ , and  $HB$  is open in  $G$  ?. Pick  $f \in R(X)^\times$  s.t. the zero locus of  $f$  is  $X \setminus X^\circ$ , then  $f$  is a  $P(X)$ -eigenfunction, but not an eigenfunction for a larger parabolic.  $f$  can be regarded as a function in  $R(G)^H$ , and the stabilizer of  $df$  in  $G$  is a Levi subgroup  $L(X) \leq P(X)$ . ?  $\lrcorner$

**Def. (29.9.1.10) [Tori  $T_X$ ].** Fix a maximal torus  $T \subset B \cap L(X)$ , and define the torus

$$T_X = L(X)/L(X) \cap H = T/T \cap H$$

whose cocharacter is  $\Lambda_X$  (29.9.1.3) ?.  $T_X$  can be identified with a subvariety of  $X^\circ$  through the orbit map  $a \mapsto x_0.a$ .  $\lrcorner$

**Prop. (29.9.1.11) [Spherical Root Systems].** The cone  $\mathfrak{a}_X^+$  is the fundamental domain for a finite reflection group  $W_X \leq \text{End}(\mathfrak{a}_X)$ , called the **little Weyl group** of  $X$ .

There is a canonical way to identify  $W_X$  with a subgroup of  $W$  normalizing  $W_{L(X)}$ .

Consider the strictly-convex negative-dual of  $\mathfrak{a}_X^+$  in  $\chi(X) \otimes \mathbb{R}$ . The generators  $\Sigma_X$  of the intersections of its extremal rays with  $\chi(X)$  (29.9.1.3) are called the **spherical roots** of  $X$ . Then they are the simple roots of a based root system with Weyl group  $W_X$ .  $\lrcorner$

*Proof:* Cf.[Knop, A Harish-Chandra homomorphism for reductive group actions]Chap6.5.  $\square$

**Def. (29.9.1.12) [Weyl Group Actions on Borel Orbits, Knop].** Cf.[SV19]P55.  $\lrcorner$

**Def. (29.9.1.13) [Center  $Z(X)$ ].** Define  $Z(X) = \text{Aut}_{G(k)}(X(k))^0$ , which is a commutative group scheme  $?$ .  $\lrcorner$

### Wavefront Spherical Varieties

**Def. (29.9.1.14) [Wavefront Spherical Varieties].** For a spherical variety  $X = H \backslash G$ ,  $\mathfrak{a}_X^+$  contains the image of the negative Weyl chamber under the natural surjective map  $\mathfrak{a} \rightarrow \mathfrak{a}_X$   $?$  Cf.[SV19]P22. And  $X$  is called a **wavefront spherical variety** if  $\mathfrak{a}_X^+$  is exactly the image of the negative Weyl chamber.  $\lrcorner$

**Def. (29.9.1.15) [Symmetric Varieties].** A **symmetric subgroup scheme** is the neutral component of the fixed point group of an involution  $\theta$  on  $G$ , and for a symmetric subgroup  $H \leq G$ ,  $X = H \backslash G$  is called a **symmetric variety**.  $\lrcorner$

**Example (29.9.1.16) [Wavefront Spherical Varieties].**

- Symmetric varieties(29.9.1.15) are wavefront spherical varieties(29.9.1.14).(Cf.[Vust. Operation de groupes reductifs dans un type de cones presque homogenes]Section1.3.)
- Whittaker cases are wavefront spherical varieties.
- Gross-Prasad cases are wavefront spherical varieties.

$\lrcorner$

*Proof:* Cf.[The Luna-Vust theory of spherical embeddings].  $\square$

### Classifying Spherical Varieties

**Prop. (29.9.1.17).** Homogeneous spherical varieties have been classified in terms of a combinatorial structure called a homogeneous spherical datum, see[Luna, Varietes spheriques de type A] and [Bravi-Pezzini, The spherical systems of the wonderful reductive subgroups].  $?$   $\lrcorner$

### Pure Inner Forms of Spherical Varieties

**Prop. (29.9.1.18).** The  $G$ -automorphism of a  $G$ -torsor is a pure inner-form of  $G$ , and there is a canonical  $G^\alpha \times G^\alpha$ -isomorphism  ${}^\alpha T \times^G T^\alpha \cong G^\alpha$ .

For a  $G$ -variety  $V \in \text{Var}/k$  and  $\alpha \in H^1(k, G)$ , the set of isomorphism classes of  $G$ -morphisms  ${}^\alpha T \rightarrow V$  is naturally bijective to the set of  $G^\alpha(k)$ -orbits on  $V^\alpha(k) \triangleq V \times^G T^\alpha$ .

In particular, the set of isomorphism classes of  $G$ -morphisms

$${}^\alpha T \rightarrow X \times X$$

is in bijection with the set of  $G^\alpha(k)$ -orbits on  $X^\alpha(k) \times X^\alpha(k)$ .  $\lrcorner$

*Proof:*  $?$   $\square$

**Def. (29.9.1.19) [Pure-Inner Forms].** A **pure inner form** of  $(G, X)$  is a cohomology class  $\alpha \in H^1(k, G)$  corresponding to a left  $G$ -torsor  $T^\alpha$  s.t. if  $G^\alpha = \text{End}_G(T^\alpha)$  is the pure inner form of  $G$  corresponding to  $\alpha$ (29.9.1.18), then  $X^\alpha = X \times^G T^\alpha$  with a right action of  $G^\alpha$  satisfies  $X^\alpha(k) \neq \emptyset$ .  $\lrcorner$

**Prop. (29.9.1.20).**  $k$ -points of the stack  $(X \times X)/G$  correspond to isomorphism classes of  $G$ -equivalent maps

$$T \rightarrow X \times X$$

where  $T$  is a right  $G$ -torsor over  $k$ . In particular, by (29.9.1.18), we have a natural bijection

$$[X \times X/G](k) = \coprod_{\alpha} (X^{\alpha}(k) \times X^{\alpha}(k))/G^{\alpha}(k)$$

where  $\alpha$  runs through all pure inner forms of  $X$ .

In fact, the pure inner forms of  $X$  are just images of the map  $H^1(k, H) \rightarrow H^1(k, G)$ , and there is a natural bijection

$$[X \times X/G](k) = \coprod_{\beta \in H^1(k, H)} (H^{\beta} \backslash G^{\beta})(k) / H^{\beta}(k),$$

where if  $H^{\beta} = \text{End}_H({}^{\beta}S)$  for an  $H$ -torsor  ${}^{\beta}S$  corresponding to  $\beta$ , then  $G^{\beta} = \text{End}_G({}^{\beta}S \times^H G)$  with a natural injection  $H^{\beta} \hookrightarrow G^{\beta}$ .  $\lrcorner$

*Proof:* Cf. [S-V17]P307.  $\square$

## 2 Dual Groups of Spherical Varieties

References are [S-V17] and [Knop-Schalke].

**Def. (29.9.2.1) [Dual Groups of Spherical Varieties, Knop/Brion/Sakellaridis-Venkatesh].**

For each  $\gamma \in \Sigma_X$  (29.9.1.11), let  $\gamma'$  be the primitive element of the intersection  $\mathbb{R}_+\gamma$  with the root lattice of  $G$ ?.

Denote  $\Delta_X = \{\gamma' : \gamma \in \Sigma_X\}$  and  $\Phi_X = W_X \cdot \Delta_X$  (29.9.1.11), then  $(\Phi_X, \Delta_X, W_X) \in \text{RootSys}$ , called the **normalized spherical root system** of  $X$ .

Assume that  $\Sigma_X \cap 2\Phi = \emptyset$ , then  $(\chi(X)^{\vee}, \Phi_X^{\vee}, \chi(X), \Phi_X) \in \text{RootDat}$ . The corresponding complex reductive group  ${}^L G_X \in \text{RedGrp}/\mathbb{C}$  with maximal torus  ${}^L T_X$  is called the **dual group** of  $X$ .  $\lrcorner$

*Proof:* ?, Cf. [S-V17]P26, 27, [Brion, On orbit closures of spherical subgroups in flag varieties].  $\square$

**Def. (29.9.2.2) [Distinguished Morphisms].** Assume that  $\Sigma_X \cap 2\Phi = \emptyset$ , a **distinguished morphism of dual groups**  ${}^L G_X(\mathbb{C}) \times \text{SL}(2) \rightarrow {}^L G(\mathbb{C})$  is a morphism that restricts to the canonical map  ${}^L T_X \subset {}^L T$ , and satisfies a condition on simple root spaces(? Cf. [SV19]Chap3.2.).

A **distinguished Arthur-morphism of dual groups**  ${}^L G_X(\mathbb{C}) \times \text{SL}(2) \rightarrow {}^L G(\mathbb{C})$  s.t. its restriction on  ${}^L G_X$  is distinguished and its restriction to  $\mathbb{G}_m \subset \text{SL}(2)$  is the map  $2\rho_{L(X)} : \mathbb{G}_m \rightarrow {}^L A$ .  $\lrcorner$

**Situation (29.9.2.3) [Hypotheses on Gaitsgory-Nadler Dual Groups, Gaitsgory--Nadler [G-N10]].** Gaitsgory-Nadler associated to any affine embedding of  $X$  a canonical reductive subgroup  ${}^L G_{X,GN} \subset {}^L G$  whose maximal torus is the image of  ${}^L T_X$  in  ${}^L T$ , and whose Weyl group is conjecturally equal to  $W_X$ ?. It is conjectured that the following hypotheses hold:

1. The image of  ${}^L G_{X,GN}$  commutes with  $2\rho_{L(X)}(\mathbb{C}^{\times}) \subset {}^L A(\mathbb{C})$ .
2. The Weyl group of  ${}^L G_{X,GN}$  equals  $W_X$ .
3. For any  $\Theta \in \Sigma_X$ , the dual group of the affine degeneration  $X_{\Theta}^a$  (Cf. [S-V17]Chap2.5.?) is naturally a subgroup of  ${}^L G_{X,GN}$ .



4.

5. If  $X_1^+$  is a spherical  $G$ -variety and  $T_1$  a torus of  $G$ -automorphisms of  $X_1^+$  with  $X_2^+ = X_1^+/T_1$ , and if  $X_1, X_2$  are affine embeddings of  $X_1^+, X_2^+$  with  $X_2 = \text{Spec } R(X_1)^{T_1}$ , then there is a canonical embedding

$${}^L G_{X_2, GN} \hookrightarrow {}^L G_{X_1, GN}$$

which restricts to the natural embedding of maximal tori  ${}^L A_{X_2, GN} \hookrightarrow {}^L A_{X_1, GN}$  corresponding to  $\chi(X_2) \hookrightarrow \chi(X_1)$ .?

┘

**Thm. (29.9.2.4) [Distinguished Morphisms Exist].** Assume that  $\Sigma_X \cap 2\Phi_G = \emptyset$ , then

- There exists at most one  $\hat{A}$ -conjugacy class of distinguished morphisms  ${}^L G_X(\mathbb{C}) \times \text{SL}(2) \rightarrow {}^L G(\mathbb{C})$  (29.9.2.2).
- Assume that the hypotheses on Gaitsgory-Nadler dual groups hold (29.9.2.3), then distinguished morphisms  ${}^L G_X(\mathbb{C}) \times \text{SL}(2) \rightarrow {}^L G(\mathbb{C})$  exist, and  ${}^L G_{X, GN}$  is independent of the affine embedding, with root system  $(\Phi_X^\vee, W_X)$ .

┘

*Proof:* Cf. [S-V17] Prop 3.4.3., 3.6.2.?

□

**Lemma (29.9.2.5).** If  $\text{rank } X = 1$ , then there is a principal map  $\text{SL}(2) \rightarrow {}^L L(X)$  commuting with  ${}^L G_{X, GN}$ .

┘

*Proof:* Cf. [S-V17] Appendix A.

□

### 3 Spherical Embeddings (Luna-Vust)

References are [L-V83, Kno91]

**Notation (29.9.3.1).**

- Let  $X$  be a wavefront spherical variety (29.9.1.14).

┘

**Def. (29.9.3.2) [Spherical Embeddings].** A **spherical embedding** of  $X$  is a normal  $G$ -variety  $\overline{X}$  containing  $X$  as an open dense  $G$ -subvariety. It is called a **simple spherical embedding** if it contains a unique closed  $G$ -orbit, and it is called a **toroidal spherical embedding** if no  $B$ -stable divisors in  $X$  contain a  $G$ -orbit of  $\overline{X}$  in its boundary.

┘

**Thm. (29.9.3.3).** For any spherical embedding  $\overline{X}$  of  $X$  (29.9.3.2), denote  $\mathcal{D}(\overline{X})$  the set of  $B$ -stable prime divisor of  $\overline{X}$ . For any  $G$ -orbit  $Y \subset \overline{X}$ , denote  $\mathcal{D}_Y(\overline{X}) \subset \mathcal{D}(\overline{X})$  the set of  $B$ -stable prime divisors containing  $Y$ , and define

$$X_0 = \overline{X} \setminus \bigcup_{D \in \mathcal{D}(\overline{X}) \setminus \mathcal{D}_Y(\overline{X})} D.$$

Then

- $X_0$  is  $B$ -stable and affine open.
- $X_0 \cap Y$  is a  $B$ -orbit in  $\overline{X}$ .
- $Y$  is the only closed orbit of  $G.X_0$ .

┘

*Proof:* Cf.[Knop]P7.?

□

**Cor. (29.9.3.4).** For any spherical embedding  $\overline{X}$  of  $X$  (29.9.3.2), the number of  $G$ -orbits in  $\overline{X}$  is finite and each orbit is a  $G$ -spherical variety. In particular,  $\overline{X}$  is covered by f.m. simple spherical embeddings of  $X$ .

Notice that there are even only f.m.  $B$ -orbits, Cf.[Knop]P7.?

┘

*Proof:* Each  $G$ -orbit  $Y$  defines an  $X_0$  as in (29.9.3.3), and there are only f.m. many such  $X_0$ ?. Then notice that  $X_0$  is a simple spherical embedding of  $X$  and  $Y$  is determined by this  $X_0$ , by (29.9.3.3).

□

**Prop. (29.9.3.5).** A simple spherical embedding  $\overline{X}$  of  $X$  (29.9.3.2) with closed  $G$ -orbit  $Y$  is determined by the pair  $(\mathcal{B}(X), \mathcal{F}(X))$  where

- $\mathcal{B}(X) = \{\rho_{v_D} \in \Lambda_X : D \in \mathcal{D}_Y(\overline{X}) \text{ is } G\text{-stable}\}.$
- $\mathcal{F}(X) = \{D \cap X : D \in \mathcal{D}_Y(\overline{X}) \text{ is not } G\text{-stable and } D \cap X \in \mathcal{D}(X)\}.$

┘

*Proof:*

□

### Classifying Spherical Embeddings

**Def. (29.9.3.6) [Spherical Colored Cones].** A **spherical colored cone** for  $X$  is a pair  $(\mathcal{C}, \mathcal{F})$  s.t.

- $\mathcal{F} \subset \mathcal{D}(X)$  (29.9.1.4).
- $\mathcal{C}$  is a f.g. cone in  $\Lambda_X$  generated by  $\{\rho_{v_D} | D \in \mathcal{F}\}$  (29.9.1.4) together with f.m. elements of  $\mathfrak{a}_X^+$  (29.9.1.3).
- $\text{Int}(\mathcal{C}) \cap \mathfrak{a}_X^+ \neq \emptyset.$

And  $(\mathcal{C}, \mathcal{F})$  is called a **strictly-convex colored cone** if  $\mathcal{C}$  is strictly-convex (i.e.  $\mathcal{C} \cap (-\mathcal{C}) = 0$ ) and  $0 \notin \rho(\mathcal{F})$ .

A **face** of a colored cone  $(\mathcal{C}, \mathcal{F})$  is a face of  $\mathcal{C}$  that is a colored cone.

┘

**Def. (29.9.3.7) [Spherical Colored Fans].** A **spherical colored fan** for  $X$  is a non-empty finite set  $\mathfrak{F}$  of spherical colored cones (29.9.3.6) s.t.

- Every face (29.9.3.6) of spherical colored cones in  $\mathfrak{F}$  is contained in  $\mathfrak{F}$ .
- Any  $v \in \mathfrak{a}_X^+$  is contained in the interior of at most one colored cone in  $\mathfrak{F}$ .

And  $\mathfrak{F}$  is called a **strictly-convex colored fan** if every colored cone in  $\mathfrak{F}$  is strictly-convex (29.9.3.6), or equivalently  $(0, \emptyset) \in \mathfrak{F}$ .

┘

**Thm. (29.9.3.8) [Classifying Simple Spherical Embeddings, Luna-Vust].**

┘

**Thm. (29.9.3.9) [Classifying Spherical Embeddings, Luna-Vust].** For any spherical embedding  $\overline{X}$  of  $X$  (29.9.3.2), define

$$\mathfrak{F}(\overline{X}) = \{(\mathcal{C}_Y(\overline{X}), \mathcal{F}_Y(\overline{X})) : Y \subset \overline{X} \text{ is a } G\text{-orbit}\},$$

then  $\mathfrak{F}(\overline{X})$  is a strictly-convex spherical colored fan (29.9.3.7), and  $Y \mapsto (\mathcal{C}_Y(\overline{X}), \mathcal{F}_Y(\overline{X}))$  induces an inclusion-reversing bijection between  $G$ -orbits on  $\overline{X}$  and  $\mathfrak{F}(X)$ , where the open orbit corresponds to  $(0, \emptyset)$ .

And  $\overline{X} \mapsto \mathfrak{F}(\overline{X})$  defines a bijection between isomorphism classes of spherical embeddings of  $X$  and strictly-convex colored fans.

┘

*Proof:* Cf.[Kno91]P11.?

□

**Thm. (29.9.3.10) [Classifying Simple Toroidal Compactifications, Luna-Vust].** To any simple toroidal spherical embedding  $\overline{X}$  of  $X$  (29.9.3.2), we associate  $\mathcal{C}(\overline{X}) \subset \mathfrak{a}_X^+$  (29.9.1.3) the cone spanned by valuations defined by  $G$ -stable divisors on  $\overline{X}$ . Then the map  $\overline{X} \mapsto \mathcal{C}(\overline{X})$  induces a bijection between isomorphism classes of simple toroidal  $G$ -compactifications of  $X$  and strictly-convex f.g. cones in  $\mathfrak{a}_X^+$ .

For any  $G$ -orbit  $Z$  in a simple toroidal embedding  $\overline{X}$ , the union of  $G$ -orbits whose closure contains  $Z$  is also a simple toroidal embedding. This induces a bijection between faces of  $\mathcal{C}(\overline{X})$  and  $G$ -orbits of  $\overline{X}$ . ┘

*Proof:* Cf.[Knop, The Luna-Vust theory of spherical embeddings]Thm3.3. □

**Cor. (29.9.3.11) [Canonical Toroidal Compactifications].** When  $\mathfrak{a}_X^+$  is itself strictly-convex, the simple toroidal compactification corresponding to  $\mathfrak{a}_X^+$  (29.9.3.10) is called the **canonical toroidal compactification** of  $X$ . This happens exactly when  $\# \text{Aut}_G(X) = \# N(H)/H < \infty$ , and it is the unique proper simple toroidal compactification of  $X$ .

If the canonical toroidal compactification of  $X$  is smooth, then it is called a **wonderful compactification** of  $X$ . ┘

*Proof:* Cf.[Brion-Pauer, Valuations des espaces homogenes spheriques] □

**Thm. (29.9.3.12) [Local Structure Theorem, Brion-Luna-Vust86].** Let  $\overline{X}$  be a simple toroidal compactification of  $X$  and denote  $\overline{X}_B$  the complement of all  $B$ -stable divisors that are not  $G$ -stable, which is an  $B$ -stable open affine subvariety of  $\overline{X}$ . Let  $Y$  be the closure of  $T_X$  in  $\overline{X}_B$  (29.9.1.10), which is the toric compactification of  $T_X$  characterized by the property that for any  $\lambda \in \Lambda_X = X^\bullet(T_X)$  (29.9.1.10),  $\lim_{t \rightarrow 0} \lambda(t) \in Y$  iff  $\lambda \in \mathfrak{a}_X^+$ .

Then the action map  $Y \times U_{P(X)} \rightarrow \overline{X}$  is an open embedding and its image  $\overline{X}_B$  meets each  $G$ -orbit in  $\overline{X}$  along its open  $B$ -orbit. In particular,  $\overline{X}$  is smooth iff  $Y$  is. ┘

*Proof:* Cf.[Brion-Luna-Vust, Espaces homogenes spheriques]. □

### Boundary Degenerations

**Def. (29.9.3.13) [Boundary Degenerations].** Let  $Z$  be a  $G$ -orbit in a smooth toroidal spherical embedding  $\overline{X}$  of  $X$ , corresponding to  $\Theta \subset \Delta_X$  (29.9.2.1)?, the open  $G$ -orbit on  $\mathcal{N}_Z \overline{X}$  will be denoted by  $X_\Theta$ , called the **boundary degeneration** of  $X$  corresponding to  $\Theta$ . ┘

**Prop. (29.9.3.14) [ $X_\Theta$  are Wavefronts].** If  $X$  is a wavefront, then for any  $\Theta \subset \Delta_X$  (29.9.2.1),  $X_\Theta$  is also a wavefront spherical variety for  $\mathcal{Z}(L_\Theta)^0 \times G$ . ┘

*Proof:* Cf.[S-V17]Prop2.7.2.?

□

**Prop. (29.9.3.15).** There is a  $B$ -equivariant isomorphism  $X_\Theta^\circ \cong X^\circ$  inducing the identity on normal bundles. ┘

*Proof:* Cf.[S-V17]P35. □

## 4 Examples

**Example (29.9.4.1) [(PGL( $n$ )  $\times$  PGL( $n$ ), PGL( $n$ )).]** For  $V \in \mathcal{V}ect^{\text{fin}}/k$ , let  $G = \text{PGL}(V) \times \text{PGL}(V)$  and  $X = \text{PGL}(V)$  with a right action of  $G$  on  $X$  given by  $A.(g, h) = g^{-1}Ah$ . ┘

## 5 Geometry over Local Fields

**Notation (29.9.5.1).**

- Let  $k \in p\text{-NField}$ .

┘

**Prop. (29.9.5.2).** Assume that  $X$  carries a positive  $G$ -eigenmeasure  $\mu$  with eigencharacter  $\mu$ . Then  $G$  acts on  $L^2(X; \mu)$  via the twisted right regular representation

$$(g \cdot \Phi)(x) = \sqrt{\eta(g)} \Phi(xg).$$

┘

*Proof:* ?

□

**Prop. (29.9.5.3) [Measures on the Boundary].** For each  $G$ -eigen volume form  $\omega$  on  $X$  and  $\Theta \subset \Delta_X$  (29.9.2.1), there is a canonical  $T_{X, \Theta} \times G$ -eigen volume form  $\omega_\Theta$  on  $X_\Theta$  with the same  $G$ -eigencharacter s.t. for any Borel subgroup  $B \leq G$ ,  $\omega|_{X^\circ}$  and  $\omega_\Theta|_{X_\Theta^\circ}$  correspond via (29.9.3.15).

┘

### Asymptotics

References are [S-V17]Chap5.

**Notation (29.9.5.4).**

- Suppose that  $Z(G)^0$  surjects onto  $Z(X)$ .

┘

**Thm. (29.9.5.5) [Finiteness of Multiplicities].**

┘

## 6 Discrete Spectra

References are [S-V17]Chap9.

**Notation (29.9.6.1).**

- Let  $k \in p\text{-NField}$ .
- Suppose that  $Z(G)^0$  surjects onto  $Z(X)$ .

┘

**Def. (29.9.6.2) [X-Discrete-Series].** Any  $f \in L^2(X)$  decomposes as  $f = \int_{\omega \in \text{Aut}(G)} f_\omega d\omega$  where  $f_\omega \in L^2(X; \omega)$ . Then  $L^2(X)_{\text{disc}}$  consists of the functions  $f$  s.t. for a.e.  $\omega \in \text{Aut}(X)$ ,  $f_\omega$  belongs to the direct sum of all irreducible subrepresentations of  $L^2(X; \omega)$ .

It is conjectured that  $L^2(X)_{\text{disc}}$  is a closed subspace of  $L^2(X)$ .

┘

**Def. (29.9.6.3) [Finiteness of Discrete Spectra].**

┘

**Def. (29.9.6.4) [Factorizable Spherical Varieties].**  $X$  is called a **factorizable spherical variety** if the Lie algebra  $\mathfrak{h}$  of the stabilizer of a generic point on  $X$  has a direct sum decomposition

$$\mathfrak{h} = (\mathfrak{h} \cap \mathcal{Z}(\mathfrak{g})) \oplus (\mathfrak{h} \cap [\mathfrak{g}, \mathfrak{g}]).$$

And  $X$  is called a **strongly-factorizable spherical variety** if for any  $\Theta \subset \Delta_X$ , the Levi variety  $X_\Theta^L$  ? is factorizable.

┘

**Prop. (29.9.6.5) [Symmetric Varieties are Strongly-Factorizable].** If  $X$  is a symmetric variety (29.9.1.15), then  $X$  is strongly-factorizable (29.9.6.4).  $\lrcorner$

*Proof:* Cf. [S-V17]P159.  $\color{red}?$   $\square$

**Conj. (29.9.6.6) [Discrete-Series Conjecture].** This can be easily checked in individual cases by the method of “unfolding”.  $\lrcorner$

## 7 Bernstein Morphisms

References are [S-V17]Chap10-11.

## 8 Plancherel Decompositions

**Def. (29.9.8.1) [Plancherel Decompositions].** A **Plancherel decomposition for spherical varieties** for  $L^2(X)$  is a set of following data:

- A positive measure  $\hat{\mu}_X$  on  $\hat{G}$  with the Mackey-Borel structure (24.7.2.12), called a **Plancherel measure for  $X$** .
- A  $\hat{\mu}$ -measurable set of non-zero  $G$ -invariant semi-norms  $\|-\|_\pi$  on the space of  $\pi$ -coinvariants

$$C_c^\infty(X)_\pi \triangleq \text{Hom}_G(C_c^\infty(X), \pi)^* \otimes \pi$$

for  $\hat{\mu}$ -a.e.  $\pi \in \hat{G}$ ,  
s.t. for any  $\Phi \in C_c^\infty(X)$ , we have

$$\|\Phi\|_{L^2}^2 = \int_{\hat{G}} \|\Phi_\pi\|_\pi^2 \hat{\mu}(\pi).$$

$\lrcorner$

**Remark (29.9.8.2).** For  $\pi \in \hat{G}$ , to give a  $G$ -invariant form on  $C_c^\infty(X)_\pi$  is equivalent to giving an equivariant map

$$M_\pi : C_c^\infty(X \times X) \rightarrow \pi \otimes \bar{\pi}.$$

$\lrcorner$

*Proof:* Cf. [S-V17]P112.  $\color{red}?$   $\square$

**Prop. (29.9.8.3).** Given a positive measure  $\hat{\mu}$  on  $\hat{G}$  with the Mackey-Borel structure (24.7.2.12) and a family  $\pi \mapsto H_\pi$  of Hermitian forms on  $C_c^\infty(X)_\pi$  (29.9.8.1) s.t.  $\Phi \mapsto H_\pi(\Phi)$  is  $\hat{\mu}$ -measurable for each  $\Phi \in C_c^\infty(X)$ , and

$$\|\Phi\|_{L^2}^2 = \int_{\hat{G}} H_\pi(\Phi_\pi) \hat{\mu}(\pi).$$

Then  $H_\pi$  are positive-semi-definite for  $\hat{\mu}$ -a.e.  $\pi \in \hat{G}$ , thus defining a Plancherel decomposition (29.9.8.1).  $\lrcorner$

*Proof:* Cf. [S-V17]P110.  $\square$

### Strongly-Tempered case

**Remark (29.9.8.4) [Tempered Matrix Coefficients are Majorized by a Spherical Matrix Coefficient].** If  $\pi_0$  is the normalized induction  $I_B^G(1)$ , and  $v_0 \in \pi_0$  is a spherical character (i.e. fixed by some compact open  $\mathcal{K} \subset G(k)$  s.t.  $G(k) = B(k)\mathcal{K}$ ), then for any compact open subgroup  $U \leq G(k)$ , there exists  $c(U) \in \mathbb{R}_+$  s.t. : For any tempered matrix coefficient  $\varphi(g) = \langle gu_1, u_2 \rangle$  where  $u_1, u_2$  are  $U$ -fixed with  $\|u_1\| = \|u_2\| = \|v_0\|$ , we have

$$|\varphi(g)| \leq c(U)(gv_0, v_0).$$

┘

*Proof:* Cf. [Almost  $L^2$  matrix coefficients]Thm2.?

□

**Def. (29.9.8.5) [Strongly-Tempered Spherical Varieties].**  $X$  is called a **strongly-tempered spherical variety** if

- $H$  is unimodular, thus  $X$  admits a  $G$ -invariant measure.
- For any  $x \in X(k)$  with  $H = \text{Stab}_G(x)$ , the restriction of any  $G$ -tempered matrix coefficient? to  $H$  is in  $L^1(H)$ .

┘

**Remark (29.9.8.6).** Gross-Prasad cases and Whittaker cases are strongly-tempered.?

If  $X$  is strongly-tempered and  $\Sigma_X \cap 2\Phi = \emptyset$ , then  ${}^L G_X = {}^L G$ ?

┘

**Thm. (29.9.8.7) [Plancherel Formula for Strongly-Tempered Spherical Varieties, Sakellaridis-Venkatesh[S-V17]].** Assume that  $X$  is a strongly-tempered spherical variety (29.9.8.5), then for any tempered representation of  $G$ , we can define a morphism

$$M_\pi : \pi \otimes \pi^\vee \rightarrow C^\infty(X \times X)$$

s.t.

$$M_\pi(v \otimes u)(x, x) = \int_{G_x} \langle \pi(h)v, u \rangle dh$$

and  $M_\pi(v \otimes u)$  vanishes outside  $(G \times G) \cdot \Delta_X$ . And denote

$$\theta_\pi : C_c^\infty(X \times X) \rightarrow \pi^\vee \otimes \pi \rightarrow \mathbb{C}$$

be the adjoint composed with the canonical pairing.

Then for any  $\Phi_1, \Phi_2 \in C_c^\infty(X)$ , we have

$$\langle \Phi_1, \Phi_2 \rangle_{L^2} = \int_{\widehat{G}} \theta_\pi(\Phi_1 \otimes \overline{\Phi_2}) d\widehat{\mu}_G(\pi).$$

where  $\widehat{\mu}_G$  is the Plancherel measure for  $G$ .

In particular,  $M_\pi(v, v)|_{\Delta_X} \geq 0$  for any tempered  $\pi \in \widehat{G}$ ,  $L^2(X)$  is tempered as a  $G$ -representation, and we have a Plancherel decomposition (29.9.8.1) for  $L^2(X)$  with plancherel measure  $\widehat{\mu}_X$  absolutely continuous w.r.t.  $\widehat{\mu}_G$ .

┘

*Proof:* Cf. [S-V17]P113.?

□

**Thm. (29.9.8.8)**[Ichino-Ikeda Type Local Conjecture, Sakellaridis-Venkatesh[S-V17]]. Suppose  $X$  is a strongly-tempered wavefront spherical variety, then for any discrete-series  $\pi \in \text{Irr}^{\text{adm}}(G(k))$ ,

$$M_\pi : v \otimes w \mapsto \int_H (\pi(h)v, w)dh$$

defines a non-degenerate Hermitian form on  $\pi_H$  (the  $H$ -coinvariants of  $\pi$ ). Moreover, if  $\sigma$  is a discrete series for the Levi subgroup of a parabolic subgroup  $P \leq G$  and  $\pi = I_P^G(\sigma)$ , then  $M_\pi$  is a non-negative Hermitian form.  $\lrcorner$

*Proof:* Cf.[S-V17]P123.  $\square$

### Scattering Theory

We can reduce the Plancherel decomposition to the understanding of discrete series for  $X$  and its boundary degenerations  $X_\Theta$ .

**Def. (29.9.8.9)** $[W_X(\Omega, \Theta)]$ . For any  $\Omega, \Theta \subset \Delta_X$ , denote  $W_X(\Omega, \Theta) = \{w \in W_X : w\Theta = \Omega\}$ (29.9.2.1).  $\lrcorner$

**Def. (29.9.8.10)**. For  $\Theta \subset \Delta_X$ , let  $c(\Theta)$  be the number of chambers in  $\mathfrak{a}_{X,\Theta}$ , which equals  $\sum_\Omega \#W_X(\Omega, \Theta)$ .  $\lrcorner$

*Proof:* Cf.[S-V17]P136.  $\square$

**Conj. (29.9.8.11)**[Scattering]. There exist canonical  $G$ -equivariant morphisms

$$\iota_\Theta : L^2(X_\Theta) \rightarrow L^2(X)$$

for each  $\Theta \subset \Delta_X$  called **Bernstein morphisms**, and  $A'_{X,\Theta} \times G$ -equivariant isomorphisms

$$S_w : L^2(X_\Theta) \rightarrow L^2(X_\Omega)$$

for  $\Omega, \Theta \subset \Delta_X$  and  $w \in W_X(\Omega, \Theta)$ (29.9.8.9), where  $A_{X,\Theta}$  acts on  $L^2(X_\Omega)$  via the isomorphism  $A_{X,\Theta} \cong A_{X,\Omega}$  induced by  $w$  and  $A'_{X,\Theta}$  is the image of  $Z(L_\Theta)^0$  in  $A_{X,\Theta}$ . And the following hold:

- $\iota_\Omega \circ S_w = \iota_\Theta$ .
- $S_{w'} \circ S_w = S_{w'w}$  for  $w \in W_X(\Omega, \Theta)$  and  $w' \in W_X(Z, \Omega)$ .
- $\iota_\Omega^* \circ \iota_\Theta = \sum_{w \in W_X(\Omega, \Theta)} S_w$ .
- The map

$$\sum_{\Theta \subset \Delta_X} \frac{\text{pr}_{\text{disc}} \iota_\Theta^*}{\sqrt{c(\Theta)}} : L^2(X) \rightarrow \bigoplus_{\Theta \subset \Delta_X} L^2(X_\Theta)_{\text{disc}} \text{ (29.9.8.10)}$$

is an isometric isomorphism onto the subspace of vectors

$$\left\{ (f_\Theta)_\Theta \in \bigoplus_{\Theta \subset \Delta_X} L^2(X_\Theta)_{\text{disc}} : S_w f_\Theta = f_\Omega, \forall w \in W_X(\Omega, \Theta) \right\}.$$

$\lrcorner$

*Proof:*

$\square$

**Thm& Conj.Cor.(29.9.8.12) [Scattering Theorem, Sakellaridis-Venkatesh[S-V17]].** This conjecture holds when the following conditions hold:

- $X$  is a wavefront spherical variety(29.9.1.14).
- All  $X_\Theta$  satisfy the discrete-series conjecture 9.4.6.?
- Generic injectivity of the map  $\mathfrak{a}_X^*/W_X \rightarrow \mathfrak{a}^*/W$  on each face.(Cf.[S-V17]Chap14.2?).

┘

*Proof:* ?

□

### Explicit Plancherel Formulae

References are [S-V17]Chap15.

**Notation(29.9.8.13).**

- Let  $X$  be strongly-factorizable(29.9.6.4)(e.g. symmetric varieties(29.9.6.5)).

┘

### Local $X$ -Distinguished Spectra

**Notation(29.9.8.14).**

- Let  $k \in p\text{-LField}$ .

┘

**Def.(29.9.8.15) [ $X$ -Distinguished Arthur-Parameters].** If  $X$  has no roots of type  $N$ (i.e.  $\Sigma_X \cap 2\Phi_G = \emptyset$ ), an  $X$ -distinguished Arthur parameter is an Arthur-parameter  $\psi : \text{WD}(k) \times \text{SL}(2) \rightarrow {}^L G(\mathbb{C})$  s.t.  $\phi|_{\text{WD}(k)}$  factors as

$$\text{WD}(k) \times \text{SL}(2) \xrightarrow{(\tilde{\phi}, \text{id})} {}^L G_X(\mathbb{C}) \times \text{SL}(2) \xrightarrow{\rho} {}^L G(\mathbb{C}) \text{ (29.9.2.2)}.$$

for some distinguished morphism  ${}^L G_X(\mathbb{C}) \times \text{SL}(2) \xrightarrow{\rho} {}^L G(\mathbb{C})$  (29.9.2.2), and  $\tilde{\phi} : \text{WD}(k) \rightarrow {}^L G_X(\mathbb{C})$  is a tempered  $L$ -parameter(29.3.1.4).

┘

**Conj.(29.9.8.16)[Plancherel Formula for Spherical Varieties, Sakellaridis-Venkatesh[S-V17]].** Situation as in(29.9.8.15), there should be a direct integral decomposition

$$L^2(X(k)) = \int_{[\psi]} \mathcal{H}_\psi d\hat{\mu}(\psi) \in \text{Rep}^{\text{uni}}(G(k))$$

where

- $[\psi]$  ranges through  ${}^L G_X(\mathbb{C})$ -conjugacy classes of  $X$ -distinguished local Arthur-parameters(29.9.8.15).
- The Hilbert space  $\mathcal{H}_\psi$  is a finite or countable(possibly empty) sum of representations belonging to the Arthur-packet corresponding to  $\psi$ .
- $\hat{\mu}$  is a natural measures for  $X$ -distinguished local Arthur-parameters modulo conjugacy that is absolutely continuous w.r.t. the natural Haar measure on the set of Arthur-parameters?.

┘

*Proof:* ?

□



**Conj.Cor.(29.9.8.17).** Situation as in(29.9.8.15),  $L^2(X(k))$  is a tempered representation of  $G(k)$  when  ${}^L G_X = {}^L G$ ?

This holds when  $X$  is strongly-tempered, by(29.9.8.7).  $\lrcorner$

**Conj.(29.9.8.18) [Plancherel Formula for Spherical Varieties(Inner-Forms), Sakellaridis-Venkatesh[S-V17]].** Situation as in(29.9.8.15), assume further that  $Z(G)$  acts faithfully on  $X$ , then there is a decomposition

$$\bigoplus_{\alpha} L^2(X^{\alpha}) = \int_{[\psi]} \mathcal{H}_{\psi} d\hat{\mu}(\psi)$$

of representations of  $G(k)$  and their inner forms, where

- $X^{\alpha}$  are pure inner forms of  $X$  (29.9.1.19).
- $[\psi]$  ranges through  ${}^L G_X(\mathbb{C})$ -conjugacy classes of  $X$ -distinguished local Arthur-parameters(29.9.8.15).
- $\hat{\mu}$  is a natural measures for  $X$ -distinguished local Arthur-parameters modulo conjugacy.
- $\mathcal{H}_{\psi}$  is a multiplicity-free direct sum of irreducible representations belonging to the Vogan-packet associated to  $[\psi]$ .(but notice that an Arthur-parameter can have several lift to an  $X$ -distinguished Arthur-parameters).
- $\mathcal{H}\psi \neq 0$  for  $\hat{\mu}$ -a.e.  $X$ -distinguished local Arthur-parameter  $\psi$ .

$\lrcorner$

*Proof:*

$\square$

**Thm& Conj.Cor.(29.9.8.19).** There are evidences for this conjecture for unramified  $L$ -parameters for  $G$ , see [Sakellaridis. On the unramified spectrum of spherical varieties] and [Sakellaridis. Spherical functions on spherical varieties].?

How to extend them to unramified groups?  $\lrcorner$

## 9 Ichino-Ikeda Conjecture

The generalized Ichino-Ikeda conjecture gives a link between the local Plancherel formula and global periods.

- Let  $k = F \in \mathbf{GField}$ .
- Denote  $[G] = A_G \backslash G(\mathbf{A}_F)^1$ .
- Assume  $X$  is reductive affine, i.e.  $H$  is reductive(29.9.0.1).
- Assume that  $Z(G)^0$  acts faithfully on  $X$ .

**Situation(29.9.9.1).** Assume that all  $\pi = \otimes_v \pi_v \in \mathrm{Irr}^{\mathrm{auto}}(G)$  appearing in this subsection satisfies

$$\mathrm{Hom}_{G(F_v)}(\pi_v, C^{\infty}(X(F_v))) \leq 1$$

for any  $v \in \Sigma_F$ .  $\lrcorner$

### Examples

## 29.10 Gan-Gross-Prasad Conjectures

References are [GGP12, Zha18, Zha19], [Background on the Gan-Gross-Prasad Conjecture, David Schwein].

**Notation (29.10.0.1).**

- Use notations on Langlands conjectured defined in [Langlands-Arthur Conjectures](#).
- Use notations on representations of  $p$ -adic reductive groups defined in [Admissible Representations of  \$p\$ -adic Reductive Groups](#).

⌋

### 1 Local Restriction Problems

References are [Zha18].

**Notation (29.10.1.1).**

- Let  $K \in \text{LField}$ .
- Let  $G \in \text{RedGrp}/K$ .
- Let  $H \leq G$  be an algebraic subgroup.

⌋

**Def. (29.10.1.2) [Local Restriction Problem].** For  $\chi \in H(K)^\vee$ ,  $\pi \in \text{Rep}^{\text{adm}}(G(K))$  is called  $(H, \chi)$ -**distinguished** if  $\text{Hom}_{H(K)}(\pi, \chi) \neq 0$ . An element of  $\text{Hom}_{H(K)}(\pi, \chi)$  is called a  $(H, \chi)$ -**functional**.

⌋

**Prop. (29.10.1.3).** For  $\chi \in H(K)^\vee$ , denote

$$C^\infty(H(K) \backslash G(K), \chi) \triangleq \{f \in C^\infty(G) \mid f(hg) = \chi(h)f(g), \forall h \in H(K), g \in G(K)\}.$$

Then there is an isomorphism

$$\text{Hom}_{H(K)}(\pi, \chi) \cong \text{Hom}_{G(K)}(\pi, C^\infty(H(K) \backslash G(K), \chi)) : \lambda \mapsto (v \mapsto I_\lambda(v)(g) = \lambda(\pi(g)v)).$$

⌋

*Proof:* If  $K$  is non-Archimedean,  $I_\lambda(v)$  is smooth because  $v$  is smooth. If  $K$  is Archimedean,  $\square$

**Def. (29.10.1.4) [Period Maps].** For  $\chi \in (H(F) \backslash H(\mathbf{A}_F))^\vee$ , denote the (twisted) **automorphic period map**:

$$P_{H, \chi} : \mathcal{A}_0(G/F) \rightarrow \mathbb{C} : \varphi \mapsto \int_{[H]} \varphi(h) \chi(h) dh.$$

(? Convergence).

Then  $(\pi, V) \subset \mathcal{A}_0(G) \in \text{Irr}^{\text{cusp}}(G)$  is called  $(H, \chi)$ -**distinguishable** if  $P_{H, \chi}(\varphi) \neq 0$  for some  $\varphi \in V$ . ⌋

**Conj. (29.10.1.5) [Period Formula Conjecture].** If  $\dim \text{Hom}_{H_v}(\pi_v, \chi_v) < \infty$  for any  $v \in \Sigma_F$ , then it is believed that the period map (29.10.1.4) is related to certain  $L$ -values:

$$\frac{|P_{H, \chi}(\varphi)|^2}{\langle \varphi, \varphi \rangle} = \frac{1}{\#S_\pi} \cdot C_H \frac{\Delta_G}{\Delta_H} \frac{L(\pi, \rho; \frac{1}{2})}{L(\pi, \text{Ad}; 1)} \prod_v l_v(\varphi_{1,v}, \overline{\varphi}_{1,v})$$

where  $S_\pi$  is a certain finite group depending only on the  $L$ -packet of  $\pi$ . The constant  $\Delta_H$  (reps.  $\Delta_G$ ) is a product of certain  $L$ -value determined by the motive (see [On the motive of a reductive group, Gross]) of reductive part of  $H$  (resp.  $G$ ). The constant  $C_H$  is a constant depending only on the choice of the local and global Haar measure on  $H(\mathbf{A}_F)$ . The representation  $\rho$  is a finite dimensional symplectic representation of  ${}^L G$ .  $\lrcorner$

*Proof:*

$\square$

**Example (29.10.1.6) [Waldspurger Periods].** Let  $E/F \in \mathbf{NField}$  be a quadratic extension and  $B$  be a quaternion algebra over  $F$ , then consider  $H = E^\times \subset G = B^\times$ .  $\lrcorner$

## 2 Local GGP Conjecture

References are [GGP12, Zha19], [Gross-Prasad conjecture and local theta correspondence, Gan-Ichino].

**Notation (29.10.2.1) [Local GGP Setting].**

- Let  $K \in \mathbf{LField}$  with  $\text{char } K \neq 2$ .
- Let  $(L, \sigma)$  be a semisimple algebraic over  $K$  with an involution s.t.  $L^\sigma = K$ . Thus one of the following holds:
  - $L$  is a quadratic extension of  $K$  with  $\text{Gal}(L/K) = \{1, \sigma\}$ .
  - $L = K \times K$  with  $\sigma : L \rightarrow L : (x, y) \mapsto (y, x)$ .
  - $L = K$  with  $\sigma = \text{id}$ .
- Let  $V \in \text{Herm}_n^\epsilon(L, \sigma)$ , which corresponds to unitary groups/general linear groups/orthogonal groups according to the above cases.
- Let  $W \leq V$  be a non-degenerate subspace s.t.
  - $\cdot(-1)^{\dim W^\perp} = -\epsilon$ ,
  - $W^\perp$  is split, i.e. there exists a decomposition  $W = X \oplus Y$  or  $W = X \oplus Y \oplus E$ , where  $X, Y$  are isotropic subspaces with  $\dim X = \dim Y$ , and  $E$  is a non-isotropic line.
- Let  $G(W) \leq G(V)$  be the subgroup of matrices acting trivially on  $W^\perp$ .
- Let  $G = G(W) \times G(V)$ .

$\lrcorner$

**Def. (29.10.2.2) [Diagonal Subgroup  $H$ ].** Situation as in (29.10.2.1), let  $P$  be a parabolic subgroup of  $G(V)$  stabilizing some complete flag of  $X$ , with unipotent radical  $U$ . Then  $G(W)$  is contained in a Levi subgroup of  $P$ . Define  $H = N \rtimes G(W)$ . Then there is an embedding  $H \subset G(W) \times G(V)$  where the first factor is the quotient  $H \rightarrow G(W)$  and the second factor is the natural embedding  $H \leq P \leq G(V)$ .

In particular, when  $\dim(W^\perp) \leq 1$ ,  $H$  is just the diagonal embedding  $G(W) \subset G(W) \times G(V)$ .  $\lrcorner$

**Notation (29.10.2.3) [Codimension 1 Setting].** Situation as in (29.10.2.1),

- Let  $e \in V$  s.t.  $\|e\| \neq 0$ , and  $W = V^\flat = e^\perp$ .

$\lrcorner$

**Def. (29.10.2.4) [Relevant Pure Inner Forms].** Situation as in (29.10.2.1), by ?, every pure inner form of  $G$  is of the form  $G' = G(W') \times G(V')$  where  $V' \in \text{Herm}_n(L, \sigma)$ ,  $W' \leq V'$  is a non-degenerate subspace s.t.  $\dim(W') = \dim(W)$ ,  $\dim(V) = \dim(V')$ . Then  $G'$  is called a **relevant pure inner form** of  $G$  if  $(W')^\perp \cong W^\perp$ .

In particular,

- If  $V$  is orthogonal or Hermitian (i.e.  $\epsilon = 1$ ), there are exactly 4 pure inner forms of  $G$ , and exactly 2 of them are relevant. And there is exactly one quasi-split relevant pure inner form.
- If  $V$  is symplectic, then there are no nontrivial pure inner forms of  $G$ , by (10.2.5.14).
- If  $V$  is skew-Hermitian, then there are exactly 4 pure inner forms of  $G$ , and 2 of them are relevant.

┘

*Proof:* ?

□

### Uniqueness of Bessel and Fourier-Jacobi Models

**Def. (29.10.2.5) [Bessel Models].** Cf. [GGP12] Section 12. ?

When  $\dim W^\perp$  is odd (i.e. the symplectic and skew-Hermitian cases), called the **distinguished representation** of  $H$ .

$\nu$  is 1-dimensional and unitary when  $\dim W^\perp$  is odd (i.e. the orthogonal and Hermitian cases), and  $\nu$  has Gelfand-Kirillov dimension  $\frac{1}{2} \dim(W/K)$ . ┘

**Def. (29.10.2.6) [Fourier-Jacobi Models].** Cf. [GGP12] Section 12. ?

When  $\dim W^\perp$  is even (i.e. the symplectic and skew-Hermitian cases), called the **distinguished representation** of  $H$ .

$\nu$  has Gelfand-Kirillov dimension  $\frac{1}{2} \dim(W/K)$ . ┘

**Thm. (29.10.2.7) [Uniqueness of Bessel and Fourier-Jacobi Models, Aizenbud-Gourevitch-Rallis-Schiffmann [AGRS]/Sun/Sun-Zhu/Waldspurger/Aizenbud-Gourevitch].**

Suppose  $\text{char } k = 0$  and  $\dim W^\perp \leq 1$ .

1. In the orthogonal case (i.e.  $G = \text{O}(W) \times \text{O}(V)$  or  $G = \text{SO}(W) \times \text{SO}(V)$ ), for any  $\pi = \pi_W \boxtimes \pi_V \in \text{Irr}^{\text{adm}}(G)$ , we have  $\dim \text{Hom}_H(\pi \otimes \nu, \mathbb{C}) = \text{Hom}_H(\pi, \nu) \leq 1$  (29.10.2.5).
2. In the unitary case (including the split case), for any  $\pi = \pi_W \boxtimes \pi_V \in \text{Irr}^{\text{adm}}(G)$ , we have  $\dim \text{Hom}_H(\pi \otimes \nu, \mathbb{C}) = \text{Hom}_H(\pi, \nu) \leq 1$  (29.10.2.5).
3. In the symplectic case, if  $k$  is non-Archimedean, for any  $\tilde{\pi} = \tilde{\pi}_W \boxtimes \pi_V \in \text{Irr}^{\text{adm}}(\widetilde{\text{Sp}}(V) \times \text{Sp}(W))$  or  $\tilde{\pi} = \pi_W \boxtimes \tilde{\pi}_V \in \text{Irr}^{\text{adm}}(\text{Sp}(V) \times \widetilde{\text{Sp}}(W))$ , we have  $\dim \text{Hom}_{\tilde{H}}(\tilde{\pi} \otimes \bar{\nu}_\psi, \mathbb{C}) \leq 1$  (29.10.2.6).
4. In the skew-Hermitian case (including the split case), if  $k$  is non-Archimedean, for any  $\pi = \pi_W \boxtimes \pi_V \in \text{Irr}^{\text{adm}}(G)$ , we have  $\dim \text{Hom}_H(\pi \otimes \bar{\nu}_{\psi_0, \mu}, \mathbb{C}) \leq 1$  (29.10.2.6).

┘

*Proof:* ?

1: For  $G = \text{O}(W) \times \text{O}(V)$ , this follows from [AGRS] in the non-Archimedean case and follows from [Sun-Zhu, Multiplicity one theorems: the Archimedean case] in the Archimedean case.

For  $G = \text{SO}(W) \times \text{SO}(V)$ , this follows from [Waldspurger, Une variante d'un résultat de Aizenbud, Gourevitch, Rallis et Schiffmann] in the non-Archimedean case and follows from [Sun-Zhu, A note on special orthogonal groups following Waldspurger] in the Archimedean case.

2: This follows from [AGRS] in the non-Archimedean case and follows from [Aizenbud-Gourevitch, Multiplicity one theorem for  $\mathrm{GL}(n+1; \mathbb{R}), \mathrm{GL}(n; \mathbb{R})$ ] and [Sun-Zhu, Multiplicity one theorems: the Archimedean case] in the Archimedean case.

3, 4: These follow from [Sun, Multiplicity one theorems for Fourier-Jacobi models].  $\square$

**Remark (29.10.2.8) [Positive Characteristic cases].** If  $\mathrm{char} k > 0$  and  $\dim W^\perp \leq 1$ , then analogous result holds for  $\mathrm{GL}(n) \times \mathrm{GL}(n+1)$ , Cf. [Aizenbud-Avni-Gourevitch, Spherical pairs over close local fields].  $\lrcorner$

**Def. (29.10.2.9) [ $H$ -Invariant Forms].** In the GGP-setting (29.10.2.1), for  $\pi \in \mathrm{Rep}^{\mathrm{adm}}(G)$ , the  $H$ -invariant forms on  $\pi$  is defined to be

$$\mathcal{P}(\pi) = \mathrm{Hom}_H(\pi, \mathbb{C}).$$

$\lrcorner$

**Def. (29.10.2.10) [Matrix Coefficients as Invariant Forms].** In the GGP-setting (29.10.2.1), if  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(G)$  is tempered, then there is an element  $I \in \mathcal{P}(\pi) \otimes \mathcal{P}(\tilde{\pi})$  (29.10.2.9) s.t.

$$I(v \otimes \tilde{v}) = \int_{H(F)} \langle \pi(h)v, \tilde{v} \rangle dh.$$

$\lrcorner$

**Thm. (29.10.2.11) [Sakellaridis-Venkatesh].** Situation as in (29.10.2.10), the non-vanishing of  $\mathcal{P}(\pi)$  is equivalent to the non-vanishing of  $I$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (29.10.2.12) [Normalized Invariant Pairing].**  $\lrcorner$

### Local GGP Conjecture

**Conj. (29.10.2.13) [Local Gan-Gross-Prasad Conjecture, Gross-Prasad [G-P94]/Gan-Gross-Prasad [GGP12]]**

In the local GGP setting (29.10.2.1), for each generic local Vogan-packet  $\Pi_\phi$  of  $G$  (29.3.4.3), there exists a relevant pure inner form  $G' = G(W') \times G(V')$  of  $G$  (29.10.2.4) and  $\pi \in \Pi_\phi \cap \mathrm{Irr}^{\mathrm{adm}}(G')$  s.t.

$$\mathrm{Hom}_{H'}(\pi \otimes \bar{\nu}; \mathbb{C}) \neq 0 \text{ (29.10.2.2)(29.10.2.5)(29.10.2.6).}$$

$\lrcorner$

*Proof:* ?  $\square$

**Thm& Conj.Cor. (29.10.2.14).**

- If  $L = K \times K$  and  $\dim W^\perp \leq 1$ , the local Gan-Gross-Prasad conjecture holds.
- If  $L = K \times K$  and  $K$  is non-Archimedean, then the local Gan-Gross-Prasad conjecture holds.
- In the special orthogonal case, [Waldspurger, Une formule intégrale reliée à la conjecture locale de Gross-Prasad], [Waldspurger, Une formule intégrale reliée à la conjecture locale de Gross-Prasad, 2] and [Mœglin-Walspurger, La conjecture locale de Gross-Prasad pour les groupes spéciaux orthogonaux: le cas général], the local Gan-Gross-Prasad conjecture holds assuming some properties of the Vogan packet.

┘

*Proof:* 1: This follows from [JSS83] and [J-S90], where the non-zero element in  $\text{Hom}_H(\pi \otimes \bar{\nu}; \mathbb{C})$  is given by the local Rankin-Selberg integral.?

2: This follows from item1 and [GGP12]Thm15.1, 16.1.?

3: ?

□

**Thm.(29.10.2.15)** [Reduction to basic cases, Gan-Gross-Prasad[GGP12]]. If  $K$  is non-Archimedean, then

┘

**Conj.(29.10.2.16)** [Refined Local Gan-Gross-Prasad Conjecture, Gan-Gross-Prasad[GGP12]]. Situation as in(29.10.2.13), we can specify the unique  $\pi$  under the Langlands-Vogan parameterization as follows: ? Cf.[GGP12]P67.

- When  $G = \text{SO}(W) \times \text{SO}(V)$ ,
- When  $G = U(W) \times U(V)$  Hermitian,
- When  $G = \widetilde{\text{Sp}}(W) \times \text{Sp}(V)$  or  $\text{Sp}(W) \times \widetilde{\text{Sp}}(V)$ ,
- When  $G = U(W) \times U(V)$  skew-Hermitian,

Notice  $\chi(-1, -1) = 1$ , and  $\chi(-1, 1) = \chi(1, -1)$  determines the relevant pure inner form acting on  $\pi_\theta(\varphi, \chi)$ ?.

┘

*Proof:* ?

□

**Prop.(29.10.2.17).** The refined local Gan-Gross-Prasad conjecture(29.10.2.16) is independent of

- $J(\theta)$ .
- Scaling of the bilinear form on  $V$ .
- The data used to define  $\nu \in \text{Irr}^{\text{adm}}(H)$ .

┘

*Proof:* Cf.[?GPP12]Chap18.

□

### Relative Trace Formula approach

### 3 Global Restriction Problems

**Notation(29.10.3.1).**

- Let  $F \in \mathbf{GField}$ .
- Let  $G \in \mathbf{RedGrp}/F$
- Let  $H \leq G$  be a connected algebraic subgroup.
- Some Jargon: Bessel case means Hermitian case, Fourier-Jacobi case means skew-Hermitian case.

┘

#### 4 Global GGP Conjecture

**Notation(29.10.4.1)[Global GGP Setting].**

- Let  $F \in \mathbf{NField}$ ,
- Let  $(E, \sigma)$  be a semisimple algebra over  $F$  with  $\text{degree} \leq 2$  s.t.  $E^\sigma = F$ . Thus one of the following holds:
  - $E$  is a quadratic extension of  $F$  with  $\text{Gal}(E/F) = \{1, \sigma\}$ .
  - $E = F \oplus F$  with  $\sigma : E \rightarrow E : (x, y) \mapsto (y, x)$ .
  - $E = F$  with  $\sigma = \text{id}$ .
- Let  $V \in \text{Herm}_n(E, \sigma)$ , which corresponds to orthogonal groups/general linear groups/unitary groups according to the above cases.
- Let  $e \in V$  with  $\|e\| \neq 0$ , and  $V^\flat = e^\perp$ .
- Let  $G = U(V) \times U(V^\flat)$ , and  $H = U(V^\flat)$  which embeds diagonally in  $G$ .
- Let  $\pi = \otimes_{v \in \Sigma_F} \pi_v \in \text{Irr}^{\text{cusp}}(G/F)$ .
- Let  $W \leq V$  be a non-degenerate subspace s.t.
  - $\cdot(-1)^{\dim W^\perp} = -\epsilon$ ,
  - $W^\perp$  is split, i.e. there exists a decomposition  $W = X \oplus Y$  or  $W = X \oplus Y \oplus E$ , where  $X, Y$  are isotropic subspaces with  $\dim X = \dim Y$ , and  $E$  is a non-isotropic line.
- Let  $G(W) \leq G(V)$  be the subgroup of matrices acting trivially on  $W^\perp$ .
- Let  $G = G(W) \times G(V)$ .

┘

**Prop.(29.10.4.2)[Normalized Invariant Pairing].** Denote  $\mathcal{P}(\pi) = \otimes_{v \in \Sigma_F} \mathcal{P}(\pi_v)$  (29.10.2.9).

If  $\mathcal{P}(\pi) \neq 0$ , then we can define a bilinear form

$$\langle \cdot, \cdot \rangle : \mathcal{P}(\pi) \otimes \mathcal{P}(\tilde{\pi}) \rightarrow \mathbb{C}$$

which is the product of local invariant pairings (29.10.2.12). By calculations in (29.10.2.12), the product is well-defined.

┘

**Conj.(29.10.4.3)[Gan-Gross-Prasad Conjecture, Gan-Gross-Prasad[GGP12]].** Let  $\pi_\chi$  be a representation in the global Vogan packet  $\Pi_\phi$  corresponding to the distinguished character  $\chi$ , then the following are equivalent:

- $\pi_\chi$  occurs with multiplicity one in  $\mathcal{A}_0(G)$ , and the linear form  $F(\nu)$  is non-zero on  $\pi_\chi$ .
- $L(\pi_\chi, R, \frac{1}{2}) \neq 0$ .

┘

*Proof:* Cf. [GGP12] Conj26.1. ?

□

**Conj.(29.10.4.4) [Refined Gan-Gross-Prasad Conjecture, Gross-Prasad/ Gan--Gross-Prasad[GGP12]/ Ichino-Ikeda[I-I10]/Harris[Har14a]].** Suppose  $\mathcal{P}(\pi) \neq 0$ , and define  $P_\pi \in \mathcal{P}(\pi)$  by

$$P_\pi(\varphi) = \int_{H(F) \backslash H(\mathbf{A}_F)} \varphi(h) dh, \quad \forall \varphi \in \pi,$$

then

$$\langle \mathcal{P}_\pi, \mathcal{P}_{\tilde{\pi}} \rangle = \frac{L(M^\vee(1); 0)}{2^\beta L(\pi, \text{Ad}; 1)} L(\pi; \frac{1}{2}),$$

where  $2^\beta = \#C_{\widehat{G}}(\text{Arthur}(\pi))$ . ┘

**Thm& Conj.Cor.(29.10.4.5).** The GGP conjecture holds for

- $V$  Hermitian, See [Unitary case](#).
- $V$  orthogonal s.t.  $\dim V = 3$  by [Waldspurger85]Prop7, See [Waldspurger's Formula\(Ichino-Ikeda for  \$n = 2\$ \)](#).
- $V$  orthogonal s.t.  $\dim V = 4$  by Ichino, Cf. [Triple Product\(Ichino-Ikeda for  \$n = 3\$ \)](#). ┘

**Conj.Cor.(29.10.4.6).**  $L(\Pi \times \Pi'; \frac{1}{2}) \neq 0$  iff the pairing  $I : \pi \otimes \pi' \rightarrow \mathbb{C}$  is non-trivial. ┘

**Def.(29.10.4.7)[Hermitian Isobaric Representations].** ? ┘

**Conj.(29.10.4.8) [Gan-Gross-Prasad Conjecture].** Let  $\Pi_n \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/E)$ ,  $\Pi_{n+1} \in \text{Irr}^{\text{cusp}}(\text{GL}(n+1)/E)$  be Hermitian, then the following are equivalent:

- $L(\Pi_n \times \Pi_{n+1}; \frac{1}{2}) \neq 0$ .
- There exist irreducible cuspidal representations  $\pi_n, \pi_{n+1}$  of  $U(V_n)(\mathbf{A}_F), U(V_{n+1})(\mathbf{A}_F)$  resp. s.t.  $\text{BC}(\pi_n) \cong \Pi_n, \text{BC}(\pi_{n+1}) \cong \Pi_{n+1}$ , and  $\varphi_n \in \pi_n, \varphi_{n+1} \in \pi_{n+1}$ , so that

$$\mathcal{P}(\varphi_n, \varphi_{n+1}) = \int_{U(V_n)(F) \backslash U(V_n)(\mathbf{A}_F)} \varphi_n(h) \varphi_{n+1}(h) dh \neq 0.$$

┘

*Proof:* □

## 5 Ichino-Ikeda Formula(Orthogonal case)

References are [\[Ike08, ?I-I10\]](#).

**Notation(29.10.5.1).**

- Let  $F \in \mathbf{GField}$  and  $\text{char } F \neq 2$ .
- Let  $(V_0, Q_0) \subset (V_1, Q_1)$  be quadratic forms over  $F$  with rank  $n, n+1$  resp. And if  $n = 2$ , we assume that  $V_0 \not\cong \mathbb{H}_F$ .
- Let  $G_i = O(V_i)$ . In particular,  $G_0 \subset G_1$ .
- Let  $(V_?, Q_?)$  be the one of  $\{(V_0, Q_0), (V_1, Q_1)\}$  that is even-dimensional, and let  $E = K_{Q_?}$  be the discriminant field([29.10.5.3](#)).
- For  $v \in \Sigma_F$ , denote  $G_{i,v} \stackrel{\Delta}{=} G_i(F_v)$ .
- Fix  $\mathcal{K}_i = \prod_{v \in \Sigma_F} \mathcal{K}_{i,v} \subset G_i(\mathbf{A}_F)$  s.t.  $[\mathcal{K}_0 : \mathcal{K}_0 \cap \mathcal{K}_1] < \infty$ .
- For each  $v \in \Sigma_F$ , fix a Haar measure  $dg_i$  on  $G_{i,v}$ .
- Let  $\pi_i = \otimes_v \pi_{i,v} \in \text{Irr}^{\text{auto}}(G/F)$ .
- Let  $\varphi_i = \otimes_v \varphi_{i,v} \in \pi_i^{\mathcal{K}_i}$  be a pure tensor.



- Let  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  s.t. for any  $v \in \Sigma_F^{\text{fin}} \setminus S$  and  $i \in \{0, 1\}$ ,
  - (U1)  $G_{i,v}$  is unramified (20.5.2.25).
  - (U2)  $\mathcal{K}_{i,v}$  is a hyperspecial maximal compact subgroup of  $G_{i,v}$ .
  - (U3)  $\mathcal{K}_{0,v} \subset \mathcal{K}_{1,v}$ .
  - (U4)  $\pi_{i,v} \in \text{Irr}^{\text{unr}}(G_{i,v})$ .
  - (U5)  $\varphi_{i,v} \in \pi_{i,v}^{\mathcal{K}_{i,v}}$ , and  $\|\varphi_{i,v}\| = 1$ .
  - (U6)  $\int_{\mathcal{K}_{i,v}} dg_{i,v} = 1$ , i.e.  $dg_{i,v}$  is the standard measure.

┘

**Def. (29.10.5.2)** [Motivic  $L$ -functions  $\Delta_{G_i}$ ]. For  $v \in \Sigma_F$ , define

$$\Delta_{G_i,v} = \begin{cases} \zeta_v(2)\zeta_v(4)\dots\zeta_v(2l) & , \dim V_i = 2l + 1 \\ \zeta_v(2)\zeta_v(4)\dots\zeta_v(2l-2)L_v(\chi_{E/F}; l) & , \dim V_i = 2l. \end{cases}$$

and  $\Delta_{G_i} = \prod_{v \in \Sigma_F^{\text{fin}}} \Delta_{G_i,v}$ . Cf. [Gross, On the motive of a reductive group].

┘

**Def. (29.10.5.3)** [Discriminant Fields]. For  $K \in \text{Field}$  and an even dimensional quadratic form  $(V, Q)$  over  $K$ , the **discriminant field** is the quadratic field  $K_Q = K(\sqrt{(-1)^{\dim V/2} \det Q})$ .

┘

**Prop. (29.10.5.4)** [Langlands Dual Groups].  $\widehat{G}_i = \begin{cases} \text{Sp}(l; \mathbb{C}) & , \dim V_i = 2l + 1 \\ \text{SO}(2l; \mathbb{C}) & , \dim V_i = 2l \end{cases}$ .

Let  $\text{Ad} : {}^L G_i \rightarrow \text{GL}(\text{Lie } \widehat{G}_i)$  be the adjoint representations.

┘

*Proof:*

□

**Def. (29.10.5.5)**. Define

$$\mathcal{P}_{\pi_1, \pi_0}(s) = \frac{L(\pi_1 \boxtimes \pi_0; s)}{L(\pi_1, \text{Ad}; s + \frac{1}{2})L(\pi_0, \text{Ad}; s + \frac{1}{2})} \text{ ?}.$$

**?** What's the properties of  $\mathcal{P}_{\pi_1, \pi_2}$ ?

┘

**Def. (29.10.5.6)** [Local Matrix Coefficients and Integrals]. Consider the matrix coefficients:

$$\Phi_{\varphi_{i,v}}(g_{i,v}) = \langle g_{i,v}(\varphi_{i,v}), \varphi_{i,v} \rangle, \quad g_{i,v} \in G_{i,v}.$$

and

$$I_v(\varphi_{1,v}, \varphi_{0,v}) = \int_{G_{0,v}} \Phi_{\varphi_{1,v}}(g_{0,v}) \overline{\Phi_{\varphi_{0,v}}(g_{0,v})} dg_{0,v}.$$

(It is convergent by (29.10.5.7)). And

$$\alpha_v(\pi_{1,v}, \pi_{2,v}) = \Delta_{G_{1,v}}^{-1} \mathcal{P}_{\pi_{1,v}, \pi_{0,v}} \left( \frac{1}{2} \right)^{-1} I_v(\pi_{1,v}, \pi_{2,v}) \text{ (29.10.5.5)}.$$

┘

**Prop. (29.10.5.7)** [Tempered case]. If  $\pi_{i,v}$  are both tempered (24.9.8.11), then the integral  $I_v(\varphi_{1,v}, \varphi_{0,v})$  is absolutely convergent, and  $I_v(\varphi_{1,v}, \varphi_{0,v}) \geq 0$  if  $\varphi_{i,v} \in \pi_{i,v}^{\mathcal{K}_{i,v}-\text{fin}}$ .

┘

*Proof:* Cf.[Ichino-Ikeda]Section4. □

**Prop. (29.10.5.8) [Unramified case].** If  $v \in \Sigma_F^{\text{fin}}$  s.t. the conditions (U1)-(U6) hold, and the integral  $I(\varphi_{1,v}, \varphi_{0,v})$  is absolutely convergent, then  $\alpha_v(\varphi_{1,v}, \varphi_{0,v}) = 1$ . ┘

*Proof:* Cf.[Ichino-Ikeda]Section5. □

**Conj. (29.10.5.9) [Ichino-Ikeda].** If  $v \in \Sigma_F^{\text{fin}}$  and  $\pi_{1,v}, \pi_{0,v}$  are both tempered, then  $\text{Hom}_{G_{0,v}}(\pi_{1,v} \otimes \bar{\pi}_{0,v}, \mathbb{C}) \neq 0$  iff  $\alpha_v(\varphi_{1,v}, \varphi_{0,v}) > 0$  for some  $\varphi_{i,v} \in \pi_{i,v}^{\mathcal{K}_{i,v}^{\text{fin}}}$  (29.10.5.6). ┘

*Proof:* □

**Remark (29.10.5.10).** It is believed that for any  $\pi_{i,v} \in \text{Irr}^{\text{adm}}(G_{i,v})$ ,  $\dim \text{Hom}_{G_{0,v}}(\pi_{1,v} \otimes \bar{\pi}_{0,v}, \mathbb{C}) \leq 1$ . ┘

*Proof:* □

**Conj. (29.10.5.11) [Ichino-Ikeda].** If  $\pi_i$  are both almost generic, then

- The integrals  $I_v(\varphi_{1,v}, \varphi_{0,v})$  is absolutely convergent and  $I(\varphi_{1,v}, \varphi_{0,v}) \geq 0$  if  $\varphi_{i,v} \in \pi_{i,v}^{\mathcal{K}_{i,v}^{\text{fin}}}$ . (This should follow from (29.10.5.7) and generalized Ramanujan conjecture (26.6.6.1)).
- $\text{Hom}_{G_{0,v}}(\pi_{1,v} \otimes \bar{\pi}_{0,v}, \mathbb{C}) \neq 0$  iff  $\alpha_v(\varphi_{1,v}, \varphi_{0,v}) > 0$  for some  $\varphi_{i,v} \in \pi_{i,v}^{\mathcal{K}_{i,v}^{\text{fin}}}$ . ┘

*Proof:* □

**Remark (29.10.5.12).** These follow from (29.10.5.7) and (29.10.5.9) if we assume the generalized Ramanujan conjecture that any almost generic representation is tempered. ┘

*Proof:* □

**Prop. (29.10.5.13) [Inner Product on Forms].** If  $\pi_i = \otimes_v \pi_{i,v} \in \text{Irr}^{\text{auto}}(G_i(\mathbf{A}_F))$  are square-integrable, then there are invariant inner forms on  $\pi_i$  s.t.

$$\langle \varphi_i, \varphi'_i \rangle = \int_{G_i(F) \backslash G_i(\mathbf{A}_F)} \varphi_i(g_i) \overline{\varphi'_i(g_i)} d\mu(g_i)$$

where  $\mu$  is a Tamagawa measure on  $G_i(\mathbf{A}_F)$  (20.5.4.5). Since  $\pi_i$  are unitary, we can also define another invariant form

$$\langle \varphi_i, \varphi'_i \rangle' = \prod_{v \in \Sigma_F} (\varphi_{i,v}, \varphi'_{i,v})_v.$$

So there exists  $C_i \in \mathbb{R}_+$  s.t.

$$\langle \varphi_i, \varphi'_i \rangle = C_i \prod_{v \in \Sigma_F} (\varphi_{i,v}, \varphi'_{i,v})_v.$$

┘

**Conj. (29.10.5.14) [Ichino-Ikeda].** If  $\pi_i = \otimes_v \pi_{i,v} \in \text{Rep}^{\text{cusp, loc. generic}}(G_i(\mathbf{A}_F))$  are square-integrable, then there exists  $\beta \in \mathbb{Z}$  s.t.

$$\frac{|\langle \varphi_1|_{G_0}, \varphi_2 \rangle|^2}{\langle \varphi_1, \varphi_1 \rangle \cdot \langle \varphi_2, \varphi_2 \rangle} = 2^\beta C_0 \Delta_{G_1} \mathcal{P}_{\pi_1, \pi_2} \left( \frac{1}{2} \right) \prod_{v \in S} \frac{\alpha_v(\varphi_{1,v}, \varphi_{0,v})}{\|\varphi_{1,v}\|^2 \cdot \|\varphi_{2,v}\|^2} \quad (29.10.5.1)(29.10.5.2)(29.10.5.5)(29.10.5.13).$$

Moreover, if  $\pi_i$  are tempered with Arthur parameter  $\psi_i : \mathcal{L}_F \rightarrow {}^L G_i$ ?, then

$$2^\beta = \frac{1}{\#\mathcal{S}_{\psi_1} \cdot \#\mathcal{S}_{\psi_2}}.$$

where  $\mathcal{S}_{\psi_i}$  is the centralizer of  $\text{Im}(\psi_i)$  in  $\widehat{G}_i$ . ┘

*Proof:* □

**Remark (29.10.5.15).** If  $\pi_i$  are tempered, then it is believed that the local  $L$ -factors  $L(\pi_{i,v}, \text{Ad}; s)$  and  $L(\pi_{1,v} \boxtimes \pi_{0,v}; s)$  are holomorphic for  $\text{Re}(s) > 0$ . In particular, the formula doesn't depend on the  $L$ -factors for bad places. ┘

*Proof:* □

**Conj. (29.10.5.16) [Non-Tempered case, Ichino-Ikeda].** If  $\pi_i = \otimes_v \pi_{i,v} \in \text{Irr}^{\text{auto}}(G_i(\mathbf{A}))$  are square-integrable s.t.  $\text{Hom}_{G_{0,v}}(\pi_{1,v} \otimes \bar{\pi}_{0,v}, \mathbb{C}) \neq 0$  for any  $v \in \Sigma_F$ , then

- The integral  $\langle \varphi_1|_{G_0}, \varphi_2 \rangle$  is convergent for any  $\varphi_i \in \pi_i$ .
- There exists  $\beta \in \mathbb{Z}$  s.t.

$$\frac{|\langle \varphi_1|_{G_0}, \varphi_2 \rangle|^2}{\langle \varphi_1, \varphi_1 \rangle \cdot \langle \varphi_2, \varphi_2 \rangle} = 2^\beta C_0 \Delta_{G_1} \mathcal{P}_{\pi_1, \pi_2} \left( \frac{1}{2} \right) \prod_{v \in S} \frac{\alpha_v(\varphi_{1,v}, \varphi_{0,v})}{\|\varphi_{1,v}\|^2 \cdot \|\varphi_{2,v}\|^2} \text{ (29.10.5.1)(29.10.5.2)(29.10.5.5)(29.10.5.13).}$$

┘

*Proof:* □

## 6 Waldspurger's Formula (Ichino-Ikeda for $n = 2$ )

**Notation (29.10.6.1).**

- Let  $F \in \mathbf{NField}$ ,  $\mathbf{A} = \mathbf{A}_F$ .
- Let  $D$  be a quaternion algebra over  $F$ .
- $E \in \mathbf{NField}$  is a quadratic extension of  $F$  contained in  $D$ .
- Let  $\mathbf{1} \neq \psi \in (\mathbf{A}_F/F)^\vee$ .
- Choose a central character  $\omega : \mathbf{C}_F \rightarrow \mathbb{C}^\times$ .
- $\pi \in \text{Irr}^{\text{cusp}}(D^\times/F, \omega)$ .
- Let  $\chi : \mathbf{C}_E \rightarrow \mathbb{C}^\times$  be a Hecke character s.t.  $\chi|_{\mathbf{I}_F} \cdot \omega = 1$ .

┘

**Def. (29.10.6.2) [Period Integrals].** The **period integral** is defined to

$$P_\chi : V_\pi \rightarrow \mathbb{C} : f \mapsto \int_{E^\times \backslash \mathbf{I}_E / \mathbf{I}_F} f(t) \chi(t) dt.$$

┘

**Def. (29.10.6.3)[Invariant Pairing].** If  $F \in \mathbf{LField}$  and  $E$  is a quadratic extension of  $F$ , then for any fixed pairing  $(-, -) : \pi \times \tilde{\pi} \rightarrow \mathbb{C}$ , there is a  $D^\times$ -invariant bilinear form

$$\alpha : \pi \otimes \tilde{\pi} \rightarrow \mathbb{C} : (f_1, f_2) \mapsto \frac{L(\chi_{E/F}; 1)L(\pi, \text{Ad}; 1)}{\zeta_F(2)L(\pi(\chi); 1/2)} \int_{E^\times/F^\times} (\pi(t)f_1, f_2)\chi(t)dt.$$

?

And the global invariant pairing is the product of the local pairings.  $\lrcorner$

**Thm. (29.10.6.4)[Waldspurger 85].** Situation as in (29.10.6.1), for any  $f_1 \in V_\pi, f_2 \in V_{\tilde{\pi}}$ ,

$$P_\chi(f_1).P_{\chi^{-1}}(f_2) = \frac{\zeta_F(2)L(\pi(\chi); 1/2)}{8L(\chi_{E/F}; 1)^2L(\pi, \text{Ad}; 1)} \alpha(f_1, f_2) \text{ (29.10.6.2)(29.10.6.3)}.$$

Where  $L(\pi, \chi; s)$  is defined via Jacquet-Langlands lifting of  $\pi$ . ?  $\lrcorner$

*Proof:*  $\square$

**Cor. (29.10.6.5)[Waldspurger].** Let  $\chi$  be a character,  $k \in 2\mathbb{Z}$  and  $\varphi \in \text{SF}_{k-1}(N, \chi^2)$ , then there exists a function  $A_\varphi$  on  $\mathbb{N}^{sc}$  s.t. for  $t \in \mathbb{N}^{sc}$ ,

$$A_\varphi(t)^2 = L(\varphi \otimes \chi_0^{-1}\chi_t, \frac{k-1}{2})\epsilon(\chi_0^{-1}\chi_t, \frac{1}{2}).$$

$\lrcorner$

*Proof:* Cf.[Explicit application of Waldspurger's theorem].  $\square$

### Relation to Ichino-Ikeda Formula

Cf.[?I-I10]Chap6.

**Prop. (29.10.6.6).** Let  $G_1 = D^\times \in \text{AlgGrp}/F$  and let  $G_0 = T$  be an anisotropic torus of  $G_1$ .

$G_1$  can be regarded as a special orthogonal group associated to a 3-dimensional quadratic space over  $F$ . And  $G_0$  can be regarded as a special orthogonal group associated to a 2-dimensional quadratic space over  $F$ .

The motivic  $L$ -function for  $G_1$  is  $\Delta_{G_1} = \zeta_F(2)$ .

There exists an exact sequence  $1 \rightarrow F^\times \rightarrow E^\times \rightarrow T \rightarrow 1$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (29.10.6.7)[Haar Measures].** For each  $v \in \Sigma_F$ , choose a Haar measure  $\mu_v$  on  $F_v^\times$  that is self-dual w.r.t.  $\psi_v$ , and let the measure  $d\nu_v = \zeta_v(1)^{-1} \frac{d\mu_v}{|v|}$ . Similarly defined  $d\nu_v$  on  $E_v^\times$ , and define the Haar measure on  $T(F_v)$  as the quotient space  $1 \rightarrow F_v^\times \rightarrow E_v^\times \rightarrow T \rightarrow 1$ .

Then the Haar measure constant  $C_0 = \Lambda(\chi_{E/F}; 1)^{-1}$  ?  $\lrcorner$

**Prop. (29.10.6.8).**  $\lrcorner$

## 7 Jacquet-Langlands-Shimizu Correspondence

**Notation (29.10.7.1).**

- Let  $F \in \mathbf{NField}$ .
- Let  $1 \neq \psi : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$  be a non-trivial additive character.
- Let  $\Pi \in \mathbf{Rep}^{\text{cusp, uni}}(\mathrm{GL}(2)/F)$ .
- Let  $D \in \mathbf{Az}_F$  be a quaternion algebra and  $\pi^D \in \mathbf{Rep}^{\text{cusp, uni}}(D^\times/F)$  corresponds to  $\pi$  via Jacquet-Langlands correspondence **?**.
- Let  $V = D$  be the 4-dimensional quadratic space over  $F$  with the non-degenerate symmetric bilinear form  $(x, y) = \mathrm{tr}_{D/F}(x\bar{y})$ .
- Let  $f = \otimes'_v f_v \in \pi$ .
- Let  $S(V(\mathbf{A}_F))$  be the subspace of  $\mathcal{S}(V(\mathbf{A}_F))$  consisting of functions that corresponds to polynomials in the mock model at every Archimedean place.
- Let  $\varphi = \otimes'_v \varphi_v \in S(V(\mathbf{A}_F))$ .
- Let  $W_{\bar{\psi}, f} = \otimes'_v W_{\bar{\psi}_v, f_v}$  be the Whittaker function of  $f$  s.t.  $W_{\bar{\psi}_v, f_v}(1) = 1$  for a.e.  $v \in \Sigma_F$  **(26.6.4.3)(26.6.4.2)**.

┘

**Prop. (29.10.7.2).** Let  $t$  be the involution on  $D^\times \times D^\times$  defined by

$$t(x, y) = (\bar{y}^{-1}, \bar{x}^{-1}).$$

Then there is an exact sequence

$$1 \rightarrow \mathbb{G}_m \rightarrow (D^\times \times D^\times) \rtimes \langle t \rangle \xrightarrow{\rho} \mathrm{GO}(V) \rightarrow 1,$$

where  $\rho(h_1, h_2)x = h_1 x h_2^{-1}$  and  $\rho(t)x = \bar{x}$  for  $h_1, h_2 \in D^\times$  and  $x \in V$ .

┘

**Def. (29.10.7.3) [Theta Maps].** There is an  $D^\times(\mathbf{A}_F)$ -equivariant surjective map

$$\theta : \pi \otimes S(V(\mathbf{A}_F)) \rightarrow \pi^D \otimes (\pi^D)^\vee.$$

Moreover, there exists  $D_{F_v}^\times$ -equivariant surjective maps

$$\theta_v : \pi_v \otimes S(V(\mathbf{A}_F)) \rightarrow \pi_v^D \otimes (\pi_v^D)^\vee$$

s.t.  $\theta = \otimes'_v \theta_v$ .

┘

*Proof:* **?** Cf. [Shimizu, Theta series and automorphic forms on  $\mathrm{GL}(2)$ ] and [Harris-Kudla91, The central critical value of a triple product  $L$ -function] Section 7. □

**Def. (29.10.7.4) [Invariant Pairings].** Define a  $D^\times(\mathbf{A}_F)$ -invariant bilinear pairing  $B : \pi^D \otimes (\pi^D)^\vee \rightarrow \mathbb{C}$  by:

$$B(\varphi \otimes \varphi') = \int_{\mathbf{I}_F D^\times(F) \backslash D^\times(\mathbf{A}_F)} \varphi(h) \varphi'(h) dh.$$

**?** Convergence.

┘

**Def. (29.10.7.5) [Waldspurger].** For each  $v \in \Sigma_F$ , define a map  $\tilde{B}_v : \pi_v \otimes S(V(F_v)) \rightarrow \mathbb{C}$  (29.10.8.3) defined by

$$\tilde{B}_v(f_v, \Phi_v) = \int_{\text{Unip}(2; F_v) \backslash \text{SL}(2; F_v)} \omega(g_v) \Phi_v(1) W_{\psi_v, f_v}^-(g_v) dg_v.$$

┘

*Proof:* This integral is convergent by [Waldspurger85]Lemma5.?

□

**Def. (29.10.7.6) [Unramified Places].**  $v \in \Sigma_F^{\text{fin}}$  is called an **unramified place** if

- $\psi_v$  is unramified.
- $\pi_v$  is unramified.
- $f_v$  is  $\text{GL}(2; \mathcal{O}_{E_v})$ -invariant.
- $W_{f_v}^{\psi_v}(1) = 1$ .
- $\varphi_v = \mathbf{1}_{\text{Mat}(2; \mathcal{O}_{E_v})}$ .
- $\text{Vol}(\text{SL}(2; \mathcal{O}_{F_v}); dg_v) = 1$ .

In particular, a.e.  $v \in \Sigma_F^{\text{fin}}$  is unramified.?

┘

**Prop. (29.10.7.7) [Waldspurger].** If  $v \in \Sigma_F^{\text{fin}}$  is unramified in the sense of (29.10.7.6), then

$$\tilde{B}_v(f_v, \varphi_v) = \zeta_v(2)^{-1} L_v(\pi_v, \text{Ad}; 1) (29.10.7.5).$$

┘

*Proof:* Cf. [Waldspurger85]Lemma5.?

□

**Prop. (29.10.7.8) [Waldspurger].**

$$B(\theta(f, \Phi)) = 2\zeta^S(2)^{-1} L^S(\pi, \text{Ad}; 1) \prod_{v \in S} \tilde{B}_v(f_v, \Phi_v) (29.10.7.3)(29.10.7.5)(29.10.7.4),$$

where  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  is any finite set containing all the primes of  $F$  not unramified in the sense of (29.10.7.6).

┘

*Proof:* Cf. [Waldspurger85]Prop5.?

□

**Prop. (29.10.7.9) [Invariant Pairing for  $\pi^D$ ].** For each  $v \in \Sigma_F$ , there exists a  $D^\times(F_v)$ -invariant pairing  $B_v : \pi_v^D \otimes (\pi_v^D)^\vee \rightarrow \mathbb{C}$  s.t.  $\tilde{B}_v = B_v \circ \theta_v$  (29.10.7.5)(29.10.7.3).

┘

*Proof:* Cf. [Waldspurger85]Lemma6.?

□

## 8 Triple Product (Ichino-Ikeda for $n = 3$ )

Cf. [Ich08] and [Harris-Kudla, Central critical value of a triple product  $L$ -function].

**Notation (29.10.8.1) [Global Triple Product Setting].**

- Let  $F \in \mathbf{NField}$ .
- Let  $1 \neq \psi : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$  be a non-trivial additive character.
- Let  $E \in \mathbf{CRing}/F$  be an étale algebra with  $\dim_F E = 3$ .

- Let  $\psi^E = \psi \circ \text{tr}_{E/F} : E \backslash \mathbf{A}_E \rightarrow \mathbb{C}^\times$  be the non-trivial additive character.
- Use notations on  $\text{GL}(n)$  defined in [General Linear Groups](#).
- Let  $\Pi \in \text{Rep}^{\text{cusp}, \text{uni}}(\text{GL}(2)/E)$  s.t. the central character is trivial on  $\mathbf{I}_F$ .
- Let  $D \in \text{Az}_F$  be a quaternion algebra and  $\Pi^D \in \text{Rep}^{\text{cusp}, \text{uni}}(D_E^\times/E)$  corresponds to  $\Pi$  via Jacquet-Langlands correspondence [?](#).
- Let  $V = D$  be the 4-dimensional quadratic space over  $F$  with the non-degenerate symmetric bilinear form  $(x, y) = \text{tr}_{D/F}(x\bar{y})$ .
- Let  $W = E^{\oplus 2}$  be the 6-dimensional symplectic space over  $F$  with the non-degenerate anti-symmetric bilinear form  $\langle (x_1, x_2), (y_1, y_2) \rangle = \text{tr}_{E/F}(x_1 y_2 - x_2 y_1)$ .
- Choose a basis  $\{e_1, e_2, e_3, e'_1, e'_2, e'_3\}$  of  $W$  over  $F$  as follows:
  - If  $E = F \times F \times F$ , define

$$e_1 = ((1, 0, 0), (0, 0, 0)), \quad e_2 = ((0, 1, 0), (0, 0, 0)), \quad e_3 = ((0, 0, 1), (0, 0, 0)),$$

$$e'_1 = ((0, 0, 0), (1, 0, 0)), \quad e'_2 = ((0, 0, 0), (0, 1, 0)), \quad e'_3 = ((0, 0, 0), (0, 0, 1)).$$

- If  $E = F' \times F$  where  $F'/F$  is a quadratic extension, fix some  $\delta, \delta' \in F'$  s.t.

$$\text{tr}_{F'/F}(\delta) = \text{tr}_{F'/F}(\delta') = 0, \quad \text{tr}_{F'/F}(\delta\delta') = 1,$$

and define

$$e_1 = ((1, 0), (0, 0)), \quad e_2 = ((\delta, 0), (0, 0)), \quad e_3 = ((0, 1), (0, 0)),$$

$$e'_1 = \left( (0, 0), \left( \frac{1}{2}, 0 \right) \right), \quad e'_2 = ((0, 0), (\delta', 0)), \quad e'_3 = ((0, 0), (0, 1)).$$

- If  $E \in \text{Field}$ , fix some  $\delta_2, \delta_3, \delta'_2, \delta'_3 \in E$  s.t.

$$\text{tr}_{E/F}(\delta_i) = \text{tr}_{E/F}(\delta'_i) = 0, \quad \text{tr}_{E/F}(\delta_i \delta'_j) = \delta_{ij} \text{ [?](#),$$

and define

$$e_1 = (1, 0), \quad e_2 = (\delta_2, 0), \quad e_3 = (\delta_3, 0),$$

$$e'_1 = (0, \frac{1}{3}), \quad e'_2 = (0, \delta'_2), \quad e'_3 = (0, \delta'_3).$$

Then they satisfy:  $\langle e_i, e_j \rangle = \langle e'_i, e'_j \rangle = 0$  and  $\langle e_i, e'_j \rangle = \delta_{ij}$ . And there is an isomorphism  $F^{\oplus 3} \cong E$  defined by

$$(x_1, x_2, x_3) \mapsto \begin{cases} (x_1, x_2, x_3) & , E = F \times F \times F \\ (x_1 + \delta x_2, x_3) & , E = F' \times F, F'/F \in \text{GalField} \\ (x_1 + \delta_2 x_2 + \delta_3 x_3) & , E \in \text{Field} \end{cases}$$

- Let  $\mathbf{G} = \{g \in \text{Res}_{E/F}(\text{GL}(2)) \mid \nu(g) \in \mathbb{G}_m\}$ . Then  $\mathbf{G}$  preserves the symplectic form on  $W$  [?](#). Using the basis  $\{e_1, e_2, e_3, e'_1, e'_2, e'_3\}$  of  $W = E^{\oplus 2}$ , we can regard  $\mathbf{G}$  as a subgroup of  $\text{GSp}(3)$ .  $\perp$

**Prop. (29.10.8.2) [Weil Representations].**  $G(\mathrm{Sp}(3) \times \mathrm{O}(V))(\mathbf{A}_F)$  acts on  $\mathcal{S}(V(\mathbf{A}_F)^3)$  via the Weil representation w.r.t.  $\psi$  and  $G(\mathrm{SL}(2) \times \mathrm{O}(V))(\mathbf{A}_E)$  acts on  $\mathrm{Sch}(V(\mathbf{A}_E))$  via the Weil representation w.r.t.  $\psi \circ \mathrm{tr}_{E/F}$  ?

This two actions are compatible on the subgroup  $\{(g, h) \in \mathbf{G}(\mathbf{A}_F) \times \mathrm{O}(V)(\mathbf{A}_F) | \nu(g) = \nu(h)\}$  via the embedding  $\mathbf{G} \subset \mathrm{GSp}(3)$ .  $\lrcorner$

**Def. (29.10.8.3).** Let  $S(V(\mathbf{A}_F)^n)$  be the subspace of  $\mathcal{S}(V(\mathbf{A}_F)^n)$  consisting of functions that corresponds to polynomials in the mock model at every Archimedean place.  $\lrcorner$

**Def. (29.10.8.4) [Measures].**

- Let  $db = \prod_i db_i$  be the Tamagawa measure on  $\mathbf{A}_F$ .
- Let  $dh = \prod_i dh_i$  be the Tamagawa measure on  $\mathbf{A}_E$ .
- Let  $dg = \prod_i dg_i$  be the Tamagawa measure on  $\mathbf{I}_F \backslash \mathbf{G}(\mathbf{A}_F)$ .
- Let  $dg = \prod_i dg_i$  be the Tamagawa measure on  $\mathbf{I}_F \backslash \mathrm{SL}(2; \mathbf{A}_F)$ .
- Let  $dh = \prod_i dh_i$  be the Tamagawa measure on  $\mathbf{I}_F \backslash D^\times(\mathbf{A}_F)$ .

$\lrcorner$

**Def. (29.10.8.5) [Invariant Pairing  $I_v$ ].** For  $v \in \Sigma_F$ , let  $B : \Pi_v^D \otimes (\Pi_v^D)^\vee \rightarrow \mathbb{C}$  be the  $D^\times(E_v)$ -invariant pairing defined in (29.10.7.9), and define

$$B_v^\flat = \zeta_E(2) L_v(\Pi_v, \mathrm{Ad}; 1)^{-1} B_v.$$

In particular, if

$$B(\phi \otimes \phi') = \int_{\mathbf{I}_F D^\times(F) \backslash D^\times(\mathbf{A}_F)} \phi(h) \phi'(h) dh.$$

is the  $D^\times(\mathbf{A}_E)$ -invariant pairing defined in (29.10.7.4), then we have

$$B = 2^{\#\pi_0(\mathrm{Spec} E)} \zeta_E(2)^{-1} L(\Pi, \mathrm{Ad}; 1).$$

by (29.10.7.8) ?.

Then we can defined  $I_v \in \mathrm{Hom}_{D^\times(F_v) \times D^\times(F_v)}(\Pi^D \otimes (\Pi^D)^\vee, \mathbb{C})$  by

$$I_v(\varphi_v \otimes \varphi'_v) = \frac{\zeta_{F_v}(2)}{\zeta_{E_v}(2)} \cdot \frac{L_v(\Pi_v, \mathrm{Ad}; 1)}{L_v(\Pi_v, r; \frac{1}{2})} \cdot \int_{F_v^\times \backslash D^\times(F_v)} B_v^\sharp(\Pi_v^D(x_v) \varphi_v \otimes \varphi'_v) d^\times x_v.$$

? Convergence.  $\lrcorner$

**Thm. (29.10.8.6)** [Harris-Kudla[H-K91, H-K02]/Ichino[Ich08]]. As element in  $\mathrm{Hom}_{D^\times(\mathbf{A}_F) \times D^\times(\mathbf{A}_F)}(\Pi^D \otimes (\Pi^D)^\vee; \mathbb{C})$ , we have

$$I = 2^{-\#\pi_0(\mathrm{Spec} E)} \cdot \frac{\zeta_E(2)}{\zeta_F(2)} \cdot \frac{L(\Pi, r; \frac{1}{2})}{L(\Pi, \mathrm{Ad}; 1)} \cdot \prod_{v \in \Sigma_F} I_v.$$

$\lrcorner$

*Proof:*

$\square$



### Local Trilinear Forms

**Prop. (29.10.8.7)[Local Triple Zeta Integrals].** The local zeta integral of Garrett and Shapiro-Rallis is defined by

$$Z_v(W_{f_v}^{\psi_v^E}, F_{\varphi_v}; s) = \int_{F_v^\times \mathbf{U}_0(F_v) \backslash \mathbf{G}(F_v)} F_{\varphi_v}(\eta g_v; s) W_{f_v}^{\psi_v^E}(g_v) dg_v,$$

┘

where  $\mathbf{U}_0 = \{u(b) | b \in \text{Res}_{E/F} \mathbf{G}_a, \text{tr}_{E/F}(b) = 0\}$  and  $\eta$  is a representative for the open orbit of  $\mathbf{G}$  in  $P \backslash \text{GSp}(3)$ . We can take  $\eta \in \text{Sp}_3(\mathbb{Z})$  as follows (in the basis  $(e_1, e_2, e_3, e'_1, e'_2, e'_3)$  of (29.10.8.1):

- If  $E = F \times F \times F$ , take

$$\eta = \begin{bmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

- If  $E = F' \times F$  where  $F'/F$  is a quadratic extension, take

$$\eta = \begin{bmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

- If  $E \in \text{Field}$ , take

$$\eta = \begin{bmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

*Proof:* ?

□

**Def. (29.10.8.8)[Unramified Places].**  $v \in \Sigma_F^{\text{fin}}$  is called an **unramified place** if

- $E_v$  is unramified.
- $\psi_v$  is unramified.
- $\Pi_v$  is unramified.
- $f_v$  is  $\text{GL}(2; \mathcal{O}_{E_v})$ -invariant.
- $W_{f_v}^{\psi_v^E}(1) = 1$ .
- $\varphi_v = \mathbf{1}_{\text{Mat}(2; \mathcal{O}_{E_v})}$ .
- $\text{Vol}(\mathcal{O}_{F_v}^* \backslash \mathbf{G}(\mathcal{O}_{F_v}); dg_v) = 1$ .

In particular, a.e.  $v \in \Sigma_F^{\text{fin}}$  is unramified?.

┘

**Prop. (29.10.8.9) [Unramified Zeta Integrals].** If  $v \in \Sigma_F^{\text{fin}}$  is unramified in the sense of (29.10.8.8), then

$$Z_v(W_{f_v}^{\psi_v}, F_{\varphi_v}; s) = \zeta_{F_v}(2s+2)^{-1} \zeta_{F_v}(4s+2)^{-1} L_v(\Pi_v, r; s + \frac{1}{2}).$$

┘

*Proof:* Cf.[Shapiro-Rallis, Rankin triple  $L$ -functions]Thm3.1.?

□

### Relation to Ichino-Ikeda Formula

**Prop. (29.10.8.10).** Ichino's theorem (29.10.8.6) implies Ichino-Ikeda conjecture for  $\text{SO}(3) \times \text{SO}(4)$  by [L-packets for inner forms of  $\text{SL}(n)$ , Hiraga-Saito]Theorem 4.13, Remark 4.20? ┘

*Proof:* ?

□

## 9 Unitary case

### 10 Orbital Integrals

**Def. (29.10.10.1) [Orbital Integrals].** Let  $F \in \mathbf{NField}$  and  $G \in \mathcal{AlgGrp}/F$  with two subgroups  $H_1, H_2 \subset G$ , let  $H_{1,2} = H_1 \times H_2$ , and let  $H_{1,2}$  acts on  $G$  on the right by  $g.(h_1, h_2) = h_1^{-1}gh_2$ . Then when  $\gamma \in G(F)/H_{1,2}(F)$  is nice?, we can define  $H_{1,2,\gamma}$  to be the stabilizer of (a lift of)  $\gamma$ , then the **orbital integral** of  $f$  w.r.t.  $\gamma$  is defined to be

$$\text{Orb}(\gamma; f) = \text{Vol}([H_{1,2,\gamma}]) \int_{H_{1,2}(\mathbf{A}_F)/H_{1,2,\gamma}(\mathbf{A}_F)} f(h_1^{-1}\gamma h_2) dh_1 dh_2.$$

┘

*Proof:* It is invariant w.r.t the lift of  $\gamma$ , because of the property of the Tamagawa measure?. □

**Def. (29.10.10.2) [Relative Trace Formula].** Situation as in (29.10.10.1), a **relative trace formula** for the triple  $(G, H_1, H_2)$  is a formula of the form

$$\text{RTF}_{(G, H_1, H_2)} : \sum_{\pi} \text{I}_{\pi}(f) + \dots = \sum_{\gamma \in H_1(F) \backslash G(F)_{\text{rs}}/H_2(F)} \tau(\gamma) \text{Orb}(\gamma; f) + \dots$$

Cf.[Periods, Cycles and L-Functions, Wei Zhang]P15.?

┘

**Remark (29.10.10.3).** This formula may be connected to Arthur-Selberg formulas. ┘

## 11 Fundamental Lemmas

**Conj. (29.10.11.1) [Fundamental Lemma, Jacquet-Rallis].** For  $p \in \text{Prime}_{\geq 3}$ , the characteristic function  $\mathbf{1}_{G'(\mathcal{O}_{F_0})} \in C_c^{\infty}(G'(F))$  transfers? to the pair of functions  $(\mathbf{1}_{G(\mathcal{O}_F)}, 0) \in C_c^{\infty}(G(F)) \times C_c^{\infty}(G^{\flat}(F))$ .

See also [Gross-Zagier formula and arithmetic fundamental lemma, Wei Zhang]P10. ┘

*Proof:* Cf.[On the Gross-Prasad conjecture for unitary groups, jacquet-Rallis]. □

**Thm& Conj.Cor. (29.10.11.2) [Yun-Gordon].** The fundamental lemma holds for  $p$  sufficiently large. ┘

*Proof:* Cf.[The fundamental lemma of Jacquet-Rallis in positive characteristics, Yun]. □

## 12 Rallis Inner Product Formula

### Local Theory

**Notation (29.10.12.1) [Inner Product Formula Setting].**

- Let  $K \in \text{LField}$ .
- Let  $(G, H)$  be one of the following:
  - $G = O(V)$  where  $V$  is a quadratic space of dimension  $2n + 1$  over  $K$ , and  $H = \text{Sp}(K^{2n})$  with the standard symplectic form.
  - $G = U(V)$  where  $V$  is a Hermitian space of dimension  $2n$  over a quadratic field extension  $L$  of  $K$ , and  $H = U(L^{2n})$  with the standard skew Hermitian form.

┘

**Prop. (29.10.12.2) [Theta Lifting].** Embed  $H \times G$  into the metaplectic group  $\text{Mp}(V^{2n})$ , and let  $R$  be the Weil representation of  $\text{Mp}(V^{2n})$  realized on the space  $\mathcal{S}(V_{\mathbf{A}}^n)$  of Schwartz functions ?.

For any irreducible representation  $\pi$  of  $H$ , we consider the space of linear forms

$$\mathcal{P}(\pi) = \bigoplus_{\sigma} \text{Hom}_{H \times G}(\pi \times R, \sigma),$$

then by Howe duality ?,  $\dim \mathcal{P}(\pi) \leq 1$ . In particular, this is non-vanishing for at most one such  $\sigma$ , called the **theta lifting** of  $\pi$ . ┘

*Proof:* □

**Prop. (29.10.12.3) [Local Inner Products].** Suppose  $\pi$  is tempered, then by [Theta correspondences for close unitary groups, Li-Harris-Sun2011], this space is non-vanishing iff some matrix coefficients is non-vanishing: For some  $f \in \pi \otimes R, \tilde{f} \in \tilde{\pi} \otimes \tilde{R}$ ,

$$\int_H \langle hf, \tilde{f} \rangle dh \neq 0.$$

We may normalize the integral s.t. it takes value 1 in the spherical case:

$$m(f, \tilde{f}) = \frac{L(M; 1)}{L(\pi; 1/2)} \int_H \langle hf, \tilde{f} \rangle dh,$$

where

$$L(M; 1) = \prod_{i=1}^n L(i; \eta_{E/F}^i)$$

in the unitary case, and

$$L(M; 1) = \zeta(1/2) \prod_{i=1}^n \zeta(2i)$$

in the orthogonal case. Thus  $m \in \text{Hom}_{H^2 \times \Delta(G)}(\pi \otimes \tilde{\pi} \otimes R \otimes \tilde{R}, \mathbb{C})$ .

Then we can define a bilinear form

$$\mathcal{P}(\pi) \otimes \mathcal{P}(\tilde{\pi}) \rightarrow \mathbb{C} : \langle \ell, \tilde{\ell} \rangle = \frac{\langle \ell(f), \tilde{\ell}(\tilde{f}) \rangle}{m(f, \tilde{f})}.$$

when  $\pi$  is unitary and tempered, this pairing is positive definite. ┘

*Proof:* □

### Inner Product Formula

**Notation(29.10.12.4).** The setting is similar to the local case, but with  $L/K$  replaced by number fields. ? ⌋

**Thm.(29.10.12.5)[Rallis Inner Product Formula].** Let  $\pi \in \text{Irr}^{\text{cusp}}(H), \sigma \in \text{Irr}^{\text{cusp}}(G)$ , define  $\mathcal{P}(\pi)$  as the product of the local data, then we can define an element  $P_\pi \in \mathcal{P}(\pi)$  as follows

$$P_\pi(f \otimes \varphi)(g) = \int_{[H]} \theta_\varphi(g, h) f(h) dh,$$

where  $\theta_\varphi$  is the theta series

$$\theta_\varphi(g) = \sum_{x \in V^m} R(g)\varphi(x), \quad g \in \text{Mp}((V \otimes W)_{\mathbf{A}_F}).$$

Then

$$\langle P_\pi, P_{\tilde{\pi}} \rangle = \frac{L(\pi; 1/2)}{2L(M; 1)} \text{ (29.10.12.3).}$$

⌋

*Proof:* Cf.[Rallis S. Injectivity properties of liftings associated to Weil representations], [Kudla S, Rallis S. A regularized Siegel-Weil formula: The first term identity], [Li J-S. Non-vanishing theorems for the cohomology of certain arithmetic quotients], [Ichino A. A regularized Siegel-Weil formula for unitary groups], [Ichino A. On the Siegel-Weil formula for unitary groups]. ? □

## 30 | Langlands Reciprocity

### 30.1 Langlands' Reciprocity Conjectures

*“Motives can be divided according to a tetrachotomy. The first form are the Tate (and potentially Tate) motives, whose automorphy was known to Riemann and Hecke. The second form are the motives (conjecturally) associated to automorphic representations which are discrete series at infinity and thus amenable to the Taylor-Wiles method. The third form are the motives (conjecturally) associated to automorphic representations which are at least seen by some flavour of cohomology, either by the Betti cohomology of locally symmetric spaces or the coherent cohomology of Shimura varieties (possibly in degrees greater than zero) which are amenable in principle to the modified Taylor-Wiles method. The fourth form consist of the rest, which (besides a few that can be accessed by cyclic base change) are a complete mystery.”*

–F. Calegari

References are [B-G14, Har14].

#### Notation (30.1.0.1).

- Let  $p \in \text{Prime}$ .
- Let  $\iota_p : \overline{\mathbb{Q}_p} \cong \mathbb{C}$  be an isomorphism.
- Let  $F \in \text{NField}$ .
- Let  $G \in \text{RedGrp}/F$ .
- Let  $T \subset B \subset G_{\overline{F}}$ , and let  $\delta \in X^*(T) \otimes \mathbb{C}$  be the half sum of positive roots.
- Use notations on automorphic representations defined in [Adelic Automorphic Representations](#).
- Recall that  $\text{H-T}_\tau(\rho)$  contains an integer  $i$  with multiplicity  $\dim_{\overline{\mathbb{Q}_\ell}}(V \otimes_{\tau, K} \mathbb{C}_K(i))^{\text{Gal}_K}$ .

┘

#### 1 Compatible Systems

**Prop. (30.1.1.1) [Compatible System of Representations (Taylor)].** For  $E \in \text{NField}$ ,  $n \in \mathbb{Z}_+$ , an  $E$ -rational  $n$ -dimensional **weakly compatible system of Galois representations** of  $\text{Gal}_F$  over  $E$  is a collection

$$\mathcal{R} = \{R_w\}_{w \in \Sigma_E},$$

where  $R_w$  are semisimple  $w$ -adic Galois representations

$$R_w : \text{Gal}_F \rightarrow \text{GL}(n; E_w)$$

that satisfy the following conditions:

1. There is a multiset of integers  $\mathrm{H-T}(\mathcal{R})$  s.t. for any  $v \in \Sigma_F^{\mathrm{fin}}$  and  $w \in \mathrm{Hom}(E, \overline{F_v})$ , the restriction  $R_w|_{\mathrm{Gal}_{F_v}}$  is deRham and  $\mathrm{H-T}_v(R_w) \triangleq \mathrm{H-T}(R_w|_{\mathrm{Gal}_{F_v}}) = \mathrm{H-T}(\mathcal{R})$ .
2. There exists a finite set  $S \subset \Sigma_F^{\mathrm{fin}}$  s.t. if  $v \notin S$  and  $w \notin \mathrm{Hom}(E, \overline{F_v})$ , then  $\mathrm{WD}_v(R_w)$  is unramified.
3. For a.e.  $v \in \Sigma_F^{\mathrm{fin}}$ , there is an F-semisimple Weil-Deligne representation  $\mathrm{WD}_v(\mathcal{R}) \in \mathfrak{wd}_E^n(W_{F_v})$  s.t. for all  $w \notin \mathrm{Hom}(E, \overline{F_v})$ ,

$$\mathrm{WD}_v(R_w)^{\mathrm{s.st}} \sim \mathrm{WD}_v(\mathcal{R}) \otimes E_w.$$

Moreover,

- $\mathcal{R}$  is called a **strongly compatible system of adic representations** if condition3 holds for any  $v \in \Sigma_F^{\mathrm{fin}}$ .
- $\mathcal{R}$  is called **odd/irreducible/geometric** if each  $R_w$  is(30.1.2.1).
- $\mathcal{R}$  is called **pure of weight**  $w \in \mathbb{Z}$  if for a.e.  $v \in \Sigma_F^{\mathrm{fin}}$  and eigenvalues  $\alpha_i$  of  $\mathrm{WD}_v(\mathcal{R})$ ,  $\alpha_i \in \overline{\mathbb{Q}}$  is a Weil- $\|v\|$ -integer of weight  $w$ . For  $\emptyset \subset_{\mathrm{fin}} S \subset \Sigma_F^{\mathrm{fin}}$ ,  $\mathcal{R}$  is called **S-pure** of weight  $w$  if the above holds for any  $v \in \Sigma_F^{\mathrm{fin}} \setminus S$ .
- $\mathcal{R}$  is called a **genuinely geometric system** if there exists an  $X \in \mathrm{SmProj}/F$  and  $i \in \mathbb{N}, j \in \mathbb{Z}$  and a subspace

$$W \subset H_{\mathrm{Betti}}(X, E(j))$$

s.t. for any  $w \in \Sigma_E^{\mathrm{fin}}$ ,  $W \otimes_{E,v} \overline{\mathbb{Q}_\ell}$  is  $\mathrm{Gal}_F$ -invariant and realizes  $R_w$ .

┘

**Conj.(30.1.1.2).**

- Any weakly compatible system is strongly compatible(30.1.1.1).
- Any irreducible strongly compatible system  $\mathcal{R}$  is genuinely geometric and pure of weight  $\frac{2}{\dim \mathcal{R}} \sum_{h \in \mathrm{H-T}(\mathcal{R})} h$ .

┘

*Proof:* ?

□

## 2 Geometric Galois Representations

References are [Clo90, B-G14].

**Def.(30.1.2.1) [Geometric Representations, Fontaine-Mazur[F-M95]].**  $(\rho, V) \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_F)$  is called a **geometric representation** if it satisfies

- For a.e.  $v \in \Sigma_F^{\mathrm{fin}}$ ,  $\rho_v$  is unramified.
- For any  $v \in S_F(p)$ , the representation  $\rho_v$  is deRham(25.2.4.10).

It is called a **genuine geometric representation** if it is isomorphic to the subquotient of some  $H_{\mathrm{et}}^r(X_{\overline{F}}, \mathbb{Q}_p)(m)$  where  $X \in \mathrm{SmPrpr}/F$  and  $m \in \mathbb{Z}$ . Genuine geometric representations are geometric by?.

┘

**Prop.(30.1.2.2) [Geometric Representations and Induction].** Let  $L/F \in \mathrm{NField}$ , then the set of geometric representations(30.1.2.1) are stable under  $\mathrm{res}_F^L$  and  $\mathrm{Ind}_F^L$ .

Moreover, if  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{fd}}(\mathrm{Gal}_K)$  is pure of weight  $w$ , then  $W = \mathrm{Ind}_F^L(V)$  is also pure of weight  $w$ .

┘

*Proof:*

□

**Conj. (30.1.2.3)[Families of Geometric Representations].** If  $\rho : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(n; \overline{\mathbb{Q}_p})$  be a geometric Galois representation(30.1.2.1), then  $\rho$  is a part of a weakly compatible system(30.1.1.1). ┘

*Proof:*

□

**Def. (30.1.2.4) [Algebraic Galois-Representations].**  $(\rho, V) \in \text{Rep}_{\overline{\mathbb{Q}_p}}^{\text{fd}}(\text{Gal}_F)$  is called an **algebraic Galois-representation** if

$$P_{v,X}(T) = \det(1 - \text{Frob}_v T | H^i(X, \mathbb{Q}_p)) \in \overline{\mathbb{Q}}[T].$$

And it is moreover called **pure of weight**  $w$  if for a.e.  $v \in \Sigma_F$ , eigenvalues of  $\rho(\text{Frob}_v)$  are all Weil-integers of weight  $w$ . And  $w$  is called the **motivic weight** of  $V$ . ┘

**Def. (30.1.2.5)[Modular Galois Representations].** A continuous irreducible representation  $\text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2, \overline{\mathbb{F}_p})$  or  $\text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2; K)$ , where  $K \in p\text{-NField}$  is called a **modular Galois representation** if it arises from a newform as in(30.1.3.23). ┘

**Def. (30.1.2.6)[Regular Algebraic Galois-Representations, Clozel[Clo90]].**  $(\rho, V) \in \text{Rep}_{\overline{\mathbb{Q}_p}}^{\text{fd}}(\text{Gal}_F)$  is called a **regular algebraic(geometric) Galois-representation** if it is geometric(30.1.2.1) and  $\text{H-T}_{\tau}(\rho)$  consists of distinct elements for each embedding  $\tau : F \rightarrow \overline{\mathbb{Q}_p}$ . (Recall that  $\text{H-T}_{\tau}(\rho)$  contains an integer  $i$  with multiplicity  $\dim_{\overline{\mathbb{Q}_\ell}}(V \otimes_{\tau, K} \mathbb{C}_K(i))^{\text{Gal}_K}$ ). ┘

**Def. (30.1.2.7) [Algebraic Weil-Representations].** For  $\tau \in \Sigma_F^\infty$  and admissible Weil-representation  $\phi : W_{F_\tau} \rightarrow {}^L G(\mathbb{C})$ , the restriction of  $\phi$  to  $W_{\mathbb{C}}$  via an embedding  $\sigma : F_\tau \rightarrow \mathbb{C}$  corresponds to a homomorphism  $\mathbb{C}^\times \rightarrow \widehat{T}(\mathbb{C})$ , which is of the form  $z \mapsto \sigma(z)^{\lambda_\sigma} \overline{\sigma(z)}^{\lambda_{\text{co}\sigma}}$  s.t.  $\lambda_\sigma - \lambda_{\text{co}\sigma} \in X_*(\widehat{T})$ . Then  $\phi$  is called

- an  **$L$ -algebraic Weil-representation** if  $\lambda_\sigma \in X^*(T)$ .
- a  **$C$ -algebraic Weil-representation** if  $\lambda_\sigma - \delta \in X^*(T)$ (30.1.0.1).

These notions are independent of the choice of  $\sigma$  and Borel  $B \leq G$ . ┘

**Thm. (30.1.2.8) [Fontaine].** Any  $\rho \in \text{Rep}_{\overline{\mathbb{Q}_\ell}}^{\text{pot. st}}(I_K)$  comes from geometry. ┘

*Proof:* Cf.[Fontaine, Galois Rep]P12.?

□

**Prop. (30.1.2.9) [Algebraic Weil Representations and Galois Representations].** ┘

### Galois Representations attached to Elliptic Curves

References are [Serre, Abelian  $\ell$ -Adic Representations and Elliptic Curves].

**Thm. (30.1.2.10) [Serre].** For any  $F \in \text{NField}$  and  $E \in \text{EllCur}/F$  without CM, the Galois-representation

$$\rho_{E, \mathbf{A}} : \text{Gal}_F \rightarrow \text{GL}(2; \mathbf{A}_{F, f})$$

attached to  $E$  has open image. ┘

*Proof:* Cf.[Ser72].?

□

### Automorphy of Galois Representations

**Def. (30.1.2.11) [Admissible Representations attached to Galois Representations].** For a geometric  $p$ -adic Galois representation (30.1.2.1)  $\rho : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}}_p)$  and an isomorphism  $\iota_p : \overline{\mathbb{Q}}_p \cong \mathbb{C}$ , we can associate a  $\Pi(\rho) = \otimes'_v \Pi_v(\rho) \in \text{Irr}^{\text{adm}}(\text{GL}(n)/F)$  s.t.

- For any  $v \in \Sigma_F^{\text{fin}} \setminus S_F(p)$ ,  $\Pi_v(\rho)$  corresponds to the Weil-Deligne representation attached to the restriction of  $\rho$  to the Weil group  $W_v$ .
- For any  $v \in S_F(p)$ , we can also associate a Weil-Deligne representation  $\text{WD}_v(\rho)$  via Fontaine's  $D_{\text{pst}}$  functor ?.
- For any  $v \in \Sigma_F^\infty$ , we can also define Hodge-Tate numbers and thus an irreducible  $(\mathfrak{g}_v, \mathcal{K}_v)$ -module  $\Pi_v$ . ?

┘

**Conj. (30.1.2.12) [Automorphy of Galois Representations, Clozel [Clo90]/Fontaine-Mazur [F-M95]].**

For any irreducible geometric  $p$ -adic Galois representation (30.1.2.1)  $\rho : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}}_\ell)$ ,

- $w(\rho) = \frac{2}{\dim \rho} \sum_{a \in \text{H-T}(\rho)} a \in \mathbb{Z}$ ,
- $m_i(\rho) = m_{w(\rho)-i}(\rho)$  for all  $i$ .
- $\Pi(\rho) \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  (30.1.2.11).
- $\Pi_v(\rho)$  is  $C$ -algebraic (30.1.3.2).

┘

*Proof:* ?

Emerton and Kisin proved this for almost all 2-dimensional odd-representations of  $\text{Gal}_{\mathbb{Q}}$ , based on Serre's conjecture over  $\mathbb{Q}$ . Cf. [Fontaine-Mazur conjecture for  $\text{GL}(2)$ , Kisin], [Local-Global Compatibility in the  $p$ -adic Langlands Programme for  $\text{GL}(2)_{\mathbb{Q}}$ , Emerton].  $\square$

**Remark (30.1.2.13).** It follows from proper base change (9.4.3.1) and (9.4.6.36) (27.7.2.1) that any such genuine geometric representation satisfies the requirement.

This conjecture is very strong, for example, the étale cohomology of smooth proper varieties are known to satisfy many good properties, like Weil conjecture, and Fontaine-Mazur conjecture implies that those properties can be derived via linear algebra data.

This conjecture holds if the image of the representation is finite, by [Henniart, Representations  $\ell$ -adiques abeliennes] ?.

The local version of this conjecture is known to be false.  $\square$

**Def. (30.1.2.14) [Modular Galois Representations].** Let  $A$  be a  $\mathbb{Z}_p$ -algebra and  $\rho : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2; A)$  be a representation, then  $\rho$  is called a **modular Galois representation** if there exists some  $N > 0$  and a homomorphism  $\text{pr} : T'(N) \rightarrow A$  (where  $T'(N) = \mathbb{Z}[T_\ell, \langle d \rangle] \subset \text{End}(S_2(\Gamma_1(N)))$ ,  $\ell \in \mathbf{P} \setminus \{p\}$ ,  $d \in (\mathbb{Z}/(N))^*$ ) s.t.  $\rho$  is unramified outside  $Np$ , and for  $\ell \nmid pN$ ,

$$\text{tr}(\rho(\text{Frob}_\ell)) = \text{pr}(T_\ell), \quad \det(\rho(\text{Frob}_\ell)) = \text{pr}(\langle \ell \rangle) \ell.$$

?

┘

**Conj. (30.1.2.15) [Langlands-Fontaine-Mazur].** Any geometric representation of  $\text{Gal}_F$  (30.1.2.1) is automorphic (30.1.2.12).  $\square$

*Proof:*

$\square$



### 3 Galois Representation attached to Algebraic Automorphic Representations

#### Algebraic Automorphic Representations

**Def. (30.1.3.1) [Algebraic Automorphic Representations].** For  $v \in \Sigma_F^\infty$ , an irreducible representation  $\pi_v \in \text{Irr}^{\text{adm}}(G(F_v))$  is called an  **$L$ -algebraic representation** (resp. a  **$C$ -algebraic representation**) if the Weil-representation associated to it by Langlands-correspondence is so (30.1.2.7).

And  $\Pi \in \text{Irr}^{\text{auto}}(G/F)$  is called an  **$L$ -algebraic representation** (resp. a  **$C$ -algebraic representation**) if  $\Pi_v$  is so, for any  $v \in \Sigma_F^\infty$ .  $\perp$

**Example (30.1.3.2) [Algebraic Representations for  $\text{GL}(n)$  and Infinite Types].**  $\Pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  is  $L$ -algebraic if  $\Pi(\frac{n-1}{2})$  is  $C$ -algebraic (30.1.3.1).

For such  $\Pi$ , we can associate an **infinite type**  $p \in (\mathbb{Z}^n / \text{Sym}_n)^{\text{Hom}(F, \mathbb{C})}$  as follows: Denote  $\Pi' = \Pi(\frac{n-1}{2})$ . If  $\tau$  is a real place of  $F$ , then  $\Pi'_\tau$  is associated to a character of  $W_{\mathbb{R}}$ , whose restriction to  $W_{\mathbb{C}} = \mathbb{C}^\times$  is of the form

$$(z^{p_1} \bar{z}^{p_2}, z^{p_2} \bar{z}^{p_1}, \dots, (z\bar{z})^{p'_{2r+1}}, \dots, (z\bar{z})^{p'_n}),$$

and we define  $p_\tau = (p_1, p_2, \dots, p'_{2r+1}, \dots, p'_n)$ . And if  $\tau$  is a complex place of  $F$ , then  $\Pi'_\tau$  is associated to a character of  $W_{\mathbb{C}} = \mathbb{C}^\times$  of the form

$$(z^{p_1} \bar{z}^{q_1}, z^{p_2} \bar{z}^{q_2}, \dots, z^{p_n} \bar{z}^{q_n}),$$

then we define  $p_\tau = (p_1, \dots, p_n)$  (and it is clear that  $p_{\mathbb{C} \circ \tau} = (q_1, \dots, q_n)$ ).

And being  $C$ -algebraic is identical to Clozel's notion of regular algebraic [Clo90], and is equivalent to having infinite type  $(p_1, \dots, p_n)$  s.t.  $p_i \neq p_j$  for any  $i \neq j$ , which is also equivalent to having the same infinitesimal character as an irreducible algebraic representation of  $\text{GL}(n)/F_v$ .  $\perp$

*Proof:* By Harish-Chandra isomorphism (Cf. [Kna02] 5.43),  $Z(U(\mathfrak{gl}_n))$  is identified with  $S(\mathfrak{h})^W$ , where  $\mathfrak{h} = \text{Res}_{\mathbb{C}/\mathbb{R}}(\text{Lie}(T)) \otimes \mathbb{C}$  in the complex case and  $\text{Lie}(T) \otimes \mathbb{C}$  in the real case. In both cases  $W \cong \text{Sym}_n$  acts diagonally on  $\mathbb{C}^n$  or  $\mathbb{C}^n \times \mathbb{C}^n$ .

Then Cf. [B-G14] Section 2.3.  $\square$

**Def. (30.1.3.3) [Arithmetic Automorphic Representations].**  $\Pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$  is called

- an  **$L$ -arithmetic representation** if there exists  $E \in \text{NField}$  s.t.  $\pi_v$  has Satake-parameters defined over  $E$  for a.e.  $v \in \Sigma_F^{\text{fin}}$  (29.3.2.12).
- a  **$C$ -arithmetic representation** if there exists  $E \in \text{NField}$  s.t.  $\pi_v$  is defined over  $E$  for a.e.  $v \in \Sigma_F^{\text{fin}}$  (29.3.2.12).

$\perp$

**Prop. (30.1.3.4) [Restriction of Algebraic and Arithmetic Representations].**  $\Pi \in \text{Irr}^{\text{auto}}(G/F)$  is  $L$ -algebraic (resp.  $C$ -algebraic /  $L$ -arithmetic /  $C$ -arithmetic) (30.1.3.1) (30.1.3.3) iff its corresponding  $\Pi \in \text{Irr}^{\text{auto}}(\text{Res}_{F/\mathbb{Q}}(G)/\mathbb{Q})$  is.  $\perp$

*Proof:* Cf. [B-G14] Sec 2.3.  $\square$

**Conj. (30.1.3.5) [Clozel [Clo90] / Buzzard-Gee [B-G14]].**

- $\Pi \in \text{Irr}^{\text{auto}}(G/F)$  is  $L$ -algebraic (30.1.2.7) iff it is  $L$ -arithmetic (30.1.3.3).
- $\Pi \in \text{Irr}^{\text{auto}}(G/F)$  is  $C$ -algebraic (30.1.2.7) iff it is  $C$ -arithmetic (30.1.3.3).

┘

*Proof:*

□

**Conj. Cor. (30.1.3.6).** For  $G = \mathrm{GL}(n)$ , these follow from Clozel's conjecture (30.1.3.17) by [Clo90] Prop 3.1(3). ?

┘

**Thm. (30.1.3.7) [Rationality for Algebraic Automorphic Representations, Clozel [Clo90]].**

Suppose  $\Pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  is  $C$ -regular (regular algebraic) with infinite type  $p \in (\mathbb{Z}^n)^{\mathrm{Hom}(F, \mathbb{C})}$  (30.1.3.2), then  $\Pi_f$  has a moduli field  $\mathbb{Q}(\Pi) \in \mathrm{NField}$ . And for each  $\sigma \in \mathrm{Aut}(\mathbb{C})$ ,  $\pi_f^\sigma$  is the finite part of some  $\pi^\sigma \in \mathrm{Alg}^0(\mathrm{GL}(n)/F)$ , whose infinite type is  $p^\sigma$ .

┘

*Proof:* Cf. [Clo90] P112. ?

□

**Prop. (30.1.3.8) [Algebraic Representations under Central  $\mathbb{G}_m$ -Extensions].** If  $1 \rightarrow \mathbb{G}_m \rightarrow G' \rightarrow G \rightarrow 1$  is a central extension of reductive groups over  $F$ , then for any  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$ , the pullback of  $\pi$  to  $G'$  is also an automorphic representation by (26.6.1.29). Then it is an  $L$ -algebraic (resp.  $L$ -arithmetic, resp.  $C$ -algebraic, resp.  $C$ -arithmetic) if  $\pi$  is.

┘

*Proof:* ?

□

**Prop. (30.1.3.9) [Algebraic Representations and Functoriality].** Let  $G, G' \in \mathrm{RedGrp}/F$  and  $r : {}^L G \rightarrow {}^L G' \in \mathrm{AlgGrp}/\bar{\mathbb{Q}}$ . Let  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$  and suppose the functorial lifting  $\pi'$  of  $\pi$  (29.3.5.1) exists, then

- $\pi'$  is  $L$ -algebraic if  $\pi$  is  $L$ -algebraic (30.1.3.1).
- $\pi'$  is  $L$ -arithmetic if  $\pi$  is  $L$ -arithmetic (30.1.3.3).

┘

*Proof:* Cf. [B-G14].

□

**Prop. (30.1.3.10) [Adjoining a Twisting Element].** There exists a canonical central  $\mathbb{G}_m$ -extension

$$1 \rightarrow \mathbb{G}_m \rightarrow \tilde{G} \rightarrow G \rightarrow 1$$

s.t.  $\tilde{G}$  has a canonical twisting element  $c$ . Moreover, there is a canonical bijection between the set of splitting of this extension with the set of twisting elements in  $G$ .

┘

*Proof:* Cf. [B-G14]. ?

□

**Def. (30.1.3.11) [Buzzard-Gee  $C$ -Groups, Buzzard-Gee [B-G14]].** The Buzzard-Gee  $C$ -group of  $G$  is defined to be the Langlands  $L$ -group  ${}^L \tilde{G}$  of the central  $\mathbb{G}_m$ -extension  $\tilde{G}$  (30.1.3.10).

┘

**Def. (30.1.3.12) [Cohomological Automorphic Representations].** For  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$  and  $v \in \Sigma_F^\infty$ ,  $(\pi_v, V_v)$  is called a **cohomological admissible representation** (for  $\xi$ ) if there exists  $(\xi, U) \in \mathrm{Irr}(G(F_v))$  and  $i \in \mathbb{N}$  s.t.

$$H^i(\mathfrak{g}_v, \mathcal{K}_v; U \otimes V_v) \neq 0.$$

And  $\pi$  is called a **cohomological automorphic representation** if  $\pi_v$  is cohomological for any  $v \in \Sigma_F^\infty$ .

┘

**Prop.(30.1.3.13)** [Cohomological Automorphic Representations are Regular  $C$ -Algebraic].  
If  $\pi \in \text{Irr}^{\text{auto}}(G/F)$  is cohomological, then it is regular  $C$ -algebraic.  $\lrcorner$

*Proof:* Cf.[B-W00]P15.  $\square$

**Prop.(30.1.3.14)** [Cuspidal Cohomological Unitary Automorphic Representations are  $C$ -Arithmetic]. Any cohomological unitary representations  $\pi \in \text{Irr}^{\text{cusp}}(G/F)$  is  $C$ -arithmetic(30.1.3.3).  $\lrcorner$

*Proof:* Cf.[Blasius-Rogawski, Zeta functions of Shimura varieties]Section2.3?  $\square$

**Prop.(30.1.3.15)** [Clozel]. Any regular  $C$ -algebraic automorphic representation of  $\text{GL}(n)$ (30.1.3.1) is cohomological after possibly twisting a quadratic character.  $\lrcorner$

*Proof:* Cf.[Clo90]Prop3.14.?  $\square$

### Galois Representations attached to Algebraic Automorphic Forms

**Conj.(30.1.3.16)**[Automorphic Galois Representations(weak version), Buzzard-Gee[B-G14]].  
If  $\pi \in \text{Irr}^{\text{auto}}(G/F)$  is  $L$ -algebraic(30.1.3.1), then for any  $p \in \text{Prime}$  and  $\iota_p : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$ , there exists a continuous geometric Galois representation

$$\rho_\pi = \rho_{\pi, \iota_p} : \text{Gal}_F \rightarrow {}^L G(\overline{\mathbb{Q}_p})$$

s.t.

- The composition of  $\rho_\pi$  with the natural projection  ${}^L G(\overline{\mathbb{Q}_p}) \rightarrow \text{Gal}_F$  is the identity map.
- For a.e.  $v \in \Sigma_F^{\text{fin}}$ ,  $\rho_\pi|_{W_{F_v}}$  is unramified and  $\iota \circ \rho_\pi(\sigma_v)$  is  $\widehat{G}(\mathbb{C})$ -conjugate to  $c(\pi_v)$ .
- For  $v \in S_F(p)$ , the Hodge-Tate character of  $\rho_\pi|_{\text{Gal}_{F_v}}$  can be read off  $\pi$ (? Cf.[B-G14]P156).
- If  $v \in \Sigma_F^{\mathbb{R}}$ , then  $\iota \circ \rho_\pi(c_v)$  is  $\widehat{G}(\mathbb{C})$ -conjugate to the element  $\alpha_v = \lambda_{\sigma_v}(i)\lambda_{\tau_v}(i)r_{\pi_v}(j)$ ?  $\lrcorner$

*Proof:*  $\square$

**Conj.Cor.(30.1.3.17)**[Automorphic Galois Representations for  $\text{GL}(n)$ , Clozel[Clo90]]. For  $p \in \text{Prime}$  and any  $C$ -algebraic  $\Pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$ (30.1.3.1),

- $\Pi_f$  is defined over a number field  $E \in \text{NField}$ (i.e.  $\Pi_f^\sigma \cong \Pi_f$  for any  $\sigma \in \text{Aut}(\mathbb{C}/E)$ ).
- There exists an irreducible geometric  $p$ -adic Galois representation

$$\rho_{\ell, \Pi} : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}_p})$$

s.t.

$$\Pi(\rho_{p, \Pi}) \cong \Pi\left(\frac{1-n}{2}\right)\text{(30.1.2.11)}.$$

- Moreover, such  $\{\rho_{p, \Pi}\}$  form a strongly compatible system  $\mathcal{R}$  of representations of  $\text{Gal}_F$ .

Such representations  $\rho_{p, \Pi}$  and  $\bar{\rho}_{p, \Pi}$ (semi-simplified reduction modulo  $p$ ) are called **automorphic Galois representations**.  $\lrcorner$

*Proof:*  $\square$

**Remark (30.1.3.18).** Notice that this can imply conjecture(30.1.3.16) by(30.1.3.20).  $\lrcorner$

**Prop. (30.1.3.19)[Algebraic Representations in the same Parabolic Induction].** If  $\pi, \pi'$  are subquotients of a common parabolic induction  $I_{P(\mathbf{A}_F)}^{G(\mathbf{A}_F)}\sigma$ (26.6.1.27), then  $\pi, \pi'$  has the same infinitesimal character. In particular,  $\pi$  is  $L$ -algebraic(resp.  $C$ -algebraic) iff  $\pi'$  is.

Moreover, if  $\pi$  is  $L$ -algebraic and the conjecture(30.1.3.16) holds for  $\pi$ , then it also holds for  $\pi'$ .

$\lrcorner$

*Proof:* Cf.[B-G14]P178?  $\square$

**Cor. (30.1.3.20)[Nearly Isomorphic Representations].** If  $G = \mathrm{GL}(n)$  and  $\pi, \pi' \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$  are nearly isomorphic(i.e.  $\pi_v \cong \pi'_v$  for a.e.  $v \in \Sigma_F$ ), then  $\pi, \pi'$  has the same infinitesimal character.

Suppose further that  $\pi$  is  $L$ -algebraic and the conjecture(30.1.3.16) holds for  $\pi$ , then it also holds for  $\pi'$ .

Moreover, for a general  $G$ , if the functorial lifting conjecture is true for  $G$ , then the same holds for  $G$ .  $\lrcorner$

*Proof:* Cf.[B-G14]P178?  $\square$

**Prop. (30.1.3.21)[Galois Representations and Functoriality].** Let  $G \rightarrow G' \in \mathrm{RedGrp}/F$  be quasi-split,  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(G/F)$ . If  $r : {}^L G \rightarrow {}^L G' \in \mathrm{AlgGrp}/\overline{\mathbb{Q}}$  is a homomorphism respecting the projection to  $\mathrm{Gal}_F$  and functorial lift  $\pi' \in \mathrm{Irr}^{\mathrm{auto}}(G'/F)$  exists(29.3.5.1), then the conjecture(30.1.3.16) holds for  $\pi'$  if it holds for  $\pi$ .  $\lrcorner$

*Proof:* Cf.[B-G14]P177?. This is easy.  $\square$

### Modular Forms( $n = 2$ case)

**Notation (30.1.3.22).**

- Let  $p \in \mathrm{Prime}$ .
- Let  $\lambda \in S_{K_{\mathbf{f}}}(p)$ .
- Let  $\Lambda_p = \mathbb{Z}_p[[T]]$ .
- Let  $(F, \mathcal{O}_F) \in \mathrm{NField}$  be totally real.
- Let  $d = [F : \mathbb{Q}]$ .

$\lrcorner$

**Thm. (30.1.3.23)[Galois Representations attached to Cuspidal Newforms, Eischler-Shimura/Deligne/Deligne-Serre/Shimura/Rogawski-Tunnell/Mazur-Wiles/Wiles[Wil88]].** For any  $\lambda \in \Sigma_{K_{\mathbf{f}}}^{\mathrm{fin}}$  that is ordinary for  $\mathbf{f}$ (26.4.2.8), there exists a continuous representation

$$\rho_{\mathbf{f}, \lambda} : \mathrm{Gal}_F \rightarrow \mathrm{GL}(2; \mathcal{O}_{\mathbf{f}, \lambda})$$

s.t. for any  $v \in \Sigma_{K_{\mathbf{f}}}^{\mathrm{fin}} \setminus S_{K_{\mathbf{f}}}(\mathbf{c}(\|\lambda\|))$ , we have

$$\mathrm{tr}(\sigma_v | \rho_{\mathbf{f}, \lambda}) = c(v, \mathbf{f}), \quad \det(\sigma_v | \rho_{\mathbf{f}, \lambda}) = \psi(v).$$

Moreover,

- If  $k = 1$ , then  $\# \mathrm{Im}(\rho_{\mathbf{f}, \lambda}) < \infty$ , and lifts to a 2-dimensional complex Artin representation.

- If  $\lambda$  is ordinary for  $\mathbf{f}$ , then  $\rho_{\mathbf{f},\lambda}|_{\text{Gal}_v}$  is equivalent to the form  $\begin{bmatrix} \varepsilon_1 & * \\ & \varepsilon_2 \end{bmatrix}$ , where  $\varepsilon_2$  is unramified and  $\varepsilon_2(\sigma_v) = \alpha_v(\mathbf{f})$  (26.4.2.8).

┘

*Proof:* ?

The construction of the ordinary  $\lambda$ -adic Galois representations depends upon Hida's method of studying  $p$ -adic families of modular forms.  $\square$

### Polarized case

**Notation (30.1.3.24).**

- Let  $F \in \mathbf{NField}$  be a CM-field or a totally real field.
- Let  $F^+$  be the maximal totally real subfield of  $F$ .
- Let  $\mathbf{c} \in \text{Gal}(F/F^+)$  be the complex multiplication element (If  $F = F^+$ , set  $\mathbf{c} = \text{id}$ ).
- Let  $d = [F^+ : \mathbb{Q}_0]$ .

┘

**Def. (30.1.3.25) [The Group  $\mathcal{G}_n$ ].** For  $n \in \mathbb{Z}_+$ , define  $\mathcal{G}_n = (\text{GL}(n) \times \text{GL}(1)) \rtimes \{1, \mathbf{c}\}$  where  $\mathbf{c}$  acts on  $\text{GL}(n) \times \text{GL}(1)$  via

$$(g, a)^{\mathbf{c}} = (ag^{-t}, a).$$

And we denote  $\mathcal{G}_n^0 = \text{GL}(n) \times \text{GL}(1) \trianglelefteq \mathcal{G}_n$ .

Let  $\nu : \mathcal{G}_n \rightarrow \text{GL}(1)$  be the multiplier map which sends  $(g, a) \in \mathcal{G}_n^0$  to  $a$  and sends  $\mathbf{c}$  to  $-1$ .

For  $m, n \in \mathbb{Z}_+$ , define a homomorphism of algebraic groups

$$\mathcal{G}_n \times \mathcal{G}_m \rightarrow (\mathcal{G}_n/\mathcal{G}_n^0) \times (\mathcal{G}_m/\mathcal{G}_m^0) \rightarrow \{\pm 1\}$$

which sends  $(\mathbf{c}, 1)$  and  $(1, \mathbf{c})$  to  $-1$ . And we denote  $(\mathcal{G}_n \times \mathcal{G}_m)^+$  the kernel of this homomorphism.  $\perp$

**Prop. (30.1.3.26).** For  $m, n \in \mathbb{Z}_+$ , there is a homomorphism of algebraic groups

$$\otimes : (\mathcal{G}_n \times \mathcal{G}_m)^+ \rightarrow \mathcal{G}_{mn}$$

which maps  $(g, a) \times (g', a')$  to  $(g \otimes g', aa')$  and maps  $\mathbf{c} \times \mathbf{c}$  to  $\mathbf{c}$ .

And there is a homomorphism of algebraic groups

$$I : \mathcal{G}_n \rightarrow \text{GSp}(2n) : (g, a) \mapsto \begin{bmatrix} g & 0 \\ 0 & ag^{-t} \end{bmatrix}.$$

┘

**Prop. (30.1.3.27) [Representations valued in  $\mathcal{G}_n$ ].** Suppose

- $\Gamma \in \mathbf{Grp}$  with a normal subgroup  $\Delta \leq \Gamma$  of index  $\leq 2$ ,
- If  $\Gamma \neq \Delta$ , choose  $\gamma_0 \in \Gamma \setminus \Delta$ .
- $A \in \mathbf{CRing}$  and  $\rho : \Gamma \rightarrow \mathcal{G}_n(A)$  is a homomorphism with  $\Delta = r^{-1}\mathcal{G}_n^0(A)$ .

Then we write  $\rho^\natural : \Delta \xrightarrow{\rho} \mathcal{G}_n^0(A) \rightarrow \text{GL}(n; A)$ . Conversely,

- If  $\rho : \Gamma \rightarrow \text{GSp}(2n; A)$  is a homomorphism with multiplier  $\mu$ , then it gives a homomorphism  $\hat{\rho}_\Delta : \Gamma \rightarrow \mathcal{G}_{2n}(A)$  which sends  $\delta \in \Delta$  to  $(\rho(\delta), \mu(\delta))$ , and sends  $\gamma \in \Gamma \setminus \Delta$  to  $(\rho(\gamma)J_{2n}^{-1}, -\mu(\gamma))\mathbf{c}$ .

- If  $\rho : \Gamma \rightarrow \mathrm{GO}(n; A)$  is a homomorphism with multiplier  $\mu$ , then it gives a homomorphism  $\hat{\rho}_\Delta : \Gamma \rightarrow \mathcal{G}_n(A)$  which sends  $\delta \in \Delta$  to  $(\rho(\delta), \mu(\delta))$ , and sends  $\gamma \in \Gamma \setminus \Delta$  to  $(\rho(\gamma), \mu(\gamma))\mathbf{c}$ .

If  $\Gamma = \mathrm{Gal}_F$  and  $\Delta = \mathrm{Gal}_{F^+}$ , then a representation  $r : \mathrm{Gal}_{F^+} \rightarrow \mathcal{G}_n(A)$  (resp. to  $\mathrm{GSp}(2n; A)$ , resp. to  $\mathrm{GO}(n; A)$ ) is called a **totally odd representation** if it maps each complex conjugation to  $-1$  (resp.  $-1$ , resp.  $1$ ).  $\square$

**Def. (30.1.3.28) [Polarized Galois-Representations].** For  $\ell \in \mathrm{Prime}$  and  $\iota_\ell : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$  an isomorphism, an  $\ell$ -adic (resp. mod- $\ell$ ) **polarized Galois-representation** is a pair  $(\rho, \mu)$  where

- $\rho : \mathrm{Gal}_F \rightarrow \mathrm{GL}(n; \overline{\mathbb{Q}_\ell})$  (resp.  $\rho : \mathrm{Gal}_F \rightarrow \mathrm{GL}(n; \overline{\mathbb{F}_\ell})$ ) for some  $n \in \mathbb{Z}_+$ ,
- $\mu : \mathrm{Gal}_{F^+} : \mathrm{Gal}_{F^+} \rightarrow \overline{\mathbb{Q}_\ell}^\times$  (resp.  $\mu : \mathrm{Gal}_{F^+} \rightarrow \overline{\mathbb{F}_\ell}^\times$ ),

s.t. there exists some  $\tau \in \Sigma_{F^+}^\infty$  and  $\epsilon_\tau \in \{\pm 1\}$  and a non-degenerate  $\epsilon_\tau$ -orthogonal form  $\langle -, - \rangle_\tau$  on  $\overline{\mathbb{Q}_\ell}^{\oplus n}$  (resp.  $\overline{\mathbb{F}_\ell}^{\oplus n}$ ) satisfying

$$\langle \rho(\sigma)x, \rho(\mathbf{c}_\tau \sigma \mathbf{c}_\tau)y \rangle_\tau = \mu(\sigma) \langle x, y \rangle_\tau$$

for each  $x, y \in \mathrm{Gal}_F$ . And if  $F \neq F^+$ , we require further that  $\epsilon_\tau = -\mu(\mathbf{c}_\tau)$ .

And a polarized representation  $(\rho, \mu)$  is called a **totally odd representation** if  $\epsilon_\tau = 1$  for all  $\tau \in \Sigma_{F^+}^\infty$ .  $\square$

**Prop. (30.1.3.29) [Polarized Representations and  $\mathcal{G}_n$ ].** Situation as in (30.1.3.28),

- If  $(\rho, \mu)$  is an  $\ell$ -adic or mod- $\ell$ -polarized Galois-representation, then the desired  $\epsilon_{\tau'}$ -orthogonal form  $\langle -, - \rangle_{\tau'}$  exists for each  $\tau' \in \Sigma_{F^+}^\infty$ , because we can just choose

$$\epsilon_{\tau'} = \mu(\mathbf{c}_\tau \mathbf{c}_{\tau'}) \epsilon_\tau, \quad \langle x, y \rangle_{\tau'} = \langle x, r(\mathbf{c}_\tau \mathbf{c}_{\tau'})y \rangle_\tau, \quad \text{where } x, y \in \mathrm{Gal}_F.$$

- If  $F \neq F^+$ , then  $(\rho, \mu)$  is  $\ell$ -adic (resp. mod- $\ell$ ) polarized iff there exists a continuous homomorphism  $\tilde{\rho} : \mathrm{Gal}_{F^+} \rightarrow \mathcal{G}_n(\overline{\mathbb{Q}_\ell})$  (resp.  $\tilde{\rho} : \mathrm{Gal}_{F^+} \rightarrow \mathcal{G}_n(\overline{\mathbb{F}_\ell})$ ) (30.1.3.25) with  $\tilde{\rho}^\natural = \rho$  and multiplier  $\mu$  (30.1.3.27) (30.1.3.25).
- If  $F = F^+$ , then  $(\rho, \mu)$  is a polarized iff  $\rho$  factors through  $\mathrm{GSp}(n; \overline{\mathbb{Q}_\ell})$  (if  $\mu(\mathbf{c}_\tau) = -\epsilon_\tau$ ) or  $\mathrm{GO}(n; \overline{\mathbb{Q}_\ell})$  (if  $\mu(\mathbf{c}_\tau) = \epsilon_\tau$ ) with multiplier  $\mu$ . A similar assertion holds for  $\overline{\mathbb{Q}_\ell}$  replaced by  $\overline{\mathbb{F}_\ell}$  if  $\ell \geq 3$ .

In particular, the two definitions of totally odd representations in (30.1.3.27) and (30.1.3.28) are compatible.  $\square$

*Proof:* 2: The existence of the pairing is equivalent to the existence of a matrix  $B \in \mathrm{GL}(n; A)$  s.t.  $\rho^t B \rho^c = \mu B$  and  $B_t = \epsilon_\tau B$ , and then we can define  $\tilde{\rho}$  to be equal to  $\rho$  on  $\mathrm{Gal}_{F^+}$ , while maps  $\mathbf{c}$  to  $(B^{-1}, \epsilon_\tau)\mathbf{c}$ .

3: If  $(\rho, \mu)$  is polarized, we can define a pairing  $\langle x, y \rangle'_\tau = \langle x, \rho(\mathbf{c}_\tau)y \rangle_\tau$ , then  $\langle \rho(\sigma)x, \rho(\sigma)y \rangle'_\tau = \mu(\sigma) \langle x, y \rangle'_\tau$ , and we have

$$\langle y, x \rangle'_\tau = \langle y, \rho(\mathbf{c}_\tau)x \rangle_\tau = \epsilon_\tau \langle \rho(\mathbf{c}_\tau)x, y \rangle_\tau = \epsilon_\tau \langle \rho(\mathbf{c}_\tau)x, \rho(\mathbf{c}_\tau)y \rangle_\tau = \epsilon_\tau \mu(\mathbf{c}_\tau) \langle x, y \rangle'_\tau,$$

and the assertion follows.  $\square$

**Prop. (30.1.3.30).** If  $F/\mathbb{Q} \in \mathrm{GalField}$  and  $V, V' \in \mathrm{Irr}_{\overline{\mathbb{Q}_\ell}}(\mathrm{Gal}_F)$  are geometric representations s.t.  $L(V; s) = L(V'; s)$ , then  $V' \cong V$ .  $\square$

*Proof:* This is because  $L(V; s)$  determines  $\mathrm{WD}_v(V)$  for all  $v$  that is unramified, thus also determines  $V$  by Chebotarev's density theorem **?**.  $\square$

**Example (30.1.3.31) [Polarized Galois-Representations].**

- Any 1-dimensional representation is polarized.
- $V_p(A)$  of an Abelian variety is polarized of weight 1.
- Representations attached to a classical modular eigenform in  $S_{2k}(\Gamma_0(N))$  is polarized of weight  $2k - 1$ .
- For an irreducible polarized representation of  $\text{Gal}_{\mathbb{Q}}$  of dimension 2, its weight is odd iff  $V$  is odd.

┘

*Proof:* ?

□

**Def. (30.1.3.32) [Polarized Automorphic Representations].**  $\Pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$  is called a **polarized automorphic representation** if there exists a Hecke-character  $\chi$  of  $F^+$  s.t.

- $\chi_{\tau}(-1)$  is independent of  $\tau \in \Sigma_F^{\infty}$  chosen.
- $\pi^c \cong \pi^{\vee} \otimes (\chi \circ \text{Nm}_{F/F^+} \circ \det)$ .

And when  $F \neq F^+$ , we require further that  $\chi_{\tau}(-1) = (-1)^n$  for each  $\tau \in \Sigma_F^{\infty}$ .

And  $\Pi$  is called a **RAESDC representation** if it is regular algebraic?? essentially self-dual and cuspidal.

┘

**Def. (30.1.3.33) [Polarized Archimedean Weights].** For  $\Omega \in \text{Field}^0$  s.t.  $\overline{\Omega} = \Omega$ , denote

$$(\mathbb{Z}^n)^{\text{Hom}(F, \Omega), +} \subset (\mathbb{Z}^n)^{\text{Hom}(F, \Omega)}$$

the subset consisting of elements  $a$  s.t.  $a_{\tau,1} \geq a_{\tau,2} \geq \dots \geq a_{\tau,n}$ . And for each  $w \in \mathbb{Z}$ , denote

$$(\mathbb{Z}^n)_w^{\text{Hom}(F, \Omega)} \subset (\mathbb{Z}^n)^{\text{Hom}(F, \Omega)}$$

the subset consisting of elements  $a$  s.t.  $a_{\tau,i} + a_{\tau \circ c, n+1-i} = w$  for each  $\tau \in \text{Hom}(F, \Omega)$  and  $i \in [n]_+$ .

An element  $a \in (\mathbb{Z}^n)^{\text{Hom}(F, \Omega), +}$  is called **extremely regular** if it satisfies the following: For some  $\tau \in \Sigma_F^{\infty}$ , the subsets of the set  $\{a_{\tau,i} + n - i\}_{i \in [n]_+}$  are pairwise different.

For  $a \in (\mathbb{Z}^n)^{\text{Hom}(F, \Omega), +}$ , denote  $\Xi_a \in \text{Irr}(\text{GL}(n)^{\text{Hom}(F, \mathbb{C})})$  the algebraic representation which is the tensor product over  $\tau$  of the irreducible representations of  $\text{GL}(n)$  with highest weight  $a_{\tau}$ . And a regular algebraic  $\Pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$  (30.1.3.2) is said to have weight  $a$  if  $\Pi_{\infty}$  has the same infinitesimal character as  $\Xi_a^{\vee}$ ?

┘

**Def. (30.1.3.34) [Ordinary Automorphic Components].** Suppose  $\Pi \in \text{Irr}^{\text{auto}}(\text{GL}(n)/F)$  is a regular algebraic (30.1.3.2) automorphic representation with weight  $a \in (\mathbb{Z}^n)^{\text{Hom}(F, \Omega), +}$ . For  $\ell \in \text{Prime}$  and  $\iota_{\ell} : \overline{\mathbb{Q}_{\ell}} \cong \mathbb{C}$  an isomorphism,  $\Pi$  is called  **$\iota$ -ordinary** if for each  $v \in S_F(\ell)$ , there is some  $b \in \mathbb{Z}_+$  s.t.  $(\iota^{-1} \pi_v)^{\text{Iw}(v^{b,b})} \neq 0$ . (Cf. [Potential automorphy and change of weight, Gee-Geraghty-Taylor] P537?).

Then for any algebraic Hecke-character  $\chi$  of  $F$  (19.4.1.30),  $\Pi$  is  $\iota$ -ordinary iff  $\Pi \otimes (\chi \circ \det)$  is  $\iota$ -ordinary.

And if  $\Pi$  has weight 0 and  $\Pi_v$  is Steinberg for each  $v \in S_F(\ell)$ , then  $\Pi_v$  is  $\iota$ -ordinary.

┘

*Proof:* Cf. [Geraghty, Modularity lifting theorems for ordinary Galois representations] Lemma 5.1.5.

□

**Thm& Conj. Cor. (30.1.3.35)** [Clozel [Clo91]/Kotwitts [Kot92]/Harris-Taylor [H-T01]/Taylor-Yoshida [TY07]/Bellaïche-Chenevier [B-C11]/Shin [Shi11]/Clozel-Harris-Labesse [CHL11]/Caraiani [Car12], [Car14]/Clozel [Clo13]/Chenevier-Harris [?C-H14]].



If  $(\Pi, \chi) \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  is polarized(30.1.3.32) and regular algebraic of weight  $a \in (\mathbb{Z}^n)^{\text{Hom}(F, \Omega), +}$ (30.1.3.2)(30.1.3.33), then for any  $\ell \in \text{Prime}$  and  $\iota_\ell : \overline{\mathbb{Q}_p} \cong \mathbb{C}$ , there exists a continuous semi-simple representation

$$\rho_{\ell, \iota_\ell}(\Pi) : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}_\ell})$$

and an integer  $w \in \mathbb{Z}_+$  s.t.

- $(\rho_{\ell, \iota_\ell}(\Pi), \rho_{\ell, \iota_\ell}(\chi)(1-n))$  is a totally odd, polarized  $\ell$ -adic representation(30.1.3.28).
- For any  $v \in \Sigma_F^{\text{fin}} \setminus S_F(\ell)$ , we have  $\text{WD}_\ell(\rho_{\ell, \iota_\ell}(\Pi)|_{\text{Gal}_{F_v}})^{F\text{-s.st}} = \iota^{-1} \text{rec}_n(\Pi_v(\frac{1-n}{2}))$ , and these associated Weil-Deligne representations are pure of weight  $w$ .
- For any  $v : F \hookrightarrow \overline{\mathbb{Q}_\ell}$ ,  $\rho_\Pi|_{\text{Gal}_{F_v}}$  is deRham and Hodge-Tate regular(25.2.2.4), with

$$\text{H-T}_v(\rho_{\ell, \iota_\ell}(\Pi)) = \{a_{\iota \circ v, 1+n-1}, a_{\iota \circ v, 2+n-2}, \dots, a_{\iota \circ v, n}\}, \quad \text{H-T}_{v \circ c}(\rho_{\ell, \iota_\ell}(\Pi)) = w - \text{H-T}_v(\rho_{\ell, \iota_\ell}(\Pi)).$$

- For any  $v \in S_F(\ell)$  that  $\Pi_v$  is Iwahori-spherical, we have  $\text{WD}_\ell(\rho_{\ell, \iota_\ell}(\Pi)|_{\text{Gal}_{F_v}})^{F\text{-s.st}} = \iota_\ell^{-1} \text{rec}_n(\Pi_v(\frac{1-n}{2}))$ . In particular,  $\rho_{\ell, \iota_\ell}(\Pi)$  is semi-stable at  $v$ , and if  $\Pi_v$  is unramified, then it is crystalline at  $v$ .
- If  $v \in S_F(\ell)$  and  $\Pi$  is  $\iota$ -ordinary at  $v$ (30.1.3.34), then  $\rho_{\ell, \iota_\ell}(\Pi)|_{\text{Gal}_{F_v}}$  is ordinary(25.2.7.4). Conversely, if  $\rho_{\ell, \iota_\ell}(\Pi)$  is ordinary and  $\pi$  has level potentially prime to  $\ell$ , then  $\pi$  is  $\iota$ -ordinary.

Such a totally odd polarized  $\ell$ -adic Galois-representation  $(\rho_{\ell, \iota_\ell}(\Pi), \rho_{\ell, \iota_\ell}(\chi)(1-n))$  or its reduction modulo  $\ell$  is called a **polarizedly automorphic Galois-representation**. And it is called a **ordinarily polarizedly automorphic Galois-representation** if moreover  $\pi$  is  $\iota$ -ordinary(30.1.3.34), etc.. By [Clo90]Thm3.13?, being polarizedly or ordinarily polarized automorphic doesn't depend on  $\iota_\ell$  chosen.  $\perp$

*Proof:* Cf.[LGGT]Thm2.1.1.?.  $\square$

**Prop.(30.1.3.36) [Potential Abelian Extensions and Cyclic Base Change].** Let  $\Pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  s.t.  $\Pi^\vee \cong \Pi^c$ ,  $\Pi(\frac{n-1}{2})$  is regular  $C$ -algebraic(30.1.3.1) and  $\Pi_x$  is supercuspidal, then

- If there exists  $\Sigma \in \text{Rep}^{\text{alg}, n}(W_F)$  s.t.  $\Sigma|_{W_{F_v}} = \text{rec}(\Pi_v)$  for  $v \in \Sigma_F^\infty$  and a.e.  $v \in \Sigma_F^{\text{fin}}$ , then it holds for any  $v \in \Sigma_F$ .
- If  $F'/F$  is a cyclic extension of prime degree s.t. there is only one place  $x'$  over  $x$ , and the cyclic base change  $\Pi'$  of  $\Pi$  to  $F'$  is potentially Abelian(29.3.5.11) and supercuspidal at  $x'$ , then  $\Pi$  is potentially Abelian.  $\perp$

*Proof:* Cf.[Sch13]Thm13.5?.  $\square$

**Thm.(30.1.3.37) [Non-Galois Automorphic Induction, Scholze].**

- Let  $F_0^3/F_0^2/F_0^1$  be totally real fields and  $2|[F_0 : \mathbb{Q}]$ . Denote  $n = [F_0^2/F_0^1]$ .
- Suppose  $F_0^3/F_0^1$  is a solvable Galois extension.
- Let  $\mathcal{K}$  be an imaginary quadratic field and  $F^i = F_0^i \mathcal{K}$ .
- Let  $x \in \Sigma_{F^1}^{\text{fin}}$  that is split over  $F_0^1$  and inert in  $F^3$ .
- Let  $\chi$  be a Hecke character of  $F^2$  s.t.



- $\bar{\chi} = \chi \circ \mathbf{c}$ .
- For any  $v \in \Sigma_{F^2}^\infty$ ,  $\chi_v(z) = z^{p_v} \bar{z}^{-p_v}$ , and  $p_v \neq p_{v'}$  if  $v \neq v'$ .
- The stablizer of  $\chi_x \circ \text{Nm}_{F^3/F^2}$  in  $\text{Gal}(F^3/F^1)$  is equal to  $\text{Gal}(F^3/F^2)$ .

Then there exists a potentially Abelian (29.3.5.11)  $\Pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F^1)$  that is associated to  $\text{Ind}_{F^1}^{F^2} \chi$  via Langlands correspondence, and  $\Pi$  is conjugate self-dual,  $\Pi(\frac{n-1}{2})$  is  $C$ -algebraic, and  $\Pi_x$  is supercuspidal.  $\lrcorner$

*Proof:* Cf. [Sch13]P712. ?  $\square$

### Non-Polarized case

**Remark (30.1.3.38) [ $p$ -adic Approximations:  $n$  is even case].** Some Galois representations associated to  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  cannot be constructed via  $\ell$ -adic cohomology of Shimura varieties attached to unitary groups, especially the case when  $n \in 2\mathbb{Z}$ .

The missing representations can be constructed by taking limits on eigenvarieties: For  $n = 2$ , Cf. [Wil88] using Hida family and for  $n > 2$  and non-ordinary case Cf. [Chenevier] using eigenvarieties. The program is completed in [C-H13]. A patching lemma is needed to descent a Galois representation down a solvable extension.  $\lrcorner$

**Thm& Conj. Cor. (30.1.3.39).** Let  $F \in \mathbf{NField}$  be CM or totally real and  $\Pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$  be regular algebraic at each archimedean place of  $F$  (30.1.3.2). Then for any  $\ell \in \mathbf{Prime}$ , there exists a semisimple Galois representation

$$\rho_{\ell, \Pi} : \text{Gal}_F \rightarrow \text{GL}(n; \overline{\mathbb{Q}}_\ell)$$

s.t. for any  $v \in \Sigma_F^{\text{fin}} \setminus S_F(\ell)$  that  $\Pi_v$  is unramified,  $\Pi_v(\frac{1-n}{2}) = \Pi_v(\rho_{\ell, \Pi})$  (30.1.3.17).  $\lrcorner$

*Proof:* Cf. [On the Rigid Cohomology of Certain Shimura Varieties, Harris-Taylor-Thorne] and [On torsion in the cohomology of locally symmetric varieties, Scholze] ?.  $\square$

### Beyond Shimura Varieties

It is believed that most Galois representations associated to non-polarized automorphic representations cannot be obtained from cohomology of Shimura varieties, and no one knows how to show that they arise from cohomology of algebraic varieties.

## 4 Potentially Automorphy

References are [Potential automorphy and change of weight, Lamb-Gee-Geraghty-Taylor].

**Def. (30.1.4.1) [Potential Cuspidal Automorphic Compatible Systems].** A compatible system of representations  $\mathcal{R} = \{R_{v,w}\}$  of  $\text{Gal}_F$  over some number field  $E$  (30.1.1.1) is called **strongly potentially cuspidal automorphic** if there exists some cuspidal automorphic forms  $\omega$  for  $\text{GL}(d)/F$  that are eigenforms for the appropriate Hecke operators, with eigenvalues in  $E$ , s.t. the corresponding Hecke eigenvalues of  $\omega$  are equal to that of  $\mathcal{R}$ .  $\lrcorner$

**Def. (30.1.4.2) [Strongly Potential Cuspidal Automorphic Galois Representations].** For  $F \in \mathbf{NField}$ , a Galois representation  $(V, \rho) \in \text{Rep}_{F_v}^{\text{fd}}(\text{Gal}_F)$  is called strongly **potentially cuspidal automorphic** if there exists a compatible system of strongly potentially cuspidal automorphic Galois representations  $\mathcal{R} = \{R_{v,w}\}$  of  $\text{Gal}_F$  over some number field  $E$  (30.1.4.1) s.t. there exists some  $w \in \Sigma_F^{\text{fin}}$  dividing  $v$  together with an isomorphism  $R_{v,w} \otimes_{E_w, w} \overline{F}_v \cong \rho \otimes \overline{F}_v$ .  $\lrcorner$

**Prop.(30.1.4.3) [Jointly Potentially Automorphy for Elliptic Curves].** Let  $F \in \mathbf{NField}$  and  $E_1, E_2, E_3, E_4 \in \mathbf{EllCur}/F$ ,  $A \in \mathbf{AbVar}^2/F$ , then

- If  $F$  is totally real or imaginary CM, then the two compatible systems  $\{\mathrm{Sym}^{k_1} H^1(E_{1,\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  and  $\{\mathrm{Sym}^{k_2} H^1(E_{2,\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  are jointly strongly potentially automorphic for any  $k_1, k_2 \in \mathbb{N}$ .
- If  $F$  is totally real or imaginary CM, then the two compatible systems  $\{\mathrm{Sym}^{k_1} H^1(E_{1,\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  and  $\{\mathrm{Sym}^{k_2} H^1(E_{2,\overline{F}}, \overline{\mathbb{Q}}_\ell) \otimes \mathrm{Sym}^{k_3} H^1(E_{3,\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  are jointly strongly potentially automorphic for any  $k_1 \in \mathbb{N}$ ,  $k_2 \in \{0, 1, 2\}$ ,  $k_3 \in \{0, 1\}$ .
- If  $F$  is totally real or imaginary CM, then the two compatible systems  $\{\mathrm{Sym}^{k_1} H^1(E_{1,\overline{F}}, \overline{\mathbb{Q}}_\ell) \otimes \mathrm{Sym}^{k_3} H^1(E_{3,\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  and  $\{\mathrm{Sym}^{k_2} H^1(E_{2,\overline{F}}, \overline{\mathbb{Q}}_\ell) \otimes \mathrm{Sym}^{k_4} H^1(E_{4,\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  are jointly strongly potentially automorphic for any  $k_1, k_2 \in \{0, 1, 2\}$ ,  $k_3, k_4 \in \{0, 1\}$ .
- if  $F$  is totally real, then the two compatible systems  $\{\mathrm{Sym}^{k_1} H^1(A_{\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  and  $\{\mathrm{Sym}^{k_2} H^1(A_{\overline{F}}, \overline{\mathbb{Q}}_\ell)\}$  are jointly strongly potentially automorphic for any  $k_1, k_2 \in \{0, 1, 2, 3, 4\}$ .  $\lrcorner$

*Proof:* 1: Cf.[Li-Zhang2022]P47. ?

2, 3 follows from 1 and the langlands functoriality  $\mathrm{GL}(2) \times \mathrm{GL}(2) \rightarrow \mathrm{GL}(4)$  and  $\mathrm{GL}(2) \times \mathrm{GL}(3) \rightarrow \mathrm{GL}(6)$ .

4 follows from the proof of [Abelian surfaces over totally real fields are potentially modular]Thm9.3.1. ?  $\square$

### Potential Modularity

**Remark(30.1.4.4).** A potential modularity result is a result that says certain representation of  $\mathrm{Gal}_{\mathbb{Q}}$  comes from geometry when restricting to  $\mathrm{Gal}_F$  for some  $F \in \mathbf{NField}$ . Cf.[Remarks on a Conjecture of Fontaine and Mazur, Taylor].  $\lrcorner$

### Dwork Family

#### without self-duality conditions

#### Applications to $L$ -Functions

**Def.(30.1.4.5) [Strongly Automorphic Compatible Systems].** Let  $L/F \in \mathbf{NField}$  be a Galois extension,  $V = \{V_\ell\}$  be a compatible system of semisimple  $\ell$ -adic  $\mathrm{Gal}_F$ -representations,  $\emptyset \subsetneq S \subset \mathbf{Prime}$ , then  $V$  is called  **$S$ -strongly automorphic over  $L$**  if for any subextension  $L/L'/F$  with  $\mathrm{Gal}(L/L')$  solvable, we have

- $V|_{\mathrm{Gal}_{L'}}$  is automorphic.
- If  $\pi = \boxplus \pi_i$  is an isobaric automorphic representation of  $\mathrm{GL}(n)/L'$  associated to  $V|_{\mathrm{Gal}_{L'}}$  and  $V|_{\mathrm{Gal}_{L'}} = \oplus V_i$  is the corresponding decomposition of compatible systems, then  $V_{i,\ell}$  is irreducible for any  $\ell \in S$ .

More generally, we can define  **$S$ -strongly potentially automorphic representations**. And  $V$  is called **potentially strongly automorphic** if it is  $S$ -strongly potentially automorphic for some  $S$  of Dirichlet density 1.

And if  $W = \{W_\ell\}$  is another compatible system, we can also define the notion of **jointly  $S$ -strongly potentially automorphic** and **jointly strongly potentially automorphic** pairs of representations.  $\lrcorner$

**Prop. (30.1.4.6).** Let  $F \in \mathbf{NField}$  and  $V = \{V_\ell\}, W = \{W_\ell\}$  are two compatible system of semisimple  $\ell$ -adic  $\mathrm{Gal}_F$ -representations, and  $\emptyset \subsetneq S \subset \mathbf{Prime}$ . Assume that  $V, W$  are jointly  $S$ -strongly potentially automorphic(30.1.4.5), then  $L(V; s)$  has a meromorphic extension to all  $s \in \mathbb{C}$ , and for any  $\ell \in S$ ,

$$\mathrm{rank}(V_\ell \otimes W_\ell)^{\mathrm{Gal}_F} = -\mathrm{ord}_{s=1} L(V \otimes W; s).$$

┘

*Proof:* Cf.[Li-Zhang2022].

□

## 5 Conjectures

**Prop. (30.1.5.1)[Properties of Étale Cohomologies].** For  $X \in \mathbf{SmProj}^d/F$ ,

- (E5): If  $v \notin \Sigma_F^p$  and  $X$  has good reduction at  $v$ , then  $H_{\text{ét}}^i(X, \mathbb{Q}_p)|_{\mathrm{Gal}_{F_v}} \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{ur}}(\mathrm{Gal}_{F_v})$ , and

$$P_{v,X}(T) = \det\left(1 - \mathrm{Frob}_v T | H_{\text{ét}}^i(X, \mathbb{Q}_p)\right) \in \mathbb{Z}[T]$$

is independent of  $p$  prime to  $v$ . And all roots of  $P_{v,X}(T)$  in  $\mathbb{C}$  has absolute value  $\|v\|^{-i/2}$ .

- (E6): Genuine geometric representations are geometric. And for  $v \in \Sigma_F^p$ , if  $X$  has good reduction at  $v$ ,  $H_{\text{ét}}^i(X, \mathbb{Q}_p)|_{\mathrm{Gal}_{F_v}}$  is even crystalline.
- (E7): There is a cycle map

$$\eta_\ell : \mathrm{CH}^i(X) \rightarrow (H_{\text{ét}}^{2i}(X, \mathbb{Q}_\ell)(i))^{\mathrm{Gal}_K}.$$

and for  $P \in X(K)$ ,  $\eta_\ell(P) \neq 0 \in (H_{\text{ét}}^{2d}(X, \mathbb{Q}_p)(d))^{\mathrm{Gal}_K}$ .

┘

*Proof:* ?

□

And there are some open conjectures:

- (EC3)[Tate's Conjecture]:  $\eta_\ell$  is surjective. There are variant of this conjecture:  
There is a decomposition  $H^i(X(\mathbb{C}), \overline{\mathbb{Q}}) = \oplus M_j$  as  $\overline{\mathbb{Q}}$ -vector spaces s.t.
  - For any  $\ell \in \mathbf{P}$  and an embedding  $\iota : \overline{\mathbb{Q}} \hookrightarrow \mathbb{Q}_\ell$ ,  $M_j \otimes_{\overline{\mathbb{Q}}, \iota} \overline{\mathbb{Q}}_\ell$  is a minimal  $\mathrm{Gal}_{G_{\mathbb{Q}_p}}$ -stable  $\overline{\mathbb{Q}}_\ell$ -space.
  - There are Weil-Deligne representations  $\mathrm{WD}_p(W_j)$  over  $\overline{\mathbb{Q}}$  s.t. for any  $\ell \in \mathbf{P}$  and an embedding  $\iota : \overline{\mathbb{Q}} \hookrightarrow \mathbb{Q}_\ell$ ,

$$\mathrm{WD}_p(M_j) \otimes_{\overline{\mathbb{Q}}, \iota} \overline{\mathbb{Q}}_\ell \cong \mathrm{WD}_p(M_j \otimes_{\overline{\mathbb{Q}}, \iota} \overline{\mathbb{Q}}_\ell).$$

- There are motivic weights  $\mathrm{H-T}(M_j)$  s.t. for any  $\ell \in \mathbf{P}$  and  $\iota : \overline{\mathbb{Q}} \hookrightarrow \mathbb{Q}_\ell$ ,

$$\mathrm{H-T}(M_j \otimes_{\overline{\mathbb{Q}}, \iota} \overline{\mathbb{Q}}_\ell) = \mathrm{H-T}(M_j).$$

and for any  $\iota : \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$ ,

$$\dim_{\mathbb{C}}(M_j \otimes_{\overline{\mathbb{Q}}, \iota} \mathbb{C}) \cap H^{a, i-a}(X(\mathbb{C}), \mathbb{C})$$

equals the multiplicity of  $a$  in  $\mathrm{H-T}(M_j)$ .

**Prop. (30.1.5.2)[Total Hodge-Tate Weights].** For a geometric representation  $V \in \text{Rep}_{\mathbb{Q}_p}(\text{Gal}_F)$ , for each  $v \in \Sigma_F^v$ , there are Hodge-Tate weights associated to  $v$ . For  $k \in \mathbb{Z}$ , define

$$m_k(V) = \sum_{v \in \Sigma_F^p} [F_v : \mathbb{Q}_p] m_k(V|_{\text{Gal}_{F_v}}),$$

called the **total Hodge-Tate weights** of  $V$ . Then it satisfies:

- $\sum_{k \in \mathbb{Z}} m_k(V) = [F : \mathbb{Q}] \cdot \dim V$ .
- For a subfield  $F_0 \subset F$ , if  $W = \text{Ind}_{\text{Gal}_F}^{\text{Gal}_{F_0}}(V)$ , then  $m_k(V) = m_k(W)$ .?
- If  $V$  is pure of weight  $w$ , then

$$w[F : \mathbb{Q}] \dim V = 2 \sum_{k \in \mathbb{Z}} m_k \cdot k.$$

┘

*Proof:* 1 is clear.

2: ?

3: By item2 and (30.1.2.2), it suffices to show for  $F = \mathbb{Q}$ . Secondly, it suffices to show for  $\det V$ , it has motivic weight  $w \dim V$  and for each  $v \in \Sigma_F^p$  the unique Hodge-Tate weights equal to the sum of Hodge-Tate weights of  $V$ . Then we may assume it has weight 0 by Tate twist because Hodge-tensoring by  $\mathbb{Q}_p(k)$  increases both sides by  $-2k$ .

Then by Sen's theory,  $V|_{\text{Gal}_{\mathbb{Q}_p}}$  is potentially unramified. Let  $\chi$  be the Hecke character attached to  $V$ , then its kernel contains an open subgroup  $U_p \subset \mathbb{Z}_p^\times$ , and also by small circle argument, it contains an open subgroup  $U^p \subset \prod_{\ell \neq p} \mathbb{Z}_\ell^\times$ , and it also contains  $\mathbb{R}_+^\times$  by (19.7.3.29). But notice  $\mathbb{I}_{\mathbb{Q}}^*/\mathbb{Q}^\times U_p U^p \mathbb{R}^\times$  is finite. Thus it is an Artin representation, and has motivic weight 0 (30.1.2.4), □

**Prop. (30.1.5.3)[Symmetry of Hodge-Tate Weights].** If the Tate conjecture<sup>5</sup> is true, then for any genuine geometric representation  $V$  pure of weight  $w$ ,

- $m_k = m_{w-k}$ .
- If  $w \in 2\mathbb{Z}$ , let  $m_{w/2}^\pm(V)$  be defined by

$$m_{w/2}^+(V) + m_{w/2}^-(V) = m_{w/2}(V)$$

$$m_{w/2}^+(V) - m_{w/2}^-(V) = (-1)^{w/2} (\dim V^{c=\text{id}} - \dim V^{c=-\text{id}}).$$

Then  $m_{w/2}^\pm(V) \in \mathbb{N}$ .

- For a subfield  $F_0 \subset F$ ,  $m_{w/2}^\pm(V) = m_{w/2}^\pm(\text{Ind}_{\text{Gal}_F}^{\text{Gal}_{F_0}}(V))$ .

┘

## 30.2 Breuil-Mezard Conjecture

### 30.3 Mod $p$ Representations and Modularity Lifting Theorems

References are [K-W09b, K-W09c].

**Notation (30.3.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $(K, \mathfrak{p}, \varpi_K, \kappa) \in p\text{-LField}$ .
- Let  $F \in \text{NField}$ .
- Use notations on automorphy of Galois-representations defined in [Langlands' Reciprocity Conjectures](#).

┘

#### 1 Global Mod $p$ Representations

**Def. (30.3.1.1) [Mod- $p$  Representations of Serre-Type].** A 2-dimensional **mod- $p$  representation of Serre-Type** is an odd representation  $\bar{\rho} \in \text{Irr}_{\mathbb{F}_p}^{\text{alg}, 2}(\text{Gal}_{\mathbb{Q}})$ .

┘

**Def. (30.3.1.2) [Artin Conductor and Weights of  $\bar{\rho}$ ].**  $k(\bar{\rho}), N(\bar{\rho})$ . Cf. [Serre, Modular Representations, Section 1.2, 2] **?**.

┘

**Def. (30.3.1.3) [Levels].** The level  $N$  is the product of Artin conductors of  $\rho$  away from  $p$  (Cf. [Modern Number Theory] P329).

┘

**Def. (30.3.1.4) [Weights].** The weight  $k$  is determined by the ramification properties of  $\rho$  at  $p$ . Cf. [Ser75] (2.2.4), (2.3.2), (2.4.5), (2.4.8) and (2.4.9). **?**

For any  $\bar{\rho}$ , there is always a twist of  $\bar{\rho}$  by characters s.t. in formulas (2.2.4), (2.3.2), (2.4.5), (2.4.8)

and (2.4.9) we can take  $a = 0$ . In particular, we can assume that  $k(\bar{\rho}) \leq \begin{cases} p+1 & , p \geq 3 \\ 4 & , p = 2 \end{cases}$  **?**

The weight reduction is due to [Edixhoven: weight in Serre's conjectures on modular forms], and the level reduction is due to [Ribet, On modular representations of  $\text{Gal}_{\mathbb{Q}}$  arising from modular forms] and [Ribet, Report on mod  $\ell$  representations of  $\text{Gal}_{\mathbb{Q}}$ ]. **?**

┘

**Thm. (30.3.1.5) [Special Liftings of Modular Representations, Khare-Wintenberger/Gee-Snowden].** Assume  $\bar{\rho} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2, \overline{\mathbb{F}_p})$  is a smooth irreducible Galois representation, and

- $\text{Im}(\bar{\rho})$  is non-solvable if  $p = 2$ ,
- $\bar{\rho}|_{\mathbb{Q}(\zeta_p)}$  is irreducible if  $p > 2$ .

Then there exists a lift  $\rho : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2; E)$  for some  $E \in p\text{-NField}$  s.t.

- If  $p = 2$  and  $k(\bar{\rho}) = 2$ , then  $\rho$  is a minimal crystalline lift **?** with  $\text{H-T}(\rho) = \{0, 1\}$ .
- If  $p = 2$  and  $k(\bar{\rho}) = 4$ , then
- If  $p > 2$ , then  $\rho$  is a minimal crystalline lift **?** with  $\text{H-T}(\rho) = \{0, k(\bar{\rho}) - 1\}$ .

And if  $p > 2$ , we can also choose  $\rho$  s.t. :

- $\text{H-T}(\rho) = \{0, 1\}$ , any inertia type  $\tau_{\ell}$  at primes  $\ell \in S \setminus \{p\}$ , and unramified outside  $S$ . And at  $p$  it is crystalline if  $k(\bar{\rho}) = 2$  and Steinberg if  $k(\bar{\rho}) = p + 1$ .

┘

*Proof:* **?**

□

**Thm. (30.3.1.6) [Extending to a Family of Representations].** Let  $F \in \mathbf{NField}$ ,  $v \in S_F(p)$  and  $\rho : \mathrm{Gal}_{\mathbb{Q}} \rightarrow \mathrm{GL}(2; F_v)$  an odd irreducible continuous Galois representation ramified at f.m. places and deRham at  $p$  with  $H\text{-T}(\rho) = \{0, a\}$  where  $a \in \mathbb{Z}_+$ , s.t.

- $\bar{\rho}|_{\mathrm{Gal}_{\mathbb{Q}}(\mu_p)}$  is irreducible.
- If  $p = 2$ , then  $\mathrm{Im}(\bar{\rho})$  is solvable.

Then  $\rho$  is a part of a rank 2 almost strictly compatible system of Galois representations? ┘

*Proof:* ? □

## 2 Level-Raising and Level-Lowering

References are [Rib84, Rib90a].

**Prop. (30.3.2.1) [ $p$ -New Hecke Rings].** Let  $N \in \mathbb{Z}_+$  and let  $\mathbf{T}_N$  be the commutative subring of  $\mathrm{End}(\mathrm{SF}_2(N, \mathbf{1}))$  generated by Hecke operators  $\{\mathbf{T}_n\}_{n \in \mathbb{Z}_+}$ . Then for any maximal ideal  $\mathfrak{m}$  of  $\mathbf{T}_N$ , there exists a semisimple continuous representation  $\rho_{\mathfrak{m}} : \mathrm{Gal}_{\mathbb{Q}} \rightarrow \mathrm{GL}(2; \mathbf{T}_N/\mathfrak{m})$ , unique up to isomorphism s.t.

$$\mathrm{tr}_{\rho_{\mathfrak{m}}}(\sigma_p) = T_p \pmod{\mathfrak{m}}, \quad \det \rho_{\mathfrak{m}}(\sigma_p) = p \pmod{\mathfrak{m}}.$$

for any  $p \in \mathbf{Prime} \setminus S(\ell N)$ .

Notice that because the weight is 2, we have  $\mathrm{tr}(\rho_{\mathfrak{m}}(\mathfrak{c})) = 0$  and  $\det(\rho_{\mathfrak{m}}(\mathfrak{c})) = -1$ . For  $N|N' \in \mathbb{Z}_+$ ,  $T_N$  is naturally a subring of  $T_{N'}$ .

For  $p \in S(N)$ ,  $\rho_{\mathfrak{m}}$  is said to be a  **$p$ -new representation** if  $\mathfrak{m}$  comes from a maximal ideal of the quotient ring  $\mathbf{T}_{p\text{-new}}(N)$  of  $\mathbf{T}_N$ , which is the commutative subring of  $\mathrm{End}(\mathrm{SF}_2(N, \mathbf{1})^{p\text{-new}})$  generated by Hecke operators  $\{\mathbf{T}_n\}_{n \in \mathbb{Z}_+}$ . ┘

*Proof:* Cf. [Rib90] Section 5. □

**Thm. (30.3.2.2) [Level-Raising for Modular Forms].** Let  $N \in \mathbb{Z}_+$ ,  $p \in \mathbf{Prime} \setminus S(N)$  and  $\mathfrak{m}$  is a maximal ideal of  $\mathbf{T}_{Np}$  containing  $T_p - (p+1)$  or  $T_p + (p+1)$ , then  $\rho_{\mathfrak{m}}$  is  $p$ -new of level  $pN$  (30.3.2.1).

In particular, for any  $\ell \in \mathbf{Prime}$  and  $\rho \in \mathrm{Irr}_{\mathbb{F}_{\ell}}^{\mathrm{alg}, 2}(\mathrm{Gal}_{\mathbb{Q}})$  that is modular of level  $N$  (i.e. a base change of  $\rho_{\mathfrak{m}}$  for some maximal ideal  $\mathfrak{m}$  of  $\mathbf{T}_N$  (30.3.2.1)), there are infinitely many  $p \in \mathbf{Prime}$  s.t.  $\rho$  is  $p$ -new of level  $pN$ . ┘

*Proof:* ? Cf. [Rib90a].

For the second assertion, by Chebotarev density theorem, there are infinitely many  $p \in \mathbf{Prime}$  s.t.  $\rho(\sigma_p)$  is conjugate to  $\rho(\mathfrak{c})$ , which has trace 0 and determinant  $p = -1$  (30.3.2.1). Thus the assertion follows from the definition of  $\rho_{\mathfrak{m}}$ . □

### Level-Lowering

**Def. (30.3.2.3) [Finite Flat Representations].** For  $k \in \mathbf{Field}^{\mathrm{fin}}$  with  $\mathrm{char} k \neq 2$  and  $\rho \in \mathrm{Irr}_k^{\mathrm{alg}, 2}(\mathrm{Gal}_{\mathbb{Q}})$ ,  $\rho$  is called a **finite flat representation at  $p$**  for a prime  $p \in \mathbf{Prime}$  if there exists a  $k$ -vector scheme  $H \in \mathrm{Sch}^{\mathrm{fin}, \flat}/\mathbb{Z}_p$  s.t.  $\rho_p$  is isomorphic to the  $\mathrm{Gal}_{\mathbb{Q}_p}$ -action on  $k$ -vector space  $H(\mathbb{Q}_p)$ . ┘

**Prop. (30.3.2.4).** Situation as in (30.3.2.3), If  $p \neq \mathrm{char} k$ , then  $\rho$  is finite flat at  $p$  iff it is unramified at  $p$ . ┘

*Proof:* Cf.[Sur les representations modulaires de degree 2 de  $\text{Gal}_{\mathbb{Q}}$ , Serre]Section2.8.  $\square$

**Conj.(30.3.2.5) [Level-Lowering Conjecture, Serre].** Suppose  $k \in \mathbf{Field}^{\text{fin}}$  with  $\text{char } k \neq 2$  and  $\rho \in \text{Irr}_k^{\text{alg},2}(\text{Gal}_{\mathbb{Q}})$  is modular of level  $N \in \mathbb{Z}_+$  and  $p \in S(N)$  s.t.  $\text{ord}_p(N) = 1$ , then  $\rho$  is finite flat at  $p$ (30.3.2.3) if  $\rho$  is of level  $N/p$ .

And Serre conjectured that if  $\rho$  is finite flat at  $p$  then  $\rho$  of of level  $N/p$ .  $\lrcorner$

**Thm& Conj.Cor.(30.3.2.6) [Mazur-Ribet[Rib84]].** Suppose  $\rho \in \text{Irr}_k^{\text{alg},2}(\text{Gal}_{\mathbb{Q}})$  is modular of level  $N \in \mathbb{Z}_+$  and  $p \in S(N)$  s.t.  $\text{ord}_p(N) = 1$  s.t.  $\rho$  is finite flat at  $p$ . If one of the following holds:

- $p \not\equiv 1 \pmod{\ell}$ .
- $(N, \ell) = 1$ .

Then  $\rho$  is modular of level  $N/p$ .  $\lrcorner$

*Proof:* Cf.[Rib84] or [Fermat's last theorem, Chap7].?  $\square$

### 3 Serre's Modularity Conjecture over $\mathbb{Q}$

References are [Ser75, Kis09, Kis09c, K-W09b, K-W09c, D-P23], [Serre's conjectures, lecture, Ribet-Stein] and [Diamond, Taylor-Wiles construction and multiplicity one(1997)], [Fujiwara, Deformation rings and Hecke algebras in the totally real case], [Khare-Wintenberger, Serre's conjecture for 2-dimensional mod  $p$  representations of  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ ], [Edixhoven, Serre's Conjecture].

**Notation(30.3.3.1).**

- For  $N \in \mathbb{Z}_+$ , let  $Q(N)$  be the maximal prime factor of  $N$ (and  $Q(1) = 1$ ).
- Let  $p \in \mathbf{Prime}$ ,  $q \in p^{\mathbb{Z}_+}$ ,
- Let  $\bar{\rho} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2; \overline{\mathbb{F}}_p)$  be a smooth representation.

$\lrcorner$

**Def.(30.3.3.2) [Locally Good Dihedral Representations].**  $\ell \in \mathbf{Prime} \setminus \{p\}$  is called a **good dihedral prime** for  $\bar{\rho}$  if

- $\bar{\rho}|_{I_\ell}$  is of the form  $\begin{bmatrix} \psi & \\ & \psi^\ell \end{bmatrix}$ , where  $\psi$  is a character of  $I_\ell$  is of order a power of an odd prime  $t$ , s.t.  $t|\ell + 1$ , and  $t > \max\left(Q\left(\frac{N(\bar{\rho})}{q^2}\right), 5, p\right)$ .

- $q \equiv 1 \pmod{8}$ , and  $q \equiv 1 \pmod{r}$  for any  $r \leq \max\left(Q\left(\frac{N(\bar{\rho})}{q^2}\right), p\right) \in \mathbf{Prime}$ .

If there exists a good dihedral prime  $q$  for  $\bar{\rho}$ , then  $\bar{\rho}$  is called **locally good-dihedral** or  $q$ -dihedral.

$\lrcorner$

**Thm.(30.3.3.3) [Serre's Modularity Conjecture[Ser75, Ser87b], Rohrlich-Tunnell[R-T97]/Kisin/Khare-Wintenberger[K-W09b, K-W09c]].** Any  $\bar{\rho} \in \text{Irr}_{\overline{\mathbb{F}}_p}^{\text{alg},2}(\text{Gal}_{\mathbb{Q}})$  of Serre-type(30.3.1.1) is modular(30.1.2.5).

Moreover, the corresponding  $f$  is in  $\text{SF}_{k(\bar{\rho})}(N(\bar{\rho}); \chi)$  where  $\chi$  is determined by

$$\det(\bar{\rho})(\sigma_\ell) \equiv \chi(\ell)\ell^{k-1} \pmod{\mathfrak{p}}, \quad \ell \nmid Np.$$

$\lrcorner$



*Proof:* Cf.[Khare-Wintenberger]Thm9.1.?

If  $p = 2$ , then the projective image of  $\bar{\rho}$  must be dihedral by(24.2.1.1). The modularity in this case follows from[R-T97].?

If  $p > 2$  and  $\text{Im}(\bar{\rho})$  is solvable, then by Dickson's classification of solvable subgroups of  $\text{PGL}(2; \overline{\mathbb{F}}_p)$ , it is isomorphic to a cyclic group/dihedral group/ $\text{Alt}_4$  or  $\text{Sym}_4$ ?. In all cases,  $\bar{\rho}$  lifts to an odd representation into  $\text{GL}(2; \mathbb{C})$ . And then the modularity follows from(31.1.1.16). Thus we assume that  $\text{Im}(\bar{\rho})$  is non-solvable.

It suffices to consider only modular forms  $f$  of weight  $k \geq 2$ , because by [Deligne-Serre, Formes modulaires de poids 1]Section6.9, any modular form of weight 1 is congruent to modular forms of weight  $\geq 2$ .?  $\square$

**Lemma (30.3.3.4).** Serre's modularity conjecture(30.3.3.3) is true for the following two cases:

- $p \neq 2$  and  $2 \nmid N(\bar{\rho})$ .
- $p = 2$  and  $k(\bar{\rho}) = 2$ .

┘

*Proof:*  $(D_0)$ (30.3.3.7) follows from the truth of  $(L_r)$ (30.3.3.6) for each  $r$ . Then the assertion follows from(30.3.3.7) and  $(D_0)$ .  $\square$

**Lemma (30.3.3.5).** Let  $\rho : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}(2, \mathcal{O})$  be a continuous odd irreducible representation s.t.

- $\bar{\rho}$  has non-solvable image, and  $\bar{\rho}$  is modular.
- $\rho$  is of weight 2, a.e. unramified, and potentially crystalline at  $p$ .

Then  $\rho$  is modular.

┘

*Proof:* Cf.[Khare-Wintenberger]Hypothesis H.?

$\square$

**Lemma (30.3.3.6).** For  $r \in \mathbb{Z}_+$ , consider the following statements:

$(L_r)$  : If  $\bar{\rho}$  is of Serre-type(30.3.3.3) and satisfies

- $\bar{\rho}$  is locally good-dihedral(30.3.3.2),
- $k(\bar{\rho}) = 2$  or  $p \neq 2$ ,
- $N(\bar{\rho})$  is odd and divisible by at most  $r$  different primes.

Then  $\bar{\rho}$  is modular.

$(W_r)$  : If  $\bar{\rho}$  is of Serre-type and satisfies

- $\bar{\rho}$  is locally good-dihedral(30.3.3.2),
- $k(\bar{\rho}) = 2$ ,
- $N(\bar{\rho})$  is odd and divisible by at most  $r$  different primes.

Then  $\bar{\rho}$  is modular.

Then

- $(W_1)$  is true.
- (Killing ramification in weight 2)( $L_r$ ) implies  $(W_{r+1})$ .
- (Reduction to weight 2)( $W_r$ ) implies  $(L_r)$ .

In particular, by induction spirally, all  $(L_r), (W_r)$  are true.

┘

*Proof:* Cf.[Khare-Wintenberger]Thm3.1, 3.2, 3.3.?

2: As  $L_r$  clearly implies  $W_r$ , it suffices to consider the case  $N(\bar{\rho})$  has exactly  $r+1$  different primes. Suppose  $q \in \mathbf{P}$ ,  $\bar{\rho}$  is  $q$ -dihedral, and  $s \in \mathbf{P}$ ,  $s|N(\bar{\rho})$ ,  $s \neq q$ .  $\square$

**Lemma (30.3.3.7) [Level-Rising].** For  $r \in \mathbb{N}$ , if the following:

$(D_r)$   $\bar{\rho}$  is modular If  $\bar{\rho}$  is of Serre-type and satisfies

- $\bar{\rho}$  is locally good-dihedral,
- $p \in \mathbf{P} \setminus \{2\}$ ,
- $2^{r+1} \nmid N(\bar{\rho})$ .

is true, then we have:  $\bar{\rho}$  is modular if  $\bar{\rho}$  is of Serre-type and satisfies

- $k(\bar{\rho}) = 2$  or  $p \neq 2$  or  $r \neq 0$ ,
- $2^{r+1} \nmid N(\bar{\rho})$ ,

┘

*Proof:* Cf. [Khare-Wintenberger]Thm3.4. ?

□

**Thm. (30.3.3.8) [Compatible Lifting].** Suppose  $\bar{\rho}$  is of Serre-type s.t.

- (if  $p = 2$ )  $\bar{\rho}$  has non-solvable image,
- (if  $p \neq 2$ )  $k(\bar{\rho}) \in [2, p+1]$  and  $\bar{\rho}|_{\mathbb{Q}(\mu_p)}$  is absolutely irreducible.

Then  $\bar{\rho}$  lifts to an  $E$ -rational almost strictly compatible, irreducible, odd Galois representation of  $\text{Gal}_{\mathbb{Q}}$  satisfying the following:

- 

┘

### Level and Weights in Serre's Conjecture

**Thm. (30.3.3.9) [Weight and Level aspect of Serre's Conjecture, Ribet, Edixhoven, Coleman, Voloch, Gross].**

┘

*Proof:* ?

□

### Applications

#### 4 Dwork's Family

**Notation (30.3.4.1).**

- Let  $n \in \mathbb{Z}_{\geq 2}$ .

┘

**Def. (30.3.4.2) [Dwork Family].** For  $N \in \mathbb{Z}_{\geq 2}$ , let  $T_0 = \mathbb{P}^1 \setminus \{\infty \cup \mu_N\} \in \text{Sch}/\mathbb{Z}[\frac{1}{N}]$  with coordinate  $t$ . And let  $Y \subset \mathbb{P}^{N-1} \times T_0$  be the projective family of hypersurfaces defined by the following equation:

$$X_1^N + \dots + X_N^N = NtX_1 \dots X_N.$$

Then  $\text{pr} : Y \rightarrow T_0$  is smooth of relative dimension  $N-2$ . Let  $\mu_N \xrightarrow{\Delta} \mu_N^N = H$  and  $H_0 = \{(\xi_1, \dots, \xi_N) : \xi_1 \dots \xi_N = 1\} \subset H$ .

Over  $\mathbb{Z}[\frac{1}{N}, \zeta_N]$ , there is an action of  $H$  on  $Y$  given by

$$(\xi_1, \dots, \xi_N)(X_1, \dots, X_N, t) = (\xi_1 X_1, \dots, \xi_N X_N, (\xi_1 \dots \xi_N)^{-1} t)$$

that induces an action of  $H_0$  on each fiber of  $Y$  over  $T_0$ .

┘

**Def. (30.3.4.3) [Dwork Family, 2].**

┘

**Def. (30.3.4.4)**[Self-Dual Avoidance Characters, Lie Qian]. Notation as in(30.3.4.2), let  $\chi : H_0 \rightarrow \mu_N$  be a character of the form

$$\chi((\xi_1, \dots, \xi_N)) = \prod_{i=1}^N \xi_i^{a_i}$$

where

$$(a_1, \dots, a_N) = \begin{cases} (1, 2, 4, 5, \dots, \frac{N-n+2}{2}, \frac{N+n-2}{2}, \dots, N-5, N-4, N-3, 0, 0, \dots, 0) & , n \in 2\mathbb{Z}_+ + 1 \\ (1, 2, 3, 4, 5, 7, 8, \dots, \frac{N-n+3}{2}, \frac{N+n-3}{2}, \dots, N-5, N-4, 0, 0, \dots, 0) & , n \in 2\mathbb{Z}_+ + 2 \\ (1, 2, \dots, \frac{N-3}{2}, 0, 0, 0, \frac{N+3}{2}, \dots, N-2, N-1) & , n = 2 \end{cases}$$

These characters are well-defined because  $N | \sum_{i=1}^N a_i$ .

In each case, there are  $n+1$  zero values in  $(a_i)$ , and we can let  $\{b_1, \dots, b_{n+1}\}$  be the  $n$  residue classes in  $\mathbb{Z}/(N)$  s.t.  $b_i \neq -a_j$  for any  $j \in \{1, \dots, N\}$ . In particular,

$$\{b_1, \dots, b_n\} = \begin{cases} \{1, 2, \frac{N-n+4}{2}, \frac{N-n+6}{2}, \dots, \frac{N+n-6}{2}, \frac{N+n-4}{2}, N-3\} & , n \in 2\mathbb{Z}_+ + 1 \\ \{1, 2, 3, \frac{N-n+5}{2}, \frac{N-n+7}{2}, \dots, \frac{N+n-7}{2}, \frac{N+n-5}{2}, N-6\} & , n \in 2\mathbb{Z}_+ + 2 \\ \{\frac{N-1}{2}, \frac{N+1}{2}\} & , n = 2 \end{cases}$$

┘

**Prop. (30.3.4.5).** Situation as in(30.3.4.4), if  $n > 2$  and  $\alpha \in (\mathbb{Z}/(N))^*$  s.t.  $\{\alpha b_1, \dots, \alpha b_n\} = \{b_1, \dots, b_n\}$ , then

┘

*Proof:* Cf.[?Qia23]P1254.

□

## 5 Serre's Modularity Conjecture over Totally Real Fields

**Thm. (30.3.5.1)** [Lie Qian]. Suppose  $F$  is a CM field,  $F^{\text{av}}/F$  a finite extension, and  $n \in \mathbb{Z}_{\geq 2}, s \in \mathbb{Z}_+, \ell \in \text{Prime}_{\geq 3}$ . Let

$$\bar{\rho} : \text{Gal}_F \rightarrow \text{GL}(n; \mathbb{F}_{\ell^s})$$

be a continuous semisimple representation, then there exists a finite CM extension  $F'/F$  linearly disjoint from  $F^{\text{av}}$  over  $F$  s.t.  $\bar{\rho}|_{\text{Gal}_{F'}}$  is ordinarily automorphic(30.1.3.35).

┘

*Proof:* Cf.[Lie Qian]P1239. This extends results of [A family of Calabi-Yau Varieties and Potential Automorphy, 2] to the non-self-dual case.

□

## 6 Taylor-Wiles-Kisin Methods

**Notation (30.3.6.1).**

- Let  $F/F_0$  be a CM extension with  $\text{Gal}(F/F_0) = \{1, \mathfrak{c}\}$ .

┘

**Def. (30.3.6.2)**[Polarized Representations].

┘

## 7 Modularity-Lifting

**Thm. (30.3.7.1) [Modularity-lifting for irreducible representations of  $GL(2)$ , Kisin[Kis09, Kis09c]/Paškūnas(16)].** Let  $k \in \mathbb{Z}_{\geq 2}$  and  $\rho : \text{Gal}_{\mathbb{Q}} \rightarrow GL(2; \overline{\mathbb{Q}}_{\ell})$  be a continuous odd Galois representation ramified at f.m. places and satisfy:

- $\rho|_{\text{Gal}_{\mathbb{Q}_p}}$  is de Rham with  $H\text{-}T(\rho) = \{0, k-1\}$ .
- $\bar{\rho}$  is modular(30.1.2.5).
- If  $p > 2$ ,  $\bar{\rho}|_{\text{Gal}_{\mathbb{Q}}\left(\sqrt{(-1)^{(p-1)/2}p}\right)}$  is irreducible.
- If  $p = 2$ ,  $\text{Im}(\bar{\rho})$  is non-solvable.

Then  $\rho$  matches the representation of a weight  $k$  modular form.  $\lrcorner$

*Proof:* ? Cf.[simplified proof of Serre’s conjecture]Thm1.4, 1.5.

For  $p > 2$ , this is??P642.

For  $p = 2$ , this follows from [On 2-dimensional 2-adic Galois representations of local and global fields, Paskunas]Theorem 1.1 and [On the modularity of 2-adic potentially semi-stable deformation rings, Tung]Theorem A. ?  $\square$

**Thm. (30.3.7.2) [Modularity-Lifting for Reducible Representations, Skinner-Wiles/Pan].** Let  $k \in \mathbb{Z}_{\geq 2}$  and  $\rho : \text{Gal}_{\mathbb{Q}} \rightarrow GL(2; \overline{\mathbb{Q}}_{\ell})$  be a continuous odd irreducible geometric Galois representation that satisfies:

- $\rho|_{\text{Gal}_{\mathbb{Q}_p}}$  is de Rham with  $H\text{-}T_p(\rho) = \{0, k-1\}$  and  $\bar{\rho}^{\text{ss}} \cong \bar{\chi}_1 \oplus \bar{\chi}_2$  where  $\chi_i$  are characters.
- If  $p = 3$ ,  $\frac{\bar{\chi}_1}{\bar{\chi}_2}|_{\text{Gal}_{\mathbb{Q}_p}} \neq \bar{\chi}_{\text{cycl}}^{\pm 1}$ .

Then  $\rho$  matches the representation of a weight  $k$  modular form.  $\lrcorner$

*Proof:* Cf.[Skinner-Wiles, Residually reducible representations and modular forms]P6. and [Fontaine-Mazur conjecture in the residually reducible case, Pan]Thm1.0.2.  $\square$

**Thm. (30.3.7.3) [Modularity Lifting for  $GL(2)$ , Taylor-Wiles/Khare-Wintenberger].** Assume  $p \in \text{Prime}$  and  $\bar{\rho} : \text{Gal}_{\mathbb{Q}} \rightarrow GL(2, \overline{\mathbb{F}}_p)$  is a smooth modular Galois representation, and

- $\text{Im}(\bar{\rho})$  is non-solvable if  $p = 2$ ,
- $\bar{\rho}|_{\mathbb{Q}(\zeta_p)}$  is absolutely irreducible if  $p > 2$ .

Then  $\rho$  is also modular if:

- $(p = 2)$  and  $\rho$  is an odd lift of  $\bar{\rho}$  to a 2-adic representation that is
  - a.e. unramified,
  - crystalline of weight 2 at 2, or semistable of weight 2 at 2 and  $k(\bar{\rho}) = 4$ ,
- $(p > 2)$  and  $\rho$  is a lift of  $\bar{\rho}$  to a  $p$ -adic representation that is crystalline of weight  $k \in [2, p+1]$  at  $p$ , or pot.semistable of weight 2 at  $p$ ,

$\lrcorner$

*Proof:* Cf.[Khare-Wintenberger 2]. ?  $\square$

**Def. (30.3.7.4) [Adequate Subgroups, Thorne[Tho12]].** For  $\ell \in \text{Prime}$  and  $n \in \mathbb{Z}_+$ , a subgroup  $H \leq GL(n; \overline{\mathbb{F}}_{\ell})$  are called an **adequate subgroup** if the following conditions hold:

- $H^1(H, \overline{\mathbb{F}}_{\ell}) = 0$  and  $H^1(H, {}_n(\overline{\mathbb{F}}_{\ell})) = 0$ .
- $H^0(H, {}_n(\overline{\mathbb{F}}_{\ell})) = 0$ , in particular,  $\ell \nmid n$ .

- The elements of  $H$  with order coprime to  $\ell$  spans  $\text{Mat}(n, \overline{\mathbb{F}_\ell})$  over  $\overline{\mathbb{F}_\ell}$ .

⌋

**Prop. (30.3.7.5) [Criterion for Adequate Subgroups].** Let  $\ell \in \text{Prime}$ ,  $n \in \mathbb{Z}_+$ . Suppose  $H \leq \text{GL}(n; \overline{\mathbb{F}_p})$  is a finite subgroup s.t. the tautological representation of  $H$  is irreducible, and denote  $H_0 \leq H$  the subgroup generated by all elements of  $\ell$ -power order, and let  $d$  be the maximal dimension of an irreducible  $\overline{\mathbb{F}_p}$ -representation of  $H_0$ . If  $\ell \geq 2(d+1)$ , then  $H$  is adequate.

⌋

*Proof:* Cf. [Tho12] Appendix Theorem9.

□

**Thm. (30.3.7.6) [Automorphic lifting in the ordinary case, Geraghty [Ger10]/Thorne [?Tor12]].** Let  $F$  be a CM field or a totally real field,  $n \in \mathbb{Z}_+$ , and  $\ell \in \text{Prime}_{\geq 3}$ . Suppose  $(\rho, \mu)$  is an  $n$ -dimensional geometric polarized (30.1.3.28)  $\ell$ -adic representation of  $\text{Gal}_F$  s.t.

- The reduction  $\bar{\rho}$  is irreducible and  $\bar{\rho}(\text{Gal}_{F(\mu_\ell)}) \leq \text{GL}(n; \overline{\mathbb{F}_\ell})$  is adequate (30.3.7.4).
- $\mu_\ell \not\in F$ .
- $\rho$  is ordinary at all primes above  $\ell$  (30.1.3.34).
- $(\bar{\rho}, \bar{\mu})$  is ordinarily automorphic?

Then  $(\rho, \mu)$  is ordinarily automorphic. And if moreover  $(\bar{\rho}, \bar{\mu})$  is (potentially) crystalline, then  $(\rho, \mu)$  is ordinarily automorphic of (potentially) level prime to  $\ell$ .

⌋

*Proof:* Cf. [Tho12] Thm9.1.. The totally real case follows from the CM case by base change? □

**Thm. (30.3.7.7) [Automorphy Lifting Theorem, Taylor-Wiles/Kisin/Barnet.Lamb-Gee-Geraghty-Taylor [LGGT]].** Suppose  $n \in \mathbb{Z}_+$ ,  $p > 2(n+1)$  and  $\zeta_\ell \notin K$ . Let  $\rho : \text{Gal}_F \rightarrow \text{GL}(n; K)$  is a geometric Galois representation (30.1.2.1) that  $\bar{\rho}$  is automorphic (30.1.3.17), together with

- $\rho^c \cong \rho^\vee \otimes \mu$  where  $\mu$  is a totally odd  $\ell$ -adic character of  $\text{Gal}_{F_0}$ .
- $\bar{\rho}_{\text{Gal}_{F(\mu_\ell)}}$  is absolutely irreducible.
- For any  $v \in S_F(p)$ ,  $\rho|_{\text{Gal}_{F_v}}$  is potentially diagonalizable (i.e. pot. crystalline and can be deformed in a moduli space of crystalline representations to a sum of characters) and Hodge-Tate regular.

Then  $\rho$  is also automorphic (30.1.3.17).

⌋

*Proof:* ? Cf. [Automorphy for some  $\ell$ -adic lifts of automorphic mod  $\ell$  Galois representations, 2, Taylor2008] and [Automorphy for some  $\ell$ -adic lifts of automorphic mod  $\ell$  Galois representations, Clozel-Harris-Taylor2008], [A family of Calabi-Yau varieties and potential automorphy, Harris-Barron-Taylor2010], [Potential automorphy and change of weight, Barnet.Lamb-Gee-Geraghty-Taylor2014].

□

## 8 Open Questions

**Remark (30.3.8.1).** Current techniques have the following restrictions: If  $(\rho, V) : \text{Gal}_\mathbb{Q} \rightarrow \text{GL}(n; \overline{\mathbb{Q}_\ell})$  is a geometric representation, then

- There should be a character  $\mu : \text{Gal}_\mathbb{Q} \rightarrow \overline{\mathbb{Q}_\ell}^\times$  and a bilinear form  $(-, -)$  on  $V$  s.t.

$$(\rho(\sigma)v_1, \rho(\sigma)v_2) = \mu(\sigma)(v_1, v_2), \quad (v_1, v_2) = \mu(c)(v_2, v_1).$$

- $\rho$  needs to be Hodge-Tate regular.

- Either  $\rho, \rho'$  are both ordinary or crystalline with  $H-T(\rho) = H-T(\rho')$  and diameter of  $H-T(\rho)$  is small compared to  $\ell$ .
- If  $n \geq 3$ , then Ihara's conjecture(30.3.8.2) is hard to prove, Cf.[On modular curves over finite fields]lemma3.2 and [Congruence relations between modular forms, Ribet]Theorem4.1?

┘

**Conj. (30.3.8.2)[Ihara's Conjecture].** Let  $G/\mathbb{Q}$  be a unitary group that is isomorphic to  $GL(n)$  after base change to an imaginary quadratic field  $\mathcal{K}$ . Let  $\ell \in \mathbf{Prime}$  that is large compared to  $n$  and  $p_1, p_2 \in \mathbf{Prime} \setminus \{\ell\}$  be two different primes s.t.  $G(\mathbb{Q}_{p_i}) \cong GL(n; \mathbb{Q}_{p_i})$ . Let  $U$  be an open compact subgroup of  $G(\mathbf{A}^{p_1 p_2})$  and consider the representation of  $GL(n; \mathbb{Q}_{p_1}) \times GL(n; \mathbb{Q}_{p_2})$  on

$$C^\infty(G(\mathbb{Q}) \backslash G(\mathbf{A})/U; \overline{\mathbb{F}_\ell}) = C^\infty((G(\mathbb{Q}) \cap U) \backslash (GL(n; \mathbb{Q}_{p_1}) \times GL(n; \mathbb{Q}_{p_2})); \overline{\mathbb{F}_\ell}).$$

Suppose  $\pi_1 \times \pi_2$  is a subrepresentation with  $\pi_1$  generic, then  $\pi_2$  is also generic.

┘

*Proof:*

□

### Ihara Avoidance

## 30.4 Shimura-Taniyama-Weil Conjecture(Modular Abelian Variteis)

*“Cubum autem in duos cubos, aut quadrato-quadratum in duos quadrato-quadratos, et generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.”*

–P. de Fermat

References are [Ring-theoretic properties of certain Hecke algebras, Taylor-Wiles], [Rib04, R-S11], [Darmon-Diamond-Taylor, Fermat’s last theorem], <http://virtualmath1.stanford.edu/~conrad/modseminar/>.

### 1 Abelian Varieties attached to Modular Forms

Cf[R-S11]Chap14.

**Prop. (30.4.1.1).** For a newform  $f$  of level  $N$ ,  $A_f \subset J_1(N)$  is of GL(2)-type(30.4.2.2). ┘

*Proof:* It has action by  $K_f$ ? □

**Remark (30.4.1.2).** The action of ┘

**Thm. (30.4.1.3) [Eichler-Shimura].** If  $f \in \text{SF}_2^{\text{eigen}}(\Gamma_0(N); \mathbb{Z})$ , then there exists an elliptic curve  $E/\mathbb{Q}$  s.t.  $L(E, s) = L(f, s)$ (31.1.6.2)(29.5.4.10). ┘

*Proof:* □

### 2 Modular Abelian Varieties

**Def. (30.4.2.1) [Modular Abelian Varieties].**  $A \in \text{AbVar}/\mathbb{Q}$  is called a **modular Abelian variety** if there exists a surjective map  $J_1(N) \rightarrow A$  for some  $N \in \mathbb{Z}_+$ . ┘

#### GL(2)-Type Abelian Varieties

**Prop. (30.4.2.2) [GL(2)-Type Abelian Varieties].** If  $A \in \text{AbVar}/\mathbb{Q}$  and  $E \in \text{Field}$  s.t.  $E \subset \text{End}(A)_{\mathbb{Q}}$ , then  $[E : \mathbb{Q}] \mid \dim A$ . In particular,  $\text{End}(E/\mathbb{Q}) = \mathbb{Z}$  for any  $E \in \text{EllCur}/\mathbb{Q}$ .

$A \in \text{AbVar}/\mathbb{Q}$  is called an **Abelian variety of GL(2)-type** if  $\text{End}(A)_{\mathbb{Q}}$  contains a number field of degree  $\dim A$ . ┘

*Proof:*  $E$  acts faithfully on the tangent space  $\text{Lie}(A/\mathbb{Q})$ , so  $\text{Lie}(A/\mathbb{Q})$  is an  $E$ -vector space thus has dimension divisible by  $[E : \mathbb{Q}]$ . □

**Def. (30.4.2.3) [Power Abelian Varieties].** The power of a GL(2)-Abelian variety is also of GL(2)-type. ┘

**Prop. (30.4.2.4) [Primitive Abelian Varieties of GL(2)-Type].** Let  $A \in \text{AbVar}/\mathbb{Q}$ , then  $A/\mathbb{Q}$  is primitive(i.e. not a power) iff  $A/\mathbb{Q}$  is simple iff  $\text{End}(A/\mathbb{Q})$  is a number field of degree  $\dim A$ . ┘

**Thm. (30.4.2.5) [Abelian Varieties of GL(2)-Type are Modular, Ribet[Rib04]/Khare-Wintenberger[K-W09b, K-W09c].** If  $A \in \text{AbVar}/\mathbb{Q}$  is of GL(2)-type, then  $A$  is modular(30.4.2.1). ┘

*Proof:* [Rib04]Thm4.4. proved that this follows from Serre’s conjecture?. □

### Q-Curves

**Def. (30.4.2.6) [Q-Curves].**  $E \in \text{EllCur}/\overline{\mathbb{Q}}$  is called a **Q-curve** if  $E^\sigma$  is isogenous to  $E$  for any  $\sigma \in \text{Gal}_{\overline{\mathbb{Q}}}$ . ┘

**Def. (30.4.2.7) [Modular Elliptic Curves over  $\overline{\mathbb{Q}}$ ].**  $E \in \text{EllCur}/\overline{\mathbb{Q}}$  is called a **modular elliptic curve** if there is a surjective map  $J_1(N) \rightarrow E$  for some  $N \in \mathbb{Z}_+$ . ┘

**Thm. (30.4.2.8) [Ribet].** If  $E \in \text{EllCur}/\overline{\mathbb{Q}}$  is modular(30.4.2.7), then either  $E$  is a Q-curve or  $E$  has CM. ┘

*Proof:* Cf.[Rib04]Section5. □

**Thm. (30.4.2.9) [Q-Curves and GL(2)-Type Abelian Varieties].** Suppose  $E \in \text{EllCur}/\overline{\mathbb{Q}}$  has no CM, then  $E$  is a Q-curve(30.4.2.6) iff there is a simple Abelian variety  $A \in \text{AbVar}/\mathbb{Q}$  of GL(2)-type s.t.  $E$  is a simple factor of  $A_{\overline{\mathbb{Q}}}$ . ┘

*Proof:* Cf.[Rib04]Thm6.1. ?

This is accomplished finding a model for  $E$  over a Galois extension  $F/\mathbb{Q}$ , restricting scalars down to  $\mathbb{Q}$  to obtain an abelian variety  $B = \text{Res}_{F/\mathbb{Q}}(E)$ , and using Galois cohomology computations (mainly in  $H^2$ s) to find the required  $A$  of GL(2)-type inside  $B$ . □

**Cor. (30.4.2.10)** [Ribet[Rib04]/Ellenberg-Skinner[E-S01]/Khare-Wintenberger[K-W09b, K-W09c]]. Let  $E \in \text{EllCur}/\overline{\mathbb{Q}}$  be a Q-curve, then  $E$  is modular over  $\mathbb{Q}$ (30.4.2.7). ┘

*Proof:* This follows from(30.4.2.8) and(30.4.2.5)(30.4.2.9). □

### 3 Elliptic Curves over $\mathbb{Q}$

**Prop. (30.4.3.1).** If  $E \in \text{EllCur}/\mathbb{Q}$  is modular(30.4.2.1), then there is a non-constant map  $X_0(N) \rightarrow E$ . ┘

**Thm. (30.4.3.2) [Mazur].** If  $E \in \text{EllCur}/\mathbb{Q}$  is a modular(30.4.2.1) if  $E_{\mathbb{C}}$  is a quotient of  $X_1(N)$  for some  $N \in \mathbb{Z}_+$ . ┘

*Proof:* [Mazur, Number theory as gadfly]. □

**Prop. (30.4.3.3) [Modular Elliptic Curves].** For  $E \in \text{EllCur}/\mathbb{Q}$ , the following are equivalent:

- $E$  is modular(30.4.2.1).
- there exists a normalized Hecke eigenform  $f \in S_2(\Gamma_0(N_E))$  with integral coefficients s.t.  $L(E; s) = L(f, s)$ .
- For some(every)  $p \in \text{Prime}$ ,  $\rho_{E,p}$  is modular.
- $E$  has a Weil parametrization(30.4.4.1).
- $E$  is isogenous to the modular Abelian variety  $A_f$  associated to some newform  $f \in S_2(\Gamma_0(N_E))$ . ┘

*Proof:* □

**Lemma (30.4.3.4).** For  $E \in \text{EllCur}/\mathbb{Q}$  and  $N \in \mathbb{Z}_+$ , there is a surjection  $X_0(N_E) \rightarrow E$  iff there is a surjection  $\text{Jac}_0(N) \rightarrow E$ . ┘



*Proof:* If  $\varphi : X_0(N) \rightarrow E$  is non-zero, then  $\varphi_* : \text{Jac}_0(N) \rightarrow \text{Jac}(E) \cong E$  is nonzero, by (20.9.10.10). If  $\Phi : J_0(N) \rightarrow E$  is surjective, then  $X_0(N) \xrightarrow{\text{Abel}} J_0(N) \rightarrow E$  is also surjective, by looking at the Tate module?  $\square$

**Lemma (30.4.3.5) [Langland-Tunnell].** If  $E \in \text{EllCur}/\mathbb{Q}$  is semistable and  $E[\ell]$  is irreducible for some odd prime  $\ell$ , then  $E[\ell]$  is modular.  $\lrcorner$

*Proof:*  $\square$

**Lemma (30.4.3.6) [Taylor-Wiles].** If  $E \in \text{EllCur}/\mathbb{Q}$  is semistable and  $\ell$  is an odd prime s.t.  $E[\ell]$  is irreducible and modular, then  $T_\ell$  is modular.  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj.Cor. (30.4.3.7) [Modularity of Elliptic Curves over  $\mathbb{Q}$ , Shimura--Taniyama[S-T61]/Weil[Wei67], Wiles[Wil95]/Taylor-Wiles[T-W95]/Breuil-Conrad-Diamond-Taylor/Khare-Wintenberger[K-W09b, K-W09c].** Any  $E \in \text{EllCur}/\mathbb{Q}$  is modular (30.4.3.3).  $\lrcorner$

*Proof:* Cf.[On the modularity of elliptic curves over  $\mathbb{Q}$ : wild 3-adic exercises, C. Breuil, B. Conrad, F. Diamond, and R. Taylor].

The proof by Serre conjecture doesn't rely on [BCDT]?  $\square$

**Thm. (30.4.3.8) [ $T = R$  Theorem, Wiles].** The modular deformation ring  $T$  is isomorphic to the Galois deformation ring  $R_{\bar{\rho}}$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (30.4.3.9) [Elliptic Curves and Modular Forms].** For  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$  and sign of functional equation  $w_E$ , and has L-function  $L(E, s) = \sum c_n n^{-s}$ . Let  $f_E(\tau) = \sum c_n e^{2\pi i \tau}$  be the Mellin transform of  $L(E, s)$ , then

- $f_E$  is a cuspidal modular form on  $S_2(\Gamma_0(N))$ .
- $f_E$  is a normalized Hecke eigenfunction, and satisfies  $f[w_N]_2 = -w_E f(\tau)$ , where  $w_E$  is the sign of functional equation of  $E$  (31.1.6.4).
- Let  $\omega$  be an invariant differential form on  $E$ , then there exists a finite morphism  $\varphi : X_0(N) \rightarrow E$  over  $\mathbb{Q}$  s.t.  $\varphi^*(\omega)$  is a multiple of the differential form on  $X_0(N)$  represented by  $f(\tau)d\tau$ .  $\lrcorner$

*Proof:*  $\square$

## 4 Weil Parametrizations

References are [Edixhoven, Manin Constant].

**Def. (30.4.4.1) [Weil Parametrizations].** For  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N = N_E$  and a Néron differential  $\omega_E$  (21.1.1.3), a **Weil parametrization** is a minimal map?  $\varphi : X_0(N) \rightarrow E$  over  $\mathbb{Q}$  s.t.  $\varphi^* \omega_E = 2\pi i c_E f_E(z) dz$  for some normalized newform  $f_E \in \text{SF}_2^{\text{new}}(\Gamma_0(N))$ , where  $\omega_E$  is the Néron differential of  $E$ .

Then  $c_E \in \mathbb{Z}^\times$ , called the **Manin constant** of  $E$ . After possibly changing  $\varphi$  to  $[-1] \circ \varphi$ , we may assume  $c_E \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* ? □

**Conj. (30.4.4.2)** [Manin[Man71, Man72]]. For any  $E \in \text{EllCur}/\mathbb{Q}$ ,  $c_E = 1$ . ┘

*Proof:* Let  $\mathcal{E} \in \text{Sch}$  be a Néron model for  $E$ , then there is a natural map  $\varphi : X_0(N)_{\text{sm}} \rightarrow \mathcal{E}$ . By [Katz-Mazur]Thm8.11.10, the formal completion of  $X_0(N)$  along the unramified cusp  $\infty$  is  $\text{Spf}(\mathbb{Z}[[q]])$ , so  $\infty \in X_0(N)_{\text{sm}}(\mathbb{Z})$ , and the  $q$ -expansion of  $\varphi^*\omega_E$  is just the pullback to  $\text{Spf}(\mathbb{Z}[[q]])$ . Thus  $c_E \in \mathbb{Z}_+$ . ? □

**Thm& Conj. Cor. (30.4.4.3)** [Manin-S.Dyer/Mazur[Maz78]]. For any  $E \in \text{EllCur}/\mathbb{Q}$ ,  $c_E \in \mathbb{Z}[\frac{1}{2m}]^*$ , where  $m$  is the largest square dividing  $N_E$ . In particular, if  $E \in \text{EllCur}/\mathbb{Q}$  is semistable, then  $c_E \in 2^{\mathbb{Z}_+}$ . ┘

*Proof:* Cf.[Maz78]P144. □

**Thm& Conj. Cor. (30.4.4.4)** [Raynaud]. If  $E \in \text{EllCur}/\mathbb{Q}$  is semistable, then  $\log(c_E)$  is bounded. ┘

*Proof:* ? □

**Prop. (30.4.4.5)** [Heights and Weil Parametrization]. Let  $E \in \text{EllCur}/\mathbb{Q}$  with a Weil parametrization  $\varphi_E : X_0(N) \rightarrow E$  and corresponding normalized modular form  $f_E$  and Manin constant  $c_E$ , then

$$\frac{1}{2} \log \left( \frac{\deg(\varphi_E)}{[\text{SL}(2, \mathbb{Z}) : \Gamma_0(N)]} \right) = h_{\text{Fal}}(E/\mathbb{Q}) + \log(4\pi^2 \|f_E\|_{\text{Pet}}) + \log |c_E|.$$

┘

*Proof:* let  $\alpha_E$  be a Néron differential of  $E$ , then  $\varphi_E^*\alpha_E = 2\pi i c_E f_E(z) dz$ , so

$$\begin{aligned} 4\pi^2 |c_E|^2 \|f_E\|_{\text{Pet}}^2 &= \frac{1}{[\text{SL}(2, \mathbb{Z}) : \Gamma_0(N)]} \frac{i}{2} \int_{X_0(N)} \varphi_E^*\alpha_E \wedge \overline{\varphi_E^*\alpha_E} \\ &= \frac{\deg(\varphi_E)}{[\text{SL}(2, \mathbb{Z}) : \Gamma_0(N)]} \frac{i}{2} \int_{E^{\text{an}}} \alpha_E \wedge \overline{\alpha_E} \\ &= \frac{\deg(\varphi_E)}{[\text{SL}(2, \mathbb{Z}) : \Gamma_0(N)] H_{\text{Fal}}^2(E)} \quad (21.1.5.1) \end{aligned}$$

□

**Cor. (30.4.4.6)** [Lower Bounded for  $\deg(\varphi_E)$ ]. For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_\varepsilon \in \mathbb{R}_+$  such that: Suppose  $E \in \text{EllCur}/\mathbb{Q}$  with a Weil parametrization  $\varphi_E : X_0(N) \rightarrow E$  and associated normalized cusp form  $f_E$  and Manin constant  $c_E = 1$ , then

$$\deg(\varphi_E) \geq C_\varepsilon \max(|c_4(E)|^{1/4}, |c_6(E)|^{1/6})^{2-\varepsilon}.$$

┘

*Proof:* Use (30.4.4.5). Firstly, by (21.1.5.13),  $h_{\text{Fal}}(E/\mathbb{Q}) \geq (1 - \varepsilon) \log \max(|c_4(E)|^{1/4}, |c_6(E)|^{1/6}) + O_\varepsilon(1)$ . Next,  $\|f_E\| \geq e^{-4\pi}/(4\pi)$  by (26.3.1.9) and the fact  $f_E$  is normalized. Then the assertion follows. □

Modularity over Totally Real Fields

**Thm. (30.4.4.7)**[Modularity of Elliptic Curves over Real Quadratic Fields, Freitas-Hung-Siksek]. Any elliptic curve defined over a real quadratic field is modular.  $\square$

*Proof:* ? Cf. [FHS15].

The proof uses a “3-5-7” trick.  $\square$

**5 Paramodular Conjecture**

**Conj. (30.4.5.1)**[Paramodular Conjecture, Brumer-Kramer]. If  $A \in \mathcal{AbVar}^2/\mathbb{Q}$  with  $\text{End}(A_{\overline{\mathbb{Q}}}) = \mathbb{Z}$  and conductor  $N$ , then  $L(A; s) = L(f; s)$  for some weight 2 paramodular newform  $f$  of level  $N$  with rational Hecke eigenvalues.

Conversely, for any weight 2 paramodular newform  $f$  of level  $N$  with rational Hecke eigenvalues, there exists  $A \in \mathcal{AbVar}^2/\mathbb{Q}$  with  $\text{End}(A_{\overline{\mathbb{Q}}}) = \mathbb{Z}$  and conductor  $N$  or  $A \in \mathcal{AbVar}^4/\mathbb{Q}$  with QM.  $\square$

*Proof:*

$\square$

### 30.5 Langlands-Kottwitz Method

References are [Lan79a, Lan79b, Lan83, Zhu20], [Kottwitz, Shimura Varieties and  $\lambda$ -adic Representations] and [Kottwitz, Points on some Shimura varieties over finite fields].

#### Notation(30.5.0.1).

- Let  $G \in \mathcal{R}\text{edGrp}/\mathbb{Q}$ .
- Assume  $[G, G]$  is simply-connected.
- Assume  $A(G_{\mathbb{R}}) = A(G)_{\mathbb{R}}$  (this holds for any Hodge-type Shimura variety, but may fail for Abelian types ?).
- Let  $p \in \mathbf{Prime}$ . Let  $q \in p^{\mathbb{Z}_+}$ .
- Let  $\sigma_q \in \text{Gal}_{\mathbb{F}_q}$  be the geometric Frobenius.
- Assume  $G$  is unramified at  $p$ .
- Fix  $L(\xi) \in \mathbf{NField}$  and  $\xi \in \text{Irr}_{L(\xi)}(G)$ .
- Let  $\chi \in X^{\bullet}(Z_G)$  be the central character of  $\xi$ .
- Use notations on étale cohomologies over finite fields defined in [Cohomologies over Finite Fields](#).
- Use notations on stable trace formulae defined in [Stable Trace Formulae](#).

⌋

#### Notation(30.5.0.2)[Unitary PE-Structures].

- Let  $E/F$  be a CM-field.
- Let  $\text{Gal}(E/F) = \{1, \mathbf{c}\}$ .
- Let  $\varepsilon \in E$  s.t.  $\mathbf{c}(\varepsilon) = -\varepsilon$ .
- Let  $(V, [-, -]) \in \text{HermSpa}_{E/F}$  be anisotropic.
- Define  $\langle x, y \rangle = \text{tr}_{E/\mathbb{Q}}(\varepsilon[x, y])$  where  $x, y \in V$ , which is a  $\mathbb{Q}$ -bilinear  $\mathbb{Q}$ -valued non-degenerate alternating form on  $V$ .
- Let  $C = \text{End}_E(V)$  and let  $*$  be the involution on  $C$  defined by

$$\langle cx, y \rangle = \langle x, c^*y \rangle, \text{ where } x, y \in V, c \in C.$$

- Let  $h : \mathbb{C} \rightarrow \text{End}_{E \otimes \mathbb{R}}(V \otimes \mathbb{R})$  be an  $\mathbb{R}$ -algebra homomorphism s.t.  $\langle h(z)x, y \rangle = \langle x, h(\mathbf{c}z)y \rangle$  and  $\langle x, y \rangle = \langle x, h(\mathbf{i})y \rangle$  is positive-definite symmetric form on  $V \otimes \mathbb{R}$ .
- Let  $E$  be the reflex field of  $h$ , and  $\mathfrak{p} \in S_E(p)$ .

In particular,  $(E, \mathbf{c}, V, \langle -, - \rangle, h)$  is a rational PE-structure. And we denote  $G = \text{Res}_{F/\mathbb{Q}} \text{GU}(V, [-, -]) \in \mathcal{R}\text{edGrp}/\mathbb{Q}$ .

⌋

#### 1 Honda-Tate Theory

References are [Kot92]Section10.

**Def.(30.5.1.1)[Virtual Abelian Varieties up to  $p$ -prime Isogenies].** For  $n \in \mathbb{Z}_+$ , the  $\mathbb{Z}_{(p)}$ -category  $\mathcal{V}_n^p$  of **virtual Abelian varieties up to  $p$ -prims isogenies** is defined as follows:

- Its object are  $A = (\bar{A}, u)$  where  $\bar{A} \in \mathcal{AbVar}_{\overline{\mathbb{F}_p}}$  and  $u : (\varphi^n)^*(\bar{A}) \cong \bar{A}$  is an isomorphism in  $(\mathcal{AbVar}_{\overline{\mathbb{F}_p}}) \otimes \mathbb{Z}_{(p)}$ .
- Its morphisms are  $f : (\bar{A}_1, u_1) \rightarrow (\bar{A}_2, u_2)$  where  $f : \bar{A}_1 \rightarrow \bar{A}_2 \in (\mathcal{AbVar}_{\overline{\mathbb{F}_p}}) \otimes \mathbb{Z}_{(p)}$  s.t.  $f u_1 = u_2 (\varphi^n)^*(f)$ .

Also denote  $\mathcal{V}_n \triangleq \mathcal{V}_n^p \otimes \mathbb{Q}$ . ┘

**Def. (30.5.1.2) [Frobenius-Endomorphisms].** For  $A = (\bar{A}, u) \in \mathcal{V}_n^p$  (30.5.1.1), define the **Frobenius-endomorphism**

$$\pi_A : \bar{A} \xrightarrow{\text{Fr}} (\varphi^n)^* \bar{A} \xrightarrow{u} \bar{A} \in \text{End}(\bar{A}) \otimes \mathbb{Z}_{(p)}.$$

Then for any  $(\bar{A}_1, u_1), (\bar{A}_2, u_2) \in \mathcal{V}_n^p$ , a map  $f : \bar{A}_1 \rightarrow \bar{A}_2 \in \text{Hom}(\bar{A}_1, \bar{A}_2) \otimes \mathbb{Z}_{(p)}$  is a morphism in  $\mathcal{V}_n^p$  iff  $f \pi_{A_1} = \pi_{A_2} f$ . In particular,

$$\text{End}_{\mathcal{V}_n^p}(A) = Z_{\text{End}(\bar{A}) \otimes \mathbb{Z}_{(p)}}(\pi_A).$$
┘

**Def. (30.5.1.3) [Polarizations].** For  $A = (\bar{A}, u) \in \mathcal{V}_n$  (30.5.1.1) and  $c \in \mathbb{Q}$ , a  **$c$ -polarization**  $\lambda : \bar{A} \rightarrow \bar{A}^\vee$  s.t.  $(\pi_A)^*(\lambda) = c\lambda$ .  $A$  is said to be  **$c$ -polarizable** if it admits a  $c$ -polarization. The full subcategory of  $\mathcal{V}_n$  consisting of  $c$ -polarizable elements is denoted by  $\mathcal{V}_{n,c}$ . ┘

**Prop. (30.5.1.4).** If  $A \in \mathcal{AbVar}/\mathbb{F}_{p^n}$ , then  $A$  defines an element of  $\mathcal{V}_n$  (30.5.1.1), and a  $p^n$ -polarization is the same as a polarization of  $A$ . ? ┘

**Def. (30.5.1.5) [ $c$ -Numbers].** For  $c \in \mathbb{Q}_+$ , a  **$c$ -number** is a number  $\pi \in \overline{\mathbb{Q}}$  s.t.

- $|\pi_A|_v \geq 0$  for any embedding  $v : \overline{\mathbb{Q}} \rightarrow \overline{\mathbb{Q}_p}$ .
  - $|\pi_A|_v = c^{1/2}$  for any  $c : \overline{\mathbb{Q}} \rightarrow \mathbb{C}$ .
- ┘

**Thm. (30.5.1.6) [Kottwitz[Kot92]].** For  $c \in \mathbb{Q}$  and  $n \in \mathbb{Z}_+$ ,  $\mathcal{V}_{n,c}$  (30.5.1.3) is non-empty iff  $c > 0$  and  $v_p(c) = n$ , and in this case, we have:

- $\mathcal{V}_{n,c}$  is a semi-simple Artinian Noetherian Abelian category (4.8.3.17).
  - For each simple object  $A \in \mathcal{V}_{n,c}$ ,  $\text{End}_{\mathcal{V}_{n,c}}(A)$  is a division algebra with center  $\mathbb{Q}(\pi_A)$  (30.5.1.2). And its Hasse-invariant is given by explicit formula ?
  - For any simple object  $A \in \mathcal{V}_{n,c}$  and any embedding  $v : \mathbb{Q}(\pi_A) \rightarrow \overline{\mathbb{Q}}$ ,  $v(\pi_A)$  is a  $c$ -number (30.5.1.5). And  $A \mapsto \pi_A$  induces a bijection between the set of isomorphism classes of simple objects in  $\mathcal{V}_{n,c}$  to the set of  $\text{Gal}_{\mathbb{Q}}$ -orbits of  $c$ -numbers (30.5.1.5).
- ┘

*Proof:* ? [Kot92]section10. □

### Endomorphism Structures

**Def. (30.5.1.7) [Virtual Abelian Varieties with Endomorphism].** For  $c \in \mathbb{Q}, n \in \mathbb{Z}_+$  and  $F \in \text{NField}$ , define  $\mathcal{V}_{n,c,F}$  the category s.t.

- Its objects are pairs  $(A, \iota)$  where  $A \in \mathcal{V}_{n,c}$  and  $\iota : F \rightarrow \text{End}(A) \in \mathbb{R}ing/\mathbb{Q}$ .
- Its morphisms are morphisms in  $\mathcal{V}_{n,c}$  compatible with the  $F$ -action.

┘

**Prop. (30.5.1.8) [Classifying Polarized Virtual Abelian Varieties with Endomorphism Structures].** Situation as in (30.5.1.7), if  $\mathcal{V}_{n,c}$  is non-empty (30.5.1.3), then  $\mathcal{V}_{n,c,F}$  (30.5.1.7) is a semi-simple Artinian Noetherian Abelian category (4.8.3.17), and  $A \mapsto \pi_A$  induces a bijection between the set of isomorphism classes of simple objects in  $\mathcal{V}_{n,c,F}$  to the set of  $\text{Gal}_F$ -orbits of  $c$ -numbers (30.5.1.5). ┘

*Proof:* Cf. [Kot92] 10.13. ?

□

**Def. (30.5.1.9) [Virtual Abelian varieties with Polarization and Endomorphism].** For  $c \in \mathbb{Q}$ ,  $n \in \mathbb{Z}_+$  and  $F \in \text{NField}$ , denote  $\mathcal{V}_{n,c,F,\text{polar}}$  the category s.t.

- Its object are triples  $(A, \iota, \lambda)$  where  $(A, \iota) \in \mathcal{V}_{n,c,F}$  (30.5.1.7), and  $\lambda : A \rightarrow A^\vee$  is a  $c$ -polarization (30.5.1.3) s.t.  $\lambda$  is compatible with  $\iota$  in the sense of (? Cf. [Zhu20] P10).
- Its morphisms are morphisms in  $\mathcal{V}_{n,c,F}$  that preserves the polarization  $\lambda$  up to a scalar.

┘

## 2 Virtual Abelian Varieties with PEL Structures

**Def. (30.5.2.1) [Virtual Abelian Varieties with PEL Structures].**  $\mathcal{S}_{\mathcal{K}^p}(\overline{\kappa(\mathfrak{p})})$ , ? Cf. [Zhu20] P6. ┘

## 3 Kottwitz-Triples

References are [Kottwitz, Shimura varieties and  $\lambda$ -adic representations].

**Notation (30.5.3.1).**

- Assume  $(G/A(G))(\mathbb{R})$  has an elliptic maximal torus (e.g. if  $G$  admits Shimura variety??).
- Assume  $G_{\mathbb{Q}_p}$  is quasi-split.
- Let  $\mathcal{N} : \mathbb{S} \rightarrow G_{\mathbb{R}}$  be a  $G(\mathbb{R})$ -conjugacy class of group homomorphisms.
- let  $E \in \text{NField}$  be the field of definition of the conjugacy class  $\mathcal{N}$ .
- Let  $\mathfrak{p} \in S_E(p)$ .

┘

**Def. (30.5.3.2) [Kottwitz-Triples].** For  $r \in \mathbb{Z}_+$ , a **Kottwitz-triple** of degree  $r$  for  $G$  w.r.t.  $\{\mu\}$  is a triple  $(\gamma_0, \gamma, \delta)$  where

- $\gamma_0 \in G(\mathbb{Q})$  is semi-simple, and becomes elliptic in  $G(\mathbb{R})$  (10.4.2.25).
- $\gamma \in G(\mathbf{A}_f^p)$ .
- $\delta \in G(\mathbb{Q}_{p^{rf(\mathfrak{p})}})$  (28.6.1.1).

s.t.

- $\gamma_0$  is stably conjugate to  $\gamma$  (10.5.1.3).
- $\gamma_0$  and  $\mathfrak{N}_\sigma(\delta)$  (10.5.1.14) are stably conjugate.
- $\kappa_{G_{\mathbb{Q}_p}}(\delta) = (\mu_h)_*(-1) \in \pi_1(G)_{\text{Gal}_{\mathbb{Q}_p}}$  (28.6.1.7), where  $\mu_h \in X_\bullet(G)$  is the Hodge-cocharacter associated to an arbitrary  $h \in \{h\}$  (28.1.1.3).

Two Kottwitz-triples  $(\gamma_0, \gamma, \delta), (\gamma'_0, \gamma', \delta')$  of the same degree are called **equivalent** if  $\gamma_0 \sim_{\text{st}} \gamma$ ,  $\delta \sim_\varphi \delta'$ , and  $\gamma_v \sim \gamma'_v$  for all  $v \in \text{Prime} \setminus \{p\}$ .

Let  $\text{KT}_p(G, \mathcal{N}; j)$  denote a set of representatives for the equivalence classes of degree- $j$  Kottwitz-triples for  $G$ . And denote  $\text{KT}(G, \mu) = \coprod_{j \in \mathbb{Z}_+} \text{KT}_p(G, \mu; j)$ . ┘

**Def. (30.5.3.3) [Kottwitz-Invariants].** For each  $(\gamma_0, \gamma, \delta) \in \text{KT}_p(G, \mu)$  (30.5.3.2), Kottwitz defined  $\alpha(\gamma_0, \gamma, \delta) \in \mathfrak{K}(\gamma_0)^\vee$  (29.2.1.2) called the **Kottwitz-invariant** of  $(\gamma_0, \gamma, \delta)$ , as follows:

- For each  $v \in \text{Prime} \setminus \{p\}$ , the difference between  $\gamma_v$  and  $\gamma_0$  in  $G(\mathbb{Q}_v)$  is measured by a class  $\alpha_v \in \ker(H^1(\mathbb{Q}_v, I_0) \rightarrow H^1(\mathbb{Q}_v, G))$ , which is equivalent to a map

$$\alpha_v : Z(\hat{I}_0)^{\text{Gal}_{\mathbb{Q}_v}} \rightarrow \mathbb{C}^\times$$

whose restriction to  $(Z(\hat{I}_0)^{\text{Gal}_{\mathbb{Q}_v}})^0 Z(G)^{\text{Gal}_{\mathbb{Q}_v}}$  is trivial. ? Cf. [Harris, Stable Trace Formula] Section 3.

- By Steinberg's theorem, there exists  $c \in G(\check{\mathbb{Q}}_p)$  s.t.  $c\gamma c^{-1} = \mathfrak{N}_\sigma(\delta)$  ?, and we can define  $\alpha_p = \kappa_G(c^{-1}\delta\varphi(c)) \in \pi_1(I_0)_{\text{Gal}_{\mathbb{Q}_p}}$ , which doesn't depend on  $c$  chosen.
- [?Kot90] P167 ? defined an element  $\alpha_\infty \in X^\bullet(Z(\hat{I}_0)^{\text{Gal}_{\mathbb{R}}})$  whose restriction to  $Z(G)^{\text{Gal}_{\mathbb{R}}}$  is  $(\mu_h)_*(1)$  (30.5.3.2).

Thus  $\alpha = \prod_{v \in \Sigma_{\mathbb{Q}}} \alpha_v$  is a character of  $\bigcap_{v \in \Sigma_F} (Z(\hat{I}_0)^{\text{Gal}_{F_v}} Z(\hat{G}))$  that is trivial on  $Z(G)$ , thus defines an element  $\alpha(\gamma_0, \gamma, \delta) \in \mathfrak{K}(\gamma_0)^\vee$  by (29.2.1.2).

When  $\gamma_0$  varies in a stable conjugacy class, the groups  $\mathfrak{K}(\gamma_0)$  are canonically identified and these elements  $\alpha(\gamma_0, \gamma, \delta)$  are compatible with these identifications.

?

⌋

*Proof:* Cf. [?Kot90] Section 2.

□

**Prop. (30.5.3.4) [Inner Forms associated to Kottwitz-Triples].** For each  $(\gamma_0, \gamma, \delta) \in \text{KT}_p(G, \mu)$  (30.5.3.2) s.t.  $\alpha(\gamma_0, \gamma, \delta) = 1$  (30.5.3.3), denote  $I_0 = Z_G(\gamma_0)^0 \in \text{RedGrp}/\mathbb{Q}$ , then there exists an inner form  $I \in \text{RedGrp}/\mathbb{Q}$  of  $I_0$  s.t.

- $I(\mathbb{R})$  is compact modulo center.
- For each  $v \in \Sigma_{\mathbb{Q}}^{\text{fin}} \setminus \{p\}$ , we have  $I_{\mathbb{Q}_v} \cong (I_0)_{\mathbb{Q}_v} \in \text{RedGrp}/\mathbb{Q}_v$ .
- $I_p \cong J_\delta \in \text{RedGrp}/\mathbb{Q}_p$  (28.6.1.23).

⌋

*Proof:* Cf. [?Kot90] Section 3. ?

□

### Kottwitz-Triples of Virtual Abelian Varieties

**Def. (30.5.3.5) [PEL-type Abelian Varieties].** For a rational PE-structure as in (30.5.0.2) and  $c \in \mathbb{Q}$ ,  $n \in \mathbb{Z}_+$ , let  $\mathcal{V}_{n,c,E,\text{polar}}^G$  be the full sub-category of  $\mathcal{V}_{n,c,E,\text{polar}}$  consisting of objects  $(A, \iota, \lambda)$  s.t.

- There is an  $\mathbf{A}_f^p$ -isomorphism

$$(H_1(\bar{A}, \mathbf{A}_f^p), (-, -)_\lambda) \cong (V, \langle -, - \rangle) \otimes \mathbf{A}_f^p$$

which is  $E$ -invariant and preserves the alternating form up to a scalar.

- There is a  $\mathbb{Q}_{p^n}$ -linear isomorphism

$$(H_1(A/\mathbb{Z}_{p^n})[\frac{1}{p}], (-, -)_\lambda) \cong (V, \langle -, - \rangle) \otimes \mathbb{Q}_{p^n}.$$

⌋

*Proof:* Cf.[Zhu20]P10.?

□

**Prop.(30.5.3.6) [Geometric Kottwitz-Triples].** For any  $c \in \mathbb{Q}, n \in \mathbb{Z}_+$  and  $(A, \iota, \lambda) \in \mathcal{V}_{n,c,E,\text{polar}}^G$  (30.5.3.5), we can translate  $\pi_A^{-1}$  to an element  $\gamma \in G(\mathbf{A}_f^p)$ , and translate  $\varphi_{\text{crys}}$  to a  $\varphi$ -linear automorphism on  $V \otimes \mathbb{Q}_{p^n}$  of the form  $\delta \circ \varphi$  for some  $\delta \in G(\mathbb{Q}_{p^n})$ . The conjugacy class of  $\gamma$  and the  $\varphi$ -conjugacy of  $\delta$  is independent of the isomorphisms in (30.5.3.5)?. And we know the similitude  $\nu(\gamma) = c^{-1}$ ?

And we denote  $\mathcal{V}_{n,c}^{(G,h^{-1})}$  the full sub-category of  $\mathcal{V}_{n,c,E,\text{polar}}^G$  consisting of objects whose associated  $\delta$  as above satisfies  $\kappa_{G_{\mathbb{Q}_p}}(\delta) = (\mu_h)_*(-1) \in \pi_1(G)_{\text{Gal}_{\mathbb{Q}_p}}$  (28.6.1.7), where  $\mu_h \in X_\bullet(G)$  is the Hodge cocharacter associated to  $h: \mathbb{S} \rightarrow G_{\mathbb{R}}$ .

Then for any  $(A, \iota, \lambda) \in \mathcal{V}_{n,c}^{(G,h^{-1})}$ , the stable-conjugacy class of  $\gamma$  contains some  $\gamma_0 \in G(\mathbb{Q})$ , and  $(\gamma, \gamma_0, \delta)$  is a  $\mathbb{F}_{p^n}$ -Kottwitz-triple for  $G$  (30.5.3.2), called the **geometric Kottwitz-triples** associated to  $(A, \iota, \lambda)$ .  $\lrcorner$

*Proof:* Cf.[Kot92]Section14 and [Zhu20]P10.  $\square$

**Thm.(30.5.3.7) [Characterizing geometric Kottwitz-Triples, Kottwitz92, Lemma18.1].**

$(\gamma_0, \gamma, \delta) \in \text{KT}_p(G, \mathcal{N})$  (30.5.3.2) is geometric (30.5.3.6) iff

- The similitude factor  $\nu(\gamma_0)$  equals  $c^{-1}$ .
- $\alpha(\gamma_0, \gamma, \delta) = 1$  (30.5.3.3).
- There is a lattice  $\Lambda \subset V \otimes \mathbb{Q}_{p^n}$  s.t.  $\Lambda \subset (\delta\sigma)\Lambda$ .

□

*Proof:* Cf.[Zhu20]P11.  $\square$

**Prop.(30.5.3.8) [From Virtual Abelian Varieties to Kottwitz-Triples].** For each  $c \in \mathbb{Q}$  and  $n \in \mathbb{Z}_+$ , denote by  $\mathcal{I}_{n,c}$  the isomorphism classes of elements in  $\mathcal{V}_{n,c}^{(G,h^{-1})}$  (30.5.3.6), and  $\mathcal{I}_n \triangleq \coprod_{c \in \mathbb{Q}} \mathcal{I}_{n,c}$ .

Let  $(\gamma_0, \gamma, \delta) \in \text{KT}_p(G, \mathcal{N}; j)$  be geometric (30.5.3.6), then  $(\gamma_0, \gamma, \delta)$  is associated to exactly  $\#\text{III}(I_0/\mathbb{Q})$ -many non-isomorphic elements  $(A_i, \iota_i, \lambda_i) \in \mathcal{V}_{f(\mathfrak{p})}^{(G,h^{-1})}$ .  $\lrcorner$

*Proof:* Cf.[Kot92]Section17 and P441.?

□

#### 4 Unstabilized Langlands-Kottwitz Formulas

**Def.(30.5.4.1).** For a neat compact open subgroup  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$ , denote  $\mathcal{H}_{\mathcal{K}^p, \xi} \triangleq \mathcal{H}_{\mathcal{K}^p}(G(\mathbf{A}_f^p); L(\xi))$ .

For  $j \in \mathbb{Z}_+$  and  $f^{p,\infty} \in \mathcal{H}_{\mathcal{K}^p, \xi}$ , defined the Euler-characteristic

$$\text{Tr}(j, f^{p,\infty}) \triangleq \text{Tr} \left( \sigma_{\mathfrak{p}}^j \times f^{p,\infty} \middle| \left[ H_{\text{ét},c}^*((\mathcal{S}_{\mathcal{K}^p})_{\overline{E}}, \mathcal{L}_{\mathcal{K}^p}) \right] \right) \cong \text{Tr} \left( \sigma_{\mathfrak{p}}^j \times f^{p,\infty} \middle| \left[ H_{\text{ét},c}^*((\mathcal{S}_{\mathcal{K}^p})_{\overline{E}}, \mathcal{L}_{\mathcal{K}^p}) \right] \right) \in L(\xi)_{\lambda}.$$

And for  $g \in G(\mathbf{A}_f^p)$ , denote  $\text{Tr}(j, g) \triangleq \text{Tr}(j, \mathbf{1}_{\mathcal{K}^p g^{-1} \mathcal{K}^p})$ .  $\lrcorner$

**Prop.(30.5.4.2).** Situation as in (30.5.4.1), denote  $\mathcal{K}_g^p = \mathcal{K}^p \cap g \mathcal{K}^p g^{-1}$ , then there is a correspondence

$$\begin{array}{ccc} & (\mathcal{S}_{\mathcal{K}_g^p})_{\overline{\kappa(\mathfrak{p})}} & \\ \text{Fr}_{\mathcal{S}_{\mathcal{K}^p/\kappa(\mathfrak{p})}}^j \circ [\cdot 1] \swarrow & & \searrow [\cdot g] \\ (\mathcal{S}_{\mathcal{K}^p})_{\overline{\kappa(\mathfrak{p})}} & & (\mathcal{S}_{\mathcal{K}^p})_{\overline{\kappa(\mathfrak{p})}} \end{array} \quad (30.5.0.2)$$



and a cohomological correspondence(9.6.1.7).

$$u(j, g) : [\cdot 1]^* \left( \mathrm{Fr}_{\mathcal{S}_{\mathcal{K}^p}/\kappa(\mathfrak{p})}^j \right)^* \mathcal{L}_{\mathcal{K}^p} \cong [\cdot 1]^* \mathcal{L}_{\mathcal{K}^p} \cong \mathcal{L}_{\mathcal{K}_g^p} \cong [\cdot g]^* \mathcal{L}_{\mathcal{K}^p} \text{ (28.5.1.8)}.$$

where  $u(j, g)_1, u(j, g)_2$  are finite étale.

Then we have

$$\mathrm{Tr}(j, g) = \mathrm{Lef}(u(j, g)) = \mathrm{Tr} \left( u(j, g)^* \left| \left[ H_{\mathrm{ét}, c}^* \left( (\mathcal{S}_{\mathcal{K}^p})_{\overline{\kappa(\mathfrak{p})}}, \mathcal{L}_{\mathcal{K}^p} \right) \right] \right. \right) \text{ (9.6.1.14)(28.5.1.8)}.$$

┘

*Proof:* This is basically the definition of the Hecke action and Galois action on cohomology of Shimura varieties, by using the unramified comparison isomorphism(9.4.6.36).  $\square$

### Fixed Points

**Prop. (30.5.4.3) [Lefschetz-Verdier Trace Formula].** Situation as in(30.5.4.2), by(9.6.1.25), when  $j$  is sufficiently large,

$$\mathrm{Fix}(j, g) = \left\{ x \in (\mathcal{S}_{\mathcal{K}^p})_{\overline{\kappa(\mathfrak{p})}} : u(j, g)_1(x) = u(j, g)_2(x) \right\} \in \mathrm{Sch}/\overline{\kappa(\mathfrak{p})}$$

is Artinian, and we have

$$\mathrm{Lef}(u(j, g)) = \sum_{x \in |\mathrm{Fix}(j, g)|_0} \mathrm{tr}(u(j, g)_x) \text{ (9.6.1.14)(9.6.1.19)},$$

┘

**Prop. (30.5.4.4) [Fixed Points and Virtual Abelian Varieties].** Situation as in(30.5.4.3), denote  $n = j \deg(\mathfrak{p})$ . Then any  $x \in \mathrm{Fix}(j, g)$ (30.5.4.3) is represented by an object  $(\overline{A}, \iota, \lambda, \overline{\eta}) \in \mathcal{S}_{\mathcal{K}^p}(\kappa(\mathfrak{p}))$ (30.5.2.1) s.t. there is an isomorphism

$$u : (\overline{A}, \iota, \lambda, \overline{\eta}) \cong (\varphi^n)^*(\overline{A}, \iota, \lambda, \overline{\eta}) \in \mathcal{S}_{\mathcal{K}^p}(\overline{\kappa(\mathfrak{p})}). \text{ ?}$$

Thus  $X = (\overline{A}, \iota, \lambda, u) \in \mathcal{V}_{n, c, E, \mathrm{polar}} \text{ (30.5.1.9)}$ . In fact,  $X \in \mathcal{V}_{n, c}^{(G, h^{-1})} \text{ (30.5.3.6)}$ , and the isomorphism class of  $X$  only depends on  $x$ . So we get a map

$$\mathbf{f} : \mathrm{Fix}(j, g) \rightarrow \mathcal{I}_{f(\mathfrak{p})j} \text{ (30.5.3.8)}.$$

┘

*Proof:* Cf.[Kot92]P430-431. ?  $\square$

**Prop. (30.5.4.5) [From Fixed Points to Virtual Abelian Varieties].** Situation as in(30.5.4.4), for  $x \in |\mathrm{Fix}(j, g)|_0$ , let  $(\gamma_0, \gamma, \delta) \in \mathrm{KT}_p(G, \mathcal{N}, j)$  be the Kottwitz-triple associated to  $\mathbf{f}(x)$ (30.5.4.4)(30.5.3.6), then

$$\mathrm{Orb}(\gamma_0, \gamma, \delta) = \# \mathbf{f}^{-1}(\mathbf{f}(x)) \text{ (29.2.4.2)}.$$

Moreover,  $\mathrm{tr}(u(j, g)_x) = \mathrm{tr}(\xi(\gamma_0)) \text{ (30.5.4.3)}$ .

┘

*Proof:* Cf.[Kot92]Section16.?

□

**Thm. (30.5.4.6) [Unstabilized Langlands-Kottwitz Formulaa, Kottwitz].** For a neat compact open subgroup  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  and  $g \in G(\mathbf{A}_f^p)$ , let  $f^{p,\infty}$  be the characteristic function of  $\mathcal{K}^p g^{-1} \mathcal{K}^p$ , then for  $j$  sufficiently large, we have

$$\begin{aligned} \frac{\text{Tr}(j, f^{p,\infty})}{\#\text{III}(\mathbb{Q}, G)} &= \text{Tr}'(j, f^{p,\infty}) \\ &\triangleq \sum_{\substack{(\gamma_0, \gamma, \delta) \in \text{KT}_p(G, \mu; j) \\ \alpha(\gamma_0, \gamma, \delta) = 1}} c(\gamma_0, \gamma, \delta) \text{Orb}_\gamma(f^p) \text{TOrb}_\delta(\phi_{j \deg(p)}) \text{tr}(\xi(\gamma_0)) \end{aligned} \quad (9.8.3.28)(29.2.4.2)(29.2.4.3)(30.5.3.3),$$

and there are only f.m. non-zero terms. ⌋

*Proof:* It follows from(30.5.4.2)(30.5.4.3) that when  $j$  is sufficiently large, we have

$$\text{Tr}(j, g) = \sum_{x \in |\text{Fix}(j, g)|_0} \text{tr}(u(j, g)_x),$$

which by(30.5.4.5) equals

$$\sum_{(A, \iota, \lambda) \in \mathcal{I}_n} \text{Orb}(\gamma_0, \gamma, \delta) \text{tr}(\xi(\gamma_0)) = \sum_{(A, \iota, \lambda) \in \mathcal{I}_n} c(\gamma_0, \gamma, \delta) \text{Orb}_\gamma(f^p) \text{TOrb}_\delta(\phi_n) \text{tr}(\xi(\gamma_0)),$$

which by(30.5.3.8) equals

$$\sum_{(\gamma_0, \gamma, \delta) \text{ geometric}} \#\text{III}(I_0/\mathbb{Q}) \# \ker(\text{III}(I_0/\mathbb{Q}) \rightarrow \text{III}(G/\mathbb{Q})) c(\gamma_0, \gamma, \delta) \text{Orb}_\gamma(f^p) \text{TOrb}_\delta(\phi_n) \text{tr}(\xi(\gamma_0)).$$

By(29.2.4.4), only geometric  $\mathbb{F}_{p^{j \deg(p)}}$ -Kottwitz-triples  $(\gamma_0, \gamma, \delta)$  contributes to  $\text{Tr}'(j, f^{p,\infty})$ , so it suffices to prove that

$$\#\text{III}(I_0/\mathbb{Q}) \# \ker(\text{III}(I_0/\mathbb{Q}) \rightarrow \text{III}(G/\mathbb{Q})) = \#\text{III}(G/\mathbb{Q}).$$

But  $\text{III}(I_0/\mathbb{Q}) \rightarrow \text{III}(G/\mathbb{Q})$  is a homomorphism of Abelian groups( ? Cf.[Kot84]), so it suffices to prove  $\text{III}(I_0/\mathbb{Q}) \rightarrow \text{III}(G/\mathbb{Q})$  is surjective, which follows from[Kot92]Section7?.

The fact that there are only f.m. non-zero terms can also be proven directly analogous to Prop8.2. of[Kot86].? □

### Abelian-Type Shimura-Varieties

It is shown in [L-R87] that the Langlands-Rapoport conjecture describing hyperspecial reduction of Shimura varieties imply the unstabilized Langlands-Kottwitz formula. Langlands-Rapoport conjecture has been proved by Kisin in [Mod  $p$  points on Shimura varieties of abelian type], and the construction of trace formula is carried out in [stable trace formula for certain Shimura varieties of abelian type, Kisin-Shin-Zhu].

## 5 Stabilization of Langlands-Kottwitz Formulae

References are [Kottwitz, Shimura Varieties and  $\lambda$ -adic Representations].

**Notation (30.5.5.1).**

- Notice when  $G = \mathrm{GU}(V)$ , the maximal split central torus of  $G_{\mathbb{R}}$  is split over  $\mathbb{Q}$ .?
- Notice when  $G = \mathrm{GU}(V)$ ,  $[G, G]$  is simply-connected and  $[H, H]$  is simply-connected for any elliptic endoscopic triple  $(H, s, \eta)$  over  $\mathbb{Q}$  or  $\mathbb{Q}_v$  for each  $v \in \Sigma_{\mathbb{Q}}$  (29.2.1.4).? ┘

### Langlands-Shelstad Transfers

**Def. (30.5.5.2) [Archimedean Langlands-Shelstad Transfer].** Let  $G \in \mathcal{R}\mathrm{edGrp}/\mathbb{Q}$  and  $(H, s, \eta) \in \mathfrak{E}_{\infty}(G)$  (29.2.1.4).

Fix the choice of  $(j, B_{G,H})$  as above?, and fix a Haar measure on  $H(\mathbb{R})$ , define

$$f_{H,\infty} \triangleq (-1)^{q(G_{\mathbb{R}})} \langle \mu_{T_G}, s \rangle_j \sum_{\varphi \in \Phi_H(\varphi_{\xi}^*)} \det(\omega_*(\varphi_H)) f_{\varphi_H}.$$

┘

**Def. (30.5.5.3) [Adelic Langlands-Shelstad Transfer away from  $p, \infty$ ].** Let  $G \in \mathcal{R}\mathrm{edGrp}/\mathbb{Q}$  and  $(H, s, \eta) \in \mathfrak{E}(G)$  (29.2.1.4). For  $\mathcal{K}^p \leq G(\mathbf{A}_f^p)$  and  $f^{p,\infty} \in \mathcal{H}_{\mathcal{K}^p}(G(\mathbf{A}_f^p))$ , suppose

- An elliptic maximal torus of  $G_{\mathbb{R}}$  comes from  $H_{\mathbb{R}}$  (29.2.1.27),
- $H$  is unramified over  $\mathbb{Q}_p$ ,
- Fix a Haar measure on  $G(\mathbf{A}_f^p)$  neutralized by  $\mathcal{K}^p$ ,
- Fix a Haar measure on  $H(\mathbf{A}_f^p)$  s.t. its product with the chosen Haar measure on  $H(\mathbb{R})$  (30.5.5.2) and the Haar measure on  $H(\mathbb{Q}_p)$  neutralized by a hyperspecial maximal subgroup is equal to the Tamagawa-measure on  $H(\mathbf{A})$  (20.5.4.5).

Define  $f_H^{p,\infty}$  to be the Langlands-Shelstad transfer of  $f^{p,\infty}$  to  $H(\mathbf{A}_f^p)$  as in (29.2.1.25). ┘

**Def. (30.5.5.4) [Langlands-Shelstad Transfer at  $p$ ].** ┘

### Stabilized Langlands-Kottwitz Formulae

**Thm. (30.5.5.5) [Stabilized Langlands-Kottwitz Formulae, Kottwitz].** Situation as in (30.5.4.6), for  $j$  sufficiently large, we have

$$\mathrm{Tr}'(j, f^{p,\infty}) = \sum_{(H,s,\eta) \in \mathfrak{E}(G)} i(G, H) \mathrm{STr}_{\mathrm{ell}}^{H,*}(f_H) (29.2.1.4)(29.2.1.8)(29.2.5.1).$$

where for each  $(H, s, \eta) \in \mathfrak{E}(G)$ ,  $f_H = 0$  unless an elliptic maximal torus of  $G_{\mathbb{R}}$  comes from  $H_{\mathbb{R}}$  (29.2.1.27) and  $H$  is unramified over  $\mathbb{Q}_p$ , in which case

$$f_H = f_{H,\infty} f_{H,p} f_H^{p,\infty} \in \mathcal{T}(H(\mathbf{A}), \mathbb{C}) (30.5.5.2)(30.5.5.3)(30.5.5.4).$$

(Notice that  $f_{H,\infty}$  is not compactly supported, but it has compact support modulo center and transforms under the split component of the center by a character, so  $\mathrm{STr}_{\mathrm{ell}}^{H,*}(\Phi^H)$  (29.2.5.1) is still definable?). ┘

*Proof:* ? Cf.[Kottwitz90]Thm7.2.

By(30.5.4.6) and(29.2.4.3), we have

$$\mathrm{Tr}'(j, f^{p,\infty}) = \sum_{\substack{(\gamma_0, \gamma, \delta) \in \mathrm{KT}_p(G, \mathcal{N}, j) \\ \alpha(\gamma_0, \gamma, \delta) = 1}} \frac{\tau(G) \# \mathfrak{K}(I_0)}{\mathrm{Vol}(A_G \backslash I(\mathbb{R}))} \mathrm{Orb}_\gamma(f^p) \mathrm{TOrb}_\delta(\phi_n) \mathrm{tr}(\xi(\gamma_0)) \quad (30.5.3.2)(30.5.3.3).$$

Denote  $e(\delta) \triangleq e(I_{\mathbb{Q}_p})$ ,  $e(\gamma_0) \triangleq e(I_{\mathbb{R}})$ ,  $e(\gamma) \triangleq \prod_{v \in \Sigma_{\mathbb{Q}}^{\mathrm{fin}} \setminus \{p\}} e(I_v)$  (29.2.4.5)(30.5.3.4), then

$$e(\delta)e(\gamma_0)e(\gamma) = e(I) = 1$$

(? Cf.[?Kot83]). Then

$$\begin{aligned} \mathrm{Tr}'(j, f^{p,\infty}) &= \tau(G) \sum_{(\gamma_0, \gamma, \delta)} \frac{\# \mathfrak{K}(I_0)}{e(\gamma_0) \mathrm{Vol}(A_G \backslash I(\mathbb{R}))} \mathrm{Orb}_\gamma(f^p) \mathrm{TOrb}_\delta(\phi_n) \mathrm{tr}(\xi(\gamma_0)) \\ &= \tau(G) \sum_{\gamma_0} \sum_{\kappa \in \mathfrak{K}(I_0/\mathbb{Q})} \frac{\mathrm{tr}(\xi(\gamma_0))}{e(\gamma_0) \mathrm{Vol}(A_G \backslash I(\mathbb{R}))} \sum_{(\gamma, \delta)} e(\gamma) e(\delta) \mathrm{Orb}_\gamma(f^{p,\infty}) \mathrm{TOrb}_\delta(\phi_n) \langle \alpha(\gamma_0, \gamma, \delta), \kappa \rangle \quad (29.2.1.2), \end{aligned}$$

where

- $\gamma_0$  runs through a set of representatives for the  $\mathbb{R}$ -elliptic stable-conjugacy classes in  $G(\mathbb{Q})$ .
- $\gamma$  runs through a set of representatives for the conjugacy classes in  $G(\mathbf{A}_f^p)$  that are stably conjugate to  $\gamma_0$ .
- $\delta$  runs through a set of representatives for the  $\sigma$ -conjugacy classes of  $G(\mathbb{Q}_{p^n})$  s.t.  $\mathfrak{N}_\sigma(\delta)$  is conjugate to  $\gamma_0$ .

And this summation has only f.m. non-zero terms by(30.5.4.6) and(29.2.1.2). □

## Unitary Shimura Varieties

For unitary Shimura varieties, Cf.[Construction of automorphic Galois representations, 1, Clozel-Harris-Labesse](4.3.5).

## 6 Destabilization of Langlands-Kottwitz Formulae

Destabilizing the stabilized Langlands-Kottwitz formula(30.5.5.5) means expanding the right hand side of(30.5.5.5) in terms of automorphic representations, and then rewriting everything in terms of  $G$  alone, so that the endoscopic groups  $H$ s no longer appear. In this way, one would obtain a description of the cohomology of the Shimura variety in terms of the conjectural relation between Galois representations and automorphic representations

[?Kot90]Chap9-10 outlined a way to destabilize the Langlands-Kottwitz formula, assuming Arthur's multiplicity conjectures(29.8.1.1) and some related hypotheses on the stable trace formula. Cf.[Morel, On the cohomology of certain noncompact Shimura varieties], [Galois representations arising from some compact Shimura varieties, Shin], [Construction of automorphic Galois representations, 1, Clozel-Harris-Labesse], [stabilization of the Frobenius-Hecke traces on the intersection cohomology of orthogonal Shimura varieties, Yihang Zhu].

## 7 beyond Compact Hyperspecial cases

In the non-compact case, the intersection cohomology of the Baily–Borel compactification turns out to be the correct object to compare with Arthur–Selberg trace formulas. Cf. [Morel, On the cohomology of certain noncompact Shimura varieties] and [stabilization of the Frobenius–Hecke traces on the intersection cohomology of orthogonal Shimura varieties, Yihang Zhu], [Galois representations arising from some compact Shimura varieties, Shin], [geometry and cohomology of some simple Shimura varieties, Harris–Taylor].

## 30.6 Cohomology of Unitary Shimura Varieties

References are [Sch15], [On the Generic Part of the Cohomology of Compact Unitary Shimura Varieties, Caraiani-Scholze], [On the Generic Part of the Cohomology of Non-Compact Unitary Shimura Varieties, Caraiani-Scholze].

### 1 Perfectoid Siegel Spaces

References are [?, Torsion in the Cohomology of Locally Symmetric Varieties, Scholze]

Geometric points of  $\mathcal{F}l$  are in bijection with principally polarized  $p$ -divisible groups  $G/\mathcal{O}_C$  with a trivialization of their Tate module.

On geometric points, the Hodge-Tate map is the map sending an Abelian variety over  $\mathcal{O}_C$  to its associated  $p$ -divisible group.

### 2 Hodge-Tate Period Map

Given a  $(G, X)$  be a Shimura datum?, there is a flag variety we can define an **Borel embedding**

$$\beta : X \rightarrow \text{Flag}_{G,\mu}(\mathbb{C}) : [\mu] \in X \mapsto \text{Fil}^\bullet(\mu).$$

where  $\mu$  is the  $X$  is defined over a number field  $E$ , called the reflex field of  $X$ , where  $\text{Fil}^p(\mu)$  for any representation is the direct sum of all subspaces of the type  $(p', q')$  with  $p' \geq p$ .

Choose a Hodge cocharacter  $\mathbb{G}_m \rightarrow G_{\mathbb{C}}$ .

Then  $P_\mu^{std}$  can be defined to be the subgroup of  $G$  stabilizing  $\text{Fil}^\bullet(\mu)$ , which is a parabolic subgroup.?. And dually, we can define  $\text{Fil}_\bullet(\mu)$  and  $P_\mu$  the stabilizer of  $\text{Fil}_\bullet(\mu)$ .

$M_\mu = C_G(\mu)$  is the Levi component of both parabolics.

Let  $Fl_{G,\mu}^{std}(\mathbb{C}) = G(\mathbb{C})/P_\mu^{std}(\mathbb{C})$ , then there is a Borel embedding  $X \hookrightarrow Fl_{G,\mu}^{std}(\mathbb{C})$ .

**Example (30.6.2.1) [Siegel Varieties].** A Shimura variety of Hodge Type can be regarded as a moduli space for Abelian varieties with certain Hodge-tensors?.  $\lrcorner$

For the Siegel variety  $V$ , the flag variety  $Fl_{\tilde{G},\tilde{\mu}}$  parametrizes totally isotropic subspaces  $W \subset V$ ?. [Ana-Scholze]P11.

For  $p \in \mathbf{P}$ , consider compact open subgroups of the form  $K = K^p \times K_p \subset G(\mathbf{A}^{p\infty}) \times G(\mathbb{Q}_p)$ .

Fix a place  $\mathfrak{p} \in \Sigma_E^p$ , let  $\mathcal{F}\mathcal{L}_{G,\mu}$  be the adic space associated with  $\mathcal{F}l_{G,\mu} \times_E E_{\mathfrak{p}}$ ?.

**Prop. (30.6.2.2) [Hodge-Tate Period Map].** For a Shimura datum of Hodge type  $(G, X)$  and any sufficiently small tame level  $K^p \subset G(\mathbf{A}^{p\infty})$ , there exists a perfectoid space  $\mathcal{S}_{K^p}$  s.t.

$$\mathcal{S}_{K^p} \sim \varprojlim_{K_p} (S_{K^p K_p} \otimes_E E_{\mathfrak{p}})^{\text{ad}},$$

and there is a  $G(\mathbb{Q}_p)$ -equivariant **Hodge-Tate period map**

$$\pi_{\text{H-T}} : \mathcal{S}_{K^p} \rightarrow \mathcal{F}\mathcal{L}_{\tilde{G},\tilde{\mu}}$$

which is equivariant w.r.t the natural Hecke action? of  $G(\mathbf{A}^{p\infty})$  on the inverse system  $\mathcal{S}_{K^p}$  and the trivial action of  $G(\mathbf{A}^{p\infty})$  on  $\mathcal{F}\mathcal{L}_{\tilde{G},\tilde{\mu}}$ .  $\lrcorner$

*Proof.* Cf. [Scholze, coh of symmetric spaces] or [S-Weinstein]  $\square$

**Remark (30.6.2.3).** This Hodge-Tate map is a  $p$ -adic analogue of the Borel embedding, and on closed points it maps an Abelian variety to its Hodge-Tate filtration of its Tate module  $?$ .  $\lrcorner$

**Prop. (30.6.2.4) [Hodge-Tate Period Map].**

- The Hodge-Tate period map (30.6.2.2) factors through  $\mathcal{GL}_{G,\mu}$  and the resulting map

$$\pi_{\text{H-T}} : \mathcal{S}_{K^p} \rightarrow \mathcal{GL}_{G,\mu}$$

is independent of the choice of the embedding of Shimura data.

- For any  $\mu$  in the given conjugacy class defined over a finite extension of  $E$ , the tensor functor

$$\begin{aligned} f_p : \text{Rep}(M_\mu) &\xrightarrow{\text{pr}^*} \text{Rep}(P_\mu) \rightarrow \{G(\mathbb{Q}_p) - \text{equivariant vector bundles on } \mathcal{FL}_{G,\mu}\} \\ &\xrightarrow{\pi_{\text{H-T}}^*} \{G(\mathbb{Q}_p) - \text{equivariant vector bundles on } \mathcal{S}_{K^p}\} \end{aligned}$$

is isomorphic to the tensor functor

$$f_\infty : \text{Rep}(M_\mu) \xrightarrow{\text{pr}^*} \text{Rep}(P_\mu^{\text{std}}) \rightarrow \{\text{automorphic bundles on } S_K\} \rightarrow \{G(\mathbb{Q}_p) - \text{equivariant vector bundles on } \mathcal{S}_K\}$$

and this isomorphism is independent of the choice of Siegel embedding, and also equivariant for the Hecke action of  $G(\mathbf{A}^{p\infty})$ .  $\lrcorner$

### 3 Newton Stratification on the Flag Variety

### 4 Geometry of Newton Strata and Igusa Varieties

**Def. (30.6.4.1) [ $\mathcal{J}_{\text{Mant}}$ ].** The **Igusa variety**  $\mathcal{J}_{\text{Mant}}$  is defined to be the projective system of  $\overline{\mathbb{F}}_p$ -schemes  $(\mathcal{J}_{\text{Mant}, K^p, m}^b)_{K^p \rightarrow 0, m \in \mathbb{Z}_+}$ .  $\lrcorner$

**Prop. (30.6.4.2) [Igusa and Manton Varieties].** The perfect scheme  $\text{Ig}^b$  is the perfection of  $\mathcal{J}_{\text{Mant}}^b$ , via the natural map  $\text{Ig}^b \rightarrow \mathcal{J}_{\text{Mant}}^b$ .  $\lrcorner$

### 5 Cohomology of Igusa Varieties

We need to compute the alternating sum of cohomology groups  $[H(\mathcal{Ig}^b, \overline{\mathbb{Q}}_\ell)]$  as a virtual representation of  $G(\mathbf{A}^{p\infty}) \times J_b(\mathbb{Q}_p)$ .

It is enough to work with the classical object  $\mathcal{J}_{\text{Mant}}^b$ , because of (30.6.4.2) and the fact perfection doesn't change the étale topos  $?$ . And by Poincaré duality, it suffices to compute  $[H_c(\mathcal{Ig}^b, \overline{\mathbb{Q}}_\ell)]$ . Sug Woo Shin has a formula expressing this as a sum of stable orbital integrals for  $G$  and its elliptic endoscopic groups  $?$ .

**Notation (30.6.5.1).**

- Let  $F = F^+ \mathcal{K}$ , where  $F^+$  is a totally real field and  $\mathcal{K}$  is an imaginary quadratic field.
- Let  $G/\mathbb{Q}$  be a unitary similitude group preserving an alternating Hermitian form  $\langle -, - \rangle$  on  $V \in \text{Vect}_F^n$ .
- Let  $(G, [h])$  be the Shimura datum with reflex field  $E$ .

- Let  $\mathrm{Spl}_{F/F^+/\mathbb{Q}} = \{p \in \mathbf{P} : \forall v \in S(p) \subset \Sigma_{F^+}, v \text{ splits in } F\}$ .
- Let  $p \in \mathbf{P}$  be unramified in  $F$  and splits in  $\mathcal{K}$ .
- Let  $\mathfrak{p} \in S_E(p)$ .
- Let  $\mathcal{K} = \mathcal{K}^p \mathcal{K}_p \in G(\mathbf{A}^{p\infty}) \times G(\mathbb{Q}_p)$  be a compact open subgroup where  $K_p$  is hyperspecial<sup>?</sup>. The fact that  $p$  is unramified in  $F$  means that  $S_K$  has good reduction with integral model  $\mathcal{S}_K/\mathcal{O}_{E_p}$ .
- If  $G$  be a topological group, denote  $\mathrm{Irr}^{\mathrm{adm}}(G)$  the set of isomorphism classes of irreducible admissible representations of  $G$ , and  $\mathrm{Groth}(G)$  the corresponding Grothendieck group.<sup>?</sup>

┘

**Notation (30.6.5.2).** For  $F \in \mathbf{NField}$ , define  $\ker^1(F, G) = \ker(H^1(F, G) \rightarrow \prod_{v \in \Sigma_F} H^1(F_v, G))$ . ┘

Let  $K \in p\text{-}\mathbf{LField}$ , and  $G$  is an unramified group over  $K$ . Choose a hyperspecial group  $\mathcal{K} \subset G(K)$ , and a Haar measure of  $G(K)$  s.t.  $\mathrm{Vol}(\mathcal{K}) = 1$ . Let  $\mathcal{H}^{\mathrm{ur}}(G(K))$  be the subspace of  $C_c^\infty(G(K))$  consisting of bi- $\mathcal{K}$ -invariant functions.

**Def. (30.6.5.3) [Setup].** Suppose some technical conditions:

- all  $p \in \mathbf{Prime}$  ramified in  $F$  are contained in  $\mathrm{Spl}_{F/F^+/\mathbb{Q}}$ .
- $G$  is quasi-split in all finite places.

┘

**Remark (30.6.5.4).** These follows from [Compact Unitary at Ramified Split Places, SS13]Chap10.<sup>?</sup> ┘

**Def. (30.6.5.5) [Endoscopic Triples].** For notation of Langlands dual, Cf. [Borel, Automorphic L-functions, 79]<sup>?</sup>. An **endoscopic triple** for  $G$  is a triple  $(H, s, \eta)$  where

- $H/F$  is a quasi-split connected reductive group,
- $s \in Z(\widehat{H})$ ,
- $\eta : \widehat{H} \rightarrow \widehat{G}$  is an embedding of complex Lie groups.
- $\eta(\widehat{H}) = Z_{\widehat{G}}(\eta(s))^0$ .
- The  $\widehat{G}$ -conjugacy classes of  $\eta$  is fixed by  $\mathrm{Gal}_F$ .
- 

Cf. [Stable Trace Formula for Igusa Varieties, Shin].

┘

## 6 Torsion in the Cohomology of Unitary Shimura Varieties



## Part V

# Cycles and $L$ -Values



# 31 | Arithmetic Theory of $L$ -Functions

## 31.1 $L$ -Functions Attached to Motives

*“In short, nineteenth century number theory showed that much, if not all, of number theory is reflected by properties of  $L$ -functions.”*

– W. W. J. Hulsbergen.

References are [Conjectures in Arithmetic Algebraic Geometry, Hulsbergen, 1992], [Galois Representations, ICM, Taylor].

**Notation(31.1.0.1).**

- Use notations on Galois representations defined in ??.
- Use notations on Langlands conjectures defined in [Langlands-Arthur Conjectures](#).
- Let  $F \in \mathbf{GField}$ .
- Fix an additive character  $0 \neq \psi = \otimes'_{v \in \Sigma_F} \psi_v : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$ .
- Let  $p \in \mathbf{Prime}$ .
- Let  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p\text{-}\mathbf{LField}$ . Let  $q = \#\kappa$ .
- Let  $\sigma_K \in \mathrm{Gal}_K$  be a geometric Frobenius element.

┘

### 1 Artin $L$ -Functions

References are [Functional Equation of the Artin  $L$ -Functions, Langlands], [[Del73](#), [Neu99](#), [GGP12](#)].

**Def.(31.1.1.1)[ $L$ -Factors for Quasi-characters].** For  $v \in \Sigma_F$  and quasi-character  $\chi_v$  of  $F_v$ , the local  $L$ -factors for  $v \in \Sigma_F$  and  $s \in \mathbb{C}$  are defined to be

$$L(\chi_v; s) = \begin{cases} (1 - \chi_v(\varpi) \|v\|^{-s})^{-1} & , v \in \Sigma_F^{\mathrm{fin}}, \chi_v \text{ unramified} \\ 1 & , v \in \Sigma_F^{\mathrm{fin}}, \chi_v \text{ ramified} \\ L_{\mathbb{R}}(s + \nu + \epsilon)(\text{12.6.1.19}) & , v \in \Sigma_F^{\mathbb{R}}, \chi_v(x) = |x|^\nu \mathrm{sgn}(x)^\epsilon, \nu \in \mathbb{C}, \epsilon \in \{0, 1\} \\ L_{\mathbb{C}}(s + \nu + \frac{|k|}{2})(\text{12.6.1.19}) & , v \in \Sigma_F^{\mathbb{C}}, \chi_v(x) = |x|^\nu e^{ik \arg(x)}, \nu \in \mathbb{C}, k \in \mathbb{Z} \end{cases}.$$

And the local  $\epsilon$ -factors are defined to be

$$\epsilon_v(\chi_v, \psi_v; s) = \begin{cases} \text{Cf. [Tate's Thesis] ?} & , v \in \Sigma_F^{\mathrm{fin}} \\ i^\epsilon & , v \in \Sigma_F^{\mathbb{R}} \\ i^{|k|} & , v \in \Sigma_F^{\mathbb{C}} \end{cases}$$

By([24.6.2.1](#)), these are all the representations.

┘

**Thm. (31.1.1.2)[Root Numbers for Weil Representations, Langlands/Deligne[Del73]].** Let  $\mathbf{1} \neq \psi_K \in K^\vee$ , then there exists a unique way to assign to each separable extension  $L/K$  a complex number  $\lambda(L/K, \psi_K)$  and to each  $(\rho, V) \in \mathcal{R}\text{ep}^{\text{fd}}(W_L)$  a number  $\epsilon(\rho, \psi_K \circ \text{tr}_{L/K})$  s.t.

- If  $\chi$  is a quasi-character of  $W_L$ , then  $\epsilon(\chi, \psi_K \circ \text{tr}_{L/K})$  agrees with local factors defined in (31.1.1.1).
- For  $\rho_1, \rho_2 \in \mathcal{R}\text{ep}^{\text{alg,fin}}(W_L)$ ,

$$\epsilon(\rho_1 \oplus \rho_2, \psi_K \circ \text{tr}_{L/K}) = \epsilon(\rho_1, \psi_K \circ \text{tr}_{L/K}) \epsilon(\rho_2, \psi_K \circ \text{tr}_{L/K})$$

- If  $K \subset_{\text{fin}} L$  and  $\rho \in \mathcal{R}\text{ep}^{\text{alg,fin}}(W_L)$ , then

$$\epsilon(\text{Ind}_K^L \rho, \psi_K) = \lambda(L/K, \psi_K)^{\dim V} \epsilon(\rho, \psi_K \circ \text{tr}_{L/K}).$$

┘

*Proof:* ? Cf.[Functional Equation for Artin  $L$ -Functions, Langlands] [Paskunas. Unicity of types for supercuspidal representations of  $\text{GL}_N$ ] and [Tate, Number Theoretical Background](not compatible).  
□

**Def. (31.1.1.3)[ $L, \epsilon$ -Factors for Weil-Deligne Representations, Deligne].** Let  $\mathbf{1} \neq \psi_K \in K^\vee$  and  $\rho = (\rho_0, N) \in \mathfrak{w}\mathfrak{d}(W_K)$ , define

- the **conductor(Weil-Deligne Representation)**

$$c(\rho) = c(\rho_0) + \dim(V^{I_K} / \ker(N)^{I_K}),$$

where  $c(\rho_0) \in \mathbb{N}$  is the Artin conductor of  $\rho_0$  (25.1.2.5).

- the local Artin  $L$ -factor

$$L(\rho; s) = \det\left(1 - q^{-s} \sigma_K | \ker(N)^{I_K}\right)^{-1}.$$

- the local  $\epsilon$ -factor

$$\epsilon(\rho, \psi_K; s) = q^{-(c(\rho) + n(\psi_K) \dim V)(s - \frac{1}{2})} \det\left(-q^{-s} \sigma_K | V^{I_K} / \ker(N)^{I_K}\right) \epsilon(\rho_0, \psi_K) \text{ (31.1.1.2)}.$$

Then they satisfy the following:

- For a quasi-character  $\chi$  of  $W_K$ ,  $\epsilon(\chi, \psi_K; s)$  agrees with local factors defined in (31.1.1.1).
- If  $K \subset_{\text{fin}} L$  and  $\rho \in \mathfrak{w}\mathfrak{d}(W_L)$ , then

$$\epsilon(\text{Ind}_K^L \rho, \psi_K; s) = \lambda(L/K, \psi_K)^{\dim V} \epsilon(\rho, \psi_K \circ \text{tr}_{L/K}; s) \text{ (31.1.1.2)}.$$

And we denote  $w(\rho, \psi_K) = \epsilon(\rho, \psi_K, 1/2)$ , called the **Weil-Deligne root numbers**.

┘

*Proof:* ? [Getz]P321 defined some different  $L$ -factor. □

**Cor. (31.1.1.4).**  $\epsilon(\rho(s), \psi_K; 0) = q^{-(c(\rho) + n(\psi_K) \dim V)s} \epsilon(\rho, \psi; 0)$ . ┘

**Def. (31.1.1.5)[Archimedean  $L$ -Factors].**

- For  $v \in \Sigma_F^{\mathbb{C}}$  and  $\rho = \chi_1 \oplus \dots \times \chi_n \in \mathcal{R}\text{ep}^{n, \text{ss}}(W_{F_v})$  where  $\sigma_i$  are quasi-characters of  $\mathbb{C}^\times$ , define

$$L_v(\rho; s) = \prod_i L(\sigma_i; s), \quad \epsilon_v(\rho, \psi_v; s) = \prod_i \epsilon_v(\rho, \psi_v; s) \text{ (31.1.1.1)}.$$

- For  $v \in \Sigma_F^{\mathbb{R}}$  and  $\rho = \sigma_1 \oplus \dots \oplus \sigma_n \in \text{Rep}^{n, \text{ss}}(W_{F_v})$  where  $\dim \sigma_i \leq 2$ , define

$$L_v(\rho; s) = \prod_i L_v(\sigma_i; s), \quad \epsilon_v(\rho, \psi_v; s) = \prod_i \epsilon_v(\sigma_i, \psi_v; s) \quad (31.1.1.1),$$

where

- If  $\sigma_i = \chi_{\nu, \epsilon}$  (24.6.2.4), define  $L_v(\sigma_i; s), \epsilon_v(\sigma_i; s)$  as in (31.1.1.1).
- If  $\sigma_i = \text{Ind}_{W_{\mathbb{C}}}^{W_{\mathbb{R}}} \chi_{\nu, k}$  (24.6.2.5), define  $L_v(\sigma_i; s) = L(\chi_{\nu, m}; s) = L_{\mathbb{C}}(s + \nu + \frac{k}{2})$  (31.1.1.1), and  
How about  $\epsilon$ -factors?

By (24.6.2.5), these are all representations of  $W_{F_v}$  with semisimple images.

┘

### (Conjugate)Self-Dual Weil-Deligne Representations

**Prop. (31.1.1.6) [(Conjugate)Self-Dual Weil Deligne Root Numbers].**

- If  $V \in \mathfrak{wd}(W_K)$  is self-dual with  $\det(V) = 1$ , then  $w(V; \psi_K)$  is independent of  $\psi_K$ , and satisfies  $\epsilon(V)^2 = 1$ .
- If  $V \in \mathfrak{wd}(W_K)$  is conjugate-self-dual and  $\psi_K^\sigma = \psi_K^{-1}$ , then  $\epsilon(V, \psi_K)^2 = 1$ . And if  $V = W + W^{\sigma_V}$ , then  $\epsilon(V, \psi_K) = 1$ .

┘

*Proof:* Cf. [GGP12] Prop 5.1.?

□

**Prop. (31.1.1.7) [(Conjugate)Orthogonal Weil Deligne Root Numbers].**

- If  $(\rho, V) \in \mathfrak{wd}(W_K)$  is orthogonal with  $\det(V) = 1$ , then  $w(V; \psi_K)$  is independent of  $\psi_K$ , and satisfies  $\epsilon(V)^2 = 1$ . And  $\epsilon(V) = 1$  iff  $\rho : W \rightarrow \text{SO}(V)$  lifts to a spin representation  $\rho : W \rightarrow \text{Spin}(V)$ .
- If  $V \in \mathfrak{wd}(W_K)$  is conjugate-orthogonal and  $\psi_K^\sigma = \psi_K^{-1}$ , then  $w(V; \psi_K)$  is independent of  $\psi_K$ , and satisfies  $\epsilon(V, \psi_K) = 1$ .

┘

*Proof:* Cf. [GGP12] Prop 5.2.?

□

### Artin L-Functions

**Def. (31.1.1.8) [Local Artin L-Factors, Artin [Art31]].** For  $L/F \in \text{GalField}$  and  $\rho \in \text{Rep}^{\text{fd}}(\text{Gal}(L/F))$  with character  $\chi$ , the local  $L$ -factors (31.1.1.3) are:

$$L_v(F, \rho; s) = \begin{cases} L(\rho|_{W_{F_v}}; s) & (31.1.1.3) \quad , v \in \Sigma_F^{\text{fin}} \\ L_{\mathbb{C}}(s)^{\chi(1)} & , v \in \Sigma_F^{\mathbb{C}} \\ L_{\mathbb{R}}(s)^{n_+} \cdot L_{\mathbb{R}}(s+1)^{n_-} & , v \in \Sigma_F^{\mathbb{R}} \quad (31.1.1.1) \end{cases}$$

where in the case  $v \in \Sigma_F^{\mathbb{R}}$ , choose  $w|v$  and  $c_w \in \text{Gal}(L/F)$  be the generator of  $\text{Gal}(L_w/F_v)$ , then

$$n_+ = \frac{\chi(1) + \chi(c_w)}{2}, \quad n_- = \frac{\chi(1) - \chi(c_w)}{2}.$$

┘

**Prop. (31.1.1.9) [Functoriality of Artin  $L$ -Factors].** Let  $F \in \mathbf{GField}$  and  $v \in \Sigma_F$ , then the Artin  $L$ -factors (31.1.1.8) satisfy the following functorial properties:

- For  $L/F \subset L'/F \in \mathbf{GalField}^{\text{fin}}$  and  $\rho \in \mathbf{Rep}^{\text{fd}}(\mathbf{Gal}(L/F))$ , we have  $L_v(L'/F, \rho; s) = L_v(L/F, \rho; s)$ . In particular, the  $L$ -function doesn't depend on  $L$ , as long as  $\rho$  is trivial on  $\mathbf{Gal}_L$ .
- $L_v(F, \mathbf{1}; s) = L_v(\mathbf{1}; s)$  (31.1.1.1).
- For  $\rho, \rho' \in \mathbf{Rep}^{\text{fd}}(\mathbf{Gal}_F)$ ,  $L_v(F, \rho \oplus \rho'; s) = L_v(F, \rho; s) L_v(F, \rho'; s)$ .
- If  $F \subset_{\text{fin}} M$ ,  $\rho \in \mathbf{Rep}^{\text{fd}}(\mathbf{Gal}_M)$ , then  $L_v(F, \text{Ind}_M^F(\rho); s) = \prod_{w|v} L_w(M, \rho; s)$ .

□

*Proof:* 1, 2, 3 are clear.

For 4: Let  $L$  be a finite extension of  $M$  s.t.  $\rho$  is trivial on  $\mathbf{Gal}_L$ . Denote  $G = \mathbf{Gal}(L/F)$ ,  $H = \mathbf{Gal}(L/M)$ , and for any  $\mathfrak{p} \in \Sigma_F^{\text{fin}}$ , let  $\{\mathfrak{q}_1, \dots, \mathfrak{q}_r\} = S(\mathfrak{p}) \subset \Sigma_M^{\text{fin}}$ , and take  $\mathfrak{P}_i \in \Sigma_L^{\text{fin}}$  over  $\mathfrak{q}_i$ . Let  $G_i, I_i$  be the decomposition and inertia groups of  $\mathfrak{P}_i/\mathfrak{p}$ , then  $H_i = G_i \cap H$ ,  $I'_i = I_i \cap H$  are the decomposition and inertia groups of  $\mathfrak{P}_i/\mathfrak{q}_i$ . Denote  $f_i$  the inertia degree of  $\mathfrak{q}_i/\mathfrak{p}$ . Let  $(\rho, W) \in \mathbf{Rep}^{\text{fd}}(\mathbf{Gal}(L/M))$ ,  $(\rho', V) = \text{Ind}_F^M(\rho) \in \mathbf{Rep}(\mathbf{Gal}(L/F))$ , then it suffices to show that

$$\det(1 - \varphi_{\mathfrak{P}_1/\mathfrak{p}} T | V^{I_1}) = \prod_{i=1}^r \det(1 - \varphi_{\mathfrak{P}_i/\mathfrak{p}}^{f_i} T^{f_i} | W^{I'_i}).$$

If  $\mathfrak{P}_i = \tau_i^{-1}(\mathfrak{P}_1)$ , then we can take  $\varphi_{\mathfrak{P}_i/\mathfrak{p}} = \tau_i^{-1} \varphi_{\mathfrak{P}_1/\mathfrak{p}} \tau_i$ ,  $I_i = \tau_i^{-1} I_1 \tau_i$ , then

$$\det(1 - \varphi_{\mathfrak{P}_i/\mathfrak{p}}^{f_i} T^{f_i} | W^{I'_i}) = \det(1 - \varphi_{\mathfrak{P}_1/\mathfrak{p}}^{f_i} T^{f_i} | W^{I_1 \cap \tau_i H \tau_i^{-1}}),$$

and  $f_i = (G_i : H_i I_i) = (G_1 : (G_1 \cap \tau_i H \tau_i^{-1}) I_1)$ .

Let  $\sigma_{ij}$  be a representative of  $G_1/(G_1 \cap \tau_i H \tau_i^{-1})$ , then because  $G/H = \cup_i G_i/H_i$ ,  $\{\sigma_{ij} \tau_i\}_{ij}$  is a representative of  $G/H$ . Then

$$V = \oplus_i \bigoplus_j \sigma_{ij} \tau_i W,$$

and  $V_i = \bigoplus_j \sigma_{ij} \tau_i W = \text{Ind}_{G_1 \cap \tau_i H \tau_i^{-1}}^{G_1}(\tau_i W)$  is a  $G_1$ -module. So

$$\det(1 - \varphi_{\mathfrak{P}_1/\mathfrak{p}} T | V^{I_1}) = \prod_{i=1}^r \det(1 - \varphi_{\mathfrak{P}_1/\mathfrak{p}} T | V_i^{I_1})$$

Then it suffices to prove for each  $i$ ,

$$\det(1 - \varphi_{\mathfrak{P}_1/\mathfrak{p}} T | V_i^{I_1}) = \det(1 - \varphi_{\mathfrak{P}_i/\mathfrak{p}}^{f_i} T^{f_i} | W^{I'_i}).$$

Notice

$$V_i^{I_1} = \left( \text{Ind}_{G_1 \cap \tau_i H \tau_i^{-1}}^{G_1}(\tau_i W) \right)^{I_1} = \text{Ind}_{H_i}^{G_i} W^{I'_i} \text{ (24.1.6.46)} = \bigoplus_{i=0}^{f_i-1} \varphi_{\mathfrak{P}_i/\mathfrak{p}}^i W^{I'_i},$$

so the assertion follow easily from this. □

**Def. (31.1.1.10) [Artin  $L$ -Functions].** Let  $L/F \in \mathbf{GField}$  be Galois, and  $(\rho, V) \in \mathcal{R}\mathrm{ep}(\mathrm{Gal}(L/F))$ , the **Artin  $L$ -function** of  $\rho$  is defined to be the Euler product of local  $L$ -factors (31.1.1.8)

$$L(F, \rho; s) = \prod_{v \in \Sigma_F^{\mathrm{fin}}} L_v(F, \rho; s).$$

This  $L$ -function also only depends on  $F$  but not  $L$ , as we will see in (31.1.1.11).

And also define the **completed Artin  $L$ -function** as the function

$$\Lambda(F, \rho; s) = \prod_{v \in \Sigma_F} L_v(F, \rho; s).$$

┘

**Prop. (31.1.1.11) [Functoriality of Artin  $L$ -Functions].** Let  $F \in \mathbf{GField}$ , then the  $L$ -functions satisfy the following functorial properties:

- $L(F, \mathbf{1}; s) = \zeta(F; s)$  (35.5.3.1).
- For  $\rho, \rho' \in \mathcal{R}\mathrm{ep}^{\mathrm{fd}}(\mathrm{Gal}_F)$ ,  $L(F, \rho \oplus \rho'; s) = L(F, \rho; s)L(F, \rho'; s)$ .
- If  $F \subset_{\mathrm{fin}} E$ ,  $\rho \in \mathcal{R}\mathrm{ep}^{\mathrm{fd}}(\mathrm{Gal}_E)$ , then  $L(F, \mathrm{Ind}_E^F(\rho); s) = L(E, \rho; s)$ .
- If  $F \subset_{\mathrm{fin}} E$ ,  $\rho \in \mathcal{R}\mathrm{ep}^{\mathrm{fd}}(\mathrm{Gal}_F)$ , then  $L(E, \mathrm{res}_E^F(\rho); s) = L(F, \rho \otimes \mathrm{Ind}_E^F(\mathbf{1}))$ .

┘

*Proof:* These follow from (31.1.1.9). For item 4, use the projection formula  $\mathrm{Ind}_E^F(\mathrm{res}_E^F \rho) \cong \rho \otimes \mathrm{Ind}_E^F(\mathbf{1})$  (24.1.2.14). □

**Cor. (31.1.1.12).** For  $L/F \in \mathbf{GalField}^{\mathrm{fin}}$ ,

$$\zeta(L; s) = \zeta(F; s) \cdot \left( \prod_{\mathbf{1} \neq \rho \in \mathrm{Irr}(\mathrm{Gal}(L/F))} L(F, \rho; s)^{\chi_\rho(\mathbf{1})} \right).$$

┘

**Prop. (31.1.1.13) [Artin  $L$ -Functions and Weber  $L$ -Functions].** Let  $L/F \in \mathbf{GField}$  be Abelian with conductor  $\mathfrak{f}$ , and  $\chi \in (\mathrm{Gal}(L/F))^\vee$ . Composing with the Artin symbols (19.7.4.26), we get a Hecke character  $\tilde{\chi} : \mathbf{C}_F/\mathbf{C}^\mathfrak{f} \cong J^\mathfrak{f}/P^\mathfrak{f} \rightarrow \mathrm{Gal}(L/F) \xrightarrow{\chi} \mathbf{C}^\times$ , which is a Dirichlet character. Then we have

$$L(F, \chi; s) = L(F, \tilde{\chi}; s) \prod_{v \in S} \frac{1}{1 - \|v\|^{-s} \chi(\sigma_v)} \quad (35.5.2.2),$$

where  $S = \{\mathfrak{p} \in S_f(\mathfrak{f}) \mid \chi(I_{\mathfrak{p}}) = 1\}$ . Moreover, if  $\chi$  is injective, then  $S = \emptyset$ . ? ┘

*Proof:*  $S$  are the primes that are ramified in  $L/F$  and  $\mathbf{C}(\chi)^{I_{\mathfrak{p}}} \neq 0$ . □

### Functional Equations

**Prop. (31.1.1.14) [Functional Equations, Brauer [Bra47]].** By Brauer theorem (24.1.3.36) and (24.1.3.36), any Artin  $L$ -function (31.1.1.10) can be written as products and inverses of Weber  $L$ -functions (35.5.2.2) and f.m.  $L$ -factors. In particular, they can be extended meromorphically to all  $s \in \mathbb{C}$  and satisfies a functional equation (35.5.2.5).

More explicitly, for  $L/F \in \mathbf{GalField}^{\text{fin}}$  and  $(\rho, V) \in \mathcal{R}\text{ep}^{\text{fd}}(\text{Gal}(L/F))$ , the completed Artin  $L$ -function satisfies a functional equation

$$\Lambda(F, \rho; s) = \epsilon(F, \rho; s) \Lambda(F, \rho^\vee; 1-s) \quad (31.1.1.3),$$

where

$$\epsilon(\rho) = w(\rho) \left( |d_F|^{\dim \rho} \|\mathfrak{f}(F, \rho)\| \right)^{1/2-s} \quad (25.1.4.1)$$

where  $|w(\rho)| = 1$ , called the **Artin root number**.

The factorization of  $w(\rho)$  awaits Deligne's work [Del73].? ┘

*Proof:* Cf.[Algebraic Number Theory, Neukirch].? □

**Conj. (31.1.1.15) [Holomorphy of Artin  $L$ -Functions, Artin].** If  $F \in \mathbf{GField}$  and  $(\rho, V) \in \mathcal{R}\text{ep}^{\text{fd}}(\text{Gal}_F)$  satisfies  $V^{\text{Gal}_F} = 0$ , then  $L(F, \rho; s)$  is an entire function w.r.t.  $s \in \mathbb{C}$ . ┘

*Proof:* □

**Thm& Conj.Cor. (31.1.1.16).** By (29.5.4.6), the Artin holomorphy conjecture is implied by the Artin-Langlands conjecture (29.3.5.4). In particular, by (29.3.5.5) and (29.3.5.9), it is true if either of the following holds:

- $\dim \rho = 1$ .
  - $\text{Im}(\rho)$  is nilpotent.
  - $F = \mathbb{Q}$ ,  $\dim \rho = 2$  and  $\text{Im}(\rho)$  is solvable.
  - $F = \mathbb{Q}$ ,  $\dim \rho = 2$  and  $\text{Im}(\rho) \cong \text{Alt}_5$ , and under mild conditions.
  - $F$  is totally real and  $\rho$  is odd.
- ┘

### Artin-Weil $L$ -Functions

Cf.[On the Functional Equation of the Artin  $L$ -Functions, Langlands].

## 2 Galois $L$ -Functions

**Def. (31.1.2.1) [Galois  $L$ -Functions].** Given an embedding  $\iota : \overline{\mathbb{Q}_p} \hookrightarrow \mathbb{C}$ , for any  $V \in \mathcal{R}\text{ep}_{\overline{\mathbb{Q}_p}}^{\text{fd}}(\text{Gal}_F)$ , define the global  $L$ -function as

$$L(\iota V; s) = \prod_{v \in \Sigma_F^{\text{fin}}} L(\iota \text{WD}_p(V); s) \quad (31.1.1.3). \quad \text{?}$$

In particular, for  $v \in S_F(p)$ , the local factor is

$$L_v(\iota V; s) = \det \left( \mathbf{1} - \sigma_{\text{crys}}^{\deg v} \|v\|^{-s} \Big| D_{\text{crys}}(V|_{\text{Gal}_{F_v}}) \right)^{-1}.$$

┘

**Prop. (31.1.2.2).** By Chebotarev density theorem,  $L(\iota V; s)$  determines  $\text{WD}_p(V)$  and thus  $V$  up to  $F$ -semisimplification. ┘



*Proof:* This follows from (29.3.1.12).  $\square$

**Prop. (31.1.2.3).** Situation as in (31.1.2.1),

- $L(V(n); s) = L(V; s + n)$ .
- If  $F/F_0 \in \mathbf{NField}$ ,  $W = \text{Ind}_{F/F_0}(V)$ , then for any  $\emptyset \subset_{\text{fin}} S_0 \subset \Sigma_{F_0}^{\text{fin}}$  with  $S = S_F(S_0)$ , we have  $L_S(V; s) = L_{S_0}(W; s)$ .
- If  $\text{Ram}_F(V) \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$  and  $V$  is  $S$ -pure of weight  $w$ , then  $L_S(V; s)$  converges absolutely for  $\text{Re } s > w/2 + 1$ , and  $L(V; s)$  is a meromorphic function without zeros for  $\text{Re } s > w/2 + 1$ .
- If  $V$  is pure of weight  $w$ , then  $L(V; s)$  is meromorphic for  $\text{Re } s > w/2 + 1$  and has no zeros there.  $\lrcorner$

*Proof:* 1 is easy.

2:  $\color{red}{?}$  Cf. [Fontaine-Perrin-Riou].

3: Compare with the Dedekind zeta function (29.5.2.1), as in (29.5.2.2).

4: clear.  $\square$

**Conj. (31.1.2.4) [Holomorphy].** Situation as in (31.1.2.1), if  $V$  is geometric and pure of weight  $w$ , then it is totally pure. In particular, it is holomorphic for  $\text{Re } s > w/2 + 1$ .  $\lrcorner$

*Proof:*  $\square$

**Conj. (31.1.2.5) [Meromorphic Extension].** If  $V$  is geometric and pure of weight  $w$ , then

- $L(V; s)$  admits a meromorphic continuation to all  $s \in \mathbb{C}$ , and essentially bounded on vertical strips.
- $L(V; s) \neq 0$  for  $\text{Re}(s) \geq w/2 + 1$ .
- If  $V$  is irreducible, then  $L(V; s)$  has no poles, except when  $V \cong \mathbb{Q}_p(n)$ , then it has a simple pole at  $s = -n + 1$ .  $\lrcorner$

*Proof:*  $\square$

**Remark (31.1.2.6).** This is true when  $V$  is automorphic, by (29.5.4.6) (29.5.3.3) and (35.4.4.1).  $\lrcorner$

**Conj. (31.1.2.7) [Grand Riemann Hypothesis].** If  $V$  is geometric and pure of weight  $w$ , then for any embedding  $\iota : \overline{\mathbb{Q}}_p \hookrightarrow \mathbb{C}$ ,  $L(\iota V; s)$  (31.1.2.1) has no zeros  $s$  with  $\text{Re } s > (w + 1)/2$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (31.1.2.8) [Completed L-Functions].** If  $V$  is geometric and pure of weight  $w$ , define

$$L_{\infty}(V; s) = \Gamma_{\mathbb{R}}(s - w/2)^{m_{w/2}^{+}} \cdot \Gamma_{\mathbb{R}}(s - w/2 + 1)^{m_{w/2}^{-}} \cdot \prod_{k \in \mathbb{Z}, k < w/2} \Gamma_{\mathbb{C}}(s - k)^{m_k}.$$

and define the **root number at  $\infty$**

$$w_{\infty}(V) = i^{m_{w/2}^{-}} \cdot \left( \prod_{k < w/2} i^{m_k} \right)^{2w+1}$$

and the Artin conductor  $\mathfrak{f}(\rho)$  (25.1.4.1), and the completed  $L$ -function

$$\Lambda(V; s) = L(\iota V; s) L_\infty(V; s).$$

and the **root number**

$$w(\iota \rho) = w_\infty(\rho) \prod_{p \in \mathbf{P}} w(\mathrm{WD}_p(\rho); \psi_p)$$

where  $\iota \psi_p(x) = e^{-2\pi i x}$ . ┘

**Prop. (31.1.2.9).** If  $V \in \mathrm{Rep}_{\overline{\mathbb{Q}}_\ell}(\mathrm{Gal}_{\mathbb{Q}})$  is geometric and pure of weight  $w$ , then

- $L_\infty(V(n); s) = L_\infty(V; s + n)$ .
- $L_\infty(V(n); s)$  has a pole of order  $\sum_{0 \leq k < w/2} m_k + m_{w/2}^+$  at  $s = 0$ .
- If Tate's conjecture (30.1.5.1) holds, then  $m_{w/2}^\pm \geq 0$ , so  $L_\infty(V(n); s)$  has no zeros. ┘

*Proof:* Only 1 deserves a proof: It suffices to prove for  $n = 1$ . If  $w \notin 2\mathbb{Z}$  this is clear. If  $w \in 2\mathbb{Z}$ , notice Tate twists change weights by 2, so it is also odd, and  $(-1)^{w'/2} = -(-1)^{w/2}$ . But also the twist changes the action of  $c$  by  $-1$  in (30.1.5.3), so the assertion follows. □

**Conj. (31.1.2.10) [Functional Equations].** If  $V$  is geometric and pure of weight  $w$ , then

$$\Lambda(V; s) = w(V) N(V)^{(w+1)/2-s} \Lambda(V^\vee; 1-s).$$

where  $N(V)$  is the conductor of  $V$ ? ┘

*Proof:* □

**Remark (31.1.2.11).** This is true for  $V$  automorphic, by Jacquet-Shalika method. ┘

**Conj. Cor. (31.1.2.12).** For a geometric representation polarized of weight  $w$  (30.1.3.28), there should be a functional equation

$$\Lambda(V; 1 + w - s) = \varepsilon(V; s) \Lambda(V; s).$$

*Proof:* □

### 3 Hasse-Weil $L$ -Functions

References are [Zeta Functions in Algebraic Geometry, Mustata].

#### Zeta-Functions over Finite Fields

**Def. (31.1.3.1) [Zeta-Functions].** Let  $q \in p^{\mathbb{Z}^+}$  and  $X \in \mathrm{Sch}^{\mathrm{ft}}/\mathbb{F}_q$ , then its **zeta-function** is defined to be the power series

$$Z(X; T) = \exp \left( \sum_{n \geq 1} \#X(\mathbb{F}_{q^n}) \frac{T^n}{n} \right) \in \mathbb{Q}[[T]].$$

Notice that we can get the information of number of rational points of  $X$  by the formula

$$\#X(\mathbb{F}_{q^n}) = \frac{1}{(n-1)!} \frac{\partial^n}{\partial T^n} \log Z(X; T)|_{T=0}.$$

┘

**Prop. (31.1.3.2)[Euler Product].** there is an Euler product formula for  $Z(X; T)$  by (9.6.2.11)(9.6.2.12):

$$Z(X; T) = \prod_{x \text{ closed in } X} \frac{1}{1 - T^{\deg(x)}}.$$

┘

**Thm. (31.1.3.3) [Rationality of Zeta Functions, Dwork].** Let  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}_+}$ , and  $X \in \text{Sch}^{\text{ft}}/\mathbb{F}_q$ , then  $Z(X; T) \in \mathbb{Q}(T)$ . ┘

*Proof:* Cf. [Zeta Functions in Algebraic Geometry, Mustata] ?. Does this follows from Weil conjecture? □

**Thm. (31.1.3.4) [Weil Conjecture, Weil [Wei49], Weil/Deligne [Del74]/Laumon].** Let  $p \in \mathbf{P}$ ,  $q \in p^{\mathbb{Z}_+}$ , and  $X \in \text{SmPrprVar}^n/\mathbb{F}_q$ , then

1. (Riemann Hypothesis):  $Z(X; T)$  is of the form

$$Z(X; T) = \frac{P_1(X; T)P_3(X; T) \dots P_{2n-1}(X; T)}{P_0(X; T)P_2(X; T) \dots P_{2n}(X; T)}$$

s.t.

- $P_0(X; T) = 1 - T$ ,
- $P_{2n}(X; T) = 1 - q^n T$ ,
- For  $0 \leq i \leq 2n$ ,  $P_i(X; T) \in \mathbb{Z}[T]$ ,
- For  $0 \leq i \leq 2n$ ,  $P_i(X; T) = \prod_{j \leq k} (1 - \alpha_{ij} T)$ , where each  $\alpha_{ij} \in \text{Weil}(q^{i/2})$  is a  $q$ -Weil number of weight  $i$  (19.3.1.16).
- $\deg(P_i(X; T))$  is called the  $i$ -th **Betti number** of  $X$ , and  $\deg P_i(X; T) = \deg P_{2n-i}(X; T)$ .

2. (Functional Equation):

$$Z(X; \frac{1}{q^n T}) = \pm q^{n\chi/2} T^\chi Z(X; T),$$

where  $\chi$  is a Euler character  $\Delta \cdot \Delta = \sum_{i=0}^{2n} (-1)^i \deg P_i(X; T)$ . Notice that  $n\chi/2 \in \mathbb{Z}$ , and the sign is positive if  $n$  is odd.

3. If  $R \subset \mathbb{C}$  is a f.g.  $\mathbb{Z}$ -algebra,  $\mathfrak{p} \in \text{Spec}(R)$  s.t.  $R/\mathfrak{p} \cong \mathbb{F}_q$ , and there exists  $\mathcal{X} \in \text{SmPrpr}/R$  s.t.  $\mathcal{X}_{\kappa(\mathfrak{p})} \cong X$ , then  $\deg(P_i(X; T))$  equals the  $i$ -th Betti number of  $\mathcal{X}_{\mathbb{C}}^{\text{an}}$ .
4.  $\chi(X) = \Delta_X \cdot \Delta_X$ .

┘

*Proof:* 1: It follows from trace formula (9.6.2.13) that

$$Z(X; T) = \frac{P_1(X; T)P_3(X; T) \dots P_{2n-1}(X; T)}{P_0(X; T)P_2(X; T) \dots P_{2n}(X; T)} = \frac{P(T)}{Q(T)} \in \mathbb{Q}_\ell(T),$$

where the roots  $\alpha_{ij}$  of  $P_i(X; T)$  are algebraic integers satisfying  $|\iota(\alpha_{ij}^{-1})| = q^{-i/2}$  by Deligne's purity theorem (9.6.6.5).

Then  $Z(X; T) \in \mathbb{Q}[[T]] \cap \mathbb{Q}_\ell(T) = \mathbb{Q}(T)$  by (10.7.1.19). Thus  $P_i(X; T) \in \mathbb{Q}[T]$  because it is in  $\overline{\mathbb{Q}}[T]$  and also it is stable under  $\text{Gal}_{\mathbb{Q}}$ , as their roots are distinguished by their  $\iota$ -values.

Moreover, by Euler product(31.1.3.2),  $Z(X; T) \in \mathbb{Z}[[T]]$ , so it follows from(10.7.1.20) that  $P(T), Q(T) \in \mathbb{Z}[T]$ .

Notice by Poincaré duality(9.4.6.48),  $H_{\text{ét}}^{2n}(X; \mathbb{Q}_\ell(n)) \cong H_{\text{ét}}^0(X; \mathbb{Q}_\ell)^\vee$ , thus  $F_X^*$  acts on  $H_{\text{ét}}^0(X; \mathbb{Q}_\ell)$  by id and acts on  $H_{\text{ét}}^{2n}(X, \mathbb{Q}_\ell)$  by  $q^n$ . Thus  $P_0(X; T) = 1 - T$ ,  $P_{2n}(X; T) = 1 - q^n T$ .

2: Apply(3.5.15.1) to  $H^i = H_{\text{ét}}^i(X; \mathbb{Q}_\ell)$ , trace map and perfect Poincaré pairings(9.4.6.48) and  $\varphi_i = q^{-i/2} F_X^*$ . Notice  $\varphi_{2n} = \text{id}$  by item2. Now(3.5.15.1) says  $\varphi_i^{-1} = \varphi_{2d-i}^t$ , so

$$\left\{ \frac{q^{i/2}}{\alpha_{ij}} \right\}_j = \left\{ \frac{\alpha_{2n-i,j}}{q^{(2n-i)/2}} \right\}_j$$

where we are counting multiplicity. Thus for  $i \neq n$ , we can assume  $\alpha_{ij}\alpha_{2n-i,j} = q^n$ , and for  $i = n$ , suppose there are  $N_\pm$  many  $\alpha_{n,j} = \pm q^{n/2}$ , and the other  $\alpha_{n,j}$ s come in pair:  $\alpha_{n,j}\alpha_{n,b_n-j} = q^n$ . Then

$$Z(X; \frac{1}{q^n T}) = \prod_i (1 - \alpha_{ij}/(q^n T))^{(-1)^{i+1}} = \prod_i (1 - \frac{1}{\alpha_{2n-i,j} T})^{(-1)^{i+1}} = Z(X; T) \cdot q^{n\chi/2} \cdot T^\chi (-1)^{N_+}.$$

Moreover, if  $n$  is odd, then the intersection form on  $H^n(X)$  is alternating, thus  $\chi$  is even, and  $n\chi/2 \in \mathbb{Z}$ . Moreover, in this case,  $N_+ = 1$  because  $\varphi_n$  is a symplectic matrix.

3: These follow from the trace formula(9.6.2.13) and(9.4.6.36).

4: By(22.1.2.2),

$$\Delta_X \cdot \Delta_X = (\gamma_{X \times X}(\Delta_X), \gamma_{X \times X}(\Delta_X)) = \sum_{i=0}^{2d} \text{tr}(\text{id}|H_{\text{ét}}^i(X)) = \chi(X).$$

□

**Cor. (31.1.3.5) [Invariance of Characteristic Polynomial with  $\ell$ ].** It follows from the proof above that  $P_i$  is determined by  $X$ , and it is also the characteristic polynomial of  $\text{Fr}_{\overline{X}}$  on  $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$  for any  $\ell \in \mathbf{P} \setminus \text{char } k$ , so the latter is independent of  $\ell$  chosen. ┘

### Points Counting

**Thm. (31.1.3.6) [Deligne's Estimate].** ┘

*Proof:* □

**Cor. (31.1.3.7) [Weil Bound].** For  $X \in \text{Cur}^{\text{sm}, \text{prpr}}/\mathbb{F}_q$ , by Weil conjecture(31.1.3.4),

$$|\#X(\mathbb{F}_{q^n}) - q^n - 1| = \left| \sum_j \alpha_{1j}(X) \right| \leq 2g(X)q^{n/2}.$$

┘

**Thm. (31.1.3.8) [Ihara/Drinfeld-Vladut[D-V83]].** For  $p \in \text{Prime}$ ,  $q \in q^{\mathbb{Z}^+}$ , let  $A(q) = \overline{\lim}_{g(X) \rightarrow \infty} \frac{\#X(\mathbb{F}_q)}{g(X)}$ , where the limit is taken over all smooth proper curves over  $\mathbb{F}_q$ . Then  $A(q) \leq \sqrt{q} - 1$ .

And if  $q \in (\mathbb{Z}_+)^2$ , then  $A(q) = \sqrt{q} - 1$ . ┘

*Proof:* Let  $\alpha_{1j}$  as in (31.1.3.4),  $\omega_i = \alpha_{1i}q^{-1/2}$ , then  $|\omega_i| = 1$  and inversion induces an automorphism of  $\{\omega_i\}$ . Then by (31.1.3.7), for any  $r \in \mathbb{Z}_+$ ,

$$N = \#X(\mathbb{F}_q) \leq \#X(\mathbb{F}_{q^r}) = q^r + 1 - q^{r/2} \sum_i \omega_i^r.$$

And notice

$$0 \leq |1 + \omega_i + \dots + \omega_i^r|^2 = (r+1) + \sum_{j=1}^r (r+1-j)(\omega_i^j + \omega_i^{-j}),$$

So

$$\begin{aligned} 2g(X)(r+1) &\geq - \sum_i \sum_{j=1}^r (r+1-j)(\omega_i^j + \omega_i^{-j}) = -2 \sum_{j=1}^r (r+1-j) \left( \sum_i \omega_i^j \right) \\ &\geq N \sum_{j=1}^r (r+1-j) q^{-j/2} - \sum_{j=1}^r (r+1-j)(\alpha^j + \alpha^{-j}) \end{aligned}$$

Thus

$$\frac{N}{g(X)} \leq \left( \sum_{j=1}^r \frac{r+1-j}{r+1} q^{-j/2} \right)^{-1} \cdot \left( 1 + \frac{1}{g} \sum_{j=1}^r \frac{r+1-j}{r+1} (\alpha^j + \alpha^{-j}) \right)$$

Taking limit  $g \rightarrow \infty$ ,

$$\frac{N}{g(X)} \leq \left( \sum_{j=1}^r \frac{r+1-j}{r+1} q^{-j/2} \right)^{-1}.$$

And taking limit  $r \rightarrow \infty$  gives

$$\frac{N}{g(X)} \leq \left( \sum_{j=1}^{\infty} q^{-j/2} \right)^{-1} = \sqrt{q} - 1.$$

For the last assertion, See [Ihara].?

□

**Thm. (31.1.3.9) [Lang-Weil Estimate, Lang-Weil [L-W54]].** For any  $n, r \in \mathbb{N}, d \in \mathbb{Z}_+$ , there exists  $a(n, d, r) \in \mathbb{Z}_+$  s.t. for any  $k \in \mathbf{Field}^{\text{fin}}$  and any  $X \subset \mathbb{P}_k^n \in \mathbf{Var}/k$  of dimension  $r$  and degree  $d$ , we have

$$|\#X(k) - \#k^r| \leq (d-1)(d-2)\#k^{r-\frac{1}{2}} + a(n, d, r)\#k^{r-1}.$$

┘

*Proof:* This can be proved by trace formula and estimating the dimensions of the cohomology groups, Cf. [Bombieri, exponential sums, 2].? or [Mustata, L-functions in algebraic geometry]. □

### Hasse-Weil L-Functions

**Prop. (31.1.3.10) [Hasse-Weil L-Functions].** Let  $F \in \mathbf{GField}$  and  $X \in \mathbf{SmPrprVar}/F$ , the **Hasse-Weil L-function** of  $X$  is defined to be

$$L(X, s) = \prod_{v \in \Sigma_F^0} Z(\mathcal{X}_v, q^{-s}),$$

where for  $v$  s.t.  $\mathcal{X}_v$  has good reduction,  $Z(\mathcal{X}_v, T) = Z(\mathcal{X}_{k_v}, T)$  in (31.1.3.1), and for bad places  $v$ , it needs to be defined otherwise.?

┘

**Prop. (31.1.3.11).** Let  $l_1, \dots, l_N$  be linear forms in  $r$ -variables with rational coefficients, then

$$\sum_{x \in \mathbb{Z}^r} \frac{1}{l_1(x) \dots l_N(x)} \in \mathbb{Q}\pi^N$$

if it is convergent. ┘

*Proof:* □

**Def. (31.1.3.12) [Hasse-Weil L-Functions].** For  $X \in \mathbf{SmPrprVar}/\mathbb{Q}$  and  $\ell \in \mathbf{Prime}$ , there is some  $N \in \mathbb{Z}_+, \ell | N$  s.t.  $X$  has a model  $\mathcal{X}/\mathbb{Z}[\frac{1}{N}]$ . Choose an embedding. Then we can define a partial zeta function

$$\zeta_N(X; s) = \prod_{p \nmid N} \left( \prod_{x \in |X_p|_0} (1 - p^{-s \deg(x)})^{-1} \right),$$

then by Grothendieck-Lefschetz formula,

$$\zeta_N(X; s) = \prod_{i=0}^{2 \dim X} L_N(\iota H^i(X, \overline{\mathbb{Q}}_\ell); s)^{(-1)^i}.$$

Thus it is natural to define the **Hasse-Weil L-function**

$$\zeta(X; s) = \prod_{i=0}^{2 \dim X} L(\iota H^i(X, \overline{\mathbb{Q}}_\ell); s)^{(-1)^i}.$$

and the completed L-function

$$\Lambda(X; s) = \prod_{i=0}^{2 \dim X} \Lambda(\iota H^i(X, \overline{\mathbb{Q}}_\ell); s)^{(-1)^i}.$$

┘

**Example (31.1.3.13).**

- $L(\mathrm{Spec} \mathbb{Q}; s) = \zeta(s)$ .
- For  $E \in \mathbf{EllCur}/\mathbb{Q}$ ,

$$L(E; s) = \frac{\zeta(s)\zeta(s-1)}{L(h^1 E; s)} \quad (31.1.6.2).$$

┘

**Conj. (31.1.3.14) [Functional Equations].** Motivated by Poincaré duality and (31.1.2.10), situation as in (31.1.3.12),  $\zeta(X; s)$  should have a meromorphic extension to the whole plane, and satisfies a functional equation of the form

$$\Lambda(X; s) = \epsilon(s) \Lambda(X, 1 + \dim X - s).$$

┘

*Proof:* □

**Remark (31.1.3.15).** This is true for elliptic curves over  $\mathbb{Q}$ , by [BCDT]. ┘

## 4 Motivic L-Functions

References are [Del79, Zag94].

**Def. (31.1.4.1) [Motivic L-Functions].** Use the fact the Grothendieck motives over a finite field is a Tannakian category ?, we can define  $L$ -functions for any motives over a number field. Such an  $L$ -function is called a **motivic L-function**. ┘

**Conj. (31.1.4.2) [Properties].** The motivic L-functions are conjectured to satisfy the following properties:

**Algebraicity:** There are Dirichlet series expansions:  $L(s) = \sum_{n \geq 1} a_n n^{-s}$  for  $\text{Re}(s)$  large, where  $\{a_n\} \in F$  for some number field  $F$ .

**Euler Product:** There are Euler product expansions:  $L(s) = \prod_{p \in \text{Prime}} \Phi_p(p^{-s})$  where  $\max_{p \in \text{Prime}} \{\deg \Phi_p\} < \infty$ . In particular,  $n \mapsto a_n$  is multiplicative.

**Functional Equation:** There is a  $\gamma$ -factor

$$\gamma(s) = A^s \cdot \prod_i \Gamma\left(\frac{s + m_i}{2}\right)$$

where  $A \in \mathbb{C}^*$ ,  $m_i \in \mathbb{Z}$ , s.t.  $\zeta(s) = \gamma(s)L(s)$  satisfies a functional equation

$$\zeta(s) = w\zeta(h - s)$$

where  $w \in \{\pm 1\}$  called the **sign of functional equation** and  $h \in \mathbb{Z}_+$ . And in this way,  $L(s)$  extends to a meromorphic function on  $\mathbb{C}$  with only f.m. poles.

**Local Riemann Hypothesis:** The zeros of  $\Phi_p(p^{-s})$  lie on the line  $\text{Re}(s) = \frac{k-1}{2}$ .

**Riemann Hypothesis:** The zeros of  $L(s)$  are either integers or lie on the line  $\text{Re}(s) = \frac{k-1}{2}$ .

**Special Values:** A **critical point** for  $L(s)$  is an integer  $m \in \mathbb{Z}$  s.t. neither  $m$  nor  $h - m$  is a pole of  $L(s)$ . Then for a critical point  $m$  for  $L(s)$ , the **critical value**  $L(m) = A(m)\Omega(m)$ , where  $A(m)$  is a reasonable algebraic number and  $\Omega(m) \in \mathbb{P}$  is a reasonable period number.

**Central Special Values:** If  $h = 2m$ ,  $m \in \mathbb{Z}$ , then  $m$  is called a **central value** of  $L(s)$ , and in this case,  $A(m)$  is a square times a simple factor.

**More...** ┘

*Proof:* □

**Conj. (31.1.4.3) [Big Automorphy Conjecture].** Every motivic  $L$ -function comes from a automorphism representation. ┘

*Proof:* □

## 5 0-Dimensional Motives

## 6 Abelian Motives

### Elliptic Curves

**Def. (31.1.6.1) [L-Factors].** ┘

**Def. (31.1.6.2) [Motivic  $L$ -Function of Elliptic Curves].** Let  $F \in \mathbf{GField}$  and  $E \in \mathbf{EllCur}/F$ , the motivic  $h^1$   $L$ -function is defined to be

$$L(h^1 E, s) = \prod_{v \in \Sigma_F^0} L(h^1 E_v, s)$$

where  $L(E_v, s)$  is defined to be

- $L(E_v, s) = (1 - a_v q_v^{-s} + q_v^{1-2s})^{-1}$  as in (31.1.6.1), where  $a_v = q_v + 1 - \#\tilde{E}_v(\kappa_v)$ , if  $E$  has good reduction at  $v$ .
- $(1 - q_v^{-s})^{-1}$  if  $E$  has split multiplicative reduction at  $v$ .
- $(1 + q_v^{-s})^{-1}$  if  $E$  has non-split multiplicative reduction at  $v$ .
- 1 if  $E$  has additive reduction at  $v$ .

Notice that in all cases we have  $L(E_v, 1) = q_v / \#\tilde{E}_{\text{sm}}(\kappa_v)$ . ┘

**Prop. (31.1.6.3) [Tate-Faltings].** If  $E, E' \in \mathbf{EllCur}/\mathbb{Q}$  s.t.  $L_E(s) = L_{E'}(s)$ , then  $E, E'$  are isogenous. ┘

*Proof:* Cf. [Milne, Elliptic Curves, Thm5.4.1]. □

**Thm& Conj. Cor. (31.1.6.4) [Analytic Continuation].** If  $F = \mathbb{Q}$  and  $E \in \mathbf{EllCur}/\mathbb{Q}$ , there exists  $w_E \in \{\pm 1\}$  called the **sign of functional equation of  $E$**  s.t.

$$(L(E), L_{\mathbb{C}}(s), N_E, w_E)$$

is an  $\frac{1}{s}$ -shifted entire analytic  $L$ -function of degree 2 (29.5.1.1). In other words,

$$\Lambda(E, s) = N_E^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s)$$

extends to an entire function, and satisfies the functional equation:

$$\Lambda(E, s) = w_E \Lambda(E, 2 - s).$$

┘

*Proof:* By modularity theorem (30.4.3.7),  $L(E; s) = L(f; s)$  for some normalized Hecke eigenform  $f \in \mathbf{SF}_2(\Gamma_0(N_E))$ . So the assertion follows from (29.5.4.13). Notice  $g = f[w_N]_k = -w_E f$  (30.4.3.9). □

**Remark (31.1.6.5) [Root Numbers].** If  $E$  is semistable, then  $w_E = -\mu(N) a_N(E) = -(-1)^m$ , where  $m$  is the number of finite places over which  $E$  has split multiplicative reduction. ┘

## 7 Zeta Functions of Shimura Varieties

## 8 Tamagawa-Number Conjecture (Bloch-Kato)

References are [A note on Height Pairings, Tamagawa Numbers and the Birch and Swinnerton-Dyer Conjecture, Bloch].



**Conj. (31.1.8.1) [Tamagawa-Number Conjecture].** Let  $F \in \mathbf{NField}$  and  $G \in \mathcal{AlgGrp}/F$ , suppose that  $G(F)$  is discrete in  $G(\mathbf{A}_F)$ , then

$$\tau(G) = \frac{\#\mathrm{Pic}(G)_{\mathrm{tor}}}{\#\mathrm{III}(G)}.$$

Moreover,  $\mathrm{ord}_{s=1} L(s) \leq 0$ , and  $r = 0$  iff  $\mathrm{Vol}(G(\mathbf{A}_F)/G(F)) < \infty$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (31.1.8.2) [Bloch].** The Tamagawa-number conjecture(31.1.8.1) implies that BSDT conjecture.  $\lrcorner$

*Proof:* [Bloch, Height Pairing and].  $\square$

### Equivariant Tamagawa Number Conjecture

## 9 Others

### Invariants of Moduli Spaces

**Prop. (31.1.9.1).** Volumes and Euler characteristics of moduli spaces are often expressible by special values of  $\zeta$ -functions.  $\lrcorner$

*Proof:*  $\square$

**Def. (31.1.9.2) [Witten  $\zeta$ -Functions].** Because of the appearance in physics(Verlinde formula), people are interested in certain moduli spaces of vector bundles on curves.

Witten gave a formula expressing the volume of these moduli spaces in terms of special values of

**Witten  $\zeta$ -functions:**

Let  $\mathfrak{g}$  be a semisimple f.d. Lie algebra, define

$$\zeta_{\mathfrak{g}}(s) = \sum_{\rho \in \mathrm{Rep}(\mathfrak{g})} \frac{\dim(\rho)^s}{?}$$

$\lrcorner$

**Prop. (31.1.9.3).** A consequence of Witten's formula(31.1.9.2) is:

$$\zeta_{\mathfrak{g}}(2m) \in \mathbb{Q}\pi^{2rm}, m \in \mathbb{Z}_+$$

where  $r$  is the number of positive roots of  $\mathfrak{g}$ .  $\lrcorner$

### Multiple $\zeta$ -Values

Cf. [Zag94].

## 31.2 Euler Systems and Kolyvagin Systems

Cf. [Kol90, Rub96, Rub00, Kat04] and [Scholl].

**Notation (31.2.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $F \in \text{NField}$ .
- For each modulus  $\mathfrak{m}$  of  $F$ , let  $F(\mathfrak{m})$  be the maximal  $p$ -extension of  $F$  inside the ray class field  $F_{\mathfrak{m}}$ , and  $\Gamma_{\mathfrak{m}} = \text{Gal}(F(\mathfrak{m})/F(1))$ .
- Fix  $T \in \text{Rep}_{\mathbb{Z}_p}^{\text{fd}}(\text{Gal}_F)$ ,  $V = T \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ ,  $W = V/T = T \otimes_{\mathbb{Z}_p} (\mathbb{Q}_p/\mathbb{Z}_p)$ .
- Suppose  $T$  is unramified outside f.m. places of  $F$ .
- WARNING: notations in this section may subject to change. ?

┘

### 1 Local Cohomologies

This subsection should be moved to somewhere else.

**Prop. (31.2.1.1).** If  $K \in p\text{-Field}$ ,  $\ell \in \text{Prime} \setminus \{p\}$ , and  $V \in \text{Rep}_{\mathbb{Q}_{\ell}}^{\text{fd}}(\text{Gal}_K)$ , then

$$\dim H_{\text{ur}}^1(K, V) = V^{\text{Gal}_K}, \quad \dim \left( H^1(K, V) / H_{\text{ur}}^1(K, V) \right) = \dim H^2(K, V).$$

┘

*Proof:* Cf. [Rubin, P5]. □

**Prop. (31.2.1.2).** Suppose  $K \in \text{Field}$  and  $K_{\infty}/K$  is an infinite  $p$ -extension,  $T \in \text{Rep}_{\mathbb{Z}_p}^{\text{fg}}(\text{Gal}_K)$ , then

$$\varprojlim_{K \subset_{\text{fin}} L \subset K_{\infty}} T^{\text{Gal}_L} = 0,$$

where the transition maps are norm maps. ┘

*Proof:* Define  $T_0 = \varinjlim_{K \subset_{\text{fin}} L \subset K_{\infty}} T^{\text{Gal}_L} \subset T$ , so  $T_0$  is also f.g. over  $\mathbb{Z}_p$ . Then  $T_0 = T^{\text{Gal}_{L_0}}$  for some  $K \subset_{\text{fin}} L_0$ . Then

$$\varprojlim_{K \subset_{\text{fin}} L \subset K_{\infty}} T^{\text{Gal}_L} = \varprojlim_{L_0 \subset_{\text{fin}} L \subset K_{\infty}} T_0 = 0.$$

□

**Prop. (31.2.1.3).** If  $\ell \in \text{Prime} \setminus \{p\}$ ,  $\mathbb{Q}_{\ell} \subset_{\text{fin}} K$ , and  $K_{\infty}$  is the unique  $\mathbb{Z}_p$ -extension of  $K$ . Suppose  $\{c_L\} \in \varprojlim_{K \subset_{\text{fin}} L \subset K_{\infty}} H^1(L, T)$ , then for any  $L$ ,  $c_L \in H_{\text{ur}}^1(L, T)$ . ┘

*Proof:*  $K_{\infty}/K$  is unramified ?, so for any  $K \subset_{\text{fin}} L \subset K_{\infty}$ , there is an exact sequence

$$0 \rightarrow H_{\text{ur}}^1(L, T) \rightarrow H^1(L, T) \rightarrow H^1(I_K, T)^{\text{Gal}_K}.$$

But  $H^1(I_K, T)$  is f.g. over  $\mathbb{Z}_p$  by [Rubin, B.2.7] ?, so taking limit of these exact sequences, we get

$$\varprojlim_{K \subset_{\text{fin}} L \subset K_{\infty}} H_{\text{ur}}^1(L, T) = \varprojlim_{K \subset_{\text{fin}} L \subset K_{\infty}} H^1(L, T)$$

by (31.2.1.2). □

**Cor. (31.2.1.4).** Let  $F_\infty$  be an  $\mathbb{Z}_p^d$ -extension of  $F$ ,  $F \subset_{\text{fin}} L \subset F_\infty$ , and  $v \in \Sigma_L^p$  has infinite decomposition group in  $\text{Gal}(F_\infty/F)$ . Suppose

$$\{c_L\} \in \varprojlim_{F \subset_{\text{fin}} L \subset F_\infty} H^1(L, T),$$

then  $(c_L)_v \in H_{\text{ur}}^1(F_\lambda, T)$ . In particular, this applies to  $L = F$ .  $\lrcorner$

*Proof:* Cf.[Rubin, P155].  $\square$

**Cor. (31.2.1.5).** If  $\Sigma \subset \Sigma_F$  is a finite set of places containing all primes of  $F$  s.t.  $T$  is ramified, and all primes over  $p$ , and all primes whose decomposition group in  $\text{Gal}(F_\infty/F)$  is finite, then

$$\varprojlim_{F \subset_{\text{fin}} L \subset F_\infty} H^1(L, T) = \varprojlim_{F \subset_{\text{fin}} L \subset F_\infty} H^1(F_\Sigma/L, T).$$

$\lrcorner$

*Proof:* Cf.[Rubin, P155].  $\square$

## 2 Selmer Groups

References are [B-K90, F-PR94].

**Def. (31.2.2.1) [Selmer Local Conditions].** For  $F \in \mathbf{NField}$ ,  $v \in \Sigma_F^p$ , denote

$$H_f^1(F_v, V) = H_{\text{ur}}^1(F_v, V) = \ker \left( H^1(F_v, V) \rightarrow H^1(F_v^{\text{ur}}, V) \right) = H^1(F_v^{\text{ur}}/F_v, V^{I_v}) \quad (9.8.1.14),$$

And for places of  $F$  over  $p$ , fix a choice of invariant subspaces  $H_s^1(F, V) \subset H^1(F, V)$ .

Then we can define for any  $v \in \Sigma_{\text{cycl}_{p,n}(\mathbb{Q})}$ ,

$$H_f^1(F_v, W) \subset H^1(F_v, W), \quad H_f^1(F_v, T) \subset H^1(F_v, T).$$

the image and inverse image of  $H_f^1(F_v, V)$  under the maps on cohomologies induced by the exact sequence

$$0 \rightarrow T \rightarrow V \rightarrow W \rightarrow 0.$$

Next, for any  $v \in \Sigma_F$ , denote

$$H_s^1(F_v, T) = H^1(F_v, T)/H_f^1(F_v, T)$$

$$\text{loc}_v^s(T) : H^1(F, T) \rightarrow H_s^1(F_v, T)$$

and similarly for  $W$ .  $\lrcorner$

**Prop. (31.2.2.2).** For  $v \in \Sigma_F^p$ ,

- $H_f^1(F_v, W) = H_{\text{ur}}^1(F_v, W)_{\text{div}}$ .
- $H_{\text{ur}}^1(F_v, T) \subset H_f^1(F_v, T)$  has finite index, and  $H_s^1(F_v, T)$  is torsion-free.
- Let  $\mathcal{W} = W^{I_v}/(W^{I_v})_{\text{div}}$  be finite, then there are natural isomorphisms

$$H_f^1(F_v, W)/H_{\text{ur}}^1(F_v, W) \cong \mathcal{W}/(\text{Frob}_v - 1)W, \quad H_{\text{ur}}^1(F_v, T)/H_f^1(F_v, T) \cong \mathcal{W}^{\text{Frob}_v=1}.$$

- If  $T$  is unramified at  $v$ , then

$$H_f^1(F_v, W) = H_{\text{ur}}^1(F_v, W), \quad H_{\text{ur}}^1(F_v, T) = H_f^1(F_v, T)$$

⌋

*Proof:* 1, 2, 4 follow from 3. For 3, Cf.[Rubin, P6] ?.

□

**Def. (31.2.2.3) [Selmer Groups].** Situation as in (31.2.2.1), define

$$\text{Sel}(F, W) = \ker \left( H^1(F, W) \xrightarrow{\text{loc}^s(W)} \bigoplus_{v \in \Sigma_F} H_s^1(F_v, W) \right).$$

Moreover, for  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$ , we can define

$$\text{Sel}^S(F, W) = \ker \left( H^1(F, W) \rightarrow \bigoplus_{v \notin S} H_s^1(F_v, W) \right).$$

$$\text{Sel}_S(F, W) = \ker \left( \text{Sel}^\Sigma(F, W) \rightarrow \bigoplus_{v \in S} H^1(F_v, W) \right).$$

⌋

**Prop. (31.2.2.4).** When  $V$  is unramified a.e., we have  $\dim H_{\mathcal{L}}^1(\text{Gal}_F, V) < \infty$  for any Selmer structure  $\mathcal{L}$ .

⌋

*Proof:* ? This is because of the fact  $\text{Gal}_{F,S}$  is  $p$ -finite and then apply the proposition of Tate. ? □

### Bloch-Kato Selmer Groups

**Conj. (31.2.2.5) [Change of Partial Bloch-Kato Selmer Groups, Fontaine-Perrin.Riou].** For  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$  and  $v \in \Sigma_F^{\text{fin}} \setminus S$ , we have

$$\dim H_{f, S \cup \{v\}}^1(\text{Gal}_F, V) = \dim H_{f, S}^1(\text{Gal}_F, V) + \dim \left( H_g^1(\text{Gal}_F, V) / H_f^1(\text{Gal}_F, V) \right).$$

⌋

*Proof:* ? Cf.[Fontaine-Perrin.Riou]Chap2.3.3.1.

□

**Def. (31.2.2.6) [Hypothesis].** Situation as in (31.2.2.1), assume for all the time that the choices  $H_f^1(F_p, V)$  and  $H_f^1(F_p, V^D)$  satisfy:

- $H_f^1(\text{cycl}_{p,n}(\mathbb{Q}_p), V)$  and  $H_f^1(\text{cycl}_{p,n}(\mathbb{Q}_p), V^D)$  are orthogonal complements under the cup product pairing

$$H^1(\text{cycl}_{p,n}(\mathbb{Q}_p), V) \times H^1(\text{cycl}_{p,n}(\mathbb{Q}_p), V^D) \rightarrow H^2(\text{cycl}_{p,n}(\mathbb{Q}_p), \mathbb{Q}_p(1)) = \mathbb{Q}_p. ?$$

- For  $m \geq n$ ,

$$\text{cor}_{\text{cycl}_{p,n}(\mathbb{Q})}^{\text{cycl}_{p,m}(\mathbb{Q}_p)} H_f^1(\text{cycl}_{p,m}(\mathbb{Q}), V) \subset H_f^1(\text{cycl}_{p,n}(\mathbb{Q}_p), V).$$

$$\text{res}_{\text{cycl}_{p,m}(\mathbb{Q})}^{\text{cycl}_{p,n}(\mathbb{Q}_p)} H_f^1(\text{cycl}_{p,n}(\mathbb{Q}), V) \subset H_f^1(\text{cycl}_{p,m}(\mathbb{Q}_p), V).$$

┘

**Prop. (31.2.2.7) [Selmer Group of Elliptic Curves].** If  $T = T_p(E)$ ,  $W = E[p^\infty]$  where  $E \in \text{EllCur}/\mathbb{Q}$ , and define

$$H_f^1(\mathbb{Q}_{n,p}, V) = \text{Im} \left( E(\mathbb{Q}_{n,p}) \otimes \mathbb{Q}_p \hookrightarrow H^1(\mathbb{Q}_{n,p}, V) \right),$$

then  $\text{Sel}(\mathbb{Q}_n, W) = \text{Sel}^{p^\infty}(E/\mathbb{Q}_n)$  (21.1.8.7)(21.1.8.3) is the classical Selmer group. ┘

*Proof:* For  $v \in \Sigma_{\mathbb{Q}_n}^p$ , by (21.3.4.16),  $\#E(\mathbb{Q}_{n,v})[p^\infty] < \infty$ , so  $E(\mathbb{Q}_{n,v}) \otimes (\mathbb{Q}_p/\mathbb{Z}_p) = 0$ . And also by (31.2.1.1),

$$\dim H_f^1(\mathbb{Q}_{n,v}, V_p(E)) = \dim V_p(E)^{\text{Gal}_{\mathbb{Q}_{n,v}}} = 0,$$

because  $\#E(\mathbb{Q}_{n,v})[p^\infty] < \infty$ . So

$$H_f^1(\mathbb{Q}_{n,v}, W) = \text{Im} \left( E(\mathbb{Q}_{n,v}) \otimes (\mathbb{Q}_p/\mathbb{Z}_p) \rightarrow H^1(\mathbb{Q}_{n,v}, W) \right)$$

for any place  $v$ . Thus the assertion follows from the definitions (31.2.2.1)(21.1.8.3). ┘

**Def. (31.2.2.8) [Variants of Bloch-Kato Selmer Groups].**  $H_g^1, H_e^1$ . ┘

### Jensen's Conjecture

**Prop. (31.2.2.9).** Let  $F \in \mathbf{NField}$ ,  $V$  be a  $p$ -adic geometric representation of  $\text{Gal}_{F,S}$ , then the following are equivalent:

- The natural map  $H^1(\text{Gal}_{F,S}, V^*(1)) \rightarrow \prod_{v \in S} H^1(\text{Gal}_v, V^*(1))$  is injective.
- 

$$\begin{aligned} \dim H^1(\text{Gal}_{F,S}, V) &= [F : \mathbb{Q}] \dim V + \dim H^0(\text{Gal}_{F,S}, V) - \dim H^0(\text{Gal}_{F,S}, V^*(1)) \\ &\quad + \sum_{v \in S} \dim H^0(\text{Gal}_{F_v}, V^*(1)) - \sum_{v \in \Sigma_F^\infty} \dim H^0(\text{Gal}_{F_v}, V). \end{aligned}$$

•

$$\dim H^2(\text{Gal}_{F,S}, V) = \sum_{v \in S} \dim H^0(\text{Gal}_{F_v}, V^*(1)) - \dim H^0(\text{Gal}_{F,S}, V^*(1)).$$

Moreover, in item2 and item3, LHS ≤ RHS always holds. ┘

*Proof:* Cf. [Belliache, introduction to conjecture of Bloch-Kato]P52. ┘

**Conj. (31.2.2.10) [Jannsen].** Situation as in (31.2.2.9), if  $V$  is pure of weight  $w \neq -1$ , then the statements of item1,2,3 hold. ┘

*Proof:*

┘

**Prop. (31.2.2.11).** Jannsen's conjecture for  $V = \mathbb{Q}_p$  is equivalent to Leopoldt's conjecture. ┘

*Proof:* ?

┘

### Selmer Groups over $F_\infty$

**Def. (31.2.2.12).** Define  $\Lambda_p$ -modules [33.1](#)

$$\begin{aligned} \text{Sel}(F_\infty, W) &= \varinjlim_{F \subset_{\text{fin}} F' \subset F_\infty} \text{Sel}(F', W), \quad X_\infty = \text{Hom}(\text{Sel}(F_\infty, W^D), \mathbb{Q}_p/\mathbb{Z}_p), \\ H_\infty^1(F, T) &= \varprojlim_{F \subset_{\text{fin}} F' \subset F_\infty} H^1(F', T), \quad H_{\infty, s}^1(F_p, T) = \varprojlim_{F \subset_{\text{fin}} F' \subset F_\infty} H_s^1(F', T). \end{aligned}$$

where the transition maps are induced by restriction and corestriction maps [\(31.2.2.6\)](#). Then they are f.g.  $\Lambda_p$ -modules [?](#).

Similarly define

$$\text{Sel}_{S(p)}(F_\infty, W) = \varinjlim_{F \subset_{\text{fin}} F' \subset F_\infty} \text{Sel}_{S(p)}(F', W), \quad X_{\infty, S(p)} = \text{Hom}(\text{Sel}_{S(p)}(F_\infty, W^D), \mathbb{Q}_p/\mathbb{Z}_p).$$

┘

**Prop. (31.2.2.13).** There is an exact sequence

$$0 \rightarrow H_{\infty, s}^1(F_p, T)/\text{loc}_{\infty, S(p)}^s(H_\infty^1(F, T)) \rightarrow X_\infty \rightarrow X_{\infty, S(p)} \rightarrow 0.$$

┘

*Proof:* Cf. [\[Rubin, P29\]](#). [?](#)

□

### Poitou-Tate Duality

**Prop. (31.2.2.14) [Poitou-Tate Duality].** Let  $m \in \mathbb{Z}_+$  and  $\emptyset \subset_{\text{fin}} \Sigma_0 \subset_{\text{fin}} \Sigma \subset \Sigma_F$ , then

- There are exact sequences

$$\begin{aligned} 0 \rightarrow \text{Sel}^{\Sigma_0}(F, W[m]) \rightarrow S^{\Sigma}(F, W[m]) &\xrightarrow{\text{loc}_{\Sigma, \Sigma_0}^s} \bigoplus_{v \in \Sigma \setminus \Sigma_0} H_s^1(F_v, W[m]) \\ 0 \rightarrow \text{Sel}_{\Sigma}(F, W[m]) \rightarrow S^{\Sigma_0}(F, W[m]) &\xrightarrow{\text{loc}_{\Sigma, \Sigma_0}^f} \bigoplus_{v \in \Sigma \setminus \Sigma_0} H_f^1(F_v, W[m]) \end{aligned}$$

- $\text{Im}(\text{loc}_{\Sigma, \Sigma_0}^s)$  and  $\text{Im}(\text{loc}_{\Sigma, \Sigma_0}^f)$  are orthogonal w.r.t. the pairing  $\sum_{v \in \Sigma \setminus \Sigma_0} \langle -, - \rangle_v$ .
- There is an isomorphism

$$\text{Sel}_{\Sigma_0}(F, W^D[m])/\text{Sel}_{\Sigma}(F, W^D[m]) \cong \text{Hom}_{\mathbb{Z}_p}(\text{Coker}(\text{loc}_{\Sigma, \Sigma_0}^s), \mathbb{Z}_p/(m)).$$

┘

*Proof:* 1 is clear, and 3 follows from 2. For 2, Cf. [\[Rubin, P17\]](#). [?](#)

□

**Remark (31.2.2.15).** When  $\Sigma$  is large, we can make  $\text{Sel}_{\Sigma}(F, W^D[m]) = 0$ , then  $\#\text{Sel}_{\Sigma_0}(F, W^D[m]) = \#\text{Coker}(\text{loc}_{\Sigma, \Sigma_0}^s)$ . So if we can construct enough element in  $S^{\Sigma}(F, W[m])$ , then we bound  $\text{Sel}_{\Sigma_0}(F, W^D[m])$ . And this can be done by Kolyvagin's derivative construction [4](#) applied to Euler classes.

┘

**Cor. (31.2.2.16).** There is an isomorphism

$$\text{Sel}(F, W^D)/\text{Sel}_{S(p)}(F, W^D) \cong \text{Hom}_{\mathbb{Z}_p}(\text{Coker}(\text{loc}_{S(p)}^s), \mathbb{Q}_p/\mathbb{Z}_p).$$

┘

### 3 Euler Systems

**Def. (31.2.3.1) [Euler Systems].** Suppose  $F \in \mathbf{NField}$ ,  $p \in \mathbf{P}$  and  $F_\infty/F$  is an  $\mathbb{Z}_p^d$ -extension s.t. no finite place of  $F$  splits completely (This is satisfied if  $F_\infty$  contains the cyclotomic  $\mathbb{Z}_p$ -extension of  $F$ ?). Suppose

- $\mathcal{K}$  is an Abelian extension of  $F$  containing  $F_\infty$  and  $F(\mathfrak{p})$  for any  $\mathfrak{p} \in \Sigma_F^{\text{fin}}$ ,
- $\mathcal{N}$  is an ideal of  $F$  divisible by  $p$  and all finite places that  $T$  is ramified (31.2.0.1).

Then an **Euler system** for  $(T, \mathcal{K}, \mathcal{N})$  is a collection of cohomology classes

$$\mathbf{c} = \{\mathbf{c}_{F'} \in H^1(F', T) : F \subset_{\text{fin}} F' \subset \mathcal{K}\}$$

satisfying

$$\text{cor}_{F''/F'}(\mathbf{c}_{F''}) = \left( \prod_{\mathfrak{q} \in \Sigma(F''/F')} P_{\mathfrak{q}}(\|\mathfrak{q}\|^{-1} \text{Frob}_{\mathfrak{q}}^{-1}) \right) (\mathbf{c}_{F'}),$$

where  $\Sigma(F''/F')$  is the set of finite places of  $F$  not dividing  $\mathcal{N}$  that is ramified in  $F''$  but not in  $F'$ .

And an Euler system for  $(T, F_\infty)$  is any Euler system for such  $(T, \mathcal{K}, \mathcal{N})$ .  $\lrcorner$

**Example (31.2.3.2) [Euler Systems for  $\mathbb{Q}$ ].** For  $N \in \mathbb{Z}_+$ , let  $\mathcal{R}(N)$  be the set of square-free integers  $r$  s.t.  $(r, N) = 1$ , and  $\mathbb{Q}_\infty$  be the cyclotomic  $\mathbb{Z}_p$ -extension of  $\mathbb{Q}$ , then an Euler system  $\mathbf{c}$  for  $(T, \mathbb{Q}_\infty)$  is a collection of cohomology classes

$$\mathbf{c}_{\mathbb{Q}_n(\mu_r)} \in H^1(\mathbb{Q}_n(\mu_r); T)$$

s.t. for any  $r \in \mathcal{R}(N)$ ,  $\ell \in \mathbf{P}$ ,  $r\ell \in \mathcal{R}(N)$  and  $m \geq n \in \mathbb{N}$ ,

$$\text{cor}_{\mathbb{Q}_n(\mu_{r\ell})/\mathbb{Q}_n(\mu_r)}(\mathbf{c}_{\mathbb{Q}_n(\mu_{r\ell})}) = P_\ell(\ell^{-1} \text{Frob}_\ell^{-1})(\mathbf{c}_{\mathbb{Q}_n(\mu_r)}).$$

$\lrcorner$

**Def. (31.2.3.3).** If  $\mathbf{c}$  is an Euler system, denote  $\mathbf{c}_{F,\infty}$  the corresponding element in  $H_\infty^1(F, T)$ , and the ideal

$$\text{ind}_{\Lambda_p}(\mathbf{c}) = \{\varphi(\mathbf{c}_{F,\infty}) : \varphi \in \text{Hom}_{\Lambda_p}(H_\infty^1(F, T), \Lambda_p)\} \subset \Lambda_p.$$

and also

$$\text{ind}_{\mathbb{Z}_p}(\mathbf{c}) = \sup\{n : \mathbf{c}_F \in p^n H^1(F, T) + H^1(F, T)_{\text{tor}}\}.$$

$\lrcorner$

### Twisting by Characters

**Prop. (31.2.3.4) [Twisting Cohomology Groups].** For any extension of number fields  $L/F \in \mathbf{NField}$ , denote

$$H_\infty^1(L, T) = \varprojlim_{F \subset_{\text{fin}} F' \subset F_\infty} H^1(F'L, T), \quad L_\infty = LK_\infty.$$

Then for any character  $\rho : \text{Gal}(L_\infty/K) \rightarrow \mathbb{Z}_p^*$  and  $S(p) \subset \Sigma \subset \Sigma_F$ , there are  $\text{Gal}_K$ -isomorphisms

$$H_\infty^1(L, T)(\rho) \cong H_\infty^1(L, T(\rho)), \quad \text{Sel}_\Sigma(L_\infty, W)(\rho) \cong \text{Sel}_\Sigma(L_\infty, W(\rho)).$$

$\lrcorner$

*Proof:* Cf.[Rubin, P91].? □

**Prop. (31.2.3.5).** Suppose  $\mathbf{c}$  is an Euler system for  $(T, \mathcal{K}, \mathcal{N})$ , and  $\rho : \text{Gal}(F_\infty/F) \rightarrow \mathbb{Z}_p^*$  is a character with fixed field  $L$  and conductor  $\mathfrak{f}$ , then for any field  $F \subset_{\text{fin}} F' \subset \mathcal{K}$ , denote by  $\mathbf{c}_{F'}^\rho \in H^1(F, T(\chi))$  the image of  $\mathbf{c}_{F'L}(\rho)$  under the map

$$H^1(F'L, T)(\rho) \xrightarrow{\text{cor}} H^1(F', T(\rho)).$$

Then  $\{\mathbf{c}_{F'}^\rho\}$  form an Euler system for  $(T(\rho), \mathcal{K}, \mathfrak{f}^{p^\infty}\mathcal{N})$ . ┘

*Proof:* Cf.[Rubin, P30, 93].? □

#### 4 Kolyvagin Derivatives

**Def. (31.2.4.1) [Kolyvagin Primes].** Suppose  $F \subset_{\text{fin}} F' \subset F_\infty, m \in \mathbb{Z}_+,$  define  $\mathcal{R}_{F',m} \subset \mathcal{R}(\mathcal{N})$  to be the set of products of primes  $\mathfrak{q}$  of  $F$  s.t.

- $m|[F(\mathfrak{q}) : F(1)],$
  - $P_{\mathfrak{q}}(|\mathfrak{q}|^{-1})/m \in \mathbb{Z}_p,$
  - $\mathfrak{q}$  splits completely in  $F'(1)/F.$
- ┘

**Def. (31.2.4.2) [Kolyvagin Derivatives].** For any  $\mathfrak{q} \in \Sigma_F^p,$   $\Gamma_{\mathfrak{q}} = \text{Gal}(F(\mathfrak{q})/F(1))$  is canonical isomorphic to a cyclic group with generator  $\sigma_{\mathfrak{q}}$  (? , [Rubin, P62]). Define

$$D_{\mathfrak{q}} = \sum_{i=0}^{\#\Gamma_{\mathfrak{q}}-1} i\sigma_{\mathfrak{q}}^i \in \mathbb{Z}[\Gamma_{\mathfrak{q}}],$$

and for any  $\mathfrak{r} \in \mathcal{R}(\mathcal{N}),$  define

$$D_{\mathfrak{r}} = \prod_{\mathfrak{q}|\mathfrak{r}} D_{\mathfrak{q}} \in \mathbb{Z}[\Gamma_{\mathfrak{r}}].$$

And if  $F \subset_{\text{fin}} L \subset F_\infty,$  fix an element  $N_{L(1)/L} \in \mathbb{Z}[\text{Gal}(L(\mathfrak{r})/L)]$  whose restriction to  $\text{Gal}(F(1)/F)$  is  $\sum_{\gamma \in \text{Gal}(F(1)/F)} \gamma,$  then define

$$D_{\mathfrak{r},L} = N_{L(1)/L} D_{\mathfrak{r}}.$$
┘

**Def. (31.2.4.3) [Kolyvagin Systems].** Suppose  $\mathbf{c}$  is an Euler system,  $F \subset_{\text{fin}} L \subset F_\infty, m \in \mathbb{Z}_+,$  and  $\mathfrak{r} \in \mathcal{R}_{L,m},$  define

$$\kappa_{L,\mathfrak{r},M} = \delta_L(\mathbf{d}(D_{\mathfrak{r},L} x_{F(\mathfrak{r})})) \in H^1(F, W[m]).$$
┘

*Proof:* ? . Cf.[Rubin, P66]. □

**Cor. (31.2.4.4) [Properties of  $\kappa_{L,\mathfrak{r},m}$ ].**

- $\kappa_{L,1,m}$  is the image of  $\mathbf{c}_L$  in  $H^1(L, W[m]).$
  - The restriction of  $\kappa_{L,\mathfrak{r},m}$  to  $L(\mathfrak{r})$  is the image of  $D_{\mathfrak{r},L} \mathbf{c}_{L(\mathfrak{r})}$  in  $H^1(L(\mathfrak{r}), W[m]).$
  - $\kappa_{L,1,m}$  is compatible with  $m.$
- ┘

*Proof:* Cf.[Rubin, P67].? □



### Local Properties

**Thm. (31.2.4.5).** Suppose  $F \subset_{\text{fin}} F' \subset F_\infty, m \in \mathbb{Z}_+,$  and  $\mathfrak{r} \in \mathcal{R}_{F',m},$  then

$$\kappa_{F,\mathfrak{r},m} \in \text{Sel}^{S(pv)}(F', W[m]).$$

┘

*Proof:* Cf.[Rubin, P67].

□

**Thm. (31.2.4.6) [Ramification of Kolyvagin Systems].** Suppose  $F \subset_{\text{fin}} F' \subset F_\infty, m \in \mathbb{Z}_+,$  and  $\mathfrak{r}\mathfrak{q} \in \mathcal{R}_{F,m},$  then

$$\text{loc}_{\mathfrak{q}}^s(\kappa_{F,\mathfrak{r}\mathfrak{q},m}) = \phi_{\mathfrak{q}}^{fs}(\kappa_{F,\mathfrak{r},m}).$$

┘

*Proof:* Cf.[Rubin, P68].

□

### 5 $p$ -adic $L$ -Functions for Elliptic Curves

**Thm. (31.2.5.1) [ $p$ -adic  $L$ -Functions].** Let  $E \in \text{EllCur}/\mathbb{Q}, p \in \mathbf{P}.$  Suppose  $E \in \text{EllCur}/\mathbb{Q}$  has good ordinary reduction or multiplicative reduction at  $p.$  Let  $\alpha \in \mathbb{Z}_p^*, \beta = p/\alpha \in p\mathbb{Z}_p^*$  be the eigenvalues of  $\text{Frob}_p$  on  $T_p(E)$  if  $E$  has good ordinary reduction at  $p,$  or  $(\alpha, \beta) = (1, p)$  or  $(-1, -p)$  if  $E$  has split or non-split multiplicative reduction.

Then there exists  $c_E \in \mathbb{Z}_+$  independent of  $p$  and a  $p$ -adic  $L$ -function  $\mathcal{L}_E \subset c_E^{-1}\Lambda_p$  s.t. for any character  $\chi$  of  $\text{Gal}(\text{cycl}_p(\mathbb{Q})/\mathbb{Q})$  of finite order,

$$\chi(\mathcal{L}_E) = \begin{cases} (1 - \alpha^{-1})^2 L(E; 1)/\Omega_E & , \chi = 1 \& E \text{ has good reduction at } p. \\ (1 - \alpha^{-1}) L(E; 1)/\Omega_E & , \chi = 1 \& E \text{ has multiplicative reduction at } p. \\ \alpha^{-n} \tau(\chi) L(E, \chi^{-1}; 1)/\Omega_E & , \mathfrak{c}(\chi) = p^n > 1 \end{cases}$$

And for any  $N \in \mathbb{Z}_+,$  we can define

$$\mathcal{L}_{E,N} = \prod_{q|N, q \neq p} P_\ell(\ell^{-1} \text{Frob}_\ell^{-1}) \mathcal{L}_E \in \Lambda_p.$$

┘

*Proof:* Cf.[Mazur-Tate-T]?

□

### Coleman Map

**Def. (31.2.5.2) [Coleman Map].** Suppose  $E \in \text{EllCur}/\mathbb{Q}$  has good ordinary reduction or multiplicative reduction at  $p,$  then there is an injective  $\Lambda_p$ -module map

$$\text{Col}_\infty : H_{\infty,s}^1(\mathbb{Q}_p, T_p(E)) \hookrightarrow \Lambda_p$$

s.t.

- For any element  $z = \{z_n\} \in H_{\infty,s}^1(\mathbb{Q}_p, T_p(E))$  and any non-trivial character  $\chi$  of  $G_n = \text{Gal}(\text{cycl}_{p,n}(\mathbb{Q})/\mathbb{Q})$  of conductor  $p^m,$

$$\chi(\text{Col}_\infty(z)) = \alpha^{-m} \tau(\chi) \sum_{\gamma \in G_n} \chi^{-1}(\gamma) \exp_{\omega_E}^*(z_n^\gamma).$$

- If  $\chi_0$  is the trivial character, then

$$\chi_0(\text{Col}_\infty(z)) = (1 - \alpha^{-1})(1 - \beta^{-1})^{-1} \exp_{\omega_E}^*(z).$$

- If  $E$  has split multiplicative reduction at  $p$ , then the image of  $\text{Col}_\infty$  is contained in the augmentation ideal  $\mathcal{J}_p$  of  $\Lambda_p$ .

┘

*Proof:* Cf.[Rubin short, Appendix] ?.

□

### Local Cohomology Groups

**Prop. (31.2.5.3).** Let  $K \in p\text{-Field}$  and  $E \in \text{EllCur}/K$  with a Néron differential  $\omega_E$  (21.1.1.3), then  $\text{Tgt}^*(E/K) = K\omega_E$ , and  $\text{Tgt}(E/K) = K\omega_E^*$ . There exists an exponential map

$$\exp_E : \text{Tgt}(E/K) \cong E(K) \otimes \mathbb{Q}_p$$

and the following diagram is commutative:

$$\begin{array}{ccc} \text{Tgt}(E/K) & \xrightarrow{\exp_E} & E(K) \otimes \mathbb{Q}_p \\ \omega_E^* \uparrow & & \uparrow = \\ K & \xleftarrow{\log_E} \widehat{E}(\mathfrak{m}_K) \otimes \mathbb{Q}_p \xrightarrow{\cong} & E_1(K) \otimes \mathbb{Q}_p \end{array}$$

where all the maps are isomorphisms.

┘

*Proof:* ?

□

**Prop. (31.2.5.4).** By the definition and local Tate pairing,

$$\text{Hom}(E(\mathbb{Q}_{n,p}), \mathbb{Q}_p) = \text{Hom}(H_f^1(\mathbb{Q}_{n,p}, V_p(E)), \mathbb{Q}_p) \cong H_s^1(\mathbb{Q}_{n,p}, V_p(E)).$$

So we can define a dual exponential map

$$\exp_E^* : H_s^1(\mathbb{Q}_{n,p}, V) \rightarrow \text{Tgt}^*(E/\mathbb{Q}_{n,p})$$

and its composite with  $\omega_E^*$ :

$$\exp_{\omega_E}^* : H_s^1(\mathbb{Q}_{n,p}, V) \rightarrow \text{Tgt}^*(E/\mathbb{Q}_{n,p})/\omega_E \cong \mathbb{Q}_{n,p}.$$

More explicitly, for  $z \in H_s^1(\mathbb{Q}_{n,p}, V_p(E))$ ,  $x \in E(\mathbb{Q}_{n,p})$ ,

$$\text{tr}_{\mathbb{Q}_{n,p}/\mathbb{Q}_p}(\lambda_E(x) \exp_{\omega_E}^*(z)) = \langle x, z \rangle_{\mathbb{Q}_{n,p}}.$$

┘

**Prop. (31.2.5.5).**

$$\exp_{\omega_E}^*(H_s^1(\mathbb{Q}_p, T)) = \frac{1}{p}[E(\mathbb{Q}_p) : E_1(\mathbb{Q}_p) + E(\mathbb{Q}_p)_{\text{tor}}]\mathbb{Z}_p.$$

┘

*Proof:* By duality, if  $\lambda_E(\mathbb{Q}_p) = p^{-a}\mathbb{Z}_p$ , then  $\exp_{\omega_E}^*(H_s^1(\mathbb{Q}_p, T)) = p^a\mathbb{Z}_p$ . But  $\lambda_E(E_1(\mathbb{Q}_p)) = p\mathbb{Z}_p$ , so the assertion follows. □

## 6 Bounding Selmer Groups

**Notation (31.2.6.1).**

- Fix an Euler system  $\mathbf{c}$  for  $(T, F_\infty)$  (31.2.3.2).  $\mathbf{c}_F \in \text{Sel}^{S(p)}(F, T)$  by (31.2.1.4) and (31.2.2.2).
- Suppose the choices of Selmer local conditions satisfy hypothesis (31.2.2.6).

┘

### Hypothesis on Representations

**Def. (31.2.6.2) [Hypothesis].**  $T$  satisfies the hypothesis

**Hypothesis Hyp** $(F_\infty; T)$  if:

- There exists  $\tau \in \text{Gal}_{F(1)F(\mu_{p^\infty})}$  s.t.  $T/(\tau - 1)T$  is free of rank 1 over  $\mathbb{Z}_p$ , and
- $T/(p) \in \text{Irr}_{\mathbb{F}_p}(\text{Gal}_{F_\infty})$ .

**Hypothesis Hyp** $(F_\infty; V)$  if:

- There exists  $\tau \in \text{Gal}_{F(1)F(\mu_{p^\infty})}$  s.t.  $\dim_{\mathbb{Q}_p} V/(\tau - 1)V = 1$ , and
- $V \in \text{Irr}_{\mathbb{Q}_p}(F_\infty)$ .

**Hypothesis Hyp** $(F_\infty/F)$  if:

- $\text{rank}_{\mathbb{Z}_p}(\text{Gal}_{F_\infty/F}) > 1$  or
- neither  $T$  nor  $T(-1)$  is trivial, or
- $F$  is imaginary quadratic, or
- $F$  is totally real and Leopoldt's conjecture holds for  $F$ .

┘

**Prop. (31.2.6.3).** Suppose  $E \in \text{EllCur}/\mathbb{Q}$  without CM, then

- $T = T_p(E)$  satisfies the hypothesis  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}); V)$  (31.2.6.2), and  $\#H^1(\mathbb{Q}(E[p^\infty])/\mathbb{Q}, E[p^\infty]) < \infty$ .
- If  $\rho_{E,p} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{Aut}(T_p)$  is surjective, then  $T_p(E)$  satisfies the hypothesis  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}), T)$  (31.2.6.2), and  $H^1(\mathbb{Q}(E[p^\infty])/\mathbb{Q}, E[p^\infty]) = 0$ .

┘

*Proof:* By Weil pairing,

$$\rho_{E,p}(\text{Gal}_{\mathbb{Q}(\mu_{p^\infty})}) = \rho_{E,p}(\text{Gal}_{\mathbb{Q}}) \cap \text{SL}(2, \mathbb{Z}_p).$$

When  $\rho_{E,p} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{Aut}(T_p)$  is surjective, then  $\overline{\rho_{E,p}}$  is surjective thus irreducible, and also we can take  $\rho_{E,p}(\tau) = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$ . So  $T_p(E)$  satisfies the hypothesis  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}), T)$ .

And because  $E$  has no CM, by [Serre, Galois Property Points Elliptiques, (1972) Cor1 of Thm3],  $\rho_{E,p}(\text{Gal}_{\mathbb{Q}}) \subset \text{GL}(2, \mathbb{Z}_p)$  is open. So  $\rho_{E,p}|_{\text{Gal}_{\mathbb{Q}(\mu_{p^\infty})}}$  is irreducible, and we can take  $\rho_{E,p}(\tau) = \begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}$  for some  $x \neq 0$ . So  $T_p(E)$  satisfies the hypothesis  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}), V)$ .

Finally, the cohomology group can be calculated directly **?**.

□

### Bounding Selmer Groups over $F$

**Thm. (31.2.6.4).** Suppose  $V$  satisfies  $\text{Hyp}(F; V)$  (31.2.6.2), and  $V$  is not the trivial representation.

- If  $\mathbf{c}_F \notin H^1(F, T)_{\text{tor}}$ , then  $\# \text{Sel}_{S(p)}(F, W^D) < \infty$ .
- If  $\text{loc}_p^s(\mathbf{c}_Q) \neq 0$  and  $[H_s^1(\mathbb{Q}_p, T) : \mathbb{Z}_p \text{loc}_p^s(\mathbf{c}_Q)] < \infty$ , then  $\# \text{Sel}(\mathbb{Q}, W^D) < \infty$ .

(If  $V$  is trivial, this is true iff the Leopoldt's conjecture holds).  $\lrcorner$

*Proof:* Cf. [Rubin, P24]. ?

2: It follows from (31.2.2.16) that

$$[\text{Sel}(\mathbb{Q}, W^D) : \text{Sel}_p(\mathbb{Q}, W^D)] = [H_s^1(\mathbb{Q}_p, T) : \mathbb{Z}_p \text{loc}_p^s(\text{Sel}^p(\mathbb{Q}, T))] \leq [H_s^1(\mathbb{Q}_p, T) : \mathbb{Z}_p \text{loc}_p^s(\mathbf{c}_Q)].$$

So the assertion follows from item1.  $\square$

**Cor. (31.2.6.5).** Suppose  $V$  satisfies  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}); V)$  (31.2.6.2),  $\text{loc}_p^s \mathbf{c}_Q \neq 0$ , and  $\text{rank}_{\mathbb{Z}_p} H_s^1(\mathbb{Q}_p, T) = 1$ , then  $\# \text{Sel}^{p^\infty}(\mathbb{Q}, W^D) < \infty$ .  $\lrcorner$

*Proof:* This is because  $H_s^1(\mathbb{Q}_p, T)$  is torsion-free.  $\square$

**Thm. (31.2.6.6).** Suppose  $T$  satisfies  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}); T)$  (31.2.6.2),  $p > 2$ . Let  $\Omega = \mathbb{Q}(W)\mathbb{Q}((\mathbb{Z}_p^*)^{1/p^\infty})$ , where  $\mathbb{Q}(W)$  is the minimal extension of  $\mathbb{Q}$  that  $\text{Gal}_{\mathbb{Q}(W)}$  acts trivially on  $W$ .

- Then

$$\# \text{Sel}_p(\mathbb{Q}, W^D) \leq p^{\text{ind}_{\mathbb{Z}_p}(\mathbf{c})} \# \left( H^1(\Omega/\mathbb{Q}, W) \cap \text{Sel}^p(\mathbb{Q}, W) \right) \# \left( H^1(\Omega/\mathbb{Q}, W) \cap \text{Sel}_p(\mathbb{Q}, W^D) \right)$$

- If  $\text{loc}_p^s(\mathbf{c}_Q) \neq 0$  (31.2.2.1), then

$$\# \text{Sel}(\mathbb{Q}, W^D) \leq [H_s^1(\mathbb{Q}_p, T) : \mathbb{Z}_p \text{loc}_p^s(\mathbf{c}_Q)] \# \left( H^1(\Omega/\mathbb{Q}, W) \cap \text{Sel}^p(\mathbb{Q}, W) \right) \# \left( H^1(\Omega/\mathbb{Q}, W^D) \cap \text{Sel}_p(\mathbb{Q}, W^D) \right).$$

Notice  $\# H^1(\Omega/\mathbb{Q}, W) \# H^1(\Omega/\mathbb{Q}, W^D) < \infty$  when  $V$  is irreducible and  $\dim V > 2$ , by [Rubin, P162]. ?  $\lrcorner$

*Proof:* Cf. [Rubin, P24]. ?

2: Notice  $H^1(\mathbb{Q}, T)/\text{Sel}^p(\mathbb{Q}, T) \hookrightarrow \bigoplus_{v \in \Sigma_Q^p} H_s^1(\mathbb{Q}_v, T)$  is torsion-free, so  $\mathbf{c}_Q \in p^n H^1(\mathbb{Q}, T) + H^1(\mathbb{Q}, T)_{\text{tor}}$  implies  $\mathbf{c}_Q \in p^n \text{Sel}^p(\mathbb{Q}, T) + H^1(\mathbb{Q}, T)_{\text{tor}}$ , which implies  $\text{loc}_p^s(\mathbf{c}_Q) \in p^n \text{loc}_p^s(\text{Sel}^p(\mathbb{Q}, T))$ . Then the assertion follows from (31.2.2.16) and item1.  $\square$

### Bounding Selmer Groups over $F_\infty$

**Prop. (31.2.6.7).** Suppose  $V$  satisfies  $\text{Hyp}(F_\infty, V)$  with  $\tau$ . Define  $Z$  (resp.  $Z^D$ ) to be the maximal  $\text{Gal}_{F_\infty}$ -stable submodule of  $(\tau - 1)W$  (resp.  $W^D$ ), and

$$a_\tau = [W^\tau : (W^\tau)_{\text{div}}] \cdot \max(\#Z, \#Z^D).$$

Then  $a_\tau < \infty$ . And if  $T$  satisfies  $\text{Hyp}(F_\infty, T)$ , then  $a_\tau = 1$ .  $\lrcorner$

*Proof:* Cf. [Rubin, P98]. ?  $\square$

**Thm. (31.2.6.8).** Suppose  $V$  satisfies  $\text{Hyp}(F_\infty; V)$  and  $\text{Hyp}(F_\infty/F)$  (31.2.6.2),  $\mathbf{c}_{F, \infty} \notin H_\infty^1(F, T)_{\Lambda_p - \text{tor}}$ , then

- (Weak Leopoldt Conjecture)  $X_{\infty, S(p)}$  is a torsion  $\Lambda_p$ -module.
- There exists  $t \in \mathbb{N}$  s.t.  $\text{char}(X_{\infty, S(p)}) | p^t \text{ind}_{\Lambda_p}(\mathbf{c})$ .
- If moreover  $T$  satisfies  $\text{Hyp}(F_{\infty}; T)$  (31.2.6.2), then  $\text{char}(X_{\infty, S(p)}) | \text{ind}_{\Lambda_p}(\mathbf{c})$ .

┘

*Proof:* 1: [Rubin, Thm2.3.2]. ?

2,3: Cf. [Rubin, P101] ? . Given item1, there exists an injective pseudo-isomorphism

$$\bigoplus_{i=1}^r \Lambda_p / (f_i) \rightarrow X_{\infty, S(p)},$$

where  $f_{i+1} | f_i$ . Then by (31.2.6.7), it suffices to show that  $\text{char}(X_{\infty, S(p)}) | a_{\tau}^{5r} \text{ind}_{\Lambda_p}(\mathbf{c})$ .

Suppose  $h \in \Lambda_p$  satisfies (31.2.6.14), and a finite set of places  $\Sigma_F^{\infty} \cup S(p) \cup \text{Ram}(T) \subset \Sigma \subset \Sigma_F$ , then by (31.2.6.15),

$$h^r a_{\tau}^{5r} \mathbf{c}_L \in \text{char}(X_{\infty, S(p)}) \cdot H^1(F_{\Sigma}/L, T)_{\text{lf}}.$$

Then by taking limit over  $L$  ( ? [Rubin, P101]),

$$h^r a_{\tau}^{5r} \mathbf{c}_{F, \infty} \in \text{char}(X_{\infty, S(p)}) \cdot H^1(F, T)_{\text{lf}}.$$

Then by the definition of  $\text{ind}_{\Lambda_p}(\mathbf{c})$  and the fact  $h$  is prime to  $\text{char}(X_{\infty, S(p)})$  (31.2.6.14), we get  $\text{char}(X_{\infty, S(p)}) | a_{\tau}^{5r} \text{ind}_{\Lambda_p}(\mathbf{c})$ .  $\square$

**Cor. (31.2.6.9).** Suppose  $V$  satisfies  $\text{Hyp}(\text{cycl}_p(\mathbb{Q}); V)$  (31.2.6.2),  $\text{loc}_{p, \infty}^s(\mathbf{c}_{F, \infty})$  is non-torsion over  $\Lambda_p$ , and  $H_{\infty, s}^1(F_p, T) / \Lambda_p \cdot \text{loc}_{p, \infty}^s(\mathbf{c}_{F, \infty})$  is torsion, define

$$\mathcal{L} = \text{char} \left( H_{\infty, s}^1(F_p, T) / \Lambda_p \cdot \text{loc}_{p, \infty}^s(\mathbf{c}_{F, \infty}) \right),$$

then

- (Weak Leopoldt Conjecture)  $X_{\infty}$  is a torsion  $\Lambda_p$ -module.
- There exists  $t \in \mathbb{N}$  s.t.  $\text{char}(X_{\infty}) | (p^t \mathcal{L})$ .
- If  $T$  satisfies  $\text{Hyp}(F_{\infty}; T)$  (31.2.6.2), then  $\text{char}(X_{\infty}) | \mathcal{L}$ .

┘

*Proof:* By (31.2.6.8),  $X_{\infty, S(p)}$  is torsion over  $\Lambda_p$ , so by (31.2.2.13) and the hypothesis,  $X_{\infty}$  is also torsion over  $\Lambda_p$ , and

$$\text{char}(X_{\infty}) = \text{char}(X_{\infty, S(p)}) \text{char} \left( H_{\infty, s}^1(F_p, T) / \text{loc}_{p, \infty}^s(H_{\infty}^1(F, T)) \right).$$

Notice the hypothesis also implies  $\text{loc}_{p, \infty}^s(H_{\infty}^1(F, T))$  has rank 1 over  $\Lambda_p$ , so by the definition of  $\text{ind}_{\Lambda_p}(\mathbf{c})$ ,

$$\text{ind}_{\Lambda_p}(\mathbf{c}) | \text{char} \left( \text{loc}_{p, \infty}^s(H_{\infty}^1(F, T)) / \Lambda_p \cdot \text{loc}_{p, \infty}^s(\mathbf{c}_{F, \infty}) \right).$$

Thus the assertion follows.  $\square$

**Proof over  $F_\infty$** 

**Lemma (31.2.6.10).** Let  $\rho : \text{Gal}(F_\infty/F) \rightarrow \Lambda_p$  be a character, then theorem (31.2.6.8) for  $T$  and  $\mathbf{c}$  are equivalent to the theorem for  $T(\rho)$  and  $\mathbf{c}^\rho$ .  $\lrcorner$

*Proof:* The hypothesis  $\text{Hyp}(F_\infty; V)$ ,  $\text{Hyp}(F_\infty/F)$  and  $\text{Hyp}(F_\infty; T)$  depends only on  $T$  as a  $\text{Gal}_{F_\infty}$ -module, so they are not affected. And the rest follows from the fact everything is twisted by  $\rho$ , by (31.2.3.4).  $\square$

**Prop. (31.2.6.11).** There are elements  $z_1, \dots, z_r \in X_{\infty, S(p)}$  and ideals  $\mathfrak{g}_1, \dots, \mathfrak{g}_r \in \Lambda_p$  s.t. for  $1 \leq k \leq r$ ,

- $z_k \in \text{ev}^*(\tau \text{Gal}_{\Omega_\infty})$ .
- $a_\tau \mathfrak{g}_k \in f_k \Lambda_p$ , and  $\mathfrak{g}_k \subset \mathfrak{g}_{k+1}$ .
- There is a split exact sequence

$$0 \rightarrow \sum_{i=1}^{k-1} \Lambda_p z_i \rightarrow \sum_{i=1}^k \Lambda_p z_i \rightarrow \Lambda_p / \mathfrak{g}_k \rightarrow 0.$$

- $a_\tau(X_{\infty, S(p)} / \sum_{i=1}^r \Lambda_p z_i)$  is pseudo-null.  $\lrcorner$

*Proof:* Cf. [Rubin, P98].  $\square$

**Def. (31.2.6.12) [Selmer Sequences and Kolyvagin Sequences].** Fix a sequence  $(z_1, \dots, z_r)$  for  $X_{\infty, S(p)}$  as in (31.2.6.11), and let  $Z_\infty \subset X_{\infty, S(p)}$  be the submodule they generate. Then for  $0 \leq k \leq r$ ,  $F \subset_{\text{fin}} L \subset F_\infty$ , then

- A **Selmer sequence** of length  $k$  is a sequence  $(\sigma_1, \dots, \sigma_k)$  s.t. Cf. [Rubin, P99]. ?
- A **Kolyvagin sequence** of length  $k$  for  $L$  and  $M$  is a sequence  $(\mathfrak{Q}_1, \dots, \mathfrak{Q}_k)$  of primes of  $L$  ?  
Cf. [Rubin, P99].

Let  $\Pi(k, L, M)$  be the set of Kolyvagin sequences of length  $k$  for  $F$  and  $M$ , and define

$$\Psi(k, L, M) = \sum_{\pi \in \Pi(k, L, M)} \sum_{\psi \in \text{Hom}(\Lambda_p \kappa_{L, \tau(\pi), M}, \Lambda_{L, M})} \psi(\kappa_{L, \tau(\pi), M}) \subset \Lambda_{L, M}$$

$\lrcorner$

**Lemma (31.2.6.13).** Suppose  $G \in \mathcal{A}b^{\text{fin}}$ ,  $R \in \mathcal{C}\text{Ring}$  is a PID, and  $B$  is a f.g.  $R[G]$ -module without  $R$ -torsion. Suppose  $f \in R[G]$  is a non-zero-divisor,  $b \in B$ , and

$$\{\psi(b) : \psi \in \text{Hom}_{R[G]}(B, R[G])\} \subset fR[G],$$

then  $b \in fB$ .  $\lrcorner$

*Proof:* Cf. [Rubin, P101].  $\square$

**Prop. (31.2.6.14).** Situation as in (31.2.6.8), there exists  $h \in \Lambda_p$  relatively prime to  $\text{char}(X_{\infty, S(p)})$ , and for each  $F \subset_{\text{fin}} L \subset F_\infty$  a number  $N_L \in p^{\mathbb{Z}^+}$ , s.t. if  $M \in p^{\mathbb{Z}^+}$ ,  $N_L | M$ ,  $0 \leq k \leq r$ , then

$$ha_\tau^5 \Psi(k, L, MN_L) \subset f_{k+1} \Psi(k+1, L, M) \text{ (31.2.6.12).}$$

$\lrcorner$

*Proof:* Cf.[Rubin, P100] ?. □

**Cor. (31.2.6.15).** Situation and notation as in (31.2.6.14), if  $F \subset_{\text{fin}} L \subset F_\infty$  and finite set of places  $\Sigma_F^\infty \cup S(p) \cup \text{Ram}(T) \subset \Sigma \subset \Sigma_F$ , then

$$h^r a_\tau^{5r} \mathbf{c}_L \in \text{char}(X_{\infty, S(p)}). H^1(F_\Sigma/L, T)_{\text{lf}}.$$

┘

*Proof:* It follows from (31.2.6.14) and induction that

$$h^r a_\tau^5 \Psi(0, L, MN_L^r) \subset \left( \prod_{i=1}^r f_i \right) \Psi(r, L, M) \subset \text{char}(X_{\infty, S(p)}) \Lambda_{L, M}.$$

By (31.2.6.13), it suffices to show that for any  $\psi \in \text{Hom}(H^1(F_\Sigma/L, T), \Lambda_L)$ ,  $h^r a_\tau^{5r} \psi(\mathbf{c}_L) \in \text{char}(X_{\infty, S(p)}) \Lambda_L$ . For this, notice by (31.2.4.4),  $\kappa_{L, 1, MN_L^r}$  is the image of  $\mathbf{c}_L$  under the map

$$H^1(F_\Sigma/L, T)/(MN_L^r) \hookrightarrow H^1(F_\Sigma/L, W[MN_L^r]) \hookrightarrow H^1(L, W[MN_L^r]),$$

so we can define  $\bar{\psi}$  to be the map

$$\Lambda_{L, MN_L^r} \kappa_{L, 1, MN_L^r} \subset H^1(F_\Sigma/L, T)/(MN_L^r) \xrightarrow{\psi} \Lambda_{L, MN_L^r} \rightarrow \Lambda_{L, M}.$$

Then by definition  $\bar{\psi}(\kappa_{L, 1, MN_L^r}) \in \Psi(0, L, MN_L^r) \Lambda_{L, M}$ , and  $\bar{\psi}(\kappa_{L, 1, MN_L^r}) \equiv \psi(\mathbf{c}_L) \pmod{M}$ . Thus we get  $\psi(\mathbf{c}_L) \in \Psi(0, L, MN_L^r)$  by noticing  $M$  can be arbitrarily large. □

## 7 Euler System for Elliptic Curves(Kato)

**Thm. (31.2.7.1) [Kato].** Suppose  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$ , and  $p \in \text{Prime}$ . Then there exists  $D, D \not\equiv 1 \pmod{p}$ ,  $(DD', 6pN) = 1$ , and  $r_E \in \mathbb{Z}_+$  independent of  $p$ , and an Euler system  $\bar{\mathbf{c}} = \bar{\mathbf{c}}(D, D')$  for  $T_p(E)$  (31.2.3.2):

$$\{\bar{\mathbf{c}}_{\mathbb{Q}_n(\mu_r)} \in H^1(\mathbb{Q}_n(\mu_r), T_p(E))\}_{r \in \mathcal{R}(NpDD'), n \in \mathbb{N}},$$

s.t. for any character  $\chi \in \text{Gal}(\mathbb{Q}_n/\mathbb{Q})^\vee$ ,

$$\sum_{\gamma \in \text{Gal}(\mathbb{Q}_n/\mathbb{Q})} \chi(\gamma) \exp_{\omega_E}^*(\text{loc}_p^s(\bar{\mathbf{c}}_{\mathbb{Q}_n}^\gamma)) = r_E DD' (D - \chi^{-1}(D))(D' - \chi^{-1}(D')) L_{Np}(E, \chi; 1) / \Omega_E.$$

┘

*Proof:* Take  $r = p^m$  and take corestriction to  $\mathbb{Q}_n$ . But we need to verify that  $\exp_{\omega_E}^* \circ \text{loc}_p^s$  equals the dual exponential map defined in (33.4.3.2). □

**Cor. (31.2.7.2).** Situation as in (31.2.7.1), suppose that  $E$  has good ordinary reduction or multiplicative reduction at  $p$ , then there is an Euler system  $\mathbf{c}$  for  $T_p(E)$  s.t.

- $\exp_{\omega_E}^*(\text{loc}_p^s(\mathbf{c}_{\mathbb{Q}})) = r_E L_{Np}(E; 1) / \Omega_E$ .
- $\text{Col}_\infty(\text{loc}_{p, \infty}^s(\mathbf{c}_{\mathbb{Q}, \infty})) = r_E \mathcal{L}_{E, N}$  (31.2.5.1).

┘

*Proof:* Let  $\sigma_D, \sigma_{D'} \in \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q})$  denote the automorphism  $\zeta \mapsto \zeta^D, \zeta \in \mu_{p^\infty}$ , then since  $D, D' \nmid 1 \pmod{p}$ ,  $(D - \sigma_D)(D' - \sigma_{D'}) \in \Lambda_p$  is invertible. Let  $\rho_{D,D'} \in \mathbb{Z}_p[[\text{Gal}_{\mathbb{Q}}]]$  be any element that restricts to  $(D - \sigma_D)^{-1}(D' - \sigma_{D'})^{-1}$ , then we can define

$$\mathbf{c}_{\mathbb{Q}_n(\mu_r)} = (DD')^{-1} \rho_{D,D'} \bar{\mathbf{c}}_{\mathbb{Q}_n(\mu_r)},$$

which is also an Euler system, and by (31.2.7.1) and a change of variable, we get

$$\sum_{\gamma \in \text{Gal}(\mathbb{Q}_n/\mathbb{Q})} \chi(\gamma) \exp_{\omega_E}^*(\text{loc}_p^s(\mathbf{c}_{\mathbb{Q}_n}^\gamma)) = r_E L_{Np}(E, \chi; 1)/\Omega_E.$$

Then if  $\chi$  is the trivial character, this gives equation 1. And for any  $\chi \in \text{Gal}(\mathbb{Q}_\infty/\mathbb{Q})^\vee$ ,

$$\begin{aligned} \chi \left[ \text{Col}_\infty(\text{loc}_{p,\infty}^s(\mathbf{c}_{\mathbb{Q},\infty})) \right] &= \\ \left\{ \begin{array}{ll} (1 - \alpha^{-1})(1 - \beta^{-1})^{-1} \exp_{\omega_E}^*(\text{loc}_p^s(\mathbf{c}_{\mathbb{Q}})) = r_E L_{Np}(E; 1)/\Omega_E & , \chi = 1 \\ \alpha^{-n} \tau(\chi) \sum_{\gamma \in \text{Gal}(\mathbb{Q}_n/\mathbb{Q})} \chi^{-1}(\gamma) \exp_{\omega_E}^*(\text{loc}_p^s(\bar{\mathbf{c}}_{\mathbb{Q}_n}^\gamma)) = r_E L_{Np}(E; \chi^{-1}, 1)/\Omega_E & \mathbf{c}(\chi) = p^n > 1 \end{array} \right. \\ &= \chi(r_E \mathcal{L}_{E,N}). \end{aligned}$$

Thus  $\text{Col}_\infty(\text{loc}_{p,\infty}^s(\mathbf{c}_{\mathbb{Q},\infty})) = r_E \mathcal{L}_{E,N}$ . □

### Bounding Selmer Groups

**Thm. (31.2.7.3).** Suppose  $E \in \text{EllCur}/\mathbb{Q}$  without CM. Notation as in (31.2.7.1),

- If  $E$  has good ordinary reduction or non-split multiplicative reduction at  $p$ , then  $X_\infty(E[p^\infty])$  is f.g. torsion over  $\Lambda_p$ , and there exists  $t \in \mathbb{N}$  s.t.  $\text{char}(X_\infty) | (p^t \mathcal{L}_{E,N})$ . And if  $\rho_{E,p}$  is surjective with  $p \nmid r_E \prod_{q|N, q \neq p} \ell_q(q^{-1})$ , then  $\text{char}(X_\infty) | (\mathcal{L}_E)$ .
- If  $E$  has split multiplicative reduction at  $p$ , then similar results hold with  $\text{char}(X_\infty)$  replaced by  $\mathcal{J} \text{char}(X_\infty)$ .

┘

*Proof:* Because  $\text{Col}_\infty$  is injective (31.2.5.2), by (31.2.7.2),

$$\mathcal{L} = \text{char} \left( H_{\infty,/\text{Sel}}^1(\mathbb{Q}_p, T)/\Lambda_p \cdot \text{loc}_{p,\infty}^s(\mathbf{c}_{\mathbb{Q}}) \right) \Big| \text{char} \left( \text{Im}(\text{Col}_\infty)/(\text{Col}_\infty(\text{loc}_{p,\infty}^s(\mathbf{c}_{\mathbb{Q}}))) \right) \Big| (r_E \mathcal{L}_{E,N})$$

So item 1 follows from (31.2.6.9). Notice the hypothesis are satisfied by (31.2.6.3). Item 2 also follows, by noticing that  $\text{Im}(\text{Col}_\infty) \subset \mathcal{J}_p$  (31.2.5.2). □

**Cor. (31.2.7.4) [Greenberg].** Let  $E \in \text{EllCur}/\mathbb{Q}$  without CM, then there exists  $M_E \in \mathbb{Z}_+$  s.t. if  $p \in \mathbf{Prime}$  is good ordinary for  $E$  and  $p \nmid M_E$ , then  $X_\infty(E)$  (31.2.2.12) has no non-zero finite submodules.

┘

*Proof:* Cf. [Rubin short, P363] and [Greenberg, Iwasawa theory for  $p$ -adic Representations] ?. □

**Cor. (31.2.7.5).** Let  $E \in \text{EllCur}/\mathbb{Q}$  without CM,  $p \in \mathbf{P}$  is a good place for  $E$  and  $p \nmid 2r_E M_E \prod_{q|N} \ell_q(q^{-1})$  (31.2.7.1)(31.2.7.4), and  $\rho_{E,p}$  is surjective, then

$$\# \text{III}(E)[p^\infty] \Big| \frac{L(E; 1)}{\Omega_E}.$$

┘



*Proof:* If  $p$  is good supersingular for  $E$ , then  $p \nmid \#\tilde{E}(\mathbb{F}_p)$ , and Cf.[Rubin short, P362]?

If  $p$  is good ordinary for  $E$ , then by (31.2.7.4), we may assume  $X_\infty(E) \subset \prod_i \Lambda_p/(f_i^{n_{ij}})$ , and then

$$\mathrm{Sel}(\mathrm{cycl}_p(\mathbb{Q}), E[p^\infty])^{\mathrm{Gal}(\mathrm{cycl}_p(\mathbb{Q})/\mathbb{Q})} = (X_\infty(E))^{\mathrm{Gal}(\mathrm{cycl}_p(\mathbb{Q})/\mathbb{Q})} \subset \chi_0\left(\prod_i \Lambda_p/(f_i^{n_{ij}})\right)$$

and the RHS has cardinality  $\chi_0(\mathrm{char}(X_\infty(E)))$ , which by (31.2.7.3) divides

$$\chi_0(\mathcal{L}_E) = (1 - \alpha^{-1})^2 \prod_{q|N} \ell_q(q^{-1}) L(E; 1) / \Omega_E \quad (31.2.5.1).$$

Also, one show by proof of Mazur control theorem that

$$\mathrm{Sel}(\mathbb{Q}, E[p^\infty]) \rightarrow \mathrm{Sel}(\mathrm{cycl}_p(\mathbb{Q}), E[p^\infty])^{\mathrm{Gal}(\mathrm{cycl}_p(\mathbb{Q})/\mathbb{Q})}.$$

is injective with cokernel of order divisible by  $(1 - \alpha^{-1})^2$ ?. Thus the assertion follows.  $\square$

**Cor. (31.2.7.6).** Suppose  $E \in \mathrm{EllCur}/\mathbb{Q}$  without CM, and  $L(E; 1) \neq 0$ , then  $\#E(\mathbb{Q}) < \infty$ ,  $\#\mathrm{III}(E) < \infty$ .  $\lrcorner$

*Proof:* This is proven by Kolyvagin before, but Kato proved it again as follows:

For each  $q|N$ ,  $\ell_q(q^{-1}) \neq 0$ , so  $L_{Np}(E; 1) \neq 0$ , and then by (31.2.7.2),  $\mathrm{loc}_p^s(\mathbf{c}_\mathbb{Q}) \neq 0$ , then it follows from (31.2.6.5) that  $\mathrm{Sel}^{p^\infty}(E/\mathbb{Q}) < \infty$ . Notice the hypothesis is satisfied by (31.2.5.5).

But then by Serre's theorem,  $\rho_{E,p}$  is surjective for a.e.  $p$ , so by (31.2.7.5),  $\mathrm{III}(E)[p^\infty] = 0$  for a.e.  $e$ . Thus the assertion follows.  $\square$

**Thm. (31.2.7.7) [Kato].** For  $E \in \mathrm{EllCur}/\mathbb{Q}$ , if  $F/\mathbb{Q}$  is an Abelian extension and  $\chi \in \mathrm{Gal}(F/\mathbb{Q})^\vee$ , and  $L(E, \chi; 1) \neq 0$ , then  $\#E(F)^\chi < \infty$ ,  $\#\mathrm{III}(E_L)^\chi < \infty$ .  $\lrcorner$

*Proof:* We only prove for the case  $E$  without CM. For the CM case, Cf.[Coates-Wiles77], [Rubin, The Iwasawa Main conjecture for Imaginary Quadratic Fields] or [Rubin-Wiles, Mordell-Weil Groups of Elliptic Curves over Cyclotomic Fields].?

The proof is similar to that of (31.2.7.6):  $\chi(\mathrm{char}(X_\infty(E)))$  is a non-zero multiple of  $L(E, \chi; 1)$ , so  $\#\mathrm{Sel}(\mathrm{cycl}_p(\mathbb{Q}), E[p^\infty])^\chi < \infty$ . Then use a variant of (31.2.7.5) to bound  $\#\mathrm{III}(E)$ .  $\square$

**Cor. (31.2.7.8) [Kato].** For  $E \in \mathrm{EllCur}/\mathbb{Q}$  and  $F/\mathbb{Q}$  an Abelian extension,  $p \in \mathbf{P}$ ,  $E(\mathbb{Q}(\mu_{p^\infty}))$  is a f.g. Abelian group.  $\lrcorner$

*Proof:* It follows from [Roh84]? that  $L(E, \chi, 1) \neq 0$  for a.e. character  $\chi$  of finite order of  $\mathrm{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q})$ . ?  $\square$

**Def. (31.2.7.9) [ $r_p(E/F)$ ].** For  $F \in \mathrm{NField}$ ,  $E \in \mathrm{EllCur}/F$ ,  $p \in \mathbf{P}$ , define

$$r_p(E/F) = \sum_{v \in S(p), E \text{ has potential supersingular reduction at } v} [F_v : \mathbb{Q}_p].$$

$\lrcorner$

**Conj. (31.2.7.10).** For  $F \in \mathrm{NField}$ ,  $E \in \mathrm{EllCur}/F$ ,  $p \in \mathbf{P}$ , let  $\Gamma = \mathrm{Gal}(\mathrm{cycl}_p(F)/F) \cong \mathbb{Z}_p$ , then

$$\mathrm{rank}_{\mathrm{Frac}(\mathbb{Z}_p(\Gamma))} [\mathrm{Sel}(E/\mathrm{cycl}_p(F)) \otimes_{\mathbb{Z}_p(\Gamma)} \mathrm{Frac}(\mathbb{Z}_p(\Gamma))] = r_p(E/F) \quad (31.2.7.9)$$

$\lrcorner$

*Proof:*

□

**Conj. Cor. (31.2.7.11).** For  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$ ,  $p \in \mathbf{P}$ ,

$$\mathrm{rank}_{\mathrm{Frac}(\Lambda_p)}[\mathrm{Sel}(E/\mathrm{cycl}_p(F)) \otimes_{\Lambda_p} \mathrm{Frac}(\Lambda_p)] \geq r_p(E/F) \quad (31.2.7.9)$$

┘

*Proof:* Cf. [Coates, Galois of Elliptic Curves, P19].

□

**Cor. (31.2.7.12).** For  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$ ,  $p \in \mathbf{P}$ , if  $E$  has potential good ordinary reduction at all places dividing  $p$ , and  $\mathrm{Sel}(E/F)$  is finite, then  $\mathrm{Sel}(E/\mathrm{cycl}_p(F))^\vee$  is  $\mathbb{Z}_p(\Gamma)$ -torsion.

┘

*Proof:*

□

**Cor. (31.2.7.13).** If  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $\mathrm{rank}_{\mathrm{an}}(E) = 0$ , then  $\mathrm{Sel}(E/\mathrm{cycl}_p(\mathbb{Q}))^\vee$  is  $\Lambda_p$ -torsion.

┘

*Proof:* This follows from (31.4.2.2) and (31.2.7.12).

□

**Cor. (31.2.7.14).** Suppose  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $p \in \mathbf{P}$  is of good ordinary reduction for  $E$ , then

$$\mathrm{Sel}(\mathbb{Q}(\mu_{p^\infty}), E)^\vee$$

is torsion over  $\Lambda_p$ .

┘

*Proof:* Cf. [Kato].

□

**Prop. (31.2.7.15) [Euler Characteristic of  $\mathrm{Sel}(E/\mathrm{cycl}_p(F))$ ].** Let  $p \in \mathbf{P} \setminus \{2\}$ ,  $F \in \mathbf{NField}$ , and  $E \in \mathbf{EllCur}/F$  with good ordinary reduction over all places  $v \in S(p)$ . Suppose  $\mathrm{Sel}(E/F)$  is finite, then  $\mathrm{Sel}(E/F)$  has finite  $\Gamma$ -Euler characteristic, and

$$\chi(\Gamma, \mathrm{Sel}(E/\mathrm{cycl}_p(F))) = \rho_p(E/F) = \left| \frac{\#\mathrm{III}(E/F)[p^\infty]}{(\#E(F)[p^\infty])^2} \cdot \prod_{v \in \Sigma_F^{\mathrm{fin}}} c_v \cdot \prod_{v \in S(p)} (\#\tilde{E}_v(\kappa_v))^2 \right|_p^{-1}$$

┘

*Proof:* Cf. [Coates, Galois cohomology of Elliptic Curves, P28].

□

## 8 Examples

**Example (31.2.8.1).** There are three elliptic curves of conductor 11 over  $\mathbb{Q}$ , namely

$$A_0 : y^2 + y = x^3 - x^2 - 10x - 20$$

$$A_1 : y^2 + y = x^3 - x^2$$

$$A_2 : y^2 + y = x^3 - x^2 - 7820x - 263580.$$

And they are all isogenous.

┘

*Proof:*

□

### 31.3 Special Values of $L$ -Functions

*“Mathematical science is in my opinion an indivisible whole, an organism whose vitality is conditioned upon the connection of its parts.”*

–*D. Hilbert*

References are [Del79], [Conrad BSD notes], [Remarks on special values of  $L$ -functions, Scholl], [nLab]. [Introduction to conjecture of Bloch and Kato, Bellaïche].

#### 1 Deligne’s Conjecture

**Prop. (31.3.1.1).** If  $X \in \text{SmProj}/\mathbb{Q}$  and  $M = h^i(X)(m)$ , then  $M^\vee \cong h^i(X)(i - m)$ , i.e. it is polarized of weight  $i - 2m$  ?. ┘

*Proof:* ? Let  $\eta \in h^2$  be a hyperplane section, then

$$H^{2d}(X)(d) \cong H^{2d}(\mathbb{P}^N)(d) = (\eta(1))^{\otimes d} \cong 1.$$

There is a pairing

$$h^i(X)(m) \times h^{2d-i}(X)(d - m) \rightarrow h^{2d}(X)(d),$$

and also an isomorphism

$$h^i(X)(i - m) \xrightarrow{\cup \eta^{d-i}} h^{2d-i}(X)(d - m).$$

□

**Def. (31.3.1.2) [Critical Values].** For  $M \in \text{Mot}(\mathbb{Q})$ ,  $m \in \mathbb{Z}$  is called a **critical value** for  $M$  if neither  $L_\infty(M; s)$  nor  $L_\infty(M^\vee; 1 - s)$  has a pole at  $s = m$ .  $M$  is called a **critical motive** if 0 is critical for  $M$ . ┘

**Prop. (31.3.1.3).**  $M$  is critical iff the Betti realization satisfies

- $\text{Real}_{\text{Betti}}(M)^{p,q} = 0$  unless  $p = q$  or  $p < 0 \leq q$  or  $q < 0 \leq p$ .
- $c$  acts on  $\text{Real}_{\text{Betti}}(M)^{p,p}$  by  $-1$  if  $p \geq 0$ , and  $1$  if  $p < 0$ .

┘

#### Deligne’s Periods

**Prop. (31.3.1.4) [Deligne’s Periods].** We assume that  $M \in \text{Mot}(\mathbb{Q})$  is pure of weight  $w$ , and  $c = \pm 1$  on  $M^{w/2, w/2}$  if  $w \in 2\mathbb{Z}$ . Notice these hypothesis are satisfied when  $M$  is critical.

Define

$$d(M) = \dim_{\mathbb{Q}} \text{Real}_{\text{Betti}}(M), \quad d^\pm(M) = \dim_{\mathbb{Q}} \text{Real}_{\text{Betti}}(M)^{c=\pm 1},$$

And also the eigenvalue decomposition of  $c^*$  on  $\text{Real}_{\text{Betti}}(M)$ :

$$\text{Real}_{\text{Betti}}(M) = \text{Real}_{\text{Betti}}(M)^+ \oplus \text{Real}_{\text{Betti}}(M)^-.$$

Then it follows from (22.1.3.18) that

$$I_{\text{dR}}(\text{Real}_{\text{Betti}}(M)^+ \otimes \mathbb{R}) \subset \text{Real}_{\text{dR}}(M), \quad I_{\text{dR}}(\text{Real}_{\text{Betti}}(M)^- \otimes \mathbb{R}) \subset i \cdot \text{Real}_{\text{dR}}(M).$$

There are filtration steps  $\text{Fil}^\pm \subset \text{Real}_{\text{dR}}(M)$  of the deRham filtration s.t.

$$\text{Fil}^\pm \text{Real}_{\text{dR}}(M) \otimes \mathbb{C} = I_{\text{dR}}(\oplus_{p>q} \text{Real}_{\text{Betti}}(M)^{p,q} \oplus (\text{Real}_{\text{Betti}}(M)^{w/2,w/2})^{c=\pm 1}).$$

Then we can define

$$\text{Real}_{\text{dR}}(M)^\pm = \text{Real}_{\text{dR}}(M) / \text{Fil}^\mp \text{Real}_{\text{dR}}(M).$$

Since  $\text{Real}_{\text{Betti}}(M)^{p,q} = \overline{\text{Real}_{\text{Betti}}(M)^{q,p}}$ , the space  $\text{Real}_{\text{Betti}}(M)^\pm$  lies (anti)diagonally, so the following maps

$$I^\pm : \text{Real}_{\text{dR}}(M)^\pm \otimes \mathbb{C} \subset \text{Real}_{\text{dR}}(M) \otimes \mathbb{C} \xrightarrow{I_{\text{dR}}} \text{Real}_{\text{dR}}(M) \otimes \mathbb{C} \rightarrow \text{Real}_{\text{dR}}(M)^\pm \otimes \mathbb{C}$$

are isomorphisms.

Then because of the  $\mathbb{Q}$ -structure on  $\text{Real}_{\text{dR}}(M)^\pm$  and  $\text{Real}_{\text{dR}}(M)^\pm$ , we have that

$$c_{\text{Del}}^\pm(M) = \det(I^\pm) \in \mathbb{C}^\times / \mathbb{Q}^\times$$

are well-defined, called the **Deligne periods** of  $M$  (if both sides are  $\{0\}$ , let  $c_{\text{Del}}^\pm(M) = 1$ ). In fact, from the argument above, we see that  $I^\pm$  are real, so in fact  $c_{\text{Del}}^\pm \in \mathbb{R}^\times / \mathbb{Q}^\times$ .  $\lrcorner$

**Conj. (31.3.1.5) [Deligne].** Let  $M \in \text{Mot}(\mathbb{Q})$  be critical, then

$$L(M; s) \sim_{\mathbb{Q}} c_{\text{Del}}^+(M) \in \widehat{\mathbb{P}} \text{ (31.3.1.4)}.$$

$\lrcorner$

*Proof:* Cf. [Deligne, Values of L-functions] for the assertion that  $c_{\text{Del}}^+(M) \in \widehat{\mathbb{P}}$ .  $\square$

**Example (31.3.1.6) [Tate Modules].** For the Tate module  $\mathbb{Q}(r)$ ,  $c^* = (-1)^r$  on  $\text{Real}_{\text{Betti}}(\mathbb{Q}(n))$ , so it follows from (22.1.3.17) that

$$c^\varepsilon(\mathbb{Q}(r)) = \begin{cases} (2\pi i)^r & , \varepsilon = (-1)^r \\ 1 & , \varepsilon = (-1)^{r-1} \end{cases}.$$

And  $\mathbb{Q}(r)$  is critical iff  $L_\infty(\mathbb{Q}) = \Gamma_{\mathbb{R}}(s)$  is holomorphic at both  $r, 1-r$ , and this is the case when  $r \in 2\mathbb{Z}_+$  or  $r \in 1 - 2\mathbb{Z}_+$ . The Deligne's conjecture is true in this case, by (31.3.5.3).  $\lrcorner$

## General Coefficients

### 2 Beilinson-Bloch Conjecture

References are [Bei85, Blo00], [Beilinson, Higher regulators of modular curves], [Iwasawa theory for the symmetric square of an elliptic curve, Flach], [Nekovar, Beilinson's Conjecture, in Motives, 537-570], [Algebraic K-Theory and Special Values of L-Functions: Beilinson's Conjecture].

**Notation (31.3.2.1).**

- Denote  $m = i + 1 - n$ .
- We assume that
  - the Euler product for  $L(h^i(X); s)$  converges absolutely for  $\text{Re}(s) > i/2 + 1$ .
  - $L(h^i(X); s)$  meromorphically extends to the whole plane, and possible poles can occur only when  $i$  is even and  $s = i/2 + 1$ .

- $L(h^i(X); i/2 + 1) \neq 0$ .
- $\Lambda(h^i(X); s)$  has a functional equation

$$\Lambda(h^i(X); i + 1 - s) = \varepsilon(h^i(X); s) \Lambda(h^i(X); s).$$

┘

**Prop. (31.3.2.2).** By (16.11.6.8), if  $m < \frac{i+1}{2}$ , there is an exact sequence

$$0 \rightarrow \text{Fil}^n H_{\text{Betti}}^i(X, \mathbb{R}) \rightarrow H_{\text{Betti}}^i(X, \mathbb{R}(n-1))^{\mathbf{c}^* = (-1)^{n-1}} \rightarrow H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)) \rightarrow 0.$$

┘

**Prop. (31.3.2.3).** If the functional equation and

$$\dim_{\mathbb{R}} H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)) = \begin{cases} \text{ord}_{s=m} L(h^i(X), s) & , m < \frac{i}{2} \\ \text{ord}_{s=m} L(h^i(X), s) - \text{ord}_{s=m+1} L(h^i(X), s) & , m = \frac{i}{2} \end{cases}.$$

Notice if  $m < i/2$ ,  $H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)) = \{0\}$  iff  $h^i(X)(m)$  is critical (31.3.1.2).

┘

*Proof:* Cf. [Schneider's notes, P5]. ?

□

**Prop. (31.3.2.4) [Beilinson's Periods].** If  $h^i(X)(m)$  is critical, then  $H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)) = 0$ , and the determinant  $c_{\text{Bei}}$  of the isomorphism

$$\text{Fil}^n H_{\text{Betti}}^i(X, \mathbb{R}) \rightarrow H_{\text{Betti}}^i(X, \mathbb{R}(n-1))^{\mathbf{c}^* = (-1)^{n-1}}$$

satisfies  $c_{\text{Bei}}(h^i(X)(m)) = c_{\text{Del}}(h^i(X)(m)) \in \mathbb{R}^\times / \mathbb{Q}^\times$ .

┘

*Proof:* Cf. [Deligne's conjecture, in Rapoport's notes, P42] ?

□

### Beilinson's Regulator Maps

**Remark (31.3.2.5) [Beilinson's Regulator].** If  $m < (i+1)/2$ , then by (31.3.2.2), there is an isomorphism

$$\det \text{Fil}^n H_{\text{Betti}}^i(X, \mathbb{R}) \otimes_{\mathbb{R}} \det H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)) \rightarrow \det H_{\text{Betti}}^i(X, \mathbb{R}(n-1))^{\mathbf{c}^* = (-1)^{n-1}},$$

and  $\text{Fil}^n H_{\text{Betti}}^i(X, \mathbb{R})$  and  $H_{\text{Betti}}^i(X, \mathbb{R}(n-1))$  has natural  $\mathbb{Q}$ -structures. So if  $\det H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n))$  has a  $\mathbb{Q}$ -structure, then we can construct a  $c_{\text{Bei}}(h^i(X)(m)) \in \mathbb{R}^\times / \mathbb{Q}^\times$ .

┘

**Def. (31.3.2.6) [Beilinson's Regulator].** There is a regulator map (22.4.0.4) (22.4.0.6)

$$\begin{array}{ccc} (K_{2n-i-1}(X_{\mathbb{Z}})_{\mathbb{Q}})^{(n)} & & (K_{2n-i-1}(X_{\mathbb{R}})_{\mathbb{Q}})^{(n)} \\ \parallel & & \parallel \\ r_X^{i+1} : H_{\text{Mot}}^{i+1}(X, \mathbb{Q}(n))_{\mathbb{Z}} & \longrightarrow & H_{\text{Mot}}^{i+1}(X, \mathbb{Q}(n))_{\mathbb{R}} \xrightarrow{\text{ch}_{2n-i-1}} H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)). \end{array}$$

┘

**Conj. (31.3.2.7) [Beilinson].**

- If  $w < -2$ , then
  - $r_{\mathbb{R}} : H_{\text{Mot}}^{i+1}(X, \mathbb{Q}(n))_{\mathbb{Z}} \otimes \mathbb{R} \rightarrow H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n))$  (31.3.2.6) is an isomorphism.
  - the determinant of the isomorphisms (31.3.2.2)

$$\begin{aligned} & \det \text{Fil}^n H_{\text{Betti}}^i(X, \mathbb{R}) \otimes_{\mathbb{R}} \det \left( H_{\text{Mot}}^{i+1}(X, \mathbb{Q}(n))_{\mathbb{Z}} \otimes \mathbb{R} \right) \xrightarrow{\text{id} \otimes \det(r_{\mathbb{R}})} \\ & \det \text{Fil}^n H_{\text{Betti}}^i(X, \mathbb{R}) \otimes_{\mathbb{R}} \det H_{\text{Del}}^{i+1}(X/\mathbb{R}, \mathbb{R}(n)) \rightarrow \\ & \det H_{\text{Betti}}^i(X, \mathbb{R}(n-1))^{\mathbf{c}^* = (-1)^{n-1}} \end{aligned}$$

equals  $L^{\dagger}(M; 0) \in \mathbb{R}^{\times}/\mathbb{Q}^{\times}$ .

- If  $w = -2$ , then  $s = 0$  may be a pole, and Tate conjecture predicts ? that

$$-\text{ord}_{s=0} L(M; 0) = \dim_{\mathbb{Q}} N^{n-1}(X),$$

where  $N^{n-1}(X) = \text{CH}^{n-1}(X)/\text{CH}^{n-1}(X)_0$ , and  $\text{CH}^{n-1}(X)_0$  is the subgroup of homologically trivial cycle. In this case, there is also a cycle class map

$$r' : N^{n-1}(X) \rightarrow H_{M_{\infty}}^{i+1}(X_{\mathbb{R}}, \mathbb{R}(n)) \text{ ?,$$

and

- $(r \otimes r')_{\mathbb{R}} : (C_1)_{\mathbb{R}} \oplus N^{n-1}(X)_{\mathbb{R}} \rightarrow H_{M_{\infty}}^{i+1}(X_{\mathbb{R}}, \mathbb{R}(n))$  is an isomorphism.
- $L^{\dagger}(M, 0) = \det(r \otimes r') \in \mathbb{R}^{\times}/\mathbb{Q}^{\times}$ .

┘

**Def. (31.3.2.8) [Polylogarithm].** For  $s \in \mathbb{Z}_+$ , the power series

$$\text{Li}_s(z) = \sum_{n \geq 1} \frac{z^n}{n^s}$$

converges absolutely for  $|z| < 1$  and can be analytically continued to a multi-valued function on  $\mathbb{C} \setminus \{1\}$ , called the **Polylogarithm function**. ┘

*Proof:* In fact,  $\text{Li}_1(z) = -\log(1-z)$ , and  $\text{Li}_{s+1}(z) = \int_0^z \text{Li}_s(t) \frac{dt}{t}$ . □

**Def. (31.3.2.9) [Bloch-Wigner Dilogarithm].** The function

$$D(z) = \text{Im} \left( \text{Li}_2(z) + \log |z| \log(1-z) \right), \quad z \in \mathbb{C} \text{ (31.3.2.8)}$$

is a real-valued function on  $\mathbb{C}$ , called the **Bloch-Wigner dilogarithm**. ┘

*Proof:* Cf. [Blo00]P44. □

### Height Pairings

**Conj. (31.3.2.10) [Beilinson].** If  $w = -1$ , then  $M$  must be critical by drawing diagram, and

- There is a non-degenerate **Beilinson height paring**

$$h : \text{CH}^n(X)_0 \otimes \text{CH}_0^{\dim X + 1 - n} \rightarrow \mathbb{R},$$

- and  $\text{ord}_{s=0} L(M; s) = \dim_{\mathbb{Q}} \text{CH}^n(X)_0$ .
- $L^{\dagger}(M; 0) = c_{\text{Del}}^+(M) \det(h) \in \mathbb{R}^{\times}/\mathbb{Q}^{\times}$ .

┘

*Proof:* □

### 3 Bloch-Kato/Fontaine-Perrion.Riou Conjecture

Cf.[Fontaine-Perrion.Riou]?

**Conj. (31.3.3.1) [Bloch-Kato].** If  $F \in \mathbf{NField}$  and  $(\rho, V) \in \mathbf{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_F)$  is a pure geometric representation, then the  $L$ -function  $L(V; s)$  (31.1.2.1) has a meromorphic extension to all  $s \in \mathbb{C}$ , and

$$\text{ord}_{s=0} L(V; s) = \dim H_f^1(\text{Gal}_F, V^*(1)) - \dim H^0(\text{Gal}_F, V^*(1)).$$

┘

*Proof:*

□

**Remark (31.3.3.2).** The lower bound on Selmer group asks to construct certain classes, which often takes input from automorphic representations.

The upper bound on Selmer group is usually done by constructing Euler systems.

┘

**Conj. Cor. (31.3.3.3).** Situation as in (31.3.3.1), if  $w(V) \geq 0$  and the meromorphic extension conjecture holds for  $V^*(1)$  (31.1.2.5), then  $H_g^1(\text{Gal}_F, V) = 0$ .

┘

*Proof:*  $V^*(1)$  is pure of weight  $-2 - w \leq -2$ , so  $L(V^*(1); 0) \neq 0$ , unless  $V \cong \mathbb{Q}_p$ , in which case  $\dim H^0(\text{Gal}_F, V) = 1$ . As  $w(V) \geq 0$ , we have  $H_f^1(\text{Gal}_F, V) = H_g^1(\text{Gal}_F, V)$ ?, so this is a conjectural corollary.

□

**Remark (31.3.3.4).** Using the motivic interpretation of  $H_g^1(\text{Gal}_F, V)$  (31.3.3.11), this is compatible with Grothendieck's theory of weights: If  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  is a non-trivial extension and  $M', M''$  are pure motives, then  $w(M') < w(M'')$ .

┘

**Conj. Cor. (31.3.3.5).** Situation as in (31.3.3.1),  $H_g^1(\text{Gal}_F, \text{Ad}V) = 0$ .

┘

*Proof:* ? This is known for most  $V$  attached to modular forms (Weston and Kisin), and also for some polarized  $V$  attached to automorphic forms.

□

**Prop. (31.3.3.6) [Stability Conduction for Inductions].** Let  $F/F_0 \in \mathbf{NField}$  and  $(\rho, V) \in \mathbf{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_F)$  is a pure geometric representation, then the Bloch-Kato conjecture holds for  $V$  (31.3.3.1) iff it holds for  $\text{Ind}_{F/F_0} V \in \mathbf{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_{F_0})$ .

┘

*Proof:* ? Firstly, the induction is also geometric of the same weight by (30.1.2.2).

□

**Thm. (31.3.3.7) [Stability Conduction for Duals].** Situation as in (31.3.3.1), and assume the meromorphic extension conjecture (31.1.2.5) and functional equation conjecture (31.1.2.10) holds for  $V$ , and also Predictions 1.2 and 1.3?, then the Bloch-Kato conjecture for  $V$  is equivalent to the Bloch-Kato conjecture for  $V^*(1)$ .

┘

*Proof:* ? Cf.[Bellaïche, intro to Bloch and Kato].

□

**Cor. (31.3.3.8).** Situation as in (31.3.3.7), under the same hypothesis, if  $w(V) \neq -1$ , then the lower bound for Selmer group holds.

┘

*Proof:* If  $w \geq 0$ , then by (31.1.2.5),  $\text{ord}_{s=0} L(V; s) + \dim H^0(\text{Gal}_F, V^*(1)) = 0$ , so the inequality clearly holds. If  $w \leq -2$ , we consider  $V^*(1)$ , which also has weight  $\geq 0$ , and the inequality for  $V^*(1)$  implies the inequality for  $V$  by the theorem (31.3.3.7).  $\square$

**Conj. (31.3.3.9) [Bloch-Kato for Partial  $L$ -Functions].** Situation as in (31.3.3.1), for any  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$ , we have

$$\text{ord}_{s=0} L_S(V; s) = \dim H_{f,S}^1(\text{Gal}_F, V^*(1)) - \dim H^0(\text{Gal}_F, V^*(1)).$$

┘

*Proof:*

$\square$

**Prop. (31.3.3.10) [Fontaine-Perrin-Riou].** Situation as in (31.3.3.9), for any two  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}, \emptyset \subset_{\text{fin}} S' \subset \Sigma_F^{\text{fin}}$ , if the conjecture on the change of partial Selmer group (31.2.2.5) holds for  $V$ , and the  $F$ -semisimplicity conjecture (9.4.6.41) holds for  $V$ , then the Bloch-Kato conjecture holds for  $L_S(V; s)$  iff it holds for  $L_{S'}(V; s)$ .  $\square$

*Proof:* It suffices to show for  $S' = S \cup \{v\}$  for some  $v \in \Sigma_F^{\text{fin}} \setminus S$ . In this case, the change on the Selmer group side is the dimension of the  $\sigma_v$ -invariant space of  $V$ , and the change in the  $L$ -function side is the order of the local  $L$ -factor, which is the arithmetic dimension of eigenvalue 0 of  $\sigma_v$ , they are equal when  $\sigma_v$  acts semisimply on  $V$ .  $\square$

### Bloch-Kato Conjecture for Mixed Motives

**Conj. (31.3.3.11) [Bloch-Kato for Mixed Motives].** It is conjectured that there is an isomorphism

$$\text{Ext}_{\text{MMot}_F}^1(\mathbb{Q}, M) = H_g^1(\text{Gal}_F, M_p) \text{ (31.2.2.8).?}$$

where  $\text{MMot}_F$  is not defined yet but the LHS can be defined using the  $K$ -theory of  $M$ .  $\square$

*Proof:*

$\square$

## 4 Mahler's Measure and Special Values

References are [Deligne Periods of Mixed Motives, K-Theory and the Entropy of certain  $\mathbb{Z}^n$ -Actions, Deninger], [Mahler's measure and special values of  $L$ -functions, Boyd].

## 5 $\mathbb{A}^1$ Case

**Thm. (31.3.5.1) [Beilinson-Bloch-Kato for Artin Representations].** If  $F \in \mathbf{NField}$  and  $\rho$  is an Artin representation of  $\text{Gal}_F$ , then the Bloch-Kato conjecture holds for  $\rho(n)$  for each  $n \in \mathbb{Z}$ .  $\square$

*Proof:* The lower bounds for Selmer groups hold because  $w(\rho(n)) \neq -1$ . For the upper bound, use Soulé's theorem relating Selmer group to  $K$ -theory, and use Borel's calculation of  $K$ -groups (31.3.5.2)?  $\square$



**Cor. (31.3.5.2)** [ $K$ -Groups of Integral Rings, Borel1972]. For  $F \in \mathbf{NField}$  and  $m \in \mathbb{Z}_{>1}$ , let

$$d_m = \text{ord}_{s=1-m} \zeta_F(s) = \begin{cases} r_2 & , m \in 2\mathbb{Z} \\ r_1 + r_2 & , m \in 2\mathbb{Z} + 1 \end{cases}.$$

Then

$$\dim_{\mathbb{Q}} K_{2m-1}(\mathcal{O}_F)_{\mathbb{Q}} = d_m$$

┘

*Proof:*

□

**Prop. (31.3.5.3).** For  $k \in \mathbb{Z}_+$ ,  $\zeta(-2k) = 0$  and  $\zeta(1-2k) = -\frac{B_{2k}}{2k}$ . And

$$\zeta(2k) = \frac{(-1)^{k+1} B_{2k}}{2(2k)!} (2\pi)^{2k} = \frac{(2\pi i)^{2k}}{2(2k-1)!} \left( -\frac{B_{2k}}{2k} \right).$$

In particular,

$$\begin{aligned} \zeta(2) &= \frac{\pi^2}{6}, & \zeta(4) &= \frac{\pi^4}{90}, & \zeta(6) &= \frac{\pi^6}{945}, & \zeta(8) &= \frac{\pi^8}{9450}. \\ \zeta(0) &= -\frac{1}{2}, & \zeta(-1) &= -\frac{1}{12}, & \zeta(-3) &= \frac{1}{120}, & \zeta(-5) &= -\frac{1}{252}. \end{aligned}$$

┘

*Proof:* This follows easily from (35.5.3.2) and the definition of  $B_k$  (10.7.1.12). The last assertion follows easily this together with (35.5.3.1). □

**Prop. (31.3.5.4)** [Leibniz Formula for  $\pi$ ].

$$L(\chi_{-4}; 1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}.$$

┘

*Proof:* Use the arctan integration formula. This also follows from (35.5.2.6). □

**Prop. (31.3.5.5)** [Comparison of Beilinson and Bloch's Regulator Maps]. Cf. [Rapoport, Comparison].

┘

### Lichtenbaum Conjecture

**Conj. (31.3.5.6)** [Bloch Conjecture for Fields]. For  $F \in \mathbf{NField}$ , mapping the higher  $K$ -groups  $K_{2i-1}(F)$  to a lattice with covolume  $\zeta_F(m)$  via higher regulator maps. And the regular maps are made from  $m$ -th polylogarithm functions (31.3.2.8), so  $\zeta_F(m)$  can be expressed by combinations of  $m$ -th polylogarithm functions of elements in  $F$ .

For example:

$$\zeta_{\mathbb{Q}(\sqrt{5})}(3) = \frac{24}{25\sqrt{5}} \text{Li}_3(1) [\text{Li}_3(\alpha) + \text{Li}_3(-\alpha) + \frac{1}{2}(\log(\alpha))^3 - \frac{\pi^2}{6} \log(\alpha)], \alpha = \frac{\sqrt{5}-1}{2}.$$

This makes the numerical calculation of higher  $K$ -groups possible.

┘

*Proof:* ?

□

**Remark (31.3.5.7).**  $m = 2$  case is proved by Bloch-Suslin-Merkuriev,  $m = 3$  case is proved by Goncharov.  $\lrcorner$

**Thm& Conj.Cor. (31.3.5.8) [Mazur-Wiles/Wiles[Wil90]].** If  $M/F \in \mathbf{NField}$  are totally real and  $p \in \mathbf{Prime}_{\geq 3}$  s.t.  $p \nmid [M : F]$ , then for any  $p$ -adic character  $\chi$  of  $\mathrm{Gal}(M/F)$ , there exists a  $p$ -adic unit  $c$  s.t.

$$\#(K_2(\mathcal{O}_M) \otimes \mathbb{Z}_p[\chi])^\chi = cw_{2,\chi}(M)L(\chi^{-1}, -1)^{[\mathcal{O}_\chi : \mathbb{Z}_p]}.$$

 $\lrcorner$ 

*Proof:* Cf.Cf.[Wil90]P499.  $\square$

**Thm& Conj.Cor. (31.3.5.9) [Lichtenbaum Conjecture for Totally Real Fields, Wiles].** If  $F \in \mathbf{NField}$  is totally real and  $p \in \mathbf{Prime}_{\geq 3}$ , then for any  $n \in 2\mathbb{N} + 1$ , there exists a  $p$ -adic unit  $c$  s.t.

$$\zeta_F(-n) = c \frac{\#H^2(\mathcal{O}_F[\frac{1}{p}], \mathbb{Z}_p(n+1))}{\#H^1(\mathcal{O}_F[\frac{1}{p}], \mathbb{Z}_p(n+1))}.$$

 $\lrcorner$ 

*Proof:* Cf.[Wil90]P499.  $\square$

## 6 Modular Form case

**Thm. (31.3.6.1) [Parity Conjecture for Modular Forms].** Kato proved the upper bound for rank of Selmer groups for modular forms.  $?$

Because the lower bound is known unless  $w(V) = -1$  by (31.3.3.8), we assume  $w(V) = -1$ .

In this case, Nekovar[On the parity of ranks of Selmer groups. 2] proved that if  $f$  is ordinary at  $p$ ,

$$\dim H_f^1(\mathrm{Gal}_F, V) \equiv \mathrm{ord}_{s=0} L(V; s) \pmod{2}.$$

using his theory of Selmer complexes.

And when  $f$  is associated to elliptic curves, this is generalized to the supersingular case by [B. D. Kim] and [Dokchitser-Dokchitser].  $?$

Another direct construction of the extension in  $H_f^1$  without the ordinary condition is given by [Skinner-Urban2006] and [Bellaïche-Chenevier09].  $?$   $\lrcorner$

*Proof:*  $?$   $\square$

**Def. (31.3.6.2) [Elliptic Dilogarithm].** Let  $E \in \mathbf{EllCur}/\mathbb{C}$  with complex parametrization  $\mathbb{C}^\times/q^\mathbb{Z}$ , define the **elliptic dilogarithm**:

$$D_q : \mathbb{C}^\times/q^\mathbb{Z} \rightarrow \mathbb{R} : z \mapsto \sum_{n \in \mathbb{Z}} D(q^n z) \quad (31.3.2.9).$$

And it extends linearly to a map  $Z_1(E) \rightarrow \mathbb{R}$ .  $\lrcorner$

**Prop. (31.3.6.3) [Bloch].** Let  $E \in \mathbf{EllCur}/\mathbb{C}$ , define

$$F : R(E)^\times \otimes R(E)^\times \rightarrow \mathbb{R} : f \otimes g \mapsto D_q(\sum m_i n_j [b_j - a_i]),$$

where  $f, g \in R(E)^*$  with divisor  $(f) = \sum m_i a_i, (g) = \sum n_j b_j$ . Then  $F$  is additive, and

$$D(f \otimes (1 - f)) = 0.$$

In particular,  $F$  factors through  $R(E)^* \otimes R(E)^* / \{f \otimes (1 - f)\} = K_2(R(E))$ .

Thus, there is a composition  $r_E : K_2(E) \rightarrow K_2(R(E)) \rightarrow \mathbb{R}$ , called the (higher) **regulator map**.

┘

*Proof:*

□

**Thm. (31.3.6.4) [Beilinson-Bloch].** Let  $E \in \text{EllCur}/\mathbb{Q}$  with complex parametrization  $E(\mathbb{C}) \cong \mathbb{C}^\times / q^{\mathbb{Z}}$ , then there exists a  $\text{Gal}_{\mathbb{Q}}$ -invariant divisor  $P$  on  $E$  s.t.  $D_q(P) \sim_{\mathbb{Q}^\times} L(E, 2)/\pi$  (31.3.6.2). ┘

*Proof:*

□

**Thm. (31.3.6.5) [Weak Beilinson Conjecture, Beilinson-Bloch].** Let  $E \in \text{EllCur}/\mathbb{Q}$ . Situation as in (31.3.2.1), we consider the motive  $h^1(E), i = 1, m = 1, n = 2$ . Then  $H_{\text{Betti}}^1(E, \mathbb{R}(1))^{c^*=-1} \cong H_{\text{Del}}^2(E/\mathbb{R}, \mathbb{R}(n))$  by (31.3.2.2).

Then there exists  $\Phi \in H_{\text{Mot}}^2(E_{\mathbb{Z}}, \mathbb{Q}(2))$  and  $\Phi \in H_{\text{Betti}}^1(E, \mathbb{Q}(1))^{c^*=-1}$  s.t.

$$r_{\text{Del}}(\Phi) = L^{\vee}(E, 0)\Phi.$$

┘

*Proof:* Cf. [Beilinson, Higher regulators of modular curves] and [Beilinson's Conjecture for elliptic curves with complex multiplication, Deninger-Wingberg].

□

## 7 Shimura Curve case

**Notation (31.3.7.1).**

- Use notation as in [Modular Curves](#).
- Denote  $\Omega^1(M^c) = \varinjlim_K \Omega^1(M_K^c)$ .

┘

**Prop. (31.3.7.2).** Denote  $\{\mathcal{O}^*(M_K), \mathcal{O}^*(M_K)\}$  the  $\mathbb{Q}$ -subspace of  $H_{\text{Mot}}^2(M_K, \mathbb{Q}(2))$  generated by the symbols  $\{u, v\}$  with  $u, v \in \mathcal{O}^*(M_K) \otimes \mathbb{Q} = H_{\text{Mot}}^1(M_K, \mathbb{Q}(1))$ .

Denote

$$\mathcal{Q}_K = H_{\text{Mot}}^2(M_K^c, \mathbb{Q}(2)) \cap \{\mathcal{O}^*(M_K), \mathcal{O}^*(M_K)\},$$

$$\mathcal{P}_K = \cup K' \subset K \text{Nm}_{K'/K}(\mathcal{Q}_{K'}) \subset H_{\text{Mot}}^2(M_K^c, \mathbb{Q}(2)).$$

┘

**Prop. (31.3.7.3).** By [Langlands, Modular Forms and l-adic Representations], The action of  $GL(\mathbf{A}_f)$  on  $\Omega^1(M^c)$  gives rise to a decomposition

$$\Omega^1(M) = \bigoplus_{\pi} V_{\pi}.$$

┘

*Proof:* ? □

**Thm. (31.3.7.4) [Beilinson's Conjecture for Modular Curves].**

- $r_{\mathbb{R}}(\mathcal{P}_K)(31.3.2.7) = H_{\text{Del}}^2(M_K^c/\mathbb{R}, \mathbb{R}(2)) = H_{\text{Betti}}^1(M_K^c(\mathbb{R}), \mathbb{R}(1))$ , thus giving a  $\mathbb{Q}$ -structure on  $H_{\text{Del}}^2(M_K^c, \mathbb{R}(2))$ .
- Let  $g$  be the genus of  $M_K^c$ , then

$$\det r_{M_K^c}(\mathcal{P}_K) = L^{(g)}(H^1(M_K^c), 0) \cdot \det H_{\text{Betti}}^1(M_K^c(\mathbb{R}), \mathbb{Q}(1)).$$

- $\mathcal{P}_K \subset H_{\text{Mot}}^2(M_K^c, \mathbb{Q}(2))_{\mathbb{Z}}$ .

┘

*Proof:* This follows from (31.3.7.5), Cf.[Scholl, P8]. ? □

**Thm. (31.3.7.5).**

- For  $u, v \in \mathcal{O}^*(M_K) \otimes \overline{\mathbb{Q}}$ ,

$$\int_{M_K(\mathbb{C})} \log |u| \overline{\log v} \wedge \omega \in \overline{\mathbb{Q}} \cdot (2\pi i c^+(\pi) L^{\vee}(\pi^{\vee}, 0)) \in \overline{\mathbb{Q}} \otimes \mathbb{C}.$$

- For some choice of  $K$ ,  $\omega \in V_{\pi}^K$  and  $u, v \in \mathcal{O}^*(M_K) \otimes \overline{\mathbb{Q}}$ , the integral is non-zero.

┘

*Proof:* Cf.[Scholl, Thm1.3.2]. ? □

**Thm. (31.3.7.6).** For  $\chi$  s.t.  $L(\pi \otimes \chi; 1) \in (\overline{\mathbb{Q}} \otimes \mathbb{C})^{\times}$ ,

$$c^+(\pi) L^{\vee}(\pi^{\vee}; 0) \sim_{\overline{\mathbb{Q}}^{\times}} \frac{L(\pi; 2) L(\pi \otimes \chi; 1)}{L(\omega_{\pi} \chi; 2)}.$$

┘

*Proof:* Cf.[Scholl, Thm2.3]. ? □

**Prop. (31.3.7.7).** Situation as in (31.3.7.5),

$$\int_{M(\mathbb{C})} \log |u| \overline{\log v} \wedge \omega = \int_{M_K(\mathbb{C})} \mathcal{E}_{\varphi}(z, g; 1) \overline{\eta}_{\xi} \wedge \omega.$$

┘

### Triple Product

**Conj. (31.3.7.8) [Beilinson-Bloch Height Pairing for Triple Product Diagonal Cycles].** If

$\epsilon(F; 2) = -1$ , and let  $X$  be the Shimura curve associated to an Eichler order of reduced discriminant  $N$  ? in the indefinite quaternion algebra  $B$  ramified at the set of finite places  $p \in \text{Prime}$  s.t.  $\epsilon_p(F; 2) = -1$ , then

$$L^{\vee}(F; 2) = \frac{(F, F)}{2\pi N 2^t} \left\langle \Delta_F^0, \Delta_F^0 \right\rangle_{\text{B-B}},$$

where  $\Delta_F$  is the  $F$ -eigencomponent of the Gross-Schoen cycle and  $\Delta$ . ┘

## 31.4 Birch-S.Dyer Conjecture

*“This remarkable conjecture relates the behaviour of a function  $L$ , at a point where it is not at present known to be defined, to the order of a group  $\text{III}$ , which is not known to be finite.”*

–J. Tate

References are [Lectures on the Conjecture of Birch and Swinnerton-Dyer, Gross], [BSD65, Sil99, Wil06, BST21], <http://virtualmath1.stanford.edu/~conrad/BSDseminar/>, [Gross, Kolyvagin’s work on modular elliptic curves(1989)]. [Skinner, Lectures on the Iwasawa theory of elliptic curves, notes, 2018].

### Notation(31.4.0.1).

- Use notations on arithmetics of Abelian varieties defined in ??.
- Use notations on Galois Cohomologies of Abelian varieties defined in [Galois-Cohomologies of Abelian Varieties](#).

⌋

## 1 Statements

### Elliptic Curves

**Conj.(31.4.1.1)[Birch-S.Dyer Formula, [BSD65]].** Let  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$ , and  $L(E; s)$  be the  $L$ -function of  $E$ (31.1.6.2), then

- $\#\text{III}(E) < \infty$ ,  $\text{rank}(E/F) = \text{rank}_{\text{an}}(E/F) = r$ , and
- 

$$L^{\dagger}(E; 1) = \frac{2^{r_2} \Omega_E \left( \prod_{v \in \Sigma_F^{\text{fin}}} c_v \right) \left( \prod_{v \in \Sigma_F^{\text{fin}}} \left| \frac{\omega}{\omega_{E,v}} \right|_v \right)}{\sqrt{|d_K|}} \cdot \#\text{III}(E/K) \cdot \frac{\text{Reg}(E/K)}{(\#E(K)_{\text{tor}})^2},$$

where

- $r_2$  is the number of complex places of  $F$ ,
- $\text{Reg}(E/F)$  is the regulator of the Neron-Tate height pairing  $E(F)_{\text{lf}} \times E(F)_{\text{lf}} \rightarrow \mathbf{R}$ (21.1.3.7),
- $c_v$  are the local Tamagawa numbers,
- $\omega$  is a non-zero global exterior differential form,
- $\Omega = \prod_{v \in \Sigma_F^{\infty}} \int_{E(F_v)} |\omega|_v$ ,
- $\omega_{E,v}$  are the Néron differentials on  $E_v$ .

⌋

*Proof:*

□

**Conj.Cor.(31.4.1.2)[Rank Part of the B-S.D Conjecture].** For  $E \in \mathbf{EllCur}/\mathbf{Q}$  and  $r \in \mathbf{N}$ , the following are equivalent:

- $\text{rank}(E/\mathbf{Q}) = r$  and  $\#\text{III}(E/\mathbf{Q}) < \infty$ .
- $\text{rank}_p(E/\mathbf{Q}) = r$  for any  $p \in \mathbf{Prime}$ .
- $\text{rank}_p(E/\mathbf{Q}) = r$  for some  $p \in \mathbf{Prime}$ .

- $\text{rank}_{\text{an}}(E/\mathbb{Q}) = r$ .

┘

**Conj. (31.4.1.3) [ $p$ -Parts of the Birch-S.Dyer Conjecture].**

┘

### Abelian Varieties

**Conj. (31.4.1.4) [Functional Equation Conjecture].** Let  $F \in \mathbf{NField}$ ,  $A \in \mathbf{AbVar}^g/F$ , then

- the zeta function  $Z(A, s)$  ? extends to an entire function on  $s \in \mathbb{C}$ , and satisfies the functional equation

$$Z(A, s) = w_A Z(A, 2 - s),$$

where  $w_A = \pm 1$  is called the **sign of functional equation of  $A$** , and

- $w_A = w = \prod_{v \in \Sigma_F} w_v$ , where  $w_v$  is the local root number of  $A/K$  at  $v$ .

┘

*Proof:*

□

**Conj. (31.4.1.5) [Birch-S.Dyer-Tate].** Let  $F \in \mathbf{NField}$ ,  $A \in \mathbf{AbVar}^g/F$ , let  $L(A, s)$  be the L-function of  $A$  (31.1.6.2), then

- $L(A, s)$  extends to an entire function on  $s \in \mathbb{C}$ ,  $\#\text{III}(A) < \infty$ ,  $\text{rank}(A/F) = \text{rank}_{\text{an}}(A/F) = r$ ,
- 

$$L^\dagger(A; 1) = \frac{2^{gr_2} \Omega_A \left( \prod_{v \in \Sigma_F^{\text{fin}}} c_v \right) \left( \prod_{v \in \Sigma_F^{\text{fin}}} \left| \frac{\omega}{\omega_{A,v}} \right|_v \right)}{\sqrt{|d_K|}^g} \cdot \#\text{III}(A/F) \cdot \frac{\text{Reg}(A/F)}{\#A(F)_{\text{tor}} \# \hat{A}(F)_{\text{tor}}},$$

where

- $r_2$  is the number of complex places of  $F$ ,
- $\text{Reg}(A/F)$  the regulator of the Neron-Tate height pairing  $A(F)_{\text{lf}} \times \hat{A}(F)_{\text{lf}} \rightarrow \mathbb{R}$  (21.1.3.7),
- $c_v$  the local Tamagawa numbers ?,
- $\omega$  a non-zero global  $g$ -form,
- $\Omega_A = \prod_{v \in \Sigma_F^\infty} \int_{A(F_v)} |\omega|_v$ ,
- $\omega_{A,v}$  are Néron differentials on  $A_v$ .

┘

*Proof:*

□

**Remark (31.4.1.6).**

- The BSDT conjecture has been verified numerically for some elliptic curves over number fields, some Jacobians of genus 2 curves and (up to square) a few Jacobians of higher genus curves.
- BSDT conjecture implies BSD conjecture (31.4.1.1), as any  $E \in \mathbf{EllCur}/K$  is an Abelian variety of dimension 1, and  $E \cong \hat{E}$  canonically.

┘

**Prop. (31.4.1.7).** If the BSDT conjecture holds for all Abelian varieties over  $\mathbb{Q}$ , then it holds for all Abelian varieties over any number fields.

┘

*Proof:* Cf.[The Arithmetic of Abelian Varieties, Milne].?

□

**Conj.Cor.(31.4.1.8)[Parity Conjecture].** Let  $F \in \mathbf{NField}$  and  $A \in \mathbf{AbVar}/F$ , then  $\text{rank}_{\text{an}}(A/F) \equiv \text{rank}(A/F) \pmod{2}$ . This is a consequence of BSDT conjecture(31.4.1.5).

┘

*Proof:*

□

**Thm& Conj.Cor.(31.4.1.9)[Nekovar].** The parity conjecture holds for elliptic curves over a totally real number field  $\mathcal{K}$  if  $\#\text{III}(E/\mathcal{K}) < \infty$ .

┘

*Proof:*

□

## 2 Low Rank cases

References are [Kolyvagin's Conjecture And Patched Euler Systems In Anticyclotomic Iwasawa Theory, Naomi] and <http://www.math.columbia.edu/~chaoli/docs/KolyvaginConjecture.html>.

### Rank 0 case

**Prop.(31.4.2.1).** If  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $p \in \mathbf{Prime}$ ,

$$\text{rank}(E/\mathbb{Q}) = 0, \#\text{III}[p^\infty] < \infty \implies \text{rank}_p(E/\mathbb{Q}) = 0.$$

┘

**Thm& Conj.Cor.(31.4.2.2) [Coates-Wiles[C-W77]/Gross-Zagier[G-Z86]/Rubin[Rub87]/ Kolyvagin[Kol90]/Kato[Kat93, Kat04]].** For  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $p \in \mathbf{Prime}$ ,

$$\text{rank}_{\text{an}}(E/\mathbb{Q}) = 0 \implies \text{rank}(E/\mathbb{Q}) = 0 \quad \& \quad \#\text{III}[p^\infty] < \infty.$$

┘

*Proof:* ? Cf.Gross-Zagier(31.6.2.2) and Kolyvagin(33.5.3.11).

□

### Rank 1 case

References are [Weil Zhang, Selmer groups and the divisibility of Heegner points (2013)] and [the Birch and Swinnerton-Dyer Formula for Elliptic Curves of Analytic Rank One].

**Prop.(31.4.2.3).** If  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $p \in \mathbf{Prime}$ ,

$$\text{rank}(E/\mathbb{Q}) = 1, \#\text{III}[p^\infty] < \infty \implies \text{rank}_p(E/\mathbb{Q}) = 1.$$

┘

**Thm& Conj.Cor.(31.4.2.4)[Gross-Zagier[G-Z86]/Kolyvagin[Kol90]/Kolvaagin-Logachëv[K-L90]/ Bump-Friedberg-Hoffstein/Murty-Murty].** For  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $p \in \mathbf{Prime}$ ,

$$\text{rank}_{\text{an}}(E/\mathbb{Q}) \leq 1 \implies \text{rank}(E/\mathbb{Q}) = \text{rank}_{\text{an}}(E/\mathbb{Q}) \quad \& \quad \#\text{III}[p^\infty] < \infty.$$

┘

*Proof:* ? Cf.Gross-Zagier(31.6.2.2) and Kolyvagin(33.5.3.11).

3 follows from(33.5.3.15).

□

$p$ -Converse Theorems

**Thm. (31.4.2.5)** [Rank 0  $p$ -Converse Theorem for CM Curves, Rubin[Rub91]/ Burungale-Tian[B-T20, B-T22]]. If  $E \in \text{EllCur}/\mathbb{Q}$  has CM and  $p \in \text{Prime}$ , then

$$\text{rank}_p(E/\mathbb{Q}) = 0 \implies L(E; 1) \neq 0.$$

┘

*Proof:* ?

□

**Thm. (31.4.2.6)** [Rank 0  $p$ -Converse Theorem, Skinner-Urban[S-U14]/ Skinner[Ski16]/Wan[?]]. Let  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$  and  $p \in \text{Prime}$  s.t.

- $E$  has good ordinary reduction at  $p$ ,
- $\bar{\rho}_{E,p} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{Aut}(E[p])$  is absolutely irreducible,
- there exists  $\ell \in \text{Prime} \setminus \{p\}$  s.t.  $v_{\ell}(N) = 1$  and  $\bar{\rho}_{E,p}$  is ramified at  $\ell$ ,

then

$$\text{rank}_p(E/\mathbb{Q}) = 0 \implies \text{rank}_{\text{an}}(E/\mathbb{Q}) = 0$$

┘

*Proof:* ?

The proof is given by an extensive generalization of the methods employed by Wiles in his proof of the Iwasawa main conjecture for totally real fields: Eisenstein congruences and Galois representations.

□

**Remark (31.4.2.7).** A rank 0  $p$ -converse theorem for supersingular  $p$  has been established by Wan. ┘

**Thm. (31.4.2.8)** [Rank 1  $p$ -Converse Theorem for CM Curves, Burungale-Tian[B-T20]/ Burungale-Castella-Skinner-Tian[BCST]]. Let  $E \in \text{EllCur}/\mathbb{Q}$  be CM with conductor  $N$  and  $p \in \text{Prime}$  s.t.  $p \nmid 6N$ , then

$$\text{rank}_p(E/\mathbb{Q}) = 1 \implies \text{rank}_{\text{an}}(E/\mathbb{Q}) = 1.$$

┘

*Proof:* ?

□

**Remark (31.4.2.9).** It seems that they can generalize to CM curves over totally real fields. ? ┘

**Thm. (31.4.2.10)** [Rank 1  $p$ -Converse Theorem, Skinner-Urban[S-U14]/ Zhang[Zha14a]/Skinner[Ski16]]. Let  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$  and  $p \in \text{Prime}$  s.t.

- $E$  has good ordinary reduction at  $p$ , and  $p \nmid 6N$ .
- $\bar{\rho}_{E,p} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{Aut}(E[p])$  is absolutely irreducible,
- there exists  $\ell \in \text{Prime}$  s.t.  $v_{\ell}(N) = 1$  and  $\bar{\rho}_{E,p}$  is ramified at  $\ell$ ,

then

$$\text{rank}_p(E/\mathbb{Q}) = 1 \implies \text{rank}_{\text{an}}(E/\mathbb{Q}) = 1.$$

┘



*Proof:* ? Cf.[Skinner, Lectures on the Iwasawa theory of elliptic curves, notes, 2018].

The proof is given by an extensive generalization of the methods employed by Wiles in his proof of the Iwasawa main conjecture for totally real fields: Eisenstein congruences and Galois representations.  $\square$

**Remark(31.4.2.11).** Question: is it possible to remove the absolutely irreducible condition in the converse theorems(the Eisenstein primes)?, maybe by using [Greenberg and V. Vatsal, On the Iwasawa invariants of elliptic curves2000]. Cf.[CGLS].  $\lrcorner$

**Remark(31.4.2.12) [Open Questions in the  $p$ -Converse Theorems].** The following cases of  $p$ -converse theorems are still unknown:

- primes  $p$  of additive reduction.
- cases where  $E[p]$  is reducible or has small Galois images.
- $p \leq 3$ , especially  $p = 2$ .

 $\lrcorner$ 

*Proof:* ?  $\square$

### Refined B-S.D Conjecture

**Thm. (31.4.2.13)[CM case, Rubin[Rub91]/Kobayashi[Kob13]].** Let  $E \in \text{EllCur}/\mathbb{Q}$  with CM by an imaginary quadratic field  $\mathcal{K}$  and with conductor  $N$ . Then:

- The B-S.D formula(31.4.1.1) holds up to an element in  $\mathbb{Z}[\frac{1}{\#\mathcal{O}_{\mathcal{K}}^*}]^*$  if  $\text{rank}_{\text{an}}(E/\mathbb{Q}) = 0$ .
- The B-S.D formula(31.4.1.1) holds up to an element in  $\mathbb{Z}[\frac{1}{\#\mathcal{O}_{\mathcal{K}}^* \cdot N}]^*$  if  $\text{rank}_{\text{an}}(E/\mathbb{Q}) = 1$ .

 $\lrcorner$ 

*Proof:* ?  $\square$

**Thm. (31.4.2.14)[General case, Kato[Kat04]/Skinner-Urban[S-U14]/Zhang[Zha14a]/Skinner[Ski16]/Jetchev-Skinner-Wan[JSW17]].** Let  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$  and  $p \in \text{Prime}_{\geq 5}$ . Suppose that

- $E$  has good ordinary reduction at  $p$ .
- $\bar{\rho}_{E,p} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{Aut}(E[p])$  is absolutely irreducible,
- there exists  $\ell \in \text{Prime} \setminus \{p\}$  s.t.  $v_{\ell}(N) = 1$  and  $\bar{\rho}_{E,p}$  is ramified at  $\ell$ ,

then the  $p$ -part of the B-S.D formula(31.4.1.1) holds if  $\text{rank}_{\text{an}}(E/\mathbb{Q}) \leq 1$ .  $\lrcorner$

*Proof:* ?

See also [Congruences between modular forms and the Birch and Swinnerton-Dyer conjecture] and [On the  $p$ -part of the Birch-Swinnerton-Dyer formula for multiplicative primes, Castella].  $\square$

**Thm. (31.4.2.15).** If  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$  satisfies  $\text{rank}_{\text{an}}(E/\mathbb{Q}) = 1$ ,  $p \in \mathbf{P} \setminus \{2, 3\}$  and  $\bar{\rho}_{E,p}$  is surjective and ramifies at any  $\ell \in \mathbf{P}$  s.t.  $v_{\ell}(N) = 1$ , then the  $p$ -part of the BSD formula(31.4.1.3) holds.  $\lrcorner$

*Proof:* Cf, [Weil Zhang, Selmer groups and the divisibility of Heegner points (2013)]. or <http://www.math.columbia.edu/~chaoli/docs/KolyvaginConjecture.html>.  $\square$

### Statistical results

**Cor. (31.4.2.16) [Bhargava-Skinner-Zhang].** When the isomorphism classes in  $\mathbf{EllCur}/\mathbb{Q}$  are ordered by the statistical height (35.9.2.1),

- At least 16%  $E$  has  $\text{rank}(E) = \text{rank}_{\text{an}}(E) = 0$ .
- At least 20%  $E$  has  $\text{rank}(E) = \text{rank}_{\text{an}}(E) = 1$ .
- At least 66%  $E$  has  $\text{rank}(E) = \text{rank}_{\text{an}}(E) \in \{0, 1\}$  and  $\#\text{III}(E) < \infty$ .

┘

*Proof:* Use 5-adic converse theorems. ?

□

### 3 over Function Fields

**Thm. (31.4.3.1) [Artin-Tate].** For  $F \in \mathbf{FField}$  and  $E \in \mathbf{EllCur}/F$ , analogue of the BSD conjecture (31.4.1.1) holds for  $E$  iff  $\text{III}(E/F)$  is finite.

┘

*Proof:* ? Cf. [Milne, Arithmetic Duality Theorems]Cor9.7.

□

### 4 Motivic B-S.D Conjectures

**Prop. (31.4.4.1) [Motivic B-S.D conjectures].** For  $E \in \mathbf{EllCur}/\mathbb{Q}$ , consider the motive  $h^1(E)$ , then

$$L(E, s) = Z(h^1(E), s).$$

More generally, if we consider L-functions of the form

$$L_3(E, s) = \prod_p L(E_{\mathbb{F}_{p^3}}, s).$$

Then there exists a motive  $M \in M_{\text{num}}(\mathbb{Q})$  s.t.

$$h^1(E) \otimes h^1(E) \otimes h^1(E) = 3h^1 E(-1) \oplus M,$$

and

$$L_3(E, s) = Z(M, s).$$

┘

*Proof:*

□

### 5 Mazur-Tate Conjecture

References are [Mazur-Tate Conjecture, Tony Feng].

**Conj. (31.4.5.1) [Mazur-Tate].** Let  $E \in \mathbf{EllCur}/\mathbb{Q}$  and  $M \in \mathbb{Z}_+$  is not divisible by any  $p^2$  where  $p \in \mathbf{Prime}$  and  $E$  has split multiplicative reduction at  $p$ . They used modular symbol to find an element  $\theta \in \mathbb{Q}[(\mathbb{Z}/(M))^*/\{\pm 1\}]$ . Let  $R \subset \mathbb{Q}$  be a subring that contains all the coefficients of  $\theta$  and that the order of the torsion groups of  $E(\mathbb{Q})$  are invertible in  $R$ . Let  $I \subset R[(\mathbb{Z}/(M))^*/\{\pm 1\}]$  be the augmentation ideal.

Let  $r = \text{rank } E(\mathbb{Q})$  and  $r'$  the number of primes  $p \in \mathbf{Prime}$  dividing  $M$  s.t.  $E$  has split multiplicative reduction at  $p$ , then  $\theta \in I^{r+r'}$ , and there is a formula relating the image of  $\theta \in I^{r+r'}/I^{r+r'+1}$  with  $\#\text{III}(E)$ .

┘

## 6 Nagao's Conjecture

**Conj. (31.4.6.1) [Statistical B-S.D Conjecture].** For any  $E \in \mathbf{E11Cur}/\mathbb{Q}$ , there exists  $C_E \in \mathbb{R}$  s.t.

$$\prod_{p \in \mathbf{Prime}, p \leq X, p \nmid \mathfrak{D}_E} \frac{N_p}{p} \sim C_E (\log X)^{\text{rank}(E/\mathbb{Q})}, \quad X \rightarrow \infty.$$

┘

*Proof:* ? What's the relation to BSD Conjecture? □

**Thm. (31.4.6.2).** For any  $E \in \mathbf{E11Cur}/\mathbb{Q}$ , if the Riemann hypothesis holds for  $L(E; s)$ , then there is a sequence  $x_n \in [2^n, 2^{n+1}]$  s.t.

$$\lim_{n \rightarrow \infty} \frac{1}{\log x_n} \sum_{p \in \mathbf{Prime}, p < x_n} \frac{a_p \log p}{p} = -\text{rank}_{\text{an}}(E/\mathbb{Q}) + \frac{1}{2}.$$

┘

**Cor. (31.4.6.3) [Kim-Murty[K-M23]].** For any  $E \in \mathbf{E11Cur}/\mathbb{Q}$ , if the limit

$$\lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{p \in \mathbf{Prime}, p < X} \frac{a_p \log p}{p}$$

exists, then the Riemann hypothesis for  $L(E; s)$  is true, and the limit equals  $-\text{rank}_{\text{an}}(E/\mathbb{Q}) + \frac{1}{2}$ . ┘

*Proof:* ? □

### 31.5 Artin $L$ -Functions: Stark Conjectures

References are [Sta77], [Tat84].

#### 1 Stark Conjecture

**Notation (31.5.1.1).**

- Let  $F \in \mathbf{NField}$ .

┘

**Conj. (31.5.1.2) [Stark].** Let  $L/F \in \mathbf{NField}$  and  $\rho \in \text{Rep}^{\text{fd}}(\text{Gal}(L/F))$  with character  $\chi$  s.t.  $\rho$  is trivial on  $\text{Gal}_L$ . Let  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  and  $S_L \subset \Sigma_L$  the set of places of  $L$  over  $S$ . Let  $Y$  be the free module on  $S$  and  $X \subset Y$  be the submodule of elements  $\sum n_w w$  s.t.  $\sum n_w = 0$ .

Let  $r_\chi = \dim \text{Hom}_{\text{Gal}(L/F)}(\rho^*, X_{\mathbb{C}})$ .

Let  $U$  be the set of  $S_L$ -units in  $L$  and define

$$\lambda : U_{\mathbb{R}} \rightarrow X_{\mathbb{R}} : u \mapsto \sum_w \log |u|_w w,$$

which is an isomorphism by the unit theorem (19.4.1.20).

For any isomorphism  $f : X_{\mathbb{Q}} \cong U_{\mathbb{Q}}$ , define the **Stark regulator**  $R(\chi, f)$  to be the determinant of the endomorphism of  $\text{Hom}(\rho^*, X_{\mathbb{C}})$  induced by  $\lambda_{\mathbb{C}} \circ f_{\mathbb{C}}$ . Then:

- There exists some  $A(\chi, f) \in \mathbb{C}^\times$  s.t. when  $s \rightarrow 0$ ,

$$L_S(\chi; s) \sim \frac{R(\chi, f)}{A(\chi, f)} s^{r(\chi)} \quad (31.1.1.10).$$

- For any  $\sigma \in \text{Aut}(\mathbb{C})$ , we have  $\sigma(A(\chi, f)) = A(\chi^\sigma, f)$ .

┘

**Thm. (31.5.1.3) [Stark].** The Stark conjecture is true if  $F = \mathbb{Q}$  or  $F$  is an imaginary quadratic field. ┘

*Proof:*

□

#### Rational case

**Thm& Conj. Cor. (31.5.1.4) [Tate1984].** The Stark conjecture is true if  $\text{Im}(\chi) \in \mathbb{Q}$ . ┘

*Proof:*

?

□

#### Rank 0 case

**Prop. (31.5.1.5) [Siegel].** Let  $L/F \in \mathbf{NField}$  be an Abelian extension and  $\emptyset \subset_{\text{fin}} S \subset \Sigma_L^{\text{fin}}$ ,  $\sigma \in \text{Gal}(L/F)$ , then the value of the partial  $L$ -function (35.5.2.1) satisfies  $\zeta_S(F, \sigma; 0) \in \mathbb{Q}$ . ┘

*Proof:* Cf. [On evaluation of zeta functions of totally real algebraic number fields at non positive integers, Shintani]. ?

□

**Thm& Conj.Cor.(31.5.1.6)[Stark Conjecture, Rank 0 case].** If  $r(\chi) = 0$ , then the Stark conductor  $R(\chi, f) = 1$ , and the Stark conjecture is equivalent to:

$$\alpha(L_S(\chi; 0)) = L_S(\chi^\alpha; 0)$$

for any  $\alpha \in \text{Aut}(\mathbb{C})$ . Using a refinement of Brauer's theorem ?,

$$L(\chi; s) = \sum_{\sigma \in \text{Gal}(L/F)} \chi(\sigma) \zeta_S(\sigma; s),$$

and  $\zeta_S(\sigma; s) \in \mathbb{Q}$  (31.5.1.5), so the assertion is true. ┘

### Rank 1 case

**Def.(31.5.1.7)**  $[e_\chi]$ . Let

$$e_\chi = \frac{\chi(1)}{\#G} \sum_{\sigma \in G} \chi(\sigma^{-1}) \sigma \in \mathbb{C}[G]$$

be the projection operator that maps any representation to its  $\chi$ -component. ┘

**Def.(31.5.1.8)** **[Stark Units]**. ┘

### Abelian case

**Conj.(31.5.1.9)** **[Refined Stark Conjecture  $\text{St}(L/F, S)$ , Abelian case]**. Let  $L/F \in \mathbf{NField}$  be an Abelian extension and  $\Sigma_F^\infty \subset_{\text{fin}} S \subset \Sigma_F$  satisfying some conditions ?. Then for any  $w \in \text{Hom}(F, \mathbb{R})$ , there exists  $\varepsilon(K, S, w) \in \mathcal{O}_{L, S_L}$  (31.5.1.2) s.t. for any  $\sigma \in G$ ,

$$\sigma(\varepsilon(K, S, w)) = \exp(-2\zeta_S^\lambda(\sigma; 0)) \text{ (35.5.2.1)}$$

┘

## 2 Brumer-Stark Conjecture

**Conj.(31.5.2.1)** **[Brumer-Stark Conjecture, B-S( $L/F, T$ )]**. Let  $L/F \in \mathbf{NField}$  be an Abelian extension with Galois group  $G$  and  $v \in \Sigma_F^{\text{fin}}$  splits completely in  $L$ . Let  $\{v\} \cup \Sigma_F^\infty \cup \text{Ram}_L(F) \subset_{\text{fin}} S \subset \Sigma_F$  and  $T = S \setminus \{v\}$ . Define

$$\theta_T(0) = \sum_{\chi \in G^\vee} L(\chi; 0) e_{\bar{\chi}} \in \mathbb{C}[G] \text{ (31.5.1.7),$$

then for any  $\mathfrak{A} \in \text{Ideal}(\mathcal{O}_F)$ , we have

$$\mathfrak{A}^{\#\mu(L)\theta_T} = (\alpha)$$

where  $\alpha \in L$  satisfies some conditions on the absolute values  $|\alpha|_w$  ( $w \in T$ ) and  $L(\alpha^{1/\#\mu(L)})/L$  is Galois and Abelian. ┘

**Remark(31.5.2.2).** Brumer-Stark conjecture is non-trivial for function fields. ┘

## 3 Rubin-Stark Conjecture

Rubin-Stark conjecture is a version of Stark's conjecture over  $\mathbb{Z}$ .

#### 4 Gross-Stark Conjecture

Gross-Stark conjecture is a  $p$ -adic analogy of Stark's conjecture for  $s = 0$ .

#### 5 Serre-Stark Conjecture

Serre-Stark conjecture is a  $p$ -adic analogy of Stark's conjecture for  $s = 1$ .

## 31.6 Gross-Zagier Formula and AGGP Conjecture

References are [Gross-Zagier formula and arithmetic fundamental lemma, Wei Zhang], [Gross-Zagier Formula notes], [Heegner Points and Representation Theory, Gross], [Gross-Zagier Revisited, Conrad]. [Moduli of Elliptic Curves, Parson], [Heegner points and derivatives of L-series, Gross-Zagier]. <http://math.columbia.edu/~yihang/GZSeminar.html>.

### 1 Heegner Points

**Def. (31.6.1.1) [Heegner Conditions].** Let  $E \in \text{EllCur}/\mathbb{Q}$  and  $\mathcal{K}$  an imaginary quadratic field with  $(\text{disc}(\mathcal{K}), N_E) = 1$  and associated quadratic character  $\eta_{\mathcal{K}} : (\mathbb{Z}/(\text{disc}(\mathcal{K})))^{\times} \rightarrow \{\pm 1\}$ . Then the **Heegner condition** is the hypothesis that every prime factor of  $N_E$  splits in  $\mathcal{K}$ .  $\lrcorner$

**Prop. (31.6.1.2).** In the Heegner situation (31.6.1.1), the sign of functional equation for  $E$  (31.1.6.4)  $w_E = -1$   $\color{red}{?}$ , thus  $\text{rank}_{\text{an}}(E/\mathbb{Q})$  is odd and the BSD conjecture (31.4.1.1) predicts  $\text{rank}(E/\mathbb{Q})$  is odd, so at least it has a non-torsion rational point.  $\lrcorner$

**Def. (31.6.1.3) [Heegner Points].** In the Heegner situation (31.6.1.1), for any  $n \in \mathbb{Z}_+$  s.t.  $(n, N_E) = 1$ ,

$$\mathbb{C}/\mathcal{O}_{\mathcal{K},n} \rightarrow \mathbb{C}/(\mathcal{O}_{\mathcal{K},n} \cap \mathcal{O}_{\mathcal{K},N_E})^{-1}$$

is an isogeny of degree  $N$  of complex elliptic curves, thus defines a point  $x_n \in X_0(N)(\mathbb{C})$ , by moduli characterization, called a **Heegner point**. Then by theory of complex multiplication  $\color{red}{?}$ ,  $x_n$  is defined over the ring class field  $\mathcal{K}_n$  corresponding to  $\mathcal{O}_{\mathcal{K},n}$  (19.7.4.37).  $\lrcorner$

*Proof:*

$\square$

**Def. (31.6.1.4) [Heegner Points of Elliptic Curves].** In the Heegner situation (31.6.1.1), define  $y_n = \varphi_E(x_n) \in E(\mathcal{K}_n)$  (31.6.1.3), where  $\varphi_E$  is the modular parametrization  $\varphi_E : X_0(N) \rightarrow E$  over  $\mathbb{Q}$  (30.4.3.7). Also define  $y_{\mathcal{K}} = \text{tr}_{\mathcal{K}_1/\mathcal{K}}(y_1) \in E(\mathcal{K})$ , called the (principal) **Heegner point of  $E$** .  $\lrcorner$

**Prop. (31.6.1.5).** The Heegner point  $y_{\mathcal{K}} \in E(\mathcal{K})$  (31.6.1.4) is uniquely defined up to sign and torsion. In particular,  $\langle y_{\mathcal{K}}, y_{\mathcal{K}} \rangle_{\text{N-T}}$  is well-defined.  $\lrcorner$

*Proof:*

$\square$

### 2 Gross-Zagier Formula

**Thm. (31.6.2.1) [Gross-Zagier [G-Z86]].** In the Heegner situation (31.6.1.1),

$$L'(E/\mathcal{K}; 1) = \frac{\int_{E(\mathbb{C})} \omega \wedge \bar{\omega}}{|d_{\mathcal{K}}|^{1/2}} \cdot \frac{1}{c^2} \cdot \langle y_{\mathcal{K}}, y_{\mathcal{K}} \rangle_{\text{N-T}} \quad (31.6.1.5).$$

$\lrcorner$

*Proof:* Cf. [GZ86] 1.(6.5).  $\color{red}{?}$

$\square$

**Cor. (31.6.2.2).** If  $\text{rank}_{\text{an}}(E/\mathcal{K}) = 1$ , then the Heegner point  $y_{\mathcal{K}}$  is non-torsion, by (21.1.3.5). And if  $\text{rank}_{\text{an}}(E/\mathcal{K}) \geq 1$ ,  $y_{\mathcal{K}}$  is torsion.  $\lrcorner$

**Def. (31.6.2.3) [Gross-Zagier curve].** The elliptic curve  $E \in \text{EllCur}/\mathbb{Q}$  defined by

$$E : -139y^2 = x^3 + 10x^2 - 20x + 8$$

has conductor  $N = 37 \cdot 139^2$  and sign of functional equation  $\epsilon_E = -1$ . And by Gross-Zagier formula,  $\text{ord}_{s=1} L(E; s) = \text{rank}(E) = 3$ .  $\lrcorner$

*Proof:* Cf. [I-K04]P541.  $\square$

### 3 AGGP Conjecture

AGGP conjecture has applications to the Beilinson-Bloch conjecture.

#### Special Cycles

**Def. (31.6.3.1) [Special Pairs].** A **special pair** of Shimura datum is a homomorphism of Shimura datum  $(H, X_H) \rightarrow (G, X_G)$  s.t.

- $H \rightarrow G$  is injective and the pair  $(H, G)$  is spherical.
- $\dim_{\mathbb{C}} X_H = \lfloor \frac{\dim_{\mathbb{C}} X_G}{2} \rfloor$ .

$\lrcorner$

**Def. (31.6.3.2) [Special Cycles].** For a special pair of Shimura datum  $(H, X_H) \rightarrow (G, X_G)$  and open subgroups  $K_H \subset H(\mathbf{A}_f), K_G \subset G(\mathbf{A}_f)$ , there is a finite morphism

$$\delta_{K_H, K_G} : \text{Sh}_{K_H}(H) \rightarrow \text{Sh}_{K_G}(G) \text{ ?}.$$

And then  $z_{K_H, K_G} = \delta_{K_H, K_G, *} [\text{Sh}_{K_H}(H)] \subset \text{CH}^*(\text{Sh}_{K_G}(G))$  is called a **special cycle**. If  $K_H = K_G \cap H(\mathbf{A}_f)$ , then denote  $z_{K_G} \stackrel{\Delta}{=} z_{K_H, K_G}$ .  $\lrcorner$

**Remark (31.6.3.3).** Note that the definition is different from that of [Kudla, Special cycles on unitary Shimura varieties] and [Kudla-Rapoport-Yang, Modular forms and special cycles on Shimura curves]. ?  $\lrcorner$

**Def. (31.6.3.4) [Gross-Zagier Special Pairs].** If  $F' \rightarrow \text{Mat}(2; \mathbb{Q})$  is an embedding where  $F'$  is an imaginary quadratic number field, then it defines a morphism

$$H = \text{res}_{F'/\mathbb{Q}} \mathbb{G}_m \hookrightarrow \text{GL}(2)_{\mathbb{Q}},$$

and  $H_{\mathbb{R}} \cong \mathbb{S}$  (7.1.5.33). In particular, we have a canonical map of special pairs  $(H, X_H) \rightarrow (G, X_G)$ , where  $\dim X_H = 0, \dim G = 1$ . ?

Generally, we can consider the case that  $E/F \in \text{CMField}$  and  $B/F$  is a quaternion algebra that is indefinite at one infinite place and positive definite at other infinite places, together with an embedding  $E \rightarrow B$ . Then we have an embedding  $E^{\times} \rightarrow B^{\times}$ , and the corresponding embedding  $\text{Sh}_{E^{\times}} \subset \text{Sh}_{B^{\times}}$  is called the **CM points** of the Shimura curve. ?  $\lrcorner$



### Restriction Problem

**Prop. (31.6.3.5).** For a special pair  $H \subset G$  (31.6.3.1), we can consider the projective limit of Chow groups  $\mathrm{CH}_*(\mathrm{Sh}_G)$ , which has an action of  $G(\mathbf{A}_F)$ . Then  $\mathrm{Sh}_H$  defines a cycle  $[\mathrm{Sh}_H] \in \mathrm{CH}_{\dim \mathrm{Sh}_H}(\mathrm{Sh}_G)$ . The mapping

$$\ell_H : \mathrm{CH}^{\dim \mathrm{Sh}_H}(\mathrm{Sh}_G)_0 \rightarrow \mathbb{C} : z \mapsto \langle z, [\mathrm{Sh}_H] \rangle_{\mathrm{B-B}}.$$

is called the **arithmetic period pairing**. Then the problem is to decide whether the restriction of  $\ell_H$  to  $\mathrm{CH}^{\dim \mathrm{Sh}_H}(\mathrm{Sh}_G)_0[\pi]$  is non-vanishing for  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(G)$ .  $\lrcorner$

**Notation (31.6.3.6) [AGGP Setting].**

- Let  $F \in \mathbf{NField}$  be totally real,
- Let  $E$  be  $F \times F$  or a quadratic CM extension of  $F$ .
- Let  $\mathbb{V}$  be an incoherent Hermitian space over  $\mathbf{A}_E$  (not of the form  $V \otimes_E \mathbf{A}_E$  for some Hermitian space  $V$  over  $E$ ).
- Assume  $\mathbb{V}$  is positive definite at all Archimedean places of  $F$ .
- Let  $e \in \mathbb{V}$  be an element with norm in  $F^\times$ .
- Let  $\mathbb{V}^\flat$  be the orthogonal complement of  $e$ .
- Let  $G = U(\mathbb{V}) \times U(\mathbb{V}^\flat)$ ,  $H = U(\mathbb{V}^\flat)$  be algebraic groups over  $\mathbf{A}_F$ , and  $H$  embeds in  $G$  diagonally.  $\lrcorner$

**Prop. (31.6.3.7) [Chow Group of Shimura Varieties].**  $\mathbb{G}$  defines a projective system  $X_K$  of Shimura varieties parametrized by open compact subgroups  $K$  of  $\mathbb{G}(\mathbf{A}_F^f)$ . Let  $X$  be the projective limit with an action of  $\mathbb{G}(\mathbf{A}_F^f)$ , and  $\mathrm{Ch}^*(X)$  the direct limit of  $\mathrm{Ch}^*(X_K)$ .

Then there is a decomposition

$$\mathrm{Ch}^*(X)_{\mathbb{C}} = \bigoplus_{\pi \in \mathcal{A}(G)} \pi \otimes \mathrm{Ch}^*(\pi),$$

where  $\mathrm{Ch}^*(\pi) = \mathrm{Hom}_{\mathbb{G}}(\pi, \mathrm{Ch}^*(X)_{\mathbb{C}})$ .

Notice that if  $\pi$  is cuspidal, we conjecture that  $\mathrm{Ch}^*(\pi)$  takes value in  $\mathrm{Ch}^*(X)_{\mathbb{C}}^{00}$  ?. And Beilinson-Bloch conjectured that  $\dim_{\mathbb{C}} \mathrm{Ch}^*(\pi) = \mathrm{ord}_{s=1/2} L(\pi; s)$ , and there is a positive definite height pairing

$$\langle \cdot, \cdot \rangle : \mathrm{Ch}^*(\pi) \otimes \mathrm{Ch}^*(\tilde{\pi}) \rightarrow \mathbb{C}.$$

Dually, if we consider  $\mathrm{Ch}_*(X)$  the projective limit of  $\mathrm{Ch}_*(X_K) = \mathrm{Ch}^{\dim X - *}(X_K)$ , then

$$\mathrm{Ch}_*(X) = \prod_{\pi \in \mathcal{A}(\mathbb{G})} \pi^\vee \otimes \mathrm{Ch}_*(\pi^\vee).$$

and  $\mathrm{Ch}_*(\pi^\vee) = \mathrm{Ch}^{\dim X - *}(\tilde{\pi})$ .  $\lrcorner$

*Proof:*

$\square$

**Conj. (31.6.3.8) [Arithmetic Gan-Gross-Prasad Conjecture].** In the AGGP setting,  $\mathbb{H}$  defines a projective system  $Y_K$  of sub-Shimura varieties of  $X_K$  with projective limit  $Y$ , which has component  $Y_\pi \in \pi^\vee \otimes \mathrm{Ch}^*(\tilde{\pi})$ .

Because  $Y$  is invariant in  $\mathbb{H}^?$ , we may replace  $\pi^\vee$  by  $\mathcal{P}(\pi) = (\pi^\vee)^\mathbb{H}$  (29.10.4.2).  
 Suppose  $\mathcal{P}(\pi) \neq 0$ , then

$$\langle Y_\pi, Y_{\widetilde{\pi}} \rangle_{\text{B-B}} = \frac{L(M^\vee(1); 0)}{2^\beta L(\pi, \text{ad}; 1)} L^\vee(\pi; 1/2),$$

where  $2^\beta = \#C_{\widehat{G}}(\text{Arthur}(\pi))$ , and the pairing is induced from Beilinson-Bloch height pairing (31.6.3.7) and the normalized invariant pairing (29.10.4.2).  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj.Cor. (31.6.3.9).** The AGGP conjecture is true when

- $\mathbb{V}$  is unitary and  $\dim \mathbb{V} = 2$  by Yuan-Zhang-Zhang.
- $\mathbb{V}$  is orthogonal and  $\dim \mathbb{V} = 4$  by [YZZ, Triple product L-series and Gross-Kudla-Schoen cycles].  $\lrcorner$

*Proof:*  $?$   $\square$

### Relative Trace Formula Approach

### Arithmetic Theta Lifting Approach

## 4 Special Divisors on Unitary Shimura Varieties

### Kudla-Rapoport Cycles

**Def. (31.6.4.1)[Kudla-Rapoport Cycles].** Let  $\mathcal{M}$  be the integral RSZ Shimura variety as in (28.11.3.5) and  $(A_0, A, \bar{\eta}) \in \mathcal{M}(S)$  with  $S$  connected, then  $\text{End}_F^0(A_0) = F$ , and  $\text{Hom}_{\mathcal{O}_F}(A_0, A)$  becomes a Hermitian  $\mathcal{O}_F$ -module with the pairing

$$\langle x, y \rangle = \lambda_0^{-1} \circ y^\vee \circ \lambda \circ x \in \text{End}_F^0(A_0) \cong F.$$

Then for any  $\xi \in F_0^\times$  and  $\mu \in V(F_{0,\mathfrak{d}})/K_{\mathfrak{d}}$ , we can consider the moduli problem  $\mathcal{Z}(\xi, \mu)$  over  $\mathcal{O}_F[\mathfrak{d}^{-1}]$  which associates to each  $S$  the tuple  $(A_0, A, \bar{\eta}, \mu)$  where

- $(A_0, A, \bar{\eta}) \in \mathcal{M}(S)$  (28.11.3.5),
- $u \in \text{Hom}_{\mathcal{O}_F}(A_0, A)[\mathfrak{d}^{-1}]$  s.t.  $(u, u) = \xi$  and  $\bar{\eta}(u) \in \mu$ .

Then forgetting  $u$  induces a morphism  $\mathcal{Z}(\xi, \mu) \rightarrow \mathcal{M}$  that is finite and unramified, and it is étale locally a Cartier divisor. The pushforward of  $\mathcal{Z}(\xi, \mu)$  inside  $\mathcal{M}$  is also denoted by  $\mathcal{Z}(\xi, \mu)$ .  $?$

Moreover, for  $\varphi_{\mathfrak{d}} \in \mathcal{S}(V_{\mathfrak{d}})^{K_{\mathfrak{d}}}$ , denote

$$\mathcal{Z}(\xi, \varphi_{\mathfrak{d}}) = \sum_{\mu \in V(F_{0,\mathfrak{d}})/K_{\mathfrak{d}}} \varphi_{\mathfrak{d}}(\mu) \mathcal{Z}(\xi, \mu).$$

$\lrcorner$

*Proof:* Cf.[Weil Representations and Arithmetic Fundamental Lemma]Prop7.3.  $?$   $\square$

**Def. (31.6.4.2) [Generating Series of KR-Cycles].** Let  $Z(\xi, \varphi)$  be the generic fiber of the Kudla-Rapoport cycles defined in (31.6.4.1), and also define

$$Z(0, \varphi) = -\varphi(0)c_1(\omega) \in \text{Ch}^1(M),$$

where  $\omega$  is the automorphic line bundle( Cf.[Kudla, Algebraic cycles on Shimura varieties of orthogonal type]).

Then the **generating series** of KR cycles on  $\mathrm{SL}(2; \mathbf{A}_{F_0})$  is defined to be

$$Z(h, \varphi) = Z(0, \omega(h_f)\varphi)W_0^{(n)}(h_\infty) + \sum_{\xi \in F_{0,+}} Z(\xi, \omega(h_f)\varphi)W_\xi^{(n)}(h_\infty),$$

where  $\omega$  is the Weil representation and  $W_\xi^{(n)}$  is the weight  $n$  Whittaker function.( Cf. [Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang]P9. )

### Green Functions

**Def. (31.6.4.3).**

### Modularity

References are [Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang].

**Thm. (31.6.4.4).** The generating series  $Z(h, \varphi)$  (31.6.4.2) lies in  $\mathcal{A}_{\mathrm{hol}}(\mathrm{SL}(2; \mathbf{A}_F), K_{\mathrm{SL}(2)}, n)_{\overline{\mathbb{Q}}} \otimes \mathrm{Ch}^1(M_{\tilde{K}})$ , where  $K_{\mathrm{SL}(2)} \subset \mathrm{SL}(2; \mathbf{A}_F)$  is a compact open subgroup that fixes  $\varphi$  under the Weil representation.

⌋

*Proof:* Cf.[Weil Representations and Arithmetic Fundamental Lemma]Prop8.1. )

□

**Cor. (31.6.4.5)[Almost Modularity of Special Divisors].** With the Shimura variety  $M = M_{\tilde{K}}$  with integral model  $\mathcal{M} = \mathcal{M}_{\tilde{K}}$  over  $\mathrm{Spec} \mathcal{O}_F[\mathfrak{d}^{-1}]$  defined in [Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang]Sec2, endow the automorphic line bundle  $\omega$  with Petersson metrics  $\|\cdot\|_{\mathrm{Pet}}$  at Archimedean places, so that we can define admissible extensions as in (23.1.4.4).

Suppose  $\varphi \in \mathcal{S}(V(\mathbf{A}_{f,F}))^K$  that is  $K_{\mathrm{SL}(2)}$ -invariant under the Weil representation, and  $z \in \mathcal{Z}_1(\mathcal{M})_{\mathrm{deg}=0}$ , then the function

$$\mathrm{SL}(2; \mathbf{A}_F) \rightarrow \mathbb{R} : h \mapsto (Z(h, \varphi), z)^{\mathrm{adm}} \text{ (31.6.4.4)}$$

lies in  $\mathcal{A}_{\mathrm{hol}}(\mathrm{SL}(2; \mathbf{A}_F), K_{\mathrm{SL}(2)}, n) \otimes_{\overline{\mathbb{Q}}} \mathbb{R}_S$ .

⌋

*Proof:* Cf. [Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang]P9. )

□

**Def. (31.6.4.6)[Arithmetic Liftings of  $Z(h, \varphi)$ ].** For any  $\xi \in F_{0,\xi}$ , we can endow the integral K-R cycle  $\mathcal{Z}(h, \varphi)$  with the automorphic Green function  $\mathcal{G}^B(\xi, \varphi)$  to define an arithmetic K-R divisor

$$\widehat{\mathcal{Z}}^B(\xi, \varphi) = (\mathcal{Z}(\xi, \varphi), (\mathcal{G}_v^B(\xi, \varphi))_v) \in \widehat{\mathrm{Ch}}^{1, \mathrm{adm}}(\mathcal{M}).$$

And also

$$\widehat{\mathcal{Z}}^B(0, \varphi) = -\varphi(0)c_1(\widehat{\omega}) \in \widehat{\mathrm{Ch}}^{1, \mathrm{adm}}(\mathcal{M}),$$

where  $\widehat{\omega}$  any extension of the automorphic line bundle  $\omega$  to a line bundle on the integral model  $\mathcal{M}$ , endowed with its Petersson metrics.

Alternatively we can define a lifting  $\widehat{\mathcal{Z}}^K(\xi, \varphi)$  with Kudla's Green functions, Cf. [Arithmetic Fundamental Lemma Conjecture over a general p-Adic Field, Mihatsch-Zhang]P19.

⌋

## 5 CM Cycles on Unitary Shimura Varieties

References are [Mihatsch-Zhang]Section5,

**Def. (31.6.5.1).** Let  $F[t]_{\deg=n}^\circ$  denote the set of monic polynomials  $\alpha$  of degree  $n$  s.t.  $\alpha(0)t^n\bar{\alpha}(t^{-1})$ .  $\lrcorner$

**Def. (31.6.5.2) [Big Fat CM Cycles].** Let  $\mathcal{M}$  be the integral RSZ Shimura variety as in(28.11.3.5). And let  $\alpha \in F[t]_{\deg=n}^\circ$  and  $\mu \in K_{\mathfrak{d}} \backslash U(V)(F_{0,\mathfrak{d}})/K_{\mathfrak{d}}$ , the **big fat CM cycle**  $\mathcal{C}(\alpha, \mu) \rightarrow \mathcal{M}$  is  $\lrcorner$

## 6 Triple Product

Cf.[G-K92] and [G-S95].

**Conj. (31.6.6.1).** We have the formula

$$L^\vee(F; 2) = \frac{(F, F)_{\text{Pet}}}{2^{t+1}\pi N} \langle \Delta_F, \Delta_F \rangle_{\text{B-B}}.$$

$\lrcorner$

## 7 Arithmetic Fundamental Lemma

References are [Arithmetic Fundamental Lemma, Wei Zhang].

**Conj. (31.6.7.1)[Arithmetic Fundamental Lemma, Zhang/Mihatsch-Zhang/Zhang].** The semi-Lie-algebraic version of the arithmetic fundamental lemma is the following: Let  $\mathbb{V}_n$  as in(28.6.3.33), then whenever  $(\gamma, u') \in (S_n \times V'_n(F_0)_{rs})$  matches  $(g, u) \in (U(\mathbb{V}_n) \times \mathbb{V}_n)(F_0)_{rs}$ (31.6.7.5), we have

$$\partial \text{Orb}((\gamma, u'), \mathbf{1}_{(S_n \times V'_n)(\mathcal{O}_{F_0})}) = -\text{Int}(g, u) \cdot \log q.$$

$\lrcorner$

### Group-Theoretic Setup

**Notation (31.6.7.2)[General Linear Groups].** Situation as in(31.6.3.6), let

- $\Phi$  be a CM-type of  $E/F$ , and  $\varphi_0 \in \Phi$
- 

$$H'_1 = \text{res}_{E/F} \text{GL}(n-1), \quad G' = \text{res}_{E/F}(\text{GL}(n-1) \times \text{GL}(n))$$

and  $H'$  embeds in  $G'$  diagonally. And

$$H'_2 = \text{GL}(n-1) \times \text{GL}(n)$$

embeds in  $G'$  in the obvious way.

- Let

$$S_n = \{\gamma \in \text{res}_{E/F} \text{GL}(n) \mid \gamma\bar{\gamma} = \mathbf{1}\}.$$

- For  $\alpha \in F[X]_{\deg=n}^0$ , let  $S_n(\alpha)$  denote the subscheme consisting of  $\gamma$  with characteristic  $\alpha$ .
- Let  $V'_n = F^n \times (F^n)^*$  which is a Hermitian space under the tautological pairing.
- $\text{GL}(n; F)$  acts naturally on  $S_n \times V'_n$ .
- For  $\xi \in F$ , let  $V'_{n,\xi}$  denote the subscheme of  $V'_n$  consisting of elements  $(u_1, u_2^*)$  s.t.  $u_2^*(u_1) = \xi$ .

┘

**Notation (31.6.7.3) [Hermitian Spaces].**

- Let  $V \in \text{Herm}_n(E/F)$ ,
- Let  $U(V)$  acts naturally on  $U(V) \times V$ .

┘

**Def. (31.6.7.4) [Regular Elements].** Notation as in (31.6.7.2), if  $F$  is a local or global field, there is a bijection

$$\mathcal{O}(G'(F)) \triangleq H'_1(F) \backslash G'(F) / H'_2(F) \cong \text{GL}(n-1; F) \backslash \backslash S_n(F) \triangleq \mathcal{O}(S_n(F)).$$

In particular, the notion of regularity is defined for  $\mathcal{O}(G'(F))$ , (Cf. [Regular formal moduli spaces and arithmetic transfer conjectures]P1092).

┘

*Proof:* Cf. [Arithmetic Fundamental Lemma, Wei Zhang]P213.

□

**Prop. (31.6.7.5) [Semi-Lie-Algebraic Orbit Matching].** There is a bijection of orbits of regular semisimple elements ?:

$$\coprod_V [(U(V) \times V)(F)]_{rs} \cong [(S_n \times V'_n)(F)]_{rs},$$

where

$$[(U(V) \times V)(F)]_{rs} = (U(V) \times V)(F)_{rs} // U(V)(F), \quad [(S_n \times V'_n)(F)]_{rs} = (S_n \times V'_n)(F)_{rs} // \text{GL}(n; F).$$

┘

*Proof:* Cf. [Weil Representation and Arithmetic Fundamental Lemma, Wei Zhang]2.2.

□

### Local Orbital Matching

**Notation (31.6.7.6).**

- Let  $L/K$  be a quadratic extension of local fields,
- Let  $\eta : K^\times \rightarrow \{\pm 1\}$  be the quadratic character associated to  $L/K$  by class field theory.
- Let  $W$  be a non-degenerate Hermitian space of dimension  $n-1$ ,
- Let  $V = W \oplus Lu$  where  $\langle u, u \rangle = 1$ .
- Let  $G = G_W = U(W) \times U(V)$ , and  $H = U(W)$  which diagonally embeds into  $G$ ,

┘

**Def. (31.6.7.7) [Regular Elements and Transfers].** Notation as in (31.6.7.6), there is an bijection

$$H(K) \backslash G(K) / H(K) \cong U(W)(K) \backslash \backslash U(V)(K),$$

where  $U(W)$  acts on  $U(V)$  by conjugation. Then  $(\delta_1, \delta_2) \in U(W)(K) \times U(V)(K)$  is called **regular element** if  $\delta_1^{-1}\delta_2 \in U(V)(K)$  is regular in the sense of [arithmetic fundamental lemma, Wei Zhang]Sec2?.

Then  $\gamma = (\gamma_1, \gamma_2) \in G'(K)$  (31.6.7.2) is a **transfer** of  $\delta = (\delta_1, \delta_2) \in G_W(K)$  if the image  $\nu(\gamma_1^{-1}\gamma_2) \in S_n(K)$  and  $\delta_1^{-1}\delta_2 \in U(V)(K)$  are transfers of each other.

┘

**Prop. (31.6.7.8) [Local Unitary Orbital Integrals].** For regular element (31.6.7.7)  $\delta \in G(K)$  and  $f \in C_c^\infty(G(K))$ , the orbital integral

$$O(\delta, f) = \int_{H(K)} \int_{H(K)} f(h_1^{-1} \delta h_2) dh_1 dh_2$$

converges absolutely. ┘

*Proof:* □

**Def. (31.6.7.9) [Smooth Transfer of Functions].**  $f' \in C_c^\infty(G'(K))$  are said to be the **smooth transfer** of a tuple  $(f_W)$  where  $f_W \in C_c^\infty(G_W(K))$  if for any  $W$  and  $\delta \in G_W(K)$  which transfers to a regular element  $\gamma \in G'(F)$  (31.6.7.7), we have

$$O(\delta, f_W) = \Omega(\gamma) O(\delta', f'; 0) \text{ (31.6.7.8) (31.6.7.17).}$$

where  $\Omega(\gamma)$  is the transfer factor ?.

$f' \in C_c^\infty(G'(K))$  is called **pure of type  $W$**  if it is a smooth transfer of a tuple  $(f_W)$  where  $f_{W'} = 0$  unless  $W' \cong W$ . ┘

**Conj. (31.6.7.10) [Existence of Smooth Transfers].** There is a transfer factor  $\Omega$  s.t. smooth transfer (31.6.7.9) always exist. ┘

**Def. (31.6.7.11) [Semi-Lie-Algebraic Linear Orbital Integral].** For  $(\gamma, u') \in (S_n \times V'_n)(K)_{rs}$ ,  $\Phi' \in \mathcal{S}((S_n \times V'_n)(K))$  and  $s \in \mathbb{C}$ , define the **linear orbital integral**

$$\text{Orb}((\gamma, u'), \Phi'; s) = \int_{\text{GL}(n)(K)} \Phi'(h \cdot (\gamma, u')) |\det(h)|^s \eta(h) dh.$$

And also the special values

$$\text{Orb}((\gamma, u'), \Phi') = \text{Orb}((\gamma, u'), \Phi'; 0), \quad \partial \text{Orb}((\gamma, u'), \Phi'; s) = \frac{\partial}{\partial s} \Big|_{s=0} \text{Orb}((\gamma, u'), \Phi'; s).$$

**Def. (31.6.7.12) [Local Semi-Lie-Algebraic Linear Orbital Integral].** For  $(g, u) \in (U(V) \times V)(K)_{rs}$  and  $\Phi \in \mathcal{S}((U(V) \times V)(K))$ , define the **unitary orbital integral**

$$\text{Orb}((g, u), \Phi) = \int_{U(V)(K)} \Phi(h \cdot (g, u)) dh.$$

**Def. (31.6.7.13) [Semi-Lie-Algebraic Smooth Transfer].**  $\Phi' \in \mathcal{S}((S_n \times Vn')(K))$  are said to be the **smooth transfer** of a tuple  $(\Phi_V)$  where  $\Phi_V \in \mathcal{S}((U(V) \times V)(K))$  if for any Hermitian space  $V$  and  $(\delta, u') \in (S_n \times Vn')(K)$  which matches  $(\gamma, u) \in (U(V) \times V)(K)_{rs}$  (31.6.7.7), we have

$$O((g, u), \Phi_V) = \omega((\gamma, u')) O((\gamma, u'), \Phi') \text{ (31.6.7.8) (31.6.7.17).}$$

where  $\omega((\gamma, u'))$  is the transfer factor ( ? Cf. [Weil Representations and Arithmetic Fundamental Lemma, Wei Zhang] Def(2.13). )

And it **partially transfers** to  $(\Phi_V)$  if the above equality holds for any  $(\delta, u') \in (S_n(\alpha) \times Vn')(K)$  which matches  $(\gamma, u) \in (U(V)(\alpha) \times V)(K)_{rs}$ , and

$$O((\gamma, u'), \Phi') = 0$$

for any other  $(\delta, u') \in (S_n(\alpha) \times Vn')(K)$ . ┘

### Global Unitary Orbital Integrals

**Def. (31.6.7.14) [Notation on Global Hermitian Spaces].** For a  $E/F$ -Hermitian space  $W$  and an incoherent Hermitian space  $W$ ,

- Let  $C_c^\infty(G'(\mathbf{A}_F))^W$  be the subspace of functions pure of type  $W$  (31.6.7.9).
- Denote  $\Sigma(W, W) = \{v \in \Sigma_F : W_v \cong W_v\}$ .
- For any  $v \in \Sigma_F$ , there is a unique  $W(v) \in \text{Herm}_n(E/F)$  s.t.  $\Sigma(W(v), W) = \{v\}$ .
- Each regular element  $\gamma \in G'(F)$  is a transfer of a  $\delta \in G_W(F)$  for some Hermitian space  $W$ . This Hermitian space is called the **type** of  $\gamma$ , denoted by  $W(\gamma)$ , and also denote  $\Sigma(\gamma, W) = \Sigma(W(\gamma), W)$ .
- Let  $\mathbb{G} = \mathbb{G}_W, \mathbb{H} = \mathbb{H}_W$  (31.6.7.6).

┘

**Prop. (31.6.7.15) [Kernel Functions].** For  $f' \in C_c^\infty(G'(\mathbf{A}_F))$ , Wei defined a kernel function (Cf. [Arithmetic Fundamental Lemma] P212)

$$K_{f'}(x, y) \in C(G'(\mathbf{A}_F) \times G'(\mathbf{A}_F)).$$

┘

**Prop. (31.6.7.16) [Jacquet-Rallis Distribution].** There is a complex-variable-indexed distribution on  $G'(\mathbf{A}_F)$ :

$$f' \mapsto I(f'; s) = \int_{[H'_1]} \int_{Z_{H'_2}(\mathbf{A}_F) \backslash [H'_2]} K_{f'}(h_1, h_2) \eta(h_2) |h_1|^s dh_1 dh_2.$$

This integral is absolutely convergent if  $f' = \otimes_v f'_v$  is decomposable and  $f'_v$  is regularly supported for some place  $v$ .

┘

*Proof:* Cf. [Arithmetic Fundamental Lemma, Wei Zhang] P213-214.

□

**Prop. (31.6.7.17) [Decomposition as General Linear Orbital Integrals].** There is a decomposition of the Jacquet-Rallis distribution

$$I(f'; s) = \sum_{\gamma \in O(G'(F))_{\text{reg}}} O(\gamma, f'; s),$$

where

$$O(\gamma, f'; s) = \int_{H'_1(\mathbf{A}_F)} \int_{H'_2(\mathbf{A}_F)} f'(h_1^{-1} \gamma h_2) |h_1|^s \eta(h_2) dh_1 dh_2 = \prod_v O(\gamma, f'_v; s).$$

┘

*Proof:* Cf. [Arithmetic Fundamental Lemma, Wei Zhang] P215.

□

**Prop. (31.6.7.18) [Properties of Global Orbital Integrals].** Let  $f' = \otimes_v f'_v \in C_c^\infty(G'(\mathbf{A}_F))^W$ , then

- For any regular element  $\gamma \in G'(F)$ , we have

$$\text{ord}_{s=0} O(\gamma, f'; s) \geq \#\Sigma(\gamma, W).$$

- If  $f'_v$  is regularly supported for some  $v$ , then  $I(f'; 0) = 0$ , and

$$I(f'; 0) \triangleq \left. \frac{\partial}{\partial s} \right|_{s=0} I(f'; s) = \sum_{v \text{ nonsplit}} I'_v(f'; 0)$$

where

$$I'_v(f'; 0) = \sum_{W, \Sigma(W, \mathbb{W})=\{v\}} \sum_{\gamma \in \mathcal{O}(G'(F))_{\text{reg}}, W(\gamma)=W} O(\gamma, f'_v; 0) \cdot O^v(\gamma, f'_v; 0) \quad (31.6.7.14)$$

where

$$O^v(\gamma, f'_v; 0) = \left. \frac{\partial}{\partial s} \right|_{s=0} O(\gamma, f'_v; s).$$

┘

*Proof:* Cf. [Arithmetic Fundamental Lemma, Wei Zhang]P217. □

### Global Jacquet-Rallis Linear Orbital Integrals

**Def. (31.6.7.19) [Global Jacquet-Rallis Linear Orbital Integral].** For each  $v|\infty$ , fix the Archimedean partial Gaussian test function  $\Phi'_v \in \mathcal{S}((S_n \times V')(F_v))_{rs}$  constructed in [Weil Representation of Arithmetic Fundamental Lemma, Wei Zhang]Sec12.4<sup>?</sup>, then for  $h \in \text{SL}(2; \mathbf{A}_F)$  and  $\Phi' = \otimes_v \Phi'_v \in \mathcal{S}((S_n \times V')(\mathbf{A}_F))$ , the global regularized **Jacquet-Rallis linear orbital integral** (relative to  $\alpha$ ) is defined to be

$$\mathbb{J}(h, \Phi'; s) = \int_{[\text{GL}(n)]} \left( \sum_{(\gamma, u') \in (S_n(\alpha) \times V')(F)} r(g) \omega(h) \Phi'(\gamma, u') \right) \deg(g)^s \eta(g) dg.$$

And also denote

$$\mathbb{J}(h, \Phi') = \mathbb{J}(h, \Phi'; 0).$$

┘

**Thm. (31.6.7.20).** The function  $\mathbb{J}(h, \Phi'; s)$  is smooth in  $(h, s)$ , entire in  $s$ , and left invariant under  $\text{SL}(2; F)$ . ┘

*Proof:* Cf. [Weil Representation of Arithmetic Fundamental Lemma, Wei Zhang]Thm12.14. □

**Prop. (31.6.7.21) [Expansion of the Jacquet-Rallis Linear Orbital Integral  $\mathbb{J}$ ].** There is an expansion of  $\mathbb{J}(h, \Phi'; s)$  (31.6.7.19) according to orbits:

$$\begin{aligned} \mathbb{J}(h, \Phi'; s) &= \sum_{(\gamma, u') \in [(S_n(\alpha) \times V')(F)], u' \neq 0} \text{Orb}((\gamma, u'), \omega(h) \Phi'; s) \\ &= \text{Orb}((\gamma, 0_{\pm}), \omega(h) \Phi'; s) + \sum_{(\gamma, u') \in [(S_n(\alpha) \times V')(F)]_{rs}} \text{Orb}((\gamma, u'), \omega(h) \Phi'; s) \end{aligned}$$

where for  $(\gamma, u') \in [(S_n(\alpha) \times V')(F)]_{rs}$ ,

$$\text{Orb}((\gamma, u'), \omega(h) \Phi'; s) = \prod_v \text{Orb}((\gamma, u'), \Phi'_v; s) \quad (31.6.7.12).$$

In particular, for  $\xi \in F^\times$ , the  $\xi$ -th Fourier coefficients of  $\mathbb{J}(h, \Phi'; s)$  w.r.t.  $\text{SL}(2)$  is equal to

$$\sum_{(\gamma, u') \in [(S_n(\alpha) \times V'_\xi)(F)]_{rs}} \text{Orb}((\gamma, u'), \omega(h) \Phi'; s).$$

┘



*Proof:* Cf.[Weil Representation of Arithmetic Fundamental Lemma, Wei Zhang]Thm12.14.  $\square$

**Def. (31.6.7.22).** Denote

$$\text{Orb}((\gamma, u'), \omega(h)\Phi') = \text{Orb}((\gamma, u'), \omega(h)\Phi'; 0) \text{ (31.6.7.21)}.$$

┘

**Prop. (31.6.7.23) [Derivatives of Jacquet-Rallis Orbital Integral].** There is a decomposition

$$\partial J(h, \Phi') = \partial \text{Orb}(0_{\pm}, \omega(h)\Phi') + \sum_v \partial J_v(h, \Phi')$$

where

$$\partial J_v(h, \Phi') = \sum_{(\gamma, u') \in [(S_n(\alpha) \times V')(F)], u' \neq 0} \partial \text{Orb}((\gamma, u'), \omega(h)\Phi'_v) \cdot \text{Orb}((\gamma, u'), \omega(h)\Phi'v).$$

And also  $\partial \text{Orb}(0_{\pm}, \omega(h)\Phi')$  is defined in [Weil Representation of Arithmetic Fundamental Lemma, Wei Zhang]Sec12.7.  $\textcolor{red}{?}$ .  $\square$

*Proof:*  $\square$

### Global Intersections

**Remark (31.6.7.24).** We want to study the global Beilinson-Bloch pairing in the AGGP conjecture. For this, let

$$J(f) = \langle R(f)[\text{Sh}(\mathbb{H})]_0, [\text{Sh}(\mathbb{H})]_0 \rangle_{\text{B-B}}, \quad f \in C_c^\infty(\mathbb{G}),$$

then there is a decomposition

$$J(f) = \sum_v J_v(f).$$

┘

**Thm. (31.6.7.25) [Special cases of  $J_v(f)$ ].** In the case that  $F = \mathbb{Q}$  and  $v$  has good reduction, we can calculate the main term  $J_v^m(f)$  for some specific  $f$ : Suppose  $f = \otimes_v f_v$  satisfies

- $f_p = \mathbb{1}_{K'_{p,0}}$ ,
- For some  $v \neq p$ ,  $f_v$  is regularly supported,
- $f_\infty = \mathbb{1}_{\mathbb{G}(\mathbf{A}_\infty)}$ .

Then

- If  $p$  is inert,

$$J_p^m(f) = 2 \sum_{\delta \in \mathcal{O}(G_{W(p)}(\mathbb{Q})_{\text{reg}}} O(\delta, f^p) O^*(\delta, f_p).$$

- If  $p$  is split, then  $J_p^m(f) = 0$ .

┘

*Proof:* Cf.[Arithmetic Fundamental Lemma, Wei Zhang]P224-227.  $\square$

## 8 Arithmetic Transfer Lemma

**Conj.(31.6.8.1)**[Arithmetic Transfer Lemma, Rapoport-Smithling-Zhang]. ┘

*Proof:* [Rapoport, B. Smithling, W. Zhang, On the arithmetic transfer conjecture for exotic smooth formal moduli spaces], [Regular formal moduli spaces and arithmetic transfer conjectures, Rapoport, B. Smithling, W. Zhang,]. □

## 9 Archimedean Arithmetic Transfer Lemma

## 10 AGGP over Function Fields

References are [Shtukas and the Taylor expansion of  $L$ -functions, Yun-Zhang].

**Remark(31.6.10.1).** The formula relates arbitrary order central derivative of the base change  $L$ -function of an unramified automorphic representation of  $\mathrm{PGL}(2)$  over a function field to the self-intersection number of a certain algebraic cycle on the moduli stack of Shtukas. ┘

## 11 Arithmetic Inner Product Formula

**Conj.(31.6.11.1).** ┘

*Proof:*

□

**Thm& Conj.Cor.(31.6.11.2).** The arithmetic inner product formula conjecture holds for  $m = 1$ , by [Kudla S, Rapoport M, Yang T. Modular Forms and Special Cycles on Shimura Curves] and [Liu Y. Arithmetic theta lifting and  $L$ -derivatives for unitary groups 2]. ┘

*Proof:*

□

## 31.7 Colmez Conjecture

**Notation (31.7.0.1).**

- Use the notion of Faltings height defined in [?](#).

┘

**Thm. (31.7.0.2) [Colmez].** Let  $(E, \Phi)$  be a CM number field and  $S(E, \Phi)$  the set of complex Abelian varieties of CM-type  $(\mathcal{O}_E, \Phi)$ , then for any two  $A, A' \in S(E, \Phi)$ ,  $h_{\text{st}}(A) = h_{\text{st}}(A')$ . This common value is denoted by  $h_{\text{st}}(E, \Phi)$ .

┘

*Proof:* Cf. [Periodes des varietes abeliennes a multiplication complexe]2.2.10.(2). [?](#)

□

**Conj. (31.7.0.3) [Colmez].** Let  $(E, \Phi)$  be a CM number field, then

$$h_{\text{st}}(E, \Phi) = \sum_{\rho} c_{\rho, \Phi} \cdot \left( \frac{L^{\vee}(\rho; 0)}{L(\rho; 0)} + \frac{\log f_{\rho}}{2} \right),$$

where  $\rho$  ranges over irreducible representations of  $\text{Gal}_E$  for which  $L(\rho; 0)$  does not vanish,  $f_{\rho}$  is the Artin conductor of  $\rho$ , and  $c_{\rho, \Phi} \in \mathbb{Q}$ .

┘

*Proof:* [?](#) Cf. [Periodes des varietes abeliennes a multiplication complexe]2.2.11.

□

### Chowla-Selberg Formulas

**Def. (31.7.0.4) [Epstein Zeta Functions].** For a positive-definite quadratic form  $Q(X, Y) = aX^2 + bXY + cY^2 \in \mathbb{Z}[X, Y]$ , the **Epstein zeta function**  $\zeta_Q(s)$  is defined to be

$$\zeta_Q(s) = \frac{1}{2} \sum_{(m, n) \in \mathbb{Z}^2 \setminus (0, 0)} \frac{1}{Q(m, n)^{-s}}.$$

┘

**Thm. (31.7.0.5) [Chowla-Selberg Formula].** For a positive-definite quadratic form  $Q(X, Y) = aX^2 + bXY + cY^2 \in \mathbb{Z}[X, Y]$  with discriminant  $d = b^2 - 4ac < 0$ ,

$$\begin{aligned} Z_Q(s) &= a^{-s} \zeta(2s) + a^{-s} \sqrt{\pi} \frac{\Gamma(s - 1/2)}{\Gamma(s)} \zeta(2s - 1) k^{1-2s} \\ &\quad + \frac{4a^{-s} k^{-s+1/2}}{\pi^{-s} \Gamma(s)} \sum_{n \in \mathbb{Z}_+} n^{s-1/2} \sigma_{1-2s}(n) K_{s-1/2}(2\pi n \frac{\sqrt{d}}{2a}) \cos(n\pi b/a). \end{aligned}$$

┘

*Proof:*

□

**Prop. (31.7.0.6) [Chowla-Selberg Formula].** Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $D$  and  $f \in \mathbb{Z}_+$ . For any proper ideals  $\mathfrak{a} \in \text{Cl}(\mathcal{O}_{\mathcal{K}, f})$  ([19.3.1.54](#)), denote

$$F(\mathfrak{a}) = \Delta(\mathfrak{a}) \Delta(\mathfrak{a}^{-1}),$$

then

$$\prod_{\mathfrak{a} \in \text{Pic}(\mathcal{O}_{K,f})} F(\mathfrak{a}) = \left( \frac{2\pi}{f^2 |d_K|} \right)^{12 \text{cl}(\mathcal{O}_{K,f})} \prod_{a=1}^{|d_K|-1} \left( \Gamma\left(\frac{a}{|d_K|}\right)^{6\chi(a)\#\mu(K)} \right)^{\text{cl}(\mathcal{O}_{K,f})/\text{cl}(K)} \cdot \left( \prod_{p \in \text{Prime}, p|f} p^{12e(p)} \right)^{\text{cl}(\mathcal{O}_{K,f})},$$

where  $\chi(a) = \left(\frac{d_K}{a}\right)$  and

$$e(p) = \frac{(1-p^{-n})(1-\chi(p))}{(1-p^{-1})(p-\chi(p))}.$$

Notice that  $6e(p) \text{cl}(\mathcal{O}_{K,f}) \in \mathbb{Z}$  by (19.3.3.7).

In particular, for  $f = 1$ ,

$$\prod_{\mathfrak{a} \in \text{Cl}(K)} F(\mathfrak{a}) = \left( \frac{2\pi}{|d_K|} \right)^{12 \text{cl}(K)} \prod_{a=1}^{|d_K|-1} \Gamma\left(\frac{a}{|d_K|}\right)^{6\chi(a)\#\mu(K)},$$

┘

*Proof:* Cf. [N-T91] and [Elliptic Functions, Weil].

□

## 1 Averaged Colmez Conjecture

References are [On Faltings heights of abelian varieties with complex multiplication, Xinyi Yuan].

**Thm& Conj.Cor.(31.7.1.1) [Averaged Colmez Conjecture, Andreatta-Goren-Howard-Per-**  
**a/Yuan-Zhang].** Let  $E \in \text{NField}$  be a CM field, then

$$\sum_{\Phi} h_{\text{st}}(E, \Phi) = \sum_{\Phi} \left( \sum_{\rho} c_{\rho, \Phi} \cdot \left( \frac{L^{\vee}(\rho; 0)}{L(\rho; 0)} + \frac{\log f_{\rho}}{2} \right) \right).$$

where  $\Phi$  ranges through all CM-types of  $E$ , and the notations are as in (31.7.0.3).

┘

*Proof:* ?

This is a consequence of the Colmez conjecture (31.7.0.3).

□

**Cor.(31.7.1.2) [Heights and Discriminants].** For  $g \in \mathbb{Z}_+$ , there exists  $c_g \in \mathbb{R}_+$  s.t.  $\lim_{g \rightarrow \infty} c_g = 0$ ,  
s.t. for any primitive CM-type  $(E, \Phi)$  of degree  $2g$ ,

$$h_{\text{st}}(E, \Phi) \leq |\text{disc}(E)|^{c_g}.$$

┘

*Proof:* Using (21.1.5.5) and (31.7.1.1),

$$h_{\text{st}}(E, \Phi) \leq -(2^g - 1)cg + \sum_{\Phi} \left( \sum_{\rho} c_{\rho, \Phi} \cdot \left( \frac{L^{\vee}(\rho; 0)}{L(\rho; 0)} + \frac{\log f_{\rho}}{2} \right) \right).$$

To bound each term in the RHS, notice first  $f_{\rho} \leq |\text{disc}(E)|$  ? , and then

□

## 32 | Kudla's Program

### 32.1 Kudla's Program

**Notation (32.1.0.1).**

- Use notations on Theta correspondence defined in [Theta Correspondence](#).

┘

#### 1 Siegel-Weil Formula for Symplectic Groups

References are [\[Wei65, K-R88, K-R88a\]](#) and [Kudla-Rallis, A regularized Siegel-Weil formula: the first term identity].

**Notation (32.1.1.1).**

- Use notations on symplectic Theta lifting defined in .

┘

**Def. (32.1.1.2) [Siegel Eisenstein Series].** For any  $\Phi \in C^\infty(G(\mathbf{A}_F))^{\mathcal{K}\text{-fin}}$  s.t.

$$\Phi(n \cdot m(a)g) = \chi_V(a)|a|^{m/2},$$

we can define  $\Phi(g; s) = \Phi(g)|a(g)|^{s-s_0}$  and

$$\text{Eis}(g, \Phi; s) = \sum_{\gamma \in P(F) \backslash \text{Sp}(n; F)} \Phi(\gamma g; s).$$

In particular, for  $\phi \in S(V^n(\mathbf{A}_F))$  and  $g \in G(\mathbf{A}_F)$ , denote

$$\theta_\varphi^{(s)}(g) = |A(g)|_{\mathbf{A}}^{s - (\frac{m}{2} - \varrho_n)} \theta_\varphi(g, 1) \text{??} (26.8.5.11).$$

and the **Siegel Eisenstein series**

$$\text{Eis}(g; \theta_\varphi^{(s)}) = \sum_{\gamma \in P(F) \backslash \text{Sp}(n; F)} \theta_\varphi^{(s)}(\gamma g),$$

which is absolutely convergent for  $\text{Re}(s) > \varrho_n$  and has a meromorphic continuation to all  $s \in \text{Re}(\mathbb{C})$ .

┘

*Proof:* ?

□

**Prop. (32.1.1.3) [int( $-, \Phi$ ) has no Cuspidal Components].** For any cusp form  $f$  on  $G(\mathbf{A}_F)$ , we have  $\langle \text{Int}(-, \varphi), f \rangle_{\text{Pet}} = 0$ .

┘

*Proof:* Cf.[K-R88]Thm2.7.?

□

**Prop. (32.1.1.4) [Constant Terms of Eisenstein Series].** Situation as in (32.1.1.2),

$$\text{Eis}_0(g, \Phi; s) = \sum_{r=0}^n \text{Eis}_0^r(g, \Phi; s),$$

where

$$\text{Eis}_0^r(g, \Phi; s) = \sum_{\gamma \in Q_r \backslash \text{GL}(n)} \Phi_r(\gamma g; s), \quad \Phi_r(g; s) = \int_{N'_{n-r}(\mathbf{A}_F)} \Phi(w_r n g; s) dn.$$

In particular,

$$\text{Eis}_0^0(g, \Phi; s) = \Phi_0(g; s) = \Phi(g; s), \quad \text{Eis}_0^n(g, \Phi; s) = \Phi_n(g; s).$$

┘

*Proof:* Cf.[K-R88]Lem2.4.?

□

**Prop. (32.1.1.5).** Situation as in (32.1.1.2),  $\text{Eis}(g, \Phi; s)$  and  $\text{Eis}_0(g, \Phi; s)$  have the same set of poles. ┘

*Proof:* Cf.[K-R88]Lem3.4.?

□

**Prop. (32.1.1.6) [Holomorphy of Eisenstein Series].** If  $r = 0$  or  $m - r > 2\varrho_n$ , then  $\text{Eis}(g; f_\Phi^{(s)})$  (32.1.1.2) is holomorphic at  $s = s_0$ . ┘

*Proof:* By (32.1.1.5), it suffices to prove the constant term

$$\text{Eis}_0(g, \Phi; s) = \Phi(g; s) + \sum_{r=1}^{n-1} \text{Eis}_0^r(g, \Phi; s) + \Phi_n(g; s) \quad (32.1.1.4)$$

is holomorphic at  $s = s_0$ . If  $r = 0$ , this is done in [K-R88]?

If  $m - r > 2\varrho_n$ , ?  
?

□

**Prop. (32.1.1.7).** There exists  $c \in \mathbb{C}$  s.t.

$$I^*(g, \Phi) = I(g, \Phi) - c \text{Eis}(g; f_\Phi^{(s)})$$

satisfies  $I_\beta^*(g, \Phi) = 0$  for any  $\beta \in \text{Sym}_n(F)$  s.t.  $\det(\beta) \neq 0$ . ┘

*Proof:* Cf.[Rallis, *L-Functions and the Oscillator Representation*].

□

**Thm. (32.1.1.8) [Convergent Siegel-Weil Formula, Weil[Wei65]/Kudla-Rallis[K-R88, K-R88a]].**

For any  $\Phi \in S(V^n(\mathbf{A}_F))^{\mathcal{K}\text{-fin}}$  and  $g \in G(\mathbf{A}_F)$

- $m > 2n$ , then  $\text{Eis}(g; f_\Phi^{(s)})$  is absolutely convergent at  $s = s_0$ ,
- If  $r = 0$  or  $m - r > 2\varrho_n$ , then  $\text{Eis}(g; f_\Phi^{(s)})$  is holomorphic at  $s = s_0$ .

And in both cases, we have

$$\text{Eis}(g; f_\Phi^{(s)}) \Big|_{s=s_0} = \kappa \text{Int}(g; \Phi) (32.1.1.2) (26.8.5.12) \text{ where } \kappa = \begin{cases} 1 & , m > 2\varrho_n \\ 2 & , m \leq 2\varrho_n \end{cases}.$$

┘

*Proof:* ?

If  $m > 2n$ , then  $\text{Eis}(g; f_\Phi^{(s)})$  converges absolutely at  $s = s_0$  by (32.1.1.2), and this formula is proved by Weil.

If  $r = 0$  or  $m - r > 2\varrho_n$ , then  $\text{Eis}(g; f_\Phi^{(s)})$  is holomorphic at  $s = s_0$  by (32.1.1.6).

Take  $v \in \Sigma_F^{\text{fin}}$  s.t. the Witt index  $r_v$  of  $V_v$  equals  $r$ . Thus  $m = 2r_v + r$  where  $r \in \{0, 2, 4\}$  by (19.5.5.7). □

## 2 Siegel-Weil Formula for Unitary Groups

References are [Ich07].

**Notation (32.1.2.1).**

- Let  $E/F \in \mathbf{NField}$  be a CM field.
- Let  $\mathbf{1} \neq \psi : F \backslash \mathbf{A}_F \rightarrow \mathbb{C}^\times$ .
- Let  $\chi_{E/F}$  be the quadratic Hecke character associated to  $E/F$ .
- Let  $V \in \text{HermSpa}^m(E/F)$ .
- Let  $r$  be the Witt index of  $V$  (19.5.2.5).
- Let  $\chi$  be the Hecke character of  $E$  s.t.  $\chi|_{\mathbf{I}_F} = \chi_{E/F}^m$ .
- Let  $G = \text{U}(n, n)_F$  be the unitary group over  $\mathcal{O}_F$  (10.5.3.19).
- Let  $P_{\text{Sie}} \subset G$  be the Siegel parabolic subgroup.
- Let  $H = \text{U}(V)$  be the unitary group of  $V$ .
- Let  $S(V^n(\mathbf{A}_F))$  be the subspace of  $\mathcal{S}(V^n(\mathbf{A}_F))$  consisting of functions that corresponds to polynomials in the Fock model ?.
- Let  $\omega_{\psi, \chi}$  be the Weil representation of  $G(\mathbf{A}_F) \times H(\mathbf{A}_F)$  acting on  $\mathcal{S}(V^n(\mathbf{A}_F))$ . ┘

**Prop. (32.1.2.2).** If  $m/2 \leq n < m - r$ , then  $\text{Eis}(g; f_\Phi^{(s)})$  (32.1.1.2) is holomorphic at  $s = s_0$  for any  $\Phi \in S(V^n(\mathbf{A}_F))$  and  $g \in G(\mathbf{A}_F)$ . ┘

*Proof:* Cf. [Ichino, A regularized Siegel-Weil formula for unitary groups] Lemma 8.2. ? □

**Prop. (32.1.2.3).** Suppose  $m/2 \leq n < m - r$ , then for any  $\Phi \in S(V^n(\mathbf{A}_F))$  and  $g \in G(\mathbf{A}_F)$ , denote

$$A(g, \Phi) = \text{Eis}\left(g; f_\Phi^{(s)}\right) \Big|_{s=\frac{m-n}{2}} - \text{Int}(g; \Phi) \text{ (26.8.5.12)}, \quad (\star)$$

and for each  $\beta \in \text{Herm}_n(E/F)$ , let

$$A_\beta(g, \Phi) = \int_{U(F) \backslash U(\mathbf{A}_F)} A(n g, \Phi) \overline{\psi_\beta(n)} dn$$

be the  $\beta$ -th Fourier coefficient of  $A(g, \Phi)$ , where  $\psi_\beta\left(\begin{bmatrix} \mathbf{1}_n & b \\ & \mathbf{1}_n \end{bmatrix}\right) = \psi(\text{tr}(\beta b))$ . Then  $A_\beta(g, \Phi) = 0$  unless  $\det(\beta) \neq 0$ . ┘

*Proof:* Cf.[Ichino, A regularized Siegel-Weil formula for unitary groups]Prop6.2?  $\square$

**Thm. (32.1.2.4)**[Unitary Siegel-Weil Formula, Weil[Wei65]/Ichino[Ich07]]. Suppose  $m/2 \leq n < m - r$ , then for any  $\Phi \in S(V^n(\mathbf{A}_F))$  and  $g \in G(\mathbf{A}_F)$ , we have

$$\text{Eis} \left( g; f_{\Phi}^{(s)} \right) \Big|_{s=\frac{m-n}{2}} = \text{Int}(g; \Phi)(32.1.2.2)(26.8.5.12).$$

For the case  $m/2 \leq n \leq m$ , see[Ichino, A regularized Siegel-Weil formula for unitary groups]?.

For the case  $n < m/2$ , see[Wei65].  $\perp$

*Proof:* Suppose  $\Phi = \otimes_{v \in \Sigma_F} \Phi_v \in S(V^n(\mathbf{A}_F))$ , and take  $v \in \Sigma_F$  s.t. the Witt index  $r_v$  of  $V_v$  equals  $r$ (which exists by(19.5.6.4)). Let  $\mathcal{R}_v \subset \mathcal{A}(G/F)$  be the subspace spanned by the residues  $\text{Res}_{s=\frac{m-n}{2}} \text{Eis}(g; f^{(s)})$ , where  $f^{(s)}$  is a holomorphic section of  $I_P^G(\chi; s)$ ? of the form

$$f^{(s)} = f_v^{(s)} \otimes \bigotimes_{w \in \Sigma_F \setminus \{v\}} f_{\Phi_w}^{(s)}$$

Let

$$A_v : S(V_v^n) \rightarrow \mathcal{A}(G/F), \quad \bar{A}_v : S(V_v^n) \rightarrow \mathcal{A}(G/F) / \mathcal{I}_v(0)$$

be the intertwining operators defined in(32.1.2.8), it suffices to prove that  $A_v = 0$ .

For  $v \in \Sigma_F^{\text{fin}}$ ,  $\bar{A}_v = 0$  by???. Then, since  $A_v$  factors through  $R(V_v, \chi_v)$ (32.1.2.8), and  $R(V_v, \chi_v)$  and  $\mathcal{I}_v(0)$  cannot share a same  $\mathcal{K}_v$ -type in common, we get  $A_v = 0$ .

For  $v \in \Sigma_F^{\infty}$ ,  $\square$

### Local Theory

**Prop. (32.1.2.5)**[Invariant Distribution Theorem, Rallis]. Situation as in(26.8.5.16), we have

$$S(V^n)_{H(F_v)} \cong R_v(V, \chi).$$

$\perp$

*Proof:* ? Cf.[Rallis, On the Howe duality conjecture], [Moeglin-Vigneras-Waldspurger, Correspondances de Howe sur un corps p-adique]P70, [Degenerate principal series and local theta correspondence]Thm4.1,  $\square$

**Prop. (32.1.2.6)**. Situation as in(26.8.5.16), if  $v \in \Sigma_F^{\text{fin}}$  and  $m$  is even, then  $I_v(\chi; s)$  is multiplicity free as a representation of  $\mathcal{K}_v$ . And if  $m = 2n$ , then  $R_v(V, \chi)$  is irreducible and non-singular.  $\perp$

*Proof:* ? Cf.[Degenerate principal series representations for  $U(n, n)$ ]P292 and Thm1.2., and [Weil-Siegel formula, 2. The isotropic convergent case, Kudla-Rallis]Prop3.2.  $\square$

**Prop. (32.1.2.7)**. Situation as in(26.8.5.16), if  $\tau \in \Sigma_F^{\infty}$ , then  $I_{\tau}(\chi; s)$  is multiplicity free s a representation of  $\mathcal{K}_{\tau}$ . And if  $\text{sgn}(V) = (p, q)$  satisfies  $\max(p, q) > n$ , then  $\nabla f_{\lambda_0}^{(\frac{m-n}{2})}$  generates  $R_{\tau}(V, \chi)$ .  $\perp$

*Proof:* ? [some degenerate principal series representations of  $U(n, n)$ ] and [Ich07].  $\square$



**Prop. (32.1.2.8) [Kudla-Rallis].** Consider the  $\mathcal{K}_v$ -intertwining map

$$A_v : S(V_v^n) \rightarrow \mathcal{A}(G/F) : \Phi_v \mapsto A \left( g, \Phi_v \otimes \bigotimes_{w \neq v} \Phi_w^0 \right) (\star).$$

Then  $A_v$  induces a map

$$\overline{A}_v : S(V_v^n) \rightarrow \mathcal{A}(G/F) / \mathcal{I}_v(0) \text{ ?}$$

which is  $G(F_v)$ -intertwining if  $v \in \Sigma_F^{\text{fin}}$  and  $(\mathfrak{g}(F_v), \mathcal{K}_v)$ -intertwining if  $v \in \Sigma_F^\infty$ .

Moreover,  $A_v$  factors through the quotient  $R(V_v, \chi_v)$  by (32.1.2.5) and the definitions.  $\lrcorner$

*Proof:* Cf. [Kudla-Rallis, Weil-Siegel formula, 2, the isotropic convergent case] Cor2.3. ?  $\square$

### 3 Modularity Conjecture

#### Siegel Modular Forms

#### Modularity Conjecture

**Thm. (32.1.3.1) [Modularity Conjecture, Kudla].**  $\lrcorner$

**Remark (32.1.3.2).** This theorem is proven by [Gross-Kohen-Zagier1987] for modular curves, and [Hirzebruch-Zagier1976] for Hilbert modular surfaces, and [Borcherds1999] for Shimura varieties of orthogonal type.  $\lrcorner$

*Proof:*  $\square$

#### W. Zhang's Work

References are [Wei Zhang's thesis].

## 32.2 Arithmetic Siegel-Weil Formula

References are [ChaoLi's notes].

### 1 Represented by Quadratic Forms

**Thm. (32.2.1.1) [Jacobi].** For  $n \in \mathbb{Z}_+$ , let  $r(n) = \#\{(x, y) \in \mathbb{Z} | x^2 + y^2 = n\}$ , then  $r(n) \neq 0$  iff  $2 | v_p(n)$  for any  $p \equiv 3 \pmod{4}$ .

Moreover,

$$r(n) = 4 \left( \#\{d \in \mathbb{Z}_+, d|n, d \equiv 1 \pmod{4}\} - \#\{d \in \mathbb{Z}_+, d|n, d \equiv 3 \pmod{4}\} \right).$$

┘

*Proof:* It is clear that

$$\sum_{n \in \mathbb{Z}_+} r(n) q^n = \theta^2(26.8.2.2).$$

□

## 32.3 Kudla-Rapoport Conjecture

**Notation (32.3.0.1).**

- Use notations on Rapoport-Zink spaces defined in [Rapoport-Zink Spaces and Mixed-Characteristic Shtukas](#).

┘

### 1 Local Kudla-Rapoport Conjecture

**Prop. (32.3.1.1) [Hermitian Spaces].** Non-degenerate Hermitian  $F/F_0$  spaces  $\mathbb{V}$  are classified up to isomorphism by dimension  $n$  and discriminant  $\text{disc}(\mathbb{V}) = (-1)^{\binom{n}{2}} \det(\mathbb{V}) \in F_0^\times / \text{Nm}_{F/F_0} F^\times$ .

$\mathbb{V}$  is called **split** if  $\text{disc}(\mathbb{V}) = 1$ , and **non-split** otherwise.

┘

*Proof:* [Kudla-Rapoport Cycles and Derivatives of Local Densities, Li-Zhang]P712.

□

**Prop. (32.3.1.2) [Lattices in Hermitian Spaces].** Let  $\mathbb{V}$  be a Non-degenerate Hermitian  $F/F_0$  space and  $L \subset \mathbb{V}$  a full  $\mathcal{O}_F$ -lattice. Then  $L$  is called an **integral lattice** if  $L \subset L^\vee$ .

For an integral lattice  $L$ , define its **fundamental invariants** to be the unique sequence of integers  $(a_1, \dots, a_n)$  s.t.  $0 \leq a_1 \leq \dots \leq a_n$  and  $L^\vee/L \cong \bigoplus_{i=1}^n \mathcal{O}_F/\varpi^{a_i}$ . Its **valuation** is defined to be  $\text{val}(L) = \sum a_i$ , and its **type**  $t(L)$  is defined to be the number of nonzero terms in  $(a_1, \dots, a_n)$ .

┘

**Prop. (32.3.1.3) [Vertex Lattices].**

┘

### Local Kudla-Rapoport Cycles

**Def. (32.3.1.4) [Kudla-Rapoport Cycles].** For each  $u \in \mathbb{V}_n$ , there is a **Kudla-Rapoport cycle**  $\mathcal{Z}(u) \subset \mathcal{N}_n$  which is the locus where the quasi-homomorphism  $u : \mathbb{E} \rightarrow \mathbb{X}_n$  lifts to a homomorphism from  $\mathcal{E}$  to the universal objects over  $\mathcal{N}_n$ . (Cf. [Special Cycles on Unitary Shimura Varieties, Kudla-Rapoport]).

┘

**Def. (32.3.1.5) [Intersection Numbers on RZ Spaces].** For a pair  $(g, u) \in (U(\mathbb{V}_n) \times \mathbb{V}_n)(F_0)_{rs}$ , the **intersection number** is defined to be

$$\text{int}(g, u) = \chi(\mathcal{N}_n \times \mathcal{N}_n, \mathcal{O}_{\Gamma_g} \otimes_{\mathcal{O}_{\mathcal{N}_n \times \mathcal{N}_n}}^L \mathcal{O}_{\Delta_{\mathcal{Z}(u)}}).$$

┘

### 2 Global Kudla-Rapoport Conjecture



## 33 | $p$ -adic $L$ -Functions

### 33.1 Classical Iwasawa Theory

References are [Was97, Hid93].

The Iwasawa theory of fields mostly study the action of the Galois group on the ideal group/-class/group/units.

**Notation(33.1.0.1).**

- Let  $p \in \text{Prime}$ .
- Let  $L \in p\text{-NField}$ .

┘

#### 1 $p$ -adic Analysis

**Prop.(33.1.1.1)[Interpolating Series, Mahler[Mah58]].** For any continuous function  $f : \mathbb{Z}_p \rightarrow \mathcal{O}_L$ , by taking different, we have:

$$f(x) = \sum_{n \in \mathbb{N}} a_n(f) \binom{x}{n}$$

for each  $x \in \mathbb{N}$ , where

$$a_n(f) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(k), \quad n \in \mathbb{N}.$$

Then  $\lim_{n \rightarrow \infty} |a_n(f)| = 0$ , and conversely, given any sequence  $\{a_n(f)\}_{n \in \mathbb{N}} \in \mathcal{O}_L$  s.t.  $\lim_{n \rightarrow \infty} |a_n(f)| = 0$ , the function  $f(x) = \sum_{n \in \mathbb{N}} a_n(f) \binom{x}{n}$  is continuous function from  $\mathbb{Z}_p$  to  $\mathcal{O}_L$ .

And we have  $\|f\|_p = \sup_{n \in \mathbb{N}} |a_n|$ . ┘

*Proof:* Because the function  $\binom{x}{n}$  is continuous and maps  $\mathbb{N}$  to  $\mathbb{N}$ , we have  $|\binom{x}{n}|_p \leq 1$  for each  $x \in \mathbb{Z}_p$ . Then it suffices to show that  $\lim_{n \rightarrow \infty} |a_n(f)| = 0$ .

For  $y \in \mathbb{N}$ , let  $f_y(x) = f(x + y)$ , then

$$f_y(x) = \sum_{m \in \mathbb{N}} a_m(f) \binom{x+y}{m} = \sum_{n \in \mathbb{N}} \binom{x}{n} \sum_{k \in \mathbb{N}} a_{n+k}(f) \binom{y}{k}.$$

Thus

$$\sum_{k \in \mathbb{N}} a_{n+k}(f) \binom{y}{k} = a_n(f_y) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(k+y).$$

As  $f$  is uniformly continuous, for any  $s \in \mathbb{Z}_+$ , there exists  $t \in \mathbb{Z}_+$  s.t.  $|f(x + p^t) - f(x)| \leq p^{-s}$  for any  $x \in \mathbb{Z}_p$ . Taking  $y = p^t$ , we get

$$\begin{aligned} a_{n+p^t}(f) &= - \sum_{j=1}^{p^t-1} \binom{p^t}{j} a_{n+j}(f) + \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(k + p^t) - a_n(f) \\ &\quad - \sum_{j=1}^{p^t-1} \binom{p^t}{j} a_{n+j}(f) + \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} (f(k + p^t) - f(k)) \end{aligned}$$

So

$$|a_{n+p^t}(f)| \leq \max(p^{-1}|a_{n+1}(f)|, \dots, p^{-1}|a_{n+p^t-1}(f)|, p^{-s}).$$

Notice  $s$  is arbitrary. From this, we can easily show that  $\lim_{n \rightarrow \infty} |a_n(f)| = 0$ .  $\square$

### $p$ -adic Gamma Function

**Def. (33.1.1.2) [ $p$ -adic Gamma Functions].** For  $k \in \mathbb{Z}_+$ , define

$$\Gamma_p(k) = (-1)^k \prod_{j \in [k-1]_+} j,$$

then  $\Gamma_p$  extends uniquely to a continuous function  $\Gamma_p : \mathbb{Z}_p \rightarrow \mathbb{Z}_p^*$ .  $\lrcorner$

*Proof:* It suffices to show that if  $k, k' \in \mathbb{Z}_+$  satisfies  $k \equiv k' \pmod{p^N}$  for  $N \in \mathbb{Z}_{\geq 2}$ , then  $\Gamma_p(k) \equiv \Gamma_p(k') \pmod{p^N}$ . Because then we can define  $\Gamma_p$  by continuity on  $\mathbb{N}$ , and it take values in  $\mathbb{Z}_p^*$ .

We may assume  $k' = k + p^N$ . Then it suffices to show that

$$(-1)^{p^N} \prod_{0 < j < p^N, p \nmid j} j \equiv 1 \pmod{p^N}.$$

We try to pair  $j$  with  $j^{-1} \pmod{p^N}$ . If  $p \neq 2$ , then  $j^2 = 1$  has only two solutions  $j \in \{1, p^N - 1\}$ , and the modular equation holds. If  $p = 2$ , then by (2.6.1.3),  $j^2 = 1$  has only four solutions  $j \in \{1, 2^N - 1, a, 2^N - a\}$ , where  $a \in [2^N]_+$  and  $a \notin \{1, 2^N - 1\}$ . Then the assertion also holds.  $\square$

**Prop. (33.1.1.3).**

$$\frac{\Gamma_p(s+1)}{\Gamma_p(s)} = \begin{cases} -s & , s \in \mathbb{Z}_p^* \\ -1 & , s \in p\mathbb{Z}_p \end{cases}.$$

$\lrcorner$

*Proof:* As both sides are continuous functions on  $\mathbb{Z}_p$ , it suffices to check on the dense subset  $\mathbb{Z}_+$ . But then this follows from the definition of  $\Gamma_p$  (33.1.1.2).  $\square$

**Prop. (33.1.1.4).** For  $s = s_0 + ps_1 \in \mathbb{Z}_p$  where  $s_0 \in [p]_+$  and  $s_1 \in \mathbb{Z}_p$ , we have

$$\Gamma_p(s)\Gamma_p(1-s) = (-1)^{s_0}.$$

$\lrcorner$

*Proof:* By continuity it suffices to check for  $s \in \mathbb{Z}_+$ . For  $s = 1$  this holds because  $\Gamma_p(1) = -1$  and  $\Gamma_p(0) = 1$  by (33.1.1.3). And by (33.1.1.3),

$$\frac{\Gamma_p(s+1)\Gamma_p(1-(s+1))}{\Gamma_p(s)\Gamma_p(1-s)} = \begin{cases} -1 & , s \in \mathbb{Z}_p^* \\ 1 & , s \in p\mathbb{Z}_p \end{cases}.$$

Thus the assertion holds by induction on  $s$ .  $\square$

**Prop. (33.1.1.5) [ $p$ -adic Gauss-Legendre Multiplication Formula].** Let  $s = s_0 + ps_1 \in \mathbb{Z}_p$  where  $s_0 \in [p]_+$  and  $s_1 \in \mathbb{Z}_p$ , then for any  $m \in \mathbb{Z}_{\geq 2} \setminus (p)$ , we have

$$\frac{\prod_{h=0}^{m-1} \Gamma_p(\frac{s+h}{m})}{\Gamma_p(s) \prod_{h=1}^{m-1} \Gamma_p(\frac{h}{m})} = m^{1-s_0} (m^{-(p-1)})^{s_1}.$$

$\lrcorner$

*Proof:* For fixed  $m$ , both sides are continuous in  $s$ , so it suffices to check for  $s \in \mathbb{Z}_+$ . For  $s = 1$  this is true. Denote the LHS by  $f(s)$ , then

$$\frac{f(s+1)}{f(s)} = \frac{\Gamma_p(s)\Gamma_p(s/m+1)}{\Gamma_p(s+1)\Gamma_p(s/m)} = \begin{cases} 1/m & s \in \mathbb{Z}_p^* \\ 1 & s \in p\mathbb{Z}_p \end{cases}.$$

And it is easy to check that the same holds true for the RHS. Thus the assertion holds by induction.  $\square$

### $p$ -adic Distributions and Measures

**Def. (33.1.1.6) [ $p$ -adic Distributions and Measures].** For  $X \in \mathcal{P}\text{rof}$  and  $L \in p\text{-NField}$ , a  $L$ -valued distribution on  $X$  (4.4.4.11) is called a ( $L$ -valued)  **$p$ -adic distribution**. A  **$p$ -adic measure** on  $X$  is a  $p$ -adic distribution  $\mu$  on  $X$  s.t. there exists  $B \in \mathbb{R}_+$  that  $|\mu(U)|_L \leq B$  for any compact open  $U \subset X$ .

The space of  $L$ -valued measures on  $X$  is denoted by  $\text{Meas}(X; L)$ , and also the subspace of  $\mathcal{O}_L$ -valued measures on  $X$  is denoted by  $\text{Meas}(X; \mathcal{O}_L)$ . Then

$$\text{Meas}(X; \mathcal{O}_L) = \text{Meas}(X; \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} \mathcal{O}_L \quad \text{Meas}(X; L) = \text{Meas}(X; \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} L.$$

We can also define a norm on  $\text{Meas}(X; L)$  by  $\|\mu\|_p = \sup_{U \subset X} |\mu(U)|_L$ .  $\lrcorner$

**Prop. (33.1.1.7) [Distributions on  $\mathbb{Q}_p$ ].** Let  $p \in \text{Prime}$  and  $X \subset \mathbb{Q}_p$  be a compact open subset. If  $\mu$  is a function from the family of subsets of  $X$  of the form  $a + (p^n)$  to  $\mathbb{C}_p$  that satisfies: Whenever  $a + (p^n) \in X$ , then

$$\mu(a + (p^n)) = \sum_{b=0}^{p-1} \mu(a + bp^n + (p^{n+1})).$$

Then  $\mu$  extends uniquely to a  $p$ -adic measure on  $X$  (4.4.4.12).  $\lrcorner$

**Prop. (33.1.1.8) [Katz].** For any bounded sequence  $\{b_n\}_{n \in \mathbb{N}} \subset L$ , there is a unique measure  $\mu \in \text{Meas}(\mathbb{Z}_p, L)$  s.t.

$$\int_{\mathbb{Z}_p} \binom{x}{n} d\mu(x) = b_n,$$

and  $\|\mu\|_p = \sup_n |b_n|$ .  $\lrcorner$

**Cor. (33.1.1.9).** Each measure  $\mu \in \text{Meas}(\mathbb{Z}_p, L)$  is determined by its value on the functions  $x^m$  for  $m \in \mathbb{N}$ .  $\square$

**Prop. (33.1.1.10) [ $p$ -adic Integrations].** Suppose  $X \in \text{Prof}$  and  $L \in p\text{-NField}$ , and  $\mu$  a  $p$ -adic measure (33.1.1.6) on  $X$ . Let  $f : X \rightarrow L$  is a continuous function, then the Riemann sum

$$S_{N, \{x_{a,N}\}} = \sum_{a \in \mathbb{Q}_p/p^N \mathbb{Z}_p, x_{a,N} \in a + (p^N) \subset X} f(x_{a,N}) \mu(a + (p^N))$$

converges to a limit in  $\mathbb{Q}_p$  when  $N \rightarrow \infty$ , denoted by  $\int_X f d\mu$ .  $\square$

*Proof:*  $\square$

### Bernoulli Measures

**Def. (33.1.1.11) [Bernoulli Distributions].** Let  $k, n \in \mathbb{N}$ . For  $a \in [p^n - 1]$ , define  $\mu_{B,k}(a + (p^n)) = p^{n(k-1)} B_k(\frac{a}{p^n})$  (10.7.1.13), then  $\mu_{B,k}$  extends to a distribution on  $\mathbb{Z}_p$ . For example,

$$\mu_{B,0} = \mu_{\text{Haar}}, \quad \mu_{\text{Mazur}}(a + (p^n)) \triangleq \mu_{B,1}(a + (p^n)) = \frac{a}{p^n} - \frac{1}{2}, \quad \mu_{B,2}(a + (p^n)) = p^n \left( \frac{a^2}{p^{2n}} - \frac{a}{p^n} + \frac{1}{6} \right)$$

$\square$

*Proof:* It suffices to check the hypothesis of (33.1.1.7) that for any  $a \in \mathbb{Z}_p$ ,

$$\mu_{B,k}(a + (p^n)) = \sum_{b=0}^{p-1} \mu_{B,k}(a + bp^n + (p^{n+1})),$$

or equivalently, if we denote  $\alpha = a/p^n$ ,

$$B_k(p\alpha) = p^{k-1} \sum_{b=0}^{p-1} B_k\left(\alpha + \frac{b}{p}\right).$$

The RHS is just the  $k!$  times the  $T^k$ -coefficient in the expansion of

$$p^{k-1} \sum_{b=0}^{p-1} \frac{e^{(\alpha+b)T/p}}{e^T - 1} = \frac{p^{k-1} T e^{\alpha T}}{e^T - 1} \cdot \frac{e^T - 1}{e^{T/p} - 1} = p^k \frac{(T/p) e^{(p\alpha)T/p}}{e^{T/p} - 1} = p^k \sum_{j \in \mathbb{N}} B_j(p\alpha) \frac{(T/p)^j}{j!},$$

which is just  $B_k(p\alpha)$ .  $\square$

**Prop. (33.1.1.12) [Bernoulli Measure].** For  $k \in \mathbb{Z}_+$  and  $m \in \mathbb{Z}_+ \setminus (p)$ , define the (regularized) **Bernoulli measure**  $\mu_{k,m}$  to be

$$\mu_{k,m}(U) \triangleq \mu_k(U) - m^{-k} \mu_k(mU), \quad U \subset \mathbb{Z}_p.$$

Then  $\mu_{k,m} \in \text{Meas}(\mathbb{Z}_p, \mathbb{Q}_p)$ , and satisfies:

- $\mu_{1,m} \in \text{Meas}(\mathbb{Z}_p, \mathbb{Z}_p)$ .
- For any  $N \in \mathbb{N}$  and  $a \in [p^N - 1]$ , if  $d_k$  is the common denominator for the coefficients of  $B_k(X)$ , then

$$d_k \mu_{k,m}(a + (p^N)) \equiv d_k k a^{k-1} \mu_{1,m}(a + (p^N)) \pmod{p^N}.$$

and both sides are in  $\mathbb{Z}_p$ .



┘

*Proof:* For  $k = 1, N \in \mathbb{N}$  and  $a \in [p^N - 1]$ ,

$$\mu_{1,m}(a + (p^N)) = \frac{a}{p^N} - \frac{1}{2} - \frac{1}{m} \left( \left\{ \frac{am}{p^N} \right\} - \frac{1}{2} \right) = \frac{1}{m} \left\lfloor \frac{am}{p^N} \right\rfloor + \frac{1/m - 1}{2} \in \mathbb{Z}_p.$$

Thus

2: By (10.7.1.13),  $B_k(X) = X^k - \frac{k}{2}X^{k-1} + \dots$ , and for any compact open subset  $U \subset \mathbb{Z}_p$ ,

$$d_k \mu_{k,m}(a + (p^N)) = p^{N(k-1)} \left( d_k B_k \left( \frac{a}{p^N} \right) - d_k B_k \left( \left\{ \frac{am}{p^N} \right\} \right) \right),$$

so except for the first two terms of  $B_k(X)$ , the other terms are divisible by  $p^N$ . Thus

$$\begin{aligned} d_k \mu_{k,m}(a + (p^N)) &\equiv d_k p^{N(k-1)} \left( \frac{a^k}{p^{Nk}} - \frac{1}{m^k} \left\{ \frac{am}{p^N} \right\}^k - \frac{k}{2} \left( \frac{a^{k-1}}{p^{N(k-1)}} - \frac{1}{m^k} \left\{ \frac{am}{p^N} \right\}^{k-1} \right) \right) \pmod{p^N} \\ &\equiv d_k \left( \frac{a^k}{p^N} - \frac{1}{m^k} \left( \frac{(am)^k}{p^N} - k(am)^{k-1} \left\lfloor \frac{am}{p^N} \right\rfloor \right) - \frac{k}{2} \left( a^{k-1} - \frac{1}{m^k} (am)^{k-1} \right) \right) \pmod{p^N} \\ &= d_k k a^{k-1} \left( \frac{1}{m} \left\lfloor \frac{am}{p^N} \right\rfloor + \frac{1/m - 1}{2} \right) \pmod{p^N} \\ &= d_k k a^{k-1} \mu_{1,m}(a + (p^N)) \pmod{p^N}. \end{aligned}$$

Finally we see that  $\mu_{k,m}$  is a measure because  $|\mu_{k,m}(U)| \leq \max(|\frac{1}{d_k}|_p, 1)$  by item 2. □

**Cor. (33.1.1.13).** If  $X \subset \mathbb{Z}_p$  is a compact open subset, and  $m \in \mathbb{Z}_{\geq 1} \setminus (p)$ , then  $\int_X \mu_{k,m} = k \int_X x^{k-1} \mu_{1,m}$  (33.1.1.12). ┘

*Proof:* This follows formally from (33.1.1.12) by taking limit  $N \rightarrow \infty$ . □

**Cor. (33.1.1.14).** Let  $m = \begin{cases} 1+p & , p \geq 3 \\ 5 & , p = 2 \end{cases}$ , then

$$\int_{\mathbb{Z}_p^*} x^{-1} \mu_{1,m} \equiv \begin{cases} -1 \pmod{p} & , p \geq 3 \\ 2 \pmod{4} & , p = 2 \end{cases}.$$

┘

*Proof:* Let  $g(x) = 1/x \pmod{p}$ , then  $g \equiv 1/x \pmod{p}$ , or if  $g(x) = 1/x \pmod{p^2}$ , then  $g \equiv 1/x \pmod{p^2}$ . Then the assertion follows easily from the equations

$$\mu_{1,m}(a + (p)) = \frac{1}{m} \left\lfloor \frac{am}{p} \right\rfloor + \frac{1/m - 1}{2}.$$

$$\mu_{1,m}(a + (p^2)) = \frac{1}{m} \left\lfloor \frac{am}{p^2} \right\rfloor + \frac{1/m - 1}{2}.$$

□

**Cor. (33.1.1.15).** Let  $k \in \mathbb{Z}_+, m \in \mathbb{Z}_+ \setminus (p)$ , then

- For any compact open subset  $U \subset \mathbb{Z}_p$ , we have

$$\mu_{k,m}(pU) = p^{k-1}\mu_{k,m}(U).$$

- $\mu_{B,k}(\mathbb{Z}_p) = B_k$ .
- $\mu_{k,m}(\mathbb{Z}_p^*) = (1 - m^{-k})(1 - p^{k-1})B_k$ .

┘

*Proof:* 1: This follows from the definition of  $\mu_{B,k}$  (33.1.1.11) and  $\mu_{k,m}$  (33.1.1.12)

2: This follows from the definition of  $\mu_{B,k}$  (33.1.1.11).

3: This follows the definition of  $\mu_{k,m}$  and item1, item2. □

**Cor. (33.1.1.16) [Kummer's Congruence, Kummer/Clausen-von-Staudt].** For  $k, k' \in \mathbb{Z}_+$ ,

- If  $p - 1 \nmid k$ , then  $|B_k/k|_p \leq 1$ .
- If  $p - 1 \nmid k$ ,  $N \in \mathbb{N}$  and  $k \equiv k' \pmod{(p-1)p^N}$ , then

$$(1 - p^{k-1}) \frac{B_k}{k} \equiv (1 - p^{k'-1}) \frac{B_{k'}}{k'} \pmod{p^{N+1}}.$$

- If  $p \geq 3$  and  $p - 1 | k$  or  $p = 2$  and  $k \in 2\mathbb{Z} \cup \{1\}$ , then  $pB_k \equiv -1 \pmod{p}$ .

┘

*Proof:* 1: We must have  $p \geq 3$ . Let  $m \in [p-2]_+$  s.t.  $m$  is the generator of the cyclic group  $(\mathbb{Z}/(p))^*$ . In particular,  $m^{-k} - 1 \in \mathbb{Z}_p^*$ . This is true for  $k = 1$  because  $B_1 = 1/2$  and  $p \geq 3$ , and if  $k \geq 2$ , then the hypothesis implies that  $m^k \not\equiv 1 \pmod{p}$ . Thus, by (33.1.1.15) and (33.1.1.13),

$$\left| \frac{B_k}{k} \right|_p = \left| \int_{\mathbb{Z}_p^*} x^{k-1} \mu_{1,m} \right|_p \leq 1 \text{ (33.1.1.12).}$$

2: We must have  $p \geq 3$ . Let  $m \in [p-2]_+$  s.t.  $m$  is the generator of the cyclic group  $(\mathbb{Z}/(p))^*$ . It suffices to prove that

$$\frac{1}{1 - m^{-k}} \int_{\mathbb{Z}_p^*} x^{k-1} d\mu_{1,m} \equiv \frac{1}{1 - m^{-k'}} \int_{\mathbb{Z}_p^*} x^{k'-1} d\mu_{1,m} \pmod{p^{N+1}}.$$

Then this follows from the fact that  $x^k \equiv x^{k'} \pmod{p^N}$  for any  $x \in \mathbb{Z}_p^*$ .

3: If  $k = 1$  the assertion is true. Now assume  $k \geq 2$ . If  $p \geq 3$ , let  $m = 1 + p$ , then

$$pB_k = \frac{-kp}{m^{-k} - 1} (1 - p^{k-1})^{-1} \cdot \int_{\mathbb{Z}_p^*} x^{k-1} d\mu_{1,m}.$$

if  $d = v_p(k)$ , then  $m^{-k} - 1 = (1 + p)^{-k} - 1 \equiv -kp \pmod{p^{d+2}}$ . Thus  $\frac{-kp}{m^{-k} - 1} \equiv 1 \pmod{p}$ . Thus

$$pB_k \equiv \int_{\mathbb{Z}_p^*} x^{k-1} d\mu_{1,m} \equiv \int_{\mathbb{Z}_p^*} x^{-1} d\mu_{1,m} \pmod{p}.$$

And if  $p = 2$ , let  $m = 5$ , then  $m^{-k} - 1 = (1 + 2^2)^{-k} - 1 \equiv -kp^2 \pmod{p^{d+3}}$ , and then

$$2B_k \equiv \frac{1}{2} \int_{\mathbb{Z}_p^*} x^{k-1} d\mu_{1,m} \equiv \int_{\mathbb{Z}_2^*} x^{-1} d\mu_{1,m} \pmod{2}.$$

The final assertion follows from (33.1.1.14). □

### Iwasawa Algebras

References are [Was97]Chap13.

**Prop. (33.1.1.17) [Iwasawa Algebras].** For  $G \in \text{ProfAb}$ , there is an isomorphism

$$\text{Meas}(G; \mathcal{O}_L) \cong \lim_{H \leq G \text{ open}} \mathcal{O}_L[G/H].$$

And if we endow  $\text{Meas}(G; \mathcal{O}_L)$  with the convolution operation (12.10.2.1)

$$(\mu * \lambda)(\varphi) = \int_G \left( \int_G \varphi(x+y) d\lambda(y) \right) d\mu(x)$$

and the RHS with the natural algebra structure, then this isomorphism is an algebra isomorphism.

Then  $\Lambda(G; L) \triangleq \text{Meas}(G; \mathcal{O}_L)$  is called the **Iwasawa algebra** of  $G$  (with values in  $\mathcal{O}_L$ ).

And the  $\mathcal{O}_L$ -valued Iwasawa algebra of  $\mathbb{Z}_p$  is called **The Iwasawa algebra**, denoted by  $\Lambda_L$ .  $\lrcorner$

*Proof:* The map is given by

$$\mu \mapsto \lambda = \left( \lambda_H = \sum_{a \in G/H} \mu(aH)[aH] \right)_H.$$

To show this is an algebra map, notice that

$$(\mu * \lambda)(aH) = \int_G \left( \int_G \chi_{aH}(x+y) d\lambda(y) \right) d\mu(x)$$

and  $\mu * \lambda$  is clearly mapped to the element

$$\left( \left( \sum_{a \in G/H} \mu(aH)[aH] \right) \cdot \left( \sum_{a \in G/H} \lambda(aH)[aH] \right) \right)_H$$

□

**Def. (33.1.1.18) [Iwasawa Algebra].** For any topological generator  $\gamma$  of  $\mathbb{Z}_p$ , there is a non-canonical isomorphism

$$\Lambda_L \cong \mathcal{O}_L[[T]] : \gamma \mapsto T + 1 \quad (33.1.1.17).$$

□

*Proof:* It suffices to prove for  $L = \mathbb{Q}_p$ .

Let  $\Gamma_n = \Gamma/\Gamma^{p^n}$ , then it can be easily shown that  $\mathbb{Z}_p[\Gamma_n] \cong \mathbb{Z}_p[T]/((T+1)^{p^n} - 1)$ . Then it suffices to show that

$$\mathbb{Z}_p[[T]] \cong \varprojlim \mathbb{Z}_p[T]/((T+1)^{p^n} - 1).$$

For this, denote  $P_n(T) = (T+1)^{p^n} - 1$ , then  $P_n(T)$  is distinguished (19.1.7.6), and  $P_n(T) | P_{n+1}(T)$ , and

$$P_{n+1}(T)/P_n(T) = (1+T)^{p^n(p-1)} + (1+T)^{p^n(p-2)} + \dots + (1+T)^{p^n} + 1 \in (p, T).$$

Thus  $P_n(T) \in (p, T)^{n+1}$ . By  $p$ -adic Weierstrass preparation theorem (19.1.7.7), for any  $f(T) \in \mathbb{Z}_p[[T]]$  and  $n \in \mathbb{N}$ ,  $f(T) = q_n(T)P_n(T) + f_n(T)$ , where  $f_n(T)$  is a polynomial with degree  $< p^n$ .

And  $f_{n+1}(T) - f_n(T) = (q_n - q_{n+1}P_{n+1}/P_n)P_n$ , so by (19.1.7.9),  $f_{n+1}(T) \equiv f_n(T) \pmod{P_n(T)}$  as polynomials. Then  $(f_0, f_1, \dots, f_n, \dots) \in \varprojlim \mathbb{Z}_p[T]/(P_n(T))$ .

Now if  $f_k = 0$  for each  $k$ , then  $P_k(T) | f(T)$  for each  $k$ , and then  $f \in \bigcup_k (p, T)^{k+1} = 0$ . And conversely, given any  $(f_i) \in \varprojlim \mathbb{Z}_p[T]/(P_n(T))$ ,  $f(T) = \lim f_k(T)$  exists in  $\mathbb{Z}_p[[T]]$ , and it can be shown that  $f(T)$  is mapped to  $(f_i)$ . Thus we are done.  $\square$

**Def. (33.1.1.19) [Pseudo-Isomorphisms].** A homomorphism between  $\Lambda_p$ -modules are called a **pseudo-isomorphism** if its kernel and cokernel are all finite. This is an equivalence relation.  
 $\lrcorner$

**Lemma (33.1.1.20).** If  $f, g \in \Lambda_p$  are relatively prime (3.2.3.4), then

- The natural map  $\Lambda_p/(fg) \rightarrow \Lambda_p/(f) \oplus \Lambda_p/(g)$  is an injection with finite cokernel.
- There exists an injection  $\Lambda/(f) \oplus \Lambda/(g) \rightarrow \Lambda_p/(fg)$  with finite cokernel.

$\lrcorner$

*Proof:* Cf. [Mazur Control Theorem, P9].  $\square$

**Thm. (33.1.1.21) [Classification of Modules over  $\Lambda_p$ ].** For  $M \in \text{Mod}_{\Lambda_p}^{\text{fg}}$ , there exists  $r, s, t \in \mathbb{N}$ ,  $n_i, m_j \in \mathbb{Z}_+$  and  $f_j$  distinguished and irreducible, s.t.  $M$  is pseudo-isomorphic to

$$\Lambda_p^{\oplus r} \oplus \left( \bigoplus_{i \in [s]_+} \Lambda_p/(p^{n_i}) \right) \oplus \left( \bigoplus_{j \in [t]_+} \Lambda_p/(f_j(T)^{m_j}) \right).$$

$\lrcorner$

*Proof:* Cf. [Washington, P272] or [Mazur Control Theorem, P11].  $\square$

**Cor. (33.1.1.22).** If  $M \in \text{Mod}_{\Lambda_p}^{\text{tor,fg}}$ , then there exists  $r, s, t \in \mathbb{N}$ ,  $n_i, m_j \in \mathbb{Z}_+$  and  $f_j$  distinguished and irreducible, s.t.  $M$  is pseudo-isomorphic to

$$\left( \bigoplus_{i \in [s]_+} \Lambda_p/(p^{n_i}) \right) \oplus \left( \bigoplus_{j \in [t]_+} \Lambda_p/(f_j(T)^{m_j}) \right).$$

Moreover, the ideal  $(\prod_i f_i) \subset \Lambda_p$  is well-defined, called the **characteristic ideal of  $B$** , denoted by  $\text{char}_p(M)$ .  $\lrcorner$

*Proof:*

$\square$

**Prop. (33.1.1.23).** If  $X \in \text{Mod}_{\Lambda_p}^{\text{fg}}$  and  $\#X_\Gamma < \infty$ , then  $X$  is torsion over  $\Lambda_p$ .  $\lrcorner$

*Proof:* Cf. [Washington].  $\square$

## 2 $p$ -adic Eisenstein Series

## 3 Coleman Map

The Coleman map is important as it puts the construction of the  $p$ -adic zeta function into a larger and more conceptual framework. Even more importantly, it provides an important bridge between analytic objects and algebraic structures.

## 4 $q$ -Primary Units

**Notation (33.1.4.1).**

- Let  $p, q \in \text{Prime}_{\geq 3}$  s.t.  $p \not\equiv 1 \pmod{q}$ .
- Let  $F = \mathbb{Q}(\zeta_p)$ .
- Let  $G = \text{Gal}(F/\mathbb{Q})$ .

- Let  $R = \mathbb{F}_q[G]$  which is isomorphic to a product of fields.
- Let  $N = \sum_{\sigma \in G} \sigma \in R$  be the norm element.

┘

**Prop. (33.1.4.2) [Units modulo  $q$ ].**  $\mathbb{Z}[\zeta_p]^*/(\mathbb{Z}[\zeta_p]^*)^q$  is a cyclic module over  $R$ , and  $\text{Ann}(\mathbb{Z}[\zeta_p]^*/(\mathbb{Z}[\zeta_p]^*)^q) = (N, 1 - \mathfrak{c})$  (33.1.4.1). ┘

*Proof:* Cf. [Catalan's Conjecture, Bilu] P16?

Let  $E = \mathbb{Z}[\zeta_p]^*$ ,  $\Omega$  the group of roots of unity of  $K$ , and let  $\overline{E} = E/\Omega$ . Because each root of unity is a  $q$ -th power, we have  $E/E^q \cong \overline{E}/\overline{E}^q$ . ┘

**Def. (33.1.4.3) [Cyclotomic Units].** Let  $C$  be the subgroup of units of  $\mathbb{Q}(\zeta_p)$  generated by  $\zeta_p$  and  $\frac{1-\zeta_p^k}{1-\zeta_p}$  for  $k \in \mathbb{Z} \setminus (p)$ .

Then  $C$  is a full rank subgroup of  $\mathbb{Z}[\zeta_p]^*$ . ┘

*Proof:* ┘

**Def. (33.1.4.4) [ $q$ -Primary Units].** A number  $\gamma \in \mathbb{Z}[\zeta_p]$  are called  **$q$ -primary** if there exists  $\nu \in \mathbb{Z}[\zeta_p]$  s.t.  $\gamma \equiv \nu^q \pmod{q^2} \mathbb{Z}[\zeta_p]$ . ┘

**Prop. (33.1.4.5).** If  $p > q$ , then not all the cyclotomic units in  $\mathbb{Q}(\zeta_p)$  (33.1.4.3) are  $q$ -primary. ┘

*Proof:* We show that in fact  $1 + \zeta_p^q = (1 - \zeta_p^{2q})/(1 - \zeta_p^q)$  is not  $q$ -primary: Suppose  $1 + \zeta_p^q \equiv \nu^q \pmod{q^2}$  for some  $\nu$ , then  $(1 + \zeta_p^q - \nu)^q \in (q)$ . But because  $q$  is unramified in  $\mathbb{Q}(\zeta_p)$  by (19.3.2.9),  $(q)$  is a radical ideal in  $\mathbb{Z}[\zeta_p]$ . So  $1 + \zeta_p^q - \nu \in (q)$ , which in turn implies that  $(1 + \zeta_p^q)^q \equiv \nu^q \pmod{q^2} \equiv 1 + \zeta_p^q \pmod{q^2}$ .

Then this implies that  $f(\zeta_p) \equiv 0 \pmod{q}$ , where  $f(X) = ((1 + X)^q - 1 - X^q)/q \in \mathbb{Z}[X]$  is a monic polynomial. By Galois conjugacy, this implies that each primitive  $p$ -th roots of unit is a root of  $f$ . This implies that  $p - 1 \leq \deg(f) = q - 1$ , contradiction. ┘

**Prop. (33.1.4.6) [Annihilators of Units].** Let  $E = \mathbb{Z}[\zeta_p]^*$ ,  $C_q \subset C$  be the group of  $q$ -primary units, then the groups

$$E/CE^q, \quad C/C_q \cong CE^q/C_qE^q, \quad C_q/(C_q \cap E^q) \cong C_qE^q/E^q$$

are all cyclic, and the annihilators

$$I_1 = \text{Ann}(E/CE^q), \quad I_2 = \text{Ann}(C/C_q), \quad I_3 = \text{Ann}(C_q/(C_q \cap E^q))$$

are pairwise coprime, and satisfy

$$I_1 I_2 I_3 = (N, 1 - \mathfrak{c}).$$

┘

*Proof:* The first assertion holds because  $C \cap C_qE^q = C_q$ : This holds by definition of  $C_q$ .

They are all cyclic because they are all subquotients of the cyclic  $R$ -module  $E/E^q$  (33.1.4.2) and  $R$  is a PID.

The last assertion follows from (33.1.4.2) and general properties of annihilators as  $R$  is a product of fields. ┘

## 33.2 $p$ -adic $L$ -Functions

References are [Kob84] and [ $p$ -adic  $L$ -Functions and  $p$ -adic Representations, Perrin-Riou].

**Notation (33.2.0.1).**

- Use notations on Iwasawa theory defined in [Classical Iwasawa Theory](#).

┘

### 1 $p$ -adic Hecke $L$ -Functions

References are [Fontaine's rings and  $p$ -adic  $L$ -functions, Colmez], [ $p$ -adic  $L$ -Functions, Iwasawa] and [Mazur-Tate-Teitelbaum, On  $p$ -adic analogues of the conjectures of Birch and Swinnerton-Dyer] and [Hid93].

**Prop. (33.2.1.1) [Local Group Rings].** For  $G \in \text{Ab}^{\text{fin}}$ ,  $R = \mathbb{Z}_p[G]$  is a complete semi-local ring, so  $R = \prod_{\mathfrak{m} \in |\text{Spec } R|_0} R_{\mathfrak{m}}$ .

Because  $\mathbb{Z}_p[\mathbb{Z}/(p^n)]$  is a local ring, for any maximal ideal  $\mathfrak{m}$  of  $R$ , there exists a  $\overline{\mathbb{Q}_p}$ -valued character of  $G$  of order prime to  $p$  s.t. the associated ring homomorphism factors through  $R_{\mathfrak{m}}$ , unique up to conjugacy.

The normalization  $\tilde{R}$  of  $R$  equals  $\prod_{\sigma \in \pi_0(\text{Spec } R)} R_{\sigma}$ .

┘

### Kubota-Leopoldt $p$ -adic Zeta Functions

**Def. (33.2.1.2) [Kubota-Leopoldt  $p$ -adic Zeta Function, Iwasawa].** For  $k \in \mathbb{Z}_+$ , define

$$\zeta_p(1-k) \triangleq (1-p^{k-1})\zeta(1-k) = (1-p^{k-1})\left(-\frac{B_k}{k}\right).$$

We will see in [\(33.2.1.3\)](#) that this extends to a locally analytic function on  $\mathbb{Z}_p^*$ .

┘

**Def. (33.2.1.3) [ $p$ -adic Mazur-Mellin Transform].** For  $s_0 \in [p-2]$  and  $s \in \mathbb{Z}_p$  s.t.  $s \neq 0$  if  $s_0 = 0$ , define

$$\zeta_{p,s_0}(s) = \frac{1}{m^{-s_0-(p-1)s} - 1} \int_{\mathbb{Z}_p^*} x^{s_0+(p-1)s-1} d\mu_{1,m}(x),$$

where  $m \in \mathbb{Z}_{\geq 2} \setminus (p)$ , then this  $\zeta_{p,s_0}$  is a continuous function on  $\mathbb{Z}_p^*$  that is independent of  $m$  chosen.

┘

*Proof.* It is clear that when  $x \in \mathbb{Z}_p^*$  and  $k, k' \in \mathbb{Z}_p$ ,  $N \in \mathbb{Z}_+$  satisfies  $k' \equiv k \pmod{(p-1)p^N}$ , then  $|x^k - x^{k'}|_p \leq p^{-N-1}$ . Thus the integral  $\int_{\mathbb{Z}_p^*} x^{s_0+(p-1)s-1} d\mu_{1,m}$  is a continuous function in  $s$ . Thus  $\zeta_{p,s_0}(s)$  is a continuous function in  $s$ .

To show that the integral is independent of  $m$  chosen, notice if  $m \in \mathbb{Z}_{\geq 2} \setminus (p)$  is another integer, then whenever  $s_0 + (p-1)s = k \in \mathbb{Z}$ , the two definitions of  $\zeta_{p,s_0}(s)$  both equal

$$\frac{1}{m^{-k} - 1} \int_{\mathbb{Z}_p^*} x^{k-1} d\mu_{1,m}(x) = \frac{1}{k(m^{-k} - 1)} \int_{\mathbb{Z}_p^*} d\mu_{k,m}(x) = (1-p^{k-1})\left(-\frac{B_k}{k}\right) = \zeta_p(1-k),$$

by [\(33.1.1.13\)](#) and [\(33.1.1.15\)](#).

□

**Interpolation at Dirichlet characters****Deligne-Ribet  $p$ -adic  $L$ -Functions**

[Values of Abelian  $L$ -functions at Negative Integers over Totally Real Fields, Deligne-Ribet] defined  $p$ -adic  $L$ -functions for totally real fields.

**Prop. (33.2.1.4) [Klingen/Siegel].** Let  $M/F \in \mathbf{NField}$  be a cyclic extension of totally real fields. For any  $\sigma \in \text{Gal}(M/F)$ ,  $L_F(\sigma; s)$  is the partial zeta function of  $F$ , then  $L_F(\sigma; 1 - n) \in \mathbb{Q}$ .  $\lrcorner$

*Proof:* ?  $\square$

**Thm. (33.2.1.5) [ $p$ -adic  $L$ -Functions for Totally Real Fields, Deligne-Ribet].** Situation as in (33.3.2.7), there exists a  $p$ -adic  $L$ -function  $L_p(\psi; s)$  s.t.

- $L_p(\psi; s)$  is continuous for  $s \in \mathbb{Z}_p \setminus \{1\}$  and even at  $s = 1$  if  $\psi \neq 1$ .
- For  $n \in \mathbb{Z}_+$ ,

$$L_p(\psi; 1 - n) = L(\psi\omega^{-n}; 1 - n) \prod_{v \in S_F(p)} (1 - \psi\omega^{-n}(\mathfrak{p}_v) \|v\|^{n-1}).$$

where  $\omega$  is the Teichmüller character.

- If  $\psi$  is a 1-dimensional character, denote  $H_\psi(T) = \psi(\gamma)(1 + T) - 1$  if  $\psi$  is of Greenberg type W(33.3.2.8) and  $H_\psi(T) = 1$  otherwise. Then there exists  $G_\psi(T) \in \mathbb{Z}_p[\psi][[T]]$  s.t.

$$L_p(\psi; 1 - s) = G_\psi(u^s - 1)/H_\psi(u^s - 1),$$

where  $u = \chi_{\text{cycl}}(\gamma)$ . Moreover, if  $\chi$  is a character of  $\text{Gal}_F$  of Greenberg type W(33.3.2.8),

$$G_{\psi\chi}(T) = G_\psi(\chi(\gamma)(1 + T) - 1), \quad H_{\psi\chi}(T) = H_\psi(\chi(\gamma)(1 + T) - 1).$$

$\lrcorner$

*Proof:* ?  $\square$

**Prop. (33.2.1.6) [ $p$ -adic  $L$ -Functions of Components].** Let  $H/F \in \mathbf{NField}$  be an Abelian extension with  $H \cap F_\infty = F$ . Then

$$\text{Gal}(H_\infty/F) \cong \text{Gal}(K/F) \times \text{Gal}(F_\infty/F)$$

and

$$\mathbb{Z}_p[[\text{Gal}(H_\infty/F)]] \cong R[[T]] \text{ where } R = \mathbb{Z}_p[\text{Gal}(H/F)].$$

Assume  $H$  is either totally real or CM, and let  $\text{Ram}_F(K/F) \subset_{\text{fin}} S \in \Sigma_F^{\text{fin}}$ . Suppose  $\mathfrak{m}$  is an even component of  $R$ (33.2.1.1). If the basic character of  $\mathfrak{m}$  is not trivial, then there exists  $G_{\mathfrak{m},S}(T) \in R_{\mathfrak{m}}[[T]]$  s.t.

$$\psi(G_{\mathfrak{m},S}(T)) = G_{\psi,S}(T) \in \mathbb{Z}_p[\psi][[T]]$$

for each character  $\psi \in \mathfrak{m}$ . And if the basic character of  $\mathfrak{m}$  is trivial, then we can take an ideal  $\mathfrak{c}$  of  $S$  that is prime to  $S$  and  $p$ , and let  $H_{\mathfrak{m},\mathfrak{c}}(T) \in R[[T]]$  be the power series corresponding to  $1 - \left(\frac{c}{L/K}\right)$ . Then there exists  $G_{\mathfrak{m},\mathfrak{c},S} \in R_{\mathfrak{m}}[[T]]$  s.t.

$$\frac{\psi(G_{\mathfrak{m},\mathfrak{c},S}(T))(u^s - 1)}{\psi(H_{\mathfrak{m},\mathfrak{c},S}(T))(u^s - 1)} = L_{p,S}(\psi; 1 - s)$$

for each character  $\psi \in \mathfrak{m}$ .

If  $\zeta_p(s)$  has no pole at  $s = 1$  (i.e. the Leopoldt conjecture fails), let  $R_\infty/F_\infty$  be the composite of all  $\mathbb{Z}_p$ -extension of  $F$  with  $\text{Gal}(R_\infty/F_\infty) = \mathbb{Z}_p\{\gamma_1, \dots, \gamma_t\}$ , then there is an isomorphism

$$\Lambda_t = \mathbb{Z}_p[[\text{Gal}(R_\infty/F)]] \cong \mathbb{Z}_p[[Y_1, \dots, Y_t, T]],$$

then there is a power series  $G(Y_1, \dots, Y_t, T) \in \Lambda_t$  s.t.

$$G(\psi(\gamma_1) - 1, \dots, \psi(\gamma_t) - 1, \psi(\gamma)u^s - 1) = L_p(\psi; s)$$

for each character  $\psi$  of finite order of  $\text{Gal}(R_\infty/F)$ .  $\lrcorner$

*Proof:* ? Cf. [Serre, Sur le residu de la fonction zeta  $p$ -adique d'un corps de nombre].  $\square$

**Thm. (33.2.1.7) [ $p$ -adic Artin Conjecture, conjectured by Greenberg [Gre83], Greenberg [Gre83]/Wiles [Wil90]].** For  $p \in \text{Prime}_{\geq 3}$ ,  $G_\psi(T) \in \mathbb{Z}_p[\psi][[T]][\frac{1}{p}]$ .  $\lrcorner$

*Proof:* Cf. [Gre83], where it is proven that this follows from (33.3.2.9).  $\square$

## 2 Leopoldt's Conjecture

**Conj. (33.2.2.1) [Leopoldt].** For  $F \in \text{NField}$  and  $p \in \text{Prime}$ , the natural map

$$\mathcal{O}_K^* \otimes \mathbb{Z}_p \rightarrow \prod_{v|p} \mathcal{O}_{K_v}^*$$

is injective.

Equivalently ?, the  $p$ -adic regulator for  $F$  doesn't vanish.  $\lrcorner$

*Proof:*  $\square$

**Thm. (33.2.2.2) [Leopoldt Conjecture for Abelian Extensions].** If  $F$  is an Abelian extension of  $\mathbb{Q}$  or an Abelian extension of an imaginary quadratic field, then the Leopoldt conjecture (33.2.2.1) holds for  $F$ .  $\lrcorner$

*Proof:* ? [A. Brumer, On the units of algebraic number fields].  $\square$

**Prop. (33.2.2.3) [Leopoldt Conjecture for Totally Real Fields].** Colmez [Residu en  $s=1$  des fonctions zeta  $p$ -adiques] proved a  $p$ -adic class number formula for totally real field  $F$ , so for these fields, the non-vanishing of the  $p$ -adic regulator is equivalent to their  $p$ -adic Dedekind zeta functions having a simple pole at  $s = 1$ .  $\lrcorner$

*Proof:* ?.  $\square$

## 3 Modular Forms

References are [Fontaine's rings and  $p$ -adic  $L$ -functions, Colmez].



#### 4 Perrin-Riou's *p*-adic *L*-Functions

References are [*p*-adic *L*-Functions and *p*-adic Representations, Perrin-Riou].

**Notation(33.2.4.1)[Notations on Iwasawa Algebras].**

- Let  $p \in \text{Prime}$ ,
- Let  $G_\infty = \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \cong \Gamma \times \Delta_p$ , where  $\Gamma \cong \mathbb{Z}_p$  and  $\Delta = \text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$ .
- Let  $\chi_{\text{cycl}}$  be the cyclotomic character of  $G_\infty$ .
- Let  $\Lambda_\Gamma = \mathbb{Z}_p[[\Gamma]]$ ,  $\Lambda = \mathbb{Z}_p[G_\infty] = \Lambda_\Gamma \otimes \mathbb{Z}_p[\Delta]$ .
- Let  $\iota \in \text{Aut}_{\mathbb{Z}_p}(\Lambda)$  be the involution operator s.t.  $\iota(\tau) = \tau^{-1}$ .
- Let  $\text{Twist} \in \text{Aut}_{\mathbb{Z}_p}(\Lambda)$  be the twisting operator s.t.  $\text{Twist}(\tau) = \chi(\tau)\tau$ .
- Denote  $2\pi i = \log([\varepsilon^p]) \in B_{\text{crys}}$ , where  $\varepsilon$  is a generator of  $(\mathbb{Z}_p(1))^*$ .
- Let  $\varphi \in \text{Aut}_{\mathbb{Z}_p}(\mathbb{Z}_p[[T]])$  be the operator s.t.  $\varphi(T) = (1+T)^p - 1$ .
- Let  $\psi \in \text{End}_{\mathbb{Z}_p}(\mathbb{Z}_p[[T]])$  be the unique linear map s.t. for any  $g \in \mathbb{Z}_p[[T]]$ ,

$$\varphi(\psi(g))(T) = \frac{1}{p} \sum_{\zeta \in \mu_p} g(\zeta(1+T) - 1).$$

- $\mathbb{Z}_p[[T]]^{\psi=0}$  is equipped with a continuous action of  $G_\infty$  by

$$\tau(1+T) = (1+T)^{\chi_{\text{cycl}}(\tau)}.$$

In this way,  $\mathbb{Z}_p[[T]]^{\psi=0}$  is a free  $\Lambda$ -module of rank 1 with basis  $(1+T)$ .

- $\mathbb{Z}_p[[T]]^{\psi=0}$  is equipped with a bijective derivation  $D = (1+T)\frac{\partial}{\partial T}$ , and for  $h \in \Lambda$ ,

$$D(h(1+T)) = \text{Twist}(h) \cdot (1+T).$$

┘

**Notation(33.2.4.2)[Notations on Galois Representations].**

- Let  $F \in \text{NField}$  that is unramified over  $p$ .
- Let  $F_n = F(\mu_{p^{n+1}})$ ,  $F_\infty = F(\mu_{p^\infty})$ ,  $\Gamma_n = \text{Gal}(F_\infty/F_n)$ ,  $\text{Gal}(F_\infty/F) \cong \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) = G_\infty$ .
- Let  $V \in \text{Rep}_{\mathbb{Q}_p}^{\text{fd}}(\text{Gal}_F)$  be a pseudo-geometric representation of  $\text{Gal}_F$  that is crystalline at  $S_F(p)$ .
- For  $v \in S_F(p)$ ,  $D_v(V) = (B_{\text{crys},v} \otimes_{\mathbb{Q}_p} V)^{\text{Gal}_{F_v}}$  is the filtered  $\varphi$ -module over  $\mathbb{F}_p$  of dimension  $\dim V$ .
- Define  $D_{F,p}(V) = D_p(\text{Ind}_{F/\mathbb{Q}} V) \cong \bigoplus_{v \in S_F(p)} D_v(V)$ .
- For any  $v \in S_F(p)$ , there is a  $\mathbb{Q}_p$ -linear **Bloch-Kato exponential map**

$$\exp_{F_v,V} : D_v(V) \rightarrow H^1(F_v, V).$$

When 1 is not an eigenvalue of  $\varphi$  on  $D_v(V)$ ,  $\exp_{F_v,V}$  is an isomorphism

$$\exp_{F_v,V} : t_{V,v} = D_v(V)/\text{Fil}^0 D_v(V) \cong H_f^1(F_v, V),$$

whose inverse is denoted by  $\log_{V,v}$ .

- When 1 is not an eigenvalue of  $\varphi$  on  $D_{F,p}(V)$ , denote

$$\log_V = \bigoplus_v \log_{V,v} : D_{F,p}(V) \rightarrow D_p(V)/\text{Fil}^0 D_p(V).$$

┘

## 5 Cyclotomic $p$ -adic $L$ -Functions

Coates and Perrin-Riou formulate a very general conjecture about the existence and properties of  $p$ -adic  $L$ -functions in [Coates-Perrin-Riou,  $p$ -adic  $L$ -functions attached to motives over  $\mathbb{Q}$ ]. The one-variable  $p$ -adic  $L$ -functions which they consider can be viewed as those corresponding to the cyclotomic deformations of motives.

## 6 Anticyclotomic $p$ -adic $L$ -Functions for Elliptic Curves

References are [Heegner points,  $p$ -adic  $L$ -functions, and the Cerednik-Drinfeld uniformization].

**Notation (33.2.6.1).**

- Let  $E \in \mathbf{EllCur}/\mathbb{Q}$  with conductor  $N$  associated to  $f \in \mathbf{SF}(\Gamma_0(N))$ .
- Let  $\mathcal{K}$  be an imaginary quadratic field s.t.  $(\text{disc}(\mathcal{K}), N) = 1$ .
- Let  $p \in \mathbf{Prime}$  be inert in  $\mathcal{K}$ .
- 

┘

**Situation (33.2.6.2).** Suppose

- $E$  has multiplicative reduction at  $p$ .
- $E$  has semistable reduction at all primes inert in  $\mathcal{K}$ .
- $w(E/K) = -1$ .

┘

**Def. (33.2.6.3).** Situation as in (33.2.6.2), write  $N = N^+ N^- p$  where  $N^+$  is divisible only by primes split in  $\mathcal{K}$ , and  $N^-$  is divisible only by primes inert in  $\mathcal{K}$ . By [GZ86]P71,  $\#S(N^-)$  is odd ?.

┘

## 7 Herbrand-Ribet Theorem

**Thm. (33.2.7.1) [Herbrand].** Let  $p \in \mathbf{Prime}$ ,  $Y = \text{Cl}(\mathbb{Q}(\mu_p))[p]$ , then  $\text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q}) = (\mathbb{Z}/(p))^\times$  acts on  $Y$  as a  $\mathbb{F}_p$ -space by conjugation. Let  $Y^j$  be the subspace of  $Y$  that  $(\mathbb{Z}/(p))^\times$  acts by character  $a \mapsto a^j$ . Then for  $1 < j < p-1$  an odd integer, if  $Y^j \neq 1$ , then the Bernoulli number  $B_{p-j}$  has numerator divisible by  $p$ .

┘

*Proof:* Cf.[Eisenstein Ideals, Mazur, P52].

□

### 33.3 Iwasawa Main Conjecture

References are [Wil90, S-U14].

#### 1 Iwasawa Main Conjectures

**Remark (33.3.1.1).** The Iwasawa main conjecture generally have two ingredients: the characteristic ideal of the dual of the Selmer group and a related  $p$ -adic  $L$ -function, ideally both in some  $\Lambda$ . The Iwasawa main conjecture essentially says that the  $p$ -adic  $L$ -function should control the size of the Selmer groups of the Galois representation it is attached to.  $\lrcorner$

#### 2 Fields case

**Notation (33.3.2.1).**

- Let  $p \in \text{Prime}$ .
- Let  $F \in \text{NField}$ .
- Let  $F_\infty$  be the cyclotomic  $\mathbb{Z}_p$ -extension of  $F$ .
- Let  $\gamma \in \text{Gal}(F_\infty/F)$  be a topological generator and  $u = \chi_{\text{cycl}}(\gamma) \in \mathbb{Z}_p^*$ .

$\lrcorner$

**Prop. (33.3.2.2) [ $\chi$ -Components of Iwasawa Modules].** If  $H/F \in \text{NField}$  s.t.  $H \cap F_\infty = F$ ,  $S_F(p) \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$  and  $M_\infty(S)$  the maximal Abelian pro- $p$ -extension of  $H_\infty$  that is unramified outside  $S_{H_\infty}(S)$ , denote

$$Y(S) = \text{Gal}(M_\infty(S)/H_\infty), \quad W(S) = Y(S) \otimes \overline{\mathbb{Q}_p}.$$

Let  $R = \mathbb{Z}_p[\text{Gal}(H/F)]$ , then  $Y(S)$  is a module over  $R[[T]]$ . For maximal ideal  $\mathfrak{m}$  of  $R$ , denote  $Y_{\mathfrak{m}}(S) = Y(S) \otimes_R R_{\mathfrak{m}}$ . Then for each character  $\chi \in \text{Gal}(H/F)^\vee$  belonging to the component  $\mathfrak{m}$  (33.2.1.1) of  $R$ , we have

$$W(S)^\chi = Y_{\mathfrak{m}}(S) \otimes_{R_{\mathfrak{m}}, \chi} \overline{\mathbb{Q}_p}.$$

And we denote  $f_{\chi, S}(T)$  the characteristic polynomial of  $\gamma - 1$  acting on  $W(S)^\chi$ .  $\lrcorner$

**Prop. (33.3.2.3) [Invariant Properties of  $f_\chi(T)$ ].** Situation as in (33.3.2.2), in fact the construction can be extended to any Artin character  $\chi$  of  $\text{Gal}(H/F)$ . Then

- If  $H'/H$  be an extension s.t.  $H' \cap F_\infty = F$ , then any Artin character  $\chi$  of  $\text{Gal}(H/F)$  extends to an Artin character of  $\text{Gal}(H'/F)$  by restriction, and

$$f_{\chi, S}(T) = f_{\chi', S_{H'}(S)}(T).$$

- If  $H/F$  is Galois and  $L/F$  is a subfield of  $H/F$ , then any Artin character  $\chi$  of  $\text{Gal}(H/L)$  extends to an Artin character  $\chi'$  of  $\text{Gal}(H/F)$  by induction. Then for any  $S_L(p) \subset_{\text{fin}} S \subset \Sigma_L^{\text{fin}}$ , we have

$$f_{\chi', S}(T) = f_{\chi, S_H(S)}(T).$$

$\lrcorner$

*Proof:* ? Cf. [Gre83].  $\square$

### Stickelberger's Ideal

**Def. (33.3.2.4) [Stickelberger's Ideal].** Let  $M \in \mathbf{NField}$  contained in  $\mathbb{Q}(\zeta_m)$  for some  $m \in \mathbb{Z}_+$  minimum. Denote  $G = \text{Gal}(M/\mathbb{Q})$  and define the **Stickelberger's element**

$$\theta = \theta_M = \sum_{1 \leq a \leq m-1, (a,m)=1} \frac{a}{m} \cdot \sigma_a^{-1} \in \mathbb{Q}[G].$$

The **Stickelberger's ideal** is the ideal of  $\mathbb{Z}[G]$  defined by  $I(M) = \mathbb{Z}[G] \cap \theta_M \mathbb{Z}[G]$ . ┘

**Prop. (33.3.2.5) [Stickelberger's Ideal for Cyclotomic Fields].** Let  $m \in \mathbb{Z}_+$   $G = \text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$  and let  $I' \subset \mathbb{Z}[G]$  be ideal generated by  $\{\sigma_c - c : c \pmod{m}, (c, m) = 1\}$ , then the Stickelberger's Ideal (33.3.2.4)  $I_{\mathbb{Q}(\zeta_m)} = I' \theta_{\mathbb{Q}(\zeta_m)}$ .

In other words, it is generated by the elements of the form

$$\Theta_c = \sum_{a=1}^{p-1} [ac/p] \sigma_a^{-1}, \quad c \in [p-1]_+.$$

┘

*Proof:*

□

**Thm. (33.3.2.6) [Stickelberger].** Let  $F \in \mathbf{NField}$  contained in  $\mathbb{Q}(\zeta_m)$  for some  $m \in \mathbb{Z}_+$  minimum. Then the Stickelberger's ideal  $I(F)$  annihilates  $\text{Cl}(F)$ . ┘

*Proof:* Cf. [Was97]P94. □

### Iwasawa Main Conjecture for Totally Real Fields

References are [Wil90].

**Notation (33.3.2.7) [Totally Real Artin Representations].**

- Let  $F$  be totally real.
- Let  $\psi$  be an Artin character of  $\text{Gal}_F$  s.t. the fixed field  $F_\psi$  is totally real.
- Denote  $\mathbb{Z}_p[\psi] \triangleq \mathbb{Z}_p[\text{Im}(\psi)]$ .

┘

**Def. (33.3.2.8) [Greenberg Types].** An Artin character  $\psi$  of  $\text{Gal}_F$  is said to be of **Greenberg type S** if  $F_\psi \cap F_\infty = F$ . It is said to be of **Greenberg type W** if  $\psi$  is 1-dimensional character and  $F_\psi \subset F_\infty$ . ┘

**Thm. (33.3.2.9) [Iwasawa Main Conjecture for Totally Real Fields, Mazur-Wiles [M-W84]/Wiles [Wil86, ?WI190]].** Let  $\chi$  be an Artin character of  $\text{Gal}_F$  of Greenberg type S (33.3.2.8) s.t. the fixed field  $F_\chi$  is a CM field. Let  $H = F_\chi(\mu_{p^{1+\delta_{2,p}}})$ , and  $L_\infty/H_\infty$  the maximal unramified pro- $p$  extension unramified away from  $S_{H_\infty}(p)$ .

Consider  $X = \text{Gal}(L_\infty/H_\infty)$  as a module over  $\text{Gal}(H_\infty/F)$ , then  $\dim X \otimes \overline{\mathbb{Q}_p} < \infty$  by [On  $\mathbb{Z}_\ell$ -extensions of algebraic number fields, Iwasawa] ?. Let  $f_\chi(T)$  be the characteristic polynomial of  $\gamma - 1$  acting on the  $\chi$ -part of  $X \otimes \overline{\mathbb{Q}_p}$ .

Suppose  $\chi$  is odd, and  $p \geq 3$  or  $F = \mathbb{Q}$ , then  $f_\chi(T)$  differs from the formal power series  $G_{\chi^{-1}\omega}(u(1+T)^{-1} - 1)$  (33.2.1.5) by a unit in  $\mathbb{Z}_p[\chi][[T]]^*$ . ┘

*Proof:* ?

By[?Iwasanwiantslbow], it suffices to prove that  $n_\alpha(\chi) \geq m_\alpha(\chi)$  for each  $\alpha \in \overline{\mathbb{Q}_p}$ . There are three case:

- $\alpha + 1 \notin \mu_{p^\infty}$  and  $\alpha \neq u - 1$  if  $\chi = \omega$ .
- $\alpha + 1 \in \mu_{p^\infty}$ .
- $\chi = \omega$  and  $\alpha = u - 1$ .

Case 1 is done in [Wil90]P516.?

Case 2 is done by taking a  $p$ -adic limit  $\chi\rho_n \rightarrow \chi$  s.t.  $m_\alpha(\chi\rho_n) = 0$  for each  $n$ . It is crucial to analyze the  $\mu$ -invariant  $\mu(G_\chi(T))$ .

Case 3 is true unless  $m_{u-1}(\omega) > 0$ , or equivalently  $\zeta_p(s)$  has no pole at  $s = 1$ . In this case, Leopoldt's conjecture is false and there exists an  $\mathbb{Z}_p^2$ -extension  $R_\infty/F$  by work of Colmez(33.2.2.3). Suppose  $F_\infty \subset R_\infty$ , and let  $M_\infty/R_\infty$  be the maximal Abelian pro- $p$ -extension that is unramified outside primes over  $p$ . Then there is an isomorphism

$$\Lambda_Y \triangleq \mathbb{Z}_p[[\text{Gal}(R_\infty/F)]] \cong \mathbb{Z}_p[[Y, T]].$$

Then we take a  $p$ -adic limit  $\chi\rho_n \rightarrow \chi$  where  $\rho_n$  are characters of this anticyclotomic extension of  $F$ .  $\square$

**Cor.(33.3.2.10)[An equivariant Formulation].** Let  $\chi$  be an Artin character of  $\text{Gal}_F$  of Greenberg type S(33.3.2.8) s.t. the fixed field  $F_\chi$  is a CM field. Let  $H = F_\chi(\mu_{p^{1+\delta_{2,p}}})$ , and  $L_\infty/H_\infty$  the maximal unramified pro- $p$  extension.

For  $S_F(p) \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$ , let  $M_\infty(S)$  be the maximal Abelian pro- $p$ -extension of  $H_\infty$  that is unramified outside  $\cup S_{H_\infty}(S)$ , and denote  $h_{\chi,S}(T)$  the characteristic polynomial of  $\gamma - 1$  acting on  $W(S)^\chi$ (33.3.2.2).

Suppose  $\chi$  is even, then  $h_{\chi,S}(T)$  differ from the formal power series  $G_{\chi,S}(T)$ (33.2.1.5) by a unit in  $\mathbb{Z}_p[\chi][[T]]^*$ .

Then thoeorem(33.3.2.9) for an odd  $\chi$  is equivalent to this statement for  $\chi^{-1}\omega$  and each  $S_F(p) \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$ .  $\lrcorner$

*Proof:* Cf.[Greenberg, on  $p$ -adic Artin  $L$ -Functions].?  $\square$

**Prop.(33.3.2.11)[Iwasawa's Analytic Class Number Formula].** Situation as in(33.3.2.9), for  $\alpha \in \overline{\mathbb{Q}_p}$ , if we denote

$$n_\alpha(\chi) = \text{ord}_{T=\alpha} f_\chi(T), \quad m_\alpha(\chi) = \text{ord}_{T=\alpha} G_{\chi^{-1}\omega}(u(1+T)^{-1} - 1),$$

then

$$\sum_\alpha \sum_{\chi \in \text{Gal}(H/F)^\vee} n_\alpha(\chi) = \sum_\alpha \sum_{\chi \in \text{Gal}(H/F)^\vee} m_\alpha(\chi).$$

$\lrcorner$

*Proof:* ?  $\square$

**Prop.(33.3.2.12).** Situation as in(33.3.2.11), Any  $\alpha$  s.t.  $\alpha + 1 \in \mu_{p^\infty}$  and  $m_\alpha(\chi) \neq 0$  are called a **trivial zero**. For such a trivial zero  $\alpha = \zeta - 1$ , let  $\rho$  be an Artin character of  $F$  of Greenberg type W s.t.  $\rho(\gamma) = \zeta$ , and denote

$$s_{\zeta-1}(\chi) = \#\{v \in S_F(p) : \chi\rho(\sigma_v) = 1\},$$

then

$$m_{\zeta-1}(\chi) = 0 \iff s_{\zeta-1}(\chi) = 0.$$

Moreover it is conjectured that  $m_{\zeta-1}(\chi) = s_{\zeta-1}(\chi)$ . Notice that it is proven that  $m_{\zeta-1}(\chi) = s_{\zeta-1}(\chi) = n_{\zeta-1}(\chi)$  for  $F = \mathbb{Q}$  by [On the behavior of  $p$ -adic  $L$ -functions at  $s = 0$ , Ferrero-Greenberg] and [On a certain  $l$ -adic representation, Greenberg].  $\lrcorner$

*Proof:* it follows from (33.2.1.5) that

$$G_{\chi^{-1}\omega}(u\zeta^{-1} - 1) = G_{\chi^{-1}\rho^{-1}\omega}(u - 1) = L_p(\chi^{-1}\rho^{-1}\omega; 0) = L(\chi^{-1}\rho^{-1}; 0) \prod_{v \in S_F(p)} (1 - \chi\rho(\sigma_v)).$$

And the assertion follows.  $\square$

**Thm. (33.3.2.13)** [ $\mu$ -Invariants, Ferrero-Washington/Wiles[Wil90]]. Situation as in (33.3.2.9), if  $p > 3$  and  $\chi$  is an odd character of  $\text{Gal}_F$  of order prime to  $p$ , then

$$\mu(X^\chi) = \mu \left( G_{\chi^{-1}\omega}(u(1+T)^{-1} - 1) \right).$$

Notice that it is conjectured that both sides vanish, which is true for  $F = \mathbb{Q}$ .  $\lrcorner$

*Proof:* ? Cf.[Wil90]P531.  $\square$

### Iwasawa Main Conjecture for Imaginary Quadratic Fields

References are [Rubin, “main conjectures” of Iwasawa theory for imaginary quadratic fields].

## 33.4 Kato's Euler Systems

Cf. [Kat04] and [Scholl], [Chao Li's Notes].

### 1 Kato-Siegel Functions

**Thm. (33.4.1.1) [Kato-Siegel-Functions].** For  $d \in \mathbb{Z}$  prime to 6, there exists a unique rule  $\vartheta_d$  that associates each elliptic curve  $E/S$  a section  $\vartheta_d^{(E/S)} \in \mathcal{O}^*(E \setminus E_S[d])$ . satisfying:

- $\text{div}(\vartheta_d^{(E/S)}) = d^2(e_E) - E_S[d]$ .
- For a base change  $g : E'/S' \rightarrow E/S$ ,  $g^*\vartheta_d^{(E/S)} = \vartheta_d^{(E'/S')}$ .
- If  $\alpha : E/S \rightarrow E'/S$  is an isogeny and  $(\deg \alpha, d) = 1$ , then  $\alpha_*\vartheta_d^{(E/S)} = \vartheta_d^{(E'/S)}$ .
- $\vartheta_{-d} = \vartheta_d, \vartheta_1 = 1$ . And if  $d = mc$  where  $m, c \in \mathbb{Z}_+$ , then

$$[m]_*\vartheta_d = \vartheta_c^{m^2}, \quad \vartheta_c \circ [m] = \vartheta_d/\vartheta_m^{c^2}.$$

In particular,  $[d]_*\vartheta_d = 1$ .

- If  $\tau \in \mathcal{H}$  and  $E_\tau/\mathbb{C} = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ , then

$$\vartheta_d^{(E_\tau/\mathbb{C})}(u) = (-1)^{\frac{d-1}{2}} \Theta(u, \tau)^{d^2} \Theta(du, \tau)^{-1},$$

where

$$\Theta(u, \tau) = q^{\frac{1}{12}}(t^{1/2} - t^{-1/2}) \prod_{n \in \mathbb{Z}_+} (1 - q^n t)(1 - q^n t^{-1}), \quad q = e^{2\pi i \tau}, t = e^{2\pi i u}.$$

In fact, this should be the case over the Tate curve. ┘

*Proof:* Cf. [Scholl, P384] ? □

### Modular Eisenstein Series

**Def. (33.4.1.2) [Eisenstein Series].** Let  $S \in \text{Sch}_{\text{int}}, E \in \text{EllCur}/S, d, N \in \mathbb{Z}_+, (N, d) = 1$ , and  $x \in E(S)$  is a torsion section of exact order  $N$ . If  $x$  is disjoint with  $E_S[d]$ , then  $\vartheta_d(x) = x^*\vartheta_d \in \mathcal{O}^*(S)$ . This is the case if  $N$  is invertible on  $S$  or  $N$  is not a prime power.

In this case, define

$$\text{dlog}_v \vartheta_d \in \Gamma(E \setminus E_S[d], \Omega_{E/S}^1)$$

and

$${}_d\text{Eis}(E/S, x) = x^*\text{dlog}_v \vartheta_d \in \Gamma(S, x^*\Omega_{E/S}^1) = \Gamma(S, \omega_{E/S}).$$

is a modular form of weight 1 (28.12.1.1).

Of course, we can also define  $\text{dlog} \vartheta_d(x) \in \Gamma(S, \Omega_S^1)$ . ┘

**Def. (33.4.1.3) [Eis( $E/S, x$ )].** Situation as in (33.4.1.2), by (33.4.1.1), for  $d, d' \in \mathbb{Z}$  prime to 6,

$$(\vartheta_d)^{d'^2} \cdot \vartheta_{d'} \circ [d] = (\vartheta_{d'})^{d^2} \cdot \vartheta_d \circ [d'] = \vartheta_{dd'},$$

so

$$d'^2 \text{dlog}_v \vartheta_d - [d']^* \text{dlog}_v \vartheta_d = d^2 \text{dlog}_v \vartheta_{d'} - [d]^* \text{dlog}_v \vartheta_{d'}.$$

And

$$d'^2 {}_d\text{Eis}(E/S, x) - d' {}_d\text{Eis}(E/S, d'x) = d^2 {}_{d'}\text{Eis}(E/S, x) - d {}_{d'}\text{Eis}(E/S, dx).$$

In particular, if  $d \equiv 1 \pmod{N}$  and  $d \neq \pm 1$ , then

$$\text{Eis}(E/S, x) = \frac{1}{d^2 - d} {}_d\text{Eis}(E/S, x) \in \Gamma(S[\frac{1}{d(d-1)}], \omega)$$

is independent of  $d$ . And because  $d$  is arbitrary, we can in fact glue the definition to a section

$$\text{Eis}(E/S, x) \in \Gamma(S[\frac{1}{2N}], \omega).$$

┘

**Cor. (33.4.1.4).** For any  $d \in \mathbb{Z}$  prime to 6,

$${}_d\text{Eis}(E/S, x) = d^2 \text{Eis}(E/S, x) - d \text{Eis}(E/S, dx) \in \Gamma(S[\frac{1}{2N}], \omega).$$

┘

**Example (33.4.1.5) [Eisenstein Series Over  $\mathbb{C}$ ].** If  $E = \mathbb{C}/(\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2)$ , and  $x = \frac{a_1}{N}\omega_1 + \frac{a_2}{N}\omega_2 \in E[N] \setminus \{O\}$ , then

$$\text{Eis}(E/\mathbb{C}, x) = \left( \sum_{m_i \in \frac{a_i}{N} + \mathbb{Z}} \frac{1}{(m_1\omega_1 + m_2\omega_2) |m_1\omega_1 + m_2\omega_2|^{2s}} \right) \Big|_{s=0} du.$$

┘

*Proof:* Cf. [Scholl, P389].

□

**Example (33.4.1.6) [Eisenstein Series over Tate Curves].** Consider the Tate curve  $\text{Tate}(q)$  over  $\mathbb{Z}[\mu_N]((q^{1/N}))$ , there is a level  $N$  structure

$$(\mathbb{Z}/(N))^2 \cong \text{Tate}(q)_{\mathbb{Z}[\mu_N]((q^{1/N}))}[N] : (a_1, a_2) \mapsto \zeta_N^{a_1} q^{a_2/N}.$$

Then if  $x = \zeta_N^{a_1} q^{a_2/N}$ ,  $0 \leq a_1, a_2 < N$ , then

$$\text{Eis}(\text{Tate}(q)/\mathbb{Z}[\mu_N]((q^{1/N})), x) = \begin{cases} \left[ \frac{a_2}{N} - \frac{1}{2} - \sum_{n \in \mathbb{Z}_+} \left( \sum_{d \in \mathbb{Z}, d|n, n/d \equiv a_2 \pmod{N}} \text{sgn}(d) \zeta_N^{a_1 d} \right) q^{n/N} \right] \text{dlog } t & a_2 \neq 0 \\ \text{?} & a_2 = 0 \end{cases}$$

Similarly, we can calculate that in  $\Omega_{\mathbb{Z}[\mu_N]((q^{1/N}))/\mathbb{Z}}^1$ ,

$$\text{dlog } \vartheta_d(x) \equiv \frac{{}_d\text{Eis}(\text{Tate}(q)/\mathbb{Z}[\mu_N]((q^{1/N})), x)}{\text{dlog } t} (a_1 \text{dlog } \zeta_N + a_2 \text{dlog}(q^{1/N})) \pmod{N}.$$

┘

*Proof:* Cf. [Scholl, P390].

□



## 2 Norm Relations

Suppose  $\ell \in \mathbf{P}$ , and  $d, d' \in \mathbb{Z}$  s.t.  $(6\ell, dd') = 1$ . Denote  $\vartheta = \vartheta_d^{(E/S)}$  and  $\vartheta' = \vartheta_{d'}^{(E/S)}$ .

**Prop. (33.4.2.1).** If  $S = Y_H$  is a modular curve of level prime to  $\ell$ , and  $z, z'$  are torsion sections of  $E/S(\ell)$  whose projections onto  $E_S[\ell]$  are linearly independent, then

$$\mathrm{Nm}_{S(\ell)/S}\{\vartheta(z), \vartheta'(z')\} = (1 - T_\ell \circ \langle \ell \rangle_* + \ell \langle \ell \rangle_*)\{\vartheta(\ell z), \vartheta'(\ell z')\}.$$

┘

*Proof:* Cf.[Scholl, P399] ?.

□

**Prop. (33.4.2.2).** Suppose  $Y_H$  is a modular curve of level prime to  $\ell$  and  $z, z'$  are two torsion sections of  $E/Y_H(\ell^m, \ell^n)$  with  $m, n \in \mathbb{Z}_{\geq 2}$ . If the projections of  $z, z'$  into  $E_S[\ell]$  are linearly independent, then

$$\mathrm{Nm}_{Y_H(\ell^m, \ell^n)/Y_H(\ell^{m-1}, \ell^{n-1})}\{\vartheta(z), \vartheta(z')\} = \{\vartheta(\ell z), \vartheta'(\ell z')\}.$$

┘

*Proof:* Cf.[Scholl, P400] ?.

□

**Prop. (33.4.2.3).** Suppose  $Y_H$  is a modular curve of level prime to  $\ell$  and  $z, z'$  are two torsion sections of  $E/Y_H(\ell^m, \ell^n)$  with  $m, n \in \mathbb{Z}_{\geq 2}$ . If the projections of  $z, z'$  into  $E_S[\ell]$  are linearly independent, then

$$\mathrm{tr}_{Y_H(\ell^m, \ell^n)/Y_H(\ell^{m-1}, \ell^{n-1})}\left({}_d\mathrm{Eis}(z) \cdot {}_{d'}\mathrm{Eis}(z')\right) = \ell^{-2} {}_d\mathrm{Eis}(\ell z) \cdot {}_{d'}\mathrm{Eis}(\ell z')$$

$$\mathrm{tr}_{Y_H(\ell^m, \ell^n)/Y_H(\ell^{m-1}, \ell^{n-1})}\left(\mathrm{KS}\left({}_d\mathrm{Eis}(z) \cdot {}_{d'}\mathrm{Eis}(z')\right)\right) = \ell^2 \mathrm{KS}\left(\ell^{-2} {}_d\mathrm{Eis}(\ell z) \cdot {}_{d'}\mathrm{Eis}(\ell z')\right)$$

┘

*Proof:* Cf.[Scholl, P401] ?.

□

## 3 Dual Exponentials

**Notation (33.4.3.1).**

- Let  $p \in \mathbf{P}$ ,  $\mathbb{Q}_p \subset_{\mathrm{fin}} K$ .
- Let  $K_n = K(\mu_{p^n})$ , and  $\mathfrak{d}_n = \mathfrak{d}_{K_n/K}$ .

┘

**Def. (33.4.3.2) [Bloch-Kato Dual Exponential Maps].** For any  $V \in \mathrm{Rep}_{\mathbb{Q}_p}^{\mathrm{dR}}(\mathrm{Gal}_K)$ , the **dual exponential map**

$$\exp_p^* : H^1(K, V) \rightarrow D_{\mathrm{dR}}^0(V)$$

is defined to be

$$H^1(K, V) \rightarrow H^1(K, B_{\mathrm{dR}}^0 \otimes_{\mathbb{Q}_p} V) \stackrel{(25.2.5.5)}{=} H^1(K, \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_K D_{\mathrm{dR}}(V))) \cong D_{\mathrm{dR}}^0(V).$$

where the last isomorphism follows from Tate's computation (25.2.1.12) and the isomorphisms  $B_{\mathrm{dR}}^j/B_{\mathrm{dR}}^{j+1} \cong \mathbb{C}_K(j)$ .

┘

**Example (33.4.3.3).** In case that  $X \in \text{SmPrpr}/K$  and  $V = H^1(X, \mathbb{Q}_p(1))$ , if  $D_{\text{dR}}^1(V) = 0$ , then the Hodge-Tate decomposition

$$\mathbb{C}_K \otimes_{\mathbb{Q}_p} V \cong \otimes_{i \in \mathbb{Z}} \mathbb{C}_K(-i) \otimes_K \text{gr}^i D_{\text{dR}}(V),$$

so the the dual exponential map is just

$$H^1(K, V) \rightarrow H^1(K, \mathbb{C}_K \otimes_K V) \cong H^1(K, \mathbb{C}_K) \otimes_K D_{\text{dR}}^0(V) \xleftarrow{\cup \log \chi_{\text{cycl}}} D_{\text{dR}}^0(V).$$

┘

**Cor. (33.4.3.4).** If  $Y$  be a smooth  $\mathcal{O}_K$ -scheme, which is a complement in a smooth proper  $\mathcal{O}_K$ -scheme  $X$  of a divisor  $Z$  with relatively normal crossings, define

$$H_{\text{dR}}^i(Y/\mathcal{O}_K) = H^i(X, \Omega_{X/\mathcal{O}}^i(\log Z)),$$

and  $V = H_{\text{ét}}^1(Y_{\overline{K}}, \mathbb{Q}_p(1))$ .

Then  $V$  is also de Rham, and the deRham comparison also holds ?, so

$$\begin{aligned} D_{\text{dR}}^{-1}(V) &= D_{\text{dR}}(V) = H_{\text{dR}}^1(Y/\mathcal{O}_K) \otimes_{\mathcal{O}_K} K, \\ D_{\text{dR}}^0(V) &= H^0(X, \Omega_{X/\mathcal{O}_K}^1(\log Z)) \otimes_{\mathcal{O}_K} K = \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) \otimes_{\mathcal{O}_K} K, \\ D_{\text{dR}}^1(V) &= 0. \end{aligned}$$

Thus by the same reasoning as in (33.4.3.3), the dual exponential map is known if we can compute the projection

$$\text{pr}_1 : \mathbb{C}_K \otimes_{\mathbb{Q}_p} V \cong \mathbb{C}_K \otimes_{\mathcal{O}_K} H^0(X, \Omega_{X/\mathcal{O}_K}^1(\log Z)) \bigoplus \mathbb{C}_K(1) \otimes_{\mathcal{O}_K} H^1(X, \mathcal{O}_X) \rightarrow \mathbb{C}_K \otimes_{\mathcal{O}_K} H^0(X, \Omega_{X/\mathcal{O}_K}^1(\log Z)),$$

which is the inversion by  $p$  of the limit of the map

$$H_{\text{ét}}^1(Y_{\overline{K}}, \mu_{p^n}) \cong H^0(Y_{\overline{K}}, \mathcal{O}_K^*/p^n) \xleftarrow{\cong} H^0(Y_{\mathcal{O}_{\overline{K}}}, \mathcal{O}^*/p^n) \xrightarrow{\text{dlog}} H^0(X, \Omega_{X/\mathcal{O}_K}^1(\log Z)) \otimes \mathcal{O}_{\overline{K}}/p^n.$$

┘

*Proof:* Cf. [Scholl, P408]. ?

□

### Explicit Reciprocity Law

**Thm. (33.4.3.5).** If  $2 \neq p \in \mathbf{P}$ ,  $K/\mathbb{Q}_p$  is unramified, then there exists  $c \in \mathbb{N}$  s.t. for any  $n \in \mathbb{Z}_+$ , the following diagram

$$\begin{array}{ccc} K_2(Y_{\mathcal{O}_{K_n}}) \otimes \mu_{p^n}^{-1} & \xrightarrow{c_2} & H^2(Y_{K_n}, \mu_{p^n}) \\ \downarrow \text{dlog} & & \downarrow \text{H-S} \\ H^0(X_{\mathcal{O}_{K_n}}, \Omega_{X_{\mathcal{O}_{K_n}}/\mathcal{O}_K}^2(\log Z))(-1) & & H^1(K_n, H^1(Y_{\overline{K}}, \mu_{p^n})) \\ \parallel & & \downarrow \text{pr}_1 \pmod{p^{n-1}} \\ \Omega_{\mathcal{O}_{K_n}/\mathcal{O}_K}^1(-1) \otimes \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) & & H^1(K_n, \mathcal{O}_{\overline{K}}/p^{n-1}) \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) \\ \text{dlog}(\zeta_{p^n}) \otimes [\zeta_{p^n}]^{-1} \mapsto 1 \downarrow \cong & & \uparrow \cup \frac{1}{p^n} \log \chi_{\text{cycl}} \\ \mathcal{O}_{K_n}/\mathfrak{d}_n \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) & \longrightarrow & \mathcal{O}_{K_n}/p^{n-1} \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) \end{array}$$

where H-S is the Hochschild-Serre boundary map defined as in (11.1.4.15), dlog is defined in (11.1.4.12),  $\text{pr}_1$  is defined in (33.4.3.4). ┘

*Proof:* Cf. [Scholl, P410].  $\square$

**Remark (33.4.3.6).**  $\color{red}{?}$  The assumption that  $K/\mathbb{Q}_p$  is unramified is not essential for the proof. The case  $p = 2$  is excluded because we don't know the compatibility of the trace map in this case.  $\lrcorner$

**Cor. (33.4.3.7) [Kato's Explicit Reciprocity Law].** If  $2 \neq p \in \mathbf{P}$ ,  $K/\mathbb{Q}_p$  is unramified, then for any  $m \in \mathbb{N}$ , the following diagram is commutative:

$$\begin{array}{ccc}
 \varprojlim_n (K_2(Y_{\mathcal{O}_{K_n}})) \otimes \mu_{p^n}^{-1} & \xrightarrow{\text{H-Soc}_2} & \varprojlim_n H^1(K_n, H^1(Y_{\overline{K}}, \mu_{p^n})) \\
 \downarrow \text{dlog} & & \downarrow \text{cor} \\
 \varprojlim_n H^0(X_{\mathcal{O}_{K_n}}, \Omega_{X_{\mathcal{O}_{K_n}}/\mathcal{O}_K}^2(\log Z))(-1) & & \varprojlim_{n \geq m} H^1(K_m, H^1(Y_{\overline{K}}, \mu_{p^n})) \\
 \parallel & & \parallel \\
 \varprojlim_n \Omega_{\mathcal{O}_{K_n}/\mathcal{O}_K}^1(-1) \otimes \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) & & H^1(K_m, H^1(Y_{\overline{K}}, \mathbb{Z}_p(1))) \\
 \downarrow \text{dlog}(\zeta_{p^n}) \otimes [\zeta_{p^n}]^{-1} \mapsto 1 \cong & & \downarrow \text{exp}^* \\
 \mathcal{O}_{K_n}/\mathfrak{d}_n \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) & \xrightarrow{\{\frac{1}{p^n} \text{tr}_{\mathcal{O}_{K_n}/\mathcal{O}_{K_m}}\}_{n \geq m}} & K_m \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K)
 \end{array}$$

$\lrcorner$

*Proof:* It follows from (33.4.3.4) that there is a commutative diagram

$$\begin{array}{ccc}
 H^1(K_m, H^1(Y_{\overline{K}}, \mathbb{Q}_p(1))) & \xrightarrow{\text{pr}_1} & H^1(K_m, \mathbb{C}_K \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K)) \\
 \downarrow \text{exp}^* & & \parallel \\
 K_m \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K) & \xrightarrow{\cup \log \chi_{\text{cycl}}} & H^1(K_m, \mathbb{C}_K) \otimes_{\mathcal{O}_K} \text{Fil}^1 H_{\text{dR}}^1(Y/\mathcal{O}_K)
 \end{array}$$

And also for any  $n \geq m$ , there exists a commutative diagram (Cf. [Scholl, P416]  $\color{red}{?}$ )

$$\begin{array}{ccc}
 \mathcal{O}_{K_n} & \xrightarrow{\frac{1}{p^n} \log \chi_{\text{cycl}}} & H^1(K_n, \mathcal{O}_{\mathbb{C}_K}) \\
 \downarrow \frac{1}{p^{n-m}} \text{tr}_{\mathcal{O}_{K_n}/\mathcal{O}_{K_m}} & & \downarrow \text{cor} \\
 \mathcal{O}_{K_m} & \xrightarrow{\frac{1}{p^m} \log \chi_{\text{cycl}}} & H^1(K_m, \mathcal{O}_{\mathbb{C}_K})
 \end{array}$$

Thus the assertion follows from (33.4.3.5) by taking limit and inverting  $p$ .  $\color{red}{?}$  How to check the compatibility of the limits?  $\square$

## 4 Rankin-Selberg Convolution

### Notations

**Def. (33.4.4.1).** For  $t \in \mathbb{Q}_p$  with  $v_p(t) < 0$ , denote

$$\phi_p^0 = \chi_{\mathbb{Z}_p \times \mathbb{Z}_p}, \quad \phi_p^{1,t} = \chi_{(t+\mathbb{Z}_p) \times \mathbb{Z}_p}, \quad \phi_p^{2,t} = \chi_{(t+\mathbb{Z}_p) \times \mathbb{Z}_p^*} - p^{-1} \chi_{(t+p^{-1}\mathbb{Z}_p) \times \mathbb{Z}_p^*}.$$

And for any  $\varphi \in C^\infty(\mathbb{Z}_p \times \mathbb{Z}_p)$ , denote  $\varphi^{\text{tr}}(x, y) = \varphi(y, x)$ .

For  $\varphi \in \mathcal{S}(\mathbf{A}_f^2)$  and  $\delta \in \mathbf{I}_f$ , denote  $[\delta]\varphi(\underline{x}) = \varphi(\delta^{-1}\underline{x})$ .  $\lrcorner$

### Results

**Notation (33.4.4.2) [Setup].**

- Let  $F \in S_{k+l}(\Gamma(1))$ , generating an irreducible  $\pi = \otimes' \pi_p$ , and whose Whittaker function  $A(g) = \prod A_p(g_p)$  factorizes, with  $A_p(1) = 1$  for each  $p$ . ?
- Let  $\varepsilon$  be the character of  $GL(2, \mathbf{A}_f)$  on  $A^*$ . ?
- Let  $M, R, N \in \mathbb{Z}_+$ ,  $(R, N) = 1$ ,  $M|N$ .
- If  $p \nmid N$ , then  $A_p$  is assumed to be  $\mathcal{K}_p$ -invariant.
- Let  $\lambda : \mathbf{I}_f/\mathbb{Q}_+^\times \rightarrow \mathbb{C}^\times$  be unramified outside  $T$ .
- Let  $N = \prod_{p \in S} p^{\nu_p}$ ,  $M = \prod_{p \in S} p^{\mu_p}$ ,  $R = \prod_{p \in T} p^{\nu_p}$ , where if  $p|N$ , then  $\nu_p > \mu_p \geq 0$ ,  $\nu_p \geq 2$ , and  $A_p$  is  $\mathcal{K}_1(p^{\nu_p})$ -invariant. And if  $p|R$ , then  $\nu_p \geq \mathfrak{c}_p(\lambda)$ .
- Let  $y \in \mathbb{Z}$ , and define  $t = 1/NR$ ,  $t' = 1/MR \in \mathbf{I}_f$ .

┘

**Def. (33.4.4.3) [Eisenstein Series].** For  $\varphi \in \mathcal{S}(\mathbf{A}_f^2)$ , define the Eisenstein series

$$E_{k,s}(\varphi)(\tau, g) = \sum_{0 \neq \underline{m} \in \mathbb{Q}^2} \frac{(g\varphi)(\underline{m})}{(m_1 + m_2\tau)^k |m_1 + m_2\tau|^{2s}}.$$

And  $E_k(\varphi) = E_{k,0}(\varphi, 1)$ .

┘

**Def. (33.4.4.4).** Denote  $\varphi = \varphi_{t,t'}$ ,  $\varphi' = \varphi'_{t,t'} \in \mathcal{S}(\mathbf{A}_f^2)$  s.t. the local components are given by (33.4.4.1)

$$\begin{cases} \phi_p^0 & , p \nmid NR \\ \phi_p^{1,t_p}, (\phi_p^{1,t'_p})^{\text{tr}} & , p|R \\ \phi_p^{1,t_p}, \phi_p^{2,t'_p} & , p|N \end{cases}$$

┘

**Def. (33.4.4.5) [Inner Products].**

┘

**Thm. (33.4.4.6) [Rankin-Selberg Equation].** ?

$$\begin{aligned} \langle E_{l,s}(\varphi) E_k(\varphi') \otimes \bar{\lambda}, F \rangle &= \frac{i^{k-1} \Gamma(k+l+s-1)}{2^{k+2l+2s-4} (k-1)!} \cdot \frac{N^{l+2s-2} R^{k+l+2s}}{\#GL(2, \mathbb{Z}/(R))} \prod_{p|R} \lambda_p(MN t_p t'_p) \prod_{p|N} (1 - \frac{1}{p^2})^{-1} \\ &\quad \times L_{NR}(\pi \otimes \lambda; l+k+s-1) \left( L_N(\pi; l+s, \frac{y}{M}) + (-1)^k \lambda(-1) L_N(\pi; l+s, -\frac{y}{M}) \right) \end{aligned}$$

┘

*Proof:* Cf. [Scholl, P446] ?

□

**Cor. (33.4.4.7).** If  $k = l = 1$ ,  $s = 0$ , then

$$\begin{aligned} \langle E_1(\varphi) E_1(\varphi') \otimes \lambda, F \rangle &= 2N^{-1} \prod_{p|N} (1 - \frac{1}{p^2})^{-1} \cdot \frac{R^2}{\#GL(2; \mathbb{Z}/(R))} \prod_{p|R} \lambda_p(MN t_p t'_p)^{-1} \\ &\quad \times L_{NR}(\pi \otimes \lambda; 1) \left( L_N(\pi; 1, \frac{y}{M}) - \lambda(-1) L_N(\pi; 1, -\frac{y}{M}) \right) \end{aligned}$$

┘

## 5 Euler System for Elliptic Curves(Kato)

**Notation(33.4.5.1).**

- Fix  $p \in \mathbf{P}$ .
- Fix  $N \in \mathbb{Z}_+$  not divisible by  $p$ , and fix  $d, d' \in \mathbb{Z}_{\geq 2}$  prime to  $6Np$ .
- Denote

$$\mathcal{R}'_p = \mathcal{R}(Npdd'), \quad \mathcal{R}_p = \mathcal{R}'_p \times p^{\mathbb{Z}_+}.$$

┘

### General Procedure to produce Euler system for Modular Curves

**Def.(33.4.5.2) [Setup].** Suppose for each  $r \in \mathcal{R}_p$ , there are points  $z_r, z'_r \in \mathcal{E}^{\text{uni}}(Y(Nr)) \cong (\mathbb{Z}/(Nr))^2$  satisfying:

- If  $r, rs \in \mathcal{R}_p$ , then  $sz_{rs} = z_r, sz'_{rs} = z'_r$ . So we have inverse limits

$$(z_r)_r, (z'_r)_r \in \varprojlim_{r \in \mathcal{R}_p} \mathcal{E}^{\text{uni}}[Nr] \cong \mathbb{Z}_p^2 \times \prod_{\ell \nmid NpDD'} \mathbb{Z}/(\ell).$$

In particular, there exists  $e \in \mathbb{Z}_p^*$  s.t. if  $r = r_0 p^m \in \mathcal{R}_p$ , the Weil pairing

$$e_{Nr}(z_r, z'_r) = \zeta_{p^m}^{er_0^{-1}} \times (\text{prime to } p \text{ roots of } 1). \text{?}$$

- for any  $r \in \mathcal{R}_p$ , the points  $Nz_r, Nz'_r$  generate  $\mathcal{E}^{\text{uni}}[r]$ .
- If  $r = p^m$ , then the orders of  $z_r, z'_r$  are not powers of  $p$ .

┘

**Prop.(33.4.5.3).** By(33.4.1.2), item3 in the setup(33.4.5.2) ensures that  $z_r, z'_r$  doesn't meet the zero section of  $\mathcal{E}^{\text{uni}}/Y(Nr)$  in any characteristic. So  $\vartheta_d(z_r), \vartheta_d(z'_r)$  actually belongs to  $\mathcal{O}^*(Y(Nr))$ .

Thus for  $r \in \mathcal{R}_p$  we can define

$$\tilde{\sigma}_r = \{\vartheta_d(z_r), \vartheta_d(z'_r)\} \in K_2(Y(Nr)),$$

$$\sigma_r = \text{Nm}_{Y(Nr)/Y(N)\otimes\mathbb{Q}(\mu_r)} \tilde{\sigma}_r \in K_2(Y(N) \otimes \mathbb{Q}(\mu_r)).$$

Then they satisfy:

- For  $r \in \mathcal{R}_p$ ,  $\text{Nm}_{\mathbb{Q}(\mu_{rp})/\mathbb{Q}(\mu_r)} \sigma_{rp} = \sigma_r$ .
- If  $\ell \in \mathbf{P}$ ,  $r, r\ell \in \mathcal{R}_p$ , then

$$\text{Nm}_{\mathbb{Q}(\mu_{r\ell})/\mathbb{Q}(\mu_r)} \sigma_{r\ell} = (1 - T_\ell \langle \ell \rangle_* \otimes \text{Frob}_\ell^{-1} + \ell \langle \ell \rangle_* \otimes \text{Frob}_\ell^{-2}) \sigma_r.$$

┘

*Proof:* If  $\ell \nmid Nr$ , then

$$\text{Nm}_{Y(Nr)/Y(N)\mathbb{Q}(\mu_r)} \circ T_{\ell, Y(Nr)} = (T_{\ell, Y(N)} \otimes \text{Frob}_\ell) \circ \text{Nm}_{Y(Nr)/Y(N)\mathbb{Q}(\mu_r)},$$

$$\text{Nm}_{Y(Nr)/Y(N)\mathbb{Q}(\mu_r)} \circ \langle \ell \rangle_{Y(Nr)*} = (\langle \ell \rangle_{Y(Nr)*} \otimes \text{Frob}_\ell^{-2}) \circ \text{Nm}_{Y(Nr)/Y(N)\mathbb{Q}(\mu_r)},$$

because  $T_\ell$  acts on the constant fields by  $\text{Frob}_\ell$ , and  $\langle \ell \rangle_*$  acts by  $\text{Frob}_\ell^{-2}$ . So the theorem follows from  $\square$

**Def. (33.4.5.4).** Denote  $\mathbb{T}_{p,N} = H^1(\overline{Y(N)}, \mathbb{Z}_p(1))$ , then for  $r = r_0 p^m \in \mathcal{R}_p$ , there are homomorphisms

$$\begin{array}{c}
 \varprojlim_{n \geq m} K_2(Y(N)_{\mathbb{Q}(\mu_{r_0 p^n})}) \otimes \mu_{p^n}^{-1} \\
 \downarrow \text{AJ} \\
 \varprojlim_{n \geq m} H^1(\mathbb{Q}(\mu_{r_0 p^n}), H^1(Y(N)_{\overline{\mathbb{Q}}}, \mu_{p^n})) \\
 \downarrow \text{cor} \\
 \varprojlim_{n \geq m} H^1(\mathbb{Q}(\mu_{r_0 p^m}), H^1(Y(N)_{\overline{\mathbb{Q}}}, \mu_{p^n})) \\
 \parallel \\
 H^1(\mathbb{Q}(\mu_r), \mathbb{T}_{p,N})
 \end{array}$$

By (33.4.5.3), the family  $\{\sigma_{r_0 p^n} \otimes [\zeta_{p^n}]^{-1}, n \geq m\}$  is an element of the first group above, so we get elements

$$\xi_r = \xi_r(Y(N); p, d, d', z, z') \in H^1(\mathbb{Q}(\mu_r), \mathbb{T}_{p,N})$$

And they satisfy:

- For each  $r \in \mathcal{R}_p$ ,  $\text{cor}_{\mathbb{Q}(\mu_{rp})/\mathbb{Q}(\mu_r)} \xi_{rp} = \xi_r$ .
- If  $\ell \in \mathbf{P}$ ,  $r, r\ell \in \mathcal{R}_p$ , then

$$\text{cor}_{\mathbb{Q}(\mu_{r\ell})/\mathbb{Q}(\mu_r)} \xi_{r\ell} = (1 - \ell^{-1} T_\ell \langle \ell \rangle_* \text{Frob}_\ell^{-1} + \ell^{-1} \langle \ell \rangle_* \text{Frob}_\ell^{-2}) \xi_r.$$

┘

*Proof:* The operators  $T_\ell$  and  $\langle \ell \rangle$  acts functorially on the cohomology groups, and  $\text{Frob}_\ell$  acts as  $\ell$  on  $\mu_{p^n}$ , so by the twisting, we get the desired formula.  $\square$

**Prop. (33.4.5.5).** For any  $N \in \mathbb{Z}_{\geq 3}$ , in  $\Omega_{Y(N)^{\text{ord}}/\mathbb{Z}} \cong \Omega_{Y(N)^{\text{ord}}/\mathbb{Z}[\mu_N]}^1 \otimes \Omega_{\mathbb{Z}[\mu_N]/\mathbb{Z}}^1$ ,

$$\text{dlog}\{\vartheta_d(z), \vartheta_{d'}(z')\} = \frac{1}{N} \text{KS}_{Y(N)}(d\text{Eis}(z) \cdot d'\text{Eis}(z')) \otimes \text{dlog } e_N(z, z').$$

┘

*Proof:* By  $q$ -expansion principle, it suffices to check this at cusps. If  $z = \zeta_N^{a_1} q^{a_2/N}$ ,  $z' = \zeta_N^{b_1} q^{b_2/N}$ , then  $e_N(z, z') = \zeta_N^{a_2 b_1 - a_1 b_2}$ , and  $\Omega_{\mathbb{Z}[\mu_N]/\mathbb{Z}}^1$  is generated by  $\text{dlog}(\zeta_N)$  and killed by  $N$ , so the assertion follows immediately from the description of  $\text{KS}$  (28.12.1.4) and (33.4.1.6).  $\square$

**Prop. (33.4.5.6).** For any  $r = r_0 p^m \in \mathcal{R}_p$ , define

$$\widetilde{\omega}_r = \frac{1}{Nr} \text{KS}_{Y(Nr)}(d\text{Eis}(z_r) \cdot d'\text{Eis}(z'_r)) \in H^0(X(Nr), \Omega^1(\log \text{cusp})),$$

$$\omega_r = \text{tr}_{X(Nr)/X(N)_{\mathbb{Q}(\mu_r)}} \in H^0(X(N)_{\mathbb{Q}(\mu_r)}, \Omega^1(\log \text{cusp})).$$

Then

$$\exp_p^* \xi_r = \frac{e}{r} \omega_r \text{ (33.4.5.2) (33.4.5.4)}.$$

┘

*Proof.* It follows from (33.4.5.5) that

$$\mathrm{dlog} \tilde{\sigma}_r = \tilde{\omega}_r \otimes \mathrm{dlog} e_{Nr}(z_r, z'_r) \in H^0(X(Nr)^{\mathrm{ord}}, \Omega_{X(Nr)/\mathbb{Z}}(\log \mathrm{cusp})),$$

so after taking  $p$ -adic completion, by (33.4.5.2),

$$\mathrm{dlog} \tilde{\sigma}_r = \frac{e}{r_0} \tilde{\omega}_r \otimes \mathrm{dlog}(\zeta_{p^m}) \in H^0(X(Nr)^{\mathrm{ord}}, \Omega_{X(Nr)_{\mathbb{Z}_p}/\mathbb{Z}}(\log \mathrm{cusp})).$$

Taking traces to  $X(N)_{\mathbb{Q}(\mu_r)}$ , and use the compatibility (11.1.4.12), then

$$\mathrm{dlog} \sigma_r = \frac{e}{r_0} \omega_r \otimes \mathrm{dlog}(\zeta_{p^m}).$$

Then it follows from Kato's explicit reciprocity law (33.4.3.7),

$$\exp_p^* \xi_r = \lim_{n \rightarrow \infty} \frac{e}{r_0 p^n} \mathrm{tr}_{\mathbb{Z}_p[\mu_{p^n}]/\mathbb{Z}_p[\mu_{p^m}]}(\omega_{r_0 p^n}).$$

But by (33.4.2.3),

$$\mathrm{tr}_{Y(Nr_0 p^n)/Y(Nr_0 p^m)} \tilde{\omega}_{r_0 p^n} = p^{n-m} \tilde{\omega}_{r_0 p^m},$$

so

$$\mathrm{tr}_{\mathbb{Z}_p[\mu_{p^n}]/\mathbb{Z}_p[\mu_{p^m}]}(\omega_{r_0 p^n}) = p^{n-m} \omega_r,$$

which implies the desired equation.  $\square$

### Elliptic Curves

**Def. (33.4.5.7) [Setup].**

- Suppose  $E \in \mathrm{EllCur}/\mathbb{Q}$  has with conductor  $N_E$  and a Weil parametrization  $\varphi_E : X_0(N_E) \rightarrow E$ .
- Let  $\omega_E$  be a differential on  $E/\mathbb{Q}$  s.t.  $\varphi_E^* \omega_E$  is a newform on  $X_0(N_E)$ . ?
- Let  $\Omega_E^+, \Omega_E^-$  be the real and imaginary periods of  $\omega_E$ . ?
- Let  $N \in \mathbb{Z}_+, N_E | N$ , and  $p \in \mathbf{P}$  be prime to  $2N$ .

┘

**Prop. (33.4.5.8).** Situation as in (33.4.5.7), there are maps

$$f : H^1(X(N)_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(1)) \rightarrow H^1(Y(N)_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(1)) = \mathbb{T}_{p,N}$$

and

$$H^1(X(N)_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(1)) \xrightarrow{\varphi_{E,N*}} H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(1)) = T_p(E).$$

By the proof of the Manin-Drinfeld theorem ?, there is an idempotent  $\Pi_N^{\mathrm{cusp}}$  in the Hecke algebra with rational coefficients s.t. for any  $p$ , it induces a left inverse to  $f$  as above. Thus for some  $h_E \in \mathbb{Z}_+$  independent of  $p$ , we can define a map

$$h_E \varphi_{E,N*} \circ \Pi_{N,*}^{\mathrm{cusp}} : \mathbb{T}_{p,N} \rightarrow T_p(E).$$

┘

**Prop. (33.4.5.9)**[Associated Euler System for Elliptic Curves]. For  $r \in \mathcal{R}_p$ (33.4.5.2), define

$$\xi_r(E) = (h_E \varphi_{E,N*} \circ \Pi_{N*}^{\text{cusp}}) \xi_r(Y(N)) \in H^1(\mathbb{Q}(\mu_r), T_p(E)) \text{ (33.4.5.4),}$$

then  $\{\xi_r(E)\}_r$  is an Euler system for  $(T_p(E), \mathcal{K}, Ndd'r)$ (31.2.3.2).  $\lrcorner$

*Proof:* This follows from(33.4.5.4) and the fact  $\langle \ell \rangle = 1, T_\ell = a_\ell$  on  $T_p(E)$  for  $\ell \in \mathbf{P} \cap \mathcal{R}_p$  by definition and the fact  $f_E \in S_2(\Gamma_0(N))$ .?  $\square$

**Prop. (33.4.5.10).** On differentials,  $\Pi_N^{\text{cusp}}$  is identity on cusp forms and annihilates Eisenstein forms?. Denote

$$\omega_r^{\text{cusp}} = \Pi_N^{\text{cusp}}(\omega_r) \in H^0(X(N)_{\mathbb{Q}(\mu_r)}, \Omega^1).$$

Then it follows from(33.4.5.6) and the functoriality of  $\exp_p^*$  that

$$\exp_p^* \xi_r(E) = \frac{eh_E}{r} \varphi_{E,N*}(\omega_r^{\text{cusp}}) \text{ (33.4.5.8).}$$

$\lrcorner$

### The Calculation

**Notation (33.4.5.11).**

- Use the setup(33.4.4.2), where  $M, N$  is defined in(33.4.5.12),  $R = r = r_0 p^m$  is any number in  $\mathcal{R}_p$ , and  $\lambda : \mathbf{I}_f / \mathbb{Q}_+^\times \rightarrow \mathbb{C}^\times$  is a character with conductor dividing  $r$ .
- In particular, by(33.4.4.2)(33.4.4.4), there are elements  $t, t' \in \mathbf{I}_f$ , and  $\varphi, \varphi' \in \mathcal{S}(\mathbf{A}_f^2)$ .
- Let  $\varepsilon = \lambda(-1) \in \{\pm 1\}$ .
- Fix  $d, d' \in \mathbb{Z}_{\geq 2}$  with  $(dd', 6pN_E) = 1, d \equiv d' \equiv 1 \pmod{M}$ .
- Denote

$$\delta \in \mathbf{I}_f : \delta_p = \begin{cases} d & , p \mid Nr \\ 1 & , \text{otherwise} \end{cases}, \quad d\varphi = d^2\varphi - d[\delta]\varphi,$$

and similarly define  $\delta'$  and  $d'\varphi'$ .  $\lrcorner$

**Prop. (33.4.5.12).** For any  $\alpha \in \mathbb{Q}$ ,

$$L_{MN_E}(E, 1; \alpha) - \varepsilon L_{MN_E}(E, 1; -\alpha)$$

is a rational multiple of  $\Omega_E^{-\varepsilon}$ , and one can find  $\alpha$  with denominator prime to any integer for which the expression is non-zero, by [Shimura, On the periods of modular forms]?.

Then we can find  $y/M$ .? Cf.[Scholl, P454].

And also define

$$N = \prod_{p \mid MN_E} p^{\nu_p}, \quad \nu_p = \max(2, v_p(N_E), v_p(M) + 1).$$

$\lrcorner$



**Prop. (33.4.5.13) [The function  $\varphi$ ].** Choose  $z_r \in \mathcal{E}^{\text{uni}}(Y(Nr))$  to be the point which in complex coordinates is

$$\frac{1}{Nr} \in \left(\frac{1}{Nr}\mathbb{Z} + \frac{\tau}{Nr}\mathbb{Z}\right)/(\mathbb{Z} + \tau\mathbb{Z}) \cong (\mathbb{Z}/(Nr))^2.$$

Then for  $\ell \in \mathbf{P}$ ,  $r, r\ell \in \mathcal{R}_p$ ,  $\ell z_{r\ell} = z_r$ , and

$${}_d\text{Eis}(z_r) = E_1({}_d\varphi)du.$$

┘

*Proof:* ? Calculate using (33.4.4.3) and (33.4.1.5) and the definition of  $\varphi$  (33.4.4.4). □

**Prop. (33.4.5.14) [The function  $\varphi'$ ].** There exists f.m. constants  $b_j \in N^{-1}\mathbb{Z}$  independent of  $r$  and points  $z'_{r,j} \in \mathcal{E}^{\text{uni}}(Y(Nr))$ , s.t.

$$\sum_j b_j \cdot {}_{d'}\text{Eis}(z'_{r,j}) = E_1({}_{d'}\varphi')du.$$

Moreover, for  $\ell \in \mathbf{P}$ ,  $r, r\ell \in \mathcal{R}_p$ ,  $\ell z_{r\ell,j} = z_{r,j}$ , and  $z'_{r,j} - z'_{r,i}$  are  $N$ -torsion. And in complex coordinates,

$$Nz'_{r,j} = (-Nt'(\bmod \widehat{\mathbb{Z}}))\tau \in \left(\frac{1}{r}\mathbb{Z} + \frac{1}{r}\tau\right)/(\mathbb{Z} + \tau\mathbb{Z}) \cong (\mathbb{Z}/(r))^2.$$

So

$$e_{Nr}(z_r, z'_{r,j}) = \zeta_{p^m}^{-(Mr_0)^{-1}} \times (\text{prime to } p \text{ roots of } 1),$$

so  $e_{z,z'_j} = -M^{-1}$  (33.4.5.2). ┘

*Proof:* Cf. [Scholl, P456] ?. □

**Thm. (33.4.5.15) [Final Construction and Calculation].** Let  $\xi_{r,j}(E)$  be the Euler system for  $T_p(E)$  associated to  $(z_r, z_{r,j})$  (33.4.5.9) in the setup (33.4.5.7), then define a new Euler system

$$c_r = \sum_j b_j \xi_{r,j}(E) \in H^1(\mathbb{Q}(\mu_r), T_p(E)).$$

Then this Euler system satisfies the requirement of (33.4.5.16). ┘

*Proof:* Let  $\tilde{\omega}_{r,j}$  be the differential associated to  $(z_r, z_{r,j})$  (33.4.5.6), then because the Kodaira-Spencer map takes  $(d\log t)^2$  to  $d\log q$ , so it takes  $du^2$  to  $(2\pi i)^{-1}d\tau$ . Thus

$$\sum_j b_j \tilde{\omega}_{r,j} = \frac{(2\pi i)^{-1}}{Nr} E_1({}_d\varphi) E_1({}_{d'}\varphi') d\tau.$$

Then for any character  $\lambda : \text{Gal}(\mathbb{Q}(\mu_r)/\mathbb{Q}) \cong (\mathbb{Z}/(r))^* \rightarrow \mathbb{C}^\times$ ,

$$\begin{aligned} \sum_{\gamma \in \text{Gal}(\mathbb{Q}(\mu_r)/\mathbb{Q})} \lambda(\gamma) \exp_p^* c_r^\gamma &= \sum_j b_j \sum_{\gamma \in \text{Gal}(\mathbb{Q}(\mu_r)/\mathbb{Q})} \lambda(\gamma) \exp_p^* \xi_{r,j}(E)^\gamma \\ (33.4.5.10) &= \sum_j b_j \sum_{x \in (\mathbb{Z}/(r))^*} \lambda(x) \frac{-M^{-1}h_E}{r} \iota_x \varphi_{E,N*}(\omega_{r,j}^{\text{cusp}}) \end{aligned}$$

where  $\iota_x : \mathbb{Q}(\mu_r) \hookrightarrow \mathbb{C}$  is the injection that maps  $\zeta_r$  to  $e^{2\pi i x/r}$ . Then it equals

$$\begin{aligned}
& \frac{-M^{-1}h_E}{r} \sum_j b_j \frac{\int_{E^{\text{an}}} \sum_{x \in (\mathbb{Z}/(r))^*} \lambda(x) \iota_x \varphi_{E,N*}(\omega_{r,j}^{\text{cusp}}) \wedge \bar{\omega}_E}{\int_{E^{\text{an}}} \omega_E \wedge \bar{\omega}_E} \omega_E \\
&= \frac{-h_E}{Mr \Omega_E^+ \Omega_E^-} \left( \sum_j b_j \int_{Y(N)^{\text{an}} \times (\mathbb{Z}/(r))^*} \lambda(x) \iota_x \omega_{r,j}^{\text{cusp}} \wedge \varphi_{E,N}^* \bar{\omega}_E \right) \omega_E \\
&= \frac{-h_E}{Mr \Omega_E^+ \Omega_E^-} \left( \sum_j b_j \int_{Y(N)^{\text{an}} \times (\mathbb{Z}/(r))^*} \lambda(x) \iota_x \omega_{r,j} \wedge \varphi_{E,N}^* \bar{\omega}_E \right) \omega_E \\
&= \frac{-h_E}{Mr \Omega_E^+ \Omega_E^-} \left( \sum_j b_j \int_{Y(Nr)} (\lambda \circ e_{Nr}) \tilde{\omega}_r \wedge \varphi_{E,N}^* \bar{\omega}_E \right) \omega_E \\
&\stackrel{?}{=} \frac{-h_E \# \text{GL}(2, \mathbb{Z}/(Nr))}{MNr^2 \Omega_E^+ \Omega_E^-} \langle E_1(d\varphi) E_1(d'\varphi') \otimes \lambda, F \rangle \omega_E
\end{aligned}$$

where  $e_{Nr} : Y_{Nr}(\mathbb{C}) \rightarrow \mu_N(\mathbb{C})$  is the Weil pairing.

Using (33.4.4.7) and the facts  $\mathfrak{c}(\lambda)|r$  and  $d \equiv d' \equiv 1 \pmod{M}$ , it equals?

$$\frac{C_{E,M}}{\Omega_E^+ \Omega_E^-} DD'(D - \chi^{-1}(D))(D' - \chi^{-1}(D')) L_{rMN_E}(E, \chi; 1) \left( L_{MN_E}(\pi; 1, \frac{y}{M}) - \lambda(-1) L_{MN_E}(\pi; 1, -\frac{y}{M}) \right) \omega_E$$

Where  $C_{E,M}$  only depends on  $E$  and  $M$ . (Notice  $N$  only depends on  $E$  and  $M$  (33.4.5.12)).

Then by the choice of  $y/M$  (33.4.5.12), depending on  $\lambda(-1) = \pm 1$ , it equals

$$C_{E,M}^{\pm} DD'(D - \chi^{-1}(D))(D' - \chi^{-1}(D')) \frac{L_{rMN_E}(E, \chi; 1)}{\Omega_E^{\pm}} \omega_E,$$

where  $C_{E,M}^{\pm}$  only depends on  $E$  and  $M$ . □

### Conclusions

**Thm. (33.4.5.16) [Kato].** Situation as in (33.4.5.7), then there exists  $M \in \mathbb{Z}_+$  prime to  $p$  s.t. for any  $d, d' \in \mathbb{Z}_{\geq 2}$  with  $(dd', 6pN_E) = 1$ ,  $M|d-1$ ,  $M|d'-1$ , there is an Euler system

$$\bar{\mathfrak{c}} = \{ \bar{\mathfrak{c}}_r(E, p, d, d') \in H^1(\mathbb{Q}(\mu_r), T_p(E)), \quad r = r_0 p^m, m \geq 1, r_0 \in \mathcal{R}(pMN_E) \}.$$

satisfying: For any  $r \in \mathcal{R}_p$  and  $\chi : \text{Gal}(\mathbb{Q}(\mu_r)/\mathbb{Q}) \cong (\mathbb{Z}/(r))^* \rightarrow \mathbb{C}^*$ , with  $\chi(-1) = \pm 1$ ,

$$\sum_{\gamma \in \text{Gal}(\mathbb{Q}(\mu_r)/\mathbb{Q})} \chi(\gamma) \exp_p^* \bar{\mathfrak{c}}_r^\gamma = C_E^{\pm} DD'(D - \chi^{-1}(D))(D' - \chi^{-1}(D')) L_{rMN_E}(E, \chi; 1) / \Omega_E^{\pm}.$$

┘

*Proof:* This follows from (33.4.5.15). □

## 33.5 Iwasawa Theory for Elliptic Curves

References are [Coates] and [Greenberg] and [LECTURES ON THE IWASAWA THEORY OF ELLIPTIC CURVES, Skinner].

**Notation (33.5.0.1).**

- Let  $p \in \text{Prime}$ .

┘

### 1 Mazur's Control Theorem

**Thm. (33.5.1.1) [Mazur's Control Theorem].** For  $F \in \mathbf{NField}$ ,  $E \in \mathbf{EllCur}/F$ ,  $p \in \text{Prime}$ , and  $E$  has good ordinary reduction over all primes lying over  $p$ . Assume  $F_\infty = \cup_n F_n$  is a  $\mathbb{Z}_p$ -extension of  $F$ , then the natural map

$$\text{Sel}_p(E/F_n) \rightarrow \text{Sel}_p(E/F_\infty)^{\text{Gal}(F_\infty/F_n)}$$

has finite kernels and cokernels of bounded orders as  $n \rightarrow \infty$ .

┘

*Proof:* Cf. [Mazur's Control Theorem for Elliptic Curves].

□

### 2 Anticyclotomic Iwasawa Conjecture for Modular Forms

**Notation (33.5.2.1).**

- Let  $N^- \in \mathbb{Z}_+$  be square free with  $d_0(N^-)$  odd.
- Let  $N^+ \in \mathbb{Z}_+$  s.t.  $(N^+, N^-) = 1$ .
- Let  $p \in \text{Prime}$  s.t.  $(p, N^+ N^-) = 1$ , and denote  $N = p N^+ N^-$ .
- Let  $B \in \mathbf{AzQ}$  be the unique definite quaternion algebra ramified exactly at primes dividing  $N^-$ .
- Let  $R$  be an Eichler  $\mathbb{Z}[\frac{1}{p}]$ -order of level  $N^+$  in  $B$ ?
- Let  $\mathcal{K} \in \mathbf{NField}$  be an imaginary quadratic field.
- Let  $\mathcal{K}_\infty/\mathcal{K}$  be the anticyclotomic extension with  $\Gamma_\infty = \text{Gal}(\mathcal{K}_\infty/\mathcal{K})$ .

┘

**Def. (33.5.2.2) [Bruhat-Tits Building].** Let  $\mathcal{T}$  be the Bruhat-Tits tree of  $B_p^\times/\mathbb{Q}_p^\times$ , whose vertices are maximal  $\mathbb{Z}_p$ -orders in  $B$ , and two vertices are adjacent if their intersection is an Eichler order of level  $p$ .

┘

### 3 Kolyvagin's Work

References are [Gro89], [Finiteness of  $E(\mathbb{Q})$  and  $\text{III}(E/\mathbb{Q})$  for a subclass of Weil curves, Kolyvagin], [Finiteness of the Shafarevich-Tate group and the group of rational points for some modular abelian varieties, Kolyvagin-Yu] and [Euler Systems, Kolyvagin].

**Def. (33.5.3.1) [Kolyvagin Primes].** In the Heegner situation (31.6.1.1), for  $p \in \text{Prime}$ , a **Kolyvagin prime**  $\ell$  for  $E$  is a prime  $\ell \in \text{Prime} \setminus S(Np\text{disc}(\mathcal{K}))$  s.t.  $\ell$  is inert in  $\mathcal{K}$  and  $p|\ell+1, p|a_{\ell,E}$ . In particular,  $\ell$  is unramified in  $\mathcal{K}(E[p])/\mathbb{Q}$  by (21.1.2.8) and  $\text{Frob}_\ell = c$  as conjugacy classes in  $\text{Gal}(\mathcal{K}(E[p])/\mathbb{Q})$ . By Chebotarev density theorem, the set of Kolyvagin primes has positive Dirichlet density.

For a Kolyvagin prime  $\ell$ ,  $M(\ell) = \min\{v_p(\ell+1), v_p(a_{\ell,E})\}$  is called the **Kolyvagin index** of  $\ell$ .

A product of distinct Kolyvagin primes is called a **Kolyvagin number** for  $E$ . The set of Kolyvagin numbers is denoted by  $\Lambda_E$ . For a Kolyvagin number  $n$ , define the **Kolyvagin index** of  $n$  as  $M(n) = \min_{\ell|n} \{M(\ell)\}$ .  $\lrcorner$

**Prop. (33.5.3.2).** In the Heegner situation (31.6.1.1), let  $\ell$  be a Kolyvagin prime for  $E$ , then  $\lrcorner$

*Proof:*  $\square$

**Def. (33.5.3.3) [Kolyvagin Derivative].** In the situation (33.5.3.1), for  $\ell \in \Lambda_E \cap \text{Prime}$ ,

$$\text{Gal}(\mathcal{K}_\ell/\mathcal{K}_1) \cong \text{Pic}(\mathcal{O}_{K,\ell})/\text{Pic}(\mathcal{O}_K) \cong (\mathcal{O}_K/(\ell))^\times/(\mathbb{Z}/(\ell))^\times \cong \mathbb{Z}/(\ell+1) = \langle \sigma \rangle.$$

Define the **Kolyvagin derivative**  $D_\ell = \sum_{i=1}^\ell i\sigma^i \in \mathbb{Z}[\text{Gal}(\mathcal{K}_\ell/\mathcal{K}_1)]$ , which satisfies  $(\sigma-1)D_\ell = \ell+1 - \text{tr}_\ell$ .  $\lrcorner$

**Prop. (33.5.3.4).** In the situation (33.5.3.1), for  $\ell \in \Lambda_E \cap \text{Prime}$  and Heegner point  $y_\ell \in E(F_\ell)$ ,  $D_\ell y_\ell \in E(F_\ell)/p^M E(F_\ell) \subset H^1(F_\ell, E[p^M])$  is invariant under the action of  $\text{Gal}(\mathcal{K}_\ell/\mathcal{K}_1)$ .

More generally, for  $n \in \Lambda_E$ , denote

$$D_n = \prod_{\ell \in S(n)} D_\ell \in \text{Gal}(\mathcal{K}_n/\mathcal{K}_1) \cong \prod_{\ell \in S(n)} D_\ell.$$

Then for  $M \in [M(n)]_+$ ,  $D_n y_n \in H^1(\mathcal{K}_n, E[p^M])$  is invariant under the action of  $\text{Gal}(\mathcal{K}_\ell/\mathcal{K}_1)$ , thus descends to an element in  $H^1(F_1, E[p^M]) \cong H^1(F_n, E[p^M])^{\text{Gal}(F_n, F_1)}$  by Hochschild-Serre spectral sequence (9.8.1.14) and the fact  $E[p^M](K_n) = 0$ ?  $\lrcorner$

*Proof:* By (33.5.3.3), it suffices to show that  $(\sigma-1)D_\ell y_\ell = (\ell+1 - \text{tr}_\ell)y_\ell \in p^M E(F_1)$ : This follows from the definition (33.5.3.1) and the fact  $\text{tr}_\ell y_\ell$  is just the Hecke operator action  $T_\ell$  on  $y_1$ , which maps to  $a_\ell y_1 \in E(F_1)$ ?  $\square$

**Def. (33.5.3.5) [Kolyvagin Systems].** In the situation (33.5.3.1), for  $n \in \Lambda_E$  and  $M \in [M(n)]_+$ , define the **Kolyvagin cohomology classes**

$$c_M(n) = \sum_{s \in \text{Gal}(F_1/F)} s D_n y_n \in H^1(F, E[p^M]), \quad M \leq M(n).$$

The collection of cohomology classes (in different groups)

$$\kappa = \{c_M(n) \in H^1(F, E[p^M]) | n \in \Lambda_E, M \in \mathbb{Z}_+, M \leq M(n)\}$$

is called a **Kolyvagin system** of  $E$ .  $\lrcorner$

**Def. (33.5.3.6) [ $p$ -Divisibility].** In the situation (33.5.3.1), for  $r \in \mathbb{N}$ , denote  $\Lambda_{E,r}$  to be the subset of  $\Lambda_E$  consisting of numbers with exactly  $r$  prime factors (33.5.3.1), and define  $\mathcal{M}_r \in [0, \infty]$  to be the supremum of integers  $x \in \mathbb{Z}_+$  s.t.  $p^x | c_M(n)$  for any  $n \in \Lambda_{E,r}$ ,  $M \leq M(n)$ .  $\lrcorner$

**Prop. (33.5.3.7) [Vanishing Orders of Kolyvagin Systems].** In the situation (33.5.3.1),  $\mathcal{M}_0 \geq \mathcal{M}_1 \geq \dots \geq 0$ .

In particular, we can define  $\mathcal{M}_\infty = \lim_{r \rightarrow \infty} \mathcal{M}_r$ . Clearly,  $\mathcal{M}_\infty = \infty$  iff  $\kappa = 0$ . The **vanishing order of  $\kappa$**  is the minimal  $r$  s.t.  $\mathcal{M}_r \neq \infty$ , denoted by  $\text{ord } \kappa$ .  $\lrcorner$

*Proof:*

□

**Prop. (33.5.3.8).** Situation as in (33.5.3.6),  $\mathcal{M}_0$  is just the  $p$ -divisibility of the Heegner point  $y_K$ . In particular,  $\mathcal{M}_0 < \infty$  iff  $y_K$  is non-torsion? .

┘

**Thm. (33.5.3.9) [Kolyvagin].** Let  $\text{ord } \kappa < \infty$ , then

- $\text{Sel}^{p^\infty}(E/F)$  is contained in the subgroup of  $H^1(F, E[p^\infty])$  generated by  $\kappa$ .
- $\text{rank}_p^{w_E(-1)^{\text{ord } \kappa+1}}(E/F) = \text{ord } \kappa + 1$ ,  $\text{rank}_p^{w_E(-1)^{\text{ord } \kappa}}(E/F) = \text{ord } \kappa - d$  and  $d \in 2\mathbb{N}$ .
- Let

$$\widetilde{\text{III}}(E/F)[p^\infty]^{\text{ord } \kappa+1} = \left(\bigoplus_{i \geq 1} \mathbb{Z}/(p^{a_i})\right)^2, \quad \widetilde{\text{III}}(E/F)[p^\infty]^{\text{ord } \kappa} = \left(\bigoplus_{i \geq 1} \mathbb{Z}/(p^{b_i})\right)^2 \quad (21.1.8.7),$$

where  $(a_i), (b_i)$  is non-increasing, then

$$a_i = \mathcal{M}_{\text{ord } \kappa+2i-1} - \mathcal{M}_{\text{ord } \kappa+2i}, \quad b_{d+i} = \mathcal{M}_{\text{ord } \kappa+2i-2} - \mathcal{M}_{\text{ord } \kappa+2i-1}.$$

In particular,  $\#\widetilde{\text{III}}(E/F)[p^\infty]^{\text{ord } \kappa+1} \geq p^{\mathcal{M}_{\text{ord } \kappa} - \mathcal{M}_\infty}$ , with equality iff  $\text{ord } \kappa = 0$ .

┘

*Proof:* ?

We only prove the following: If  $p \in \text{Prime}_{\geq 3}$  s.t.  $\bar{\rho}_{E,p}$  is surjective and  $p$  doesn't divide  $y_K$  in  $E(K)$ , then  $\text{rank } E(K) = 1$  and  $\text{III}(E/K)[p] = 0$ . For this, notice that  $E(K)[p] = 0$  otherwise  $\#\bar{\rho}_{E,p}(\text{Gal}_K) \leq p(p-1)$ . Thus by the exact sequence (21.1.8.4), it suffices to show that  $\text{Sel}(E/K)^p$  is cyclic and generated by  $y_K$ . □

**Conj. (33.5.3.10) [Gross-Zagier [G-Z86]].** If  $y_K \in E(K)$  is non-torsion, then  $\text{rank}(E/\mathbb{Q}) = 1$  and

$$\#\text{III}(E/K) = \left( \frac{[E(K) : \mathbb{Z}.y_K]}{c_E \prod_{p \in S(N_E)} m_p} \right)^2 \quad (30.4.4.1), \quad \text{where } m_p = [E(\mathbb{Q}_p) : E^0(\mathbb{Q}_p)].$$

In particular, by Cassels-Tate pairing (21.1.8.12),  $\text{III}(E/K)$  is annihilated by  $[E(K) : \mathbb{Z}.y_K]$ .

┘

**Conj. Cor. (33.5.3.11) [Kolyvagin].** If  $\text{ord } \kappa = 0$ , i.e.  $y_K \in E(K)$  is non-torsion (33.5.3.8), then  $d = 0$ , and  $\text{rank}(E/\mathbb{Q}) = 1$ ,  $\#\text{III}(E/K) < \infty$ , by (33.5.3.9) and (33.5.3.8).

┘

### Kolyvagin's Conjecture

**Conj. (33.5.3.12) [Kolyvagin].** For  $p \in \text{Prime}_{\geq 3}$  s.t.  $\bar{\rho}_{E,p}$  is surjective, we have  $\kappa \neq \{0\}$ , i.e.  $\mathcal{M}_\infty < \infty$  (33.5.3.7).

┘

*Proof:*

□

**Thm& Conj. Cor. (33.5.3.13) [Zhang].** Let  $E \in \text{EllCur}/\mathbb{Q}$  with conductor  $N$  and  $p \geq 5$  is good ordinary and  $\bar{\rho}_{E,p}$  is surjective and ramifies at every prime  $\ell \in \text{Prime}$  s.t.  $v_\ell(N) = 1$ , then  $\mathcal{M}_\infty = 0$ . In particular, Kolyvagin's conjecture (33.5.3.12) is true.

┘

**Remark (33.5.3.14).** There are results that generalize this.

┘

**Cor. (33.5.3.15).** Under the hypothesis of (33.5.3.13), if  $\text{rank}_p(E/\mathbb{Q}) = 1$ , then  $r_{\text{an}}(E/\mathbb{Q}) = r(E/\mathbb{Q}) = 1$ , and  $\#\text{III}(E/\mathbb{Q}) < \infty$ .

┘

*Proof:* Cf. [Wei Zhang].

□

#### 4 $p$ -adic BSD Conjecture

**Conj. (33.5.4.1) [ $p$ -adic BSD Conjecture, Mazur-Tate-Teitelbaum[MTT86]].** If  $E \in \text{EllCur}/\mathbb{Q}$  has ordinary or non-split multiplicative reduction at  $p$ , then the  $p$ -adic  $L$ -function

$$r = \text{ord}_{s=1} L_p(E; s) \geq \text{rank } E(\mathbb{Q}),$$

and the leading term of the expansion of  $L_p(E; s)$  at  $s = 1$  is an expression involving  $|\text{III}(E)|_p$  and a  $p$ -adic regulator  $\text{Reg}_p(E)$ . When  $E$  has split multiplicative reduction, it is necessary to replace  $r$  by  $r + 1$ .  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj. Cor. (33.5.4.2) [Kato].** If  $E \in \text{EllCur}/\mathbb{Q}$  has ordinary or non-split multiplicative reduction at  $p$ , then  $p$ -adic  $L$ -function

$$r = \text{ord}_{s=1} L_p(E; s) \geq \text{rank } E(\mathbb{Q}).$$

And if the equality holds, then  $\#\text{III}(E)[p^\infty] < \infty$  and  $\mathcal{R}\text{ep}_p(E) \neq 0$ .  $\lrcorner$

*Proof:*  $\color{red}{?}$   $\square$

**Thm. (33.5.4.3) [Mazur-Tate-Teitelbaum Conjecture, Greenberg-Stevens].** If  $E \in \text{EllCur}/\mathbb{Q}$  has split multiplicative reduction at  $p$  (so  $E_{\mathbb{Q}_p}$  is a Tate curve and  $L_p(E; 1) = 0$ ), we have

$$L_p^\lambda(E; 1) = \frac{\log_p(q)}{\text{ord}_p(q)} \frac{L(E; 1)}{\Omega}$$

where  $q$  is a period of  $E_{\mathbb{Q}_p}$  and  $\Omega$  is the period of  $E_{\mathbb{R}}$ .  $\lrcorner$

*Proof:*  $\square$

#### 5 $\text{GL}(2)$ case

References are [Iwasawa Main Conjectures for  $\text{GL}(2)$ , Skinner-Urban].

**Notation (33.5.5.1).**

- Let  $p \in \text{Prime}$ .
- Let  $\mathcal{K}$  be an imaginary quadratic field s.t.  $p$  splits in  $\mathcal{K}$ .
- Let  $\mathbb{Q}_\infty$  be the unique  $\mathbb{Z}_p$ -extension of  $\mathbb{Q}$ .
- Let  $\mathcal{K}_\infty^+ = \mathcal{K}\mathbb{Q}_\infty$  be the cyclotomic  $\mathbb{Z}_p$ -extension of  $\mathcal{K}$ , and  $\mathcal{K}_\infty^-$  be the anti-cyclotomic extension of  $\mathcal{K}$ . So  $\mathcal{K}_\infty^+ \cap \mathcal{K}_\infty^- = \mathcal{K}$ .
- Denote  $\mathcal{K}_\infty = \mathcal{K}_\infty^+ \cdot \mathcal{K}_\infty^-$ .

$\lrcorner$

## 33.6 Bloch-Kato Conjecture in the GGP Setting

### 1 $\mathrm{SO}(2) \times \mathrm{SO}(3)$

References are [Yifeng Liu].

**Notation (33.6.1.1).**

- Let  $p \in \mathbf{Prime}$ .
- Let  $F \leq \overline{\mathbb{Q}}$  be a real quadratic field with  $\mathrm{Gal}(F/\mathbb{Q}) = \{\mathrm{id}, \theta\}$ .
- Let  $E \in \mathbf{EllCur}/\mathbb{Q}$ , and let  $\sigma \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/\mathbb{Q})$  be associated to  $E$ .
- Let  $A \in \mathbf{EllCur}/F$ , and let  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/F)$  be associated to  $A$ .
- $V_p(A) \otimes V_p(A^\theta)$  has a natural action of  $\Gamma_{\mathbb{Q}}$ , and this  $\mathrm{Gal}_{\mathbb{Q}}$ -module is denoted by  $\mathrm{As} V_p(\mathbb{Q}_p)$ . ┘

**Notation (33.6.1.2) [Assumption].** Assume that

- Neither  $E$  or  $A$  has CM.
- $(N, M\mathrm{disc}(F)) = 1$ .
- If we write  $N = N^+ N^-$  where  $N^+$  is the product of prime factors that split in  $F$ , then  $N^-$  is square-free. ┘

**Thm. (33.6.1.3).** If  $E_1, E_2 \in \mathbf{EllCur}/\mathbb{Q}$  with conductor  $N_1, N_2$  resp. s.t.

- $(N_1, N_2) = 1$ .
- $E_1$  has multiplicative reduction at some  $p \in \mathbf{Prime}$
- $E_2$  has no CM.
- $L(E_1 \times \mathrm{Sym}^2 E_2; 2) \neq 0$ .

Then for a.e.  $p \in \mathbf{Prime}$ , we have

$$\dim H_f^1(\mathbb{Q}, V_p(E_1) \otimes \mathrm{Sym}^2 V_p(E_2)(-1)) = 0.$$
┘

### Cohomologies

- $\sigma \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/\mathbb{Q})$  of weight 2 and  $N(\sigma) = N = N^+ N^-$ .
- $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(2)/F)$  of weight  $(2, 2)$  and  $N(\pi) = \mathfrak{M} \in \mathrm{Ideal}^\times(\mathcal{O}_F)$ .

**Prop. (33.6.1.4).**

$$H_\pi^2(X_{\mathfrak{M}, \overline{\mathbb{Q}}}, \mathbb{Q}_p) \cong {}^\# V_\pi.$$
┘

*Proof:* Cf. [Yifeng Liu]P732. □

### Galois Cohomology Arguments

**Notation (33.6.1.5).**

- Let  $T \in \text{Mod}_{\mathbb{Z}_p}$ .

┘

**Def. (33.6.1.6) [ $p$ -divisibility].** For  $t \in T$ , define  $\text{ord}_p(t; T) = \sup_{n \in \mathbb{N}} (t \in p^n T)$ .

┘

**Def. (33.6.1.7) [Reducibility Depth].** If  $G \in \text{Grp}$  and  $T \in \cup_{n \in \mathbb{Z}_+} \text{Rep}(G, \mathbb{Z}/(p^n))$ , then the **reduction depth** of  $T$  is defined to be the smallest integer  $n_T \in \mathbb{N}$  s.t.

- If  $T' \subset T$  is a  $G$ -stable submodule that is not contained in  $pT$ , then  $p^{n_T} T \subset T'$ .
- For any  $m \in \mathbb{Z}_+$ , the group  $\text{Hom}_G(T/(p^m), T/(p^m))/\mathbb{Z}_p.\text{id}$  is annihilated by  $p^{n_T}$ .

┘

**Lemma (33.6.1.8).** Let  $V \in \text{Rep}^{\text{fd}}(\text{Gal}_{\mathbb{Q}}, \mathbb{Q}_p)$  be absolutely irreducible and a.e. unramified with a stable lattice  $\Gamma \subset V$ , then there exists  $n(T) \in \mathbb{Z}_+$  s.t.  $T/(p^n)$  has reducibility depth  $\leq n(T)$  (33.6.1.7) for any  $n \in \mathbb{Z}_+$ .

┘

*Proof:* Cf. [Yifeng Liu]P764.

□



### 33.7 $p$ -adic Gross-Zagier Formulae

References are [Heegner points and a  $p$ -adic Gross-Zagier formula, notes],

#### 1 Anticyclotomic $L$ -Functions for Elliptic Curves

References are [Heegner points,  $p$ -adic  $L$ -functions, and the Cerednik-Drinfeld uniformization].

**Thm. (33.7.1.1) [Bertolini-Darmon].** Let  $w = \begin{cases} 1 & , p \text{ split multiplicative for } E \\ 1 & , p \text{ non-split multiplicative for } E \end{cases}$ , then

$$\Phi_{\text{Tate}}(\mathcal{L}_p^\lambda(E/\mathcal{K})) = \pm(\alpha_{\mathcal{K}} - w\overline{\alpha_{\mathcal{K}}}).$$

┘

*Proof:* ?

□

**Cor. (33.7.1.2).**  $L_p^\lambda(E/\mathcal{K}) \neq 0$  iff  $\alpha_{\mathcal{K}} - w\alpha_{\mathcal{K}}$  has finite order in  $E(\mathcal{K})$ .

┘

**Cor. (33.7.1.3).** If  $L_p^\lambda(E/\mathcal{K}) \neq 0$ , then  $\alpha_{\mathcal{K}}$  has finite order in  $E(\mathcal{K})$ , and (by Kolyvagin's work)  $\text{rank}(E/\mathcal{K}) = 1$ ,  $\#\text{III}(E/\mathcal{K}) < \infty$ .

┘

#### 2 Triple Product $L$ -Functions

References are [Darmon-Rotger, Diagonal cycles and Euler systems I: A  $p$ -adic Gross-Zagier formula].

#### 3 $p$ -adic Heights

### 33.8 $p$ -adic Beilinson-Bloch-Kato Conjectures

#### 1 Basics

##### Stabilizations

There are two actions on the space of  $p$ -adic modular forms,

$$U(f)(q) = \sum_{n=1}^{\infty} a_{np} q^n, \quad V(f)(q) = \sum_{n=1}^{\infty} a_n q^{np}.$$

And

$$UV = \text{id}, \quad f^{[p]} = (1 - VU)(f) = \sum_{p \nmid n} a_n q^n.$$

If  $f$  is ordinary, then the two  $p$ -eigenvalues  $\alpha_p$  and  $\beta_p$  satisfies  $\alpha_p$  is a  $p$ -unit. And

$$f^{(p)}(q) = f(q) - \beta_p f(q^p)$$

satisfies  $U(f^{(p)}) = \alpha_p f^{(p)}$ , called the **ordinary stabilization** of  $f$ .

##### Hida Family

**Def. (33.8.1.1) [Hida Family].** Define the **weight space**

$$\Omega = \text{Hom}(\mathbb{Z}_p^*, \mathbb{C}_p^\times).$$

A point in  $\Omega$  is called arithmetic if it is of the form  $\nu_{k,\varepsilon} : x \mapsto x^{k-1}\varepsilon(x)$ , where  $k \in \mathbb{Z}_{\geq 2}$  and  $\varepsilon$  is a finite order character of  $\mathbb{Z}_p^*$ .

A **Hida family** of tame character  $\chi : (\mathbb{Z}/(N))^* \rightarrow \mathbb{C}_p^\times$  is a pair  $(\Omega_\varphi, \varphi)$  s.t.

- $\Omega_\varphi$  is a rigid analytic space equipped with a finite morphism  $\kappa : \Omega_\varphi \rightarrow \Omega$ .
- $\varphi = \sum_{n \in \mathbb{Z}_+} a_n(\varphi) q^n$  is a formal  $q$ -series with coefficients in the ring  $\mathcal{A}(\Omega_\varphi)$  of analytic functions on  $\Omega_\varphi$ . s.t. for any  $x \in \Omega_\varphi$  with  $\kappa(x) = \nu_{k,\varepsilon}$ , the restriction of  $\varphi$  to  $x$  is the  $q$ -expansion of a normalized eigenform in  $\text{SF}_k(Np^s, \chi\varepsilon\omega^{1-k})$ , where  $\omega$  is the Teichmüller character.

┘

**Def. (33.8.1.2) [Hida Family of Ordinary Modular Forms].** Let  $p \nmid N$  and  $f$  ordinary at  $p$ , there is a Hida families of over-convergent  $p$ -adic modular forms

$$\mathbf{f} : \Omega_f \rightarrow \mathbb{C}_p[[q]]$$

where  $\kappa : \Omega_f \rightarrow \Omega$  is a finite rigid analytic coverings of suitable subsets of the  $\mathbb{Z} \subset \Omega$  as  $k \mapsto (x \mapsto x^k)$ , and the image of  $\mathbb{Z}_{\geq 2}$  is called the set of **classical points**.

┘

**Prop. (33.8.1.3) [Hida Family of Weight 1 Modular Forms].** Let  $g$  be a weight 1 modular form with one of the ordinary  $p$ -stabilizations  $g_\alpha$ , then there exists a Hida family  $\underline{g}$  of tame character  $\chi$  that specializes to  $g_\alpha$  at some point  $y_0 \in \underline{\Omega}_g$  with  $\kappa(y_0) = \nu_{1,1}$ .

┘

*Proof:*

□

**Prop. (33.8.1.4) [Hida Family of Eisenstein Series].** There is a Hida family of Eisenstein series  $\underline{E}(1, \chi^{-1})$  with tame character  $\chi$  s.t. its specialization at  $\nu_{\ell, \varepsilon}$  equals the classical Eisenstein series  $E_\ell(1, \chi^{-1} \varepsilon \omega^{1-k})$ .

And it's involution  $\underline{E}^*(1, \chi^{-1})$  s.t. its specialization at  $\nu_{\ell, \varepsilon}$  equals the over-convergent Eisenstein series  $E_{2-\ell}(1, \chi^{-1} \varepsilon^{-1} \omega^{\ell-1})$ .  $\lrcorner$

*Proof:*  $\square$

### Abel-Jacobi Maps

**Def. (33.8.1.5) [Triple Product Kuga-Sato Varieties].** Let  $W = \mathcal{E}^{k-2} \times \mathcal{E}^{\ell-2} \times \mathcal{E}^{m-2}$  be the triple product Kuga-Sato varieties.

Then there is a unique way to embed  $\mathcal{E}^r$  to  $W$ , where  $r = (k + \ell + m - 6)/2$ .

There is a modification of this embedding that gives a homologically trivial cycle

$$\Delta_{k, \ell, m} \in \mathrm{CH}^{r+2}(W)_0.$$

$\lrcorner$

**Def. (33.8.1.6) [Abel-Jacobi Maps].** There is a natural  $p$ -adic Abel-Jacobi map

$$\mathrm{AJ}_p : \mathrm{CH}^{r+2}(W)_0(\mathbb{Q}_p) \rightarrow \mathrm{Fil}^{r+2} H_{\mathrm{dR}}^{2r+3}(W/\mathbb{Q}_p)^\vee.$$

$\lrcorner$

**Def. (33.8.1.7) [Unit Root Subspaces].** For  $f \in \mathrm{SF}_r(\Gamma_1(N))$  that is ordinary at  $p$ , there is a cohomology class

$$\omega_f \in \mathrm{Fil}^{r-1} H_{\mathrm{dR}}^{r-1}(\mathcal{E}^{r-2}/\overline{\mathbb{Q}}) \subset \mathrm{Fil}^{r-1} H_{\mathrm{dR}}^{r-1}(\mathcal{E}^{r-2}/\mathbb{C}_p)$$

And there is a unique unit-root element  $\eta_f^{u-r} \in H_{\mathrm{dR}}^{r-1}(\mathcal{E}^{r-2}/\mathbb{C}_p)$  s.t.  $\langle \omega_f, \eta_f^{u-r} \rangle = 1$ , and Frobenius acts on  $\eta_f^{u-r}$  as a multiplication by a  $p$ -adic unit.  $\lrcorner$

### $p$ -adic $L$ -Functions

#### 2 [Darmon-Rotger14]

A  $p$ -adic Gross-Zagier formula relating:

the image under the  $p$ -adic Abel-Jacobi map of certain generalized Gross-Kudla-Schoen cycles in the product of three Kuga-Sato varieties,

and

the special value of the Harris-Tilouine  $p$ -adic  $L$ -function attached to the Garrett-Rankin triple convolution of three Hida families of modular forms, at a point lying outside its region of interpolation.

This  $L$ -function  $L(f, g, h; s)$  satisfies a functional equation relating its values at  $s$  and  $k + \ell + m - 2 - s$  with sign of functional equation  $\varepsilon = \prod_{v \nmid N_\infty} \varepsilon_v$ , where we assume that

Hyp: For  $v \in \Sigma_F$  and  $v|N$ , we have  $\varepsilon_v = 1$ . This hypothesis is used to avoid having to deal with Shimura curves associated with a quaternion division algebra.

**Thm. (33.8.2.1) [ $p$ -adic Gross-Zagier Formula].** Let  $(x, y, z) \in \Sigma_{bal} \subset \Omega_f \otimes \Omega_g \otimes \Omega_h$ , let

$$(f, g, h) = (\mathbf{f}_x, \mathbf{g}_y, \mathbf{h}_z), \quad (k, \ell, m) = (\kappa(x), \kappa(y), \kappa(z)), \quad c = (k + \ell + m - 2)/2, \quad k = \ell + m - 2 - 2t,$$

then  $\eta_f^{u-r} \otimes \omega_g \otimes \omega_h \in \text{Fil}^{r+2} H_{\text{dR}}^{2r+3}(W/\mathbb{Q}_p)$ , and

$$\mathcal{L}_p^f(\mathbf{f}, \mathbf{g}, \mathbf{h})(x, y, z) = \frac{(-1)^t}{t!} \frac{\mathcal{E}(f, g, h)}{\mathcal{E}_0(f)\mathcal{E}_1(f)} \cdot \text{AJ}_p(\Delta_{k,\ell,m})(\eta_f^{u-r} \otimes \omega_g \otimes \omega_h).$$

┘

*Proof:* Cf.[DR14]Thm5.1. □

### 3 [Bertolini-Darmon-Rotger14]

A  $p$ -adic Beilinson formula relating:

the image under the the  $p$ -adic syntomic regulator(Bloch-Kato logarithm) on  $K_1$  of a certain Beilinson-Flach element in the product of two modular curves

and

the special value at  $(2, 2)$  of Hida's  $p$ -adic Rankin  $L$ -series attached to two ordinary families of modular forms interpolating  $f$  and  $g$ .

**Notation (33.8.3.1).** Let  $f \in \text{SF}_k(N, \chi_f)$ ,  $g \in \text{SF}_\ell(N, \chi_g)$  be normalized newforms, and  $k \leq \ell$ .

When  $k < \ell$ , fix an  $j \in [\frac{k+\ell-1}{2}, k-1]$ , and let  $t = k-1-j \geq 0$ ,  $m = k-\ell-2t \geq 1$ .

Denote  $\chi = \chi_f^{-1} \chi_g^{-1}$ . ┘

**Def. (33.8.3.2) [Eisenstein Series].** When  $k < \ell$ , the Eisenstein series is defined to be

$$E_{m,\chi}(z) = \frac{1}{2} L(\chi; 1-m) + \sum_{n=1}^{\infty} \sigma_{m-1,\chi}(n) q^n$$

is holomorphic of weight  $m$  and character  $\chi$ . ┘

**Def. (33.8.3.3) [Shimura-Maass Derivatives].** The differential operator  $\delta_m = \frac{1}{2\pi i} (\frac{\partial}{\partial z} + \frac{im}{2y})$  maps  $\text{MF}_m^{nh}(N, \chi)$  to  $\text{MF}_{m+2}^{nh}(N, \chi)$ . ┘

**Prop. (33.8.3.4).** Define

$$\Xi(f, g, j) = \delta_m^t(E_{m,\chi}) \cdot g \in \text{SF}_k^{nh}(N, \chi_f^{-1}).$$

Then

$$L(f \otimes g; j) = C(k, \ell, j) \langle f^c(z), \Xi(f, g, j) \rangle_{\text{Pet}}.$$

┘

**Cor. (33.8.3.5) [Rational Part].** Denote

$$L^{\text{alg}}(f \otimes g; j) = C(k, \ell, j)^{-1} \frac{\langle f^c(z), \Xi(f, g, j) \rangle_{\text{Pet}}}{\langle f^c(z), f^c(z) \rangle} = \frac{\langle f^c(z), \Xi(f, g, j) \rangle}{\langle f^c(z), f^c(z) \rangle} \in K_{f,g}.$$

┘

de Rham Cohomologies

**Def. (33.8.3.6).** Let  $\mathcal{E} \rightarrow Y_1(N)$  be the universal elliptic curve, and  $\omega$  be the sheaf of relative differentials on  $Y_1(N)$ . Denote  $\mathcal{L} = R^1\pi_*(\mathcal{O}_{\mathcal{E}} \rightarrow \Omega_{\mathcal{E}/Y_1(N)}^1)$  and  $\mathcal{L}^r = \text{Sym}^r \mathcal{L}$ . There is a Gauss-Manin connection  $\nabla$  on  $\mathcal{L}^r$ .

Then there is an exact sequence

$$0 \rightarrow H^0(X_1(N), \omega^r \otimes \Omega_{X_1(N)}^1) \rightarrow H_{\text{dR}}^1(X_1(N), \mathcal{L}_r, \nabla) \rightarrow H^0(X_1(N), \omega^{-r}) \rightarrow 0$$

s.t.  $H^0(X_1(N), \omega^r \otimes \Omega_{X_1(N)}^1)$  is isotropic w.r.t. the perfect Poincaré pairing on  $H_{\text{dR}}^1(X_1(N), \mathcal{L}_r, \nabla)$ .

┘

**Prop. (33.8.3.7).** Let  $r = k - 2$ , then

$$\eta_f^{a-h} = \frac{\overline{f^c(z)} d\bar{z}}{\langle f^c(z), f^c(z) \rangle_{\text{Pet}}} \in H_{\text{dR}}^1(X_1(N)_{\mathbb{C}}, \mathcal{L}_r, \nabla),$$

and its image in  $H^0(X_1(N)_{\mathbb{C}}, \omega^{-r})$  is denoted by  $\eta_f$ . Then  $\eta_f$  is in fact contained in  $H^0(X_1(N), \omega^{-r})$ .

┘

**Cor. (33.8.3.8).**

$$L^{\text{alg}}(f \otimes g; j) = \langle \eta_f, \Xi(f, g, j)^{\text{hol}} \rangle.$$

┘

**Prop. (33.8.3.9).** This exact sequence

$$0 \rightarrow H^0(X_1(N), \omega^r \otimes \Omega_{X_1(N)}^1) \rightarrow H_{\text{dR}}^1(X_1(N), \mathcal{L}_r, \nabla) \rightarrow H^0(X_1(N), \omega^{-r}) \rightarrow 0$$

when extended to  $\mathbb{C}_p$  has a canonical unit-root splitting s.t. the Frobenius acts on the complement by  $p$ -adic units. Let  $\eta_f^{u-r}$  be the inverse image in this unit-root subspace.

┘

Beilinson-Flach Elements

Let  $X = X_1(N)$ ,  $S = X_1(N) \times X_1(N)$ . Then  $H_{\text{Mot}}^3(S, \mathbb{Q}(2)) = \text{CH}^2(S, 1)_{\mathbb{Q}}$ , the latter has a definition as the homology of the **Gersten complex**

$$K_2(R(S)) \xrightarrow{\partial} \oplus_{Z \subset S} R(Z)^{\times} \xrightarrow{\text{div}} \oplus_{P \in S} \mathbb{Z}.$$

Then for the diagonal cycle  $\Delta \subset S$  and  $u \in \mathcal{O}(\Delta_{Y_1(N)}^{\times})$ , there exists a sum  $\theta_u$  of horizontal and vertical cycles s.t.

$$\text{div}(\theta_u) = \text{div}((\Delta, u)).$$

Then we define  $\mathbf{BF}_u = \Delta_u = \theta_u - (\Delta, u)$ , which is determined up to horizontal and vertical cycles.

Regulators

**Prop. (33.8.3.10) [Complex Regulators].** There is a regulator map

$$\text{Reg}_{\mathbb{C}} : H_{\text{Mot}}^3(S_{\mathbb{C}}, \mathbb{Q}(2)) \rightarrow (\text{Fil}^1 H_{\text{dR}}^2(S/\mathbb{C}))^{\vee}$$

s.t. the class  $\omega = \sum (Z_i, u_i)$  is sent to

$$\text{Reg}_{\mathbb{C}}(\omega) = \frac{1}{2\pi i} \sum \int_{Z_i^{\text{sm}}} \omega \log |u_i|.$$

┘

*Proof:*

□

**Prop. (33.8.3.11).**

$$\mathrm{Reg}_{\mathbb{C}}(\Delta_{u_{\chi}})(\omega_g \otimes \eta_f^{n-h}) = (-2i)[\Gamma_0(N) : \Gamma_1(N)\{\pm 1\}] \frac{\langle f^c(z), \log |u_{\chi}(z)| \cdot g(z) \rangle_{\mathrm{Pet}}}{\langle f^c(z), f^c(z) \rangle}.$$

┘

*Proof:* Cf. [Bertolini-Darmon-Rotger14] Prop 3.2. ?

□

**Prop. (33.8.3.12) [ $p$ -adic Regulators].** Let  $F \in \mathbf{NField}$  be the field cut out by the Dirichlet characters, and embed  $F$  to  $\mathbb{C}_p$  and let the completion be  $K$ . Let  $\mathcal{X}/\mathcal{O}_K$  be the Deligne-Rapoport smooth model of  $X_1(N)$ , and  $\mathcal{S} = \mathcal{X} \times \mathcal{X}$ . Then there is a  $p$ -adic syntomic map

$$\mathrm{Reg}_p : H_{\mathrm{Mot}}^3(S_K, \mathbb{Q}(2)) \rightarrow (\mathrm{Fil}^1 H_{\mathrm{dR}}^2(S/K))^{\vee}.$$

┘

*Proof:*

□

**Prop. (33.8.3.13).**

$$\mathrm{Reg}_p(\Delta_{u, \chi})(\omega_g \otimes \eta_f^{u-r}) = \langle u_f^{u-r}, \xi \rangle$$

where  $\xi \in H_{\mathrm{dR}}^1(X)$  is explicit, and the pairing on  $H_{\mathrm{dR}}^1(X)$  is induced by Poincaré pairing.

┘

*Proof:*

□

### Ordinary Projections

**Prop. (33.8.3.14).** Let  $e_{\mathrm{ord}}$  be Hida's ordinary projector on  $H_{\mathrm{dR}}^1(X_K, \mathcal{L}_r, \nabla)$ . Then

$$L^{\mathrm{alg}}(f \otimes g; j) = \langle \eta_f^{u-r}, e_{\mathrm{ord}} \Xi(f, g, j)^{\mathrm{hol}} \rangle.$$

Moreover,

$$\Xi(f, g, j)^{\mathrm{ord}} \triangleq e_{\mathrm{ord}} \Xi(f, g, j)^{\mathrm{hol}} = e_{\mathrm{ord}} \left( \left( q \frac{\partial}{\partial q} \right)^t (E_{m, \chi}) \cdot g \right).$$

┘

**Def. (33.8.3.15).**

$$\Xi(f, g, j)^{\mathrm{ord}, p} \triangleq e_{\mathrm{ord}} \Xi(f, g, j)^{\mathrm{hol}} = e_{\mathrm{ord}} \left( \left( q \frac{\partial}{\partial q} \right)^t (E_{m, \chi})^{[p]} \cdot g \right).$$

┘

### $p$ -adic Rankin $L$ -Functions

**Prop. (33.8.3.16) [ $p$ -adic Rankin  $L$ -function].** Let  $\mathbf{f}, \mathbf{g}$  be Hida families of ordinary  $p$ -adic modular forms of tame level  $N$  indexed by weight variables  $k$  and  $\ell$  valued in suitable compact open subsets of  $\mathbb{Z}/(p-1) \times \mathbb{Z}_p$ . Assume  $j = k - 1 - \ell$  belongs to the same residue class modulo  $(p-1)$ .

Then whenever  $k < \ell$  and  $j \in [\frac{k+\ell-1}{2}, k-1]$ , the  $p$ -adic modular form  $\Xi(f_k, g_{\ell}, j)^{\mathrm{ord}, p}$  is defined, and can be extended analytically to  $U_f \times U_g \times \mathbb{Z}_p$ .

Thus

$$\mathcal{L}_p(\mathbf{f}, \mathbf{g})(k, \ell, j) = \frac{1}{\mathcal{E}^*(f_k)} \left\langle \eta_{f_k}^{u-r}, \Xi(f_k, g_\ell, j)^{\text{ord}, p} \right\rangle_{k, X}$$

extends to an analytic function on  $U_f \times U_g \times \mathbb{Z}_p$ , called the *p*-adic **Rankin *L*-function** of  $\mathbf{f}$  and  $\mathbf{g}$ .

It satisfies that for any classical  $k, \ell, j$ ,

$$\mathcal{L}_p(\mathbf{f}, \mathbf{g})(k, \ell, j) = \frac{\mathcal{E}(f, g, j)}{\mathcal{E}^*(f_k) \mathcal{E}(f_k)} L^{\text{alg}}(f_k \otimes g_\ell, j).$$

┘

**Cor. (33.8.3.17) [A Variant].** If  $g$  is not ordinary, then we can just fix  $g$  and get a *p*-adic function

$$\mathcal{L}_p(\mathbf{f}, g)(k, \ell).$$

┘

### Beilinson's Conjectures

**Thm. (33.8.3.18) [Beilinson Formula].** If  $f, g$  are modular forms of weight 2 and  $\chi = \chi_f^{-1} \chi_g^{-1} \neq 1$ , then

$$\frac{L(f \otimes g; 2)}{\langle f^c(z), f^c(z) \rangle} = \frac{8i\pi^3}{N^2[\Gamma_0(N) : \Gamma_1(N)\{\pm 1\}]} \tau(\chi^{-1}) \text{Reg}_{\mathbb{C}}(\Delta_{u_\chi})(\omega_g \otimes \eta_f^{a-h}).$$

┘

*Proof:*

□

**Thm. (33.8.3.19) [*p*-adic Beilinson Formula].** Let  $p \in \text{Prime}$ ,  $(N, p) = 1$ ,  $f, g$  are weight 2 modular forms s.t.  $f$  is ordinary at  $p$ . Then we can extend the *p*-stabilization of  $f$  to a Hida family  $\mathbf{f}$ . Then

$$\mathcal{L}_p(\mathbf{f}, g)(2, 2) = \frac{\mathcal{E}(f, g, j)}{\mathcal{E}^*(f_k) \mathcal{E}(f_k)} \cdot \text{Reg}_p(\Delta_{u_\chi})(\omega_g \otimes \eta_f^{u-r}).$$

And if  $g$  is also ordinary at  $p$ , then the LHS can be replaced by  $\mathcal{L}_p(\mathbf{f}, \mathbf{g})(2, 2, 2)$ .

┘

*Proof:* Cf. [Bertolini-Darmon-Rotger14] Thm4.2. ?

□

### 4 [Bertolini-Darmon14]

What to recover Beilinson's result without using explicit reciprocity law.

### 5 [Lei-Loeffler-Zerbes14]

### Generalized Beilinson-Flach Elements

Defined a compatible system of cohomology classes (Generalized Beilinson-Flach elements)

$${}_c \Xi_{m, N, j} \in \text{CH}^2(Y_1(N)^2 \otimes \mathbb{Q}(\mu_m), 1)$$

for  $m \in \mathbb{Z}_+$ ,  $N \in \mathbb{Z}_{\geq 5}$ ,  $j \in \mathbb{Z}/(m)$  s.t. for  $m = 1$ , this reduces to the Beilinson-Flach element constructed in [Bertolini-Darmon-Rotger14].

### Bounding the Strict Selmer group

Prove (under some technical hypotheses) a finiteness theorem for the strict Selmer group of a product of modular forms when the associated  $p$ -adic Rankin–Selberg  $L$ -function is nonvanishing at  $s = 1$ .

### Perrin-Riou’s Conjecture

**Conj. (33.8.5.1).** ┘

### 6 [Bertolini-Darmon-Rotger15]

**Thm. (33.8.6.1).** Let  $\rho$  be an odd irreducible 2-dimensional Artin representation with image in  $L$  and cut out  $H$ , and  $E \in \text{EllCur}/\mathbb{Q}$ , then with some coprime conditions,

$$L(E/K, \psi; 1) \neq 0 \implies E(H)^\psi = 0.$$

┘

*Proof:* □

Sketch: Firstly  $\psi$  corresponds to a weight 1 cusp form  $g$ .

By Kolyvagin’s strategy, it can be proven that if for any embedding  $L \hookrightarrow \mathbb{C}_p$ , the singular residue map

$$\partial_p : H^1(\mathbb{Q}, V_f \otimes V_g) \rightarrow H_{\text{sing}}^1(\mathbb{Q}_v, V_f \otimes V_g)$$

is surjective, then  $E(H)^\psi = 0$ . We can choose  $p$  ordinary for  $g$  s.t.  $\alpha_p(g) \neq \beta_p(g)$ . Then we can consider the  $p$ -stabilization  $g_\alpha$ , and it suffices to prove the following:

**Lemma (33.8.6.2).** There exists a global class

$$\kappa(f, g_\alpha) \in H^1(\mathbb{Q}, V_f \otimes V_g)$$

s.t.

- $\partial_p(\kappa(f, g_\alpha))$  generates  $H_{\text{sing}}^1(\mathbb{Q}_p, V_f \otimes V_g)$ .
- $\kappa(f, g_\alpha) = 0$  iff  $L(f \otimes g; 1) = 0$ .

To show item2, the strategy is to make  $\kappa$  a family of cycles, then compare them at weight 2 points.

┘

### Rigid Cohomologies and Syntomic Regulators

**Prop. (33.8.6.3).** Let  $(N, p) = 1$  and  $s \in \mathbb{N}$ , let  $\Sigma_s = X_1(Np^s) \setminus Y_1(Np^s)$ , and let  $K/\mathbb{Q}_p$  be a finite extension s.t.  $X_1(Np^s)$  has a semisimple model and closed points of  $\Sigma_s$  are defined over  $K$ , then there is a Gysin exact sequence

$$0 \rightarrow H_{\text{ét}}^1(X_1(Np^s)_{\overline{\mathbb{Q}_p}}, \mathbb{Q}_p) \rightarrow H_{\text{ét}}^1(Y_1(Np^s)_{\overline{\mathbb{Q}_p}}, \mathbb{Q}_p) \rightarrow \mathbb{Q}_p(-1)^{\oplus(\#\Sigma_s-1)} \rightarrow 0,$$

which when applied  $D_{st}$  implies an exact sequence of filtered Frobenius monodromy modules

$$0 \rightarrow H_{\text{dR}}^1(X_1(Np^s)/K) \rightarrow H_{\text{dR}}^1(Y_1(Np^s)/K) \rightarrow K(-1)^{\oplus(\#\Sigma_s-1)} \rightarrow 0.$$

┘



*Proof:*

□

**Prop. (33.8.6.4)**[Analytic Description of Syntomic Regulators]. For  $s \in \mathbb{Z}_+$ ,

$$\mathrm{Reg}_p^{\mathrm{fg}}(\mathbf{BF}_s)(\eta_f \otimes \omega_{g^w}) = \mathcal{G}(\chi\chi_p) \frac{\alpha_f^{s-1}}{1 - \chi^{-1}(p)\alpha_f^{-1}\alpha_g} \times \left\langle \eta_f^w, e_{\mathrm{ord}}(g^{[p]} \cdot E_0^{[p]}(1, \chi^{-1}\chi_p^{-1})) \right\rangle_{X_0(Np)}.$$

┘

*Proof:* Cf.[Bertolini-Darmon-Rotger15]Thm2.10.

□

### Generalized Beilinson-Flach Elements and $p$ -adic $L$ -Functions

**Lemma (33.8.6.5).**

$$\mathcal{L}_p(f, \underline{g})(y) = \left\langle \eta_f^w, e_{\mathrm{ord}}(g_y^{[p]} \cdot E_{2-\ell}^{[p]}(1, \chi^{-1}\varepsilon^{-1}\omega^{\ell-1})) \right\rangle_{X_0(Np^s)}.$$

┘

*Proof:* Cf.[Bertolini-Darmon-Rotger15]P594. ?

□

**Thm. (33.8.6.6).** Let  $\varepsilon$  be a character of  $\mathbb{Z}_p^*$  of conductor  $p^s$ , and  $y \in \Omega_{\underline{g}}$  with  $\kappa(y) = \nu_{2,\varepsilon}$ . Thus  $g_y \in \mathrm{SF}_2(Np^s, \chi\varepsilon\omega^{-1})$  and

$$\mathrm{Reg}_p^{\mathrm{fg}}(\mathbf{BF}_s)(\eta_f \otimes \omega_{g^w}) = \mathcal{G}(\chi\varepsilon\omega^{-1}) \cdot \frac{\alpha_f^{s-1}}{1 - \chi(p)^{-1}\alpha_f^{-1}\alpha_g} \cdot \mathcal{L}_p(f, \underline{g})(y).$$

┘

*Proof:* This follows from (33.8.6.5) and (33.8.6.4).

□

## 7 [Darmon-Rorger16]

**Thm. (33.8.7.1).** Let  $\rho_1, \rho_2$  be two odd irreducible 2-dimensional Artin representation and  $\rho = \rho_1 \otimes \rho_2$  with image in  $L$  and cut out  $H$ , and  $E \in \mathrm{El1Cur}/\mathbb{Q}$ , then

$$L(E, \rho; 1) \neq 0 \implies E(H)^\rho = 0.$$

┘

*Proof:* Cf.[Darmon-Rorger16]Thm6.7., 6.13.

□

**Cor. (33.8.7.2).** If  $E \in \mathrm{El1Cur}/\mathbb{Q}$ ,  $K$  is a quadratic field. Given a ray class character  $\psi : \mathrm{Gal}(H/K) \rightarrow \mathbb{C}^\times$ , then when  $(E, K)$  satisfies some mild non-vanishing hypothesis, we have

$$L(E/K, \psi; 1) \neq 0 \implies E(H)^\psi \neq 0.$$

┘

### Generalized Gross-Kudla-Schoen Cycles

**Thm. (33.8.7.3).** The generated Gross-Kudla-Schoen cycle  $\kappa(f, g_\alpha, h_\alpha)$  is crystalline at  $p$  iff  $L(E, \rho; 1) \neq 0$ .

┘

*Proof:*

□

## 8 Kings-Loeffler-Zerbes17

construct three-variable  $p$ -adic families of Galois cohomology classes attached to Rankin convolutions of modular forms, and prove an explicit reciprocity law relating these classes to critical values of  $L$ -functions. As a consequence, we prove finiteness results for the Selmer group of an elliptic curve twisted by a 2-dimensional odd irreducible Artin representation when the associated  $L$ -value does not vanish

## 9 Unitary Groups

Iwasawa theory for the product of two unitary groups, generalizing phenomena observed in studying elliptic curves twisted by imaginary quadratic Hecke characters.

# 34 | Probability

## 34.1 Probability Theory

*“At a purely formal level, one could call probability theory the study of measure spaces with total measure one, but that would be like calling number theory the study of strings of digits which terminate.”*

–T. Tao

References are [Dudley, Real Analysis and Probability], [Fol99] and [Mathematical Methods for Financial Markets].

**Notation (34.1.0.1).**

- Use notations on real analysis defined in [Measure Theory and Functions on  \$\mathbb{R}^n\$](#) .
- Let  $(\Omega, \mathcal{F}, \mathbf{P})$  be a probability space [\(12.3.1.8\)](#).

┘

### 1 Basics

**Def. (34.1.1.1) [Events].** Any measurable space  $A \in \mathcal{M}$  is called an **event** in  $\mathcal{M}$ , and its **probability of the event** is defined to be  $\mathbf{P}(A)$ .

┘

**Def. (34.1.1.2) [Independent Events].** Two events [\(34.1.1.1\)](#)  $A$  and  $B$  are called **independent events** if  $\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$ . Similarly, we can define jointly-independent events.

┘

**Def. (34.1.1.3) [Random Variables].** A measurable function [\(12.3.1.26\)](#)  $X$  from  $(\Omega, \mathcal{F}, \mathbf{P})$  to a measurable space  $(S, \mathcal{A})$  is called a **random variable** on  $\Omega$ .

┘

**Def. (34.1.1.4) [Independent Variables].** Suppose  $X : (\Omega, \mathcal{F}, \mathbf{P}) \rightarrow (S, \mathcal{A})$  and  $Y : (\Omega, \mathcal{F}, \mathbf{P}) \rightarrow (T, \mathcal{B})$  are two random variables, then  $X, Y$  is called a pair of **independent variables** if for any event  $U \in \mathcal{A}$  and  $V \in \mathcal{B}$ ,  $X^{-1}(U)$  and  $Y^{-1}(V)$  are independent events [\(34.1.1.2\)](#), i.e.

$$\mathbf{P}(X \in U, Y \in V) = \mathbf{P}(X \in U) \cdot \mathbf{P}(Y \in V).$$

Similarly, we can define jointly-independent variables.

If  $\mathcal{S}$  is a set of random variable, then variables in  $S$  are called **independent variables** if any subset of variables in  $\mathcal{S}$  are jointly-independent.

┘

**Def. (34.1.1.5) [Distribution Functions].** Suppose  $X$  is a random variable on  $\Omega$  [\(34.1.1.3\)](#) valued in  $(\mathbb{R}, \text{dm})$ , the **distribution function** of  $X$  is defined to be the non-decreasing function  $F : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$  s.t.

$$F(x) = \mathbf{P}(X \geq x).$$

┘

**Def. (34.1.1.6) [Expected Values, Variances and Standard Deviations].** Suppose  $X$  is a random variable on  $\Omega$  (34.1.1.3) valued in  $(\mathbb{R}, \text{dm})$ . Then

- If  $\int_{\Omega} X d\mathbf{P}$  exists, it is called the **expected value** of  $X$ , denoted by  $\mathbf{E}(X)$ .
- The **variance** of  $X$  is defined to be

$$\text{var}(X) \triangleq \sigma^2(X) \triangleq \begin{cases} \mathbf{E}(X - \mathbf{E}(X))^2 = \mathbf{E}(X^2) - (\mathbf{E} X)^2 & , \mathbf{E}(X^2) < \infty \\ +\infty & , \mathbf{E}(X^2) = \infty \end{cases}.$$

- The **standard deviation** of  $X$  is defined to be  $\sigma(X) = \sqrt{\text{var}(X)} \in [0, \infty]$ .

Notice by Cauchy's inequality, if  $\mathbf{E}(X^2) < \infty$ , then  $\mathbf{E}(X) < \infty$ , so the definition makes sense.  $\lrcorner$

**Def. (34.1.1.7) [Covariances and Correlation Coefficients].** Suppose  $X, Y$  are real-valued random variables on  $\Omega$ , then

- If  $\mathbf{E}(X) < \infty, \mathbf{E}(Y) < \infty$ , the **covariance** of  $X$  and  $Y$  is defined to be

$$\text{cov}(X, Y) \triangleq \mathbf{E}((X - \mathbf{E}(X))(Y - \mathbf{E}(Y))) = \mathbf{E}(XY) - \mathbf{E}(X) \cdot \mathbf{E}(Y).$$

- If  $\mathbf{E}(X^2) < \infty$  and  $\mathbf{E}(Y^2) < \infty$ , and  $\text{cov}(X) \neq 0, \text{cov}(Y) \neq 0$ , then the **correlation coefficient** between  $X$  and  $Y$  is defined to be

$$r(X, Y) = \frac{\text{cov}(X, Y)}{\sigma(X)\sigma(Y)}.$$

Notice  $r(X, Y) \in [-1, 1]$  if it is defined.  $\lrcorner$

*Proof:*  $r(X, Y) \in [-1, 1]$  by Cauchy inequality because  $L^2(\Omega)$  is a Hilbert space.  $\square$

**Prop. (34.1.1.8) [Independence and Covariances].** Suppose  $X, Y$  are real-valued random variables on  $\Omega$  that are independent, then  $\text{cov}(X, Y) = 0$ .  $\lrcorner$

*Proof:* Cf. [Dudley, P252].  $\square$

**Prop. (34.1.1.9).** Suppose  $(\Omega, \mathcal{M}, P)$  is a probability space and  $X, Y$  are real-valued random variables on  $\Omega$ , then

$$\text{var}\left(\sum_i X_i\right) = \sum_i \text{var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{cov}(X_i, X_j).$$

In particular, if  $X_i$  are independent or pairwise-independent, by (34.1.1.8),

$$\text{var}(X_1 + \dots + X_n) = \sum_i \text{var}(X_i).$$

$\lrcorner$

*Proof:* We may first assume that  $\mathbf{E}(X_i) = 0$  for each  $i$ , then

$$\text{var}\left(\sum_i X_i\right) = \mathbf{E}\left(\left(\sum_i X_i\right)^2\right) = \mathbf{E}\left(\sum_i X_i^2 + 2 \sum_{i < j} X_i X_j\right) = \sum_i \mathbf{E}(X_i^2) + 2 \sum_{i < j} \mathbf{E}(X_i X_j).$$

So the assertion is easy.  $\square$

### Moment Generating Functions

**Def. (34.1.1.10) [Moment Generating Functions].** The **moment generating function** of a real-valued random variable  $X$  is defined to be the function  $\varphi(t) : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$  s.t.

$$\varphi(t) = \mathbf{E}(e^{tX})$$

if it converges. ┘

**Prop. (34.1.1.11).** moment generating function of the sum of independent random variables is just the product of the individual moment generating functions. ┘

**Prop. (34.1.1.12) [Moment Generating Functions and Moments].** If a real-valued random variable  $X$  has a moment generating function  $\varphi(t)$ , then for any  $n \in \mathbb{N}$ ,

$$\varphi^{(n)}(0) = \mathbf{E}(X^n).$$

┘

*Proof:* ? □

**Prop. (34.1.1.13) [Distribution Determined by Moment Generating Functions].** If two real-valued random variables  $X_1, X_2$  has the same moment generating function, then  $X_1$  and  $X_2$  has the same distribution function. ┘

*Proof:* ? □

### Special Random Variables

**Def. (34.1.1.14) [Normal Distributions].**  $\mathcal{N}(\mu, \sigma^2)$ . ┘

## 2 Convergence of Laws and Central Limit Theorems

**Thm. (34.1.2.1) [Central Limit Theorem].** ┘

### Special Random Variables

**Prop. (34.1.2.2) [Normal Approximation of Binomial Probabilities].** Let  $X$  be the random variable with parameter  $(n, p)$  that represents the number of successes in  $n$  independent trials when each trial is a success with probability  $p$ , then for  $n$  large, by the central limit theorem (34.1.2.1),

$$\frac{X - np}{\sqrt{np(1-p)}}$$

will approximately be a standard normal random variable. ┘

**Remark (34.1.2.3).** This normal approximation will in general be quite good for values of  $n$  satisfying  $np(1p) \geq 10$ . ┘

**Prop. (34.1.2.4) [Poisson Approximation of Binomial Probabilities].** Let  $X$  be the random variable with parameter  $(n, p)$  that represents the number of successes in  $n$  independent trials when each trial is a success with probability  $p$ , then ? ┘

### 3 Conditional Expectations

### 4 Stochastic Processes

References are [Revuz-Yor, Continuous Martingales and Brownian Motion].

**Def. (34.1.4.1) [Measurable Time Process].** A **measurable time process** on  $\Omega$  is a family of random variables  $X = (X_t)_{t \in \mathbb{R}_{\geq 0}}$  s.t. the map  $\Omega \times \mathbb{R} \rightarrow \mathbb{R} : (\omega, t) \mapsto X_t(\omega)$  is  $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_{\geq 0})$ -measurable.

$X$  is called a **continuous time process** if for a.e.  $\omega \in \Omega$ , the map  $\mathbb{R}_{\geq 0} \rightarrow \mathbb{R} : t \mapsto X_t(\omega)$  is continuous.

$X$  is called a **cadlag time process** if for a.e.  $\omega \in \Omega$ , the map  $\mathbb{R}_{\geq 0} \rightarrow \mathbb{R} : t \mapsto X_t(\omega)$  is continuous on the right with limits on the left.

$X$  is said to be **increasing** if  $X$  is a.e. right continuous,  $X_0 = 0$ , and  $X_s \leq X_t$  a.e. for any  $s < t \in \mathbb{R}_{\geq 0}$ .  $\perp$

**Def. (34.1.4.2) [Filtered Probability Spaces].** Given a probability space  $(\Omega, \mathcal{F}, \mathbf{P})$ , a **filtration** on this space is a family  $\mathbf{F} = (\mathcal{F}_t)_{t \in \mathbb{R}_{\geq 0}}$  of  $\sigma$ -algebras on  $\Omega$  s.t.  $\cup_t \mathcal{F}_t \subset \mathcal{F}$ , and

- $\mathcal{F}_s \subset \mathcal{F}_t$  whenever  $s < t$ .
  - For any  $t \in \mathbb{R}_{\geq 0}$ ,  $\mathcal{F}_t = \cap_{u>t} \mathcal{F}_u$ .
  - $\mathcal{F}_0$  contains all  $(\mathbf{P}, \mathcal{F})$ -negligible sets of  $\mathcal{F}$ .
- $\perp$

**Def. (34.1.4.3) [F-adapted Processes].** Let  $(\Omega, \mathcal{F}, \mathbf{F}, \mathbf{P})$  be a filtered probability space, then a measurable time process (34.1.4.1)  $X = (X_t)$  is said to be **F-adapted** if for any  $t \in \mathbb{R}_{\geq 0}$ ,  $X_t$  is  $\mathcal{F}_t$ -measurable.  $\perp$

**Def. (34.1.4.4) [F-Progressively Measurable Processes].** Let  $(\Omega, \mathcal{F}, \mathbf{F}, \mathbf{P})$  be a filtered probability space, then a measurable time process  $X = (X_t)$  is said to be **progressively measurable** w.r.t.  $\mathbf{F}$  if for any  $t \in \mathbb{R}_{\geq 0}$ , the map

$$\Omega \times [0, t] \rightarrow \mathbb{R} : (\omega, s) \mapsto X_s(\omega)$$

is  $\mathcal{F}_t \times \mathcal{B}([0, t])$  measurable.  $\perp$

**Prop. (34.1.4.5).** Any right continuous  $\mathbf{F}$ -adapted time process  $X$  is  $\mathbf{F}$ -progressively measurable.

Any  $\mathbf{F}$ -progressively measurable process  $X$  is  $\mathbf{F}$ -adapted.

If  $X$  is  $\mathbf{F}$ -progressively measurable, then

$$\mathbf{E}(\int_0^\infty X_t dt) = \int_0^\infty \mathbf{E}(X_t) dt.$$
 $\perp$

*Proof:* ?  $\square$

**Def. (34.1.4.6) [Processes with the same Law].** Let  $X = (X_t), Y = (Y_t)$  be two processes on  $\Omega$ , then they are said to have the same law, denoted by  $X \stackrel{\text{law}}{=} Y$ , if for any  $n \in \mathbb{Z}_+$  and  $(t_1, \dots, t_n) \in \mathbb{R}_{\geq 0}$ ,

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{\text{law}}{=} (Y_{t_1}, \dots, Y_{t_n}).$$
 $\perp$

**Def. (34.1.4.7) [Modifications of Processes].** Let  $X, Y$  be two processes on  $\Omega$ , then  $X$  is called a **modification** of  $Y$  if for any  $t \in \mathbb{R}_{\geq 0}$ ,  $\mathbf{P}(X_t = Y_t) = 1$ . And  $X$  is called **indistinguishable** from  $Y$  if  $\{\omega : X_t(\omega) = Y_t(\omega), \forall t\}$  is measurable and  $\mathbf{P}(X_t = Y_t, \forall t) = 1$ .  $\lrcorner$

**Thm. (34.1.4.8) [Continuous Modifications, Kolmogorov].** If  $X = (X_t)$  is a time process on  $\Omega$  satisfies: there exists  $C, p, \varepsilon \in \mathbb{R}_+$  s.t. for any  $t, s \in \mathbb{R}_{\geq 0}$ ,

$$\mathbf{E}(|X_t - X_s|^p) \leq C|t - s|^{1+\varepsilon}.$$

Then this process admits a modification  $(\tilde{X}_t)$  which is a.e. continuous (34.1.4.1).  $\lrcorner$

*Proof:* Cf. [Stochastic Differential Equations and Diffusion Processes 1989] P20.  $\square$

**Prop. (34.1.4.9) [Processes Convergences].** A sequence of processes  $(Z^n)_{n \in \mathbb{Z}_+}$  **converges in  $L^2(\Omega \times [0, T])$**  to a process  $Z$  if  $\mathbf{E} \int_0^T |Z_s^n - Z_s|^2 ds$  converges to 0.

And it is said to **converge uniformly on compact subsets in probability** to  $Z$  if for any  $t \in \mathbb{R}_+$ ,  $\sup_{0 \leq s \leq t} |Z_s^n - Z_s|$  converges to 0 in probability.  $\lrcorner$

## 5 Martingales

**Thm. (34.1.5.1) [No-Arbitrage Implies Semi-Martingale].** Let  $S$  be an adapted cadlag process that is locally bounded and satisfies the no-free-lunch-with-vanishing-risk property for simple integrands, then  $S$  is a semi-martingale.  $\lrcorner$

*Proof:* Cf. [Mathematical Methods for Financial Markets] P86.  $\square$

### Markov Processes

**Def. (34.1.5.2) [Markov Kernels].** Let  $(X, \mathcal{A}), (Y, \mathcal{B})$  be measurable spaces, a **Markov kernel** from  $(X, \mathcal{A})$  to  $(Y, \mathcal{B})$  is a map  $\kappa : X \times \mathcal{B} \rightarrow [0, 1]$  s.t.

- For any  $B \in \mathcal{B}$ , the map  $x \mapsto \kappa(x, B)$  is  $\mathcal{A}$ -measurable.
  - For any  $x \in X$ , the map  $B \mapsto \kappa(x, B)$  is a probability measure on  $(Y, \mathcal{B})$ .
- $\lrcorner$

**Prop. (34.1.5.3).** Let  $(X, \mathcal{A})$  be a measurable space and  $P(x, dy)$  is a Markov kernel on  $X$ . Then  $P$  acts both on the the space of (integrable) measurable functions on  $X$  and the space of measures (of finite mass) on  $X$  by:

$$(P\varphi)(x) = \int_X \varphi(y) P(x, dy), \quad (P\mu)(A) = \int_X P(x, A) d\mu(x).$$

In particular,

$$\mu(P\varphi) = P(\mu)(\varphi).$$
 $\lrcorner$

**Thm. (34.1.5.4) [Harris Ergodic Theorem].** Let  $(X, \mathcal{A})$  be a measurable space with a Markov kernel  $P(x, dy)$  on  $X$ . Suppose

- There exists a measurable function  $V : X \rightarrow \mathbb{R}_+$  and  $K \in \mathbb{R}_{\geq 0}, \gamma \in (0, 1)$  s.t.

$$PV(x) \leq \gamma V(x) + K.$$

- There exists  $\alpha \in (0, 1)$ ,  $\varepsilon \in \mathbb{R}_+$  and a probability measure  $\nu$  on  $X$  s.t.

$$\inf_{x \in X: V(x) \leq \frac{2K}{1-\gamma} + \varepsilon} P(x, -) \geq \alpha \nu(-).$$

Then  $P$  admits a unique invariant measure  $\mu_*$  (under the action defined in (34.1.5.3)) s.t.  $\int_X V(x) d\mu_*(x) < \infty$ , and there exists  $C \in \mathbb{R}_+$  and  $\lambda \in (0, 1)$  s.t.

$$\|P^n \varphi - \mu_*(\varphi)\|_V \leq C \lambda^n \|\varphi\|_V,$$

for any integrable measurable function  $\varphi$  on  $X$ , where  $\|\varphi\|_V = \left\| \frac{\varphi(x)}{1+V(x)} \right\|_\infty$ .  $\lrcorner$

*Proof:* Use notations as in (34.1.5.5), for the existence of the invariant measure  $\mu_*$  s.t.  $\mu_*(V) < \infty$ , denote the measures  $\mu_n = P^n \delta_x$ , then  $\mu_n$  is a Cauchy sequence w.r.t. the metric  $\rho_\beta$  (34.1.5.5). Since the space of measures with finite total mass is clearly complete w.r.t.  $\rho_\beta$ , there exists a measure  $\mu_*$  s.t.  $\mu_n \rightarrow \mu_*$ . Then because  $P$  is a contraction, we see  $P\mu_* = \mu_*$ . And  $\mu_*(V) < \infty$ .

Then we have  $\|\varphi\|_1 \leq \|\varphi\|_\beta \leq \beta^{-1} \|\varphi\|_1 \leq \beta^{-1} \|\varphi\|_V$ . So

$$\|P^n \varphi\|'_1 \leq \|P^n \varphi\|'_\beta \leq \lambda^n \|\varphi\|'_\beta \leq \lambda^n \beta^{-1} \|\varphi\|_V.$$

or equivalently:

$$|P^n \varphi(x) - P^n \varphi(y)| \leq \lambda^n \beta^{-1} \|\varphi\|_V (2 + V(x) + V(y))$$

for a.e.  $x, y \in X$ . Thus

$$|P^n \varphi(x) - \mu_*(\varphi)| \leq \int_X |P^n \varphi(x) - P^n \varphi(y)| d\mu_*(y) \leq \lambda^n \beta^{-1} \|\varphi\|_V (2 + V(x) + \mu_*(V)).$$

$\square$

**Lemma (34.1.5.5).** Under the assumption of (34.1.5.4), there exists  $\lambda \in (0, 1)$  and  $\beta \in (0, 1]$  s.t.

- $\|P\varphi\|'_\beta \leq \lambda \|\varphi\|'_\beta$  for any measurable function  $\varphi$  where

$$d_\beta(x, y) = \begin{cases} 0 & , x = y \\ 2 + \beta(V(x) + V(y)) & , x \neq y \end{cases}, \quad \|\varphi\|'_\beta = \sup_{x \neq y \in X} \frac{\varphi(x) - \varphi(y)}{d_\beta(x, y)},$$

- $\rho_\beta(P\mu_1, P\mu_2) \leq \lambda \rho_\beta(\mu_1, \mu_2)$  for any two probability measures  $\mu_1, \mu_2$  on  $X$ , where

$$\|\varphi\|_\beta = \left\| \frac{\varphi(x)}{1 + \beta V(x)} \right\|_\infty, \quad \rho_\beta(\mu_1, \mu_2) = \sup_{\varphi: \|\varphi\|_\beta \leq 1} \int_X \varphi(x) d(\mu_1 - \mu_2)(x).$$

$\lrcorner$

*Proof:* Firstly we claim that  $\|\varphi\|'_\beta = \min_{c \in \mathbb{R}} \|\varphi + c\|_\beta$ : It is obvious that  $\|\varphi\|'_\beta \leq \|\varphi\|_\beta$ , so  $\|\varphi\|'_\beta \leq \inf_{c \in \mathbb{R}} \|\varphi + c\|_\beta$ .

Conversely, if  $\|\varphi\|'_\beta \leq 1$ , define  $c = \inf_x (1 + \beta V(x) - \varphi(x))$ . Because for any  $x, y \in X$ ,

$$\varphi(x) \leq |\varphi(y)| + |\varphi(x) - \varphi(y)| \leq |\varphi(y)| + 2 + \beta V(x) + \beta V(y),$$

we have  $1 + \beta V(x) - \varphi(x) \geq -1 - \beta V(y) - |\varphi(y)|$ , so  $c \in \mathbb{R}$ . And it is clear that

$$1 + \beta V(x) \geq \varphi(x) + c = \min_y (\varphi(x) + 1 + \beta V(y) - \varphi(y)) \geq \min_y (1 + \beta V(y) - \|\varphi\|'_\beta \cdot d_\beta(x, y)) \geq -(1 + \beta V(x)).$$



Now we prove the assertion: For item1: Suppose  $\|\varphi\|'_\beta = 1 = \|\varphi\|_\beta$ , we need to show that  $|P\varphi(x) - P\varphi(y)| \leq \lambda(2 + \beta(V(x) + V(y)))$  for  $x \neq y$ . Denote

$$R = \frac{2K}{1-\gamma} + \varepsilon, \quad \gamma_0 = \gamma + 2K/R \in (0, 1), \quad \gamma_1 = (2 + \beta R \gamma_0)/(2 + \beta R) \in (\gamma_0, 1).$$

Take  $\beta = \alpha_0/K$  for some  $\alpha_0 \in (0, \alpha)$  and  $\gamma_2 = \max(1 - (\alpha - \alpha_0), \gamma_0)$ , then if  $V(x) + V(y) < R$ , then we can define a Markov transformation  $\tilde{P}(x, dy) = \frac{1}{1-\alpha}P(x, dy) - \frac{\alpha}{1-\alpha}d\nu(y)$ . Then

$$\begin{aligned} |P\varphi(y) - P\varphi(x)| &= (1 - \alpha)|\tilde{P}\varphi(x) - \tilde{P}\varphi(y)| \\ &\leq 2(1 - \alpha) + (1 - \alpha)\beta(\tilde{P}V(x) + \tilde{P}V(y)) \\ &\leq 2(1 - \alpha) + \beta(PV(x) + PV(y)) \\ &\leq 2(1 - \alpha) + \beta\gamma(V(x) + V(y)) + 2\beta K \\ &\leq 2(1 - (\alpha - \alpha_0)) + \beta\gamma(V(x) + V(y)) \\ &\leq \gamma_2 d_\beta(x, y). \end{aligned}$$

And if  $V(x) + V(y) \geq R$ , then by the hypothesis,

$$\begin{aligned} |P\varphi(y) - P\varphi(x)| &\leq 2 + \beta(PV(x) + PV(y)) \\ &\leq 2 + \lambda\beta(V(x) + V(y)) + 2\beta K \\ &\leq 2 + \beta\gamma_0(V(x) + V(y)) \\ &\leq 2\gamma_1 + \beta\gamma_1(V(x) + V(y)) \\ &\leq \gamma_1 d_\beta(x, y). \end{aligned}$$

And item2 follows from item1 because of (34.1.5.3) and the fact  $\rho_\beta$  is dual to  $\|\cdot\|'_\beta$  by what we proved at the beginning.  $\square$

### Finite Markov Processes

**Def. (34.1.5.6) [Markov Chain].** A (finite discrete) **Markov chain** is a discrete stochastic process  $\{X_t\}_{t \in \mathbb{N}}$  with the same phase space  $S \in \text{finSet}$  s.t. for any  $t$ ,

$$\mathbf{P}(X_{t+1} = S_j | X_t = S_{i_t}, X_{t-1} = S_{i_{t-1}}, \dots, X_0 = S_{i_0}) = \mathbf{P}(X_{t+1} = S_j | X_t = S_{i_t}).$$

This property is called the **Markov property** or **memoryless property**.  $\lrcorner$

**Prop. (34.1.5.7) [Markov Chains and Stochastic Matrices].** For any Markov chain  $\{X_t\}_{t \in \mathbb{N}}$ , the transition matrices form a family of stochastic matrices (3.5.14.9)  $\{p_{ij}(t)\}_{t \in \mathbb{N}}$ . And it is called a **stationary stochastic matrix** if  $p_{ij}(t)$  is independent of  $t$ .

Conversely, any family of stochastic matrices  $\{p_{ij}(t)\}_{t \in \mathbb{N}}$  defines a Markov chain.  $\lrcorner$

**Thm. (34.1.5.8) [Limiting Behavior of Irreducible Markov Chains].** Let  $X = \{X_t\}_{t \in \mathbb{N}}$  be a stationary finite Markov chain with (right) transition matrix  $P$  on a phase space  $S$  of size  $n$ . Suppose  $P$  is irreducible (3.5.14.4).

- The limit probability distribution exists for any initial distribution iff  $\lim_{k \rightarrow \infty} P^k$  exists.

- If  $P$  is primitive, by (3.5.14.6), if  $\varpi$  is the Perron vector (3.5.14.2) for  $P$ , then the limit of the probability distribution is just

$$\frac{\varpi(\mathbf{e}/n)^t}{(\mathbf{e}/n)^t \varpi} p(0) = \varpi \mathbf{e}^t p(0) = \varpi,$$

which is independent of the initial distribution  $p(0)$ . In this case,  $X$  is called an **aperiodical Markov chain**.

- If  $P$  is imprimitive, then the same argument as above but using (3.5.14.14) shows that the Cesàro limit

$$\lim_{k \rightarrow \infty} \frac{\mathbf{1} + P + P^2 + \dots + P^{k-1}}{k} = \mathbf{e} \varpi^t,$$

and

$$\lim_{k \rightarrow \infty} \frac{p^t(0) + p^t(1) + \dots + p^t(k-1)}{k} = \varpi^t.$$

In this case,  $X$  is called an **periodical Markov chain** (with period equal to the index of imprimitivity of  $P$ ).

- Whether or not  $P$  is primitive, the components of  $\varpi$  determines the expected value of long-time frequency of  $S_i$  appearing in the Markov sequence.

Thus  $\varpi^t$  is called the **stationary distribution vector** of this Markov chain.  $\lrcorner$

**Thm. (34.1.5.9) [Limiting Behavior of Reducible Markov Chains].** Let  $X = \{X_t\}_{t \in \mathbb{N}}$  be a stationary finite Markov chain with (right) transition matrix  $P$  on a phase space  $S$  of size  $n$ . Suppose  $P$  is reducible (3.5.14.4) and the states are ordered s.t.  $P$  is in the canonical form

$$P = \begin{bmatrix} T_{11} & T_{12} \\ & T_{22} \end{bmatrix}$$

and if  $\varpi_j$  are Perron vectors for  $P_{jj}$  ( $r+1 \leq j \leq m$ ), then  $\mathbf{1} - T_{11}$  is non-singular, and

•

$$\lim_{k \rightarrow \infty} \frac{\mathbf{1} + P + \dots + P^{k-1}}{k} = \begin{bmatrix} 0 & (1 - T_{11})^{-1} T_{12} E \\ 0 & E \end{bmatrix}$$

where  $E = \text{diag}(\mathbf{e} \varpi_{r+1}^t, \dots, \mathbf{e} \varpi_m^t)$ .

- The limit probability distribution exists for any initial distribution iff  $\lim_{k \rightarrow \infty} P^k$  exists, which is the case when  $P_{r+1, r+1}, \dots, P_m$  are all primitive. In which case,

$$\lim_{k \rightarrow \infty} P^k = \begin{bmatrix} 0 & (1 - T_{11})^{-1} T_{12} E \\ 0 & E \end{bmatrix}.$$

$\lrcorner$

*Proof:*

$\square$

## Lévy Processes

Lévy processes are generalizations of random walks to continuous time.

### Brownian Motion

**Remark (34.1.5.10).** The motion of a Brownian particle in  $\mathbb{R}^N$  with a potential field  $U$  is described by a stochastic process  $q = (q_1(t), \dots, q_N(t)) \in \mathbb{R}^N$ . (Which means that for each time  $t$ , there is a random variable  $q(t) \in \mathbb{R}^N$  s.t. the dependence in  $t$  is regular in some sense.)

All in all, the density of the probability that the motion with restrict  $q(a) = \underline{a}, q(b) = \underline{b}$  is a smooth curve  $\gamma : [a, b] \rightarrow \mathbb{R}^N$  is propositional to

$$e^{-S(\gamma)/\kappa}, \quad S(\gamma) = \int_a^b \left( \frac{1}{2} \dot{\gamma}^2 + U(\gamma) \right) dt$$

where  $\kappa$  is called the **diffusion coefficient**. ┘

**Remark (34.1.5.11).** It follows from Kolmogorov's theorem that the stochastic process is determined by the **correlations functions**:

$$\text{cor}(j_1, \dots, j_n; t_1, \dots, t_n) = \mathbf{E} \left( q_{j_1}(t_1) \cdot \dots \cdot q_{j_n}(t_n) \right).$$

┘

*Proof:* ?

□

### Simple Random Walks

#### 6 Brownian Motions

#### 7 Statistics

References are [INTRODUCTION TO PROBABILITY AND STATISTICS FOR ENGINEERS AND SCIENTISTS].

**Def. (34.1.7.1) [Inferential Statistics].** **Inferential statistics** is the art of learning from data.

To be able to draw logical conclusions from data, we usually make some assumptions about the probabilities of obtaining the different data values. The totality of these assumptions is referred to as a **probability model for the data**. ┘

**Def. (34.1.7.2) [Samples].** A set of independent random variables having a common distribution  $F$  is called a **sample** from the distribution  $F$ .

And a **statistic** is a random variable whose value is determined by the the sample data(variables).

┘

**Def. (34.1.7.3) [Sample Means, Sample Variances and Sample Standard Deviations].** Let  $n \in \mathbb{Z}_{\geq 2}$  and  $X_1, \dots, X_n$  be a random sample from a distribution  $F$  with mean  $\mu$  and variance  $\sigma$ ,

- Define the statistic

$$\overline{X} = \frac{\sum_{i=1}^{n-1} X_i}{n}$$

to be the sample mean.

- Define the statistic

$$S^2 = \frac{\sum_{i=1}^{n-1} (X_i - \bar{X})^2}{n-1},$$

called the **sample variance**, and define the statistic  $S = \sqrt{S^2}$ , called the **sample standard deviation**. ┘

**Prop. (34.1.7.4).**

$$\mathbf{E}(\bar{X}) = \mu, \quad \text{var}(\bar{X}) = \sigma^2/n, \quad \mathbf{E}(S^2) = \sigma^2.$$
┘

### Parameter Estimation

**Def. (34.1.7.5) [Statistical Estimates].** If the samples  $\{X_1, \dots, X_n\}$  are drawn from a distribution  $F_\theta$  depending on an unknown parameter  $\theta \in \mathbb{R}$ , any statistic (34.1.7.2) used to estimate the value of an unknown parameter  $\theta$  is called an **statistical estimator** of  $\theta$ . The observed value of the estimator is called the **statistical estimate**. ┘

**Def. (34.1.7.6) [Maximal Likelihood Estimates].** Situation as in (34.1.7.5), a **maximal likelihood estimate** is a estimate  $\hat{\theta}$  depending on  $X_1, \dots, X_n$  s.t.  $\hat{\theta}(x_1, \dots, x_n)$  is the value maximizing  $P_{F_\theta}(X_i = x_i)$ . And the function  $P_{F_\theta}(X_i = x_i)$  is called the **likelihood function** of  $\theta$ . ┘

**Def. (34.1.7.7) [Interval Estimates for Mean Values].** If the distribution is a normal distribution with an unknown mean  $\mu$  and known variance  $\sigma^2$ , then for  $\alpha \in (0, 1)$ , an  **$\alpha$ -interval estimate** of a the mean  $\mu$  is an interval  $I \subset \mathbb{R}$  s.t.  $P(\mu \in I : X) \geq 1 - \alpha$ .  $\alpha$  is called the **confidence level**. ┘

**Remark (34.1.7.8) [Interpreting the Confidence Level  $\alpha$ ].** The unknown mean is not a random variable, but the confidence level  $\alpha$  can be interpreted as follows: If you can do sampling arbitrarily many times and calculate the  $\alpha$ -confidence intervals, then the frequency that the unknown mean  $\mu$  is contained in the  $\alpha$ -confidence intervals is  $\alpha$ . ┘

*Proof:* ? □

### Hypothesis Testing

**Def. (34.1.7.9) [Statistical Hypotheses].** A **statistical hypothesis** is usually a statement about a set of parameters of a population distribution which is not known whether or not it is true. The statement is called the **null hypothesis**.

The **alternative hypothesis** is a hypothesis that is disjoint from the null hypothesis, usually just the back of the null hypothesis. ┘

**Def. (34.1.7.10) [Level of Significance].** The objective of a statistical test of a null hypothesis ( $H_0$ ) is not to explicitly determine whether or not it is true but rather to determine if its validity is consistent with the resultant data. We say the data **rejects** ( $H_0$ ) if we think it is inconsistent with the hypothesis, and we say the data **accepts** ( $H_0$ ) if we think it is consistent with the hypothesis.

The **level of significance** of a test is a value  $\alpha \in (0, 1)$  s.t. whenever the null hypothesis is true, its probability of being rejected is never greater than  $\alpha$ . ┘

**Def. (34.1.7.11) [ $p$ -Values of the Test].** When the distribution is a normal distribution having an unknown mean  $\mu$  and a known variance  $\sigma^2$ , and the null hypothesis is  $\mu = \mu_0$ , then the  **$p$ -value of a data** of a test is the probability that a data drawn from a normal distribution having mean value  $\mu_0$  and variance  $\sigma^2$  would be further than this data.

Then this data accepts  $\mu = \mu_0$  if its  $p$ -value is greater than the level of significance  $\alpha$ , and rejects otherwise. ┘

**Def. (34.1.7.12) [Critical Regions].**  $\{C_n \in \mathbb{R}^n\}$  is called a system of **critical regions** for the test if the rule of the test is designed s.t. the data  $(X_1, \dots, X_n)$  accepts  $(H_0)$  iff  $(X_1, \dots, X_n) \subset C_n$ . ┘

**Prop. (34.1.7.13) [Case of unknown variance: the  $t$ -test].** ┘

*Proof:* ? □

### Regression

### Analysis of Variance

### Goodness of Fit Tests

**Def. (34.1.7.14) [Goodness of Fit Tests].** Statistical tests that determine whether a given probabilistic mechanism is appropriate are called **goodness of fit tests**. ┘

### Nonparametric Hypothesis Tests

### Quality Control

## 34.2 Random Matrices

## 35 | Analytic Theory of $L$ -Functions

### 35.1 Inequalities and Probabilistic Number Theory

References are [I-K04, Ten15].

**Notation (35.1.0.1).**

- Use notations on probability theory defined in [Probability Theory](#).

┘

#### 1 Tauberian Theorems

**Thm. (35.1.1.1) [Wiener].** If  $Y$  is a closed translation-invariant space of  $L^1(\mathbb{R}^n)$  s.t.

$$Z(Y) \triangleq \cap_{f \in Y} \{s \in \mathbb{R}^n : \widehat{f}(s) = 0\} = \emptyset,$$

then  $Y = L^1(\mathbb{R}^n)$ .

┘

*Proof:* Cf. [Rud91]P228.

□

**Cor. (35.1.1.2).** If  $\varphi \in L^1(\mathbb{R}^n)$  and  $Y$  is the smallest closed translation-invariant subspace of  $L^1(\mathbb{R}^n)$  containing  $\varphi$ , then  $Y = L^1(\mathbb{R}^n)$  iff  $\widehat{\varphi}(t) \neq 0$  for any  $t \in \mathbb{R}^n$ .

┘

*Proof:* Notice  $Z(Y) = \cap_{f \in Y} \{s \in \mathbb{R}^n : \widehat{f}(s) = 0\} = \{t \in \mathbb{R}^n : \widehat{\varphi}(t) = 0\}$ .

□

**Thm. (35.1.1.3) [Karamata].** Let  $A(t) : \mathbb{R}_+ \rightarrow \mathbb{R}_{\geq 0}$  be a non-decreasing function s.t.

$$F(\sigma) = \int_0^\infty e^{-\sigma t} dA(t)$$

converges for all  $\sigma \in \mathbb{R}_+$ . Suppose  $c \in \mathbb{R}_{\geq 0}, w \in \mathbb{R}_+$  s.t.

$$\lim_{\sigma \rightarrow 0^+} F(\sigma) \sigma^w = c,$$

then we have

$$\lim_{X \rightarrow \infty} \frac{A(X)}{X^w} = \frac{c}{\Gamma(w+1)}.$$

┘

*Proof:* Cf. [Ten15].

□

**Prop. (35.1.1.4).** If  $f : \mathbb{R}_{\geq 1} \rightarrow \mathbb{R}$  is a non-decreasing function and  $\int_1^\infty \frac{f(t)-t}{t^2} dt$  converges, then  $f(x) \sim x, x \rightarrow \infty$ .

┘

*Proof:* Let  $F(x) = \int_1^x \frac{f(t)-t}{t^2} dt$ , then the hypothesis implies that for any  $\lambda > 1$  and  $\varepsilon > 0$ ,  $|F(\lambda x) - F(x)| < \varepsilon$  for  $x$  large.

Suppose there exists  $\lambda \in \mathbb{R}_{>1}$  and a sequence  $(x_n)_{n \in \mathbb{Z}_+}$  s.t.  $\lim_{n \rightarrow \infty} x_n = \infty$ , and  $f(x_n) \geq \lambda x_n$  for each  $n$ , then

$$F(\lambda x_n) - F(x_n) = \int_{x_n}^{\lambda x_n} \frac{f(t)-t}{t^2} dt \geq \int_{x_n}^{\lambda x_n} \frac{\lambda x_n - t}{t^2} dt = \int_1^\lambda \frac{\lambda - t}{t^2} dt = C$$

where  $C$  is a positive constant independent of  $n$ . This clearly contradicts the statement above.

A similar statement shows that there are no  $\lambda \in \mathbb{R}_{>1}$  and sequences  $(x_n)_{n \in \mathbb{Z}_+}$  s.t.  $\lim_{n \rightarrow \infty} x_n = \infty$ , and  $f(x_n) \leq \lambda^{-1} x_n$  for each  $n$ . So  $f(x) \sim x, x \rightarrow \infty$ .  $\square$

### Transcendental Tauberian Theorems

**Thm. (35.1.1.5) [Ikehara-Delange Tauberian theorem, Ikehara[Ike31]/ Wiener[Wie32]/ Delange[Del54]/ Kable[Kab08]].** Let  $F(s) = \sum_{n \in \mathbb{Z}_+} a_n n^{-s}$  be a Dirichlet series with  $a_n \in \mathbb{R}_{\geq 0}$  s.t.

$$F(s) = \frac{1}{(s-a)^{\omega+1}} \sum_{j \in [q]} g_j(s) \log^j \left( \frac{1}{s-a} \right) + g(s).$$

where  $\omega \in \mathbb{R}, q \in \mathbb{N}$ , and  $g_i(s), g(s)$  are holomorphic on a nbhd of the half-plane  $\{s \in \mathbb{C} : \operatorname{Re}(s) \geq a\}$  and  $g_q(a) \neq 0$ . Then

- If  $\omega \notin \mathbb{Z}_-$ , for any  $X \in \mathbb{R}_+$ , we have

$$\mathcal{M}_a(X) \sim \frac{g_q(a)}{a\Gamma(\omega+1)} X^a \log^\omega X \log_{(2)}^q X \text{ (35.1.2.2)}.$$

- If  $\omega = -m - 1$  for some  $m \in \mathbb{N}$  and  $q \in \mathbb{Z}_+$ , we have

$$\mathcal{M}_a(X) \sim (-1)^m m! \frac{q g_q(a)}{a} \frac{X^a \log_{(2)}^{q-1} X}{\log^{m+1} X}.$$

┘

*Proof:* ?

$\square$

## 2 Averages of Arithmetic Functions

**Notation (35.1.2.1).**

- Let  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  be a function.

┘

**Def. (35.1.2.2) [Partial Sums].** Define  $\mathcal{M}_f : \mathbb{R}_{\geq 1} \rightarrow \mathbb{C}$ :

$$\mathcal{M}_f(X) = \sum_{n \in [X]_+} f(n).$$

┘



**Prop. (35.1.2.3).** If  $f : \mathbb{R}_+ \rightarrow \mathbb{C}$  is a  $C^1$  function, then for any  $a < b \in \mathbb{R}_+$ , we have

$$\sum_{a < n \leq b} f(n) = \int_a^b (f(t) + \psi(t)f'(t)) dt + \frac{1}{2}(f(b) - f(a)).$$

where  $\psi(t) = t - \lfloor t \rfloor - \frac{1}{2}$ . ┘

**Prop. (35.1.2.4).** If  $f$  is real-valued, monotone and continuous, then for any  $X \in \mathbb{R}_{\geq 1}$ , we have

$$\left| \mathcal{M}_f(X) - \int_1^X f(y) dy \right| \leq |f(X)| + |f(1)|.$$

┘

**Cor. (35.1.2.5).** For  $k \in \mathbb{Z}_+$  and any  $X \in \mathbb{R}_{\geq 1}$ ,

$$\sum_{n \in [X]_+} \log^k n = X P_k(\log X) + O(\log^k X), \quad P_k(X) = \sum_{m \in [k]} (-1)^{k-m} \frac{k!}{m!} X^m.$$

And

$$\sum_{n \in [X]_+} \log^k(X/n) = k!X + O(\log^k X).$$

┘

*Proof:* For the second equation, notice that

$$\int_1^X \log^k(X/t) dt = -\log^k X + k \int_1^X \log^{k-1}(X/t) dt$$

□

**Cor. (35.1.2.6).** For any  $\varepsilon \in \mathbb{R}_+$  and  $X \in \mathbb{R}_{\geq 1}$ ,

$$\left| \sum_{n \in [X]_+} \frac{1}{n^{1+\varepsilon}} - \varepsilon^{-1}(1 - X^{-\varepsilon}) \right| \leq 1 + X^{-1-\varepsilon}.$$

In particular,

$$\left| \zeta(1+\varepsilon) - \varepsilon^{-1} \right| \leq 1.$$

┘

**Prop. (35.1.2.7)[Estimates for Decreasing Functions].** If  $f$  extends to a  $C^1$ , non-increasing function  $f : \mathbb{R}_{\geq 1} \rightarrow \mathbb{R}_+$  and  $\lim_{x \rightarrow \infty} f(x) = 0$ , then for  $X \in \mathbb{R}_{\geq 1}$ ,

$$\left| \mathcal{M}_f(X) - \int_1^X f(y) dy - \gamma_f \right| \leq f(X), \quad \gamma_f = f(1) + \int_1^\infty \{y\} df(y).$$

┘

*Proof:*

$$\begin{aligned}
& \mathcal{M}_f(X) - \int_1^X f(y) dy \\
&= \int_1^2 (f(1) - f(y)) dy + \int_2^3 (f(2) - f(y)) dy + \dots + \int_{[X]}^X (f([X]) - f(y)) dy \\
&= \int_0^1 (f(1) - f(1 + \{y\})) d\{y\} + \int_0^1 (f(2) - f(2 + \{y\})) d\{y\} + \dots + \int_0^{\{X\}} (f([X]) - f([X] + \{y\})) d\{y\} \\
&= f(1) - f(X) + \int_1^X \{y\} df(y) \\
&= \gamma_f - \int_X^\infty \{y\} df(y) - f(X)
\end{aligned}$$

□

**Cor. (35.1.2.8) [Stieltjes Constant].** For  $k \in \mathbb{N}$  and  $X \in \mathbb{R}_+$ ,

$$\left| \sum_{n \in [X]_+} \frac{\log^k n}{n} - \frac{1}{k+1} \log^{k+1}(X) - (-1)^k k! \gamma_k \right| \leq \frac{\log^k X}{X},$$

where  $\gamma_k$  is called the **Stieltjes-constant**. For  $k = 0$ , this is called the **Euler's constant**

$$\gamma_0 = 1 - \int_1^\infty \{y\} y^{-2} dy \approx 0.57215.$$

┘

**Prop. (35.1.2.9) [Summation by Parts].** If  $f, g : \mathbb{Z}_+ \rightarrow \mathbb{R}$  where  $g$  extends to a  $C^1$ -function on  $\mathbb{R}_{\geq 1}$ , then by (12.3.5.7), for each  $X \in \mathbb{R}_{\geq 1}$ , we have

$$\mathcal{M}_{fg}(X) = \mathcal{M}_f(X)g(X) - \int_1^X \mathcal{M}_f(t)g'(t)dt.$$

In particular,

$$\sum_{n \in [X]_+} f(n) \log \frac{X}{n} = \int_1^X \mathcal{M}_f(t) t^{-1} dt.$$

┘

**Thm. (35.1.2.10) [Estimates for Convolution Functions].** If  $f = g * h : \mathbb{Z}_+ \rightarrow \mathbb{R}$  where  $h$  extends to a monotone and bounded continuous function on  $\mathbb{R}_{\geq 0}$ , then for any  $X \in \mathbb{R}_+$ ,

$$\mathcal{M}_f(X) = \int_0^X F(y) dy + O(\mathcal{M}_{|g|}(X)), \quad F(y) = \sum_{m \in [y]_+} \frac{g(m)}{m} h\left(\frac{y}{m}\right).$$

┘

*Proof:*

$$\mathcal{M}_f(X) = \sum_{m \in [X]_+} g(m) \mathcal{M}_h(X/m).$$

After replacing  $\mathcal{M}_h(y)$  by  $\int_0^y h(t)dt + O(1)$  (35.1.2.4), we get

$$\mathcal{M}_f(X) = \int_0^X \sum_{m \in [X]_+} \frac{g(m)}{m} h\left(\frac{y}{m}\right) dy + O\left(\sum_{m \in [X]_+} |g(m)|\right) = \int_0^X F(y) dy + O(\mathcal{M}_{|g|}(X)).$$

□

**Cor. (35.1.2.11).** For  $X \in \mathbb{R}_{\geq 1}$ , we have

$$\sum_{n \in [X]_+} \frac{\phi(n)}{n} = \frac{6}{\pi^2} X + O(\log X).$$

┘

*Proof:* Because  $\frac{\phi(n)}{n} = \frac{\mu(n)}{n} * \mathbf{1}$ , we can apply (35.1.2.10):

$$F(y) = \sum_{m \in [y]_+} \frac{\mu(m)}{m^2} = \frac{1}{\zeta(2)} + O\left(\frac{1}{y + 1/2}\right).$$

And

$$\int_0^X F(y) dy = \frac{X}{\zeta(2)} + O\left(\int_0^X \frac{1}{y + 1/2} dy\right) = \frac{X}{\zeta(2)} + O(\log X).$$

□

**Cor. (35.1.2.12).** There exists  $C \in \mathbb{R}_+$  s.t. for any  $\delta \in \mathbb{R}_+$ , when  $Y \in \mathbb{R}_+$  is large,

$$\#\{n \leq X : \phi(n) \geq \delta n\} \geq \left(\frac{6}{\pi^2} - \delta\right)X - C \log Y.$$

┘

*Proof:* This follows easily from (35.1.2.11). □

### Summation Formulae

**Prop. (35.1.2.13) [Dirichlet's Hyperbola method].** If  $f = g * h : \mathbb{Z}_+ \rightarrow \mathbb{C}$ , then for any  $X \in \mathbb{R}_+$ ,

$$\mathcal{M}_f(X) = \sum_{a \in [\sqrt{X}]_+} g(a) \mathcal{M}_h\left(\frac{X}{a}\right) + \sum_{b \in [\sqrt{X}]_+} h(b) \mathcal{M}_g\left(\frac{X}{b}\right) - \mathcal{M}_g(\sqrt{X}) \mathcal{M}_h(\sqrt{X}).$$

We can use this to estimate  $\mathcal{M}_f(X)$ . ┘

**Def. (35.1.2.14)  $[\psi_k(x)]$ .** For any  $k \in \mathbb{N}$ , denote  $\psi_k(x) = B_k(\{x\})$ , where  $B_k(X)$  is the Bernoulli-polynomial (10.7.1.13), then each  $\psi_k$  is continuous and periodic of period 1 by (10.7.1.14), and we have

$$\sum_{k \in \mathbb{N}} \psi_k(x) \frac{T^k}{k!} = \frac{T e^{T\{x\}}}{e^T - 1}.$$

When  $k \geq 2$ , we have the absolutely convergent series expansion

$$\psi_k(x) = -k! \sum_{n \in \mathbb{Z}^\times} \frac{e^{2\pi i n x}}{(2\pi i n)^k}.$$

by (10.7.1.13) and (12.11.5.4). And when  $k = 1$ , we have the expansion

$$\psi_1(x) = - \sum_{n \in \mathbb{Z}_+} \frac{\sin(2\pi nx)}{\pi n}$$

which converges pointwise and boundedly away from  $x \in \mathbb{Z}$  by (12.11.5.3). In particular, for any  $N \in \mathbb{Z}_+$  and  $x \in \mathbb{R}$ , we have

$$\psi_1(x) = - \sum_{n \in [N]_+} \frac{\sin(2\pi nx)}{\pi n} + O\left(\frac{1}{1 + \{x\}N}\right).$$

┘

*Proof:* Compute the Fourier-coefficients. The last assertion follows from Abel-summation.  $\square$

**Thm. (35.1.2.15) [Euler-Maclaurin Formula].** Let  $a < b \in \mathbb{Z}$  and  $k \in \mathbb{Z}_+$ . Suppose  $f \in C^k([a, b])$ , then we have

$$\sum_{n \in \mathbb{Z}, a < n \leq b} f(n) = \int_a^b \left( f(x) - \frac{(-1)^k}{k!} \psi_k(x) f^{(k)}(x) \right) dx + \sum_{l \in [k]_+} \frac{(-1)^l}{l!} \left( f^{(l-1)}(b) - f^{(l-1)}(a) \right) B_l \quad (35.1.2.14).$$

where  $B_l$  are the Bernoulli-numbers (10.7.1.12).

In particular, we have

$$\sum_{n \in \mathbb{Z}, a < n \leq b} f(n) = \int_a^b \left( f(x) + \left( \{x\} - \frac{1}{2} \right) f'(x) \right) dx + \frac{1}{2} (f(b) - f(a)).$$

┘

*Proof:* We prove this by induction. For  $k = 1$ , the assertion follows by integration by part. And from  $k$  to  $k + 1$  we use the properties of Bernoulli-polynomials to get that

$$\begin{aligned} \int_a^b \frac{(-1)^k}{k!} \psi_k(x) f^{(k)}(x) dx &= \int_a^b \frac{(-1)^k}{(k+1)!} f^{(k)}(x) d(\psi_{k+1}(x)) \\ &= \int_a^b \frac{(-1)^{k+1}}{(k+1)!} \psi_{k+1}(x) f^{(k+1)}(x) dx - \frac{(-1)^{k+1}}{(k+1)!} (f^{(k)}(b) - f^{(k)}(a)). \end{aligned}$$

$\square$

**Cor. (35.1.2.16).** For each  $N \in \mathbb{Z}_+$  and  $k \in \mathbb{N}$ , we have

$$\sum_{n \in [N]_+} n^k = \frac{N^{k+1}}{k+1} + \sum_{l \in [k]_+} \frac{(-1)^l}{l!} k(k-1) \dots (k-l+2) B_l N^{k-l+1} = \sum_{i \in [k]} (-1)^i \binom{k+1}{i} B_i N^{k+1-i}.$$

┘

**Prop. (35.1.2.17).** If  $a < b \in \mathbb{Z}$  and  $f \in C^1([a, b])$ , then we have

$$\sum_{n \in [a, b] \cap \mathbb{Z}} f(n) - \frac{1}{2} (f(a) + f(b)) = \sum_{n \in [-N, N] \cap \mathbb{Z}} \int_a^b f(x) dx + O\left(\int_a^b \frac{|f'(x)|}{1 + N\{x\}} dx\right).$$

┘

### 3 Average of Multiplicative Functions

**Prop. (35.1.3.1).** If  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}_{\geq 0}$  is multiplicative (2.6.1.19), then for each  $X \in \mathbb{R}_+$ , we have

$$\mathcal{M}_f(X) \leq X \sum_{n \in [X]_+} \frac{f(n)}{n} \leq X \prod_{p \in \text{Prime} \leq X} \left( \sum_{k \in \mathbb{N}} f(p^k) p^{-k} \right).$$

And if  $f$  is non-decreasing on prime powers, then  $h = \mu * f$  is non-negative, in fact  $h(1) = 1$  and  $h(p^k) = f(p^k) - f(p^{k-1})$ . Writing  $f = h * \mathbf{1}$ , we get

$$\begin{aligned} \mathcal{M}_f(X) &= \sum_{m \in [X]_+} h(m) \left\lfloor \frac{X}{m} \right\rfloor \leq X \sum_{m \in [X]_+} \frac{h(m)}{m} \\ &\leq X \cdot \prod_{p \in \text{Prime} \leq X} \left( \sum_{k \in \mathbb{N}} h(p^k) p^{-k} \right) \\ &= X \cdot \prod_{p \in \text{Prime} \leq X} \left( \sum_{k \in \mathbb{N}} f(p^k) p^{-k} \right) \left( 1 - \frac{1}{p} \right) \end{aligned}$$

┘

**Cor. (35.1.3.2).** Situation as in (35.1.3.1), if  $f$  is supported on square-free numbers, then we can take  $h : \mathbb{Z}_+ \rightarrow \mathbb{R}$  to be the multiplicative function s.t. for each  $p \in \text{Prime}$  and  $k \in \mathbb{Z}_+$ , we have

$$h(p^k) = \begin{cases} \max(f(p) - 1, 0) & , k = 1 \\ 0 & , k \geq 2 \end{cases}.$$

Then we have  $f \leq h * \mathbf{1}$ , and

$$\mathcal{M}_f(X) \leq X \cdot \prod_{p \in \text{Prime} \leq X} \left( 1 + \frac{\max(f(p) - 1, 0)}{p} \right).$$

┘

**Prop. (35.1.3.3).** Suppose  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}_{\geq 0}$  satisfies:

- $f$  is completely sub-multiplicative, i.e.  $f(mn) \leq f(m)f(n)$  for  $m, n \in \mathbb{Z}_+$ .
- There exists  $c \in \mathbb{R}_+$  s.t.  $\sum_{m \in [X]_+} f(m) \Lambda(m) \leq cX$  for any  $X \in \mathbb{R}_{\geq 2}$  (By prime number theorem (35.5.7.18), this can be understood as  $f(p)$  being bounded by  $c$  on average).

Then we have

$$\mathcal{M}_f(X) \leq \frac{3cX}{\log X} \sum_{n \in [X]_+} \frac{f(n)}{n} \leq 3cd \frac{X}{\log X} \prod_{p \in \text{Prime} \leq X} \left( 1 + \frac{f(p)}{p} \right), \text{ where } d = \sum_{n \in [X]_+} \frac{f(n)^2}{n^2}.$$

┘

*Proof:* It follows from the hypotheses that

$$\sum_{n \in [X]_+} f(n) \log n = \sum_{nm \leq X} f(nm) \Lambda(m) \leq \sum_{nm \leq X} f(m) f(n) \Lambda(m) \leq cX \sum_{n \in [X]_+} \frac{f(n)}{n}.$$

Thus it follows by partial summation (35.1.2.9) that

$$\mathcal{M}_f(X) \leq \frac{3cX}{\log X} \sum_{n \in [X]_+} \frac{f(n)}{n} \leq 3cd \frac{X}{\log X} \prod_{p \in \text{Prime} \leq X} \left( 1 + \frac{f(p)}{p} \right).$$

□

### Wirsing' Formula

**Thm. (35.1.3.4) [Wirsing's Formula, Wirsing[Wir67]].** If  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}$  is multiplicative and there exists  $\kappa \in \mathbb{R}_{>-1/2}$  s.t.

- $\sum_{n \in [X]_+} \Lambda_f(n) = \kappa \log X + O(1)$  for  $X \in \mathbb{R}_+$  (By Merten's theorem (35.5.7.8), this can be understood as  $f(p)$  being about  $\kappa p^{-1}$  on average).
- $\mathcal{M}_{|f|}(X) \ll \log^{|\kappa|} X$  for  $X \in \mathbb{R}_+$ .

Then for  $X \in \mathbb{R}_+$ , we have

$$\mathcal{M}_f(X) = C(f) \log^\kappa(X) + O(\log^{|\kappa|-1} X),$$

where

$$C(f) = \frac{1}{\Gamma(\kappa + 1)} \prod_{p \in \text{Prime}} \left(1 - \frac{1}{p}\right)^\kappa \sum_{k \in \mathbb{N}} f(p^k).$$

**?** Why  $C(f)$  converges. ⌋

*Proof:* Firstly we notice

$$\prod_{p \in \text{Prime} \cap [X]_+} \left(1 - \frac{1}{p}\right)^\kappa \sum_{k \in [\log_p X]_+} f(p^k) = O\left(\log X^{-\kappa} \mathcal{M}_{|f|}(X^r)\right),$$

Because  $f \cdot L = f * \Lambda_f$ , we have

$$\sum_{n \in [X]_+} f(n) \log n = \sum_{d \in [X]_+} f(d) \sum_{m \in [X/d]_+} \Lambda_f(m) = \sum_{d \in [X]_+} f(d) \left( \kappa \log \frac{X}{d} + O(1) \right).$$

Then we get

$$(\kappa + 1) \sum_{n \in [X]_+} f(n) \log n = \kappa \mathcal{M}_f(X) \log X + O(\log^{|\kappa|} X).$$

Using the equation

$$\sum_{n \in [X]_+} f(n) \log \frac{X}{n} = \int_1^X \mathcal{M}_f(y) y^{-1} dy,$$

we see that

$$\Delta(X) \triangleq \mathcal{M}_f(X) \log X - (\kappa + 1) \int_2^X \mathcal{M}_f(y) y^{-1} dy = O(\log^{|\kappa|} X).$$

So

$$\int_2^X \Delta(y) y^{-1} \log^{-\kappa-2}(y) dy = \int_2^X \mathcal{M}_f(y) y^{-1} \log^{-\kappa-1}(y) dy - (\kappa + 1) \int_2^X y^{-1} \log^{-\kappa-2}(y) \left( \int_2^y \mathcal{M}_f(u) u^{-1} du \right) dy$$

$$(\text{change order of integration}) = \log^{-\kappa-1}(X) \int_2^X \mathcal{M}_f(u) u^{-1} du.$$

Plug in this back to the definition of  $\Delta(X)$  we see that

$$\mathcal{M}_f(X) = \log^\kappa(X) \int_2^X (-\Delta(y)) d\left(\log^{-\kappa-1}(y)\right) + \frac{\Delta(X)}{\log X}.$$

It can be checked that this integral is absolutely convergent, so we have

$$\mathcal{M}_f(X) = (c(f) + r_f(X)) \log^\kappa(X),$$

where

$$c(f) = - \int_2^\infty \Delta(y) d(\log^{-\kappa-1}(y)), \quad r_f(X) = \int_X^\infty (\Delta(y) - \Delta(X)) d(\log^{-\kappa-1}(y)).$$

It follows from the fact that  $\Delta(X) = O(\log^{|\kappa|}(X))$  that we see

$$r_f(X) = O(\log^{|\kappa|-\kappa-1}(X)).$$

To conclude, it suffices to show that

$$c(f) = \frac{1}{\Gamma(\kappa+1)} \prod_{p \in \text{Prime}} \left(1 - \frac{1}{p}\right)^\kappa \sum_{k \in \mathbb{N}} f(p^k).$$

For this, notice that  $L(f; s)$  is absolutely convergent for  $\text{Re}(s) > 0$ , and for these  $s$ , we have

$$\begin{aligned} L(f; s) &= \int_1^\infty y^{-s} d\mathcal{M}_f(y) \\ &= - \int_1^\infty \mathcal{M}_f(y) d(y^{-s}) \\ &= - \int_0^\infty \mathcal{M}_f(e^t) d(e^{-st}) \\ &= - \int_0^\infty \left(c(f) + O(t^{|\kappa|-\kappa-1})\right) t^\kappa d(e^{-st}) \end{aligned}$$

which equals  $\left(c(f) + O(s^{\kappa+1-|\kappa|})\right) s^{-\kappa} \Gamma(\kappa+1)$  when  $s \rightarrow 0_+$ . So

$$c(f) \Gamma(\kappa+1) = \lim_{s \rightarrow 0_+} (\zeta(s+1)^{-\kappa} L(f; s)) = \prod_{p \in \text{Prime}} \left(1 - \frac{1}{p}\right)^\kappa \left(1 + f(p) + f(p^2) + \dots\right) \text{ ?},$$

and the assertion follows.  $\square$

**Cor. (35.1.3.5)**[Wirsing's Formula, Wirsing[Wir62]]. If  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  is a non-negative multiplicative function s.t.

- There exists  $\gamma_1 \in \mathbb{R}_+$ ,  $\gamma_2 \in (1, 2)$  s.t.  $f(p^k) \leq \gamma_1 \gamma_2^k$  for each  $k \in \mathbb{Z}_+$ .
- There exists  $\tau \in \mathbb{R}_{\geq 0}$  s.t.  $\sum_{p \in \text{Prime} \cap [X]_+} f(p) \sim \tau \frac{X}{\log X}$  for  $X \rightarrow \infty$ . Then when  $x \rightarrow \infty$ , we have

$$\mathcal{M}_f(X) \sim \frac{e^{-c\tau}}{\Gamma(\tau)} \frac{X}{\log X} \prod_{p \in \text{Prime} \cap [X]_+} \left(1 + \frac{f(p)}{p} + \frac{f(p^2)}{p^2} + \dots\right).$$

┘

*Proof:* Cf.[Wir62]. ?  $\square$

**Cor. (35.1.3.6)**[Forbidden Prime Divisors]. If  $\mathcal{P} \subset \text{Prime}$  is a set of primes and  $\kappa \in [0, 1/2)$  s.t. for each  $X \in \mathbb{R}_+$  we have

$$\sum_{p \in \text{Prime}_{\leq X}} \frac{\log p}{p} = \kappa \log X + O(1),$$

then the number  $S(\mathcal{P}, X)$  of integers in  $[X]_+$  having no prime divisors in  $\mathcal{P}$  satisfies

$$S(\mathcal{P}, X) = c(\mathcal{P}) \frac{X}{\log^\kappa X} \left( 1 + O\left(\frac{1}{\log^{1-2\kappa} X}\right) \right),$$

where

$$c(\mathcal{P}) = \frac{1}{\Gamma(1-\kappa)} \prod_{p \in \text{Prime}} \left( 1 - \frac{f(p)}{p} \right) \left( 1 - \frac{1}{p} \right)^{-\kappa}.$$

┘

*Proof:* Let  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}$  be the completely multiplicative function s.t.  $f(p) = \mathbf{1}_{\mathcal{P}}(p)$  for  $p \in \text{Prime}$ , and let  $g = (\mu \cdot f) * \mathbf{1}$ , then

$$g(n) = \prod_{p \in S(n)} (1 - f(p)),$$

and

$$S(\mathcal{P}, X) = \mathcal{M}_g(X) = \sum_{d \in [X]_+} \mu(d) f(d) \left\lfloor \frac{X}{d} \right\rfloor = X \sum_{d \in [X]_+} \mu(d) \frac{f(d)}{d} + O(\mathcal{M}_f(X))$$

Wirsing's formula (35.1.3.4) applied to  $\mu(n)f(n)/n$  implies that

$$\sum_{d \in [X]_+} \mu(d) \frac{f(d)}{d} = c(\mathcal{P}) \log^{-\kappa} X + O(\log^{\kappa-1} X)$$

where

$$c(\mathcal{P}) = \frac{1}{\Gamma(1-\kappa)} \prod_{p \in \text{Prime}} \left( 1 - \frac{f(p)}{p} \right) \left( 1 - \frac{1}{p} \right)^{-\kappa}.$$

And also

$$\mathcal{M}_f(X) = O\left(X \log^{\kappa-1}(X)\right)$$

by (35.1.3.3), noticing that

$$\sum_{p \in \text{Prime} \leq X} \frac{f(p)}{p} \leq \kappa \log_{(2)} X$$

by partial summation. Thus the assertion follows. □

**Prop. (35.1.3.7) [Landau1908].** For  $X \in \mathbb{R}_+$ , we have

$$B(X) = \#\{n \in [X]_+ : n = a^2 + b^2 \text{ for some } a, b \in \mathbb{Z}_+\} = C \frac{X}{\sqrt{\log X}} + O\left(\frac{X}{\log^{3/4} X}\right),$$

where

$$C = \frac{1}{\sqrt{2}} \prod_{p \equiv 3 \pmod{4}} \left( 1 - \frac{1}{p^2} \right)^{-1/2}.$$

┘



*Proof:* Cf.[Topics in Number Theory, 2, LeVeque]P261<sup>?</sup>. We only prove the asymptotic.

For  $n \in \mathbb{Z}_+$ , let  $b(n) = \begin{cases} 1 & , n = a^2 + b^2 \text{ for some } a, b \in \mathbb{Z}_+ \\ 0 & , \text{otherwise} \end{cases}$ , then  $B(X) = \mathcal{M}_b(X)$ , and we have

$$L(b; s) = \frac{1}{1-2^{-s}} \prod_{p \equiv 1 \pmod{4}} \frac{1}{1-p^{-s}} \prod_{p \equiv 3 \pmod{4}} \frac{1}{1-p^{-2s}}.$$

So

$$L(b; s)^2 = \zeta(s) L(\chi_{-4}; s) \frac{1}{1-2^{-s}} \prod_{p \equiv 3 \pmod{4}} \frac{1}{1-p^{-2s}}.$$

So  $g(s) = (s-1)L(b; s)^2$  is holomorphic and non-vanishing on a nbhd of the half plane  $\{s \in \mathbb{C} : \operatorname{Re}(s) \geq 1\}$  by (35.5.2.5), and

$$g(1) = \frac{\pi}{2} \prod_{p \equiv 3 \pmod{4}} \left(1 - \frac{1}{p^2}\right)^{-1}$$

by (31.3.5.4). So we get

$$B(X) \sim C \frac{X}{\sqrt{\log X}}$$

by the Tauberian theorem (35.1.1.5). □

$\tau(n)$

**Thm. (35.1.3.8)** [Dirichlet Hyperbola Problem for  $\tau(n)$ , Dirichlet/Voronoi(1903)/Iwaniec-Mozzochi(1988) /Huxley(2003)]. For  $X \in \mathbb{R}_{\geq 1}$ ,

$$|\mathcal{M}_\tau(X) - X \log X - (2\gamma_0 - 1)X| = O(X^{131/416})$$

and

$$|\mathcal{M}_\tau(X) - X \log X - (2\gamma_0 - 1)X| \leq 4\sqrt{X}.$$

┘

*Proof:* <sup>?</sup>

In order to estimate the number of positive integral points under the hyperbola  $\{xy = X\}$ , usual estimates of the formula  $\mathcal{M}_\tau = \sum_{n \in [X]_+} \tau(n) = \sum_{n \in [X]_+} \lfloor \frac{X}{n} \rfloor$  using (35.1.2.10) only gives

$$\mathcal{M}_\tau(X) = X \log X + O(X).$$

But if we use Dirichlet's hyperbola method (35.1.2.13), we get

$$\mathcal{M}_\tau(X) = 2 \sum_{m \leq \sqrt{X}} \lfloor \frac{X}{m} \rfloor - \lfloor \sqrt{X} \rfloor^2,$$

then by (35.1.2.8) we get

$$|\mathcal{M}_\tau(X) - X \log X - (2\gamma_0 - 1)X| \leq 4\sqrt{X}.$$

□

**Cor. (35.1.3.9).** For any  $X \in \mathbb{R}_{\geq 1}$ , we have  $\mathcal{M}_\tau(X) \leq X \log(3X)$ . ┘

*Proof:* This holds by (35.1.3.8) if  $X \geq 18$ . And for the rest, we can check one by one. □

**Prop. (35.1.3.10).** For any  $X \in \mathbb{R}_{\geq 1}$ ,

$$\sum_{n \in [X]_+} \frac{\tau(n)}{n} = P(\log X) + O(1)$$

for some quadratic polynomial  $P$  with leading term  $\frac{1}{2}X^2$ . ┘

*Proof:* Cf. <https://terrytao.wordpress.com/2014/11/23/254a-notes-1-elementary-multiplicative-numbers/#cheap-upper>. □

**Lemma (35.1.3.11)** [Landreau[Lan89]]. For any  $\theta \in \mathbb{R}_+$  and  $k, n \in \mathbb{Z}_+$ , we have

$$\tau^k(n) \leq 2^{2k/\theta} \max_{d|n, d \leq n^\theta} \tau(d)^{2k/\theta}.$$

┘

*Proof:* Write  $n = qm$  where  $q$  has all prime factors  $\geq n^{\theta/2}$ , and  $m$  has all prime factors  $< n^{\theta/2}$ . Then we can write  $m = n_1 \dots n_r$  where  $n_1, \dots, n_{r-1} \in (n^{\theta/2}, n^\theta]$ , and  $n_r \in [n^\theta]_+$  s.t.  $n_{r-1}n_r > n^\theta$  (if  $r \geq 2$ ). So  $r < 2/\theta$ , and

$$\tau(n) \leq 2^{2/\theta} \tau(n_1) \dots \tau(n_r) \leq 2^{2/\theta} \tau(n_i)^{2/\theta}$$

for some  $n_i$ . □

**Thm. (35.1.3.12)** [van.der.Corput[Cor39]/Erdős[Erd52]]. For any polynomial  $P : \mathbb{Z} \rightarrow \mathbb{Z}$  and  $m \in \mathbb{Z}_+$ ,  $X \in \mathbb{R}_+$ , we have

$$X \ll \sum_{n \in [X]_+} \tau^m(P(n)) \ll X \log^{O(1)} X.$$

And if  $P$  is irreducible, then

$$\sum_{n \in [X]_+} \tau(P(n)) \sim X \log X.$$

And also

$$\mathcal{M}_{\tau^2(n)}(X) \sim \frac{\pi^2}{36} X \log^3 X.$$

┘

*Proof:* Cf. <https://terrytao.wordpress.com/2011/07/23/erdos-divisor-bound/#mcc>.

The assertion about  $\tau^2(n)$  follows from Tauberian theorem (35.1.1.5) because  $L(\tau^2; s) = \zeta(2s)\zeta(s)^{-4}$ . □

**Cor. (35.1.3.13)** [Hooley[Hoo63]]. For  $X \in \mathbb{R}_+$ , we have

$$\sum_{n \in [X]_+} \tau(n^2 + 1) = \frac{3}{\pi} X \log X + O(X).$$

┘

*Proof:*

$$\sum_{n \in [X]_+} \tau(n^2 + 1) = 2 \sum_{n \in [X]_+} \sum_{d \leq n, d|n^2+1} 1 = 2 \sum_{d \in [X]_+} \sum_{d \leq n \leq X, d|n^2+1} 1 = 2 \sum_{d \in [X]_+} \left( \frac{X}{d} \rho(d) + O(\rho(d)) \right)$$

where  $\rho(d)$  is the number of solutions in  $\mathbb{Z}/(d)$  of the equation  $X^2 + 1 = 0$ .  $\rho(d)$  is multiplicative, and its  $L$ -series is given by

$$\begin{aligned} L(\rho; s) &= \left(1 + \frac{1}{2^s}\right) \prod_{p \in \text{Prime}_{\geq 3}, p \equiv 1 \pmod{4}} \left(1 + \frac{2}{p^s} + \frac{2}{p^{2s}} + \dots\right) \\ &= \left(1 + \frac{1}{2^s}\right) \prod_{p \in \text{Prime}_{\geq 3}, p \equiv 1 \pmod{4}} \frac{1 + p^{-s}}{1 - p^{-s}} \\ &= \zeta(s) \frac{L(\chi_{-4}; s)}{\zeta(2s)} \end{aligned}$$

Denote

$$\frac{L(\chi_{-4}; s)}{\zeta(2s)} = \sum_{n \in \mathbb{Z}_+} \frac{b(n)}{n^s} = L(b; s),$$

then  $|b(n)| \leq 1$ , and  $\rho = \mathbf{1} * b$ . And we have:

$$\mathcal{M}_b(X) = \sum_{t^2 l \leq X} \mu(t) \chi_{-4}(l) = \sum_{t \in [\sqrt{X}]_+} O(1) = O(\sqrt{X}).$$

(where the first equality follows from the fact that  $b = \chi_{-4} * \mu(\sqrt{\cdot})$ ). So

$$\begin{aligned} \sum_{n \in [X]_+} \frac{\rho(n)}{n} &= \sum_{a \in [\sqrt{X}]_+} \frac{1}{a} \sum_{\sqrt{X} < m \leq X/a} \frac{b(m)}{m} + \sum_{a \in [\sqrt{X}]_+} \frac{b(a)}{a} (\log(X/a) + O(1)) \\ (\text{partial summation}) &= \sum_{a \in [\sqrt{X}]_+} \frac{1}{a} \cdot O(1/X^{1/4}) + \sum_{a \in [\sqrt{X}]_+} \frac{b(a)}{a} (\log(X/a) + O(1)) \\ &= L(b; 1)(\log X + O(1)) + L^\wedge(b; 1) \\ &\quad + O\left(\log X \left| \sum_{m \geq \sqrt{X}} \frac{b(m)}{m} \right| + \left| \sum_{m \geq \sqrt{X}} \frac{b(m)}{m} \log m \right| + \frac{\log X}{X^{1/4}}\right) \\ (\text{partial summation}) &= L(b; 1) \log X + O(1) \\ (31.3.5.4) &= \frac{3}{2\pi} \log X + O(1). \end{aligned}$$

Similarly,

$$\begin{aligned} \mathcal{M}_\rho(X) &= \sum_{a \in [\sqrt{X}]_+} \sum_{\sqrt{X} < m \leq X/a} b(m) + \sum_{a \in [\sqrt{X}]_+} b(a)(X/a + O(1)) \\ (\text{partial summation}) &= O(X^{3/4}) + O(X^{3/4}) + O(\sqrt{X}) \\ &= O(X^{3/4}) \end{aligned}$$

And the theorem follows.  $\square$

**Thm. (35.1.3.14).** For  $k \in \mathbb{Z}_{\geq 2}$ , denote  $\tau_k \triangleq \mathbf{1}^{*k}$ , then for each  $X \in \mathbb{R}_+$ , we have

- There exists  $C_{k,l} \in \mathbb{R}_+$  s.t.  $\mathcal{M}_{\tau_k^l}(X) \leq C_{k,l} X \log^{k^l-1} X$ .
- There exists some  $C_k \in \mathbb{R}_+$  s.t. for  $X \in \mathbb{R}_+$ ,

$$\mathcal{M}_{\tau_k(n)^2/n}(X) \sim C_k \log^{k^2}(X).$$

In particular, for each  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_{k,\varepsilon} \in \mathbb{R}_+$  s.t.  $\tau_k(n) \leq C_{k,\varepsilon} n^\varepsilon$ . ┘

*Proof:* 1 follows from (35.1.3.1). Cf. [I-K04]P23 ?. □

**Prop. (35.1.3.15).** For  $X \in \mathbb{R}_+$  and  $k \in \mathbb{N}$ ,

$$\sum_{n \in [X]_+} \frac{\tau^k(n)}{\phi(n)} \ll \log^{k+1} X.$$

┘

*Proof:* ? □

$\sigma(n)$

**Thm. (35.1.3.16) [Walfisz(1963)].** For  $X \in \mathbb{R}_+$ ,

$$\mathcal{M}_\sigma(X) = \frac{\pi^2}{12} X^2 + O(X \log^{2/3} X).$$

┘

*Proof:* ? We only prove that  $\mathcal{M}_\sigma(X) = \frac{\pi^2}{12} X^2 + O(X \log X)$ :

$$\begin{aligned} \mathcal{M}_\sigma(X) &= \sum_{m,d \in \mathbb{Z}_+, md \leq X} m \\ &= \frac{1}{2} \sum_{d \in [X]_+} \left\lfloor \frac{X}{d} \right\rfloor \left( \left\lfloor \frac{X}{d} \right\rfloor + 1 \right) \\ &= \frac{1}{2} \sum_{d \in [X]_+} \frac{X^2}{d^2} + O \left( X \sum_{d \in [X]_+} \frac{1}{d} \right) \\ &= \frac{\pi^2}{12} X^2 + O(X \log X) \text{ (35.1.2.8)}. \end{aligned}$$

□

$\mu(n)$

**Prop. (35.1.3.17) [ $\mu(n)$  and Prime Number Theorem].**  $\mathcal{M}_\mu(X) \sim X$  for  $X \in \mathbb{R}_+$  is equivalent to  $\psi(X) \sim X$ , i.e. the crude prime number theorem (35.5.7.18). ┘

*Proof:* Cf. [I-K04]P32. □

**Prop. (35.1.3.18).** For any  $X \in \mathbb{R}_+$ ,

$$\left| \sum_{n \in [X]_+} \frac{\mu(n)}{n} \right| \leq 1.$$

┘

*Proof:* Because  $\delta = 1 * \mu$ , we have

$$\sum_{d \in [X]_+} \mu(d) \lfloor \frac{X}{d} \rfloor = 1,$$

and then estimating the integral parts gives the result. □

## 4 Densities and Distributions

### Densities

**Def. (35.1.4.1) [Density of Places].** For  $F \in \mathbf{NField}$  and  $T \subset \Sigma_F^{\text{fin}}$ ,  $T$  is said to have

- **natural density**  $\delta \in [0, 1]$  if

$$\lim_{X \rightarrow \infty} \frac{\#\{v \in \Sigma_F^{\text{fin}} : v \in T, \|v\| \leq X\}}{\#\{v \in \Sigma_F^{\text{fin}} : \|v\| \leq X\}} = \delta.$$

- **polar density**  $m/n \in \mathbb{Q}$  if

$$\zeta_{F,T} = \prod_{v \in T} \frac{1}{1 - \|v\|^{-s}}$$

satisfies  $\zeta_{F,T}^n$  extends to a nbhd of  $s = 1$  with a pole of order  $m$ .

- **Dirichlet-density** or **analytic density**  $\delta \in [0, 1]$  iff

$$\lim_{s \rightarrow 1^+} \frac{\sum_{v \in T} \frac{1}{\|v\|^s}}{\log \frac{1}{s-1}} = \delta.$$

┘

**Prop. (35.1.4.2).** If the polar density exists, then so does the Dirichlet density, and they are equal. If the natural density exists, then so does the Dirichlet density, and they are equal. ┘

*Proof:* Cf. [Mil20]P194. □

**Def. (35.1.4.3) [Density of Ideals].** For  $A \subset \mathbb{Z}_+$ ,  $A$  is said to have

- **natural density**  $d(A) \in [0, 1]$  if

$$\lim_{X \rightarrow \infty} \frac{\#\{n \in A \cap [X]_+\}}{X} = d(A).$$

In any case, we can define the **upper natural density**

$$\bar{d}(A) = \overline{\lim}_{X \rightarrow \infty} \frac{\#\{n \in A \cap [X]_+\}}{X}.$$

- **logarithmic density**  $m/n \in \mathbb{Q}$  if

$$\lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{n \in A \cap [X]_+} \frac{1}{n} = \delta(A).$$

- **analytic density**  $\delta(A) \in [0, 1]$  iff

$$\lim_{s \rightarrow 1^+} (s-1) \sum_{n \in A \cap [X]_+} \frac{1}{n^s} = \delta(A).$$

┘

**Prop. (35.1.4.4) [Logarithmic Density and Natural Density].** For  $A \subset \mathbb{Z}_+$ , if  $A$  has natural density, then it also has a logarithmic density, and these two densities are equal. ┘

*Proof:* If we denote  $A(X) = \#A \cap [X]_+$ , then

$$\sum_{n \in A \cap [X]_+} \frac{1}{n} = \frac{A(X)}{X} + \int_1^X \frac{A(t)}{t^2} dt.$$

From this the assertion follows easily. □

**Prop. (35.1.4.5) [Logarithmic Density and Analytic Density].** For  $A \subset \mathbb{Z}_+$ ,  $A$  has logarithmic density iff it has analytic density, and in this case, the two densities are equal. ┘

*Proof:* For  $X \in \mathbb{R}_{\geq 1}$  and  $\sigma \in \mathbb{R}_{>1}$ , by Abel summation,

$$(\sigma-1) \sum_{n \in A} \frac{1}{n^\sigma} = (\sigma-1)^2 \int_1^\infty \frac{\sum_{n \in A \cap [t]_+} 1/n}{t^\sigma} dt.$$

If  $\sum_{n \in A \cap [t]_+} 1/n = (\delta + o(1)) \log t$ , then for each  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_\varepsilon \in \mathbb{R}_+$  s.t.

$$\left| (\sigma-1)^2 \int_1^\infty \frac{\sum_{n \in A \cap [t]_+} 1/n}{t^\sigma} dt - \delta \right| \leq \varepsilon + (\sigma-1)C_\varepsilon$$

Thus

$$\lim_{s \rightarrow 1^+} (s-1) \sum_{n \in A \cap [X]_+} \frac{1}{n^s} = \delta.$$

Conversely, if

$$\lim_{s \rightarrow 1^+} (s-1) \sum_{n \in A \cap [X]_+} \frac{1}{n^s} = \lim_{h \rightarrow 0^+} h \int_0^\infty e^{-ht} dL(e^t) = \delta.$$

where  $L(X) = \sum_{n \in A \cap [X]_+} 1/n$ , then by Karamata theorem (35.1.1.3),

$$\lim_{t \rightarrow \infty} \frac{L(e^t)}{t} = \delta$$

□

### Limiting Distributions

**Def. (35.1.4.6)[Limiting Distributions].** A real arithmetic function  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}$  is said to have **limit law**  $F$  where  $F$  is a real non-decreasing NBV(12.3.3.24) with  $F(\infty) = 1$  if the discrete measures

$$\nu_N(f) = \frac{1}{N} \sum_{n \leq N} \delta_{f(n)}$$

converges to  $\mu_F$ (12.3.3.25). Equivalently, if we denote

$$F_N(x) = \frac{1}{N} \# \{n \in [N]_+, f(n) \leq x\},$$

then

- $F(x) = \lim_{N \rightarrow \infty} F_N(x)$  exists for all  $x$  belonging to an open dense subset  $E \subset \mathbb{R}$ .
- $\lim_{x \in E, x \rightarrow -\infty} F(x) = 0, \lim_{x \in E, x \rightarrow +\infty} F(x) = 1$ .

Because if this is the case then we can define  $F$  by right limits. ┘

**Prop. (35.1.4.7).** For  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}$ , if for any  $\varepsilon \in \mathbb{R}_+$ , there is a function  $a_\varepsilon(n) : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$  s.t.

- $\lim_{\varepsilon \rightarrow 0} \lim_{T \rightarrow \infty} \bar{d}\{n \in \mathbb{Z}_+ | a_\varepsilon(n) > T\} = 0$ .
- $\lim_{\varepsilon \rightarrow 0} \bar{d}\{n \in \mathbb{Z}_+ : |f(n) - f(a_\varepsilon(n))| > \varepsilon\} = 0$ .
- For any  $a \in \mathbb{Z}_+$ , the natural density  $d(a, \varepsilon) = d\{n \in \mathbb{Z}_+ | a_\varepsilon(n) = a\}$ (35.1.4.3) exists.

Then  $f$  has a limit law(35.1.4.6). ┘

*Proof:* Select, as  $\eta \rightarrow 0+$ , two functions  $\varepsilon = \varepsilon(\eta) \rightarrow 0+$  and  $T = T(\varepsilon(\eta)) \rightarrow +\infty$ , s.t.

$$\bar{d}\{n \in \mathbb{Z}_+ | a_\varepsilon(n) > T(\varepsilon)\} \leq \eta, \quad \bar{d}\{n \in \mathbb{Z}_+ : |f(n) - f(a_\varepsilon(n))| > \varepsilon\} < \eta.$$

Define

$$F(x, \eta) = \sum_{a \leq T(\varepsilon), f(a) \leq x} d(a, \varepsilon), \quad F(x) = \overline{\lim}_{\eta \rightarrow 0} F(x, \eta),$$

then  $F(x)$  is non-decreasing. And for any continuous point  $x$  of  $F$ ,

$$F_N(x) \leq \frac{1}{N} \sum_{n \leq N, a_\varepsilon(n) \leq T(\varepsilon), f(a_\varepsilon(n)) \leq x + \varepsilon} 1 + \frac{1}{N} \sum_{a_\varepsilon(n) > T(\varepsilon)} 1 + \frac{1}{N} \sum_{n \leq N, |f(n) - f(a_\varepsilon(n))| > \varepsilon} 1.$$

where  $\eta$  is arbitrary. So by the hypothesis,

$$\overline{\lim}_{N \rightarrow \infty} F_N(x) \leq \overline{\lim}_{\eta \rightarrow 0+} (F(x + \varepsilon(\eta), \eta) + 2\eta) = F(x).$$

Symmetrically, we can also get

$$\underline{\lim}_{N \rightarrow \infty} F_N(x) \leq \overline{\lim}_{\eta \rightarrow 0+} (F(x + \varepsilon(\eta), \eta) + 2\eta) = F(x).$$

So we have  $\lim_{N \rightarrow \infty} F_N(x) = F(x)$ .

Next, if  $x > \max\{f(a) | a \leq T(\varepsilon)\} + \varepsilon$ , then  $f(n) > x$  implies either  $a_\varepsilon(n) > T(\varepsilon)$  or  $|f(n) - f(a_\varepsilon(n))| > \varepsilon$ , so the corresponding limit density  $F(x) > 1 - \eta$ . So we get  $F(\infty) = 1$ . Similarly we get  $F(-\infty) = 0$ . Thus by(35.1.4.6), we see  $f$  has a limit law. □

## 5 Distributions of Additive Functions

References are [Ten15]Chap3.3.

**Thm. (35.1.5.1)**[Turán-Kubilius Inequality, Turán[Tur36]/Kubilius[Kub55]]. There exists  $\kappa \in \mathbb{R}_+$  s.t. if  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  is an additive function, then for  $X \in \mathbb{R}_+$ , we have

$$\frac{1}{X} \sum_{n \in [X]_+} |f(n) - E(X)|^2 \leq \kappa D^2(X),$$

where

$$E(X) = \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} f(p^\alpha) p^{-\alpha} (1 - p^{-1}), \quad D(X) = \left( \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} |f(p^\alpha)|^2 p^{-\alpha} \right)^{1/2}.$$

*Proof:* Firstly we can reduce to the case that  $f$  is real-valued and non-negative: Denote  $f = f_1 - f_2 + i(f_3 - f_4)$  where  $f_i$  are real-valued and non-negative, then the inequalities for  $f_i$  implies the inequalities for  $f$ .

Then

$$\begin{aligned} \mathcal{M}_f(X) &= \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} f(p^\alpha) \left( \left[ \frac{X}{p^\alpha} \right] - \left[ \frac{X}{p^{\alpha+1}} \right] \right) \\ &= X E(X) + O \left( \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} |f(p^\alpha)| \right) \\ (\text{Cauchy}) &= X E(X) + O \left( D(X) \left( \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} p^\alpha \right)^{1/2} \right) \\ (35.5.7.14) &= X E(X) + X O \left( \frac{D(X)}{\sqrt{\log X}} \right). \end{aligned}$$

And

$$\begin{aligned} \mathcal{M}_{f^2}(X) &= \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} \sum_{q \in \text{Prime}, \beta \in [\log_q X]_+} f(p^\alpha) f(q^\beta) \sum_{n \in [X]_+, v_p(n) = \alpha, v_q(n) = \beta} 1 \\ &= X \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} \sum_{q \in \text{Prime}, \beta \in [\log_q X]_+, q \neq p} f(p^\alpha) p^{-\alpha} f(q^\beta) q^{-\beta} (1 - p^{-1})(1 - q^{-1}) \\ &\quad + X \sum_{p \in \text{Prime}, \alpha \in [\log_p X]_+} f^2(p^\alpha) p^{-\alpha} (1 - p^{-1}) + O \left( \sum_{p \neq q \in \text{Prime}, \alpha, \beta \in \mathbb{Z}_+, p^\alpha q^\beta \leq X} |f(p^\alpha) f(q^\beta)| \right) \\ &= X E(X)^2 + O(X D^2(X)) \end{aligned}$$

where the last estimation follows from Cauchy-Schwarz inequality and the fact

$$\sum_{p \neq q \in \text{Prime}, \alpha, \beta \in \mathbb{Z}_+, p^\alpha q^\beta \leq X} p^\alpha q^\beta = \frac{X^2}{\log X} O \left( \sum_{p \in \text{Prime}, \alpha \in [\log_p X/2]_+} p^{-\alpha} \right) = O(X^2).$$



Thus

$$\begin{aligned} \sum_{n \in [X]_+} |f(n) - E(X)|^2 &= \mathcal{M}_{f^2}(X) - 2\mathcal{M}_f(X)E(X) + XE(X)^2 \\ &= O\left(XD^2(X) + \frac{X}{\sqrt{\log X}}E(X)D(X)\right) \\ &= O\left(XD^2(X)\right) \end{aligned}$$

where the last estimation follows from the estimation

$$E^2(X) = O(\log XD^2(X)),$$

which is true by Cauchy-Schwarz inequality.  $\square$

**Cor. (35.1.5.2)** [Hardy-Ramanujan[H-R18]]. There exists  $\kappa \in \mathbb{R}_+$  s.t. for any  $X \in \mathbb{R}_{\geq 3}$ , we have

$$\sum_{n \leq X} \left| \omega(n) - \log_{(2)} X \right|^2 \leq \kappa X \log_{(2)}(X).$$

In particular, for  $z \in \mathbb{R}_+$ ,

$$\#\{n \in [X]_+, |\omega(n) - \log_{(2)}(X)| > z\sqrt{\log_{(2)}(X)}\} \leq \frac{\kappa}{z^2} X.$$

┘

**Thm. (35.1.5.3)** [Fundamental Theorem of Probabilistic Number Theory, Erdős-Kac]. The probabilistic distribution of

$$\frac{\omega(n) - \log_{(2)}(X)}{\sqrt{\log_{(2)}(X)}}$$

is the standard distribution.  $\square$

*Proof:* Cf. [Erdős-Kac, The Gaussian law of errors in the theory of additive number theoretic functions].  $\square$

## 6 Large Sieve method

### Bilinear Forms

**Prop. (35.1.6.1)** [Duality Principle]. Let  $(\varphi(m, n)) \in \mathbb{C}$  and suppose there exists  $\Delta \in \mathbb{R}_+$  s.t. for any  $(\beta_n) \subset \mathbb{C}$  we have

$$\sum_m \left| \sum_n \beta_n \varphi(m, n) \right|^2 \leq \Delta \|\beta\|^2,$$

then for any  $(\alpha_n)$ , we also have

$$\sum_n \left| \sum_m \alpha_m \varphi(m, n) \right|^2 \leq \Delta \|\alpha\|^2.$$

┘

*Proof:*

$$\begin{aligned} \left( \sum_n \left| \sum_m \alpha_m \phi(m, n) \right|^2 \right)^2 &= \left( \sum_n \sum_m \alpha_m \phi(m, n) \left( \sum_k \overline{\alpha_k \phi(k, n)} \right) \right)^2 \\ &\quad (\text{Cauchy}) \leq \|\alpha\|^2 \sum_m \left| \sum_n \phi(m, n) \left( \sum_k \overline{\alpha_k \phi(k, n)} \right) \right|^2 \\ &\leq \Delta \|\alpha\|^2 \sum_n \left| \sum_k \alpha_k \phi(k, n) \right|^2 \end{aligned}$$

So the assertion follows.  $\square$

**Lemma (35.1.6.2) [Averaging Exponential Sums].** For  $(x_m) \in \mathbb{R}$  and  $(\alpha_m) \in \mathbb{C}$ , we have

$$\int_{-Y}^Y \left| \sum_m \alpha_m e^{2\pi i x_m y} \right|^2 dy \leq 5Y \sum_{2Y|x_{m_1} - x_{m_2}| < 1} |\alpha_{m_1} \alpha_{m_2}|.$$

┘

*Proof:* Cf. [I-K04]P171 ?.  $\square$

**Thm. (35.1.6.3) [Bilinear Exponential Sums].** Let  $(x_m), (y_n) \in \mathbb{R}$  s.t.  $|x_m| \leq X, |y_n| \leq Y$ , then for any  $(\alpha_m), (\beta_n) \in \mathbb{C}$ , we have

$$\left| \sum_m \sum_n \alpha_m \beta_n e^{2\pi i x_m y_n} \right|^2 \leq 25(XY + 1) \left( \sum_{2Y|x_{m_1} - x_{m_2}| < 1} |\alpha_{m_1} \alpha_{m_2}| \right) \left( \sum_{2X|y_{n_1} - y_{n_2}| < 1} |\beta_{n_1} \beta_{n_2}| \right).$$

┘

*Proof:* Let  $\varepsilon = (4X)^{-1}$  and  $\gamma_m = \frac{\pi x_m \alpha_m}{\sin(2\pi \varepsilon x_m)}$ , then  $|\gamma_m| \leq \pi X |\alpha_m|$ . We have

$$e^{2\pi i xy} = \frac{\pi x}{\sin(2\pi \varepsilon x)} \int_{y-\varepsilon}^{y+\varepsilon} e^{2\pi i xt} dt,$$

so

$$\begin{aligned} \left| \sum_m \sum_n \gamma_m \beta_n e^{2\pi i x_m y_n} \right|^2 &= \left| \sum_n \beta_n \int_{y_n-\varepsilon}^{y_n+\varepsilon} \gamma_m e^{2\pi i x_m y} dy \right|^2 \\ &\leq \left| \int_{-Y-\varepsilon}^{Y+\varepsilon} \left( \sum_{|y_n-y| < \varepsilon} |\beta_n| \right) \left| \sum_m \gamma_m e^{2\pi i x_m y} \right| dy \right|^2 \\ &\leq \left( \int_{-Y-\varepsilon}^{Y+\varepsilon} \left( \sum_{|y_n-y| < \varepsilon} |\beta_n| \right)^2 dy \right) \left( \int_{-Y-\varepsilon}^{Y+\varepsilon} \left| \sum_m \gamma_m e^{2\pi i x_m y} \right|^2 dy \right) \\ (35.1.6.2) &\leq 2\varepsilon \left( \sum_{|y_{n_1} - y_{n_2}| < 2\varepsilon} |\beta_{n_1} \beta_{n_2}| \right) 5(Y + \varepsilon)(\pi X)^2 \left( \sum_{2Y|x_{m_1} - x_{m_2}| < 1} |\alpha_{m_1} \alpha_{m_2}| \right). \end{aligned}$$

Here  $10\varepsilon(Y + \varepsilon)(\pi X)^2 = \frac{5\pi^2}{2}(XY + \frac{1}{4}) < 25(XY + 1)$ , completing the proof.  $\square$

**Cor. (35.1.6.4) [Well-Spaced Bilinear Exponential Sums].** Let  $(x_m), (y_n) \in \mathbb{R}$  s.t.  $|x_m| \leq X, |y_n| \leq Y$  and for any  $m_1 \neq m_2, n_1 \neq n_2$ , we have  $2|x_{m_1} - x_{m_2}| \geq A, 2|y_{n_1} - y_{n_2}| \geq B$ , then for any  $(\alpha_m), (\beta_n) \subset \mathbb{C}$ , we have

$$\left| \sum_m \sum_n \alpha_m \beta_n e^{2\pi i x_m y_n} \right| \leq 5 \sqrt{\max(X, B^{-1}) \cdot \max(Y, A^{-1}) + 1} \cdot \|\alpha\| \|\beta\|.$$

┘

**Cor. (35.1.6.5) [Bilinear Exponential Sums for Smooth Functions].** Let  $F, G \in \mathbb{R}_+, M, N \in \mathbb{Z}_+$  and  $f, g$  be real functions on  $[M, 2M], [N, 2N]$  resp. s.t.  $f \ll F, g \ll G$  and  $|f'| \geq FM^{-1}, |g'| \geq GN^{-1}$ , then for any  $(\alpha_m), (\beta_n) \in \mathbb{C}$ , we have

$$\left| \sum_m \sum_n \alpha_m \beta_n e^{2\pi i f(m)g(n)} \right| \ll \sqrt{\frac{(FG + M)(FG + N)}{FG}} \cdot \|\alpha\| \|\beta\|.$$

┘

### Additive Large Sieve Methods

**Prop. (35.1.6.6) [Hilbert's Inequality, Schur[Sch11]/Montgomery-Vaughan[M-V74]].** Suppose  $\delta \in \mathbb{R}_+$  and  $(\lambda_r) \subset \mathbb{R}$  s.t.  $|\lambda_r - \lambda_s| \geq \delta$  for any  $r \neq s$ , then for any  $(z_r) \in \mathbb{C}$ , we have

$$\left| \sum_{r \neq s} \frac{z_r \bar{z}_s}{\lambda_r - \lambda_s} \right| \leq \frac{\pi}{\delta} \sum_r |z_r|^2.$$

┘

*Proof:* By Cauchy inequality, it suffices to prove that

$$\sum_r \left| \sum_{s \neq r} \frac{z_s}{\lambda_r - \lambda_s} \right|^2 \leq \frac{\pi^2}{\delta^2} \sum_r |z_r|^2.$$

The LHS equals

$$\begin{aligned} \sum_s \sum_t \bar{z}_s z_t \sum_{r \neq s, r \neq t} \frac{1}{(\lambda_r - \lambda_s)(\lambda_r - \lambda_t)} &= \sum_s |z_s|^2 \sum_{r \neq s} \frac{1}{(\lambda_r - \lambda_s)^2} + \sum_{s \neq t} \frac{\bar{z}_s z_t}{\lambda_s - \lambda_t} \sum_{r \neq s, r \neq t} \left( \frac{1}{\lambda_r - \lambda_s} - \frac{1}{\lambda_r - \lambda_t} \right) \\ &= \sum_s |z_s|^2 \sum_{r \neq s} \frac{1}{(\lambda_r - \lambda_s)^2} + \sum_{s \neq t} \frac{\bar{z}_s z_t}{\lambda_s - \lambda_t} \left( \sum_{r \neq s} \frac{1}{\lambda_r - \lambda_s} - \sum_{r \neq t} \frac{1}{\lambda_r - \lambda_t} \right) \\ &\quad + 2 \sum_{s \neq t} \frac{\bar{z}_s z_t}{(\lambda_r - \lambda_s)^2} \\ &\leq 3 \sum_s |z_s|^2 \sum_{r \neq s} \frac{1}{(\lambda_r - \lambda_s)^2} + \sum_{s \neq t} \frac{\bar{z}_s z_t}{\lambda_s - \lambda_t} \left( \sum_{r \neq s} \frac{1}{\lambda_r - \lambda_s} - \sum_{r \neq t} \frac{1}{\lambda_r - \lambda_t} \right). \end{aligned}$$

Now we can use the idea of extremal, when  $\sum_r \left| \sum_{s \neq r} \frac{z_s}{\lambda_r - \lambda_s} \right|^2 / \sum_r |z_r|^2$  takes its maximum, we can assume  $(z_r)$  is an eigenvector for the Hermitian matrix

$$A = (a_{rs})_{rs}, \text{ where } a_{rs} = \begin{cases} \frac{i}{\lambda_r - \lambda_s} & , r \neq s \\ 0 & , r = s \end{cases}.$$

Thus there exists  $\nu \in \mathbb{R}$  s.t.

$$\sum_{r \neq s} \frac{z_r}{\lambda_r - \lambda_s} = i\nu z_s.$$

In this case,

$$\begin{aligned} \sum_{s \neq t} \frac{\bar{z}_s z_t}{\lambda_s - \lambda_t} \sum_{r \neq s} \left( \frac{1}{\lambda_r - \lambda_s} - \sum_{r \neq t} \frac{1}{\lambda_r - \lambda_t} \right) &= \sum_{r \neq s} \frac{\bar{z}_s}{\lambda_r - \lambda_s} \sum_{t \neq s} \frac{z_t}{\lambda_s - \lambda_t} - \sum_{r \neq t} \frac{z_t}{\lambda_r - \lambda_t} \sum_{s \neq t} \frac{\bar{z}_s}{\lambda_s - \lambda_t} \\ &= -i \sum_{r \neq s} \frac{|z_s|^2}{\lambda_r - \lambda_s} + i \sum_{r \neq s} \frac{|z_t|^2}{\lambda_r - \lambda_t} = 0. \end{aligned}$$

And we finish because by hypothesis

$$\sum_{r \neq s} \frac{1}{(\lambda_r - \lambda_s)^2} \leq \frac{2}{\delta^2} \zeta(2) = \frac{\pi^2}{3\delta^2}.$$

□

**Cor. (35.1.6.7).** Suppose  $\delta \in \mathbb{R}_+$  and  $\{\alpha_r\} \subset \mathbb{R}/\mathbb{Z}$  s.t.  $d(\alpha_r, \alpha_s) \geq \delta$  if  $r \neq s$ , then for any  $(z_r) \subset \mathbb{C}$ , we have

$$\left| \sum_{r \neq s} \frac{z_r \bar{z}_s}{\sin(\pi(\alpha_r - \alpha_s))} \right| \leq \delta^{-1} \sum_r |z_r|^2.$$

┘

*Proof:* We take a set of representatives of  $\alpha_r \in [0, 1]$ , and apply (35.1.6.6) to the double-indexed set of numbers  $z_{m,r} = (-1)^m z_r$  and  $\lambda_{m,r} = \alpha_r + m$  with  $m \in [K]_+$ , where  $K \in \mathbb{Z}_+$  is large. Then we get

$$\left| \sum_{(r,m) \neq (s,n)} (-1)^{m-n} \frac{z_r \bar{z}_s}{m - n + \alpha_r - \alpha_s} \right| \leq \frac{\pi K}{\delta} \sum_r |z_r|^2.$$

We can replace the condition  $(r, m) \neq (s, n)$  by  $r \neq s$  because if  $r = s$ , the summation cancels out. So we get

$$\left| \sum_{r \neq s} (-1)^{m-n} z_r \bar{z}_s \sum_{k=-K}^K \left( 1 - \frac{|k|}{K} \right) \frac{(-1)^k}{k + \alpha_r - \alpha_s} \right| \leq \frac{\pi}{\delta} \sum_r |z_r|^2.$$

We get the desired result by taking  $K \rightarrow \infty$ , and (12.4.3.12). □

**Cor. (35.1.6.8).** Suppose  $\delta \in \mathbb{R}_+$  and  $\{\alpha_r\} \subset \mathbb{R}/\mathbb{Z}$  s.t.  $d(\alpha_r, \alpha_s) \geq \delta$  if  $r \neq s$ , then for any  $\{z_r\} \subset \mathbb{C}$  and  $x \in \mathbb{R}$ , we have

$$\left| \sum_{r \neq s} z_r \bar{z}_s \frac{\sin(2\pi x(\alpha_r - \alpha_s))}{\sin(\pi(\alpha_r - \alpha_s))} \right| \leq \delta^{-1} \sum_r |z_r|^2.$$

┘

*Proof:* This follows from applying (35.1.6.7) twice for  $z'_r = z_r e^{2\pi i x \alpha_r}$  and  $z'_r = z_r e^{-2\pi i x \alpha_r}$ . □

**Thm. (35.1.6.9)** [Large Sieve method for Exponentials, Davenport-Halberstam[D-H66a]/ Selberg/ Montgomery-Vaughan[M-V74]]. Suppose  $\delta \in \mathbb{R}_+$  and  $\{\alpha_r\} \in \mathbb{R}/\mathbb{Z}$  s.t.  $d(\alpha_r, \alpha_s) \geq \delta$  if  $r \neq s$ , then for any  $M, N \in \mathbb{Z}_+$  and  $(a_n)_{M < n \leq M+N} \in \mathbb{C}$ , we have

$$\sum_r \left( \sum_{M < n \leq M+N} a_n e^{2\pi i \alpha_r n} \right)^2 \leq (\delta^{-1} + N - 1) \sum_n |a_n|^2.$$

For higher dimension we have: Suppose  $\delta \in \mathbb{R}_+$  and  $\boldsymbol{\alpha}_r \in (\mathbb{R}/\mathbb{Z})^d$  s.t.  $\|\boldsymbol{\alpha}_r - \boldsymbol{\alpha}_s\|_\infty \geq \delta$  if  $r \neq s$ , then for any  $\mathbf{M} \in \mathbb{Z}^d$ ,  $N \in \mathbb{Z}_+$  and  $(a_n)_{\mathbf{M}_i < n_i \leq \mathbf{M}_i + N} \in \mathbb{C}$ , we have

$$\sum_r \left( \sum_{\mathbf{M}_i < n_i \leq \mathbf{M}_i + N} a_n e^{2\pi i \boldsymbol{\alpha}_r \cdot \mathbf{n}} \right)^2 \leq ((N-1)^d + c_d \delta^{-d}) \sum_{\mathbf{M}_i < n_i \leq \mathbf{M}_i + N} |a_n|^2$$

for some  $c_d \in \mathbb{R}_+$  depending on  $d$ . ┘

*Proof:* Firstly we prove the assertion with  $(\delta^{-1} + N - 1)$  replaced by  $\delta^{-1} + N$ . By duality, it suffices to prove that

$$\sum_{\mathbf{M}_i < n_i \leq \mathbf{M}_i + N} \left( \sum_r z_r e^{2\pi i \boldsymbol{\alpha}_r \cdot \mathbf{n}} \right)^2 \leq (N + \delta^{-1}) \sum_r |z_r|^2.$$

By squaring out, the diagonal terms contribute  $N \sum_r |z_r|^2$ , and the off-diagonal terms yield

$$\sum_{r \neq s} z_r \bar{z}_s \sum_{M < n \leq M+N} e^{2\pi i n(\alpha_r - \alpha_s)} = \sum_{r \neq s} z_r \bar{z}_s e^{2\pi i (M + \frac{N+1}{2})(\alpha_r - \alpha_s)} \frac{\sin(\pi N(\alpha_r - \alpha_s))}{\sin(\pi(\alpha_r - \alpha_s))},$$

which is bounded by  $\delta^{-1} \sum_r |z_r|^2$  by (35.1.6.8).

To prove the original inequality, take  $K \in \mathbb{Z}_+$  large, then  $\{\frac{\alpha_r + k}{K}\}_{k \in [K]_+, r}$  satisfies the hypothesis with  $\delta$  changed to  $\delta/K$ , thus

$$\begin{aligned} \sum_r \sum_{k \in [K]_+} \left( \sum_{KM + K + 1 < n \leq KM + KN} a_n e^{2\pi i \frac{\alpha_r + k}{K} n} \right)^2 &= K \sum_r \left( \sum_{M < n \leq M+N} a_n e^{2\pi i \alpha_r n} \right)^2 \\ &\leq (\delta^{-1} K + NK - K + 1) \sum_n |a_n|^2. \end{aligned}$$

Let  $K \rightarrow \infty$ , we get the desired assertion.

? For the second assertion, it suffices to prove a higher-dimensional version of (35.1.6.6):

$$\left| \sum_{r \neq s} (z_r \bar{z}_s) \prod_{i=1}^d \frac{1}{\lambda_{r,i} - \lambda_{s,i}} \right| \leq c_d \frac{\pi^d}{\delta^d} \sum_r |z_r|^2.$$

□

**Cor. (35.1.6.10)** [Large Sieve for Rational Exponentials]. For any  $Q, M, N \in \mathbb{Z}_+$  and  $(a_n)_{M < n \leq M+N} \in \mathbb{C}$ , we have

$$\sum_{q \in [Q]_+} \sum_{a \in (\mathbb{Z}/(q))^*} \left| \sum_{M < n \leq M+N} a_n e^{2\pi i \frac{an}{q}} \right|^2 \leq (Q^2 + N - 1) \sum_n |a_n|^2.$$

┘

**Cor. (35.1.6.11)** [Large Sieve for Rational Exponentials in Arithmetic Progressions]. For any  $k, l, Q, M, N \in \mathbb{Z}_+$  and  $(a_n)_{M < n \leq M+N} \in \mathbb{C}$ , we have

$$\sum_{q \in [Q]_+, (q, k)=1} \sum_{a \in (\mathbb{Z}/(q))^*} \left| \sum_{n \equiv l \pmod{k}} a_n e^{2\pi i \frac{an}{q}} \right|^2 \leq (Q^2 + k^{-1}N) \sum_{n \equiv l \pmod{k}} |a_n|^2.$$

┘

*Proof:*

□

### Multiplicative Large Sieve Inequalities

**Prop. (35.1.6.12).** For  $q, N \in \mathbb{Z}_+$  and  $(a_n) \in \mathbb{C}$ , we have

$$\sum_{\chi \in \text{Diri}(q)} \left| \sum_{n \in [N]_+} a_n \chi(n) \right|^2 \leq (q + N) \sum_n |a_n|^2.$$

┘

*Proof:*

$$\begin{aligned} \sum_{\chi \in \text{Diri}(q)} \left| \sum_{n \leq N} a_n \chi(n) \right|^2 &= \sum_{\chi \in \text{Diri}(q)} \sum_{n \in [N]_+} \sum_{m \in [N]_+} a_n \bar{a}_m \chi(n) \bar{\chi}(m) \\ &= \sum_{n, m \in [N]_+, n \equiv m \pmod{q}} q a_n \bar{a}_m \\ &\leq q \sum_{n, m \in [N]_+, n \equiv m \pmod{q}} \frac{|a_n|^2 + |a_m|^2}{2} \\ &\leq (q + N) \sum_n |a_n|^2. \end{aligned}$$

□

**Thm. (35.1.6.13)** [Bombieri-Davenport[B-D69]]. For  $Q, M, N \in \mathbb{Z}_+$  and any  $(a_n)_{M < n \leq M+N}$ , we have

$$\sum_{rs \leq Q, (r, s)=1} \frac{s}{\phi(rs)} \sum_{\chi \in \text{Diri}'(s)} \left| \sum_{M < n \leq M+N} a_n \chi(n) c_r(n) \right|^2 \leq (Q^2 + N - 1) \|\alpha\|^2 \text{ (35.3.6.8)}.$$

In particular,

$$\sum_{q \leq Q} \frac{q}{\phi(q)} \sum_{\chi \in \text{Diri}'(q)} \left| \sum_{M < n \leq M+N} a_n \chi(n) \right|^2 \leq (Q^2 + N - 1) \|\alpha\|^2.$$

┘

*Proof:* For  $q \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}'(q)$ , by (35.3.6.11), for any  $n \in \mathbb{Z}$ , we have

$$\tau(\chi)\bar{\chi}(n) = \sum_{a \in (\mathbb{Z}/(q))^*} \chi(a) e^{2\pi i \frac{an}{q}}.$$

And if  $\chi$  is induced from a  $\chi_s \in \text{Diri}'(s)$  with  $q = sr$  and  $(r, s) = 1$ , then for any  $n \in \mathbb{Z}$ , we have

$$\sum_{a \in (\mathbb{Z}/(q))^*} \chi(a) e^{2\pi i \frac{an}{q}} = \sum_{b \in (\mathbb{Z}/r)^*} \sum_{c \in (\mathbb{Z}/s)^*} \chi(bs + cr) e^{2\pi i (\frac{b}{r} + \frac{c}{s})n} = \bar{\chi}(n) \chi(r) c_r(n) \quad (35.3.6.8).$$

Thus if we denote  $S(\alpha) = \sum_n a_n e^{2\pi i \alpha n}$ , then

$$\sum_n a_n \bar{\chi}(n) c_r(n) = \frac{\bar{\chi}_s(n)}{\tau(\chi_s)} \sum_{a \in (\mathbb{Z}/(q))^*} \chi(a) S\left(\frac{a}{q}\right),$$

and

$$\begin{aligned} \sum_{rs \leq Q, (r,s)=1} \frac{s}{\phi(rs)} \sum_{\chi \in \text{Diri}'(s)} \left| \sum_{M < n \leq M+N} a_n \chi(n) c_r(n) \right|^2 &\leq \sum_{q \leq Q} \frac{1}{\phi(q)} \sum_{\chi \in \text{Diri}(q)} \left| \sum_{a \in (\mathbb{Z}/(q))^*} \chi(a) S\left(\frac{a}{q}\right) \right|^2 \\ &= \sum_{q \leq Q} \sum_{a \in (\mathbb{Z}/(q))^*} \left| \sum_{N < n \leq N+M} a_n e^{2\pi i \frac{an}{q}} \right|^2 \\ (35.1.6.10) &\leq (Q^2 + N - 1) \sum_n |a_n|^2 \end{aligned}$$

□

### Large-Sieve Bounds

**Thm. (35.1.6.14) [Large-Sieve Bounds for Number Fields, Gallagher[Gal73]].** Let  $F \in \text{NField}$ ,  $d = [F : \mathbb{Q}]$  and  $\Lambda$  a torsion-free  $\mathcal{O}_F$ -module of f.t. with rank  $n$ , and  $\|\cdot\|$  a norm on  $\Lambda \otimes \mathbb{R}$ , then there exists a constant  $c = c(\Lambda, \|\cdot\|) \in \mathbb{R}_+$  satisfying the following: If  $X \subset \Lambda$  is a subset s.t. there exists  $N \in \mathbb{R}_{\geq 1}$ ,  $Q \in \mathbb{R}_+$ , and for each  $v \in \Sigma_F^{\text{fin}}$  with  $\|v\| \leq Q$ , there exists  $w_v \in [0, 1]$  s.t.

- There exists  $x_0 \in \Lambda \otimes \mathbb{R}$  s.t.  $\|x - x_0\| \leq N/2$  for each  $x \in X$ .
- For each  $v \in \Sigma_F^{\text{fin}}$  with  $\|v\| \leq Q$ , the image  $[X]_v$  of  $X$  under the projection  $\Lambda \rightarrow \Lambda/\mathfrak{p}_v \Lambda$  satisfies

$$\#[X]_v \leq (1 - w_v) \# \Lambda/\mathfrak{p}_v \Lambda.$$

Then

$$\#X \leq c \max([N]^{nd} + Q^{2n})/L(Q), \text{ where } L(Q) = L(Q, \{w_v\}) = \sum_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)^b, \|\mathfrak{a}\| \leq Q} \prod_{v \in S(\mathfrak{a})} \frac{w_v}{1 - w_v}.$$

And we also have

$$L(Q) \geq L'(Q) = \sum_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F), \|\mathfrak{a}\| \leq Q} w_{\mathfrak{a}},$$

where  $w_{\mathfrak{a}}$  is the strongly multiplicative function on  $\text{Ideal}^\times(\mathcal{O}_F)$  defined by  $w$ .

In fact, if  $F = \mathbb{Q}$ ,  $\Lambda = \mathbb{Z}^n$  with the usual norm  $\|x\|_\infty$  on  $\Lambda$ , then we can take replace  $c \max([N]^{nd} + Q^{2n})$  by  $([N]^n + c_d Q^{2n})$  where  $c_d$  comes from (35.1.6.9). Notice that  $c_1 = 1$ .  $\square$

*Proof:* Consider a function  $T_\Lambda = \Lambda_{\mathbb{R}}^\vee / \Lambda^\vee$  s.t.  $f(x) = \sum_{\lambda \in X} e^{2\pi i \langle \lambda, x \rangle}$ , then  $\|f\|_\infty = f(0) = \#X = \|f\|_2^2$ . So the lemma (35.1.6.15) applied to  $f$  gives the assertion.  $\square$

**Lemma (35.1.6.15).** Situation as in (35.1.6.14), if  $f$  is a function on  $T_\Lambda = \Lambda_{\mathbb{R}}^\vee / \Lambda^\vee$  s.t.  $\text{Supp}(f) \subset X$ , then there exists  $c \in c(\Lambda, \|\cdot\|) \in \mathbb{R}_+$  s.t.

$$\|f\|_\infty^2 \leq c \max(N^{nd} + Q^{2n}) \|f\|_2^2 / L(Q).$$

In fact, if  $F = \mathbb{Q}$ ,  $\Lambda = \mathbb{Z}^n$  with the usual norm  $\|x\|_\infty$  on  $\Lambda$ , then we can take replace  $c \max(\lfloor N \rfloor^{nd} + Q^{2n})$  by  $(\lfloor N \rfloor^n + c_d Q^{2n})$  where  $c_d$  comes from (35.1.6.9). Notice that  $c_1 = 1$ .  $\lrcorner$

*Proof:* It suffices to bound  $f(0)$ , by translation invariance of  $\text{Supp}(f^\vee)$ . Define

$$T_Q = \bigcup_{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F), \|\mathfrak{a}\| \leq Q} T_\Lambda[\mathfrak{a}],$$

then  $T_Q$  is  $c_1 Q^{-2/d}$ -spaced for some  $c_1 = c_1(\Lambda, \|\cdot\|) \in \mathbb{R}_+$ : There exists  $\mathfrak{b}_1, \dots, \mathfrak{b}_n \in \text{Ideal}^\times(\mathcal{O}_F)$  s.t.  $\Lambda^\vee \cong \mathfrak{b}_1 \oplus \dots \oplus \mathfrak{b}_n$ , and then

$$\Lambda^\vee \otimes \mathbb{R} \cong (F \otimes \mathbb{R})^{\oplus n}.$$

Now we can choose a norm on  $\Lambda^\vee \otimes \mathbb{R}$  as

$$\|(x_1, \dots, x_n)\| = \max_{\sigma: F \rightarrow \mathbb{C}, i \in [n]_+} |\sigma(x_i)|.$$

And from here it is clear that  $T_Q$  is  $c_1 Q^{-2/d}$ -spaced for some  $c_1$ .

Thus

$$\sum_{x \in T_Q} |f(x)|^2 \leq c \max(\lfloor N \rfloor^{nd}, Q^{2n}) \|f\|_2^2$$

for some  $c \in c(\Lambda, \|\cdot\|) \in \mathbb{R}_+$ , by (35.1.6.9).

But

$$\sum_{x \in T_Q} |f(x)|^2 \geq \sum_{\substack{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)^\flat \\ \|\mathfrak{a}\| \leq Q}} \sum_{x \in T[\mathfrak{a}] \text{ primitive}} |f(x)|^2.$$

And because  $T[\mathfrak{a}]^\vee = \Lambda / \mathfrak{a}\Lambda$ , the restriction of  $f$  to  $T[\mathfrak{a}]$  has its spectrum contained in the image of  $X$  in  $\Lambda / \mathfrak{a}\Lambda$ . Thus

$$\sum_{x \in T[\mathfrak{a}] \text{ primitive}} |f(x)|^2 \geq |f(0)|^2 \prod_{v \in S(\mathfrak{a})} \frac{w_v}{1 - w_v}$$

by (35.1.6.16), and the assertion follows.  $\square$

**Lemma (35.1.6.16) [Harmonic Analysis on Finite Abelian Fields].** Let  $G_1, \dots, G_m \in \mathcal{A}b^{\text{fin}}$  and  $\Omega_i \subset G_i^\vee$ , and denote  $w_i = \#\Omega_i / \#G_i$ . Suppose  $f$  is a function on  $G = G_1 \times \dots \times G_m$  that is a linear combination of characters of the form  $\chi_1 \times \dots \times \chi_m$  with  $\chi_i \in G_i^\vee \setminus \Omega_i$ , then we have

$$|f(e)|^2 \leq \left( \prod_i \frac{1 - w_i}{w_i} \right) \sum_{x_i \in G_i \setminus \{e\}} |f(x_1, \dots, x_m)|^2.$$

$\lrcorner$



*Proof:* Use induction on  $m$ : If  $m = 1$ , write  $f = \sum_{\chi \in G_i^\vee \setminus \Omega_i} a_\chi \chi$ , then

$$|f(0)|^2 = \left| \sum_{\chi \in G^\vee} a_\chi \right|^2 \leq (1 - \omega) \#G \left( \sum_{\chi \in G^\vee} |a_\chi|^2 \right) = (1 - \omega) \sum_{g \in G} |f(g)|^2.$$

Thus the assertion follows.

Now if  $m \geq 2$  and the assertion holds true for any smaller  $m$ . For each  $x_m \in G_m$ , denote  $f_{x_m}(x_1, \dots, x_{m-1}) \triangleq f(x_1, \dots, x_m)$ , then by induction hypothesis we have

$$\sum_{x_i \in G_i \setminus \{e\}} |f_{x_m}(x_1, \dots, x_{m-1})|^2 \geq \left( \prod_{i=1}^{m-1} \frac{w_i}{1 - w_i} \right) |f_{x_m}(e)|^2$$

Thus

$$\sum_{x_i \in G_i \setminus \{e\}} |f(x_1, \dots, x_{m-1}, x_m)|^2 \geq \left( \prod_{i=1}^{m-1} \frac{w_i}{1 - w_i} \right) \sum_{x_m \in G_m \setminus \{e\}} |f(e, \dots, e, x_m)|^2 \geq \left( \prod_{i=1}^m \frac{w_i}{1 - w_i} \right) f(e).$$

□

**Cor. (35.1.6.17) [1-Dimensional Large Sieve Bounds, Montgomery].** Suppose

- $\mathcal{M} \subset \mathbb{Z}_+$  is a subset contained in an interval of length  $N \geq 1$ .
- $\mathcal{P} \subset \text{Prime}$  are finite set of primes.
- For each  $p \in \mathcal{P}$ , let  $\Omega_p \subset \mathbb{Z}/(p)$  be a properly-contained subset.

$$\mathcal{S}(\mathcal{M}, \mathcal{P}, \Omega) \triangleq \{m \in \mathcal{M} : m \pmod{p} \notin \Omega_p \forall p \in \mathcal{P}\}.$$

Then for each sequence of complex numbers  $(a_n)_{n \in \mathbb{Z}_+} \in \mathbb{C}$ , denote

$$Z = \sum_{n \in \mathcal{S}(\mathcal{M}, \mathcal{O}, \Omega)} a_n,$$

then

$$|Z|^2 \leq \frac{N + Q^2}{H(Q)} \|a\|^2, \text{ where } H(Q) = \sum_{q \in [Q]_+} h(q),$$

where  $h$  is a multiplicative function supported on square-free numbers s.t.  $h(p) = \frac{\#\Omega_p}{p - \#\Omega_p}$  for  $p \in S(\mathcal{P})$  and  $h(p) = 0$  for  $p \notin S(\mathcal{P})$ .

In particular, we have  $\#\mathcal{S}(\mathcal{M}, \mathcal{P}, \Omega) \leq \frac{N + Q^2}{H(Q)}$ . ┘

**Remark (35.1.6.18).** See [I-K04]P181 for a direct proof. ┘

## 7 Probabilistic Models for Primes

See <https://terrytao.wordpress.com/2015/01/04/254a-supplement-4-probabilistic-models-and-heuristics/> [sieve method, Ford]

## 35.2 Sieve Theory

References are [Ser97, I-K04, H-W08].

**Notation (35.2.0.1).**

- Use notations on elementary number theory defined in [Elementary Number Theory](#).

┘

### 1 Basics

**Def. (35.2.1.1) [Sifted Sums].** For  $f : \mathbb{Z}_+ \rightarrow \mathbb{R}_+$ ,  $\mathbf{P} \subset \mathbf{Prime}$  and  $z, X \in \mathbb{R}_{\geq 1}$ , define

$$\mathbf{P}(z) = \prod_{p \in \mathbf{P} \cap [z-1]_+} p.$$

If  $\mathbf{P}$  is not specified, then  $\mathbf{P} = \mathbf{Prime}$  by default. Then consider the sifted sum

$$\mathcal{M}_f(X|z) = \sum_{n \in [X]_+, (n, \mathbf{P}(z))=1} f(n),$$

┘

**Prop. (35.2.1.2) [Legendre Formula].** Situation as in [\(35.2.1.1\)](#),

$$\mathcal{M}_f(X|z) = \sum_{n \in [X]_+} \left( \sum_{d|n, d|\mathbf{P}(z)} \mu(d) \right) f(n) = \sum_{d|\mathbf{P}(z)} \mu(d) A_d(X), \text{ where } A_d(X) = \sum_{n \in [X]_+, d|n} f(n).$$

┘

**Remark (35.2.1.3) [Heuristics].** Situation as in [\(35.2.1.2\)](#), heuristically, for each  $d|\mathbf{P}(z)$ , we should have

$$A_d(X) = g(d) \mathcal{M}_f(X) + r_d(X),$$

where  $g : \mathbb{Z}_+ \rightarrow [0, 1]$  is multiplicative and  $r_d(X)$  is relatively small. Thus we have

$$\mathcal{M}_f(X|z) \sim V(z)X, \text{ where } V(z) = \prod_{p|\mathbf{P}(z)} (1 - g(p)).$$

┘

### Parity Barrier

### 2 Combinatorial Sieves

**Prop. (35.2.2.1) [Burn's Combinatorial Sieves].** Burn modified the range of  $d$  in the summation in Legendre's formula [\(35.2.1.2\)](#) by two sets  $\mathcal{D}^+$  and  $\mathcal{D}^-$  to get an upper and a lower bound.

We write prime decompositions in descending order, and choose a family of parameters  $(y_m)_{m \in \mathbb{Z}_+} \in \mathbb{Z}_+$ , and define

$$\mathcal{D}^+ = \{d \in \mathbb{Z}_+ | d = p_1 \dots p_r, p_{2k-1} < y_{2k-1}, \forall k \in \mathbb{Z}_+\},$$

$$\mathcal{D}^- = \{d \in \mathbb{Z}_+ | d = p_1 \dots p_r, p_{2k} < y_{2k}, \forall k \in \mathbb{Z}_+\}.$$

And for each multiplicative function  $g : \mathbb{Z}_+ \rightarrow [0, 1]$ , define

$$V^\pm(z) = \prod_{d \in \mathcal{D}^\pm, d|P(z)} \mu(d)g(d),$$

then we have

$$V^-(z) \leq V(z) \leq V^+(z), \text{ where } V(z) = \prod_{p|P(z)} (1 - g(p)).$$

In particular, if the heuristic in (35.2.1.3) holds, then

$$V^-(z)\mathcal{M}_f(X) + R^-(X|z) \leq \mathcal{M}_f(X|z) \leq V^+(z)\mathcal{M}_f(X) + R^+(X|z)$$

where  $R^\pm(z) = \sum_{d|P(z), d \in \mathcal{D}^\pm} \mu(d)r_d(X)$ . ┘

*Proof:* We have the formula

$$V(z) = 1 - \sum_{p \in P(z)} g(p)V(p),$$

so by iterating this down and use inclusion-exclusion, we get

$$V(z) = V^+(z) - \sum_{n \in 2\mathbb{Z}_+ - 1} V_n(z) = V^-(z) + \sum_{n \in 2\mathbb{Z}_+} V_n(z),$$

where

$$V_n(z) = \sum_{\substack{y_n \leq p_n < \dots < p_1 < z \\ p_m < y_m, \forall m < n \text{ s.t. } m \equiv n \pmod{2}}} g(p_1 \dots p_n) V(p_n).$$

The two inequalities follow from plugging in the inequalities (which are special cases where  $g \equiv 1$ )

$$\sum_{d|n, d|P(z), d \in \mathcal{D}^-} \mu(d) \leq \sum_{d|n, d|P(z)} \mu(d) \leq \sum_{d|n, d|P(z), d \in \mathcal{D}^+} \mu(d).$$

to the Legendre formula (35.2.1.2). □

### $\beta$ -Stopping Criterion

**Def. (35.2.2.2) [Iwaniec's Combinatorial Sieve ( $\beta$ -Stopping Criterion)].** Situation as in (35.2.2.1), Iwaniec modifies the sets  $\mathcal{D}^\pm$  by making  $(y_m)$  to be depending on  $p_1, \dots, p_{m-1}$ , i.e. take a parameter  $(y, \beta)$  where  $y \in \mathbb{Z}_{\geq z}$  and  $\beta \in \mathbb{R}_{>1}$ , and define

$$y_1 = y, \quad y_m = \left( \frac{y}{p_1 \dots p_m} \right)^{1/\beta},$$

i.e. he wants to impose the condition  $p_1 \dots p_{m-1} p_m^{\beta+1} < y$ . It is clear that  $\mathcal{D}^+ \cup \mathcal{D}^- \subset [y-1]_+$ . ┘

**Thm. (35.2.2.3) [Iwaniec's Combinatorial Sieve, Iwaniec (1980)].** Situation as in (35.2.2.1), suppose there exists  $\kappa, K \in \mathbb{R}_+$  s.t. the multiplicative function  $g$  satisfies:  $g(\text{Prime}) \subset [0, 1)$  and for any  $w \leq z \in \mathbb{R}_+$  we have

$$\prod_{p \in \text{Prime}, w \leq p \leq z} (1 - g(p))^{-1} \leq K \left( \frac{\log z}{\log w} \right)^\kappa.$$

Then we can choose Iwaniec's combinatorial sieve parameters  $y \in \mathbb{Z}_+, \beta = 9\kappa + 1$  as in (35.2.2.2). Then for any  $z \in [y^{1/\beta}]_+$ , we have

$$V^+(z) < (1 + e^{\beta-s} K^{10}) V(z), \quad V^-(z) > (1 - e^{\beta-s} K^{10}) V(z),$$

where  $s = \log y / \log z \geq \beta$ . This  $\kappa$  is called a **sieve dimension** of this combinatorial sieve. ┘

*Proof:* Notation as in the proof of (35.2.2.1),

$$V_n(z) \leq \sum_{\substack{y_n \leq p_n < \dots < p_1 < z \\ p_1 \dots p_{m-1} p_m^\beta < y, \forall m < n}} g(p_1 \dots p_n) V(p_n).$$

For such sequence  $y_n \leq p_n < \dots < p_1 < z$ ,  $p_1^{n+\beta} \geq y$ , so  $n \geq s - \beta$ . And by induction we have

$$p_1 \dots p_m < y^{1 - \left(\frac{\beta-1}{\beta}\right)^m}$$

for any  $m < n$ , and then

$$p_n \geq (y/p_1 \dots p_m)^{\frac{1}{\beta+1}} \geq y^{\frac{1}{\beta+1} \left(\frac{\beta-1}{\beta}\right)^{n-1}} \geq y^{\frac{1}{\beta} \left(\frac{\beta-1}{\beta}\right)^n} \geq z^{\left(\frac{\beta-1}{\beta}\right)^n}.$$

So if we set  $z_n \triangleq z^{\left(\frac{\beta-1}{\beta}\right)^n}$ , then

$$V_n(z) \leq \sum_{z_n \leq p_n < \dots < p_1 < z} g(p_1 \dots p_n) V(p_n) \leq \frac{V(z_n)}{n!} \left( \sum_{z_n \leq p < z} g(p) \right)^n \leq \frac{V(z_n)}{n!} \left( \log \frac{V(z_n)}{V(z)} \right)^n.$$

The hypothesis implies that

$$\frac{V(z_n)}{V(z)} \leq K \left( \frac{\beta}{\beta-1} \right)^{\kappa n} < K e^{n/9},$$

so

$$V_n(z) \leq \frac{K}{n!} \left( \frac{k}{9} + \log K \right)^n e^{n/9} V(z) \leq \frac{K}{n!} \left( \frac{k}{9} \right)^n e^{n/9} K^9 V(z) \leq e^{-1} \left( \frac{e^{1+1/9}}{9} \right)^n K^{9+1} V(z) \leq e^{-n-1} K^{10} V(z).$$

by Stirling's inequality (12.6.1.18). Notice also that  $p_1^{n+\beta} \geq y$ , so

$$\sum_{n \geq s-\beta} V_n(z) \leq \frac{e^{\beta-s}}{e-1} K^{10} V(z).$$

□

**Cor. (35.2.2.4).** Situation as in (35.2.1.2), suppose the heuristic (35.2.1.3) holds and there exists  $\kappa, K \in \mathbb{R}_+$  s.t. the multiplicative function  $g$  satisfies:  $g(\text{Prime}) \subset [0, 1)$  and for any  $w \leq z \in \mathbb{R}_+$  we have

$$\prod_{p \in \text{Prime}, w \leq p \leq z} (1 - g(p))^{-1} \leq K \left( \frac{\log z}{\log w} \right)^\kappa.$$

Then

$$\begin{aligned} \mathcal{M}_f(X|z) &\leq (1 + e^{\beta-s} K^{10}) V(z) \mathcal{M}_f(X) + R(X|z^s), \\ \mathcal{M}_f(X|z) &\geq (1 - e^{\beta-s} K^{10}) V(z) \mathcal{M}_f(X) - R(X|z^s), \end{aligned}$$

for  $\beta = 9\kappa + 1$  and any  $s \in \mathbb{R}_{\geq \beta}$ , where

$$R(X|z^s) = \sum_{d|P(z), d < z^s} |r_d(X)| \quad (35.2.1.3).$$

In particular, if  $y(X)$  is a function of  $X$  s.t.  $\lim_{X \rightarrow \infty} y(X) = \infty$  and  $R(X|y(X))$  is relatively small, then for any  $s \in \mathbb{R}_{> \beta}$ , we can take  $z(X) = y(X)^{1/s} > 1$ , then

$$\frac{\mathcal{M}_f(X)}{\log^\kappa z(X)} \ll_{\kappa, K, s} \mathcal{M}_f(X) V(z(X)) \ll_{\kappa, K, s} \mathcal{M}_f(X|z(X)) \ll_{\kappa, K, s} V(z(X)) \mathcal{M}_f(X).$$

┘

*Proof:* The remainder term are bounded by the fact  $\mathcal{D}^+ \cup \mathcal{D}^- \subset [z^s - 1]_+$  (35.2.2.2).  $\square$

**Cor. (35.2.2.5).**

$$\pi_2(X) \ll \frac{X}{\log^2 X}, \quad \pi_2(X, X^{1/20}) \gg \frac{X}{\log^2 X}.$$

$\perp$

### 3 Selberg's Sieve

**Prop. (35.2.3.1) [Setup for Selberg's Upper Bound Sieve [Sel52]].** Suppose

- $\mathcal{A} = (a_n)_{n \in \mathbb{Z}_+} \subset \mathbb{R}_{\geq 0}$  is a sequence of non-negative numbers,
- $P \in \mathbb{Z}_+^b$ ,
- $g : \mathbb{Z}_+ \rightarrow \mathbb{R}_{\geq 0}$  is multiplicative s.t.  $g(p) < 1$  for each  $p \in \text{Prime}$ ,

s.t.

$$\mathcal{A}_d \triangleq \sum_{d|n} a_n = g(d)X + r_d(\mathcal{A})$$

for each  $d|P$ . Then for each sequence of real numbers  $(\rho_d)_{d \in \mathbb{Z}_+} \in \mathbb{R}$  s.t.  $\rho_1 = 1$ , we have

$$\begin{aligned} S(\mathcal{A}; P) &\triangleq \sum_{n \in \mathbb{Z}_+, (n, P)=1} a_n \leq \sum_{n \in \mathbb{Z}_+} a_n \left( \sum_{d \in \mathbb{Z}_+, d|P} \rho_d \right)^2 \\ &= \sum_{d_1, d_2 \in \mathbb{Z}_+, d_1 | P, d_2 | P} \rho_{d_1} \rho_{d_2} \mathcal{A}_{[d_1, d_2]} = GX + R(\mathcal{A}; P), \end{aligned}$$

where

$$G = \sum_{d_1, d_2 \in \mathbb{Z}_+, d_1 | P, d_2 | P} g([d_1, d_2]) \rho_{d_1} \rho_{d_2}, \quad R(\mathcal{A}; P) = \sum_{d_1, d_2 \in \mathbb{Z}_+, d_1 | P, d_2 | P} \rho_{d_1} \rho_{d_2} r_{[d_1, d_2]}(\mathcal{A}).$$

$\perp$

**Thm. (35.2.3.2) [Selberg's Upper Bound Sieve, van.Lint-Richert [L-R65]].** Situation as in (35.2.3.1), suppose

- $g$  is supported on square-free numbers, and  $g(p) > 0$  for each  $p \in S(P)$ .
- $D \in \mathbb{R}_{\geq 1}$  s.t.  $\rho_d = 0$  for  $d \geq \sqrt{D}$ .

Let  $h : \mathbb{Z}_+ \rightarrow \mathbb{R}_+$  be the multiplicative function s.t.

$$h(p^k) = \begin{cases} \frac{g(p)}{1-g(p)} & , k = 1 \\ 0 & , \text{otherwise} \end{cases}.$$

Then we have

$$S(\mathcal{A}; P) \leq H(D)^{-1} X + R(\mathcal{A}; P),$$

where

$$H(D) = \sum_{\substack{d \in \mathbb{Z}_+, d \leq \sqrt{D} \\ d|P}} h(d), \quad R(\mathcal{A}; P) = \sum_{d \in \mathbb{Z}_+, d|P} \lambda_d r_d(\mathcal{A}),$$

where

$$\lambda_d = \sum_{\substack{d_1, d_2 \in \mathbb{Z}_+ \\ [d_1, d_2] = d}} \rho_{d_1} \rho_{d_2}, \quad \rho_l = \frac{\mu(l)h(l)}{g(l)H} \sum_{\substack{d \in \mathbb{Z}_+, d < \sqrt{D}/l \\ (d, l) = 1}} h(d).$$

Moreover, we have  $|\rho_d| \leq 1$  and  $|\lambda_d| \leq \tau_3(d) \leq 3^{\#S(d)}$  for each  $d \in \mathbb{Z}_+$ .  $\lrcorner$

**Remark (35.2.3.3).** The Selberg's sieve yields good results for sifting problems when the number of residue classes to be excluded is large.  $\lrcorner$

*Proof:*

$$\begin{aligned} G &= \sum_{abc|P} g(abc) \rho_{ac} \rho_{bc} \\ &= \sum_{c|P} g(c) \sum_{(a,b)=1, ac|P, bc|P} g(a)g(b) \rho_{ac} \rho_{bc} \\ &= \sum_{c|P} g(c) \sum_{cd|P} \mu(d)g(d)^2 \left( \sum_{cdm|P} g(m) \rho_{cdm} \right)^2 \\ &= \sum_{d|P} h(d)^{-1} \left( \sum_{d|m} g(m) \rho_m \right)^2. \end{aligned}$$

If we denote  $\xi_d = \mu(d) \sum_{d|m} g(m) \rho_m$  for each  $d \in \mathbb{Z}_+$ , then we get

$$G = \sum_{d|P} h(d)^{-1} \xi_d^2,$$

and by Möbius-inversion?? we have

$$\rho_l = \frac{\mu(l)}{g(l)} \sum_{l|d} \xi_d.$$

In particular, we have  $\sum_{d|P} \xi_d = 1$ , and  $\xi_d = 0$  for  $d \geq \sqrt{D}$ .

Then it follows from Cauchy's formula that  $GH(D) \geq 1$ , and the equality holds if  $\xi_d = h(d)H^{-1}$  for  $d < \sqrt{D}$ , in which case

$$\rho_l = \frac{\mu(l)h(l)}{g(l)H} \sum_{\substack{d \in \mathbb{Z}_+, d < \sqrt{D}/l \\ (d, l) = 1}} h(d).$$

Then the inequality follows from (35.2.3.1). To show that  $|\rho_l| \leq 1$ , notice

$$\begin{aligned} H &= \sum_{k|l} \sum_{d < \sqrt{D}, (d, l) = k} h(d) \\ &= \sum_{k|l} h(k) \sum_{m < \sqrt{D}/k, (m, l) = 1} h(m) \\ &\geq \left( \sum_{k|l} h(k) \right) \sum_{m < \sqrt{D}/l, (m, l) = 1} h(m) \\ &= \mu(l) \rho_l H, \end{aligned}$$

so we get  $0 \leq \mu(l) \rho_l \leq 1$ . And  $|\lambda_d| \leq \tau_3(d)$  follows trivially.  $\square$

**Cor. (35.2.3.4) [Main Terms].** Situation as in (35.2.3.2), suppose  $A_1, A_2, L, \kappa \in \mathbb{R}_+$  s.t.

- $\prod_{p \in \text{Prime}_{< \sqrt{D}}} p \mid P$ .
- $0 \leq g(p) \leq 1 - A_1$  for each  $p \in \text{Prime}$ .
- For any  $2 \leq w \leq y \in \mathbb{R}_+$ , we have

$$-L \leq \sum_{p \in \text{Prime}, w \leq p \leq y} g(p) \log p - \kappa \log \frac{y}{w} \leq A_2.$$

- $\sum_{p \in \text{Prime}} g(p)^2 \log p < \infty$ .

Then we have

$$H(D) = C \frac{\log^\kappa \sqrt{D}}{\Gamma(\kappa + 1)} + O_{\kappa, A_1, A_2} \left( C[(L + 1) \log^{\kappa-1} X + (L + 1)^\kappa] \right), \text{ where } C = \prod_{p \in \text{Prime}} (1 - g(p))^{-1} \left(1 - \frac{1}{p}\right)^\kappa.$$

┘

*Proof:* In this case,

$$H(D) = \sum_{d \in \mathbb{Z}_+, d < \sqrt{D}} h(d),$$

so we can apply Wirsing's formula (35.1.3.4) to  $h$ ? Cf. [Sieve method, Ford]P46.

□

**Cor. (35.2.3.5) [Remainder Terms].** Situation as in (35.2.3.2), suppose

- $g(p)p \geq 1$  for each  $p \in S(P)$ .
- $|r_d(\mathcal{A})| \leq g(d)d$  for each  $d \mid P$ .
- There exists  $c \in \mathbb{R}_+$  s.t.  $\sum_{p \in \text{Prime}, y \leq p \leq x} g(p)p \leq cX$  for each  $X \in \mathbb{R}_{\geq 2}$ .
- $\sum_{p \in \text{Prime}_{\leq X}} g(p)^2 p \log p = o(X)$  for  $X \in \mathbb{R}_{\geq 2}$ .

then we have

$$S(\mathcal{A}; P) \leq H(D)^{-1} X + O_c \left( \frac{D}{\log^2 D} \right).$$

┘

*Proof:* The hypotheses imply that  $g([d_1, d_2])[d_1, d_2] \leq g(d_1)g(d_2)d_1d_2$ , so

$$R(\mathcal{A}; P) \leq \left( \sum_{d < \sqrt{D}} |\rho_d| g(d)d \right)^2 \leq \left( \frac{1}{H(D)} \sum_{m < \sqrt{D}} h(m)\sigma(m) \right)^2 \quad (35.2.3.2).$$

Then it follows from (35.1.3.3) applied to  $h\sigma$  that

$$\sum_{m \in \mathbb{Z}_+, m < \sqrt{D}} h(m)\sigma(m) \ll_c \frac{\sqrt{D}}{\log D} \sum_{m \in \mathbb{Z}_+, m < \sqrt{D}} h(m)\sigma(m)m^{-1},$$

And we also have

$$\sum_{m \in \mathbb{Z}_+, m < \sqrt{D}} h(m)\sigma(m)m^{-1} \leq H(D) \sum_{m \in \mathbb{Z}_+} h(m)m^{-1} \ll_c H.$$

So  $R(\mathcal{A}; P) = O_c(D \log^{-2} D)$ .

□

#### 4 Summation over Primes

**Def. (35.2.4.1) [Summation over Primes].** For  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$ , denote

$$\psi(f; X) = \sum_{n \in [X]_+} \Lambda(n) f(n),$$

And if  $f$  decays rapidly, denote  $\psi(f) = \lim_{X \rightarrow \infty} \psi(f; X)$ . ┘

**Thm. (35.2.4.2) [Explicit Formula].** For  $f : \mathbb{R}_+ \rightarrow \mathbb{C}$  that is smooth with compact support, then

$$\psi(f) = \mathcal{M}(f)(1) - \sum_{n \in \mathbb{Z}_+} \mathcal{M}(f)(-2n) - \sum_{\rho \in \text{Zero}'(\zeta)} \mathcal{M}(f)(\rho).$$

┘

*Proof:* ? Cf. [I-K04]P337. The proof is based on a version of the Chebotarev density theorem, and uses a two-dimensional large sieve inequality together with results of Deuring, Hurwitz and Masser-Wüstholz. □

**Prop. (35.2.4.3).** Situation as in (35.2.4.1), by (2.6.2.11),  $\Lambda = \mu * \log$ , so

$$\psi(f; X) = - \sum_{d \in \mathbb{Z}_+} (\mu(d) \log d) A_d(X), \text{ where } A_d(X) = \sum_{n \in [X]_+, d|n} f(n).$$

┘

**Remark (35.2.4.4) [Heuristics].** Situation as in (35.2.4.3), similar to (35.2.1.3), heuristically, there exists  $D \in \mathbb{Z}_+$  s.t. the summand for  $d > D$  is small, and for  $d \leq D$ , we should have

$$A_d(X) = g(d) \mathcal{M}_f(X) + r_d(X),$$

where  $g : \mathbb{Z}_+ \rightarrow [0, 1]$  is multiplicative and  $r_d(X)$  is relatively small. Thus we have

$$\psi(f; X) \sim H \mathcal{M}_f(X), \text{ where } H = - \sum_{d \in \mathbb{Z}_+} (\mu(d) \log d) g(d) = \prod_{p \in \text{Prime}} \frac{1 - g(p)}{1 - p^{-1}}.$$

┘

**Remark (35.2.4.5) [Vinogradov's Method].** Situation as in (35.2.4.1), many times we can use inclusion-exclusion method to write

$$\psi(f) = \psi_1(f) + \psi_2(f),$$

where

$$\psi_1(f) = \sum_{d \in [D]_+} \alpha_d A_d(X), \quad \psi_2(f) = \sum_{M < m \leq N, mn \in [X]_+} \beta_m \gamma_n a_{mn}.$$

$\psi_1(f)$  can be handled because we can show that  $A_d(X)$  are small, and  $\psi_2(f)$  can be handled by large sieve methods Large Sieve method. ┘



**Useful Identities for  $\Lambda(n)$  and  $\mu(n)$** 

**Prop. (35.2.4.6)** [Linnik's Identity, Linnik[[Lin63](#)]]. For  $n, k \in \mathbb{Z}_+$ , denote

$$\Lambda^{\vee}(n) = \frac{\Lambda(n)}{\log n} = \begin{cases} a^{-1} & , n^{1/a} \in \mathbf{Prime} \\ 0 & \text{otherwise} \end{cases}$$

and

$$\tau'_k(n) = \#\{n_1, \dots, n_k \in \mathbb{Z}_{\geq 2}, n_1 \dots n_k = n\},$$

then

$$\Lambda^{\vee}(n) = - \sum_{k \in \mathbb{Z}_+} \frac{(-1)^k}{k} \tau'_k(n).$$

In particular, for  $X \in \mathbb{R}_{\geq 2}$  and  $z \in [2, \sqrt{X}]$ , any sequence  $(a_n)_{n \in \mathbb{Z}_+} \in \mathbb{C}$ , we have

$$\sum_{\substack{p \in \mathbf{Prime}, \nu \in \mathbb{Z}_+ \\ p^\nu \leq X, p > z}} \frac{a_{p^\nu}}{\nu} = - \sum_{k \leq K} \sum_{\substack{n \in [X]_+ \\ (n, \mathbf{P}(z))=1}} \tau'_k(n) a_n.$$

In the RHS, the  $k$ -th summand can be tackled using GL( $k$ )-methods. ┘

*Proof:* Because  $\zeta(s) = \sum_{n \in \mathbb{Z}_+} n^{-s} = \prod_{p \in \mathbf{Prime}} (1 - p^{-s})^{-1}$ , we have

$$\log \zeta(s) = \sum_{n \in \mathbb{Z}_{\geq 2}} \Lambda^{\vee}(n) n^{-s}.$$

But also

$$\log \zeta(s) = \log(1 - (1 - \zeta(s))) = - \sum_{k \in \mathbb{Z}_+} \frac{(-1)^k}{k} \left( \sum_{n \geq 2} n^{-s} \right)^k = - \sum_{n \in \mathbb{Z}_{\geq 2}} \sum_{k \in \mathbb{Z}_+} \frac{(-1)^k}{k} \tau'_k(n) n^{-s}.$$

So the assertion follows. □

**Prop. (35.2.4.7)** [Heath.Brown's Identities, Heath.Brown[[HB82](#)]]. For  $K, n \in \mathbb{Z}_+, z \in \mathbb{R}_{\geq 1}$ ,

$$\Lambda_{\leq 2z^K} = - \sum_{k \in [K]_+} (-1)^k \binom{K}{k} (\mu_{\leq z})^{*k} * \mathbf{1}^{*(k-1)} * \text{Log}.$$

$$\mu_{\leq z^K} = - \sum_{k \in [K]_+} (-1)^k \binom{K}{k} (\mu_{\leq z})^{*k} * \mathbf{1}^{*(k-1)}.$$

┘

*Proof:* 1: Let  $M(s) = \sum_{m \in [z]_+} \mu(m) m^{-s}$ , then

$$\zeta(s) M(s) = 1 + \sum_{n \in \mathbb{Z}_+, n > z} a_z(n) n^{-s}$$

for some sequence  $(a_z(n)) \in \mathbb{Z}$ , because  $\zeta * \mu = \delta$  ([2.6.2.3](#)). So

$$\frac{\zeta^{\vee}}{\zeta}(s) (1 - \zeta(s) M(s))^K = \frac{\zeta^{\vee}}{\zeta}(s) = \frac{\zeta^{\vee}}{\zeta}(s) + \sum_{k \in [K]_+} (-1)^k \binom{K}{k} \zeta^{\vee}(s) \zeta^{k-1}(s) M^k(s).$$

Notice the Dirichlet series for the LHS vanishes for  $n \leq 2z^K$ , so the assertion follows by comparing the two Dirichlet series.

2: The proof is similar to the above.  $\square$

**Prop. (35.2.4.8) [Vaughan's Identities, Vaughan[Vau77]].** For  $y, z \in \mathbb{R}_+$ ,

$$\Lambda_{>z} = \mu_{\leq y} * \text{Log} - \mu_{\leq y} * \Lambda_{\leq z} * \mathbf{1} + \mu_{>y} * \Lambda_{>z} * \mathbf{1}.$$

$$\mu - \mu_{\leq y} - \mu_{\leq z} = -\mu_{\leq y} * \mu_{\leq z} * \mathbf{1} + \mu_{>y} * \mu_{>z} * \mathbf{1}.$$

┘

*Proof:* 1: By (2.6.2.11),  $\Lambda = \mu * \text{Log}$ , so

$$\mu * \Lambda = \mu * \Lambda_{\leq z} + \mu_{\leq y} * \Lambda - \mu_{\leq y} * \Lambda_{\leq z} + \mu_{>y} * \Lambda_{>z}.$$

And then the assertion follows by convolution with  $\mathbf{1}$  and use the identities  $\mathbf{1} * \mu = \delta$ ,  $\mathbf{1} * \Lambda = \text{Log}$ .

2: Use similar argument as above.  $\square$

**Prop. (35.2.4.9) [Montgomery-Vaughan Identity(1981)].** Denote  $G_2(s) = \sum_{d \leq V} \mu(d) d^{-s}$ , then it follows from the identity

$$\frac{1}{\zeta(s)} = 2G(s) - G(s)^2 \zeta(s) + \left( \frac{1}{\zeta(s)} - G(s) \right) (1 - \zeta(s)G(s))$$

that

$$\mu(n) = 2\mu(n)\mathbf{1}_{[V]_+}(n) - \sum_{\substack{d \cdot m = n \\ d \leq V, e \leq V}} \mu(d)\mu(e) - \sum_{\substack{d \cdot k = n \\ d > V, k > V}} \mu(d) \left( \sum_{e|k, e \leq V} \mu(e) \right).$$

In particular, for any  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  and  $V \leq N \in \mathbb{Z}_+$ , we have

$$\sum_{n=1}^N \mu(n)f(n) = T_0 + T_1 + T_2,$$

where

$$T_0 = 2 \sum_{n=1}^V \mu(n)f(n), \quad T_1 = - \sum_{m \leq V^2} b_m \sum_{n \leq N/m} f(mn), \quad \text{where } b_m = \sum_{d \cdot e = m, d \leq V, e \leq V} \mu(d)\mu(e),$$

$$T_2 = - \sum_{V < m \leq N/V} \mu(m) \sum_{V < n \leq N/m} c_n f(mn), \quad \text{where } c_n = \sum_{d|n, d \leq V} \mu(d).$$

They satisfy

$$T_0 \ll \sum_{n \leq V} |f(n)|.$$

$$T_1 \ll \sum_{r \leq V^2} d(r) \left| \sum_{k \leq N/r} f(rk) \right|.$$

$$T_2 \ll N^{1/2} \log^{5/2} N \cdot \max_{V \leq M \leq N/V} \max_{j \leq N/M} \sqrt{\sum_{k \leq N/M} \left| \sum_{\substack{M < m \leq 2M \\ m \leq N/k, m \leq N/j}} f(mj) \overline{f(mk)} \right|}.$$

┘

*Proof:* ?  $\square$

### Vinogradov's method

**Prop. (35.2.4.10) [O. Ramaré's Identities].** For  $X \in \mathbb{R}_{\geq 4}$  and  $1 \neq n \in [X]_+$ , define

$$\tau(n|\sqrt{X}) = \#\{p \in \mathbf{Prime} : p|n, p \leq \sqrt{X}\}, \quad \beta_n(X) = \frac{1}{1 + \tau(n|\sqrt{X})}.$$

Then for any  $n \in \mathbb{Z}_+$  that is square-free, we have

$$\sum_{\substack{p \in \mathbf{Prime} \\ p|n, p \leq \sqrt{X}}} \beta_{n/p}(X) = 1 - \mathbf{1}_{\mathbf{Prime} \cap (\sqrt{X}, X]}(n).$$

In particular, for any  $(a_n) \in \mathbb{C}$  and  $z \in [2, \sqrt{X}]$ , we have

$$\sum_{\sqrt{X} < p \leq X} a_p = \sum_{\substack{n \in \mathbb{Z}_+ \text{ square-free} \\ 1 < n \leq X}} a_n - \sum_{\substack{p \in \mathbf{Prime} \leq \sqrt{X}, m \in \mathbb{Z}_+ \text{ square-free} \\ p \nmid m, mp \leq X}} \beta_m(X) a_{mp}.$$

and

$$\sum_{\sqrt{X} < p \leq X} a_p = \sum_{\substack{n \in \mathbb{Z}_+ \text{ square-free} \\ 1 < n \leq X, (n, \mathbf{P}(z))=1}} a_n - \sum_{\substack{p \in \mathbf{Prime}, m \in [X]_+ \text{ square-free} \\ (m, \mathbf{P}(z))=1, z \leq p \leq \sqrt{X}, p \nmid m, mp \leq X}} \beta_m(X) a_{mp}.$$

┘

*Proof:* If  $n \in \mathbf{Prime}$  and  $\sqrt{X} < n \leq X$ , then clearly the LHS vanishes. On the other hand, if  $n$  has a prime divisor smaller than  $\sqrt{X}$ , then then clearly the LHS equals 1.  $\square$

**Thm. (35.2.4.11) [Vinogradov's method].** Notation as in (35.2.4.10),

$$\begin{aligned} \sum_{p \in \mathbf{Prime} \leq X} a_p &= \sum_{d|\mathbf{P}(z), d \leq \sqrt{X}} \mu(d) A_d(X) - \sum_{\substack{m \in [X]_+ \text{ square-free} \\ (m, \mathbf{P}(z))=1}} \beta_m(X) \sum_{\substack{p \in \mathbf{Prime} \\ z \leq p \leq \sqrt{X}, p \leq X/m}} a_{mp} \\ &\quad + \varepsilon(X, z) \sqrt{X} \|\mathbf{a}\mathbf{1}_{[X]_+}\|. \end{aligned}$$

where  $A_d(X)$  is given in (35.2.1.2), and

$$|\varepsilon(x, z)| \leq \frac{2}{\sqrt{z}} + \frac{1}{2^{20}} \exp\left(-\frac{\log X}{2 \log z}\right) \log(8X)^{21}.$$

In application, when  $(a_n)$  is very oscillating, it is natural to take  $z = \exp(\sqrt{\log X})$  and get

$$\left| \sum_{p \in \mathbf{Prime} \leq X} a_p \right| \leq \sum_{d \in [\sqrt{X}]_+} |A_d(X)| + \sum_{m \in [X]_+} \left| \sum_{\substack{p \in \mathbf{Prime} \\ z \leq p \leq \sqrt{X}, p \leq X/m}} a_{mp} \right| + \varepsilon(X, z) \sqrt{X} \|\mathbf{a}\mathbf{1}_{[X]_+}\|.$$

┘

*Proof:* Based on Ramaré's Identity (35.2.4.10), we relax the restrictions that  $n$  is square-free and  $p \nmid m$ , then the error term is bounded by

$$\sum_{n \in [X]_+, \exists p \in \mathbf{Prime}_{\geq z} p^2 | n} |a_n| \leq \|\mathbf{a}\mathbf{1}_{[X]_+}\| \left( \sum_{p \in \mathbf{Prime}_{\geq z}} X p^{-2} \right)^{1/2} \leq X^{1/2} z^{-1/2} \|\mathbf{a}\mathbf{1}_{[X]_+}\|$$

Then we also add the terms  $a_1$  and  $\{a_p\}_{p \in \text{Prime}_{\leq \sqrt{X}}}$ , which is bounded by

$$\sum_{n \in [\sqrt{X}]_+} |a_n| \leq \sqrt{X} \left\| \mathbf{a} \mathbf{1}_{[X]_+} \right\|.$$

Next we use Legendre's formula (35.2.1.2) to get

$$\left| \sum_{p \in \text{Prime}_{\leq X}} a_p - \sum_{d | \mathbb{P}(z)} \mu(d) A_d(X) + \sum_{\substack{m \in [X]_+ \text{ square-free} \\ (m, \mathbb{P}(z)) = 1}} \beta_m(X) \sum_{\substack{p \in \text{Prime} \\ z \leq p \leq \sqrt{X}, p \leq X/m}} a_{mp} \right| \leq (X^{1/4} + X^{1/2} z^{-1/2}) \left\| \mathbf{a} \mathbf{1}_{[X]_+} \right\|.$$

Then add the restriction that  $d \leq \sqrt{X}$ , then the error term is bounded by

$$\begin{aligned} \sum_{d \in \mathbb{Z}_{> \sqrt{X}}, d | \mathbb{P}(z)} |A_d(X)| &\leq \sum_{d \in \mathbb{Z}_{> \sqrt{X}}, d | \mathbb{P}(z), n \leq X/d} |a_{dn}| \\ (35.1.3.9) &\leq \sqrt{\sum_{n \in [X]_+} |a_n|^2} \left( \sum_{d \in \mathbb{Z}_{> \sqrt{X}}, d | \mathbb{P}(z), n \leq X/d} \tau(dn) \right)^{1/2} \\ (35.1.3.8) &\leq \sqrt{X \log(3X) \sum_{n \in [X]_+} |a_n|^2} \left( \sum_{d \in \mathbb{Z}_{> \sqrt{X}}, d | \mathbb{P}(z)} \frac{\tau(d)}{d} \right)^{1/2} \\ (35.2.4.12) &\leq \exp\left(-\frac{\log X}{\log z}\right) (\log z + 1)^{e^3}. \end{aligned}$$

Then the assertion follows. □

**Lemma (35.2.4.12) [Rankin's Trick].** For  $X, z \in \mathbb{R}_{\geq 1}$ ,

$$\sum_{d | \mathbb{P}(z), d > \sqrt{X}} \frac{\tau(d)}{d} \leq \exp\left(-\frac{\log X}{\log z}\right) (\log z + 1)^{e^3}.$$

┘

*Proof:* For any  $\varepsilon \in \mathbb{R}_+$ , we have

$$\begin{aligned} \sum_{d | \mathbb{P}(z), d > \sqrt{X}} \frac{\tau(d)}{d} &\leq X^{-\varepsilon} \sum_{d | \mathbb{P}(z)} \tau(d) d^{2\varepsilon-1} \\ &\leq X^{-\varepsilon} \prod_{p \in \text{Prime}_{< z}} (1 + 2p^{2\varepsilon-1}) \\ &\leq X^{-\varepsilon} \prod_{p \in \text{Prime}} (1 + p^{-1-\varepsilon})^{2z^{3\varepsilon}} \\ (35.1.2.6) &\leq X^{-\varepsilon} (1 + \varepsilon^{-1})^{2z^{3\varepsilon}} \end{aligned}$$

The assertion follows by taking  $\varepsilon = 1/\log z$ . □

### Prime-Summation Inequalities

**Thm. (35.2.4.13).** There exists  $\kappa \in \mathbb{R}_+$  s.t. : For any  $\mathcal{A} = (a_n)_{n \in \mathbb{Z}_+} \subset \mathbb{C}$  be a sequence of complex numbers s.t.  $|a_n| \leq \tau(n)$  for each  $n \in \mathbb{Z}_+$ , and  $X \in \mathbb{R}_{\geq e^{12}}, \varepsilon \in [\log^{-1} X, 1/12]$ , if

$$\sum_{d \in [X^{1/2-\varepsilon}]} \left| \sum_{dm \leq X} a_{dm} \right| \leq \frac{X}{\log^2 X},$$

and

$$\sum_{(m, P(w))=1} \left| \sum_{\substack{p \in \text{Prime} \cap [X/m]_+ \\ \leq p < X^{1/3-\varepsilon}}} p^{it} a_{pm} \right| \leq \frac{X}{\log^4 X}$$

for each  $t \in \mathbb{R}$ , where  $w = X^{\varepsilon/10 \log(2) X}$  and  $P(w) = \prod_{p \in \text{Prime}_{<w}}$ , then we have

$$\sum_{p \in \text{Prime} \cap [X]_+} a_p \leq \kappa \varepsilon \frac{X}{\log X}.$$

┘

*Proof:* Cf. [I-K04]P351 ?.

□

## 5 Numbers Represented by Homogenous Forms

**Thm. (35.2.5.1) [Fermat/Jacobi].** For  $n \in \mathbb{Z}_+$ , let  $r_2(n) = \#\{(x, y) \in \mathbb{Z} | x^2 + y^2 = n\}$ , then  $r(n) \neq 0$  iff  $2|v_p(n)$  for any  $p \equiv 3 \pmod{4}$ . Moreover,

$$r_2(n) = 4 \cdot \#\{d \in \mathbb{Z}_+, d|n, d \equiv 1 \pmod{4}\} - \#\{d \in \mathbb{Z}_+, d|n, d \equiv 3 \pmod{4}\}.$$

┘

*Proof:*  $1 + \sum_{n \in \mathbb{Z}_+} r_2(n) e^{2\pi i n \tau} = \theta^2(\tau) \in \text{MF}_1(\Gamma_0(4), \chi_4)$  by (26.3.1.15), and also

$$G_{4, \chi_4} = 1 + \sum_{n \in \mathbb{Z}_+} \left( \sum_{d \in \mathbb{Z}_+, d|n} \chi_4(d) \right) e^{2\pi i n \tau} \in \text{MF}_1(\Gamma_0(4), \chi_4) \text{ (26.3.4.10)}.$$

But  $\dim \text{MF}_1(\Gamma_1(4)) \leq 1$  by (26.3.3.17), so we can compare coefficients. □

**Thm. (35.2.5.2) [Lagrange].** For  $n \in \mathbb{Z}_+$ , let  $r_4(n) = \#\{(x, y) \in \mathbb{Z} | x^2 + y^2 = n\}$ , then  $r_4(n)$  equals 8 times the sum of positive divisors of  $n$  that are not multiples of 4.

In particular, any  $n \in \mathbb{Z}_+$  is a sum of four squares. ┘

*Proof:* Calculate  $\theta^4(\tau) \in \text{MF}_2(\Gamma_0(4))$ , where  $\dim \text{MF}_2(\Gamma_0(4)) = 2$ . ? Cf. [Modular Forms 1-2-3]P27.

□

### 35.3 Character Sums

References are [I-K04], [Kats's work] and [Weil conjectures and Fourier transform].

#### 1 Van der Corput's method

**Prop. (35.3.1.1).** If  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  is smooth and  $a < b \in \mathbb{R}$  s.t.  $|f'(x)| \leq 1 - \theta$  and  $f''(x) \neq 0$  on  $[a, b]$ , then

$$\sum_{n \in \mathbb{Z} \cap (a, b)} g(n) e^{2\pi i f(n)} = \int_a^b g(x) e^{2\pi i f(x)} dx + O(G\theta^{-1}),$$

where  $G = |g(b)| + \int_a^b |g'(t)| dt$ . ┘

*Proof:* Cf. [I-K04]P206. □

#### 2 Vinogradov's method

#### 3 Incomplete Sums

**Prop. (35.3.3.1).** For any  $\alpha \in \mathbb{R} \setminus \mathbb{Z}$  and  $M \in \mathbb{Z}, N \in \mathbb{Z}_+$ , we have

$$\left| \sum_{M < n \leq M+N} e^{2\pi i n \alpha} \right| \leq \min(N, \frac{1}{2\{\{\alpha\}\}}).$$

*Proof:*

$$\left| \sum_{M < n \leq M+N} e^{2\pi i n \alpha} \right| = \left| \frac{\sin(N\alpha\pi)}{\sin(\alpha\pi)} \right| \leq \frac{1}{2\{\{\alpha\}\}}.$$

□

**Prop. (35.3.3.2) [Completing Method].** Suppose  $F : \mathbb{Z} \rightarrow \mathbb{C}$  has period  $q$  and  $G(x)$  is a nice analytic function that decays to 0 rapidly as  $x \rightarrow \infty$ , denote

$$S_{F,G} = \sum_{n \in \mathbb{Z}} F(n) G(n),$$

then by Poisson summation (Cf. [I-K04]P69?),

$$S_{F,G} = \frac{1}{q} \sum_{a \in \mathbb{Z}} S_F\left(\frac{a}{q}\right) G^\vee\left(\frac{a}{q}\right),$$

where

$$S_F\left(\frac{a}{q}\right) = \sum_{n \in \mathbb{Z}/(q)} F(n) e^{2\pi i \frac{an}{q}}.$$

In particular,

$$S_F(M, N) = \sum_{M < n \leq M+N} F(n) = \frac{1}{q} \sum_{a \in \mathbb{Z}/(q)} \lambda\left(\frac{a}{q}\right) S_F\left(\frac{a}{q}\right), \text{ where } \lambda\left(\frac{a}{q}\right) = \sum_{M < n \leq M+N} e^{-2\pi i \frac{an}{q}}$$

by orthogonality. ┘

**Cor. (35.3.3.3).** Notation as in (35.3.3.2), by (35.3.3.1),

$$\left| S_F(M, N) - \frac{N}{q} S(0) \right| \leq \sum_{0 < |a| < \frac{q}{2}} \frac{1}{2|a|} \left| S\left(\frac{a}{q}\right) \right|.$$

┘

**Thm. (35.3.3.4) [Pólya-Vinogradov Inequality, Pólya[?P\ 'o118]/Vinogradov[Vin18]].** For any  $q \in \mathbb{Z}_{\geq 2}$  and  $\chi \in \text{Diri}(q) \setminus \mathbf{1}_q$  and  $M, N \in \mathbb{Z}_+$ , we have

$$\left| \sum_{M < n \leq M+N} \chi(n) \right| \leq 6\sqrt{q} \log q.$$

┘

*Proof:* Cf. [I-K04]P324. □

#### 4 Katz's Trigonometric Sums

**Def. (35.3.4.1) [Generalized Trigonometric Sums].** For  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}}$ ,  $\mathbf{1} \neq \psi \in \mathbb{F}_q^{\vee}$ , and  $h : X \rightarrow \mathbb{A}_{\mathbb{F}_q}^1 \in \text{Sch}^{\text{fg}}/\mathbb{A}_{\mathbb{F}_q}^1$ , the generalized **trigonometric sum** is defined to be the exponential sum

$$\sum_{x \in X(\mathbb{F}_q)} \psi(h(x)).$$

┘

**Thm. (35.3.4.2) [Katz].** Let  $h : X \rightarrow \mathbb{A}^1 \in \text{Sch}^{\text{fg}}/\mathbb{A}^1$ , then there exists  $C \in \mathbb{R}_+$  s.t. for any  $p \in \mathbf{P}$  and  $q \in p^{\mathbb{Z}_+}$ ,  $\psi \neq \mathbf{1} \in \chi(\mathbb{F}_q)$ ,

$$\left| \sum_{x \in A(\mathbb{F}_q)} \psi(h(x)) \right| \leq Cq^{N+\frac{1}{2}},$$

where  $N$  is the supremum of the dimension of the fiber of  $h_{\mathbb{C}}$ .

And if moreover the generic fiber  $h_{\bar{\eta}}$  is geo.irreducible or has dimension  $< N$ , then for any  $p \in \mathbf{P}$  and  $q \in p^{\mathbb{Z}_+}$ ,  $\psi \neq \mathbf{1} \in \chi(\mathbb{F}_q)$ ,

$$\left| \sum_{x \in A(\mathbb{F}_q)} \psi(h(x)) \right| \leq Cq^N.$$

┘

*Proof:* Cf. [K-W01]P228. □

#### 5 Gauss-Sums

**Def. (35.3.5.1) [Gauss-Sums and Heilbronn-Sums].** For  $N \in \mathbb{Z}_+$  and  $f(X) \in \mathbb{Z}[X]$ , a **Gauss sum** is a summation

$$G_N(f) = \sum_{n=1}^N e^{2\pi i f(n)/N}.$$

A special case is the summation

$$G_N(a; k) = G_p(aX^k) = \sum_{n=1}^p e^{2\pi i a n^k / p}$$

where  $k \in \mathbb{Z}_+$ ,  $a \in \mathbb{Z} \setminus (p)$ .

And a **Heilbronn-sum** is the summation of the form

$$H_p(a) = \sum_{n=1}^N e^{2\pi i \frac{a n^p}{p^2}}, \text{ where } p \in \text{Prime}.$$

Notice if  $p \in \text{Prime}$  and  $k_0 = \gcd(k, p-1)$ , then  $G_p(a, k) = G_p(a, k_0)$ . Thus for  $G_p(a; k)$ , we always assume  $k|p-1$ .  $\perp$

**Prop. (35.3.5.2).** If  $a \in \mathbb{Z}$  and  $N_1, N_2 \in \mathbb{Z}_+$  with  $(N_1, N_2) = 1$ , then

$$G_N(a; k) = G_{N_1; k}(aN_2^{k-1}; k)G_{N_2}(aN_1^{k-1}; k).$$

$\perp$

*Proof:* Writing any  $n \pmod{N}$  as  $n \equiv n_1 N_2 + n_2 N_1 \pmod{N}$ .  $\square$

**Prop. (35.3.5.3) [Dirichlet].** For any  $N \in \mathbb{Z}_+$ , we have

$$G_N(1; 2) = \frac{1 + i^{-N}}{1 - i} \sqrt{N}.$$

$\perp$

*Proof:* Firstly, if  $N$  is even

$$G_N(1; 2) = \sum_{n \pmod{N}} e^{2\pi i \frac{n^2}{N}} = \sum_{n \pmod{N}} e^{2\pi i \frac{(n+N/2)^2}{N}} = i^m G_N(1; 2),$$

so  $G(m) = 0$  if  $m \equiv 2 \pmod{4}$ , so the assertion holds for  $N \equiv 2 \pmod{4}$ . Now assume  $N \not\equiv 2 \pmod{4}$ , then

$$G_{N^3}(1; 2) = N G_N(1; 2)$$

because we can write  $a \pmod{N^3}$  as  $a \equiv r + sN^2 \pmod{N^3}$  when  $N$  is odd and  $a \equiv r + s\frac{N^2}{2} \pmod{N^3}$  when  $4|N$ .

Next,

$$G_N(1; 2) = 2 \sum_{n \in [\frac{N}{2}]_+} e^{2\pi i \frac{n^2}{N}} + O(1) = 2 \sum_{n \in [\frac{N}{4}]_+} e^{2\pi i \frac{n^2}{N}} + 2i^N \sum_{n \in [\frac{N}{4}]_+} e^{2\pi i \frac{(n+\{N/2\})^2}{N}} + O(1) \quad (35.3.1.1).$$

And notice

$$\sum_{n \in [\frac{N}{2}]_+} e^{2\pi i \frac{n^2}{N}} = \int_0^{N/4} e^{2\pi i \frac{x^2}{N}} dx = \frac{1+i}{4} \sqrt{N} \quad (12.4.6.5),$$

and similarly for  $\sum_{n \in [\frac{N}{2}]_+} e^{2\pi i \frac{(n+\{N/2\})^2}{N}}$ , so

$$G_N(1; 2) = \frac{1 + i^{-N}}{1 - i} \sqrt{N} + O(1).$$

But this holds for  $N^{3^k}$  for any  $k \in \mathbb{Z}_+$  with the same error term, so the assertion follows.  $\square$



**Cor. (35.3.5.4).** For  $p \in \text{Prime}_{\geq 3}$ ,  $a, b, c \in \mathbb{Z}$  s.t.  $(a, p) = 1$ , we have

$$\sum_{t=0}^{p-1} e^{2\pi i \frac{at^2+bt+c}{p}} = e^{2\pi i \frac{c-b^2[4a]^{-1}}{p}} \left(\frac{a}{p}\right) \frac{1-i^p}{1-i} \sqrt{p}, \text{ where } \epsilon_p = \begin{cases} 1 & , p \equiv 1 \pmod{4} \\ i & , p \equiv 3 \pmod{4} \end{cases}.$$

In particular,

$$G_p(a) = \left(\frac{a}{p}\right) \frac{1-i^p}{1-i} \sqrt{p}.$$

┘

*Proof:* By completing square, it suffices to prove for  $b = c = 0$ . But then

$$G_p(a) = \sum_{t=0}^{p-1} e^{2\pi i \frac{at^2}{p}} = \sum_{b \pmod{p}} \left(1 + \left(\frac{b}{p}\right)\right) e^{2\pi i \frac{ab}{p}} = \left(\frac{a}{p}\right) G_p(1) = \left(\frac{a}{p}\right) \frac{1+i^{-p}}{1-i} \sqrt{p}.$$

□

**Prop. (35.3.5.5).** For  $p \in \text{Prime}_{\geq 3}$ ,  $n \in \mathbb{Z}_+$ ,  $a \in \mathbb{Z}$ ,

$$\left| \sum_{t=1}^{p^{n+1}} e^{2\pi i \left[ \frac{at^2}{p} + \frac{ht}{p^{n+1}} \right]} \right| = \begin{cases} p^{n+\frac{1}{2}} & , p^n | h, p \nmid a \\ 0 & , p^n \nmid h \end{cases}.$$

┘

*Proof:*

$$\sum_{t=1}^{p^{n+1}} e^{2\pi i \left[ \frac{at^2}{p} + \frac{ht}{p^{n+1}} \right]} = \sum_{t=1}^p e^{2\pi i \left[ \frac{at^2}{p} + \frac{ht}{p^{n+1}} \right]} \left( \sum_{r=1}^{p^n} e^{2\pi i \frac{hr}{p^n}} \right).$$

Then the assertion follows from (35.3.5.4).

□

**Prop. (35.3.5.6) [Burgess].** For  $p \in \text{Prime}$ ,  $s \in \mathbb{Z}_+$  and  $b_1, \dots, b_s \in \mathbb{F}_p$ ,

$$\left| \sum_{x \in \mathbb{F}_p} \left( \frac{\prod_i (x + b_i)}{p} \right) \right| \leq s^2 \sqrt{p},$$

unless  $s \in 2\mathbb{Z}_+$  and up to permutation  $b_1 = b_2, b_3 = b_4, \dots, b_{s-1} = b_s$ .

┘

*Proof:* This is clear for  $s = 1$ , and for  $s \geq 2$ , use Weil conjecture on the the hyperelliptic curve

$$y^2 = \prod_i (x + b_i).$$

□

**Prop. (35.3.5.7) [Quadratic Legendre Symbol Sum].** For  $p \in \text{Prime}_{\geq 3}$ ;

$$\sum_{x=0}^{p-1} \left( \frac{x^2 + 1}{p} \right) = -1.$$

┘

*Proof:* The sum is equivalent modulo  $p$  to  $\sum_{x=0}^{p-1}(x^2 + ax + b)^{\frac{p-1}{2}}$ , which by (2.6.1.29) is equivalent to  $-1$  modulo  $p$ . Now it can not be  $p-1$ , because otherwise there is a solution for  $p|x^2 + 1$ , and then it can be calculated directly.  $\square$

**Prop. (35.3.5.8) [Multiplicative Character Bound, Burgess].** For any  $\beta \in \mathbb{R}_{>1/4}$ , there exists  $C(\beta), e(\beta) \in \mathbb{R}_+$  s.t. for any  $\chi : \mathbb{F}_p^\times \rightarrow \mathbb{C}^\times$  and  $x_0 \in \mathbb{Z}$ ,

$$\left| \sum_{x=x_0}^{x_0+N-1} \chi(x) \right| \leq C(\beta) N^{1-e(\beta)}.$$

┘

*Proof:* ?

□

**Cor. (35.3.5.9).** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C(\varepsilon) \in \mathbb{R}_+$  s.t. for any  $p \in \mathbf{Prime}$ , any interval of length  $C(\varepsilon)p^{1/4+\varepsilon}$  contains a square and a non-square.  $\square$

**Thm. (35.3.5.10) [Weil].** For  $p \in \mathbf{Prime}$  and  $f(X) \in \mathbb{F}_p[X]$  with  $\deg(f) = k$ ,

$$|G_p(f)| \leq k\sqrt{p}.$$

┘

*Proof:* Cf. [Larry Guth]P56, and a weaker bound is found in [Larry Guth]P62.  $\square$

### Monomial Gauss Sums and Heilbronn Sums

**Conj. (35.3.5.11).**

$$|G_p(a, k)| \leq \min \left( (k-1)\sqrt{p}, (1+\eta)\sqrt{2kp \log(kp)} \right),$$

where  $\eta \rightarrow 0$  as  $k$  and  $p/k$  both tend to infinity.  $\square$

*Proof:*

□

**Prop. (35.3.5.12).**

$$|G_p(a, k)| \leq \min(p, (k-1)\sqrt{p}).$$

┘

*Proof:*  $|G_p(a, k)| \leq p$  is trivial. For the second assertion, notice

$$G_p(a, k) = \sum_{\chi \in \text{Diri}(p), \chi^k = 1, \chi \neq 1} \chi^{-1}(a) \tau(\chi) \text{ (35.3.6.8)},$$

and use the fact  $|\tau(\chi)| = \sqrt{p}$  for each  $\chi$ .  $\square$

**Thm. (35.3.5.13) [Heath-Brown-Konyagin].** Situation as in (35.3.5.1),

$$G_p(a, k) = O(\min(k^{5/8}p^{5/8}, k^{3/8}p^{3/4})).$$

┘

*Proof:* Cf. [B-K20]P223.  $\square$

**Lemma (35.3.5.14).** Situation as in (35.3.5.1), if  $h = (p-1)/k$ , and

$$A(h) = \{(x_1, \dots, x_4) \in \mu_h(\mathbb{Z}_p)^4 \mid x_1 + x_2 = x_3 + x_4\},$$

then

$$|G_p(a, k)| \leq \min(k^{4/5}, p^{1/8})(\#A(h))^{1/4}.$$

┘

*Proof:* Cf. [B-K20]P224.

□

**Lemma (35.3.5.15).** Situation as in (35.3.5.14), if  $h < p^{2/3}$ , then  $\#A(h) = O(h^{5/2})$ .

┘

*Proof:* Cf. [B-K20]P225.

□

**Thm. (35.3.5.16) [Heath.Brown-Konyagin].** Situation as in (35.3.5.1),

$$\sum_{r=1}^p |H_p(a + rp)|^4 = O(p^{7/2}),$$

in particular,

$$H_p(a) = O(p^{7/8}).$$

┘

*Proof:*

□

**Lemma (35.3.5.17).** Situation as in (35.3.5.1), if

$$f(X) = X + \frac{X^2}{2} + \dots + \frac{X^{p-1}}{p-1} \in \mathbb{Z}_p[X],$$

and

$$B = \{(x_1, x_2) \in \mathbb{Z}_p^2 \mid f(x_1) = f(x_2)\},$$

then

$$\sum_{r=1}^p |H_p(a + rp)|^4 = O(p^3 + p^2 \#B).$$

┘

*Proof:* Cf. [B-K20]P224.

□

**Lemma (35.3.5.18).** Situation as in (35.3.5.17), then  $\#B = O(p^{3/2})$ .

┘

*Proof:* Cf. [B-K20]P225.

□

## Gauss Sum over Finite Fields

### 6 Dirichlet Characters

**Def. (35.3.6.1) [Dirichlet Character, Dirichlet(1837)].** For  $F \in \mathbf{NField}$  and  $\mathfrak{m} \in \mathbf{Ideal}^\times(\mathcal{O}_F)$ , a **Dirichlet character modulo  $\mathfrak{m}$**  is a map  $\chi : (\mathcal{O}_F/\mathfrak{m})^* \rightarrow \mathbb{C}^\times$ .

The **conductor of Dirichlet character** of  $\chi : (\mathcal{O}_F/\mathfrak{m})^* \rightarrow \mathbb{C}^\times$  is defined to be the maximal modulus  $\mathfrak{m}' \supset \mathfrak{m}$  s.t.  $\chi$  factors through the quotient map  $(\mathcal{O}_F/\mathfrak{m})^* \rightarrow (\mathcal{O}_F/\mathfrak{m}')^*$ .  $\chi$  is called a **primitive Dirichlet character** if  $c(\chi) = \mathfrak{m}$ .

The set of Dirichlet characters modulo  $\mathfrak{m}$  is denoted by  $\mathbf{Diri}(\mathfrak{m})$ , and the set of primitive Dirichlet characters modulo  $\mathfrak{m}$  is denoted by  $\mathbf{Diri}'(\mathfrak{m})$ .

For each  $\mathfrak{m} \in \mathbf{Ideal}^\times(\mathcal{O}_F)$ , there is a **principal Dirichlet character  $\mathbf{1}_\mathfrak{m} : (\mathcal{O}_F/\mathfrak{m})^* \rightarrow \mathbb{C}^\times$**  with image  $\{1\}$ .  $\lrcorner$

**Prop. (35.3.6.2).** For coprime elements  $\mathfrak{m}_1, \mathfrak{m}_2 \in \mathbf{Ideal}^\times(\mathcal{O}_F)$ ,  $(\mathbb{Z}/\mathfrak{m}_1\mathfrak{m}_2)^* \cong (\mathbb{Z}/\mathfrak{m}_1)^* \times (\mathbb{Z}/\mathfrak{m}_2)^*$  via projection. Thus any  $\chi \in \mathbf{Diri}(\mathfrak{m}_1\mathfrak{m}_2)$  is uniquely a product  $\chi = \chi_1 \cdot \chi_2$  where  $\chi_i \in \mathbf{Diri}(\mathfrak{m}_i)$ .  $\lrcorner$

**Def. (35.3.6.3) [Quadratic Dirichlet Characters].** A **quadratic Dirichlet character** or a **real Dirichlet character** is a Dirichlet character with image  $\{\pm 1\}$ .  $\lrcorner$

**Def. (35.3.6.4) [Kronecker Characters].** If  $D \in \mathbb{Z}$  is a fundamental discriminant(19.3.1.47), denote  $\chi_D \in \mathbf{Diri}(|D|)$  the character  $\chi_D(m) = \left(\frac{D}{m}\right)$ (2.6.1.27), called the **Kronecker character**.

Notice that  $\chi_D$  are distinct primitive quadratic Dirichlet characters, and every primitive quadratic Dirichlet character is a Kronecker character.  $\lrcorner$

*Proof:* To verify that  $\chi_D$  is a Dirichlet character, it suffices to verify for each  $D \in \{-4, 8, -8\} \cup \left\{\left(\frac{-1}{p}\right)p\right\}_{p \in \mathbf{Prime}_{\geq 3}}$ . It follows from quadratic reciprocity(19.7.4.34) that

$$\chi_{-4}(n) = \begin{cases} 0 & , n \in (2) \\ 1 & , n \equiv 1 \pmod{4} \\ -1 & , n \equiv -1 \pmod{4} \end{cases}, \quad \chi_8(n) = \begin{cases} 0 & , n \in (2) \\ 1 & , n \equiv 1, -1 \pmod{8} \\ -1 & , n \equiv 3, -3 \pmod{8} \end{cases}$$

$$\chi_{-8}(n) = \chi_{-4}(n)\chi_8(n) = \begin{cases} 0 & , a \in (2) \\ 1 & , a \equiv 1, 3 \pmod{8} \\ -1 & , a \equiv -1, -3 \pmod{8} \end{cases}.$$

So they are Dirichlet characters. And for  $p \in \mathbf{Prime}_{\geq 3}$ ,  $\chi\left(\frac{-1}{p}\right)_p$  is a Dirichlet character by quadratic reciprocity(19.7.4.34).

To show that every primitive quadratic Dirichlet character is a Kronecker character, it suffices to notice that  $(\mathbb{Z}/(2^r))^\times \cong \mathbb{Z}/2$  or  $\mathbb{Z}/2 \times \mathbb{Z}/2^{r-2}$ , and  $(\mathbb{Z}/(p^r))^* \cong \mathbb{Z}/((p-1)p^{r-1})$  for  $p \in \mathbf{Prime}_{\geq 3}$ .  $\square$

**Prop. (35.3.6.5).** If  $m \in \mathbb{Z}_+$  and  $\chi \in \mathbf{Diri}'(m)$ , then for any  $a, b \in \mathbb{Z}$ , we have

$$\frac{1}{m} \sum_{x \in \mathbb{Z}/(m)} \chi(ax + b) = \begin{cases} \chi(b) & , m|a \\ 0 & , m \nmid a \end{cases}.$$

$\lrcorner$

*Proof:* By multiplicity, we can assume  $m$  is a prime power  $p^r$ . Denote the LHS by  $S$ , then it can be verified that  $\chi(1 + m_1 t)S = S$  for any  $t \in \mathbb{Z}$ , where  $m_1 = m(a, m)/(a^2, m)$ . If  $S \neq 0$ , then  $\chi$  is periodic of period  $m_1$ . But  $\chi$  is primitive, thus  $m_1 = m$ , showing that  $m|a$ . And in this case it is clear that  $S = \chi(b)$ .  $\square$

**Prop. (35.3.6.6).** If  $a, m \in \mathbb{Z}_+$  and  $(a, m) = 1$ , then

$$\sum_{\chi \in \text{Diri}'(m)} \chi(a) = \sum_{d|(a-1, m)} \phi(d) \mu\left(\frac{m}{d}\right).$$

┘

*Proof:* Use Möbius inversion and orthogonality of characters.  $\square$

**Prop. (35.3.6.7) [Real Primitive Characters].** Cf. [I-K04]P49.  $\perp$

### Gauss-Sums for Hecke-Characters

**Def. (35.3.6.8) [Gauss-Sum for Hecke-Characters].** For  $F \in \mathbf{NField}$ , a modulus  $\mathfrak{m}$  for  $F$  and a Hecke character  $\chi$  with Dirichlet component  $\chi_0$  of conductor  $\mathfrak{m}$  (19.4.1.22), the **Gauss-sum** of  $\chi$  is defined to be

$$\tau(\chi) = \frac{1}{\chi_\infty(\gamma)\chi(\mathfrak{c})} \sum_{\alpha \in \mathfrak{c}/\mathfrak{c}\mathfrak{m}} \chi_0(\alpha) e^{2\pi i \text{tr}_{F/\mathbb{Q}}(\alpha/\gamma)},$$

where  $\gamma \in \mathcal{O}_F^\times$  and  $\mathfrak{c} \in \text{Ideal}^\times(\mathcal{O}_F)$  are chosen s.t.  $(\mathfrak{c}, \mathfrak{m}) = 1$  and  $\mathfrak{c}\mathfrak{d}_F\mathfrak{m} = (\gamma)$ . The definition is independent of  $(\mathfrak{c}, \gamma)$  chosen.

In particular, if  $F = \mathbb{Q}$  and  $\mathfrak{m} = (N)$  for  $N \in \mathbb{Z}_+$ , then for  $a \in \mathbb{Z}$ , define

$$\tau_a(\chi) \triangleq \sum_{n \in \mathbb{Z}/(N)} \chi(n) e^{2\pi i \frac{an}{N}}, \quad \tau(\chi) \triangleq \tau_1(\chi).$$

In particular,  $c_N(a) \triangleq \tau_a(\mathbf{1}_N)$  is called the **Ramanujan sum**, and by Möbius inversion on  $N$ , we have

$$\text{Kloos}(a, 0, \text{mod } N) = \sum_{r \in (\mathbb{Z}/(N))^*} e^{2\pi i \frac{ar}{N}} = \sum_{d|(N, n)} = \mu\left(\frac{N}{(N, n)}\right) \frac{\phi(N)}{\phi(N/(N, n))}.$$

┘

**Prop. (35.3.6.9).** For  $F \in \mathbf{NField}$  and a Hecke character  $\chi$  for  $F$  with Dirichlet component  $\chi_0$  of conductor  $\mathfrak{m}$ , we have

$$\tau(\chi) = \chi(\mathfrak{a}) \tau_{\chi_0}(\beta)$$

for some  $(\mathfrak{a}, \mathfrak{m}) = 1$  and  $\beta \in (\mathfrak{d}\mathfrak{m})^{-1}$ .  $\perp$

*Proof:* Cf. [I-K04]P61. ?  $\square$

**Prop. (35.3.6.10).** For  $\chi \in \text{Diri}(N)$ ,  $\tau(\overline{\chi}) = \chi(-1) \overline{\tau(\chi)}$ .  $\perp$

**Lemma (35.3.6.11) [Fourier-Expansions].** For  $\chi \in \text{Diri}'(N)$  and  $m \in \mathbb{Z}$ , we have  $\sum_{n \pmod{N}} \chi(n) e^{2\pi i \frac{nm}{N}} = \overline{\chi(m)} \tau(\chi)$  (35.3.6.8).  $\perp$

*Proof:* If  $(m, N) = 1$ , then this follows from

$$\tau(\chi) = \sum_{n(\bmod N)} \chi(mn) e^{2\pi i mn/N} = \chi(m) \sum_{n(\bmod N)} \chi(n) e^{2\pi i mn/N}.$$

If  $(m, N) \neq 1$ , then we need to show the LHS is 0. Let  $m = dM, N = dN_1$ . Because  $\chi$  is primitive character mod  $N$ , there is some  $c \equiv 1 \pmod{N_1}$  that  $\chi(c) \neq 1$ , otherwise  $\chi$  is defined mod  $N_1$ . Notice

$$\sum_{n(\bmod N)} \chi(n) e^{2\pi i mn/N} = \sum_{r(\bmod N_1)} \left( \sum_{n(\bmod N), n \equiv r(\bmod N_1)} \chi(n) \right) e^{2\pi i rM/N_1}$$

But  $r \mapsto cr$  is a permutation of  $\{n(\bmod N), n \equiv r(\bmod N_1)\}$ , thus

$$\sum_{n(\bmod N), n \equiv r(\bmod N_1)} \chi(n) = \sum_{n(\bmod N), n \equiv r(\bmod N_1)} \chi(cn) = \chi(c) \sum_{n(\bmod N), n \equiv r(\bmod N_1)} \chi(n)$$

which means this sum vanishes.  $\square$

**Prop. (35.3.6.12).** For  $\chi \in \text{Diri}'(N)$ ,  $|\tau(\chi)|^2 = N$  (35.3.6.8). And if  $\chi \in \text{Diri}(N)$  is induced from  $\chi^* \in \text{Diri}(N^*)$ , then

$$\tau(\chi) = \mu\left(\frac{N}{N^*}\right) \chi^*\left(\frac{N}{N^*}\right) \tau(\chi^*).$$

This also holds for Hecke characters.  $\lrcorner$

*Proof:* If  $\chi \in \text{Diri}'(N)$ , for any  $m$ , by (35.3.6.11), we have

$$|\tau(\chi)|^2 = \left| \sum_{n(\bmod N)} \chi(n) e^{2\pi i nm/N} \right|^2 = \sum_{(n_1, n_2, N)=1} \chi(n_1) \overline{\chi(n_2)} e^{2\pi i (n_1 - n_2)m/N},$$

summing over  $m \in (\mathbb{Z}/(N))^*$ , we get

$$\phi(N) |\tau(\chi)|^2 = \sum_{m(\bmod N)} \sum_{(n_1, n_2, N)=1} \chi(n_1) \overline{\chi(n_2)} e^{2\pi i (n_1 - n_2)m/N} = \sum_{n_1 \equiv n_2(\bmod N), (n_1, n_2, N)=1} N = \phi(N) N$$

if  $\chi \in \text{Diri}(N)$  is induced from  $\chi^* \in \text{Diri}(N^*)$ , then by inclusion-exclusion principle,

$$\tau(\chi) = \sum_{a \in (\mathbb{Z}/N)^*} \chi^*(a) e^{2\pi i \frac{a}{N}} = \sum_{d|m} \mu(d) \chi^*(d) \sum_{a \in (\mathbb{Z}/(N/d))^*} \chi^*(a) e^{2\pi i \frac{ad}{N}}.$$

$\square$

**Cor. (35.3.6.13) [Fourier Expansion of Dirichlet Characters].** From (35.3.6.12) (35.3.6.11) and (35.3.6.10), for  $\chi \in \text{Diri}'(N)$ , we get

$$\chi(n) = \frac{\chi(-1)\tau(\chi)}{N} \sum_{m(\bmod N)} \overline{\chi(m)} e^{2\pi i \frac{mn}{N}}.$$

$\lrcorner$

**Cor. (35.3.6.14).** If  $\chi \in \text{Diri}'(N)$  is quadratic, by (35.3.6.12) and (35.3.6.10), we have  $\tau(\chi)^2 = \chi(-1)N$ .

$\lrcorner$

**Prop. (35.3.6.15).** For  $N_1, N_2 \in \mathbb{Z}_+$  s.t.  $(N_1, N_2) = 1$  and  $\chi_i \in \text{Diri}(N_i)$ , we have

$$\tau(\chi_1 \chi_2) = \chi_1(N_2) \chi_2(N_1) \tau(\chi_1) \tau(\chi_2).$$

This also holds for Hecke characters. ┘

*Proof:*

$$\tau(\chi_1 \chi_2) = \sum_{n_1 \pmod{N_1}} \sum_{n_2 \pmod{N_2}} \chi_1(N_2 n_1) \chi_2(N_1 n_2) e^{2\pi i \frac{N_1 n_2 + N_2 n_1}{N_1 N_2}} = \chi_1(N_2) \chi_2(N_1) \tau(\chi_1) \tau(\chi_2).$$

□

**Prop. (35.3.6.16) [Gauss Sum of Quadratic Characters].** If  $N \in 2\mathbb{N} + 1$  is square-free and  $\chi \in \text{Diri}'(N)$  is the unique principal quadratic character (i.e.  $\chi = \left(\frac{\cdot}{N}\right)$ ), then

$$\tau(\chi) = \frac{1 + i^{-N}}{1 - i} \sqrt{N}.$$

And also

$$\tau(\chi_{-4}) = 2i, \quad \tau(\chi_8) = 2\sqrt{2}, \quad \tau(\chi_{-8}) = 2\sqrt{2}i \text{ (35.3.6.4)}.$$

┘

*Proof:* By (35.3.6.15) and quadratic reciprocity (19.7.4.34), it suffices to prove for  $N \in \text{Prime}_{\geq 3}$ . But in this case,  $\tau(\chi) = G_p(1; 2) = \frac{1+i^{-p}}{1-i} \sqrt{p}$  (35.3.5.3). □

**Prop. (35.3.6.17) [Gauss Sum for Quadratic Imaginary Fields].** Let  $\mathcal{K}$  be an imaginary quadratic field,  $N \in \mathbb{Z}_+$  s.t.  $(N, \text{disc}(\mathcal{K})) = 1$ , and  $\chi \in \text{Diri}'(N)$ , then  $\psi = \text{Nm} \circ \chi$  is a Hecke character of  $\mathcal{K}$  with  $\psi_\infty = 1$  and conductor  $(N)$ . Then

$$\tau(\psi) = \chi_{\mathcal{K}}(N) \chi(|\text{disc}(\mathcal{K})|) \tau(\chi)^2.$$

┘

*Proof:* ? □

**Prop. (35.3.6.18) [Finite Fourier Transform].** Let  $\psi$  be a non-trivial character on a finite field  $\mathbb{F}_q$ , then let  $g_\psi = \sum_{x \in \mathbb{F}_q} \psi(x^2)$ , then

- $|g_\psi|^2 = q$ .
- $g_\psi(a) = \left(\frac{a}{\mathbb{F}_q}\right) g_\psi(1)$ .

┘

*Proof:* Use Fourier transform on  $\mathbb{F}_q$  ? □

## 7 Kloosterman Sums

References are [K-W01].

**Def. (35.3.7.1) [Generalized Kloosterman Sums].** Let  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}_+}$ ,  $m \in \mathbb{Z}_+$ , and  $\psi \in \chi(\mathbb{F}_q)$ , then the generalized **Kloosterman sum** is defined to be

$$\text{Kloos}_m(q, \psi, a) = \sum_{x_1, \dots, x_m \in \mathbb{F}_q^\times, x_1 \dots x_m = a} \psi(x_1 + \dots + x_m) \in \mathbb{C}.$$

And  $\text{Kloos}_2(q, \psi, a)$  is denoted by  $\text{Kloos}(q, \psi, a)$ . ┘

**Thm. (35.3.7.2) [Weil/Deligne].**  $|\text{Kloos}_m(q, \psi, a)| \leq \min(mq^{\frac{m-1}{2}}, q^{m/2})$ . ┘

*Proof:* Let  $X_0 = \text{Spec } \mathbb{F}_q[X_1, \dots, X_m]/(X_1 \dots X_m = a)$  and  $h : X \rightarrow \mathbb{A}_{\mathbb{F}_q}^1 : (x_1, \dots, x_m) \mapsto x_1 + \dots + x_m$ . Let  $\mathcal{L}(\psi^{-1})$  be the Artin-Schreier sheaf on  $\mathbb{A}_{\mathbb{F}_q}^1$  (9.4.6.31), then by (9.4.6.33), for any  $x \in X(\mathbb{F}_q)$ , the Frobenius action  $F_{x_0}^*$  at  $x_0$  on  $h^*(\mathcal{L}(\psi^{-1}))$  is multiplication by  $\psi(h(x))$ .

Then it follows from (9.6.1.26) that

$$\text{Kloos}_m(q, \psi, a) = \text{tr}(F_X^*) = \text{tr}(F_X^* | [\text{R}\Gamma(X, h^*\mathcal{L}(\psi^{-1}))]).$$

Thus, to show that  $|\text{Kloos}_m(q, \psi, a)| \leq mq^{\frac{m-1}{2}}$ , it suffices to show the following:

•

$$\dim H_{\text{ét},c}^k(X_{\overline{\mathbb{F}_q}}, h^*(\mathcal{L}(\psi^{-1}))) = m\delta_{k,m-1}.$$

- $H_{\text{ét},c}^{m-1}(X_{\overline{\mathbb{F}_q}}, h^*(\mathcal{L}(\psi^{-1})))$  is pure of weight  $m-1$ .

For this, Cf. [Deligne, Applications de la formule des traces aux sommes trigonometriques, in SGA4.5] Thm 7.4. ?

For the second inequality, Cf. [K-W01] P227. ? □

### Classical Kloosterman Sums

**Def. (35.3.7.3) [Kloosterman Sums].** For  $m \in \mathbb{Z}_+$  and  $a, b \in \mathbb{Z}$ , a classical **Kloosterman sum** is defined to be the summation

$$\text{Kloos}(a, b, \text{ mod } m) = \sum_{r \in (\mathbb{Z}/m)^*} e^{2\pi i \frac{ar+b[r^{-1}]}{m}} \in \mathbb{C}.$$

┘

**Prop. (35.3.7.4) [Properties of Kloosterman Sums].**

- $\text{Kloos}(a, b, \text{ mod } m) = \text{Kloos}(b, a, \text{ mod } m)$ .
- $\text{Kloos}(ac, b, \text{ mod } m) = \text{Kloos}(a, bc, \text{ mod } m)$  if  $(c, m) = 1$ .
- If  $m_1, m_2 \in \mathbb{Z}_+$ ,  $(m_1, m_2) = 1$ , and choose  $n_1, n_2 \in \mathbb{Z}$  s.t.  $n_1 m_1 \equiv 1 \text{ mod } m_2$  and  $n_2 m_2 \equiv 1 \text{ (mod } m_1)$ , then

$$\text{Kloos}(a, b, \text{ mod } m_1 m_2) = \text{Kloos}(n_2 a, n_2 b, \text{ mod } m_1) \text{Kloos}(n_1 a, n_1 b, \text{ mod } m_2).$$

- If  $p \in \text{Prime}$  and  $p|m$ , then

$$\text{Kloos}(pa, pb, pm) = p \text{Kloos}(a, b, m).$$

┘



**Lemma (35.3.7.5).** If  $p \in \text{Prime}$ ,  $m, n, u, v \in \mathbb{Z}_+$  s.t.  $m/2 \leq n < m$ , and  $\gcd(u, v, p) = 1$ , then

$$|\text{Kloos}(u, v, \text{mod } p^m)| \leq Cp^n,$$

$$\text{where } C = \begin{cases} 4 & , p = 2 \text{ \& } m - n \geq 3 \\ 2 & , \text{otherwise} \end{cases}.$$

*Proof:* In this case, for  $s \in \mathbb{Z} \setminus (p)$  and  $t \in \mathbb{Z}$ , it is easy to verify that

$$[s + p^n t]^{-1} \equiv [s^{-1}] - p^n [s^{-1}]^2 t \pmod{p^m}.$$

So

$$\begin{aligned} |\text{Kloos}(u, v, \text{mod } p^m)| &= \left| \sum_{s \in (\mathbb{Z}/p^n)^*} e^{2\pi i \frac{us + v[s^{-1}]}{p^m}} \sum_{t=1}^{p^{m-n}} e^{2\pi i \frac{(u-v[s^{-1}]^2)t}{p^{m-n}}} \right| \\ &\leq p^{m-n} \#\{s \in (\mathbb{Z}/p^n)^* | us^2 \equiv v \pmod{p^{m-n}}\} \\ &= p^m \#\{s \in (\mathbb{Z}/p^{m-n})^* | us^2 \equiv v \pmod{p^{m-n}}\} \\ &\leq Cp^n \end{aligned}$$

$$\text{where } C = \begin{cases} 4 & , p = 2 \text{ \& } m - n \geq 3 \\ 2 & , \text{otherwise} \end{cases}.$$

**Thm. (35.3.7.6)** [Weil[Wei48]/Estermann[Est61]].

$$|\text{Kloos}(u, v, \text{mod } m)| \leq \sigma_0 \left( \frac{m}{(u, v, m)} \right) \sqrt{\gcd(u, v, m)m}. \quad (35.3.7.3)$$

*Proof:* As the assertion is multiplicative in  $m$ , by (35.3.7.4), it suffices to show for  $m = p^k$ , where  $p \in \mathbf{P}, k \in \mathbb{Z}_+$ . If  $\gcd(a, b, p^k) = p^k$ , then the assertion is trivial. And if  $\gcd(a, b, p^k) < p^k$ , then it follows from (35.3.7.4) that we can reduce to the case  $\gcd(a, b, p^k) = 1$ .

If  $p = 2$  and  $k = 1$ , the assertion is trivial, and if  $p = 2$  and  $k \geq 2$ , then the assertion follows from the lemma (35.3.7.5).

If  $p \geq 3$ , and  $2|k$ , then the assertion follows from (35.3.7.5).

So suppose  $p \geq 3$  and  $2 \nmid k$ . If  $k = 1$  and  $p \nmid uv$ , this is just theorem of Weil (35.3.7.2). And if  $k = 1$  and  $p|uv$ , this is also trivial, because  $\text{Kloos}(u, v, \text{mod } p) \in \{0, 1\}$ .

So suppose  $m > 1$  and denote  $n = \frac{k-1}{2}$ . In this case, for  $s \in \mathbb{Z} \setminus (p)$  and  $t \in \mathbb{Z}$ , it can be verified that

$$[(s + p^n t)^{-1}] \equiv [s^{-1}] - p^n [s^{-1}]^2 t + p^{2n} [s^{-1}]^3 t^2 \pmod{p^k}.$$

So

$$\text{Kloos}(u, v, \text{mod } p^k) = \sum_{s \in (\mathbb{Z}/(p^n))^*} e^{2\pi i \frac{us + v[s^{-1}]}{p^k}} \left( \sum_{t \in (\mathbb{Z}/(p^{n+1}))^*} e^{2\pi i \left[ \frac{(u-v[s^{-1}]^2)t}{p^{n+1}} + \frac{v[s^{-1}]^3 t^2}{p} \right]} \right).$$

It follows from (35.3.5.5) that

$$\sum_{t \in (\mathbb{Z}/p^{n+1})^*} e^{2\pi i \left[ \frac{(u-v[s^{-1}]^2)t}{p^{n+1}} + \frac{v[s^{-1}]^3 t^2}{p} \right]} = \begin{cases} p^{n+\frac{1}{2}} & , us^2 \equiv v \pmod{p^n}, p \nmid v \\ 0 & , us^2 \not\equiv v \pmod{p^n} \end{cases}.$$

So

$$\text{Kloos}(u, v, \text{mod } p^k) = p^{n+\frac{1}{2}} \#\{s \in (\mathbb{Z}/p^n)^* | us^2 \equiv v \pmod{p^n}\} \leq 2p^{n+\frac{1}{2}} = 2p^{\frac{m}{2}}.$$

## 8 Irrational Trigonometric Sums

**Conj. (35.3.8.1).** Given  $k \in \mathbb{Z}_+$ , then for any  $N \in \mathbb{Z}_+$ ,

$$\sum_{n=1}^N e^{2\pi i \sqrt{2} n^k} = \tilde{O}(N^{1/2}).$$

┘

*Proof:*

□

**Def. (35.3.8.2).** For  $\alpha \in \mathbb{R}$  and  $D, N \in \mathbb{Z}_+$ , define

$$R_{D,N}(\alpha) = \sum_{d \in \mathbb{Z}_+, d \leq D} \min(N, \{\{d\alpha\}\}^{-1}).$$

Then this  $R_{D,N}$  satisfies:

- $D \leq R_{D,N}(\alpha) \leq DN$ .
- If  $\alpha$  is  $(C, \mu)$ -Diophantine, then  $R_{D,N}(\alpha) \leq C^{-1} DN^{\mu-2} \log N$ .
- $R_{D,N}(\sqrt{2}) \sim D \log N$ .

┘

*Proof:*

□

**Thm. (35.3.8.3)[Weyl].** Let  $P(X) = \alpha X^k + \sum_{j=1}^{k-1} \alpha_j X^j \in \mathbb{R}[X]$ ,  $N, N_1, N_2 \in \mathbb{Z}_+$ ,  $N_1 < N_2 \leq N_1 + N$ , then for each  $\delta \in \mathbb{R}_+$ , there exists  $C(\delta) \in \mathbb{R}_+$  s.t.

$$\left| \sum_{n=N_1}^{N_2} e^{2\pi i P(n)} \right|^{2^{k-1}} \leq C(\delta) N^\delta N^{2^{k-1}-k} R_{k!N^{k-1}, N}(\alpha) \text{ (35.3.8.2)}.$$

In particular, if  $\alpha$  is  $\mu$ -Diophantine, then by (35.3.8.2), for each  $\delta \in \mathbb{R}_+$ , there exists  $C'(\delta) \in \mathbb{R}_+$  s.t.

$$\left| \sum_{n=N_1}^{N_2} e^{2\pi i P(n)} \right|^{2^{k-1}} \leq C'(\delta) N^{\delta+\mu-2} N^{2^{k-1}-1}.$$

┘

*Proof:* Cf.[Larry Guth]P19.

□

**Thm. (35.3.8.4)[Irrational Trigonometric Sums, Vinogradov].** Let  $P(X) = \sum_{j=1}^k \alpha_j X^j \in \mathbb{R}[X]$ , then there exists  $C_k \in \mathbb{R}_+$  s.t. for  $N \in \mathbb{Z}_+$ ,

$$\left| \sum_{n=1}^N e^{2\pi i P(n)} \right| \leq C_k N^{1 - \frac{1}{10k^2 \log k}},$$

unless there exists  $\frac{a}{q} \in \mathbb{Q}$  s.t.  $1 \leq q \leq N$  and

$$\left| \alpha_k - \frac{a}{q} \right| \leq \frac{1}{N^{k-1}}.$$

┘

*Proof:* Cf.[Larry Guth]P44.?

□

**Waring's Problem**

For  $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}/\mathbb{Z})^k$ , denote

$$W_{k,N}(\alpha) = \sum_{n=1}^N e^{2\pi i P_\alpha(n)}, \quad P_\alpha(X) = \alpha_k X^k + \dots + \alpha_1 X.$$

For  $l \in \mathbb{Z}_+$ , define the linear operator

$$T_l : (\mathbb{R}/\mathbb{Z})^k \rightarrow (\mathbb{R}/\mathbb{Z})^k : P_\alpha(X+l) - P_\alpha(X) = P_{T_l(\alpha)}(X).$$

Notice that for any  $s \in \mathbb{Z}_+$ ,

$$\int_{(\mathbb{R}/\mathbb{Z})^k} |W_{k,N}(\beta)|^{2s} d\beta = J_{s,k}(N) \triangleq \#\{(a_1, \dots, a_s, b_1, \dots, b_s) \in [N]_+^{2s} \mid \sum_i a_i^j = \sum_i b_i^j, \forall j \in [1, k]\}.$$

**Lemma (35.3.8.5) [W-BDG].** For  $s, k, N \in \mathbb{Z}_+$  and  $\varepsilon \in \mathbb{R}_+$ , there exists  $C(k, \varepsilon) \in \mathbb{R}_+$  s.t.

$$J_{k,N}(N) \leq C(k, \varepsilon) N^\varepsilon (N^s + N^{2s - \binom{k+1}{2}}).$$

┘

*Proof:*

□

**Lemma (35.3.8.6) [Vinogradov].** For  $s, k, N \in \mathbb{Z}_+$ , there exists  $C(k) \in \mathbb{R}_+$  s.t.

$$J_{k,N}(N) \leq C(k) N^{2s - \binom{k+1}{2}} + k^{2 - \frac{s}{k^2 \log k}}.$$

┘

*Proof:*

□

Denote

$$\varphi_{s,k} : \mathbb{R}^s \rightarrow \mathbb{R}^k : (a_1, \dots, a_s) \mapsto \left( \sum_i a_i, \sum_i a_i^2, \dots, \sum_i a_i^k \right),$$

Then we have:

**Lemma (35.3.8.7).** For  $\gamma \in \mathbb{R}_+$ , a sequence  $(a_1, \dots, a_n) \in \mathbb{R}$  is said to be in  $WS_\gamma$  if  $|a_i - a_j| > \gamma$  for any  $i \neq j$ . Then there exists  $C(k, \gamma) \in \mathbb{R}_+$  s.t. for any rectangular  $R$  with lengths greater than  $1/N$ , we have:

$$\#\left\{ (a_1, \dots, a_k) \in \left(\frac{\mathbb{Z}}{N}\right)^k \cap [0, 1]^k \cap WS_\gamma \mid \varphi_{s,k}(\underline{a}) \in R \right\} \leq C(k, \gamma) N^k \text{Vol}(R).$$

┘

*Proof:* Cf. [Larry Guth]P49.?

□

### 35.4 Riemann Hypothesis and Non-vanishing Theorems

*“Why are numbers beautiful? It’s like asking why is Beethoven’s Ninth Symphony beautiful. If you don’t see why, someone can’t tell you. I know numbers are beautiful. If they aren’t beautiful, nothing is.”*

–P. Erdős

References are [Ivic, Riemann zeta-function. The theory of the Riemann zeta-function with applications(1985)].

**Notation(35.4.0.1).**

- Let  $F \in \mathbf{GField}$ .
- Use notations on automorphic representations defined in [Adelic Automorphic Representations](#).
- Use notations on automorphic  $L$ -functions defined in [Automorphic  \$L\$ -Functions](#).
- Use notations on  $L$ -functions and primes numbers defined in ??.

┘

#### 1 Generalized Riemann Hypothesis

**Conj.(35.4.1.1) [Grand Riemann Hypothesis].** For any  $n \in \mathbb{Z}_+$ ,  $F \in \mathbf{NField}$  and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$ , any zero  $s \in \mathbb{C}$  of the principal series  $\Lambda(\pi; s)$  has real part  $\text{Re}(s) = \frac{1}{2}$ . ┘

**Conj.Cor.(35.4.1.2) [Generalized Riemann Hypothesis].** For any  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}'(N)$  and any zero  $s \in \mathbb{C}$  of  $\Lambda(\chi; s)$  has real part  $\text{Re}(s) = \frac{1}{2}$ . ┘

*Proof:* ?

□

**Conj.Cor.(35.4.1.3) [Riemann Hypothesis, Riemann1859].**

$$\text{Zero}(\zeta) \subset -2\mathbb{Z}_+ \cup \left\{ \frac{1}{2} + i\mathbb{R} \right\} \text{ (35.5.3.1).}$$

┘

*Proof:*

□

**Remark(35.4.1.4) [Numerical Verifications, Platt].** For any  $N \leq 300000$  and  $\chi \in \text{Diri}(N)$ , any zero  $s \in \mathbb{C}$  of  $L(\chi; s)$  s.t.  $\text{Re}(s) \in (0, 1)$  and  $\text{Im}(s) \leq \begin{cases} \frac{10^8}{N} & , N \in 2\mathbb{Z}_+ \\ \max(\frac{10^8}{N}, \frac{7.5 \cdot 10^7}{N} + 200) & , N \in 2\mathbb{Z}_+ - 1 \end{cases}$  satisfies  $\text{Im}(s) = \frac{1}{2}$ . ┘

*Proof:* Cf.[Platt, Numerical computations concerning GRH].

□

#### 2 Application to Distribution of Primes

**Thm.(35.4.2.1) [Riemann-Mangoldt Formula].** Define

$$f(X) = \pi_0(X) + \frac{1}{2}\pi_0(X^{\frac{1}{2}}) + \frac{1}{3}\pi_0(X^{\frac{1}{3}}) + \dots,$$

then by Möbius inversion,

$$\pi_0(X) = \sum_n \frac{\mu(n)}{n} f(X^{\frac{1}{n}}) = f(X) - \frac{1}{2}f(X^{\frac{1}{2}}) + \frac{1}{3}f(X^{\frac{1}{3}}) - \dots$$

Then there is a **Riemann-Mangoldt formula**

$$f(X) = \text{li}(X) - \sum_{\rho} \text{li}(X^{\rho}) - \log 2 + \int_X^{\infty} \frac{dt}{t(t^2 - 1) \log t}.$$

*Proof:* ?

**Thm. (35.4.2.2) [Riemann's Explicit Formula].**

**Def. (35.4.2.3).** For  $X \in \mathbb{R}_+$ , define  $\text{Err}(X) = \psi(X) - X$  (2.6.1.32).

**Prop. (35.4.2.4).** The Riemann conjecture is equivalent to the following:

- $|\pi(X) - \text{Li}(X)| = \tilde{O}(X^{1/2})$ .
- $\text{Err}(X) = \tilde{O}(X^{1/2})$  (35.4.2.3).

*Proof:* This should follow from the Riemann-Mangoldt formula (35.4.2.1) ?.

**Thm. (35.4.2.5) [Vinogradov].**  $\text{Err}(X) = O(Xe^{-C(\log X)^{3/5}(\log \log X)^{-1/5}})$ .

*Proof:* We only prove that there exists  $c \in \mathbb{R}_+$  s.t.  $\text{Err}(X) = O(Xe^{-c \log^{1/2} X})$ :

Cf. [Larry Guth]. The assertion follows from (35.5.3.16). The same argument should work if we use the bigger zero-free region.  $\square$

### 3 Linnik-Type Problems

References are [least quadratic non-residue].

**Prop. (35.4.3.1) [Linnik Problem].** There exists  $C \in \mathbb{R}_+$  s.t. if  $L(f; s), L(g; s)$  are two analytic  $L$ -functions with the same degree  $d$  and gamma factor (29.5.1.1) s.t.

- $L(f \otimes \bar{f}; s)$  and  $L(f \otimes \bar{g}; s)$  exist and the latter is entire (29.5.1.5).
- The GRH holds for both  $f \otimes \bar{f}$  and  $f \otimes \bar{g}$ .
- The local parameters of  $f \otimes \bar{f}$  and  $f \otimes \bar{g}$  at ramified places are all of modulus  $\leq 1$  (29.5.1.1).

Then there exists  $p \in \text{Prime}$  s.t.  $p \leq C(d \log(N(f)N(g)))^2$  unramified for both  $f$  and  $g$  s.t.  $L_p(f; s) \neq L_p(g; s)$ .  $\square$

*Proof:* Cf. [I-K04] P118 ?.

**Thm& Conj. Cor. (35.4.3.2) [Linnik Problem for Dirichlet Characters].** For  $N \in \mathbb{Z}_{\geq 2}, \chi \in \text{Diri}(N) \setminus \mathbf{1}_N$ , denote  $\mathfrak{p}_{\min}(\chi) \in \text{Prime}$  to be the minimal prime s.t.  $\chi(p) \neq 1$ . Then

- $\mathfrak{p}_{\min}(\chi) \ll \log^2 N$  if GRH holds for  $\chi$  (35.4.1.2).
- (Vinogradov's conjecture)  $\mathfrak{p}_{\min}(\chi) \ll_{\varepsilon} N^{\varepsilon}$  for any  $\varepsilon \in \mathbb{R}_+$  if Lindelöff-hypothesis holds for  $\chi$  (35.11.0.10).

- $\mathfrak{p}_{\min}(\chi) \ll_{\varepsilon} N^{\frac{1}{4\sqrt{\varepsilon}} + \varepsilon}$  for any  $\varepsilon \in \mathbb{R}_+$ , by Burgess-inequality? ?

┘

*Proof:* Cf.[数论导引, 华罗庚] and [least quadratic non-residue, notes]. □

**Thm& Conj.Cor.(35.4.3.3) [Linnik[Lin41]].** For any  $\varepsilon \in (0, 1)$ , there exists  $C_{\varepsilon} \in \mathbb{Z}_+$  s.t. for each  $N \in \mathbb{Z}_+$ , the number of primes  $p \in [N]_+$  s.t.  $\mathfrak{p}_{\min}(p) > N^{\varepsilon}$  (35.4.3.2)(the least non-quadratic residue) is bounded by  $C_{\varepsilon}$ .

In particular, there are at most  $O(N^{\varepsilon})$ -many primes  $p \in [N]_+$  s.t.  $\mathfrak{p}_{\min}(p) \geq p^{\varepsilon}$ , and Vinogradov's conjecture(35.4.3.2) fails for a set of primes of density 0. ┘

*Proof:* Let  $\mathcal{P}_{\varepsilon}$  be the set of primes in  $[\sqrt{N}]_+$  s.t.  $\mathfrak{p}_{\min}(p) > N^{\varepsilon}$ . Then we apply large sieve method(35.1.6.17) with

- $\mathcal{M} = [N]_+$ ,
- For each  $p \in \mathcal{P}_{\varepsilon}$ , let  $\Omega_p = \{n \in \mathbb{Z}/(p) \mid (\frac{n}{p}) = -1\}$ .

Then the sieved set  $\mathcal{S}(\mathcal{M}, \mathcal{P}, \Omega)$  contains  $\mathcal{Z}_{\varepsilon} = \{n \in [N]_+ : S(n) \subset [N^{\varepsilon}]_+\}$ . Thus by(35.1.6.17), we have

$$\#\mathcal{Z}_{\varepsilon} \leq 2NH(\sqrt{N})^{-1}.$$

Notice that

$$H(\sqrt{N}) \geq \sum_{p \in \mathcal{P}_{\varepsilon}} \frac{\#\Omega_p}{p - \#\Omega_p} \geq \#\mathcal{P}_{\varepsilon}/3.$$

So we get  $\#\mathcal{P}_{\varepsilon}\#\mathcal{Z}_{\varepsilon} \leq 6N$ . Finally we bound  $\#\mathcal{Z}_{\varepsilon}$ :  $\mathcal{Z}_{\varepsilon}$  contains all numbers  $n \in [N]_+$  of the form  $n = mp_1 \dots p_k$  where  $k = [\varepsilon^{-1}]$  and  $N^{\alpha} < p_1 \leq p_2 \dots \leq p_k \leq N^{\beta}$  are primes, where  $\alpha = 1/(1 + [\varepsilon^{-1}])$ ,  $\beta = 1/([\varepsilon^{-1}] + 1/2)$ . Then

$$\begin{aligned} \#\mathcal{Z}_{\varepsilon} &\geq \sum_{N^{\alpha} < p_1 \leq p_2 \dots \leq p_k \leq N^{\beta}} \left\lfloor \frac{X}{p_1 \dots p_k} \right\rfloor \\ &\gg_{\varepsilon} X \sum_{N^{\alpha} < p_1 \leq p_2 \dots \leq p_k \leq N^{\beta}} \frac{1}{p_1 \dots p_k} \\ &= X \left( \sum_{N^{\alpha} < p \leq N^{\beta}} \frac{1}{p} \right)^k \\ (35.5.7.8) &\gg X \log^k(\beta/\alpha) = X \log^{[\varepsilon^{-1}]} \left( \frac{[\varepsilon^{-1}] + 1}{[\varepsilon^{-1}] + 1/2} \right). \end{aligned}$$

So  $\#\mathcal{P}_{\varepsilon}$  is bounded by a constant depending on  $\varepsilon$ . □

### Deuring-Heilbronn Phenomenon(Repulsion)

#### 4 Non-vanishing on Boundary Lines

**Thm.(35.4.4.1) [Prime Number Theorem, Jacquet-Shalika[J-S76]].** If  $F \in \mathbf{NField}$ ,  $n \in \mathbb{Z}_+$  and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n)/F)$ , then  $L(\pi; s) \neq 0$  if  $\text{Re}(s) = 1$ . ┘

*Proof:* Cf.[J-S76] or [Sarnak, Nonvanishing of  $L$ -functions on  $\text{Re}(s) = 1$ ].? ?

For  $n = 1$ , Cf.[J-S76]Chap7.

Thus we assume that  $n \geq 2$ .

The proof uses the Eisenstein series for the standard parabolic subgroup  $P_{1,n-1}$ (26.9.1.1). □

**Cor. (35.4.4.2).** If  $f \in \text{SF}^{\text{new}}(\Gamma_0(N), \chi)$ , then  $L(f; \frac{k-1}{2} + s) \neq 0$  if  $\text{Re}(s) = 1$ .  $\lrcorner$

**Cor. (35.4.4.3) [Prime Number Theorem for Hecke  $L$ -Functions].** If  $F \in \text{NField}$  and  $\chi \neq 1$  is a Hecke character on  $F$ , then  $L(\chi; 1 + it) \neq 0$  for any  $t \in \mathbb{R}$ .  $\lrcorner$

*Proof:* Cf. [GTM186, P289]. ?

We prove for  $F = \mathbb{Q}$  and  $\chi = 1$ : If  $\zeta$  has a zero at  $x + iy$  for  $y \neq 0$ , then by (35.5.2.5),  $\zeta(s)$  has a simple pole at  $s = 1$  and holomorphic at  $s = x + 2iy$ . But by (35.4.4.4)

$$\lim_{x \rightarrow 1^+} |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| \geq 1,$$

contradiction.  $\square$

**Lemma (35.4.4.4) [Mertens].** For  $x, y \in \mathbb{R}$  with  $x > 1$ ,  $|\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| \geq 1$ .  $\lrcorner$

*Proof:* For  $s = x + iy$ ,

$$\log |\zeta(s)| = - \sum_{p \in \text{Prime}} \log |1 - p^{-s}| = - \sum_{p \in \text{Prime}} \text{Re} \log(1 - p^{-s}) = \sum_{p \in \mathbf{P}} \sum_{n \in \mathbb{Z}_+} \frac{\text{Re}(p^{-ns})}{n},$$

so

$$\log |\zeta(x + iy)| = \sum_{p \in \text{Prime}} \sum_{n \in \mathbb{Z}_+} \frac{\cos(ny \log p)}{np^{nx}}.$$

So

$$\begin{aligned} \log |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| &= \sum_{p \in \mathbf{P}} \sum_{n \in \mathbb{Z}_+} \frac{3 + 4 \cos(ny \log p) + \cos(2ny \log p)}{np^{nx}} \\ &= \sum_{p \in \mathbf{P}} \sum_{n \in \mathbb{Z}_+} \frac{2(\cos(ny \log p) + 1)^2}{np^{nx}} \geq 0 \end{aligned}$$

$\square$

**Cor. (35.4.4.5) [Prime Number Theorem for Artin  $L$ -Functions].** Artin  $L$ -functions are non-vanishing on the line  $\text{Re}(s) = 1$ .  $\lrcorner$

*Proof:* This follows from the fact that Artin  $L$ -functions are quotients of products of Hecke  $L$ -functions.  $\square$

**Thm. (35.4.4.6) [Prime Number Theorem for Rankin-Selberg  $L$ -Functions, Shalika [Sha81]].** If  $F \in \text{NField}$ ,  $n_1, n_2 \in \mathbb{Z}_+$  and  $\pi_i \in \text{Irr}^{\text{cusp}}(\text{GL}(n_i)/F)$ , then the Rankin-Selberg  $L$ -function (29.5.5.1)  $L(\pi_1 \times \pi_2; s) \neq 0$  for  $\text{Re}(s) = 1$ .  $\lrcorner$

*Proof:*  $\square$

**Remark (35.4.4.7).** See also Langlands-Shahidi method ([Sha81] and [Kim-Shahidi, Functorial products for  $\text{GL}(2) \times \text{GL}(3)$  and the symmetric cube for  $\text{GL}(2)$ ]) or the Rankin-Selberg method ([Rankin-Selberg Convolutions] and [Remarks on Rankin-Selberg convolutions]). ?  $\lrcorner$

### Upper Bounds

**Prop. (35.4.4.8) [Louboutin].** Let  $N \in \mathbb{Z}_{\geq 2}$  and  $\chi \in \text{Diri}(N)$ , then

$$|L(\chi; 1)| \leq \begin{cases} \frac{1}{2} \log N + 0.009 & , \chi(-1) = 1 \\ \frac{1}{2} \log N + 0.716 & , \chi(-1) = -1 \end{cases}.$$

┘

*Proof:* Cf. [S. Louboutin, Majorations explicites de  $|L(1, \chi)|$ ].

□

**Remark (35.4.4.9).** It is easy to show that  $|L(\chi; 1)| \leq \log(N) + C$ :

$$\begin{aligned} |L(\chi; 1)| &\leq \left| \sum_{n \leq N} \frac{\chi(n)}{n} \right| + \left| \sum_{n > N} \frac{\chi(n)}{n} \right| \\ &\leq \sum_{n \leq N} \frac{1}{n} + \left| \int_N^\infty \left( \sum_{N < n \leq t} \chi(n) \right) \frac{dt}{t^2} \right| \\ &\leq 1 + \int_1^{N-1} \frac{dt}{t} + \frac{1}{N} \max_X \left| \sum_{N < n \leq t} \chi(n) \right| \\ &< \log N + 1 + \frac{\varphi(N)}{N} \end{aligned}$$

┘

### Lower Bounds

**Thm. (35.4.4.10) [Zhang].** If  $D \in \mathbb{Z}_{\geq 2}$  and  $\chi \in \text{Diri}'(D)$  which is a real character, then there exists an absolute effectively computable constant  $c_1 > 0$  s.t.

$$L(\chi; 1) > c_1 (\log D)^{-2022}$$

┘

*Proof:* Cf. [Yitang Zhang, Discrete mean estimates and the Landau-Siegel zero].

□

**Cor. (35.4.4.11).** For  $D \in \mathbb{Z}_{\geq 2}$  and  $\chi \in \text{Diri}'(N)$  quadratic, there exists an absolute effectively computable constant  $c_1 > 0$  s.t. for any  $\sigma > 1 - c_2 (\log D)^{-2024}$ , we have  $L(\sigma, \chi) \neq 0$ .

┘

*Proof:* ?

□

**Cor. (35.4.4.12) [Landau(36)/Siegel[Sie35]].** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\kappa_\varepsilon \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_{\geq 2}$ ,  $\chi \in \text{Diri}'(N)$  quadratic, there exists  $\kappa_\varepsilon \in \mathbb{R}_+$  s.t.

$$L(\chi; 1) \geq \frac{\kappa_\varepsilon}{D^\varepsilon}.$$

If  $\varepsilon \geq 1/2$ , this constant  $\kappa_\varepsilon$  can be made effective. In particular, There exists  $\kappa_\varepsilon \in \mathbb{R}_+$  s.t. any real zero  $\beta$  of  $\chi$  satisfies

$$\beta \leq 1 - \frac{\kappa_\varepsilon}{N^\varepsilon}.$$

If  $\varepsilon \geq 1/2$ , this constant  $\kappa_\varepsilon$  can be made effective.

┘

*Proof:* There is a direct proof of this theorem in [Gol74] and [A simple proof of Siegel's theorem via Mellin's transform] or [I-K04]P122. ?

□



### Applications

**Prop. (35.4.4.13)[Constructing Non-cuspidal Representations].** Let  $\sigma \in \text{Irr}^{\text{cusp}}(\text{GL}(2)/F)$  and  $\mu$  a Hecke character of  $F$ , then  $\pi = \sigma \times \mu \in \text{Irr}^{\text{auto}}(\text{GL}(3)/F)$  is irreducible.

Then  $\pi$  is not a cuspidal representation.  $\lrcorner$

*Proof:* If  $\pi$  is a cuspidal representation, then by  $L(\pi \otimes \chi; s)$  is entire for any Hecke character  $\chi$  of  $F$ . But

$$L(\pi \otimes \mu^{-1}; s) = L(\sigma \otimes \mu^{-1}; s)L(1; s),$$

and  $L(\sigma \otimes \mu^{-1}; 1) \neq 0$  by (35.4.4.1), contradiction.  $\square$

## 5 Zero-Free Regions

**Def. (35.4.5.1)[Standard Zero-Free Regions].** A **standard Zero-Free Region** is the region in  $\mathbb{C}$  of the form

$$\Omega_c = \left\{ \sigma + it \in \mathbb{C} : \sigma \geq 1 - \frac{c}{\log(|t| + 3)} \right\}$$

for some  $c \in \mathbb{R}_+$ .  $\lrcorner$

**Lemma (35.4.5.2)[Goldfeld-Hoffstein-Liehman[H-L94]].** There exists  $c \in \mathbb{R}_+$  s.t. if  $f$  is an analytic  $L$ -function of degree  $d$  s.t.

- $\text{Re}(\Lambda_f(n)) \geq 0$  for  $n \in \mathbb{Z}_+$  s.t.  $(n, N(f)) = 1$ .
- For any prime  $p \in \text{Prime}$  ramified for  $f$ , the local parameters satisfy  $|\alpha_i(p)| \leq p/2$ .

Then  $\text{ord}_{s=1} L(f; s) \leq 0$ , and  $L(f; s)$  has at most  $-\text{ord}_{s=1} L(f; s)$ -many real zeros  $s$  s.t.

$$s \geq 1 - \frac{c}{d(r+1) \log C(f)}.$$

$\lrcorner$

*Proof:* Cf.[I-K04]P105<sup>?</sup>.  $\square$

**Thm. (35.4.5.3)[Exceptional Zeros].** There exists  $c \in \mathbb{R}_+$  s.t. if  $f$  is an analytic  $L$ -function s.t.

- The Rankin-Selberg convolutions  $L(f \otimes f; s)$  and  $L(f \otimes \bar{f}; s)$  exist, where the latter has a simple pole at  $s = 1$  and the former is entire unless  $f = \bar{f}$ .
- At any ramified prime  $p$  the local parameters for  $f$  satisfies  $|\alpha_i(p)|^2 \leq p/2$ .

Then  $L(f; s)$  has no zeros in the standard region

$$\Omega_c = \left\{ \sigma + it \in \mathbb{C} : \sigma \geq 1 - \frac{c}{d^4 \log(C(f)(|t| + 3))} \right\}$$

except possibly one simple real zero  $\beta_f < 1$  called the  $c$ -**exceptional zero**, in which case  $f$  is self-dual.  $\lrcorner$

*Proof:* <sup>?</sup>  $\square$

### Exceptional Characters

**Prop. (35.4.5.4).** For  $N \in \mathbb{Z}_+$ ,  $\chi \in \text{Diri}(N) \setminus \mathbf{1}_N$  and  $\sigma \in [0, 1]$ ,  $k \in \mathbb{Z}_+$ , we have

$$L^{(k)}(\chi; \sigma) \ll_k N^{1-\sigma} \log^{k+1} N.$$

□

*Proof:* Use Abel summation

$$\sum_{n \in [X]_+} \chi(n) n^{-s} = \mathcal{M}_\chi(X) \frac{\log^k X}{X^\sigma} - \int_1^X \mathcal{M}_\chi(t) (\log^k t) t^\sigma dt$$

and the fact that  $\mathcal{M}_\chi(X) \leq \min(N, X)$ . □

**Thm. (35.4.5.5) [Landau(36)].** There exists  $c \in \mathbb{R}_+$  s.t. if  $N_1, N_2 \in \mathbb{Z}_{\geq 2}$  and  $\chi_i \in \text{Diri}'(N_i)$  are quadratic s.t.  $\chi_1 \neq \chi_2$ , and  $\chi_1, \chi_2$  has real zeros  $\beta_1, \beta_2 \in \mathbb{R}_+$ , then

$$\min(\beta_1, \beta_2) \leq 1 - \frac{c}{\log(N_1 N_2)}.$$

□

*Proof:* Consider the  $L$ -function of degree 4:

$$L(f; s) = \zeta(s) L(\chi_1; s) L(\chi_2; s) L(\chi_1 \chi_2; s)$$

which has conductor at most  $(N_1 N_2)^2$ . Since  $\chi_1$  are non-trivial and distinct,  $L(f; s)$  has a simple pole at  $s = 1$ . Moreover,

$$-\frac{L^\vee(f; s)}{L(f; s)} = \sum_{n \in \mathbb{Z}_+} (1 + \chi_1(n))(1 + \chi_2(n)) \Lambda(n) n^{-s}$$

has non-negative coefficients. Then the assertion follows from (35.4.5.2). □

**Cor. (35.4.5.6) [Exceptional Characters].** There exists  $c \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}(N)$ ,  $L(\chi; s)$  doesn't has a zero in the standard region

$$\Omega_c = \left\{ \sigma + it \in \mathbb{C} : \sigma \geq 1 - \frac{c}{\log(N(|t| + 3))} \right\},$$

with possibly one  $c$ -exceptional quadratic character  $\chi \in \text{Diri}(N)$  and it has a unique simple root

$$1 - \frac{c}{\log(3N)} \leq \beta_\chi < 1$$

for each  $N$ . □

*Proof:* This follows from (35.4.5.3). The uniqueness of the exceptional quadratic character  $\chi \in \text{Diri}(N)$  follows from Landau's theorem (35.4.5.5). □

## 6 Dirichlet Theorems

References are [J-S81a].

**Thm. (35.4.6.1)** [Jaquet-Shalika[J-S81a]]. Suppose  $F \in \mathbf{NField}$ ,  $n, n' \in \mathbb{Z}_+$  and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$ ,  $\pi' \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n')/F)$ . Let  $\Sigma_F^\infty \subset_{\mathrm{fin}} S \subset \Sigma_F$  be a set of places s.t.  $\pi, \pi'$  are unramified outside  $S$ . For  $\sigma \in \mathbb{R}_{>1}$ , define

$$l(\pi \times \pi'; \sigma) = \sum_{v \notin S} \sum_{n \in \mathbb{Z}_+} \frac{\overline{\mathrm{tr}(c_v(\pi)^n)} \mathrm{tr}(c_v(\pi')^n)}{n \|v\|^{n\sigma}},$$

then

- $l(\pi \times \pi'; \sigma)$  is absolutely convergent,
- $L_S(\pi \times \pi'; \sigma) = \exp(l(\pi \times \pi'; \sigma))$ .
- 

$$\lim_{\sigma \rightarrow 1^+} \left( \log \frac{1}{\sigma - 1} \right)^{-1} l(\pi \times \pi'; \sigma) = \delta_{\pi, \pi'}.$$

┘

*Proof:* Cf. [J-S81a]P805?

□

**Cor. (35.4.6.2)** [Dirichlet Theorem for Cuspidal Automorphic Forms, Jaquet-Shalika[J-S81a]]. Let  $r \in \mathbb{Z}_+$  and  $\pi_i \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n_i)/F)$  for  $i \in [r]_+$  be pairwise distinct, and let  $\Sigma_F^\infty \subset_{\mathrm{fin}} S \subset \Sigma_F$  be a set of places s.t. each  $\pi_i$  is unramified outside  $S$ . Suppose  $c_1, \dots, c_r \in \mathbb{C}$  s.t.

$$\sum_{1 \leq i \leq r} c_i \mathrm{tr}(c_v(\pi)^m) = 0$$

for any  $v \in \Sigma_F^{\mathrm{fin}} \setminus S$  and  $m \in \mathbb{Z}_+$ , then we have  $c_1 = \dots = c_r = 0$ . ┘

*Proof:* Write each  $\pi_i = \pi'_i(t_i)$  where  $\pi' \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n_i)/F)$  is unitary and  $t_i \in \mathbb{R}$ ?. We may assume  $t_1 \leq t_2 \leq \dots \leq t_r$ . Then it follows from (35.4.6.1) that

$$\sum_{1 \leq i \leq r} c_i l(\pi'_1 \times \pi'_i; \sigma + t_i - t_1) = 0.$$

Taking  $\sigma \rightarrow 1^+$  gives  $\sum_{1 \leq i \leq r} c_i \delta_{\pi_1, \pi_i} = 0$ , which cannot be true unless  $c_1 = \dots = c_r = 0$ . □

## 7 Non-Negativity of Central Values

**Conj. Cor. (35.4.7.1)** [Non-Negativity of Central Values]. For  $F \in \mathbf{NField}$  and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  that is self-dual,  $L(\pi; s) \in \mathbb{R}$  for  $s \in \mathbb{R}$ , and  $\lim_{s \rightarrow +\infty} L(\pi; s) = 1$ . So GRH(35.4.1.1) would imply that  $L(\frac{1}{2}; s) \geq 0$ . ┘

*Proof:*

□

**Thm. (35.4.7.2)** [Guo]. For  $F \in \mathbf{NField}$  and a self-dual  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{PGL}(2)/F)$ ,

$$L(\pi \otimes \chi; \frac{1}{2}) \geq 0.$$

for any quadratic ray class character  $\chi$ . ┘

*Proof:* Cf.[Guo, on the positivity of the central value of automorphic  $L$ -functions for  $\mathrm{GL}(2)$ ]. The proof uses relative trace formula.  $\square$

**Thm. (35.4.7.3) [Rankin-Lapid[Lap03]].** Let  $F \in \mathbf{NField}$  and  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  be self-dual. Then  $\pi$  is either symplectic or orthogonal, i.e. exactly one of the partial  $L$ -functions  $L_S(\pi, \wedge^2; s)$  and  $L_S(\pi, \mathrm{Sym}^2; s)$  has a pole at  $s = 1$ .

Then if  $\pi \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(n)/F)$  is orthogonal and  $\pi' \in \mathrm{Irr}^{\mathrm{auto}}(\mathrm{GL}(2m)/F)$  is symplectic, we have  $L(\pi \times \pi'; \frac{1}{2})$ .  $\square$

*Proof:* Cf.[Lap03].  $\square$

**Cor. (35.4.7.4).** Let  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$  be orthogonal and  $\tau \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{SO}(2m+1)/F)$  be generic, then  $L(\pi \times \tau; \frac{1}{2}) \geq 0$ .  $\square$

*Proof:* By(29.7.1.1), this is equivalent to (35.4.7.3).  $\square$

**Cor. (35.4.7.5) [Harris-Kudla[H-K91]/Watson/Lapid[Lap03]].** Let  $\pi_i \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F, \mathbf{1})$  for  $i = 1, 2, 3$ , then  $L(\pi_1 \times \pi_2 \times \pi_3; \frac{1}{2}) \geq 0$ .  $\square$

*Proof:* The original proof uses the period integral formula(Ichino-Ikeda).?  $\square$

### Mollification(Selberg)

References are [Iwaniec-Sarnak, The non-vanishing of central values of automorphic  $L$ -functions and Landau-Siegel zeros] and [Kowalski-Michel-VanderKam, Non-vanishing of high derivatives of automorphic  $L$ -functions at the center of the critical strip].

**Thm. (35.4.7.6) [Iwaniec-Sarnak/Kowalski-Michel-VanderKam].** For  $k \in \mathbb{Z}_{\geq 2}$ ,  $\epsilon \in \{\pm 1\}$  and  $N \in \mathbb{Z}_+$  square-free and large, roughly half of the newforms  $f \in \mathrm{SF}_k(\Gamma_0(N))$  has sign of equation  $\epsilon_f = \epsilon$ . And

•

$$\lim_{N \rightarrow \infty} \frac{\#\{f \in \mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N)) | \epsilon_f = 1, L(f; \frac{1}{2}) \geq (\log N)^{-2}\}}{\#\{f \in \mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N)) | \epsilon_f = 1\}} \geq \frac{1}{2}.$$

•

$$\lim_{N \rightarrow \infty} \frac{\#\{f \in \mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N)) | \epsilon_f = -1, L(f; \frac{1}{2}) \neq 0\}}{\#\{f \in \mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N)) | \epsilon_f = -1\}} \geq \frac{1}{2}.$$

•

$$\lim_{N \rightarrow \infty} \frac{1}{\#\mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N))} \sum_{f \in \mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N))} \mathrm{ord}_{s=1/2} L(f; s) \leq 1.2.$$

$\square$

*Proof:* ?  $\square$

**Remark (35.4.7.7).** Any improvement of item1 replacing  $1/2$  with any  $c > 1/2$  would resolve the Landau-Siegel lacuna.  $\square$

**Thm. (35.4.7.8) [Luo].** Let  $k \in \mathbb{Z}_+$  and  $N \in \mathbb{Z}_+$ , and let  $f \in \mathrm{SF}_k(\Gamma_0(N))$ . Let  $\{\phi_i\}_{i \in \mathbb{Z}}$  be an orthonormal basis of Maass forms where  $\Delta \phi_i = (\frac{1}{4} + t_i^2) \phi_i$  and  $t_1 \leq t_2 \leq \dots$ , then

$$\lim_{t \rightarrow \infty} \frac{\#\{t_i \leq t | L(f \otimes \phi_i; \frac{1}{2} + it_i)\}}{\#\{t_i \leq t\}} > 0.$$

$\square$

*Proof:* Cf.[Luo, Nonvanishing of  $L$ -values and the Weyl law2001].  $\square$

## 8 Density Theorems

**Prop. (35.4.8.1).** For an analytic  $L$ -function  $f$  (29.5.1.1), we have  $\text{Zero}(\Lambda(f; s)) \subset \{z \in \mathbb{C} | 0 \leq \text{Re}(s) \leq 1\}$ , and also

$$\sum_{\rho \in \text{Zero}(L(f; s)) \setminus \{0\}} \frac{1}{|\rho|^{1+\varepsilon}} < \infty$$

for any  $\varepsilon \in \mathbb{R}_+$ . ┘

*Proof:* The Euler-product expansion of  $L(f; s)$  converges for  $\text{Re}(s) > 1$ , and  $L_\infty(f; s)$  is non-vanishing for  $\text{Re}(s) > 1$ , so there are no zeros  $\rho$  with  $\text{Re}(\rho) > 1$ . By functional equation, there are no zeros  $\rho$  with  $\text{Re}(\rho) < 1$ . The last assertion follows from (12.4.3.19). □

**Prop. (35.4.8.2).** For an analytic  $L$ -function  $f$  (29.5.1.1) with pole of order  $r$  at  $s = 1$ , there exists constants  $a, b \in \mathbb{C}$  s.t.

$$(s(1-s))^r \Lambda(f; s) = e^{a+bs} \prod_{\rho \in \text{Zero}(L(f; s)) \setminus \{0,1\}} \left(1 - \frac{s}{\rho}\right) e^{s/\rho}.$$

Taking logarithmic differential we get

$$\frac{1}{2} \log N(f) + \frac{L'_\infty(f; s)}{L_\infty(f; s)} - b + \frac{r}{s} + \frac{r}{s-1} - \sum_{\rho \in \text{Zero}(L(f; s)) \setminus \{0,1\}} \left( \frac{1}{s-\rho} + \frac{1}{\rho} \right) = -\frac{L'(f; s)}{L(f; s)} = \sum_{n \in \mathbb{Z}_+} \Lambda_f(n) n^{-s}.$$

And both sides converges uniformly and absolutely on compact subsets away from zeros and poles. ┘

*Proof:* This follows from Hadamard's factorization theorem for entire functions of finite order. □

**Prop. (35.4.8.3).** Situation as in (35.4.8.2), then

•

$$\text{Re}(b(f)) = - \sum_{\rho \in \text{Zero}(f) \setminus \{0,1\}} \text{Re}(\rho^{-1}).$$

- There exists  $c \in \mathbb{R}_+$  s.t. for any  $T \in \mathbb{R}$ , we have

$$\sum_{\rho \in \text{Zero}(L(f; s))} \frac{1}{1 + (T - \text{Im}(\rho))^2} \ll \log C(f; iT).$$

- For any  $s \in \mathbb{C}$  s.t.  $-\frac{1}{2} \leq \text{Re}(s) \leq 2$ , we have

$$\frac{L'(f; s)}{L(f; s)} + \frac{r}{s} + \frac{r}{s-1} - \sum_{|s+\kappa_i| < 1} \frac{1}{s+\kappa_i} - \sum_{\rho \in \text{Zero}(L(f; s)), |\rho-s| < 1} \frac{1}{s-\rho} \ll \log C(f; s).$$

┘

*Proof:* 1: Take logarithmic differential of the functional equation  $(s(1-s))^r \Lambda(f; s) = \epsilon(f)(s(1-s))^r \Lambda(\bar{f}; 1-s)$ , we get

$$2 \text{Re}(b(f)) = - \sum_{\rho \in \text{Zero}(L(f; s)) \setminus \{0,1\}} \left( \frac{1}{s-\rho} + \frac{1}{1-s-\bar{\rho}} + \frac{1}{\rho} + \frac{1}{\bar{\rho}} \right).$$

When  $s \notin \text{Zero}(f)$ , we have  $(s - \rho)^{-1} + (1 - s - \bar{\rho})^{-1} \ll_s |\rho|^{-2}$ , and also  $\rho^{-1} + \bar{\rho}^{-1} \ll |\rho|^{-2}$ , so we can arrange the sums by (35.4.8.1). So the assertion follows because the  $s$ -term cancels, because if  $\rho \in \text{Zero}(L(f; s))$ , then  $1 - \bar{\rho}$  is also a zero of  $L(f; s)$ , by functional equation.

2: Let  $T \in \mathbb{R}_{\geq 2}$  and  $s = 3 + iT$ , then we have  $|\Lambda_f(n)| \leq \deg(f)n \log n$ , and then

$$\left| \frac{L^\vee(f; s)}{L(f; s)} \right| \leq \deg(f) \zeta^\vee(2) \ll \log C(f).$$

And by Stirling's formula (35.11.0.3) we have ?

$$\frac{1}{2} \log N(f) + \frac{L_\infty^\vee(f; s)}{L_\infty(f; s)} \ll \log C(f; s).$$

Also for  $\rho = \beta + i\gamma \in \text{Zero}(L(f; s))$  we have

$$\frac{2}{9 + (T - \gamma)^2} < \text{Re} \left( \frac{1}{s - \rho} \right) < \frac{3}{4 + (T - \gamma)^2}.$$

Then we can we can take the real part of the equation

$$\frac{1}{2} \log N(f) + \frac{L_\infty^\vee(f; s)}{L_\infty(f; s)} - b + \frac{r}{s} + \frac{r}{s-1} - \sum_{\rho \in \text{Zero}(L(f; s)) \setminus \{0,1\}} \left( \frac{1}{s - \rho} + \frac{1}{\rho} \right) = -\frac{L^\vee(f; s)}{L(f; s)} \quad (35.4.8.2)$$

and item1 to conclude.

3: Cf. [I-K04]P103 ?.

□

**Def. (35.4.8.4) [Zero-Counting Functions].** For an analytic  $L$ -function  $f$  and  $T \in \mathbb{R}_+$ , denote  $N(T, f)$  the number of zeros of  $L(f; s)$  in the region  $\{\sigma + it | \sigma \in [0, 1], t \in [-T, T]\}$ . ┘

**Prop. (35.4.8.5).** If  $f$  is an analytic  $L$ -function of degree  $d$ , then for  $T \in \mathbb{R}_+$ ,

$$N(T, f) = \frac{T}{\pi} \log \left( \frac{N(f) T^d}{(2\pi e)^d} \right) + O(\log C(f; iT)) \quad (35.4.8.4).$$

┘

*Proof:* Cf. [I-K04]P104 ?

□

## 9 Hilbert-Pólya Conjecture

References are [Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, Connes] and [R-S96, K-S99].

### Montgomery-Odlyzko Law

**Conj. (35.4.9.1)** [Riemann/Montgomery [Mon73]/Odlyzko [Id187]/Rudnick-Sarnak [R-S96]].

Let  $\{\frac{1}{2} + i\gamma_i\}_{i \in \mathbb{Z}^\times}$  be the zeros of  $\zeta(s)$  where

$$\dots \leq \gamma_{-1} \leq \gamma_1 \leq \gamma_2 \leq \dots$$

(Why are there infinitely many ?). Then  $\gamma_{-k} = -\gamma_k$ . And it is conjectured that

$$\#\{j \in \mathbb{Z}_+ | \gamma_j < T\} \sim \frac{T \log T}{2\pi}, \quad T \rightarrow \infty.$$

┘

*Proof:*

□

## 35.5 L-Functions and Theory of Primes

References are [I-K04, H-W08, Ten15]. [Some results in the additive prime-number theory, Long-Keng Hua], [Larry Guth's notes]

**Notation(35.5.0.1).**

- Let  $F \in \mathbf{NField}$ .
- For  $n \in \mathbb{Z}_+$ , let  $p_n$  be the  $n$ -th smallest prime number.
- Use notations on elementary number theory defined in [Elementary Number Theory](#).

┘

### 1 Representing Primes(Hardy-Littlewood)

**Conj.(35.5.1.1)**[Hardy-Littlewood].  $\#\{n^2 + 1\}_{n \in \mathbb{Z}} \cap \mathbf{Prime} = \infty$ .

┘

*Proof:*

□

**Thm& Conj.Cor.(35.5.1.2).** There exists an explicit  $C \in \mathbb{R}_+$  s.t. for any  $X, Y \in \mathbb{R}_{\geq 2}$ , we have

$$\#\{n^2 + 1\}_{n \in (X, X+Y]} \cap \mathbf{Prime} \leq C \frac{Y}{\log Y} (1 + o(1)).$$

┘

*Proof:* Cf.[Applications of The Selberg Sieve, Oh].

□

**Prop.(35.5.1.3).** If  $\underline{X} = (X_1, \dots, X_n)$  and  $f(\underline{X}) \in \mathbb{C}[\underline{X}]$  only takes values in  $\mathbf{Prime}$  at non-negative integer points, then  $f$  is constant.

┘

*Proof:*

□

**Prop.(35.5.1.4).** There is an integral polynomial of degree 25 in 26 variables that its only positive values at integer points are primes.

┘

*Proof:* Cf.[J. P. Jones, D. Sato, H. Wada and D. Wiens, "Diophantine representation of the set of prime numbers," Amer. Math. Monthly, 83 (1976) 449-464.].

□

**Prop.(35.5.1.5)** [Landau]. For  $n \in \mathbb{Z}_+$ , let  $f(n)$  be the cardinality of numbers in  $[n]_+$  that can be written as the form  $x^2 + y^2$ , then  $f(n) = O(\frac{n}{\sqrt{\log n}})$ .

┘

*Proof:* ?

□

### Representing Primes by Quadratic Forms

References are [Primes of the form  $x^2 + ny^2$ , Cox].

**Prop.(35.5.1.6)** [Quadratic Forms]. For  $D \in \mathbb{Z}_{<0}$  and  $D \equiv 0, 1 \pmod{4}$ , the equivalence classes of quadratic forms  $AX^2 + BXY + CY^2$  with discriminant  $D$  is in bijection with the class group  $\text{Cl}(\mathbb{Q}(\sqrt{D}))$ .

┘

**Thm. (35.5.1.7) [Representing Primes].** For any  $n \in \mathbb{Z}_+$ , there is a monic irreducible polynomial  $f_n(T) \in \mathbb{Z}[T]$  of degree  $h(-4n)$  s.t.:

For  $p > 2 \in \text{Prime}$  dividing neither  $n$  nor the discriminant of  $f_n(T)$ ,

$$p \in \{x^2 + ny^2 | x, y \in \mathbb{Z}\} \iff \left(\frac{-n}{p}\right) = 1 \vee \{x \in \mathbb{Z} | f_n(x) = 0 \in \mathbb{F}_p\} \neq \emptyset.$$

Moreover,  $f_n(T)$  is the minimal polynomial of a real algebraic integer  $\alpha$  s.t.  $L = K(\alpha)$  is ring class field of the order  $\mathbb{Z}[\sqrt{-n}] \subset \mathcal{O}_{\mathbb{Q}(\sqrt{-n})}$  (19.7.4.37), and any such  $f_n(T)$  is of this form.  $\lrcorner$

*Proof:* Cf. [Cox,  $p = x^2 + ny^2$ ] P163. ?

□

**Cor. (35.5.1.8).** Let  $p \in \text{Prime}$ , then

•

$$\exists(x, y \in \mathbb{Z}) p = x^2 + y^2 \iff p \equiv 1 \pmod{4}.$$

•

$$\exists(x, y \in \mathbb{Z}) p = x^2 + 2y^2 \iff p \equiv 1, 3 \pmod{8}.$$

•

$$\exists(x, y \in \mathbb{Z}) p = x^2 + 3y^2 \iff p = 3 \text{ or } p \equiv 1 \pmod{3}.$$

•

$$\exists(x, y \in \mathbb{Z}) p = x^2 + 5y^2 \iff p \equiv 1, 9 \pmod{20}.$$

•

$$\exists(x, y \in \mathbb{Z}) p = x^2 + 27y^2 \iff \{p \equiv 1 \pmod{3}\} \wedge \{\exists a, p | a^3 - 2\}.$$

•

$$\{\exists(x, y \in \mathbb{Z}) p = x^2 + 27y^2\} \text{ or } \{\exists(x, y \in \mathbb{Z}) p = 4x^2 + 2xy + 7y^2\} \iff p \equiv 1 \pmod{3}.$$

□

*Proof:*

□

**Prop. (35.5.1.9).** Let  $ax^2 + bxy + cy^2$  be a primitive definite quadratic form of discriminant  $D < 0$ , and let  $\mathcal{S} = \{p \in \text{Prime} | p = ax^2 + bxy + cy^2, \exists x, y \in \mathbb{Z}\}$ , then  $\mathcal{S}$  has Dirichlet-density

$$\delta(\mathcal{S}) = \begin{cases} \frac{1}{\text{cl}(D)} & , ax^2 + bxy + cy^2 \text{ is properly equivalent to its opposite} \\ \frac{1}{\text{cl}(D)} & , \text{otherwise} \end{cases}.$$

□

*Proof:* Cf. [Cox13] P170.

□



### Higher Polynomials

**Conj. (35.5.1.10) [Hardy-Littlewood Conjecture N].** There are infinitely many primes  $p \in \mathbf{Prime}$  of the form  $p = x^3 + y^3 + z^3$ .

Moreover, for  $X \in \mathbb{R}_+$ , if  $P(X)$  is the number of triples  $(x, y, z)$  s.t.  $x^3 + y^3 + z^3 \in \mathbf{Prime} \cap [0, X]$ , then

$$P(X) \sim \Gamma\left(\frac{4}{3}\right)^3 \cdot \prod_{q \in \mathbf{Prime}, q \equiv 1 \pmod{3}} \left(1 - \frac{A_q}{q^2}\right) \cdot \frac{X}{\log X},$$

where  $A_q \equiv 1 \pmod{3}$  and  $4q = A_q^2 + 27B_q^2$ . ┘

*Proof:* □

**Thm. (35.5.1.11) [Heath-Brown].** There are infinitely many primes  $p \in \mathbf{Prime}$  of the form  $p = x^3 + 2y^3$ ,  $x, y \in \mathbb{Z}_+$ . ┘

*Proof:* □

**Thm. (35.5.1.12) [Iwaniec-Friedlander].** There are infinitely many primes  $p \in \mathbf{Prime}$  of the form  $p = x^2 + y^4$ ,  $x, y \in \mathbb{Z}_+$ . ┘

*Proof:* □

## 2 Dirichlet-Weber Class Number L-Functions

References are [I-K04, Mil20].

**Def. (35.5.2.1) [Partial Class Group L-Functions].** If  $\mathfrak{m}$  is a modulus for  $F$  (19.7.4.6) and  $\mathcal{R} \in \text{Cl}_{\mathfrak{m}}(F)$  (19.7.4.8) be a single ideal class, define the **partial L-function**

$$\zeta(F, \mathcal{R}; s) = \sum_{\mathfrak{a} \in \mathcal{R}} \frac{1}{\|\mathfrak{a}\|^s}.$$

┘

**Prop. (35.5.2.2) [Weber's L-Functions].** If  $\mathfrak{m}$  is a modulus for  $F$  (19.7.4.6),  $\chi : \text{Cl}^{\mathfrak{m}}(F) \rightarrow \mathbb{C}^\times$  a Dirichlet character for  $F$  which corresponding to a Hecke character of  $F$  via global class field theory (19.7.4.8), then the **Weber L-function** attached to  $\chi$  is defined to be the corresponding  $L$ -function ?. By (19.7.4.17)(31.1.1.1), it is just

$$L(F, \chi; s) = \prod_{\mathfrak{p} \nmid \mathfrak{m}} \frac{1}{1 - \chi(\mathfrak{p}) \|\mathfrak{p}\|^{-s}} = \sum_{(\mathfrak{a}, \mathfrak{m})=1} \frac{\chi(\mathfrak{a})}{\|\mathfrak{a}\|^s} = \sum_{\mathcal{R} \in \text{Cl}_{\mathfrak{m}}(F)} \chi(\mathcal{R}) \zeta(F, \mathcal{R}; s).$$

Thus by (29.5.3.4),  $L(F, \chi; s)$  is analytic if  $\chi \neq \mathbf{1}$ , and has a simple pole at  $s = 1$  if  $\chi = \mathbf{1}$ . ┘

**Cor. (35.5.2.3).** Situation as in (35.5.2.1),  $\zeta(F, \mathcal{R}; s)$  is analytic for  $\text{Re}(s) > 1 - 1/|\mathfrak{m}|$  except a simple pole at  $s = 1$ , where it has residue  $g_{\mathfrak{m}}$  depending only on  $\mathfrak{m}$ . ┘

*Proof:* This follows from (35.5.2.2) and finite Fourier transform on  $\text{Cl}_{\mathfrak{m}}(F)$ . □

**Def. (35.5.2.4) [Dirichlet  $L$ -Functions, Dirichlet-Satzes[D-S37]].** For  $F = \mathbb{Q}$ , the characters of  $\text{Cl}_m(\mathbb{Q})$  are just Dirichlet characters  $\chi$ (19.4.1.24), thus the Weber  $L$ -function is

$$L(\chi; s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

where  $\text{Re } s > 1$ . Called the **Dirichlet  $L$ -function** attached to  $\chi$ .  $\lrcorner$

**Thm. (35.5.2.5) [Dirichlet's  $L$ -Functions].** Let  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}'(N)$  with  $\chi(-1) = (-1)^\epsilon$  where  $\epsilon \in \{0, 1\}$ , then

$$\Lambda(\chi; s) = L_{\mathbb{R}}(s + \epsilon)L(\chi; s),$$

is the Hecke  $L$ -function attached to the Hecke character corresponding to  $\chi$ (19.4.1.24)(29.5.3.3). Thus  $L(\chi; s)$  can be extended to a meromorphic function for all  $s \in \mathbb{C}$ , and it has simple poles at  $s = 1$  when  $\chi = \mathbf{1}$ , and analytic otherwise. When  $\chi = \mathbf{1}$ ,  $L(\mathbf{1}; s) = \zeta(s)$ , and by(35.5.3.3),

$$\text{res}_{s=1} \zeta(s) = 1.$$

Moreover, there are functional equations

$$\Lambda(\chi; s) = (-i)^\epsilon \tau(\chi) N^{-s} \Lambda(\bar{\chi}; 1-s) \text{(35.3.6.8)},$$

In other words,

$$\epsilon(\chi; s) = (-i)^\epsilon \tau(\chi) N^{-s}, \quad w(\chi) = (-i)^\epsilon \frac{\tau(\chi)}{\sqrt{N}}.$$

In particular, if  $\chi$  is a quadratic character, then  $w(\chi) = 1$ , by(35.3.6.16).  $\lrcorner$

*Proof:* For the calculation of  $\epsilon(\chi; s)$ , Cf.[Bum97]P10 or [I-K04]P84. ?  $\square$

**Thm. (35.5.2.6) [Special Values of Dirichlet's  $L$ -Functions].** Let  $N \in \mathbb{Z}_{\geq 2}$  and  $\chi \in \text{Diri}'(N)$ , then

$$\begin{aligned} L(\chi; 1) &= \frac{-\tau(\chi)}{N} \sum_{n \in \mathbb{Z}/(N)} \chi^{-1}(n) \log(1 - e^{-2\pi i n/N}) \\ &= \begin{cases} \frac{\tau(\chi)}{N^2} \pi i \sum_{n \in \mathbb{Z}/(N)} \chi^{-1}(n) n & , \chi(-1) = -1 \\ \frac{-\tau(\chi)}{N} \sum_{n \in \mathbb{Z}/(N)} \chi^{-1}(n) \log |1 - e^{-2\pi i n/N}| & , \chi(-1) = 1 \end{cases} \end{aligned}$$

$\lrcorner$

*Proof:* Use Fourier transform ?.  $\square$

### Hecke $L$ -Functions of Imaginary Quadratic Fields

**Notation (35.5.2.7).**

- Let  $\mathcal{K}$  be an imaginary quadratic field with discriminant  $D \in \mathbb{Z}_-$ .

$\lrcorner$

**Thm. (35.5.2.8) [Hecke  $L$ -Functions of Imaginary Quadratic Fields].** Let  $\mathfrak{m}$  be a modulus for  $\mathcal{K}$ , and  $\chi$  a Hecke-character of  $\mathcal{K}$  with  $\chi_0$  of conductor  $\mathfrak{m}$  (19.4.1.22) and associated ideal class map  $\psi : J^{\mathfrak{m}} \rightarrow \mathbb{C}^{\times}$ . Suppose  $\chi_{\infty}(z) = e^{k \arg z}$ , then

$$\Lambda(\chi; s) = L_{\mathbb{C}}(s + \frac{|k|}{2}) L(\chi; s)$$

is the Hecke  $L$ -function attached to  $\chi$ . Thus  $L(\chi; s)$  can be extended to a meromorphic function for all  $s \in \mathbb{C}$ , and it has simple poles at  $s = 1$  when  $\chi = \mathbf{1}$ , and analytic otherwise. When  $\chi = \mathbf{1}$ ,  $L(\mathbf{1}; s) = \zeta_{\mathcal{K}}(s) = \zeta(s) L(\chi_{\mathcal{K}}, s)$ , and

$$\operatorname{res}_{s=1} \zeta_{\mathcal{K}}(s) = \frac{2 \operatorname{cl}(\mathcal{K})}{\#\mathcal{K}_{\operatorname{tor}}^{\times} \sqrt{|\operatorname{disc}(\mathcal{K})|}} \pi.$$

Moreover, there are functional equations

$$\Lambda(\chi; s) = i^{-k} \frac{\tau(\chi)}{\|\mathfrak{m}\|^{1/2}} (|\operatorname{disc}(\mathcal{K})| \cdot \|\mathfrak{m}\|)^{1/2-s} \Lambda(\bar{\chi}; 1-s) \text{ (35.3.6.8)}.$$

In other words,

$$\epsilon(\chi; s) = i^{-k} \frac{\tau(\chi)}{\|\mathfrak{m}\|^{1/2}} (|\operatorname{disc}(\mathcal{K})| \cdot \|\mathfrak{m}\|)^{1/2-s}, \quad w(\chi) = i^{-k} \frac{\tau(\chi)}{\|\mathfrak{m}\|^{1/2}}.$$

┘

*Proof:* ?

□

**Prop. (35.5.2.9) [Class Group  $L$ -Functions].** For  $\chi \in \operatorname{Cl}(\mathcal{K})^{\vee}$ , we have

$$L(\mathcal{K}, \chi; s) = \sum_{\mathfrak{a} \in \operatorname{Ideal}(\mathcal{O}_{\mathcal{K}})^{\times}} \chi(\mathfrak{a}) \|\mathfrak{a}\|^{-s}$$

which converges absolutely for  $\operatorname{Re}(s) > 1$ . And we denote

$$\Lambda(\mathcal{K}, \chi; s) =$$

for any primitive ideal  $\mathfrak{a} = [a, \frac{b+\sqrt{D}}{2}]$ , we have

$$\sum_{\chi \in \operatorname{Cl}(\mathfrak{K})^{\vee}} \chi(\mathfrak{a}) L(\mathcal{K}, \chi; s) = \operatorname{cl}(\mathcal{K}) \sum_{\mathfrak{b} \sim \mathfrak{a}} \|\mathfrak{b}\|^{-s} = \frac{\operatorname{cl}(\mathcal{K})}{\#\operatorname{tor}(\mathcal{K}^{\times})} a^{-s} \sum_{\alpha \in \mathfrak{a}^{-1}} |\alpha|^{-2s}.$$

┘

### 3 Riemann's Zeta-Functions

**Def. (35.5.3.1) [Riemann's Zeta-Functions].**  $\zeta(F; s) = L(F, \mathbf{1}; s)$  (35.5.2.2) is called the **Riemann's  $L$ -function** for  $F$ . Also denote

$$\zeta(s) = \zeta(\mathbb{Q}; s), \quad \Lambda(s) = \Lambda(\mathbb{Q}, \mathbf{1}; s) = L_{\mathbb{R}}(s) \zeta(s) \text{ (12.6.1.19)}.$$

Notice this is compatible with that of (2.6.2.2).

By (35.5.2.5),  $\zeta(s)$  extends to a meromorphic function for all  $s \in \mathbb{C}$  with a simple pole at  $s \in \{0, 1\}$  and  $\text{res}_{s=1} \zeta(s) = 1$ . And there is a functional equation

$$\Lambda(s) = \Lambda(1-s),$$

which is equivalent

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

┘

*Proof:* The last equation follows from the two equations

$$\Gamma\left(\frac{s}{2}\right) \Gamma\left(\frac{s+1}{2}\right) = 2^{1-s} \sqrt{\pi} \Gamma(s) \quad (12.6.1.11)$$

and

$$\Gamma\left(\frac{1-s}{2}\right) \Gamma\left(\frac{1+s}{2}\right) = \frac{\pi}{\cos\left(\frac{s}{2}\right)} \quad (12.6.1.7).$$

□

**Prop. (35.5.3.2) [Riemann Zeta-Function as Mellin Transform].**

$$(s-1)\zeta(s) = L\left(\frac{t}{e^t-1}; s-1\right) \quad (12.5.4.3).$$

This is an integral representation of Riemann zeta-function.

┘

*Proof:* The classical definition of  $\Gamma(s)$  (12.6.1.1) implies that

$$n^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty e^{-nt} t^{s-1} dt.$$

In particular,

$$\zeta(s) = \sum_{n \in \mathbb{Z}_+} n^{-s} = \frac{1}{\Gamma(s)} \sum_{n \in \mathbb{Z}_+} \int_0^\infty e^{-nt} t^{s-1} dt = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t}{e^t-1} t^{s-2} dt.$$

Thus the assertion follows.

□

**Thm. (35.5.3.3) [Class Number Formula, Dirichlet].** By (29.5.3.4) (19.4.2.20),

$$\text{res}_{s=1} \zeta(F; s) = \frac{2^{r_1} (2\pi)^{r_2}}{\#F_{\text{tor}}^\times \sqrt{|\text{disc}(F)|}} \text{cl}(F) \text{Reg}(F),$$

where  $\text{Reg}(F)$  is the regulator of  $F$  (20.8.1.7). In particular,

$$\text{res}_{s=1} \zeta_{\mathbb{Q}}(s) = 1.$$

┘

**Prop. (35.5.3.4) [Dirichlet's Class Number Formula for Quadratic Fields].** Let  $\mathcal{K} = \mathbb{Q}(\sqrt{D})$  be a quadratic field with discriminant  $D$ , and  $\chi_D$  (35.3.6.4) the real Dirichlet character associated to  $K$ , and  $\epsilon = 0$  (resp. 1) if  $\mathcal{K}$  is real (resp. complex), then

1. The root number of  $\chi_D$  (29.5.4.7) is given by

$$w(\chi_D) = \frac{\tau(\chi)}{\sqrt{|D|}} (-1)^\epsilon \in \{\pm 1\},$$

2. if  $D < 0$ , then

$$\text{cl}(\mathcal{K}) = \frac{\#\mu(\mathcal{K})\sqrt{|D|}}{2\pi} L(\chi_D; 1) = \frac{\#\mu(\mathcal{K})i}{2|D|} \cdot \frac{\tau(\chi_D)}{\sqrt{|D|}} \cdot \sum_{a \in \mathbb{Z}/(D)} \chi(a) = \frac{\#\mu(\mathcal{K})w(\chi_D)i}{2D} \cdot \sum_{a \in \mathbb{Z}/(D)} \chi(a)a.$$

3. If  $D > 0$ , then

$$\log \varepsilon = \frac{\sqrt{|D|}}{2 \text{cl}(\mathcal{K})} L(\chi_D; 1)$$

where  $\varepsilon$  is the fundamental unit of  $\mathcal{K}$ . ┘

*Proof:* For 1: Cf. [GTM 218]P305-306, Ex.14-15. ?

2, 3 follows from (35.5.3.3)(31.1.1.12)(35.5.2.5)(35.5.2.6) and item1. □

**Prop. (35.5.3.5) [Kronecker's First Limit Formula].**

$$\zeta(s) = \frac{1}{s-1} + \gamma_0 + O(s-1).$$

*Proof:* Cf. [Kronecker's First Limit Formula, Revisited]. This can be deduced from (26.9.2.4). ? □

**Remark (35.5.3.6).** This is related to the averaged Colmez's conjecture ? ┘

### Estimates

**Thm. (35.5.3.7).** For  $s \in \mathbb{C}$  with  $\text{Re}(s) > 0$ ,

$$\left| \zeta(s) - \frac{1}{s-1} \right| \leq \frac{|s|}{\text{Re}(s)}.$$

*Proof:* Cf. [Larry Guth]P87. □

**Prop. (35.5.3.8).** For any  $\varepsilon > 0$ , there exists constant  $C_\varepsilon > 0$  s.t.  $|\zeta(s)|^{-1} \leq C_\varepsilon |t|^\varepsilon$ , where  $s = \sigma + it$ ,  $\sigma \geq 1$ ,  $|t| \geq 1$ . ┘

*Proof:* □

**Prop. (35.5.3.9).** There exists  $C \in \mathbb{R}_+$  s.t. if  $s = \sigma + it \in \mathbb{C}$  where  $\sigma > 1/4$  and  $|t| > 1$ , then  $|\zeta(s)| \leq C|s|$ . ┘

*Proof:* Cf. [Larry Guth]P92. □

**Prop. (35.5.3.10).** Given  $\varepsilon \in \mathbb{R}_+$ , then there exists  $C(\varepsilon) \in \mathbb{R}_+$  s.t. for any  $s \in \mathbb{C}$  with  $\operatorname{Re}(s) > \varepsilon$ ,

$$\left| \frac{\zeta'}{\zeta}(s) - \sum_{\rho \in \operatorname{Zero}(\zeta)} \frac{1}{s - \rho} \right| \leq C(\varepsilon) \log(2 + |\tau|).$$

┘

*Proof:* Cf.[Larry Guth]P87.?

□

**Cor. (35.5.3.11).** For  $\varepsilon \in \mathbb{R}_+$  and  $\tau \in \mathbb{R}$  with  $|\tau| > 1$ ,

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(1 + \varepsilon + i\tau)\right) \leq O(\log |\tau|).$$

And if  $\rho = 1 - \delta + i\tau$  is a zero of  $\zeta(s)$ , then

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(1 + \varepsilon + i\tau)\right) \leq -\frac{1}{\varepsilon + \delta} + O(\log |\tau|).$$

┘

*Proof:* These follow from the fact that for a zero  $\rho$  of  $\zeta$  and  $s \in \mathbb{C}$  s.t.  $\operatorname{Re}(s)$ , we have  $\operatorname{Re}\left(-\frac{1}{s-\rho}\right) < 0$ .

□

**Thm. (35.5.3.12) [Vinogradov].** There exists  $c, C \in \mathbb{R}_+$  s.t. if  $\beta, \tau \in \mathbb{R}$  satisfies  $\beta > 1 - \frac{c}{\log^{2/3}(2+|\tau|)}$ , then

$$|\zeta(\beta + i\tau)| \leq C \log(2 + |\tau|).$$

┘

*Proof:* Cf.[Larry Guth]P101.

□

### Density Theorems

**Prop. (35.5.3.13).** Let  $g \in C^\infty(\mathbb{R})$  s.t.  $e^{\beta x} g(x) \in \mathcal{S}$  for any  $\beta \in \mathbb{R}_+$ , then for any  $\beta \in \mathbb{R}_{>1}$ ,

$$\sum_{n \in \mathbb{Z}_+} \Lambda(n) g(\log n) = -\frac{1}{2\pi i} \int_{\operatorname{Re}(z)=\beta} \frac{\zeta'}{\zeta}(s) \mathcal{L}(g)(-s) ds.$$

┘

*Proof:* Cf.[Larry Guth]P79.

□

**Cor. (35.5.3.14).** Situation as in (35.5.3.13), for  $\sigma \in (0, 1)$ ,  $T \in \mathbb{R}_+$ , denote  $\operatorname{Box}(\sigma, T)$  the open rectangular  $\{s = x + iy | \sigma < x < \beta, |y| < T\}$ , and the arc

$$\gamma_{\sigma, T} = (\beta, -T) \longrightarrow (\sigma, -T) \longrightarrow (\sigma, T) \longrightarrow (\beta, T).$$

Then by residue formula,

$$\sum_{n \in \mathbb{Z}_+} \Lambda(n) g(\log n) = \int e^x g(x) dx - \sum_{\rho \in \operatorname{Zero}(\zeta) \cap \operatorname{Box}(\sigma, T)} \int e^{\rho x} g(x) dx - \frac{1}{2\pi i} \int_{\gamma_{\sigma, T}} \frac{\zeta'}{\zeta} \mathcal{L}g(-s) ds.$$

┘

**Prop. (35.5.3.15) [Bounding Numbers of Zeros].** For  $\varepsilon$ , there exists  $C(\varepsilon) \in \mathbb{R}_+$  s.t. for any  $T \in \mathbb{R}_+$ ,

$$\#\operatorname{Zero}(\zeta) \cap \operatorname{Box}(\varepsilon, T) \leq C(\varepsilon) T \log T \text{ (35.5.3.14)}.$$

┘

*Proof:* Cf.[Larry Guth]P87.

□

### Zero-Free Regions

**Thm. (35.5.3.16) [Zero-Free Region for  $\zeta(s)$ , Vinogradov].** There exists  $c \in \mathbb{R}_+$  s.t. if  $\beta, \tau \in \mathbb{R}$  and

$$\beta > \frac{c}{\log^{2/3}(2 + |\tau|) \log \log |\tau|},$$

then  $\zeta(\beta + i\tau) \neq 0$ . In particular,  $\zeta(s)$  is non-vanishing on a standard zero-free region (35.4.5.1).  $\lrcorner$

**Remark (35.5.3.17).** The barrier for improving this bound lies in the barrier of improving estimates for trigonometric sums (35.3.8.4).  $\lrcorner$

*Proof:* Cf. [Larry Guth]P97? and P101.  $\square$

**Lemma (35.5.3.18).** For  $\sigma \in \mathbb{R}_{>1}$  and  $\tau \in \mathbb{R}$ ,

$$\operatorname{Re} \left( 3 \frac{-\zeta'}{\zeta}(\sigma) + 4 \frac{-\zeta'}{\zeta}(\sigma + i\tau) + \frac{-\zeta'}{\zeta}(\sigma + 2i\tau) \right) \geq 0.$$

$\lrcorner$

*Proof:* It follows from (2.6.2.11) that the LHS equals

$$\sum_{n \in \mathbb{Z}_+} \frac{\Lambda(n)}{n^\sigma} (3 + 4 \cos(\tau \log n) + \cos(2\tau \log n)) = \sum_{n \in \mathbb{Z}_+} \frac{\Lambda(n)}{n^\sigma} 2(1 + \cos(\tau \log n))^2 \geq 0.$$

$\square$

## 4 Models for Primes

See <https://terrytao.wordpress.com/2015/01/04/254a-supplement-4-probabilistic-models-and-heur>

## 5 Sums over Arithmetic Progressions

**Thm. (35.5.5.1).** For  $c_1 < c_2 \in (0, 1)$ ,  $k \in \mathbb{Z}_+$ ,  $X \in \mathbb{R}_+$  and  $Y \in [X^{c_2}, X]$ ,  $q \in [X^{c_1}]_+$ ,  $a \in \mathbb{Z}_+$  s.t.  $(a, q) = 1$ , we have

$$\sum_{n \in [X, X+Y], n \equiv a \pmod{q}} \tau^k(n) \ll_{c_1, c_2, k} \frac{Y}{q} \log^{O_{c_1, c_2, k}(1)} X.$$

And also

$$\sum_{n \in [X, X+Y], n \equiv a \pmod{q}} \tau(n) \sim \frac{Y}{q} \log X.$$

$\lrcorner$

*Proof:* By (35.1.3.11), if  $\theta = c_2 - c_1$ , then for any  $n \in [X + Y]_+$ , we have

$$\tau(n) \ll \max_{d|n, d \leq (X+Y)^\theta} \tau(d)^{2/\theta}.$$

So

$$\sum_{n \in [X, X+Y], n \equiv a \pmod{q}} \tau(n) \ll \sum_{d \in [(X+Y)^\theta]_+} \tau(d)^{2/\theta} \sum_{n \in [X, X+Y], n \equiv a \pmod{q}, d|n} 1 \ll \max_{d \in [(X+Y)^\theta]_+} \tau(d)^{2/\theta} \frac{Y}{qd}$$

because  $qd \leq X^{c_2} \ll Y$ . And the last term is bounded by  $\frac{Y}{q} \log^{2^{2/\theta}+1} X$  by (35.1.3.10).

? For further refinement, Cf. <https://terrytao.wordpress.com/2011/07/23/erdos-divisor-bound/>.  $\square$

## 6 Prime Number Theorems

**Def. (35.5.6.1) [General Prime-Counting Functions].** For an analytic  $L$ -function  $f$ , denote

$$\psi(f; X) = \sum_{n \in [X]_+} \Lambda_f(n).$$

In particular, if  $L(f; s) = \zeta(s)$ ,  $\psi(f; X) = \psi(X)$  (2.6.1.32).  $\lrcorner$

**Thm. (35.5.6.2) [General Prime Number Theorem].** Let  $f$  be an analytic  $L$ -function of degree  $d$  s.t.

- There exists  $c \in \mathbb{R}_+$  s.t.  $L(f; s)$  has at most one  $c$ -exceptional zero (35.4.5.3).
- For any  $X \in \mathbb{R}_+$ , we have

$$\sum_{n \in [X]_+} |\Lambda_f(n)|^2 \ll X d^2 \log^2(XC(f)).$$

with implied constant absolute.

Then for  $X \in \mathbb{R}_+$ , we have

$$\psi(f; X) + \text{ord}_{s=1}(f)X + \frac{X^{\beta_f}}{\beta_f} \ll \frac{X}{\exp\left(\frac{c}{d^4} \frac{\log X}{\sqrt{\log X} + 3 \log N(f)}\right)} (d \log(XC(f)))^4 \quad (35.5.6.1).$$

$$\psi(f; X) + \text{ord}_{s=1}(f)X + \frac{X^{\beta_f}}{\beta_f} \ll \sqrt{C(f)} \frac{X}{\exp\left(\frac{c}{2d^4} \sqrt{\log X}\right)} \quad (35.5.6.1).$$

with implied constants absolute. Notice that the conditions hold if Rankin-Selberg convolution  $L(f \otimes \bar{f}; s)$  exists and has a simple pole at  $s = 1$  (Cf. [I-K04]P110).  $\lrcorner$

*Proof:* Cf. [I-K04]P111 ?.

The idea is to use Mellin transform and integrate on the zero-free region.

The second inequality follows from the first one (Cf. [I-K04]P111).  $\square$

**Cor. (35.5.6.3) [Prime Number Theorem for Dirichlet Characters].** There exists effective constants  $c, \kappa \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Dir}'(N)$ , we have

$$\left| \psi(\chi; X) - \delta_{\chi, 1_N} X + \frac{X^{\beta_\chi}}{\beta_\chi} \right| \leq \kappa \log^4 N \frac{X}{\exp\left(\frac{c \log X}{\sqrt{\log X} + \log N}\right)}.$$

$\lrcorner$

*Proof:* The conditions hold by (35.4.5.6) and trivial estimation.  $\square$

**Thm. (35.5.6.4) [Explicit Formula with Ramanujan Conjecture].** Let  $f$  be an analytic  $L$ -function of degree  $d$  s.t. the generalized Ramanujan conjecture holds (35.8.0.1). Then for  $X \in \mathbb{R}_{\geq 2}$ ,  $T \in [1, X]$ , we have

$$\psi(f; X) + \text{ord}_{s=1}(f)X + \sum_{\substack{\rho \in \text{Zero}(L(f; s)) \setminus \{0\} \\ |\text{Im}(\rho)| \leq T}} \frac{X^\rho - 1}{\rho} \ll \frac{X}{T} \log X \log(X^d C(f)).$$

with implied constant absolute.  $\lrcorner$



*Proof:* ? Cf. [I-K04]P112. □

**Cor. (35.5.6.5) [Explicit Formula for Dirichlet Characters].** For  $N \in \mathbb{Z}_+$ ,  $\chi \in \text{Diri}(N)$  and  $X \in \mathbb{R}_{\geq 2}$ ,  $T \in [1, X]$ , we have

$$\psi(\chi; X) - \delta_{\chi, 1_N} X + \sum_{\substack{\rho \in \text{Zero}(L(\chi; s)) \setminus \{0\} \\ |\text{Im}(\rho)| \leq T}} \frac{X^\rho - 1}{\rho} \ll \frac{X}{T} \log^2(XN).$$

┘

*Proof:* if  $\chi \in \text{Diri}'(N)$ , this follows from (35.5.6.4), as  $L(\chi; s)$  trivially satisfies the Ramanujan conjecture. In general, if  $\chi$  is induced from  $\chi' \in \text{Diri}'(N')$  where  $N' | N$ , then

$$|\psi(\chi; X) - \psi(\chi'; X)| \leq \sum_{\substack{p \in \text{Prime}, a \in \mathbb{Z}_+ \\ p^a \leq X, p | N}} \log p \ll \log X \log N,$$

and  $L(\chi; s) = L(\chi'; s) \prod_{p \in \text{Prime}, p | N, p \nmid N'} (1 - p^{-s})$ , so the contribution of  $\rho$  differ by at most

$$\sum_{p \in S(N)} \sum_{l \in \mathbb{Z}^\times, |\rho_{p,l}| \leq T} \left| \frac{X^{\rho_{p,l}} - 1}{\rho_{p,l}} \right| \ll \log^2(XN), \text{ where } \rho_{p,l} = \frac{2\pi i l}{\log p}.$$

Thus the assertion follows. □

## 7 Primes in Arithmetic Progressions

**Def. (35.5.7.1) [Prime Counting Functions in Arithmetic Progressions].** For  $N \in \mathbb{Z}_+$ ,  $a \in (\mathbb{Z}/(N))^*$  and  $X \in \mathbb{R}_+$ , define

$$\pi(N, a; X) \triangleq \#\{p \in \text{Prime} : p \leq X, p \equiv a \pmod{N}\}.$$

And  $\pi(1, 1; X)$  is also denoted by  $\pi(X)$ .

Similarly, define analogues of Chebyshev functions (2.6.1.32) as

$$\vartheta(N, a; X) \triangleq \sum_{p \in \text{Prime}, p \leq X, p \equiv a \pmod{N}} \log p,$$

and

$$\psi(N, a; X) \triangleq \sum_{n \in \mathbb{Z}_+, n \leq X, n \equiv a \pmod{N}} \Lambda(n),$$

Notice that  $\psi(N, a; X) - \vartheta(N, a; X)$  is bounded by a constant. ┘

**Def. (35.5.7.2) [Normalized Prime Counting Function].** The **normalized prime counting function**  $\pi_0$  is defined to be the function  $\pi_0 : \mathbb{R} \rightarrow \mathbb{R}_+$ :

$$\pi_0(X) \triangleq \lim_{h \rightarrow 0} \frac{1}{2} (\pi(X+h) + \pi(X-h)).$$

┘

**Def. (35.5.7.3) [Discrepancy Functions].** For  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$ ,  $X \in \mathbb{R}_+$  and  $N, a \in \mathbb{Z}$  s.t.  $(a, N) = 1$ , define the **discrepancy function**

$$\mathcal{D}_f(N, a; X) = \sum_{n \in [X]_+, n \equiv a \pmod{N}} f(n) - \frac{1}{\phi(N)} \sum_{n \in [X]_+, (n, N) = 1} f(n)$$

and

$$\text{Err}_f(N; X) = \max_{a \in (\mathbb{Z}/(N))^*} |\mathcal{D}_f(N, a; X)|.$$

It follows from orthogonality of Dirichlet characters that

$$\text{Err}_f(N; X) \leq \max_{\chi \in \text{Diri}(N) \setminus \mathbf{1}_N} \left| \sum_{n \in [X]_+} f(n) \chi(n) \right| \leq \|f \mathbf{1}_{[X]_+}\| N^{1/2}.$$

And it also follows by trivial bound that

$$\text{Err}_f(N; X) \leq \frac{\max_{n \in [X]_+} |f(n)|}{\phi(N)} N.$$

┘

**Cor. (35.5.7.4) [Discrepancy for Monotone Functions].** If  $a < b \in \mathbb{Z}_+$  and  $f : [a, b] \rightarrow \mathbb{C}$  is a monotone function, then for any  $N \in \mathbb{Z}_+$  and  $X \in \mathbb{R}_{\geq b}$ , we have

$$\text{Err}_f(N; X) \leq |f(a)| + |f(b)| \text{ (35.5.7.3)}.$$

┘

*Proof:* This follows from elementary consideration. □

**Cor. (35.5.7.5) [Discrepancy for Convolutions].** If  $g, h : \mathbb{Z}_+ \rightarrow \mathbb{C}$  are arithmetic functions, then for any  $X \in \mathbb{R}_{\geq b}$  and  $a, N \in \mathbb{Z}_+$  s.t.  $(a, N) = 1$ , we have

$$|\mathcal{D}_{g*h}(N, a; X)| \leq \sum_{d \in [X]_+, (d, N) = 1} |g(d)| |\mathcal{D}_h(a/d, N; X/d)|.$$

┘

**Cor. (35.5.7.6) [Discrepancy for Products].** If  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  is an arithmetic function and  $g : \mathbb{R}_{\geq 1} \rightarrow \mathbb{C}_+$  is smooth, then for any  $X \in \mathbb{R}_{\geq b}$  and  $a, N \in \mathbb{Z}_+$  s.t.  $(a, N) = 1$ , we have

$$|\mathcal{D}_{fg}(N, a; X)| = \mathcal{D}_f(N, a; X) g(X) - \int_{1.5}^X \mathcal{D}_f(N, a; t) g'(t) dt.$$

┘

**Lemma (35.5.7.7).** For  $X \in \mathbb{R}_{\geq 2}$ ,

$$\prod_{p \in \text{Prime}, p \leq X} p \leq 4^X.$$

┘

*Proof:* ? Cf. [Ten15]P13. □

**Thm.(35.5.7.8)**[Mertens' Theorem, Mertens[Mer74]]. For  $X \in \mathbb{R}_{\geq 1}$ ,

1.

$$-1 - \log 4 \leq \left( \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p} \right) - \log X + 1 \leq \log 4, \quad \sum_{n \in [X]_+} \frac{\Lambda(n)}{n} = \log X + O(1).$$

2.

$$\sum_{p \in \text{Prime}, p \leq X} \frac{1}{p} = \log_{(2)} X + M + O\left(\frac{1}{\log X}\right),$$

where  $M \approx 0.26$  is called the **Meissel-Mertens constant**.

3.

$$\sum_{p \in \text{Prime}, p \leq X} \log\left(1 - \frac{1}{p}\right) + \log_{(2)} X + \gamma_0 = O\left(\frac{1}{\log X}\right).$$

┘

**Remark(35.5.7.9)**[Rosen, A generalization of Mertens' theorem]. has extended this to general number fields. ┘

*Proof:* 1: An easy estimation of the off-diagonal terms shows that the former implies the latter. For the former, notice that

$$\begin{aligned} \log(\lfloor X \rfloor!) &= \sum_{p \in \text{Prime}} v_p(\lfloor X \rfloor!) \log p \\ (2.6.1.33) &> X \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p} - \sum_{p \in \text{Prime}, p \leq X} \log p \\ (35.5.7.7) &\geq X \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p} - X \log 4. \end{aligned}$$

But also by Stirling's inequality(12.6.1.16),

$$X \log X - X + 1 \leq \log(\lfloor X \rfloor!) \leq X \log X - X + 1 + \log X \leq X \log X,$$

so the upper bound follows. For the lower bound, notice that

$$\begin{aligned} \log(\lfloor X \rfloor!) &= \sum_{p \in \text{Prime}} v_p(\lfloor X \rfloor!) \log p \\ (2.6.1.33) &< X \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p} + X \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p(p-1)} \\ &\leq X \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p} + X \sum_{n \in \mathbb{Z}_{\geq 2}} \frac{\log m}{m(m-1)} \\ &\leq X \sum_{p \in \text{Prime}, p \leq X} \frac{\log p}{p} + X \log 4. \end{aligned}$$

So the lower bound follows.

2: Denote  $\theta(X) = \sum_{p \in \text{Prime} \cap [X]_+} \frac{\log p}{p} - \log X = O(1)$ , then It follows from summation by part(35.1.2.9) and item1 that

$$\sum_{p \in \text{Prime} \cap [X]_+} \frac{1}{p} = 1 + O\left(\frac{1}{\log X}\right) + \int_2^X \frac{\log t + \theta(t)}{t \log^2 t} dt = \log_{(2)}(X) + 1 - \log_{(2)}(2) + \int_2^X \frac{\log t + \theta(t)}{t \log^2 t} dt + O\left(\frac{1}{\log X}\right).$$

3: It suffices to prove that

$$\sum_{p \in \text{Prime}} \left( \log \left( 1 - \frac{1}{p} \right) + \frac{1}{p} \right) = M - \gamma_0.$$

This summation is absolutely convergent and equals

$$\lim_{\delta \rightarrow 0^+} \sum_{p \in \text{Prime}} \left( \log \left( 1 - \frac{1}{p^{1+\delta}} \right) + \frac{1}{p^{1+\delta}} \right) = \lim_{\delta \rightarrow 0^+} (g(\delta) - h(\delta)),$$

where

$$g(\delta) = \sum_{p \in \text{Prime}} \frac{1}{p^{1+\delta}} + 2^\delta \log \delta, \quad h(\delta) = \log \zeta(1 + \delta) + 2^\delta \log \delta.$$

Notice  $\lim_{\delta \rightarrow 0^+} h(\delta) = 0$  by (35.5.3.3), and for  $g(\delta)$ , by item2 and summation by part (35.1.2.9), we have

$$\begin{aligned} \sum_{p \in \text{Prime}} \frac{1}{p^{1+\delta}} &= \lim_{X \rightarrow \infty} \frac{\log_{(2)} X + M + O(1/\log X)}{X^\delta} + \delta \int_2^\infty \frac{\log_{(2)} t + M + O(1/\log t)}{t^{1+\delta}} dt \\ &= M + \delta \int_2^\infty \frac{\log_{(2)} t}{t^{1+\delta}} dt + O(1) \cdot \delta \int_2^{\exp(\frac{1}{\sqrt{\delta}})} \frac{1}{t} dt + O(1) \cdot \delta \cdot \sqrt{\delta} \int_{\exp(\frac{1}{\sqrt{\delta}})}^\infty \frac{1}{t^{1+\delta}} dt \\ (t = e^{u/\delta}) &= M + \int_{\delta \log 2}^\infty e^{-u} \log \left( \frac{u}{\delta} \right) du + O \left( \sqrt{\delta} + \frac{\sqrt{\delta}}{e\sqrt{\delta}} \right) \\ &= M + \int_{\delta \log 2}^\infty e^{-u} \log u du - 2^\delta \log \delta + O \left( \sqrt{\delta} + \frac{\sqrt{\delta}}{e\sqrt{\delta}} \right) \end{aligned}$$

So  $\lim_{\delta \rightarrow 0^+} g(\delta) = M - \gamma_0$  by (12.6.1.13). □

**Prop. (35.5.7.10)** [Chebyshev's Estimates, Chebyshev1852/Nair1982].

- For  $X \in \mathbb{R}_{\geq 1}$ ,  $\vartheta(X) \leq \log 4 \cdot X$  (2.6.1.32).
- For  $X \in \mathbb{R}_{\geq 7}$ ,  $\psi(X) = \log(\text{lcm}\{n \in [X]_+\}) \geq (\log 2) \lfloor X \rfloor$  (2.6.1.32).

┘

*Proof:* For  $m \in \mathbb{Z}_+$ ,

$$2^{2m+1} = (1+1)^{2m+1} \geq 2 \binom{2m+1}{m} \geq 2 \prod_{m+1 < p \leq 2m+1, p \in \text{Prime}} p,$$

so

$$\vartheta(2m+1) - \vartheta(m+1) \leq m \log 4.$$

Also notice that  $\vartheta(2m) = \vartheta(2m-1)$ . From this the assertion follows by induction.

2: For  $1 \leq m \leq n \leq \mathbb{Z}_+$ ,  $m \binom{n}{m} \mid \text{lcm}\{k \in [n]_+\}$  because for each  $p \in \text{Prime}$ ,  $v_p(m \binom{n}{m}) \leq \lfloor \log_p(n) \rfloor = v_p(\text{lcm}\{k \in [n]_+\})$  by (2.6.1.34). From which we get

$$n(2n+1) \binom{2n}{n} \mid \text{lcm}\{k \in [n]_+\},$$

In particular, we get  $\text{lcm}\{k \in [2n+1]_+\} \geq n4^n$  for  $n \in \mathbb{Z}_+$ , and the assertion follows. □

**Prop. (35.5.7.11)**[Chebyshev]. For  $X \in \mathbb{R}_{\geq 1}$ , we have

$$\psi(X) = \sum_{k \in \mathbb{Z}_+} \vartheta(X^{1/k}) = \vartheta(X) + O(\sqrt{X}).$$

$$\pi(X) = \frac{\vartheta(X)}{\log X} + O\left(\frac{X}{\log^2 X}\right).$$

In particular,

$$\pi(X) \sim \frac{X}{\log X} \iff \psi(X) \sim X \iff \vartheta(X) \sim X.$$

(35.5.7.1). ┘

*Proof:* 1 follows from (35.5.7.10).

2: By Abel summation,

$$\begin{aligned} \pi(X) &= \frac{\vartheta(X)}{\log X} + \int_2^X \frac{\vartheta(t)}{t \log^2 t} dt \\ (35.5.7.10) &= \frac{\vartheta(X)}{\log X} + O\left(\int_2^X \frac{1}{\log^2 t} dt\right) \\ &= \frac{\vartheta(X)}{\log X} + O\left(\int_2^X \left(\frac{1}{\log^2 t} - \frac{2}{\log^3 t}\right) dt\right) \\ &= \frac{\vartheta(X)}{\log X} + O\left(\frac{X}{\log^2 X}\right). \end{aligned}$$

□

**Lemma (35.5.7.12).** Define the function  $\Phi(s) = \sum_{p \in \text{Prime}} p^{-s} \log p$ , then  $(\mathcal{L}\vartheta(e^t))(s) = \Phi(s)/s$ . Thus  $\Phi(s)$  is holomorphic on  $\text{Re}(s) > 1$  by (12.5.4.4) and (35.5.7.10), and  $\Phi(s) - \frac{1}{s-1}$  extends to meromorphic functions on  $\text{Re}(s) > 1/2$  and holomorphic on  $\text{Re}(s) \geq 1$ . ┘

*Proof:* By (2.6.1.2) there are infinitely many primes. For  $n \in \mathbb{Z}_+$ , let  $\mathbf{p}_n$  be the  $n$ -th smallest prime number, then the function  $\vartheta(e^t)$  is constant on  $(\log \mathbf{p}_n, \log \mathbf{p}_{n+1})$ , and

$$\int_{\log \mathbf{p}_n}^{\log \mathbf{p}_{n+1}} e^{-st} \vartheta(e^t) dt = \vartheta(\mathbf{p}_n) \int_{\log \mathbf{p}_n}^{\log \mathbf{p}_{n+1}} e^{-st} dt = \vartheta(\mathbf{p}_n) \frac{\mathbf{p}_n^{-s} - \mathbf{p}_{n+1}^{-s}}{s}.$$

Thus

$$(\mathcal{L}\vartheta(e^t))(s) = \int_0^\infty e^{-st} \vartheta(e^t) dt = \frac{1}{s} \sum_{n \in \mathbb{Z}_+} \vartheta(\mathbf{p}_n) (\mathbf{p}_n^{-s} - \mathbf{p}_{n+1}^{-s}) = \frac{1}{s} \sum_{n \in \mathbb{Z}_+} \mathbf{p}_n^{-s} \log \mathbf{p}_n = \frac{\Phi(s)}{s}.$$

For the last assertion, notice

$$-\frac{\zeta'(s)}{\zeta(s)} = (-\log \zeta(s))' = \left( \sum_{p \in \text{Prime}} \log(1 - p^{-s}) \right)' = \sum_{p \in \text{Prime}} \frac{\log p}{p^s - 1} = \Phi(s) + \sum_{p \in \text{Prime}} \frac{\log p}{p^s(p^s - 1)},$$

And  $\sum_{p \in \text{Prime}} \frac{\log p}{p^s(p^s - 1)}$  is absolutely convergent and holomorphic for  $\text{Re}(s) > 1/2$ , and  $\frac{\zeta'(s)}{\zeta(s)}$  has a simple pole with residue 1 by (35.5.3.1). So the assertion follows. □

**Thm. (35.5.7.13)**[Estimates for  $n$ -th Prime Number]. For  $n \in \mathbb{Z}_{\geq 6}$ ,

$$n(\log n + \log \log n - 1) \leq p_n \leq n(\log n + \log \log n).$$

┘

*Proof:* [https://en.wikipedia.org/wiki/Prime\\_number\\_theorem#Approximations\\_for\\_the\\_nth\\_prime\\_number.](https://en.wikipedia.org/wiki/Prime_number_theorem#Approximations_for_the_nth_prime_number.) ? □

**Prop. (35.5.7.14).** For  $X \in \mathbb{R}_{>17}$ ,

$$\frac{X}{\log X} < \pi(X) < 1.25506 \frac{X}{\log X}.$$

┘

*Proof:* [ROSSER and L. SCHOENFELD, Approximate formulas for some functions of prime numbers, Ill. J. Math. 6(1962), 64-94.]Thm2 ? □

**Cor. (35.5.7.15)**[Bertrand's Postulate, Chebyshev]. For any  $X \in \mathbb{R}_{\geq 1}$ , there is a rational prime in  $(X, 2X]$ . ┘

*Proof:* When  $X \geq 16$ ,  $4 \log 2 \leq \log X$ , so

$$\pi(2X) - \pi(X) \geq \frac{2X}{\log(2X)} - 1.25506 \frac{X}{\log X} \geq 0.344 \frac{X}{\log X} \geq \frac{4 \cdot 0.344}{\log 2} > 1$$

And for  $X \in (0, 16)$ , we can verify by hand. □

**Thm. (35.5.7.16)**[Brun-Titchmarsh Inequality, Titchmarsh[Tit30]/Montgomery-Vaughan[M-V73]]. For any  $X, Y \in \mathbb{R}_+$ ,  $N \in \mathbb{Z}_+$ ,  $a \in (\mathbb{Z}/(N))^*$  s.t.  $N < Y$ , we have

$$\pi(N, a; X + Y) - \pi(N, a; X) \leq \frac{2Y}{\phi(N) \log(Y/N)}.$$

┘

*Proof:* ? We only prove

$$\pi(N, a; X + Y) - \pi(N, a; X) < \frac{2Y}{\phi(N) \log(Y/N)} + O\left(\frac{Y}{N \log^2(Y/N)}\right).$$

We use the Selberg's sieve(35.2.3.2) to the arithmetic progression  $\mathcal{A} \subset (X, X + Y]$  and  $P$  is the product of primes in  $\mathbf{Prime}_{\leq \sqrt{Y}} \setminus S(N)$ , then

$$\pi(N, a; X + Y) - \pi(N, a; X) \leq S(\mathcal{A}; P) + \sqrt{Y}/N.$$

And the sieve condition of(35.2.3.4)(35.2.3.5) are satisfied with  $D = Y/N$ ,  $g(d) = d^{-1}$  when  $(d, N) = 1$ ,  $\kappa = 1$ , and  $X = Y/N$ , i.e. for each  $d \in \mathbb{Z}_+$  s.t.  $(d, N) = 1$ , we have

$$\mathcal{A}_d = \frac{Y}{dN} + r_d(\mathcal{A}), \text{ where } |r_d(\mathcal{A})| \leq 1.$$

So

$$H(D) > \frac{\phi(N)}{2N} \log D,$$

and we get the desired result. □

**Remark(35.5.7.17).** If the constant 2 can be replaced by  $2 - \varepsilon$  for some  $\varepsilon \in \mathbb{R}_+$ , then there are no Siegel-Landau zeros. ? ┘

**Thm.(35.5.7.18)** [Prime Number Theorem, conjectured by Legendre/Gauss, Hadamard[Had96]/Vallée-Poussin(1896, 1899)/Page[Pag35]/Bombieri[Bom62]/Wirsing[Wir62, Wir64]/Korobov[Kor58]/Vinogradov[Vin65]/Diamond-Steinig[D-S70]/Siegel[Sie35a]/Walfisz[Wal36]].

1. For  $N \in \mathbb{Z}_+$ ,  $\chi \in \text{Diri}'(N)$  and  $X \in \mathbb{R}_{\geq 2}$ , we have

$$\psi(\chi; X) - \delta_{\chi, 1_N} X \ll_A \frac{X}{\exp(A\sqrt{\log X})} \quad (35.5.6.1),$$

$$\sum_{p \in \text{Prime} \cap [X]_+} \chi(p) - \delta_{\chi, 1_N} \text{Li}(X) \ll_A \frac{X}{\exp(A\sqrt{\log X})},$$

$$\sum_{n \in [X]_+} \chi(n) \mu(n) \ll_A \frac{X}{\exp(A\sqrt{\log X})},$$

for any  $A \in \mathbb{R}_+$  and  $X \in \mathbb{R}_{\geq 2}$  with  $N \leq \log^A X$ .

2. For any  $N, a \in \mathbb{Z}_+$ ,  $a \in \mathbb{Z}_+$  s.t.  $(N, a) = 1$  and  $X \in \mathbb{R}_{\geq 2}$ , we have

$$\psi(N, a; X) - \frac{1}{\phi(N)} X \ll_A \frac{X}{\log^A X}.$$

$$\pi(N, a; X) - \frac{1}{\phi(N)} \text{Li}(X) \ll_A \frac{X}{\log^A X},$$

$$\mathcal{M}_\mu(N, a; X) \ll_A \frac{X}{\log^A X},$$

3. For  $N = 1$ , we have the stronger: There exists  $C \in \mathbb{R}_+$  s.t. for any  $X \in \mathbb{R}_{\geq 3}$ ,

$$\psi(X) - X \ll \left( \frac{X}{\exp \left( C \log^{3/5} X \log_{(2)}^{-1/5} X \right)} \right).$$

$$\pi(X) - \text{Li}(X) \ll \left( \frac{X}{\exp \left( C \log^{3/5} X \log_{(2)}^{-1/5} X \right)} \right).$$

$$\mathcal{M}_\mu(X) \ll \left( \frac{X}{\exp \left( C \log^{3/5} X \log_{(2)}^{-1/5} X \right)} \right).$$

┘

**Remark(35.5.7.19).** The barrier for improving the bound of the error term lies in the barrier to improve zero-free regions for  $L(\chi; s)$ . ┘

*Proof:* 1: For  $\Lambda(n)$ , the assertion follows from (35.5.6.3) and Siegel's estimation for  $\beta_\chi$  (35.4.4.12).  
For  $\pi(\chi; X)$ , this follows from (35.5.7.11) and partial summation:

$$\begin{aligned}\pi(\chi; X) &= \frac{\psi(\chi; X)}{\log X} - \int_2^X \psi(\chi; t) \left( \frac{1}{\log t} \right)' dt \\ &= \frac{X}{\log X} - \int_2^X \frac{1}{\log^2 t} dt + O\left(\int_2^X \frac{1}{\log^{A+2} t} dt\right) \\ &= \text{Li}(X) + O\left(\frac{X}{\log^{A+1} X}\right).\end{aligned}$$

For  $\mu(n)$  ? Cf. [I-K04]P124.

2: If  $N \leq \log^{A+1} X$ , this follows from orthogonality of Dirichlet-characters and item1( ? Cf. [Ten15]P395). And for  $N > \log^{A+1} X$ , the assertions trivially follows from ??.

3: ? Cf. [I-K04]P227.

□

**Cor. (35.5.7.20)** [Dirichlet's Theorem on Prime in Arithmetic Progression, Dirichlet(1847)].

For any  $N \in \mathbb{Z}_{\geq 2}$  and  $a \in \mathbb{Z}$  s.t.  $(a, N) = 1$ , we have

$$\#\text{Prime} \cap (a + N\mathbb{Z}) = \infty.$$

┘

**Cor. (35.5.7.21).** For any  $N \in \mathbb{Z}_+$ ,  $A \in \mathbb{R}_+$ , we have

$$-\sum_{m \in [X/N]_+, (m, N)=1} \frac{\mu(m)}{m} \log(Nm) = \frac{N}{\phi(N)} \left(1 + O_A\left(\tau(N) \log^{-A} X\right)\right)$$

for each  $X \in \mathbb{R}_{\geq 2}$ .

┘

*Proof:* ? Cf. <https://math.stackexchange.com/questions/255262/prove-that-sum-n-1-infty-mun-frac-log-nn-1>. □

**Def. (35.5.7.22)** [Restrictions on  $N$ ]. Let  $\mathcal{B}$  be the set of numbers  $B \in (0, 1)$  s.t. there exists  $D_B \in \mathbb{N}$  s.t. when  $X \in \mathbb{R}_+$  is large, there exists a set  $\mathcal{D}_B(X)$  of at most  $D_B$  many integers bigger than  $\log X$ , s.t. for any  $Y \in \mathbb{R}_+$ ,  $N \in \mathbb{Z}_+$ ,  $N \leq \min(X^B, Y/X^{1-B})$  and  $a \in (\mathbb{Z}/(N))^*$ ,

$$\pi(Y; N, a) \geq \frac{\pi(Y)}{2\phi(N)},$$

unless  $N$  divides one of the numbers in  $\mathcal{D}_B(X)$ .

Also denote  $B_{\pi,1}$  the supremum of numbers  $B$  s.t. there exists  $D_B \in \mathbb{N}$  s.t. when  $X \in \mathbb{R}_+$  is large, there exists a set  $S_B(X)$  of at most  $D_B$  many integers bigger than  $\log X$  s.t. for any  $N \in \mathbb{Z}_+$ ,  $N \leq X^B$ ,

$$\pi(X; N, 1) \geq \frac{\pi(X)}{2\phi(N)},$$

unless  $N$  divides one of the numbers in  $S_B(X)$ . Then it is clear that  $(0, B_{\pi,1}) \subset \mathcal{B}$ .

┘

**Prop. (35.5.7.23).**  $(0, 5/12) \in \mathcal{B}$ .

┘



*Proof:* [Alford-Granville-Pomerance, Carmichael numbers].  $\square$

**Prop. (35.5.7.24).** For any  $A \in \mathcal{A}^?$  and  $\varepsilon, \delta \in \mathbb{R}_+$ , there exist numbers  $\eta_{\varepsilon, \delta}, D_{\varepsilon, \delta}$  s.t. for any  $X$  sufficiently large, there is a set  $\mathcal{D}_{\varepsilon, \delta}(x)$  of integers in of cardinality  $\leq D_{\varepsilon, \delta}$ , s.t. every element of  $\mathcal{D}_{\varepsilon, \delta}$  is bigger than  $\log X$ , and at most one of them is bigger than  $X^{\eta_{\varepsilon, \delta}}$ , and

$$\left| \vartheta(Y; N, a) - \frac{y}{\phi(d)} \right| \leq \varepsilon \frac{y}{\phi(d)}$$

for any  $N \in \mathbb{Z}_+, Y \in \mathbb{R}_+, a \in (\mathbb{Z}/(N))^*$  s.t.  $N$  is not divisible by elements of  $\mathcal{D}_{\varepsilon, \delta}(x)$ , and  $1 \leq d \leq \min(X^{1/A-\delta}, Y/X^{1-1/A+\delta})$ .  $\lrcorner$

*Proof:* Cf. [AGP94]P712.  $?$   $\square$

**Prop. (35.5.7.25).** Suppose  $B \in \mathcal{B}(35.5.7.22)$ , then for any  $X$  large and  $L \in \mathbb{Z}_+$  s.t.  $L$  is square-free not divisible by any prime exceeding  $X^{(1-B)/2}$ , and

$$\sum_{p \in S(L)} \frac{1}{p} \leq (1-B)/32,$$

there exists  $k \in \mathbb{Z}_+, k \leq X^{1-B}, (k, L) = 1$  s.t.

$$\#\{d \in \mathbb{Z}_+ : d|L, dk+1 \leq X, dk+1 \in \text{Prime}\} \geq \frac{2^{-D_B-2}}{\log X} \#\{d \in \mathbb{Z}_+ : d|L, 1 \leq d \leq X^B\}.$$

*Proof:* Cf. [AGP94]P715.  $?$   $\square$

### Elliott-Halberstam Conjecture(Averaged GRH for Dirichlet Series)

**Conj. (35.5.7.26) [Montgomery].** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_\varepsilon \in \mathbb{R}_+$  s.t. for any  $a, N \in \mathbb{Z}_+, (a, N) = 1$  and  $X \in \mathbb{R}_+$ , we have

$$|\mathcal{D}_\Psi(N, a; X)| \leq C_\varepsilon (N^{-1/2} X^{1/2+\varepsilon}) \quad (35.5.7.3).$$

Notice that this doesn't follow from the generalized Riemann hypothesis.  $\lrcorner$

*Proof:*  $\square$

**Conj. Cor. (35.5.7.27) [Level of Distributions of Primes, Elliott-Halberstam].** For  $N \in \mathbb{Z}_+$  and  $X \in \mathbb{R}_+$  and any  $\theta \in (0, 1)$  and  $A \in \mathbb{R}_+$ , there exists a constant  $C_{A, \theta} \in \mathbb{R}_+$  s.t.

$$\sum_{1 \leq N \leq X^\theta} \text{Err}_\pi(X; N) \leq C_{A, \theta} \frac{X}{\log^A X} \quad (35.5.7.3).$$

This is a refinement of the averaged form of the prime number theorem (35.5.7.18). If the assertion is true for  $\theta$ , then we say that the **primes have level of distribution  $\theta$** .

An equivalent formulation is

$$\sum_{1 \leq N \leq X^\theta} \text{Err}_\Psi(X, N) \leq C_{A, \theta} \frac{X}{\log^A X} \quad (35.5.7.3).$$

$\lrcorner$

*Proof:*

□

**Thm& Conj. Cor. (35.5.7.28).** The Elliott-Halberstam conjecture (35.5.7.27) is true for any  $\theta \in (0, 1/2)$ , by Bombieri-Vinogradov (35.5.7.31). ┘

**Lemma (35.5.7.29) [Siegel-Walfisz type Inequalities].** If  $N \in \mathbb{Z}_+$  and  $(\beta_n)_{n \in [N]_+}$ ,  $\Delta \in (0, 1]$  satisfies

$$\text{Err}_\beta(q; N) \leq \|\beta\| N^{1/2} \Delta^9,$$

then for any  $r \in \mathbb{Z}_{\geq 2}$ ,  $s \in \mathbb{Z}_+$  and  $\mathbf{1}_r \neq \chi \in \text{Diri}(r)$ , we have

$$\left| \sum_{n \in \mathbb{Z}_+, (n, s)=1} \beta_n \chi(n) \right| \leq \|\beta\| N^{1/2} \Delta^3 \tau(s) r^{1/3}.$$

┘

*Proof:* By Legendre formula (35.2.1.2), for any  $K \in \mathbb{Z}_+$ ,

$$\begin{aligned} \sum_{n \in \mathbb{Z}_+, (n, s)=1} \beta_n \chi(n) &= \sum_{k \in \mathbb{Z}_+, k|s} \mu(k) \sum_{n \in \mathbb{Z}_+, k|n} \beta_n \chi(n) \\ &= \sum_{k \in [K]_+, k|s} \mu(k) \sum_{l \in \mathbb{Z}_+, l|k} \mu(l) \sum_{n \in \mathbb{Z}_+, (n, l)=1} \beta_n \chi(n) + \sum_{k \in \mathbb{Z}_{>K}, k|s} \mu(k) \sum_{n \in \mathbb{Z}_+, k|n} \beta_n \chi(n) \\ &= \sum_{k \in [K]_+, k|s} \mu(k) \sum_{l \in \mathbb{Z}_+, l|k} \mu(l) \sum_{a \in (\mathbb{Z}/(lr))^*} \chi(a) \mathcal{D}_\beta(s, a; N) + \sum_{k \in \mathbb{Z}_{>K}, k|s} \mu(k) \sum_{n \in \mathbb{Z}_+, k|n} \beta_n \chi(n) \end{aligned}$$

The first term is bounded by

$$\|\beta\| N^{1/2} \Delta^9 \sum_{k \in [K]_+, k|s} \sum_{l \in \mathbb{Z}_+, l|k} \phi(lr) \leq \|\beta\| N^{1/2} \Delta^9 K \phi(r) \tau(s).$$

And by Cauchy's inequality, the second term is bounded

$$\|\beta\| \sum_{k \in \mathbb{Z}_{>K}, k|s} N^{1/2} k^{-1/2} \leq \|\beta\| N^{1/2} K^{-1/2} \tau(s).$$

Then the assertion follows by taking  $K = \Delta^{-6} \phi(r)^{-2/3}$ . □

**Thm. (35.5.7.30) [general Bombieri-Vinogradov Theorem (Type II Estimates)].** If  $M, N \in \mathbb{Z}_+$  and  $(\alpha_m)_{m \in [M]_+}$ ,  $(\beta_n)_{n \in [N]_+}$ ,  $\Delta \in (0, 1]$  s.t. for any  $q \in \mathbb{Z}_+$ , we have a Siegel-Walfisz type inequality (35.5.7.29):

$$\text{Err}_\beta(q; N) \leq \|\beta\| N^{1/2} \Delta^9.$$

Then for any  $Q \in \mathbb{R}_+$  we have

$$\sum_{q \in [Q]_+} \text{Err}_{\alpha * \beta}(q; MN) \ll \|\alpha\| \|\beta\| \left( \Delta^{9/7} M^{1/2} N^{1/2} + M^{1/2} + N^{1/2} + Q \right) \log^2 Q.$$

┘

*Proof:* By orthogonality of Dirichlet characters, for any  $a \in (\mathbb{Z}/(q))^*$ ,

$$D_{\alpha*\beta}(q, a; MN) = \frac{1}{\phi(q)} \sum_{\chi \in \text{Diri}(q) \setminus \mathbf{1}_q} \bar{\chi}(a) \left( \sum_m \alpha_m \chi(m) \right) \left( \sum_n \beta_n \chi(n) \right),$$

so

$$\sum_{q \in [Q]_+} \text{Err}_{\alpha*\beta}(q; MN) \leq \sum_{s \in [Q]_+} \frac{1}{\phi(s)} \sum_{1 < r \in [Q]_+} \frac{1}{\phi(r)} \sum_{\chi \in \text{Diri}'(r)} \left| \sum_{(m,s)=1} \alpha_m \chi(m) \right| \left| \sum_{(n,s)=1} \beta_n \chi(n) \right|.$$

For  $R \in \mathbb{R}_+$ , we can bound the sums for  $r \leq R$  using (35.5.7.29), to get the bound

$$\sum_{r \in [R]_+} \leq \|\alpha\| \|\beta\| M^{1/2} N^{1/2} \Delta^3 \sum_{s \in [Q]_+} \frac{\tau(s)}{\phi(s)} \sum_{r \in [R]_+} r^{1/3} \ll \|\alpha\| \|\beta\| M^{1/2} N^{1/2} \Delta^3 R^{4/3} \log^2 Q \quad (35.1.3.15).$$

And for the sums for  $r > R$ , we split the range into dyadic segments  $[2^k + 1, 2^{k+1}]$  and apply Cauchy's inequality and the large sieve inequality (35.1.6.13) to get the bound

$$\begin{aligned} \sum_{r \in \mathbb{Z}_{>R}} &\ll \|\alpha\| \|\beta\| \sum_{P \in 2^{\mathbb{N}} \cap [R/2, Q]} \left( 16P^2 + 4(M+N) + P^{-2} \right)^{1/2} \log Q \quad (35.1.3.15) \\ &\ll \|\alpha\| \|\beta\| \left( Q + M^{1/2} + N^{1/2} + M^{1/2} N^{1/2} R^{-1} \right) \log^2 Q. \end{aligned}$$

Then the assertion follows by taking  $R = \Delta^{-9/7}$ .  $\square$

**Cor. (35.5.7.31)** [Bombieri-Vinogradov Theorem, Linnik[Lin41]/Renyi[Ren48]/Roth[Rot65]/Barban[Bar66]/Vinogradov[Vin65]/Bombieri[Bom65]/Gallagher[Gal70]/Motohashi[Mot76]].

For any  $A \in \mathbb{R}_+$ , there exists  $C_A \in \mathbb{R}_+$  s.t.

$$\sum_{1 \leq N \leq \sqrt{X}/\log^{2A+7} X} \text{Err}_{\Lambda}(N; X) \leq C_A \frac{X}{\log^A X}$$

And the same assertion holds for  $\text{Err}_{\text{Prime}}(N; X)$  or  $\text{Err}_{\mu}(N; X)$ .  $\lrcorner$

*Proof:* By Vaughan's identity (35.2.4.8),

$$\Lambda = \Lambda_1^{\sharp} - \Lambda_2^{\sharp} + \Lambda^{\flat} + \Lambda_{\leq X^{1/5}}, \text{ where } \Lambda_1^{\sharp} = \text{Log} * \mu_{\leq X^{1/5}}, \quad \Lambda_2^{\sharp} = \Lambda_{\leq X^{1/5}} * \mu_{\leq X^{1/5}} * \mathbf{1}, \quad \Lambda^{\flat} = \Lambda_{> X^{1/5}} * \mathbf{1} * \mu_{> X^{1/5}}.$$

So for any  $a, N \in \mathbb{Z}_+$  s.t.  $(a, N) = 1$ , we have

$$\mathcal{D}_{\Lambda}(N, a; X) = \mathcal{D}_{\Lambda_1^{\sharp}}(N, a; X) - \mathcal{D}_{\Lambda_2^{\sharp}}(N, a; X) + \mathcal{D}_{\Lambda^{\flat}}(N, a; X) + \mathcal{D}_{\Lambda}(N, a; X^{1/5})$$

For  $Q \in [\sqrt{X}]_+$ , we bound their contribution to  $\sum_{N \in [Q]_+} \text{Err}_{\Psi}(N; X)$  separately.

- By (35.5.7.3),  $\mathcal{D}_{\Lambda}(N, a; X^{1/5})$  can be bounded trivially by  $X^{1/5} \log X / \phi(N)$ . So it contribute  $O(X^{1/5} \log^2 X)$  to the final result, by (35.1.3.15).
- By (35.5.7.4),  $\mathcal{D}_{\Lambda_1^{\sharp}}(N, a; X) = O(\log X)$ , so it contributes  $O(X^{2/5} \log X)$  to the final result, by (35.5.7.5).

- By (35.5.7.5), the contribution of  $\mathcal{D}_{\Lambda_2^\sharp}(N, a; X)$  is bounded by

$$\sum_{N \in [Q]_+} \sum_{d \in [X]_+} \left| \mu_{\leq X^{1/5}} * \Lambda_{\leq X^{1/5}}(d) \right| \ll Q \left| \mathcal{M}_{|\mu|}(X^{1/5}) \right| \left| \mathcal{M}_\Lambda(X^{1/5}) \right| \ll X^{9/10}.$$

- For  $\Lambda^b$ , the restriction  $n = lm \leq X$  makes the convolution interconnected, so we replace  $\Lambda^b(n)$  with

$$\Lambda_{L,M}^b(n) \triangleq \sum_{\substack{L, M \in (1+\delta)^{\mathbb{N}} \cap (X^{1/5}, X^{4/5}) \\ LM < X}} \sum_{\substack{l \in ((1+\delta)^{-1}L, L] \\ m \in ((1+\delta)^{-1}M, M] \\ lm = n}} (\mu_{>X^{1/5}} * \mathbf{1})(l) \Lambda(m), \text{ where } \delta \in (X^{-1/5}, 1).$$

Their difference lies in the area  $n = lm$  and  $m \in ((1+\delta)^{\lfloor \log_{1+\delta} X^{1/5} \rfloor}, X^{1/5}]$ , or the area  $n \in ((1+\delta)^{\lfloor \log_{1+\delta} X \rfloor}, X]$ . Estimating these trivially ? we get

$$\mathcal{D}_{\Lambda^b}(N, a; X) = \sum_L \sum_M \mathcal{D}_{\Lambda_{L,M}^b}(N, a; X) + O(\delta N^{-1} X \log X).$$

The contribution of each  $\mathcal{D}_{\Lambda_{L,M}^b}(N, a; X)$  can be bounded by the general Bombieri-Vinogradov theorem (35.5.7.30) (where the hypothesis is satisfied for  $\beta = \Lambda$  with  $\Delta = \log^{-A} X$  by Siegel-Walfisz theorem (35.5.7.18)), which gives the bound

$$\left\| (\mu_{>X^{1/5}} * \mathbf{1}) \mathbf{1}_{((1+\delta)^{-1}L, L]} \right\| \left\| \Lambda \mathbf{1}_{((1+\delta)^{-1}M, M]} \right\| \Delta^{9/7} X^{1/2} \log^2 X \ll \Delta^{9/7} \delta X \log^{5/2} X \quad (35.1.3.8)$$

if we take  $Q = \Delta^{9/7} \sqrt{X}$ . Summing over  $(L, M)$  (there are  $O(\lceil \delta^{-2} \log^2 X \rceil)$ -many terms), we get the bound  $O(\delta^{-1} \Delta^{9/7} \log^2 X + \delta) X \log^{5/2} X$ . Taking  $\delta = \Delta^{9/14} \log X$ , we get the bound  $\Delta^{9/14} X \log^{7/2} X$ .

Thus we get

$$\sum_{1 \leq N \leq \sqrt{X} \log^{-9/7A} X} \text{Err}_\Lambda(N; X) \ll X \log^{7/2-9/14A} X.$$

For  $\mu(n)$ , use Vaughan's identity for  $\mu$  (35.2.4.8), and the rest is totally analogous to the above.

For  $\mathbf{1}_{\text{Prime}}$ , use (35.5.7.6). □

**Remark (35.5.7.32).** generalizing the Bombieri-Vinogradov theorem is closed related to density theorems for Dirichlet series Density Theorems. ? ┘

**Thm& Conj. Cor. (35.5.7.33) [Modified Bombieri-Vinogradov Theorem beyond  $1/2$ ].** During Zhang's proof of the twin prime conjecture, he proved a modified version (excluding  $x^\omega$ -large prime factors in  $N$ ) version beyond Bombieri-Vinogradov theorem for  $\theta = 1/2 + \omega$ . ┘

*Proof:*

□

**Thm. (35.5.7.34) [general Barban-Davenport-Halberstam Theorem].** If  $f : \mathbb{Z}_+ \rightarrow \mathbb{C}$  is an arithmetic function and  $X \in \mathbb{R}_+, \Delta \in (0, 1]$  s.t.

$$\text{Err}_f(N, a; X) \leq \left\| f \mathbf{1}_{[X]_+} \right\| X^{1/2} \Delta^9,$$

for any  $a, N \in \mathbb{Z}_+$  s.t.  $(a, N) = 1$ . Then we have

$$\sum_{q \in [Q]_+} \sum_{a \in (\mathbb{Z}/(q))^*} |D_f(q, a; X)|^2 \leq \|f \mathbf{1}_{[X]_+}\|^2 (\Delta X + Q) \log^2 Q$$

for any  $Q \in \mathbb{R}_+$ . ┘

*Proof:* By orthogonality of Dirichlet characters,

$$\sum_{a \in (\mathbb{Z}/(q))^*} |D_f(q, a; X)|^2 = \frac{1}{\phi(q)} \sum_{a \in (\mathbb{Z}/(q))^*} \left| \sum_{\chi \in \text{Diri}(q) \setminus \mathbf{1}_N} \overline{\chi(a)} \sum_{n \in [X]_+} f(n) \chi(n) \right|^2 = \frac{1}{\phi(q)} \sum_{\chi \in \text{Diri}(q) \setminus \mathbf{1}_N} \left| \sum_{n \in [X]_+} f(n) \chi(n) \right|^2.$$

So

$$\sum_{q \in [Q]_+} \sum_{a \in (\mathbb{Z}/(q))^*} |D_f(q, a; X)|^2 \leq \sum_{s \in [Q]_+} \frac{1}{\phi(s)} \sum_{1 < r \in [Q]_+} \frac{1}{\phi(r)} \sum_{\chi \in \text{Diri}'(r)} \left| \sum_{n \in [X]_+} f(n) \chi(n) \right|^2.$$

By the proof of (35.5.7.30), this is bounded by

$$\|f\|^2 \left( X \Delta^3 R^2 + Q + X^{1/2} + X R^{-1} \right) \log^2 Q$$

for any  $R \in \mathbb{R}_+$ . By taking  $R = \Delta^{-1}$ , we get the desired assertion. □

**Cor. (35.5.7.35)** [Barban[Bar66]/Davenport-Halberstam[D-H66]]. For any  $A \in \mathbb{R}_+$ , there exists  $C_A \in \mathbb{R}_+$  s.t.

$$\sum_{1 \leq N \leq X / \log^{2A+7} X} \sum_{a \in (\mathbb{Z}/(N))^*} (\mathcal{D}_\Lambda(N, a; X))^2 \leq C_A \frac{X^2}{\log^A X}.$$

And similar assertion holds for  $\Lambda$  replaced by  $\mu$ . ┘

*Proof:* This follows from the general Barban-Davenport-Halberstam theorem (35.5.7.34), where the conditions with  $\Delta = \log^{-A} X$  are satisfied by (35.5.7.18) and (35.1.3.17). □

**Thm. (35.5.7.36)** [Bombieri-Friedlander-Iwaniec[BFI86]]. For any  $A \in \mathbb{R}_+$ , there exists  $C_A \in \mathbb{R}_+$  and  $B_A \in \mathbb{R}_+$  s.t.

$$\sum_{1 \leq N \leq \sqrt{X} \log^A X} \sum_{a \in (\mathbb{Z}/(N))^*} |\mathcal{D}_\Lambda(N, a; X)| \leq C_A \frac{X}{\log^3 X} \cdot (\log_{(2)} X)^{B_A} \text{ (2.6.1.32)}.$$

*Proof:* ? ┘

### Titchmarsh Divisor Problem

**Thm. (35.5.7.37).** For  $X \in \mathbb{R}_+$ ,

$$\sum_{p \in \text{Prime} \leq X} \tau(p-1) \sim \frac{\zeta(2)\zeta(3)}{\zeta(6)} X.$$

*Proof:* Cf. [I-K04]P420. ? □

## Divisors in Arithmetic Progressions

### Green-Tao Theorem

**Thm. (35.5.7.38)[Green-Tao].** Prime contains arithmetic progressions of any length. In fact, if  $A \subset P$  is a subset with positive natural upper density, then  $A$  contains arithmetic progressions of any length.

┘

*Proof:* Cf.[Green-Tao]. Notice this is a consequence of the Erdős-Turán conjecture(37.6.2.8).  $\square$

### Sophie-Germain Primes

**Def. (35.5.7.39) [Sophie-Germain Primes].** A **Sophie-Germain prime** is a prime  $p \in \text{Prime}$  s.t.  $2p + 1$  is also a prime.  $\square$

**Conj. (35.5.7.40)[Sophie-Germain Primes Conjecture].** Let  $C_2$  be the twin-prime constant?, then

$$\#\{p \in \text{Prime} \mid p \leq m, 2p + 1 \in \text{Prime}\} \sim 2C_2 \frac{X}{\log X}, \quad X \rightarrow \infty.$$

┘

*Proof:*  $\square$

## 8 Special Factorizations

**Def. (35.5.8.1).** For  $Y, Y \in \mathbb{R}_+$ , let  $\pi(X|Y)$  be the cardinality of primes  $p$  s.t.  $p \leq X$ , and  $p - 1$  is free of prime divisors exceeding  $Y$ .  $\square$

**Def. (35.5.8.2).** Let  $E_\pi$  denote the supremum of numbers  $E \in (0, 1)$  s.t. there exists  $\gamma_1(E) \in \mathbb{R}_+$  s.t.  $\pi(X|X^{1-E}) \geq \gamma_1(E)\pi(X)$  for  $X \in \mathbb{R}_+$  sufficiently large.  $\square$

**Prop. (35.5.8.3).**  $E_\pi > 0$ , and if  $E \in (0, 1)$  has the property stated in(35.5.8.2), then so does  $E + \varepsilon$  for some  $\varepsilon \in \mathbb{R}_+$ .  $\square$

*Proof:* For the first assertion, Cf.[Erdos]?,  
For the second assertion, for  $X$  large,

$$\begin{aligned} \pi(X|X^{1-E'}) &\geq \pi(X|X^{1-E}) - \sum_{p \in \text{Prime}, x^{1-E'} \leq p < x^{1-E}} \pi(X; p, 1) \\ (35.5.8.2) &\geq \gamma_1(E)\pi(X) - \sum_{p \in \text{Prime}, x^{1-E'} \leq p < x^{1-E}} \frac{2X}{\varphi(p) \log(X/p)} \\ (35.5.7.14) &\geq \gamma_1(E) \frac{X}{\log X} - \sum_{p \in \text{Prime}, x^{1-E'} \leq p < x^{1-E}} \frac{2X}{E(p-1) \log X} \\ &= \frac{X}{\log X} \left( \gamma_1(E) - \frac{2}{E} \sum_{p \in \text{Prime}, x^{1-E'} \leq p < x^{1-E}} \frac{1}{p-1} \right) \\ (35.5.7.8) &= \frac{X}{\log X} \left( \gamma_1(E) - \frac{2}{E} \log \frac{1-E}{1-E'} + o(1) \right) \end{aligned}$$

This shows that when  $E'$  is sufficiently closed to  $E$ , when  $X$  is large,  $E'$  also satisfies the property stated in(35.5.8.2).  $\square$

**Conj. (35.5.8.4).**  $E_\pi = 1$ . ┘

*Proof:* □

**Prop. (35.5.8.5) [Friedlander].**  $1 - (2\sqrt{e})^{-1} \leq E_\pi$ . ┘

*Proof:* [FRIEDLANDER, Shifted primes without large prime factors]. □

**Prop. (35.5.8.6).**  $B_{\pi,1} \leq E_\pi$  (35.5.7.22). ┘

*Proof:* Cf. [AGP94]P720<sup>?</sup>. □

### Wieferich's Primes

**Def. (35.5.8.7) [Wieferich's Prime].** A **Wieferich's prime** is a prime  $p \in \mathbf{Prime}$  s.t.  $2^{p-1} \equiv 1 \pmod{p^2}$ . More generally, for  $a \in \mathbb{Z}_+$ , a **Wieferich's prime in base  $a$**  is a prime  $p \in \mathbf{Prime}$  s.t.  $a^{p-1} \equiv 1 \pmod{p^2}$ . ┘

**Thm. (35.5.8.8) [Wieferich's Criterion, Silverman].** If the abc conjecture is true, then for any  $\alpha \in \mathbb{Q}^* \setminus \{0, 1\}$ , there exists  $C(\alpha) \in \mathbb{R}_+$  s.t.

$$\{p \in \mathbf{Prime} : p < X, \alpha^{p-1} \not\equiv 1 \pmod{p^2}\} \geq C(\alpha) \log X$$

for  $X$  large. ┘

*Proof:* Let  $\alpha = a/b$ . We may assume that  $(a, b) = 1$  and  $a > b \geq 1$ . The strategy is use the cyclotomic polynomial to find the order of  $\alpha$  modulo  $p$ .

For  $n \in \mathbb{Z}_+$ , let  $U_n$  be the product of primes  $p \in \mathbf{Prime}$  s.t.  $v_p(\Phi_n(a, b)) = 1$ . And let  $\Phi_n(a, b) = U_n V_n$ . Then:

**Lemma (35.5.8.9).** Suppose  $p \in \mathbf{Prime}$ ,  $p \nmid abn$ ,  $p | U_n$ , then  $\text{ord}_p(\alpha) = n$ , and  $\alpha^{p-1} \not\equiv 1 \pmod{p^2}$ . ┘

*Proof:* By property of cyclotomic polynomials,  $\text{ord}_p(\alpha) = n$ . Also because  $v_p(U_n) = 1$ , we have  $a^n \not\equiv 1 \pmod{p^2}$ . Then  $a^{p-1} = (a^n)^{\frac{p-1}{n}}$  also  $\not\equiv 1 \pmod{p^2}$ . □

**Lemma (35.5.8.10).** For  $X \in \mathbb{R}_+$ ,

$$\#\{p \in \mathbf{Prime} : p < X, \alpha^{p-1} \not\equiv 1 \pmod{p^2}\} \geq \#\{n \in \mathbb{Z}_+ : n \leq \log_a X, U_n > abn\}.$$

┘

*Proof:* For  $n \in \mathbb{Z}_+ : n \leq \log_a X, U_n > abn$ , because  $U_n$  is square-free, we may choose  $\mathbf{p}_n | U_n$  that is coprime to  $abn$ . Then by (35.5.8.9),  $\alpha^{\mathbf{p}_n-1} \not\equiv 1 \pmod{\mathbf{p}_n^2}$ . Moreover,  $\mathbf{p}_n \leq U_n < a^n \leq X$ .

Moreover,  $\mathbf{p}_n$  are different for different  $n$ , because  $n = \text{ord}_{\mathbf{p}_n}(\alpha)$  (35.5.8.9). Thus the assertion follows. □

**Lemma (35.5.8.11).** If the abc-conjecture is true, then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C(\alpha) \in \mathbb{R}_+$  s.t.

$$V_n \leq C(\varepsilon) a^{\varepsilon n}.$$

┘

*Proof:* Define similarly  $u_n$  be the product of primes  $p \in \mathbf{Prime}$  s.t.  $v_p(a^n - b^n) = 1$ . And let  $a^n - b^n = u_n v_n$ , then clearly  $U_n | u_n$ . So it suffices to show for  $u_n$ .

Because  $a^n - b^n - u_n v_n = 0$ , by abc-conjecture, for any  $\varepsilon \in \mathbb{R}_+$ ,

$$a^n = \max\{a^n, b^n, u_n v_n\} \leq C(\varepsilon) \left( \prod_{p|abu_n v_n} p \right)^{1+\varepsilon} \leq C'(\varepsilon) (u_n \sqrt{v_n})^{1+\varepsilon} \leq C'(\varepsilon) (a^n / \sqrt{v_n})^{1+\varepsilon}.$$

Which is equivalent to that  $v_n \leq C(\varepsilon) a^{\varepsilon n}$  (by shrinking  $\varepsilon$ ).  $\square$

Now by (3.3.2.12) and (35.5.8.11),

$$U_n \geq |\Phi_n(a, b)| / |V_n| \geq e^{c\varphi(n)} / (c_1(\varepsilon) a^{\varepsilon n}).$$

Then if

$$c\varphi(n) - \varepsilon n \log(n) - c_2(\varepsilon) \geq \log n,$$

we will have  $U_n \geq abn$ . This holds if  $\varphi(n)$  is bigger than a fraction of  $n$ . In view of (35.1.2.12), this is true for a positive portion of  $n$ .

Then by (35.5.8.10),  $\#\{p \in \mathbf{Prime} : p < X, \alpha^{p-1} \not\equiv 1 \pmod{p^2}\}$  is bounded below by  $\Omega(\log X)$ .  $\square$

**Thm. (35.5.8.12) [Wieferich's Criterion for Elliptic Curves, Silverman].** If the abc conjecture is true, then for any  $E/\mathbb{Q}$  with  $j(E) \notin \{0, 1728\}$  and  $P \in E(\mathbb{Q})$  a point of infinite order, there exists  $C(E, P) \in \mathbb{R}_+$  s.t.

$$\{p \in \mathbf{Prime} : p < X, [N_p]P \not\equiv O \pmod{p^2}\} \geq C(E, P) \sqrt{\log X}$$

for  $X$  large, where  $N_p = \#\mathcal{E}(\mathbb{F}_p)$ .  $\lrcorner$

*Proof:* Cf. [Sil88]P228.  $\square$

### Carmichael Numbers

**Def. (35.5.8.13) [Carmichael Numbers].**  $n \in \mathbb{Z}_+$  is called a **Carmichael number** if  $n \notin \mathbf{Prime}$  and  $n | a^n - a$  for any  $a \in \mathbb{Z}$ . For  $X \in \mathbb{R}_+$ , let  $\text{Car}(X)$  denote the cardinality of Carmichael numbers not exceeding  $X$ .  $\lrcorner$

**Thm. (35.5.8.14) [Korselt].** For  $n \in \mathbb{Z}_+$ ,  $n | a^n - a$  for all  $a \in \mathbb{Z}$  iff  $n$  is square-free and  $p-1 | n-1$  for any  $p \in \mathbf{Prime}, p | n$ .  $\lrcorner$

*Proof:* Suppose  $n$  is a Carmichael number. Then  $n$  is square-free: For  $p | n$ , let  $n = p^k n'$ , then  $(1+p)^{p^k n'} \equiv 1 + p \pmod{p^k}$ , but  $p^{k+1} | (1+p)^{p^k n'} - 1$ , so  $p^k | p$ , and  $k = 1$ . And let  $g$  be a primitive root for  $p$ , then  $g^n \equiv g \pmod{n}$ , thus  $p-1 | n-1$ .  $\square$

The converse is obvious.  $\square$

**Cor. (35.5.8.15).** Any Carmichael number  $n \in \mathbb{Z}_+$  is odd, and has at least three prime divisors.  $\lrcorner$

*Proof:* If  $n$  is even, because  $n$  is square-free (35.5.8.14),  $n$  has an odd prime divisor  $p \in \mathbf{Prime}$ , so  $p-1 | n-1$ , contradiction. If  $n = pq$ , then  $p-1 | pq-1, q-1 | pq-1$ , thus  $p-1 | q-1 | p-1$ , and  $p = q$ , contradiction.  $\square$

**Thm. (35.5.8.16) [Alford-Granville-Pomerance].** For each  $E \in \mathcal{E}(35.5.8.2)$  and  $B \in \mathcal{B}(35.5.7.22)$ , then  $\text{Car}(X) \geq X^{EB}$  for  $X$  sufficiently large.  $\lrcorner$



*Proof:* Because of (35.5.8.3), it suffices to show that  $\text{Car}(X) \geq X^{EB-\varepsilon}$  for any  $\varepsilon \in \mathbb{R}_+$  and  $X$  sufficiently large. □

**Cor. (35.5.8.17)** [Alford-Granville-Pomerance/Harman].  $(0, 1 - (2\sqrt{e})^{-1}) \subset \mathcal{E}(35.5.8.5)$ , and  $(0, 5/12) \in \mathcal{B}$  by (35.5.7.23), so  $\text{Car}(X) > X^{2/7}$  for  $X$  sufficiently large. In fact, Harman proved that  $\text{Car}(X) > X^{1/3}$  for  $X$  sufficiently large. ┘

## 9 Prime Ideals

**Def. (35.5.9.1)** [Prime Ideal Counting Functions]. For  $F \in \mathbf{NField}$ , denote

$$\pi(X; F) = \#\{\mathfrak{p} \in \Sigma_F^{\text{fin}} : \|\mathfrak{p}\| \leq X\}.$$
┘

**Thm. (35.5.9.2)** [Prime Ideal Theorem, Landau]. For  $F \in \mathbf{NField}$ , there exists  $c_F \in \mathbb{R}_+$  s.t.

$$\pi(X; F) = \text{Li}(X) + O\left(\frac{X}{\exp(c_F \sqrt{\log X})}\right).$$
┘

*Proof:* □

## 10 Others

**Prop. (35.5.10.1)**. For any  $p \in \mathbf{Prime}_{\geq 5}$ , there exists  $P \in \mathbf{Prime}, \ell \in \mathbf{Prime}_{\geq 3}, m, r \in \mathbb{Z}_+$  s.t.  $P > p$ ,  $\ell^r \parallel (P - 1)$ ,  $\ell^r = 2m + 1$ , and

$$\frac{P}{p} \leq \frac{2m+1}{m+1} - \frac{m}{m+1} \cdot \frac{1}{p}.$$
┘

*Proof:* Cf. [Serre's Modularity Conjecture, the Level One Case]P582. □

**Conj. (35.5.10.2)** [Artin's Conjecture on Primitive Roots]. For any  $n \in \mathbb{Z}_+ \setminus (\mathbb{Z}_+)^2$ , there exists  $A(n) > 0.35 \in \mathbb{R}_+$  s.t.

$$\#\{p \in \mathbf{Prime} \mid p \leq X, \text{ord}_p(n) = p - 1\} \sim A(n) \frac{X}{\log X}, \quad X \rightarrow \infty$$
┘

*Proof:* □

**Prop. (35.5.10.3)** [Fouvry]. There exists  $C \in \mathbb{R}_+$  s.t. when  $X \in \mathbb{R}_+$  is large,

$$\#\{\ell \in \mathbf{Prime} \mid \ell \leq X, P_{\max}(\ell - 1) \geq \ell^{0.668}\} \geq C \frac{X}{\log X}.$$
┘

*Proof:* Cf.[Fouvry, Théorème de Brun-Titchmarsh; application au théorème de Fermat], and [Baker and Harman, The Brun-Titchmarsh Theorem on average].  $\square$

**Cor. (35.5.10.4).** When  $n \in \mathbb{Z}_+$  is large, there exists  $r \in \mathbb{Z}_+$  s.t.  $r \leq \log^3 n$  and  $\text{ord}_r(n) > \log^2 n$ .  $\lrcorner$

*Proof:* Notice by (35.5.10.3), for any  $\varepsilon \in \mathbb{R}_+$ , when  $n$  is large, there exists at least  $\Omega(\log^{3+\frac{\varepsilon}{2}} n)$  primes  $\ell \in \mathbf{Prime}$  s.t.  $P_{\max}(\ell - 1) \geq \ell^{2/3}$  and  $\log^2 n < \ell < \log^{3+\varepsilon} n$ . Then  $\text{ord}_\ell(n) \geq \log^2 n$  or  $\ell | n^k + 1$  for some  $k \leq \log^{1+\frac{\varepsilon}{3}} n$ . The latter case happens only for  $O(\log^{3+\frac{2\varepsilon}{3}} n)$  many  $\ell$ . So we can find some  $\ell < \log^{3+\varepsilon} n$  s.t.  $\text{ord}_\ell(n) \geq \log^2 n$ .  $\square$

## 35.6 Gaps between Primes

References are [I-K04, H-W08, Ten15].

**Notation (35.6.0.1).**

- Let  $F \in \mathbf{NField}$ .
- For  $n \in \mathbb{Z}_+$ , let  $p_n$  be the  $n$ -th smallest prime number.
- Use notations on Elementary Number Theory defined in [Elementary Number Theory](#).

┘

### 1 Small Gaps between Primes

References are [GPY09, Zha14, May15].

**Def. (35.6.1.1) [Admissible Sets].** An **admissible set** of numbers  $H = \{h_1, \dots, h_k\} \subset \mathbb{N}$  s.t. for any  $p \in \mathbf{Prime}$ , there exists  $a_p \in \mathbb{Z}$  s.t.  $a_p \not\equiv h_i \pmod{p}$  for any  $i$ .

┘

**Example (35.6.1.2) [Admissible Sets].**

- The set  $H = \{0, 2, 6, 8, 12\}$  is an admissible set [\(35.6.1.1\)](#) with cardinality 5.
- The set  $H = \{0, 10, 12, 24, 28, 30, 34, 42, 48, 52, 54, 64, 70, 72, 78, 82, 90, 94, 100, 112, 114, 118, 120, 124, 132, 138, 148, 154, 168, 174, 178, 180, 184, 190, 192, 202, 204, 208, 220, 222, 232, 234, 250, 252, 258, 262, 264, 268, 280, 288, 294, 300, 310, 322, 324, 328, 330, 334, 342, 352, 358, 360, 364, 372, 378, 384, 390, 394, 400, 402, 408, 412, 418, 420, 430, 432, 442, 444, 450, 454, 462, 468, 472, 478, 484, 490, 492, 498, 504, 510, 528, 532, 534, 538, 544, 558, 562, 570, 574, 580, 582, 588, 594, 598, 600\}$  is an admissible set [\(35.6.1.1\)](#) with cardinality 105.

┘

**Conj. (35.6.1.3) [Prime  $k$ -Tuples Conjecture, Hardy-Littlewood].** Let  $H = \{h_1, \dots, h_k\} \subset \mathbb{N}$  be an admissible set [\(35.6.1.1\)](#), and let  $\pi(X; H)$  denotes the number of integers  $n \in [X]_+$  s.t.  $\{n - h_1, \dots, n - h_k\} \subset \mathbf{Prime}$ , and for each  $p \in \mathbf{Prime}$ , let  $m(p)$  be the cardinality of the image of  $H$  in  $\mathbb{Z}/(p)$ , then we have

$$\pi(X; H) \sim B \frac{X}{\log^k X}, \text{ where } B = \prod_{p \in \mathbf{Prime}} \left(1 - \frac{m(p)}{p}\right) \left(1 - \frac{1}{p}\right)^{-k}.$$

In particular, there exists infinitely many  $n \in \mathbb{Z}_+$  s.t.  $\{n + h_1, \dots, n + h_k\} \subset \mathbf{Prime}$ .

┘

*Proof:*

□

**Conj. Cor. (35.6.1.4) [Density of Twin Primes].** If  $m \in 2\mathbb{Z}_+$ , then uniformly in  $X \in \mathbb{R}_+$ , we have

$$\#\{p \in [X]_+ \mid p, p + m \in \mathbf{Prime}\} \sim C \left( \prod_{p \in S(m) \setminus \{2\}} \frac{p-1}{p-2} \right) \frac{X}{\log^2 X},$$

where

$$C = 2 \prod_{p \in \mathbf{Prime}_{\geq 3}} \left(1 - \frac{1}{(p-1)^2}\right),$$

called the **twin prime constant**. This is compatible with the heuristic in [\(35.2.4.4\)](#).

┘

*Proof:*

□

**Thm& Conj.Cor.(35.6.1.5) [Selberg's Upper Bound on Prime  $k$ -Tuples].** Situation as in (35.6.1.3), we have

$$\pi(X; H) \leq 2^k k! B \frac{X}{\log^k X} \left( 1 + O_k \left( \frac{\log_{(2)} X}{\log X} \right) \right).$$

┘

*Proof:* Use Selberg's sieve (35.2.3.2). When  $p$  is sufficiently large, we have  $m(p) = k$ , so this is a  $k$ -dimensional sieve, and the sieve conditions (35.2.3.4) (35.2.3.5) are satisfied with

- $P = \prod_{p \in \text{Prime}_{\leq \sqrt{X}}} p$ .
- $g(p) = m(p)p^{-1}$ .
- $D = X / \log^{k-1} X$ .

(Notice the condition  $|r_d(\mathcal{A})| \leq g(d)d$  is only satisfied for  $d$  large, but the proof of (35.2.3.5) can be easily modified to go through). Thus the assertion follows. □

**Thm& Conj.Cor.(35.6.1.6) [Selberg's Upper Sieve for Twin Primes].** If  $m \in 2\mathbb{Z}_+$ , then uniformly in  $X \in \mathbb{R}_{\geq 3}$  we have

$$\#\{p \in [X]_+ | p, p+m \in \text{Prime}\} \leq C \left( \prod_{p \in S(m) \setminus \{2\}} \frac{p-1}{p-2} \right) \frac{X}{\log^2 X} \left( 4 + O \left( \frac{\log_{(2)} X}{\log X} \right) \right),$$

where  $C$  is the twin prime constant (35.6.1.4). ┘

*Proof:* Cf. [Sieve method, Ford] P50 ?.

□

**Remark (35.6.1.7).** Bombieri-Friedlander-Iwaniec showed that 4 can be changed to 3.5 but with error term non-uniform in  $k$ . And if we assume the Elliott-Halberstam conjecture, then we can change 4 to  $2 + o(1)$ . ┘

**Thm& Conj.Cor.(35.6.1.8).** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C(\varepsilon) \in \mathbb{R}_+$  s.t.

$$\#\{p \in \text{Prime}_{\leq X} | p+2 \text{ not divisible by primes } \leq X^\varepsilon\} \leq C(\varepsilon) \frac{X}{(\log X)^2}$$

┘

*Proof:* ?

□

**Thm& Conj.Cor.(35.6.1.9) [Chen [Che73]].** There exists  $C \in \mathbb{R}_+$  s.t. for  $X \in \mathbb{R}_{\geq 3}$ ,

$$\#\{p \in \text{Prime}_{\leq X} | p+2 \in \text{Prime} \text{ or } p+2 = p_1 p_2, p_1, p_2 \in \text{Prime}_{> X^{3/11}}\} \geq C \frac{X}{(\log X)^2}$$

┘

*Proof:* ?

□

**Prop. (35.6.1.10) [Sieberg's Sieve Method].** Given an admissible set (35.6.1.2)  $H = \{h_1, \dots, h_k\} \subset \mathbb{N}$ , for any  $N \in \mathbb{Z}_+$  and  $\rho \in \mathbb{R}_+$ , consider the sum

$$S(N, \rho) = \sum_{N \leq n < 2N} \left( \sum_{i=1}^k \mathbf{1}_{\text{Prime}}(n + h_i) - \rho \right) \omega(n),$$

where  $\omega_n$  are chosen non-negative weights.

If we can show that  $S(N, \rho) > 0$  for any  $N$ , then there exists infinitely many  $n$  s.t. there are at least  $\lfloor \rho + 1 \rfloor$  many primes in  $n + H$ .

This procedure only works if we can find the weights  $\omega_n$  that is suitably concentrated on integers  $n$  s.t.  $n + h_i \in \text{Prime}$ , but at the same time is simple enough that we can calculate this expectation unconditionally.  $\lrcorner$

**Prop. (35.6.1.11) [GPY's Sieve Weights, Goldston-Pintz-Yildirim [GPY09]].** Situation as in (35.6.1.10), GPY's sieve weights are chosen s.t.

$$\omega(n) = \left( \sum_{d \mid \prod_{i=1}^k (n+h_i), d < R} \mu(d) \left( \log \frac{R}{d} \right)^{k + \lfloor \sqrt{k} \rfloor} \right)^2.$$

where  $R$  is usually chosen to be  $R = N^b$  for  $b \in (0, \frac{1}{2})$ .  $\lrcorner$

**Prop. (35.6.1.12) [Maynard's Sieve Weights].** Situation as in (35.6.1.10), Maynard's sieve weights are chosen s.t.

$$\omega(n) = \left( \sum_{d_i \mid n+h_i} \lambda_{d_1, \dots, d_k} \right)^2,$$

and  $\lambda_{d_1, \dots, d_k}$  are chosen real numbers.

Then we choose  $\textcolor{red}{?}$ .  $\lrcorner$

*Proof:* Cf. [Small Gaps between Primes]P388.  $\textcolor{red}{?}$   $\square$

**Prop. (35.6.1.13) [Maynard [May15]].** Suppose the primes have level of distribution  $\theta \in \mathbb{R}_+$  (35.5.7.27).

Let  $H = \{h_1, \dots, h_k\}$  be an admissible set of cardinality  $k$ . Let  $I_k(F)$  and  $J_k^{(m)}(F)$  be defined as in (35.6.1.12) and let  $\mathcal{S}_k$  be the set of continuous functions  $F$  supported on  $\{(x_1, \dots, x_k) \in [0, 1]^k, \sum_i x_i \leq 1\}$  with  $I_k(F) \neq 0, J_k^{(m)}(F) \neq 0$  for any  $m$ . Define

$$M_k = \sup_{F \in \mathcal{S}_k} \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}, \quad r_k = \lfloor \frac{\theta M_k}{2} \rfloor,$$

then there exists infinitely many  $n \in \mathbb{Z}_+$  s.t. at least  $r_k$  of numbers  $\{n + h_i\}$  are primes. In particular,

$$\varliminf_{n \rightarrow \infty} (p_{n+r_k-1} - p_n) \leq \max_{1 \leq i, j \leq k} (h_i - h_j).$$

$\lrcorner$

*Proof:* Cf. [Small Gaps between Primes]P389.  $\textcolor{red}{?}$   $\square$

**Lemma (35.6.1.14) [Maynard [May15]].** Let  $k \in \mathbb{Z}_+$  and  $M_k$  defined as in (35.6.1.13), then

- $M_5 > 2$ .
- $M_{105} > 4$ .
- If  $k$  is sufficiently large, then  $M_k > \log k - 2 \log \log k - 2$ .

┘

*Proof:* Cf.[Small Gaps between Primes]P390.?

□

**Thm& Conj.Cor.(35.6.1.15)** [A Positive Portion of Prime  $k$ -Tuple Conjectures are True, Maynard[May15]]. For any  $m \in \mathbb{Z}_+$ , there exists  $C(m) \in \mathbb{R}_+$  s.t. when  $r$  is large, for any set  $A = \{a_1, \dots, a_r\} \subset \mathbb{N}$  of cardinality  $r$ , at least a  $C(m)$ -portion of  $m$ -subsets of  $A$  are admissible and satisfies the prime  $k$ -tuples conjecture(35.6.1.3). ┘

*Proof:* Use(35.6.1.13) and a counting argument: Let  $k = \lceil 1000m^2e^{4m} \rceil$ , then each admissible subset of cardinality  $k$  contains a  $m$ -subset satisfying the prime  $k$ -tuple conjecture. Notice for any set  $A$  of cardinality  $r$ , we can choose an admissible subset  $A'$  of cardinality

$$s = r \prod_{p < k} \left(1 - \frac{1}{p}\right).$$

So by counting argument, there exists at least

$$\frac{\binom{s}{k}}{\binom{s-m}{k-m}} \gg_m s^m \gg_m r^m$$

many  $m$ -subsets of  $A'$  of cardinality  $m$ . And there are  $\binom{r}{m} \leq r^m$  many  $m$ -subsets of  $A$ , so the assertion follows. □

**Thm. (35.6.1.16)**[Goldston-Pintz-Yildirim[GPY09]/Zhang[Zha14]/Maynard[May15]/Polymath[Pol14]].

$$\varliminf_{n \rightarrow \infty} (\mathbf{p}_{n+1} - \mathbf{p}_n) \leq 246.$$

┘

*Proof:* ?

We only prove  $\varliminf_{n \rightarrow \infty} (\mathbf{p}_{n+1} - \mathbf{p}_n) \leq 600$  using(35.6.1.13). By Bombieri-Vinogradov(35.5.7.31),  $\theta$  can be  $1/2 - \varepsilon$  for any  $\varepsilon \in \mathbb{R}_+$ . So by(35.6.1.14), we can choose  $\theta M_k/2 > 1$ . Then by(35.6.1.13),

$$\varliminf_{n \rightarrow \infty} (\mathbf{p}_{n+1} - \mathbf{p}_n) \leq \max_{1 \leq i, j \leq 105} (h_i - h_j),$$

where  $H = \{h_1, \dots, h_{105}\}$  is any admissible set(35.6.1.1). By(35.6.1.2), we can choose  $H$  s.t.  $\max_{1 \leq i, j \leq 105} (h_i - h_j) = 600$ . Thus the assertion follows. □

**Thm. (35.6.1.17)**[Maynard[May15]/Polymath[Pol14a]]. There exists  $C \in \mathbb{R}_+$  s.t. for any  $m \in \mathbb{Z}_+$ ,

$$\varliminf_{n \rightarrow \infty} (\mathbf{p}_{n+m} - \mathbf{p}_n) \leq Ce^{3.815m}.$$

┘

*Proof:* ?

We only prove  $\lim_{n \rightarrow \infty} (\mathfrak{p}_{n+m} - \mathfrak{p}_n) \leq Cm^3 e^{4m}$ , using the same proof as that of (35.6.1.16). By Bombieri-Vinogradov (35.5.7.31) and (35.6.1.14),

$$\frac{\theta M_k}{2} \geq \left(\frac{1}{4} - \varepsilon\right)(\log k - 2 \log_{(2)}(k) - 2) > m$$

for  $k \geq 1000m^2 e^{4m}$ .

Take  $X = \lceil 1000m^2 e^{4m} \rceil$  and take the admissible set  $H = \{\mathfrak{p}_{\pi(X)+1}, \dots, \mathfrak{p}_{\pi(X)+X}\}$ , then this set has diameter  $\mathfrak{p}_{\pi(X)+X} - \mathfrak{p}_{\pi(X)+1} \ll X \log X \ll m^3 e^{4m}$  (35.5.7.13). Thus by (35.6.1.13), the assertion follows.  $\square$

**Thm& Conj.Cor. (35.6.1.18)** [Maynard[May15]/Polymath[Pol14]]. If the Elliott-Halberstam conjecture (35.5.7.27) holds, then

$$\lim_{n \rightarrow \infty} (\mathfrak{p}_{n+1} - \mathfrak{p}_n) \leq 12.$$

$$\lim_{n \rightarrow \infty} (\mathfrak{p}_{n+2} - \mathfrak{p}_n) \leq 600.$$

And assuming the generalized Elliott-Halberstam conjecture ?, we have

$$\lim_{n \rightarrow \infty} (\mathfrak{p}_{n+1} - \mathfrak{p}_n) \leq 6.$$

which due to the parity phenomenon seems to be the best result that can be obtained using a first moment method.  $\lrcorner$

*Proof:* The proof is the same as that of (35.6.1.16).

For the last assertion ? .  $\square$

## 2 Large Gaps between Primes

**Thm. (35.6.2.1)** [Chebyshev]. For any  $n \in \mathbb{Z}_+$ , there is always a prime  $p \in \text{Prime}$  s.t.  $n < p \leq 2n$ .  $\lrcorner$

*Proof:* ?  $\square$

**Conj. (35.6.2.2)** [Cramér].

$$\overline{\lim}_{n \rightarrow \infty} \frac{\mathfrak{p}_{n+1} - \mathfrak{p}_n}{\log^2 \mathfrak{p}_n} = 1.$$

In particular, the largest gap between consecutive primes  $\leq X$  is  $\Theta((\log X)^2)$ .  $\lrcorner$

*Proof:*  $\square$

**Conj.Cor. (35.6.2.3)** [Oppermann]. For any  $n \in \mathbb{Z}_{\geq 2}$ ,

$$\pi(n^2 - n) < \pi(n^2) < \pi(n^2 + n).$$

$\lrcorner$

**Thm& Conj.Cor. (35.6.2.4)** [Bounds on Gaps between Primes, Montgomery/Huxley/Baker-Harman-Pintz]. There exists an effective constant  $x_0 \in \mathbb{R}_+$  s.t. for any  $X \in \mathbb{R}_{>x_0}$ ,  $\text{Prime} \cap [X - X^{0.525}, X] \neq \emptyset$ .

In particular,  $\mathfrak{p}_{n+1} - \mathfrak{p}_n \ll \mathfrak{p}_{n+1}^{0.525}$ .  $\lrcorner$

*Proof:* Cf.[The difference between consecutive primes, 2].? □

**Thm& Conj.Cor.(35.6.2.5)** [Large Gaps between Primes, Backlund/Brauer-Zeitz/Westzynthius/Erdős/Rankin/Schönhage/-Maier-Pomerance/Pintz/Ford-Green-Konyagin-Maynard-Tao]. There exist  $C \in \mathbb{R}_+$  s.t. when  $X \in \mathbb{R}_+$  is large,

$$\max_{p_n \leq X} (p_{n+1} - p_n) \geq C \frac{\log X \log_{(2)}(X) \log_{(4)}(X)}{\log_{(3)}(X)}.$$

┘

*Proof:* Cf.[Long Gaps between Primes, 2018]. □



## 35.7 Circle Method(Hardy-Ramanujan)

References are [H-L23, I-K04, Nat96] and <https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method-and-vinogradovs-theorem/#saw>.

**Remark (35.7.0.1) [Equidistribution Principle].** An exponential sum  $\sum_n e^{2\pi i f(n)}$  or a bilinear sum  $\sum_m \sum_n a_m b_n e^{2\pi i f(m,n)}$  involving a structured phase function  $f$  should exhibit non-trivial cancellation, unless there are obvious algebraic reason that such a cancellation cannot occur.

There are rigorous version called Ratner's theorem, Cf. <https://terrytao.wordpress.com/2007/09/29/ratners-theorems/>. ?

**Prop. (35.7.0.2).** For  $f, g, h : \mathbb{Z}_+ \rightarrow \mathbb{C}$  and  $\theta \in \mathbb{R}$ , denote

$$\mathcal{F}f(\theta) = \sum_{n \in \mathbb{Z}_+} f(n) e^{-2\pi i n \theta}.$$

Then

$$\|f\|^2 = \int_{\mathbb{R}/\mathbb{Z}} |\mathcal{F}f(\theta)|^2 d\theta.$$

And for any  $N \in \mathbb{Z}_+$ , we have

$$\sum_{a,b,c \in \mathbb{Z}_+, a+b+c=N} f(a)g(b)h(c) = \int_{\mathbb{R}/\mathbb{Z}} \mathcal{F}f(\theta) \mathcal{F}g(\theta) \mathcal{F}h(\theta) e^{2\pi i N \theta} d\theta.$$

*Proof:*

**Cor. (35.7.0.3).** Situation as in (35.7.0.2), we have

$$\left| \sum_{a,b,c \in \mathbb{Z}_+, a+b+c=N} f(a)g(b)h(c) \right| \leq \|f\|^2 \|g\|^2 \sup_{\theta} |\mathcal{F}h(\theta)|.$$

**Remark (35.7.0.4).** Usually one estimates the major arc using Siegel-Walfisz type estimates, and estimates the minor arc by turning it into a bilinear sum.

### 1 Exponential Sums over Primes(Vinogradov)

**Def. (35.7.1.1) [W-Trick].** For  $N \in \mathbb{Z}_+$ ,

$$\Lambda(N) \approx \frac{W}{\phi(W)} \mathbf{1}_{(n,W)=1} = \sum_{q|W} \sum_{a \in (\mathbb{Z}/(q))^*} \frac{\mu(q)}{\phi(q)} e^{2\pi i \frac{an}{q}}, \text{ where } W = \prod_{p \in \text{Prime} \cap [w(N)]_+} p,$$

where  $w(N)$  is a slowly-growing function.

This trick is related to Green-Tao's work relating linear patterns in **Prime**. ?

**Minor Arcs**

**Prop. (35.7.1.2) [Exponential Sums for Minor Arcs].** If  $q \in \mathbb{Z}_+, a \in \mathbb{Z}$  s.t.  $(a, q) = 1$  and  $\alpha \in \mathbb{R}_+$  s.t.  $\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{q^2}$ , then

- For any  $M \in \mathbb{Z}_+$  and  $X \in \mathbb{R}_+$ , we have

$$\sum_{m \in [M]_+} \left| \sum_{n \in [X/m]_+} e^{2\pi i \alpha m n} \right| \leq 6(M + Xq^{-1} + 2q) \log(3qM).$$

- For any sequence of intervals  $(I_m)_{m \in \mathbb{Z}} \subset \mathbb{Z}_+$  and  $M, N \in \mathbb{Z}_+$ , we have

$$\sum_{|m| \leq M} \left| \sum_{n \in [N]_+, n \in I_m} e^{2\pi i \alpha m n} \right| \leq 4(MNq^{-1} + M + N + 2q) \log(3q).$$

┘

*Proof:* If  $q = 1$  the assertion clearly holds. So we assume that  $q \geq 2$ .

1: Using (35.3.3.1), we have

$$\sum_{|m| \leq M} \left| \sum_{n \in [X/m]_+} e^{2\pi i \alpha m n} \right| \leq \sum_{|m| \leq M} \min \left( \frac{X}{m}, \frac{1}{2\{\{\alpha m\}\}} \right).$$

Notice that for any  $\lfloor q/2 \rfloor$ -many consecutive integers  $\{m, \dots, m + \lfloor q/2 \rfloor\}$ ,  $\{\{\alpha m\}\}$  are all distinct and  $\frac{1}{2q}$ -spaced, so

$$\sum_{n \in [M]_+} \min \left( \frac{X}{m}, \frac{1}{2\{\{\alpha m\}\}} \right) \leq (1 + 3Mq^{-1}) \sum_{l \in [\frac{q}{2}]_+} ql^{-1} + 3Xq^{-1} \sum_{l \in [\frac{2M}{q}]_+} l^{-1} \leq 6(M + Xq^{-1} + 2q) \log(3qM).$$

2; Using (35.3.3.1), we have

$$\sum_{|m| \leq M} \left| \sum_{n \in [N]_+, n \leq x(m)} e^{2\pi i \alpha m n} \right| \leq \sum_{|m| \leq M} \min \left( N, \frac{1}{2\{\{\alpha m\}\}} \right).$$

Notice that for any  $\lfloor q/2 \rfloor$ -many consecutive integers  $\{m, \dots, m + \lfloor q/2 \rfloor\}$ ,  $\{\alpha m\}$  are all distinct and  $\frac{1}{2q}$ -spaced. so

$$\sum_{|m| \leq M} \min \left( N, \frac{1}{2\{\{\alpha m\}\}} \right) \leq (1 + 6Mq^{-1})(N + \sum_{l \in [q]_+} ql^{-1}) \leq 4(MNq^{-1} + M + N + q) \log(3q)$$

□

**Thm. (35.7.1.3) [Bilinear Forms for Minor Arcs (Type II Estimates)].** If  $q \in \mathbb{Z}_+, a \in \mathbb{Z}$  s.t.  $(a, q) = 1$  and  $\theta \in \mathbb{R}_+$  s.t.  $\left| \theta - \frac{a}{q} \right| \leq \frac{1}{q^2}$ , then for any  $M, N \in \mathbb{Z}_+, (\alpha_m)_{m \in [M]_+}, (\beta_n)_{n \in [N]_+}$  and any interval  $I \subset [1, MN]$ , we have

$$\sum_{mn \in I} \alpha_m \beta_n e^{2\pi i m n} \ll \|\alpha\| \|\beta\| (MNq^{-1} + M + N + 2q)^{1/2} \log^{1/2}(3q)$$

┘

*Proof:* By Cauchy's inequality,

$$\begin{aligned}
 \left| \sum_{mn \in I} \alpha_m \beta_n e^{2\pi i mn} \right|^2 &\leq \|\alpha\|^2 \sum_m \left| \sum_{mn \in I} \beta_n e^{2\pi i mn} \right|^2 \\
 &= \|\alpha\|^2 \sum_{n_1, n_2} \left| \beta_{n_1} \beta_{n_2} \sum_{mn_1 \in I, mn_2 \in I} e^{2\pi i (n_1 - n_2)m} \right| \\
 &\leq \|\alpha\|^2 \sum_{n_1} |\beta_{n_1}|^2 \sum_{n_2} \left| \sum_{mn_1 \in I, mn_2 \in I} e^{2\pi i (n_1 - n_2)m} \right| \\
 (35.7.1.2) \quad &\ll \|\alpha\|^2 \|\beta\|^2 (MNq^{-1} + M + N + 2q) \log(3q).
 \end{aligned}$$

□

**Cor. (35.7.1.4) [Bilinear Forms for Minor Arcs].** Let  $q \in \mathbb{Z}_+$ ,  $a \in \mathbb{Z}$  s.t.  $(a, q) = 1$  and  $\alpha \in \mathbb{R}_+$  s.t.  $\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{q^2}$ . Suppose  $M, N \in \mathbb{Z}_+$ ,  $(\alpha_m)_{m \in \mathbb{Z}_+}, (\beta_n)_{n \in \mathbb{Z}_+}$  satisfies:

$$\|\alpha \mathbf{1}_X\| \leq X\Delta, \quad \|\beta \mathbf{1}_X\| \leq X\Delta,$$

for any  $X$ . Then we have

$$\sum_{\substack{m, n \in \mathbb{Z}_+ \\ m > M, n > N, mn \leq X}} \alpha_m \beta_n e^{2\pi i \alpha mn} \ll \left( \frac{X}{\sqrt{M}} + \frac{X}{\sqrt{N}} + \frac{X}{\sqrt{q}} + \sqrt{q} \right) \log(3q) \log^2 X \Delta.$$

for any  $X \in \mathbb{R}_+$ . ┘

*Proof:* We divide  $[M, X] \times [N, X]$  into dyadic intervals  $[2^{i-1}, 2^i] \times [2^{j-1}, 2^j]$ , where  $2^i \geq M, 2^j \geq N$  and  $2^{i+j} \leq 4X$ . Then on each of these, the contribution is bounded by (35.7.1.3) by

$$X^{1/2} \left( \frac{X}{q} + \frac{X}{2^i} + \frac{X}{2^j} + 2q \right)^{1/2} \log(3q) \Delta.$$

Summing them together, the assertion follows. □

**Thm. (35.7.1.5) [Exponential Sums over Primes for the Minor Arc, Vinogradov [Vin37]].** If  $q \in \mathbb{Z}_+, a \in \mathbb{Z}$  s.t.  $(a, q) = 1$  and  $\alpha \in \mathbb{R}_+$  s.t.  $\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{q^2}$ , then there exists  $C_\alpha \in \mathbb{R}_+$  s.t. for  $X \in \mathbb{R}_+$ , we have

$$\begin{aligned}
 \left| \mathcal{E}_\Lambda(\alpha; X) \triangleq \sum_{n \in [X]_+} e^{2\pi i \alpha n} \Lambda(n) \right| &\leq C_\alpha \left( q + q^{1/2} X^{1/2} + q^{-1/2} X + X^{4/5} \right) \log(3qX) \log^2 X, \\
 \left| \mathcal{E}(\alpha; X) \triangleq \sum_{p \in \text{Prime} \cap [X]_+} e^{2\pi i \alpha p} \right| &\leq C_\alpha \left( q + q^{1/2} X^{1/2} + q^{-1/2} X + \frac{X}{e^{1/2} \sqrt{X}} \right), \\
 \left| \mathcal{E}_\mu(\alpha; X) \triangleq \sum_{n \in [X]_+} \mu(n) e^{2\pi i \alpha n} \right| &\leq C_\alpha \left( \frac{X}{\sqrt{q}} + \frac{X}{\log^A X} + q \right) \log(3qX) \log^3 X.
 \end{aligned}$$

┘

*Proof:* For  $\Lambda(n)$ , by Vaughan's identity (35.2.4.8), for  $M = X^{2/5}$ , we have

$$\mathcal{E}_\Lambda(\alpha; X) = \sum_{\substack{l, m \in \mathbb{Z}_+ \\ lm \leq X \\ m \leq M}} \mu(m) \log(l) e^{2\pi i \alpha lm} - \sum_{\substack{l, m, n \in \mathbb{Z}_+ \\ lmn \leq X \\ m, n \leq M}} \mu(m) \Lambda(n) e^{2\pi i \alpha lmn} + \sum_{\substack{l, m, n \in \mathbb{Z}_+ \\ lmn \leq X \\ m, n > M}} \mu(m) \Lambda(n) e^{2\pi i \alpha lmn} + O(M).$$

By (35.7.1.2), the first term is bounded by

$$\log X \sum_{m \in [M]_+} \left| \sum_{mn \leq X} e^{2\pi i \alpha mn} \right| \ll (Xq^{-1} + X^{2/5} + q) \log(3qX) \log X.$$

Similarly by (35.7.1.2), the second term is bounded by

$$\sum_{\substack{l, m, n \in \mathbb{Z}_+ \\ mk \leq X \\ m \leq M}} \mu(m) \left( \sum_{n|k, n \leq M} \Lambda(n) \right) e^{2\pi i \alpha mk} \ll \log X \sum_{m \in [M]_+} \left| \sum_{mk \leq X} e^{2\pi i \alpha mk} \right| \ll (Xq^{-1} + X^{2/5} + q) \log(3qX) \log X.$$

And by (35.7.1.4), the third term is bounded by

$$\sum_{\substack{m, k \in \mathbb{Z}_+ \\ m, k > M, mk \leq X}} \mu(m) \left( \sum_{n|k, n \geq M} \Lambda(n) \right) e^{2\pi i \alpha mk} \ll (X^{3/5} + Xq^{-1} + q)^{1/2} X^{1/2} \log^3 X.$$

Thus the assertion follows.

For  $\mu(n)$  the proof is similar (Cf. <https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method-and-vinogradovs-theorem/#star>).

Using Vaughan's identity (35.2.4.8), for  $M = \log^{2A} X$ , we have

$$\mathcal{E}_\Lambda(\alpha; X) = - \sum_{\substack{l, m, n \in \mathbb{Z}_+ \\ lmn \leq X \\ m, n \leq M}} \mu(m) \mu(n) e^{2\pi i \alpha lmn} + \sum_{\substack{l, m, n \in \mathbb{Z}_+ \\ lmn \leq X \\ m, n > M}} \mu(m) \mu(n) e^{2\pi i \alpha lmn} + O(M).$$

By (35.7.1.2), the first term is bounded by

$$\sum_{k \in [M^2]_+} \left( \sum_{m, n \in [M]_+, mn=k} \mu(m) \mu(n) \right) \sum_{l \in [X/k]_+} e^{2\pi i \alpha kl} \ll 6(\log X + Xq^{-1} + 2q) \log(3q \log X) \log X$$

And by (35.7.1.4), the second term is bounded by

$$\left( \frac{X}{\log^A X} + \frac{X}{\sqrt{q}} + \sqrt{q} \right) \log(3q) \log^{3/2} X.$$

For  $\mathcal{E}(\alpha; X)$ , use Abel summation ?.

□

**Thm. (35.7.1.6) [Minor Arcs Exceptional Sums are Small].** Let  $X \in \mathbb{R}_{\geq 2}$ ,  $\delta \in \mathbb{R}$ ,  $\theta \in \mathbb{S}^1$  and  $I \subset [1, X]$  is an interval s.t.

$$\left| \sum_{n \in \mathbb{Z}_+ \cap I} \Lambda(n) e^{2\pi i \theta n} \right| \geq \delta X$$

or

$$\left| \sum_{n \in \mathbb{Z}_+ \cap I} \mu(n) e^{2\pi i \theta n} \right| \geq \delta X,$$

or

$$\left| \sum_{p \in \text{Prime} \cap I} e^{2\pi i \theta p} \right| \geq \delta \frac{X}{\log X},$$

then either  $X \ll \delta^{-O(1)}$ , or there exists  $q \in [\delta^{-2} \log^{O(1)} X]_+$  s.t.  $\{\{q\theta\}\} \leq \delta^{-5} \frac{\log^{O(1)} X}{X}$ .  $\lrcorner$

*Proof:* ? Cf. <https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method-prop18>.  $\square$

### Major Arcs

**Thm. (35.7.1.7) [Major Arcs for  $\mu$ ].** For  $X \in \mathbb{R}_{\geq 2}$ ,  $\theta \in \mathbb{S}^1$ ,  $q \in \mathbb{Z}_+$  and any interval  $I \subset [1, X]$ , we have

$$\sum_{n \in \mathbb{Z}_+ \cap I} \mu(n) e^{2\pi i \theta n} \ll_A (q + X \{\{q\theta\}\}) \frac{X}{\log^A X}$$

for any  $A \in \mathbb{R}_+$ .  $\lrcorner$

*Proof:* It follows from Siegel-Walfisz (35.5.7.18) that for any  $a \in \mathbb{Z}$  we have

$$\sum_{n \in \mathbb{Z}_+ \cap I} \mu(n) e^{2\pi i \frac{an}{q}} \ll_A q \frac{X}{\log^A X}.$$

Suppose  $|q\theta - a| = \{\{q\theta\}\} = \alpha$ , then it follows from Abel summation that

$$\left| \sum_{n \in \mathbb{Z}_+ \cap I} \mu(n) e^{2\pi i \theta n} \right| \ll_A q \frac{X}{\log^A X} \left( 1 + 2\pi \frac{\alpha}{q} X \right).$$

$\square$

**Cor. (35.7.1.8) [Davenport(37)].** For any  $\theta \in \mathbb{S}^1$ ,  $X \in \mathbb{R}_{\geq 2}$  and any  $A \in \mathbb{R}_+$ , we have

$$\mathcal{E}_\mu(\theta; X) \ll_A \frac{X}{\log^A X} \quad (35.7.1.5).$$

$\lrcorner$

*Proof:* For  $Q \in \mathbb{Z}_+$ , there exists  $q \in [Q]_+$  and  $a \in \mathbb{Z}_+$  s.t.  $(a, q) \neq 1$  s.t.  $\left| \theta - \frac{a}{q} \right| \leq \frac{1}{qQ}$ . We use (35.7.1.7) when  $q \leq X/Q$  and (35.7.1.5) when  $X/Q < q \leq Q$ , to get

$$\mathcal{E}_\mu(\theta; X) \ll Q^{-1} X^2 \log^{-5A} X + \frac{X}{\log^A X} + Q^{1/2} X^{1/2} \log^3 X + Q^{1/2} \log^3 X.$$

Then by choosing  $Q = X \log^{-4A} X$ , the assertion follows.

Cf. [Davenport, some infinite series involving arithmetical functions, 2] or <https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method-and-vinogradovs-theorem-fab>.  $\square$

**Thm. (35.7.1.9) [Major Arcs for  $\Lambda$ ].** For any interval  $I \subset [1, X]$ ,  $\theta \in \mathbb{S}^1$  and  $R \in \mathbb{R}_+$ , if we denote

$$\Lambda_R^\sharp(n) = - \sum_{d \in [R]_+, d|n} \mu(d) \log(d), \quad \Lambda_R^\flat = \Lambda - \Lambda_R^\sharp,$$

Then for any  $A \in \mathbb{R}_+$ , we have

$$\sum_{n \in I} \Lambda_R^\flat(n) e^{2\pi i \theta n} \ll_A \frac{X \log X}{\log^A R}.$$

The same holds for  $\Lambda_R^\sharp$  replaced by

$$\Lambda_R(n) = \sum_{d \in [R]_+, d|n} \mu(d) \log \left( \frac{R}{n} \right)$$

or more generally

$$\Lambda_{R,\eta}(n) = \log R \sum_{d|n} \mu(d) \eta \left( \frac{\log d}{\log R} \right)$$

where  $\eta : \mathbb{R} \rightarrow \mathbb{R}$  is a smooth function s.t.  $\eta(t) = 1 - t$  for  $t \leq 1/2$  or  $t \geq 1$ . ┘

*Proof:* The LHS equals

$$- \sum_{m \in [X/R]_+} \sum_{d > R, dm \in I} \mu(d) \log(d) e^{2\pi i m d \theta}.$$

It follows from Davenport's theorem (35.7.1.8) and Abel summation that

$$\sum_{n \in [X]_+} \mu(n) \log(n) e^{2\pi i \theta n} X \ll_A \frac{X}{\log^A X}$$

uniformly in  $\theta$ . Thus

$$\begin{aligned} - \sum_{m \in [X/R]_+} \sum_{d > R, dm \in I} \mu(d) \log(d) e^{2\pi i m d \theta} &\ll_A \sum_{m \in [X/R]_+} X m^{-1} \log^{-A}(X m^{-1}) \\ &\ll_A X \log^{-A} R \sum_{m \in [X/R]_+} m^{-1} \\ &\ll_A \frac{X \log X}{\log^A R}. \end{aligned}$$

For  $\Lambda_{R,\eta}$ , Cf. [https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-meth?](https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method/)  
#leth? □

**Cor. (35.7.1.10) [Bounding  $\Lambda_R^\sharp$ ].** Situation as in (35.7.1.9), we also have

$$|\Lambda_R^\sharp(n)| \leq \tau(n) \log R.$$

In particular, for  $X \in \mathbb{R}_+$ , we have

$$\sum_{n \in [X]_+} \Lambda_R^\sharp(n)^2 \ll \log^2 R X \log^3 X, \quad \sum_{n \in [X]_+} \Lambda(n)^2 \ll X \log^5 X$$

by (35.1.3.12). ┘

**Thm.(35.7.1.11) [Another Major Arc Approximation for  $\Lambda$ ].** Let  $X \in \mathbb{R}_{\geq 2}$  and an interval  $I \subset [1, X]$ . For any  $\theta \in \mathbb{S}^1$  of the form  $\theta = \frac{a}{q} + \frac{\xi}{q}$  where  $a \in \mathbb{Z}_+$  and  $(a, q) = 1$ . Then for any  $A \in \mathbb{R}_+$ , we have

$$\sum_{n \in I} \Lambda(n) e^{2\pi i \theta n} - \frac{\mu(q)}{\phi(q)} \int_I e^{2\pi i \frac{\xi t}{q}} dt \ll_A (q + X|\xi|) \frac{X}{\log^A X}.$$

┘

*Proof:* Cf. <https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method-#lethP24>. □

**Thm.(35.7.1.12).** For  $X \in \mathbb{R}_+$ ,  $\theta \in \mathbb{S}^1 \setminus \{1\}$  and any  $A \in \mathbb{R}_+$ , we have

$$\sum_{n \in [X]_+} e^{2\pi i \theta \sqrt{n}} \Lambda(n) \ll_{\theta} X^{5/6} \log^4 X.$$

┘

*Proof:* ? Use the same method as in (35.7.1.8). □

## 2 Goldbach's Problems

References are [H-L23, Nat96, I-K04] and <https://terrytao.wordpress.com/2015/03/30/254a-notes-8-the-hardy-littlewood-circle-method-and-vinogradovs-theorem/>.

**Def.(35.7.2.1) [Goldbach's Functions].** For  $k \in \mathbb{Z}_{\geq 2}$ , define  $G_k : \mathbb{Z}_+ \rightarrow \mathbb{R}$ :

$$G_k(n) = \sum_{n_i \in \mathbb{Z}_+, \sum_{i=1}^k n_i = n} \Lambda(n_1) \cdots \Lambda(n_k).$$

┘

**Def.(35.7.2.2) [Singular Series for Goldbach's Functions].** For  $N \in 2\mathbb{Z}_+$ , denote

$$\mathfrak{S}_2(N) \triangleq \sum_{d|N} \frac{\mu^2(d)d}{\phi^2(d)} \sum_{c \in \mathbb{Z}_+, (c,d)=1} \frac{\mu(c)}{\phi^2(c)},$$

then

$$\mathfrak{S}_2(N) = 2 \prod_{p \in \text{Prime}_{\geq 3}} \left(1 - \frac{1}{(p-1)^2}\right) \cdot \prod_{p \in \text{Prime}_{\geq 3}, p|N} \left(\frac{p-1}{p-2}\right)$$

and is a non-zero unbounded function.

And for  $N \in 2\mathbb{Z}_+ + 1$ , denote

$$\mathfrak{S}_3(N) \triangleq \frac{1}{2} \sum_{c,d \in \mathbb{Z}_+, (d,cN)=1} \frac{\mu(c)\mu^2(d)d}{\phi^2(c)\phi^3(d)},$$

then

$$\mathfrak{S}_3(N) = \frac{1}{2} \prod_{p \in \text{Prime}} \left(1 + \frac{1}{(p-1)^3}\right) \prod_{p \in \text{Prime}, p|N} \left(1 - \frac{1}{p^2 - 3p + 3}\right)$$

and is a non-zero bounded function. ┘

*Proof:*

$$\begin{aligned}
 \mathfrak{G}_2(N) \prod_{p \in \text{Prime}_{\geq 3}} \left(1 - \frac{1}{(p-1)^2}\right)^{-1} &= \sum_{2|d|N} \frac{\mu^2(d)d}{\phi^2(d)} \prod_{p|d, p \geq 3} \left(1 - \frac{1}{(p-1)^2}\right)^{-1} \\
 &= 2 \prod_{p \in S(N) \setminus \{2\}} \left(1 + \frac{p}{(p-1)^2} \left(1 - \frac{1}{(p-1)^2}\right)^{-1}\right) \\
 &= 2 \prod_{p \in \text{Prime}_{\geq 3}, p|N} \left(\frac{p-1}{p-2}\right).
 \end{aligned}$$

$$\begin{aligned}
 \mathfrak{G}_3(N) \prod_{p \in \text{Prime}_{\geq 3}} \left(1 - \frac{1}{(p-1)^2}\right)^{-1} &= \sum_{d \in 2\mathbb{Z}_+ + 1, (d, N)=1} \frac{\mu^2(d)d}{\phi^3(d)} \prod_{p|d, p \geq 3} \left(1 - \frac{1}{(p-1)^2}\right)^{-1} \\
 &= \prod_{p \in \text{Prime}_{\geq 3}, p \nmid N} \left(1 + \frac{p}{(p-1)^3} \left(1 - \frac{1}{(p-1)^2}\right)^{-1}\right) \\
 &= \prod_{p \in \text{Prime}_{\geq 3}, p \nmid N} \frac{p^2 - 3p + 3}{(p-1)(p-2)}.
 \end{aligned}$$

And the assertion follows.  $\square$

**Conj. (35.7.2.3)** [Goldbach1742/Hardy-Littlewood[H-L23]]. For any  $m \in \mathbb{Z}_{\geq 2}$ ,  $2m$  is a sum of two prime numbers.

By Möbius-randomness law ?, Hardy-Littlewood even conjectures that the number  $R_2(2m)$  of representations of  $2m$  satisfies

$$R_2(2m) \sim \mathfrak{G}_2(2m) \frac{2m}{\log^2(2m)} \quad (35.7.2.2).$$

Or more precisely, for any  $A \in \mathbb{R}_+$ , there exists  $C_A \in \mathbb{R}_+$  s.t.

$$|G_2(2m) - \mathfrak{G}_2(2m)2m| \leq C_A \frac{m}{\log^A m} \quad (35.7.2.1).$$

┘

*Proof:*

$\square$

**Thm& Conj. Cor. (35.7.2.4)** [Chen[Che73]]. For any  $m \in \mathbb{Z}_{\geq 3}$ , there exists  $p \in \text{Prime}_{\geq 3}$  and  $q \in \mathbb{Z}_+$  s.t.  $q$  is a prime or a product of two primes, and  $2m = p + q$ .

In fact, if  $R'_2(2m)$  denote the number of such pairs  $(p, q)$ , then

$$R'_2(2m) \gg \mathfrak{G}_2(2m) \frac{2m}{\log^2 m}.$$

┘

*Proof:* ?

$\square$



**Lemma(35.7.2.5)[The Main Contributions].** Fix some  $z \in \mathbb{R}_{\geq 2}$  and  $N \in 2\mathbb{Z}_+ + 2$ , if we denote

$$G_2^{\sharp\sharp}(N) = \sum_{n_1+n_2=N} \Lambda_z^{\sharp}(n_1)\Lambda_z^{\sharp}(n_2) \quad (35.7.1.9),$$

then

$$G_2^{\sharp\sharp}(N) = \mathfrak{G}_2(N)N + O_A\left(N \log^{-A}(z) + \tau(N)Nz^{-1/3} + z^3\right) \quad (35.7.2.2).$$

Heuristically, this implies that the main contribution to  $G_2(N)$  comes from  $\Lambda_z^{\sharp}$ . □

*Proof:*

$$\begin{aligned} G_2^{\sharp\sharp}(N) &= \sum_{m_1, m_2 \in [z]_+} \mu(m_1)\mu(m_2) \log(m_1) \log(m_2) \sum_{l_1, l_2 \in \mathbb{Z}_+, l_1 m_1 + l_2 m_2 = N} 1 \\ &= \sum_{m_1, m_2 \in [z]_+} \mu(m_1)\mu(m_2) \log(m_1) \log(m_2) \left( \frac{N}{[m_1, m_2]} + O(1) \right) \\ &= N \sum_{m_1, m_2 \in [z]_+} \frac{\mu(m_1)\mu(m_2)}{[m_1, m_2]} \log(m_1) \log(m_2) + O(z^2 \log^2 z) \end{aligned}$$

And we have

$$\begin{aligned} \sum_{m_1, m_2 \in [z]_+} \frac{\mu(m_1)\mu(m_2)}{[m_1, m_2]} \log(m_1) \log(m_2) &= \sum_{d|N} \frac{\mu^2(d)}{d} \sum_{\substack{m_1, m_2 \in [z/d]_+ \\ (m_1, m_2)=1 \\ (m_1 m_2, d)=1}} \frac{\mu(m_1 m_2)}{m_1 m_2} \log(dm_1) \log(dm_2) \\ &= \sum_{d|N} \frac{\mu^2(d)}{d} \sum_{\substack{m_1, m_2 \in [z/d]_+ \\ (m_1 m_2, d)=1}} \sum_{c|(m_1, m_2)} \mu(c) \frac{\mu(m_1 m_2)}{m_1 m_2} \log(dm_1) \log(dm_2) \\ &= \sum_{d|N} \frac{\mu(d)}{d} \sum_{c \in [z/d]_+} \frac{\mu(cd)}{c^2} \left( \sum_{m \in [z/cd]_+, (m, cd)=1} \frac{\mu(m)}{m} \log(cdm) \right)^2 \end{aligned}$$

For  $c > \sqrt{z}/d$ , this is bounded by

$$\sum_{d|N} \sum_{c > \sqrt{z}/d} c^{-2} d^{-1} \log^4 z \leq 2\tau(N)z^{-1/2} \log^4 z,$$

and for  $c \leq \sqrt{z}/d$ , it follows from prime number theorem(35.5.7.21) that

$$- \sum_{m \in [z/cd]_+, (m, cd)=1} \frac{\mu(m)}{m} \log(cdm) = \frac{cd}{\phi(cd)} \left( 1 + O\left(\tau(cd) \log^{-A} z\right) \right),$$

so we can estimate by bounding  $\phi(n)$  and  $\tau(n)$  for example by(35.1.3.15) and(35.5.7.8) that

$$\begin{aligned} \sum_{d|N} \sum_{c \in [z^{1/2}/d]_+} \frac{\mu(cd)}{c^2} \left( \sum_{m \in [z/cd]_+, (m, cd)=1} \frac{\mu(m)}{m} \log(cdm) \right)^2 &= \sum_{d|N} \sum_{c \in [z^{1/2}/d]_+} \mu(d)\mu(cd)d\phi^{-2}(cd) \\ &\quad + O_A\left(\log^{4-A} z + \tau(N)z^{-1/3}\right) \\ &= \mathfrak{G}_2(N) + O_A\left(\log^{-A} z + \tau(N)z^{-1/3}\right) \quad (35.7.2.2), \end{aligned}$$

and the assertion follows. □

**Lemma (35.7.2.6) [The Minor Contributions].** For any  $(u_k)_{k \in \mathbb{Z}_+}, (v_m)_{m \in \mathbb{Z}_+} \in \mathbb{C}$  and  $z \in \mathbb{R}_{\geq 2}, A \in \mathbb{R}_+$ , we have

$$\sum_{l+m+n=N} u_l v_m \Lambda_z^b(n) \ll_A \log^{-A} z \|u\| \|v\| N \log N, \text{ where } N \in \mathbb{Z}_+.$$

┘

*Proof:* For any  $\alpha \in \mathbb{R}$ , it follows from Davenport's inequality (35.7.1.8) and partial summation that

$$\sum_{n \in [N]_+} \mu(n) \log(n) e^{2\pi i \alpha n} = O_A \left( \frac{X}{\log^A X} \right).$$

If we write  $S^b(\alpha; N) = \sum_{n \in [N]_+} \Lambda_z^b(n) e^{2\pi i \alpha n}$ , then we have

$$\begin{aligned} |S^b(\alpha; N)| &\leq \sum_{l \in [N/z]_+} \left| \sum_{m \in \mathbb{Z}_+, z < m \leq N/l} \mu(m) \log(m) e^{2\pi i \alpha l m} \right| \\ &\ll_A \sum_{l \in [N/z]_+} \frac{N}{l} \log^{-A}(N/l) \\ &\leq_A N \log N \log^{-A} z. \end{aligned}$$

And we have

$$\sum_{l+m+n=N} u_l v_m \Lambda_z^b(n) = \int_0^1 \left( \sum_{l \in [N]_+} u_l e^{2\pi i \alpha l} \right) \left( \sum_{m \in [N]_+} v_m e^{2\pi i \alpha m} \right) S^b(\alpha; N) e^{-2\pi i \alpha N} d\alpha.$$

Then the assertion follows from Schwarz's inequality and Parseval's formula.  $\square$

**Thm& Conj. Cor. (35.7.2.7) [Goldbach's Conjecture for almost all Even Numbers, Tchu-dakov [Tch37]/Estermann [Est38]/Van.der.Corput [Cor39]/Montgomery-Vaughan [M-V75]].**

For any  $A \in \mathbb{R}_+$ , there exists  $C_A \in \mathbb{R}_+$  s.t. for any  $X \in \mathbb{R}_{\geq 2}$ , we have

$$\sum_{N \in 2\mathbb{Z}_+ \cap [X]_+} (G_2(N) - \mathfrak{G}_2(N)N)^2 \leq C_A \frac{X^3}{\log^{2A} X}.$$

In particular, for  $A, B \in \mathbb{R}_{\geq 4}$ , there exists  $C_{A,B} \in \mathbb{R}_+$  s.t. for any  $X \in \mathbb{R}_{\geq 2}$ , there are at most  $C_{A,B} X \log^{-A}(X)$ -many  $m \in [X]_+$  s.t.

$$|G_2(2m) - \mathfrak{G}_2(2m)2m| \geq B \frac{X}{\log^A X} \quad (35.7.2.2).$$

In particular, all but at most  $O_A(X \log^{-A} X)$ -many even numbers in  $[X]_+$  are sums of two primes.  $\square$

*Proof:* For any sequence  $(c_N)_{N \in \mathbb{Z}_+} \in \mathbb{C}$  supported on odd integers, we have

$$\sum_{N \in [X]_+} c_N G_2(N) = \sum_{N \in (2\mathbb{Z}_+ + 1) \cap [X]_+} c_N \left( G_2^{\sharp\sharp}(N) + 2G_2^{\sharp b}(N) + G_2^{bb}(N) \right).$$

And by (35.7.2.5) applied to  $z = X^{1/4}$ , we see

$$\sum_{N \in [X]_+} c_N G_2^{\sharp\sharp}(N) = \sum_{N \in (2\mathbb{Z}_+ + 1) \cap [X]_+} c_N G_2(N) + O_A \left( \|c\| X^{3/2} \log^{-A} X \right).$$

And (35.7.2.6) applied to  $z = X^{1/4}$  implies that

$$\sum_{N \in [X]_+} c_N (2G_2^{\sharp\sharp}(N) + G_2^{bb}(N)) = O_A \left( \|c\| \left( 2 \left\| \Lambda_{X^{1/4}}^{\sharp} \right\| + \left\| \Lambda_{X^{1/4}}^b \right\| \right) X \log^{-A} X \right) = O_A \left( \|c\| X^{3/2} \log^{-A} X \right) \quad (35.7.1.10).$$

So we get

$$\sum_{N \in [X]_+} c_N G_2(N) = \sum_{N \in [X]_+} c_N \mathfrak{G}_2(N) N + O_A \left( \|c\| X^{3/2} \log^{-A} X \right).$$

The assertion follows by applying this to  $c_N = G_2(N) - \mathfrak{G}_2(N)N$ .  $\square$

**Cor. (35.7.2.8)[Ternary Goldbach's Function, Vinogradov [Vin37]].** For any  $A \in \mathbb{R}_+$ , there exists  $C_A \in \mathbb{R}_+$  s.t. for any  $N \in 2\mathbb{Z}_+ + 5$ , we have

$$\left| G_3(N) - \mathfrak{G}_3(N) N^2 \right| \leq C_A \frac{N^2}{\log^A N}, \quad \left| R_3(N) - \mathfrak{G}_3(N) \frac{N^2}{\log^3 N} \right| \leq C_A \frac{N^2}{\log^A N} \quad (35.7.2.2),$$

where  $R_3(N)$  is the number of ternary Goldbach-representations of  $N$ . In particular, any sufficiently large odd number  $N$  is a sum of three primes.  $\lrcorner$

*Proof:* By the proof of (35.7.2.7), we see that for any sequence  $(c_N)_{N \in \mathbb{Z}_+} \in \mathbb{C}$  supported on odd integers, we have

$$\sum_{N \in [X]_+} c_N G_2(N) = \sum_{N \in [X]_+} c_N \mathfrak{G}_2(N) N + O_A \left( \|c\| X^{3/2} \log^{-A} X \right).$$

Apply this to

$$\begin{aligned} G_3(N) &= \sum_{n \in [N-1]_+} \Lambda(n) G_2(N-n) \\ &= \sum_{n \in [N-1]_+} \Lambda(n) \mathfrak{G}_2(N-n)(N-n) + O_A(N^2 \log^{-A} N) \quad (35.5.7.18) \\ (35.7.2.2) &= \sum_{(c,d)=1} \frac{\mu(c)\mu^2(d)d}{\phi^2(c)\phi^2(d)} \sum_{n < N, n \equiv N \pmod{d}} \Lambda(n)(N-n) + O_A(N^2 \log^{-A} N). \end{aligned}$$

If  $d \nmid N$ , then

$$\sum_{n < N, n \equiv N \pmod{d}} \Lambda(n)(N-n) = \frac{N^2}{2\phi(d)} + O_A(N^2 \log^{-A} N)$$

by prime number theorem (35.5.7.18) and partial summation. And if  $d = p \in S(N)$ , then this summation is bounded by  $N \log N$ , and the contribution is smaller than  $N \log^2 N$ . Hence we get

$$G_3(N) = \frac{N^2}{2} \sum_{c,d \in \mathbb{Z}_+, (c,dN)=1} \frac{\mu(c)\mu^2(d)d}{\phi^2(c)\phi^3(d)} + O_A(N^2 \log^{-A} N) = \mathfrak{G}_3(N) N^2 + O_A \left( \frac{N^2}{\log^A N} \right).$$

The proof for  $R_3(N)$  is similar.  $\square$

**Thm& Conj.Cor. (35.7.2.9)** [Ternary Goldbach Conjecture, Hardy-Littlewood[H-L23]/Shnirelman(1930)/Vinogradov(1937)/Borozdkin(1956)/Zinoviev[Zin97]/Deshouillers-Effinger-te.Riele-Zinoviev[DERZ]/Ramaré[Ram95]/Tao[Tao14]/Helfgott[Hel13]]. For any  $k \in \mathbb{Z}_{\geq 4}$ ,  $2k + 1$  is a sum of three odd primes.  $\lrcorner$

*Proof:* ? Helfgott[Hel13] proved this for  $2k + 1 \geq 10^{27}$ , and the rest is verified by computers. The proof uses the circle method and the large sieve.

Notice the result for  $k$  sufficiently large follows from(35.7.2.8).  $\square$

### Number of Representations by Primes

**Prop. (35.7.2.10)**[Selberg's Upper Bound Sieve for Representations]. Uniformly for  $N \in 2\mathbb{Z}_+ + 2$ , we have

$$\#\{p_1, p_2 \in \text{Prime} : N = p_1 + p_2\} \leq C \left( \prod_{p \in S(N) \setminus \{2\}} \frac{p-1}{p-2} \right) \frac{N}{\log^2 N} \left( 4 + O \left( \frac{\log_{(2)} N}{\log N} \right) \right).$$

$\lrcorner$

*Proof:* The proof is similar to(35.6.1.6).?  $\square$

### Waring-Goldbach Problems

**Thm. (35.7.2.11)**[Lilu Zhao].

$$\begin{aligned} H(1) &\leq 3, & H(2) &\leq 5, & H(3) &\leq 9, & H(4) &\leq 13, \\ H(5) &\leq 21, & H(6) &\leq 32, & H(7) &\leq 46, & H(8) &\leq 63. \end{aligned}$$

$\lrcorner$

*Proof:*

$\square$

## 3 Partition Numbers

## 4 Waring's Problems

**Def. (35.7.4.1)** [Waring's Problem]. For  $k \geq 1$ , let  $g(k) \in \mathbb{Z}_+$  be the smallest integer s.t. every positive integer can be written as a sum of at most  $g(k)$  positive  $k$ -th powers.  $\lrcorner$

**Prop. (35.7.4.2)**. Situation as in(35.7.4.1),  $g(k) \geq \lfloor 2^k \rfloor + \lfloor (\frac{3}{2})^k \rfloor - 2$ .  $\lrcorner$

*Proof:* This is because the number  $\lfloor (\frac{3}{2})^k \rfloor 2^k - 1$  requires  $\lfloor (\frac{3}{2})^k \rfloor - 1$  powers  $2^k$  and  $2^k - 1$  powers  $1^k$ .  $\square$

**Thm. (35.7.4.3)**[Dickson-Niven-Pillai/Rubugunday]. Situation as in(35.7.4.1),

$$g(k) = \begin{cases} 2^k + \lfloor (\frac{3}{2})^k \rfloor - 2, & \lfloor (\frac{3}{2})^k \rfloor - (\frac{3}{2})^k \geq (\frac{3}{4})^k \\ \text{?}, & \text{otherwise} \end{cases}.$$

In particular,  $g(1) = 1, g(2) = 4, g(3) = 9, g(4) = 19$ .  $\lrcorner$

*Proof:* ?

$g(2) = 4$  follows from (19.5.8.4) and (19.5.8.3).  $\square$

**Conj. (35.7.4.4) [Mahler].** Situation as in (35.7.4.1),  $g(k) = 2^k + \lfloor (\frac{3}{2})^k \rfloor - 2$  any  $k$ .  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj.Cor. (35.7.4.5).** By (20.2.5.11)(35.7.4.3), this is true for  $k$  sufficiently large. But it is not known for how large  $k$  this is true, eventually due to the ineffectiveness of the proof of Roth's theorem (20.2.3.8).  $\lrcorner$

### Asymptotic Waring's Problem

References are [Waring's Problem: a survey, Vaughan-Wooley] and [Translation invariance, exponential sums, and Waring's problem, Wooley].

**Def. (35.7.4.6) [Asymptotic Waring's Problem].** For  $k \geq 1$ , let  $G(k) \in \mathbb{Z}_+$  be the smallest integer s.t. every sufficiently large integer can be written as a sum of at most  $G(k)$  positive  $k$ -th powers.

And let  $G_1(k) \in \mathbb{Z}_+$  be the smallest integer s.t. a set of integers of natural density 1 can be written as a sum of at most  $G(k)$  positive  $k$ -th powers.

Clearly  $G_1(k) \leq G(k) \leq g(k)$  (35.7.4.1).  $\lrcorner$

**Def. (35.7.4.7) [Warning Numbers].** For  $k, s, N \in \mathbb{Z}_+$ , denote  $r_{k,s}(N)$  the number of  $s$ -tuples  $(x_1, \dots, x_s) \in \mathbb{Z}_+$  s.t.

$$N = x_1^k + \dots + x_s^k.$$

$\lrcorner$

**Prop. (35.7.4.8).** Given  $s \in \mathbb{Z}_+$ , we have: For any  $N \in \mathbb{Z}_+$ ,

$$r_{1,s}(N) = \binom{N-1}{s-1} = \frac{N-1}{s-1} + O(N^{s-2}).$$

$\lrcorner$

**Thm. (35.7.4.9) [Small Values, Davenport [Dav39]/Hardy-Littlewood [H-L25]/Vaughan [Vau86]/Wooley [Woo96].**

- $g(2) = G(2) = G_1(2) = 4$ .
- $G_1(3) = 4$ .
- $G_1(4) = 15$ .
- $G_1(8) = 32$ .
- $G_1(16) = 64$ .
- $G_1(32) = 128$ .

$\lrcorner$

*Proof:* ?

$\square$

**Thm. (35.7.4.10)** [Upper Bounds for  $G(k)$ , Hardy-Littlewood[H-L22, H-L25]/ Vinogradov[Vin35, Vin47, Vin59]/Tong[Ton57]/ Chen[Che58]/Karatsuba[Kar85]/Vaughan[Vau89]/Wooley[Woo99, Woo95]]. There exists  $C \in \mathbb{R}_+$  s.t. for any  $k \in \mathbb{Z}_+$ , we have

$$G(k) \leq k \left( \log k + \log_{(2)} k + 2 + C \frac{\log_{(2)} k}{\log k} \right) \quad (35.7.4.6).$$

and

$$G(k) \leq k (3 \log k + 11).$$

And when  $k \geq 4000$ , we have

$$G(k) \leq k (2 \log k + 2 \log_{(2)} k + 12).$$

┘

*Proof:* ?

□

### Vinogradov's Mean Value Problem

**Thm. (35.7.4.11)** [general Vinogradov's Mean Value Problem]. Let  $N \in \mathbb{Z}_+$  and  $S = \{x_1, \dots, x_N\}$  where  $x_i \in (i-1, i]$ . For  $s, n \in \mathbb{Z}_+$ , let  $J_{s,n}(X)$  be the number of solutions to the following inequalities:

$$\left| X_1^i + \dots + X_s^i - X_{s+1}^i - \dots - X_{2s}^i \right| \leq N^{i-n}, \text{ where } i \in [n]_+.$$

with  $X_i \in X$ . Then we have

$$J_{s,n}(S) \ll_{\varepsilon, s} N^{s+\varepsilon} + N^{2s - \frac{n(n+1)}{2} + \varepsilon},$$

where the implied constant doesn't depend on  $S$ .

┘

*Proof:* Let  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$  be a positive-valued Schwartz function with positive-valued Fourier transform s.t.  $\mathcal{F}\varphi(\xi) \geq 1$  for any  $\xi \in \mathbb{R}^n$  s.t.  $|\xi_i| \leq 1$ . Then it follows from (35.7.5.2) with  $a_i = 1$  and  $t_i = X_i/N$  and the rapidly decaying property of  $\varphi$  that

$$\left( \frac{1}{\text{Vol}(\mathbb{B}(0; N^n))} \int_{\mathbb{R}^n} \varphi_{N^n}(\mathbf{x}) \left| \sum_{i \in [N]_+} e^{2\pi i(x_1 t_i + x_2 t_i^2 + \dots + x_n t_i^n)} \right|^{2s} d\mathbf{x} \right)^{\frac{1}{2s}} \ll_{n,s,\varepsilon} N^{1/2+\varepsilon} + N^{1 - \frac{n(n+1)}{4s} + \varepsilon}.$$

Notice

$$\begin{aligned} & \int_{\mathbb{R}^n} \varphi(\mathbf{x}/N^n) \left| \sum_{i \in [N]_+} e^{2\pi i(x_1 t_i + x_2 t_i^2 + \dots + x_n t_i^n)} \right|^{2s} d\mathbf{x} \\ &= \sum_{X_1, \dots, X_{2s} \in S} N^{n^2} \mathcal{F}\varphi(N^{n-1} Z_1, N^{n-2} Z_2, \dots, Z_n), \text{ where } Z_i = X_{s+1}^i + \dots + X_{2s}^i - X_1^i - \dots - X_s^i. \\ &\geq N^{n^2} J_{s,n}(S) \end{aligned}$$

Thus the assertion follows.

□

## 5 Decoupling

References are [Proof of the  $L^2$  decoupling conjecture, Bourgain-Demeter] and [Proof of the main conjecture in Vinogradov's Mean Value Theorem for degrees higher than three, Bourgain-Guth].

**Thm. (35.7.5.1) [Restriction Estimates, Bourgain-Demeter-Guth].** For  $n \in \mathbb{Z}_{\geq 2}$  and an interval  $J \subset [0, 1]$ , define the **extension operator**  $E_J = E_J^{(n)}$  s.t. for any measurable function  $g : [0, 1] \rightarrow \mathbb{C}$ ,

$$E_J g : \mathbb{R}^n \rightarrow \mathbb{C} : \mathbf{x} \mapsto \int_J g(t) e^{2\pi i(t x_1 + t^2 x_2 + \dots + t^n x_n)} dt.$$

Let  $E \in \mathbb{R}_{\geq 100n}$  and  $B \subset \mathbb{R}^n$  be a ball of radius  $r$ , define a weight function

$$w_B(\mathbf{x}) = w_{B,E}(\mathbf{x}) = \left(1 + \frac{|\mathbf{x} - \mathbf{c}|}{r}\right)^{-E}$$

on  $\mathbb{R}^n$ . Then for any  $\delta \in (0, 1]$ ,  $\varepsilon \in \mathbb{R}_+$  and  $B \subset \mathbb{R}^n$  a ball of radius  $\delta^{-n}$ , we have

$$\|E_{[0,1]} g\|_{L^{n(n+1)}(w_B)} \ll_{n,E,p,\varepsilon} \delta^{-\varepsilon} \left( \sum_{J_i \in \mathcal{J}_\delta} \|E_{J_i} g\|_{L^{n(n+1)}(w_B)}^2 \right)^{1/2}.$$

where  $\mathcal{J}_\delta$  is a 2-overlapping covering of  $[0, 1]$  by intervals of radius  $\delta$ , where the implied constant is independent of  $\delta, B, g$ .  $\perp$

*Proof:* Cf. [Proof of the main conjecture in Vinogradov's Mean Value Theorem for degrees higher than three, Bourgain-Demeter-Guth].  $\square$

**Cor. (35.7.5.2) [Discrete Restriction Estimates, Bourgain-Demeter-Guth].** Situation as in (35.7.5.1), for  $N \in \mathbb{Z}_+$  and for each  $i \in [N]_+$ , choose  $t_i \in (\frac{i-1}{N}, \frac{i}{N}]$ , then for each  $R \gg N^n$  and each ball  $B \subset \mathbb{R}^n$  of radius  $R$ , and any sequence  $(a_i)_{i \in [N]_+} \in \mathbb{C}$  and  $p \in \mathbb{R}_{\geq 2}$ ,  $\varepsilon \in \mathbb{R}_+$ , we have

$$\left( \frac{1}{\text{Vol}(B)} \int \left| \sum_{i \in [N]_+} a_i e^{2\pi i(x_1 t_i + x_2 t_i^2 + \dots + x_n t_i^n)} \right|^p w_B(\mathbf{x}) d\mathbf{x} \right)^{1/p} \ll_{n,\varepsilon} \left( N^\varepsilon + N^{\frac{1}{2}(1 - \frac{n(n+1)}{p}) + \varepsilon} \right) \|a_i\|_2.$$

where the implied constant doesn't depend on  $N, R$  or  $(a_i)$ .  $\perp$

*Proof:* By Hölder inequality and the trivial bound for  $L^\infty$ , it suffices to prove for  $p = n(n+1)$ . Let  $\mathcal{B}$  be a finitely-overlapping covering of  $B$  with balls of radius  $N^n$ , then we have

$$w_B \ll_{n,E} \sum_{B' \in \mathcal{B}} w_{B'} \ll_{n,E} w_B.$$

where the implied constant doesn't depend on  $N$  and  $R$ .

Apply (35.7.5.1) to every ball in  $\mathcal{B}$  and sum up, we get

$$\|E_{[0,1]} g\|_{L^{n(n+1)}(w_B)} \ll_{n,E,\varepsilon} N^\varepsilon \left( \sum_{J_i \in \mathcal{J}_{N^{-1}}} \|E_{J_i} g\|_{L^{n(n+1)}(w_B)}^2 \right)^{1/2},$$

where  $\mathcal{J}_{N^{-1}}$  is the standard partition of  $[0, 1]$  into  $N$  intervals of length  $N^{-1}$ .

Then we apply this inequality to

$$g_\tau = \frac{1}{2\tau} \sum_{i \in [N]_+} a_i \mathbf{1}_{[t_i - \tau, t_i + \tau]}, \text{ where } \tau \in \mathbb{R}_+,$$

and let  $\tau \rightarrow 0$ , then the assertion follows.  $\square$

## 6 limitations of Circle methods

See <https://terrytao.wordpress.com/2012/05/20/heuristic-limitations-of-the-circle-method/>.



## 35.8 Generalized Ramanujan Conjecture

GRC will imply a bound for the Fourier coefficients of a cusp form. And as a consequence, it will imply that  $|L(F; 1/2)| \leq C_\varepsilon C(F)^\varepsilon$  for any  $\varepsilon \in \mathbb{R}_+$ , thus give results in subconvexity problems.

There are partial results towards GRC.

References are [B-B11].

**Conj. (35.8.0.1) [Generalized Ramanujan-Selberg Conjectures].** An analytic  $L$ -function  $L(f; s)$  of degree  $d$  (29.5.1.1) is said to satisfy the generalized **Ramanujan conjecture** or the **Ramanujan-Petersson conjecture** if for any prime  $p \in \text{Prime}$  unramified for  $f$  (29.5.1.1), we have  $|\alpha_i(p)| = 1$  for  $i \in [d]_+$  (29.5.1.1).

An analytic  $L$ -function  $L(f; s)$  of degree  $d$  is said to satisfy the generalized **Ramanujan conjecture** if  $\text{Re}(\kappa_i) = 0$  for  $i \in [d]_+$ .  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (35.8.0.2).** If  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(2; \mathbb{Q}))$  with unitary central character, then  $L(\pi; s)$  is an analytic  $L$ -function of degree 2 over  $F$  with local parameters  $\{\alpha_1(p), \alpha_2(p)\}$  for  $p \in \text{Prime}$  and local parameters  $\{\kappa_1, \kappa_2\}$  at  $\infty$ . Then  $|\alpha_1(p)\alpha_2(p)| = 1$  for  $v \in \Sigma_F$ .

If  $L(\text{Sym}^n(f) \otimes \text{Sym}^n(\bar{f}); s)$  exists, then

$$\max(|\alpha_1(p)|, |\alpha_2(p)|) \leq p^{\frac{1}{2n}}.$$

In particular, if  $L(\text{Sym}^n(f) \otimes \text{Sym}^n(\bar{f}); s)$  exists for each  $n \in \mathbb{Z}_+$ , then generalized Ramanujan conjecture (35.8.0.1) holds for  $f$ . Similarly, generalized Selberg conjecture also holds for  $f$ .  $\lrcorner$

*Proof:* If  $L(\text{Sym}^n(f) \otimes \text{Sym}^n(\bar{f}); s)$  exists, then it is self-dual by strong multiplicity one. And then by examining its local parameters, we see the largest parameters  $\max(|\alpha_1(p)|^{2n}, |\alpha_2(p)|^{2n}) < p$ . And then the assertion follows.

For the Selberg conjecture ?.  $\square$

**Thm. (35.8.0.3) [Ramanujan Bounds, Luo-Rudnick-Sarnak [LRS99]].** For any  $F \in \text{NField}$  and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n; F))$  with unitary central character, then for each  $i \in [n]_+$ , we have

$$|\log_{\|v\|} \alpha_i(\pi_v)| \leq \frac{1}{2} - \frac{1}{m^2 + 1}, \quad v \in \Sigma_F^{\text{fin}}, \quad |\text{Re}(\kappa_j(\pi_v))| < \frac{1}{2} - \frac{1}{m^2 + 1}, \quad v \in \Sigma_F^\infty.$$

$\lrcorner$

*Proof:* ? Cf. [B-B11].  $\square$

**Thm. (35.8.0.4) [Ramanujan Bounds for GL(2), Kim-Sarnak/Blomer-Brumley [B-B11]].** For any  $F \in \text{NField}$  and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(n; F))$  with unitary central character, then for each  $i \in \{1, 2\}$ , we have

$$|\log_{\|v\|} \alpha_i(\pi_v)| \leq \begin{cases} \frac{7}{64} & , n = 2 \\ \frac{5}{14} & , n = 3 \\ \frac{9}{22} & , n = 4 \end{cases}, \quad v \in \Sigma_F^{\text{fin}}, \quad |\text{Re}(\kappa_j(\pi_v))| < \begin{cases} \frac{7}{64} & , n = 2 \\ \frac{5}{14} & , n = 3 \\ \frac{9}{22} & , n = 4 \end{cases}, \quad v \in \Sigma_F^\infty.$$

$\lrcorner$

*Proof:* Cf. [B-B11].  $\square$

## 1 Kuznetsov Trace Formula

References are [arithmetic and spectral analysis of Poincare series, Cogdell-Shapiro].

**Thm. (35.8.1.1) [Density Theorem for Selberg Conjecture, Iwaniec].** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_\varepsilon \in \mathbb{R}_+$  s.t. for any  $r \in [0, 1/2]$ , we have

$$\# \left\{ \lambda \leq \frac{1}{4} - r^2 \mid \lambda \in \text{Spec}(\Delta|_{L_0^2(\Gamma_0(N) \backslash \mathcal{H})}) \right\} \leq C_\varepsilon \text{Vol}(\Gamma_0(N) \backslash \mathcal{H})^{1-4r+\varepsilon}.$$

┘

*Proof:* ? Cf. [Iwaniec, Small eigenvalues of Laplacian for  $\Gamma_0(N)$ ]. The proof uses Kuznetsov's trace formula. □

**Cor. (35.8.1.2) [Selberg].** For any  $N \in \mathbb{Z}_+$  and any eigenvalue  $\lambda$  of the Laplacian on  $L_0^2(\Gamma_0(N) \backslash \mathcal{H})$ , we have  $\lambda \geq 3/16$ . ┘

### Linnik-Selberg Conjecture

The Kuznetsov trace formula can be used to capture cancellations in Kloosterman sums beyond Weil's bound.

## 35.9 Arithmetic Statistics

**Notation (35.9.0.1).**

- Use notations on global fields defined in [Global Fields](#).
- Use notations on arithmetic of Abelian varieties defined in [Arithmetic of Abelian Varieties](#).

┘

### 1 Fields

**Thm. (35.9.1.1) [Davenport-Heilbronn].** For  $\xi, \eta \in \mathbb{R}$ , let  $N_3(\xi, \eta)$  be the isomorphism classes of cubic number fields  $\mathcal{K}$  s.t.  $d_{\mathcal{K}} \in (\xi, \eta)$ . Then

$$\lim_{X \rightarrow \infty} \frac{N_3(0, X)}{X} = \frac{1}{12\zeta(3)}, \quad \lim_{X \rightarrow \infty} \frac{N_3(-X, 0)}{X} = \frac{1}{4\zeta(3)}.$$

┘

*Proof:* Cf.[On the density of discriminants of cubic fields, Davenport-Heilbronn]. □

**Thm. (35.9.1.2) [Bhargava].** For  $\xi, \eta \in \mathbb{R}$ , let  $N_n(\xi, \eta)$  be the isomorphism classes of cubic fields  $\mathcal{K}$  s.t.  $d_{\mathcal{K}} \in (\xi, \eta)$ . Then for  $n = 4$  or  $5$ ,

$$\lim_{X \rightarrow \infty} \frac{N_n(-X, X)}{X}$$

exists. ┘

*Proof:* Cf.[The density of discriminants of quartic rings and fields], [The density of discriminants of quintic rings and fields]. □

### 2 Elliptic Curves

References are [\[P-R12, BKLPR, PPVW\]](#).

**Def. (35.9.2.1) [Statistical Heights].** Every  $E \in \text{EllCur}/\mathbb{Q}$  is isomorphic to the projective closure of a unique affine curve  $y^2 = x^3 + Ax + B$  s.t.  $4A^3 + 27B^2 \neq 0$ , and there are no  $p \in \text{Prime}$  with  $p^4|A$  and  $p^6|B$ .

Then we can define the **statistical height** of  $E$  to be  $\text{ht}_0(E) = \max(4|A^3|, 27B^2)$ . ┘

**Prop. (35.9.2.2) [Elliptic Curves Bounded by Heights].** For  $H \in \mathbb{R}_+$ , denote  $\text{EllCur}_{\leq H}/\mathbb{Q}$  to be the isomorphism classes of elliptic curves over  $\mathbb{Q}$  with statistical height bounded by  $H$ . Then

$$\#\text{EllCur}_{\leq H}/\mathbb{Q} = (2^{4/3}3^{-3/2}\zeta(10)^{-1} + o(1))H^{5/6}.$$

┘

*Proof:* Cf.[Brumer, The average rank of elliptic curves]Lemma4.3. □

**Def. (35.9.2.3) [Statistical Density w.r.t. Heights].** For any set  $S \subset \text{EllCur}/\mathbb{Q}$ , define the **density** of  $S$  to be

$$\lim_{H \rightarrow \infty} \frac{\#S \cap (\text{EllCur}_{\text{ht}_0 \leq H}/\mathbb{Q})}{\#(\text{EllCur}_{\text{ht}_0 \leq H}/\mathbb{Q})}$$

if this limit exists. ┘

**Prop. (35.9.2.4).** For  $F \in \mathbf{GField}$ ,  $\mathbf{P}_{\text{ht}_0}(E \in \mathbf{EllCur} / F : E(F)_{\text{tor}} = 0) = 1$ . ┘

*Proof:* Cf. [BKLPR]Lemma5.7. ? □

**Thm. (35.9.2.5)** [Almost all Elliptic Curves have Surjective Galois-Representations, Duke [Duk97]]. For each  $X \in \mathbb{R}_+$ , let  $\mathbf{EllCur}'_{\leq X}$  be the set of elliptic curves over  $\mathbb{Q}$  with statistical height bounded by  $X$  s.t.  $\rho_{E, \mathbf{A}} : \text{Gal}_{\mathbb{Q}} \rightarrow \prod_{p \in \text{Prime}} \text{GL}(2; E[p])$  is non-surjective, then

$$\lim_{X \rightarrow \infty} \frac{\#\mathbf{EllCur}'_{\leq X}}{\#\mathbf{EllCur}_{\leq X}} = 0.$$

┘

*Proof:* Cf. [Duk97]. ? □

### Statistics for $\text{rank}(E/\mathbb{Q})$

**Conj. (35.9.2.6)** [Minimalist Conjecture, Katz-Sarnak [K-S99]]. The distribution of elliptic curves ordered by conductors satisfies

$$\mathbf{P}_{\text{rank}}(r) = \begin{cases} 0.5 & , r = 0 \\ 0.5 & , r = 1 . \\ 0 & , r \geq 2 \end{cases}$$

In particular,  $\mathbf{E}_{\text{conductor}}(r) = 0.5$ . ┘

*Proof:* □

**Thm& Conj. Cor. (35.9.2.7)** [Bhargava-Shankar].  $\mathbf{E}(r) \leq 0.99$ , if it exists. ┘

*Proof:* □

### Quadratic Twists and Goldfeld's Conjecture

Cf. [Gol79].

**Conj. (35.9.2.8)** [Goldfeld]. For  $E \in \mathbf{EllCur} / \mathbb{Q}$ , consider its twists  $E^{(D)}$  as in (21.3.6.3), then

$$\lim_{X \rightarrow \infty} \frac{\#\{d \in \mathbb{Z}^\times : d \text{ is square-free}, r_{\text{an}}(E^{(d)}) = r, |d| < X\}}{\#\{d \in \mathbb{Z}^\times : d \text{ is square-free}, |d| < X\}} = \begin{cases} 0.5 & r = 0 \\ 0.5 & r = 1 . \\ 0 & r \geq 2 \end{cases}$$

┘

*Proof:* □

**Conj. Cor. (35.9.2.9)** [Weak Goldfeld Conjecture]. For  $E \in \mathbf{EllCur} / \mathbb{Q}$ ,

$$\liminf_{X \rightarrow \infty} \frac{\#\{d \in \mathbb{Z}^\times : d \text{ is square-free}, r_{\text{an}}(E^{(d)}) = r, |d| < X\}}{\#\{d \in \mathbb{Z}^\times : d \text{ is square-free}, |d| < X\}} > 0.$$

┘

**Thm& Conj.Cor.(35.9.2.10) [Kriz-Li].** If  $E \in \text{EllCur}/\mathbb{Q}$  has a rational 3-isogeny, then the weak Goldfeld conjecture(35.9.2.9) is true.

And for the sextic family  $\{y^2 = x^3 + d\}$ , the proportion is  $\geq 1/6$  for  $r = 0$  and  $\geq 1/6$  for  $r = 1$ .  $\lrcorner$

*Proof:* [Daniel Kriz and Chao Li. Goldfeld's conjecture and congruences between Heegner points].  
□

**Thm& Conj.Cor.(35.9.2.11)[Tian-Yuan-Zhang, Smith].** For the sextic family  $\{y^2 = x^3 + d\}$ , the proportion is  $\geq 1/6$  for  $r = 0$  and  $\geq 1/6$  for  $r = 1$ .  $\lrcorner$

*Proof:* [Ye Tian, Xinyi Yuan, and Shou-Wu Zhang. Genus periods, genus points and congruent number problem], [Smith. The congruent numbers have positive natural density]. □

### Statistics for $\text{III}(E/\mathbb{Q})$

**Prop.(35.9.2.12) [Heuristics for  $\text{III}(E/\mathbb{Q})$ ].** For  $E \in \text{EllCur}/\mathbb{Q}$ , denote  $\text{III}_0(E/\mathbb{Q}) = \begin{cases} \text{III}(E/\mathbb{Q}) & , \text{rank}(E) = 0 \\ 0 & , \text{rank}(E) \geq 1 \end{cases}$ . Then if we assume the B-S.D Conjecture for  $E$  and the generalized Riemann hypothesis for  $L(E; s)$ , we have

$$\frac{\mathbf{E}_{E \in \text{EllCur}_{\text{ht}_0 \leq H}}(\text{III}_0(E/\mathbb{Q}))}{H^{1/12}} = 1.$$

$\lrcorner$

*Proof:* Firstly, by the B-S.D conjecture(31.4.1.1), we have(notice if  $\text{rank}(E) = 0$ , then  $\text{Reg}(E/\mathbb{Q}) = 1$ )

$$L(E; 1) = \frac{\text{III}_0(E/\mathbb{Q}) \cdot \Omega \cdot \prod_{p \in \text{Prime}} c_p}{\#E(\mathbb{Q})_{\text{tor}}^2}$$

And

•

□

## 3 Asymptotic Density of Rational Points

References are [Ser97, HB02].

**Conj.(35.9.3.1) [Serre[Ser97]].** For  $F \in \text{NField}$  with  $[F : \mathbb{Q}] = d$  and any integral projective scheme  $V \in \text{Sch}/F$  of degree  $\geq 2$ , there exists  $C \in \mathbb{R}_+$  and  $\gamma \in (0, 1)$  s.t.

$$\#\{x \in V(F), H(x) \leq X\} \leq CX^{d \dim V} \log^\gamma X.$$

$\lrcorner$

*Proof:*

□

**Thm& Conj.Cor.(35.9.3.2) [Uniform Dimension Growth Conjecture, Conjectured by Heath.Brown[HB83]/ Serre[Ser08], Cohen/Serre[Ser97]/Heath.Brown[HB02]/ Brown-ing-Heath.Brown-Salberger[BHBS]/Salberger[Sal07, Sal23]].** For any integral projective scheme  $V \subset \mathbb{P}^N$  of degree  $d \geq 2$  and  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_{d,N,\varepsilon} \in \mathbb{R}_+$  s.t.

$$\#\{x \in V(F), H(x) \leq X\} \leq C_{d,N,\varepsilon} X^{d \dim V + \varepsilon}.$$

$\lrcorner$

*Proof:* ?

□

**Lemma (35.9.3.3).** If  $Z \in \text{Sch}^{\text{ft,sep}}/\mathcal{O}_F$  with  $Z_F \in \text{Var}/F$  and  $f : Z \rightarrow \mathbb{A}^n \in \text{Sch}/\mathcal{O}_F$  is a finite morphism of degree  $r \geq 2$  and  $M = f(Z(\mathcal{O}_F))$ , then there exists  $F_f/F \in \text{GalField}^{\text{fin}}$  and a constant  $c_f \in (0, 1)$  s.t. if  $v \in \text{Spl}_F(F_f)$ , then the image  $M_v$  of  $M$  in  $\mathbb{A}^n(\kappa_v)$  satisfies:

$$\#M_v \leq (1 - \frac{1}{r!}) \|v\|^n + O(\|v\|^{n-1/2}).$$

┘

*Proof:*  $R(Z)$  is a finite separable field extension of  $F(T_1, \dots, T_n)$ , and let  $\Omega = \widetilde{R(Z)}/F(T_1, \dots, T_n)$ . Denote  $G = \text{Gal}(\Omega/F(T_1, \dots, T_n))$  and  $H = \text{Gal}(\Omega/R(Z))$ .

Let  $F_f$  be the normalization of  $F$  in  $\Omega$ , and let  $Z_\Omega$  be the normalization of  $Z$  in  $\Omega$ . Because finite étale is fiber-wise, there exists an open subset  $U \subset \mathbb{A}^n$  s.t. the preimages  $U_Z \subset Z$  and  $U_\Omega \subset Z_\Omega$  is finite étale over  $U$ .

For  $v \in \text{Spl}_F(F_f)$ , we show that the image of  $U_Z(\kappa(v))$  in  $U(\kappa(v))$  has cardinality  $\leq (1 - 1/r!) \|v\|^n + O(\|v\|^{n-1/2})$ . We may base change to  $F_f$  as  $v$  is split. Thus  $(U_Z)_{F_f}$  and  $(U_\Omega)_{F_f}$  are both geo.irreducible over  $F_f$ , and we can use the Lang-Weil estimate (31.1.3.9)(7.12.5.5). As  $U_\Omega/U_Z$  is finite étale with Galois group  $H$ , the image of  $U_\Omega(\kappa(v))$  in  $U_Z(\kappa(v))$  has cardinality  $\frac{1}{\#H} \|v\|^n + O(\|v\|^{n-1/2})$  (because étale implies these local extension are unramified). Similarly, the image of  $U_\Omega(\kappa(v))$  in  $U(\kappa(v))$  has cardinality  $\frac{1}{\#G} \|v\|^n + O(\|v\|^{n-1/2})$ . Thus the image of  $U_\Omega(\kappa(v))$  in  $U(\kappa(v))$  has cardinality at most

$$\left(1 + \frac{1}{\#G} - \frac{1}{\#H}\right) \|v\|^n + O(\|v\|^{n-1/2}) = \left(1 - \frac{r-1}{\#G}\right) \|v\|^n + O(\|v\|^{n-1/2}) \leq \left(1 - \frac{1}{r!}\right) \|v\|^n + O(\|v\|^{n-1/2}).$$

□

**Cor. (35.9.3.4).** If  $F \in \text{NField}$  and  $n \in \mathbb{Z}_+$ , then for any subset  $M \subset \mathbb{P}^n(F)$  which is a thin set in  $\mathbb{P}^n$  (20.1.2.1) and  $\emptyset \subset_{\text{fin}} T \subset \Sigma_F^{\text{fin}}$ , there exists  $\emptyset \subset_{\text{fin}} S \subset \Sigma_F^{\text{fin}}$  s.t.  $S \cap T = \emptyset$ , and the reduction of  $M$  to

$$\prod_{v \in S} \mathbb{P}^n(\kappa_{v_i})$$

is not surjective.

┘

*Proof:* By (20.1.2.3), it suffice to prove for  $M$  of type 1 or type 2. If  $M$  is of type 1, then the projection of  $M$  to  $\mathbb{P}^n(\kappa_v)$  is not surjective for a.e.  $v \in \Sigma_F^{\text{fin}}$ . If  $M$  is of type 2, then up to type 1 thin sets, we may assume that  $M = \mathcal{O}_F \cap f(Z(F))$  for some finite morphism  $f : Z \rightarrow \mathbb{A}^n \in \text{Sch}^{\text{ft,sep}}/F$  of degree  $\geq 2$ . By changing  $Z$ , we may also assume that  $M = f(Z(\mathcal{O}_F))$  for some finite morphism  $f : Z \rightarrow \mathbb{A}^n \in \text{Sch}^{\text{ft,sep}}/\mathcal{O}_F$  of degree  $\geq 2$  (by writing  $R(Z)$  as a finite field extension of  $F(T_1, \dots, T_n)$  and make the defining function monic). By the same reasoning as in (20.1.2.3), we may assume that  $Z_F \in \text{Var}/F$ . Then the assertion follows from (35.9.3.3) and Chebotarev density theorem (19.7.5.2). □

**Prop. (35.9.3.5) [Distribution of Integral Points on Affine Thin Sets, Cohen].** If  $F \in \text{NField}$  with  $[F : \mathbb{Q}] = d$  and  $n \in \mathbb{Z}_+$ , then for any subset  $M \subset \mathcal{O}_F^n$  which is a thin set in  $\mathbb{A}_F^n$  (20.1.2.1), there exists  $C \in \mathbb{R}_+$  and  $\gamma \in (0, 1)$  s.t. for any  $X \in \mathbb{R}_+$ , we have

$$\#\{x \in M, \overline{|x|} \leq X\} \leq CX^{(n-1/2)d} \log^\gamma X \quad (20.2.1.13).$$

┘

*Proof:* The assertion is trivial for type 1 thin sets. So up to type 1 thin sets, we may assume that  $M = \mathcal{O}_F \cap f(Z(F))$  for some finite morphism  $f : Z \rightarrow \mathbb{A}^n \in \text{Sch}^{\text{ft,sep}}/F$  of degree  $\geq 2$ . By changing  $Z$ , we may also assume that  $M = f(Z(\mathcal{O}_F))$  for some finite morphism  $f : Z \rightarrow \mathbb{A}^n \in \text{Sch}^{\text{ft,sep}}/\mathcal{O}_F$  of degree  $\geq 2$  (by writing  $R(Z)$  as a finite field extension of  $F(T_1, \dots, T_n)$  and make the defining function monic). By the same reasoning as in (20.1.2.3), we may assume that  $Z_F \in \text{Var}/F$ .

By the large sieve inequality (35.1.6.14), for any  $X \in \mathbb{R}_+$ , we have

$$\#\{x \in M, \overline{|x|} \leq X\} \leq c_{F,n} X^{nd} / L(X^{d/2}),$$

where

$$L(Q) = \sum_{\substack{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)^\flat \\ \|\mathfrak{a}\| \leq Q}} \prod_{v \in S(\mathfrak{a})} \frac{w_v}{1 - w_v}, \text{ where } 1 - w_v = \frac{\#M_v}{\|v\|^n}.$$

and  $c_{F,n}$  depends only on  $F$  and  $n$ . Then it suffices to prove that  $L(Q) \geq Q / \log^\gamma Q$  for some  $\gamma \in (0, 1)$ . Notice

$$L(Q) \geq L'(Q) \triangleq \sum_{\substack{\mathfrak{a} \in \text{Ideal}^\times(\mathcal{O}_F)^\flat \\ \|\mathfrak{a}\| \leq Q}} \prod_{v \in S(\mathfrak{a})} c_v, \text{ where } c_v = \begin{cases} \frac{\|v\|^n}{\#M_v} - 1 & , v \in \text{Spl}_F(F_f) \\ 0 & , v \notin \text{Spl}_F(F_f) \end{cases}.$$

By (35.9.3.3), for any  $v \in \text{Spl}_F(F_f)$ ,

$$c_v \geq c + O(\|v\|^{-1/2}), \text{ where } c = \frac{1}{r! - 1}.$$

Then we can choose  $c'_v \in \mathbb{R}_+$  for each  $v \in \text{Spl}_F(F_f)$  s.t.  $c_v \geq c'_v$  and  $c'_v = c + O(\|v\|^{-1/2})$ .

Define  $L$ -functions

$$f_1(s) = \prod_{v \in \text{Spl}_F(F_f)} (1 + c'_v \|v\|^{-s}), \quad f_2(s) = \prod_{v \in \text{Spl}_F(F_f)} (1 - \|v\|^{-s})^{-c},$$

then  $f_1(s) \geq f_2(s)g(s)$ , where  $g(s) = \exp(h(s))$  and  $h(s)$  is an  $L$ -function that converges for  $\text{Re}(s) > 1/2$ . Thus by Chebotarev density theorem (19.7.5.4),

$$f_1(s) \sim \kappa_1 f_2(s) \sim \kappa_2 \zeta_{F_f}^{c/[F_f:F]} \sim \kappa_3 \frac{1}{(s-1)^{c/[F_f:F]}}, \text{ where } \kappa_i \in \mathbb{C}^\times, s \rightarrow 1.$$

Thus by Ikehara-Delange Tauberian theorem (35.1.1.5), we have

$$L'(Q) \geq \sum_{n \in [Q]_+} a_n(f_1) \sim \frac{\kappa_3}{\Gamma(c/[F_f:F])} \frac{Q}{\log^{1-c/[F_f:F]} Q}.$$

□

**Cor. (35.9.3.6) [Distribution of Integral Points on Affine Varieties].** If  $F \in \text{NField}$  with  $[F : \mathbb{Q}] = d$  and  $N \in \mathbb{Z}_+$ , then for any integral closed subscheme  $V \subset \mathbb{A}_F^n$  that is not a linear subspace, there exists  $C \in \mathbb{R}_+$  and  $\gamma \in (0, 1)$  s.t. for any  $X \in \mathbb{R}_+$ , we have

$$\#\{x \in V(\mathcal{O}_F), \overline{|x|} \leq X\} \leq CX^{d(\dim V - 1/2)} \log^\gamma X \text{ (20.2.1.13)}.$$

┘

*Proof:* By Noetherian normalization (5.2.4.22), we can find  $\dim(V)$ -many coordinates s.t. the projection onto these coordinate is a finite map, and the hypothesis implies that  $\text{pr}(V(K))$  is a thin set (20.1.2.1), because there are no rational sections. Thus the assertion follows from (35.9.3.5).  $\square$

**Cor. (35.9.3.7) [Distribution of Rational Points on Projective Thin Sets].** If  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = d$  and  $n \in \mathbb{Z}_+$ , then for any thin subset  $M \subset \mathbb{P}^n(F)$  (20.1.2.1), there exists  $C \in \mathbb{R}_+$  and  $\gamma \in (0, 1)$  s.t. for any  $X \in \mathbb{R}_+$ , we have

$$\#\{x \in M, H(x) \leq X\} \leq CX^{d(n+1/2)} \log^\gamma X.$$

┘

*Proof:* Consider

$$M' = \{(x_0, \dots, x_n) \in \mathcal{O}_F^{\oplus(n+1)} \setminus 0 \mid \overline{(x_0, \dots, x_n)} \in M\}.$$

Then by (20.2.1.14), there exists  $c = c(F, n)$  s.t.

$$\#\{x \in M, H(x) \leq X\} \leq \#\{x' \in M', \lceil x' \rceil \leq cX\}.$$

Thus the assertion follows from (35.9.3.5).  $\square$

**Cor. (35.9.3.8) [Distribution of Rational Points on Projective Varieties].** If  $F \in \mathbf{NField}$  with  $[F : \mathbb{Q}] = d$  and  $N \in \mathbb{Z}_+$ , then for any integral projective variety  $V$  of degree  $\geq 2$ , there exists  $C \in \mathbb{R}_+$  and  $\gamma \in (0, 1)$  s.t. for any  $X \in \mathbb{R}_+$ , we have

$$\#\{x \in V(F), H(x) \leq X\} \leq CX^{d(\dim V + 1/2)} \log^\gamma X.$$

┘

*Proof:* By Noetherian normalization (5.2.4.22), we can find  $\dim V$ -many coordinates s.t. the projection onto these coordinate is a finite map, and the hypothesis implies that  $\text{pr}(V(K))$  is a thin set (20.1.2.1), because there are no rational sections. Thus the assertion follows from (35.9.3.7).  $\square$

### Manin's Conjecture



## 35.10 Sato-Tate Conjecture

### 1 Modular Forms

References are [M-M10].

**Thm. (35.10.1.1) [Chebotarev-Sato-Tate for Modular Forms].** For  $f \in \text{SF}_k^{\text{new}}(\Gamma_0(N))$ , suppose that the  $\ell$ -adic Galois representation  $\rho_{\ell,f}$  attached to  $f$  is not dihedral, then when  $p \in \text{Prime}$  varies,  $a_p/2p^{\frac{k-1}{2}}$  are equidistributed in  $[-1, 1]$  with density measure  $\sqrt{1-t^2}dt$ .  $\square$

*Proof:* ?  $\square$

**Cor. (35.10.1.2) [Chebotarev-Sato-Tate [Tat65b], Clozel-Harris-Shepherd-Barron-Taylor].** Let  $F \in \text{NField}$  be totally real and  $E \in \text{EllCur}/F$ . For any  $v \in \Sigma_F^{\text{fin}}$ , define  $a_v = q_v + 1 - \#\tilde{E}_v(\kappa_v)$ , then by Weil conjecture,  $|a_v| \leq 2\sqrt{q_v}$ . Let  $a_v/2\sqrt{q_v} = \cos \theta_v$ ,  $0 \leq \theta \leq \pi$ . Then for any finite solvable Galois extension  $E/F$  and any conjugacy class  $C$  of  $\text{Gal}(E/F)$ , the density of  $v \in \Sigma_F^{\text{fin}}$  s.t.  $(v, E/F) \in C$  and  $\theta_v \in [\alpha, \beta]$  for  $0 \leq \alpha < \beta \leq \pi$  has density

$$\frac{2\#C}{\pi} \int_{\alpha}^{\beta} \sin^2 \theta d\theta.$$

Remark: We can also consider the joint distribution of the angles for two non-isogenous elliptic curves  $E_1, E_2$ , then by Tauberian theorems in [Serre, Abelian  $\ell$ -adic representations and elliptic curves], this amounts to prove prime number theorem for the tensor product of the symmetric powers of  $E_1$  and  $E_2$ . ?  $\square$

*Proof:* [Clozel-Harris-Taylor, Automorphy for some  $\ell$ -adic lifts of automorphic mod  $\ell$  representations], [A family of Calabi-Yau varieties and potential automorphy] ?.

By (3.4.1.3) and Weierstrass approximation theorem, it suffices to show that for any  $k \in \mathbb{Z}_+$ , we can find  $m, n \in \mathbb{N}$  s.t.  $m \neq n, m+n = k$ , and that

$$\lim_{X \rightarrow \infty} \frac{1}{\pi(X)} \sum_{p \leq X, p \in \text{Prime}} S_m(\cos \theta_p) S_n(\cos \theta_p) = 0.$$

By (35.10.1.3), it suffices to show the prime number theorem for  $L(\text{Sym}^m(E) \otimes \text{Sym}^n(E); s)$ .

It follows from (35.10.1.4) that  $\text{Sym}^m(E)$  and  $\text{Sym}^n(E)$  are jointly automorphic over a totally real Galois extension  $E/F$ . Then we can use [Arthur-Clozel] theory of base change to show that for any subfield  $F' \subset F$ , if  $\text{Gal}(F'/F)$  is solvable, then  $\text{Sym}^m(E)$  and  $\text{Sym}^n(E)$  are automorphic over  $F'$ . ?

Now use Brauer induction to write

$$1_{\text{Gal}(E/F)} = \sum_i a_i \text{Ind}_{H_i}^{\text{Gal}(E/F)} \psi_i$$

where  $a_i \in \mathbb{Z}$ ,  $H_i \subset \text{Gal}(E/F)$  are nilpotent subgroups of  $G_i$ ,  $\psi_i \in H_i^{\vee}$ . Then by Frobenius reciprocity and invariance of  $L$ -function under inductions, we have

$$\begin{aligned} L(\text{Sym}^m(E); s) &= \prod_i L(\text{Sym}^m(E) \otimes \text{Ind}_{H_i}^{\text{Gal}(E/F)} \psi_i; s)^{a_i} \\ &= \prod_i L(\text{Ind}_{H_i}^{\text{Gal}(E/F)} (\text{Sym}^m(E)|_{L^{H_i}} \otimes \psi_i); s)^{a_i} \end{aligned}$$

$$= \prod_i L(\mathrm{Sym}^m(E)|_{L_{H_i}} \otimes \psi_i; s)^{a_i}$$

Now by Jacquet-Shahidi(35.4.4.6), each term on the RHS is analytic and non-vanishing on  $\mathrm{Re}(s) \geq 1$ , thus so is the LHS.  $\square$

**Lemma (35.10.1.3) [Ikehara].** For  $m < n \in \mathbb{N}$ , if the prime number theorem holds for  $L(\mathrm{Sym}^m(E) \otimes \mathrm{Sym}^n(E); s)$  (i.e. holomorphic and non-vanishing for  $\mathrm{Re}(s) \geq 1$ ), then

$$\lim_{X \rightarrow \infty} \frac{1}{\pi(X)} \sum_{p \leq X, p \in \text{Prime}} S_m(\cos \theta_p) S_n(\cos \theta_p) = 0.$$

┘

*Proof:*

$\square$

**Lemma (35.10.1.4) [Potential Automorphy].** For any  $E \in \mathrm{EllCur}/\mathbb{Q}$  and  $m \in \mathbb{Z}_+$ , the symmetric power  $\mathrm{Sym}^m(H^1(E, \mathbb{Q}_\ell))$  is strongly potentially cuspidal automorphic(30.1.4.2).  $\square$

*Proof:* ? Cf.[Maz08]P223.

To show this fact, we need to use a Galois deformation theorem and a prime hopping method applied to a compatible family of  $n+1$  dimensional representations occurring as subquotients of the compatible family of Galois representations attached to the middle dimensional cohomology of  $Y_{t_0}$  where

$$Y_t : X_0^{n+1} + \dots + X_n^{n+1} = t(n+1)X_0X_1 \dots X_n.$$

$\square$

**Cor. (35.10.1.5) [Birch].** Given  $p \in \mathbf{P}$ , for any  $E \in \mathrm{EllCur}/\mathbb{F}_p$ , define  $a_v(E) = p+1 - \#E(\mathbb{F}_p)$  and  $a_v(E)/2\sqrt{p} = \cos \theta_p(E)$ ,  $0 \leq \theta \leq \pi$ . Then the distribution density of

$$\{\theta_p(E)\}_{E \in \mathrm{EllCur}/\mathbb{F}_p}$$

tends to  $\frac{2}{\pi} \sin^2 \theta$  for  $p \rightarrow \infty$  when  $p \rightarrow \infty$ .  $\square$

*Proof:* ?  $\square$

**Cor. (35.10.1.6).** Let  $\tau(n)$  be the Ramanujan's  $\tau$ -function, then  $\{\tau(p)/2p^{11/2}\}_{p \in \text{Prime}}$  are equidistributed in  $[-1, 1]$  w.r.t. the measure  $\frac{2}{\pi} \sqrt{1-t^2} dt$ .  $\square$

*Proof:* Cf.[A family of Calabi-Yau varieties and potential automorphy, 2, Lamb-Geraghty-Harris-Taylor]P32.  $\square$

### Independence of the Distributions

References are [Harris, Potential automorphy of odd-dimensional symmetric powers of elliptic curves].

## 2 Sato-Tate Groups

References are [Sato-Tate Groups, Kedlaya].

### More Serre-Tate Conjectures

**Remark (35.10.2.1).** For  $f \in \mathrm{SF}_k^{\mathrm{new}}(\Gamma_0(N))$  with associated automorphic representation  $\Pi$ , the Sato-Tate theorem (35.10.1.1) is equivalent to the assertion that  $\rho_{\ell,f}(\sigma_p)$  is equidistributed in the space of conjugacy classes of  $\mathrm{SU}(2)$ .

For a general motive  $M$ ,  $\mathrm{SU}(2)$  is replaced by the derived subgroup of the compact real form of the Mumford-Tate group  $\mathrm{MT}(M)$ . And to prove the Sato-Tate conjecture for  $M$ , one needs to prove that  $\sigma \circ \rho_{\ell,f}$  is potentially automorphic for any irreducible  $\overline{\mathbb{Q}}_\ell$ -representation  $\sigma$  of  $\mathrm{MT}(M)^\vee$ . But in general, this is not Hodge-Tate regular if  $\mathrm{MT}(M)^\vee$  has  $\mathrm{rank} > 1$ .  $\lrcorner$

### 35.11 Subconvexity Problems

References are [I-S00, M-V06].

Subconvexity problem investigates the growth of absolute values of  $L$ -functions along the central line.

**Notation (35.11.0.1).**

- Let  $F \in \mathbf{NField}$ .
- Let  $n \in \mathbb{Z}_+$ .
- Let  $\pi \in \mathrm{Irr}^{\mathrm{cusp}}(\mathrm{GL}(n)/F)$ .

┘

**Def. (35.11.0.2) [Analytic Conductors].** For an analytic  $L$ -function  $(L(f; s), L_\infty(f; s), N(f))$  of degree  $d \in \mathbb{Z}_+$  (29.5.1.1), the **analytic conductor** for  $L(f; s)$  is defined to be the function

$$C_\infty(f; s) = \prod_{i=1}^d (3 + |\kappa_i + s|), \quad C(f; s) = N(f)C_\infty(f; s).$$

And also denote  $C(f) = C(f; 0)$ .

┘

**Prop. (35.11.0.3) [Bounding Gamma Factors].** If  $f$  is an analytic  $L$ -function of degree  $d$  with local parameters  $\{\kappa_i\}$  at  $\infty$ , then for  $-1/2 \leq \mathrm{Re}(s) < 2$ , when  $s = \sigma + it$  and  $\lim |s| \rightarrow \infty$ , we have

$$|L_\infty(f; s)| \prod_i |s + \kappa_i| \sim C_\infty(f; s)^{\frac{1}{2}(\kappa + \sigma + 1)} \exp \left( -\frac{\pi}{4} \sum_i |t + \mathrm{Im}(\kappa_i)| + C_d \right),$$

where  $\kappa = (\sum_i \mathrm{Re}(\kappa_i))/d$  and  $C_d$  is a constant depending on  $d$ . And

$$\frac{L_\infty(1-s)}{L_\infty(s)} \sim C_\infty(f; s)^{\frac{1}{2}-\sigma} \prod_j \frac{s + \kappa_j}{1 - s + \kappa_j}.$$

┘

*Proof:* This follows from Stirling's formula (12.6.1.16). □

**Prop. (35.11.0.4) [Polynomial Growth on Vertical Strips].** Any analytic  $L$ -function  $L(f; s)$  has uniformly polynomial growth on vertical strips  $a \leq \mathrm{Re}(s) \leq b$  for any  $a < b \in \mathbb{R}$ . ┘

*Proof:* For any  $\varepsilon > 0$ ,  $L(f; s)$  is bounded on  $\{s \in \mathbb{C} : \mathrm{Re}(s) > 1 + \varepsilon\}$ , and then it is also polynomially bounded on the half-plane  $\{s \in \mathbb{C} : \mathrm{Re}(s) < \varepsilon\}$  by functional equation and (35.11.0.3). Then the assertion now follows from the Phragmen-Lindelöf principle (12.4.5.7). □

**Thm. (35.11.0.5) [Approximate Functional Equations].** If  $L(f; s)$  is an analytic  $L$ -function of degree  $d$  and  $G(u)$  is any even function that is holomorphic and bounded on the vertical strip  $-4 < \mathrm{Re}(s) < 4$  with  $G(0) = 1$ , then for any  $X \in \mathbb{R}_+$  and  $s \in \mathbb{C}$  s.t.  $0 \leq \mathrm{Re}(s) \leq 1$ , we have

$$L(f; s) = \sum_{n \in \mathbb{Z}_+} \frac{a_f(n)}{n^s} V_s \left( \frac{n}{X \sqrt{N(f)}} \right) + \Upsilon(f; s) \sum_{n \in \mathbb{Z}_+} \frac{\overline{a_f(n)}}{n^{1-s}} V_{1-s} \left( \frac{nX}{\sqrt{N(f)}} \right) + R,$$

where  $V_s(y) \in C^\infty(\mathbb{R}_+)$  is defined by

$$V_s(y) = \frac{1}{2\pi i} \int_{\operatorname{Re}(s)=3} y^{-u} G(u) \frac{L_\infty(f; s+u)}{L_\infty(f; s)} \frac{du}{u},$$

and

$$\Upsilon(f; s) = w(f) N(f)^{\frac{1}{2}-s} \frac{L_\infty(f; 1-s)}{L_\infty(f; s)}.$$

$$R = (\operatorname{Res}_{u=1-s} + \operatorname{Res}_{u=-s}) \left( \frac{\Lambda(f; s+u)}{N(f)^{s/2} L_\infty(f; s)} \frac{G(u)}{u} X^u \right),$$

which vanishes when  $f$  is entire.  $\lrcorner$

*Proof:* Consider the integral

$$I(X, f; s) = \frac{1}{2\pi i} \int_{\operatorname{Re}(s)=3} X^u \Lambda(f; s+u) G(u) \frac{du}{u},$$

which is absolutely convergent because  $\Lambda(f; s)$  decays rapidly on vertical strips by (35.11.0.3) and (35.11.0.4). Plug in functional equation into  $I(X^{-1}, \bar{f}, 1-s)$  and move the integration to the line  $\operatorname{Re}(s) = -3$  using Residue theorem, we get

$$\Lambda(f; s) = I(X, f, s) + w(f) I(X^{-1}, \bar{f}, 1-s) + N(f)^{s/2} L_\infty(f; s) R. \quad (\star)$$

Expanding  $\Lambda(f; s)$  into absolutely convergent Dirichlet series, we get

$$\begin{aligned} I(X, f, s) &= N(f)^{s/2} \sum_{n \in \mathbb{Z}_+} \frac{\lambda_f(n)}{n^{-s}} \frac{1}{2\pi i} \int_{\operatorname{Re}(s)=3} L_\infty(f; s+u) \left( \frac{X \sqrt{N(f)}}{n} \right)^u G(u) \frac{du}{u} \\ &= N(f)^{s/2} L_\infty(f; s) \sum_{n \in \mathbb{Z}_+} \frac{\lambda_f(n)}{n^{-s}} V_s \left( \frac{n}{X \sqrt{N(f)}} \right), \end{aligned}$$

And  $V_s(y)$  is smooth because it is absolutely convergent. We can do the same for  $I(X^{-1}, \bar{f}, 1-s)$ , and the the assertion follows from  $(\star)$  above.  $\square$

**Cor. (35.11.0.6) [Central Value Formulae].** For  $f \in \operatorname{SF}_2^{\text{new}}(\Gamma(N))$  and  $X \in \mathbb{R}_+$ , we have

$$L(f; \frac{1}{2}) = \sum_{n \in \mathbb{Z}_+} \frac{a_f(n)}{\sqrt{n}} \exp \left( -\frac{2\pi n}{X \sqrt{N}} \right) + w(f) \sum_{n \in \mathbb{Z}_+} \frac{\overline{a_f(n)}}{\sqrt{n}} \exp \left( -\frac{2\pi n X}{\sqrt{N}} \right).$$

$\lrcorner$

*Proof:* In (35.11.0.5), take  $G(u) = 1$ , then  $L_\infty(f; s) = L_{\mathbb{C}}(s)$ , and

$$V_s(y) = e^{-2\pi y}.$$

$\square$

**Cor. (35.11.0.7) [A Choice of  $G(u)$ ].** Situation as in (35.11.0.5), suppose the local parameters  $\{\kappa_i\}$  of  $f$  at  $\infty$  and  $s \in \mathbb{C}$  s.t.  $0 \leq \operatorname{Re}(s) \leq 1$  with  $\operatorname{Re}(s + \kappa_i) \geq 3\alpha \in \mathbb{R}_+$  for any  $i \in [d]_+$ . For  $A \in \mathbb{Z}_+$ , if we take  $G(u) = \cos(\frac{\pi u}{4A})^{-4dA}$ , then the derivatives of  $V_s(y)$  satisfies:

$$y^a V_s^{(a)}(y) \ll_{\alpha, a, A, d} \left(1 + \frac{y}{\sqrt{C_\infty(f)}}\right)^{-A},$$

$$y^a V_s^{(a)}(y) = \delta_{a,1} + O_{\alpha, a, A} \left( \left( \frac{y}{\sqrt{C_\infty(f)}} \right)^\alpha \right).$$

┘

*Proof:* For  $s, u \in \mathbb{C}$  with  $\operatorname{Re}(s) = \sigma > 0$  and  $\operatorname{Re}(u) = \beta > -\sigma$ , we have

$$\frac{\Gamma(s+u)}{\Gamma(s)} \ll \frac{|s+u|^{\sigma+\beta-\frac{1}{2}}}{|s|^{\sigma-\frac{1}{2}}} \exp\left(\frac{\pi}{2}(|s| - |s+u|)\right) \ll_{\sigma, \beta} (|s| + 3)^\beta e^{\pi|u|/2}.$$

Hence

$$\frac{L_\infty(f; s+u)}{L_\infty(f; s)} \ll_{\alpha, \beta} C_\infty(f)^{\beta/2} e^{\pi d|u|/2}.$$

Now

$$y^a V_s^{(a)}(y) = \frac{1}{2\pi i} \int_{u=3} y^{-u} G(u) (-u)^a \frac{L_\infty(f; s+u)}{L_\infty(f; s)} \frac{du}{u}.$$

Moving the integration to the line  $\operatorname{Re}(u) = \beta = -\alpha$  or  $\operatorname{Re}(u) = A$ , we get the desired assertion. ?

□

**Thm. (35.11.0.8) [Convexity Bound].** If  $L(f; s)$  is an analytic  $L$ -function of degree  $d$ , then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\kappa_{f, \varepsilon} \in \mathbb{R}_+$  s.t. for any  $s = \sigma + it \in \mathbb{C}$  with  $0 \leq \sigma \leq 1$ , we have

$$\left| \left( \frac{s+1}{s-1} \right)^{\operatorname{ord}_{s=1} L(f; s)} L(f; s) \right| \leq \kappa_{f, \varepsilon} C(f; s)^{\frac{1-\sigma}{2} + \varepsilon}.$$

In particular, for  $\sigma = 1/2$ , we have

$$|L(f; s)| \leq \kappa_{f, \varepsilon} C(f; s)^{\frac{1}{4} + \varepsilon}.$$

And if  $f$  satisfies the generalized Ramanujan-Selberg conjecture (35.8.0.1), then there exists  $\kappa_d \in \mathbb{R}_+$  s.t. for  $\sigma = 1/2$ , we have

$$|L(f; s)| \leq \kappa_d C(f; s)^{1/4} \log(C(f; s))^{d-1}.$$

┘

*Proof:* The desired bound holds for  $\sigma = 1 + 2\varepsilon$  because  $L(f; s)$  is absolutely convergent there. And it also holds for  $\sigma = -2\varepsilon$  by functional equation and (35.11.0.3). Thus it holds for all  $\sigma$  by Phragmen-Lindelöf (12.4.5.7).

If the generalized Ramanujan-Selberg conjecture holds, use the approximate Functional equation ?. □

**Cor. (35.11.0.9).** For Dirichlet characters, the Ramanujan-Selberg conjecture trivially holds, so there exists  $\kappa \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_+$  and  $\chi \in \operatorname{Dir}'(N)$ , we have

$$|L(\chi; s)| \leq \kappa(N|s|)^{1/4}.$$

┘

Lindelöff Hypothesis

**Conj. (35.11.0.10) [Lindelöff Hypothesis].** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\kappa_{\pi,\varepsilon} \in \mathbb{R}_+$  s.t.

$$\left| L\left(\pi; \frac{1}{2} + it\right) \right| \leq \kappa_{\pi,\varepsilon} C(\pi; t)^\varepsilon \text{ (35.11.0.2)}$$

for any  $t \in \mathbb{R}$ . ┘

*Proof:* □

**Remark (35.11.0.11).** The Lindelöff hypothesis follows from GRH. ? ┘

**Conj. Cor. (35.11.0.12) [Subconvexity].** There exists  $\delta \in \mathbb{R}_+$  and  $\kappa_{\pi,\delta} \in \mathbb{R}_+$  s.t.

$$\left| L\left(\pi; \frac{1}{2} + it\right) \right| \ll \kappa_{\pi,\delta} C(\pi; t)^{\frac{1}{4}-\delta}.$$

*Proof:* ? □

**Thm& Conj. Cor. (35.11.0.13) [Convexity Bound].** It follows from the functional equation and the Phragmen-Lindelöff principle that for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $\kappa_{\pi,\varepsilon} \in \mathbb{R}_+$  s.t.

$$\left| L\left(\pi; \frac{1}{2} + it\right) \right| \leq \kappa_{\pi,\varepsilon} C(\pi; t)^{\frac{1}{4}+\varepsilon}$$

for any  $t \in \mathbb{R}$ . ┘

*Proof:* ? Cf. [Upper and lower bounds at s=1 for certain Dirichlet series with Euler product]. □

**1 Dirichlet  $L$ -Functions**

**Thm. (35.11.1.1) [Weyl21].**

$$\left| \zeta\left(\frac{1}{2} + it\right) \right| \ll (|t| + 1)^{\frac{1}{6}+\varepsilon}$$

for any  $t \in \mathbb{R}$ . ┘

*Proof:* ? This is proved using Weyl's bound on exponential sums of polynomial functions. □

**Thm. (35.11.1.2) [Subconvexity in the Level aspect, Burgess [Bur63a]].** For  $t \in \mathbb{R}$  and  $\varepsilon \in \mathbb{R}_+$ , there exists  $\kappa_{\varepsilon,t} \in \mathbb{R}_+$  s.t.

$$\left| L\left(\chi; \frac{1}{2} + it\right) \right| \leq \kappa_{\varepsilon,t} N^{\frac{3}{16}+\varepsilon}$$

for any  $N \in \mathbb{Z}_+$  and  $\chi \in \text{Diri}(N)$ . ┘

*Proof:* ? The proof proceeds by estimating the sum

$$S = \sum_{N < n \leq N+H} \chi(n)$$

for certain  $N$  and  $H$ . □

**Thm. (35.11.1.3)** [Subconvexity in the Level aspect, Conrey-Iwaniec[?C-I00]]. For any  $t \in \mathbb{R}$ , there exists  $A_t \in \mathbb{N}$  s.t. for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $C_\varepsilon \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{N}_+$  and real non-principal character  $\chi \in \text{Diri}(N)$ , we have

$$|L(\chi; \frac{1}{2} + it)| \leq C_\varepsilon (1 + |t|)^{A_t} N^{1/6+\varepsilon}.$$

┘

*Proof:* Cf.[Conrey-Iwaniec, Cubic moment of central values of automorphic  $L$ -functions].? □

## 2 $\text{GL}(2)$

**Thm. (35.11.2.1)** [Subconvexity in the Level aspect, Duke-Friedlander-Iwaniec[DFI93, DFI94, ?DFI02]/ Kowalski-Michel-VanderKam[KMV02]]. For any  $F \in \mathbb{N}\text{Field}$ , there exists  $\delta \in \mathbb{R}_+$  s.t. for any  $\chi \in \text{Irr}^{\text{auto}}(\text{GL}(1)/F)$ ,  $\pi_1, \pi_2 \in \text{Irr}^{\text{auto}}(\text{GL}(2)/F)$  and  $s \in \mathbb{C}$  with  $\text{Re } s = 1/2$ , we have

$$L(\pi_1(\chi); s) \ll_{s, \pi_1, \chi, \infty} (\text{Nm}_{F/\mathbb{Q}} \mathfrak{c}(\chi))^{1/2-\delta},$$

$$L(\pi_1; s) \ll_{s, \pi_1, \infty} (\text{Nm}_{F/\mathbb{Q}} \mathfrak{c}(\pi_1))^{1/4-\delta},$$

$$L(\pi_1 \times \pi_2; s) \ll_{s, \pi_1, \pi_2, \infty} (\text{Nm}_{F/\mathbb{Q}} \mathfrak{c}(\pi_1))^{1/2-\delta}.$$

┘

*Proof:* ? □

**Cor. (35.11.2.2).** For any  $F \in \mathbb{N}\text{Field}$ , there exists  $\delta \in \mathbb{R}_+$  s.t. for any  $\rho : \text{Gal}_F \rightarrow \text{GL}(2; \mathbb{C})$  a modular Artin representation and  $s \in \mathbb{C}$  with  $\text{Re}(s) = 1/2$ , we have

$$L(\rho; s) \ll_{F, s} \text{Nm}_{F/\mathbb{Q}}(\mathfrak{f}_\rho)^{1/4-\delta}.$$

┘

*Proof:* ? □

**Cor. (35.11.2.3).** For any  $F \in \mathbb{N}\text{Field}$ , there exists  $\delta \in \mathbb{R}_+$  s.t. for any Abelian or cubic extension  $L/F \in \mathbb{N}\text{Field}$  and  $s \in \mathbb{C}$  with  $\text{Re}(s) = 1/2$ , we have

$$\zeta_L(s) \ll_{F, s} |\text{disc}(L)|^{1/4-\delta}.$$

┘

*Proof:* ? □

**Cor. (35.11.2.4).** For any  $F \in \mathbb{N}\text{Field}$  and  $\pi \in \text{Irr}^{\text{cusp}}(\text{GL}(2)/F)$ , there exists  $\delta \in \mathbb{R}_+$  s.t. for any Abelian or cubic extension  $L/F \in \mathbb{N}\text{Field}$  and  $s \in \mathbb{C}$  with  $\text{Re}(s) = 1/2$ , we have

$$L(\text{BC}_{L/F}(\pi); s) \ll_{F, \pi, s} |\text{disc}(L)|^{1/4-\delta}.$$

┘

*Proof:* ? □



### Period Approach to Subconvexity

#### 3 Bounds on Fourier Coefficients

**Thm. (35.11.3.1)** [Iwaniec[Iwa87]/Duke[Duk88]]. Let  $F(z) = \sum_{n \in \mathbb{Z}_+} a(n) e^{2\pi i n z}$  be a holomorphic cusp form of weight  $k = \ell + \frac{1}{2}$  for  $\Gamma_0(4N)$  with

$$\int_{\widehat{\Gamma}_0(4N) \backslash \mathcal{H}} |F(z)|^2 y^k \frac{dx dy}{y^2} = 1,$$

then

$$a(n) \ll_{k,\varepsilon} n^{k/2-1/4+\varepsilon}.$$

Moreover, if  $k \geq 5/2$  and  $n$  is square-free, then

$$a(n) \ll_{k,\varepsilon} n^{k/2-2/7+\varepsilon}.$$

┘

*Proof:*

□

#### 4 Moments

References are [Integral moments of  $L$ -functions, Conrey-Farmer-Keating-Rubinstein-Snaith] and [Random Matrix Theory and  $L$ -Functions at  $s = 1/2$ , Keating-Snaith], [Random Matrix Theory and  $\zeta(1/2 + it)$ , Keating-Snaith], [Random matrix theory and  $L$ -functions at  $s=1/2$ , Keating-Snaith].

**Def. (35.11.4.1) [Moments of  $L$ -Functions]**. An important problem in analytic number theory is to gain an understanding of the moments

$$\mathbf{M}_\zeta(k; T) = \int_0^T \left| \zeta\left(\frac{1}{2} + it\right) \right|^{2k} dt$$

for  $k \in \mathbb{R}_+$ .

┘

**Conj. (35.11.4.2) [Conjectured by Keating-Saith/Conrey-Keating-Snaith]**.  $\mathbf{M}_\zeta(k; T) \sim C_k T (\log T)^{k^2}$  for some specific  $C_k \in \mathbb{R}_+$ .

┘

*Proof:*

□

### Moments in Families

**Conj. Cor. (35.11.4.3) [Moments for Families of Dirichlet Series]**. For any  $k \in \mathbb{N}$ , there exists  $C_k \in \mathbb{R}_+$  s.t. for  $N \in \mathbb{Z}_+$ ,

$$\sum_{\chi \in \text{Diri}'(N)} \left| L\left(\chi; \frac{1}{2}\right) \right|^{2k} \sim C_k q \log^{k^2} q.$$

┘

*Proof:*

□

**Conj.Cor.(35.11.4.4).** For any  $k \in \mathbb{N}$ , there exists  $C_k \in \mathbb{R}_+$  s.t. for  $X \in \mathbb{R}_+$ ,

$$\sum_{|d| \leq X} L(\chi_d; \frac{1}{2})^k \sim C_k X \log^{k(k+1)/2} X.$$

┘

*Proof:*

□

**Conj.Cor.(35.11.4.5).** For any  $k \in \mathbb{N}$  and a modular new form  $f$ , there exists  $C_k \in \mathbb{R}_+$  s.t. for  $X \in \mathbb{R}_+$ ,

$$\sum_{|d| \leq X} L(f \otimes \chi_d; \frac{1}{2})^k \sim C_k X \log^{k(k-1)/2} X.$$

┘

*Proof:*

□

### Lower Bounds

References are [Lower bounds for moments of  $L$ -functions: symplectic and orthogonal examples, Rudnick-Soundararajan].

**Thm.(35.11.4.6)**[Moments for Families of Dirichlet Series, Rudnick-Soundararajan[R-S05]].

For  $k \in \mathbb{N}$ , there exists  $C_k \in \mathbb{R}_+$  s.t. for any  $q \in \text{Prime}$ ,

$$\sum_{\mathbf{1} \neq \chi \in \text{Diri}(q)} \left| L(\chi; \frac{1}{2}) \right|^{2k} \geq C_k q \log^{k^2} q.$$

┘

*Proof:* Denote  $x = q^{1/(2k)}$  and  $A(\chi) = \sum_{n \in [x]_+} \chi(n) / \sqrt{n}$ . Define

$$S_1 = \sum_{\mathbf{1} \neq \chi \in \text{Diri}(q)} L(\chi; \frac{1}{2}) A(\chi)^{k-1} \overline{A(\chi)}^k, \quad S_2 = \sum_{\mathbf{1} \neq \chi \in \text{Diri}(q)} |A(\chi)|^{2k},$$

then it suffices to show that

$$S_2 \ll q \log^{k^2} q \ll S_1,$$

because then by Hölder' inequality,

$$\sum_{\mathbf{1} \neq \chi \in \text{Diri}(q)} \left| L(\chi; \frac{1}{2}) \right|^{2k} \geq \frac{|S_1|^{2k}}{S_2^{2k-1}} \gg q \log^{k^2} q.$$

Notice that for  $l \in \mathbb{Z}_+$ ,  $A(\chi)^l = \sum_{n \in [x^l]_+} \chi(n) d_l(n; x) / \sqrt{n}$  where  $d_l(n; x)$  is the number of ways to write  $n$  as a sum of an ordered set of  $l$  numbers in  $[x]_+$ .

Because  $A(\mathbf{1}) = O(\sqrt{x})$ , we have

$$S_2 = \sum_{\chi \in \text{Diri}(q)} A(\chi)^{2k} + O(x^k)$$

$$\begin{aligned}
&= \sum_{m,n \leq x^k} \frac{d_k(m; x) d_k(n; x)}{\sqrt{mn}} \sum_{\chi \in \text{Diri}(q)} \chi(m) \bar{\chi}(n) \\
&\text{(orthogonality)} = \phi(q) \frac{d_k(n; x)^2}{n} + O(\sqrt{q}) \\
&\ll \phi(q) \frac{d_k(n)^2}{n} + O(\sqrt{q}) \\
&\text{(35.1.3.14)} = C_k \log^{k^2} y + q \log^{k^2} q.
\end{aligned}$$

For  $S_1$ , by (35.11.4.7), for any  $\mathbf{1} \neq \chi \in \text{Diri}(q)$  and  $X \in \mathbb{R}_+$ , we have

$$L(\chi; \frac{1}{2}) = \sum_{n \in [X]_+} \frac{\chi(n)}{\sqrt{n}} + O\left(\frac{\sqrt{q} \log q}{\sqrt{X}}\right).$$

Then we choose  $X = q \log^4 q$  and obtain

$$\begin{aligned}
S_1 &= \sum_{\mathbf{1} \neq \chi \in \text{Diri}(q)} \sum_{n \in [X]_+} \frac{\chi(n)}{\sqrt{n}} A(\chi)^{k-1} \overline{A(\chi)}^k + O\left(\frac{1}{\log q} \sum_{\mathbf{1} \neq \chi \in \text{Diri}(q)} |A(\chi)|^{2k-1}\right) \\
&= \sum_{\chi \in \text{Diri}(q)} \sum_{n \in [X]_+} \frac{\chi(n)}{\sqrt{n}} A(\chi)^{k-1} \overline{A(\chi)}^k + O\left(\frac{S_2}{\log q} + q \log^2 q\right) \\
&\text{(orthogonality)} = \phi(q) \sum_{b \in [x^k]_+} \sum_{a \in [x^{k-1}]_+, n \in [X]_+, an \equiv b \pmod{q}} \frac{d_{k-1}(a; x) d_k(b; x)}{\sqrt{abn}} + O\left(q \log^{k^2-1} q\right)
\end{aligned}$$

The main contribution comes from the terms with  $an = b$ : These terms sum to

$$\begin{aligned}
&\phi(q) \sum_{b \in [x^k]_+} \frac{d_k(b; x)}{b} \sum_{a \in [x^{k-1}]_+, n \in [X]_+, an=b} + O\left(q \log^{k^2-1} q\right) \\
&\geq \phi(q) \sum_{b \in [x^k]_+} \frac{d_k(b; x)}{b} \sum_{a \in [x^{k-1}]_+, n \in [x]_+, an=b} + O\left(q \log^{k^2-1} q\right) \\
&= \phi(q) \sum_{b \in [x^k]_+} \frac{d_k(b; x)^2}{b} + O\left(q \log^{k^2-1} q\right) \\
&\text{(35.1.3.14)} \geq \phi(q) \sum_{b \in [x]_+} \frac{d_k(b)^2}{b} + O\left(q \log^{k^2-1} q\right) \geq q \log^{k^2} q.
\end{aligned}$$

And for the off-diagonal terms, we write  $an = b + ql$ , where  $l \in [Xx^{k-1}/q]_+ = [x^{k-1} \log^4 q]_+$ , then these terms is bounded by

$$\begin{aligned}
&q \sum_{b \in [x^k]_+} \frac{d_k(b; x)}{\sqrt{b}} \sum_{l \in [x^{k-1} \log^4 q]_+} \frac{1}{\sqrt{ql}} \sum_{an=b+ql} d_{k-1}(a; x) + O\left(q \log^{k^2-1} q\right) \\
&\ll q^{\frac{1}{2}+\varepsilon} x^{\frac{k-1}{2}+\frac{k}{2}} \ll \frac{q}{\log q},
\end{aligned}$$

because  $\sum_{an=b+ql} d_{k-1}(a; x) \leq d_k(b+ql) \ll (ql)^\varepsilon$  for any  $\varepsilon \in \mathbb{R}_+$  (2.6.3.2).

Thus  $S_1 \gg q \log^{k^2} q$  is proved, and the assertion follows.  $\square$

**Lemma (35.11.4.7)[Integral Representation of  $L(\chi; s)$ ].** Let  $q \in \mathbb{Z}_{\geq 2}$ ,  $\chi \in \text{Diri}(q) \setminus \mathbf{1}_q$  and  $X \in \mathbb{R}_+$ , then for any  $s \in \mathbb{C}$  s.t.  $\text{Re}(s) > 1$ , integration by parts gives

$$L(\chi; s) = \sum_{n \in [X]_+} \frac{\chi(n)}{n^s} + \int_X^\infty \frac{1}{y^s} d \left( \sum_{X < n \leq y} \chi(n) \right) = \sum_{n \in [X]_+} \frac{\chi(n)}{n^s} + s \int_X^\infty \frac{\sum_{X < n \leq y} \chi(n)}{y^{s+1}} dy.$$

By Pólya-Vinogradov inequality (35.3.3.4),  $|\sum_{X < n \leq y} \chi(n)| \leq 6\sqrt{q} \log q$ , so the integration extends to an analytic function to  $s \in \mathbb{C}$  s.t.  $\text{Re}(s) > 0$ .  $\lrcorner$

### Upper Bounds

References are [Soundararajan, Moments of the Riemann zeta function].

**Thm. (35.11.4.8)[Moments of Zeta Functions].**  $M_\zeta(1; T) \ll T \log T$  (35.11.4.1).  $\lrcorner$

*Proof:* Cf. [I-K04]P194.  $\square$

**Thm. (35.11.4.9)[Moments for Families of Dirichlet Series].**

$$\sum_{q \leq Q} \sum_{\chi \in \text{Diri}'(q)} \left| L(\chi; \frac{1}{2} + it) \right|^8 \ll Q^2(t^2 + 1) \log^{17}(Q(|t| + 2)).$$

$\lrcorner$

*Proof:* For  $\chi \in \text{Diri}'(q)$  and  $s \in \mathbb{C}$  s.t.  $\text{Re}(s) > 1$ , we have:

$$L(\chi; s)^4 = \sum_{n \in \mathbb{Z}_+} \chi(n) \tau_4(n) n^{-s}.$$

Using approximate functional equation for  $L(\chi; s)^4$  (35.11.0.5) with chosen  $G(u)$  (35.11.0.7), we get

$$L(\chi; \frac{1}{2} + it)^4 = \sum_{n \in \mathbb{Z}_+} \frac{\chi(n) \tau_4(n)}{n^{1/2+it}} V_{1/2+it}(\frac{n}{q^2}) + w_\chi^4 q^{-4it} \frac{L_\infty(\chi; \frac{1}{2} + it)}{L_\infty(\chi; \frac{1}{2} - it)} \sum_{m \in \mathbb{Z}_+} \frac{\bar{\chi}(m) \tau_4(m)}{m^{1/2-it}} V_{1/2-it}(\frac{n}{q^2}),$$

where  $L_\infty(\chi; s) = \pi^{-2s} \Gamma(\frac{s+\kappa}{2})^4$  for  $\kappa = \frac{1-\chi(-1)}{2}$ .

Using the fact  $\tau_4(n) = n^\varepsilon$  and the fact  $V_s(y)$  decay rapidly (35.11.0.7), I can only truncate the summation at  $N = Q^{2+\varepsilon}(t^2 + 1)$  to get the bound

$$\sum_{q \leq Q} \sum_{\chi \in \text{Diri}'(q)} \left| L(\chi; \frac{1}{2} + it) \right|^8 \ll_\varepsilon Q^{2+\varepsilon}(t^2 + 1) \log^{17}(Q(|t| + 2))$$

$\square$

**Thm. (35.11.4.10)[Moments for Families of Modular Forms].** Let  $\mathcal{F}$  be a basis of Hecke eigenforms for  $\text{SF}_2(N)$ , then

$$\sum_{f \in \mathcal{F}} \left| L(f; \frac{1}{2} + it) \right|^8 \ll N(t^2 + 1) \log^{17}(N(|t| + 2)).$$

$\lrcorner$

*Proof:* Cf. [I-K04]P194.  $\square$

## 5 Equidistributions

References are [M-V06].

Subconvexity bounds can be used to prove equidistribution results.

### Quadratic Forms and Heegner Points

**Thm. (35.11.5.1)** [Linnik[Lin55, Lin60]/Skubenko[Sku62]/Duke[Duk88]].

- For  $d \in \mathbb{Z}_+$  and  $d \not\equiv 0, 1, 4 \pmod{8}$ , when  $d \rightarrow -\infty$ , the set  $\mathcal{G}_d$  which is the projection of the set  $V_d = \{(x, y, z) \in \mathbb{Z}^3, a^2 + b^2 + c^2 = d\}$  onto the unit sphere is equidistributed in  $\mathbb{S}^2$  w.r.t. the Lebesgue measure.
- If  $d \in \mathbb{Z}_+$  is square-free and  $d \equiv 0, 1 \pmod{4}$ , then for any convex subset  $\Omega \subset F$  with smooth boundary (where  $F$  is the standard fundamental domain of  $\Gamma(1)$ ), there exists  $\delta, \kappa \in \mathbb{R}_+$  s.t. :
  - For  $d \rightarrow -\infty$ ,

$$\left( \frac{\#\Lambda_d \cap \Omega}{\#\Lambda_d} - \text{Vol}(\Omega) \right) \leq \kappa |d|^{-\delta},$$

- For  $d \rightarrow +\infty$ ,

$$\left( \frac{\sum_{C \in \Lambda_d} \text{length}(C \cap \Omega)}{\sum_{C \in \Lambda_d} \text{length}(C)} - \text{Vol}(\Omega) \right) \leq \kappa |d|^{-\delta}.$$

┘

*Proof:* ?

□

**Conj. (35.11.5.2)** [Sparse Equidistribution of Heegner Points]. For any  $\eta \in \mathbb{R}_+$ , suppose for each fundamental discriminant  $d < 0$  we are given a Heegner point  $z_{0,d}$  and a subgroup  $G_d \subset \text{Cl}(d)$  with  $\#G_d \geq |d|^\eta$ , then for  $d \rightarrow -\infty$ , the sequence

$$\mathcal{H}'_d = G_d \cdot z_{0,d}$$

is equidistributed on  $Y_0(1)$ .

┘

*Proof:*

□

**Thm& Conj. Cor. (35.11.5.3)** [Harcos-Michel[H-M06]]. There exists  $\eta \in (0, 1)$  s.t. the conjecture (35.11.5.2) holds.

┘

*Proof:*

□

**Conj. (35.11.5.4)** [Mixing Conjecture for Heegner Points]. Suppose for each fundamental discriminant  $d < 0$ , an element  $\sigma_d \in \text{Cl}(d)$  (19.3.3.1) is given s.t. the minimal norm of the integral ideal representing the class  $\sigma_d$  tends to  $\infty$  as  $|d| \rightarrow \infty$ , then the sequence

$$\mathcal{H}''_d = \{(z, \sigma_d z) : z \in \mathcal{H}_d\}$$

is equidistributed on  $Y_0(1) \times Y_0(1)$ .

┘

*Proof:*

□

### Integers Representable by a Quadratic Form(Hilbert's 11-th Problem)

For  $F \in \mathbf{NField}$ , we ask the question of when an algebraic integer  $\nu$  is representable by a given quadratic form  $f(x_1, \dots, x_n)$ .

For  $n = 2$ , this is solved via class field theory, so we assume that  $n \geq 3$ .

The question of when a given  $\nu$  is representable by a form in the genus of  $f$  is solved by the Siegel's mass formula<sup>?</sup>, which gives the number of solutions in terms of local data, via the product of local masses.

If  $n \geq 3$  and  $f$  is indefinite at an Archimedean place of  $K$ , then Kneser's result [Darstellungsmasse indefiniter quadratischer Formen1961] on the class numbers and the spinor genus of  $f$ <sup>?</sup> shows that essentially there are only one class in the genus of  $f$ .<sup>?</sup>

**Thm. (35.11.5.5)[Siegel's Mass Formula].**

┘

*Proof:*   <sup>?</sup>

□

## 35.12 Correlations of $L$ -Functions(Second Moment method)

### 1 Chowla's Conjecture

**Conj. (35.12.1.1)[Chowla].** Let  $\lambda$  be the Liouville function, then for any  $k \in \mathbb{Z}_+$  and distinct numbers  $h_1, \dots, h_k \in \mathbb{N}$ , we have

$$\sum_{1 \leq n \leq X} \lambda(n + h_1) \dots \lambda(n + h_k) = o(X)$$

as  $X \rightarrow \infty$ .

┘





## Part VI

# Miscellaneous



## 36 | Computational Number Theory

### 36.1 Computational Number Theory

References are [Princeton Companion] and [Pi and the AGM. A study in Analytic Number theory and computational complexity].

#### 1 Primality Testing

**Algo. (36.1.1.1) [AKS Primality Proving, Agrawal-Kayal-Saxena [AKS04]].** Consider the following algorithm:

**Input:** An integer  $n \in \mathbb{Z}_{\geq 2}$

**Output:** PRIME or COMPOSITE

1. **if**  $n = a^b$  for some  $a \in \mathbb{Z}_+, b \in \mathbb{Z}_{\geq 2}$  **then return** COMPOSITE;
2. Find the smallest number  $r$  s.t.  $\text{ord}_r(n) > \log^2 n$ ;
3. **if**  $1 < \gcd(a, n) < n$  for some  $a \leq r$  **then return** COMPOSITE;
4. **if**  $n \leq r$  **then return** PRIME;
5. **for**  $a \leftarrow 1$  **to**  $\lfloor \sqrt{\varphi(r) \log n} \rfloor$  **do**  
     **if**  $(X + a)^n \not\equiv X^n + a \pmod{(X^r - 1, n)}$  **then return** COMPOSITE;
6. **return** PRIME;

Then for any input  $n \in \mathbb{Z}_+$ , this algorithm returns PRIME iff  $n \in \text{Prime}$ . And its asymptotic time complexity is  $\tilde{O}(\log^{15/2} n)$ .

In particular, deciding the language  $\text{Prime} \subset \mathbb{N}$  is in the class  $P(2.7.4.1)$ . ┘

**Remark (36.1.1.2) [Lenstra-Pomerance, Primality testing with gaussian periods].** has an improvement that has complexity  $\tilde{O}(\log^6 n)$ . ┘

*Proof:* To prove the algorithm is correct: Notice if  $n$  is prime, then it can only return prime. And the converse is proved in (36.1.1.3).

To analyze the asymptotic time complexity, we use [Modern Computer Algebra].

The first step of the algorithm takes asymptotic time  $\tilde{O}(\log^3 n)$ .

For the second step, use (35.5.10.4). To find such an  $r$ , we check for each  $r < \log^{3+\varepsilon}$  and each  $k < \log^2 n$  whether  $r | n^k - 1$ . For each  $r$ , the complexity is  $\tilde{O}(\log^2 n \log r)$ , so the asymptotic complexity is  $\tilde{O}(\log^5 n)$ .

The third step takes asymptotic time  $O(\log^6 n)$ .

The fourth step takes asymptotic time  $O(\log n)$ .

In step 5, we need to verify  $\lfloor \sqrt{\varphi(r) \log n} \rfloor$  equations, each equation requires  $O(\log n)$  multiplications of degree  $r$  polynomials with coefficients of size  $O(\log n)$ . So the asymptotic time is

$$\tilde{O}(r \log^2 n \sqrt{\varphi(r) \log n}) \leq \tilde{O}(r^{3/2} \log^3 n) \leq \tilde{O}(\log^{15/2} n).$$

Thus the total asymptotic time is  $\tilde{O}(\log^{15/2} n)$ .  $\square$

**Lemma (36.1.1.3).** If the AKS algorithm in (36.1.1.1) returns **PRIME**, then  $n$  is a prime.  $\lrcorner$

*Proof:* If it returns at step 4, then step3 implies that  $n$  is a prime. So we assume that it returns at step6. Since  $\text{ord}_r(n) > 1$ , there exists  $p|n$  s.t.  $\text{ord}_r(p) > 1$ . By step3,  $p > r$  and  $(r, n) = 1$ .

Let  $G$  be the subgroup of  $(\mathbb{Z}/r)^*$  generated by  $n$  and  $p$ , with  $\#G > \log^2 n$ . And denote  $\zeta = \zeta_{\text{ord}_r(p)}$  and let  $\mathcal{G}$  be the subgroup of  $\mathbb{F}_p(\zeta)$  generated by  $\zeta, \zeta + 1, \dots, \zeta + \lfloor \sqrt{\varphi(r) \log n} \rfloor$ . Then

$$\begin{aligned} \# \mathcal{G} &\stackrel{(36.1.1.4)}{\geq} \binom{\#G + \lfloor \sqrt{\varphi(r) \log n} \rfloor}{\#G - 1} \\ (\because \#G > \sqrt{\#G \log n}) &\geq \binom{\lfloor \sqrt{\varphi(r) \log n} \rfloor + 1 + \lfloor \sqrt{\#G \log n} \rfloor}{\lfloor \sqrt{\#G \log n} \rfloor} \\ &\geq \binom{2\lfloor \sqrt{\#G \log n} \rfloor + 1}{\lfloor \sqrt{\#G \log n} \rfloor} \\ (12.6.1.20) &> 2^{\lfloor \sqrt{\#G \log n} \rfloor + 1} \\ &\geq n^{\sqrt{\#G}}. \end{aligned}$$

But then by (36.1.1.4),  $n \in p^{\mathbb{Z}_+}$ . So in view of step1,  $n = p \in \mathbf{Prime}$ .  $\square$

**Lemma (36.1.1.4).** Let  $G, \mathcal{G}$  be the group defined in the proof of (36.1.1.3), then

$$\# \mathcal{G} \geq \binom{\#G + \sqrt{\varphi(r) \log n}}{\#G - 1}.$$

And if  $n \notin p^{\mathbb{Z}_+}$ , then  $\# \mathcal{G} \leq n^{\sqrt{\#G}}$ .  $\lrcorner$

*Proof:* By the hypothesis,  $n$  passed step4, so any  $0 \leq a \leq \sqrt{\varphi(r) \log n}$ ,

$$(X + a)^n \equiv X^n + a \pmod{X^r - 1, p},$$

so

$$(X + a)^{\frac{n}{p}} \equiv X^{\frac{n}{p}} + a \pmod{X^r - 1, p}.$$

More generally, for any  $m \in I = \{(\frac{n}{p})^i p^j : i, j \in \mathbb{N}\}$  and any polynomial

$$f(X) \in P = \left\{ \prod_{a=0}^{\lfloor \sqrt{\#G \log n} \rfloor} (X + a)^{e_a} : e_a \in \mathbb{N} \right\},$$

we have

$$f(X)^m \equiv f(X^m) \pmod{X^r - 1, p}.$$

Then we claim that any two polynomials in  $P$  with degree  $< \#G$  will be distinct elements in  $\mathbb{F}_p(\zeta)$ : If  $\bar{f} = \bar{g}$ , then for each  $m \in G$ ,  $f(\zeta^m) \equiv g(\zeta^m)$ , so  $f$  and  $g$  are equal on  $\#G$  values. This forces  $f \equiv g \pmod{p}$ . But this is not possible as  $i \not\equiv j \pmod{p}$  for  $i, j \leq \lfloor \sqrt{\#G \log n} \rfloor < \sqrt{r \log n} < r < p$ .

Then by combinatorics,  $\#\mathcal{G} \geq \binom{\#G + \sqrt{\varphi(r) \log n}}{\#G - 1}$ .

For the upper bound, consider the subset  $I' = \{(\frac{n}{p})^i p^j : 0 \leq i, j \leq \sqrt{\#G}\}$ . If  $n \notin p^{\mathbb{Z}_+}$ , then  $\#I' > \#G$ , so there are two elements  $m_1, m_2 \in I'$  s.t.  $r | m_1 - m_2$ . Then  $f(\zeta)^{m_1} = f(\zeta^{m_1}) = f(\zeta^{m_2}) = f(\zeta)^{m_2}$  for any  $f(X) \in P$ . Then  $f(\zeta)$  is a root of the polynomial  $X^{m_1} - X^{m_2}$ , which has degree  $\leq n^{\sqrt{\#G}}$ . In view of the fact  $\mathcal{G}$  is the values of polynomials in  $P$  at  $\zeta$ , we have  $\#\mathcal{G} \leq n^{\sqrt{\#G}}$ .  $\square$

**Algo. (36.1.1.5) [Elliptic Curve Primality Proving (ECP), Goldwasser-Kilian-Atkin].** An **ECP primality certificate** of  $p \in \text{Prime}$  is a tuple  $(p, A, B, x_1, y_1, q) \in \mathbb{Z}$  s.t.

- $E : y^2 = x^3 + Ax + B$  is an Elliptic curve over  $\mathbb{Q}$ ,
- $P = (x_1, y_1, 1) \in E(\mathbb{Q})$ ,
- $(\Delta(E), p) = 1$ ,
- $q \in \text{Prime}$ ,
- $q > (p^{1/4} + 1)^2$ ,
- $p | z(qP)$ .

Then consider the following algorithm that either generates a series of primality certificates that verifies  $p$  is prime or return that  $p$  is not a prime:  $\lrcorner$

Cf. [Sutherland]L12, P6. or [Cox, Primes of the form]P297.

**Remark (36.1.1.6).** This ECP primality testing is a practically faster deterministic algorithm than AKS(36.1.1.1).  $\lrcorner$

**Lemma (36.1.1.7) [Goldwasser-Kilian].** Let  $E \in \text{EllCur}/\mathbb{Q}$  and  $M, N \in \mathbb{Z}_+$  s.t.  $M > (N^{1/4} + 1)^2$  and  $(N, \Delta(E)) = 1$ .

In particular, any ECP primality certificate(36.1.1.5) implies that  $p$  is a prime.  $\lrcorner$

*Proof:*  $\square$

**Algo. (36.1.1.8) [Miller-Rabin probabilistic Primality Proving].** Cf. [Sutherland]L12.

This is the more efficient and practical way of primality proving. And it can be astronomically convincing.  $\lrcorner$

## 2 Factorizing Integers

**Algo. (36.1.2.1) [Buhler-Lenstra-Pomerance].** There is an algorithm to factor a  $n$ -bit number  $N$  in  $O(e^{cn^{1/3} \log^{2/3} n})$  times.  $\lrcorner$

## 3 Others

**Remark (36.1.3.1).** There are not yet a polynomial algorithm that gives for a prime  $p \in \mathbf{P}$  a primitive root  $a \in (\mathbb{Z}/(p))^*$ .  $\lrcorner$



# 37 | Combinatorics

## 37.1 Combinatorics

### 1 Polytopes

**Def. (37.1.1.1) [Polytopes].** A  **$n$ -polytope** is a compact subset of  $\mathbb{R}^n$  for some  $n \in \mathbb{Z}$  with boundaries given by polytopes of smaller dimensions. A **polygon** is a 2-polytope. A **polyhedron** is a 3-polytope.  $\lrcorner$

**Def. (37.1.1.2) [Tetrahedra].** A **tetrahedron** is an 3-simplex in  $\mathbb{R}^3$ .  $\lrcorner$

**Def. (37.1.1.3) [Orthohedra].** An **orthohedron** is a tetrahedron isomorphic to the convex hull of the four points

$$\{(0, 0, 0), (x, 0, 0), (x, y, 0), (x, y, z)\} \in \mathbb{R}^3.$$

$\lrcorner$

**Prop. (37.1.1.4) [Dihedral Angles of Orthohedra].**

- Each orthohedron has dihedral angles  $(\alpha, \beta, \gamma, \pi/2, \pi/2, \pi/2)$ .
- A rectangular solid can be cut into 6 orthohedra sharing a common diagonal. And for any  $\alpha, \beta, \gamma \in (0, \pi), \alpha + \beta + \gamma = \pi$ , there is a rectangular solid that can be cut into 6 orthohedra s.t. the dihedral angles along the diagonal are  $(\alpha, \beta, \gamma, \alpha, \beta, \gamma)$ .

$\lrcorner$

*Proof:* Only 2 needs a proof, and this is verified by brutal force calculations.  $\square$

### 2 Hilbert's 3rd Problem

Cf. [Sch13b].

**Def. (37.1.2.1) [Scissors Congruence].** A **dissection of a polytope**  $P$  is an expression  $P = \cup_i^n P_i$  s.t. the interiors of  $P_i$  are disjoint.

Two polytopes  $P, Q$  are called **scissors-congruent** if there are dissections  $P = \cup_i^n P_i, Q = \cup_i^n Q_i$  s.t.  $P_i \cong Q_i$  for  $1 \leq i \leq n$ . Scissors congruence relations are denoted by  $\sim_s$ .  $\lrcorner$

**Thm. (37.1.2.2) [Wallace-Bolyai-Gerwien].** Any two polygons  $P, Q$  in  $\mathbb{R}^2$  are scissors-congruent.  $\lrcorner$

*Proof:* Find a line  $l$  that is not parallel to any line generated by vertices of  $P$  and  $Q$ , cut  $P, Q$  through lines parallel to  $l$  and passing through some vertices of  $P, Q$ . Then we get some trapezoids. Then we cut them into right triangles.

Now we transform any right triangle  $P$  with vertices  $\{A = (-0, 0), B = (0, 2b), C = (0, c)\}$  to a triangle triangle of height 2: Now consider another right triangle with vertices  $\{A = (-0, 0), X = (0, 2), Y = (0, bc)\}$ . Then  $BY//CX$ , thus  $BYC$  and  $BYX$  are triangles of the same base and height.

Then  $BYC \sim_s BYX$ : cut them into parallelograms with the same base and height, then it is easy to see any two such parallelograms are scissor-congruent.

Now all the triangles has a side of length 2, we can transform them into rectangles with a side of length 1. Then clearly  $P, Q$  are transformed together.  $\square$

**Thm. (37.1.2.3) [Zylev].** If  $P, A, B$  are polytopes,  $A \subset P, B \subset P, P \setminus A \sim_s P \setminus B$ , then  $A \sim_s B$ .  $\lrcorner$

*Proof:* Let

$$F = A \coprod \coprod_{k=1}^n P_k = B \coprod \coprod_{k=1}^n Q_k, P_k \sim_s Q_k$$

We can dissect them even further s.t.  $\text{Vol}(P_k) < \frac{1}{2} \text{Vol}(A)$ . Then we use induction on  $n$  to show that  $A \sim_s B$ :  $n = 0$  is trivial. For  $n > 0$ , because of the volume bound,  $\text{Vol}(A \setminus Q_n) > \text{Vol}(Q_n)$ , thus we can find disjoint  $T_1, \dots, T_n \subset A \setminus Q_n$  s.t.  $T_k \sim_s P_k \cap Q_n$ . Now define

$$F' = F \setminus Q, P'_k = (P_k \setminus Q_n) \cup T_k \sim_s P_k, \quad A' = F' \setminus \bigcup_{k=1}^{n-1} P'_k = (A \setminus \bigcup_{k=1}^{n-1} T_k) \bigcup_{k=1}^{n-1} Q_k$$

Then by induction,  $A' \sim_s B' = B$ . But  $A' \sim_s A$  by changing  $T_k$  with  $Q_k$ .  $\square$

### Dehn-Sydler Theorem

**Prop. (37.1.2.4) [Tetrahedra].** Any polyhedron has a dissection into tetrahedra.  $\lrcorner$

*Proof:* By choose a sufficiently small lattice dissection of  $\mathbb{R}^3$ , it suffices to consider three cases:

- A cube.
- A cube cut by one hypersurfaces.
- A cube cut by two hypersurfaces.
- A cube containing a single vertex.

All cases are straightforward.  $\square$

**Def. (37.1.2.5) [Dehn Invariants].** For a polyhedron  $P$ , define the **Dehn invariant** of  $P$  as

$$\Delta(P) = \sum_i l_i [\alpha_i] \in \mathbb{R} \otimes (\mathbb{R}/\mathbb{Z}\pi)$$

where the sum is over all edges  $e_i$ , of  $P$ ,  $l_i = l(e_i)$ , and  $\alpha_i$  is the dihedral angles of  $e_i$ .  $\lrcorner$

**Thm. (37.1.2.6) [Dehn].** If two polyhedra in  $\mathbb{R}^3$  are scissors-congruent, then they have the same volume and Dehn invariants.  $\lrcorner$

*Proof:* The volumes are the same by the invariance properties of Lebesgue measure. To show the Dehn invariants are the same, notice the dissections of a polytope  $P$  may create new edges, but the sum of angles around a new edge is  $2\pi$  for if it is contained in the interior of  $P$ , or  $\pi$  if it is in the surface of  $P$ .  $\square$

**Def. (37.1.2.7) [Simple Prisms].** A **simple prism** is a polyhedron that is isomorphic to an affine translate of  $T \times I \subset \mathbb{R}^2 \times \mathbb{R}$ , where  $T \subset \mathbb{R}^2$  is a triangle.  $\lrcorner$



**Prop. (37.1.2.8).** A simple prism  $P$  is scissors-congruent to  $[0, 1]^2 \times [0, \text{Vol}(P)]$ .  $\lrcorner$

*Proof:* Place the three edges  $P_1Q_1, P_2Q_1, P_3Q_3$  that is the image of  $\text{Vertices}(T) \times I$  vertically, and they have length  $l$ . Then we can talk about the **heights** of  $P_i, Q_i$ . Suppose  $h(P_1) \geq h(P_2) \geq h(P_3)$ . If  $h(P_3) \geq h(Q_1)$ , then we can cut  $P$  along height  $h(Q_1)$  make  $P$  scissors-congruent to a perfect simple prism  $P' = T' \times I'$ . If  $h(P_3) < h(Q_1)$ , then we can cut  $P$  along the plane  $P_2P_3Q_1$  to make  $P \sim_s P'$ , where for  $P'$ ,  $h(P'_3) - h(Q'_1)$  is larger. And if this process goes on, every 2 transform makes  $h(P_3) - h(Q_1)$  bigger by at least  $l$ , so eventually  $h(P_3) \geq h(Q_1)$ , and we can transform it to a perfect simple prism.

Now if  $P = T \times I$  is a perfect simple prism, (37.1.2.2) shows  $T$  is scissors equivalent to  $[0, 1] \times [0, \text{Vol}(T)]$ , and then it suffices to show  $[0, \text{Vol}(T)] \times I$  is scissors equivalent to  $[0, \text{Vol}(P)]$ , which is by (37.1.2.2) again.  $\square$

**Def. (37.1.2.9) [pseudo-prisms].** A **pseudo-prisms** is a convex polyhedra  $(OPQRS)$ , where

- $OPQ$  is an isosceles with  $OP = PQ$ ,
- $PR, QS$  is orthogonal to the plane  $OPQ$ ,
- $l(QS) = 2l(PR)$ .

Notice a pseudo-prism  $(OPQRS)$  is scissors-congruent to the perfect simple prism  $OPQ \times PR$ .  $\lrcorner$

**Def. (37.1.2.10).** Denote

- $\mathcal{P}$  the free Abelian group generated by isomorphism classes of all polyhedra.
- $\mathcal{E}$  the subgroup of  $\mathcal{P}$  generated by the dissection relations.
- $\mathcal{F}$  is the subgroup of  $\mathcal{P}$  generated by  $\mathcal{E}$  and all the simple prisms (37.1.2.7).
- $\mathcal{V} = \mathcal{P}/\mathcal{F}$ .

Then clearly two polyhedra  $P, Q$  are scissors-congruent iff  $[P] = [Q] \in \mathcal{P}/\mathcal{E}$ .  $\lrcorner$

**Prop. (37.1.2.11) [R-Structures].**  $\mathbb{R}$  acts on the set of polyhedra by scaling, and this action is additive and multiplicative, and stabilizes  $\mathcal{F}$ , thus inducing an  $\mathbb{R}$ -structure on  $\mathcal{V}$ .  $\lrcorner$

*Proof:* By (37.1.2.4),  $\mathcal{P}$  is generated by tetrahedra. So it suffices to show that for  $\lambda_1, \lambda_2 \in \mathbb{R}_+$  and a tetrahedron  $T$ ,  $[\lambda_1 T] + [\lambda_2 T] - [(\lambda_1 + \lambda_2)T] \in \mathcal{F}$ . For this, use geometry, notice  $(\lambda_1 + \lambda_2)T \setminus (\lambda_1 T \cup \lambda_2 T)$  consists of two simple prisms, where these two smaller tetrahedra aligned along an edge of  $T$ .  $\square$

**Prop. (37.1.2.12) [Orthohedra].**  $\mathcal{V}$  is generated by orthohedra (37.1.1.3).  $\lrcorner$

*Proof:* By (37.1.2.4), it suffices to show that for any tetrahedron  $T$ ,  $[T]$  is a linear combinations of orthohedra. For this, take the center  $I$  of the inscribed circle of  $T$ , and take its projections  $I_1, I_2, I_3, I_4$  to the faces of  $T$ , then they form 24 orthohedra (may be degenerate ones). Thus  $T$  is a linear combinations of those non-degenerate orthohedra.  $\square$

**Prop. (37.1.2.13).** A simple primes  $P$  has Dehn invariant  $\Delta(P) = 0$  (37.1.2.14) and (37.1.2.8), so by (37.1.2.8) again,  $\Delta$  factors through  $\Delta : \mathcal{V} \rightarrow \mathbb{R} \otimes (\mathbb{R}/\mathbb{Z}\pi)$ , which is  $\mathbb{R}$ -linear with  $\mathbb{R}$ -structure on  $\mathcal{V}$  given in (37.1.2.11).  $\lrcorner$

**Thm. (37.1.2.14) [Dehn-Sydler].** Two polyhedra in  $\mathbb{R}^3$  are scissors-congruent iff they have the same volume and Dehn invariants.  $\lrcorner$

*Proof:* One direction is shown in (37.1.2.6). For the other, it suffices to show this map  $\Delta : \mathcal{V} = \mathcal{P}/\mathcal{F} \rightarrow \mathbb{R} \otimes (\mathbb{R}/\mathbb{Z}\pi)$  (37.1.2.13) is injective, because in this way, if two polyhedra has the same volume and Dehn invariants, then they are scissors equivalent to two prisms with the same volume, hence scissors-congruent, by (37.1.2.8).

Take a good function  $\varphi : \mathbb{R} \rightarrow \mathcal{V}$  (37.1.2.24).  $\varphi$  vanishes on  $\mathbb{Z}\pi$ , thus extends to an  $\mathbb{R}$ -linear map  $\Phi : \mathbb{R} \otimes (\mathbb{R}/\mathbb{Z}\pi) \rightarrow \mathcal{V}$ , and for an orthohedron  $T$ ,  $\Phi(\Delta([T])) = [T]$  by (37.1.2.15), and orthohedra generate  $\mathcal{V}$  by (37.1.2.12), so  $\Phi \circ \Delta = \text{id}_{\mathcal{V}}$ . Thus  $\Delta$  is injective.  $\square$

### Homological Arguments

**Def. (37.1.2.15) [Good Functions].** A map  $\varphi : \mathbb{R} \rightarrow \mathcal{V}$  is called a **good function** if

- $\varphi$  is additive and  $\varphi(\pi) = 0$ .
- For any orthohedron  $T$ ,  $[T] = \sum_{i=1}^6 l_i \varphi(\alpha_i)$ , where the notation is the same as in (37.1.2.5).

┘

**Def. (37.1.2.16)  $[T(a, b)]$ .** For  $a, b \in (0, 1)$ , define  $a' = \sqrt{a^{-1} - 1}$ ,  $b' = \sqrt{b^{-1} - 1}$ , and define  $T(a, b)$  to be the orthohedron with vertices

$$\{(0, 0, 0), (a', 0, 0), (a', a'b', 0), (a', a'b', b')\} \in \mathbb{R}^3.$$

Notice  $T(a, b) \cong T(b, a)$ .

The order of the vertices is also important. It is always assumed to be written in this order.  $\square$

**Prop. (37.1.2.17).** For  $\alpha, \beta \in \mathbb{R}$ ,  $T(\sin^2(\alpha), \sin^2(\beta))$  has three edges with dihedral angle  $\pi/2$ , and three edges with lengths  $\cot(\alpha), \cot(\beta), \cot(\gamma)$  and dihedral angles  $(\alpha, \beta, \pi/2 - \gamma)$  resp., where  $\sin^2(\gamma) = ab, \gamma \in (0, \pi/2)$ . In particular,

$$\Delta(T(\sin^2(\alpha), \sin^2(\beta))) = \cot(\alpha) \otimes \alpha + \cot(\beta) \otimes \beta + \cot(\gamma) \otimes (\pi/2 - \gamma).$$

┘

*Proof:* Brutal force calculation.  $\square$

**Prop. (37.1.2.18).** For  $a, b, c \in \mathbb{R}_+$ ,

$$a[T(\frac{a+b}{a+b+c}, \frac{a}{a+b})] + b[T(\frac{a+b}{a+b+c}, \frac{b}{a+b})] = a[T(\frac{a+c}{a+b+c}, \frac{a}{a+c})] + c[T(\frac{a+c}{a+b+c}, \frac{c}{a+c})]$$

┘

*Proof:* This follows from the different ways of cutting the tetrahedron with vertices

$$\{O = (0, 0, 0), X = (\sqrt{bc}, 0, 0), Y = (0, \sqrt{ac}, 0), Z = (0, 0, \sqrt{ab})\} \in \mathbb{R}^3.$$

In fact, for this tetrahedron, the plane orthogonal to  $YZ$  passing through  $X$  cut this to two orthohedra, and similar does the plane orthogonal to  $XZ$  passing through  $Y$ . These two cutting give the assertion above.  $\square$

**Def. (37.1.2.19) [Homological Functions].** A **homological function** is a function  $h : (0, 1) \rightarrow \mathcal{V}$  s.t.

- For  $a, b \in (0, 1)$ ,  $[T(a, b)] = h(a) + h(b) - h(ab)$ .
- If  $a, b \in (0, 1)$ ,  $a + b = 1$ , then  $ah(a) + bh(b) = 0$ .

┘

**Prop. (37.1.2.20).** For  $V \in \text{Vect}/\mathbb{R}$ , if  $g : \mathbb{R}_+ \rightarrow V$ , define  $\delta(g)(a, b) = g(a) + g(b) - g(ab)$ . And if  $F : \mathbb{R}_+^2 \rightarrow V$ , define  $\delta(F)(a, b, c) = F(a, b) - F(a, c) + F(ab, c) - F(ac, b)$ .

Then if  $F : (0, 1) \rightarrow V$  is symmetric and  $\delta(F) = 0$ , then  $F = \delta(f)$  for some  $f : (0, 1) \rightarrow V$ . ┘

*Proof:* Cf.[Sch13b]. □

**Prop. (37.1.2.21).** Let  $G : \mathbb{R}_+^2 \rightarrow V$  be a symmetric function satisfying

- $G(\lambda a, \lambda b) = \lambda G(a, b), \lambda, a, b \in \mathbb{R}_+$ ,
- $G(a, b) + G(a + b, c) = G(a, c) + G(a + c, b)$ .

Then there exists a homomorphism  $g : \mathbb{R}_+^\times \rightarrow V$  s.t. when  $a + b = 1$ ,

$$G(a, b) = ag(a) + bg(b).$$

┘

*Proof:* Cf.[Sch13b]. □

**Prop. (37.1.2.22)[Sydler].** Let  $F : (0, 1)^2 \rightarrow \mathcal{V} : (a, b) \mapsto [T(a, b)]$ , then  $F$  is symmetric, and  $\delta(F) = 0$ .

┘

*Proof:* We check that  $[T(a, b)] + [T(ab, c)] = [T(a, c)] + [T(ac, b)]$ . This is done by geometry:

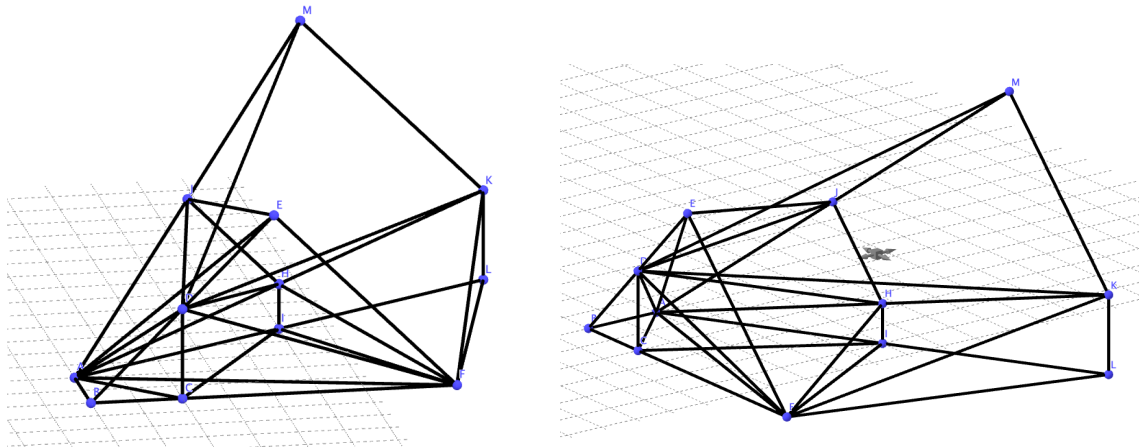
Put  $T(a, b), T(a, c)$  together:  $T(a, b) \cong ABCD, T(a, c) \cong ABEF$ , where  $BDE$  is collinear. Let  $H$  be the center of the sphere  $S$  passing through  $ACDEF$ , and the projection of  $H$  on the planes  $ABC, ABD$  is denoted by  $I, J$  resp. Let  $K, L, M$  be the intersection of  $AH, AI, AJ$  with the sphere  $S$  resp.

Then it can be verified that  $T(ab, c) \cong ANMK, T(ac, b) \cong AFLK$ . Let  $P = ABDFHI + ADHJ$ , then

$$P - (AICH D) - (FICH D) + (DJMH K) = T(a, b) + T(ab, c).$$

$$P + (DJEHF) - (AJEH F) + (FILHK) = T(a, c) + T(ac, b).$$

□



**Cor. (37.1.2.23).** Homological functions(37.1.2.19) exist. ┘

*Proof:* By (37.1.2.20), there exists  $f : (0, 1) \rightarrow \mathcal{V}$  s.t.  $F(a, b) = [T(a, b)] = f(a) + f(b) - f(ab)$ . Then we want to find  $h = f - g$ , where  $g : (0, 1) \rightarrow \mathcal{V}$  is a homomorphism s.t. when  $a + b = 1$ ,  $ah(a) + bh(b) = 1$ , which is equivalent to

$$ag(a) + bg(b) = af(a) + bf(b).$$

Let

$$G : \mathbb{R}_+^2 \rightarrow \mathcal{V} : (a, b) \mapsto af\left(\frac{a}{a+b}\right) + bf\left(\frac{b}{a+b}\right),$$

then  $G(\lambda a, \lambda b) = \lambda G(a, b)$ ,  $\lambda, a, b \in \mathbb{R}_+$ , and  $G(a, b) + G(a + b, c) = G(a, c) + G(a + c, b)$ . In fact, this boils down to (37.1.2.18). Thus by (37.1.2.21), we truly can find such a function  $g$ .  $\square$

**Prop. (37.1.2.24).** Good functions (37.1.2.15) exist.  $\lrcorner$

*Proof:* Take a homological function  $h$  (37.1.2.23), define  $\varphi : \mathbb{R} \rightarrow \mathcal{V}$ :

$$\varphi\left(\frac{n\pi}{2}\right) = 0, \quad n \in \mathbb{Z}, \quad \varphi(\alpha) = \tan(\alpha)h(\sin^2(\alpha)).$$

We show this is a good function:

Firstly we show:

**Lemma (37.1.2.25).**  $\varphi(\pi/2 - \alpha) = -\varphi(\alpha)$ ,  $\varphi(\alpha + \pi/2) = \varphi(\alpha)$ .  $\lrcorner$

*Proof:* If  $\alpha = n\pi/2$ , the assertion holds. So we assume  $\alpha \neq n\pi/2$  and let  $\beta = \pi/2 - \alpha$ ,  $a = \sin^2(\alpha)$ ,  $b = \sin^2(\beta)$ , then  $a + b = 1$ , so by definition (37.1.2.19),

$$\begin{aligned} 0 &= 2ah(a) + abh(b) = 2\sin^2(\alpha)\cot(\alpha)\varphi(\alpha) + 2\sin^2(\beta)\cot(\beta)\varphi(\beta) \\ &= \sin(2\alpha)\varphi(\alpha) + \sin(2\beta)\varphi(\beta) = \sin(2\alpha)(\varphi(\alpha) + \varphi(\beta)) \end{aligned}$$

So the assertion holds as  $\sin(2\alpha) \neq 0$ .

The second equality follows from the first.  $\square$

Next, we show  $[T] = \sum_{i=1}^6 l_i \varphi(\alpha_i)$  for a orthohedron  $T$ : By scaling, it suffices to prove for  $T$  of the form  $T(\sin^2(\alpha), \sin^2(\beta))$  (37.1.2.16). Thus by (37.1.2.19) and (37.1.2.17),

$$\begin{aligned} [T(\sin^2(\alpha), \sin^2(\beta))] &= h(\sin^2(\alpha)) + h(\sin^2(\beta)) - h(\sin^2(\alpha)\sin^2(\beta)) \\ &= \cot(\alpha)\varphi(\alpha) + \cot(\beta)\varphi(\beta) + \cot(\gamma)\varphi(\pi/2 - \gamma) = \sum_{i=1}^6 l_i \varphi(\alpha_i) \end{aligned}$$

Finally, to show additivity of  $\varphi$ , notice by the lemma above, it suffices to show  $\varphi(\alpha) + \varphi(\beta) = \varphi(\alpha + \beta)$  for  $\alpha, \beta \in (0, \pi/2)$ . In this case, let  $\gamma = \pi - \alpha - \beta \in (0, 2\pi)$ , then use (37.1.1.4) to find a rectangular solid that cuts along a diagonal to 6 orthohedra  $T_i$  s.t. the dihedral angles along the diagonals are  $(\alpha, \beta, \gamma, \alpha, \beta, \gamma)$ . Then

$$0 = [R] = \sum_{k=1}^6 [T_i] = 2l(\varphi(\alpha) + \varphi(\beta) + \varphi(\gamma)) + \sum_{k=1}^6 l_k(\varphi(\theta_{k1}) + \varphi(\theta_{k2}))$$

where  $l$  is the length of the diagonal,  $\theta_{k_i}$  are the dihedral angles along the edges of the rectangular solid that is connected to the diagonal, and  $\theta_{k1} + \theta_{k2} = \pi/2$ . Thus by the lemma above,  $\varphi(\alpha) + \varphi(\beta) = -\varphi(\gamma) = \varphi(\alpha + \beta)$ .  $\square$

### 3 Permutohedra and Associahedra

References are [Permutohedra, Associahedra, and Beyond, Postnikov].

**Def. (37.1.3.1)[Permutohedra].** For  $(x_1, \dots, x_n) \in \mathbb{R}^n$ , the **permutohedron** of  $(x_1, \dots, x_n)$  is defined to be

$$P_n(x_1, \dots, x_n) = \text{Hull}((x_{\sigma(1)}, \dots, x_{\sigma(n)})).$$

$P_n(n-1, n-2, \dots, 0)$  is called the  $(n-1)$ -dimensional **regular permutohedron**. ┘

**Prop. (37.1.3.2) [Radon].** For  $x_1 \geq x_2 \geq \dots \geq x_n \in \mathbb{R}$ , then for any  $\mathbf{t} = (t_1, \dots, t_n) \in \mathbb{R}^n$ ,  $\mathbf{t} \in P_n(x_1, \dots, x_n)$  iff

$$\sum_i t_i = \sum_i x_i$$

and for any  $S \subset [n]_+$ ,

$$\sum_{i \in S} t_i \leq \sum_{i=1}^{\#S} x_i.$$

┘

*Proof:*

□

**Prop. (37.1.3.3) [General Type Permutohedras].** If  $x_1 > x_2 > \dots > x_n \in \mathbb{R}$ , then there is a description of the  $d$ -dimensional faces of  $P_n(x_1, \dots, x_n)$ . Cf.[Permutohedra, Associahedra, and Beyond, Postnikov]Proposition 2.6. ? ┘

**Def. (37.1.3.4)[Zonotope].** A **zonotope** is a polytope that is a Minkowski sum of several line segments. ┘

**Def. (37.1.3.5)[Graphical Zonotopes].** For  $n \in \mathbb{Z}_+$  and a graph  $\Gamma$  on  $[n]_+$ , the **graphical zonotope**  $Z_\Gamma$  is defined to be the Minkowski of line segments

$$Z_\Gamma = \sum_{(i,j) \in \Gamma} [\mathbf{e}_i, \mathbf{e}_j] = \text{Newton} \left( \prod_{(i,j) \in \Gamma} (X_i - X_j) \right).$$

┘

**Prop. (37.1.3.6).** For  $n \in \mathbb{Z}_+$ , the  $(n-1)$ -dimensional regular permutohedron(37.1.3.1) is the graphical zonotope for the complex graph  $K_n$  on  $[n]_+$ (37.1.3.5). ┘

*Proof:* The regular permutohedron is the newton polytope of the determinant of the Vandermonde matrix  $\det \text{Van}(X_1, \dots, X_n) = \prod_{1 \leq i < j \leq n} (X_i - X_j)$ . □

**Prop. (37.1.3.7)[Volume of Graphical Zonotopes, Stanley].** For  $n \in \mathbb{Z}_+$  and  $\Gamma$  a simple graph on  $[n]_+$ ,

$$\text{Vol}(Z_\Gamma) = \# \{ \text{spanning tree of } \Gamma \}.$$

┘

*Proof:* Cf.[Stanley, R. P. Enumerative Combinatorics, vol. 1]Exercise 4.32. □

### Generalized Permutohedra

**Def. (37.1.3.8) [Generalized Permutohedra].** A **generalized permutohedra** is a polytope deformation of a permutohedra  $P_n(x_1, \dots, x_n)$  (37.1.3.1) s.t. all the the angles of the edges are preserved.  $\lrcorner$

### Stasheff Associahedra

Cf.[Operads in algebra, topology and physics, Stasheff]Section1.6.

### Lattice Points

**Def. (37.1.3.9) [Number of Lattice Points].** If  $x_1, \dots, x_n \in \mathbb{Z}$ , denote  $N_n(x_1, \dots, x_n) = P_n(x_1, \dots, x_n) \cap \mathbb{Z}^n$ . And define the **Ehrhart polynomial** as

$$E_{P_n(x_1, \dots, x_n)}(t) = N_n(tx_1, \dots, tx_n) \in \mathbb{Z}[t].$$

$\lrcorner$

## 4 Spherical Programming

### Minimizing Discrete Energy on the Sphere

**Prop. (37.1.4.1) [Equal-Area Partition of Spheres with Small Diameters, Leopardi[Leo09]].**

For  $d \in \mathbb{Z}_+$ , there exists  $C_d \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_+$ , there exists a partition of the unit sphere into  $N$  closed regions with piecewise smooth boundaries s.t. each partition has diameter at most  $C_d N^{-1/d}$ .  $\lrcorner$

*Proof:*

$\square$

### Spherical $t$ -Designs

**Def. (37.1.4.2) [Spherical  $t$ -Designs].** Let  $d, t \in \mathbb{Z}_+$  and  $\mathbb{S}^d \subset \mathbb{R}^{d+1}$  be the unit sphere with the Lebesgue measure  $\mu$  normalized s.t.  $\mu(\mathbb{S}^d) = 1$ . Then a **spherical  $t$ -design** of size  $N \in \mathbb{Z}_+$  for  $\mathbb{S}^d$  is a set pf points  $\{X_1, \dots, X_N\} \subset \mathbb{S}^d$  s.t.

$$\int_{\mathbb{S}^d} P(x) d\mu(x) = \frac{1}{N} \sum_{i=1}^d P(x_i)$$

for any polynomial  $P \in \mathbb{R}[X_0, X_1, \dots, X_d]$  s.t.  $\deg P \leq t$ .  $\lrcorner$

**Thm. (37.1.4.3) [Delsarte-Goethals-Seidel[[DGS77](#)]/Seymour-Zaslavsky[[S-Z84](#)]/Korevaar-Meyers[[K-M93](#)]/Bondarenko-Radchenko-Viazovska[[BRV13](#)]].** For any  $d \in \mathbb{Z}_+$ , there exists  $C_d \in \mathbb{R}_+$  s.t. for any  $t \in \mathbb{Z}_+$ , if we denote by  $N(d, t)$  the minimal cardinality of a spherical  $t$ -design for  $\mathbb{S}^d$ , then

$$C_d t^d \geq N(d, t) \geq \begin{cases} \binom{d+k}{d} + \binom{d+k-1}{d} & , t \in 2\mathbb{Z} \\ 2 \binom{d+k}{d} & , t \in 2\mathbb{Z} + 1 \end{cases}.$$

In fact, for any  $N \geq C_d t^d$ , there exists a spherical  $t$ -design for  $\mathbb{S}^d$  with  $N$  points. And for  $d = 1$ , the lower bound is satisfied:  $N(1, t) = t + 1$ , by taking the regular  $(t + 1)$ -gon.  $\lrcorner$

*Proof:* **?**

For the upper bound: Consider the inner space  $\mathcal{P}_t$  consisting of polynomial of degree  $\leq d$  s.t.  $\int_{\mathbb{S}^d} P(x) d\mu(x) = 0$ , then for any  $x \in \mathbb{S}^d$ , there exists  $G_x \in \mathcal{P}_t$  s.t.  $\langle P_x, Q \rangle = Q(x)$  for any  $Q \in \mathcal{P}_t$ . Then the set of points  $X = \{x_1, \dots, x_N\} \subset \mathbb{S}^d$  forms a spherical  $t$ -design iff  $G_{x_1} + \dots + G_{x_N} = 0$ .

Define the convex subset  $\Omega \subset \mathcal{P}_t$ :

$$\Omega = \left\{ P \in \mathcal{P}_t : \int_{\mathbb{S}^d} \|\nabla_S P(x)\| d\mu(x) < 1 \right\}, \quad \nabla_S P(x) = \nabla \left( P \left( \frac{x}{|x|} \right) \right).$$

Then it suffices to show that for any  $N \geq C_d t^d$ , there exists a function:  $F : \mathcal{P} \rightarrow (\mathbb{S}^d)^N$  s.t.  $\sum_{i=1}^N P(F_i(P)) > 0$  for any  $P \in \partial\Omega \subset \mathcal{P}_t$ : If such  $F$  exists, we can consider

$$f : \mathcal{P}_t \rightarrow \mathcal{P}_t : P \mapsto \sum_{i=1}^N G_{F_i(P)},$$

then  $\langle P, f(P) \rangle > 0$  for any  $P \in \partial\Omega$ , and there exists a  $P$  s.t.  $f(P) = 0$  by [Brouwer Degree Theorem](#).

The construction of  $F(P)$  is given by choosing points from the equal-area partition of  $\mathbb{S}^n$  [\(37.1.4.1\)](#) and let these point flow along the vector field  $\nabla_S P$  **?**, Cf. [\[BRV13\]](#).  $\square$

**Prop. (37.1.4.4) [Nearly Spherical Design].** For any  $d \in \mathbb{Z}_+$ , there exists  $r_d \in \mathbb{R}_+$  s.t. for any  $t \in \mathbb{Z}_+$  and any equal-area partition  $\mathcal{R} = \{R_1, \dots, R_t\}$  of  $\mathbb{S}^d$  of diameter bounded by  $r_d/t$ , if we choose any  $x_i \in \mathcal{R}$ , then the following inequality holds for any polynomial  $P \in \mathbb{R}[X_0, X_1, \dots, X_d]$  s.t.  $\deg P \leq t$ :

$$\frac{1}{2} \int_{\mathbb{S}^d} |P(x)| d\mu(x) \leq \frac{1}{N} \sum_{i=1}^N P(x_i) \leq \frac{3}{2} \int_{\mathbb{S}^d} |P(x)| d\mu(x).$$

$\lrcorner$

*Proof:* Cf. [\[Mhaskar-Narcowich-Ward, Spherical Marcinkiewicz-Zygmund inequalities and positive quadrature\]](#) Thm3.1.  $\square$

**Cor. (37.1.4.5).** For any  $d \in \mathbb{Z}_+$ , there exists  $r_d \in \mathbb{R}_+$  s.t. for any  $t \in \mathbb{Z}_+$  and any equal-area partition  $\mathcal{R} = \{R_1, \dots, R_t\}$  of  $\mathbb{S}^d$  of diameter bounded by  $r_d/t$ , if we choose any  $x_i \in \mathcal{R}$ , then the following inequality holds for any polynomial  $P \in \mathbb{R}[X_0, X_1, \dots, X_d]$  s.t.  $\deg P \leq t$ :

$$\frac{1}{3} \int_{\mathbb{S}^d} \|\nabla_S P(x)\| d\mu(x) \leq \frac{1}{N} \sum_{i=1}^N \|\nabla_S P(x_i)\| \leq \frac{3}{2} \int_{\mathbb{S}^d} \|\nabla_S P(x)\| d\mu(x).$$

where  $\nabla_S P(x) = \nabla \left( P \left( \frac{x}{|x|} \right) \right)$ .  $\lrcorner$

*Proof:* For  $x \in \mathbb{S}^d$ ,

$$\nabla_S P(x) = (P_1(x), \dots, P_{d+1}(x)), \quad P_j(x) = \frac{\partial}{\partial x_i} P(x) - x_j \sum_{k=1}^{d+1} x_k \frac{\partial}{\partial x_k} P(x)$$

is a polynomial of degree  $\leq t + 1$ . Then we can use [\(37.1.4.4\)](#) on each  $P_j(x)$  together with Cauchy's inequality.  $\square$

## 5 Spherical Packing

References are [Sphere Packings, Lattices and Groups, Conway-Sloane].

**Def. (37.1.5.1) [Spherical Packing].** For  $d \in \mathbb{Z}_+$ , a **spherical packing** in dimension  $d$  is a placing of unit spheres in  $\mathbb{R}^d$ .

The problem of spherical packing is to find the greatest packing density.  $\lrcorner$

**Def. (37.1.5.2) [Lattice Packings].** A **lattice packing** is a sphere packing s.t. the center of spheres form a complete lattice in  $\mathbb{R}^d$ .

A **periodic packing** is a sphere packing s.t. the spheres are centered on the points of f.m. translates of a complete lattice.  $\lrcorner$

**Def. (37.1.5.3) [Upper Density of Packings].** For any sphere packing  $\mathcal{P}$  in  $\mathbb{R}^d$ , the upper density of  $\mathcal{P}$  is defined to be

$$\Delta = \overline{\lim}_{r \rightarrow \infty} \sup_{p \in \mathbb{R}^d} \frac{\text{Vol}(B(p, r) \cap \mathcal{P})}{\text{Vol}(B(p, r))}.$$

$\lrcorner$

**Prop. (37.1.5.4) [Uniformly Dense Packing].** For a sphere packing  $\mathcal{P}$  in  $\mathbb{R}^d$ , if the limit

$$\Delta = \lim_{r \rightarrow \infty} \frac{\text{Vol}(B(p, r) \cap \mathcal{P})}{\text{Vol}(B(p, r))}$$

exists for some  $p \in \mathbb{R}^d$ , then it exists for any  $p$ , and the limit is equal.

A **uniformly dense packing** is a packing  $\mathcal{P}$  that the limit is uniform in  $p$ .  $\lrcorner$

**Prop. (37.1.5.5).** The maximal upper density is achieved by a uniformly dense packing (37.1.5.4).  $\lrcorner$

*Proof:* ?  $\square$

**Prop. (37.1.5.6).** If  $\mathfrak{P} \subset \mathbb{R}^d$  is a uniformly dense packing, then for any compact subset  $R \subset \mathbb{R}^d$  that is the closure of its interior, and for any  $p \in \mathbb{R}^d$ , we have

$$\Delta = \lim_{r \rightarrow \infty} \frac{\text{Vol}(rR \cap \mathcal{P})}{\text{Vol}(rR)}.$$

$\lrcorner$

*Proof:*  $\square$

**Prop. (37.1.5.7) [Approximation by Periodic Packings].** Suppose  $\Delta_d$  is the maximal density of sphere packings in  $\mathbb{R}^d$ , then for any  $\varepsilon \in \mathbb{R}_+$ , there exists a periodic packing with density  $\geq \Delta - \varepsilon$ .  $\lrcorner$

*Proof:* By (37.1.5.5), it suffices to show for a uniformly dense packing  $\mathcal{P}$  with maximal density  $\Delta$ . Take an arbitrary lattice  $\Lambda$  with fundamental parallelogram  $R$ , then by (37.1.5.6),

$$\Delta = \lim_{r \rightarrow \infty} \frac{\text{Vol}(rR \cap \mathcal{P})}{\text{Vol}(rR)}.$$

Then we can take  $r$  large enough, and take all spheres with center in  $rR$  and their translations by  $r\Lambda$ . This density can be arbitrarily close to  $\Delta$ .  $\square$



### Upper Bounds

**Thm. (37.1.5.8).** One can prove that periodic packings come arbitrarily close to the greatest packing density, so when proving upper bounds it suffices to consider periodic packings.  $\lrcorner$

*Proof:*  $\square$

**Def. (37.1.5.9) [Admissible Functions].** A  $f \in L^1(\mathbb{R}^d)$  is called an **admissible function** if  $|f(x)|$  and  $|\text{Fou}f(x)|$  are both bounded by a constant times of  $(1 + |x|)^{-d-\delta}$  for some  $\delta \in \mathbb{R}_+$ .  $\lrcorner$

**Thm. (37.1.5.10) [Cohen-Elkies Linear Programing Bounds].** Let  $f \neq 0 \in L^1(\mathbb{R}^d)$  be a real admissible function and  $\mathcal{F}f(0) \neq 0$ , s.t.

$$f(x) \leq 0, \forall |x| \geq 1, \quad \mathcal{F}f(x) \geq 0, \forall x.$$

Then the density of spherical packing of dimension  $d$  is bounded above by

$$\frac{f(0)}{\mathcal{F}f(0)} \text{Vol } D_d(0, \frac{1}{2}) = \frac{f(0)}{\text{Fou}f(0)} \frac{\pi^{d/2}}{2^d \Gamma(\frac{d}{2} + 1)}.$$

$\lrcorner$

*Proof:*  $\square$

### Lower Bounds

**Prop. (37.1.5.11) [Sphere Packing in Dimension 1].** The maximal density of the sphere packing in dimension 1 is  $\Delta_1 = 1$ .  $\lrcorner$

**Thm. (37.1.5.12) [Sphere Packing in Dimension 2, Toth].** The maximal density of the sphere packing in dimension 2 is  $\Delta_2 = \frac{\pi}{\sqrt{12}} \approx 0.90690$ , and it is attained by the hexagonal lattice.  $\lrcorner$

*Proof:*  $\square$

**Thm. (37.1.5.13) [Kepler Conjecture, Hales].** The maximal density of the sphere packing in dimension 3 is  $\Delta_3 = \frac{\pi}{\sqrt{18}} \approx 0.74048$ , and it is attained by the face-centered cubic packing.  $\lrcorner$

*Proof:* [Hales, A proof of the Kepler conjecture].?  $\square$

**Thm. (37.1.5.14) [Sphere Packing in Dimension 8, Cohen-Elkies-Viazovska].** The maximal density of the sphere packing in dimension 8 is  $\Delta_8 = \frac{\pi^4}{384} \approx 0.25367$ . And up to rotation, this density is only attained by the  $E_8$ -lattice.  $\lrcorner$

*Proof:* Taking  $f(x) = g(\sqrt{2}x)$  where  $g$  is as in lemma(37.1.5.15), then  $f$  satisfies the condition in(37.1.5.10), and  $\frac{f(0)}{\text{Fou}f(0)} = 2^4$ , thus the density is bounded above by  $\Delta_8 = \frac{\pi^4}{384}$ . However, the density of the  $E_8$ -lattice packing is just  $\Delta_8$ ?.

For the uniqueness, for any periodic packing  $\bigcup_{i=1}^N (\Lambda + v_i)$ , we may assume that  $\text{Vol}(\mathbb{R}^n/\Lambda) = N$ . Because this is a densest packing, it implies that the packing sphere size is the same as that of  $\Lambda_8$ . Then the Poisson summation formula

$$\sum_{1 \leq j, k \leq N} \sum_{x \in \Lambda} g(x + v_j - v_k) = \frac{1}{N} \sum_{t \in \Lambda^D} \text{Fou}g(t) \left| \sum_j e^{2\pi i \langle v_j, t \rangle} \right|^2.$$

Then the conditions on  $g$  in (37.1.5.15) imply that each  $|x + v_i - v_j|^2 \in 2\mathbb{Z}$ . This implies that the lattice generated by centers of this periodic packing is an integral even lattice, which has covolume at least 1. But then it must have covolume 1 because it contains  $\Lambda$ , and also it has the same packing radius as  $E_8$  because it is even. So the periodic packing is in fact an even lattice packing. Then by ?, it is isomorphic to  $E_8$ .  $\square$

**Lemma (37.1.5.15).** There exists a radial function  $g : \mathbb{R}^8 \rightarrow \mathbb{R}$  s.t.

- $g(x) \leq 0$  for any  $\|x\| \geq 2$ .
- $\text{Foug}(x) \geq 0$  for any  $x$ .
- $g(0) = \mathcal{F}g(0) = 1$ .
- $g(x) \neq 0, \mathcal{F}g(x) \neq 0$  for any  $x$  s.t.  $\|x\|^2 \notin 2\mathbb{Z}$ .

Notice that Poisson summation w.r.t. the self-dual lattice  $\Lambda_8$  will imply that  $g(\lambda) = \text{Foug}(\lambda) = 0$  for any  $\lambda \in \Lambda^*$ .  $\lrcorner$

*Proof:* Cf. [Viazovska, Sphere Packing in Dimension 8]P994. ?  $\square$

## 6 Probabilistic Methods

References are [The probabilistic method, Alon-Spencer].

## 37.2 Graph Theory I

References are [Die17].

### Notation (37.2.0.1).

- For simplicity, we usually use graph to denote a simple graph (37.2.1.1) (unless specified otherwise), and a multi-graph to denote a graph with (possibly) multi-edges or loops.
- For a multi-graph  $G$ ,
  - let  $V(G)$  denote the set of vertices of  $G$ ,
  - let  $E(G)$  denote the set of edges of  $G$ ,
  - let  $\vec{E}(G)$  denote the set of triples  $(e, x, y)$  where  $e \in E(G)$ , and  $e \cap V(G) = \{x, y\}$ ,
  - let  $v(G)$  denote the number of vertices,
  - let  $e(G)$  denote the number of edges,
  - let  $\chi(G) = v(G) - e(G)$  denote the Euler characteristic of  $G$ ,
  - for  $v \in V(G)$ , let  $\deg(v)$  denote the degree of  $v$  in  $G$ ,
  - let  $d(G)$  (or  $t(G)$ ) denote the average degree of vertices in  $G$ , which equals  $2e(G)/v(G)$  when  $G$  is finite,
  - let  $\delta(G)$  denote the minimum degree of vertices of  $G$ ,
  - let  $\Delta(G)$  denote the maximum degree of vertices of  $G$ ,
  - for  $U \subset V(G)$ , let  $N(U)$  denote the set of vertices connected to  $U$ , and let  $E(U)$  denote the set of edges adjacent to  $U$ .
  - let  $\omega(G)$  denote the maximal cardinality of a clique in  $G$  (37.2.5.1),
  - let  $\alpha(G)$  denote the maximal cardinality of an independent subset of  $G$ ,
  - for two disjoint non-empty subset  $X, Y \subset V(G)$ , let  $e(X, Y)$  denote the number of edges between vertices in  $X$  and vertices in  $Y$ , and  $d(X, Y) = \frac{e(X, Y)}{\#X\#Y}$  the density between  $X$  and  $Y$ .
  - for  $U \subset G$ ,  $G(U)$  is the induced subgraph on  $U$ ,
  - let  $\gamma(G)$  denote the chromatic number of  $G$ ,
  - let  $\kappa(G)$  denote the number of spanning trees of  $G$  (37.2.1.23),
  - let  $g(G)$  denote the girth of  $G$ ,
  - let  $h(G)$  denote the expanding constant of  $G$  (37.3.10.1).

┘

### 1 Basics

**Def. (37.2.1.1) [Graphs].** A **graph**  $X$  is a CW-complex of dimension  $\leq 1$  with a given presentation  $(X^0, X^1 = X)$  (4.13.3.1). the elements in the set of  $X^0$  are called **vertices** of  $X$ , and the 1-cells are called **edges** of  $X$ .

A **simple graph** is a graph without loops and multi-edges.

For a graph  $X$ , a **subgraph** is a sub CW-complex of  $X$ . And if  $V' \subset V(X)$  is a subset, the **induced subgraph** of  $X$  is defined to be the subgraph  $X' \subset V(X)$  whose vertices are just  $V'$ , and two vertices in  $V'$  are connected by an edge iff they are connected by an edge in  $X$ . ┘

**Def. (37.2.1.2)[Claws].** A **claw graph** is any graph isomorphic to the graph  $G = (V(G), E(G))$  where  $V(G) = \{1, 2, 3, 4\}$ ,  $E(G) = \{12, 13, 14\}$ .  $\lrcorner$

**Def. (37.2.1.3)[Wheel Graphs].** For  $n \in \mathbb{Z}_+$ , a **Wheel graph**  $\text{Wheel}_n$  is defined to be the graph  $C_{n-1} * \text{pt}$ .  $\lrcorner$

### Degrees of Vertices

**Def. (37.2.1.4)[Complete Graphs].** A **complete graph** is a simple graph that each two vertices is connected by an edge. For  $r \in \mathbb{Z}_+$ , a complete graph of size  $r$  is denoted by  $K^r$ .  $\lrcorner$

**Def. (37.2.1.5)[Complement Graph].** For  $G \in \text{Graph}^{\text{simple}}$  the **complement graph**  $G^c \in \text{Graph}^{\text{simple}}$  is a simple graph s.t.  $V(G^c) = V(G)$ , and two distinct vertices in  $G^c$  are adjacent if they are not adjacent in  $G$ .  $\lrcorner$

**Def. (37.2.1.6)[Kernels].** For any oriented graph  $H \in \text{Graph}$ , a **kernel of graph** of  $H$  is an independent set  $U \subset V(H)$  s.t. for any  $v \in V(H) \setminus U$ , there exists an edge  $vu \in \vec{E}(G)$  with  $u \in U$ .  $\lrcorner$

**Prop. (37.2.1.7).** For any  $G \in \text{Graph}^{\text{fin}}$  with  $E(G) \neq \emptyset$ , there is a subgraph  $H$  of  $G$  s.t.  $2\delta(H) \geq t(H) \geq t(G)$ .  $\lrcorner$

*Proof:* If there is a vertex  $v \in V(G)$  s.t.  $\deg(v) < \frac{1}{2}t(G) = e(G)/v(G)$ , then we delete  $v$ , and get  $G_1 = G \setminus \{v\}$ . Do the same for  $G_1$ , and so on. In this way, it is easy to see that  $t(G)$  is increasing. Because  $G$  is finite, this process stops at some point. But because  $t(\text{pt}) = 0 < G$ , the stopping graph  $H$  cannot be empty. And this graph  $H$  satisfies  $2\delta(H) \geq t(H) \geq t(G)$ .  $\square$

**Def. (37.2.1.8)[Groupies].** A **groupie** in a graph  $G$  is a vertex  $P \in V(G)$  s.t. if  $r(P)$  is the summation of degree of vertices of points  $Q$  adjacent to  $P$ , then  $r(P) \geq \frac{2e(G)}{v(G)} \deg P$ .  $\lrcorner$

**Prop. (37.2.1.9).** Any  $G \in \text{Graph}$  has a groupie.  $\lrcorner$

*Proof:* For  $P \in V(G)$ , let  $r(P)$  is the summation of degree of vertices of points  $Q$  adjacent to  $P$ . then

$$\sum_{P \in V(G)} r(P) = \sum_{Q \in V(G)} \deg(Q)^2 \geq (\sum \deg(P))^2 / v(G) = t(G)^2 v(G) = t(G) \sum \deg(P).$$

So there must be some  $P \in V(G)$  that  $r(P) \geq t(G) \deg P$ .  $\square$

### Paths and Circuits

**Def. (37.2.1.10)[Paths, Circuits and Cycles].** For  $G \in \text{Graph}$ , a **walk in an (oriented)graph** of length  $k \in \mathbb{Z}_{\geq 2}$  is a sequence  $(v_1, e_1, \dots, e_{k-1}, v_k)$  s.t.

- $v_j \in V(G)$  for any  $j \in [k]_+$ ,
- $e_i = v_i v_{i+1} \in E(G)$  for  $i \in [k-1]_+$ .

A **simple path** is a path without repetition vertices.

A **circuit in a graph** of length  $k \in \mathbb{Z}_{\geq 2}$  is a sequence  $(v_1, e_1, \dots, e_{k-1}, v_k, e_k)$  s.t.

- $v_j \in V(G)$  for any  $j \in [k]_+$ ,
- $e_i = v_i v_{i+1} \in E(G)$  for  $i \in [k-1]_+$ ,

- $e_k = v_k v_1 \in E(G)$ .

A **simple circuit** is a circuit without repetition vertices.

A **cycle in a graph**(resp. ordered cycle) is an  $\mathbb{Z}/(k) \times \mathbb{Z}/(2)$ (resp.  $\mathbb{Z}/(k)$ )-orbit of simple circuits.

The **girth**  $g(G)$  of  $G$  is the minimal length of a circuit in  $G$ .  $\lrcorner$

**Prop. (37.2.1.11) [Minimal Degree and Large Cycles].** For any  $G \in \mathbf{Graph}$ ,  $G$  contains a path of length  $\delta(G)$ . And if  $\delta(G) \geq 2$ , it contains a cycle of length  $\geq \delta(G) + 1$ .  $\lrcorner$

*Proof:* Consider the largest simple path, then the neighbourhood of the endpoints are all on this path, thus it has length  $\geq \delta(G)$ . And consider its neighbors on this path, we get a simple circuit.  $\square$

**Def. (37.2.1.12) [Distances].** For any  $G \in \mathbf{Graph}$  and  $x, y \in V(G)$ , the **distance between vertices**  $d_G(x, y)$  is defined to be the infimum of lengths of paths in  $G$  with endpoints  $x$  and  $y$ .

The maximal distances between two vertices in  $G$  is called the **diameter of graph** of  $G$ , denoted by  $\text{diam}(G)$ .  $\lrcorner$

**Def. (37.2.1.13) [Radius].** For  $G \in \mathbf{Graph}$ , the **radius** of  $G$  is defined to be

$$\text{rad}(G) = \min_{x \in V(G)} \max_{y \in V(G)} d_G(x, y).$$

Clearly  $\text{rad}(G) \leq \text{diam}(G) \leq 2\text{rad}(G)$ .  $\lrcorner$

**Prop. (37.2.1.14).** For  $G \in \mathbf{Graph}$ , either  $g(G) = \infty$  or  $g(G) \leq 2\text{diam}(G) + 1$ .  $\lrcorner$

**Prop. (37.2.1.15) [Radius, Maximal Degrees and Sizes].** For  $k, d \in \mathbb{Z}_+$ ,  $G \in \mathbf{Graph}$ , if  $\text{rad}(G) \leq k$ ,  $\Delta(G) \geq d \geq 3$ , then

$$v(G) \leq \frac{d}{d-2}(d-1)^k.$$

*Proof:* Take a central point  $z \in G$  and let  $D_i$  be the set of vertices with distances  $i$  from  $z$ . Then  $\#D_0 = 1$ ,  $\#D_i \leq d(d-1)^{i-1}$  for  $i \in \mathbb{Z}_+$ . Thus we get the desired result by counting.  $\square$

**Prop. (37.2.1.16) [Average Degree, Girth and Sizes, Alon-Hoory-Linial2002].** For  $g, d \in \mathbb{Z}_+$ ,  $G \in \mathbf{Graph}$ , if  $d(G) \geq d \geq 2$  and  $g \leq g(G) < \infty$ , then

$$v(G) \geq \begin{cases} 2 \sum_{i=0}^{\lfloor \frac{g}{2} \rfloor - 1} (d-1)^i & , 2|g \\ 1 + d \sum_{i=0}^{\lfloor \frac{g}{2} \rfloor - 1} (d-1)^i & , 2 \nmid g \end{cases}.$$

*Proof:* ?  $\square$

**Cor. (37.2.1.17).** If  $d(G) \geq 3$ , then  $g(G) < 2 \log v(G)$ .  $\lrcorner$

**Def. (37.2.1.18) [Large Girth Families].** For  $k \in \mathbb{Z}_{\geq 3}$ , a family of  $k$ -regular finite simple graphs  $(X_n) \in \mathbf{Graph}^{\mathbb{Z}_+}$  is called a family of **large girth graphs** if there exists  $C \in \mathbb{R}_+$  s.t.  $\lim_{n \rightarrow \infty} v(X_n) = \infty$  and  $g(X_n) \geq C \log_{k-1} v(X_n)$ .  $\lrcorner$

**Prop. (37.2.1.19) [Large Girth and Large Connectivity Graphs].** For any  $k \in \mathbb{Z}_+$ , there exists a finite graph  $H \in \mathbf{Graph}^{\text{simple}}$  with  $g(H) > k$  and  $\kappa(H) > k$ .  $\lrcorner$

*Proof:* This follows from (37.3.2.9) and (37.2.4.5).  $\square$

**Def. (37.2.1.20) [Prime circuits].** A **prime circuit** in an unoriented graph  $G$  is a circuit that is not a non-trivial multiple of another cycle, and doesn't contain a segment of the form  $\dots, x, y, x, \dots$  for  $x, y \in V(G)$ .  $\lrcorner$

**Def. (37.2.1.21) [Primitive Graphs].** For a connected graph  $G \in \mathbf{Graph}$ , the gcd of lengths of circuits in  $G$  is called the **period of graph**  $G$ .  $G$  is called a **primitive graph** if it has period 1.

If  $G$  has period  $v$ , then it can be decomposed into  $G_1 \amalg \dots \amalg G_v$  s.t. each edge is between some  $G_i$  to  $G_{i+1}$ .  $\lrcorner$

**Def. (37.2.1.22) [Holes].** For a simple graph, a **hole of graph**  $H \subset G$  is an induced subgraph that is isomorphic to a cycle of length  $\geq 4$ . An **antihole of a graph** is an induced subgraph of  $G$  whose complement is a graph in  $G^c$ .  $\lrcorner$

### Trees and Forests

**Def. (37.2.1.23) [Trees].** A **tree** is a graph  $T$  s.t.  $H_1(T) = 0$ . A tree in a graph  $X$  is called a **maximal tree** if it contains all vertices of  $X$ . A **forest** is a graph that is a disjoint union of trees.  $\lrcorner$

**Prop. (37.2.1.24).** Any tree  $T$  is contractible, and any vertex  $v$  is a deformation retract of  $T$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (37.2.1.25) [Maximal Trees].** Every connected graph contains a maximal tree, and any tree in a connected graph is contained in a maximal tree.

A tree  $T$  in a graph  $X$  is maximal iff  $X^0 \subset T$ .  $\lrcorner$

*Proof:* Cf. [Hat02]P84.  $\square$

**Prop. (37.2.1.26) [ $\pi_1$  of Graphs].** For a connected graph  $X$  with a maximal tree  $T$ ,  $\pi_1(X)$  is a free group with basis the classes  $[f_\alpha]$  corresponding to the edges  $e_\alpha \in X \setminus T$ .  $\lrcorner$

*Proof:* The quotient map  $X \rightarrow X/T$  is a homotopy equivalence by (4.13.7.7). And the quotient space  $X/T$  has only one vertex, thus is a wedge sum of circles corresponding to  $e_\alpha \in X \setminus T$ . So we are done by (4.13.5.29).  $\square$

**Prop. (37.2.1.27) [Covering Spaces of Graphs].** Every covering space  $\pi : \tilde{X} \rightarrow X$  of a graph is also a graph, with vertices and edges lifts of vertices and edges of  $X$ .  $\lrcorner$

*Proof:* Let the sets  $\pi^{-1}(v)$  be vertices of  $\tilde{X}$ , where  $v$  are vertices of  $X$ . And if we write  $X$  as a quotient space of  $X^0 \cup_\alpha I_\alpha$ , each  $I_\alpha$  can be lifted to maps to  $\tilde{X}$ , and we let these be edges of  $\tilde{X}$ . The topology of  $\tilde{X}$  coming from quotient topology of this is the same as the original topology, because they has the same basis open sets, because  $\pi$  is a local homeomorphism.  $\square$

**Def. (37.2.1.28) [Path Trees].** For  $G \in \mathbf{Graph}$ , the **path tree** of  $G$  at  $u$  is defined to be the universal covering of  $G$  at  $u$  consisting of simple paths at  $u$ .  $\lrcorner$

**Prop. (37.2.1.29) [Spencer's Lemma].** If there is a triangulation of a plane polygon  $P$ , for arbitrary 3-color numbering  $(\{0, 1, 2\})$  of the vertices, if the number of edges on the boundary with color  $(0, 1)$  is odd, then there is a triangle with vertices of pairwise different colors.  $\lrcorner$

*Proof:* In fact the number of those triangles with vertices of pairwise different colors is odd. In fact, the number of  $(0, 1)$ -edges on a triangle is odd iff its vertices has pairwise different colors. But the sum of numbers of  $(0, 1)$ -edges on the triangles are odd, by hypothesis, thus the result.  $\square$

### Contractions and Minors

**Def. (37.2.1.30) [Subdivisions and Topological Minors].** A **subdivision** of a graph  $X$  is any graph obtained from  $X$  by ‘subdividing’ some or all of its edges by drawing new vertices on those edges.

For graphs  $X, Y$ , if  $Y$  contains a subgraph that is isomorphic to a subdivision of  $X$ , then  $X$  is called a **topological minor graph** of  $Y$ .  $\lrcorner$

**Def. (37.2.1.31) [Contractions and Minors].** For any graph  $G$ , divide  $V(G)$  into connected subsets  $V_{\alpha_1}, \dots, V_{\alpha_k}$ , and let  $X$  be the graph with

$$V(X) = \{\alpha_1, \dots, \alpha_k\}, \quad E(X) = \{(\alpha_i, \alpha_j) \mid i \neq j, E(V_{\alpha_i}, V_{\alpha_j}) \neq \emptyset\}.$$

Then  $X$  is called a **contraction graph** of  $G$ .

For graphs  $X, Y$ , if  $X$  is isomorphic to a contraction of a subgraph of  $Y$ , then  $X$  is called a **minor graph** of  $Y$ . We sometimes use the notation  $X = Y/U$  to denote a minor of  $Y$  that only contracts one connected subset  $U$ .  $\lrcorner$

**Prop. (37.2.1.32).** Any topological minor is a minor. And if  $X$  is a minor of a graph  $G$  with  $\Delta(X) \leq 3$ , then  $X$  is also a topological minor of  $G$ .  $\lrcorner$

*Proof:* The first assertion is easy. For the last assertion, consider each  $V_x$  for  $x \in X$ , then it suffices to show that for any three vertices  $x, y, z$  in a connected graph, there is a vertex with separate paths to them without intersection. For this, firstly find a shortest path  $W$  from  $x$  to  $y$ . If this path contains  $z$ , then  $z$  is the vertex we want. If this path doesn’t contain  $z$ , then find a shortest path  $W'$  from  $z$  to  $W$ , then the endpoint of  $W'$  is the vertex we want.  $\square$

## 2 Spectral Graph Theory

**Def. (37.2.2.1) [Companion Matrices].** For a finite multi-graph  $G$  with  $\#V(G) = n$ , given an ordering of the vertices, the

- **adjacency matrix**  $A_G$  of  $G$  is defined to be the  $n \times n$  matrix  $(a_{ij})$  with  $a_{ij} = \alpha(v_i, v_j)$ , the number of edges between  $v_i$  and  $v_j$ .
- **valence matrix**  $D_G$  of  $G$  is defined to be the  $n \times n$  diagonal matrix  $\text{diag}(\deg(v_1), \dots, \deg(v_n))$ .
- **Laplacian matrix**  $L_G$  of  $G$  is defined to be the  $A_G - D_G$ .

It is clear that  $M_G, D_G, L_G$  are all real symmetric. In particular, they have real spectrums.  $\lrcorner$

**Def. (37.2.2.2) [Spectrums of Graphs].** For any finite multi-graph  $G \in \mathbf{Graph}$ , denote

$$\lambda_0 \geq \lambda_1 \geq \dots \geq \lambda_{v(G)}$$

be the set of eigenvalues of  $M_G$ . And denote  $\rho(G) = \max |\lambda_i(G)|$  the **spectrum radius** of  $G$ .  $\lrcorner$

**Prop. (37.2.2.3) [Spectra of Adjacency Matrices].** For  $k, n \in \mathbb{Z}_+$ , if  $G \in \mathbf{Graph}$  is  $k$ -regular and  $n = v(G) \leq n$ , then

- $k = \lambda_0 \geq \lambda_1 \geq \dots \geq \lambda_{n-1} \geq -k$ .
- $k > \mu_1$  iff  $G$  is connected.

$\lrcorner$

*Proof:* 1: Clearly  $(1, \dots, 1)^t$  is an eigenvector with eigenvalue  $k$ . To show that  $|\lambda_i| \leq k$ , just choose a coordinate with largest absolute value.

2: This is also easy by taking a maximum coordinate.  $\square$

**Prop. (37.2.2.4) [Spectrum of Bipartite Graphs].** For  $k, n \in \mathbb{Z}_+$ , if  $G \in \mathbf{Graph}$  is connected,  $k$ -regular and  $n = v(G) \leq n$ , then the following are equivalent:

- $G$  is bipartite.
- $\lambda_i + \lambda_{n-1-i} = 0$ .
- $\mu_{n-1} = -k$ .

┘

*Proof:* 1  $\rightarrow$  2: Suppose  $G = G^+ \amalg G^-$ . If  $v$  is an eigenvector of  $A_G$  with eigenvalue  $\mu$ , then by multiplying the coordinates of  $v$  corresponding to  $v \in G^-$ , we get an eigenvector of  $A_G$  with eigenvalue  $-\mu$ .

2  $\rightarrow$  3 follows from (37.2.2.3).

3  $\rightarrow$  1: similarly, choose a coordinate of the eigenvector with eigenvalue  $-k$ . Then

$$v_x = -\frac{1}{k} \sum_{yx \in E(G)} v_y.$$

So it can be shown that  $v_y = -v_x$  if  $xy \in E(G)$ . By this way and the fact  $G$  is connected, we get a bipartition of  $V(G)$ . □

### $L_G$ and Spectral Gaps

**Prop. (37.2.2.5).** For any finite graph  $G \in \mathbf{Graph}$ ,  $L_G$  is positive-semidefinite, as for any  $\mathbf{x} \in \mathbb{R}^n$ ,

$$\mathbf{x}^T L_G \mathbf{x} = \sum_{1 \leq i < j \leq n} \alpha(v_i, v_j) (x_i - x_j)^2.$$

And if  $G$  is connected, then  $L_G$  has nullity 1, whose eigenspace is spanned by  $\mathbf{x} = (1, 1, \dots, 1)^t$ . ┘

**Prop. (37.2.2.6).** Let  $G$  be a finite connected multi-graph with  $\#V(G) = n$  and diameter  $d = d(G)$ , then the smallest eigenvalue  $\lambda_1$  of  $L_G$  satisfies  $\lambda_1 \geq 1/(nd)$ . ┘

*Proof:* Let  $w$  be the unit eigenvector belong to  $\lambda_1$ , then there exists  $i \in [n]_+$  s.t.  $|w_i| \geq 1/\sqrt{n}$ . We may assume that  $w_1 > 1/\sqrt{n}$ . Since  $w$  is orthogonal to  $(1, \dots, 1)^t$ , we have  $\sum w_i = 0$ . So there exists some  $w_k < 0$ . Let  $v_1, v_2, \dots, v_k$  be a minimal path from  $v_1$  to  $v_k$ , then  $k-1 \leq d$ , and

$$\lambda_1 = \mathbf{w}^t L_G \mathbf{w} = \sum_{1 \leq i < j \leq n} \alpha(v_i, v_j) (w_i - w_j)^2 \geq \sum_{i=1}^{k-1} (w_i - w_{i+1})^2 \geq \frac{1}{k-1} (w_1 - w_k)^2 \geq \frac{1}{nd}.$$

□

**Prop. (37.2.2.7) [Bigg].** For  $k$ -regular finite graph  $G \in \mathbf{Graph}$  and any cofactor  $M$  of  $L_G$ ,  $\det(M) = \det(\text{tree}(G))$ . ┘

*Proof:* Cf. [Biggs, Algebraic Graph Theory] Thm6.3. ? □



### Spectral Radius

**Prop. (37.2.2.8).** If  $X \in \mathbf{Graph}$  is connected, then for any  $x \in X$ ,

$$\rho(X) = \limsup_{s \rightarrow \infty} \sqrt[s]{t_s(x)}$$

where  $t_s(x)$  is the number of closed paths starting from  $x$ . ┘

*Proof:* Cf. [Discrete Groups, Expanding Graphs and Invariant Measures] Chap4. ? □

**Prop. (37.2.2.9).** The spectrum radius of a  $(c, d)$ -regular tree is  $\sqrt{c-1} + \sqrt{d-1}$ . ┘

*Proof:* ? Cf. [Walk generating functions and spectral measures of infinite graphs] and [Spectra of regular graphs and hypergraphs and orthogonal polynomials]. □

**Thm. (37.2.2.10) [Greenberg/Cioabă].** Let  $X$  be a connected, infinite graph with  $\Delta(X) < \infty$ . Given  $\varepsilon \in \mathbb{R}_+$ , there exists  $c(X, \varepsilon) \in \mathbb{R}_+$ ,  $g(X, \varepsilon) \in \mathbb{Z}_+$  such that

- for any finite graph  $Y \in \mathbf{Graph}^{\text{simple}}$  covered by  $X$ ,

$$\#\{\lambda \in \text{Spec}(Y) \mid \lambda \geq \rho(X) - \varepsilon\} \geq c(\varepsilon, k)v(Y).$$

- for any finite graph  $Y \in \mathbf{Graph}^{\text{simple}}$  covered by  $X$  s.t.  $g(Y) \geq g$ ,

$$\#\{\lambda \in \text{Spec}(Y) \mid \lambda \leq -\rho(X) + \varepsilon\} \geq c(\varepsilon, k)v(Y).$$

┘

*Proof:* Cf. [Eigenvalues of graphs and a simple proof of a theorem of Greenberg]. ? □

**Cor. (37.2.2.11) [Alon-Boppana/Serre].** For any  $\varepsilon \in \mathbb{R}_+$  and  $k \in \mathbb{Z}_{\geq 3}$ , there exists a  $c(\varepsilon, k) \in \mathbb{R}_+$  s.t. for  $k$ -regular connected finite graph  $G \in \mathbf{Graph}^{\text{simple}}$ ,

$$\#\{\lambda \in \text{Spec}(G) \mid \lambda \geq (2 - \varepsilon)\sqrt{k-1}\} \geq c(\varepsilon, k)v(G).$$

┘

*Proof:* This follows from (37.2.2.10) and (37.2.2.9). □

### 3 Matching, Covering and Packing

**Def. (37.2.3.1) [Matchings].** For  $G \in \mathbf{Graph}$ , a **matching on graph** is a subset  $M \subset E(G)$  s.t. no two edges in  $M$  have a common vertex (nor loops).

For  $A \subset V(G)$ , a **matching for  $A$**  is a matching  $M \subset E(G)$  s.t. each  $v \in A$  is incident with some  $e \in M$ . ┘

**Def. (37.2.3.2) [Vertex Coverings of Edges].** For  $G \in \mathbf{Graph}$ , a **vertex covering of edges** of  $G$  is a subset  $U \subset V(G)$  s.t. each  $e \in E(G)$  is incident with at least one vertex in  $U$ . ┘

**Thm. (37.2.3.3) [König1931].** For a bipartite graph  $G \in \mathbf{Graph}^{\text{fin}}$ , the maximal cardinality of a matching in  $G$  equals the minimal cardinality of a vertex covering of edges of  $G$ . ┘

*Proof:* It suffices to prove that for a matching  $M \subset E(G)$ , we can find a vertex covering of cardinality  $\#M$ .

Let  $G = G_1 \amalg G_2$  where  $G_1, G_2$  are independent subsets, define an **augmentation path** as a simple path in  $G$  that starts with a vertex in  $G_1$  and alternate between edges in  $E(G) \setminus M$  and  $M$ . Notice an augmentation path ending in a vertex other than vertices of  $M$  will result in a larger path, so any augmentation path ends in vertices of  $M$ .

Now from each matching in  $M$  we choose one of its vertex to form a set  $U = U_1 \amalg U_2$  where a vertex of  $M \cap G_2$  is in  $U_2$  iff it is the endpoint of an augmentation path. Then we claim  $U$  is a vertex covering of edges of  $G$ : It suffices to show that if  $a \in G_1, b \in G_2$  is adjacent, then  $a \in U_1$  or  $b \in U_2$ : Suppose  $a \notin U_1$ , if  $a$  is unmatched, then  $ab$  is an augmentation path, so  $b \in U_2$ . If  $a$  is matched with  $b'$ , because  $a$  is not in  $U_1$ , there is an augmentation path  $P$  ending in  $b'$ . If  $b \in P$ , then  $b \in U$ . If  $b \notin P$ , then  $Pb'ab$  is an augmentation path ending in  $b$ , so also  $b \in U$ .  $\square$

**Cor. (37.2.3.4).** If  $G \in \mathbf{Graph}^{\text{fin}}$  is a  $k$ -regular bipartite graph where  $k \in \mathbb{Z}_+$ , then  $G$  has a 1-factor (full matching).  $\lrcorner$

**Cor. (37.2.3.5) [Petersen 1891].** Let  $G \in \mathbf{Graph}^{\text{fin}}$  be a  $2k$ -regular graph where  $k \in \mathbb{Z}_+$ , then  $G$  has a 2-factor.  $\lrcorner$

*Proof:* By (37.2.4.12), there is an Euler tour  $v_0v_1 \dots v_l$ . Make  $G$  an oriented graph using this Euler tour, and we can construct a 2-covering  $G'$  by replacing each vertex  $v \in V(G)$  by a pair of vertices  $v^-, v^+$ , and replacing each vertex  $\vec{e} = v_i v_{i+1}$  by the edge  $v_i^+ v_{i+1}^-$ . Then this graph is bipartite and  $k$ -regular. Then by (37.2.3.4),  $G'$  has a 1-factor. The projection of this 1-factor in  $G$  is a 2-factor.  $\square$

**Cor. (37.2.3.6) [Marriage Theorem, Hall 1935].** For any finite bipartite graph  $G \in \mathbf{Graph}^{\text{fin}}$  such that  $G = G_1 \amalg G_2$  where  $G_1, G_2$  are independent sets. Then for any  $A \subset G_1$ ,  $G$  contains a matching for  $A$  iff  $\#N(S) \geq \#S$  for any  $S \subset A$ .  $\lrcorner$

*Proof:* Using König's theorem (37.2.3.3), it suffices to consider the case  $A = G_1$  and show that for any vertex cover  $S = S_1 \amalg S_2$  of edges of  $G$ ,  $\#S \geq \#A$ . Suppose for the contrary that  $\#S < \#A$ , then  $\#N(A \setminus S_1) \geq \#A - \#S_1 > \#S_2$ , so there is a vertex in  $G_2 \setminus S_2$  connected to  $S_2$ , contradicting the fact that  $S$  is a vertex cover of edges of  $G$ .  $\square$

### Stable Matchings

**Def. (37.2.3.7) [Stable Matchings].** For any finite bipartite graph  $G \in \mathbf{Graph}^{\text{fin}}$ , a **family of preferences** on  $E(G)$  is a family  $\{\leq_v\}_{v \in V(G)}$  where each  $\leq_v$  is a linear ordering on  $E(v)$ .

Given a finite simple bipartite graph  $G \in \mathbf{Graph}^{\text{fin}}$  with a family of preferences, a **stable matching** on  $G$  is a matching  $M$  on  $G$  s.t. for any  $e \in E(G) \setminus M$ ,  $\exists f \in M$  s.t.  $e, f$  has a common vertex  $v$  and  $e \leq_v f$ .  $\lrcorner$

**Thm. (37.2.3.8) [Stable Mapping Algorithm, Gale-Shapley [G-S62]].** Any bipartite graph  $G \in \mathbf{Graph}^{\text{fin}}$  with a family of preferences has a stable matching (37.2.3.7). And there is an algorithm to do so.  $\lrcorner$

*Proof:* Let  $G = G_1 \amalg G_2$  where  $G_1, G_2$  are independent subsets. The algorithm to find a stable matching works as follows: Each vertex in  $G_1$  propose a matching to its favorite vertex in  $G_2$ , then each vertex in  $G_2$  declines all the proposals it receives except for its favorite one. After this, each vertex in  $G_1$  that is currently declined propose a matching to its favorite vertex in  $G_2$  among those

it hasn't proposed before (if there is any proposals that can be made), and each vertex in  $G_2$  declines all the proposals it receives except for its favorite one (among all proposed formerly proposed). And the process goes on, until no new proposals can be made.

The process will stop otherwise each round a new propose will be made, and there are at most  $\#G_1 \cdot \#G_2$  many possible proposes, contradiction. Then we prove that in this stage, the proposed matching is stable: If  $u \in G_1, v \in G_2$  is not a match and  $u$  is not matched or prefers  $v$  than its current match, then  $u$  must have proposed to  $v$  before and got declined. So  $v$  must be currently matched and it prefers this match than  $u$ . So  $e = uv$  is not a threat to this matching. Thus we proved that this matching is stable.  $\square$

**Prop. (37.2.3.9) [Gale-Shapley Algorithms are Optimal for Proposers].** If  $G = G_1 \amalg G_2 \in \mathbf{Graph}^{\text{fin}}$  is a bipartite graph with  $G_1, G_2$  independent with a family of preferences, then for the assignment of the Gale-Shapley algorithm that  $G_1$  proposes, every vertex in  $G_1$  is at least as well off under this assignment as he would under any other stable assignment (unmatched vertices are considered the worst cases).  $\lrcorner$

*Proof:* We use induction with the procedure to prove that whenever  $A \in G_2$  rejects  $\alpha \in G_1$ ,  $(A, \alpha)$  can never be matched in any stable matching: Consider the time that  $A$  rejects  $\alpha$ , then  $A$  accepts a  $\beta \in G_1$ . By procedure,  $A$  is considered by  $\beta$  as the best, except for those who rejected him, which is by induction hypothesis those can never be matched with  $\beta$  in any stable matching.

Now suppose in some stable matching  $(A, \alpha)$  is matching,  $\beta$  is matched with some  $B$ . Then  $(A, \beta)$  is an alternative match that show this matching is unstable, contradiction. This shows that  $(A, \alpha)$  can never be matched, and the induction is completed.  $\square$

### Matching Polynomials

**Def. (37.2.3.10) [Matching Polynomials].** Let  $G \in \mathbf{Graph}$  with  $n = v(G) < \infty$ . For  $i \in \mathbb{N}$ , let  $m_i$  be the number of matchings in  $G$  consisting of  $i$  edges, then the **matching polynomial** of  $G$  is defined to be

$$\mu_G(T) = \sum_{i \in \mathbb{N}} (-1)^i m_i T^{n-2i}.$$

$\lrcorner$

**Thm. (37.2.3.11) [Roots of Matching Polynomials, Heilmann-Lieb/Godsil].** For any  $G \in \mathbf{Graph}$ , the roots of the matching polynomial  $\mu_G(X)$  are real, and have absolute value at most  $\rho(P(G, u))$ , where  $P(G, u)$  is the path tree of  $G$  at  $u$  (37.2.1.28).

In particular, the roots have absolute values at most  $\rho(T)$ , where  $T$  is the universal cover of  $G$ , which is bounded by  $2\sqrt{\Delta(G) - 1}$ .  $\lrcorner$

*Proof:* Cf. [Algebraic Combinatorics, Godsil]. ?

For the last assertion, notice  $P(G, u)$  is an induced subgroup of  $T$ , so so clearly  $\rho(P(G, u)) \leq \rho(T)$ .

$\square$

## 4 Connectivity

**Def. (37.2.4.1) [ $s$ -Connected Graphs].** For  $s \in \mathbb{Z}_+$ , a graph  $G \in \mathbf{Graph}$  is called  **$k$ -connected** if it is connected after deleting any  $k - 1$  vertices.

The maximal  $k \in \mathbb{Z}_+$  s.t.  $G$  is  $k$ -connected is denoted by  $\kappa(G)$ .  $\lrcorner$

**Def. (37.2.4.2) [Edge-Connected Graphs].** The maximal  $k \in \mathbb{Z}_+$  s.t.  $G$  is  $k$ -edge-connected is denoted by  $\lambda(G)$ .  $\lrcorner$

**Def. (37.2.4.3) [Toughness].** For  $t \in \mathbb{R}_+$ , a graph  $G \in \mathbf{Graph}$  is called a  $t$ -tough graph if for any separator  $S \subset V(G) \cup E(G)$ ,  $G \setminus S$  has at most  $\#S/t$  connected components.  $\lrcorner$

**Prop. (37.2.4.4).** For any finite graph  $G \neq \text{pt} \in \mathbf{Graph}^{\text{simple}}$ ,  $\kappa(G) \leq \lambda(G) \leq \delta(G)$ .  $\lrcorner$

*Proof:* The second inequality is trivial. For the first, let  $F$  be a minimal edge separating set of  $G$ . If there is a vertex  $v \in V(G)$  that is not incident to any edge in  $F$ , let  $C$  be the connected component of  $G \setminus F$  containing  $v$ , then each edge in  $F$  connects  $C$  to another connected component, by minimality. But the vertices in  $C$  that is incident to any edge in  $F$  separates  $v$  from  $G \setminus C$ , thus  $\kappa(G) \leq \lambda(G)$ .

If every vertex  $v \in V(G)$  is incident to an edge in  $F$ , then the neighborhoods  $w$  of  $v$  s.t.  $vw \notin F$  are incident to different edges in  $F$  by minimality. Thus  $\deg(v) \leq \lambda(G)$ . As the neighborhoods of  $v$  separates  $G$  if  $G \neq \{v\} \cup N(v)$ , if  $G$  is not complete, then we get  $\kappa(G) \leq \lambda(G)$ . But when  $G$  is complete,  $\kappa(G) = \lambda(G) = v(G) - 1$ .  $\square$

**Thm. (37.2.4.5) [Minimal Degree and Connectivity, Mader1972].** For  $k \in \mathbb{Z}_+$ , any  $G \in \mathbf{Graph}$  with  $d(G) \geq 4k$  has a  $(k+1)$ -connected subgraph  $H$  s.t.  $d(H) \geq d(G) - 2k$ .  $\lrcorner$

*Proof:* Cf.[Diestel]P13. ?  $\square$

**Def. (37.2.4.6) [( $x, y$ )-Cuts].** For a finite graph  $G$  and  $x, y \in V(G)$  with  $d(x, y) \geq 2$ , an  $(x, y)$ -cut in  $G$  is a set  $S \in V(G) \setminus \{x, y\}$  s.t.  $G \setminus S$  has no  $(x, y)$ -paths.  $\lrcorner$

**Thm. (37.2.4.7) [Menger].** Let  $G$  be a finite graph and  $x, y \in V(G)$  s.t.  $d(x, y) \geq 2$ . Denote  $\kappa_G(x, y)$  the minimum size of an  $(x, y)$ -cut in  $G$  and  $\lambda_G(x, y)$  a maximum number of internally-disjoint  $(x, y)$ -paths in  $G$ , then  $\kappa_G(x, y) = \lambda_G(x, y)$ .  $\lrcorner$

*Proof:*  $\square$

**Def. (37.2.4.8) [( $x, U$ )-Fans].** For a finite graph  $G$  and a subset  $U \subset V(G)$  and  $x \in V(G) \setminus U$ , an  $(x, U)$ -fan of size  $k$  is a set of  $k$ -paths from  $x$  to  $U$  s.t. any two of them share only  $x$ .  $\lrcorner$

**Prop. (37.2.4.9) [Fan Lemma, Dirac].** For  $k \in \mathbb{Z}_+$  and  $G \in \mathbf{Graph}$ ,  $\kappa(G) \geq k$  iff  $v(G) \geq k+1$ , and for any subset  $U \subset V(G)$  and each  $x \in V(G) \setminus U$ ,  $G$  has an  $(x, U)$ -fan of size  $\min(k, \#U)$ .  $\lrcorner$

*Proof:* If  $G$  has at least  $k+1$  vertices and for any subset  $U \subset V(G)$  with  $|U| \geq k$  and each vertex  $x \notin U$ ,  $G$  has an  $(x, U)$ -fan of size  $k$ , we show  $G$  is  $k$ -connected: If there is a  $(k-1)$ -vertex cut  $S$ , let  $A$  be a connected component of  $G \setminus S$ , and  $U = V(G) \setminus A$ . Then  $\#U \geq k$ , and for any  $x \in A$ , there are no  $(x, U)$ -cut of size  $k$ , because each path from  $x$  to  $U$  must intersect with  $S$ .

Conversely, if  $G$  is  $k$ -connected and  $U \subset V(G)$ ,  $\#U \geq k$ . Let  $G' = G * \{y\}$ , then  $G'$  is also  $k$ -connected. Then by Menger's theorem(37.2.4.7),  $\lambda_G(x, y) = \kappa_G(x, y) \geq k$ . Let  $P_1, \dots, P_k$  be internally-disjoint  $(x, y)$ -paths in  $G'$ , then after deleting  $y$ , they form an  $(x, U)$ -fan of size  $k$ .  $\square$

## Hamilton Cycles

**Def. (37.2.4.10) [Eulerian Tours].** For a connected graph  $G \in \mathbf{Graph}^{\text{fin}}$ , an **Eulerian tour** is a circuit on  $G$  that traverses each edge of  $G$  exactly once. An **Eulerian graph** is a finite connected graph that admits an Euler tour.  $\lrcorner$

**Def. (37.2.4.11) [Hamilton Cycles].** For a graph  $G \in \mathbf{Graph}^{\text{fin}}$ , a **Hamilton-cycle** is a cycle in  $G$  that contains each vertex of  $G$  exactly once. Similarly, a **Hamilton-path** is a simple path in  $G$  that traverse each vertex of  $G$  exactly once.  $G$  is called a **Hamiltonian graph** if it has a Hamiltonian path.

$G$  is called **Hamiltonian-connected** if each pair of vertices  $x \neq y \in G$  is the endpoint of a Hamiltonian path in  $G$ .  $\lrcorner$

**Prop. (37.2.4.12) [Euler1736].** A connected graph  $G \in \mathbf{Graph}^{\text{fin}}$  is an Eulerian graph iff  $2 \mid \deg(v)$  for any  $v \in V(G)$ .  $\lrcorner$

*Proof:* One direction is clear. For the other, we use induction on  $v(G)$ . If  $v(G) = 0$ , this is true. For  $v(G) > 0$ , we can clearly find a walk with no repeated edges. Let  $W$  be such a walk with maximum length. Then we claim  $W$  is a Euler tour: Suppose not, then  $G' = G \setminus E(W)$  also satisfies the hypothesis that  $2 \mid \deg(v)$  for any  $v \in V(G)$ . As  $G$  is connected there is an edge  $e \in E(G')$  that is incident to a vertex in  $W$ . Then by induction hypothesis, the component of  $G'$  containing  $e$  has an Eulerian tour, which in concatenating with  $W$  makes a larger walk, contradiction.  $\square$

**Thm. (37.2.4.13) [Asratian-Khachatrian1990].** A connected graph  $G \in \mathbf{Graph}^{\text{fin}}$  has a Hamiltonian cycle if  $v(G) \geq 3$ , and

$$\deg(u) + \deg(w) \geq \#N(u) \cup N(v) \cup N(w)$$

for any induced path  $uvw$  (i.e.  $uw \notin E(G)$ ).  $\lrcorner$

*Proof:* The hypothesis is in fact equivalent to that

$$\#N(u) \cup N(w) \geq \#N(v) \setminus N(\{u, w\}).$$

As  $\{u, w\} \subset N(v) \setminus N(\{u, w\})$ , this implies that  $G$  has simple circuits. Let  $C$  be a longest simple circuit in  $G$ . Suppose  $G$  is not Hamiltonian, then there exists  $u \in G \setminus C$  with  $V = N(u) \cap C \neq \emptyset$ . Let  $V^+ = \{v^+ \mid v \in V\}$  be the set of successors of  $V$  in  $G$ , then by the maximality of  $C$ ,  $V \cap V^+ \neq \emptyset$ , and  $V^+ \cup \{u\}$  is an independent set, and no two of them have a common neighborhood outside  $C$ . Thus for any  $v \in V$ , we have

$$\#N(u) \cap N(v^+) \geq \#N(v) \setminus N(\{u, v^+\}) \geq \#N(v) \cap V^+ + 1.$$

Thus

$$e(V, V^+) = \sum_{v \in V} \#N(v) \cap V^+ \leq \sum_{v \in V} (\#N(u) \cap N(v^+) - 1) = e(V, V^+) - \#V,$$

contradiction.  $\square$

**Cor. (37.2.4.14) [Dirac1952].** For any finite simple graph  $G \in \mathbf{Graph}^{\text{simple}}$  with  $n = v(G) \geq 3$  and  $\delta(G) \geq n/2$  has a Hamilton cycle.  $\lrcorner$

**Thm. (37.2.4.15) [Chvátal-Erdős].** If  $k \in \mathbb{Z}_+$  and a simple graph  $G \in \mathbf{Graph}^{\text{fin}}$  satisfies  $v(G) \geq 3$ ,  $i(G) \leq \kappa(G)$ , then  $G$  has a Hamiltonian-cycle.  $\lrcorner$

*Proof:* The hypothesis implies that  $G$  is not a tree thus contains a simple path. Let  $k = \kappa(G)$ , and take a longest simple path  $C$ , if there exists  $x \in V(G) \setminus C$ , then by fan lemma (37.2.4.9), there exists an  $(x, U)$ -fan of size  $\min(k, \#C)$ , with vertices  $x_1, \dots, x_r$ . Let  $y_i$  be the successor of  $x_i$  in  $C$ , then  $xy_i \notin E(G)$  for each  $i$ , otherwise there is a longer simple path. This implies that  $\#C > k$ . Now for the set  $\{x, y_1, \dots, y_r\}$ , the hypothesis  $\kappa(G) = k$  implies that there is a path  $y_i y_j$ . Then we can find a cycle of larger size, contradiction.  $\square$

**Cor. (37.2.4.16).** If  $k \in \mathbb{Z}_+$  and a simple graph  $G \in \mathbf{Graph}^{\text{fin}}$  satisfies  $v(G) \geq 3, i(G) \leq \kappa(G) + 1$ , then  $G$  has a Hamiltonian-cycle.  $\lrcorner$

*Proof:* The graph  $G$  satisfies the hypothesis of (37.2.4.15) and has a Hamiltonian cycle. Thus  $G$  has an Hamiltonian-path.  $\square$

**Cor. (37.2.4.17).** If  $k \in \mathbb{Z}_+$  and a simple graph  $G \in \mathbf{Graph}^{\text{fin}}$  satisfies  $v(G) \geq 3, i(G) \leq k$ , and  $G$  is  $k$ -connected, then  $G$  is Hamiltonian-connected.  $\lrcorner$

*Proof:* The proof is similar to that of (37.2.4.15): For any vertices  $x \neq y \in G$ , just choose the longest path from  $x$  to  $y$ .  $\square$

**Prop. (37.2.4.18)[Tutte1956].** Any 4-connected Tutte planer graph has a Hamilton cycle.  $\lrcorner$

*Proof:*  $\square$

**Conj. (37.2.4.19) [Toughness Conjecture, Chvátal].** There exists  $t \in \mathbb{R}_+$  s.t. every  $t$ -tough graph (37.2.4.3) has a Hamilton-cycle.  $\lrcorner$

*Proof:*  $\square$

## 5 Complete Graphs

### Cliques and Independent Sets

**Def. (37.2.5.1)[Cliques].** For  $G \in \mathbf{Graph}$ , a **clique** in  $G$  is a subgraph  $H \subset G$  that is a complete graph. An **independent set** in a graph is a set of vertices that no two of them are adjacent.

For any graph  $G$ , let  $\omega(G)$  be the size of the maximal clique in  $G$ , and let  $\alpha(G) = \omega(G^c)$  be the size of the maximal independent set of  $G$ .  $\lrcorner$

**Def. (37.2.5.2)[Simplicial Cliques].** For  $G \in \mathbf{Graph}$ , a **simplicial clique** in  $G$  is a clique  $K$  in  $G$  s.t.  $N(k) \setminus K$  is a clique in  $G \setminus K$  for any  $k \in K$ .  $\lrcorner$

**Prop. (37.2.5.3).** If  $G \in \mathbf{Graph}$  is claw-free and  $K$  is a simplicial clique in  $G$ , then for any  $k \in K$ ,  $N(k) \setminus K$  is also a simplicial clique.  $\lrcorner$

*Proof:* By definition  $N(k) \setminus K$  is a clique. If for some  $k \in K$  and  $n \in N(k) \setminus K$ , the set  $H = N(n) \setminus (N(k) \cup K)$  is not a clique, then there exists two non-adjacent vertices  $x, y \in H$ . Then the graph induced by  $\{x, y, n, k\}$  is a claw, contradiction.  $\square$

**Prop. (37.2.5.4)[Erdős].** For any  $k, l \in \mathbb{Z}_+$  and a graph  $G$  of size  $(k-1)(l-1)+1$ , either  $G$  contains a complete  $k$ -graph, or any no-circuits ordering of the complement graph  $G^c$  contains a directed path of  $l$ -vertices.  $\lrcorner$

*Proof:* Consider any ordering on  $G^c$ . then we can put the vertices in a matrix by the following rule: The first row are vertices with no inward order, and for any  $k \geq 1$ , the row  $k$  are vertices not in the first  $k$  row and has an inward edge from some vertex in the row  $k$ .

Then there are no edges between vertices in the same row, and we are done if some row has  $k$  vertices. If each row has less than  $k$  vertices, then there are at least  $l$  row, which will give us a directed path of  $l$ -vertices, by our rule of construction.  $\square$

**Prop. (37.2.5.5) [Turán/Ajtai-Komlós-Szemerédi].** If  $G$  is a multi-graph with  $\omega(G) = 2$  (37.2.5.1) and  $v(G) \leq n, d(G) \leq t$ , where  $n, t \in \mathbb{R}_+$ , then

$$\alpha(G) \geq \max\left(\frac{n}{100t} \log t, \frac{n}{t+1}\right).$$

┘

*Proof:* Cf.[note on Ramsey Numbers].

□

**Prop. (37.2.5.6).** Let  $G$  be a graph with  $n = \#V(G)$ ,  $e = \#E(G)$ , and  $h$  be the number of triangles in  $G$ . Suppose  $p \in (0, 1)$  with  $pn \geq 3$ , then there exists an induced subgraph  $G' \subset G$  s.t.

$$n' > np/2, \quad e' < 3ep^2, \quad h' < 3hp^3.$$

┘

*Proof:*

□

**Prop. (37.2.5.7) [Ajtai-Komlós-Szemerédi].** Let  $\varepsilon \in \mathbb{R}_+$  and  $G$  be a graph with  $n = v(G)$ ,  $t = d(G)$ ,  $h$  the number of triangles in  $G$ . Suppose  $h < nm^{2-\varepsilon}$  and  $t \leq m - 1$ , where  $m \in \mathbb{Z}_{>122/\varepsilon}$ , then

$$\alpha(G) > \frac{\varepsilon}{4800} \cdot \frac{n}{m} \cdot \log m.$$

┘

*Proof:* Cf.[note on Ramsey Numbers].

□

### Independence Polynomials

**Def. (37.2.5.8) [Independence Polynomials].** For  $G \in \text{Graph}^{\text{fin}}$ , the **independence polynomial** is defined to be

$$I(G) = \sum_{A \subset V(G), A \text{ independent}} X^{\#A} \in \mathbb{Z}[X].$$

┘

**Prop. (37.2.5.9).** For any  $G \in \text{Graph}^{\text{fin}}$  and  $v \in V(G)$ ,  $I(G) = I(G \setminus \{v\}) + XI(G \setminus N(v))$ .

┘

**Prop. (37.2.5.10) [Independence Polynomials of Line Graphs are Real-Rooted].** If  $G \in \text{Graph}^{\text{fin}}$  is claw-free, then  $I(G)$  is real-rooted. In particular, the independence polynomial of a line graph is real-rooted.

┘

*Proof:* Use induction on  $v(G)$ . The case  $v(G) = 0$  is trivial, and if  $v(G) > 0$ , take  $v \in V(G)$ , then  $I(G) = I(G \setminus \{v\}) + XI(G \setminus N(v))$ . So to prove  $I(G)$  is real-rooted, it suffices to prove that  $I(G \setminus \{v\})$  and  $XI(G \setminus N(v))$  has a common interlacing polynomial (3.3.3.14).

For this we use another induction on  $v(G)$ . The case  $v(G) = 1$  is trivial, so suppose  $v(G) \geq 2$ . If  $N(v) \setminus \{v\} = \emptyset$ , then  $I(G \setminus N(v)) = I(G \setminus \{v\})$ , so the assertion is true. And if  $N(v) \setminus \{v\} \neq \emptyset$ , let  $n \in N(v) \setminus \{v\}$ , then

$$I(G \setminus \{v\}) = I(G \setminus \{u, v\}) + XI(G \setminus (u \cup N(u))),$$

so by (3.3.3.14), it suffices to show that every pair of the polynomials

$$XI(G \setminus (u \cup N(u))), \quad XI(G \setminus (v \cup N(v))), \quad I(G \setminus \{u, v\})$$



has a common interlacing. The two pairs  $(XI(G \setminus (u \cup N(u))), I(G \setminus \{u, v\}))$  and  $(XI(G \setminus (v \cup N(v))), I(G \setminus \{u, v\}))$  are compatible by induction hypothesis. And for the pair  $(I(G \setminus (u \cup N(u))), I(G \setminus (v \cup N(v))))$ , let  $H = G \setminus (\{u, v\} \cup N(u) \cup N(v))$ , then it is easy to show that  $N_H(u), N_H(v)$  are both simplicial cliques in  $H$  (37.2.5.2). So we can use lemma (37.2.5.11) to show that the pair  $(I(G \setminus (u \cup N(u))), I(G \setminus (v \cup N(v))))$  has a common interlacing.  $\square$

**Lemma (37.2.5.11).** If  $G \in \mathbf{Graph}^{\text{fin}}$ , and  $I(H)$  is real-rooted for every induced subgroup  $H$ , then

- For any two simplicial cliques  $K, L \subset V(G)$  (37.2.5.2), the two polynomials  $I(G \setminus K), I(G \setminus L)$  have a common interlacing polynomial.
- For any simplicial clique  $K \subset V(G)$ , the two polynomials  $I(G), XI(G \setminus K)$  have a common interlacing polynomial.

┘

*Proof:* The proof use induction on  $v(G)$  for both assertions. The case  $v(G) = 0$  is trivial. For  $v(G) > 0$ , we may assume  $K \neq \emptyset$ , and denote  $H = G \setminus (K \cup L)$ . Then

$$\begin{aligned} I(G \setminus L) &= I(H) + \sum_{v \in K \setminus L} XI(H \setminus N_H(v)), \\ I(G \setminus K) &= I(H) + \sum_{v \in L \setminus K} XI(H \setminus N_H(v)). \\ I(G) &= I(G \setminus K) + \sum_{v \in K} XI(G \setminus N(v)). \end{aligned}$$

So it suffices to show that every pair of the polynomials

$$\{XI(H \setminus N_H(v))\}_{v \in K \cup L}, \quad I(H)$$

has a common interlacing polynomial (3.3.3.14). These from from the induction hypothesis and (37.2.5.3). For assertion 2, it suffices to show that every pair of the polynomials

$$I(G \setminus K), XI(G \setminus K), \{XI(H \setminus N_H(v))\}_{v \in K}$$

has a common interlacing polynomial. The pair  $(I(G \setminus K), XI(G \setminus K))$  follows from the hypothesis, The pairs  $(I(G \setminus K), XI(H \setminus N_H(v)))$  have common interlacings by induction hypothesis and (37.2.5.3). The pairs  $(XI(G \setminus K), XI(H \setminus N_H(v)))$  have common interlacings by induction hypothesis and (37.2.5.3). The family  $\{XI(H \setminus N_H(v))\}_{v \in K}$  have common interlacings by induction hypothesis and (37.2.5.3).  $\square$

## 6 Planer Graphs

**Def. (37.2.6.1) [Planer Graphs].** A **planer graph** is a graph  $G \in \mathbf{Graph}$  that is embeddable in  $\mathbb{R}^2$ . ┘

**Prop. (37.2.6.2) [Plane Graphs].** When can a graph be embedded in  $\mathbb{R}^2$ ? ? ┘

*Proof:*  $\square$

**Prop. (37.2.6.3) [Euler].** Any finite planer graph  $G \in \mathbf{Graph}^{\text{simple}}$  satisfies  $e(G) < 3v(G) - 6$ . ┘

*Proof:*  $\square$

**Prop. (37.2.6.4) [Maximally Plane Graphs are Triangulations].** For any finite planer graph  $G \in \mathbf{Graph}^{\text{simple}}$  with  $v(G) \geq 3$ ,  $G$  is maximally planer iff it is a planer triangulation. ┘

*Proof:* Cf. [Diestel]P96.  $\square$



## 7 Flows

**Def. (37.2.7.1) [Circulations].** A **circulation** on a finite multi-graph  $G$  is a function  $f : \vec{E}(G) \rightarrow H$  where  $H \in \mathcal{Ab}$ , satisfying the following:

- For any  $(e, x, y) \in \vec{E}(G)$ ,  $f(e, x, y) = -f(e, y, x)$ ,
- (**Kirchhoff's law**) For any  $x \in V(G)$ ,

$$\sum_{(e,x,y) \in \vec{E}(G)} f(e, x, y) = 0.$$

┘

**Def. (37.2.7.2).** For a finite multi-graph  $G$  and a function  $f : \vec{E}(G) \rightarrow H$  where  $H \in \mathcal{Ab}$ , for any two subsets  $X, Y \subset V(G)$ , denote

$$f(X, Y) = \sum_{(e,x,y) \in \vec{E}(G), x \in X, y \in \bar{Y}} f(e, x, y).$$

┘

**Def. (37.2.7.3) [Networks and Flows].** A **network** is a quadruple  $N = (G, s, t, c)$  where  $G$  is a finite multi-graph,  $s \neq t \in V(G)$ , and  $c : \vec{E}(G) \rightarrow \mathbb{N}$  a map called the **capacity function**. Then a **flow** on a network  $N$  is a function  $f : \vec{E}(G) \rightarrow \mathbb{R}$  satisfying the following:

- For any  $(e, x, y) \in \vec{E}(G)$ ,  $f(e, x, y) = -f(e, y, x)$ ,
- For any  $x \in V(G) \setminus \{s, t\}$ ,

$$\sum_{(e,x,y) \in \vec{E}(G)} f(e, x, y) = 0.$$

- $f(\vec{e}) \leq c(\vec{e})$  for any  $\vec{e} \in \vec{E}(G)$ .

┘

**Def. (37.2.7.4) [Flow-Cuts].** For a flow  $f$  on a network  $N = (G, s, t, c)$ , a **flow-cut** is a pair  $(S, \bar{S})$  where  $S \subset V(G)$ ,  $\bar{S} = V(G) \setminus S$ ,  $s \in S$ ,  $t \in \bar{S}$ , and  $c(S, \bar{S}) = \sum_{(e,x,y) \in \vec{E}(G), x \in S, y \in \bar{S}} c(e, x, y)$  is called the **capacity of the cut**.

┘

**Prop. (37.2.7.5) [Total Values of Flows].** Given a flow  $f$  on a network  $N = (G, s, t, c)$  for any flow-cut  $(S, \bar{S})$ ,  $f(S, \bar{S}) = f(s, V(G))$ , called the **total value of the flow**  $f$ , denoted by  $|f|$ .

It is clear that  $|f| \leq c(S, \bar{S})$  for any flow-cut  $S$ .

┘

**Thm. (37.2.7.6) [Maximum Value and Minimum Capacity].** For any network  $N = (G, s, t, c)$ , the maximal total value of a flow on  $N$  equals the minimum value  $c(S, \bar{S})$  for any flow-cut  $S$ . Moreover, such a maximal flow can be integral-valued.

┘

*Proof:* Let  $f$  be a flow with maximal total value. Let  $S$  be the set of all vertices  $v \in V(G)$  s.t. there exists a path  $x_0 e_0 x_1 e_1 \dots x_{k-1} e_{k-1} x_k$  from  $s$  to  $v$  with  $f(e_i, x_i, x_{i+1}) < c(e_i, x_i, x_{i+1})$  for any  $i \in [k-1]$ .

If  $t \in S$ , suppose  $W$  is a shortest such path from  $s$  to  $t$ , and  $\varepsilon = \min_{\vec{e} \in W} (c(\vec{e}) - f(\vec{e}))$ , then we can modify  $f$  to add a value  $\varepsilon$  along the orientation of  $W$ . In this way, we get a flow with bigger  $|f| = f(s, V)$ , contradiction.

So  $t \notin S$ . Thus  $(S, \bar{S} = V(G) \setminus S)$  is a flow-cut for  $N$ , and by the definition of  $S$ ,  $f(\vec{e}) = c(\vec{e})$  for any  $\vec{e} \in \vec{E}(S, \bar{S})$ . Thus  $|f| = f(S, \bar{S}) = c(S, \bar{S})$ .  $\square$

## 8 Extremal Graph Theory

**Prop. (37.2.8.1) [Rademacher].** For any  $n \in \mathbb{Z}_+$ , any graph  $G$  with  $\#V(G) = 2n, \#E(G) = n^2 + 1$  contains at least  $n$  triangles.  $\lrcorner$

*Proof:*

□

**Thm. (37.2.8.2) [Turán1941].** For any  $r, n \in \mathbb{Z}_+, r > 1$ , any  $G \in \mathbf{Graph}^{\text{simple}^n}$  not containing a subgraph  $K^r$  with  $e(G) = \text{Xtrm}(n, K_r)$  is isomorphic to  $\text{Tur}^{r-1}(n)$ .  $\lrcorner$

*Proof:* Cf.[Graph Theory, Diestel]P175.  $\square$

**Thm. (37.2.8.3) [Erdős-Stone1946].** Given any  $r, s \in \mathbb{Z}_+, r \geq 2$  and  $\varepsilon \in \mathbb{R}_+$ , then for  $n \in \mathbb{Z}_+$  sufficiently large, any graph  $G \in \mathbf{Graph}^{\text{simple}}$  with  $v(G) = n$  and  $e(G) \geq t_{r-1}(n) + \varepsilon n^2$  contains  $K_s^r$  as a subgraph.  $\lrcorner$

*Proof:* Cf.[Graph Theory, Diestel]P178. ?  $\square$

### Szemerédi's Regularity Lemma

**Def. (37.2.8.4) [ $\varepsilon$ -Regular Pairs].** For  $G \in \mathbf{Graph}^{\text{simple}}$ , two disjoint non-empty subsets  $A, B \subset V(G)$  are called a  $\varepsilon$ -regular pair if for any  $X \subset A, Y \subset B$  s.t.  $\#X > \varepsilon\#A, \#Y > \varepsilon\#B$ , we have

$$|d(X, Y) - d(A, B)| < \varepsilon.$$

$\lrcorner$

**Thm. (37.2.8.5) [Szemerédi's Regularity Lemma].** For any  $\varepsilon \in \mathbb{R}_+$  and  $m \in \mathbb{Z}_+$ , there exists  $M(\varepsilon, m), N(\varepsilon, m) \in \mathbb{Z}_+$  satisfying the following: For any graph  $G \in \mathbf{Graph}^{\text{simple}}$  with  $v(G) \geq N(\varepsilon, m)$ , there is a partition of  $V(G)$  into  $k + 1$  classes

$$V(G) = V_0 \amalg V_1 \amalg \dots \amalg V_k$$

where  $V_0$  is called the **exceptional set**, s.t.

- $m \leq k \leq M(\varepsilon, m)$ ,
- $\#V_0 < \varepsilon v(G)$ ,
- $\#V_1 = \dots = \#V_k$ ,
- All but at most  $\varepsilon k^2$  pairs of  $(V_i, V_j), 1 \leq i < j \leq k$  are  $\varepsilon$ -regular pairs(37.2.8.4).

In other words, in some sense, every graph can be approximated by random graphs.  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (37.2.8.6)[Szemerédi's Regularity Lemma, alternative forms].** For any  $\varepsilon \in \mathbb{R}_+$ , there exists  $M(\varepsilon) \in \mathbb{Z}_+$  satisfying the following: For any graph  $G \in \mathbf{Graph}^{\text{simple}}$ ,  $V(G)$  can be partitions into

$$V(G) = V_1 \amalg \dots \amalg V_k$$

s.t.

- $\#V_i \leq \lceil \varepsilon v(G) \rceil$  for any  $i$ ,
- $|\#V_i - \#V_j| \leq 1$  for any  $1 \leq i < j \leq k$ ,
- All but at most  $\varepsilon k^2$  pairs of  $(V_i, V_j), 1 \leq i < j \leq k$  are  $\varepsilon$ -regular pairs(37.2.8.4).

$\lrcorner$

*Proof:* ?  $\square$

## 37.3 Graph Theory II

References are [Die17].

**Notation(37.3.0.1).**

- Use references as in [Graph Theory I](#).

┘

### 1 Random Graphs

**Prop.(37.3.1.1)[Small Random Independent Sets].** Let  $k \in \mathbb{Z}_+$  and let  $G_n$  be a random graph of size  $n$  with each edge appearing with probability  $p = 8k^2/n$ , then

$$\lim_{n \rightarrow \infty} \mathbf{P}(\alpha(G_n) \geq \frac{k}{2n}) = 0.$$

┘

*Proof:*

$$\mathbf{P}(\alpha(G_n) \geq n/(2k)) \leq \binom{n}{n/(2k)} (1 - 16k^2/n)^{\binom{n}{2}} \leq 2^n e^{-p(\frac{n}{2k})^2/2} \leq 2^n e^{-n},$$

which tends to 0 as  $n$  tends to  $\infty$ .

□

### Signed Adjacency Matrices

References are [MSS15].

**Prop.(37.3.1.2)[Signed Adjacency Matrices and Matching Polynomials, Godsil-Gutman].**

For any  $G \in \mathbf{Graph}$  and a choice of signing  $s \in \{\pm 1\}^{E(G)}$  of the edges of  $G$ , then we can consider the signed adjacency matrices  $A_{s,G}$  of  $G$  and the corresponding characteristic polynomials  $f_s(G)$ . Then

$$\mathbf{E}_{s \in \{\pm 1\}^{E(G)}} f_s(G) = \mu_G(X).$$

┘

*Proof:*

$$\begin{aligned} \mathbf{E}_{s \in \{\pm 1\}^{E(G)}} f_s(G) &= \mathbf{E} \left( \sum_{\sigma \in \text{Sym}([n]_+)} \text{sgn}(\sigma) \prod_{i=1}^n (xI - A_{s,G})_{i,\sigma(i)} \right) \\ &= \sum_{k=0}^n x^{n-k} (-1)^k \sum_{S \subset [n]_+, \#S=k} \sum_{\pi \in \text{Sym}(S)} \mathbf{E} \left( \text{sgn}(\pi) \prod_{i \in S} (A_s)_{i,\pi(i)} \right) \\ &= \sum_{k \in 2\mathbb{N}, 0 \leq k \leq n} x^{n-k} \sum_{S \subset [n]_+, \#S=k} \sum_{\pi \in \text{Matching}(S)} (-1)^{k/2} \\ &= \mu_G(x) \end{aligned}$$

□

**Prop. (37.3.1.3) [Random Adjacency matrices, Marcus-Spielman-Srivastava].** For any  $G \in \text{Graph}^{\text{simple}}$  with an arbitrary ordering of  $E(G)$ , and any  $p_1, \dots, p_{e(G)} \in [0, 1]$ , the expected value of the signed adjacency characteristic polynomial as in (37.3.1.2):

$$P(X) = \sum_{s \in \{\pm 1\}^{E(G)}} \prod_{s_i=1} p_i \prod_{s_i=-1} (1-p_i) f_s(X) \in \mathbb{R}[X]$$

is real-rooted.  $\lrcorner$

*Proof:* Denote  $d = \Delta(G)$ . For any  $s \in \{\pm 1\}^{E(G)}$  and  $u \neq v \in V(G)$ , denote by  $s_{u,v}$  the value on the edge  $(u, v)$ , and define  $a_{u,v}^\pm = e_u \pm e_v$ ,  $L_{u,v}^\pm = a_{u,v}^\pm a_{u,v}^t$ . Then

$$d\mathbf{1} - A_s = \sum_{(u,v) \in E} a_{u,v}^{s_{u,v}} (a_{u,v}^{s_{u,v}})^t + D$$

where  $D = \text{diag}(d - d_1, \dots, d - d_{e(G)})$ .

Now

$$P(X + d) = \sum_{s \in \{\pm 1\}^{E(G)}} \prod_{s_i=1} p_i \prod_{s_i=-1} (1-p_i) \det \left( X\mathbf{1} + D + \sum_{(u,v) \in E(G)} a_{u,v}^{s_{u,v}} (a_{u,v}^{s_{u,v}})^t \right),$$

which is real-rooted by (3.3.3.10).  $\square$

**Cor. (37.3.1.4) [Signed Adjacency Characteristic polynomials are Interlacing, Marcus-Spielman-Srivastava].** For any  $G \in \text{Graph}^{\text{simple}}$  with an arbitrary ordering of  $E(G)$ , the signed adjacency characteristic polynomials (37.3.1.2) form an interlacing family (3.3.3.16).  $\lrcorner$

*Proof:* By (3.3.3.14), it suffices to show that for any  $k \in [e(G) - 1]_+$  and  $s_1, \dots, s_k \in \{\pm 1\}$ ,  $\lambda \in \mathbb{R}_{\geq 0}$ , the polynomial

$$\lambda f_{s_1, \dots, s_k, 1}(X) + (1 - \lambda) f_{s_1, \dots, s_k, -1}(X)$$

is real-rooted. But this follows from (37.3.1.3).  $\square$

**Thm. (37.3.1.5) [Minimal Eigenvalues of Signed Adjacency Characteristic Polynomials, Marcus-Spielman-Srivastava [MSS15]].** For any connected graph  $G \in \text{Graph}^{\text{simple}}$  with universal covering  $T$ , there exists a signed adjacency characteristic polynomial  $f_{s,G}$  of  $G$  (37.3.1.2) with all roots bounded above by  $\rho(T)$ .  $\lrcorner$

*Proof:* This follows from (37.3.1.4)(3.3.3.18) and (37.3.1.2)(37.2.3.11).  $\square$

## Second Moment Method

### 2 Coloring

**Def. (37.3.2.1) [Colorings].** For  $G \in \text{Graph}$  and  $k \in \mathbb{Z}_+$ , a  $k$ -vertex coloring of  $G$  is a map  $c : V(G) \rightarrow [k]_+$  s.t.  $c(u) \neq c(v)$  for any edge  $e = (u, v) \in E(G)$ .

A graph  $G \in \text{Graph}$  with a  $k$ -vertex coloring is called  $k$ -colorable. The infimum of  $k \in \mathbb{Z}_+$  s.t.  $G$  is  $k$ -colorable is denoted by  $\gamma(G)$ , called the **chromatic number** of  $G$ .

Similarly, we can define **edge-colorings** of a graph as a vertex coloring of the line graph  $L(G)$ , and define the **edge-chromatic number** of  $G$ , denoted by  $\gamma'(G)$ .  $\lrcorner$

**Def. (37.3.2.2) [Chromatic Numbers].** For  $G \in \text{Graph}$ , the **chromatic number**  $\gamma(G) \in \mathbb{Z}_+ \cup \{\infty\}$  is the minimal number  $n \in \mathbb{Z}_+$  s.t.  $V(G)$  can be partitioned into  $n$  disjoint subsets  $V = V_1 \amalg V_2 \amalg \dots \amalg V_n$  s.t. if  $x, y \in V_i(G)$  for some  $i$ , then  $e(x, y) = 0$ .  $\lrcorner$

**Def. (37.3.2.3) [Partite Graphs].** For  $r \in \mathbb{Z}_{\geq 2}$ , an  **$r$ -partite graph** is a graph  $G$  s.t.  $V(G)$  has an  $r$ -partition  $V = V_1 \amalg \dots \amalg V_r$  s.t. vertices in a same partition class are not adjacent. 2-partite graphs are also called **bipartite graphs**.

A maximal  $r$ -bipartite graph  $K_s * K_s * \dots * K_s$  is denoted by  $K_s^r$ .  $\lrcorner$

**Prop. (37.3.2.4) [Bipartite Graphs and Odd Cycles].** A graph  $G \in \text{Graph}$  is bipartite iff it contains no odd cycles.  $\lrcorner$

*Proof:* One direction is clear. For the other, let  $T$  be a spanning tree of  $G$ , with a root  $t$ . Then we can define a tree order on  $V(G)$ . Now  $V(G)$  is partitions into two classes according to whether  $v \in V(G)$  satisfies  $d(t, v)$  is even.

This is truly a bipartite partition, because if an odd-distance vertex is adjacent to an even-distance vertex, then we get an odd cycle.  $\square$

**Prop. (37.3.2.5) [Chromatic Numbers and Independent Sets].** For a finite graph  $G \in \text{Graph}^{\text{simple}}$ ,  $v(G) \leq i(G)\gamma(G)$ .  $\lrcorner$

*Proof:* Let  $V(G) = V_1 \amalg \dots \amalg V_{\gamma(G)}$  be a coloring of  $G$ , then each  $V_i$  is an independent set. Thus  $\#V_i(G) \leq i(G)$ .  $\square$

**Prop. (37.3.2.6) [Chromatic Numbers and Minimal Degree].** For any  $k \in \mathbb{Z}_+$ , any  $k$ -chromatic graph  $G \in \text{Graph}^{\text{fin}}$  has a  $k$ -chromatic subgraph  $H$  with  $\delta(H) \geq k - 1$ .  $\lrcorner$

*Proof:* Cf. [Diestel]P123.  $\square$

**Cor. (37.3.2.7).** For any  $G \in \text{Graph}^{\text{fin}}$ ,  $G$  contains a cycle of length at least  $v(G)/\gamma(G)$ .  $\lrcorner$

*Proof:* This follows from (37.3.2.6) and (37.2.1.11).  $\square$

**Prop. (37.3.2.8) [Coloring Numbers and Spectrums].** For  $k \in \mathbb{Z}_{\geq 2}$  and a connected  $k$ -regular graph  $G \in \text{Graph}^{\text{simple}}$  with  $n = v(G) < \infty$ , we have

$$i(G) \leq \frac{n}{k} \max \left( |\lambda_1(G)|, |\lambda_{v(G)-1}(G)| \right).$$

In particular, by (37.3.2.5),

$$\gamma(G) \geq \frac{k}{\max \left( |\lambda_1(G)|, |\lambda_{v(G)-1}(G)| \right)}.$$

$\lrcorner$

*Proof:* Let  $F \subset V(G)$  be a maximal independent set of  $G$ . Take  $f \in \ell^2(V(G))$  s.t.

$$f(x) = \begin{cases} \#V(G) \setminus F & , x \in F \\ -\#F & , x \notin F \end{cases}.$$

Then  $\|f\| = \#F \cdot \#V(G) \setminus F \cdot \#V(G) \leq i(G)n^2$ . And for any  $x \in F$ ,  $(Af)(x) = -ki(G)$ . So  $\|Af\|^2 \geq k^2 i(G)^3$ .

Since  $\sum_{v \in V(G)} f(v) = 0$ ,  $\|Af\| \leq \max(|\mu_1|, |\mu_{n-1}|) \|f\|$ , so  $ki(G)^{3/2} \leq \max(|\mu_1|, |\mu_{n-1}|) \cdot ni(G)^{1/2}$ , which gives the desired result.  $\square$

**Thm. (37.3.2.9)** [Large Girth, Large Chromatic and Large minimal Degree Graphs, Erdős].

For any  $k \in \mathbb{Z}_+$ , there exists a finite graph  $G \in \text{Graph}^{\text{simple}}$  with

$$g(H), \gamma(H), \delta(H) > k.$$

┘

*Proof:* Assume  $k \geq 3$ . Fix  $\varepsilon \in (0, 1/k)$  and  $n$  large, and consider the random graph  $G_n$  of size  $n$  with each edge appearing with probability  $p = n^{\varepsilon-1}$ . Then the expected value of number  $X$  of cycles with length  $\leq k$  is

$$\mathbf{E}(X) = \sum_{i=3}^k \frac{n(n-1) \dots (n-i+1)}{2i} p^i \leq \frac{1}{2} \sum_{i=3}^k n^i p^i \leq \frac{k-2}{2} n^{k\varepsilon}.$$

Thus

$$\mathbf{P}(X \geq n/2) \leq \mathbf{E}(X)/(n/2) \leq (k-2)n^{k\varepsilon-1}.$$

So when  $n$  is large,  $\mathbf{P}(X \geq n/2) \leq 1/2$ . And notice  $n^{\varepsilon-1} \geq 16k^2/n$  for  $n$  large, so by (37.3.1.1), when  $n$  is large,  $\mathbf{P}(i(G_n) \geq n/(2k)) \leq 1/2$ . Thus there is a graph  $G$  with  $v(G) = n$ ,  $i(G) < n/(2k)$ , and has less than  $n/2$  cycles of length  $\leq k$ .

Delete a vertex from each cycle of length  $\leq k$  in  $G$ , and let  $H$  be the remaining graph, then  $v(H) \geq n/2$ ,  $g(H) > k$ , and  $\gamma(H) \geq v(H)/i(H) \geq \frac{1}{2}n/i(G) > k$  (37.3.2.5).

Finally, we can make  $\delta(G)$  large by restricting to a subgraph  $H$ , by (37.3.2.6). □

**Thm. (37.3.2.10)** [Vizing[Viz64]]. For any finite simple graph  $G \in \text{Graph}^{\text{simple}}$ ,  $\Delta(G) \leq \gamma'(G) \leq \Delta(G) + 1$ . ┘

*Proof:*  $\Delta(G) \leq \gamma'(G)$  is clear. For the other inequality, use induction on  $e(G)$ : The case  $e(G) = 0$  is trivial, and for  $e(G) > 0$ , for any  $e = (x, y) \in E(G)$ , there is a color of  $G \setminus \{e\}$ , and if there is a color  $\alpha$  that is missing at  $x$  and a color  $\beta$  that is missing at  $y$ , then the maximal  $(\beta, \alpha)$ -alternating path from  $y$  will end in  $x$ , otherwise we can switch  $\alpha, \beta$  in the color of this maximal path, and color  $e$  with  $\alpha$ .

Now let  $xy_0 \in E(G)$ , and let  $c_0$  be the coloring of  $G \setminus \{xy_0\}$ , and let  $\alpha$  be a color missing at  $x$  in the coloring  $c_0$ , and  $c(1)$  be the color missing at  $y_0$  in the coloring  $c_0$ . Then there may exists  $xy_1 \in E(G)$  s.t.  $c_0(xy_1) = c(1)$ . Continuing this way, we find a maximal  $k \in \mathbb{Z}_+$  s.t. there exists  $xy_0, xy_1, \dots, xy_k \in E(G)$  s.t.  $c(i) \triangleq c_0(xy_i)$  is a color missing at  $y_{i-1}$  in the coloring  $c_0$ .

Then we can define colorings  $c_i$  of  $G \setminus \{xy_i\}$  for each  $i$ :  $c_i(e) = \begin{cases} c(i+1) & , e = xy_k, k \in [i] \\ c_0(e) & \text{otherwise} \end{cases}$ .

Let  $\beta$  be a color missing at  $y_k$  in the coloring  $c_0$ , and consider the  $(\beta, \alpha)$ -alternating path  $P$  from  $y_k$  ending at  $x$ . By maximality of  $k$ , this path ends with  $y_k, \dots, y_i, x$  for some  $i \in [k]$ . Then  $\beta = c(i+1)$ , which is missing at  $y_i$  in the coloring  $c_0$  (thus also missing at  $y_i$  in the coloring  $c_k$ ). Then in the coloring  $c_i$ , the  $(\beta, \alpha)$ -path from  $y_i$  is just the reverse of  $P$  ending at  $y_k$ , and then cannot go further, contradiction. □

**Prop. (37.3.2.11)** [König1916]. For any bipartite graph  $G \in \text{Graph}^{\text{fin}}$ ,  $\gamma'(G) = \Delta(G)$ . ┘

*Proof:* Use induction on  $e(G)$ . The case  $e(G) = 0$  is trivial. For  $e(G) > 0$ , take  $e = (x, y) \in G$ , and choose a  $\Delta(G)$ -coloring of  $G' = G \setminus \{e\}$ , which is possible by induction hypothesis. Then there exists  $\alpha, \beta \in [\Delta(G)]_+$  s.t.  $x$  is not incident with an  $\alpha$ -edge and  $y$  is not incident with an  $\beta$ -edge. If  $\alpha$  can be chosen to be  $\beta$ , then a coloring of  $G$  is possible.

If  $\alpha \neq \beta$ , then  $x$  has an  $\beta$ -edge. Then we can extend this edge to a maximal  $(\alpha, \beta)$ -alternating path  $W$  in  $G'$ . This path must be a simple path, and it cannot contain  $y$ , because in this way, it must end with an  $\alpha$ -edge, making an odd cycle in  $G$ , contradicting the fact that  $G$  is bipartite. So we can now swap all the colors of edges in  $W$ , and then color  $e$  with  $\beta$ . This is a  $\Delta(G)$ -coloring of  $G$ .  $\square$

**Prop. (37.3.2.12)** [Csaba-Kühn-Lo-Osthus-Treglown2016]. There exists  $n_0 \in \mathbb{Z}_+$  s.t. for any  $n \in \mathbb{Z}_{n_0}$  and  $d \in \mathbb{Z}_{\geq n/2}$ , any  $d$ -regular graph  $G \in \mathbf{Graph}$  with  $v(G) = n$  satisfies  $\gamma'(G) = \Delta(G)$ .  $\lrcorner$

*Proof:*  $\square$

### Tutte Polynomial

**Def. (37.3.2.13)** [Tutte Polynomials]. The **Tutte polynomial** of a multi-graph  $G$  is defined to be  $T_G(q, v) = ?$ .  $\lrcorner$

**Prop. (37.3.2.14)**. For any  $G \in \mathbf{Graph}$  and  $q \in \mathbb{Z}_+$ , the number of ways of coloring vertices of  $G$  with no two vertices of the same color connected is just  $T_G(q, -1)$ .  $\lrcorner$

*Proof:* Cf.[The Knot Book, Adams]P233.  $\square$

### List Colorings

**Def. (37.3.2.15)** [List Colorings]. For  $k \in \mathbb{Z}_+$ , a graph  $G \in \mathbf{Graph}$  is called  **$k$ -choosable** iff for any family of subsets  $S_v \subset \mathbb{N}$  indexed by  $V(G)$ , there is a map  $c : V(G) \rightarrow \mathbb{N}$  s.t.  $c(v) \in S_v$  for any  $v \in V(G)$ , and  $c(u) \neq c(v)$  for any any edge  $e = (u, v) \in E(G)$ . The infimum of  $k \in \mathbb{Z}_+$  s.t.  $G$  is  $k$ -choosable is denoted by  $\text{ch}(G)$ , called the **choice number** of  $G$ .

Similarly, we can define **edge-list-colorings** of a graph as a vertex coloring of the line graph  $L(G)$ , and define the **edge-choice number** of  $G$ , denoted by  $\text{ch}'(G)$ .

Clearly,  $\text{ch}(G) \geq \gamma(G)$ ,  $\text{ch}'(G) \geq \gamma'(G)$ .  $\lrcorner$

**Prop. (37.3.2.16)** [Bondy-Boppana-Siegel]. If  $H \in \mathbf{Graph}^{\text{fin}}$  be an oriented graph s.t. every induced subgraph has a kernel(37.2.1.6), and  $\{S_v\}_{v \in V(H)}$  is a family of lists(37.3.2.15) s.t.  $\deg^+(v) < \#S_v$  for any  $v \in V(H)$ , then  $H$  can be colored from the list  $S_v$ .  $\lrcorner$

*Proof:* Use induction on  $v(H)$ .  $v(H) = 0$  is trivial, and for  $v(H) > 0$ , take  $\alpha \in \cup_v S_v$ , and let  $H'$  be the subgraph induced by vertices  $v \in V(H)$  s.t.  $\alpha \in S_v$ . Then  $H'$  has a kernel  $U \neq \emptyset$ . Now color  $U$  with  $\alpha$ . Then the graph  $H \setminus U$  can be colored from the list  $\{S'_v\}_{v \in V(H)}$ , where  $S'_v = S_v \setminus \{\alpha\}$ , by induction hypothesis and the definition of a kernel. Thus  $H$  can also be colored from the list  $S_v$ .  $\square$

**Conj. (37.3.2.17)** [List Coloring Conjecture]. For any  $G \in \mathbf{Graph}$ ,  $\text{ch}'(G) = \gamma'(G)$ .  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj.Cor. (37.3.2.18)** [Bipartite Graphs, Galvin[Gal95]]. For any bipartite graph  $G \in \mathbf{Graph}$ ,  $\text{ch}'(G) = \gamma'(G) = \Delta(G)$ .  $\lrcorner$

*Proof:* The second equality follows from(37.3.2.11). For the first, it suffices to show that  $\text{ch}'(G) \leq \gamma'(G)$ . Let  $G = X \amalg Y$  where  $X, Y$  are independent subset of  $G$ . Assume  $k = \gamma'(G) < \infty$ , and let  $c$  be a  $k$ -coloring.

We define an orientation on  $L(G)$  as follows: For any two adjacent edges  $e, e' \in E(G)$  s.t.  $c(e) < c(e')$ , assign  $e < e'$  if  $e, e'$  meet in  $X$ , and  $e > e'$  if  $e, e'$  meet in  $Y$ . In this way,  $\deg^+(e) > k$  for any  $e \in E(G)$ .

Notice this orientation also defines a linear preference on  $E(G)$  (37.2.3.7), thus any induced subgraph of  $L(G)$  has a kernel by the stable matching theorem (37.2.3.8). Now it follows from (37.3.2.16) that  $L(G)$  is  $k$ -choosable.  $\square$

### Planer Graphs

**Thm. (37.3.2.19) [4-Coloring Problem, Appel-Haken1977].** Every planer graph is 4-colorable.  $\lrcorner$

*Proof:* ? Cf. [N. Robertson, D. Sanders, P.D. Seymour and R. Thomas1977].  $\square$

**Thm. (37.3.2.20) [3-Coloring Theorem, Grötzsch1959].** Every Planer graph not containing a triangle is 3-colorable.  $\lrcorner$

*Proof:*  $\square$

**Thm. (37.3.2.21) [Alon1993].** For any  $k \in \mathbb{N}$ , there exists  $f(k) \in \mathbb{N}$  s.t. any graph  $G \in \mathbf{Graph}$  with  $d(G) \geq f(k)$  satisfies  $\text{ch}(G) \geq k$ .  $\lrcorner$

*Proof:*  $\square$

**Thm. (37.3.2.22) [5-Choice Theorem, Thomassen[Tho94]].** Any planer graph is 5-choosable.  $\lrcorner$

*Proof:* It suffices to prove for finite maximally planer graphs  $G \in \mathbf{Graph}^{\text{simple}}$ . By (37.2.6.4),  $G$  is a graph triangulation, so this follows from the following excellently tailored lemma:

**Lemma (37.3.2.23).** Let  $G$  be a planer graph s.t.  $v(G) \geq 3$ , its interiors is triangularized, and its boundary is a cycle  $C = v_1 v_2 \dots v_k v_1$ . Then given any family of subsets  $S_v \subset \mathbb{N}$  indexed by  $V(G)$  s.t.  $\#S_v \leq 5$ , and  $\#S_v \leq 3$  if  $v \in C$ , and given  $c_1 \in S_{v_1}, c_2 \in S_{v_2}$ , there is a map  $c : V(G) \rightarrow \mathbb{N}$  s.t.  $c(1) = c_1, c(2) = c_2, c(v) \in S_v$  for any  $v \in V(G)$ , and  $c(u) \neq c(v)$  for any any edge  $e = (u, v) \in E(G)$ .  $\lrcorner$

*Proof:* Use induction on  $v(G)$ . The case  $v(G) = 3$  is trivial. And for  $v(G) \geq 4$ , if there is an arc  $vw$  in  $G$ , then we can use induction hypothesis on the two parts divided by  $vw$  and get the desired result for  $G$ . If instead there are no arcs in  $G$ , let  $v_1, u_1, u_2, \dots, u_m, v_{k-1}$  be the neighborhoods of  $v_k$  with their clockwise orientation around  $v_k$ . Now choose two elements  $a, b \in S_{v_k}$  distinct to  $c_1$ , and delete  $a, b$  from  $S_{u_1}, \dots, S_{u_m}$ . Then the resulting family is a family for the graph  $G' = G \setminus \{v_k\}$  that satisfies the induction hypothesis, thus there is a choice function  $c : V(G) \setminus \{v\} \rightarrow \mathbb{N}$ . Now either  $a$  or  $b$  are different from  $c(v_{k-1})$ , so we can define  $c(v_k)$  to be this number.  $\square$

**Remark (37.3.2.24).** This is strict because Voigt(1993) constructed a plane graph of order 238 that is not 4-choosable.  $\lrcorner$

## 3 Graph labeling

### Super Edge-Magic Graphs

References are [ELNR].



**Def. (37.3.3.1) [(Super)Edge-Magic Labeling].** For  $G \neq \emptyset \in \text{Graph}^{\text{fin}}$ , an **edge-magic labeling** of  $G$  is a bijection  $f : V(G) \cup E(G) \cong [v(G) + e(G)]_+$  satisfying the following: there exists  $s \in \mathbb{Z}_+$  s.t.  $f(uv) + f(u) + f(v) = s$  for any  $uv \in E(G)$ . And moreover it is called a **super edge-magic labeling** if  $f(V(G)) = [v(G)]_+$ . The number  $s$  is called the **magic number** of the labeling.

And  $G$  is called **edge-magic/super edge-magic** if there exists an edge-magic/super edge-magic labeling on  $G$ .  $\lrcorner$

**Prop. (37.3.3.2).** If  $G \neq \emptyset \in \text{Graph}^{\text{fin}}$  is super edge-magic, then  $e(G) \leq 2v(G) - 3$ .  $\lrcorner$

*Proof:* Considering the extreme values of the labeling of vertices and edges, the magic number must satisfy

$$1 + 2 + (v(G) + e(G)) \leq s \leq v(G) + (v(G) - 1) + (v(G) + 1),$$

thus the assertion follows.  $\square$

**Prop. (37.3.3.3) [Kotzig-Rosa].** For  $n \in \mathbb{Z}_{\geq 3}$ , the circle graph  $C_n$  is edge-magic.  $\lrcorner$

*Proof:*  $\square$

**Prop. (37.3.3.4) [Enomoto-Llado-Nakamigawa-Ringel].** For  $n \in \mathbb{Z}_{\geq 3}$ , the circle graph  $C_n$  is super edge-magic iff  $2|n + 1$ .  $\lrcorner$

*Proof:* If  $f$  is a super edge-magic labeling with magic number  $s$ , then

$$sn = n(n + 1) + \frac{(3n + 1)n}{2},$$

so  $2s = 5n + 3$  is even.

If  $2|n + 1$ , and  $V(C_n) = \{v_0, \dots, v_{n-1}\}$ , with  $E(C_n) = \{v_{i-1(\text{mod } n)}v_i\}$ . Then we can define

$$f(v_i) = \begin{cases} \frac{i+2}{2}, & 2|i \\ \frac{n+i+2}{2}, & 2 \nmid i+1 \end{cases}.$$

$$f(v_{i-1(\text{mod } n)}v_i) = 2n - i, \quad 0 \leq i \leq n - 1.$$

Then it is easy to check that this is a super edge-magic labeling with magic number  $\frac{5n+3}{2}$ .  $\square$

**Prop. (37.3.3.5) [Kotzig-Rosa].** For  $n \in \mathbb{Z}_+$ , the complete graph  $K_n$  is edge-magic iff  $n \in \{1, 2, 3, 5, 6\}$ , and it is super edge-magic iff  $n \leq 3$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (37.3.3.6) [Kotzig-Rosa/Enomoto-Lladó-Nakamigawa-Ringel].** For  $m, n \in \mathbb{Z}_+$ , the complete bipartite graph  $K_{m,n}$  is edge-magic. And it is super edge-magic iff  $n = 1$  or  $m = 1$ .  $\lrcorner$

*Proof:* To show it is edge-magic, Cf.[Kotzig-Rosa] ?. And it is easy to show that  $K_{1,n}$  is super edge-magic ?.

Suppose  $2 \leq m \leq n$ , then  $mn \leq 2(m + n) - 3$  by (37.3.3.2), so  $(m - 2)(n - 2) \leq 1$ . Hence  $m = 2$  or  $m = n = 3$ . It is straightforward to check that  $K_{3,3}$  doesn't admit a super edge-magic labeling ?. And  $K_{2,2} = C_4$  is not super edge-magic by (37.3.3.4), so we only need to consider  $n \geq 3$ .

In this case, Suppose  $K_{m,n}$  has a super edge-magic labeling  $f$  with magic number  $s$ , then by the proof of (37.3.3.2),  $s = 3n + 5$  or  $s = 3n + 6$ . By changing the labeling by

$$f'(v) = n + 3 - f(v), \quad f'(uv) = 6n + 11 - f(uv),$$

we may assume that  $s = 3n + 6$ .

Denote  $V(K_{2,n}) = \{x_1, \dots, x_{n+2}\}$  with  $f(x_i) = i$ , then  $x_1x_2$  is not an edge. Let  $A$  be the part containing  $x_1, x_2$ , and  $B$  the part containing  $x_3$ , then Considering the edge  $3n + 2$ , we see that  $x_1x_3$  is an edge. Then  $x_1x_4$  is not an edge because  $1 + 4 = 2 + 3$ . So  $x_4 \in B$ . Now the only possibility of the edge  $3n$  is  $x_1x_5$ , so  $x_5 \in B$ . But  $2 + 5 = 4 + 3$ , contradiction.  $\square$

**Prop. (37.3.3.7) [Enomoto-Llado-Nakamigawa-Ringel].** For  $n \in \mathbb{Z}_{\geq 4}$ , the Wheel graph (37.2.1.3)  $\text{Wheel}_n$  is not super edge-magic. And if  $4|n$ , then  $\text{Wheel}_n$  is not edge-magic.  $\lrcorner$

*Proof:* Because  $e(\text{Wheel}_n) = 2n - 2 > 2v(\text{Wheel}_n) - 3 = 2n - 3$ ,  $\text{Wheel}_n$  is not super edge-magic by (37.3.3.2).

If  $4|n$ , and  $f$  is an edge-magic labeling with magic number  $s$ . Suppose  $V(G) = \{v_0, \dots, v_n\}$  with  $\deg(v_0) = n - 1$ . Then

$$2(n - 1)s = \sum_{i=1}^{3n-2} i + (n - 2)f(v_0) + 2 \sum_{j=1}^{n-1} f(v_j).$$

But then the parity is wrong, contradiction.  $\square$

**Conj. (37.3.3.8) [Kotzig-Rosa].** Every finite tree is (super)edge-magic.  $\lrcorner$

*Proof:*  $\square$

**Conj. (37.3.3.9) [Kotzig-Rosa].** For  $n \in \mathbb{Z}_+ \setminus \{4\}$ ,  $W_n$  is edge-magic.  $\lrcorner$

*Proof:*  $\square$

**Conj. (37.3.3.10) [Kotzig-Rosa].** Suppose  $G \in \text{Graph}^{\text{fin}}$ ,  $\#G = n + m$  and  $G$  contains a complete graph  $K_n$ , then if  $n \gg m$ ,  $G$  is not edge-magic.  $\lrcorner$

*Proof:*  $\square$

## 4 Cayley Graphs

**Def. (37.3.4.1) [Cayley Graphs].** For  $G \in \mathcal{G}\text{rp}$  and  $S \subset G$  s.t.  $k \triangleq \#S < \infty$  and  $S = S^{-1}$ , the **Cayley graph** is a graph  $\text{Cay}(G, S) \in \text{Graph}$  s.t.

$$V(\text{Cay}(G, S)) = G, \quad E(\text{Cay}(G, S)) = \{(x, y) \in G^2 | y^{-1}x \in S\}.$$

Then

- $\text{Cay}(G, S)$  is a  $k$ -regular.
- $\text{Cay}(G, S)$  is connected iff  $S$  generates  $G$ .
- If there is a homomorphism  $\chi : G \rightarrow \mathbb{Z}/(2)$  s.t.  $\chi(S) = \{1\}$ , then  $\text{Cay}(G, S)$  is bipartite. The converse is also true if  $S$  generates  $G$ .  $\lrcorner$

*Proof:* 3: One direction is clear. For the other direction, if  $S$  generates  $G$ , by item2,  $G$  is connected, so we can use the bipartition to define  $\chi$  using the parity of distances.  $\square$

## 5 Perfect Graphs

**Def.(37.3.5.1) [Perfect Graphs].** A **Perfect graph** is a finite simple graph  $G$  s.t. for any induced subgraph  $H \subset G$ , the maximal cardinality of a clique of  $H$  equals the chromatic number of  $H$ .  $\lrcorner$

**Def.(37.3.5.2) [Berge Graphs].** A finite simple graph  $G$  is called a **Berge graph** if it doesn't contain holes or antiholes(37.2.1.22) of odd cardinality.  $\lrcorner$

**Thm.(37.3.5.3) [Perfect Graph Theorem, Chudnovsky-Robertson-Seymour-Thomas].** A simple finite graph  $G$  is perfect iff it is Berge.  $\lrcorner$

*Proof:*

$\square$

## 6 Triangulated Graphs

## 7 Interval Graphs

## 8 Graph Minors

**Conj.(37.3.8.1) [Self-Minor Conjecture, Seymour].** Every countably-infinite graph is a proper minor of itself(i.e. the minor is not an identity).  $\lrcorner$

*Proof:*

$\square$

**Thm& Conj.Cor.(37.3.8.2) [Graph Minor Theorme, Robertson-Seymour 1986–2004].** The minor relation on the set of finite graphs is a quasi-well-ordering.  $\lrcorner$

*Proof:* Clearly there cannot be an infinite decreasing minor sequence of finite graphs, so by(2.2.3.19), it suffices to show that there cannot be an infinite set of non-comparable finite graphs. This follows from the self-minor conjecture, because by considering the disjoint union of the graphs.

Alternatively, **?**

$\square$

**Lemma(37.3.8.3) [Kruskal1960].** The root-preserving topological minor relation on the set of finite rooted trees is a quasi-well-ordering.  $\lrcorner$

*Proof:* Clearly there cannot be an infinite decreasing topological minor sequence of finite rooted trees, so by(2.2.3.19), it suffices to show that there cannot be an infinite set of non-comparable finite rooted trees.

Suppose the contrary, and we can choose a bad sequence  $(T_i)_{i \in \mathbb{N}}$  of finite rooted trees s.t. for any  $n \in \mathbb{N}$ ,  $T_n$  is a tree of minimum order s.t.  $(T_0, T_1, \dots, T_n)$  extends to a bad sequence. Let  $r_n$  be root of  $T_n$ , and let  $A_n$  be the set of trees of the forest  $T_n \setminus \{r_n\}$ . Then we claim that  $A = \cup_n A_n$  is quasi-well ordered by the root-preserving topological minor relation:

Let  $(T^k)$  be a sequence of trees in  $A$ , and find a smallest  $n \in \mathbb{N}$  s.t.  $T^k \in A_n$  for some  $k$ . Then  $T^k \subsetneq T_n$ , and

$$T_0, \dots, T_{n-1}, T^k, T^{k+1}, \dots$$

is not a bad sequence by the choice of  $T_n$ . Let  $(T, T')$  be an ascending pair, then  $T$  cannot be in  $(T_0, \dots, T_{n-1})$ , as then we will have  $T \leq T' \leq T_{n'}$ , but  $n' \geq n$ , so  $(T, T_{n'})$  is an ascending pair in  $(T_i)_{i \in \mathbb{N}}$ , contradiction. Thus  $(T, T')$  is an ascending pair in  $(T^k)$ , which is what we want to prove.

Now by(2.2.3.20), the sequence  $(A_n)_{n \in \mathbb{N}} \in [A]^{<\aleph_0}$  has an ascending pair  $A_i \leq A_j$  where  $i < j$ . Let  $f : A_i \rightarrow A_j$  be injective with  $T \leq f(T)$  for any  $T \in A_i$ , then we get a root-preserving injection  $T_i \rightarrow T_j$ . Thus  $T_i \leq T_j$ , contradiction.  $\square$

## 9 Lifts of Graphs

**Def.(37.3.9.1) [Double Covers of Graphs].** For any simple graph  $G \in \text{Graph}^{\text{simple}}$ , the **double covering** of  $G$  is the graph  $\tilde{G} = (\tilde{V}, \tilde{E})$  where  $\tilde{V} = V(G) \times \{\pm 1\}$ , and

$$\tilde{E} = \{(v_1, \varepsilon_1, v_2, \varepsilon_2) \in \tilde{V}(G)^2 \mid (v_1, v_2) \in E(G), \varepsilon_1 \neq \varepsilon_2\}.$$

It is a bi-partite graph, and  $\tilde{G} \rightarrow G$  is a covering map.  $\lrcorner$

**Def.(37.3.9.2) [Signing of a 2-Lifting].** For any simple graph  $G \in \text{Graph}^{\text{simple}}$ , the **signed 2-lift** of  $G$  with signing  $s : E(G) \rightarrow \{\pm 1\}$  is the graph  $\tilde{G}_s = (\tilde{V}_s, \tilde{E}_s)$  s.t.  $\tilde{V}_s = E(G) \times \{\pm 1\}$ , and

$$\tilde{E}_s = \{(v_1, \varepsilon_1, v_2, \varepsilon_2) \in \tilde{V}_s(G)^2 \mid (v_1, v_2) \in E(G), \varepsilon_1 \begin{cases} = \varepsilon_2 & , s(v_1, v_2) = 1 \\ \neq \varepsilon_2 & , s(e(v_1, v_2)) = -1 \end{cases}\}.$$

It is a bi-partite graph, and  $\tilde{G} \rightarrow G$  is a covering map.  $\lrcorner$

**Thm.(37.3.9.3) [Spectrums of Signed 2-Lifts, Bilu-Linial].** For any simple graph  $G \in \text{Graph}^{\text{simple}}$  with signing  $s : E(G) \rightarrow \{\pm 1\}$ , the spectrum of the signed 2-lift of  $G$  w.r.t.  $s$  is the sum of the spectrum of  $G$  and the spectrum of the signed adjacency matrix  $A_{s,G}$  of  $G$  (37.3.1.2).  $\lrcorner$

*Proof:* It is not hard to see that the adjacency matrix of  $\tilde{G}_s$  is  $\tilde{A} = \begin{bmatrix} A_1 & A_2 \\ A_2 & A_1 \end{bmatrix}$ , where  $A_1 = (V(G), s^{-1}(1))$ , and  $A_2 = (V(G), s^{-1}(-1))$ ,  $A = A_1 + A_2$ ,  $A_s = A_1 - A_2$ .

Then it is easy to see that if  $v$  is an eigenvector of  $A$  with eigenvalue  $\lambda$ , then  $(v, v)$  is an eigenvector of  $\tilde{A}$  with eigenvalue  $\lambda$ , and if  $u$  is an eigenvector of  $A_s$  with eigenvalue  $\mu$ , then  $(u, -u)$  is an eigenvector of  $\tilde{A}$  with eigenvalue  $\mu$ . As these vectors are linearly independent, the result follows.  $\square$

## 10 Expander Graphs

References are [DSV03] and [Entropy waves, the zig-zag graph product, and new constant-degree expanders].

**Def.(37.3.10.1) [Expanding Constants].** For  $G \in \text{Graph}$ , the **expanding constant** or **Cheeger constant** of  $G$  is defined to be

$$h(G) = \inf_{F \subset V(G), \#F < \infty} \left( \frac{e(F, V(G) \setminus F)}{\min(\#F, \#(V(G) \setminus F))} \right).$$

$\lrcorner$

**Prop.(37.3.10.2) [Cheeger – Buser Inequality, Dodziuk/Alon-Milman].** Let  $k \in \mathbb{Z}_{\geq 2}$  and  $G \in \text{Graph}^{\text{simple}}$  be a finite connected  $k$ -regular graph with  $n \triangleq v(G)$ , then

$$\frac{k - \lambda_1}{2} \leq h(X) \leq \sqrt{2k(k - \lambda_1)} \quad (37.2.2.2).$$

$\lrcorner$

*Proof:* Take an arbitrary orientation of  $G$ , and consider the the coboundary functor  $d : \ell^2(V(G)) \rightarrow \ell^2(E(G)) : df(e) = f(e^+) - f(e^-)$ , and its adjoint  $d^* : \ell^2(E(G)) \rightarrow \ell^2(V(G))$ . Then  $d^*d = L_G =$

$k\mathbf{1} - A_G$ . The spectrum of  $L_G$  is  $0, k - \lambda_1, \dots, k - \lambda_{n-1}$ . So for any  $f \in \ell^2(V(G))$  s.t.  $\sum_{v \in V(G)} f(v) = 0$ , we have  $|df|_2^2 = \langle L_G f, f \rangle \geq (k - \lambda_1)|f|_2^2$ .

Now for any  $F \subset V(G)$ , consider  $f \in \ell^2(V) : f(x) = \begin{cases} \#(V \setminus F) & , x \in F \\ -\#F & , x \notin F \end{cases}$ . Then  $\sum_{v \in V(G)} f(v) = 0$ , and  $|f|_2^2 = n\#F\#(V \setminus F)$ . And

$$df(e) = \begin{cases} \pm n & , e \in E(F, V \setminus F) \\ 0 & , \text{otherwise} \end{cases}.$$

So  $|df|_2^2 = n^2 e(F, V \setminus F)$ . The the inequality above applied to  $f$  implies that

$$e(F, V \setminus F) \geq (k - \lambda_1) \frac{\#F \cdot \#V \setminus F}{n} \geq \frac{k - \lambda_1}{2} \min(\#F, \#V \setminus F).$$

This proves the lower bound for  $h(F)$ .

For the upper bound: For any  $f \in \ell^2(V(G))$ , define  $B_f = \sum_{e \in E(G)} |f(e^+) - f(e^-)|^2$ , then

**Lemma(37.3.10.3).**

- If  $\beta_r > \beta_{r-1} > \dots > \beta_0$  are the values of  $f$ , and let  $L_i = f^{-1}([\beta_i, \infty))$ , then  $B_f = \sum_{i=1}^r e(L_i, V(G) \setminus L_i)(\beta_i^2 - \beta_{i-1}^2)$ .
- $B_f \leq \sqrt{2k}|df|_2|f|_2$ .
- If  $\#\text{Supp}(f) \leq n/2$ , then  $B_f \geq h(G)|f|_2^2$ .

□

*Proof:* 1: This is clear.

2:

$$B_f = \sum_{e \in E(G)} |f(e^+) + f(e^-)| \cdot |f(e^+) - f(e^-)| \leq \sqrt{\sum_{e \in E(G)} (f(e^+) + f(e^-))^2} \cdot \sqrt{\sum_{e \in E(G)} (f(e^+) - f(e^-))^2} \leq \sqrt{2k}|df|_2|f|_2.$$

3: The hypothesis implies that  $\#L_i \leq n/2$  for any  $i$ , thus  $e(L_i, V(G) \setminus L_i) \geq h(G)\#L_i$ . Thus

$$B_f = \sum_{i=1}^r e(L_i, V(G) \setminus L_i)(\beta_i^2 - \beta_{i-1}^2) \geq h(G) \sum_{i=1}^r \#L_i(\beta_i^2 - \beta_{i-1}^2) = h(G)|f|_2^2.$$

□

Now consider an eigenvalue  $g$  for  $L_G$  with eigenvector  $k - \lambda_1$ , and let  $V^+ = \{x \in V | g(x) > 0\}$ ,  $f = \max(g, 0)$ . We may assume  $0 < \#V^+ \leq n/2$ . Then for  $x \in V^+$ ,

$$L_G f(x) = kg(x) - \sum_{y \in V^+} A_{xy}g(y) \leq kg(x) - \sum_{y \in V} A_{xy}g(y) = (k - \lambda_1)g(x).$$

So

$$|df|_2^2 = \langle L_G f, f \rangle = \sum_{x \in V^+} L_G f(x)g(x) \leq (k - \lambda_1) \sum_{x \in V^+} g(x)^2 \leq (k - \lambda_1)|f|_2^2.$$

Combining this and the lemma above, we get

$$h(x)|f|_2^2 \leq B_f \leq \sqrt{2k}|df|_2|f|_2 \leq \sqrt{2k(k - \lambda_1)}|f|_2^2,$$

which implies  $h(x) \leq \sqrt{2k(k - \lambda_1)}$ .

□

**Def. (37.3.10.4) [Expander Graphs].** For  $k \in \mathbb{Z}_{\geq 2}$ , a family of  $k$ -regular finite simple graphs  $(X_n) \in \text{Graph}^{\mathbb{Z}_+}$  is called a family of **expander graphs** if  $\lim_{n \rightarrow \infty} v(X_n) = \infty$ , and  $\inf_n h(X_n) > 0$ , or equivalently by (37.3.10.2),  $\inf_n (k - \lambda_1(X_n)) > 0$ .  $\lrcorner$

**Prop. (37.3.10.5) [Almost All Graphs are Expanders].** For  $k \in \mathbb{Z}_{\geq 3}$ , almost all  $k$ -regular graphs are expanders.  $\lrcorner$

*Proof:* Cf. [Pinsker, On the complexity of a concentrator]. ?  $\square$

### Ramanujan Graphs

References are [DSV03] and [L-S88].

**Def. (37.3.10.6) [Ramanujan Graphs].** A **Ramanujan graph** is a finite connected graph  $G \in \text{Graph}^{\text{fin}}$  with universal covering  $T$  s.t. for any eigenvalue  $\lambda_i$  of  $G$ , either  $|\lambda_i| = \lambda_0$ , or  $|\lambda_i| \leq \rho(T)$ .

In particular, by (37.2.2.9), for any  $c, d \in \mathbb{Z}_{\geq 2}$ , a finite connected  $(c, d)$ -bi-regular graph has trivial eigenvalues  $\pm\sqrt{cd}$ , so it is a **Ramanujan bi-regular graph** iff all its non-trivial eigenvalues  $\lambda_i$  satisfy  $|\lambda_i| \leq \sqrt{c-1} + \sqrt{d-1}$ .  $\lrcorner$

**Cor. (37.3.10.7).** For a  $k$ -regular non-bi-partite Ramanujan graph  $G$ ,  $\gamma(G) \geq \frac{k}{2\sqrt{k-1}} \geq \frac{\sqrt{k}}{2}$ , by (37.3.2.8).  $\lrcorner$

**Conj. (37.3.10.8) [Bilu-Linial].** For any  $c, d \in \mathbb{Z}_{\geq 3}$  and any  $(c, d)$ -bi-regular graph  $G$ , there is a signing of a 2-lift (37.3.9.2) such that the new eigenvalues all have absolute value  $\leq \sqrt{c-1} + \sqrt{d-1}$ .  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj. Cor. (37.3.10.9) [Marcus-Spielman-Srivastava].** For any  $c, d \in \mathbb{Z}_{\geq 3}$  and any  $(c, d)$ -bi-regular graph  $G$ , there is a signing of a 2-lift such that the new eigenvalues are all bounded above by  $\sqrt{c-1} + \sqrt{d-1}$ .  $\lrcorner$

*Proof:* The universal cover  $T$  of  $G$  is a  $(c, d)$ -bi-regular tree, which has spectrum radius  $\rho(T) = \sqrt{c-1} + \sqrt{d-1}$  by (37.2.2.9). Thus the assertion follows from (37.3.1.5) and (37.3.9.3).  $\square$

**Cor. (37.3.10.10) [Lubotzky-Phillips-Sarnak/Margulis/Chiu/Lubotzky/Marcus-Spielman-Srivastava [MSS15]].** For any  $c, d \in \mathbb{Z}_{\geq 3}$ , there exists an infinite family of  $(c, d)$ -bi-regular Ramanujan graphs.  $\lrcorner$

*Proof:* This follows from consecutively using (37.3.10.9) on any  $(c, d)$ -bi-regular Ramanujan graph  $G$ , for example a complete  $(c, d)$ -bi-regular graph (because its spectrum is  $\{\sqrt{cd}, -\sqrt{cd}, 0, 0, \dots, 0\}$ ).  $\square$

### Explicit Constructions

**Def. (37.3.10.11) [ $S_{p,q}$ ].** Let  $p \neq q \in \text{Prime}_{\geq 3}$  s.t.  $q \geq 2\sqrt{p}$ , then  $\mathbb{H}(\mathbb{F}_q) \cong \text{Mat}(2; \mathbb{F}_q)$ . Fix such an isomorphism  $\psi_p$ .

Let  $S_p \subset \mathbb{H}(\mathbb{Z})$  be a distinguished set of elements of norm  $p$ ,  $\#S_p = p+1$  (? Cf. [Sarnak, Ramanujan Graph book] P68). Define

$$S_{p,q} = \psi_q(S_p) \subset \text{PGL}(2; \mathbb{F}_q),$$

then  $\#S_{p,q} = p+1$ , and  $S_{p,q} \subset \text{PSL}(2; \mathbb{F}_q)$  iff  $\left(\frac{p}{q}\right) = 1$ .  $\lrcorner$

*Proof:* Cf.[Sarnak, Ramanujan Graph book]P113.?

For the last assertion, notice  $x \in \text{GL}(2; \mathbb{F}_q)$  mapsto  $\text{PSL}(2; \mathbb{F}_q)$  iff  $\det(x) \in \mathbb{F}_q^2$ , so  $S_{p,q} \subset \text{PSL}(2; \mathbb{F}_q)$  iff  $\left(\frac{p}{q}\right) = 1$ .  $\square$

**Thm. (37.3.10.12)** $[X^{p,q}]$ . Let  $p \neq q \in \text{Prime}_{\geq 3}$  s.t.  $q \geq 2\sqrt{p}$ , we can define

$$X^{p,q} = \begin{cases} \text{Cay}(\text{PSL}(2; \mathbb{F}_q), S_{p,q}) & , \left(\frac{p}{q}\right) = 1 \\ \text{Cay}(\text{PGL}(2; \mathbb{F}_q), S_{p,q}) & , \left(\frac{p}{q}\right) = -1 \end{cases}.$$

Then  $X^{p,q}$  is a  $(p+1)$ -regular connected Ramanujan graph, and for fixed  $p$ , this forms a large girth family(37.2.1.18). In fact,

- If  $\left(\frac{p}{q}\right) = -1$ , then  $X^{p,q}$  is bipartite,  $v(X^{p,q}) = q(q^2 - 1)$ , and  $g(X^{p,q}) \geq 4\log_p q - \log_p 4$ .
- If  $\left(\frac{p}{q}\right) = 1$ , then  $X^{p,q}$  is not bipartite,  $v(X^{p,q}) = q(q^2 - 1)/2$ , and  $g(X^{p,q}) \geq 2\log_p q$ .

┘

*Proof:*

$\square$

## 11 Potential Theory on Graphs

### Riemann-Roch on Graphs

References are [B-N07].

**Conj. (37.3.11.1)**. Is there is a Serre duality for graphs?

Is there an analogy of Hirzebruch – Riemann – Roch theorem for higher-dimensional simplicial complexes?

Is there a relation between Riemann-Roch theorems for tropical varieties/toric varieties/graphs?

┘

*Proof:*

$\square$

## 12 Zeta Functions

**Def. (37.3.12.1)** $[\text{Ihara Zeta Functions}]$ . For a finite unoriented graph  $G \in \text{Graph}^{\text{fin}}$ , the **Ihara zeta function** is defined to be

$$Z(u, G) = \prod_{p \in P} (1 - u^{\text{length}(p)})^{-1},$$

where  $P$  is the set of primes circuits in  $G$ (37.2.1.20).

Similarly we can define zeta function  $Z^o(u, G)$  of an oriented graph.

WARNING: the zeta function of a graph and its duplication oriented graph, because of the primality condition.  $\square$

**Prop. (37.3.12.2).** For a finite oriented graph  $G \in \mathbf{Graph}^{\text{fin}}$  and  $m \in \mathbb{Z}_+$ , let  $N_m$  be the number of circuits in  $G$  of length  $m$  with an endpoint, then

$$Z^o(u, G) = \exp\left(\sum_{m \in \mathbb{Z}_+} \frac{N_m}{m} u^m\right) = \det(1 - uM_G)^{-1}.$$

In particular, it is a rational function.  $\lrcorner$

*Proof:* The first equation is easy by taking log. For the second equation, because  $N_m = \text{tr}(M_G^m)$ , let the eigenvalues of  $M_G$  be  $\lambda_1, \dots, \lambda_n$ , then using an triangulation of  $M_G$ ,

$$\exp\left(\sum_{m \in \mathbb{Z}_+} \frac{N_m}{m} u^m\right) = \exp\left(\sum_{m \in \mathbb{Z}_+} \frac{1}{m} \sum_{i=1}^n \lambda_i^m u^m\right) = \prod_{i=1}^n \exp(-\log(1 - \lambda_i u)) = \det(1 - uM_G)^{-1}.$$

□

**Cor. (37.3.12.3).** For  $G \in \mathbf{Graph}^{\text{fin}}$ ,  $G$  is bipartite iff  $Z(u, G)$  is an even function.  $\lrcorner$

*Proof:* Since the vertices  $v \in V(G)$  s.t.  $\deg(v) = 1$  doesn't affect the result, we can discard them. Then the assertion follows easily from (37.3.2.4).  $\square$

**Prop. (37.3.12.4) [Edge Zeta Functions].** For a finite unoriented graph  $G \in \mathbf{Graph}^{\text{fin}}$ , consider its line graph  $G_L$  where

$$V(G_L) = \vec{E}(G), \quad E(G_L) = \{(e_1, e_2) \in \vec{E}(G) \times \vec{E}(G), e_1 \neq \bar{e}_2, t(e_1) = s(e_2)\}.$$

Then it is easy to see that

$$Z(u, G) = Z^o(u, G_L).$$

□

**Thm. (37.3.12.5) [Bass].** For a finite connected unoriented graph  $G \in \mathbf{Graph}^{\text{fin}}$  s.t.  $\deg(v) \geq 2$  for any  $v \in G$ , we have

$$Z(u, G) = (1 - u^2)^{\chi(G)} \det(\mathbf{1} - uA + u^2(D - \mathbf{1}))^{-1}.$$

□

*Proof:* Cf. [Zeta functions of finite graphs].  $\square$

**Cor. (37.3.12.6) [Regular Graphs].** Let  $q \in \mathbb{Z}_+$  and  $G \in \mathbf{Graph}$  a finite connected  $k$ -regular graph, then

$$Z(u, G) = (1 - u^2)^{(1-q)v(G)/2} \det((1 + qu^2)\mathbf{1} - uA)^{-1}.$$

In particular, by (37.2.2.3),

•

$$\text{Pole}(Z(u, G)) \subset [q^{-1}, 1] \cup [-1, -q^{-1}] \cup \{|z| = q^{-1/2}\}.$$

- $u = -q^{-1}$  is a pole iff  $G$  is bipartite.
- $\text{Pole}(Z(u, G)) \subset \{|z| = q^{-1/2}\} \cup \{\pm 1, \pm q^{-1}\}$  iff  $G$  is a Ramanjan graph (37.3.10.6).

□



**Prop. (37.3.12.7) [Northshield].** For a finite connected oriented graph  $G \in \mathbf{Graph}^{\text{fin}}$  s.t.  $\deg(v) \geq 2$  for any  $v \in G$ ,

$$\lim_{u \rightarrow 1} \frac{Z(u, G)}{(1-u)^{\chi(G)-1}} = \frac{2^{\chi(G)-1}}{\chi(G) \text{tree}(G)}.$$

┘

*Proof:* Let  $f(u) = \det(\mathbf{1} - uA + u^2(D - \mathbf{1})|\ell^2(V(G)))$ , then by (37.3.12.5), it suffices to prove that

$$f'(1) = -2\chi(G) \text{tree}(G).$$

Let  $M(u) = \mathbf{1} - uA + u^2(D - \mathbf{1})$ , and let  $M_k$  be the result of  $M$  derivating the  $k$ -th row. Then  $M_k(1) = M(1) + (\deg(v_k) - 2)e_{kk}$ . So

$$\frac{\partial}{\partial u} \det(M(u))|_{u=1} = \sum_k [\det M(1) + (\deg(v_k) - 2) \det \widetilde{M}_{kk}],$$

where  $M_{kk}$  is the  $(k, k)$ -th cofactor of  $M(1)$ . But by (37.2.2.7),  $\det \widetilde{M}_{kk} = \text{tree}(G)$ , so the result follows.  $\square$

### 13 Applications

**Thm. (37.3.13.1) [Monsky].** A square cannot be divided into  $m$  triangles of the same area, where  $m$  is odd.  $\square$

*Proof:* Choose a 3-coloring on  $\mathbb{R}^2$ : By (19.1.2.19) there can be an extended 2-adic valuation  $|\cdot|_2$  on  $\mathbb{R}$  that extends the 2-adic valuation on  $\mathbb{Q}$ , then color a point  $(x, y)$

- 0 if  $|x|_2 < 1, |y|_2 < 1$ ,
- 1 if  $|x|_2 \geq 1$  and  $|x|_2 \geq |y|_2$ .
- 2 if  $|y|_2 \geq 1$  and  $|y|_2 > |x|_2$ .

Then there are two things:

1. The coloring is invariant under translation by vectors represented by a point of color 0.
2. The valuation of the area of a triangle with vertices of 3 different colors is bigger than 1, because we can assume one vertex is the origin, and then its area is  $\frac{1}{2}|x_1y_2 - x_2y_1|$ , which, because the coloring, must have 2-adic valuation bigger than 1.

Now back to the question, let the square be placed as a unit square, then its area is 1, and it has exactly one  $(0, 1)$ -edge, so by Spencer's lemma (37.2.1.29), it has a triangle with vertices of pairwise different colors, so its area  $A$  has valuation  $> 1$ , but the total area is  $mA = 1$  that has valuation 1, so  $|m| < 1$ , which means that  $m$  is even.  $\square$

**Prop. (37.3.13.2) [Hindman's Theorem].** Whenever  $\mathbb{N}$  is colored with f.m. colors, one can find an infinite subset  $A \subset \mathbb{N}$  and a color  $c$  that whenever  $F \subset A$  is finite, the color of the sum of numbers in  $F$  is colored  $c$ .  $\square$

*Proof:* Cf. [W. W. Comfort, Ultrafilters: some old and some new results, Bull. Amer. Math. Soc. 83 (1977) 417–455.]  $\square$

### 37.4 Combinatorial Set Theory

**Def. (37.4.0.1) [Notations].** For a set  $S$ , let  $[S]^r$  be the set of subsets of  $S$  of order  $r$ . Let  $\kappa, \lambda$  be cardinals, we write  $\kappa \rightarrow (\lambda)_s^r$  as a shorthand for: for any set  $S$  with  $|S| = \kappa$  and every partition of  $[S]^r$  into  $s$  classes, there exists a subset  $H \subset S$  that  $[H]^r$  is in the same class, and  $|H| \geq \lambda$ .  $\lrcorner$

**Prop. (37.4.0.2) [Ramsey's Theorem for  $\aleph_0$ ].** For any positive natural number  $r, s$  if we color the  $r$ -subsets of a set with cardinality  $\aleph_0$  into  $s$  families, then there is a subset of cardinal  $\aleph_0$  that all its  $r$ -subsets are colored the same.  $\lrcorner$

*Proof:* Cf.[Graph Theory, Thm9.1.2].  $\square$

**Cor. (37.4.0.3).** Every infinite linearly ordered set contains a subset isomorphic to  $(\mathbb{N}, <)$  or  $(\mathbb{N}, >)$ .  $\lrcorner$

*Proof:* Choose a well ordering of it. Then consider this new ordering and the original ordering. Then there is an infinite set that is compatible with the original ordering, or converse. Then its initial segment of order type  $\omega_0$  satisfies the requirement.  $\square$

**Def. (37.4.0.4) [Weakly Compact Cardinals].** An **weakly compact cardinal** is an uncountable cardinal  $\kappa$  that  $\kappa \rightarrow (\kappa)_s^r$  (37.4.0.1) for any  $r, s \in \mathbb{Z}_+$ .  $\lrcorner$

**Prop. (37.4.0.5).** Weakly compact cardinals are strongly inaccessible.  $\lrcorner$

*Proof:* Cf.[Jec03]P224.  $\square$

#### Trees

**Def. (37.4.0.6).** A **tree** is a partial ordered set  $T$  that there is a minimal element  $r$  and for each  $x$ ,  $\{y \in T | y < x\}$  is finite and linearly ordered.

A tree is called **of finite branched** for each  $x$ , there is a finite set  $\{y_1, \dots, y_r\}$  is  $T$  that  $y_i > x$  and if  $z > x$ , then  $z \geq y_i$  for some  $i$ .  $\lrcorner$

**Def. (37.4.0.7) [Height].** For any node  $x$ ,  $\{y \in T | y < x\}$  is a well-ordered set, which is isomorphic to an ordinal by (2.2.4.7), it is called the **height** of  $x$ .  $T_\alpha$  denotes the set of all nodes of  $T$  of order  $\alpha$ . The least  $\alpha$  that  $T_\alpha \neq \emptyset$  is called the **height** of  $T$ .

A **branch** is a maximal chain in  $T$ , its **length** is its ordinal. The length is always smaller than the height of the tree. If it equals the height of the tree, it is called **cofinal**.  $\lrcorner$

**Def. (37.4.0.8).** A **subtree** is a subset  $T'$  of  $T$  that if  $x \in T', y < x$ , then  $x \in T'$ .

An **antichain** of a tree  $T$  is a subset  $A \subset T$  that any two elements in  $A$  are incomparable.  $\lrcorner$

**Def. (37.4.0.9).** A **path** through  $T$  is a morphism of ordering from  $\omega$  to  $T$ .  $\lrcorner$

**Lemma (37.4.0.10) [König's Lemma].** If  $T$  is an infinite finite branching tree, then there is a path through  $T$ .  $\lrcorner$

*Proof:* Use recursion to choose for each  $n$  an element that has infinite successors.  $\square$

**Def. (37.4.0.11).** An **Aronszajn tree** is a tree of height  $\kappa$  and all its level sets are at most countable, but has no branches of length  $\kappa$ .  $\lrcorner$

**Prop. (37.4.0.12).** An **Aronszajn tree** of height  $\omega_1$  exists.  $\lrcorner$

*Proof:* Cf.[Set Theory Jech P228].  $\square$

## 1 Finite Sets

**Thm. (37.4.1.1)** [Dirichlet's Box Principle]. ┘

*Proof:* □

## 2 Finite Ramsey's Theory

**Thm. (37.4.2.1)** [Finite Ramsey's Theorem]. For any  $\alpha, k, n_1, \dots, n_k \in \mathbb{Z}_+, n_i \geq \alpha$ , there exists a minimal  $R(\alpha, k; n_i) \in \mathbb{Z}_+$  s.t. if we divide the  $\alpha$ -subsets of a set with cardinality  $R(\alpha, k; n_i)$  into  $k$  groups  $C_1, \dots, C_k$ , then there is some  $1 \leq i \leq k$  and a subset  $S$  of cardinality  $n_i$  s.t. all the  $\alpha$ -subset of  $S$  is in  $C_i$ . Moreover,

$$R(\alpha, k; n_i) \leq R(\alpha - 1, k; m_j = R(\alpha, k; n_1, \dots, n_{j-1}, n_j - 1, n_{j+1}, \dots, n_k)) + 1.$$
┘

*Proof:* We use double induction on  $(\alpha, k)$  in lexicographical order. If  $\alpha = 1$ , clearly  $R(1, k; n_i) = \sum_{i=1}^k (n_i - 1) + 1$ . And if  $k = 1$ , then  $R(\alpha, 1; n_1) = n_1$ .

Suppose  $\alpha > 1, k > 1$ , if some  $n_i = \alpha$ , then  $R(\alpha, k; n_i) = R(\alpha, k - 1; n_1, \dots, \hat{n}_i, \dots, n_k)$ , thus we are reduced to smaller  $k$ . So we can assume that  $n_i > \alpha$  for each  $i$ . In this case, we prove that

$$R(\alpha, k; n_i) \leq R(\alpha - 1, k; m_j = R(\alpha, k; n_1, \dots, n_{j-1}, n_j - 1, n_{j+1}, \dots, n_k)) + 1.$$

And this will finish the induction process.

To prove this, notice that if we have a set  $X$  with cardinality equal to the RHS, and let  $x \in X$ , then we can divide the  $(\alpha - 1)$  subsets of  $X \setminus \{x\}$  into  $k$  groups  $C'_1, \dots, C'_k$  s.t. an  $(\alpha - 1)$ -set  $S$  has cardinality  $\alpha - 1$  belongs to  $C'_i$  iff  $S \cup \{x\}$  belongs to  $C_i$ . Then by definition, there exists some  $i$  and some subset  $Y$  of  $X \setminus \{x\}$  of cardinality  $R(\alpha, k; n_1, \dots, n_{i-1}, n_i - 1, n_{i+1}, \dots, n_k)$  s.t. all  $(\alpha - 1)$ -subset of  $Y$  belongs to  $C'_i$ . Then by definition, there either exists some  $j \neq i$  and some subset  $Z_1$  of  $Y$  of cardinality  $n_j$  s.t. all the  $\alpha$ -subset of  $Z_1$  is in  $C_j$ , in which case the assertion is satisfied; or exists some subset  $Z_2$  of  $Y$  of cardinality  $n_i - 1$  s.t. all the  $\alpha$ -subset of  $Z_2$  is in  $C_i$ . Then  $Z_2 \cup \{x\}$  satisfies the assertion. □

**Prop. (37.4.2.2)** [Szekeres]. For any  $a, b \in \mathbb{Z}_{\geq 2}$ ,  $R(2; a, b) \leq \binom{a+b-2}{a-1}$ . ┘

*Proof:* Use induction on  $a + b$ : If  $a = 2$  or  $b = 2$ , this is easy. And for  $a, b \geq 3$ , by (37.4.2.1),

$$R(2; a, b) \leq R(2; a - 1, b) + R(2; a, b - 1) \leq \binom{a+b-3}{a-2} + \binom{a+b-3}{a-1} = \binom{a+b-2}{a-1}$$
□

**Prop. (37.4.2.3)** [Erdős]. For  $k \in \mathbb{Z}_{\geq 3}$ ,

$$2^{k/2} < R(2; k, k) \leq \binom{2k-2}{k-1} < 4^{k-1}.$$
┘

*Proof:* If  $N < 2^{k/2}$ , then the number of different graphs of  $N$  vertices equals  $2^{N(N-1)/2}$ , and the number of different graphs containing a complete  $k$ -graph is less than

$$\binom{N}{k} 2^{N(N-1)/2 - k(k-1)/2} < \frac{N^k}{k!} 2^{N(N-1)/2 - k(k-1)/2} < \frac{2^{N(N-1)/2}}{2},$$

because  $2^{\frac{k}{2}+1} < k!$  for  $k \geq 3$ . So there exists a graph  $G$  that neither  $G$  and its complement graph  $G'$  contains a complete  $k$ -graph. Thus  $2^{k/2} < R(2; k, k)$ .

The second inequality follows from (37.4.2.2).  $\square$

**Prop. (37.4.2.4) [Erdős].** There exists  $C \in \mathbb{R}_+$  s.t.

$$R(2; 3, x) \geq Cx^2 / \log^2 x$$

for any  $x \in \mathbb{Z}_+$ .  $\lrcorner$

*Proof:* [Graph theory and Probability 2, Erdős].  $\square$

**Prop. (37.4.2.5) [Ajtai-Komlós-Szemerédi].** For any  $k \in \mathbb{Z}_{\geq 2}$ , there exists  $C_k \in \mathbb{R}_+$  s.t. for  $x \in \mathbb{Z}_+$  sufficiently large,

$$R(2; k, x) \leq 5000 \frac{x}{\log \left( R(2; k-1, x)^{\frac{1}{k-2}} \right)} \cdot R(2; k-1, x).$$

In particular, there exists  $C'_k \in \mathbb{R}_+$  s.t.

$$R(2; k, x) \leq C'_k x^{k-1} / \log^{k-2} x.$$

$\lrcorner$

*Proof:* Cf. [note on Ramsey numbers]. ?

$k = 2$  is trivial. For  $k \in \mathbb{Z}_{\geq 3}$ , let  $\varepsilon \in (\frac{0.96}{k-2}, \frac{1}{k-2})$ . Suppose  $G$  is a graph with

$$n = \#V(G) \geq 5000 \frac{x}{\log \left( R(2; k-1, x)^{\frac{1}{k-2}} \right)} \cdot R(2; k-1, x)$$

and  $\omega(G) < k$  and  $\alpha(G) \geq x$ . Let  $h(G)$  be the number of triangles in  $G$ .

Suppose  $\deg(P) \geq R(2; k-1, x)$  for some  $P \in V(G)$ , then  $\omega(G) \geq k$  or  $\alpha(G) \geq x$ , contradiction. So  $\deg(P) < R(2; k-1, x)$  for any  $P \in V(G)$ , which implies that  $t = 2\#E(G)/n < R(2; k-1, x)$ .

If  $h(G) < nR(2; k-1, x)^{2-\varepsilon}$ , when  $x$  is large,  $R(2; k, x)$  is large, so by (37.2.5.7),

$$\alpha(G) > \frac{\varepsilon}{4800} \cdot \frac{n}{R(2; k-1, x)} \cdot \log R(2; k-1, x) \geq x,$$

contradiction.

Thus when  $x$  is large, we can assume that  $h(G) \geq nR(2; k-1, x)^{2-\varepsilon}$ , thus some point  $P \in V(G)$  belongs to at least  $R(2; k-1, x)^{2-\varepsilon}/3$  triangles. Because  $\deg(P) < R(2; k-1, x)$ , there exists a point  $Q$  adjacent to  $P$  and  $\deg(Q) > 2R(2; k-1, x)^{1-\varepsilon}/3 > R(2; k-2, x)$  by hypothesis on  $\varepsilon$ . Then we can find a clique of size  $k-2$  that is adjoint to  $P$  and  $Q$ , contradiction.  $\square$

### Erdős-Hajnal Conjecture

**Conj. (37.4.2.6) [Erdős-Hajnal].** For any finite graph  $H \in \text{Graph}^{\text{fin}}$ , there exists  $\delta_H \in \mathbb{R}_+$  s.t. for any graph  $G \in \text{Graph}^{\text{fin}}$ , if  $H$  is not an induced subgroup of  $G$ , then  $\max(\omega(G), i(G)) \geq v(G)^{\delta_H}$ .  $\lrcorner$

*Proof:*  $\square$

### 3 Intersection Theorems

References are [EKR61] and [G-M16].

**Def. (37.4.3.1) [Shifting Operators].** Let  $n \in \mathbb{Z}_+$  and  $\mathcal{A} \subset \mathcal{P}([n]_+)$  be a family of sets, then for  $i \neq j \in [n]_+$ , the **shifting of sets**  $S_{ij}(\mathcal{A})$  is defined to be  $S_{ij}(\mathcal{A}) = \{S_{ij}(A) | A \in \mathcal{A}\}$ , where

$$S_{ij}(A) = \begin{cases} \{i\} \cup A \setminus \{j\} & , j \in A, i \notin A, \{i\} \cup A \setminus \{j\} \notin \mathcal{A} \\ A & , \text{otherwise} \end{cases}.$$

If  $i < j$ , then  $S_{ij}$  is called a **left shift**, and it is called a **right shift** otherwise.  $\lrcorner$

**Prop. (37.4.3.2).** Let  $n \in \mathbb{Z}_+$  and  $\mathcal{A} \subset \mathcal{P}([n]_+)$  be a family of sets, then for any  $i \neq j \in [n]_+$ ,  $S_{ji} \circ S_{ij}(\mathcal{A}) = S_{ji}(\mathcal{A})$ .  $\lrcorner$

**Prop. (37.4.3.3) [Generating Sets].** For any  $k \leq n \in \mathbb{Z}_+$  and  $\mathcal{B} \subset \mathcal{P}([n]_+)$ , let  $\lrcorner$

### Erdős-Ko-Rado Theorem

**Thm. (37.4.3.4) [Complete Erdős-Ko-Rado Theorem, Ahlswede-Khachatrian].** For  $t \leq k \leq n \in \mathbb{Z}_+$  and  $i \in [k - t]$ , let

$$\mathcal{F}_i = \{F \subset [n]_+, \#F = k, \#F \cap [t + 2i]_+ \geq t + i\}.$$

And consider

$$M(n, k, t) = \max\{\#\mathcal{A} | \mathcal{A} \subset \mathcal{P}([n]_+), (\forall A \in \mathcal{A}) \#A = k, (\forall A_1, A_2 \in \mathcal{A}) \#A_1 \cap A_2 \geq t\}.$$

Then

- If  $(k - t + 1)(2 + \frac{t-1}{r+1}) < n < (k - t + 1)(2 + \frac{t-1}{r})$  for some  $r \in \mathbb{N}$  (where  $\frac{t-1}{0} = \infty$ ), then  $M(n, k, t) = \#\mathcal{F}_r$ . And the maximum is attained only at  $\mathcal{F}_r$  up to permutations.
- If  $n = (k - t + 1)(2 + \frac{t-1}{r+1})$  for some  $r \in \mathbb{N}$ , then  $M(n, k, t) = \#\mathcal{F}_r = \#\mathcal{F}_{r+1}$ , and the maximum is attained only at  $\mathcal{F}_r$  or  $\mathcal{F}_{r+1}$  up to permutations (but notice when  $t = 1$ , i.e. when  $n = 2k$ , there can be multiple  $r$  s.t. this inequality holds).  $\lrcorner$

*Proof:* Let  $I(n, k, t)$  denote the set of families

$$\{\mathcal{A} | \mathcal{A} \subset \mathcal{P}([n]_+), (\forall A \in \mathcal{A}) \#A = k, (\forall A_1, A_2 \in \mathcal{A}) \#A_1 \cap A_2 \geq t\}.$$

**Lemma (37.4.3.5).** For any  $\mathcal{A} \in I(n, k, t)$ ,  $\overline{\mathcal{A}} \in I(n, n - k, n - 2k + t)$ , and for any  $A \in \mathcal{A}, B \in \overline{\mathcal{A}}$ ,  $\lrcorner$

To determine the families  $\mathcal{A}$  that attains the maximal cardinality, we use left shifts (37.4.3.1). Because any family can be left-shifted to a left compressed family by f.m. left-shifts, it suffices to show that if  $S_{ij}(\mathcal{A}) = \mathcal{F}_r$ , then  $\mathcal{A}$  is isomorphic to  $\mathcal{F}_r$  by permutation. It is clear that if  $i, j \leq t + 2r$  or  $i, j > t + 2r$ , then  $\mathcal{A} = \mathcal{F}_r$ .  $\square$

**Cor. (37.4.3.6) [Erdős-Ko-Rado/Wilson].** For  $t \leq k \leq n \in \mathbb{Z}_+$  s.t.  $n \geq (t+1)(k-t+1)$ , we have

$$M(n, k, t) = \binom{n-t}{k-t}.$$

Moreover, if  $n \geq (t+1)(k-t+1) + 1$ , then the equality is attained at the family  $\mathcal{A} = \{F \subset [n]_+ \mid \#F = n, [t]_+ \subset F\}$  up to permutations.  $\lrcorner$

**Cor. (37.4.3.7) [2k-Intersection Theorem].** For  $k \in \mathbb{Z}_+$ , we have  $M(2k, k, 1) = \binom{2k-1}{k-1}$ , and the maximum is attained only at one of the families  $\mathcal{F}_r$  up to permutations, where  $r \in [k-1]$ .  $\lrcorner$

**Cor. (37.4.3.8) [4m-Conjecture].** For  $m \in \mathbb{Z}_+$ ,  $M(4m, 2m, 2) = \frac{1}{2} \left( \binom{4m}{2m} - \binom{2m}{m}^2 \right)$ , and the maximality is achieved only at the family  $\mathcal{A} = \{F \subset [4m]_+ \mid \#F = 2m, \#F \cap [2m]_+ \geq m+1\}$   $\lrcorner$

**Thm. (37.4.3.9) [Non-trivial Intersecting Families, Hilton-Milner [H-M67]].** For  $k, n \in \mathbb{Z}_+, n \geq 2k$  and any system  $\mathcal{S} = \{S_1, \dots, S_v\} \subset \mathcal{P}([n]_+)$  s.t.

- $\#S_i = k$  for each  $i$ ,
- $\#S_i \cap S_j \neq \emptyset$  for any  $1 \leq i, j \leq v$ .
- $\bigcap_i S_i = \emptyset$ ,

we have  $v \leq \binom{n-1}{k-1} - \binom{n-k-1}{k-1} + 1$ .  $\lrcorner$

*Proof:* Cf. [Erdős-Ko-Rado theorem]P15.  $\square$

**Conj. (37.4.3.10) [Erdős-Ko-Rado].** For  $k \leq n \in \mathbb{Z}_+$  and any system  $\mathcal{S} = \{S_1, \dots, S_v\} \subset \mathcal{P}([n]_+)$  s.t.  $\#S_i \cap S_j \geq k$  for any  $1 \leq i, j \leq v$ , find the largest value of  $v$ .  $\lrcorner$

*Proof:*  $\square$

**Prop. (37.4.3.11) [Erdős-Ko-Rado].** For  $n \in \mathbb{Z}_{\geq 2}$  and any system  $\mathcal{S} = \{S_1, \dots, S_v\} \subset \mathcal{P}([n]_+)$  s.t.  $S_i \cap S_j \cap S_k \neq \emptyset$  for any  $1 \leq i, j, k \in v$ , then we have  $v \leq 2^{n-1}$ , with equality iff  $\mathcal{S}$  consists of all subsets containing a fixed element.  $\lrcorner$

*Proof:* The inequality is clear. Consider the smallest  $p \in [v]_+$  s.t.

$$S_1 \cap S_2 \dots \cap S_p \in \{S_1, \dots, S_v\}.$$

If  $p = n$ , then they have a common element. If  $p < n$ , then

$$\{S_1 \cap S_2 \dots \cap S_{p+1}, S_1, \dots, S_v\}$$

is another class s.t. any two of them has non-empty intersection. This implies that  $v+1 < 2^{n-1}$ .  $\square$

**Prop. (37.4.3.12) [Sperner].** For  $n \in \mathbb{Z}_+$ , any system  $\{S_1, \dots, S_v\}$  of subsets of  $[n]_+$  s.t. no set  $S_i$  contains another  $S_j$ , then  $v \leq \binom{n}{\lfloor n/2 \rfloor}$ .  $\lrcorner$

*Proof:*  $\square$

### Intersection Families of Permutations

## 37.5 Partition Theory

References are [The Theory of Partitions, Andrews] and [A-G88].

**Def. (37.5.0.1) [Partitions].** For  $n \in \mathbb{Z}_+$ , a **partition** of  $n$  is a tuple  $(a_1, \dots, a_k) \in \mathbb{Z}_+^{\mathbb{N}}$  s.t.

$$a_1 \leq a_2 \leq \dots \leq a_k, \quad \sum_i a_i = n.$$

denoted by  $\lambda = (a_1, \dots, a_k) \vdash n$ .

The number of partitions of  $n$  is denoted by  $\text{Par}(n)$ . For convenience, denote  $\text{Par}(0) = 1$ , and  $P(\alpha) = 0$  for  $\alpha \in \mathbb{R} \setminus \mathbb{N}$ . Clearly,

$$\sum_{n \in \mathbb{N}} \text{Par}(n) X^n = \prod_{k \in \mathbb{Z}_+} \frac{1}{(1 - X^k)} \in \mathbb{Z}[[X]].$$

┘

**Def. (37.5.0.2) [Dominance of Partitions].** For  $n \in \mathbb{Z}_+$  and  $\lambda = (\lambda_1, \dots, \lambda_m) \vdash n, \mu = (\mu_1, \dots, \mu_l) \vdash n$ , we say  $\lambda$  is a **dominant partition** of  $\mu$ , denoted by  $\lambda \preceq \mu$  if

$$\sum_{i \leq k} \lambda_i \leq \sum_{j \leq k} \mu_j$$

for any  $k \in \mathbb{Z}_+$ .

┘

**Def. (37.5.0.3) [Lexicographic Order of Partitions].** For  $n \in \mathbb{Z}_+$  and  $\lambda = (\lambda_1, \dots, \lambda_m) \vdash n, \mu = (\mu_1, \dots, \mu_l) \vdash n$ , we denote  $\lambda < \mu$  if for some  $i$ ,  $\lambda_j < \mu_j$  for any  $j < i$  and  $\lambda_i < \mu_i$ .

It is clear that if  $\lambda \preceq \mu$ , then  $\lambda < \mu$ .

┘

**Def. (37.5.0.4) [Partition Permutations].** For  $n \in \mathbb{Z}_+$  and  $\lambda = (\lambda_1, \dots, \lambda_l) \vdash n$ , denote the **partition permutation number**  $\lambda! = (\lambda_1)! \dots (\lambda_l)!$ .

┘

### 1 Young Tableaux

**Def. (37.5.1.1) [Young Tableaux and Young Subgroups].** Suppose  $n \in \mathbb{Z}_+$  and  $\lambda = (\lambda_1, \dots, \lambda_l) \vdash n$ , then

- The **Ferrers diagram** of  $\lambda$  is an array of  $n$  dots having  $l$  left-justified rows with row  $i$  containing  $\lambda_i$  dots for  $i = 1, \dots, l$ .
- The **Young tableaux** of  $\lambda$  is an array  $t^\lambda$  obtained by replacing the dots of the Ferrers diagram of  $\lambda$  with the numbers  $1, 2, \dots, n$  bijectively.
- The **Young subgroup** of  $\lambda$  is

$$\text{Sym}_\lambda = \text{Sym}_{[1, \dots, \lambda_1]} \times \text{Sym}_{[\lambda_1+1, \dots, \lambda_2]} \times \dots \times \text{Sym}_{[n-\lambda_l+1, \dots, n]} \leq \text{Sym}_n.$$

┘

**Def. (37.5.1.2) [Young Tabloids].** Two Young tableaux of the same size are called **row equivalent** if they contains the same row sets  $R_1, \dots, R_l$ . An equivalent class of Young tableaux of shape  $\lambda$  is called a  $\lambda$ -**tabloid**.

┘

**Prop. (37.5.1.3) [Dominance Lemma].** Let  $n \in \mathbb{Z}_+$ ,  $\lambda, \mu \vdash n$  and  $t^\lambda, s^\mu$  be tableaux of shape  $\lambda$  and  $\mu$  resp.. If for any  $i$ , the elements in row  $i$  of  $s^\mu$  appear in different columns in  $t^\lambda$ , then  $\lambda \succcurlyeq \mu$ .  $\lrcorner$

*Proof:* The hypothesis implies that we can sort the entries in each column of  $t^\lambda$  s.t. the elements of rows  $1, 2, \dots, i$  of  $s^\mu$  all occur in the first  $i$  rows of  $t^\lambda$ . This exactly means that  $\sum_{k \leq i} \lambda_i \geq \sum_{k \leq i} \mu_k$ .  $\square$

**Def. (37.5.1.4) [Row and Column Stabilizers].** Let  $n \in \mathbb{Z}_+$  and  $t$  is a Young tableaux of size  $n$  with row sets  $R_1, \dots, R_l$  and column sets  $C_1, \dots, C_k$ , then define the **row stabilizer** and **column stabilizer** groups as

$$R_t = \text{Sym}_{R_1} \times \dots \times \text{Sym}_{R_l} \subset \text{Sym}_n, \quad C_t = \text{Sym}_{C_1} \times \dots \times \text{Sym}_{C_k} \subset \text{Sym}_n.$$

 $\lrcorner$ 

## 2 Enumerations

**Thm. (37.5.2.1) [Ramanujan's most Beautiful Identity].**

$$\sum_{n \in \mathbb{N}} \text{Par}(5n+4)X^n = 5 \prod_{k \in \mathbb{Z}_+} \frac{(1 - X^{5k})^5}{(1 - X^k)^6} \in \mathbb{Z}[[X]].$$

 $\lrcorner$ 

*Proof:*

 $\square$ 

**Prop. (37.5.2.2) [Euler].** For  $n \in \mathbb{N}$ , let  $\text{Par}(\mathcal{O}, n)$  be the number of partitions into odd numbers, and  $\text{Par}(\mathcal{D}, n)$  be the number of partitions into distinct parts. Then

$$\text{Par}(\mathcal{O}, n) = \text{Par}(\mathcal{D}, n).$$

 $\lrcorner$ 

*Proof:*

$$\sum_{n \in \mathbb{N}} \text{Par}(\mathcal{D}, n)q^n = \prod_{n \in \mathbb{Z}_+} (1 + q^n) = \prod_{n \in \mathbb{Z}_+} \frac{1 - q^{2n}}{1 - q^n} = \prod_{n \in \mathbb{Z}_+} \frac{1}{1 - q^{2n-1}} = \sum_{n \in \mathbb{N}} \text{Par}(\mathcal{O}, n)q^n.$$

 $\square$ 

**Prop. (37.5.2.3) [Graphical Duality].** For any  $n, m \in \mathbb{Z}_+$ , the number of partitions with at most  $m$  parts equals the number of partitions where no part exceeds  $m$ . This follows easily from the graphical representation of partitions.  $\lrcorner$

**Def. (37.5.2.4) [Dyson's Rank and Crank].** For  $n \in \mathbb{N}$  and a partition  $\lambda$  of  $n$ , denote

- $\sigma(\lambda) = n$ .
- $\#(\lambda)$  the number of parts of  $\lambda$ .
- $l(\lambda)$  the largest part of  $\lambda$ ,
- $\omega(\lambda)$  the number of 1s in  $\lambda$ ,
- $\mu(\lambda)$  the number of parts of  $\lambda$  larger than  $\omega(\lambda)$ .



- **Dyson's rank** of  $\lambda$  is defined as the largest part minus the number of parts.
- **Dyson's crank** of  $\lambda$  is defined as

$$c(\lambda) = \begin{cases} l(\lambda) & , \omega(\lambda) = 0 \\ \mu(\lambda) - \omega(\lambda) & , \omega(\lambda) > 0 \end{cases}.$$

┘

### Restricted Partitions

### 3 Congruences between Partition Functions

**Def. (37.5.3.1)[Vector Partitions and Vector-Cranks].** For  $n \in \mathbb{Z}_+$ , a **vector partition** of  $n$  is a triple  $\bar{\lambda} = (\lambda_1, \lambda_2, \lambda_3)$  of partitions s.t.

$$\sigma(\bar{\lambda}) = \sigma(\lambda_1) + \sigma(\lambda_2) + \sigma(\lambda_3) = n,$$

and  $\lambda_1$  is a partition into distinct parts.

For any vector partition, define

- $\omega(\bar{\lambda}) = (-1)^{\#(\lambda_1)}$ .
- $r(\bar{\lambda}) = \#(\lambda_2) - \#(\lambda_3)$ .

And for any  $m, k \in \mathbb{Z}$ , the counting functions

$$N_V(m; n) = \sum_{\sigma(\bar{\lambda})=n, r(\bar{\lambda})=m} \omega(\bar{\lambda}),$$

$$N_V(k, m; n) = \sum_{t=-\infty}^{\infty} N_V(mt + k; n).$$

┘

**Prop. (37.5.3.2).** For  $n \in \mathbb{Z}_+, m \in \mathbb{Z}$ , changing  $\lambda$  and  $\lambda_3$  implies that

$$N_V(m; n) = N_V(m, n)$$

and

$$N_V(k, m; n) = N_V(-k, m; n).$$

And it is easy to see that it has a generating function

$$\sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{N}} N_V(m; n) z^m q^n = \prod_{n \in \mathbb{Z}_+} \frac{(1 - q^n)}{(1 - zq^n)(1 - z^{-1}q^n)}.$$

In particular, by letting  $z = 1$ , we see that  $\sum_{m \in \mathbb{Z}} N_V(m; n) = \text{Par}(n)$ .

┘

**Thm. (37.5.3.3)[Vector-Crank Theorem, Garvan].** For  $n \in \mathbb{Z}_+$  and any  $i \in \mathbb{N}$ ,

$$N_V(i, 5; 5n + 4) = \frac{\text{Par}(5n + 4)}{5}.$$

$$N_V(i, 7; 7n + 5) = \frac{\text{Par}(7n + 5)}{7}.$$

$$N_V(i, 11; 11n + 6) = \frac{\text{Par}(11n + 5)}{11}.$$

┘

*Proof:* Cf.[Generalizations of Dyson's rank], [New combinatorial interpretations of Ramanujan's partition congruences mod 5,7 and 11].  $\square$

**Thm. (37.5.3.4) [Cranks and Vector-Cranks].** For  $n \in \mathbb{Z}_+, m \in \mathbb{Z}$ , the number of partitions  $\lambda$  of  $n$  with  $c(\lambda) = m$  is equal to  $N_v(m; n)$  (37.5.3.1).  $\lrcorner$

*Proof:* Cf,[A-G88].?  $\square$

**Thm. (37.5.3.5) [Ramanujan Congruences].** For any  $n \in \mathbb{Z}_+$ ,

$$5 \mid \text{Par}(5n + 4),$$

$$7 \mid \text{Par}(7n + 5),$$

$$11 \mid \text{Par}(11n + 6).$$

$\lrcorner$

*Proof:* These follow from (37.5.3.3) and (37.5.3.4). Alternatively 1 also follows from (37.5.2.1).  $\square$

**Prop. (37.5.3.6) [Ahlgren-Boylan].** A **Ramanujan congruence** is a congruence property of the partition function of the form: For any  $n \in \mathbb{Z}_+$ ,

$$p \mid \text{Par}(pn + r)$$

where  $p \in \text{Prime}, r \in \mathbb{Z}$ . Then the only Ramanujan congruences are those three listed in (37.5.3.5).  $\lrcorner$

*Proof:*  $\square$

**Prop. (37.5.3.7) [Atkin-O'Brien].** For any  $n \in \mathbb{Z}_+$ ,

$$13 \mid \text{Par}(11^3 \cdot 13n + 237).$$

$\lrcorner$

**Thm. (37.5.3.8) [Schinzel/Ono].** For any  $p \in \text{Prime} \setminus \{3\}$  and  $k \in \mathbb{Z}_+$ , a positive portion of primes  $\ell \in \text{Prime}$  satisfy

$$p \mid \text{Par}\left(\frac{p^k \ell^3 n + 1}{24}\right).$$

$\lrcorner$

*Proof:* Cf.[Distribution of the partition function modulo m].  $\square$

**Prop. (37.5.3.9) [Ono].** For any  $p \in \text{Prime}_{\geq 5}$ , there exists  $N(p), P(p) \in \mathbb{Z}_+, N(p), P(p) \leq 8(m^3 - 2m - 1)$  s.t. for any  $i \in \mathbb{Z}_{\geq N(p)}$ ,

$$\text{Par}\left(\frac{p^i n + 1}{24}\right) \equiv \text{Par}\left(\frac{p^{P(p)+i} n + 1}{24}\right) \pmod{p}.$$

$\lrcorner$

*Proof:* Cf.[Distribution of the partition function modulo m, Ono].  $\square$

**Conj. (37.5.3.10)**[Newmann]. For any  $m \in \mathbb{Z}_+$  and  $r \in \mathbb{Z}$ , there are infinitely many  $n \in \mathbb{Z}_+$  s.t.

$$\text{Par}(n) \equiv r \pmod{m}.$$

┘

*Proof:*

□

**Thm& Conj. Cor. (37.5.3.11)**[Schinzel/Ono/Ahlgren-Boylan]. For any  $p \in \text{Prime}_{\geq 5}$  and  $r \in \mathbb{Z}_+$ , there exists  $C(p, r) \in \mathbb{R}_+$  s.t.

$$\#\{n \in \mathbb{Z}_+ : n \leq X, \text{Par}(n) \equiv r \pmod{p}\} \geq \begin{cases} C(p, r)\sqrt{X}/\log X & , 1 \leq r \leq p-1 \\ C(p)X & , r=0, p \geq 5 \\ C(p)\sqrt{X} & , p=2 \end{cases}.$$

┘

*Proof:* Cf.[Distribution of the partition function modulo m, Ono], [Arithmetic properties of the partition function], [On the parity of additive representation functions, Nicolas-Ruzsa-Sárközy]. □

**Remark (37.5.3.12).** What about  $p = 3$ ?

┘

## 4 Compositions

**Prop. (37.5.4.1).** For  $n, m \in \mathbb{Z}_+$ , there are  $\binom{n-1}{m-1}$  compositions of  $n$  with exactly  $m$  parts.

┘

*Proof:* Let the number be denoted by  $c(m, n)$ , then

$$\sum_{n \in \mathbb{N}} c(m, n)q^n = (q + q^2 + q^3 + \dots)^m = \frac{q^m}{(1-q)^m} = \sum_{n \in \mathbb{N}} \binom{n-1}{m-1} q^n.$$

□

## 5 Asymptotics

**Prop. (37.5.5.1)**[Erdős]. For  $n, m \in \mathbb{Z}_+$ , let  $\text{Par}_m(n)$  be the number of partitions of  $n$  with exactly  $m$  parts, then

$$\text{Par}_m(n) \sim \frac{1}{m!} \binom{n-1}{m-1}, \quad n \rightarrow \infty,$$

provided that  $m = o(n^{1/3})$ .

┘

*Proof:* Cf.[Theory of Partitions, Andrews]P56.

□

## 37.6 Additive Combinatorics

### 1 Zero-Sum Sets

References are [Zero-sum problems in finite abelian groups: a survey].

**Def. (37.6.1.1) [Davenport Numbers].** For  $G \in \mathcal{Ab}^{\text{fin}}$ , define the **Davenport number**  $D(G)$  to be the smallest positive integer  $s$  s.t. any sequence of elements of  $G$  of length  $s$  has a non-empty subsequence whose elements product to  $1 \in G$ .  $\lrcorner$

**Thm. (37.6.1.2) [Emde-Kruyswijk/Meshulam].** For  $G \in \mathcal{Ab}^{\text{fin}}$  and let  $m$  be the largest order of an element of  $G$ , then

$$D(G) \leq \lceil m(1 + \log \frac{\#G}{m}) \rceil.$$

$\lrcorner$

*Proof:* Let  $n \geq m(1 + \log \frac{\#G}{m})$  and let  $g_1, \dots, g_n$  be a sequence of elements in  $G$ .  $\square$

**Prop. (37.6.1.3).** For  $G \in \mathcal{Ab}^{\text{fin}}$  and  $r > t \geq D(G)$ , any sequence of elements of  $G$  of cardinality  $r$  contains at least  $\binom{r}{t} / \binom{r}{D(G)-1}$  many distinct sequences of length at most  $t$  and at least  $t - D(G) + 1$  s.t. its elements product to  $1 \in G$ .  $\lrcorner$

*Proof:*  $\square$

### Zero-Sum Sets of Prescribed Sizes

**Thm. (37.6.1.4) [Restricted Sum-Sets and Polynomials, Alon-Nathanson-Ruzsa[ANR94]].** For  $p \in \text{Prime}$ ,  $n \in \mathbb{Z}_+$ ,  $h(X_1, \dots, X_n) \in \mathbb{F}_p[X_1, \dots, X_n]$  and non-empty subsets  $A_1, \dots, A_n \subset \mathbb{F}_p$ , denote the **restricted sum-set**

$$\sum(A_1, \dots, A_n; h) = \{a_1 + \dots + a_n \mid a_i \in A_i, h(a_1, \dots, a_n) \neq 0\}.$$

Denote  $c_i = \#A_i - 1$  and  $m = \sum_{i \in [n]} c_i - \deg(h)$ . If the coefficient of  $\prod_{i \in [n]} X_i^{c_i}$  in

$$(X_1 + \dots + X_n)^m h(X_1, \dots, X_n)$$

is nonzero in  $\mathbb{F}_p$ , then  $m < p$  and we have

$$\#\sum(A_1, \dots, A_n; h) \geq m + 1.$$

And the same holds for  $\mathbb{F}_{p^r}$  for any  $r \in \mathbb{Z}_+$ .  $\lrcorner$

**Cor. (37.6.1.5) [Cauchy-Davenport].** If  $A, B \subset \mathbb{F}_p$  are non-empty, then  $\#(A + B) \geq \min\{p, \#A + \#B - 1\}$ .  $\lrcorner$

*Proof:* This follows from (37.6.1.4) by taking  $h = 1$ .  $\square$

**Cor. (37.6.1.6) [Distinct Sum-Sets, Alon-Nathanson-Ruzsa[ANR96]].** For  $p \in \text{Prime}$ ,  $n \in \mathbb{Z}_+$  and non-empty subsets  $A_1, \dots, A_n \subset \mathbb{F}_p$  s.t.  $\#A_i \neq \#A_j$  for any  $i \neq j$ . Then

$$\#\{a_1 + \dots + a_n \mid a_i \in A_i, a_i \neq a_j, \forall i < j \in [n]_+\} \geq \min \left( p, \sum_{i \in [n]} \#A_i - \frac{(n-1)(n+2)}{2} \right).$$

And for arbitrary  $A_1, \dots, A_n \subset \mathbb{F}_p$ , we get a similar lower bound for the distinct sum-set by shrinking  $A_i$ . And this bound is optimal, by choosing  $A_i = \{1, \dots, \#A_i\}$ .  $\lrcorner$

*Proof:* It suffices to prove this for  $m = \sum_{i \in [n]_+} \#A_i - \binom{k+2}{2} < p$ . Denote  $c_i = \#A_i - 1$ , then this follows from (37.6.1.4), by computing that the coefficient of  $\prod_{i \in [n]_+} X_i^{c_i}$  in

$$(X_1 + \dots + X_n)^m \prod_{i > j \in [n]_+} (X_i - X_j)$$

is

$$\frac{m!}{c_1! \cdots c_n!} \prod_{i > j \in [n]_+} (c_i - c_j).$$

Cf. [ANR96] Lemma 3.1 ?.

□

**Cor. (37.6.1.7)** [Distinct Sum-Sets, Conjectured by Erdős-Heilbronn(64), Silva-Hamidoune[S-H94]]. If  $p \in \text{Prime}$  and  $A \subset \mathbb{F}_p$  is a non-empty subset, then for  $s \in [\#A]_+$ , we have

$$\#\{a_1 + \dots + a_s \mid a_i \in A, a_i \neq a_j, \forall i \neq j \in [s]_+\} \geq \min(p, s\#A - s^2 + 1).$$

┘

*Proof:* This follows from (37.6.1.6) by taking  $A_1 = A, A_2 = A \setminus \{a_1\}, A_3 = A \setminus \{a_1, a_2\}$  and so on. □

**Cor. (37.6.1.8).** Any element of  $\mathbb{F}_p$  is a sum of two squares, as there are  $\frac{p+1}{2}$  squares. ┘

**Cor. (37.6.1.9).** If  $p \in \text{Prime}$  and  $A \subset \mathbb{F}_p$  has cardinality  $\lfloor \sqrt{4p-7} \rfloor + 1$ , then every element of  $\mathbb{F}_p$  is a sum of subsets of  $A$  of cardinality  $\lfloor \frac{\#A}{2} \rfloor$ . ┘

**Cor. (37.6.1.10).** If  $p \in \text{Prime}, n \in \mathbb{Z}_+$  and  $A_1, \dots, A_n \subset \mathbb{F}_p$  are non-empty subsets, then

$$\#\{a_1 + \dots + a_n \mid a_i \in A_i, \prod_i a_i \neq 1\} \geq \min(p, \sum_i \#A_i - 2k + 1).$$

┘

*Proof:* Cf. [ANR96] ?.

□

**Cor. (37.6.1.11).** If  $p \in \text{Prime}, n \in \mathbb{Z}_+$  and  $A_1, \dots, A_n \subset \mathbb{F}_p$  are non-empty subsets s.t.  $\#A_i \geq n$  for each  $i \in [n]_+$ , then

$$\#\{a_1 + \dots + a_n \mid a_i \in A_i, a_i a_j \neq 1, \forall i < j \in [n]_+\} \geq \min(p, \sum_i \#A_i - (k+1)^2 + 1).$$

┘

*Proof:* Cf. [ANR96] ?.

□

**Cor. (37.6.1.12).** If  $p \in \text{Prime}$  and  $A, B \subset \mathbb{F}_p$  are non-empty subsets s.t.  $\#A > \#B$ , then for any  $g \in \mathbb{F}_p$ , we have

$$\#\{a + b \mid a \in A, b \in B, ab \neq g, a \neq b\} \geq \min(p, \#A + \#B - 4).$$

┘

*Proof:* Cf. [ANR96] ?.

□

**Cor. (37.6.1.13) [Erdős-Ginzburg-Ziv].** For any  $2n - 1$  elements  $(a_i)$  in  $\mathbb{Z}/(n)$ , there exists  $I \subset \{1, 2, \dots, 2n - 1\}$  with  $\#I = n$  and  $\sum_{i \in I} a_i = 0$ .  $\lrcorner$

*Proof:* We use induction on  $n$ . If  $n \in \mathbf{Prime}$ , then this follows from Cauchy-Davenport(37.6.1.5). If this is true for  $m$ , then for  $p \in \mathbf{Prime}$  and  $n = mp$ , given any  $2n - 1$  numbers, we can find pairwise disjoint subsets  $I_1, \dots, I_{2m-1}$  of  $\{1, \dots, 2pm - 1\}$  where  $\#I_i = p$  and  $\sum_{j \in I_i} a_j \equiv 0 \pmod{p}$  for any  $1 \leq i \leq 2m - 1$ . Define

$$A_i = \left( \sum_{j \in I_i} a_j \right) / p,$$

then we can use the induction hypothesis for  $m$  and  $A_i$  to get the desired assertion.  $\square$

## 2 Arithmetic Progressions

### Arithmetic Progressions in $\mathbb{Z}_+$

**Thm. (37.6.2.1)[Salem-Spencer/Behrend].** Given  $\varepsilon \in \mathbb{R}_+$ , for  $N \in \mathbb{Z}_+$  sufficiently large, there exists a subset  $S \subset \mathbb{Z}/(N)$  that avoids three-term arithmetic sequences, with

$$\#S > \frac{N}{e^{(2\sqrt{\log 4 + \varepsilon})\sqrt{\log N}}}.$$

$\lrcorner$

*Proof:* For this  $N$ , let  $n = \lfloor \sqrt{\frac{2\log N}{\log 2}} \rfloor$ , and  $d \in \mathbb{Z}_+$  s.t.  $(2d - 1)^n \leq N < (2d + 1)^n$ . Then  $d \geq 2$ . Then for any  $k \leq n(d - 1)^2$ , consider the set  $S_k(n, d)$  of all positive integers of the form

$$A = a_1 + a_2(2d - 1) + \dots + a_n(2d - 1)^{n-1},$$

where  $0 \leq a_i \leq d - 1$ , and  $\sum_i a_i^2 = k$ .

It follows from triangular inequality that the image of any such  $S_k(n, d)$  in  $\mathbb{Z}/(N)$  avoids three-term arithmetic sequences, and by Dirichlet's box principle, there exists some  $1 \leq k \leq n(d - 1)^2$  s.t.

$$\#S_k(n, d) \geq (d^n - 1)/n(d - 1)^2 \geq d^{n-2}/n \geq \frac{(N^{\frac{1}{n}} - 1)^{n-2}}{n(2 + \varepsilon)^{n-2}} = \frac{N^{1-2/n}}{n(2 + \varepsilon)^{n-2}}.$$

So for  $N$  large,

$$\#S_k(n, d) \geq N^{1 - \frac{2}{n} - \frac{\log n}{\log N} - \frac{(n-2)\log(2+\varepsilon)}{\log N}} > N^{1 - \frac{2\sqrt{\log 4 + 3\varepsilon}}{\sqrt{\log N}}}.$$

$\square$

**Thm. (37.6.2.2) [Erdős-Turán Conjecture, Szemerédi(1974)/Furstenberg(1977)/Gowers].**

For any  $k \in \mathbb{Z}_+$  and  $\delta \in \mathbb{R}_+$ , if  $N \in \mathbb{Z}_+$  and  $N \geq 2^{2^{(\delta-1)2^{-2k+9}}}$ , then any subset  $A \subset [N]_+$  with  $\#A \geq \delta N$  contains an arithmetic sequence of length  $k$ .  $\lrcorner$

*Proof:* ?  $\square$

**Cor. (37.6.2.3)[van der Waerden/Gowers].** For  $k \in \mathbb{Z}_{\geq 3}$  and  $X \in \mathbb{Z}_+$ , if  $A_k(X)$  be the maximal set of a subset of  $[X]_+$  that avoids  $k$ -term arithmetic sequences, then

$$A(X) = O \left( \frac{X}{\left( \log_{(2)} X \right)^{\frac{1}{2^{k+9}}}} \right).$$

$\lrcorner$

**Thm. (37.6.2.4) [Shelah/Roth/Bloom-Sisask].** There exists  $c \in \mathbb{R}_+$  s.t. for any  $X \in \mathbb{Z}_+$ , if  $A(X)$  be the maximal set of a subset of  $[X]_+$  that avoids three-term arithmetic sequences, then

$$A(X) = O\left(\frac{X}{\log^{1+c} X}\right).$$

┘

*Proof:* ? We only give Roth's proof that  $A(X) = O(\frac{X}{\log_{(2)} X})$ :

Let  $b(X) = A(2^{4^X})/2^{4^X}$ , then  $b(X)$  is decreasing by (37.6.2.5), and (37.6.2.6) applied to  $m = 2^{4^X}$  implies that if  $\delta = \frac{1}{2^{4^{X+1}\eta}}$  where  $0 < \eta < 1/2$ , then

$$b(X)^2 < C[b(X)\delta + b(X)^2\delta^2 + (\delta^{-1}b(X) + 1)(b(X) - b(X+1) + \frac{1}{2^{4^x}})].$$

Then notice for  $X$  large, we can choose  $\delta = \delta(X) = \frac{b(X)}{2C_1} < \frac{1}{2}$  where  $C_1 > \min(C, 1)$  is a constant: it suffices to verify that  $\frac{1}{2^{4^{X+1}\cdot\frac{1}{2}}} < \frac{b(X)}{2C}$ , or equivalently  $C < 2^{3\cdot 4^X - 2} A(2^{4^X})$ , which is clearly true for  $X$  large.

Thus by the fact  $b(X) < 1$ ,

$$\begin{aligned} b(X)^2 &< C_1[b(X)\delta + b(X)^2\delta^2 + (\delta^{-1}b(X) + 1)(b(X) - b(X+1) + \frac{1}{2^{4^x}})] \\ &\leq b(X)^2(\frac{1}{2} + \frac{1}{4C_1}) + C_1(\delta^{-1}b(X) + 1)(b(X) - b(X+1) + \frac{1}{2^{4^x}}) \\ &< \frac{3}{4}b(X)^2 + C_1(2C_1 + 1)(b(X) - b(X+1) + \frac{1}{2^{4^x}}) \end{aligned}$$

Thus there exists a constant  $C_2$  s.t. for  $X$  large,

$$b(X)^2 < C_2(b(X) - b(X+1) + \frac{1}{2^{4^x}}).$$

which implies there exists a constant  $C_3 > C_2$  that for  $X$  large,

$$Xb(2X)^2 \leq \sum_{k=X}^{2X-1} b(X)^2 < C_3(b(X) - b(2X) + \frac{2C_3}{X}).$$

Hence whenever  $2Xb(2X) > 4C_3$ ,

$$2Xb(2X) < \frac{1}{4C_3} 4X^2b(2X)^2 < (Xb(X) - Xb(2X) + 2C_3) < Xb(X).$$

From this that the fact  $b(X)$  is decreasing (37.6.2.5) it is clear that  $Xb(X)$  is bounded, i.e.  $b(X) = O(\frac{1}{X})$ . And clearly this together with (37.6.2.5) implies the desired assertion.  $\square$

**Lemma (37.6.2.5).** For any  $X, Y \in \mathbb{Z}_+$ ,  $A(X)$  equals the maximal number of elements without three-term arithmetic sequences that can be selected from any  $X$ -term arithmetic sequence.  $\frac{A(XY)}{XY} \leq \frac{A(X)}{X}$ , and  $\frac{A(X)}{X} \leq \frac{X+Y}{X} \frac{A(Y)}{Y}$ .  $\square$

*Proof:* The first assertion is easy. Thus  $A(X+Y) \leq A(X) + A(Y)$ , and

$$A(X) \leq A\left(\left[1 + \left\lfloor \frac{X}{Y} \right\rfloor\right]Y\right) \leq \frac{X+Y}{X} A(Y).$$

$\square$

**Lemma (37.6.2.6) [Hardy-Littlewood Method].** For  $X \in \mathbb{Z}_+$ , let  $a(X) = \frac{A(X)}{X}$ , then for any  $m \in 2\mathbb{Z}_+$ , if  $\delta = \frac{1}{m^4\eta}$  where  $0 < \eta < 1/2$ , there exists constant  $C > 0$

$$a(m)^2 < C[a(m)\delta + a(m)^2\delta^2 + (\delta^{-1}a(m) + 1)(a(m) - a(m^4) + m^{-1})].$$

┘

*Proof:* Let  $\{u_1, \dots, u_{A(m^4)}\}$  be a maximal subset of  $\{m^4\}$  without three-term arithmetic sequences, and let  $\{2v_1, \dots, 2v_V\}$  be the set of even integers among  $u_k$ . Then by (37.6.2.5),

$$A(m^4) \leq m^4 a(m), \quad V \leq A\left(\frac{m^4}{2}\right) \leq \frac{m^4}{2} a(m), \quad V \geq A(m^4) - A\left(\frac{m^4}{2}\right) \geq m^4 a(m^4) - \frac{m^4}{2} a(m). \quad (\star)$$

Define

$$f_1(\alpha) = \sum_{k=1}^{A(m^4)} e^{2\pi i \alpha u_k}, \quad f_2(\alpha) = \sum_{k=1}^V e^{2\pi i \alpha v_k},$$

$$F_1(\alpha) = a(m) \sum_{n=1}^{m^4} e^{2\pi i \alpha n}, \quad F_2(\alpha) = a(m) \sum_{n=1}^{m^4/2} e^{2\pi i \alpha n},$$

then by the above,  $|f_i(\alpha)| \leq m^4 a(m)$ ,  $|F_i(\alpha)| \leq m^4 a(m)$ . And

$$f_i(\alpha) - F_i(\alpha) = O\left(m^4[a(m) - a(m^4)] + m^3\right) \quad (\star\star)$$

Using (37.6.2.7), if  $q = 1$ , this is true for  $i = 1$ , and for  $i = 2$ , use the lower bound for  $V$  above. On the other hand, if  $q$  cannot be chosen to be 1, then  $S' = 0$ , and it suffices to show that  $F_r(\alpha) = O(m^3)$ , which is true by (35.3.3.1) and the fact  $\{\{\alpha\}\} \geq 1/\sqrt{2N}$  (because  $q$  cannot be chosen to be 1).

Thus

$$\begin{aligned} f_1(\alpha)f_2(-\alpha)^2 - F_1(\alpha)F_2(-\alpha)^2 &= f_1(\alpha)(f_2(-\alpha) + F_2(-\alpha))(f_2(-\alpha) - F_2(-\alpha)) + F_2(-\alpha)^2(f_1(\alpha) - F_1(\alpha)) \\ &= O\left([m^4 a(m)]^2(m^4[a(m) - a(m^4)] + m^3)\right). \end{aligned}$$

and by (35.3.3.1), if  $0 < \eta < \{\{\alpha\}\} < 1/2$ , then

$$f_1(\alpha) = O\left(a(m)\eta^{-1} + m^4[a(m) - a(m^4)] + m^3\right). \quad (\star\star\star)$$

Then for any  $0 < \eta < 1/2$ ,

- The hypothesis implies that  $u_h = v_k + v_l$  iff  $k = l$  and  $u_h = 2v_k$ , so by  $(\star)$ ,

$$\int_{-\eta}^{1-\eta} f_1(\alpha)f_2^2(-\alpha)d\alpha = V = O(m^4 a(m)).$$

- By  $(\star\star\star)$ ,

$$\int_{-\eta}^{1-\eta} f_1(\alpha)f_2^2(-\alpha)d\alpha < \left[\max_{\eta < \alpha < 1-\eta} f_1(\alpha)\right] \int_0^1 f_2^2(-\alpha)d\alpha = O\left([a(m)\eta^{-1} + m^4[a(m) - a(m^4)] + m^3] m^4 a(m)\right)$$



- By  $(\star\star)$  and (35.3.3.1),

$$\begin{aligned} \int_{-\eta}^{\eta} f_1(\alpha) f_2^2(-\alpha) d\alpha &= \int_{-\eta}^{\eta} F_1(\alpha) F_2^2(-\alpha) d\alpha + O\left(\eta[m^4 a(m)]^2 (m^4[a(m) - a(m^4)] + m^3)\right) \\ &= \int_{-1/2}^{1/2} F_1(\alpha) F_2^2(-\alpha) d\alpha + O(a(m)^3 \eta^{-2}) + O\left(\eta[m^4 a(m)]^2 (m^4[a(m) - a(m^4)] + m^3)\right) \\ &= a(m)^3 m^8 / 4 + O(a(m)^3 \eta^{-2}) + O\left(\eta[m^4 a(m)]^2 (m^4[a(m) - a(m^4)] + m^3)\right) \end{aligned}$$

And these three estimates give the desired assertion.  $\square$

**Lemma (37.6.2.7).** Let  $M \in \mathbb{Z}_+$  and  $A = \{u_1, \dots, u_U\}$  be a subset of  $[M]_+$  without three-term arithmetic sequences. For any  $\alpha \in \mathbb{R}$ , by Dirichlet's box principle, there exists  $h, q \in \mathbb{Z}$  s.t.

$$\alpha = \frac{h}{q} + \beta, \quad (h, q) = 1, 1 \leq q \leq \sqrt{M}, \quad q|\beta| < 1/\sqrt{M}.$$

Thus for any  $m \in \mathbb{Z}_+, m < M$ , we can define

$$S = S(\alpha) = \sum_{k=1}^U e^{2\pi i \alpha u_k}, \quad S' = S'(\alpha, q, h, m) = \frac{A(m)}{m} \frac{1}{q} \left( \sum_{r=1}^q e^{2\pi i \frac{r h}{q}} \right) \left( \sum_{n=1}^M e^{2\pi i \beta n} \right).$$

Then

$$|S - S'| = M \frac{A(m)}{m} - U + O(m\sqrt{M}),$$

and  $S' = 0$  unless  $q = 1$ .  $\perp$

*Proof:* Firstly, notice

$$S = \frac{1}{mq} \sum_{r=1}^q \sum_{n=1}^M \sum_{n \leq u_k \leq n+mq, u_k \equiv r \pmod{q}} e^{2\pi i \alpha u_k} + O(mq),$$

because the coefficient for  $e^{2\pi i \alpha u_k}$  is 1 unless  $u_k \leq mq$ , which is compensated by the error term.

Then notice

$$e^{2\pi i \alpha u_k} = e^{2\pi i \frac{r h}{q}} e^{2\pi i \beta u_k} = e^{2\pi i \frac{r h}{q}} e^{2\pi i \beta n} + O(mq|\beta|),$$

and the number of  $k$  s.t.  $u_k \equiv r \pmod{q}$  and  $n \leq u_k \leq n + mq$  is at most  $A(m)$  for each  $n, r$ , by (37.6.2.5), which we denote by  $A(m) - D(n, m, q, r) \leq A(m)$ , then

$$S = S' - \frac{1}{mq} \sum_{r=1}^q e^{2\pi i \frac{r h}{q}} \sum_{n=1}^M e^{2\pi i \beta n} D(n, m, q, r) + O(Mmq|\beta|) + O(mq).$$

But this estimate is also true for  $\alpha \in \mathbb{Z}$ , so

$$U = M \frac{A(m)}{m} - \frac{1}{mq} \sum_{r=1}^q \sum_{n=1}^M D(n, m, q, r) + O(mq),$$

then combining these two estimates and the facts  $q \leq \sqrt{M}, q|\beta| \leq 1/\sqrt{M}$ , the assertion follows.  $\square$

**Conj. (37.6.2.8)**[Erdős-Turán]. Let  $S \subset \mathbb{Z}_+$  s.t.

$$\sum_{n \in S} \frac{1}{n} = \infty,$$

then  $S$  contains arithmetic progressions of any length.  $\lrcorner$

*Proof:*

□

**Thm& Conj.Cor. (37.6.2.9)**. Let  $S \subset \mathbb{Z}_+$  s.t.

$$\sum_{n \in S} \frac{1}{n} = \infty,$$

then  $S$  contains a 3-term arithmetic progression.  $\lrcorner$

*Proof:* If  $S$  doesn't contain a 3-term arithmetic progression, then for each  $X \in \mathbb{R}_+$ ,

$$F(X) = \#S \cap [X]_+ = O\left(\frac{X}{\log^{1+c} X}\right)$$

by (37.6.2.4). Thus by Abel summation,

$$\sum_{n \in \#S \cap [X]_+} \frac{1}{n} \ll 1 + \frac{F(X)}{X} + \int_2^X \frac{F(t)}{t^2} dt \leq 2 + \int_1^\infty \frac{t}{t \log^{1+c} t} dt = O(1),$$

contradicting the hypothesis.  $\square$

### Arithmetic Progressions in Finite Fields

**Lemma (37.6.2.10)**. For  $n \in \mathbb{Z}_+$ ,  $d \in \mathbb{R}_+$ , let  $S_{n,d}$  be the subspace of all polynomials in  $\mathbb{F}_q[X_1, \dots, X_n]$  with total degree at most  $d$  and each variable has degree at most  $q-1$ . Let  $\alpha, \beta \in \mathbb{F}_q$ .

Suppose  $P \in S_{n,d}$  satisfies  $P(\alpha a + \beta b) = 0$  for every pair  $(a, b) \in A^2$  unless  $a = b$ , then

$$\#\{a \in A \mid P((\alpha + \beta)a) \neq 0\} \leq 2 \dim S_{n,d/2}.$$

$\lrcorner$

*Proof:* Let

$$P(\alpha x + \beta y) = \sum c_{m,m'} m(x) m'(y)$$

where the sum is over pairs of monomials  $(m, m')$  s.t.  $\deg(mm') \leq d$ . Then

$$P(\alpha x + \beta y) = \sum_{\deg m \leq d/2} m(x) F_m(y) + \sum_{\deg m \leq d/2} m(y) G_m(x).$$

Then the  $\#A \times \#A$  matrix  $\left[ P(\alpha a + \beta b) \right]_{a,b \in A}$  has rank at most  $2 \dim S_{n,d/2}$ . And the hypothesis means that this matrix is diagonal, so at most  $2 \dim S_{n,d/2}$  many diagonal entries are non-zero.  $\square$

**Thm. (37.6.2.11).** For  $p \in \text{Prime}$ ,  $q \in p^{\mathbb{Z}^+}$ , and  $\alpha, \beta, \gamma \in \mathbb{F}_q$  that are not all zero s.t.  $\alpha + \beta + \gamma = 0$ . Let  $A$  be a subset of  $\mathbb{F}_q^n$  s.t. the equation

$$\alpha a_1 + \beta a_2 + \gamma a_3 = 0$$

has no solutions  $(a_1, a_2, a_3) \in A^3$  except when  $a_1 = a_2 = a_3$ . Then

$$\#A \leq 3 \dim S_{n, (q-1)n/3} \text{ (37.6.2.10)}.$$

┘

*Proof:* WLOG, assume that  $\gamma \neq 0$ . Let  $d \in [0, (p-1)n]$ , then the space  $V$  of polynomials in  $S_{n,d}$  that vanishes on the complement of  $-\gamma A$  has dimension at least  $\dim S_{n,d} - q^n + \#A$ .

Let  $P \in V$  be a function with maximal support  $\Sigma \subset \mathbb{F}_q^n$ , then  $\#\Sigma \geq \dim V$ , otherwise there is a non-zero polynomial  $Q \in V$  vanishing on  $\Sigma$ , thus  $\Sigma \subsetneq \text{Supp}(P+Q)$ .

Let  $S(A) = \{\alpha a_1 + \beta a_2 | a_1 \neq a_2 \in A\} \subset \mathbb{F}_q$ , then by hypothesis,  $S(A) \cap \{-\gamma A\} = \emptyset$ . Thus  $P$  vanishes on  $S(A)$ . Then by (37.6.2.10),  $P(-\gamma a) \neq 0$  for at most  $2 \dim S_{n,d/2}$  many points of  $a \in A$ . Thus

$$\dim S_{n,d} - q^n + \#A \leq \dim V \leq \#\Sigma \leq 2 \dim S_{n,d/2}.$$

So

$$\#A \leq 2 \dim S_{n,d/2} + (q^n - \dim S_{n,d}) = 2 \dim S_{n,d/2} + \dim S_{n, (q-1)n-d}.$$

So if we take  $d = \frac{2}{3}(q-1)n$ , then we get the desired assertion. □

**Cor. (37.6.2.12) [Ellenberg-Gijswijt].** For  $p \in \mathbf{P}$ ,  $q \in p^{\mathbb{Z}^+}$ , and let  $A$  be a subset of  $(\mathbb{F}_q)^n$  containing no three-term arithmetic progressions, then there exists some calculable constant  $C_q < q$  s.t.  $\#A = O(C_q^n)$ .

In particular,  $C_3 < 2.756$ . ┘

*Proof:* Take  $\alpha = \beta = 1, \gamma = -2$ , and use the fact (Cf. [E-G17] P342. ?)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \frac{1}{q^n} \dim S_{n, (q-1)n/3} \right) = -I\left(\frac{2}{3}(q-1)\right) = -\sup_{\theta \in \mathbb{R}} \left\{ \frac{2}{3}(q-1)\theta - \log \left( \frac{1}{q} (1 + e^\theta + \dots + e^{(q-1)\theta}) \right) \right\} < 0.$$

□

### Generalized Arithmetic Progressions and Freiman's Theorem

References are [Ruz99].

**Def. (37.6.2.13) [Generalized Arithmetic Progressions].** A generalized arithmetic progression of dimension  $d$  in a commutative group  $G$  is a subset of the form

$$P(q_1, \dots, q_d, l_1, \dots, l_d; a) = \{a + x_1 q_1 + \dots + x_d q_d, 0 \leq x_i \leq l_i\}.$$

And the size of this arithmetic progression is defined to be  $\prod_{i=1}^d (l_i + 1)$ .

A generated arithmetic progression is called **proper** if its size equals its cardinality. ┘

**Def. (37.6.2.14)[Freiman Homomorphisms].** Let  $G_1, G_2$  be commutative groups and  $A_1 \subset G_1, A_2 \subset G_2$ , then a map between sets  $\varphi : A_1 \rightarrow A_2$  is called a **Freiman homomorphism** of order  $r$  if for any  $x_1, \dots, x_r, y_1, \dots, y_r \in A_1$ , the equation

$$x_1 + \dots + x_r = y_1 + \dots + y_r$$

implies that

$$\varphi(x_1) + \dots + \varphi(x_r) = \varphi(y_1) + \dots + \varphi(y_r).$$

┘

**Prop. (37.6.2.15).** Let  $G_1, G_2$  be commutative groups and  $A_1 \subset G_1, A_2 \subset G_2$  be Freiman isomorphic of order  $r$ . Suppose  $r = r'(k + l)$  where  $r', k, l \in \mathbb{N}$ , then the sets  $[k]A - [l]A$  and  $[k]A' - [l]A'$  are also Freiman isomorphic of order  $r'$ . ┘

*Proof:* This is clear. □

**Prop. (37.6.2.16).** Let  $G \in \mathcal{Ab}$  and  $A, B$  are subsets of  $G$  s.t.  $\#B = n, \#(A + B) \leq \alpha n$ . Then for any  $k, l \in \mathbb{Z}_+$ ,

$$\#([k]A - [l]A) \leq \alpha^{k+l} n.$$

┘

*Proof:* Cf.[Arithmetical progressions and the number of sums, Rusza]Lemma3.3. ? □

**Lemma (37.6.2.17) [Freiman Homomorphism Preserves Generalized Arithmetic Progressions].** Let  $G, G'$  be commutative groups, then any Freiman map of order 2 maps a generalized arithmetic progression in  $G$  to a generalized arithmetic progression of the same dimension in  $G'$ . ┘

*Proof:* Let  $\varphi : G \rightarrow G'$  be a Freiman homomorphism and  $P = P(q_1, \dots, q_d, l_1, \dots, l_d; a) \subset G$  is a generalized arithmetic progression. Then we claim that  $\varphi(P) = P' = P(q'_1, \dots, q'_d, l_1, \dots, l_d, a')$ , where

$$a' = \varphi(a), \quad q'_i = \varphi(a + q_i) - \varphi(a).$$

Then we need to show that  $\varphi(a + x_1 q_1 + \dots + x_d q_d) = a' + x_1 q'_1 + \dots + x_d q'_d$ . The proof is by induction on  $r = x_1 + x_2 + \dots + x_d$ . The equation holds for  $r = 1$ . And for  $r \geq 2$ , denote  $x = a + x_1 q_1 + \dots + x_d q_d$ , then there exists  $1 \leq i, j \leq d$  s.t.  $x - q_i - q_j \in P$ . Because

$$x + (x - q_i - q_j) = (x - q_i) + (x - q_j),$$

the Freiman property implies that

$$\varphi(x) = \varphi(x - q_i) + \varphi(x - q_j) - \varphi(x - q_i - q_j) = a' + x_1 q'_1 + \dots + x_d q'_d.$$

□

**Lemma (37.6.2.18).** Let  $A \subset \mathbb{Z}$  be a subset of cardinality  $n \in \mathbb{Z}_+$ , and  $r \in \mathbb{Z}_{\geq 2}$ , and  $N = \#([r]A - [r]A)$ . Then for any  $m > 2r(N - 1)$ , there exists a set  $A' \subset A$  s.t.  $\#A' \geq n/r$  and  $A'$  is Freiman isomorphic of order 2 to a subset of  $\mathbb{Z}/(m)$ . ┘

*Proof:* Cf.[Arithmetical progressions and the number of sums, Rusza] ? □

**Lemma (37.6.2.19).** Let  $p \in \text{Prime}$  and  $A \subset \mathbb{Z}/(p)$  s.t.  $\#A = \beta p$ , then there exists a proper generalized arithmetic progression (37.6.2.13)  $P = P(q_1, \dots, q_k, l_1, \dots, l_k, 0) \subset [2]A - [2]A$  s.t.  $\#P = \|P\| > (4k)^{-k}p$ , and  $k \leq \beta^{-2}$ .  $\lrcorner$

*Proof:* Cf. [Ruz99]P383.  $\color{red}?$   $\square$

**Thm. (37.6.2.20) [Freiman's Theorem for Torsion-Free Abelian Groups, Freiman-Ruzsa-Chang].** For any  $\alpha \in \mathbb{R}_+$ , there exists  $d, C \in \mathbb{R}_+$  s.t. if  $A, B$  be finite sets in a torsion-free commutative group  $G$  s.t.  $\#A = \#B = n$ ,  $\#(A + B) \leq \alpha n$ , then  $A$  is contained in a generalized arithmetic progression of dimension at most  $d$  and size at most  $Cn$ .  $\lrcorner$

*Proof:* The subgroup of  $G$  generated by  $A$  is isomorphic to  $\mathbb{Z}^v$  for some  $v \in \mathbb{Z}_+$ . Choose an isomorphism, which gives elements of  $A$  coordinates. We can find  $A_1 \subset \mathbb{Z}$  that is Freiman isomorphic to  $A$  of order 8: In fact we can take  $t$ -adic base

$$(a_1, \dots, a_v) \mapsto a_1 + ta_2 + \dots + t^{v-1}a_v,$$

where  $t > 16H$  and  $H$  is the maximal absolute value of coordinates of elements of  $A$ .

Then by (37.6.2.16),  $\#([2]A_1 - [2]A_1) = \#([2]A - [2]A) \leq \alpha^4 n$ . By Chebyshev's theorem (35.6.2.1), we can find a prime  $p \in \text{Prime}$  s.t.  $16\#([2]A - [2]A) < p \leq 32\alpha^4 n$ . Then (37.6.2.18) implies that there is a subset  $A_2 \subset A_1$  that is Freiman isomorphic of order 8 to a subset  $T \subset \mathbb{Z}/(p)$ , and  $\#A_2 \geq n/8$ .

Then by (37.6.2.19), we can find a  $k$ -dimensional proper arithmetic progression  $P \subset [2]T - [2]T$  of cardinality  $\geq \delta n$  (37.6.2.19).

Because  $T$  is Freiman isomorphic of order 8 to a subset of  $A$ , by (37.6.2.15),  $[2]T - [2]T$  is Freiman isomorphic of order 2 to a subset of  $[2]A - [2]A$ . Then by (37.6.2.17), the image  $P'$  of  $P \subset [2]T - [2]T$  is also a  $k$ -dimensional proper arithmetic progression.

Select a maximal subset  $\{a_1, \dots, a_s\}$  of  $A$  s.t. the sets  $P' + a_i$  are pairwise disjoint. Then

$$s \leq \frac{[3]A - [2]A}{\|P'\|} \leq \alpha^5 n / \delta n = \alpha^5 / \delta \text{ (37.6.2.16)}.$$

By the maximality of  $\{a_1, \dots, a_s\}$ , we have  $A \subset \{a_1, \dots, a_s\} + P' - P'$ .  $P' - P'$  is also a  $k$ -dimensional arithmetic progression, and

$$\|P' - P'\| \leq 2^k \|P'\| \leq 2^k \#([2]A - [2]A) \leq 2^k \alpha^4 n.$$

Then  $A$  is contained in a  $k + s$ -dimensional arithmetic progression of size  $\leq 2^{k+s} \alpha^4 n$ . Because  $s$  and  $k$  are bounded by  $\alpha$  (See (37.6.2.19)), the assertion follows.  $\square$

**Thm. (37.6.2.21) [Polynomial Freiman-Ruzsa Conjecture, Gowers-Green-Manner--Tao(2023)].** Let  $A \subset \mathbb{F}_2^n$  s.t.  $\#(A + A) \leq K\#A$ , then  $A$  can be covered by  $2K^{12}$  translates of a subspace  $H$  of  $\mathbb{F}_2^n$  of cardinality at most  $\#A$ .  $\lrcorner$

*Proof:*  $\color{red}?$   $\square$

**Cor. (37.6.2.22) [Freiman's Theorem for Torsion Abelian Groups, Ruzsa].** Let  $r \in \mathbb{Z}_{\geq 2}$  and  $G \in \text{Ab}$  s.t. each element of  $G$  has exponent at most  $r$ . Suppose  $A, B$  are subsets of  $G$  and  $\#A = \#B = n$ ,  $\#(A + B) \leq \alpha n$ , then  $A$  is contained in a subgroup  $H$  of  $G$  s.t.

$$\#H \leq \alpha^2 r^{\alpha^4} n.$$

Moreover, Ruzsa conjectured that  $\alpha^2 r^{\alpha^4}$  can be replaced by  $r^{C\alpha}$  for some  $C \in \mathbb{R}_+$ .  $\lrcorner$

*Proof:* Let  $B = \{b_1, \dots, b_k\}$  be a maximal collection of elements in  $[2]A - A$  s.t. the sets  $b_i - A$  are all disjoint. Then for any  $x \in [2]A - A$ ,  $x - a_1 = b_i - a_2$  for some  $a_1, a_2 \in A$ , which means  $x \in b_i + (A - A)$ . Hence  $[2]A - A \subset B + A - A$ . Then

$$kn = \# \cup (b_i - A) \leq \#([2]A - [2]A) \subset \alpha^4 n \text{ (37.6.2.16),}$$

which implies that  $k \leq \alpha^4$ .

By induction we can prove that  $[j+1]A - A \subset [j]B + A - A$  for any  $j \in \mathbb{Z}_+$ .

Let  $H, I$  be the subgroups generated by  $A, B$  resp., then

$$[j+1]A - A \subset I + A - A$$

for any  $j \in \mathbb{Z}_+$ , and also we have

$$H = \cup([j]A - A)$$

because the order of elements in  $G$  are bounded. So

$$H \subset I + A - A.$$

Now  $I$  is generated by  $k$  elements of order bounded by  $r$ , so  $\#I \leq r^k \leq r^{\alpha^4}$ . So it follows that  $\#H \leq \alpha^2 r^{\alpha^4} n$ .  $\square$

### Squares in Arithmetic Progressions

**Conj. (37.6.2.23) [Rudin].** There exists  $c_1 \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_+$ , there exists at most  $c_1 \sqrt{N}$  many squares in any arithmetic progression of integers of length  $N$ .  $\lrcorner$

*Proof:*  $\square$

**Thm& Conj. Cor. (37.6.2.24) [Bombieri-Zannier].** There exists  $c_1, c_2 \in \mathbb{R}_+$  s.t. for any  $N \in \mathbb{Z}_+$ , there exists at most  $c_1 N^{3/5} (\log N)^{c_2}$  many squares in  $\lrcorner$

*Proof:* Cf. [squares in arithmetic progressions, 2, Bombieri-Zannier]. ?  $\square$

### 3 Littlewood-Offord Problem

**Prop. (37.6.3.1) [Littlewood-Offord Problem, Erdős].**

- If  $\{x_i\}_{1 \leq i \leq n}$  is a set of real numbers with  $|x_i| \geq 1$ , then for any  $r \in \mathbb{Z}_+$ , the number of sums  $\sum \pm x_k$  which lies in the interior of any interval of length  $2r$  doesn't exceed  $r \binom{n}{\lfloor n/2 \rfloor}$ .
- If  $\{x_i\}_{1 \leq i \leq n}$  is a set of complex numbers with  $|x_i| \geq 1$ , then for any  $r \in \mathbb{Z}_+$ , the number of sums  $\sum \pm x_k$  which lies in the interior of an arbitrary circle of radius  $r$  is  $O(r \binom{n}{\lfloor n/2 \rfloor}) \leq O(r 2^n / \sqrt{n})$ .  $\lrcorner$

*Proof:* Cf. [Erdős, On a lemma of Littlewood and Offord. Bull. Amer. Math. Soc. 51 (1945), 898–902.]  $\square$

#### 4 Sidon's Problem

**Prop. (37.6.4.1)** [Erdős-Turán[E-T41]]. For  $n \in \mathbb{Z}_+$ , let  $\Phi(n)$  be the maximal number of subset  $\{a_1, \dots, a_X\}$  of  $[n]_+$  s.t.  $\{a_i + a_j\}_{i \leq j}$  are pairwise different. Then

$$\frac{1}{\sqrt{2}} \leq \liminf_{n \rightarrow \infty} \frac{\Phi(n)}{\sqrt{n}} \leq \limsup_{n \rightarrow \infty} \frac{\Phi(n)}{\sqrt{n}} \leq 1.$$

┘

*Proof:* Let  $p \in \mathbf{P} \setminus \{2\}$  and  $a_k = 2pk + u(k)$ ,  $k = 1, 2, \dots, p-1$ , where  $1 \leq u(k) \leq p-1$ ,  $u(k) \equiv k^2 \pmod{p}$ . Then  $a_k < 2p^2$  for each  $k$ , and

$$a_i + a_j = a_k + a_l \iff (i, j) = (k, l),$$

because in this case

$$i + j = k + l, \quad i^2 + j^2 \equiv k^2 + l^2 \equiv q.$$

Thus  $\Phi(2p^2) \geq p-1$ , and because there are infinitely primes,  $\liminf_{n \rightarrow \infty} \frac{\Phi(n)}{\sqrt{n}} \geq \frac{1}{\sqrt{2}}$ .

For the other direction, for any such set  $S = \{a_1, \dots, a_X\}$  of  $[n]_+$  s.t.  $\{a_i + a_j\}_{i \leq j}$  are pairwise different. Choose  $1 \leq m < n$ , then for each  $u \in \mathbb{Z}$ , let  $A_u = \#[u-m, u] \cap S$ , then

$$\sum_{u=1}^{m+n} A_u = mx,$$

and the number of triples  $(i, j, n)$  s.t.  $a_i, a_j \in A_u, a_j > a_i$  is

$$\sum_{u=1}^{m+n} \frac{1}{2} A_u (A_u - 1) \geq \frac{1}{2} (m+n) \frac{mx}{m+n} \left( \frac{mx}{m+n} - 1 \right).$$

But by the hypothesis, there are at most

$$\sum_{r=1}^{m-1} (m-r) = \frac{1}{2} m(m-1)$$

such triples. Thus

$$\frac{1}{2} mx(mx - m - n) \leq \frac{1}{2} m(m-1)(m+n),$$

and whence

$$x < \frac{n}{m} + \sqrt{n + m + \frac{n^2}{m^2}}.$$

Taking  $m = \lfloor \sqrt{n} \rfloor$ , we get  $x = \sqrt{n} + O(n^{\frac{1}{4}})$ , thus  $\limsup_{n \rightarrow \infty} \frac{\Phi(n)}{\sqrt{n}} \leq 1$ . □

**Prop. (37.6.4.2)** [Infinite Sidon Sets]. For any infinite Sidon sequence (37.6.4.1)  $S$ , we have

$$\limsup_{n \rightarrow \infty} \frac{\#[n]_+ \cap S}{\sqrt{n}} = 0.$$

┘

*Proof:* ? □

**Prop. (37.6.4.3)** [Infinite Sidon Sets, Ruzsa[Ruz98]]. There exists an infinite Sidon sequence s.t.

$$\#[n]_+ \cap S > n^{\sqrt{2}-1+o(1)}$$

for  $n$  large. ┘

*Proof:* □

## 5 Sums and Products

**Conj. (37.6.5.1)** [Erdős-Szemerédi]. For  $n \in \mathbb{Z}_+$ , denote

$$g(n) = \min(\max(\#(A + A), \#(A \cdot A)) : A \subset \mathbb{C}, \#A = n).$$

Then it is conjectured that for any  $\varepsilon \in \mathbb{R}_+$ , when  $n$  is large,  $g(n) \geq n^{2-\varepsilon}$ . ┘

**Thm& Conj. Cor. (37.6.5.2)** [Erdős-Szemerédi/Elekes/Solymosi]. Let  $g(n)$  be as in (37.6.5.1), then there exists  $C \in \mathbb{R}_+$  s.t. if  $n \geq 2$ ,

$$g(n) \geq C \frac{n^{14/11}}{(\log n)^{3/11}}.$$

┘

*Proof:* We only prove that  $g(n) \geq 0.23n^{5/4}$ , for the stronger result, Cf.[On the number of sums and products, Solymosi] ?.

Let  $A = \{a_1, \dots, a_n\}$ . and define  $n^2$  linear functions on  $\mathbb{R}$ :

$$f_{j,k}(x) = a_j(x - a_k).$$

Then each  $f_{j,k}$  maps at least  $n$  points in  $A + A$  into  $A \cdot A$ : In fact,  $f_{j,k}(a_k + a_i) = a_i a_j$ . Thus  $\Gamma(f_{j,k}) \subset \mathbb{R}^2$  contain at least  $n$  points from  $(A + A) \times (A \cdot A)$ . As  $n \leq 10\sqrt{\#(A + A) \cdot \#(A \cdot A)}$ , it follows from (37.9.3.3) that

$$n^2 \leq 100(\#(A + A) \cdot \#(A \cdot A))^2 / (n - 1)^3$$

from which the assertion follows. □

**Prop. (37.6.5.3)** [Erdős -Szemerédi]. Let  $g(n)$  be as in (37.6.5.1), then there exists  $c \in \mathbb{R}_+$  s.t.

$$g(n) < \frac{n^2}{e^{c \log n / \log_{(2)} n}}.$$

┘

*Proof:* Cf.[Sums and Products of Integers]. □



### Finite Fields

**Thm. (37.6.5.4) [Bourgain-Katz-Tao/Bourgain-Konyagin].** Let  $p \in \text{Prime}$ , then for  $\delta \in \mathbb{R}_+$  and any  $A \subset \mathbb{F}_p$  s.t.  $\#A < q^{1-\varepsilon}$ , there exists  $\delta = \delta(\varepsilon) \in \mathbb{R}_+$  and  $C \in \mathbb{R}_+$  s.t.

$$\max(\#(A + A), \#(A \cdot A)) \geq C(\delta)(\#A)^{1+\delta}.$$

┘

*Proof:* Cf.[BGK06]

□

**Lemma (37.6.5.5).** For  $p \in \mathbf{P}$ , and  $A \subset \mathbb{F}_p$ , denote

$$I(A) = \{a_1(a - 2 - a_3) + a_4(a_5 - a_6) : a_1, \dots, a_6 \in A\}.$$

Then

$$\#I(A) \geq \begin{cases} p/2 & , \#A \geq \sqrt{p} \\ 0.1(\#A)^{3/2} & , \#A \leq \sqrt{p} \end{cases}.$$

┘

*Proof:* Cf.[BGK06]P381.

□

**Thm. (37.6.5.6).** For any  $C_1, C_2 \in \mathbb{R}_+$ , there exists  $C \in \mathbb{R}_+$  s.t. for any  $p \in \mathbf{P}_{\geq 3}, q \in p^{\mathbb{Z}_+}$ , if  $A \subset \mathbb{F}_q$  with  $C_1 q^{1/2} \leq \#A \leq C_2 q^{7/10}$ , then

$$\max(\#(A + A), \#(A \cdot A)) \geq C \frac{(\#A)^{3/2}}{q^{1/4}}.$$

┘

*Proof:* Cf.[Sum-product Estimates in Finite Fields via Kloosterman Sums].

□

### Finite Kakeya Sets

**Def. (37.6.5.7) [Kakeya Sets].** Let  $p \in \text{Prime}, q \in p^{\mathbb{Z}_+}, n \in \mathbb{Z}_+$ , then  $K \subset \mathbb{F}_q^n$  is called a **Kakeya set(finite)** if for any  $x \in \mathbb{F}_q^n$ , there exists  $y \in \mathbb{F}_q^n$  s.t.  $L_{y,x} = \{y + \mathbb{F}_q x\} \subset K$ .

┘

**Prop. (37.6.5.8).** Let  $p \in \text{Prime}, q \in p^{\mathbb{Z}_+}, n \in \mathbb{Z}_+$  and  $K \subset \mathbb{F}_q^n$  is a Kakeya set, then for any  $\varepsilon \in \mathbb{R}_+$ , there exists  $c_{n,\varepsilon} \in \mathbb{R}_+$  s.t.  $\#K \geq c_{n,\varepsilon} q^{n-\varepsilon}$ .

┘

*Proof:* Notice that if  $K \subset \mathbb{F}_q^n$  is a Kakeya set, then  $K^m \subset (\mathbb{F}_q)^{mn}$  is also a Kakeya set. Thus it suffices to show for  $\varepsilon = 1$  (Because the  $n = 1$  case is trivial). □

**Prop. (37.6.5.9) [Finite Kakeya Conjecture, Dvir-Alon-Tao[Dvi09]].** Let  $p \in \text{Prime}, q \in p^{\mathbb{Z}_+}, n \in \mathbb{Z}_+$  and  $K \subset \mathbb{F}_q^n$  is a Kakeya set(37.6.5.7), then

$$\#K \geq \binom{n+q-1}{q-1}.$$

┘

*Proof:* If  $K$  is a Kakeya set and  $\#K < \binom{n+q-1}{q-1}$ , then there exists a non-zero polynomial  $f \in \mathbb{F}_q[X_1, \dots, X_n]$  of degree  $d \leq q-1$  that vanishes on the whole set  $K$ . Let  $g$  be the degree  $d$  part of  $f \neq 0$ . For any  $y \in \mathbb{F}_q^n$ , there exists  $z \in \mathbb{F}_q^n$  s.t.  $z + \mathbb{F}_q y \in K$ . Thus

$$P_{y,z}(t) = f(z + ty) = 0$$

for any  $t \in \mathbb{F}_q$ . This implies that  $f(z + ty) = 0$  as a polynomial of  $t$ . But the leading coefficient is just  $g(y)$ , so  $g(y) = 0$  for any  $y \in \mathbb{F}_q^n$ , contradiction. □

## 6 Others

**Thm. (37.6.6.1).** Given  $n \in \mathbb{Z}_+$  and a set  $M \subset (\mathbb{Z}_+)^n$ , there exists a finite subset  $S \subset M$  s.t. for any  $m \in M$ , there exists  $s \in S$  s.t.  $s_i \leq m_i$  for each  $1 \leq i \leq n$ .  $\lrcorner$

*Proof:* Use induction on  $n$ . The case  $n = 1$  is trivial, and for the induction process, choose randomly an  $m \in M$ . Then an element  $n$  is not “divisible” by  $m$  iff some  $n_i < m_i$ . Notice there are only f.m. such  $n_i$ . For each of this case, consider the subset  $M'_{i,n_i} \subset (\mathbb{Z}_+)^{n-1}$  with the  $i$ -th coordinate discarded, then by induction there exists a finite subset  $S'_{i,n_i}$  s.t. each element in  $M'_{i,n_i}$  is divisible by some element in  $S'_{i,n_i}$ . Then  $S$  being  $\{m\}$  with the collection of  $S'_{i,n_i}$  just satisfies the requirement.  $\square$

**Prop. (37.6.6.2)[Erdős].** For any  $k, l \in \mathbb{Z}_+$  and  $a_1 < a_2 < \dots < a_{(k-1)(l-1)+1} \in \mathbb{Z}_+$ , there either exists  $k$  of them no one dividing the other, or there exists  $l$  of them each of a multiple of the previous one.  $\lrcorner$

*Proof:* This follows immediately from (37.2.5.4).  $\square$

## 37.7 Word Theory

### 1 Sturmian Words

**Def. (37.7.1.1).** Given any word  $w$  on an alphabet  $\mathcal{A}$ , for any  $k \in \mathbb{Z}_+$ , let  $W_k(w)$  be the set of  $k$ -words appearing in  $A$ . ┘

**Prop. (37.7.1.2) [Morse-Hedlund].** For any infinite word  $w$  over an alphabet  $\mathcal{A}$ ,

- If  $w$  is eventually periodic, then there exists  $C \in \mathbb{R}_+$  s.t.  $\#W_k(w) \leq C$  for any  $k \in \mathbb{Z}_+$ .
- If  $w$  is not eventually periodic, then  $\#W_{k+1}(w) \geq \#W_k(w) + 1$  for any  $k \in \mathbb{Z}_+$ . In particular,  $\#W_k(w) \geq k + 1$ . ┘

*Proof:* □

**Cor. (37.7.1.3).** Given any infinite word  $w$  over a finite alphabet  $\mathcal{A}$  that is not eventually periodic, then for any  $n \in \mathbb{Z}_+$ , there exists finite words  $V, W$  of length  $n$ , and  $a, a', b, b' \in \mathcal{A}$  s.t. each of  $a \amalg W, a' \amalg W, V \amalg b, V \amalg b'$  appears in  $w$  infinitely often. ┘

*Proof:* This follows from the fact  $\#W_{k+1}(w) \geq \#W_k(w) + 1$  and the fact  $\#\mathcal{A} < \infty$ . □

**Def. (37.7.1.4) [Sturmian Words].** For  $\theta, \rho \in \mathbb{R}$  s.t.  $\theta \in (0, 1) \setminus \mathbb{Q}$ , define

$$s_n = \lfloor (n+1)\theta + \rho \rfloor - \lfloor n\theta + \rho \rfloor, \quad s'_n = \lceil (n+1)\theta + \rho \rceil - \lceil n\theta + \rho \rceil, \quad n \in \mathbb{Z}_+.$$

And define the infinite words

$$s_{\theta, \rho} = s_1 s_2 s_3 \dots, \quad s'_{\theta, \rho} = s'_1 s'_2 s'_3 \dots$$

These are called **Sturmian words** with slope  $\theta$  and intercept  $\rho$ . ┘

**Prop. (37.7.1.5).** An infinite word  $w$  on alphabet  $\{0, 1\}$  is a Sturmian word iff  $\#W_k(w) = k + 1$  for any  $k \in \mathbb{Z}_+$ . ┘

*Proof:* Cf. [Distributions modulo one] P225. ? □

### 2 Thue-Morse Words

Cf. [Distributions modulo one].

## 37.8 Algebraic Combinatorics

References are [Combinatorial Nullstellensatz, Alon].

### 1 Combinatorial Nullstellensatz

**Lemma (37.8.1.1) [Combinatorial Nullstellensatz, Alon[Alo99]].** Let  $K \in \text{Field}$ ,  $f \in K[X_1, \dots, X_n]$ , and  $S_1, \dots, S_n \subset K$  be nonempty finite subsets, with  $g_i(X_i) = \prod_{s \in S_i} (X_i - s)$ . If  $f$  vanishes at the common zeros of  $g_i$ , then there are polynomials  $h_i \in F[X_1, \dots, X_n]$  s.t.  $\deg(h_i) \leq \deg(f) - \#S_i$  and

$$f = \sum_{i \in [n]_+} h_i g_i.$$

┘

*Proof:* The proof is very simple, just replace terms of  $f$  by lower degree terms, using equation of  $g_i$ , then we get a polynomial that has degree in  $x_i$  smaller than  $\#S_i$  and vanish on  $S_1 \times \dots \times S_n$ , so it must be 0, as easily checked.  $\square$

**Thm. (37.8.1.2) [Combinatorial Nullstellensatz, Alon[Alo99]].** If  $K \in \text{Field}$  and  $f \in K[X_1, \dots, X_n]$  is a polynomial of degree  $n$ , if  $\prod X_i^{t_i}$  is a highest-degree term of  $f$  (i.e.  $\deg(f) = \sum_i t_i$ ) and  $S_i$  are subsets of  $K$  s.t.  $\#S_i > t_i$ , then there are some  $s_i \in S_i$  that  $f(s_1, \dots, s_n) \neq 0$ .  $\square$

*Proof:* We may assume  $\#S_i = t_i + 1$ . By combinatorial Nullstellensatz (37.8.1.1), if no such  $s_i$  exist, then there are  $h_i$  with  $\deg(h_i) \leq \deg(f) - \#S_i$  s.t.  $f = \sum_{i \in [n]_+} h_i \prod_{s \in S_i} (x_i - s)$ .

The term  $\prod X_i^{t_i}$  needs to appear on the RHS, so by degree restriction it must be some term of  $h_i$  times  $x_i^{t_i+1}$ , which is impossible.  $\square$

### 2 Stanley Depth

References are [Stanley decompositions and partitionable simplicial complexes] and [Linear Diophantine equations and local cohomology, Stanley].

### 3 Combinatorial Hodge Theory

**Def. (37.8.3.1) [Monomial Ideals and Stanley Spaces].** Let  $K \in \text{Field}$ ,  $n \in \mathbb{Z}_+$ ,  $\underline{X} = \{X_1, \dots, X_n\}$  and  $S = K[\underline{X}]$ , a **monomial ideal** of  $S$  is an ideal generated by monomials.

For  $k \in [n]$ , a **Stanley space** of dimension  $k$  is a  $K$ -subspace of  $S$  of the form  $\underline{X}^\alpha \cdot K[\underline{Z}]$ , where  $\underline{Z} \subset \underline{X}$  and  $\#\underline{Z} = k$ .

A **Stanley decomposition** of a monomial ideal  $I \in \text{Ideal}(S)$  is a finite  $K$ -space decomposition of  $I$  into a direct sum of Stanley spaces.  $\square$

**Def. (37.8.3.2) [Stanley Depth].** For any Stanley decomposition of a monomial ideal  $I$ , the minimal dimension of Stanley spaces contained in this Stanley decomposition is called the **Stanley depth**. And the Stanley depth of  $I$  is defined to be the maximal Stanley depth of any Stanley decomposition of  $I$ , denoted by  $\text{Sdepth}(I)$ .

Also if  $I^c$  is  $K$ -subspace generated by all monomials not in  $I$ , then  $S = I \oplus I^c$ , and we can also define the Stanley depth of  $I^c$ , denoted by  $\text{Sdepth}(S/I)$ .

More generally, we can define Stanley depth for any  $\mathbb{Z}^n$ -graded  $S$ -module  $M$ .  $\square$

**Conj. (37.8.3.3) [Stanley Conjecture].** For any f.g.  $\mathbb{Z}^n$ -graded  $S$ -module  $M$ , we have  $\text{depth}(M) \leq \text{Sdepth}(M)$ .  $\square$

## 37.9 Discrete Geometry

### 1 Convex Polygons

**Prop. (37.9.1.1).** For  $n \in \mathbb{Z}_{\geq 4}$  and any  $n$  points in  $\mathbb{R}^2$  in general position, they form a convex  $n$ -gon iff any 4 points of them form a convex quadrilateral.  $\lrcorner$

*Proof:* We use induction on  $n$ .  $n = 4$  is trivial. For  $n \geq 5$ , let the points be  $\{P_1, \dots, P_n\}$ , then by induction hypothesis, any  $n - 1$  points of them form a convex  $(n - 1)$ -gon. So for any  $1 \leq k \leq n$ ,  $\{X_1, \dots, \widehat{X_k}, \dots, X_n\}$  is a convex  $(n - 1)$ -gon, and  $X_k$  is not contained in any sub- $(n - 2)$ -gon of  $X_1 \dots \widehat{X_k} \dots X_{n-1}$ . But any convex  $(n - 1)$  is a sum of its sub- $(n - 2)$ -gons, because we can use equations of the form  $a_1 P_1 + a_2 P_2 = a_3 P_3 + a_4 P_4$ ,  $a_i > 0$  to eliminate some  $P_i$ . Because  $k$  is arbitrary, this means  $P_1 \dots P_n$  is convex.  $\square$

**Thm. (37.9.1.2) [Erdős-Szekeres].** For any  $n \in \mathbb{Z}_{\geq 2}$ , there exists a minimal positive integer  $N(n)$  s.t. for any  $ES(n)$  points in  $\mathbb{R}^2$  in general position, there exists a subset of  $n$  points forming a convex  $n$ -gon. Moreover,

$$2^{n-2} + 1 \leq ES(n) \leq \binom{2n-4}{n-2} + 1.$$

$\lrcorner$

*Proof:* We first prove that  $ES(4) = 5$ : Given 5 points in  $\mathbb{R}^2$  in general position, we need to find 4 points that form a convex quadrilateral. If the convex hull of these 5 points is a quadrilateral or a pentagon, we are done, and if the convex hull is a triangle  $ABC$ , and  $D, E$  are inside  $ABC$ , then suppose  $BC$  lies in the same side of the line  $\overline{DE}$ , then  $BDEC$  form a convex quadrilateral.

For general  $n$ , in fact  $ES(n) \leq R(4, 2; 5, n)$  (37.4.2.1), because for  $R(4, 2; 5, n)$  many points  $X \subset \mathbb{R}^2$ , divide the 4-subsets of  $X$  into two classes depending on either they form a convex quadrilateral or not, then either there exists a 5-subset whose 4-subsets are all non-convex, or there exists an  $n$ -subset whose 4-subsets are all convex. The first case is not possible by the fact  $ES(4) = 5$ , so the second case is true. Thus we get a convex  $n$ -gon by (37.9.1.1).

For the lower bound, it suffices to construct  $2^{n-2}$  points with no  $n$  points forming a convex  $n$ -gon: Cf. [Art of Counting, Erdős] P680. ?

For the upper bound for  $ES(n)$ , Cf. [Erdős-Szekeres] (or [Holmsen-Mojarrad-Pach-Tardos], where they proved that  $ES(n) \leq 2^{n+O(\sqrt{n \log n})}$ ).  $\square$

**Conj. (37.9.1.3) [Erdős-Szekeres].** For any  $n \in \mathbb{Z}_{\geq 2}$ ,  $ES(n) = 2^{n-2} + 1$ .  $\lrcorner$

*Proof:*  $\square$

**Remark (37.9.1.4).** This conjecture has been proven for  $n \leq 6$ .  $\lrcorner$

### 2 Crossing Numbers of Graphs

**Thm. (37.9.2.1) [Leighton].** For any simple graph  $G$  with  $n$  vertices and  $e \geq 4n$  edges drawn in  $\mathbb{R}^2$ ,  $\text{Cr}(G)$  (the number of crossing of  $G$ ) is at least  $e^3/(100n^2)$ .  $\lrcorner$

*Proof:* Cf. [Ajtai, M.; Chvátal, V.; Newborn, M. M.; Szemerédi, E. Crossing-free subgraphs].  $\square$

### 3 Incidences

**Thm. (37.9.3.1) [Szemerédi-Trotter].** If  $n, t \in \mathbb{Z}_+$ , then for any  $\mathcal{P} \subset \mathbb{R}^2$ ,  $\#\mathcal{P} = n$  and a family  $\mathcal{L}$  of lines in  $\mathbb{R}^2$  s.t.  $\#\mathcal{L} = t$ , the numbers of incidences  $\#I(\mathcal{P}, \mathcal{L})$  between points in  $\mathcal{P}$  and lines in  $\mathcal{L}$  is  $\leq 22(nt)^{2/3} + 4n + t$ .  $\lrcorner$

*Proof:* We may assume that any line in  $\mathcal{L}$  is incident to at least one point in  $\mathcal{P}$ . Draw a graph  $G$  in  $\mathbb{R}^2$  with  $V(G) = \mathcal{P}$ , and join two points with a straight segment if they are consecutive points on one of the lines in  $\mathcal{L}$ . Then  $G$  is a simple graph, and the crossing number of  $G$  is at most  $t^2$ . Also  $\#I(\mathcal{P}, \mathcal{L}) = t + E(G)$ .

Then it follows from (37.9.2.1) that either  $4n > \#I(\mathcal{P}, \mathcal{L}) - t$ , or  $t^2 \geq \text{Cr}(G) \geq (\#I(\mathcal{P}, \mathcal{L}) - t)^3 / (100n^2)$ , from these the assertion follows.  $\square$

**Cor. (37.9.3.2).** There exists  $C \in \mathbb{R}_+$  s.t. if  $n \in \mathbb{Z}_+$  and  $\mathcal{P} \subset \mathbb{R}^2$  is a set of  $n$  points, and  $\mathcal{L}$  is family of lines determined by  $\mathcal{P}$ , then there exists a point  $P \in \mathcal{P}$  belong to at most  $Cn$  lines in  $\mathcal{L}$ .  $\lrcorner$

*Proof:* Cf. [S-T83].  $\square$

**Thm. (37.9.3.3) [Szemerédi-Trotter].** There exists  $C \in \mathbb{R}_+$  s.t. if  $n, k \in \mathbb{Z}_+$ ,  $2 \leq k \leq 10\sqrt{n} + 1$ ,  $\mathcal{P} \subset \mathbb{R}^2$ ,  $\#\mathcal{P} = n$ , and  $\mathcal{L}$  is a family of lines in  $\mathbb{R}^2$  each containing at least  $k$  points from  $\mathcal{P}$ , then  $\#\mathcal{L} \leq 100n^2 / (k-1)^3$ .

(The constant  $C$  can be taken to be 3 by [Clarkson et al. Combinatorial complexity bounds for arrangements of curves and surfaces].)  $\lrcorner$

*Proof:* Let  $t = \#\mathcal{L}$ . Draw a graph  $G$  in  $\mathbb{R}^2$  with  $V(G) = \mathcal{P}$ , and join two points with a straight segment if they are consecutive points on one of the lines in  $\mathcal{L}$ . Then  $G$  is a simple graph, and the crossing number of  $G$  is at most  $t^2$ . Also  $E(G) \geq t(k-1)$ .

Then it follows from (37.9.2.1) that either  $4n > t(k-1)$ , or  $t^2 \geq \text{Cr}(G) \geq t^3(k-1)^3 / (100n^2)$ , from these the assertion follows.  $\square$

**Remark (37.9.3.4).** The assertion is tight for  $2 \leq k \leq \sqrt{n}$ : Take  $\mathcal{P}$  to be the  $\sqrt{n} \times \sqrt{n}$  integer grid. Cf. [Beck, On the lattice property of the plane and some problems of Dirac, Motzkin, and Erdős in combinatorial geometry]  $\textcolor{red}{?}$ .  $\lrcorner$

### 4 Distance Problems

**Thm. (37.9.4.1) [Erdős/Guth-Katz].** For  $n \in \mathbb{Z}_{\geq 2}$ , let  $f(n)$  be the minimal number of distinct distances generated by  $n$  distinct points in  $\mathbb{R}^2$ , then

$$f(n) = \Theta\left(\frac{n}{\log n}\right).$$

$\lrcorner$

*Proof:* Consider the points  $(x, y)$  where  $1 \leq x, y \leq \sqrt{n}$ , then the distances between them are all of the form  $\sqrt{a^2 + b^2}$ , where  $1 \leq a^2 + b^2 < 2n$ . Then by (35.5.1.5), there are  $O(\frac{n}{\log n})$  many points.

Conversely, Cf. [Guth-Katz]  $\textcolor{red}{?}$   $\square$

**Thm. (37.9.4.2) [Solymosi].** There exists  $C \in \mathbb{R}_+$  s.t. for any  $n \in \mathbb{Z}_{\geq 2}$  and any  $n$  distinct points in  $\mathbb{R}^2$ , one of them determine at least  $Cn^{6/7}$  different distances from the others.  $\lrcorner$

*Proof:* Cf. [Solymosi, Distinct distances in the plane].  $\square$

**Conj. (37.9.4.3) [Erdős].** For  $n \in \mathbb{Z}_{\geq 2}$ , the number of distinct distances determined by the vertices of a convex  $n$ -gon in  $\mathbb{R}^2$  is at least  $\lfloor \frac{n}{2} \rfloor$ , with equality iff this is a regular  $n$ -gon.  $\lrcorner$

*Proof:*  $\square$

**Thm. (37.9.4.4) [Erdős/Spencer-Szemerédi-Trotter].** For  $n \in \mathbb{Z}_{\geq 3}$  and  $r \in \mathbb{R}_+$ , let  $U(n)$  be the maximal number of times the unit distance occur among  $n$  points in  $\mathbb{R}^2$ , then there exists  $C \in \mathbb{R}_+$  s.t.

$$n^{1+C/\log \log n} < U(n) < \min(n^{3/2}, 6n^{5/4}).$$

$\lrcorner$

*Proof:* To prove  $U(n) < n^{3/2}$ : This is clearly true for  $n \leq 3$ , so we assume  $n \geq 4$ . Given  $n$  points  $\{P_1, \dots, P_n\}$ , and for  $1 \leq i \leq n$ , suppose there are  $x_i$  points with distance  $r$  from  $P_i$ . We may assume that  $x_1 \geq x_2 \geq \dots \geq x_n$ . Then for any  $i \neq j$ , the  $x_i$  points at distance  $r$  from  $P_i$  can contain at most two points with distance  $r$  from  $P_j$ , so for each  $1 \leq j \leq n$ ,

$$\sum_{i=1}^j (x_i - 2i + 2) \leq n.$$

Denote  $\lfloor \sqrt{n} \rfloor = a$ ,  $\{\sqrt{n}\} = \varepsilon$ , then

$$x_1 + x_2 + \dots + x_a \leq n + a(a-1) = 2n - 2\varepsilon\sqrt{n} + \varepsilon^2 - \sqrt{n} + \varepsilon < 2n - 2\varepsilon\sqrt{n}.$$

Thus  $x_a < \frac{1}{a}(2n - 2\varepsilon\sqrt{n}) = 2\sqrt{n}$ , and

$$\sum x_i < 2n - 2\varepsilon\sqrt{n} + (n-a)2\sqrt{n} = 2n^{3/2}.$$

To prove  $U(n) < 6n^{5/4}$ : Let  $U$  be the number of times the unit distance occur. Draw a graph  $G$  in  $\mathbb{R}^2$  with vertices the given points, and two points are connected by an arc in unit circle if they are consecutive points on the unit circle centered at one of the points. Then  $G$  is a multigraph. Noticing two points are connected by at most two arcs, so minimally delete arcs so that each pair is connected by at most one arc, we get a graph  $G$  satisfies  $E(G) \geq U - n/2$  and  $\text{Cr}(G) \leq 2n^2$ . Then by (37.9.2.1), either  $U - n/2 \leq E(G) < 4n$ , or  $2n^2 \geq \text{Cr}(G) \geq (U - n/2)^3/(100n^2)$ , from these the assertion follows.

For the lower bound, consider the set of points  $(x, y)$ ,  $0 \leq x, y \leq \sqrt{n}$ . ? Cf. [Erd46].  $\square$

**Prop. (37.9.4.5).** For  $n \geq 3$ , let  $a_{\min}(n)$  and  $a_{\max}(n)$  be the maximal number of times the minimal and maximal distance of distinct  $n$  points in  $\mathbb{R}^2$  can be achieved, then

$$a_{\max}(n) = n, \quad a_{\min}(n) \leq 3n - 6, \quad |3n - a_{\min}(n)| = \Theta(\sqrt{n}).$$

$\lrcorner$

*Proof:* For  $a_{\max}(n)$ , notice  $a_{\max}(n) \geq n$  because it can easily achieve  $n$  times. To prove  $a_{\max}(n) \leq n$ , we use induction on  $n$ : if the given points are  $\{P_1, \dots, P_n\}$ , and  $d(P_1, P_2)$  and  $d(P_3, P_4)$  are both maximal, then by an easy argument,  $P_1P_2 \cap P_3P_4 \neq \emptyset$ . Now connect any two points with maximal distance, and consider two cases: if each point is connected to at most 2 points, then there are at most  $n$  edges; and if some point  $P_1$  is connected to 3 points  $P_2, P_3, P_4$ , where  $\angle P_2P_1P_4$  is acute, and  $P_3$  are between  $P_2, P_4$ , then it is clear  $P_3$  cannot connect to any other  $P_i$ , so we can omit  $P_3$  and use induction on  $n$ .

For  $a_{\min}(n)$ , we first prove that  $a_{\min}(n) \leq 3n - 6$ ?: Connect two points if they have minimal distance, then clearly these edges don't intersect. Thus the resulting graph is planar, and then has at most  $3n - 6$  edges by Euler's theorem (37.2.6.3).

To show  $|3n - a_{\min}(n)| = \Theta(\sqrt{n})$ , use triangular constellation? □

**Conj. (37.9.4.6) [Borsuk].** For  $k \in \mathbb{Z}_+$ , each subset  $S \subset \mathbb{R}^k$  of diameter 1 can be decomposed into  $k + 1$  subsets s.t. each subset has diameter  $< 1$ . ┘

*Proof:* □

## 5 Rational Distances

References are [Sol00], [A-E45] and [Erd45a].

**Def. (37.9.5.1) [Integral Sets].** An **integral set** in  $\mathbb{R}^2$  is a point set s.t. the distances between the elements in  $S$  are all integers. ┘

**Prop. (37.9.5.2) [Anning-Erdős].** For any  $n \in \mathbb{Z}_+$ , we can find a planar integral set of cardinality  $n$  in general position. ┘

*Proof:* let  $\alpha = \arcsin(3/5)$ , then  $\alpha/\pi$  is non-rational. Consider the  $n$  points  $P_k = e^{ik\alpha}$ ,  $0 \leq k \leq n-1$ , then it is clear that all the distances between the points are rational. Thus by scaling, we can find  $n$  points with integral distances. □

**Lemma (37.9.5.3).** If a triangle  $T$  has integer side-lengths  $a \leq b \leq c$ , then the minimal height  $m$  of it is at least  $\sqrt{a - \frac{1}{4}}$ . ┘

*Proof:* As  $a, b, c \in \mathbb{Z}_+$ ,  $a + b \geq c + 1$ . The minimal height  $m$  corresponds to the side with length  $c$ , so it follows from Heron's formula? that

$$m^2 = a^2 - \left(\frac{c^2 + a^2 - b^2}{2c}\right)^2.$$

Clearly larger  $c$  gives smaller height, so we may assume  $c = a + b - 1$ . Then

$$\frac{c^2 + a^2 - b^2}{c} = c + \frac{a+b}{c}(a-b) \leq c + a - b = 2a - 1.$$

And the assertion follows. □

**Lemma (37.9.5.4) [Erdős].** In a planar integral set  $H$ , if there exists two point-pairs  $(A, B)$  and  $(C, D)$  in general position (i.e. the lines  $AB$  and  $CD$  are distinct, and the bisector lines are distinct), then  $\#H \leq 4d(A, B) \cdot d(C, D)$ . ┘

*Proof:* This is because for any  $d \in \mathbb{Z}$ , the set  $\{d(P, A) - d(P, B) = d\}$  is a hyperbola, and similar for  $C, D$ . The condition implies that the hyperbolas for any two such distance pairs are different, so they intersect at at most 2 points. So there are at most  $4d(A, B) \cdot d(C, D)$  many points. □

**Prop. (37.9.5.5) [Anning-Erdős/Solymosi].** If  $H$  is a planar infinite integral set, then the points of  $H$  lie on a line.

In fact, if  $A, B$  are two points in a planar integral set  $H$  with  $d(A, B) = d > 1$ , and  $\#H \geq d^3$ , then there are at most  $2d - 1$  points of  $H$  not on the line  $AB$ . ┘



*Proof:* The first assertion follows from (37.9.5.4). For the second assertion, Cf. [Sol00]P339. ?  $\square$

**Cor. (37.9.5.6) [Minimal distances of Integral Sets].** If  $H$  is a planer integral set in general position and  $\#H \geq 4$ , then the minimal distance between points in  $H$  is at least  $(\#H + 1)^{1/3}$ .  $\lrcorner$

*Proof:* If there are two points  $A, B$  with  $d(A, B) = 1$ , then by triangular inequality, any other points must be in equal distances with  $A$  and  $B$ . So it follows from (37.9.5.3) that there are at most points.  $\square$

In other cases, it follows from the proposition that  $\#H \leq d(A, B)^3 - 1$ .  $\square$

**Conj. (37.9.5.7) [Erdős-Ulam].** There doesn't exist a dense subset  $S \subset \mathbb{R}^2$  s.t. all the distances between points in  $S$  are rational.  $\lrcorner$

*Proof:* It follows from <https://terrytao.wordpress.com/2014/12/20/the-erdos-ulam-problem-varieties-of-general-type-and-the-bombieri-lang-conjecture/#more-7930> that this conjecture is a consequence of the Bombieri-Lang conjecture (21.7.10.1). ?  $\square$

**Lemma (37.9.5.8).** If  $H$  is a planer integral set with  $\#H \geq 4$  with diameter  $\Delta$  and having at most  $n/2$  colinear points, then there exists two point-pairs  $(A, B)$  and  $(C, D)$  in general position (as in (37.9.5.4)) s.t.  $d(A, B), d(C, D) \leq 12\Delta/\sqrt{n}$ .  $\lrcorner$

*Proof:* Call a point  $P \in H$  single if there is at most one other point in  $H$  with distance smaller than  $12\Delta/\sqrt{n}$  with  $P$ . Then a method of volume counting implies that there are less than  $n/2$  single points.

Remove all the single points, then in the remaining set  $H'$ , any point  $P$  has two points  $Q_1, Q_2 \in H'$  s.t.  $d(P, Q_i) \leq 12\Delta/\sqrt{n}$ . ? Then we can find two point-pairs in general position. ?  $\square$

**Thm. (37.9.5.9) [Minimal Diameters of Integral Sets].** There exists  $c \in \mathbb{R}_+$  s.t. for any  $n \in \mathbb{Z}_+$ , any planer integral set with cardinality  $n$  has diameter at least  $cn$ .  $\lrcorner$

*Proof:* If there are more than  $n/2$  points in a single line, then the diameter is at least  $n/2$ . Suppose this is not the case, then it follows from (37.9.5.8) that

$$n \leq 4\left(\frac{\Delta}{\sqrt{n}}\right)^2,$$

and the assertion follows.  $\square$

## 37.10 Game Theory

### 1 Nash Equilibrium

References are [Non-corporative Games, Nash].

**Def. (37.10.1.1) [Games].** A finite normal-form (mathematical) **game** is a tuple  $(N, A, O, \mu, u)$ , where

- $N$  is a finite set of players,  $n = \#N$ .
- $A = (A_1, \dots, A_n)$ , where each  $A_i$  is a finite set of actions (or **pure strategies**).
- $O$  is a set of outcomes.
- $\mu : A \rightarrow O$  determines the outcome as a function of the action profile.
- $u = (u_1, \dots, u_n)$  where  $u_i : O \rightarrow \mathbb{R}$  is the real-valued utility function for player  $i$ .

┘

**Def. (37.10.1.2) [Mixed Strategies].** In a mathematical game (37.10.1.1)  $(N, A, u)$ , each player may randomize his actions, this is called a **mixed strategy**. The **support** of a mixed strategy  $s_i$  for player  $i$  is the set of pure strategies that  $s_i$  has positive probability.

Give a mixed strategy profile  $s = (s_1, \dots, s_n)$ , the **expected utility** for player  $i$  is

$$u_i(s) = \sum_{a \in A} u_i(a) \prod_j s_j(a_j).$$

┘

**Def. (37.10.1.3) [Best Response].** In a mathematical game (37.10.1.1)  $(N, A, u)$ , given any mixed strategy profile  $s^i = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$ , a **best response** of player  $i$  to this strategy profile is a mixed strategy  $s_i$  s.t.  $u_i(s_i, s^i) \geq u_i(s'_i, s^i)$  for any mixed strategy  $s'_i$ .

┘

**Def. (37.10.1.4) [Nash-Equilibrium].** A strategy profile  $s = (s_1, \dots, s_n)$  is a **Nash equilibrium** if for each player  $i$ ,  $s_i$  is a best response (37.10.1.3) to  $s^i = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$ .

┘

**Thm. (37.10.1.5) [Nash1951].** Each finite normal-form mathematical game (37.10.1.1) has a Nash-equilibrium.

┘

*Proof:* Let  $S$  be the set of mixed strategy profiles. Given a mixed strategy profile  $s$ , for all  $i$  and  $a_i \in A_i$ , define

$$\varphi_{i,a}(s) = \max(0, u_i(a, s^i) - u_i(s)),$$

and define the function  $f : S \rightarrow S$  by

$$f(s)_i(a) = \frac{s_i(a) + \varphi_{i,a}(s)}{1 + \sum_{b \in A_i} \varphi_{i,b}(s)}.$$

Then with the natural topology,  $S$  is homeomorphic to  $\mathbb{D}^1$  and  $f$  is continuous. So by [Brouwer Fixed-Point Theorem](#),  $f$  has a fixed point  $s$ .

Now we show this  $s$  is a Nash-equilibrium: By linearity of expected utility (37.10.1.2), for each  $i$ , there exists at least one  $a \in A_i$  s.t.  $u_{i,a}(s) \leq u_i(s)$ . Then  $\varphi_{i,a}(s) = 0$ , and because  $f(s) = s$ ,  $\varphi_{i,b}(s) = 0$  for each  $b \in A_i$ . This implies that  $u_i(s'_i, s^i) \leq u_i(s)$  for any mixed strategy  $s'_i$ . As this is true for any  $i$ , we know  $s$  is a Nash-equilibrium. □

**Prop. (37.10.1.6) [Symmetries in Games].**

┘

*Proof:*

□

**Def. (37.10.1.7) [Solutions of Games].**

┘

### Finite Deterministic Competitive Games

**Thm. (37.10.1.8).** For any finite deterministic competitive game with complete information, one of the players has a winning strategy.  $\lrcorner$

*Proof:*  $\square$

## 2 Greedoids

**Def. (37.10.2.1) [Greedoids and Antimatroids with Repetitions].** For a set of symbols  $\Sigma$ , a language  $L$  on  $\Sigma$  (2.7.2.5) is called

- **left-hereditary** if  $\alpha, \beta \in \text{String}(\Sigma)$  and  $\alpha\beta \in L$ , then  $\alpha \in L$ .
- **permutable** if  $\alpha, \beta \in L$  and  $\text{length}_x(\alpha) = \text{length}_x(\beta)$  for any  $x \in A$ , then  $\beta\alpha \in L \iff \alpha\beta \in L$ .
- **locally-free** if  $\alpha \in L$  and  $x \neq y \in \Sigma$  with  $\alpha x \in L, \alpha y \in L$ , then  $\alpha xy \in L$ .
- **strongly exchangeable** if for any  $\alpha, \beta \in L$ , there is a subword  $\alpha'$  of  $\alpha$  s.t.  $\beta\alpha' \in L$ , and  $\text{length}_x(\beta\alpha') = \max(\text{length}_x(\beta), \text{length}_x(\alpha))$  for any  $x \in \Sigma$ .
- **augmentary** if  $\alpha, \beta \in L$  and  $\text{length}(\alpha) > \text{length}(\beta)$ , then there exists  $x \in \alpha$  s.t.  $\beta x \in L$ .

A **greedoid** is a left-hereditary and augmentary language on a set of symbols. An **antimatroids with repetition** is a left-hereditary locally-free permutable language on a set of symbols.  $\lrcorner$

**Prop. (37.10.2.2) [Ranks of Greedoids].** In a greedoids, any two maximal words have the same length, called the **rank of this language** (which is defined to be  $\infty$  if no maximal words exist).  $\lrcorner$

**Def. (37.10.2.3) [Flats].** In a left-hereditary language  $L$  on a set of symbols  $\Sigma$ , a **flat of words** is an equivalent class of words where two words  $\alpha, \beta \in L$  are called equivalent if for any  $\gamma \in \text{String}(\Sigma)$ ,  $\alpha\gamma \in L \iff \beta\gamma \in L$ .

A flat  $f$  is called a **subflat** of a flat  $g$  if any word in  $f$  can be extended to a word in  $g$ .  $\lrcorner$

**Prop. (37.10.2.4) [Strongly Exchangeable Languages].** Any strongly exchangeable language  $L$  on a set of symbols  $\Sigma$  is locally-free and permutable. Conversely, any left-hereditary locally-free and permutable language is strongly exchangeable.  $\lrcorner$

*Proof:* It is easy to show that a strongly exchangeable language is locally-free and permutable. Conversely, for  $\alpha, \beta \in L$ , let  $\alpha'$  be the subword of  $\alpha$  consisting of those  $x \in \Sigma$  that are preceded by at least  $\text{length}_x(\beta)$  many occurrence of  $x$ , then we show that  $\beta\alpha' \in L$ : We use induction on  $\sum_x \max(\text{length}_x(\alpha), \text{length}_x(\beta))$ . Let  $\alpha''$  be the longest prefix of  $\alpha'$  s.t.  $\beta\alpha'' \in L$ . Suppose for the contrary that  $\alpha'' \neq \alpha'$ , and let  $x$  be the letter in  $\alpha'$  following  $\alpha''$ . Then  $\text{length}_x(\beta\alpha'') < \text{length}_x(\alpha)$ . Then we can write  $\alpha = \alpha_1 x \alpha_2$ , where  $\text{length}_x(\alpha_1) = \text{length}_x(\beta\alpha'')$ .

By the definition of  $\alpha'$ ,  $\text{length}_y(\alpha_1) \leq \text{length}_y(\beta\alpha'')$  for any  $y \in \Sigma$ , so  $\sum_y \max(\text{length}_y(\alpha_1), \text{length}_y(\beta\alpha'')) < \sum_y \max(\text{length}_y(\alpha), \text{length}_y(\beta))$ . Then by induction hypothesis, there exists  $\gamma \in \text{String}(\Sigma)$  s.t.  $\alpha_1\gamma \in L$ , and  $\text{length}_y(\alpha_1\gamma) = \text{length}_y(\beta\alpha'')$  for any  $y \in \Sigma$ . Clearly  $x \notin \gamma$ , so we can use locally-freeness repeatedly to show that  $\alpha_1\gamma x \in L$ . Now permutability implies that  $\beta\alpha''x \in L$ , contradicting the maximality of  $\alpha''$ .  $\square$

**Cor. (37.10.2.5) [Antimatroids with Repetitions are Greedoids, Björner-Lovász-Shor].** Any antimatroids  $L$  with repetitions on a set of symbols  $\Sigma$  is a greedoids. And if  $L$  has finite rank (37.10.2.2), for any two words  $\alpha, \beta \in L$ ,  $\bar{\alpha}$  is a subflat of  $\bar{\beta}$  (37.10.2.3) iff  $\text{length}_x(\alpha) \leq \text{length}_x(\beta)$  for any  $x \in \Sigma$ . In particular,  $\alpha, \beta$  belongs to the same flat iff  $\text{length}_x(\alpha) = \text{length}_x(\beta)$  for any  $x \in \Sigma$ .  $\lrcorner$

*Proof:* By (37.10.2.4), any antimatroids with repetitions is strongly-exchangeable, thus is clearly augmentary.

If  $\text{length}_x(\alpha) \leq \text{length}_x(\beta)$  for any  $x \in \Sigma$ , then by strongly exchangeable property, there exists  $\gamma \in \text{String}(\Sigma)$  s.t.  $\text{length}_x(\alpha\gamma) = \text{length}_x(\beta)$  for any  $x \in \Sigma$ . Thus by permutability,  $\alpha$  is a subflat of  $\beta$ . Conversely, let  $\alpha\gamma$  be equivalent to  $\beta$ , and extend  $\alpha\gamma$  to a maximal word  $\alpha\gamma\delta$ , then  $\beta\delta$  is also in  $L$ , and by strong exchangeability,  $\text{length}_x(\alpha\gamma) = \text{length}_x(\beta)$  for any  $x \in \Sigma$ . In particular,  $\text{length}_x(\alpha) \leq \text{length}_x(\beta)$  for any  $x \in \Sigma$ .  $\square$

### 3 Firing Games

References are [ALS91].

**Def. (37.10.3.1) [Firing Games].** Given a finite set  $A$ , a **firing game** is a legal sequence of elements of  $A$ . where for any finite legal sequence  $(a_1, \dots, a_k)$  of  $A$ , there is a rule to determine which element  $a_{k+1} \in A$  can be chosen s.t.  $(a_1, \dots, a_{k+1})$  is also legal. A firing game terminates for a legal sequence  $(a_1, \dots, a_k)$  if no such  $a_{k+1} \in A$  can be found.  $\lrcorner$

**Def. (37.10.3.2) [Monotone Firing Games].** A **monotone firing game** is a firing game on a set  $A$  satisfying the following: For any  $k, l \in \mathbb{N}$  and  $c \in A$ , if  $X = (a_1, \dots, a_k), Y = (b_1, \dots, b_l)$  be two legal sequences s.t.

$$\#\{i | a_i = x\} \leq \#\{j | b_j = x\}$$

for any  $x \in A$ , and

$$\#\{i | a_i = c\} = \#\{j | b_j = c\},$$

then  $(b_1, \dots, b_l, c)$  is legal if  $(a_1, \dots, a_k, c)$  is legal.  $\lrcorner$

**Thm. (37.10.3.3) [Thorup].** For a monotone firing game on a set  $A$ , any element of  $A$  appears the same number of times in any terminating sequence legal sequence.  $\lrcorner$

*Proof:* It is easy to check that the set of legal sequences form a antimatroids with repetitions on  $A$  (37.10.2.1), so the assertion follows from (37.10.2.5).

Alternatively, let  $X = (a_1, \dots, a_m), Y = (b_1, \dots, b_n)$  be two legal terminating firing sequences. It suffices to prove that for any element  $c \in A$ ,  $c$  appears in  $Y$  no more than it does in  $X$ . If this is not true, consider the smallest  $k \in \mathbb{N}$  s.t.

$$\#\{i | a_i = a_{k+1}, i \leq k\} = \#\{j | b_j = a_{k+1}\}.$$

Then we can compare the two sequences  $(a_1, \dots, a_k)$  and  $(b_1, \dots, b_n)$ , and use the minimality of  $k$  to conclude that  $(b_1, \dots, b_n, c)$  is also a legal sequence, contradicting the fact that  $Y$  is terminating.  $\square$

**Lemma (37.10.3.4).** The possible records of games form a locally-free permutable left-hereditary language on the set  $V(G)$ .  $\lrcorner$

#### Chip-Firing Games

**Def. (37.10.3.5) [Chip-Firing Games].** Given a simple finite graph  $G$  and a number  $a_i$  to each of its vertex  $v_i$  called the **initial distribution of chips**, a **chip-firing game** on this graph is sequence of **chip-firings** which means if  $a_i \geq \deg(v_i)$  for some  $i$ , then we can decrease  $a_i$  by  $\deg(v_i)$ , and increase  $a_j$  by 1 for each  $v_j$  that is connected to  $v_i$ .  $\lrcorner$

**Thm. (37.10.3.6) [Björner-Lovász-Shor].** Given a simple finite graph  $G$  and initial distribution of chips (37.10.3.5), either every game can be continued infinitely, or every game terminates under after the same number of moves, and the number of times a given vertex is fired is the same in each vertex.  $\perp$

*Proof:* This is clearly a monotone firing game, so the assertion follows from (37.10.3.3).  $\square$

**Lemma (37.10.3.7) [Tardos/Björner-Lovász-Shor].** Let  $G$  be a connected simple finite graph. Then:

- If a chip-firing game is infinite, then every node is fired infinitely often.
- If a chip-firing game terminates, then there is a node that is not fired at all.

$\perp$

*Proof:* 1: There is a vertex that is fired infinitely often, so the vertices connected to it receives infinitely many chips, so they must fire infinitely often too. In this way, because  $G$  is connected, every vertex is fired infinitely often.

2: Suppose each vertex is fired. Consider the vertex  $v$  s.t. at the end of the game,  $v$  has been idle for the longest time. Then after it fired the last time, all of its neighbors fired at least once. Thus  $v$  has  $\deg(v)$  chips, so it can be fired again, contradiction.  $\square$

**Prop. (37.10.3.8).** Let  $G$  be a connected simple finite graph with  $n = \#V(G)$  and  $e = \#E(G)$ , and  $\sum_i a_i = N$ .

- If  $N > 2e - n$ , then the game is infinite.
- If  $e \leq N \leq 2e - n$ , then there is an initial configuration of chips with infinite games and also an initial configuration of chips with finite games.
- If  $N < e$ , then the game is finite.

$\perp$

*Proof:* 1: If  $N > 2e - n$ , then there is always a vertex  $v$  with at least  $\deg(v)$  chips.

2: We can always place  $\deg(v_i) - 1$  chips at vertex  $v_i$ , so in this way we can place  $2e - n$  chips with a finite game. On the other hand, if we have  $e$  chips, then we can choose an arbitrary acyclic orientation of vertices of  $G$ , and place  $\deg^+(v_i)$  chips at vertex  $v_i$ . Then for this configuration the game is infinite: There is a source  $v$ , so we can fire  $v$ , and then after reverting the orientation of all edges from  $v$ , we get another acyclic orientation of  $G$ , and the chips are also placed by  $\deg^+(v_i)$  at vertex  $v_i$ . Then this process can go on forever.

3: Choose any acyclic orientation of  $G$ , and let  $T = \sum_i \max(0, a_i - \deg^+(v_i))$ . Call a vertex deficient if  $a_i - \deg^+(v_i) > 0$ . Then along with any game, we can modify the orientation of  $G$  so that the quantity  $T$  never increases. And moreover, when the set of deficient vertices  $v_i$  changes,  $T$  decreases. In this way, we can show that the game is finite: A deficient vertex can never be fired, so an infinite game must change the set of deficient vertices infinitely often, contradiction.

To show the monotone property of  $T$ , when a vertex  $v$  is fired, reverse the orientation of all the edges leaving  $v$ , then this is also an acyclic orientation, but for  $T$ , the term at  $v$  decreases by  $\deg^-(v)$ , and all the terms at the vertices with an edge going to  $v$  increases by at most 1. Moreover, if such an vertex with an edge going to  $v$  is deficient, then in fact  $T$  decreases. If none of these are deficient, then the set of deficient sets doesn't change.  $\square$

**Prop. (37.10.3.9).** Let  $G$  be a connected simple finite graph with  $n = \#V(G)$ , and let  $\lambda_1$  be the least non-zero eigenvalue of the Laplacian matrix  $L_G$  (37.2.2.1), then any terminating chip-firing game with  $N$  chips has at most  $\sqrt{2nN}/\lambda_1$  steps.  $\perp$

*Proof:* Suppose we start with  $a_i$  chips on vertex  $v_i$  and after  $s$  steps, we end up with  $b_i$  chips on vertex  $v_i$ . Let  $x_i$  be the number of times that  $v_i$  was fired. By (37.10.3.7), we may assume  $x_n = 0$ . Then clearly  $Lx = \mathbf{a} - \mathbf{b}$ .

Let

$$L_G = \sum_{i=1}^{n-1} \lambda_i v_i v_i^t$$

where  $v_1 \leq v_1 \leq \dots \leq v_{n-1}$  are the eigenvalues of  $L_G$ , and  $v_1, \dots, v_{n-1}$  are the corresponding orthogonal eigenvectors of unit length. Denote

$$L' = \sum_{i=1}^{n-1} \frac{1}{\lambda_i} v_i v_i^t, \quad v_n = \frac{1}{\sqrt{n}}(1, \dots, 1)^t,$$

then

$$L' L_G = \sum_{i=1}^{n-1} v_i v_i^t = \mathbf{1} - v_n v_n^t = \mathbf{1} - \frac{1}{n} \mathbf{J}.$$

Hence  $e_n^t L' L_G = e_n^t - \frac{1}{n}(1, \dots, 1)^t$ , and

$$s = (1, \dots, 1) \cdot \mathbf{x} = n(e_n^t - e_n^t L' L) \mathbf{x} = -n e_n^t L' L \mathbf{x} = \sum_{i=1}^{n-1} -\frac{n}{\lambda_i} (e_n^t \lambda_i) (v_i^t (\mathbf{a} - \mathbf{b})).$$

Thus

$$s \leq \frac{n}{\lambda_1} \sum_{i=1}^{n-1} |e_n^t v_i| \cdot |v_i^t (\mathbf{a} - \mathbf{b})| \leq \frac{n}{\lambda_1} \sqrt{\sum_{i=1}^{n-1} (e_n^t v_i)^2} \cdot \sqrt{\sum_{i=1}^{n-1} (v_i^t (\mathbf{a} - \mathbf{b}))^2} \leq \frac{n}{\lambda_1} |\mathbf{a} - \mathbf{b}| \leq \sqrt{2} \frac{nN}{\lambda_1}.$$

□

## 38 | Others

### 38.1 Financial Mathematics

#### 1 Financial Terminologies

##### Generals

**Def. (38.1.1.1) [Social Relations].** A **social relation** is the fundamental unit of analysis within the social sciences, and describes any voluntary or involuntary social association, connection, or affiliation between two or more individuals within and/or between groups. ┘

**Def. (38.1.1.2) [Rights].** **Rights** are legal, social, or ethical principles of freedom or entitlement, that is, rights are the fundamental normative rules about what is allowed of people or owed to people according to some legal system, social convention, or ethical theory. ┘

**Def. (38.1.1.3) [Obligations].** **Obligation** is a course of action that a person or entity is required to take, whether legal or moral. Obligations are constraints that limit freedom. ┘

**Def. (38.1.1.4) [Contract].** A **contract** is an agreement that specifies certain legally enforceable rights(38.1.1.2) and obligations(38.1.1.3) pertaining to two or more mutually agreeing entities.

A **claim** is a demand for enforcement of a right(38.1.1.2) provided by law or a legal contract. ┘

**Def. (38.1.1.5) [Institutions].** An **institution** is a humanly devised structure of rules and norms that shape and constrain individual behavior. ┘

**Def. (38.1.1.6) [Infrastructure].** **Infrastructure** is the set of facilities and systems that serve a country, city, or other area, and encompasses the services and facilities necessary for its economy, households and firms to function. ┘

**Def. (38.1.1.7) [Corporations and Companies].** A **company**, is a legal entity representing an association of people, whether natural, legal or a mixture of both, with a specific objective.

A **corporation** is an organization authorized by the state to act as a single entity and recognized as such in law for certain purposes.

One big difference between companies and corporations is that corporations are legally separate from their owners. ┘

**Def. (38.1.1.8) [Goods and Services].** **Goods** are items that are usually tangible, e.g. pens or apples. **Services** are activities provided by other people, e.g. by teachers or barbers. ┘

**Def. (38.1.1.9) [Economic Values].** **Economic value** is a measure of the benefit provided by a good or service(38.1.1.8) to a financial entity. ┘

**Def. (38.1.1.10) [Assets].** An **asset** is anything(tangible or intangible) that can be used to produce positive economic value(38.1.1.9). ┘

**Def. (38.1.1.11) [Commodities].** a **commodity** is an economic good(38.1.1.8), usually a resource, that specifically has full or substantial fungibility: that is, the market treats instances of the good as equivalent or nearly so with no regard to who produced them. ┘

**Def. (38.1.1.12) [Payments and Money].** A **payment** is the voluntary tender of money or its equivalent or of things of value by one person or entity to another in exchange for goods or services(38.1.1.8) provided by them, or to fulfill a legal obligation(38.1.1.3)/philanthropy desire. The person or entity making the payment is called the **tbfpayer**, while the **payee** is the person or entity receiving the payment. ┘

An item or verifiable record that is generally accepted as payment for goods or services is called **money**. ┘

**Def. (38.1.1.13) [Trade].** **Trade** is the transfer of goods or services(38.1.1.8) from one person or entity to another, often in exchange for money as a payment(38.1.1.12). ┘

**Def. (38.1.1.14) [Loan and Debt].** A **loan** is the transfer of money(38.1.1.12) by one person or entity(called the **lender** or **creditor**) to another with an agreement to pay it back. The recipient, or borrower of a loan is called a **debtor**, and is said to incur a **debt**. ┘

### Finance

**Def. (38.1.1.15) [Currency].** A **currency** is a standardization of money(38.1.1.12) in any form, in use or circulation as a medium of exchange. ┘

**Def. (38.1.1.16) [Cash].** **Cash** is money(38.1.1.12) in the physical form of currency(38.1.1.15). ┘

**Def. (38.1.1.17) [Financial Institutions].** A **financial institution** is an institution(38.1.1.5) that provides service as an intermediary for different types of financial monetary transactions. ┘

**Def. (38.1.1.18) [Financial Markets].** A **financial market** is a composition of systems, financial institutions(38.1.1.17), procedures, social relations(38.1.1.1) or infrastructures(38.1.1.6) whereby parties can trade(38.1.1.13). ┘

**Def. (38.1.1.19) [Banks].** A **bank** is a financial institution(38.1.1.17) that accepts deposits(38.1.1.20) from the public while simultaneously making loans. Lending activities can be directly performed by the bank or indirectly through capital markets ┘

**Def. (38.1.1.20) [Deposits].** A **deposit** is the act of placing cash(38.1.1.16)(or cash equivalent) with some entity, most commonly with a financial institution(38.1.1.17), e.g. a bank(38.1.1.19). ┘

The deposit is a credit for one person or entity who placed it, and it may be taken back(withdrawn) in accordance with the terms agreed at time of deposit, transferred to another person or entity, or used for a purchase at a later date. ┘

**Def. (38.1.1.21) [Financial Assets].** A **financial asset** is a non-physical asset(38.1.1.10) whose value is derived from a contractual claim(38.1.1.4), e.g. bank deposits(38.1.1.20), bonds(38.1.1.28), and participations in companies' share capital. ┘

**Def. (38.1.1.22) [Financial Liabilities].** A **financial liability** is a measure of the financial obligations(38.1.1.3) that a financial entity owes. For example, debt(38.1.1.14) is a kind of liability. ┘



**Def. (38.1.1.23) [Corporate Liabilities].** **Corporate liability** is the financial liability(38.1.1.22) of a company as a legal person can be held liable for the acts and omissions of the natural persons it employs and, in some legal systems, for those of other associates and business partners. ┘

**Def. (38.1.1.24) [Equities].** **Equity** is the measure of ownership interest in an asset(38.1.1.10) that may be offset by debts or other liabilities(38.1.1.22). Equity equals the value(38.1.1.9) of the asset minus the liabilities(38.1.1.22). ┘

**Def. (38.1.1.25) [Non-Financial Assets].** A **non-financial asset** is an asset(38.1.1.10) that cannot be traded(38.1.1.13) on the financial markets(38.1.1.18) and whose value is derived by its physical net worth rather than from a contractual claim, as opposed to a financial asset(38.1.1.21). ┘

**Def. (38.1.1.26) [Securities].** A **security** is a tradable(38.1.1.13) financial asset(38.1.1.21). The term commonly refers to any form of financial instrument. ┘

**Def. (38.1.1.27) [Securitization].** **Securitization** is the financial practice of pooling various types of contractual liabilities(38.1.1.22) and selling their related cash flows to third party investors as securities(38.1.1.26). ┘

**Def. (38.1.1.28) [Bonds].** A **bond** is a type of security(38.1.1.26) under which the issuer owes the holder a debt(38.1.1.14), and is obliged to provide cash flow to the creditor(38.1.1.14). ┘

**Def. (38.1.1.29) [Exchanges].** An **exchange**, **bourse**, **trading exchange** or **trading venue** is an organized financial market(38.1.1.18) where tradable securities(38.1.1.26), commodities(38.1.1.11), foreign exchange, futures, and options(38.1.1.35) contracts are bought and sold. ┘

**Def. (38.1.1.30) [Orders].** An **order in an exchange** is an instruction to buy or sell a security(38.1.1.26) on an exchange.

A **market order** is a buy or sell order to be executed immediately at the current market prices.

A **limit order** is an order to buy at no more than a specific price, or to sell at no less than a specific price. ┘

### Stocks

**Def. (38.1.1.31) [Stocks].** ┘

**Example (38.1.1.32) [Stocks].** NYSE, AMEX, NASDAQ: ? ┘

**Def. (38.1.1.33) [Penny Stocks].** A **penny stock** is a stock with a low nominal share price that typically trades in the over-the-counter market?, often an OTC Bulletin Board or Pink Sheets quoted stock. ? ┘

### Financial Derivatives

**Def. (38.1.1.34) [Financial Derivatives].** A **financial derivative** is a contract that derives its value from the performance of an **underlying entity**, e.g. an asset, a stock or interest rate. ┘

**Def. (38.1.1.35) [Options].** An **option** is a security(38.1.1.26) giving the right to buy or sell an asset, subject to certain conditions, within a specified period of time. The last day on which the option may be exercised is called the **expiration date** or **maturity date**.

An **American option** is one that can be exercised at any time up to the date the option expires. A **European option** is one that can be exercised only on a specified future date.

The price that is paid for the asset when the option is exercised is called the **exercise price** or **striking price**. ┘

**Def. (38.1.1.36) [Call Options and Sell Options]**. The simplest kind of option (38.1.1.35) is one that gives the holder the right to buy (resp. sell) a single share of common stock, called a **call option** (resp. **put option**). ┘

**Def. (38.1.1.37) [Financial Warrants]**. A **financial warrant** is an option (38.1.1.35) that is a liability (38.1.1.22) of a corporation. ┘

**Prop. (38.1.1.38)**. Almost all corporate liabilities (38.1.1.23) can be realized as combinations of options. ┘

*Proof:* ┘

### Liquidity Risks

**Def. (38.1.1.39) [Market Liquidity]**. **market liquidity** of an asset in a financial market (38.1.1.18) is the feature whereby an individual or firm can quickly purchase or sell this asset without causing a drastic change in the asset's price.

The trader initiating the transaction is said to **demand liquidity**, and the other party (counterparty) to the transaction **supplies liquidity**. ┘

**Def. (38.1.1.40) [Bid-Ask Spread]**. The **bid-ask spread** or the **bid-offer spread** or **buy-sell spread** of an asset is the difference between the prices quoted <sup>?</sup> for an immediate sale (ask) and an immediate purchase (bid) for stocks, futures contracts, options, or currency pairs.

The bid-ask spread is one indicator of an asset's liquidity and of the size of the transaction cost. If the bid-ask spread is 0, then it is called a **frictionless asset**. ┘

**Def. (38.1.1.41) [Market-Makers]**. A **market-maker** or a **liquidity provider** is a company (38.1.1.7) or an individual that quotes both a buy and a sell price in a tradable asset held in inventory, hoping to make a profit on the bid-ask spread (38.1.1.40). ┘

**Def. (38.1.1.42) [Transaction Cost]**. Two elements comprise almost all of the transaction cost: brokerage fees and bid-ask spreads. ┘

**Def. (38.1.1.43) [Average True Range (ATR)]**. **Average true range** or **ATR** was introduced by J. Welles Wilder in his 1978 book *New Concepts In Technical Trading Systems*. ATR is a measure of volatility for a stock or index. ┘

**Prop. (38.1.1.44)**. Theory and empirical evidence suggest that investors require higher return on assets with lower market liquidity to compensate them for the higher cost of trading these assets. ┘

*Proof:* Cf. [Yakov Amihud and Haim Mendelson, *Asset Pricing and the Bid-Ask Spread*. 1986]. ┘

**Prop. (38.1.1.45)**. risk-averse investors require higher expected return if the asset's market-liquidity risk is greater. ┘

*Proof:* Cf. [Viral Acharya and Lasse Heje Pedersen, "Asset pricing with liquidity risk." *Journal of Financial Economics* 77, 2005]. ┘

### Portfolios

**Def. (38.1.1.46) [Portfolios].** ┘

**Def. (38.1.1.47) [Long and Short Positions].** An investor is said to have a **long position** at time  $t$  on an asset  $S$  if  $\pi_t^S \geq 0$ , and is said to have a **short position** at time  $t$  on an asset  $S$  if  $\pi_t^S < 0$ . ┘

## 2 Macroeconomy

### Keynes' Ideas

**Remark (38.1.2.1).** Keynes' idea is roughly the following:

- Economies can and often do suffer from an overall lack of demand, which leads to involuntary unemployment.
  - The economy's automatic tendency to correct shortfalls in demand, if it exists at all, operates slowly and painfully.
  - Government policies to increase demand, by contrast, can reduce unemployment quickly.
  - Sometimes increasing the money supply won't be enough to persuade the private sector to spend more, and government spending must step into the breach.
- ┘

postulates of the classical theory are applicable to a special case only and not to the general case, the situation which it assumes being a limiting point of the possible positions of equilibrium. Moreover, the characteristics of the special case assumed by the classical theory happen not to be those of the economic society in which we actually live, with the result that its teaching is misleading and disastrous if we attempt to apply it to the facts of experience.

## 3 Asset Pricing

**Notation (38.1.3.1).**

- Let  $(\Omega, \mathcal{F}, \mathbf{P})$  be a probability space.
- ┘

**Remark (38.1.3.2).** We study a financial market where assets (stocks) are traded (38.1.1.13) in continuous time. We assume that there are  $d$  assets, and that the prices  $S^i, i = 1, \dots, d$  of these assets are modeled as semi-martingales with respect to a reference filtration  $\mathbf{F}$ .

These assets are called **risky assets** or **securities**. We shall also assume that there is an  $\mathbf{F}$ -adapted riskless asset (also called the savings account) with dynamics

$$dS_t^0 = S_t^0 r_t dt, \quad S_0^0 = 1.$$

where  $r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_+$  is called the **interest rate**. ┘

**Def. (38.1.3.3) [Zero-Coupon Bonds (ZC)].** A **zero-coupon bond** is an asset that delivers one monetary unit at time  $T$ . Clearly, the dynamics of the ZC's price  $P_T$  is

$$\begin{cases} dP_T(t) &= r_t P_T(t) dt, & t \leq T \\ P_T(T) &= 1 \end{cases}.$$
┘

### Semi-Martingale Stock Price Processes

**Thm. (38.1.3.4) [Stock Price Processes are Semi-Martingales].** ┘

*Proof:* □

**Thm. (38.1.3.5) [Fundamental Theorem of Asset Pricing].** For an  $\mathbb{R}^d$ -valued semi-martingale  $S = (S_t)_{t \in [0, T]}$ , the following are equivalent:

- There exists a probability measure  $\mathbf{Q}$  equivalent to  $\mathbf{P}$  under which  $S$  is a sigma-martingale.
- $S$  doesn't permit a free lunch with vanishing risk.

┘

*Proof:* □

### Option Pricing

**Prop. (38.1.3.6).** The current price of a option is equal to the price of the stock minus the price of a pure discount bond that matures on the same date as the option, with a face value equal to the striking price of the option (38.1.1.35). ┘

## 4 Quantitative Trading

### Basics

**Def. (38.1.4.1) [Systematic Trading].** **systematic trading** is the act of trading following clearly defined rules that can be defined mathematically and tested empirically. ┘

**Def. (38.1.4.2) [Statistical Arbitrage].** The key underlying assumption of statistical arbitrage is that the prices of financial instruments are driven by consistent rules, which can be discovered from their historical behavior and applied to the future. ┘

**Def. (38.1.4.3) [Alphas].** An **alpha** is an automated predictive model that describes, or decodes, some market relations.

Alpha comes from data. A change in information should in turn produce a change in the alpha.

┘

**Prop. (38.1.4.4) [Good Qualities of Alphas].**

- The idea and expression are simple.
- The expression/code is elegant.
- It has a good in-sample Sharpe ratio.
- It is not sensitive to small changes in data or parameters.
- It works in multiple universes.
- It works in different regions.

┘

**Def. (38.1.4.5) [Information Ratios and Information Coefficients].** The **information ratio** of an alpha (38.1.4.3) is the ratio of excess returns over a benchmark to the variability of those returns.

The **information coefficient** of an alpha measures the correlation between the predicted and actual values. ┘

**Prop.(38.1.4.6)[Ways to Find Alphas].**

- Analyze the variables in the data.
- Get an idea of the price response to the change you want to model.
- Come up with a mathematical expression that translates this change into stock positions.
- Test the expression.
- If the result is favorable, submit the alpha.

┘

**Prop.(38.1.4.7)[Ways to Avoid Overfit].**

- Look at the performance of their alphas on data that was not used in alpha construction(out-of-sample data) and comparing it with the data used to improve the alpha(in-sample data).

┘

**Prop.(38.1.4.8)[Cutting Losses].** Cutting losses means abandoning rules that no longer work. Instead of following a particular security, we apply trend following to the accumulated profit and loss of a strategy as a whole, which may consist of many individual alphas on many securities.

┘

**Prop.(38.1.4.9)[Failure of Alphas].** There are several way to tell that a strategy isn't working:

- A drawdown exceeds the typical drawdowns observed previously.
- The strategy's Sharpe ratio falls significantly.
- Rules that were initially observed in historical simulation are no longer valid in live trading.

┘

**Prop.(38.1.4.10)[Good Qualities of Quant Researchers].**

- Work Hard.
- Set ambitious long-term targets but attainable weekly goals.
- Prioritize based on risk and reward.
- Stay curious.
- Perform value-added work, and automate.
- Make sensible changes, and beware of overfitting.
- Has synergistic teams.

┘

**Def.(38.1.4.11)[Long-Volatility Programs].** A **long-volatility program** is an investing strategy that tends to benefit from increasing volatility and/or persistent directional trends. Often associated with strategies. Long-volatility program is usually employed by commodity trading advisors from the managed futures industry?.

┘

**Data****Prop.(38.1.4.12)[Types of Data].**

- Prices and volumes. Technical analysis or regression models may be built based on this data.
- Fundamentals. By automating the analysis of key metrics for each company, you can build alphas that typically have very low turnover.

- Macroeconomic data, such as GDP numbers and employment rates, that have market-wide effects upon their release.
- Text, such as Federal Open Market Committee minutes, company filings, papers, journals, news, or social media.
- Multimedia, notably relevant videos or audio. There are mature techniques to process such data –for example, converting audio into text that can be used to build models.

┘

**Prop.(38.1.4.13)[Common Ways to get Data].**

- Literature.
- 

┘

### Stocks

**Prop.(38.1.4.14).** The empirical results strongly suggest that trend following on stocks does offer a positive mathematical expectancy.

┘

*Proof:* Cf.[Does Trend Following Work on Stocks?, Wilcox-Crittenden].

□

### Back Testing

There are several parameters that you would need to monitor when analyzing a strategy's performance and risk. Some important metrics/ratios are mentioned below:

Total Returns (CAGR) –Compound Annual Growth Rate (CAGR) is the mean annual growth rate of an investment over a specified period of time longer than one year.

Hit Ratio –Order to trade ratio.

Average Profit per Trade –Total profit divided by the total number of trades

Average Loss per Trade –Total loss divided by the total number of trades

Maximum Drawdown –Maximum loss in any trade

The Volatility of Returns –Standard deviation of the “returns”

Sharpe Ratio –Risk-adjusted returns, i.e. excess returns (over risk-free rate) per unit volatility or total risk.

**Fact(38.1.4.15)[Factors that may influence effects of Back-Testing].** There are several factors that may be influencing the accuracy of back-testing:

- Survivorship bias: A phenomenon where poorly performing stocks, having been delisted, are not reflected in a current sample or database. This results in overestimations of what past performance would have been. To neutralize this bias, you need to include historical data for all stocks that were delisted.
- Corporate actions: Corporate actions, including cash dividends, splits, mergers, spin-offs, stock dividends, reverse splits, etc may influence stock prices. To neutralize this influence, you need to proportionately back adjust for these actions.
- Realistic investable universe: Penny stocks and illiquid stocks may be untradable. To neutralize this influence, you need to set minimum stock price filter was used to avoid penny stocks, and set a minimum daily liquidity filter was used to avoid illiquid stocks.

┘

## 5 Momentum Strategies

**Remark (38.1.5.1).** The idea of trend-following strategy is that people have FOMO (Fear Of Missing Out), so the trends emerge as a result of human emotion and the desire to follow the crowd since a group of people can base their trading decision on an emotion triggered by an event. ┘

### Scalping Strategies

#### Smart Order Routing Strategies

#### Market-Making Strategies

#### Time-Weighted Average Price Strategies

#### Volume-Weighted Average Price Strategies

#### Seasonality Trading Strategies

#### Machine-Learning Based Strategies

#### Option-Trading Strategies

## 6 Arbitrage Strategies

### Pairs Trading Strategies

**Remark (38.1.6.1)** [Mean-Reversing Strategies]. ┘

## 38.2 Information Theory

### 1 Cryptography

References are [Algebraic aspects of cryptography] and [Computational Complexity].

Cryptography is the study of “mathematical” systems for solving two kinds of security problems: privacy and authentication.

**Def. (38.2.1.1) [Protocols].** A **protocol** is a multi-party algorithm, defined by a sequence of steps, specifying the actions required of two or more parties in order to achieve a specified objective. ┘

**Def. (38.2.1.2) [Cryptosystems and Keys].** Key cryptosystems use cyphers whose algorithm can be revealed to everyone without compromising the security of the cryptosystem.

In such cryptosystems, a set of specific parameters, called the **keys**, is supplemented together with the plaintext as an input to the cypher. The security of this cryptosystem depends entirely on the secrecy of the keys.

Mathematically, a **cryptographic system** is a family of injective maps

$$\{S_k\}_{k \in K} : \mathcal{P} \rightarrow \mathcal{C},$$

where the parameter space  $K$  is called the space of **keys**. ┘

**Remark (38.2.1.3) [Goals of a Cryptosystem].** For practical uses, a cryptosystem must be easy to computationally simple, but also computing the key  $k$  from the function  $S_k$  must be computationally infeasible. ┘

**Def. (38.2.1.4) [Key Establishment Protocols].** A **key establishment protocol** is a protocol whereby a key becomes available to two or more parties, for subsequent cryptographic applications.

The key establishment protocol is usually done over a reliable and secure channel. Once the key is established, subsequent communications involve sending cryptograms over a public channel which is vulnerable to total passive interception. ┘

**Def. (38.2.1.5) [Threats to Cryptosystems].** There are several kinds of threats to a cryptosystem:

- A **ciphertext only attack** is a cryptanalytic attack in which the cryptanalyst possesses only ciphertext.
- A **known plaintext attack** is a cryptanalytic attack in which the cryptanalyst possesses a substantial quantity of corresponding plaintext and ciphertext.
- A **chosen plaintext attack** is a cryptanalytic attack in which the cryptanalyst can submit an unlimited number of plaintext messages of his own choosing and examine the resulting cryptograms.

┘

**Prop. (38.2.1.6) [Known Plaintext Attacks and One-way Functions].** A cryptosystem (38.2.1.2)  $\{S_k\}_{k \in K} : \mathcal{P} \rightarrow \mathcal{C}$  which is secure against a known plaintext attack can be used to produce a one-way function  $K \rightarrow \mathcal{C}$ , by restricting to a fixed  $p \in \mathcal{P}$ . ┘

**Def. (38.2.1.7) [Digital Authentications(Signature)].** In order to digitally sign a document, each user must be able to produce a message whose authenticity can be checked by anyone, but which could not have been produced by anyone else, even the recipient. ┘



### Public Key Cryptosystems, Diffie-Hellman Scheme

References are [AAG99] and [D-H76].

**Algo. (38.2.1.8) [Public Key Cryptosystem, Diffie-Hellman [D-H76]].** A **public key cryptosystem** with signature works as follows:

- Each user A finds an encryption method  $E_A$  and a decryption method  $D_A$  of data, such that

$$E_A \circ D_A = D_A \circ E_A = \text{id},$$

and both  $E_A$  and  $D_A$  are easy to compute, but it is hard or ineffective to know  $D_A$  from  $E_A$ .

- Each user A reveals  $E_A$ , and keeps  $D_A$  as secret.
- If user A wants to send message  $M$  to user B, he sends the message  $E_B(D_A(M))$  to user B.
- If user B receives a message  $S$  from user A, he gets the plaintext  $E_A(D_B(S))$ .

In this way, one-way authentication is also achieved.  $\perp$

*Proof:* The security of this system depends on the fact that it is ineffective to compute  $M$  from  $E_B(D_A(M))$ .

User B can prove the message is sent from user A because he has the texts  $M$  and  $D_A(M)$ , which is possible only because user A sent him the text  $D_A(M)$ , as it is hard to compute  $D_A$ .

User B cannot forge the message because he cannot compute  $D_A(M)$  from the message  $M$ .  $\square$

**Prop. (38.2.1.9) [Cryptoanalysis].** The cryptanalytic difficulty of a system whose encryption and decryption operations can be done in P time cannot be greater than NP.  $\perp$

**Algo. (38.2.1.10) [RSA Public Key cryptosystem, Rives-Shami-Adleman [RSA78]].** The (Rives-Shami-Adleman) **RSA public key cryptosystem** works as follows:

- Chooses two large primes  $p, q$  using primality testing, and calculate  $N = pq$ .
- Encode data in series  $(a_i)$  of positive integers less than  $N$ .
- Then user A chooses randomly an integer  $2 \leq e \leq \varphi(N) - 2$  s.t.  $(e, p) = (e, q) = 1$ , and use the extended Euclid algorithm to find the inverse

$$d = (\text{rem } ()e^{-1}(\text{mod } \varphi(N))).$$

- User A reveals  $(N, e)$ , which is called the **Public RSA key** of user A.
- The encryption method to user A is mapping a series of positive integers  $(a_i)$  less than  $N$  to  $E_A(a_i) = (\text{rem } ()a_i^e(\text{mod } N))$ .
- The decryption method of user A is changing a series of positive integers  $(s_i)$  less than  $N$  to  $D_A(s_i) = (\text{rem } ()s_i^d(\text{mod } N))$ .
- Then use Diffie-Hellman cryptosystem scheme as described in (38.2.1.8).  $\perp$

*Proof:* It is easy to show that  $D_A \circ E_A = E_A \circ D_A = \text{id}$ . By (38.2.1.8), the security of this cryptosystem depends on the difficulty of calculating  $d$ . It is shown that if  $d$  is known, then we can factor  $n$  effectively  $\color{red}{?}$ , which is assumed to be difficult.  $\square$

**Algo. (38.2.1.11) [Discrete Logarithm Key Establish Protocol, Diffie-Hellman [D-H76]].** Choose a  $p \in \mathbf{P}$  and a primitive root  $g$  for  $p$ , then there is a key establish protocol as follows:

1. User A choose a random chosen  $n \in \mathbb{Z}_+, n < p$ , user B choose a random chosen  $m \in \mathbb{Z}_+, n < p$ .
2. User A sends  $g^n(\text{mod } p)$  to user B, and user B sends  $g^m(\text{mod } p)$  to user A.
3. User A calculates  $(g^n)^m(\text{mod } p)$ , and user B calculates  $(g^m)^n(\text{mod } p)$ . Then user A and user B get a same element  $s = g^{mn}(\text{mod } p)$ .
4. Now User A can use this element  $s$  to send user B messages 1 bit a time.

┘

*Proof:* The security of this cryptosystem is based on the difficulty of computing discrete logarithm over finite fields ?. Because it is not feasible to determine any of  $g^{mn}(\text{mod } p), g, m, n$  from  $p, g^n(\text{mod } p)$  and  $g^m(\text{mod } p)$ . □

**Algo.(38.2.1.12)** [An Algebraic Key Establish Protocol, Anshel-Andhel-Goldfeld]. Let  $U, V$  be monoids and

$$\beta : U \times U \rightarrow V, \gamma_0, \gamma_1 : U \times V \rightarrow V$$

be feasibly computable functions satisfying the following properties:

- $\beta(x, y_1 y_2) = \beta(x, y_1) \beta(x, y_2)$ .
- $\gamma_1(x, \beta(y, x)) = \gamma_2(y, \beta(x, y))$ .
- Given  $y_1, y_2, \dots, y_k \in U$  and  $\beta(x, y_1), \dots, \beta(x, y_k)$ , it is in general infeasible to determine the element  $x$ .

Then there is a key establish protocol as follows:

1. Publicly assign users A, B elements  $S = \{s_1, \dots, s_m\}$  and  $T = \{t_1, \dots, t_n\}$ .
2. User A choose some  $a \in S$  and transmit the elements  $\beta(a, t_i)$  to user B.
3. User B chose some  $b \in T$  and transmit the elements  $\beta(b, t_i)$  to user A.
4. Then item1 guarantees that user A can calculate  $\beta(b, a)$  and thus  $\gamma_1(a, \beta(b, a))$ . Similarly user B can calculate  $\beta(a, b)$  and thus  $\gamma_2(b, \beta(a, b))$ . Then notice

$$\kappa = \gamma_1(a, \beta(b, a)) = \gamma_2(b, \beta(a, b))$$

is an established key.

But to extract an identical element, there are two cases: If there is a feasible algorithm to put every word in  $U$  in canonical form, then they get an identical element. Otherwise there are no canonical form algorithm but has a fast word identifying algorithm, then user A can choose a random word  $\tau$  other than  $\kappa$  and then choose either send  $\tau$  or  $\kappa$  to user B. Then user B can determine if this is  $\kappa$  or not, and receives a bit in this way.

┘

*Proof:* The security of this cryptosystem is ? □

**Remark(38.2.1.13).** An example is given by  $U = V = G \in \text{Grp}$  and

$$\beta(x, y) = x^{-1}yx, \quad \gamma_1(u, v) = u^{-1}v, \quad \gamma_2(u, v) = v^{-1}u.$$

Then

$$\kappa(a, b) = [a, b].$$

┘

**Algo.(38.2.1.14) [ECDH(Elliptic Curve Diffie-Hellman) Key Establishment Protocol, Koblitz-Miller].** This protocol is based on hardness of computing discrete logarithm on elliptic curves  $E$  over a finite field. It is totally analogous to the discrete logarithm key establish protocol. For details, Cf.[Kob87].

Another subtle point is that we need to embed plaintext into  $E(\mathbb{F}_q)$ , and find a point of  $E(\mathbb{F}_q)$  whose order is divisible by a large prime.  $\perp$

**Algo.(38.2.1.15) [SIDH(Supersingular Isogeny Diffie-Hellman) Cryptosystem, Jao-Feo].** WARNING: The SIDH protocol has been cracked. SIDH is based on isogenies of a supersingular elliptic curve  $E/\mathbb{F}_{p^2}$ .  $\perp$

### Probabilistic Cryptosystems

**Algo.(38.2.1.16) [NTRU, Hoffstein-Pipher-Silverman[HPS98]].** The NTRU cryptosystem depends on three integer parameters  $(N, p, q)$  s.t.  $(p, q) = 1$ , and four sets of integer polynomials of degree  $N - 1$ :  $\mathcal{L}_f, \mathcal{L}_g, \mathcal{L}_\varphi, \mathcal{L}_m$  (These parameters must satisfy conditions as in(38.2.1.17)). It works as follows:

- Encode data in functions in  $\mathcal{L}_m$ .
- User A randomly choose two polynomials  $f, g \in \mathcal{L}_g$ , s.t.  $f$  must be invertible in both  $\mathbb{F}_p[X]/(X^N - 1)$  and  $\mathbb{F}_q[X]/(X^N - 1)$ , with inverses  $F_p$  and  $F_q$ .
- User A reveals the polynomial  $h = F_q g \in \mathbb{F}_q[X]/(X^N - 1)$ , which is his **public key**.
- The encryption method of User A is probabilistic: Choose a random function  $\varphi \in \mathcal{L}_\varphi$ , and maps a function  $m \in \mathcal{L}_m$  to  $E_A(m) = p\varphi h + m \in \mathbb{F}_q[X]/(X^N - 1)$ .
- The decryption method of User A is: Given a  $e \in \mathbb{F}_q[X]/(X^N - 1)$ , compute  $a = fe$ , and makes the coefficients of  $a$  in  $[-\frac{q}{2}, \frac{q}{2}]$ , then  $D_A(e) = aF_p \in \mathbb{F}_p[X]/(X^N - 1)$ .  $\perp$

*Proof:* Firstly the method works because  $a \equiv p\varphi g + fm \pmod{(q, X^N - 1)}$ , and the conditions on the parameters(38.2.1.17) implies that  $a = p\varphi g + fm \in \mathbb{Z}[X]/(X^N - 1)$  because  $q$  is large compared to  $p, g, f, m$ . Thus  $aF_p = m \in \mathbb{F}_p[X]/(X^N - 1)$ .

For security of this cryptosystem, Cf.[NTRU: A Ring-Based Public Key Cryptosystem].?  $\square$

**Remark(38.2.1.17)[Parameter Selections].** Cf.[NTRU: A Ring-Based Public Key Cryptosystem].?  $\perp$

### Distributed Data Structures

**Def.(38.2.1.18)[Secret Sharing].** The goal of **secret sharing** is to share a common secret in such a way that all of a group of people together can reconstruct the secret but any proper subset of them cannot do so.  $\perp$

**Prop.(38.2.1.19)[Secret Sharing via Lagrange Interpolation].** For  $n \in \mathbb{Z}_{\geq 2}$ , if we want to share a secret among  $n$  peoples, we can choose a (large)field  $k \in \mathbf{Field}$ , and encode the information in a polynomial  $f \in k[X]$  of degree  $n - 1$ . Then we can choose distinct numbers  $a_1, \dots, a_n \in k$ , and give each user  $i$  the number  $f(a_i)$ . In this way, it is easy to see that only all of them together can reconstruct the original information, through Lagrange interpolation.

The same method can be modified to store data among users that are robust to failure of some of them.  $\perp$

## 2 Quantum Cryptography

**Algo.(38.2.2.1) [Ekert's Key Establish Protocol].** The **Ekert's key establish protocol** is the following:

- A source emits a sequence of pair of entangled singlet half-spin particles to users A and B resp..
- For each particle, user A randomly choose from the three axis:  $0, \pi/4, \pi/2$  to measure the spin of the particle.
- For each particle, user B randomly choose from the three axis:  $\pi/4, \pi/2, 3\pi/4$  to measure the spin of the particle.
- User A and user B publish their data of axis of measurements, and determine the group particles that their axis of measurements are the same.
- Then they can use Bell's inequality on the group of particles with different axis of measurements as in [Eke91] to determine if there are eavesdroppers<sup>?</sup>.
- Finally, for the group of particles that they measured with the same axis of measurements, their result are opposite. Thus they get a sequence of same keys.

┘

*Proof:* The security of this protocol follows from the no-cloning theorem(18.4.1.14) and also cannot measure a particle without disturbing the purity of the entanglement. Cf.[Neilson, Quantum Informations].  $\square$

## 3 Signal Theory

### Recovering Sparse Signals

References are [Stable signal recovery from incomplete and inaccurate measurements, Tao], [Decoding by linear programming, Tao].

**Def.(38.2.3.1) [Signal Recovering Problem].** Suppose we have a signal  $x_0 \in \mathbb{R}^m$  whose support has small cardinality, we want to recover  $x_0$  from  $n$ -measurements

$$y_k = (x_0, a_k)$$

where  $a_k \in \mathbb{R}^m$  are called **test signals**. Equivalently, we want to recover  $x_0$  from  $Ax_0$  where  $A \in \text{Mat}(n \times m; \mathbb{R})$ .  $\perp$

**Def.(38.2.3.2) [Imperfect Signal Recovering Problem].** Suppose we have a signal  $x_0 \in \mathbb{R}^m$  whose support has small cardinality, we want to recover  $x_0$  from  $Ax_0 + e$  where  $A \in \text{Mat}(n \times m; \mathbb{R})$  and  $e \in \mathbb{R}^n$  is some unknown perturbation with  $\|e\|_2 \leq \varepsilon$ .  $\perp$

**Def.(38.2.3.3) [ $k$ -restricted Isometry Constants].** For  $A \in \text{Mat}(n \times m; \mathbb{R})$  and  $k \in [m]_+$ , the  $k$ -restricted isometry constant  $\delta_k(A)$  is defined to be the smallest quantity  $\delta$  s.t. if  $T \subset [m]_+$  and  $\#T \leq k$ , then the matrix  $A_T$  (obtained from  $A$  by subtracting columns corresponding to  $T$ ) satisfies

$$(1 - \delta)\|c\|_2^2 \leq \|A_T c\|_2^2 \leq (1 + \delta)\|c\|_2^2$$

for any coefficient sequences  $(c_j)_{j \in T}$ .  $\perp$

**Example(38.2.3.4) [Matrices with Small Isometry Constants].** • Random matrices with i.i.d. entries.

- Fourier ensemble.
- General orthogonal measurement ensembles.

┘

*Proof:* Cf.[Near-optimal signal recovery from random projections and universal encoding strategies].  $\square$

**Thm.(38.2.3.5) [Exact Decoding by Linear Programing].** In the signal recovering problem(38.2.3.1), if  $x_0 \in \mathbb{R}^m$ ,  $A \in \text{Mat}(n \times m; \mathbb{R})$  and  $k \in \mathbb{Z}_+$  satisfies:

- The support  $T_0$  of  $x_0$  satisfies  $\#T_0 \leq k$ .
- $\delta_k(A) + \delta_{2k}(A) + \delta_{3k}(A) < 1$ (38.2.3.3).

Then  $x_0$  attains the minimal  $\|x\|_1$  among  $x \in \mathbb{R}^m$  that satisfies the constraints  $Ax = y$ .  $\square$

*Proof:* Cf.[Decoding by linear programming, Tao].?  $\square$

**Thm.(38.2.3.6) [Stable Signal Recovery].** In the imperfect signal recovering problem(38.2.3.2), if  $x_0 \in \mathbb{R}^m$ ,  $A \in \text{Mat}(n \times m; \mathbb{R})$ ,  $e \in \mathbb{R}^n$  and  $k \in \mathbb{Z}_+$  satisfies:

- The support  $T_0$  of  $x_0$  satisfies  $\#T_0 \leq k$ .
- $\delta_{3k}(A) + 3\delta_{4k}(A) < 2$ (38.2.3.3).

Then the solution  $x^\sharp$  to the convex programing problem:

$$\min \left( \|x\|_1 : \|Ax - y\|_2 \leq \|e\|_2 \right)$$

satisfies

$$\|x^\sharp - x_0\| \leq C_k \|e\|_2,$$

where  $C_k$  only depends on  $\delta_{4k}(A)$ .  $\square$

**Remark(38.2.3.7).** From a certain viewpoint, recovering via the convex programing problem is using a priori information about the nature of the underlying image, i.e., that it is sparse in some known orthogonal basis, to overcome the shortage of data.  $\square$

*Proof:* Cf.[Decoding by linear programming, Tao].?  $\square$

**Thm.(38.2.3.8)[Stable Signal Recovery of Largest Entries].** Situation as in(38.2.3.6), if  $x_0 \in \mathbb{R}^m$  is arbitrary and  $x_{0,k} \in \mathbb{R}^m$  is the truncated vector corresponding to the  $k$  largest vectors in  $x_0$ (in absolute value), then under the same hypothesis, Then the solution  $x^\sharp$  to the convex programing problem:

$$\min \left( \|x\|_1 : \|Ax - y\|_2 \leq \|e\|_2 \right)$$

satisfies

$$\|x^\sharp - x_0\| \leq C_{1,k} \|e\|_2 + C_{2,k} \frac{\|x_0 - x_{0,k}\|_1}{\sqrt{k}},$$

where  $C_{1,k}$  and  $C_{2,k}$  only depends on  $\delta_{4k}(A)$ .  $\square$

*Proof:* Cf.[Decoding by linear programming, Tao].?  $\square$

## 4 Error Correction

**Def. (38.2.4.1) [Error Corrections].** **Error correction** are data structure designations that are used to eliminate the perturbations in the process of transmitting the data.  $\lrcorner$

**Prop. (38.2.4.2) [Error Correction via Lagrange Interpolation].** We can divide the data into  $n$  part and encode them in a polynomial  $f$  of degree  $n - 1$  over a (large)field  $k \in \mathbf{Field}$ . In this way, we can transmit the data  $(u_i, f(u_i))$  for any  $u_i \in k$ . Then sending  $n + l$  such pairs will be robust to the failure of losing  $l$  of them, by recovering the information via Lagrange's interpolation.  $\lrcorner$

## 38.3 French to English Dictionary

### 1 A

- a: have
- aide: help
- Ajoutons: Let's add
- apres: after
- alors: then
- ans: years
- avoir: have
- aussi: also
- aux: to the

### 2 B

- bornee: bounded.

### 3 C

- carre: square
- cas: case
- cet: this
- chaleureusement: warmly
- choix: choice
- classifiant: classifying
- connexe: connected
- considere: consider
- corp: field
- courbe: curve

### 4 D

- dans: in
- d'apres: according to
- de: of, than
- deduit: deduced
- definie: defined
- demi: semi
- des: of
- divise: divide
- du: of
- d'une: of a

### 5 E

- en: in
- encore: again
- enonce: state
- entire: integer
- est: is
- et: and
- etabli: established
- être: be

### 6 F

- facon: way
- fournit: provide

### 7 G

- générale: general
- grandes: large

**8 H**

- 

**9 I**

- |                    |                          |                                     |
|--------------------|--------------------------|-------------------------------------|
| • il y a: there is | • introduite: introduced | • indépendamment: inde-<br>pendence |
|--------------------|--------------------------|-------------------------------------|

**10 J**

- Je: I

**11 K**

- 

**12 L**

- |           |                 |              |
|-----------|-----------------|--------------|
| •         | • les: the      | • lues: read |
| • la: the | • lignes: lines |              |
| • le: the | • lorsque: when |              |

**13 M**

- |             |                                    |                      |
|-------------|------------------------------------|----------------------|
| • ma: my    | • Mentionnons: Let us men-<br>tion | • modulaire: modular |
| • mais: but |                                    | • munie: equipped    |

**14 N**

- |                  |                |            |
|------------------|----------------|------------|
| •                | • notee: noted | • nous: we |
| • nombre: number | • notons: note |            |

**15 O**

- |          |                  |          |
|----------|------------------|----------|
| • on: we | • opere: operate | • ou: or |
|----------|------------------|----------|

**16 P**

- |   |                    |             |
|---|--------------------|-------------|
| • | • par: by, through | • peut: can |
|---|--------------------|-------------|



- peuvent: can
- plus: more
- poids: weights
- pour: for, to
- premier: prime
- prefere: prefer
- preuve: proof
- principales: main
- presenterons: introduce
- prolonge: extended
- proprieties: properties

## 17 Q

- que: that
- qui: who

## 18 R

- rationalite: rationality
- reel: real
- recement: recently
- remercie: thank
- renvoyant: returning
- reposant: relaxing
- reste: rest

## 19 S

- 
- sa: his
- sait: know
- sont: are
- ses: its, his, her
- schema: scheme
- strategie: strategy
- son: his
- sont: are
- suit: follows
- suivante: following
- sur: on, about

## 20 T

- temperees: tempered
- texte: text
- theoreme: theorem
- these: thesis
- toroidale: toroidal
- tout: every, all
- traiter: treat

## 21 U

- 
- un: a
- une: a

## 22 V

- variante: variant
- variete: variety
- vectoriel: vector
- voir: see

23 W

•

24 X

•

25 Y

•

26 Z

•

# Notations

This chapter consists of definitions and notations that are used often. These notations will be used throughout this book without referring, so if you are confused about some notation, you might find it here.

Let's call Book/Chapter/Section/Subsection/... **classes**. Sometimes in the beginning of a class, there are notation assignments. They influence only the propositions contained in the same class. The notations in this chapter should be regarded as the notation for the whole book.

When they contradict(which I will try to avoid from happening), they have ascending priorities(e.g. notations assigned in the beginning of a chapter will be overrode by the notation assigned in the beginning of a section in this chapter). In particular, any notation assignment has higher priorities than notation assignments in this chapter.

## 27 General

### Notation(38.3.27.1).

- Use “;” instead of “,” to separate condition/hypothesis with results/conclusions.
- $\in$  is an abbreviation for “in” or “be/is/are in”, depending on the grammar syntax. So I will use this symbol more casually than people usually do.
- $i$  is a chosen square root of  $-1$  in  $\mathbb{C}$ . It also means any element in a ring  $R$  when there is a chosen map  $Z(i) \rightarrow R$ .
- Use  $\epsilon$  to denote an element or a function valued in  $\{\pm 1\}$ .
- a.e. is an abbreviation for “all but finitely many”.
- e.g. is an abbreviation for “for example”.
- f.m. is an abbreviation for “finitely many”.
- FTSOC is an abbreviation for “for the sake of contradiction”.(Seldom used).
- i.e. is an abbreviation for “which means”.
- n.n. is an abbreviation for “not necessarily”.
- resp. is an abbreviation for “respectively”.
- s.t. is an abbreviation for “such that”.
- “sufficiently large” means “an integer greater than a given integer  $N_0$  that only depends on the parameters defined in this particular situation”.
- WLOG is an abbreviation for “without loss of generality”.

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## 28 Set Theory

**Notation (38.3.28.1) [Sets].** For  $X \in \mathsf{Set}$ ,

- Use `\mrs` to denote a set-theoretic object.
- Use  $\#X$  instead of  $|X|$  (obsolete) to denote the cardinality of a set  $X$ .
- $\mathcal{P}(X)$  is the power set of  $X$ .
- To avoid confusion with other structures on sets, use bijection of sets instead of isomorphism of sets to indicate an isomorphism in  $\mathsf{Set}$ .
- Use  $U \subset X$  to indicate  $U$  is a subset of  $X$ . Use  $U \subsetneq X$  to indicate  $U$  is a proper subset of  $X$ .
- Use `\dutchcal` capital to represent a property on a set.
- If  $\mathcal{P}$  is a property on  $X$ , then use  $\{x \in X : \mathcal{P}(x)\}$  or  $\{x \in X | \mathcal{P}(x)\}$  to indicate the subset of  $X$  defined by the property  $\mathcal{P}$ , depending on which avoids confusion better.
- If  $S \subset \mathcal{P}(X)$ , we say a property  $\mathcal{P}$  holds for elements of  $Y$  sufficiently large if it holds for all elements of  $Y$  containing a fixed subset  $S_0 \subset X$ . Notice this definition is compatible with the definition of numbers being sufficiently large, by taking  $X = \aleph_0$  and  $Y = \mathbb{N} \subset \mathcal{P}(\aleph_0)$ .
- Use `\mathsf{mathtt}` capital to represent a set that doesn't form a category with good properties. For example, `GField` represents the set of global fields.
- If  $f : X \rightarrow X$  is a self-map and  $S \subset X$ , then  $f$  is said to fix  $S$  or stabilize  $S$  if  $f(S) \subset S$ .
- For  $n \in \mathbb{Z}_+$ , use  $\underline{a}^n$  to denote an  $n$ -tuple of elements  $(a_1, \dots, a_n)$ . And the superscript  $n$  can be omitted sometimes.
- the Kronecker delta function  $\delta^X(i, j) : X^2 \rightarrow \mathbb{N}$  is the function:  $\delta^X(i, j) = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$ . It is often denoted by  $\delta_{i,j}^S$  or  $\delta_{ij}^S$ , or when  $S$  is clear in the situation, just  $\delta_{i,j}$  or  $\delta_{ij}$ .

┘

**Notation (38.3.28.2) [Common Sets].**

- For  $X \in \mathbb{R}_+$ ,  $[X]$  is the set  $\mathbb{N} \cap [0, [X]]$ .
- For  $X \in \mathbb{R}_+$ ,  $[X]_+$  is the set  $\mathbb{Z}_+ \cap [0, [X]]$ .
- $\mathbb{Z}$  is the ring of integers.
- $\mathbb{Z}_+ \subset \mathbb{Z}$  is the commutative multiplicative monoid of positive integers.
- $\mathbb{Z}_- \subset \mathbb{Z}$  is the set of negative integers.
- $\mathbb{N} \subset \mathbb{Z}$  is the commutative multiplicative monoid of non-negative integers.
- $\mathbb{Q}$  is the field of rational numbers.
- $\mathbb{R}$  is the complete ordered valued field of real numbers.
- $\mathbb{R}_+$  is the ordered commutative multiplicative monoid of positive real numbers.
- $\mathbb{R}_-$  is the set of negative real numbers.
- $\mathbb{C}$  is the complete valued field of complex numbers.

┘

## 29 Categories

**Notation (38.3.29.1) [Categories].**

- Use `\mathscr` capital for symbols representing a category.
- Use `pr` for morphisms that look like projections.

┘

**Notation (38.3.29.2) [Common Categories].** To avoid the set-theoretical issue that the class of all sets is not a set, we fix a cardinal  $\kappa$ , which will be given in prior in each situation, and define

- $\mathbf{Set}$  to be the category of sets with cardinality  $< \kappa$ .
- $\mathbf{Grp}$  to be the category of groups with cardinality  $< \kappa$ .
- $\mathbf{Ab}$  to be the category of Abelian groups with cardinality  $< \kappa$ .
- $\mathbf{Grp}^{\text{fin}}$  to be the category of finite groups.  $\mathbf{Ab}^{\text{fin}}$  to be the category of finite Abelian groups.
- $\mathbf{CAlg}$  to be the category of commutative unital rings with cardinality  $< \kappa$ .
- $\mathbf{CAlg}^p$  to be the category of commutative unital rings of characteristic  $p$  with cardinality  $< \kappa$ . This is a legal syntax if  $p \in \mathbf{Prime} \cup \{0\}$ .
- $\mathbf{Field}$  to be the category of fields (3.2.1.3) with cardinality  $< \kappa$ .
- $\mathbf{Field}^p$  to be the category of fields of characteristic  $p$  (3.2.1.4) with cardinality  $< \kappa$ . This is a legal syntax if  $p \in \mathbf{Prime} \cup \{0\}$ .
- $\mathbf{Top}$  to be the category of topological spaces with cardinality  $< \kappa$ .
- $\mathbf{TopGrp}$  to be the category of topological groups with cardinality  $< \kappa$ .
- $\mathbf{Cat}$  to be the category of categories with cardinality  $< \kappa$ .
- $\mathbf{Diff}$  to be the category of smooth manifolds.
- $\mathbf{Diff}^n$  to be the category of smooth manifolds of dimension  $n$ . This is a legal syntax if  $n \in \mathbb{Z}_+$ .
- In generally, whenever we define the category of some objects, we mean the category of objects with cardinality  $< \kappa$ . This applies to the whole book.

┘

**Remark (38.3.29.3).** If the cardinal  $\kappa$  in (38.3.29.2) is chosen to be too small, some construction in the categories won't be possible. For example, if you take  $\kappa = 2$ , then  $\{\emptyset\} \coprod \{\emptyset\}$  is not definable in  $\mathbf{Set}$ . If you take  $\kappa = \aleph$ , then  $\coprod_{\aleph} \{\emptyset\}$  is not definable in  $\mathbf{Set}$ .

So usually  $\kappa$  is taken to be a large enough strongly inaccessible cardinal (whose existence depends on the large cardinal axiom). But in specific cases we can take  $\kappa$  to be smaller to show some proposition is invariant of the axiom of large cardinals, for example the Weil conjecture (Deligne's theorem).

┘

**Notation (38.3.29.4) [Categories].** For  $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}$ ,  $A \in \mathcal{C}$ ,

- $\mathcal{C}/A$  or  $\mathcal{C}_{/A}$  is the slice category of objects over  $A$ ,  $\mathcal{C}_{A/}$  is the slice category of objects under  $A$ .
- $\mathbf{Func}(\mathcal{C}, \mathcal{D})$  is the category of functors from  $\mathcal{C}$  to  $\mathcal{D}$ .
- $\mathcal{C}^{\text{op}}$  is the category with the arrows reversed.

┘

**Notation (38.3.29.5) [Monoidal Categories].** For a monoidal category  $(\mathcal{C}, \otimes)$

- $\mathcal{C}^{\text{opp}}$  is the monoidal category with the same underlying category and the tensor product  $\otimes$  reversed.

┘

### Homological Algebra

**Notation(38.3.29.6)[Homological Algebras].**

- Use  $\hat{\bullet}$  capital for symbols representing a complex over an Abelian category.
- Use  $K$ -groups instead of  $K$ -groups.

┘

## 30 Topology

**Notation(38.3.30.1)[Topological Spaces].**

- $\text{cntd}$  is an abbreviation for “connected”.
- For a metric space  $X$  and  $x \in X, \delta \in \mathbb{R}_+$ , denote  $U(x, \delta) = \{y \in X : d(x, y) < \delta\}, \mathbb{D}(x, \delta) = \{y \in X : d(x, y) \leq \delta\}$ .
- For  $X \in \mathcal{T}\text{op}$ , let  $X_0$  to denote the closed points of  $X$ .

┘

**Notation(38.3.30.2)[Common Spaces].**

- For  $n \in \mathbb{Z}_+$ ,  $\mathbb{S}^n$  is the unit sphere  $\{x \in \mathbb{R}^n \mid \|x\| = 1\}$ .
- For  $n \in \mathbb{Z}_+$ ,  $\mathbb{D}^n$  is the unit disk  $\{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ .  $\mathbb{D}^2$  is also denoted by  $\mathbb{D}$ .
- For  $n \in \mathbb{Z}_+$ ,  $\mathbb{I}^n$  is the unit cube  $\{x = (x_i) \in \mathbb{R}^n \mid |x_i| \leq 1\}$ .  $\mathbb{I}^1$  is also denoted by  $\mathbb{I}$ .
- For  $n \in \mathbb{Z}_+, a < b \in \mathbb{R}_{\geq 0}$ ,  $\mathbb{D}^n(a, b)$  is the annulus  $\{x \in \mathbb{R}^n \mid a < \|x\| < b\}$ . Similarly, we can define  $\mathbb{D}^n[a, b], \mathbb{D}^n(a, b], \mathbb{D}^n[a, b]$ .
- For  $K \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ ,  $\mathbb{S}_K^\infty$  is the unit sphere in  $K^\infty$ .

┘

## 31 Algebras

**Notation(38.3.31.1)[Groups].** For  $G \in \mathcal{G}\text{rp}$ ,

- $G_{\text{ab}}$  is the abelianization of  $G$ .
- Use  $H \leq G$  to indicate that  $H$  is a subgroup of  $G$ .
- Use  $H \trianglelefteq G$  to indicate that  $H$  is a normal subgroup of  $G$ .

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**Notation(38.3.31.2)[Free Groups].**

- For  $n \in \mathbb{Z}_+$ ,  $F_n$  is the free group generated by  $[n - 1]$ .

┘

**Notation(38.3.31.3)[Abelian Groups].** For  $G \in \mathcal{A}\text{b}$ ,

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┘

**Notation(38.3.31.4)[Linear Algebras].** For  $R \in \mathcal{C}\mathcal{A}\text{lg}$ ,  $m, n \in \mathbb{Z}_+$ ,

- Use `\mathhtt{v}` lowercase to denote a vector in  $R^n$ .
- For  $V \in \mathcal{V}ect/R$ , use  $V^*$  to denote the dual space of  $V$ .
- $\text{Mat}(m \times n; R)$  is the group of  $m \times n$ -matrices with coefficients in  $R$ .  $\text{Mat}(m \times n; R)$  is also denoted by  $\text{Mat}(n, R)$ .
- $\text{GL}(n; R)$  is the group of invertible elements in  $\text{Mat}(n; R)$ .
- For  $A \in \text{Mat}(m \times n; R)$ ,  $A^t \in \text{Mat}(n \times m; R)$  is the transpose of  $A$ .
- For  $A \in \text{GL}(n; R)$ ,  $A^{-1} \in \text{GL}(n; R)$  is the inverse of  $A$ .  $A^{-t}$  denotes  $(A^{-1})^t = (A^t)^{-1}$ .

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**Prop. (38.3.31.5) [Algebras].** For  $R \in \mathcal{R}ing$ ,

- $\text{Mod}_R$  is the category of left  $R$ -modules.
- $\text{Alg}_R$  is the the category of  $R$ -algebras.
- $\text{naAlg}_R$  is the category of non-associative  $R$ -algebras.
- $\text{Alg}_R^{\text{asso}}$  is the category of associative  $R$ -algebras.
- $\text{Ring}_R$  is the category of associative unital  $R$ -algebras.
- $\mathcal{CRing}_R$  is the category of commutative unital  $R$ -algebras.

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**Notation (38.3.31.6) [Lie Algebras].**

- Use `\mathfrak{g}` (aka `\mfk`) lowercase for symbols representing a Lie algebra.

┘

## 32 Commutative Algebras

**Notation (38.3.32.1) [Fields].** For  $k \in \mathcal{F}ield$ ,

- $\text{char } k$  is the characteristic of  $k$ .
- $\bar{k}$  is a fixed algebraic closure of  $k$ .
- $k^{\text{sep}}$  is the separable closure of  $k$  in  $\bar{k}$ .
- $k^{\text{perf}}$  is the perfect closure of  $k$  in  $\bar{k}$ .
- For a Galois extension  $k'/k$ , use  $\text{Gal}(k'/k)$  instead of  $\text{Gal}_{k'/k}$  or  $G(k'/k)$  or  $G_{k'/k}$  to denote the Galois group of  $k'/k$ .
- $\text{Gal}_k = \text{Gal}(k^s/k)$ .
- $c \in \text{Gal}(\mathbb{C}/\mathbb{R})$  is the complex conjugation. For  $a \in \mathbb{C}$ ,  $c(a)$  is also denoted by  $\bar{a}$ .
- For  $p \in \mathcal{P}rime$ ,  $\mathbb{F}_p$  is the a fixed finite field of order  $p$ .
- For  $p \in \mathcal{P}rime$ ,  $\mathbb{F}_{p^r}$  is the fixed finite field of order  $p^r$  contained in  $\bar{\mathbb{F}}_p$ .
- $\mu(k)$  is the group of roots of unity in  $k$ .

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**Notation (38.3.32.2) [Commutative Algebras].**

- Use `\mathfrak{p}` lowercase for a first prime ideals. Use `\mathfrak{P}` uppercase for a prime ideal that is over a given prime ideal.

- Let  $R \in \mathcal{CRing}$  and  $f$  a non-zero-divisor in  $R$ , then use  $R[\frac{1}{f}]$  instead of  $R_f$  to denote the localization ring at  $f$ .

┘

**Notation(38.3.32.3)[Integral Domains].** For  $R \in \mathcal{CRing}$ ,

- Use the phrase “let  $(R, K)$  be an integral domain” to indicate that  $R$  is an integral domain with fraction field  $K$ .
- $R^\times$  is the commutative monoid of non-zero elements in  $R$ .
- $R^*$  is the commutative group of units of  $R$ .

┘

**Notation(38.3.32.4)[Local Rings].**

- Use “let  $(R, K, \mathfrak{m}, k)$  be an integral local ring” to indicate that  $R$  is a local ring with maximal ideal  $\mathfrak{m}$ ,  $\kappa(\mathfrak{m}) = k$ , and  $\text{Frac}(R) = K$ .
- DVR is the synonym for “discrete valuation ring”.
- CDVR is the synonym for “complete discrete valuation ring”.

┘

**Notation(38.3.32.5)[Dedekind Domains].**

- Use `\mathcal{d}` uppercase for symbols representing a Dedekind domain.?
- Use  $\mathfrak{D}$  to denote differentials of extensions of Dedekind domains.
- Use  $\mathfrak{d}$  instead of  $\delta$  to denote discriminants of extensions of Dedekind domains.

┘

**Notation(38.3.32.6)[Extensions of Dedekind Domains].** Let  $(\mathcal{O}_K, K)$  be a Dedekind domain and  $K \subset_{\text{fin}} L \in \text{Field}$ , then

- $\mathfrak{D}_{L/K} \in \text{Ideal}(\mathcal{O}_L)$  denote the different of  $L/K$ .
- $\mathfrak{d}_{L/K} \in \text{Ideal}(\mathcal{O}_K)$  denote the discriminant of  $L/K$ .

┘

### 33 Algebraic Geometry

**Notation(38.3.33.1)[Sheaves].**

- Use `\mathcal{S}` capital for symbols representing a sheaf of sets or a complex of sheaves of sets.
- Use `\mathscr{S}` capital for symbols representing a sheaf of  $\infty$ -categories.

┘

**Notation(38.3.33.2)[Schemes].** Let  $(X, \mathcal{O}_X) \in \text{Sch}$ ,

- For brevity, we can use  $X$  to denote the the scheme  $(X, \mathcal{O}_X)$ .
- Use  $|X|$  to denote the underlying topological space, and  $|X|_0$  to denote the closed points of  $|X|$ .
- Use `\mathbf{S}` capital for a relative scheme.
- Use `\mathcal{M}` capital for symbols representing an integral model.
- Use  $\kappa(x)$  to denote the residue field of a point  $x \in X$  instead of  $k(x)$ .



- Use  $R(x)$  instead of  $K(X)$  to denote the function field of an integral scheme  $X$ .
- Use  $\mathcal{O}_X(D)$  instead of  $\mathcal{L}(D)$  to denote the line bundle associated to a Cartier divisor  $D$ .

┘

**Notation(38.3.33.3)[Categories of Schemes].** Let  $X \in \text{Sch}$ ,

- $\text{Sch}$  is the category of schemes.
- $\text{NSch}$  is the category of locally Noetherian schemes.
- For  $R \in \mathcal{CAlg}$ , use  $\text{Sch}/R$  to denote  $\text{Sch}/\text{Spec } R$ .
- $\text{Sch}_{\text{int}}$  is the category of integral schemes.
- $\text{Sch}^{\text{ft}}/X$  is the category of schemes of f.t. over  $X$ .
- $\text{Sch}^{\text{loc.ft}}/X$  is the category of schemes locally of f.t. over  $X$ .
- $\text{Sch}^{\text{fét}}/X$  is the category of schemes finite étale over  $X$ . In general, use superscript to denote relative properties w.r.t.  $X$ , and superscript to denote absolute properties.
- Use  $\backslash \text{underline}$  for a functor on the category  $\text{Sch}/X$ .

┘

**Notation(38.3.33.4)[Morphisms of Schemes].**

- Use  $i$  for a closed immersion and  $j$  for an open immersion.
- For  $R \in \mathcal{CAlg}$ ,  $S \in \mathcal{CAlg}_R$ ,  $X \in \text{Sch}/R$ ,  $X \times_{\text{Spec } R} \text{Spec } S$  is denoted by  $X \otimes_R S$  or  $X_S$ .

┘

**Notation(38.3.33.5)[Modules on Schemes].** Let  $(X, \mathcal{O}_X) \in \text{Sch}$ ,

- $\text{Mod}(\mathcal{O}_X)$  is the category of  $\mathcal{O}_X$ -modules.
- $\mathcal{QCoh}(X)$  is the category of  $\mathcal{Qco}$   $\mathcal{O}_X$ -modules.
- $\mathcal{Coh}(X)$  is the category of coherent  $\mathcal{O}_X$ -modules.
- $\mathcal{Coh}^{\text{free}}(X)$  is the category of locally free sheaves on  $X$ .
- For  $k \in \mathbb{N}$ ,  $\mathcal{Coh}^{\leq k}(X)$  is the category of coherent  $\mathcal{O}_X$ -modules with  $\dim \text{Supp}(\mathcal{F}) \leq k$ .
- $D(X) = D(\mathcal{O}_X) = D(\text{Mod}(\mathcal{O}_X))$ .
- $D_{\mathcal{QCoh}}(X) = D_{\mathcal{QCoh}(X)}(\text{Mod}(\mathcal{O}_X))$ .
- $D_{\mathcal{Coh}}(X) = D_{\mathcal{Coh}(X)}(\text{Mod}(\mathcal{O}_X))$ .
- $\mathcal{D}(X) = \mathcal{D}(\mathcal{O}_X) = \text{Mod}_{\mathcal{O}_X}(\text{Sh}(X; \mathcal{D}(\mathbb{Z})))$ .
- $\mathcal{D}_{\mathcal{QCoh}}(X) = \mathcal{D}_{\mathcal{QCoh}(X)}(\text{Mod}(\mathcal{O}_X))$ .
- $\mathcal{D}_{\mathcal{Coh}}(X) = \mathcal{D}_{\mathcal{Coh}(X)}(\text{Mod}(\mathcal{O}_X))$ .

┘

**Notation(38.3.33.6)[Group Schemes].**

- $\mu$  is the group scheme  $X \mapsto \mu(\Gamma(X, \mathcal{O}_X))$ .

┘

**Notation(38.3.33.7)[Formal Schemes].**

- Use  $\mathfrak{m}k$  uppercase(aka.  $\backslash \text{mfk}$ ) for symbols representing a formal scheme that is usually not representable.

┘

### 34 Analysis

**Notation (38.3.34.1) [Numbers].**

- $\pi$  represents the smallest  $x \in \mathbb{R}_+$  s.t.  $e^{ix} + 1 = 0$ .
- $\gamma_0$  is the Euler's constant.

┘

**Notation (38.3.34.2) [Functions].** Suppose  $S \in \text{Set}$ .

- Use  $\backslash \text{mdc}$  to denote a function-theoretic space.
- For any positive-valued function  $f : S \rightarrow \mathbb{R}_+$ , I use  $O(f)$  in an equation to denote that the difference of the other part of the equation is bounded by  $Cf$ , where  $C \in \mathbb{R}_+$  is a constant that doesn't depend on any variables appearing in this equation.
- For any positive-valued function  $f : S \rightarrow \mathbb{R}_+$ , I use  $\tilde{O}(f)$  in an equation to denote that the difference of the other part of the equation is bounded by  $Cf \cdot (\log f)^k$ , where  $C, k \in \mathbb{R}_+$  is a constant that doesn't depend on any variables appearing in this equation.
- For any functions  $f, g : S \rightarrow \mathbb{C}$ , we use  $f \sim g$  or  $f = \Theta(g)$  to denote  $\{f = O(|g|) \text{ and } g = O(|f|)\}$ .

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**Notation (38.3.34.3) [Real Analysis].**

- Let  $\mathbb{D}$  be the unit disk  $\mathbb{D}(0, 1) \subset \mathbb{R}^2$  (38.3.30.1).
- $\sqrt{\phantom{x}} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  is inverse of the function  $\mathbb{R}_+ \rightarrow \mathbb{R}_+ : x \mapsto x^2$ .
- For a differentiable function  $f$ , write  $f^\wedge$  instead of  $f'$  to denote the derivative of  $f$ . For  $p \in \mathbb{Z}_+$  and a  $p$ -th differentiable function  $f$ , write  $f^{(p)}$  to denote the  $p$ -th derivative of  $f$ .
- Use  $\mathcal{S}$  to denote the space of Schwartz-Bruhat functions.
- For  $n \in \mathbb{Z}_+$ , the function  $\log_{(n)}(X) : \mathbb{R}_+ \rightarrow \mathbb{R}$  is defined inductively as follows:  $\log_{(1)}(X) = \log X$ , and  $\log_{(k+1)}(X) = \log_{(k)}(X)$  for any  $k \in \mathbb{Z}_+$ . Notice that it is defined only for  $X \in \mathbb{R}_+$  sufficiently large.

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**Notation (38.3.34.4) [Complex Analysis].**

- For  $\alpha \in \mathbb{D}$  (38.3.34.3), denote  $\psi_\alpha(z) = \psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z} \in \mathcal{M}(\mathbb{C})$ .
- For  $z \in \mathbb{C} \setminus (-\mathbb{R}_+)$ ,  $\log z$  stands for the the principle branch of  $\log z$ , and for any  $n \in \mathbb{Z}_+$ ,  $\sqrt[n]{z}$  stands for  $\exp\left(\frac{1}{n} \log z\right)$ .
- For  $n \in \mathbb{Z}_+$  and  $\xi \in \mathbb{C}^n$ ,  $e_\xi \in (\mathbb{C}^n)^\vee$  is the multiplicative character

$$e_\xi : \mathbb{C}^n \rightarrow \mathbb{C}^* : x \mapsto \exp(2\pi i \langle x, \xi \rangle).$$

Similarly, if  $V \in \text{Vect}^n / \mathbb{C}$  and  $\xi \in V^*$ , then  $e_\xi \in V^\vee$  is the multiplicative character

$$e_\xi : \mathbb{C}^n \rightarrow \mathbb{C}^* : x \mapsto \exp(2\pi i \langle x, \xi \rangle).$$

•

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**Notation (38.3.34.5) [Special Functions].**

- $\Gamma(s) \in \mathcal{M}(\mathbb{C})$  is the Gamma function.
- $\zeta(s) \in \mathcal{M}(\mathbb{C})$  is the Riemann zeta function.
- $\wp(\tau) \in \mathcal{M}(\mathbb{C})$  is the Weierstrass  $\wp$  function.

┘

**Notation(38.3.34.6)[Fourier Transforms].**

- Use  $f^\vee$  or  $\mathcal{F}f$  to denote the Fourier transform of a function  $f$  on a locally compact Abelian group.
- Sometimes, use  $f^\wedge$  to denote the inverse Fourier transform, which is just  $(x \mapsto -x) \circ f^\vee$ .

┘

**Notation(38.3.34.7)[Functional Analysis].**

- Use `dutchcal`(aka `\mdc`) capital to denote a subset of the space of continuous functions on a topological space.
- Use  $\|v\|$  to denote the norm of a vector in a normed space.
- Use  $|x|$  to denote the norm of  $x$  in a normed field. Or it can also denote the norm applied to a function with values in a normed field. For example, for a function  $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ , we can defined its  $L^2$ -norm  $\|\Phi\|_2$ , and also its absolute value  $|\Phi|$ , so

$$\|\Phi\|_2 = \left( \int_{\mathbb{R}} |\Phi|^2 dx \right)^{\frac{1}{2}}$$

┘

**Notation(38.3.34.8)[Harmonic Analysis].** Let  $G \in \text{LCG}$ .

- A left Haar measure on  $G$  is denoted by  $\mu_G$ , and a right Haar measure on  $G$  is denoted by  $\rho_G$ .

┘

**35 Geometry****Notation(38.3.35.1).**

- Use  $\Gamma$  instead of  $\Lambda$  for a lattice.

┘

**36 Differential Geometry****Notation(38.3.36.1)[Smooth Manifolds].** For  $M \in \text{Diff}$ ,

- Let  $\mathfrak{X}(M)$  denote the space of smooth vector fields on  $M$ .

┘

**Notation(38.3.36.2)[Lie Groups].**

- For  $\theta \in \mathbb{R}/(2\pi)$ , denote  $\kappa_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ .

┘

### 37 Algebraic Number Theory

**Notation(38.3.37.1)[Numbers].**

- For  $X \in \mathbb{R}$ , denote

$$\lfloor X \rfloor = \sup\{n \in \mathbb{Z} | n \leq X\}, \quad \lceil X \rceil = \inf\{n \in \mathbb{Z} | n \geq X\}, \quad \{X\} = X - \lfloor X \rfloor, \quad \{\{X\}\} = \inf\{|X - n| : n \in \mathbb{Z}\}.$$

- **Prime** is the set of rational primes.
- $\mathbb{P}$  is the algebra consisting of periods.
- Don't use “#” as an abbreviation of “number” or “numbers”.
- $\square$  is an abbreviation of “square” or “a square number”. This is obsolete and I strongly suggest not using this notation, for one reason it doesn't specify square of what numbers, and for the other reason it may be confused with unrecognizable codes.
- For  $n \in \mathbb{Z}^\times$ , use  $\mathbb{Z}/(n)$  instead of  $\mathbb{Z}/n$  or  $\mathbb{Z}/n\mathbb{Z}$ .
- Use  $p$  to denote a prime number.
- Use  $\ell$  instead of  $l$  or  $q$  to denote a second prime number when  $p$  is present.

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**Notation(38.3.37.2)[Common Functions].** For  $n \in \mathbb{Z}_+$ ,

- $\phi(n) = \#(\mathbb{Z}/(n))^*$ .

┘

**Notation(38.3.37.3)[Number Theory].**

- Use  $\varpi$  instead of  $\pi$  to denote a uniformizer in a valuation ring to avoid confusion with the frequently used  $\pi$ .
- In number theory/arithmetic geometry, try to use  $F$  to denote a global field,  $K$  a CDVR and  $k$  a finite field or general field.
- For  $p \in \mathbf{Prime}$ , use  $\overline{\mathbb{Q}_p}$  instead of  $\overline{\mathbb{Q}}_p$  to denote the fixed algebraic closure of  $\mathbb{Q}_p$ .
- For  $p \in \mathbf{Prime}$ ,  $\mathbb{Q}_{p^n}$  is the unique unramified extension of  $\mathbb{Q}_p$  of the degree  $n$  contained in  $\overline{\mathbb{Q}_p}$ , and  $\mathbb{Z}_{p^r} = \mathcal{O}_{\mathbb{Q}_{p^n}}$ .
- $\{\zeta_n, n \in \mathbb{Z}_+\}$  is a compatible system of roots of unity in  $\overline{\mathbb{Q}}$ , and  $\zeta_4 = i$ .

┘

**Notation(38.3.37.4)[Global Fields].** For  $E/F \in \mathbf{GField}$ ,

- $F_0$  is the constant field of  $F$ .
- Let  $\tilde{F}/E$  to denote the normal closure of  $F$  over  $E$ . If  $E = \mathbb{Q}$ , just write  $\tilde{F}$ .
- Use  $\mathbf{GField}$  to denote the set of global fields. Use  $\mathbf{FField}$  to denote the set of global function fields, use  $\mathbf{NField}$  to denote the set of global number fields.  $\mathbf{GField}/F$  is the set of global fields over  $F$ .
- $\mathcal{O}_F$  is the ring of integers of  $F$ .
- $\Sigma_F$  is the set of places of  $F$ .  $\Sigma_F^{\text{fin}}$  is the set of finite places of  $F$ .  $\Sigma_F^\infty$  is the set of infinite places of  $F$ .  $\Sigma_F^{\mathbb{R}}$  is the set of real places of  $F$ ,  $\Sigma_F^{\mathbb{C}}$  is the set of complex places of  $F$ .
- For a finite extension  $L/F$  and  $v \in \Sigma_F^{\text{fin}}$ ,  $\Sigma_L^v$  is the set of finite places over  $v$ .

- For  $x \in \mathcal{O}_F$ , let  $\Sigma_F^x \subset \Sigma_F^{\text{fin}}$  denote the set of finite places dividing  $x$ .
- For  $v \in \Sigma_F^{\text{fin}}$ ,  $(R_v, \mathfrak{m}_v, k_v)$  is the valuation ring of  $v$ , and  $\|v\| = \#k_v$ .
- $\text{Unr}_{E/F}, \text{Ram}_{E/F}, \text{Spl}_{E/F}$  is the set of finite places of  $F$  unramified, ramified, splitting in  $E$  resp..  $\text{Spl}_{E/F/F_0}$  is the set of finite places of  $F_0$  s.t. every place of  $F$  over  $p$  splits in  $E$ .

┘

**Notation(38.3.37.5)[Local Fields].**

- Use **LField** to denote the set of local fields.
- Use  $p$ -**LField** to denote the set of  $p$ -adic local fields.
- Use  $p$ -**FField** to denote the set of  $p$ -adic number fields.
- Use  $p$ -**NField** to denote the set of  $p$ -adic number fields.
- Use **ArchField** to denote the set  $\{\mathbb{R}, \mathbb{C}\}$ .
- Use  $(K, \mathcal{O}_K, \varpi_K, \kappa) \in p$ -**LField** to denote that  $K$  is a  $p$ -adic local field with ring of integrals  $\mathcal{O}_K$ ,  $\varpi_K$  is a uniformizer of  $K$ , and  $\kappa$  is the residue field of  $K$ .

For  $K \in p$ -**LField**,

- Let  $\sigma_K \in \text{Gal}_K$  be a geometric Frobenius of  $K$ . Which is defined up to  $I_K$ .

┘

**Notation(38.3.37.6)[Analysis on Adeles and Ideles].**

- For  $F \in \mathbf{GField}$ ; denote  $\mathbf{A}_F$  the Adele group of  $F$ .
- For  $F \in \mathbf{GField}$ ; denote  $\mathbf{C}_F$  the idele class group of  $F$ .
- For  $K \in p$ -**Field** and  $V \in \mathcal{V}ect/K$ ;  $\mathcal{S}(V)$  is the space of Bruhat-Schwartz Functions on  $V$ .
- For  $F \in \mathbf{GField}$ , and  $V$  a finite free module over  $\mathbf{A}_F$ ;  $\mathcal{S}(V)$  is the space of Bruhat-Schwartz functions on  $V$ .

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**38 Arithmetic Geometry****Notation(38.3.38.1)[Perfectoid Spaces].**

- Use old `\mathcal{a}`(aka. `\mathfrak{a}`) to denote adic spaces.

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**39 Representation Theory****Notation(38.3.39.1)[Representations].** Let  $G \in \mathcal{TopGrp}$ ,  $L \in \mathcal{TopCRing}$ ,

- $\text{Rep}_L(G)$  is the category of continuous representations of  $G$  with coefficients in  $L$ .  $\text{Rep}_{\mathbb{C}}(G)$  is also denoted by  $\text{Rep}(G)$ .
- $\text{Irr}_L(G)$  is the category of irreducible representations of  $G$  with coefficients in  $L$ .  $\text{Irr}_{\mathbb{C}}(G)$  is also denoted by  $\text{Irr}(G)$ .
- $\text{Rep}^{\text{alg}}(G)$  is the category of smooth complex representations of  $G$ .  $\text{Irr}^{\text{alg}}(G)$  is the category of irreducible smooth complex representations of  $G$ .
- For  $(\rho, V) \in \text{Rep}_L^{\text{alg}}(G)$ , let  $(\rho^{\vee}, V^{\vee})$  denote the contragradient representation:  $\rho^{\vee}(g) = \rho(g)^{-t}$ .

- Let  $\widehat{G}$  be the set of unitary representations of  $G$ , and let  $G^\vee \subset \widehat{G}$  be the subset of characters of  $G$  (i.e. 1-dimensional representations of  $G$ ). ? Try not to use  $\widehat{G}$ , because this may be confused with the Langlands dual group or the completion.

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**Notation (38.3.39.2).**

- Use D-modules instead of  $D$ -modules.

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## 40 $L$ -Functions

**Notation (38.3.40.1) [ $L$ -Functions].**

- Use  $L$ -functions instead of L-functions.
- Use  $\zeta$  for a zeta-function, i.e. a function that is usually a rational function in the indeterminate  $T$ .
- Use  $L$  for an  $L$ -function, i.e. a function that is usually a meromorphic function in  $s \in \mathbb{C}$  with Euler products.

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## 41 Combinatorics

**Notation (38.3.41.1).**

- Let  $\delta(n) : \mathbb{Z}_+ \rightarrow \mathbb{Z}$  be the function s.t.  $\delta = \delta_{n,1}$ .

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**Notation (38.3.41.2) [Combinatorics Numbers].**

- For  $m \leq n \in \mathbb{N}$ ,

$$n! = \begin{cases} 1 & , n = 0 \\ \prod_{k=1}^n k & , n > 0 \end{cases}, \quad \binom{n}{m} = \frac{n!}{(n-m)!m!}$$

- For  $m \leq n \in \mathbb{N}$  and  $v \in \mathbb{N}$ ,

$$[n]_v = \begin{cases} 1 & , n = 0 \\ v^{n-1} + v^{n-2} + \dots + v + 1 & , n > 0 \end{cases}, \quad [n]!_v = \begin{cases} 1 & , n = 0 \\ \prod_{k=1}^n [k]_v & , n > 0 \end{cases}, \quad \left[ \begin{matrix} n \\ m \end{matrix} \right]_v = \frac{[n]!_v}{[n-m]!_v [m]!_v}.$$

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## 42 Physics

**Notation (38.3.42.1) [SI System of Units].**

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**Notation (38.3.42.2) [Heaviside-Lorentz System of Units].**

- The Heaviside-Lorentz system of units can be obtained from the SI system of units by setting  $\epsilon_0 = \mu_0 = 1$ .

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